Microsoft Dynamics 365 Supply Chain Management documentation

Discover how to make the most of Dynamics 365 Supply Chain Management with training, docs, and videos covering product capabilities and how-to articles. Learn how to build an adaptable, resilient supply chain that automatically reacts to challenges using real-time visibility, agile planning, and advanced insights.

GET STARTED
Welcome to Dynamics 365 Supply Chain Management

WHAT'S NEW
What's new

TRAINING
Training

CONCEPT
Troubleshoot

Implement

Guidance

- Security, compliance, and privacy
- Implementation guide
- Implementation tools and samples
- Migrate from on-premises to online
- End-to-end business processes

Try/Buy/Deploy

- Try Dynamics 365 Supply Chain Management
- Buy Dynamics 365 Supply Chain Management
- Deploy Dynamics 365 Supply Chain Management

Set up

- Microsoft Dynamics Lifecycle Services user guide
- Project onboarding
- Set up the Warehouse management module

Administer and develop

Manage environments

- Microsoft Dynamics Lifecycle Services resources
- Continuous delivery
- Upgrades, updates, and hotfixes

Manage data storage

- Data management overview
- Data import and export
- Database movement operations
- Data entities overview

System administration

- System administration overview
- Process automation
- Batch processing overview

Organization administration

- Organization administration overview
- Organizational hierarchies
- Workflow system
- Number sequences overview
- Global address book overview

Manage user access

- Role-based security
- Create users
- Assign users to security roles

Integrate with other products

- Microsoft Office
- Microsoft Power BI
- Microsoft Power Platform
- Microsoft Azure Data Lake
- Third-party apps

Develop integration

- Integration between finance and operations apps and third party apps
- Data management REST API
- Service endpoint overview
- Dual-write overview
- Design principles and best practices for data entities

Platform development

- Develop and customize your environment
- Deploy and access development environments
- Application stack and server architecture

Extend and customize

- Extensibility overview
- Application extensibility roadmap
- Write extensible code
- Integrate with third-party manufacturing execution systems
- Extend warehouse mobile processes

Developer reference

- API references
- X++ language reference
- Open Data Protocol (OData)
- Dual-write mapping reference

Developer tools

- Development tools in Visual Studio
- Version control, metadata search, and navigation
- Build automation using Azure

Develop the user interface

- User interface development home
- General form guidelines
- Build extensible controls

Use

Warehouse management

- Warehouse management overview
- Warehouse configuration overview
- Warehouse handling of inbound loads for purchase orders
- Release outbound orders to the warehouse
- Cycle counting

Warehouse Management mobile app

- Install the mobile app
- Set up mobile devices for warehouse work
- Mobile device user accounts
- Mobile device user settings

Inventory management

- Inbound operations
- Quality assurance
- Inventory activities
- Outbound operations
- Inventory Visibility service

Transportation management

- Transportation management overview
- Transportation management engines
- Plan freight transportation routes
- The load building workbench
- Reconcile freight

Landed cost

- Landed cost overview
- Manage voyages
- Track inbound voyages and shipping container journeys
- Estimate and manage landed costs
- Landed cost reports

Product information management

- Product information overview
- Product configuration
- Product lifecycle states
- Hazardous materials
- Bills of materials and formulas

Engineering change management

- Engineering change management overview
- Release product structures
- Manage changes to engineering products
- Engineering change management feature walkthrough
- Engineering versions and engineering product categories

Procurement and sourcing

- Procurement and sourcing home page
- Procurement and sourcing overview
- Requests for quotation (RFQs)
- Purchase orders
- Supply risk assessment

Master planning

- Master planning home page
- Demand Driven Material Requirements Planning (DDMRP)
- Demand planning home page
- Set up master plans
- Manage planned orders

Manufacturing and production

Asset management

Service management

- Production process overview
- Bills of materials and formulas
- Lean manufacturing
- Catch weight product processing with warehouse management
- How workers use the production floor execution interface

- Assets
- Maintenance requests
- Work orders
- The Asset Management mobile app
- Sensor Data Intelligence

- Service management overview
- Service tasks
- Service objects
- Fulfill service agreements
- Service orders

Sales and marketing

- Sales and marketing overview
- Order promising
- Sales agreements
- Sales returns
- Customer portal

Unified pricing management

- Unified pricing management overview
- Price attributes
- Price component codes
- Price structures
- Discounts

Rebate management

- Rebate management overview
- Rebate deals
- Process, review, and post rebates
- Manage deductions

Cost accounting

- Cost accounting overview
- Cost accounting terminology
- Overhead calculation
- Dimension hierarchy

Cost management

- Cost management overview
- Inventory costing FAQ
- Standard cost conversion overview
- Costing versions overview
- Global Inventory Accounting

Reporting

- Information access and reporting
- Reporting and analytics with Power BI
- Electronic reporting (ER)
- Electronic reporting components

Related products

Dynamics 365 Finance

Monitor your global financial operations in real time, predict outcomes, and make data-driven decisions to drive...

Dynamics 365 Commerce

Dynamics 365 Commerce delivers a comprehensive omnichannel solution that unifies back-office, in-store,...

Dynamics 365 Human Resources

Connect people and operations data to optimize workforce costs and take care of your employees.

Microsoft Power Platform

Dynamics 365 Intelligent Order Management

Intelligent Order Management is a suite of capabilities to help your organization coordinate, standardize, and optimize...

Build, automate workflows, analyze data, rapidly design, configure, and publish modern websites, and build intelligen...

Dynamics 365 Project Operations

Project Operations connects sales, resourcing, project management, and finance teams in a single application ...

Welcome to Dynamics 365 Supply Chain Management

09/10/2025

This article provides an overview of help articles and other resources for Supply Chain Management.

What's new and in development

Go to Release plans for Dynamics 365, Power Platform, and Cloud for Industry to see what new features are released and what new features are in development.

Core concepts and tasks

Select a feature area to learn more about it.

- Asset management
- Cost accounting
- Cost management
- Inventory management
- Sensor Data Intelligence (preview)
- Master planning
- Demand planning
- Unified pricing management
- Procurement and sourcing
- Product information management
- Engineering change management
- Production control
- Rebate management
- Sales and marketing
- Service management
- Transportation management
- Landed cost
- Warehouse management

Videos

Check out the following collections of videos to learn more about Supply Chain Management, finance and operations apps, and other related technologies. Topics include feature introductions, event presentations, tech talks, customer stories, and more.

- Supply Chain Management videos on the Microsoft Dynamics 365 YouTube channel
- Supply Chain Management videos on the Microsoft Dynamics 365 Community YouTube channel

Blogs

The following blogs provide information about Supply Chain Management and related technologies:

- Supply Chain Management blog articles
- Dynamics AX Manufacturing R&D Team Blog (archive)

E-learning courses and training

Many online training courses for Supply Chain Management are available on the Microsoft Learn Training site.

Related documentation

Development and administration for Supply Chain Management and other finance and operations apps

For information about how to develop and administer Supply Chain Management and other finance and operations apps, go to Development and administration for finance and operations apps.

Other finance and operations apps

For information about other finance and operations apps, see the following documentation:

- Dynamics 365 Finance.
- Dynamics 365 Commerce
- Dynamics 365 Human Resources

What's new or changed in finance and operations apps home page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Application releases

To see what's new or changed in each release of a finance and operations app, see the following topics:

- **Finance:** What's new or changed in Dynamics 365 Finance
- **Human Resources:** What's new or changed in Microsoft Dynamics 365 Human Resources
- **Supply Chain Management:** What's new or changed in Dynamics 365 Supply Chain Management
- **Commerce:** What's new or changed in Dynamics 365 Commerce

Platform updates

To see what's new or changed in the Platform updates for finance and operations apps, see the following article:

- What's new or changed in Platform updates

Lifecycle Services releases

To see what's new or changed in Lifecycle Services, see the following article:

- What's new or changed in Lifecycle Services

Last updated on 04/24/2026

What's new or changed in Dynamics 365 Supply Chain Management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Releases of Dynamics 365 Supply Chain Management

To see what's new or changed in each release of Dynamics 365 Supply Chain Management, see the following articles.

Expand table

Version	Build number	Auto-update availability	Learn more
10.0.47	10.0.2527	April 2026	What's new or changed in Dynamics 365 Supply Chain Management 10.0.47
10.0.46	10.0.2428	February 2026	What's new or changed in Dynamics 365 Supply Chain Management 10.0.46
10.0.45	10.0.2345	October 2025	What's new or changed in Dynamics 365 Supply Chain Management 10.0.45
10.0.44	10.0.2263	July 2025	What's new or changed in Dynamics 365 Supply Chain Management 10.0.44
10.0.43	10.0.2177	April 2025	What's new or changed in Dynamics 365 Supply Chain Management 10.0.43
10.0.42	10.0.2095	February 2025	What's new or changed in Dynamics 365 Supply Chain Management 10.0.42
10.0.41	10.0.2015	October 2024	What's new or changed in Dynamics 365 Supply Chain Management 10.0.41
10.0.40	10.0.1935	June 2024	What's new or changed in Dynamics 365 Supply Chain Management 10.0.40

Version	Build number	Auto-update availability	Learn more
10.0.39	10.0.1860	April 2024	What's new or changed in Dynamics 365 Supply Chain Management 10.0.39
10.0.38	10.0.1777	February 2024	What's new or changed in Dynamics 365 Supply Chain Management 10.0.38
10.0.37	10.0.1725	November 2023	What's new or changed in Dynamics 365 Supply Chain Management 10.0.37
10.0.36	10.0.1695	September 2023	What's new or changed in Dynamics 365 Supply Chain Management 10.0.36
10.0.35	10.0.1627	July 2023	What's new or changed in Dynamics 365 Supply Chain Management 10.0.35
10.0.34	10.0.1591	June 2023	What's new or changed in Dynamics 365 Supply Chain Management 10.0.34
10.0.33	10.0.1549	April 2023	What's new or changed in Dynamics 365 Supply Chain Management 10.0.33
10.0.32	10.0.1515	March 2023	What's new or changed in Dynamics 365 Supply Chain Management 10.0.32
10.0.31	10.0.1406	February 2023	What's new or changed in Dynamics 365 Supply Chain Management 10.0.31
10.0.30	10.0.1362	November 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.30
10.0.29	10.0.1326	September 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.29
10.0.28	10.0.1264	July 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.28
10.0.27	10.0.1227	June 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.27
10.0.26	10.0.1192	April 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.26
10.0.25	10.0.1149	March 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.25
10.0.24	10.0.1084	February 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.24

Version	Build number	Auto-update availability	Learn more
10.0.23	10.0.1037	January 2022	What's new or changed in Dynamics 365 Supply Chain Management 10.0.23
10.0.22	10.0.995	November 2021	What's new or changed in Dynamics 365 Supply Chain Management 10.0.22
10.0.21	10.0.960	October 2021	What's new or changed in Dynamics 365 Supply Chain Management 10.0.21
10.0.20	10.0.886	August 2021	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.20
10.0.19	10.0.837	June 2021	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.19
10.0.18	10.0.793	May 2021	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.18
10.0.17	10.0.761	April 2021	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.17
10.0.16	10.0.689	February 2021	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.16
10.0.15	10.0.644	January 2021	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.15
10.0.14	10.0.605	November 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.14
10.0.13	10.0.569	October 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.13
10.0.12	10.0.507	August 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.12
10.0.11	10.0.464	July 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.11
10.0.10	10.0.420	May 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.10
10.0.9	10.0.383	April 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.9
10.0.8	10.0.319	February 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.8

Version	Build number	Auto-update availability	Learn more
10.0.7	10.0.283	January 2020	What's new or changed in Dynamics 365 Supply Chain Management version 10.0.7
10.0.6	10.0.234	November 2019	What's new or changed in Dynamics 365 Supply Chain Management 10.0.6

Releases before November 2019

To see what's new or changed in releases before November 2019, see the following articles.

Expand table

Release	Version	Build number	Availability	Learn more
Microsoft Dynamics 365 Finance and Operations	10.0.5	10.0.197	October 2019	What's new or changed in Dynamics 365 Finance and Operations version 10.0.5 (October 2019)
Microsoft Dynamics 365 Finance and Operations	10.0.4	10.0.136	July 2019	What's new or changed in Dynamics 365 Finance and Operations version 10.0.4 (July 2019)
Microsoft Dynamics 365 Finance and Operations	10.0.3	10.0.107	June 2019	What's new or changed in Dynamics 365 Finance and Operations version 10.0.3 (June 2019)
Microsoft Dynamics 365 Finance and Operations	10.0.2	10.0.80	May 2019	What's new or changed in Dynamics 365 Finance and Operations version 10.0.2 (May 2019)
Microsoft Dynamics 365 Finance and Operations	10.0.1	10.0.51	April 2019	What's new or changed in Dynamics 365 Finance and Operations version 10.0.1 (April 2019)
Microsoft Dynamics 365 Finance and Operations	10.0	10.0.8	April 2019	What's new or changed in Dynamics 365 Finance and Operations version 10.0 (April 2019)

Release	Version	Build number	Availability	Learn more
Microsoft Dynamics 365 Finance and Operations	8.1.3	8.1.227	January 2019	What's new or changed in Dynamics 365 Finance and Operations version 8.1.3 (January 2019)
Microsoft Dynamics 365 Finance and Operations	8.1.2	8.1.195	December 2018	What's new or changed in Dynamics 365 Finance and Operations version 8.1.2 (December 2018)
Microsoft Dynamics 365 Finance and Operations	8.1.1	8.1.170	October 2018	What's new or changed in Dynamics 365 Finance and Operations version 8.1.1 (October 2018)
Microsoft Dynamics 365 Finance and Operations	8.1	8.1.136	October 2018	What's new or changed in Dynamics 365 Finance and Operations version 8.1 (October 2018)
Microsoft Dynamics 365 Finance and Operations	8.0	8.0.30, 8.0.35	April 2018	What's new or changed in Dynamics 365 Finance and Operations version 8.0 (April 2018)
Microsoft Dynamics 365 Finance and Operations, Enterprise edition	7.3	7.3.11971.56116	December 2017	What's new or changed in Dynamics 365 Finance and Operations, Enterprise edition 7.3
Microsoft Dynamics 365 Finance and Operations, Enterprise edition	July 2017	7.2.11792.56024	June 2017	What's new or changed in Dynamics 365 Finance and Operations, Enterprise edition (July 2017)
Microsoft Dynamics 365 for Operations	1611	7.1.1541.3036	November 2016	What's new or changed in Dynamics 365 for Operations version 1611 (November 2016)
Microsoft Dynamics AX	7.0.1	7.0.1265.23014	May 2016	What's new or changed in Dynamics AX application version 7.0.1 (May 2016)
Microsoft Dynamics AX	7.0	7.0.1265.3015	February 2016	What's new or changed in Dynamics AX 7.0 (February 2016)

Last updated on 03/23/2026

What's new or changed in Dynamics 365 Supply Chain Management 10.0.49 (September 2026)

This article lists features that are either new or changed in Microsoft Dynamics 365 Supply Chain Management version 10.0.49. This version has a build number of 10.0.2790 and is available on the following schedule:

- **Preview of release:** July 2026
- **General availability of release (self-update):** September 2026
- **General availability of release (auto-update):** October 2026

Features included in this release

The following table lists the features that are included in this release. We might update this article to include features that were added to the build after this article was originally published.

Expand table

Module or feature area	Feature	More information	Enabled by
Accounts payable	Use financial tags in procurement documents	Use financial tags in procurement documents	Feature management: *Enable financial tags for purchase order invoicing*
Master planning	Closed for pickup policy on the transport calendar	Calendars and master planning	Feature management: *Consider open for pickup on transport calendar for planned transfer orders with Planning Optimization*
Master planning	Consider sub-BOM and sub-route in supply forecast reduction	Master planning with supply forecasts	Feature management: *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization*
Master planning	Filter derived requirements by coverage time fence with Planning Optimization	Coverage time fences	Feature management: *Filter derived requirements by*

Module or feature area	Feature	More information	Enabled by
			coverage time fence with Planning Optimization
Procurement and sourcing	Manage supplier relationships with Supplier Engagement	Supplier Engagement overview (preview)	Feature management: *(Preview) Supplier Engagement*
Procurement and sourcing	RFQ bid–driven fixed asset field updates	Purchase requisition overview	Enabled by default
Production control	Enforce same-batch selection for raw materials in production and batch orders	Reserve the same batch for a production order (preview)	Feature management: *(Preview) Use same batch selection for raw materials in production*
Warehouse management	Import inbound ASNs as despatch advice messages	Import inbound ASNs as despatch advice messages	Enabled by default
Warehouse management	Prevent multiple owners in warehouse locations	Prevent multiple owners in warehouse locations	Enabled by default
Warehouse management	Source system inventory status mapping	Source system inventory status mapping	Enabled by default

Feature enhancements added in this release

The following table lists the feature enhancements that were added in this release. Each of these enhancements provides an incremental improvement to an existing feature. Because they're only enhancements, they're not listed in the release plan and might not be mentioned elsewhere in the documentation. To ensure that these enhancements don't conflict with your existing customizations or preferences, most of these features are turned off by default, so you must turn them on in Feature management if you want to use them (unless otherwise noted).

Expand table

Module	Feature name	More information
Inventory and warehouse management	*CAPA case HTML note editor*	Enables the rich HTML note editor on the **CAPA Case Detail** page, replacing the read-only viewer. This feature lets you create and edit formatted notes directly on the CAPA case log.

Module	Feature name	More information
Inventory and warehouse management	*(Preview) Optimized inventory transaction consolidation*	Consolidate financially updated inventory transactions from closed periods into summary records to reduce table size and improve system performance. Consolidation is also a prerequisite for archiving inventory transactions with Dataverse long term retention.
Procurement and sourcing	*Synchronize establishment across the direct delivery chain*	Enables a purchase parameter and an origin prompt to control whether the system synchronizes establishments across the direct delivery chain between sales orders and purchase orders.
Production control	*Prevent shop floor activity after daily registrations are calculated*	Blocks workers from making additional shop floor registrations (such as clock-in, clock-out, job start, or job stop) after the system calculates time registrations for the day. The system displays the following error message: "You can't register any activities now because your registrations are already processed for today. Please contact your supervisor for assistance." When a worker receives this message, their supervisor should go to the **Approve** page and select **Undo calculation** for today for the affected worker.
Production control	*The feature enables batch auto-scale and priority based scheduling for message processor queues.*	Previously, the message queue setup for *Number of processor tasks* was limited to the maximum batch threads for a single batch server. When batch auto-scale is enabled, this feature makes it possible to utilize available capacity beyond a single batch. With priority based scheduling, you can also link message processor queues and batch server groups, use scheduling priority, and use maximum concurrency.
Sales and marketing	*Unified pricing management change tracking*	Enables automatic price change tracking for global Unified pricing management rules, discounts, and trade agreements. This feature writes change tracking rows to `RetailPriceChangeTracking` to notify Azure Product Search of price changes.

Preview features that are now generally available

The following table lists the features that were introduced as public preview features in a previous version and became generally available in version 10.0.49. As with other new features, most of these features are turned off by default, so you must turn them on in Feature management if you want to use them.

Expand table

Module	Feature name	More information
Inventory and warehouse management	*Optional linking of test instruments to maintainable assets*	Manage test instrument calibration with Asset Management What's new or changed in Dynamics 365 Supply Chain Management 10.0.46 (December 2025)
Master Planning	*Keep supply for confirmed demand in Planning Optimization*	Keep supply for confirmed demand in Planning Optimization What's new or changed in Dynamics 365 Supply Chain Management 10.0.48 (June 2026)
Master Planning	*Optimize confirmed dates for CTP line changes*	Optimize confirmed dates for CTP line changes What's new or changed in Dynamics 365 Supply Chain Management 10.0.48 (June 2026)
Master Planning	*Soft issue margins with Planning Optimization*	Soft issue margin What's new or changed in Dynamics 365 Supply Chain Management 10.0.48 (June 2026)
Product information management	*Configure Product Lifecycle State business process policies without Engineering change management.*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.48 (June 2026)
Sales and marketing	*Enable lookup based search for Sales External Item Identifier field*	Use external item identifiers to add products to orders What's new or changed in Dynamics 365 Supply Chain Management 10.0.45 (September 2025) This feature went directly from preview to mandatory in version 10.0.49.
Warehouse management	*(Production Ready Preview) Dynamic work classification*	Dynamic work classification What's new or changed in Dynamics 365 Supply Chain Management 10.0.48 (June 2026)

Module	Feature name	More information
		This feature went directly from preview to on by default in version 10.0.49.

Features turned on by default in this release

The following table lists the features that became turned on by default in version 10.0.49. You can still turn these features off in Feature management if necessary.

Expand table

Module	Feature name	More information
Asset Management	*Add maintenance checklist template lines to work order lines*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.47 (March 2026)
Asset Management	*Remarks can be associated with each change in lifecycle states*	Introduction to assets Introduction to work orders What's new or changed in Dynamics 365 Supply Chain Management 10.0.46 (December 2025)
Inventory and warehouse management	*Acceptance sampling*	Acceptance sampling What's new or changed in Dynamics 365 Supply Chain Management 10.0.45 (September 2025)
Inventory and warehouse management	*Configure quality associations by product dimensions*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.47 (March 2026)
Inventory and warehouse management	*Sample management*	Sample management overview Enable and configure sample management What's new or changed in Dynamics 365 Supply Chain Management 10.0.46 (December 2025)
Inventory and warehouse management	*View calibration tools for test instruments*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.46 (December 2025)

Module	Feature name	More information
Warehouse management	*(Production Ready Preview) Dynamic work classification*	Dynamic work classification What's new or changed in Dynamics 365 Supply Chain Management 10.0.48 (June 2026) This feature went directly from preview to on by default in version 10.0.49.

Features becoming mandatory in this release

The following table lists the features that became mandatory in version 10.0.49. These features still appear in Feature management but you can no longer turn them off.

Expand table

Module	Feature name	More information
Asset Management	*Aggregated material availability check*	Material availability check for work orders What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Asset Management	*Credit limit check on work order dispatch*	Bill for maintenance on customer-owned assets What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Inventory and warehouse management	*Advanced quality management*	Advanced quality management overview What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Inventory and warehouse management	*Inventory Visibility integration with ATP*	Track time-series inventory in Inventory Visibility What's new or changed in Dynamics 365 Supply Chain Management 10.0.41 (September 2024)
Inventory and warehouse	*Inventory Visibility integration with reservation offset*	Inventory Visibility reservations

Module	Feature name	More information
management		What's new or changed in Dynamics 365 Supply Chain Management 10.0.22 (November 2021)
Inventory and warehouse management	*Inventory Visibility transaction integration*	Sync external inventory adjustments through Inventory Visibility What's new or changed in Dynamics 365 Supply Chain Management 10.0.41 (September 2024)
Inventory and warehouse management	*Enable warehouse items in Inventory Visibility*	Enable Inventory Visibility for Commerce Inventory Visibility support for WMS items What's new or changed in Dynamics 365 Supply Chain Management 10.0.26 (May 2022)
Production control	*Dispense management*	Advanced quality management overview What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Sales and marketing	*Recalculate delivery dates upon creation of direct delivery*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.40 (June 2024)
Sales and marketing	*Enable lookup based search for Sales External Item Identifier field*	Use external item identifiers to add products to orders What's new or changed in Dynamics 365 Supply Chain Management 10.0.45 (September 2025) This feature went directly from preview to mandatory in version 10.0.49.
Shared AP and AR	*Consolidate vendor invoices for the same vendor in vendor rebate deals*	Process, review, and post rebates What's new or changed in Dynamics 365 Supply Chain Management 10.0.43 (March 2025)
Shared AP and AR	*Enable posting of vendor rebate outputs to purchase order vendors*	Rebate management posting setup What's new or changed in Dynamics 365

Module	Feature name	More information
		Supply Chain Management 10.0.43 (March 2025)

Features removed from feature management

The following table lists features that were removed from Feature management in version 10.0.49. These features are no longer listed in Feature management and are now a permanent part of Supply Chain Management.

Expand table

Module	Feature name	More information
Asset Management	*Print work order document attachments*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.41 (September 2024)
Asset Management	*Adjust forecast hours to match schedule hours when work orders are dispatched*	Dispatch work orders What's new or changed in Dynamics 365 Supply Chain Management 10.0.43 (March 2025)
Asset Management	*Role-based access control for work order lifecycle stages*	Work order lifecycle states What's new or changed in Dynamics 365 Supply Chain Management 10.0.43 (March 2025)
Inventory and warehouse management	*Correct item that is not visible in released products form manually*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.40 (June 2024)
Inventory and warehouse management	*Using unit of measure and unit quantity in inventory journals.*	N/A
Inventory and warehouse management	*Cascade transfer order header date change to lines*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.35 (August 2023)
Procurement and sourcing	*Synchronize updates of the Requested ship and receipt dates on intercompany sales and purchase order lines*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.38 (February 2024)

Module	Feature name	More information
Procurement and sourcing	*Choose whether to consolidate sales orders for direct deliveries*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.41 (September 2024)
Production control	*Production batch balancing enhanced onhand grid capability*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.32 (April 2023)
Sales and marketing	*Calculate and push prices, discounts and totals for selective sales orders and sales quotations when integrated with Dynamics 365 Sales*	Calculate and push prices, discounts, and totals from Supply Chain Management to Sales What's new or changed in Dynamics 365 Supply Chain Management 10.0.34 (July 2023)
Sales and marketing	*Copy Supply Chain Management sales quotation data to sales orders synced from Dynamics 365 Sales*	Copy Supply Chain Management sales quotation data to sales orders synced from Sales What's new or changed in Dynamics 365 Supply Chain Management 10.0.34 (July 2023).
Sales and marketing	*Replace alternative item defaults on sales lines*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.39 (April 2024)

Related information

Platform updates for finance and operations apps

Microsoft Dynamics 365 Supply Chain Management 10.0.49 includes platform updates. Learn more in Platform updates for version 10.0.49 of Finance and Operations apps (September 2026).

For information about the bug fixes included in each of the updates that are part of version 10.0.49, sign in to Microsoft Dynamics Lifecycle Services and view the KB article.

What's new or changed in the Warehouse Management mobile app

The Warehouse Management mobile app is updated regularly following a different schedule than Supply Chain Management. For information about the latest versions and features, go to What's

new or changed in the Warehouse Management mobile app.

What's new or changed in Demand planning

Demand planning in Dynamics 365 Supply Chain Management is updated regularly following a different schedule than Supply Chain Management. For information about the latest versions and features, go to What's new or changed in Demand planning.

What's new or changed in Planning Optimization

Microsoft updates Planning Optimization monthly. However, based on business requirements, the product team occasionally releases other updates between the scheduled releases. For information about the latest versions and features, go to Planning Optimization release process and release history.

Dynamics 365 2026 release wave 1 plan

Wondering about upcoming and recently released capabilities in any of our business apps or platform?

Check out the Microsoft Dynamics 365: 2026 release wave 1 plan. It captures all the details, end to end, top to bottom, in a single document that you can use for planning.

Removed and deprecated Supply Chain Management features

The Removed or deprecated features in Dynamics 365 Supply Chain Management article describes features that are removed or scheduled to be removed or deprecated for Supply Chain Management.

- A *removed* feature is no longer available in the product.
- A *deprecated* feature isn't in active development and might be removed in a future update.

Before Microsoft removes any feature from the product, it announces the deprecation notice in the Removed or deprecated features in Dynamics 365 Supply Chain Management article 12 months prior to the removal.

For breaking changes that only affect compilation time but are binary compatible with sandbox and production environments, the deprecation time is less than 12 months. Typically, these changes are functional updates that need to be made to the compiler.

Last updated on 07/29/2026

What's new or changed in Dynamics 365 Supply Chain Management 10.0.48 (June 2026)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article lists features that are either new or changed in Microsoft Dynamics 365 Supply Chain Management version 10.0.48. This version has a build number of 10.0.2645 and is available on the following schedule:

- **Preview of release:** April 2026
- **General availability of release (self-update):** June 2026
- **General availability of release (auto-update):** July 2026

Features included in this release

The following table lists the features that are included in this release. We might update this article to include features that were added to the build after this article was originally published.

Expand table

Module or feature area	Feature	More information	Enabled by
Master planning	Keep supply for confirmed demand in Planning Optimization	Keep supply for confirmed demand in Planning Optimization (Preview)	Feature management: *(Preview) Keep supply for confirmed demand in Planning Optimization*
Master planning	Optimize confirmed dates for CTP line changes	Optimize confirmed dates for CTP line changes (preview)	Feature management: *(Preview) Optimize confirmed dates for CTP line changes*

Module or feature area	Feature	More information	Enabled by
Master planning	Soft issue margins with Planning Optimization (preview)	Soft issue margin (preview)	Feature management: *Soft issue margins with Planning Optimization*
Procurement and sourcing	Manage the downstream impact of PO changes with Procurement Agent impact analysis	Impact analysis overview (production-ready preview)	Feature management: • *(Production ready preview) Agent management* • *(Production-ready preview) Procurement Agent - Impact analysis*
Procurement and sourcing	Try out the *updates from vendors* feature without a complete email setup	Try out the *updates from vendors* feature without a complete email setup (production-ready preview)	Enabled by default
Warehouse management	Automate dynamic work classification with Power FX	Dynamic work classification (production-ready preview)	Feature management: *(production-ready preview) Dynamic work classification*
Warehouse management	Enable precise serial and batch capture in cluster picking	Cluster picking strategy	Enabled by default

Feature enhancements added in this release

The following table lists the feature enhancements that were added in this release. Each of these enhancements provides an incremental improvement to an existing feature. Because they're only enhancements, they're not listed in the release plan. However, to ensure that these enhancements don't conflict with your existing customizations or preferences, each enhancement is turned off by default (unless otherwise noted).

Some of these features aren't visible on your system until you turn them on in feature management by using the name listed here for the *Feature management name* (additional configuration might also be needed). Features that don't show a feature management name are visible by default as of this version of Supply Chain Management, but typically add a new configuration option that you need to set to use the new functionality.

Expand table

Feature enhancement	Description
Module: Inventory and warehouse management **Feature management name:** *Flexible sampling plan - Usability enhancements*	Improves the user experience for defining flexible sampling plans by moving the **Last level** assignment from the header to a checkbox directly in the lines grid. This change makes the action more intuitive and provides contextual confirmation dialogs when users set, change, or clear the last level.
Module: Inventory and warehouse management **Feature management name:** *Quality quick results entry enhancements*	Enhances the information display on the quick results entry page by adding information such as test descriptions and test plan notes. Now you can view test-related information and associated notes directly within the quick results entry workflow.
Module: Product information management **Feature management name:** *Configure Product Lifecycle State business process policies without Engineering change management.*	Lets you configure product lifecycle state business process policies when engineering change management configuration keys aren't enabled. As a result, you can manage product behavior (such as purchasing, sales, and production) based on lifecycle states without relying on engineering change management. If you do use engineering change management, then you don't need this feature because engineering change management provides the required lifecycle-based business process rules. This setup offers flexibility to tailor operations according to a product's lifecycle stage. For example, you can choose to allow a new product to be purchased with a warning that indicates the item is still a prototype; or, for discontinued or end-of-life products, you can choose to allow sales of existing inventory while blocking new purchases or production.
Module: Production control **Feature management name:** *Automatically stop and resume production jobs for indirect activities in production floor execution*	When a worker starts an indirect activity in the production floor execution interface, any production jobs currently in progress are automatically stopped and remembered. When the indirect activity ends, the previously active production jobs are automatically restarted, eliminating the need for workers to manually search for and restart their jobs.
Module: Sales and marketing **Feature management name:** *Enable non-attribute based rebate*	Allows rebate management deals that aren't based on attributes to be considered when using Unified pricing management.

Feature enhancement	Description
management deals for Unified pricing scenario	
Module: Warehouse management **Feature management name:** *(Production ready preview) Optimized 3D packing containerization algorithm*	Introduces a new optimized 3D packing algorithm to warehouse containerization processes.

Features turned on by default in this release

The following table lists the features that became turned on by default in version 10.0.48. You can still turn these features off in Feature management if necessary.

Expand table

Module	Feature name	More information
Master planning	*Keep pegging between approved planned orders for Planning Optimization*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.47 (March 2026)

Related information

Platform updates for finance and operations apps

Microsoft Dynamics 365 Supply Chain Management 10.0.48 includes platform updates. Learn more in Platform updates for version 10.0.48 of Finance and Operations apps (June 2026).

For information about the bug fixes included in each of the updates that are part of version 10.0.48, sign in to Microsoft Dynamics Lifecycle Services and view the KB article.

What's new or changed in the Warehouse Management mobile app

The Warehouse Management mobile app is updated regularly following a different schedule than Supply Chain Management. For information about the latest versions and features, go to What's new or changed in the Warehouse Management mobile app.

What's new or changed in Demand planning

Demand planning in Dynamics 365 Supply Chain Management is updated regularly following a different schedule than Supply Chain Management. For information about the latest versions and features, go to What's new or changed in Demand planning.

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For breaking changes that only affect compilation time, but are binary compatible with sandbox and production environments, the deprecation time is less than 12 months. Typically, these changes are functional updates that need to be made to the compiler.

Last updated on 05/27/2026

What's new or changed in Dynamics 365 Supply Chain Management 10.0.47 (March 2026)

This article lists features that are either new or changed in Microsoft Dynamics 365 Supply Chain Management version 10.0.47. This version has a build number of 10.0.2527 and is available on the following schedule:

- **Preview of release:** January 2026
- **General availability of release (self-update):** March 2026
- **General availability of release (auto-update):** April 2026

Features included in this release

The following table lists the features that are included in this release. We might update this article to include features that were added to the build after this article was originally published.

Expand table

Module or feature area	Feature	More information	Enabled by
Production Control	Automate production reporting with electronic sensors	Production auto report scenario (preview)	Feature management: *(Preview) Sensor Data Intelligence - Auto report progress*
Sales and marketing	Calculate prices for external systems through the pricing calculation API	Calculate prices for external systems through the pricing calculation API	Enabled by default
Sales and marketing	Manage pricing rules using a generic base currency	*Coming soon*	Enabled by default
Sales and marketing	Optimize pricing data import and export workflows	Pricing data import and export entities	Feature management: *Unified pricing management pricing rule performance enhancement*

Module or feature area	Feature	More information	Enabled by
Sales and marketing	Pay-by-link payment with the Dynamics 365 Payment Connector for Adyen	Pay-by-link payment with the Dynamics 365 Payment Connector for Adyen (preview)	Feature management: • *Enable unified payments experience in POS* • *Enable Payments Notification feature* • *Enable Pay By Link Payment feature* • *Enable asynchronous payments for sales orders*
Sales and marketing	Simulate prices based on sales order line attributes	*Coming soon*	Enabled by default
Warehouse management	Improve picking efficiency with wrist-mounted scanning devices	Haptic feedback through external wearable devices	Enabled by default

Feature enhancements added in this release

The following table lists the feature enhancements that were added in this release. Each of these enhancements provides an incremental improvement to an existing feature. Because they're only enhancements, they're not listed in the release plan. However, to ensure that these enhancements don't conflict with your existing customizations or preferences, each enhancement is turned off by default (unless otherwise noted).

Some of these features aren't visible on your system until you turn them on in feature management by using the name listed here for the *Feature management name* (additional configuration might also be needed). Features that don't show a feature management name are visible by default as of this version of Supply Chain Management, but typically add a new configuration option that you need to set to use the new functionality.

Expand table

Feature enhancement	Description
Module: Asset Management **Feature management name:** *Add maintenance*	Add maintenance checklist lines from a predefined template directly to work order lines.

Feature enhancement	Description
checklist template lines to work order lines	
Module: Inventory and warehouse management **Enhancement:** *Consolidate inventory transactions* **Feature management name:** *(None)*	Inventory transaction consolidation is a prerequisite for the archiving and long-term retention of inventory transactions. During consolidation, detailed inventory transactions are replaced with summarized receipt and issue transactions. This change significantly reduces the size of the `InventTrans` table, delivering storage savings and improved system performance while preserving the original source transactions in an archive.
Module: Inventory and warehouse management **Feature management name:** *Configure quality associations by product dimensions*	Allows administrators to configure quality associations by product dimensions such as configuration, color, and size. It ensures that quality checks can be applied at a more granular level for specific product variants.
Module: Master Planning **Feature management name:** *(Preview) Exclude blocked transactions from planning if inventory status isn't a coverage plan dimension*	Excludes blocked transactions from planning when inventory status isn't set as a coverage plan dimension. When this feature is turned on, the **Net requirements** page no longer shows blocked transactions as *On-hand* and *Inventory journal* entries that offset each other by inventory blocking. This new behavior is more consistent with that of other planning scenarios. This feature only affects Planning Optimization, not the deprecated planning engine.
Module: Master Planning **Feature management name:** *Consider receipt margins when applying action*	Causes the system to consider receipt margins when applying actions and to adjust receipt and delivery dates accordingly. For a purchase or a production order, receipt and delivery date can be set in the past. However, for a transfer order, the receipt date must always be after the ship date. In situations where scheduling backwards from the requirement date results in a transfer order with a ship date that's in the past, the system applies the receipt margin only partially or not at all because transfer orders can't have ship dates in the past.
Module: Transportation management **Enhancement:** *Performance-optimized pro number generation*	If your high-volume processes (such as release to warehouse) run slowly when you're using pro numbers, the new *Performance optimized* pro number generation option can help improve throughput. Learn more in Transportation management number sequence.

Feature enhancement	Description
Feature management name: *(None)*	
Module: Warehouse management **Enhancement:** *Archive load related records with long-term retention* **Feature management name:** *(None)*	Adds the ability to archive or remove historical load and work-related data. This ability helps reduce index size, improve long-term database health, and can accelerate scenarios such as refreshing sandbox environments from production. The existing archiving framework is extended to support the following tables: • `WHSLOADTABLE` • `WHSLOADLINE` • `WHSSHIPMENTTABLE` • `WHSWORKTABLE` • `WHSWORKLINE` • `WHSWORKINVENTTRANS` • `WHSWORKTRANS`
Module: Warehouse management **Enhancement:** *Hand-free glove-mounted scanning support* **Feature management name:** *(None)*	The Warehouse Management mobile app now supports wrist-mounted, hands-free glove scanners, enabling faster and more ergonomic warehouse operations. Developed in collaboration with ProGlove and validated on Mark 2, Mark Basic, and LEO devices, the solution connects glove scanners to an Android-powered device running the native Warehouse Management mobile app, providing real-time feedback and intent-based scanning. Hands-free scanning reduces unnecessary motion, lowers cognitive load, and improves throughput and accuracy in high-volume picking and packing scenarios. The capability is device-agnostic, so you can choose supported glove scanners based on your operational needs. Requires Warehouse Management mobile app version 4.0.29.0 or higher.
Module: Warehouse management **Enhancement:** *License plate consolidation performance improvements* **Feature management name:** *(None)*	Enhances how warehouse inventory transactions are used during license plate consolidation, which improves system performance.
Module: Warehouse management **Enhancement:** *Packed container picking performance enhancements*	Enhances how warehouse inventory transactions are used when picking from a packed container, which improves system performance.

Feature enhancement	Description
Feature management name: *(None)*	
Module: Warehouse management **Enhancement:** *Warehouse Management mobile app support for Microsoft Entra Conditional Access* **Feature management name:** *(None)*	The Warehouse Management mobile app now supports Microsoft Entra Conditional Access, enabling organizations to apply modern, enterprise-grade security policies that control how and when users can sign in. IT administrators can enforce requirements such as multifactor authentication, device trust validation, and risk- or location-based access controls. The solution makes use of brokered authentication and aligns with the same Zero Trust security standards used across Microsoft 365 and Dynamics 365 applications. Microsoft Entra Conditional Access ensures secure access for the right user, on the right device, under the right conditions, without impacting warehouse productivity. Learn more in Use Microsoft Entra Conditional Access with the Warehouse Management mobile app.
Module: Warehouse management **Feature management name:** *Remove Show Closed Button*	Removes the **Show closed** button from warehouse related forms. Instead, you can now use filters and saved views directly on the form. This feature is enabled by default.

Preview features that are now generally available

The following table lists the features that were introduced as public preview features in a previous version and became generally available in version 10.0.47. As with other new features, most of these features are turned off by default, so you must turn them on in Feature management if you want to use them.

Expand table

Module	Feature name	More information
Inventory and warehouse management	*Sample management*	Sample management overview Enable and configure sample management What's new or changed in Dynamics 365 Supply Chain Management 10.0.46 (December 2025)
Other	Immersive Home	Immersive Home overview
System administration	Agent management	Enable agent management

Features turned on by default in this release

The following table lists the features that became turned on by default in version 10.0.47. You can still turn these features off in Feature management if necessary.

Expand table

Module	Feature name	More information
Accounts receivable	*Approved customer list*	Advanced quality management overview What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Asset Management	*Aggregated material availability check*	Material availability check for work orders What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Asset Management	*Credit limit check on work order dispatch*	Bill for maintenance on customer-owned assets What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Inventory and warehouse management	*Advanced quality management*	Advanced quality management overview What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Organization administration	*Electronic signature improvements*	Advanced quality management overview What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Production control	*Dispense management*	Advanced quality management overview What's new or changed in Dynamics 365 Supply Chain Management 10.0.44 (June 2025)
Shared AP and AR	*Rebate processing use multiple batch tasks*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.45 (September 2025)

Features becoming mandatory in this release

The following table lists the features that became mandatory in version 10.0.47. These features still appear in Feature management but you can no longer turn them off.

Expand table

Module	Feature name	More information
Asset Management	*Adjust forecast hours to match schedule hours when work orders are dispatched*	Dispatch work orders What's new or changed in Dynamics 365 Supply Chain Management 10.0.43 (March 2025)
Asset Management	*Print work order document attachments*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.41 (September 2024)
Asset Management	*Role-based access control for work order lifecycle stages*	Work order lifecycle states What's new or changed in Dynamics 365 Supply Chain Management 10.0.43 (March 2025)
Inventory and warehouse management	*Dimension history for documents report data clean up*	What's new or changed in Dynamics 365 Supply Chain Management 10.0.40 (June 2024)
Warehouse management	*Warehouse management only mode*	Warehouse management only mode overview

Related information

Platform updates for finance and operations apps

Microsoft Dynamics 365 Supply Chain Management 10.0.47 includes platform updates. Learn more in Platform updates for version 10.0.47 of Finance and Operations apps (March 2026).

For information about the bug fixes included in each of the updates that are part of version 10.0.47, sign in to Microsoft Dynamics Lifecycle Services and view the KB article.

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Removed and deprecated Supply Chain Management features

The Removed or deprecated features in Dynamics 365 Supply Chain Management article describes features that are removed or scheduled to be removed or deprecated for Supply Chain Management.

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Before Microsoft removes any feature from the product, it announces the deprecation notice in the Removed or deprecated features in Dynamics 365 Supply Chain Management article 12 months prior to the removal.

For breaking changes that only affect compilation time, but are binary compatible with sandbox and production environments, the deprecation time is less than 12 months. Typically, these changes are functional updates that need to be made to the compiler.

Last updated on 07/23/2026

Removed or deprecated features home page

09/03/2025

Removed or deprecated features in finance and operations apps

The following topics provide information about removed or deprecated features in finance and operations apps:

- Removed or deprecated features in Dynamics 365 Commerce
- Removed or deprecated features in Dynamics 365 Finance
- Removed or deprecated features in Dynamics 365 Supply Chain Management
- Removed or deprecated features in Dynamics 365 Human Resources
- Removed or deprecated features in Dynamics 365 Project Operations
- Removed or deprecated platform features
- Removed or deprecated features in previous releases

Deprecations in the Microsoft Power Platform

The following article provides information about deprecations in the Microsoft Power Platform:

- Important changes (deprecations) coming in Power Apps, Power Automate, and customer engagement apps

Removed or deprecated features in Dynamics 365 Supply Chain Management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article is updated as new removed or deprecated features are documented for Dynamics 365 Supply Chain Management.

- A *removed* feature is no longer available in the product.
- A *deprecated* feature isn't in active development and might be removed in a future update.

Use this list to help you consider these removals and deprecations for your own planning.

> **Note**
>
> For detailed information about objects in finance and operations apps, see the Technical reference reports. You can compare the different versions of these reports to learn about objects that changed or were removed in each version of finance and operations apps.

Features removed or deprecated in the Supply Chain Management 10.0.49 release

Global Inventory Accounting

Expand table

Reason for deprecation or removal	The Global Inventory Accounting (GIA) solution is being retired to bring modernized inventory costing capability with improved architecture.

Replaced by another feature?	Yes, a multi-ledger costing solution will serve as GIA's successor, providing comparable capabilities along with significant improvements. Learn more in the blog post See what's next in financial operations from Microsoft Dynamics 365.
Product areas affected	Dynamics 365 Finance and Dynamics 365 Supply Chain Management
Deployment option	Cloud and on-premises
Status	Deprecated. GIA will reach end of support on July 1, 2027. If you're using GIA, review your usage and take the appropriate action based on your scenario: • If you aren't using GIA in a production environment, no action is required. • If you only installed or enabled GIA in sandbox or test environments, review whether you're actively using it. If you don't need GIA, disable or remove the feature and avoid new implementations that are based on it. • If you're actively using GIA, or if dependencies on it exist, on a production environment, assess your current usage and dependencies. Begin planning a transition to supported costing capabilities, and validate changes in a test environment before July 2027. We recommend that you engage with Microsoft or your implementation partner for transition planning support.

Follow-up questions in the Copilot sidecar (preview)

Expand table

Reason for deprecation or removal	Microsoft decided to discontinue development on this preview feature.
Replaced by another feature?	No
Product areas affected	Microsoft finance and operations applications
Deployment option	Cloud and on-premises
Status	Removed. This preview feature was never made generally available and is now completely removed from all finance and operations applications.

Features removed or deprecated in the Supply Chain Management 10.0.48 release

Asset Management mobile app

Expand table

Reason for deprecation or removal	Microsoft released the Asset Management mobile app source code, so the app is now open source and you can freely customize it. Previously, Power Apps Studio didn't preserve canvas-app customizations across official updates.
Replaced by another feature?	Yes. The app is now available as an open-source sample.
Product areas affected	Supply Chain Management – Asset Management
Deployment option	Cloud and on-premises
Status	Deprecated as of June 1, 2026. The Asset Management mobile app is now available on GitHub as an open-source sample at microsoft/scmsamples-EnterpriseAssetManagement. It provides the same functionality as the original app and is freely customizable. The existing app remains available for installation through Dynamics 365 apps in the Power Platform admin center until November 30, 2026, with critical fixes from Microsoft during this transition period. After that date, the app is no longer distributed through the admin center. Existing installations continue to work in customer tenants. Learn more in Asset Management mobile app open-source release.

Features removed or deprecated in the Supply Chain Management 10.0.46 release

Data Integrator templates for Dynamics 365 Field Service

Expand table

Reason for deprecation or removal	This functionality is scheduled to be replaced by a simpler, more integrated Field Service experience.

Replaced by another feature?	Yes
Product areas affected	Microsoft finance and operations applications
Deployment option	Cloud and on-premises
Status	Deprecated. The Data Integrator templates for Dynamics 365 Field Service will no longer be available after February 28, 2027. This integration works by synchronizing work orders, agreements, inventory, and projects between Field Service and Supply Chain Management. Instead, look to a simpler, more integrated Field Service experience. Learn more in Connect Field Service to finance and operations and Link work orders to projects for unified financials.

Features removed or deprecated in the Supply Chain Management 10.0.45 release

Inventory On-hand mobile app (preview)

Expand table

Reason for deprecation or removal	Microsoft decided to discontinue development and support for this preview feature.
Replaced by another feature?	Yes
Product areas affected	Supply Chain Management – Inventory management
Deployment option	Cloud and on-premises
Status	Removed. The *Inventory On-hand mobile app* preview (previously available on the Power Platform Admin Center) was never made generally available. It's now provided as a sample app that you can download from the scmsamples-InventoryOnHand repository on GitHub and customize as needed. Learn more in Inventory On-hand mobile app.

Features removed or deprecated in the Supply Chain Management 10.0.44 release

Inquire into inventory with Copilot (preview)

Expand table

Reason for deprecation or removal	Microsoft decided to discontinue development on this preview feature.
Replaced by another feature?	No
Product areas affected	Supply Chain Management – Inventory management
Deployment option	Cloud and on-premises
Status	Removed. This preview feature was never made generally available and is now completely removed from the Inventory Visibility service.

SHA1 hashing in the InventDim table

Expand table

Reason for deprecation or removal	To ensure that data in the `InventDim` table is unique and easy to search, the system hashes some of its data. The system previously used SHA1 hashing, which is now outdated, so it's replaced by SpookyHash.
Replaced by another feature?	Yes. SHA1 hashing is replaced by SpookyHash.
Product areas affected	Supply Chain Management – Inventory
Deployment option	Cloud and on-premises
Status	As of Supply Chain Management version 10.0.44, the system uses SpookyHash instead of SHA1 to hash data in the `InventDim` table. When you upgrade to version 10.0.44, the system automatically schedules a batch job that converts all `InventDim` data hashed using SHA1 to instead use SpookyHash. Version 10.0.44 can work with both SHA1 and SpookyHash data, but support for SHA1 will be removed in a future release. Extensibility was never supported for this part of the application, but if you have external code that

	uses SHA1 when creating, updating, or searching data in the `InventDim` table, you must stop using that code before upgrading to Supply Chain Management 10.0.44 because it will corrupt your data.

Rename item number (preview)

Expand table

Reason for deprecation or removal	The item number field is a primary key that's used across multiple systems. The *(Preview) Rename item number* feature allowed users to edit the item number field, but it could cause data inconsistencies in multi-system and apps-integration scenarios, leading to data integrity and data corruption problems. The feature was previously in preview and was never made generally available for use in production environments.
Replaced by another feature?	No
Product areas affected	Supply Chain Management – Product information management
Deployment option	Cloud and on-premises
Status	The *(Preview) Rename item number* feature was removed from the feature management workspace in Dynamics 365 Supply Chain Management version 10.0.44 and is no longer available. The capability is completely removed for all customers. If you need to rename an item number, we recommend that you use alternate methods, such as deleting and creating a new item, renaming attributes, or renaming search names, product names, and other item-related fields.

Features removed or deprecated in the Supply Chain Management 10.0.43 release

The "Work creation number" number sequence is removed

Expand table

Reason for deprecation	The *Work creation number* number sequence could sometimes create problems when the same prefixes were used for other number sequences.

or removal	
Replaced by another feature?	Yes, the *Work creation number* number sequence is replaced by the *Work creation ID* number sequence. The *Work creation ID* number sequence is designed to avoid conflicts with other number sequences.
Product areas affected	Supply Chain Management – Warehouse management
Deployment option	Cloud and on-premises
Status	Previously, there were two number sequences that were used to generate values for the **Work creation number** field for work records (*Work creation ID* and *Work creation number*). The newer *Work creation ID* number sequence has been available as a replacement for *Work creation number* for some time. For all currently available versions of Supply Chain Management (not just version 10.0.43 and later), you should always use the *Work creation ID* number sequence to generate work creation identification numbers. Although the *Work creation number* sequence is still present in the demo data, it's no longer used by the system and production data doesn't include the *Work creation number* number sequence. You can set up the *Work creation ID* number sequence on the **Warehouse management parameters** page.

"Adjustment out" mobile device menu items must now use process guide

Expand table

Reason for deprecation or removal	The process guide framework offers enhanced extensibility support and allows for easier changes in the feature behavior. The framework is now required for all mobile device menu items that use the *Adjustment out* work creation process.
Replaced by another feature?	No. The process guide framework is required for all mobile device menu items that use the *Adjustment out* work creation process. The ability to turn off this option is being removed from the product.
Product areas affected	Supply Chain Management – Warehouse management
Deployment option	Cloud and on-premises
Status	The **Use process guide** setting for mobile device menu items that use the *Adjustment out* work creation process is mandatory starting in Supply Chain Management version 10.0.45. Approximately one year after the release of version 10.0.45, the non-process guide implementation is no longer supported and might eventually be removed from the

	product. Learn more in Set up mobile device menu items for adjustment in and adjustment out.

"Spot cycle counting" mobile device menu items must now use process guide

Expand table

Reason for deprecation or removal	The process guide framework offers enhanced extensibility support and allows for easier changes in the feature behavior. The framework is now required for all mobile device menu items that use the *Spot cycle counting* work creation process.
Replaced by another feature?	No. The process guide framework is required for all mobile device menu items that use the *Spot cycle counting* work creation process. The ability to turn off this option is being removed from the product.
Product areas affected	Supply Chain Management – Warehouse management
Deployment option	Cloud and on-premises
Status	The **Use process guide** setting for mobile device menu items that use the *Spot cycle counting* work creation process is enabled by default in Supply Chain Management version 10.0.45, and it's mandatory in version 10.0.47. Approximately one year after the release of version 10.0.47, the non-process guide implementation is no longer supported and might eventually be removed from the product. Learn more in Set up mobile devices for warehouse work.

Features removed or deprecated in the Supply Chain Management 10.0.42 release

Register material consumption on the production floor execution interface (WMS-enabled) (preview)

Expand table

Reason for deprecation or removal	The *Register material consumption on the production floor execution interface (WMS-enabled)* feature (previously in preview) is replaced by the *Register material consumption as complete and edit dimensions on the production floor execution interface* feature.

Replaced by another feature?	Yes. The *Register material consumption on the production floor execution interface (WMS-enabled)* feature (previously in preview) is replaced by the *Register material consumption as complete and edit dimensions on the production floor execution interface* feature, which also adds the ability to use the production floor execution interface to register material consumption for WMS-enabled items.
Product areas affected	Supply Chain Management – Production control
Deployment option	Cloud and on-premises
Status	Removed. As of Supply Chain Management version 10.0.41, if you never enabled the old *Register material consumption on the production floor execution interface (WMS-enabled)* feature, then you see only the newer *Register material consumption as complete and edit dimensions on the production floor execution interface* feature in feature management. If you previously enabled the old feature, you can still use it, but you'll soon be contacted by Microsoft Support with instructions on how to replace it with the newer feature.

Features removed or deprecated in the Supply Chain Management 10.0.41 release

Pricing management module (preview)

Expand table

Reason for deprecation or removal	The Pricing management module (previously in preview) is replaced by the new Unified pricing management module.
Replaced by another feature?	Yes, the Pricing management module is replaced by the new Unified pricing management module.
Product areas affected	Supply Chain Management – Pricing management
Deployment option	Cloud and on-premises
Status	Deprecated and will be removed in a future release. It is replaced by the Unified pricing management module, which offers nearly all of the same functionality as the deprecated Pricing management module, plus several important enhancements. Both the deprecated pricing management module and the Unified pricing management module use similar or

	identical navigation paths in the Supply Chain Management user interface, so only one of these modules can be active at a time. As of Supply Chain Management version 10.0.41, if you never enabled the deprecated pricing management module in your environment, only the Unified pricing management module is available to you. If you enabled the deprecated pricing management module, you can continue to use it until it's removed, but we recommend that you transition to the Unified pricing management module as soon as possible to take advantage of the new features and enhancements that it offers. For help transitioning from the deprecated pricing management module to the Unified pricing management module, contact Microsoft Support.

Job card terminal

Expand table

Reason for deprecation or removal	The job card terminal is replaced by the new production floor execution interface.
Replaced by another feature?	Yes, the job card terminal is replaced by the new production floor execution interface.
Product areas affected	Supply Chain Management – production control
Deployment option	Cloud and on-premises
Status	Deprecated. Approximately one year after the release of version 10.0.41, the job card terminal is no longer supported and might be removed from the product.

Inventory transactions support for internal warehouse operations

Expand table

Reason for deprecation or removal	Using inventory transactions to track on-hand inventory for internal warehouse operations has well-known performance problems.
Replaced by another feature?	Yes. Warehouse-specific inventory transactions, which have been available since version 10.0.32, replace the older inventory transactions for tracking internal warehouse operations.

Product areas affected	Supply Chain Management – Warehouse management
Deployment option	Cloud and on-premises
Status	Supported until version 10.0.40. As of version 10.0.41, inventory transactions are deprecated for tracking on-hand inventory for internal warehouse operations. Existing customers can continue using this scenario after that version, but new features and bug fixes for this scenario are only implemented for warehouse-specific inventory transactions. Approximately one year after the release of version 10.0.41, support for this scenario is removed and all customers are required to move to warehouse-specific inventory transactions for tracking on-hand inventory for internal warehouse operations.

Release to warehouse page

Expand table

Reason for deprecation or removal	The **Release to warehouse** page has performance problems.
Replaced by another feature?	Yes. The **Release to warehouse** page is split into two new release to warehouse pages, which together provide equivalent functionality with significantly improved performance. The new pages are **Release sales orders to warehouse** and **Release transfer orders to warehouse**.
Product areas affected	Supply Chain Management – Release to warehouse
Deployment option	Cloud and on-premises
Status	Deprecated. The **Release to warehouse** page is now hidden in the app, but you can enable it if necessary by contacting Microsoft Support. We strongly recommend that you instead use the new pages (**Release sales orders to warehouse** and **Release transfer orders to warehouse**) because they provide equivalent functionality with significantly improved performance. The **Release to warehouse** page is removed from the product one year after the release of Supply Chain Management version 10.0.41.

Features removed or deprecated in the Supply Chain Management 10.0.40 release

Scale unit capability for Supply Chain Management

Expand table

Reason for deprecation or removal	The scale unit capability for Microsoft Dynamics 365 Supply Chain Management has been paused for new customers since July 2022 and is now deprecated.
Replaced by another feature?	Yes. Warehouse management only mode replaces some of the functionality planned for scale units and adds many new architectural and integration possibilities.
Product areas affected	Supply Chain Management – Warehouse management Supply Chain Management – Production control
Deployment options	Cloud and on-premises
Status	The scale unit capability for Microsoft Dynamics 365 Supply Chain Management has been paused for new customers since July 2022. The capability is formally deprecated as of Supply Chain Management version 10.0.40 and will be completely removed for all customers one year after the release of that version.

Features removed or deprecated in the Supply Chain Management 10.0.39 release

Load planning workbench

Expand table

Reason for deprecation or removal	The **Load planning workbench** page has performance issues.
Replaced by another feature?	Yes. The **Load planning workbench** page has been split into two new workbench pages, which together provide equivalent functionality with significantly improved performance. The new pages are **Inbound load planning workbench** and **Outbound load planning workbench**.
Product areas affected	Supply Chain Management – Warehouse management Supply Chain Management – Transportation management

Deployment option	Cloud and on-premises
Status	Deprecated. The **Load planning workbench** page is now hidden in the app, but you can enable it if necessary by contacting Microsoft Support. We strongly recommend that you instead use the new pages (**Inbound load planning workbench** and **Outbound load planning workbench**) because they provide equivalent functionality with significantly improved performance. The **Load planning workbench** page will be completely removed from the product one year after the release of Supply Chain Management version 10.0.39.

Features removed or deprecated in the Supply Chain Management 10.0.37 release

Service-based authentication methods for the Warehouse Management mobile app

Expand table

Reason for deprecation or removal	Support for service-based authentication methods (certificate and shared secret) is being removed to improve security.
Replaced by another feature?	Yes. Service-based authentication is being replaced by user-based authentication (device code flow), which is more secure.
Product areas affected	Supply Chain Management – Warehouse management
Deployment option	Cloud and on-premises
Status	Removed. As of July 15, 2024, Microsoft discontinued support for using service-based authentication methods (certificate and shared secret) to connect the Warehouse Management mobile app to Supply Chain Management. Service-based authentication has been replaced by user-based authentication (device code flow). Administrators must update all devices to use user-based authentication before July 15, 2024. For more information about device code flow, see User-based authentication FAQ and User-based authentication. Mass deployment with Microsoft Intune isn't yet supported for user-based authentication, but we expect to add support for it soon. For more information and the latest news about

	mass deploying to mobile devices for user-based authentication, see Mass deploy the mobile app with user-based authentication.

Features removed or deprecated in the Supply Chain Management 10.0.35 release

Integration with Outlook for contacts, appointments, and tasks

Expand table

Reason for deprecation or removal	Exchange web service API is transitioning from SOAP to REST.
Replaced by another feature?	No replacement is planned. We recommend that customers integrate with Dynamics 365 Sales.
Product areas affected	Supply Chain Management – sales and marketing
Deployment option	Cloud and on-premises
Status	Deprecated. As of December 1, 2023, Microsoft will discontinue support for synchronizing contacts, appointments, and tasks between Supply Chain Management and Outlook. In addition, security updates will no longer be provided for these features. This functionality is being discontinued because the existing integration is based on SOAP. However, Exchange Web Service (EWS) is transitioning from SOAP to REST and will support only REST going forward. There are no plans to add REST support for Outlook integration in Supply Chain Management. As of May 1, 2024, it will no longer be possible to synchronize the following types of records between Supply Chain Management and Outlook: • Contacts • Appointment activities • Task activities To prepare for this change, redesign any extensions that you've made to any of these capabilities to remove the dependency. To support customer relations management scenarios, consider integrating Dynamics 365 Sales with Outlook and/or using dual-write to synchronize contacts for customer

	accounts between Supply Chain Management and Dynamics 365 Sales. (Learn more in Integrated customer master.)

Features removed or deprecated in the Supply Chain Management 10.0.29 release

Stock transfer orders that have tax on the transfer price

Expand table

Reason for deprecation or removal	The Stock transfer orders that have tax on the transfer price functionality is being replaced by the Stock transfer orders for India functionality.
Replaced by another feature?	Yes, the Stock transfer orders that have tax on the transfer price functionality is being replaced by the Stock transfer orders for India functionality.
Product areas affected	Supply Chain Management – inventory
Deployment option	Cloud and on-premises
Status	Removed. The *Stock transfer orders that have tax on the transfer price* functionality has been removed as of October 2023. Customers must instead use the improved functionality, *Stock transfer orders for India*. Learn more in Stock transfer orders for India.

Features removed or deprecated in the Supply Chain Management 10.0.19 release

Job card device

Expand table

Reason for deprecation or removal	The job card device is replaced by the production floor execution interface.
Replaced by another	Yes, the job card device is replaced by the production floor execution interface.

feature?	
Product areas affected	Supply Chain Management – production control
Deployment option	Cloud and on-premises
Status	Unsupported. As of April 2022, the job card device is no longer supported and customers must use the new production floor execution interface.

Features removed or deprecated in the Supply Chain Management 10.0.18 release

Supply Chain Management- Warehousing (the warehouse app)

Expand table

Reason for deprecation or removal	Effective April 2021, *Supply Chain Management – Warehousing* (the warehouse app) is deprecated and won't be supported after April 2022. It's now replaced by the *Warehouse Management mobile app*, which was released with version 10.0.17 of Supply Chain Management. The new app is a complete replacement but uses same underlying framework, which makes migration easy. For more information about the new Warehouse Management mobile app, see Warehouse Management mobile application and Install the Warehouse Management mobile app.
Replaced by another feature?	Yes, replaced by the new Warehouse Management mobile app.
Product areas affected	Supply Chain Management – warehouse app
Deployment option	Cloud and on-premises
Status	Removed. As of April 2022, the old warehouse app is no longer supported and has been removed from the Microsoft Store and Google Play store. Customers must now use the new Warehouse Management mobile app instead.

Features removed or deprecated in the Supply Chain Management 10.0.15 release

Internet Explorer 11 support for Dynamics 365 is deprecated

Expand table

Reason for deprecation or removal	Effective August 2021, Microsoft Internet Explorer 11 is no longer supported for use with Dynamics 365 products.
Replaced by another feature?	We recommend that customers transition to Microsoft Edge.
Product areas affected	All Dynamics 365 products
Deployment option	All
Status	Removed. Support for Internet Explorer 11 was removed in August 2021.

Use of built-in Supply Chain Management master planning engine for manufacturing scenarios

Expand table

Reason for deprecation or removal	To enhance performance and minimize the SQL database load during master planning runs, the built-in Supply Chain Management master planning engine is being replaced by Planning Optimization. Planning Optimization allows for fast planning runs that can be performed even during office hours. This enables planners to react immediately to changes in demand or planning parameters.
Replaced by another feature?	Yes, Planning Optimization will replace the existing built-in Supply Chain Management master planning engine.
Product areas affected	Supply Chain Management – Master planning
Deployment option	Cloud only. Planning Optimization is not supported with on-premises deployments.
Status	Deprecated. On April 1, 2022, support for manufacturing scenarios was discontinued for the built-in master planning engine for Supply Chain Management. As of that date, Microsoft stopped all active development on manufacturing scenarios for the built-in planning engine, stopped releasing new features, and only released critical bug fixes. Since that date, all companies that require support for manufacturing scenarios have been required to use Planning Optimization for their master planning calculations. Learn more in Deprecated master planning overview.

	Since April 2022, only companies with on-premises deployments of Supply Chain Management have been able to continue using the built-in master planning engine for manufacturing scenarios. However, as of March 2023, Microsoft has now fully discontinued all support for the built-in master planning engine for all types of deployments. Hereafter, Microsoft will only provide support for critical blocking issues (which result in no planned orders being created or the continuous failure of built-in master planning). The built-in master planning engine is now referred to as the *deprecated master planning engine*.

Features removed or deprecated in the Supply Chain Management 10.0.11 release

Use of built-in Supply Chain Management master planning engine for distribution scenarios

Expand table

Reason for deprecation or removal	To enhance performance and minimize the SQL database load during master planning runs, the built-in Supply Chain Management master planning engine is being replaced by Planning Optimization. Planning Optimization allows for fast planning runs that can be performed even during office hours. This enables planners to react immediately to changes in demand or planning parameters.
Replaced by another feature?	Yes, Planning Optimization will replace the existing built-in Supply Chain Management master planning engine.
Product areas affected	Supply Chain Management – Master planning
Deployment option	Cloud only. Planning Optimization isn't supported for on-premises deployments.
Status	Removed. On April 1, 2021, support for distribution scenarios was discontinued for the built-in master planning engine for Supply Chain Management. Since then, customers running distribution scenarios have been required to use Planning Optimization for master planning calculations. Learn more in Deprecated master planning overview. Since April 2021, only companies with on-premises deployments of Supply Chain Management have been able to continue using the built-in master planning engine for distribution scenarios. However, as of March 2023, Microsoft has now fully discontinued all support for the built-in master planning engine for all types of deployments. Hereafter, Microsoft will only provide support for critical blocking issues (which result in no planned

	orders being created or the continuous failure of built-in master planning). The built-in master planning engine is now referred to as the *deprecated master planning engine*.

Previous announcements about removed or deprecated features

To learn more about features that have been removed or deprecated in previous releases, see Removed or deprecated features in previous releases.

Last updated on 06/24/2026

Removed or deprecated features in previous releases

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> This article is no longer updated. To see a current list of features that are removed or deprecated from Finance and Operations apps, search for **"Removed or deprecated features"** content that relates to the app you're using.

This article describes features that Microsoft removed or deprecated from Dynamics 365 for Finance and Operations and previous releases of that product.

- A *removed* feature is no longer available in the product.
- A *deprecated* feature isn't in active development and might be removed in a future update.

Use this list to help you consider these removals and deprecations for your own planning.

You can find detailed information about objects in Finance and Operations apps in the Technical reference reports. To learn about objects that changed or were removed in each version of Finance and Operations apps, compare the different versions of these reports.

Finance 10.0.7 with Platform update 31

Chinese voucher types without Account groups selection

Expand table

Reason for deprecation or removal	Changed to the feature with account groups selection.

Replaced by another feature?	Yes
Product areas affected	Application
Deployment option	All
Status	Deprecated: By December 1, 2020, Microsoft plans to no longer support Chinese voucher types set up without Account groups selection.

Finance and Operations 10.0.6 with Platform update 30

DimensionHash.getHash(str _message)

Expand table

Reason for deprecation/removal	Windows is deprecating the use of SHA1, as documented in Windows Enforcement of SHA1 Certificates.
Replaced by another feature?	Yes
Product areas affected	Application
Deployment option	All
Status	Deprecated: By April 1, 2020, developers must use the platform APIs found in the class **HasFunction**.

Hash.ComputeSHA1Hash(string message)

Expand table

Reason for deprecation/removal	Windows is deprecating the use of SHA1, as documented in Windows Enforcement of SHA1 Certificates.
Replaced by another feature?	Yes

Product areas affected	Platform
Deployment option	All
Status	Deprecated: By April 1, 2020, developers must use the platform APIs found in the class **HasFunction**.

FormDateTimeControl.setUtcString()

Expand table

Reason for deprecation/removal	We're retiring the **setUtcString()** method, because a better replacement method is available.
Replaced by another feature?	Yes
Product areas affected	Platform
Deployment option	All
Status	Deprecated: By October 1, 2020, we plan to no longer support the **setUtcString()** method. Developers should use the **setUtcDateTime()** method instead.

Block list report (IT) – Feature reference IT-00001

Expand table

Reason for deprecation or removal	Not legally required.
Replaced by another feature?	No
Product areas affected	Italian localization
Deployment option	All
Status	Deprecated: By October 1, 2020, support ends for this report.

Domestic tax report – Feature reference IT-00003

Expand table

Reason for deprecation or removal	Not legally required.
Replaced by another feature?	No
Product areas affected	Italian localization
Deployment option	All
Status	Deprecated: By October 1, 2020, support ends for the **Domestic tax report – Feature reference IT-00003**.

October 2019 deprecation announcement

Flowchart diagrams in Business process modeler

Expand table

Reason for deprecation/removal	Microsoft is deprecating the flowchart diagrams component in Business process modeler (BPM), because the legacy design caused low usage.
Replaced by another feature?	No
Areas affected	Business process modeler
Status	Deprecated: The flowchart diagrams component in BPM is expected to be removed in 2020. The following functionality is unavailable: All flowcharts are read-only and unavailable for editing. The shape properties that are associated with flowchart activities are also unavailable. These flowcharts include both the default flowcharts that are automatically generated and customized flowcharts that are modified based on those default flowcharts. The process steps are read-only and unavailable for editing. The legacy fit/gap analysis feature is unavailable. Therefore, no gap list is automatically created or available for export. **Note:** This feature was previously deprecated and replaced by Microsoft Azure DevOps integrations. The version history of the flowchart is unavailable.

Finance and Operations 10.0.5 with Platform update 29

US Payroll tax updates

Expand table

Reason for deprecation/removal	Microsoft is retiring tax updates for the US Payroll functionality due to low usage and enhanced functionality that's now offered via strategic integrations.
Replaced by another feature?	Yes
Product areas affected	Payroll
Deployment option	All
Status	Deprecated: By July 31, 2024, Microsoft plans to no longer provide tax updates to US Payroll customers. The functionality remains in the product, but Microsoft won't provide enhancements to keep the functionality up to date. Microsoft will evaluate any product defects on a case-by-case basis.

> **Note**
>
> This change affects the original discontinuation date of October 1, 2021. For more information, see Tax updates being retired for US Payroll feature in Microsoft Dynamics 365 for Finance and Operations.

Data management staging clean up

Expand table

Reason for deprecation/removal	Doesn't meet the core requirements for scheduling periodic cleanup.
Replaced by another feature?	Yes, the job history cleanup feature. It meets the scenarios holistically.
Product areas affected	Data management
Deployment option	All

Status	Deprecated: Target timeframe for removal is December 2020.

Finance and Operations 10.0.4 with Platform update 28

France: FEC Accounting data export in XML

Expand table

Reason for deprecation/removal	Replaced by TXT format. You can access the **French FEC audit file** through **General ledger** > **Periodic tasks** > **Data export**.
Replaced by another feature?	Yes
Product areas affected	General ledger
Deployment option	All
Status	Deprecated. Target timeframe for removal is July 2020.

Legacy navigation bar

Expand table

Reason for deprecation/removal	Header alignment with other Dynamics and Office products. For more information, see Updated navigation bar that aligns with the Office header.
Replaced by another feature?	Starting in Platform update 24, a restyled navigation bar that features search was introduced.
Product areas affected	Web client
Deployment option	All
Status	Deprecated: Starting in April 2020, the legacy navigation bar isn't available. Until that point, customers can revert to the legacy navigation bar through the **Client performance options** page.

Finance and Operations 10.0.2 with Platform update 26

Legacy default action behavior

Expand table

Reason for deprecation/removal	The legacy behavior for default actions in grids results in an unexpected column having the default action link after grid columns are reordered through personalization. The new sticky default action feature corrects this. For more information, see Sticky default actions in grids.
Replaced by another feature?	Starting in Platform update 21, a feature for "sticky default actions" was introduced. You can enable this feature on the **Client performance options** page.
Product areas affected	Grids in the web client
Deployment option	All
Status	Deprecated: Starting in April 2020, sticky default actions are the default behavior, without a mechanism to revert to the legacy behavior.

Legacy "is one of" filtering experience

Expand table

Reason for deprecation or removal	The "is one of" filtering experience underwent a redesign in Platform update 22. This change aims to make the updated version the only "is one of" filtering experience.
Replaced by another feature?	Starting in Platform update 22, an improved "is one of" filtering experience is available on the **Client performance options** page. For more information, see Optimized is one of filtering experience.
Product areas affected	Web client
Deployment option	All
Status	Deprecated: Starting in April 2020, the improved "is one of" experience is the default behavior, without a mechanism to revert to the legacy behavior.

Parameter to enable sales orders with multiple project contract funding sources

Support for creating project-based sales orders where the project contract has multiple funding sources is enabled by using the **Project management parameters** setting **Allow sales orders for project with multiple funding sources**. By default, this parameter isn't enabled.

Expand table

Reason for deprecation or removal	The functionality is always enabled after the parameter is removed.
Replaced by another feature?	No. The functionality to support project-based sales orders with multiple funding sources is always enabled.
Product areas affected	The **Allow sales orders for projects with multiple funding sources** parameter will be removed. The following methods are modified when the parameter is removed: **ctrlSalesOrderTable** method in **ProjStatusType** class, **validate** method for **ProjId** field, and **run** method in **SalescreateOrder** form. The following methods are deprecated when the parameter is removed: **IsSalesOrderAllowedForMultipleFundingSources** in **ProjTable** table file, **IsAllowSalesOrdersForMultipleFundingSourcesParamEnabled** method in **ProjTable** table file, **AllowSalesOrdersForMultipleFundingSources** data field in **ProjParameters** form and **ProjParameterEntity** files, **IsAssociatedToMultipleFundingSourcesContract** private method in **ProjTable** table file.
Deployment option	All
Status	Deprecation is planned for the April 2020 release wave.

Legacy workflow reports for tracking and instance status

Expand table

Reason for deprecation or removal	The legacy workflow reports for tracking and instance status are deprecated because they're no longer referenced from the navigation. The report names are WorkflowWorkflowInstanceByStatusReport and WorkflowWorkflowTrackingReport.
Replaced by another feature?	Use the workflow history form instead.

Product areas affected	Web client
Deployment option	All
Status	Deprecated: Target timeframe for removal is April 2020.

Finance and Operations 10.0.1 with Platform update 25

Deprecated APIs and potential breaking changes

Deriving from internal classes is deprecated

Expand table

Reason for deprecation/removal	Before Platform update 25, you could create a class or table that derives from an internal class or table that's defined in another package or module. This practice isn't safe. As of Platform update 25, the compiler displays a warning.
Replaced by another feature?	The compiler warning is replaced by an error in Platform update 26. This change is backward compatible at runtime, which means that you can deploy Platform update 25 or newer on any sandbox or production environment without needing to modify custom code. This change only affects development and compile time.
Product areas affected	Visual Studio development tools
Deployment option	All
Status	Deprecated: The warning becomes a compilation error in Platform update 26.

Overriding internal methods is deprecated

Expand table

Reason for deprecation/removal	Before Platform update 25, you could override an internal method in a derived class that's defined in another package or module. This practice isn't safe. As of Platform update 25, the compiler displays a warning.
Replaced by another feature?	This warning is replaced by a compile error in Platform update 26. This change is backward compatible at runtime, which means that you can deploy Platform update 25 or newer on any sandbox or production environment without needing to modify custom code. This change only affects development and compile time.
Product areas affected	Visual Studio development tools
Deployment option	All
Status	Deprecated: The warning becomes a compilation error in Platform update 26.

Finance and Operations 10.0.0 with Platform update 24

Renaming released products

Expand table

Reason for deprecation or removal	When you use the **Rename primary key** function to change the ItemId of a released product, the system updates only direct foreign key references. Any other references to the released product, such as references from production orders, keep the old ItemId. This inconsistency can block business processes.
Replaced by another feature?	No.
Product areas affected	Product information management
Deployment option	All
Status	Removed as of Finance and Operations 10.0.0 with Platform update 24.

Finance and Operations 8.1.3 with Platform update 23

SQL Server Reporting Services ReportViewer Control

Customers can use the **Export** action provided by the embedded SQL Server Reporting Services (SSRS) ReportViewer control to download documents produced by Finance and Operations applications. This HTML-based presentation of the report offers users a non-paginated preview of the document.

Expand table

Reason for deprecation or removal	The non-paginated nature of the HTML-based preview experience doesn't deliver fidelity with the physical documents that Finance and Operations ultimately produces. By fully embracing PDF as the standard format for business documents, users can take advantage of a modern viewing experience with improved performance when producing application reports.
Replaced by another feature?	Going forward, Finance and Operations renders reports as PDF documents by default.
Product areas affected	This change doesn't impact customer scenarios where reports are distributed electronically or sent directly to printers.
Deployment option	All
Status	Deprecated: A removal date hasn't been set for this feature. The functionality to automatically preview application reports by using an embedded PDF viewer is planned for the May 2019 Platform update.

Client KPI controls

A developer can model embedded key performance indicators (KPIs) in Visual Studio, and an end user can customize them.

Expand table

Reason for deprecation or removal	The native client controls used to define KPIs have low customer uptake and rely on a developer to add trackable metrics.
Replaced by another feature?	PowerBI.com service delivers world-class tooling for defining and managing KPIs based on data from external sources. In an upcoming release, we plan to enable you to embed solutions hosted on PowerBI.com in application workspaces.

Product areas affected	This update prevents developers from introducing new KPI controls in Visual Studio designer.
Deployment option	All
Status	Deprecated: A removal date hasn't been set for this feature.

Deprecated APIs and future breaking changes

Field groups containing invalid field references

Expand table

Reason for deprecation or removal	Table metadata definitions can have field groups that contain invalid field references. If you deploy these definitions, they can cause runtime failures in Financial Reporting and SQL Server Reporting Services (SSRS). This issue is currently categorized as a *compiler warning* rather than an *error*, which means that you can create and deploy the deployable package without fixing the issue. To fix this issue: 1. Remove the invalid field reference from the table field group definition. 2. Recompile. 3. Ensure you address any warnings or errors.
Replaced by another feature?	This warning will be replaced by a compile error in the future.
Product areas affected	Visual Studio development tools
Deployment option	All
Status	Deprecated: The warning is a compile-time error with platform updates for version 10.0.11 of Finance and Operations apps.

Complete list

To access the full list of APIs that are being deprecated, see Deprecation of methods and metadata elements.

Finance and Operations 8.1 with Platform update 20

Batch transfer rules for subledger journal account entries

The General ledger parameters currently include a synchronous transfer mode that is deprecated. This mode is replaced by asynchronous and scheduled batch only, which already exist as options for transfer. For more information, see the General Ledger Parameters – Batch transfer rules blog.

Expand table

Reason for deprecation/removal	The synchronous option causes a performance impact on the system.
Replaced by another feature?	Use asynchronous and scheduled batch options instead of synchronous.
Product areas affected	General Ledger, Accounts payable, Accounts Receivable, Procurement, Expense
Deployment option	All
Status	Deprecated: Target timeframe for removal is the 10.0 version.

Electronic reporting for Russia

Feature for configuring .txt and .xml file formats of declarations.

Expand table

Reason for deprecation/removal	Replaced by Electronic reporting.
Replaced by another feature?	Yes.
Product areas affected	General Ledger
Deployment option	All
Status	Removed as of Finance and Operations 8.1 with Platform update 20.

Financial reports generator for Russia

A tool for setting up data collection for accounting and tax reports, and to export data to XLS and DOC report templates. The removed functional parts include exporting data to XLS and DOC report templates, queries, and fixed requisites.

Expand table

Reason for deprecation/removal	Removed parts are replaced by Electronic reporting.
Replaced by another feature?	Yes. Use the Financial reports setup user interface to set up data collection rules by GL accounts or tax registers. Use Electronic reporting to export data to various file types, and to configure fixed requisites and query-like data collection rules.
Product areas affected	General ledger.
Deployment option	All
Status	Removed as of Finance and Operations 8.1 with Platform update 20.

Integration with external providers for sending electronic reporting through communication channels for Russia

A feature that exports generated electronic files of declarations to a folder for further sending to official providers of electronic reporting, and imports state back.

Expand table

Reason for deprecation/removal	Replaced by electronic messages configurable feature.
Replaced by another feature?	Yes.
Product areas affected	General Ledger, Tax
Deployment option	All
Status	Removed as of Finance and Operations 8.1 with Platform update 20.

Profit tax register wizard

Feature for creating templates for new profit tax registers. This feature creates X++ objects for new registers, which you can use as templates for adding the appropriate calculation logic.

Expand table

Reason for deprecation/removal	Feature isn't compatible with the Finance and Operations extensibility model.
Replaced by another feature?	No
Product areas affected	Tax
Deployment option	All
Status	Removed as of Finance and Operations 8.1 with Platform update 20.

Payroll and Human Resources for Russia

Russian country/region specific module for managing staff administration information, timesheet details for employees, payroll accounting, and creating pay statements.

Expand table

Reason for deprecation/removal	Payroll isn't included in the global strategic focus of the Dynamics 365 portfolio. Partners and ISVs are best positioned to provide payroll functionality that is compliant with local regulations and tax updates.
Replaced by another feature?	No
Product areas affected	Russian Payroll and Human Resources Management
Deployment option	All
Status	Deprecated: Target timeframe for the functionality to be removed is one of future updates of the 10.0 version.

Finance and Operations 8.0 with Platform update 15

This release doesn't remove or deprecate any features. Platform update 15 is cumulative and contains new or changed features from Platform update 13, Platform update 14, and Platform

update 15.

Finance and Operations, Enterprise edition 7.3 with Platform update 12

Personalized product recommendations

Starting February 15, 2018, retailers can't display personalized product recommendations on a point of sale (POS) device. For more information, see Product recommendations overview.

Expand table

Reason for deprecation or removal	Microsoft is removing the current version of the product recommendation service to redesign this feature with a better algorithm and newer retail-oriented capabilities.
Replaced by another feature?	No. However, after spring 2018, Microsoft plans to bring back this feature to leverage a new recommendation service.
Product areas affected	Personalized product recommendations in POS.
Deployment option	All
Status	Removed as of February 15, 2018. This change affects customers running Dynamics 365 for Operations 1611 and later.

Extension of the list of Electronic reporting (ER) functions

The product no longer supports introducing custom functions to use in the ER expression builder. For more information, see Extend the list of Electronic reporting (ER) functions. Due to changes in the ER APIs, the API to call built-in functions from the ER expression builder is internal and can't be extended.

Expand table

Reason for deprecation or removal	Code sealing initiative

Replaced by another feature?	None. Whenever you need a new built-in function, submit a new extension request to the ER framework team. As a temporary workaround while the requested function is under development by the ER team, you can program the required logic as a method of a custom application class. You can access this method in an ER expression as a property of the added ER data source of the **Application\Class** type that refers to that custom application class.
Product areas affected	Electronic reporting framework
Deployment option	All
Status	Removed as of Finance and Operations, Enterprise edition 7.3.

Inventory by item group and Inventory by inventory dimension aging reports

Finance and Operations no longer supports these two reports. Instead, use the **Inventory aging** report to improve the user experience.

Expand table

Reason for deprecation	Duplicate functionality
Replaced by another feature?	Yes. The two reports have been replaced by the **Inventory aging** report.
Product areas affected	Inventory management, Cost management
Deployment option	All
Status	Deprecated: The product removes the menu items for the two reports in version 7.3. However, the product retains the code for the reports. The plan is to remove the code in a future release.

Power BI content packs available on Marketplace

Product updates in Microsoft Power BI deprecate the **Cost management**, **Financial performance**, and **Retail channel performance** content packs that are available on the Microsoft Marketplace site. Finance and Operations also deprecates system administration forms that you use to deploy these content packs to PowerBI.com.

Expand table

Reason for deprecation/removal	Product updates in Microsoft Power BI.
Replaced by another feature?	Analytical applications which allow for solution integrations at the database level replace the **Cost management**, **Financial performance**, and **Retail channel performance** content packs that are available on the Marketplace site. For more information about analytical applications, see Embedded Power BI in workspaces.
Product areas affected	Cost management, Finance, and Retail
Deployment option	Cloud only (Integration with PowerBI.com is not supported in on-premises deployments.)
Status	Deprecated: Target timeframe for the functionality removal is Q2 2018.

Standard UI in data management workspace

The standard UI in data management is the legacy UI. It's the default UI that users see when they visit the data management workspace.

Expand table

Reason for deprecation or removal	Investments are focused on providing new user experiences in the new UI.
Replaced by another feature?	The new UI called *Enhanced views* replaces the old UI.
Product areas affected	Data management workspace
Deployment option	All
Status	Deprecated: Target timeframe for removal is Q2 2018.

Excise, sales tax, service tax for India

These taxes are subsumed into Indian GST.

Expand table

Reason for removal or deprecation	These taxes are subsumed into Indian GST.
Replaced by another feature?	Indian GST
Product areas affected	Tax
Deployment option	All modules
Status	Deprecated: A removal date isn't set for this feature.

File Validation Utility (FVU) for India

Expand table

Reason for removal or deprecation	Lack of customer usage
Replaced by another feature?	No
Product areas affected	Indian withholding tax
Deployment option	All modules
Status	Deprecated: A removal date hasn't been set for this feature.

TDS/TCS certificate for India

Users can download this certificate from the government portal.

Expand table

Reason for removal or deprecation	Lack of customer usage
Replaced by another feature?	No
Product areas affected	Indian withholding tax
Deployment option	All modules

Status	Deprecated: A removal date hasn't been set for this feature.

Export/import (EXIM) incentive scheme for India

Expand table

Reason for removal or deprecation	Lack of customer usage
Replaced by another feature?	No
Product areas affected	Import and export
Deployment option	All modules
Status	Deprecated: A removal date hasn't been set for this feature.

Dynamics 365 for Retail 7.2

Personalized product recommendations

Starting February 15, 2018, retailers can't display personalized product recommendations on a point of sale (POS) device. For more information, see Product recommendations overview.

Expand table

Reason for deprecation or removal	Microsoft is removing the current version of the product recommendation service to redesign this feature with a better algorithm and newer retail-oriented capabilities.
Replaced by another feature?	No. However, after spring 2018, Microsoft plans to bring back this feature to leverage a new recommendation service.
Product areas affected	Personalized product recommendations in POS.
Deployment option	All
Status	Removed as of February 15, 2018. This change affects customers running Dynamics 365 for Retail 7.2 and later.

Finance and Operations, Enterprise edition July 2017 with Platform update 8

Currency conversion for accounting and reporting currencies

Currency conversion for accounting and reporting currencies was introduced when the euro was introduced.

Expand table

Reason for deprecation or removal	Limited usage and addition of the Copy legal entity functionality as a replacement.
Replaced by another feature?	No, but the Copy legal entity and Configurations features were added to make it easier to move to a company that has changing core requirements.
Product areas affected	Financial management
Status	Deprecated: A removal date hasn't been set for this feature.

Warehouse mobile devices portal

Warehouse mobile devices portal (WMDP) was a standalone component that you intended to self-deploy on-premises. Finance and Operations no longer supports this component. A native app that improves the user experience replaces the functionality of WMDP.

Expand table

Reason for deprecation or removal	Duplicate functionality.
Replaced by another feature?	Yes. Finance and Operations - Warehousing replaces this feature. For more information about setup and prerequisites, see Install and configure the Warehousing app overview.
Product areas affected	Warehouse management, Transportation management
Deployment option	Warehouse mobile devices portal (WMDP) was a standalone component that you

	intended to self-deploy on-premises.
Status	Deprecated: Target timeframe for the functionality to be removed is Q4 2019.

Advanced bank reconciliation matching rule for manual matching

Use this matching rule to select and mark a bank document when you manually match documents in the reconciliation worksheet.

Expand table

Reason for deprecation/removal	Limited usage.
Replaced by another feature?	No. Use column filtering capabilities to find documents for reconciliation.
Product areas affected	Cash and bank management
Deployment option	All
Status	Removed as of July 2017.

Dynamics 365 for Operations 1611 with Platform update 3

AEB payment formats for Spain

Use the Consejo Superior Bancario payment formats to send remittance files to the bank for customer payments and vendor payments. The Asociación Española de Banca determines the content of these formats. It covers Cuaderno 19, 32, 58, 34.

Expand table

Reason for deprecation/removal	The payment formats are no longer used.

Replaced by another feature?	Yes, ISO20022 Credit transfer and Direct debit payment formats for Spain
Product areas affected	Accounts payable, Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Bank payments transfer for Lithuania

Use the Payment transfer (LT) export format to generate and print bank payment transfers for Lithuania. In 2005, the Lithuanian market started using LITAS, the unified electronic banking system.

Expand table

Reason for deprecation/removal	These payment formats are no longer used.
Replaced by another feature?	Yes, ISO20022 Credit transfer payment format for Lithuania
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

BBS Direkte Remittering payment formats for Norway

BBS Direkte Remittering payment formats include customer payment collection export (direct debit) and return message import.

Expand table

Reason for deprecation/removal	These payment formats are no longer used.
Replaced by another feature?	Use the AvtaleGiro customer payment format for Norway to generate direct debit messages. Return message import will be implemented in future releases.
Product areas affected	Accounts payable, Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Chart of Accounts tool for Spain

Use this tool when you need to make major changes to a chart of accounts in Spain. You can import a new chart of accounts in Microsoft Excel or text format, and you can also import financial statements.

Expand table

Reason for deprecation or removal	Limited usage
Replaced by another feature?	No
Product areas affected	General ledger
Status	Deprecated: A removal date hasn't been set for this feature.

Dom80 payment format for Belgium

Legacy Belgian payment format for payment collection (direct debit).

Expand table

Reason for deprecation or removal	The payment format is no longer used.
Replaced by another feature?	Yes, ISO 20022 Direct debit payment format for Belgium
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

DTA/EZAG payment formats for Switzerland

The DTA and EZAG formats are part of the ESR system because they support the reference number. Since the reference number isn't mandatory, you can use these formats to process any vendor payments. Companies that use these formats typically have a bank account at a bank other than PostFinance.

Expand table

Reason for deprecation/removal	These payment formats are no longer used.
Replaced by another feature?	Yes, ISO 20022 Credit transfer payment format for Switzerland
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

EDIFACT-DIRDEB payment format for Austria

The EDIFACT-DIRDEB payment format supports payment collection through direct debit.

Expand table

Reason for deprecation or removal	The payment format is no longer used.
Replaced by another feature?	Yes, ISO 20022 Direct debit payment format for Austria
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

EDIVAT for Belgium

EDIVAT is an obsolete Belgian standard for electronic declaration via secure mail. Dynamics AX 2012 retains the read-only solution to enable access to the historical data.

Expand table

Reason for deprecation or removal	The functionality is no longer used.
Replaced by another feature?	No
Product areas affected	General ledger
Status	Deprecated: A removal date hasn't been set for this feature.

eGiro EDIFACT CREMUL payment import format for Norway

eGiro is based on the international UN EDIFACT CREMUL (Multiple Credit Advice Message) standard that banks use for automatic posting of customer payments. In Dynamics AX, you implement eGiro as a customer payment import format.

Expand table

Reason for deprecation or removal	The payment format is no longer used.
Replaced by another feature?	Yes, the ISO20022 Camt.054 notification import.
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

External inventory for Poland

Evidence of goods that you take from a vendor for sales without purchase. Goods that you handle in external inventory don't affect standard inventory, and you can sell and then purchase them automatically. This process creates real inventory movements.

Expand table

Reason for deprecation or removal	Replaced by another feature
Replaced by another feature?	Yes, the core Inbound consignment functionality
Product areas affected	Accounts payable, Inventory management
Status	Deprecated: A removal date hasn't been set for this feature.

Financial reports generator for Eastern Europe

A tool that you use to set up data collection for accounting and tax reports, and to export data to XLS and DOC report templates.

Expand table

Reason for deprecation or removal	Limited usage

Replaced by another feature?	No. The tool will be replaced by Electronic reporting configurations in future releases.
Product areas affected	General Ledger
Status	Deprecated: A removal date hasn't been set for this feature.

Import of customer payment transactions for Finland

You can select an import format for Finnish payments to import customer payment transactions from an external file that the bank provides.

Expand table

Reason for deprecation or removal	The payment format is no longer used.
Replaced by another feature?	Yes, the ISO20022 Camt.054 notification import.
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Import of payment transactions into a general ledger journal for Finland

A format that's specific to Finland is used to import accounting transactions into the general ledger.

Expand table

Reason for deprecation or removal	The payment format is no longer used.
Replaced by another feature?	Yes, the ISO20022 Camt.053 bank statement import by using Advanced Bank Reconciliation.
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Integration with Isabel synchronized (CIS) for Belgium

Isabel is the framework for electronic banking in Europe and is a de facto standard in Belgium.

Expand table

Reason for deprecation or removal	Integration with Isabel client is discontinued.
Replaced by another feature?	No. The payment formats that are no longer used are replaced by ISO 20022 Credit transfer payment format for Belgium.
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

Modifications in the chart of accounts and accounting rules for Spain

This feature is used for changes in the chart of accounts and accounting rules in Spain. It maps accounts to help transform the old chart of accounts into the new chart of accounts, and compares the previous fiscal year with the new fiscal year, even if they were posted to different account numbers.

Expand table

Reason for deprecation or removal	Limited usage
Replaced by another feature?	No
Product areas affected	General ledger
Status	Deprecated: A removal date hasn't been set for this feature.

Pagamento Fornittori vendor payment format

Legacy Italian payment format for credit transfers.

Expand table

Reason for deprecation or removal	The payment format is no longer used.
Replaced by another feature?	Yes, ISO 20022 Credit transfer payment format for Italy
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

Payment export formats for Estonia

The Telehansa and Teleservice formats are used for bank payment export.

Expand table

Reason for deprecation/removal	These payment formats are no longer used.
Replaced by another feature?	Yes, ISO 20022 Credit transfer payment format for Estonia
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

Payment file archive for Norway

When you generate payment files, the file archive automatically archives all created files, even files that you previously wrote or read.

Expand table

Reason for deprecation or removal	Replaced by another feature
Replaced by another feature?	Yes, Electronic reporting archived jobs
Product areas affected	Accounts payable, Accounts receivable, Organization administration
Status	Deprecated: A removal date hasn't been set for this feature.

Payment import formats for Estonia

Use the Telehansa and TeleTeenus formats for bank payment import.

Expand table

Reason for deprecation or removal	The payment formats are no longer used.
Replaced by another feature?	Yes, the ISO20022 Camt.054 bank notification import.
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Payroll information in Human Resources

Human Resources Payroll information

Expand table

Reason for deprecation or removal	Core Payroll and Human Resources pages now provide this functionality.
Replaced by another feature?	**Benefits**, **Earnings**, and other related pages that were previously in US Payroll have been reconfigured. They're now part of the core Human Resources configuration to help support external payroll processing. Use the **Human Resources 1** > **Payroll** configuration key to access this functionality.
Product areas affected	Human Resources, Payroll
Status	Removed as of Dynamics 365 for Operations version 1611.

Performance management goal workflow

Performance management includes goal management and integration with performance reviews.

Expand table

Reason for deprecation or removal	Performance management was redesigned, and the number of goal pages was reduced to simplify the process.

Replaced by another feature?	No. Managers can see goals through the Manager Self Service portal, and they can change and view the goals.
Product areas affected	Human capital management
Status	Removed as of Dynamics 365 for Operations version 1611.

Postgirot and Postgirot Utland payment formats for Sweden

Postgirot and Postgirot Utland payment formats for Sweden.

Expand table

Reason for deprecation/removal	These payment formats are no longer used.
Replaced by another feature?	Yes, by the ISO 20022 credit transfer payment format for Sweden.
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

Radio frequency identifier

Radio Frequency Identification (RFID) is a data-collection technology that uses electronic tags to store identification data and a no-line-of-sight requirement reader to capture the identification data.

Expand table

Reason for deprecation or removal	Low customer usage and a limited feature set.
Replaced by another feature?	No
Product areas affected	Inventory management
Status	Removed as of Dynamics 365 for Operations 1611.

Report about state invoices numbering for Latvia

Latvian legislation provides specific rules about the numbering of sales invoices. By using this functionality, you can assign specific numbers to sales invoices based on the user or user group. You can then generate a report or an XML file. You can also print a report about invoice numbers that are used.

Expand table

Reason for deprecation/removal	You no longer need to maintain the state invoice numbering. You no longer need to report about used invoice numbers.
Replaced by another feature?	No
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Set up the names of the manager and general accountant of a company for Lithuania

You can specify the names of the manager and the general accountant of a company in the company information and use them in different local report printouts.

Expand table

Reason for deprecation/removal	Replaced by another feature
Replaced by another feature?	Yes, use the setup of officials for the same purpose.
Product areas affected	Accounts payable, Accounts receivable, Cash and bank management
Status	Deprecated: A removal date hasn't been set for this feature.

Shipping carrier interface

Expand table

Reason for deprecation/removal	Duplicate functionality

Replaced by another feature?	Partially replaced by Transportation management
Product areas affected	Sales and marketing, Inventory management
Status	Removed as of Dynamics 365 for Operations version 1611.

Telepay payment formats for Norway

Telepay payment formats include vendor payment export (credit transfer) and customer payment collection (direct debit).

Expand table

Reason for deprecation/removal	The payment formats are no longer used.
Replaced by another feature?	Yes, ISO 20022 Credit transfer payment format and AvtaleGiro customer payment format for Norway, as well as pain.002 and camt.054 bank notification return files import.
Product areas affected	Accounts payable, Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Vendor payment export formats for Finland

Two formats for exporting payments are available for Finland. Use LM02 (FI) for domestic payments, and use LUM2 (FI) for foreign payments.

Expand table

Reason for deprecation/removal	These payment formats are no longer used.
Replaced by another feature?	Yes, ISO 20022 Credit transfer payment format for Finland
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

Warehouse management II

Expand table

Reason for deprecation or removal	The Warehouse management II solution (WMS II) that was available in the **Inventory management** module duplicates functionality that's in the **Warehouse management** module that was released in Dynamics AX 2012 R3.
Replaced by another feature?	The **Warehouse management** module that was released in AX 2012 R3, Dynamics AX 2012 R3 CU8, and Dynamics AX 2012 R3 CU9 replaces the Warehouse management II features. The new module has more advanced features and more flexible warehouse management processes than Warehouse management II.
Product areas affected	Inventory management, Sales and marketing, Procurement and sourcing
Status	Removed as of Dynamics 365 for Operations version 1611.

Worker reminders in Human Resources

Human Resources Payroll information

Expand table

Reason for deprecation/removal	Low usage
Replaced by another feature?	No
Product areas affected	Human resources
Status	Removed as of Dynamics 365 for Operations version 1611

Workflow for creating goals

A workflow for managing the creation of employee goals was one of several workflows that helped coordinate the performance management process.

Expand table

Reason for deprecation/removal	Performance management has been completely redesigned in Finance and Operations.

Replaced by another feature?	The redesigned Performance management feature gives more control over the content of the goals, the measurements that are used to track progress, and the attachment of supporting documentation. Goals can be stored as templates and then reused. This feature can help you set up additional goals for your employees more quickly.
Product areas affected	Human capital management
Status	Removed as of Dynamics 365 for Operations version 1611.

Dynamics AX 7.0

Ability to cancel changes to a vendor invoice

Expand table

Reason for deprecation or removal	Performance enhancement
Replaced by another feature?	No
Product areas affected	Accounts payable
Status	Removed as of Dynamics AX 7.0.

AIF, AxD, and AxBC integrations

In Application Integration Framework (AIF), you can exchange data with external systems through business logic that you expose as services. Dynamics AX includes services that are based on documents and .NET Business Connector (AxBC). You create a document by using XML. The XML includes header information that you add to create a *message* that you can transfer into or out of Dynamics AX. Examples of documents include sales orders and purchase orders. However, you can represent almost any entity, such as a customer, by using a document. Services that are based on documents use the **Axd <Document>** classes.

Expand table

Reason for deprecation or removal	The architecture of AIF and AxDs couldn't scale to a cloud service. There were performance problems around bulk import.

Replaced by another feature?	This feature is replaced by the Data Import/Export framework, which supports recurring bulk import and export. For AxBC, use the actual tables.
Product areas affected	AxDs, AxBCs, and AIF
Status	Removed as of Dynamics AX 7.0.

Billing code rate scripts

Use billing scripts to calculate billing rates for billing codes. You need to custom develop these scripts by using C# or Visual Basic. In the current version of Dynamics AX, billing code rate scripts aren't supported.

Expand table

Reason for deprecation or removal	Dynamics AX doesn't support custom C# or Visual Basic scripts.
Replaced by another feature?	No
Product areas affected	Public sector, Accounts receivable
Status	Removed as of Dynamics AX 7.0.

BOMs without BOM versions

When you disable the **BOM versions** configuration key, the system hides bill of materials (BOM) versions in all forms and enforces a one-to-one relationship between released products and BOMs. In the current version of Dynamics AX, you can't disable the **BOM versions** configuration key.

Expand table

Reason for deprecation or removal	Using a configuration key to control BOM versions doesn't scale in a cloud environment.
Replaced by another feature?	No
Product areas affected	Product information management, Inventory management
Status	Removed as of Dynamics AX 7.0.

Brazilian Bordero

Specific method of payment for Brazilian companies

Expand table

Reason for deprecation or removal	Brazilian localization no longer supports the Brazilian Bordero method of payment.
Replaced by another feature?	No
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

Brazilian Sintegra statement

Federal tax statement for ICMS tax

Expand table

Reason for deprecation or removal	This statement is no longer applicable in some Brazilian states.
Replaced by another feature?	No. Use the Generic Electronic reporting tool to configure the statement if required under specific situations.
Product areas affected	Fiscal books
Status	Deprecated: A removal date hasn't been set for this feature.

Brazilian SCAN contingency mode for NF-e

Use the (SCAN) contingency environment to generate, export, and import the status of a Nota Fiscal eletrônica (NF-e) when the environment of Secretaria da Fazenda (SEFAZ) isn't available.

Expand table

Reason for deprecation/removal	This method of contingency is no longer applicable in all Brazilian states

Replaced by another feature?	No
Product areas affected	Accounts receivable
Status	Deprecated: A removal date hasn't been set for this feature.

Business Analyzer

This mobile application lets users review key business metrics.

Expand table

Reason for deprecation/removal	This functionality has been replaced by another feature.
Replaced by another feature?	The Monitor financial performance content pack for Microsoft Power BI includes key financial metrics that were previously available in Business Analyzer.
Product areas affected	General ledger
Status	Deprecated: The use of Business Analyzer is deprecated.

Business statistics

The setup of business statistics inquiries that you can use to analyze the performance of the organization.

Expand table

Reason for deprecation or removal	Legacy approach to business intelligence (BI), low customer usage, and a limited feature set.
Replaced by another feature?	New BI solutions for the current version of Dynamics AX.
Product areas affected	Procurement and sourcing, Accounts payable, Sales and marketing, Accounts receivable.
Status	Removed as of Dynamics AX 7.0.

Change document date function in Invoice approval journal

Expand table

Reason for deprecation or removal	Low usage.
Replaced by another feature?	Yes. You can change the document date on the posted vendor transaction.
Product areas affected	Accounts payable.
Status	Removed as of Dynamics AX 7.0.

ClieOp03 payment format for the Netherlands

Expand table

Reason for deprecation or removal	This format is no longer applicable in the Netherlands, because it's replaced by SEPA functionality.
Replaced by another feature?	SEPA payments export.
Product areas affected	All modules.
Status	Deprecated: A removal date hasn't been set for this feature.

Compliance Center

The Compliance Center was an enterprise portal site for managing the documentation requirements for compliance initiatives that are related to the Sarbanes-Oxley law.

Expand table

Reason for deprecation or removal	Lack of customer usage. Microsoft SharePoint includes the same capability that was available in the Compliance Center.
Replaced by another feature?	No

Product areas affected	Compliance and internal controls
Status	Removed as of Dynamics AX 7.0.

Connector for Microsoft Dynamics

Use this tool to integrate key data from Microsoft Dynamics CRM to Dynamics ERP applications.

Expand table

Reason for deprecation or removal	This functionality is replaced by another feature.
Replaced by another feature?	Dataverse
Product areas affected	Connector for Dynamics
Status	Removed as of Dynamics AX 7.0.

Container unit and multi dimension on-hand

Expand table

Reason for deprecation/removal	Duplicate functionality
Replaced by another feature?	Yes. Since AX 2012, this functionality has been replaced by the consolidated batch orders feature set. This feature set includes the consolidated on-hand view.
Product areas affected	Product information management, Production control, Inventory management, Sales and marketing
Status	Removed as of Dynamics AX 7.0.

Cue group metadata

Expand table

Reason for deprecation/removal	Cue groups displayed one or more Cues in the FactBox area. There was limited uptake, and there were also performance concerns. A record change in a parent form caused one query per Cue in the Cue group.
Replaced by another feature?	No
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0.

Cue metadata

Expand table

Reason for deprecation/removal	Cue metadata was limited to count or sum information.
Replaced by another feature?	Tile metadata was introduced to provide more flexibility for modeling. For example, you can model current counts, navigation, and key performance indicators (KPIs). Count tile metadata is the direct replacement of the Cue metadata.
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0

Danish check format

Expand table

Reason for deprecation or removal	Support for the Danish check format layout has been discontinued, and DK localization no longer includes the report.
Replaced by another feature?	No
Product areas affected	All modules
Status	Deprecated: A removal date hasn't been set for this feature.

Data partitions

Data partitions provide a logical separation of data in the Dynamics AX database.

Expand table

Reason for deprecation or removal	Data partitions were introduced in Dynamics AX 2012 R2 to enable data isolation. In a common scenario, a company has subsidiaries, and the data from one subsidiary shouldn't be visible to another subsidiary, even though the same IT department manages both subsidiaries. However, extra scripts and management overhead throughout the program were required to create new partitions, populate them with data, and back up partition data. In the cloud, where you have access to platform as a service (PaaS) database services (Microsoft Azure SQL Database), it's much more efficient to use a database as the isolation container than to do isolation in the program. Regardless of whether data partitioning is required for subsidiaries, for multiple tenants, or just for scale, Finance and Operations can handle these scenarios better through multiple instances.
Replaced by another feature?	Customers using data partitions must use multiple instances of Finance and Operations if database level separation is a critical issue.
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0.

Database and file share storage for attachments

Dynamics AX 2012 allowed storage of attachments in the database and in file shares. Both of those options aren't supported anymore.

Expand table

Reason for deprecation or removal	File share storage isn't supported anymore because cloud-hosted environments can't communicate with local file shares. Database storage is deprecated in favor of Azure Blob storage. Azure Blob storage is equivalent to storage in the database, as you can only access documents through Finance and Operations client forms. This storage option provides the added benefit of not negatively affecting the performance of the database. Blob storage is the default storage mechanism for Document Management and works immediately.
Replaced by another	Database storage is deprecated in favor of Azure Blob storage.

feature?	
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0.

Delimitation

Expand table

Reason for deprecation or removal	No use of the functionality was found.
Replaced by another feature?	No
Product areas affected	Time and attendance
Status	Removed as of Dynamics AX 7.0.

Desktop client

Expand table

Reason for deprecation or removal	The Dynamics AX client experience was redesigned to improve usability across multiple platforms and devices.
Replaced by another feature?	The new web client is based on the desktop Form metadata and programming model that are modified to provide a rich web platform.
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0.

Direct database connection

In Dynamics AX 2012 R3, Retail Modern POS can connect directly to the Channel DB in a similar way to Enterprise POS. This connection is in addition to the standard communication method where Retail Modern POS communicates through Retail Server.

Expand table

Reason for deprecation or removal	Direct database connectivity requires lower security protocols. Use it primarily to achieve the highest levels of performance. Due to the performance and security enhancements that occurred in Finance and Operations, this functionality now causes more problems than it solves.
Replaced by another feature?	No. Only standard Retail Server communication is now supported.
Product areas affected	Channel DB and Retail Modern POS
Status	Removed as of Dynamics AX 7.0.

Dutch SWIFT MT940

Expand table

Reason for deprecation or removal	Use generic functionality instead of localized functionality.
Replaced by another feature?	Yes, Advanced bank reconciliation functionality.
Product areas affected	All modules
Status	Deprecated: A removal date hasn't been set for this feature.

eBilanz (XBRL for Germany)

This functionality provided eXtensible Business Reporting Language (XBRL) output that's intended specifically for the German eBilanz taxonomy.

Expand table

Reason for deprecation or removal	Lack of customer usage
Replaced by another feature?	This feature isn't replaced by another feature. However, multiple specialized XBRL packages that provide rich XBRL functionality are available for the German market.

Product areas affected	Management Reporter
Status	Deprecated: A removal date hasn't been set for this feature.

Enterprise Portal client

Expand table

Reason for deprecation or removal	A single client platform is provided.
Replaced by another feature?	The new web client is based on the desktop form metadata and programming model. This model is modified to provide a rich web platform.
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0.

Environmental sustainability

Expand table

Reason for deprecation or removal	Low customer usage and a limited feature set
Replaced by another feature?	No
Product areas affected	Compliance and internal controls, Accounts payable
Status	Removed as of Dynamics AX 7.0.

Form ActiveX and managed host controls

Expand table

Reason for deprecation/removal	The ActiveX and managed host controls are based on the deprecated desktop client.

Replaced by another feature?	The extensible control framework supports building new controls that are based on HTML, CSS, and JavaScript, and is a first-class control in the Microsoft Visual Studio Tooling environment.
Product areas affected	All modules.
Status	Removed as of Dynamics AX 7.0.

Generate prenotes by using a batch

You can't use a batch to generate prenotes, but a user can still generate them.

Expand table

Reason for deprecation/removal	No form exists to persist and display the resulting prenote file when you generate it by using a batch.
Replaced by another feature?	You can still generate prenotes, and you have control over the location where the file is saved.
Product areas affected	Accounts payable, Accounts receivable, Cash and bank management
Status	Removed as of AX 7.0.

German DTAUS payment export and account statement import (totals and transactions)

Expand table

Reason for deprecation/removal	This format is no longer applicable in Germany, because it's replaced by Single Euro Payments Area (SEPA) functionality.
Replaced by another feature?	Yes, this functionality is replaced by SEPA payment export and advanced bank reconciliation functionality for importing account statements.
Product areas affected	All modules
Status	Deprecated: A removal date hasn't been set for this feature.

German DTAZV payment format in domestic currency

Expand table

Reason for deprecation/removal	The format is no longer applicable in Germany, because SEPA functionality replaced it.
Replaced by another feature?	SEPA payments export
Product areas affected	Accounts payable
Status	Deprecated: A removal date hasn't been set for this feature.

German MT940 import

Expand table

Reason for deprecation or removal	Use generic functionality instead of localized functionality.
Replaced by another feature?	Yes, Advanced bank reconciliation functionality.
Product areas affected	All modules
Status	Deprecated: A removal date hasn't been set for this feature.

German XML EU Sales list

Expand table

Reason for deprecation/removal	The XML format for German EU Sales List reporting is no longer supported. You can only use the ELMA5 text file format to submit the EU Sales List report to the German Tax Office.
Replaced by another feature?	No
Product areas affected	Tax
Status	Deprecated: A removal date hasn't been set for this feature.

GL SSRS reports

Reports that include the following menu items are removed: **Summary trial balance**, **Detailed trial balance**, **Chart of accounts**, **Audit trail**, **Balances**, and **Balance list**.

Expand table

Reason for deprecation or removal	Financial Microsoft SQL Server Reporting Services (SSRS) reports are replaced by Management Reporter capabilities and default reports.
Replaced by another feature?	Management Reporter (labeled **Financial reporting** in the current version of Dynamics AX)
Product areas affected	General ledger
Status	Removed as of Dynamics AX 7.0.

InfoPart and FormPart metadata

Expand table

Reason for deprecation or removal	InfoPart and FormPart metadata enabled the creation of FactBoxes for two different clients.
Replaced by another feature?	InfoPart metadata, which was a simplified form definition, is converted into a Form by upgrade tooling. FormPart metadata, which referenced a Form, is replaced by a more direct reference that is created by upgrade tooling.
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0.

Main account list page

A list of accounts for the legal entity and related balance information.

Expand table

Reason for deprecation or removal	The **Trial balance** list page shows balance information by account and dimension.

Replaced by another feature?	The **Main accounts** list page contains the same list of accounts. It also provides a more streamlined grid view.
Product areas affected	General ledger
Status	Removed as of Dynamics AX 7.0.

Malaysia and Singapore bank cash flow report

This feature lets you print a cash flow report that shows transactions and details of the cash inflows and outflows for a specific date range for selected bank accounts.

Expand table

Reason for deprecation or removal	You can get the same information from the **Inquiry bank transaction**.
Replaced by another feature?	The **Inquiry bank transaction**.
Product areas affected	Cash and bank management.
Status	Deprecated: A removal date hasn't been set for this feature.

Mexican CFD electronic invoice

This feature enabled the generation of Mexican electronic invoices by using the Comprobante Fiscal Digital (CFD) method, where the company signs the invoice by requesting the related authorization from the government. This feature also provides a monthly report that includes all electronic invoices that were issued in the period.

Expand table

Reason for deprecation or removal	The method is no longer applicable. The tax authorities deprecated the generation of electronic invoices by using the CFD method and replaced it with the Comprobante Fiscal Digital a través de Internet (CFDI) method, where the signing is delegated to the third-party provider (PAC). The monthly report was removed, and an inquiry option lets users inquire about historical transactions.
Replaced by another	No

feature?	
Product areas affected	Accounts receivable, Project
Status	Deprecated: A removal date hasn't been set for this feature.

Mexico realized and unrealized VAT

Dynamics AX 2012 managed unrealized value-added tax (VAT) by using Mexico-specific functionality for unrealized tax.

Expand table

Reason for deprecation or removal	Duplicate functionality
Replaced by another feature?	Yes, standard conditional sales tax functionality provided by Core replaces this functionality.
Product areas affected	Tax
Status	Deprecated: A removal date hasn't been set for this feature.

Microsoft Outlook integration

Expand table

Reason for deprecation or removal	Microsoft Exchange Server integration replaces this functionality.
Replaced by another feature?	Yes
Product areas affected	Sales and marketing
Status	Removed as of Dynamics AX 7.0.

Private blocking of inventory and warehouse management journals

The inventory and warehouse journals no longer support the ability to mark a journal as private for a selected user. They support only blocking journals as private for user groups and blocking during editing.

Expand table

Reason for deprecation or removal	No use of the functionality was found.
Replaced by another feature?	No
Product areas affected	Inventory management
Status	Removed as of Dynamics AX 7.0.

Product builder

Use product builder to dynamically configure items from a sales order, purchase order, production order, sales quotation, project quotation, or item requirement. Based on a product model that has modeling variables, select values to meet the customer requirements and get a unique product variant that has a BOM and route.

Expand table

Reason for deprecation/removal	Product builder exposed X++ code to end users and isn't supported in the current version of Dynamics AX. Microsoft removed it to avoid duplicate maintenance efforts on overlapping, sizeable codebases.
Replaced by another feature?	Yes. Constraint-based configuration was introduced in Dynamics AX 2012. Microsoft announced the deprecation of product builder in future versions. Select constraint-based configuration technology on the product masters to enable the configuration. To learn more, see Product configuration overview.
Product areas affected	Product information management, Sales and marketing
Status	Removed as of Dynamics AX 7.0.

Production Floor app

This app runs on tablet devices that use Windows 8.1 RT and Windows 8.1 Pro.

Expand table

Reason for deprecation/removal	With the change to a web-based client, you can deliver similar functionality through the native Dynamics AX 7.0 client. The Job Card Device provides a production floor user interface that is optimized for touch and tablet form factors.
Replaced by another feature?	Yes. The Job Card Device, which is a native part of Dynamics AX 7.0.
Product areas affected	Production control
Status	Deprecated: A removal date from the Microsoft store hasn't yet been set for this feature.

Rename product dimension

This feature lets you change the name of one of the three standard product dimensions (size, color, or style) to a name that better suits your business requirements. Renaming includes all the labels where the product dimension name is used.

Expand table

Reason for deprecation/removal	The current version of Dynamics AX doesn't support label changes at run time.
Replaced by another feature?	No
Product areas affected	Product information management
Status	Removed as of Dynamics AX 7.0.

Retail Server connectivity using HTTP

In Dynamics AX 2012 R3, the Retail Server could function by using HTTP communication (non-secured). This option was in addition to the standard communication by using HTTPS.

Expand table

Reason for deprecation/removal	Due to new security requirements, only secured communication by using TLS 1.2 (or above, as available) is now supported. The self-service installer automatically configures the computer for this communication.

Replaced by another feature?	No. Only standard HTTPS communication is now supported.
Product areas affected	Retail Server
Status	Removed as of Dynamics AX 7.0.

Role Center pages

Expand table

Reason for deprecation or removal	Role Center pages were built on the deprecated Enterprise Portal platform. The current version of Dynamics AX replaces Enterprise Portal with the new web client platform.
Replaced by another feature?	The new Workspace form pattern provides users with a process-centered design that provides easy access to commonly used tasks within that process.
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0

Sales tax jurisdictions

Expand table

Reason for deprecation or removal	Low customer usage and a limited feature set
Replaced by another feature?	No
Product areas affected	US sales tax
Status	Removed as of Dynamics AX 7.0.

Sites Services

By using Sites Services, you can build websites that extend your business processes to the Internet without IT support.

Expand table

Reason for deprecation or removal	Use the new capabilities in the Microsoft Azure infrastructure that Dynamics AX uses, such as Azure sites.
Replaced by another feature?	No
Product areas affected	HR recruiting, Case management, Request for quotes, Vendor registration, Collaborative workspaces for opportunities and campaigns
Status	Removed as of Dynamics AX 7.0.

SSAS demand forecasting strategy

Expand table

Reason for deprecation/removal	The new cloud architecture can't support the design of this feature.
Replaced by another feature?	Azure Machine Learning demand forecasting strategy
Product areas affected	Master planning
Status	Removed as of Dynamics AX 7.0.

Vendor invoice pool excluding posting details

Expand table

Reason for deprecation/removal	Low usage. This functionality has been replaced by the Invoice journal that has workflow functionality.
Replaced by another feature?	Workflow capabilities of the Invoice journal.
Product areas affected	Accounts payable
Status	Removed as of Dynamics AX 7.0.

Virtual company accounts

Dynamics AX no longer supports the virtual companies feature. By using the virtual companies feature, you could set up tables that a set of companies shared. For a description of the feature, see Company accounts and Virtual company accounts. The feature works by grouping tables into collections that you assign to virtual companies, which are groups of existing "real" companies. You create queries so that all the companies in the virtual company can access the data in the tables of the associated table collections.

Expand table

Reason for deprecation/removal	- You must set up virtual companies before storing data in the tables. Retrofitting virtual companies onto an existing implementation is very difficult. - Because the current version of Dynamics AX uses so much data normalization, it can be difficult to know what to add to the table collections. For example, it's difficult to know which tables to share. You must add all the tables referenced from tables that are in a virtual company. Because of table normalization, even simple master data that is spread across multiple tables must be part of the virtual company. Any mistake that you make here causes functional problems. - When a table is part of a virtual company, it loses information about the origin of the data, and only the virtual company is recorded.
Replaced by another feature?	Use global tables to make tables accessible from all companies. Currently, there is no replacement.
Product areas affected	All modules
Status	Removed as of Dynamics AX 7.0.

Windows 8 tablet app

The Windows 8 tablet app provided functionality for expense entry and approval.

Expand table

Reason for deprecation or removal	Finance and Operations is compatible with tablets. The tablet app is no longer required.
Replaced by another feature?	No.
Product areas affected	Expense management
Status	Removed: This functionality is only available for Dynamics AX 2012 R3.

Workplanner

Expand table

Reason for deprecation or removal	Low usage
Replaced by another feature?	No, but the **Profile relation** page, which you open from the **Profile groups** page, supports the same business scenario as the deprecated **Workplanner** page.
Product areas affected	Time and attendance
Status	The code isn't removed. However, the form, JmgWorkPlanner, wasn't migrated.

X++ financial statements

Expand table

Reason for deprecation or removal	This functionality is replaced by another feature.
Replaced by another feature?	Management Reporter (labeled **Financial reporting** in the current version of Dynamics AX)
Product areas affected	General ledger
Status	Removed as of Dynamics AX 2012

Last updated on 03/18/2026

Capabilities that were postponed from Dynamics AX 2012

This article lists features from Microsoft Dynamics AX 2012 that Microsoft postponed delivery of. In the following table, the **Current status** column indicates whether the feature is available since the version 7.0.

For a detailed list of the release date for each version, see Software lifecycle policy and cloud releases.

Expand table

AX 2012 feature that Microsoft postponed	Description	Current status (as of February 2019)
Absence management in Human resources	This feature doesn't include functionality for entering absence transactions. Additionally, it doesn't include functionality for approving absence transactions as a manager. However, you can access setup capabilities that are required for integration with other modules through the **Human Resources 2** configuration key.	Implemented in Dynamics 365 Human Resources
Alerts	Alerts help users keep track of data changes in the system.	Implemented in Platform update 15
Bank payment order for Latvia and Lithuania	You can print a payment order for Latvia and Lithuania. This feature will be available in a future update.	Not implemented
Bankgirot AP return format for Sweden	The Bankgirot return format is used to import bank return messages. This feature will be available in a future update.	Not implemented
Client drag-and-drop	The web client controls have application programming interfaces (APIs) for drag-and-drop operations, but these APIs are based on the deprecated desktop client technology. They require a redesign so that they work on the new web client platform. APIs that support drag-and-drop operations will be reviewed for inclusion in a future update.	Not implemented
Client right-to-left (RTL) layout	RTL layout is now supported.	Implemented in Platform update 2

AX 2012 feature that Microsoft postponed	Description	Current status (as of February 2019)
Cost accounting	The **Cost accounting** module is designed to meet the requirements of internal costs and profitability reports at multiple organizational levels. To define the cost object level, the module depends on a correct mapping of financial dimensions. The module lets you perform advanced allocations of cost origin from expenditures that are registered in the general ledger or budget. It also lets you compare realized costs and budgeted costs.	Implemented in version 1611
Customer self-service (CSS)	CSS lets you create approved customer records. It also allows users to view selected product catalogs, order items, and view the status of invoices. Additionally, CSS lets you create and follow return orders.	Not implemented
Customizable help articles	The ability to create customized help articles hasn't yet been implemented. Custom task guides and custom field help are available. This feature will be available in a future update.	Not implemented
Employee self-service (ESS)	ESS shows employees several tiles that have task-related and career-related information on a single page. Employees can view pending work items and select links that open pages where they can take action on their tasks. ESS pages also show employees the status of their certifications, when their next performance reviews are scheduled, skills, goals, and compensation information, and other information, such as balances for vacation and sick time. Employees can also access a company directory from their ESS page.	Implemented in version 1611
External questionnaire and recruiting functionality	Functionality for externally posting questionnaires and open jobs will be added to Human Resources in a future update.	External questionnaire functionality isn't implemented. Recruiting functionality is available in Microsoft Dynamics 365 Talent: Attract.
Fiscal printers for Poland	Integration with Polish fiscal printers enables the required information to be sent to the fiscal printer in the correct format during invoice posting. Examples of Polish fiscal printers include the Posnet Thermal and Elzab Omega printer types. This feature will be available in a future update.	Not implemented
General budget reservations	The General budget reservations document is sometimes referred to as a commitment. Public sector entities often use this	Implemented in version 8.1

AX 2012 feature that Microsoft postponed	Description	Current status (as of February 2019)
	document to set aside or earmark budgeted funds so that they're not available for other purposes.	
Graphics tab on the **Fixed asset value model** and **Depreciation book profile** pages	The chart shows the depreciation, accumulated depreciation, and net book value over time. Users can select the **Data** tab to view more detailed information than the chart shows. This chart will be redesigned in a future update.	Not implemented
Intelligent Data Management Framework (IDMF)	IDMF is an add-on tool that lets system administrators optimize performance. IDMF assesses the health of the application, analyzes current usage patterns, and helps reduce database size.	Not implemented
Microsoft Project client integration	The Microsoft Project client is integrated with projects.	Implemented in version 7.2 (July 2017 update)
Procurement site	In previous versions, the Employee self-service procurement site lets you enter requisitions for employees, view the status of an order (created, received, or receipt confirmed), and request onboarding of a new vendor. You could configure different procurement catalogs to show on the site depending on policy. You could also design procurement catalogs by adding new nodes. In the current version, procurement catalog capabilities are reduced and are used only to limit the products that can be ordered for an organization. The structure is always based on the Procurement categories hierarchy. Additionally, on the procurement site the employee could approve a vendor invoice and confirm receipts in relation to the requisitions and derived purchase orders.	Not implemented
Secure global address book	The ability to help secure the global address book by legal entity and address book isn't available. This feature will be available in a future update.	Not implemented
Specifications for Electronic reporting (ER) payment formats	Currently, you must enter the payment format specifications manually. In a future update, you can select payment format specifications from a list. The following payment specifications are currently supported per payment format.	Not implemented
US Payroll	US Payroll provides gross-to-net processing for employees in the United States. In Payroll, you can set up, enter, and maintain all payroll records and transactions.	Implemented in version 1611

AX 2012 feature that Microsoft postponed	Description	Current status (as of February 2019)
Vendor collaboration (Vendor Portal)	Dynamics AX 2012 provided vendor portal capabilities through Enterprise Portal. Financial and Operations also provides these capabilities. In version 7.1 (also known as Dynamics 365 for Operations 1611), a vendor can view and respond to purchase orders. In version 7.3, the vendor can view and respond to RFQs. Vendors can also view and edit selected information from the vendor record such as addresses, contact information, and contact persons, and they can upload documents in relation to their certifications.	Implemented in version 7.3
Vendor requests - external request to become a new vendor	Dynamics AX 2012 provided the ability for an anonymous user to sign up to be a vendor in the system, which could lead to a vendor request for adding a new vendor to the vendor master. In version 7.3, you can import the anonymous request from a prospective vendor via an entity (Data Management/OData), which can lead to inviting the vendor - or the vendor's contact person - to register more details about the prospective vendor. The information provided is included in a new vendor request that can be reviewed and approved via a workflow process. An approval of the vendor request leads to creation of a new vendor account.	Implemented in version 7.3
Vendor requests in general	Dynamics AX 2012 had a concept of vendor requests that served various purposes related to updating vendor-related information, such as requesting new procurement categories for the vendor, internal employees requesting new vendors, or requesting to add a vendor to another company. Only the vendor's request to be added as a vendor was implemented in version 7.3.	Not implemented
[Russia] Tax registers	Legal entities can use registers to disclose their revenues and expenses. The registers track revenue and expense data from the time that primary documents, such as sales invoices and delivery notes, are first entered by using the calculation of cost prices for production. The data from the registers is used to confirm the declared profit of the legal entity. This functionality includes the following features: Current period incomes, Tax expenses, Other expenses of current period, Unrealized expenses of current period, Other unrealized expenses, Accounts receivable debt – inventory, Bad debts reserve calculation, Bad debts reserve movement, Accounts receivable movement, Procedure for writing-off AR bad debts, Accounts payable debt - inventory, Accounts payable debt movement, Procedure for writing-off AP bad debts, Goods cost calculation, FA object information, IA object information, FA depreciation, IA depreciation, FA/IA sale, Depreciation bonus recovery	Implemented in version 8.1.3

AX 2012 feature that Microsoft postponed	Description	Current status (as of February 2019)
[Russia] Electronic export/import format for Client-Bank interface and reconciliation procedure	Electronic formats for export of outgoing payments, and import of incoming payments.	Implemented in version 8.1.3
[Russia] VAT declaration	Electronic format of VAT declaration.	Implemented in version 10.0.1
[Russia] Cash flow management	The functionality that obtains a cash flow forecast and performs an analysis, manages payments on a daily basis by using payment schedule journals, controls the company's cash position, and maintains the company's cash flows with centralized control.	Implemented in version 10.0.1
[Russia] Accounting reporting format	Electronic format of the following accounting reports: BalanceSheet, IncomeStatement, CashFlow, EquityStatement, TargetUsageMoney.	Implemented in version 10.0.1
[Russia] Assessed tax reporting	Assessed tax declaration.	Implemented in version 10.0.1
[Russia] Land tax reporting	Land tax declaration. Creation of land tax declaration by separate divisions.	Implemented in version 10.0.1
[Russia] Transport tax reporting	Transport tax declaration.	Implemented in version 10.0.1
[Russia] Indirect tax return (VAT and Excise) on import of goods	Indirect (withholding) tax return (VAT and Excise) on import of goods from state members of Customs union.	Implemented in version 10.0.1
[Russia] Journal of Alcohol sales in Retail	Daily Alcohol journal.	Implemented in version 10.0.1
[Russia] Optional posting of transfer orders to General ledger	Option to post or not post transactions to General ledger when posting a transfer order.	Implemented in version 8.1.2
[Russia] Inventory owner	Inventory dimension used to track owner of inventory, including consignment stock, bailment, tolling, and similar arrangements.	Implemented in version 10.0.1

AX 2012 feature that Microsoft postponed	Description	Current status (as of February 2019)
[Russia] AP/AR - Third-party miscellaneous charges	Register third-party miscellaneous charges and allocate them by the following regimes: inclusion into cost of purchased goods (allocation to invoice lines from other vendors), and redrawing to other parties (reallocation to other expense accounts).	Implemented in version 8.1.1
[Russia] Goods in transit from vendor	Register goods in transit from vendor by using a special posting profile with item type "purchased items en route." Create Act of inventory holdings en route (INV-6).	Implemented in version 8.1.2
[Russia] Goods in transit - sales to customer with postponed passing of property	Post sales invoice with postponed property transfer: no customer debts posted, all outgoing taxes posted, items transferred to transit warehouse. Register passing of property with posting debts and items sale from transit warehouse.	Implemented in version 8.1.2
[Russia] Bailment - accounting at bailee side	Account for inventory receipt for bailment as required by the law and generate primary form MX-1. Account for inventory return from bailment and generate primary form MX-3. Calculate bailment costs from bailee side.	Implemented in version 8.1.2
[Russia] Bailment - accounting at owner side	Account for inventory transfer to bailment and inventory return from bailment on goods owner side under bailment service contract.	Implemented in version 8.1.2
[Russia] Localization of Process Industries solution	Basic localization in two areas: correspondence of accounts for all new general ledger postings, and functional coexistence of Process Industries features and Russian country/region context (no issues when both Process Industries and Russian country/region context are enabled).	Implemented in version 10.0.1
[Russia] Alcohol sales declarations: Application 6, 7, 8 for wholesale. Applications 11, 12 for retail	Keep track of alcoholic beverage types, including producers, units of measure, licenses for retail and wholesale trade. Prepare data for alcoholic beverages activities, including printing declarations and exporting them in XML format through e-reporting.	Implemented in version 10.0.1

Last updated on 08/21/2026

End of mainstream support for Microsoft Dynamics AX 2009, Dynamics AX 2012, Dynamics AX 2012 R2, and Dynamics AX 2012 R3

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Mainstream support for Dynamics AX 2009 Service Pack 1 (SP1), Dynamics AX 2012, and Dynamics AX 2012 R2 ended on October 9, 2018. Security hotfixes were provided for those three versions through the extended support period, which ended on April 12, 2022. For more information, see Microsoft Lifecycle Policy.

Mainstream support for Dynamics AX 2012 R3 ended on October 12, 2021. Only security hotfixes continue to be provided through the extended support period that ends on January 10, 2023. For more information, see Microsoft Lifecycle Policy.

Upgrade to the latest version of finance and operations apps, such as Dynamics 365 Finance, Supply Chain Management, Commerce, and Project Operations:

- Dynamics AX 2009 Service Pack 1 customers should use the migration tool that's available.
- Dynamics AX 2012 and Dynamics AX 2012 R2 customers should upgrade to finance and operations apps through Dynamics AX 2012 R3 using the upgrade tool that's available. For more upgrade information, see Upgrade from AX 2012 to finance and operations apps.

Frequently asked questions

When does the mainstream support for Dynamics AX 2009 Service Pack 1, Dynamics AX 2012, and Dynamics AX 2012 R2 end?

Mainstream support ended on October 9, 2018.

When does the mainstream support for Dynamics AX 2012 R3 end?

Mainstream support ended on October 12, 2021.

Was the information of the end date of the mainstream support for Dynamics AX 2009 Service Pack 1, Dynamics AX 2012, Dynamics AX 2012 R2, and Dynamics AX 2012 R3 available before?

Yes, it's always been publicly available on the Microsoft Support Lifecycle site at support.microsoft.com.

Can customers on Premier Extended Hotfix Support or on Unified Support Advanced and Performance Levels get a nonsecurity hotfix or regulatory update?

No. Neither nonsecurity hotfixes nor regulatory updates are available for the Dynamics AX products during the Extended Support phase of the product lifecycle (Dynamics AX 2009 SP1, Dynamics AX 2012, Dynamics AX 2012 R2, or Dynamics AX 2012 R3).

While the ability to request a nonsecurity hotfix for select products is included with Unified Support Advanced and Performance Levels, Microsoft determined that it can't provide nonsecurity hotfixes with a *commercially reasonable* effort for these products. As a result, Microsoft doesn't accept requests for nonsecurity hotfixes or regulatory updates.

I knew about the regulatory change before October 9, 2018, but it has the law enforcement date after October 9, 2018. Will I still get a regulatory update for Dynamics AX 2009 Service Pack 1, Dynamics AX 2012, and Dynamics AX 2012 R2?

No, Microsoft only provides regulatory updates for Dynamics AX 2009 Service Pack 1, Dynamics AX 2012, and Dynamics AX 2012 R2 for regulatory changes with the law enforcement dates on or earlier than October 9, 2018.

A customer or partner can already download a fix through Lifecycle Services and inspect the code by installing it into a

test Dynamics AX 2012 R3 environment. Is there any difference with the approach that you proposed?

No, there's no difference.

How are binary hotfixes handled for Dynamics AX 2009 Service Pack 1, Dynamics AX 2012, Dynamics AX 2012 R2, and Dynamics AX 2012 R3?

If a hotfix is needed for a part of the system where Microsoft doesn't provide the source code and it's not a security bug, Microsoft doesn't provide the hotfix.

Can I access information about upcoming legislation changes in supported countries/regions?

Yes, all legislation changes regardless of source (such as vendors, Microsoft research, or the localization community) are stored in the Lifecycle Service alerting project. To access the Lifecycle Services alerting project, follow these steps:

1. **Sign up**: Send an email request to join the localization community (under NDA) at DynRegW@microsoft.com.
2. **Access**: Sign in to the Lifecycle Services project **Regulatory Alerts - Worldwide** (available only for companies with NDA or individuals who signed up for the Insider Program).
3. **Alerting guide**: Inform Microsoft about country/region regulation alerts and track the status of regulatory features. For more information, see Submit alerts about country/region-specific regulatory features.

The Lifecycle Services alerting project includes all identified and reported legislation changes (alerts) but not actual Microsoft plans.

Can I get visibility into Microsoft plans for releases of new regulatory features?

Yes, in Lifecycle Services Issue Search, Microsoft publishes information about upcoming regulatory updates for Dynamics 365 Finance, Supply Chain Management, Commerce, and Project Operations.

Are details about released regulatory features available?

Yes, for larger features, Microsoft publishes this information in Globalization resources.

Can I view and try actual regulatory features in Dynamics 365 before a release?

Yes, you can try these features if you purchase cloud licenses for Dynamics 365. All service updates are available for preview through the Shared Asset Library in Lifecycle Services. For regulatory changes specified by authorities early, the features are available in Preview before the law-enforced date (or latest date for first filing).

Can I access the code and configurations for regulatory features before a release?

Yes, starting with version 10.0.26, Microsoft makes preview packages for every service update available to all customers through the Shared Asset Library in Lifecycle Services, under **Software deployable package**.

Configurations are available in the Global repository. For more information, see Download ER configurations from the Global repository of Configuration service.

Last updated on 03/26/2026

Service description for finance and operations apps

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the **Request access to Finance and Operations Viva Engage Community** form and choose the community you want to join.

Finance and operations apps are enterprise resource planning (ERP) software as a service (SaaS) offerings that are built on and for Microsoft Azure. The finance and operations service provides organizations with ERP functionality that supports their unique requirements and helps them adjust to constantly changing business environments, without requiring that they manage infrastructure. Finance and operations apps can include one or more of the following solution areas:

- Dynamics 365 Finance
- Dynamics 365 Human Resources
- Dynamics 365 Supply Chain Management
- Dynamics 365 Commerce
- Dynamics 365 Project Operations

Together with business intelligence, infrastructure, compute, and database services, these apps enable organizations to run industry-specific and operational business processes. Customers, supported by their implementation partner, determine the configuration of the business application logic that best suits their unique business processes. Functionality and business processes can be augmented or extended through one or a combination of the following solutions:

- Built in personalization experience
- Microsoft Power Platform tools
- Visual Studio–based finance and operations software development kit (SDK) and Azure DevOps build automation
- Independent software vendor (ISV) solutions from Marketplace

Based on requirements, customers choose their solution approach. They work with their implementation partner to define, develop, and test their solution by using the tools and best practices that are provided in Microsoft Dynamics Lifecycle Services. There are four common scenarios:

- Standard finance and operations apps "out of the box" configuration (no extensions)
- Finance and operations apps configuration that includes one or more ISV solutions
- Finance and operations apps configuration that includes one or more customer-specific extensions
- Finance and operations apps configuration that includes a combination of customer-specific extensions and one or more ISV solutions

Organizations can match their business growth by easily adding users and business processes through a simple, transparent subscription model. For more information, see the Dynamics 365 Licensing Guide.

Operating model

The operating model of finance and operations apps defines specific roles and responsibilities for the customer, implementation partner, and Microsoft throughout the lifecycle of the service. For more information, see Cloud operations and servicing.

Customer activities

Customers work with their partner and Microsoft FastTrack following the Dynamics 365 Implementation Guide, the Success by Design framework, and the tools and best practice templates provided in Lifecycle Services to implement their solution. Common activities include:

- User identity and security management
- Define, develop, and operate business processes
- Define, develop, test, and operate extensions
- Monitor and manage nonproduction deployments
- Manage application updates and validate extensions
- Manage ISV solutions and third party integrations

Microsoft responsibilities

Microsoft manages the finance and operations service by deploying, actively monitoring, and servicing customer sandbox and production environments in the Microsoft SaaS subscription. This management includes allocating the required system infrastructure to run the service and proactively communicating with customers about the service's health. Responsibilities include:

Infrastructure management

- Security and isolation

- Operating systems and virtualization
- Servers, storage, and networking
- Data center power, networking, cooling

Application platform management

- 24/7 application monitoring and notifications
- Diagnostics, platform updates, patches, service updates
- Application routing, load balancing, site replication
- Environment provisioning and management
- Database management, high availability (HA)/disaster recovery (DR), scale, operations
- Compute deployment, scale up, scale down

System configuration

Finance and operations apps scale according to the transaction volume and user load. Each customer implementation produces a unique solution that consists of the following elements:

- **Data composition** – A unique set of parameters that control behavior, the layout of the organization, the structure of master data (such as financial and inventory dimensions), and the granularity of transaction tracking.
- **Extension and configuration** – Extension mechanisms that use code extensions, ISV solutions, and unique configurations that include workflows, integrations, and report configurations.
- **Usage patterns** – A unique combination of online and batch usage, together with the ability to integrate with upstream and downstream systems for unified data flow and the ability to differentiate based on the information views that customers use in their business processes.

Microsoft configures customer production environments that are sized to handle the transaction volumes and user concurrency. Microsoft is responsible for the following tasks:

- Correctly allocating the resources of production environments, based on the customer's profiling information in the Lifecycle Services Subscription estimator
- Continually monitoring and diagnosing the service availability of production environments
- Analyzing and troubleshooting system performance issues with finance and operations apps

To ensure that an implementation is configured for high performance, customers must complete these tasks:

- Provide accurate usage information about the finance and operations implementation in the Lifecycle Services Subscription estimator.
- Build and test extensions for performance and scale.
- Appropriately test data configurations for performance.
- Ensure scalability by doing performance testing) before go-live.

Onboarding and implementation

The following table shows typical onboarding and implementation events.

Expand table

Request	Expected Microsoft action	Expected customer or implementation partner action
Initial offer purchase	Microsoft creates a Lifecycle Services project after the purchase of the offer, based on an event that the customer triggers.	Go through the Enterprise Agreement (EA) or Cloud Solution Provider (CSP) commercial process. The partner creates a tenant for the customer, if applicable.
Add-on purchase	Microsoft grants the customer access to the add-on that the customer selects during implementation.	Not applicable
Implementation planning and analysis	Microsoft provides relevant tools in Lifecycle Services, such as Business process modeler (BPM) and interoperability with Azure DevOps.	Do project planning, set up Azure DevOps, complete system onboarding, and set up admin accounts.

For more information, see Onboarding an implementation project.

Globalization

Finance and operations apps are served from several Azure regions around the world. Finance and operations apps provide functionality to support different countries or regions and native languages. For more information, see Localization and regulatory features.

Country or region-specific considerations

- Customers in regulated industry or commercial organizations that do business with entities in France that require local data residency should review Finance and operations in France.

- Customers that have operations in China should review Azure China Playbook and Finance and operations operated by 21Vianet in China.
- Customers that have operations in Russia should review the Russian personal data localization law.

Privacy laws and regulations

For finance and operations apps, Microsoft acts as a processor. As a data processor, finance and operations apps provide processes and features that help customers comply with privacy obligations as a data controller. For more information, see Privacy overview.

Environment and data management

This section describes some typical environment and data management events that occur in the service.

Environment and data management events for production instances

Lifecycle Services provides self-service tools and database movement operations that you use to perform environment and data management tasks. Here are some examples:

Event: Requesting a production instance

- Complete the Go-live readiness review, and submit it to the Microsoft FastTrack team.
- Complete the Lifecycle Services Subscription estimator before you request a production instance.
- Complete all the implementation tasks that are specified in the Lifecycle Services methodology.

Event: Copying a sandbox database to a production instance

- Perform this operation for a mock go-live or an actual go-live cutover.

Event: Maintenance mode

1. Turn on maintenance mode in Lifecycle Services.
2. Complete the required maintenance.
3. Turn off maintenance mode in Lifecycle Services.

Environment and data management events for nonproduction instances

Lifecycle Services provides self-service provisioning and database movement operations that you use to perform environment and data management tasks. Here are some examples:

Event: Deploying a new sandbox instance

- Make sure that you plan all required instances and purchase the applicable add-on offers.
- Run the deployment process in Lifecycle Services.
- Complete all tasks that are specified in the Lifecycle Services checklists.

> **Note**
>
> A development sandbox environment is a virtual machine (VM) that is deployed to the customer's Azure subscription and managed by the customer.

Event: Copying a golden configuration database from sandbox to production

- Perform this operation for a mock go-live or an actual go-live cutover.

Event: Restoring a production point-in-time backup to a nonproduction instance

- Run the Refresh database option in Lifecycle Services.
- Post-copy: Delete or obfuscate sensitive data via scripts, adjust environment-specific application configurations (such as integration endpoints), and enable or add users. Apply these changes by using a data package.

Event: Nonproduction instance database point-in-time restore

- Accept that the process can't be undone.
- Run the point-in-time restore operation in Lifecycle Services.

Event: Copying a Tier 2 sandbox database to a development sandbox for troubleshooting and debugging

- Run the database export operation in Lifecycle Services on the Tier 2 sandbox environment.
- Import and update the database in the development environment.

Data backup and retention

The SaaS subscription automatically backs up databases for finance and operations environments. For production environments, automatic backups are available for 28 days, unless Microsoft initiates a rollback. For sandbox (Tier 2+) environments, automatic backups are available for seven days. If a failure occurs during any planned maintenance update, Microsoft can roll back the production environment.

For more information about automatic backups, see Automated backups - Azure SQL Database & SQL Managed Instance.

Service activity responsibilities

The following table describes some typical scenarios and activities for the service. It also indicates whether Microsoft, the customer, or both Microsoft and the customer are responsible for the activities.

Expand table

Activity	Responsibility of Microsoft	Responsibility of the customer
Provisioning initial tenants		
Size the projected load in Lifecycle Services by using the Subscription estimator tool, and request that specific environments be provisioned.		X
Provision all production instances and nonproduction instances.	X	
Validate the deployed production instances and nonproduction instances.		X
Service updates		
Apply service updates to designated nonproduction and production instances.	X	
Manually apply service updates from Lifecycle Services to sandbox instances. Define, develop, test the update, and provide the code update package back to Lifecycle Services.		X
Request and schedule extension updates be applied to the production instance.		X
Create a code and data backup for the production instance before any updates are applied.	X	

Activity	Responsibility of Microsoft	Responsibility of the customer
If there's any failure, roll back the production instance to the code and data backup.	X	
Data management (backup, restore, and update)		
Back up the database.	X	
Determine high availability and a disaster recovery plan.	X	
Monitor performance of the production instance database.	X	
Tune the production instance database for performance.	X	
Perform production instance database point-in-time refresh to a nonproduction instance.		X
Updating the infrastructure		
Schedule regular infrastructure updates.	X	
Scaling up and down (users, storage, and instances)		
Purchase additional users and nonproduction add-ons.		X
Update usage changes in the Lifecycle Services Subscription estimator tool.		X
Report any significant performance issues that affect use of the service.		X
Proactively manage the resources that are required for the applicable service.	X	
Investigate and troubleshoot incidents.	X	
Security (user access)		
Provide user access to the service.		X
Provide Lifecycle Services project access for the management and operation of instances that were deployed through Lifecycle Services.		X
Monitoring production instances		
Monitor production instances 24/7.	X	
Proactively notify the customer about incidents that involve the production instance.	X	

Activity	Responsibility of Microsoft	Responsibility of the customer
Managing and monitoring nonproduction instances		
Manage nonproduction instances by using Lifecycle Services.		X
Monitor nonproduction instances.		X

Service update strategy

In accordance with the software lifecycle policy, finance and operations apps follow the Microsoft Modern Lifecycle Policy, which covers products that are continuously serviced and supported.

Microsoft releases four service updates to finance and operations apps every year in the following months:

- February
- April
- July
- October

> **Note**
>
> April and October are major feature release waves. You can pause an individual service update for up to one update cycle.

For more information, see the following topics:

- One Version service updates overview
- Configure service updates through Lifecycle Services
- Pause service updates through Lifecycle Services
- Get notified about service updates through Lifecycle Services
- Regression testing tools to validate service updates
- Dynamics 365 roadmap and release waves

Security and administrative access

Administrative access to a finance and operations production environment is strictly controlled and logged. Handle customer data in accordance with the Microsoft Online Services Terms.

Customer administrative access

The customer's tenant administrator can access production instances or nonproduction instances, as described in the following table:

Expand table

Environment type	Purpose	Level of customer access
Nonproduction Tier 1 sandbox	A nonproduction environment that customers deploy for development, demonstration, or training purposes.	A Tier 1 sandbox (also referred to as a cloud-hosted environment) is a customer-managed VM that you deploy to your Azure subscription from Lifecycle Services. Because it's a VM in your Azure subscription, you have full administrative access to the environment via Remote Desktop.
Nonproduction Tier 2 (or higher) sandbox	A nonproduction environment that customers deploy for user acceptance testing, integration testing, training, staging, or any other preproduction scenario.	Tier 2 and higher sandboxes are deployed to the finance and operations SaaS subscription. You grant access to Azure SQL databases that are associated with the nonproduction environment via just-in-time access. Remote Desktop access isn't available.
Production	A production environment is deployed when the project is ready for initial go-live.	Production environments are deployed to the SaaS subscription. All access is through the browser, service endpoints, or Lifecycle Services.

Microsoft administrative access

The following table shows the different levels of access for different Microsoft administrators:

Expand table

Administrator	Customer data
Operations responses team (Limited to key personnel only)	Access is granted through a support ticket. Access is audited and limited to the duration of the support activity.
Microsoft Customer Support Services (CSS)	No direct access. Customer can use screen sharing to work with support staff to debug issues.
Engineering	No direct access. The Operations response team can use screen sharing to work with engineering to debug issues.
Others in Microsoft	No access.

Monitoring

Microsoft provides an extensive toolset to monitor and diagnose production instances. Microsoft monitors production environments 24 hours a day, 7 days a week. For more information, see Production support and monitoring.

Expand table

Microsoft responsibilities	Customer responsibilities
• Monitor availability of the service. • Continuously monitor and alert through health metrics and watchdogs for critical components such as Application Object Server (AOS), Batch, Data Import/Export Framework (DIXF), Commerce, and Management Reporter. • Monitor for performance degradation that infrastructure services cause (such as Microsoft Entra [Microsoft Entra] and Azure SQL). • If Microsoft determines that a single process or batch job is causing aberrations, Microsoft terminates that process or job after communication with the customer.	• Monitor changes to application configurations and extensions that can cause functional and performance problems. • Use the monitoring tools to diagnose application errors and user-reported performance problems. • Inform Microsoft if you expect system load beyond projected peak usage. • If the applicable service is unavailable in the production instance, use Lifecycle Services to report a production outage.

By submitting support requests online through Lifecycle Services, you enable Microsoft to deliver fast and deep technical expertise in the most effective and efficient manner. Although a phone option is available, use it only if the online option isn't available. For more information, see Phone support options.

Incident management

Microsoft responds to and fixes incidents based on severity levels. Microsoft can change the incident severity levels during the initial assessment of the incident, and as more information about the impact and scope becomes available. If the incident is mitigated, the incident severity remains unchanged.

For more information about severity levels, see this severity table.

Business continuity through high availability and disaster recovery

Microsoft provides business continuity and disaster recovery for production instances of finance and operations apps in the event of Azure region–wide outages. For more information, including service Recovery Time Objective (RTO) and Recovery Point Objective (RPO), see Business continuity and disaster recovery.

- **High availability** – High availability (HA) functionality provides ways to prevent downtime that a failure of a single node in an Azure datacenter causes. Each service's cloud architecture uses Azure availability sets for the compute tier to prevent single-point-of-failure events. Azure SQL HA features provide HA for databases.

- **Disaster recovery** – Azure disaster recovery features protect each service against outages that broadly affect an entire Azure datacenter. Here are some of these features:
  - Azure SQL active-geo replication for the primary database (business database).
  - Geo-redundant copies of Azure Blob Storage (which contains document attachments) in other Azure regions.
  - Secondary region for the Azure SQL and Azure Blob Storage replications.

If you use disaster recovery to recover your production instance, you and Microsoft meet your incident management responsibilities.

Microsoft regularly examines its disaster recovery plans and procedures through System and Organization Controls (SOC) audits. Compliance audits attest to the technical and procedural process of Microsoft's disaster recovery, including Dynamics 365 finance and operations apps. SOC compliance audit reports and all other Compliance Reports are available on Microsoft Trust Center Compliance Offerings.

Finance and operations support offerings

Technical support is available in markets where finance and operations services are offered. Support experiences are provided in Lifecycle Services or finance and operations apps. Here are some examples:

- Issue search in Lifecycle Services
- Integrated technical support in finance and operations apps
- Cloud-powered support in Lifecycle Services

Microsoft offers finance and operations customers three support plans: Premier, Professional Direct, and the support that is included in the subscription. The level of support differs per plan. The following table shows a comparison of the three plans.

Expand table

Support feature	Premier	Professional Direct	Subscription
Unlimited break/fix incident submission	Yes	Yes	Yes
24/7 access via Lifecycle Services	Yes	Yes	Yes
Incident response time	Less than one hour	Less than one hour	Next business day
Advisory hours	Pools are acquired per agreement.	No	No
Dedicated support account manager	Yes	No	No
Dedicated support engineer	Engaged under a separate agreement	No	No

For more information, see Support overview.

Process to engage support

If incidents involve finance and operations apps, customers can submit support tickets to Microsoft through Lifecycle Services. CSS handles the incidents, based on the customer's support plan and the severity of the incident as designated by CSS.

Service level agreement

Microsoft is committed to an availability rate of 99.9 percent per month of the service. If Microsoft doesn't achieve and maintain the service level for the applicable service as described in the service level agreement (SLA), the customer might be eligible for a credit toward a portion of its monthly service fees for that service. For information about how to initiate a service credit, see the "Claims" section of the SLA.

Important resources

- **Trust Center** – Get information about where your finance and operations data is stored, plus additional information about privacy, compliance, and security procedures.
- **Licensing terms and documentation** – Quickly access licensing terms, conditions, and supplemental information that is relevant to the use of products and services that are

licensed through Microsoft volume licensing and Microsoft commercial licensing programs.

- **Microsoft Lifecycle Policy** – This page provides consistent and predictable guidelines for the availability of support throughout the life of a product.
- **Licensing guide** – Use this guide to learn more about how to license Dynamics 365.
- **Customer support** – Get industry-leading support for your Dynamics 365 apps.
- **Dynamics Lifecycle Services** – Manage your application lifecycle, and move towards predictable, repeatable, high-quality implementations.
- **Dynamics 365 Implementation Guide** - The Dynamics 365 Implementation Guide documents time-tested Success by Design principles and provides prescriptive guidance to architect, build, test, and deploy Dynamics 365 solutions.

Definitions

Azure region

A geographical region where one or more Azure datacenters exist. Examples include US and Europe.

Business process modeler (BPM)

A tool in Lifecycle Services that helps complete a fit-gap analysis for a given implementation by using business process definitions from American Productivity & Quality Center (APQC) that are supported in finance and operations apps.

Cloud solution provider

A partner that is part of the Microsoft Cloud Solution Provider (CSP) program, and that provides customers with value-added cloud services, support, one invoice, and customer management at scale.

Customer

A business entity that consumes finance and operations apps and is represented by a tenant in Office 365.

Development environment

A nonproduction sandbox environment that you use to develop extensions. Deploy this environment to your own Azure subscription from Lifecycle Services. You can also use this environment for demonstration, training, or other testing tasks. It's also known as a Tier 1 sandbox.

Downtime

Any period when users can't sign in or access their active tenant because of a failure in the unexpired platform or the service infrastructure, as Microsoft determines from automated health monitoring and system logs. Downtime doesn't include scheduled downtime, the unavailability of service add-on features, the inability to access the service because of your modifications to the service, or periods where the scale unit capacity is exceeded.

Implementation partner

The partner that you select to customize, configure, implement, and manage your finance and operations solutions.

Incident

An issue that you encounter while using the finance and operations service, and that you submit a ticket for via Lifecycle Services.

Microsoft Customer Support Services (CSS)

The Microsoft global support team that's dedicated to providing quality service for finance and operations apps.

Microsoft Dynamics Lifecycle Services

The administrative portal for lifecycle management of finance and operations apps from trial, to implementation, to post-production management and support. For more information, see Lifecycle Services Resources.

Nonproduction instance

Any of the following instances of a service that you use to validate extensions and perform other development tasks:

- **Sandbox Tier 1** – Developer instance (customer-hosted)

- **Sandbox Tier 2** – Standard Acceptance Testing instance
- **Sandbox Tiers 3–5** – Add-on sandboxes

For more information about Tiers 2 through 5, see Selecting the correct Tier-2 or higher environment.

Production instance

A finance and operations environment that you use to manage your "live" daily transactions and business processes.

Sandbox environment

A nonproduction environment that you use for demonstration, training, user acceptance testing, validation of extensions, and other testing tasks.

Service

Any of the core services that are included in finance and operations apps.

Service level agreement (SLA) for Microsoft online services

The SLA applies to Microsoft online services. For more information, see Service Level Agreements.

Service update

Microsoft services finance and operations environments on consistent basis through service updates. Customers set their own service update calendar, based on their business needs. For more information, see One Version service updates.

Success by Design

Success by Design is the framework that systematically guides an implementation through a series of assessments at critical stages. It helps ensure optimal architecture, security, performance, and user experience for a Dynamics 365 solution.

User

A single person who uses finance and operations environments and is associated with a customer's tenant.

Last updated on 03/28/2026

Before you buy

New to Dynamics 365 Finance, Dynamics 365 Supply Chain Management, or Dynamics 365 Commerce? Use the following step-by-step guidance whether you're still evaluating or ready to make a purchase.

Step one: Try out finance and operations free for 30 days

You can try Dynamics 365 for Finance, Dynamics 365 Supply Chain Management, or Dynamics 365 Commerce for 30 days through a simple email signup. The trial version of finance and operations applications includes Getting started task guides that provide step-by-step instructions that allow you to view specific scenarios in action. You can explore the product and exercise scenarios, but you can't customize it. Demo data is included to ease the use of the product and to make the experience more meaningful. A reminder email is sent three days before the trial expires. For more information, see Sign up for preview subscriptions.

Step two: Choose a deployment option

You can now deploy finance and operations applications in the cloud or on-premises. Cloud deployments offer an ERP service that Microsoft fully manages, while on-premises deployments are deployed locally within a customer's data center.

Consider the following factors when you choose on-premises as a deployment option:

- Regulatory and compliance needs that aren't available in the cloud certifications.
- Disconnected business process with intermittent internet connectivity required to access Microsoft Dynamics Lifecycle Services for application lifecycle management.
- Comparison of cloud and on-premises features.
- System requirements for on-premises deployments.

> **Important**
>
> On-premises deployments aren't supported on any public cloud infrastructure, including Microsoft Azure. However, they're supported to run on Microsoft Azure Stack HCI and Microsoft Azure Stack Hub.

For more information, see Deployment options.

Step three: Buy and manage a subscription

To explore subscription options, go to the Dynamics 365 pricing page. This page includes several different plans to fit your organization's needs.

You can buy a subscription in many ways:

- Buy through a Cloud solution provider (cloud only).
- Buy through a partner, and use volume licensing (cloud or on-premises).
- Buy through a partner from the Dynamics price list (on-premises only).

Buy through a Cloud solution provider (cloud-only)

A Microsoft Cloud Solution Provider can work closely with you to understand the needs of your business or organization. Use the Microsoft Partner Center portal to find a partner to fit your needs.

Buy through a Dynamics partner (on-premises)

You must work with a partner to purchase Finance + Operations (on-premises). For more information, see Buy Finance + Operations (on-premises).

Buy through volume licensing (cloud or on-premises)

If your organization has 250 or more Dynamics 365 users, you might be interested in a Volume licensing agreement.

In volume licensing, finance and operations applications are available through:

- Enterprise Agreement
- Enterprise Agreement Subscription
- Enrollment for Education Solutions (under the Campus and School Agreement)
- Microsoft Products and Services Agreement (MPSA)

Choose your support option

Microsoft provides flexible, industry-leading support, services, and resources that enable you to quickly address technical issues and maximize return on your Dynamics 365 investment. Choose a plan that best meets your business requirements.

For more resources, see:

- Dynamics 365 Support
- Quick Start Guide for Microsoft Dynamics Cloud Service Support Benefits (PDF)

Step four: Learn about FastTrack and plan your deployment

Microsoft FastTrack for Dynamics 365 is a customer success service designed to help you move to Dynamics 365 smoothly and confidently, so you can realize business value faster. When you participate in the FastTrack program, you receive guidance on best practices and how to plan for successful rollouts. You also learn ways to enable new users and expand capabilities – all at your own pace. Additionally, you have access to Microsoft engineering resources committed to making your experience with Dynamics 365 a success. For more information, see Microsoft FastTrack.

If you're upgrading from Dynamics AX 2012 or migrating from AX 2009

If you're upgrading from Microsoft Dynamics AX 2012 or migrating from Microsoft Dynamics AX 2009, you might be eligible for a longer trial. Contact daxcf@microsoft.com for more information.

Last updated on 06/26/2026

Sign up for preview subscriptions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to subscribe to the preview or partner offer and deploy an environment. The subscription that you create gives you a Microsoft Online Services test tenant and a Microsoft Dynamics 365 Lifecycle Services project where you can deploy an environment. This article also helps you set up additional users in your Microsoft Online Services tenant and gain experience with service administration capabilities. Here are the skills that you learn:

- Subscribe and create a new Microsoft Online test tenant.
- Navigate to Lifecycle Services projects.
- Use various features of Lifecycle Services.
- Add users to Microsoft Entra and the client.
- View resources in your subscription email.

Key terms

- **Microsoft Online Services tenant** – A tenant is the group of all subscriptions and users for your organization. You create the tenant at the same time as your first subscription in Microsoft Online Services.
- **Subscription** – A subscription gives you an online cloud environment and experience. It also lets you see how customizations that you develop can be deployed to the cloud.
- **Microsoft Entra ID** – The cloud environment includes Microsoft Entra ID. Microsoft Entra helps you manage users, groups, security roles, and licenses for online applications, much as you manage them for on-premises environments.
- **Users** – You manage users of the services that your organization subscribes to in Microsoft Entra ID. You can add any users in your tenant and assign them to security roles.

- **Developers and administrators** – Developers and administrators are users who also have access to Lifecycle Services that lets them manage projects and environments. These users are also end users.

- **Organizational account** – Users receive Microsoft Entra credentials. These credentials are separate from other desktop or corporate credentials. The Microsoft Entra credentials are used to sign in to Microsoft 365 and other Microsoft cloud services. Users sign in by using their organizational account.

  > **ⓘ Important**
  >
  > For this release, don't use any existing credentials that are associated with other online services, such as Microsoft 365 or Microsoft Dynamics CRM Online.

- **Microsoft account** – Microsoft accounts were formerly known as Passport accounts or Windows Live ID accounts. Currently, you can't use Microsoft accounts with finance and operations applications, Microsoft Dynamics 365 Commerce, or other Microsoft Online Services. However, Microsoft accounts are still required for Microsoft Connect and other Microsoft Business Solutions sites, such as Information Source and Microsoft Dynamics Community. You continue to use your Microsoft account to access these services.

- **Microsoft 365 admin center** – Microsoft 365 admin center is the subscription management portal that Microsoft 365 provides for administrators. Use Microsoft 365 admin center to provide management functions for users and subscriptions.

- **Environments** – You can deploy as many single instances of a virtual machine (VM) as you require. These instances are called *environments*.

Prerequisites

1. You receive an email that invites you to participate in the preview.
2. If your company has an organizational account with Microsoft Online Services and you're signed in, sign out before you continue. Alternatively, use **InPrivate Browsing** mode.
3. If you're not sure whether you're signed in, delete your browser cookies, and then close your browser before you continue.

Subscribe

> **Important**
>
> Only one person (tenant administrator) in an organization must perform this task. If you're not the person who is subscribing to this release, wait until your organization is signed up and you receive your user credentials. Then continue with the procedure.

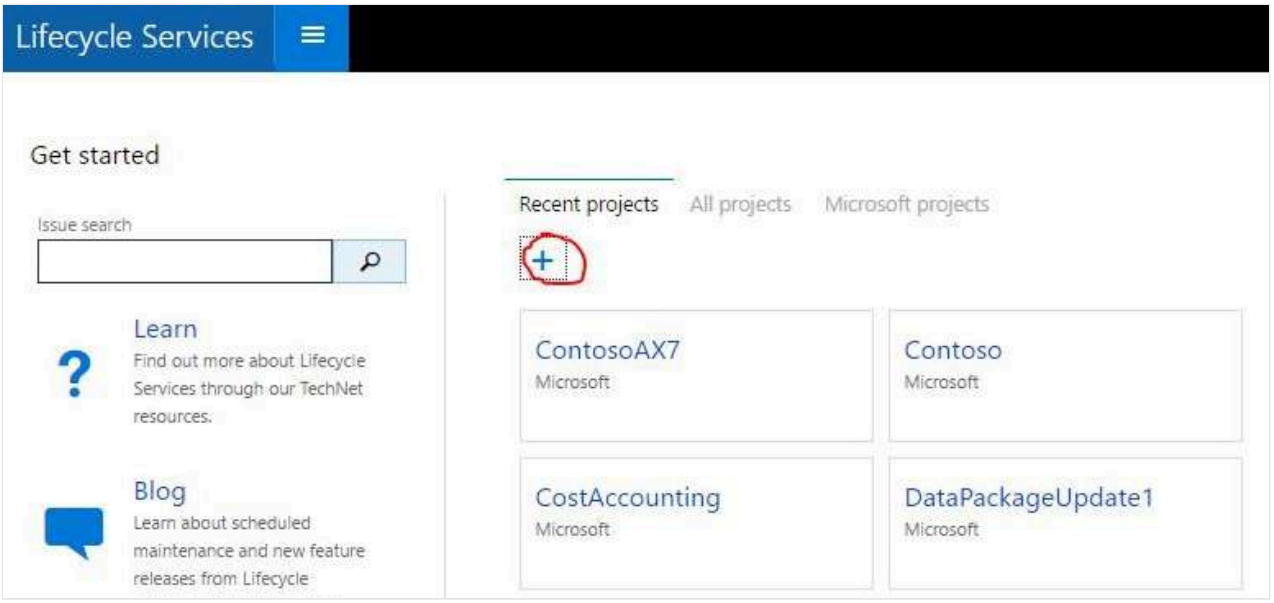
1. Finance and operations applications and Retail are available only to existing Microsoft Dynamics 365 channel partners and customers who are currently enrolled in the Business Ready Enhancement Plan (BREP) service plan. To subscribe, visit PartnerSource Business Center.
2. On the **Account setup** page, in the **Country or region** field, select the country or region.
3. Follow the wizard and prompts to complete the sign-up, until you reach the last step.

Start a new project in Lifecycle Services

> **Note**
>
> Prospect projects are available only in Global Lifecycle Services (https://lcs.dynamics.com). The preview license isn't available in Local Lifecycle Services. For more information about Lifecycle Services in local geographies, see Dynamics 365 Finance, Supply Chain Management, and Commerce in local geographies.

To use Lifecycle Services to manage your environments, you must create a new project.

1. Go to https://lcs.dynamics.com/Logon/Index.
2. Select **Sign in**.
3. Sign in by using the account that you used to subscribe.
4. Select the plus sign (+) to create a new project.

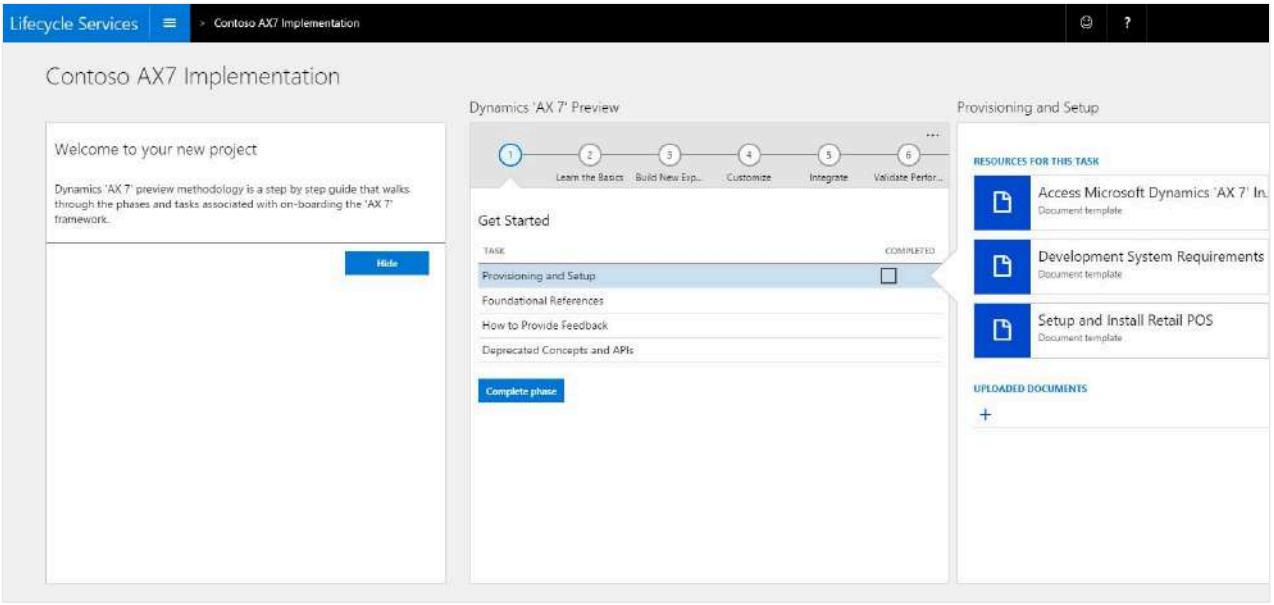


5. Select the project type.

6. Enter the project information, and then select **Create**.

   If you plan to evaluate Commerce, be sure to select **Microsoft Dynamics 365 Commerce** in the **Product name** field.

   The new project for managing your instance is created.



Add users to Lifecycle Services

You're already set up as a user of your Lifecycle Services project. If you add other Microsoft 365 users, you must add them to this project. Other administrators and developers can then deploy their own environments. These Lifecycle Services users are team members who actively work on

the implementation. Don't confuse them with end users. Start on the project page in Lifecycle Services.

1. Scroll to the right, and then, in the **More tools** section, select the **Project users** tile.
2. In the upper left, select the plus sign (+) to add a new user.
3. In the **Email** field, enter the email address of the user that you're adding. This email address should be the Microsoft 365 organization email address that you created earlier.
4. In the **Project role** field, select **Project Owner**.
5. Select **Invite**.
6. Repeat steps 2 through 5 for all users in your organization.

Deploy environments

Deploy environments to an existing Azure subscription.

> **Note**
>
> Each developer of an environment must deploy their own system to Azure. However, only the first project user needs to set up the Azure subscription for deployment.

You can create environments in two ways:

- Deploy to Microsoft cloud services (Azure).
- Download a local virtual hard disk (VHD).

Start on the project page in Lifecycle Services.

1. In the **Environments** section, select the plus sign (+). The **Microsoft Azure setup** dialog box appears.

2. Enter your Azure subscription ID. You can find your Azure subscription ID in Azure Portal (https://ms.portal.azure.com/), under **Settings** in the lower left.

3. Select **Next**.

4. Download the Azure Management Certificate to a local folder on your computer, and then upload it to Azure Management Portal by going to **Settings** > **Management Certificates**. This certificate enables Lifecycle Services to communicate with Azure on your behalf.

5. Return to Lifecycle Services, and select **Next**.

6. Select the Azure region to deploy in. The **West US** region has the fastest deployments, but it's important that you select a data center that is close to where you plan to use this system.

7. Select **Connect**.

8. In the list of available topologies, select the topology to deploy. You can select either the **Download** link to download the VHD or **Next** to deploy on Azure. Azure is the preferred path.

9. Enter the name of the environment.

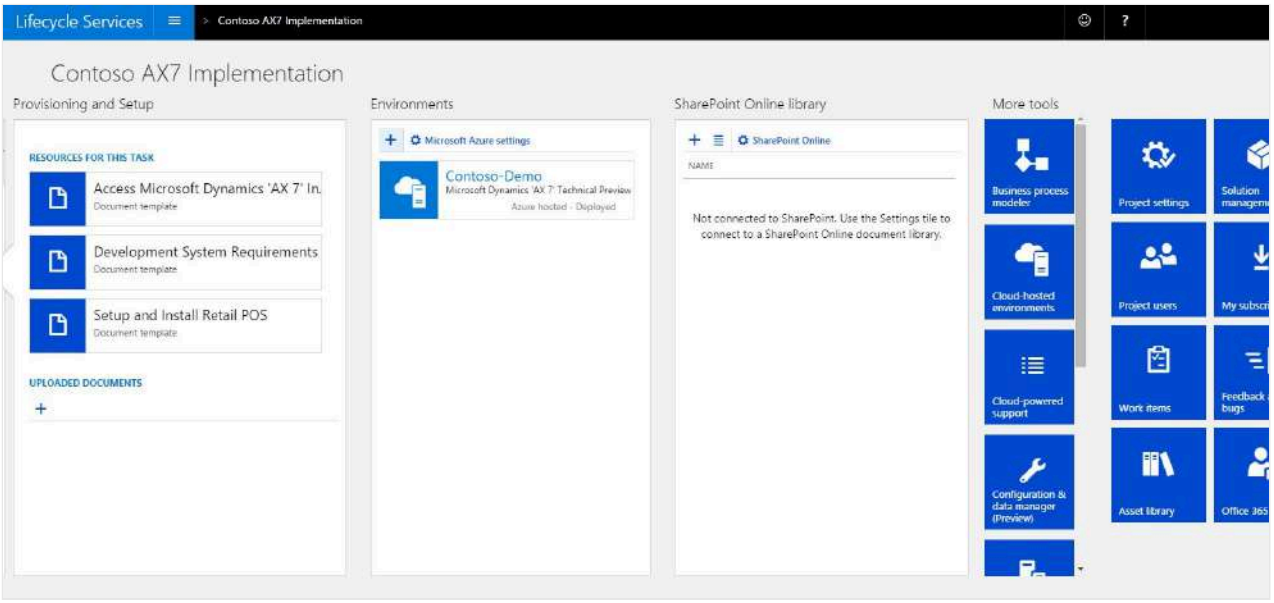
10. Read the pricing and licensing terms, and then select the check box to indicate that you understand them.

11. Select **Next**.

12. Confirm the details, and then select **Deploy**.

    > **Note**
    >
    > Developers and administrators who use their own environments must sign in and repeat these steps.

    After you deploy your environment, it's available in the **Environments** section.



13. Select the environment to view details about the deployment status. The first deployment requires a few hours, but each subsequent deployment is much faster.

14. When the deployment status changes to **Deployed**, select **Login** to connect to the client, or select the name of the VM to connect to the development machine by using Remote Desktop. After the deployment is completed, you can find the base URL and the information that you require to connect to the environment via Remote Desktop.

Use the features of Lifecycle Services

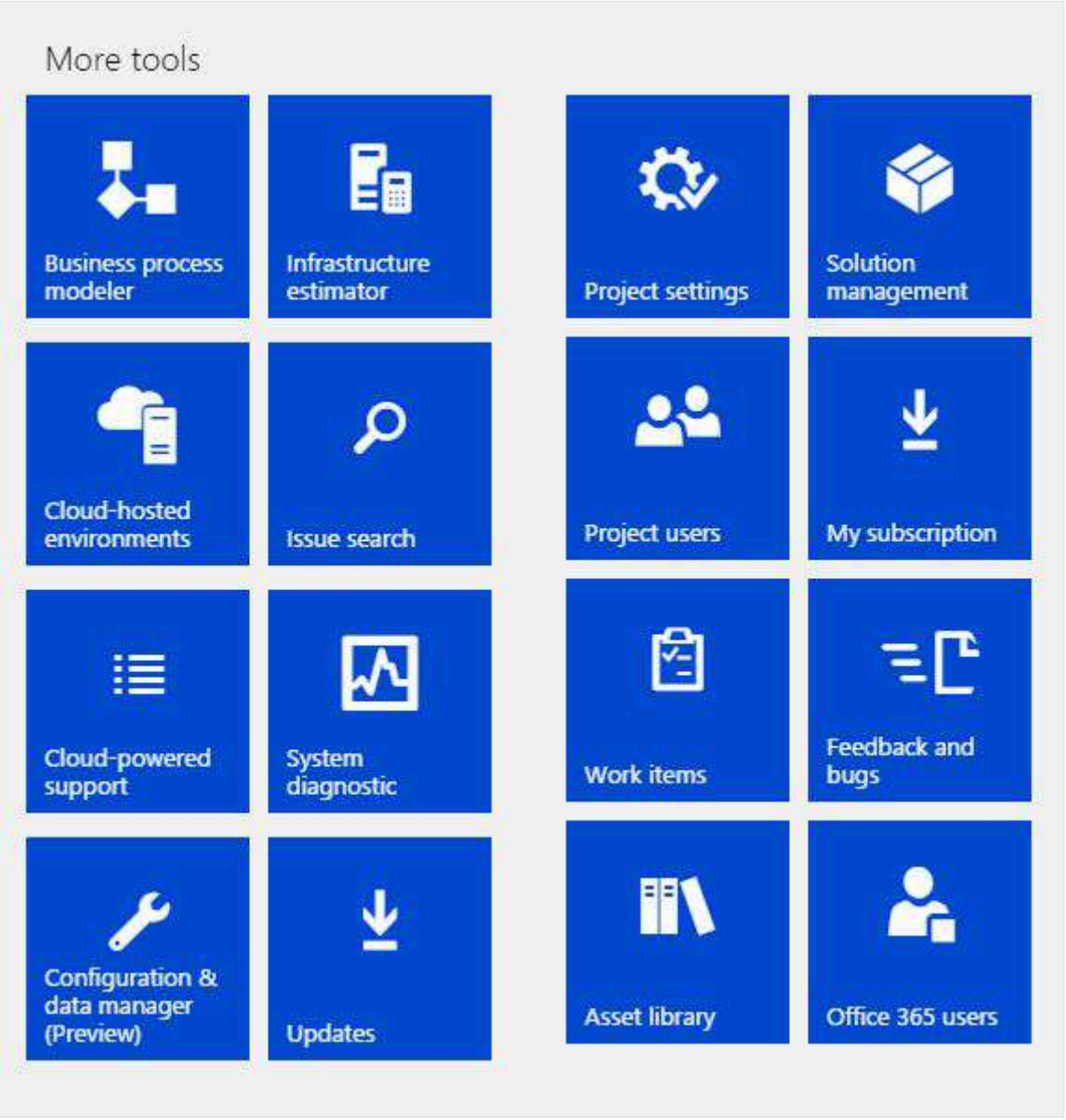
Lifecycle Services is the starting point for performing online administrative activities. Here are some of these activities:

- Deploy VMs on Azure.
- Access materials.
- Access downloads of tools and resources.

Explore the Lifecycle Services project

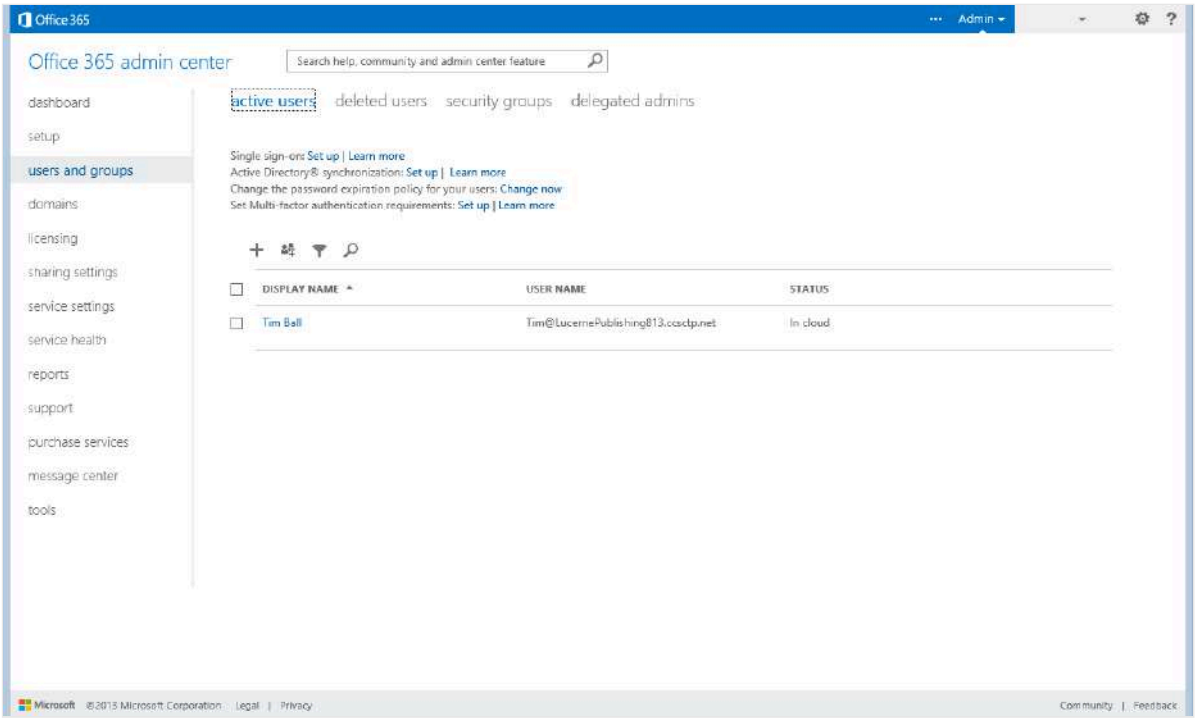
1. Review the methodology, and complete the tasks and phases as you progress through the life cycle. The phases and task information lets you view tools and resources that are available throughout your enterprise resource planning (ERP) experience.

2. Scroll to the right, and review the tiles.



The available tiles include various tools and services in Lifecycle Services. They also include the following additional tiles:

- **My subscription** – The Microsoft 365 subscription management portal is where you can view and work with your online subscriptions. By selecting **User and Groups** in the left navigation section of the page, you can also manage your online users.
- **Feedback and bugs** – This tile opens the **General Feedback** page in Microsoft Connect. Use this page to record bugs, and to design change requests, feature requests, and suggestions.
- **Microsoft 365 users** – This tile opens the **Users and groups** page in Microsoft 365 admin center. You can add, update, and remove users, reset passwords, and assign licenses for other services.



Last updated on 04/01/2026

One Version service updates overview

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

The following set of topics provide information that is related to service updates for Microsoft Dynamics 365 for Finance and Operations version 8.1 (October 2018) and later. This is applicable for cloud releases only.

- Service update availability – This article provides information about the release cadence and release process.
- Software lifecycle policy and cloud releases – This article provides information about the service updates, availability, and end of service.
- One Version service updates FAQ – This article answers questions about the update process, tools, planning, and Retail service updates.

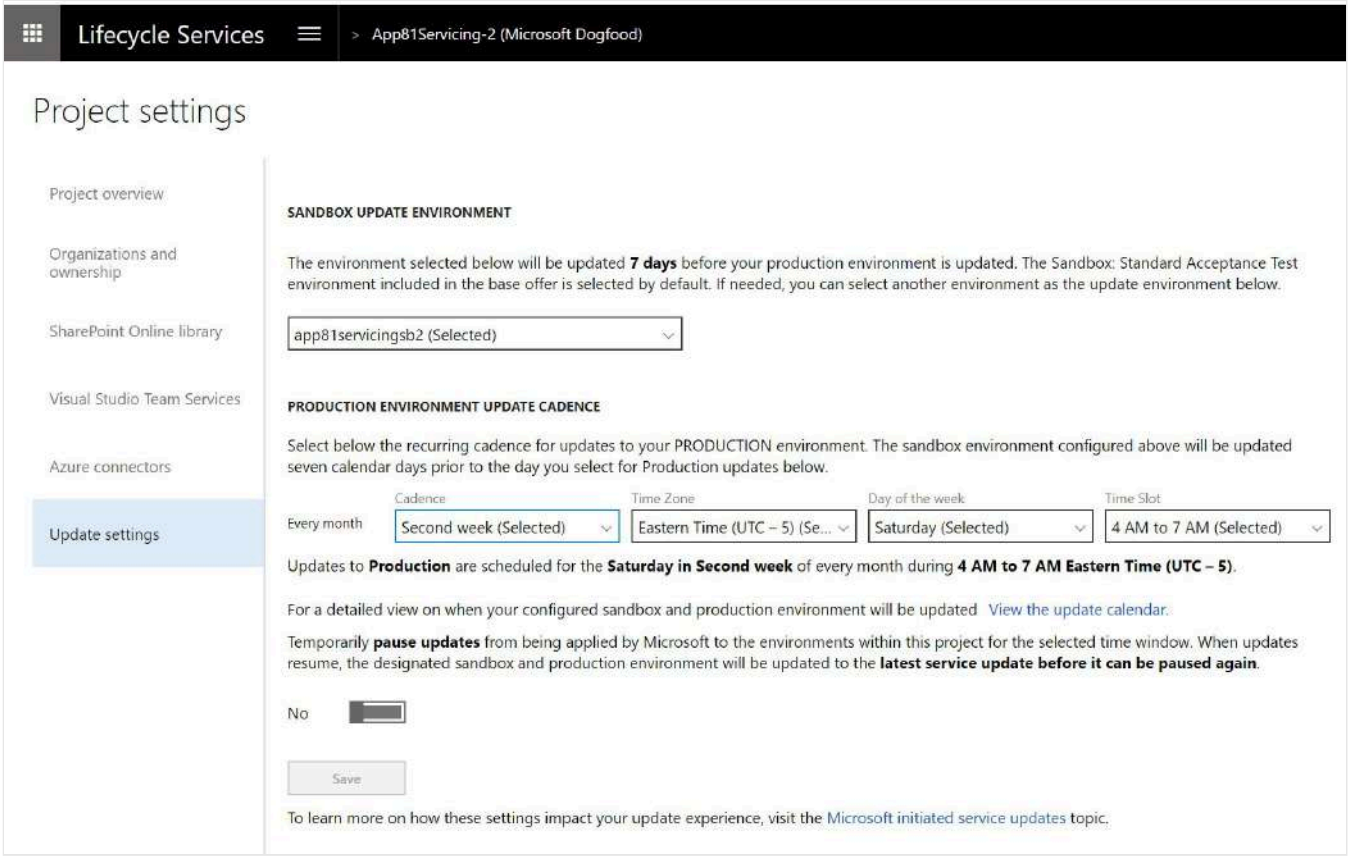
The experience for service updates consists of four distinct steps:

1. Configure
2. Notice
3. Update
4. Validate

The rest of this article describes each step and provides links to related topics.

Configure

Customers can select a maintenance window, based on their business constraints. In Microsoft Dynamics Lifecycle Services, use the fields in the **Production environment update cadence** section on the **Update settings** tab of the **Project settings** page, as shown in the following image. A calendar of upcoming updates is available to help you plan ahead.

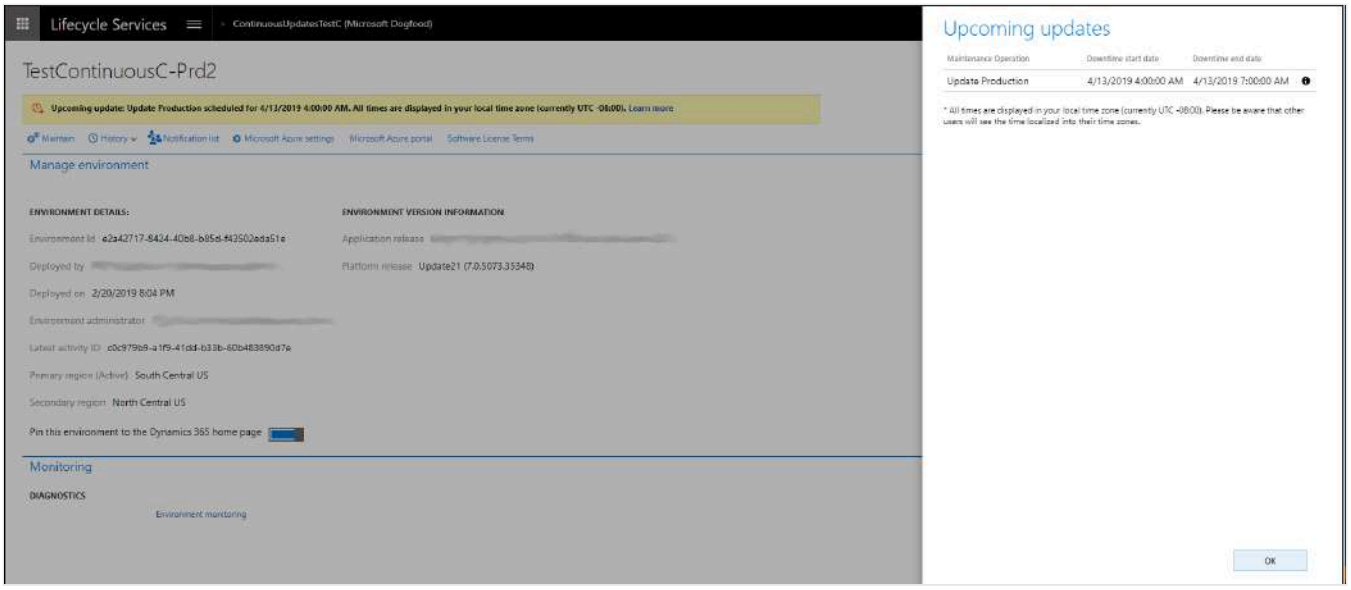


Users must opt in to new features and turn them on. All updates are applied first to the user acceptance testing (UAT) environment and then to the production environment. Therefore, customers have time to do any validation that is required. Customers can select the environment that is updated. They can also pause for a maximum of one consecutive service update.

Notice

Release plans will be available to help you plan ahead and understand what is changing. You can learn about upcoming features up to three months in advance. The What's new topics provide details about the updates for specific months.

Additionally, a notification email will be sent five days in advance, and a notification will appear in Lifecycle Services just before an update, as shown in the following illustration.

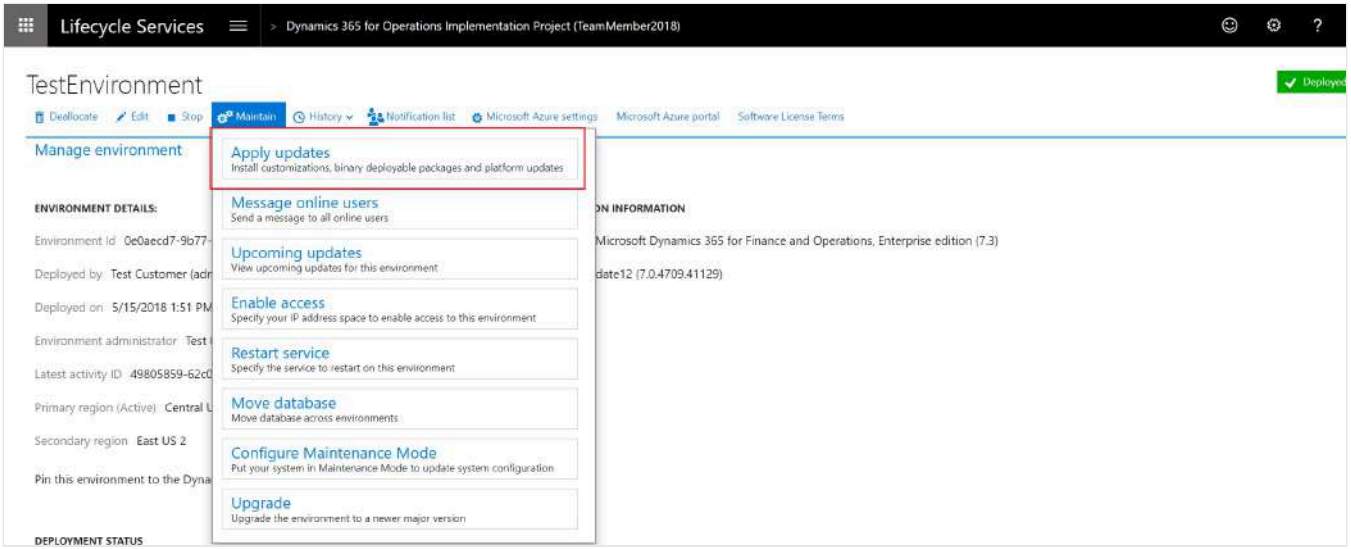


Update

After notifications have been sent, Microsoft will apply the update (**auto update**) during the designated maintenance window. After this operation is completed, a notification email will be sent to indicate the status of the update. Customers will also be able to **self-update** by using the standard update experience in Lifecycle Services. For more information, see Apply updates to cloud environments.

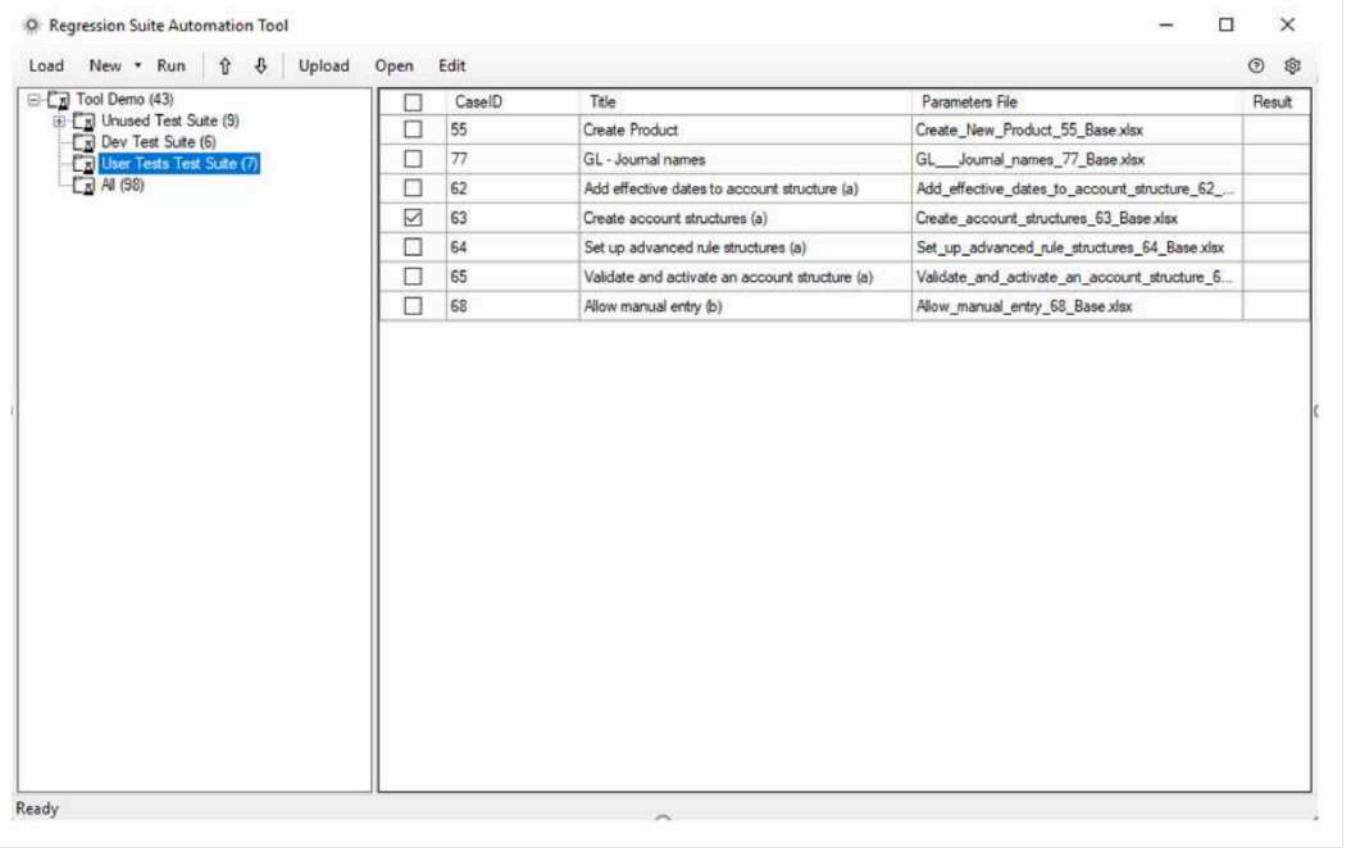
Customers who participate in the First release program will have an opportunity to update their sandbox environment and other environments before general availability. To sign up for the First release program, go to https://aka.ms/FirstReleaseFnO.

Customers will also be able to **self-update** by using the standard update experience in Lifecycle Services, as shown in the following illustration.



Validate

After an update is completed in the UAT environment, a basic business process test can be executed to validate the environment. To support this effort, a no-code automation test tool for business process testing is available, as shown in the following illustration. For more information, see Create and automate user acceptance tests.



Some customers have both external data integrations and internal data integrations. We recommend that these customers use the Data task automation tool for testing.

Last updated on 03/07/2026

One Version service updates FAQ

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This FAQ is intended to provide clarity about the service updates, processes, and tools that you can use to prepare for the change. We continue to add information to this article as required.

For more information about One Version service updates, see One Version service updates overview.

> **Important**
>
> The number of service updates that are released annually is now reduced from seven to four. This change affects the preview availability, general availability (self-update), and end-of-service dates for version 10.0.38, and it went into full effect as of version 10.0.39 (the "April" 2024 release).

What's changing with the new release cadence?

The following changes are being implemented:

- Service updates are released only in February (December self-update), April, July, and October. The May, August, and November releases are no longer available for self-update or autoupdate.
- The lifecycle of each release is substantially extended. It now lasts 404 to 414 days.
- The preview period is extended. Every release now includes one scheduled update to the preview build.
- The autoupdate and feedback phase of the First Release program is extended by two weeks.
- The servicing window of every release is substantially extended. It now lasts 186 to 214 days, and there's improved overlap between releases.
- The maximum number of consecutive updates that can be paused is reduced from three to one. However, because release durations are extended, the same minimum of two

service updates per year is maintained.
- We're introducing another autoupdate window for each service update, starting with version 10.0.39. For more information, see the FAQ for autoupdate section in this article.

Is the change from the maximum of three pauses to one already in effect?

Yes. Beginning with 10.0.39 (the April'24 release) you're only able to pause a maximum of one update. See the Targeted release schedule for GA (self-update) and autoupdate dates by version.

The following table shows the allowed pauses through the transition based on your installed versions, starting 10.0.35, 10.0.36, and 10.0.37, which are noncompliant since the release of 10.0.39. For more information about how to pause service updates, see Pause service updates through Lifecycle Services).

> **Note**
>
> If your Lifecycle Services project has environments older, noncompliant versions like 10.0.35, 10.0.36, or 10.0.37, pausing the upcoming autoupdate (for example, 10.0.39 the May 2024 autoupdate) isn't supported. To comply, update all environments to the current version (10.0.39) in the next autoupdate window. You can also self-update to a compliant version (for example, 10.0.38 as of April 22, 2024) to pause an upcoming autoupdate. Microsoft can't make exceptions for projects with environments on noncompliant versions. Upgrade to a supported version to use the pause functionality.

Expand table

On 10.0.35	On 10.0.36	On 10.0.37	On 10.0.38	On 10.0.39
10.0.36 release can be paused.	10.0.37 release can be paused.	10.0.38 release can be paused.	10.0.39 release can be paused.	10.0.40 release can be paused
10.0.37 release can be paused.	10.0.38 release can be paused.	Must take 10.0.39 release in one of the two autoupdate windows.	Must take 10.0.40 release in one of the two autoupdate windows.	Must take 10.0.41 release in one of the two autoupdate windows.
10.0.38 release can be paused.	Must take 10.0.39 release autoupdate windows for			

On 10.0.35	On 10.0.36	On 10.0.37	On 10.0.38	On 10.0.39
	pausing not applicable.			
Must take 10.0.39 release autoupdate windows for pausing not applicable.				

What can I expect with the new (autoupdate) cadence?

Beginning version 10.0.39, the service update autoupdate window is divided into two windows that are separated by approximately a four-week gap. This change provides customers with greater flexibility in scheduling their autoupdates. Autoupdate Window one closely resembles the historical approach of OneVersion service updates. Autoupdate Window two allows you to schedule your update for four weeks later. Autoupdates continue to function as before, with UAT sandbox updates occurring seven days prior to production. Note that if your LCS project has environments on older, noncompliant versions like 10.0.35, 10.0.36, or 10.0.37, pausing the upcoming autoupdate (for example, 10.0.39 May autoupdate) isn't supported. To comply, update all environments to the current version (10.0.39) in the next autoupdate window. You can also self-update to a compliant version (for example, 10.0.38 as of April 22, 2024) to pause an upcoming autoupdate. Microsoft can't make exceptions for projects with environments on noncompliant versions. Upgrade to a supported version to use the pause functionality.

Does the new release schedule affect when I can schedule autoupdates?

Yes, with version 10.0.39, there are two autoupdate windows to choose from for every service update. Customers can then select a weekend for the second autoupdate, which commences one month after the first autoupdate instance. There isn't a change in how autoupdates are scheduled in Microsoft Dynamics Lifecycle Services and when those autoupdates occur. The only change is which service updates are released each year.

As an example, let's say you have opted for the 10.0.39 ("April") release through autoupdate. Microsoft makes this release generally available for self-update by all customers on March 15, 2024. If you've enabled autoupdates through Lifecycle Services, you'll start receiving production updates two weeks after the public availability date, which is March 15. This occurs during the first autoupdate window, starting either on April 5, April 12, depending on your chosen configuration. Alternatively, if you've selected the second autoupdate window, your updates begins on May 3, May 10. Opting for the second window gives you four more weeks

between general availability and the final broadcast weekend, extending beyond the standard six-week timeframe.

In the example above, if a customer opts out of both autoupdate windows for the 10.0.39 release, they can only opt out of the first autoupdate window for the next release, 10.0.40 ("July"). This is because skipping a release is no longer possible due to a previous pause under the new pause policy. For more information about how to pause service updates, see Pause service updates through Lifecycle Services (LCS).

> **Important**
>
> Customers are explicitly required to pause both autoupdate windows to pause a release if they are eligible to do so, based on the pause policy. Pausing the first autoupdate window will not auto pause the second window.

When does the new service update release cadence take effect?

Version 10.0.38 (the "February" release) is revised to act as a transition release. Version 10.0.39 (the "April" release) is the first service update that's released under the new cadence.

Is the published 10.0.38 schedule affected by the new service update release cadence?

Yes. To enable version 10.0.38 to act as a transition release, some release milestones were adjusted for alignment with the new release cadence.

- Preview availability was pushed out by two weeks, from October 13 to October 27, 2023.
- An update to the preview was added and released on November 3, 2023.
- Customers who participate in the First Release program receive their autoupdates a month earlier, in December 2023.
- General availability for self-update occurs a month earlier, on December 22, 2023.
- There's no change to the general availability broadcast (autoupdate) dates.
- The servicing window is extended from April 12 to August 9, 2024.

How does this change affect pausing updates for noncompliant versions like 10.0.35, 10.0.36, and 10.0.37?

If your Lifecycle Services project has any environment (default or sandbox or production) on noncompliant versions like 10.0.35, 10.0.36, or 10.0.37, pausing 10.0.39 isn't supported. To

comply, update all environments to the current version (10.0.39) in the next autoupdate window. You can also self-update all environments to 10.0.38 to pause 10.0.39. See the autoupdate policy change for details.

Is it possible to stop the 10.0.39 autoupdate from happening even though I'm on a version that isn't compliant (see related question above)?

At this time Microsoft can't make exceptions for projects that have environments on versions that are out of compliance. Upgrade to a supported version to use the pause functionality.

Can the updates be delayed? What's the policy?

Yes, customers can pause, delay, or opt out of an update by using the update settings in Lifecycle Services projects. As of the April 2024 autoupdate, customers can choose to pause one update. Before April 2024, the number of pauses that are available to a customer depends on that customer's release version relative to the latest version. For more information, see **Is the change from the maximum of three pauses to one already in effect** in this FAQ. For more information on the twice autoupdate window we're introducing starting 10.0.39 see **What can I expect with the new (autoupdate) cadence** in this FAQ.

For information about how to pause an update, see Pause service updates through Lifecycle Services (LCS).

How does the timing for general availability of a release work?

The release package is made generally available to all customers for self-update before autoupdates. The timing of the package release for self-update relative to the production autoupdates varies. To determine the timing of self-update and autoupdates for upcoming releases, see Targeted release schedule (dates subject to change).

Sandbox updates are always scheduled one week before the update. Production autoupdates for releases are scheduled for the first, second, and third weeks of the month. Updates are received during the selected week based on the configuration set up in Lifecycle Services.

> **ⓘ Important**
>
> Starting with version 10.0.39, customers can choose between two autoupdate windows that occur four weeks apart for every service update. There isn't a change in how the

broadcast occurs between the first and second windows. See **What can I expect with the new (autoupdate) cadence** in this FAQ.

Customers can always choose to apply the update earlier than the suggested times in Lifecycle Services, or at a time that's more convenient. If a customer is already on the latest version, the automatic update is canceled.

Service updates

What product versions are affected by service updates?

Expand table

Version	Description
10	All customers are scheduled for automatic service updates. The updates are combined application and platform updates. As of February 2024, you can pause only one update before you're required to take the next update. For information about how to pause an update, see Pause service updates through Lifecycle Services (LCS).

What do the service updates contain?

Service updates contain application and platform changes that are critical to the service, including regulatory updates. Service updates are backward compatible, and new experiences are configurable. The update is represented by a single version.

> **Note**
>
> Add-in components, Power Apps, and Dataverse are updated independently of the service update package and process for finance and operations apps.

What's a regulatory update?

A regulatory update is a new feature, or a change to an existing feature, that's required by law, usually for a specific country or region. A regulatory update is always required by a specific law enforcement date (LED) and should be enabled by that date or earlier.

How can I determine what's changed in a service update?

The "What's new or changed" documentation is the primary source for the details of each service update. The release plans are the primary source of information for new features and changes for a future release. Features are also documented on Microsoft Learn as needed.

> **Note**
>
> To view the specific fixes included in a major release version, such as Dynamics 365 Finance 10.0.XX: Go to LCS, search for **version 10.0.XX**. This displays **Version 10.0.XX of Finance and Operations apps**. The fixes included in that release are listed.

What's the upcoming schedule of updates?

Each year, four service updates are released. You have the option to apply them at a time that's convenient for you, or you can let Microsoft automatically apply them, based on the selected maintenance window. You're required to use an update that's no more than one update behind the current update.

To view the targeted release schedule for upcoming releases, see Service update availability.

Are there any major differences between the updates?

There are two major updates each year: the April update and the October update. New experiences can be enabled in these updates. Major updates don't require code or data upgrade. Breaking changes are communicated 12 months in advance, so that customers can plan accordingly. Breaking changes are introduced only during major updates.

What does it mean for an update to be backward compatible?

Backward compatibility covers binary and functional compatibility.

- **Binary compatibility** means that you can apply an update in any runtime environment without having to recompile, reconfigure, or redeploy customizations. It also means that, in a development environment at design time, X++ public and protected APIs and metadata aren't modified or deleted. If Microsoft must break compatibility by removing obsolete APIs, the change is communicated 12 months in advance and follows a deprecation schedule.
- **Functional compatibility** refers to the user experience. All new experiences are available on an opt-in basis for a 12-month period.

Backward compatibility doesn't include non-X++/metadata APIs. Microsoft reserves the right to update versions of any dependencies that the product uses, and to remove dependencies,

without early warning. Microsoft doesn't commit itself to maintaining backward compatibility of dependent software libraries unless this commitment is expressly stated.

For more information about deprecation guidelines, and deprecated methods and metadata elements, see Deprecation of methods and metadata elements.

What's the process for deprecation?

The deprecation notice is announced in the product documentation 12 months before Microsoft removes a feature from the app.

For breaking changes that only affect compilation time, but that are binary compatible with sandbox and production environments, the deprecation time is less than 12 months. Typically, these changes are functional updates that must be made to the compiler.

If I'm not doing active development/recompilation of my code, how can I learn if there's a deprecated feature that affects me?

Deprecated features are documented for each release. For more information, see Removed or Deprecated features.

Can I update just the platform? Similarly, can I update just the application?

No, only combined application and platform update packages are released for both service updates and quality updates.

Service updates for on-premises deployments

Both cloud deployments and on-premises deployments have the policy and schedule for service updates. For example, the option to delay one service update applies to both types of deployment. However, the process for applying each of these updates remains slightly different. For more information, see Apply updates to on-premises deployments.

Process

How does Microsoft ensure the quality of releases?

Ensuring the quality of releases is a fundamental principle that's enabled through a series of progressive, rigorous, automated validations, as described in Service update availability.

Can I select the day and time of the update?

You can configure the day and maintenance time windows in Lifecycle Services. The service update, which is based on your update settings, starts within 15 minutes. If you opt in to receive Lifecycle Services notifications, you receive an email that includes update instructions. You can select the designated Tier-2/user acceptance testing (UAT) sandbox for the update. You have seven calendar days to do testing and validation before the production environment is updated. All other sandboxes are automatically updated on the same day as the production environment. For more information, see Configure service update.

You can optionally apply the update earlier to all environments through Lifecycle Services. The production-ready deployable package is available to all customers via the Action Center in Lifecycle Services by the general availability (self-update) date. For the targeted release schedule of upcoming releases, see Service update availability.

A service update was applied to the environment. In Lifecycle Services, what does the number on the tile for this environment represent?

Microsoft automatically applies the same service update to all customers. Microsoft continues to service the update until the end-of-service date for that release is reached. In Lifecycle Services, the available updates tile for the environment represents the cumulative quality update package that's available to be applied to your environment. There are two numbers on the tile. The top number is the release version, and the bottom number is the build number of the latest quality update package. The build number is always larger than the number of the service update that was applied to your environment, either through self-update or autoupdate. Because Microsoft automatically applies the same version to all customers, you're responsible for applying the cumulative hotfix package if it's required.

> ⓘ **Note**
>
> If your environment version is earlier than the latest release version, a tile appears under **Available updates** to remind you to apply the latest service update by using self-update.

How do I update to the latest version?

You can self-update to the latest version by using the tile on the **Environment details** page. After an update is released by Microsoft, the tile shows the available update. You can choose to apply the update by going through the update experience in your sandbox and production environments. For more information, see Apply updates to cloud environments and Apply updates to on-premises deployments.

How do I update the production environment to the same version after Microsoft updates the sandbox environment?

When Microsoft updates a sandbox environment, the package that's used for the update is saved in the project's Asset library. The name of the package is prefixed with the words "Service Update." Because the package was already applied to the sandbox environment, you can mark it as a release candidate. You can then go to the production environment and schedule to apply the package, just as you might schedule any other update.

What's the expected downtime during an autoupdate?

The expected downtime for a successful update is approximately 15 minutes. However, Microsoft asks for three hours of downtime in case issues occur while the update is being applied.

Can I delay an update?

Yes, you can pause, delay, or opt out of an update by using the update settings in Lifecycle Services projects. However, this option is unavailable if any of your sandbox and production environments are more than one version behind the latest available update. After the delay, Microsoft schedules and automatically applies an update. The update experience for a delayed update incurs more downtime.

Can I delay an update for longer than the one permitted skipped service update because of seasonal activity or other business reasons?

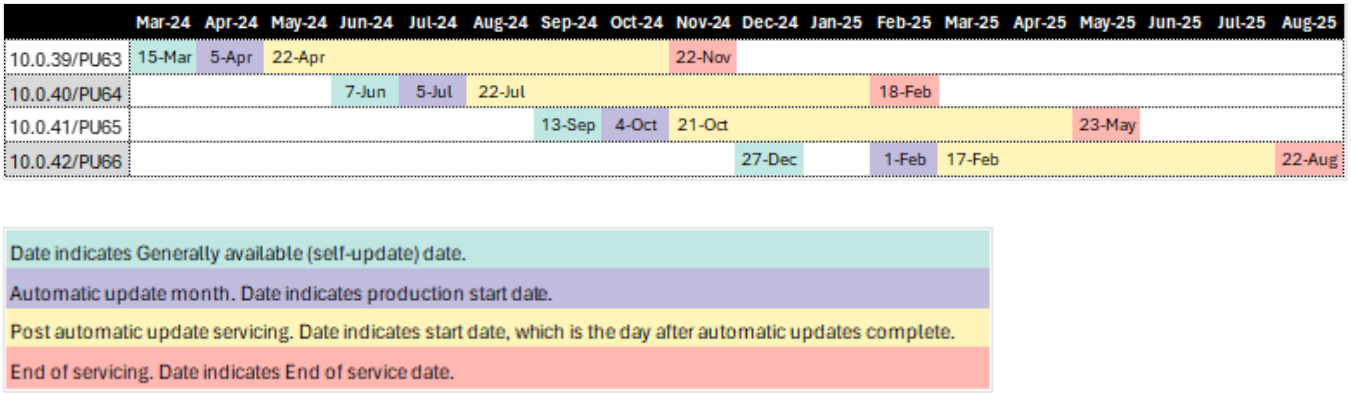
No, after your environment version is more than one version older than the latest version, Microsoft automatically applies the latest service update to the default Tier-2 sandbox. Seven days later, the update is applied to all other sandbox environments and production environments that are also more than one version older than the latest version. You can pause only one update before you must take the next service update. Therefore, in effect, a minimum of two service updates are required annually. For example, a customer is running version 10.0.39 and chooses to pause update 10.0.40. In this case, service update 10.0.41 is

automatically applied first to the Tier-2 sandbox environment, and then later to all other sandbox environments and production environments.

What does it mean for a release to be "in service"?

A release is a service update version that has been made available to customers. A release is "in service" from the day when it's made available to customers for production use through its end-of-service date. For release milestones by release, see Targeted release schedule (dates subject to change).

A release in post-update servicing reaches its end-of-service date about one month after autoupdates are completed for the latest version. Therefore, customers who chose to pause have time to complete their required update before the servicing window for their version is closed. The following illustration shows the staggered release rollout and servicing model.



> **Important**
>
> The Support team at Microsoft can't accept cases for issues that are reported from versions that are at their end of service. You must first update to the latest service update and then apply the latest quality update. At that point, if the issue persists, you can report it.

What happens to an environment that's running a finance and operations app version that's no longer supported?

For environments that are running a finance and operations apps version that's no longer supported, a warning message appears at the top of the **Environment details** page in Lifecycle Services.

For all Microsoft-managed environments, and for sandbox and production environments in on-premises implementation projects, some Lifecycle Services functionality might not be available

when an environment is running a finance and operations apps version that's no longer supported. This functionality includes the ability to complete the following actions:

- Enable maintenance mode.
- Use all capabilities that are provided for moving databases within an environment or across environments.
- Enable firewall access to SQL Server databases.
- Download Regression suite automation tool (RSAT) certificates.
- Regenerate RSAT certificates.

After you apply a service update for a supported version, this functionality becomes available in the affected environment.

How do the automatic updates affect my Microsoft-managed additional sandbox environments in my Lifecycle Services implementation project?

All additional sandbox environments are updated during the same update window as your production environment, and they're updated to the same release version that's used for the production update. The update is also applied to additional sandbox environments that are on versions that haven't reached their end of service.

What happens during an update when my additional sandboxes are on different versions than the default sandbox and production environments, which are scheduled to receive the latest service update?

All environments are updated to the current version that's being used for autoupdates.

What if the default sandbox environment is manually updated to the same version as, or a newer version than, the automatic update version?

Automatic updates for the production environment and all additional sandbox environments are updated to the current version that's being used for autoupdates.

Updates for the default sandbox environment are canceled.

What if the default sandbox environment is manually updated to an older version than the automatic update version?

The default sandbox environment, the production environment, and all additional sandbox environments are updated to the current version that's being used for autoupdates.

What if the production environment is manually updated before the production environment email is sent?

Automatic updates for the production environment and all additional sandbox environments are canceled.

What if the production environment is manually updated after the production environment email is sent?

Automatic updates for the production environment are canceled, but all additional sandbox environments are updated to the current version that's being used for autoupdates.

What if I find an issue during the sandbox update?

If you find an issue while you're doing validations in a sandbox environment, you can request to skip the update directly through Lifecycle Services, by providing a valid support ticket number and a business justification. For more information, see Pause service updates through Lifecycle Services (LCS).

What if I find a critical issue during sandbox testing, but I can't pause the production automatic update?

Critical issues should always be submitted to the Support team via Lifecycle Services as soon as they're identified. The Support team works with you to fix the critical issue.

How much time do I have for validation?

You have seven calendar days for validation after the update is applied to your sandbox environment. If you need more time, you can access the deployable package via the Action Center in Lifecycle Services and apply it to your environments. In this way, you get more time to test the update before a production roll-out.

What happens when the service update is completed?

After the service update is applied by Microsoft, you receive a notification that indicates whether the update was successful, or whether it couldn't be applied. An update might not be applied for the following reasons:

- **Pending Package Sign off** – If a package is pending sign off, Microsoft won't apply the service update to production.
- **Deployment Failure** – If there was a deployment failure, the environment is rolled back to the original state.

Can an environment be brought back to its previous state if there are issues after a service update is applied?

As with other code promotions, rollbacks can't be done after a service update is applied. Microsoft uses flighting and features to help mitigate the risk of updates and enable rapid enabling and disabling of new functionality. For more informatoin, see Feature management and flighting.

If there's a failure, can I reschedule the update to be automatically applied?

No, you can't reschedule the update. However, you can apply the package when it's convenient, just as you might schedule any other update.

Will critical hotfixes be automatically applied to my sandbox/production environment during automatic update?

The service update that's generally available to all customers for self-update and autoupdate contains hotfixes and new functionality. If a critical issue is reported and fixed after the service update was applied, you can pull the latest cumulative quality update from the tile in Lifecycle Services.

How do my ISVs stay current?

Service updates to customer environments are backward compatible, and no action by the independent software vendors (ISVs) is required. ISVs develop on the minimum required platform release that their code depends on. Breaking changes have a 12-month lead time, so that ISVs can include them and do validation. Microsoft recommends that ISVs take advantage of the preview release for each service update. In this way, they can get early access to the platform code and validate their solutions against the update before it's made generally available. Microsoft encourages both ISVs and customers to join the Preview Early Access

community on Microsoft Viva Engage. This Viva Engage group provides a forum where participants can receive preview and release-related announcements, and collaborate with others in the finance and operations apps community.

What about new features?

All new features are available on an opt-in basis for a 12-month period. They don't require any change management until you enable them.

Are batch jobs suspended during a service update?

Batch jobs are suspended during the maintenance windows and resume when the maintenance is completed.

For information about how to pause an update, see Pause service updates through Lifecycle Services (LCS).

Tools

How can I get early access to updates that haven't yet been released?

As of version 10.0.26, the preview package for all service updates is made available to all customers through the Shared asset library in Lifecycle Services, under **Software deployable package**. Preview packages can be deployed to development or test environments. However, they can't be used in production environments. You agree to the program terms at the time of installation. Sign-up for access to preview packages (formerly known as the Preview Early Access Program [PEAP]) is no longer required.

To be in the first group of customers to take service updates to production, you can join the First Release program. While you're enrolled in First Release, Microsoft keeps your system current with the latest updates. First Release customers receive updates to all sandbox and production environments, and are always updated before general availability.

Is any tooling available to support testing of the latest release?

UAT is typically required before you take a Microsoft application update or apply custom code and configurations in your production environment. The Regression suite automation tool (RSAT) is available now and significantly reduces the time and cost of UAT.

RSAT lets functional power users record business tasks by using the Task recorder and then convert them into a suite of automated tests, without having to write source code. Test libraries are stored and distributed in Lifecycle Services by using the Business process modeler (BPM) libraries. They're fully integrated with Azure DevOps for test execution, reporting, and investigation. Test data parameters are decoupled from test steps and stored in Excel data files.

How can I test and validate that the integrations continue to work?

Data task automation lets you easily repeat many types of data tasks and validate the outcome of each task. You can also do automated testing of data entities by using task outcome validation. For more information, see Data task automation.

Preparing for One Version

How can I log an extensibility request?

Extensibility requests can be logged in Lifecycle Services. For more information, see Extensibility requests.

What does "end of service" mean?

Microsoft doesn't provide any fixes for issues on versions that have reached their end of service. If you encounter an issue on a version that has reached its end of service, you must apply the latest update and then report the issue if it persists. For end-of-service dates by version, see Service update availability.

All environments continue to be operated by Microsoft. In addition, all automatic processes that involve your environments, such as monitoring or self-healing, continue as-is for supported versions.

Are individual hotfixes supported?

Individual hotfixes aren't supported after version 8.1. To apply a fix, you must update to the latest cumulative update available. Critical fixes are also applied via cumulative update and are available through the Lifecycle Services servicing experience.

Will you notify me about critical hotfixes that are released for the monthly update that I'm on?

For every preview and general availability release, a searchable summary list of customer-reported issues is published to Issue search in Lifecycle Services. To find the list by title search, use the pattern "Version 10.0.xx of Finance and Operations apps." You can sign up to be notified when an open issue is resolved.

Commerce service updates

What options are available to help minimize the impact on my Commerce cloud components?

Commerce cloud components require the same downtime as Dynamics 365 headquarters. In an upcoming release, the Retail Cloud Scale Unit (RCSU) is available to reduce and further schedule updates to your deployment. For more information about RCSU, see the published release information on our documentation and release notes sites.

Will there be options to take individual hotfixes for my Commerce solution components?

All fixes and updates for Commerce components are cumulative.

What are the maintenance downtime requirements that might affect channel operations?

For retailers that have a business need for redundancy, Modern POS offline capability enables core point of sale (POS) operations to be available while the system is disconnected from the internet or the cloud environment is being updated. In addition, stores that use Commerce Scale Unit continue to operate and support core POS operations during cloud maintenance windows. For more information, see Online and offline point of sale (POS) operations.

When will I have to update my in-store components?

To maintain support, all in-store components must be running released software that's less than one year old. Customers are responsible for updating self-hosted components (such as components that are installed in stores or in privately managed datacenters). They're also responsible for ensuring that the installed versions of those components are actively supported.

Will there continue to be backward compatibility for the in-store components?

Updates to components that are hosted in the cloud continue to preserve backward compatibility with component versions that are self-hosted by the retailer for 12 months after the release date for that version. (These components include components that are installed in stores or in privately managed datacenters: Modern POS, Commerce Scale Unit, or Hardware Station.) Self-hosted components don't have to be updated at the same time as cloud-hosted components. They can be updated on a separate cadence, so that there's time to roll updates out to stores.

What options are available for updating in-store components across my organization?

Customers can choose to manually update self-hosted components at each store, or they can use mass update tools such as Microsoft System Center Configuration Manager and Microsoft Intune.

What options do I have to slowly enable new functionality across my channels?

Microsoft provides several mechanisms for progressively rolling out and enabling functional enhancements across stores, devices, and users.

- **Screen layout designer** – Most visual elements in POS are configured and centrally managed by an administrative user in the customer organization. Therefore, new POS operations don't automatically appear in POS unless they're explicitly configured for inclusion in corresponding screen layouts. Screen layouts are configured by using Screen layout designer. They can be specific to a store or a POS device. For more information, see Screen layouts for the point of sale (POS).
- **Functionality profiles, POS permissions, and Commerce parameters** – Significant elements of functionality in POS are typically configurable by the user. These elements can be configured through functionality profiles, POS permissions, Commerce parameters, or other controls that allow for device-level, register-level, store-level, or user-level control over functionality in applicable scenarios.
- **Modern POS and Commerce Scale Unit** – Because Modern POS and Commerce Scale Unit are self-hosted by the retailer, topologies that include either of these components enable updates to be rolled out at a separate (and slower) cadence and in a more granular fashion than cloud-only topologies.

Last updated on 03/05/2026

Manage finance and operations updates and your custom code lifecycle

This article describes application lifecycle use cases for finance and operations implementations. It's focused on the following scenarios:

- Managing your source code development branches
- Applying the next version of a Microsoft service update
- Applying a new version of your custom code

This article applies to Microsoft Dynamics 365 Finance, Dynamics 365 Supply Chain Management, Dynamics 365 Human Resources, Dynamics 365 Commerce, and Dynamics 365 Project Operations.

The main goal is to show how to complete the following tasks:

- Stay up to date and manage Microsoft service updates (or quality updates) for finance and operations apps (including Dynamics 365 Commerce) in incremental phases, independently of the lifecycle of your own customization. This approach simplifies the update process, and reduces the cost and risk of regressions that are associated with all-in-one upgrade projects.
- Take advantage of source code branches for version control of your custom code. By using version control, you can isolate the rollout of critical changes and hotfixes from the development of new features and capabilities.

This article doesn't explain how to use the different tools in Azure DevOps and Microsoft Dynamics Lifecycle Services (LCS). Instead, it's focused on processes and best practices. The Apply the next version of a Microsoft service update and Apply a new version of your custom code sections contain both an overview of the phases and the steps of the process.

This article includes the following sections:

- Environments
  - Environments that run your current release
  - Environments that run the next version of your custom code
- Manage source code branches
- Apply the next version of a Microsoft service update
  - Backward compatibility of Microsoft updates
    - Runtime compatibility
    - Design-time compatibility

    - Phase 1: Update the finance and operations implementation
        - Track 1: Update your runtime environments
            - Update Test 1
            - Update UAT
            - Update Prod
        - Track 2: Update your development environments
        - Error situations
            - Case 1
            - Case 2
            - Case 3
    - Phase 2: Update CSU to version 10.0.11
        - Phase 2 prerequisites
        - Track 1: Update your CSU runtime environments
            - Update Test 1
            - Update UAT
            - Update Prod
        - Track 2: Update your development environments
    - Phase 3: Update POS to version 10.0.11
        - Phase 3 prerequisites
        - Update your Commerce development environment
        - Option 1: Store component updates that include only runtime changes
        - Option 2: Store component updates that include runtime and custom changes
        - Update Process
            - Update Test 1
            - Update UAT
            - Update Prod
- Apply a new version of your custom code
    - Create a hotfix of your code
    - Update your custom code from release N to release N+1
- Upload, update, and deploy store components

Environments

This section describes the collection of finance and operations environments that the application lifecycle management (ALM) scenarios in this article rely on. This configuration is typical for organizations that have implementations that rely on custom code (extensions). This custom code includes customizations that independent software vendors (ISVs) provide.

Environments that run your current release

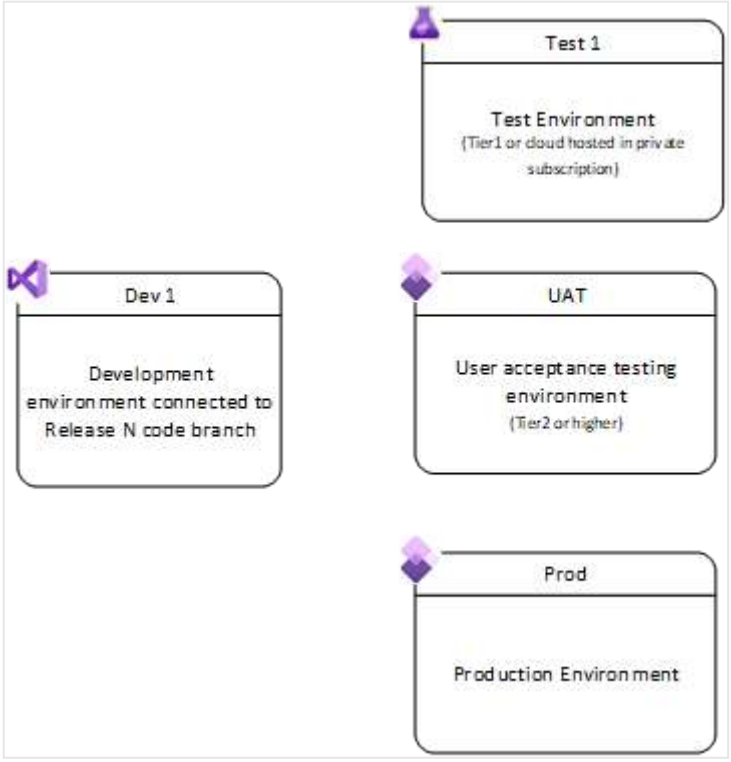
The following environments run your current release:

- **Dev 1** – A development environment that runs the same version of finance and operations apps as the production environment. The Dev 1 environment uses Azure DevOps for version control of custom code. It's connected to the current release branch of your custom code. For more information, see the Manage source code branches section.
  - Many options are available for development environments, both cloud and on-premises. For more information, see Deploy and access development environments.
  - For build automation, use the new Azure DevOps Hosted Agents functionality. For more information, see Build automation that uses Microsoft-hosted agents and Azure Pipelines.

- **Test 1** – A Tier-1 test environment that you use for functional and configuration testing. The Test 1 environment runs the same version of finance and operations apps as the production environment. It also runs the latest release version of your custom code extensions.

- **UAT** – A pre-production environment that you use for user acceptance testing. The UAT environment is a Tier-2 (Standard Acceptance Test) or higher environment. It runs the same version of finance and operations apps as the production environment. It also runs the latest release version of your custom code extensions. You typically connect this environment to a copy of the production database.

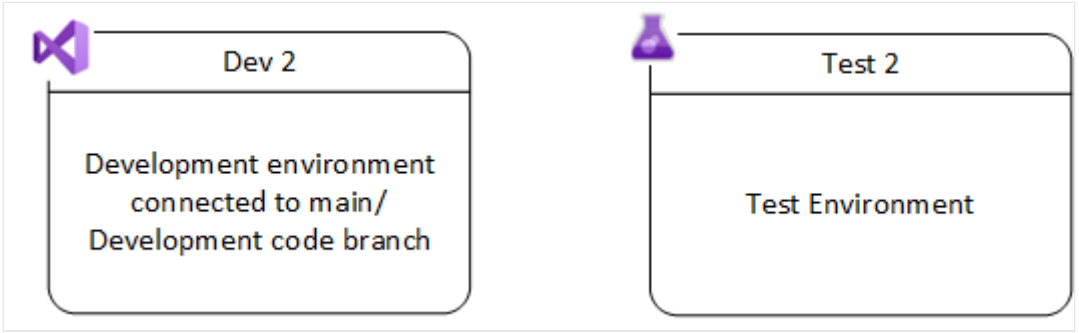
- **Prod** – Your live production environment that runs on your production database.



Environments that run the next version of your custom code

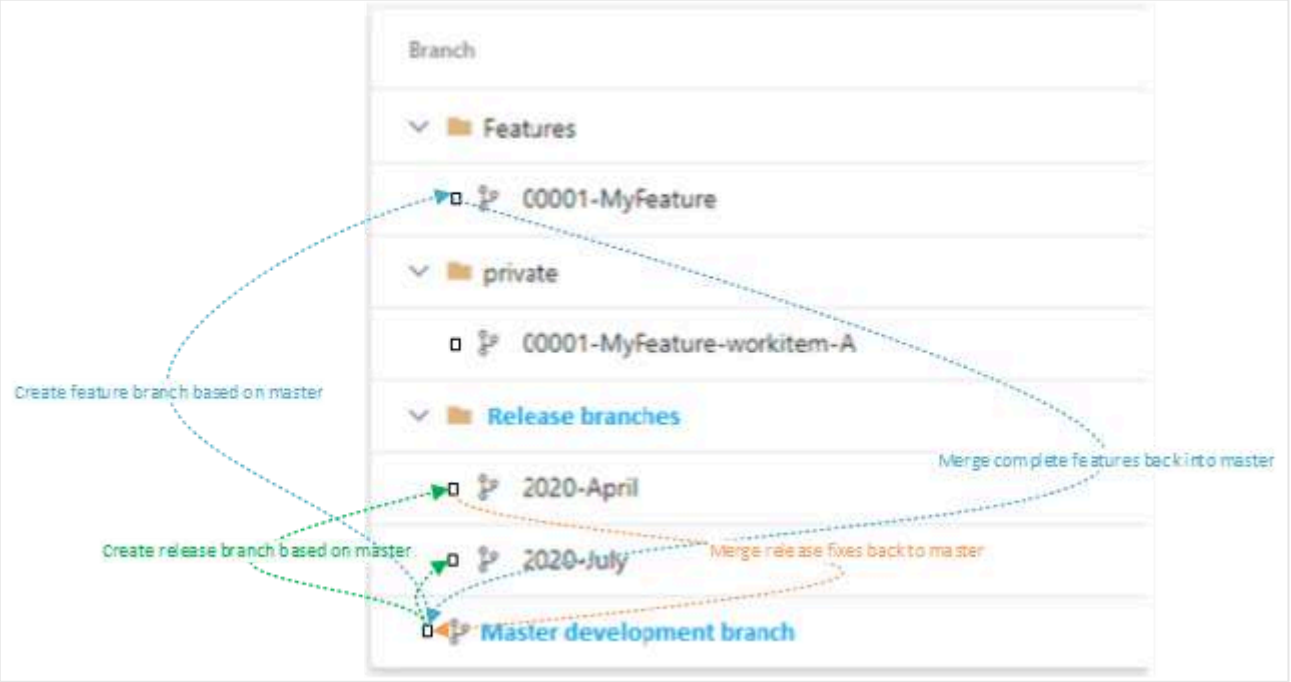
The following environments run the next version of your custom code:

- **Dev 2** – A development environment that you use for development of the next version of your custom code extensions. It uses Azure DevOps for version control of custom code. It's connected to the development branch (**main** branch) of your custom code. For more information, see the Manage source code branches section.
- **Test 2** – A functional test environment that you use for testing of the next version of your custom code extensions.



Manage source code branches

It's important to follow best practices when you manage branches of custom code. By following these practices, you help minimize cost and ensure the quality of your releases and updates.



The **main** branch (development branch) contains the latest functioning version of the next release of your code.

When you work on new features, create a new feature branch from the **main** branch. Then, when you complete the feature work, integrate the feature branch back into the **main** branch.

The release branches contain the code base of your official releases. In the preceding illustration, the assumption is that you have only one release branch, **release/2020-April**. **2020-April** isn't a build. It's a source code branch. Because you probably create hotfixes for your release, you make changes and produce builds from this release branch.

- Don't use release branches to develop new features. Use them only for critical fixes or changes that are required in your live environment.
- After you make a change in a release branch, integrate the branch back into the **main** branch. By using this approach, you ensure that your next release also contains the fix.
- Among the example environments that were described earlier, the Dev 1 environment connects to the **release/2020-April** branch.

When you're ready to release a new version of your custom code, create a new release branch from the **main** branch. For this example, create a new release branch from **main** and name it **2020-July**.

You might have private branches that individual developers work in while they work on a specific work item that's based on a specific branch of your code. Merge private branches back into their parent branch when the work is completed. For more information, see Learn about branching strategies for Team Foundation Version Control (TFVC) and how to select an effective strategy.

Apply the next version of a Microsoft service update

By using a phased approach, you help maximize the efficiency of service updates. Each phase updates one component of your implementation.

1. **Phase 1** – Update your finance and operations environments.

   Your current version of Commerce Scale Unit (CSU) and Point of Sale (POS) works correctly with the new finance and operations update. For example, version 10.0.7 of CSU is compatible with version 10.0.11 of finance and operations apps.

2. **Phase 2** – Update CSU.

3. **Phase 3** – Update POS.

When you take a Microsoft update, you don't need to update your custom code to the next version. By taking Microsoft updates without bundling them with custom code updates, you help simplify the update process. You also help reduce the cost and risk of regressions that are associated with all-in-one upgrade projects.

Backward compatibility of Microsoft updates

It's important that you understand what Microsoft means by *backward compatibility of service updates*, so that you have context for the next sections of this article. Service and quality updates are *runtime* backward-compatible. However, they're not always *design-time* (*compile-time*) backward-compatible.

Runtime compatibility

All Microsoft updates are intended to be runtime backward-compatible. This compatibility covers both binary compatibility and functional compatibility. Runtime compatibility means that customizations that exist in production and sandbox environments continue to work after Microsoft service updates are deployed to those environments. Those updates include service updates and quality updates. Runtime compatibility also means that Microsoft updates are backward-compatible with customizations that were compiled on an earlier platform.

Binary compatibility is backward only. You can compile a customization on an older application version and platform version, and deploy it to an environment that's running a later version. However, you can't deploy code to an environment that's running a version that is earlier than the version that you compiled on.

> **Important**
>
> Runtime compatibility covers personalizations, so they continue to work after Microsoft service updates are deployed to these environments. Because of this, reimporting all personalizations after a service update (or at any other time) is unnecessary and highly discouraged.

Design-time compatibility

Design-time (compile-time) backward compatibility means that developers can apply updates to their development environments and successfully compile their code without having to make any changes.

Microsoft aims for design-time compatibility. However, some updates might include changes that aren't compatible at design time, but that are binary-compatible. Therefore, after an update is applied, new errors or warnings might occur when you compile your code. Here are some examples of these changes:

- Microsoft makes an enumeration extensible.
- Microsoft marks an API as obsolete or internal.
- Microsoft introduces a new compiler error to help prevent unsafe coding practices.

All these changes might require work on your solution. Design-time breaking changes that are binary-compatible don't require a 12-month deprecation notice. These breaking changes are documented for each service update. For more information, see What's new and changed in Platform updates.

Phase 1: Update the finance and operations implementation

This section summarizes the process that you use to update your finance and operations implementation to the latest service update. An update from version 10.0.7 to version 10.0.11 is used as an example.

This phase is divided into two tracks. These tracks can occur in parallel.

- **Track 1** – Update your runtime environments.
- **Track 2** – Update your development environments.

After you complete track 1, you're live on version 10.0.11, unless you encounter one of the error situations that are described in the Error situations section.

Track 1: Update your runtime environments

By completing track 1, you essentially complete your finance and operations update to version 10.0.11, because your production environment is live on version 10.0.11. You don't have to recompile your custom code as part of this track.

Update Test 1

The Test 1 environment runs version 10.0.7 together with the latest released version of your custom extension. Although an update of Test 1 isn't a prerequisite in this flow, it's recommended, because it adds another level of functional verification and predictability. Complete this update before you update the UAT environment. The sooner that you update Test 1 and validate on it, the more predictable your "real" update (that is, the update of the UAT and Prod environments) is.

1. Apply version 10.0.11 of finance and operations apps to Test 1.
2. Sign off on functional scenarios. You can use the Regression Suite Automation Tool to automate user acceptance testing on test and UAT environments.
3. If you encounter regressions, see the Error situations section.

Update UAT

The UAT environment runs version 10.0.7 together with a released version of your custom extension. The UAT environment is the same as the Prod environment.

1. Apply version 10.0.11 of finance and operations apps to UAT.

   You might configure your UAT environment so that Microsoft automatically updates it. However, you can always pull the update as soon as it's available.

2. Complete user acceptance testing, and sign off.

3. If you encounter regressions, see the Error situations section.

Update Prod

1. Apply version 10.0.11 of finance and operations apps to Prod.

   You might configure your Prod environment so that Microsoft automatically updates it. However, you can always pull the update as soon as it's available.

2. Sign off.

Track 2: Update your development environments

The purpose of track 2 is to update the Dev 1 environment to version 10.0.11. Dev 1 is your **main** development environment that is connected to the current release branch of your custom code. It runs version 10.0.7 of finance and operations apps. By completing track 2, you ensure that Dev 1 runs version 10.0.11 together with your latest release, and that it's ready for any future hotfixes that are required for your code.

1. Apply version 10.0.11 of finance and operations apps to Dev 1.
2. Compile your custom code, and run tests.
3. Make any required changes to your custom code.
4. Check code changes into the release branch.
5. If you have more than one development environment, follow these steps for each environment:
   a. Apply version 10.0.11 of finance and operations apps to the development environment.
   b. Sync and compile your latest custom code from the target code branch.

Error situations

Case 1

During track 1, you find a bug that requires a Microsoft quality update. If Microsoft releases the update in a timely manner, use the new Microsoft update instead of the original service update to complete track 1.

Case 2

During track 1, you find a bug that requires a Microsoft service update. However, Microsoft can't release the update in a timely manner.

As a workaround for this issue, Microsoft proposes a change to your custom code.

1. Modify your code in Dev 1 (in the release branch), and test your updates.
2. Check code changes into the release branch.
3. Create a new deployable package from the release branch.
4. Apply the new deployable package to Test 1 and/or UAT.

> ⓘ **Note**

You can deploy custom code that is compiled on version 10.0.7 to any runtime environment that's running version 10.0.7 or later. Therefore, you don't need to update Dev 1 to version 10.0.11 yet. However, you might already update Dev 1 as part of track 2.

Case 3

During track 1, you find a bug that requires a change to your custom code. This bug might be in either your code or the ISV code.

1. Modify your code in Dev 1 (in the release branch), and test your updates.
2. Check code changes into the release branch.
3. Create a new deployable package from the release branch.
4. Apply the new deployable package to Test 1 and/or UAT.

Phase 2: Update CSU to version 10.0.11

This section summarizes the process that you use to update your Commerce Scale Units to the latest service update. An update from release 10.0.7 (Commerce version 9.17) to release 10.0.11 (Commerce version 9.21) is used as an example.

Phase 2 prerequisites

Before you update your CSUs, update the Commerce headquarters environments (in the finance and operations app) to the same release or a later release. For this example, use release 10.0.11.

This phase is divided into two tracks. These tracks can occur in parallel.

- **Track 1** – Update your CSU runtime environments.
- **Track 2** – Update your development environments.

After you complete track 1, you're live on release 10.0.11 (Commerce version 9.21), unless you encounter one of the error situations that were described for phase 1.

Track 1: Update your CSU runtime environments

When you complete track 1, you complete your CSU update to version 10.0.11. Your production environment is live on version 10.0.11. You don't need to recompile your custom code as part of this track.

Update Test 1

The software as a service (SaaS) components of Commerce aren't currently supported in tier-1 environments (development/test environments). A copy of Retail Server runs locally in each tier-1 environment, and you deploy both Microsoft code for Retail Server and retail customizations through the previous system. You apply application binary packages and retail deployable packages against the infrastructure as a service (IaaS) instance.

Update UAT

The UAT environment runs CSU that corresponds to release 10.0.7, together with the same version of your retail extension that the production environment is running.

1. Update the Commerce Scale Unit. Select **9.21 (10.0.11)** as the target version. For more information, see Phase 2: Update CSU to version 10.0.11.
2. Complete user acceptance testing, and sign off.
3. If you encounter regressions, see the Error situations section.

Update Prod

1. Update the Commerce Scale Unit. Select **9.21 (10.0.11)** as the target version. For more information, see Phase 2: Update CSU to version 10.0.11.
2. Sign off.

Track 2: Update your development environments

1. Get the latest version of the Retail software development kit (SDK).
   a. Apply the finance and operations service update for release 10.0.11 to Dev 1.
   b. Get the updated version of the Retail SDK from **%ServiceDrive%\RetailSDK\Update\<newest directory>**.

2. In your **main** (development) branch, update the Retail artifacts from the new version of the Retail SDK.

3. Compile. The new version should be backward-compatible and shouldn't require any changes to your code.

4. Commit the change that includes the Retail SDK update.

5. Make any required changes to your custom code.

6. Commit code changes to the target branch.

7. Optional: If you have more than one development environment, merge the latest changes from step 4, and compile the latest custom code.

Phase 3: Update POS to version 10.0.11

This section summarizes the process that you use to update your store components, such as Modern POS (Store Commerce in the newer versions) and Hardware Station, to the latest service update. This section uses an update from release 10.0.7 (Commerce version 9.17) to release 10.0.11 (Commerce version 9.21) as an example.

Unlike updates for Commerce headquarters (in the finance and operations app) and CSU, updates for store components are delivered in the same packages. After you update Commerce headquarters and CSU, you have the following options:

- **Option 0 (no operation is required)** – Leave the store components in their previous release if the version is supported and in-policy.
- **Option 1** – Update the store components runtime (Microsoft code) so that it matches the same release as CSU.
- **Option 2** – Update the store components runtime (Microsoft code) and customization together.

In the rest of this section, the assumption is that you want or need to update the store components (option 1 or option 2).

After you complete this phase, you're live on release 10.0.11 (Commerce version 9.21) for store components, unless you encounter one of the error situations that are described for phase 1.

Phase 3 prerequisites

Phase 3 has the following prerequisites:

- You updated the Commerce headquarters components (in the finance and operations app) to the same release or a later release before you updated the CSUs. In this example, the version is 10.0.11.
- You updated the CSUs to the same release as, or a later release than, the store components. In this example, the release is 10.0.11.

Update your Commerce development environment

Follow the steps in the Track 2: Update your development environments section.

Option 1: Store component updates that include only runtime changes

This option generates a new retail deployable package that contains your store components. It includes changes only from the Microsoft code.

1. Update your release branch, which has the code that is currently used in the production environment, to the Retail SDK that matches your target release. In this example, the version is 10.0.11 (9.21). Follow the steps in the Track 2: Update your development environments section.
2. Generate a new build for the retail deployable package. This build contains the same set of customizations that is currently in the production environment, plus version 10.0.11 (9.21) of the Microsoft code.

Option 2: Store component updates that include runtime and custom changes

This option generates a new retail deployable package that contains your store components. It includes changes from both the Microsoft code and customizations.

1. Update your release branch, which has the code that is currently used in the production environment, to the Retail SDK that matches your target release. In this example, the version is 10.0.11 (9.21). Follow the steps in the Track 2: Update your development environments section.
2. Commit changes.
3. Update custom code for store components, or update references to ISV-updated components.
4. Generate a new build for the retail deployable package. This build contains updated custom code plus version 10.0.11 (9.21) of the Microsoft code.

Update process

Update Test 1

The SaaS components of Commerce aren't currently supported in development or test one-box environments. A copy of Retail Server runs locally in each development or test one-box environment, and the deployment of both Microsoft code for Retail Server and retail customizations is done through the previous system, where application binary packages and retail deployable packages are applied against the IaaS instance.

The Test 1 environment runs version 10.0.11 of Commerce headquarters and a local version of CSU from previous phases. An update of Test 1 isn't a prerequisite in this flow. Because there's no CSU in this environment, this step is mostly for verification.

1. Upload, update, and deploy the store components. For more information, see the Upload, update, and deploy store components section.
2. Sign off on functional scenarios.

3. If you encounter regressions, see the Error situations section.

Update UAT

Clients that point to the UAT environment run applications (for example, Modern POS) for release 10.0.7 (the same version that is currently running in the production environment).

1. Upload, update, and deploy the store components. For more information, see the Upload, update, and deploy store components section.
2. Sign off on functional scenarios.
3. If you encounter regressions, see the Error situations section.

Update Prod

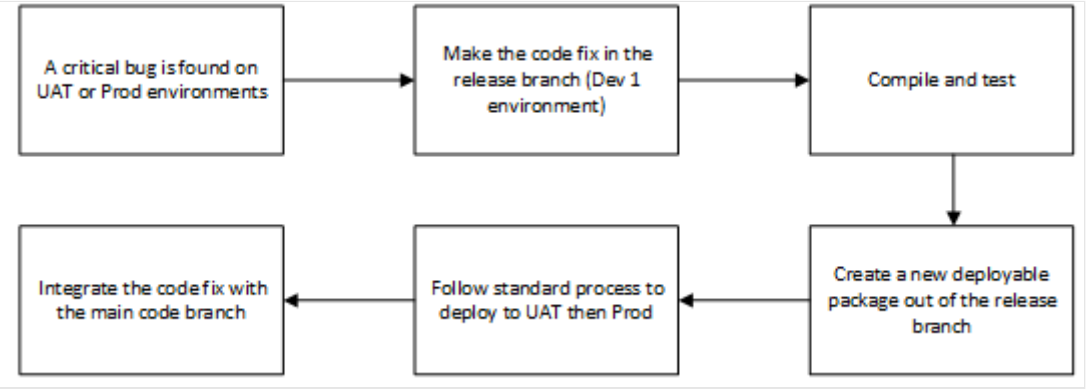
1. Upload, update, and deploy the store components.
2. Sign off.

Apply a new version of your custom code

This section describes the recommended flow for two use cases that require that you update your UAT and production environment with a new build of your custom extension.

Create a hotfix of your code

When you find a critical bug in the UAT or production environment, fix the bug in the release branch (not in the **main** or development branch). Then, use the standard process to apply a deployable package to UAT and production.

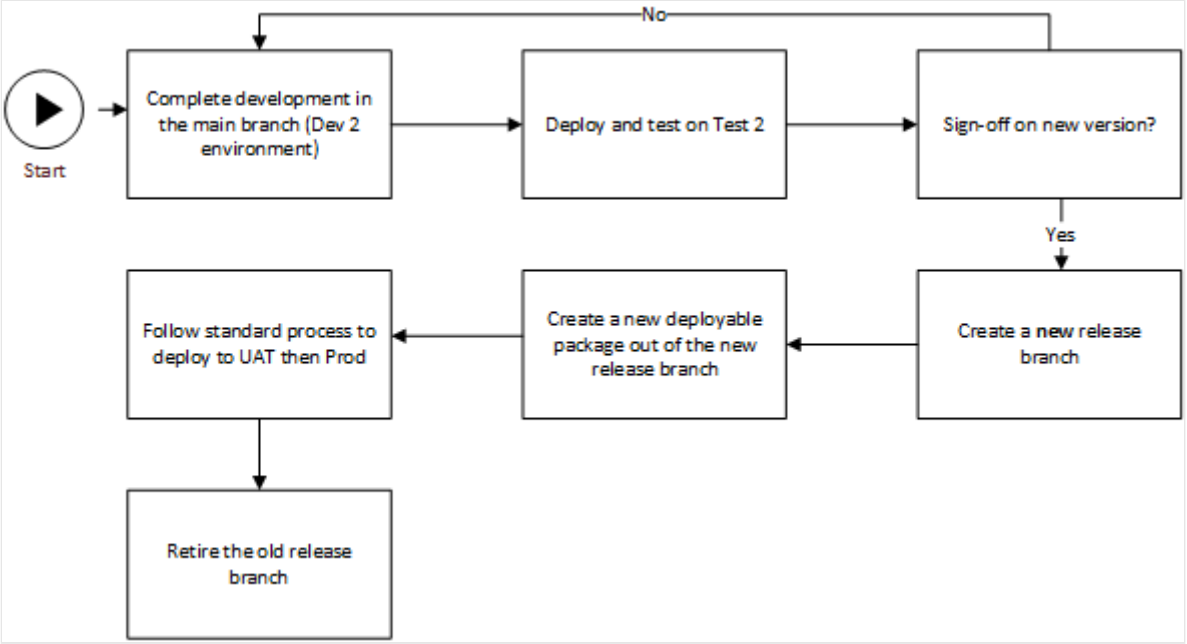


1. In Dev 1, make the code fix in the release branch. If the required fix is in ISV code, ask the ISV to send you a new build of the current release, not the next version of the solution.
2. Compile and test.
3. Create a new deployable package from the release branch.

4. Deploy on Test 1, and sign off if sign-off is required.
5. Use the standard process to deploy custom code first to UAT and then to production.
6. Integrate the code fix with your **main** code branch.

Update your custom code from release N to release N+1

When you're ready to release the next version of your custom code, use the following process to create and deploy the new release.



1. In the Dev 2 environment, or another environment that is connected to the **main** branch, complete development in the **main** branch.
2. Deploy and test on Test 2.
3. Repeat steps 1 through 2 until you get sign-off on the new version.
4. Create a **new** release branch.
5. Create a new deployable package from the **new** release branch.
6. Use the standard process to deploy custom code first to UAT and then to production.
7. Retire the old release branch.

Upload, update, and deploy store components

1. Upload the retail deployable package to LCS.

2. Update Application Object Server (AOS) with the latest client.

3. Upload the retail deployable package to your Asset library:

    - **Older deployments (earlier than version 10.0.11):** Software deployable package
    - **New deployments (version 10.0.11 and later):** Retail self-service installers

4. Update Commerce headquarters with the new self-service installers for store components:

    - **Older deployments (earlier than version 10.0.11):** Deploy the retail deployable package to Commerce headquarters via LCS.
    - **New deployments (version 10.0.11 and later):** In Commerce headquarters, on the **Commerce parameters** form, on the **Channel deployment** tab, select **Check for package updates**. This update brings in the packages that are available on the **Retail Self-service package** tab in the LCS Asset library.

5. Assign the client versions to your devices.

6. Download the installers for the desired client type and device.

7. Install in target device.

8. Test and validate.

Last updated on 03/18/2026

Software lifecycle policy and cloud releases

This article outlines the lifecycle and support policies for the finance and operations online service.

Modern Lifecycle Policy

The finance and operations online service is covered by the Modern Lifecycle Policy. The Modern Lifecycle Policy covers products and services that Microsoft continuously services and supports. For more information about this policy, see Modern Lifecycle Policy. Licensed customers must stay current with updates to the finance and operations online service in accordance with the following servicing and system requirements:

- Customers who purchase subscriptions of finance and operations apps experience continuous updates that Microsoft manages. Customers can postpone one consecutive service update.
- Platform versions maintain backward compatibility with the application versions that the platform release supports within the application support lifecycle. For more information about platform versions, see What's new or changed in Platform updates.
- Critical fixes and noncritical updates are handled in the following way:
  - **Critical fixes** – Critical fixes include security fixes and any fixes that are required to adhere to the availability service level agreement (SLA) that the service supports. The latest service update provides critical fixes. In addition, to help protect customers and the online service, Microsoft might apply critical fixes directly to a customer's environment. If a critical fix must be applied, Microsoft notifies the customer about the required downtime window (if there's any downtime) and applies the fix to the applicable environment. The critical fix updates the system to the latest update version.
  - **Non-critical updates** – Service updates and quality updates provide noncritical updates.

> **Important**
>
> Microsoft doesn't investigate, troubleshoot, or provide any fixes to issues on versions that reached their end of service. For end-of-service dates by release, see Targeted release schedule (dates subject to change).

The Microsoft Support team can't take cases that are related to versions that reached their end of service. You must first update to the latest service update and then apply the latest quality update. At that point, if the issue persists, you can report it. For more information, see What happens to an environment that's running a finance and operations app version that's no longer supported.

Microsoft operates all environments. All automatic processes that involve your environments, such as monitoring or self-healing, are as-is for supported versions.

Dates and versions for application and platform releases

For details about dates and versions for releases, see the What's new or changed in finance and operations apps home page.

Note

- Service updates are cumulative and might include updates for some or all of the following components: platform, application, Financial Reporting, Commerce, and operating system updates.
- You can pause, delay, or opt out of a service update by using the update settings in Microsoft Dynamics Lifecycle Services projects. However, this option isn't available if any of your sandbox and production environments are more than one version behind the latest available update. For more information, see Can the updates be delayed? What's the policy?

Downloadable virtual hard drive releases

For finance and operations apps, updates to the downloadable virtual hard drive (VHD) are released twice per year, concurrently with the major releases in April and October. Because the VHD is bundled with the latest service update, its release always follows the release of the associated service update for production use. You can expect the VHD update to be released after the generally available (self-update) date of the service update. For finance and operations release dates, see Targeted release schedule (dates subject to change).

Use of the VHDs is subject to the Software license terms.

Last updated on 06/26/2026

Software Lifecycle Policy for Microsoft Dynamics 365 Finance + Operations (on-premises)

08/20/2025

This article outlines the lifecycle and support policies for Microsoft Dynamics 365 Finance + Operations (on-premises) releases.

Modern Lifecycle Policy

Finance + Operations (on-premises) software is covered by the Modern Lifecycle Policy. The Modern Lifecycle Policy rules products and services that are serviced and supported continuously. For more information about this policy, see Modern Lifecycle Policy.

Licensed customers must regularly update their deployment of the Finance + Operations (on-premises) software. The servicing policy requires that you maintain the Software Assurance (SA) or the Enhancement Plan and it requires that you deploy the update releases provided through the continuous service updates. To check for the most up to date status on the lifecycle of the product, you can check the lifecycle page Dynamics 365 for Finance + Operations (on-premises).

> **Note**
>
> If you want to use the Fixed Support Lifecycle Policy (5+5), you must downgrade to Microsoft Dynamics AX 2012 R3. If you instead lapse on your SA or the Enhancement Plan, you'll only be eligible for the perpetual license rights to Microsoft Dynamics AX 2012 R3 and must uninstall Microsoft Dynamics 365 for Finance + Operations (on-premises).

On-premises software update policies

As a customer, you are in full control of your on-premises deployments and must actively follow this policy. You are in control of installing updates in your on-premises environments.

Microsoft supports your deployment of Finance + Operations (on-premises) software only if you keep the deployed software current according to this policy.

Critical fixes and noncritical updates are handled in the following way:

- **Critical fixes** – Critical fixes include security fixes and any fixes that are required to support reliability and availability. Critical fixes are made available in the latest platform update version.
- **Non-critical updates** – Customers must update to the most current Finance + Operations platform and financial reporter version to deploy noncritical updates.

Finance + Operations (on-premises) release dates

Continuous service updates

Since November 2018, on-premises service updates are released continuously. For more information about version numbers and availability dates, see Software lifecycle policy and cloud releases. For more information about One Version service updates, see One Version service updates FAQ.

Every service update release has a specific expiration date after which the release isn't entitled to receive support.

Continuous combined releases (Application and Platform)

With Continuous service updates, application and platform are released together. The One Version schedule specifies the dates for early access, public release, and expiration of the release.

Expand table

Product	Deployment options	Application version	Release or expiration dates
Finance + Operations	Base deployment + Service update	10.0.49	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.48	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.47	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.46	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.45	One Version schedule

Product	Deployment options	Application version	Release or expiration dates
Finance + Operations	Base deployment + Service update	10.0.44	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.43	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.42	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.41*	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.40*	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.39*	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.38*	One Version schedule
Finance + Operations	Service update	10.0.37*	One Version schedule
Finance + Operations	Service update	10.0.36*	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.35*	One Version schedule
Finance + Operations	Service update	10.0.34*	One Version schedule
Finance + Operations	Service update	10.0.33*	One Version schedule
Finance + Operations	Base deployment + Service update	10.0.32*	One Version schedule
Finance + Operations	Service update	10.0.31*	One Version schedule
Finance + Operations	Service update	10.0.30*	One Version schedule

* Expired version. All customers must be on the latest version of Finance + Operations according to the Modern Lifecycle Policy.

Older application releases

> **Note**
>
> Older platform releases passed the expiration date are no longer supported. Extension requests can't be logged since January 1, 2019.
>
> Customers who aren't following the Modern Lifecycle Policy and the Continuous service updates may not be entitled to receive support after the expiration date of the deployed application version. Expired versions are removed from Lifecycle Services at some point where after servicing and deployment of such versions from Lifecycle Services may not be possible.

Expand table

Product	Application version	Public availability	Expiration date
Finance + Operations (on-premises)	10.0 (10.0.8)	Mar 2019	Jun 2019
Finance + Operations (on-premises)	8.1 (8.1.136)	Nov 2018	Apr 2019
Finance + Operations, Enterprise edition (on-premises)	7.3 (7.3.11971)	Mar 2018	Apr 2020*
Finance + Operations, Enterprise edition (on-premises)	July 2017 (7.2.11792)	Jun 2017	Apr 2019

* All customers must be on the latest version of Finance + Operations. The deadline for this requirement was April 30, 2019. In the transition to Continuous service updates, we gave exemptions for customers who had unfulfilled extension requests that were submitted to Microsoft. Those customers could stay on version 7.3 until April 2020. For more information, see One Version service updates FAQ.

Older platform releases

> **Note**
>
> Older platform releases passed the expiration date are no longer supported. Extension requests can't be logged since January 1, 2019.

Expand table

Platform release	Build Number	Availability	Expiration date
Platform update 20	7.0.5030	November 2018	February 2019
Platform update 15	7.0.4841	June 2018	September 2018
Platform update 12	7.0.4709.41182	March 2018	June 2018
Platform update 11	7.0.4679.35176	October 2017	April 2018
Platform update 10	7.0.4641.16233	August 2017	April 2018
Platform update 9	7.0.4612.35162	August 2017	April 2018
Platform update 8	7.0.4565.1612	July 2017	April 2018

Service update availability

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft is committed to delivering predictable service updates. These service updates are made generally available for self-deployment before Microsoft automatically applies them. The timing of the package release for self-update relative to the production autoupdates varies. Customers can choose from two autoupdate windows that are four weeks apart for each service update. Organizations can select the update window that better accommodates their validation process and operational schedules. To determine the timing of self-update and autoupdates for upcoming releases, including the second autoupdate window see, Targeted release schedule (dates subject to change). To learn more about twice autoupdate window options, see One Version service updates FAQ.

Customers can take up to four service updates per year and are required to take a minimum of two per year. Customers can choose to pause one update at a time. A pause of a service update can apply to the designated user acceptance testing (UAT) sandbox environment, the production environment, or both environments. After the pause window ends, if the customer hasn't self-updated to a supported service update, Microsoft automatically applies the latest update, based on the configuration in Microsoft Dynamics Lifecycle Services. To learn more about how to pause service updates, see Pause service updates through Lifecycle Services.

> **Note**
>
> Service updates are provided four times annually. Autoupdates occur in February, April, July, and October.

Targeted release schedule (dates subject to change)

> **Important**

Microsoft releases four service updates annually, in February, April, July, and October. The maximum number of consecutive updates that can be paused is one and taking a minimum of two service updates per year is required. For answers to common questions about how these changes affect the release process, see One Version service updates FAQ.

> **Note**
>
> A sandbox autoupdate occurs seven days before the production update.

In the following table:

- An asterisk (*) in the "Release version" column denotes a major release.
- The "autoupdate schedule production start date" column refers to the autoupdate schedule that's configured in the update settings in Lifecycle Services. See Configure service updates through Lifecycle Services.
- The "End of service" column indicates the date when no new cumulative service updates are provided.

Expand table

Release version	Preview availability	Preview latest possible update	General availability (self-update)	First autoupdate schedule for production start date	Second autoupdate schedule for production start date	End of service
CY27Q2: 10.0.51*	January 25, 2027	February 15, 2027	March 12, 2027	April 2, 2027	April 30, 2027	November 19, 2027
CY27Q1: 10.0.50	October 23, 2026	November 17, 2026	December 22, 2026	January 31, 2027	February 28, 2027	August 20, 2027
CY26Q4: 10.0.49*	July 27, 2026	August 17, 2026	September 11, 2026	October 2, 2026	November 1, 2026	May 21, 2027
CY26Q3: 10.0.48	April 24, 2026	May 11, 2026	June 5, 2026	July 3, 2026	July 31, 2026	February 16, 2027
CY26Q2: 10.0.47*	January 26, 2026	February 16, 2026	March 13, 2026	April 3, 2026	May 1, 2026	November 20, 2026
CY26Q1: 10.0.46	October 24, 2025	November 17, 2025	December 26, 2025	February 1, 2026	March 1, 2026	August 21, 2026

Release version	Preview availability	Preview latest possible update	General availability (self-update)	First autoupdate schedule for production start date	Second autoupdate schedule for production start date	End of service
CY25Q4: 10.0.45*	July 28, 2025	August 8, 2025	September 12, 2025	October 3, 2025	October 31, 2025	May 22, 2026

> **Note**
>
> The Software lifecycle policy applies to customers who are enrolled in the First Release program and to the date when the service update is made generally available.
>
> ### Release naming convention as of the 10.0.38 release
>
> The first half of the release label refers to the calendar year and quarter when the auto update production start date is scheduled. The second part is the product version as it appears in Lifecycle Services. An asterisk (*) at the end of the label indicates a major release. For example: **CY25Q4: 10.0.45*** is product version 10.0.45 that's made available for autoupdate in the fourth quarter of 2025. It's a major update (the "October" release).
>
> Previews and preview updates are available as a deployable package in the Shared asset library in Lifecycle Services. For more information, see One Version service updates FAQ.

Service update overview

Service updates are continuous, touchless updates that provide new features and functionality. They eliminate the need to do expensive upgrades every few years. Because service updates maintain backward compatibility, there's no need to "merge your code." We recommend that you use tools such as the Regression suite automation tool (RSAT) for regression testing.

You're in control and manage how your organization receives these updates. For example, you can sign up for the First Release program so that your organization receives updates first. You can manually apply the updates to any of your environments (self-update). Alternatively, you can remain on the default release schedule and receive the autoupdates when you schedule them by using Lifecycle Services.

Service updates contain both application changes and platform changes that are critical to the service, including regulatory updates. For more information, see Service updates.

Release processes

The Dynamics 365 team designs, develops, and validates each new release. Extensive testing is done on various test topologies. A compatibility checker also runs tests to ensure backward compatibility.

All customers who take advantage of the preview have early access to the upcoming service update. The preview service update is used to validate customizations, learn about new features, and provide feedback to Microsoft. During the preview phase, customers must deploy the service update in a development/test environment. The preview release can't be used in production. After release, customers can download the package from the Shared Asset Library in Lifecycle Services. Customers must agree to the program terms at the time of installation. Sign up for access to preview packages is no longer required. One scheduled update to the preview build is included as standard with every release.

The First Release program is available to all finance and operations apps customers. There are two ways to participate:

- **Microsoft Managed Update**s: Fill out the First Release Program: Microsoft Dynamics 365 Finance and Operations applications form and receive a service update that Microsoft automatically deploys, first in a UAT sandbox and then in production after seven days. It isn't possible to pause these updates.
- **Customer Initiated Updates**: Customers can pull the First Release build into their UAT sandbox from Lifecycle Services for testing and into production, without autodeployment from Microsoft.

Typically, the First Release build is the same as the generally available (GA) build. However, if critical issues arise that can't be fixed with a hot fix, a new First Release build is declared, and customers are informed through various communication channels, including Viva Engage and Lifecycle Services. Microsoft restarts the sandbox and production environments and deploy any newly declared First Release build for customers that receive Microsoft managed updates. Customers that initiate updates should deploy newly declared First Release builds that are available on Lifecycle Services.

The service update is made generally available through the Action Center in Lifecycle Services. When the service update is available, customers can manually apply it to all environments, including production. If the service update isn't applied to the designated sandbox or production environment, Microsoft automatically applies it, based on the update settings for the Lifecycle Services project. To learn more, see Configure service updates through Lifecycle Services.

Last updated on 06/04/2026

Apply updates to cloud environments

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to use Microsoft Dynamics Lifecycle Services to automatically apply updates to cloud environments.

> **Important**
>
> Apply updates by using deployable packages. Applying updates causes system downtime. All relevant services stop, and you can't use your environments while the package is being applied. Plan accordingly.

Supported environments

All customer-managed and Microsoft-managed environments deployed through Lifecycle Services are supported. For more information about self-service environments, see Update an environment.

> **Note**
>
> If you have a build environment, you can only use Lifecycle Services to apply binary updates and data upgrade packages. You can't use Lifecycle Services to apply an application deployable package.

For other environments (listed in the following section), you must use Remote Desktop Protocol (RDP) to connect to the environment and install from the command line. For information about manual package deployment, see Install deployable packages from the command line.

- Local development environments (Downloadable virtual hard disk [VHD])
- Multibox dev/test environments in Microsoft Azure (Partner and trial projects)

Key concepts

Before you begin, you should understand *deployable packages*, *runbooks*, and the *AXInstaller*. A deployable package is a unit of deployment that you can apply in any environment. A deployable package can be a binary update to the platform or other runtime components, an updated application (AOT) package, or a new application (AOT) package. The AXInstaller creates a runbook that enables installing a package. Learn more in Packages, runbooks, and the AXUpdateInstaller in depth at the end of this article.

Supported package types

- **AOT deployable package** – A deployable package that's generated from application metadata and source code. You create this deployable package in a development or build environment.
- **Application and Platform Binary update package** – A deployable package that contains dynamic-link libraries (DLLs), other binaries, and metadata that the platform and application depend on. Microsoft releases this package. You can get it from the **All binary updates** tile in Lifecycle Services.
- **Platform update package** – A deployable package that contains DLLs, other binaries, and metadata that the platform depends on. Microsoft releases this package. You can get it from the **Platform binary updates** tile in Lifecycle Services.
- **Commerce deployable package** – A combination of various packages that are generated after the Commerce code is combined.
- **Merged package** – A package that you create by combining one package of each type. For example, you can merge one binary update package and one AOT package, or one AOT package and one Commerce deployable package. You merge the packages in the Asset library for the project in Lifecycle Services.

> **Note**
>
> You can't include a binary package and a Commerce deployable package in the same merged package.
>
> For information about how to download an update from Lifecycle Services and what you see in the tiles based on your environment version, see Download updates from Lifecycle

Services.

If your environment is on an application version 8.1 or later, the **Platform Update package** doesn't apply to your environment. Starting with 8.1 and later releases, **Application and Platform Binary update package** applies since application and platform are combined into a single cumulative package and is released by Microsoft. You don't apply granular X++ hotfixes and get all application and platform updates together. On the **Environment details** page, clicking on **View detailed version information** doesn't show details on the granular hotfixes or KBs applied as there's no way to apply them.

Prerequisite steps

- **Make sure that the package you want to apply is valid** - When you upload a package to the Asset library, the system doesn't analyze it. If you select the package, the package status appears in the right pane as **Not Validated**. A package must pass validation before you can apply it in an environment by using the following procedures. The Asset library updates the status of the package to indicate whether the package is valid. Validation helps ensure that production environments aren't affected by packages that don't meet the guidelines.

  There are three types of validations:
  - Basic package format validations
  - Platform version checks
  - Types of packages

- **Make sure that you apply the package in a sandbox environment before you apply it in the production environment.** To help ensure that the production environment is always in a good state, test the package in a sandbox environment before you apply it in the production environment. Therefore, before you request that the package be applied in your production environment, make sure that it is applied in your sandbox environment by using the automated flows.

- **If you want to apply multiple packages, create a merged package that you can apply first in a sandbox environment and then in the production environment.** Application of a single package in an average environment requires about five hours of downtime. To avoid more hours of downtime when you must apply multiple packages, create a single combined package that contains one package of each type. If you select a binary package and an application deployable package in the Asset library, a **Merge** button becomes

available on the toolbar. By selecting this button, you can merge the two packages into a single package and therefore reduce the total downtime by half.

- **Make sure that you apply the application binary update package to your dev/build environment AFTER you apply it to your sandbox and production environment** - If you apply the application binary package on your dev/build environment and this action raises the platform build version to be higher than your target sandbox or production environment, you're blocked from applying any AOT packages that are produced from this dev/build environment. To apply AOT packages produced from a dev/build environment, your dev/build instance must be equal to or lower than your target environments.

Apply a package to a nonproduction environment by using Lifecycle Services

> ⓘ **Note**
>
> For self-service type environments, see Update an environment.

Before you begin, verify that you uploaded the deployable package to the Asset library in Lifecycle Services.

1. For a binary update, upload the package directly to the Asset library. For information about how to download an update from Lifecycle Services, see Download updates from Lifecycle Services. For an application (AOT) deployable package that results from an X++ hotfix, or from application customizations and extensions, create the deployable package in your development or build environment, and then upload it to the Asset library.
2. Open the **Environment details** view for the environment where you want to apply the update.
3. Select **Maintain** > **Apply updates** to apply an update.
4. Select the package to apply. Use the filter at the top to find your package.
5. Select **Apply**. The status in the upper-right corner of the **Environment details** view changes from **Queued** to **In Progress**, and an **Environment updates** section now shows the progress of the package. You can refresh the page to check the status.
6. Continue to refresh the page to see the status updates for the package application request. When the package is applied, the environment status changes to **Deployed**, and the servicing status changes to **Completed**.

Apply a package to a production environment by using Lifecycle Services

In a production environment, schedule a downtime for when you want the update to be applied. For self-service type environments, see Update an environment.

> **Important**
>
> An important prerequisite for applying a package to a production environment is that you successfully apply the package to at least one sandbox environment in the same project.

Troubleshoot package deployment failures

If package deployment fails, see Troubleshoot package application issues.

Applying updates and extensions

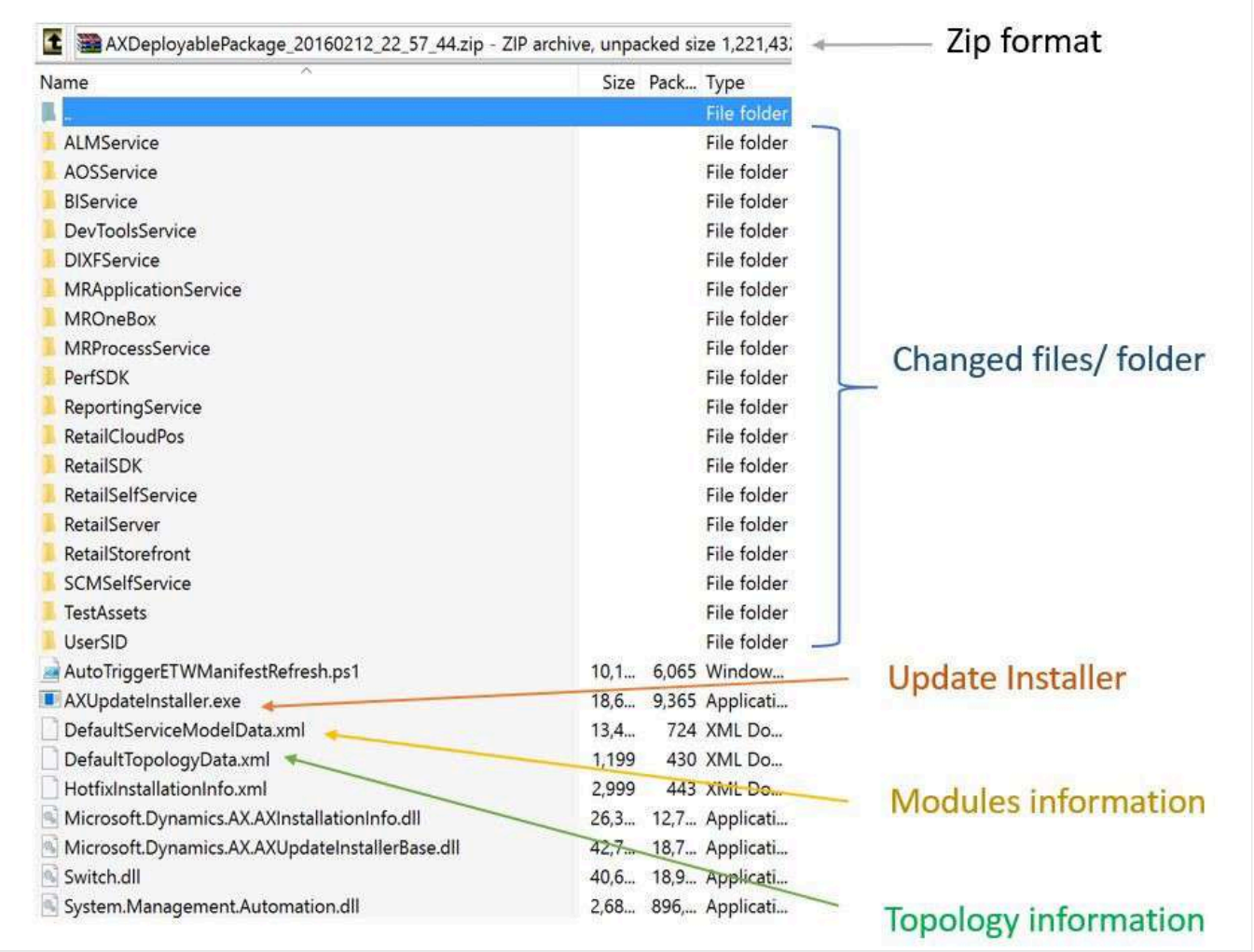
If you're updating a Tier-2 Sandbox or Production environment on application version 8.1.2.x or newer and you initialized Cloud Scale Unit, you also need to update Commerce channel components. For more information, see Update Retail Cloud Scale Unit.

If you're using components such as Modern POS, after you apply updates and extensions in your environment, you must also update your in-store components. For more information, see Configure, install, and activate Modern POS (MPOS).

Packages, runbooks, and the AXUpdateInstaller in depth

Use deployable packages, runbooks, and the AXUpdateInstaller to apply updates.

Deployable package – A deployable package is a unit of deployment that you can apply in an environment. A deployable package can be a binary update to the platform or other runtime components, an updated application (AOT) package, or a new application (AOT) package.



Runbook – The deployment runbook is a series of steps that the system generates to apply the deployable package to the target environment. Some steps are automated, and some steps are manual. AXUpdateInstaller lets you run these steps one at a time and in the correct order.

AXUpdateInstaller – When you create a customization package from Microsoft Visual Studio or a Microsoft binary update, the installer executable is bundled together with the deployable package. The installer generates the runbook for the specified topology. The installer can also run steps in order, according to the runbook for a specific topology.

More information

Install deployable packages from the command line

Last updated on 04/03/2026

Apply updates to on-premises deployments

This article explains how to apply supported updates to Dynamics 365 Finance + Operations (on-premises). All updates to on-premises environments are done through Microsoft Dynamics Lifecycle Services.

Search for and download updates

For more information about how to find the updates that you can apply to your on-premises environment, see Issue search in Lifecycle Services. For information about how to download updates from the tiles in the **Updates** section of the **Environment details** page in Lifecycle Services, see Download updates from Lifecycle Services.

Update an on-premises deployment

You can apply updates to an on-premises environment either during deployment or after the deployment is completed.

While an on-premises environment is being deployed, you can select to deploy a custom package in the **Advanced** settings. For more information about how to apply customizations or application X++ updates, see Develop and deploy custom models to on-premises environments.

> **Note**
>
> Merged packages, which include platform/application code and custom code, are restricted to use during quality updates only. They can't be utilized when updating to a newer service update. For instance, if you are updating from Dynamics 365 Finance + Operation (on-premises) version 10.0.41 to 10.0.42, the service update must be applied before using a merged package.

To apply updates to an on-premises environment after it's deployed, follow these steps:

1. In Lifecycle Services, open the **Environment details** page for the environment.
2. Under **Maintain**, select **Apply updates**.

> **Note**

The environment must also have the latest version of the local agent available in Lifecycle Services. For more information, see Update the local agent.

Apply platform update package updates through Lifecycle Services

You can use the following steps to apply platform update packages.

> **Important**
>
> The application of updates requires downtime for your environment. Therefore, no business transactions can be performed in the environment during the update. When you complete the following steps, verify that the system isn't being used, and that an official downtime notice has been communicated to all system users.

Prerequisites

- Before you begin, complete a full backup of the Management Reporter (MR), Microsoft Dynamics AX, and Microsoft SQL Server Reporting Services (SSRS) databases. Although the code is restored through Lifecycle Services, the database must be manually restored to help ensure that no data loss occurs.
- Update the local agent to the latest version. For more information, see Update the local agent.
- Depending on the type of update, complete the following steps to generate a deployable package:
  - **Platform update packages** – Download or save the update directly to the Asset library in Lifecycle Services by following the steps in Download updates from Lifecycle Services.
  - **Customizations** – Follow the steps in Develop and deploy custom models to on-premises environments.

Update a sandbox environment

1. In Lifecycle Services, open the on-premises implementation project, and then, on the project menu, select **Asset library**. In the Asset library, ensure that the platform update package or custom deployable package appears on the **Software deployable packages** tab.

2. In the on-premises implementation project, open the **Environment details** page of the environment to update.

3. Under **Maintain**, select **Apply updates**. A dialog box shows the updates that were uploaded to the Asset library. Note that only packages that are marked as **Valid** in the Asset library appear.

4. Select the update, and then select **Prepare** to prepare your on-premises environment for servicing.

   > ⓘ **Note**
   >
   > While the environment is being prepared for servicing, its state is **Deployed**. However, the **Deployment status** field shows the progress of the preparation. During preparation, actions such formatting the package and downloading the package are performed.
   >
   > Because the environment isn't directly touched during preparation, there is no downtime at this point. Users can continue to use the system during preparation.

5. After preparation is completed, **Abort** and **Update Environment** buttons appear. To start to apply the update, select **Update environment**. If preparation fails, see the Resolve a failed update application section later in this article.

6. In the confirmation message that appears, select **Yes**. The servicing operation now begins on the environment. This point is the start of the downtime on your environment.

   The environment state is changed from **Deployed** to **Deploying**.

   After the update is completed, the environment state is changed back to **Deployed**. If application of the update fails, the environment state is changed to **Failed**. For information about what to do if package application fails, see the Resolve a failed update application section later in this article.

   > ⓘ **Note**
   >
   > During the update, the database is synchronized to update any changes to table definitions. This action can take some time for larger databases. Additionally, all reports are deployed again to the SSRS nodes.

7. Open the **History** and **Environment details** pages to view the operations that were performed on the environment. You can view a record of major actions that were performed on the environment, such as deployments, servicing, and rollbacks.

Update a production environment

Before you update a production environment, you must successfully complete the package application update on a sandbox environment.

1. In the project for the sandbox environment that you applied the package to, open the Asset library, and then, on the **Software deployable packages** tab, select the package, and mark it as a **Release candidate**.
2. On the **Environment details** page, under **Maintain**, select **Apply updates**. The dialog box that appears shows only packages that are marked **Release candidate**.
3. Select the **Release candidate** package to apply to the production environment.

The rest of the update flow is the same as the flow for a sandbox environment.

Resolve a failed update application

When preparation fails, the environment state is **Deployed**. When the application of an update fails, the environment state is **Failed**. The first step is to determine why there is a failure. The location of the logs varies, depending on the stage where the failure occurred:

- **Preparation stage** – If the operation fails during the **Preparation** stage, the logs are uploaded to Lifecycle Services. In the log files, select **Download logs** to download the log files. If the package has any merge issues, the error is included in the log file.
- **Deploying stage** – If the operation fails during the **Deploying** stage, the logs are located in the on-premises environment. Sign in to the environment, and then access the logs and event viewer.

For more information about how to use the troubleshooting logs, see Troubleshoot on-premises deployments.

After you review the logs and determine the cause of the failure, complete one of the following operations to restore the environment to a healthy state. No actions can be performed on an environment that is in a **Failed** state. The environment must first be restored to a healthy state.

- **Retry failed operation** – If update application fails, select **Retry** to recover from the failed operation.

- **Abort failed operation** – Because there is no change made to the on-premises environment, if the preparation fails, you have the option to cancel the operation. Select **Abort** to cancel the preparation.

- **Roll back the update** – To roll back the update that failed, select **Rollback**. Before you start the rollback, you must restore the database to the last known good state. When you select **Rollback**, the environment is restored to the last known good state. The environment state is then changed to **Preparation**, then to **Deploying**, and then to either **Deployed** or **Failed**.

  > ⓘ Note
  >
  > The **Rollback** button doesn't roll back the database. You're responsible for restoring the database to the last known backup that was made before update application. This step is critical to help guarantee that there is no data loss.

- **Refresh the state** – If update application fails during the **Preparation** stage, the failure is on the Lifecycle Services side, and update application hasn't yet started. Therefore, the on-premises environment is in a good state. To restore the Lifecycle Services environment state to **Deployed**, on the project dashboard page, select **Refresh**.

- **Delete and redeploy an environment** – If the retry and rollback options don't work, you must delete and redeploy the environment. To delete the environment, on the project dashboard page, select **Delete**. You then see the option to configure the environment.

  > ⓘ Important
  >
  > This option should **not** be used on a production environment. However, it can be used on a sandbox deployment to restore the environment to a healthy state.
  >
  > Because this option requires that you do a fresh deployment of the environment, you lose any updates that were previously applied. Any customizations and binary updates must be reapplied to the environment.

Last updated on 04/13/2026

Configure service updates through Lifecycle Services

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

In Microsoft Dynamics Lifecycle Services, you can specify how and when you receive service updates from Microsoft for your environments.

> **Important**
>
> This feature is available only to customers who are using **version 8.1 and later** or are using **version 7.3**, and who are **not** part of the First release program. Microsoft is working to make the feature available to First release customers. For customers who are on versions 7.1, 7.2, or 8.0, you can take the update manually by using the regular servicing flows.

Only users (customers or partners) assigned to the **Project owner** role in Lifecycle Services can configure updates. Additionally, you can configure updates only for **implementation projects**.

Follow these steps to change your update settings.

> **Note**
>
> These settings apply only to service updates. They have no effect on the operating system–level security updates that are applied to your environments every month.

1. In Lifecycle Services, in your implementation project, open the **Project settings** page.

   This page has a new tab named **Update settings**.

2. On the **Update settings** tab, set the following configuration options as you require:

   - **Default Sandbox Update environment** – Select an alternate sandbox environment that should be updated before the production update.

     By default, Microsoft first updates the Tier 2 Standard Acceptance Test (sandbox) environment that is included in the base offer. It then applies the update to the production environment. If you purchased Tier 2 and higher sandbox add-on environments and want a different sandbox to be updated, use this field to change the default environment to an alternate environment.

     > **Important**
     >
     > The environment that you select here's updated seven calendar days before the update cadence that you select for the production environment.

   - **Production environment update cadence** – Select a recurring cadence for updates to your production environment. The sandbox environment that you select in the **Default Sandbox Update environment** field is updated seven calendar days before the selected cadence. The remaining additional sandboxes are updated on the same cadence as the production environment. The following options are available:
     - **Select the cadence** – Select whether to receive updates in the first, second, or third week of the month.
     - **Select one of the three time-zones** – Select the time zone that the production environment should be updated in: Eastern Time (UTC – 5), Hong Kong Time (UTC + 8), or Greenwich Mean Time (UTC + 0).
     - **Select a day of the week:** Select the day in the week when you want to receive updates.
     - **Select a time slot:** Select the time slot when you want to receive updates.

     > **Important**
     >
     > Currently, only a few options are available for the day of the week and time slot options. Microsoft adds more options soon, such as weekdays for customers.

     > If the above time slots don't meet your needs, you always have the option to do a self-update at a time that is convenient to you by taking the update and

> applying it to your environments by using the regular servicing flows.

3. When you finish setting the configuration options, select **Save**.

After you set the update environment and update cadence, Microsoft generates an update calendar for the next six months. This calendar shows exactly when the configured sandbox and production environments are updated. Therefore, you know when to expect each update. To view the calendar, select **View the update calendar**.

> **Important**
>
> After you save the settings, you can change them at any time. However, if there's an ongoing rollout, the new settings aren't used to update the existing rollout timings. Instead, they start to be used in the next rollout. An ongoing rollout is defined by the 14-day period between the date when the email notification about the update of the sandbox environment is sent and the date when the production environment is updated.

For more information about how to pause updates to configured sandbox and production environments, see Pause service updates through Lifecycle Services.

For more information about One Version and Microsoft-managed service updates, see One Version service updates FAQ.

Canceled updates

A scheduled update can be canceled for various reasons. Here are some common reasons that cause Microsoft to cancel a scheduled update.

- An error is found during update preparation. The update preparation process starts approximately four hours before the update to ensure that the environment is in a healthy state. If the environment is in a failed state or maintenance mode, the scheduled update is canceled before it starts.
- An error is found while updating the environment. If there are problems during the update, the scheduled update is canceled and the environment rolls back to the previous state.
- The environment is already running on the latest version. There's no need to apply the update again, so the scheduled update is canceled before it starts.
- The target environment isn't found. If the designated sandbox is deleted or the production environment isn't deployed, the scheduled update is canceled before it starts.
- You're enrolled in the First Release program. The First Release program has a different release cadence, so previously scheduled updates are canceled.

You can find the canceled updates through the **View recent canceled updates** in the update settings. It shows all canceled updates, if any, within the last two scheduled updates.

Additional sandbox environments

In addition to the default sandbox environment and the production environment, if you deploy additional sandbox environments in your Lifecycle Services implementation project, the One Version service update also auto-updates the additional sandbox environments.

Update additional sandbox environments

You update the additional sandbox environments on the same cadence as your production environment, based on your update settings. If you have additional sandbox environments, you schedule them for update together.

If you don't deploy the production environment, none of the additional sandbox environments are auto-updated.

If you update the production environment **before** you send the email about the production update, the production environment and all additional sandbox environments **aren't** updated. If you update the production environment **after** you send the email about the production update, the production environment update **is** canceled, but additional sandbox environments are updated.

If there's any update failure on the production environment or any of the additional sandbox environments, it doesn't interfere with the remaining updates. That is, if the production update fails, the additional sandbox update continues. Similarly, if you have additional sandbox environments and one of them fails during the update, the others continue.

You can self-update any of your environments prior to the scheduled service update. If you self-update to the latest version on an environment, the scheduled service update doesn't apply an update on that environment again.

Change the default sandbox environment

If you already have selected a default sandbox environment, and want to change it with one of the additional sandbox environments, the change will take effect in the next scheduled service update.

Pause service updates

Additional sandbox environments can't be paused independently. However, if you pause the default sandbox environment, then both the production environment and additional sandbox environments will be paused automatically. If you pause the production environment, then additional sandbox environments will also be paused.

The current pause service updates policy still applies.

Email notifications

If your additional sandbox environments are scheduled for an update, you'll receive reminder email notifications 6 days prior to the scheduled service update. After the update completes, you'll also receive email notification regarding the update result for each environment.

Last updated on 03/06/2026

Preservicing and post-servicing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

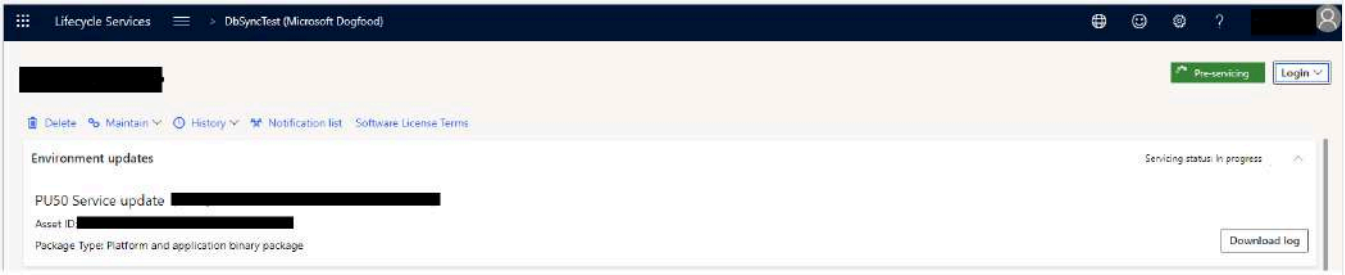
This article explains new changes that Microsoft introduced for the servicing process.

Microsoft introduced new steps for the servicing pipeline preservicing and post-servicing to enable database synchronization precheck and index creation on tables to occur outside offline servicing. Therefore, the environment remains up and running, and accessible so that users can perform regular activities. However, servicing can't be done, because all servicing operations are restricted during preservicing and post-servicing. The main benefit of the change is it helps reduce the overall servicing downtime.

Preservicing

For service updates, the preservicing step runs when you trigger the servicing operation. During this step, Microsoft Dynamics Lifecycle Services shows the environment state as **Pre-servicing**. This status indicates that the environment is online and accessible, but no other service operations are allowed.

The preservicing step identifies the predefined set of errors that cause servicing to fail. Precheck failure immediately triggers a soft rollback, without point-in-time recovery (PITR), and the environment is brought back to a deployed state. By fetching precheck errors from the environment history logs, you can apply the required mitigation before you retrigger the update.

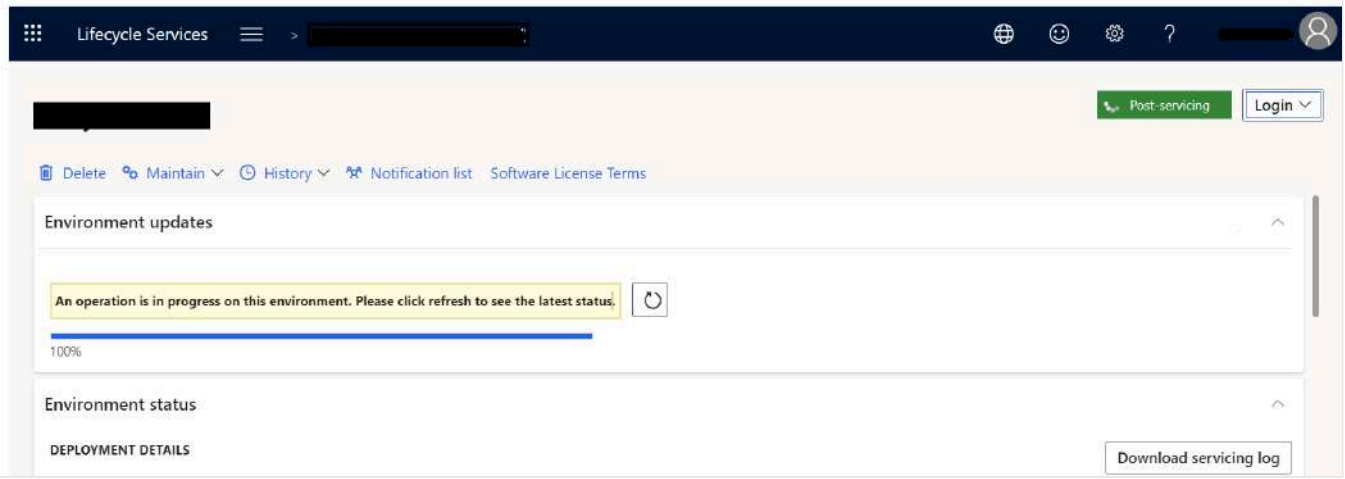


When pre-servicing succeeds, the environment state is updated to **Servicing**. During the servicing step, the system goes offline and remains inaccessible until the environment state is

updated to **Post-servicing**. After the servicing step is completed, email notifications are sent to environment admins.

Post-servicing

While the post-processing step is occurring, after offline servicing is completed, Lifecycle Services shows the environment state as **Post-servicing**. During this time, the index creation and modification that weren't done during offline servicing steps are done in online mode. The environment remains accessible so that users can perform regular activities. However, performance might be degraded for the changes that are involved in the update package. During post-servicing, users can't cancel existing service requests or trigger new service requests. There's an option to download the logs to check the process.



If any failure occurs during the post-servicing step, Lifecycle Services shows the environment state as **Post-servicing failed**. During this time, the environment remains accessible so that users can perform regular activities. However, performance might be degraded for the specific table where the index change failed. If the issue isn't resolved in 24 hours, contact Microsoft Support.

Recommended best practices for reduced servicing downtime

Restrict table schema changes to support faster schema synchronization. Eventually, this synchronization can be done online. All the schema change operations that are described in this section either require that an index be dropped and recreated, or require a considerable amount of downtime and therefore a longer servicing window.

Field property changes

- You change the **Scale** property for a field of the **Real number** type.
- You change the **Size** property from **(Memo)** to a fixed size *n*, or from a fixed size *n* to **(Memo)**, for a field of the **String** type.

Index-related changes

- You change the **Primary Index** property of a table.
- You change the **Clustered Index** property of a table.
- You add or remove fields from the columnstore index.
- You add or remove fields from the primary index.

Common failures

This section provides troubleshooting and mitigation guidance for the most common issues faced during preservicing or that are caused due to custom integrations. Review the logs of a service action from the environment's Lifecycle Services portal. Go to the environment history and select target service action to download the logs. After you download the logs, in zip format containing multiple files, locate the file that starts with "dbsyncadhoc_precheck_error". This file provides precheck failure logs. If there are multiple files with the same name, select any one for troubleshooting.

Precheck servicing can fail for various reasons. The following list describes common issues. If you see any of the following errors in the error file, the servicing deployment doesn't succeed unless you fix either the package or database issues. See the following list for the most common reasons for failures and recommended actions to perform before retrying servicing.

- Error message contains **"Database execution failed: The CREATE UNIQUE INDEX statement terminated because a duplicate key was found for the object name 'Table name' and the index name 'Index name' ".** This error occurs when a user attempts to create a unique index, but more than one row in the table contains duplicate values. If there are duplicates, the Dbsync fails and the error logs contain the details about the failed index.

Recommended action: Connect to the environment's database axDB and remove duplicate values in the unique index columns from [table name] found in error logs and the retry.

- Error message contains **"Package deployment failed with error like 'Converting Field XXX of Type 'AxTableFieldXXX' to 'AxTableFieldXXX' is not supported. Drop the original field, sync the table and add new field with same name if needed'**. This error occurs when a user creates a table and changes the column type to one that isn't supported.

Recommended action: Specific field type conversion isn't supported and the table in question is a customer created table. Fix the issue by converting the field type to the original type and then retry the servicing.

- Error message contains **"The enum value 'XXX' on the extension 'YYY.ZZZ' already exists. This extension can't be applied to the base element."** This error occurs if an ISV or a user adds an enumeration value to an extensible enum, and then a subsequent addition to the same extensible enumeration with the same name from the Microsoft-shipped application collides with it. The naming guidelines require that ISV and customer extensions are prefixed in order to avoid this type of collision. For more information, see Naming guidelines for extensions.

Recommended action: If this error occurs, rename the customer or ISV enumeration value to follow naming guidelines and avoid collision with the new application value.

FAQ

What are the most common errors that cause pre-servicing to fail?

The following list shows some common errors that cause pre-servicing to fail. Review the environment history logs before you retrigger the update.

- The `CREATE UNIQUE INDEX` statement terminated because of a duplicate key.
- Drop the original field, sync the table.
- Extension can't be applied to the base element.
- Missing dependencies.

Last updated on 11/26/2025

Pause service updates through Lifecycle Services

This article explains how to pause updates to your sandbox and production cloud environments by using Microsoft Dynamics Lifecycle Services. This article doesn't apply to on-premises environments.

Microsoft updates your configured sandbox and production environments to the latest service update that Microsoft releases. Microsoft notifies you about upcoming updates to your environments via email and through notifications in Lifecycle Services. At that point, if you can't proceed with the update for some reason, you can pause it through Lifecycle Services.

> **Important**
>
> As of February 19, 2024, the maximum number of consecutive updates that you can pause is reduced from three to one. However, because the service update cadence is also reduced, from seven to four releases, the same minimum of two service updates per year is maintained. For more information, see One Version service updates FAQ. For information about how to change the configured sandbox environment and set the production update cadence, see Configure service updates through Lifecycle Services.
>
> Microsoft is giving customers more flexibility in their update schedules. Beginning with the 10.0.39 release, customers can choose from two auto-update windows that are four weeks apart for each service update. Organizations can select the update window that better accommodates their validation process and operational schedules. For more information about how this change affects pause policy, see the **Who can pause service updates** section later in this article.

Who can pause service updates?

Only users (customers or partners) assigned the **project owner** role in Lifecycle Services can pause updates. You can pause updates only for **implementation projects**.

Staying current with service updates helps guarantee that customers always run on the latest set of fixes that Microsoft releases, so they have the best service experience. Therefore, Microsoft doesn't allow updates to be paused indefinitely.

Starting February 19, 2024, the maximum number of consecutive updates that you can pause is reduced from three to one. Therefore, customers must take a minimum of two service updates annually.

You can't use Lifecycle Services to pause a release if any sandbox and production environments in your project are more than two updates behind the latest release. This rule applies even in the case of two autoupdate windows now available with 10.0.39 release. For example, if Microsoft's latest released version is 10.0.39, customers with version 10.0.38 can pause one or both autoupdate windows to completely pause the 10.0.39 release. Customers with version 10.0.37 can pause only the first autoupdate window and must take the 10.0.39 release in the second update window. If environments are two releases behind the latest release (older than 10.0.37 in this example), none of the autoupdate windows are applicable as no pause is allowed.

If any sandbox or production environment is too far behind, you receive the following error message if you try to pause updates in Lifecycle Services:

> One or more sandbox and/or production environment isn't compliant with Microsoft's service update policy. All your environments need to be compliant before you can pause updates.

> **ⓘ Important**
>
> To pause a release, customers must pause both autoupdate windows if they're eligible to do so, based on the pause policy. Pausing the first autoupdate window doesn't automatically pause the second window.

What updates can I pause?

If you decide to pause updates, you have these options:

- Pause updates only to your production environment.
- Pause updates to both your sandbox environment and your production environment.
- Pause updates to other sandbox environments by pausing updates to your production environment.

> **ⓘ Note**
>
> If you pause updates to the production environment, all updates to other sandbox environments are paused too.

You can pause a maximum of one consecutive update at a time. For example, if you're using version 10.0.39, you can pause the update to version 10.0.40. However, you can't pause the update to version 10.0.41. You must take at least two updates per year.

> **Important**
>
> There's no way to pause more than one consecutive service update by using Lifecycle Services. Microsoft grants additional pauses beyond the standard pause allowance only when it confirms an update-blocking Microsoft product bug or regression and no viable workaround exists.
>
> Customers are expected to test service updates in sandbox environments as early as possible and report potential Microsoft issues immediately to support. Microsoft must have sufficient time to investigate and validate the issue before an exception can be considered. Requests submitted too close to a scheduled or mandatory update date are rejected if there's insufficient time to complete the investigation.
>
> An open support request alone doesn't qualify for an additional pause. If Microsoft determines that a viable workaround is available, the update isn't paused.
>
> Issues in third-party solutions, ISV applications, partner customizations, or other non-Microsoft components aren't eligible for additional pause approvals. Customers are responsible for working with the applicable solution provider to obtain a fix and validate compatibility before the scheduled update date.
>
> If you pause updates to a sandbox environment, updates are automatically paused for the corresponding production environment because Microsoft always updates configured sandbox environments before production environments.

Can I pause updates to my additional sandbox environments only?

No, you can't pause updates to additional sandbox environments only.

What if the update to the default sandbox environment is paused?

If you pause the update to the default sandbox environment, the updates to the production environment and all additional sandbox environments are also paused.

How do I pause updates?

To pause updates, follow these steps:

1. In Lifecycle Services, in your implementation project, open the **Project settings** page.

   This page has a new tab named **Update settings**.

2. In the **Pause Updates** section, on the **Update settings** tab, select **Pause upcoming update**.

3. In the dialog box that appears, select whether you want to pause updates to your production environment only, or to both your sandbox environment and your production environment.

4. Select **Next**.

5. Select your reason for pausing updates. If you select **Issue found during validations**, enter a valid support ticket number. You can add any other details that help Microsoft understand why you want to pause updates.

6. When you finish, select **Confirm**.

You can also edit an existing pause to cancel it so that updates resume. To edit a pause, select **Pause upcoming update**. The maximum of one update pause still applies. Therefore, you can't edit the pause to extend its duration past one update.

Whenever you pause an update or edit an existing pause, a notification appears at the top of the **Update settings** tab. This notification shows what version was paused. All stakeholders (the project owner and environment manager) also receive an email to inform them that a service update for the selected environments was paused. If someone cancels an existing pause and resumes updates, the notification disappears, and an email is sent to inform the stakeholders the paused update resumed.

> **Important**
>
> You can pause an update through Lifecycle Services up to four hours before the start of the downtime window.

You can cancel a pause and resume updates up to seven days before the start of the downtime date. If there are fewer than seven days before your downtime starts, you can't cancel a pause.

What happens after the pause duration expires?

Cumulative service updates help guarantee that customers always run on the latest set of fixes that Microsoft releases, so they get the best service experience. Therefore, you can't pause updates indefinitely.

You can cancel a pause and resume updates in two ways:

- Someone manually cancels a pause, as explained in the previous section.
- The duration that you set for the pause expires, and updates to the configured environments automatically resume.

In both cases, the system sends an email to inform the stakeholders.

For more information about service updates, see One Version service updates FAQ.

Last updated on 08/04/2026

Get notified about service updates through Lifecycle Services

This article explains how you can stay up to date about service updates from Microsoft.

Microsoft uses service updates to update your configured sandbox and production environments to the latest released version. Microsoft notifies you about upcoming updates to your environments by email and through notifications in Microsoft Dynamics Lifecycle Services.

Here are the different types of notifications that you receive:

- **Notification when an update is available:** When a new release is available, Microsoft surfaces a notification in your implementation projects' action center. You can then save that update in your projects' asset library if you want to apply the update to your environments before Microsoft does an automatic update. When Microsoft does an automatic update, it also saves a copy of the update in your projects' asset library.
- **Notification that is sent five days before the update:** Microsoft notifies you five days before it updates your environment. After you configure your update cadence, you receive notifications about upcoming updates five days before they occur. These notifications take three forms:
    - **Email notification:** Project owners, environment managers, and users who are listed as additional stakeholders for an environment are notified by email about the upcoming update.
    - **Notification bar on the environment details page:** A notification that appears on the environment details page in Lifecycle Services informs the customer about the upcoming update.
    - **Upcoming update reflects the update:** On the environment details page in Lifecycle Services, select **Maintain** > **Upcoming Update** to open a dialog box that contains details about the upcoming update.

      > **ⓘ Important**
      >
      > Notifications for Product quality updates (PQU) aren't delivered this way. For more information on PQUs, see:

[Proactive quality updates overview]

[Release schedule for proactive quality updates]

- **Notification that is sent one hour before the update:** One hour before the start of the downtime window, users in the application receive a notification. This notification asks users to save their work, because the environment is taken down for an update.
- **Notification that is sent when the update is completed:** After Microsoft finishes updating your configured environment, it notifies you by email about the outcome of the update. This email is always sent, regardless of whether the update was successfully applied. It's sent to project owners, environment managers, and users who are listed as the additional stakeholders for the environment. If Microsoft can't start the update for some reason, the email includes a reason to explain why the update wasn't started.

After you receive a notification, if you can't proceed with the update for some reason, you can pause it. For more information about how to pause updates to configured sandbox and production environments, see Pause service updates through Lifecycle Services.

For more information about One Version and Microsoft-managed service updates, see One Version service updates FAQ.

Last updated on 06/26/2026

Proactive quality updates overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about proactive quality updates (PQUs). It explains what they are and why they're used, and describes some of their benefits.

What are PQUs?

PQUs are cumulative builds of hotfixes that Microsoft delivers with near-zero downtime. PQUs follow a push model, where Microsoft applies updates to a Microsoft Dynamics Lifecycle Services environment in the background with minimal impact on customers. Microsoft deploys every PQU region by region, following a "safe deployment process" that tracks problems found within each region during deployment. The safe deployment process helps identify and fix problems before the PQU is deployed to more regions. PQUs are fully automated and contain important bug fixes and high-priority product features that are ready after the service update is generally available.

Why is Microsoft introducing PQUs?

Before the introduction of PQUs, most customers didn't receive fixes for service problems until they deployed their next service update. Therefore, businesses could be unnecessarily affected by service problems that a fix was already available for. PQUs get fixes to customers more quickly.

What investments is Microsoft making to enable safe deployments of PQUs?

Microsoft invested in the following technologies and processes to make the PQU process nondisruptive:

- **Higher-quality concise payloads** – The changes that PQU payloads contain are limited to bug fixes and regulatory changes that flighting supports. Microsoft applies extra diligence

when backporting changes into a PQU, to further reduce the risk of unintended regressions.

- **Safe deployment rollout process** – Deployment of PQUs follows a safe deployment process. Microsoft first delivers each PQU to a single region that contains a small group of customers who have the highest tolerance for risk. Then, if no regressions are identified, the process continues through a broader group of customers, based on geographies (stations), until all customers are using the new version. Microsoft uses this deployment model across its online services to help build confidence in an update as it's delivered to more customers, and to help minimize the impact if unforeseen problems are detected.
- **Fallback via flighting** – Flighting is used to enable or disable changes that PQUs introduce. If a change must be turned off after a PQU deployment, the flight system can turn it off to help reduce the impact.

Additional resources

- Release schedule for proactive quality updates
- Proactive quality updates FAQ

Last updated on 03/28/2026

Release schedule for proactive quality updates

This article provides the release schedule for proactive quality updates (PQUs).

The detailed schedule for each PQU train and its corresponding build app version number is published five days before the start of the PQU train. Five days before the start of release train, Microsoft sends a notification to customers who use the environments that receive the update.

Station 1 is the first release station. It consists of customers who opt in to have their sandbox environments receive the update before all other stations. More details are published after this functionality is made generally available in May 2023.

To determine when your Microsoft Dynamics Lifecycle Service environment is updated, find the region and the station-to-region mapping. Cross-reference the region and station-to-region mapping information with the upcoming PQU train for a service update.

For information about the maintenance windows for each region, see What are the planned maintenance windows by region?

Station-to-region mapping

Expand table

Station	Regions
Station 1	Only for opted-in environments
Station 2	Canada Central, Canada East, France Central, India Central, Norway East, Switzerland West
Station 3	France South, India South, Norway West, Switzerland North, South Africa North, Australia East, UK South, UAE North, Japan East, Australia South East, South East Asia
Station 4	East Asia, UK West, Japan West, Brazil South, North Europe, East US, UAE Central
Station 5	West Europe, Central US, West US
Station	DoD, Government Community Cloud, China

Station	Regions
6	

> **Important**
>
> 1. The PQU build for every train is typically published to Lifecycle Services after the change cutoff date that shows in the following high-level PQU train schedule. In this way, customers can download and self-apply the build as needed, to frontload any testing requirements.
> 2. Standard performance test (tier-4) sandbox environments, Premium performance test (tier-5) sandbox environments, and production environments receive PQUs on weekends. If a PQU of production environment, Standard performance test (tier4) sandbox, or Premium performance test (tier5) sandbox didn't complete on the scheduled weekend then the operation gets rescheduled to the next weekend.
> 3. Standard performance test (tier-4) sandbox environments, Premium performance test (tier-5) sandbox environments opted-in for the First release program (station 1) receive updates on the following weekend of the station 1 schedule.

High-level PQU train schedule

The following table shows the high-level train schedule. It includes a description of the service update that pertains to the train, the cutoff date after which new changes are no longer accepted, and the development duration for the PQU train.

Expand table

PQU release train	Change cutoff date	PQU train duration	Status	Application Version	Platform Version
10.0.46 PQU-1	February 4, 2026	February 9, 2025 to March 14, 2026	Completed	10.0.2428.95	7.0.7778.45
10.0.46 PQU-2	March 4, 2026	March 9, 2026 to April 11, 2026	Completed	10.0.2428.114	7.0.7778.56
10.0.46 PQU-3	April 1, 2026	April 6, 2026 to May 9, 2026	Completed	10.0.2428.139	7.0.7778.76
10.0.46 PQU-4	April 29, 2026	May 4, 2026 to June 6, 2026	Completed	10.0.2428.169	7.0.7778.104

PQU release train	Change cutoff date	PQU train duration	Status	Application Version	Platform Version
10.0.46 PQU-5	May 27, 2026	June 1, 2026 to July 5, 2026	Completed	10.0.2428.188	7.0.7778.118
10.0.46 PQU-6	June 24, 2026	June 29, 2026 to August 1, 2026	Completed	10.0.2428.205	7.0.7778.137
10.0.46 PQU-7	July 22, 2026	July 27, 2026 to August 29, 2026	Completed	10.0.2428.220	7.0.7778.145
10.0.46 PQU-8	August 21, 2026	August 31, 2026 to October 3, 2026	In-Progress	10.0.2428.236	7.0.7778.149
10.0.47 PQU-1	April 8, 2026	April 8, 2026 to May 2, 2026	Completed	10.0.2527.78	7.0.7858.54
10.0.47 PQU-2	April 22, 2026	April 22, 2026 to May 16, 2026	Canceled	-	-
10.0.47 PQU-3	May 6, 2026	May 6, 2026 to May 30, 2026	Completed	10.0.2527.109	7.0.7858.96
10.0.47 PQU-4	May 20, 2026	May 20, 2026 to June 13, 2026	Completed	10.0.2527.116	7.0.7858.102
10.0.47 PQU-5	June 3, 2026	June 3, 2026 to June 27, 2026	Completed	10.0.2527.130	7.0.7858.111
10.0.47 PQU-6	June 17, 2026	June 17, 2026 to July 11, 2026	Completed	10.0.2527.142	7.0.7858.121
10.0.47 PQU-7	July 1, 2026	July 1, 2026 to July 25, 2026	Completed	10.0.2527.152	7.0.7858.132
10.0.47 PQU-8	July 15, 2026	July 15, 2026 to August 8, 2026	Completed	10.0.2527.160	7.0.7858.134
10.0.47 PQU-9	July 29, 2026	July 29, 2026 to August 22, 2026	Completed	10.0.2527.174	7.0.7858.145
10.0.47 PQU-10	August 12, 2026	August 12, 2026 to September 5, 2026	Completed	10.0.2527.187	7.0.7858.152
10.0.47 PQU-11	August 26, 2026	August 26, 2026 to September 19, 2026	In-Progress	10.0.2527.197	7.0.7858.163
10.0.47 PQU-12	September 9, 2026	September 9, 2026 to October 3, 2026	In-Progress	10.0.2527.208	7.0.7858.166

PQU release train	Change cutoff date	PQU train duration	Status	Application Version	Platform Version
10.0.47 PQU-13	September 23, 2026	September 23, 2026 to October 17, 2026	Not Started		
10.0.47 PQU-14	October 7, 2026	October 7, 2026 to October 31, 2026	Not Started		
10.0.47 PQU-15	October 21, 2026	October 21, 2026 to November 14, 2026	Not Started		
10.0.47 PQU-16	November 4, 2026	November 4, 2026 to November 28, 2026	Not Started		
10.0.47 PQU-17	November 25, 2026	November 25, 2026 to December 19, 2026	Not Started		
10.0.48 PQU-1	July 8, 2026	July 8, 2026 to August 1, 2026	Completed	10.0.2645.81	7.0.7996.74
10.0.48 PQU-2	July 22, 2026	July 22, 2026 to August 15, 2026	Completed	10.0.2645.90	7.0.7996.81
10.0.48 PQU-3	August 5, 2026	August 5, 2026 to August 29, 2026	Completed	10.0.2645.99	7.0.7996.88
10.0.48 PQU-4	August 19, 2026	August 19, 2026 to September 12, 2026	In-Progress	10.0.2645.111	7.0.7996.102
10.0.48 PQU-5	September 2, 2026	September 2, 2026 to September 26, 2026	In-Progress	10.0.2645.124	7.0.7996.111
10.0.48 PQU-6	September 16, 2026	September 16, 2026 to October 10, 2026	Not Started		
10.0.48 PQU-7	September 30, 2026	September 30, 2026 to October 24, 2026	Not Started		
10.0.48 PQU-8	October 14, 2026	October 14, 2026 to November 7, 2026	Not Started		
10.0.48 PQU-9	October 28, 2026	October 28, 2026 to November 21, 2026	Not Started		
10.0.48 PQU-10	November 11, 2026	November 11, 2026 to December 5, 2026	Not Started		
10.0.48 PQU-11	November 25, 2026	November 25, 2026 to December 19, 2026	Not Started		

PQU release train	Change cutoff date	PQU train duration	Status	Application Version	Platform Version
10.0.48 PQU-12	December 9, 2026	December 9, 2026 to January 2, 2027	Not Started		
10.0.48 PQU-13	December 23, 2026	December 23, 2026 to January 16, 2027	Not Started		
10.0.48 PQU-14	January 6, 2027	January 6, 2027 to January 30, 2027	Not Started		
10.0.48 PQU-15	January 20, 2027	January 20, 2027 to February 13, 2027	Not Started		
10.0.48 PQU-16	February 3, 2027	February 3, 2027 to February 27, 2027	Not Started		
10.0.48 PQU-17	February 24, 2027	February 24, 2027 to March 20, 2027	Not Started		
10.0.49 PQU-1	October 7, 2026	October 7, 2026 to October 31, 2026	Not Started		
10.0.49 PQU-2	October 21, 2026	October 21, 2026 to November 14, 2026	Not Started		
10.0.49 PQU-3	November 4, 2026	November 4, 2026 to November 28, 2026	Not Started		
10.0.49 PQU-4	November 18, 2026	November 18, 2026 to December 12, 2026	Not Started		
10.0.49 PQU-5	December 2, 2026	December 2, 2026 to December 26, 2026	Not Started		
10.0.49 PQU-6	December 16, 2026	December 16, 2026 to January 9, 2027	Not Started		
10.0.49 PQU-7	December 30, 2026	December 30, 2026 to January 23, 2027	Not Started		
10.0.49 PQU-8	January 13, 2027	January 13, 2027 to February 6, 2027	Not Started		
10.0.49 PQU-9	January 27, 2027	January 27, 2027 to February 20, 2027	Not Started		
10.0.49 PQU-10	February 10, 2027	February 10, 2027 to March 6, 2027	Not Started		

PQU release train	Change cutoff date	PQU train duration	Status	Application Version	Platform Version
10.0.49 PQU-11	February 24, 2027	February 24, 2027 to March 20, 2027	Not Started		
10.0.49 PQU-12	March 10, 2027	March 10, 2027 to April 3, 2027	Not Started		
10.0.49 PQU-13	March 24, 2027	March 24, 2027 to April 17, 2027	Not Started		
10.0.49 PQU-14	April 7, 2027	April 7, 2027 to May 1, 2027	Not Started		
10.0.49 PQU-15	April 21, 2027	April 21, 2027 to May 15, 2027	Not Started		
10.0.49 PQU-16	May 5, 2027	May 5, 2027 to May 29, 2027	Not Started		
10.0.49 PQU-17	May 26, 2027	May 26, 2027 to June 19, 2027	Not Started		

> **Note**
>
> Any new finance and operations apps environment that is provisioned after August 17th, 2023 is automatically signed up to receive PQUs per the schedule as applicable. Canceled* - PQU will occur only on Station-1. Releases for other stations have been canceled due to the holiday deployment freeze, and the build will be available for manual uptake. For environments configured with a weekday update schedule, updates are applied on the selected weekday following the published rollout schedule.

[NEW] Proactive quality update upcoming 10.0.46 Release-8 train schedule

App version: 10.0.2428.236

Platform version: 7.0.7778.149

Unified Environment Provisioning Application Version: 10.0.46.10

Expand table

Stations	Upcoming sandbox schedule	Upcoming production schedule
Station 1	August 31 to September 3, 2026	N/A
Station 2	September 7 to September 10, 2026	September 19 to September 20, 2026
Station 3	September 8 to September 11, 2026	September 19 to September 20, 2026
Station 4	September 14 to September 17, 2026	September 26 to September 27, 2026
Station 5	September 21 to September 24, 2026	October 3 to October 4, 2026
Station 6	September 22 to September 25, 2026	October 3 to October 4, 2026

Proactive quality update upcoming 10.0.47 Release-11 train schedule

App version: 10.0.2527.197

Platform version: 7.0.7858.163

Unified Environment Provisioning Application Version: 10.0.47.12

Expand table

Stations	Upcoming sandbox schedule	Upcoming production schedule
Station 1	August 26 to August 29, 2026	N/A
Station 2	August 31 to September 3, 2026	September 5 to September 6, 2026
Station 3	August 31 to September 3, 2026	September 5 to September 6, 2026
Station 4	September 7 to September 10, 2026	September 12 to September 13, 2026
Station 5	September 14 to September 17, 2026	September 19 to September 20, 2026
Station 6	September 14 to September 17, 2026	September 19 to September 20, 2026

[NEW] Proactive quality update upcoming 10.0.47 Release-12 train schedule

App version: 10.0.2527.208

Platform version: 7.0.7858.166

Unified Environment Provisioning Application Version: 10.0.47.13

Expand table

Stations	Upcoming sandbox schedule	Upcoming production schedule
Station 1	September 9 to September 12, 2026	N/A
Station 2	September 14 to September 17, 2026	September 19 to September 20, 2026
Station 3	September 14 to September 17, 2026	September 19 to September 20, 2026
Station 4	September 21 to September 24, 2026	September 26 to September 27, 2026
Station 5	September 28 to October 1, 2026	October 3 to October 4, 2026
Station 6	September 28 to October 1, 2026	October 3 to October 4, 2026

Proactive quality update upcoming 10.0.48 Release-4 train schedule

App version: 10.0.2645.111

Platform version: 7.0.7996.102

Unified Environment Provisioning Application Version: 10.0.48.5

Expand table

Stations	Upcoming sandbox schedule	Upcoming production schedule
Station 1	August 19 to August 22, 2026	N/A
Station 2	August 24 to August 27, 2026	August 29 to August 30, 2026
Station 3	August 24 to August 27, 2026	August 29 to August 30, 2026
Station 4	August 31 to September 3, 2026	September 5 to September 6, 2026
Station 5	September 7 to September 10, 2026	September 12 to September 13, 2026
Station 6	September 7 to September 10, 2026	September 12 to September 13, 2026

[NEW] Proactive quality update upcoming 10.0.48 Release-5 train schedule

App version: 10.0.2645.124

Platform version: 7.0.7996.111

Unified Environment Provisioning Application Version: 10.0.48.6

Expand table

Stations	Upcoming sandbox schedule	Upcoming production schedule
Station 1	September 2 to September 5, 2026	N/A
Station 2	September 7 to September 10, 2026	September 12 to September 13, 2026
Station 3	September 7 to September 10, 2026	September 12 to September 13, 2026
Station 4	September 14 to September 17, 2026	September 19 to September 20, 2026
Station 5	September 21 to September 24, 2026	September 26 to September 27, 2026
Station 6	September 21 to September 24, 2026	September 26 to September 27, 2026

Important

At least five days in advance, Microsoft updates the preceding schedule and send a notification for the set of environments that are scheduled to receive these quality updates. The preceding schedule is applicable only to environments that are notified about an upcoming update. For information on the dark hours for each region, see What are the planned maintenance windows by region?.

For each region group, or *station*, where a quality update is currently scheduled to roll out, the schedule shows a range of four days. Quality updates start with only sandbox environments. Then, as the percentage of successfully deployed sandboxes increases, deployment to production environments begins with advance notifications to customers.

Quality updates always occur in a rolling manner that enables us to target a set of environments per schedule and complete all the sets by the end of the fourth day for a station. However, this doesn't mean that an environment update spans four days. It just means that we can't predetermine which set of environments is updated on a given day within the four-day range. All updates are done during dark hours, with near-zero downtime. Updates definitively end within the dark-hour window of a given region.

More information

- Proactive quality updates overview
- Proactive quality updates FAQ

Last updated on 09/08/2026

Proactive quality updates (PQU) - FAQ

This article provides answers to frequently asked questions about proactive quality updates (PQUs), new biweekly proactive quality updates, and the optional weekday update schedule.

> **Note**
>
> The weekday update schedule option isn't supported in sovereign cloud environments at this time.

What is the biweekly cadence for PQU?

The biweekly cadence introduces a more frequent release cycle for proactive quality updates.

Starting with version **10.0.47**, PQUs move from the previous 28-day schedule to a **two-week cadence**, helping customers receive fixes and improvements sooner while maintaining a predictable update rhythm.

Each biweekly cycle includes:

- A release cut of the latest platform and application updates
- Deployment to all stations - Trail, OnlineDev, Sandbox, and Live environments

You can view the upcoming rollout calendar here: PQU Release Calendar.

Why is the rollout schedule changing from 28 days to a biweekly cadence?

This change ensures that customers receive fixes and improvements faster. A biweekly cadence helps:

- Reduce waiting time for important fixes.
- Provide more frequent validation cycles in Sandbox.
- Improve service stability through faster delivery.
- Keep customer environments closer to the most reliable build.

What does "weekday schedule" mean for PQU?

Previously, Live environments received PQUs only during weekend maintenance windows. By using the new enhancement, you can now opt to receive updates on weekdays as well.

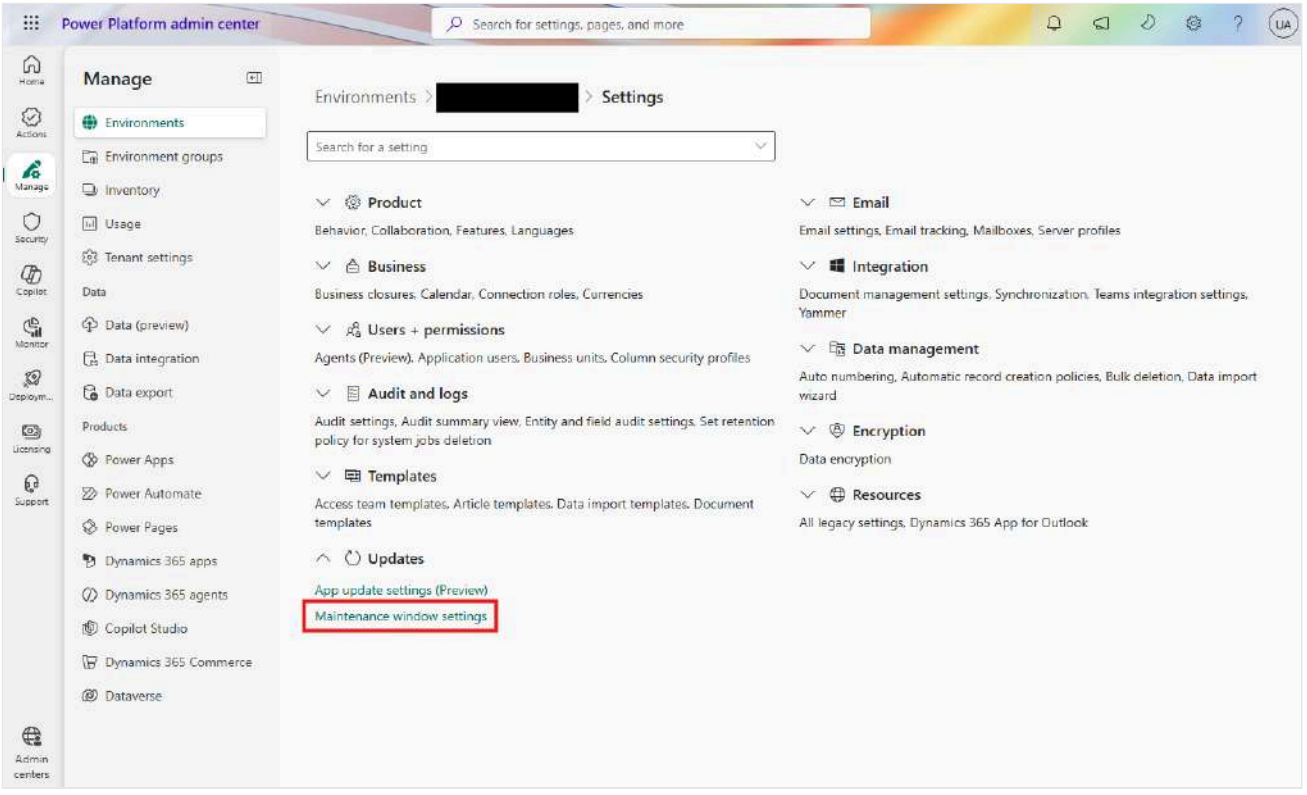
Benefits of combining the biweekly cadence with weekday scheduling include:

- Greater flexibility in selecting maintenance windows.
- More buffer time to validate PQU updates in Sandbox.
- Better alignment with internal release and testing cycles.
- Improved control over update timing.

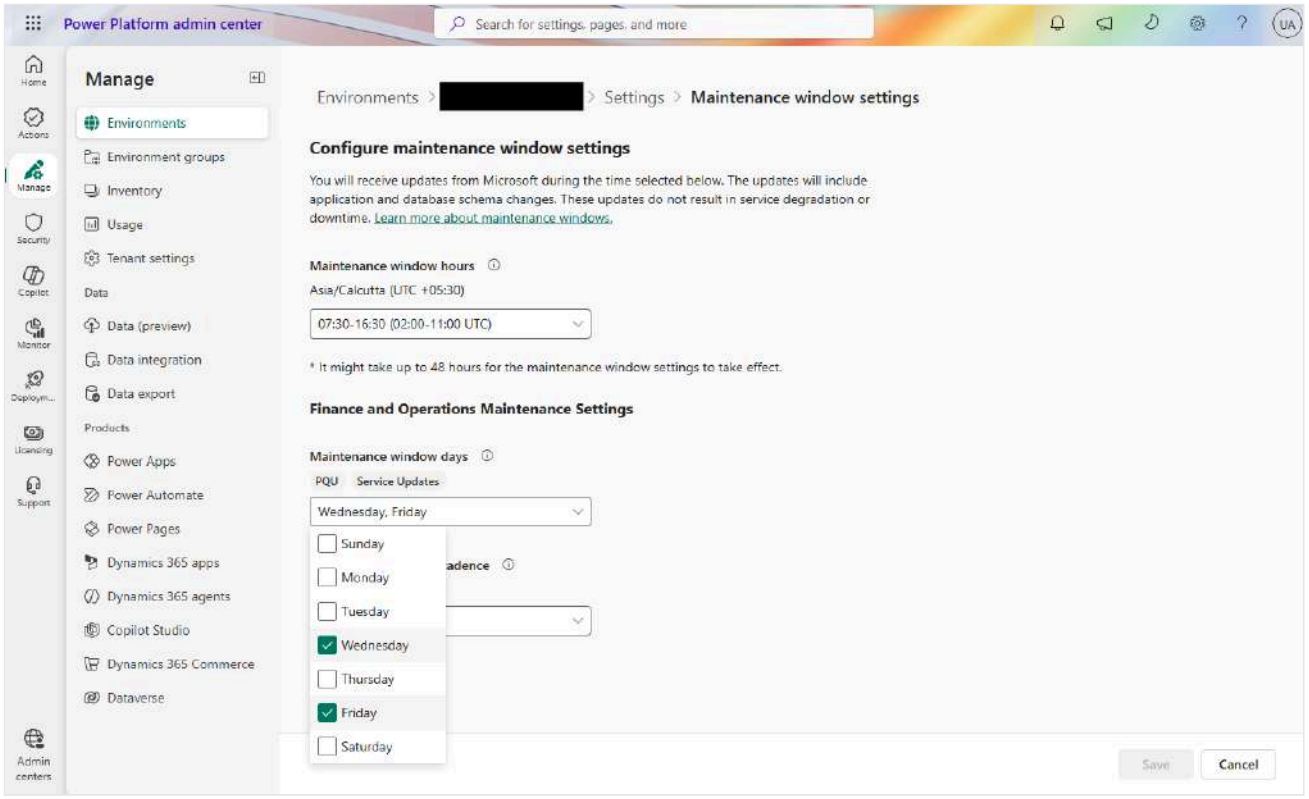
How do I configure my production environment to receive PQU updates on weekdays?

Follow these steps in the Power Platform Admin Center (PPAC):

1. Sign in to PPAC.
2. Select your production environment.
3. Select **Settings**.
4. In the **Updates** section, select **Maintenance Settings**.



5. Choose your preferred days to receive PQUs by updating **Maintenance window days** under **Finance and Operations Maintenance Settings** section.



6. Select **Save**.

> **Note**
>
> Production environments always receive updates at least five days after sandbox environments. This process remains unchanged under the biweekly cadence.

Which day should I choose for receiving PQU updates?

Choose any day that aligns with your organization's internal operations and testing cycles.

> **Tip**
>
> Selecting multiple weekdays ensures greater flexibility and reduces the risk of delays if one maintenance window is missed.

What's the minimum timeline between sandbox and production under the biweekly cadence?

The process always maintains a minimum five-day gap between when the PQU is applied to sandbox (UAT) and when it becomes eligible for live (production).

You expect this version difference. It doesn't affect your ability to deploy custom packages, as long as you follow compatibility best practices.

Can customers request exclusions from the new biweekly cadence?

No. Exclusions aren't supported. The standardized biweekly cadence ensures consistent servicing, predictable timelines, and compliance with platform servicing requirements.

How does the biweekly cadence work? Can you provide an example?

Yes. Consider this example: If PQU-1 rolls out to Station-2 on April 13–14, 2026: For customers using the default weekend-only schedule:

- PQU applied on April 13 → Live receives the update on April 18, 2026 (sandbox + 5 days)
- PQU applied on April 14 → Live receives the update on April 19, 2026 (sandbox + 5 days)

For customers who enabled weekday updates: Live receives the update on the next available selected weekday, starting after the required 5-day minimum window. Example: For a PQU applied on April 13, the earliest eligible date is April 18, 2026, and the system selects the next chosen weekday from your maintenance settings.

How are the maintenance windows handled for customers who have one finance and operations apps instance but are active in multiple time zones?

Aside from the currently supported maintenance windows, there are no special schedules for cases where an instance of finance and operations apps exists, because the rollout of PQUs involves a minimally disruptive manner that involves near-zero downtime.

Can customers delay, reschedule, or pause a PQU?

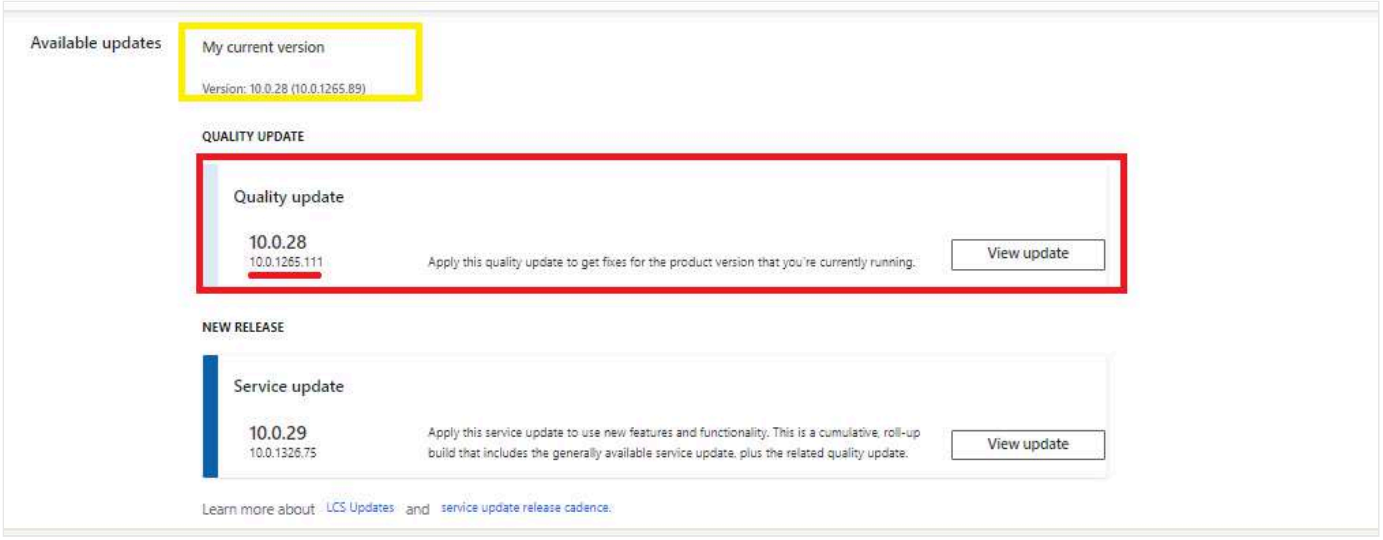
No. PQUs help ensure that fundamentals such as security, privacy, reliability, availability, and performance continuously improve for customers.

How do I know what set of changes a PQU includes?

To identify the changes that a PQU includes, follow these steps. This example uses the 10.0.28 PQU train and related app version 10.0.1265.89.

1. In Microsoft Dynamics Lifecycle Services, open the **Environment details** page for your sandbox environment.
2. In the **Available Updates** section, select **View Update** for the latest quality update build.
3. Export the build to a comma-separated values (CSV) or Excel file.
4. In the exported file, filter for and select the build version that's less than or equal to build number 10.0.1265.89. You can now review the delta payload.

> **Note**
>
> You must export to a CSV or Excel file before you update the environment. Otherwise, you can follow the preceding steps for an environment that has a similar configuration but doesn't have the update installed.



What's the process if a critical issue is found after a PQU?

A critical issue is one or more events that cause multiple customers to have a degraded experience with one or more of Microsoft's services. These problems can cause unplanned downtime, including unavailability, performance degradation, and interference with service management. If an issue with a PQU causes customer impact, Microsoft stops the rollout of that PQU until it can communicate and fix the problem. Typically, the next PQU includes the fix that's required to resume the rollout.

If the problem affects only one customer environment, contact Microsoft support to open a support ticket. Based on the justification, Microsoft stops the rollout of the PQU to other customer environments in that project until the issue is mitigated.

How do I see notifications about a PQU?

You can see notifications for PQUs in the message center in Microsoft 365 admin center. For more information, see Track new and changed features in the Microsoft 365 Message center.

Can customers manually apply hotfix updates from Lifecycle Services?

Yes. To ensure ongoing parity in the way that hotfixes work, customers can still apply hotfix updates to their environments in Lifecycle Services. PQUs are cumulative builds and continue to be available on Lifecycle Services as they're published. PQUs are published to Lifecycle Services for manual application according to the change cutoff schedule.

Hotfixes that are deployed as part of a PQU go through a rigorous safe deployment practice before the update is deployed. Because PQUs are more reliable, and involve minimal disruption and reduced downtime, use a PQU instead of manually applying an update.

Can customers proactively install a PQU ahead of the schedule?

Yes. You can proactively install a PQU. Microsoft skips the update if the environment's current build version is equal to or more than the PQU that's being deployed. If you apply the PQU manually the customers AOT package will also be promoted.

Under what circumstances does the system skip a PQU?

- If a service update is scheduled within seven days of a scheduled PQU, the system skips the PQU. For example, a PQU is scheduled on January 28 for a production environment, and a service update is scheduled on February 4 for the same environment. In this case, the system skips the PQU on January 28.
- If a sandbox environment has the same build version as the scheduled PQU, or if it has a later version, the system skips the PQU.
- If a production environment has the same build version as the scheduled PQU, or if it has a later version, the system skips the PQU.
- If a sandbox environment has the same build version or a later version because of a PQU or a manual update to the production environment, the production environment still receives the scheduled version of the PQU.
- If any Lifecycle Services environment or project is under exemption, the system skips PQUs.
- The system skips PQUs for a production environment if the update failed or was skipped in other sandbox environments in the Lifecycle Services project.

If an environment has an upcoming scheduled action, and a PQU is scheduled within the same maintenance window, does the environment still receive the PQU?

If there's a conflict with a prescheduled action, such as a point-in-time restore (PITR), the system reschedules the PQU for the next available maintenance window (within four days).

Can you restore an environment to its previous state if there are problems after a PQU is applied?

As with other code promotions, you can't roll back after a PQU is applied. For information about how flighting can help mitigate an issue, see What investments is Microsoft making to enable safe deployments of PQUs?

What's the guidance for customers who are subject to U.S. Food and Drug Administration (FDA) regulatory requirements and "good practice" (GxP) quality regulatory requirements?

The plan for customers who are subject to U.S. Food and Drug Administration (FDA) validation and regulations is still evolving. Expect more updates in this space soon. Until Microsoft makes those updates, customers who are subject to FDA regulations and "good practice" (GxP) quality regulations are exempt from the PQU process. Customers who are subject to these regulations and don't have an exemption must open a support ticket and provide justification that's related to the regulatory requirement for FDA/GxP to receive an exemption for their Lifecycle Services projects. If Microsoft already confirmed an exemption for these customers, no further action is required at this time. For more information about FDA and GxP regulations, see Microsoft Azure GxP Offering.

What's the guidance for customers who are subject to Sarbanes-Oxley (SOX) requirements?

Microsoft commissions a full System and Organization Controls (SOC) 1 Type II and SOC 2 Type II examination of Dynamics 365 online services two times each year. Dynamics 365 Finance, Dynamics 365 Project Operations, Dynamics 365 Human Resources, and other Dynamics 365 online services are detailed in the Microsoft Azure (including Dynamics 365) SOC 1 Type II report. This report and others are available in the Microsoft Service Trust Portal. Customers can use the Microsoft Azure (including Dynamics 365) SOC 1 Type II report when they're pursuing their own financial compliance requirements, such as Sarbanes-Oxley (SOX).

Which versions of service updates are supported for PQUs?

The PQU process includes all environments on a supported version.

Typically, finance and operations apps deployments that include Retail components require more work, and Retail Modern Point of Sale (MPOS) must be redeployed. How do these PQUs affect the Retail SDK?

Because the nature of the hotfix itself doesn't change in the PQU payload, you don't currently anticipate any additional impact on Retail components.

Are PQUs applied to customer-managed environments (also known as cloud-hosted environments)?

Customer-managed environments are out of scope for PQUs, because they're outside the purview of Microsoft.

Are there any integration issues with Dataverse?

There are no known integration issues for the use of PQUs with Dataverse.

What's the purpose of Station 1 or the First Release Program for PQUs?

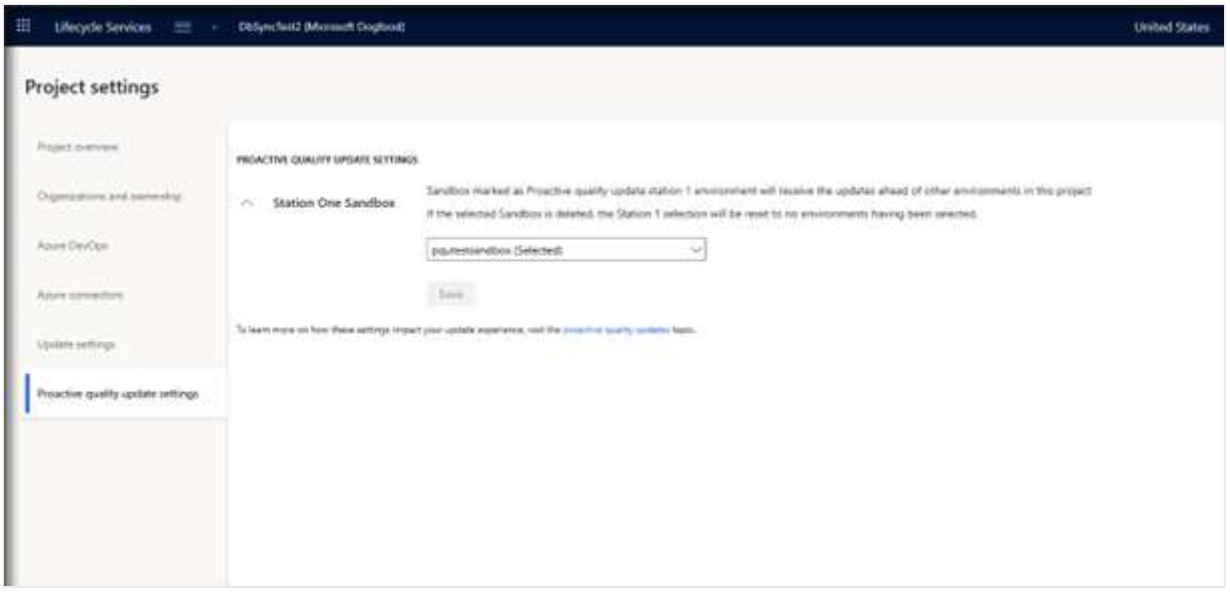
The First Release Program (FRP) for PQUs (also known as Station 1) gives all finance and operations apps customers an opportunity to receive PQUs ahead of the other stations. This program is optional and is open to all customers. Customers can take advantage of it by signing up a sandbox environment of their choice to receive PQUs. They can then provide early feedback so that Microsoft can address it before the PQUs are rolled out to the other stations. Customers who participate in the FRP for PQUs are the first select group of customers to receive a PQU in a sandbox environment of their choice. They have the benefit of a shorter feedback cycle and fast response before the rollout to other stations.

How do I join the First Release Program for PQUs?

The program is always open to all finance and operations apps customers. Only sandbox environments qualify for selection. Anyone who has a **Project owner** security role for the Lifecycle Services project can select a sandbox environment in the project.

To select a sandbox environment for the FRP for PQUs, follow these steps:

1. In Lifecycle Services, select the project.
2. Select **Project Settings** for the project.
3. On the **Proactive quality update settings** tab, in the drop-down list, select a sandbox environment. Then select **Save**.



Can customers revert or change the selection for Station 1?

After a selection, you can modify it at any time. However, if a PQU is already scheduled, the new selection isn't honored for the upcoming PQU. The selection is honored for subsequent PQUs.

What does Microsoft expect from customers who participate in the First Release Program for PQUs?

1. Participants provide feedback about the PQU and receive direct support from Microsoft Dynamics product teams for any problems that the PQUs encounter.
2. If a critical problem is found, participants might have to take an additional PQU on short notice.
3. Participants shouldn't delay or pause a PQU on the sandbox environment that they selected to receive a PQU in Station 1 or the FRP.
4. If there's a critical problem, participants should open a support ticket with Microsoft Support to get faster response and action.

How soon does a sandbox environment receive a PQU after it's selected for Station 1?

To learn when the group of sandbox environments in the FRP receive a PQU, see the release schedule for Station 1.

More information

If you have a question that isn't covered in this article, or if you need clarification about any answer in this article, open a support case.

Additional resources

- Proactive quality updates overview
- Release schedule for proactive quality updates

Last updated on 08/10/2026

Data task automation

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Data task automation makes it easy to repeat many types of data tasks and validate the outcome of each task. Data task automation is very useful for projects that are in the implementation phase. For example, you can automate the creation and configuration of data projects. You can also automate the configuration and triggering of import/export operations, such as the setup of demo data and golden configuration data, and other tasks that are related to data migration. You can also create automated testing of data entities by using task outcome validation.

> **Important**
>
> Data task automation isn't currently supported for on-premises environments. The user who executes data task automation must be in the same tenant as the application environment and Lifecycle Services project.

Use the following approach for data task automation.

1. Identify the data-related tasks that benefit from automation.

   Review your configuration management plan and data migration plan to identify potential data tasks for automation. Also identify data entity test cases.

2. Define tasks.

   Define tasks in an XML manifest. Keep your manifest under source control as part of configuration management in your application lifecycle management (ALM) strategy.

3. Put the data packages that are related to data task automation in the Shared asset library of Microsoft Dynamics Lifecycle Services. You can also use a specific Lifecycle Services project as you require.

Data task automation manager can consume packages from any sandbox and production environment that is related to the Lifecycle Services project.

> **Important**
>
> - The user account that runs Data task automation manager must have access to Lifecycle Services and to the Lifecycle Services project that the manifest references for data packages.
> - Although you can run data task automation in any environment in the cloud, don't run any import/export tasks that use integration application programming interfaces (APIs) in a production environment. Use data task automation that involves integration APIs only for automated testing.

4. Run the data tasks, and then review the outcomes.

   Data task automation manager provides the success or failure outcome for each task. It also provides insights into the reason why a task failed.

   > **Important**
   >
   > Although you can run data task automation in any environment in the cloud, don't run any import/export tasks that use integration APIs in a production environment. Use data task automation that involves integration APIs only for automated testing.

The following video is a 55-minute TechTalk that walks you through an early release of Task automation framework.

Task manifest

Define a task in an XML manifest. This section describes the manifest. For guidance about how to name and design the manifest, see the "Best practices for manifest design" section later in this article.

Manifest root

The <**TestManifest**> element is the root of the manifest. All other elements are children of this element.

XML

```
<?xml version='1.0' encoding='utf-8'?>
<TestManifest name='Data management demo data set up'>
    <SharedSetup />
        <JobDefinition ID='ImportJobDefinition_1' />
        <EntitySetup ID='Generic' />
    </SharedSetup>
    <TestGroup />
</TestManifest>
```

Expand table

Element	Element Cardinality	Attributes	Attribute description
<TestManifest>	1..1	name	The *name* helps to identify the purpose of the manifest.

Shared setup

The **Shared setup** section defines general task parameters and behaviors for all tasks in the manifest.

Expand table

Parent element	Element	Element Cardinality	Attributes	Attribute description
<TestManifest>	<SharedSetup>	1..1	-	This element takes no attributes.

Data files

<DataFile> elements define the data packages and data files that the tasks in the manifest use. The data files must be either in the Lifecycle Services asset library of your Lifecycle Services project or in the Shared asset library.

XML

```
<DataFile ID='SharedSetup' name='Demo data-7.3-100-System and Shared'
assetType='Data package' lcsProjectId=''/>
<DataFile ID='FinancialsHQUS' name='Demo data-7.3-200-Financials-HQUS'
assetType='Data package' lcsProjectId=''/>
<DataFile ID='FinancialsPICH' name='Demo data-7.3-200-Financials-PICH'
assetType='Data package' lcsProjectId=''/>
<DataFile ID='FinancialsPIFB' name='Demo data-7.3-200-Financials-PIFB'
assetType='Data package' lcsProjectId=''/>
```

Expand table

Parent element	Element	Element Cardinality	Attributes	Attribute description
<SharedSetup>	<DataFile>	1..n	-	-
	<DataFile>	-	ID	
	<DataFile>	-	name	Name of the asset that represents the data file.
	<DataFile>	-	assetType	The asset type in Lifecycle Services asset library that stores the data file. This is the asset type name as shown in Lifecycle Services asset library.
	<DataFile>	-	lcsProjectId	The Lifecycle Services project that has the data file in its asset library. If the project ID is specified as '' then, it indicates the Shared asset library.

Data project definition

The **<JobDefinition>** element defines the data project definition. A manifest can contain more than one job definition.

XML

```
<JobDefinition ID='ImportJobDefinition_1'>
    <Operation>Import</Operation>
    <ConfigurationOnly>No</ConfigurationOnly>
    <Truncate></Truncate>
    <Mode>Import async</Mode>
    <BatchFrequencyInMinutes>1</BatchFrequencyInMinutes>
    <NumberOfTimesToRunBatch >2</NumberOfTimesToRunBatch>
    <UploadFrequencyInSeconds>1</UploadFrequencyInSeconds>
    <TotalNumberOfTimesToUploadFile>1</TotalNumberOfTimesToUploadFile>
    <SupportedDataSourceType>Package</SupportedDataSourceType>
    <ProcessMessagesInOrder>No</ProcessMessagesInOrder>
    <PreventUploadWhenZeroRecords>No</PreventUploadWhenZeroRecords>
    <UseCompanyFromMessage>Yes</UseCompanyFromMessage>
    <LegalEntity>DAT</LegalEntity>
    <PackageAPIExecute>true</PackageAPIExecute>
    <PackageAPIOverwrite>false</PackageAPIOverwrite>
    <PackageAPIReexecute>false</PackageAPIReexecute>
    <DefinitionGroupID>TestExport</DefinitionGroupID>
    <PackageName>TestExportPackage</PackageName>
</JobDefinition>
```

Expand table

Parent element	Element	Element cardinality	Attribute	Description
<SharedSetup>	<JobDefinition>	1..n	ID	Use the job definition ID in the tasks to reference the definition for the data project.
<JobDefinition>	<Operation>	1..1	-	Specify the operation to perform by using the following values: - Import - Export
	<Truncate>	1..1	-	This Boolean field has possible values of Yes or No. This field is applicable only when operation is set to *Import*.
	<Mode>	1..1	-	The mode specifies the method to use for the operation. The possible values are: - Import async - Export async - Recurring batch: This value uses the enqueue API. Dequeue API isn't supported yet. Package API supports both export and import.
	<ConfigurationOnly>	0..1	-	This Boolean field has possible values of Yes or No. Set this value to Yes if the task is only to configure the data project but not to perform the specified operation.
	<BatchFrequencyInMinutes>	1..1	-	This value specifies the frequency in which the batch must

Parent element	Element	Element cardinality	Attribute	Description
				be scheduled. This value is applicable only when mode is set to *recurring batch*.
	<NumberOfTimesToRunBatch>	1..1	-	Set a limit to how many times the scheduled batch should run. This value is applicable only when mode is set to *recurring batch*.
	<UploadFrequencyInSeconds>	1..1	-	This value controls the rate at which a file is uploaded to the recurring batch job for import. Use this value only for automated testing of recurring integrations in non-production environments. This value is applicable only when mode is set to *recurring batch* and operation is set to *Import*.
	<TotalNumberOfTimesToUpload>	1..1		This value controls the total number of times the file should be uploaded to the recurring batch. Use this value only for automated testing of recurring integrations in non-production environments. This value is applicable only when mode is set to *recurring batch* and operation is set to *Import*.
	<SupportedDataSoureType>	1..1		Use this value to specify if a file is being sent to the recurring

Parent element	Element	Element cardinality	Attribute	Description
				batch or a package. This value is only applicable when mode is set to 'recurring batch'.
	<ProcessMessagesInOrder>	1..1		This Boolean field has possible values of Yes or No. This field is applicable only when mode is set to *recurring batch* and operation is *Import*.
	<PreventUploadWhenZeroRecords>	1..1		This Boolean field has possible values of Yes or No. This field is applicable only when mode is set to *recurring batch* and operation is *Export*.
	<UseCompanyFromMessage>	1..1		This Boolean field can be set to Yes or No. This field is applicable only when mode is set to *recurring batch* and operation is *Import*.
	<LegalEntity>	1..1		Use this value to specify the legal entity in which the import/export job must be executed.
	<PackageAPIExecute>	1..1		Refer to package API documentation to understand this parameter. This Boolean field takes 'true' or 'false'.
	<PackageAPIOverwrite>	1..1		Refer to package API documentation to understand this parameter. This Boolean field takes 'true' or 'false'.

Parent element	Element	Element cardinality	Attribute	Description
	<PackageAPIReexecute>	1..1		Refer to package API documentation to understand this parameter. This Boolean field takes 'true' or 'false'.
	<DefinitionGroupID>	1..1		Refer to package API documentation to understand this parameter. This string field.
	<PackageName>	1..1		Refer to package API documentation to understand this parameter. This string field.

Entity setup

The **Entity setup** section defines the characteristics of an entity that a task in the manifest uses. You can include more than one definition, with each definition corresponding to an entity that the tasks in the manifest use.

XML

```
<EntitySetup ID='Generic'>
    <Entity name='*'>
        <SourceDataFormatName>Package</SourceDataFormatName>
        <ChangeTracking></ChangeTracking>
        <PublishToBYOD></PublishToBYOD>
        <DefaultRefreshType>Full push only</DefaultRefreshType>
        <ExcelWorkSheetName></ExcelWorkSheetName>
        <SelectFields>All fields</SelectFields>
        <SetBasedProcessing></SetBasedProcessing>
        <FailBatchOnErrorForExecutionUnit>No</FailBatchOnErrorForExecutionUnit>
        <FailBatchOnErrorForLevel>No</FailBatchOnErrorForLevel>
        <DisableEntity>No</DisableEntity>
        <SkipStaging>Yes</SkipStaging>
        <ParallelProcessing>
            <Threshold></Threshold>
            <TaskCount></TaskCount>
        </ParallelProcessing>
        <MappingDetail StagingFieldName='RoundingRulePrices' AutoGenerate='Yes'
AutoDefault='No' DefaultValue='' IgnoreBlankValues='No' TextQualifier='No'
UseEnumLabel='No'/>
```

```
        </Entity>
</EntitySetup>
```

Expand table

Parent element	Element	Element cardinality	Attribute	Description
<SharedSetup>	<EntitySetup>	1..n	ID	An identification that tasks use to reference an entity definition.
<EntitySetup>	<Entity>	1..1	name	The entity element is identified by the entity's name. To facilitate easy manifest definition, this element also supports * as a wildcard which means all entities used in a task. This support comes in very handy when using data packages with hundreds of entities in a task.
<Entity>	<SourceDataFormatName>	1..1	-	The file format to use for the entity.

Parent element	Element	Element cardinality	Attribute	Description
	<ChangeTracking>	1..1	-	A Boolean field with possible values of Yes or No. It enables or disables change tracking on the entire entity.
	<PublishToBYOD>	1..1	-	A Boolean field with possible values of Yes or No.
	<DefaultRefreshType>	1..1	-	Sets the default refresh rate on the entity. The possible values are *Incremental push only* or *Full push*.
	<ExcelWorkSheetName>	1..1	-	Specifies the worksheet to use for the entity.
	<SelectFields>	1..1	-	Specifies the fields to include in the template for an export operation.
	<SetBasedProcessing>	1..1	-	A Boolean field with possible values of Yes or No. It enables or disables set

Parent element	Element	Element cardinality	Attribute	Description
				based processing on an entity.
	<FailBatchOnErrorForExecutionUnit>	1..1	-	A Boolean field with possible values of Yes or No. It enables or disables failure at execution unit level on an entity.
	<FailBatchOnErrorForLevel>	1..1	-	A Boolean field with possible values of Yes or No. It enables or disables failure at execution level on an entity.
	<DisableEntity>	1..1	-	A Boolean field with possible values of Yes or No. It enables or disables an entity in a data project.
	<SkipStaging>	1..1	-	A Boolean field with possible values of Yes or No. It skips staging table for an entity during exports.

Parent element	Element	Element cardinality	Attribute	Description
	<ParallelProcessing>	1..1	-	Defines the parallel processing set up for an entity. The task deletes these settings if they already exist at the beginning of the task and it deletes the created settings at the end of its execution.
<ParallelProcessing>	<Threshold>	1..1	-	Specifies the threshold for the parallel processing rule.
	<TaskCount>	1..1	-	Specifies the number of parallel tasks to use for parallel processing.
<Entity>	<MappingDetail>	0..n	-	Allows you to configure the *auto generate*, *auto default*, and other settings on the mapping for an entity.
	<MappingDetail>	-	StagingFieldName	Identifies the entity column for which to

Parent element	Element	Element cardinality	Attribute	Description
				specify the settings.
	<MappingDetail>	-	AutoGenerate	A Boolean field with possible values of Yes or No for enabling or disabling auto generate option.
	<MappingDetail>	-	AutoDefault	A Boolean field with possible values of Yes or No for enabling or disabling auto default option.
	<MappingDetail>	-	DefaultValue	The default value to use if auto defaulting is enabled.
	<MappingDetail>	-	IgnoreBlankValues	A Boolean field with possible values of Yes or No for enabling or disabling this option.
	<MappingDetail>	-	TextQualifier	A Boolean field with possible values of Yes or No for enabling or disabling this option.

Parent element	Element	Element cardinality	Attribute	Description
	<MappingDetail>	-	UseEnumLabel	A Boolean field with possible values of Yes or No for enabling or disabling this option.

Test groups

Use test groups to organize related tasks in a manifest. A manifest can contain more than one test group.

XML

```
<TestGroup name='Set up Financials'>
    <TestCase Title='Import shared set up data package' ID='3933885' RepeatCount='1'
TraceParser='off' TimeOut='20'>
        <DataFile RefID='SharedSetup' />
        <JobDefinition RefID='ImportJobDefinition_1' />
        <EntitySetup RefID='Generic' />
    </TestCase>

    <TestCase Title='Import financials for HQUS' ID='3933886' RepeatCount='1'
TraceParser='off' TimeOut='20'>
        <DataFile RefID='FinancialsHQUS' />
        <JobDefinition RefID='ImportJobDefinition_1'>
            <LegalEntity>HQUS</LegalEntity>
        </JobDefinition>
        <EntitySetup RefID='Generic' />
    </TestCase>

    <TestCase Title='Import financials for PICH' ID='3933887' RepeatCount='1'
TraceParser='off' TimeOut='20'>
        <DataFile RefID='FinancialsPICH' />
        <JobDefinition RefID='ImportJobDefinition_1'>
            <LegalEntity>PICH</LegalEntity>
        </JobDefinition>
        <EntitySetup RefID='Generic' />
    </TestCase>

    <TestCase Title='Import financials for PIFB' ID='3933888' RepeatCount='1'
TraceParser='off' TimeOut='20'>
        <DataFile RefID='FinancialsPIFB' />
        <JobDefinition RefID='ImportJobDefinition_1'>
            <LegalEntity>PIFB</LegalEntity>
        </JobDefinition>
```

```
        <EntitySetup RefID='Generic' />
    </TestCase>
</TestGroup>
```

Expand table

Parent element	Element	Element Cardinality	Attributes	Description
<TestManifest>	<TestGroup>	1..n	-	-
	<TestGroup>	1..1	Name	The name for the group that identifies its functional reason.
<TestGroup>	<TestCase>	1..n	-	Defines the task. The task can refer to the shared set up to inherit task parameters and task behavior. The task can also override parameters and behavior at its level, which simplifies the management of the manifest.
	<TestCase>	-	Title	The title for the task.
	<TestCase>	-	ID	The ID for the task. This value can be alphanumeric with a maximum character limit of 10.
	<TestCase>	-	RepeatCount	A placeholder for a future functionality. However, you must specify this value as *1*.
	<TestCase>	-	TraceParser	A placeholder for a future functionality. However, you must specify this value as *off*.
	<TestCase>	-	Timeout	The maximum duration that the task automation manager monitors a task. If the task is still active beyond the timeout specified, the manager proceeds to the next task in the manifest.
<TestCase>	<DataFile>	1..n	-	Defines the file or data package to use by the task. This element can reference an already declared file or a data package in the shared section of the manifest. A task can have more than one data file specified for recurring batch import scenarios only. For other

Parent element	Element	Element Cardinality	Attributes	Description
				scenarios, even if you specify more than one file, the task uses the first file.
	<JobDefinition>	1..1	-	Defines the data project to use by the task. This element can reference an already declared job definition in the shared section of the manifest. The task can override elements of job definition to a new value that you define in the shared set up.
	<EntitySetup>	1..1	-	Defines the entity set up for entities used by the task. This element can reference an already declared entity set up in the shared section of the manifest. The task can override elements of entity setup to a new value that you define in the shared set up.

Best practices for manifest design

You can define a manifest in many ways. Consider the following pointers when you design a manifest.

Granularity

Determine the granularity of your manifest as a functional decision. Your team must decide whether it wants to manage many manifests or manage changes in a single manifest.

- Start with as many manifests as your team thinks you logically need. Later, when the team actually starts to run the manifests, it might find that it uses fewer manifests than it expected, and it might want to merge them. In this case, you can merge manifests.
- Consider separation of duties. For example, you might have one manifest for the setup of demo data and another manifest for the setup of the golden configuration for your environment. By using this approach, you make sure that team members use only the manifests that they're supposed to use.
- Consider user access to Lifecycle Services. For example, larger and globally distributed implementation teams might have multiple instances of the application or multiple Lifecycle Services projects.

Inheritance

The manifest schema supports inheritance of common elements that apply to all tasks in the manifest. A task can override a common element to define a unique behavior. The **Shared setup** section minimizes repetition of configuration elements by reusing elements as much as possible. This approach keeps the manifest concise and clean, improving maintenance and readability.

Source control

Store manifests that all members of an implementation team must use in source control in the Application Object Tree (AOT). This approach not only provides the benefits of source control, but also enables a process to distribute or make manifests available to all users in a consistent manner. This approach also enables configuration management for data projects that are related to data management, if manifests are used for configuration.

Number of job definitions and entity definitions

For most use cases, one job definition in a manifest is enough, because you can use inheritance to change the behavior at the task level. This principle also applies to entity definitions.

Validations

Data task automation manager performs validations based on the setup of a task. If a task fails, you can quickly determine the reason for the failure by viewing the validations after the task completes. The level of information that Data task automation manager provides is optimized to facilitate initial discovery. For detailed investigation, you must look at the corresponding data project and its execution details.

The following data validations are currently supported:

- **Job status** – This validation checks whether the job was successful.
- **Batch status** – This validation checks whether the batch was successful.
- **Message status** – If the test is about integrations, this validation checks the message status.
- **Truncation** – If truncation is enabled, this validation checks whether truncation occurred.
- **Skip staging** – If **Skip staging** is enabled on a test, this validation checks whether staging was skipped.

Example 1: Configuration management for data projects

Use the <**ConfigurationOnly**> element to create configuration tasks for data projects. When you set ConfigurationOnly to Yes, the data projects are only created but not executed. This setting allows you to manage data projects across environments in an automated manner.

XML

```
<?xml version='1.0' encoding='utf-8'?>
<TestManifest name='Data management demo data set up'>
    <SharedSetup>
        <DataFile ID='SharedSetup' name='Demo data-7.3-100-System and Shared'
assetType='Data package' lcsProjectId=''/>
        <DataFile ID='FinancialsHQUS' name='Demo data-7.3-200-Financials-HQUS'
assetType='Data package' lcsProjectId=''/>
        <DataFile ID='FinancialsPICH' name='Demo data-7.3-200-Financials-PICH'
assetType='Data package' lcsProjectId=''/>
        <DataFile ID='FinancialsPIFB' name='Demo data-7.3-200-Financials-PIFB'
assetType='Data package' lcsProjectId=''/>

        <JobDefinition ID='ImportJobDefinition_1'>
            <ConfigurationOnly>Yes</ConfigurationOnly>
            <Operation>Import</Operation>
            <Truncate>No</Truncate>
            <Mode>Import async</Mode>
            <BatchFrequencyInMinutes>1</BatchFrequencyInMinutes>
            <NumberOfTimesToRunBatch >2</NumberOfTimesToRunBatch>
            <UploadFrequencyInSeconds>1</UploadFrequencyInSeconds>
            <TotalNumberOfTimesToUploadFile>1</TotalNumberOfTimesToUploadFile>
            <SupportedDataSourceType>Package</SupportedDataSourceType>
            <ProcessMessagesInOrder>No</ProcessMessagesInOrder>
            <PreventUploadWhenZeroRecords>No</PreventUploadWhenZeroRecords>
            <UseCompanyFromMessage>Yes</UseCompanyFromMessage>
            <LegalEntity>DAT</LegalEntity>
        </JobDefinition>

        <EntitySetup ID='Generic'>
            <Entity name='*'>
                <SourceDataFormatName>Package</SourceDataFormatName>
                <ChangeTracking>No</ChangeTracking>
                <PublishToBYOD>No</PublishToBYOD>
                <DefaultRefreshType>Full push only</DefaultRefreshType>
                <ExcelWorkSheetName></ExcelWorkSheetName>
                <SelectFields>All fields</SelectFields>
                <SetBasedProcessing>No</SetBasedProcessing>

<FailBatchOnErrorForExecutionUnit>No</FailBatchOnErrorForExecutionUnit>
                <FailBatchOnErrorForLevel>No</FailBatchOnErrorForLevel>
                <FailBatchOnErrorForSequence>No</FailBatchOnErrorForSequence>
                <ParallelProcessing>
```

```
                    <Threshold></Threshold>
                    <TaskCount></TaskCount>
                </ParallelProcessing>
            </Entity>
        </EntitySetup>
    </SharedSetup>

    <TestGroup name='Set up import jobs for Financials'>
        <TestCase Title='Set up import job for shared set up data package'
ID='3933885' RepeatCount='1' TraceParser='off' TimeOut='20'>
            <DataFile RefID='SharedSetup' />
            <JobDefinition RefID='ImportJobDefinition_1' />
            <EntitySetup RefID='Generic' />
        </TestCase>

        <TestCase Title='Set up import job for financials HQUS' ID='3933886'
RepeatCount='1' TraceParser='off' TimeOut='20'>
            <DataFile RefID='FinancialsHQUS' />
                <JobDefinition RefID='ImportJobDefinition_1'>
                    <LegalEntity>HQUS</LegalEntity>
                </JobDefinition>
                <EntitySetup RefID='Generic' />
        </TestCase>

        <TestCase Title='Set up import job for financials PICH' ID='3933887'
RepeatCount='1' TraceParser='off' TimeOut='20'>
            <DataFile RefID='FinancialsPICH' />
                <JobDefinition RefID='ImportJobDefinition_1'>
                    <LegalEntity>PICH</LegalEntity>
                </JobDefinition>
                <EntitySetup RefID='Generic' />
        </TestCase>

        <TestCase Title='Set up import job for financials PIFB' ID='3933888'
RepeatCount='1' TraceParser='off' TimeOut='20'>
            <DataFile RefID='FinancialsPIFB' />
                <JobDefinition RefID='ImportJobDefinition_1'>
                    <LegalEntity>PIFB</LegalEntity>
                </JobDefinition>
                <EntitySetup RefID='Generic' />
        </TestCase>
    </TestGroup>
</TestManifest>
```

Example 2: Automated setup of demo data

The following manifest shows the setup of demo data for three legal entities when the demo data packages are stored in the Shared asset library. The difference in this example from the previous example is the actual execution of the data projects to set up the demo data. For consistency of the manifest, don't use the `ConfigurationOnly` option or set it to `No`.

XML

```xml
<?xml version='1.0' encoding='utf-8'?>
<TestManifest name='Data management demo data set up'>
    <SharedSetup>
        <DataFile ID='SharedSetup' name='Demo data-7.3-100-System and Shared'
assetType='Data package' lcsProjectId=''/>
        <DataFile ID='FinancialsHQUS' name='Demo data-7.3-200-Financials-HQUS'
assetType='Data package' lcsProjectId=''/>
        <DataFile ID='FinancialsPICH' name='Demo data-7.3-200-Financials-PICH'
assetType='Data package' lcsProjectId=''/>
        <DataFile ID='FinancialsPIFB' name='Demo data-7.3-200-Financials-PIFB'
assetType='Data package' lcsProjectId=''/>

        <JobDefinition ID='ImportJobDefinition_1'>
                <Operation>Import</Operation>
                <Truncate></Truncate>
                <Mode>Import async</Mode>
                <BatchFrequencyInMinutes>1</BatchFrequencyInMinutes>
                <NumberOfTimesToRunBatch >2</NumberOfTimesToRunBatch>
                <UploadFrequencyInSeconds>1</UploadFrequencyInSeconds>
                <TotalNumberOfTimesToUploadFile>1</TotalNumberOfTimesToUploadFile>
                <SupportedDataSourceType>Package</SupportedDataSourceType>
                <ProcessMessagesInOrder>No</ProcessMessagesInOrder>
                <PreventUploadWhenZeroRecords>No</PreventUploadWhenZeroRecords>
                <UseCompanyFromMessage>Yes</UseCompanyFromMessage>
                <LegalEntity>DAT</LegalEntity>
        </JobDefinition>

        <EntitySetup ID='Generic'>
            <Entity name='*'>
                <SourceDataFormatName>Package</SourceDataFormatName>
                <ChangeTracking></ChangeTracking>
                <PublishToBYOD></PublishToBYOD>
                <DefaultRefreshType>Full push only</DefaultRefreshType>
                <ExcelWorkSheetName></ExcelWorkSheetName>
                <SelectFields>All fields</SelectFields>
                <SetBasedProcessing></SetBasedProcessing>

<FailBatchOnErrorForExecutionUnit>No</FailBatchOnErrorForExecutionUnit>
                <FailBatchOnErrorForLevel>No</FailBatchOnErrorForLevel>
                <FailBatchOnErrorForSequence>No</FailBatchOnErrorForSequence>
                <ParallelProcessing>
                    <Threshold></Threshold>
                    <TaskCount></TaskCount>
                </ParallelProcessing>
            </Entity>
        </EntitySetup>
    </SharedSetup>

    <TestGroup name='Set up Financials'>
        <TestCase Title='Import shared set up data package' ID='3933885'
RepeatCount='1' TraceParser='off' TimeOut='20'>
            <DataFile RefID='SharedSetup' />
            <JobDefinition RefID='ImportJobDefinition_1' />
```

```
            <EntitySetup RefID='Generic' />
        </TestCase>

        <TestCase Title='Import financials for HQUS' ID='3933886' RepeatCount='1'
TraceParser='off' TimeOut='20'>
            <DataFile RefID='FinancialsHQUS' />
            <JobDefinition RefID='ImportJobDefinition_1'>
                <LegalEntity>HQUS</LegalEntity>
            </JobDefinition>
            <EntitySetup RefID='Generic' />
        </TestCase>

        <TestCase Title='Import financials for PICH' ID='3933887' RepeatCount='1'
TraceParser='off' TimeOut='20'>
            <DataFile RefID='FinancialsPICH' />
            <JobDefinition RefID='ImportJobDefinition_1'>
                <LegalEntity>PICH</LegalEntity>
            </JobDefinition>
            <EntitySetup RefID='Generic' />
        </TestCase>

        <TestCase Title='Import financials for PIFB' ID='3933888' RepeatCount='1'
TraceParser='off' TimeOut='20'>
            <DataFile RefID='FinancialsPIFB' />
            <JobDefinition RefID='ImportJobDefinition_1'>
                <LegalEntity>PIFB</LegalEntity>
            </JobDefinition>
            <EntitySetup RefID='Generic' />
        </TestCase>
    </TestGroup>
</TestManifest>
```

Last updated on 04/03/2026

Delivering ISV solutions using One Version

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

Thanks to One Version, new updates are now automatically broadcast, downtime is minimal, and customers enjoy the benefits of staying current with recent features and fixes without having to go through expensive upgrades.

By using feature management, customers control when new features are applied. Therefore, as an independent software vendor (ISV) partner, you can innovate together with Microsoft to take advantage of new features without having to handle the waiting times that come with long release cycles. When all your customers run on current versions, you have fewer versions to maintain. You can focus instead on building quality into the solutions that you provide for your customers.

The process of servicing current versions is also more seamless and safer than it was in earlier versions. Previously, patching required that individual fixes be combined and merged into a customer environment.

Extensibility allows for deployment of side-by-side solutions that give customers more choices about how they configure their solutions.

In the One Version model, customer user acceptance testing (UAT) environments and production environments are updated several times a year. It's critical that updates don't cause problems. However, Microsoft acknowledges that both technical problems and functional problems might arise when environments are updated.

- Technical problems include breaking changes in application programming interfaces (APIs) that customizations in your solutions use.
- Functional problems that customers experience can be caused by the untimely introduction of new features. Microsoft puts any new functionality that might affect existing processes under feature management. In this way, customers can control when they adopt new functionality. Therefore, they have time to validate, document, and train their users about the new features.
- Functional problems might also be unintended changes that cause functional regressions.

Preventing technical and functional problems is difficult and requires close coordination between Microsoft and you as an ISV partner. The Microsoft goal is that you adopt practices that resemble Microsoft practices. This article explains how you and Microsoft can achieve this goal together. Over the next several months, Microsoft releases new practices and tools to help you. This article is updated as the tools and practices evolve.

This article includes the following sections:

- Servicing customers
- Compatibility
  - Runtime compatibility
  - Design-time compatibility
- Developing new releases
  - Designing for extensibility
  - Data upgrade
  - Feature exposure
- Branches and builds
- Testing
- Deploying updates
- ISV solutions as part of One Version automated deployment
- Should I release binaries or source code?

Servicing customers

Dynamics 365 finance and operations apps run on Microsoft Azure. Therefore, they're a solution that runs as a service. Microsoft services companies 24/7, either proactively from alerts that report unusual behavior, or from support tickets that customers or their partners submit. Microsoft has a range of tools to help support the services that are running. These tools

include usage data that the services collect. To help safeguard customer data, Microsoft is also careful about who can access customer systems.

When Microsoft analyzes an issue, it might determine that the issue is related to your ISV solution. Microsoft reports this type of issue to you, so that you can follow up offline.

Companies can opt out of updates for two consecutive service updates before the next service update is applied to their environments. Therefore, at any time, Microsoft can expect that companies are running one of the last three monthly updates.

When Microsoft resolves an issue that requires a code fix, it generally includes that code fix in the next monthly update. However, critical issues that are reported, such as production outage, might require that a fix be provided for the version that customers are currently running.

Similar policies apply to your ISV solution, and you might also have to provide a code update. For your solution to be binary-compatible with all your customers, it must be built on the oldest platform release that you want to support. All new updates that Microsoft releases are intended to be binary backward-compatible. This compatibility gives you the option of maintaining only one servicing version of your solution that is based on the oldest of the three most recent updates. Therefore, you must maintain just one released solution. You can then use that solution to update all your customers, regardless of which of the three most recent updates they're running. As your customers adopt new Microsoft updates, you can rebase your maintained solution to a newer release to remain current with the three most recent updates.

This recommendation applies to servicing and maintaining your released solution. You use a different approach to develop new releases of your solution. For more information, see the Developing new releases section of this article.

Compatibility

Microsoft diligently tries to guarantee compatibility with existing customizations. To achieve this goal, Microsoft uses strict practices in its engineering processes, together with tool and automation support that helps identify API contracts that are unintentionally broken. Telemetry lets Microsoft engineers determine customizations that reference or extend a Microsoft API.

Updates to finance and operations apps that Microsoft applies to customer environments are intended to be functionally compatible and binary-compatible with existing customizations. This compatibility covers not only APIs, but also functionality and the user experience. Customers must explicitly opt in to all new experiences.

Any deprecation or breaking change in binary or functional compatibility is announced 12 months in advance. Therefore, you have enough time to align your customizations with an alternative design. You must pay attention to the monthly updates to Microsoft

documentation, and you must review the APIs that are marked as obsolete (deprecated) or internal. In this way, you can manage changes in a timely manner.

The following sections define and describe the aspects of backward compatibility: runtime compatibility and design-time compatibility.

Runtime compatibility

All new updates are runtime backward-compatible. This compatibility covers both binary compatibility and functional compatibility. Runtime compatibility means that customizations in production and sandbox environments continue to work after new updates are deployed in those environments. These updates include both service updates and quality updates.

Runtime compatibility also means that changes Microsoft makes to the platform are backward-compatible with customizations that were compiled on an earlier platform.

Binary compatibility is backward only. You can compile a customization on an older platform and deploy it to a customer environment that has been updated to a later version. You can't deploy code compiled on a later version than the one running on the environment you deploy to.

Design-time compatibility

Design-time backward compatibility (that is, compile-time compatibility) means that developers can apply updates to their development environments and successfully compile their code without making any changes.

You must be aware of how APIs in your solution are used in your customers' implementations, and how you can use these APIs without causing breaking changes. As part of this effort, pay careful attention to what is changed and rely on engineering best practices. For examples of changes that you should avoid, see Breaking changes.

Try to meet a bar that resembles the bar that Microsoft sets. In that way, both you and Microsoft can help avoid creating regressions.

All Microsoft updates are binary (runtime) compatible, and Microsoft also aims for design-time compatibility. However, one category of required changes isn't design time–compatible but remains binary-compatible. After an update is applied, new errors or warnings might occur when your code is compiled. Here are some examples of these changes:

- Microsoft makes an enumeration extensible.
- Microsoft marks an API as obsolete or internal.
- Microsoft introduces a new compiler error to help avoid unsafe coding practices.

All these changes might require work on your solution.

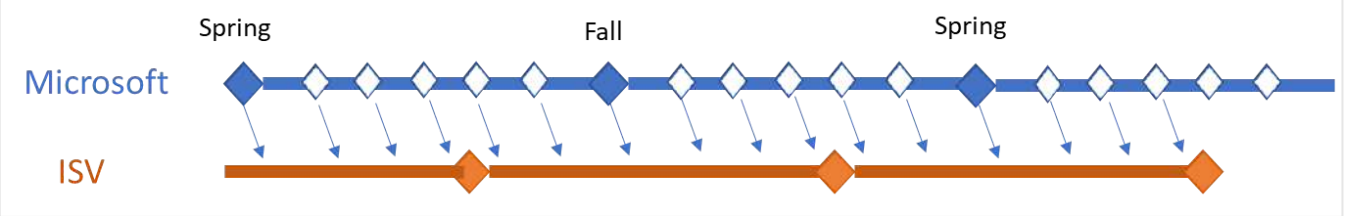
Design-time breaking changes that are binary-compatible don't require a 12-month deprecation notice.

Developing new releases

Together, One Version and the fact that the finance and operations apps run as a service provide a great vehicle for collecting feedback. Feedback is useful, because it helps Microsoft decide which new features it should add to upcoming updates. Historically, the Microsoft approach was to release major versions that include many new features. However, the new model encourages a different approach. Microsoft now releases a series of continuous updates that gradually build on the available capabilities of the system. In many cases, one update contains an initial small feature that Microsoft then enriches in later updates. In some cases, Microsoft must provide staging for new features, and must use feature exposure to hide the new features or control modifications to them.

Follow a similar approach for your ISV solutions. You benefit from quicker integration and extension of new standard features.

As the following illustration shows, the frequency of your new releases can be independent of Microsoft releases. Consider adopting a strategy for source code branching, as described in the Branches and builds section of this article.



The quality of every update that you release is essential. Although testing helps guarantee quality, you must also build quality during the design and implementation phases. In the One Version model, consider some new dimensions, as described in the following sections.

Designing for extensibility

To design your solution for extensibility, consider not only how you customize by extending the standard application, but also how you enable customization of your ISV solutions by your customers and partners.

Make sure that customizations are additive instead of intrusive, and follow the guidance on the Extensibility home page.

Don't be too creative about the way that you build your customization. Otherwise, you might extend an API that's questionable and increase the risk that later updates break your solution. Instead, log an extensibility request, and ask that Microsoft creates a more explicit API that is more resilient to breakage.

Design solutions that are extensible. For inspiration, see Write extensible code.

Design for backward compatibility to avoid breaking customer implementations. A good strategy is to be explicit about what you offer for hooking and wrapping extension code. The way that you decorate your methods gives you lots of control over which methods you enable extensions for. For more information, see Attributes that make methods extensible.

Data upgrade

Earlier versions of Dynamics 365 Finance and Operations supported different types of data upgrade jobs. These jobs aren't supported in the current version. Microsoft made this change to minimize downtime while updating a production environment.

Database synchronization still runs during upgrade. It supports basic operations such as adding new tables, fields, and indexes.

To prevent downtime, Microsoft is introducing new ways of driving data upgrade that are run asynchronously. For example, data upgrade is sometimes triggered when a feature is turned on through a feature flag. This new approach for data upgrade differs significantly from earlier approaches and becomes available in upcoming updates. Documentation resources will also be available.

Feature exposure

In the One Version model, updates are managed by Microsoft and pushed to customer environments. Pushed updates should not require that customers adjust to functional changes, or that they train their users about new or changed features every month. Pushed updates also should not cause customers to delay updates to their environment.

Feature management is a new concept that puts customers in control of deciding when new or changed features are used. Customers can review, validate, and document new or changed features before they are adopted. They can also train users before new or changed processes are adopted, to reduce the impact on daily operations.

Feature management will be released in upcoming monthly updates.

You should consider using feature management with your ISV solution to let customers control when new features are adopted.

Branches and builds

As an ISV, you should plan on a minimum of two source code branches: a servicing branch and a development branch.

Servicing branch

The servicing branch is used to produce bug fixes for your ISV solution. As the ISV, you specify the frequency of releases from the branch and distribution of the releases. The expectation is that these releases from the servicing branch will be binary cumulative releases.

The base Microsoft version that you use to build your ISV solution should match the oldest version that customers use with the solution. In the One Version model, that version starts with version 8.1 and is a maximum of three months old.

Development branch

Use the development branch to develop new capabilities for the ISV solution. As the ISV, you determine the frequency of releases from the development branch. You don't need to synchronize your releases with the monthly Microsoft releases. The best approach might be to decouple your release schedule from the Microsoft release schedule and deliver releases less often. A quarterly or biannual cadence is a good starting point.

The base Microsoft version in the development branch should be either the latest released version that is available or the released version that you plan to use for servicing when your new release goes out. The goals are that you innovate together with Microsoft by staying as current as is feasible, and that your development model allows for uptake of recent feature work.

Testing

Microsoft includes several checks and balances in its development process to help guarantee functional and binary compatibility. ISV solutions must be validated with each Microsoft release to help guarantee this compatibility. Validate this compatibility during the Preview phase of each release.

Provide quick turnaround for feedback, so that you have time to fix any problems before the monthly updates are deployed in customer environments.

Test automation is important for quick validation of new updates. Microsoft plans to release the test framework and libraries to support you as you build your test automation.

Microsoft has an extensive suite of tests that support its validation. You, as an ISV, should create your own suites of automated tests.

In addition to the SysTest automation framework that's aimed at developers, the Regression Suite Automation Tool (RSAT) enables automation of business processes without requiring that code be written. Functional users can use the RSAT to record their critical tests and automate part of their UAT. You can also use the RSAT as you start to build your test automation.

Recently, Microsoft released the Acceptance test library resources and accompanying libraries. This framework is aimed at developers and lets them build tests that are more comprehensive than unit tests. The libraries that accompany the framework help make it a seamless way to build suites of tests.

Currently released products – Testing binary and functional compatibility

Test the currently released product that the servicing branch maintains for binary and functional compatibility. Run your suite of automated developer tests, automated functional tests, and manual tests for your ISV solution on an environment that has the new version from Microsoft and your existing ISV solution. Because this test run is testing for binary and functional compatibility, don't recompile the ISV solution.

If the testing is successful, this step validates that a customer installation of the current version of your ISV solution doesn't need to be updated when Microsoft broadcasts the new release to the customer.

If the testing isn't successful, immediately notify Microsoft through the Preview communication process. This process uses Microsoft Viva Engage and an issue notification process. The issue requires either a fix from Microsoft or a fix in your ISV solution. A fix in your solution might, in turn, require that customers be updated from the servicing branch. In both cases, Microsoft must know about the issue, so that it can become more proactive in its processes for future releases.

Currently released products – Testing design-time compatibility

Next, test design-time compatibility for the currently released product that's maintained in the servicing branch. To perform this testing, compile the solution against a deployment of the new Microsoft release. Although Microsoft aims to minimize design-time compatibility issues, they might occur in some situations. One example is when Microsoft makes an enumeration extensible, and the solution uses it in a manner that assumes an underlying integer

representation (for example, it uses the enumeration value in a logical comparison or mathematical function). Although the code continues to work in a customer deployment because the underlying values are maintained, a compiler error is generated and addressed in future releases. Another example of a design-time compatibility issue is when Microsoft introduces a new compiler error to protect against unsafe coding patterns. For more information about categories of design-time compatibility issues, see Breaking changes.

Run your suite of automated developer tests, automated functional tests, and manual tests on an environment that has the new version from Microsoft and your compiled ISV solution.

If the testing is successful, this step validates that your ISV solution doesn't need to be updated even if source code is supplied to the customer and the customer recompiles the ISV solution.

If the testing isn't successful, and the issue isn't one of the categories that are described in Breaking changes, you, as the ISV, must immediately notify Microsoft through the Preview communication process. This process uses Microsoft Viva Engage and an issue notification process. The issue requires either a fix from Microsoft or a fix in your ISV solution. A fix in your solution might, in turn, require that customers be updated from the servicing branch. In both cases, Microsoft must know about the issue, so that it can become more proactive in its processes for future releases.

Currently released products – Updating the base build

As Microsoft updates your customers to new releases, periodically update the base build so that it matches the oldest version that customers use when running your ISV solution.

Solutions that are in development

Validate your new solution development on either the latest released version or the released version that you plan to use for servicing when your new release goes out. However, in both cases, consider doing validation on the most current version. This validation helps with early discovery of issues or uptake work that you must do.

If an unexpected break occurs, then you, as the ISV, must immediately notify Microsoft through the Preview communication process. This process uses Microsoft Viva Engage and an issue notification process.

Deploying updates

For Microsoft standard platform and application updates, One Version servicing includes a process for automated updates of customer environments. However, this automation isn't

currently available for ISV solutions. For more information, see ISV's as part of One Version service updates.

Manually update ISV solutions, and control your release cadence. The binary backward compatibility allows for safe updates of the standard platform and application.

The update process includes database synchronization (for example, the addition of new fields and indexes).

ISV solutions as part of One Version automated deployment

Although Microsoft doesn't currently plan to release ISV solutions as part of the One Version automated deployment process, this option might become available. However, Microsoft must first align engineering processes to make this option feasible.

Here are some areas where alignment is required:

- **Feature management** – The user must be able to control when a new feature is turned on.
- **Backward compatibility and compliance** – Compliance with API customization usage is required.
- **Feature deprecation** – Advanced notice about deprecation of features or APIs must be provided.
- **Test automation suite**
- **Testing during the preview phase**
- **ISV solution sign-off and upload**
- **Automated deployment scripts**
- **Zero downtime** – Deployment of updates must be instantaneous.
- **Data migration without downtime**
- **Support for on-call duty, for servicing of critical production issues**

Should I release binaries or source code?

Binary compatibility is supported, as long as you don't recompile. We recommend that you don't compile your ISV solution in customer environments. Instead, deploy precompiled binaries that you prepare and validate. You can create your solution binaries from your servicing branch, based on an earlier version, when this approach is practical.

If an implementation partner or customer compiles your solution in an updated environment, new warnings and errors might occur, as mentioned in the Design-time compatibility section of

this article. Therefore, implementation partners shouldn't compile your solution. However, this recommendation doesn't mean that you shouldn't share your source code to help support debugging, for example. Consider taking steps to avoid compilation of your code, so that implementation partners aren't exposed to design-time issues.

Last updated on 03/06/2026

Create and automate user acceptance tests

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

Use Task recorder and Business process modeler (BPM) to create user acceptance test libraries. Task recorder is a powerful tool to record test cases and organize them by business process using BPM. As a Microsoft partner, use BPM to distribute test libraries to your customers via Lifecycle Services and Lifecycle Services solutions. If you're a customer, use BPM to author and distribute test libraries across different projects and teams.

Because you can synchronize BPM with Azure DevOps (formerly known as Visual Studio Team Services), you can automatically create test cases (including test steps) in your Azure DevOps project. Azure DevOps can then serve as your test configuration and test management tool where you can create targeted test plans and test suites, manage the execution of tests, and investigate results. For more information about testing with Azure DevOps, see What are test plans, test suites, and test cases?

This article walks through the process of creating and executing acceptance test suites to use for manual or automated testing.

Create a Scenario Acceptance Testing BPM library

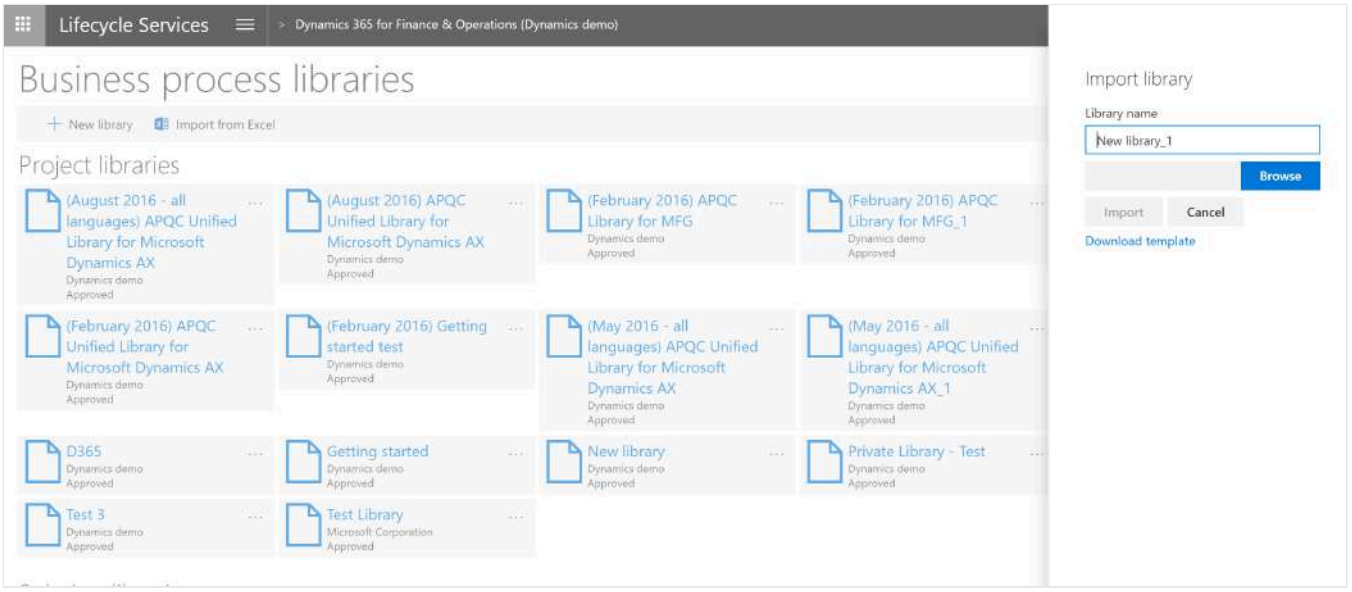
BPM is a great Lifecycle Services tool to describe a hierarchy of business processes and user tasks. Lifecycle Services also allows Microsoft partners and customers to author and distribute

BPM libraries across Lifecycle Services projects via the Asset library. This section describes how to take advantage of BPM to define your acceptance test library.

Create a BPM library

You can create a Business process modeler (BPM) library in several ways. For more information about how to create libraries in BPM, see Create, edit, and browse Business process modeler (BPM) libraries.

For illustration purposes, this article uses a library that contains common business processes, such as create an expense report and approve order requests. The article uses the Excel import functionality to create the library.



Record test cases and save to BPM

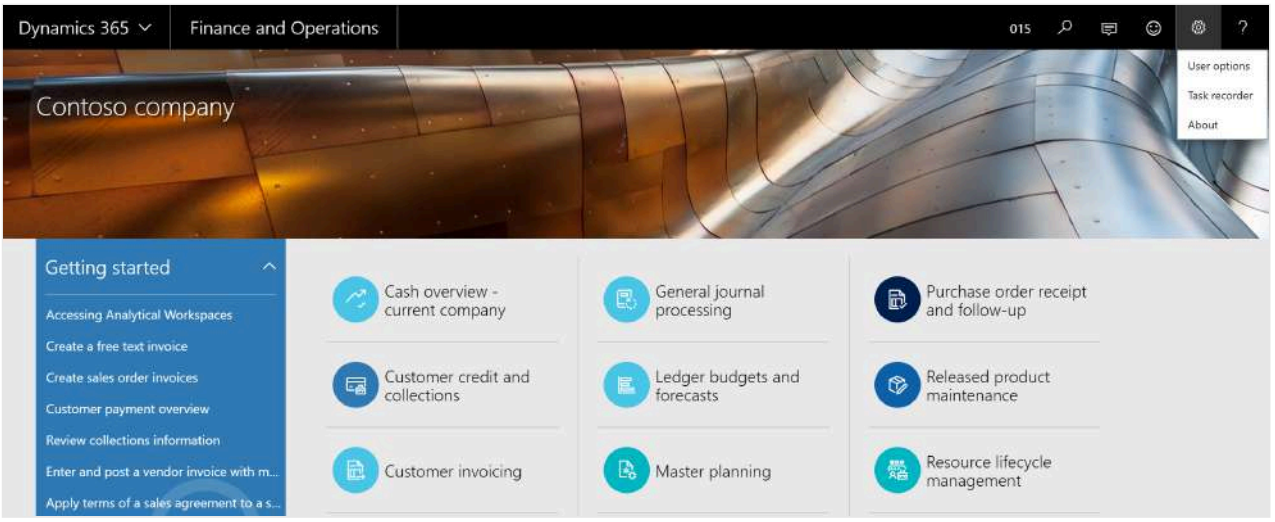
After you create a BPM library, use Task recorder to create your test cases and then upload the cases to BPM. You can use several methods to do this.

If you're using a BPM library that already has all of the necessary task recordings (test cases) attached, you can skip this step. Otherwise, follow the instructions in this section to create new task recordings.

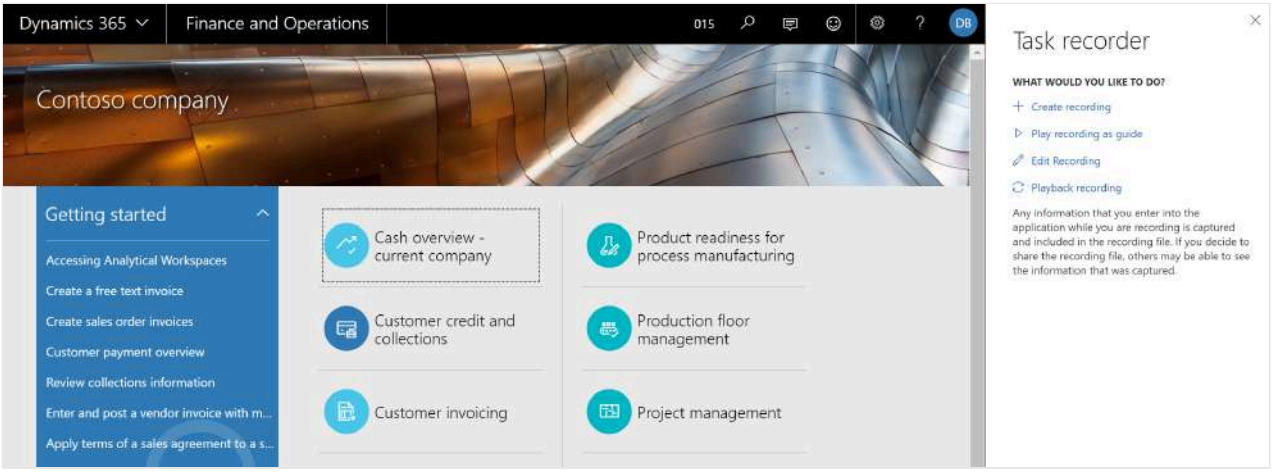
Create and save a new task recording

> **Note**
>
> Starting in January 2026, you can only create, edit, or use Task Recordings in Sandbox and Production environments. Cloud-hosted environments no longer have this capability.

1. Open the client and sign in.
2. Select the company that you want to use while recording.
3. Go to **Settings** > **Task recorder**.



4. Select **Create a new recording**.
5. Enter a name for the recording, and then select **Start**. Recording begins the moment that you select **Start**.
6. When the recording is complete, in the Task recorder pane, select **Stop**.
7. To save the task recording to an attached BPM, select **Save to Lifecycle Services**.



8. Select the library that you want to save the recording to, and then select **Save**. Otherwise, select **Save to Disk** and follow the steps in the next section, "Upload an AXTR file to BPM."

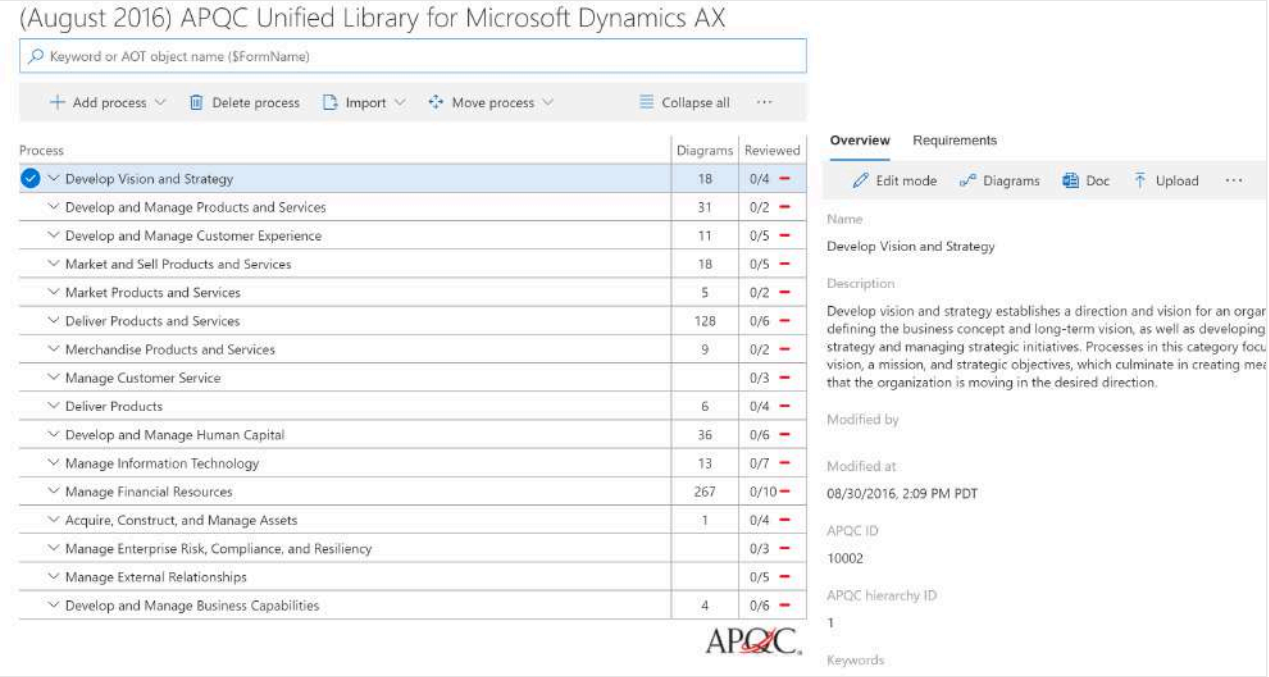
> ⓘ **Note**

To enable the effective execution of your tests using automation tools, make sure all of your task recordings start on the main dashboard of your application. For end-to-end processes that are performed by more than one user, divide your task recordings into user-specific tasks. This approach simplifies the maintenance of test cases and allows you to execute test cases in the context of security roles, which is a best practice.

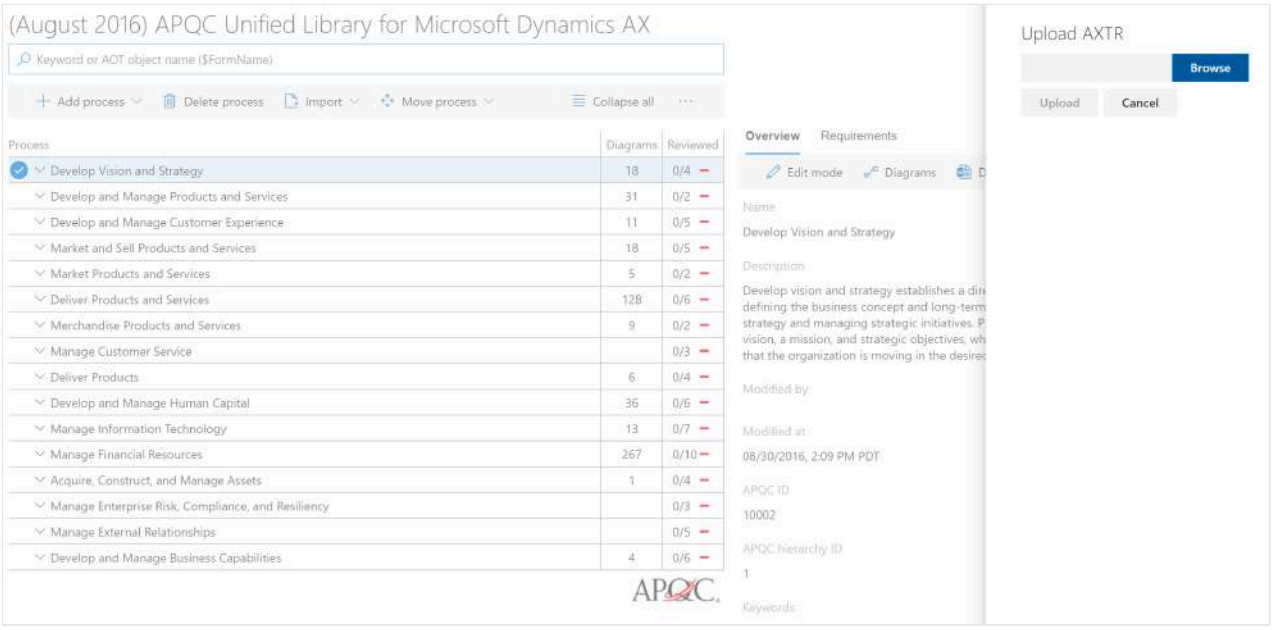
Upload an AXTR file to BPM

If you save your recordings (AXTR files) to disk, follow these steps to upload them to BPM.

1. In Lifecycle Services, in your project, on the **Business process libraries** page, select the library to upload the task recording to.
2. Select **Author and edit** and in the lines, locate and select the process to upload the task recording to.
3. In the right pane, select **Upload**.



4. Select **Browse** to find and select the file to upload, and then select **Upload**.



Save an existing task recording to BPM

> **Note**
>
> Starting in January 2026, you can only create, edit, or use Task Recordings in Sandbox and Production environments. Cloud-hosted environments no longer have this capability.

1. To attach an existing task recording, sign in to the client.
2. Go to **Settings** > **Task recorder**.
3. Select **Edit Task Recording** and attach the file by either saving directly to Lifecycle Services or downloading the AXTR and then uploading to BPM.

Guidelines for recording test cases

Follow these guidelines when authoring and recording your test cases, especially if you're planning to automate test execution. The process and tools described in this article apply to business process acceptance tests. They aren't meant to replace component and unit testing that is typically owned by developers.

- Author a limited number of test cases that, when combined, cover complete end-to-end processes.
- Focus on business processes that have been customized.
- An individual test case (recording) should cover one or two business tasks only, typically executed by one person. This simplifies task recording maintenance. Don't combine a complete end-to-end business process such as "Procure to Pay" or "Order to Cash" into one large task recording. For example, instead of having RFQ > Purchase Order > Product Receipt > Vendor Invoice > Vendor Payment as one test case, divide the process into

three or four test cases. You'll have the opportunity to combine these tests into an ordered test suite later.
- A test case should have at least one validation. Try to validate critical fields that cover the impact of other fields. For example: Validation of totals on sales or purchase orders cover the unit price/quantity/discount/tax ...etc.
- Avoid printing a report in a test case. If a test case needs to print a report, it should be selected on screen.
- 80+% of test cases should be of transactions or source documents. Master data should be limited to up to 20% of test cases only.

Synchronize and configure your test plan in Azure DevOps

An acceptance test library is your starting point. It typically contains all test cases (task recordings) of a particular application organized by business process. During a particular test pass, you usually don't need to execute all test cases. What test cases you select depends on the phase of your implementation or the nature of the update you're planning to apply to your production environment. Azure DevOps enables you to organize your test cases in test plans and test suites. A test plan contains one or more test suites (A subset of your test library); test cases can belong to more than one test suite.

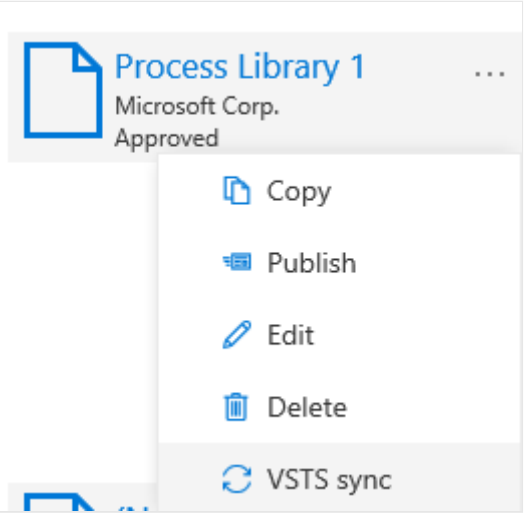
Once you have selected your acceptance testing BPM library, synchronize it with Azure DevOps and create your test plan and test suites.

Sync with Azure DevOps

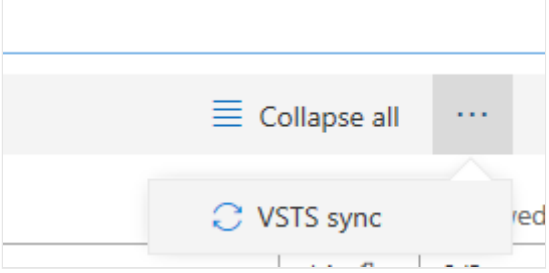
Synchronize your BPM library with your Azure DevOps project. For more information, see Synchronize BPM libraries with Azure DevOps.

After configuration is complete, synchronize the BPM library with an Azure DevOps project.

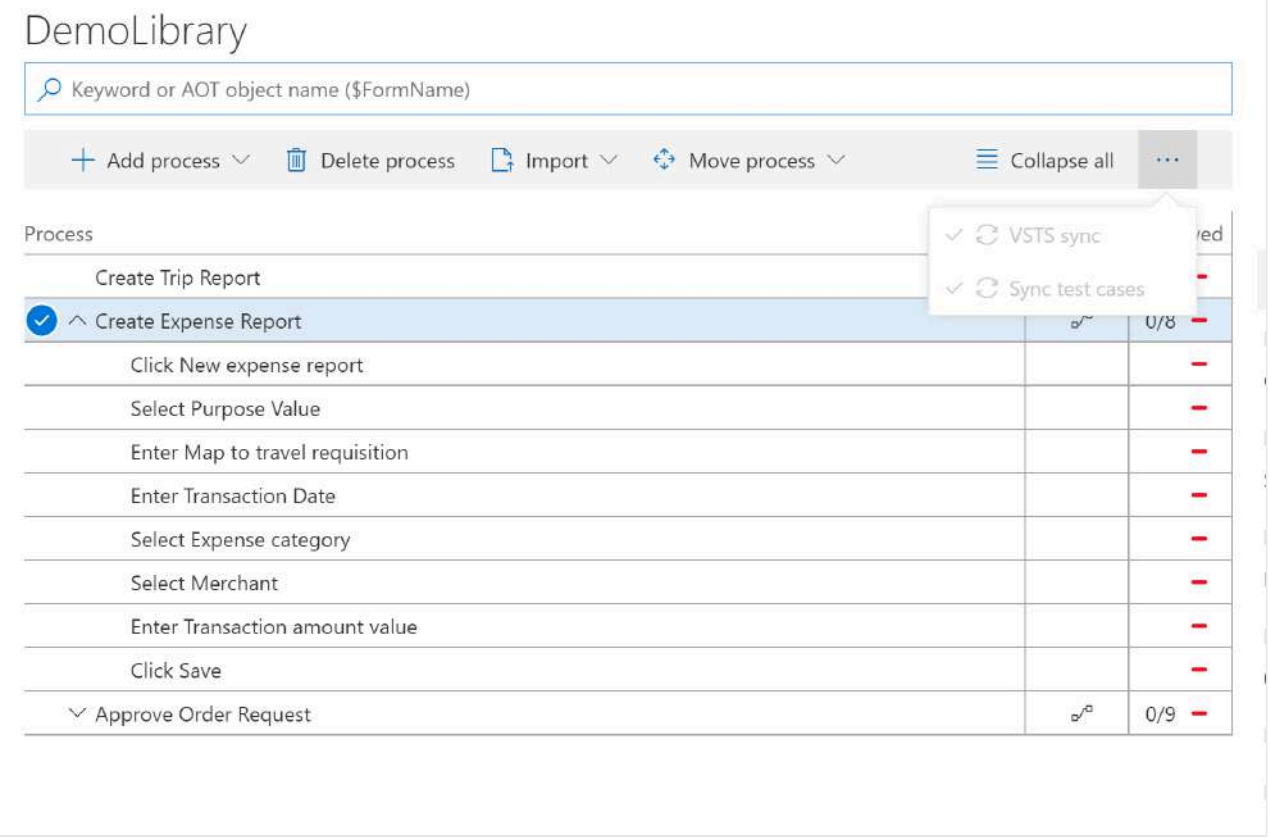
1. On the **Business process libraries** page, on the tile for the library that you want to synchronize, select the ellipsis button (...), and then select **Azure DevOps sync**.



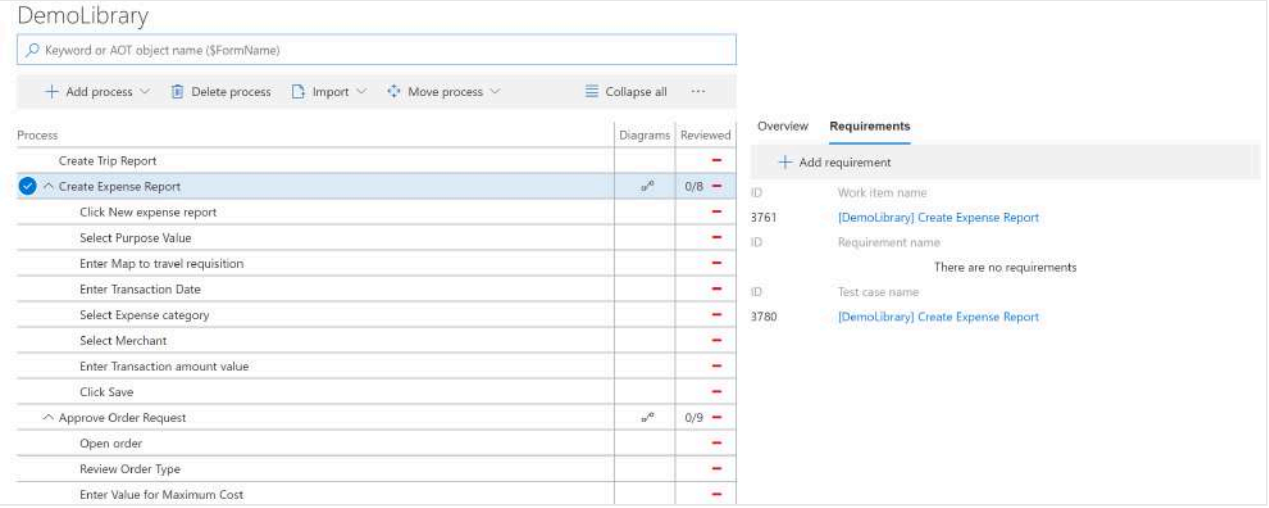
You can also start Azure DevOps synchronization from the toolbar in a BPM library. Select the ellipsis button (...), and then select **Azure DevOps sync**.



2. After Azure DevOps synchronization is complete, select the ellipsis button (...), and then select **Sync test cases**.



3. When this step is complete, your task recordings become test cases in Azure DevOps, and a link appear under the **Requirements** tab.

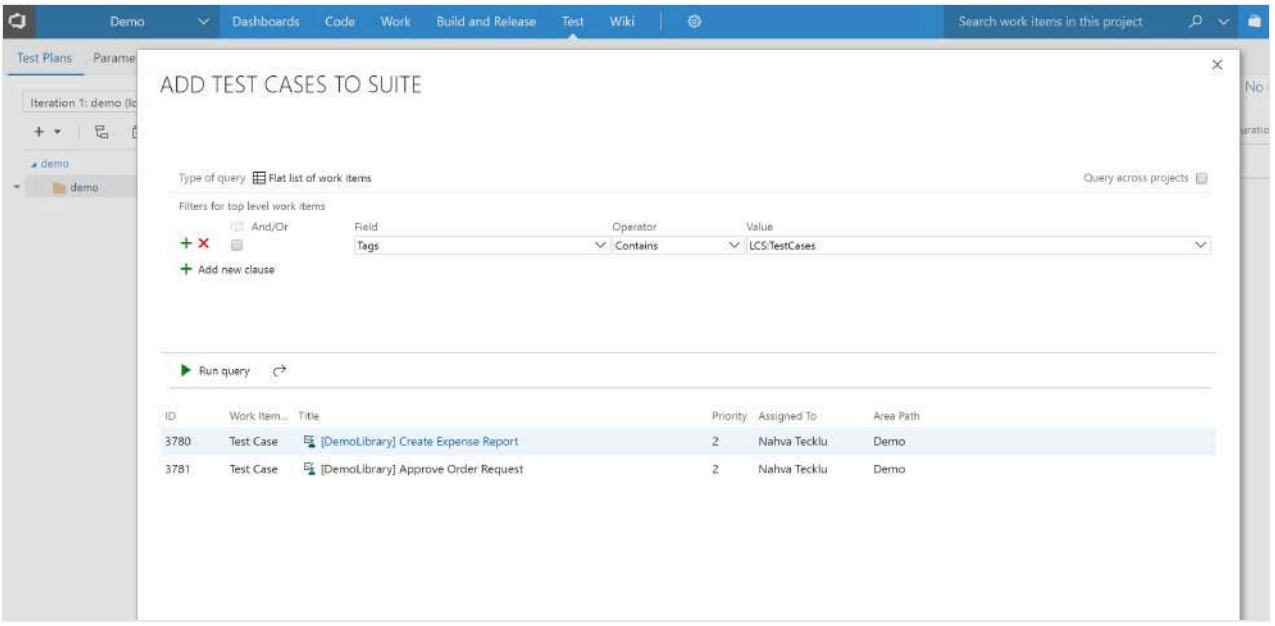


In addition to the test steps, the task recording XML file is attached to the Azure DevOps test case. This file is needed if you want to automate test execution.

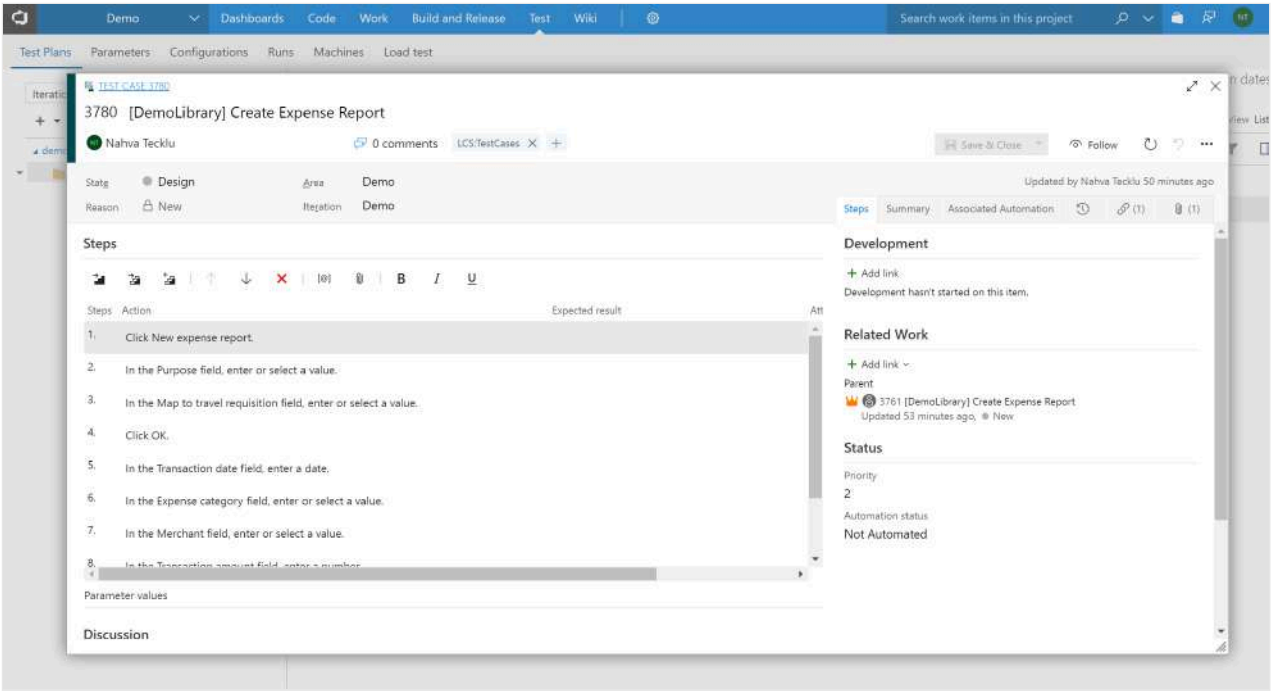
Create a test suite in Azure DevOps

Next, you need to create a test plan and test suite in Azure DevOps. This allows you to execute an ordered suite of test cases and easily manage, investigate, and track the results.

1. Sign in to Azure DevOps and select the project and test plan that you want to test in.
2. On the toolbar, select **Test** > **Test Plans**.
3. In the left pane, select **+**, and then select **Static suite**.
4. Enter a name for the suite.
5. Select **Add existing** and query the tag **Lifecycle Services:Test Cases**.
6. Select **Run** > **Add test cases**.



7. Select the test case to view details and the attached XML file.

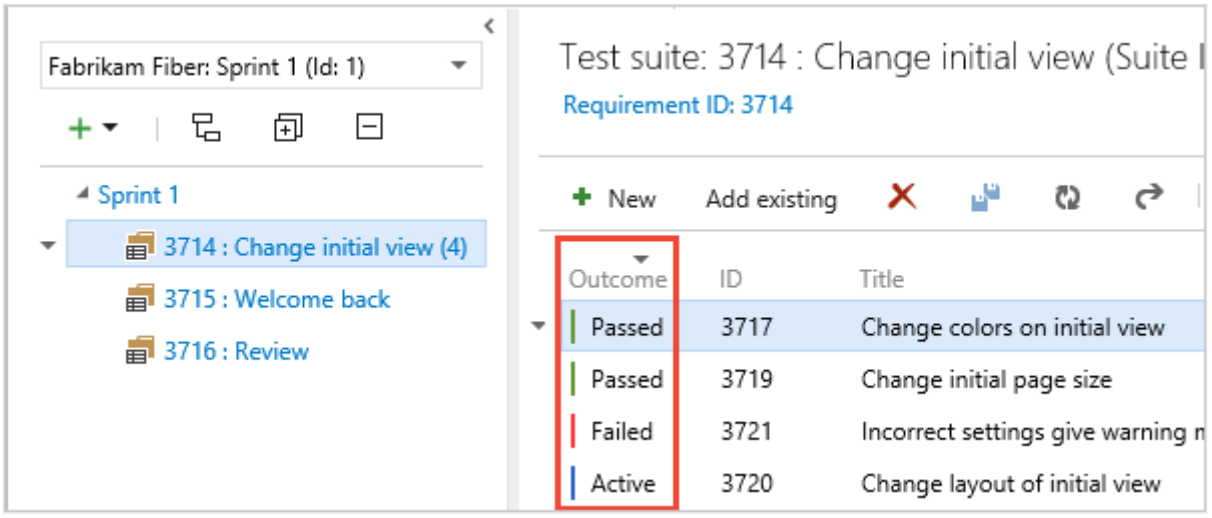


> **Note**
>
> This example shows how to create one comprehensive acceptance test suite with all test cases added. Instead, you should create various test suites under the same test plan and then use custom queries to add specific test cases to a test suite. A test case can belong to more than one test suite.

Execute your tests

Run manual test cases

After you have a test suite, you are ready to use it for regression testing after updates have been made to your application in a sandbox or test environment. You can run the test cases in your test suite manually or play the task recordings that are part of the test suite and use Azure DevOps to mark the test cases as passed or failed.



Azure DevOps also provides a tool, **Test Runner**, to manage manual test case execution. For more information about using Test Runner, see Run manual tests.

We recommend that you take advantage of Azure DevOps as it provides a rich set of management features not only for testing, but result management and mitigation.

Run automated test cases

The platform for finance and operations provides developers with tools to author test cases based on task recordings and use Azure DevOps to manage the automated execution of these test cases.

Developers can use the build and test automation capabilities of **build and test** environments. For details, see the Continuous delivery home page.

Functional power users can automate the execution of their test cases using the **Regression suite automation tool**. For more information, download the tool and read the Regression suite automation tool.

Investigate test runs

Once an automated run is complete, on the Azure DevOps toolbar, select **Test** > **Runs** (or **Test Plans** > **Runs**) to investigate your test run. Select the desired test run to investigate test case failures and errors. You can also go to your test suite in Azure DevOps to see the latest results associated with your test cases. For more information on testing and test management in Azure DevOps, see the Azure DevOps documentation.

Last updated on 03/06/2026

Deploy a demo environment

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to deploy a demo environment on Microsoft Azure by using Microsoft Dynamics Lifecycle Services. This article applies to deploying a demo environment for:

- Dynamics 365 Finance
- Dynamics 365 Supply Chain Management
- Dynamics 365 Human Resources
- Dynamics 365 Commerce

Prerequisites

Before you begin your deployment, make sure you meet the following prerequisites:

- Verify that you have an Azure subscription and that you're a co-administrator on it.
- Verify that you have access to an Lifecycle Services project and permissions to deploy an environment.
- Verify that you connected your Azure subscription to your Lifecycle Services project by using the information in the Complete the Azure Resource Manager (ARM) onboarding process article.

Deploy a demo environment

Use this procedure to deploy a demo environment on Azure by using Lifecycle Services.

1. In Lifecycle Services, open your project. In the **Environments** section, select the plus sign (+).
2. Select the Azure environment topology, and then select **Demo**.
3. Select a topology.

- For finance and operations, select the most recent Azure Resource Manager (ARM) topology for finance and operations.
- For Commerce, select **Dynamics 365 for Commerce - Demo**.

4. In the **Deploy environment** dialog box, enter the name of the environment. This name should be unique in the Azure subscription. To make environments easy to identify, consider forming an acronym by using the user's name and the topology.

5. Select the size of the virtual machine (VM). You must use **Ev5-series sizes** for finance and operations workloads. Use **Ev5**. If you experience allocation failures, see the Azure troubleshooting guide.

6. Set the **Instances** field to 1.

   > **Note**
   >
   > The size of the VM and the number of instances affect the cost of your subscription. For more information, see Azure pricing.

7. Select **Advanced settings** to add customizations to your deployment. For the demo environment, keep the default settings.

8. Agree to the licensing and pricing terms, and then select **Next**.

9. In the **Confirm** message box, select **Deploy**.

10. Open the **Cloud hosted environments** page to view the status of the deployment. After the deployment is successfully completed, the environment is ready.

Sign in to your demo environment

To sign in to your demo environment, complete the following steps:

1. In Lifecycle Services, open the **Cloud-hosted environments** page, and select the demo environment that you just deployed.

2. Scroll to the right. In the **Environment details** pane, under **Cloud services**, select the appropriate link:

   - **Log on to finance and operations**
   - **Log on to Commerce**

Last updated on 04/03/2026

Deployment options

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can deploy finance and operations apps in the cloud or on-premises. Cloud deployments offer an ERP service that Microsoft fully manages, while on-premises deployments are deployed locally within a customer's data center.

> **Important**
>
> On-premises deployments aren't supported on any public cloud infrastructure, including Azure. However, they support running on Microsoft Azure Stack HCI and Microsoft Azure Stack Hub.

The following table provides a comparison of the capabilities provided by the two deployment options.

Dynamics 365 deployment options

Capability	Cloud	On-premises
Infrastructure and data location	• Full Microsoft managed cloud service • Microsoft managed data centers	• Customer or partner managed infrastructure • Disconnected data center • Local data residency
Data trustee	• Microsoft	• Customer
Application lifecycle management (ALM)	• Managed by Microsoft • Customer access to ALM and telemetry using Lifecycle Services (LCS)	• Managed by customer or partner with cloud-based ALM and telemetry using LCS
Cloud capabilities	• High availability and disaster recovery included • Sandbox environments	• Automated deployment and telemetry using LCS
Intelligence and analytics	• Author and publish Power BI reports • Ready-made Analytical workspaces • Pinning tiles and Reports from PowerBI.com	• Author and publish PowerBI reports
Updates, health monitoring	• Provided through LCS	• Provided through LCS
Licensing	• Subscription	• License with Software Assurance/Business Ready Enhancement Plan, or subscription

Why cloud

Cloud deployments provide a cloud service that you can easily scale up or down as needed, as well as data centers that Microsoft fully manages. You can significantly shorten the time spent implementing finance and operations apps, require fewer customizations, and lower the costs of IT hardware and infrastructure.

Cloud deployments include high availability, disaster recovery, sandbox environments, and application lifecycle management combined with cloud-based systems of intelligence, infrastructure, compute, and database services in a single offering. When needed, data failover in the cloud, automated deployment and continuous updates, and elastic compute capacity are available. A cloud deployment also provides data aggregation, financial reporting, and intelligence.

The cloud service provides customers with the greatest value, the broadest range of functionality, the best application lifecycle experience, the easiest and broadest integration with Microsoft Azure services, the best option for business insights and intelligence, and the most value for customers' technology investments.

Why on-premises

By using an on-premises deployment, you can leverage your existing data center investments. You can also configure your enterprise preferences to meet the regulatory and compliance needs of your business, comply with data sovereignty rules in regions where there are no Azure Data Centers, or ensure business continuity in areas with limited public infrastructure.

Your business data and processes are disconnected from the cloud and are stored and run locally in your data center or your partner's data center. Some connectivity is required for system management and updates, which Microsoft Dynamics Lifecycle Services, a cloud-based application lifecycle management service, enables. You might store customer data that relates to the configuration and application customization in the cloud.

For customers who choose to run finance and operations apps in their own data center, the on-premises deployment option has a similar user interface and application functionality as other deployment options. However, you must take on the following responsibilities:

- Stand up your own infrastructure.
- Configure your own high-availability and disaster recovery solutions.
- Stand up sandbox environments.

- Manage your infrastructure, including scheduling operating system updates.

The additional costs to deploy and manage these capabilities might lead to higher deployment costs and a greater total cost of ownership (TCO). Partners and customers can use Lifecycle Services to access tools for deploying the finance and operations apps and updates. Unlike the cloud deployment option, Advanced Analytics and Azure Machine Learning aren't included in the on-premises deployment option.

Last updated on 04/03/2026

Comparison of cloud and on-premises features

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows a comparison of features available in cloud versus on-premises versions for the following applications:

- Dynamics 365 Finance
- Dynamics 365 Supply Chain Management
- Dynamics 365 Commerce
- Dynamics 365 Human Resources

The article also includes information about the development and administration features.

The following tables list the application areas. The tables show cloud and on-premises support for each feature. When specific features differ from the overall area, the tables list those features on a separate line in the Feature column.

All Dynamics 365 finance and operations apps

Expand table

Area	Feature	Cloud	On-premises
AI and Copilot features	Generative Help and Guidance	Yes	No
	Any copilot capabilities	Yes	No
	Summary generations	Yes	No
	AI Agents	Yes	No

Dynamics 365 Finance

Expand table

Area	Feature	Cloud	On-premises
Compliance and certifications		Yes	Yes
	SOC 1 Type 1 certification	Yes	No
Data management and integration		Yes	Yes
	Export data to your own data warehouse	Yes	Yes
	Enable the export of incremental updates to a data entity	Yes	Yes
	Data integrations	Yes	Yes
Document management		Yes	Yes
Financial management		Yes	Yes
Help		Yes	No
Human resources		Yes	Yes
Intelligence		Yes	Yes
	Electronic reporting (ER)	Yes	Yes
	ER: Integration with Lifecycle Services	Yes	No
	ER: Integration with SharePoint	Yes	No
	ER: Integration with Regulatory Configuration Services (RCS)	Yes	No
	ER: Uses local file system as storage of ER configurations accessible via ER repositories	No	Yes
	Integration with Power BI.com	Yes	No
	Integration with Power BI Desktop	No	Yes
	Analytical workspaces	Yes	No

Area	Feature	Cloud	On-premises
	Intelligent business process: Recommendations	Yes	No
	Authoring Power BI reports with OData using Power BI desktop or Excel PowerQuery tools	Yes	No
	SQL Server Reporting Services (SSRS) supports scaling out	Yes	Yes
	Telemetry is transferred into the cloud	Yes	No
Lifecycle services		Yes	Yes
	Configurable business processes	Yes	No
Localizations		Yes	Yes
Mobile app, workspaces, and platform		Yes	Yes
Office integration		Yes	Yes
Organization administration		Yes	Yes
Payroll		Yes	Yes
	Direct deposit	Yes	No
Project management and accounting		Yes	Yes
Security		Yes	Yes
Service management		Yes	Yes
Web client		Yes	Yes
	Task recorder - Save or load task recordings from the BPM library	Yes	No
Support		Yes	Yes
	Access to Support via the Help & Support menu	Yes	No

Area	Feature	Cloud	On-premises
	Business events	Yes	Yes (either internet connectivity is required or custom endpoints must be implemented to send or receive business events within intranet)

Dynamics 365 Supply Chain Management

Expand table

Area	Feature	Cloud	On-premises
Asset management		Yes	Yes
Compliance and certifications		Yes	Yes
	SOC 1 Type 1 certification	Yes	No
Cost accounting		Yes	Yes
	Cost accounting content pack for Power BI	Yes	No
	Cost accounting workspace for mobile app	Yes	No
Cost management		Yes	Yes
	Cost management content pack for Power BI	Yes	No
Data management and integration		Yes	Yes
	Configuration-driven extension	Yes	No
	Export data to your own data warehouse	Yes	Yes
	Enable the export of incremental updates to a data entity	Yes	Yes
	Data integrations	Yes	Yes
Document management		Yes	Yes
Help		Yes	No
Intelligence		Yes	Yes
	Electronic reporting (ER)	Yes	Yes

Area	Feature	Cloud	On-premises
	ER: Integration with LCS	Yes	No
	ER: Integration with SharePoint	Yes	No
	ER: Integration with Regulatory Configuration Services (RCS)	Yes	No
	ER: Uses local file system as storage of ER configurations accessible via ER repositories	No	Yes
	Integration with PowerBI.com	Yes	No
	Integration with Power BI Desktop	No	Yes
	Analytical workspaces	Yes	No
	Intelligent business process: Recommendations	Yes	No
	Authoring Power BI reports with OData using Power BI desktop or Excel PowerQuery tools	Yes	No
	SQL Server Reporting Services (SSRS) supports scaling out	Yes	Yes
	Telemetry is transferred into the cloud	Yes	No
Inventory management		Yes	Yes
Lifecycle services		Yes	Yes
	Configurable business processes	Yes	No
Localizations		Yes	Yes
Manufacturing		Yes	Yes
Master planning and forecasting		Yes	Yes
	Planning optimization	Yes	No
	Demand forecasting	Yes	No
Mobile app, workspaces, and platform		Yes	Yes
Office integration		Yes	Yes
Organization administration		Yes	Yes

Area	Feature	Cloud	On-premises
Procurement and sourcing		Yes	Yes
	Punch-out to external catalog from purchase requisition	Yes	No
	Purchase spend analysis Power BI reports	Yes	No
Product information management		Yes	Yes
Product master data		Yes	Yes
Production		Yes	Yes
	Production performance Power BI reports	Yes	No
Project management and accounting		Yes	Yes
Sales		Yes	Yes
	Sales and profitability performance Power BI reports	Yes	No
Security		Yes	Yes
Service management		Yes	Yes
Supply chain management		Yes	Yes
Transportation management		Yes	Yes
Vendor collaboration		Yes	No
Warehouse management		Yes	Yes
	Mobile warehouse app	Yes	Yes
	Warehousing Power BI reports	Yes	No
Web client		Yes	Yes
	Task recorder - Save or load task recordings from the BPM library	Yes	No
Support		Yes	Yes
	Access to Support via the Help & Support menu	Yes	No

Dynamics 365 Commerce

To see a list of capabilities that are available in on-premises deployments, see Commerce capabilities that are available in on-premises deployments.

Dynamics 365 Human Resources

This table applies to the Human Resources stand-alone version.

Expand table

Area	Feature	Cloud	On-premises
All Human Resources areas	All Human Resources features	Yes	No

> **Note**
>
> For the comparison of cloud vs. on-premises of a migrated Human resources environment on the merged infrastructure, see Dynamics 365 Finance.

Development and administration features

Expand table

Area	Feature	Cloud	On-premises
Build and test		Yes	Yes
Extensibility		Yes	Yes
Monitoring and telemetry		Yes	Yes
Platform compatibility		Yes	Yes
Servicing		Yes	Yes
	Servicing environments	Yes	No
Trace Parser		Yes	Yes
PerfTimer		Yes	Yes*
Upgrade		Yes	Yes

Area	Feature	Cloud	On-premises
	Upgrade	Yes	No
	Upgrade and support for previous versions	Yes	No
Visual Studio development		Yes	Yes

* In on-premises environments, PerfTimer only shows results for the client.

Last updated on 03/09/2026

Cloud deployment overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To work with Microsoft to deploy finance and operations apps in the cloud, you need to understand the environment and subscription that you're deploying to, who can perform which tasks, and the data and customizations that you need to manage. Sign up for the Full Microsoft FastTrack for Dynamics 365 to help speed your deployment and implementation. This program provides training and consulting to help you realize business value faster. For more information, see Microsoft FastTrack. If you choose to use the Essentials FastTrack program instead, use the Implementation Project Methodology in Microsoft Dynamics 365 Lifecycle Services to help you manage your implementation project.

Customer lifecycle, subscriptions, and environment types

Microsoft assumes that all customers follow a lifecycle similar to the following for all cloud deployments, and therefore need different environment topologies at each phase.

- Evaluate
- Develop customizations, if needed.
- Curate a "golden configuration" environment that contains only module configurations without master or transactional data. This environment serves as the baseline for your data migration testing and eventual go live.
- Install and test customizations and partner solutions on a tier-1 sandbox (Development or test environment).
- Test customizations, partner solutions, and data configuration on a tier-2 sandbox environment.
- Deploy customizations and data configurations to a production environment with high availability.

At some phases of a project, you might have all of the environments live at once. For more information about the default licenses and tiers that are available, see the Dynamics 365 Licensing Guide.

You might notice the terms cloud hosted or Microsoft subscriptions. A *cloud hosted subscription* means that the customer or partner brings their own Azure subscription and deploys finance and operations apps to it, for evaluation and development purposes only. The customer or partner pays for the resources deployed to their Azure subscription based on the Azure price list. A *Microsoft subscription* means that the customer purchases finance and operations licenses, which then allow them to deploy environments to an Azure subscription that Microsoft manages. Therefore, the customer has no separate Azure billing.

Each Enterprise offer includes two environments by default:

- One Tier 2 sandbox (multibox environment) for user acceptance testing (UAT).
- One production environment with high availability (HA).

You can purchase additional environments as add-ons. For information about licensing and what is included in Microsoft Dynamics 365, see the Dynamics 365 Licensing Guide.

Here's how the lifecycle maps to the available environments. If you already have environments deployed in your Lifecycle Services project, you can find the Environment Type and Environment Sub type on each environment's details page.

Expand table

Lifecycle phase	Environment tier	Subscription	Environment types	Environment sub-type
Evaluation and analysis	Tier 1 sandbox	Cloud hosted	Customer-managed	Demo
Customize	Tier 1 sandbox	Cloud hosted or VHD	Customer-managed	Develop
Golden configuration	Tier 1 sandbox	Cloud hosted	Customer-managed	Develop
User acceptance testing (UAT)	Tiers 2-5 sandbox	Microsoft	Microsoft-managed or self-service	Not applicable
Go live	Production	Microsoft	Microsoft-managed or self-service	Not applicable

Tiers 2-5 can be purchased to increase performance of the environment. The higher the tier, the more compute and database capacity is reserved for your use. For more information about Self-service environment types, see Self-service deployment overview.

> **Important**
>
> Starting in November 2020, Microsoft no longer manages tier 1 sandbox environments. For demo, build, and develop purposes, you can deploy the tier 1 environments on a customer's Azure subscription directly from Lifecycle Services.

Environment lifecycle operations

Users with the Environment Administrator or Project Owner roles in Lifecycle Services can perform various lifecycle operations on their environments. These operations often involve downtime on the environment until the task finishes. You can find each of these operations under or next to the **Maintain** button on each environment details page.

Expand table

Lifecycle operation	Description	Learn more
Apply software	Install Microsoft updates, ISV solutions, or your own customization packages.	Apply updates to cloud environments
Enable access	Allow list your IP for Remote Desktop or database access.	See the Remote Desktop section later in this article.
Restart services	Restart components of your environment.	Restart environment services
Move database	Full data lifecycle management.	Database movement operations
Maintenance mode	Change configuration with only admin access.	Maintenance mode
Upgrade	Upgrade code and data from 7.x to the latest version.	Process for moving to the latest update
Deallocate	Turn off an environment that isn't being used, or to troubleshoot a failed action.	Not applicable
Start	Turn on an environment for use.	Not applicable
Delete	Delete an environment that you previously deallocated.	Not applicable

Security and compliance

Finance and operations apps are PA-DSS 3.1 certified, which means that all communications between components are secured automatically.

All finance and operations front-end virtual machines in Microsoft Azure are configured during deployment to accept only TLS 1.2.

> **ⓘ Important**
>
> Customers who have administrator access to Microsoft-managed sandboxes, including any add-on sandboxes purchased, must follow these guidelines:
>
> - By default, automatic Windows update is enabled for all Tier 1 - 5 sandboxes and you shouldn't disable it. This setting ensures that any time Microsoft pushes security or critical infrastructure updates to your environment, your environment receives the latest set of updates and is updated each month with the operating system fixes that Microsoft releases.
> - Don't change admin passwords on these environments. Microsoft flags environments when admin passwords are changed. Microsoft reserves the right to reset the admin password, and it will do so.
> - Don't add new user accounts to any Microsoft-managed VM. Microsoft reserves the right to remove the newly added user accounts without providing notice.
>
> Finance and operations isn't covered by a FedRAMP ATO at this time. If you provision finance and operations in the United States, all customer data at rest is stored in data centers located in the United States. Finance and operations doesn't support any other Dynamics 365 US Government or Microsoft 365 GCC compliance attributes (for example, access by US screened personnel, and support for CJIS and IRS 1075).

Remote Desktop

Microsoft-managed environments

> **⚠ Warning**
>
> Microsoft is removing the use of Remote Desktop by customers and partners. Each environment first loses administrator access but still allows non-administrator access to

the virtual machines. After this step, all access is removed. For each step of this phased removal, an email notification is sent to the notification list set up for each environment. Microsoft removes all Remote Desktop access by November 2020.

To connect to virtual machines (VMs) through Microsoft Remote Desktop (RDP), you need to complete additional setup. This setup applies to all Microsoft-managed environments, including Tier 1 through Tier 5 sandboxes and add-ons. To connect to Tier 1 through Tier 5 sandbox environments, you must explicitly enable access (allow list) from your organization's IP address space. A Lifecycle Services user with access to the **Environment** page (**Maintain** > **Enable Access**) can enter the IP address space that you use to connect to the virtual machines through Remote Desktop. Access rules are either a single IP address (example: 10.10.10.10) or an IP address range (example: 192.168.1.0/24). You can add multiple entries at once as a semicolon (;) separated list (example: 10.10.10.10;20.20.20.20;192.168.1.0/24). These entries configure the Azure Network Security Group that is associated with your environment's virtual network. For more information, see Security rules.

ⓘ Important

You need to ensure that RDP endpoints are secured through explicit IP allow list rules as mentioned earlier. The IP allow list rules must adhere to the following conditions.

- Don't use asterisk or zero in IP allow list rules.
- Don't use wide IP address ranges.
- Restrict IP address ranges to your organization's CORPNET.
- If you use computers outside your organization's CORPNET (such as a home office) to connect to sandbox environments, add only the specific IP addresses of the computers used to connect to the sandbox environments.
- Don't add Azure Datacenter IP address ranges.
- Don't add public IP addresses, such as a coffee shop location.
- Remove IP allow list rules when not in use. Periodic review of environment IP allow list rules is recommended.

Microsoft runs periodic tests on the Microsoft-managed environments to validate that the environments are sufficiently restricted. Microsoft reserves the right to remove any IP address allow list rules that violate the preceding guidelines, immediately and without providing notice.

Customer-managed/Tier-1 environments

By default, Remote Desktop is enabled for all environments that aren't managed by Microsoft. Restrict access to any environments that belong to your subscriptions. You can restrict access by configuring Network Security Group rules on the environments directly in Azure portal.

Windows Remoting (WinRM)

Windows Remoting (WinRM) is disabled on all environments. Although you can enable WinRM on environments that belong to your subscriptions through the Azure portal, don't enable it.

> ⚠ **Warning**
>
> Microsoft doesn't grant exceptions to enable WinRM for any Microsoft-managed environments.

Availability

The guaranteed uptime for finance and operations apps is 99.9%. Planned downtime occurs once a month and lasts no longer than eight hours. Because the work completed during the downtime doesn't always take eight hours, the estimated amount of time that your environments are down is always communicated. For more information, see Get support for finance and operations apps or Lifecycle Services.

High-availability features

To ensure service availability, all production environments use default Azure high availability (HA) features. HA functionality provides ways to avoid downtime caused by the failure of a single node within a datacenter, and DR features protect against outages that broadly impact an entire datacenter. Azure availability sets prevent single-point-of-failure events. For more information about Azure availability sets, see Use availability zones to protect from datacenter level failures. High availability for databases is supported through Azure SQL. For more information, see Overview of business continuity with Azure SQL Database.

Database backup retention

For Microsoft-managed or self-service Tiers 2-5 environments, Azure SQL takes automated backups every few minutes. You can restore these backups by using Lifecycle Services point-in-

time restore capability up to 14 days in the past. For production type environments, you can restore backups up to 28 days in the past. For Tier 1 or customer-managed environments, database backups aren't automatic and need to be taken manually as often as required.

Disaster recovery features

Production environments are configured with Azure disaster recovery support that includes the following features:

- Azure SQL active-geo replication is configured for the finance and operations database of the production environment. For more information about SQL replication, see Compare geo-replication with failover groups.
- Geo-redundant copies of Azure blob storage (containing document attachments) in other Azure regions. For more information, see Azure Storage redundancy.
- Same secondary region for the Azure SQL and Azure blob storage replication.

Only primary data stores are supported by replication. The Financial reporting services and Entity store database use transformed data from the primary database and must be generated after the recovery site is set up and the finance and operations service starts.

Service availability in Azure regions

You can deploy finance and operations apps to a subset of Microsoft Azure datacenters by using Dynamics Lifecycle Services. Azure is generally available in datacenters and geographical locations around the world. By using finance and operations apps, you can specify the region or datacenter where you want to store your customer data. Microsoft might replicate data to other regions for data durability, but it doesn't replicate or move customer data outside the geographical location. For more details, see Service description for finance and operations apps.

> **ⓘ Important**
>
> Regardless of where customer data is stored, Microsoft doesn't control or limit the locations from which customers or their end-users can access it. For more information, see the following articles:
>
> - Dynamics 365 Finance, Supply Chain Management, and Commerce in local geographies

- Dynamics 365 Finance, Supply Chain Management, and Commerce in US Government Community Cloud (GCC)
- Dynamics 365 Finance, Supply Chain Management, and Commerce operated by 21Vianet in China

Frequently asked questions

Why does the status display 'Maintenance' on my environment in Lifecycle Services?

To provide the best experience and performance, Microsoft performs maintenance operations on your environment. During some of these maintenance operations, your environment status might display one of the following statuses:

- Preparing for maintenance
- Prepared for maintenance
- Maintenance in progress

While your environment is in this state and until the status returns to 'Deployed', you can't perform any lifecycle operations, such as package applications. There's no impact to finance and operations apps. Users can continue with normal operations without any service interruption. You receive an email notification before any maintenance operation puts your environment in this state.

How do I connect to the SQL database on my sandbox environment?

To connect to the SQL database in your sandbox environment, follow the steps in Enable just-in-time access.

How do I access a development instance?

For information about how to access development instances, configure on-premises development VMs, and find configuration settings for developers and administrators, see Deploy and access development environments.

How do I deploy a demo environment?

A demo environment includes only Microsoft demo data. Use a demo environment to explore default features and functionality. For more information, see Deploy a demo environment.

How do I move my customizations between environments?

To move customizations from a development environment to a sandbox or production environment, see Create deployable packages of models.

Can I bring my own domain name?

You can bring your own domain name if it's running Microsoft Entra ID, and the administrator of your Microsoft Entra instance enables the finance and operations apps within their Microsoft Entra ID. This configuration is usually done through the office email, after you buy a license. When you click the link to accept the offer, Microsoft Entra ID is set up for you.

Can I add guest Microsoft Entra accounts as users?

You can add guest Microsoft Entra accounts if you correctly configure them within Microsoft Entra and enable the finance and operations apps within your Microsoft Entra.

Why am I no longer able to see the Private AOS machines in one or more of my Tier 2 through Tier 5 sandbox environments?

The Private AOS VMs were part of your environment configuration as they secured communication between the AOS and BI machines. With recent updates, all communication between AOS and BI machines is secure directly and no longer needs the intermediary Private AOS machines. Therefore, Microsoft is in the process of rolling out removal of the Private AOS machines. As the removal process rolls out in batches, you might notice that only some of your environments have the Private AOS machines removed. This change doesn't impact functionality or security in any way and is transparent to you.

Why can't I use Remote Desktop to connect to one or more of my Tier 1 through Tier 5 Microsoft-managed sandbox environments?

Microsoft-managed Tier 1 through Tier 5 sandbox environments require Remote Desktop management endpoints to be restricted to specific IP address sets (allow list). Microsoft regularly validates that the environments are sufficiently restricted. Microsoft reserves the right

to remove any IP address allow list rules that violate the preceding guidelines without notice. You might not be able to use Remote Desktop to connect to your environment for one of these reasons:

- Your current IP address isn't in the allow list.
- Your IP address changed from the IP address listed in the allow list.
- Microsoft deleted the rule containing your IP address from the allow list because it violated a guideline.

To regain access to the environment, add the IP address of the computer you're connecting from. To do this, complete the steps in the Remote Desktop section earlier in this article.

When does the availability of reduced regions go into effect for new onboarding?

Starting August 1, 2020, onboard new projects for finance and operations to the following regions:

- East US
- West US
- Central US

  > **Important**
  >
  > Central US is no longer an option for self-service migrations starting April 1, 2021.

- East US2
- West US2
- West Central US
- North Central US
- South Central US

This change doesn't affect any environments that store their data in the deprecated regions before August 2020. In the near future, there's a transition plan to move customers in the deprecated regions into the reduced regions.

I can't redeploy an environment after deleting it because the environment slot is missing

This issue happens when your license expires. You no longer have the minimum required licenses to get an environment slot. Review your subscription status and reactivate the expired license to enable the redeployment.

What does it mean when some deployment regions options are marked as Data resident region

If you select a region that isn't **Data resident**, you see a warning that the selected region for environment deployment isn't the same as where Lifecycle Services stores the project data.

For example, when you use the global Lifecycle Services endpoint, the project data is stored in the United States. The available regions that show Data Resident include:

- brazilsouth
- uscentral
- useast
- uswest

> **Note**
>
> Brazil South is only included in this list due to business and disaster recovery scenarios. Environments deployed to Brazil South might fail over to US Central in the event of a disaster. Environments deployed in US regions only fail over to other US regions in the event of a disaster.

For information about migrating Lifecycle Services project data to a different geography, see Project migration manager.

Last updated on 04/03/2026

System requirements for cloud deployments

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article lists the system requirements for the current version of finance and operations apps. Verify that your system meets or exceeds the minimum network, hardware, and software requirements.

Supported web browsers

Users can access the apps by using the most recent versions of these popular browsers:

- Microsoft Edge (recommended: Chromium-based Microsoft Edge)
- Google Chrome
- Apple Safari
- Internet Explorer 11 (deprecated, not recommended)

To find the latest release for each web browser, go to the software manufacturer's website.

> **Note**
>
> For optimal performance and an optimal experience, use the latest version of a modern browser, especially Microsoft Edge.
>
> **Version 10.0.21 and later:** Users of old versions of Microsoft Edge and Google Chrome (version 83 and earlier) receive prompts to update their browser to the latest version.

Internet Explorer deprecation

Support for Internet Explorer 11 was deprecated in December 2020, and support for the browser ended in August 2021. For more information, see Internet Explorer deprecation announcement.

Starting in version 10.0.20, users accessing finance and operations apps with Internet Explorer see notifications about the end of support for that browser. Before August 17, 2021, Internet Explorer users saw an informational message that Internet Explorer support was soon ending. After that date, Internet Explorer users saw a warning that support officially ended. Organizations should keep these notifications on unless Internet Explorer is mandated for your users. If Internet Explorer is mandated, you can choose to suppress these notifications by disabling the **Internet Explorer end-of-support notifications** feature and relying on internal processes for migrating your user base to Microsoft Edge or another modern browser.

Starting in version 10.0.25, the use of Internet Explorer 11 is blocked in finance and operations apps. If your organization wants to block Internet Explorer earlier, and you're using version 10.0.21 or later, contact Microsoft Support.

To prepare organizations and users for the upcoming block of Internet Explorer, in January 2022, Internet Explorer users started to receive a non-dismissible error message that states that Internet Explorer support will soon be blocked. This error message is **not** controlled by the **Internet Explorer end-of-support notifications** feature. Customers must contact Microsoft Support if this message must be suppressed for their organization.

Special considerations

- To enable Task Recorder to capture screenshots and include them in Microsoft Word documents that it generates, you must install a prerelease Chrome extension.
- The Workflow Editor and Report Designer for Financial reporting start as ClickOnce applications. They require a 64-bit-compatible operating system. Only Microsoft Edge and Internet Explorer (on a supported version of Microsoft Windows) support ClickOnce applications out of the box. If you're using Chrome, you must install a ClickOnce extension, such as Meta4 to use ClickOnce applications. If you use Chrome in incognito mode, make sure that the ClickOnce extension is also enabled for incognito mode.
- To preview PDF files, use browsers such as Microsoft Edge (latest publicly available version) on Windows 10, or Google Chrome (latest publicly available version) on Windows 10, Windows 8.1, Windows 8, Windows 7, or Google Nexus 10 tablet.

Network requirements

- The app is designed for networks that have a latency of 250–300 milliseconds (ms) or less. This latency is the latency from a browser client to the Microsoft Azure datacenter that hosts the app. Test network latency at AzureSpeed.com.
- Bandwidth requirements for the app depend on your scenario. Most typical scenarios require a bandwidth that is more than 50 kilobytes per second (KBps). However, you need

more bandwidth for scenarios that have high payload requirements, such as scenarios that involve workspaces or extensive customization.

In general, the app is optimized for the internet. The number of round trips from a browser client to the Azure datacenter is very small, and the whole payload is compressed.

> ⚠ **Warning**
>
> Don't calculate bandwidth requirements from a client location by multiplying the number of users by the minimum bandwidth requirements. The concurrent usage of a given location is very difficult to calculate. If you're concerned about bandwidth requirements, use a preview version of the app.

.NET Framework requirements

The app requires the Microsoft .NET Framework version 4.6.2 for all ClickOnce applications, such as the document routing agent. For installation instructions, see Install the .NET Framework for developers.

Supported Microsoft Office applications

The following Microsoft Office applications are supported in the cloud:

- To run the Microsoft Excel and Word add-ins, you must have Microsoft Office 2016 for Windows installed. For more information about version requirements, see Troubleshoot the Office integration.
- To view documents that are generated by the Export to Excel or Export to Word functionality, you must have Microsoft Office 2007 or later installed.

Requirements for development on local VMs

For information about the requirements for development on local virtual machines (VMs), see VM that is running locally.

Database collation

The only supported collation for application databases in the cloud is **SQL_Latin1_General_CP1_CI_AS**. Make sure that your SQL Server and database collations in

development environments are set to this collation. Also ensure that any configuration environments that you publish to Sandbox use this same collation.

Related information

Get evaluations copies

Last updated on 03/09/2026

On-premises deployment home page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can deploy Dynamics 365 Finance + Operations (on-premises). When you choose an on-premises deployment type, the system requirements, hardware sizing, and functionality differ from a cloud deployment. This article provides links to content that contains information specific to on-premises deployments.

Get started

- On-premises deployment overview
- Plan and prepare for on-premises deployments
- System requirements for on-premises deployments
- Microsoft Dynamics 365 Finance + Operations (on-premises), Microsoft Dynamics 365 Finance, and Microsoft Dynamics 365 Supply Chain Management supported software
- Hardware sizing requirements for on-premises environments
- Buy Finance + Operations (on-premises)
- Comparison of cloud and on-premises features

Onboard

- Set up on-premises projects in Lifecycle Services
- Configure the infrastructure scripts for your Finance + Operations (on-premises) deployment
- Set up and deploy on-premises environments (Application 10.0.32 and later)
- Set up and deploy on-premises environments (Platform update 41 through 55)
- Install network printer devices in on-premises environments
- Configure SQL Server Reporting Services for on-premises deployments
- Develop and deploy custom models to on-premises environments

Work in your on-premises deployment

- Configure high availability for SQL Server Reporting Services (SSRS) nodes
- Configure document management
- Import Electronic reporting (ER) configurations
- Document generation, publishing, and printing in on-premises deployments
- Configure proxies for on-premises environments
- Set up technical support for finance and operations apps
- Client internet connectivity
- Apply updates to on-premises deployments
- Redeploy on-premises environments
- Reuse the same AD FS instance for multiple environments

Commerce

- Commerce capabilities that are available in on-premises deployments
- Installation steps for Retail channel components in an on-premises environment
- Configure, install, and activate Modern POS (MPOS)
- Configure and install Commerce Scale Unit

Upgrade

- In-place upgrade process for on-premises environments

Other resources

- Troubleshoot on-premises deployments
- Scripts for resolving issues in on-premises environments
- Certificate rotation
- On-premises diagnostics
- Removed or deprecated features for finance and operations
- Software lifecycle policy and on-premises releases

Last updated on 04/02/2026

System requirements for on-premises deployments

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article lists the system requirements for the current version of Microsoft Dynamics 365 Finance + Operations (on-premises) deployments. Verify that the system that you're working with meets or exceeds the minimum network, hardware, and software requirements.

> **Important**
>
> Dynamics 365 Finance + Operations (on-premises) is not supported on any public cloud infrastructure, including Microsoft Azure Cloud services. However, it is supported to run on Microsoft Azure Stack HCI and Microsoft Azure Stack Hub.

Network requirements

Dynamics 365 Finance + Operations (on-premises) can work on networks that use Internet Protocol Version 4 (IPv4) or Internet Protocol Version 6 (IPv6). Consider the network environment when you plan your system, and use the following guidelines.

Network response time

The following table lists the minimum network requirements for the connection between the web browser and Application Object Server (AOS), and for the connection between AOS and the database in an on-premises system.

Expand table

Value	Web browser to AOS	AOS to database
Bandwidth	50 kilobytes per second (KBps) per user	100 megabytes per second (MBps)
Latency	Less than 250–300 milliseconds (ms)	Less than 1 ms (local area network [LAN] only). AOS and the database must be co-located.

- Finance + Operations is designed for networks that have a latency of 250–300 milliseconds (ms) or less. This latency is the latency from a browser client to the datacenter that hosts Finance + Operations.
- Bandwidth requirements depend on your scenario. Typical scenarios require a bandwidth of more than 50 KBps between the browser and the server. However, we recommend higher bandwidth for scenarios that have high payload requirements, such as scenarios that involve workspaces or extensive customization. The specific amount of bandwidth depends on use.

Deployments where AOS and the Microsoft SQL Server database are in different datacenters aren't supported. AOS and the SQL Server database must be co-located.

In general, Finance + Operations is optimized to reduce browser-to-server round trips. The number of round trips from a browser client to the datacenter is either zero or one for each user interaction, and the payload is compressed.

> ⚠ **Warning**
>
> Don't calculate bandwidth requirements from a client location by multiplying the number of users by the minimum bandwidth requirements. The concurrent usage of a given location is very difficult to calculate. We recommend that you use a real-life simulation against a non-production environment as the best gauge of performance for your specific case.

LAN environments

In LAN environments, Microsoft Remote Desktop in Microsoft Windows Server isn't required in order to connect to Finance + Operations. However, Remote Desktop might be required for servicing operations on the virtual machines (VMs) that make up the server deployments.

WAN environments

In wide area network (WAN) environments, Remote Desktop in Windows Server isn't required in order to connect to Finance + Operations.

Internet connectivity requirements

Finance + Operations doesn't require internet connectivity from user workstations. However, some features won't be available if there is no internet connectivity.

Expand table

Browser client	An intranet scenario without internet connectivity is a design point for the on-premises deployment option. Some features that require cloud services won't be available, such as Help and Task guide libraries in Microsoft Dynamics Lifecycle Services.
Server	The orchestrator node types must be able to communicate with Lifecycle Services for deployment and servicing operations. The on-premises browser-based client doesn't require internet access.
Telemetry	Telemetry data might be lost if there are long interruptions in connectivity. Interruptions in connectivity to Lifecycle Services don't affect the on-premises application functionality.

Telemetry data transfer to the cloud

Most telemetry data is stored locally and can be accessed by using Event Viewer in Microsoft Windows. A small subset of telemetry events are transferred to the Microsoft telemetry pipeline in the cloud for diagnostics. Customer data and user-identifiable data aren't part of the telemetry data that is sent to Microsoft. VM names are sent to Microsoft to help with environment management and diagnostics from the Lifecycle Services portal.

Domain requirements

Consider the following domain requirements when you install Finance + Operations:

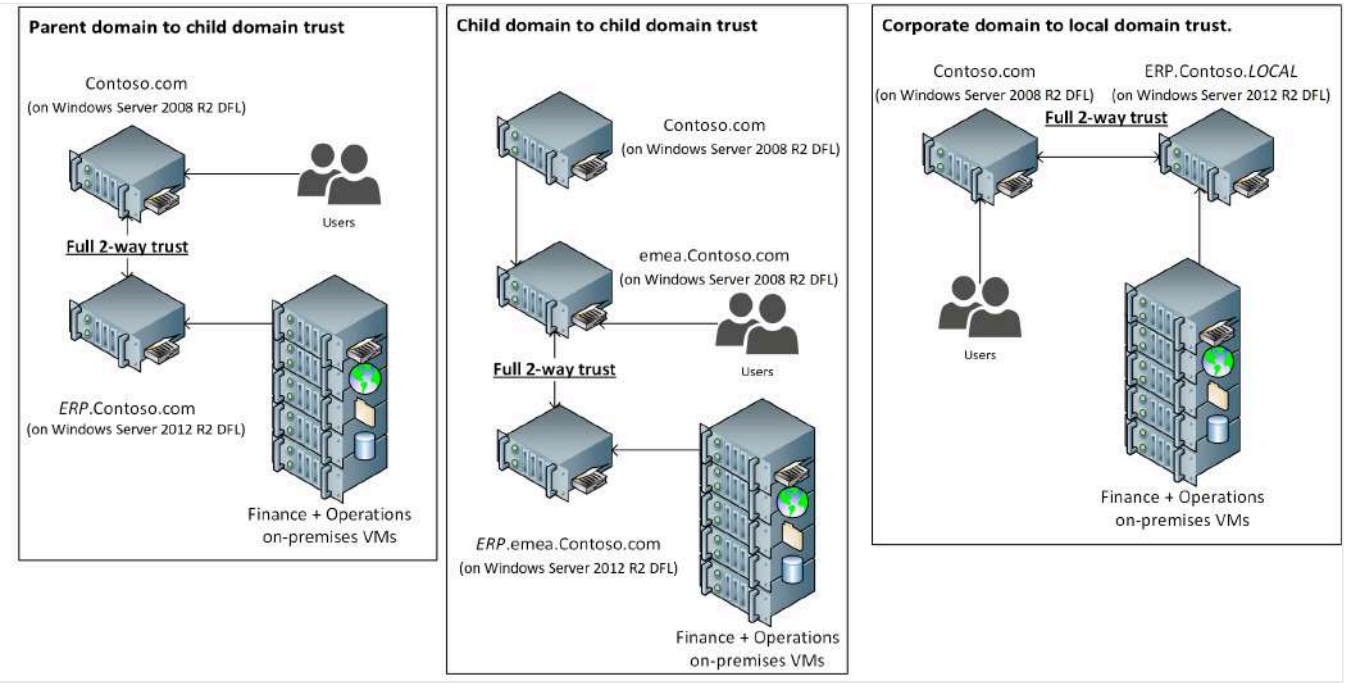
- VMs that host Finance + Operations components must belong to an Active Directory domain. Active Directory Domain Services (AD DS) must be configured in native mode.
- VMs that run Finance + Operations components must have access to each other. This access is configured in AD DS.
- The domain controller must be Microsoft Windows Server 2012 R2 or later, and the domain functional level must be 2012 R2 or more.

Full 2-way trust

For compatibility with corporate domain controllers on Windows Server 2008 R2 domain functional level (DFL), a full 2-way trust between the Windows Server 2008 R2 DFL user domain and the Windows Server 2012 R2 DFL Finance + Operations service domain is supported.

This means that users of the Finance + Operations (on-premises) application will come from the Windows Server 2008 R2 DFL domain, and the resources and service accounts hosting the Finance + Operations (on-premises) infrastructure and services will come from the Windows Server 2012 R2 DFL domain.

Examples for a full 2-way trust setup could be.



Known limitations with using the full 2-way trust setup

- Import of security groups from the Windows Server 2008 R2 user domain is not supported.

Hardware requirements

This section describes the hardware that is required in order to run Finance + Operations.

The actual hardware requirements vary, based on the system configuration, the data composition, and the features that you decide to use. Here are some of the factors that can affect the choice of appropriate hardware:

- The number of transactions per hour

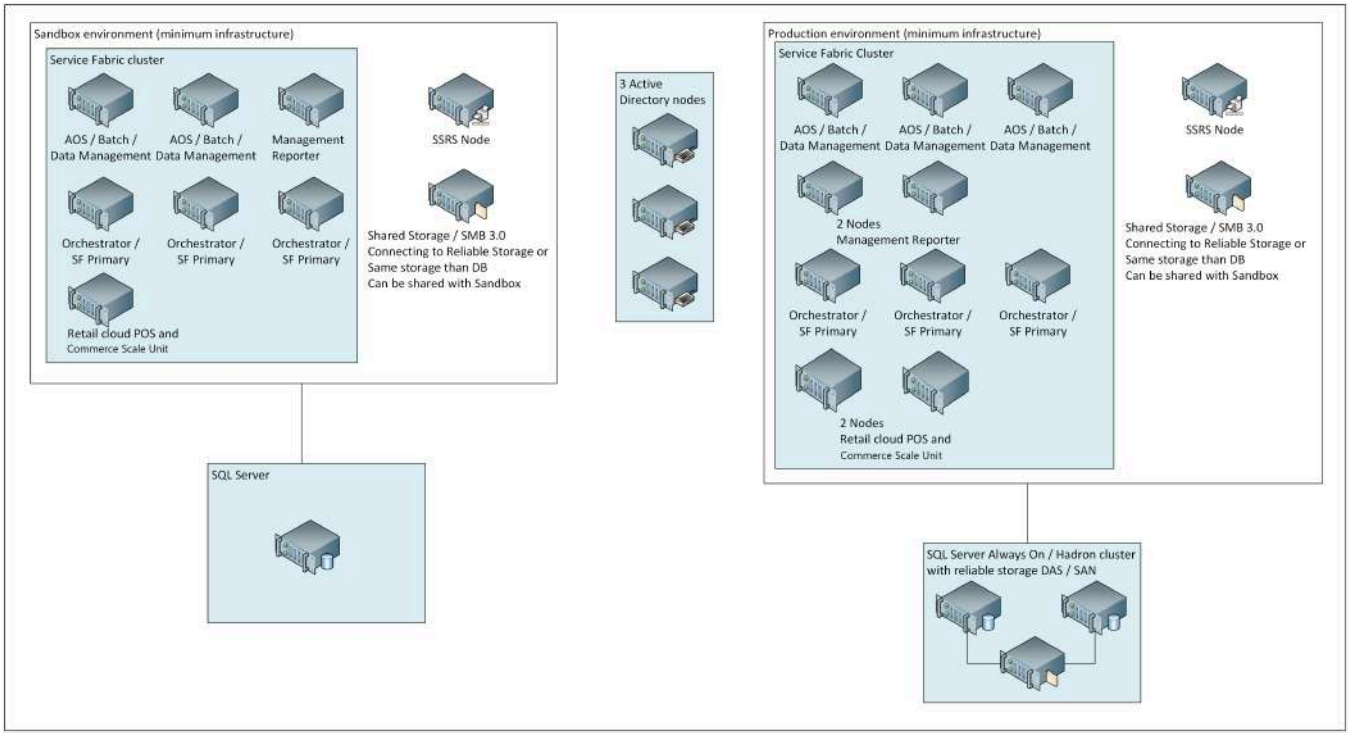
- The number of concurrent users
- Integration loads

Minimum infrastructure requirements

Finance + Operations uses Service Fabric to host the AOS, Batch, Data management, SQL reporting, Management reporter, and Environment orchestrator services.

SQL Server must have a high-availability HADRON setup that has at least two nodes for production use.

The following illustration shows the minimum number of nodes that is recommended for your Service Fabric cluster.



Processor and RAM requirements

The following tables list the number of processors and the amount of random-access memory (RAM) that are required for each role that is required in order to run this deployment option. For more information, see the recommended minimum requirements for a Service Fabric standalone cluster in Plan and prepare your Service Fabric cluster.

> **Note**
>
> If other Microsoft software is installed on the same computer, the system must also comply with the hardware requirements for that software. If other server applications are

installed on the same computer as AOS, we recommend that you limit those server applications 1 gigabyte (GB) of RAM.

Sizing by role and topology type

Expand table

Topology	Role (node type)	Recommended processor cores	Recommended memory (GB)
Production	AOS, Data management, Batch	8	24
	SSIS	4	16
	Management Reporter	4	16
	SQL Server Reporting Services	4	16
	Orchestrator	4	16
	SQL Server	8	32
Sandbox	AOS, Data management, Batch	4	24
	SSIS	4	16
	Management Reporter	4	16
	SQL Server Reporting Services	4	16
	Orchestrator	4	16
	SQL Server	8	32

Minimum sizing estimates for production and sandbox deployments

Expand table

Topology	Role	Number of instances
Production	AOS (Data management, Batch)	3
	SSIS	2

Topology	Role	Number of instances
	Management Reporter	2
	SQL Server Reporting Services	1
	Orchestrator**	3
	SQL Server	2
Sandbox	AOS, Data management, Batch	2
	SSIS	1
	Management Reporter	1
	SQL Server Reporting Services	1
	Orchestrator	3
	SQL Server	1
Summary for production and sandbox topologies		*22*

* The recommended hardware sizes are general guidelines only. Actual hardware requirements may vary based on specific customer processes, customizations, transaction volumes, and workloads. We strongly recommend that customers conduct thorough testing of any proposed configurations as part of a comprehensive performance evaluation exercise.

** Orchestrator is designated as the primary node type and will also be used to run the Service Fabric services.

Initial estimates for the back-end SQL Server and AD DS

I was unable to convert this HTML table due to its complexity (use of rowspan and colspan attributes). You can try converting it with another AI model.

Expand table

	Role	VMs/instances	Cores	Total cores	Memory per instance (GB)	Total memory (GB)
Shared infrastructure	SQL Server*	2	8	16	32	64

	Role	VMs/instances	Cores	Total cores	Memory per instance (GB)	Total memory (GB)
	File server/Storage area network/Highly available storage	The back-end storage must be based on solid-state drives (SSDs) on a runtime storage area network (SAN). Size and input/output operations per second (IOPS) throughput is based on the size of the workload.				
	Active Directory	3	4	12	16	48
Summary for shared infrastructure		*5*		*28*		*112*

* SQL Server sizes are highly dependent on workloads. For more information, see Hardware sizing requirements for on-premises environments. Separate SQL Server machines for sandbox and production environments must be used. However, SQL Server can be shared in all sandbox environments.

Storage

- **AOS & SSIS** – Finance + Operations uses a Server Message Block (SMB) 3.0 share to store unstructured data. For more information, see Storage Spaces Direct in Windows Server 2016.
- **SQL** – The following options are viable:
  - A highly available SSD setup
  - A SAN that is optimized for online transaction processing (OLTP) throughputs
  - High-performance direct-attached storage (DAS)
- **SQL Server and data management IOPS** – The storage for both data management and SQL Server should have at least 2,000 IOPS. Production IOPS depends on many factors. For more information, see Hardware sizing requirements for on-premises environments.
- **VM IOPS** – Each VM should have at least 100 write IOPS.

Virtual host requirements

When you set up the virtual hosts for an environment, see the guidelines in Plan and prepare your Service Fabric cluster and Describing a service fabric cluster. Each virtual host should have enough cores for the infrastructure that is being sized. Multiple advanced configurations are

possible, where SQL Server resides on physical hardware but everything else is virtualized. If SQL Server is virtualized, the disk subsystem should be a fast SAN or the equivalent. In all cases, make sure that the basic setup of the virtual host is highly available and redundant. In all cases, when virtualization is used, no VM snapshots should be taken.

> **Important**
>
> Do not use Dynamic Memory for your virtual hosts.

Finance + Operations falls under Microsoft's standard support policy regarding operation on non-Microsoft virtualization platforms – specifically VMware. For more information, read Support policy for Microsoft software. In short, we support our products in this environment, but if we are asked to investigate an issue, we may ask the customer to first reproduce the problem without the virtualization platform or on the Microsoft virtualization platform.

Software requirements for all server computers

The following software must be present on a computer before any Finance + Operations components can be installed:

- The Microsoft .NET Framework. See Deployment setup for version information.
- Service Fabric

For more information about Service Fabric, see Plan and prepare your Service Fabric cluster.

> **Note**
>
> For supported versions, see Microsoft Dynamics 365 Finance + Operations (on-premises), Microsoft Dynamics 365 Finance, and Microsoft Dynamics 365 Supply Chain Management supported software.

Software requirements for database servers

For the hardware requirements for SQL Server, see Hardware and Software Requirements for Installing SQL Server. |

- Only 64-bit versions of SQL Server are supported.
- Only **SQL_Latin1_General_CP1_CI_AS** is valid for the server and database collation. For more information about how to select a collation for a SQL Server database, see the SQL

Server documentation.

- In a production environment, we recommend that you install the latest cumulative update (CU) for the version of SQL Server that you're using.

Software requirements for Application Object Server (AOS)

- [Optional] SQL Server Integration Services (SSIS)
  - If you are using a dedicated SSIS node, you don't need to install SSIS on the AOS nodes.

Software requirements for Reporting Server (BI)

- SQL Server Reporting Services (SSRS)

Software requirements for SQL Server Integration Services (SSIS)

- SQL Server Integration Services (SSIS)

Software requirements for client computers

Users can access Finance + Operations by using the most recent versions of these popular browsers:

- Microsoft Edge (recommended: Chromium-based Edge)
- Google Chrome
- Apple Safari

> **Note**
>
> For optimal performance and an optimal experience, we recommend that you use the latest version of a modern browser, especially Microsoft Edge.

Special considerations

- To enable Task Recorder to capture screenshots and include them in Microsoft Word documents that are generated, you must install a pre-release Chrome extension.
- The Workflow Editor and Report Designer for Financial reporting are started as ClickOnce applications. They require a 64-bit-compatible operating system. Only Microsoft Edge

and Internet Explorer (on a supported version of Microsoft Windows) support ClickOnce applications out of the box. If you're using Chrome, you must install a ClickOnce extension, such as Meta4 to use ClickOnce applications. If you use Chrome in incognito mode, make sure that the ClickOnce extension is also enabled for incognito mode.
- To preview PDF files, we recommend that you use browsers such as Microsoft Edge (latest publicly available version) on Windows 10, or Google Chrome (latest publicly available version) on Windows 10, Windows 8.1, Windows 8, Windows 7, or Google Nexus 10 tablet.

Software requirements for Active Directory Federation Services

Active Directory Federation Services (AD FS) on Windows Server is required.

The domain controller must be Windows Server 2012 R2 or later, and the domain functional level must be 2012 R2 or more. For more information about domain functional levels, see the following pages:

- What Are Active Directory Functional Levels
- Understanding Active Directory Domain Services Functional Levels
- Full 2-way trust

Supported Microsoft Office applications

The following Microsoft Office applications are supported in on-premises deployments:

- To run the Microsoft Excel and Microsoft Word add-ins, you must have Microsoft Office 2016 for Windows (or newer) installed. For more information about version requirements, see Troubleshoot the Office integration.
- To view documents that are generated by the Export to Excel or Export to Word functionality, you must have Microsoft Office 2007 or later installed.

Hardware and software requirements for Commerce components

Currently, Finance + Operations doesn't include the Commerce components.

Last updated on 04/10/2026

Buy Finance + Operations (on-premises)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft Dynamics 365 Finance + Operations (on-premises) offers various license types to best suit the needs of your organization. To better understand how Finance + Operations (on-premises) is licensed, work with your partner, who can access the Licensing guide. When you're ready to purchase licenses for your organization, work with your partner to follow the steps that are outlined in this article.

> **Important**
>
> On-premises environments aren't supported on any public cloud infrastructure, including Azure. However, they support running on Microsoft Azure Stack HCI and Microsoft Azure Stack Hub.

Purchase a Dynamics 365 cloud subscription license

Thanks to the dual use rights that are mentioned in the Licensing guide, you can use a Dynamics 365 cloud subscription license to deploy Finance + Operations (on-premises).

For information about how to purchase a subscription, see Buy and manage a subscription. After you have a subscription, follow these steps to create your on-premises implementation project.

Get started with Lifecycle Services

When you purchase a subscription, you assign it to your Microsoft Online Services ID. You need the Microsoft Online Services ID to provision and register Microsoft Dynamics 365 applications. Use the Microsoft Online Services ID to provision a Microsoft Dynamics Lifecycle Services project that contains the necessary artifacts for on-premises environments. Lifecycle Services is the service where you provision on-premises environments, create and upload business

process models, make hotfixes available, enter and manage support cases, validate subscriptions, and so on.

Before you can create your Lifecycle Services project and begin creating on-premises environments, follow the Provisioning guide to obtain an on-premises license.

If you have an existing Microsoft Online Services trial or paid subscription, you already have a Microsoft Online Services ID that you created at the time of sign-up. When you sign in to Lifecycle Services, select to use this account for sign-in if you want to use the same Microsoft Entra tenant for the on-premises environment.

After you sign in to Lifecycle Services, a cloud project is automatically provisioned for you. However, you need an on-premises project. To create a new on-premises project, follow the Start a new Lifecycle Services project guide and select the **On-premises implementation** project type. The Lifecycle Services project lets you deploy an on-premises environment. For more information about how to get started with your Lifecycle Services project, see Set up on-premises projects in Lifecycle Services.

Purchase a Dynamics 365 Finance + Operations (on-premises) license

Purchase client access licenses

To run on-premises environments, you must obtain the proper number of client access licenses (CALs) for your organization per the licensing guide. The CAL you purchase for an individual user determines the functionality that the user has the rights to use. You can purchase user CALs from the Microsoft Volume Licensing Service Center.

Purchase server licenses

You need a server license for every server that runs Finance + Operations (on-premises). After you purchase your server licenses, work with your partner to download a license file from the PartnerSource Business Center. Keep this license file handy, because you use the details that it contains when you set up your Lifecycle Services project.

Partners can download a customer's Finance + Operations (on-premises) license file from the PartnerSource Business Center by using these steps:

1. Sign in to the PartnerSource Business Center.
2. Enter the customer name or account number in the **Find A Customer** field, and then select **Search**.

3. Select the company name of the customer. The **Customer Summary** page opens.
4. Under **Registered Products**, select **Registration Keys**.
5. Select **version 07** in the **Request and Display License Keys For Version** field.
6. Select **Display License Keys**.
7. On the **Request License Keys** page, select **Download Current License/Registration Key**.
8. Select **Save As** in the **File Download** dialog box, select the folder where you want to download the license file to in the **Save As** dialog box, and then select **Save**.

> **Note**
>
> If you can't see registration keys in PartnerSource Business Center, make sure that **Can See Registration Keys** is set to **Yes** in your PartnerSource Business Center profile.

Get started with Lifecycle Services

To purchase Finance + Operations (on-premises), you must have a Microsoft Online Services ID. Use the Microsoft Online Services ID to provision a Lifecycle Services project that contains the necessary artifacts for on-premises environments. In Lifecycle Services, you can provision on-premises environments, create and upload business process models, make hotfixes available, enter and manage support cases, submit license key serial number activations, and more.

You need a Microsoft Online Services ID to provision and register Finance + Operations (on-premises) into entity-owned hardware and environment. See the Provisioning guide to complete the provisioning and registration process. If a Microsoft Online Services ID already exists, the tenant administrator must complete the process. If you're creating a Microsoft Online Services ID for the first time, you're the tenant administrator.

If you have an existing Microsoft Online Services trial or paid subscription, you already have a Microsoft Online Services ID that you created at the time of sign-up. Choose to sign in with this account if you want to use this same Microsoft Entra ID tenant for the on-premises environment. For details, see the Provisioning guide.

After you sign in to Lifecycle Services, a project is automatically provisioned for you. The Lifecycle Services project lets you deploy an on-premises environment. For more information about how to get started with your Lifecycle Services project, see Set up on-premises projects in Lifecycle Services.

Last updated on 03/28/2026

Help system

Summarize this article for me

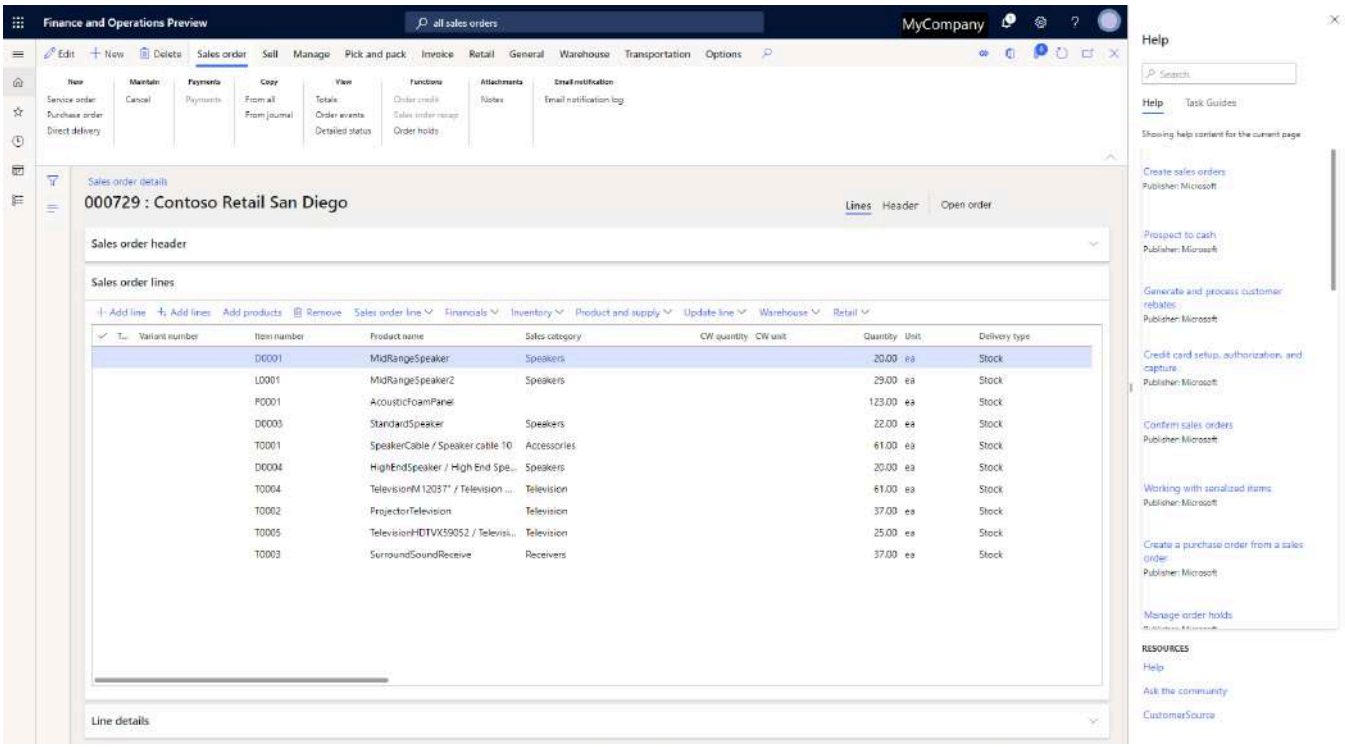
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Customers using Human Resources, the functionality noted in this article is currently available in both the stand-alone Dynamics 365 Human Resources and the merged Finance infrastructure. Navigation might be different than noted while we make updates. If you need to find a specific page, you can use Search.

Users of the following apps can access context-sensitive Help and other content that is based on the same Help system:

- Dynamics 365 Commerce
- Dynamics 365 Finance
- Dynamics 365 Human Resources
- Dynamics 365 Supply Chain Management

In all these apps, you can access product-specific Help from the **Help** pane.



Help on Microsoft Learn

Microsoft publishes content for the Dynamics 365 apps on the Microsoft Learn site. Get an overview at Microsoft Dynamics 365 documentation. This site offers the following features:

- **Access to the most up-to-date content** – The site gives Microsoft a faster and more flexible way to create, deliver, and update product documentation. Therefore, you have easy access to the latest technical information.
- **Content that is written by experts** – Content on the site is open to contributions by community members both inside and outside Microsoft.

You can find content on Microsoft Learn by using any search engine. For the best results, we recommend that you use a site search, such as **site:learn.microsoft.com dynamics 365 "search term"**.

Find other content

If you land on an article that doesn't answer your question, you can find other content in two ways on Microsoft Learn:

- Use the **Filter by title** field in the table of contents on the left side of the website

  This way, you can filter by words used in the titles of the various articles. For example, type *invoice*, and then choose one of the offered links, such as Apply a payment schedule to the invoice journal or Vendor invoices overview.

Underneath each link, you can see where that article is in the table of contents. That way, you can easily see if the article is in the *Accounts payable* section or in the section for *accounts receivable*, for example.

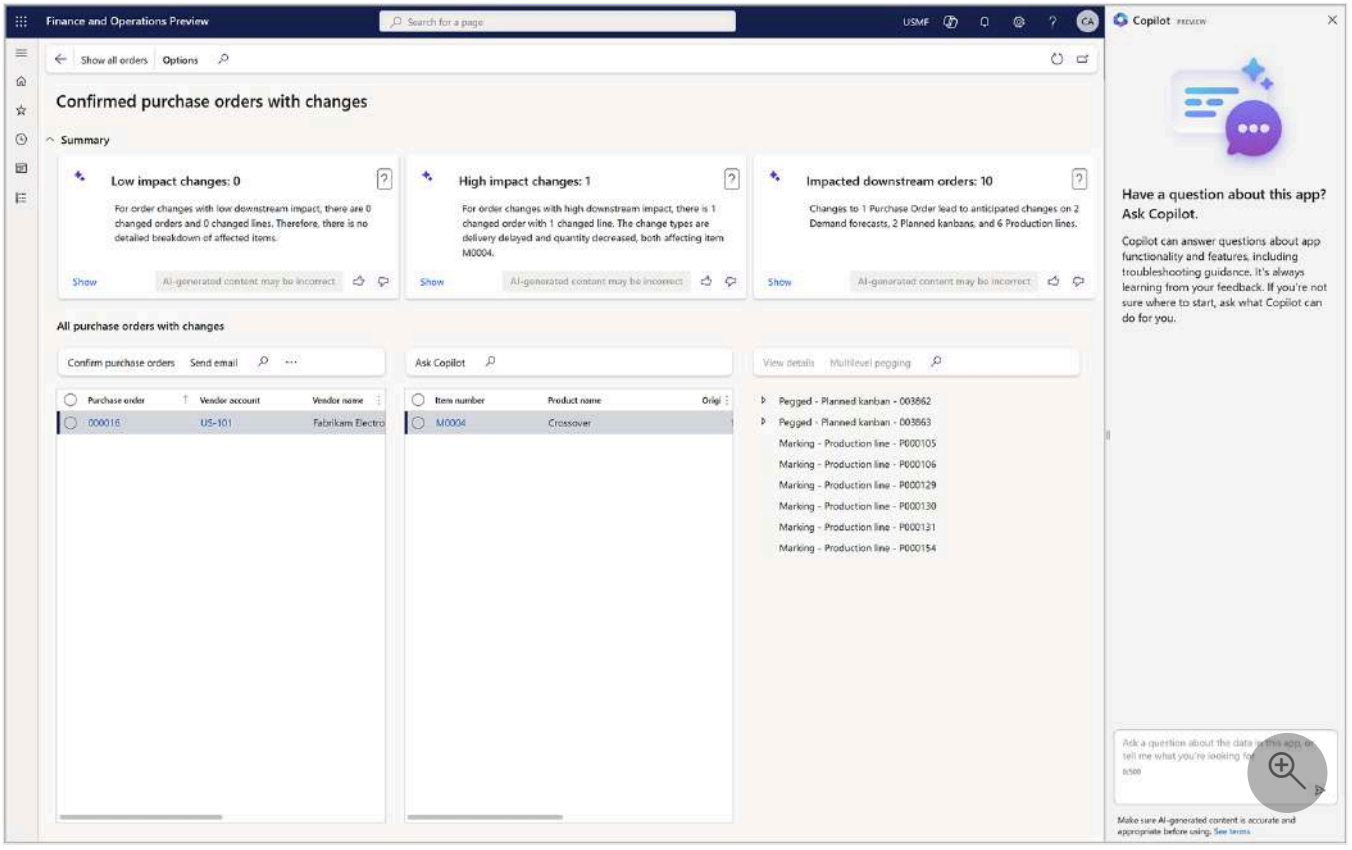
- Use the **Search** field in the top-right corner of the website

  This way, you can search for any term, and you can choose to search for that term in the documentation for a different Dynamics 365 app, for example. The site then searches all the content for that term or expression. Search for *invoice* again, and scroll through the list of hundreds of articles that uses that word somewhere in the text.

Generative help and guidance with Copilot

Generative help and guidance with Copilot provides a conversational guide that's always nearby and ready to help. Users can ask questions in natural language. Copilot then identifies the relevant content in the documentation and composes a targeted response to the question.

After generative help and guidance with Copilot is enabled for your system, users can open the Copilot sidecar by using the **Copilot** button at the top of the page.



For more information, see Generative help and guidance with Copilot.

Get notified about changes through an RSS feed

To subscribe to an RSS feed of all updates made to the content on Microsoft technical documentation across the finance and operations apps, use the following link:

RSS feed

> **Note**
>
> The RSS feed returns a list of the 100 topics most recently updated. The list is sorted by date, but it can take up to a week before the most recently updated articles make it to the list.

Alternatively, you can subscribe to an RSS feed by app:

- Commerce
- Finance
- Human Resources
- Supply Chain
- Talent

Leave us feedback

On the *learn.microsoft.com* website, each documentation article encourages you to provide feedback. We've changed the mechanism a couple of times based on, well, feedback. Currently, we ask you to choose the **Feedback** action below the article's title. You can give the article a rating, and write a comment if you want to. The feedback goes straight to the article's author and the team that owns the docs.

Contribute to the documentation

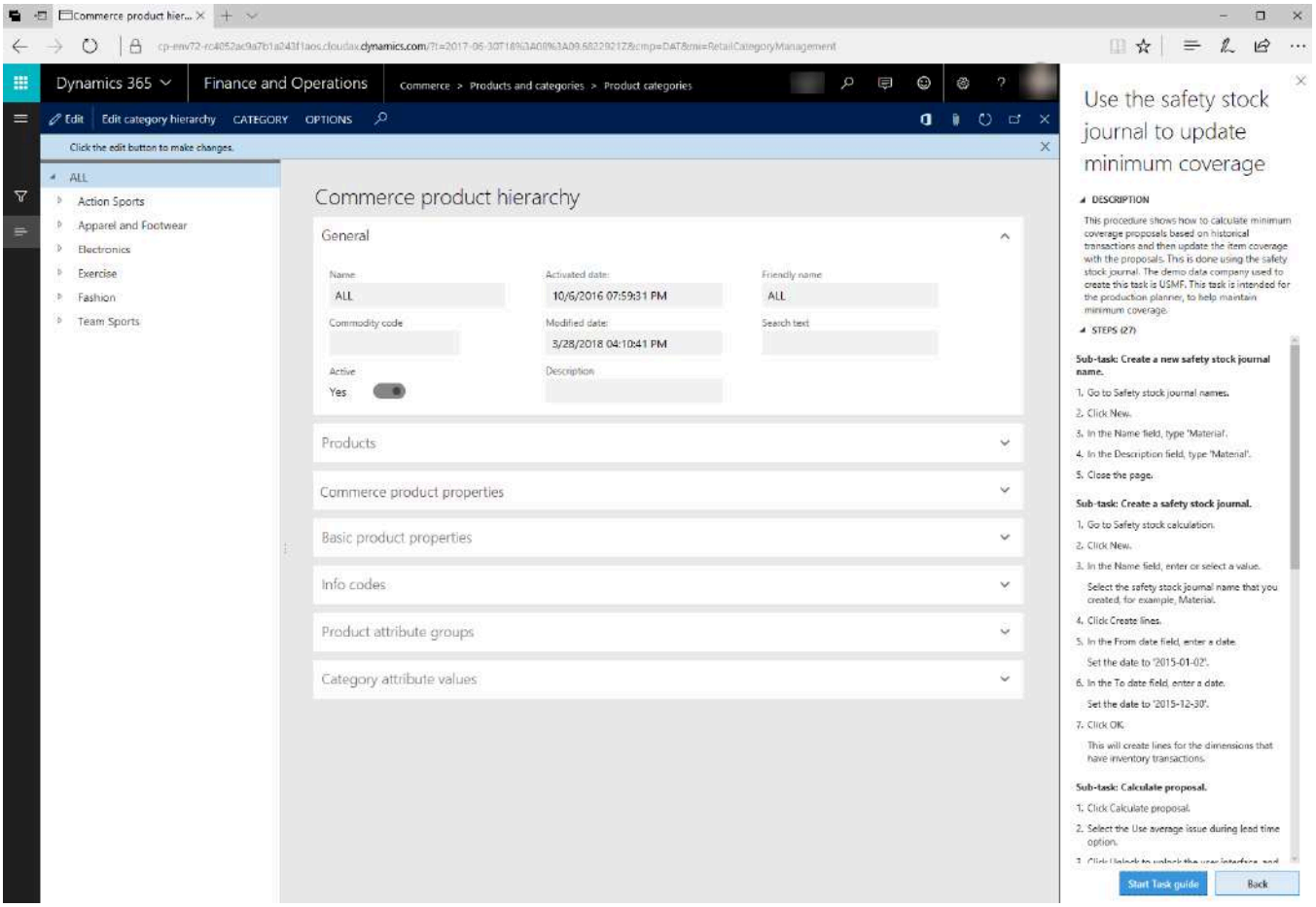
You can contribute and make edits to the documentation. To get started, select the **Edit** button (pencil symbol) on an article. Learn more in the article Contribute to Dynamics 365 documentation.

> **Note**
>
> We only accept contributions to our English content at this time.

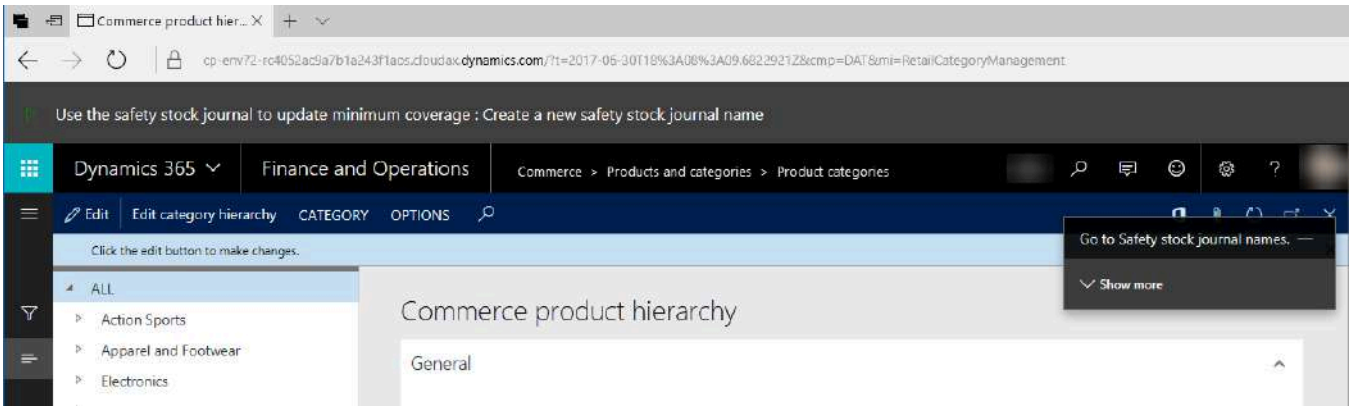
Task guides

A task guide is a controlled, guided, interactive experience that leads you through the steps of a task, or business process. You can open (play) a task guide from the **Help** pane. When you first select a task guide, the **Help** pane will show the step-by-step instructions for the task. Localized task guides are available.

Microsoft released task guide libraries for product versions through the December 2017 release of Dynamics 365 Finance and Operations. The Accessing task guides from the Help pane section of this article explains how to find the correct task guides for your product.



To begin the guided, interactive experience, select **Start task guide** at the bottom of the **Help** pane. A black pointer shows you where to go first. Follow the instructions that appear in the user interface (UI), and enter data as directed.



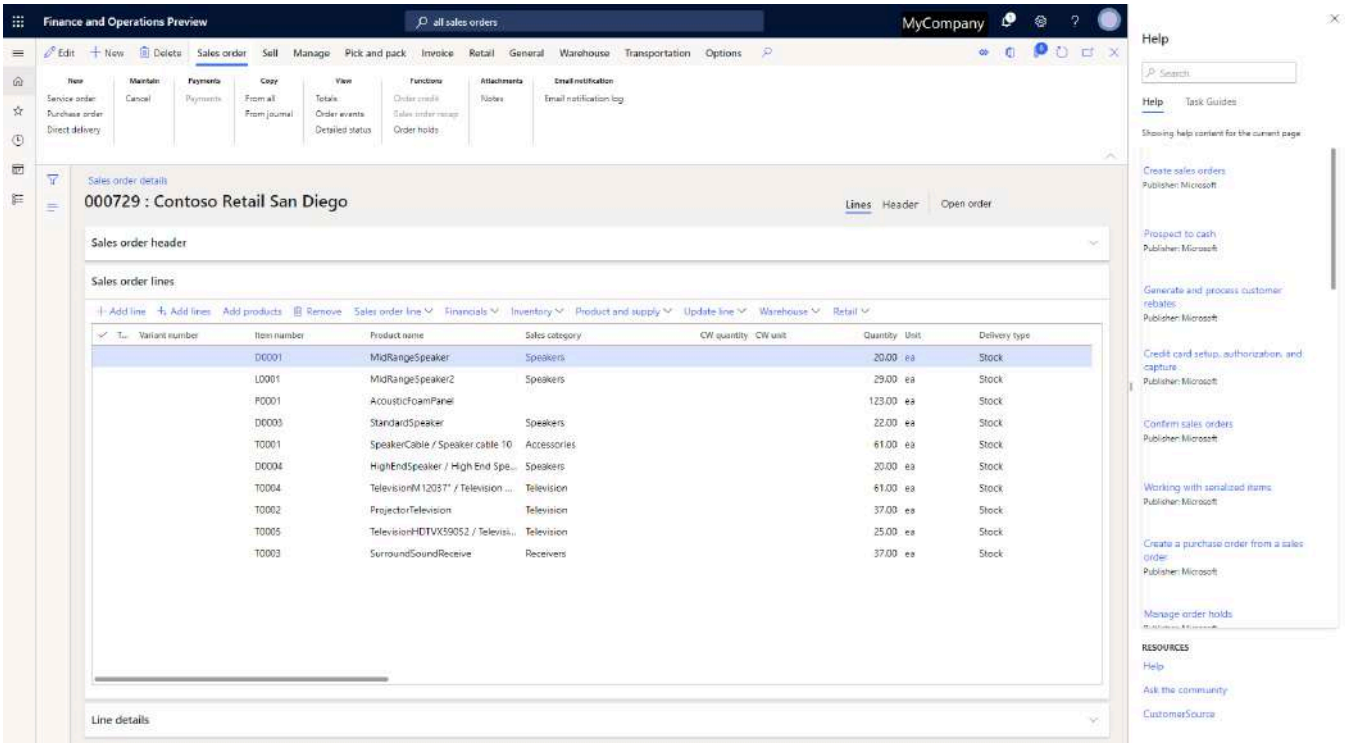
> **ⓘ Important**
>
> The data that you enter when you play a task guide is real. If you're in a production environment, the data will be entered in the company that you're currently using.

You can use Task recorder to create your own custom task guides. For more information, see Create documentation or training with Task Recorder.

In-product Help

Some fields have field descriptions to help users get unblocked when they are uncertain about the data that the field contains, for example. Additionally, the in-product **Help** pane provides context-sensitive access to content that can help users get started, get unblocked, and learn more.

To access Help content, select the **Help** button (**?**), and then select **Help**. Alternatively, press **Ctrl+Shift+?**. In both cases, the **Help** pane appears. From the **Help** pane, you can access conceptual topics or task guides that are relevant to the area of the product that you're currently in.



Accessing Help topics from the Help pane

From the **Help** pane, you can access topics that apply to the client. When you first open the **Help** pane, the **Help** tab shows the topics that apply to the page that you're currently on. If no

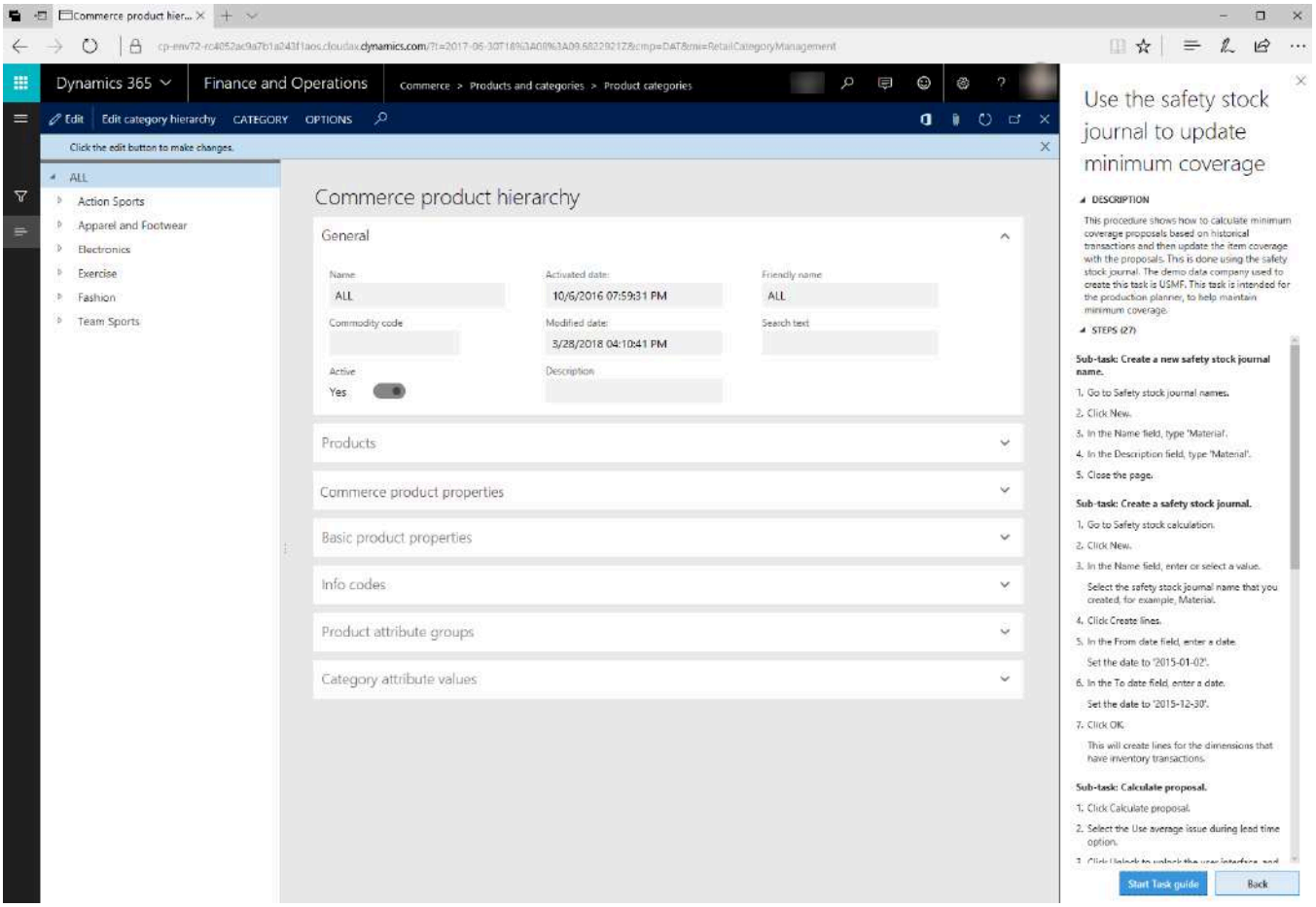
topics are found, you can enter keywords to refine your search. When you select an article in the **Help** pane, it's opened on a new tab in your browser.

> **Important**
>
> This section does not apply to Dynamics 365 Human Resources. The Help system for Human Resources is automatically connected to task guides for the product. Also, you cannot create custom task guides for Human Resources.

Accessing task guides from the Help pane

Before you can access task guides from the **Help** pane, a system admin must configure some settings on the **System parameters** page in Finance, Supply Chain Management, or Commerce. For more information, see Adding task guides.

After a System administrator has completed these steps, you can open the **Help** pane and select the **Task guides** tab. You'll now see the task guides that apply to the page that you're currently on. If no task guides are found, you can enter keywords to refine your search. After you select a task guide in the **Help** pane, the **Help** pane shows the step-by-step instructions, and you can play the task guide.



Where are the translated task guides for Microsoft libraries?

Translated task guides are released in libraries that have "All languages" in the title. To view localized task guide Help, make sure that you're connected to an appropriate library. Each user can change the language that a task guide appears in by changing the language settings under **Options** > **Preferences**.

- If a task guide has been translated, when you open that task guide all the text of the task guide will appear in your selected language.
- If a task guide has not yet been translated, when you open it, only the text of the controls will appear in your chosen language.

Creating custom Help

You can create Help for your users by creating custom task guides or connecting your own website to the **Help** pane. For more information, see the following topics:

- Task recorder resources
- Custom Help overview

Additional resources

The following table lists our websites. Sites that have an asterisk (*) next to the name require that you sign in by using an account that is associated with a service plan.

Expand table

Site	Description
Microsoft Dynamics 365 documentation	This site hosts or links to all product documentation for Dynamics 365.
Microsoft Learn training	This site is the free Microsoft eLearning site.
Microsoft Dynamics Lifecycle Services (LCS) *	This site provides a cloud-based collaborative workspace that customers and partners can use to manage projects from pre-sales to implementation and operations. It's useful in all phases of an implementation.
Previous versions	This site hosts content from previous releases.
Dynamics Community	This site hosts blogs, forums, and videos.
Microsoft.com/dynamics365	This site provides evaluation and sales information.

Site	Description
The Dynamics 365 guidance hub	This site hosts or links to resources that can help with Dynamics 365 implementation projects.

Last updated on 03/10/2026

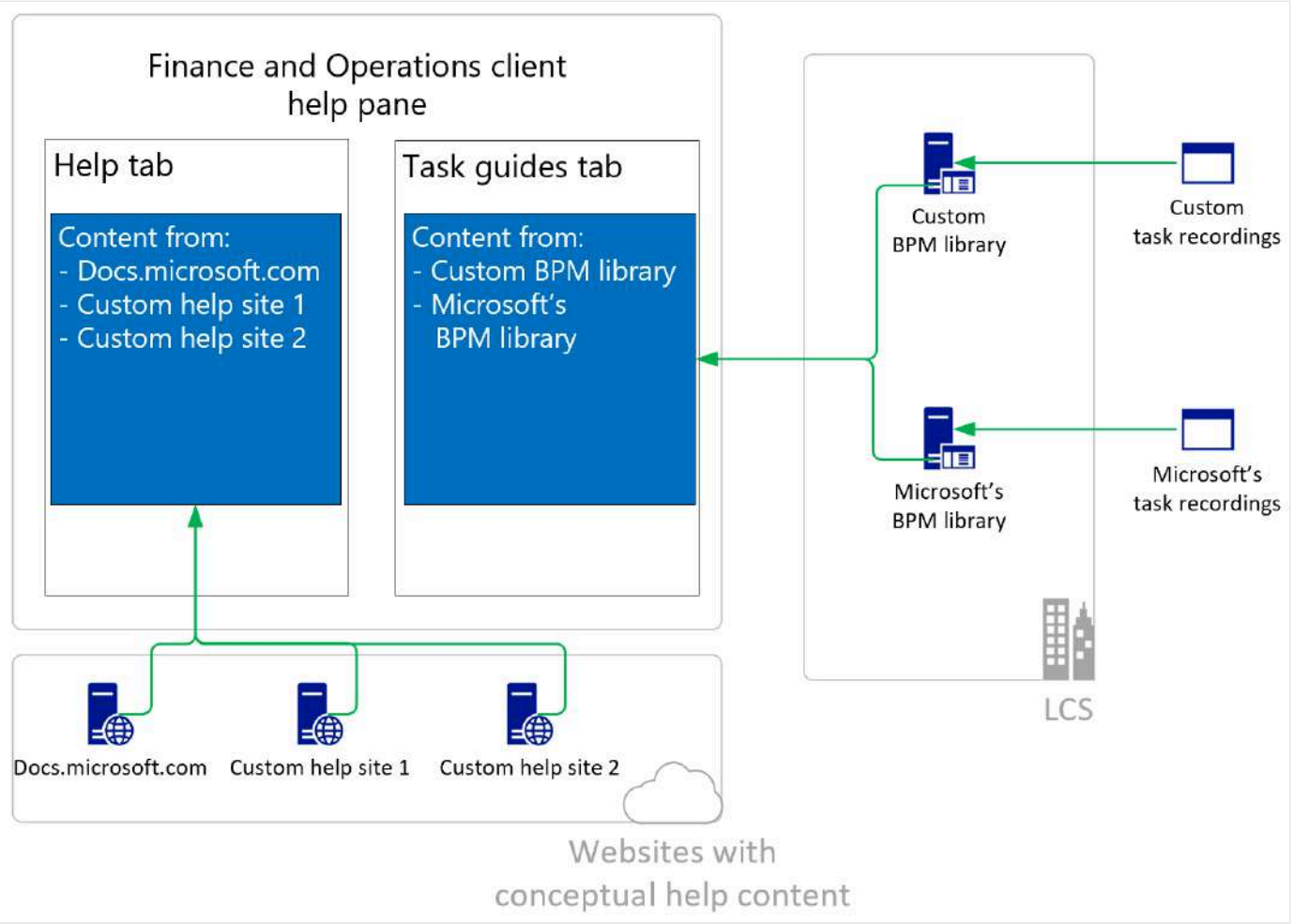
Configure the Help experience for finance and operations apps

> **Important**
>
> Customers using Human Resources, the functionality noted in this article is currently available in both the stand-alone Dynamics 365 Human Resources and the merged Finance infrastructure. Navigation might be different than noted while we make updates. If you need to find a specific page, you can use Search.

This article provides an overview of the components of the Help system for solutions with Dynamics 365 Finance, Dynamics 365 Supply Chain Management, Dynamics 365 Commerce, or Dynamics 365 Human Resources. The article also explains how to connect these components.

Help architecture

Finance and operations apps include conceptual overviews and other articles that are published to the Microsoft Learn site. You can access this content from the in-product **Help** pane. The following illustration shows the parts of the Help system.



The in-product Help system pulls articles from Microsoft Learn and other connected websites. It also pulls in task guides that are stored in Business process modeler (BPM) in Microsoft Dynamics Lifecycle Services.

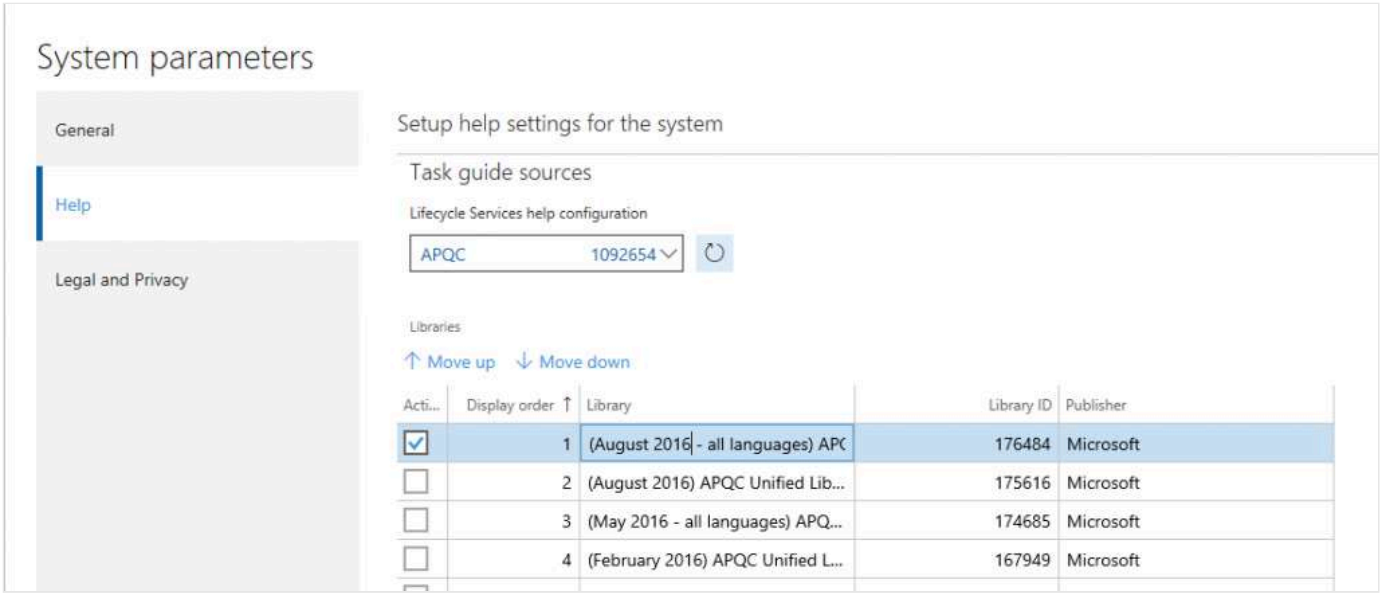
Adding task guides

> ⓘ Note
>
> The **Task guides** tab isn't currently available in Human Resources or Commerce. However, the task guides in the Getting Started experience in Human Resources remain available to cover basic functionality. For both Human Resources and Commerce, procedural Help is available on the Microsoft Dynamics 365 documentation site.

On the **System parameters** page, system admins can configure access to the relevant task guide libraries for an implementation.

> ⓘ Note

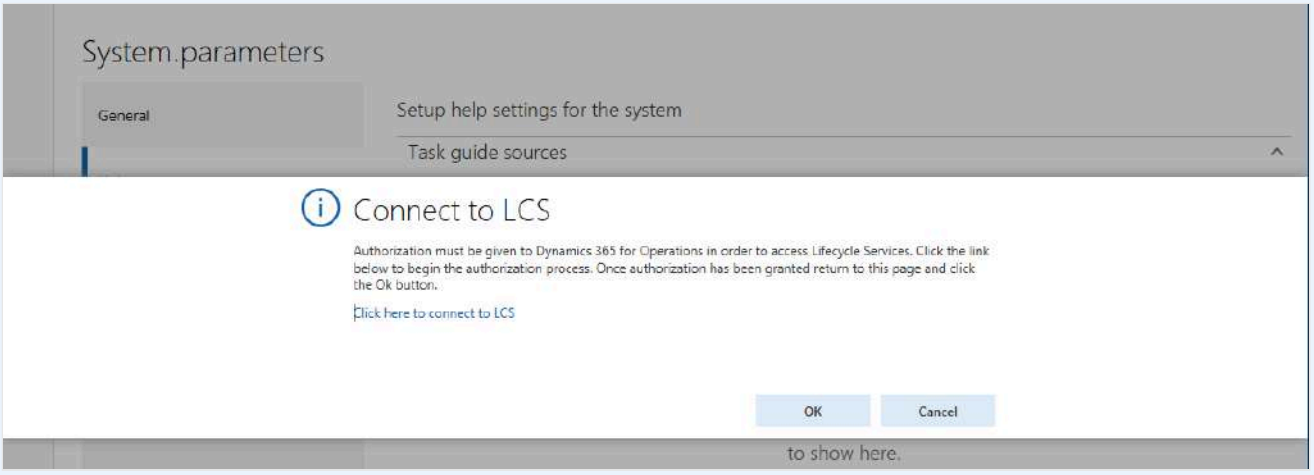
- To configure Help, you must sign in by using an account in the same tenant as the tenant where the app is deployed.
- You can't connect an Lifecycle Services library from an instance of the app that runs on a local virtual hard drive (VHD).



To configure task guides for a solution, follow these steps on the **System parameters** page.

Important

The first time that you open the **Help** tab, you must connect to Lifecycle Services. Be sure to select the link in the middle of the form, wait for the connection, close the dialog box, and then select **OK** to get to the **System Parameters** page.



1. Select the Lifecycle Services project to connect to.
2. Select the BPM libraries (within the selected project) to retrieve task recordings from.

3. Set the display order of the BPM libraries. The display order defines the order in which task recordings from the libraries appear in the **Help** pane.

After you complete these steps, you can open the **Help** pane and select the **Task guides** tab. You see the task guides that apply to the page that you're currently on in finance and operations apps. If no task guides are found, you can enter keywords to refine your search.

Showing translated task guides

Translated task guides were first released in the May 2016 APQC Unified Library and in the Getting Started library. To view localized task guide Help, make sure that your Dynamics 365 solution is connected to the May 2016 library. Users can change the language that a task guide appears in by changing the language settings under **Options** > **Preferences**.

> ⓘ **Note**
>
> Although many task guides are translated, the client doesn't currently show the translated task guide names. Additionally, in the May 2016 library, translations are available only for the task guides that were released in February 2016. Microsoft releases an updated library that includes more translations.
>
> - If a task guide is translated, when you open that task guide all the text of the task guide appears in your selected language.
> - If a task guide isn't yet translated, when you open it, only some of the text (the text of the controls) appears in your selected language.

Adding custom Help

You can use task guides to create custom Help, or you can connect a custom Help website to the **Help** pane.

Create custom Help by using task guides

You can create custom Help for the supported apps by creating task recordings that reflect your implementation and then saving them to a Business process library in Lifecycle Services. You can't create custom task guides for Human Resources.

If you're a partner, and you promote a library to a corporate library and include it in a solution, it becomes available to your customers. You can also make a copy of the APQC Unified Library,

then open the task recordings in the copy, edit them, and save your changes. For more information, see Task recorder resources.

Connect a custom Help site

Finance and operations apps are rarely used in their out-of-box form. Instead, the solution is customized and extended to fit the organization's needs. You can also customize and extend the Help experience. For example, you can add custom Help to the in-product **Help** pane.

Microsoft provides a toolkit to help you deploy and connect custom Help to the **Help** pane. Learn more at Custom Help overview.

See also

Help system
Custom Help overview
Task recorder resources
Create documentation or training with Task Recorder
Custom Help GitHub repository

Last updated on 04/13/2026

Custom Help overview

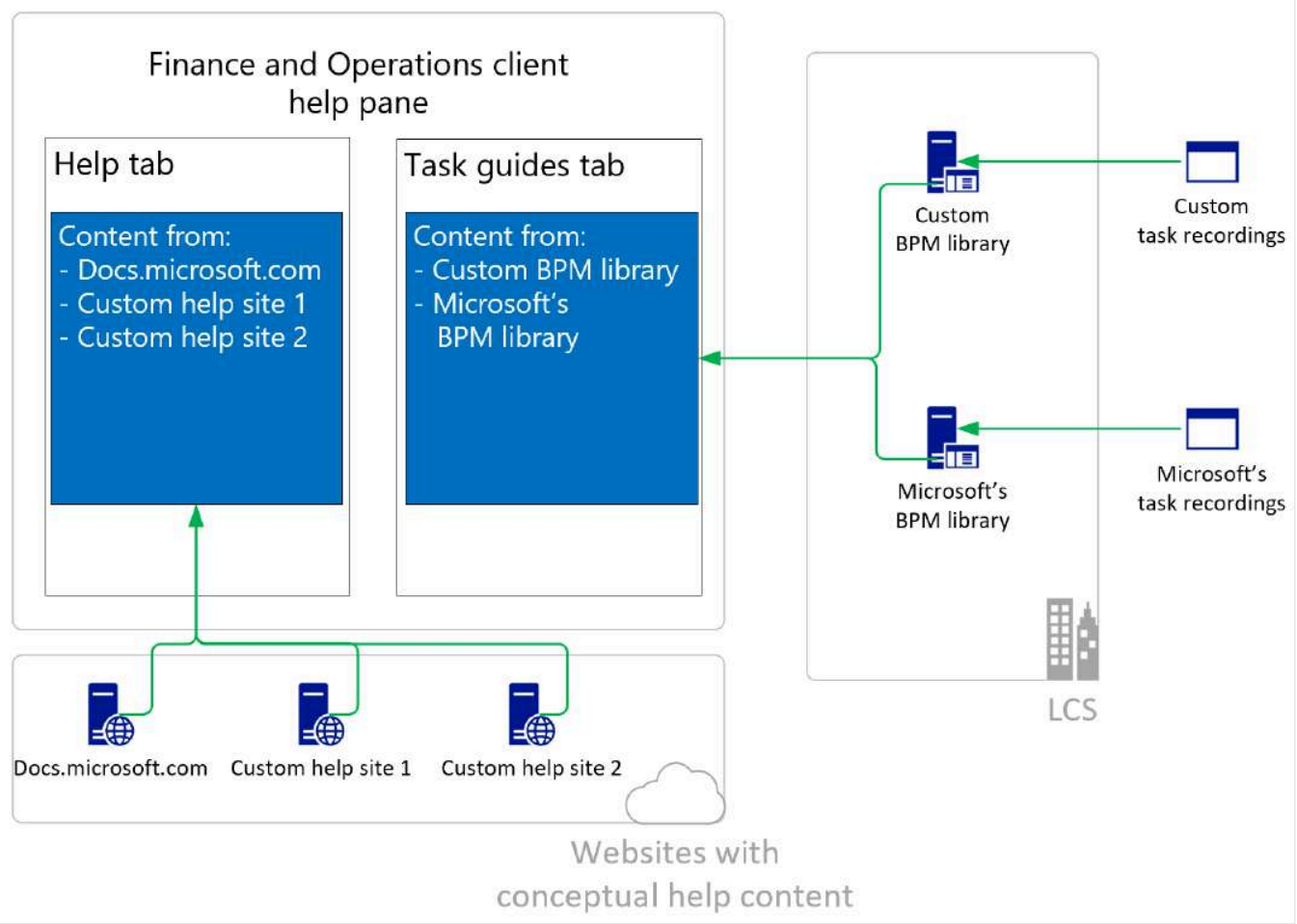
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Customers using Human Resources, the functionality noted in this article is currently available in both the stand-alone Dynamics 365 Human Resources and the merged Finance infrastructure. Navigation might be different than noted while we make updates. If you need to find a specific page, you can use Search.

Organizations often customize and extend finance and operations apps to fit their needs. If your solution is based on Microsoft Dynamics 365 Finance, Dynamics 365 Supply Chain Management, or Dynamics 365 Commerce, you can connect solution-specific and customer-specific Help content to the Help pane in the finance and operations client. This article describes the main steps and decision points.

> **Note**
>
> Users of finance and operations apps can create custom task guides to supplement conceptual content that describes the functionality of their solution. Microsoft, partners, and an organization itself can provide these conceptual descriptions, which are also referred to as Help. For more information, see Help system.

The following illustration, and this article in general, use the term *Help* for conceptual descriptions that either include or exclude how-to guides. The term *task guides* refers to in-product task guides.



Custom Help content

Custom Help content typically comes from one of three sources:

- Microsoft documentation repositories (repos)

  Microsoft's content in the various GitHub repos is optimized for the learn.microsoft.com site and the tools that are used for this site. It's not intended to be customized directly but to be supplemented by articles on your local website. However, depending on your solution, you might need a copy of Microsoft's content.

- Existing customized Dynamics AX content

  You can convert Dynamics AX custom Help content so that you can use it in Dynamics 365.

- HTML files that you create specifically for your solution

  Learn more about the metadata that must be added to your HTML files for context-sensitive Help and search to work correctly in the Metadata requirements for custom Help files section.

Process

The end-to-end process depends on the customer's solution and the users' expectations. A typical process involves the following steps:

1. Create the custom Help content.
2. Publish the content on a website.
3. Index the content by using a search service.
4. Connect the custom **Help** pane to the website and the search service.

Microsoft provides sample tools and scripts that can help you convert content and configure the Help pane to include your custom Help. Find the samples at https://github.com/microsoft/dynamics365f-o-custom-help/ .

The following table outlines the main objectives that admins typically have for configuring the Help experience.

Expand table

Objective	Learn more
I want to give my users a customized in-product Help experience that reflects their actual solution.	Go to the Custom Help websites section of this article and Create documentation or training with Task Recorder.
I want to add my content to the **Help** pane.	Go to Connect a custom Help website to the Help pane.
I want to reuse content that was originally created for Dynamics AX.	Go to the Reuse Dynamics AX 2012 content section.
I want to reuse content in one language for another, related language.	Go to the Language and locale descriptors in Help article.
I want to contribute to the Microsoft Help content.	Go to Contribute to the Help.

Custom Help websites

Before the product can connect to your Help content, you must customize the in-product **Help** pane so that it shows your content. For more information, see Connect a custom Help website to the Help pane.

The following conditions must be met:

- Your content must be available on a website.

You can deploy your content to an existing website, or you can set up a dedicated website to host your content. The website can be private or public, but don't require users to sign in to access your content.

- A search service must index your content.

  Learn more in the Generate JSON files for the search service section.

- The content must include the right metadata properties.

  Learn more in the section.

Metadata requirements for custom Help files

To ensure context-sensitive Help and full-text search return results, include the following metadata in your topics.

Expand table

Property	Description
title	The value for full-text search from the **Help** pane.
description	The value for full-text search from the **Help** pane.
ms.search.form	The value contains the Application Object Tree (AOT) name of a page and is used for context-sensitive search from the **Help** pane.
ms.locale	The value indicates the language of the article. It's mapped against the current browser locale when the **Help** pane searches the content. You can configure language fallback for the target custom Help website. For more information, see Language and locale descriptors in the product and in Help.
ms.search.scope	The value determines which client the Help article shows in. You can specify one or more values. Values include **Core**, **Operations**, **Retail**, and **Human Resources**.

The following table describes the values that you can specify for the **ms.search.scope** property. You can specify one or more values. The values determine which client a Help article shows in.

Expand table

Value	Description
Core	If this value is present, the article appears in the **Help** pane. Otherwise, the article doesn't appear in the **Help** pane. This value is set for the part of the Microsoft content that must always be available in the **Help** pane, for all users across all supported Dynamics 365 solutions.

Value	Description
Operations	This value applies to solutions that are based on Finance or Supply Chain Management.
Retail	This value applies to solutions that are based on Commerce.
Human Resources	This value applies to solutions that are based on Dynamics 365 Human Resources.

Non-required metadata

The following properties are reserved for future use:

- **ms.search.region** – Eventually, this property might be used to limit the content that is shown to content that is tagged either as global or for the region of the implementation.
- **ms.search.validFrom** – Eventually, this property might be used to limit the content that is shown to content for a product that was released on a given date or earlier.
- **ms.dyn365.ops.version** – Eventually, this property might be used to limit the content that is shown to content for a specific version of a product or earlier.
- **ms.search.industry** – Eventually, this property might be used to limit the content that is shown to content for a specific industry.

Generate JSON files for the search service

The **ConvertHtmlToJson** tool transforms HTML files into JSON files. You can then add the JSON files to the Microsoft Azure Search service, which generates context-sensitive links to your Help content. Find the tool at https://github.com/microsoft/dynamics365f-o-custom-help/.

The JSON files include metadata that the indexer uses to identify the form and language that the target Help page is intended for.

Here's the syntax for running ConvertHtmlToJson.exe.

```
ConvertHtmlToJson.exe --h <path> -j <path> --v <true|false>
```

The following table explains the parameters.

Expand table

Parameter	Description
h	Specify the path of the HTML files to process.
j	Specify the folder to save the JSON files to. The folder must already exist.
v	Set this parameter to **true** to turn on verbose logging. Otherwise, set it to **false**.

The following example generates JSON files without verbose logging.

```
ConvertHtmlToJson.exe --h D:\D365-Operations\d365F-O\supply-chain\de -j D:\D365-Operations\json\supply-chain\de
```

Reuse Dynamics AX 2012 content

If you have content from Dynamics AX 2012, you can reuse it for Dynamics 365 Finance, Dynamics 365 Supply Chain Management, and Dynamics 365 Commerce. However, you must first transform the HTML files so that you can use them in the custom Help environment.

The **run_ax2012.ps1** Windows PowerShell script transforms AX 2012 HTML files so that you can use them in the custom Help environment. The script makes the following changes to the AX 2012 HTML files:

- Replaces the **Microsoft.Help.F1** metadata name with **ms.search.form**.
- Replaces the **Title** metadata name with **title**.
- Changes the file name extension from **.htm** to **.html**.
- Adds the following metadata.

HTML

```
<meta name="ms.search.region" content="Global" />
<meta name="ms.search.scope" content="Operations, Core" />
<meta name="ms.dyn365.ops.version" content="AX 7.0.0" />
<meta name="ms.search.validFrom" content="2016-05-31" />
<meta name="ms.search.industry" content="cross" />
```

Run the script

Run the following command from a Command Prompt window, or directly in Windows PowerShell.

PowerShell

```
PowerShell.exe -File run_ax2012.ps1 "Original file location" "New file location"
```

The following metadata is used, or it's reserved so that it can be used during indexing.

HTML

```
<meta name="title" content="Title of file" />
<meta name="ms.locale" content="locale" />
<meta name="ms.search.form" content="FormAOTName" />
<meta name="description" content="Description of file" />
<meta name="ms.search.region" content="Global" />
<meta name="ms.search.scope" content="Operations, Core" />
<meta name="ms.dyn365.ops.version" content="AX 7.0.0" />
<meta name="ms.search.validFrom" content="2016-05-31" />
<meta name="ms.search.industry" content="cross" />
```

For a description of the required metadata, go to the Metadata requirements for custom Help files section.

Convert HTML content to Markdown

To convert your content to Markdown, use a third-party tool. Microsoft doesn't provide a tool that can convert HTML to Markdown.

After you've converted your content to Markdown, use an open-source tool to generate content for your website. Markdown gives you access to a wide range of open-source tools. For more information, go to Contribute to the Help.

See also

Connect a custom Help website to the Help pane
Language and locale descriptors in the product and in Help
Configure the Help experience for finance and operations apps
Help system

Last updated on 03/28/2026

View and export field descriptions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Customers using Human Resources, the functionality noted in this article is currently available in both the stand-alone Dynamics 365 Human Resources and the merged Finance infrastructure. Navigation might be different than noted while we make updates. If you need to find a specific page, you can use Search.

This article describes how to view field descriptions and how to use the **Field descriptions** page to export descriptions.

Some of the more complex fields have field descriptions. These descriptions appear when you hover over a field. You can also view and export descriptions on the **Field descriptions** page.

Not all pages have field descriptions. They're available only for the more complex fields, not where the use of the field is obvious. Therefore, only the complex pages have a lot of field descriptions. The remaining pages either have a few field descriptions or none.

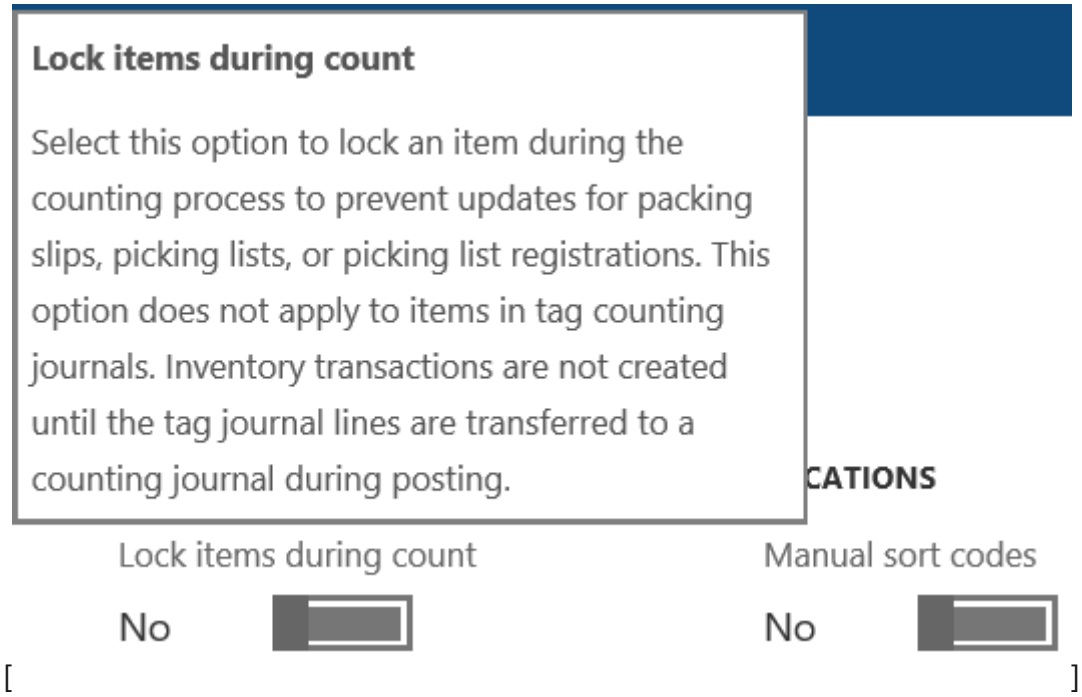
If you have access to the development environment, you can add new field descriptions and customize existing descriptions. For example, you can add company-specific information to a field description. For more information, see Customize field descriptions.

See field descriptions in the user interface

You can view field descriptions by hovering over a field. If no description is available, you see the field name when you hover over the field.

> **Note**
>
> In Dynamics AX 7.0 (February 2016), you can view field descriptions only on the **Field descriptions** page.

The following illustration shows the field description that appears when you hover over the **Lock items during count** field.



[ ]

Use the Field descriptions page to view and export field help

The **Field descriptions** page lets you view and export field descriptions. You can see the descriptions that are available for one page at a time.

View the descriptions for a page

To view the descriptions for a page, follow this step.

- In the **Select a page** field, type the name of the page. Alternatively, select the arrow to open a list of all the pages, and then browse or filter the list.

You can use either the name of the page that appears in the user interface (UI) (for example, **Customers**) or the code name (AOT name) that's available when you right-click a page (for example, **CustTable**).

For information about the various ways to filter the list of pages, see the "Searching for a page" section later in this article.

If you set the **Include fields without a description** option to **Yes**, the page shows all the fields, even if they don't have a field description.

Export the descriptions for a page

To export the descriptions for a page, follow these steps:

1. In the **Select a page** field, select a page.
2. Select the **Open in Microsoft Office** button in the upper-right corner, and then select **FieldDescriptionTmp**.

Searching for a page

You can search for a page in the **Select a page** field in several ways. In many cases, you need to select the arrow in the **Select a page** field to open the drop-down list, and then select from a filtered list of pages.

- Type part of the name, and then open the drop-down list to select from a filtered list of pages.
- Open the drop-down list, and then select either the **Page name** heading at the top of the list or the **Page AOT name** heading. A dialog box appears, where you can use advanced filtering options, such as **Page name begins with**.
- Type the full name of the page. When you use this option, it's best to open the drop-down list and see what else is in the list, even if field descriptions are shown.
  - If there's a single exact match to the name, the field descriptions for that page are shown.
  - If there are multiple exact matches, no descriptions are shown. You need to open the drop-down list and select the page that you want.
  - If the name that you typed is part of the name of another page, you see the descriptions for your page. However, if you open the drop-down list, you see additional pages that contain that name.

For example, no descriptions are shown when you type **Counting** in the **Select a page** field. You open the drop-down list, and see that there are two pages that have the name **Counting** and several pages that contain the word "Counting" in the name. If you select the page that has the AOT name **InventJournalCount**, the field descriptions are shown for that page. However, if you open the drop-down list again, the list now contains all pages that have "InventJournalCount" as part of their AOT name.

Troubleshooting

This section provides information to help you troubleshoot issues that you might encounter when you use field descriptions.

I can't find a field description

The product team is in the process of adding descriptions for the more complex fields. If you require help for a particular field, add a comment for this article.

The field description isn't helpful

Add a comment for this article. If you can, describe the additional information that you require.

I can't find a field on the Field descriptions page

To show all the fields on a page, set the **Include fields without a description** option to **Yes**. Select the **Select a page** field to verify that you selected the correct page. If the name that you typed is part of another field name, you might have selected the page that has the longer name.

I can't find a page on the Field descriptions page

For information about the various ways to find pages, see the "Searching for pages" section earlier in this article. If you type the exact name of the page, the field descriptions might not show if more than one page has the same name. Select the arrow in the **Select a page** field to open a filtered list of the available pages.

Additional resources

Customize field descriptions

Last updated on 03/17/2026

Task recorder resources

> **Important**
>
> Customers using Human Resources, the functionality noted in this article is currently available in both the stand-alone Dynamics 365 Human Resources and the merged Finance infrastructure. Navigation might be different than noted while we make updates. If you need to find a specific page, you can use Search.

This article describes how to use Task recorder to record business processes.

> **Important**
>
> Starting in January 2026, you can only create, edit, or use Task Recordings in Sandbox and Production environments. Cloud-hosted environments no longer support this capability.

Overview

Task recorder

Task recorder for finance and operations apps is a utility that lets users record business processes for several different use cases. Here are some examples:

- Step-by-step guided tours of a specific business process in the application itself
- Documentation of a business process as a Microsoft Word document that can optionally include screenshots
- Regression tests for a business process
- Automatic playback of a business process in the application

Task recorder for finance and operations apps offers high responsiveness, a flexible extensibility application programming interface (API), and seamless integration with consumers of business process recordings. Task recorder also integrates with the Business process modeler (BPM) tool in Microsoft Dynamics Lifecycle Services (LCS), so that users can continue to organize their recordings. However, users can no longer produce business process diagrams from recordings.

Task recorder can automatically generate application regression tests from business process recordings and play back previously recorded processes. These features also include test-specific gestures that let users take full advantage of Task recorder.

Architecture

Task recorder can record user actions in the client with exact fidelity, because every control is instrumented to notify Task recorder about the execution of user actions. The control notifies Task recorder that an event occurred and passes all the relevant information about the corresponding user action in real time. From this information, Task recorder can capture the type of user action (for example, a button select, value entry, or navigation) and any data that is related to the user action (for example, the input data value and type, form context, or record context). Task recorder persists the information with enough detail to ensure that a playback of the recording can run the recorded actions exactly as they were performed by the user.

Basic configuration

Every finance and operations app includes task recorder. After users open the client for the first time, they can immediately start recording business processes.

Start a new recording

The following steps show how to use Task recorder to start a new recording.

1. Open the product, and sign in. It's a good practice to refresh the browser before each new recording. A refresh creates a new user session and restarts Task recorder. Therefore, it provides the most stable recording experience.

2. Select the company that you want to use while recording. If this is your first time using Task recorder, you can follow along as this tutorial creates a sample recording based on a Fleet Management business process. You need to load the Fleet demo data to follow along:
   a. Go to **Dashboard** > **Fleet Management** > **Fleet setup**.
   b. Select **Load demo data**.
   c. When the data finishes loading, select **Close**.
   d. Go back to the **Dashboard** by selecting the product name in the navigation bar.

3. Go to **Settings** > **Task recorder**.

4. The **Task recorder** pane opens. Select the **Close** button (**X**) in the upper-right corner to close the **Task recorder** pane before you begin a new recording. You can reopen the pane by

following the previous steps.

5. Select **Create recording**.

6. Enter a name for the recording and select **Start**. Recording begins the moment **Start** is selected. For the Fleet example in this tutorial, use the name "Create a new rental reservation."

   While you're recording, you can select the **Close** button (**X**) in the upper-right corner to hide the **Task recorder** pane without stopping the recording. You can reopen the pane by selecting the **Task recorder** button that appears at the top of the page. This button appears only while recording is in progress.

   > ⓘ **Note**
   >
   > If the Saved views feature is turned on, create recordings by using either published views or the standard view, to ensure that recordings work reliably for users.

7. Task recorder enters **recording mode**. The pane shows information and controls that are related to the process of recording.

Now you're ready to record a business process by using Task recorder. If you're following this guide as a first-time user, you might complete the following Fleet Management scenario as an example. Otherwise, you can record your own application scenario.

Record a Fleet Management scenario

1. In the **Task recorder** pane, select **Start subtask**.
2. Set **Name** to "Create a new rental customer". Leave the **Comment** field blank.
3. Select **OK**. The task is added to the step list.
4. Go to **Dashboard** > **Fleet Management** > **Reservation Management**.
5. Go to **All customers** under the **Summary** tab.
6. In the Action Pane, select **New**.
7. Enter a first and last name for the customer.
8. Select **Save**.
9. In the **Task recorder** pane, select **End subtask**.
10. Return to the **Reservation Management** workspace by selecting the browser back button twice.

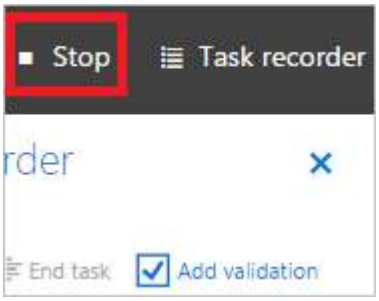
11. In the **Task recorder** pane, select **Start subtask**. Name the task "Rent a vehicle to the new customer". Select **OK**.
12. Select (+) **Rental** under **Summary**.
13. Under **Information**, select a "1975 Litware McKinley" as the vehicle.
14. Under **Information**, set the customer to the one just created.
15. Expand the **Discounts** section.
16. Select **Add** under **Discounts** and add the Frequent Customer discount. Select **OK**.
17. In the Action Pane, select **Start Rental**.
18. Set the return date to some date in the future.
19. Select **OK**.
20. In the **Task recorder** pane, select **End subtask**.
21. Select **Stop** at the top of the page.

Recording a business process

After you start your recording, you can perform your business process just as you would typically perform it by using the web client. As you interact with the product, new steps are added to the step list in the **Task recorder** pane. In this section, you learn about other actions that you can perform while you're recording a business process, to take full advantage of Task recorder's capabilities.

Stop

Use **Stop** to end the recording session. Before you select **Stop**, make sure that the recording is complete, because this action isn't reversible. When you select **Stop**, you're taken to the download options screen.



Start/End subtask

Start/End subtask lets you specify the beginning and end of a set of grouped steps in a recording. Select the **Start subtask** button to add a "Subtask" step to the end of the current list of recorded steps. The subtask includes all steps that you perform from this point until you select

the **End subtask** button. When you select the **End subtask** button, an "End subtask" step is also added to the list of recorded steps.

> **Note**
>
> You must start a subtask before you perform or record the steps that you want to include in the task. Then, after you perform or record all the steps that you want to include in the task, you must end the subtask.

Subtasks are purely an organization tool, and consumers of business process recordings can interpret the task groupings in useful ways. Because tasks can be nested inside other tasks, they provide the flexibility to organize long and complex business processes.

Delete/Restore step

Delete/Restore step enables a user to remove steps from the recording, or undo the removal of a step from the recording. You must first select the step in the Steps list that you want to delete/restore, and then select the **Delete/Restore step** button.

> **Note**
>
> The behavior of the **Delete** button changes when you play back a recording. In playback mode, a deleted step can't be restored after playback has passed the point where it would have run the deleted step. For example, you load a recording that contains three steps, and then you delete step 2 before you start playback. You can restore step 2 only as long as playback hasn't run step 3. After you start playback, and playback has "skipped" step 2 (because you deleted it) and run step 3, you won't be able to restore step 2. Because step 2 wasn't run and therefore wasn't recorded, it can't be retroactively added back into the recording at its previous position.

Add developer placeholder

Add developer placeholder lets the user add a placeholder step to the list of recorded steps. This placeholder step doesn't appear when the task guide is viewed, and it isn't run during maintenance of a recording. It's used only by the Regression suite automation tool (RSAT) or the X++ code generator that enables an X++ test to be created from a task recording. When the code generator creates an X++ test, it automatically adds a method stub to the generated code. The developer can then add X++ code into this method stub. The automated code calls the

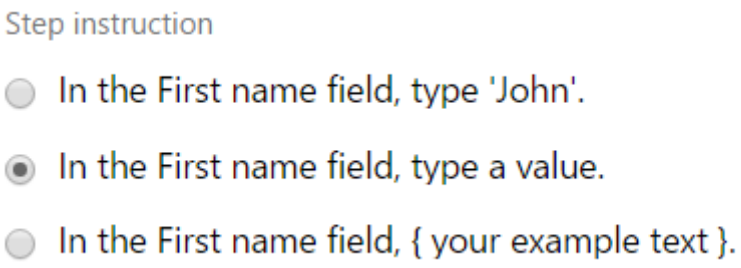
validation when the generated test is run at the point in the recording where this placeholder was added.

Enriching steps in a recording

There are various options for enriching a step in a recording. For example, you can adjust the text that is associated with a step and add information about a specific step. This section describes the step enrichment capabilities that are available. To access these options, select the **Edit step** button on a specific step of a recording.

Step instruction

The **Step instruction** is the primary text that is displayed for this step in the task guide. There are usually 2-3 alternative options for step instructions, and they appear in the following order when editing the annotation.



This image shows the annotation options for changing a step.

- **Preferred value instruction** This type of instruction directs the user to enter the same data that was used when the step was recorded. *Example*: In the First name field, enter 'John'.
- **Example value label** This type of instruction directs the user to enter their own data, indicating that the data that was used when the step was recorded was only *Example* data. *Example*: In the First name field, enter a value.

  If users select the **See more** button on this step when they play the recording as a task guide, they're able to see the data that was used when the step was recorded. This recorded data value is labeled as an *Example* data value.

  > ⓘ **Note**
  >
  > Steps that aren't related to fields, such as selecting buttons, opening forms, or selecting records from a lookup, don't set *Example value label* as an option when annotating.

- **User-supplied value label** This step instruction contains placeholder text, which the author can fill in with their own text. For steps which have an **Example value label** option, the placeholder allows substituting the text which normally specifies the data to enter. This is useful for scenarios where neither the **Preferred value label** nor the **Example value label** sufficiently express the data that should be used for this step.
  - *Example label*: In the First name field, enter *{your example text}*.
  - *Example label after supplying the placeholder text*: In the First name field, enter the customer's name.

  For steps which don't have an **Example value label** option, the placeholder allows substituting all of the label text. Steps associated with buttons, for example, don't have **Example value labels**, so you may replace the entire label text with your own text.
  - *Example label before replacement*: Select Post.
  - *Example label after replacement*: To post the order, select Post.

Titles and notes

Titles and notes provide places for user-specified text to be associated with a step in a task guide.

- **Title** – The title lets you specify the text that appears above the step instruction for this step in the task guide. The title a good place to put text that you want users to read before they complete the action that is indicated by the step instruction.
- **Note** – You can use a note to specify text that appears in the expandable section of the popup for this step in the task guide. A note is a good place to put optional reading material or other information that might be useful to users, but that they aren't required to read to complete the action that is indicated by the step instruction.

Change recorded values

Starting in version 10.0.12, you can adjust the values that are recorded in basic input controls (for example, simple text, numeric, date, and picklist fields), without having to re-record those steps. Lookup controls and reference groups aren't currently supported.

Hide from task guide

The **Hide this step** option lets the author prevent specific steps from appearing in the task guide. This option is useful for hiding steps that are required for the task recording to run in playback mode, but that shouldn't be seen by users. Examples of these steps include copy steps, system-

generated steps, and data clean-up steps. If you hide a subtask, all the steps that are recorded inside that subtask are also hidden.

Using control gestures

The basic recording capability lets a user record an end-to-end business process by using Task recorder, but without adding overhead to the process. In some circumstances, more advanced recording features can be used to create even richer business process recordings. Each of the following gestures is found under the **Task recorder** option on the shortcut menu (also known as a right-click menu or context menu) for a control and causes a step to be added to the recording. If the gesture isn't supported for a control, it doesn't appear on the shortcut menu for that control.

Copy

The **Copy** gesture lets you copy the value for the current control to the Task recorder "clipboard." That value can then be used later as part of a **Paste** or **Validate** gesture. Because values from multiple controls might have to be pasted, the Task recorder clipboard maintains a list of all control values that are copied in the recording.

Paste

The **Paste** gesture lets you paste a value from a previous **Copy** gesture in the same recording. The Task recorder Paste function works like the standard paste function that users might be familiar with, but it has an additional benefit when it's used during recordings. Because Task recorder will replay the recorded **Copy** and **Paste** commands during playback, if the copied control has a different value than it had during recording, Task recorder pastes the current value instead of the value that the copied control had during recording. This feature is useful in scenarios where the copied control has a value that can change between environments (for example, recID values or number sequences).

There's an additional benefit from using the **Copy** and **Paste** gestures when test code is generated. For any control where the value is set via the **Paste** command, Task recorder doesn't have to create a parameterized input variable for that control's value, because it's based on another control's value. This feature can be useful in scenarios where an entity such as a customer is created, and an identifier for that entity is frequently entered during the recording. Instead of manually re-entering the customer name or ID throughout the scenario, and causing Task recorder to generate a parameterized input variable for each entry, the user can copy the customer name or ID one time, and then repeatedly paste it. In this case, Task recorder generates

a single parameterized input variable to represent the customer name or ID. This feature can make it much easier to change the input data for a generated test.

Validate

The **Validate** gesture lets you insert a step that validates the value of the targeted control. This gesture always uses equality to validate the control value. *Validations aren't currently run during recording playback*. Instead, they're run only when the generated test code is run. Two kinds of validation are available:

- **Current value validation** captures the targeted control's value at the time of recording and use it to generate an assertion in the test code. In the list of validation options on the shortcut menu, **Current value** is always first.
- **Reference value validation** uses the value of a previously copied control when generating an assertion in the test code. This allows creating assertions that are resilient to changes in the data, since the value isn't hardcoded into the test code. In the list of validation options on the shortcut menu, **Reference value validation** follows the format [AOT name of copied control: current copied value].

Additional options are available in version 10.0.13 and later. Here are some examples:

- **Enabled/Disabled** validates that the targeted control's state is enabled (or disabled), and then uses that validation step to generate an assertion in the test code.
- **Read-only/Editable** validates that the targeted control's state is read-only (or editable), and then uses that validation step to generate an assertion in the test code.

Add info step

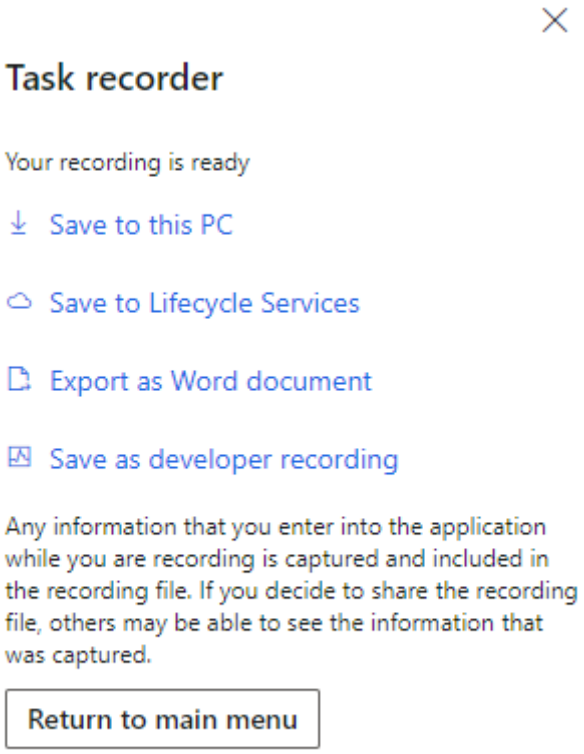
The **Add info step** gesture lets you insert a step and supply your own text for it. This feature is useful primarily for creating task guides. An **informational step** (or **info step** for short) is a task guide step where the instruction text for the step is user-specified. Info steps are useful for describing actions that are a part of the scenario but must occur outside the client. For example, a scenario might require the user to search for item inventory or check an email for information.

You can specify where an info step should appear in the task guide. The info step can point to a control on the page, if the step is associated with the control. Alternatively, the info step can appear in the upper right of the page, if the step is external to the client, or if it's an explanation that applies to the whole page.

> **Note**
>
> Because info steps are manually specified steps and aren't automatically recorded by Task recorder when the user takes an action on a control, the info step doesn't have the capability to automatically progress when a user completes the step in the task guide. Because the info step isn't associated with taking an action in the client, there's no action for a task guide to detect that the user has completed in order to automatically progress to the next step.

Options after a recording is completed

After you select **Stop** to end your recording session, several options are shown so that you can save the files that are related to the completed recording. Select **Save to this PC**, and save the task recording package to your desktop. You'll use this file later.



Save to this PC

One option after you finish your recording is to download the task recording package (an .axtr file) to your computer. This file can be loaded later via the **Task recorder** pane, so that the recording can be played as a task guide or edited.

Save to Lifecycle Services (LCS)

When you save your recording to an LCS library, it's published on the specified business process in a BPM library. If the selected LCS library is set as a Help library, you're able to find the task guide for the recording by searching the **Help** menu.

> ⓘ **Note**
>
> To be able to save a recording to an LCS library, the user must be in the Microsoft Entra tenant that the environment was deployed from.

Export as Word document

The Microsoft Word document for your recording contains the recorded steps and any screenshots that were captured.

Save as developer recording

The raw recording file (developer recording) is useful for developer scenarios, such as test code generation and scenarios where RSAT is used.

Playing back a recording

The **playback** functionality of Task recorder can automatically run the steps of an existing recording by using the pages and values that were originally recorded. Playback mode can be used to update an existing recording if changes were made to the underlying application, and those changes altered the business process steps that are required for the scenario. It's important to remember that, in this mode, Task recorder simultaneously re-records the steps and plays them back. When the playback is completed, a new recording is produced that reflects both the steps that were run from the existing recording and any new steps that the user manually performed. Any steps that aren't run either by the user or automatically by Task recorder aren't included in this new recording.

To play back an existing recording, follow these steps:

1. Refresh the browser tab.

   > ⓘ **Note**
   >
   > It's a good practice to refresh the browser before each new recording.

2. Open the **Task recorder** pane.
3. Select **Playback recording**.
4. Select **Open from this PC** to load a recording from a previously downloaded Task recorder package (.axtr file).
    - If you're reading this guide for the first time and following along, choose the "Create a new rental reservation" file that you downloaded previously.
5. Select **Start**.

When you play back a recording, additional actions are available in the **Task recorder** pane.

Play next pending step

Play next pending step runs the next step in the recording. This action is useful because it gives you more control over the playback speed when you want to analyze the effects of a single step. This action has a side-effect that it's important to be aware of. When you select **Play next pending step**, any open lookups, drop-down dialog boxes, or Action Pane tabs might be dismissed, because this action removes focus from those elements. For these situations, we recommend that you use **Play all pending steps** instead.

Play all pending steps

Play all pending steps begins sequential execution of the remaining steps in the recording, and continues until either playback is paused or all steps are run. During playback, the **Play all pending steps** button is replaced by a **Pause** button that can be used to pause playback. If playback can't successfully run a step for any reason (for example, because it can't find a button that has been renamed), Task recorder skips that step, and playback is automatically paused. In this way, the user has an opportunity to replace the obsolete step by completing the new steps in the client. Task recorder records the new steps and ignore the step that was skipped. The user can then select **Play all pending steps** to continue playback for the remaining steps. After the recording is completed, the user can download the updated recording. This recording contains all the steps of the original recording, but excludes any skipped steps and includes any new steps.

Play to selected step

Play to selected step behaves like **Play all pending steps**, but it lets you play only a subset of the steps instead of all the steps. In the list, select the step that you want playback to stop at, and

then select **Play to selected step**. Task recorder begins to run the steps in the list and stops after it runs the step that you selected.

Editing a recording

Although you can edit a recording through the playback functionality, there's also a mode that lets you make simple edits to a recording without having to replay the whole recording. To access this feature, select **Edit recording** after you open the **Task recorder** pane. You can use this feature to make the following edits:

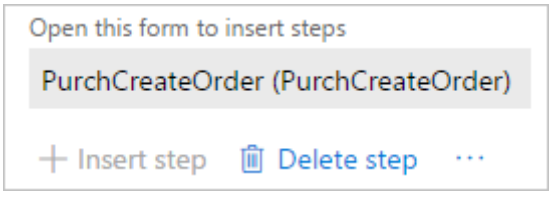
- Insert steps into a recording without re-recording the whole file.
- Move steps under a subtask without re-recording the whole file.
- Adjust the name and description of the recording.

Insert steps without re-recording the entire file

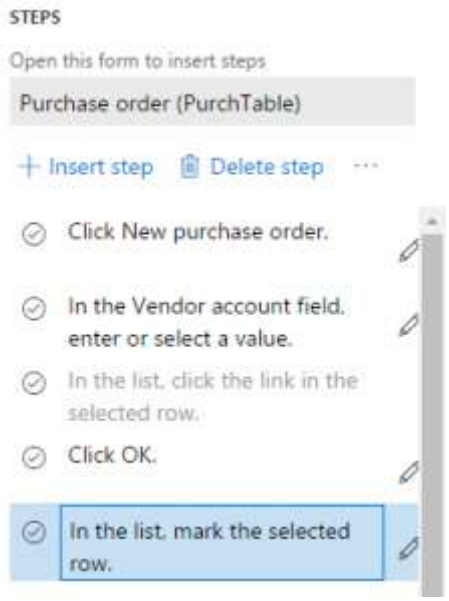
You can add a step anywhere in a task recording without playing back or re-recording the whole file.

1. Select the step after which you want the new step to be inserted. Make sure the step is highlighted.

   In order for task recorder to insert a step, you must have the correct page open. The correct page is the page on which the new step occurs. Task recorder has a mechanism that determines what the active page is, and disables the functionality if the correct page isn't open.

   When you're on the correct page, **Insert step** becomes available.



2. Select **Insert step**.

   When you select **Insert step**, Task recorder switches to recording mode. Any action that is performed in the user interface (UI) is now recorded and inserted into the recording as steps.

3. Select **Stop**.

   Recording mode is stopped, and you can now continue to edit the recording. For example, you can repeat this process to insert steps in other places in the recording, or you can move subtasks as described in the next section.

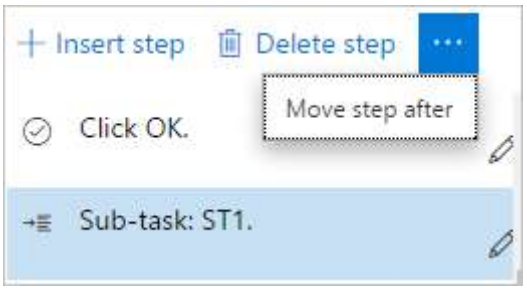
4. When you finish editing the task recording, select **Done editing**, and then select one of the options to save or publish the recording.

Move steps under a subtask without re-recording the entire file

You can move steps under a subtask without playing back or re-recording the entire file. You can also move the subtask step or the end subtask step if you want to group an existing block of steps.

1. Select the step or subtask step that you want to move. Make sure that the step is highlighted.

2. Select **Move step after**. To access this command, you might have to select the ellipsis (**...**) button.



3. Select the step or subtask step that you want to move the step or subtask step after. Task recorder moves the step.

4. To move the end subtask step, select it, select **Move step after**, and then select the step that you want the end subtask step to be after.

   If you want the first step in the task guide to be within a subtask, create a subtask step as the second step, and then move the first step into it. You can add or move as many steps or subtasks as needed.

5. When you've finished editing the task recording, select **Done editing**, and then select one of the options to save or publish the recording.

Adjust the recording name and description

You can adjust values of the **Recording name** and **Recording description** fields. If you want to see more steps in the Task recorder editing pane, you can also collapse the section that shows the recording name and description.

Playing a task guide

A **task guide** is a user-focused experience that lets the user follow a guided step-by-step set of instructions to complete a business scenario by using a task recording. The user is instructed to

complete each step through an animated popup prompt that moves across the page and point to the UI element that the user should interact with. The prompt also tells the user how to interact with the element. For example, it might state, "Select here" or "In this field, enter data." Each step that the user is instructed to complete is based on a step originally recorded in the task recording. Because the task recording file contains the data that describes the step that was originally recorded, the task guide can automatically determine when the user completes the step as expected. It then automatically moves on to the next step.

> **Note**
>
> One way that the task guide determines that a user completes a step is by detecting when the value in a field has changed. Although the task guide doesn't require that a specific value be set, it does require that the field value be changed in order to determine that the step was completed. The user must change the field value, and then press the **Tab** key or select in an area outside the UI element. Only at that point does the client detect that the field value has changed, and it can then proceed to run any required application validation or business logic. Therefore, before the task guide can determine that the step was completed by the user, it relies on the client to detect that the field value has changed.

What can a task guide allow a user to do?

When a user is completing a task guide, the client behaves in the same manner, with the same data, security, and validation rules as it does when the user isn't completing a task guide. There's no difference of behavior in the client that would allow a user to take an action that they can't otherwise take when they're not completing a task guide. When a user is completing a task guide:

- Any data the user enters is subject to the same data validation rules as when not playing the task guide.
- Any data the user enters may be saved, and the user may modify data according to the same restrictions and rules as when not playing the task guide.
- Any security mechanisms the user encounters behave the same as when the user isn't playing the task guide.
- Any forms or controls the user accesses are subject to the same security and access mechanisms as when the user isn't playing the task guide.

The "On-rails" feature of task guides

By default, when a user begins a task guide, they're placed "on-rails". These "rails" prevent the user from *selecting* on elements other than the element the task guide is pointing to. When the user tries to select on something outside of the UI element that the task guide is pointing to, the task guide popup becomes animated to let the user know that they can't progress until they complete the current step of the task guide.

While a user is prohibited from *selecting* on other elements, the user isn't prevented from tabbing through the other controls on the form, and the user isn't prevented from using keyboard shortcuts. This is by design, as the "on-rails" feature is designed for and targeted at first-time users, who are expected to primarily use the mouse as they become familiar with the application.

More advanced or experienced users can turn off the "on-rails" feature when they complete a task guide. At any point during the task guide, these users can turn off the rails by selecting the **Unlock** button that appears on the Task recorder toolbar at the top of the page. This button can also be used to restore the rails at any point during the task guide. In some situations, the task guide might automatically turn off the "on-rails" feature. When the rails are turned off, the user can select UI elements just as they do when the task guide isn't running. The "on-rails" feature might be automatically turned off in the following situations:

- The user is being directed to go to a page by using the navigation pane or navigation search.
    - Because the user can use either entry point, the task guide doesn't point to a specific entry point, and it doesn't prevent the user from using either entry point.
- The task guide enters an error state (see the next section for a list of error states).
- The task guide is showing an info step.

Error detection

An *error state* occurs when the task guide isn't able to point to the UI element that is associated with the current step because the UI element isn't visible on the screen. When the task guide detects that the current step requires the user to interact with a UI element that isn't visible, then the task guide popup moves to the upper-right side of the screen. These causes of an error state can be simplified into two categories.

The control isn't visible on the form

This error state usually occurs when the user has opened or closed the incorrect tab, FastTab, collapsible section, FactBox, or pop-out menu.

Because the UI element that is needed for the current step is somewhere on the current form, but it isn't visible on the screen, the task guide popup moves to the upper-right side of the screen while displaying the same instruction that informs the user of the action they need to take.

Because the task guide can't find the UI element on the screen, the user must manually determine what is causing the UI element to be hidden and then make the element visible on the screen. The task guide popup automatically detects that the UI element is visible and repositions itself so that it's pointing at the now-visible element.

The control isn't on the form

This error state usually occurs when the user is on the wrong form, either by navigating to the wrong form or by leaving the correct form.

Because the UI element isn't visible on the screen, the task guide popup moves to the upper-right side of the screen. In addition, when the task guide detects the user is on the wrong form, the task guide popup text changes to inform the user of the form they should navigate to.

In some cases, the task guide popup doesn't mention the form by name. This is because the user may need to navigate to a dynamic form. A dynamic form is a form that isn't modeled, frequently known as a runtime-generated form. These sorts of forms don't have a proper name. Some examples of runtime-generated forms include simple and custom lookups. The way for a user to navigate to a lookup form is to reopen the lookup.

Next step and Previous step

The **Next step** and **Previous step** buttons appear in the task guide popup and let a user manually control the flow of the task guide. When these buttons are selected, the task guide proceeds to the next or previous step. The task guide doesn't verify that the user completes a step before it goes to the next or previous step.

The task guide **never** automatically completes any step for the user, even when the **Next step** and **Previous step** buttons are used. Use of these buttons can cause an error state if the previous or next step refers to a UI element that isn't on the current page. When the user is completing an info step, the only way to proceed is to use the **Next step** button. This action is required because an info step doesn't represent an action that was recorded on any UI element. Because no action was recorded in the task recording, the task guide doesn't have the necessary information to determine what action it should expect the user to complete.

The See more button

When the **See more** button is selected, the task guide popup expands to show additional information that is related to the step. The additional information is often optional reading material that isn't required for the user to successfully complete the step. The following information might be included:

- An **Example** value
  - The Example value is the value that was originally used when the task recording was created.
  - Example values appear only for steps that use nonlookup fields. These fields include text fields, number fields, date fields, combo boxes, and check boxes.
- A **Note**
  - A Note may contain scenario-specific information that helps provide context to the user about the current step of the task guide.

Taking screenshots in Task recorder

By using a **prerelease** Chromium browser extension that works for both the new (Chromium-based) Microsoft Edge browser and Google Chrome, Task recorder can take screenshots of the browser as a user records a business process. After the user completes the recording, Task recorder can use these screenshots to generate Microsoft Word documents. To turn on this functionality, follow these steps to install the prerelease Chromium extension that enables Task recorder to take screenshots during recording.

1. Download the **FMLabTaskRecorderScreenshot** folder that contains the extension from GitHub, at https://github.com/Microsoft/FMLab.

2. **On-premises deployments only:** Adjust the manifest for the extension so that it matches the following code. Replace <hostname> with the base URL for your environment.

   JSON

   ```
   ...
   "content_scripts": [
       {
           "matches": ["https://*.dynamics.com/*", "<hostname>"],
           "js": ["screenshot.js"]
       }
       ...
   ```

3. **21Vianet in China deployments only:** Adjust the manifest for the extension so that it matches the following code. Replace **.com** with **.cn**

JSON

```
...
"content_scripts": [
    {
        "matches": ["https://*.dynamics.cn/*"],
        "js": ["screenshot.js"]
    }
    ...
```

4. US Government Community Cloud (GCC) **deployments only:** Adjust the manifest for the extension so that it matches the following code. Replace **dynamics.com** with **microsoftdynamics.us**

JSON

```
...
"content_scripts": [
    {
        "matches": ["https://*.microsoftdynamics.us/*"],
        "js": ["screenshot.js"]
    }
    ...
```

5. **Multiple deployments:** Add additional string values to the matches clause, separating them with a comma.

JSON

```
...
"content_scripts": [
    {
        "matches": ["https://*.dynamics.com/*",
"https://*.microsoftdynamics.us/*"],
        "js": ["screenshot.js"]
    }
    ...
```

6. Open the latest Microsoft Edge browser or Google Chrome.
7. Select **Settings and more** > **Extensions** in Microsoft Edge (or **Customize and control Google Chrome** > **More tools** > **Extensions** in Google Chrome).
8. Select **Developer mode**.
9. Select **Load unpacked extension**.

10. Browse to the folder that contains the Task recorder extension by using the path **FMLab-master** > **FMLab** > **TaskRecorderScreenshot**, and then select **Select Folder**.

11. Make sure that **Enabled** is selected so that extension is turned on.

12. Restart the browser.

Task recorder then takes screenshots of the tab where the client is running. If needed, you can regenerate the screenshots by playing the task recording again.

Task recorder doesn't capture screenshots from other tabs or of the user's desktop.

Generating tests from a recording

After a business process recording is completed using Task recorder, a developer can import the raw developer recording file (.xml file) into Visual Studio to create an X++ test. The import tool generates a human-readable X++ test from the recording, and translates any control gestures, validations, or tasks into the appropriate test code.

Import a recorded test

1. Open Visual Studio by using the finance and operations development tools.

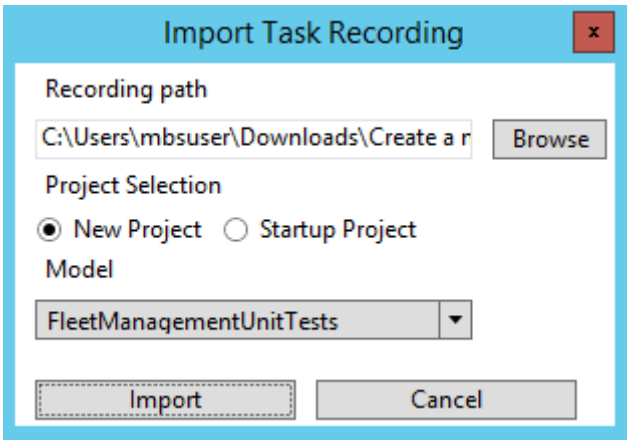
2. Go to **Dynamics 365** > **Addins** > **Import task recording**.

3. In the **Import task recording** menu, use the **Browse** button to locate a previously downloaded recording file.

4. Optionally, choose to have the generated test code be added to the startup project. This requires that a solution containing a project is set as the startup project. This places the generated X++ test into the same model as the project.

5. If you're creating a new project, select the model for the project. The generated X++ test is put in this model. For the generated test to be successfully built, the model must have references to the **TestEssentials** model.

6. Select **Import**.



7. In the **New Project** dialog box, provide a name for the project.

8. After the project is created, the user can open and inspect the generated code.

9. To run the test, build the project.

10. Go to **Test** > **Windows** >**Test Explorer**.

Appendix

Controls that are known to have incomplete support for Task recorder

- Table
- Filter pane, which is the filter that pops out from the left side
  - When you add filters to the filter pane, the steps are delayed. The steps don't get recorded until the user selects "Apply" on the Filter pane.
- Enhanced previews
  - No planned support for recording gestures inside of enhanced previews. While you're recording, enhanced previews are disabled.
- No extensible controls are supported out of the box, except Segmented Entry.
  - Extensible control owners need to individually build support for Task recorder.

Controls that can be recorded, but have limited support for the Copy/Paste/Validate gestures

- Date/Time
  - Doesn't support copy/pasting "Never" as a value.
- Image
  - No ability to copy/paste/validate an image value.

- Filter pane
  - Copy/Paste works, but the UI doesn't show the pasted data. You can proceed as if it pasted correctly.
- Message box
  - You can't validate the text in the message box.

Controls that are known to have incomplete support for being used in a task guide

- Quick Filter, which is the filter control that appears above lists.
  - Doesn't support displaying a "generic value" during the task guide. Currently displays the value that was used during recording.
- Filter pane, which is the filter that pops out from the left side
  - The task guide doesn't point to the individual elements within the Filter pane that need to be selected on.

Last updated on 08/04/2026

Create documentation or training with Task Recorder

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Customers using Human Resources, the functionality noted in this article is currently available in both the stand-alone Dynamics 365 Human Resources and the merged Finance infrastructure. Navigation might be different than noted while we make updates. If you need to find a specific page, you can use Search.

This article explains what Task recorder and task guides are, how to create task recordings, and how to customize Microsoft task guides and include them in your Help.

> **Important**
>
> You can record your own task guides for Dynamics 365 Human Resources, but you can't save them to a Business Process Modeler (BPM) library or open them from the Help pane. You can save them locally or to a network location, and then open and replay them by using Task recorder.

Learn about Task recorder

Task recorder is a tool that you can use to record actions that you take in the product user interface (UI). When you use Task recorder, it captures all of the events that you perform in the UI that are executed against the server—including adding values, changing settings, and removing data. The steps that you record are collectively called a *task recording*. You can use task recordings in many ways:

- **Play task recordings as task guides** - Task guides are an integral piece of the Help experience. A task guide is a controlled, guided, interactive experience through the steps of a business process. The user is instructed to complete each step by way of a pop-up prompt, which animates across the UI and points to the UI element that the user should interact with. The pop-up prompt also provides information about how to interact with the element, such as **Click here** or **In this field, enter a value.** A task guide runs against the user's current data set and the data that is entered is saved in the user's environment.
- **Save task recordings as Word documents** - By saving task recordings as Word documents, you can easily produce printable training guides.

You can create your own task recordings, play task recordings provided by Microsoft, or modify Microsoft-provided task recordings to reflect your configuration. For more information about Task recorder, see Task recorder.

Plan your task recording

Whether you're creating a new task recording or basing your recording on a Microsoft task recording, keep the following information in mind.

- Plan your recording like you would a video.
- Walk through the business process once or twice without recording it to understand the steps.
- When you walk through the process before you record, notice where you use shortcut keys or the **Enter** key, so that you can avoid using them during the actual recording.
- Do you want to group steps together into subtasks? Subtasks visually set apart sections of a process. For example, if you're creating a recording for creating and releasing a product, you might want to group together the steps that are required to create a product, and then group together the steps that are required to release the product. Subtasks make longer processes easier to read.
- Do you want to add annotations? See **Understand the different types of annotations** below for more information.
- What values do you add in the various fields as you complete the steps of the business process?

Write your description and annotations ahead of time

- At the beginning of each task recording, there's a description field where you can enter an introduction to the recording. Write and save the description ahead of time in a separate document so you can copy and paste it into the task recording when you are recording. Cutting and pasting the text makes the recording process go more quickly and smoothly.

- For each step in a task recording, you can create annotations. During playback of a task guide, annotations appear as notes above or below the text for the step. When viewed as text in the Help pane, annotations appear as text inline in the step. Write and save your annotations in a separate document. When you're recording the task recording, cut and paste the annotations in from that document.

Understand the different types of annotations All annotations are optional. Only add them if they provide helpful information to the user.

- **Title**: A title annotation appears before the step text that task recorder automatically generates. In the task guide, the title annotation appears above the automatically generated text. Use this type of annotation to explain why the user is doing the step or to give context.
- **Notes:** A notes annotation appears after the step text that task recorder automatically generates. In the task guide, it'll only be visible if the user clicks **Show more** in the task guide bubble. Use this type of annotation to describe anything that a user needs to know to complete the step.
- **Info step**: Create these annotations by right-clicking on a control or anywhere on a form, and then selecting **Task recorder** > **Add info step**. Info steps appear as a numbered step at whatever point you insert it, even though no action was recorded in the UI. You can add a form-level info step or an info step associated with a control. When an info step is associated with a page, the task guide **bubble** appears someplace on the page, with no pointer, when the task guide is played. When an info step is associated with a control, the task guide **bubble** points to the control when the task guide is played. In the Help pane, an info step annotation appears as a numbered step with whatever text you entered. Use info steps to prepare the user for the next steps, to describe steps that need to be done outside of the application, or to refer to other recordings.

Determine how long to make your recording

- Users either read or play the recording from start to finish, so don't combine steps or tasks that are better done separately.
- Try not to record a long scenario that spans multiple sub-processes. For example, **Operate in-store customer service desk** is too broad; break it up into shorter tasks such as **Accept returns** and **Add to gift card**.
- If a task can be carried out as part of several different business processes, create a separate recording for it, and you can refer to it in the other recordings.
- If the process involves multiple tasks that the person likely does all at once, you can keep the tasks in one recording, for example, **Set up and assign functionality profiles**.
- If it's something someone does once (such as configuration) and then another task that they can do immediately afterward but might do repeatedly, and on its own, break them up into two task recordings.

Decide where, in the UI, to start a recording The page that you're on when you start recording a task recording affects which page the task guide is displayed for. For example, if you want your task recording to be listed in the Help pane when a user clicks Help on the **General ledger parameters** page, start your recording on the **General ledger parameters** page. **Save recordings as .axtr files** When you're done creating or editing a task recording, you see several options for how you want to download or save the recording. You can download the file as a task recording package (.axtr), download it as a raw recording file (.xml), download it as a Word document, or save the file to an LCS library. Always save your task recording as a task recording package file (.axtr). This practice helps make maintenance of the file easier if procedures or annotations need to change later. If you want to download the file as a Word document, also save it as a task recording package file.

Create your task recording

For detailed walk-through steps, see Task recorder resources.

Copy and customize Microsoft's task recordings

You can download and edit Microsoft's task recordings to use them for your own help documentation or training materials. To download a Microsoft task recording, follow these steps:

1. Open Task recorder. Task recorder is located in the **Settings** menu.
2. In the Task recorder pane, select **Maintain a recording**.
3. Under **Where is the recording**, select **It is in an LCS library**.
4. Select **Select the LCS library**.
5. Select the Microsoft global library.
6. In the tree, select the business process library node that the task recording is associated with.
7. Select **OK**.
8. Select **Start**.
9. Step through the recording, changing any steps as you go to re-record it. **Note**: If you only need to change the text of a recording, you can open the recording in **Edit a recording's annotations** mode, and then save it.
10. After the recording plays to the end, select **Stop** in the task recorder bar at the top of the screen.
11. Choose how you want to save the task recording.

Additional resources

Help system

Connect the Help system

Task Recorder

Last updated on 03/11/2026

Videos

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

On the Microsoft Dynamics 365 YouTube channel, you can find videos created by Microsoft that demonstrate a wide range of business solutions for Dynamics 365 products. This section lists the "how-to" videos for finance and operations apps that are hosted on the channel.

> **Note**
>
> Some of the videos listed were published under a previous product name, but they're still applicable.

Videos for finance and operations development and administration

Data integration

- Synchronize a work order between Dynamics 365 for Field Service and finance and operations apps

Data management

- Use data entities and data packages

Development

- Optimization advisor
- Setting up a development branch and build
- Setting up a development machine
- Azure DevOps integration with Lifecycle Services

Intelligence

- How to edit an embedded report in an analytical workspace
- How to embed PowerBI.com reports in finance and operations apps
- Bring your own database (BYOD) to finance and operations apps
- How to use cost management Power BI content in Dynamics 365

Lifecycle Services

- Asset library in Lifecycle Services
- Cloud-hosted environments
- Creating support tickets from Dynamics 365 for Operations
- Deploying environments
- Getting started with Lifecycle Services
- Deploying code to a sandbox environment
- Deploying code to a production environment
- Implementation projects in Lifecycle Services
- Manage the code upgrade and tool process
- Managing business process libraries in Lifecycle Services
- Methodologies in Lifecycle Services
- Request a production environment
- Refreshing data in a production environment
- Uptake a new platform release after Platform update 3
- Use telemetry to monitor key performance counters in Dynamics Lifecycle Services

Regression suite automation tool (RSAT)

- How to use task recorder to create a test case for the Regression suite automation tool (RSAT)
- How to create a test plan in Azure DevOps to use with the Regression suite automation tool (RSAT)
- How to use the Regression suite automation tool (RSAT)
- The improved Excel experience in Regression Suite Automation Tool (RSAT) 2.0

Videos for Dynamics 365 Finance

Business document management

- Create a new business document using Business Document Management

Customize the app

- Add custom fields
- Embed Power Apps

Financial reporting

- Find the version of Report designer

Help system

- How to contribute to the Microsoft Dynamics 365 documentation

Office integration

- Create an Excel template for header and line patterns

Organization administration

- Document management

Revenue recognition

- How to use revenue recognition

Tax engine

- Tax engine overview

Videos for Dynamics 365 Supply Chain Management

Business document management

- Create a new business document using Business Document Management

Costs

- Cost control mobile workspace
- Get started with Cost accounting
- Use Excel for cost analysis

Customer portal

- Overview of the Customer portal template
- Invite customers to register and use your customer portal

Help system

- How to contribute to the Microsoft Dynamics 365 documentation

Master planning

- Master planning setup wizard

Office integration

- Create an Excel template for header and line patterns

Organization administration

- Document management

Procurement and sourcing

- Approve purchase orders on a mobile device
- Onboard a new vendor

Product information management

- Change management capabilities

Production control

- Batch balancing
- Enhancements to the production order release process
- Visual scheduling with Gantt chart for production and batch orders

Warehouse management

- Release production picking to the warehouse in batch
- Use warehouse template to copy configuration

FastTrack Tech Talks

FastTrack Tech Talks focus on providing technical depth and best practices that give customers and partners detailed knowledge that's specific to the subject areas.

Last updated on 11/20/2025

User interface elements

Article • 01/03/2025

This article describes the user interface (UI) elements used in the app. Before users can navigate the interface, it's important to know the names and functions of the elements that make up the interface.

- **Action Pane** - The bar beneath the navigation bar. Here, you can select tabs to change records shown in the page. You can edit and save the records here.
- **FactBox** - You can see information and follow the activities of certain records in this pane.
- **FactBox pane** Here, you can scroll through different aspects of a record to view in the FactBox.
- **Filter pane** - On some pages, you can select **Show filters** to open this pane. It allows you to narrow the results visible to you on the page.
- **Navigation bar** - The bar at the top of the interface. It contains the **Dynamics 365 portal**, **Search**, **company picker**, **Action center**, **Settings**, **Help & Support**, and the user profile.
- **Navigation list** - On some pages, you can scroll through this pane to find a specific record. When selected, the details of the record will appear in the page.
- **Navigation pane** - The left-most pane. From here, you can find any page in the product.
- **Page** - The central focus of the interface. Selections made on the other UI components will affect what records are shown here.
- **Pane** - The right-most pane. This will open in some cases when aspects of a record need to be changed and saved.
- **Tab** - When referring to the Action Pane, it's a menu of options that appears when you select a given option in the Action Pane.

Tabs, fields, and sections

A *tab* is a selection made on the page that opens a different aspect of a record on the same page. Often, it will allow you to change certain *fields*, or UI elements that allow typed input.

A *FastTab* is a tab with the added benefit of allowing multiple tabs to be visible at the same. You can expand a FastTab by selecting the downward-pointing arrow on the right end of it.

A *section* is similar to a tab. The word "section" is often used to describe any area of a page that organizes a specific category of information. "Summary", "Orders and favorites", and "Links" are all examples of sections.

Dialogs and drop-down menus

A *dialog* is a pane that opens when certain selections are made to change or create a record. Dialogs contain fields that allow you to enter typed input. Sometimes, a given field will allow you to select a downward facing arrow that opens a list of options to choose from. This is called a *drop-down menu*.

In some cases, a dialog will open near a given button when you select it. This is called a *drop-down dialog*.

Notifications

Certain changes to the objects you oversee will appear as *notifications*. Notifications may notify you when a specific customer's information has been changed, or it may alert you when the system can't accept inputs you've added in certain fields. You can learn how to customize what you receive notifications about in the Alerts overview.

Notifications appear in a variety of ways.

- **Feature callout** - This will appear next to a field, tab, or other button to offer an explanation of what the feature is used for.
- **Action center** - A box that contains the notification will appear next to the Action center button on the navigation bar. You can see details about the notification by selecting **Action center**.
- **Message bar** - This will appear beneath the Action Pane.
- **Message box** - This will appear over the interface and must be interacted with before you can continue to use the product.

Toolbars, grids, and lists

A *toolbar* contains tools, such as the ability to add fields or remove records. Sometimes, a toolbar will appear on the page above a *grid*. This area, grid, is a name given to rows of records with various columns of data. Not all grids have toolbars above them.

A *list* is the name given to a collection of records that you can scroll through. You can bring these records into the page by selecting them. Often, this will open a grid.

Feedback

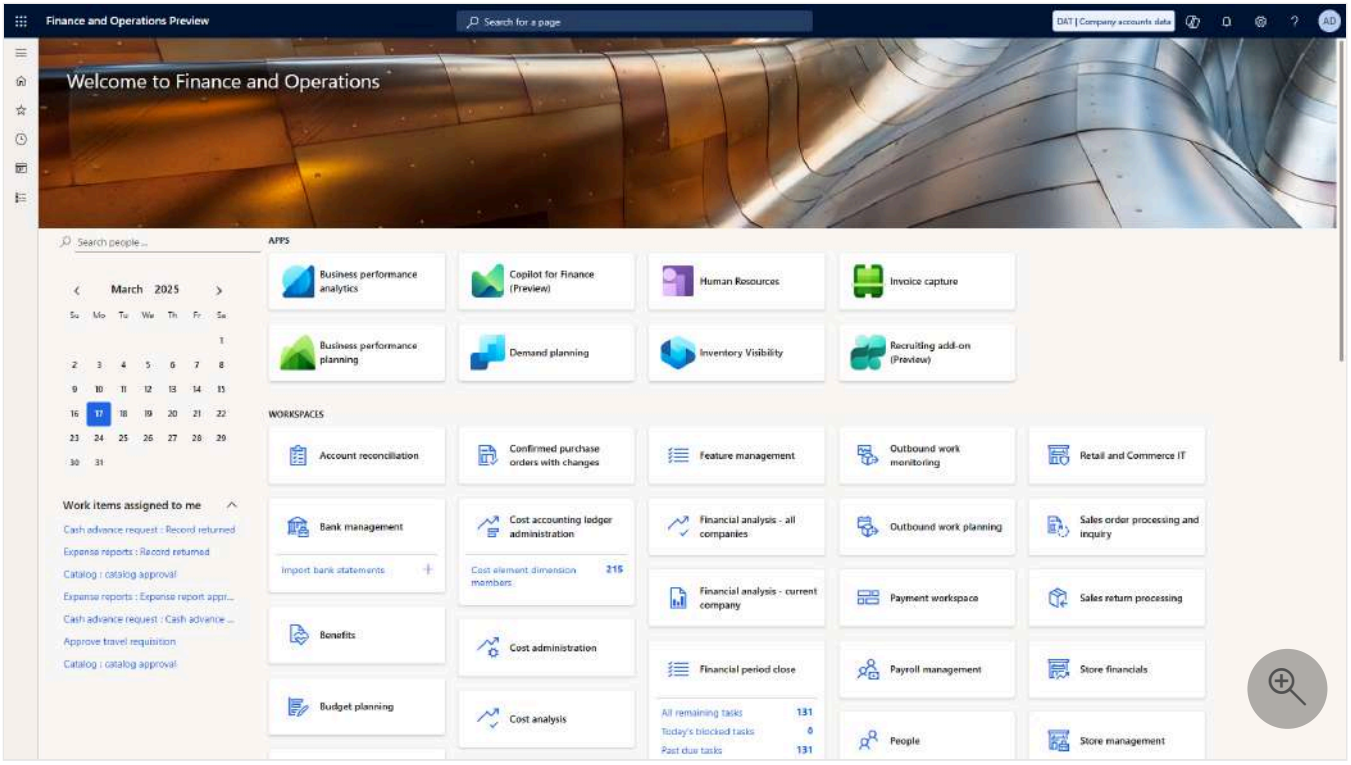
Was this page helpful? Yes No

Provide product feedback

Default dashboard

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the purpose and use cases for the default dashboard, its content (such as app tiles and workspaces), and how to personalize it.



The default dashboard is a dashboard page that is set as each user's initial page by default. It comes preconfigured with a set of useful components that help you navigate to workspaces and other useful pages in the application. You'll only see components, such as workspaces and apps, that you have permission to access.

The default dashboard can be customized to show additional information or hide components. You can also personalize it to match your own preferences.

Out-of-the-box components in the default dashboard

The components that are included by default on the default dashboard are:

- A page header representing the current legal company.
- An **Apps** area that provides quick access to apps.
- A **Workspaces** area that provides quick access to workspaces.
- A search field for finding people.
- A calendar where you can set the current session date.
- A list of workflow work items assigned to you.

Page header

The page header presents, by default, the name of the current legal entity and the default dashboard image. You can change the dashboard image to a different banner or a logo.

Related apps

The **Apps** area shows tiles for finance and operations apps available to you so you can launch them quickly. It can also show links to documentation about how to set up the application.

Workspaces

The **Workspaces** area shows one tile for each workspace that's available to you. Workspace tiles can also highlight information that you have pinned to the dashboard. To pin information to the dashboard, open the relevant workspace, right-click on a tile in the workspace, select **Personalize: <tile name>** and then select the **Pin to dashboard** checkbox (clear this checkbox to remove the information from your Immersive Home).

People search

The **Search people** field helps human resource workers quickly find and access details about personnel in the **People** workspace of the Dynamics 365 Human Resources app.

Session date picker

The session date picker lets you quickly change the current session date, which is relevant for many financial transaction functions. The picker component is a quick alternative to using the Session date and time module.

Assigned workflow items

The **Work items assigned to me** area presents a view of all work items that were generated and assigned to you from a workflow. The list provides an overview and links to the relevant pages where you can take action.

Learn more in the Workflow system overview.

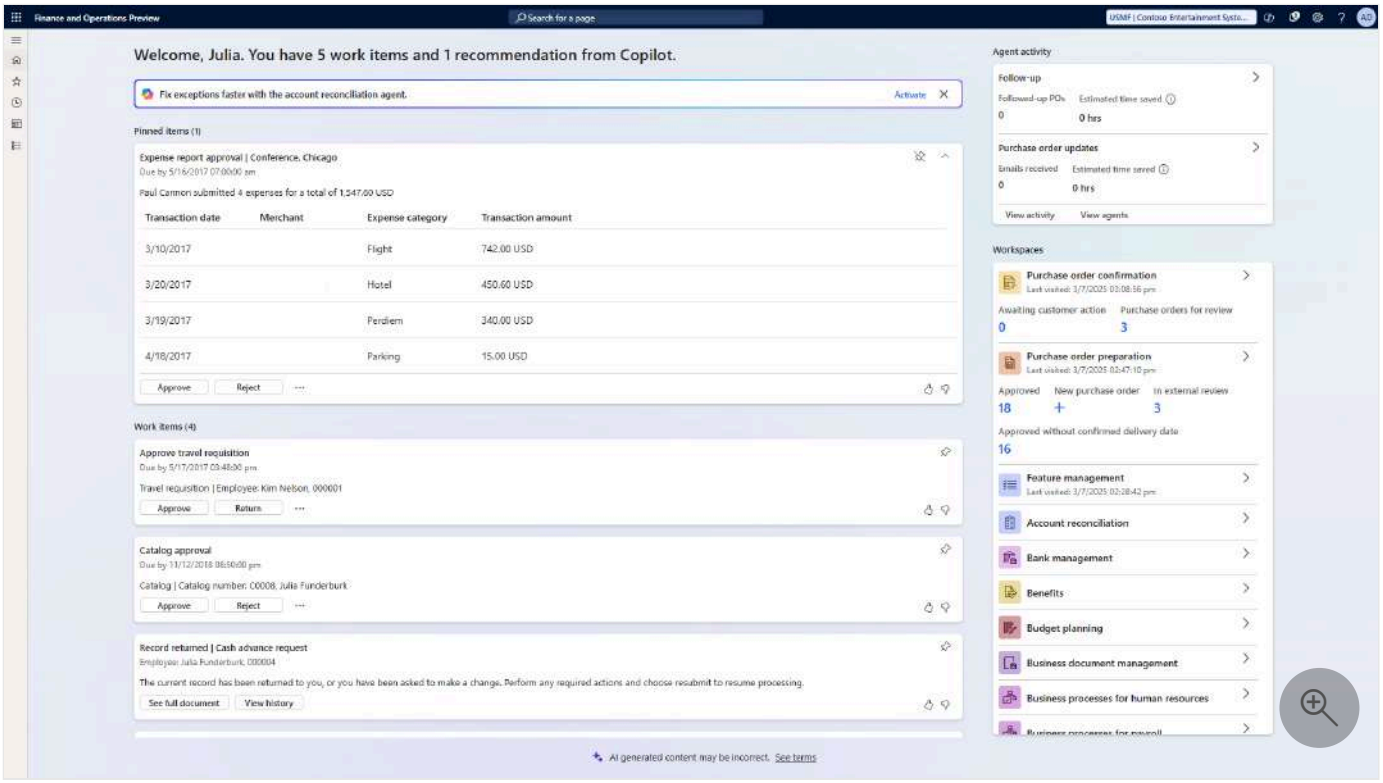
Last updated on 07/11/2025

Immersive Home overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes Immersive Home, how it helps you work hand in hand with AI agents and focus on the most important work items assigned to you.

Immersive Home is a modern, AI-first landing page that adapts to your most important work, aids decision making, and reduces the need to navigate elsewhere to complete common tasks. It's a modern option for the initial page in finance and operations apps and links to tools for enabling, monitoring, teaching, and interacting with AI agents.



Prerequisites

System requirements

To use Immersive Home, your system must meet the following requirements:

- You must be running version 10.0.44 or later of your Microsoft finance and operations apps.
- The feature that is named *Immersive Home* must be turned on in feature management.

Required user privileges

Each user that requires access to Immersive Home must be assigned the *View immersive home* privilege, which is typically assigned through the duty with the same name. By default, this duty is assigned to the *System User* security role.

The components of Immersive Home

Immersive Home consists of a new library of controls for adaptive experiences in Microsoft business applications following the Fluent design patterns. Adaptive experiences aim not only to modernize user experiences but also to bring dynamic approaches that move away from static forms toward experiences that bring relevant work directly to your attention.

Immersive Home features a greeting and work summary, suggestions related to AI agents, and a central focus on work items. It also includes agent activity overviews and a ranked view of spaces, including the classic workspaces in finance and operations apps.

Greeting and summary

The greeting at the top of the page welcomes you by name and provides a concise summary of relevant work items and suggestions for what to address. This provides a quick overview of the immediate tasks at hand.

Suggestions area

The suggestion area presents relevant callouts, such as to suggest that you enroll an agent skill into service for a specific task or provide additional instructions to make an agent's service in a task more beneficial.

Each suggestion typically provides an action that guides you to follow the suggestion and take the necessary steps.

The work items area

The **Work items** area hosts a list of priority-ranked activity cards. Each card reflects a step that is part of an action plan to complete a task. Multiple similar work items can be collapsed into one activity card, allowing you to move the action plan forward for multiple tasks of the same kind. Examples include expense approvals, catalog requests that need approval, cash forward requests awaiting approval, and other workflow-generated work items.

Other examples include activities from agent-generated action plans, such as when an agent prepares a reminder email for a vendor (asking them to confirm purchase orders) that you need to review it before sending. The **Work items** area might also announce that an agent identified a vendor email that references a purchase order, and prepares a resulting confirmation or change request ready for your review.

In most cases, the activity card allows you to complete an action without navigating away from Immersive Home. To achieve this, action cards show relevant information that allows you to take action, which you can expand as needed. Alternatively, you can navigate to the activity-specific experience for more complex tasks.

If you have activities that you need to work on for some time and track until completion, you can add them to the **Pinned items** list.

Once you complete an activity, it disappears from Immersive Home.

Activity cards show:

- The type of activity as the title
- A description summary with relevant information
- Action buttons to complete the activity
- A navigation link to the activity-related task or workspace

> **Note**
>
> Activity cards include a feedback mechanism. You can use it to share feedback about your experience using Immersive Home directly with Microsoft (provided an administrator enables user feedback for Copilot and related experiences).
>
> Learn more about how the Microsoft in-product feedback system works in **Learn about Microsoft feedback for your organization**.

Learn more about how to enable user feedback for Copilot and related experiences in finance and operations apps in Enable enhanced user feedback for Copilot and related experiences.

The agent activity area

The **Agent activity** area is only shown when AI agents are enabled in your environment. For each activated agent, this area shows a section where the agent presents its key activity metrics. This could be the number of cases processed, number of emails sent, or metrics such as estimated time saved.

Typically, agents provide a dedicated task space where you can go into more detail on what the agent has been doing, work hand in hand with the agent, and configure or teach the agent. The chevron at the top right of each agent activity card navigates to the specific agent task space.

At the bottom of the **Agent activity** area, you can navigate to agent lifecycle management-related views. The **View activity** button opens a holistic tracing list of all agent activities. The **View agents** button opens a repository of all currently running agents and the library of all agents available for activation.

The workspaces area

The **Workspaces** area shows a list of ranked tiles for each workspace. The ranking considers how recently you last used a workspace. Favorite workspaces rank at the top.

Workspace tiles highlight information that you pin to the dashboard. To pin information to the dashboard, open the relevant workspace, right-click on a tile in the workspace, select **Personalize: <tile name>** and then select the **Pin to dashboard** checkbox (clear this checkbox to remove the information from your Immersive Home).

Related information

Responsible AI FAQ for Immersive Home in finance and operations apps

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Last updated on 05/08/2026

Set the preferred initial page for each user

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set the initial page shown to a user when they open the application. The initial page can be different for each user. Each user can choose their own initial page at any time, and admins can set the initial page for any user.

By default the initial page is set to the default dashboard.

To set the initial page for another user (as system administrator), follow these steps:

1. Go to **System administration** > **Users** > **Users**.
2. In the list, find and select the user you want to set up.
3. On the Action Pane, select **User options**.
4. Open the **Preferences** tab.
5. On the **Startup** FastTab, set the **Initial page** field to your preferred initial page.
6. Select **Save**.

To set your own initial page (as a user), follow these steps:

1. Open the **Settings** menu (gear icon) in the top navigation bar.
2. Select **User options**.
3. Open the **Preferences** tab.
4. On the **Startup** FastTab, set the **Initial page** field to your preferred initial page.
5. Select **Save**.

Last updated on 03/05/2026

Accessibility features

This article describes the functionality that is designed to help users who have various disabilities use this app. For example, there are features for people who use sight-assistive technologies such as Microsoft Windows Narrator.

Windows Narrator and keyboard-only access

Every field and control has a label and a description of applicable shortcuts. A screen reader can read the label and description.

Shortcuts for the most frequently performed actions

For most users, everyday system use involves lots of data entry and keyboard interaction. To enhance the user experience, we have created shortcuts to help you "jump" around the screen and shortcuts for specialized actions. For more information, see Keyboard shortcuts.

Navigation search

Any page that is accessed by using the Navigation pane menu, the left-most pane, is also available from the **Search** box. Press Alt+G to move focus to the **Search** box, and then type the name or description of the page.

For more information, see Navigation search.

> **Note**
>
> You can navigate directly to top-level pages only. Secondary pages rely on information or context from their parent page.

Action search for keyboard-only users or for heads-down data entry

Every action that is provided on a page can be accessed from a keyboard, via the tab sequence. Information about the tab sequence is provided later in this article. To run actions more directly, you can use the action search functionality.

Example

You want to run the **Email notification log** action that appears in the **Email notification** group on the **Sales order** tab on the Action Pane.

One option is to use your keyboard. Press Ctrl+F6 to move focus to the Action Pane, and then press Tab repeatedly to move through all the tabs and actions, until the **Email notification log** action has focus.

However, you can also run the action more directly. From anywhere on the page, press Ctrl+Apostrophe (') to show the search box for actions.

In the search box, type words that describe the action. The action is made available to you, and you can run it directly. For example, by typing **email**, **notific** (a partial word), or **log**, you can "jump" to the Email notification log functionality.

When you've finished, you can press Ctrl+Apostrophe again to return focus to the field that you were working with before you ran the action search.

For more information, see Action search.

Tab sequence

In everyday system use, not every field is required in order to perform typical tasks. Therefore, by default, the tab sequence is "optimized." Tab stops are set only on those fields that are essential for typical scenarios.

However, you might find that some fields that you often use to perform tasks aren't included in the default tab sequence. In this case, if you use Windows Narrator, you can use Windows Narrator's keyboard actions to access those fields and inspect their content. Alternatively, you can turn on the **Enhanced tab sequence** option on the **Options** page. This option makes all editable and read-only fields part of the tab sequence. You can then use page personalization to create a custom tab sequence and omit fields that don't have to be part of the tab sequence. For more information about personalization, see Personalize the user experience.

Form patterns

Almost 90 percent of the pages in the app are based on a small set of patterns. These patterns are referred to as *form patterns*. Each form pattern is used to provide the actions that are most often performed on the page. A form pattern helps guarantee familiarity and ease of

understanding, because frequently used actions and data are always presented in the same location on different pages. Because of the small number of form patterns, users can easily learn the system, regardless of the number of pages in it, and can confidently use it after they recognize the form patterns.

To learn more about form patterns, see Form styles and patterns.

Responsive layout

The product is designed to work on various devices and form factors, from the smallest screens to large screens that have the highest resolution. Our responsive layout engine lets users zoom in to a magnification level of 200 percent (or, in some scenarios, more than 200 percent).

On smartphones and other small screens, the controls and the form layout will responsively adapt to ensure that the core data is favored. These responsive behaviors can also include reducing the number of columns in groups and tabs to a single column, hiding shell elements, and collapsing the Action Pane.

Guidance to help developers and customers incorporate accessible thinking in their customizations

To learn more about Microsoft best practices for enabling accessibility, see Accessibility in forms, products, and controls.

Last updated on 06/26/2026

Feature management overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Features are added and updated in every release. The Feature management experience provides a workspace to view a list of features delivered in each release. You can then use the workspace to view feature documentation, and to enable or disable features.

The Feature management workspace

You can open the **Feature management** workspace by selecting the appropriate tile on the dashboard. You see a page that shows a list of features for all releases supported by the Feature management experience.

The feature list includes the following information:

- **Name** – A description of the feature that was added.
- **Status** – A symbol indicates whether a feature is turned on (check mark), is turned off (blank), scheduled to be turned on (clock), is mandatory (lock), requires attention before you turn it on (warning symbol), or can't be turned on (X). The setting that's shown is used for all legal entities. When a feature has been turned on, it's still controlled by security. Therefore, the feature is only available to users who have access to it based on their security role. It's also only available in legal entities that the user has access to.
- **Enable date** – The date when the feature was turned on or scheduled to be turned on.
- **Added** – The date when the feature was added to your environment. This date is automatically entered when you update your environment during the monthly release cycles.
- **Feature state** – The current lifecycle state of the feature: **Preview**, **Released** (shown as blank), **On by default**, and **Mandatory**. The states are covered in more details later in this article.
- **Module** – The module that is affected by the new feature.
- **Type** – The type of the feature and can be **Feature** or **Hotfix**. A **Hotfix** is a subtype of a feature, and has a shorter life-cycle.

When you select a feature, more information appears in the details pane to the right of the feature list. At the top of the pane, you see the feature name, the date when the feature was

added, the module that's affected by the feature, and a **Learn more** link. Select this link to view the documentation for the feature. If documentation isn't available, you're taken to a temporary page. The details pane also includes a **Comments** field where you can add your own comments about the feature.

The **Feature management** workspace also has several tabs, each of which shows a list of features.

- **New** – This tab shows all features that have been added since the last monthly update. If you've skipped any monthly updates, the tab shows all the new features that have been added since the last time that you updated. The newest features appear at the top of the list. The total number of new features is also shown on a tile at the top of the page.
- **Not enabled** – This tab shows all features that aren't turned on. The newest features appear at the top of the list. In addition, a tile at the top of the page shows the total number of new features that are currently off.
- **Scheduled** – This tab shows all features scheduled to be turned on in the future. The features that have the earliest scheduled date appear at the top of the list. In addition, a tile at the top of the page shows the total number of scheduled features.
- **All** – This tab shows all features. The newest features appear at the top of the list.

Feature recommendation notifications

Starting in Dynamics 365 finance and operations version 10.0.35, users may start to see notifications informing them about recommended features. Users can review a recommended feature and request it be enabled by an administrator. The request triggers a notification to be sent to administrators. They can assess the suggested feature and decide if it should be enabled for their organization.

Feature states

Features can transition between several states, from being introduced in Feature management to eventually becoming mandatory in the product. This section describes the valid feature states.

Preview features (optional)

Product teams can decide to initially start a new feature as a preview feature. Preview features aren't enabled by default, and they're optional. The owning product team updates features to released after they've completed a successful preview period.

> **Note**
>
> Preview features are subject to specific preview terms and conditions.

Released features (optional)

The **Feature state** column for these features is blank. Features that are initially added as released aren't turned on by default, and enabling them is optional. Features that are updated from preview keep their enablement status.

On by default features (optional)

Features that are updated to **On by default** are turned on by default, but can be disabled. After features that can be disabled have been in the **Released** state for at least six months, they're expected to move to this state in the next major release. Features that transition to **On by default** are expected to be communicated in the What's new article for the release. The update is initiated by the owning product team.

> **Note**
>
> Because these features are enabled automatically, it's important that you determine whether your organization is ready to uptake these features, or if more time is required. If more time is required, it might be necessary to temporarily disable these features. The transition of a feature to **On by default** is typically done in the major release before the feature is targeted to become **Mandatory**. At that point, you don't have the option to disable the feature.

Mandatory

Mandatory is the expected final state for features. It indicates that the features are turned on, and you can't disable them without contacting Microsoft. Optional features are expected to become mandatory after two major releases. Critical features can, by exception, be introduced as mandatory.

Example of expected feature lifecycle

Features that can be disabled, and that were added as released and optional before or as part of the April release, are expected to transition to **On by default** in the following October release. They're then expected to become **Mandatory** in April of the following year.

Features that can't be disabled, and that were added as released and optional before or as part of the April release, are expected to transition to **Mandatory** in April of the following year.

Hotfix states

Hotfixes have a shorter transition through the states than features. Hotfixes are introduced by Product quality updates, and eventually become fully integrated in the product and no longer appear in Feature management. For each hotfix, use the **Learn more** link to get detailed information about availability. This section describes the valid hotfix states.

Released (optional)

The **Feature state** column is blank. Hotfixes are initially added as released, and aren't turned on by default, and enabling them is optional.

Fully integrated

The hotfix doesn't appear in Feature management, the hotfix is turned on, and can't be disabled without contacting Microsoft.

Example of expected hotfix lifecycle

Hotfixes are added in a Product quality update for the April release, are expected to be **Fully integrated** in the next upcoming release. For versions already released, the hotfix is provided in Product quality updates, and remains as **Released**. For example, a hotfix released in August, is available in Product quality updates for the April and July releases, and **Fully integrated** in the October release.

Enable a feature

If a feature hasn't been turned on, an **Enable now** button appears in the details pane. You can use this button to enable the feature.

Some features can't be disabled after you enable them. If the feature that you're trying to turn on can't be enabled, you receive a warning. Select **Cancel** to cancel the operation and leave the feature disabled. However, if you select **Enable**, you won't be able to disable it later.

Some features display a message that provides additional information before you enable them. These features are indicated by a yellow warning symbol. You should read the additional

information carefully, to ensure that you understand what happens when the feature is enabled. However, you can still select **Enable** to enable the feature.

Some features display a message that the feature can't be enabled until an action is taken. These features are indicated by a red X symbol. You must take the actions described in the description before the feature is enabled. For example, if you can't use a feature until a configuration key is disabled, then you must disable the configuration key first and then return to Feature management to enable the feature.

After a feature is enabled, a message appears below the **Learn more** link in the details pane. This message states either that the feature was enabled or indicates the future date when the feature is scheduled to be enabled. It appears every time that you select the feature in the feature list.

Features scheduled to be enabled in the future appear on the **Scheduled** tab. A batch process enables them at midnight on the specified date, based on the time zone that's represented by the system date.

Reschedule a feature

If a feature is scheduled to be enabled in the future, a **Schedule** button appears in the details pane. Use this button to change the **Enable date** value to a different date.

1. Select the scheduled feature to reschedule, and then, in the details pane, select **Schedule**.
2. In the dialog box that appears, in the **Enable date** field, specify the new date when the feature should be enabled.
3. Select **Enable** to reschedule the feature or **Disable** to cancel the schedule.

Disable a feature

If a feature has been enabled, a **Disable** button appears in the details pane. Use this button to disable the feature. The **Disable** button isn't available if the feature can't be disabled.

After a feature is disabled, a message appears below the **Learn more** link in the details pane. This message states that the feature hasn't been enabled. It appears every time that you select the feature in the feature list. Features that haven't been enabled appear on the **Not enabled** tab.

Features that must be enabled

Sometimes, a critical feature is delivered that must be enabled automatically when you do an update. These features are enabled automatically on the date that is specified in the **Enable date** field. For these features, a message appears below the **Learn more** link in the details pane. This message either states that the feature was enabled or indicates the future date when the feature will be enabled. It appears every time that you select the feature in the feature list.

Enable all features

You can enable all features by selecting the **Enable all** button.

When you select **Enable all**, an option appears where you must provide the following information:

- A list of all features that require confirmation before they can be enabled. If you want to enable the features in the list, select **Yes** for the **Enable features requiring confirmation** button.
- A list of all features that can't be enabled will be shown. Those features won't be enabled.

All features that can be enabled are enabled. If a feature is already scheduled to be enabled in the future, the schedule doesn't change.

Enable all features automatically

If you want to automatically enable all new features, you can use the drop-down list under the workspace title to change what occurs when new features are added.

- Select **Enable new features automatically** to automatically enable all new features when they're added to your environment.
- Select **Do not enable new features automatically** if all applicable new features should be off by default when they're added to your environment.

When you enable all features automatically, it enables all of the features that would be enabled when you click the **Enable all** button. It won't enable the features that require confirmation or the features that can't be enabled until an action is taken.

Check for updates

Features are added to your environment after each update. However, you can manually check for updates by clicking on the **Check for updates** button. Any feature that was added to the system after the update is added to the list of features. For example, if a flighted feature is enabled after a release, then you can check for updates and the feature is added to your list.

Assigning roles

The **Feature management** workspace can be opened by system administrators, and also by users who are assigned to the Feature manager role or the Feature viewer role. These two roles were created to support the Feature management experience. Users in the Feature manager role can turn any feature on or off. They can also update the **Comments** field for the feature. Users in the Feature viewer role can only view the **Feature management** workspace. They can't turn features on or off.

The Feature manager role and Feature viewer role don't override the existing security that a user has. They just control whether the user can turn features on and off. They don't provide access to the features themselves.

Features that use configuration keys

If a feature uses a configuration key, but the configuration key isn't turned on, the **Feature management** workspace doesn't show the feature in the list of available features. After you turn on the configuration key, update the feature list by using the **Check for update** menu item. The feature then appears in the feature list.

If you turn off the configuration key, the feature isn't removed from the feature list.

Data entities

A data entity that is named **Feature management** lets you export the Feature management settings from one environment and then import them into another environment. This entity updates only existing features. The business logic in the entity also helps guarantee that the same rules that are used on the **Feature management** workspace are applied when the import is done. For example, you can't override a mandatory feature setting by removing the date during import.

The following examples describe what occurs when you use the **Feature management** entity to import data.

- If you change the value of the **Enabled** field to **Yes**, the feature is enabled, and the **Enable date** field is set to the current date.
- If you change the value of the **Enabled** field to **No** or leave the **EnableDate** field blank, the feature is disabled, and the **Enable date** field is cleared. You can't disable a mandatory feature or a feature that can't be disabled after it's enabled.
- If you change the value of the **EnableDate** field to a future date, the feature is scheduled for that date.

- If you change the value of the **Enabled** field to **Yes** and change the value of the **EnableDate** field to a future date, the feature is scheduled for that date.
- If you change the value of the **Enabled** field to **No**, but you also change the value of the **EnableDate** field to a future date, the feature is scheduled for that date.
- If a feature is enabled, and you add an **EnableDate** field that is set to a future date, the feature remains enabled. To reschedule the feature, you must change the value of the **Enabled** field to **No**.

Feature management and flighting

Feature management lets you control the features that are delivered in each release. Flighting lets Microsoft teams release features to a limited number of customers, so that those features can be tested and validated without affecting all customers. **Feature management** doesn't control the flighting of any features.

Using Feature management to turn on ISV features or custom features

Feature management is currently unavailable for features from independent software vendors (ISVs) and custom features. However, Microsoft is adding more functionality to enhance **Feature management**. After those enhancements are completed, Microsoft makes **Feature management** available to all features and provide instructions for updating your features to use it.

Frequently asked questions (FAQ)

When are features added, removed, or changed?

Features are added, removed, and changed through code changes by the owning product teams. Environments must be updated to receive those changes.

Does a feature become mandatory automatically?

No, a feature doesn't become mandatory automatically. The owning product team must make a code change.

Why isn't there a specific 'mandatory-enabled date'?

Update release timing is variable, environment update timing is variable, and customers can opt to skip some updates. As a result, specific dates are difficult to determine.

Where's the documentation for features that are mandatory?

This documentation comes from each Dynamics 365 application team. Often, these features are mentioned in Updates to client feature states or Removed or deprecated features.

Is there an in-product notification or signal that a feature is going to be mandatory-enabled?

A notification mechanism related to making a feature mandatory doesn't exist today.

Do features ever get enabled without the customer knowing about it?

Yes, features can be enabled without the customer's knowledge in the following situations:

- A feature is moved to **On by default**. In this state, the feature can be disabled.
- A feature is updated to **Mandatory**. This change only occurs in combination with a major release. Critical features might, by exception, be moved to **Mandatory** at any update.

What is feature flighting and how does it relate to feature management?

Feature flights are real-time on or off switches that Microsoft controls. They're separate from the customer control provided by Feature Management.

- Private preview features aren't listed in **Feature management** until they're flighted on. In production, the customer needs to agree to be part of a special program for that to occur.
- Public preview and released (generally available) features are listed in **Feature management** unless they're flighted off. Flighting a feature off is considered as a last resort option for product teams if a critical issue is found and would usually be a per customer operation.

Do features ever get flighted off without the customer knowing about it?

Yes, if a feature is impacting the functioning of an environment that doesn't have a functional impact then they can be enabled by default.

How can feature enablement be checked in code?

Use the **isFeatureEnabled** method on the **FeatureStateProvider** class, passing it an instance of the feature class. Example:

X++

```
if
(FeatureStateProvider::isFeatureEnabled(BatchContentionPreventionFeature::instance()
))
```

How can feature enablement be checked in metadata?

The **FeatureClass** property can be used to indicate that some metadata is associated with a feature. The class name used for the feature should be used, such as **BatchContentionPreventionFeature**. This metadata is visible only in that feature. The **FeatureClass** property is available on menus, menu items, enum values, and table/view fields.

What is a feature class?

Features in **Feature management** are defined as *feature classes*. A feature class **implements IFeatureMetadata** and uses the feature class attribute to identify itself to the **Feature management** workspace. There are numerous examples of feature classes available that can be checked for enablement in code using the **FeatureStateProvider** API and in metadata using the **FeatureClass** property. Example:

X++

```
[ExportAttribute(identifierStr(Microsoft.Dynamics.ApplicationPlatform.FeatureExposur
e.IFeatureMetadata))]
internal final class BankCurrencyRevalGlobalEnableFeature implements
IFeatureMetadata
```

What is the IFeatureLifecycle implemented by some feature classes?

IFeatureLifecycle is a Microsoft-internal mechanism for indicating the feature lifecycle stage. Features can be:

- `PrivatePreview` - Needs a flight to be visible.
- `PublicPreview` - Shown by default but with a warning that the feature is in preview.
- `Released` - Fully released.

Last updated on 10/24/2025

Client FAQ

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides answers to frequently asked questions about the finance and operations client.

Why aren't symbols loaded?

The security settings on your browser might prevent the symbols from being loaded correctly. To resolve this issue, try the following steps:

- If you're experiencing this issue in Internet Explorer, click **Tools**, and then click **Internet Options**. In the Internet Options dialog box, on the **Privacy** tab, click **Custom level**, and make sure the **Font download** option is selected.
- Otherwise, you might have to add the app site to the list of trusted sites.

I miss the ribbon from Dynamics AX 2012. Can I keep Action Pane tabs open all the time?

We are planning to implement this feature soon. Users will then be able to choose to keep the tabs on Action Panes open all the time. Otherwise, the tabs will be collapsed when they aren't being used, to gain more screen space for the page.

Why do I sometimes see different shortcut menus when I right click?

If you right-click in an editable field (or if text is selected), the browser's shortcut menu is displayed. This menu gives you access to the **Cut**, **Copy**, and **Paste** commands. We can't embed these commands into the shortcut menus because, for security reasons, browsers don't allow us to programmatically access the system clipboard.

If you right-click a field label or the value of a read-only control, you'll see the shortcut menu.

To make keyboard access easier, we plan to implement a keyboard shortcut in the future that will open the shortcut menu.

Where is the View details functionality?

The **View details** option is available in a couple of ways:

- If a control has **View details** capabilities, and if the control has a value, that value is displayed as a hyperlink. You can click the hyperlink to open a page that contains additional details.
- **View details** is also an option on shortcut menus. For more information about when shortcut menus are displayed when you right-click, see the previous section.

Last updated on 07/11/2025

Action search

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the action search functionality. Action search will help you find and run actions on a page.

Introduction

Pages primarily expose commands on Action Panes, both the standard Action Pane that appears at the top of a page and the toolbars that appear in various sections of the page. In previous versions, a Key Tips feature let you quickly access any button on an Action Pane by pressing the Alt key and then a series of letters.

The action search feature replaces Key Tips, which are no longer available. This feature lets you quickly search for and run a button from any visible Action Pane.

Using action search

To use the action search feature, follow these steps:

1. On the Action Pane, click in the **Action search** field. (The **action search** field contains a magnifying glass icon.)
2. Type all or part of the name of the button that you want to run. You can also search by using words from the button's "path." (For more information, see the next section of this article.) Typically, a button appears near the top of the results list after you've typed two to four characters.
3. Find and run the button in the results list (by using your mouse or keyboard).

After the button is run, the focus is returned to your last position on the page, so that you can continue to work.

You can also start action search by pressing Ctrl+/ or Alt+Q. Press the keyboard shortcut again to return the focus to your last position on the page.

Understanding the results list

Often, you must know both the location and the context of a button to fully understand the purpose of that button. Therefore, the results list shows additional information to help you understand exactly which buttons appear in the list. In particular, the "path" of the button is shown. This path might include the labels of the following UI elements, as relevant:

- Action Pane tab
- Button group
- Menu button (if the button is inside a menu button)
- Menu separator (if the button is inside a named group inside a menu button)
- Group or tab on the page (for example, the name of a FastTab)

For example, you typed **tot** in the **Action search** field and are now examining the results list. The first entry, for a button that is named **Totals**, is highlighted. A button path of **Sales order** > **View** is also shown. The **Sales order** part of the path corresponds to the **Sales order** tab on the Action Pane, and the **View** part of the path corresponds to the **View** group on that tab. Similarly, the path of the **Total discount** button (**Sell** > **Calculate**) informs you that this button is located in the **Calculate** group on the **Sell** tab of the Action Pane. Therefore, this information helps you understand exactly which button will be triggered by action search (if you select that button in the results list).

In the previous example, action search showed results from the standard Action Pane at the top of a page. However, action search also shows results from visible toolbars that are in other places on the page. For example, you're searching for the **On-hand inventory** button that is on the **Sales order lines** FastTab. In this case, the button path in the results list (**Sales order lines** > **Inventory** > **View**) informs you that this button is under the **View** heading on the **Inventory** menu button on the **Sales order lines** FastTab.

> ⓘ **Note**
>
> There are some buttons that do not show up in Action search. These include drop dialog buttons and buttons from subforms.

Action search vs. Navigation search

Whereas action search is intended to find and run actions on a page, there is a separate search mechanism for finding and navigating to pages. For more information about that feature, see the Navigation search article.

Last updated on 03/10/2026

Advanced filtering and query syntax

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the filtering and query options that are available when you use the Advanced filter/sort dialog or the **matches** operator in the filter pane or grid column header filters.

Advanced query syntax

Expand table

Syntax	Character description	Description	Example
value	Equal to the value that is entered	Type the value to find.	**Smith** finds "Smith".
!*value* (exclamation point)	Not equal to the value that is entered	Type an exclamation point and then the value to exclude.	**!Smith** finds all values except "Smith".
from-value..to-value (double period)	Between the two values that are separated by double periods	Type the from-value, then two periods, and then the to-value.	**1..10** finds all values from 1 through 10. However, in a string field, **A..C** finds all values that start with "A" and "B", and values that are exactly equal to "C". For example, this query won't find "Ca". To find all values from "A" *through* "*C*", type **A..D**.
..*value* (double period)	Less than or equal to the value that is entered	Type two periods and then the value.	**..1000** finds any number that is less than or equal to 1000, such as "100", "999.95", and "1,000".
value.. (double period)	Greater than or equal to the value	Type the value and then two periods.	**1000..** finds any number that is greater than or

Syntax	Character description	Description	Example
	that is entered		equal to 1000, such as "1,000", "1,000.01", and "1,000,000".
>*value* (greater than sign)	Greater than the value that is entered	Type a greater than sign (>) and then the value.	**>1000** finds any number that is greater than 1000, such as "1000.01", "20,000", and "1,000,000".
<*value* (less than sign)	Less than the value that is entered	Type a less than sign (<) and then the value.	**<1000** finds any number that is less than 1000, such as "999.99", "1", and "-200".
*value** (asterisk)	Starting from the value that is entered	Type the starting value and then an asterisk (*).	**S*** finds any string that starts with "S", such as "Stockholm", "Sydney", and "San Francisco".
**value* (asterisk)	Ending with the value that is entered	Type an asterisk and then the ending value.	***east** finds any string that ends with "east", such as "Northeast" and "Southeast".
value (asterisk)	Containing the value that is entered	Type an asterisk, then a value, and then another asterisk.	***th*** finds any string that contains "th", such as "Northeast" and "Southeast".
? (question mark)	Having one or more unknown characters	Type a question mark at the position of the unknown character in the value.	**Sm?th** finds "Smith" and "Smyth".
value,*value* (comma)	Matching the values that are separated by commas	Type all your criteria, and separate them by using commas.	**A, D, F, G** finds exactly "A", "D", "F", and "G". **10, 20, 30, 100** finds exactly "10, 20, 30, 100".
"" (two double quotes)	Matching a blank value	Type two consecutive double quotes to filter for blank values in that field.	Two consecutive double quotes ("") finds rows with no value for the current column.
(Finance and operations query) (finance and	Matching a defined query	Type a query as an SQL statement	**((AccountNum LIKE "US*") && (DirPartyTable.Name**

Syntax	Character description	Description	Example
operations query between parentheses)		between parentheses using the finance and operations query language.	**LIKE "Cont*"))** as an example of syntax for a filter condition on a field from the root datasource as well as a field from a different datasource (for the All customers page)
T	Today's date	Type **T**.	**T** matches today's date.
(methodName(parameters)) (**SysQueryRangeUtil** method between parentheses)	Matching the value or range of values that are specified by the parameters of the **SysQueryRangeUtil** method	Type a **SysQueryRangeUtil** method that has parameters that specify the value or range of values.	1. Click **Accounts receivable** > **Invoices** > **Open customer invoices**. 2. Press Ctrl+Shift+F3 to open the **Inquiry** page. 3. On the **Range** tab, click **Add**. 4. In the **Table** field, select **Open customer transactions**. 5. In the **Field** field, select **Due date**. 6. In the **Criteria** field, enter **(yearRange(-2,0))**. 7. Click **OK**. The list page is updated and lists the invoices that match the criterion that you entered. For this example, invoices that were due in the previous two years are listed. See the table in the next section for additional details about **SysQueryRangeUtil** date methods, and several examples.

Advanced date queries that use SysQueryRangeUtil methods

Expand table

Method	Description	Example
Day (_relativeDays=0)	Find a date relative to the session date. Positive values indicate future dates, and negative values indicate past dates.	• **Tomorrow** – Enter **(Day(1))**. • **Today** – Enter **(Day(0))**. • **Yesterday** – Enter **(Day(-1))**.
DayRange (_relativeDaysFrom=0, _relativeDaysTo=0)	Find a range of dates relative to the session date. Positive values indicate future dates, and negative values indicate past dates.	• **Last 30 days** – Enter **(DayRange(-30,0))**. • **Previous 30 days and next 30 days** – Enter **(DayRange(-30,30))**.
GreaterThanDate (_relativeDays=0) GreaterThanUtcDate (_relativeDays=0)	Find all dates after the specified relative date.	• **More than 30 days from now** – Enter **(GreaterThanDate(30))**.
GreaterThanUtcNow ()	Find all date/time entries after the current time.	• **All future date/times** – Enter **(GreaterThanUtcNow())**.
LessThanDate (_relativeDays=0) LessThanUtcDate (_relativeDays=0)	Find all dates before the specified relative date.	• **Less than seven days from now** – Enter **(LessThanDate(7))**.
LessThanUtcNow ()	Find all date/time entries before the current time.	• **All past date/times** – Enter **(LessThanUtcNow())**.
MonthRange (_relativeFrom=0, _relativeTo=0)	Find a range of dates, based on months relative to the current month.	• **Previous two months** – Enter **(MonthRange(-2,0))**. • **Next three months** – Enter **(MonthRange(0,3))**.
YearRange (_relativeFrom=0, _relativeTo=0)	Find a range of dates, based on years relative to the current year.	• **Next year** – Enter **(YearRange(0, 1))**. • **Previous year** – Enter **(YearRange(-1,0))**.

Last updated on 07/11/2025

Configure and filter workspaces

Article • 01/24/2025

This article provides an overview about how to configure and filter workspaces.

Configuring a workspace

You can change the appearance and behavior of some workspaces by updating settings that apply to the whole workspace. When a workspace can be configured, the Action Pane includes a button that instructs you to click it to make configuration changes.

When you click the button, a dialog appears, where you can modify the predefined settings for the workspace. The specific settings that you see in this dialog vary by workspace, and depend on the specific controls and business data that are available in the workspace.

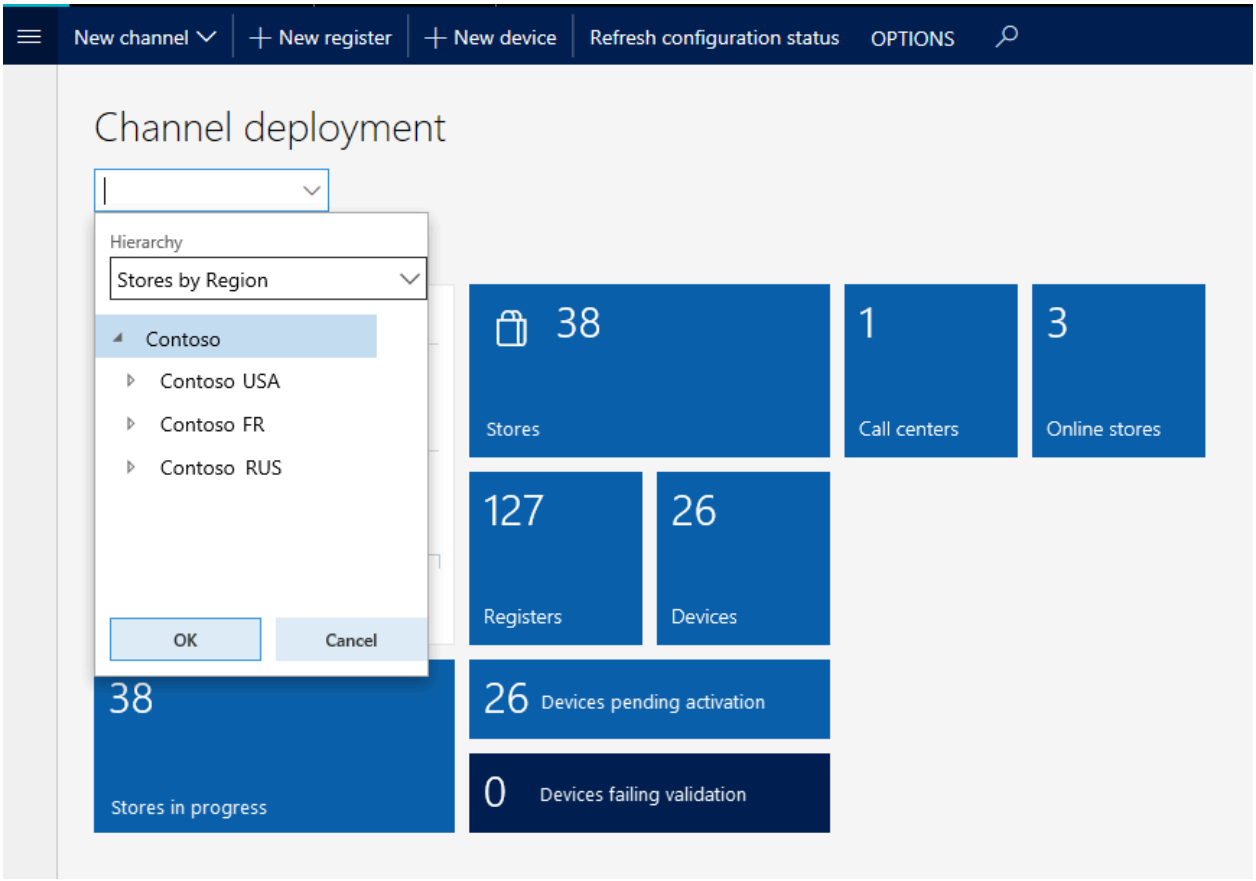
Filtering a workspace

Many workspaces let you filter the content that appears in them. The controls that are available might let you filter all the content in the workspace or only the content in a specific section of the workspace. The filters on workspaces can be lookups, combo boxes, free-form text fields, or other types of controls. However, every type of filter has the same effects, as described in the following sections.

Workspace-wide filters

You can filter the whole workspace by using a workspace-wide filter. A workspace-wide filter appears in the upper-left corner of the workspace. When you select a specific value in the drop-down list, the contents of the workspace are filtered based on that selection.

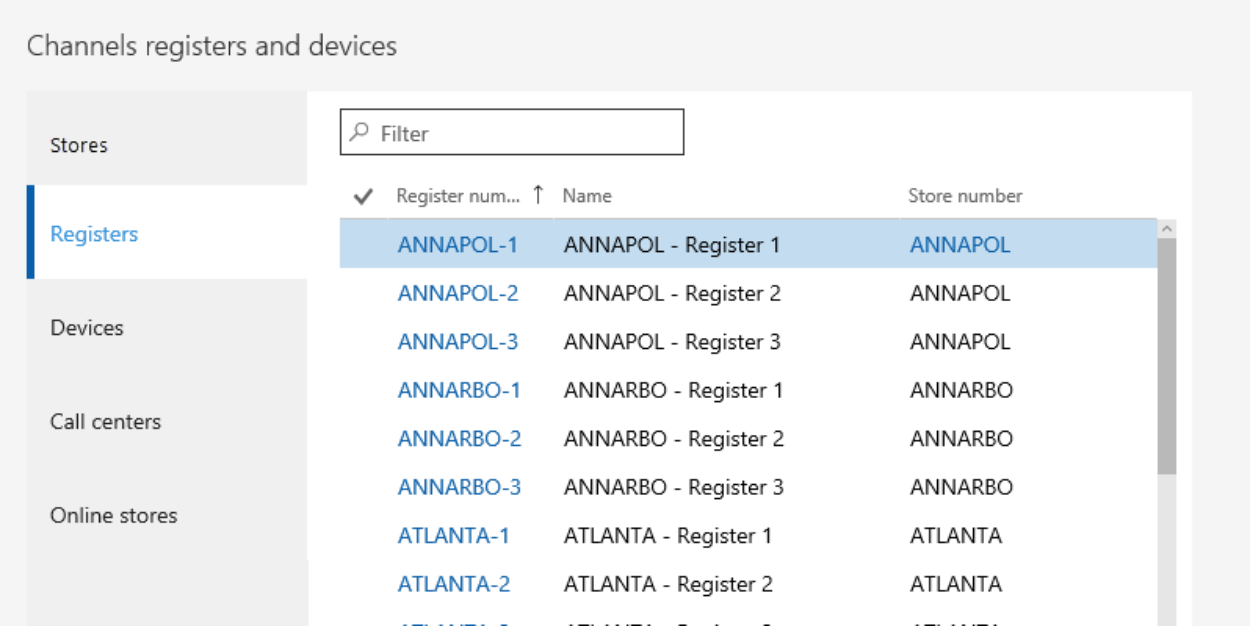
When you click to open the filter, you're presented with several options.



Select an option to filter the workspace based on that option.

Workspace section filters

If individual sections of the workspace have filters, you can filter each section separately. In the following illustration, the filter (the field that contains the text "Filter") is an example of a free-form text field filter.



As with a workspace-wide filter, select or enter a value in the field to filter the contents of the section.

Feedback

Was this page helpful? Yes No

Provide product feedback

Show pages side-by-side using the Open in new window feature

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to display pages side by side.

You may want to view multiple pages side by side to complete tasks quickly. As an example, you might want to validate or enter lines in more than one journal. Typically, to validate or enter lines in more than one journal you would have to go back and forth between the page that displays a list of journals, and the page that displays lines for a given journal. However, the **Open in new window** feature enables you to display these pages side-by-side so that you can perform your tasks quickly.

Clicking the **Open in new window** icon opens the lines page in a new, pop-up browser, and then navigates the original browser back in history to the page that displayed the list of journals. You can then display both pages side by side. After viewing a journal, you can change the selected journal on the journal list page, and the lines page in the pop-up window will automatically display the lines of the newly selected journal.

The dynamic linking and refreshing happens due to the relations that exist between the data that is backing these pages. If the system is not aware of the relation between the data, the pop-up window will not refresh automatically in response to a change in the window it originated from.

Some pages have multiple views such as the Grid view, Header view, and Details view. The **Open in new window** icon causes the entire page to open in the new browser window. Therefore, you cannot keep two views of the same page side by side using the **Open in new window** feature. Almost all such pages have a navigation list that you can use to switch between records and achieve a similar experience.

Before using the **Open in new window** feature, you should configure your browser's pop-up blocker to allow pop-ups from the URL of the site. As an example, you could allow pop-ups from "*.dynamics.com".

The **Open in new window** feature is only available when there is more than one page open in the window. Also, the pop-up window automatically closes when there are no more pages

open (that is, when you close the last page in that window). The system also closes open pages when you navigate to a different area in the application. Therefore, if you have pop-up windows open and navigate to a different area in the application, the pop-up windows close automatically because the system closed the pages in those windows.

The top bar in the pop-up windows displays information about the company the page was opened in and is read-only. The pop-up windows also rely on the main browser window. If the main window is closed or refreshed, all open pop-up windows will become read only. If this situation occurs, you can still view the information in these windows, but you will not be able to interact with it.

Last updated on 07/11/2025

Keyboard shortcuts

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Keyboard shortcuts can help you quickly and efficiently enter data in finance and operations apps.

> **Note**
>
> The keyboard shortcuts described here refer to the United States keyboard layout. Keys on other keyboard layouts might not correspond exactly to the keys on a US keyboard.

Some of the shortcuts on this page are *key chords*, meaning they require two consecutive sets of key combinations (separated by a comma) that need to be independently pressed to trigger the required action. For example, the shortcut "Alt+M,A" requires the user to first press "Alt+M", release the keys, and then press "A".

Finding a shortcut

Users can discover currently available shortcuts directly from the user interface. Simply right-click on a control and select **View shortcuts**. This opens a dialog box showing the shortcuts you can use based on where you are on the page. Alternatively, you can press Alt+Shift+K to open this dialog box (and press Alt+Shift+K a second time to see all currently available shortcuts for the page).

Action shortcuts

Expand table

To do this	Press
Open action search	Ctrl+' or Alt+Q

To do this	Press
Move to the standard Action Pane	Alt+M, A, or Ctrl+F6
Open a tab on the Action Pane or a menu	Enter or Space or Alt+Down arrow
Move to next/previous option in a menu	Down arrow / Up arrow
Close a tab on the Action Pane or a menu	Esc
Simulate a right-click	Shift+F10
Open the context menu	Ctrl+F10
Execute the default button on a page/dialog box	Alt+Enter
Click a button or tile	Enter or Space
View refresh information for a count tile	Alt+Up arrow
View currently available shortcuts	Alt+Shift+K

Date picker shortcuts

Expand table

To do this	Press
Open the date picker	Alt+Down arrow
Move between dates in the date picker	Ctrl+Arrow keys
Move to the next/previous month	Page down / Page up
Move to the next/previous year	Ctrl+Shift+Page down / Ctrl+Shift+Page up
Move to today	Ctrl+Home
Pick today's date	T
Clear the selected date	C
Pick Never (or the max date)	N

FactBox shortcuts

Expand table

To do this	Press
Open the FactBox pane (or move focus to the FactBox pane if it is already open)	Alt+M, B, or Ctrl+F2
Close the FactBox pane (with focus in the FactBox pane)	Esc
Move to the next/previous FactBox (with focus in the FactBox pane)	Alt+Shift+Down arrow / Alt+Shift+Up arrow
Move to the <n>th FactBox (with focus in the FactBox pane)	Alt+<n> (<n> = 1-9)
Expand a FactBox (with focus on the FactBox header)	Space or Enter
Collapse the current FactBox	Alt+0

Filtering shortcuts

Expand table

To do this	Press
Open grid filtering for the current column	Ctrl+G
Close grid filtering for the current column	Esc
Open the Filter pane (or switch focus between the Filter pane and the main page if the Filter pane is already open)	Alt+M, F, or Ctrl+F3
Close the Filter pane (with focus in the Filter pane)	Esc
Open advanced filtering/sort	Ctrl+Shift+F3
Apply the Quick Filter (when focus is in Quick Filter)	Enter
Apply the grid filter (when focus is in the filter field of the column header)	Enter

Page shortcuts

Expand table

To do this	Press
Create a new record	Alt+N
Delete a record	Alt+Del or Alt+F9

To do this	Press
Save record	Alt+S or Ctrl+S
Revert (restore)	Ctrl+Shift+F5
Data refresh	Shift+F5
Move to the visible first field on the page	Ctrl+Shift+F
Toggle edit mode	F2
Attach a document	Ctrl+Shift+A
Export to Excel	Ctrl+Shift+E
Move to the previous record (outside a grid)	Ctrl+Up arrow
Move to the next record (outside a grid)	Ctrl+Down arrow
Move to the first record (outside a grid)	Ctrl+Home
Move to the last record (outside a grid)	Ctrl+End
Close the page (select the **Back** button) **Note:** If any popup window (tooltip, date picker, or drop-down control) is open, the first use of the **Esc** key will dismiss it.	Esc
Close the page with explicit save	Shift+Esc
Close the page discarding any unsaved changes	Alt+Shift+Q

Page navigation shortcuts

Expand table

To do this	Press
Move to the next/previous field	Tab / Shift+Tab
Move to the next/previous tab	Alt+Shift+Right arrow / Alt+Shift+Left arrow
Move to the <n>th tab	Alt+Shift+<n> (<n> = 1-9)
Move to the next/previous FastTab	Alt+Shift+Down arrow / Alt+Shift+Up arrow
Move to the <n>th FastTab	Alt+<n> (<n> = 1-9)

To do this	Press
Move to the next/previous blade (vertical tab)	Alt+Shift+Right arrow / Alt+Shift+Left arrow
Move to the <n>th blade (vertical tab)	Alt+Shift+<n> (<n> = 1-9)
Expand a FastTab (with focus on the FastTab header)	Space or Enter
Collapse the current FastTab	Alt+0
Switch to grid view	Ctrl+Shift+G
Switch to details view	Ctrl+Shift+D
Switch to header view	Ctrl+Shift+H
Switch to lines view	Ctrl+Shift+L

Grid shortcuts

Expand table

To do this	Press
Vertical scrolling	Mouse wheel
Horizontal scrolling	Shift + Mouse wheel
Move to the next/previous column	Tab / Shift+Tab
Move to the next/previous row	Down arrow / Up arrow
Move to the next/previous row without selecting **Note:** This shortcut applies to multi-select scenarios only.	Ctrl+Up arrow / Ctrl+Down arrow
Select/clear the current row **Note:** This shortcut applies to multi-select scenarios only.	Ctrl+Space / Ctrl+Click
Add the next/previous row to the selected set **Note:** This shortcut applies to multi-select scenarios only.	Shift+Space
Add a range of rows to the selected set **Note:** This shortcut applies to multi-select scenarios only.	Shift+Click
Go to the next/previous page of data	Page up / Page down

To do this	Press
Create a new row at the bottom of the grid	Down arrow (from the last row)
Move to the first record	Ctrl+Home
Move to the last record	Ctrl+End
Select or clear all rows	Ctrl+Shift+M
Move to the first marked row	Alt+Shift+M, F
Move to the next marked row	Alt+Shift+M, L
Move to the previous marked row	Alt+Shift+M, P
Moved to the last marked row	Alt+Shift+M, N
Execute the default action in a grid **Note:** This shortcut is enabled when focus is on a cell containing a hyperlink and all cells in that column have hyperlinks.	Enter
Toggle focus between the selected row and the header row	Alt+T,H or Alt+Shift+H
Make the current column larger/smaller (with focus in the header row)	Right arrow / Left arrow
Move the current column to the next/previous position (with focus in the header row)	Ctrl+Shift+Right arrow / Ctrl+Shift+Left arrow
Open grid filtering for the current column (with focus in the header row)	Enter
Resize the current column to fit the contents (with focus in the header row)	A (or double click on the column sizing handle)
Open the grid options menu*	Alt+T, O
Change the type of calculated value for the current column*	Alt+T, V
Show/hide the grid footer*	Alt+T, F
Group/ungroup the current column*	Alt+T, G
Freeze/unfreeze the current column*	Alt+T, P
Go to the row in an invalid or paused state (when using fast data entry)*	Alt+T, A
Revert changes in a newly created but unsaved row*	Ctrl+Shift+F5
Delete the current row*	Alt+Del

> **Note**
>
> Shortcuts that are marked with an asterisk (*) are available only in version 10.0.32 and later.

Input control shortcuts

Expand table

To do this	Press
Open the hyperlink	Ctrl+Enter
Open the hyperlink (when focus is on a read-only control) **Note:** This shortcut only applies when the **Standardize keyboard interaction for combo box and lookup controls** feature is enabled.	Enter
Enter the session date in a date field	D
Enter the current date in a date field	T
Open lookup, combo box, date picker, drop dialog box	Alt+Down arrow Enter (when the **Standardize keyboard interaction for combo box and lookup controls** feature is enabled).
Close lookup, combo box, date picker, drop dialog box	Esc
Move focus into a lookup (when the lookup is already open)	Alt+Down arrow
Open the control's enhanced preview	Alt+Up arrow
Select text in the current field	Ctrl+A
Enter/leave the text area in an HTML editor control	Alt+Down arrow / Alt+Up arrow
Switch focus between the text area and the toolbar in an HTML editor control	F6

Messaging shortcuts

Expand table

To do this	Press
Go to the Message Center	Ctrl+Shift+F7
Go to the Message Bar	Ctrl+F7

Navigation shortcuts

Expand table

To do this	Press
Go to the dashboard	Alt+Shift+Home
Move to the navigation bar	Alt+M, N, or Alt+Shift+F1
Move to the company picker	Ctrl+Shift+O
Search for a page	Ctrl+/ or Alt+G
Open the help pane	Ctrl+?
Open the trace parser	Alt+Shift+T
Toggle the navigation pane between open, pinned open, and closed	Alt+F1
Add/remove page as a favorite (with focus on a page in the navigation pane)	Shift+F
Move to the standard Action Pane	Alt+M, A, or Ctrl+F6
Move to the Filter pane (which may include opening it)	Alt+M, F, or Ctrl+F3
Move focus to the page content (with focus in the Filter pane)	Alt+M, M, or Ctrl+F3
Move to the navigation list (which may include opening it)	Alt+M,S or Ctrl+F8
Move focus to the page content (with focus in the navigation list)	Alt+M,M or Ctrl+F8
Close the navigation list (with focus in the navigation list)	Esc
Move to the main page content (with focus in another pane)	Alt+M,M
Move to the FactBox pane (which may include opening it)	Alt+M,B, or Ctrl+F2
Move to the next feature callout	Alt+M,C

Note

The **Streamline tabbing behavior in full-page forms** feature, introduced in version 10.0.26, allows users to tab through all regions of a page without requiring the mouse or a dedicated keyboard shortcut to move between regions. You can still use the existing keyboard shortcuts (such as Alt+M key chord shortcuts) to move quickly between the regions on a page.

Personalization shortcuts

Expand table

To do this	Press
Transition the page into personalization mode	Ctrl+Shift+P
Use the Select tool (when in personalization mode)	S
Open the selected control's quick personalization dialog box (when using the Select tool)	Space or Enter
Use the Move tool (when in personalization mode)	M
Select the current control as the one to move (when using the Move tool and no control has been selected to move yet)	Space or Enter
Clear the control to move (when using the Move tool)	Esc
Move the selected control to the next position (when using the Move tool)	Tab or Right arrow or Down arrow
Move the selected control to the previous position (when using the Move tool)	Shift+Tab or Left arrow or Up arrow
Use the Hide tool (when in personalization mode)	H
Switch whether the current control is visible or hidden (when using the Hide tool)	Space or Enter
Use the Skip tool (when in personalization mode)	K
Switch whether the current control is in the tab sequence (when using the Skip tool)	Space or Enter
Use the Edit tool (when in personalization mode)	E
Switch whether the current control is editable or read-only (when using the Edit tool)	Space or Enter
Use the Summary tool (when in personalization mode)	U

To do this	Press
Switch whether the current control is a summary field in the current fast tab (when using the Summary tool)	Space or Enter
Use the Add tool (when in personalization mode)	A
Select the control whose container is used to insert the new fields (when using the Add tool)	Space or Enter
Import a personalization (when in personalization mode)	I
Export a personalization (when in personalization mode)	X
Clear this page's personalizations (when in personalization mode)	Ctrl+C
Move focus between the personalization toolbar and the page (when in personalization mode)	T
Exit personalization mode (when in personalization mode)	Esc

Segmented entry shortcuts

Expand table

To do this	Press
Open the drop-down list (when the drop-down list is closed)	Alt+Down arrow
Move focus into the input field for the current segment in the drop-down list (when the drop-down list is already open)	Alt+Down arrow
Close the drop-down list	Alt+Up arrow
Close/open the right portion of the drop-down list	Alt+Left arrow / Alt+Right arrow
Switch between "Show valid" and "Show all" modes	Alt+W
Select the value from drop-down list and move to the next segment	Enter
Move to the next/previous control on the page (when focus is in input control)	Tab / Shift+Tab
Move to the next/previous input field in the flyout (when focus is in the drop-down list)	Tab / Shift+Tab
Move up/down a row in the lookup	Up arrow / Down arrow
Move up/down a page in the lookup	Page up / Page down

To do this	Press
Move to the top/bottom of the lookup	Home / End

Task recorder shortcuts

Expand table

To do this	Press
Stop the recording (while recording)	Alt+R,S
Toggle the visibility of the Task recorder pane (while recording)	Alt+R,T
Toggle Rails mode (while playing a task guide)	Alt+R,L
Go to the previous step (while playing a task guide)	Alt+R,P
Go to the next step (while playing a task guide)	Alt+R,N
Toggle focus between the page and the animated pop-up prompt (while playing a task guide)	Alt+R,F
Expand/collapse the animated pop-up prompt (while playing a task guide)	Alt+R,C
Show more/less information in the animated pop-up prompt (while playing a task guide)	Alt+R,M

Focus management

Focus management describes how user focus is handled in an application. Finance and operations apps have specialized focus behaviors that are intended to optimize user productivity, especially for data entry scenarios. Specialized focus behaviors include the following:

- When a page is loaded, the first editable input control on the page has focus and can accept user input immediately. If there is no editable input control on the page, the fallback option for initial focus is the first input control on the page followed by the first, focusable control on the page.
- Typically when a button is selected, the action is processed and focus may return to the page. When this happens, focus tends to return to the button control that was selected. This is true in finance and operations apps **except** when the button is located in the Action Pane or on a toolbar. In these cases, when the action completes, focus returns to the last control *outside* of the Action Pane (or toolbar) that had focus before the button was selected.

- When a user opens a column header dropdown, the filter field receives focus initially. This facilitates an optimal flow for filtering by allowing the user to immediately type a value and hit Enter to apply the filter.

Additional resources

Keyboard shortcuts for missing account analysis

Last updated on 03/10/2026

Change the banner or logo

> **Note**
>
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The following procedure lists the steps that system administrators can use to update the banner or logo image that is displayed for a legal entity. The demo data company used to create this procedure is USMF.

1. Go to **Navigation pane** > **Modules** > **Organization administration** > **Organizations** > **Legal entities**.
2. In the list on the left, select the legal entity for which you want to update the banner or logo. If it is already selected, go to the next step.
3. Expand **Dashboard image** tab.
4. Select **Change**.

   The ideal resolution for a banner image is 1920 x 281 pixels. The ideal width for a logo image is 350 pixels.
5. Select **Save**.
6. Go to **Navigation pane** > **Modules** > **Common** > **Common** > **Default dashboard**. You should see the new banner or logo image on the dashboard.

   You may need to change your company, using the company picker, to the one you uploaded the banner for.

Last updated on 07/11/2025

Navigation search

Summarize this article for me

> **Note**
>
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This article explains how to use the search functionality to navigate to pages.

The application includes many areas and pages to help you perform various tasks. To quickly find the pages that you need to complete your tasks, use the navigation search feature.

To use this feature, click the **Search** icon to display the **Search** box. You can then type one or more words in the box. The system instantly searches for relevant pages in the application that match the words that you entered. For example, you could type "vendor invoice" as the input, and then the system displays results that match that input.

> **Note**
>
> The **Search** box helps you find and navigate to pages. The **Search** box doesn't help you find specific data or actions.

Quickly navigate to a particular page

The navigation search feature also serves as a great way for you to quickly navigate to a particular page. For example, if you are an accounts payable clerk who frequently uses the **Payment journal** page, you could enter "payment journal" in the **Search** box. Because the input is an exact match for the page title, the page is listed at the top of the search results, and you can quickly navigate to it.

The search results list displays the page title and the navigation path and helps you differentiate between two or more similar pages in the results.

When you search for a page, your input is matched against the page title andd its navigation path. For example, if you enter "receivable" in the **Search** box, you see results for the pages available to you in the Accounts receivable area – even though the page titles do not include the word "receivable."

Quickly navigate to a page based on the technical form name

The navigation search functionality also includes a much-requested feature for power users: the ability to quickly navigate to a page based on the technical form name. Many users are so familiar with the system that they know the exact form names they work with. If you are one of these users, you can enter **form:** followed by the name of the form you are looking for. For example, if you enter **form: vendinvoice**, the search results show all pages where the form name starts with **vendinvoice**.

Administration and security

From an administration and security perspective, the navigation search functionality only surfaces two types of results:

- Pages that are enabled in the current configuration (via configuration keys).
- Pages that the user has access to based on the user's role.

The list of search results is limited to 10 items. If you do not find what you're looking for in the results, you should try refining or updating the input.

Development

From a development perspective, the navigation search functionality is easy to use because the delay between the deployment of menu items and their appearance in search results is minimal. As long as the menu items are linked to from either the navigation pane or the dashboard, they will automatically become searchable.

Last updated on 03/10/2026

Personalize the user experience

> **Note**
>
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This article explains how you can personalize the app and covers the following subjects:

- **System-wide options** – These personalization options are made on a setup page and are available to all users. Examples include the color theme and time zone.
- **Restricted personalization access** – At this access level, user actions that are associated with typical page usage are automatically saved by the app and restored the next time that you visit the page. For example, the app stores the width of grid columns if you adjust them, and the expanded or collapsed state of FastTabs.
- **Full personalization access** – At this access level, users have access to all personalization capabilities in the app. In particular, they have access to the **Personalization** toolbar.
- **Sharing personalizations** – Users who have full personalization access can export their page personalizations and share them with other users.
- **Administration of personalizations** – Privileged users can access the **Personalization** administration page to manage all personalizations at an organizational level.

System-wide options for the current user

The **User options** page contains several system-wide settings for the current user. These options are available to all users, even users who aren't given any access to personalization. To open the **User options** page, select the **Settings** button on the navigation bar, and then select **User options**. The **User options** page has four tabs that contain various user settings:

- **Visual** – Select a color theme and the default size of elements on pages.
- **Preferences** – Select default values that are used every time that you open the system. These values include the default company, the initial page, and the default view/edit mode. (The view/edit mode determines whether a page is locked for viewing or opened for editing every time that you open it.) This tab also includes options for the language, the time zone, and date, time, and number formats. Finally, this tab includes several miscellaneous preferences that vary from release to release.
- **Account** – View or adjust your user name and other account-related options.
- **Workflow** – Select workflow-related options.

In addition to changing your user settings, you can also view and delete your usage data and personalizations from the **User options** page. To see your usage data, select **Usage data** on the Action Pane. On the **Personalization** tab, you can view and manage the personal changes that you made to pages in the system. On this tab, you can also reset feature callouts (the pop-up windows that introduce new system features). You are then alerted again about previously encountered features.

Restricted personalization access (formerly implicit personalizations)

At the **restricted personalization access** level, user actions that are associated with typical page usage are automatically saved by the app and restored the next time that you visit the page. No explicit save action is required.

Here's a list of the actions that fall under typical page usage and are covered by restricted personalization access:

- **Grid column widths** – You can adjust the width of a column in a grid by selecting the sizing bar to the left or right of the column header, and then sliding it left or right until the column is the desired width. The app stores the width that you set for a column. Then, the next time that you open that page, the column will be resized to that width.
- **Grid footer and column totals** – *(Available only when the new grid control is turned on)* You can decide whether a total should be shown at the bottom of any numeric column in a grid, and whether the grid footer should be visible. The app stores these preferences and applies them the next time that you open the page. For more information, see Grid capabilities.
- **FastTabs** – Some pages have expandable sections that are known as *FastTabs*. The app stores information about the FastTabs that you expanded or collapsed. The next time that you open the page, the same FastTabs will be either expanded or collapsed, based on your last interaction with the page. In some cases, you can help improve system performance by collapsing a FastTab, because the app doesn't have to retrieve the information for FastTabs until they're expanded. As is explained later in this article, you can also change the order of the FastTabs on a page.
- **FactBoxes** – Some pages have a **Related information** pane that shows read-only information that is related to the current subject of the page. Each section in the **Related information** pane is known as a *FactBox*. You can expand or collapse the **Related information** pane, and you can also expand or collapse individual FactBoxes. The app stores these preferences. The next time that you open the page, the **Related information** pane and the individual FactBoxes will be either expanded or collapsed, based on your last interaction with the page. In some cases, you can help improve system performance

by collapsing the **Related information** Pane or a FactBox, because the app doesn't have to retrieve the information for FactBoxes until they're expanded.

- **Action Panes** – An *Action Pane* appears near the top of most pages. The Action Pane contains buttons for many of the actions that you can perform on the current page. These buttons are often organized on tabs. You can *pin* the whole Action Pane open, or you can collapse it by default. The next time that you open the page, the Action Pane will be either open or collapsed, based on your last interaction with the page. If you pinned the Action Pane open, the last tab that you were using is displayed.
- **QuickFilters** – A *QuickFilter* appears above many grids. The QuickFilter lets you filter the grid based on a single column that you select. The app stores the column that you filtered on. Then, the next time that you open that page, the grid will use that same column for filtering by default. However, you can still select a different column to filter the grid on.
- **Column header filters** – When you filter a grid by using *column header filters*, you can change the filter operator as you require to find the data that you want. For example, you can change the operator from **begins with** to **is exactly**. Every time that you use a column header filter and change the filter operator, the app stores the change. Then, the next time that you filter on that column, the filter operator will be restored.
- **Navigation pane** – You can open the *navigation pane* by selecting the **Expand the navigation pane** button in the upper left of any page. (This button is sometimes referred to as the ***Menu*** *button*, *hamburger*, *hamburger menu*, or *hamburger button*.) You can pin the navigation pane open, or you can collapsed it by default. After you pin the navigation pane open, the app will keep it open until you collapse it.

Full personalization access (formerly explicit personalizations)

At the **full personalization access** level, users have access to all the personalization capabilities that the app provides. Because different people and companies have different needs when they interact with the app, especially in terms of utilized fields, personalization provides tools that let users and organizations tailor the way that information is ordered and interacted with in the app. These capabilities are key to providing simplified, optimized experiences in the app that are tailored to you and your organization.

For pages with saved views, an explicit save is required to persist any user changes to the user experience for the current view. For pages without saved views, these changes are saved automatically.

The following sections cover the extent of personalization capabilities that are available to users at the **full personalization access** level. Here are some of these capabilities:

- Shortcut menu options

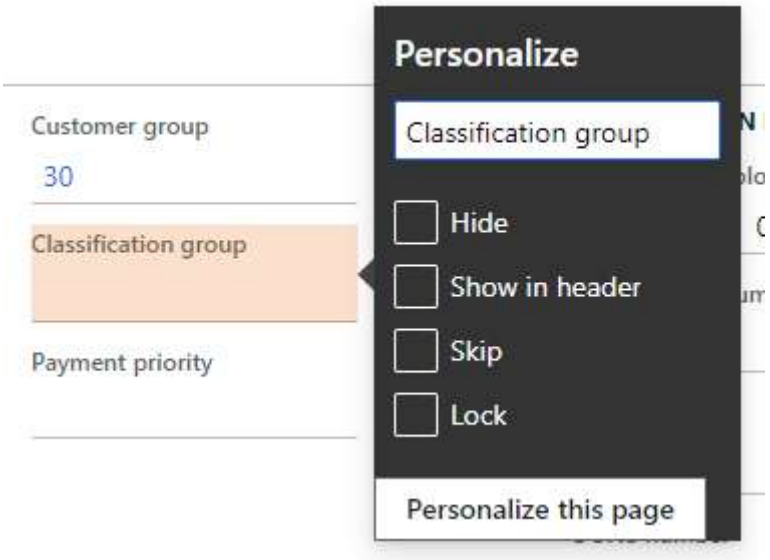
- The **Personalization** toolbar
- Adding tiles, lists, and links to workspaces
- Adding a summary from a workspace to a dashboard
- Personalizing the dashboard

Shortcut menu options

Shortcut menus provide one way to change a page's interface so that it better meets your requirements or the requirements of your organization. (A shortcut menu is also known as a *right-click menu* or a *context menu*.)

Some of the most typical and important changes that can be made to a page are available directly as options on a shortcut menu. For example, if you want to add or hide columns in a grid, just right-click a column header, and then select **Insert columns** or **Hide this column**.

Additionally, the most basic types of personalizations are available by right-clicking an element and then selecting **Personalize**. Not all elements on your page can be personalized. When you use this personalization method, the element's *property window* appears.



You can use the property window to personalize an element in the following ways:

- Change the element's label.
- Hide the element so that it isn't shown on the page. The data in the field isn't deleted or modified. The information just isn't shown on the page any longer.
- Include the information in the FastTab's summary section (if the element is on a FastTab).
- Skip the field so that it never receives focus when you tab through the page.
- Prevent data in the field from being edited (for any record).
- Designate a field to be required for data entry. If no value is entered in this field, it appears with a red border and an asterisk to indicate this state.

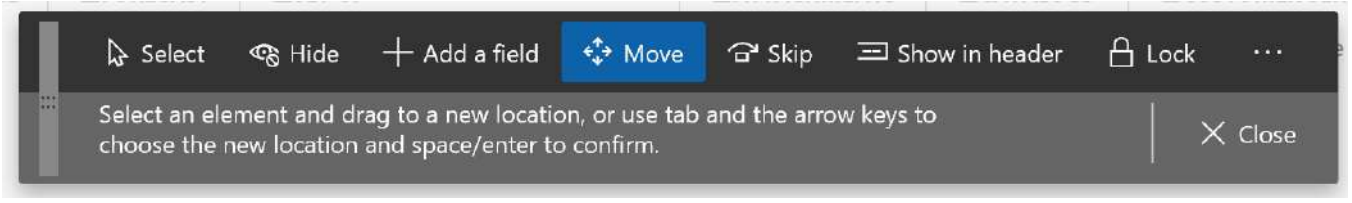
The property window might include other personalization capabilities, depending on the element. For example, the property window for a tile might let you promote that tile to a dashboard, and property windows for elements on the default dashboard might let you create a new custom workspace.

Personalization toolbar

If you want to make multiple changes to a page, or changes that aren't available through other mechanisms (for example, if you want to reorder elements), you can use the **Personalization** toolbar. To open the **Personalization** toolbar, follow one of these steps:

- Select **Ctrl+Shift+P** from any element on the page.
- Select **Personalize this page** in an element's property window.
- Select **Personalize this page** in the **Personalize** group on the **Options** tab of any page's Action Pane.
- Select the **Settings** button (the gear symbol) on the navigation bar, and then select **Personalize**.

[



]

Navigate the page

When the **Personalization** toolbar is open, the underlying page is read-only (in other words, you can't edit data), but it's still interactive. Specifically, you can expand or collapse the **Related information** pane, switch tabs, and expand or collapse sections, just as you would usually perform those actions on the page. To apply a personalization to a collapsible section or tab (for example, to hide a FastTab), you just have to select the button that appears next to that section or tab when it gains keyboard focus or when you hover over it.

Personalization tools

The following tools are available on the **Personalization** toolbar:

- Use the **Select** tool to select and change the properties of an element. To use this tool, select the **Select** button on the toolbar, and then select the desired element. The element's property window appears, where you can change any of the properties of that

element. You can repeat the process for other elements that can be personalized on the page. Some personalization properties might not be available in some scenarios. For example, you can't lock a field that is required.

- Use the **Hide** tool to hide an element on the page. To use this tool, select the **Hide** button on the toolbar, and then select the element to hide. When you use the **Hide** tool, all elements that are currently hidden are made visible, but they're shown in a shaded container. You can then make an element visible by selecting it. To see how the page looks when elements are hidden, switch to another personalization tool or close the personalization toolbar.

- Use the **Add fields** tool to add fields to your page. When you use this tool, you can only add fields that are part of the page definition. For information about how to create new fields that aren't part of the current page definition, see Create and work with custom fields. After you select the **Add fields** button on the toolbar, you must first select the grid or section where you want to add a field. A dialog shows the list of fields that are related to the selected grid or section. In the dialog, select one or more fields to add from either the **Recommended fields** or **All fields** list. After you select the desired fields, select **Update**. To remove a field that you previously added, repeat the process, but clear the selection of the field in the dialog.

  The **Recommended fields** list shows fields that were previously added by other users in your organization. This list of fields is updated based on the recurrence frequency of the **Recommendation batch job**. A similar experience exists when adding new filter fields using the **Filter** pane on a page.

- Use the **Move** tool to move an element to a different location in the current group of elements. You can't move an element outside its parent group. To use this tool, select the **Move** button on the toolbar, and then select the element to move. When you select an element, the app determines the locations where the element is allowed to be moved. These locations are known as *drop zones*. As you drag the element around in the current group, each drop zone is shown as a colored, bold line next to the area where the element can be dropped.

- Use the **Skip** tool to remove an element from the page's keyboard tab sequence. When you select the **Skip** button on the toolbar, all elements that are currently skipped are shown in a shaded container. You can interactively remove or add fields to the tab sequence.

- Use the **Show in header** tool when you want a field to appear in the FastTab's summary section. When you select the **Show in header** button on the toolbar, all fields that are selected as summary fields are shown in a shaded container. You can interactively add

fields to the FastTab summary and remove fields from the summary by selecting the fields.

- Use the **Require** tool to designate an element as required for data entry. When you select the **Require** button on the toolbar, all elements that are personalized to make them required are shown in a shaded container. You can then make them not required again.
- Use the **Lock** tool to mark an element as either editable or noneditable. When you select the **Lock** button on the toolbar, all elements that are currently noneditable are shown in a shaded container. You can then make them editable again. Some fields are required and can't be made noneditable. A padlock symbol appears next to those fields.
- Use the **Add an app from Power Apps** tool to embed an app that was created by using Microsoft Power Apps into the page. This option is available only when the Saved views feature is turned off.
- Use the **Add an app** button to embed an app into the page. This app can either be an app created from Microsoft Power Apps or a [third-party](embed-website.md] app.
- Use the **Clear** tool to reset the page to its default state. This action applies to the current view for pages that support saved views and to all personalizations on the current page for pages that don't support saved views. You can't undo this action, so only use this tool if you're sure that you want to reset the view or page.
- Use the **Import** tool to load a personalization from a file that you or someone else previously created. The imported personalizations then become a view on the page. If the view already exists, you have the option to skip the import, replace the current view that has the same name, or rename the imported view.
- Use the **Export** tool to save the current view for the page to a file with which you can share your personalizations with other users. Users can then import the file that contains your personalizations for the page.
- Select the **Close** button to close the **Personalization** toolbar and return the page to its previous interactive state.

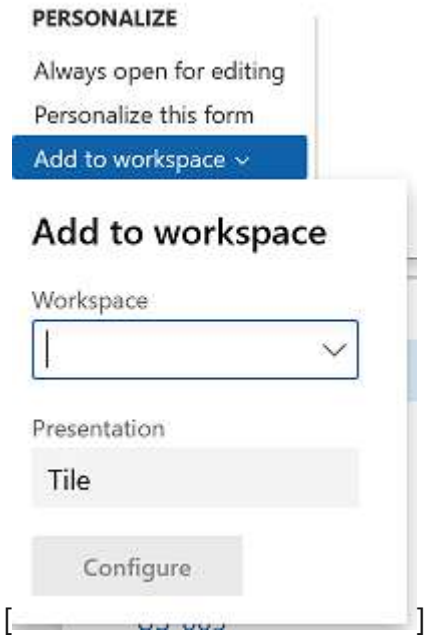
When you use the **Personalization** toolbar, you must explicitly save personalizations to a view for them to be persisted.

In some cases, when you select a tool, a padlock symbol appears next to an element. This symbol indicates that you can't modify the element properties that are related to the selected tool, because changes to those properties can prevent the page from working correctly.

Add tiles, lists, and links to a workspace

For some pages that include lists, the **Add to workspace** personalization feature is available in the **Personalize** group on the **Options** tab of the Action Pane. This feature lets you push relevant information from the current list to a specific workspace. The information that appears in the workspace can be based on either the whole list, or a filtered and sorted version of the list. You can also specify whether the information appears in the workspace as a list, a summary tile that can show the number of items in the list, or a link.

> **Note**
>
> If the Saved views feature is turned on, the content that you push to a workspace is directly linked to a view. The view's query is used to retrieve data into the workspace, and the corresponding tile or link in the workspace opens the page to that view, so that the view's query and personalizations are applied to it. If the view is updated, the corresponding workspace elements will be adjusted to the new view definition.



- To add a list to a workspace, first sort or filter the list on the page so that it displays the information as you want it to appear in the workspace. Save it to a view if the current page has a large view selector. After you select **Add to workspace**, select a workspace and then select **List** in the **Presentation** field. After you select **Configure**, select the columns that should appear in the list and specify the label to be used for the list in the workspace.
- To add a tile to a workspace, first filter the list on the page so that it displays the data that should be summarized or that you want quick access to (and save it to a view if the current page has a large view selector). After you select **Add to workspace**, select a workspace and then select **Tile** in the **Presentation** field. After you select **Configure**, you can specify the label that should be used for the tile in the workspace, choose whether

the tile should show a count, and select from one of four available tile sizes. (You can change the tile size directly from the workspace later.).

- To add a link to a workspace, first filter the list on the page so that it displays the data that you're interested in (and save it to a view if the current page has a large view selector). After you select **Add to workspace**, select a workspace and then select **Link** in the **Presentation** field. After you select **Configure**, you can specify the label that should be used for the link and also optionally a label for the section on the workspace where this link is to be placed. If that section doesn't exist, a new section is created.

> ⓘ **Note**
>
> If the **Saved views for workspaces** feature is enabled, when you configure your list, tile, or link you also have to select the workspace view to which you want to add the element. The available workspace views appear in the **Workspace options** section of each **Configure** dialog.

After you add a list, tile, or link to a workspace, you can open that workspace and rearrange the elements within it as you want.

Promote a tile from a workspace to a dashboard

Some workspaces contain count tiles (that is, tiles that have numbers on them), and you might want those tiles to appear on your dashboard too. In a workspace, right-click a count tile, select **Personalize**, and then, in the tile's property window, select **Pin to dashboard**. The next time that you open and refresh the dashboard, the count will appear below the navigation tile for that workspace. You can select that count to go directly to the data that it represents.

Change the size of a tile

Users can modify the size of any non-KPI tile via personalization. In a workspace, right-click a tile, and then select **Personalize**. In the tile's property window, select the desired size from the **Tile size** options. After you select a size, the tile size adjusts immediately. If the **Saved views support for workspaces** feature is enabled, you can save this personalization to a workspace view.

Personalize your dashboard

The dashboard is often the first page that you see when you open the app. It can be personalized like any other page in the system, by using the same mechanisms that are described earlier in this article.

One unique personalization capability that is available on the dashboard is the ability to add tiles.

- If the **Full-page apps** feature is turned off, you add a new tile by right-clicking an element on the dashboard and then selecting **Add a workspace**. A new workspace tile is created at the bottom of the dashboard. You can rename this new workspace tile to whatever you want.
- If the **Full-page apps** feature is turned on, you add a new tile by right-clicking an element on the dashboard and then selecting **Add an app**. In the dialog, select whether you want to add a tile for a new workspace or a tile that has content from Power Apps or a website. Then follow the steps to configure the option that you selected. A new tile is then created at the bottom of the dashboard.

Share personalizations

After you personalize a page, there are a few methods that you can use to share your personalizations with other users. In the following list, the methods are arranged in order from most recommended to least recommended.

1. Publish views to users.
2. Copy views or personalizations to users.
3. Export and import views or personalizations.

Publish views to users

If the page supports saved views, the best way to share personalizations with other users is to publish the view to users who have one or more security roles. For more information, see Publishing views.

Copy views or personalizations to users

If the page doesn't support views, the recommended way to share personalizations is to copy them between users. This method is available only to privileged users (for example, system administrators) who can look up a specific user's personalization in the system (including the users' personal views) and copy the configuration to other users.

To copy personalizations between users, follow these steps:

1. Go **System administration** > **Setup** > **Personalization**.

2. To copy personal views, follow these steps:
   a. Select **Personal views**.

b. Select the desired views in the list.
c. Select **Copy to users**.
d. Select the users to distribute the views to.

3. To copy personalizations on pages that don't support views, follow these steps:
   a. Select **User settings**.
   b. Select the user who has the personalization that you want to distribute.
   c. Select **Manage all personalizations**.
   d. Select the desired personalizations in the list.
   e. Select **Copy to users**.
   f. Select the users to distribute the personalizations to.

Export and import views or personalizations

Another way to share personalizations is via export and import. Individual users, or an admin who acts on their behalf, can use this method to export their personalizations or views, and then give the exported file to other users to import. Alternatively, users can give their exported personalizations to a user who has admin privileges, and that user can then use the **Personalization** administration page to apply the personalization file to many users at the same time.

> **Important**
>
> As personalizations persist across updates, re-importing all personalizations after a service update or at any other time is unnecessary and highly discouraged.

Export views or personalizations

In general, you can export one of your own views or personalizations by opening the appropriate page, opening the **Personalization** toolbar, and then selecting **Export**. For more information about the toolbar, see the Personalization toolbar. Alternatively, you can go to **Settings** > **User options** > **Personalization** to view a list of all your personalizations in the system. From there, you can select the views or personalizations to export, and then select **Export**.

Additionally, admins can export other users' personalizations by following these steps.

1. Go to **System administration** > **Setup** > **Personalization**.
2. On the **Users** tab, select the desired user.
3. Find and select the view or personalization that you're interested in.
4. Select **Export**.

Import views or personalizations

To import a view or personalization, you can just open the **Personalization** toolbar and select **Import**. Additionally, admins can import a file and immediately give it to one or more users.

If saved views are enabled, follow these steps:

1. Go to **System administration** > **Setup** > **Personalization**.
2. On the Action Pane, select **Import views** > **User views**.
3. Select the import mode:
    - **Select specific users** – Give the view or personalization to selected users.
    - **Import as-is** – Import the view or personalization to the same user who exported it.
4. Select **Browse**, and find and select the personalization to import.
5. Select **Next**.
6. If you selected **Select specific users** in step 3, select the users to import the personalization to.
7. Select **Import**.
8. Resolve conflicts as required.

Administer personalizations

The **Personalization** page is the central hub for managing personalizations at an organizational level. For more information, see Managing views globally.

Service updates and personalization

Personalization data doesn't need to be cleared or reset when moving to a new version of the product. Personalizations and saved views data continue to work across updates. Exceptions include changes to the form itself that invalidate a personalization (such as a removed control) or changes to the form's base query that changes its data source subtree so that it no longer matches the data source subtree associated with the view query.

Personalize inventory dimensions

When you personalize the setup of inventory dimensions on a page, consider the settings that have been created by using the **Display dimension** option. For example, you use

personalization to hide a column for the Batch number inventory dimension, but the column appears the next time that the page is opened. This behavior occurs because the **Dimension display** settings control the inventory dimension columns that are shown. The **Dimension display** settings apply across all pages and override any personalized setup of inventory dimension fields on individual pages.

Therefore, in the preceding example, if you don't want the column for the Batch number inventory dimension to appear on a page, you must clear that dimension as part of the **Display dimensions** option for that page.

Last updated on 03/26/2026

Saved views

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Personalization plays an important role in allowing users and organizations to optimize the user experience to meet their needs. For more information on personalization, see Personalize the user experience.

Traditional personalization allows users to only have one set of personalizations per page. *Saved view* features expand on personalization in several important ways:

- Views permit users to have multiple named sets of personalizations per form that they can quickly switch between as needed. This feature allows a user to create multiple optimized views of a page, where each view is tailored to fit the needs of performing a particular business task.
- Views created for particular page types can also include user-added filters or sorts, which allows users to quickly return to commonly filtered datasets. For more information, see What pages support views.
- You can publish views to users in specific security roles and specific legal entities. Therefore, any user who has a specified role and access to a specified legal entity can access and use that view, even if that user doesn't have permission to personalize. This publish capability lets organizations define corporate, standard views that are optimized for their business. For more information, see the Managing personalizations at an organizational level with views section.
- Unlike traditional personalization, views aren't automatically saved when a user performs personalizations or filters a list. Explicit saves are required to give users the flexibility to create a view before or after the changes that are associated with that view are made. This requirement also ensures that view definitions aren't unintentionally changed by filters or personalizations that aren't intended for long-term use. Items that the system automatically stores as part of typical page usage (for example, column widths, or the expanded or collapsed state of sections) are saved per view.
- You can add views to workspaces as tiles, lists, or links. Therefore, a filtered data set can be surfaced in a workspace, and users can associate a set of personalizations that is relevant to that data set with a tile or link.

Switching between views

After you make views available for an environment, the top of any page that supports views includes a collapsed view selector control that shows the name of the current view.

The view selector comes in two size variations:

- **Large view selectors** – Pages that prominently feature a list use a larger view selector. Most importantly, the larger view selector indicates the pages where the view can include user-defined filters and sorts. Because filters and sorts are included in the views, the larger selector size is also warranted as the view names is often the best description of the data that is shown on the screen and the expectation is that users switch between views more often on these page types. Grouping in a grid can also be saved to views on a page with large view selectors.
- **Small view selectors** – All other full-screen pages (except workspaces and the dashboard) use a smaller view selector that appears next to the page caption. Views on these pages include only personalizations, not user-defined filters. On these pages, the caption or record title is often the most important information at the top of the page. The smaller size of the view selector also reflects the lower frequency of view switching that is expected on these pages.

If you select the view name, the view selector opens and shows the list of available views for the page in up to two sections. The first section shows any views that are specific to the current legal entity, and the second section shows views that are available to all legal entities. The first section is only visible if there are legal entity–specific views for the page.

- **Standard view** – The **Standard** view is the out-of-box view of the page, where no personalizations are applied.
- **Personal views** – The views without padlocks represent your personal views. These views are ones that you created or that an administrator provides to you.
- **Locked views** – Some views, such as the **Standard** view and any views that are published to your role, have a padlock symbol next to them in the view selector. This symbol indicates that you can't edit those views. However, changes that reflect page usage are automatically saved. These changes include changes to the width of a grid column, and changes to the expanded or collapsed state of a FastTab. Nevertheless, if you have personalization privileges, you can use the **Save as** action to make a personal view that is based on a locked view.
- **New views** – Published views that you didn't open have a spark symbol to the left of the view name.

To switch to a different view, open the view selector and then select the view that you want to load.

Creating and modifying views

Unlike traditional personalization, the system doesn't automatically save views when you personalize the page, apply a filter to a list, or sort a list. You must take explicit action to save these changes to a view. This requirement gives you the flexibility to create a view before or after you make the changes that are associated with that view. It also ensures that one-time filters or personalizations don't unintentionally change view definitions. Typical page usage items, such as column widths or the expanded or collapsed state of sections, automatically save to the current view, even for locked views.

To ensure that the system knows the current state of the view, when you start to change a view by personalizing or filtering it, an asterisk (*) appears next to the current view name. This symbol indicates that you're looking at an unsaved, modified version of that view.

If you want to save those changes, follow these steps:

1. Select the view name to open the view selector.
2. To modify the existing view, select **Save**. This action isn't available for locked views.
3. To create a new view:
   a. Select **Save as**.
   b. In the **Save view as** pane, enter a name and, optionally, a description for the view.
   c. If you want this view to be your default view, select **Pin as default**. For more information about default views, see Changing the default view.
   d. (Optional) Select whether you want this view to be available for just a subset of legal entities. By default, views are saved as global views.
   e. Select **Save**.

Change the default view

The default view is the view that the system tries to open when you first open the page. Set the default view to the view that you expect to use most often.

To change the default view for a page, follow these steps:

1. Switch to the view that you want to use as the default.
2. Select the view name to open the view selector.
3. Select **More** and then **Pin as default**.

> ⓘ **Note**

- If the view is specific to one or more legal entities, it becomes the default view for those legal entities.
- If the view is global, it becomes the default view for legal entities that don't have a legal-entity specific default view defined.

Alternatively, when you create a new view (by using the **Save as** action), make that new view the default view by setting the **Pin as default** option before you save the view.

> ⚠ **Warning**
>
> In some cases, the query that is associated with the default view doesn't run when you first open a page. For example, if you open the page through a tile, the tile's query runs, regardless of the query that is associated with the default view. Additionally, if you open a page that has a **standard** view that already has a defined query, the original query runs instead of the default view's query. In this case, you receive an informational message when the view loads with an embedded action that lets you load the default view's query directly. If you switch views after the page loads, the view query runs as expected.

Managing personal views

The **Manage my views** dialog box provides basic maintenance capabilities for your personal views and the order of views in the view selector. To open this page, select the view name to open the view selector drop-down menu, select **More**, and then select **Manage my views**.

The **My views** section of the **Manage my views** dialog shows the available views for the page in sections. Any views that are specific to the current legal entity are displayed in their own section. The **Global views** section is always displayed so you can manage the available views for the page in all legal entities.

For a list of available views for that page, the following set of actions are available.

- **Change the default view** – Use the **Pin as default** action to make the currently selected view the default view for the page. When you take this action on a view in the **Global views** section, you can choose to make the view the default view for either the current legal entity or all legal entities.
- **Reorder your views** – Use the **Move up** and **Move down** actions to rearrange your views in a specific order.
- **Rename a view** – Use the **Rename** action to change the name of the currently selected personal view. This action is turned off for locked views.

- **Delete a view** – Use the **Delete** action to permanently delete the currently selected view from the page. There's no way to recover a view after you remove it.

Any changes you make in this dialog box take effect after you select the **Update** button.

Managing personalizations at an organizational level with views

To help you understand how saved views help improve management of personalizations at an organizational level, this section describes some differences in personalization management for pages that do and don't support saved views.

When views aren't available, administrators can apply a set of personalizations for a page to a user or a group of users via the **Personalization** page. However, administrators can't prevent users from further personalizing the page, which means the organization can't ensure that its users have a consistent user interface on that page. If new users are hired into an organization, administrators must manually load a set of personalizations for each new user because there's no automatic mechanism for specifying that a certain set of personalizations should be available for users in that role.

> ⓘ **Note**
>
> If the administrator applies personalizations to a user without personalization rights, the personalizations aren't applied and don't impact the user interface.

With saved views, organizational management of personalizations is easier because you can publish views to groups of users. After a view is published, any user who has one of the defined security roles and access to one of the specified legal entities can see and use the view, even if that user doesn't have access to personalization. These published views can't be edited by users, although users with personalization rights can create their own personal views starting from a published view definition. If new users are assigned to roles in legal entities to which views were published, they automatically see the views associated with their roles and legal entities. No additional action is required by the administrator. Likewise, if users change roles in an organization or are given access to different legal entities, they might no longer be able to access the views that were previously published to them. Again, no additional action is required by the administrator.

You can easily distribute updates to a published view to users by republishing the view to the appropriate security roles and legal entities.

The publish capability allows organizations to define corporate standard views that are optimized for their business, targeted at users in specific security roles.

Publishing views

During the publishing process, assign views to one or more security roles for one or more legal entities. Any user who has access to a legal entity and is assigned to one of those roles can access and use the views. However, the user can't edit the views. By default, system admins have access to the **Publish** action in the view selector drop-down menu. However, other trusted users in your organization can also be given access to view publishing via the new **Saved views administrator** role.

To publish a view, follow these steps:

1. Create and save a personal copy of the view that you want to publish.
2. With that view currently loaded, select the view name to open the view selector drop-down menu.
3. Select the **More** button and then select **Publish**. The **Publish** dialog box opens.
4. Enter a name for the view. The name that you enter is the name that users who receive this view see in their view selectors. The names of published views for a page must be unique. No duplicate names are allowed, even if the list of roles or legal entities that the views are applied to differ.
5. (Optional) Add translations for your view name in as many languages as your organization requires by selecting the **Translations** button next to the **Name** field. The view name is then shown to users in their current language. You can also set the default language to specify the translation that is shown to users who are running languages that no translation is defined for.
6. (Optional) Enter a description for the view, so that users who receive this view can understand its purpose.
7. Determine whether the view should be published as the default view for the selected users. When you make a view the default view, users see it the next time that they open the target page. However, users can still change their default view after publishing occurs.

   > ⓘ **Note**
   >
   > Be aware of the following behavior when you publish a view as the default view:

- If you publish the view as the default for all legal entities, this action overrides any legal entity default view a user previously set.
- If you publish the view to a subset of legal entities, the default view for those legal entities changes for every targeted user.
- If a user has roles where multiple views are published as the default view, the last view that you published is used as the user's default view.
- Publishing doesn't work for role assignments made using Microsoft Entra groups.

8. Add the security roles that correspond to the users who are being targeted by this view.
9. Determine whether you want to publish the view to the child roles of each security role that you select. If you do, select the **Include child roles** checkbox in the row for the appropriate security roles. This checkbox isn't available for roles that don't have child roles.
10. Add the legal entities that this view should be available for.
11. Select **Publish**.

In some environments, it might take up to an hour before users see the published view.

> ⓘ **Note**
>
> Published workspace views don't appear in the view selector for the original workspace. Instead, published workspace views follow the pattern of published embedded apps and appear as new tiles on the dashboard.

Modifying a published view

After you publish a view, you might want to change it. Although you can't make live changes to a published view, because these views are locked for editing for all users (including publishers), you can republish a view to update it.

If the changes that you want to make to a published view only involve the publish parameters (the name and description of the view, or the security roles the view is published to), follow these steps:

1. Switch to the published view for the parameters that you want to update.
2. On the view selector drop-down menu, select **Republish**. If you're using version 10.0.12 or earlier, you must select **Publish** and then **Yes** to update the existing view.

3. Update the name, description, security roles, and legal entities for the view.
4. Select **Publish**. If you originally selected this published view as the default view, it becomes the default view for users again after you republish it.

If the changes to the published view involve modifications of the personalizations or filters that are associated with the view, follow these steps:

1. Load the published view that you want to change.
2. Make the required changes to the local draft.
3. On the view selector drop-down menu, select **Republish**.
4. Select **Yes** to indicate that you want to publish the view together with its unsaved changes.
5. Adjust any publishing parameters that require adjustment, and then select **Publish**.

Managing published views

Like managing personal views, the **Manage my views** dialog box gives users with publish privileges basic maintenance capabilities over that page's published views (in addition to their own personal views). To open this page, select the view name to open the view selector drop-down menu, select **More**, and then select **Manage my views**.

Although all users have a **My views** tab that shows their personal views, users who have publish privileges also have an **Organization views** tab that shows all the published and unpublished views for that page. Because several users might be publishing views, it's important that you can manage the full list of published views, even if you're not the user who published a given view.

For the list of all published views for the page, the following set of actions are available:

- **Republish** – Use the **Republish** action to republish a view after publishing parameters (name, description, security roles, or legal entities) change.
- **Publish** – Use the **Publish** action to publish a view that is currently unpublished.
- **Unpublish** – Use the **Unpublish** action to make a view inactive. The view is still available in the system, but users don't see it in the view selector until the view is published again.
- **Save as personal** – Use the **Save as personal** action to create a personal draft copy of the published view. This capability can help you understand the contents of a view that wasn't published to you or that wasn't published. You can also use it to edit and then republish a view.
- **Delete** – Use the **Delete** action to permanently delete a published or unpublished view. This action also removes the view for all users in the system. The removal of published views takes effect after the **Save** button is selected. After a view is deleted, it can't be recovered.

Managing views globally

Although some management capabilities are surfaced on every page, as indicated in this article, **system administrators** and **saved view administrators** can manage views more holistically for the system via the **Personalization** page. In particular, this page has the following sections and capabilities:

- **Published views** – This section lists all views that are published for your organization. From here, you can republish a view after you adjust the security roles or legal entities that the view targets. You can also export, delete, or unpublish views. Use the **Save as personal** action to create a personal copy of a view, so that you can update the view or gain a better understanding of its contents.
- **Unpublished views** – This section lists all the organization views in your system that aren't currently published. These views most often come into the system through the import capability. You can publish, export, or delete these views. The **Quick publish** action that was added in version 10.0.12 enables multiple views from this section to be published in one action, by using the existing security role and legal entity configurations. Use the **Save as personal** action to create personal copies of these views, so that you can gain a better understanding of their contents.
- **Personal views** – This section lists all views created by users in the system. From here, you can publish a personal view to the organization, or copy one or more of these views to other users. You can also export or delete these views as required.
- **User settings** – Select a user to view, or adjust the user's ability to use personalization either for the whole system or for specific pages that the user visited. You can view and interact with the user's personalizations in the system. You can also delete all personalizations for that user or reset feature callouts for the user. If you reset feature callouts, any pop-up windows that introduced new features and that the user previously dismissed appear again the next time that the user encounters those features.
- **System settings** – You can temporarily turn off personalization for all users in the system. In this case, no personalizations are applied for any user, and all pages are reset to their default state. If you turn personalization back on later, all personalizations are reapplied. You can also permanently delete all personalizations for all users in the system. Deleted personalizations can't be recovered. Therefore, before you perform this task, be sure to export any personalizations that you might want later.

Users who have access to the **Personalization** page can also import personal or organization views by using the **Import views** button on the Action Pane. For organization views, you can select **Publish immediately** to make the views available to users without an additional explicit publish.

Frequently asked questions

What pages support views?

Most pages support views, but not all. Specifically, all full-screen pages support views, except for dashboards. The **Saved views support for workspaces** feature provides view support for workspaces. Most pages that aren't full screen, such as drop-down dialogs, lookups, and enhanced previews, don't support views.

Who can publish views?

Only system admins and users assigned to the **Saved views administrator** role can publish views.

Why can't I save filters with this view?

A filter might not save with a view for a few reasons:

- The page might not support saving filters as part of the view definition. Only pages with large view selectors support saving personalizations and query modifications as part of the view.
- The page might not properly support views. It might ignore the view query completely or operate on a temporary table whose data isn't persistent.

What data do I see when I visit a page?

For pages that have small view selectors (only personalizations can be saved to the view), you see the same data as you always see when you visit the page.

For pages that have large view selectors (both personalizations and queries can be saved to the view), you typically see the data that is linked to the query associated with your default view. Two main exceptions exist:

- If you go to a page from a tile, the tile query executes regardless of the query associated with the default view. If you created that tile after views are enabled, selecting a tile opens the page with the view associated with that tile.
- If you go to a page and that entry point includes a query, the original query executes instead of the default view's query. An informational message when the view is loading alerts you when this occurs. You can also confirm by switching to this view after the page loads, as that action should allow the view query to execute regardless.

Last updated on 03/26/2026

Standard saved views for Supply Chain Management

07/10/2025

Microsoft Dynamics 365 Supply Chain Management includes several saved views that you can enable and use as needed. Some of these standard saved views are optimized and named for a specific role or task (for example, "Quality control" or "Receiving"). Others are optimized so that they include only the fields and settings that Microsoft usage statistics indicate are most often used by customers. These saved views are typically referred to as *simplified* views. This article describes the standard saved views that are available, and explains how to enable and customize them.

For complete details about how to work with saved views, including the standard saved views, after you enable them, see Saved views.

Customizing the standard saved views

You can customize the standard saved views, just as you can customize your own saved views. However, when you customize a standard saved view, we strongly recommend that you save your custom version under a new name. Otherwise, your customizations could be overwritten when you update Supply Chain Management.

For more information about how to customize and rename saved views, see Saved views.

Saved views for the Inventory management module

The following table describes the saved views available for the Inventory management module.

Expand table

Page	View name	View description
On-hand list	Financials	This simplified view lets you focus on financial information while you manage on-hand inventory.
On-hand list	Quality control	This simplified view lets you focus on quality control while you manage on-hand inventory.
On-hand list	Receiving	This simplified view lets you focus on receiving operations while you manage on-hand inventory.

Page	View name	View description
On-hand list	Shipping	This simplified view lets you focus on shipping operations while you manage on-hand inventory.
Transactions	Simplified	This simplified view lets you review inventory status without being distracted by financial information and other fields that are used less often.
Transfer orders	Shipping	This simplified view lets you focus on shipping operations while you manage transfer orders.
Transfer orders	Receiving	This simplified view lets you focus on receiving operations while you manage transfer orders.
Transfer orders	Quality control	This simplified view lets you focus on quality control while you manage transfer orders.
Transfer orders	Financials	This simplified view lets you focus on financial information while you manage transfer orders.

Saved views for the Master planning module

The following table describes the saved views available for the Master planning module.

Expand table

Page	View name	View description
Planned orders: Planned order details page	Simplified	This simplified view includes only the fields that are most often used to work with the details of a single planned order. In this way, it provides a quicker overview and a streamlined work process.
Planned orders: Planned orders list page	Simplified	This simplified view includes only the fields that are most often used to work with the list of planned orders. In this way, it provides a quicker overview and a streamlined work process.

Saved views for the Procurement and sourcing module

The following table describes the saved views available for the Procurement and sourcing module.

Expand table

Page	View name	View description
Purchase order details	Order creation	This simplified view is optimized for creating new purchase orders.
Purchase order details	Inventory management	This simplified view is optimized for performing inventory-related activities, such as following up on inventory that has been received, receiving inventory, checking net requirements, and adjusting order quantities.
Purchase order details	Financial management	This simplified view is optimized for performing finance-related activities, such as invoicing and checking prices, totals, and charges.
Purchase order details	Order approval	This simplified view is optimized for approving purchase orders.

Saved views for the Product information management module

The following table describes the saved views available for the Product information management module.

Expand table

Page	View name	View description
Released products list	Product creation	A simplified page view that only includes the fields most often used when creating products.
Released product details	Product creation	A simplified page view that only includes the fields most often used when creating products.
Released product details	Logistical information management	A simplified page view that only includes the fields most often used when managing logistical information for products.
Released product details	Purchase information management	A simplified page view that only includes the fields most often used when managing purchase information for products.
Released product details	Sales information management	A simplified page view that only includes the fields most often used when managing sales-related information for products.

Saved views for the Production control module

The following table describes the saved views available for the Production control module.

Expand table

Page	View name	View description
Production order BOM page	Simplified	This simplified view includes only the fields that are most often used. In this way, it provides a quicker overview and a streamlined work process.
Production order details page	Simplified	This simplified view includes only the fields that are most often used. In this way, it provides a quicker overview and a streamlined work process.
Production order picking list page	Simplified	This simplified view includes only the fields that are most often used. In this way, it provides a quicker overview and a streamlined work process.
Production orders list page	Simplified	This simplified view includes only the fields that are most often used. In this way, it provides a quicker overview and a streamlined work process.

Saved views for the Sales and marketing module

The following table describes the saved views available for the Sales and marketing module.

Expand table

Page	View name	View description
Packing slip journal	Journal review	This simplified view includes only the fields that are most often used to review packing slip journals.
Sales order	Order creation	This simplified view includes only the fields that are most often used to create sales orders.
Sales order	Order review	This simplified view includes only the fields that are most often used to review sales orders.
Sales quotation	Quote creation	This simplified view includes only the fields that are most often used to create sales quotations.

Saved views for the Warehouse management module

The following table describes the saved views available for the Warehouse management module.

Expand table

Page	View name	View description
All loads	Inbound processing	This simplified view includes only the fields that are most often used to process inbound loads.
All loads	Outbound processing	This simplified view includes only the fields that are most often used to process outbound loads.
All shipments	Inbound processing	This simplified view includes only the fields that are most often used to process inbound shipments.
All shipments	Outbound processing	This simplified view includes only the fields that are most often used to process outbound shipments.
All waves	Simplified	This simplified view includes only the fields that are most often used. In this way, it provides a quicker overview and a streamlined work process.
Load planning workbench	Simplified	This simplified view includes only the fields that are most often used. In this way, it provides a quicker overview and a streamlined work process.
Work details	Simplified	This simplified view includes only the fields that are most often used. In this way, it provides a quicker overview and a streamlined work process.

Grid capabilities

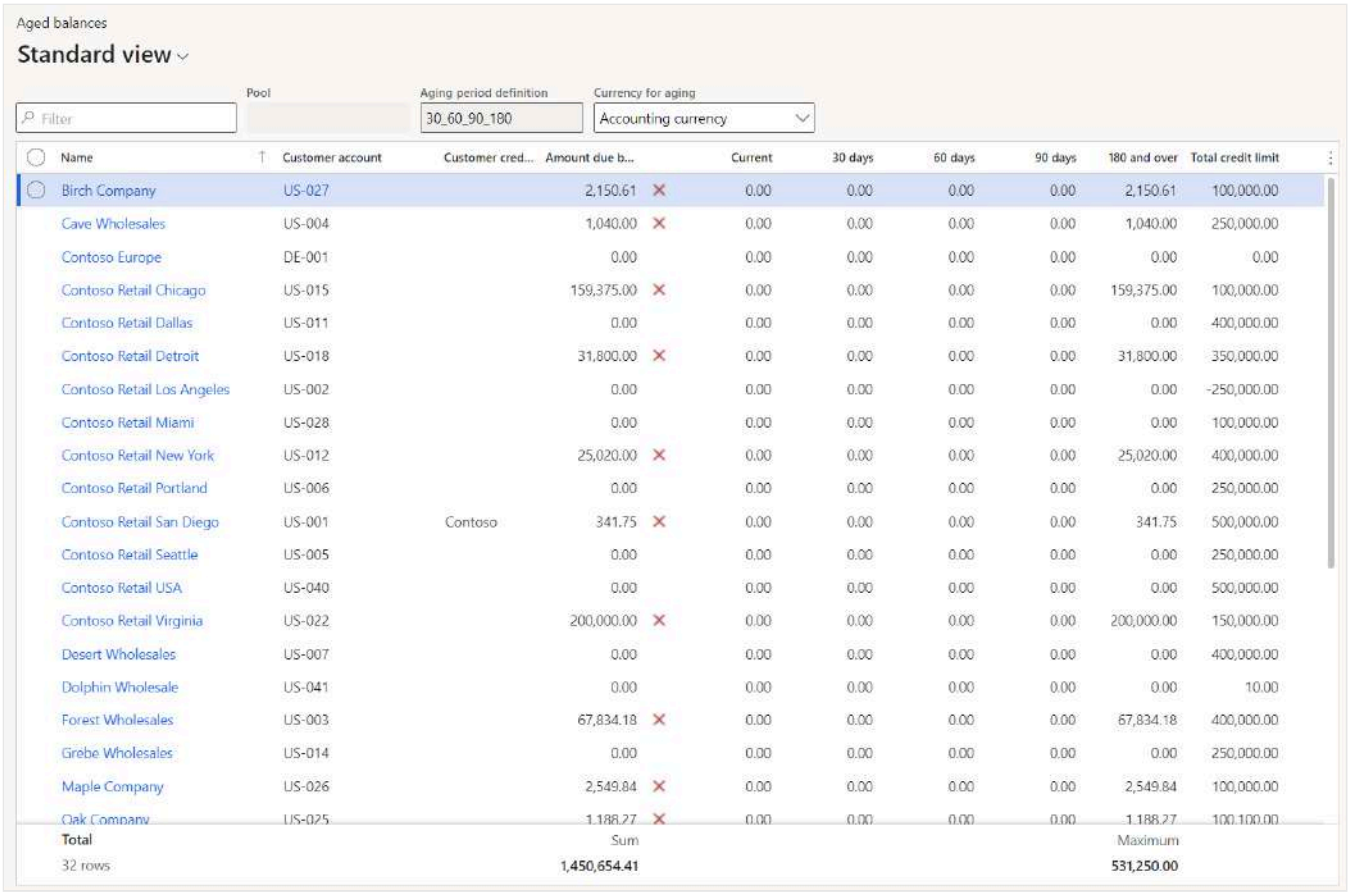
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The new grid control provides several useful and powerful capabilities that you can use to enhance user productivity, construct more interesting views of your data, and get meaningful insights into your data. This article will cover the following capabilities:

- Showing calculated values
- Typing ahead of the system
- Evaluating math expressions
- Grouping tabular data (enabled separately by using the **Grouping in grids** feature)
- Freezing columns (enabled separately by using the **Freezing columns in grids** feature)
- Autofit column width
- Stretchable columns

Showing calculated values

In finance and operations apps, users can view a calculated value for each numeric column in a grid. A footer section at the bottom of the grid shows these calculated values.



Name	Customer account	Customer cred...	Amount due b...	Current	30 days	60 days	90 days	180 and over	Total credit limit
Birch Company	US-027		2,150.61	0.00	0.00	0.00	0.00	2,150.61	100,000.00
Cave Wholesales	US-004		1,040.00	0.00	0.00	0.00	0.00	1,040.00	250,000.00
Contoso Europe	DE-001		0.00	0.00	0.00	0.00	0.00	0.00	0.00
Contoso Retail Chicago	US-015		159,375.00	0.00	0.00	0.00	0.00	159,375.00	100,000.00
Contoso Retail Dallas	US-011		0.00	0.00	0.00	0.00	0.00	0.00	400,000.00
Contoso Retail Detroit	US-018		31,800.00	0.00	0.00	0.00	0.00	31,800.00	350,000.00
Contoso Retail Los Angeles	US-002		0.00	0.00	0.00	0.00	0.00	0.00	-250,000.00
Contoso Retail Miami	US-028		0.00	0.00	0.00	0.00	0.00	0.00	100,000.00
Contoso Retail New York	US-012		25,020.00	0.00	0.00	0.00	0.00	25,020.00	400,000.00
Contoso Retail Portland	US-006		0.00	0.00	0.00	0.00	0.00	0.00	250,000.00
Contoso Retail San Diego	US-001	Contoso	341.75	0.00	0.00	0.00	0.00	341.75	500,000.00
Contoso Retail Seattle	US-005		0.00	0.00	0.00	0.00	0.00	0.00	250,000.00
Contoso Retail USA	US-040		0.00	0.00	0.00	0.00	0.00	0.00	500,000.00
Contoso Retail Virginia	US-022		200,000.00	0.00	0.00	0.00	0.00	200,000.00	150,000.00
Desert Wholesales	US-007		0.00	0.00	0.00	0.00	0.00	0.00	400,000.00
Dolphin Wholesale	US-041		0.00	0.00	0.00	0.00	0.00	0.00	10.00
Forest Wholesales	US-003		67,834.18	0.00	0.00	0.00	0.00	67,834.18	400,000.00
Grebe Wholesales	US-014		0.00	0.00	0.00	0.00	0.00	0.00	250,000.00
Maple Company	US-026		2,549.84	0.00	0.00	0.00	0.00	2,549.84	100,000.00
Oak Company	US-025		1,188.27	0.00	0.00	0.00	0.00	1,188.27	100,100.00
Total			Sum					Maximum	
32 rows			1,450,654.41					531,250.00	

In versions before 10.0.29, the total is the only supported calculated value. However, as of version 10.0.29, if your organization has enabled the **Extended grid aggregation capabilities** feature, users can select among the following four calculated values:

- Minimum
- Maximum
- Total
- Average

A single column can show only one type of calculated value. However, each column in the grid can be configured to show a different type of calculated value.

If more than one row is selected in the grid, the calculated value for the column is updated based on the values in the selected rows.

Showing the grid footer

There is a footer area at the bottom of every tabular grid in finance and operations apps. The footer can show valuable information that is related to the data that appears in the grid. Here are some examples of this information:

- The number of selected rows in the table (when you select more than one record)
- Calculated values at the bottom of configured, numeric columns (for example, grand totals)

- The number of rows in the dataset

This footer is hidden by default, but you can turn it on. To show the footer for a grid, select the **Grid options** button in the grid header, and then select the **Show footer** option. After you turn on the footer for a particular grid, that setting will be remembered until the user chooses to hide the footer. To hide the footer, select **Hide footer** on the **Grid options** menu.

Specifying columns with calculated values

Currently, no columns show calculated values by default. Instead, the setup is considered a one-time activity, like adjusting the widths of columns in grids. After you specify that you want to view a calculated value for a column, that setting will be remembered the next time that you visit the page.

There are two ways to configure a column to show a calculated value:

- Select and hold (or right-click) in the column that you want to view a calculated value for. If the **Extended grid aggregation capabilities** feature is enabled, select **View column totals**, and then select the desired calculated value. If that feature isn't enabled, select **Total this column**. This action causes three events to occur:

    1. The grid footer becomes visible.
    2. Your preference for viewing a calculated value for the column is saved.
    3. The desired calculation is initiated for the column and any other columns that you previously configured to show a calculated value. The time that is required to show the calculated values depends on the size of the dataset.

- After the footer is visible, select **Show total** (or **Select calculated value** if the **Extended grid aggregation capabilities** feature is enabled) in the footer area at the bottom of the column that you want to view a calculated value for. If there are no configured columns, that button will be available in the footer in all numeric columns.

    After at least one column is configured to show a calculated value, the **Show total** (or **Select calculated value**) button will be available only on hover or focus. The action of selecting the button just saves your preference for viewing a calculated value in the column, so that the preference is applied during future visits to the page. In the footer, this state is indicated by a dash that appears in the column. (Note that the calculated value will appear immediately if the dataset is small enough.)

If you make a mistake and no longer want to view a calculated value in a specific column, select and hold (or right-click) in the column, and then select **Hide total** (or **View column totals** > **None** if the **Extended grid aggregation capabilities** feature is enabled). Alternatively, select

Hide total (or **Hide calculated value**) in the footer in that column. This preference will also be saved for future visits to the page.

Calculating aggregate values

When you go to a page where the footer is visible and columns are already configured to show calculated values, those values might not be shown in the footer. The behavior depends on the size of the dataset on the page. If the dataset is small enough, the calculated values will be shown automatically, together with the number of rows in the dataset. If there are dashes in the footer under the columns that you configured, the dataset is too large for the system to show calculated values immediately. In this case, explicit action is required to calculate the values. To calculate the values, select the **Calculate** button in the footer. Alternatively, select and hold (or right-click) in a column that you want to view the total for, and then select **Total this column** (or **View column totals** and then the desired calculated value if the **Extended grid aggregation capabilities** feature is enabled).

If the calculation takes a long time to be completed, you can cancel the operation at any time by selecting **Cancel**. Sometimes, the dataset will be too large to calculate aggregate values (a limit that is imposed by your organization). In this case, you will instead be notified to filter your data more.

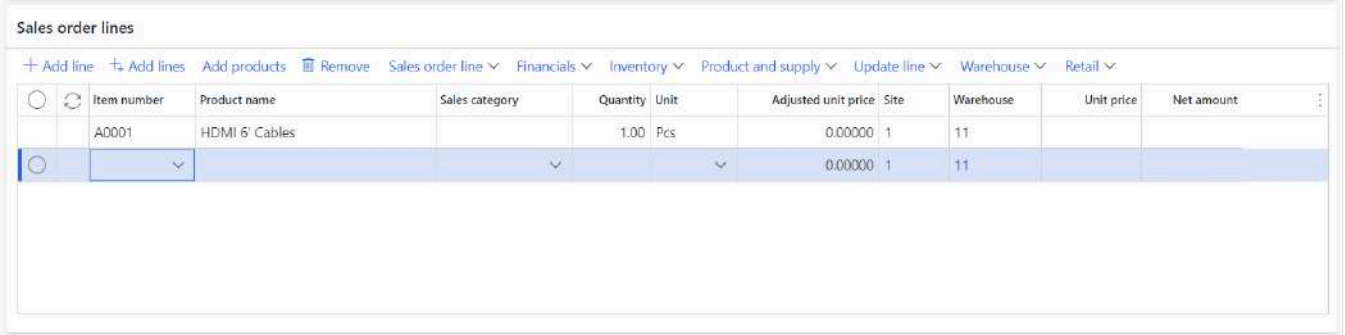
> ⓘ **Note**
>
> System administrators can modify the limit for the number of records that are available for calculating aggregates by adjusting the **Maximum number of local records for each grid** parameter on the **Client performance options** page. The default value is 25,000 records. Administrators should be careful when they adjust this value, because a value that is too large can exhaust the available memory on the user's machine. We recommend that the value not exceed 50,000 records.

Calculated values will be automatically updated as you update, delete, or create rows in the dataset.

Typing ahead of the system

In many business scenarios, the ability to quickly enter data into the system is very important. Before the introduction of the new grid control, users could change data only in the current row. Therefore, after they made changes in a row, users could not switch to a different row or create a new row until the system successfully validated the changes in the current row and (in the case of row creation) ran all the logic that is associated with the creation of a new row. To

help reduce the time that users spend waiting for these operations to be completed, and to help improve user productivity, the new grid adjusts these actions so that they are asynchronous. Users can create new rows or move to other rows to make changes while previous row validations and row creation logic are pending.



To support this new behavior, a new column for the row status has been added to the right of the row selection column when the grid is in edit mode. This column indicates one of the following statuses:

- **Blank** – No status image indicates that the row has been successfully saved by the system.
- **Processing pending** – This status indicates that the changes in the row haven't yet been saved by the server but are in a queue of changes that must be processed. Before you take action outside the grid, you must wait for all the pending changes to be processed. Additionally, the text in these rows is italicized to indicate the unsaved status of the rows.
- **Invalid state** – This status indicates that some warning or message was triggered during the processing of the row, and it might have prevented the system from saving the changes in that row. In the old grid, if the save operation was unsuccessful, you were forced back into the row to fix the issue immediately. However, in the new grid, you're notified that a validation issue was encountered, but you can decide when you want to fix any issues in the row. When you're ready to fix an issue, you can manually move focus back to the row. Alternatively, you can select the **Fix this issue** action. This action immediately moves focus back to the row that has the issue, and lets you make edits inside or outside the grid. Note that the processing of subsequent pending rows is stopped until this validation warning is resolved.
- **Paused** – This status indicates that processing by the server is paused because validation of the row triggered a pop-up dialog box that requires user input. Because the user might be entering data in some other row, the pop-up dialog box isn't immediately presented to that user. Instead, it will be presented when the user chooses to resume processing. This status is accompanied by a notification that informs the user about the situation. The notification includes a **Resume processing** action that will trigger the pop-up dialog box.

Differences when entering data ahead of the system

When you enter data ahead of the place where the system is processing, you can expect a few changes in the data entry experience:

- You will notice that there are no lookup drop-down lists, field values aren't validated after you move to a different column in the same row, and columns don't initially show default values. This behavior occurs when you create or update ahead of the system. However, after the system catches up to the place where you're currently editing, the standard experience will resume. If you made changes to a field that typically receives a default value, your changes override the default field value when the server starts to process the row.
- If you create a new row by using the **Down arrow** key, all columns in the grid will appear as editable. By default, the focus will be put in the first column in the new row. This column might not be the same column that received the initial focus in the legacy grid, or the same column that receives focus after you select a **New** button. Your organization can customize the system and change the column that receives the initial focus when the **Down arrow** key is selected. For more information, see the Specifying the column that receives the initial focus when new rows are created by using the Down arrow key section. Regardless, you can use personalization to optimize each grid for data entry. Specifically, you can reorder fields so that the first column is the column that you want to start to enter data in. You might also want to reorder fields in general for data entry, to reduce tab stops and hide any fields that aren't required for data entry in this particular view.

Editing multiple rows simultaneously (bulk edit)

In version 10.0.38 and later, users can update multiple selected rows in a single action by using the **Bulk editing in grids** feature, which can be enabled through Feature management.

To use this feature, follow these steps:

1. In the grid, select the rows that you want to update.
2. Select **Grid options** > **Edit selected rows**.
3. In the **Edit rows** dialog box that appears, select the columns that you want to update, and provide the new values. Only columns that are backed by editable fields in the grid can be selected for update. The **Edit rows** dialog box defines the order that columns will be updated in (from top to bottom). This order can affect field-level validations that are triggered during the save process. After you've finished specifying the columns and values that you want to update, select **Apply**.

As when you paste from Excel (as described in the next section), the grid shows the updated contents in italic as a preview of the data changes. Select **Discard** to cancel the bulk editing or **Save** to initiate the save process. During the save process, the system uses the **Typing ahead of the system** save model to submit the updates one row at a time. This approach ensures that all

form and field validations are run for each row. Monitor the progress and results of each validation by checking the row status at the beginning of the row.

Pasting from Excel

Users have always been able to export data from grids in finance and operations apps to Microsoft Excel by using the **Export to Excel** mechanism. However, the ability to enter data ahead of the system enables the new grid to support copying tables from Excel and pasting them directly into grids in finance and operations apps. The grid cell that the paste operation is initiated from determines where the copied table begins to be pasted in. The contents of the grid are overwritten by the contents of the copied table, except in two cases:

- If the number of columns in the copied table exceeds the number of columns that remain in the grid, starting from the paste location, the user is notified that the extra columns have been ignored.
- If the number of rows in the copied table exceeds the number of rows in the grid, starting from the paste location, the existing cells are overwritten by the pasted content, and any extra rows from the copied table are inserted as new rows at the bottom of the grid.

> ⓘ **Note**
>
> Currently, this functionality can't be used in specific situations. For example, in grids using the segmented entry control like General journal lines.

Copying from a grid in the product

Grids in finance and operations apps support a range selection capability that resembles the corresponding feature in Excel. This capability enables users to use the keyboard to select a range of cells (multiple rows and columns), which can then be copied and pasted as required.

Evaluating math expressions

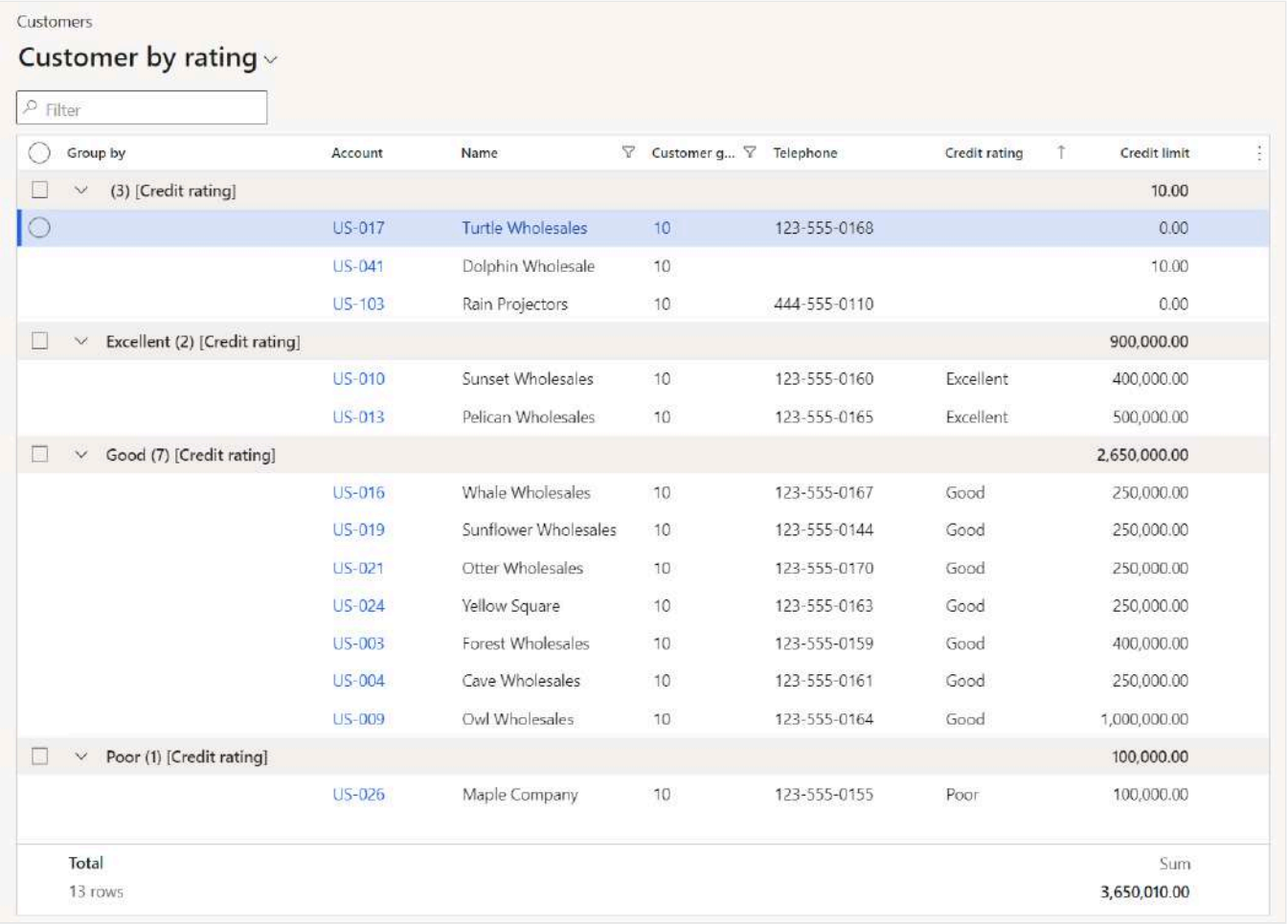
As a productivity booster, users can enter mathematical formulas in numeric cells in a grid. They don't have to do the calculation in an app outside the system. For example, if you enter **=15*4** and then press the **Tab** key to move out of the field, the system will evaluate the expression and save a value of **60** for the field.

To make the system recognize a value as an expression, start the value with an equal sign (=). For more information about the supported operators and syntax, see Supported math symbols.

> **Note**
>
> The ability to evaluate math expressions in numeric controls is available outside the grid.

Grouping tabular data

Business users often have to perform ad-hoc analysis of data. Although this analysis can be done by exporting data to Microsoft Excel and using pivot tables, the **Grouping in grids** capability lets users organize their tabular data in interesting ways within finance and operations apps. Grouping works together with the **Calculated values** capability to let you gain meaningful insights into the data by showing calculated values (for example, subtotals) at the group level.



Group by	Account	Name	Customer g...	Telephone	Credit rating	Credit limit
(3) [Credit rating]						10.00
	US-017	Turtle Wholesales	10	123-555-0168		0.00
	US-041	Dolphin Wholesale	10			10.00
	US-103	Rain Projectors	10	444-555-0110		0.00
Excellent (2) [Credit rating]						900,000.00
	US-010	Sunset Wholesales	10	123-555-0160	Excellent	400,000.00
	US-013	Pelican Wholesales	10	123-555-0165	Excellent	500,000.00
Good (7) [Credit rating]						2,650,000.00
	US-016	Whale Wholesales	10	123-555-0167	Good	250,000.00
	US-019	Sunflower Wholesales	10	123-555-0144	Good	250,000.00
	US-021	Otter Wholesales	10	123-555-0170	Good	250,000.00
	US-024	Yellow Square	10	123-555-0163	Good	250,000.00
	US-003	Forest Wholesales	10	123-555-0159	Good	400,000.00
	US-004	Cave Wholesales	10	123-555-0161	Good	250,000.00
	US-009	Owl Wholesales	10	123-555-0164	Good	1,000,000.00
Poor (1) [Credit rating]						100,000.00
	US-026	Maple Company	10	123-555-0155	Poor	100,000.00
Total						Sum
13 rows						3,650,010.00

To use this capability, right-click the column that you want to group by, and select **Group by this column**. This action will sort the data by the selected column, add a new **Group by** column to the beginning of the grid, and insert "header rows" at the beginning of each group. These header rows provide the following information about each group:

- Data value for the group
- Column name (this information is especially useful when you have multiple levels of grouping)
- Number of data rows in this group
- Calculated values for any configured column (for example, subtotals if the column is configured to show a total)

With Saved views enabled, you can save grouping as part of a view on pages that allow queries to be saved to views. For example, those with large view selectors. See the Switching between views section for more details.

Multiple levels of grouping

After you've grouped data by a single column, you can group the data by a different column by selecting **Group by this column** on the desired column. This process can be repeated until

you have 5 nested levels of grouping, which is the maximum supported depth. At this point, you will no longer be able to group by additional columns.

At any point, you can remove the grouping on any column by right-clicking that column and selecting **Ungroup**. You can also remove the grouping from all columns by selecting **Grid options** and then **Ungroup all**.

Sorting grouped data

After you group data by one or more columns, you can change the sort direction for any grouping column through the corresponding column header.

If you sort on a non-grouped column, the grouping remains intact. The data is sorted inside each group, based on the selected column.

Expanding and collapsing groups

The initial grouping of data will have all groups expanded. You can create summarized views of the data by collapsing individual groups, or you can use group expanding and collapsing to assist in navigating through the data. To expand or collapse a group, select the chevron (>) button in the corresponding group header row. Note that the expand/collapse state of individual groups is **not** saved in personalization.

Selecting and unselecting rows at the group level

In the same way that you can select (or unselect) all rows in the grid by selecting the check box at the top of the first column in the grid, you can also quickly select (or unselect) all the rows in a group by selecting the check box in the corresponding group header row. The check box in the group header row will always reflect the current selection state of rows in that group, regardless of whether all rows are selected, no rows are selected, or only some rows are selected.

Hiding column names

When grouping data, the default behavior is to show the column name in the group header row. You can choose to suppress the column name in group header rows by selecting **Grid options** > **Hide group column name**.

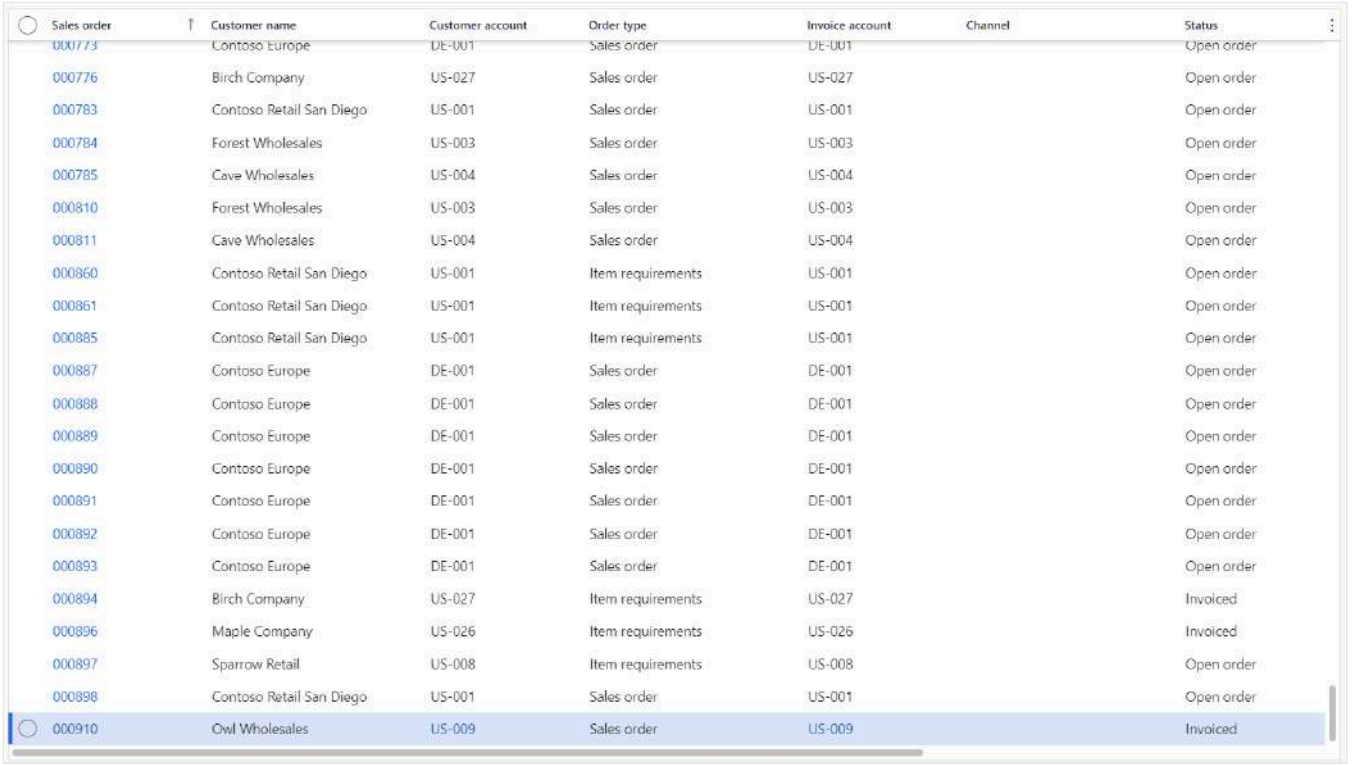
Grouping on date and time columns

When you group on Date or DateTime fields, you have the option to group by year, month, or day. The group "value" in the corresponding header row will match the format from that field. Additionally, for DateTime and Time fields, you can group by hour, minute, or second.

> **ⓘ Important**
>
> Users can't currently add a grouping column after they group on a segment of a date or time column.

Freezing columns

Some columns in a grid might be important enough for context that you don't want them to scroll out of view. Instead, you might want the values in those columns to always be visible. The **Freezing columns in grid** capability provides this flexibility to users.



Sales order	Customer name	Customer account	Order type	Invoice account	Channel	Status
000773	Contoso Europe	DE-001	Sales order	DE-001		Open order
000776	Birch Company	US-027	Sales order	US-027		Open order
000783	Contoso Retail San Diego	US-001	Sales order	US-001		Open order
000784	Forest Wholesales	US-003	Sales order	US-003		Open order
000785	Cave Wholesales	US-004	Sales order	US-004		Open order
000810	Forest Wholesales	US-003	Sales order	US-003		Open order
000811	Cave Wholesales	US-004	Sales order	US-004		Open order
000860	Contoso Retail San Diego	US-001	Item requirements	US-001		Open order
000861	Contoso Retail San Diego	US-001	Item requirements	US-001		Open order
000885	Contoso Retail San Diego	US-001	Item requirements	US-001		Open order
000887	Contoso Europe	DE-001	Sales order	DE-001		Open order
000888	Contoso Europe	DE-001	Sales order	DE-001		Open order
000889	Contoso Europe	DE-001	Sales order	DE-001		Open order
000890	Contoso Europe	DE-001	Sales order	DE-001		Open order
000891	Contoso Europe	DE-001	Sales order	DE-001		Open order
000892	Contoso Europe	DE-001	Sales order	DE-001		Open order
000893	Contoso Europe	DE-001	Sales order	DE-001		Open order
000894	Birch Company	US-027	Item requirements	US-027		Invoiced
000896	Maple Company	US-026	Item requirements	US-026		Invoiced
000897	Sparrow Retail	US-008	Item requirements	US-008		Open order
000898	Contoso Retail San Diego	US-001	Sales order	US-001		Open order
000910	Owl Wholesales	US-009	Sales order	US-009		Invoiced

To freeze a column, right-click in the column's header, and then select **Freeze column**. The first time that you complete this step, the selected column becomes the first column and will no longer scroll out of view. Any subsequent column that you freeze will be added to the right of the last frozen column. You can use the standard Move functionality to reorder frozen columns as you require. However, frozen columns can't be moved so that they appear among the set of unfrozen columns. Likewise, unfrozen columns can't be moved so that they appear among the set of frozen columns.

To unfreeze a column, right-click in the frozen column's header, and then select **Unfreeze column**.

Note that the row selection and row status columns in the new grid are always frozen as the first two columns. Therefore, when these columns are included in a grid, they will always be visible to users, regardless of the horizontal scroll position in the grid. These two columns can't be reordered.

Autofit column width

As in Excel, users can force a column to be automatically resized based on the content that is currently shown in it. Just double-tap (or double-click) the sizing handles in the column. Alternatively, put the focus in the column header, and then select the **A** key (for autofit).

Developer topics

[Developer] Opting out individual pages from using the new grid

If your organization discovers a page that has some issues utilizing the new grid, an API is available to allow an individual form to use the legacy grid control while still permitting the rest of the system to utilize the new grid control. To opt out an individual page from the new grid, add the following call post `super()` in the `run()` method for the form.

```
this.forceLegacyGrid();
```

This API will eventually be deprecated to allow for removal of the legacy grid control. However, it will remain available for at least 12 months after its deprecation is announced. If any issues require that this API be used, report them to Microsoft.

Forcing a page to use the new grid after previously opting out the grid

If you have opted out an individual page from using the new grid, you might want to later re-enable the new grid after the underlying issues were solved. To do this, you simply need to remove the call to `forceLegacyGrid()`. The change will not take effect until one of the following occurs:

- **Environment redeployment**: When an environment is updated and redeployed, the table that stores the pages that have opted out of the new grid (FormControlReactGridState) is automatically cleared.
- **Manual clearing of the table**: For development scenarios, you will need to use SQL to clear the FormControlReactGridState table and then restart the AOS. This combination of actions will reset the caching of pages that have opted out of the new grid.

[Developer] Opting individual grids out of the Typing ahead of the system capability

Some scenarios have arisen that don't lend themselves to working well with the *Typing ahead of the system* capability of the grid. (For example, some code that is triggered when a row is validated causes a datasource research to be triggered, and the research can then corrupt uncommitted edits on existing rows.) If your organization discovers such a scenario, an API is available that lets a developer opt an individual grid out of asynchronous row validation and revert to the legacy behavior.

When asynchronous row validation is disabled in a grid, users can't create a new row or move to a different existing row in the grid while there are validation issues on the current row. As a side effect of this action, tables can't be pasted from Excel into finance and operations grids.

To opt an individual grid out of asynchronous row validation, add the following call after `super()` in the `run()` method of the form.

```
<gridControl>.allowPreemptiveClient(false);
```

> ⓘ **Note**
>
> - This call should be invoked only in exceptional cases and should not be the norm for all grids.
> - We don't recommend that you toggle this API at runtime after the form loads.

[Developer] Size-to-available-width columns

If a developer sets the **WidthMode** property to **SizeToAvailable** for columns inside the new grid, those columns initially have the same width that they would have if the property were set to **SizeToContent**. However, they stretch to use any extra available width inside the grid. If the property is set to **SizeToAvailable** for multiple columns, all those columns share any extra available width inside the grid. However, if a user manually resizes one of those columns, the column becomes static. It will remain at that width and will no longer stretch to take up extra available grid width.

[Developer] Specifying the column that receives the initial focus when new rows are created by using the Down arrow key

As was discussed in the Differences when entering data ahead of the system section, if the "Typing ahead of the system" capability is enabled, and a user creates a new row by using the **Down arrow** key, the default behavior is to put the focus in the first column in the new row. This experience might differ from the experience in the legacy grid or when a **New** button is selected.

Users and organizations can create saved views that are optimized for data entry. (For example, you can reorder columns so that the first column is the one that you want to start to enter data in.) In addition, as of version 10.0.29, organizations can adjust this behavior by using the **selectedControlOnCreate()** method. This method lets a developer specify the column that should receive the initial focus when a new row is created by using the **Down arrow** key. As input, this API takes the control ID that corresponds to the column that should receive the initial focus.

[Developer] Handling grids with non-React extensible controls

When a grid is loading, if the system encounters an extensible control that is not React based, the system will force the legacy grid to render instead. When a user first encounters this situation, a message will be shown indicating the page needs to be refreshed. Afterwards, this page will load the legacy grid automatically without any further notifications to users until the next system update.

To overcome this situation permanently, extensible control authors can create a React version of the control for use in the grid. Once developed, the X++ class for the control can be decorated with the **FormReactControlAttribute** attribute to specify the location of the React bundle to load for that control. See the `SegmentedEntryControl` class as an example.

Last updated on 03/26/2026

Create and work with custom fields

Article • 12/31/2024

While there's an extensive set of fields out-of-the-box for managing a broad range of business processes, sometimes there's a need for a company to track additional information in the system. While programmers can be used to add those fields as extensions in the developer tools, the custom fields feature allows fields to be added directly from the user interface, allowing you to tailor the application to fit your business using your web browser.

Only users with special permissions have access to this feature.

This video shows how easy it's to add a custom field to a page: Adding custom fields.

Creating custom fields

After you've identified additional information to track in the application, you can create the custom field on the appropriate table and expose that new field on a page.

The following steps describe the process for creating a custom field and placing that field on a page.

1. Navigate to the page where the new field is needed.
2. Because the end goal is to expose the custom field on a form, the entry point for creating custom fields exists inside the personalization experience. Open the personalization toolbar by selecting **Options**, and then **Personalize this form**.
3. Click **Insert** and then **Field**.
4. Select the region of the form where you want to expose the new field. After selection, the **Insert fields** dialog box will display a list of existing fields that can be inserted into the selected region of the page.
5. Confirm that the field you're interested in doesn't already exist in the list. If it does, you can simply select that field in the list and click **Insert**.
6. Click the **Create new field** button above the list to initiate the process of creating a custom field. This will open the **Create new field** dialog box.

   If you don't see the **Create new field** button, you do not have the necessary permissions to use this feature.

7. In the **Create new field** dialog box, enter the following information.

    a. Select the database table where this field should be added. Note that only tables that support custom fields will appear in the drop-down list. See the section below for technical details on supported tables.

    b. Select the data type for the new field. The available data types are checkbox, date, date time, decimal, number, picklist, and text.

        - If you choose the text data type, you can also specify the maximum length of the text that can be entered in this field.
        - If you choose the picklist data type, you can also select the set of valid values for the field.

    c. Provide a name, label, and help text for the field. The name corresponds to the physical field name in the database, whereas the label and help text are the text used to represent this field in the user interface.

8. If this is the only field that you need to create for this page, click **Save**. If you need to create additional fields, click **Save and new** and go back to step 7.

> ⓘ **Note**
>
> Currently, there's a limit of **20 custom fields per table**.

9. Leaving the **Create new field** dialog box will return you to the **Insert fields** dialog box. Any custom fields that were just added will be automatically marked in the field list to be inserted into the page.
10. Click **Insert** to insert the marked fields into the selected region of the page.
11. **Optional:** Enable **Move** mode from the personalization toolbar to move the new fields to their desired location in the selected region. See Personalize the user experience for more information about how to use the various personalization capabilities to optimize a form for your personal usage.

> ⚠ **Warning**
>
> The ability to enter values in a custom field added to a page is dependent on whether the table associated with the custom field is editable or read only. When the associated table is read only, all fields linked to that table, including any custom fields, will also be read only.

Sharing custom fields with other users

After you've created a custom field and exposed it on a page, you might want to provide this updated page view that includes the new field to other users in the system. This can be accomplished in two different ways using the personalization capabilities of the product:

- The recommended route is to **publish a saved view** with the custom field added to the page to the appropriate set of users. If the saved views feature isn't enabled, the system administrator can apply the personalization to the desired users from the **Personalization** page. For more information, see Personalize the user experience.
- Alternatively, you can export your changes (called *personalizations*), send them to one or more users, and have each of those users import your changes. The **Manage** option on the personalization toolbar enables you to export and import personalizations.

Managing custom fields

Management of all the custom fields can be accomplished through the **Custom fields** page in the System administration module. This page allows users access to many capabilities, including:

- Viewing a list of all custom fields in the system.
- Limited editing of existing custom fields.
- Deleting custom fields.
- Exposing custom fields on data entities.
- Providing translations of custom field labels and help text.

Viewing all custom fields

The **Custom fields** page provides visibility to all the custom fields that have been defined in the system. Select the table that you are interested in, and the page will update to show a list of the custom fields associated with that table. Choosing a custom field from the list will allow you to view all the details about the field.

Editing custom fields

After a custom field has been created, only certain pieces of information about the custom field can be modified on the **Custom fields** page.

You *can* modify these attributes:

- Label
- Help text
- Length, for Text fields

You *can't* edit the following attributes:

- Field name
- Data type

Additionally, for picklist fields, the set of valid values for the custom field can be reordered, and new values can be added; however, existing values for the picklist field can't be removed. Click **Apply changes** when you're done editing fields for a particular table so the changes are saved.

Exposing custom fields on data entities

It may also be important to allow custom fields to be visible on data entities. Data entities are utilized in the Office integration overview feature, and for data import/export scenarios.

Follow these steps to expose a custom field on a data entity:

1. Select the custom field on the **Custom fields** page.
2. Expand the **Entities** section to view the set of relevant entities.
3. Click the **Edit** button.
4. Modify the **Enabled** field to be selected for each entity that should expose this field.
5. Click **Apply changes** to save your selections.

Allowing custom fields to be displayed in other languages

Because custom fields may need to be accessed by users in a variety of languages, the **Custom fields** page provides a mechanism to allow the label and help text for a custom field to be translated into other languages.

The following steps describe the process for translating custom fields in other languages:

1. Select the custom field on the **Custom fields** page.

2. Select the **Translations** button in the Action Pane. This will open a drop-down menu with existing translations for this field.

3. The **Language** drop-down menu shows the set of languages for which translations have already been provided.

   If you want to edit an existing translation, select the language from the menu and modify the values for the label and help text.

   Otherwise, click the **Add language** button, select the desired language from the menu, and then provide translated values for the label and help text.

4. Click **OK** when you are finished.

Deleting custom fields

When you decide that a custom field is no longer needed, a system administrator can choose to delete the field from the **Custom fields** page. To delete a custom field, select the field to delete, click **Delete**, click **Yes** to confirm the deletion, and finally click **Apply changes**.

> ⓘ **Note**
>
> This action can't be undone, and will result in the data associated with the field being permanently deleted from the database.

Appendix

Why can't I enter a value in my custom field?

If you can't type a value into the custom field when the page is in **Edit** mode, this may be because the table that the field was added to is currently read only. All fields in a table become read only if the backing table is currently configured as read only on the page.

Who can create custom fields?

Only system administrators are able to create custom fields by default. However, those power users whom the organization deems necessary can be given rights to create custom fields by a system administrator using the **Runtime customization power user**

security role. Users without this security role will not be able to create custom fields, but will still be able to see and interact with custom fields added by other users in the system.

What tables support custom fields?

For performance and technical reasons, only tables that meet the following conditions currently allow custom fields to be added.

- The table must be tagged as one of these groups:
  - Group
  - WorksheetHeader
  - Main
  - Miscellaneous
  - Parameter
  - Reference
  - TransactionHeader
- The table can't extend another table.
- The table can't be marked as a system table.
- The table can't be a temporary table.

Can I reference custom fields from the developer tools?

Custom fields can only be managed through the user interface and can't be referenced by code.

Can I transfer the custom field to the table as an extension field while keeping the values?

Custom field data can be transferred to the table as extension fields. To use the custom X++ scripts to get and map the corresponding data from a custom field, follow these steps:

1. Get the field name of the custom field.
2. Use the `.getFieldValue(_fieldName)` method to get a value on the source table.

You can use the `TableExtensionManagerFactory::CreateExtensionManager()` extension manager to create the `.GetRuntimeExtension(_tableName, SysCustomFieldConstants::ExtensionName)` extension build object on the

Microsoft.Dynamics.Ax.Xpp.MetadataExtensions library to get more information about the type/metadata of the extension field.

> **Note**
>
> Exclude records with default values from your query when transferring data for all records between a custom field and a new extension field. Use the **update statement** of the SysDa framework for better performance.

How can I move custom fields between environments?

The current recommendation for moving custom fields between environments is to manually re-create the custom fields in the target environment. To see the full list of custom fields on a particular table:

1. Go to the **Custom fields** page, select that table from the dropdown.
2. In the target environment, follow the process described earlier in this article to recreate each field.
3. Once all the fields have been created, click **Apply changes**.
4. Move all the personalizations containing custom fields by exporting those personalizations from the original environment and importing them into the target environment.

Feedback

Was this page helpful? Yes No

Provide product feedback

Embed canvas apps from Power Apps

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft Power Apps is a service that lets developers and nontechnical users build custom business apps for mobile devices, tablets, and the web without writing code. Finance and operations apps support integration with Power Apps. Canvas apps that you, your organization, or the broader ecosystem develop can be embedded into finance and operations apps to augment the product's functionality. For example, you might build a canvas app from Power Apps to supplement a finance and operations app with information that is retrieved from another system.

To learn more about embedding canvas apps, watch the short How to embed canvas apps video.

Adding an embedded canvas app from Power Apps to a page

Before you embed a canvas app from Power Apps into the client, you must find or build an app that has the desired visuals or functionality. This article doesn't include a detailed description of the process for building apps. If you're new to Power Apps, see the Power Apps documentation.

There are three ways to embed a canvas app into a finance and operations app. You can use the approach that best fits your scenario.

- Embed the canvas app into the **Power Apps** button on the standard Action Pane of a page. Apps that you add in this way appear as items on the **Power Apps** menu button, and the apps open in side panes.
- Embed the canvas app directly on an existing page as a new tab page (pivot tab, FastTab, blade, or workspace section).
- Create a new full-page experience for the canvas app from the dashboard.

When you configure your embedded canvas app, you can select a single field that you want to send as context to the app. This step enables the app to be responsive, based on the data that you're currently viewing.

> **Note**
>
> You can't use this mechanism to embed model-driven apps.

Embedding a canvas app on an existing page

The following procedure shows how to embed a canvas app on an existing page from Power Apps.

1. Go to the page where you want to embed the canvas app. This page contains any data that must be passed to the app as input.
2. Open the **Add an app from Power Apps** pane:
    - If the app will be embedded directly on the page, select **Options** > **Personalize this page** > **More**. Select **Add a page**, and then select the region on the page where you want to add the app. To embed the app into the **Power Apps** menu button, on the Action Pane select **Power Apps**. To embed the app directly on the page, select the appropriate tab, FastTab, blade, or section (if you're on a workspace). Then, in the **Add an app** pane, select **Power Apps**.
    - If the app will be accessed by using the **Power Apps** menu button, you can select the **Power Apps** menu button on the standard Action Pane and then select **Add an app**.
3. Configure the embedded app. For more information, see the Configuring a canvas app section later in this article.
4. After you confirm that the configuration is correct, select **Insert**. Remember to save the view for the change to be persisted if you're adding the app to a page that supports views.

Embedding a canvas app as a full-page experience from the dashboard

You might want to embed a canvas app from the dashboard if the app isn't related to an existing page, or if you just want to surface the app as a full-page experience inside the finance and operations app.

1. Open the dashboard.
2. Select and hold (or right-click) the page, select **Personalize**, and then select **Add a page**.
3. In the **Add a page** pane, select **Power Apps**.

4. Configure the embedded app. For more information, see the Configuring a canvas app section later in this article.
5. Select **Save** to add the app to the dashboard as a new tile.
6. Select the new tile on the dashboard, and confirm that the canvas app appears as expected.

Configuring a canvas app

When you embed a canvas app, you must set the following parameters:

- **Name** – Enter the text that should be shown for the button or tab that contains the embedded app. Often, you might want to repeat the name of the app in this field.
- **App ID** – Specify the globally unique identifier (GUID) for the canvas app that you want to embed. To retrieve this value, find the app on make.powerapps.com, and then look in the **App ID** field under **Details**.
- **Input context for the app** – You can optionally select the field that contains the data that you want to pass to the app as input. For information about how the app can access the data that is sent from finance and operations apps, see Building an app that leverages data sent from finance and operations apps.

  The current legal entity is also passed to the canvas app as context, via the **cmp** URL parameter.
- **Application size** – Select the type of app that you're embedding. Select **Thin** for apps that are built for mobile devices or **Wide** for apps that are built for tablets. This parameter ensures that enough space is allotted for the embedded app.

Sharing an embedded app

After you embed a canvas app on a page and confirmed that it's working correctly, you might want to share the app with other users in the system.

To share an embedded canvas app, follow these steps:

1. Share the canvas app in Power Apps with the appropriate users, so that they can access the app directly in Power Apps.
2. Share the personalizations that are associated with the embedded app with the desired users. You can use either of the following approaches:

- **Publish the view (Recommended):** For pages that support views, the recommended and preferred approach is to create a view that includes the embedded canvas app, and then publish that view to the desired users. This approach ensures that all users who have the security roles that are targeted by the published view see the canvas app on the page.

  You can also publish a canvas app that is embedded as a full-page experience from the dashboard. On the dashboard, select and hold (or right-click) the tile that is associated with the app, select **Personalize**, and then select **Publish page**. An experience that resembles the *Publishing views* experience is shown, and you can select the security roles and legal entities to publish to.

- For pages that don't support views, the system administrator can implement a personalization that provides the canvas app to the appropriate set of users via the **Personalization** page. Alternatively, you can export your page's personalizations and send them to one or more users. Each of those users can then import the personalization. The personalization toolbar has buttons that let you export and import personalizations.

> ⓘ **Note**
>
> If the canvas app has been shared with external users, those users can't use the embedded app inside finance and operations apps. However, they can access the app directly inside Power Apps. External users include guests and users who don't belong to the Microsoft 365 Azure Directory where the finance and operations app is deployed.

For more information about the personalization capabilities in the product and how to use them, see Personalize the user experience.

Building a canvas app that uses data that is sent from finance and operations apps

When you build a canvas app that will be embedded in a finance and operations app, one important part of the process is to use the input data from that finance and operations app. From the Power Apps development experience, the input data that is passed from a finance and operations app can be accessed by using the **Param("EntityId")** variable. The current legal entity is also passed to the canvas app via the **Param("cmp")** variable.

For example, in the OnStart function of the app, you could set the input data from finance and operations apps to a variable like this:

Power

```
If(!IsBlank(Param("EntityId")), Set(FinOpsInput, Param("EntityId")),
Set(FinOpsInput, ""));

If(!IsBlank(Param("cmp")), Set(FinOpsLegalEntity, Param("cmp")),
Set(FinOpsLegalEntity, ""));
```

Viewing a canvas app

To view an embedded canvas app on a page in finance and operations apps, just go to a page that has an embedded app. Remember that apps can be accessed by using the **Power Apps** button on the standard Action Pane. Alternatively, they can appear directly on the page as a new tab, or FastTab, or blade, or as a new section in a workspace. When users first try to load an app on a page, they are prompted to sign in. This step ensures that the users have the appropriate permissions to use the app.

Editing an embedded app

After an app is embedded on a page, you may need to make some changes to the configuration of the app. For example, you may want to modify the label that is associated with the embedded app, or a new version of the app is released and you need to update the App ID to point to the latest version.

Follow these steps to edit the configuration of an embedded app:

1. Go to the **Edit the app** pane.

    - If the embedded app is accessed through the Power Apps menu button, select and hold (or right-click) the Power Apps menu button and select **Personalize**. Select the app that you want to configure from the **Select an app** drop-down menu.
    - If the embedded app appears directly on the page, select **Options**, and then select **Personalize this page**. Using the **Select** tool, click the embedded app.
    - If the embedded app was added from the dashboard, open the dashboard, select and hold (or right-click) the tile that is associated with the canvas app, select **Personalize**, and then select **Edit page**.

2. Make the needed modifications to the app configuration, and then click **Save**.

3. Resave or republish the view. If you leave the page without explicitly saving the view, none of the actions that you performed in the **Edit website** pane are maintained. This step isn't needed for pages that don't support views.

Removing an app

After an app is embedded on a page, there are a few ways to remove it if needed:

- Go to the **Edit an app** pane using the instructions from the Editing an embedded app section earlier in this article. Confirm that the pane displays information for the embedded app that you would like to remove, and then click the **Delete** button.
- If the embedded app was added from the dashboard, open the dashboard, select and hold (or right-click) the tile that is associated with the canvas app, select **Personalize**, and then select **Remove page**.
- Because the embedded app is saved as personalization data, clearing your page's personalization also removes any embedded apps on that page. Clearing the page's personalization is permanent and can't be undone. To remove your personalizations on a page, select **Options**, select **Personalize this page**, and then select the **Clear** button. After you refresh your browser, all previous personalizations for the current view (or the entire page if the page doesn't support views) are removed. For more information about how to optimize pages using personalization, see Personalize the user experience.

Appendix

[Developer] Modeling a canvas app on a form

While this article focuses on embedding canvas apps through personalization, developers also have the option to add a canvas app to a form using the Visual Studio development experience by adding a PowerAppsHostControl to the form. The metadata properties available on the control provide the same capabilities as the personalization experience.

[Developer] Specifying where an app can be embedded

By default, users can embed apps on any page, either under the Power Apps menu button or directly on the page as a tab, FastTab, blade or as a new section in a workspace. However, if necessary developers can also configure this feature to only allow embedding of apps on certain pages by implementing the following methods:

- **isPowerAppPersonalizationEnabled** – If this method returns false for a specific page, then the Power Apps menu button isn't shown, and users aren't able to embed apps anywhere on this page, including as a tab.
- **isPowerAppTabPersonalizationEnabled** – If this method returns false for a specific page, then users aren't able to embed apps directly on the page as a tab, FastTab, or panorama

section. Users are still able to embed apps through the Power Apps menu button if embedding is allowed on the page.

The following example shows a new class with the two methods needed to configure where apps can be embedded.

powerapps

```
[ExtensionOf(classStr(FormRunConfigurationPowerAppsConfiguration))]

public final class ClassTest_Extension
{
    public static boolean isPowerAppPersonalizationEnabled(str pageName)
    {
        var result = next isPowerAppPersonalizationEnabled(pageName);
        return result;
    }

    public static boolean isPowerAppTabPersonalizationEnabled(str pageName)
    {
        var result = next isPowerAppTabPersonalizationEnabled(pageName);
        return result;
    }
}
```

Last updated on 03/26/2026

Embed third-party apps

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Many customers use a range of applications to run their businesses. Some of those applications are third-party web apps that work with finance and operations apps. To provide a more seamless user experience, you can embed third-party apps directly into your finance and operations apps (if the third-party apps support embedding). By using this approach, users can access the websites and apps they require without having to switch tabs or windows.

Use one of the following methods to embed a third-party app or website. These methods are similar to the methods that you use to embed canvas apps from Microsoft Power Apps into finance and operations apps.

- Embed the app or website on an existing page as a new tab page (pivot tab, FastTab, blade, or workspace section).
- Create a new full-page experience for the app or website from the dashboard.

Embed a website on an existing page

Use this procedure to supplement an existing page in the system with an embedded app.

1. Open the page where you want to embed the app.
2. Open the **Add an app** pane:
   a. Select **Settings** and then **Personalize** to open the **Personalization** toolbar.
   b. Select **More** > **Add an app**.
3. Select the region of the page where you want to add the app. This region must be a *tab container* for a pivot tab, FastTab, blade, or workspace section.
4. Select **Website**.
5. Configure the embedded app:

- **Name** – Enter the text that appears for the tab that contains the embedded app. Often, you want this text to be the name of the app.
- **URL** – Specify the location of the app.

> **Important**
>
> - For security reasons, the URL must use Hypertext Transfer Protocol Secure (HTTPS).
> - The app or website must be configured to allow itself to be embedded.

6. Select **Save** to embed the app on the page. The app is added as the last tab or section in the group.
7. Confirm that the app appears as expected. If the app isn't rendered, see the Troubleshooting section later in this article.
8. Open the view selector, and select either **Save** (if the app should be associated with the current view) or **Save as** (to save the app to a different view). If the page doesn't support views, you can skip this step.

Embed a website as a full-page experience from the dashboard

Use this procedure if the app that you want to embed isn't related to an existing page, or if you want a full-page experience for the app inside the finance and operations app.

1. Open the dashboard.
2. Select and hold (or right-click) on the dashboard, select **Personalize**, select **Add a page**.
3. In the **Add a page** pane, select **Website**.
4. Configure the embedded app:

   - **Name** – Enter the text that appears on the tile for the embedded app on the dashboard. Often, you want this text to be the name of the app.
   - **URL** – Specify the location of the app.

> **Important**
>
> - For security reasons, the URL must use HTTPS.

- The app or website must be configured to allow itself to be embedded.

5. Select **Save** to add the app to the dashboard as a new tile.
6. Select the new tile on the dashboard, and confirm that the app appears as expected. If the app isn't being rendered, see the Troubleshooting section later in this article.

Sharing embedded apps

After you embed an app by using one of the methods that are described in the previous sections, you might want to share the view with other users in the system. To share an embedded app, use one of the following methods:

- **Publish the view (Recommended):** If you save the embedded app to a view, publish the view to users who have the appropriate security roles in the targeted legal entities. In this case, only the desired users see the embedded app on that page. For more information about how to publish a view, see Publishing views.

  You can also publish an app that you embedded as a full-page experience from the dashboard. On the dashboard, select and hold (or right-click) the tile that is associated with the app, select **Personalize**, and then select **Publish page**. An experience that resembles the *Publishing views* experience is shown, and you can select the security roles and legal entities to publish to.

- **Copy the personalization:** For pages that don't support views, you can copy the personalization to the appropriate users. For more information, see Sharing personalizations.

Viewing embedded apps

To view an embedded app on a page in finance and operations apps, open the page that has the embedded app. On some pages, you can access embedded Power Apps by using the **Power Apps** button on the standard Action Pane. Alternatively, any embedded app might appear directly on the page as a new tab, FastTab, or blade, or as a new section in a workspace.

Editing or removing embedded apps

After you embed an app on a page, you might need to edit its configuration, such as changing the section label or the URL. Or, you might need to remove the app from the page. Use one of the following procedures to edit the configuration of an embedded app or remove it.

Apps that you embed on existing pages

1. Open the page where you embedded the app.
2. Select **Settings** and then **Personalize** to open the **Personalization** toolbar.
3. Select the **Select** tool, and then select the embedded app.
4. To edit the app, make the required changes to its configuration, and then select **Save**.

   To remove the app, select **Delete**.
5. Resave or republish the view. If you leave the page without explicitly saving the view, you don't maintain any of the actions that you performed in the **Edit website** pane.

Apps that you embed from the dashboard

1. Open the dashboard.
2. Select and hold (or right-click) the tile that's associated with the embedded app, and then select **Personalize**.
3. To edit the app, select **Edit page**. In the **Edit website** pane, make the required changes to the app configuration, and then select **Save**.

   To remove the app, select **Remove page**.

Appendix

Troubleshooting

If a website doesn't render correctly after you embed it in a finance and operations app, or if you receive an error message that states that the URL denied a connection, the website is probably configured to prevent itself from being embedded in an iframe. Follow these steps to determine whether you can embed the website.

1. Open the developer tools for the browser that you're using.
2. On the **Network** tab, find and select the response from the embedded site.
3. On the **Headers** tab, under **Response Headers**, look for **x-frame-options**. If **x-frame-options** exists in the response headers, and has a value of **DENY** or **SAMEORIGIN**, you can't currently embed the website. You need to work with the owner of the app to enable it to be safely embedded.

[Developer] Modeling a website on a form

Although this article focuses on embedding third-party apps or websites through personalization, developers can also embed them in a form by using the Visual Studio development experience. Just add a **WebsiteHostControl** control to the form. The metadata properties that are available for the control provide the same capabilities as the personalization experience.

Last updated on 03/11/2026

Find information by using lookups

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Many fields have lookups that can help you easily find the correct or desired value. Several enhancements have been added to lookups that make these controls more usable and make users more productive. In this article, you will learn about these new lookup features and receive some helpful tips to get the optimal use out of lookups in the system.

Responsive lookups

In previous versions, when interacting with a lookup control, users would have to take an explicit action to open the drop-down menu. This may have been by typing an asterisk (*) in the control to filter the lookup based on the current value of the control, clicking the drop-down button, or by using the **Alt**+**Down arrow** keyboard shortcut. Lookup controls have been modified in the following ways to better align with current web practices:

- Lookup drop-down menus now open automatically after a slight pause in typing, with the drop-down menu contents filtered based on the lookup control's value.

  Note that the old behavior of automatic opening of the dropdown after typing an asterisk (*) has been deprecated.

- After the lookup drop-down menu has opened, the following occurs:
  - The cursor stays in the lookup control (instead of focus moving into the drop-down menu) so you can continue to make modifications to the control's value. However, the user can still use the **Up arrow** and **Down arrow** to change rows in the drop-down menu, and enter to select the current row in the drop-down menu.
  - The contents of the drop-down menu adjusts after any modifications are made to the lookup control's value.

For example, consider a lookup field called **City**.

When focus is in the **City** field, you can start looking for the city that you want by typing a few letters, like "col." After you stop typing, the lookup will open automatically, filtered to those cities that begin with "col".

At this point, the cursor is still in the lookup field. If you continue typing so the value is "colum," the lookup contents adjust automatically to reflect the latest value in the control.

Even though focus is still in the lookup control, you can also use the **Up arrow** or **Down arrow** keys to highlight the row that you want to select. If you press **Enter** the highlighted row will be selected from the lookup and the control's value will be updated.

Typing in more than IDs

When entering data, it's natural for users to attempt to identify an entity, such as a customer or vendor, in terms of the name rather than an identifier representing the entity. Many (but not all) lookups now allow contextual data entry. This powerful feature allows the user to type the ID or the corresponding name into the lookup control.

For example, consider the **Customer account** field when creating a sales order. This field shows the **Account ID** for the customer, but a user would typically prefer to enter an **Account name** instead of an **Account ID** for this field when creating a sales order, such as "Forest Wholesales" instead of "US-003."

If the user started to enter an **Account ID** into the lookup control, the drop-down menu would automatically open as described in the previous section.

However, the user can now enter the beginning of an **Account name** as well. If this is detected, the **Name** column is moved to be the first column in the lookup, and the lookup is sorted and filtered based on the **Name** column.

Using grid column headers for more advanced filtering and sorting

The lookup enhancements discussed in the previous two sections greatly improve a user's ability to navigate the rows in a lookup based on a "begins with" search of the **ID** or **Name** field in the lookup. However, there are situations in which more advanced filtering (or sorting) is needed to find the correct row. In these situations, the user needs to use the filtering and sorting options in the grid column headers inside the lookup. For example, consider an employee entering a sales order line who needs to locate the right "cable" as the product. Typing "cable" into the **Item number** control isn't helpful, as there are no product names that begin with "cable."

Instead, the user needs to clear the value of the lookup control, open the lookup drop-down menu, and filter the drop-down menu using the grid column header. A mouse (or touch) user can simply click (or touch) any column header to access the filtering and sorting options for

that column. For a keyboard user, the user simply needs to press **Alt**+**Down arrow** a second time to move focus into the drop-down menu, after which the user can tab to the correct column, and then press **Ctrl**+**G** to open the grid column header drop-down menu.

After the filter has been applied, the user can find and select the row as usual.

Last updated on 07/11/2025

Change the date for a session

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to change the date for a session. By default, the current date is used when you enter and post journal entries or source documents. You can change the date for your current session. Use this feature to back-date journal entries or source documents, as necessary.

1. In the navigation pane, go to **Modules** > **Common** > **Common** > **Session date and time**.
2. In the **Date** field, enter a date.
3. Select **OK**.

Last updated on 03/11/2026

Set a user's preferred time zone

Summarize this article for me

> **Note**
>
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The following article explains how a user in the System Administrator role can set the time zone for a user.

1. Go to **Navigation pane** > **Modules** > **System administration** > **Users** > **Users**.
2. In the list, find and select the desired record.
3. Select **User options**.
4. Select the **Preferences** tab.
5. In the **Time zone** field, select an option from the drop-down list.
6. Select **Save**.

Last updated on 03/10/2026

Lifecycle Services for finance and operations apps customers

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article is for customers who sign up for the current versions of finance and operations apps. Partners who work with customers to help them move through the lifecycle of their Microsoft Dynamics 365 Lifecycle Services project also find this information useful.

Lifecycle Services workspace for the current versions of the finance and operations apps

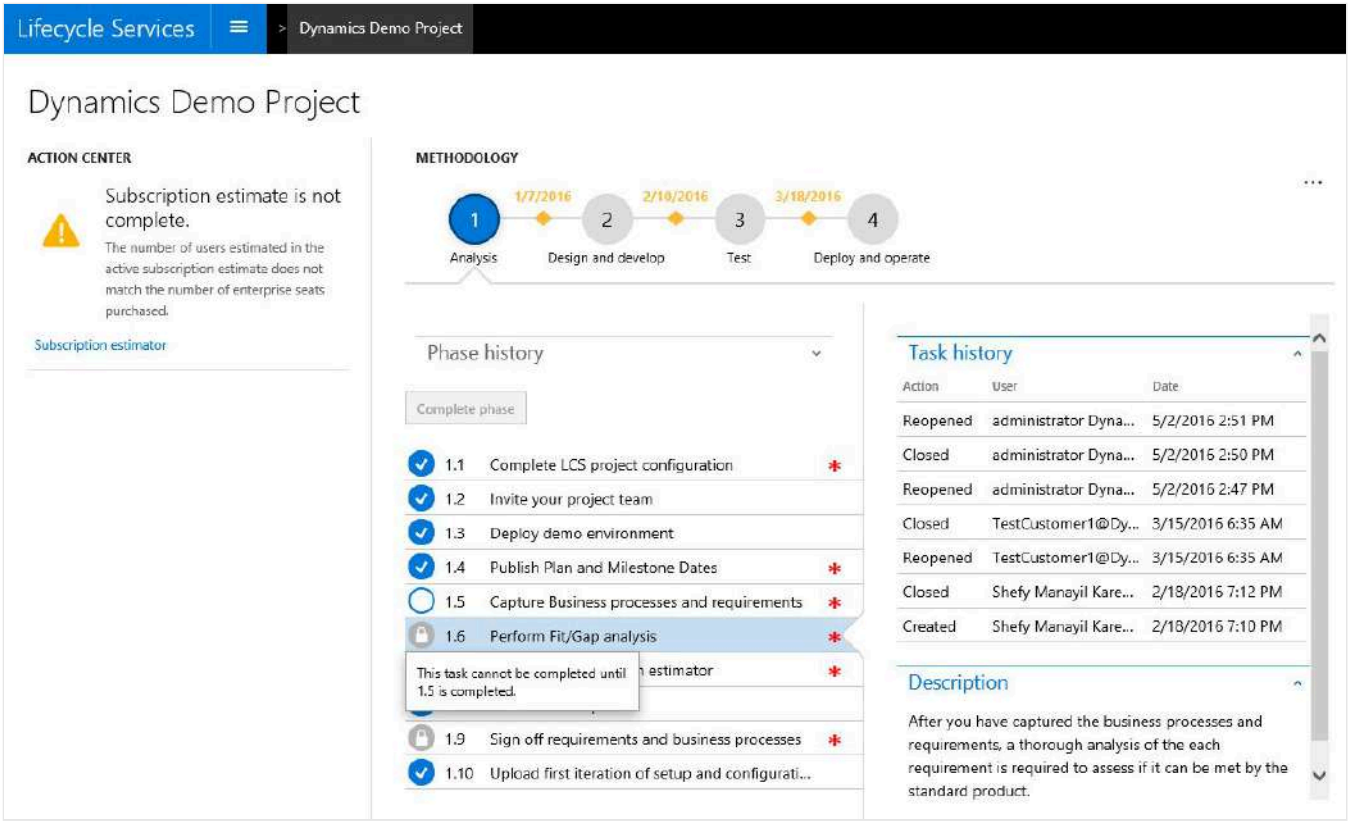
When you sign up for the current versions of finance and operations apps, your subscription includes an Implementation project workspace. After you activate the service, the tenant administrator must sign in at the preferred Lifecycle Services endpoint by using the tenant account. The project workspace is automatically created for your organization. The workspace includes the following elements:

- Enabled features, based on the offer that you selected
- Environments that Microsoft deploys and manages
- Guidance that the Action center provides to help you complete required actions
- A new methodology experience that includes tasks that lock as you move through the implementation
- A more complete history that specifies who completed each methodology phase and task

- Milestones that you can use to track critical project dates
- Various services to help you with your implementation

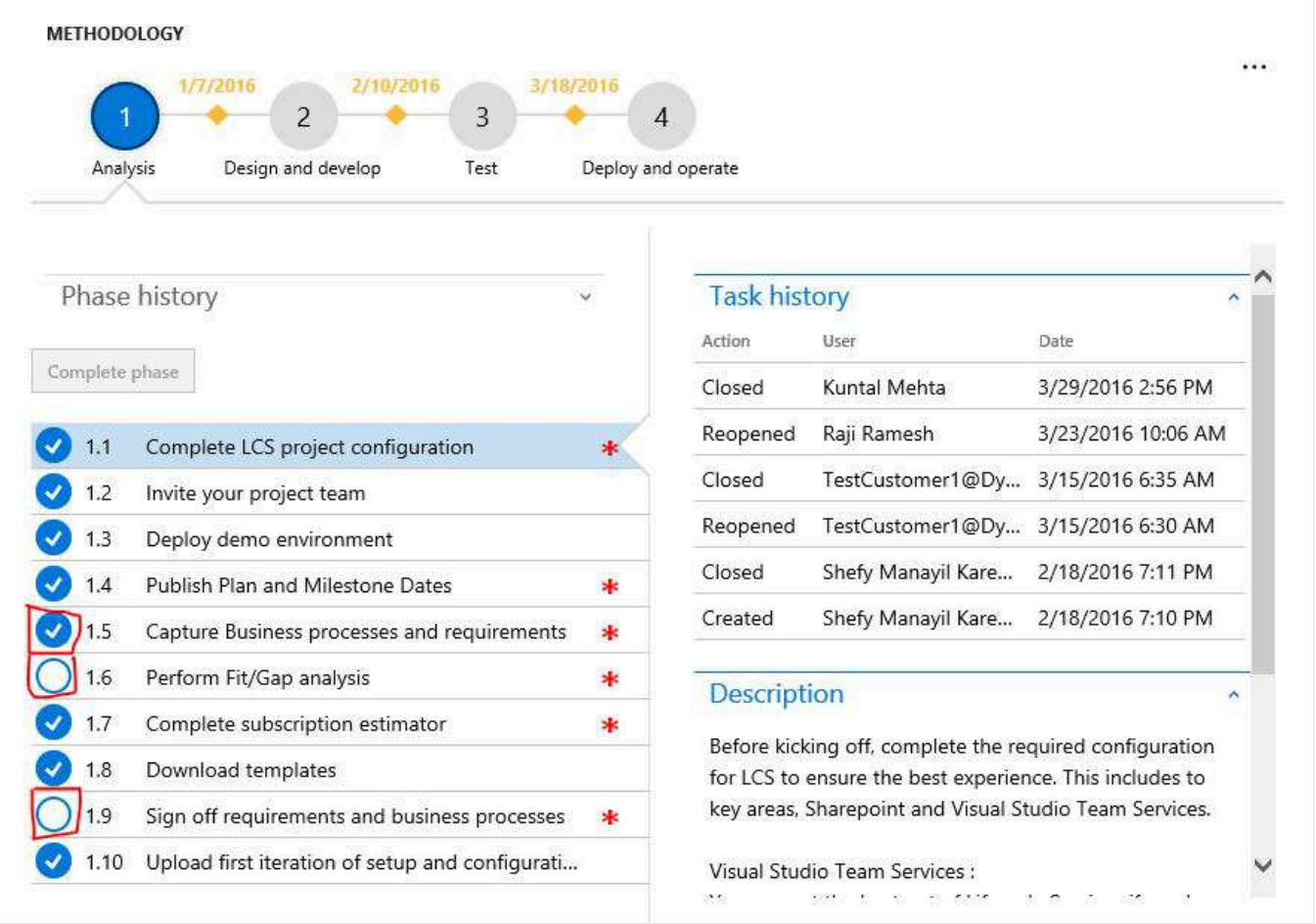
Methodologies

As a customer, you must complete the steps that the methodology outlines to gain access to the production environment. Before you can mark a phase as completed, you must complete the specified mandatory tasks. Locked tasks, such as tasks 1.6 and 1.9 in the following screenshot, unlock after you complete the required actions. To learn which actions must be completed before a specific task can be unlocked, select the lock icon for that task.



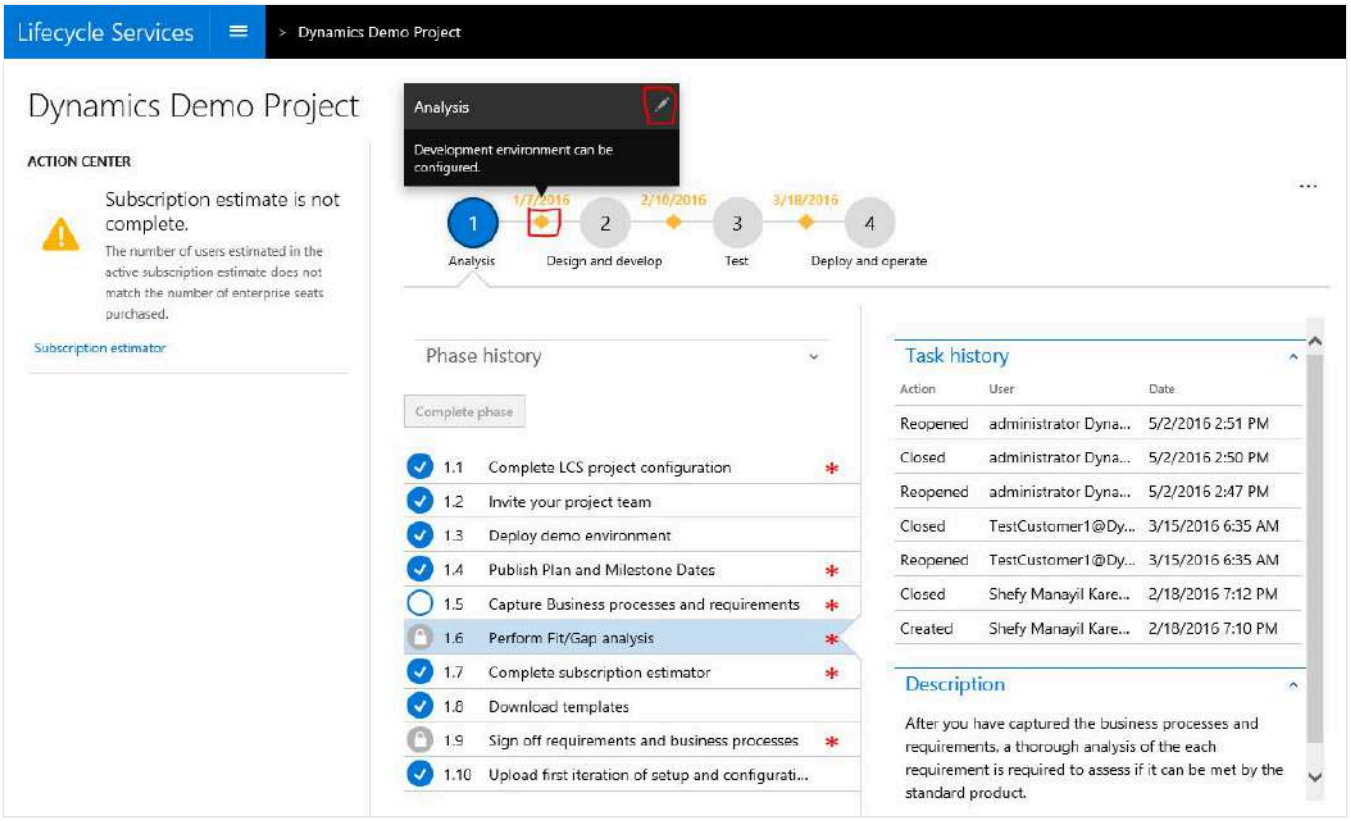
In the case of prerequisites, after you complete the required tasks, you can mark the dependent tasks as completed. For example, in the following screenshot, tasks 1.6 and 1.9 depend on task 1.5. Because task 1.5 is now completed, the two dependent tasks can be

marked as completed.

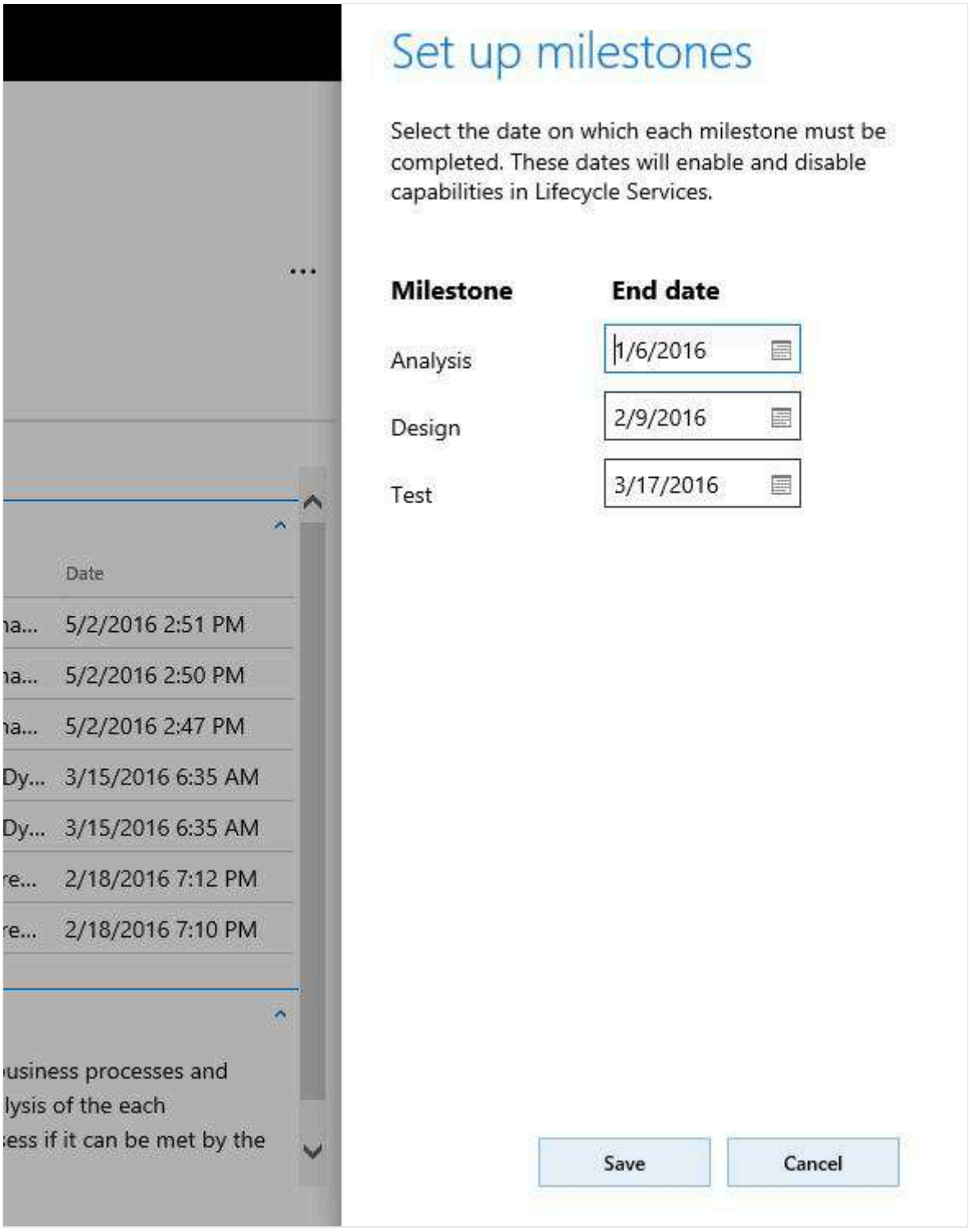


Milestones

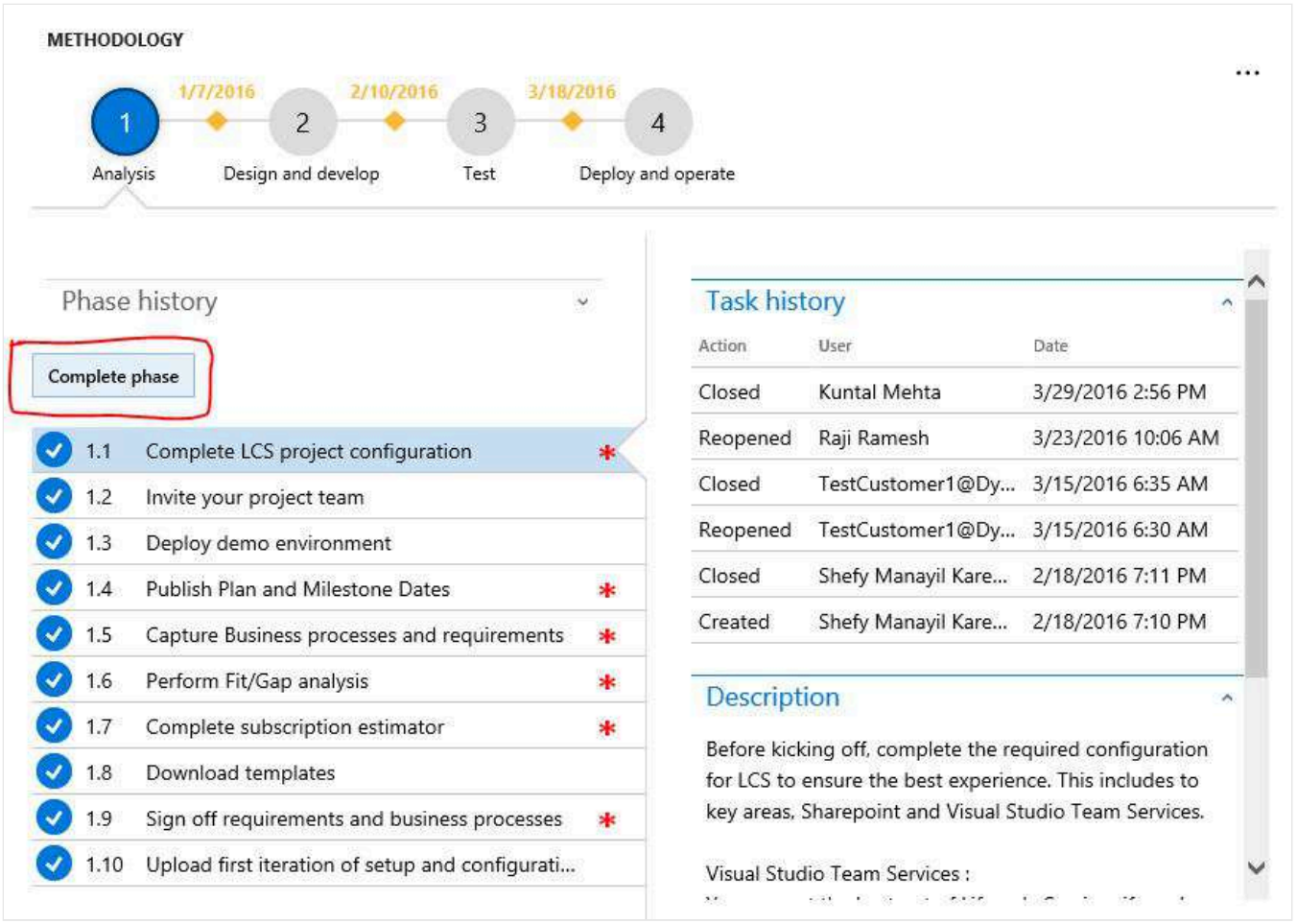
Define high-level milestones for your project. Milestones help you track the deliverables that you must complete and your progress toward the milestone goals. Color indicators help you quickly see whether you're behind schedule. For example, in the following screenshot, the milestones are yellow. To enter or update the milestone dates, select the diamond shape in the methodology, and then select the **Edit** button (pencil icon). You can change milestone dates at any time.



When you finish entering milestones, the **Publish plan and milestone** task opens, and you can mark it as completed.

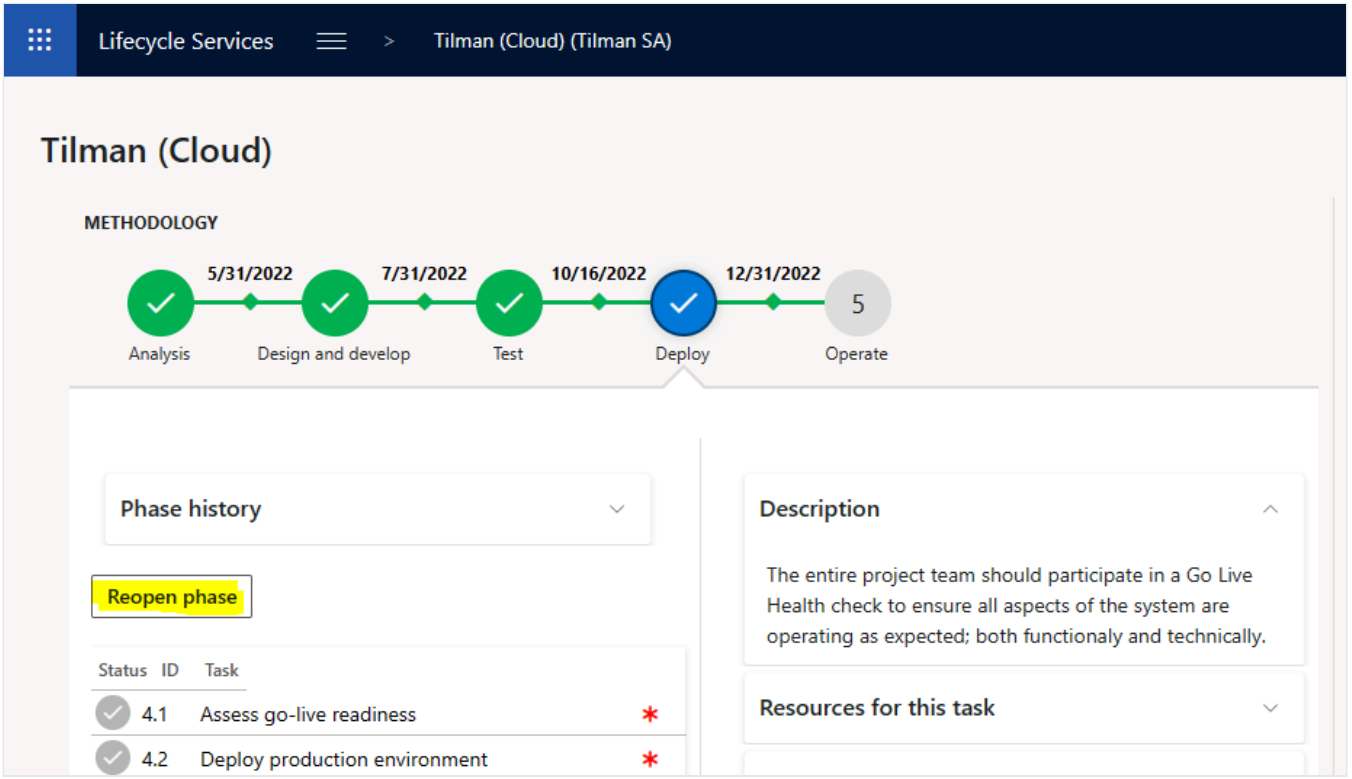


When you complete all the required tasks in a phase, select **Complete phase** to mark the phase as completed. After you mark a phase as completed, next steps become available in Lifecycle Services.



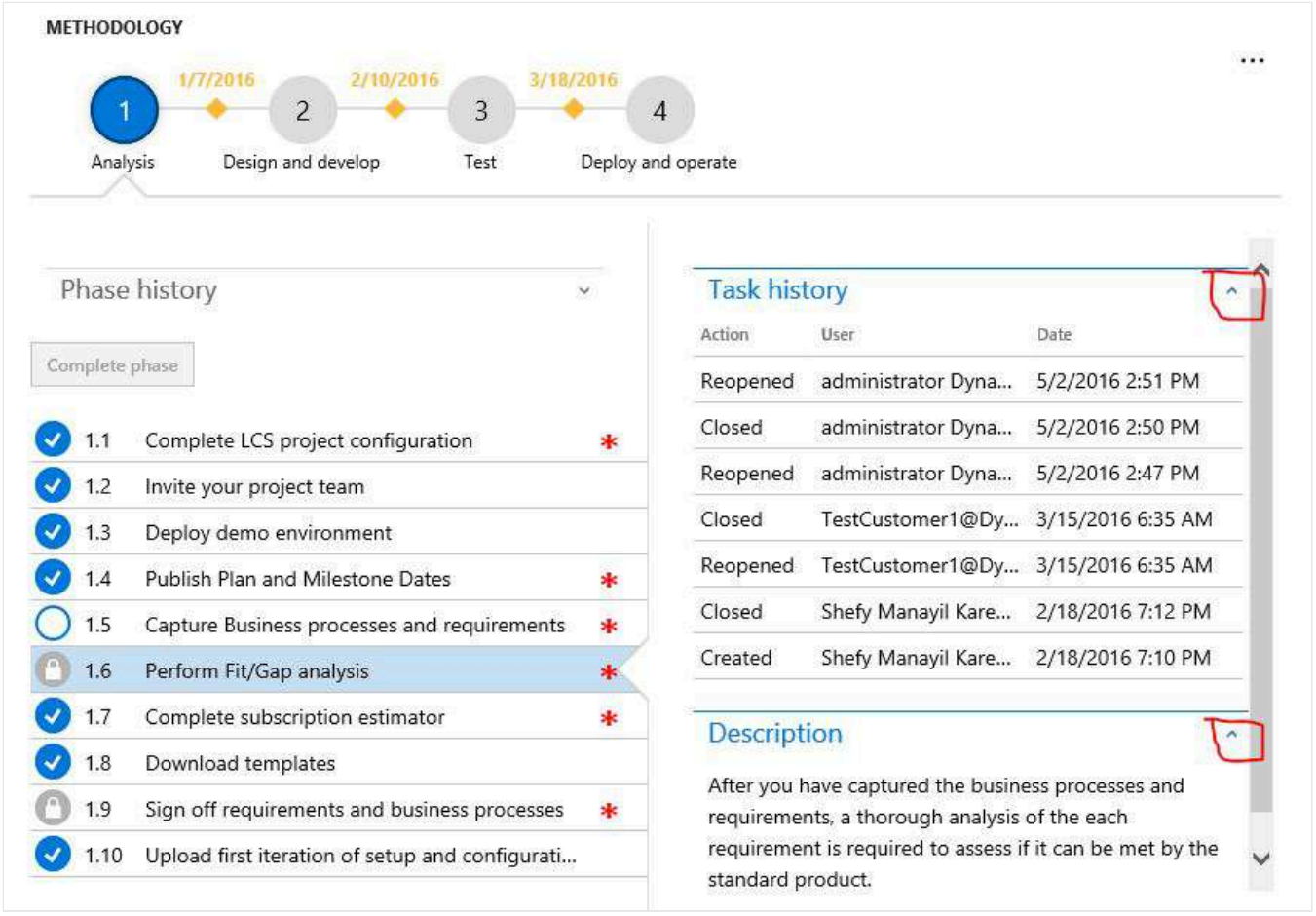
Reopen a completed milestone

If you need to make any changes to your completed milestone tasks or dates, select the **Reopen phase** button on a prior milestone. You can then make edits and complete it again.



Methodology description and history

Descriptions can help you understand what is expected of you for a specific methodology task or phase. You can expand the methodology description to learn more about each task, and then collapse the description when you've finished. The task and phase history can tell you when a task or phase was completed or reopened. If you're a project manager, this information can help you stay on top of the high-level tasks that are required for your implementations.



Subscription estimator

Use the Subscription estimator tool to evaluate your subscription requirements for the current versions of the finance and operations apps. To use Subscription estimator, download the usage profile, which is a Microsoft Excel workbook. Then, in the workbook, complete the following worksheets:

- Deployment details
- Instance Characteristics
- Retail & Commerce

After you complete the worksheets, enter the data from the summary sheet into Subscription estimator by selecting + **New estimate**. You must make one estimate the active estimate. Make sure that the estimate that you mark as active is the same as the offer that you bought through the VL or CSP channel.

New deployment experience

To provision your environment, you must complete a configuration checklist. As you make progress through the methodology, environments become available to you. Select **Configure** to add deployment information.

Because the information that you enter determines your experience, carefully review your input. After you enter all the required information, sign out is required for the deployment request. The user who completes the sign out becomes the system administrator on the instance. Verify that the correct user completes the sign out for the deployment. After the sign out is completed, the Microsoft site reliability engagement team reviews the request. After the team reviews the information that you entered, it initiates the provisioning. If the information isn't correct, the team contacts you. After the provisioning is completed, the status is updated to indicate that the environment is deployed, as shown in the following screenshot. If the provisioning takes longer than expected, the Microsoft site reliability engagement team reviews the status and takes appropriate actions. These actions might include contacting you. After the environment is provisioned, select **Full details** to open the **Detailed environment** page, where you can sign in to the system, view the monitoring status, or view relevant updates.

Last updated on 03/07/2026

Implementation lifecycle management home page

> **Note**
>
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> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

These topics describe the programs, tools, and processes related to the implementation lifecycle of your finance and operations project.

Programs

Microsoft FastTrack

Tools

Microsoft Dynamics Lifecycle Services

Lifecycle Services (LCS) for finance and operations apps customers

Processes

Onboard an implementation project

Prepare for go-live

Frequently asked questions

Go-live for implementation projects FAQ

Last updated on 06/18/2026

Finance and operations application architecture

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

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> **Important**
>
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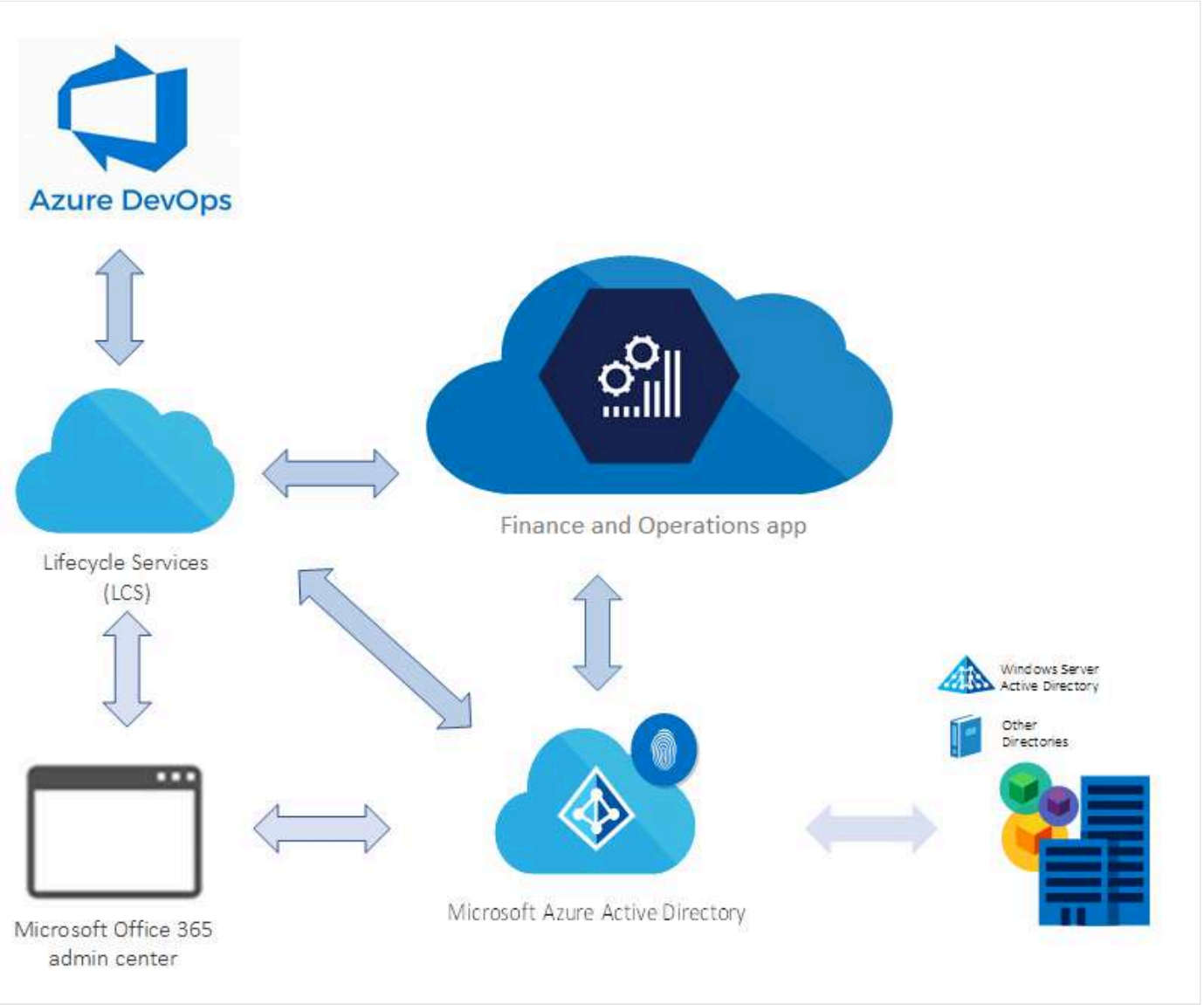
The finance and operations application cloud architecture contains all the elements that are common to all Microsoft cloud offerings, as described in Subscriptions, licenses, accounts, and tenants for Microsoft's cloud offerings. Beyond this architecture, it also includes services that automate software deployment and provisioning, operational monitoring and reporting, and seamless application lifecycle management.



The cloud architecture consists of these conceptual areas:

- **Subscription** – A subscription to finance and operations apps gives you an online cloud environment (or multiple environments) and experience.
- **Licenses** – Customers must purchase subscription licenses (SLs) for their organization, or for their affiliates' employees and on-site agents, vendors, or contractors who directly or indirectly access finance and operations apps. These apps are licensed through Microsoft Volume Licensing and the Microsoft Cloud Solution Provider (CSP) program. For more information, download the latest Microsoft Dynamics 365 Licensing Guide from Dynamics 365 pricing.
- **Tenant** – In Microsoft Entra ID (Microsoft Entra ID), a tenant represents an organization. It's a dedicated instance of the Microsoft Entra service that an organization receives and owns when it creates a relationship with Microsoft (for example, by signing up for a Microsoft cloud service, such as Azure, Microsoft Intune, or Microsoft 365). Every Microsoft Entra

tenant is distinct and separate from other Microsoft Entra tenants. For more information about Microsoft Entra tenants, see How to get a Microsoft Entra Tenant.

A tenant houses the company's user information. This information includes passwords, user profile data, permissions, and related information. The tenant also contains groups, applications, and other information that pertains to an organization and its security.

The tenant is created when customers sign up for their first subscription to any Microsoft online service, such as Microsoft 365, Microsoft Dynamics 365, or Azure. Any later subscriptions to the same online services or other online services can be grouped within the same tenant.

An organization can have multiple Microsoft Entra tenants. If there are multiple tenants, make sure that any subscriptions for finance and operations apps are associated with the correct tenant.

- **Microsoft Entra ID (Microsoft Entra ID)** – Microsoft Entra ID is the multitenant, cloud-based directory, and identity management service from Microsoft that combines core directory services, application access management, and identity protection in a single solution. For more information, see Microsoft Entra ID. Finance and operations apps use Microsoft Entra ID as the store for identity. Access to Microsoft Entra ID is provided as part of a subscription to finance and operations apps.

- **Microsoft 365 admin center** – Microsoft 365 admin center is the subscription management portal that Microsoft 365 provides for administrators. Use it to provide management functions for users (Microsoft Entra ID) and subscriptions. As part of these management functions, it provides information about service health. For more information, see About the Microsoft 365 admin center.

  > **Note**
  >
  > You don't need a Microsoft 365 license to deploy finance and operations apps. However, you might require a license for specific Office integration scenarios. For more information, see Office integration overview.

- **Microsoft Dynamics Lifecycle Services (LCS)** – LCS is a collaboration portal that provides an environment and a set of regularly updated services that can help you manage the application lifecycle of your implementations. For more information, see Lifecycle Services resources. After you purchase and activate a subscription for a finance and operations app,

an **Implementation project** workspace is provisioned in LCS when the tenant administrator signs in for the first time.

> **Note**
>
> An implementation project is an LCS project for the cloud service. As a Microsoft partner, you can also provision nonimplementation LCS projects for your own purposes. For more information, see **Lifecycle Services (LCS) for finance and operations apps partners**.

- **Finance and operations apps** – Deploy finance and operations apps through LCS. Various topologies are available: development/test/build, acceptance test, performance test, and high-availability production. For more information about the various topologies, download the latest Microsoft Dynamics 365 Licensing Guide from Dynamics 365 pricing.

- **Microsoft Azure DevOps** – Use Azure DevOps primarily for code version control, development, and to deploy a build environment. Use Azure DevOps to track support incidents, such as work items in Azure DevOps that are submitted to Microsoft through Cloud-powered support, and to integrate the Business process modeler (BPM) library hierarchy into your Azure DevOps project as a hierarchy of work items. Use Azure DevOps during code upgrade.

"Under the hood," finance and operations apps use many features of the Azure platform, such as Azure Storage, networking, monitoring, and Azure SQL Database, to name just a few. Shared services put into operation and orchestrate the application lifecycle of the environments for participants. Together, Azure functionality and LCS offer a robust cloud service.

> **Note**
>
> Although many features of the Azure platform are used, you don't need an Azure subscription to deploy finance and operations apps in the Microsoft-managed cloud. You must have an Azure subscription only if you want to deploy finance and operations apps cloud-hosted environments in your own Azure subscription.

Last updated on 06/18/2026

What is FastTrack for Dynamics 365?

Welcome to Microsoft FastTrack for Dynamics 365, our customer success service designed to help your customers implement and go live so they can realize business value faster. It's an onboarding program run by the product engineering team that offers best practices, tools, resources, and expert advice. Your customer gains valuable insights with solution implementation, onboarding, and continued education.

FastTrack for Dynamics 365 represents our strong commitment to fostering the business success of our customers. When an organization participates in the FastTrack program, they receive ongoing guidance on best practices and how to plan for successful rollouts. FastTrack program participants receive emails with implementation guidance and communications about their active FastTrack engagement. In addition to proactive guidance and ongoing workshops, FastTrack customers also learn ways to enable new users and expand capabilities—all at their own pace.

What to expect when working with FastTrack

The FastTrack solution architects have a unique role in a customer implementation. Even though we have no formal role on the project and we don't design or develop a solution for the customer, we're committed to the success of the program and the timely onboarding of users. Success by Design is our delivery framework for FastTrack customers. It provides a structure for the FastTrack solution architects to effectively engage with customers and partners in a timely and consistent manner to help them successfully onboard onto our Dynamics 365 platform.

Timely engagement

With the Success by Design framework, the FastTrack solution architects engage with project teams through a set of strategic workshops that align to key project stages with clear goals, actions, and deliverables. During the workshops, the solution architects assess if the project follows best practices, and they identify and highlight issues or potential risks that could derail the program. Timely engagement with the solution architects through workshops ensures that Microsoft experts have an opportunity to provide input into your architectural design decisions, helping you align with product direction and best practices for performance and scalability.

Information sharing

The project team, including the customer and partners, should provide the information needed to run the workshops and keep us updated on any design changes that the FastTrack solution architect needs to review. The quality and depth of the information directly impacts the recommendations from the FastTrack solution architect. The more engaged and open the team is to sharing information with the FastTrack solution architect, the more value we deliver in return.

Time investment

Running the workshops with the FastTrack solution architect requires a time investment from the project team, based on the complexity and number of apps being deployed. This can vary from project to project. Fundamentally, don't consider time spent with FastTrack on workshops as overhead. It's an opportunity to revisit your decisions with a stakeholder from the product team who can help identify potential risks and issues. FastTrack engagements help save time and effort in the long term and greatly improve your chances of success and adoption.

Communication, feedback, and survey

When working with FastTrack, you can expect:

- Email communications, conference calls, and in-person sessions with the product group.
- Feedback surveys on the product, and your experiences and interest in new capabilities.
- Invitations to exclusive product group events, advisory boards, and table talks with product managers.
- Executive engagement opportunities with senior product and engineering leaders at Microsoft.
- Requests for joining our reference program to share your success story and experiences with other customers and analysts.

For questions regarding FastTrack, reach out to your partner or Microsoft account team.

Implementation portal

Dynamics 365 Implementation Portal helps you manage implementation projects. Learn more at Dynamics 365 Implementation Portal.

Next steps

- Customer eligibility and partner qualifications

Last updated on 12/02/2025

Onboard an implementation project

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes how to onboard a finance and operations project by using Microsoft Dynamics Lifecycle Services (LCS).

Microsoft 365 Admin Center

After your organization purchases a subscription to a finance and operations app, a tenant administrator must activate the service on your organization's Microsoft Entra tenant. For more information about Microsoft Entra roles, see Understand roles in Microsoft Entra ID.

The tenant administrator must complete the following steps:

1. Open an InPrivate or Incognito browser session and go to the Microsoft 365 Admin Center.

2. Sign in by using your tenant administrator credentials.

3. Go to **Billing** > **Products & services** and confirm that there's an active subscription for the application that you want to deploy.

   > ⓘ **Note**
   >
   > If you don't see an active subscription, consult with your licensing partner to confirm the status of the subscription transaction. It's important to confirm that you purchased the subscriptions for the correct Microsoft Entra tenant. By default, all Microsoft online services run on the same Microsoft Entra tenant. The most frequent cause for onboarding delays is that subscriptions are inadvertently placed on an incorrect Microsoft Entra tenant.

4. If the subscription is active, proceed to the next step by signing in to LCS to trigger the implementation project creation flow.

5. Open another private browser tab and go to Lifecycle Services. Select **Login** to access LCS by using your current tenant administrator credentials.

   > ⓘ **Note**
   >
   > Connection endpoints might be different for Government Community Cloud (GCC) and other local cloud deployment options. For more information, see Sovereign and local cloud deployment options for Dynamics 365 Finance and Dynamics 365 Supply Chain Management.

6. Accept and confirm any other prompts displayed to complete the implementation project provisioning.

7. The tenant administrator is assigned the Project Owner security role in the provisioned implementation project.

   > ⓘ **Note**
   >
   > If the tenant administrator won't be a participant in the implementation, assign at least one additional Project Owner to the implementation project.

For an overview of LCS user management, including the security roles that you can assign to users, see Configure Lifecycle Services (LCS) security.

LCS implementation project workspace

After the tenant administrator finishes the finance and operations subscription activation and adds extra project users as needed, those team members can access the **Implementation project** workspace.

The first step to complete in LCS is **Project onboarding**. This step is required for all LCS implementation projects that are created **on or after August 22, 2019, PST**, before deploying any of the Microsoft-managed environments. You can access the **Project onboarding** feature by using the action center notification or the LCS Implementation project menu. You must be assigned to the Project owner security role to access **Project onboarding** in LCS.

To get started with LCS, see Lifecycle Services (LCS) for finance and operations apps customers.

FastTrack onboarding services

After the LCS **Implementation project** workspace is provisioned, the Microsoft FastTrack team monitors your onboarding progress. If you don't complete project onboarding within a few weeks after creating an LCS **Implementation project**, the FastTrack team sends a reminder to the project team.

For more information about the FastTrack program and the services it provides, see Microsoft FastTrack.

For more information about LCS project onboarding, see LCS project onboarding.

> **Note**
>
> All onboarding-related emails from the FastTrack team come from Dynamics 365 Onboarding (ond365@microsoft.com). Make sure your spam blocker or filter allows email from this address.

Key data to keep current in LCS

Add key project members, such as project managers, from the customer and partner teams to the LCS implementation project. Include each person's work email address. By keeping this

information current, you help us work best with you and help ensure that project members don't miss important communication from us.

Keep the milestone dates in your LCS project current. By keeping these dates up to date, you help us connect with you at different project stages. When you're closer to your go-live date, we contact you for a project Go-live assessment before we deploy your production environment.

LCS stores milestone dates in the implementation methodology. For more information, see the Methodologies section of the *LCS for Customers* article.

Last updated on 06/18/2026

Environment planning

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

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This article provides an overview of various aspects that you must consider while you plan for your project's environment. To help guarantee a successful cloud implementation, discuss and plan your environment early in the project.

Environment planning overview

To begin, consider these important concepts:

- **Environment purpose** – The reasons why the environment exists. Examples include development, system testing, user acceptance testing (UAT), and operations.
- **Environment topology** – The composition of the environment and the purpose. Examples include **Develop** and **Build and Test** for Tier-1 environments.

- **Environment tier** – The type or category of the environment. Examples include Tier-1 environments and Tier-2 environments.

For more information about the various environments and tiers, see Cloud deployment overview and download the latest *Microsoft Dynamics 365 Licensing Guide* from Dynamics 365 pricing.

Environment types

You can use the following environment types for your project:

- **Standard** – This environment is included in the standard offer and Microsoft manages it in a Microsoft subscription. Standard environments include the production environment and a Tier-2 Standard Acceptance Test environment.
- **Add-on** – The add-on environments are in a Microsoft-managed subscription that you purchase in addition to the standard offer. For example, an add-on environment might be an additional Tier-4 environment for performance testing.
- **Cloud-hosted** – You can add cloud-hosted environments that you manage in a customer or partner Microsoft Azure subscription. A cloud-hosted environment can include a Tier-1 demo environment, or it can be used for Development and Build. See Deploy and use a continuous build and test automation environment.
- **Environment image (VHD)** – These additional Tier-1 environments are hosted on-premises by using a virtual hard disk (VHD) that you can download from Microsoft Dynamics Lifecycle Services (LCS).

> **Important**
>
> In a *customer or partner Azure subscription*, the customer or partner brings its own Azure subscription, and deploys environments to that subscription for evaluation and development purposes only. The customer or partner pays for the resources that are deployed to its Azure subscription. The amount that the customer or partner pays is based on the Azure price list. By contrast, in a *Microsoft subscription*, the customer purchases licenses that allow the customer to deploy environments to an Azure subscription that Microsoft manages. Therefore, the customer has no separate Azure billing.

Tier 1 vs. Tier 2 and higher

Expand table

Tier 1	Tier 2 and higher
Single-box environment	Multi-box environment
All components are installed on the same server. These components include Application Object Server (AOS), the database, Dynamics 365 Commerce, and Management Reporter.	Components are installed on multiple servers.
Microsoft SQL Server is used.	Azure SQL Database is used.
The architecture differs from the architecture of the production environment to maximize efficiency and cost of the development team.	The architecture is the same as the architecture of the production environment, even though this type of environment has a different sizing and isn't enabled for disaster recovery.
The environment can be cloud-hosted, or it can be deployed as an environment image (VHD).	The environment can be deployed only as a standard environment or an add-on environment. It can't be cloud-hosted.
The environment isn't suitable for UAT or performance testing.	The environment is suitable for UAT and performance testing.

Standard cloud offer

The standard cloud offer includes two environments:

- **Tier 2 environment: Standard Acceptance Testing** – One Standard Acceptance Testing (UAT) instance is provided for the duration of the subscription. This instance is a non-production multiserver instance that customers can use for UAT, integration testing, and training. You can purchase additional sandbox or staging instances separately as an optional add-on.

- **Production environment** – One production instance is provided per tenant. The production multiserver instance includes disaster recovery and high availability. It provisions when the implementation approaches the Operate phase, after the required activities in the Microsoft Dynamics Lifecycle Services (LCS) methodology and a successful go-live assessment are completed. Additionally, the offer includes some file storage and database storage:
  - **File storage:** Every customer receives a certain amount of file/Azure blob cloud storage for files and binary data. You can purchase additional file/blob storage.
  - **Database storage:** Every subscription includes a certain amount of Azure SQL Database storage per customer at no extra charge. Additional storage capacity is provided at no charge as an organization increases the number of user and device service licenses. For more information about the various environments and the various types of storage, as

well as the currently included free file and storage capacity, download the latest *Microsoft Dynamics 365 Licensing Guide* from Dynamics 365 pricing.

> **Important**
>
> Microsoft promises service and data high availability as well as minimal servicing downtime guarantees as part of the Dynamics 365 software license agreement (SLA) for production environments. The SLA goals don't apply to non-production environments.

Provisioning of standard environments

You provision the standard environments at different times. The following table shows the suggested timing for the environments in the standard cloud offer.

Expand table

Environment	When does provisioning occur?	Is it self-service?
Tier-2 Standard Acceptance Test	Immediately after project onboarding is complete in LCS	Yes
Production	At production system readiness	Complete a Go-live assessment before you configure the production deployment request in LCS.

> **Important**
>
> Always deploy environments by using an **unnamed** account. This account must be from the customer domain, such as `dynadmin@customer.com` or `dynadmin@customer.onmicrosoft.com`. Use the same dedicated environment admin account on all environments. You must issue ISV licenses to the domain of the account you use for environment deployment. If you use both `dynadmin@customer.com` and `dynadmin@customer.onmicrosoft.com` on different environments, you can encounter problems with ISV licenses.

Production system readiness

You can deploy the production environment when the project is ready for the initial go-live. For more information, see Prepare for go-live.

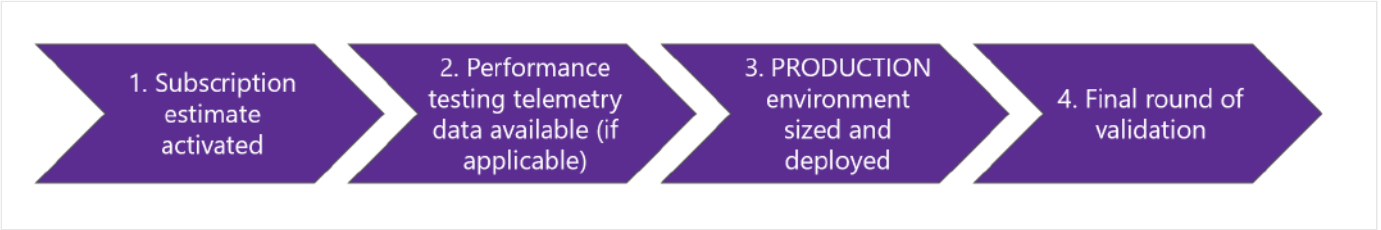
Production system readiness includes, but isn't limited to, the following conditions:

- An up-to-date subscription estimate is activated, as described in Subscription estimator in Lifecycle Services (LCS).
- Code, configuration, and data are ready for cutover.
- An engineering process is in place to manage critical fixes.
- The customer signs off on the solution and UAT.
- A cutover plan is in place.

Use the production environment to **operate** the solution, not build it. The production environment is sized to run your business. The sizing is based on the subscription estimate and diagnostic data from performance testing. After deployment, you can and should do a mock cutover and a final round of validation on the production environment. Before the final cutover, you can request a point-in-time restore to restore the production environment to a clean snapshot (maximum 28 days in the past).

To select the appropriate data center for the production environment, consider the latency from the geographic locations where the business operates. Use tools such as PsPing to test latency to Azure data centers.

The following illustration shows the environment planning process.



Additional environments

You can purchase additional environments as add-ons or deploy them as cloud-hosted environments. The following illustration shows a *sample* overview of standard and additional environments, based on the complexity of the implementation.

Environment purpose	Tier	Standard project	Medium complexity project	Complex project
Build	1	X	X	X
Test (e.g. system integration testing or UAT)	2	X	X	X
Production	Based on the **sizing**	X	X	X
Development	1	X	X	X
Golden configuration	1 or 2		X	X
Pre-production / data migration	2		X	X
Performance testing	4 or 5			X
Training	2 or 3			X
System integration testing	2 or 3			X
Other / ad-hoc	...			X

> **Important**
>
> Always deploy environments by using an **unnamed** account, such as `dynadmin@customer.com`. Assign the environments an owner who is responsible for their status and maintenance. Use the same dedicated environment admin account on all environments. After go-live, if you plan to work on new releases, get an additional Tier-2 or higher environment to support production.

Deployment considerations for development environments

For development environments, use one of the following deployment options:

- **Cloud-hosted** – You manage the environments in your Azure subscription.
- **Environment image (downloadable VHD)** – You host the environments on-premises.

> **Note**
>
> You must allocate one development environment per developer.

The following table compares the deployment options.

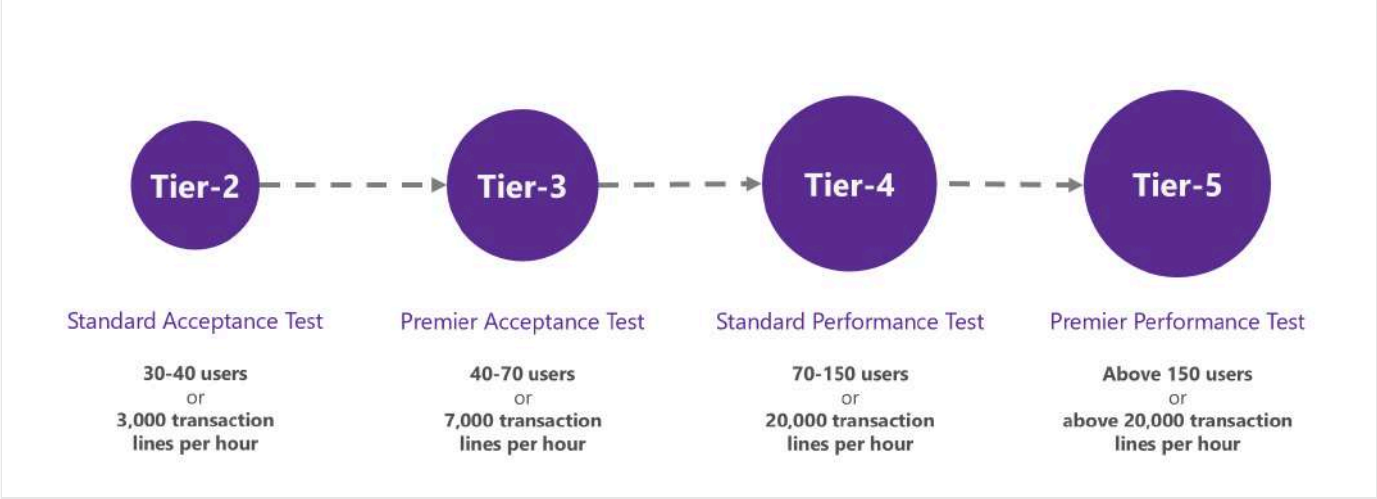
Expand table

Capability	Cloud-hosted	Environment image
Public URL	✓	Not supported

Capability	Cloud-hosted	Environment image
Integration development	✓	Extra setup is required. (For example, run the admin user provisioning tool.)
Azure DevOps	✓	Extra setup is required. (For example, rename the computer.)
Applying deployable packages from LCS	Automated	Command line runbooks (AxUpdateInstaller.exe tool)
Deploying data packages from LCS	✓	Not supported
Maintenance	Managed by you	Managed by you
Cost model	Pay as you go (If the environment is on for eight hours, you pay for eight hours.). Cost is based on selected Virtual Machine size, disk size and settings, and premium storage settings	Hardware-related
Limitations	None. You have full control over VM specs, disk size and storage settings. You have administrator access to the VM.	None

Selecting the correct Tier 2 or higher environment

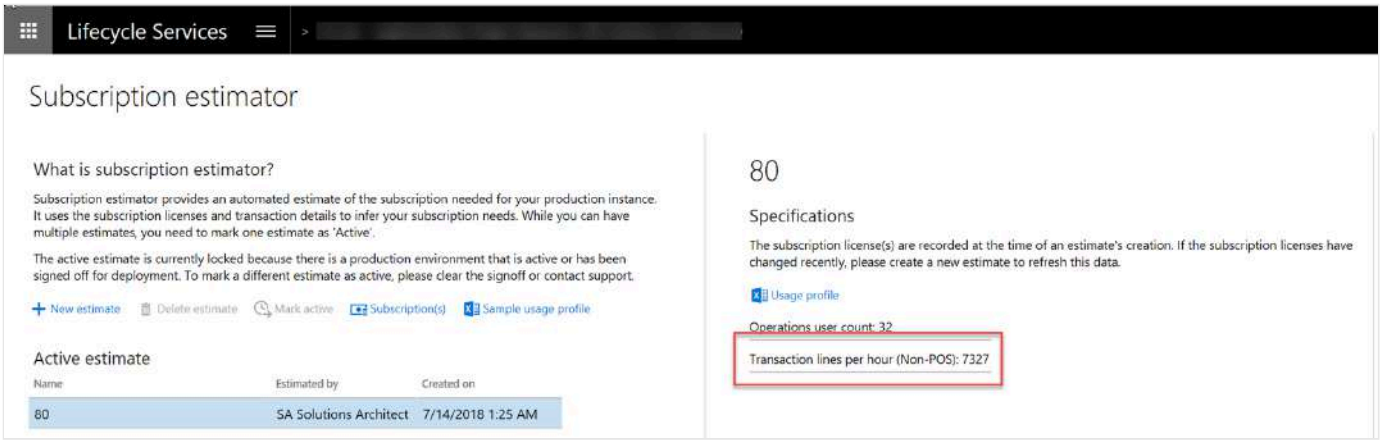
Select the correct Tier 2 or higher environment based on the purpose of the environment.



> **Important**
>
> The provided values are for reference only. Work with your implementation partner to adjust this guidance based on your specific business scenarios and factors such as type of users,

complexity, and volumes.

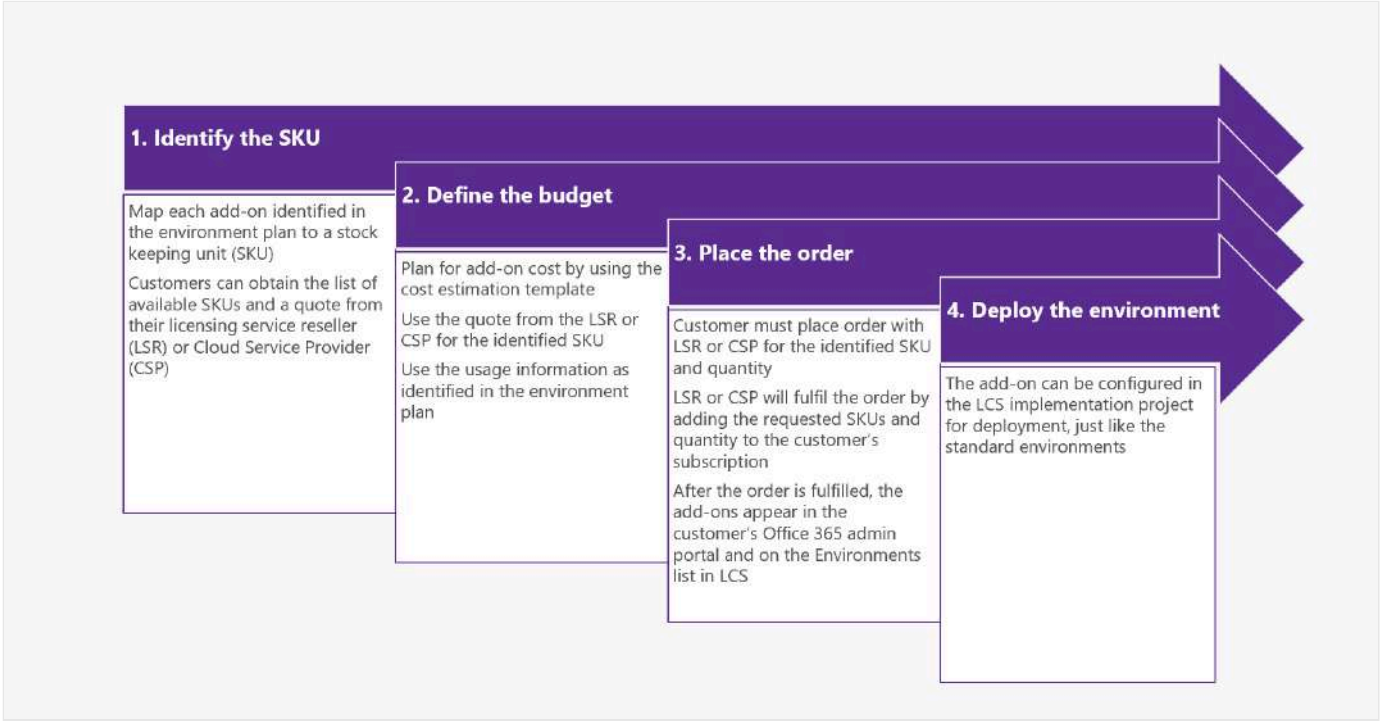
Use the LCS Subscription estimator to get an indication of the transaction lines per hour in LCS for a volume check, as shown in the following illustration.



Purchasing add-on environments

If you want to purchase add-on environments, work closely with your Cloud Solution Provider or License Service Reseller. Consider the potential lead time that occurs between the time when you place the order and the time when the environment is deployed.

The following illustration shows the process for purchasing add-on environments.



ⓘ Important

If you have a Microsoft Volume Licensing agreement, you can subscribe to add-on environments on a monthly basis through the Microsoft Products and Services Agreement (MPSA) licensing program. Alternatively, you can subscribe to them through the Microsoft Cloud Solution Provider (CSP) program. For more information about the various environments and tiers, download the latest *Microsoft Dynamics 365 Licensing Guide* from Dynamics 365 pricing.

Environments plan

Create the environments plan early in your implementation.

1. Identify the project activities that require an environment. These activities include, but aren't limited to, development of customizations and maintenance of golden configuration data.
2. Determine the *activities lifecycle* to determine the *environments lifecycle*. Here are some examples of the questions that you should ask during this step:
    - When and for how long do you require the environment?
    - Do you require the environment before or after go-live?
3. Determine the type and topology of the required environments.
4. Summarize the list of required environments in a matrix.

After you identify the environments, use the environments plan to structure the Application Lifecycle Management (ALM) flows. For example, after you finalize your environments plan, you can define the flows for building and moving the code and the data across environments.

Last updated on 06/12/2026

Prepare for go-live

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article provides guidance about how to prepare for the go-live for finance and operations apps in Lifecycle Services if you already have a project there.

To ensure that the production environment is used for live operations, Microsoft provisions the production instance when the solution is ready and after project readiness is validated as part of the Go-live Readiness Review with Microsoft. For more information about the environments in your subscription, see the Licensing guide.

> **Note**
>
> Most projects are **required** to use the **FastTrack for Dynamics 365 implementation portal** for their Go-live Readiness Review with Microsoft. If you have a Microsoft FastTrack Solution

Architect assigned to your implementation project, reach them before creating a new project to avoid duplicate projects or reviews. The Microsoft FastTrack Solution Architect can work with you on the Go-Live review and guides you on the process to follow.

For answers to common questions about go-live, see the Go-live FAQ.

The following table lists the key steps in the go-live process.

Expand table

Step	Duration/When	Who	Notes
Validation of prerequisites	A project is close to the end of the testing phase, and you're planning the go-live review.	Customer/Partner	For more information, see the Go-live Readiness Review prerequisites section.
Go-live Readiness Review with Microsoft	No later than four weeks before the go-live	Customer/Partner submits answers to review questions in the FastTrack for Dynamics 365 implementation portal. Microsoft FastTrack provides a review report.	For more information, see the Go-live Readiness Review with Microsoft section.
Go-live review workshop with Microsoft (only for eligible projects)	The workshop is part of the Go-live Readiness Review.	Microsoft FastTrack together with the project team	This step applies only to eligible projects that have FastTrack engagement. For more information about the workshop, see Go-live Readiness workshops.
Production deployment	Production deployment can be initiated as soon as the Go-live Readiness Review is completed. After deployment is triggered, production deployment takes approximately 30 minutes.	Customer/Partner triggers the deployment in Microsoft Dynamics Lifecycle Services.	After the go-live review is successfully completed, the **Configure** button is enabled for the production environment. Selection of this button triggers the production deployment.
Deployable package installation	An average of 30 minutes	Customer/Partner	Follow the instructions in Promote an update to production environments. The packages must contain all the models and binaries consolidated into

Step	Duration/When	Who	Notes
			all-in-one deployable packages.
Data migration to production	The duration/time depends on the method that is used.	Customer/Partner	
Cutover activities	The duration/time depends on the project.	Customer/Partner	
Go-live		Customer/Partner	

Go-live Readiness Review prerequisites

The project team should validate solution readiness. Before you initiate the Go-live Readiness Review with Microsoft, ensure that the following prerequisites are met. You can deploy the production environment after you successfully complete the Go-live Readiness Review with Microsoft.

- User acceptance testing (UAT) and performance testing are complete or almost complete in a Tier-2 (or higher) environment. Don't use Tier-1 environments for UAT or performance testing. For more information, see Tier-1 vs. Tier-2 and higher. For information about how to identify the correct sandbox environment tier, based on your transaction volumes, see Selecting the correct Tier-2 or higher environment.

  During the UAT phase, test all the business processes that you implemented and any customizations that you made.

  - Test cases should cover the entire scope of requirements.
  - Use migrated data for testing. This data should include master data and opening balances, even if they aren't yet final.
  - Use the correct security roles for testing, including default roles and custom roles that you assign to users.
  - Make sure that the solution complies with any company-specific and industry-specific regulatory requirements.
  - Obtain approval from the customer.

    Performance testing is a crucial part of validating the readiness of your solution. For detailed guidance about the recommended practices, review the Performance Testing

Techtalks.

- The environment version that you plan for the go-live complies with the Software lifecycle policy. If you use Commerce Scale Units (CSU), the CSU version must be in sync with the environment version. For more information, see Component dependencies.
- Add key customer team members to the Lifecycle Services project.*
- Add a generic service account that's used to deploy the production environment to Lifecycle Services.
- Purchase all licenses that are required for go-live in the correct tenant.
- After you purchase all licenses, upload and activate the final subscription estimator in Lifecycle Services. For more information, see Subscription estimator in Lifecycle Services.*
- Run the Customization analysis report (CAR) and address critical issues. For more information, see Customization Analysis Report (CAR).
- The go-live date in Lifecycle Services correctly represents the go-live date that you're targeting. This date is the date when end users start live operations, not cutover activities.*
- Complete all relevant tasks and phases in the Lifecycle Services methodology. You can deploy production only after all phases, including the Test phase, are complete*. To complete a phase in Lifecycle Services, first complete every required step in that phase. When all the steps in a phase are complete, you can complete the whole phase. You can always reopen a phase later if you must make changes. The process of completing a step has two parts:
    - Do the actual work, such as a fit-gap analysis or UAT.
    - Mark the corresponding step in the Lifecycle Services methodology as you complete it.

      Complete the steps in the methodology as you make progress with the implementation. Don't wait until the last minute.

You must complete all steps that are marked with an asterisk (*) before you submit the project for review in the portal. The project shows as blocked for review until you mark those steps as completed. When you update a setting in Lifecycle Services, it takes about 15 minutes before the status of review readiness changes.

Go-live readiness review with Microsoft

Before you deploy production environments, the implementation project in Lifecycle Services must complete a go-live readiness review with the Microsoft FastTrack for Dynamics 365 team. The main purpose of the review is to assess project readiness and help you have a smooth and successful go-live experience. Lifecycle Services project users receive an email reminder about the go-live readiness review, based on the go-live date in Lifecycle Services. For most projects, you complete the go-live readiness review in the Dynamics 365 implementation portal. To learn more, see What is the Dynamics 365 Implementation Portal?.

If a project opts in for an AI review when you submit the request for review, the review process is autonomously managed by AI. The review process completes on the same day, and the stakeholders are notified by email. Any risks or issues are highlighted and shared in the Dynamics 365 Implementation Portal. The customer stakeholders added to the project can review the risks and mitigate them, and the production slot is automatically released in Lifecycle Services within few minutes.

For projects that opt out of the AI review, the familiar manual review process follows. The manual reviews can take up to three business days for the initial report, plus more time for any risk mitigation that is required. Determine an appropriate time to start this review, so that you can meet the go-live timeline.

> ⓘ **Note**
>
> There are **four exceptions** that don't use the FastTrack for Dynamics 365 implementation portal:
>
> 1. **GCC, GCC High, and Department of Defense projects** - Projects that are in the United States (US) Government Community Cloud (GCC) or **US Government Community Cloud High (GCC High)** or **US Department of Defense (DoD)**. - Download the Go-live checklist, fill in all necessary details, and send it via email to d365fogccglr@microsoft.com to start the Go-Live process. Always include a key stakeholder from the customer and the implementation partner on the email. Microsoft FastTrack reviews the project and follows up.
> 2. **Tenant moves** - For projects that are already live, but planning to move the live solution to a new tenant if the solution is not changing, completing a new Go-live Readiness review is not necessary. Follow the steps described in Move your production environment to the new tenant to get the production slot enabled on the new tenant.
> 3. The following steps apply to Lifecycle Services projects. These steps do no apply to implementations that you manage in the Power Platform admin center.

Initiate the go-live readiness review in the portal

> **Note**
>
> If a Microsoft FastTrack Solution Architect is assigned to your implementation project, contact them before creating a new project by using the following steps in the portal. This approach helps you avoid creating duplicate projects or reviews. The Microsoft FastTrack Solution Architect can work with you on the go-live review and guide you on the process to follow.

Self-service create or join the project

Partners and customers can submit the go-live readiness review in the Dynamics 365 Implementation portal without involving Microsoft. The Onboarding Wizard in the Implementation portal simplifies the steps to create the project, add the admin and users, and submit the go-live review for self-service.

1. Create the project in Implementation portal. For more information about creating and joining the project, see Create or join a project in the Implementation Portal.
2. Ensure that the project admin on the project is from the **customer** organization. This user from the customer organization can then add more users and serve as the key participant in the go-live readiness review. For more information about adding users, see Admin section.

Initiate the review

1. Once the project is created, the project team can create and perform the go-live readiness review in the portal by using the guidance in the Portal Help article. All users who register on portal can access this article.
2. Read the prerequisites and proceed to the next step to add the customer team members and key stakeholders to the review. The customer stakeholders must belong to the same tenant linked to the project.

> **Note**
>
> If you encounter any issue with the portal, contact the portal Support team by selecting **Contact us** in the upper-right corner of the portal or sending an email to the Support team. In the email, specify the ID of your project in Lifecycle Services, and provide details that describe the issue.

Submit the review

- The project team should answer all questions in the review. The review process in the portal supports multi-user scenarios. Multiple team members can provide details for the go-live review at the same time.
- When you answer all the questions in the Go-live Readiness Review, submit the review to Microsoft by selecting **Submit for review**.
- If you meet all the prerequisites, choose the key review participants and submit the review. The key review participant must be from the customer organization and their authentication must belong to the same tenant linked to the project.
    - In this dialog box, there's an option to opt out of the AI review. If you choose this option, a manual review process follows, and the corresponding SLA (three days) applies.
    - If you don't choose to opt out, the autonomous AI review process starts so that the review completes on the same day.

> **Note**
>
> The key review participant must belong to the same tenant linked to the project. Only then can they see the **Resolve/Mitigate all risks** button in the portal if the AI review process is opted.

After you submit the review in the portal

Manual review process (opted out of AI review)

- Microsoft FastTrack reviews the project and provides a report that describes the potential risks, best practices, and recommendations for a successful go-live of the project. **The review might take up to three business days for the initial report, plus more time for any risk mitigation that is required.**
- Users who are selected as **Review participants** in the portal receive email communication that provides updates about the review.
- When all critical risks are addressed and the review is completed, Microsoft enables the production environment slot in the Lifecycle Services project. The customer or partner can then trigger the production environment deployment.

Automated review process (not opted out of AI review)

- An AI system conducts the review. When the review completes, users who are selected as **Review participants** in the portal receive an email that provides updates about the review. The review completes within a few minutes after you submit the project for review.
- The stakeholders can use the link in the email to access the Implementation Portal where they can review any potential risks, best practices, and recommendations for a successful go-live of the project. It's highly recommended to take corrective action on all the risks and issues for a successful go-live.
- Only the people who are added to the review as key participants (belonging to the same tenant) get a button with the label **Resolve/Mitigate all risks** in the Implementation Portal. If they choose that button, they confirm the mitigation of the relevant risks.
- When all critical risks are mitigated or accepted by the stakeholder in the Implementation Portal, the review is marked as completed, and the production environment slot in the Lifecycle Services project becomes available in 15 minutes. A person from the organization or the implementation partner can then choose the **Configure** button in Lifecycle Services to configure the production environment.

Production environment deployment

> **Note**
>
> Use the production environment exclusively to run your business operations. Don't use it for testing, demo, or training purposes. You can do the cutover, and mock the cutover (if a mock cutover is planned), in production. To test the solution, use an appropriate sandbox environment that includes all the elements and services that are required for testing.

Select a service account for environment deployment. For example, select a generic user account as the admin user of the environments that you deploy. If you select a named user account, you might not be able to access an environment if that user is unavailable. Here are some examples of scenarios where the admin user must access an environment:

- **First sign-in to any environment after initial deployment** – The admin user is the only user who can access the environment.
- **First sign-in to a sandbox environment after a database refresh from the production environment** – No user accounts except the admin account can sign in.

Deploy the production environment to the same datacenter where your sandbox environments are deployed, and where UAT and performance testing were done. For information about network requirements and how to do latency testing, see Network requirements.

Production deployment takes approximately 30 minutes. When deployment is completed, an email notification is sent to the environment administrator.

The production environment is sized based on the number of licenses that are allocated to the Lifecycle Services project and the transaction volumes in the Subscription estimator.

After you deploy production, the project team can apply the deployable package by following the instructions in Promote an update to production environments and then migrate the data. For data migration, prepare and validate data in a nonproduction environment and then copy the sandbox database to production.

FAQ

Who can create the project in the portal?

The customer users or partner users (on behalf of the customer) can create the project in the portal. Ensure you give the right tenant ID to link to the project.

I'm facing issues or errors while creating the project

Send an email with the details and screenshot of the error to Support team.

Why can't I submit the review?

To submit the review, you must answer all the questions, add at least one key customer stakeholder from the same tenant linked to the project, set the Subscription estimator to **Active** in the Lifecycle Services Project, and set the Lifecycle Services Phase to **Deploy** (or ahead). If you don't meet these conditions, you can't submit the review. Update these settings in the Lifecycle Services project and retry after 15 minutes.

I'm a customer and I was added as a key stakeholder in the project review. I received the email to sign off on risks but I don't see the "Resolve/Mitigate all risks" button in the portal

The button is only visible for users who belong to the same domain as that of the tenant linked to the project. The system validates the domain name (for example, @contoso.com) and compares it with the tenant linked to the project. If they don't match, the button isn't visible. If you identify the mismatch after submission of review, please reach d365fogl@microsoft.com with the details to get it updated.

I resolved the risks but the production "Configure" button isn't enabled in Lifecycle Services

It might take about 15 minutes for the button to be enabled. If the button stays disabled even after 15 minutes, send an email to d365fogl@microsoft.com for assistance.

Last updated on 06/12/2026

Go-live for implementation projects FAQ

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article lists frequently asked questions about how to go live with an implementation project.

When can I configure and request my production environment?

Typically, you deploy a production environment after all customizations are code-complete, you complete user acceptance testing (UAT), you sign off on the solution, there are no blocking issues for go-live, and you successfully complete a **Go-live Readiness Review** with Microsoft. To understand the process, see Prepare for go-live.

Use your production environment exclusively for running your business operations. Don't use it for testing or training purposes. You can perform the cutover, and if planned, you can mock the

cutover in production. To test the solution, use a sandbox environment, which is designed with the necessary elements and services for testing.

What are the prerequisites to deploy a production environment?

For a list of prerequisites, see Prepare for go-live.

What is a Go-live Readiness Review and why is it required?

The Go-live Readiness Review is aimed to assess project readiness and help you have a smooth and successful go-live. To understand the process, see Prepare for go-live.

You must complete this review for every implementation project before you can deploy a production environment.

I want to request my production environment. Who do I contact for a Go-live Readiness Review?

To understand how to initiate the Go-live Readiness Review, see the guidance in the Prepare for go-live article.

The Production button isn't available in LCS. How do I request my production environment?

The **Production** button in LCS becomes available only after you complete the **Analysis**, **Design & develop**, and **Test** phases of the LCS implementation methodology and complete a **Go-live Readiness Review** with Microsoft. For more information, see Prepare for go-live.

For more information about how to complete methodology phases in LCS, see Lifecycle Services (LCS) for finance and operations apps customers.

My sandbox environment is currently on an update that is set to expire in two months. Can I request a production environment that has the latest update?

Deploy the production environment with the same version used in the acceptance testing environment, which should be the version you plan to go live with. Don't go live on a version different from the version you tested on because it can lead to unexpected behavior and unforeseen issues that can negatively impact the go-live experience.

Go live on a version that isn't near end of service, so that you can apply quality updates for some time after the go-live, if needed. Carefully plan the version on which you're conducting the User Acceptance Test (UAT), and keep it as up-to-date as possible.

For more information, see Software lifecycle policy and cloud releases and Service update availability.

Our sandbox environments are deployed in a different datacenter from where we want our production environment to be deployed. Can I select a different datacenter for my production environment?

Deploy the production environment in the same datacenter used in the acceptance testing environment, which should be the datacenter you plan to go live with. By using the same datacenter, you help ensure that you're validating latency as well as support for all services under production-like conditions. Don't go live on a datacenter different from the one you tested on because it can lead to unexpected performance issues and unforeseen functionality gaps that can negatively impact the go-live experience.

For information that can help you select the correct datacenter, see the Network requirements section of the "System requirements" article.

How is my production environment sized?

The current user license count and the information in the subscription estimate that is active when you request the production environment determine the size of your production environment.

> **Note**
>
> If you submit a new subscription estimate, your production environment might need to be resized, depending on the number of users, the type of user licenses, and the expected peak transaction volume. After the new estimate is processed, the next servicing operation on the

environment applies the update automatically within three days. Alternatively, you can create a support ticket to request a specific time to apply the changes.

I submitted the request for a production environment, but I made a mistake. Can I still change it?

No. Production environment deployment starts as soon as you submit the request. Wait until the deployment finishes, and then log a support ticket to delete the production environment so you can redeploy it.

How long does it take to deploy my production environment?

After the go-live readiness review with Microsoft is complete, the production environment slot in LCS is enabled, and the project team (customer or partner) can trigger the environment deployment. The deployment process takes approximately 30 minutes.

What level of access do I have in my production environment? Can I sign in to the VM?

No. Access to the production environment is limited. You can't access the virtual machine (VM) or Microsoft Internet Information Services (IIS). You also can't access the database through Microsoft SQL Server Management Studio.

How often is my production database backed up?

Automatic backups protect databases. Full database backups are done weekly, differential database backups are done hourly, and transaction log backups are done every five minutes. Automatic backups are retained for 28 days.

For more information, see Learn about automatic SQL Database backups.

Can I request a copy of the backup of my production database?

No. However, you can copy your production database to your Tier 2 or higher sandbox environment. After the copy operation finishes, you can back up your sandbox environment. For

more information, see Database refresh.

My golden configuration database is in a Tier 1 sandbox environment. How can I copy and restore it to my production environment?

To copy and restore your database, follow the instructions in the article Golden configuration promotion.

> **Note**
>
> If your golden configuration is in data packages, you must manually import the data packages to the production environment.

After go-live, can I apply new code changes to the production environment?

Yes.

For more information, see Apply updates to cloud environments.

What should I do if my production environment is down?

To report a production outage, follow the process described in the article, Report a production outage.

Last updated on 06/18/2026

Submit service requests to the Dynamics Service Engineering team

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

A service request is a ticket that you use to request that the Dynamics Service Engineering (DSE) team perform a predefined set of tasks on your environments.

> **Note**
>
> Service requests are only required for environments that are **Microsoft-managed**. Most environments are self-service. For more information about environment types, see Cloud deployment overview.

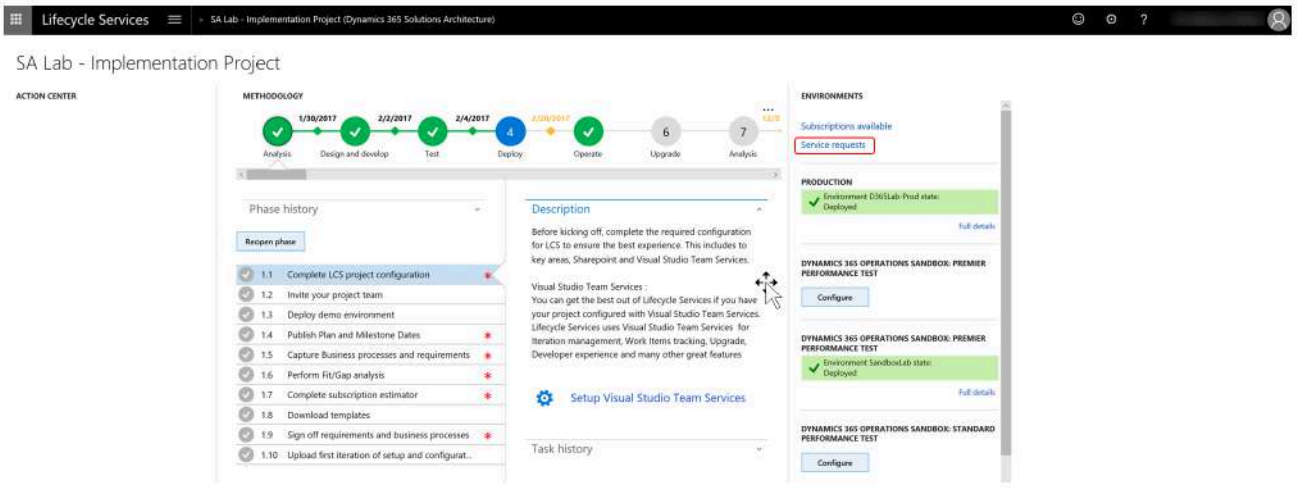
> Do not use service requests for product issues. If you encounter a situation that doesn't fit into any of the tasks that are described in this article, submit a support ticket instead. For more information about support tickets, see Get support for finance and operations apps or Lifecycle Services.

You can use Microsoft Dynamics Lifecycle Services to submit service requests directly to the DSE team. You can also view which requests have been submitted, executed, and canceled for your environments.

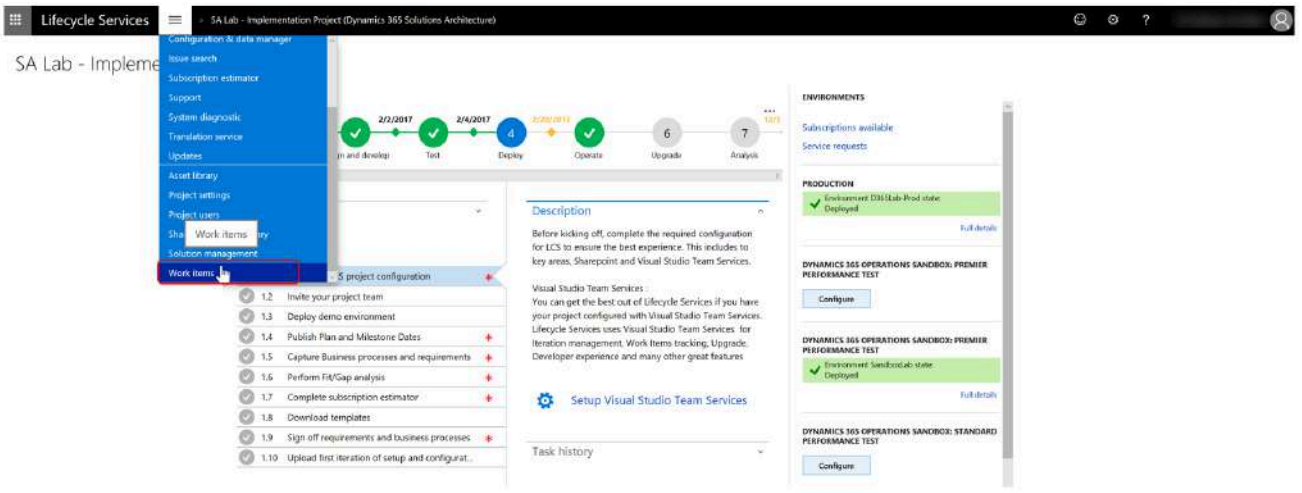
View service requests

There are two ways to view service requests:

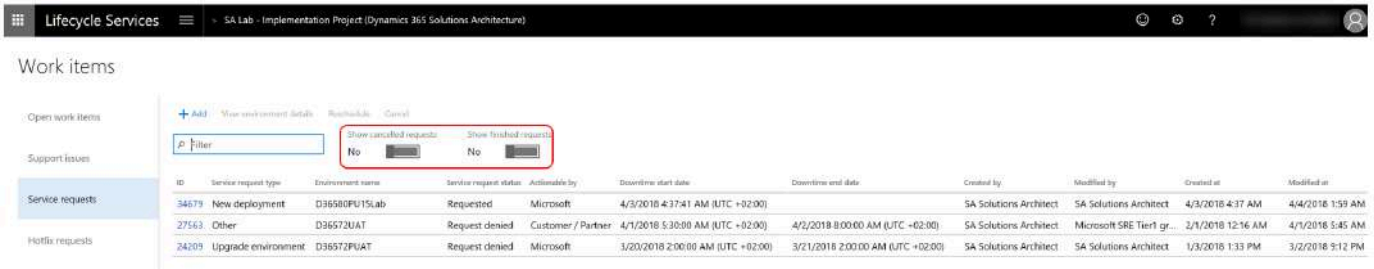
- On the project dashboard, in the **Environments** section, select **Service requests**.



- Select the **Menu** button and select **Work items**. On the **Work items** page select the **Service requests** tab.



By default, the **Service requests** tab on the **Work items** page lists all requests that are currently active and requests that have been denied. However, you can use the filter options to show canceled and finished requests too.



After you submit a request, it has a status of **Requested**. Before the DSE team acts on the request, it might ask for clarification by entering a comment in the **Comment** field. For example, you might receive a comment from the DSE team if you request deployment of a production environment, but the data center differs from the data center where your sandbox environments are deployed. Carefully review the comments, and provide any required clarification in your own

comment. To view the details of a specific request, or to submit comments for a service request, select the request ID.

If you signed up for Lifecycle Services notifications, you receive an email when the status of a service request changes or a comment is entered.

If you submit a service request to the DSE team, and the action is outside the team's scope, the service request will be denied. In this case, the reason for the denial and suggestions for further action are provided. For some typical examples of service requests that the DSE team will deny, see the "Denied service requests" section later in this article.

Create service requests

There are two ways to create a service request: automatically and on demand.

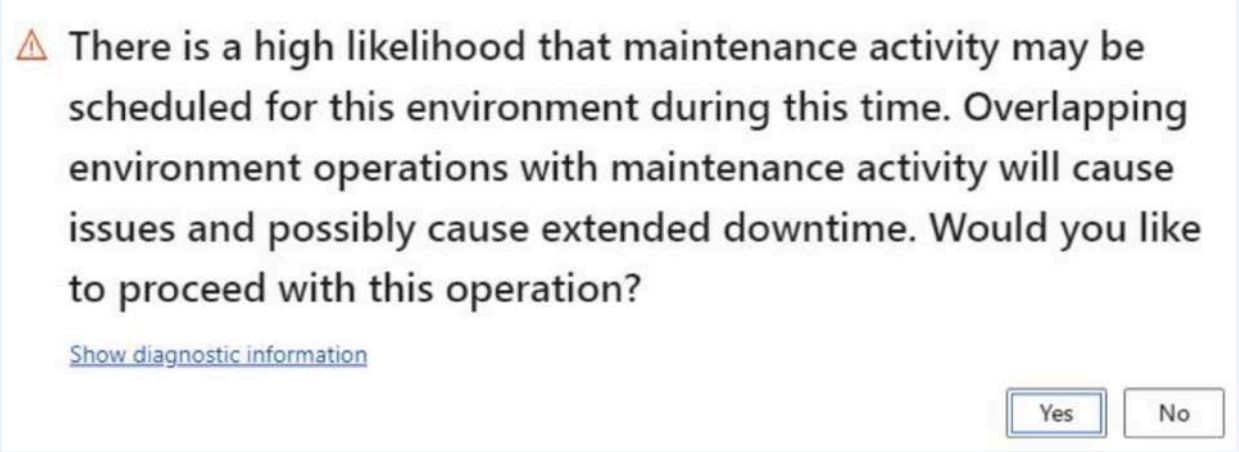
- **Automatically** – A service request is automatically created when you request deployment of an environment, or an application of a package.
- **On demand** – A service request is manually created when you enter a request for a database point-in-time restore, and some other services.

Automatically create a service request

- **Environment deployment** – To set up deployment options and submit a request to the DSE team to deploy a new environment, in the **Environments** section, select **Configure**.

- **Package application** – To apply a package to the production environment, on the **Environment details** page, select **Maintain**, select the package to apply, and then select **Schedule**. For more information, see Apply updates to cloud environments.

  > **Important**
  >
  > If your scheduled time overlaps with a planned maintenance window, you will receive the following warning message.



If you choose to continue deploying the package, the package deployment operation will be rolled-back in the event of conflict, as planned maintenance takes priority.

This restriction is appliable to **Microsoft-managed IAAS environments** only.

Create a service request on demand

Service requests that are created on demand aren't explicitly accepted by the DSE team. They will be addressed during the specified downtime window unless the DSE team has entered a comment in the request or has had to deny the request. For details, review the comments in the service request.

Microsoft frequently reviews all incoming service requests. By selecting the correct type of service request for your scenario, you help the DSE team handle the request in a timely manner.

1. On the **Work items** page, on the **Service requests** tab, select **Add**.
2. In the **Create request** dialog box, select the type of service request to create. The options on the page then reflect the specific type of request that you selected.

    - **Sandbox point-in-time restore request** – Select this request type to restore a *non-production* database to a specific point in time. For more information, see Database movement operations home page.

        > ⓘ Note
        >
        > If you need to restore a *production* database to a previous point-in-time during the cutover phase, select the **Production point-in-time restore request** type. If you need to restore a production database when you're already live in operations, submit a support ticket through Lifecycle Services.

- **Database refresh request** – Select this request type to refresh a database from a production environment to a sandbox environment, or from one sandbox environment to another. For more information, see Refresh database. *This request type is being retired on January 31, 2019.*

  > ⓘ **Note**
  >
  > If you need to refresh a database from a sandbox environment to a production environment during the cutover phase, select the **Sandbox to Production** type.

- **Sandbox to Production** - Perform a database refresh of your configuration data to a production environment during the cutover phase. For more information, see Database movement operations home page.

- **Other request** – You need to use the **Other request** type exactly as described here. If you word a request in a way that isn't clear to the DSE team, the team will enter a comment to ask for clarification, and your request will be delayed. If you use the **Other request** type for any request that isn't listed below, the request will be denied. Select this request type to request that the DSE team perform one of the following actions:

  - Turn on maintenance mode in a production environment. For more information, see Maintenance mode.

  - Tenant move of a live Production environment. Request the Microsoft Service Engineering team to move the Production database and Azure Blob Storage from the old tenant to the new tenant if you are moving tenant on a live Production environment. Make sure that you only request this service when you are ready with all prerequisites. For more details, see Move Lifecycle Services implementation projects to different Microsoft Entra tenants.

  - Define explicit Internet Protocol (IP) safe list rules in a production environment.

    > ⓘ **Note**
    >
    > Support for explicit safe list rules is deprecated for self-service environments. For more information, see Removed or deprecated platform features.

  - Request that Microsoft Power BI Embedded be activated in a sandbox environment, Standard Acceptance Test environment, or production environment if you receive

the following message: "Power BI embedded isn't enabled. Please contact your system administrator."

Commonly denied service requests

Here are some typical examples of service requests that will be denied:

- You submit a request of the **Other request** type for one of the following actions, but you should have submitted a support ticket instead:
  - You want to activate a new subscription estimate after you're live in production or after you've requested a production environment.
  - You want to reset the Financial reporting data mart in a release that is earlier than Microsoft Dynamics 365 for Finance and Operations Financial reporting release 7.2.6.0.
  - You want to restore a production database after go-live.
  - You encountered an issue after the DSE team did an application upgrade.
- You submit a request of the **Other request** type for an action that you should have requested through a different request type. Examples include a database refresh in a non-production environment.
- You submit a request of the **Other request** type for an action that you should perform yourself. Examples include a database upgrade in a development environment.

Service request types and SLAs

Expand table

Service request type	Applicable environments	Requested service	Lead time	Downtime
Environment deployment	Any	Environment deployment	Service level agreement (SLA): within two business days	
Package application	Production	Deployable package application	Five hours	Five hours
Sandbox point-in-time restore	Any Tier 2 or higher sandbox	Database point-in-time restore	Five hours	Four hours

Service request type	Applicable environments	Requested service	Lead time	Downtime
Production point-in-time restore	Production	Database point-in-time restore	Based on data volume	Based on data volume
Sandbox to Production	Tier 2 or higher sandbox to Production	Sandbox to Production	Five hours	Four hours
Other	Production	Maintenance mode	Five hours	Not applicable, because the customer indicates in the service request when the environment should be taken out of maintenance mode again
	Production	IP safe list rules	Five hours	Two hours
	Production	Power BI Embedded	Five hours	Two hours

Last updated on 04/13/2026

Subscriptions, Lifecycle Services projects, and Microsoft Entra tenants FAQ

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

When customers subscribe through a Microsoft Volume Licensing agreement or a Microsoft Cloud Solution Provider (CSP) agreement, they usually have one Microsoft Microsoft Entra tenant, one Microsoft Dynamics Lifecycle Services Implementation project, and any number of sandbox environments that are deployed to one data center of the customer's choice, and one production environment. For more information about these core concepts, see finance and operations application architecture. Although this setup works well for most projects, more advanced scenarios are sometimes required, or changes during the implementation lifecycle must be accommodated.

This article provides answers to frequently asked questions about subscriptions and licenses, Microsoft Entra tenants, and Lifecycle Services Implementation projects.

For more information, see the following topics:

- Move environments between data centers
- Move licenses between agreement types
- Move Lifecycle Services implementation projects to different Microsoft Entra tenants
- Multiple Lifecycle Services projects and production environments on one Microsoft Entra tenant

Do I have to move Microsoft Entra tenants when I move from a CSP agreement to a Volume Licensing agreement?

No. You can keep the existing Microsoft Entra tenant, but you must make sure that the Volume Licensing subscriptions are purchased against the same Microsoft Entra tenant as the CSP subscriptions.

Do I get a new Lifecycle Services Implementation project when I move from a CSP agreement to a Volume Licensing agreement?

No. The Lifecycle Services project remains the same.

Can I keep the existing Lifecycle Services Implementation project when I move to different Microsoft Entra tenant?

No. A new Lifecycle Services project is created.

How long does it take to move from a CSP agreement to a Volume Licensing agreement?

For a Volume Licensing purchase, it can take a few days for the order to process and the subscriptions to activate. Redeployment of add-on environments has a service level agreement (SLA) of two business days. It takes a few hours to deallocate and delete old add-on environments.

What if I forget to delete the existing environments before I suspend the existing subscription?

If you don't deallocate and delete the existing environments before you suspend the subscriptions, the environments stay in a **Deployed** state. However, if you try to access the full details of these environments, you receive an error message.

Can I have a CSP agreement and a Volume Licensing agreement in parallel?

Yes. However, you must maintain the minimum required number of licenses under each program.

How can I find the Tenant name and Tenant ID within Lifecycle Services?

1. Go to the project home page in Lifecycle Services.
2. In the **Environments** section, select **Subscriptions available**.
3. On the **Subscriptions available** page, you find the **Tenant name** and the **Tenant ID**.

How can I find the subscription status?

1. Go to the project home page in Lifecycle Services.
2. In the **Environments** section, select **Subscriptions available**.
3. On the **Subscriptions available** page, you see all **Service plans** available to the tenant.
4. The **Assigned date** indicates the date that service plan status changed.

How does the subscription status affect the environment?

The subscription status can affect some of the environment operations:

- **Active** - Your subscription is active. You can perform all environment operations.
- **In Grace Period** - Your subscription expired, but is still within the grace period. Renew your subscription soon. The subscription status doesn't affect your license quantity, ability to deploy a new environment, or ability to perform environment operations.
- **Suspended** - Your subscription expired and is beyond the grace period. This subscription status might reduce the license quantity, affect your ability to deploy a new environment, or affect your ability to perform environment operations.
- **Deleted** - Your subscription is deleted. This status affects your ability to deploy a new environment or perform environment operations.

Last updated on 03/28/2026

Geo to geo migration overview

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

To support data residency, Microsoft Dynamics 365 finance and operations apps and Microsoft Dynamics Lifecycle Services are generally available in specific geographies. You can migrate your workload from one geography to another. This article provides an overview of the process and the considerations.

Movement between geographies involves two separate processes:

- Finance and operations apps environment migration – Use this feature to move your finance and operations apps environments between geographies.
- Lifecycle Services project migration manager – Use this feature to move your Lifecycle Services project data between geographies.

To start planning for the migration, assess whether you want to migrate only environments, only Lifecycle Services projects, or both.

General considerations

Before migrating environments and data from one geography to another, consider the following aspects.

Feature parity

Review the availability of features in the selected target geography before you decide which geography to deploy in. If any features aren't available in the target geography, associated functionality won't work in that geography after migration. Therefore, plan the migration activity accordingly.

Dynamics 365 Commerce isn't available in all target geographies. If you enable Commerce components, your Lifecycle Services project migration isn't scheduled if you're migrating to one of the target geographies where Commerce isn't available.

Sovereign data resident geographies

Migrations into or out of a sovereign cloud environment (for example, US Government Community Cloud [GCC] and China) aren't supported.

Lifecycle Services project migration considerations

For more information, see Lifecycle Services project migration manager - Considerations.

Environment migration considerations

For more information, see Finance and operations apps environment migration - Considerations.

Last updated on 06/18/2026

Move licenses between agreement types

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

Sometimes, a customer who originally purchased subscriptions through a Microsoft Cloud Service Provider (CSP) agreement decides to change to a Microsoft Volume Licensing agreement with Microsoft after the Microsoft Dynamics Lifecycle Services Implementation project is created. The customer can make this change even after the project goes live in production.

Less often, a customer who originally purchased the subscriptions through a Volume Licensing agreement with Microsoft decides to change to a CSP agreement. In this case, the change must align with the renewal date of the Volume Licensing agreement.

Moving subscriptions from one type of agreement to another is primarily a commercial process. The technical implications for the Lifecycle Services Implementation project are minimal.

> **Note**
>
> Moving subscriptions between agreement types isn't the same as moving a Microsoft Entra tenant. If the contractual changes in the agreements require that you change the **Microsoft Entra tenant** on the subscriptions, you must also follow the process described in Move Lifecycle Services implementation projects to different Microsoft Entra tenants.

Subscriptions come with two standard environments:

- A **production** environment
- A **Tier-2 Standard Acceptance Test** environment

The movement of subscriptions between agreement types doesn't affect these environments. You see impact in Lifecycle Services only if the customer has additional **add-on environments**. In this case, the partner or customer doesn't need to take any explicit action related to the add-on environments. However, during the time when both CSP and Volume Licensing agreements are active or in a grace period, Lifecycle Services shows additional slots for the add-on environments. You should simply ignore these slots, as they disappear as soon as the redundant licensing agreement expires. You must **not** deploy these additional slots as this action would violate the licensing agreement.

Commercial activities

When a customer changes the licensing agreement through which they obtained their subscriptions, consider this change a *commercial transaction* that requires working with external parties who sell the subscriptions to Dynamics 365 services. The goal is to maintain uninterrupted access to the service during that time of transition.

Use the following steps to complete the commercial transaction.

1. The customer places the order for subscriptions under the new agreement with the Volume Licensing reseller or the CSP.

   > **Important**
   >
   > Make sure that you purchase the subscriptions against the same Microsoft Entra tenant that you used on the original agreement.

2. The customer activates the subscriptions.

3. In Microsoft 365 Admin Center, the customer verifies that both the new and the existing subscriptions are active for all user and device licenses, as well as for the add-on environment subscriptions.

4. When the new subscriptions are active, the customer requests that the Volume Licensing reseller or the CSP suspend the existing subscriptions. Typically, there's an overlap to help guarantee continuity and avoid disruption of service.

The customer only has default environments

If the customer has only the two standard environments that come with the Microsoft-managed subscription, and didn't purchase any add-on environments through the original CSP agreement or Volume Licensing agreement, no changes appear in Lifecycle Services.

The customer has add-on environments

If the customer purchased add-on environments through the original CSP agreement or Volume Licensing agreement, those environments continue to function without any disruption. For each add-on environment that the customer ordered on the old agreement, as well as on the new licensing agreement, they now see two slots in Lifecycle Services. Typically, the slot from the old agreement is deployed, while the slot from the new agreement shows up as a new undeployed **DYNAMICS 365 OPERATIONS SANDBOX** slot in **Configure** state. Customers must not deploy more environments in total than they purchased on the new licensing agreement, as the environment slots disappear when the old licensing agreement expires.

It's important that customers verify that they have the desired number of add-on environments on the new licensing agreement. Customers can verify the number of purchased add-on environments on the **Subscriptions available** page in Lifecycle Services, or in the Microsoft 365 admin center under **Billing** > **Your products**.

> ⓘ **Note**
>
> Lifecycle Services displays the aggregate number of subscriptions across all licensing agreements. During the transition period, while the old and new license agreement are active, the numbers are probably higher than the desired number of subscriptions after the old licensing agreement expires.

Last updated on 06/18/2026

Move Lifecycle Services implementation projects to different Microsoft Entra tenants

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

> **Note**
>
> Moving into or out of a sovereign cloud environment (for example, US Government Community Cloud [GCC] and China) isn't supported.

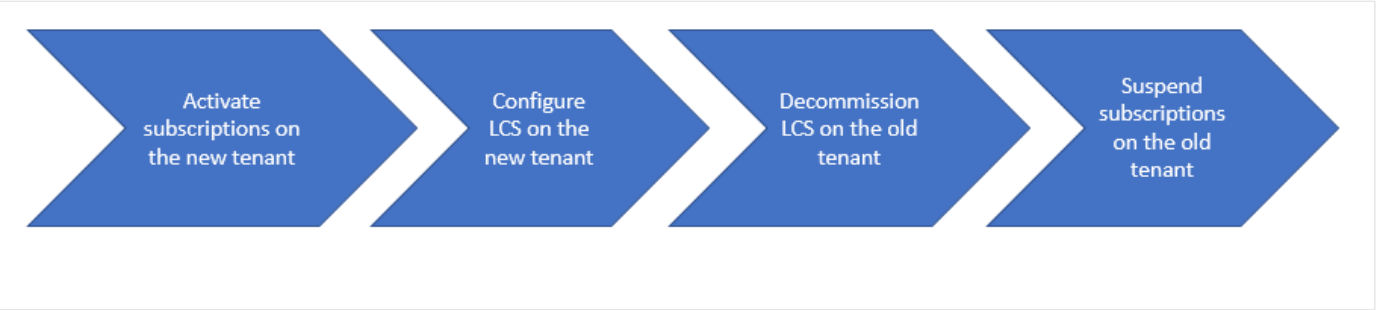
You can move your subscriptions and your Microsoft Dynamics 365 Lifecycle Services implementation project to a different Microsoft Microsoft Entra tenant. Here are some scenarios where this move might be required:

- You accidentally purchase subscriptions against the wrong Microsoft Entra tenant.

  > **Note**
  >
  > If you're a cloud service provider, and you sell subscriptions for finance and operations apps to an existing customer, you must request a reseller relationship with that customer to put the subscriptions on the customer's existing Microsoft Entra tenant. If you create a new customer record for the customer in Microsoft Partner Center, you create a new Microsoft Entra tenant for the customer.

- The customer changes the structure of the Microsoft Entra tenant after the subscription is purchased.

The process for moving your subscriptions and all related artifacts has four main steps, as shown in the following illustration.



Activate subscriptions on the new tenant

Work with your cloud service provider or volume license reseller to activate the subscriptions against the new Microsoft Entra tenant. Activate all subscriptions for users, and for add-on environments.

Cloud service provider

If you're licensed through a Microsoft Cloud Solution Provider (CSP) agreement, purchase the required subscriptions against the new tenant from your cloud service provider. If the new tenant already exists, the cloud service provider must request a reseller relationship. Alternatively, in Partner Center, the cloud service provider must create a new customer that has the desired default domain name, ***.onmicrosoft.com** (for example, **contoso.onmicrosoft.com**).

Ask the cloud service provider not to suspend the existing subscriptions at this time.

Volume Licensing

If you licensed through a Microsoft Volume Licensing agreement, call the Volume Licensing support center and ask them to remap the subscriptions from the old tenant to the new tenant. You can contact Volume Licensing Support through Microsoft 365 admin center. Request a grace period, when the subscriptions are active on both tenants. Because of customer privacy concerns, the customer must make this request. Have the following information available:

- Public customer number.
- Enrollment number.
- The current tenant domain that the subscriptions are currently provisioned on.
- The destination tenant domain that the customer wants the subscriptions provisioned under.
- A detailed explanation of why the customer must migrate its Volume Licensing subscriptions to a different tenant.

- The total number of paid subscriptions that must be moved to the new tenant, together with the subscription type and seat count.

> **Important**
>
> It's crucial that the subscriptions are active on both tenants in parallel for a few weeks, until you finish decommissioning Lifecycle Services on the old tenant.

Configure Lifecycle Services on the new tenant

On the new tenant, you get a new Lifecycle Services project that you must initiate and set up.

1. Complete the Project Onboarding wizard. For more information, see Lifecycle Services project onboarding. When completing the wizard, indicate on the **Project overview** page that you're **Moving existing Lifecycle Services project from another tenant** and provide the source Lifecycle Services project ID.
2. Fully configure Lifecycle Services. As part of this configuration, you must:
3. Upload and activate a subscription estimator. If you're already live in the source Lifecycle Services project, you need to ensure that the estimates match.
4. Add your deployable package to the asset library.
5. Update your Business process modeler (BPM) library.

> **Important**
>
> During this period, you have two parallel Lifecycle Services projects. You can verify the name and ID of the Microsoft Entra tenant that is associated with an Lifecycle Services project on the **Subscriptions available** page in Lifecycle Services.

Move your sandbox environments to the new tenant

1. Deploy the nonproduction environments in the new Lifecycle Services project.

2. Apply the required code packages to the environments. Make sure that the target is running the same application version as the source. Use All-in-one deployable packages and include any ISV licenses, if applicable.

3. Upload data to the environments. You can move the data through data packages or by restoring the database. If you restore the database, you need to remap some properties to

the new tenant.

a. This script is only required if you are performing a database restore. Before doing the export, update the placeholders with your old and new tenant details and execute the SQL command on the source environment.
   i. This takes the backup of SYSCONFIG table and updates the value of the tenant with target tenant information.
   ii. Once the export is done, you'll need to recover this table to get back the source tenant information. Use the script below:

```
  -- ============================================
   -- Script: Update AAD Tenant in SYSCONFIG
   -- Purpose: Updates the AAD tenant domain and tenant ID
   --          in the SYSCONFIG table after a tenant migration.
   -- ============================================

   DECLARE @NewTenantDomain VARCHAR(255)
   DECLARE @NewTenantId VARCHAR(255)
   DECLARE @OldTenantDomain VARCHAR(255)
   DECLARE @OldTenantId VARCHAR(255)

   SET @NewTenantDomain = '<new-tenant-domain>' -- New AAD tenant domain (e.g.,
contoso.onmicrosoft.com)
   SET @NewTenantId = '<new-tenant-id>'         -- New AAD tenant ID (GUID)
   SET @OldTenantDomain = '<old-tenant-domain>' -- Current AAD tenant domain
   SET @OldTenantId = '<old-tenant-id>'         -- Current AAD tenant ID (GUID)

   -- Back up SYSCONFIG before making changes
   PRINT 'Backing up SYSCONFIG table...'
   SELECT * INTO SYSCONFIG_BKP FROM SYSCONFIG

   -- Update tenant domain
   IF EXISTS (SELECT VALUE FROM SYSCONFIG
              WHERE CONFIGTYPE = 0 AND ID = 0 AND VALUE = @OldTenantDomain)
   BEGIN
       PRINT 'Updating tenant domain to: ' + @NewTenantDomain
       UPDATE SYSCONFIG
       SET VALUE = @NewTenantDomain
       WHERE CONFIGTYPE = 0
         AND ID = 0
         AND VALUE = @OldTenantDomain
   END
   ELSE
   BEGIN
       PRINT 'No tenant domain update required.'
   END

   -- Update tenant ID
   IF EXISTS (SELECT VALUE FROM SYSCONFIG
```

```
                WHERE CONFIGTYPE = 1 AND ID = 0 AND VALUE = @OldTenantId)
    BEGIN
        PRINT 'Updating tenant ID to: ' + @NewTenantId
        UPDATE SYSCONFIG
        SET VALUE = @NewTenantId
        WHERE CONFIGTYPE = 1
          AND ID = 0
          AND VALUE = @OldTenantId
    END
    ELSE
    BEGIN
        PRINT 'No tenant ID update required.'
    END
```

```
-- Restore SYSCONFIG to original values after export
 DELETE FROM SYSCONFIG
 INSERT INTO SYSCONFIG SELECT * FROM SYSCONFIG_BKP
 DROP TABLE SYSCONFIG_BKP
```

4. Update your user information.

5. Remove all user accounts except the admin user.

6. Fix the admin user record in USERINFO.

SQL

```
       UPDATE USERINFO
       SET SID='mysid', NETWORKALIAS='myalias/email',
NETWORKDOMAIN='https://sts.windows.net'
       WHERE ID = 'Admin'
       ```
1. Reimport all other users that have the correct security identifier (SID) and
identity provider.
1. Run the following commands to update the tenant ID in the appropriate tables. You
can verify the Microsoft Entra tenant ID that is associated with an Lifecycle Services
project on the **Subscriptions available** page in Lifecycle Services.

```sql
Update POWERBICONFIG set TENANTID = 'newtenantid' where TENANTID = 'oldtenantid'
Update PROVISIONINGMESSAGETABLE set TENANTID = 'newtenantid' where TENANTID =
'oldtenantid'
Update B2BINVITATIONCONFIG set TENANTID = 'newtenantid' where TENANTID =
'oldtenantid'
Update RETAILSHAREDPARAMETERS set TENANTID = 'newtenantid' where TENANTID =
'oldtenantid'
```

1. Fully configure the environments. As part of this step, configure the integration endpoints.

2. Perform smoke tests on the user acceptance testing (UAT) environment in the new Lifecycle Services project. These tests should focus on user sign-in, integrations, workflows, printing, reporting, and similar processes that depend on configuration and user information.

Depending on your solution and scope, you might have to perform more steps on the new Microsoft Entra tenant. These steps might include registering applications (for recurring integrations and warehouse management), adding domains, and setting up directory synchronization to enable single sign-on (SSO).

Calls to web services are allowed only from the **home** tenant for the environment. For example, the original tenant was companya.com, and integration ran as `services\@companya.com`. In this case, when you switch tenants to companyb.com, you can't use `services\@companya.com` for web service calls, even if you update **userInfo.networkdomain** to `https://sts.windows.net/companyb.com`.

> **Important**
>
> On your sandbox environments, you lose any document handling attachments that are stored in Azure Blob storage. Microsoft moves Blob storage only for production environments.

Move your production environment to the new tenant

If you don't already have a production environment deployed on the old tenant, you can skip this section.

> **Important**
>
> You can only perform tenant migration if the **source and target production environments are in the same region**. If the source environment is in a different region, you need to move the environment to the same region as the target. For more information about moving an environment, see Finance and operations apps environment migration.

> **Important**
>
> Tenant-to-tenant migration isn't supported when Customer lockbox is enabled. You must disable Customer lockbox to move an environment to another tenant. You can re-enable

Customer lockbox after the migration is completed. For more information, see [Securely access customer data by using Customer Lockbox Known Issues](#).

If you already have a production environment deployed on the old tenant, Microsoft moves your database and Azure Blob storage from your old production environment to the new one. As a prerequisite, complete the following steps after you finish moving all the sandbox environments and complete UAT. The process of moving a production environment to a new tenant requires downtime of up to 48 hours.

Before requesting the production environment, ensure that you complete all prerequisites:

1. Get all required licenses that are needed to correctly license all users on the production environment.
2. When the licenses are in place, upload a subscription estimator to the new Lifecycle Services project. It should match the subscription estimator that is active in the source Lifecycle Services project, and it must correctly reflect peak transaction volumes.
3. Send an email to Dynamics 365 FO Go-Live (d365fogl@microsoft.com) stating that your new Lifecycle Services project is ready for Microsoft to move your production database and Azure Blob Storage. To ensure that the process runs smoothly, provide the following details in the email. Copy the following list to your email, and then answer all of the information line by line.

Lifecycle Services

- Provide the Lifecycle Services IDs (number in the Lifecycle Services project URL) for source and target Lifecycle Services project.
- Confirm that the go-live date is set correctly in the target Lifecycle Services project.
- Confirm that the update schedules are set in the target Lifecycle Services project (**Lifecycle Services** > **Menu** > **Project settings** > **Update settings**).
- Confirm if you're using Azure Blob Storage for document attachments.
- Confirm that your project is identified as a tenant move in the Project Onboarding wizard.
- Confirm that your deployable package is marked as a release candidate in the target Lifecycle Services project.

  Environment region

- Confirm that the target production environment is deployed to the same region as the source production environment.

  > **ⓘ Important**
  >
  > You can only perform tenant migration if the **source and target production environments are in the same region**. If the source environment is in a different region, you need to move the environment to the same region as the target. For more information about moving an environment, see [Finance and operations apps environment migration].

  Testing

Confirm that the smoke testing is completed on the sandbox environment (Tier-2 or higher) in the target Lifecycle Services project.

Code Management

- Share which environment version you plan to deploy your new production environment.
- List the ISV solutions you're using.
- Confirm which version your old production environment is running on.
- Confirm that nonstandard code to be applied in the new production environment is exactly the same as the nonstandard code present in the old production environment to prevent database copy issues.
- Confirm if there were any non-typical actions taken on your old production environment that need to be considered on the new production environment, like installation of a custom font or environment upscale.

  Cutover and timeline

- Describe how you'll conduct your cut over.
- Confirm the dates when the source Lifecycle Services environments and project are deallocated and deleted.

1. The Dynamics 365 FO Go-Live team replies to you within two business days and a FastTrack team works with you on the assessment of the project readiness for production deployment.

2. When the tenant move assessment is successfully completed, the FastTrack team enables production environment slot in Lifecycle Services.
3. Deploy production environment in Lifecycle Services.

- You can't select the same name for the new production environment, as it's in use for your old production environment. Choose a new environment name so that a new URL is generated.
- Make sure you select the same application version that is used by your current production environment.
- In the **Production configuration** wizard, select a generic user account, not a named user, as Environment administrator.

1. After the production environment is deployed, verify that source and target environments have exactly the same code, otherwise migration fails. If necessary, install deployable packages on the target production environment.
2. Request to copy database and blob storage from the old production environment to the new production environment. Submit a support ticket requesting a copy of the database and blob storage, if applicable, from the old production environment to the new production environment. Be sure to include Lifecycle Services IDs and environment IDs from source and target projects in the support ticket.

- This process requires interaction between Microsoft and the implementing project team. Ensure that you follow the email notifications or notifications directly in the service request.
- After Microsoft completes the tenant migration activity and informs you to continue with the next steps, you need to reimport all users from the new Microsoft Entra tenant.
- You need to validate the new production environment to ensure that all business processes are successful.
- If you encounter an issue after the migration, file a support ticket.

Tear down the Lifecycle Services project on the old tenant

After the new Lifecycle Services project on the new Microsoft Entra tenant is fully functional, you must stop, deallocate, and delete the environments on the old Lifecycle Services project. When you've finished, the **Configure** button becomes available for each environment. If you already had a production environment on the old tenant, you must open a support ticket to have it deleted.

You should save any remaining artifacts from the Asset library that you might require later.

After all environments have been deleted and all artifacts saved, an Organization administrator on the old tenant must delete the Lifecycle Services project. Microsoft reserves the right to disable the customer's account and delete the customer data after the service has been suspended for an extended period.

Suspend subscriptions on the old tenant

After all the environments have been deleted, and you've saved the Lifecycle Services artifacts that you require, work with your cloud service provider or Volume Licensing Support to suspend all the licenses on the old Microsoft Entra tenant.

- **Cloud service provider** - Suspend the existing subscriptions against the old tenant.
- **Volume Licensing Support** - Call Volume Licensing Support to confirm that you've completed the work and that the subscriptions can now be suspended against the old tenant.

Last updated on 08/28/2026

Move Lifecycle Services implementation projects from on-premises to the cloud

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article explains how to move your Microsoft Dynamics 365 Finance + Operations (on-premises) environments that are hosted on your own infrastructure to the Azure cloud.

Background

Previous migrations to the cloud used SQLPackage to create and restore a .bacpac file to move the on-premises database into the cloud. That approach is no longer supported. The current approach uses the Data Migration Toolkit version 1.0.8 (or later).

Cloud subscription licenses

If you don't already have cloud subscription licenses, work with your cloud service provider or volume license reseller to get and activate the required subscriptions on your Microsoft Entra tenant. All subscriptions for users and add-on environments must be activated.

Configure Lifecycle Services cloud implementation project

If no finance and operations cloud-named user subscription licenses have previously been activated on the Microsoft Entra tenant, a new Microsoft Dynamics Lifecycle Services cloud implementation project is automatically provisioned. Otherwise, you must open a support request to have a Lifecycle Services cloud implementation project created. For more information, see Multiple LCS projects and production environments on one Microsoft Entra tenant.

After your Lifecycle Services cloud implementation project has been created, you must fully configure it. As part of this configuration, you must complete the following steps (and others):

- Add users.
- Add an Azure DevOps association.
- Add subscription estimates.
- Fill in the Asset library.
- Add Business process modeler (BPM).

Complete development and testing of updated integrations

You have to make some changes to the integration design patterns that you used for interfaces with your Finance + Operations (on-premises) environment. These changes can be substantial, and a detailed discussion of them is beyond the scope of this article. Nevertheless, you must evaluate all your interfaces and make the appropriate changes to them.

You should consider developing your updated interfaces in such a way that they can coexist in the same code base as the original interfaces. This approach simplifies management of the code lifecycle during the period of your transition from on-premises to cloud. If this approach isn't possible, you must manage a new development branch through your cloud go-live. To simplify management of this new branch during the transition period, we recommend that you freeze other code changes as much as you can. Additionally, in your detailed cut-over plan, you should carefully document the steps for inactivating your old interfaces and activating the new interfaces.

Prerequisites

1. Deploy a Tier-2 sandbox (UAT) self-service environment.

2. Apply the same code package that is applied in your on-premises production environment (or, as appropriate, in the current build from the cloud integration development branch that was discussed in the previous section). This code package

should be a single, complete deployable package that includes any independent software vendor (ISV) solutions and licenses.

> **Note**
>
> - The local business data (LBD) migration process is for finance and operations sandbox (UAT) self-service environments only. It can never be run against a production environment. A gold copy refresh is done to move the migrated data into production.
> - Make sure that you download the latest version of the Data Migration Toolkit for Dynamics 365 from Lifecycle Services.
> - Don't deploy or use the linked Power Platform environment for the migration. The Power Platform environment can be deployed and used after the data upgrade is completed.

3. Download the Data Migration Toolkit for Dynamics 365 version 1.0.8 (or later) from Lifecycle Services. In the Shared asset library, select **Model** as the asset type, and then select the model file.

4. Download and install the .NET Framework version 4.7.1 if it isn't already installed.

5. Make sure that the replication feature is installed and enabled for the source SQL Server instance. To determine whether replication is enabled, run the following SQL script.

   SQL

   ```
   -- If @installed is 0, replication must be added to the SQL Server
   installation.
   USE master;
   GO
   DECLARE @installed int;
   EXEC @installed = sys.sp_MS_replication_installed;
   SELECT @installed;
   ```

   If the replication components aren't installed, follow the steps in Install SQL Server replication to install them.

6. SQL Server authentication must be set to **SQL Server and Windows Authentication mode**. (This change requires a restart of the SQL Server service.) The toolkit uses native SQL logins only.

7. Enable and start the SQL Server Agent on the source database server.

> ⓘ Note
>
> A user should have the **DB_Owner** privilege in the source database, and should have access to the master database and the source database.

Data Migration Toolkit setup

1. Extract the zip file for the Data Migration Toolkit for Dynamics 365 version 1.0.x that you downloaded from Lifecycle Services into a folder of your choice.

   > ⓘ Note
   >
   > To ensure that the zip file is correctly extracted, select and hold (or right-click) the download, and then clear the **Unblock** checkbox.

2. In the folder where you extracted the zip file, open the **DataMigrationTool.exe.config** file, and follow these steps:

   a. Find the following line:

   ```XML
   <add key="isD365OnPrem" value="false" />
   ```

   b. Change the value from **false** to **true**, as in the example below:

   ```XML
   <add key="isD365OnPrem" value="true" />
   ```

3. Optional: If you don't want some of the source database tables to be replicated in the target database, you can specify them in the **IgnoreTables.xml** file. Likewise, if you don't want some of the functions to be replicated, you can specify them in the **IgnoreFunctions.xml** file. Additionally, if you want to put specific tables in publications outside the main publications, you can use the **SpecialTables.xml** file.

   - **Path of the IgnoreTables.xml file:** Data\IgnoreTables.xml
   - **Path of the IgnoreFunctions.xml file:** Data\IgnoreFunctions.xml
   - **Path of the SpecialTables.xml file:** Data\SpecialTables.xml

   The following examples show how to specify tables and functions in the XML files.

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<IgnoreTables>
    <Name>
        <Table>NON_AOT_TABLE1</Table>
        <Table>NON_AOT_TABLE2</Table>
        <Table>NON_AOT_TABLE3</Table>
    </Name>
</IgnoreTables>
```

> **Note**
>
> The tables that you add to the ignore list shouldn't exist in the Dynamics 365 Application Object Tree (AOT). If tables that exist in the AOT are included, an error will occur during the data upgrade. These tables won't be replicated to the target database.

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<IgnoreFunctions>
    <Name>
        <Function>if_WHSInventReserveUnionDelta</Function>
    </Name>
</IgnoreFunctions>
```

> **Important**
>
> The functions that are specified in the IgnoreFunctions.xml file won't be replicated to the target database.

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<SpecialTables>
    <Name>
        <Table></Table>
    </Name>
</SpecialTables>
```

> **Important**

Tables that are specified in the SpecialTables.xml file will be added to a dedicated publisher. The number of special table publishers is based on the **NumberOfPublishers** parameter. For more information, see the next step. Special table handling can be useful for very large tables that you might have to manually start the replication on during downtime hours.

4. Optional: To optimize the replication latency/performance, you can update the following distributor parameters in the **App.config** file:

   - **MaxBcpThreads** – By default, this parameter is set to **6**. If the machine has fewer than six cores, update the value to the number of cores. The maximum value that you can specify is **8**.
   - **NumberOfPublishers** – By default, this parameter is set to **4**. The recommendation is to use this value. However, there might be situations where you want to increase the number of publishers, to distribute smaller numbers of tables to each publisher. This change, with the manual snapshot start, lets you run smaller initial snapshots, which can be useful if you have limited maintenance windows and must split the startup of the replication over several.
   - **snapshotPostPublication** – This option will add a five-minute delay between starts of the automatic snapshot processes. This delay can help with loads on the source server. The toolkit also allows for manual snapshot starts. If you use that approach, you don't have to set this parameter.

   > ⓘ **Note**
   >
   > Don't set up or configure replication during peak times when system resource usage, memory usage, and disk IO operations are high.
   >
   > When resources are being used to the maximum extent (that is, more than 90 percent is already consumed), the replication might be delayed as the system tries to find available resources. We recommend that you start the replication during off hours, when the system resources are at minimum usage (that is, during off-peak time).
   >
   > For a go-live cutover, we recommend that you start the replication the previous weekend.

Migration process

Run the DataMigrationTool.exe application

> **ⓘ Important**
>
> Before you begin the migration process, ensure that the environment is in a **Deployed** state.

1. Run the **DataMigrationTool.exe** application.

   A console window will appear, where you can provide the cloud environment type.

   - **Public:** lcs.dynamics.com
   - **GCC:** gov.lcs.microsoftdynamics.us
   - **UAE:** [uae.lcs.dynamics.com]

2. After you enter the cloud environment, you're prompted to sign in. Provide the credentials that are used to sign in to Lifecycle Services.

3. After you're successfully authenticated, in the console window, provide the **Project-Id** value and then the **Environment-Id** value.

   To validate the given values, you must sign in by using the credentials that are used to sign in to Lifecycle Services.

   > **ⓘ Note**
   >
   > You can find the **Project-Id** and **Environment-Id** values on the **Manage environment** page in Lifecycle Services. You can also find the **Environment-Id** value on the **Environment details** page.

Complete the data replication and migration

After the validation is successful, the application presents a set of menu options that correspond to the steps in the data upgrade process. To complete the data replication and upgrade, you should perform the steps in the following order.

1. **Environment preparation: Environment setup activity**

   This step prompts you for the following information:

   - Details of the source database:
     - Source server (in the format *servername**serverinstance*)
     - Source database name
     - User name

    - Password
- IP address of the source database server (for the allowlist)
- Distribution database path (for example, **D:\SQLServer\Data**)
- Replication snapshot path (for example, **D:\SQLServer\Snapshot**)

> **Important**
>
> You must use SQL Server authentication. You can't use a domain sign-in.
>
> You must use the SQL server (NetBIOS) name, and not the fully qualified domain name (FQDN). If you have a SQL cluster, you must use the server name of the active primary node.
>
> If you have an issue connecting with the SQL Server (NetBIOS) name, ensure that the certificate used for SQL encryption has the server's name registered as an DNS entry in the Subject Alternative Names property. The SQL encryption certificate should have NetBIOS as well as FQDN entries for all SQL server nodes, and the listener if a cluster.
>
> The specified distribution database and replication snapshot paths should have enough space. We recommend that the amount of space be at least the size of the source database. If you've used compression in your Dynamics 365 database, more space will be required, because the snapshot is uncompressed. The paths should be on the local disk of the machine. (Avoid using shared paths.)
>
> We recommend that you have a static IP address for the virtual machine (VM) or machine (for the allowlist in step 1). In this way, you help prevent connection issues with the target database.

This step performs the following actions:

- It validates the connection to the source database.
- It validates the version of the Dynamics 365 LBD database.
- It authorizes the source IP address.
- It validates the target databases.

2. **Environment preparation: Prepare the target environment for the data upgrade**

   This step changes the state of the Lifecycle Services environment from **Deployed** to **Ready for replication**.

3. **Replication: Cleanup Target Database**

   This step performs the following actions:
   a. It changes the state of the Lifecycle Services environment from **Ready for replication** to **Replication in progress**.
   b. It deletes all Dynamics AX product tables, views, stored procedures, and user-defined functions in the target database.

4. **Replication: Setup Distributor**

   This step creates a distribution database under the **System Databases** folder on the source server. This distribution database is used for replication.

5. **Replication: Setup Publication for Primary Key (PK) Tables**

   This step creates publications for primary key tables under the **Replication** folder on the source server and replicates them in the target database. If any **ignore-table** entries were specified, the specified tables are exempted from replication. Any **special-table** entries that were added, these will be added to additional special tables publications.

   > ⓘ **Note**
   >
   > Snapshots must be manually started in SQL Management Studio.
   > a. Open the replication monitor, and select your SQL Server instance.
   > b. On the **Agents** tab, select and hold (or right-click) the publisher that you want to start, and select **Start**.
   >
   > Start one snapshot at a time, and wait for the replication of that snapshot to be completed. In the replication monitor, check the **Distributor to subscriber** history until you receive a message that resembles the following example: "Delivered snapshot from the \unc\server\folder."
   >
   > For more information about how to monitor the replication, see Monitor replication for the Data migration toolkit.
   >
   > You must manually start the publisher snapshots for steps 6 and 7.
   >
   > Older versions of the Data Migration Toolkit have automatic and manual starts. We recommend that you migrate to the latest version of the toolkit.

   Created publisher names: AX_PUB_PkTable_[*]

   > ⓘ **Note**

> After this replication configuration step is completed, actual data replication will occur as a SQL job that runs in the background. This job takes some time to complete. You can view the status of the replication by providing the **'rs'** option. For more information about the **'rs'** option, see the Reporting section of the application section later in this article.

6. **Replication: Setup Publication for Other Objects (functions)**

   This step creates a publication for other objects (functions) and replicates them in the target database. If you don't want some of the functions to be replicated, you can specify them in the IgnoreFunctions.xml file.

   Created publisher names: AX_PUB_OtherObjects

   > ⓘ **Note**
   >
   > The replication takes some time to be completed. You can view the replication status by providing the **'rs'** option. If there are no functions to replicate, the publication won't be created. Don't move on to the next step until the **DataReplicationStatus** property for this step is shown as completed.

7. **Cutover: Setup Publication for Non PK Tables**

   This step creates two publications: one used to replicate non-primary key tables, and the other one used to replicate locked tables.

   > ⓘ **Note**
   >
   > If there are no locked tables, then the publication won't be created.

   Publication names: AX_PUB_NoPKTable, AX_PUB_TABLE_LockedTable

   If AX Service acquires a schema lock during creation of the primary key publication, those tables are ignored from the publication. They are added to temporary tables and marked for replication during creation of the cutover publication.

   > [IMPORTANT] Don't move on to the next step until the **DataReplicationStatus** property for this step is shown as completed.

   > ⓘ **Note**

You can validate the replicated data by using the **'dv'** option. If there are mismatched tables, this step lets you create publications for them. If you want to exclude any mismatched tables for replication, close the app, and add those tables in **Data/IgnoreTables.xml**. Then rerun the app, and use the **'dv'** option.

To learn more about the **'dv'** option, see the Reporting section of the application section later in this article.

8. **Cutover: Remove replication setup**

   This step deletes all the publications that were created in the source database, the distribution database, and the replication snapshot.

   > ⓘ **Note**
   >
   > To remove the **Snapshot** folder without causing an exception, run the following script in the source database. Even if you don't run this script, you can ignore the exception message that you receive.

   ```SQL
   EXEC master.dbo.sp_configure 'show advanced options', 1
   RECONFIGURE WITH OVERRIDE
   EXEC master.dbo.sp_configure 'xp_cmdshell', 1
   RECONFIGURE WITH OVERRIDE
   ```

9. **Post-replication: Update environment state to Replicated**

   This step changes the state of the Lifecycle Services environment from **Replication in progress** to **Replication completed**.

10. **Data synchronize: Trigger transformation**

    This step triggers a servicing step in Lifecycle Services that will update the database for the cloud. When the action is successful, the state of the Lifecycle Services environment changes from **Replication completed** to **Data upgrade in progress**.

    If data migration is successful, the environment will be set to a **Deployed** status in Lifecycle Services.

11. **Data synchronize: Trigger rollback**

    This step triggers a rollback of the data upgrade. This action rolls back the data to the point before the transformation was triggered and sets the Lifecycle Services environment

state to **Replicated**. These actions will change the environment from **Failed** to the **Replicated** state.

At this point, you have only triggered the rollback. To view the rollback status, use the '**rbs**' option. To learn more about this option, see the Reporting section of the application section later in this article.

If rollback is successful, the '**rbs**' option is shown as **D365 upgrade topology(LCS) status: Replicated**.

If rollback fails, the '**rbs**' option is shown as **D365 upgrade topology(LCS) status: Failed**.

Reporting section of the application

You can use the following options to review the reports of the replication validation, replication status, data upgrade status, and rollback data upgrade status.

- **dv) Report** – Validate the replication.

  This option compares the number of tables and records in the source server database and the target server database, and then shows the report. You should use this option only after step 7 is completed.

  If there are mismatched tables, this step lets you create a publication for them. If you want to exclude any mismatched tables for replication, close the app, and add those tables in the **Data/IgnoreTables.xml** file. Then rerun the app, and use the '**dv**' option.

  You can find the report data at **output/PostValidationInfo.csv**.

- **rs) Report** – Get the replication status.

  This option shows the report of the replication process for the publications that were created. You should use this option only after step 5 is started (during the replication process for any publication).

- **rbs) Report** – Get the rollback status.

  This option shows the report of the rollback process. You should use this option only after step 11 is started.

Tooling section of the application

- **Reset-rep** – Reset the replication setup by removing all the replication configurations. Publications and the distribution database are deleted. The status of all **Replication** and

Cutover menu options is reset from **Completed** mode to **Reset** mode to help you redo the replication from the beginning.

- **Reset-all** – Reset all the menu options, and remove the replication configurations. The status of all the options is changed to **Reset**.
- **Clear** – Clear the environment setup activity. All information is cleared from the cache, such as the **project-Id** value, the **Environment-Id** value, and source database details.
- **Help** – Show the data upgrade migration options with the updated status.
- **Exit** – Close the application.
- **Set-failed** – If you want to delete the environment, and if the environment is in the **PreparingForReplication**, **ReadyForReplication**, or **Replicating & Replicated)** state, use this option to set the environment state to **Failed**. The environment can then be deleted from Lifecycle Services.

Post-migration tasks

1. If necessary, reimport all other users, and assign the appropriate security roles. Users must now be assigned to Microsoft Entra accounts.
2. Direct printing in a cloud environment is done by using the Document routing agent (DRA). Set up sandbox DRAs as described in Install the Document routing agent to enable network printing, so that regression testing can include your printing scenarios.
3. Copy document handling attachments to the cloud. Document handling attachments aren't stored in the database. If they must be preserved, you must move them separately. For more information, see the Migrate document handling attachments to your sandbox section in this article.
4. Run a complete regression test cycle. This cycle should include testing of integrations.
5. Fix any issues that are discovered during testing. For each issue, document and keep track of the correcting adjustments that you make in the sandbox, and repeat those adjustments in the on-premises source. If any change must **not** be made in the on-premises environment, because it's incompatible with the correct functioning of that environment, we recommend that you create a DMF data package for it instead of manually applying it for each iteration of the migration process.
6. Repeat steps 2 through 10 until all tests have been passed, and no further changes are being made to code or the configuration.

Repeat the migration to production

Follow all previous steps to migrate data, and then follow the instructions in Golden configuration promotion - Copy the sandbox database to production.

1. Deploy the new production environment. Note that the regular prerequisites apply. For example, you must have an active subscription estimator, complete the Lifecycle Services methodology phases before the operate phase, and complete the FastTrack readiness review. For more information, see Prepare for go-live.
2. Apply the final version of the software deployable package to production.
3. Stop making data changes to the on-premises production environment.
4. Repeat steps 3 through 6 in the Do a trial migration and resolve issues section to copy the final/up-to-date on-premises production database to the cloud sandbox.
5. Repeat step 8 in the Do a trial migration and resolve issues section to copy the final/up-to-date document handling attachments to the cloud sandbox.
6. Request a database refresh from sandbox to production. (The process is the same as the process that is used to promote a golden configuration database to production.)
7. Open a support request to have Dynamics Support Engineering copy the document handling attachments from the sandbox storage account to the production storage account and update the references in the production database's DocuValue and DocuDeletedValue tables. After the request has been completed, validate that the attachments are available for a sample of document handling records.

> ⓘ **Note**
>
> The support request for copying the document handling attachments **must be submitted 72 hours in advance** for scheduling. Requests made without advance notice cannot be honored.

8. Set up DRAs for production. If you're reusing any of the DRAs that were previously installed as part of your trial migration, remember to update their configuration so that they connect to the production URL instead of the sandbox URL.
9. Reconcile your cloud and on-premises production environments, as detailed in your cut-over plan.
10. Obtain sign-off for the go-live.
11. Activate cloud production interfaces, batch jobs, and so on.
12. Start to transact in your cloud production environment.

Migrate document handling attachments to your sandbox

Document handling attachments for Finance + Operations (on-premises) environments are stored in a file share. However, the cloud version doesn't support this file share. You can use the following procedure to copy the attachments to the Azure storage account for your sandbox environment and update the corresponding metadata in the database. For

subsequent promotion to production, you can request that Dynamics Support Engineering copy the attachments from your sandbox to production.

1. Upload a copy of the document handling attachment files from the on-premises production file share to a temporary folder on one of the sandbox instances of Application Object Server (AOS). For example, you can upload a zip file of the attachments and unpack it on the target. If you don't have remote desktop access (for example, for a self-service environment), you can use a different virtual machine (VM) instead. For reasonable conversion performance, this VM should be in the same Azure datacenter as the target sandbox. If you aren't using the AOS instance, you must add the VM to an allow list for access to the sandbox's SQL database instance.

2. Open a support request to get the name of the sandbox Azure storage account and a time-limited shared access signature (SAS) token for the documents container. Update the corresponding placeholders in the Windows PowerShell script that is run in the next step. Also update the placeholders for your temporary folder, and for your finance and operations transactional database, by using the environment details in Lifecycle Services.

3. Run the following Windows PowerShell script on the sandbox AOS instance or other VM to upload the document handling files to the storage account and create the required metadata for each file.

PowerShell

```
#Upload F&O on-prem document handling attachments to Azure storage account
#
$filesPath = "<TEMP_ATTACHMENTS_FOLDER_PATH>"
$dBHostName = "<DATABASE_SERVER>.database.windows.net"
$dBName = "<DATABASE_NAME>"
$dBUsername = "<DATABASE_USER>"
$dBPassword  = "<DATABASE_PASSWORD>"
# Note! - The storageAccountName should be set without the
".blob.core.windows.net", e.g. $storageAccountName = wxxtst4xx5bfxx
$storageAccountName = "<STORAGE_ACCOUNT_NAME>"
# Note! - The sasToken starts with "sv=20"
$sasToken = "<SAS_TOKEN>"

[Reflection.Assembly]::LoadWithPartialName("System.Security.Cryptography")
#Load crypto
$cryptoObj = [System.Security.Cryptography.SHA256]::Create()
$StorageContext = New-AzStorageContext -StorageAccountName $storageAccountName
-SasToken $sasToken
foreach ($file in Get-ChildItem $filesPath)
{
    try
    {
        $blob = (Set-AzStorageBlobContent -Context $StorageContext -Container
documents -File $file.FullName -Blob "$($file.Name)" -Force).ICloudBlob
```

```
    }
    catch
    {
        Write-Host "Could not upload $($file.Fullname) to blob"
        Write-Host $_
    }
    if($blob)
    {
        #Write-Host "Processing $($file.Fullname)..."
        #FileHash:
        $fileBytes = [System.IO.File]::ReadAllBytes($file.FullName)
        $hashBytes = $cryptoObj.ComputeHash($fileBytes)
        $encodedHash = [System.Convert]::ToBase64String($hashBytes)
        #FullFileName:
        $origFileName = (Invoke-Sqlcmd -Query "SELECT ORIGINALFILENAME FROM
DOCUVALUE WHERE FILEID = '$($file.Name)'" -ServerInstance $dBHostName -
Database $dBName -Username $dBUsername -Password $dBPassword).ORIGINALFILENAME
        if ($origFileName.Length -eq 0)
        {
            $origFileName = (Invoke-Sqlcmd -Query "SELECT ORIGINALFILENAME
FROM DOCUDELETEDVALUE WHERE FILEID = '$($file.Name)'" -ServerInstance
$dBHostName -Database $dBName -Username $dBUsername -Password
$dBPassword).ORIGINALFILENAME
        }
        if ($origFileName.Length -eq 0)
        {
            Write-Host "Missing DOCUVALUE $($file.Name)"
        }
        else
        {
            $nameBytes  = [System.Text.Encoding]::UTF8.GetBytes($origFileName)
            $encodedName =  [System.Convert]::ToBase64String($nameBytes)
            #Write-Host "Base64 encoded original filename $encodedName."
            $blob.Metadata["FileHash"] = $encodedHash
            $blob.Metadata["FileSize"] = $file.Length
            $blob.Metadata["FullFileName"] = $encodedName
            $blob.SetMetadata()
            Write-Host "Uploaded $($file.Fullname)"
        }
    }
}
```

4. In SSMS, run the following T-SQL commands to update the DocuValue and DocuDeletedValue records so that they reference the target storage location.

SQL

```
update DOCUVALUE
set ACCESSINFORMATION = replace(ACCESSINFORMATION,
'file://<SOURCE_PREFIX>/documents/',
'https://<STORAGE_ACCOUNT>.blob.core.windows.net/documents/'),
STORAGEPROVIDERID = 1
where STORAGEPROVIDERID = 4 --4 for LBD filesystem, 1 for Azure blob
```

```
and ACCESSINFORMATION like 'file://<SOURCE_PREFIX>/documents/%'

update DOCUDELETEDVALUE
set ACCESSINFORMATION = replace(ACCESSINFORMATION,
'file://<SOURCE_PREFIX>/documents/',
'https://<STORAGE_ACCOUNT>.blob.core.windows.net/documents/'),
STORAGEPROVIDERID = 1
where STORAGEPROVIDERID = 4 --4 for LBD filesystem, 1 for Azure blob
and ACCESSINFORMATION like 'file://<SOURCE_PREFIX>/documents/%'
```

5. Test a sample of the document handling attachments to make sure that they can now be accessed in the sandbox environment.

Troubleshooting

For troubleshooting information, see Troubleshoot upgrades to Dynamics 365 finance and operations self-service environments.

Learn about the replication configuration and status via SSMS

In SSMS, if Object explorer includes a **Replication** folder, the replication feature is installed on the server and available.

After step 3 of the data upgrade process is completed, you should find the publisher configured under the **Replication** folder. To learn the replication status, select and hold (or right-click) the **Replication** folder, and then select **Launch replication monitor**.

- In the replication monitor, you can view all the publishers that have been created for replication.
- On the **Snapshot** tab, you can view the status of the snapshot.
- To view the detail log/transaction, double-tap (or double-click) a grid item.
- To view the data replication to the target, on the **All subscription** tab, double-tap (or double-click) the subscription from the grid item.

Last updated on 03/06/2026

Multiple Lifecycle Services projects and environments on one Microsoft Entra tenant

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> As part of the One Dynamics One Platform work effort, some Microsoft Dynamics 365 Lifecycle Services (LCS) features have been deprecated. For more information, see Removed or deprecated platform features.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

For any new cloud project, you instantiate one Microsoft Dynamics Lifecycle Services implementation project on a Microsoft Entra tenant that provides access to one production instance. In rare cases, to handle the requirements of a specific implementation, you might require multiple production instances that run in parallel. By creating multiple Lifecycle Services projects against the same Microsoft Entra tenant, you can have multiple production instances. Here are the most common scenarios where you might need multiple production instances:

- A global implementation's requirements for data residency, latency, or data volume can't be met by one instance.

- Different business units in an organization are implementing the product separately as independent applications.

Licensing requirements

Every Lifecycle Services implementation project that runs on the same Microsoft Entra tenant must satisfy the minimum licensing requirements. For example, if there are three Lifecycle Services implementation projects on the same Microsoft Entra tenant, a customer must purchase no less than three times the minimum number of subscription licenses. Currently, the minimum license requirement is 20 full user licenses. Therefore, to run three Lifecycle Services implementation projects on the same Microsoft Entra tenant, the customer must purchase at least 60 licenses.

Because the licenses are associated with the Microsoft Entra tenant, the **Subscriptions available** page for every Lifecycle Services project shows the total number of licenses, even though a given Lifecycle Services project can use only the portion of licenses allocated to it. The allocation of licenses to Lifecycle Services projects is described in Edit the licenses allocated to projects.

For more information about licensing, download the Licensing guide.

Disadvantages of multiple Lifecycle Services projects

Here are some of the disadvantages to having multiple Lifecycle Services projects:

- Master data isn't shared.
- Intercompany transactions aren't supported.
- Integrations must be configured in each Lifecycle Services project.
- Each Lifecycle Services project requires a separate Bring your own database (BYOD) instance.
- User acceptance testing (UAT) must be done on each instance, even if the code is the same. UAT is required on each instance, because differences can occur across the Lifecycle Services projects, even if they share a code base. One source of differences can be the integration setup and BYOD configuration that must be done separately in each Lifecycle Services project and therefore must be tested in each Lifecycle Services project. There might be data variations, different application configurations per region might affect functionality, and different data centers might support a different set of Azure services.
- Microsoft Azure DevOps must be configured in each Lifecycle Services project. When customizations and code are shared, it makes sense to use the same Azure DevOps project.

Advantages of multiple Lifecycle Services projects

Having multiple Lifecycle Services projects offers several advantages. Here are some of them:

- You can select data centers for each Lifecycle Services project to provide the best latency experience.
- You can select data centers for each Lifecycle Services project to satisfy statutory requirements for data residency.
- You get more flexibility to schedule servicing operations, such as code deployments and upgrades.
- You can have different stakeholders for the different Lifecycle Services implementation projects.

Create multiple Lifecycle Services projects

Follow these steps to create another implementation project in your existing organization.

1. On the Lifecycle Services home page, select the plus sign (+) button.
2. Select the desired product for the new project.
3. Select **Implementation** as the project type.
4. Enter information about the new implementation project.
5. Select **Create** to finish creating the project. If you already have an existing implementation project in your organization, a message box might prompt you to confirm that you want to create another implementation project. If you receive an error message, your organization might not have the required total license count to support another implementation project.

The newly created implementation project is shown, and you're the owner of it.

Edit the licenses allocated to projects

Determine how you want to assign licenses across the implementation projects in your organization. Make sure that the number of licenses assigned across all projects doesn't exceed the total license count that you purchased.

After you determine the allocation, open the Subscription estimator tool for each implementation project. Edit the active subscription estimate to apply the desired license allocation for that project.

Online deployments in China sovereign cloud

If your implementation includes China deployment or rollout, see finance and operations apps - operated by 21Vianet in China. This deployment is designed to comply with regulatory requirements in China. The services include a physically separated instance of a cloud service that has a different tenant (Microsoft Entra ID) that 21Vianet operates and transacts.

In this case, a single organization exists in multiple clouds with a different tenant (Microsoft Entra ID). The previously described advantages and disadvantages of multiple Lifecycle Services projects or production environments still apply. However, the licensing requirements and request procedures differ. If you need process assistance, work with your Microsoft account executive or your implementation partners.

Last updated on 06/18/2026

Supply Chain Management troubleshooting and FAQs

Troubleshooting articles

For a large collection of troubleshooting articles, see the following site provided by Microsoft Support:

- Microsoft Dynamics 365 Supply Chain Management troubleshooting

These articles explain how to determine, diagnose, and fix issues that you might encounter when you use Supply Chain Management. The articles are organized by feature area and each article focuses on a specific issue and how to solve it.

Frequently asked questions

Collections of frequently asked questions (FAQs) are integrated into the Supply Chain Management documentation. Each of these articles includes multiple FAQs for a specific feature area.

The following FAQ articles discuss issues specific to Supply Chain Management:

- Inventory costing FAQ
- Engineering change management FAQ
- Improve master planning performance
- Improve scheduling engine performance
- Procurement FAQ
- Product-related translations FAQ
- Calculations for product configuration models FAQ
- Sales order FAQs
- Schedule sales history data cleanup
- DDMRP FAQ
- User-based authentication FAQ

The following FAQ articles discuss issues related to the platform shared by all finance and operations apps, including Supply Chain Management:

- Address books FAQ
- Client FAQ
- Electronic Invoicing FAQ
- Go-live for implementation projects FAQ

- One Version service updates FAQ
- Open in Excel experiences FAQ
- Subscriptions, LCS projects, and Microsoft Entra tenants FAQ
- Workflow FAQ

Other support resources

The following Microsoft support and community resources are also available, where you can discuss and solve issues, browse Blogs, request features, and more:

- Microsoft Dynamics 365 Blog
- Dynamics 365 Community
- Supply Chain Management ideas portal
- Dynamics 365 support

Last updated on 12/09/2025

AI summaries with Copilot

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

AI summaries with Microsoft Copilot are available on many of the most-used pages in Dynamics 365 Supply Chain Management. These summaries provide a quick overview of the most important information that's related to the page, personalized for the current user. Summaries can include information such as the number of lines on a purchase order, the number of items in a warehouse, or the number of overdue invoices for a vendor.

The information that Copilot provides depends on the current page and user context. For example, the information can vary based on the other pages that the user works with the most, and it's limited based on the user's security roles and permissions.

> **Tip**
>
> Users can provide feedback to Copilot by selecting the thumbs-up or thumbs-down buttons shown under the summaries. You can add support for written feedback by turning on enhanced user feedback for Copilot. This feedback helps improve the quality of the summaries that Copilot generates.

TechTalk video

For a video overview of all of the features described in this topic, watch Copilot Capabilities in Dynamics 365 Finance and SCM | Dynamics 365 TechTalk.

Prerequisites

Before you can use AI summaries with Copilot, your system must meet the following requirements:

- You must be running one of the following versions of Supply Chain Management:
  - Version 10.0.39 with proactive quality update 3 (PQU-3) or later
  - Version 10.0.40 with PQU-1 or later
  - Any build of version 10.0.41 or later

- Power Platform Integration must be enabled for your environment. Usually, all the components that are required to use Copilot summaries are automatically deployed when Power Platform Integration is enabled. However, if they don't work as expected, see Enable Copilot capabilities in finance and operations apps for more detailed requirements.

- To use Workload insights with Copilot in the Warehouse Management mobile app, you must be running Warehouse Management mobile app version 2.3.2.0 or later.

Turn AI summaries with Copilot on or off

Admins can control which AI summaries are shown in your system by searching for the following features in Feature management:

- Workload insights with Copilot in the Warehouse Management mobile app – *Context-aware worker summary screen in WMA*
- Product hover summaries – *Product summary when hovering on item*
- Released product details summary – *Product details summary*
- Purchase order summary – *Purchase order summary*
- Sales order summary – *Sales order summary*
- Vendor summary – *Vendor summary*

All of these features are turned on by default, but you can turn any or all of them off if you don't want to see them.

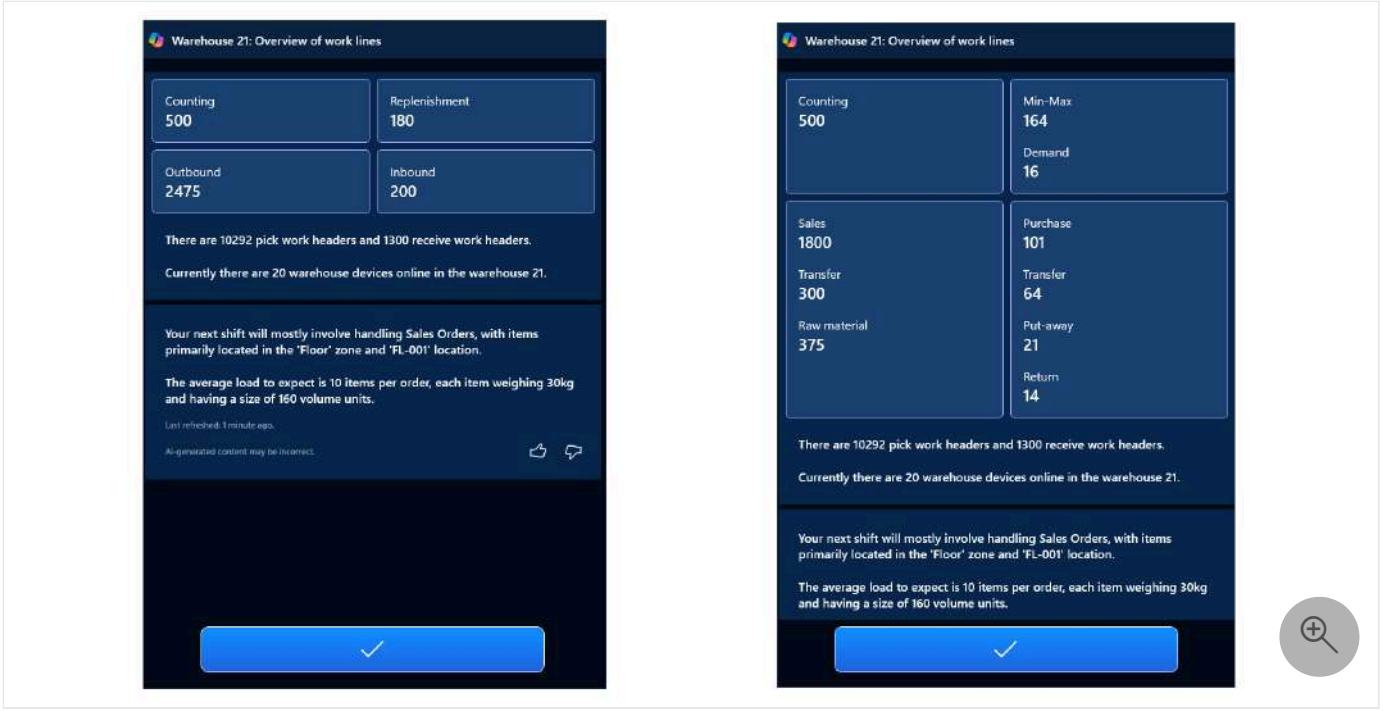
Workload insights with Copilot in the Warehouse Management mobile app

The Warehouse Management mobile app provides a *workload* page that shows work summaries and AI-generated insights to help warehouse workers better plan their shift. The information can include the following details:

- The number of pick and receive work headers or work lines.
- The number of active warehouse mobile app sessions in the warehouse.
- Insights into the available work types.
- The current workload. This information can be shown either as work headers or work lines.

- Details of the available work per work type.
- More information about the available work. Microsoft Copilot generates this information in natural language.

The following illustration shows an example of the workload page in the Warehouse Management mobile app. When workers first open the page, it shows an overview of unstarted tasks that are scheduled for the current warehouse, together with a natural-language summary that Copilot generates (left). Workers can then tap any tile to view detailed information about open work by type (right). This information helps them quickly grasp their objectives for the day.



Learn more in Workload insights with Copilot in the Warehouse Management mobile app.

Product hover summaries

On most pages that show an item number, users can hover over the item number to view a Copilot-generated summary of the item.

Released product details summary

The **Released product details** page includes a **Summary by Copilot** FastTab that shows a product summary based on the current user's most-used pages and context.

Vendor summary

The **All vendors** details page includes a **Summary by Copilot** FastTab that shows status information and insights for the selected vendor. The information that's provided can include the vendor currency, accounting currency, vendor hold status, number of purchase orders, posted vendor invoices, payment status, and more. It can also include the number of overdue vendor invoices, purchasing agreements, and active rebates agreements.

Purchase order summary

The **All purchase orders** details page includes a **Summary by Copilot** FastTab that shows an overview of the selected purchase order's status. For example, it shows the number of partially or fully confirmed, received, and invoiced lines. The FastTab also shows other selected insights, such as a summary of order lines that are overdue or nearly overdue.

Status summary for purchase orders

For purchase orders that aren't canceled, invoiced, or finalized, the **Status** section of the **Summary by Copilot** FastTab for purchase orders provides status details that apply across all order lines. The section provides insights into the progression of order lines in each of the following categories:

- **Not fully received** – Lines where the quantity hasn't been fully registered or received.
- **Fully received, not fully invoiced** – Lines where the quantity has been fully registered or received, but not yet fully invoiced.
- **Fully invoiced** – Lines where the quantity has been received and fully invoiced.

As appropriate, the **Status** section provides buttons that let you filter the order lines according to their status. Use the filters to quickly find relevant order lines so that you can take appropriate action. Each filter is shown only when at least one order line has the indicated status. The following filter buttons might be shown:

- **Not fully received** – Show only order lines where the quantity hasn't been fully registered or received.
- **Fully received, not fully invoiced** – Show only order lines where the quantity has been fully registered or received, but not yet fully invoiced.

Insight summary for purchase orders

For purchase orders that aren't canceled, received, invoiced, or finalized, the **Insights** section of the **Summary by Copilot** FastTab for purchase orders provides two more insights that apply across all order lines:

- **Overdue or nearly overdue lines** – The number of order lines that are awaiting receipt and have an expected receipt date (requested or confirmed) that's after the current date.
- **Lines without confirmed receipt dates** – The number order lines that are awaiting receipt or haven't been fully received, and don't have a confirmed receipt date.

As appropriate, the **Insights** section provides buttons that let you filter the order lines according to their status. Use the filters to quickly find relevant order lines so that you can take appropriate action. Each filter is shown only when at least one order line has the indicated status. The following filter buttons might be shown:

- **Overdue or nearly overdue lines** – Show only order lines that are overdue or nearly overdue.
- **Lines without confirmed receipt dates** – Show only order lines that are awaiting receipt or haven't been fully received, and don't have a confirmed receipt date.

Sales order summary

The **Sales order details** page includes a **Summary by Copilot** FastTab that shows an overview of the selected sales order's status. For example, it shows the number of partially or fully confirmed, received, and invoiced order lines. The FastTab also shows other selected insights, such as a summary of order lines that are overdue or nearly overdue.

Status summary for sales orders

For sales orders that aren't canceled or invoiced, the **Status** section of the **Summary by Copilot** FastTab for sales orders provides status details that apply across all order lines. The section provides insights into the progression of order lines in each of the following categories:

- **Not fully picked** – Lines where the quantity hasn't been fully picked.
- **Not fully shipped** – Lines where the quantity has been picked (or doesn't require picking) but not fully shipped.
- **Not fully invoiced** – Lines where the quantity has been completely shipped but not yet fully invoiced.

The **Status** section provides a quick overview of the progression of all sales order lines. Unlike purchase order status summaries, sales order status summaries groups order lines into categories that are forward looking. In other words, instead of focusing on what already happened, the categories focus on what is expected to happen from a fulfillment perspective.

As appropriate, the **Insights** section provides buttons that let you filter the order lines according to their status. Use the filters to quickly find relevant order lines so that you can take appropriate action. Each filter is shown only when at least one order line has the indicated status. The following filter buttons might be shown:

- **Filter to lines not fully picked** – Show only order lines that haven't been fully picked.
- **Filter to lines not fully shipped** – Show only order lines that have been picked (or don't require picking) but not fully shipped.

Insight summary for sales orders

For sales orders that aren't canceled, delivered, or invoiced, the **Insights** section of the **Summary by Copilot** FastTab for sales orders provides two more insights that apply across all order lines:

- **Overdue or nearly overdue lines** – The number of order lines that waiting to be shipped (packing slip posted) and have an expected ship date (requested or confirmed) that's after the current date.
- **Lines without confirmed ship dates** – The number of order lines that aren't fully shipped (packing slip posted) and don't have a confirmed ship date.

As appropriate, the **Insights** section provides buttons that let you filter the order lines according to their status. Use the filters to quickly find relevant order lines so that you can take appropriate action. Each filter is shown only when at least one order line has the indicated status. The following filter buttons might be shown:

- **Overdue or nearly overdue lines** – Show only order lines that are overdue or nearly overdue.
- **Lines without confirmed ship dates** – Show only order lines that aren't fully shipped (packing slip posted) and don't have a confirmed ship date.

Traceability summaries

The Traceability Add-in for Dynamics 365 Supply Chain Management lets you collect business and industry-specific data across the product value chain, which can be used for regulation audits and compliance process. The Traceability app in Power Apps provides two types of Copilot-generated summaries: activity summaries and where-used summaries.

Traceability activity summaries

Traceability collects activity events across the supply chain and product lifecycle, but it isn't always easy for users to get a quick picture of a tracked object's history. Copilot analyzes the many activity events associated with a selected tracked object and then provides a natural-language summary that helps you quickly understand the object's history. This summary is shown at the top of the full list of activity events.

Traceability where-used summaries

Traceability provides a comprehensive where-used insight report, which lets you track the usage of items across various processes. Copilot offers detailed where-used summaries at both the item-number and quantity levels, allowing for precise analysis.

Related information

- Workload insights with Copilot in the Warehouse Management mobile app
- Responsible AI FAQ for AI summaries with Copilot
- Responsible AI FAQ for Workload insights with Copilot

Last updated on 05/07/2026

Globalization Studio overview

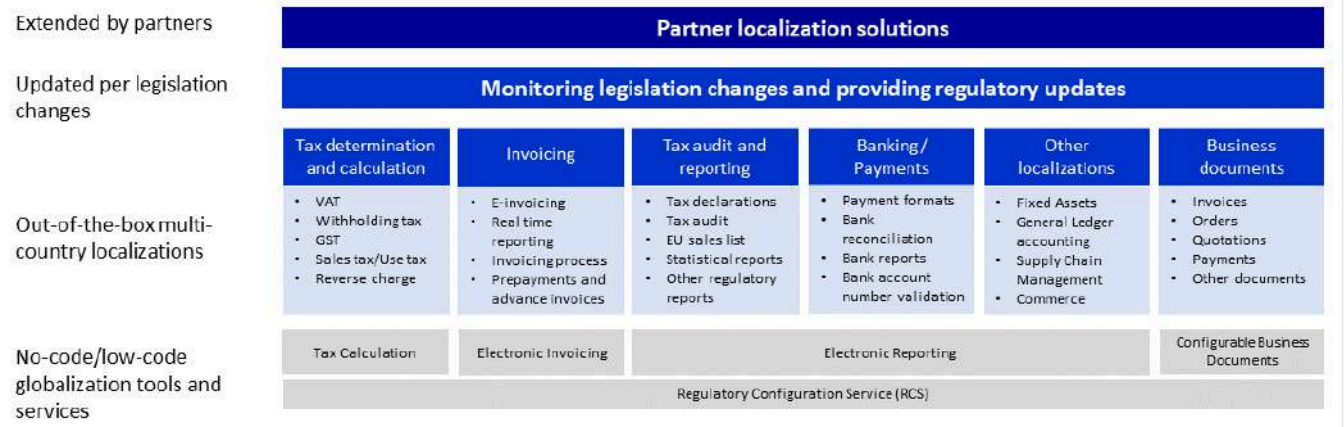
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To run their businesses globally, customers must meet multiple country/region-specific requirements for tax compliance and local business practices. This concept is referred to as *localization*. The requirements are complex and are frequently changed by local governments. They often have very tight law enforcement dates.

The Globalization Studio addresses these challenges by providing the following features:

- **No-code/low-code globalization tools and services** – These flexible, easy-to-use tools and services make localization easier for Microsoft. They also make it easier for partners and customers to create, extend, automate, and maintain the product.
- **Out-of-the-box multi-country/region localization content** – This Microsoft-provided content is continuously monitored and updated by Microsoft, and can be extended by partners.

Together, these features let you operate the solution in over 200 countries and regions, and help you meet multiple requirements for tax compliance and local business practices.



The Globalization Studio consists of the following tools and services:

- Regulatory Configuration Service (RCS) is a standalone designer and lifecycle management service for no-code and low-code globalization functionality.
- Electronic Reporting (ER) is a configurable tool that helps create and maintain regulatory electronic reporting, payment formats, and other features in a no-code or low-code format.

- Configurable business documents are built on top of the ER framework and enable business users to edit business document templates by using a Microsoft Dynamics 365 service or appropriate Microsoft Office desktop application.
- Tax Calculation is a hyper-scalable multitenant service that significantly enhances the core tax determination and calculation capabilities of Dynamics 365 finance and operations apps with a more flexible and fully configurable engine.
- Electronic Invoicing is a hyper-scalable multitenant service that extends the existing electronic invoicing capabilities to more countries and regions, and enables configurable processing of electronic invoices and configurable electronic document exchange.
- Out-of-the-box multi-country/region localizations and regulatory updates are provided by Microsoft and updated per legislation changes in the supported countries and regions.
- Partner localization solutions that partners publish on Microsoft Marketplace or provide directly.

Last updated on 03/21/2026

Globalization resources

Summarize this article for me

Local and regional deployments

If government regulations require you to store or service data differently, consider country/region requirements during deployment. Consider the following resources that might be relevant to you:

Finance and operations apps operated by 21Vianet in China

Localization and regulatory features

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the **Request access to Finance and Operations Viva Engage Community** form and choose the community you want to join.

Finance and operations apps include functionality for the countries and regions documented in the Product localization and translation availability guide. Enable this functionality based on the primary address of the active legal entity.

This article includes lists of resources that can help you:

- Learn more about developing country/region-specific solutions.
- Get country/region specific updates.
- Submit and review regulatory alerts.
- Learn how to use country/region specific functionality.

Developing localized solutions

The following resources provide guidance and information that can help developers and ISVs who are creating country/region-specific customizations or are creating a solution for a country/region that Microsoft doesn't support.

- Separation of localization models
- Apply country/region context
- Regulatory certification information in feature titles

- Classification of localization features
- Country/region Codes - ISO 3166

Regulatory updates and communication

The following resources provide information about planned and new localization features.

Regulatory updates

- Regulatory updates
- Issue search in Lifecycle Services (Updated daily)

Communication and alerts

- Regulatory watch and communication of regulatory updates
- Submit alerts about country/region-specific regulatory features

Dynamics 365 release plans

The Dynamics 365 release plans provide descriptions of new and enhanced capabilities that are planned for Dynamics 365 business applications and application platforms.

Finance and operations apps what's new

The What's new or changed in finance and operations home page lists the features that are included in specific releases of the finance and operations apps.

Electronic reporting

By using the Electronic reporting (ER) tool, you can configure formats for electronic documents that meet the legal requirements of various countries or regions. ER lets you manage these formats throughout their lifecycle. For more information, see one of the following topics:

- Electronic reporting (ER) overview
- Manage the Electronic reporting (ER) configuration lifecycle
- Create Electronic reporting (ER) configurations
- Extend the list of Electronic reporting (ER) functions
- Electronic reporting (ER) destinations
- Download Electronic reporting configurations from Lifecycle Services
- Import Electronic reporting (ER) configurations

- Configure Electronic reporting (ER) to pull data into Power BI
- Generate electronic documents and update application data by using ER

Task guides

You can access task guides from the product help pane. These guides provide a guided walk-through of key business processes. You can open a task guide to read the steps of a business process, or you can play a task guide to walk through a business process and enter data.

To find task guides, go to a page in the application and select **Help**. The help pane lists task guides that use the page. You can also use the help pane to search for task guides by title.

To learn more, see Help system.

Country/region specific help content

- Australia
- Austria
- Bahrain
- Belgium
- Bolivia
- Brazil
- Canada
- Chile
- China
- Colombia
- Costa Rica
- The Czech Republic
- Denmark
- Dominican Republic
- Ecuador
- Egypt
- Estonia
- Europe
- Finland
- France
- Germany
- Guatemala
- Hungary
- Iceland
- India

- Indonesia
- Italy
- Japan
- Kuwait
- Latin America (except Brazil and Mexico)
- Latvia
- Lithuania
- Malaysia
- Mexico
- Netherlands
- New Zealand
- Nicaragua
- Norway
- Oman
- Panama
- Paraguay
- Peru
- Poland
- Qatar
- Russia
- Saudi Arabia
- Singapore
- Spain
- Sweden
- Switzerland
- Thailand
- Türkiye
- United Kingdom
- United States
- Uruguay
- Venezuela

Last updated on 03/11/2026

Regulatory updates

This article lists the regulatory updates that are planned and released in Dynamics 365 Finance supported localizations. Delivery timelines might change, and projected functionality might be different or might not be released. For more information, see Microsoft policy.

Regulatory updates are features that support new or changed country/region-specific legislation. For more information about planned and released country/region-specific features, see Dynamics 365 and Power Platform release plans.

Microsoft strives to implement new regulatory requirements as early as possible. The actual delivery date depends on the law announcement date, availability of the requirement details from the local authorities, the availability of the validation tools, and on the size and complexity of the change.

Regulatory updates are planned for delivery in One Version service updates that are released in time for customers to update and be ready for the enforcement date (for transactional regulatory updates), or for the first mandatory reporting deadline (for regulatory updates related to reporting). Customers and partners can preview the new regulatory updates in the preview package provided for each service update.

If there are late announcement dates, late availability of requirement details or validation tools, or exceptionally large and complex changes, it might not be possible to make a regulatory update available by the General Availability date of a monthly update. In these cases, the regulatory update is shipped as hot fixes for relevant available monthly updates.

Regulatory updates that are released as part of the monthly updates are indicated by release version only. Regulatory updates that are delivered either as hot fixes or as part of a release preview can be identified through the abbreviations HF and Preview, respectively.

For the latest regulatory update plans, refer to the following table.

Expand table

Country/region	Release date	Release version	Regulatory update
Norway	August 2026	10.0.50, also available in prior versions	Regulatory update to support change to the authorization scopes used for VAT Return Submission API integrations from September 2026. Register an

Country/region	Release date	Release version	Regulatory update
			integration point in the ID-porten web portal; Prepare your environment to interoperate with ID-porten and Altinn web services.
Austria	August 2026	10.0.50, also available in prior versions	Regulatory update to support changes in VAT declaration format for Austria from July 2026. VAT declaration (Austria).
Hungary	July 2026	10.0.50, also available in prior versions	Regulatory update to support changes in VAT declaration format for Hungary from July 2026. VAT declaration (Hungary).
Lithuania	June 2026	10.0.49, also available in prior versions	Updated Lithuanian Intrastat.
France	June 2026	10.0.49, also available in prior versions	Released Electronic reporting of transactions for France (e-Reporting).
Switzerland	June 2026	10.0.49, also available in prior versions	Updated VAT declaration for Switzerland - Version 2.0.0.
Netherlands	June 2026	10.0.49, also available in prior versions	Updated EU sales list for Netherlands - Taxonomy NT20.
Malaysia	April 2026	10.0.48, also available in prior versions	Updated SST-02 declaration for Malaysia - Sections 13d, 18c (points 1, 2, 3), and subsequent sections (18d, 18e).
Brazil	April 2026	10.0.47, also available in prior versions	Regulatory update to extend NFS-e support for project-based invoicing scenarios, enabling compliant XML generation for São Paulo and Federal NFS-e formats. Includes NBS field, service location attributes, tax estimation percentages, and updates to ER configurations for fiscal document mapping.
Italy	March 2026	10.0.48, also available in prior versions	Updated Unique certification (Certificazione Unica) form aligned with the provision of January 15, 2026 (Prot. n. 9454/2026) for tax year 2025.
Lithuania	March 2026	10.0.48, also available in prior versions	New VMI VAT codes PVM58, PVM59, and PVM60 for i.SAF reporting, effective from January 1, 2026, aligned with the updated 12% VAT rate classification.

Country/region	Release date	Release version	Regulatory update
Lithuania	March 2026	10.0.48, also available in prior versions	Updated VAT declaration (FR0600) with a new field 29A to support the 12% VAT rate, effective from January 1, 2026.
Spain	March 2026	10.0.48, also available in prior versions	Updated Declaration 347 annual informative return for fiscal year 2025.
Spain	February 2026	10.0.48, also available in prior versions	Updated VAT declaration Model 303 registry design for fiscal year 2026.
Germany	January 2026	10.0.48, also available in prior versions	Support for generating the EU Sales list for Germany in XML file format.
Netherlands	December 2025	10.0.47, also available in prior versions	Regulatory update to support version 4.0 of Audit file (XML Auditfile Financieel, XAF).
Poland	December 2025	10.0.47, also available in prior versions	Regulatory update to support version (3) of VAT declaration for Poland, JPK-V7, effective from February 1, 2026.
United Kingdom	November 2025	10.0.47, also available in prior versions	Regulatory update to support Statistics on payment practices report the Reporting on Payment Practices and Performance (Amendment) Regulations 2025.
Germany	November 2025	10.0.47, also available in prior versions	EU Sales list for Germany in CSV format. EU Sales list for Germany
Indonesia	August 2025	10.0.46, also available in prior versions	Regulatory update to support generating the e-Bupot file (Withholding tax report for Indonesia) for reporting periods starting from 2025 based on the withholding tax transactions.
Indonesia	August 2025	10.0.46, also available in prior versions	Regulatory update to support generating the SPT Masa PPN (Pajak Pertambahan Nilai) VAT declaration in Excel format for reporting periods starting from 2025.
Estonia	July 2025	10.0.46, also available in prior versions	Regulatory update to support changes in VAT declaration format for Estonia 2025. VAT declaration for Estonia
Australia	May 2025	10.0.45, 10.0.44HF,	Regulatory update to support changes in Payment times reporting schema (PTRS) required for Australian

Country/region	Release date	Release version	Regulatory update
		10.0.43HF, 10.0.42HF	legal entities for periods from July 1, 2024. Payment times reporting schema
Austria	April 2025	10.0.45, also available in prior versions	Regulatory update to support changes in EU Sales list for Austria - summary report (ZM). EU sales list for Austria
Czech Republic	May 2025	10.0.45, also available in prior versions	Regulatory update to support changes in VAT declaration format for Czech Republic 2025. VAT declaration - Czech Republic
Hungary	April 2025	10.0.45, also available in prior versions	Regulatory update to support changes in VAT declaration format for Hungary 2024-2025. VAT declaration (Hungary).
Mexico	March 2025	10.0.44, 10.0.43HF, 10.0.42HF, 10.0.41HF	DIOT declaration statement regulatory update to support new rules for DIOT 2025. DIOT declaration statement
Italy	February 2025	10.0.44, 10.0.43HF, 10.0.42HF, 10.0.41HF	Unique certification - 2025
Poland	November 2024	10.0.40HF, 10.0.41HF, 10.0.42HF, 10.0.43	SAF Accounting Books Income Tax - JPK_KR_PD
Japan	September 2024	10.0.39HF, 10.0.40HF, 10.0.41HF, 10.0.42	Japan consumption Tax return report layout change - 2024 for periods after October 1, 5th year of Reiwa. Japan consumption tax report
Norway	August 2024	10.0.39HF, 10.0.40HF, 10.0.41HF, 10.0.42	Standard Audit File for Tax (SAF-T) for Norway version 1.30 required from January 1, 2025.
France	March 2024	10.0.40	VAT declaration - 2024
Japan	March 2024	10.0.37HF, 10.0.38HF, 10.0.39HF, 10.0.40	Japan consumption Tax return report layout change - 2024. Japan consumption tax report

Country/region	Release date	Release version	Regulatory update
Italy	February 2024	10.0.40	Unique certification - 2024
Spain	February 2024	10.0.40	VAT declaration - Modelo 303 - 2023. VAT declaration
Estonia	January 2024	10.0.36HF, 10.0.37HF, 10.0.38HF, 10.0.39	Estonia - VAT return form, structure and specifications change 2024. VAT declaration for Estonia
Denmark	August 2023	10.0.37	Accounting basis (Regnskab Basis) electronic report for Denmark
Switzerland	July 2023	10.0.36	VAT declaration - 2024
Poland	July 2023	10.0.36	SAF VAT Invoices - JPK-FA project advance invoice handling
Poland	July 2023	10.0.36	VAT declaration - JPK-V7 - supplementary package based on ideas
United Arab Emirates	June 2023	10.0.36	FTA VAT Audit File (AE) redesign to "SAF-T General model mapping" and SAF-T feature
France	June 2023	10.0.36	VAT declaration - 2023
Italy	June 2023	10.0.35HF, 10.0.36	Modello 770 - 2023
Denmark	May 2023	10.0.35	Standard Audit File for Tax (SAF-T) for Denmark
Germany	March 2023	10.0.31HF, 10.0.32HF, 10.0.33HF, 10.0.34HF, 10.0.35	VAT declaration - 2023
Italy	March 2023	10.0.31HF, 10.0.32HF, 10.0.33HF, 10.0.34	Yearly VAT communication - Dichiarazione Iva annuale - 2023
Italy	February 2023	10.0.30HF, 10.0.31HF, 10.0.32HF, 10.0.33	Unique certification - 2023

Country/region	Release date	Release version	Regulatory update
Brazil	February 2023	10.0.33HF	Record E115 of EFD ICMS IPI for SC becoming mandatory as of May 1, 2023
Brazil	March 2023	10.0.32HF, 10.0.33HF, 10.0.34HF	Exclusion of ICMS from the PIS/COFINS CREDITS calculation base (MP 1.159/23, IN 2121/22)-sales tax
Brazil	March 2023	10.0.32HF, 10.0.33HF, 10.0.34HF	Exclusion of ICMS from the PIS/COFINS CREDITS calculation base (MP 1.159/23, IN 2121/22)-withholding tax
Denmark	November 2022	10.0.32	VAT declaration preview report update for 2022
Italy	October 2022	10.0.32	The new type of TD28 document to communicate the purchase of goods from San Marino
Italy	February 2023	10.0.34	Changes in the SEPA Payment tracings - Urgent Wire Transfers and Foreign Payments - Address and BIC fields
Japan	March 2023	10.0.34	New Qualified Invoice System requirements in Japan from October 1, 2023
Poland	April 2023	10.0.34	Updates Payment terms in commercial transactions report (PL-00053) in accordance to D20222414L
Saudi Arabia	November 2022	10.0.32	Retail - Electronic invoicing in Saudi Arabia - Phase 2

Additional resources

- For more information about all planned and released country or region-specific regulatory updates, see Search for country/region-specific regulatory updates. (Sign-in is required.)
- For a list of the supported localizations, see the International availability guide.

Last updated on 08/19/2026

Search for country/region-specific regulatory updates

This article provides links to planned and released country/region-specific regulatory updates for Microsoft Dynamics 365 Finance. You must sign in to Microsoft Dynamics Lifecycle Services (LCS) to access this information.

Expand table

Country or region	Issue in LCS Issue search
Australia	Planned and released regulatory features for Australia
Austria	Planned and released regulatory features for Austria
Bahrain	Planned and released regulatory features for Bahrain
Belgium	Planned and released regulatory features for Belgium
Brazil	Planned and released regulatory features for Brazil
China	Planned and released regulatory features for China
Czech Republic	Planned and released regulatory features for Czech Republic
Denmark	Planned and released regulatory features for Denmark
Egypt	Planned and released regulatory features for Egypt
Estonia	Planned and released regulatory features for Estonia
Finland	Planned and released regulatory features for Finland
France	Planned and released regulatory features for France
Germany	Planned and released regulatory features for Germany
Hong Kong SAR	Planned and released regulatory features for Hong Kong SAR
Hungary	Planned and released regulatory features for Hungary
Iceland	Planned and released regulatory features for Iceland
India	Planned and released regulatory features for India
Ireland	Planned and released regulatory features for Ireland

Country or region	Issue in LCS Issue search
Italy	Planned and released regulatory features for Italy
Japan	Planned and released regulatory features for Japan
Kuwait	Planned and released regulatory features for Kuwait
Latvia	Planned and released regulatory features for Latvia
Lithuania	Planned and released regulatory features for Lithuania
Malaysia	Planned and released regulatory features for Malaysia
Mexico	Planned and released regulatory features for Mexico
Netherlands	Planned and released regulatory features for the Netherlands
New Zealand	Planned and released regulatory features for New Zealand
Norway	Planned and released regulatory features for Norway
Oman	Planned and released regulatory features for Oman
Poland	Planned and released regulatory features for Poland
Qatar	Planned and released regulatory features for Qatar
Russia	Planned and released regulatory features for Russia
Saudi Arabia	Planned and released regulatory features for Saudi Arabia
Singapore	Planned and released regulatory features for Singapore
South Africa	Planned and released regulatory features for South Africa
Spain	Planned and released regulatory features for Spain
Sweden	Planned and released regulatory features for Sweden
Switzerland	Planned and released regulatory features for Switzerland
Thailand	Planned and released regulatory features for Thailand
United Arab Emirates	Planned and released regulatory features for the United Arab Emirates
United Kingdom	Planned and released regulatory features for the United Kingdom

For more information about the globalization features that are included in monthly updates, join the DGX - GLOBALIZATION FEATURES community on Microsoft Viva Engage.

Last updated on 03/21/2026

Globalization Studio workspace features

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Regulatory Configuration Service (RCS) will be deprecated. All new RCS provisioning is stopped as of 10.0.39 GA. If provisioning is required, please register a support ticket. We will be providing tools and required support for migration from RCS to the **Globalization Studio** workspace. We plan to fully shut down RCS by August 1, 2024. For more information about migration to **Globalization Studio** workspace, see Regulatory Configuration Service merge to the Globalization Studio workspace

By using Globalization Studio, you can create Globalization features for use in the following Globalization services: Electronic Invoicing and Tax Calculation. You can:

- Define related components of a feature.
- Manage the feature lifecycle through a feature's status.

To use a Globalization feature, you must first import it from a Dataverse repository and create your own version of it. You can add Globalization features by:

- Adding a derived feature based on an existing feature that's published or shared.
- Adding a new feature that you created from scratch.

Feature component overview

Globalization features include several components:

- **Version** – This component supports feature lifecycle management. It lets you manage the status of different versions of the feature.
- **Configurations** – This component lets you manage, view, and edit related Electronic reporting (ER) formats and format mappings.
- **Setups** – This component lets users of Globalization services, such as an e-invoicing service, manage the setup of the related feature version. Therefore, it supports the flexible construction of communication and responses rules.

Last updated on 03/21/2026

Import Electronic reporting (ER) configurations from Dataverse

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

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This article explains how to import Electronic reporting (ER) configurations from Microsoft Dataverse.

Set up integration with Dataverse

To use this functionality, you need a Dataverse environment that's connected to your Dynamics 365 Finance environment. For more information, see the following articles:

- Enable Power Platform Integration
- Connect finance and operations apps with a new Microsoft Dataverse instance
- Connect finance and operations apps with an existing Microsoft Dataverse instance

You should also add a security role to access tables in Dataverse by following these steps:

1. Create a new security role as described in Create or edit a security role.
2. In that security role, add **Create, Read, Write, Delete, Append, Append to, Assign, Share** permissions for **Organization** to the following tables: **Electronic Reporting Configuration**

File, Electronic Reporting Configurations Index File, Globalization Feature File, and Globalization Features Index File.

3. Assign the created security role to users as described in Assign security roles.

Import the Globalization solution

This functionality uses the Microsoft Dataverse Web API.

Import the Globalization solution by following these steps:

1. Go to Globalization Solution for Microsoft Dynamics 365 Finance on Marketplace.
2. Select **Get it now**.
3. Fill in the required data.
4. Select an environment, and then select **Install**.

Import configurations from Dataverse

This solution adds a new type of repository for getting ER configurations in Dynamics 365 Finance. This repository is known as the Dataverse configuration repository. It provides the same user interface (UI) that Dataverse provides. Therefore, you can import single and filtered configurations.

> **Note**
>
> To use the Dataverse configuration repository, make sure that the **Dataverse repository** feature is enabled in the **Feature management** workspace of the Dynamics 365 Finance app. The feature is enabled by default since the service update 10.0.39. However, it might be disabled in your environment.

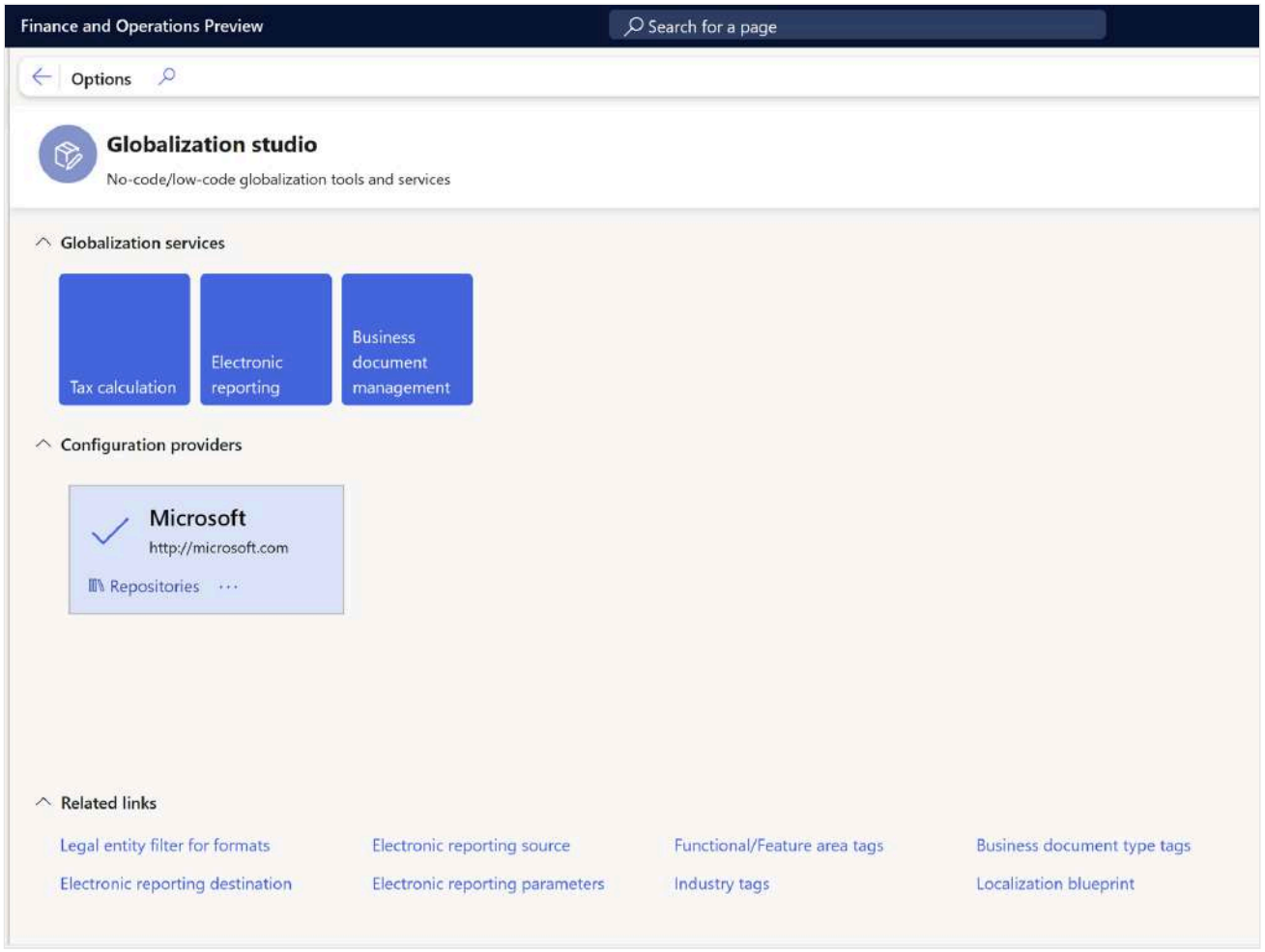
Open the Dataverse configuration repository

To open the Dataverse configuration repository, follow these steps:

1. Sign in to the Dynamics 365 Finance app by using one of the following roles:
   - Electronic reporting developer
   - Electronic reporting functional consultant
   - System administrator
2. Go to **Organization administration** > **Workspaces** > **Globalization studio**.

3. In the **Configuration providers** section, select the **Microsoft** tile.

4. On the **Microsoft** tile, select **Repositories**.



5. On the **Configuration repositories** page, in the grid, select the existing repository of the **Dataverse** type. If this repository doesn't appear in the grid, follow these steps:
   a. Select **Add** to add a repository.
   b. Select **Dataverse** as the repository type, and then select **Create repository**.
   c. If you're prompted for authorization, follow the on-screen instructions.
   d. Enter a name and description for the repository, and then select **OK** to confirm the new repository entry.
   e. In the grid, select the new repository of the **Dataverse** type.

6. Select **Open** to view the list of ER configurations for the selected repository.

Import a single configuration

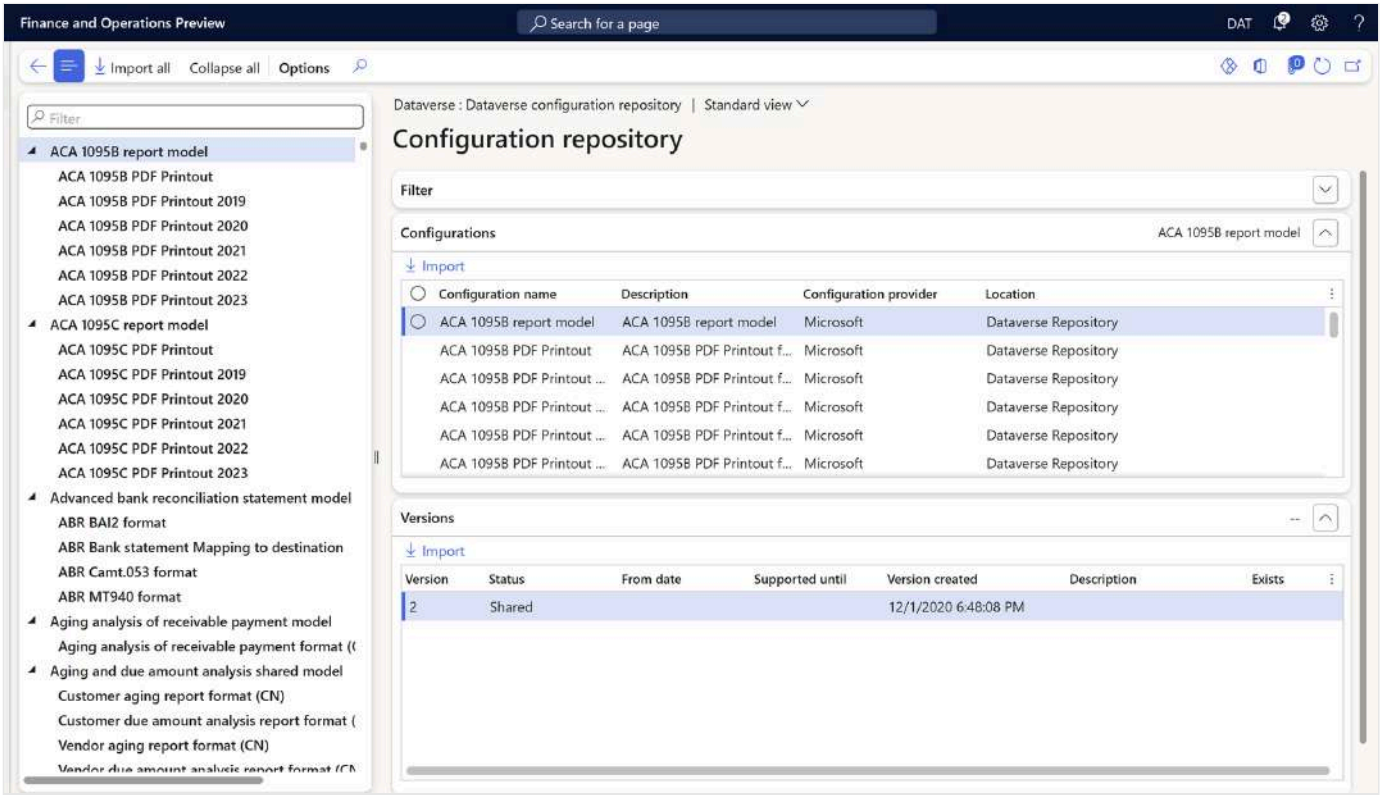
> ⓘ **Note**

It can take up to two weeks for a new ER configuration released by Microsoft or a new version of an ER configuration released by Microsoft to appear in the Dataverse configuration repository. If you expect a configuration update but don't see it in the Dataverse configuration repository, or if auto-updates are disabled in your environment, you can manually trigger the update of the Globalization solution package in your Power Platform admin center portal. To learn more about updating solutions in Power Platform admin center, see Manage Dynamics 365 apps.

To import a single configuration, follow these steps:

1. On **Configuration repository**, in the configurations tree, select the ER configuration that you want.
2. On the **Versions** FastTab, select the required version of the selected ER configuration.
3. Select **Import** to download the selected version from the Dataverse to the current Dynamics 365 Finance instance.

> **Note**
>
> The **Import** button is unavailable for ER configuration versions that are already present in the current Dynamics 365 Finance instance.



Review configuration prerequisites before import

Starting in version 10.0.47 of Dynamics 365 Finance apps, before you import an ER configuration version, you can review its prerequisites and verify that your environment meets all requirements. This process helps you identify potential issues, such as missing dependent configurations or an incompatible application version, before you start the import.

View prerequisites for a configuration version

To view the prerequisites for a specific configuration version, follow these steps:

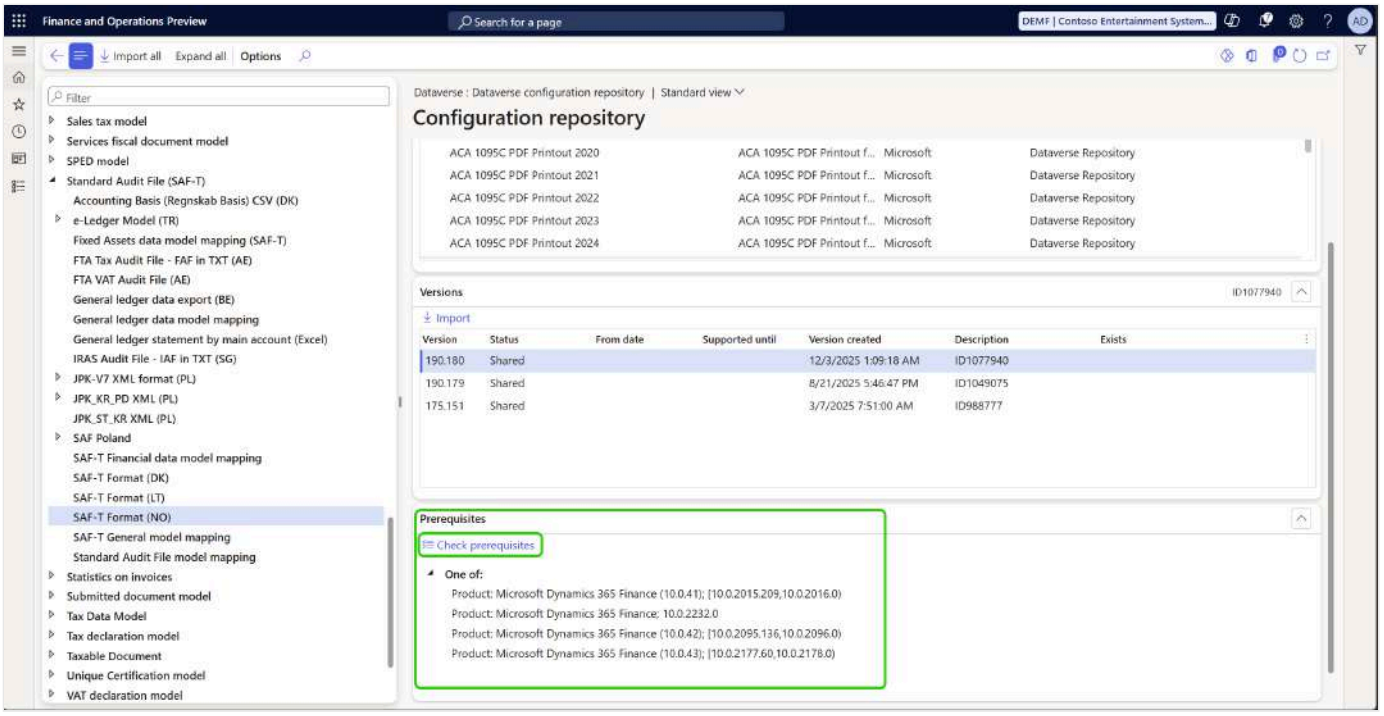
1. On **Configuration repository**, in the configurations tree, select the ER configuration that you want.

2. On the **Versions** FastTab, select the required version of the selected ER configuration.

3. On the **Prerequisites** FastTab, review the list of prerequisites for the selected configuration version.

   The **Prerequisites** FastTab shows the following types of requirements:

   - **Configuration** – Other ER configurations that must be present in the current Dynamics 365 Finance instance before you can import the selected version. Prerequisites might be organized into groups:
     - **All of** – All configurations in the group must be available.
     - **One of** – At least one configuration in the group must be available.
   - **Product** – The minimum Dynamics 365 Finance application version that is required.



Check prerequisites before import

To proactively validate that your environment meets all prerequisites for a configuration version, follow these steps:

1. On **Configuration repository**, in the configurations tree, select the ER configuration that you want.

2. On the **Versions** FastTab, select the required version of the selected ER configuration.

3. Select **Check prerequisites**.

   The system validates the following requirements:
   a. The required Dynamics 365 Finance application version is installed.
   b. All required ER configuration dependencies are available in the current instance.

   If any prerequisite isn't met, the validation results show the specific gaps and provide guidance about how to resolve them.

> **Tip**
>
> By checking prerequisites before you import, you can avoid failed imports and reduce troubleshooting time, especially in environments where configuration updates are applied regularly.

Error messages for failed imports

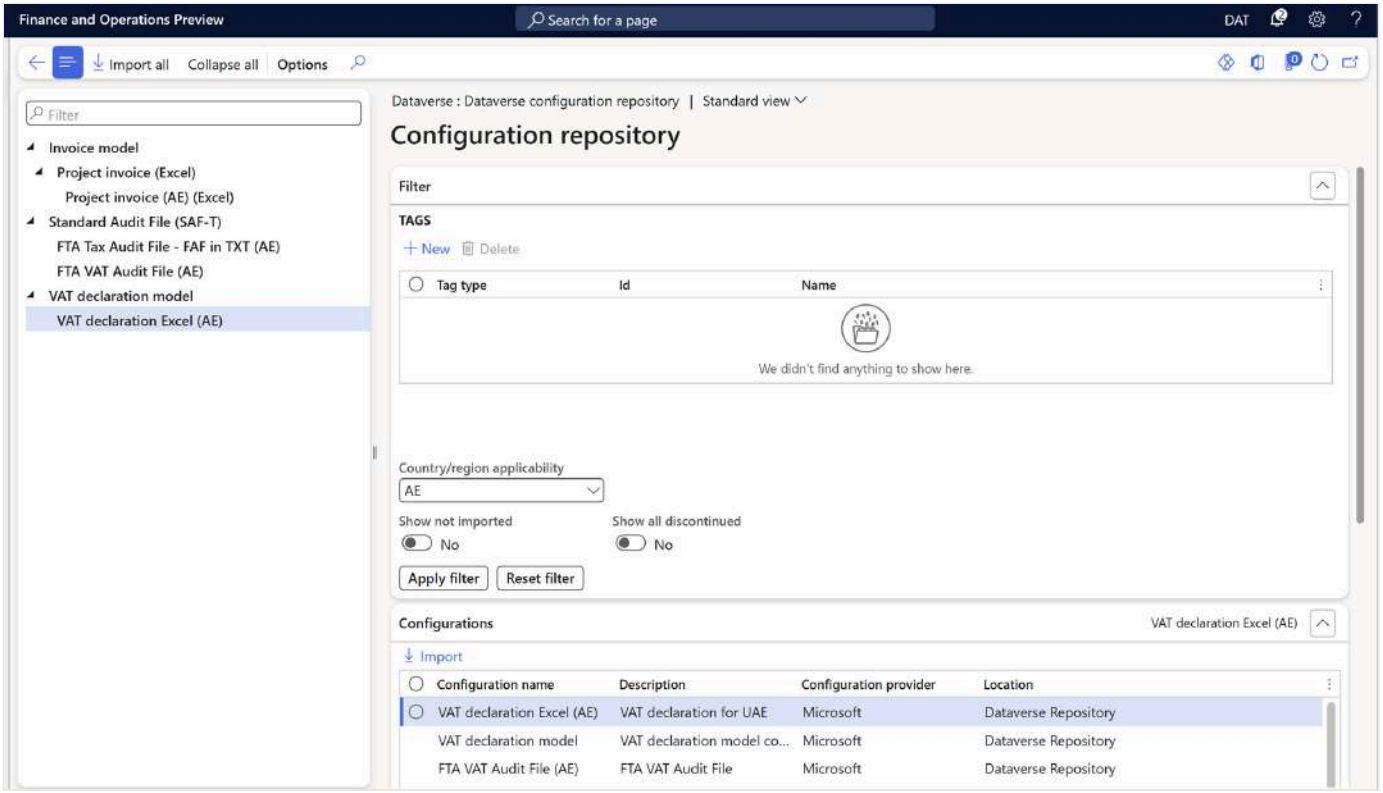
If you import a configuration version that has unmet prerequisites, the system provides an error message that explicitly identifies the missing prerequisites. The error message includes:

- The name and version of the missing prerequisite configuration.
- The required application version or platform build, if applicable.

Import filtered configurations

To import filtered configurations, follow these steps:

1. On **Configuration repository**, on the **Filter** FastTab, in the **Tags** grid, add any needed tags.
2. In the **Country/region applicability** field, select the appropriate country/region codes.
3. Select **Apply filter**.
4. The **Configurations** FastTab shows all the configurations that satisfy the specified selection conditions. Select **Import** to download the filtered configurations from Dataverse to the current Dynamics 365 Finance instance.
5. Select **Reset filter** to remove the specified selection conditions.



> **Note**
>
> - Depending on the ER settings, the system validates configurations after you import them. You might see a notification about any inconsistency problems that it finds. You

must fix these problems before you can use the imported configuration version. For more information, see the list of related resources for this article.

- You can configure ER configurations as dependent on other configurations. Therefore, when you import a selected configuration, the system might automatically import other configurations. For more information about configuration dependencies, see Define the dependency of ER configurations on other components.

Additional resources

Get started with Tax Calculation - High-level configuration

Globalization feature components

Last updated on 05/15/2026

Upload ER configurations and Globalization features as a Dataverse solution

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Regulatory Configuration Service (RCS) will be deprecated. All new RCS provisioning is stopped as of 10.0.39 GA. If provisioning is required, please register a support ticket. We will be providing tools and required support for migration from RCS to the **Globalization Studio** workspace. We plan to fully shut down RCS by August 1, 2024. For more information about migration to **Globalization Studio** workspace, see Regulatory Configuration Service merge to the Globalization Studio workspace

You can use Microsoft Dataverse solutions for Application Lifecycle Management (ALM) scenarios. For example, you can move Electronic reporting (ER) configurations to other environments or share them with other tenants via Marketplace.

You can also export and import specific versions as a file by using the export and import buttons on the page (existing functionality).

Prerequisites

- Install PowerShell 7. For more information, see Installing PowerShell on Windows.
- Install Microsoft Power Platform CLI. For more information, see What is Microsoft Power Platform CLI?
- Get the **Generate-GlobalizationDataverseSolution.ps1** script from the Shared asset library/Data package files in Microsoft Dynamics Lifecycle Services (<<Asset type:, Asset name:>>).

Step 1: Export your artifacts

To export your artifacts, follow these steps:

1. Create a folder, such as **C:\DvArtifactsSrc**.
2. Export the .xml files of any ER configurations and the .json files of any Globalization features to the new folder.

   > **ⓘ Important**
   >
   > Don't include Microsoft configurations.

Step 2: Create a Dataverse solution

To create a Dataverse solution, follow these steps:

1. Create a solution in Dataverse, and export it. For more information, see Create a solution and Export solutions.
2. Extract the new solution by using the **pac solution unpack** command. Here's an example:

   command

   ```
   pac solution unpack --zipfile C:\SampleEmptySolution.zip --folder
   C:\SampleSolutionUnpacked
   ```

   For more information, see pac solution unpack.

Step 3: Run the script

- Run the **Generate-GlobalizationDataverseSolution.ps1** script in PowerShell 7. Specify the following parameters:
  - **DvArtifactsSrcFolder** – Specify the working folder.
  - **SolutionFolder** – Specify the solution folder.
  - **ConfigurationsIndexRowName** – Specify the name of the Dataverse table record that has the index file for configurations. Use only alphanumeric characters and spaces. If you change the name later, the corresponding row in the Dataverse table has a different globally unique identifier (GUID), and index files are duplicated. If you don't have any ER configurations, you can omit this parameter. Here's an example.

    command

```
Contoso Electronic Reporting Configurations Index
```

- **FeaturesIndexRowName** – Specify the name of the Dataverse table record that has the index file for features. Use only alphanumeric characters and spaces. If you change the name later, the corresponding row in the Dataverse table has a different GUID, and index files are duplicated. If you don't have any Globalization features, you can omit this parameter. Here's an example.

  command

  ```
  Contoso Globalization Features Index
  ```

  Here's an example of the full PowerShell 7 command line.

  command

  ```
  pwsh -ExecutionPolicy RemoteSigned -File C:\Generate-
  GlobalizationDataverseSolution.ps1 -DvArtifactsSrcFolder C:\DvArtifactsSrc -
  SolutionFolder C:\SampleSolutionUnpacked -configurationsIndexRowName
  "Contoso Electronic Reporting Configurations Index" -featuresIndexRowName
  "Contoso Globalization Features Index"
  ```

Step 4: Pack the solution

- Pack the solution by using the **pac solution unpack** command. Here's an example.

  command

  ```
  pac solution pack --zipfile C:\SampleSolutionWithContents.zip --folder
  C:\SampleSolutionUnpacked --packagetype Both
  ```

  For more information, see pac solution unpack.

> **Note**
>
> Check the size of the generated .zip file. If it exceeds 90 megabytes (MB), you must split your source files into multiple Dataverse solutions.

Step 5: Import the solution into Dataverse

1. Import the solution back into Dataverse. For more information, see Import solutions.

You can also use the pac solution import command.

> **Note**
>
> After you import the solution into Dataverse, there's no way to remove it together with the content. Although you can remove the solution, all inserted table records remain and must be manually cleaned up.

2. Publish your app on Marketplace. For more information, see Publish your app on Marketplace.

Last updated on 03/21/2026

Create a Dataverse solution using Upload into repository

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Starting with version 10.0.40, instead of manually creating a Dataverse solution as described in Upload ER configurations and Globalization features as a Dataverse solution, you can use the **Upload into repository** functionality to upload one or all of your custom Electronic reporting (ER) configurations and Globalization features.

To use this functionality, you need a Dataverse environment that's connected to your Dynamics 365 Finance environment. You also need to add a security role to access tables in Dataverse by following these steps:

1. Create a new security role as described in Create or edit a security role.
2. In that security role, add **Create, Read, Write, Delete, Append, Append to, Assign, Share** permissions for the **Organization** to the following tables: **Solution, Publisher, Electronic Reporting Configuration File, Electronic Reporting Configurations Index File, Globalization Feature File, and Globalization Features Index File**.
3. Assign the created security role to users as described in Assign security roles.

The process of creating a Dataverse solution by using **Upload into repository** has three phases:

1. Create a Dataverse configuration repository.
2. Export configurations to Dataverse.
3. View configurations as a Dataverse solution.

The following sections describe each phase.

Create a Dataverse configuration repository

Create a Dataverse configuration repository for your active configuration provider.

To create a Dataverse configuration repository, follow these steps:

1. Sign in to the Dynamics 365 Finance app by using one of the following roles:

- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

2. Go to **Organization administration** > **Workspaces** > **Globalization studio**.

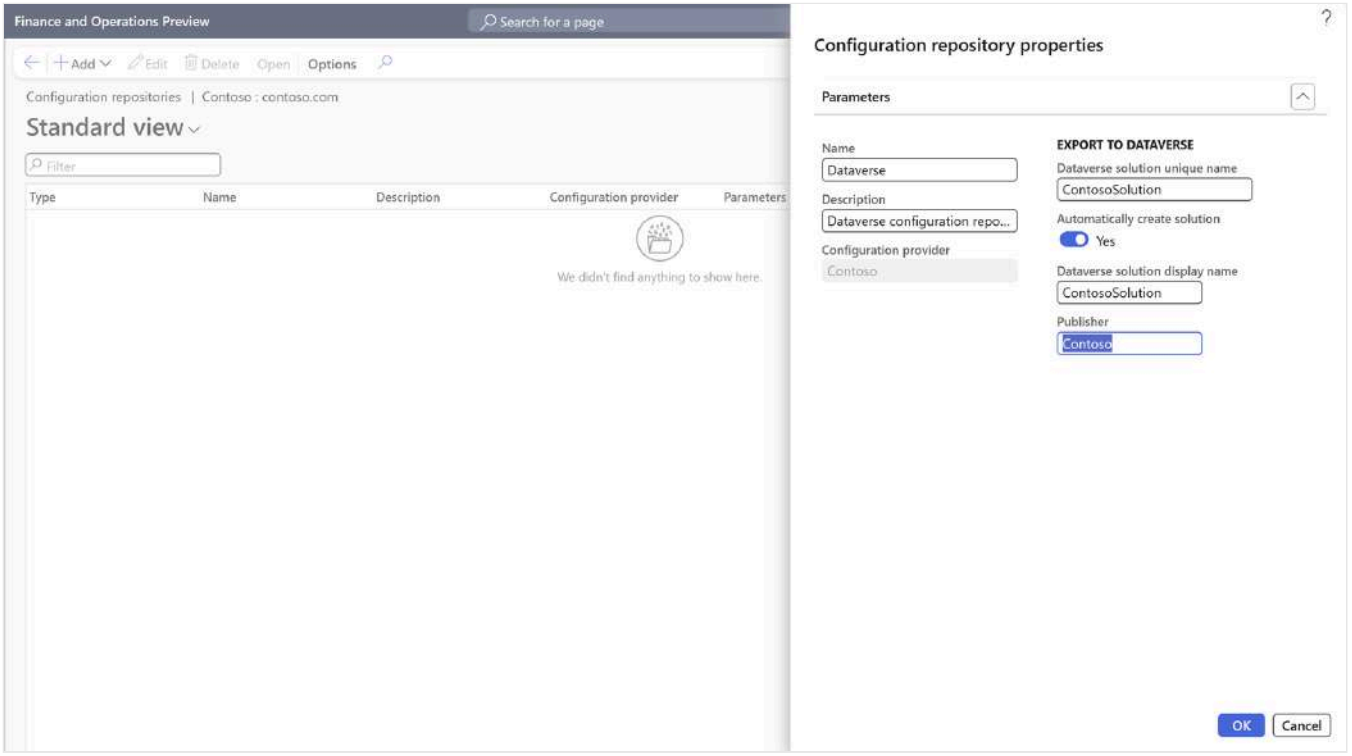
3. In the **Configuration providers** section, select the tile that has your provider (for example, Contoso).

4. On the **Contoso** tile, select **Repositories**.

5. Select **Add**.

6. In the **Configuration repository properties** dialog box, set up parameters for the Dataverse solution that is associated with the new repository.

   If you set the **Automatically create solution** option to **Yes**, the system creates the solution for you. If you leave the **Publisher** field blank, a default publisher is used.



Export configurations to Dataverse

After you create the repository, you can export configurations to Dataverse.

To export configurations to Dataverse, follow these steps:

1. To upload a single configuration version that has a status of **Completed**, use the **Upload into repository** button on the **Versions** FastTab of the **Electronic reporting**

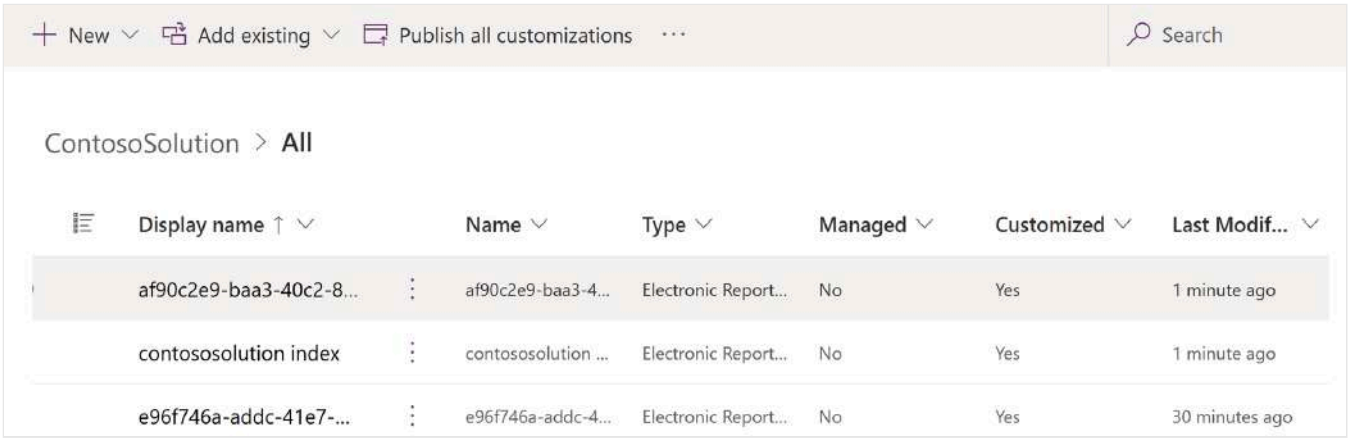
Configurations page. The system uploads all parent configuration versions to make the solution independently deployable.

2. To upload all the latest **Completed** configuration versions with the active provider, select **Upload all configurations into repository** on the **Configurations** tab on the Action Pane of the **Electronic reporting Configurations** page. Then use the **Upload into repository** dropdown dialog box.

View configurations as a Dataverse solution

If you set the **Automatically create solution** option to **Yes** when you create a Dataverse repository for your configuration provider, after you upload your configurations, a new solution appears in the Power Apps maker portal.

When you upload additional configuration versions, the system adds the new configuration versions to the same solution as the files.



Last updated on 03/19/2026

Electronic invoicing service overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic invoicing for Microsoft Dynamics 365 Finance and Dynamics 365 Supply Chain Management is a hyper-scalable multitenant service that enables configurable processing of electronic invoices and configurable electronic document exchange. You can fully configure the processing and integration rules, and the service runs the logic outside Finance and Supply Chain Management. The service mainly targets processing electronic invoices in various scenarios. However, you can configure it for other scenarios and different types of documents.

Electronic invoicing can help you achieve the following goals:

- Fast and easy adoption of country/region-specific requirements
- Standardized implementations of an electronic invoicing solution
- Enhanced traceability of document history
- A shorter implementation cycle
- Reduced total cost of ownership (TCO)
- Easily adjustable configurations that don't require code changes
- Simplified configuration packaging
- Built-in export, import, and integration, and easy extensibility in the processing of electronic invoice documents
- Easy reuse of the same export, import, and integration configurations across companies

Service availability

Currently, Finance and Supply Chain Management customers can use the electronic invoicing functionality. For more information, see the license terms and conditions for your application.

Because functionality that addresses country/region-specific requirements might be limited at different phases of the release, always review the most up-to-date documentation that highlights the coverage and scope of supported country/region-specific solutions.

You can deploy electronic invoicing in the following Azure geographies:

- United States
- Europe

- United Kingdom
- Asia
- Japan
- Switzerland
- Brazil
- United Arab Emirates
- Australia
- Canada
- France
- India
- Norway
- South Africa

> **Note**
>
> Electronic invoicing doesn't support on-premises deployments.

Feature highlights

- Out-of-box integration with Finance and Supply Chain Management
- A consistent user experience for the configuration and monitoring of the electronic invoice process for all countries and regions
- Faster, easier, and less expensive adoption of electronic invoicing solutions in new countries or regions
- Configuration of the service through the setup of Globalization features in the Globalization Studio workspace
- Transformation of business data into multiple electronic invoice formats (XML, JSON, TXT, and CSV) by using Electronic reporting (ER) format configurations
- Configurable submission of electronic invoices to external web services, including certification handling through digital signatures
- Handling of responses from web services, including configurable handling of exception messages
- Support for electronic signatures (for example, electronic signatures that use the XMLDSig signing algorithm)
- The capability to send documents to emails and store them in SharePoint
- Batch processing of electronic invoice messages
- Configurable transformation of incoming documents and processing of those documents in Finance and Supply Chain Management
- The capability to receive incoming documents from channels such as email and SharePoint

Privacy notice

Enabling and using electronic invoicing might require sending limited data, such as your organization's tax registration ID. The system transmits this data to third-party agencies that the tax authorities authorize to send electronic invoices in the predefined formats required for integration with government web services. Data imported from these external systems into this Dynamics 365 online service is subject to our privacy statement. For more information, see the "Privacy notice" section in country/region-specific feature documentation.

Last updated on 03/21/2026

Electronic invoicing components

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an overview of the administration and integration components for Electronic invoicing. It includes information about their roles in the setup and operation of Electronic invoicing.

Microsoft Azure

Set up a storage account and create secrets for an Azure Key Vault in your Azure subscription. You fully own and manage these resources.

A storage account is a mandatory component of the Electronic Invoicing service. Use it to store all electronic files that Electronic invoicing processes or generates. Here are some examples of these files:

- Business data from Dynamics 365 Finance and Dynamics 365 Supply Chain Management in a unified structure (JavaScript Object Notation [JSON] format)
- Files that are generated during file generation and transformation
- Incoming external data, such as responses from external web services
- Inbound electronic invoices

A key vault is another mandatory component for working with Electronic invoicing. Use it to store the following items:

- The shared access signature (SAS) token of your storage account. Microsoft supports a high level of security and data privacy. Therefore, store your SAS token as a secret in Azure Key Vault instead of using it directly.
- Secrets, accounts, and passwords that you use in different scenarios.
- Certificates that have secrets for digital signing and for establishing a trusted connection with external web services.

Microsoft Dynamics Lifecycle Services

Use Microsoft Dynamics Lifecycle Services to enable the Electronic invoicing add-in for your Lifecycle Services deployment project.

> **Note**
>
> Installation of add-ins in Lifecycle Services requires at least a Tier 2 environment. For more information about environment planning, see **Environment planning**.

Finance and Supply Chain Management

Currently, Microsoft supports out-of-the-box integration with Finance and Supply Chain Management as the main billing system that works with the Electronic Invoicing service to generate and submit electronic invoices (*outbound electronic documents*), and to receive electronic invoices from vendors (*inbound electronic documents*).

Finance and Supply Chain Management generate business data in a unified structure and then submit it, together with additional metadata (*context*), to Electronic invoicing. They then request the status of the documents that Electronic invoicing is processing.

The detailed results come as a log that's linked to the generated electronic files or other data that's related to the submitted documents. Depending on the scenario, the system updates Finance and Supply Chain Management data. For example, it updates the invoice status or externally provided universally unique identifiers (UUIDs) of invoices that are registered with the tax authority.

For more information about Electronic invoicing components and how to set them up, see Set up Electronic invoicing.

Last updated on 03/21/2026

Electronic invoicing configuration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an overview of the process for setting up and configuring Electronic invoicing. Complete the setup steps in the order specified here. If you skip a mandatory step, the functionality won't work correctly, and multiple failures occur during subsequent steps or when you use the functionality.

Install the add-in for Electronic invoicing microservices

The Electronic Invoicing service is a set of microservices that's hosted in Microsoft datacenters. By default, customers of Dynamics 365 Finance and Dynamics 365 Supply Chain Management don't have access to the Electronic Invoicing service. You must install it as an add-in in Microsoft Dynamics Lifecycle Services. When you install the add-in, you register your Finance or Supply Chain Management environment in the register of applications that are allowed to connect to the Electronic Invoicing service.

To register an environment, follow these steps:

1. Sign in to your Lifecycle Services account.
2. On the project dashboard, select a Lifecycle Services project.
3. In the project, on the **Environments** dashboard, select your deployed environment. The environment that you select must be running.
4. On the **Power Platform Integration** tab, in the **Environment add-ins** section, select **Install a new add-in**.
5. Select **Electronic Invoicing**.
6. In the **AAD application ID** field, enter the fixed value **091c98b0-a1c9-4b02-b62c-7753395ccabe**. This value is always fixed. Make sure that you enter only a globally unique identifier (GUID). Don't include any other symbols, such as spaces, commas, periods, or quotation marks.
7. In the **AAD tenant ID** field, enter the Microsoft Entra ID tenant ID of your Azure subscription account.
8. Review the terms and conditions, and then select the checkbox.

9. Select **Install**. After a few minutes, the status changes from **Installing** to **Installed**. You might have to refresh the page to see this change.

> ⓘ **Note**
>
> Companies usually have several Finance or Supply Chain Management environments. These environments include production environments, user acceptance testing (UAT) environments, and development (sandbox) environments. You must complete the preceding procedure for all environments that you want to connect to Electronic invoicing.

Enable Electronic invoicing integration

To enable communication between Electronic invoicing and Finance or Supply Chain Management, you must enable the **Electronic Invoicing integration** feature.

1. In the **Feature management** workspace, on the **All** tab, search for the **Electronic invoicing integration** feature. If this feature doesn't appear on the page, select **Check for updates**.
2. Select the feature, and then select **Enable now**.

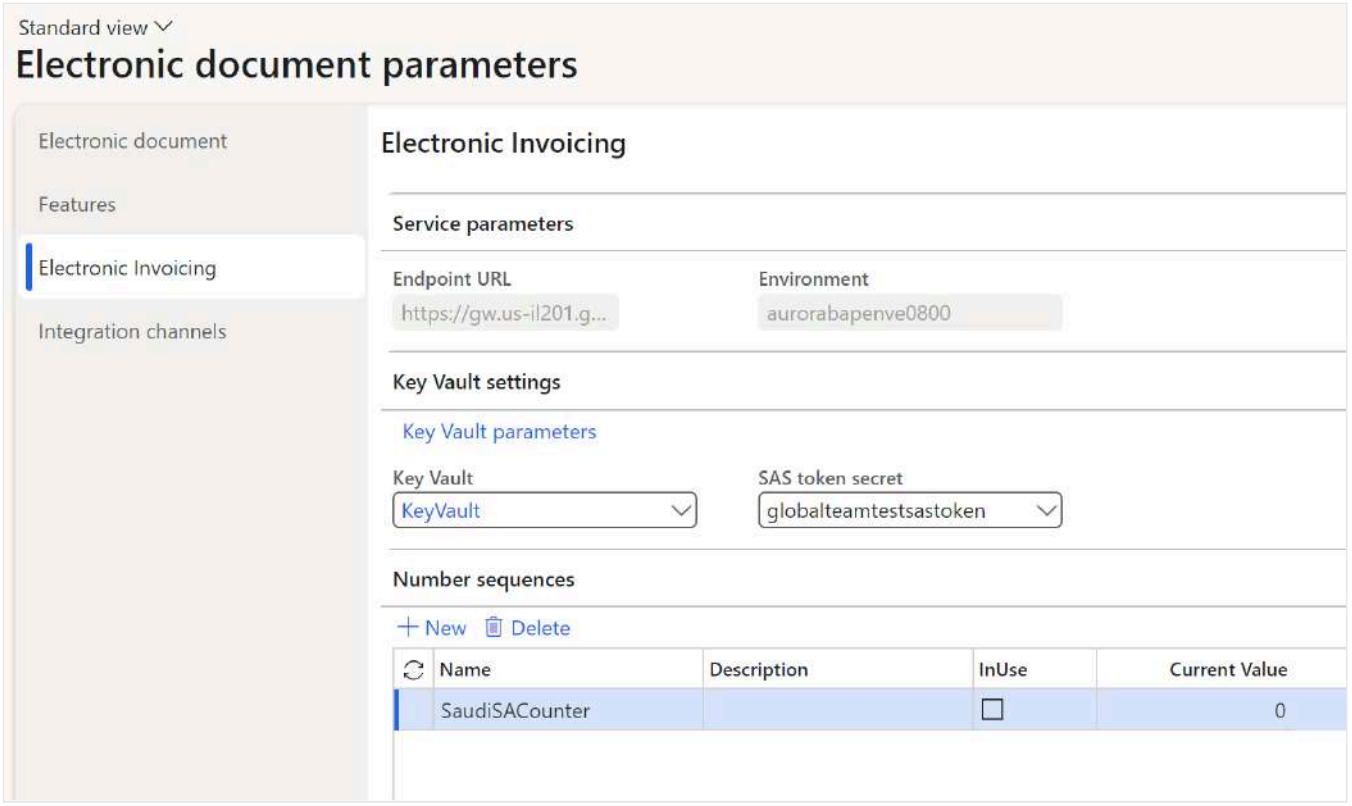
> ⓘ **Note**
>
> Before you work with Globalization features, the **E-Invoicing designer workspace** feature is enabled by default. However, after the Electronic Invoicing service add-in is installed, there's a delay of 45 minutes before the feature is enabled.

Service environment configuration

> ⓘ **Note**
>
> The following procedure is required if the Regulatory Configuration Service (RCS) experience was previously used to configure the Electronic Invoicing service. If you're doing the initial configuration of the Electronic Invoicing service through Globalization Studio, you can skip this procedure.

1. Identify one of the previously deployed service environments. This environment is the only one that you configure in the Globalization Studio and then use by the Electronic Invoicing service.
2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

3. On the **Electronic invoicing** tab, on the **Service parameters** FastTab, in the **Environment** field, enter the name of the environment that you identified in step 1.



> **Important**
>
> After the initial configuration of **Electronic document parameters**, the field for the service environment name becomes unavailable. Therefore, you must finalize the name of the relevant service environment in advance.

Configure the Azure resources for Electronic invoicing

Set up the Azure resources that Electronic invoicing requires to do its work. For more information, see Set up Azure resources for Electronic invoicing.

Configure Globalization features

Implement different scenarios for processing electronic documents by using Globalization features. A Globalization feature is a set of components that define the rules for data transformation and further processing of electronic documents, such as sending them to or receiving them from external channels. You can use Globalization features that Microsoft

provides, or you can create your own. For more information about how to work with Globalization features, see Globalization feature components.

> **Note**
>
> If your scenarios require integration with email or SharePoint to process inbound electronic documents, see **Process incoming electronic documents** for information about how to set up and use those channels.

Configure Electronic invoicing parameters

For additional configuration steps related to Electronic invoicing, see Configure Electronic invoicing parameters.

Last updated on 12/18/2025

Configure Electronic invoicing parameters

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Import ER configurations from the repository

To submit electronic documents to the Electronic invoicing service, you need to generate a message in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. This message must follow a unified structure that matches the structure of the Electronic reporting (ER) invoice model and metadata (*context*). Because ER generates the message, you can customize and flexibly adjust its structure.

Before you can submit documents to Electronic invoicing and process responses, you must import ER configurations from the repository and define them in parameters.

> **Note**
>
> Most of the ER configurations mentioned in the following procedure are automatically imported when you import related Globalization features. For more information, see Import features from the repository. If you need to reimport configurations, use the following procedure.

To set up, adjust, or review electronic documents directly in your Finance or Supply Chain Management environment, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile.
2. In the **Configuration providers** section, select the **Microsoft** configuration provider, and then select **Repositories**.
3. Select the repository of the **Dataverse** type, and then, on the Action Pane, select **Open**.

4. Import the following ER configurations:

    - **Customer invoice context model**
    - **Invoice model**
    - **Fiscal documents** (for Brazilian scenarios, if required)
    - **Retail channel data** (for Saudi Arabian retail-specific scenarios, if required)
    - **Invoices Communication Model** (for Malaysian and Indonesian batch submission scenarios, if required)
    - **Response message model**
    - **Invoice model mapping**
    - **Fiscal documents mapping** (for Brazilian scenarios, if required)
    - **Advance invoice model mapping** (for Polish scenarios, if required)
    - **Retail fiscal document mapping** (for Saudi Arabian retail-specific scenarios, if required)
    - **Pending vendor invoice model mapping** (for Danish responses scenarios, if required)
    - **Invoices communication model mapping (ID)** (for Indonesian batch submission scenarios, if required)
    - **Consolidated Invoice Model Mapping (MY)** (for Malaysian batch submission scenarios, if required)
    - **Invoice status model mapping** (for French lifecycle statuses scenarios, if required)
    - **AR response context** (for French lifecycle statuses scenarios, if required)
    - **AP response context** (for French lifecycle statuses scenarios, if required)

Set up Electronic document parameters

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

2. On the **Electronic document** tab, above the **Electronic reporting** grid, select **Add**.

3. Depending on the scenarios that you want to support, enter the following information in the grid, and then save your changes.

Expand table

Table or view name	Document context	Electronic document model mapping	Features that it's required for
CustInvoiceJour	**Mapping name:**	**Mapping name:**	• Austrian electronic

Table or view name	Document context	Electronic document model mapping	Features that it's required for
	Customer invoice context **Configuration:** Customer invoice context model	Customer Invoice **Configuration:** Invoice model mapping	invoices (AT) • Belgian electronic invoice (BE) • Danish electronic invoice (DK) • Egyptian electronic invoice (EG) • Estonian electronic invoice (EE) • Finish electronic invoice (FI) • French Chorus Pro submission (FR) • German electronic invoice (DE) • FatturaPA (IT) • Dutch electronic invoice (NL) • Norwegian electronic invoice (NO) • Polish electronic invoice (PL) • Spanish electronic invoice (ES) • PEPPOL electronic invoice • Saudi Arabian electronic invoice (SA)

Table or view name	Document context	Electronic document model mapping	Features that it's required for
			• Australian electronic invoice (AU) • New Zealand electronic invoice (NZ) • Malaysian electronic invoicing (MY) • Singaporean electronic invoice (SG)
ProjInvoiceJour	**Mapping name:** Project invoice context **Configuration:** Customer invoice context model	**Mapping name:** Project Invoice **Configuration:** Invoice model mapping	• Austrian electronic invoices (AT) • Belgian electronic invoice (BE) • Danish electronic invoice (DK) • Egyptian electronic invoice (EG) • Estonian electronic invoice (EE) • Finish electronic invoice (FI) • French Chorus Pro submission (FR) • German electronic invoice (DE) • FatturaPA (IT) • Dutch electronic

Table or view name	Document context	Electronic document model mapping	Features that it's required for
			invoice (NL) • Norwegian electronic invoice (NO) • Polish electronic invoice (PL) • Spanish electronic invoice (ES) • PEPPOL electronic invoice • Saudi Arabian electronic invoice (SA) • Australian electronic invoice (AU) • New Zealand electronic invoice (NZ) • Malaysian electronic invoicing (MY) • Singaporean electronic invoice (SG)
CzCustAdvanceInvoiceTable	**Mapping name:** Advance invoice context **Configuration:** Customer invoice context model	**Mapping name:** Advance invoice model mapping **Configuration:** Advance invoice model mapping	Polish electronic invoice (PL)
RetailTransactionFiscalTransDocumentView	**Mapping name:** Retail	**Mapping name:** Retail	Saudi Arabian electronic invoice

Table or view name	Document context	Electronic document model mapping	Features that it's required for
	fiscal document context **Configuration:** Customer invoice context model	fiscal document **Configuration:** Retail fiscal document mapping	(SA)
VendInvoiceInfoTable	**Mapping name:** Pending vendor invoice context **Configuration:** Customer invoice context model	**Mapping name:** Pending vendor invoice model mapping **Configuration:** Pending vendor invoice model mapping	• Danish electronic invoice (DK) • Chilean electronic invoice (CL)
LedgerJournalTrans	**Mapping name:** Prepayment invoice context **Configuration:** Customer invoice context model	**Mapping name:** Customer prepayments **Configuration:** Invoice model mapping	Saudi Arabian Zatca submission (SA)
VendInvoiceJour	**Mapping name:** Self invoice context **Configuration:** Customer invoice context model	**Mapping name:** Self invoice **Configuration:** Self invoice model mapping	Malaysian electronic invoicing (MY)

Table or view name	Document context	Electronic document model mapping	Features that it's required for
CustInvoiceJour ProjInvoiceJour (*In case of batch submission*)	**Mapping name:** Batch submission **Configuration:** Customer invoice context model	**Mapping name:** Invoices communication model mapping (ID) **Configuration:** Invoices communication model mapping (ID)	Indonesian electronic invoice (ID)
CustInvoiceJour (*In case of batch submission*)	**Mapping name:** Batch submission **Configuration:** Customer invoice context model	**Mapping name:** Consolidated Invoice Model Mapping (MY) **Configuration:** Consolidated Invoice Model Mapping (MY)	Malaysian electronic invoicing (MY)
LTMCustInvoiceJour	**Mapping name:** Customer invoice context **Configuration:** Customer invoice context model	**Mapping name:** Customer E-Invoice **Configuration:** Invoice model mapping LATAM	• Chilean electronic invoice (CL) • Costa Rican electronic invoice (CR) • Panama electronic invoice (PA) • Uruguay electronic invoice (UY)

Table or view name	Document context	Electronic document model mapping	Features that it's required for
LTMProjInvoiceJour	**Mapping name:** Project invoice context **Configuration:** Customer invoice context model	**Mapping name:** Project E-Invoice **Configuration:** Invoice model mapping LATAM	• Chilean electronic invoice (CL) • Costa Rican electronic invoice (CR) • Panama electronic invoice (PA) • Uruguay electronic invoice (UY)
LTMCustPackingSlipJour	**Mapping name:** Customer Packing Slip context **Configuration:** Customer invoice context model	**Mapping name:** Customer E-Invoice **Configuration:** Invoice model mapping LATAM	Chilean electronic invoice (CL)
CustInvoiceJournalDomesticB2B_FR	**Mapping name:** Customer invoice context **Configuration:** Customer invoice context model (FR)	**Mapping name:** Customer Invoice **Configuration:** Invoice model mapping (FR)	French electronic invoice (FR)
ProjInvoiceJournalDomesticB2B_FR	**Mapping name:** Project invoice context **Configuration:** Customer invoice	**Mapping name:** Project Invoice **Configuration:** Invoice model mapping (FR)	French electronic invoice (FR)

Table or view name	Document context	Electronic document model mapping	Features that it's required for
	context model (FR)		
BusinessDocumentApplicationARResponseView	**Mapping name:** Data channel context **Configuration:** AR response context	**Mapping name:** Customer Invoice status **Configuration:** Invoice status model mapping	French electronic invoice status (FR)
BusinessDocumentApplicationAPResponseView	**Mapping name:** Data channel context **Configuration:** AP response context	**Mapping name:** Vendor Invoice status **Configuration:** Invoice status model mapping	French electronic invoice status (FR)

If you derive a configuration from the configuration that's mentioned in the preceding table, define the new configuration.

> **Note**
>
> By default, configured Electronic document parameters apply to all legal entities. To activate a configuration for specific legal entities, enable the **Electronic documents configuration per legal entities** feature in Feature management. For more information, see Feature management overview.

To set up the rules to process responses from the Electronic Invoicing service, and to update Finance and Supply Chain Management data based on invoices that are processed by the service, set up response types. In most scenarios, this setup is country/region-specific. Therefore, we recommend that you follow the country/region-specific instructions. For more information, see Availability of Electronic Invoicing Service features by country or region.

Create a Key Vault reference

1. On the **Electronic document parameters** page, on the **Electronic Invoicing** tab, on the **Key Vault settings** FastTab, select **Key Vault parameters**.

2. On the **Key Vault parameters** page, select **New** to create a Key Vault reference.

3. In the **Name** field, enter the name of the Key Vault reference.

4. In the **Description** field, enter a description.

5. In the **Key Vault URI** field, paste the Key Vault URI from the key vault (`https://<your key vault>.vault.azure.net/`). For more information, see Create an Azure Key Vault in the Azure portal.

6. In the **Certificates** section, select **Add**.

7. In the **Name** field, enter the name of the storage account secret or certificate. This name should match the name of the Key Vault secret that holds the shared access signature (SAS) token of the storage account. For more information, see Create an Azure storage account in the Azure portal.

8. In the **Description** field, enter a description.

9. In the **Type** field, select either **Secret** or **Certificate**, depending on what you're configuring.

> **Note**
>
> In some scenarios, you must use public certificates that have the .cer file name extension. However, Key Vault doesn't support importing and storing certificates of this type as Key Vault certificates. In these scenarios, save the .cer file as a Base-64-encoded X.509 (.CER) string. Then, in a Key Vault secret, store the string that appears between the **BEGIN CERTIFICATE** line and the **END CERTIFICATE** line in the file. In the service environment, you should still create a reference to the Key Vault record and set the **Type** field to **Certificate**.
>
> Alternatively, use the following PowerShell script to generate a Base-64 string of the .cer certificate file.

PowerShell

```
$FilePath = ''
$Cer = New-Object -TypeName
System.Security.Cryptography.X509Certificates.X509Certificate2($FilePath)
$BinCert = $Cer.GetRawCertData()
```

```
$Base64Cert = [System.Convert]::ToBase64String($BinCert)
echo $Base64Cert
```

10. If your specific scenarios require a chain of certificates to apply digital signatures or establish a secure (Secure Sockets Layer [SSL]) connection to external web services, create a chain of certificates where the certificates are in the following order: *Root certificates* > *Intermediate certificates* > *End-user certificates*. Root certificate authorities (CAs) are a trusted source of certificates. Intermediate CA certificates are bridges that link the end-user certificates to the root CA certificates. To create and set up a chain of certificates, follow these steps:
    a. On the Action Pane, select **Chain of certificates**.
    b. Select **New** to create a chain of certificates.
    c. In the **Name** field, enter the name of the chain of certificates.
    d. In the **Description** field, enter a description.
    e. In the **Certificates** section, select **Add** to add a certificate to the chain.
    f. Use the **Up** or **Down** button to change the position of the certificate in the chain. Keep the CA root certificate at the top of the list and the end-user certificate at the bottom.
    g. Save the data, and close the **Chain of certificates** page.

11. Save the data, and close the **Key Vault parameters** page.

12. In the **Key Vault** field, select the key vault that you created in previous steps.

13. In the **Storage SAS token secret** field, select the name of the storage account secret that must be used to authenticate access to the storage account.

Configure number sequences

If your scenarios require number sequences, such as in file names, use number sequences that work across globalization features or number sequences for a specific globalization feature. After you define a number sequence, use it in variables and processing pipelines. To track the use of a number sequence, check the **Current Value** field and the selected **In use** checkbox on the **Number sequences** FastTab on the **Electronic Invoicing** tab of the **Electronic document parameters** page.

To create a number sequence, select **New** on the **Number sequences** FastTab. Then enter a name and description.

To delete a number sequence, select **Delete**.

Inactivate legacy Electronic invoicing functionality

To deactivate old (legacy) ER functionality for some features, and to activate additional functionality in Finance and Supply Chain Management for some country/region-specific scenarios, enable the corresponding feature on the **Features** tab of the **Electronic document parameters** page.

When you enable a feature in the list, the legacy functionality is deactivated for the corresponding country or region and feature name.

Last updated on 04/29/2026

Configure Azure resources for Electronic invoicing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The process for setting up Microsoft Azure resources for Electronic invoicing has three steps. The first two steps, "Create an Azure Key Vault in the Azure portal" and "Create an Azure storage account in the Azure portal," are mandatory. The third step, "Configure a SharePoint connection," is optional. These steps can be completed by using any of the following methods:

- Use the Azure portal user interface (UI).
- Run an Azure PowerShell script.
- Deploy an Azure Resource Manager (ARM) template.

If you prefer to use the Azure portal UI, continue with this article. If you prefer to use the PowerShell script or an ARM template, see Create Azure resources using Azure PowerShell or ARM templates.

Create an Azure Key Vault in the Azure portal

Create a key vault in your Azure subscription. Create a separate key vault for Electronic invoicing, and don't combine secrets with other applications. Create as many secrets and certificates as you require for your scenarios for electronic documents. You must create at least one secret to store the shared access signature (SAS) token of the Azure storage account that you create in the next step.

For information about how to complete this step, see Create an Azure Key Vault in the Azure portal.

Create an Azure storage account in the Azure portal

You own all the electronic documents and files that the Electronic Invoicing service generates or that enter the service. Store those documents and files in an Azure storage account that you

create in your Azure subscription. The service accesses your storage account by using the SAS token that's taken from your Key Vault secret.

For information about how to complete this step, see Create an Azure storage account in the Azure portal.

Configure a SharePoint connection

In some scenarios, you need to save electronic files to SharePoint and retrieve them from SharePoint. To ensure that the Electronic Invoicing service can access your SharePoint folders, configure access to SharePoint.

For information about how to complete this step, see Configure a SharePoint connection.

Last updated on 03/21/2026

Configure an Azure Key Vault in the Azure portal

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

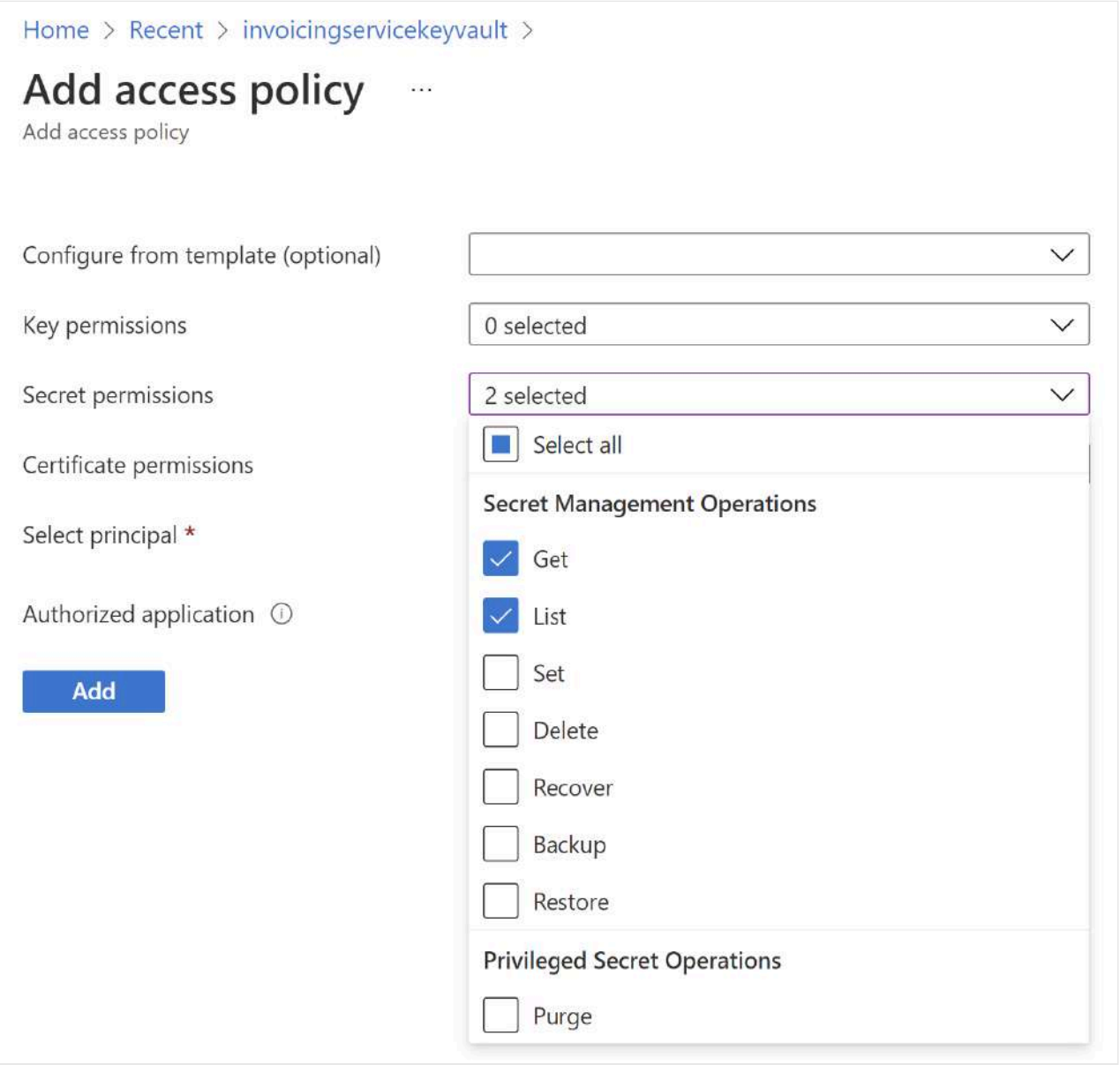
All the secrets and certificates that are used in the Electronic Invoicing service must be stored in a Microsoft Azure Key Vault. This approach helps ensure that you don't work directly with the secrets, and that the secrets are securely stored. When you must use digital signing or secure a connection to external web services, set the reference to the Key Vault secrets and certificates instead of using the secrets and certificates directly.

1. Create a key vault in the tenant that you use for the Dynamics 365 Finance environment. For more information, see Create a key vault using the Azure portal.

   Next, set up the access policy to grant the Electronic Invoicing service the correct level of secure access to the secret that you created.

2. Go to **Settings** > **Access policies**, and select **Add Access Policy**.

3. In the **Secret permissions** field, select the **Get** and **List** operations.



4. In the **Certificate permissions** field, select the **Get** and **List** operations.

5. In the **Select principal** field, select **None selected**.

6. In the **Principal** dialog box, select the principal by adding **e-Invoicing Service**.

> **Note**
>
> If **e-Invoicing Service** isn't in the list of principals in your tenant, run the following command in the Azure portal.
>
> ```
> New-AzureADServicePrincipal -AppId "ecd93392-c922-4f48-9ddf-10741e4a9b65"
> ```

7. Select **Add**, and then select **Save**.

8. On the **Overview** page, copy the value of the Domain Name System (DNS) name for the key vault. Use this value during the setup of Electronic invoicing parameters in Finance

and refer to it as the **Key Vault URI** value. For more information about how to set up Electronic invoicing parameters, see Configure Electronic invoicing parameters.

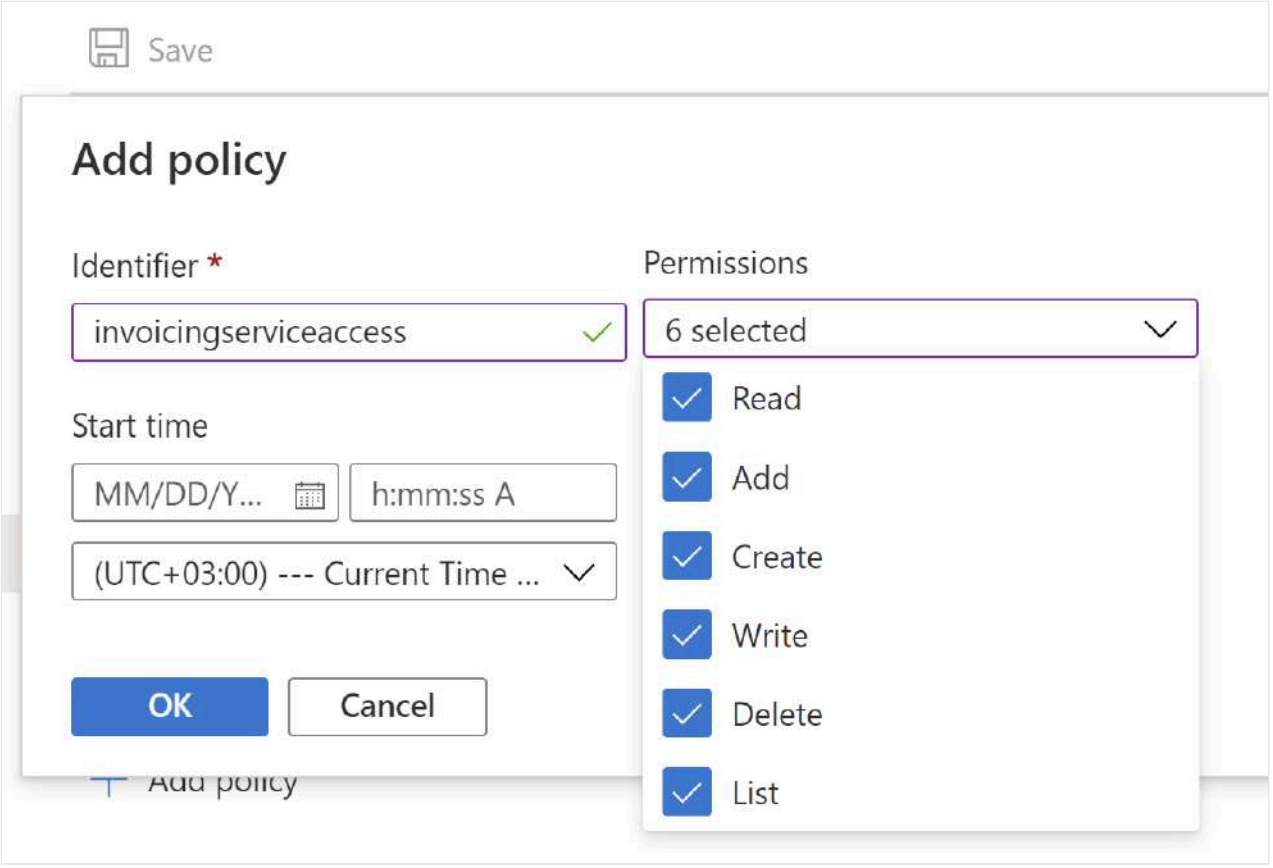
Last updated on 03/21/2026

Configure an Azure storage account in the Azure portal

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Electronic Invoicing service stores all electronic files it generates or processes in containers within your Microsoft Azure storage account. To ensure that the Electronic Invoicing service can access these containers, you must provide the shared access signature (SAS) token of the storage account to the service. To keep the token secure, don't provide the SAS token directly. Instead, store it in an Azure Key Vault and provide a Key Vault secret.

1. Open the storage account that you plan to use with the Electronic Invoicing service.
2. Go to **Settings** > **Configuration**, and make sure that the **Allow Blob public access** parameter is set to **Enabled**.
3. Go to **Data storage** > **Containers**, and create a container.
4. Enter a name for the container, and set the **Public access level** field to **Private (no anonymous access)**.
5. Open the new container, and go to **Settings** > **Access policy**.
6. Select **Add policy** to add a stored access policy.
7. Set the **Identifier** field as appropriate.
8. In the **Permissions** field, select all permissions.



9. Enter the start and end dates. The end date should be in the future.
10. Select **OK** to save the policy, and then save your changes to the container.
11. Go to **Settings** > **Shared access tokens**, and set the field values.
12. Enter the start and end dates. The end date should be in the future.
13. In the **Permissions** field, select the following permissions:
    - Read
    - Add
    - Create
    - Write
    - Delete
    - List
14. Select **Generate SAS token and URL**.
15. Copy and store the value of the **Blob SAS URL** field. You'll use this value later in this procedure and refer to it as the *shared access signature URI*.
16. Open the key vault that you intend to use with Electronic invoicing.
17. Go to **Settings** > **Secrets**, and select **Generate/Import** to create a secret.

18. On the **Create a secret** page, in the **Upload options** field, select **Manual**.
19. Enter the name of the secret. Use this name during the setup of the service in Dynamics 365 Finance and refer to it as the *key vault secret name*. For more information about how to set up Electronic invoicing parameters, see Configure Electronic invoicing parameters.
20. In the **Value** field, enter the shared access signature URI that you copied earlier.
21. Select **Create**.

Last updated on 03/21/2026

Create an Azure file share in the Azure portal

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic documents that the Electronic Invoicing service generates can be stored in a Microsoft Azure file share that is provisioned under a storage account. To ensure that Electronic invoicing can access the file share, you must provide the connection string to the Electronic Invoicing service. In addition, to ensure that the connection string is securely stored, don't provide it directly. Instead, store it in an Azure Key Vault, and provide an Azure Key Vault secret.

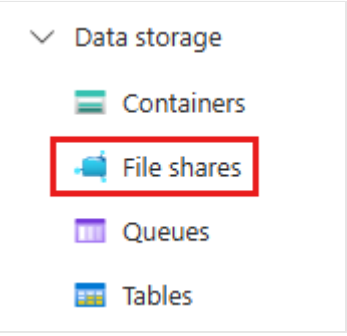
Prerequisites

Before you begin the procedures in this article, create an Azure storage account. To learn more, see Configure an Azure storage account in the Azure portal.

Create an Azure file share

To create an Azure file share, follow these steps:

1. Select the storage account from your dashboard.
2. On the service menu, under **Data storage**, select **File shares**.



3. At the top of the **File shares** page, select **File share**.
4. On the **New file share** page, in the **Name** field, enter a name for the new file share.

> **Note**
>
> File share names can contain only lowercase letters, numbers, and hyphens. However, they can't contain two or more consecutive hyphens. In addition, they must begin and end with either a lowercase letter or a number.

5. To create the Azure file share, select **Review + create**, and then select **Create**.

To get the connection string, follow these steps:

1. Go to the Azure portal with your subscription, and find the corresponding Storage account resource.
2. Go to **Security + Networking** > **Access keys**.
3. Under **key1**, copy the value in the **Connection string** field.

4. Store the **Connection string** value in a Key Vault secret. To learn how to set up Key Vault, see Configure an Azure Key Vault in the Azure portal.
5. Refer to the Key Vault secret in Key Vault parameters. To learn more, see Configure Electronic invoicing parameters.

> **Important**
>
> Don't confuse the **Connection string** value with the **Storage shared access signature (SAS)** token.

Last updated on 03/21/2026

Create a SharePoint connection

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Electronic Invoicing service can read files from Microsoft SharePoint folders and upload files to SharePoint. To ensure that the Electronic Invoicing service can access a specific SharePoint site, you must provide the site credentials to the Electronic Invoicing service. To ensure secure storage of the credentials, don't provide them directly. Instead, store them in an Azure Key Vault, and provide a Key Vault secret.

Grant access to a SharePoint folder

1. Create an app registration in the tenant where you installed the Dynamics 365 Finance environment.
   a. Sign in to the Azure portal.
   b. Go to **App registrations**.
   c. Select **New registration**.
   d. Enter a name, such as **SharePoint App for Electronic Invoicing**, and complete the registration.
   e. Select the new app registration.
   f. On the **Authentication** tab, enable the **Allow public client flows** option.
   g. On the **Certificates & secrets** tab, select **New client secret** to create a client secret.
   h. Copy the value of the secret that you created.

   Follow these guidelines:

   - Don't use the same app registration for different services.
   - Follow the password policy recommendations.
   - Set up rotation of passwords. During rotation, create a new client secret for the app registration, update the key vault, and then delete the old secret.

2. Save the **App Registration secret** and **Application (client) ID** values as two new secrets in the key vault in the setup of your Electronic invoicing parameters.

3. Add the secrets that you created to the Key Vault parameters in the setup of your Electronic invoicing parameters. For more information, see Configure Electronic invoicing

parameters.

4. In the Azure portal, grant access to SharePoint. The tenant administrator should complete this step.

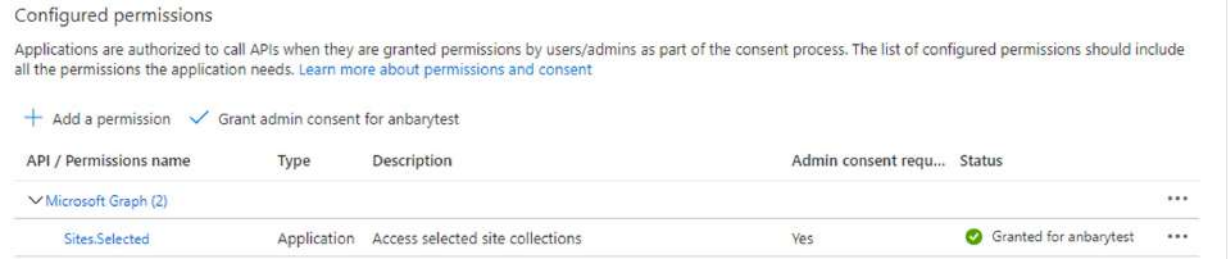
   a. Select the app registration that you created.

   b. On the **API permissions** tab, select **Add a permission**.

   c. Select **Microsoft graph (Application permissions)** > **Sites.Selected**.

   d. Select **Grant admin consent for** <***user name***>.

   e. Review the **Status** field to make sure that permissions are granted.



   f. Open Graph Explorer, and sign in.

   g. In the left pane, on the **Sample queries** tab, under **SharePoint Sites**, select **get SharePoint site based on relative path of the site**.

   h. Fill in the **{host-name}** and **{server-relative-path}** parameters. For example, fill in `<domain>.sharepoint.com` for **{host-name}** and `sites/<siteName>` for **{server-relative-path}**.

> **Note**
>
> For the default website, leave the **{server-relative-path}** parameter blank.

   i. Select **Run query**, and save the result.

   j. Configure the following query.

```
POST https://graph.microsoft.com/v1.0/sites/{site-id}/permissions
```

   In this query, **{site-id}** is the value of the **id** node from the previous query response.

   Here's the request body.

```
JSON
```

```
{
    "roles": [
        "read",
        "write"
    ],
    "grantedToIdentities": [
        {
            "application": {
                "id": "{app-id}",
                "displayName": "{app-name}"
            }
        }
    ]
}
```

In this request body, **{app-id}** is the **Application (client) ID** value, and **{app-name}** is the **Application name** value.

k. On the **Modify permissions** tab, select **Open the permissions panel**, and then select **Sites** > **Sites.FullControl.All** > **Consent**.

l. Select **Run query**.

The Electronic Invoicing service now has access to your SharePoint site.

Last updated on 03/21/2026

Configure Azure resources for Electronic invoicing

> **Note**
>
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The process for setting up Microsoft Azure resources for Electronic invoicing has three steps. The first two steps, "Create an Azure Key Vault in the Azure portal" and "Create an Azure storage account in the Azure portal," are mandatory. The third step, "Configure a SharePoint connection," is optional. These steps can be completed by using any of the following methods:

- Use the Azure portal user interface (UI).
- Run an Azure PowerShell script.
- Deploy an Azure Resource Manager (ARM) template.

If you prefer to use the Azure portal UI, continue with this article. If you prefer to use the PowerShell script or an ARM template, see Create Azure resources using Azure PowerShell or ARM templates.

Create an Azure Key Vault in the Azure portal

Create a key vault in your Azure subscription. Create a separate key vault for Electronic invoicing, and don't combine secrets with other applications. Create as many secrets and certificates as you require for your scenarios for electronic documents. You must create at least one secret to store the shared access signature (SAS) token of the Azure storage account that you create in the next step.

For information about how to complete this step, see Create an Azure Key Vault in the Azure portal.

Create an Azure storage account in the Azure portal

You own all the electronic documents and files that the Electronic Invoicing service generates or that enter the service. Store those documents and files in an Azure storage account that you

create in your Azure subscription. The service accesses your storage account by using the SAS token that's taken from your Key Vault secret.

For information about how to complete this step, see Create an Azure storage account in the Azure portal.

Configure a SharePoint connection

In some scenarios, you need to save electronic files to SharePoint and retrieve them from SharePoint. To ensure that the Electronic Invoicing service can access your SharePoint folders, configure access to SharePoint.

For information about how to complete this step, see Configure a SharePoint connection.

Last updated on 03/21/2026

Globalization feature components

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

A Globalization feature is a set of components that define the rules for data transformation in Electronic reporting (ER) configurations. These components include instructions for processing electronic documents and then sending them to or receiving them from external channels. They also include conditions that define when a feature should run for the incoming business data.

All the components depend on each other. Globalization features make it easy to create and maintain this dependency, and to support lifecycle management and versioning of the set of components.

Access Electronic invoicing feature components

In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.

Globalization features have several components. The **Electronic invoicing features** page includes a separate tab for each component.

- **Version** – This component supports lifecycle management of the feature. Use it to manage status for different versions of the feature.
- **Configurations** – Use this component to manage, view, and edit related ER format and format mapping configurations that are used in the processing pipeline.
- **Setups** – Use this component to manage the setup of the related feature version for users of Globalization services, such as an e-invoicing service. Therefore, it supports the flexible construction of communication and responses rules.
- **Tags** – Use this component to tag features that you can use in the Globalization blueprint for the reference.

Last updated on 03/21/2026

Import features from the repository

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Dataverse repository contains electronic invoicing features that you can share with your configuration provider. Microsoft shares all electronic invoicing features with all companies. You automatically get access to all the electronic invoicing features that Microsoft releases and publishes to the Dataverse repository.

To start working with electronic invoicing features that you can share with your configuration provider, import them into your Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management environment from the Dataverse repository. Then, review the feature details, such as the Electronic reporting (ER) configurations and processing pipelines.

Set up integration with Dataverse

To use this functionality, you need a Dataverse environment that's connected to your Dynamics 365 Finance environment. For more information, see the following articles:

- Enable Power Platform Integration
- Connect finance and operations apps with a new Microsoft Dataverse instance
- Connect finance and operations apps with an existing Microsoft Dataverse instance

Also, add a security role to access tables in Dataverse by following these steps:

1. Create a new security role as described in Create or edit a security role.
2. In that security role, add **Create, Read, Write, Delete, Append, Append to, Assign, Share** permissions for **Organization** to the following tables: **Electronic Reporting Configuration File, Electronic Reporting Configurations Index File, Globalization Feature File, and Globalization Features Index File**.
3. Assign the created security role to users as described in Assign security roles.

Import a feature from the Dataverse repository

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. Select **Import** to open the **Import feature from Dataverse repository** lookup.

3. Select **Synchronize** to sync your organization's Finance or Supply Chain Management environment and browse the Electronic invoicing features that are available for a specific configuration provider in the Dataverse repository. After synchronization is completed, the list of available Electronic invoicing features is updated from the Dataverse repository. This update might take a few minutes.
4. Select the feature to import, and then select **Import**.

To view the imported Electronic invoicing feature, make sure that the correct configuration provider is selected. By default, the portal filters out features that the active configuration provider created. You can adjust the filter to view features that other configuration providers created, such as Microsoft.

> **Note**
>
> It can take up to two weeks for a new Electronic invoicing feature released by Microsoft or a new version of a feature released by Microsoft to appear in the Dataverse repository. If you expect a feature update but it doesn't appear in the Dataverse repository, or if auto-updates are disabled in your environment, you can also manually trigger the update of the Globalization solution package in your Power Platform admin center portal. To learn more about how to update solutions in Power Platform admin center, see Manage Dynamics 365 apps.

You can now work with the imported feature. You can review its details and create a new feature by using the imported feature as a template.

> **Note**
>
> You can modify a feature only if the active configuration provider created it. You can create a new feature by using the original feature as a base. You can then make the required changes or set up the parameters.

Additional information

Import Electronic reporting (ER) configurations from Dataverse

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Last updated on 03/21/2026

Create Globalization features

> **Note**
>
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You can create an Electronic invoicing feature from scratch, or you can base it on an existing feature. When you create a feature from scratch, you must manually add Electronic reporting (ER) configurations and other components, such as the feature setup and application setup details. When you create a feature based on an existing feature, the new feature inherits all the base feature's configurations and parameters. You can review what's copied from the base feature to the new feature.

Create a feature based on an existing feature

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. In the **Configuration provider** field, make sure that you select your active configuration provider. By using this configuration provider, the new Electronic invoicing feature appears in the list after you create it. If you don't select your active configuration provider, the new feature is filtered out of the list.
3. Select **Add**, and then, in the dropdown dialog box, select the **Based on existing version** option.
4. Enter a name and description for the feature.
5. In the **Base feature** field, select the base version of the feature.
6. Select **Create feature**. The feature is created and has a status of **Draft**.
7. Review the feature components to determine whether updates are required:
    - Review the configurations in case you need to customize the ER formats and their binding with format mappings for the feature version.
    - Review the setup in case you need to customize the **Actions** tab, **Applicability rules** tab, or **Variables** tab for the feature version.
8. Complete the setup, and deploy the Electronic invoicing feature. For more information, see Complete and deploy a Globalization feature.

Create a feature from scratch

This section explains how to create an Electronic invoicing feature when the Global repository doesn't include a Globalization feature for your business scenarios.

To create an Electronic invoicing feature, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. Select **Add**, and then, in the dropdown dialog box, select the **New feature** option.
3. Enter a name and description for the Electronic invoicing feature.
4. Select **Create feature**. The first version of the new feature appears on the right side of the page and has a status of **Draft**.

   Next, you must add ER configurations and feature setups. Before you add new or existing ER format configurations, make sure that they exist in your Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management environment.
5. Select the Electronic invoicing feature that you're working on.
6. On the **Configurations** tab, select **Add**.
7. In the **Configurations** grid, browse to and select the format configurations that are required for the processing pipeline (for example, to generate electronic invoice files or process responses from external web services).
8. Select **OK**. You can now use the configurations in actions of the processing pipeline. For more information, see Work with configurations.
9. To add an Electronic invoicing feature setup, create it on the **Setups** tab of the **New feature** page. For more information, see Work with feature setups.
10. Complete the setup, and deploy the Electronic invoicing feature to the service environment. For more information, see Complete and deploy a Globalization feature.

Create file format configurations that are derived from the existing invoice model

You can skip this section if you don't need to create ER configurations but can reuse existing versions as a base.

This procedure shows how to create the file format configurations required for the new Electronic invoicing feature that you want to create. Create the electronic invoice file format

configuration and any other file format configurations that your new Electronic invoicing feature requires.

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. Select **Invoice model** or, for Brazilian scenarios, **Fiscal model**.
3. Select **Create configuration**, and then, in the dropdown dialog box, select the **Format based on data model Invoice** option.
4. Enter a name and description for the format configuration.
5. In the **Format type** field, select the file extension type.
6. In the **Data model definition** field, select the root structure to map your format to. For example, **InvoiceCustomer** is used for the customer invoice formats.
7. Select **Create configuration**.
8. Select the new format configuration.
9. Select **Designer**, and use the Format designer tool to configure the file layout so that it meets the file format specifications.
10. Close the page.
11. On the **Versions** tab, select **Change status** > **Complete**.

Last updated on 03/18/2026

Delete Globalization features

You can delete an Electronic invoicing feature.

To delete an Electronic invoicing feature, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. Select the feature you want to delete from the feature list.
3. Select the **Delete** button. The **Delete** button is located in the workspace toolbar and is only enabled when deletion is allowed for the selected feature.
4. Confirm the deletion when prompted.

> ⓘ **Note**
>
> If you attempt to delete a feature with deployed versions, you receive the following error: `Feature cannot be deleted as it has deployed version. Undeploy version and retry.`

Other Considerations

Feature Relationships

Consider the following before deleting a feature:

Derived Features: If other features are derived from (inherit from) the feature you want to delete, check if those relationships are affected.

- Features can have a base feature relationship tracked in the `Base` field.

Active Provider: You can only delete features owned by the active configuration provider.

- The active provider is set in the ER parameters.

Draft Versions: Features with only draft versions (not deployed) can be deleted freely.

Best Practices

- **Check Dependencies**: Before deleting, verify that no other features depend on this feature as a base.

- **Export Configuration**: Consider exporting the feature configuration to JSON as a backup before deletion.
- **Verify Deployments**: Always double-check that all deployments are properly removed.

Last updated on 02/04/2026

Work with Electronic reporting (ER) configurations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) configurations are one of the main sets of components of Electronic invoicing features. An ER configuration contains the setup of the file structure and a set of transformation rules for transforming data in two ways:

- **Export ER format configuration** – This configuration transforms data from the unified internal structure (ER model) to the electronic file format.
- **Import ER format configuration** – This configuration transforms data from the electronic file format to the unified internal structure (ER model).

In addition to generating outbound electronic files in the predefined format, you can use ER configurations to parse incoming messages from external web services as responses. This functionality helps you build configurable logic to get the results of communication with external channels, convert those results (codes, messages, and logs) to a user-readable structure, and then pass that information to the client application.

To start using ER configurations in your Electronic invoicing feature, link them to the feature and make them available on the **Configurations** tab for the current feature version. You can work with a linked ER configuration by selecting **Add**, **Delete**, **Edit**, or **View**.

Before you can add a link to an ER configuration, the configuration must exist in your Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management environment. To review the available ER configurations, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile.
2. In the **Configurations** section, select the **Reporting configurations** tile.

For information about how to create a new ER configuration or import an existing ER configuration from the Global repository, see Electronic reporting.

To adjust a linked ER configuration, select **Edit** on the **Configurations** tab for the current Electronic invoicing feature. The system automatically creates a version of the ER configuration. If the active configuration provider created the current version, the system creates a new

version. If the active configuration provider didn't create the current version, the system creates a derived version that uses your configuration provider. In both cases, you can modify the created version and then automatically make all required updates.

Last updated on 03/21/2026

Configure feature setups

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To set up the generation of electronic files and other processing steps (such as digitally signing files, submitting them to the government web service and receiving a response, or storing them), set up the processing pipeline, applicability rules, and variables for Electronic invoicing features.

Combine the setup information in the *feature setup* concept. You can create, delete, and adjust feature setups. You can also view the setup information for completed versions of Electronic invoicing features.

Create as many feature setup items as you need to define different scenarios for generating, processing, and receiving electronic files. Although these feature setup items have independent processing actions and conditions for execution, they belong to a single Electronic invoicing feature and inherit its lifecycle and deployment process.

Add a feature setup

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. On the **Electronic invoicing features** page, select an Electronic invoicing feature version that has a status of **Draft**.
3. On the **Setups** tab, select **Add**.
4. In the **Create feature setup** dropdown dialog box, in the **New** field group, select one of the following options:
    - **Custom setup** – Create a new feature setup that has empty channels, an empty processing pipeline list, an empty section for applicability rules, and a default set of variables, depending on the setup type.
    - **From feature setup** – Create a copy of another feature setup in the scope of the current Electronic invoicing feature.

5. If you selected the **Custom setup** option in the last step, enter a name and description for the feature setup item. Then, in the **Setup type** field group, select one of the following options:

    - **Processing pipeline** – Select this option to generate and process outbound electronic documents. For this setup type, the system creates an empty processing pipeline list, an empty section for applicability rules, and a default set of variables. You can't work with the channels for inbound electronic documents.
    - **Import channel** – Select this option to set up the process of receiving inbound electronic documents from one of the defined channels and passing them directly to Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management without additional actions. For this setup type, the system creates a predefined list of parameters for the data channels, an empty section for applicability rules, and a set of default variables. You can't add any actions in the processing pipeline.
    - **Import channel and processing pipeline** – This setup type resembles the **Import channel** setup type. However, before an inbound electronic document is passed to Finance or Supply Chain Management, you can set up additional actions in the processing pipeline.
    - **Export channel and processing pipeline** – You can set up additional actions in the processing pipeline for outbound electronic documents that are submitted from Finance or Supply Chain Management.

6. If you selected the **Import channel**, **Import channel and processing pipeline**, or **Export channel and processing pipeline** option in the last step, in the **Select data channel** field, select the channel to integrate with.

7. Select **Create**.

After you create a feature setup, select **Edit** to configure it.

Edit a feature setup

Depending on the type of feature setup, you can configure the process of generating outbound electronic files and submitting them to external channels, or the process of receiving inbound documents and passing them to your Finance or Supply Chain Management application.

Import channel

An *import channel* either processes the electronic file or passes it directly to Finance or Supply Chain Management. This option isn't available for feature setups of the **Processing pipeline** type.

To set up an import channel, enter the name of the channel. Then define the parameters based on the selected channel type. The import channel identifier must refer to the variable that's created specifically to identify the import channel. Finance or Supply Chain Management uses it as a reference.

On the **Attachments filter** tab, define a set of filters to get specific files from the channel. You can create several lines if you expect that files will have different file name extensions or if files must be processed differently depending on the file name. Here are the main parameters:

- **Name** – Enter the name of the file that the system refers to during processing. In the Finance and Supply Chain Management applications, the files appear in the submission log.
- **Filter** – Define the filter to select files.
- **Optional** – If you clear this checkbox, the file is required. In this case, the system shows an error message if the channels don't contain files that correspond to the filter. This parameter is applicable to emails.

If the channel is a mailbox, and an inbound email contains several attachments, you can configure rules to define how the service should handle the attachments. The service can consider each attachment a separate electronic invoice, or it can treat one attachment as a main invoice and all other attachments as additions.

Processing pipeline

A *processing pipeline* is a set of *actions* that run sequentially until they finish successfully. Use a processing pipeline to set up the process for generating electronic files and performing other steps, such as digitally signing files, submitting them to external web services and parsing the response, and receiving and processing inbound documents.

The list of actions is predefined. Use the **New** and **Delete** buttons to specify the combination of actions that logically defines the required process for submitting or receiving documents.

The order of actions is important. Use the **Up** and **Down** buttons to adjust the order.

Each action has a set of parameters that you can configure or adjust. The specific set of parameters depends on the type of action.

- **Action** – Select the type of action.
- **Action name** – Enter the name of the action. This name appears in submission logs, and in messages in Finance and Supply Chain Management applications.
- **Description** – Provide more details about the purpose of the action in the process.

- **Enable retry** – If you select this checkbox, you can select an action in the **Retry action** field and configure additional parameters on the **Retry parameters** tab.

  By using the retry functionality, you can configure the service's behavior if it can't run a specific action. For example, if the external web service that you're trying to connect to doesn't respond, you can specify how many times the system should reattempt the connection, at what interval, and from what action in the processing pipeline.

- **Export results** – Select this checkbox for *one* action in the processing pipeline. You can export the electronic file that the action produces in the Finance or Supply Chain Management application from the **Electronic documents submission log** page.

Applicability rules

Applicability rules are configurable clauses that provide context for running Electronic invoicing features through the Electronic Invoicing capability set. Business documents that Finance or Supply Chain Management submits to Electronic invoicing don't carry an explicit reference that enables the Electronic Invoicing capability set to call a specific version of an Electronic invoicing feature and a specific feature setup item to process the submission. However, when you correctly configure a business document, it contains the required elements that enable Electronic invoicing to determine which Electronic invoicing feature version and processing pipeline to select and run.

By using applicability rules, the Electronic Invoicing service can find the exact Electronic invoicing features it needs to process a submission. To find the correct features, the service matches the **Context** fields from the submitted business document against clauses from the applicability rules.

You can work with applicability rules in the following ways:

- Select **New** to add a new clause to a set of applicability rules.
- Select **Delete** to delete a clause.
- Select several records, and then use the **Group** or **Ungroup** button to create a combination of items or logical statements. For example, select the lines that you want to group, and then select **Group clause**. When you group clauses, the grid adds a new column. This column specifies the logical operator for the grouped clause. The following operators are supported:
  - Equal
  - Not equal
  - Greater than
  - Less than

- Greater than or equal to
- Less than or equal to
- Contains
- Begin with

> **Note**
>
> When you ungroup a clause, always start from the innermost grouping level.

Variables

Variables give you more flexibility when you set up a processing pipeline and the data flow among actions. You can refer to a variable in some action parameters to temporarily store data or files. Some variables pass data between the Finance or Supply Chain Management application and the Electronic Invoicing service.

The system creates some variables by default. For example, the **BusinessDocumentDataModel** variable contains business data in a JavaScript Object Notation (JSON) structure. This data comes in the call from the Finance application during the submission process.

The following types of variables are available:

- **Constant** – The variable is a container for storing temporary data that's used in processing pipeline actions.
- **From client** – The variable content is acquired from the Dynamics 365 client when the submission process runs.
- **To client** – The variable content is made available for import by the Dynamics 365 client when the submission process runs.
- **Data type** – Select the data type of the information that's stored in the variable.

Parameters

Use the **Disable business data reduction** option to optimize the structure of the business data payload that comes from the Finance or Supply Chain Management application during electronic document submission. Optimization helps reduce the data that goes into the Electronic Invoicing service for further transformation into the required electronic file. The default value is **No**.

- **Yes** – Finance or Supply Chain Management sends a JSON file of the **Invoice Model** or, for Brazil, **Fiscal Model** structure that's defined in the **Electronic reporting** module. All elements of the model are filled with business data on the application side, regardless of

the structure of the final electronic file. Models usually contain more business data than the target file structure requires, and generating this data in the application can take more time. Set this option to **Yes** in the rare event where you want to generate different output files in one Electronic invoicing feature and feature setup. A value of **Yes** is useful when you troubleshoot your scenarios, the structure of the electronic files, and completeness of the business data.

- **No** – Finance or Supply Chain Management makes the first call to the service without any business data. The purpose of this call is to get information about the Electronic reporting (ER) configuration that the processing pipeline uses for electronic file transformation. This information returns to the application, which uses it to determine the subset of business data that should be included in the JSON file of the **Invoice Model** or, for Brazil, **Fiscal Model** structure. A value of **No** helps reduce the volume of business data that the application submits to the service, because the application can submit only the business data that's required to successfully generate an electronic file.

Validate a feature setup

You can run validation to check the consistency of the whole configuration. For example, if you leave a mandatory parameter for an action blank, validation detects this inconsistency and you receive a warning message.

- On the **Feature version setup** page, on the action bar, select **Validate**.

Delete a feature setup

1. On the **Electronic invoicing features** page, select an Electronic invoicing feature version that has a status of **Draft**.
2. On the **Setups** tab, select **Delete**.
3. In the message box, select **Yes** to confirm that you want to delete the feature setup.

Last updated on 03/21/2026

Complete and deploy a Globalization feature

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic invoicing feature versions

Electronic invoicing features are versioned. When a new version is created, the version number is automatically incremented.

Electronic invoicing feature versions follow a lifecycle that has up to two statuses:

- **Draft** – When a feature version has this status, you can edit its configuration attributes and its artifacts (for example, file format configurations).
- **Completed** – This status indicates that you finished editing the feature version and don't intend to make any more updates to it. When a feature version has this status, you can no longer edit it or any of its components.

Specify an **Effective from** date for any new version of an Electronic invoicing feature. By using this date, you can define a default version that the deployment uses or a version that it overwrites.

To change the status of an Electronic invoicing feature version, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. On the left side of the **Electronic invoicing features** page, select the Electronic invoicing feature.
3. On the **Versions** tab on the right side of the page, select the version.
4. Select **Change status**, and then select **Complete**.
5. In the message box, select **Yes** to confirm the request.

You can track the status in the **Feature version status** column on the **Versions** tab.

Deploy feature versions

Use the **Deploy** command to deploy a version of an Electronic invoicing feature to the Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management environment.

1. On the left side of the **Electronic invoicing features** page, select the Electronic invoicing feature.
2. On the **Versions** tab on the right side of the page, select the version to deploy. The selected version must have a status of **Completed**.
3. Select **Deploy**, and then select the date when the feature version becomes effective.

Remove feature versions

Select **Cancel deployment** to remove a specific version of an Electronic invoicing feature from the service, if you deployed it there. The **Deployed to the service** column indicates whether a version is deployed to the service.

Rebase Electronic invoicing features

When one Electronic invoicing feature is derived from another feature, select **Rebase** to update the derived feature with the changes that you made in the original (parent) feature.

To rebase a derived version of a feature that you created, follow these steps:

1. Get the latest version of the feature by importing it from the Global repository. For more information, see Import features from the repository.
2. In the list of features, select the feature to rebase.
3. On the **Versions** tab, select **New** to create a draft version.
4. Select **Rebase**.
5. In the **Rebase** dialog box, select the version of the feature to rebase to.
6. Select **OK**.
7. Review the feature components, and make any required changes.
8. Select **Change status** to complete the rebased feature. When the rebase is completed, you can perform additional actions.

Get a specific version of Electronic invoicing features

When you create a new version of an Electronic invoicing feature, the system copies the latest feature version. To use an earlier version of the feature as the base for a new version, select the version, and then select **Get this version**. The system creates a new draft version of the feature that's a copy of the selected version.

Last updated on 03/21/2026

Usage management dashboard

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the **Usage management** dashboard to monitor your usage of the Electronic Invoicing service. This dashboard helps your organization stay compliant with monthly usage requirements. It calculates compliance by comparing your submission volume with your acquired volume of submissions.

The dashboard includes the following indicators:

- One cost indicator to monitor consumption for licensing compliance:
  - Usage per month (export)
- Three operational indicators to track usage of the Electronic Invoicing service for management purposes:
  - Usage by feature
  - Usage by environment
  - Usage per month (import)

Measurement scope

- Unit of measure:

  The **Usage management** dashboard counts the submission of business documents and imported electronic invoices.

- Organization:

  The dashboard summarizes counts by the tenant ID of your organization, regardless of the number of instances of Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management, or the number of legal entities where you enable the Electronic Invoicing service.

Indicator: Usage per month (export)

This indicator shows details about the electronic invoices that your organization issues during a defined period.

Scope

- The number of business documents that Finance and Supply Chain Management submits to the Electronic Invoicing service, regardless of the number of lines that those business documents contain.
- The number of submitted business documents that the Electronic Invoicing service processes successfully. A document is considered successfully processed when the flow of actions defined in the feature setup is completed.

> **Note**
>
> When you submit an electronic invoice to an external web service, counting is independent of the results of web service processing (accepted, rejected, or schema validation error). If the web service receives and responds to the request from the Electronic Invoicing service, the submission is considered a valid counting.

- **Exception:** The system doesn't count business documents if Finance and Supply Chain Management resubmits them to the Electronic Invoicing service. This exception applies to scenarios where you need to resubmit the same business document to change the status of the electronic invoice. For example, the system counts the issuance of a Brazilian electronic invoice (fiscal document) as valid, but it doesn't count the cancellation request for it.

Calculation

Calculate the balance as follows:

Balance = Free + Purchased – Used

Where:

- Free:

  A free volume of business documents you can submit each month. It includes a package of 100 business documents that you can submit to the Electronic Invoicing service. The free volume isn't cumulative, and you can't roll over any remaining balance to the next period.

- Purchased:

A package of 1,000 business documents that you can submit to the Electronic Invoicing service. You must acquire this package for your organization on a monthly basis. The purchased volume isn't cumulative, and you can't roll over any remaining balance to the next period.

- Used:

  The count of business document submissions to the Electronic Invoicing service according to defined criteria.

> **Important**
>
> - If your organization plans to exceed the monthly volume of 100 free submissions, purchase the peak volume to ensure that you remain compliant with the Microsoft licensing policy for the Electronic Invoicing service.
> - Currently, you must manually enter the purchased volume, according to the date of acquisition, as multiple packages of 1,000 submissions.

Indicator: Usage by feature

This indicator shows the usage of electronic invoicing features for issuing electronic invoices during a defined period.

- Calculation:

  Used = The count of how many times each electronic invoicing feature was used during the submission of business documents to the Electronic Invoicing service.

Indicator: Usage by environment

This indicator shows the usage of electronic invoicing service environments during a defined period.

- Calculation:

  Used = The count of how many times each electronic invoicing service environment was used during the submission of business documents to the Electronic Invoicing service.

Indicator: Usage per month (import)

This indicator shows the volume of imported electronic invoices by the Electronic Invoicing service into Finance and Supply Chain Management during a defined period.

- Scope:

  Electronic invoices that are imported into Finance and Supply Chain Management, regardless of the number of lines that those electronic invoices contain.

- Calculation:

  Received = The count of imported electronic invoices into Finance and Supply Chain Management.

Functions

Purchase

On the **Usage per month (export)** tab, select **Purchase** to manually register the amount of acquired submissions.

For a given date, select **New** and enter the number of **Packages** acquired for that date.

The **Quantity** is calculated as:

Quantity = Packages * 1,000

The calculated **Quantity** appears in the **Purchased** section of the **Usage per month (export)** indicator.

Update

On the action pane, select **Update** to refresh the calculation and update the data shown on the page and in the chart.

Reset history data

On the action pane, select **Reset history data** to refresh the database from where the indicators are calculated.

> ⓘ **Note**

The **Usage management** dashboard doesn't show monetary amounts. Instead, it shows only the volume of counted submissions and imported documents.

Last updated on 03/14/2026

Process incoming electronic documents

You can import and process incoming electronic documents, such as vendor electronic invoices, in two ways:

- Retrieve files from external channels and pass them directly to your connected application. In the connected application, files undergo additional transformation before you import them into the database.
- Preprocess files in the Electronic Invoicing service before passing them to your connected application.

Electronic invoicing supports two channels for incoming documents: email and Microsoft SharePoint folders.

Use two setup types to specify whether documents undergo preprocessing or are passed directly to your connected application:

- **Import channel** – The system passes the document directly to the connected application.
- **Import channel with processing pipeline** – Set up additional actions that run before the document is passed to the connected application.

For information about how to set up the scenarios for processing incoming electronic documents for the different channels, see Configure an email channel and Configure a SharePoint channel.

Last updated on 06/26/2026

Configure email channels

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

If the Electronic invoicing feature that you created imports electronic vendor invoices from attached files that you receive by email, configure an email account channel.

1. Select the Electronic invoicing feature that you created. Select the version that has a status of **Draft**.
2. On the **Setups** tab, select **Add**.
3. In the **Create feature setup** drop-down dialog box, in the **New** field group, select the **Custom setup** option.
4. In the **Setup type** field group, select the **Data channel** option.
5. In the **Select data channel** field, enter **Incoming e-mail**.
6. Select **Create**.
7. Select the line that you created, and then select **Edit**.
8. On the **Data channel** tab, in the **Parameters** section, set the following required fields.

Expand table

Field	Description
Data channel	Enter a unique name to identify the data channel. The name can have a maximum of 10 characters. The system references it in applicability rules and in connected applications during the communication process.
Server address	Enter the server address of the email account provider for Internet Message Access Protocol (IMAP). For example, the server address for the `https://outlook.live.com` provider is imap-mail.outlook.com.
Server port	Enter the number of the port that the email account provider uses. For example, the server port for the `https://outlook.live.com` provider is 993.

Field	Description
User name secret	Enter the name of the Microsoft Azure Key Vault secret that contains the ID of the email user account. Create this secret in Key Vault and set it up in your service environment.
User password secret	Enter the name of the Key Vault secret that contains the password of the email user account.
Timeout	Enter the maximum time limit, in milliseconds (ms), that the system should wait for a response. The default value is 10,000 ms (10 seconds).
Main folder	Specify the email import source, or the folder that the service should process emails from.
Archive folder	Specify the folder where processed emails should be stored. If you don't specify this folder, the system automatically creates it.
Error folder	Specify the folder that the system should move emails to if the processing fails. If you don't specify this folder, the system automatically creates it.
Max message size	Enter the maximum size, in bytes, of a single message that is processed. The default value is 20,000,000 bytes.
Max message number	Enter the maximum number of messages to process for a single action. If you don't want to limit the number of messages, set the value to **0** (zero).
From filter	Enter a string to filter by sender address. Only emails where the sender address matches the filter are processed. This field is optional. To specify multiple sender addresses, use semicolons (;) as separators.
Subject filter	Enter a string to filter by subject. Only emails where the subject matches the filter are processed. This field is optional. A simple mask such as ***smth*.ext** is supported, where each asterisk (*) represents zero or more occurrences of any character.
Date filter	Specify a date to define the maximum age, in days, of messages that are processed. This field is optional. The default value is 30 days.
Processing mode	Select one of the following options to specify whether the system can process all the attachments in an email together or whether it should process each attachment separately: • **By attachment** – The system creates a new electronic document for each attachment in the email. For example, if one email includes several files that contain e-invoice data, each file is considered a new e-invoice in the system. • **By email** – One attachment is considered a base attachment, and one electronic invoice is created in the system. Other attachments can be used as supporting files.

9. In the **Attachments filter** section, add the file filtering information. The system processes only attachments that satisfy the defined filter. For example, ***.xml** filters for attachments that have the .xml file name extension. Use the name of the attachment in Dynamics 365 Finance or Dynamics 365 Supply Chain Management during setup.

    - If you set the **Processing mode** field to **By email** in the previous step, you can add multiple filters. The name identifies the specific document.
    - If you set the **Processing mode** field to **By attachment**, you can add only one filter.

10. On the **Applicability rules** tab, review and update the criteria as required. The value of the **Channel** field must equal the value that you entered in the **Data channel** field in step 8.

11. Select **Save**, and close the page.

Last updated on 03/14/2026

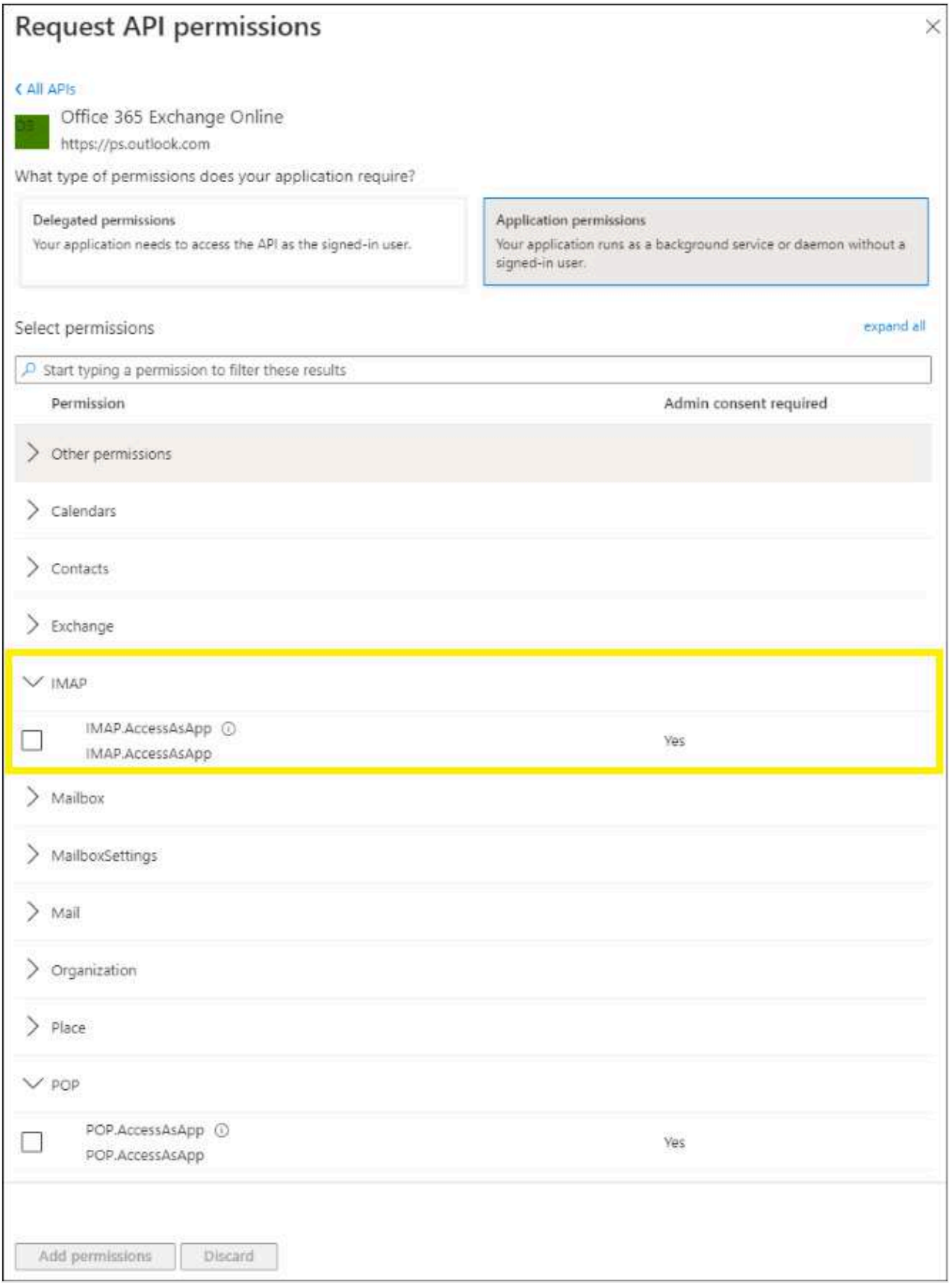
Configure an email channel for Office 365 Exchange Online

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

If you create the Electronic invoicing feature that imports electronic vendor invoices from attached files received in email, configure an email account channel.

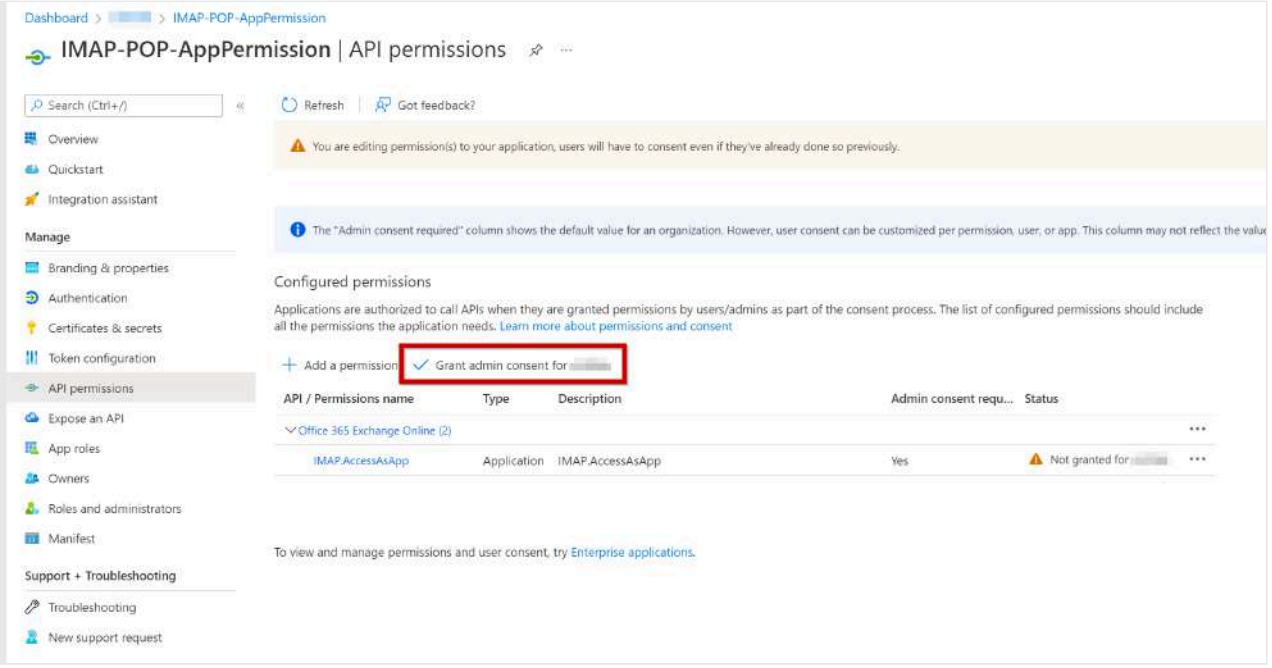
Add the IMAP permissions to your Microsoft Azure AD application

1. In the Microsoft Azure portal, select the **App registrations** service.
2. Select **New registration** to create an app registration.
3. Enter a name.
4. In the **Supported account types** section, select the **Accounts in this organizational directory only** option.
5. After the app registration is created, select **API Permissions**.
6. Select **Add permission**.
7. On the **APIs my organization uses** tab, search for **Office 365 Exchange Online**.
8. Select **Application permissions**.
9. For Internet Message Access Protocol (IMAP) access, select the **IMAP.AccessAsApp** permission.



10. Select **Add permissions**.

11. To access Exchange mailboxes through IMAP, your Azure Active Directory (Azure AD) application must get tenant admin consent for each tenant. For more information, see Introduction to permissions and consent.
    a. In the management view for your Azure AD application, select **API Permissions**.
    b. Under **Configured permissions**, select **Grant admin consent for <*name*>**.



12. Select **Overview**.
13. Copy the **Application (client) ID** and **Directory (tenant) ID** values.
14. Select **Certificates & secrets**.
15. On the **Client secrets** tab, select **New client secret**, and notice the value in the **Expiration** field. When a secret expires, it can't be used anymore, and you must create a new secret.
16. After you create a secret, copy the **Value** and save it in the Azure Key Vault that's associated with the Electronic Invoicing Service service environment.
17. Add a link to this secret in the Regulatory Configuration Service (RCS) service environment. For more information, see Customer certificates and secrets.
18. Save the **Application (client) ID** as a secret in the same key vault, and add a link to this secret in the Electronic Invoicing Service service environment in RCS.
19. In the Azure portal, select the **Enterprise Applications** service.
20. Find the application that has the same name as the app registration that you created in step 2.
21. Select **Overview**.
22. Copy the **Object ID** value.

Register service principals in Exchange

After a tenant admin consents to your Azure AD application, they must register the application's service principal in Exchange by running a series of PowerShell snippets through Exchange Online PowerShell. Some of these snippets use values that you copied in the previous procedure:

- <**ORGANIZATION_ID**> is the **Directory (tenant) ID** value that you copied.
- <**APPLICATION_ID**> is the **Application (client) ID** value that you copied.
- <**OBJECT_ID**> is the **Object ID** value that you copied.

Follow these steps to register service principals in Exchange.

1. Install the **ExchangeOnlineManagement** module.

   PowerShell

   ```
   Install-Module -Name ExchangeOnlineManagement -allowprerelease
   Import-module ExchangeOnlineManagement
   ```

   > ⓘ Note
   >
   > You only need to install the **ExchangeOnlineManagement** module once on the Exchange administrator's machine.

2. Connect to Exchange Online.

   PowerShell

   ```
   Connect-ExchangeOnline -Organization <ORGANIZATION_ID>
   ```

3. The following example shows how to register the Azure AD application's service principal in Exchange.

   PowerShell

   ```
   New-ServicePrincipal -AppId <APPLICATION_ID> -ServiceId <OBJECT_ID> -
   Organization <ORGANIZATION_ID>
   ```

   > ⓘ Note
   >
   > If you receive an error when you run the **New-ServicePrincipal** cmdlet, you probably don't have sufficient permissions in Exchange Online to perform the operation. Alternatively, you're trying to use a guest account to perform the operation.

4. The following example shows how to give the application's service principal access to one mailbox.

```
PowerShell
Add-MailboxPermission -Identity "john.smith@contoso.com" -User <OBJECT_ID> -
AccessRights FullAccess
```

Create a feature setup in RCS

1. In RCS, select the Electronic invoicing feature that you created. Select the version that has a status of **Draft**.
2. On the **Setups** tab, select **Add**.
3. In the **Create feature setup** dropdown dialog box, in the **New** field group, select the **Custom setup** option.
4. In the **Setup type** field group, select the **Data channel** option.
5. In the **Select data channel** field, enter **Incoming e-mail on Exchange**.
6. Select **Create**.
7. Select the line that you created, and then select **Edit**.
8. On the **Data channel** tab, in the **Parameters** section, set the following required fields.

Expand table

Field	Description
Data channel	Enter a unique name to identify the data channel. The name can have a maximum of 10 characters. It's referenced in applicability rules and in the Dynamics 365 Finance application during the communication process.
Server address	Enter the server address of Exchange Online. For example, the default value for Office 365 Exchange Online is **outlook.office365.com**.
Server port	Enter the number of the port that the email account provider uses. For example, the default value for Office 365 Exchange Online is **993**.
User name secret	Enter the name of the Key Vault secret that contains the ID of the email user account. You must create this secret in Key Vault and set it up in your service environment.

Field	Description
Client Id	Enter the name of the Key Vault secret that contains the **Application (client) ID** value for the app registration.
Client Secret	Enter the name of the Key Vault secret that contains the **Value** value of the app registration client secret.
Timeout	The maximum time limit, in milliseconds (ms), that the system should wait for a response. The default value is 10,000 ms (10 seconds).
Main folder	Specify the email import source, or the folder that the service should process emails from.
Archive folder	Specify the folder where processed emails should be stored. If you don't set a value, the service automatically creates a folder.
Error folder	Specify the folder that emails should be moved to if the processing fails. If you don't set a value, the service automatically creates a folder.
Max message size	Enter the maximum size, in bytes, of a single message that's processed. The default value is 20,000,000 bytes.
Max message number	Enter the maximum number of messages to process for a single action. If you don't want to limit the number of messages, set the value to **0** (zero).
From filter	Enter a string to filter by sender address. Only emails where the sender address matches the filter are processed. This field is optional. To specify multiple sender addresses, use semicolons (;) as separators.
Subject filter	Enter a string to filter by subject. Only emails where the subject matches the filter are processed. This field is optional. A simple mask such as ***smth*.ext** is supported. Here, each asterisk (*) represents zero or more occurrences of any character.
Date filter	Specify a date to define the maximum age, in days, of messages that are processed. This field is optional. The default value is 30 days.
Processing mode	Select one of the following options to specify whether you can process all the attachments in an email together, or whether you should process each attachment separately: • **By attachment** – A new electronic document is created for each attachment in the email. For example, if one email includes several files that contain e-invoice data, each file is considered a new e-invoice. • **By email** – One attachment is considered a base attachment, and one electronic invoice is created. Other attachments can be used as supporting files.

9. In the **Attachments filter** section, add the file filtering information. Only attachments that satisfy the defined filter are processed. For example, ***.xml** filters for attachments that have the .xml file name extension. You use the name of the attachment in Dynamics 365 Finance or Dynamics 365 Supply Chain Management during setup.

    - If you set the **Processing mode** field to **By email** in the previous step, you can add multiple filters. The name identifies the specific document.
    - If you set the **Processing mode** field to **By attachment**, you can add only one filter.

10. On the **Applicability rules** tab, review and update the criteria as required. The value of the **Channel** field must equal the value that you entered in the **Data channel** field in step 8.

11. Select **Save**, and close the page.

Last updated on 03/21/2026

Configure SharePoint channels

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Before you configure a Microsoft SharePoint channel to process incoming documents, complete the steps described in Configure a SharePoint connection.

After you complete those steps, you can use the SharePoint channel to import electronic vendor invoices from files that are stored in your SharePoint folders.

1. On your SharePoint site, create the following folders:
    - **Main folder** – The folder that the service processes files from.
    - **Archive folder** – The folder where processed files are stored.
    - **Error folder** – The folder that the system moves files to if the processing fails.
2. Select the Electronic invoicing feature that you created. Make sure that you select the version that has a status of **Draft**.
3. On the **Setups** tab, select **Add**.
4. In the **Create feature setup** drop-down dialog box, in the **New** field group, select the **Custom setup** option.
5. In the **Setup type** field group, select the **Data channel** option.
6. In the **Select data channel** field, select **SharePoint folder**.
7. Select **Create**.
8. Select the line that you created, and then select **Edit**.
9. On the **Data channel** tab, in the **Parameters** section, set the required fields.

Expand table

Field	Description
Data channel	Enter a unique name to identify the data channel. The name can have a maximum of 10 characters. The communication process references this name in the

Field	Description
	applicability rules and in connected applications.
SharePoint address	Enter the SharePoint URL. For example, enter `<domain>.sharepoint.com`.
Application Id	Enter the name of the Azure Key Vault secret that contains the ID of the SharePoint user account. You must create this secret in Key Vault and set it up in your service environment. The value is the **Application (client) ID** value from Configure SharePoint connection.
Application secret	Enter the name of the Key Vault secret that contains the password of the SharePoint user account. The value is the **App Registration secret** value from Configure SharePoint connection.
Site name	Enter the name of the SharePoint site.
Document library name	Enter the name of the SharePoint document library.
Timeout	Enter the maximum time, in milliseconds (ms), that the system should wait for a response. The default value is 10,000 ms (10 seconds).
Main folder	Specify the file import source, or the folder that the service processes files from.
Archive folder	Specify the folder where processed files are stored.
Error folder	Specify the folder that the system moves files to if the processing fails.
Max file size	Enter the maximum size, in bytes, of a single file that is processed. The default value is 20,000,000 bytes.
Max files number	Enter the maximum number of files to process for a single action. If you don't want to limit the number of files, set the value to **0** (zero).
File filter	Enter a string to filter by file name. This field is optional. A simple mask such as ***smth*.ext** is supported, where each asterisk (*) represents zero or more occurrences of any character. For example, enter ***.pdf** to configure the channel to read PDF files.
Custom file name	Enter the custom file name from the client.

10. On the **Applicability rules** tab, review and update the criteria as required. The value of the **Channel** field must equal the value that you entered in the **Data channel** field in the previous step.

11. Select **Save**, and close the page.

Last updated on 03/14/2026

Use the electronic invoicing service to import vendor invoices

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information to help you get started with importing vendor invoices by using the Electronic Invoicing service. It guides you through the configuration steps that you must complete in the **Globalization studio** workspace in Microsoft Dynamics 365 Finance and Dynamics 365 Supply Chain Management to receive electronic vendor invoices from vendors.

Set up vendor invoice import in the Electronic Invoicing service

To set up vendor invoice import in the **Globalization studio** workspace, follow these steps:

1. Consult the list of generally available Electronic invoicing features.
2. Select and import the Electronic invoicing feature. For more information, see Import features from the repository.
3. Create an Electronic invoicing feature for your organization. For more information, see Create Globalization features.

Configure an email account channel

Configure an email account channel if the Electronic invoicing feature you created imports electronic vendor invoices from attached files that you receive by email.

1. In the **Globalization studio** workspace, select the Electronic invoicing feature that you created. Make sure that you select the version with a status of **Draft**.

2. On the **Setups** tab, in the grid, select a feature setup, and then select **Edit**.

3. On the **Data channel** tab in the **Parameters** field group, enter the name of the channel in the **Data channel** field. The channel name can't be more than ten characters.

4. In the **Server address** field, enter the email account provider. For example, the server address for **https://outlook.live.com/** is **imap-mail.outlook.com**.

5. In the **Server port** field, enter the port used by the email account provider. For example, the server port for **https://outlook.live.com/** is **993**.

6. In the **User name secret** field, enter the name of the Key vault secret that contains the ID of the email user account. Create this secret in Azure Key Vault and set it up in your service environment.

7. In the **User password secret** field, enter the name of the Key vault secret that contains the password of the email user account.

8. (Optional) Enter values in the **From filter**, **Subject filter**, and **Date filter** fields.

9. Enter the names of the mailbox folders where mails are:

    - Imported from: **Main folder**
    - Saved after successful processing: **Archive folder**
    - Saved after not successful processing: **Error folder**

    You don't need to create these folders in the mailbox. The folders are created automatically after the first e-invoice import and processing.

10. In the **Attachments filter** field group, add the file filtering information. Only attachments that satisfy the defined filter are processed. For example, you can set up "*.xml" for attachments with an XML extension. Use the name of the attachment in Dynamics 365 Finance or Dynamics 365 Supply Chain Management during setup.

11. On the **Applicability rules** tab, review and update the criteria as necessary. The value of the **Channel** field must equal the **Data channel** value that you provided earlier. For more information, see Applicability rules.

12. Select **Save** and close the page.

Configure a Microsoft SharePoint channel

Configure a Microsoft SharePoint channel if the Electronic invoicing feature imports electronic vendor invoices from files placed in SharePoint folders.

1. In the **Globalization studio** workspace, select the Electronic invoicing feature that you created. Make sure that you select the version that has a status **Draft**.
2. On the **Setups** tab, in the grid, select a Feature setup and then select **Edit**.
3. On the **Data channel** tab, in the **Parameters** field group, select **SharePoint address** and enter the SharePoint URL.

4. Select **Server port** and enter the port used by the email account provider.
5. Select **Application Id** and enter the name of the Key vault secret that contains the ID of the SharePoint client.
6. Select **Application secret** and enter the name of the Key vault secret that contains the password of the SharePoint client.
7. Select **File filter** and review and update the mask to filter the files that contain the electronic vendor invoice to import.
8. On the **Applicability rules** tab, review and update the criteria as necessary. For more information, see Applicability rules.
9. Select **Save** and close the page.

Deploy an Electronic invoicing feature

To deploy the Electronic invoicing feature, see Complete and deploy a Globalization feature.

Set up vendor invoice import in Finance or Supply Chain Management

Complete the steps in the following section to set up a PEPPOL-type vendor invoice import as an example.

Import PEPPOL electronic vendor invoices

1. Go to the **Electronic reporting** workspace and select **Reporting configurations**.
2. Select **Customer invoice context model**, and then select **Create configuration** > **Derive from Name: Customer invoice context model, Microsoft** to create a derived configuration.
3. On the **Draft** version, select **Designer** and in the **Data model** tree, select **Map model to datasource**.
4. In the **Definitions** tree, select **DataChannel** and then select **Designer**.
5. In the **Data sources** tree, expand the **$Context_Channel** container.
6. In the **Value** field, select **Edit** > **Edit formula**, and enter the name of the channel as it's given in the configuration of the data channel for the Electronic invoicing feature in Globalization studio.
7. Select **Save** and close the page.
8. Close the page.
9. Select the derived configuration you just created from the **Customer invoice context model**, and on the **Versions** FastTab, select **Change Status** > **Completed**.
10. Go to **Organization administration** > **Setup** > **Electronic document parameters** and on the **Features** tab, make sure that **PEPPOL Global electronic invoices** is selected.

11. On the **External channels** tab, in the **Channels** field group, select **Add**.
12. In the **Channel** field, enter the data channel name and in the **Description** field, enter a description.
13. In the **Company** field, select the legal entity.
14. In the **Document context** field, select the new derived configuration from **Customer invoice context model**. The mapping description should be **Data channel context**.
15. In the **Import sources** field group, select **Add**.
16. In the **Name** field enter **Attachments filter name** and in the **Data entity name** field, select **Vendor invoice header**.
17. In the **Model mapping** field, select **Vendor invoice imprt - Import vendor invoice**.
18. Select **Save** and then close the page.

Receive electronic invoices

The Electronic Invoicing service performs two steps during invoice import from the data channels:

1. Accesses the mailbox and reads the emails.
2. Processes the emails.

To perform these two steps, the client must call the service manually for each step. However, set up a batch for receiving electronic documents.

To receive electronic invoices, follow these steps:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. Select **OK** and then close the page.

View receive logs for electronic invoices

To view the receive logs for electronic invoices, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**. If you don't see successfully processed invoices, remove the table filter.

Last updated on 03/14/2026

Electronic Invoicing service ISV last-mile connector

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The independent software vendor (ISV) last-mile connector complements the standard Electronic Invoicing service functionality when no direct integration with government electronic invoicing platforms or final electronic invoice recipients is supported out of the box. In these scenarios, Microsoft Dynamics 365 Finance is used to generate electronic documents in legally required formats. The documents are then passed to the ISV last-mile connector for further communication. In the case of incoming electronic documents, the ISV last-mile connector is used as a source of inbound documents, which are then handled by Finance.

This article explains how to configure and use the Electronic Invoicing service ISV last-mile connector.

Prerequisites

Before you begin the procedures in this article, the following prerequisites must be met:

- Your company must have a separate signed service agreement with an ISV that will provide electronic document delivery service. Additionally, you must obtain the required credentials to enable integration of the Electronic Invoicing service with the ISV last-mile connector.
- Become familiar with Electronic invoicing functionality. For more information, see Electronic invoicing overview.
- Consult the list of available country/region-specific Electronic invoicing features. The capability to use the ISV last-mile connector for submission complements this list.

Integration with Edicom

> **Note**
>
> The integration of the Electronic Invoicing service with the ISV last-mile connector is described based on **Danish electronic invoicing** example. For other countries/regions please refer to the respective section in the **Country/region-specific scenarios implemented via Edicom** chapter.

You can configure and use the Electronic Invoicing service ISV last-mile connector's integration with the Global e-Invoicing Platform that Edicom provides. For more information refer to: Edicom integration with Microsoft Dynamics 365.

To enable the integration of the Electronic Invoicing service with the ISV last-mile connector, you must obtain the following required credentials from Edicom:

- **Service ID** – The service ID number is used to identify the company to Edicom.
- **Group** – The group code is used for internal routing within the Edicom infrastructure.
- **Token** – The token is used to grant authorization to access the Edicom services.

The token that you obtain must be uploaded to the secret that you created in the Azure Key Vault that your company manages. For more information, see Customer certificates and secrets. The secret is used as a parameter in the related Electronic Invoicing feature pipeline actions.

> **Note**
>
> With this integration, you are able to submit generated e-invoices' files either in the **OIOUBL** format to the Danish electronic invoicing infrastructure, **NemHandel**, or in the **PEPPOL** format to the PEPPOL Delivery Network. The format of a submitted file will be dynamically analyzed by Edicom and routed correspondingly.

Electronic invoice submission

The following pipeline actions are introduced or updated to enable outbound document submission via the ISV last-mile connector.

- **Integrate with Edicom** – This new action submits electronic documents that were generated by using preceding actions to Edicom. You must configure the parameters for this action as described in the following table.

Expand table

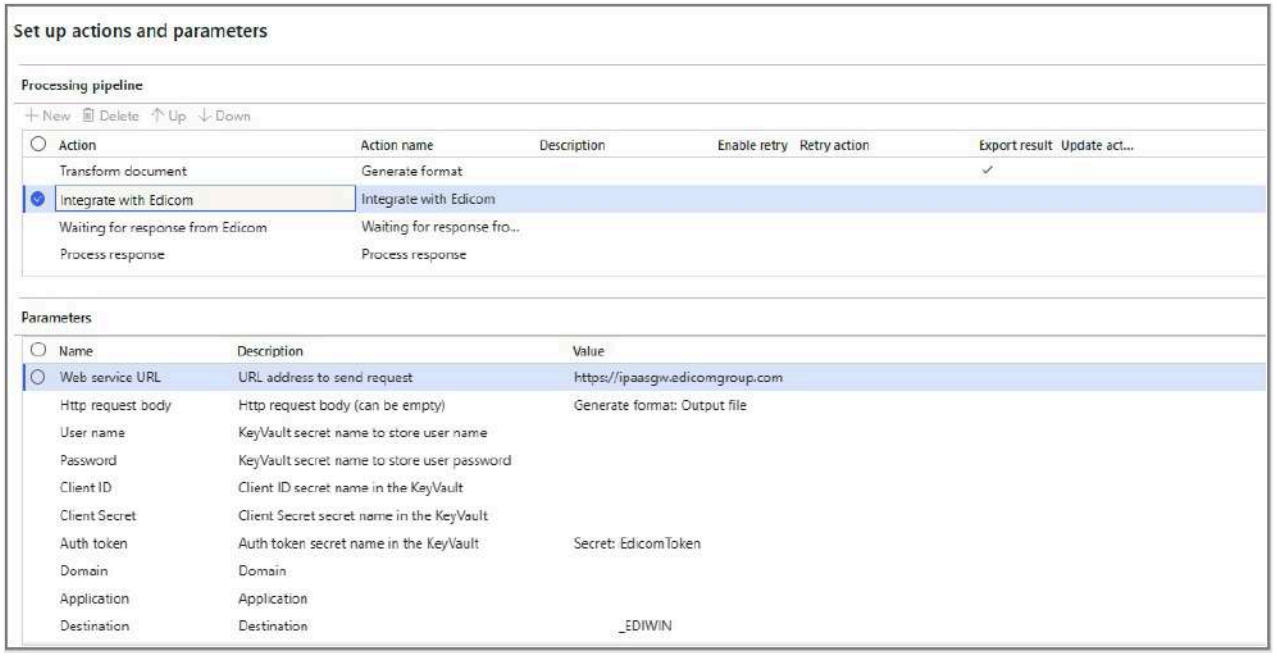
Parameter	Description
Domain	Enter the service ID number that Edicom provided.
Application	Enter the same service ID number.
Destination	Enter the service ID number concatenated with the string **_EDIWIN**. For example, if the service ID number is **123456**, enter **123456_EDIWIN**.
Group	Enter the group code that Edicom provided.
Auth token	Select the name of the secret that you created for the token that Edicom provided.

The following parameters are Edicom-specific. You can leave them set to the default values that Microsoft provides in related Globalization features.

Expand table

Parameter	Description
Web service URL	Default value for Edicom: `https://ipaasgw.edicomgroup.com`.
Http request body	Default value: **Generate format: Output file**.
Schema	Default value for Edicom: **OUTBOUND_DOCUMENT_MICROSOFT_DK**.

You can leave all remaining parameters blank.



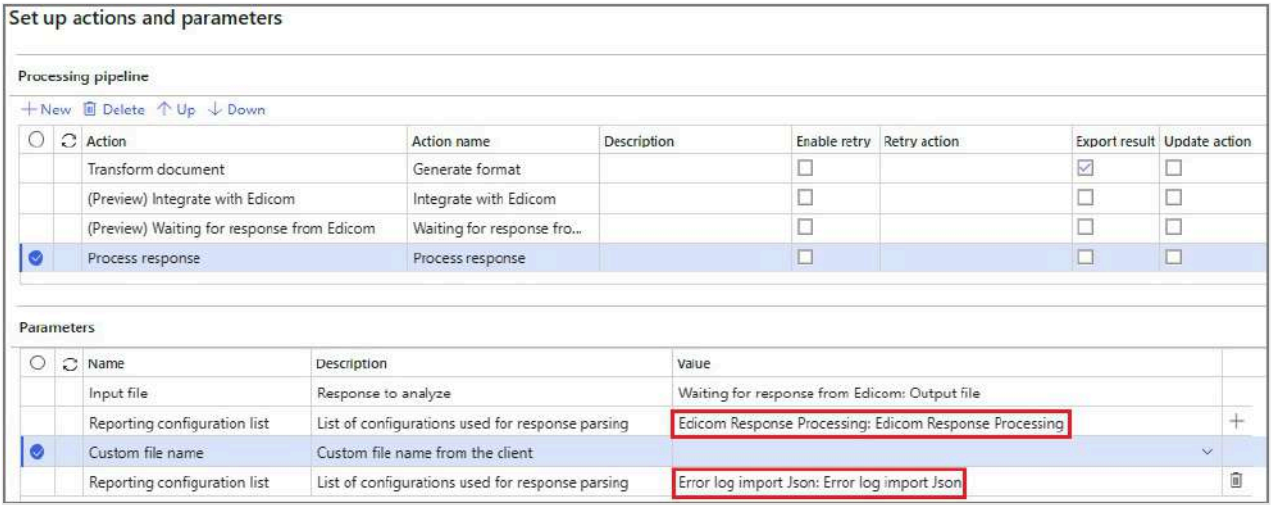
- **Waiting for response from Edicom** – This new action waits for the response from Edicom. No specific parameters must be configured.

- **Process response** – For this existing action, two new Electronic reporting (ER) configurations, **Edicom Response Processing** and **Error log import Json**, were created to handle the response that the **Waiting for response from Edicom** action receives from Edicom.

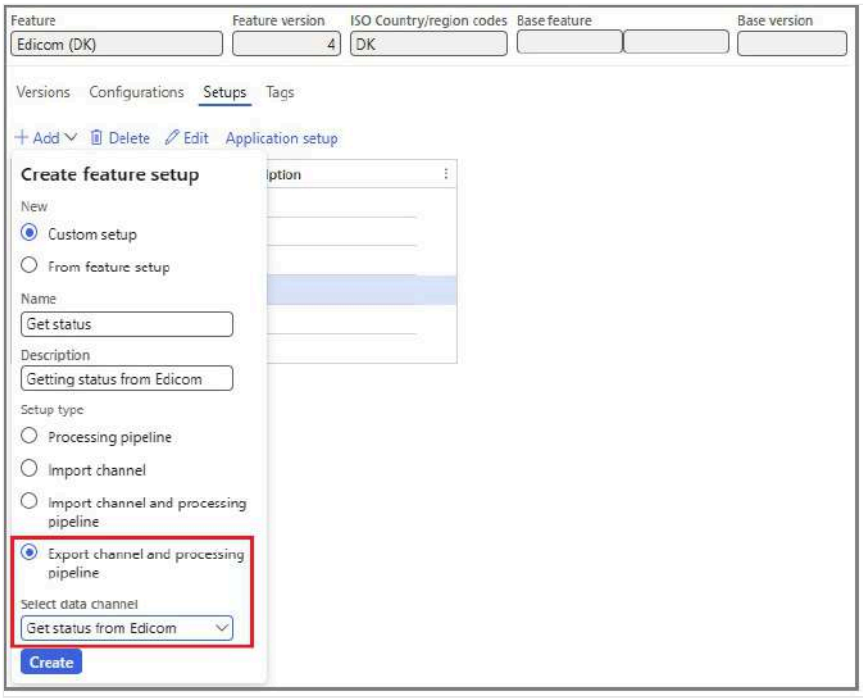
  The following parameters are Edicom-specific. You can leave them set to the default values that Microsoft provides in related Globalization features.

  Expand table

Parameter	Description
Input file	Default value for Edicom: **Waiting for response from Edicom: Output file**.
Reporting configuration list	The first new ER configuration. Default value for Edicom: **Edicom Response Processing: Edicom Response Processing**.
Reporting configuration list	The second new ER configuration. Default value for Edicom: **Error log import Json: Error log import Json**.



A new data channel type, **Get status from Edicom**, is implemented for feature setups of the **Export channel and processing pipeline** type. You must create a new feature setup where you select **Get status from Edicom** as the data channel.



Configure the export channel's parameters as described in the following table.

Expand table

Parameter	Description
Domain	Enter the service ID number that Edicom provided.
Application	Enter the same service ID number.
Data channel	Enter the name of the export channel to use. Enter the same name in each of the following places: • In the applicability rules of the parent's feature setup • In the value of the **$Context_Channel** variable in the **DataChannel** definition in the **Customer invoice context model** ER configuration that's used • At **Organization administration** > **Setup** > **Electronic document parameters** > **Integration channels** in Finance
Group	Enter the group code that Edicom provided.
Auth token	Select the name of the secret that you created for the token that Edicom provided.

You can leave the following parameters set to the default values that Microsoft provides in related Globalization features.

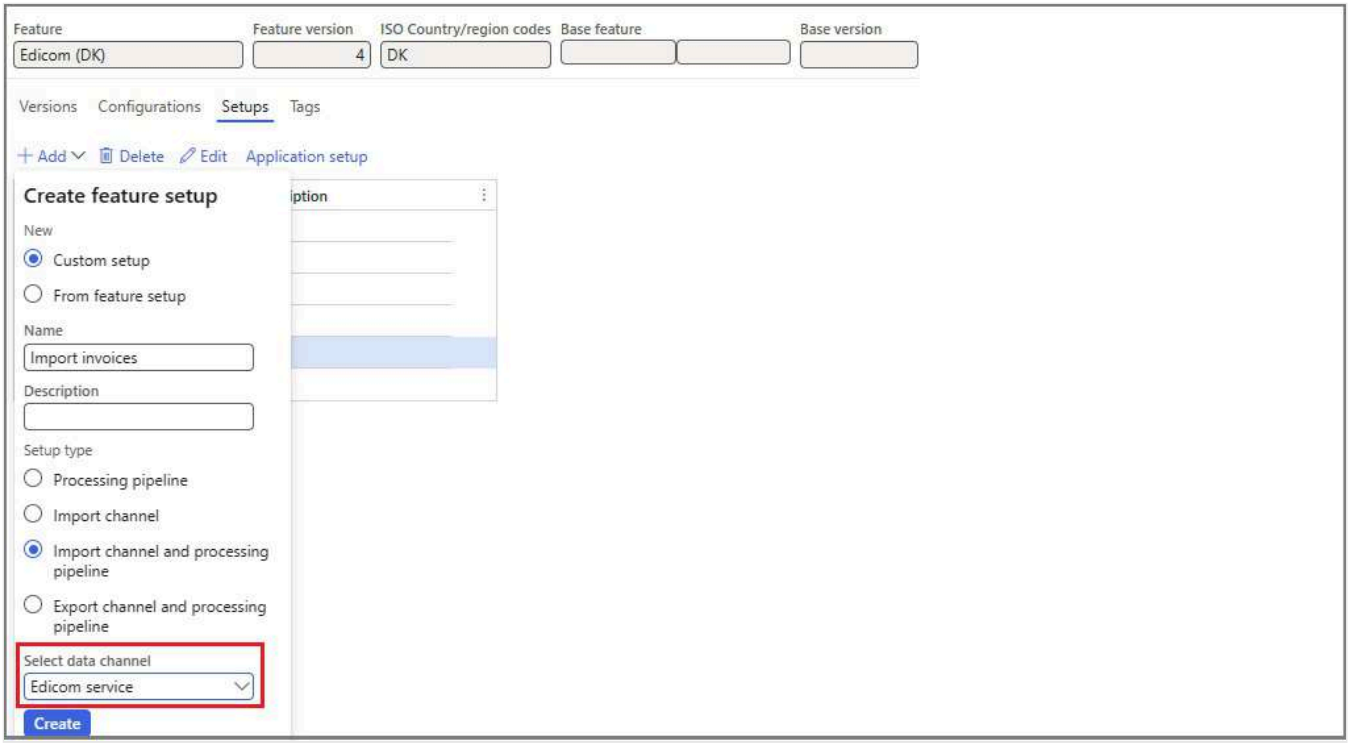
Expand table

Parameter	Description
Web service URL	Default value: `https://ipaasgw.edicomgroup.com`.
Document limit	Default value: **100**.
Schema	Default value: **IN_DOCUMENT_STATUS_MICROSOFT_DK**.

You can leave all remaining parameters blank.

Electronic invoice receipt

A new data channel type, **Edicom service**, is implemented for feature setups of the **Import channel** or **Import channel and processing pipeline** type. You must create a new feature setup where you select **Edicom service** as the data channel.



Configure the import channel's parameters as described in the following table.

Expand table

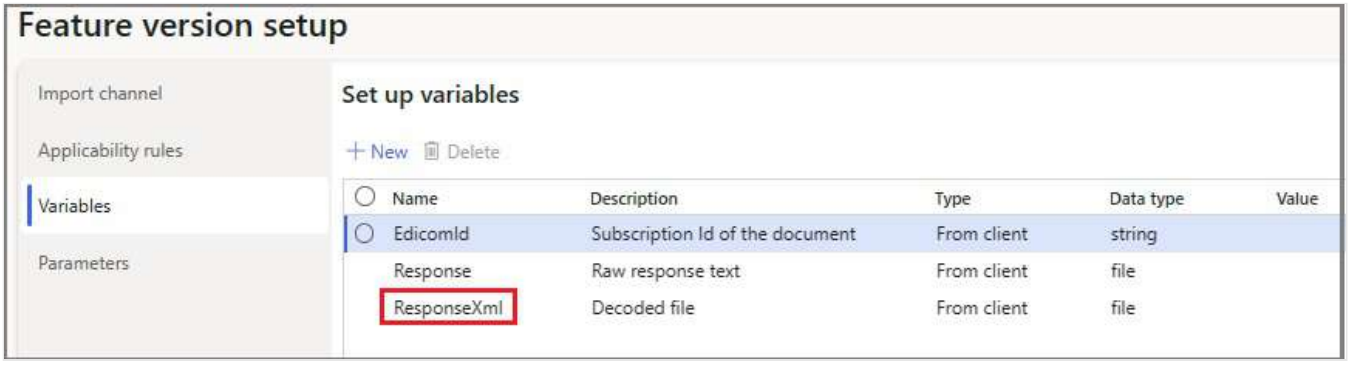
Parameter	Description
Domain	Enter the service ID number that Edicom provided.
Application	Enter the same service ID number.
Data channel	Enter the name of the import channel to use. Enter the same name in each of the following places:

Parameter	Description
	• In the applicability rules of the parent's feature setup • In the value of the **$Context_Channel** variable in the **DataChannel** definition in the derived version of the **Customer invoice context model** ER configuration that's used for the import scenario. **Important:** The derived configuration must differ from the configuration that's used for the invoice *submission* setup. • At **Organization administration** > **Setup** > **Electronic document parameters** > **Integration channels** in Finance
Group	Enter the group code that Edicom provided.
Auth token	Select the name of the secret that you created for the token that Edicom provided.

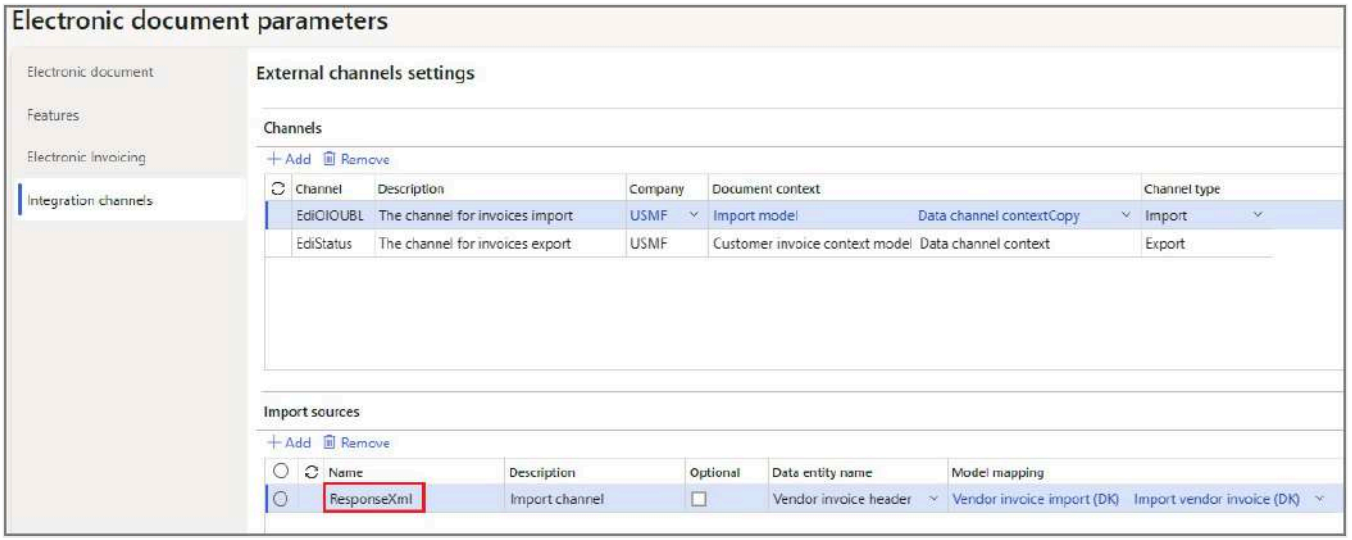
You can leave all remaining parameters set to the default values that Microsoft provides in related Globalization features.

Additionally, you must configure the following feature setup variable to use for the decoded file:

- **Name:** ResponseXml
- **Description:** Decoded file
- **Type:** From client
- **Data type:** file



The name of this variable must be used as the name of the import source for the import channel on **Integration channels** tab of the **Electronic document parameters** page in Finance.



Country/region-specific scenarios implemented via Edicom

Learn more about electronic invoicing for a specific courntry/region by selecting the appropriate article from the following table.

Expand table

Country/region	Documentation article
Belgium	Electronic invoicing for Belgium
Chile	• Get started with Electronic invoicing for Chile • Vendor electronic invoice import in Chile
Colombia	Get started with Electronic invoicing for Colombia
Costa Rica	Get started with Electronic invoicing for Costa Rica
Denmark	Use electronic invoicing for Denmark
Panama	Get started with Electronic invoicing for Panama
Paraguay	Get started with electronic invoicing for Paraguay.
Singapore	Electronic invoicing for Singapore
Uruguay	Get started with Electronic invoicing for Uruguay

Additional resources

- Electronic invoicing administration and integration components
- Electronic invoicing setup
- Work with Globalization features

Last updated on 04/17/2026

Submit electronic documents

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Submitting electronic documents creates a single point of communication between Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management and Electronic invoicing. During each submission event, communication flows in both directions:

- **From Finance or Supply Chain Management to Electronic invoicing** – Finance or Supply Chain Management sends the financial data to Electronic invoicing in a unified structure. As required, it also sends the contents of variables that were configured as part of the electronic invoicing features.
- **From Electronic invoicing to Finance or Supply Chain Management** – Depending on the electronic invoicing features, Finance or Supply Chain Management receive updates from Electronic invoicing about the processing results of invoices that were previously submitted. It also receives the contents of variables that were configured as part of the electronic invoicing features.

To submit electronic documents to Electronic invoicing from Finance or Supply Chain Management, go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**.

The starting point is a posted invoice. This invoice can come from different origins. For example, it can come from sales orders, project invoices, or free text invoices.

You can run the submission process manually or in the background.

- **Manual** – In the **Submit electronic documents** dialog box, on the **Records to include** FastTab, select **Filter** to define the range of documents that must be submitted. In the **Inquiry** dialog box, you can configure your own query to select the posted invoices that must be submitted. After you make your selection, manually start execution of the processing, and wait for it to finish running. When the processing is completed, a message in the Action center shows the number of electronic documents that are successfully submitted.
- **Background** – Background execution doesn't require that you be signed in or keep the **Submit electronic documents** dialog box open. You can schedule the process to run, and you can define the execution frequency.

> **Note**
>
> In some specific scenarios, the standard submission procedure that was described earlier only generates electronic invoices and stores them on the service side. The invoices aren't submitted. Submitting electronic invoices requires that you complete the following additional steps.

To submit the generated electronic invoices in batch mode, follow these steps:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.
2. In the **Channel** field, select the export channel, and then select **OK**. You must create the export channel in advance.

You can inquire about the results of the submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. For more information, see Work with Electronic document submission log.

Last updated on 03/14/2026

Work with Electronic document submission log

07/30/2025

In Microsoft Dynamics 365 Finance and Dynamics 365 Supply Chain Management, you can use the submission log to view the results of the processing of submissions to Electronic invoicing.

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log** to open the log.
2. In the **Document type** field, select a value to filter the type of invoices that the log shows.

There are possible submission statuses:

- **Scheduled** – Electronic invoicing received the submission from Finance or Supply Chain Management, and scheduled it for further processing by the related electronic invoicing feature.
- **Executing** – Electronic invoicing started processing according to the applicable electronic invoicing feature's pipeline.
- **Pending** – Electronic invoicing is waiting for the response from external web-services involved into the related pipeline action's processing.
- **Completed** – Electronic invoicing successfully processed the electronic invoicing feature in the way that it was configured to process it.
- **Failed** – Electronic invoicing encountered an error or was stopped by an exception while it was processing the electronic invoicing feature.
- **Canceled** – The submission has been explicitly cancelled by a user from Finance or Supply Chain Management using the **Cancel submissions** functionality. For more information, refer to Cancel submissions.

> **Important**
>
> The submission status refers to the processing status of the electronic invoicing feature by Electronic invoicing. It doesn't refer to the final status of the electronic invoice itself. For example, if an electronic invoice must be submitted to an external web service for approval, the submission status might be **Completed**, but the status of the electronic invoice might be **Rejected**. In this case, Electronic invoicing was able to successfully process the electronic invoicing feature as it was configured to process that feature. However, the electronic invoice was rejected because it didn't meet the criteria that the web service established for invoice approval.

The submission status is updated only when you send a request to the Electronic invoicing service by submitting new electronic documents, or by submitting functions and inquiries.

The submission log includes the following additional commands:

- **Functions** > **Cancel submissions** – This function enables a special submission process when the electronic invoice must be approved by an external web service. It instructs Electronic invoicing to send the web service a specific message that is intended to cancel the status of an approved electronic invoice in the web service database.

  > ⓘ **Note**
  >
  > The cancellation functionality is controlled by the **BusinessDocumentEnableCancelForCompleteSubmissionFlight** flight which is disabled by default. The flight must be intentionally enabled per request to activate the functionality.

- **Functions** > **Resubmit document** – Resubmit an electronic document that has already been submitted to Electronic invoicing. A new log is created on the **Submission details** page. In this case, no data is sent to the Electronic invoicing service. Instead, the Electronic invoicing service reruns the processing of previously submitted data.

- **Functions** > **Send related submission** – Send the same document from Finance or Supply Chain Management.

- **Inquiries** > **Submission details** – View the details of the main submission. The visualization shows the complete execution log of the actions that are configured in the electronic invoicing feature. You can download the files that are created as a result of each action during the processing. When the invoice must be approved by an external web service, you can view the status of the invoice.

  Processing data contains variables that are sent to the Electronic invoicing service or returned from it. You can review the value of these variables. For example, the **BusinessDocumentDataModel** variable contains financial data that is submitted to the Electronic invoicing service in a unified structure (JavaScript Object Notation [JSON] format). You can use this information to troubleshoot the correctness of the data that was passed for further processing.

- **Inquiries** > **Related submissions** – View the details of child submissions or the submission of related documents.

- **Electronic document** > **Download file** – For a single electronic document or a selected range of the documents, download a ZIP archive that contains generated electronic files.

Because the processing pipeline of the electronic invoicing feature might contain several actions that generate different files, set the **Export result** parameter for a specific action to ensure that the result of that action is exported.

Receive electronic documents

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In Microsoft Dynamics 365 Finance and Dynamics 365 Supply Chain Management, follow these steps to receive electronic invoices.

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. Select **OK**, and then close the page.

Use the receipt log to view the results of processing incoming documents.

To open the log, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**.

> **Note**
>
> By default, the system sets a filter for the **Submission status** value so that you see only failed receipt submissions. In-progress and successful receipt submissions aren't shown. However, you can adjust the filter to fit your needs.

There are three possible submission statuses:

- **Scheduled** – Electronic invoicing received the submission from Finance or Supply Chain Management, and is currently processing the electronic invoicing feature.
- **Completed** – Electronic invoicing successfully processed the electronic invoicing feature in the way that it was configured to process it.
- **Failed** – Electronic invoicing encountered an error or was stopped by an exception while it was processing the electronic invoicing feature.

The system updates the receipt status only when you send a request to the Electronic invoicing service to receive and process new electronic documents, or to receive and process functions and inquiries.

The receipt log includes the following additional commands:

- **Functions** > **Cancel submissions** – This function enables a special submission process when the electronic invoice must be approved by an external web service. It instructs Electronic invoicing to send the web service a specific message that is intended to cancel the status of an approved electronic invoice in the web service database.
- **Functions** > **Resubmit document** – Resubmit the request to the Electronic invoicing service to run the receipt process. A new log is created on the **Submission details** page.
- **Inquiries** > **Submission details** – View the details of the main submission. The visualization shows the complete execution log of the actions that are configured in the electronic invoicing feature. You can download the files that are created as a result of each action during the processing. When the invoice must be approved by an external web service, you can view the status of the invoice.

Processing data contains variables that are sent to the Electronic invoicing service or returned from it. You can review the value of these variables.

To view successfully received invoices, go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.

Last updated on 03/14/2026

Electronic invoicing coverage for supported and planned countries and regions

This article describes the current geographical coverage for electronic invoicing in Microsoft Dynamics 365 Finance. It also describes known plans for upcoming electronic invoicing implementations.

Electronic invoicing, or e-invoicing, is the process of sending and receiving invoices in a digital format. E-invoicing is rapidly becoming an important part of business processes and an essential part of compliance with the global and country/region-specific regulatory requirements for both buyers and suppliers.

E-invoicing has several advantages over traditional paper-based invoicing. Here are some examples:

- It reduces errors and disputes by eliminating manual data entry and validation.
- It increases cash flow and visibility by speeding up the invoice approval and payment cycle.
- It lowers operational costs by saving time, paper, postage, and storage space.
- It improves customer satisfaction and retention by offering convenience and flexibility.
- It supports environmental sustainability by reducing paper waste and carbon footprint.

Microsoft monitors changes to legislative requirements in the e-invoicing area for all supported countries and regions, and delivers regulatory updates in a timely manner. When Microsoft extends geographical coverage, the company prioritizes e-invoicing as one of the most important requirements.

The next section contains detailed information about out-of-box e-invoicing functionality for countries and regions that are already supported. It includes information about upcoming regulatory changes and plans for new country/region implementations.

Electronic invoicing coverage

> **Note**
>
> Delivery timelines for planned implementations that are mentioned in the following table might change. In addition, projected functionality might not be released at all, for various reasons. For more information, see Microsoft policy.

Expand table

Country/region	Description	Availability
Argentina	Generation of electronic documents in the predefined XML formats and direct submission through WebServices to the Argentinian Tax Authority (ARCA).	Planned for 2026 release wave 1 or later
Australia	Generation of electronic invoices for sales and project invoices and credit notes in the Australian extension of the **PEPPOL** format. Learn more: Electronic invoicing for Australia	Available
Austria	Generation of electronic invoices for sales and project invoices and credit notes in the **PEPPOL** format.	Available
Belgium	Generation of electronic invoices for sales and project invoices and credit notes in the **PEPPOL** format and submission to the PEPPOL Delivery Network via the "last-mile" third-party connection to Edicom. Import of vendor electronic invoices. Learn more: Electronic invoicing for Belgium	Available
Bolivia	Generation of electronic invoices with the right to fiscal credit, invoices without the right to fiscal credit, and adjustment documents in XML format. Direct submission to the Bolivian Tax Authority (SFV).	Planned for 2026 release wave 1 or later
Brazil	**2026 Tax Changes in Electronic Invoicing for Brazil** The update includes the revision of the NF-e format and its transition to Electronic Reporting to ensure compliance with updated tax rules and classifications. It also introduces the National (Federal) NFS-e and São Paulo city NFS-e formats for service invoices. **New:** Support for project-based service invoicing scenarios has been added. NFS-e XML can now be generated from project service invoices, including project invoice proposals and journals (Hour, Fee, Expense, Item). The implementation enables NBS codes and operation indicator codes across project-related processes, including project categories, journal lines, on-account transactions, and invoice proposals. This ensures that NFS-e XML is generated with the required service classification and operation context. All formats are generated using the Electronic Reporting framework. NF-e continues to be submitted to SEFAZ in the same way as previously (in the code in Finance).	Available

Country/region	Description	Availability
	NFS-e (Federal and São Paulo) isn't submitted out of the box, and the XML isn't digitally signed. Integration with a third-party solution or customization is required to submit and sign service fiscal documents. The solution supports only generation of the DPS (Documento Preliminar de Serviços). The official NFS-e fiscal document is issued by the municipal or federal authority after processing, and the access key must be handled via customer or partner solutions / extensions. Inbound vendor service invoices aren't currently in scope for NBS classification. Learn more: • NF-e process overview • Brazil Tax Reform setup • NFS-e service invoices • NFS-e service invoices for Projects	
Chile	Generation of electronic tax documents (DTE) in a country/region-specific format for invoices, debit notes, credit notes, export invoices, export debit notes, export credit notes, and packing slips. Import of vendor electronic invoices (inbound flow). Submission to the Chilean Tax Authority (SII) via a third-party "last mile" connection to Edicom. Learn more: • Get started with Electronic invoicing for Chile • Vendor electronic invoice import in Chile	Available
China	Invoice reporting (fapiao) support. File upload integration with Golden Tax System via BaiWang and Asino software. Learn more: Golden tax system integration	Available
Colombia	Generation of electronic invoices in the standard XML format based on UBL V2.1 and adopted by DIAN. Submission to the Colombian Tax Authority (DIAN) via a third-party "last mile" connection to Edicom. Learn more in Electronic invoicing for Colombia.	Available
Costa Rica	Generation of e-documents for invoices, credit notes, debit notes, vendor invoices (self-invoices), and payment in the country/region-specific format: version 4.4. Submission to the Costa Rican Tax Authority (Ministerio de Hacienda) via a third-party "last mile" connection to Edicom.	Available

Country/region	Description	Availability
	Learn more: Electronic invoicing for Costa Rica	
Denmark	Generation to NemHandel or to the Peppol network of electronic invoices for sales and project invoices and credit notes in the **OIOUBL** or **PEPPOL** formats and submission to NemHandel or to the PEPPOL Delivery Network via the "last-mile" third-party connection to Edicom. Import of vendor electronic invoices. Learn more: Use electronic invoicing for Denmark	Available
Dominican Republic	Generation of electronic fiscal vouchers (e-CF) in the XML format based on the UBL standard. Submission to the Dominican Republic Tax Authority (DGII) via a third-party "last mile" connection to Edicom.	Planned for 2026 release wave 1 or later
Ecuador	Generation of electronic receipts in XML format that comply with the legal and regulatory requirements of the SRI. Submission to the Ecuadorian Tax Authority (SRI) via a third-party "last mile" connection to Edicom.	Planned for 2026 release wave 1 or later
Egypt	Generation and direct submission to the Egyptian Tax Authority of electronic invoices for sales and project invoices, credit notes, and debit notes in the country/region-specific format. Learn more: Electronic invoicing in Egypt	Available
Estonia	Generation of electronic invoices for sales and project invoices and credit notes in the country/region-specific format. Learn more: Export of customers electronic invoices in Estonia	Available
Finland	Generation of electronic invoices for sales and project invoices and credit notes in the **Finvoice** format. Learn more: Export of customers electronic invoices in Finland	Available
France	Generation and submission to Chorus Pro of B2G electronic invoices for sales and project invoices and credit notes in the **PEPPOL** format. Import of vendor B2G electronic invoices. Learn more: Electronic invoicing via Chorus Pro	Available
	Generation, submission, and reception of domestic B2B electronic invoices for sales and project invoices and credit notes in the French-specific UBL-based format to/from	Available

Country/region	Description	Availability
	Edicom acting as a certified Platform Agent. Support of documents mandatory life-cycle statuses. Learn more: Electronic invoicing via EDICOM	
Germany	Generation of electronic invoices for sales and project invoices and credit notes in the **xRechnung** format. Import of vendor electronic invoices in the **xRechnung** format. Learn more: • Customer electronic invoices in Germany • Use vendor electronic invoices import in Germany	Available
Guatemala	Generation of electronic invoices in a file structured in the country/region-specific XML format. Submission to the Guatemalan Tax Authority (SAT) via a third-party "last mile" connection to Edicom.	Planned for 2026 release wave 1 or later
Hungary	Generation and submission to the governmental Real-time invoice reporting (RTIR) platform of required data in the country/region-specific format. Learn more: Online invoicing system	Available
India	Generation and submission to the governmental Invoice Registration Portal (IRP) of electronic invoices for sales and project invoices and credit notes in the country/region-specific format. Learn more: Electronic invoices in India	Available
Indonesia	Generation and file upload integration with the tax authorities' software of electronic invoices for sales and project invoices, credit notes, and debit notes in the country/region-specific format. Import of vendor electronic invoices. Learn more: Get started with electronic invoicing for Indonesia	Available
Italy	Generation and direct submission to the SDI system of electronic invoices for sales and project invoices and credit notes in the **FatturaPA** format. Import of vendor electronic invoices. Learn more: • Customer electronic invoices • Vendor electronic invoices	Available

Country/region	Description	Availability
	• Import of vendors electronic invoices • Get started with Electronic invoicing for Italy	
Malaysia	Generation and direct submission to IRBM of electronic invoices for sales, project, and self invoices, credit notes, and debit notes. Learn more: Get started with electronic invoicing for Malaysia	Available
Mexico	Generation and submission of CFDI electronic documents to PAC Interfactura. Import of vendor electronic invoices. Learn more: Electronic invoices (CFDI)	Available
Netherlands	Generation of electronic invoices for sales and project invoices and credit notes in the **UBL-OHNL** format.	Available
New Zealand	Generation of electronic invoices for sales and project invoices and credit notes in the New Zealand extension of the PEPPOL format. Learn more: Electronic invoicing for New Zealand	Available
Norway	Generation of electronic invoices for sales and project invoices and credit notes in the **EHF** format. Learn more: Customer electronic invoices in Norway	Available
Oman	Generation of electronic invoices in the PEPPOL format and submission to the PEPPOL Delivery Network via the "last mile" third party connection. This will be enforced by August 2026 for the 100 largest tax paying companies in the country/region with following adoption for other groups of taxpayers by year 2028.	Planned for 2026 release wave 2 or later
Panama	Generation of e-documents for invoices in the country/region-specific format. Submission to the Panamanian Tax Authority (DGI) via a third-party "last mile" connection to Edicom. Learn more: Electronic invoicing for Panama	Available
Paraguay	Generation of electronic documents and electronic tax documents in the standardized country/region-specific XML format. Submission to the Paraguayan Tax Authority (SET) via a third-party "last mile" connection to Edicom. Learn more in Electronic invoicing for Paraguay.	Available

Country/region	Description	Availability
Peru	Generation of electronic invoicing documents in the standard XML format based on UBL V2.1 and adopted by SUNAT, also known as CPE (Electronic Payment Vouchers). Submission to the Peruvian Tax Authority (SUNAT) via a third-party "last mile" connection to Edicom.	Planned for 2026 release wave 1 or later
Poland	Generation and direct submission to the governmental KSEF system of electronic invoices for sales, project, and advance invoices and credit notes in a country/region-specific format. Import of vendor electronic invoices. Learn more: Get started with electronic invoicing for Poland	Available
Saudi Arabia	Generation and direct submission to ZATCA of electronic invoices for customer prepayments, sales and project invoices, credit notes, and debit notes in the country/region-specific format. Learn more: Electronic invoicing in Saudi Arabia	Available
Singapore	Generation of electronic invoices in the Peppol International (PINT) format and submission to the "InvoiceNow" network via the "last mile" 3rd-party connection to Edicom. Learn more: Electronic invoicing for Singapore	Available
Spain	Generation of electronic invoices for sales and project invoices and credit notes in the **FacturaE** format.	• Available • B2B mandate is expected in 2027
Spain	Generation and submission to the governmental SII system of required data in the country/region-specific format. Learn more: Immediate Supply of Information on VAT (SII)	Available
Türkiye	Generation of e-invoices for customer invoices in the format mandated by Turkish Revenue Administration to meet e-invoicing obligations.	Available in Public preview. General availability is planned for 2025 release wave 2
United Arab Emirates	Generation of electronic invoices in the Peppol International (PINT) format and submission to the PEPPOL Delivery Network via the "last mile" third party connection. To be enforced by 01 January 2027.	Planned for 2026 release wave 2
Uruguay	Generation of electronic fiscal receipts (CFE) based on the XML standard, with a syntax defined and maintained by the	Available

Country/region	Description	Availability
	DGI. Submission to the Uruguayan Tax Authority (DGI) via a third-party "last mile" connection to Edicom. Learn more: Electronic invoicing for Uruguay	
PEPPOL	Generation of electronic invoices for sales and project invoices and credit notes in the **PEPPOL** format. This country/region-independent functionality can be used "as is" if no country/region-specific amendments are supposed, or it can be used as the basis for further modifications. Learn more: • Export of customers electronic invoices in PEPPOL BIS 3 format • Import vendor electronic invoices in PEPPOL format	Available

Additional resources

- Electronic invoicing overview
- Globalization Studio overview

Last updated on 07/22/2026

Availability of Electronic Invoicing Service features by country or region

The availability of Electronic Invoicing Service globalization features depends on the country or region where you're located. Although some features are generally available, others are still in preview.

> **Note**
>
> This article relates only to the globalization features introduced in the scope of the Electronic Invoicing Service. It doesn't cover other approaches for supported e-Invoicing capabilities for some countries or regions that are designed based on built-in X++ code and the Electronic Messaging framework. For more information, see Electronic Invoicing and Electronic messaging.

Generally available features

The following table shows the Electronic invoicing globalization features that are generally available.

Expand table

Country or region	Globalization feature name	Business documents
Australia	Electronic invoicing for Australia and New Zealand	Sales invoices and project invoices
Austria	Austrian electronic invoices (AT)	Sales invoices and project invoices
Belgium	Belgian electronic invoice (BE)	Sales invoices and project invoices
Chile	Chilean electronic invoice (CL)	Sales invoices, project invoices, and packing slips
Chile	Chilean electronic invoice (CL)	Import vendor invoices
Colombia	Colombia Electronic Invoice (CO)	Sales invoices and project invoices

Country or region	Globalization feature name	Business documents
Costa Rica	Costa Rica electronic invoice (CRI)	Sales invoices and project invoices
Denmark	Danish electronic invoice (DK)	Sales invoices and project invoices
Egypt	Egyptian electronic invoice (EG)	Sales invoices and project invoices
Estonia	Estonian electronic invoice (EE)	Sales invoices and project invoices (format generation only)
Europe	PEPPOL electronic invoice	Pan-European Public Procurement Online (PEPPOL) sales invoices and project invoices (format generation only)
Europe	PEPPOL vendor invoice import	PEPPOL import vendor invoices
Finland	Finnish electronic invoice (FI)	Sales invoices and project invoices (format generation only)
France	French Chorus Pro submission (FR)	Sales invoices and project invoices
France	French Chorus Pro vendor invoice import (FR)	Import vendor invoices
France	French electronic invoice (FR)	Sales invoices and project invoices via EDICOM
Germany	German electronic invoice (DE)	Sales invoices and project invoices (format generation only)
Indonesia	Indonesian electronic invoice (ID)	Sales invoices, project invoices, and vendor invoices
Italy	Italian FatturaPA (IT)	Sales invoices and project invoices, and vendor invoices
Malaysia	Malaysian electronic invoicing (MY)	Sales invoices, project invoices, and self invoices
Netherlands	Dutch electronic invoice (NL)	Sales invoices and project invoices
New Zealand	Electronic invoicing for Australia and New Zealand	Sales invoices and project invoices
Norway	Norwegian electronic invoice (NO)	Sales invoices and project invoices (format generation only)
Panama	Panama Electronic Invoice (PA)	Sales invoices and project invoices
Paraguay	Paraguay Electronic Invoice	Sales invoices, project invoices, and packing slips

Country or region	Globalization feature name	Business documents
	(PY)	
Poland	Polish electronic invoice (PL)	Sales invoices, project invoices, and advance invoices
Saudi Arabia	Saudi Arabian Zatca submission (SA)	Sales invoices and project invoices
Saudi Arabia	Saudi Arabian Zatca compliance check (SA)	Onboarding process
Singapore	Singaporean electronic invoice (SG)	Sales invoices and project invoices
Spain	Spanish electronic invoice (ES)	Sales invoices and project invoices
Uruguay	Uruguay Electronic Invoice (UY)	Sales invoices, project invoices, and packing slips

Important

Microsoft planned to transition NF-e/NFS-e and CFDI to the Electronic Invoicing service. For quite a while, the features and formats for NF-e/NFS-e were in preview in Brazil, and those for CFDI were in preview in Mexico. However, because of a lack of interest in adoption and transition to the new approach, Microsoft is still evaluating the timing of the switch to the Electronic Invoicing service platform in those countries or regions. The format of these documents doesn't contain the latest compliance changes that were published since the preview release. In the meantime, continue to use the built-in X++ implementations in Brazil and Mexico. **These implementations are kept up to date with the legislation.** For more information, see Brazil NF-e process overview and Electronic invoices (CFDI).

Note

Electronic invoicing capabilities for Brazil (NF-e for goods and NFS-e for services) are implemented in Dynamics 365 Finance by using Electronic Reporting (ER) configurations and country-specific localization features.

As part of the Brazil tax reform updates, the generation of NF-e and NFS-e XML formats was transitioned to the Electronic Reporting framework to support evolving tax rules and classifications.

NF-e documents continue to be submitted to SEFAZ by using existing integration mechanisms in Finance, while NFS-e (Federal and São Paulo formats) are generated as XML files that require integration or a third-party solution for submission and digital signature.

For Brazil, don't use the Electronic Invoicing Service. Instead, use the built-in localization and ER-based configurations for compliant electronic invoice generation.

Preview features

The following table shows the Electronic invoicing Service globalization features that are currently in preview.

Expand table

Country or region	Globalization feature name	Business documents
Brazil	Brazilian NF-e (BR)	Fiscal document model 55, correction letters, cancellations, and discards
Brazil	Brazilian NFS-e ABRASF Curitiba (BR)	Service fiscal documents
Brazil	Brazilian NF-e import from e-mail (BR)	Fiscal document model 55
Mexico	Mexican CFDI Interfactura (MX)	Sales invoices, packing slips, inventory transfers, payment complements, and cancellations

Last updated on 08/07/2026

Get started with Electronic invoicing for Australia and New Zealand

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Regulatory Configuration Service (RCS) will be deprecated. All new RCS provisioning is stopped as of 10.0.39 GA. If provisioning is required, please register a support ticket. We will be providing tools and required support for migration from RCS to the **Globalization Studio** workspace. We plan to fully shut down RCS by August 1, 2024. For more information about migration to **Globalization Studio** workspace, see Regulatory Configuration Service merge to the Globalization Studio workspace

This article helps you get started with Electronic invoicing for Australia and New Zealand. It guides you through the configuration steps that depend on the country or region in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. These steps complement the steps described in Set up Electronic invoicing.

After you configure Electronic invoicing, you can generate XML files of electronic invoices by using the Pan-European Public Procurement Online (PEPPOL) format specification with the Australian and New Zealand extension.

> **Note**
>
> The electronic invoicing approach described in this article is implemented by using an invoicing service that's applicable only to cloud deployments of Finance or Supply Chain Management. For on-premises deployments, see Customer electronic invoices in Australia and New Zealand.

Prerequisites

Before you begin the procedures in this article, make sure the following prerequisites are met:

- You're familiar with Electronic invoicing as described in Electronic invoicing overview.
- You complete the common part of Electronic Invoicing service configuration as described in Set up Electronic invoicing.

Country/region-specific configuration for the Electronic invoicing for Australia and New Zealand feature

Some parameters in the **Electronic invoicing for Australia and New Zealand** Electronic invoicing feature have default values. Before you deploy this feature to the service environment, review the default values, and update them as needed to better reflect your business operations.

1. Import the latest version of the **Electronic invoicing for Australia and New Zealand** Globalization feature as described in Import features from the repository.

2. Create a copy of the imported Globalization feature, and select your configuration provider for it. For more information, see Create a Globalization feature.

   > **Note**
   >
   > Microsoft provides the **Electronic invoicing for Australia and New Zealand** feature. It's ready to use and requires no additional configuration. For information about how to configure invoicing features if you must apply changes, see **Work with feature setups**. For example, you can filter specific legal entities so that they're processed in applicability rules. By default, the feature is applicable to all legal entities that have a primary address in Australia or New Zealand.

3. Always create the copy of the feature as a **Draft** version. Regardless of whether you made changes, complete and deploy the feature as described in Complete and deploy a Globalization feature.

Finance configuration

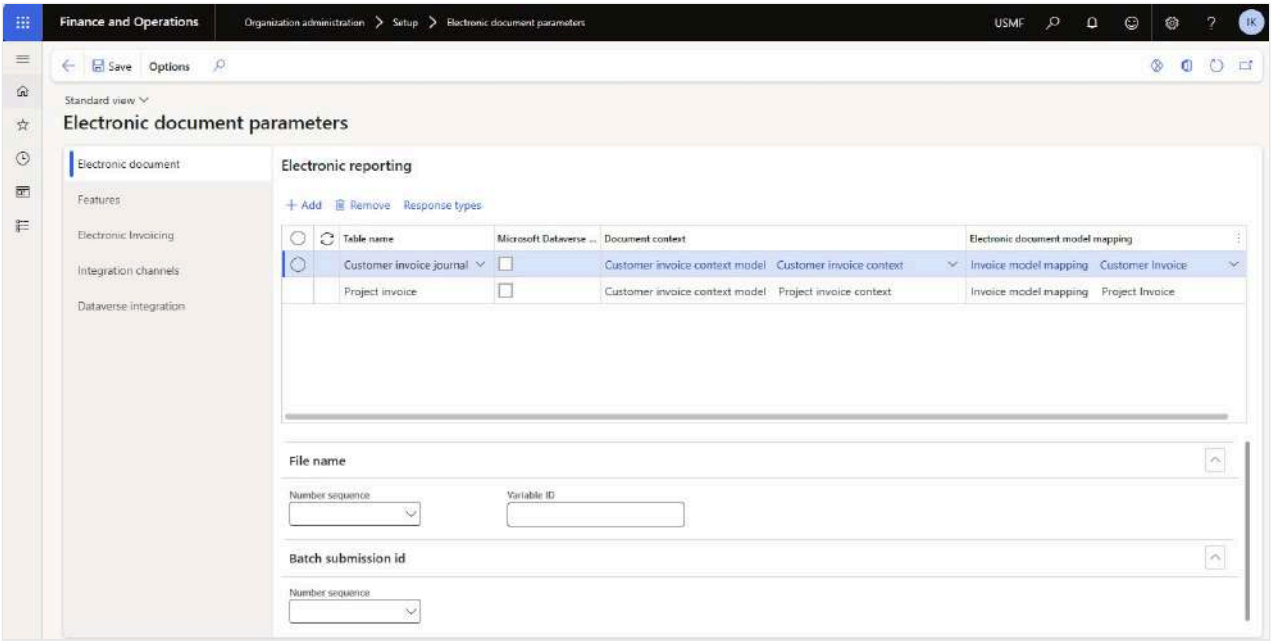
You must configure some additional parameters directly in Finance.

1. Make sure that the country/region-specific Electronic reporting (ER) configurations for the document context and electronic document model mapping that are required for

Australia and New Zealand are imported. For more information, see Set up Electronic invoicing parameters.

> **Note**
>
> After you import the **Electronic invoicing for Australia and New Zealand** Electronic invoicing feature, electronic documents are configured by default. Follow the remaining steps of this procedure if you must make changes.

2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
3. On the **Electronic document** tab, add records for the **Customer Invoice journal** and **Project invoice** table names.
4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.



5. Save your changes, and close the page.

Finance business data configuration

Prerequisites

The primary address of the legal entity must be in Australia or New Zealand.

Configure legal entity data

1. Go to **Organization administration** > **Organizations** > **Legal entities**, and select a legal entity.
2. On the **Addresses** FastTab, add a valid primary address for the legal entity.
3. On the **Tax registration** FastTab, in the **Tax registration number** field, enter a valid tax registration number for the legal entity.

Configure customer data

1. Go to **Accounts receivable** > **Customers** > **All customers**, and select a customer.
2. On the **Addresses** FastTab, add a valid address for the customer.
3. On the **Invoice and delivery** FastTab, in the **Tax exempt number** field, enter a valid tax registration number for the customer.
4. On the **Sales demographics** FastTab, in the **Primary contact** field, select a person who is the buyer's contact.

> ⓘ **Note**
>
> Define all available contact persons for this customer.

Configure units of measure

1. Go to **Organization administration** > **Setup** > **Units** > **Units**.
2. Select a unit ID in the list, and then select **External codes**.
3. On **External codes**, in the **Overview** section, enter a code in the **Code** field that corresponds to the selected unit ID.
4. In the **Value** section, enter the external code in the **Value** field to use as the units of measure code for international trade.

> ⓘ **Note**
>
> When no specific units of measure are assumed, the default value **EA** (each) is used.

Configure sales tax codes

1. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax codes**.

2. Select a sales tax code. On the Action Pane, on the **Sales tax code** tab, in the **Sales tax code** group, select **External codes**.
3. In the **Overview** section, create a line for the selected unit. Enter the sales tax code in the **External code** field that you selected in step 2.
4. In the **Value** section, enter an external code in the **Value** field to use for the selected sales tax code, according to the required codification.

Customer requisition

When you register free text invoices, invoices that are based on sales orders, or project invoices, enter a customer requisition. You can also add an optional customer reference.

Free text invoices

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. Create a new invoice, or select an existing invoice.
3. In the **Header** view, on the **Customer** FastTab, in the **References** section, enter values in the **Customer requisition** and **Customer reference** fields.

Sales orders

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Create a new sales order, or select an existing sales order.
3. In the **Header** view, on the **General** FastTab, in the **References** section, enter values in the **Customer requisition** and **Customer reference** fields.

Project invoices

1. Go to **Project management and accounting** > **Projects** > **Project contracts**.
2. Create a new project contract, or select an existing project contract.
3. On the **Funding sources** FastTab, select or create a funding source of the **Customer** type, and then select **Details**.
4. On the **Funding source details** page, on the **Other** FastTab, in the **References** section, enter default values for the contract in the **Customer requisition** and **Customer reference** fields.

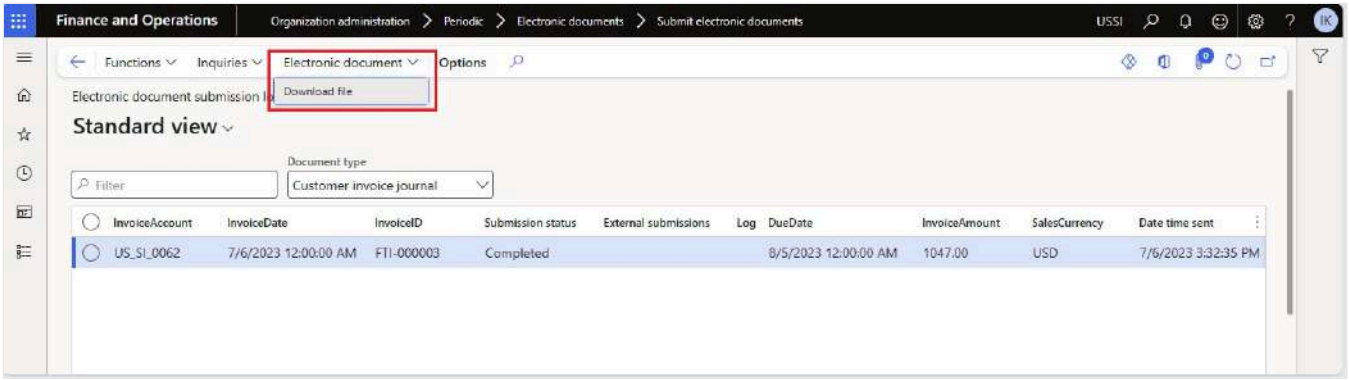
Issue electronic invoices

After you complete all the required configuration steps, you can generate electronic invoices for posted invoices. For more information about how to generate electronic invoices, see Issue

electronic invoices in Finance and Supply chain management.

You can check the results of a submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log** and selecting the required document type.

To download the XML files of electronic invoices for successfully processed invoices, select **Electronic document** > **Download file**.



Additional resources

- Electronic invoicing overview
- Customer electronic invoices in Australia and New Zealand

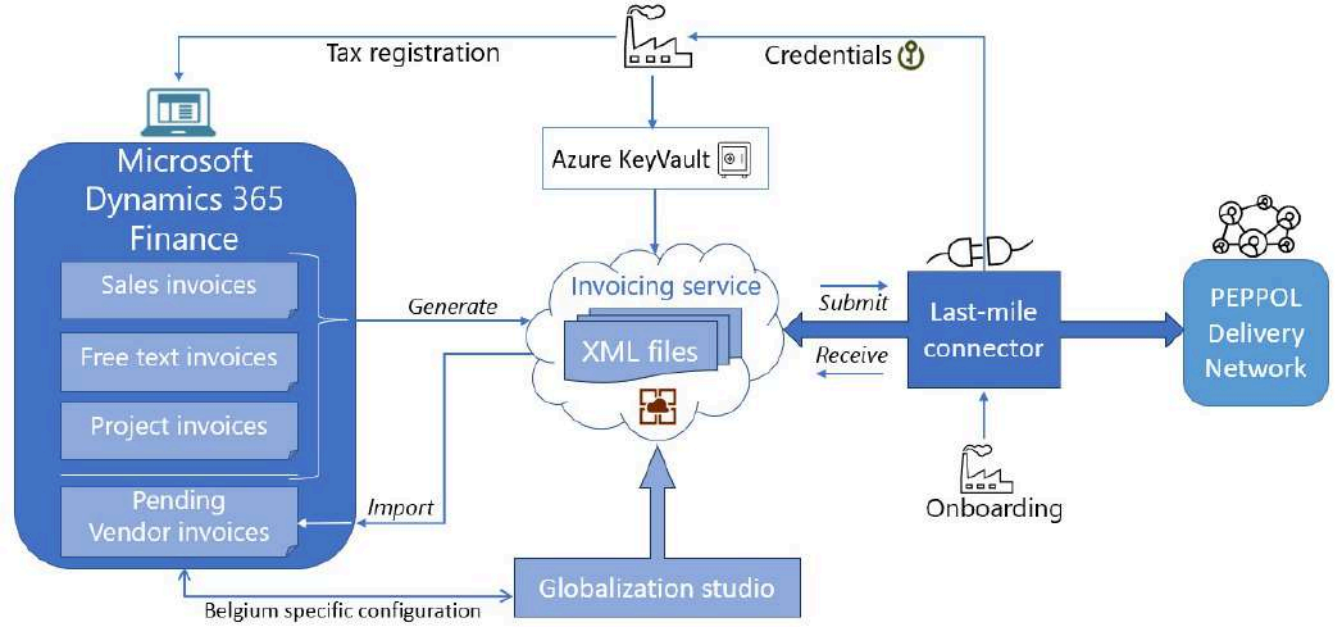
Last updated on 03/04/2026

Electronic invoicing for Belgium

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information to help you get started with Electronic invoicing for Belgium. It explains how to configure the system so that you can use the last-mile connector to generate, submit, and receive electronic invoices in the Pan-European Public Procurement Online (PEPPOL) format in Microsoft Dynamics 365 Finance.



> **Note**
>
> The electronic invoicing approach that this article describes is implemented by using an invoicing service that's applicable only to cloud deployments of Dynamics 365 Finance.

Watch the following overview of the Belgian electronic invoicing implementation in Finance.

https://learn-video.azurefd.net/vod/player?id=d21698ae-599e-4895-9a45-e19226a852ab&locale=en-us&embedUrl=%2Fdynamics365%2Ffinance%2Flocalizations%2Fbelgium%2Femea-bel-e-invoices

Prerequisites

Before you begin the procedures in this article, meet the following prerequisites:

- The company must be a registered taxpayer in Belgium.
- The company must have a signed agreement with the provider of electronic document delivery service that secures electronic document interchange in the **PEPPOL** format.
- Among the registered profile IDs, the company should have the **urn:fdc:peppol.eu:2017:poacc:billing:01:1.0** profile. Finance uses this profile for the interchange of documents in the **PEPPOL** format.
- The company must obtain, from the service provider, the required credentials to enable integration of the Electronic Invoicing service with the Electronic Invoicing service independent software vendor (ISV) last-mile connector.

  > **Note**
  >
  > The current implementation assumes **Edicom** as the Electronic Invoicing ISV last-mile connection service provider. Learn more in **e-Invoicing Integration with Microsoft Dynamics 365**).

- Become familiar with Electronic invoicing as it's described in Electronic Invoicing service overview and Electronic invoicing components.
- Complete the common part of Electronic Invoicing service configuration as described in Electronic invoicing components.

Create the Azure Key Vault configuration

Create an Azure key vault to store the required secrets that are issued for your company. Learn more in Configure Azure resources for Electronic invoicing.

In the key vault, add the secret for the token that grants authorization to access the Edicom services. You must obtain this token from Edicom before you begin, as described in the prerequisites.

Configure electronic invoicing Key Vault parameters

To configure electronic invoicing Key Vault parameters, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic invoicing** tab, in the **Key Vault settings** section, in the **Key Vault** field, select the reference to the key vault that you created in the previous section of this article.
3. In the **SAS token secret** field, select the name of the storage account secret URL that you use to authenticate access to the storage account.
4. Select **Key Vault parameters**.
5. On the **Key Vault parameters** page, in the **Certificates** section, select **Add** to create a **token** element of the **Secret** type for the secret that is described in the previous section.

   > ⓘ Note
   >
   > The value in the **Name** column should match the name of the secret that is described in the previous section.

Import the electronic invoicing feature

1. Go to **Globalization Studio**, and select the **Electronic invoicing** tile. Then import the latest version (**5** or later) of the **Belgian electronic invoice (BE)** Globalization feature as described in Import features from the repository.
2. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, make sure that the following Electronic reporting (ER) configurations are successfully imported when you import the **Belgian electronic invoice (BE)** Globalization feature:

   - Invoice model
   - Invoice model mapping
   - PEPPOL Sales e-invoice
   - PEPPOL Sales e-credit note
   - PEPPOL Project e-invoice
   - PEPPOL Project e-credit note
   - Customer invoice context model
   - Response message model
   - Edicom Response Processing
   - Error log import Json

   > ⓘ Note

If the preceding ER configurations aren't imported for some reason, manually import them as described in Import Electronic reporting (ER) configurations from Dataverse.

3. On the **Reporting configurations** tile, import the latest versions of the following ER configurations. These configurations are required for the receipt of incoming vendor invoices.

    - Vendor invoice import
    - Vendor invoice Mapping to destination
    - Import invoice context model

Configure the electronic invoicing feature

The **Belgian electronic invoice (BE)** electronic invoicing feature includes some parameters with default values. Before you deploy the electronic invoicing feature to the service, review the default values, and update them as needed to better reflect your business operations.

To review and update the configuration of the **Belgian electronic invoice (BE)** electronic invoicing feature, follow these steps:

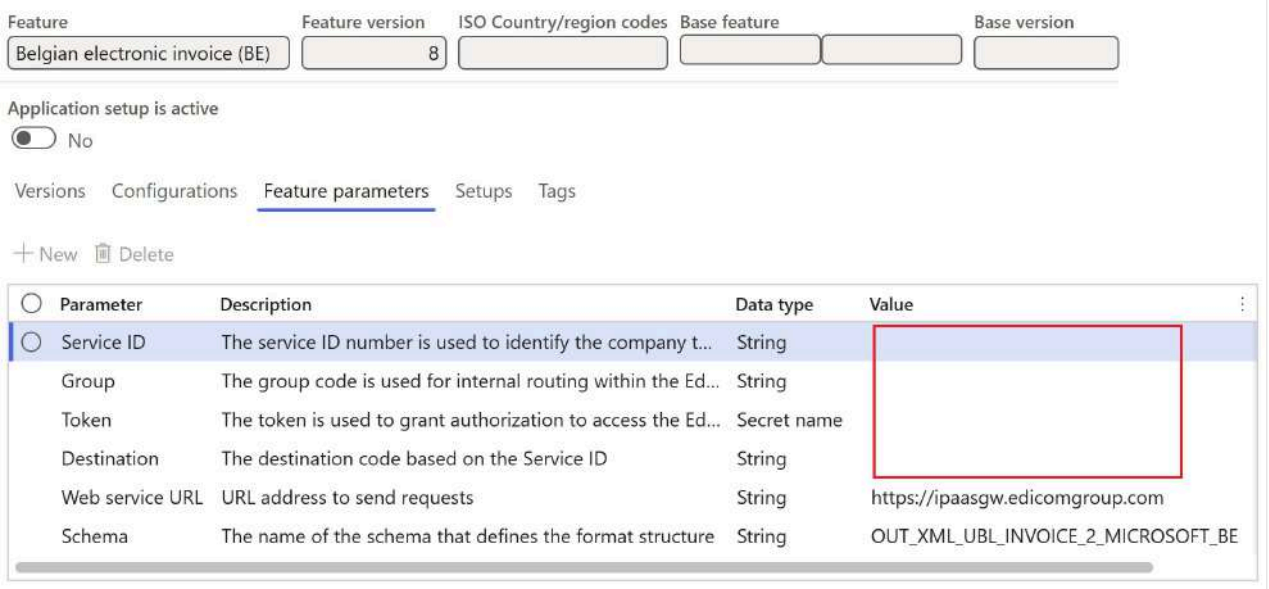
1. Go to **Globalization Studio**, and select the **Electronic invoicing** tile. Then import the **Belgian electronic invoice (BE)** Globalization feature as described in Import features from the repository.

2. Create a copy of the imported Globalization feature, and select your configuration provider for it, as described in Create Globalization features.

3. On the **Versions** tab, verify that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters. These parameters are required for interoperation with Edicom services.

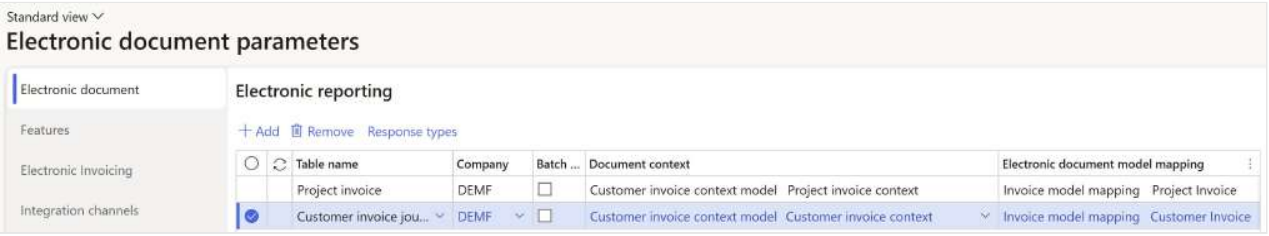
    - **Service ID** – Enter the service ID (domain) number that you obtained from Edicom. This value is used for company identification.
    - **Group** – Enter the group code. This value is used for internal routing within the Edicom infrastructure.
    - **Destination** – Enter the destination, which you construct by concatenating the service ID number and the string **_EDIWIN**. For example, if the service ID number is **123456**, enter **123456_EDIWIN**.
    - **Token** – Select the name of the token that you previously created.



5. The copy of the feature is always created as a **Draft** version. Complete and deploy the feature as described in Complete and deploy a Globalization feature.

Configure electronic document parameters

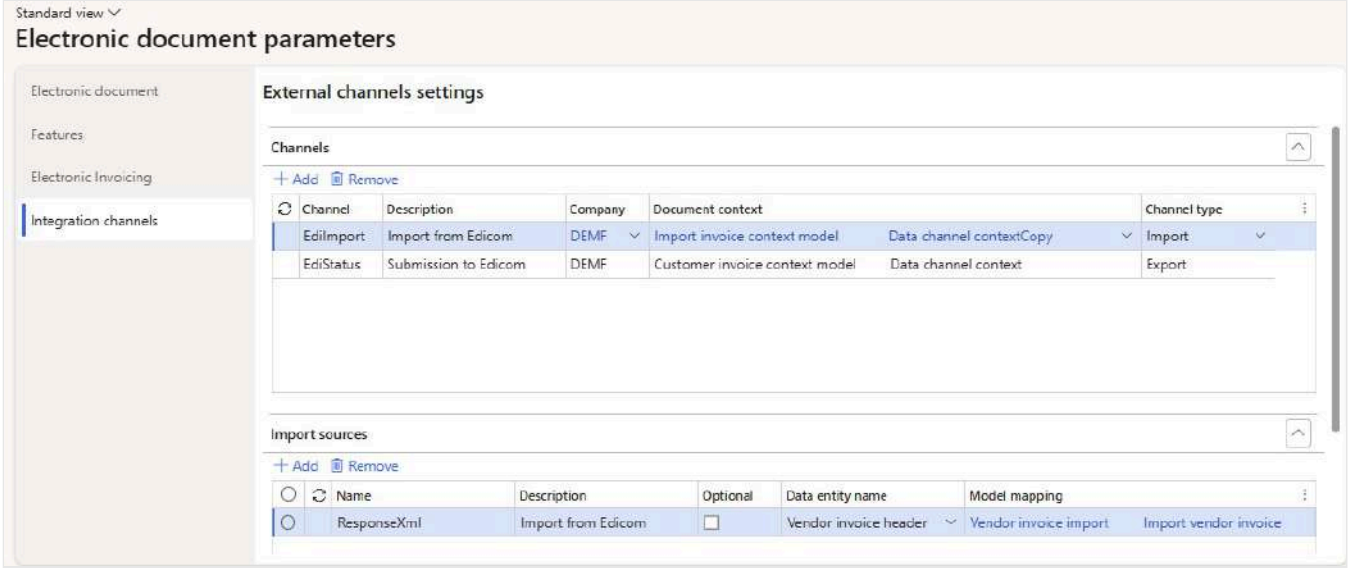
1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic document** tab, add records for the **Customer Invoice journal** and **Project invoice** table names.
3. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with Set up Electronic document parameters.



> **Note**
>
> If you create derived analogs of the previously mentioned ER configurations, use them instead of the standard ones.

4. On the **Integration channels** tab, in the **Channels** section, select **Add** to create a channel.
5. In the **Channel** field, enter **EdiStatus**. Enter the value exactly as it appears here. The system uses this channel to submit outgoing electronic invoices.
6. In the **Company** field, select a required legal entity.

7. In the **Document context** field, select the **Data channel context** mapping from the **Customer invoice context model** configuration.
8. In the **Channel type** field, select **Export**.
9. In the **Channels** section, select **Add** to create another channel.
10. In the **Channel** field, enter **EdiImport**. Enter the value exactly as it appears here. The system uses this channel to import incoming electronic invoices.
11. In the **Company** field, select a required legal entity.
12. In the **Document context** field, select the **Data channel context** mapping from the **Import invoice context model** configuration.
13. In the **Channel type** field, select **Import**.
14. In the **Import sources** section, select **Add** to create an import source.
15. In the **Name** field, enter **ResponseXml**. Enter the value exactly as it appears here.
16. In the **Data entity name** field, select the **Vendor invoice header** entity.
17. In the **Model mapping** field, select the **Import vendor invoice** mapping from the **Vendor invoice import** configuration.
18. Select **Save**, and close the page.



> **Note**
>
> If you must use other integration channels besides **EdiStatus** and **EdiImport**, you must do extra configuration of the feature and context.

Configure the address structure

To configure the structure of the postal address and define all required elements, follow these steps:

1. Go to **Organization administration** > **Global address book** > **Addresses** > **Address setup**.

2. Make sure that at least the following mandatory elements are configured:

    - Country/region code
    - Postal code
    - City name
    - Street

Configure legal entity data

Enter the address

To enter the address, follow these steps:

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity.
3. On the **Addresses** FastTab, add a valid primary address for the selected legal entity.

Verify the seller's identification

To identify a company by its value-added tax (VAT) number, follow these steps:

1. Go to **Organization administration** > **Organizations** > **Legal entities**.

2. On the **Foreign trade and statistics** FastTab, in the **INTRASTAT** section, in the **VAT exempt number export** field, make sure that a valid VAT number is entered for the legal entity.

    The VAT number is entered in the **Invoice\cac:AccountingSupplierParty\cac:Party\cbc:EndpointID** element in the electronic invoice XML file that is generated. You use it as the seller's identification during the submission process.

> ⓘ **Note**

The contact information for the legal entity is automatically retrieved from the related **Person** record that is associated with the current user in Finance.

Configure customer data

Enter the address

To enter the address, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer.
3. On the **Addresses** FastTab, add a valid address for the selected customer.

Enter the contact person

To enter the contact person, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer.
3. On the **Sales demographics** FastTab, in the **Primary contact** field, select the person who is considered as the buyer's contact.

   > ⓘ **Note**
   >
   > You must already define all available contact persons for the selected customer. Make sure that the selected contact person has a valid email address and phone number.

Verify the buyer's identification

The buyer's EndpointID determination uses the following hierarchy of built-in Registration numbers.

- If you define the Global Location Number (GLN), also known as a European article numbering (EAN), for the customer as an active Registration Number with the **EAN** Registration category, the system uses it as the customer's EndpointID and uses the **0088** constant (EAN Location Code) as the EndpointID **schemeID** attribute's value.

- If you don't define the EAN registration number, the system uses the customer's active Registration Number of the **Enterprise ID** Registration category as the customer's EndpointID and dynamically determines the EndpointID **schemeID** attribute's value depending on the country/region from the Buyer's address. For example, **0208** for Belgium, **0002** for France, etc.

- If you don't define the EAN and Enterprise ID registration numbers, the system uses the customer's active Registration Number of the **VAT ID** Registration category as the customer's EndpointID and dynamically determines the EndpointID **schemeID** attribute's value depending on the country/region from the Buyer's address. For example, **9925** - Belgium VAT number, **9957** - French VAT number, etc.

- If you don't define the VAT ID registration number, the system uses the **Tax exempt number** defined in customer's master data.

  The resulting Endpoint ID value populates the **Invoice\cac:AccountingCustomerParty\cac:Party\cbc:EndpointID** element in the generated electronic invoice XML file and is used as the buyer's identification during the submission process.

> ⓘ **Note**
>
> If the built-in registration numbers aren't sufficient, you can redefine the default schema by following the configuration steps for the **Buyer** identification in the next section. You can define any schema from the list of allowed values Electronic Address Scheme (EAS). Otherwise, skip ahead to the Configure units of measure section.

Configure identification schemas

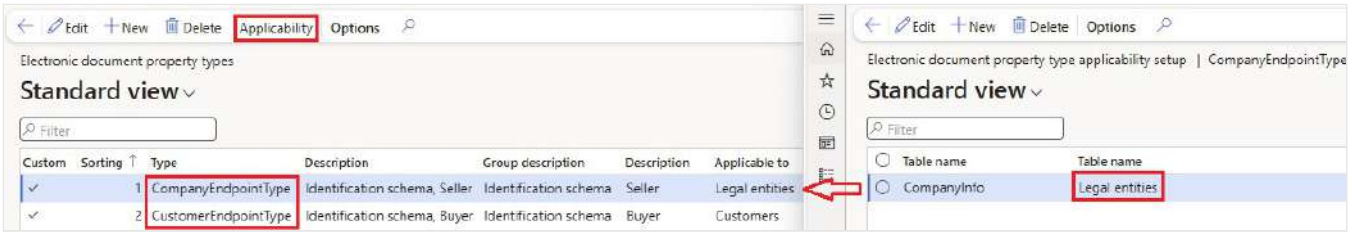
Follow the configuration steps in this section only if you need to redefine the default identification schemas for sellers and buyers.

Configure electronic document properties

To configure electronic document properties, follow these steps:

1. Go to **Accounts receivable** > **Setup** > **Electronic document property types**, and select **New**.
2. In the **Type** field, enter **CompanyEndpointType**. Enter the value exactly as it appears here. This value is used for the Seller identification schema definition.
3. Select **Applicability** to add an applicable table.

4. On the **Electronic document property type applicability setup** page, in the **Table name** field, select the **Legal entities** table name.
5. Save your changes, and return to the **Electronic document property types** page.
6. Select **New** to create another electronic document property type.
7. In the **Type** field, enter **CustomerEndpointType**. Enter the value exactly as it appears here. This value is used for the Buyer identification schema definition.
8. Select **Applicability** to add an applicable table.
9. On the **Electronic document property type applicability setup** page, in the **Table name** field, select the **Customers** table name.
10. Save your changes, and return to the **Electronic document property types** page.
11. Save your changes, and close the page.



Enter the Seller schema code

To enter the Seller schema code, follow these steps:

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity then on the Action Pane, select **Electronic document properties**.
3. In the **Value** column, enter the required Seller schema code.

> **Note**
>
> For the seller's identification, schema **9925** (Belgium VAT number) is used by default if you don't define another schema. Schema codes that you define through electronic document property types take priority over the default **9925** schema code. The value of the **EndpointID** is always retrieved from the seller's VAT exempt number export field regardless of the schema value redefined in the seller's **CompanyEndpointType** parameter.

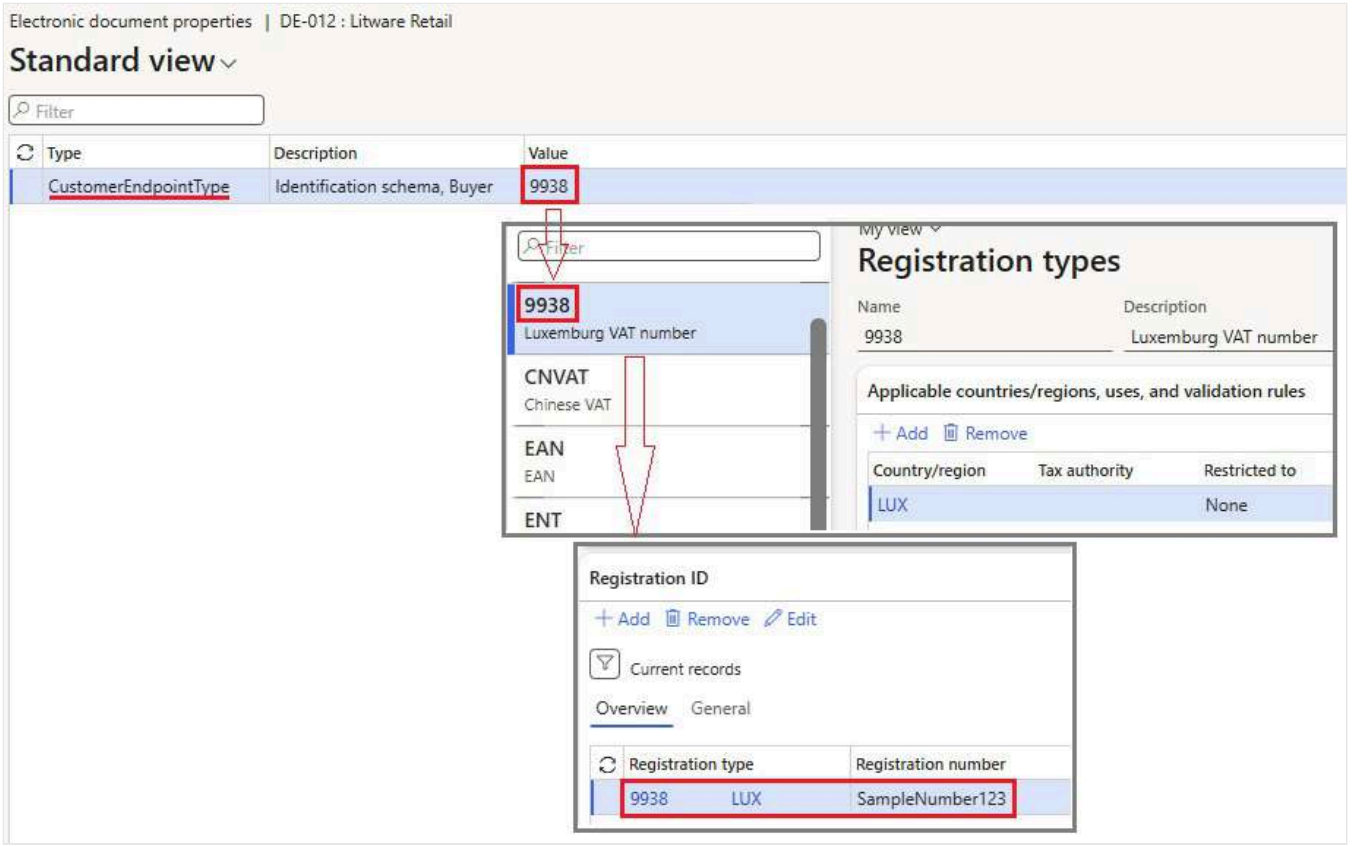
Enter the Buyer schema codes

To enter the Buyer schema codes, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.

2. Select a specific customer in the list then on the Action Pane, on the **Customer** tab, in the **Properties** group, select **Electronic document properties**.
3. In the **Value** column, enter the required Buyer schema code.

> **Note**
>
> For the buyer's identification, schema **9925** (Belgium VAT number) is used by default if you don't define another schema. Schema codes that you define through electronic document property types take priority over the other built-in **0088**, **0208**, or **9925** schema codes. The value of the **EndpointID** is retrieved from the buyer's active registration number, which you must create with the type that exactly matches the value of the buyer's **CustomerEndpointType** parameter.



Configure units of measure

To configure units of measure, follow these steps:

1. Go to **Organization administration** > **Setup** > **Units** > **Units**.
2. Select a unit ID, then select **External codes**.
3. On the **External codes** page, in the **Overview** section, in the **Code** column, enter a code that matches the selected unit ID.

4. In the **Standard code** column, select the checkbox.
5. In the **Value** section, in the **Value** field, enter the external code according to the required codification.

> **Note**
>
> For scenarios where no specific units of measure are assumed, the default value **EA** is used.

Configure sales tax codes

1. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax codes**.
2. Select a sales tax code then on the Action Pane, on the **Sales tax code** tab, in the **Sales tax code** group, select **External codes**.
3. In the **Overview** section, create a line for the selected unit. In the **External code** field, enter the sales tax code you selected in step 2.
4. In the **Value** section, in the **Value** field, enter an external code to use for the selected sales tax code, according to Duty or tax or fee category code (Subset of UNCL5305).

Buyer reference

According to PEPPOL requirements, when you register free text invoices, invoices that are based on sales orders, or project invoices, you must enter either a customer reference or a customer requisition.

Free text invoices

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. Create a new invoice, or select an existing invoice.
3. In the **Header** view, on the **Customer** FastTab, in the **References** section, enter values in the **Customer requisition** and/or **Customer reference** fields.

Sales orders

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Create a new sales order, or select an existing sales order.
3. In the **Header** view, on the **General** FastTab, in the **References** section, enter values in the **Customer requisition** and/or **Customer reference** fields.

Project invoices

1. Go to **Project management and accounting** > **Projects** > **Project contracts**.
2. Create a new project contract, or select an existing project contract.
3. On the **Funding sources** FastTab, select or create a funding source of the **Customer** type, and then select **Details**.
4. On the **Funding source details** page, on the **Other** FastTab, in the **References** section, enter default values for the contract in the **Customer requisition** and/or **Customer reference** fields. Alternatively, you can enter project-specific values in the corresponding fields on the **E-invoice** FastTab.
5. To enter customer requisition and/or reference values directly on the project invoice proposal, follow these steps:
   a. Go to **Project management and accounting** > **Projects invoices** > **Project invoice proposals**.
   b. Create a new invoice proposal, or select an existing invoice proposal.
   c. On the **Invoice proposal header** FastTab, in the **e-Invoice** section, enter values in the **Customer requisition** and **Customer reference** fields.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. Learn more about how to generate electronic invoices in Submit electronic documents.

> ⓘ **Important**
>
> In current implementations, the standard submission procedure that was described earlier only generates electronic invoices and stores them on the service side. The procedure doesn't submit the invoices. To submit electronic invoices, you need to complete the following additional steps.

To submit the generated electronic invoices, follow these steps:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.
2. In the **Channel** field, select the export channel that you previously created. Then select **OK**.

You can check the results of the submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. Learn more in Work with Electronic document submission log.

> **Note**
>
> Submitted electronic invoices are also available in the Outbound folder and its subfolders in your **Ediwin** portal. There, you can monitor further processing of the documents.

The following types of invoices are processed during the submission.

- Invoices based on Sales orders - electronic invoices of **380** type are generated.
- Credit notes based on Sales orders - electronic invoices of **381** type are generated.
- Free text invoices - electronic invoices of **380** type are generated.
- Free text credit notes - electronic invoices of **381** type are generated.
- Project invoices - electronic invoices of **380** type are generated.
- Project credit notes - electronic invoices of **381** type are generated.
- Customer prepayment invoices created using Customer prepayment invoices functionality - electronic invoices of **386** type are generated.
- Customer prepayment credit notes that created as reversals of initial prepayment invoices after a final invoice is issued - electronic invoices of **381** type are generated.

Receive incoming electronic invoices

Before you can import incoming invoices, you must configure some other parameters directly in Finance.

You must configure the following types of master data to provide a match for incoming electronic invoices:

- Vendors
- Products
- Units

Follow the configuration steps in Import vendor electronic invoices. Start from the Configure vendor data section.

Receive electronic invoices

After you complete all the required configuration steps, you can receive incoming electronic invoices in the **PEPPOL** format.

> **Note**
>
> You can review incoming electronic invoices in the Inbound folder and its subfolders in your Ediwin portal.

To receive electronic invoices, follow these steps:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. Select **OK**, and then close the page.

To view the receipt logs for processed electronic invoices, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**.

To view successfully received invoices, go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.

Additional resources

- Electronic invoicing coverage for supported and planned countries and regions

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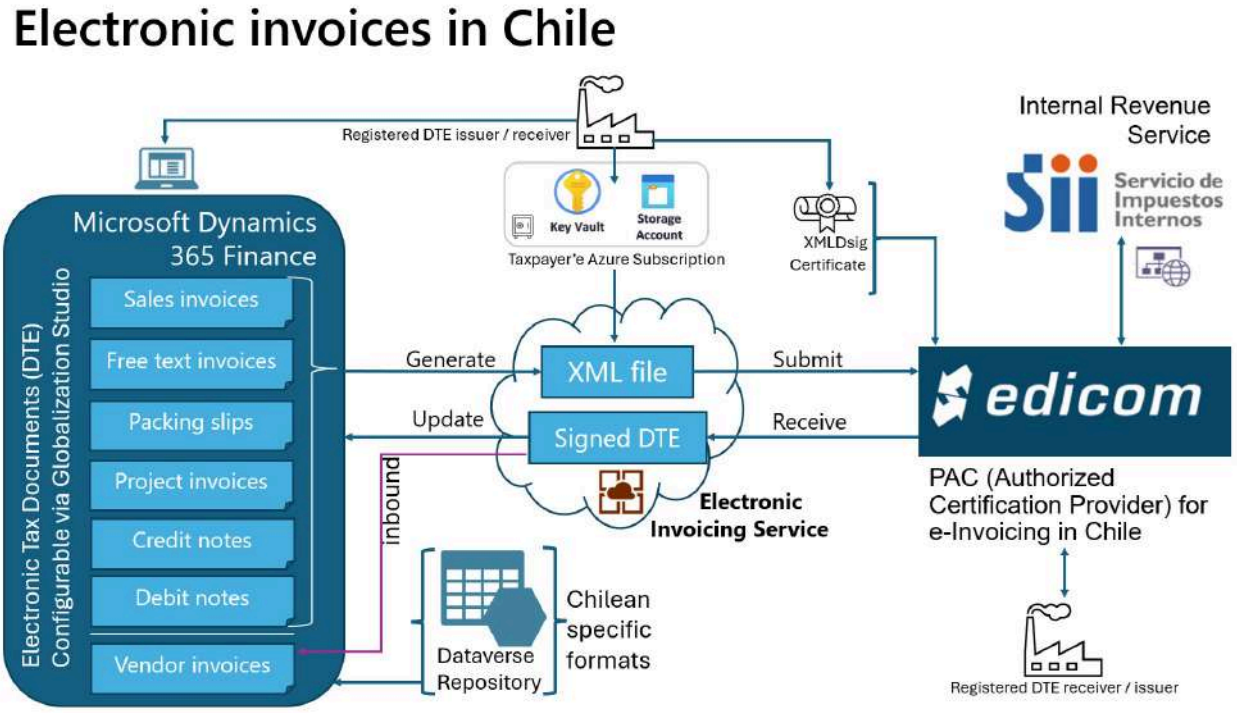
Last updated on 02/24/2026

Get started with Electronic invoicing for Chile

06/24/2025

This article provides information to help you get started with Electronic invoicing for Chile. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. These steps complement the steps that are described in Electronic invoicing setup. For the last-mile integration with the Chilean Tax Authorities, Microsoft is partnering with Edicom.

After you configure Electronic invoicing, you can generate, digitally sign, and submit the XML files of electronic invoices to the Edicom authorized certification provider (PAC) according to the regulatory requirements in Chile.



> **Note**
>
> The electronic invoicing approach that this article describes is implemented by using an invoicing service that is applicable only to cloud deployments of Finance or Supply Chain Management.

> **Important**

The new Electronic invoicing Globalization feature for Chile (outbound flow) requires that you run Finance version 10.0.40. Specifically, you must run build number 10.0.1935.60 or later. The feature can only be imported into the new Globalization Studio. It isn't supported in Regulatory Configuration Service (RCS).

Prerequisites

Before you begin the procedures in this article, the following prerequisites must be met:

1. Ensure that the settings for the Chilean legal entity are in place. For more information, see Set up legal entity and tax information for Chile.
2. Configure electronic invoice parameters for Chile.
3. Gain familiarity with and understanding of Electronic invoicing as it's described in Electronic Invoicing service overview.
4. Do the common part of Electronic Invoicing service configuration as it's described in Electronic invoicing configuration.
5. You must enable the following features in Feature management:
    - Electronic invoicing integration
    - Electronic invoicing integration resubmit document from failed action
    - Execute update actions for submitted documents
6. Make sure that the following Electronic reporting (ER) format configurations are imported. For more information, see: Import Electronic reporting (ER) configurations from Dataverse, Import Electronic reporting (ER) configurations.
    - Customer invoice context model
    - Inventory e-invoice (CL)
    - Inventory Export e-Invoice (CL)
    - E-shipping guide (CL)
    - Project e-invoice (CL)
    - Project Export e-Invoice (CL)
    - Edicom source file response import format
    - Edicom response processing (CL)
    - Edicom response error log import

    Note

These formats are based on the corresponding **LATAM** format configurations that use the **Invoice model LATAM** and **Invoice model mapping LATAM** configurations. All required additional configurations are automatically imported.

Configure the Electronic invoicing feature

The **Chilean electronic invoice (CL) "E-Invoicing for Chile: ISV last-mile connector with Edicom"** feature represents an outbound flow for issuing the following sales documents:

Expand table

Name	Code	Original name
Invoice	33	Factura Electrónica
Untaxed Invoice	34	Factura no afecta Electrónica
Packing slip (e-shipping guide)	52	Guía de Despacho Electrónica
Debit note	56	Nota de Débito Electrónica
Credit note	61	Nota de Crédito Electrónica
Export invoice	110	Factura de Exportación
Export debit note	111	Nota de Débito de Exportación
Export credit note	112	Nota de Crédito de Exportación

Some parameters of the feature are published with default values. Before you deploy the Electronic invoicing feature to the service environment, add a feature that is based on the Microsoft-provided feature, and complete the common parameters on the **Feature parameters** tab. Review the default values, and update them as required, so that they better reflect your business operations.

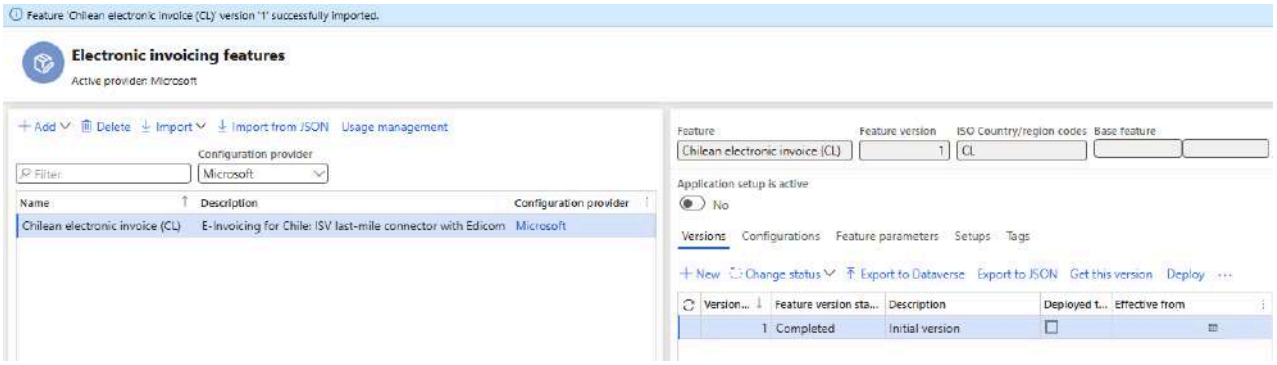
For Chile, there are at least three interactions with Edicom in the pipeline: first to submit the invoice, then to fetch the signed XML, and finally to fetch the status of the submitted invoice. Each interaction requires common parameters, such as Edicom connection details and the authentication token that Edicom provides. These common parameters are reused in the feature setup for all document types. Edicom provides the values when a company is onboarded.

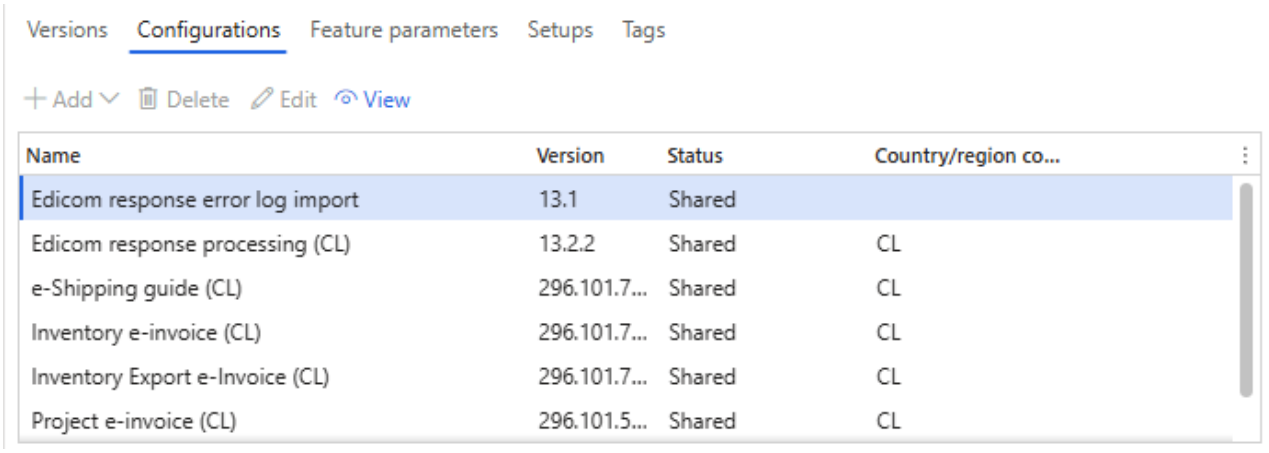
Note

The configuration of common parameters is simplified. You no longer have to go to each action and feature setup, and repeatedly specify the common connection parameters. Use of the **Feature parameters** tab is available only as of version 10.0.41.

> **Important**
>
> The **Chilean electronic invoice (CL)** feature is provided by Microsoft. Before it can be used, additional configuration is required, as described in this article. For information about how to configure invoicing features and apply changes, see Work with feature setups. For example, in addition to the connection parameters, you can filter specific legal entities so that they are processed in applicability rules. By default, the feature is applicable to all legal entities that have a primary address in Chile.

1. Import the latest version of the **Chilean electronic invoice (CL)** Globalization feature as described in Import features from the repository. The following illustrations show what the feature looks like after you import it from Dataverse.





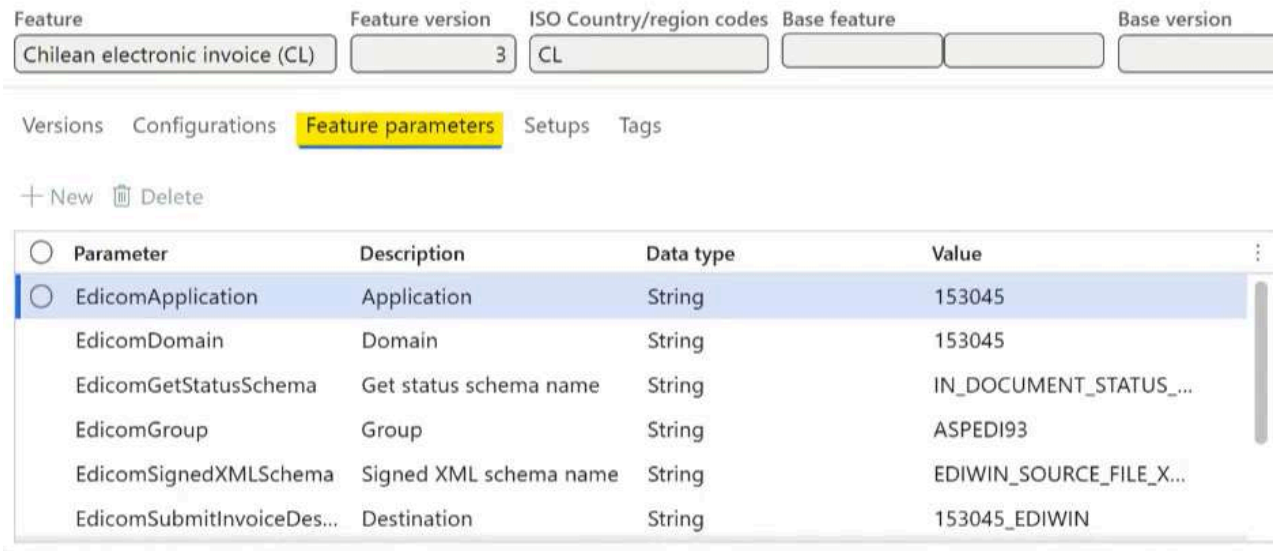
2. Create a copy of the imported Globalization feature, and select your configuration provider. For more information, see Create a Globalization feature.

3. On the **Versions** tab, confirm that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with Edicom's API:

    - Select **Application**, and then enter the service ID number that you obtained.
    - Select **Domain**, and then enter the same service ID number.
    - Select **Get status schema name**, and then enter the schema name.
    - Select **Group**, and then enter the group code that you obtained.
    - Select **Signed XML schema name**, and then enter the schema name.
    - Select **Destination name**, and then enter the service ID number concatenated with the string "_EDIWIN." For example, if the service ID number is 123456, enter **123456_EDIWIN**.
    - Select **Submit invoice schema name**, and then enter the schema name.
    - Select **Auth token**, and then select the name of the secret that you created for the token.
    - Select **Web service URL**, and confirm the web address.

    The following illustration shows these feature parameters set to the values that Edicom provided to Microsoft for testing purposes. The values that you enter will differ. Edicom provides these values to you when you're onboarded.



5. The copy of the feature is always created as a **Draft** version. Regardless of whether you made changes, you must complete, publish, and deploy the feature as described in Complete and deploy a Globalization feature.

Outbound flow pipeline

To review the processing pipeline, on the **Setups** tab, go to **Feature setup**, select the derived document type that you want, and then select **Edit**. The outbound flow consists of the following actions:

1. **Transform document** – Generate a format that can be sent to Edicom.
2. **Integrate with Edicom** – Submit the generated invoice to Edicom.
3. **Get status from Edicom for an invoice** – Fetch the signed XML from Edicom. This document might not be immediately available, because the PAC needs time to generate it.

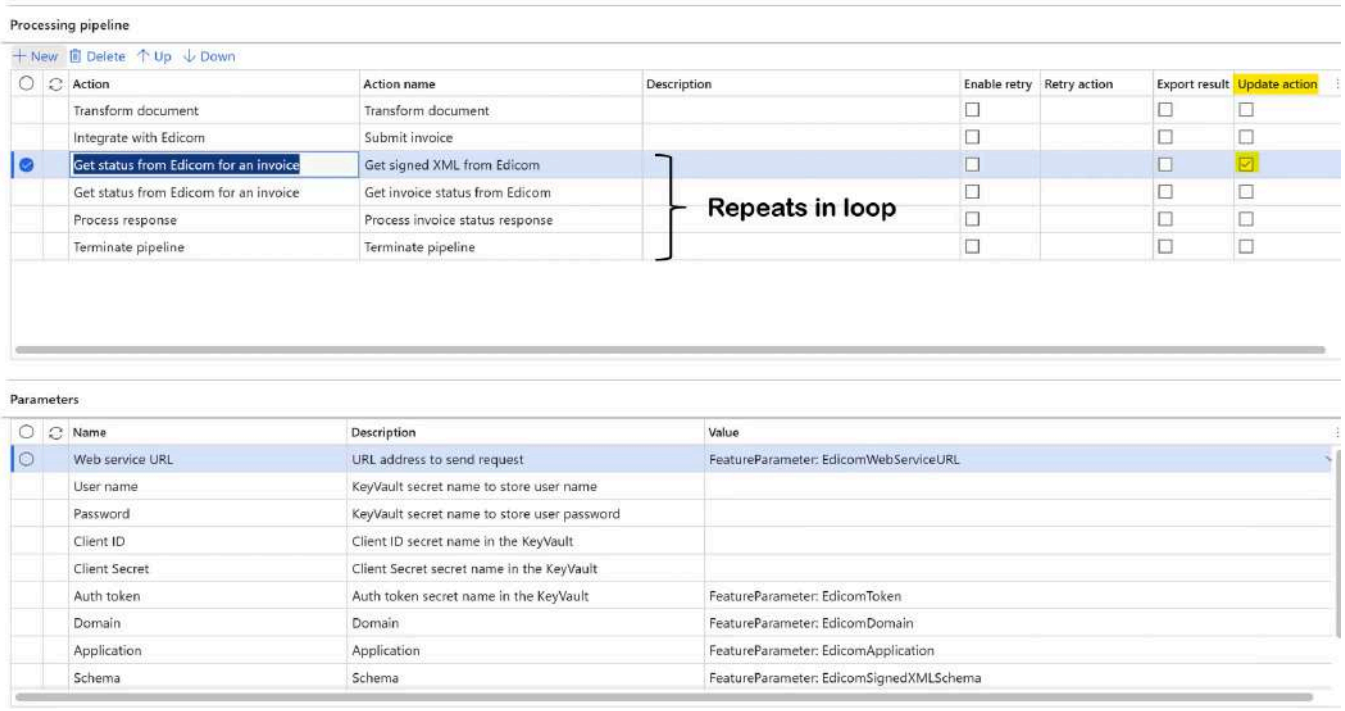
   > ⓘ **Note**
   >
   > At this point in the flow, the concept of *update actions* comes into play. Notice that the **Update action** checkbox is selected for this step. Therefore, this step and all subsequent steps run in a loop until the system determines that a terminal state has been reached.

4. **Get status from Edicom for an invoice** – Fetch the status of the submitted invoice from Edicom.
5. **Process response** – Process the received response to determine whether a terminal state has been reached.

   - If the status response indicates a failure, the pipeline is terminated, and the submission is marked as failed.
   - Even if the status response indicates successful submission to the Chilean Internal Revenue Service (SII), the pipeline can't be completed yet, because customers or buyers in Chile can reject invoices for up to eight days. During this time, the pipeline is kept on hold in a **Pending execute update action** state. If a response is received that indicates customer rejection of the invoice, it's detected during the **Process response** step, and the pipeline is marked as failed.

6. **Terminate pipeline** – Wait the specified number of days, and then terminate the pipeline. In the out-of-box default setup, the pipeline is terminated with a **Completed** status if more than nine days pass since the invoice was submitted. If there are no rejections, the **Terminate pipeline** step marks the pipeline as completed.

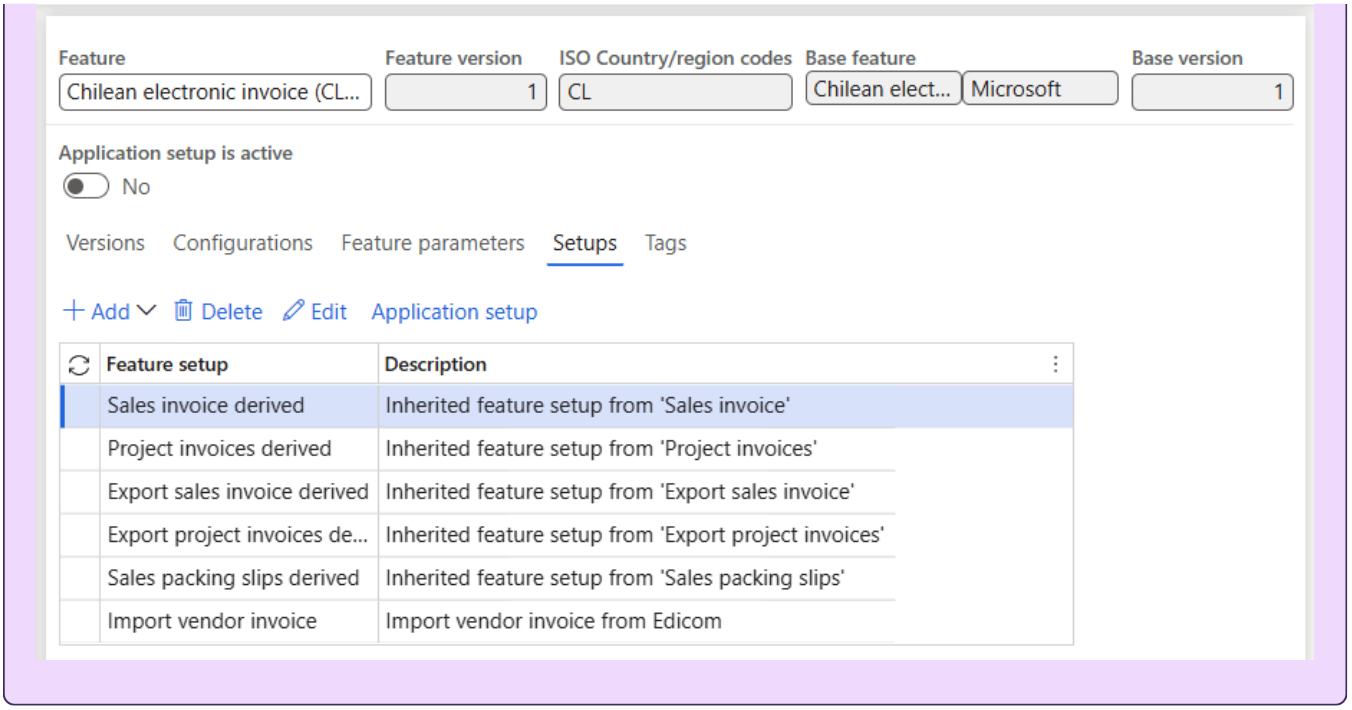
Set up actions and parameters



ⓘ Note

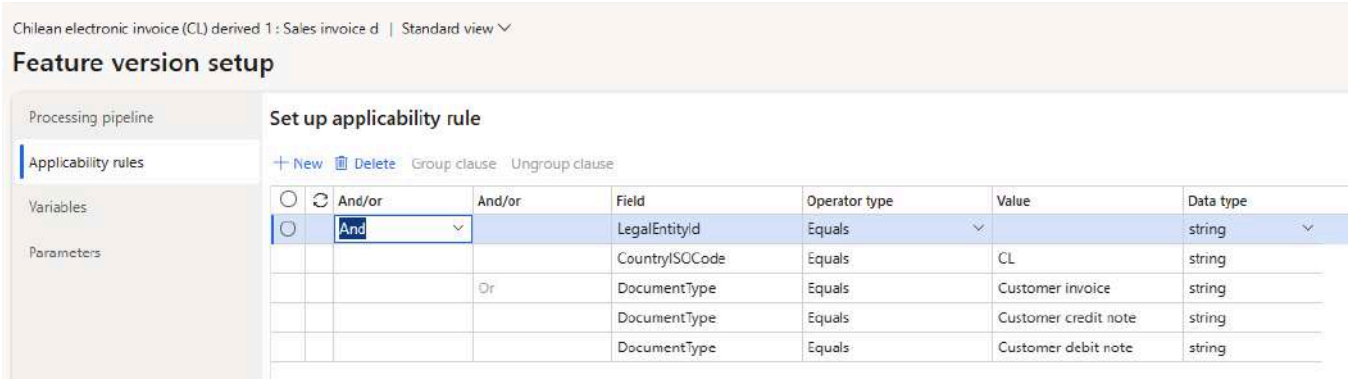
There is a similar setup for each format that is included in the **Chilean electronic invoice (CL)** feature. For Chile, there are six setups: five for sales documents (outbound flow) and one for import vendor invoices (inbound flow).

- Sales invoice derived
- Project invoices derived
- Export sales invoice derived
- Export project invoices derived
- Sales packing slips derived
- Import vendor invoice



Applicability rules

Applicability rules must be correctly configured to provide context, so that the exact Electronic invoicing Globalization feature that must run in Electronic Invoicing service can be found. Applicability rules are provided out of the box by checking the legal entity in the International Organization for Standardization (ISO) country/region code. This particular feature setup supports all three types of invoices: customer invoices, debit notes, and credit notes.

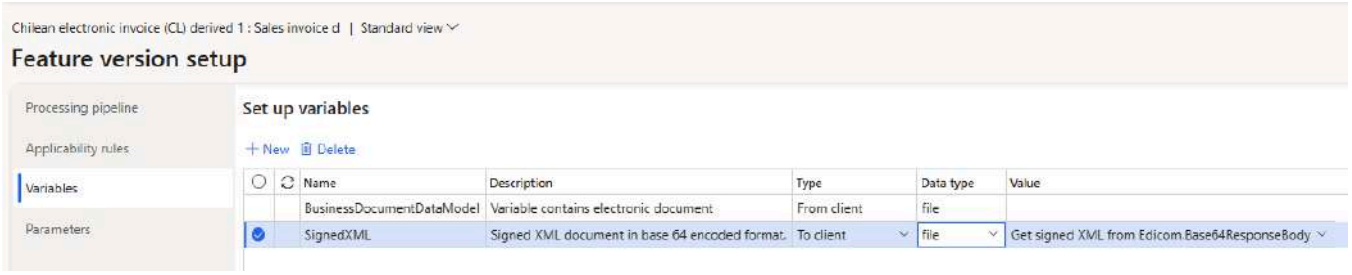


Variables

The following variables are used in the outbound data flow actions of the Chilean feature. (These variables are provided out of the box.)

- **BusinessDocumentDataModel** – The Business Document Data model variable that is received from Finance or Supply Chain Management and transformed into the format that is required for submission.
- **SignedXML** – The signed XML variable that is sent back to Finance or Supply Chain Management. This variable contains the base64-encoded response body from the **Get**

Signed XML from Edicom step. It's used in the response types to save the signed XML that is obtained from Edicom as an attachment to the invoice journal. It's also used to generate printable reports through QR codes.



After you import the **Electronic invoicing for Chile** feature that includes the out-of-box default feature setup, follow the steps in the next section to configure electronic documents.

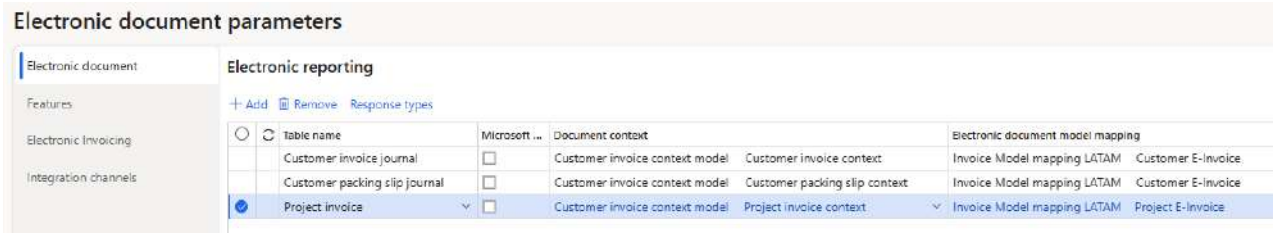
Configure electronic document parameters

1. Make sure that the country/region-specific ER configurations for the document context and electronic document model mapping that are required for Chile are imported. For more information, see Set up Electronic document parameters.

2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

3. In the **Electronic document** section, add records for the **Customer Invoice journal**, **Customer packing slip journal**, and **Project invoice** table names.

4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.



5. Save your changes, and close the page.

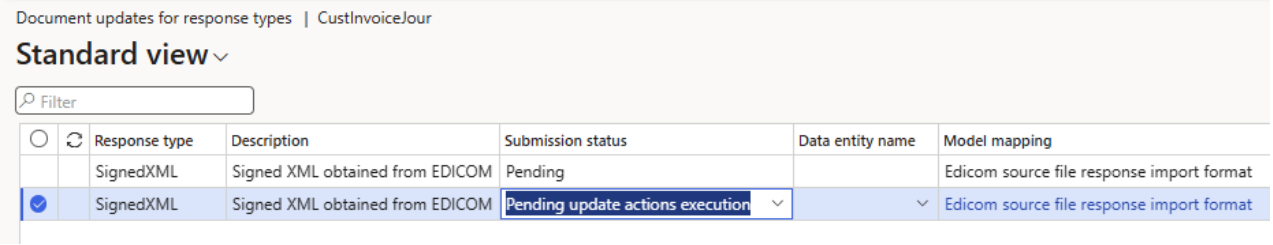
6. For each table name, follow these steps:

   a. Select **Response types**, select **New** to create a response type that you will get from the back end, and enter the following values:

      - In the **Response type** field, enter **SignedXML** (the default value). (See the Variables section of this article.)
      - In the **Description** field, enter any meaningful name. Alternatively, leave the field blank.

- In the **Submission status** field, select **Pending**.
- In the **Model mapping** field, select **Edicom source file response format**.

b. Repeat the preceding step, but set the **Submission status** field to **Pending update actions execution**. In this way, you can run a subset of the actions in the processing pipeline in a loop to continuously pull updated statuses and other information for the submitted documents from the tax authority.



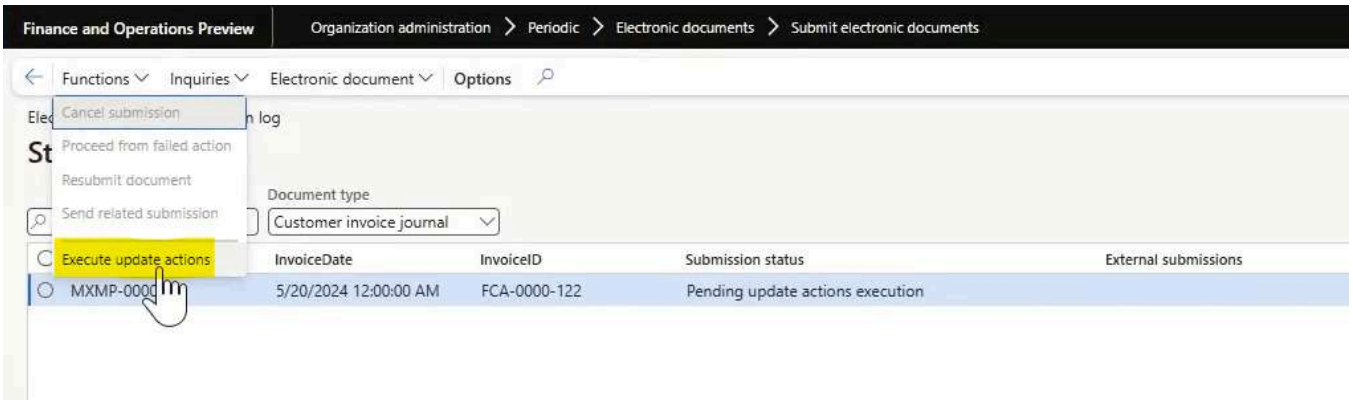
> **Note**
>
> The response includes the signed XML that is obtained from Edicom. This signed XML is stored in the system as an attachment to the corresponding invoice journal. It will eventually be used to generate printable invoices through QR codes.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information about how to generate electronic invoices, see Submit electronic documents.

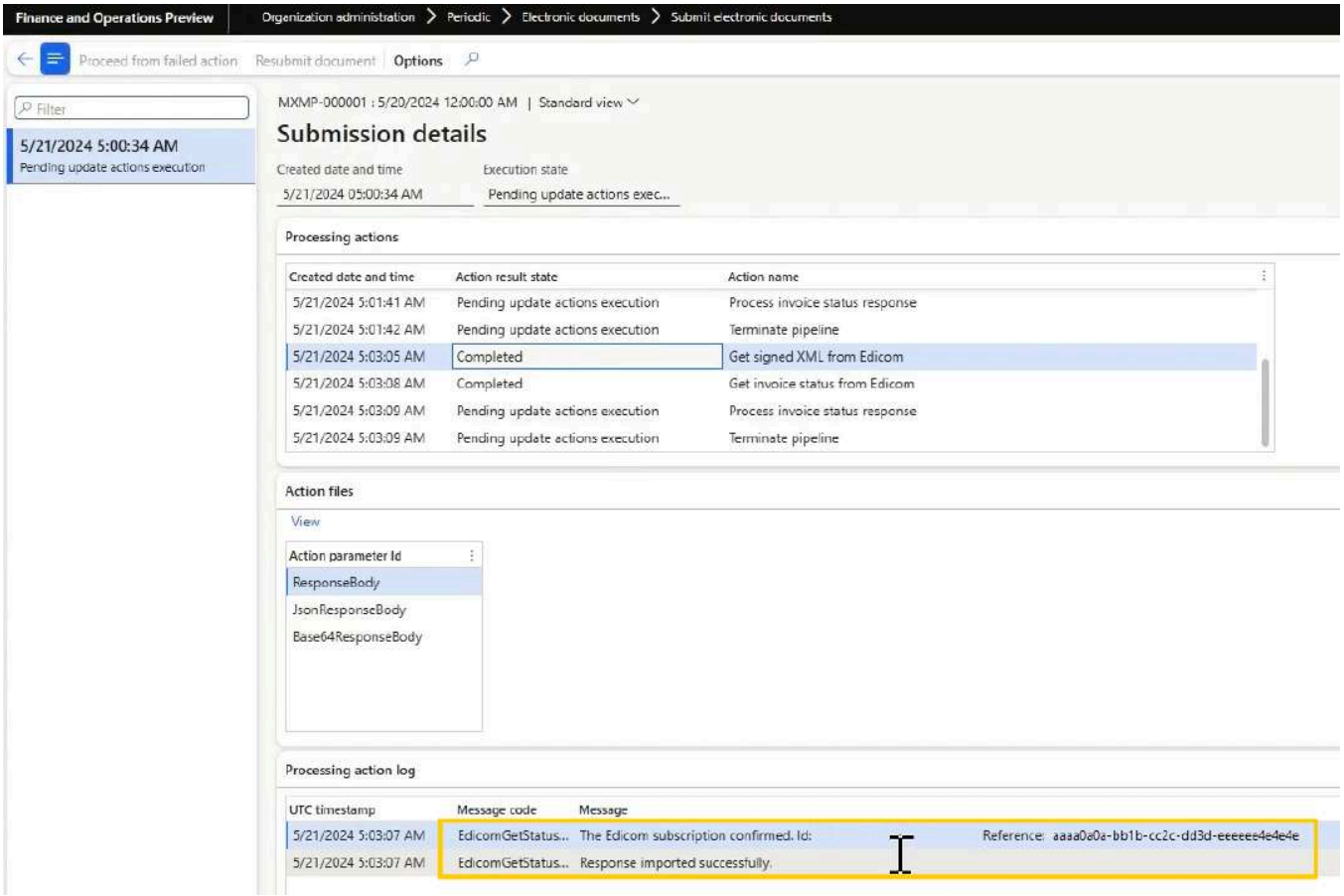
To inquire about the results of a submission, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**, and select the required document type. For more information, see Work with Electronic document submission log.

For Chile, after you submit the invoice, the submission status is set to **Pending update actions execution**. The response body is probably empty for the signed XML and the call to get the invoice status. An empty response body indicates that the XML wasn't available immediately after submission. To address the pending status, a function that is named **Execute update actions** is used. This function resumes the pipeline from the action that is marked as an update action, and then runs all subsequent actions in the pipeline in a loop. The status should change to **Executing** again. Then, after a few seconds, it reverts to **Pending update actions execution**.

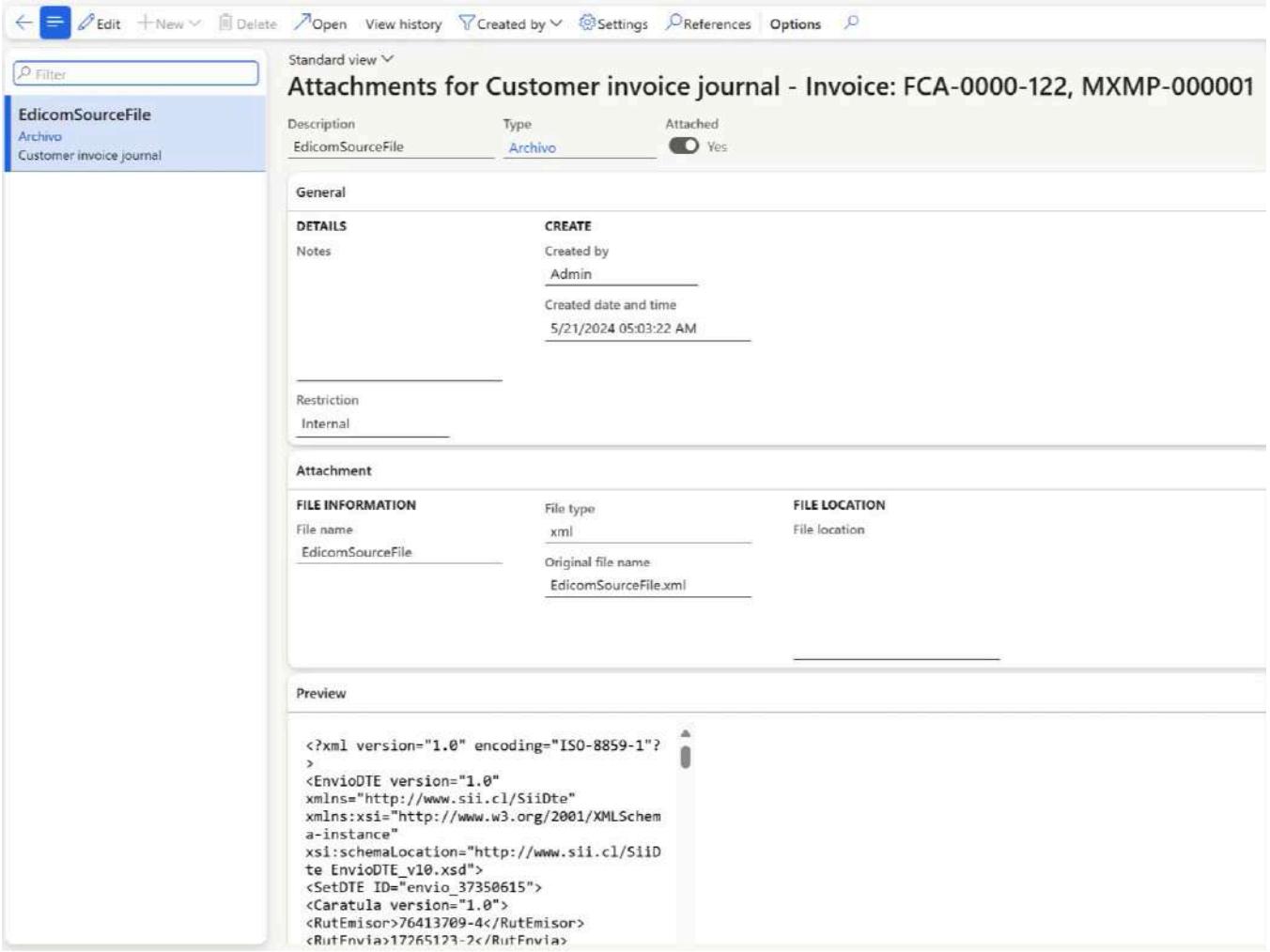


> **Note**
>
> By setting a recurrence schedule, you can configure the **Execute update actions** function to run in batch mode on a periodic basis.

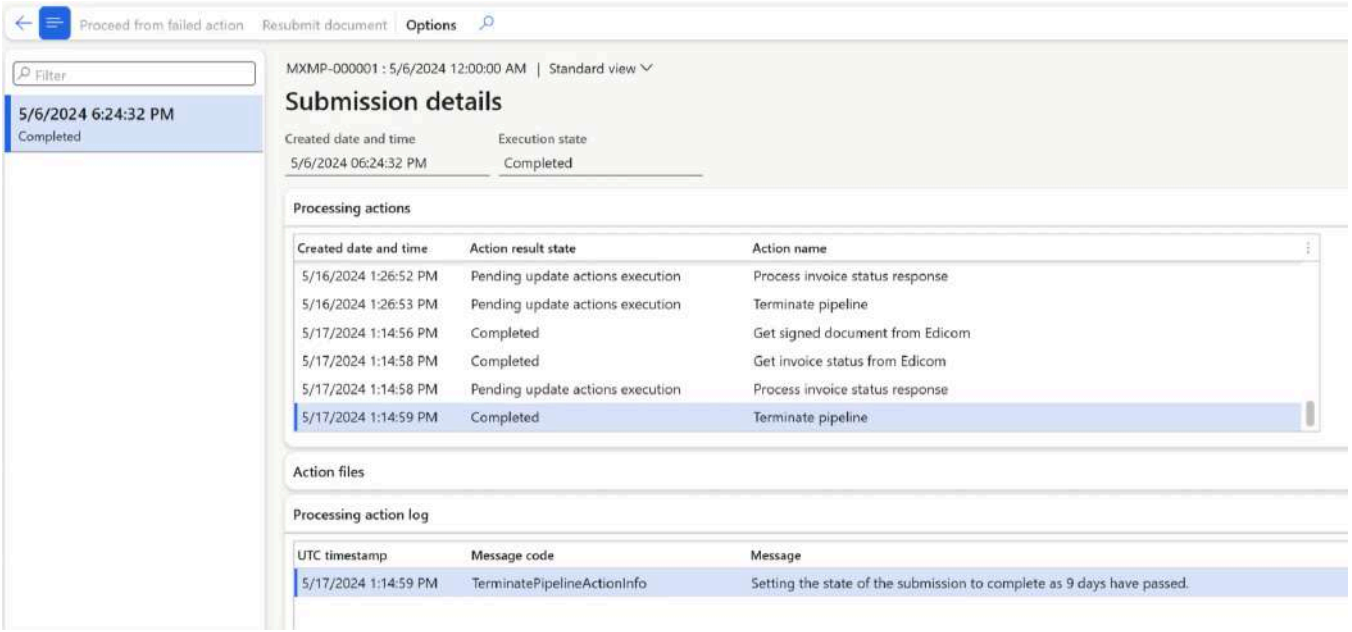
When you review the submission details, you should notice that the steps run again. This time, the signed XML is received.



As a result of the completed outbound flow, the signed XML is attached to the invoice journal as **EdicomSourceFile**.



If no business rejection or acceptance is received from the buyer after nine days, the **Terminate pipeline** action completes the pipeline.



More resources

- Electronic Invoicing service overview

- Get started with Electronic invoicing service administration
- Setting up Electronic Invoicing
- Electronic Invoicing service independent software vendor (ISV) last-mile connector
- Dynamics 365 Country expansion: localizations for LATAM countries | June 27, 2024
- Dynamics 365 Country expansion: localizations for LATAM countries | D365 FastTrack Tech Talks (youtube.com)

Vendor electronic invoice import in Chile

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure and use vendor electronic invoice import for Chile from the country-specific XML format in Microsoft Dynamics 365 Finance.

Prerequisites

Before you complete the tasks in this article, the following prerequisites must be met:

1. The primary address of the legal entity must be in Chile.
2. Ensure that the settings for the Chilean legal entity are in place. For more information, see Set up legal entity and tax information for Chile.
3. Configure electronic invoice parameters for Chile.
4. Gain familiarity with and understanding of Electronic invoicing as it's described in Electronic invoicing overview.
5. Do the common part of Electronic Invoicing service configuration as it's described in Electronic invoicing configuration.
6. The **Enable application response for vendor invoices** feature must be enabled in Feature management.
7. Ensure that the following Electronic reporting (ER) format configurations are imported:
    - Customer invoice context model
    - Inventory e-invoice (CL) XML Import
    - Vendor Invoice Mapping to Destination LATAM
    - Application response (CL)
    - Pending vendor invoice model mapping

> **Note**
>
> The ER format is based on the **Invoice model LATAM** configuration and uses the **Vendor Invoice Mapping to Destination LATAM** configuration. All additional configurations that are required are automatically imported.

Configure the Electronic invoicing feature

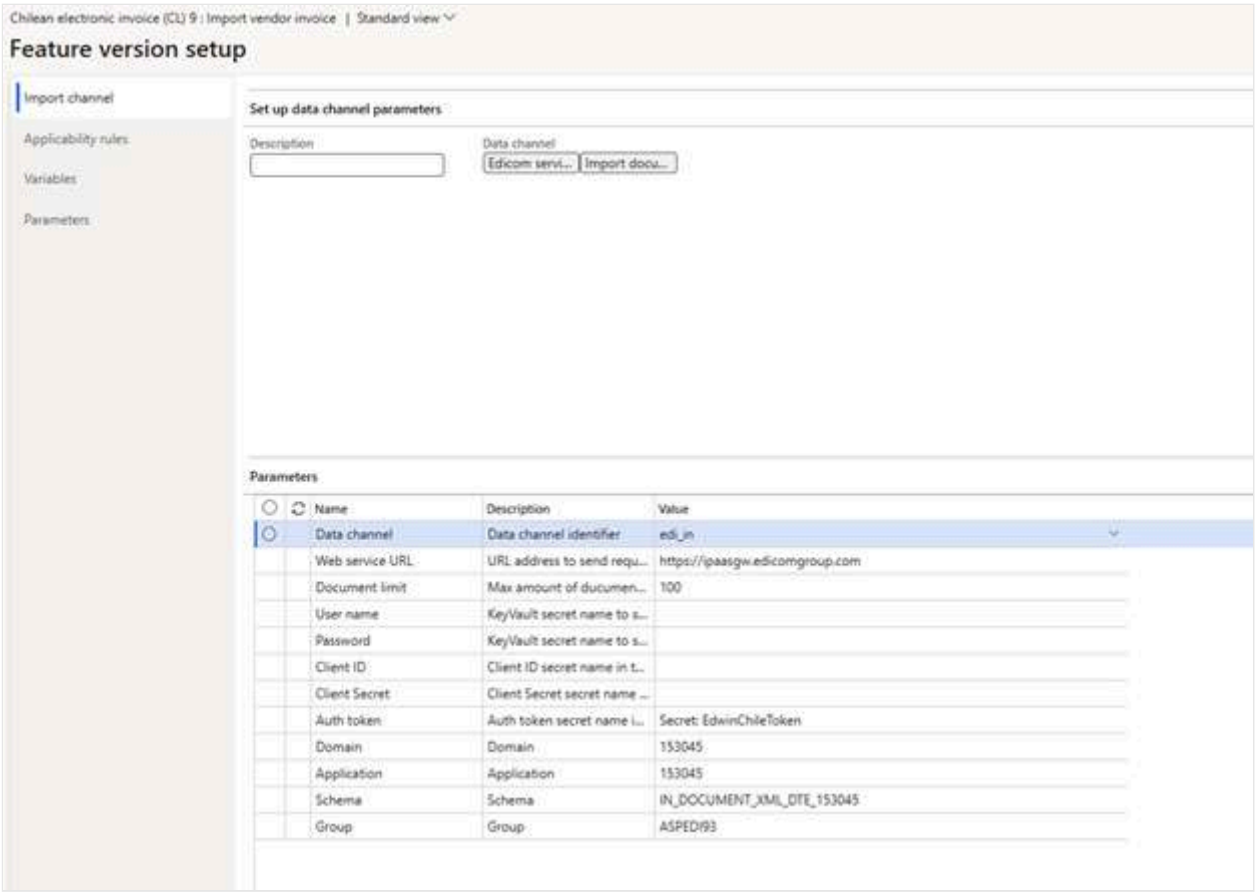
> **Important**
>
> The inbound flow of the Electronic invoicing Globalization feature for Chile requires that you run Finance version 10.0.41 or later. It can be configured only in the new Globalization Studio. It isn't supported in Regulatory Configuration Service (RCS).

Import channel

To enable an inbound flow, you must create a feature setup of the **Import channel** type for the **Chilean electronic invoice (CL) "E-Invoicing for Chile: ISV last-mile connector with Edicom"** Globalization feature.

1. Select the **Setups** tab of the derived feature.
2. Select **Add**, and then select **Custom setup**.
3. Enter a name and description for the feature setup.
4. Select **Import channel** as the setup type and **Edicom service** as the data channel.
5. Select **Create**.
6. Select the feature setup that is created for vendor invoice import.
7. Select **Edit**.
8. On the **Import channel** tab, set the import channel parameters.

   The following illustration shows the parameters for the feature setup set to the values that Edicom provided to Microsoft for testing purposes. The values that you enter will differ. Edicom provides these values to you when you're onboarded.

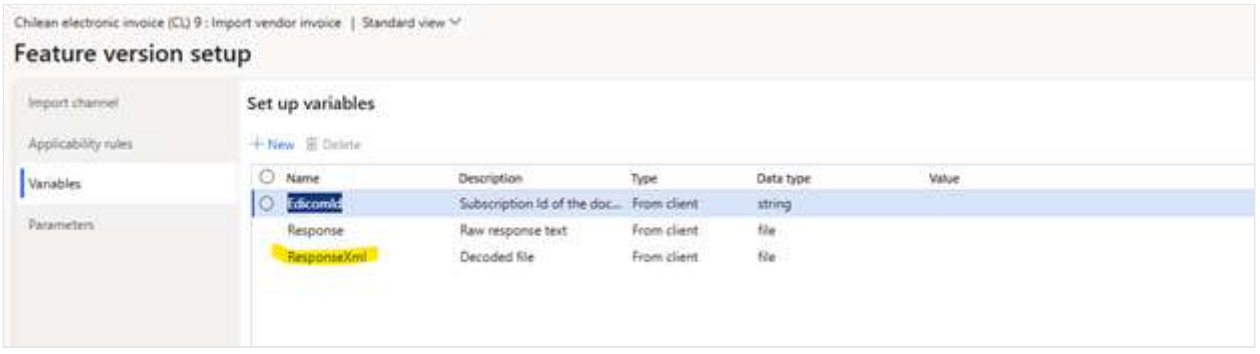


Applicability rules

Applicability rules must be correctly configured to provide context, so that the exact Electronic invoicing Globalization feature that must run in the Electronic Invoicing service can be found. Create a new applicability rule for the same channel name that you specified for the import channel configuration in the previous section. For example, add an applicability rule that has an **And** clause for the **Channel** field when its value equals the "edi_in" string.

Variables

Out of the box, variables are provided with the specific feature setup to support the `EdicomId`, `Response`, and `ResponseXML` variables, as shown in the following illustration.



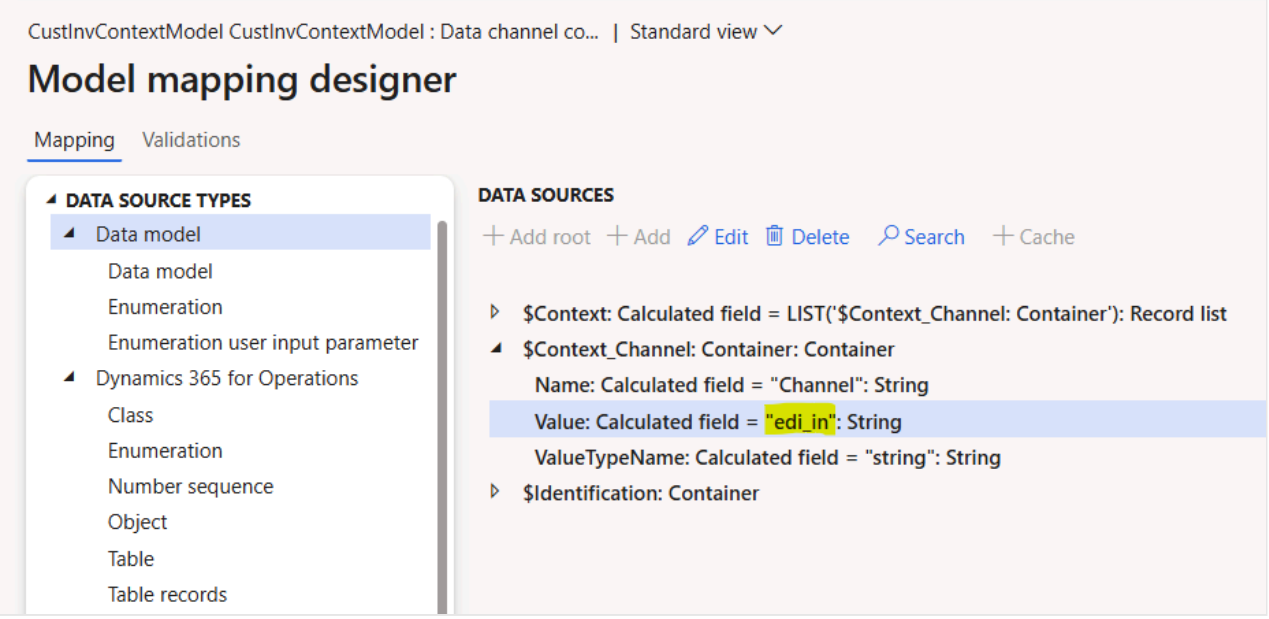
> ⓘ Note

After the feature setup is completed, save, complete, and deploy the version.

Configure ER

1. Go to the **Electronic reporting** workspace, and open **Reporting configurations**.
2. Derive a configuration from **Customer Invoice Context Model**.
3. In the new configuration, go to **Designer** > **Map model to datasource** > **Data channel** > **Designer**.
4. Point to the Data model source type under the **Mapping** tab on the left.
5. To set up the data channel context, expand the **$Context_Channel** container and update the **Value** calculated field with your channel name.
6. Save your change.

   The following illustration shows an example of the change.



7. Return to **Reporting configurations**.
8. Select the **Vendor invoice Mapping to destination LATAM** configuration.
9. Set the **Default for model mapping** option to **Yes**.

Configure parameters

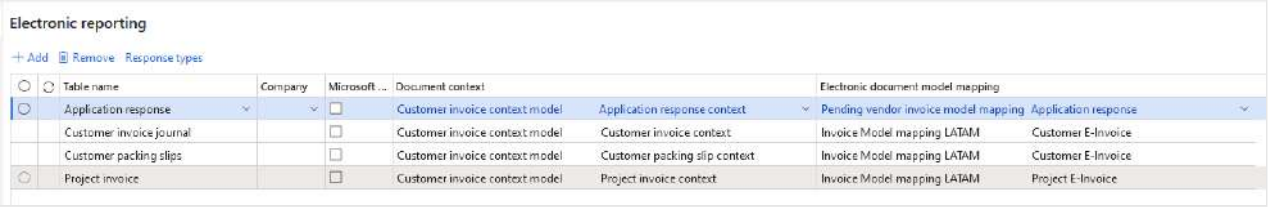
Configure electronic document parameters

1. Make sure that the country/region-specific ER configurations for the document context and electronic document model mapping that are required for Chile are imported. For more information, see Set up Electronic document parameters.

2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

3. On the **Electronic document** tab, add a record, and set the following values for it:

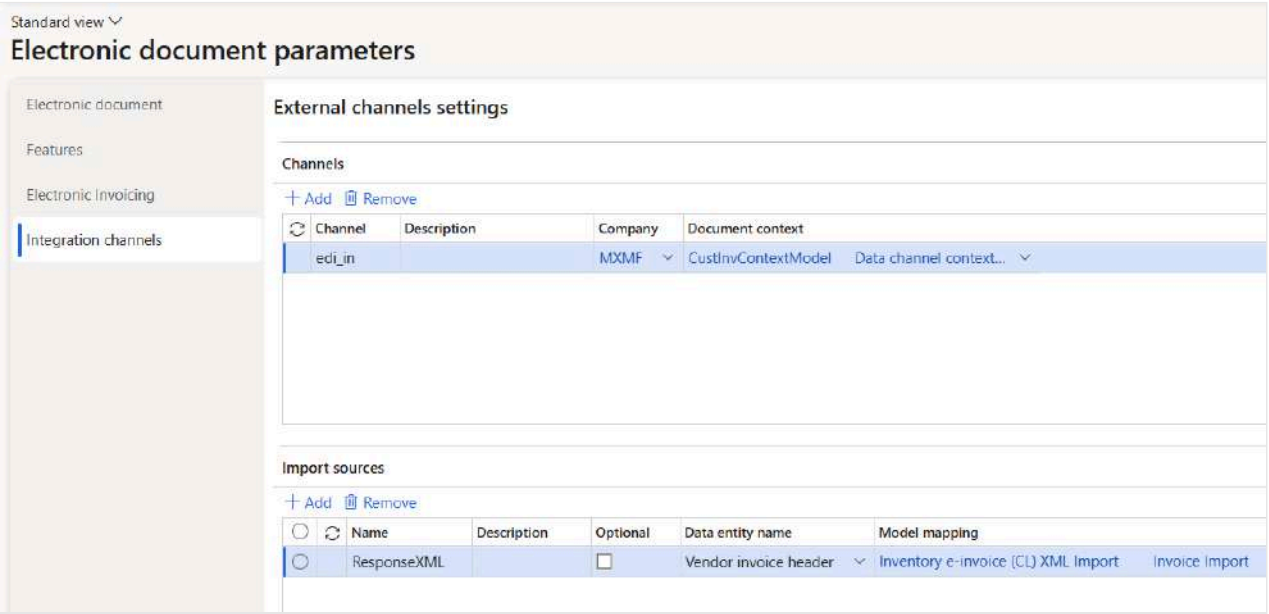
    - **Table name:** Application response
    - **Document context:** Customer invoice context model - Application response context
    - **Electronic document model mapping:** Pending vendor invoice model mapping - Application response



4. On the **Integration channels** tab, create records. Specify a channel name of the **Import** type, and specify the document context configuration for every required company.

5. For each channel, add an import source, and set the following values for it:

    - **Name:** Enter the response format.
    - **Data entity name:** Select **Vendor invoice header**.
    - **Model mapping:** Specify the model mapping for the invoice import.



6. Save your changes, and close the page.

Configure vendor data

During the import process, vendors are identified by their tax exempt number. To enable correct vendor identification, follow these steps:

1. Go to **Accounts payable** > **Vendors** > **All vendors**, and select a vendor.
2. On the **LATAM** FastTab, in the **Country identification number** field, enter a valid identification number of the country for the vendor. This number is used to identify the vendor during import, by matching it to the value of the **DTE\Documento\Encabezado\RUTEmisor** element in the import XML file.

Configure products

During the import process, products are identified by their external descriptions. These descriptions are usually vendor-specific. To enable correct product identification, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a product, and then, on the **Purchase** menu, in the **Related information** section, select **External item description**.
3. Create a new external description for the selected product.
4. In the **Account code** column, select **Table** to define an external product description for a specific vendor.
5. In the **Vendor relation** column, select a vendor.
6. In the **External item number** column, enter an external product code. This code is used to identify the product during import, by matching the code to the value of the **DTE\Documento\Detalle\NmbItem** element in the import XML file.

> ⓘ **Note**
>
> If there must be non-stocked items, the system expects that these products belong to an item model group where the **Stocked product** checkbox is cleared on the **Inventory policy** page.
>
> If no related **Non-stock** products exist, the system tries to import invoice lines by referring to a default item. The default item must be configured in the system as a released product where the code is defined exactly as **DEFAULT_ITEM**, and the product must belong to an item model group where the **Stocked product** checkbox is cleared on the **Inventory policy** page.

Import vendor electronic invoices and send application responses

To run the import vendor electronic invoices, follow these steps:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. In the **Receive electronic documents** dialog box, specify the parameters as required.
3. Select **OK** to immediately start the import process or to schedule the import to run in the background.

Import process description

Here is an overview of the steps in the import process and the order that they occur in.

1. Vendors are identified by using the country identification number that is defined in the vendor record. If no vendor matches the data that is being searched, the import process fails, and a related error message is shown.
2. Products that are used on invoice lines are identified by using an external item number, which might be vendor-specific. If no product matches the external description, the import process fails, and a related error message is shown.
3. If an incoming import file contains the information about purchase orders and its lines in the **DTE\Documento\Referencia\FolioRef** elements, the numbers are used for invoice matching with purchase orders and lines that are entered in the system.
4. If no order or line references are defined in an incoming file, the system tries to automatically match incoming vendor invoices with existing purchase orders.
5. If no purchase order is found, the system raises a warning but continues the import. It now considers products on invoice lines **Non-stock** items. The system expects that these products belong to an item model group where the **Stocked product** checkbox is cleared on the **Inventory policy** page.
6. If no related **Non-stock** products exist, the system tries to import invoice lines by referring to a default item. The default item must be configured in the system as a released product where the code is defined exactly as **DEFAULT_ITEM**, and the product must belong to an item model group where the **Stocked product** checkbox is cleared on the **Inventory policy** page. If no default item is configured in the system, the import process fails, and a related error message is shown.

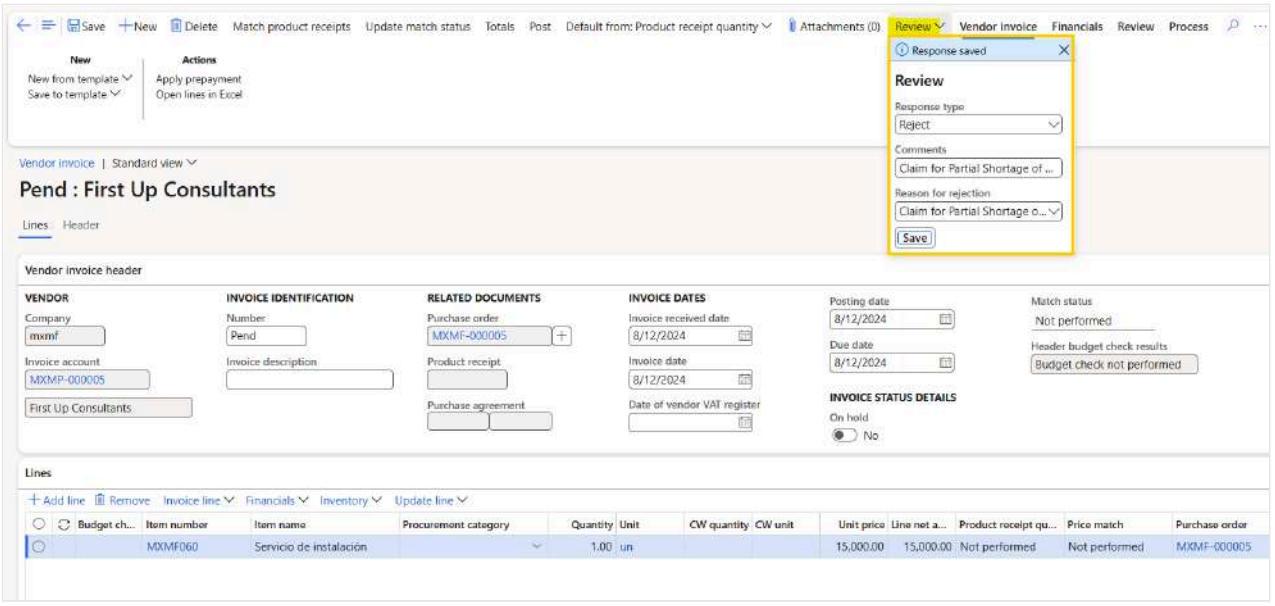
Successfully imported vendor electronic invoices are shown in the system as pending invoices. To review imported invoices, go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.

Application response

After the imported vendor electronic invoice is reviewed on the **Pending vendor invoices** page (**Accounts payable** > **Invoices** > **Pending vendor invoices**), an application response can be saved and submitted. The application response indicates either acceptance or rejection of the invoice. If the invoice is either accepted with discrepancies or rejected, the response includes comments. If the invoice is rejected, the response also includes the reason for the rejection.

To fill in the application response, follow these steps:

1. Open the **Pending vendor invoices** page in edit mode for the desired invoice.
2. On the Action Pane, select **Review**.
3. In the dropdown dialog box, in the **Response type** field, select one of the following values, as required:
    - Accept
    - Accept with discrepancies
    - Reject
4. If you selected either **Accept with discrepancies** or **Reject** as the response type, enter comments in the **Comments** field.
5. If you selected **Reject** as the response type, select one of the following values in the **Reason for rejection** field:
    - Claim to the content of the document
    - Claim for Partial Shortage of Goods
    - Claim for Total Shortage of Goods
6. Select **Save**.



7. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**.
8. On the **Records to include** FastTab, select **Filter**.
9. On the **Joins** tab, add vendor invoices to the join for the application response.
10. On the **Range** tab, add a criterion for the invoice number to the **Vendor invoices** table.
11. Submit the application response.

> **ⓘ Important**
>
> After the application response is submitted, it can't be modified.
>
> Application responses must be sent before the pending vendor invoices are posted. The application response capability isn't available for posted invoices.

Learn more

- Get started with Electronic invoicing for Chile

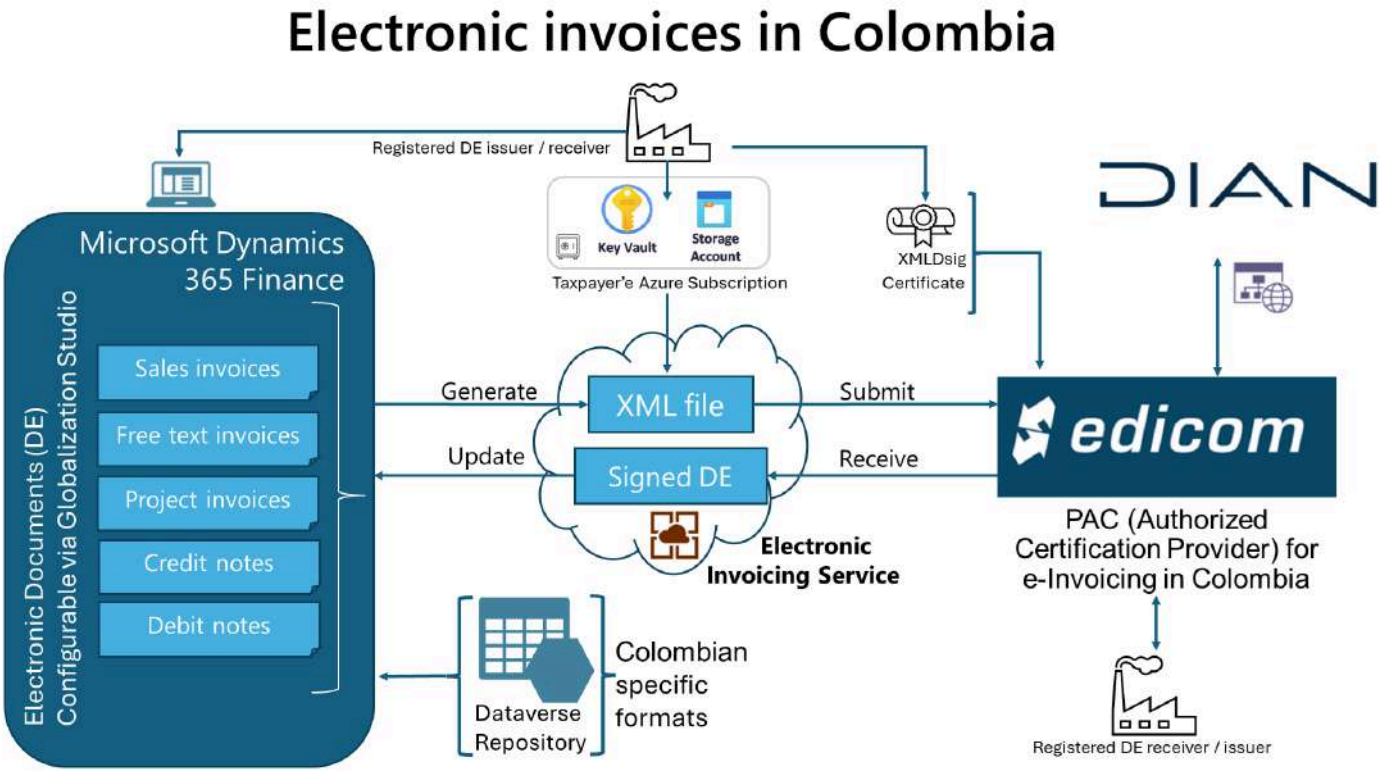
Last updated on 04/13/2026

Get started with Electronic invoicing for Colombia

- Introduction
- Prerequisites
- Configure the electronic invoicing feature
- Edicom Interactions for Colombia
  - Additional Interactions (Conditionally Triggered)
- Outbound flow pipeline
- Electronic invoice feature
  - Setups-View
  - Feature version setup
    - Processing pipeline
    - Applicability Rules and Feature Setup Scope
    - Variables
  - Setups-Application Setup
- Configure electronic document parameters
- Issue electronic invoices
- More resources

This article provides information to help you get started with Electronic invoicing for Colombia. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. These steps complement the steps that are described in Electronic invoicing setup. For the last-mile integration with the Colombian Tax Authorities, Microsoft is partnering with Edicom.

After you configure Electronic invoicing, you can generate, digitally sign, and submit the XML files of electronic invoices to the Edicom authorized certification provider (PAC) according to the regulatory requirements in Colombia.



> **Note**
>
> The electronic invoicing approach that this article describes is implemented by using an invoicing service that is applicable only to cloud deployments of Finance or Supply Chain Management.

Prerequisites

Before you begin the procedures in this article, the following prerequisites must be met.

1. Ensure that the settings for the Colombian legal entity are in place. For more information, see Set up a legal entity and tax information for Colombia.
2. Gain familiarity with and understanding of Electronic invoicing as it's described in Electronic invoicing overview.
3. Do the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.
4. You must enable the following features in Feature management:
    - Electronic invoicing integration
    - Electronic invoicing integration resubmit document from failed action
    - Execute update actions for submitted documents

5. Ensure that the following Electronic reporting (ER) format configurations are imported. For more information, see: Import Electronic reporting (ER) configurations from Dataverse, Import Electronic reporting (ER) configurations.

    - Customer invoice context model
    - Edicom response error log import
    - Edicom Response Processing CO
    - Edicom source file response import format
    - Inventory e-Invoice (CO)
    - Inventory e-DebitNote (CO)
    - Inventory e-CreditNote (CO)
    - Project e-Invoice (CO)
    - Project e-DebitNote (CO)
    - Project e-CreditNote (CO)

    > ⓘ **Note**
    >
    > These formats are based on the corresponding **LATAM** format configurations that use the **Invoice model LATAM** and **Invoice model mapping LATAM** configurations. All other required configurations are automatically imported.

[!IMPORTANT] **Colombia – Referencing another document using CUFE or CUDE**

For Colombia, whenever a document must reference another one — for example:

- Credit Notes that cancel or correct an invoice
- Invoices that cancel Credit Notes or Debit Notes
- Debit Notes that cancel or correct a document The relationship is established only through the CUFE (Unique Electronic Invoice Code) or CUDE (Unique Electronic Document Code) of the original document.

Therefore, the referenced document must have been previously submitted and approved by the DIAN, so that the CUFE or CUDE is available in the official signed XML. If the referenced document has not been approved by the DIAN, this information will not be available and cannot be included in the document being processed.

Configure the electronic invoicing feature

The **Colombia Electronic Invoice (CO) Edicom integration for Colombia**" feature represents an outbound flow to issue the following sales documents.

Expand table

Name	Code	Original name
Electronic sales invoice	01	Factura electrónica de venta
Export Invoice	02	Factura de exportación
Credit Note	91	Nota de Crédito
Debit Note	92	Nota de Débito

Some parameters of the feature are published with default values. Before you deploy the Electronic invoicing feature to the service environment, add a feature that is based on the Microsoft-provided feature, and complete the common parameters on the **Feature parameters** tab. Review the default values, and update them as required, so that they better reflect your business operations.

Edicom Interactions for Colombia

For Colombia, there are at least three core interactions with Edicom in the base flow:

1. **Invoice submission**
2. **Signed XML retrieval**
3. **Status retrieval of the submitted invoice**

Additional Interactions (Conditionally Triggered)

In addition to the base flow, up to three extra interactions may occur depending on the response from the receiver:

1. **Status 30 – Invoice reception**
   The receiver acknowledges that the invoice was received.

2. **Status 32 – Good or service reception**
   The receiver confirms receipt of the goods or services described in the invoice.

3. **One of the following may follow**:

- **Status 31 – Reclaim**
  The receiver issues a formal complaint or disagreement regarding the invoice or delivered goods/services.
- **Status 33 – Express acceptance**
  The receiver explicitly accepts the invoice and its content.
- **Status 34 – Tacit acceptance**
  If no response is received from the receiver within the legal timeframe, the sender may generate this status to indicate automatic acceptance.

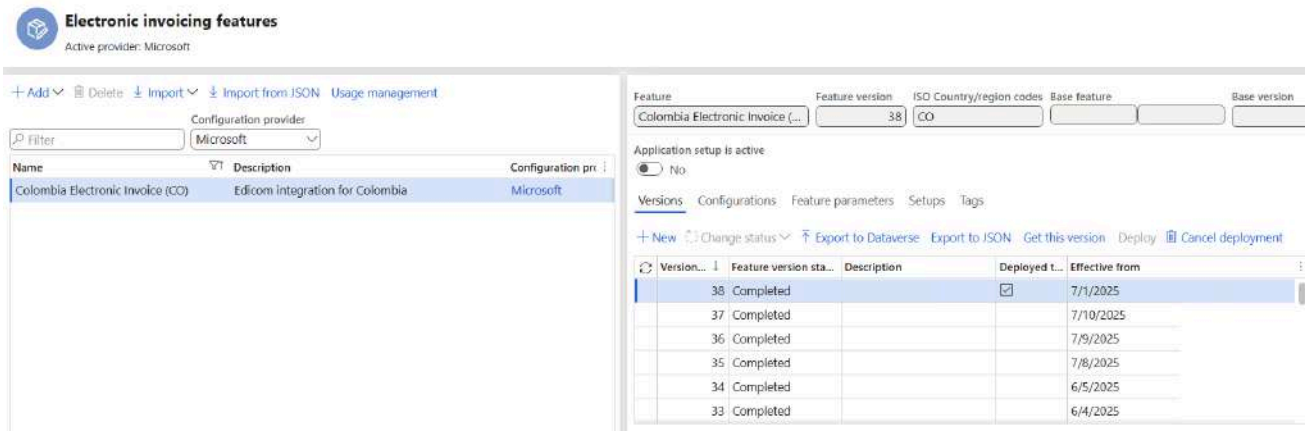
> **Note:** Status codes **30**, **31**, **32**, and **33** are generated by the **receiver**, while status **34** will be returned automatically by Edicom after 72 hours if status 31 or 33 are not received.

Each of these interactions requires common parameters, such as Edicom connection details and the authentication token provided by Edicom. These parameters are reused across the feature setup for all document types. Edicom supplies these values during the company's onboarding process.

Colombia Electronic Invoice (CO) Edicom integration for Colombia

> ⓘ **Important**
>
> The **Colombia Electronic Invoice (CO)** feature is provided by Microsoft. Before it can be used, it requires additional configuration, as described in this article. For information about how to configure invoicing features and apply changes, see **Work with feature setups**. For example, in addition to the connection parameters, you can filter specific legal entities so that they are processed in applicability rules. By default, the feature is applicable to all legal entities that have a primary address in Colombia.

1. Import the latest version of the **Colombia Electronic Invoice (CO)** Globalization feature as described in Import features from the repository. The following illustrations show what the feature looks like after you import it from Dataverse.



If you go to the **Configuration** tab, as shown in the previous screenshot, you should see a page displaying information similar to the following table:

Expand table

Name	Version	Status	Country/region code
Edicom response error log import	13.2	Completed	
Project e-CreditNote (CO)	296.141.21.22	Completed	CO
Project e-DebitNote (CO)	296.141.21.17	Completed	CO
Project e-Invoice (CO)	296.141.21.14	Completed	CO
Inventory e-CreditNote (CO)	296.142.39.27	Completed	CO
Inventory e-DebitNote (CO)	296.142.39.23	Completed	CO

> **Note**
>
> The version numbers may change if the formats have been updated. The table is for illustrative purposes only.

2. Create a copy of the imported Globalization feature, and select your configuration provider. For more information, see Create a Globalization feature.

3. On the **Versions** tab, confirm that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with Edicom's API.
   These parameters are general-purpose and will be used across the different actions in the setups, so you only need to configure them once in this section.

> **Tip**
>
> The following table provides **example values** for each feature parameter, so you can see the type of data that should be entered for each one. These values are **illustrative only** and were provided by Edicom to Microsoft for testing purposes. When you are onboarded, Edicom will provide you with your own unique values to use in your configuration.

Expand table

Parameter	Description	Data type	Value
EdicomApplication	Application	String	<ID_PROVIDED_EDICOM>
EdicomDomain	Domain	String	<ID_PROVIDED_EDICOM>
EdicomGetStatusSchema	Get status schema name	String	IN_DOCUMENT_STATUS_MICROSOFT_CO
EdicomGroup	Group	String	<GROUP_ID>
EdicomSignedXMLSchema	Signed XML schema name	String	EDIWIN-SOURCE-FILE_SIGNED_<ID_PROVIDED_EDICOM>
EdicomSubmitInvoiceDestination	Destination name	String	<ID_PROVIDED_EDICOM>_EDIWIN
EdicomSubmitInvoiceSchema	Submit invoice schema name	String	OUTBOUND_DOCUMENT_MICROSOFT_CO
EdicomToken	Auth token	Secret name	EdiwinColombiaToken
EdicomWebServiceURL	Web service URL	String	https://ipaasgw.edicomgroup.com

> **Tip**
>
> When configuring the feature parameters for Edicom integration, follow these steps:

- Select **Application**, and enter the service ID number you obtained from Edicom.

- Select **Domain**, and enter the same service ID number.
- Select **Get status schema name**, and enter the provided schema name.
- Select **Group**, and enter the group code you received.
- Select **Signed XML schema name**, and enter the appropriate schema name.
- Select **Destination name**, and enter the service ID number concatenated with "_EDIWIN". For example, if your service ID is 123456, enter **123456_EDIWIN**.
- Select **Submit invoice schema name**, and enter the schema name.
- Select **Auth token**, and choose the name of the secret you created for the token.
- Select **Web service URL**, and verify the web address is correct.

5. The copy of the feature is always created as a **Draft** version. Regardless of whether you made changes, you must complete, publish, and deploy the feature as described in Complete and deploy a Globalization feature.

Outbound flow pipeline

There are six electronic invoice formats for Colombia included in the **Colombia Electronic Invoice (CO)** feature. Each format requires its own setup and configuration. The formats are:

- **Project e-Invoice (CO)**
- **Project e-CreditNote (CO)**
- **Project e-DebitNote (CO)**
- **Inventory e-Invoice (CO)**
- **Inventory e-CreditNote (CO)**
- **Inventory e-DebitNote (CO)**

In the following sections, you will find guidance on the general configuration steps that apply to all six formats. Additionally, specific considerations and requirements for each individual format will be explained, so you can ensure that each one is configured correctly according to its unique needs.

Electronic invoice feature

Setups-View

After importing the feature, in the second screenshot, you will see five tabs: **Versions**, **Configuration**, **Feature parameters**, **Setups**, and **Tags**. Select the **Setups** tab, and then click **View**. This opens the page **Feature version setup**:

Feature version setup

These page allows you to review and configure the details of the feature setup for Colombia. On the left side of this page, you will see four sections: **Processing pipeline**, **Applicability rules**, **Variables**, and **Parameters**.

Processing pipeline

The Setup Actions and Parameters section is located on the right-hand side of the screen and is divided into two distinct grids:

Top Right – Processing Pipeline

The **Processing pipeline** grid should look like the following table:

Expand table

Action	Action name	Description	Enable retry	Retry action	Export result	Update action
Transform document	Generate invoice XML file					
Integrate with Edicom	Submit invoice					
Get status from Edicom for an invoice	Get signed XML from Edicom					**Checked**
Get status from Edicom for an invoice	Get invoice status from Edicom					
Process response	Process invoice status response					

> **Note**
>
> The **Get status from Edicom for an invoice** action is crucial in the flow, as it is marked with the **Update action** checkbox. This designation means that this specific step—and all steps following it—will repeatedly execute in a loop. The system continues this process until the invoice reaches a final, terminal status, ensuring that the latest status from Edicom is always retrieved and processed.

Bottom Right – Parameters

The lower grid shows the Parameters associated with the selected processing action. These parameters allow for fine-tuning of the selected step's behavior. Each entry includes a parameter name, description, data type, and value, enabling users to configure processing logic according to their business requirements.

Transform document parameters

Expand table

Name	Description	Value
Input file	Source file provides to the action the data to be executed.	Variable: BusinessDocumentDataModel
Direction	Direction describes which format will be used: import or export	export
Configuration	Configuration describes format which will be executed	Inventory e-Invoice (CO): Inventory e-Invoice (CO)
Configuration Integration point	Source file provides data to the reporting runtime	InvoiceCustomer
Custom file name	Custom file name from the client	

Linking the correct format in Transform document parameters

When configuring the **Transform document** action in the processing pipeline, ensure the following:

- **Configuration**: Set this to the exact name of the feature format you are configuring.
  - For example: - `Inventory e-Invoice (CO)` -> Configuration: `Inventory e-Invoice (CO)`
    - `Inventory e-DebitNote (CO)` → Configuration: `Inventory e-DebitNote (CO)`
    - `Inventory e-CreditNote (CO)` → Configuration: `Inventory e-CreditNote (CO)`
    - `Project e-Invoice (CO)` → Configuration: `Project e-Invoice (CO)`
    - `Project e-DebitNote (CO)` → Configuration: `Project e-DebitNote (CO)`
    - `Project e-CreditNote (CO)` → Configuration: `Project e-CreditNote (CO)`
- **Configuration Integration Point**:
  - Use `InvoiceCustomer` for inventory-related formats (`Inventory e-Invoice (CO)`, `Inventory e-CreditNote (CO)`, `Inventory e-DebitNote (CO)`).

- Use `InvoiceProject` for project-related formats (`Project e-Invoice (CO)`, `Project e-CreditNote (CO)`, `Project e-DebitNote (CO)`).

> **Tip**
>
> If the feature name contains "Project", use `InvoiceProject`. If it contains "Inventory", use `InvoiceCustomer`. Incorrect settings will result in empty XML or missing required data.

Integrate with Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
Http request body	Http request body (can be empty)	Generate invoice XML file: Output file
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinColombiaToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Destination	Destination	FeatureParameter: EdicomSubmitInvoiceDestination
Schema	Schema	FeatureParameter: EdicomSubmitInvoiceSchema
Duplicates	Duplicates	0
Group	Group	FeatureParameter: EdicomGroup
Custom file	Custom file name from the client	Generate invoice XML file: File name

Name	Description	Value
name		

Get signed XML from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinColombiaToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomSignedXMLSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference Id of the document submitted to Edicom	Submit invoice: ExternalID

Get invoice status from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	

Name	Description	Value
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinColombiaToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomGetStatusSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference Id of the document submitted to Edicom	Submit invoice: ExternalID

Process response parameters

Expand table

Name	Description	Value
Input file	Response to analyze	Get invoice status from Edicom: Output file
Reporting configuration list	List of configurations used for response parsing	Edicom Response Processing CO: Edicom Response Processing CO
Custom file name	Custom file name from the client	
Reporting configuration list	List of configurations used for response parsing	Edicom response error log import: Edicom response error log import

> **Note**
>
> - If the status response indicates a failure, the pipeline is terminated, and the submission is marked as failed.
> - If the response indicates successful submission to the Colombian Tax Authorities (DIAN), the process is completed when acknowledgment receipts **31, 33, or 34** are received.

Applicability Rules and Feature Setup Scope

Applicability rules must be correctly configured to provide context, so that the exact Electronic Invoicing Globalization feature that must run in the Electronic Invoicing service can be found. These applicability rules are provided out of the box by checking the legal entity's International Organization for Standardization (ISO) country/region code.

Applicability rules for Colombian electronic invoice formats

Below are the applicability rules for each electronic invoice format for Colombia. Replace `<Legal Entity ID>` with the actual ID of your legal entity.

Inventory e-Invoice (CO)

Expand table

And/Or	Field	Operator type	Value	Data type
And	CountryISOCode	Equals	CO	string
	LegalEntityId	Equals	`<Legal Entity ID>`	string
Or	DocumentType	Equals	Customer invoice	string
	DocumentType	Equals	Export Customer invoice	string

Inventory e-CreditNote (CO)

Expand table

And/Or	Field	Operator type	Value	Data type
And	CountryISOCode	Equals	CO	string
	LegalEntityId	Equals	`<Legal Entity ID>`	string
	DocumentType	Equals	Customer credit note	string

Inventory e-DebitNote (CO)

Expand table

And/Or	Field	Operator type	Value	Data type
And	CountryISOCode	Equals	CO	string
	LegalEntityId	Equals	`<Legal Entity ID>`	string

And/Or	Field	Operator type	Value	Data type
	DocumentType	Equals	Customer debit note	string

Project e-Invoice (CO)

Expand table

And/Or	Field	Operator type	Value	Data type
And	CountryISOCode	Equals	CO	string
	LegalEntityId	Equals	`<Legal Entity ID>`	string
Or	DocumentType	Equals	Project invoice	string
	DocumentType	Equals	Project Export invoice	string

Project e-CreditNote (CO)

Expand table

And/Or	Field	Operator type	Value	Data type
And	CountryISOCode	Equals	CO	string
	LegalEntityId	Equals	`<Legal Entity ID>`	string
	DocumentType	Equals	Project credit note	string

Project e-DebitNote (CO)

Expand table

And/Or	Field	Operator type	Value	Data type
And	CountryISOCode	Equals	CO	string
	LegalEntityId	Equals	`<Legal Entity ID>`	string
	DocumentType	Equals	Project debit note	string

Variables

The following variables are used in the outbound data flow actions of the Colombian feature. (These variables are provided out of the box.)

- **BusinessDocumentDataModel** – The Business Document Data model variable that is received from Finance or Supply Chain Management and transformed into the format that is required for submission.
- **SignedXML** – The signed XML variable that is sent back to Finance or Supply Chain Management. This variable contains the base64-encoded response body from the **Get Signed XML from Edicom** step. It's used in the response types to save the signed XML that is obtained from Edicom as an attachment to the invoice journal. It's also used to generate printable reports through QR codes.

Name	Description	Type	Data type	Value
BusinessDocumentDataModel	Contains the electronic document from the client	From client	file	
SignedXML	Signed XML document in base64-encoded format	To client	file	Get signed XML from Edicom.Base64ResponseBody

After you import the **Electronic invoicing for Colombia** feature that includes the out-of-box default feature setup, follow the steps in the next section to configure electronic documents.

Setups-Application Setup

To configure the mapping between Dynamics 365 Finance source tables and electronic document formats, go to the **Setups** tab within the **Electronic invoice feature** section, and then select **Application Setup**. This step lets you define how data from Finance is linked to the correct electronic invoice formats for Colombia.

> **Note**
>
> Although this configuration is not set at the general feature level, it acts globally for the selected feature setup. You do not need to repeat these steps for each individual format or feature—one configuration is sufficient for all supported document types within the feature.

The following grid defines the link between the source table in Dynamics 365, the document context model, and the mapping configuration used for electronic invoicing:

Table name	Context model / Context	Electronic document mapping	Mapping name	Description
Customer invoice journal	Customer invoice context model / Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice	Maps customer invoices from the CustInvoiceJour table to the LATAM invoice model for electronic submission.
Project invoice	Customer invoice context model / Project invoice context	Invoice Model mapping LATAM	Project E-Invoice	Maps project invoices to the LATAM invoice model for electronic submission.

> **Tip**
>
> Ensure that the Context and Mapping name match the document type. Incorrect mappings will cause errors in XML generation or submission to DIAN.

Configure electronic document parameters

To configure electronic document parameters, follow these steps:

1. Make sure that the country/region-specific ER configurations for the document context and electronic document model mapping that are required for Colombia are imported. For more information, see Set up Electronic document parameters.
2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
3. In the **Electronic document** section, add records for the **Customer Invoice journal** and **Project invoice** table names.
4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.

the Electronic Document Parameters page in Dynamics 365 Finance and Operations. Shows on the left-hand navigation pane, we can see the following options:

- Electronic Document
- Features
- Electronic Invoicing
- Integration Channels

For the Electronic Document option on the right-hand side, the section labeled Electronic Reporting is displayed.

Within Electronic Reporting, a grid is presented which lists the electronic document configurations. This grid includes important associations between:

- The source table name
- The document context model and context
- The electronic document model mapping
- The final mapping name used for generating the electronic invoice.

This configuration is essential for determining how Dynamics 365 maps internal data models to external electronic formats for submission to tax authorities or service providers.

Configure the grid as follows:

Table name	Microsoft-owned	Document context model	Document context	Electronic document model mapping	Mapping name
Customer invoice journal	No	Customer invoice context model	Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice
Project invoice	No	Customer invoice context model	Project invoice context	Invoice Model mapping LATAM	Project E-Invoice

1. Save your changes, and close the page.
2. For each table name, follow these steps:

    a. Select **Response types**, select **New** to create a response type that you will get from the back end, and enter the following values:

    - In the **Response type** field, enter **SignedXML** (the default value). (See the Variables section of this article.)
    - In the **Description** field, enter any meaningful name. Alternatively, leave the field blank.
    - In the **Submission status** field, select **Pending**.
    - In the **Model mapping** field, select **Edicom source file response format**.

    b. Repeat the preceding step, but set the **Submission status** field to **Pending update actions execution**. In this way, you can run a subset of the actions in the processing

pipeline in a loop to continuously pull updated statuses and other information for the submitted documents from the tax authority. In the standard view, you should see a table similar to the following:

Response type	Description	Submission status	Data entity name	Model mapping
SignedXML	Signed XML obtained from EDICOM	Pending		Edicom source file response import format
SignedXML	Signed XML obtained from EDICOM	Pending update actions execution		Edicom source file response import format

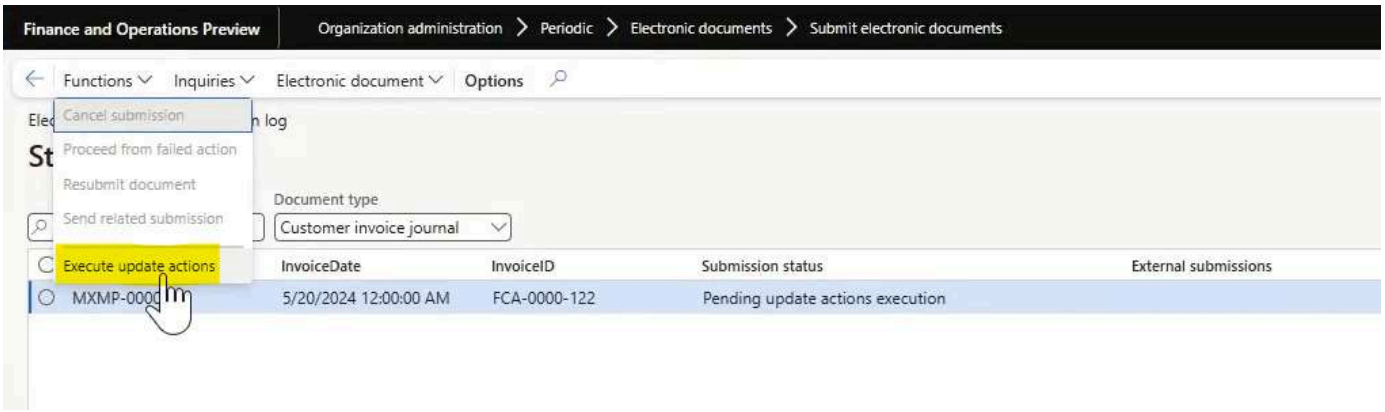
> **Note**
>
> The response includes the signed XML that is obtained from Edicom. This signed XML is stored in the system as an attachment to the corresponding invoice journal. It will eventually be used to generate printable invoices through QR codes.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information about how to generate electronic invoices, see Submit electronic documents.

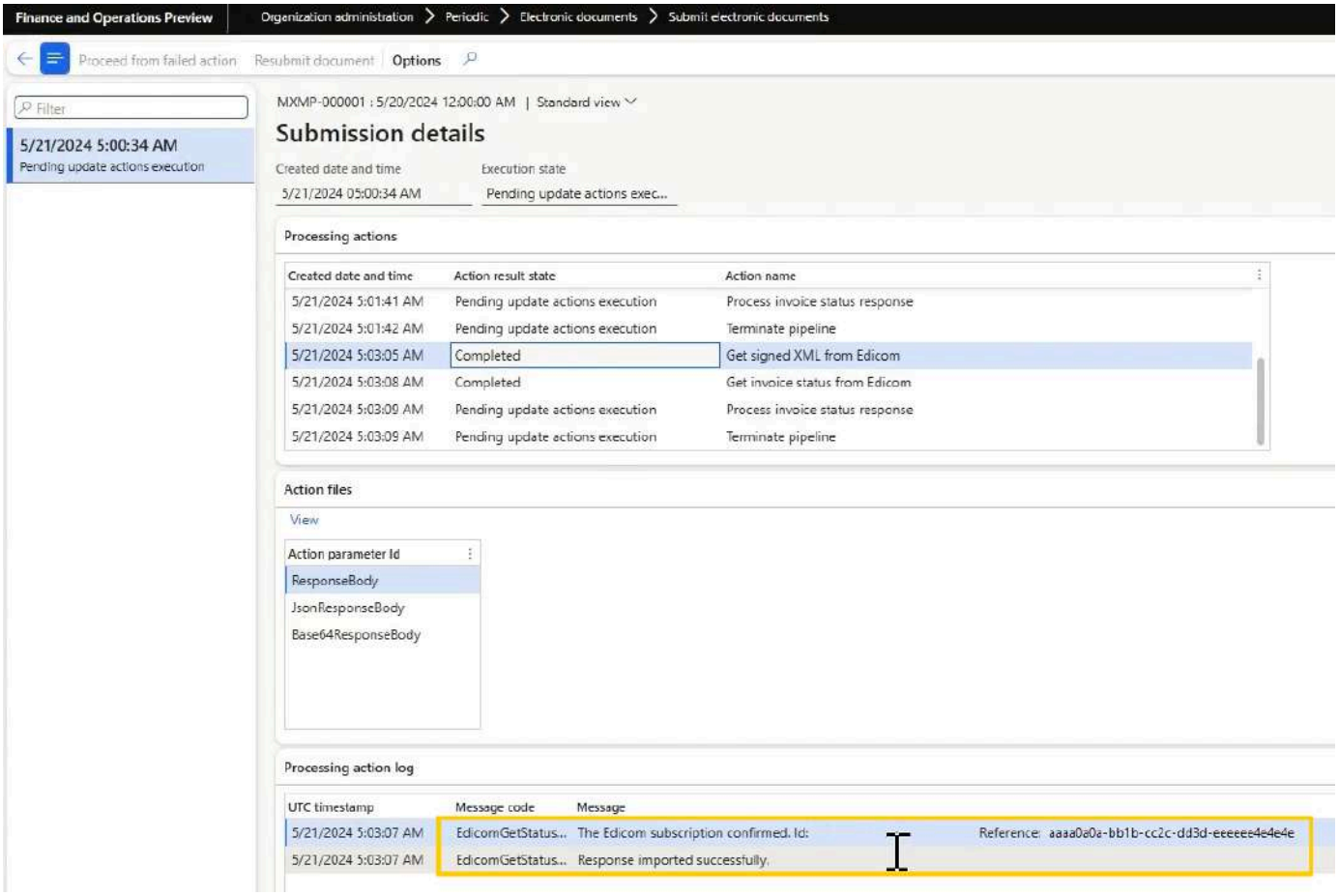
To inquire about the results of a submission, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**, and select the required document type. For more information, see Work with Electronic document submission log.

For Colombia, after you submit the invoice, the submission status is set to **Pending update actions execution**. The response body is probably empty for the signed XML and the call to get the invoice status. An empty response body indicates that the XML wasn't available immediately after submission. To address the pending status, a function that is named **Execute update actions** is used. This function resumes the pipeline from the action that is marked as an update action. It then runs all subsequent actions in the pipeline in a loop. The status should change to **Executing** again. Then, after a few seconds, it reverts to **Pending update actions execution**.

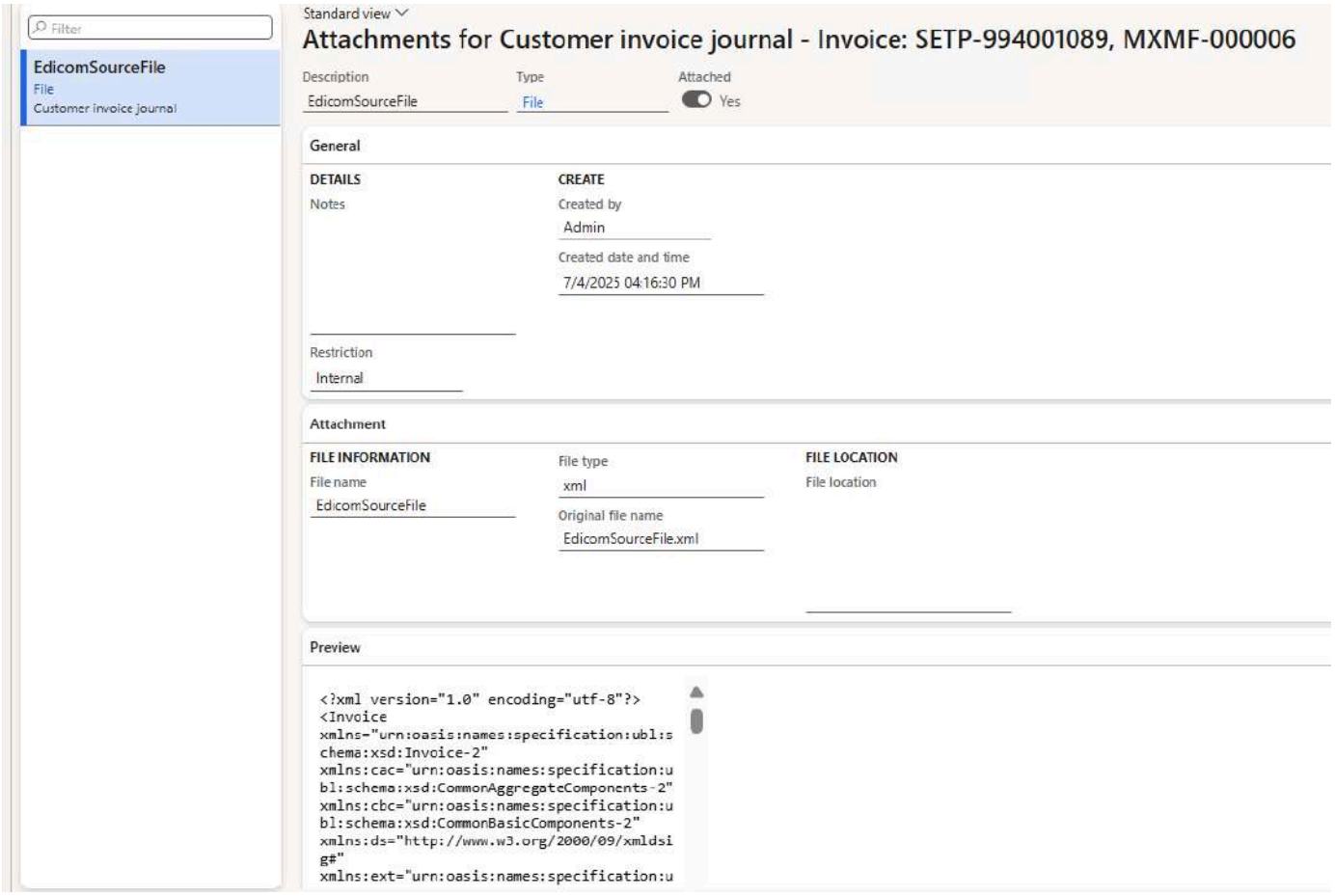


> **Note**
>
> By setting a recurrence schedule, you can configure the **Execute update actions** function to run in batch mode on a periodic basis.

When you review the submission details, you should notice that the steps run again. This time, the signed XML is received.



As a result of the completed outbound flow, the signed XML is attached to the invoice journal as **EdicomSourceFile**.



For Colombia, the **Process response** action completes the pipeline after a few minutes.

More resources

- Electronic Invoicing service overview
- Get started with Electronic invoicing service administration
- Setting up Electronic Invoicing
- Electronic Invoicing service independent software vendor (ISV) last-mile connector
- Dynamics 365 Country expansion: localizations for LATAM countries | June 27, 2024
- Dynamics 365 Country expansion: localizations for LATAM countries | D365 FastTrack Tech Talks (youtube.com)
- Finance Localization for LATAM: Update on Additional Countries | Dynamics 365 Fast Track TechTalk | Jun 23, 2025
- Finance Localization for LATAM: Update on Additional Countries | Dynamics 365 Fast Track TechTalk (youtube.com)

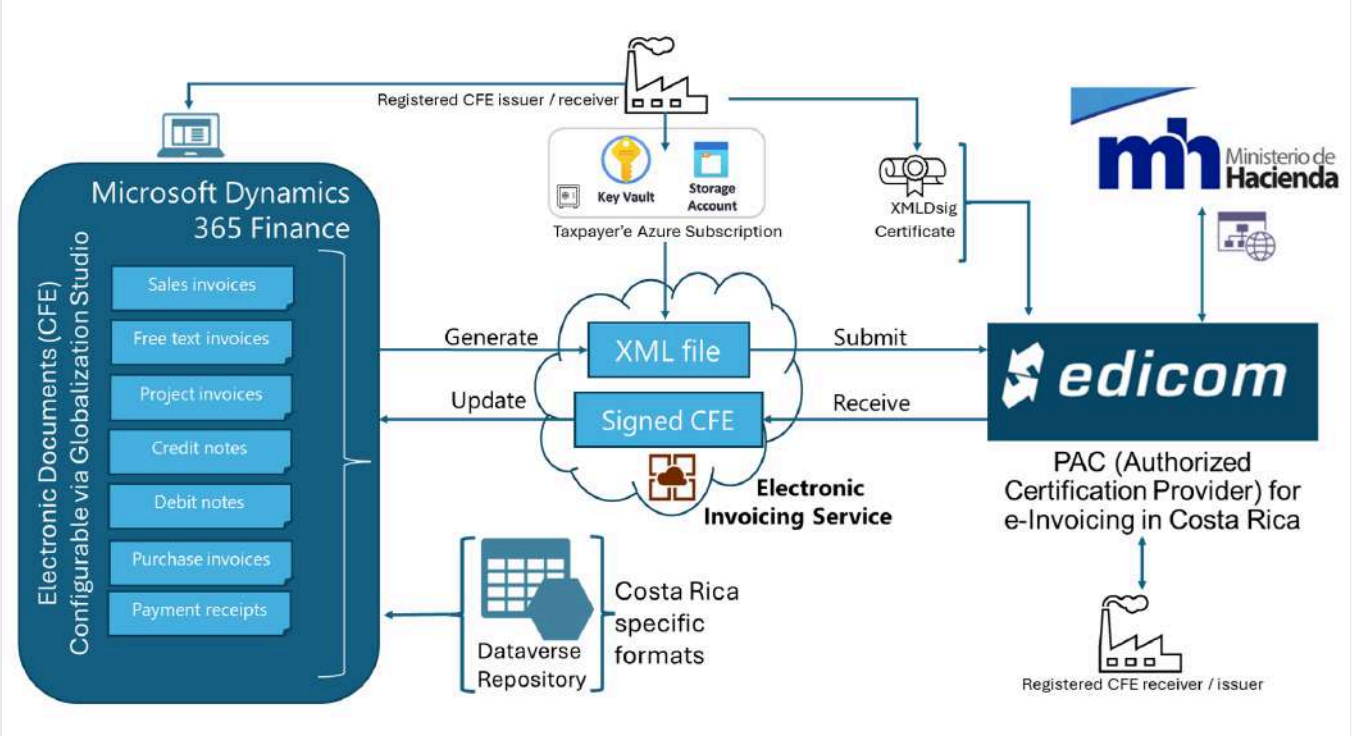
Last updated on 09/22/2025

Get started with Electronic invoicing for Costa Rica

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information to help you get started with Electronic invoicing for Costa Rica. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Microsoft Dynamics 365 Supply Chain Management. These steps complement the steps that are described in Electronic invoicing setup. For the last-mile integration with the Costa Rican Tax Authorities, Microsoft partners with Edicom.

After you configure Electronic invoicing, you can generate, digitally sign, and submit the XML files of electronic invoices to the Edicom authorized certification provider (PAC) according to the regulatory requirements in Costa Rica.



> **Note**
>
> The electronic invoicing approach that this article describes uses an invoicing service that's applicable only to cloud deployments of Finance or Supply Chain Management.

Prerequisites

Before you begin the procedures in this article, complete the following prerequisites:

1. Ensure that the settings for the Costa Rican legal entity are in place. For more information, see Set up legal entity and tax information for Costa Rica.

2. Configure electronic invoice parameters for Costa Rica.

3. Gain familiarity with and understanding of Electronic invoicing as it's described in Electronic Invoicing service overview.

4. Complete the common part of Electronic Invoicing service configuration as it's described in Electronic invoicing configuration.

5. Enable the following features in Feature management:

    - Electronic invoicing integration
    - Electronic invoicing integration resubmit document from failed action
    - Execute update actions for submitted documents

6. Make sure that the following Electronic reporting (ER) format configurations are imported. For more information, see: Import Electronic reporting (ER) configurations from Dataverse, Import Electronic reporting (ER) configurations.

    - Inventory e-invoice (CR)
    - Inventory Credit Note (CRI)
    - Inventory Debit Note (CRI)
    - Inventory Export e-Invoice (CRI)
    - Project e-invoice (CR)
    - Project Credit Note (CRI)
    - Project Debit Note (CRI)
    - Project Export e-Invoice (CRI)
    - Purchase e-invoice (CRI)
    - Payment Receipt (CRI)
    - Customer invoice context model
    - Edicom source file response import format
    - Edicom Response Processing LATAM
    - Edicom Response Processing CR
    - Edicom response error log import

    > **Note**

These formats are based on the corresponding **LATAM** format configurations that use the **Invoice model LATAM** and **Invoice model mapping LATAM** configurations. All other required configurations are automatically imported.

Country/region-specific configuration for Electronic invoicing for Costa Rica

The **Costa Rica electronic invoice (CRI) "E-Invoicing for Costa Rica: ISV last-mile connector with Edicom"** feature represents an outbound flow for issuing the following sales documents:

Expand table

Name	Code	Original Name
Invoice	01	Factura Electrónica
Debit note	02	Nota de Débito Electrónica
Credit note	03	Nota de Crédito Electrónica
Purchase invoice	08	Factura electrónica de compras
Export invoice	09	Factura de Exportación
Payment receipt	10	Recibo Electrónico de Pago

Some parameters of the feature are published with default values. Before you deploy the Electronic invoicing feature to the service environment, add a feature that is based on the Microsoft-provided feature, and complete the common parameters on the **Feature parameters** tab. Review the default values, and update them as required, so that they better reflect your business operations.

Edicom Interactions for Costa Rica

For Costa Rica, the base flow includes at least three core interactions with Edicom:

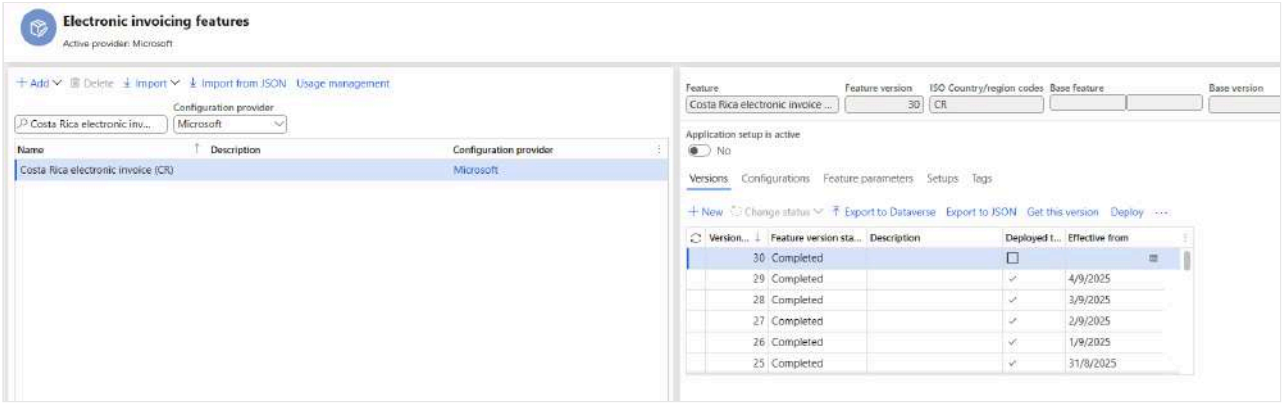
1. **Invoice submission**
2. **Signed XML retrieval**
3. **Status retrieval of the submitted invoice**

Each of these interactions requires common parameters, such as Edicom connection details and the authentication token provided by Edicom. The feature setup for all document types reuses these parameters. Edicom supplies these values during the company's onboarding process.

Costa Rica Electronic Invoice (CR) Edicom integration for Costa Rica

> **Important**
>
> Microsoft provides the **Costa Rica Electronic Invoice (CR)** feature. Before you can use it, you need to configure it as described in this article. Learn how to configure invoicing features and apply changes in Work with feature setups. For example, in addition to the connection parameters, you can filter specific legal entities so that they are processed in applicability rules. By default, the feature applies to all legal entities that have a primary address in Costa Rica.

1. Import the latest version of the **Costa Rica Electronic Invoice (CR)** Globalization feature as described in Import features from the repository. The following illustrations show what the feature looks like after you import it from Dataverse.



If you go to the Configuration tab, as shown in the previous screenshot, you see a page that displays information similar to the following table:

Expand table

Name	Version	Status	Country/region code
Edicom Response Processing LATAM	13.2.5	Completed	
Edicom response error log import	13.3	Completed	
Payment Receipt (CRI)	296.152.5	Completed	CR
Purchase e-invoice (CRI)	296.152.14	Completed	CR
Inventory Credit Note (CRI)	296.152.55.41	Completed	CR
Inventory Debit Note (CRI)	296.152.55.41	Completed	CR
Inventory e-invoice (CR)	296.152.55.46	Completed	CR
Inventory Export e-Invoice (CRI)	296.152.55.29	Completed	CR

Name	Version	Status	Country/region code
Project Credit Note (CRI)	296.152.27.22	Completed	CR
Project Debit Note (CRI)	296.152.27.21	Completed	CR
Project e-invoice (CR)	296.152.27.22	Completed	CR
Project Export e-Invoice (CRI)	296.152.27.6	Completed	CR

> **Note**
>
> The version numbers might change if the formats are updated. The table is for illustrative purposes only.

2. Create a copy of the imported Globalization feature, and select your configuration provider. For more information, see Create a Globalization feature.

3. On the **Versions** tab, confirm that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with Edicom's API.
   These parameters are general-purpose and are used across the different actions in the setups, so you only need to configure them once in this section.

   > **Tip**
   >
   > The following table provides **example values** for each feature parameter, so you can understand the type of data that you should enter for each one. These values are **illustrative only** and were provided by Edicom to Microsoft for testing purposes. When you're onboarded, Edicom provides you with your own unique values to use in your configuration.

Expand table

Parameter	Description	Data type	Value
EdicomApplication	Application	String	<ID_PROVIDED_EDICOM>
EdicomDomain	Domain	String	<ID_PROVIDED_EDICOM>
EdicomGetStatusSchema	Get status schema name	String	IN_DOCUMENT_STATUS_MICROSOFT_UY
EdicomGroup	Group	String	<GROUP_ID>

Parameter	Description	Data type	Value
EdicomSignedXMLSchema	Signed XML schema name	String	EDIWIN-SOURCE-FILE_XML_CFE_ADENDA_<ID_PROVIDED_EDICOM>
EdicomSubmitInvoiceDestination	Destination name	String	<ID_PROVIDED_EDICOM>_EDIWIN
EdicomSubmitInvoiceSchema	Submit invoice schema name	String	OUTBOUND_DOCUMENT_MICROSOFT_UY
EdicomToken	Auth token	Secret name	EdiwinCosta RicaToken
EdicomWebServiceURL	Web service URL	String	https://ipaasgw.edicomgroup.com

> **Tip**
>
> When configuring the feature parameters for Edicom integration, follow these steps:
>
> - Select **Application**, and enter the service ID number you obtained from Edicom.
> - Select **Domain**, and enter the same service ID number.
> - Select **Get status schema name**, and enter the provided schema name.
> - Select **Group**, and enter the group code you received.
> - Select **Signed XML schema name**, and enter the appropriate schema name.
> - Select **Destination name**, and enter the service ID number concatenated with "_EDIWIN". For example, if your service ID is 123456, enter **123456_EDIWIN**.
> - Select **Submit invoice schema name**, and enter the schema name.
> - Select **Auth token**, and choose the name of the secret you created for the token.
> - Select **Web service URL**, and verify the web address is correct.

5. The copy of the feature is always created as a **Draft** version. Regardless of whether you make changes, you must complete, publish, and deploy the feature as described in Complete and deploy a Globalization feature.

Outbound flow pipeline

The **Costa Rica Electronic Invoice (CR) feature** includes 10 electronic invoice formats for Costa Rica. Each format requires its own setup and configuration. The formats are:

- **Inventory e-invoice (CRI)**

- **Inventory Credit Note (CRI)**
- **Inventory Debit Note (CRI)**
- **Inventory Export e-Invoice (CRI)**
- **Payment Receipt (CRI)**
- **Project e-invoice (CRI)**
- **Project Credit Note (CRI)**
- **Project Debit Note (CRI)**
- **Project Export e-Invoice (CRI)**
- **Purchase e-invoice (CRI)**

The following sections provide guidance on the general configuration steps that apply to all 10 formats. They also explain specific considerations and requirements for each individual format, so you can ensure that each one is configured correctly according to its unique needs.

Electronic invoice feature

Setups-View

After you import the feature, the second screenshot shows five tabs: **Versions**, **Configurations**, **Feature parameters**, **Setups**, and **Tags**. Select the **Setups** tab, then select **View**. This action opens the **Feature version setup** page:

Feature version setup

This page allows you to review and configure the details of the feature setup for Costa Rica. On the left side of this page, you see four sections: **Processing pipeline**, **Applicability rules**, **Variables**, and **Parameters**.

Processing pipeline

The Setup Actions and Parameters section contain two distinct grids:

Top Right – Processing Pipeline

The **Processing pipeline** grid is similar to the following table:

Expand table

Action	Action name	Description	Enable retry	Retry action	Export result	Update action
Transform document	Generate invoice XML file					

Action	Action name	Description	Enable retry	Retry action	Export result	Update action
Integrate with Edicom	Submit invoice					
Get status from Edicom for an invoice	Get signed XML from Edicom					**Checked**
Get status from Edicom for an invoice	Get invoice status from Edicom					
Process response	Process invoice status response					

> **Note**
>
> The **Get status from Edicom for an invoice** action is crucial in the flow, as it's marked with the **Update action** checkbox. This designation means that this specific step—and all steps following it—repeatedly execute in a loop. The system continues this process until the invoice reaches a final, terminal status, ensuring that the latest status from Edicom is always retrieved and processed.

Bottom Right – Parameters

The lower grid shows the Parameters associated with the selected processing action. These parameters allow for fine-tuning of the selected step's behavior. Each entry includes a parameter name, description, data type, and value, enabling you to configure processing logic according to your business requirements.

Transform document parameters

Expand table

Name	Description	Value
Input file	Source file that provides the action with the data to execute.	Variable: BusinessDocumentDataModel
Direction	Direction describes which format is used: import or export	export
Configuration	Configuration describes the format that's executed	Inventory e-Invoice: Inventory e-Invoice
Configuration Integration point	Source file that provides data to the reporting runtime	InvoiceCustomer

Name	Description	Value
Custom file name	Custom file name from the client	

Linking the correct format in Transform document parameters

When configuring the **Transform document** action in the processing pipeline, ensure the following:

- **Configuration**: Set this value to the exact name of the feature format you're configuring.
  - For example:
    - Inventory e-invoice (CR) → Configuration: Inventory e-invoice (CR)
    - Inventory Credit Note (CRI) → Configuration: Inventory Credit Note (CRI)
    - Inventory Debit Note (CRI) → Configuration: Inventory Debit Note (CRI)
    - Inventory Export e-Invoice (CRI) → Configuration: Inventory Export e-Invoice (CRI)
    - Project e-invoice (CR) → Configuration: Project e-invoice (CR)
    - Project Credit Note (CRI) → Configuration: Project Credit Note (CRI)
    - Project Debit Note (CRI) → Configuration: Project Debit Note (CRI)
    - Project Export e-Invoice (CRI) → Configuration: Project Export e-Invoice (CRI)
    - Payment Receipt (CRI) → Configuration: Payment Receipt (CRI)
    - Purchase e-invoice (CRI) → Configuration: Purchase e-invoice (CRI)
- **Configuration Integration Point**:
  - Configuration Integration Point:
    - `InvoiceCustomer` — Inventory e-invoice (CRI), Inventory Debit Note (CRI), Inventory Credit Note (CRI), Inventory Export e-Invoice (CRI)
    - `InvoiceProject` — Project e-invoice (CRI), Project Debit Note (CRI), Project Credit Note (CRI), Project Export e-Invoice (CRI)
    - `CustomerPrepayment` — Payment Receipt (CRI)
    - `EInvoiceVendor` — Purchase e-invoice (CRI)

> **Tip**
>
> If the feature name contains "Project", use `InvoiceProject`. If it contains "Inventory," use `InvoiceCustomer`. Incorrect settings result in empty XML or missing required data.

Integrate with Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL

Name	Description	Value
Http request body	Http request body (can be empty)	Generate invoice XML file: Output file
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinCostaRicaToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Destination	Destination	FeatureParameter: EdicomSubmitInvoiceDestination
Schema	Schema	FeatureParameter: EdicomSubmitInvoiceSchema
Duplicates	Duplicates	0
Group	Group	FeatureParameter: EdicomGroup
Custom file name	Custom file name from the client	Generate invoice XML file: File name

Get signed XML from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinCosta RicaToken

Name	Description	Value
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomSignedXMLSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference ID of the document submitted to Edicom	Submit invoice: ExternalID

Get invoice status from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinCosta RicaToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomGetStatusSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference ID of the document submitted to Edicom	Submit invoice: ExternalID

Process response parameters

Expand table

Name	Description	Value
Input file	Response to analyze	Get invoice status from Edicom: Output file
Reporting configuration list	List of configurations used for response parsing	Edicom Response Processing LATAM: Edicom Response Processing LATAM
Custom file name	Custom file name from the client	
Reporting configuration list	List of configurations used for response parsing	Edicom response error log import: Edicom response error log import

> **Note**
>
> - If the status response indicates a failure, the pipeline terminates, and the submission is marked as failed.

- If the status response indicates successful submission to the Costa Rican Tax Authorities (MH), the pipeline completes automatically.

Applicability rules and feature setup scope

You must correctly configure applicability rules to provide context, so that the Electronic Invoicing service can find the exact Electronic Invoicing Globalization feature to run. The system provides these applicability rules out of the box by checking the legal entity's International Organization for Standardization (ISO) country/region code.

Applicability rules for Costa Rican electronic invoice formats

The following table lists the applicability rules for each electronic invoice format for Costa Rica. Replace `<Legal Entity ID>` with the actual ID of your legal entity.

Inventory e-invoice (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Customer invoice	string

Inventory Credit Note (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Customer credit note	string

Inventory Debit Note (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Customer debit note	string

Inventory Export e-Invoice (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Customer Export invoice	string

Project e-invoice (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Project invoice	string

Project Credit Note (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Project credit note	string

Project Debit Note (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Project debit note	string

Project Export e-Invoice (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Project Export invoice	string

Payment Receipt (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Payment	string

Purchase e-invoice (CRI)

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	CR	string
		LegalEntityId	Equals	CRMF	string
		DocumentType	Equals	Self invoice	string

Variables

The outbound data flow actions for the Costa Rican feature use the following variables. The system provides these variables.

- **BusinessDocumentDataModel** – The Business Document Data model variable that Finance or Supply Chain Management sends. The system transforms this variable into the format required for submission.
- **SignedXML** – The signed XML variable that the system sends back to Finance or Supply Chain Management. This variable contains the base64-encoded response body from the **Get Signed XML from Edicom** step. The response types use this variable to save the signed XML that the system gets from Edicom as an attachment to the invoice journal. The system also uses this variable to generate printable reports through QR codes.

Expand table

Name	Description	Type	Data type	Value
BusinessDocumentDataModel	Contains the electronic document from the client	From client	file	
SignedXML	Signed XML document in base64-encoded format	To client	file	Get signed XML from Edicom.Base64ResponseBody

After you import the **Electronic invoicing for Costa Rica** feature that includes the default feature setup, follow the steps in the next section to configure electronic documents.

Setups-Application Setup

To configure the mapping between Dynamics 365 Finance source tables and electronic document formats, go to the **Setups** tab within the **Electronic invoice feature** section, and then select **Application Setup**. This step lets you define how data from Finance is linked to the correct electronic invoice formats for Costa Rica.

> **Note**
>
> Although you don't set this configuration at the general feature level, it acts globally for the selected feature setup. You don't need to repeat these steps for each individual format or feature - one configuration is sufficient for all supported document types within the feature.

The following grid defines the link between the source table in Dynamics 365 Finance, the document context model, and the mapping configuration used for electronic invoicing:

Expand table

Table name	Context model / Context	Electronic document mapping	Mapping name	Description
Customer invoice journal	Customer invoice context model / Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice	Maps customer invoices to the LATAM invoice model for electronic submission
Project invoice	Customer invoice context model / Project invoice context	Invoice Model mapping LATAM	Project E-Invoice	Maps project invoices to the LATAM invoice model for electronic submission
Vendor invoice journal	Customer invoice context model / Self invoice context	Invoice Model mapping LATAM	Vendor E-Invoice	Maps vendor invoices to the LATAM invoice model for electronic submission
Customer transactions	Customer invoice context model / Payment invoice	Invoice Model mapping LATAM	Customer Payments	Maps customer payments to the LATAM invoice model for electronic submission

> **Tip**
>
> Ensure that the Context and Mapping name match the document type. Incorrect mappings cause errors in XML generation or submission to DGI.

Configure electronic document parameters

To configure electronic document parameters, follow these steps.

1. Make sure that you import the country/region-specific ER configurations for the document context and electronic document model mapping that are required for Costa Rica. For more information, see Set up Electronic document parameters.
2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

3. In the **Electronic document** section, add records for the **Customer Invoice journal** and **Project invoice** table names.
4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.

The Electronic Document Parameters page in Dynamics 365 Finance. Shows on the left-hand navigation pane, you can see the following options:

- Electronic Document
- Features
- Electronic Invoicing
- Integration Channels

For the Electronic Document option on the right-hand side, the section labeled Electronic Reporting is displayed.

Within Electronic Reporting, a grid is presented which lists the electronic document configurations. This grid includes important associations between:

- The source table name
- The document context model and context
- The electronic document model mapping
- The final mapping name used for generating the electronic invoice.

This configuration is essential for determining how Dynamics 365 Finance maps internal data models to external electronic formats for submission to tax authorities or service providers.

Configure the grid as follows:

Expand table

Table name	Microsoft-owned	Document context model	Document context	Electronic document model mapping	Mapping name
Customer invoice journal	No	Customer invoice context model	Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice
Customer transactions	No	Customer invoice context model	Payment invoice	Invoice Model mapping LATAM	Customer Payments
Project invoice	No	Customer invoice context model	Project invoice context	Invoice Model mapping LATAM	Project E-Invoice

Table name	Microsoft-owned	Document context model	Document context	Electronic document model mapping	Mapping name
Vendor invoice journal	No	Customer invoice context model	Vendor invoice journal	Invoice Model mapping LATAM	Vendor E-Invoice

1. Save your changes, and close the page.

2. For each table name, follow these steps:

   a. Select **Response types**, select **New** to create a response type that you get from the back end, and enter the following values:

   - In the **Response type** field, enter **SignedXML** (the default value). (See the Variables section of this article.)
   - In the **Description** field, enter any meaningful name. Alternatively, leave the field blank.
   - In the **Submission status** field, select **Pending**.
   - In the **Model mapping** field, select **Edicom source file response format**.

   b. Repeat the preceding step, but set the **Submission status** field to **Pending update actions execution**. In this way, you can run a subset of the actions in the processing pipeline in a loop to continuously pull updated statuses and other information for the submitted documents from the tax authority. In the standard view, you should see a table similar to the following:

Expand table

Response type	Description	Submission status	Data entity name	Model mapping
SignedXML	Signed XML obtained from EDICOM	Pending		Edicom source file response import format
SignedXML	Signed XML obtained from EDICOM	Pending update actions execution		Edicom source file response import format

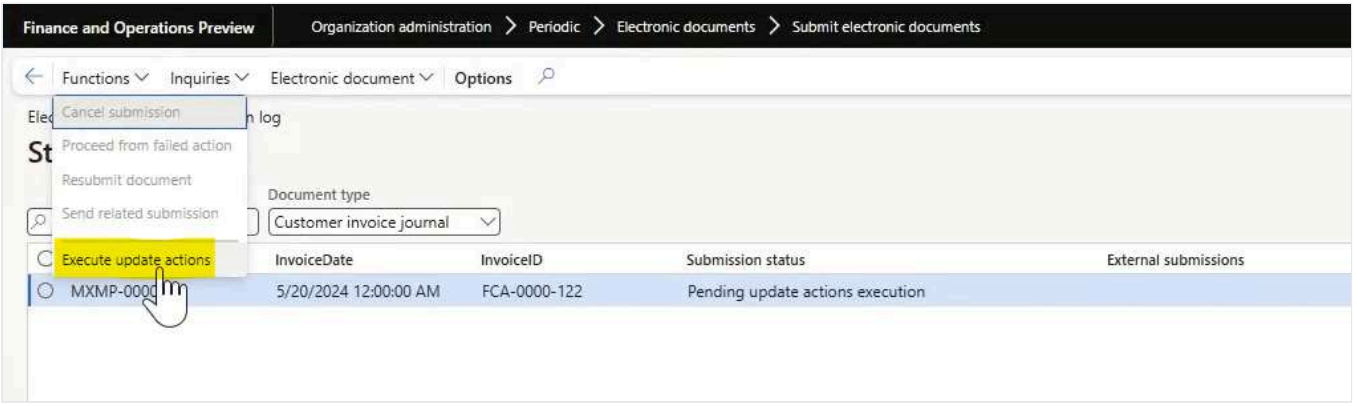
> **Note**
>
> The response includes the signed XML that you get from Edicom. The system stores this signed XML as an attachment to the corresponding invoice journal. You can use it later to generate printable invoices through QR codes.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information about how to generate electronic invoices, see Submit electronic documents.

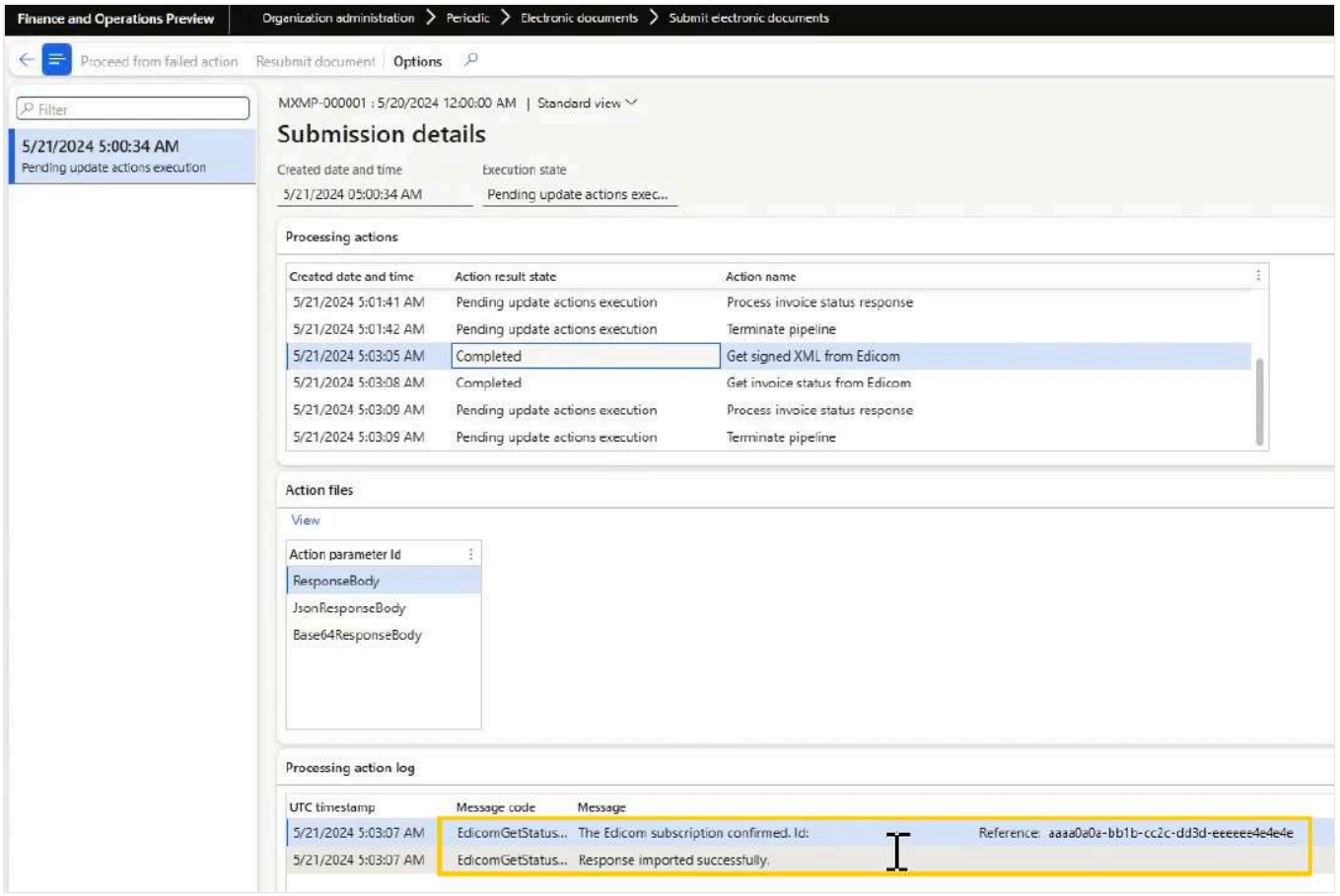
To check the results of a submission, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**, and select the required document type. For more information, see Work with Electronic document submission log.

For Costa Rica, after you submit the invoice, the submission status is set to **Pending update actions execution**. The response body is probably empty for the signed XML and the call to get the invoice status. An empty response body indicates that the XML isn't available immediately after submission. To address the pending status, use a function named **Execute update actions**. This function resumes the pipeline from the action that is marked as an update action. It then runs all subsequent actions in the pipeline in a loop. The status should change to **Executing** again. Then, after a few seconds, it reverts to **Pending update actions execution**.

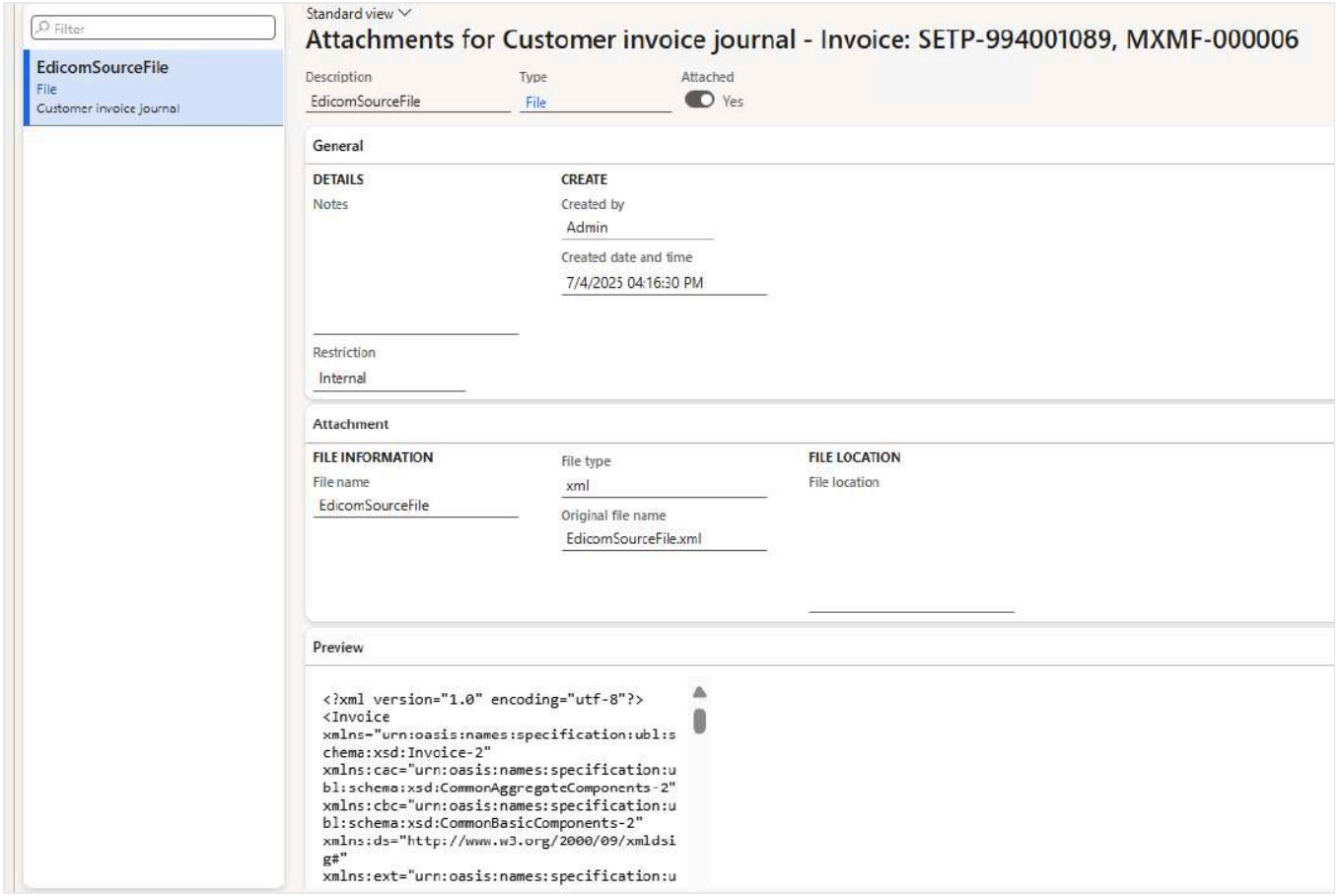


> **Note**
>
> By setting a recurrence schedule, you can configure the **Execute update actions** function to run in batch mode on a periodic basis.

When you review the submission details, you should notice that the steps run again. This time, the signed XML is received.



As a result of the completed outbound flow, the signed XML is attached to the invoice journal as **EdicomSourceFile**.



For Costa Rica, the **Process response** action completes the pipeline after a few minutes.

More resources

- Electronic Invoicing service overview
- Get started with Electronic invoicing service administration
- Setting up Electronic Invoicing
- Electronic Invoicing service independent software vendor (ISV) last-mile connector
- Dynamics 365 Country/Region expansion: localizations for LATAM Countries/Regions | June 27, 2024
- Dynamics 365 Country/Region expansion: localizations for LATAM Countries/Regions | Dynamics 365 FastTrack Tech Talks (youtube.com)
- Finance Localization for LATAM: Update on Additional Countries/Regions | Dynamics 365 Fast Track TechTalk | Jun 23, 2025
- Finance Localization for LATAM: Update on Additional Countries/Regions | Dynamics 365 Fast Track TechTalk (youtube.com)

Last updated on 12/05/2025

Use electronic invoicing for Denmark

07/30/2025

This article provides information to help you get started with Electronic invoicing for Denmark. It includes information about how to configure the system so that you can generate, submit, and receive electronic invoices in the Denmark-specific OIOUBL format and, if necessary, in Pan-European Public Procurement Online (PEPPOL) format.

Prerequisites

Before you begin the procedures in this article, the following prerequisites must be met:

- The company must be registered in the Danish Central Business Register (CVR) and in the Danish electronic invoicing infrastructure, NemHandel.
- The company must have a signed agreement with the provider of electronic document delivery service that secures electronic document interchange in OIOUBL and PEPPOL formats.
- Among the registered profile IDs, the company should have the following profiles that Microsoft Dynamics 365 Finance uses for electronic document interchange:
  - **Procurement-BilSim-1.0** – This profile is used to interchange documents in OIOUBL format.
  - **urn:fdc:peppol.eu:2017:poacc:billing:01:1.0** – This profile is used to interchange documents in PEPPOL format.
- The company must obtain, from the service provider, the required credentials to enable integration of the Electronic Invoicing service with the Electronic Invoicing service independent software vendor (ISV) last-mile connector.
- Become familiar with Electronic invoicing as it's described in Electronic Invoicing service overview and Electronic invoicing components.
- Do the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.
- In the key vault, create a secret for the token that grants authorization to access the infrastructure of the provider of electronic document delivery service, and set up Azure Key Vault as described in Configure Azure resources for Electronic invoicing.

Use the country-specific configuration for the Danish electronic invoice (DK) feature

Some parameters for the **Danish electronic invoice (DK)** electronic invoicing feature have default values. Before you deploy the feature to the service environment, review the default values, and update them as required, so that they better reflect your business operations.

To review and update the parameters for the **Danish electronic invoice (DK)** electronic invoicing feature, follow these steps.

1. Import the latest version of the **Danish electronic invoice (DK)** Globalization feature. Learn more in Import features from the repository.

> **Note**
>
> **Version 3** of the OIOUBL format is implemented starting from the **version 6** of the of the **Danish electronic invoice (DK)** Globalization feature.

1. Create a copy of the imported Globalization feature, and select your configuration provider for it. Learn more in Create Globalization features.
2. On the **Versions** tab, verify that **Draft** is selected.
3. On the **Setups** tab, in the grid, select the **Sales invoice OIOUBL** feature setup, and then select **Edit**.
4. On the **Processing pipeline** tab, in the **Processing pipeline** section, select **Integrate with Edicom**.
5. In the **Parameters** section, select **Domain**, and then enter the service ID number that you obtained.
6. Select **Application**, and then enter the same service ID number.
7. Select **Destination**, and then enter the service ID number concatenated with the string "_EDIWIN." For example, if the service ID number is **123456**, enter **123456_EDIWIN**.
8. Select **Group**, and then enter the group code that you obtained.
9. Select **Auth token**, and then select the name of the secret that you created for the token.
10. Select **Save**, and close the page.

11. Repeat steps 4 through 11 for each of the following feature setups if your business process assumes the involvement of the related types of documents:

    - Sales credit note OIOUBL
    - Project invoice OIOUBL
    - Project credit note OIOUBL
    - Sales invoice PEPPOL
    - Sales credit note PEPPOL
    - Project invoice PEPPOL
    - Project credit PEPPOL

12. On the **Setups** tab, in the grid, select the **Get status** feature setup, and then select **Edit**.

13. On the **Export channel** tab, in the **Parameters** section, select **Auth token**, and then select the name of the secret that you created for the token.

14. Select **Domain**, and then enter the service ID number that you obtained.

15. Select **Application**, and then enter the same service ID number.

16. Select **Group**, and then enter the group code that you obtained.

17. Select **Data channel**, and then enter the name of the integration channel that is configured on the **Electronic document parameters** page in Finance.

18. Select **Save**, and close the page.

Finance configuration

Some additional parameters must be configured directly in Finance.

To configure the additional parameters directly in Finance, follow these steps.

1. In the **Feature management** workspace, make sure that the **Export channels for electronic invoicing integration** feature is enabled. Learn more in Feature management overview.
2. Make sure that the country/region-specific **Document context** and **Electronic document model mapping** Electronic reporting (ER) configurations that are required for Denmark are imported. Learn more in Set up Electronic document parameters.
3. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
4. In the **Electronic document** section, add records for the **Customer Invoice journal** and **Project invoice** table names.
5. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.

6. In the **Integration channels** section, add a record for the channel that is used for electronic invoice submission in batch mode.
7. In the **Channel** column, enter **EdiStatus**. This channel name is used by default, but you can use a different channel name as you require. In this case, you must enter the same name in the value of the **$Context_Channel** variable in the **DataChannel** definition in the **Customer invoice context model** ER configuration. You must also enter it in the parameters and applicability rules of the related feature setup.
8. In the **Company** column, select a required legal entity code.
9. In the **Document context** column, refer to the **Customer invoice context model** configuration by using the **Data channel context** definition.
10. In the **Channel type** column, select **Export**.
11. Save your changes, and close the page.

Finance business data configuration

Follow the configuration steps in Customer electronic invoices in Denmark. Start from the Configure parameters section.

Seller identification

Companies that submit electronic invoices can be identified by their CVR number or their Global Location Number (GLN). The GLN is also known as a European article numbering (EAN) location number.

To identify a company by its CVR number, follow these steps.

1. Go to **Organization administration** > **Organizations** > **Legal entities**.

2. On the **Bank account information** FastTab, in the **Codes** section, in the **Routing number** field, make sure that a valid CVR number is entered for the legal entity.

   The CVR number is entered in the **Invoice\cac:AccountingSupplierParty\cac:Party\cbc:EndpointID** element in the electronic invoice XML file that is generated. It's used as the seller's identification during the submission process.

To identify a company by its GLN, follow these steps.

1. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration types**.

2. Define a new registration type for Denmark that has the name **EAN**. You must enter the name exactly as it appears here.

3. Go to **Organization administration** > **Organizations** > **Legal entities**, and select **Registration IDs** on the Action Pane.
4. On the **Registration ID** FastTab, add the **EAN** registration type that you created.
5. In the **Registration number** field, enter a valid GLN.

   The GLN is entered in the **Invoice\cac:AccountingSupplierParty\cac:Party\cbc:EndpointID** element in the electronic invoice XML file that is generated. It's used as the seller's identification during the submission process.

   > ⓘ **Note**
   >
   > The GLN has higher priority than the CVR number. If both numbers are configured at the same time, the GLN is used.

Buyer identification

1. Go to **Accounts receivable** > **Customers** > **All customers**, and select a customer.
2. On the **Invoice and delivery** FastTab, in the **EAN** field, make sure that a valid GLN is entered for the customer.

   The GLN is entered in the **Invoice\cac:AccountingCustomerParty\cac:Party\cbc:EndpointID** element in the electronic invoice XML file that is generated. It's used as the buyer's identification during the submission process.

   > ⓘ **Note**
   >
   > If no GLN is defined, the customer's tax exempt number is used.

Configure the output format type

By default, all outgoing electronic invoices are generated in OIOUBL format for all customers. However, by using configurable electronic document property types, you can configure electronic invoices so that they are generated in PEPPOL format for specific customers.

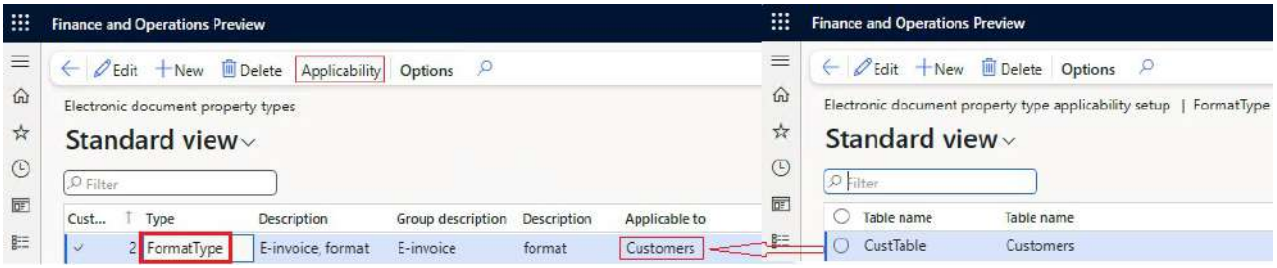
> ⓘ **Note**

Follow the configuration steps in this section only if you must also generate electronic invoices in PEPPOL format. If you generate them only in OIOUBL format, you can skip these steps.

Configure electronic document properties

To configure electronic document properties, follow these steps.

1. Go to **Accounts receivable** > **Setup** > **Electronic document property types**, and select **New**.
2. In the **Type** field, enter **FormatType**. You must enter the value exactly as it appears here.
3. Select **Applicability** to add an applicable table.
4. On the **Electronic document property type applicability setup** page, in the **Table name** field, select the **Customers** table name.
5. Save your changes, and return to the **Electronic document property types** page.

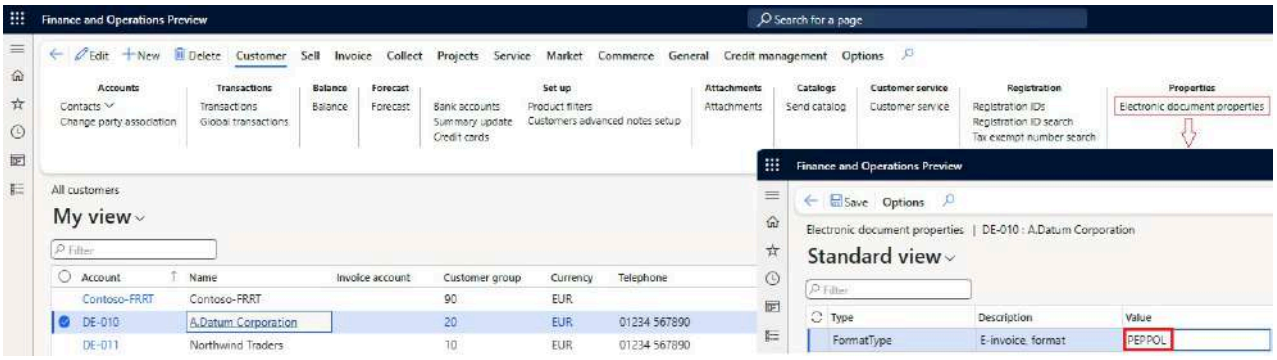


Enter the type of format

To enter the type of format for specific customers, follow these steps.

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a specific customer in the list, and then, on the Action Pane, on the **Customer** tab, in the **Properties** group, select **Electronic document properties**.
3. In the **Value** column, enter **PEPPOL**. You must enter the value exactly as it appears here.

> **Note**
>
> The system generates electronic invoices in PEPPOL format only if you enter the value **PEPPOL** here. If you enter a different value, or no value, electronic invoices are generated in the default OIOUBL format.



Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. Learn more about how to generate electronic invoices in Submit electronic documents.

> **Important**
>
> In current implementations, the standard submission procedure that was described earlier only generates electronic invoices and stores them on the service side. The invoices aren't submitted. Submission of Danish electronic invoices requires that you complete the following additional steps.

To submit the generated electronic invoices in batch mode, follow these steps.

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.
2. In the **Channel** field, select the export channel that you previously created, and then select **OK**.

You can inquire about the results of the submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. Learn more in Work with Electronic document submission log.

Receive incoming electronic invoices

To import incoming invoices in OIOUBL and PEPPOL formats, follow these additional configuration steps for the same version of the **Danish electronic invoice (DK)** electronic invoicing feature that is used for outgoing invoice submission.

1. Import the latest version of the **Danish electronic invoice (DK)** Globalization feature. Learn more in Import features from the repository.

2. Create a copy of the imported Globalization feature, and select your configuration provider for it. Learn more in Create Globalization features.
3. On the **Versions** tab, verify that **Draft** is selected.
4. On the **Setups** tab, in the grid, select **Incoming OIOUBL**, and then select **Edit**.
5. On the **Import channel** tab, in the **Parameters** section, select the **Data channel** parameter. Then, in the **Value** field, define the name of the data channel. Alternatively, leave the default value unchanged. For the OIOUBL format, the default channel name is **EdiOIOUBL**. For the PEPPOL format, it's **EdiPEPPOL**. In both cases, make a note of the value, because you must use it in later configuration steps.
6. Select the **Service ID** parameter, and then select the name of the secret that contains the service ID number.
7. Select the **Group** parameter, and then select the name of the secret that contains the group code.
8. Select the **Token** parameter, and then select the name of the secret that you created for the token.
9. On the **Applicability rules** tab, in the **Channel** field, make sure that the **Value** column contains the same import channel name that you defined earlier.
10. Select **Save**, and close the page.
11. If an import in PEPPOL format is also required, repeat steps 4 through 10 for the **Incoming PEPPOL** feature setup.
12. Complete and deploy the configured version of the **Danish electronic invoice (DK)** electronic invoicing feature.

Finance configuration

Some additional parameters must be configured directly in Finance.

To configure the additional parameters directly in Finance, follow these steps.

1. Make sure that the latest version of the **Vendor invoice import (DK)** ER configuration is imported.

   > **Note**
   >
   > The **Vendor invoice import (DK)** format configuration is used for import invoices in OIOUBL format. It's based on the parent **Vendor invoice import** format configuration that implements invoice import in PEPPOL format. The formats use the **Invoice model** and **Vendor invoice Mapping to destination** configurations. All additional configurations that are required are automatically imported.

2. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, select **Customer invoice context model**.

3. Select **Create configuration**, and then, in the dropdown dialog box, select **Derive from Name: Customer invoice context model, Microsoft** to create a derived configuration.

   > **Note**
   >
   > The derived configuration must differ from the configuration that is used for the invoice *submission* setup.

4. Open the derived configuration to edit it in the designer, and then select **Map model to datasource**.

5. Open the **DataChannel** definition to edit it in the designer.

6. In the **Data sources** tree, expand the **$Context_Channel** container.

7. In the **Value** field, select **Edit**, and then enter the import channel name.

8. Save your changes, and complete the derived configuration.

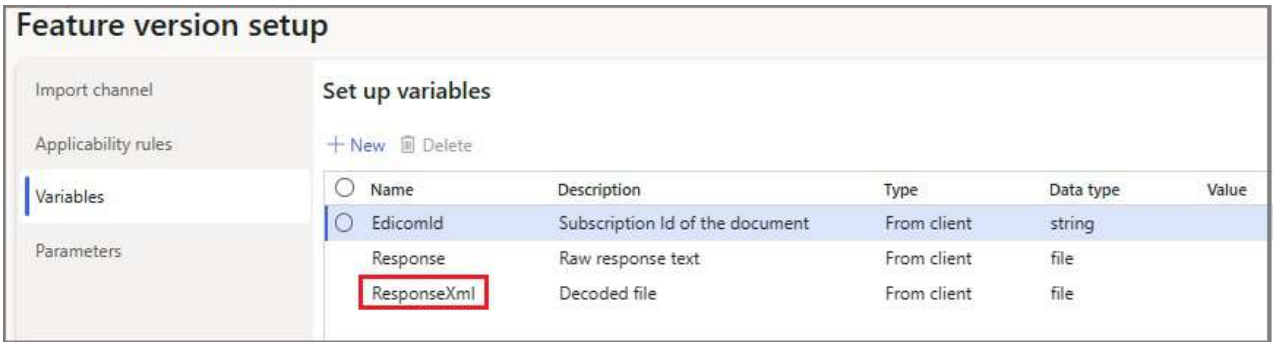
9. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

10. On the **Integration channels** tab, in the **Channels** section, in the **Channel** field, enter the import channel name that you created earlier.

11. In the **Channels** section, in the **Company** field, select a required legal entity.

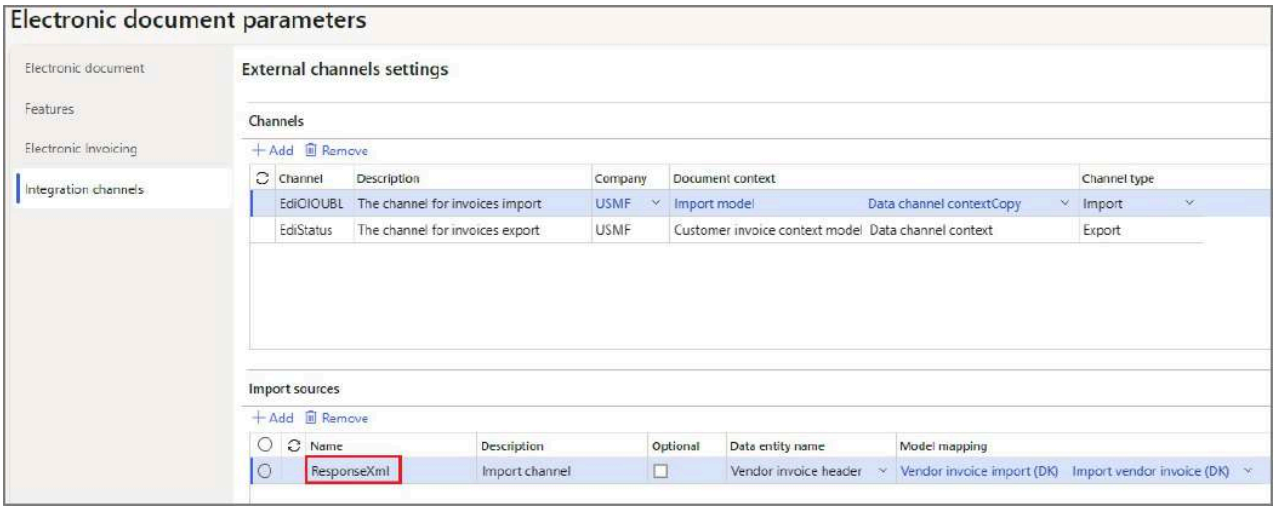
12. In the **Document context** field, select the **Customer invoice context model** configuration.

13. In the **Import sources** section, in the **Name** field, enter the same **ResponseXml** name that is used in the variable for **Decoded file** in the import feature setup.



14. In the **Data entity name** field, select **Vendor invoice header**.

15. In the **Model mapping** field, reference the **Vendor invoice import (DK)** configuration.



16. If import in PEPPOL format is also required, create another import channel, and repeat steps 3 through 15 for it. In step 15, reference the **Vendor invoice import** configuration.

17. Select **Save**, and close the page.

Configure Finance business data

You must configure the following types of master data to provide a match for incoming electronic invoices:

- Vendors
- Products
- Units

Follow the configuration steps in Vendor electronic invoice import in Denmark. Start from the Configure vendor data section.

Receive electronic invoices

To receive electronic invoices, follow these steps.

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. Select **OK**, and then close the page.

To view the receipt logs for electronic invoices, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**.

To view successfully received invoices, go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.

Send responses for received electronic invoices

After incoming vendor invoices are successfully received, you can respond to invoice issuers about the decision to accept or reject the received invoices. Rejection responses can be sent for both incoming invoice formats, OIOUBL and PEPPOL. Acceptance responses aren't assumed. If no rejection response is sent during an agreed-on time, the invoices are considered automatically accepted.

For the PEPPOL format, a receipt acknowledgement message can also be sent. This message is a technical confirmation of invoice receipt. The invoices can still be rejected later from a business perspective.

Configure responses

Responses can be configured by using electronic document property types in the same manner as format types.

Configure electronic document properties

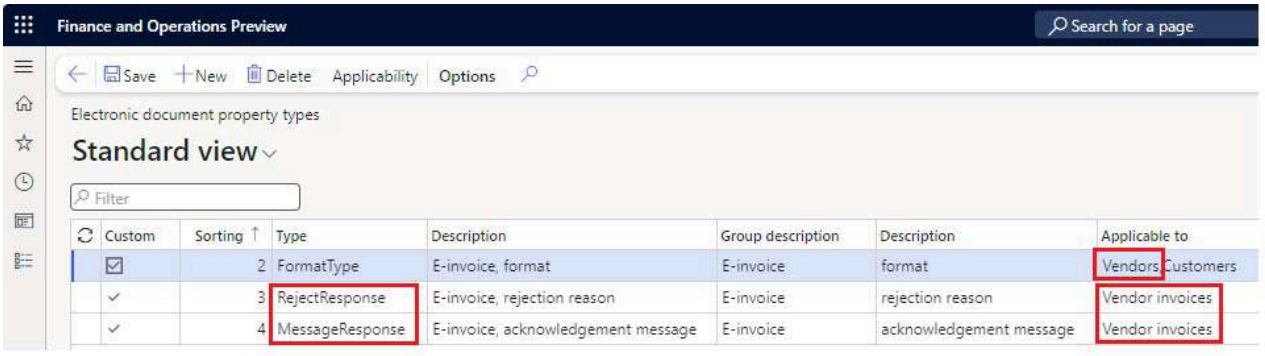
To configure electronic document properties, follow these steps.

1. Go to **Accounts receivable** > **Setup** > **Electronic document property types**, and select **New**.
2. In the **Type** field, enter **RejectResponse**. You must enter the value exactly as it appears here.
3. Select **Applicability** to add an applicable table.
4. On the **Electronic document property type applicability setup** page, in the **Table name** field, select the **VendInvoiceInfoTable** table name.
5. Save your changes, and return to the **Electronic document property types** page.
6. Select **New** to create another electronic document property type.
7. In the **Type** field, enter **MessageResponse**. You must enter the value exactly as it appears here.
8. Select **Applicability** to add an applicable table.
9. On the **Electronic document property type applicability setup** page, in the **Table name** field, select the **VendInvoiceInfoTable** table name.

10. Save your changes, and return to the **Electronic document property types** page.
11. Select the existing FormatType electronic document property type.
12. Select **Applicability** to add an applicable table.
13. On the **Electronic document property type applicability setup** page, in the **Table name** field, select the **Vendors** table name.

> **Note**
>
> The same electronic document property type is used to determine which response type is generated for a specific vendor: **OIOUBL application response** or **PEPPOL invoice level response**.

14. Save your changes, and return to the **Electronic document property types** page.



Define responses formats

To identify which response format type is used for a specific vendor, follow these steps.

1. Go to **Accounts payable** > **Vendors** > **All vendors**.
2. Select a specific vendor in the list, and then, on the Action Pane, on the **Vendor** tab, in the **Properties** group, select **Electronic document properties**.
3. In the **Value** column, enter **PEPPOL**. You must enter the value exactly as it appears here.

> **Note**
>
> The system generates invoice-level responses in PEPPOL format only if you enter the value **PEPPOL** here. If you enter a different value, or no value, application responses are generated in the default OIOUBL format.

Enter responses

To enter responses for specific pending vendor invoices, follow these steps.

1. Go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.
2. Select a specific pending vendor invoice in the list, and then, on the Action Pane, on the **Vendor invoice** tab, in the **Properties** group, select **Electronic document properties**.
3. Select the **RejectResponse** type.
4. In the **Value** column, enter the rejection reason. This text is sent back to the invoice issuer as justification for the rejection. **OIOUBL application response** or **PEPPOL invoice level response** messages are generated and sent only if you enter a non-empty value.
5. Select the **MessageResponse** type.
6. In the **Value** column, enter any text that should be used as an acknowledgement message. This text is sent back to the invoice issuer as confirmation of invoice receipt. **PEPPOL message level response** messages are generated and sent only if you enter a non-empty value.

Configure electronic invoicing feature setups

Some parameters for the **Danish electronic invoice (DK)** electronic invoicing feature have default values. Before you deploy the feature to the service environment, review the default values, and update the authorization credentials that are required for response submission via the ISV last-mile connector.

To review and update the authorization credentials, follow these steps.

1. Import the latest version of the **Danish electronic invoice (DK)** Globalization feature, **version 5** or later. Learn more in Import features from the repository.

2. Create a copy of the imported Globalization feature, and select your configuration provider for it. Learn more in Create a feature based on an existing feature.

3. On the **Versions** tab, verify that **Draft** is selected.

4. On the **Setups** tab, in the grid, select the **Application Response OIOUBL** feature setup, and then select **Edit**.

5. On the **Processing pipeline** tab, in the **Processing pipeline** section, select **Integrate with Edicom**.

6. In the **Parameters** section, select **Domain**, and then enter the service ID number that you obtained.

7. Select **Application**, and then enter the same service ID number.

8. Select **Destination**, and then enter the service ID number concatenated with the string "_EDIWIN." For example, if the service ID number is **123456**, enter **123456_EDIWIN**.
9. Select **Group**, and then enter the group code that you obtained.
10. Select **Auth token**, and then select the name of the secret that you created for the token.
11. Select **Save**, and close the page.
12. Repeat steps 4 through 11 for each of the following feature setups if your business process assumes that responses are sent in PEPPOL format:
    - Invoice Response PEPPOL
    - Message Response PEPPOL

Configure electronic document parameters

To configure electronic document parameters, follow these steps.

1. Make sure that the latest country/region-specific **Document context** and **Electronic document model mapping** ER configurations that are required for Denmark are imported. Learn more in Set up Electronic document parameters.
2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
3. In the **Electronic document** section, add a record for the **VendInvoiceInfoTable** table name.
4. Set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.
5. In the **Company** column, you can select a legal entity code to limit the applicability of this electronic document type. If you don't select a value, electronic documents of this type are processed in all legal entities.
6. Save your changes, and close the page.

Send responses

To generate and submit responses, follow these steps.

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. Learn more in Submit electronic documents.
2. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.
3. In the **Channel** field, select the export channel that you previously created, and then select **OK**.

You can inquire about the results of the submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. In the **Document type** field, select **Vendor invoices** to filter only responses submissions. Learn more in Work with Electronic document submission log.

Additional resources

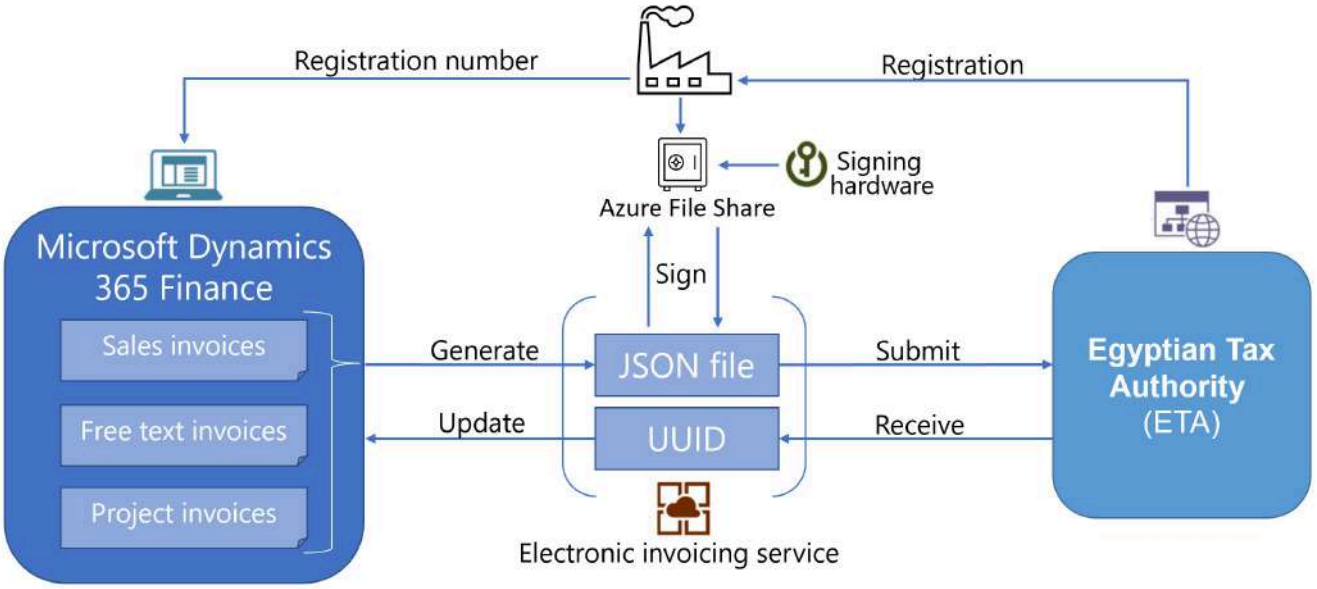
- Customer electronic invoices in Denmark
- Vendor electronic invoice import in Denmark

Electronic invoicing in Egypt

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information to help you get started with Electronic invoicing for Egypt. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. These steps complement the steps that are described in Electronic invoicing configuration.

After you configure Electronic invoicing, you can generate JavaScript Object Notation (JSON) files of electronic invoices in the Egypt-specific format and submit them to the Egyptian tax authority.



> **Note**
>
> Egyptian authorities provide the certificate for digital signing with the required key. They provide this certificate only as a hardware token device that can't be added to an Azure key vault. However, Dynamics 365 Finance enables files to be written to and read from an Azure file share, and that file share can be configured to enable external signing of electronic

documents. To learn more, go to the Egyptian tax authority's website by using the link that is provided in Egyptian eInvoicing & eReceipt SDK – Frequently asked questions.

Prerequisites

Before you begin the procedures in this article, complete the following prerequisites:

- Become familiar with Electronic invoicing as it's described in Electronic Invoicing service overview.
- Complete the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.
- The primary addresses of the legal entities must be in Egypt.
- Your company must be a registered taxpayer in Egypt and must have the required registration numbers.
- Obtain **Client ID** and **Client secret** values from the Egyptian regulatory authorities. These values serve as credentials that are used to establish a secure connection to the authorities' portal.

Azure resources configuration

Complete the common part of Azure resources configuration as described in Configure Azure resources for Electronic invoicing.

Configure an Azure file share

- Configure an Azure file share as described in Create an Azure file share in the Azure portal.
- Create the following Azure file share directories:
  - The directory that generated electronic invoices for digital signing should be written to.
  - The directory that digitally signed electronic invoices should be read from.

> **Note**
>
> We don't recommend that you create a hierarchical structure of subfolders. Instead, create the required folders at the same level, and distinguish them through their names. For example, name the folders **SalesInvoices_BeforeSigning** and **SalesInvoices_Signed**.

Configure the Azure key vault

Add the following required elements in the key vault:

- The secret for the **Client ID** value that the regulatory authorities provided.
- The secret for the **Client secret** value that the regulatory authorities provided.
- The secret for the **Connection string** value of the Azure file share that you configured in the previous section.

> **Important**
>
> Don't confuse the **Connection string** value with the **Storage shared access signature (SAS)** token that is used in the parameters in the next section.

Configure electronic invoicing Key Vault parameters

To configure electronic invoicing Key Vault parameters, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic invoicing** tab, in the **Key Vault settings** section, in the **Key Vault** field, select the reference to the key vault that you configured in the previous section.
3. In the **SAS token secret** field, select the name of the storage account secret URL that must be used to authenticate access to the storage account.
4. Select **Key Vault parameters**.
5. On the **Key Vault parameters** page, in the **Certificates** section, select **Add** to create an element of the appropriate type for each secret that is described in the previous section.
    - A **Client ID** element of the **Secret** type
    - A **Client secret** element of the **Secret** type
    - A **Connection string** element of the **Secret** type

> **Note**
>
> The values in the **Name** column should match the names of the secrets that are described in the previous section.

Configure the electronic invoicing feature

Some of the parameters from the **Egyptian electronic invoice (EG)** electronic invoicing feature are published with default values. Before you deploy the electronic invoicing feature, review the default values, and update them as required, so that they better reflect your business operation.

> ⓘ **Note**
>
> Only the parameters that are mentioned in this section require modification. If you also modify other parameters to meet your needs, you're responsible for ensuring that the feature continues to work correctly.

To configure the electronic invoicing feature, follow these steps:

1. Ensure that the latest version, **20** or later, of the **Egyptian electronic invoice (EG)** Globalization feature is imported as described in Import features from the repository.

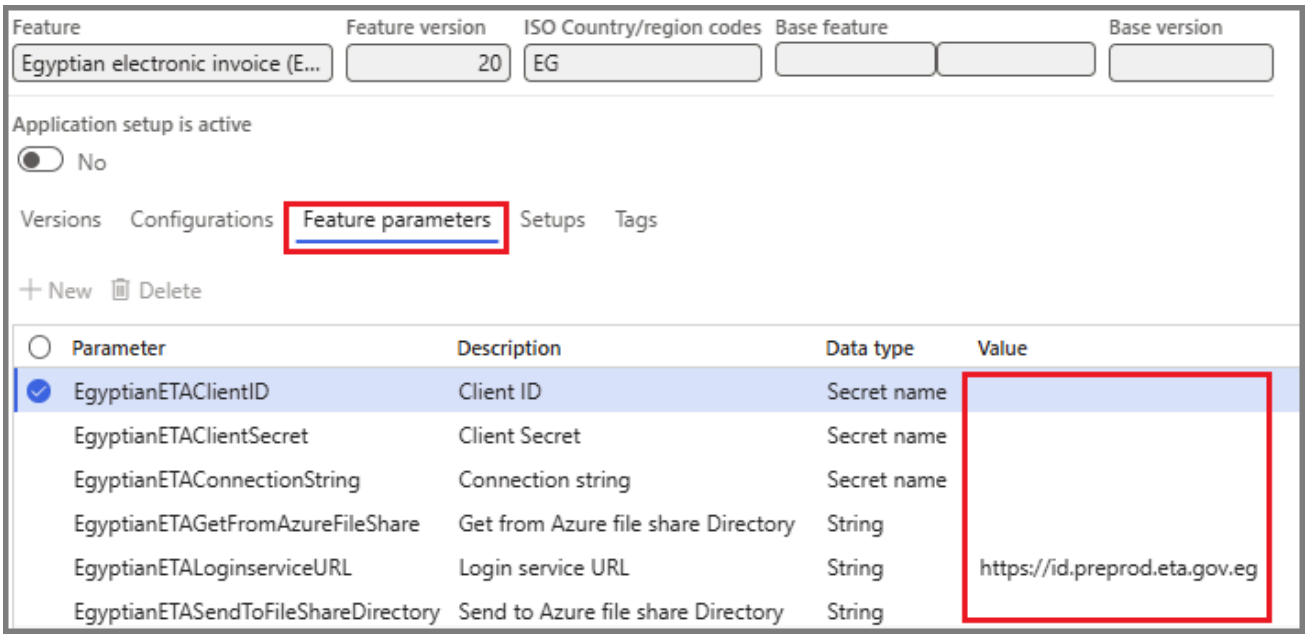
2. Create a copy of the imported Globalization feature, and select your configuration provider for it. Learn more in Create Globalization features.

3. On the **Versions** tab, verify that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters. These parameters are required for interoperation with Egyptian tax authority's services.

    - **EgyptianETAClientID** – Select the secret for the client ID that you configured in the parameters.
    - **EgyptianETAClientSecret** – Select the secret for the client secret that you configured in the parameters.
    - **EgyptianETAConnectionString** – Select the secret for the connection string to the Azure file share that you provisioned and configured in the parameters.
    - **EgyptianETASendToFileShareDirectory** – Enter the name of the Azure file share directory that invoices for digital signing should be written to.
    - **EgyptianETAGetFromAzureFileShare** – Enter the name of the Azure file share directory that digitally signed invoices should be read from.
    - **EgyptianETALoginserviceURL** – Enter the URL of the Egyptian tax authority's login service. The feature that Microsoft released points to the **pre-production** endpoint.

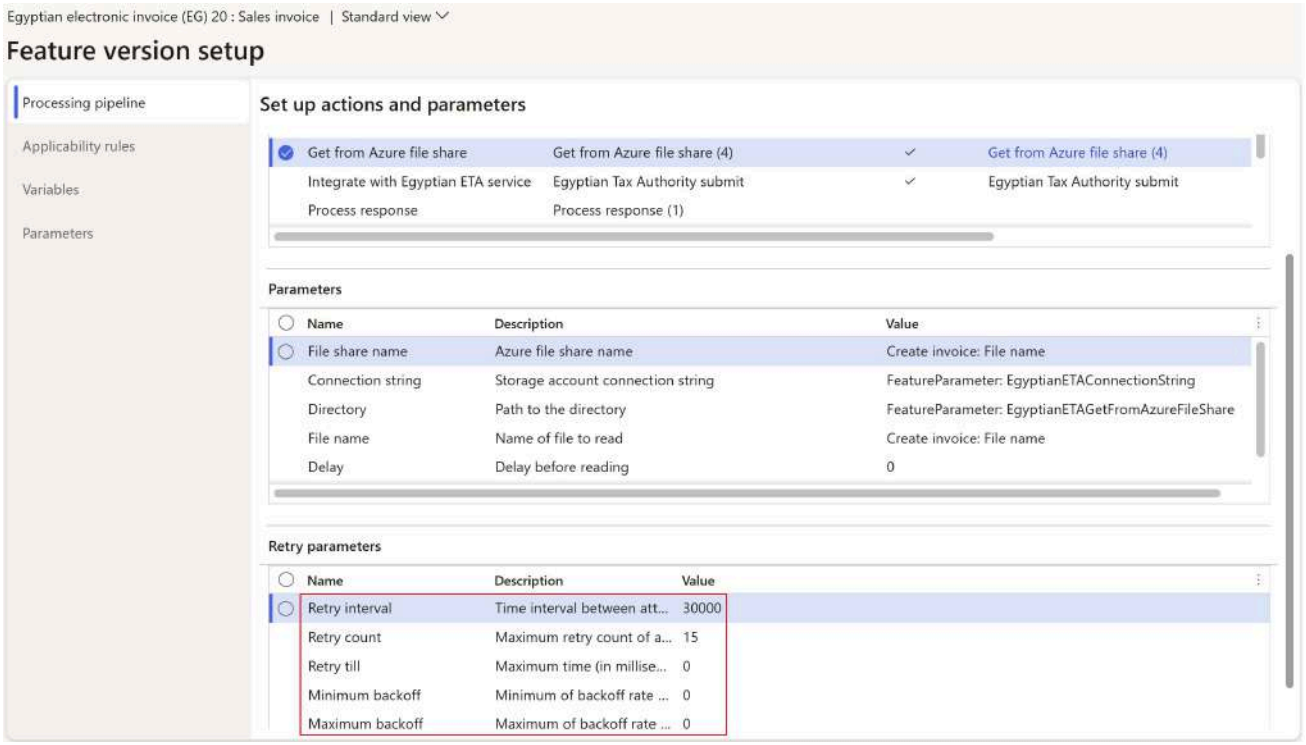
- **EgyptianETAWebserviceURL** – Enter the URL of the Egyptian tax authority's web service. The feature that Microsoft released points to the **pre-production** endpoint.



> ⓘ Note
>
> Because external signing is involved, features that Microsoft publishes repeatedly check the availability of signed invoices until a maximum limit is reached. You can reconfigure the retry parameters as you require by following the next two steps.

5. To reconfigure the retry parameters, on the **Setups** tab, select the **Sales invoice** or **Project invoice** setup, depending on the scenario. Then select **Edit**.

6. On the **Processing pipeline** tab, select the **Get from Azure file share** action, and then, in the **Retry parameters** section, configure the required values.



7. Complete your version of the **Egyptian electronic invoice (EG)** feature, and deploy it to the service. Learn more in Complete and deploy a Globalization feature.

Configure electronic document parameters

To configure electronic document parameters, follow these steps:

1. Make sure that the country/region-specific Electronic reporting (ER) configurations for the document context and electronic document model mapping that are required for Egypt are imported. Learn more in Set up Electronic document parameters.

2. In addition to the ER configurations that are mentioned in step 1, import the Egypt-specific **Egypt response data import format (EG)** configuration from the repository. Learn more in Import Electronic reporting (ER) configurations from Dataverse.

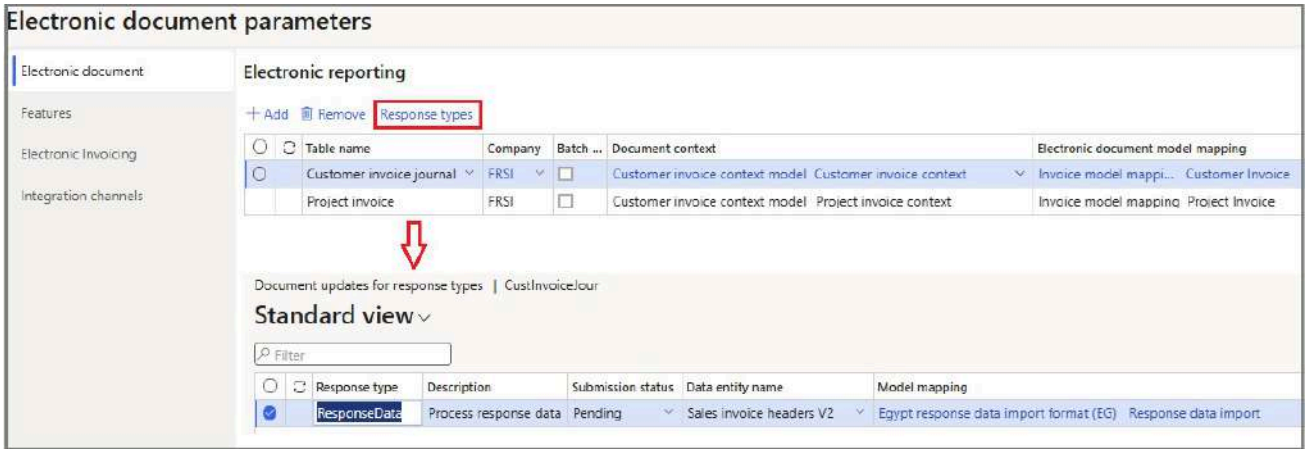
   > **Note**
   >
   > After you import the **Egyptian electronic invoice (EG)** electronic invoicing feature, electronic documents are configured by default. Follow the remaining steps of this procedure if you must make changes. Otherwise, consider them informational only.

3. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

4. On the **Electronic document** tab, add records for the **Customer Invoice journal** and **Project invoice** table names.

5. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.

6. For each table name, select **Response types**, and then follow these steps:
   a. On the **Response types** page, select **Add** to create a response type.
   b. In the **Response type** field, enter **ResponseData**.
   c. In the **Description** field, enter **Process response data**.
   d. In the **Submission status** field, select **Pending**.
   e. In the **Data entity name** field, select **SalesInvoiceHeaderV2Entity**.
   f. In the **Model mapping** field, select **Response data import**. The configuration is **Egypt response data import format (EG)**.
   g. Select **Save**, and then close the **Response types** page.



7. Select **Save**, and then close the **Electronic document parameters** page.

Finance business data configuration

Follow the configuration steps in Customer electronic invoices in Egypt.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. Learn more about how to generate electronic invoices in Submit electronic documents.

> ⓘ **Note**
>
> The submission process involves manual steps for digitally signing the generated electronic invoices files that are stored in the dedicated Azure file share folder, and then moving the

signed files to another Azure file share folder. After those manual steps are completed, the automatic submission process is resumed.

You can inquire about the results of a submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log** and selecting the required document type. Learn more in Work with Electronic document submission log.

Privacy notice

Enabling the **Egyptian electronic invoice (EG)** feature might require that limited data is sent. This data includes the organization's tax registration ID. The data will be transmitted to third-party agencies that the tax authority authorized to send electronic invoices to it in the predefined format that is required for integration with the government's web service. An administrator can enable and disable the feature by going to **Organization administration** > **Setup** > **Electronic document parameters**. On the **Features** tab, select the row that contains the **Egyptian electronic invoice (EG)** feature, and then make the appropriate selection. Data that is imported from external systems into this Dynamics 365 online service is subject to our privacy statement. For more information, see the "Privacy notice" section in country/region-specific feature documentation.

Last updated on 05/04/2026

Electronic invoicing for France

This article helps you get started with electronic invoicing for France. Set up the system to generate, submit, and receive electronic invoices and other related documents in the required format in Microsoft Dynamics 365 Finance through a certified service provider acting as an Approved Platform (*Platform Agréée* - **PA**).

> ⓘ **Note**
>
> This electronic invoicing approach uses an invoicing service that's applicable only to cloud deployments of Microsoft Dynamics 365 Finance.

Watch the overview of the French electronic invoicing implementation in Finance.

https://learn-video.azurefd.net/vod/player?id=45223d9e-5e49-42dc-895c-7d9ccb08b17a&locale=en-us&embedUrl=%2Fdynamics365%2Ffinance%2Flocalizations%2Ffrance%2Femea-fra-einv-ereport

System configuration

> ⓘ **Note**
>
> You can also find all the configuration steps described in the next chapters in the tutorial video in the **Appendix** section.

Prerequisites

Before you start, make sure you have the following prerequisites:

- The company is a registered taxpayer in France.
- The company has a signed agreement with the selected Approved Platform and obtained the credentials required for establishing a secure connection to the Approved Platform's infrastructure.

  > ⓘ **Note**

> This implementation assumes Edicom is the selected certified Approved Platform (PA). For more information, see Edicom integration with Microsoft Dynamics 365.

- Install the **Electronic invoicing add-in** as described in Install the add-in for Electronic invoicing microservices.
- Activate **Electronic invoicing integration** with Finance or Supply Chain Management as described in Enable Electronic invoicing integration.
- Configure the common part of the **Electronic document parameters**.

  > **Note**
  >
  > You need to configure the service environment only if you previously used the Regulatory Configuration Service (RCS) experience to configure the Electronic Invoicing service. Otherwise, keep the **Environment** parameter empty. The system assigns it automatically and makes it read-only. For more information, see Service environment configuration.

- In the **Feature management** workspace, on the **All** tab, enable the following features. If these features don't appear on the page, select **Check for updates**. Select the features, and then select **Enable now** for each of them. For more information, see Feature management overview.
  - **E-Invoice Document Status Lifecycle Management**
  - **E-Invoice Document Response Submission**
  - **Export channels for electronic invoicing integration**
  - **Execute update actions for submitted documents**

  And **disable** the following feature.
  - **Simultaneous globalization features completion and deployment** (*to activate the possibility of the **Application setup** copying*).

Create the Azure Key Vault configuration

Configure the common part of the Azure resources required for Electronic invoicing functioning. For more information, see Configure Azure resources for Electronic invoicing.

Add the following element to the key vault:

- Add the secret for the **token** that authorizes access to Edicom services.

Set up electronic invoicing Key Vault parameters

Set up electronic invoicing Key Vault parameters.

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic invoicing** tab, in **Key Vault settings**, in **Key Vault**, select the key vault reference you created earlier.
3. In **SAS token secret**, select the storage account secret URL used to authenticate access to the storage account.
4. Select **Key Vault parameters**.
5. On the **Key Vault parameters** page, in **Certificates**, select **Add** and create an element of the appropriate type for each secret described earlier.
    - Add the **token** element of type **Secret**.

> ⓘ **Note**
>
> Match the value in the **Name** column to the secret name described earlier.

For more information, see Create a Key Vault reference.

Synchronization of the Electronic invoicing service with Finance

After you complete all the configuration steps described in the previous chapters, validate the configuration.

1. In Dynamics 365 Finance, go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. Select the **Electronic invoicing** tab and select the **Save** menu button.
3. If the configuration is correct, the system shows the *Synchronization with the e-invoicing service was successful* information message. You can continue with the next chapters.
4. If there are synchronization errors, address the errors to achieve successful synchronization.

Import the electronic invoicing feature

1. Go to **Globalization Studio** and select the **Electronic invoicing** tile. Import the latest versions of the following globalization features as described in Import features from the repository.

    - **French electronic invoice (FR)**
    - **French electronic invoice status (FR)**

2. All the required Electronic Reporting configurations are automatically imported as a result of the globalization features import. You can review imported Electronic Reporting configurations in the **Electronic reporting** workspace, on the **Reporting configurations** tile. The full list of the required Electronic Reporting configurations can be found in the **List of Electronic Reporting configurations** section of the **Appendix** chapter.

> ⓘ **Note**
>
> If some of the configurations aren't imported, import them manually as described in Import Electronic reporting (ER) configurations from Dataverse.

> ⓘ **Important**
>
> Make sure that the **Vendor invoice Mapping to destination** and **Response message model mapping to destination (FR domestic)** Electronic Reporting configurations are marked as **Default for model mapping**.

Configure the electronic invoicing features

The **French electronic invoice (FR)** and **French electronic invoice status (FR)** features publish some parameters with default values. Before you deploy the features, review the default values and update them so they reflect your business operations.

Review and update the **French electronic invoice (FR)** feature configuration:

1. Go to **Globalization Studio** and select the **Electronic invoicing** tile. Import the globalization feature as described in Import features from the repository.

2. Copy the imported **French electronic invoice (FR)** globalization feature and select your configuration provider, as described in Create a Globalization feature.

3. On the **Versions** tab, check that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify these required **Edicom** connection and integration parameters:

    - **Domain** – Use the domain number (can be also referred to as **Service ID**) from Edicom to identify the company.
    - **Group** – Use the group code for internal routing within the Edicom infrastructure.
    - **Destination** – Construct the destination by appending **_EDIWIN** to the Domain/Service ID number. For example, if the Domain number is **123456**, enter **123456_EDIWIN**.
    - **Token** – Select the name of the token you created earlier.

5. Each copy starts as a **Draft** version. Complete and deploy the feature as described in Complete and deploy a Globalization feature.

    > ⓘ **Note**
    >
    > While deploying the feature, toggle the **Activate application setup** checkbox in the **APPLICATION** section. In this case, the system automatically preconfigures the significant part of the **Electronic document parameters** in Finance. You can review and change the preconfigured parameters, as described in the next chapter.

6. Repeat steps 2 through 5 for the **French electronic invoice status (FR)** feature.

Configure electronic document parameters

The system preconfigures electronic documents when you deploy the globalization features to the service and select the **Activate application setup** option during deployment. This section provides information about the setup, or if you decide to amend the standard system setup. If no amendment is required, go to the **Configure integration channels** chapter.

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic document** tab, add records for the following views.

    - **Customer invoice journal - Domestic B2B transactions (France)**
    - **Project invoice journal - Domestic B2B transactions (France)**
    - **Customer invoice response**
    - **Pending vendor invoice response**

3. For each electronic document, set the **Document context** and **Electronic document model mapping** fields as described in Set up electronic invoicing parameters.

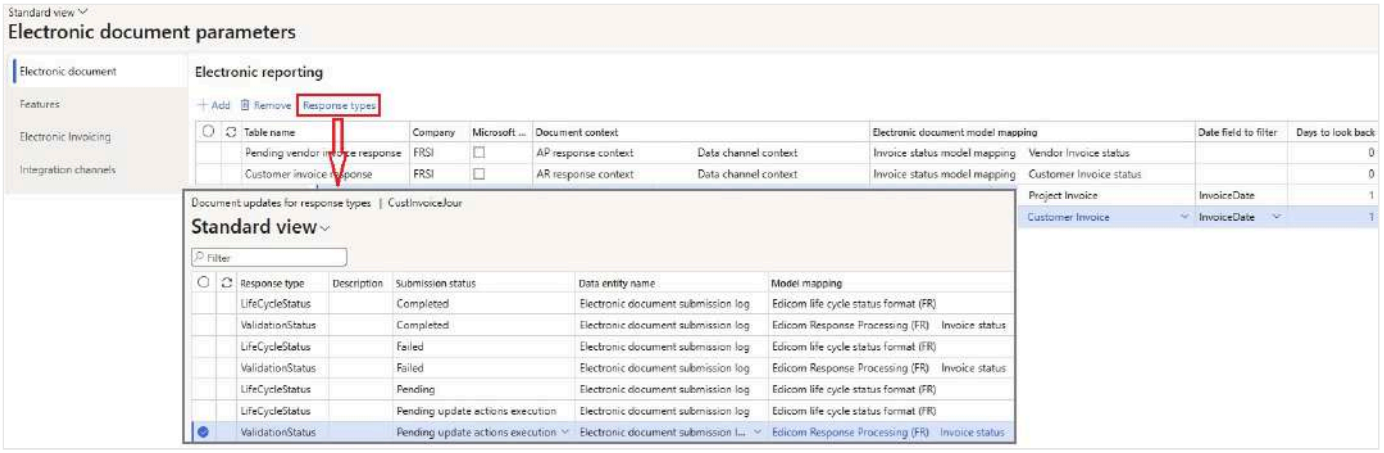
> **Note**
>
> To minimize the risk of accidental massive submissions, the system implements default filtering by document dates. In the **Date filed to filter** column, specify the exact selected table's field for filtering. In the **Days to look back** column, define the number of days to subtract from the current date to determine the earliest date for document processing. If you don't configure the **Date filed to filter** and **Days to look back** columns, the **Invoice date** equal to the current date is used by default.

1. For the **Customer invoice journal - Domestic B2B transactions (France)** and **Project invoice journal - Domestic B2B transactions (France)** electronic documents, select **Response types**.
2. Select **New** to create a new response type.
3. In the **Response type** column, enter **LifeCycleStatus**. Enter the value exactly as shown.
4. In the **Submission status** column, select the **Completed** value.
5. In the **Data entity name** column, select the **Electronic document submission log** entity.
6. In the **Model mapping** column, select the **Edicom life cycle status format (FR)** configuration.
7. Repeat steps 2 through 6 for the following values in the **Submission status** column.

    - **Failed**
    - **Pending**
    - **Pending update actions execution**

8. Select **New** to create a new response type.
9. In the **Response type** column, enter **ValidationStatus**. Enter the value exactly as shown.
10. In the **Submission status** column, select the **Completed** value.
11. In the **Data entity name** column, select the **Electronic document submission log** entity.
12. In the **Model mapping** column, select the **Edicom Response Processing (FR)** configuration and the **Invoice status** mapping name.
13. Repeat steps 8 through 12 for the following values in the **Submission status** column.

    - **Failed**
    - **Pending update actions execution**



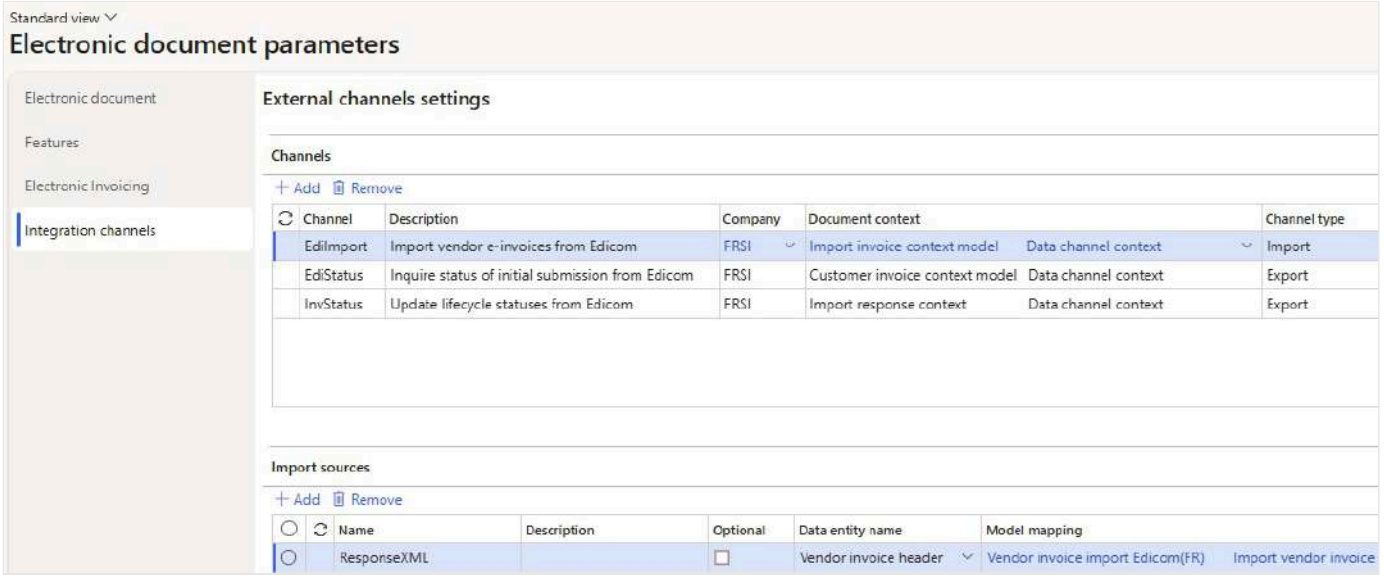
1. For the **Customer invoice response** electronic document, select **Response types**.
2. Select **New** to create a new response type.
3. In the **Response type** column, enter **Response**. Enter the value exactly as shown.
4. In the **Submission status** column, select the **Completed** value.
5. In the **Data entity name** column, select the **Electronic document submission log** entity.
6. In the **Model mapping** column, select the **Edicom Cust App Response Processing** configuration.
7. Repeat steps 2 through 6 for the **Pending** value in the **Submission status** column.
8. Select **Save**, and then close the page.
9. For the **Pending vendor invoice response** electronic document, select **Response types**.
10. Select **New** to create a new response type.
11. In the **Response type** column, enter **Response**. Enter the value exactly as shown.
12. In the **Submission status** column, select the **Completed** value.
13. In the **Data entity name** column, select the **Electronic document submission log** entity.
14. In the **Model mapping** column, select the **Edicom Vend App Response Processing** configuration.
15. Repeat steps 10 through 14 for the **Pending** value in the **Submission status** column.
16. Select **Save**, and then close the page.

> ⓘ **Note**
>
> If you created derived equivalents of the earlier Electronic Reporting configurations, use them instead of the standard configurations.

Configure integration channels

1. On the **Integration channels** tab, in the **Channels** section, select **Add** to create a new channel.

2. In the **Channel** field, enter **EdiStatus**. Enter the value exactly as shown. The system uses it to submit outgoing electronic invoices.
3. In the **Company** field, select the legal entity.
4. In the **Document context** field, select the **Data channel context** mapping from the **Customer invoice context model** configuration.
5. In the **Channel type** field, select **Export**.
6. In the **Channels** section, select **Add** to create another channel.
7. In the **Channel** field, enter **InvStatus**. Enter the value exactly as shown. The system uses it to update lifecycle statuses for previously submitted invoices.
8. In the **Company** field, select the legal entity.
9. In the **Document context** field, select the **Data channel context** mapping from the **Import response context** configuration.
10. In the **Channel type** field, select **Export**.
11. In the **Channels** section, select **Add** to create another channel.
12. In the **Channel** field, enter **EdiImport**. Enter the value exactly as shown. The system uses it to import incoming electronic invoices.
13. In the **Company** field, select a required legal entity.
14. In the **Document context** field, select the **Data channel context** mapping from the **Import invoice context model** configuration.
15. In the **Channel type** field, select **Import**.
16. In the **Import sources** section, select **Add** to create an import source.
17. In the **Name** field, enter **ResponseXml**. Enter the value exactly as shown.
18. In the **Data entity name** field, select the **Vendor invoice header** entity.
19. In the **Model mapping** field, select the **Import vendor invoice** mapping from the **Vendor invoice import Edicom (FR)** configuration.
20. Select **Save**, and then close the page.



> ⓘ Note
>
> If you use integration channels names other than **InvStatus**, **EdiStatus**, and **EdiImport**, or import source name other than **ResponseXml**, you need to make related changes in the involved context configurations and invoicing feature setups' applicability rules and variables.

Set up registration numbers

If these registration types and categories already exist and are configured accordingly, skip this procedure.

Set up SIREN number

> ⓘ Note
>
> When you generate the output files for electronic invoices, the system uses registration numbers from the **Enterprise ID (COID)** category as *Système d'identification du répertoire des entreprises* numbers (**SIREN**). If the **Enterprise ID (COID)** registration category already exists and is assigned to a registration type, skip this procedure.

To configure the **SIREN** registration number, follow these steps.

1. In Dynamics 365 Finance, go to **Organization administration** > **Global address book** > **Registration types** > **Registration types**.
2. Create a registration type.
3. In the **Country/region** field, select **FRA - France**.
4. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration categories**.
5. Create a registration category.
6. In the **Registration types** field, select the registration type that you created in step 2.
7. In the **Registration categories** field, select **Enterprise ID (COID)**.

Set up SIRET number

To configure the *Système d'identification du répertoire des établissements* (**SIRET**) registration number, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Global address book** > **Registration types** > **Registration types**.
2. Create a registration type.
3. In the **Country/region** field, select **FRA - France**.
4. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration categories**.
5. Create a registration category.
6. In the **Registration types** field, select the registration type that you created in step 2.
7. In the **Registration categories** field, select **SIRET**.

Set up VAT number

Optionally, to configure the tax-exempt number, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Global address book** > **Registration types** > **Registration types**.
2. Create a registration type.
3. In the **Country/region** field, select **FRA - France**.
4. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration categories**.
5. Create a registration category.
6. In the **Registration types** field, select the registration type that you created in step 2.
7. In the **Registration categories** field, select **VAT ID**.

Set up Branch ID

If you configure Branch IDs, the system uses them as electronic addresses for routing documents. To configure the Branch ID, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Global address book** > **Registration types** > **Registration types**.
2. Create a registration type.
3. In the **Country/region** field, select **FRA - France**.
4. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration categories**.
5. Create a registration category.
6. In the **Registration types** field, select the registration type that you created in step 2.
7. In the **Registration categories** field, select **Branch ID**.

> **Note**
>
> If you can't use branches as electronic addresses, you can alternatively configure electronic addresses through electronic document properties as described in the Configure electronic addresses chapter. In this case, only one electronic address per invoice party is supported.

Additional configuration of registration numbers

To support scenarios when companies have multiple subsidiaries with their own registration numbers such as SIREN, SIRET, VAT ID, and Branch ID as an electronic address, you need to activate and configure the **Establishments** functionality. For more information, see the following articles:

- Invoice party applicability rules for registration categories
- Establishments in France
- Registration numbers setup for France

> **Note**
>
> In the current implementation of Electronic invoicing, multiple registration numbers are supported only for Buyers.

Set up address structure

Set up the postal address structure.

1. Go to **Organization administration** > **Global address book** > **Addresses** > **Address setup**.
2. Ensure that at least the **Country/Region code** element is configured.

Configure legal entity data

Enter the address

Add the primary address.

1. Go to **Organization administration** > **Organizations** > **Legal entities**, and select a legal entity.
2. On the **Addresses** FastTab, add the primary address for the legal entity.

Seller identification

Add the registration numbers.

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. On the Action Pane, select **Registration IDs**.
3. On the **Registration ID** FastTab, select **Add**, set **Registration type** to the SIREN type you created earlier, and enter the SIREN number in the **Registration number** column.
4. Select **Add**, set **Registration type** to the SIRET type you created earlier, and enter the SIRET number in the **Registration number** column.
5. Define the Tax exempt (VAT) number whatever way is used in your company.

> **Note**
>
> If you don't define the registration number with the **VAT** registration category, the value from **Organization administration** > **Organizations** > **Legal entities** > **Foreign trade and statistics** > **INTRASTAT** > **VAT exempt number export** is used.

Configure customer data

Enter the address

To enter the address, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer.
3. On the **Addresses** FastTab, add a valid address for the selected customer.

> **Note**
>
> Each address links to one set of registration numbers, including the Branch ID (electronic address). If a customer has several electronic addresses, you need to create several location addresses to ensure a one-to-one relation.

Buyer identification

To enter the registration numbers, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.

2. On the Action Pane, on the **Customer** tab, in the **Registration** group, select **Registration IDs**.
3. Select the **Address** that the registration numbers belong to. Make sure that the **Purpose** of the selected address contains the **Delivery** and **Invoice** values.
4. On the **Registration ID** FastTab, select **Add** to create a registration ID.
5. In the **Registration type** field, select the SIREN registration type that you created earlier.
6. Select **Add**, and in the **Registration type** field, select the SIRET registration type that you created earlier.
7. Select **Add** to create another registration ID, if necessary.
8. In the **Registration number** field, enter a valid Branch ID for the specific address of the selected customer.
9. Select **Add** to create another registration ID, if necessary.
10. In the **Registration number** field, enter a valid VAT number for the selected customer.

> **Note**
>
> If you don't define the registration number with the **VAT** registration category, the value from **Accounts receivable** > **Customers** > **All customers** > **Invoice and delivery** > **SALES TAX** > **Tax exempt number** is used.

Configure electronic addresses

> **Note**
>
> Follow the steps in this section only if you don't use registration numbers of the **Branch ID** category. Electronic addresses that you configure through electronic document properties take priority and overwrite registration numbers of the **Branch ID** category when generating e-invoices XML files.

Follow these steps to configure electronic addresses for sellers or buyers that you use as **EndpointIDs** for document routing.

Configure electronic document properties

Set up electronic document properties.

1. Go to **Accounts receivable** > **Setup** > **Electronic document property types**, and select **New**.

2. In the **Type** field, enter **SellerElectronicAddress**. Enter the value exactly as shown. It's used for the **Seller** identification schema definition.
3. Select **Applicability** to add an applicable table.
4. On the **Electronic document property type applicability setup** page, in the **Table name** field, select **Legal entities**.
5. Save your changes and return to the **Electronic document property types** page.
6. Select **New** to create another electronic document property type.
7. In the **Type** field, enter **BuyerElectronicAddress**. Enter the value exactly as shown. It's used for the **Buyer** identification schema definition.
8. Select **Applicability** to add an applicable table.
9. On the **Electronic document property type applicability setup** page, in the **Table name** field, select **Customers**.
10. Save your changes, and return to the **Electronic document property types** page.
11. Save your changes, and close the page.

Enter the seller electronic address

To enter the Seller schema code, follow these steps:

1. Go to **Organization administration** > **Organizations** > **Legal entities** and select a legal entity.
2. Select **Electronic document properties** from the Action Pane.
3. In the **Value** column, enter the required seller electronic address.

Enter the buyer electronic address

To enter the Buyer schema codes, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a specific customer in the list. On the Action Pane, on the **Customer** tab, in the **Properties** group, select **Electronic document properties**.
3. In the **Value** column, enter the required buyer electronic address.

Configure the electronic address handling

For both Seller and Buyer identification, the system uses the electronic address you define as the **EndpointID** value with the **schemeID** attribute set to **0225** (FRCTC ELECTRONIC ADDRESS) by default, according to the Electronic Address Scheme (EAS).

You can change the code of the electronic address scheme to any value that better fits your business processes by configuring the **CompanyEndpointType** and **CustomerEndpointType** electronic document properties types for sellers and buyers respectively. The values defined via these electronic document properties have higher priority and overwrite the default **0225** value.

You can also control the interpretation of the Buyer's **Branch ID** value. By default, the system considers it as an entire electronic address. Alternatively, you can configure the **ElectronicAddressSuffix** electronic document property to force the system to interpret the Branch ID value only as a *SUFFIX* part of the electronic address. The whole electronic address is generated as the concatenation of the Buyer's **SIREN_**, **SIRET_**, and the **SUFFIX**.

> ⓘ **Note**
>
> To learn more about the full list of the electronic document properties used during the generation of electronic invoices XML files, refer to the following Appendix chapter List of electronic document properties.

Configure mandatory notes

According to French requirements, each individual electronic invoice must contain three mandatory **Note** elements in the header with the **following** predefined prefixes:

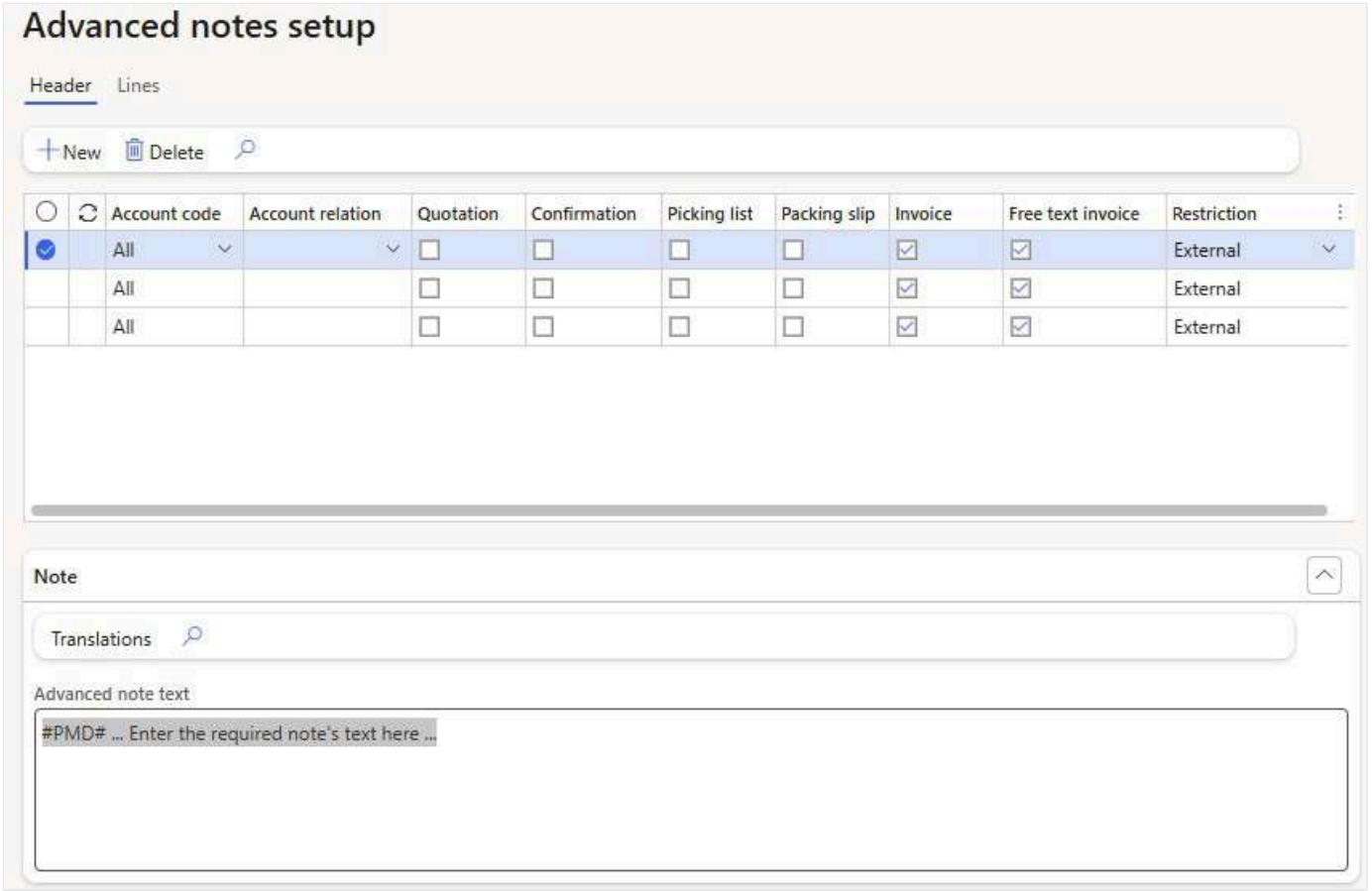
- **#PMD#** *the text of the first note related to payment/settlement information*
- **#PMT#** *the text of the second note related to payment instructions*
- **#AAB#** *the text of the third note related to payment terms*

> ⓘ **Note**

Optionally, you can add one more note with the predefined **#ADN#** prefix that is the indication for EDICOM to treat a submitted invoice that contains this note as representing **B2G** communication.

Configure mandatory notes for Sales and Free text invoices

To automate the creation of notes, see Advanced notes management. Configure at least three external header-level notes with the mandatory prefixes for the required documents, for all or selected customers. When you configure these notes, the system automatically adds them to the applicable documents upon creation.

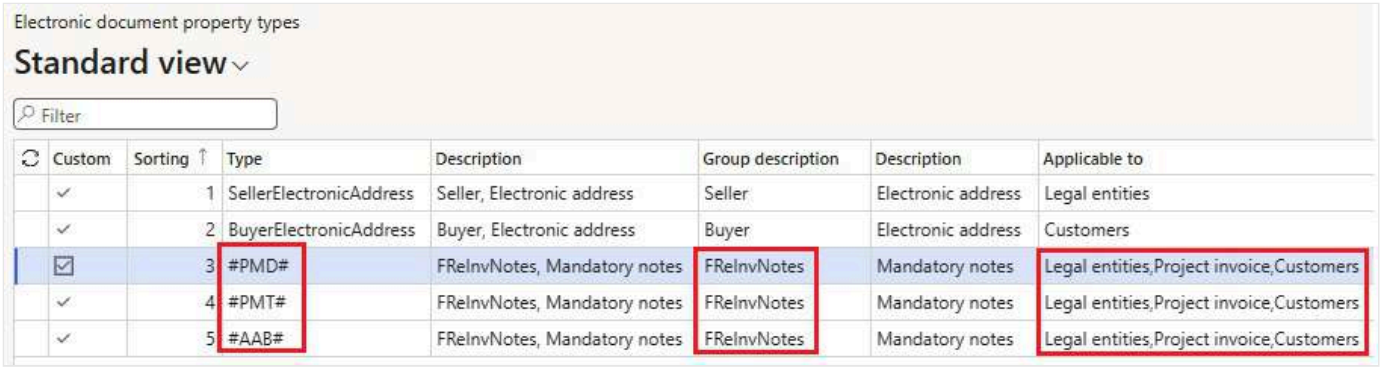


Configure mandatory notes for Project invoices

For Project invoices, the functionality of Advanced notes doesn't apply. Follow the steps described in this section to configure the mandatory notes for Project invoices.

1. Go to **Accounts receivable** > **Setup** > **Electronic document property types**, and select **New**.
2. In the **Type** field, enter the required notes codes at least, **#PMD#**, **#PMT#**, and **#AAB#**.
3. In the **Group description** field, enter **FReInvNotes**. Enter the value exactly as shown.
4. Select **Applicability** to add an applicable table.

5. On the **Electronic document property type applicability setup** page, in the **Table name** field, select **Legal entities**, **Customers**, and **Project invoices**.
6. Save your changes and return to the **Electronic document property types** page.
7. Save your changes, and close the page.



Custom	Sorting	Type	Description	Group description	Description	Applicable to
✓	1	SellerElectronicAddress	Seller, Electronic address	Seller	Electronic address	Legal entities
✓	2	BuyerElectronicAddress	Buyer, Electronic address	Buyer	Electronic address	Customers
☑	3	#PMD#	FRelnvNotes, Mandatory notes	FRelnvNotes	Mandatory notes	Legal entities,Project invoice,Customers
✓	4	#PMT#	FRelnvNotes, Mandatory notes	FRelnvNotes	Mandatory notes	Legal entities,Project invoice,Customers
✓	5	#AAB#	FRelnvNotes, Mandatory notes	FRelnvNotes	Mandatory notes	Legal entities,Project invoice,Customers

Enter the Legal Entity mandatory notes

To enter the Legal Entity mandatory notes, follow these steps.

1. Go to **Organization administration** > **Organizations** > **Legal entities** and select a legal entity.
2. Select **Electronic document properties** from the Action Pane.
3. In the **Value** column, enter the required note text for each of the mandatory notes.

Enter the Customers mandatory notes

To enter the Customers mandatory notes, follow these steps.

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a specific customer in the list. On the Action Pane, on the **Customer** tab, in the **Properties** group, select **Electronic document properties**.
3. In the **Value** column, enter the required note text for each of the mandatory notes.

Enter the Invoices mandatory notes

To enter the Invoices mandatory notes, follow these steps.

1. Go to **Project management and accounting** > **Project invoices** > **Project invoices**.
2. Select a specific project invoice in the list, and then, on the Action Pane, on the **Project invoice** tab, in the **Properties** group, select **Electronic document properties**.
3. In the **Value** column, enter the required note text for each of the mandatory notes.

> ⓘ **Note**

Invoice-level notes have higher priority than Customer-level notes and Legal Entity-level notes. Use them if you need to enter specific notes for a particular invoice. Customer-level notes have higher priority than Legal Entity-level notes. Use them when you need to have similar notes for all invoices issued to a specific customer. Use Legal Entity-level notes for the same notes for all invoices issued from the Legal Entity for all customers.

Set up units of measure

Set up units of measure.

1. Go to **Organization administration** > **Setup** > **Units** > **Units**.
2. Select a unit ID, and then select **External codes**.
3. On **External codes**, in **Overview**, enter the unit ID in the **Code** column.
4. Select the check box in the **Standard code** column.
5. In the **Value** section, enter the external code from the UNECE Recommendation 20 code list in the **Value** field.

   > **Note**
   >
   > If no specific unit of measure applies, the system uses the default value **EA**.

Configure sales tax codes

1. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax codes**.
2. Select a sales tax code. On the Action Pane, on the **Sales tax code** tab, in the **Sales tax code** group, select **External codes**.
3. In the **Overview** section, create a line for the selected unit. Enter the sales tax code from step 2 in the **External code** field.
4. In the **Value** section, enter an external code according to the Duty or tax or fee category code (Subset of UNCL5305) in the **Value** field.
5. Go to **Tax** > **Setup** > **Sales tax** > **Sales tax exempt codes**.
6. Define exempt codes that you use for nontaxable, zero-rated, or exempted operations.

> **Note**

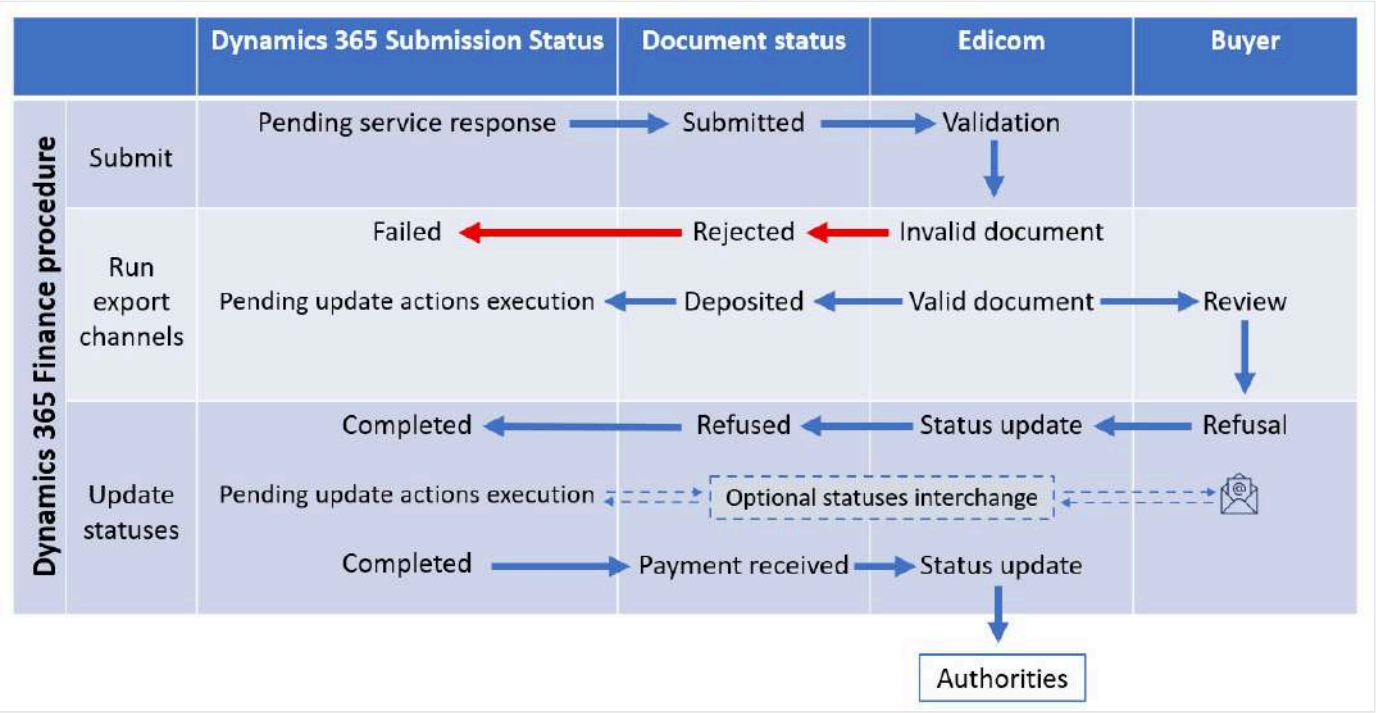
The exempt codes can have any value that your company uses internally, but the **Description** field must contain the standardized tax exemption code according to the official categorization introduced in France, such as *VATEX-FR-298SEXDECIESA*. The value from the **Description** field is used as **Tax Exemption Reason** when generating the output XML files for electronic invoices.

Issue electronic invoices

> ⓘ **Important**
>
> **Electronic invoicing scope**
>
> The scope of **E-Invoicing** individual submissions *includes* only invoices issued for customers with a nonempty **SIREN** registration number defined for the French country/region code (**FRA**). All other invoices are *excluded* from **E-Invoicing** and considered for **E-Reporting**.

After you complete the required configuration steps, generate and submit electronic invoices for posted invoices. The submission process consists of three major steps.

- **Submission of e-invoices to Edicom** - at this stage, the system generates the XMLs of e-invoices and submits them to Edicom.
- **Inquiring statuses of submitted e-invoices** - at this stage, the system inquires the initial status of submitted e-invoices from Edicom. The submission status can be **Failed** if the submitted invoice is **Rejected** by Edicom due to various reasons, or **Pending update actions execution** if the submitted invoice is successfully validated by Edicom and the system is ready for further processing.
- **Update statuses of submitted e-invoices** - at this stage, Dynamics 365 and Edicom can intercommunicate exchanging different e-invoices statuses until the concluding status is reached or due to preconfigured timeout.



Submission of e-invoices to Edicom

To start the electronic invoice submission process, go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information, see Submit electronic documents.

> **Note**
>
> During this stage, the system performs the first two actions from the **French electronic invoice (FR)** feature for the related invoice: it generates the XML file of the electronic invoice in the required format and, if the first action is successful, submits the generated XML file to Edicom.

Check the submission results at **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. For more information, see Work with Electronic document submission log. The documents can have either **Failed** submission status, if there were either Electronic Reporting run-time errors or Edicom portal is unreachable, or **Pending service response** submission status, when ready for further processing.

The following types of invoices are processed during the submission.

- Invoices based on Sales orders - electronic invoices of **380** type are generated.
- Credit notes based on Sales orders - electronic invoices of **381** type are generated.
- Free text invoices - electronic invoices of **380** type are generated.

- Free text credit notes - electronic invoices of **381** type are generated.
- Project invoices - electronic invoices of **380** type are generated.
- Project credit notes - electronic invoices of **381** type are generated.
- Customer prepayment invoices created using Customer prepayment invoices functionality - electronic invoices of **386** type are generated.

Inquiring statuses of submitted e-invoices

To check the initial status of the submitted electronic invoices from Edicom, follow these steps:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.
2. In the **Channel** field, select the **EdiStatus** export channel or the channel you created for invoices, and then select **OK**.

Check the results at **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. The documents can be in either **Failed** status if Edicom rejects them due to syntactical, or other validation errors, or **Pending update actions execution** if Edicom successfully validates them and they're ready for further processing.

> ⓘ **Note**
>
> You can also view submitted electronic invoices in the Ediwin portal in the **Outbound** folder and its subfolders, where you can monitor further processing.

Update statuses of submitted e-invoices

As part of the document lifecycle, some buyer responses for successfully submitted invoices might come later. These responses can affect the statuses of submitted documents. To keep document statuses up to date, run the following procedure.

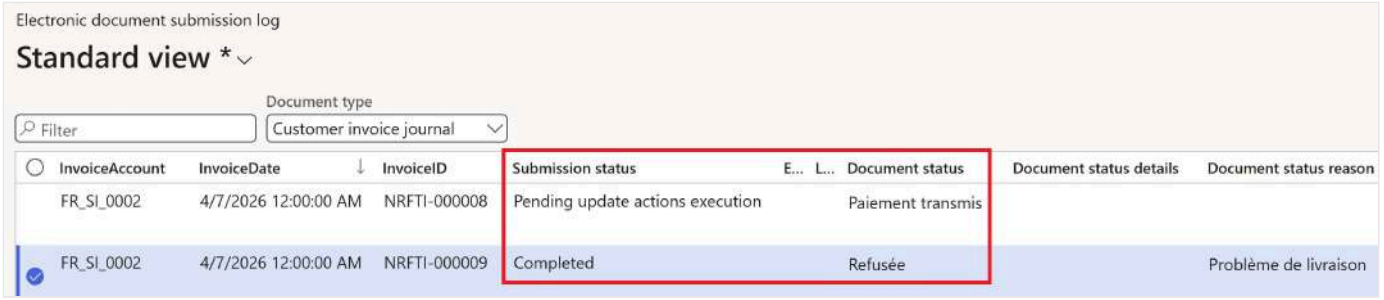
1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**.
2. Select any document and run **Functions** > **Execute update actions** from the menu.
3. The procedure runs for all documents in the **Pending update actions execution** submission status. If necessary, define additional filtering, and then select **OK**.

> ⓘ **Note**

You can also configure periodic running of this procedure in the background to avoid manual intervention.

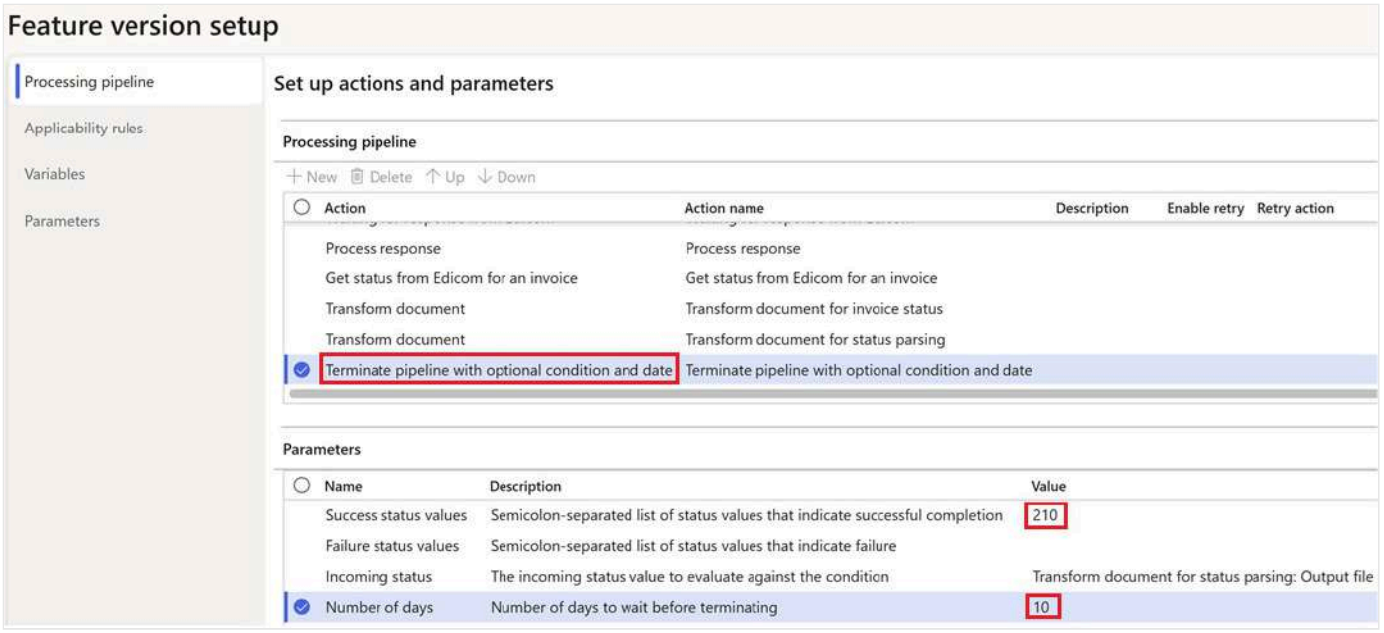
You can run the procedure multiple times to retrieve various lifecycle document statuses from Edicom until it reaches a **termination** status code or a predefined **timeout**.

If the procedure reaches any termination document status code, it updates the related document's submission status to **Completed** - this status concludes the lifecycle of the document. All other, nontermination, document statuses are for information only and keep the **Pending update actions execution** submission status intact.



You can configure the status codes for successful or unsuccessful termination and the timeout period in days in the related feature setup parameters. To configure the parameters, follow these steps.

1. In the **Globalization Studio** workspace, the **Electronic invoicing** tile, select the **French electronic invoice (FR)** globalization feature.
2. On the **Versions** tab, check that the **Draft** version is selected. Create a new draft version of the feature if necessary.
3. On the **Setups** tab, select the feature setup for a required document.
4. In the **Processing pipeline** section, select the last, **Terminate pipeline with optional condition and date** action.
5. In the **Parameters** grid, select the **Success status values** parameter. In the **Value** column, enter status codes separated by comma. These statuses are used for the pipeline processing termination when reached. To learn more about the available statuses, see the List of lifecycle status codes section of the **Appendix** chapter.
6. Select the **Number of days** parameter. In the **Value** column, enter the number of days after which the pipeline terminates regardless of the processing document status.
7. Complete and deploy the feature.



Receive incoming electronic invoices

Set more parameters in Microsoft Dynamics 365 Finance to import incoming invoices.

Configure these master data types to match incoming electronic invoices:

- vendors
- products
- units

Follow the steps in Import vendor electronic invoices, starting at Configure vendor data.

Receive electronic invoices

After you complete the configuration, receive incoming electronic invoices in the French-specific UBL-based format.

> **Note**
>
> Review incoming electronic invoices in the **Inbound** folder and its subfolders in your Ediwin portal.

Receive electronic invoices:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. Select **OK**, and then close the page.

View receipt logs for processed electronic invoices: go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**.

View successfully received invoices: go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.

Send electronic invoice responses

Some business scenarios assume sending responses for either incoming or outgoing electronic invoices. The responses result in assigning mandatory document statuses that conclude the electronic invoices lifecycle.

Send responses for Customer and Project invoices

The current implementation allows sending only *payment reception confirmation* responses to your Buyers. To enter responses, follow these steps.

1. Go to **Accounts receivable** > **Inquires and reports** > **Invoices** > **Invoice journal** for Sales and Free text invoices or to **Project management and accounting** > **Project invoices** > **Project invoices**.

2. Select a specific invoice in the list. On the Action Pane, on the **Invoice** / **Project invoice** tab, in the **Document** group, select **Add response**.

3. In the **Response code** field, select the required value from the list. The **Amount paid** field is automatically populated based on payments settled against this invoice.

4. Select **OK**, and then close the page.

   > ⓘ **Note**
   >
   > You can also use a dedicated procedure in **Organization administration** > **Periodic** > **Electronic documents** > **Generate payment responses** for mass generation of payment responses for eligible invoices. These invoices must be already submitted to EDICOM and fully paid in Dynamics 365 Finance.

5. After you manually enter or automatically generate payment responses, go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**.

6. In the **Record to include** section, make sure that the required **Customer invoice response** records are selected.

7. Select **OK** to start the responses submission process.

> ⓘ Note
>
> At this stage, the **French electronic invoice status (FR)** globalization feature, **Send AR response** feature setup runs for responses processing.

8. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.

9. In the **Channel** field, select the **InvStatus** export channel or the channel you created for responses, and then select **OK**.

10. Check the responses submission results at **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. Select **Customer invoice response** document type.

Send responses for pending vendor invoices

The current implementation only supports sending *refusal* responses to your sellers. To enter responses, follow these steps:

1. Go to **Accounts payable** > **Invoices** > **Pending vendor invoice**.

2. Select a specific invoice in the list. On the Action Pane, on the **Vendor invoice** tab, in the **Actions** group, select **Add response**.

3. In the **Response code** field, select the required value from the list.

4. In the **Reason code** field, select the required value from the list.

5. Select **OK**, and then close the page.

> ⓘ Note
>
> You can add refusal responses only to vendor invoices that you receive by using the **Receive electronic documents** procedure and that have related records in the **Electronic document receipt log** in Dynamics 365 Finance.

6. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**.

7. In the **Record to include** section, make sure that the required **Pending vendor invoice response** records are selected.

8. Select **OK** to start the responses submission process.

   > **Note**
   >
   > At this stage, the **French electronic invoice status (FR)** globalization feature, **Send AP response** feature setup runs for responses processing.

9. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.

10. In the **Channel** field, select the **InvStatus** export channel or the channel you created for responses, and then select **OK**.

11. Check the responses submission results at **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. Select **Pending vendor invoice response** document type.

Appendix

List of lifecycle status codes

The list of mandatory status codes supported in electronic invoicing for Microsoft Dynamics 365 Finance.

Expand table

Code	Status	Requirement	Description
200	Deposited	**Mandatory**	An e-invoice is transmitted to the PA, which certifies that the invoice is validated and compliant.
210	Refused	**Mandatory**	The recipient refuses the invoice.
212	Payment received	**Mandatory**	The invoice is fully paid by the recipient.
213	Rejected	**Mandatory**	The invoice is rejected by either issuer's or receiver's PA.

> **Note**
>
> The electronic invoicing functionality in Microsoft Dynamics 365 Finance isn't limited to the statuses listed in the preceding table. The system can receive any valid status from your buyers' PAs through Edicom. At the same time, the current implementation supports sending only refusal responses to your sellers and payment reception confirmation responses to your buyers. You can find the full list of statuses in the Specifications and standards for electronic invoicing.

List of Electronic Reporting configurations

- Invoice model
  - Invoice model mapping (FR)
  - UBL Sales e-invoice (FR)
  - UBL Sales e-credit note (FR)
  - UBL Project e-invoice (FR)
  - UBL Project e-credit note (FR)
  - Vendor invoice import Edicom (FR)
  - Vendor invoice Mapping to destination
  - Invoice status model mapping
  - Edicom invoice status payment confirmation format
  - Edicom invoice status reject format
- Customer invoice context model
  - Customer invoice context model (FR)
  - Import invoice context model
  - Import response context
  - AP response context
  - AR response context
- Response message model
  - Response message model mapping to destination
  - Response message model mapping to destination (FR domestic)
  - Edicom life cycle status format (FR)
  - Edicom Response status parsing format(FR)
  - Edicom Response Processing
  - Edicom Response Processing (FR)
  - Edicom response error log import

List of electronic document properties

The following configurable electronic document properties are used during the generation of electronic invoices XML files.

Expand table

Type	Applicability	Description
CompanyEndpointType	Legal entities	Defines the code of the electronic address *scheme* for **sellers**. If you set this property, it overwrites the default 0225 value.
CustomerEndpointType	Customers	Defines the code of the electronic address *scheme* for **buyers**. If you set this property, it overwrites the default 0225 value.
SellerElectronicAddress	Legal entities	Defines the electronic address for **sellers**. It has the highest priority and overwrites any other electronic address value.
BuyerElectronicAddress	Customers	Defines the electronic address for **buyers**. It has the highest priority and overwrites any other electronic address value.
#PMD#	Legal entities Customers Project invoices	Project invoices mandatory note prefix about payment/settlement information.
#PMT#	Legal entities Customers Project invoices	Project invoices mandatory note prefix about payment instructions.
#AAB#	Legal entities Customers Project invoices	Project invoices mandatory note prefix about payment terms.
#ADN#	Customers	Optional note prefix determining buyers for B2G communication.
ElectronicAddressSuffix	Legal entities	If you set any non-empty value for this parameter, the Buyer's electronic address is constructed as **SIREN_SIRET_BranchID**. Otherwise, only the Branch ID is used as the whole electronic address.

Configuration tutorial video

Watch the tutorial video about the configuration of Frecn electronic invoicing in Dynamics 365 Finance.

https://learn-video.azurefd.net/vod/player?id=848427f7-6557-451b-9e80-0d8cf891aea9&locale=en-us&embedUrl=%2Fdynamics365%2Ffinance%2Flocalizations%2Ffrance%2Femea-fra-einv-ereport

More information

- Electronic invoicing coverage by country and region

Last updated on 08/06/2026

Get started with electronic invoicing for France

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Learn more about how to get started with electronic invoicing for France. This article guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. These steps complement the steps described in Electronic invoicing configuration.

> **Important**
>
> As of September 2024, you can import all electronic invoicing Globalization features only into the **Globalization studio** workspace, because Regulatory Configuration Service (RCS) is decommissioned. For more information about migration to Globalization Studio, see Regulatory Configuration Service merge to the Globalization Studio workspace.

Country/region-specific configuration for the French Chorus Pro submission (FR) electronic invoicing feature

You need to complete some steps to configure the **French Chorus Pro submission (FR)** electronic invoicing feature. The configuration publishes some parameters with default values. Review and update these values so that they better reflect your business operations.

Prerequisites

Before you begin the procedures in this article, complete the following prerequisites:

- Become familiar with electronic invoicing. For more information, see Electronic Invoicing service overview.

- Complete the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.
- Your organization must be registered to operate with Chorus Pro. Microsoft provides integration with Chorus pro in OAuth2 Mode via an application programming interface (API). For detailed information about Chorus Pro registration and application activation, see the official documentation.

  Follow these steps to register with Chorus Pro.

  1. Go to the PISTE portal to start your registration.
  2. Register an application, and activate Open Authorization (OAuth) credentials. For more information, see Set up a client ID and a client secret.
  3. Copy and save the client ID of the OAuth credentials and the secret key. You'll use this information in later steps.
  4. Go to the Chorus Pro portal to register.
  5. Create a technical account for API access. For more information, see Creation of a technical account for API access in production.
  6. Copy the user ID of the technical account and the password. You'll use this information in later steps.

Country/region-specific configuration of the application setup for the French Chorus Pro submission (FR) electronic invoicing feature

Some of the parameters for the **French Chorus Pro submission (FR)** electronic invoicing feature are published with default values. Before you deploy the electronic invoicing feature to the service environment, add a feature that is based on the Microsoft-provided feature, and complete the common parameters on the **Feature parameters** tab. Review and update the default values as required, so that they better reflect your business operations.

For France, there are at least three interactions with Chorus Pro in the pipeline: first to submit the invoice, then to fetch the status, and finally to get the details of the submitted invoice. Each interaction requires common parameters, such as Chorus Pro connection and authentication details. You reuse these common parameters in the feature setup for all document types. Chorus Pro provides the values when a company is onboarded.

> **Note**
>
> The configuration of common parameters is simplified. You no longer have to go to each action and feature setup, and repeatedly specify the common connection parameters. The

Feature parameters tab is available only as of version 10.0.41 or later.

> **ⓘ Important**
>
> Microsoft provides the **French Chorus Pro submission (FR)** feature. Before you can use it, you must complete additional configuration, as described in this article. For information about how to configure invoicing features and apply changes, see Configure feature setups. For example, in addition to the connection parameters, you can filter specific legal entities so that they're processed in applicability rules. By default, the feature applies to all legal entities that have a primary address in France. Additionally, **out of the box**, the feature is configured to connect to the **sandbox environment of the PISTE service**.

1. Import the latest version of the **French Chorus Pro submission (FR)** globalization feature as described in Import features from the repository.
2. Create a copy of the imported globalization feature, and select your configuration provider. For more information, see Create Globalization features.
3. On the **Versions** tab, verify that the **Draft** version is selected.
4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with Chorus Pro's API:
    - In the **Client ID secret name in the KeyVault** field, select the secret name that you created for the client ID in the key vault.
    - In the **Client Secret secret name in the KeyVault** field, select the secret name you created for the client secret in the key vault.
    - In the **Login service URL** field, specify the service URL to connect to.
    - In the **Technical account login secret name in the KeyVault** field, select the secret name that you created for technical account sign-in in the key vault.
    - In the **Technical account password secret name in the KeyVault** field, select the secret name that you created for the technical account password in the key vault.
    - Select **URL address to send request**, and confirm the web address.
5. On the **Setups** tab, in the grid, select the **UBL Sales invoice derived** feature setup.
6. Select **Edit**, and then, on the **Processing pipeline** tab, in the **Processing pipeline** section, select the **Integrate with French Chorus Pro** action that has the action name **French Chorus Pro submit**.
7. On the **Processing pipeline** tab, in the **Processing pipeline** section, select the **Integrate with French Chorus Pro** action that has the action name **French Chorus Pro request**

status.

8. Review the **Parameters** sections of the pipeline's steps, and make any adjustments that are required.
9. Select **Save**, and then close the page.
10. Repeat steps 6 through 10 for the **UBL Project invoice derived**, **UBL Sales Credit Note derived**, and **UBL Project Credit Note derived** feature setups.

Finance configuration

Prerequisites

- The primary address of the legal entity must be in France.
- The following features must be enabled in **Feature management**:
  - Electronic invoicing integration
  - E-Invoicing service workspace designer
  - (France) Electronic invoicing integration with Chorus Pro

> ⓘ **Note**
>
> To track the status of submitted documents in Chorus Pro, enable two additional features in the **Feature management** workspace: **Extended document identification in submission log** and **Execute update actions for submitted documents**.

Configure legal entity data

Enter a legal entity's address

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity, and then, on the **Addresses** FastTab, add a valid French primary address for the legal entity. Make sure that the following mandatory address elements are defined:
   - Country/region code
   - ZIP/postal code
   - City
   - Building number

Enter a legal entity and a customer Siret number

Use the information in NAF codes and siret numbers and FR-00003 NAF codes and Siret numbers to set up NAF codes and Siret numbers. Alternatively, you can use registration IDs to set up Siret numbers.

Set up the project manager account

When you submit work invoices, use the **Project manager** party. Before you use it, set up the new party type as a customer account. You can then use it when you post invoices.

Follow these steps to enable the project manager for a project invoice.

1. Go to **Project management and accounting** > **Projects** > **Project contracts**.
2. On the **Funding sources** tab, in the **Project manager** field, select a customer account. Then select **Project contracts** > **Project contracts** > **Project contracts**.

> ⓘ **Note**
>
> The **Project manager** field is available only when the funding type is **Customer**.

Set up Chorus Pro electronic document parameters

You can enable SIRET numbers and service code validation before you post and invoice documents.

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Chorus Pro** tab, add the following information:

    - Web service URL
    - Login service URL
    - Client ID
    - Client secret
    - Technical user name
    - Technical user password

> ⓘ **Note**
>
> Use Azure Key Vault storage to store the secrets. For more information, see Set up the Azure Key Vault client.

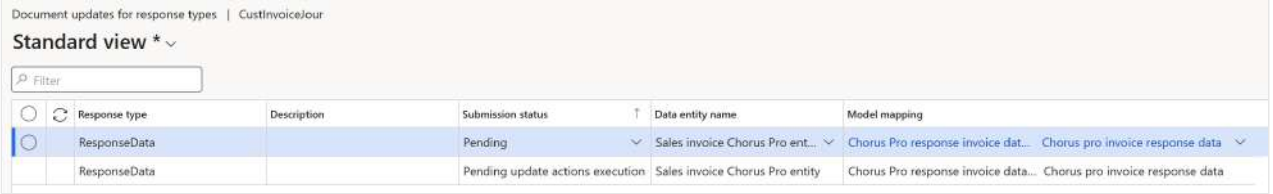
3. Enable the **SIRET and service code validation during invoice posting** parameter.
4. In the **Report format** field, select the configurable business document to use when a submission report is printed.

> **Note**
>
> To create a new submission report instead of customizing a default report, use the **DocumentSubmitted** model name and the **SubmittedInvoice** mapping name for the integration point.

To obtain the latest processing details for the invoice from Chorus Pro in the submission log, you must configure the response types in Finance.

Follow these steps to complete the configuration.

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic document** tab, add records for the **Customer Invoice journal** and **Project invoice** tables.
3. For each table name, set the **Document context** and **Electronic document model mapping** fields as specified in Set up Electronic document parameters.
4. In the **Customer Invoice journal** table name field, select **Response types**.
5. Create a new response type that has the same name that was defined for the related variable in the corresponding feature setups.
   a. In the **Submission status** field, select **Pending**.
   b. In the **Data entity name** field, select **Sales invoice Chorus Pro entity**.
   c. In the **Configuration** field, select **Chorus Pro response invoice data import format (FR)**.
6. Create another new response type that has the same name that was defined for the related variable in the corresponding feature setups.
   a. In the **Submission status** field, select **Pending update actions execution**.
   b. In the **Data entity name** field, select **Sales invoice Chorus Pro entity**.
   c. In the **Configuration** field, select **Chorus Pro response invoice data import format (FR)**.



Define electronic invoice frameworks

1. Go to **Accounts receivable** > **Setup** > **Electronic invoice frameworks**.
2. In the **Type** field, enter the electronic invoicing framework.

   If you don't configure an electronic invoice framework, the system sends all invoices as **A1 - simple invoice**. Use the following codes for the invoice framework to attribute work invoices: A4, A7, A8, A9, A10, A12, A13, A14, and A22. For more information, see Invoicing framework and transmission modes.
3. If you enable **SIRET and service code validation during invoice posting**, set up the following options to have more granular control over the validations that run in the invoice posting process:

   a. Go to **Accounts receivable** > **Setup** > **Electronic invoice frameworks**.

   b. Add the required setup lines to allow users to enable or disable validation per document and party type.

   > **Note**
   >
   > Where relevant (Project invoice proposal, Free text invoice, Sales order), the system runs **only enabled validations** after selecting the Invoicing framework and during posting when **SIRET and Service code validation during invoice posting** is enabled. When validation for the Service code is enabled, it validates **only** when the MOE/MOA SIRET structure requires it, not indiscriminately.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices. For more information about how to generate electronic invoices, see Issue electronic invoices in Finance and Supply chain management.

In France, you can add extra information before you run the standard invoice posting procedure. To add extra information for invoices, follow these steps in the applicable business process.

For sales orders or free text invoices

1. Go to **Accounts receivable** > **Orders** > **All sales orders** or **Accounts receivable** > **Invoices** > **All free text invoices**.

2. Switch to the **Header** view, and select the following information:

    - Electronic invoice framework type
    - Invoice account service code

      > ⓘ **Note**
      >
      > If you enabled the **SIRET and service code validation during invoice posting** parameter in the previous section, you get the list of available service codes from Chorus Pro.

    - Project manager

For project invoices

1. Go to **Project management and accounting** > **Project invoices** > **Project invoice proposals**.
2. Open an existing invoice proposal, or create a new one.
3. Switch to the **Header** view, and select the following information:

    - Electronic invoice framework type
    - Funding source service code
    - Project manager
    - Project manager service code

To view the submission results, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**, and then select **Electronic document** > **Generate report** to generate a submission report.

Receive electronic invoices

To import documents from Chorus Pro, go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**. After the import process finishes, you can find the received documents in the **Pending vendor invoice** workspace. You can add more attributes to these documents as needed. Some of the attributes you can add include:

- Electronic invoice framework type
- Invoice account service code
- Project manager
- Project manager service code

For more information, see Use the electronic invoicing service to import vendor invoices.

Additional resources

Electronic Invoicing service overview

Electronic invoicing configuration

Globalization feature components

Last updated on 03/14/2026

Get started with Electronic invoicing for Indonesia

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information that will help you get started with Electronic invoicing for Indonesia. It describes the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance. It guides through the steps that you must follow in Finance to export sales invoices through the service, and to review the processing results and the status of invoices.

Prerequisites

- Become familiar with electronic invoicing as it's described in Electronic Invoicing service overview.
- Do the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.

Configure Electronic invoicing features

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. Import the latest version of the **Indonesian electronic invoice (ID)** Globalization feature as described in Import features from the repository.

   > **Note**
   >
   > If the feature doesn't appear in the list, select **Synchronize**, and then repeat step 2.

Create a new version of the Indonesian electronic invoice (ID) feature

When you import the **Indonesian electronic invoice (ID)** feature from the repository, all the feature settings are imported. These settings include the configurations, actions of the processing pipeline, and applicability rules. If a new feature version is required, you can create it by using your configuration provider.

1. Create a copy of the imported Globalization feature, and select your configuration provider for it. For more information, see Create Globalization features.
2. On the **Versions** tab, verify that the **Draft** version is selected.

Update the configurations versions

Review the format configurations that are required to generate and export sales invoices.

1. In the **Globalization features** workspace, select the **Electronic Invoicing** tile.
2. On the **Electronic invoicing features** page, on the **Configurations** tab, select **Add** or **Delete** to manage the configuration versions.

When you create a new version, all configurations are inherited from the imported version of the Electronic invoicing feature. The following configurations are required to process invoices:

- Invoices issued (ID)
- eInvoice import (ID)

In the list, select a configuration version, and then select **Edit** or **View** to open the **Format designer** page, where you can edit or view the configuration.

You can review the configuration and customize it as you require. Use the **Format designer** page to edit and view the Electronic reporting (ER) format file configurations. For more information, see Create Electronic reporting (ER) configurations.

Configure the application-specific parameters

Before you begin, make sure that the **Invoices issued (ID)** ER format configuration is imported.

Follow these steps to begin the configuration.

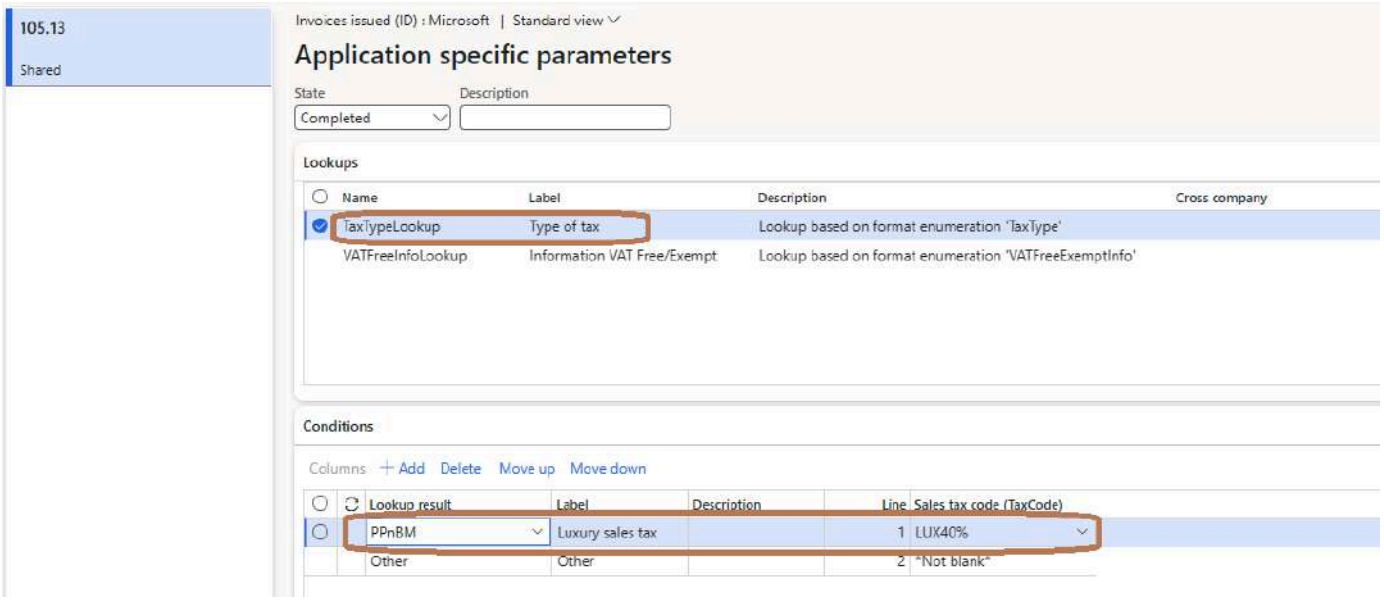
1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile, and then select the **Reporting configurations** tile.
2. On the **Configurations** page, select the **Invoices issued (ID)** format configuration.
3. On the **Configurations** menu, in the **Application specific parameters** section, selected **Setup**.

To enable the system to determine which sales tax code in Finance corresponds to the tax code for luxury goods (PPnBM) when invoices are exported, follow these steps to set the application-specific parameters for luxury sales tax.

1. In the **Lookups** grid, select the row for **TaxTypeLookup**.
2. In the **Conditions** grid, on the first line, set the **Lookup result** field to **PPnBM** and the **Sales tax code (TaxCode)** field to the sales tax code that's used for luxury tax in Finance.

   > **Note**
   >
   > You can set up several sales tax codes for PPnBM and have several lines for them in the **Conditions** grid.

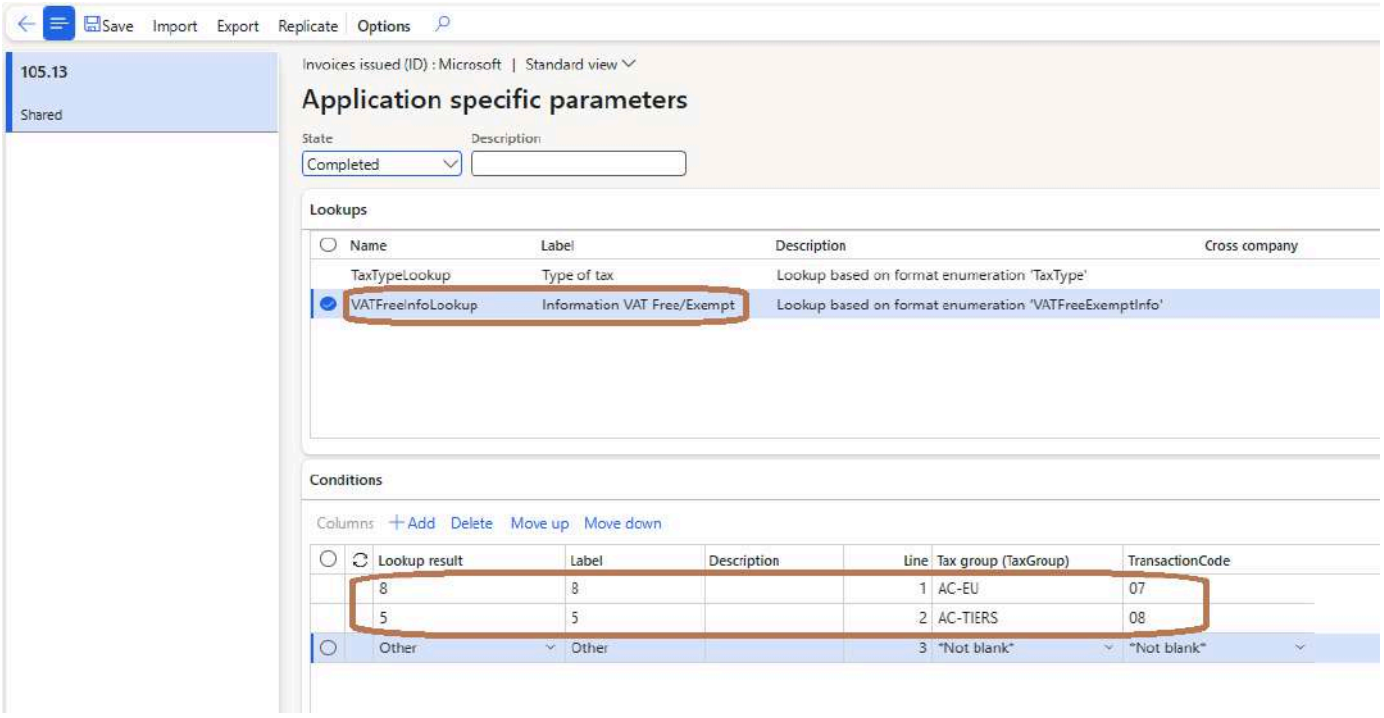
3. On the last line, set the **Lookup result** field to **Other** and the **Sales tax code (TaxCode)** field to ***Not blank***. These settings specify that all other sales tax codes should not be considered as luxury tax by the system..



To enable the system to determine which sales tax group in Finance corresponds to VAT-free reasons (transaction code 07, reasons 1 through 8) or VAT-exempt reasons (transaction code 08, reasons 1 through 5) when invoices are exported, follow these steps to set the application-specific parameters for those reasons.

1. In the **Lookups** grid, select the row for **VATFreeInfoLookup**.
2. In the **Conditions** grid, on the first line, set the **Lookup result** field from **1** through **8** for transaction code **07** and from **1** through **5** for transaction code **08**.

3. Set the **Tax group (TaxGroup)** field to the sales group that's used for exemption operations in Finance.
4. On the last line, set the **Lookup result** field to **Other** and the **Tax group (TaxGroup)** field to ***Not blank***. These settings specify that all other sales tax groups should not be considered exemption groups by the system.



After you define specific conditions for each application-specific parameter, save your changes.

Manage the Electronic invoicing feature setup

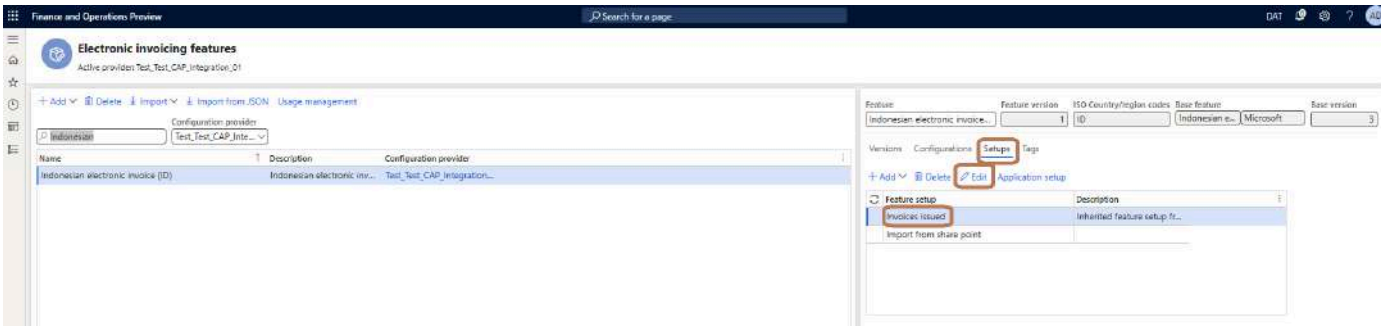
Review or configure the actions in the processing pipeline that support the sales invoice export and import scenarios.

1. In the **Globalization Studio** workspace, select the **Electronic Invoicing** tile.
2. On the **Electronic invoicing features** page, on the **Setups** tab, select **Add**, **Delete**, or **Edit** to manage the Electronic invoicing feature setup.

Configure the Sales invoice feature setup

To generate the comma-separated values (CSV) file for a sales invoice, you must set up the Sales invoice feature.

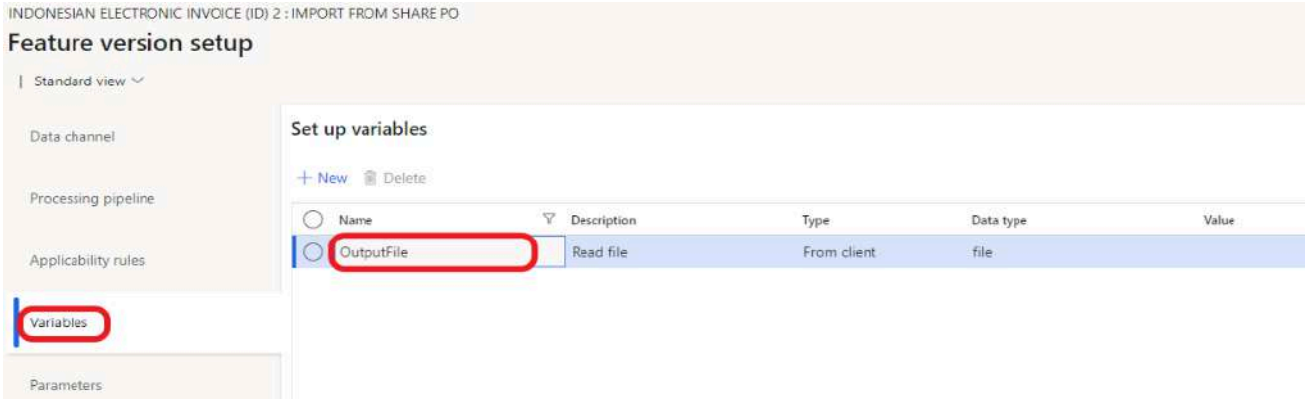
1. On the **Electronic invoicing features** page, on the **Setups** tab, in the **Feature setup** column, select the **Invoice issued** record.
2. Select **Edit** to review or configure the actions, applicability rules, and variables.



Configure the Vendor invoice feature setup

Before you configure the Vendor invoice feature setup, you should create a draft version of the feature that you'll work with for this procedure.

1. On the **Electronic invoicing features** page, on the **Setups** tab, in the **Feature setup** column, select the **Import from share point** record.
2. Select **Edit** to review or configure the actions, applicability rules, and variables.
3. On the **Feature version setup** page, on the **Data channel** tab, in the **Parameters** list, in the **Data channel** record, in the **Value** field, enter **$Context Channel** from the derived configuration.
4. Set the other parameters for SharePoint.
5. In the **Custom file name** record, set up a filter for the file names of vendor invoices.
6. On the **Applicability rules** tab, in the record that has a **Channel** field, in the **Value** field, enter **$Context Channel** from the derived configuration.
7. On the **Variables** tab, create or validate the record. From client variable **OutputFile** is included by default, which represents the **import source** to be configured in Finance.



Change the version status and deploy the feature

Complete and deploy the Electronic invoicing feature for export sales invoices and import vendor invoices.

1. On the **Electronic invoicing features** page, on the **Versions** tab, select the version of the Electronic invoicing feature that has a status of **Draft**.
2. Select **Change status** > **Complete**.
3. Select **Deploy**, and then, in the **Effective from** field, select the date when the feature should become effective.
4. Select **OK**.

For more information, see Complete and deploy a Globalization feature.

Set up Electronic invoicing integration in Finance

Import the ER data model, ER data model mapping, and context configurations for invoices

1. Sign in to your Finance environment.
2. In the **Electronic reporting** workspace, in the **Configuration providers** section, select the **Microsoft** tile. Make sure that this configuration provider is marked as **Active**. For information about how to mark a provider as active, see Create configuration providers and mark them as active.
3. Select **Repositories**.
4. Select **Global resource** > **Open**.
5. Import the following configurations:
    - Customer invoice context model
    - Invoices Communication Model
    - Vendor invoice Mapping to destination
    - Vendor invoice import (ID), Vendor invoice import XML (ID)

Enable Finance features for processing Indonesian electronic invoices

1. In the **Feature management** workspace, disable the **Electronic Invoicing integration using Microsoft Dataverse** feature if it's enabled. Then enable the **Electronic document batch submission** feature.

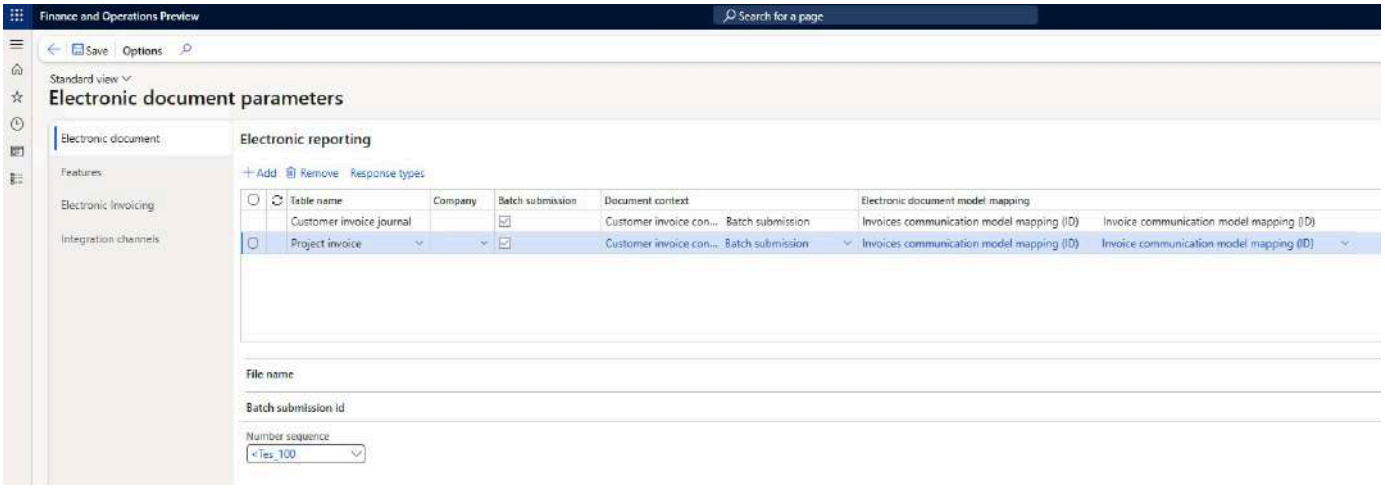
> **Important**
>
> The **Electronic document batch submission** feature can't be enabled if the **Electronic Invoicing integration using Microsoft Dataverse** feature is already enabled.

2. If you want to be able to configure Electronic document parameters on a legal entity basis, enable the **Electronic documents configuration per legal entities in Electronic Invoicing** feature.

3. To enable electronic invoice integration for Indonesia, go to **Organization administration** > **Setup** > **Electronic document parameters**. On the **Features** tab, in the row for the **Indonesian electronic invoice** feature, select the **Enable** checkbox.

Set up the processing for Indonesian electronic sales invoices

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic document** tab, select **Add**, and enter the customer invoice journal.
3. Optional: Select **Add** again, and enter the project invoice journal.
4. Select the **Batch submission** checkbox.
5. In the **Batch submission ID** section, add a number sequence. The selected number sequence should be continuous. This number sequence is used to number the invoice batches when the system exports them.
6. Select **Save**.



Set up the processing for Indonesian electronic vendor invoices

1. In the **Electronic reporting** workspace, select **Reporting configurations**.

2. Select **Customer invoice context model**, and then select **Create configuration**.

3. Select **Derive from Name: Customer invoice context model, Microsoft** to create a derived configuration.
4. In the **Draft** version, select **Designer**.
5. In the **Data model** tree, select **Map model to datasource**.
6. In the **Definitions** tree, select **DataChannel**, and then select **Designer**.
7. In the **Data sources** tree, expand the **$Context_Channel** container.
8. In the **Value** field, select **Edit**, and enter the name of the data channel. The name should have a maximum of 10 characters. It's the name that's given in the configuration of the data channel for the Electronic invoicing feature.
9. Select **Save**, and close the page.
10. Close the page.
11. Select the derived configuration that you just created from **Customer invoice context model**, and then, on the **Versions** FastTab, select **Change Status** > **Completed**.

    > ⓘ **Note**
    >
    > You can create several derived configurations that have different **$Context Channel** values. In this way, you can import vendor invoices from different sources. For example, you might want to import vendor invoices for different legal entities.

12. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
13. On the **External channels** tab, set up the import of vendor invoices.
14. On the **Channels** FastTab, select **Add**.
15. In the **Channel** field, enter **$Context Channel**.
16. In the **Description** and **Company** fields, enter a value.
17. In the **Document context** field, select the new configuration that you derived from **Customer invoice context model**. The mapping description should be **Data channel context**.
18. On the **Import sources** FastTab, select **Add**.

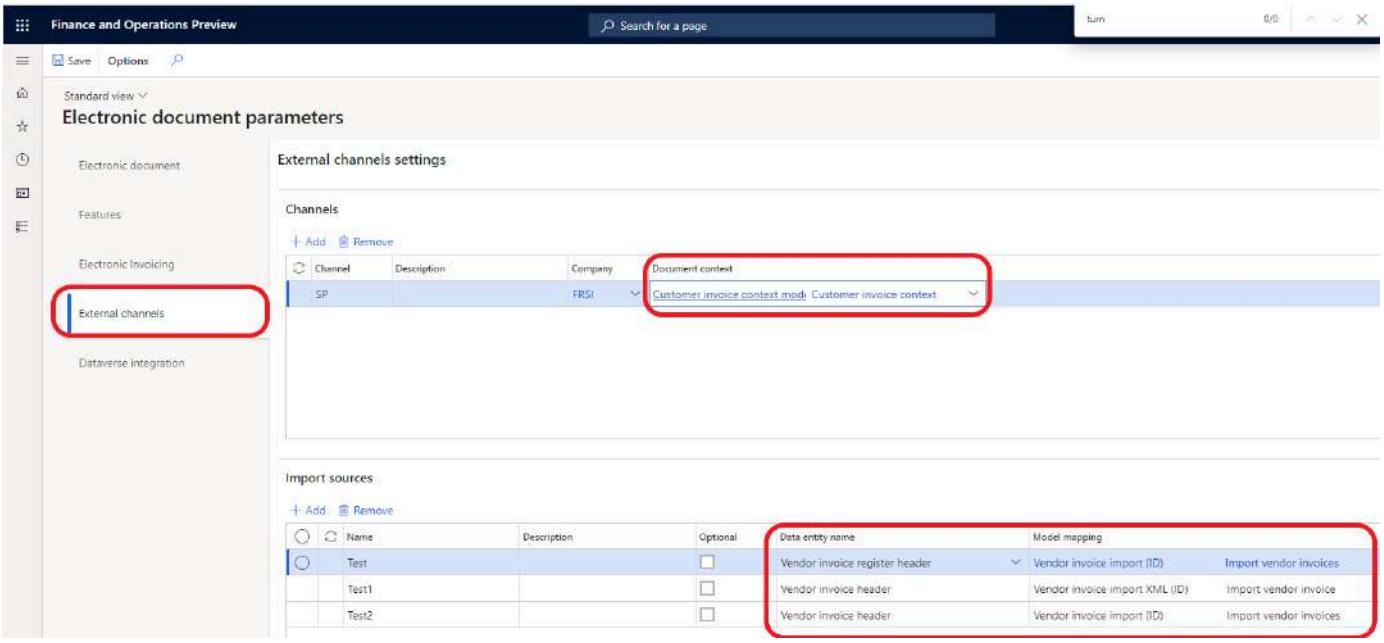
19. In the **Name** field, enter the value from the **Variables** tab on the **Feature version setup** page.

20. In the **Description** field, enter a value.

21. In **Data entity name** field, select **Vendor Invoice register header** if you want to import vendor invoices into the Invoice register. If you want to import them into pending vendor invoices, select **Vendor invoice journal**. You can have only one import source, **Vendor Invoice register header** or **Vendor invoice journal**.

22. In the **Model mapping** field, select **Vendor invoice import (ID)** to import the invoice header into the Invoice register or pending vendor invoices. Select **Vendor invoice import XML (ID)** to import the invoice header and lines into pending vendor invoices.

    > **Note**
    >
    > Before you import vendor invoices from XML files, you must set up an external item description for vendors. The system can then match an item name in the XML file with line items in sales orders.



If you must import vendor invoices into a different legal entity, for example, create a new channel record that has the new document context.

Process electronic invoices in Finance

When issued invoices or imported vendor invoices are processed through Electronic invoicing, you can complete the following tasks:

- Submit or export issued invoices, and import vendor invoices.
- View the electronic document submission and receipt logs.

Submit or export issued invoices

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**.
2. To export invoices in a batch, set the **Submit document batch** option to **Yes**. The first time that you submit a document, always set the **Resubmit documents** option to **No**. If you must resubmit a document through the service, set this option to **Yes**.
3. On the **Records to include** FastTab, select **Filter** to open the **Inquiry** dialog box, where you can build a query to select documents for export.

View submission logs

You can view the submission logs for all exported documents.

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**.

2. Select **Update status**. An invoice batch can have the following statuses:

    - Scheduled
    - Completed
    - Failed

3. On the Action Pane, select **Inquiries** > **Submission details** to view the details of the submission execution logs. The information in the logs is divided among three FastTabs:

    - **Processing actions** – This FastTab shows the execution log for the actions that are configured in the feature version. The **Status** column shows whether the action was successfully run.
    - **Action files** – This FastTab shows the intermediate files that were generated during execution of the actions. Select **View** to download and view the file.
    - **Processing action log** – This FastTab shows the results of the submission of electronic invoices.

4. On the Action Pane, select **Inquiries** > **Batch submission invoices** to view the invoices that were submitted in one batch.

Import vendor invoices and view the electronic document receipt log

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.

   > **Note**
   >
   > For the first receipt of any document, always set the **Re-import documents** option to **No**. If you must re-import a document through the service, set this option to **Yes**. This job can be run in batch mode.

2. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**.

3. On the Action Pane, select **Inquiries** > **Submission details** to view the details of the submission execution logs.

Privacy notice

Enabling the **Indonesian electronic invoice** feature might require that limited data be sent. This data includes the organization's tax registration ID. An administrator can enable or turn off the **Indonesian electronic invoice** feature by going to **Organization administration** > **Setup** > **Electronic document parameters**, and then, on the **Features** tab, selecting the rows that contain the **Indonesian electronic invoice** feature and making the appropriate selection. Data that's imported from these external systems into this Dynamics 365 online service are subject to our privacy statement. For more information, see the "Privacy notice" section in country/region-specific feature documentation.

Additional resources

- Electronic Invoicing service overview
- Get started with Electronic invoicing
- Set up Electronic invoicing

Last updated on 05/01/2026

Get started with Electronic invoicing for Italy

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Electronic invoicing for Italy might not currently support all the functions that are available for electronic invoices in Microsoft Dynamics 365 Finance and Dynamics 365 Supply Chain Management.

This article provides information to help you get started with Electronic invoicing for Italy in Finance and Supply Chain Management. It guides you through the configuration steps that you must complete to enable the generation of electronic invoices in the Italy-specific format, **FatturaPA**, and to exchange electronic invoices with the Italian Revenue Agency's e-invoicing exchange system, SDI (Sistema di Interscambio).

> **Note**
>
> The solution for direct integration with SDI supports digital signing of documents only through separately stored certificate files that can be uploaded to the key vault. The solution doesn't support the use of qualified digital signatures, which require additional hardware involvement or an online connection to certification service providers.

Prerequisites

Before you complete the steps in this article, the following prerequisites must be met:

- Obtain SDICoop service channel accreditation in the Exchange system (SDI) government portal.
- Complete the steps in Electronic invoicing configuration.

- Import the **Italian FatturaPA (IT)** Electronic invoicing feature (version 5 or later) from the repository. Learn more in Import features from the repository.
- Configure Azure resources to store required certificates, including the Certificate authority (CA) certificate and Clients certificate. Learn more in Create a Key Vault reference.
- Optional: You can use the Digital signature certificate only for non-qualified signatures.

Country/region-specific configuration for the Italian FatturaPA (IT) Electronic invoicing feature

Complete the following procedures before you deploy the feature to Finance.

This section complements the common part of Electronic Invoicing service configuration that is described in Electronic invoicing configuration.

Create a new number sequence for ProgressivoInvio

To create a new number sequence for ProgressivoInvio, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic invoicing** tab, on the **Number sequences** FastTab, select **New** to add a record.
3. Enter a name for the new sequence (for example, **ProgressivoInvio**), and then select **Save**.

Create a new chain of certificates

To create a new chain of certificates, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic invoicing** tab, on the **Key Vault settings** FastTab, select **Key Vault parameters**.
3. In the **Certificates** section, select **Add** to add the following certificates:
    - **CAentrate.cer** (and **CAEntratetest.cer** in the case of a test environment)
    - **CAActalisOV.cer** (in the case of a production environment)
    - Client identity certificate

    > ⓘ Note

    - Usually, the client identity certificate is generated while the user registers the SDICoop service channel. It's named **SDI-<*VatNumber*>**, where <*VatNumber*> is a company's 11-digit value-added tax (VAT) number.
    - The client identity certificate has an expiration date. Review this date, and arrange to obtain a new certificate when the current one is about to expire. Learn how to establish proper rotation of the certificates that are used for this integration in the Rotate certificates section of this article.

- Certificate for digital signature (optional)

4. On the Action Pane, select **Chain of certificates**.
5. Select **New**, and enter a name (for example, **SDIChainProd** or **SDIChainTest**).
6. In the **Certificates** section, add the following certificates in the order that they are listed in here:
   a. **CAentrate.cer** (and **CAEntratetest.cer** in the case of a test environment)
   b. **CAActalisOV.cer** (in the case of a production environment)
   c. Client identity certificate
7. Save your changes, and close the page.

Create a new feature

To create a new feature, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile.
2. In the **Configuration providers** section, mark your company's configuration provider as active.
3. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
4. On the **Electronic invoicing features** page, select **Add** > **Based on existing feature**.
5. Under the **Microsoft** configuration provider, select **Italian FatturaPA (IT)** as a base feature, enter a name, and then select **Create feature**.

Align Electronic reporting configurations

The following procedures must be completed for both Electronic reporting (ER) format configurations that are used in the feature: **Sales invoice (IT)** and **Project invoice (IT)**.

Set up payment method substitution

To set up payment method substitution, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile, and then select the **Reporting configurations** tile.
2. On the **Configurations** page, select the **Sales invoice (IT)** format configuration.
3. In the **Versions** section, verify that the **Draft** version is selected.
4. On the Action Pane, select **Designer**.
5. On the **Mapping** tab, select **$PaymentMethodSubstitution**, and then select **Edit** > **Edit formula**.
6. Align the formula with the payment methods that are configured in Finance.
7. Save your changes, close the formula designer, and then select **OK**.
8. Save the whole **Sales invoice (IT)** format configuration, and then close the **Designer** page.
9. In the **Versions** section, select **Change status** > **Complete** to complete the **Sales invoice (IT)** format configuration.
10. Repeat steps 2 through 9 for the **Project invoice (IT)** format configuration, as required.

Set up application-specific parameters

To set up application-specific parameters, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile, and then select the **Reporting configurations** tile.

2. On the **Configurations** page, select the **Sales invoice (IT)** format configuration.

3. On the menu, select **Configurations** > **Application specific parameters** > **Setup**.

4. In the **Lookups** section, make sure that the **List of Natura reverse charge subcategories** lookup is selected.

5. In the **Conditions** section, select **Add**.

6. Add specific conditions for each subcategory that is defined in the system.

   > ⓘ **Note**
   >
   > In the **Name** column, you can select the ***Blank*** or ***Not blank*** placeholder value instead of a specific value.

7. Change the **State** value to **Completed**, save your changes, and then close the page.

8. Repeat steps 2 through 7 for the **Project invoice (IT)** format configuration, as required.

Configure a processing pipeline for export

To configure a processing pipeline for export, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. On the **Electronic invoicing features** page, select the feature that you created based on the **Italian FatturaPA (IT)** feature.
3. On the **Setups** tab, select **Sales invoice**, and then select **Edit**.
4. In the **Processing pipeline** section, go through the actions, and set the required fields:
    - **Get next number sequence value** – In the **Parameters** section, select the **Number sequence** parameter. Then, in the **Value** field, select the number sequence that you previously defined for ProgressivoInvio.
    - **Submit** – In the **Parameters** section, in the **URL address** field, enter or select a value. In the **Certificates** field, enter or select a chain of certificates that you added earlier in this article (**SDIChainProd** or **SDIChainTest**).
    - **Sign document** – In the **Certificate name** field, specify the Digital signature certificate (optional).
5. In the **Applicability rules** section, go through the clauses, and review or set the required fields. As part of this step, review the **LegalEntityID** clause, and update it with the correct value from your legal entity.
6. Select **Validate** to ensure that all required fields are set.
7. Save your changes, and close the page.
8. On the **Setups** tab, select **Project invoices**, and then select **Edit**.
9. Repeat steps 3 through 7 for **Project invoice**, **Sales invoice PA**, and **Project invoice PA** setups, if required.

Configure a processing pipeline for import

To configure a processing pipeline for import, follow these steps:

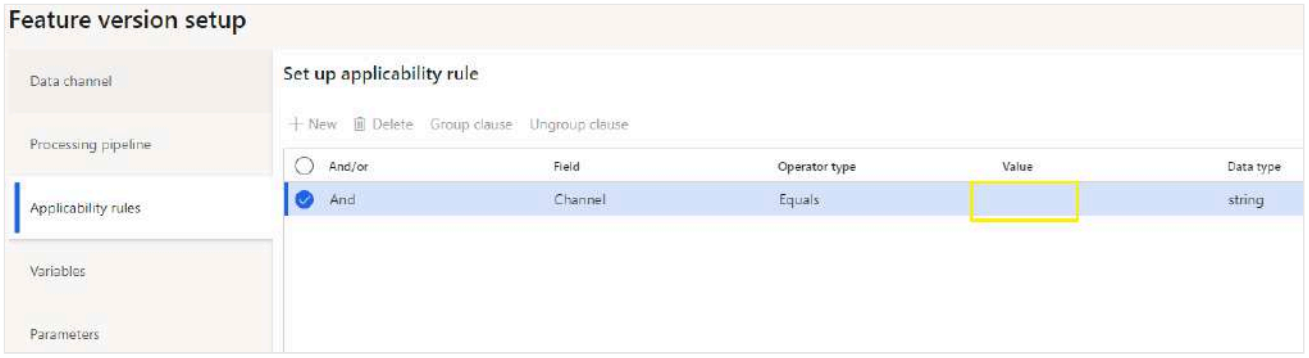
1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.

2. On the **Electronic invoicing features** page, select the feature that you created based on the **Italian FatturaPA (IT)** feature.

3. On the **Setups** tab, select **Import invoices**, and then select **Edit**.

4. On the **Import channel** tab, in the **Data channel** section, on the **Parameters** tab, in the **Data channel** field, enter a string value.

5. On the **Applicability rules** tab, set the fields for the setup. To use the default **Channel** clause, pass the value that you set for the **Data channel** field in the previous step to the **Value** field.



6. Select **Validate** to ensure that all required fields are set.

7. Save your changes, and close the page.

Deploy the feature

Complete and deploy the feature as described in Complete and deploy a Globalization feature.

Configure Electronic document parameters

To configure Electronic document parameters, follow these steps:

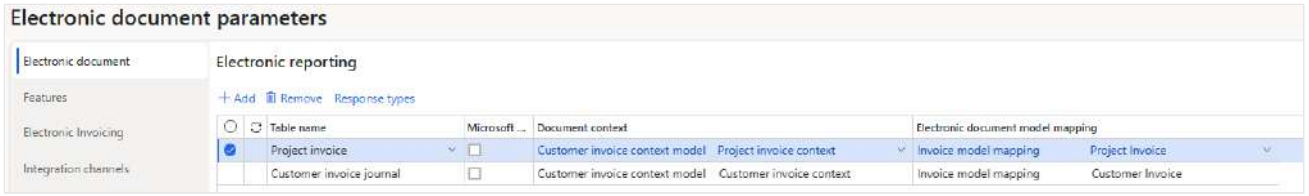
1. Import the **Customer invoice context model** (version 54 or later) and **Vendor invoice import (IT)** ER configurations from the repository. Learn more in Import Electronic reporting (ER) configurations from Dataverse.

2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

3. On the **Features** tab, find and select the **Italian electronic invoice** feature, and then select **Enable**.

4. On the **Electronic document** tab, make sure that the fields for **Customer invoice journal** and **Project invoice** are set according to the information in Set up Electronic document

parameters.



Set up vendor invoice import

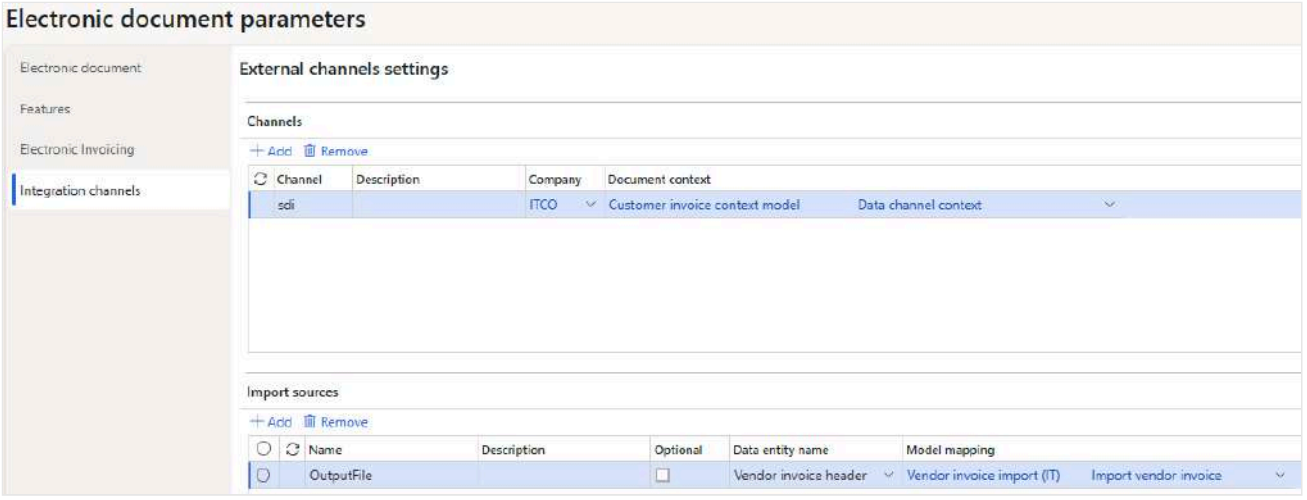
To set up vendor invoice import, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile, and then select the **Reporting configurations** tile.
2. Select **Customer invoice context model**, and then select **Create configuration**.
3. Select **Derive from Name: Customer invoice context, Microsoft** to create a derived configuration.
4. Select the derived configuration, and then select **Designer** to open it for editing.
5. Select **Map model to datasource**.
6. In the **Definitions** tree, select **DataChannel**, and then select **Designer**.
7. In the **Data sources** tree, expand the **$Context_Channel** container.
8. In the **Value** field, select **Edit**.
9. Enter the name of the data channel. The name should have a maximum of 10 characters. It should match the value of the **Data channel** parameter of the data channel for the Electronic invoicing feature.
10. Save your changes, and close the form.
11. In the **Versions** section, select **Change status** > **Complete** to complete the derived context configuration.
12. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
13. On the **Integration channels** tab, in the **Channels** section, select **Add**.
14. In the **Channel** field, enter the **$Context Channel** value.
15. Enter values in the **Description** and **Company** fields.

16. In the **Document context** field, select the new configuration that you derived from **Customer invoice context model**. The mapping name should be **Data channel context**.

17. On the **Import sources** tab, select **Add**, and set the following values:

    - **Name:** OutputFile
    - **Data entity name:** Vendor invoice header (**Data entity:** VendorInvoiceHeaderEntity)
    - **Model mapping:** Vendor invoice import (IT)



> ⓘ **Note**
>
> If you have import vendor invoices from different sources, you can create several channels and several derived configurations that have different **$Context Channel** values. For example, you might want to import vendor invoices for different legal entities.

Finance business data configuration

Follow the configuration steps in Customer electronic invoices. Start from the Electronic invoice currency section.

Proxy server setup

This section provides information to help you set up and configure the proxy service for communication between the Exchange system (SDI) and the Electronic Invoicing service.

> ⓘ **Note**
>
> This section is provided solely for illustrative purposes. It offers a high-level overview of the steps involved in the process of **Proxy server** configuration. It isn't a comprehensive guide

for design, configuration, implementation, maintenance, or ongoing support of the Proxy server. All responsibilities related to this area, including but not limited to detailed technical execution, troubleshooting, and system upkeep, lie with the customer's team. Users are advised to consult appropriate technical resources or engage qualified professionals for full implementation and support needs.

Create an app registration

To create an app registration, follow these steps:

1. Use the following Windows PowerShell script to create a self-signed certificate for service-to-service (S2S) authentication.

   > **Note**
   >
   > The certificate is set to expire in 12 months. You can extend this period by modifying the value of the **NotAfter** parameter. Learn how to establish proper rotation of the certificates that are used for this integration in the Rotate certificates section of this article.

   PowerShell

   ```
   $certOutputLocation = "C:\certs\proxytest"
   $certName = "sdiProxyClientS2SCert"
   $certPassword = "123"

   $certCerFile = Join-Path $certOutputLocation "$certName.cer"
   $certPfxFile = Join-Path $certOutputLocation "$certName.pfx"

   $securePassword = ConvertTo-SecureString $certPassword -AsPlainText -Force

   $cert = New-SelfSignedCertificate -KeyLength 2048 -KeyExportPolicy Exportable -
   FriendlyName "CN=$certName" -CertStoreLocation Cert:\CurrentUser\My -Subject
   $certName
   -Provider "Microsoft Enhanced RSA and AES Cryptographic Provider" -NotAfter (Get-
   Date).AddMonths(12)

   Export-Certificate -Cert $cert -FilePath $certCerFile -type CERT | Out-Null
   Export-PfxCertificate -Cert $cert -FilePath $certPfxFile -Password
   $securePassword | Out-Null
   ```

2. Save the .pfx certificate file to the key vault. This certificate is referred to later as the *App Registration Certificate*.

3. Sign in to the Azure portal as an administrator.

4. Create an app registration for the SDI Proxy service:

   a. Go to **App registrations**, create a registration, and set the following values for it:

      - **Name:** SDI Proxy Client
      - **Supported account types:** Accounts in this organizational directory only (Single tenant)

   b. Select **Register**, and then select the app registration that you just created.

   c. Go to **API permissions**, and select **Grant admin consent**.

   d. Go to **Certificates & secrets**, select **Upload certificate**, and upload the App Registration Certificate .cer file for S2S authentication.

   e. Go to **Enterprise applications**, and select the app that you created.

   f. Save the **Application ID** (client ID) and **Object ID** values for the app.

   g. The Invoicing Service team must grant the app access to the service. Open a support ticket that has the theme **Electronic invoicing service integration with SDI. Application allow-listing**. Provide the following information:

      - Microsoft Entra tenant ID
      - Microsoft Dynamics Lifecycle Services environment ID
      - Application ID
      - Object ID

      > **Note**
      >
      > The object ID for an application differs on the **App registrations** and **Enterprise applications** pages. Use the **Object ID** value from the **Enterprise applications** page.

Create an Azure virtual machine

To create an Azure virtual machine (VM), follow these steps:

1. In the Azure portal, go to **Virtual machines**, and select **Create new**.

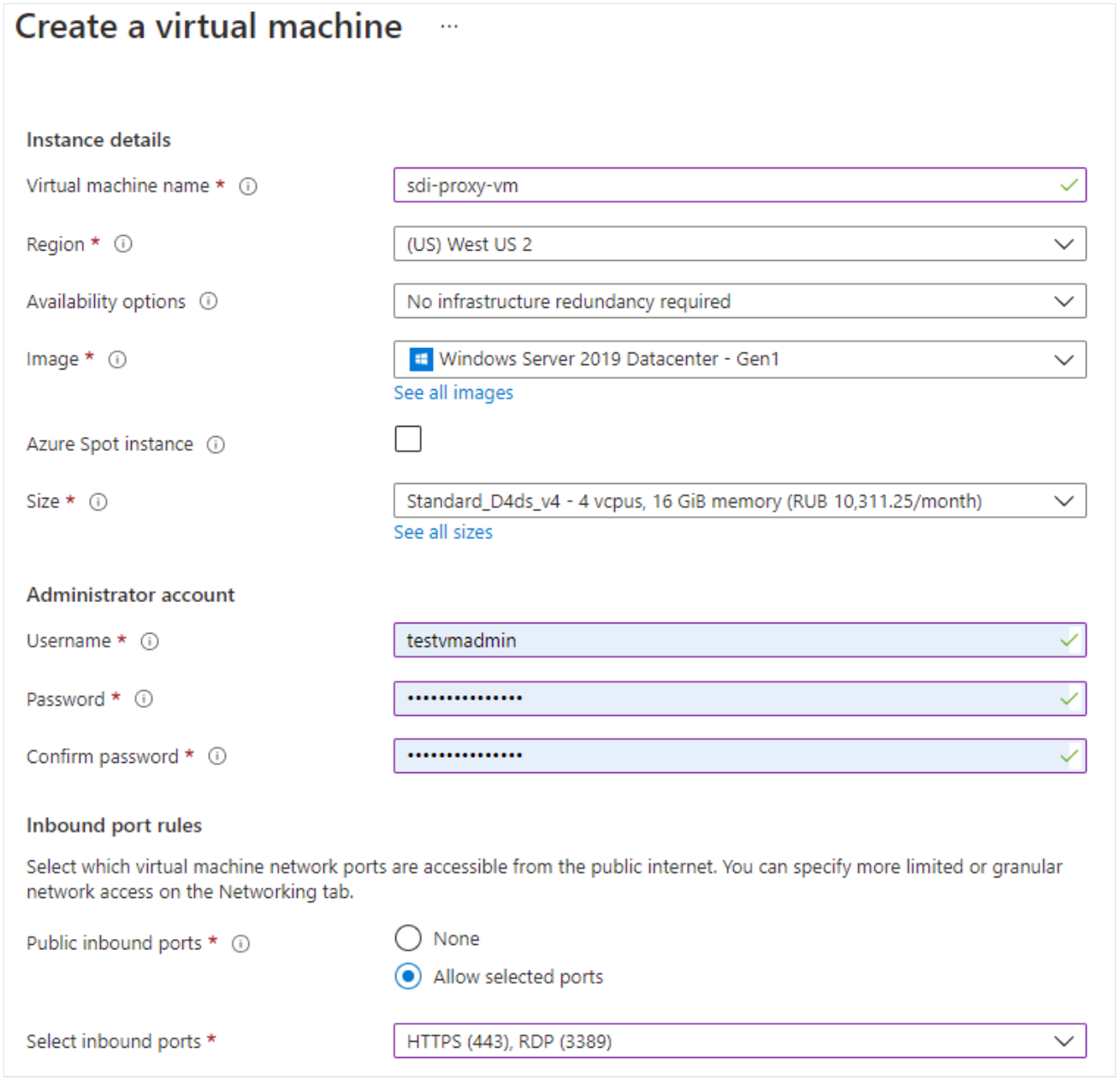
2. On the **Basics** tab, select the subscription and resource group. The values should be the subscription and resource group where your key vault and Blob storage are located.

3. Select the region. The value should be the region where your Finance environment is deployed.

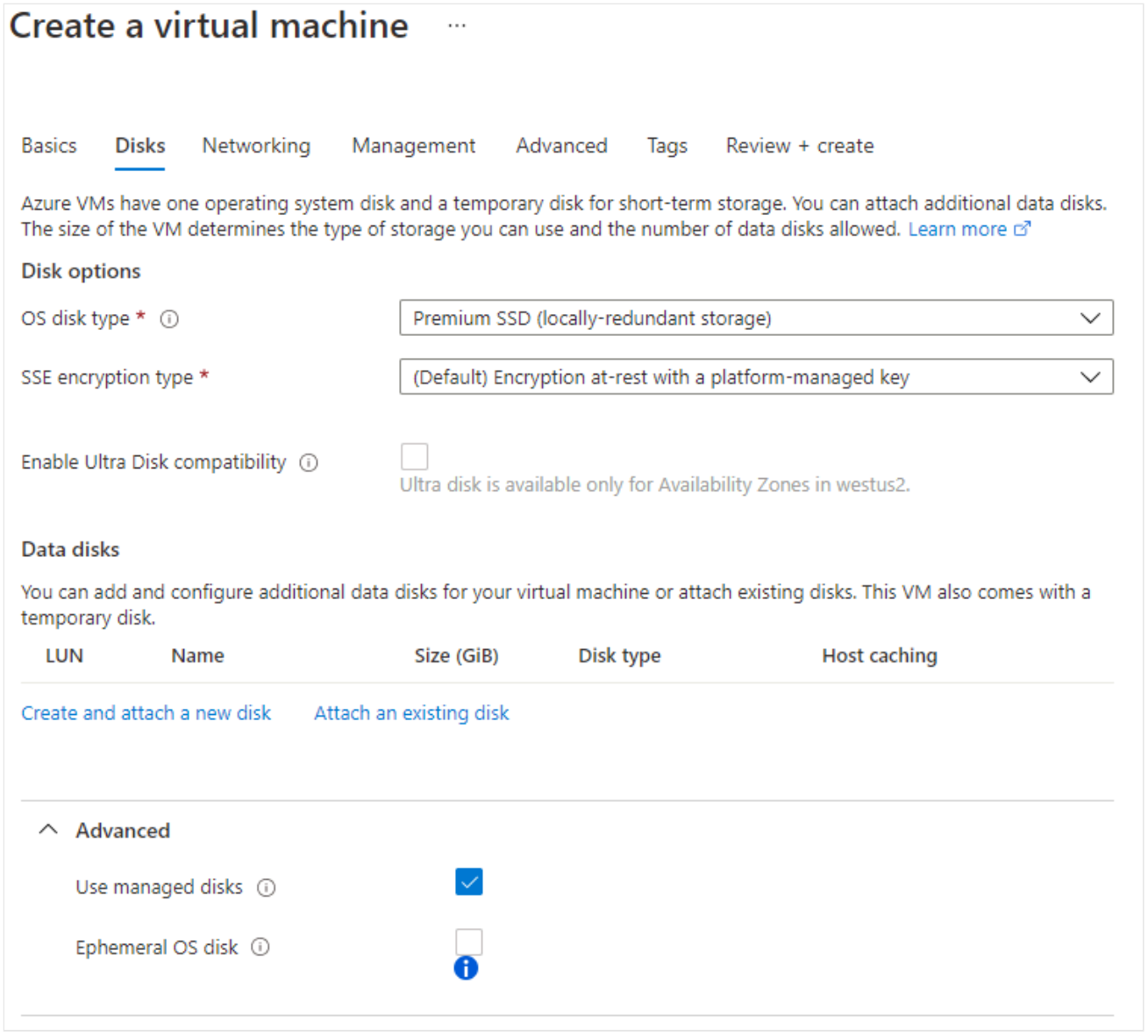
4. Add the administrator's user name and password, and save them to the key vault.

5. In the **Select inbound ports** field, select **HTTPS (443)** and **RPD (3389)**.

> ⓘ **Note**
>
> We recommend that you disable the **RDP (3389)** port when the system goes to production. You can reenable it if you must connect to the VM for troubleshooting purposes.
>
> We also recommend that you use the Windows Server 2019 Datacenter operating system (OS). The proxy application sample might have issues if it runs on another type of OS.



6. On the **Disks** tab, on the **Advanced** FastTab, select the **Use managed disks** checkbox. Leave the **Ephemeral OS disk** checkbox cleared.



7. On the **Networking** tab, under the **Public IP** field, select **Create new**.
8. In the **Create public IP address** dialog box, in the **SKU** field group, select the **Standard** option. In the **Assignment** field group, select the **Static** option.
9. On the **Management** tab, clear the **Auto-shutdown** checkbox to disable automatic shutdown.
10. Set the **Guest OS updates** field to **Manual**, and set any other policies.
11. Review and create the VM.
12. In the new VM, go to **Identity** > **System assigned**, and set the **Status** option to **On**.
13. Grant the VM access to the key vault:

    a. In the key vault, go to **Access control (IAM)** > **Role assignments**.

    b. Select **Add role assignment**, and follow these steps:

i. In the **Role** field, specify **Key Vault Secrets User**.
ii. In the **Assign access to** field, specify **Virtual machine**.
iii. In the **Subscription** field, specify your subscription.
iv. In the **Select** field, specify your VM.

c. Go to **Access policies**.

d. Select **Add Access Policy**, and follow these steps:
i. In the **Selected principal** field, select your VM.
ii. In the **Certificate** section, select **List** and **Get** permissions.
iii. In the **Secret** section, select **List** and **Get** permissions.

14. In the Azure portal, go to **Public IP addresses**, and select the IP address that was created in the VM.

15. Go to **Configuration**, and set the Domain Name System (DNS) name.

Prepare the proxy service environment

To prepare the proxy service environment, follow these steps on the machine where the proxy service is hosted.

1. Connect to the VM by using Remote Desktop Connection.

2. Open the Local Machine Certificate snap-in. Learn more in How to: View certificates with the MMC snap-in.

3. Import the **caentrate.cer** certificate into the Trusted Root Certification Authorities store.

4. Import the **CAEntratetest.cer** certificate into the Trusted Root Certification Authorities store (only for a test environment).

5. Import the **SistemaInterscambioFatturaPA.cer** certificate for production and the **SistemaInterscambioFatturaPATest.cer** certificate for testing into the Trusted People store.

6. In Control Panel, open **Turn Windows features on or off**, or go to **Server Manager** > **Add Roles and Features** for the server OS, and turn on the following Internet Information Services (IIS) features:

   - Web Management Tools
     - IIS Management Console

   - World Wide Web Services

    - Application Development Features
        - .NET Extensibility 4.7 (or 4.8)
        - ASP
        - ASP.NET 4.7 (or 4.8)
        - CGI
        - ISAPI Extensions
        - ISAPI Filters
    - Common HTTP Features
        - Default Document
        - Directory Browsing
        - HTTP Errors
        - Static Content
    - Health and Diagnostics
        - HTTP Logging
        - Tracing
    - Performance Features
        - Static Content Compression
    - Security
        - Request Filtering
- .NET Framework 4.7 Features
    - WCF Services
        - HTTP Activation

Set up the SDI Proxy service in IIS

To set up the SDI Proxy service in IIS, follow these steps:

1. In Lifecycle Services, go to the Shared asset library, and select **Data package** as the asset type.
2. Find **Electronic Invoicing Service Sdi Proxy v1.1**, and download it to the VM.
3. Configure the service:

    a. Unzip the **Electronic Invoicing Service Sdi Proxy** archive folder that you downloaded.

b. In the **src\FattureService** folder, open the **appsettings.json** file, and set the following parameters:

- **KeyVaultUri** – Specify the address of the key vault that stores the client certificate for the invoicing service.
- **TenantId** – Specify the globally unique identifier (GUID) of the customer's tenant.
- **EnvironmentId** – Specify the ID of the Lifecycle Services environment.
- **ClientId** – Specify the app ID of the intermediate services app registration in the customer's tenant.
- **ClientCertificateName** – Specify the name of the App Registration Certificate in the key vault.
- **SecurityServiceClientOptions.Endpoint** – Specify the URL of the security service.
- **SecurityServiceClientOptions.Resource** – Specify the scope to obtain the token for.
- **InvoicingServiceClientOptions.Endpoint** – Specify the endpoint of the invoicing service. (You can obtain this value from the **Endpoint URL** field in the **Service parameters** section on the **Electronic Invoicing** tab of the **Electronic document parameters** page in Finance.), after copying remove **/** at the end.
- **InvoicingServiceClientOptions.ServiceEnvironmentId** – You can leave the default value, **ServiceEnvironment**, because Microsoft plans to deprecate this field in future releases.
- **NotificationsFolder** – Specify the folder to save incoming notification files in (for example, **C:\\Files**).

> **Tip**
>
> You can copy the **InvoicingServiceClientOptions.Endpoint** and **InvoicingServiceClientOptions.ServiceEnvironmentId** values from the **Electronic Invoicing** tab of the **Electronic document parameters** page in finance and operations apps (**Organization Administration** > **Setup** > **Electronic document parameters**).

c. In the **src\Clients** folder, open the **Constants.cs** file, and set the following parameters:

- **ApiVersion** – Update the value to **2.0**.
- **InvoicingServiceResumePath** – Update the value to **/businessdocumentsubmission/resume**.
- **InvoicingServiceCreateIncomingPath** – Update the value to **/incomingdocuments/{0}**.

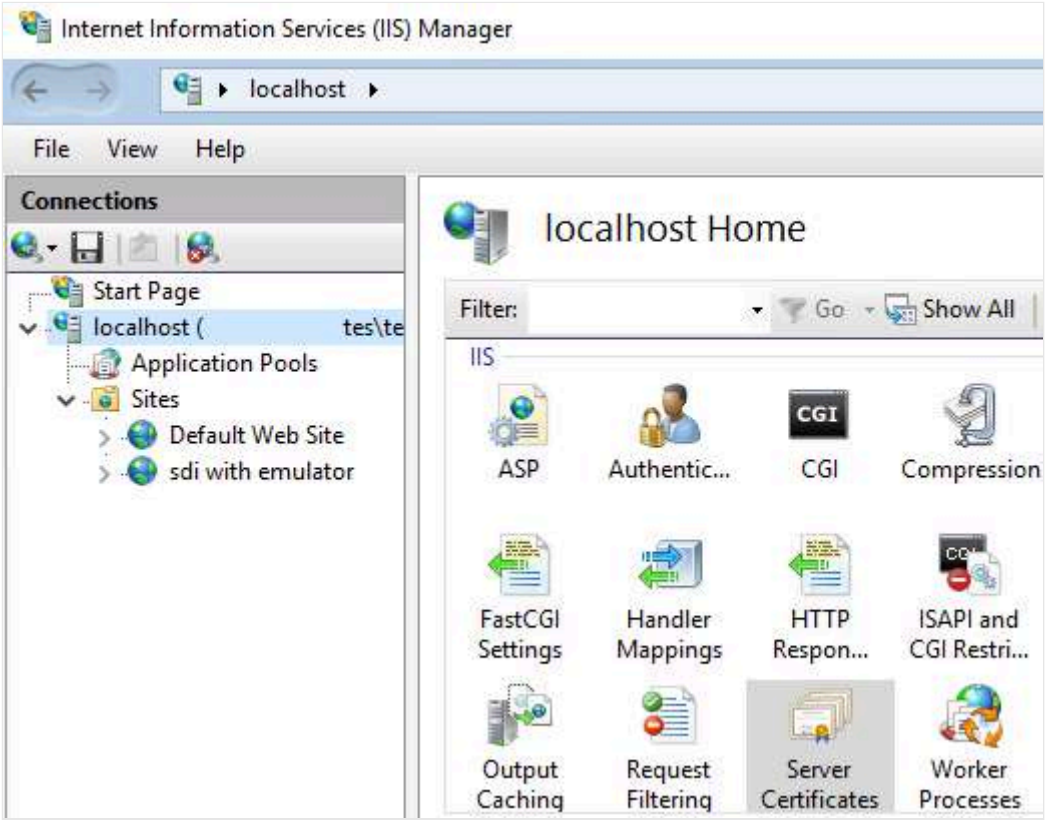
d. In the **web.config** file, find the following line, and add the thumbprint of the proxy server certificate.

```
<serviceCertificate findValue="[certificate thumbprint]"
storeLocation="LocalMachine" storeName="My" x509FindType="FindByThumbprint">
```

> **Tip**
>
> When the system goes to production, you can change some values in the web.config file to help reduce the amount of log information that is collected and help save disk space. In the **<system.diagnostics><source>** node, change the value of **switchValue** to **Critical,Error**. Learn more in Service Trace Viewer Tool (SvcTraceViewer.exe).

e. Build the solution.

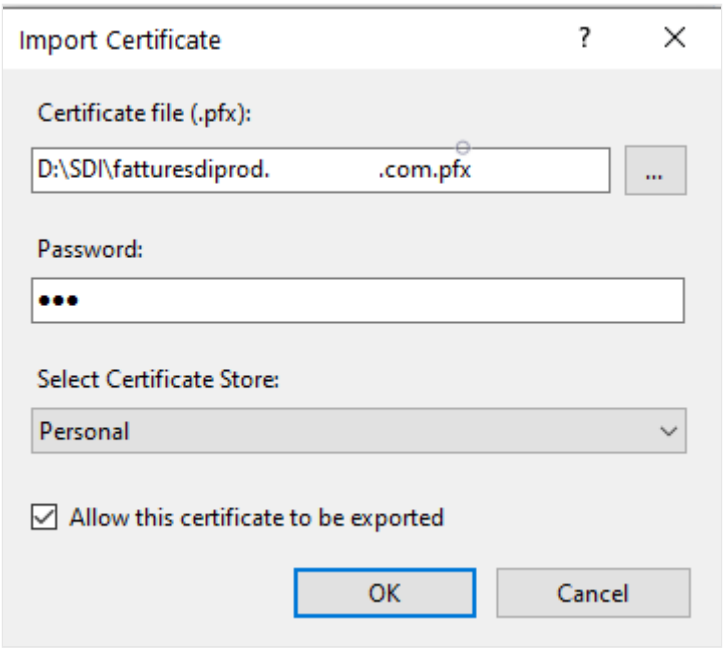
4. Open IIS Manager. In the tree on the left, remain in the root node. On the right, select **Server Certificates**.



5. Open the menu, and select **Import**.

6. In the **Import Certificate** dialog box, in the **Certificate file (.pfx)** field, specify the path of the .pfx file for the proxy server certificate. Usually, this file is generated while you register the

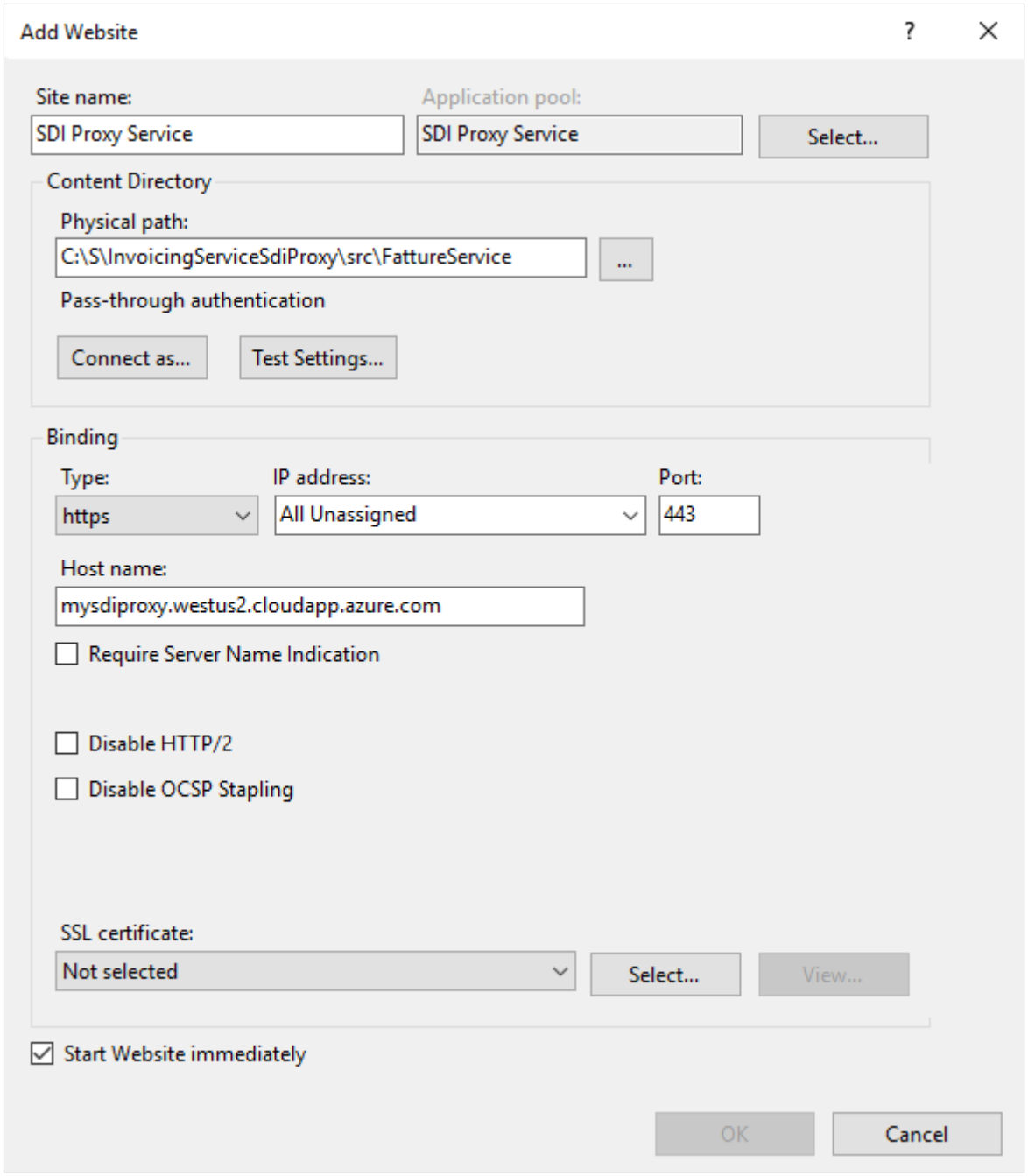
SDICoop service channel. It's named **sdiproxy.com.pfx**, where **sdiproxy.com** is a DNS that is assigned to the proxy VM.



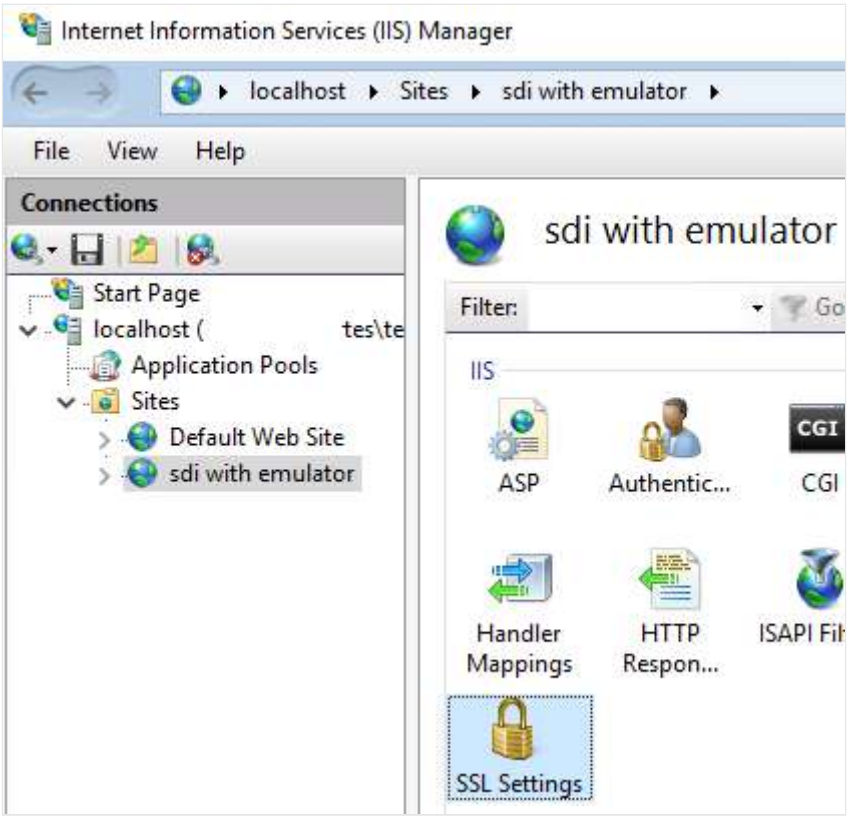
> ⓘ **Note**
>
> The server certificate has an expiration date. Review this date, and arrange to obtain a new certificate when the current one is about to expire. Learn how to establish proper rotation of the certificates that are used for this integration in the Rotate certificates section of this article.

7. Select and hold (or right-click) **Sites**, and then select **Add website**.
8. In the **Add Website** dialog box, in the **Site name** field, enter a name for the site.
9. In the **Physical path** field, point to the **src\FattureService** folder.
10. In the **Binding** section, in the **Type** field, select **https**.
11. In the **Host name** field, specify the host name.
12. Leave the **IP address** and **Port** fields set to the default values.
13. Make sure that the **Require Server Name Indication** checkbox is cleared, because SDI doesn't support that technology.
14. In the **SSL certificate** field, select the proxy server certificate that you imported.

15. In the **Application pool** field, specify a pool for the site, and make a note of its name (for example, **SdiAppPool**).



16. After you finish creating the website, open the menu for **SSL Settings**.



17. Select the **Require SSL** checkbox.

18. In the **Client certificates** field group, select the **Require** option.

19. Open **Directory Browsing**, and select **Enable**.

20. Open the Configuration Editor, and select **system.webServer** > **serverRuntime**.

21. Modify the **uploadReadAheadSize** and **maxRequestEntityAllowed** values to at least 10 megabits.

22. In any web browser, go to **serverDNS/TrasmissioneFatture.svc**. A standard page about the service must appear, or you might receive a server error, such as "403 - Forbidden." This step is required to ensure that **serverDNS** is accessible, and that it isn't hidden by a firewall or something else.

23. Create the following folders to store logs and files:

    - **C:\Logs** – Store log files here. These files can be viewed by using Service Trace Viewer Tool.
    - **C:\Files** – Store all the response files here.

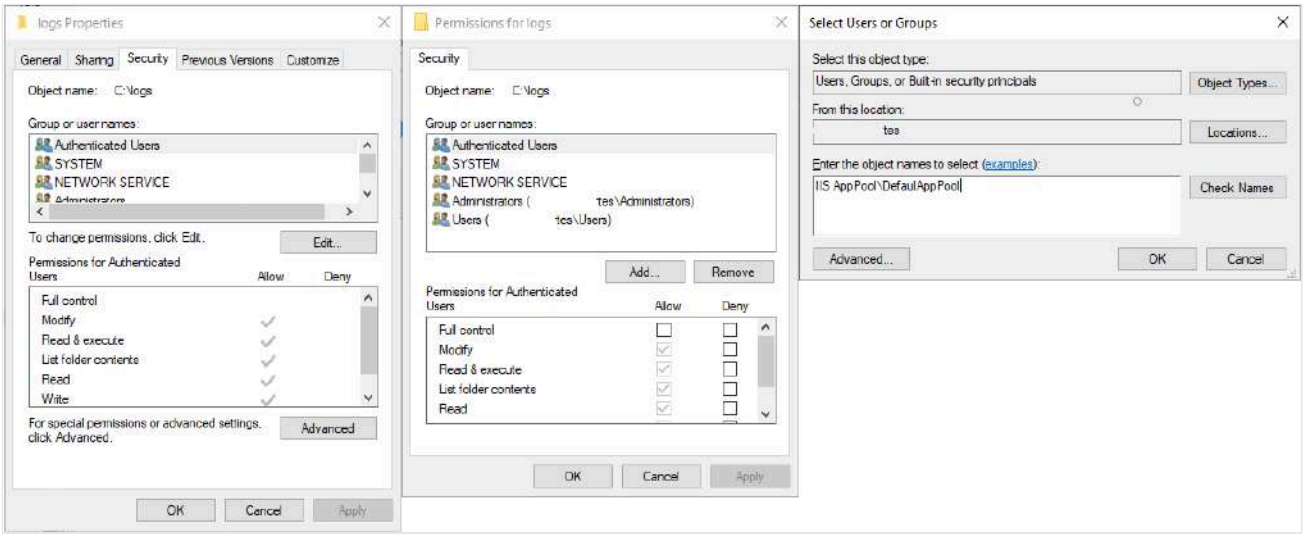
24. In File Explorer, grant **NETWORK SERVICE** and **IIS AppPool\SdiAppPool** (or **IIS AppPool\DefaultAppPool** if you're using the default pool) access to the **Logs** and **Files** folders:

    a. Select and hold (or right-click) one of the folders, and then select **Properties**.

    b. In the **Properties** dialog box, on the **Security** tab, select **Edit**.

    c. Add the users if they aren't listed.

    d. Repeat steps 1 through 3 for the other folder.



25. Enable client certificate renegotiation:

    a. Run Windows PowerShell as an administrator.

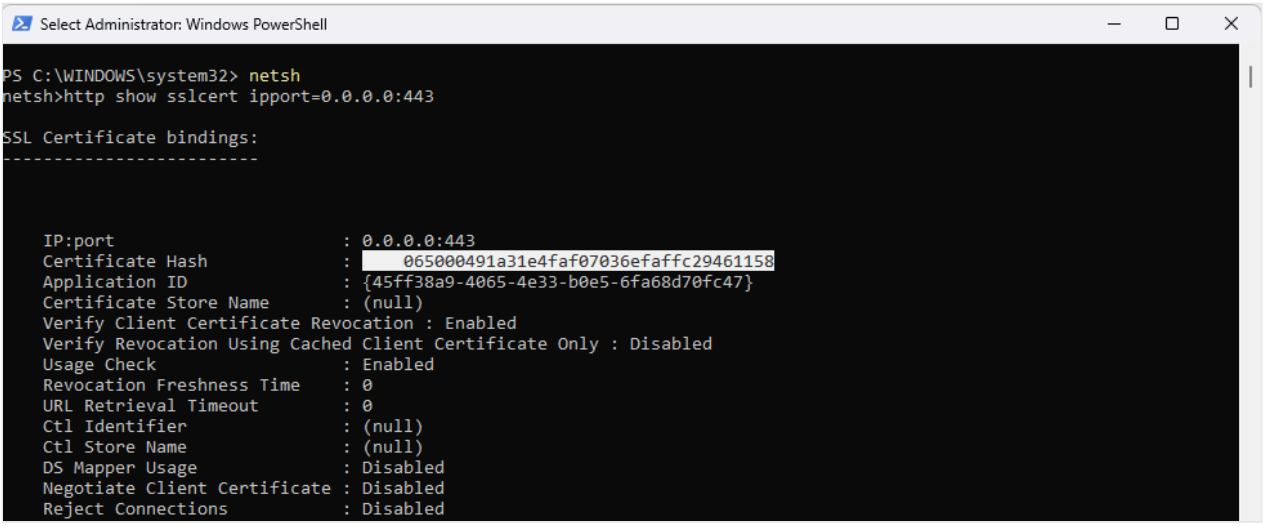
    b. Run the following command to open Network Shell (netsh).

    ```
    PowerShell
    netsh
    ```

    c. Run the following command to view the current binding parameters.

    ```
    PowerShell
    http show sslcert ipport=0.0.0.0:443
    ```

    d. Copy the **Certificate hash** value from the output, and ensure that it equals the server certificate thumbprint.



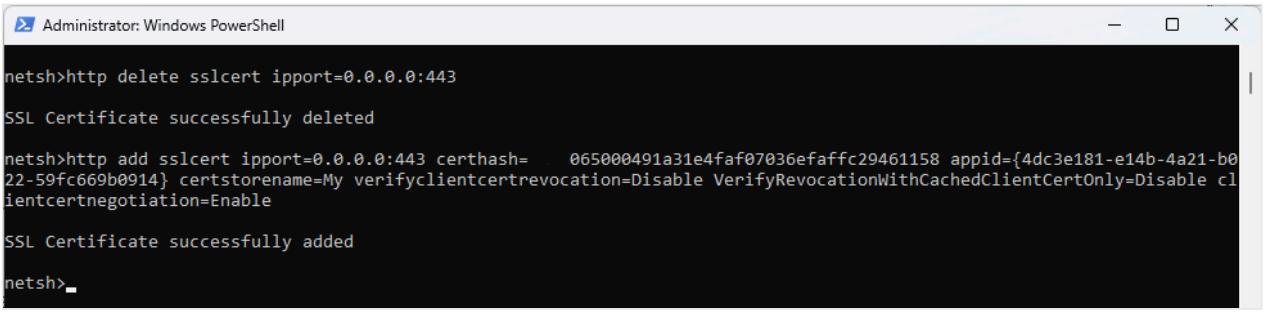
    e. Run the following command to remove the current binding.

    ```
    PowerShell
    http delete sslcert ipport=0.0.0.0:443
    ```

    f. Run the following command to add a new binding that has the same certificate but different parameters. Substitute the **CertificateThumbprint** placeholder in the command with the **Certificate hash** value from previous commands, but remove any spaces or other symbols.

    ```
    PowerShell
    ```

```
http add sslcert ipport=0.0.0.0:443 certhash=CertificateThumbprint appid=
{4dc3e181-e14b-4a21-b022-59fc669b0914} certstorename=My
verifyclientcertrevocation=Disable
VerifyRevocationWithCachedClientCertOnly=Disable clientcertnegotiation=Enable
```



g. Run the following command to restart IIS.

PowerShell

```
exit
iisreset
```

Rotate certificates

To rotate certificates, follow these steps:

1. Rotate the app registration S2S authentication certificate:
   a. Generate a new certificate by running the PowerShell script from the Create an app registration section of this article.
   b. In the Azure portal, go to **App registrations**, and find the app that was created for S2S.
   c. Go to **Certificates & secrets**, select **Upload certificate**, and upload the .cer file for S2S authentication.
   d. Go to Azure Key Vault, and select the certificate that you previously uploaded, and that was referred to as the *App Registration Certificate*. Select **New Version**.
   e. Select **Method of certificate creation** = **Import**, specify the path of the .pfx file and the password, and then select **Create**.

2. Rotate the proxy server certificate:

a. Request a new server certificate from the tax authority portal.

b. Install the .pfx certificate file into **Local Machine\Personal storage**.

c. Open IIS Manager. In the tree on the left, select the proxy website. On the right, select **Bindings**.

d. Select the existing binding, and then select **Edit**.

e. In the **SSL certificate** field, select the newly added certificate. Then select **OK**.

f. In the **web.config** file, find the following line, and add the thumbprint of the proxy server certificate.

```
<serviceCertificate findValue="[certificate thumbprint]"
storeLocation="LocalMachine" storeName="My" x509FindType="FindByThumbprint">
```

g. Complete the Enable client certificate renegotiation step in the Set up the SDI Proxy service in IIS section of this article.

3. Rotate the client identity certificate:
   a. Request a new client certificate from the tax authority portal.
   b. In the Azure portal, go to **Key Vaults**, and find the key vault that is linked to the service.
   c. In the **Certificates** section, find and select the client identity certificate. Then select **New Version**.
   d. Select **Method of certificate creation** = **Import**, specify the path of the .pfx file and the password, and then select **Create**.

Privacy notice

Enabling the **Italian electronic invoice** feature might require that limited data is sent. This data includes the organization's tax registration ID. An administrator can enable and disable the **Italian electronic invoice** feature. To disable the feature, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Features** tab, select the rows that contain the **Italian electronic invoice** feature, and then select **Disable now**.

Data that is imported from these external systems into this Dynamics 365 online service are subject to our privacy statement. For more information, go to the "Privacy notice" section in country/region-specific feature documentation.

Additional resources

- Electronic Invoicing service overview
- Electronic invoicing components
- Electronic invoicing configuration
- Customer electronic invoices
- Vendor electronic invoices

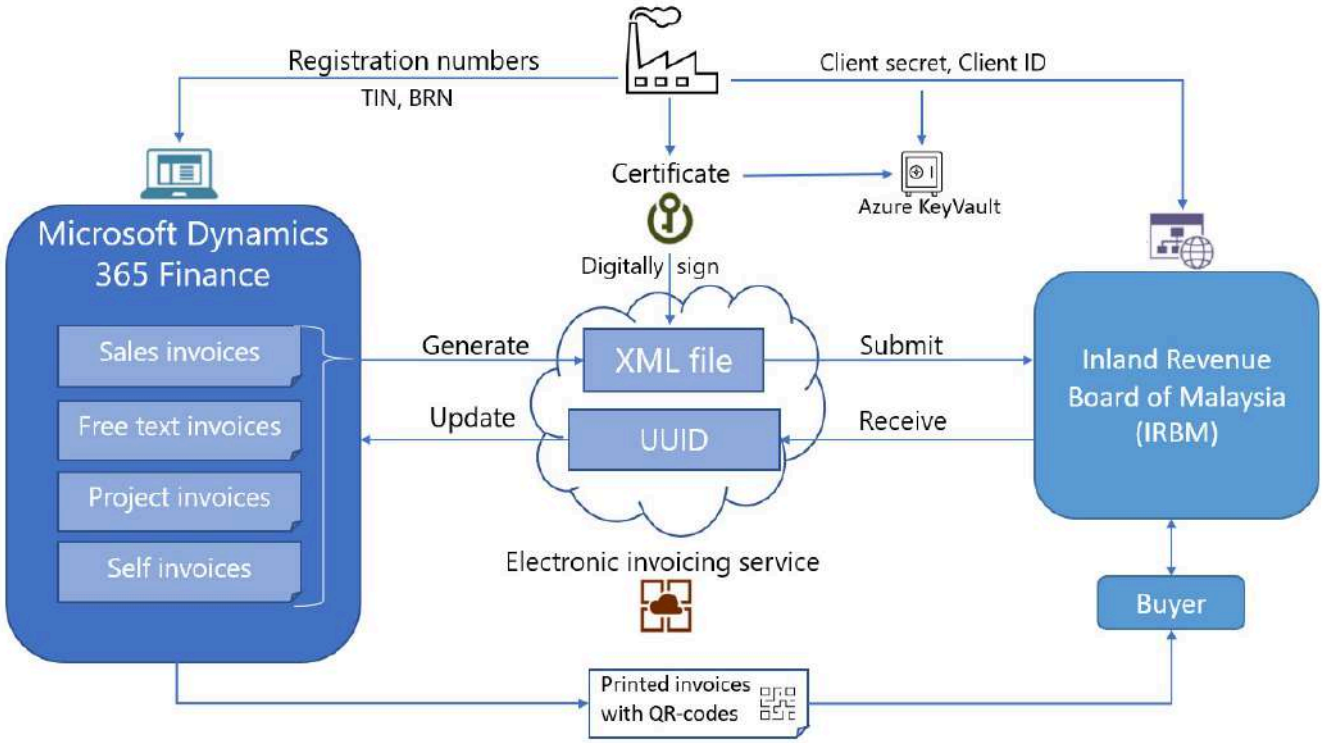
Last updated on 05/01/2026

Get started with electronic invoicing for Malaysia

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information to help you get started with electronic invoicing for Malaysia. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management.

After you configure electronic invoicing, you can generate, digitally sign, and submit the XML files of electronic invoices according to the regulatory requirements in Malaysia.



> **Note**
>
> The electronic invoicing approach that this article describes is implemented by using an invoicing service that's applicable only to cloud deployments of Finance or Supply Chain

Management.

Prerequisites

Before you begin the procedures in this article, the following prerequisites must be met:

- The primary address of the legal entity must be in Malaysia.
- Your company must be a registered taxpayer in Malaysia and must have the following registration numbers: **Tax Identification Number (TIN)** and **Business Registration Number (BRN)**.
- Obtain **Client ID** and **Client secret** values in the Inland Revenue Board of Malaysia (IRBN) (Lembaga Hasil Dalam Negeri Malaysia [LHDN]). These values serve as credentials that are used to establish a secure connection to the IRBN portal.
- Obtain a digital signature certificate from one of the Malaysian certification authorities. This certificate is used for digital signing of generated electronic invoices.
- Become familiar with electronic invoicing as it's described in Electronic Invoicing service overview.
- Do the common part of electronic invoicing service configuration as described in Electronic invoicing configuration.

Configure Azure Key Vault

Create an Azure Key Vault to store the required certificates and secrets that are issued for your company. For more information, see Configure Azure resources for Electronic invoicing.

Add the following required elements in the key vault:

- The secret for the **Client secret** value that's used to establish secure communication to IRBN.
- The secret for the **Client ID** value that's used to establish secure communication to IRBN.
- The certificate that's used for digital signing of outgoing electronic invoices.

Configure Key Vault parameters for electronic invoicing

To configure Key Vault parameters for electronic invoicing, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Setup** > **Electronic document parameters**.

2. On the **Electronic invoicing** tab, in the **Key Vault settings** section, in the **Key Vault** field, select the reference to the key vault that you created in the previous section.

3. In the **SAS token secret** field, select the name of the storage account secret URL that must be used to authenticate access to the storage account.

4. Select **Key Vault parameters**.

5. On the **Key Vault parameters** page, in the **Certificates** section, select **Add** to create new elements of the appropriate type for each certificate or secret that's described in the previous section.

    - A **Client secret** element of the **Secret** type
    - A **Client ID** element of the **Secret** type
    - A **Certificate** element of the **Certificate** type

    > ⓘ **Note**
    >
    > The values in the **Name** column should match the names of the certificates or secrets that are described in the previous section.

Configure the electronic invoicing feature

Some of the parameters from the **Malaysian electronic invoicing (MY)** electronic invoicing feature are published with default values. Before you deploy this feature to the service, review the default values, and update them as required, so that they better reflect your business operations.

To configure the electronic invoicing feature, follow these steps:

1. Import the latest version of the **Malaysian electronic invoicing (MY)** Globalization feature as described in Import features from the repository.

2. Create a copy of the imported Globalization feature, and select your configuration provider for it. For more information, see Create Globalization features.

3. On the **Versions** tab, verify that the **Draft** version is selected.

4. On the **Setups** tab, in the grid, select the **Sales invoice derived** feature setup, and then select **Edit**.

5. On the **Processing pipeline** tab, in the **Processing pipeline** section, select the **Sign document for Malaysia** action.

6. In the **Parameters** section, select the **Certificate name** parameter, and then, in the **Value** field, select the name of the digital certificate that you created.

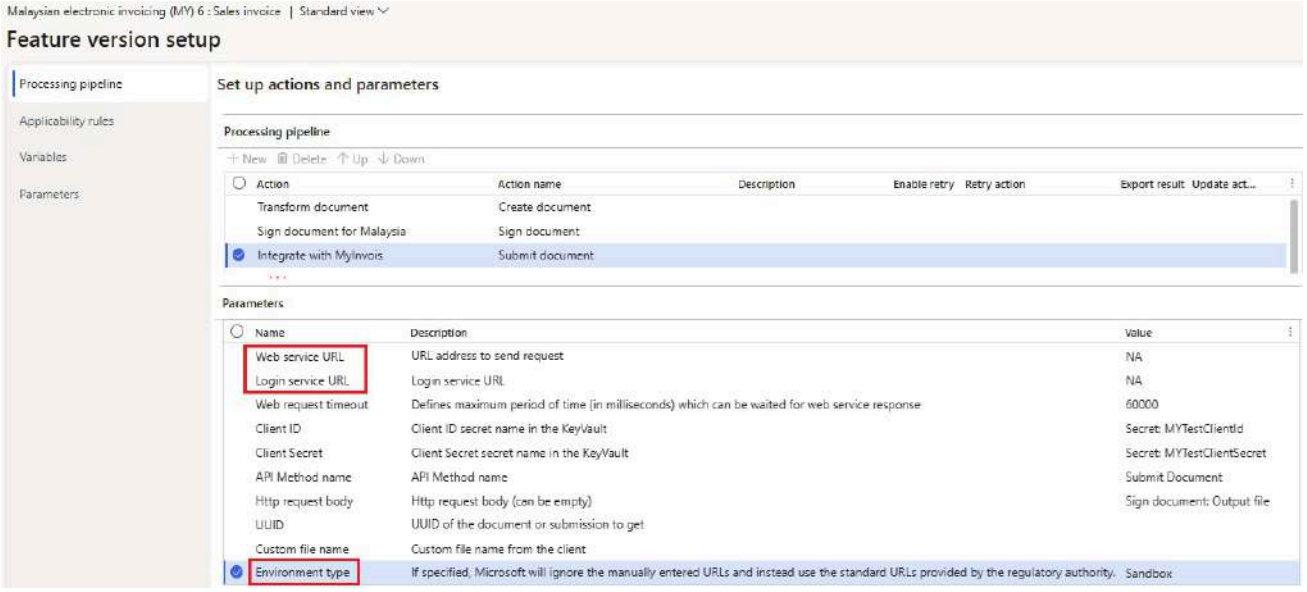
7. In the **Processing pipeline** section, select the **Integrate with MyInvois (Submit document)** action.

8. In the **Parameters** section, select the **Client secret** parameter, and then, in the **Value** field, select the name of the secret that you created.

9. Select the **Client ID** parameter, and then, in the **Value** field, select the name of the client ID that you created.

10. In the **Parameters** section, select the **Web service URL** and **Login service URL** parameters. In the **Value** fields, enter the testing or production URLs, depending on the implementation stage. Review the list of available URLs in MyInvois system environment URLs.

> **Note**
>
> Alternatively, you can leave the **Web service URL** and **Login service URL** parameters blank. Instead, you can select the **Environment type** parameter, and then, in the **Value** field, select the type of the environment, depending on the implementation stage. In this case, the required URLs are automatically constructed. If the **Web service URL** and **Login service URL** parameters are defined, they have higher priority than the **Environment type** parameter.



11. Repeat steps 7 through 10 for the **Integrate with MyInvois (Get document)** action.
12. On the **Processing pipeline** tab, in the **Processing pipeline** section, select the **Generate QR code for Malaysia** action.
13. In the **Parameters** section, select the **Environment type** parameter, and then, in the **Value** field, select the type of the environment, depending on the implementation stage.
14. Select **Save**, and close the page.
15. Repeat steps 4 through 14 for the **Project invoice derived** and **Self invoice derived** feature setups.
16. The copy of the feature is always created as a **Draft** version. Regardless of whether you made changes, complete and deploy the feature as described in Complete and deploy a Globalization feature.

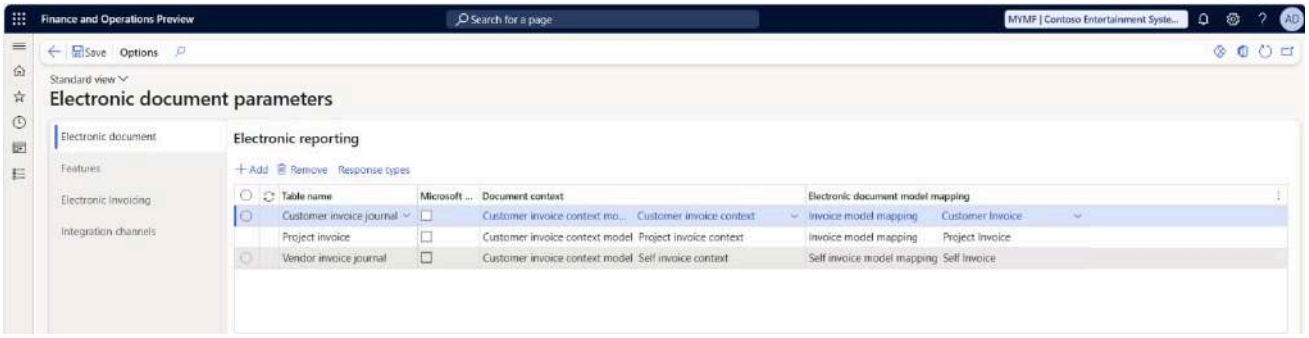
Configure electronic document parameters

To configure electronic document parameters, follow these steps:

1. Make sure that the country/region-specific electronic reporting (ER) configurations for the document context and electronic document model mapping that are required for Malaysia are imported. For more information, see Set up Electronic document parameters.

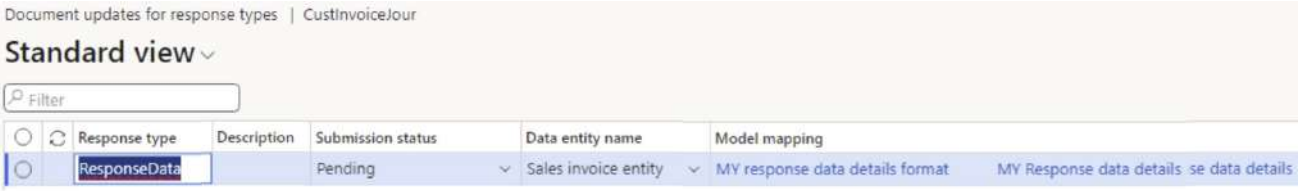
   > **Note**
   >
   > After you import the **Malaysian electronic invoicing (MY)** electronic invoicing feature, electronic documents are configured by default. Follow the remaining steps of this procedure if you must make changes. Otherwise, consider this section informational only.

2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
3. On the **Electronic document** tab, add records for the **Customer Invoice journal**, **Vendor Invoice journal**, and **Project invoice** table names.
4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.



5. For each table name, select **Response types**, select **New** to create a response type, and enter the following values:

   - In the **Response type** field, enter **ResponseData** (the default value).
   - In the **Description** field, enter any meaningful name. Alternatively, leave the field blank.
   - In the **Submission status** field, select **Pending**.
   - In the **Data entity name** field, select one of the following values, depending on the table name that you're configuring:
     - For the **Customer Invoice journal** table name, select **Sales invoice entity**.
     - For the **Project invoice** table name, select **Project invoice entity**.
     - For the **Vendor Invoice journal** table name, select **Self invoice entity**.
   - In the **Model mapping** field, select **MY response data details format**.



> **Note**
>
> **ResponseData** is the default name of the response type. If you must change it, make sure that the new name matches the name that was defined for the related variable of the **To client** type in the corresponding feature setups. To validate the variable's value, go to **Globalization Studio**, and select the **Electronic invoicing** tile. On the **Electronic invoicing features** page, verify that the **Malaysian electronic invoicing (MY)** feature is selected. On the **Setups** tab, in the grid, select the **Sales invoice derived** feature setup. Then select **Edit** or **View**, depending on the status of the feature version.

6. Select **Save**, and close the page.

> ⓘ Note
>
> The setup described here allows you to submit electronic invoices for the following **posted** source documents:
>
> - Customer invoices that are based on sales orders
> - Free text invoices
> - Project invoices
> - Self invoices that are based on vendor invoices from foreign vendors
> - Credit notes and debit notes for all the previously described invoice types
> - Refund notes, which represent credit notes for already paid invoices.

7. On the **Features** tab, select and enable the **Malaysian electronic invoice** feature.

8. Save your changes on the **Electronic document parameters** page, and then close the page.

> ⓘ Note
>
> Watch the following video about the configuration of Malaysian electronic **Self invoices**.

https://learn-video.azurefd.net/vod/player?id=012f47ed-0472-4e12-a5be-5f7a594c1cb5&locale=en-us&embedUrl=%2Fdynamics365%2Ffinance%2Flocalizations%2Fmalaysia%2Fapac-mys-e-invoices

Configure registration numbers

> ⓘ Note
>
> When the output files of electronic invoices are generated, registration numbers of the **Enterprise ID (COID)** category are used as Business Registration Numbers (BRNs). If the **Enterprise ID (COID)** registration category already exists and has been assigned to a registration type, skip this procedure.

To configure registration numbers, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Global address book** > **Registration types** > **Registration types**.
2. Create a registration type.
3. In the **Country/region** field, select **MYS - Malaysia**.
4. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration categories**.
5. Create a registration category.
6. In the **Registration types** field, select the registration type that you created in step 2.
7. In the **Registration categories** field, select **Enterprise ID (COID)**.

Configure Sales and Service Tax numbers

To configure Sales and Service Tax (SST) numbers, follow the instructions in GST/SST registration number.

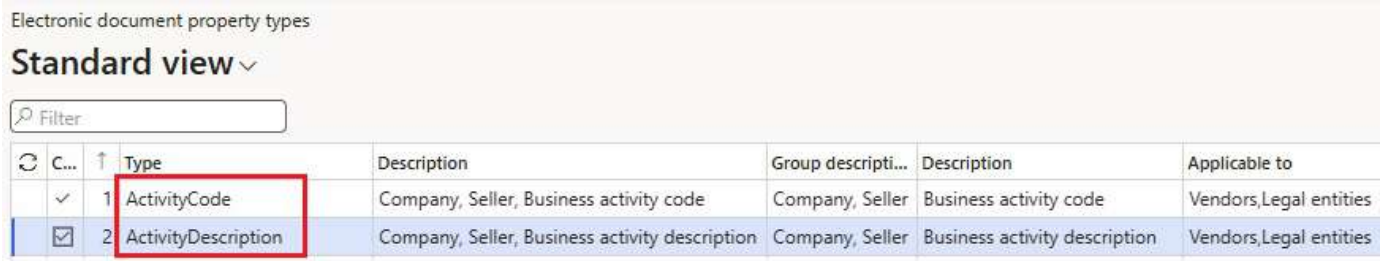
> **Note**
>
> Make sure that the configured SST registration number is in the **VAT ID** registration category.

Configure electronic document property types

To configure the electronic document property type that's required to define the taxpayer activity code, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts receivable** > **Setup** > **Electronic document property types**.
2. Select **New** to create an electronic document property type.
3. In the **Type** field, enter **ActivityCode**. You must enter the value exactly as it appears here.
4. On the Action Pane, select **Applicability**.
5. In the **Table name** field, select **CompanyInfo**.
6. Save and close the **Electronic document property type applicability setup** page.
7. Select **New** to create another electronic document property type.
8. In the **Type** field, enter **ActivityDescription**. You must enter the value exactly as it appears here.
9. On the Action Pane, select **Applicability**.
10. In the **Table name** field, select **CompanyInfo**.
11. Select **New** to create the record for another applicable table.

12. In the **Table name** field, select **VendTable**.
13. Save and close the **Electronic document property type applicability setup** page.
14. Save and close the **Electronic document property types** page.



C...		Type	Description	Group descripti...	Description	Applicable to
✓	1	ActivityCode	Company, Seller, Business activity code	Company, Seller	Business activity code	Vendors,Legal entities
☑	2	ActivityDescription	Company, Seller, Business activity description	Company, Seller	Business activity description	Vendors,Legal entities

Configure address structure

To configure the structure of the postal address to define all required elements, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Global address book** > **Addresses** > **Address setup**.

2. Make sure that all the following mandatory elements are configured:

    - Country code
    - Postal code
    - City name
    - Street
    - State code

3. For the states, on the **State/province** tab, in the **State code** field, define the official state code according to the published requirements.

Configure legal entity data

Enter the address

To enter the address, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity.
3. On the **Addresses** FastTab, add a valid primary address for the selected legal entity.

Enter the registration numbers

To enter the registration numbers, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity, and then, on the **Tax registration** FastTab, in the **Tax registration number** field, enter the company's TIN.
3. On the Action Pane, select **Registration IDs**.
4. On the **Registration ID** FastTab, select **Add** to create a registration ID.
5. In the **Registration type** field, select the BRN registration type that you created earlier.
6. In the **Registration number** field, enter a valid BRN registration number for the selected legal entity.
7. Select **Add** to create another registration ID, if required.
8. In the **Registration type** field, select the SST registration type that you created earlier.
9. In the **Registration number** field, enter a valid SST registration number for the selected legal entity.

Enter a business activity code and description

To enter a business activity code and description, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity, and then, on the Action Pane, select **Electronic document properties**.
3. Select the line where the **Type** field is set to **ActivityCode**.
4. In the **Value** field, enter the taxpayer business activity code.
5. Select the line where the **Type** field is set to **ActivityDescription**.
6. In the **Value** field, enter the taxpayer business activity description.

> **Note**
>
> The contact information for the legal entity is automatically retrieved from the related **Person** that is associated with the current user in Dynamics 365 Finance.

Configure customer data

Enter the address

To enter the address, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer.
3. On the **Addresses** FastTab, add a valid address for the selected customer.

Enter the contact person

To enter the contact person, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer.
3. On the **Sales demographics** FastTab, in the **Primary contact** field, select the person who is considered the buyer's contact.

> ⓘ **Note**
>
> All available contact persons must already be defined for the selected customer. Make sure that the selected contact person has a valid email address and phone number.

Enter the registration numbers

To enter the registration numbers, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer, and then, on the **Invoice and delivery** FastTab, in the **Tax exempt number** field, enter a valid TIN for the customer.
3. On the Action Pane, on the **Customer** tab, in the **Registration** group, select **Registration IDs**.
4. On the **Registration ID** FastTab, select **Add** to create a registration ID.
5. In the **Registration type** field, select the BRN registration type that you created earlier.
6. In the **Registration number** field, enter a valid BRN registration number for the selected customer.
7. Select **Add** to create another registration ID, if required.
8. In the **Registration type** field, select the SST registration type that you created earlier.

9. In the **Registration number** field, enter a valid SST registration number for the selected customer.

> ⓘ Note
>
> For **foreign** customers, actual tax registration numbers aren't always mandatory. If no tax registration numbers are defined for them, the following general numbers are used:
>
> - **EI00000000020** is used as the customer's **TIN**.
> - **NA** is used as the customer's **BRN** and **SST**.

Additional property configuration

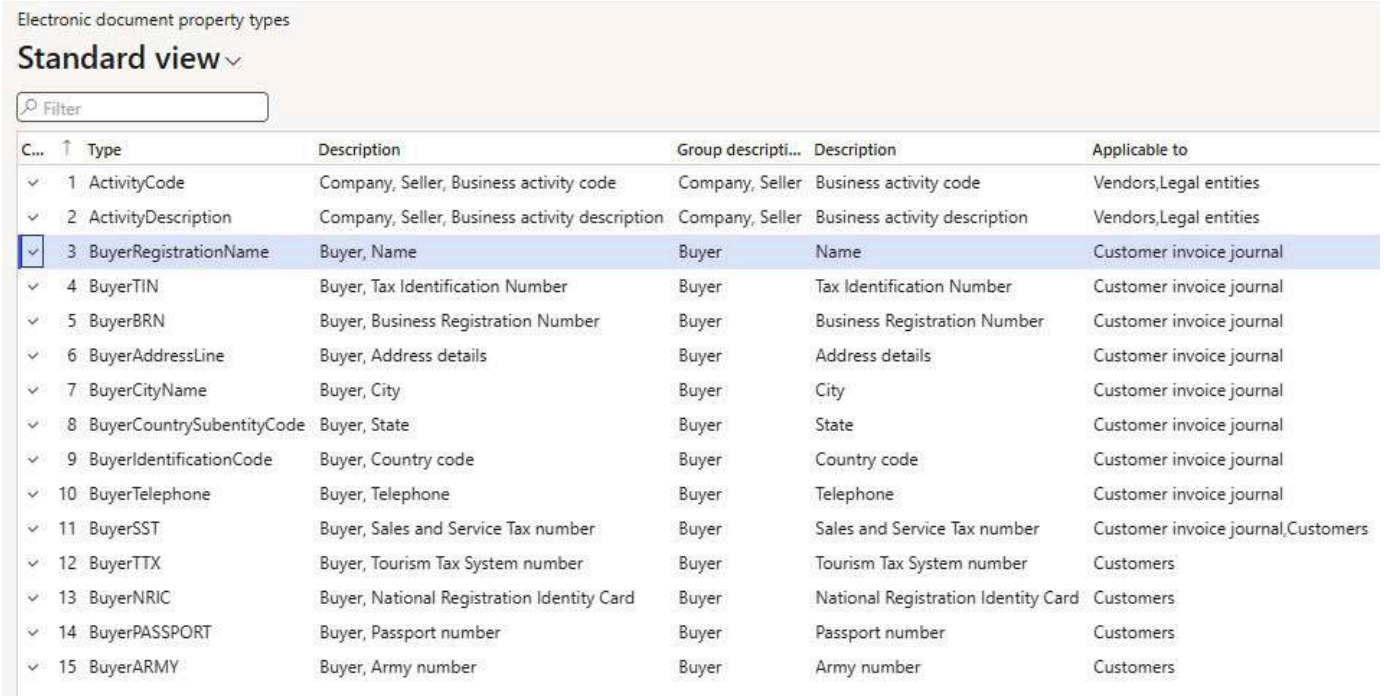
Customizable electronic document property types can be used to extend the properties of legal entities, customers, vendors, and invoice journals. Users can configure them to cover some specific scenarios through configuration only. No coding is required.

To configure additional buyer and journal properties, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts receivable** > **Setup** > **Electronic document property types**.
2. Select **New** to create an electronic document property type.
3. In the **Type** field, enter **BuyerRegistrationName**. You must enter the value exactly as it appears here.
4. On the Action Pane, select **Applicability**.
5. In the **Table name** field, select **CustInvoiceJour**.
6. To create the record for another applicable table, select **New**.
7. In the **Table name** field, select **CustTable** as required.
8. Save your changes, and close the **Electronic document property type applicability setup** page.
9. Repeat steps 2 through 8 for the remaining properties:
    - BuyerTIN
    - BuyerBRN

- BuyerAddressLine
- BuyerCityName
- BuyerCountrySubentityCode
- BuyerIdentificationCode
- BuyerTelephone
- BuyerSST
- BuyerTTX
- BuyerNRIC
- BuyerPASSPORT
- BuyerARMY

10. Save your changes, and close the **Electronic document property types** page.


Electronic document property types

Standard view

Filter

C...	Type	Description	Group descripti...	Description	Applicable to
1	ActivityCode	Company, Seller, Business activity code	Company, Seller	Business activity code	Vendors,Legal entities
2	ActivityDescription	Company, Seller, Business activity description	Company, Seller	Business activity description	Vendors,Legal entities
3	BuyerRegistrationName	Buyer, Name	Buyer	Name	Customer invoice journal
4	BuyerTIN	Buyer, Tax Identification Number	Buyer	Tax Identification Number	Customer invoice journal
5	BuyerBRN	Buyer, Business Registration Number	Buyer	Business Registration Number	Customer invoice journal
6	BuyerAddressLine	Buyer, Address details	Buyer	Address details	Customer invoice journal
7	BuyerCityName	Buyer, City	Buyer	City	Customer invoice journal
8	BuyerCountrySubentityCode	Buyer, State	Buyer	State	Customer invoice journal
9	BuyerIdentificationCode	Buyer, Country code	Buyer	Country code	Customer invoice journal
10	BuyerTelephone	Buyer, Telephone	Buyer	Telephone	Customer invoice journal
11	BuyerSST	Buyer, Sales and Service Tax number	Buyer	Sales and Service Tax number	Customer invoice journal,Customers
12	BuyerTTX	Buyer, Tourism Tax System number	Buyer	Tourism Tax System number	Customers
13	BuyerNRIC	Buyer, National Registration Identity Card	Buyer	National Registration Identity Card	Customers
14	BuyerPASSPORT	Buyer, Passport number	Buyer	Passport number	Customers
15	BuyerARMY	Buyer, Army number	Buyer	Army number	Customers

To enter the values for the previously described electronic document properties at the customer level, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer, and then, on the Action Pane, on the **Customer** tab, select **Electronic document properties**.
3. For each property that must be overridden in the electronic invoice, enter the required value in the **Value** field.

To enter the values for the electronic document properties at the invoice level, follow these steps:

1. In Dynamics 365 Finance, go to **Account receivable** > **Inquiries and reports** > **Invoices** > **Invoice journal**.

2. Select a required invoice, and then, on the Action Pane, on the **Invoice** tab, select **Electronic document properties**.
3. For each property that must be overridden in the electronic invoice, enter the required value in the **Value** field.

> ⓘ Note
>
> - Nonblank values in the previously described electronic document properties override the corresponding default values that are automatically populated from either customer or invoice master data.
> - Values must be entered for the electronic document properties before electronic invoices are submitted to the authorities.

Configure funding sources

If business processes assume that **project invoices** are issued, follow these additional configuration steps.

1. In Dynamics 365 Finance, go to **Project management and accounting** > **Projects** > **Project contracts**.
2. Create a new project contract, or select an existing project contract.
3. On the **Funding sources** FastTab, create or select a funding source of the **Customer** type, and then select **Details**.
4. On the **Funding source details** page, on the **Other** FastTab, in the **Contact** section, in the **Contact ID** field, select a valid contact person.

   > ⓘ Note
   >
   > All available contact persons must already be defined for the selected customer. Make sure that the selected contact person has a valid email address and phone number.

Configure vendor data

If business processes assume that self invoices are issued, you must also configure required vendor data.

Enter the address

To enter the address, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts payable** > **Vendors** > **All vendors**.
2. Select a vendor.
3. On the **Addresses** FastTab, add a valid address for the selected vendor.

Enter the contact person

To enter the contact person, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts payable** > **Vendors** > **All vendors**.
2. Select a vendor.
3. On the **Purchasing demographics** FastTab, in the **Primary contact** field, select the person who is considered the buyer's contact.

   > **Note**
   >
   > All available contact persons must already be defined for the selected vendor. Make sure that the selected contact person has a valid email address and phone number.

Enter the registration numbers

To enter the registration numbers, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts payable** > **Vendors** > **All vendors**.
2. Select a vendor, and then, on the **Invoice and delivery** FastTab, in the **Tax exempt number** field, enter a valid TIN for the vendor.
3. On the Action Pane, on the **Vendor** tab, in the **Registration** group, select **Registration IDs**.
4. On the **Registration ID** FastTab, select **Add** to create a registration ID.
5. In the **Registration type** field, select the BRN registration type that you created earlier.
6. In the **Registration number** field, enter a valid BRN registration number for the selected vendor.

7. Select **Add** to create another registration ID, if required.
8. In the **Registration type** field, select the SST registration type that you created earlier.
9. In the **Registration number** field, enter a valid SST registration number for the selected vendor.

> ⓘ Note
>
> For **foreign** vendors, actual tax registration numbers aren't always mandatory. If no tax registration numbers are defined for them, the following general numbers are used:
>
> - **EI00000000030** is used as the vendor's **TIN**.
> - **NA** is used as the vendor's **BRN** and **SST**.

Enter a business activity code and description

To enter a business activity code and descriptions, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts payable** > **Vendors** > **All vendors**.
2. Select a vendor, and then, on the Action Pane, on the **Vendor** tab, select **Electronic document properties**.
3. Select the line where the **Type** field is set to **ActivityCode**.
4. In the **Value** field, enter the business activity code.
5. Select the line where the **Type** field is set to **ActivityDescription**.
6. In the **Value** field, enter the business activity description.

> ⓘ Note
>
> A Malaysian Standard Industrial Classification (MSIC) code might not be applicable to foreign vendors. In this case, the fixed value **00000** is used for the activity code, and the value **NA** is used as the activity description.

Configure the application-specific parameters

This setup is required to establish a mapping between sales tax codes that are used in the system and legally required tax types.

Make sure that the following ER format configurations are imported:

- Sales invoice (MY)
- Project invoice (MY)
- Self invoice (MY)

To complete the configuration, follow these steps:

1. In Dynamics 365 Finance, go to **Globalization Studio**.
2. Select the **Electronic reporting** tile, and then select the **Reporting configurations** tile.
3. On the **Configurations** page, select the **Sales invoice (MY)** format configuration.
4. On the **Configurations** menu, in the **Application specific parameters** section, selected **Setup**.
5. In the **Lookups** section, make sure that the **$TaxTypes** lookup is selected.
6. In the **Conditions** section, select **Add** to add a condition.
7. In the **Name** column for the new condition, select the sales tax code that's defined in the application. Then, in the **Lookup result** column, select a tax type code according to the official list of tax type codes.
8. Add specific conditions for each sales tax code that's defined in the system, and save your changes.

   > **Note**
   >
   > In the **Name** column, you can select the ***Blank*** or ***Not blank*** placeholder value instead of a specific sales tax code.

9. In the **State** field, select **Completed**. Then select **Save**.
10. Repeat steps 2 through 8 for the **Project invoice (MY)** and **Self invoice (MY)** format configurations, as required.

Configure sales tax codes

To cover scenarios for non-taxable, zero-rated, or exempted operations, you must define tax exempt codes in the system.

To configure sales tax codes, follow these steps:

1. In Dynamics 365 Finance, go to **Tax** > **Setup** > **Sales tax** > **Sales tax exempt codes**.
2. Add the required exempt codes together with descriptions.
3. Save your changes, and close the page.
4. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax groups**.
5. In the **Setup** section, select the row for a required sales tax code, and then, in the **Exempt** column, mark the sales tax code as exempt.
6. In the **Exempt code** field, reference one of the exempt codes that you created in step 1.
7. Save your changes, and close the page.

Configure units of measure

To configure units of measure, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Setup** > **Units** > **Units**.
2. Select a unit ID, and then select **External codes**.
3. On the **External codes** page, in the **Overview** section, in the **Code** column, enter a code that is equal to the selected unit ID.
4. In the **Standard code** column, select the checkbox.
5. In the **Value** section, in the **Value** field, enter the external code to use as the units according to the required codification.

> **Note**
>
> For scenarios where no specific units of measure are assumed, the default value **EA** is used.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information about how to generate electronic invoices, see Submit electronic documents.

You can inquire about the results of a submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log** and selecting the

required document type. For more information, see Work with Electronic document submission log.

Configure printable invoice layouts

To enable QR code printing in full and simplified invoices, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts receivable** > **Setup** > **Forms** > **Form setup**.
2. Select **Print management**.
3. Select the **Customer invoice** report, and then, in the **Report format** field, reference the **SalesInvoice.ReportFull_MY** or **SalesInvoice.ReportSimplified_MY** layout.
4. Select the **Free text invoice** report, and then, in the **Report format** field, reference the **FreeTextInvoice.ReportFull_MY** or **FreeTextInvoice.ReportSimplified_MY** layout.

If you use project invoices, follow these steps:

1. In Dynamics 365 Finance, go to **Project management and accounting** > **Setup** > **Forms** > **Form setup**.
2. Select **Print management**.
3. Select the **Project invoices without billing rules** report, and then, in the **Report format** field, reference the **PSAProjInvoice.ReportFull_MY** or **PSAProjInvoice.ReportSimplified_MY** layout.
4. Select the **Project invoices with billing rules** report, and then, in the **Report format** field, reference the **PSAContractLineInvoice.ReportFull_MY** or **PSAContractLineInvoice.ReportSimplified_MY** layout.

If you use self invoices, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts payable** > **Setup** > **Forms** > **Form setup**.
2. Select **Print management**.
3. Select the **Vendor invoice** report, and then, in the **Report format** field, reference the **VendInvoiceDocument.ReportMY** layout.

> ⓘ **Note**
>
> The QR code that's shown on the printouts of invoices represents the URL that takes you to the official portal of **MyInvois** system, where you can find the details of the related electronic invoice.

QR codes are printed only for invoices that were successfully submitted, validated, and accepted by the **MyInvois** system.

Batch submission of electronic invoices

In addition to submitting electronic invoices individually, as described in the previous sections, you can submit posted invoices in batches as consolidated electronic invoices.

By default, all invoices that originate from retail operations can be consolidated, but they can't be submitted individually. Nevertheless, you can force specific retail-related invoices to be submitted individually.

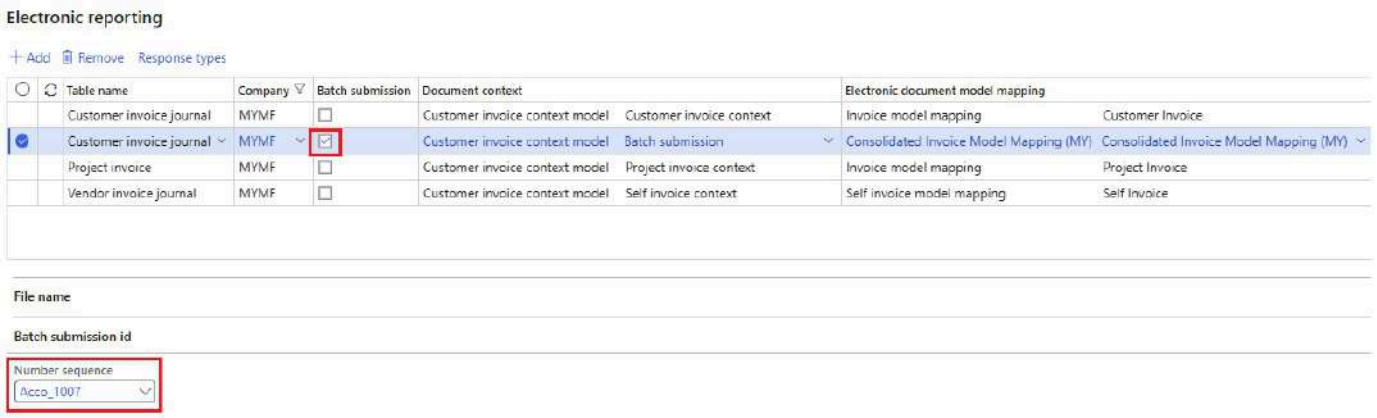
To activate batch submission of invoices, enable the **Electronic document batch submission** feature in Feature management. For more information, see Feature management overview.

Configure electronic document parameters

Before you begin to configure electronic document parameters for batch submission, import the **Consolidated Invoice Model Mapping (MY)** ER configuration from the repository. For more information, see Import electronic reporting (ER) configurations from Dataverse.

To configure electronic document parameters for batch submission, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic document** tab, on the **Electronic reporting** FastTab, add a record for the **Customer Invoice journal** table name.
3. In the **Batch submission** column, select the checkbox.
4. In the **Document context** column, select the **Batch submission** mapping name from the **Customer invoice context model** configuration.
5. In the **Electronic document model mapping** column, select the **Consolidated Invoice Model Mapping (MY)** mapping name from the **Consolidated Invoice Model Mapping (MY)** configuration.
6. On the **Batch submission id** FastTab, in the **Number sequence** field, select the number sequence that should be used to number the batches that are generated. The number sequence that you select must be continuous.



Configure the electronic invoicing feature

To configure the electronic invoicing feature, follow these steps:

1. Import the latest version of the **Malaysian electronic invoicing (MY)** Globalization feature (version **10** or later) as described in Import features from the repository.

   > **Note**
   >
   > Make sure that the **Consolidated invoice (MY)** ER format configuration is imported after you import the **Malaysian electronic invoicing (MY)** Globalization feature. If it isn't imported, manually import it. This format configuration is based on the **Invoices Communication Model** configuration and also on the **Consolidated Invoice Model Mapping (MY)** configuration that you imported in the previous section.

2. Create a copy of the imported Globalization feature, and select your configuration provider for it. For more information, see Create Globalization features.

3. On the **Versions** tab, verify that the **Draft** version is selected.

4. On the **Setups** tab, in the grid, select the **Consolidated invoice derived** feature setup, select **Edit**, and configure the following processing pipeline action parameters.

   > **Note**
   >
   > Configure these parameters just as you configured them for the Sales invoice derived feature setup.

Expand table

Processing pipeline action	Parameters to configure
Sign document for Malaysia	• Certificate name
Integrate with MyInvois (Submit document)	• Client secret • Client ID • Web service URL • Login service URL • Environment type
Integrate with MyInvois (Get document)	• Client secret • Client ID • Web service URL • Login service URL • Environment type

5. Select **Save**, and close the page.

6. Complete and deploy the feature as described in Complete and deploy a Globalization feature.

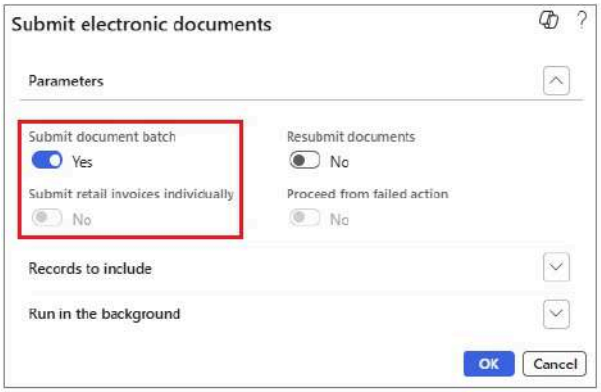
Issue consolidated electronic invoices

To generate and submit electronic invoices in batches for posted Customer invoices, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**.

2. To submit invoices in batches, set the **Submit document batch** option to **Yes**. To submit retail invoices individually, set the **Submit retail invoices individually** option to **Yes**.

> **ⓘ Important**
>
> Batch submission and individual submission are mutually exclusive and can't be run simultaneously. During a given run, only batch submissions or individual submissions can be processed.
>
> If you must submit some retail invoices individually, you must do the individual submission **before** the batch submission. All individually submitted retail invoices will be excluded from batches.



3. On the **Records to include** FastTab, select **Filter** to open the **Inquiry** dialog box, where you can build a query to select only required documents.

4. Select **OK** to start submission.

To inquire about the submission results, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**.
2. In the **Document type** field, select **Batch submission**.
3. Select a specific batch that has one of the following statuses: **Scheduled**, **Completed**, or **Failed**.
4. On the Action Pane, select **Inquiries** > **Submission details** to view the details of the submission execution logs.
5. On the Action Pane, select **Inquiries** > **Batch submission invoices** to view the invoices that were submitted in the selected batch.

> **Note**
>
> Because of restrictions that the authorities impose on the size of submitted data, the maximum number of invoices that can be included in one batch is 60. If more than 60 invoices are being processed, multiple batches are automatically generated.

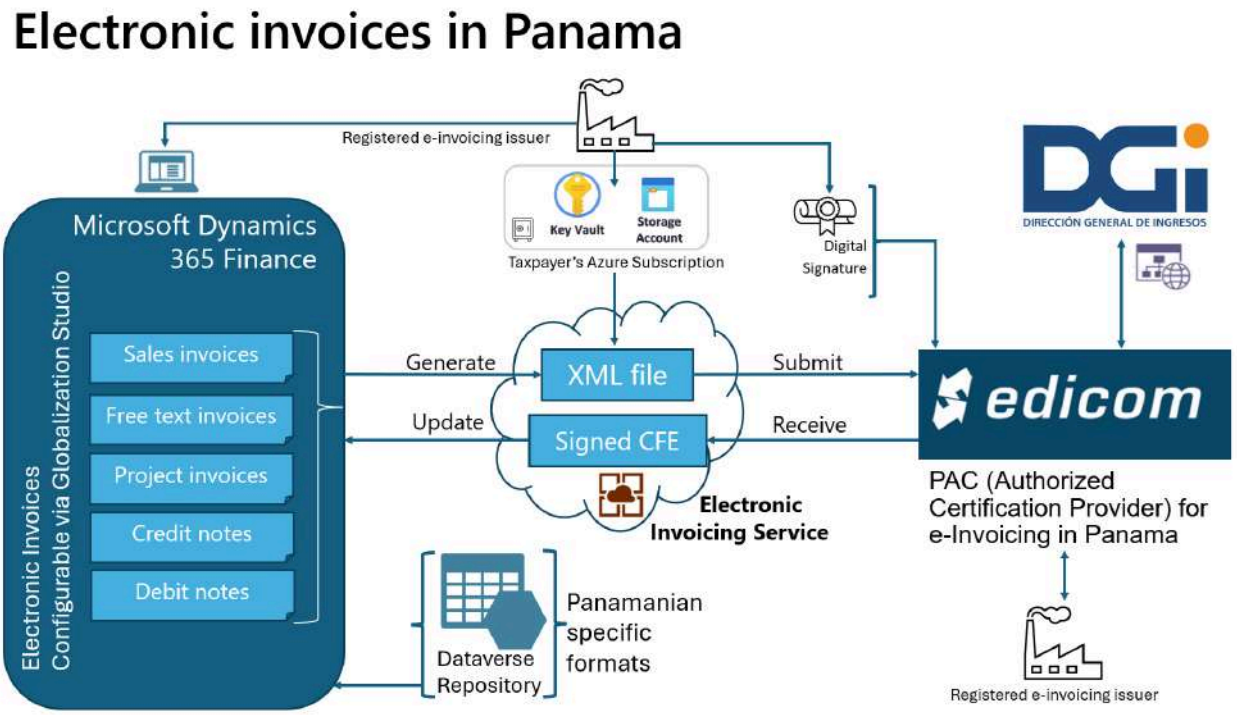
Last updated on 05/04/2026

Get started with Electronic invoicing for Panama

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information to help you get started with Electronic invoicing for Panama. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. These steps complement the steps that are described in Electronic invoicing setup. For the last-mile integration with the Panamanian Tax Authorities, Microsoft is partnering with Edicom.

After you configure Electronic invoicing, you can generate, digitally sign, and submit the XML files of electronic invoices to the Edicom authorized certification provider (PAC) according to the regulatory requirements in Panama.



> **Note**
>
> The electronic invoicing approach that this article describes is implemented by using an invoicing service that is applicable only to cloud deployments of Finance or Supply Chain Management.

> **Important**
>
> The new Electronic invoicing Globalization feature for Panama (outbound flow) requires that you run Finance version 10.0.40. Specifically, you must run build number 10.0.1935.60 or later. It can be imported only into the new Globalization Studio. It isn't supported in Regulatory Configuration Service (RCS).

Prerequisites

Before you begin the procedures in this article, the following prerequisites must be met.

1. Ensure that the settings for the Panamanian legal entity are in place. For more information, see Set up a legal entity and tax information for Panama.
2. Gain familiarity with and understanding of Electronic invoicing as it's described in Electronic invoicing overview.
3. Do the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.
4. You must enable the following features in Feature management:
    - Electronic invoicing integration
    - Electronic invoicing integration resubmit document from failed action
    - Execute update actions for submitted documents
5. Ensure that the following Electronic reporting (ER) format configurations are imported. For more information, see: Import Electronic reporting (ER) configurations from Dataverse, Import Electronic reporting (ER) configurations.
    - Customer invoice context model
    - Inventory e-invoice (PA)
    - Project e-invoice (PA)

- Edicom source file response import format
- Edicom response processing LATAM
- Edicom response error log import

> **Note**
>
> These formats are based on the corresponding **LATAM** format configurations that use the **Invoice model LATAM** and **Invoice model mapping LATAM** configurations. All other required configurations are automatically imported.

Configure the electronic invoicing feature

The **Panama electronic invoice (PA) "E-Invoicing for Panama: ISV last-mile connector with Edicom"** feature represents an outbound flow to issue the following sales documents.

Expand table

Name	Code	Original name
Internal Operation Invoice	01	Factura de operación interna
Export Invoice	03	Factura de exportación
Credit Note referring to one or more Sales Invoices	04	Nota de Crédito referente a una o varias FE
Debit Note referring to one or more Sales Invoices	05	Nota de Débito referente a una o varias FE
Generic Credit Note	06	Nota de Crédito genérica
Generic Debit Note	07	Nota de Débito genérica
Free Zone Invoice	08	Factura de Zona Franca
Foreign Operation Invoice	10	Factura de Operación Extranjera

Some parameters of the feature are published with default values. Before you deploy the Electronic invoicing feature to the service environment, add a feature that is based on the Microsoft-provided feature, and complete the common parameters on the **Feature parameters** tab. Review the default values, and update them as required, so that they better reflect your business operations.

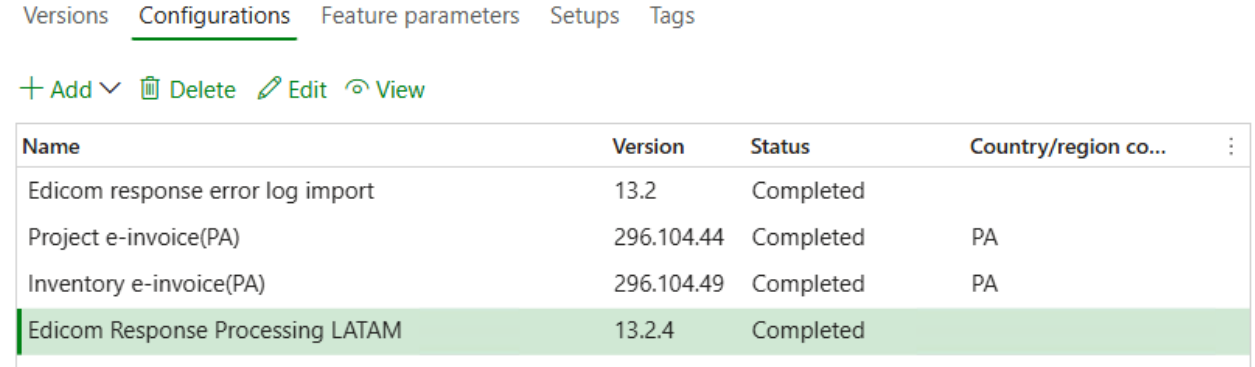
For Panama, there are at least three interactions with Edicom in the pipeline: first to submit the invoice, then to fetch the signed XML, and finally to fetch the status of the submitted invoice.

Each interaction requires common parameters, such as Edicom connection details and the authentication token that Edicom provides. These common parameters are reused in the feature setup for all document types. Edicom provides the values when a company is onboarded.

> **Note**
>
> The configuration of common parameters is now simplified. You no longer have to go to each action and feature setup, and repeatedly specify the common connection parameters. Use of the **Feature parameters** tab is available only as of version 10.0.41.

> **Important**
>
> The **Panama electronic invoice (PA)** feature is provided by Microsoft. Before it can be used, it requires additional configuration, as described in this article. For information about how to configure invoicing features and apply changes, see Work with feature setups. For example, in addition to the connection parameters, you can filter specific legal entities so that they are processed in applicability rules. By default, the feature is applicable to all legal entities that have a primary address in Panama.

1. Import the latest version of the **Panamanian electronic invoice (PA)** Globalization feature as described in Import features from the repository. The following illustrations show what the feature looks like after you import it from Dataverse.



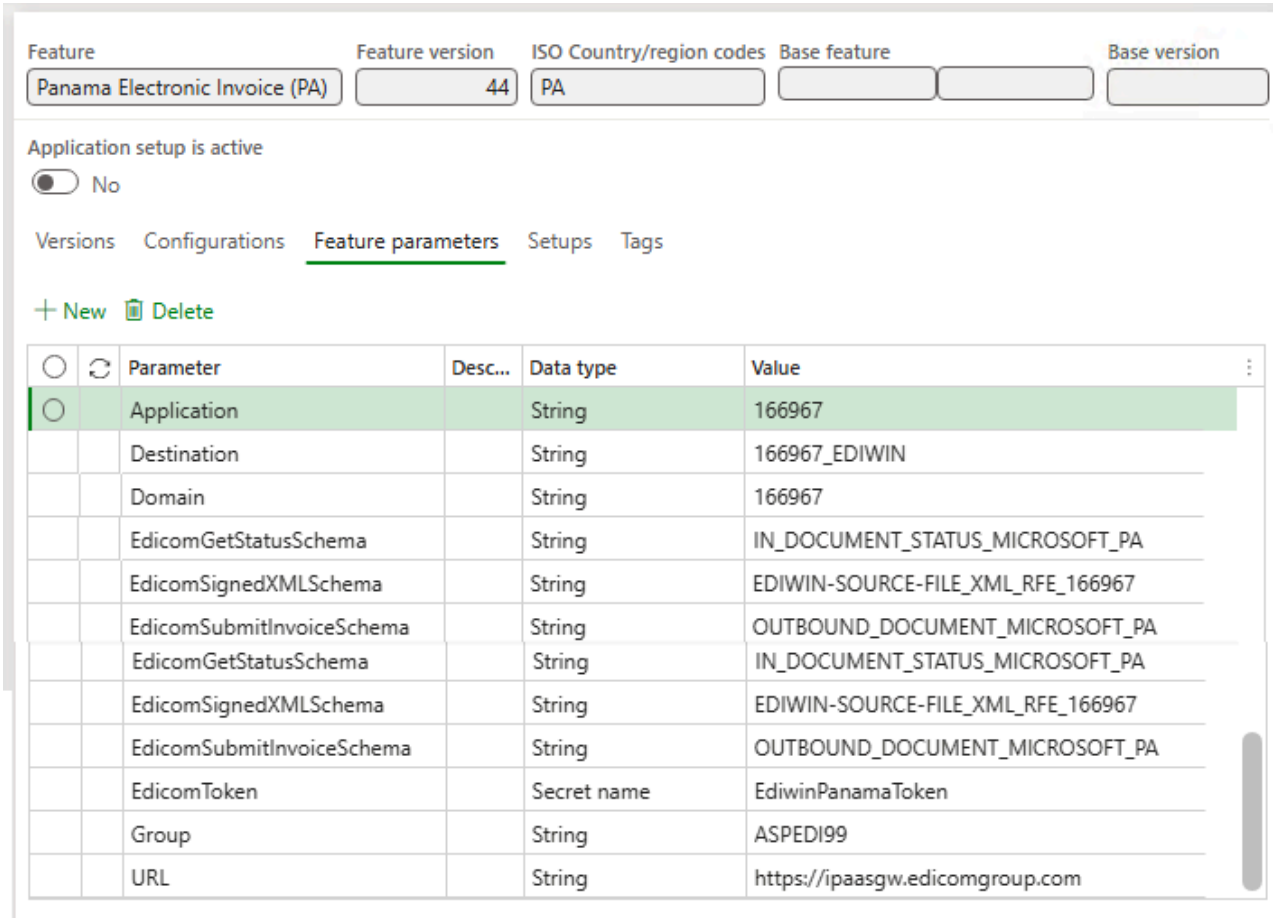
2. Create a copy of the imported Globalization feature, and select your configuration provider. For more information, see Create a Globalization feature.

3. On the **Versions** tab, confirm that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with Edicom's API:

    - Select **Application**, and then enter the service ID number that you obtained.
    - Select **Domain**, and then enter the same service ID number.
    - Select **Get status schema name**, and then enter the schema name.
    - Select **Group**, and then enter the group code that you obtained.
    - Select **Signed XML schema name**, and then enter the schema name.
    - Select **Destination name**, and then enter the service ID number concatenated with the string "_EDIWIN." For example, if the service ID number is 123456, enter **123456_EDIWIN**.
    - Select **Submit invoice schema name**, and then enter the schema name.
    - Select **Auth token**, and then select the name of the secret that you created for the token.
    - Select **Web service URL**, and check the web address.

    The following illustration shows these feature parameters set to the values that Edicom provided to Microsoft for testing purposes. The values that you enter will differ. Edicom provides these values to you when you're onboarded.



5. The copy of the feature is always created as a **Draft** version. Regardless of whether you made changes, you must complete, publish, and deploy the feature as described in Complete and deploy a Globalization feature.

Outbound flow pipeline

To review the processing pipeline, on the **Setups** tab, go to **Feature setup**, select the derived document type that you want, and then select **Edit**. The outbound flow consists of the following actions:

1. **Transform document** – Generate a format that can be sent to Edicom.
2. **Integrate with Edicom** – Submit the generated invoice to Edicom.
3. **Get status from Edicom for an invoice** – Fetch the signed XML from Edicom. This document might not be immediately available, because the PAC needs time to generate it.

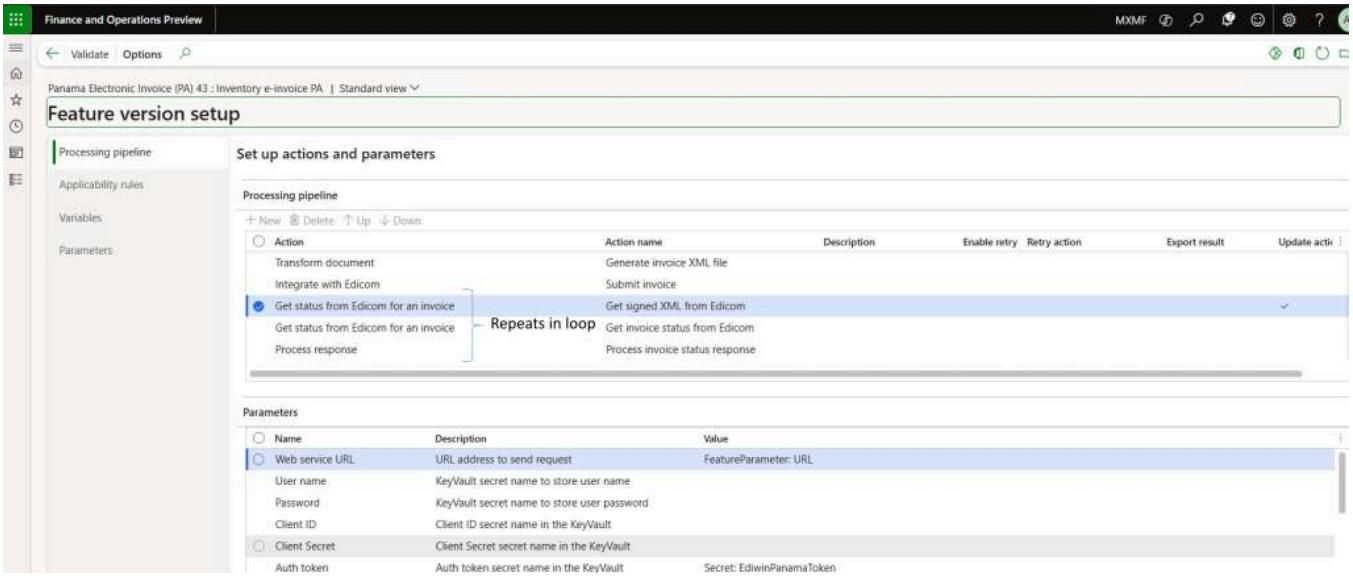
> **Note**
>
> At this point in the flow, the concept of *update actions* comes into play. Notice that the **Update action** checkbox is selected for this step. Therefore, this step and all

subsequent steps run in a loop until the system determines that a terminal state has been reached.

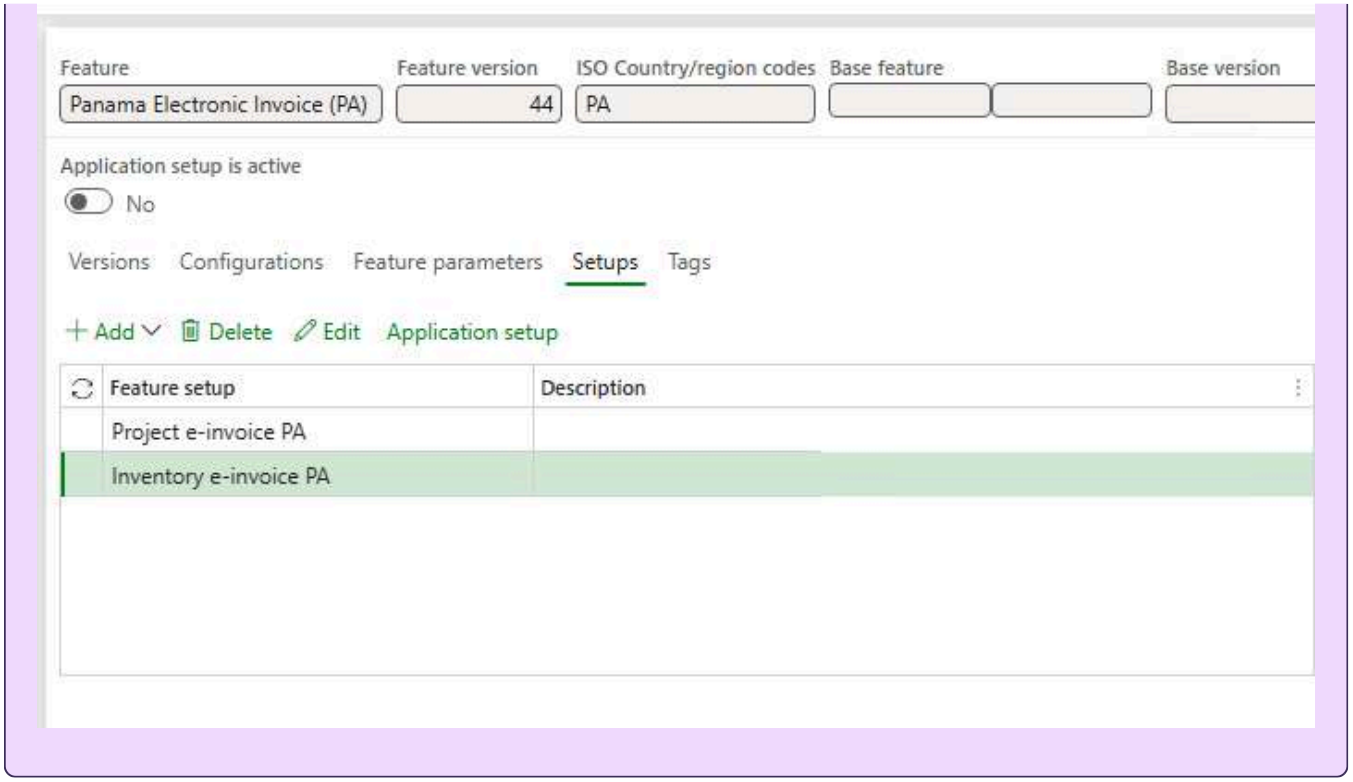
4. **Get status from Edicom for an invoice** – Fetch the status of the submitted invoice from Edicom.

5. **Process response** – Process the received response to determine whether a terminal state has been reached.

    - If the status response indicates a failure, the pipeline is terminated, and the submission is marked as failed.
    - If the response indicates successful submission to the DGI Panamanian Fiscal Authorities, the pipeline is marked as completed.

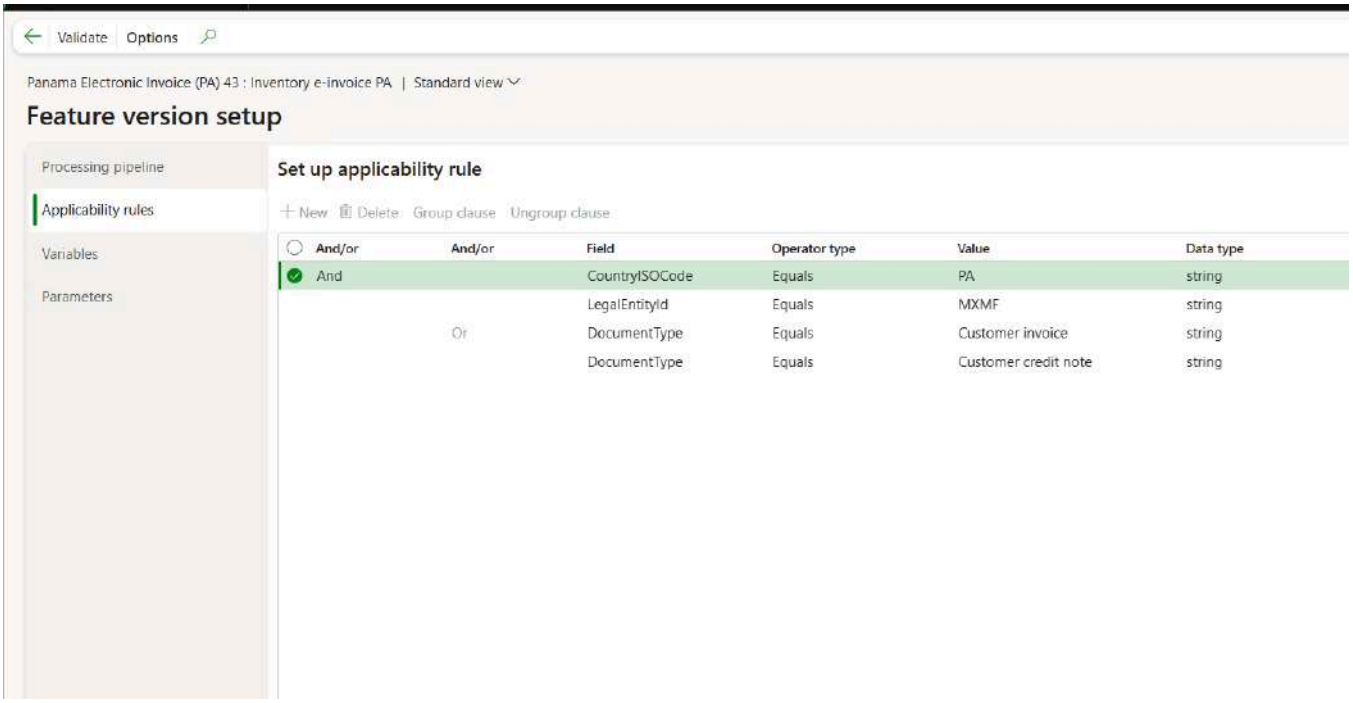


> ⓘ **Note**
>
> There is a similar setup for each format that is included in the **Panama electronic invoice (PA)** feature. For Panama, there are two setups: one for project invoices (**Project e-invoice PA**) and one for inventory and free text invoices (**Inventory e-invoice PA**).



Applicability rules

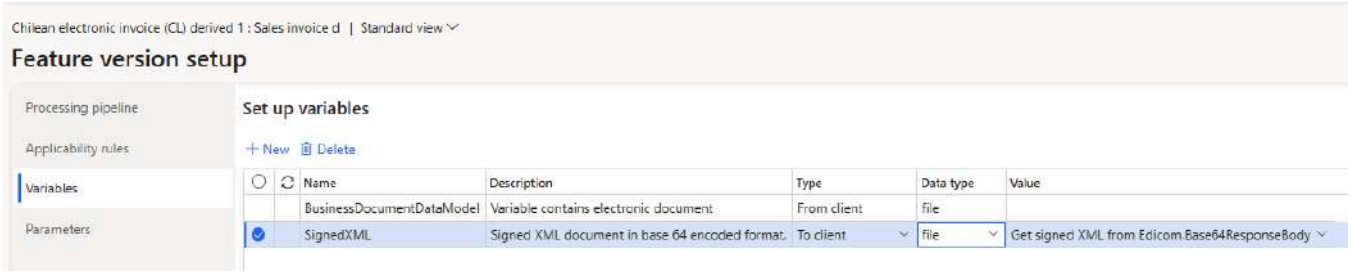
Applicability rules must be correctly configured to provide context, so that the exact Electronic invoicing Globalization feature that must run in the Electronic Invoicing service can be found. Applicability rules are provided out of the box by checking the legal entity in the International Organization for Standardization (ISO) country/region code. This particular feature setup supports all three types of invoices: customer invoices, debit notes, and credit notes.



Variables

The following variables are used in the outbound data flow actions of the Panamanian feature. (These variables are provided out of the box.)

- **BusinessDocumentDataModel** – The Business Document Data model variable that is received from Finance or Supply Chain Management and transformed into the format that is required for submission.
- **SignedXML** – The signed XML variable that is sent back to Finance or Supply Chain Management. This variable contains the base64-encoded response body from the **Get Signed XML from Edicom** step. It's used in the response types to save the signed XML that is obtained from Edicom as an attachment to the invoice journal. It's also used to generate printable reports through QR codes.



After you import the **Electronic invoicing for Panama** feature that includes the out-of-box default feature setup, follow the steps in the next section to configure electronic documents.

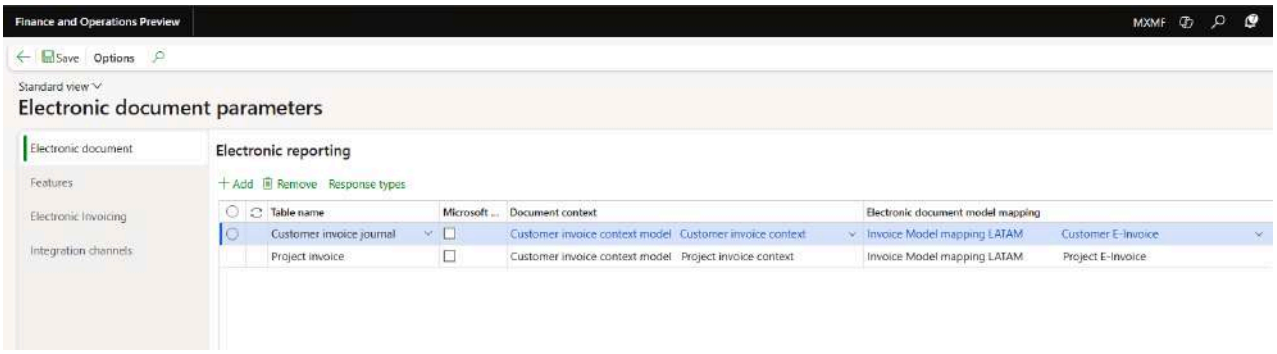
Configure electronic document parameters

1. Make sure that the country/region-specific ER configurations for the document context and electronic document model mapping that are required for Panama are imported. For more information, see Set up Electronic document parameters.

2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

3. In the **Electronic document** section, add records for the **Customer Invoice journal** and **Project invoice** table names.

4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.



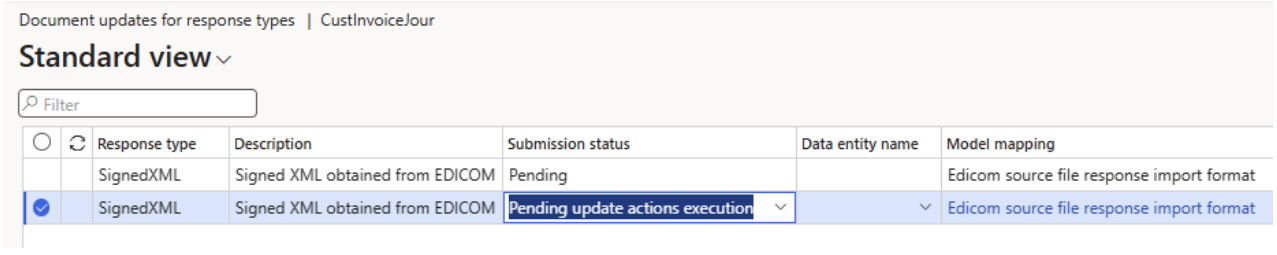
5. Save your changes, and close the page.

6. For each table name, follow these steps:

   a. Select **Response types**, select **New** to create a response type that you will get from the back end, and enter the following values:

      - In the **Response type** field, enter **SignedXML** (the default value). (See the Variables section of this article.)
      - In the **Description** field, enter any meaningful name. Alternatively, leave the field blank.
      - In the **Submission status** field, select **Pending**.
      - In the **Model mapping** field, select **Edicom source file response format**.

   b. Repeat the preceding step, but set the **Submission status** field to **Pending update actions execution**. In this way, you can run a subset of the actions in the processing pipeline in a loop to continuously pull updated statuses and other information for the submitted documents from the tax authority.



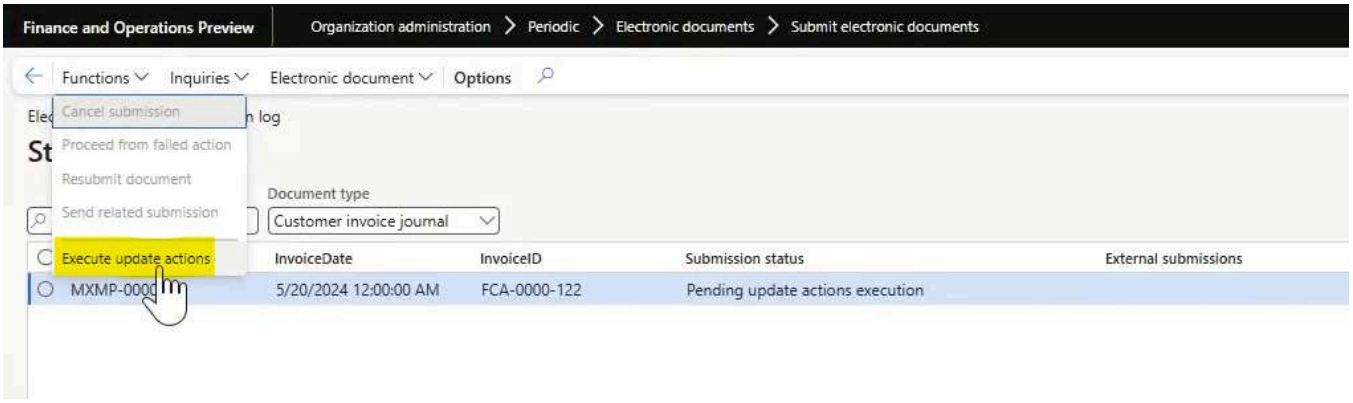
> **Note**
>
> The response includes the signed XML that is obtained from Edicom. This signed XML is stored in the system as an attachment to the corresponding invoice journal. It will eventually be used to generate printable invoices through QR codes.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information about how to generate electronic invoices, see Submit electronic documents.

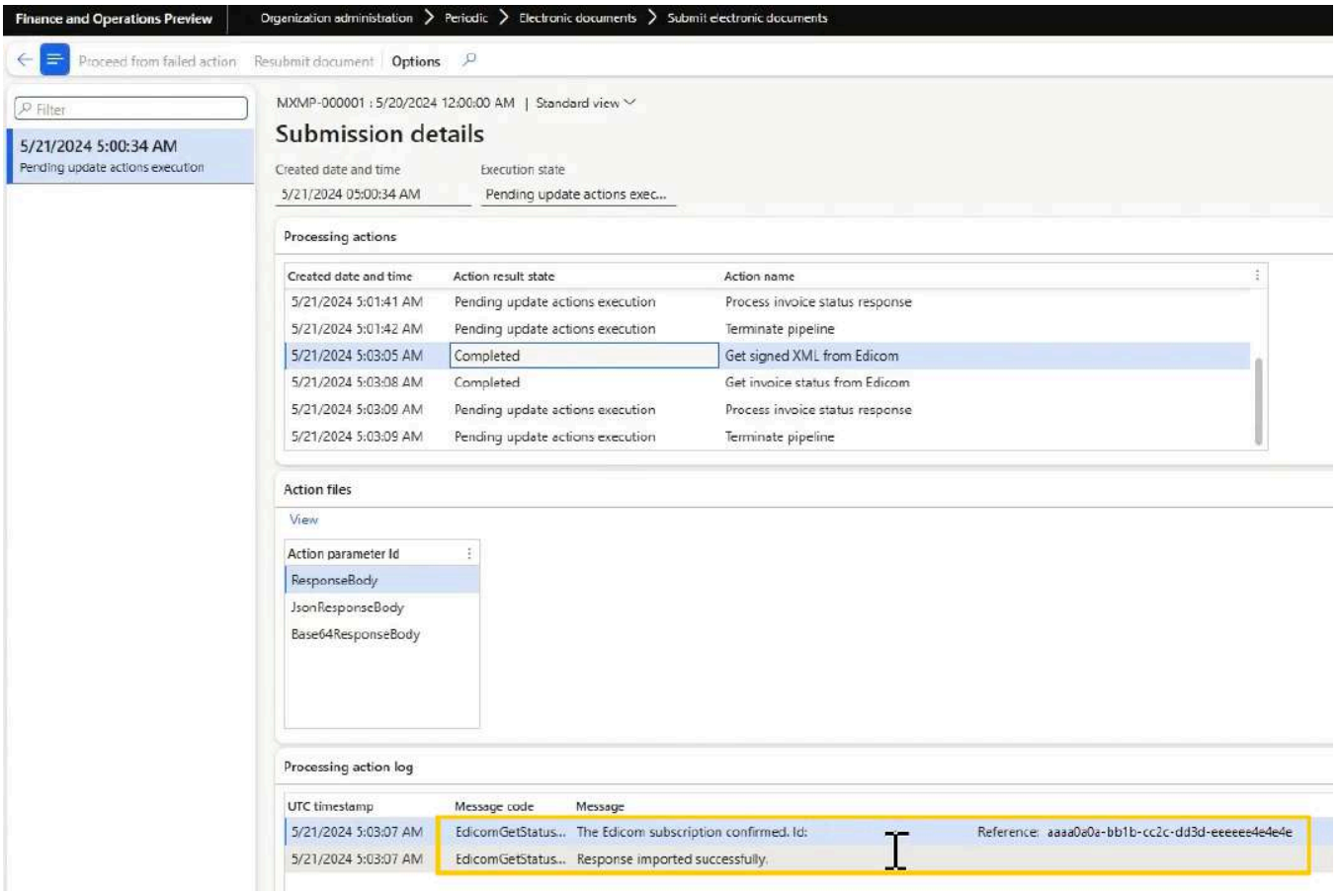
To inquire about the results of a submission, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**, and select the required document type. For more information, see Work with Electronic document submission log.

For Panama, after you submit the invoice, the submission status is set to **Pending update actions execution**. The response body is probably empty for the signed XML and the call to get the invoice status. An empty response body indicates that the XML wasn't available immediately after submission. To address the pending status, a function that is named **Execute update actions** is used. This function resumes the pipeline from the action that is marked as an update action. It then runs all subsequent actions in the pipeline in a loop. The status should change to **Executing** again. Then, after a few seconds, it reverts to **Pending update actions execution**.

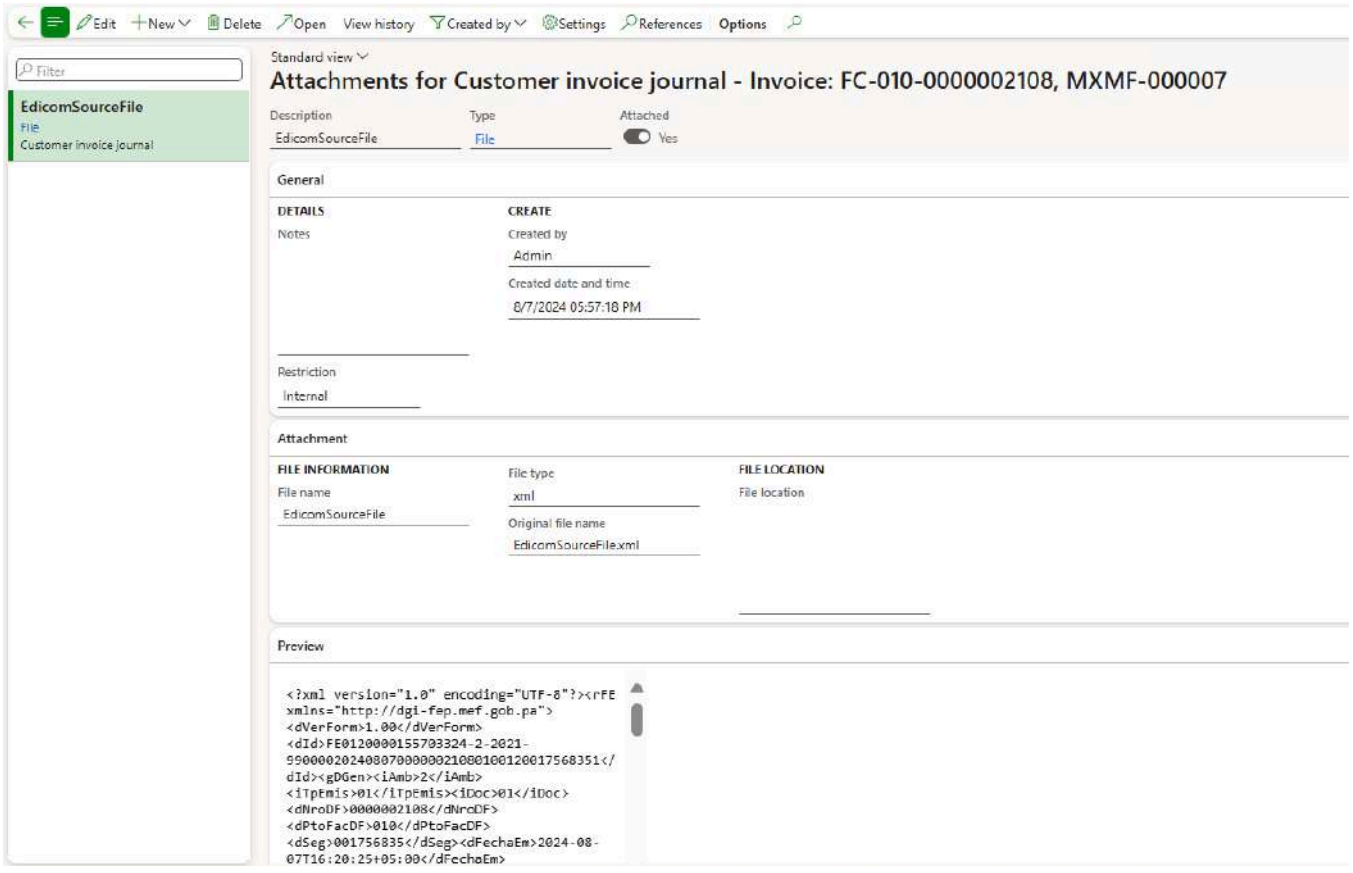


> **Note**
>
> By setting a recurrence schedule, you can configure the **Execute update actions** function to run in batch mode on a periodic basis.

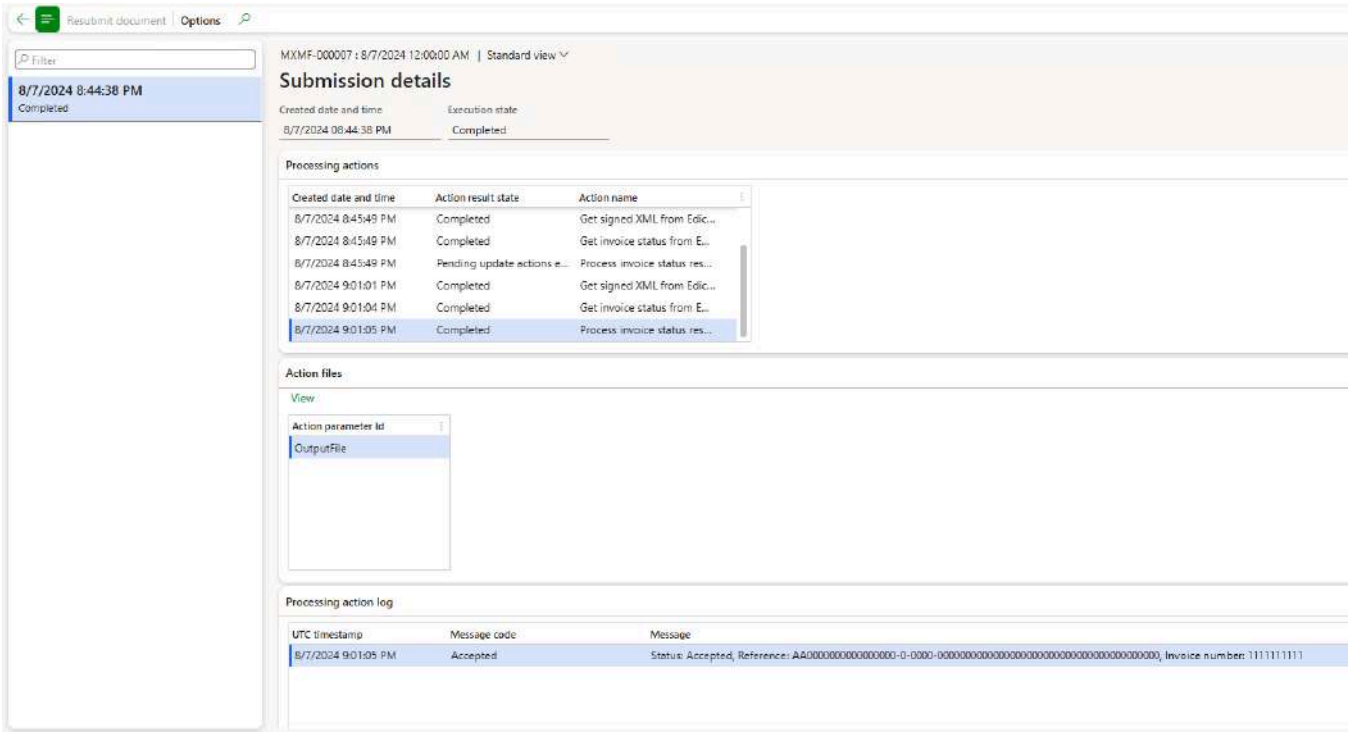
When you review the submission details, you should notice that the steps run again. This time, the signed XML is received.



As a result of the completed outbound flow, the signed XML is attached to the invoice journal as **EdicomSourceFile**.



For Panama, the **Process response** action completes the pipeline after a few minutes.



More resources

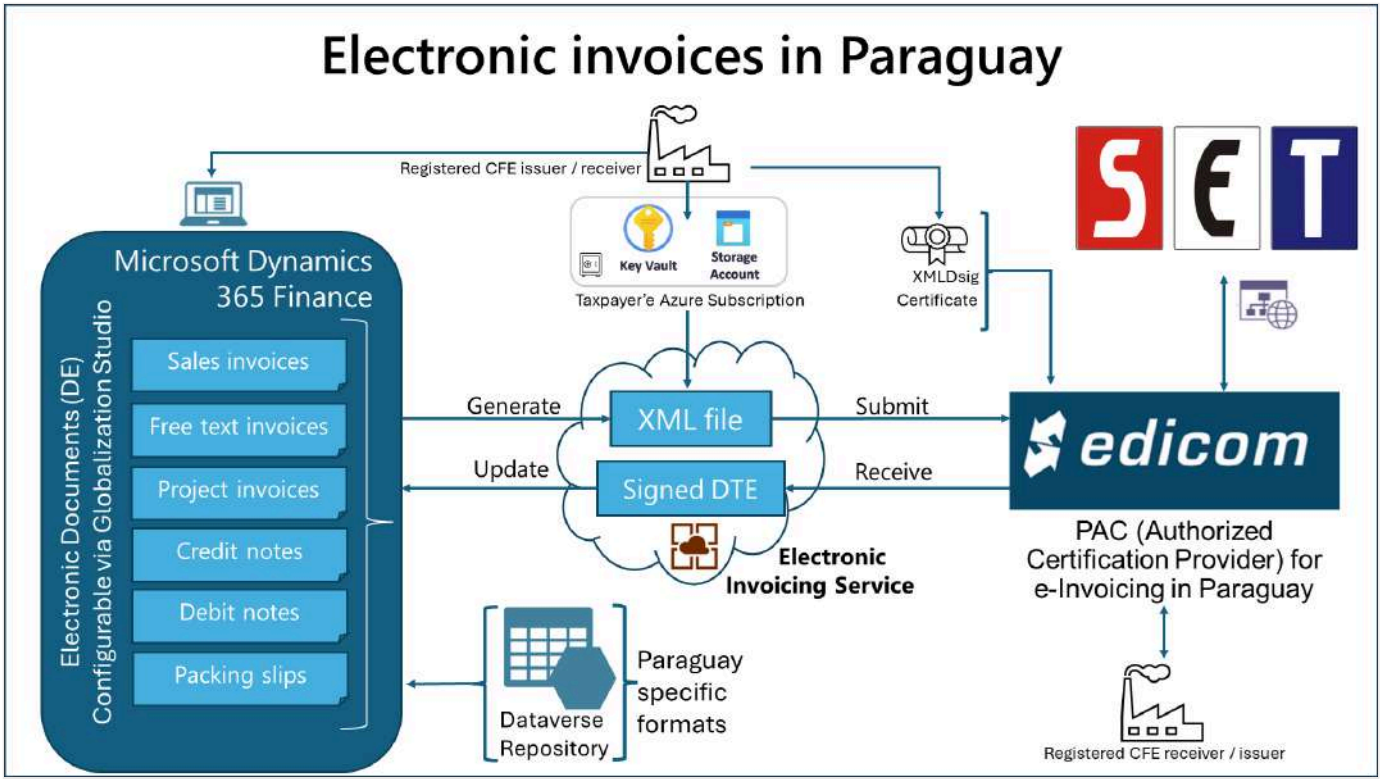
- Electronic Invoicing service overview
- Get started with Electronic invoicing service administration
- Setting up Electronic Invoicing
- Electronic Invoicing service independent software vendor (ISV) last-mile connector
- Dynamics 365 Country expansion: localizations for LATAM countries | June 27, 2024
- Dynamics 365 Country expansion: localizations for LATAM countries | D365 FastTrack Tech Talks (youtube.com)

Last updated on 04/13/2026

Get started with electronic invoicing for Paraguay

This article helps you set up electronic invoicing for Paraguay. It lists configuration steps that depend on the country or region in Microsoft Dynamics 365 Finance and Microsoft Dynamics 365 Supply Chain Management. These steps supplement Electronic invoicing setup. For last mile integration with the Paraguayan tax authority, Microsoft partners with Edicom.

After you set up electronic invoicing, generate, digitally sign, and submit XML invoice files to the Edicom authorized certification provider (PAC) according to the regulatory requirements in Paraguay.



> **Note**
>
> This electronic invoicing approach uses an invoicing service that's available only in cloud deployments of Finance or Supply Chain Management.

Prerequisites

Before you start, make sure you meet these prerequisites.

1. Set up the Paraguayan legal entity. Learn more in Set up a legal entity and tax information for Paraguay.
2. Review the Electronic invoicing overview.
3. Complete the common Electronic invoicing service configuration described in Electronic invoicing configuration.
4. Enable the following features in Feature management:
    - Electronic invoicing integration
    - Electronic invoicing integration resubmits document from failed action
    - Execute upda1. Import the following Electronic reporting (ER) format configurations. Learn more in Import Electronic reporting (ER) configurations from Dataverse and Import Electronic reporting (ER) configurations.reporting-import-ger-configurations.md).
    - Customer invoice context model
    - Edicom response error log import
    - Edicom Response Processing LATAM
    - Edicom source file response import format
    - Inventory-PY
    - PackingSlip (PY)
    - Project e-invoice (PY)

> **Note**
>
> These formats build on the corresponding LATAM format configurations, which use the **Invoice model LATAM** and **Invoice model mapping LATAM** configurations. The system imports all other required configurations automatically.

Configure the electronic invoicing feature

The **"Paraguay Electronic Invoice (PY) Edicom integration for Paraguay"** feature is an outbound flow that issues the following sales documents.

Expand table

Name	Code	Original name
Electronic sales invoice	1	Factura electrónica de venta
Credit Note	5	Nota de Crédito
Debit Note	6	Nota de Débito

The feature publishes some parameters with default values. Before you deploy the electronic invoicing feature to the service environment, add a feature that's based on the Microsoft-provided feature and complete the common parameters on the **Feature parameters** tab. Review the default values and update them as needed so they reflect your business operations.

Edicom interactions for Paraguay

Paraguay has three core interactions with Edicom in the base flow:

1. **Invoice submission**
2. **Signed XML retrieval**
3. **Status retrieval of the submitted invoice**

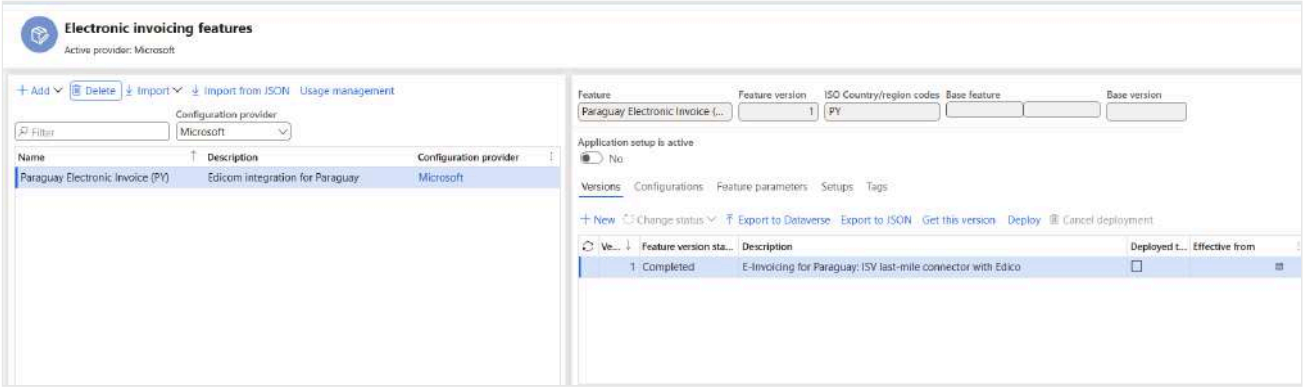
Each of these interactions requires common parameters, such as Edicom connection details and the authentication token provided by Edicom. These parameters are reused across the feature setup for all document types. Edicom supplies these values during the company's onboarding process.

Paraguay electronic invoice (PY) Edicom integration

> **Important**
>
> The **Paraguay Electronic Invoice (PY)** feature is provided by Microsoft. Before it can be used, it requires more configuration, as described in this article. Learn how to configure invoicing features and apply changes in Work with feature setups. For example, in addition to the connection parameters, you can filter specific legal entities so that they're processed in applicability rules. By default, the feature is applicable to all legal entities that have a primary address in Paraguay.

To integrate Edicom for Paraguay electronic invoices, follow these steps:

1. Import the latest version of the **Paraguay Electronic Invoice (PY)** Globalization feature as described in Import features from the repository. The following illustrations show what the feature looks like after you import it from Dataverse.



If you go to the Configuration tab, as shown in the previous screenshot, there's a page displaying information similar to the following table:

Expand table

Name	Version	Status	Country/region code
Edicom Response Processing LATAM	13.2.5	Completed	
Edicom response error log import	13.1	Completed	
Inventory (PY)	296.140.32.25	Completed	PY
PackingSlip (PY)	296.140.32.31	Completed	PY
Project e-invoice (PY)	296.140.19	Completed	PY

> **Note**
>
> The version numbers can change if the formats are updated. The table is for illustrative purposes only.

2. Create a copy of the imported Globalization feature, and select your configuration provider. Learn more in Create a Globalization feature.

3. On the **Versions** tab, confirm that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with Edicom's API.

These parameters are general purpose and apply to all actions, so configure them once here.

> **Tip**
>
> The following table provides **example values** for each feature parameter, so you can understand the type of data that should be entered for each one. These values are **illustrative only** and were provided by Edicom to Microsoft for testing purposes. When you're onboarded, Edicom provides you with your own unique values to use in your configuration.

Expand table

Parameter	Description	Data type	Value
EdicomApplication	Application	String	`<ID_PROVIDED_EDICOM>`
EdicomDomain	Domain	String	`<ID_PROVIDED_EDICOM>`
EdicomGetStatusSchema	Get status schema name	String	IN_DOCUMENT_STATUS_MICROSOFT_UY
EdicomGroup	Group	String	`<GROUP_ID>`
EdicomSignedXMLSchema	Signed XML schema name	String	`EDIWIN-SOURCE-FILE_XML_CFE_ADENDA_<ID_PROVIDED_EDICOM>`
EdicomSubmitInvoiceDestination	Destination name	String	`<ID_PROVIDED_EDICOM>`_EDIWIN
EdicomSubmitInvoiceSchema	Submit invoice schema name	String	OUTBOUND_DOCUMENT_MICROSOFT_UY
EdicomToken	Auth token	Secret name	EdiwinParaguayToken
EdicomWebServiceURL	Web service URL	String	https://ipaasgw.edicomgroup.com

> **Tip**

When configuring the feature parameters for Edicom integration, follow these steps:

- Select **Application**, and enter the service ID number you obtained from Edicom.
- Select **Domain**, and enter the same service ID number.
- Select **Get status schema name**, and enter the provided schema name.
- Select **Group**, and enter the group code you received.
- Select **Signed XML schema name**, and enter the appropriate schema name.
- Select **Destination name**, and enter the service ID number concatenated with "_EDIWIN". For example, if your service ID is 123456, enter **123456_EDIWIN**.
- Select **Submit invoice schema name**, and enter the schema name.
- Select **Auth token**, and choose the name of the secret you created for the token.
- Select **Web service URL**, and verify the web address is correct.

5. The copy of the feature always starts in a **Draft** version. Regardless of whether you made changes, you must complete, publish, and deploy the feature as described in Complete and deploy a Globalization feature.

Outbound flow pipeline

There are three electronic invoice formats for Paraguay included in the **Paraguay Electronic Invoice (PY) feature**. Each format requires its own setup and configuration. The formats are:

- **Inventory (PY)**
- **PackingSlip (PY)**
- **Project e-invoice (PY)**

The following sections provide general configuration steps that apply to all formats. They also explain specific considerations for each format.

Electronic invoice feature

Setups view

After you import the feature, you see five tabs: **Versions**, **Configuration**, **Feature parameters**, **Setups**, and **Tags**. Select **Setups**, then select **View** to open the **Feature version setup** page.

Feature version setup

Use this page to review and set up the feature for Paraguay. The page has four sections: **Processing pipeline**, **Applicability rules**, **Variables**, and **Parameters**.

Processing pipeline

The Set up Actions and Parameters section is divided into two distinct grids:

Top right – processing pipeline

The **Processing pipeline** grid is similar to the following table:

Expand table

Action	Action name	Description	Enable retry	Retry action	Export result	Update action
Transform document	Generate invoice XML file					
Integrate with Edicom	Submit invoice					
Get status from Edicom for an invoice	Get signed XML from Edicom					**Checked**
Get status from Edicom for an invoice	Get invoice status from Edicom					
Process response	Process invoice status response					

> **Note**
>
> The **Get status from Edicom for an invoice** action is crucial in the flow, as it's marked with the **Update action** checkbox. This designation means that this specific step—and all steps following it repeatedly execute in a loop. The system continues this process until the invoice reaches a final, terminal status, ensuring that the latest status from Edicom is always retrieved and processed.

Bottom right – parameters

The lower grid shows the parameters for the selected processing action. These parameters fine-tune the step's behavior. Each entry lists a name, description, data type, and value so you can set

processing logic for business needs.

Transform document parameters

Expand table

Name	Description	Value
Input file	Source file provides to the action the data to be executed.	Variable: BusinessDocumentDataModel
Direction	Direction describes which format is used: import or export	export
Configuration	Configuration describes format that is executed	Inventory e-Invoice: Inventory e-Invoice
Configuration Integration point	Source file provides data to the reporting runtime	InvoiceCustomer
Custom file name	Custom file name from the client	

Linking the correct format in Transform document parameters

When configuring the **Transform document** action in the processing pipeline, ensure the following:

- **Configuration**: Set this to the exact name of the feature format you're configuring.
  - For example:
    - `Project e-Invoice (PY)` → Configuration: `Project e-Invoice (PY)`
    - `Inventory e-Invoice (PY)` → Configuration: `Inventory (PY)`
    - `e-Packing Slip (PY)` → Configuration: `PackingSlip (PY)`
- **Configuration Integration Point**:
  - Use `InvoiceCustomer` for inventory formats (`Inventory (PY)`, `PackingSlip (PY)`).
  - Use `InvoiceProject` for project formats (`Project e-Invoice (PY)`).

> **Tip**
>
> If the feature name contains "Project," use `InvoiceProject`. If it contains "Inventory," use `InvoiceCustomer`. Incorrect settings result in empty XML or missing required data.

Integrate with Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
Http request body	Http request body (can be empty)	Generate invoice XML file: Output file
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinParaguayToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Destination	Destination	FeatureParameter: EdicomSubmitInvoiceDestination
Schema	Schema	FeatureParameter: EdicomSubmitInvoiceSchema
Duplicates	Duplicates	0
Group	Group	FeatureParameter: EdicomGroup
Custom file name	Custom file name from the client	Generate invoice XML file: File name

Get signed XML from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	

Name	Description	Value
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinParaguayToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomSignedXMLSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference ID of the document submitted to Edicom	Submit invoice: ExternalID

Get invoice status from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinParaguayToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomGetStatusSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference ID of the document submitted to	Submit invoice: ExternalID

Name	Description	Value
	Edicom	

Process response parameters

Expand table

Name	Description	Value
Input file	Response to analyze	Get invoice status from Edicom: Output file
Reporting configuration list	List of configurations used for response parsing	Edicom Response Processing LATAM: Edicom Response Processing LATAM
Custom file name	Custom file name from the client	
Reporting configuration list	List of configurations used for response parsing	Edicom response error log import: Edicom response error log import

Terminate pipeline parameters

Expand table

Name	Description	Value
Terminal state	Desired state of the pipeline when terminating	complete
Number of days	Number of days to wait before terminating	8

> **Note**
>
> - If the status response indicates a failure, the pipeline is terminated, and the submission is marked as failed.
> - If the response shows successful submission to the Paraguayan Tax Authorities (DGI), the process completes after eight days if no further responses arrive.

Applicability rules and feature setup scope

Configure applicability rules so the service can find the correct Electronic Invoicing Globalization feature. These applicability rules are provided out of the box by checking the legal entity's International Organization for Standardization (ISO) country/region code.

Applicability rules for Paraguayan electronic invoice formats

The following applicability rules for each electronic invoice format for Paraguay. Replace `<Legal Entity ID>` with the actual ID of your legal entity.

Inventory PY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	PY	string
	Or	DocumentType	Equals	Customer invoice	string
		DocumentType	Equals	Customer credit note	string
		DocumentType	Equals	Customer debit note	string

Packing slip PY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	PY	string
		DocumentType	Equals	Customer packing slip	string

Project PY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	PY	string
	Or	DocumentType	Equals	Project invoice	string
		DocumentType	Equals	Project credit note	string
		DocumentType	Equals	Project debit note	string

Variables

The following variables are used in the outbound data flow actions of the Paraguayan feature. (These variables are provided out of the box.)

- **BusinessDocumentDataModel** – The Business Document Data model variable that is received from Finance or Supply Chain Management and transformed into the format that is required for submission.
- **SignedXML** – The signed XML variable that is sent back to Finance or Supply Chain Management. This variable contains the base64-encoded response body from the **Get Signed XML from Edicom** step. It's used in the response types to save the signed XML that is obtained from Edicom as an attachment to the invoice journal. It's also used to generate printable reports through QR codes.

Expand table

Name	Description	Type	Data type	Value
BusinessDocumentDataModel	Contains the electronic document from the client	From client	file	
SignedXML	Signed XML document in base64-encoded format	To client	file	Get signed XML from Edicom.Base64ResponseBody

After you import the **Electronic invoicing for Paraguay** feature with the default setup, follow the next steps to set up electronic documents.

Setups > Application setup

To configure the mapping between Dynamics 365 Finance source tables and electronic document formats, go to the **Setups** tab within the Electronic invoice feature section, and then select **Application Setup**. This step lets you define how data from Finance is linked to the correct electronic invoice formats for Paraguay.

> **Note**
>
> Although this configuration isn't set at the general feature level, it acts globally for the selected feature setup. You don't need to repeat these steps for each individual format or feature—one configuration is sufficient for all supported document types within the feature.

The following grid defines the link between the source table in Dynamics 365 Finance, the document context model, and the mapping configuration used for electronic invoicing:

Expand table

Table name	Context model / Context	Electronic document mapping	Mapping name	Description
Customer invoice journal	Customer invoice context model / Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice	Maps customer invoices from the CustInvoiceJour table to the LATAM invoice model for electronic submission.
Project invoice	Customer invoice context model / Project invoice context	Invoice Model mapping LATAM	Project E-Invoice	Maps project invoices to the LATAM invoice model for electronic submission.

> **Tip**
>
> Ensure that the Context and Mapping name match the document type. Incorrect mappings cause errors in XML generation or submission to DGI.

Configure electronic document parameters

To configure electronic document parameters, follow these steps:

1. Import the Paraguay-specific ER configurations for the document context and electronic document model mapping. Learn more in Set up Electronic document parameters.
2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
3. In the **Electronic document** section, add records for the **Customer Invoice journal** and **Project invoice** table names.
4. For each table name, set the **Document context** and **Electronic document model mapping** fields as in step 1.

On the Electronic Document Parameters page in Dynamics 365 Finance, the navigation pane shows the following options:

- **Electronic Document**
- **Features**
- **Electronic Invoicing**
- **Integration Channels**

For the **Electronic Document** option, the **Electronic Reporting** section appears.

Within **Electronic Reporting**, the grid lists the electronic document configurations. It includes associations between:

- The source table name
- The document context model and context
- The electronic document model mapping
- The final mapping name used for generating the electronic invoice.

This configuration controls how Dynamics 365 Finance maps internal data models to external electronic formats for submission to tax authorities or service providers.

Configure the grid as follows:

Table name	Microsoft-owned	Document context model	Document context	Electronic document model mapping	Mapping name
Customer invoice journal	No	Customer invoice context model	Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice
Project invoice	No	Customer invoice context model	Project invoice context	Invoice Model mapping LATAM	Project E-Invoice

1. Save your changes, and close the page.
2. For each table name, follow these steps:

    a. Select **Response types**, then select **New** to create a response type from the back end, and enter these values:

    - In the **Response type** field, enter **SignedXML** (default). See the Variables section.
    - In the **Description** field, enter a meaningful name, or leave it blank.
    - In the **Submission status** field, select **Pending**.
    - In the **Model mapping** field, select **Edicom source file response format**.

    b. Repeat the preceding step, but set the **Submission status** field to **Pending update actions execution**. This setting lets you loop a subset of pipeline actions to pull updated statuses and other information for submitted documents from the tax authority.

Response type	Description	Submission status	Data entity name	Model mapping
SignedXML	Signed XML obtained from EDICOM	Pending		Edicom source file response import format
SignedXML	Signed XML obtained from EDICOM	Pending update actions execution		Edicom source file response import format

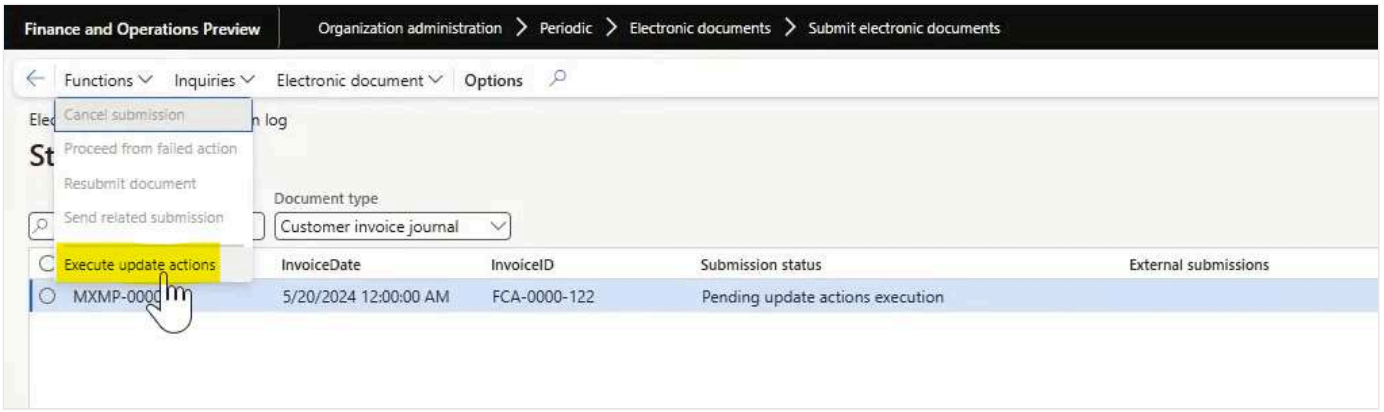
> **Note**
>
> The response includes the signed XML obtained from Edicom. The system stores this signed XML as an attachment to the corresponding invoice journal. The system uses it to generate printable invoices with QR codes.

Issue electronic invoices

After you complete the required configuration steps, generate and submit electronic invoices for posted invoices. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information, see Submit electronic documents.

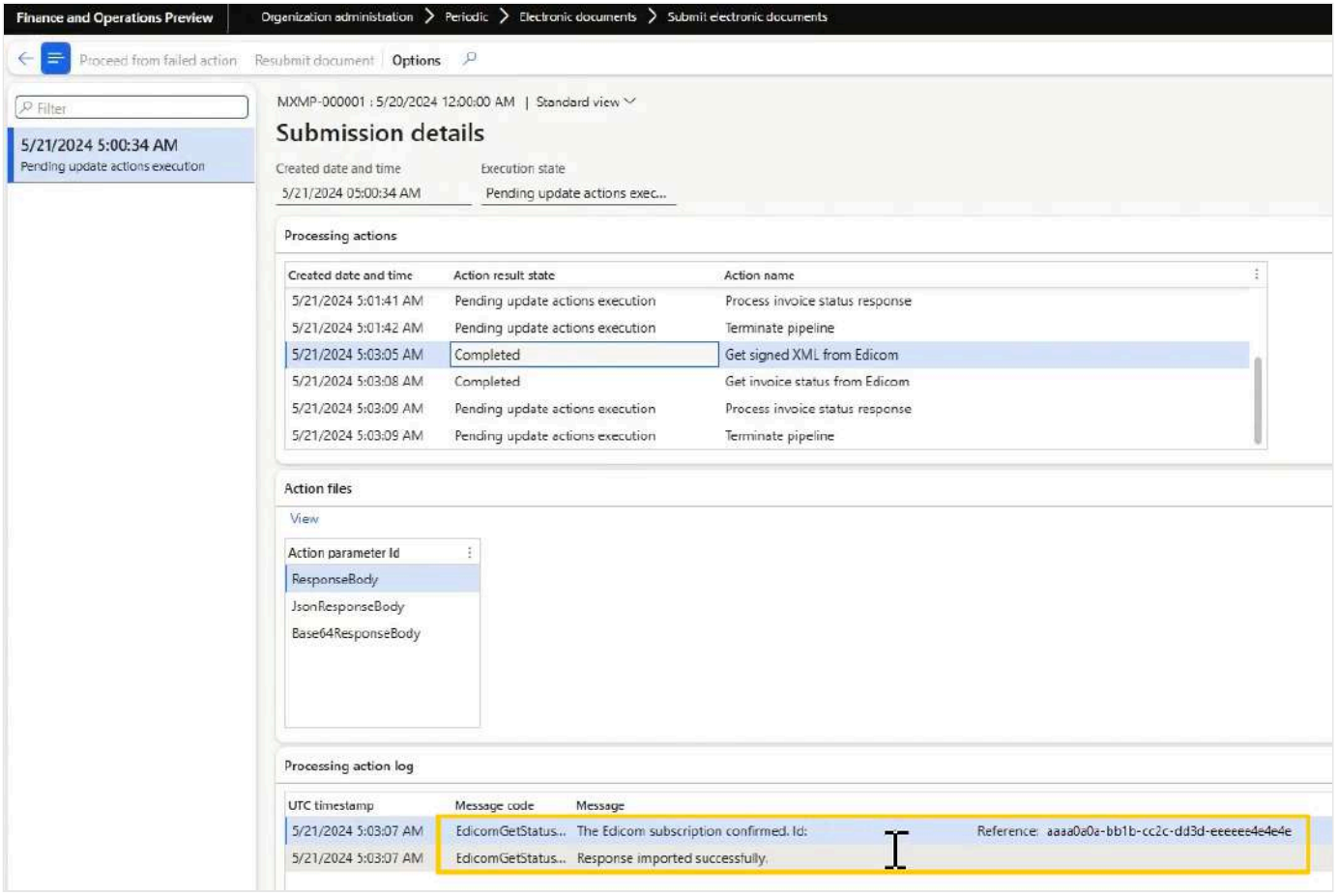
Check submission results. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**, and select the required document type. For more information, see Work with Electronic document submission log.

For Paraguay, after you submit the invoice, the submission status changes to **Pending update actions execution**. The response body might be empty for the signed XML and for the call that gets the invoice status. An empty response body indicates the XML isn't available immediately after submission. To address the pending status, use the **Execute update actions** function. This function resumes the pipeline from the action that's marked as an update action. It then runs subsequent actions in a loop. The status changes to **Executing**, and after a few seconds reverts to **Pending update actions execution**.

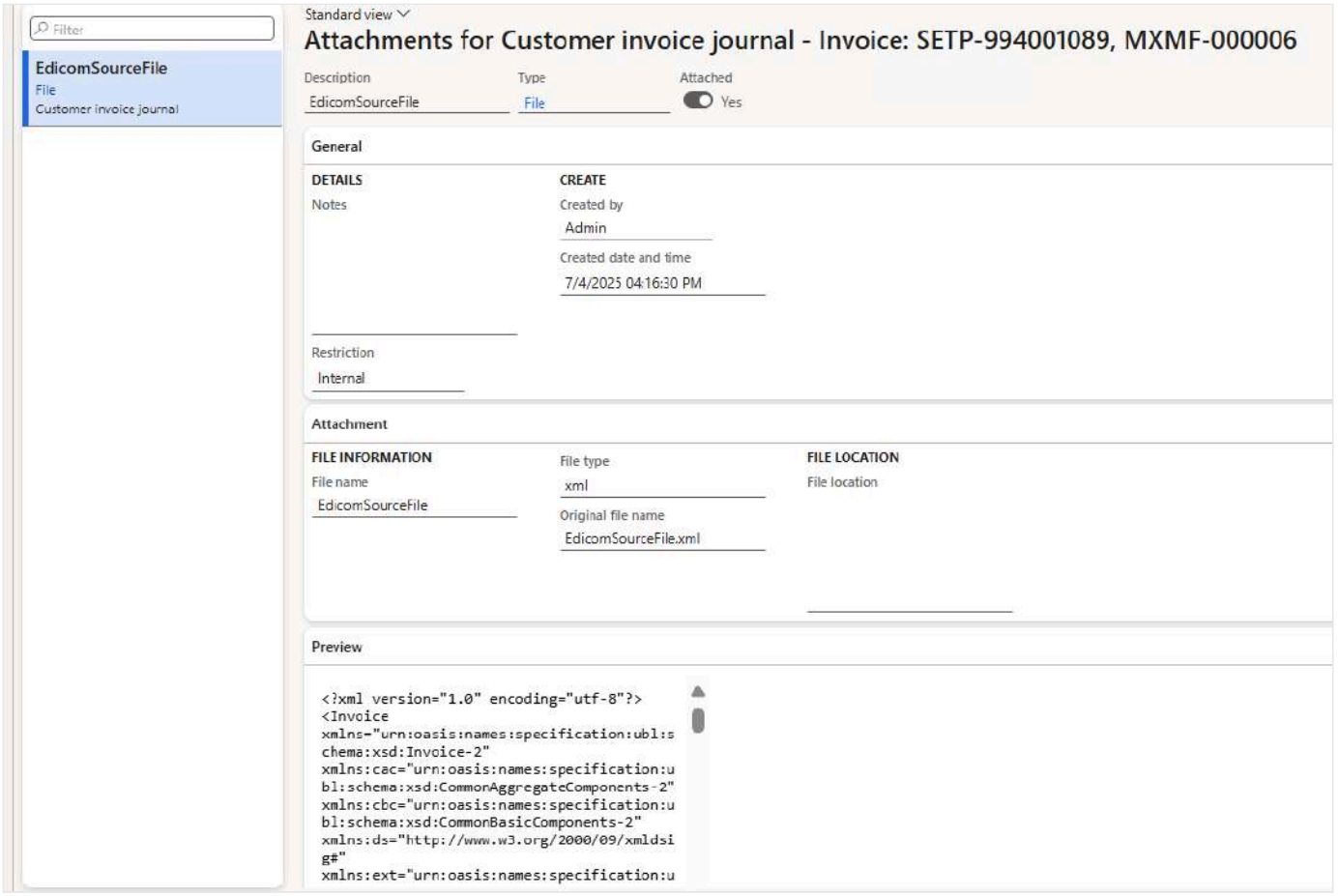


> **Note**
>
> Set a recurrence schedule to run the **Execute update actions** function in batch mode on a periodic basis.

When you review the submission details, you see that the steps run again. This time, the signed XML is received.



After the outbound flow completes, the signed XML is attached to the invoice journal as **EdicomSourceFile**.



For Paraguay, the **Process response** action completes the pipeline after a few minutes.

More resources

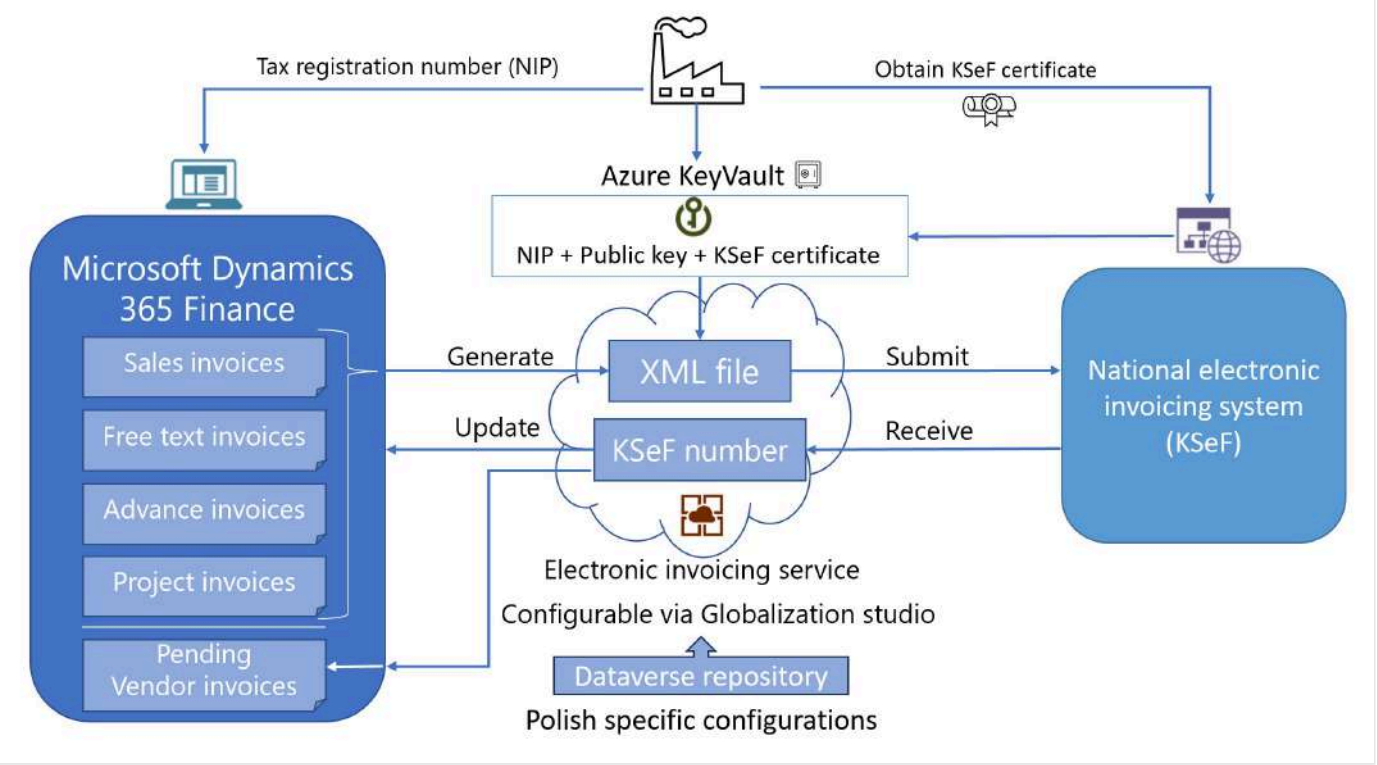
- Electronic invoicing service overview
- Get started with electronic invoicing service administration
- Set up electronic invoicing
- Electronic invoicing service independent software vendor (ISV) last mile connector
- Dynamics 365 Country/Region expansion: localizations for LATAM Countries/Regions | June 27, 2024
- Dynamics 365 Country/Region expansion: localizations for LATAM Countries/Regions | Dynamics 365 FastTrack Tech Talks (youtube.com)
- Finance Localization for LATAM: Update on more Countries/Regions | Dynamics 365 Fast Track TechTalk | Jun 23, 2025
- Finance Localization for LATAM: Update on more Countries/Regions | Dynamics 365 Fast Track TechTalk (youtube.com)

Last updated on 04/13/2026

Electronic invoicing for Poland

This article explains how to get started with electronic invoicing for Poland in Microsoft Dynamics 365 Finance.

After you configure electronic invoicing, you can submit and receive the XML files of electronic invoices according to the regulatory requirements in Poland.



> **Note**
>
> The electronic invoicing approach that this article describes uses an invoicing service that's applicable only to cloud deployments of Finance or Supply Chain Management.

Prerequisites

Before you begin the procedures in this article, complete the following prerequisites:

- The primary address of the legal entity must be in Poland.
- The legal entity must be registered as a taxpayer in Poland and must have a valid tax identification number (*Numer identyfikacji podatkowej* [NIP]).

- Sign in to the Polish National system for electronic invoicing (Krajowy System e-Faktur [KSeF]) via a trusted profile, qualified signature, or qualified seal. Obtain the certificate of the **Authentication in KSeF system** type that the Invoicing Service can use to securely communicate with KSeF. For more information, see the video from the Ministry of Finance: Applying for and Managing Certificates.

  During the initial downloading you obtain the pair of files - the **.crt* file with the certificate itself and the **.key* file for the private key.

  > **Note**
  >
  > Store the private key file securely as since it might not be downloadable during the consequent obtaining of the same certificate.

  To process further with the system configuration, you need to convert the obtained pair of files into one **.pfx* certificate file, which is then stored in your Key Vault and used as a parameter in the feature setup.

  You can use any convenient tool for the certificate conversion. As an example, the following Windows PowerShell script can be used for the conversion.

  > **Note**
  >
  > The sample script isn't supported under any Microsoft standard support program or service. The sample script is provided *AS IS* without warranty of any kind. Microsoft further disclaims all implied warranties including, without limitation, any implied warranties of merchantability or of fitness for a particular purpose. The entire risk arising out of the use or performance of the sample scripts and documentation remains with you. In no event shall Microsoft, its authors, or anyone else involved in the creation, production, or delivery of the script be liable for any damages whatsoever (including, without limitation, damages for loss of business profits, business interruption, loss of business information, or other pecuniary loss) arising out of the use of or inability to use the sample script or documentation, even if Microsoft has been advised of the possibility of such damages.

  PowerShell

  ```
  openssl pkcs12 -export -in <your-ksef-certificate>.crt -inkey <your-ksef-
  ```

```
certificate>.key -out <your-ksef-certificate>.pfx
```

Where:

- *<your-ksef-certificate>.crt* - is the certificate file obtained from KSeF the private key will be combined with
- *<your-ksef-certificate>.key* - is the private key file obtained from KSeF to combine with the certificate
- *<your-ksef-certificate>.pfx* - the output file name that is used for uploading to your Key Vault

- Obtain the **public key** these links depending on the type of environment you're configuring: test, demo, or prod. From the response, copy the value from the second *certificate* element with the *SymmetricKeyEncryption* usage type. You use it during Key Vault parameters configuration described in the next chapter. For more information, see the details provided by KSeF.

- Install the **Electronic invoicing add-in** as described in Install the add-in for Electronic invoicing microservices.

- Activate **Electronic invoicing integration** with Finance or Supply Chain Management as it's described in Enable Electronic invoicing integration.

- Configure the common part of the **Electronic document parameters**.

> **Note**
>
> Service environment configuration is required only if the Regulatory Configuration Service (RCS) experience was previously used to configure the Electronic Invoicing service. Otherwise, keep the **Environment** parameter empty. The system assigns it automatically and makes read-only. For more information, see Service environment configuration.

Create the Azure Key Vault configuration

Configure the common part of the Azure resources required for Electronic invoicing functioning. For more information, see Configure Azure resources for Electronic invoicing.

Add the following required elements in the key vault:

- The secret that holds the shared access signature (SAS) token of the storage account.

- The certificate for the obtained **KSeF certificate**.
- The secret for the **client ID**, which must equal the taxpayer's tax identification number (NIP).
- The secret for the obtained **public key**.

> ⓘ Note
>
> The value of the public key must be wrapped in the following commands.
>
> `-----BEGIN CERTIFICATE-----`
>
> ...
>
> `-----END CERTIFICATE-----`

Configure electronic invoicing Key Vault parameters

To configure electronic invoicing Key Vault parameters, follow these steps:

1. In Finance, go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic invoicing** tab, in the **Key Vault settings** section, in the **Key Vault** field, select the reference to the key vault that you created in the previous section of this article.
3. In the **SAS token secret** field, select the name of the storage account secret URL that you use to authenticate access to the storage account.
4. Select **Key Vault parameters**.
5. On the **Key Vault parameters** page, in the **Certificates** section, select **Add** to create new elements of the appropriate type for each secret that is described in the previous section.
    - The secret for the SAS token of the storage account.
    - The **certificate** element of the **Certificate** type.
    - The **Client ID** element of the **Secret** type.
    - The **Public key** element of the **Secret** type.

> ⓘ Note
>
> The values in the **Name** column should match the names of the secrets that are described in the previous section.

For more information, see Create a Key Vault reference.

Synchronization of the Electronic invoicing service with Finance

After you complete all the configuration steps described in the previous chapters, validate the configuration.

1. In Finance, go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. Select the **Electronic invoicing** tab and select the **Save** menu button.
3. If the configuration is correct, the system shows the *Synchronization with the e-invoicing service was successful* information message. You can continue with the next chapters.
4. If there are synchronization errors, address the errors to achieve successful synchronization.

Import the electronic invoicing feature

To import the electronic invoicing feature, follow these steps:

1. In Finance, go to **Globalization Studio**, and select the **Electronic invoicing** tile. Then import the latest version of the **Polish electronic invoice (PL)** Globalization feature as described in Import features from the repository.

2. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, make sure that the following Electronic reporting configurations are successfully imported as result of the **Polish electronic invoice (PL)** Globalization feature import.

    - **Invoice model**
    - **Invoice model mapping**
    - **Advance invoice model mapping**
    - **Sales e-invoice (PL)**
    - **Project e-invoice (PL)**
    - **Advance e-invoice (PL)**
    - **Customer invoice context model**
    - **Response message model**
    - **Response message model mapping to destination (PL)**
    - **KSeF response message import (PL)**

    > ⓘ **Note**

If the import process doesn't import these Electronic reporting configurations, import them manually as described in Import Electronic reporting (ER) configurations from Dataverse.

3. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, import the latest versions of the following Electronic reporting configurations required for receiving incoming vendor invoices.

    - **Vendor invoice import (PL)**
    - **Vendor invoice Mapping to destination (PL)**
    - **KSeF number message import (PL)**
    - **KSeF number model mapping to destination (PL)**

Configure application-specific parameters

> ⓘ **Note**
>
> The steps in this article apply only if you use the **Sales e-invoice (PL)** format configuration of **version 329.40** or higher.

To configure application-specific parameters (ASP), follow these steps for the **Sales e-invoice (PL)** Electronic Reporting format configuration.

1. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, select the related format configuration.
2. In the **Configurations** menu, in the **Application specific parameters** section, select **Setup**.
3. In the left pane, select the format version that you use for generating e-invoice XML files.
4. In the **Lookups** grid, make sure that the **$ZeroTaxRate** is selected.
5. In the **Conditions** grid, select **Add** to create a new row.
6. In the **Lookup result** column, select a value from the enumeration list that contains allowed values for a zero tax rate. This value is used for **P_12** XML element population.

    > ⓘ **Note**

Select the **Auto** value to let the system automatically determine which zero tax rate value to use.

7. In the **Code** column, select a related sales tax code value for which the respective **Lookup result** value is used.

   > ⓘ Note
   >
   > Instead of selecting specific sales tax codes, you can use ****Blank**** or ****Not blank**** placeholder values for the mass matching.

8. In the **Lookups** grid, make sure that the **TaxExemptsLookUp** is selected.
9. In the **Conditions** grid, select **Add** to create a new row.
10. In the **Lookup result** column, select a value from the enumeration list that contains allowed values for tax exemptions categories. This value is used for **P_19** XML block and its child elements population.
11. In the **Code** column, select a related sales tax code value for which the respective **Lookup result** value is used.

    > ⓘ Note
    >
    > Instead of selecting specific tax exemptions codes, you can use ****Blank**** or ****Not blank**** placeholder values for the mass matching.

12. In the **State** field, select the **Completed** value to indicate that the configuration is finished.
13. Select **Save**, and close the page.

Configure the import channel

To configure the import channel, follow these steps:

1. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, select the **Customer invoice context model** configuration.
2. Select **Create configuration**, and then select **Derive from Name: Customer invoice context model, Microsoft** in the dropdown dialog. This action creates a derived configuration.

3. Open the derived configuration for editing in the designer, and then select **Map model to datasource**.

4. Open the **DataChannel** definition for editing in the designer.

5. In the **Data sources** tree, expand the **$Context_Channel** container.

6. In the **Value** field, select **Edit**, and then enter the name of the data channel. Make a note of the value, because you use it in later configuration steps.

7. Save your changes and complete the derived configuration.

Configure the electronic invoicing feature

Some parameters for the **Polish electronic invoice (PL)** electronic invoicing feature have default values. Before you deploy the electronic invoicing feature to the service, review the default values, and update them as needed to better reflect your business operations.

To review and update the **Polish electronic invoice (PL)** electronic invoicing feature configuration, follow these steps:

1. In Finance, go to **Globalization Studio**, and select the **Electronic invoicing** tile. Then import the latest version of the **Polish electronic invoice (PL)** Globalization feature as described in Import features from the repository.

2. Create a copy of the imported Globalization feature, and select your configuration provider for it, as described in Create a Globalization feature.

3. On the **Versions** tab, verify that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters. You need these parameters for interoperation with Polish KSEF services.

- **EnvironmentName** – select the type of the environment, depending on the implementation stage: *Test*, *Demo*, or *Prod*.
- **PolishClientID** – select the name of the client ID that you previously created.
- **PolishImportDataChannel** – enter the name of the data channel that you previously defined.
- **PolishPublicKey** – select the name of the public key that you previously created.
- **PolishCertificate** – select the name of the certificate that you previously created.

5. On the **Setups** tab, in the grid, select the **Import vendor invoices derived** feature setup and select **Edit**.

6. On the **Import channel** tab, in the **Parameters** section, in the **Value** field for the **Start Date** parameter, enter the date starting from which the import is to be performed.

7. On the **Applicability rules** tab, in the **Set up applicability rule** section, for the line with the **Channel** field, make sure that the previously defined import channel name is defined in the **Value** field.

8. On the **Variables** tab, make a note of the **OutputFile** name, because you use it in later configuration steps.

9. Select **Save**, and close the page.

10. The copy of the feature is always a **Draft** version. Complete and deploy the feature as described in Complete and deploy a Globalization feature.

Configure electronic document parameters

To configure electronic document parameters, follow these steps:

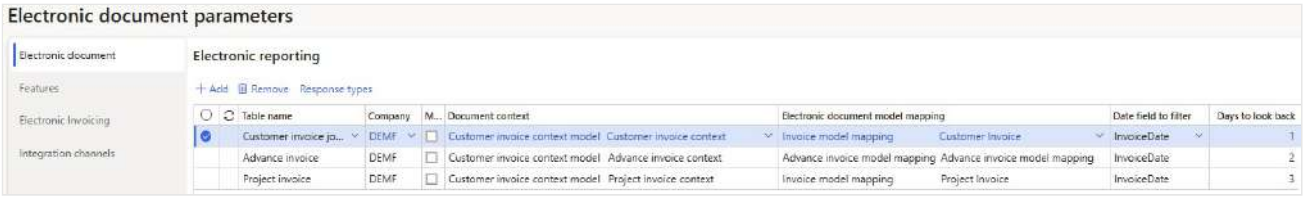
1. In Finance, go to **Organization administration** > **Setup** > **Electronic document parameters**.

2. On the **Electronic document** tab, add records for the **Customer Invoice journal**, **Project invoice**, and **Advance invoice** table names.

3. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with Set up electronic invoicing parameters.

> ⓘ **Note**

If you created the derived analogs of the mentioned above Electronic Reporting configurations, use them instead of standard ones.



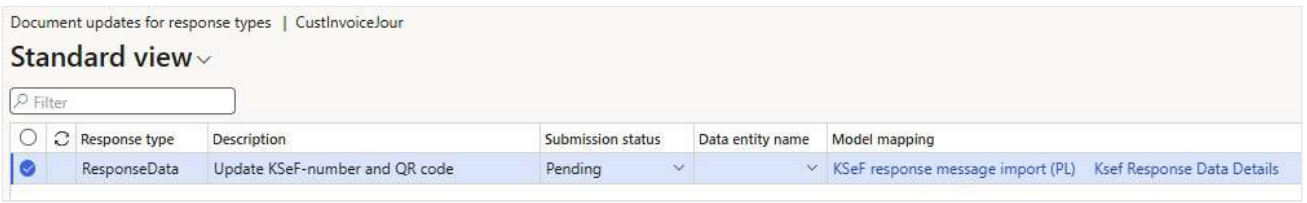
> ⓘ **Note**
>
> To minimize the risk of accidental massive submissions, the system implements default filtering by document dates. In the **Date filed to filter** column, specify the exact selected table's field for filtering. In the **Days to look back** column, define the number of days to subtract from the current date to determine the earliest date for document processing. If you don't configure the **Date filed to filter** and **Days to look back** columns, the **Invoice date** equal to the current date is used by default.

4. For the **Customer Invoice journal** table name, select **Response types**.

5. Select **New** to create a response type, and enter the following values:

   - In the **Response type** field, enter **ResponseData** (the default value).
   - In the **Submission status** field, select **Pending**.
   - In the **Model mapping** field, select **KSeF response message import (PL)**.

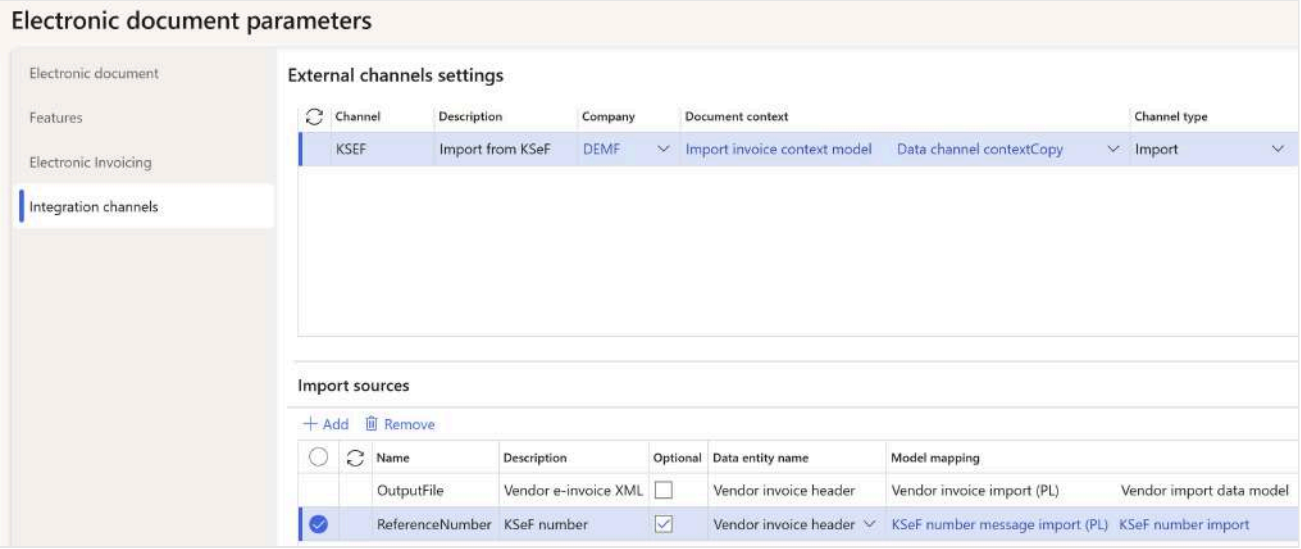
   > ⓘ **Important**
   >
   > Make sure that the **Response message model mapping to destination (PL)** Electronic Reporting configuration is marked as **Default for model mapping**.



> ⓘ **Note**

ResponseData is the default name of the response type. If you change it, make sure that the new name matches the name that you defined for the related variable of the **To client** type in the corresponding feature setups. To validate the variable's value, go to **Globalization Studio**, and select the **Electronic invoicing** tile. On the **Electronic invoicing features** page, verify that the **Polish electronic invoice (PL)** electronic invoicing feature is selected. On the **Setups** tab, in the grid, select the **Submit customer invoice derived** feature setup. Then select **Edit** or **View**, depending on the status of the feature version.

6. Repeat steps 4 and 5 for the **Project invoice** and **Advance invoice** table names.
7. On the **Integration channels** tab, in the **Channels** section, in the **Channel** field, enter the name of the data channel that you previously defined.
8. In the **Company** field, select a required legal entity. In the **Document context** field, select the context configuration that you previously created.
9. In the **Import sources** section, in the **Name** field, enter the **OutputFile** name that is actually used.
10. In the **Data entity name** field, select **Vendor invoice header**. In the **Model mapping** field, reference the **Vendor invoice import (PL)** configuration.
11. Add the **second** import source line, in the **Name** field, enter the **ReferenceNumber** value.
12. In the **Data entity name** field, select **Vendor invoice header**. In the **Model mapping** field, reference the **KSeF number message import (PL)** configuration.



13. Select **Save**, and close the page.

Configure Finance business data

Prerequisites

The legal entity's primary address must be in Poland.

Configure legal entity data

To configure the legal entity data, complete the steps in each of the following sections.

Enter a legal entity's address

To enter a legal entity's address, follow these steps:

1. In Finance, go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity. On the **Addresses** FastTab, add a valid primary address for the legal entity.

   > ⓘ **Note**
   >
   > Make sure that you define the following mandatory address elements: country/region code, ZIP/postal code, city, and building number.

Enter a legal entity's tax registration number

To enter a legal entity's tax registration number, follow these steps:

1. In Finance, go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select a legal entity. On the **Tax registration** FastTab, in the **Tax registration number** field, enter a valid tax registration number for the legal entity. This number is the seller's tax identification number (NIP).

Configure customer data

To configure the customer data, complete the steps in each of the following sections.

Enter a customer's address

To enter a customer's address, follow these steps:

1. In Finance, go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer. On the **Addresses** FastTab, add a valid address for the customer.

> ⓘ Note
>
> - For addresses in Poland, make sure that you define the following mandatory elements: country/region code, ZIP/postal code, city, and building number.
> - For foreign addresses, make sure that you define at least the following mandatory elements: country/region code and city.

Enter a customer's tax registration number

To enter a customer's tax registration number, follow these steps:

1. In Finance, go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer. On the **Invoice and delivery** FastTab, in the **Tax exempt number** field, enter a valid tax registration number for the customer. This number is the buyer's tax identification number (NIP).

Configure sales taxes

Configure the following sales tax codes related data that affect the system behavior in the event of non-taxable, zero-rated, or exempted operations:

1. Tax > Indirect taxes > Sales tax > Sales tax codes > Report setup > **Country/region type**.
2. Tax > Indirect taxes > Sales tax > Sales tax groups > Setup > **Exempt**.
3. Tax > Indirect taxes > Sales tax > Sales tax groups > Setup > **Exempt code**.
4. Tax > Indirect taxes > Sales tax > Sales tax groups > Setup > **Reverse charge**.
5. Tax > Setup > Sales tax > **Sales tax exempt codes**.

> ⓘ Note
>
> Different combinations of the preceding parameters dynamically enable and disable related XML elements in the output files of electronic invoices.

Configure extra data

You can add extra data to invoices. The system uses this data during electronic invoices XML file generation.

Configure electronic document properties

To configure electronic document properties, follow these steps:

1. In Finance, go to **Accounts receivable** > **Setup** > **Electronic document property types**.
2. Select **New** to add a property type.
3. In the **Type** field, enter the value to use as an extra data key (`Klucz`) in the resulting XML file of an electronic invoice.
4. In the **Group description** field, enter the **KSEF** value.

> **Note**
>
> Only the properties with the **KSEF** value in the **Group description** field are considered during the electronic invoice generation process. This data goes in a special section of electronic invoices headers named *DodatkowyOpis*.

1. Select **Applicability** to add an applicable table.
2. On the **Electronic document property type applicability setup** page, in the **Table name** field, select **Customer invoice journal** and **Project invoice**.
3. Add as many extra document properties as you need.
4. Save your changes and return to the **Electronic document property types** page.

1. Select **New** to add a property type.
2. In the **Type** field, enter **CorrectionType**. Enter the value exactly as shown.

> **Note**
>
> The electronic invoice generation process considers this property regardless of the value in the **Group description** field. This data is applicable only for corrected invoices and is used as

the type of correction populated in the *Faktura/Fa/TypKorekty* XML element.

1. Select **Applicability** to add an applicable table.
2. On the **Electronic document property type applicability setup** page, in the **Table name** field, select **Customer invoice journal** and **Project invoice**.
3. Save your changes and return to the **Electronic document property types** page.
4. Save your changes and close page.

Enter extra invoice data

To enter extra invoice data, follow these steps:

1. In Finance, go to **Accounts payable** > **Inquiries and reports** > **Invoice** > **Invoice journal**.
2. Select an invoice in the list. On the Action Pane, on the **Invoice** tab, in the **Properties** group, select **Electronic document properties**.
3. For all the properties related to *DodatkowyOpis* block, enter the required values. The system uses these values in the respective `Wartosc` elements in the resulting XML file of an electronic invoice.
4. For the **CorrectionType** property specifically, enter **1**, **2**, or **3**. If you don't define this property, or leave it empty, or enter any value other than 1, 2, or 3, the default value of the correction type is **1**.

> **Note**
>
> You can enter extra data for project invoices in a similar way at **Project management and accounting** > **Project invoices** > **Project invoice**.
>
> Extra data apply to the invoice header level only.

Configure the data for incoming electronic invoices matching

You must configure the following types of master data to provide a match for incoming electronic invoices:

- Vendors
- Products
- Units

Configure vendors

To configure vendors, follow these steps:

1. In Finance, go to **Accounts payable** > **Vendors** > **All vendors**, and select a vendor.
2. On the **Invoice and delivery** FastTab, in the **Tax exempt number** field, enter a valid value. The vendor's tax exempt number identifies the vendor during the import process for incoming electronic invoices. If the system doesn't find a vendor with matching data, the import process fails and shows a related error message.

Configure products

To configure products, follow these steps:

1. In Finance, go to **Product information management** > **Products** > **Released products**, and select a product.

2. On the Action Pane, on the **Purchase** tab, in the **Related information** group, select **External item description**.

3. In the **Vendor relation** field, select the vendor or vendor group that the product's external identification is being set up for.

4. In the **External item number** field, enter the identification number of the product for a specific vendor or the group of vendors. External item numbers identify the product during the import process for incoming electronic invoices. If the system doesn't find a product with matching criteria, the import process fails and shows a related error message.

   > ⓘ **Note**
   >
   > You should populate the external item number in the invoice line level **Indeks** element of incoming XML files of vendor electronic invoices.
   >
   > Symmetrically, while generating outgoing electronic invoices XML files for customers, the **Indeks** element is populated from the **External item number** defined for a specific customer in the related released product's **Sell** tab, in the **Related information** group, in the **External item description**.

Configure units

To configure units, follow these steps:

1. In Finance, go to **Organization administration** > **Setup** > **Units** > **Units**.

2. Select a unit, and then select **External codes**.
3. On the **External codes** page, in the **Overview** section, in the **Code** field, enter a code that corresponds to the selected unit.
4. In the **Value** section, in the **Value** field, enter the external code to match with the unit codes from incoming electronic invoices during the import process.

> **Note**
>
> External unit codes make sense only if incoming electronic invoices contain explicitly defined units. If incoming electronic invoices don't contain explicitly defined units, you can skip step 4.

Issue outgoing electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information about how to generate and submit electronic invoices, see Submit electronic documents.

You can check the results of a submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log** and selecting the required document type. For more information about the submission log, see Work with Electronic document submission log.

Configure printable invoice layouts

To enable QR code printing in invoices, follow these steps:

1. In Finance, go to **Accounts receivable** > **Setup** > **Forms** > **Form setup**.
2. Select **Print management**.
3. Select the **Customer invoice** report, and then, in the **Report format** field, reference the **SalesInvoice.ReportPL** layout.
4. Select the **Free text invoice** report, and then, in the **Report format** field, reference the **FreeTextInvoice.ReportPL** layout.
5. Select the **Sales advance invoice** report, and then, in the **Report format** field, reference the **CustAdvanceInvoice.ReportPL** layout.

If you use project invoices, follow these steps:

1. In Finance, go to **Project management and accounting** > **Setup** > **Forms** > **Form setup**.
2. Select **Print management**.
3. Select the **Project invoices without billing rules** report, and then, in the **Report format** field, reference the **PSAProjInvoice.ReportPL** layout.
4. Select the **Project invoices with billing rules** report, and then, in the **Report format** field, reference the **PSAContractLineInvoice.ReportPL** layout.
5. Select the **User defined project invoice** report, and then, in the **Report format** field, reference the **PSAManageInvoice.ReportPL** layout.

> **Note**
>
> The QR code that appears on the printouts of invoices represents the URL that takes you to the official portal of the **KSeF** system, where you can find the details of the related electronic invoice.
>
> QR codes are printed only for invoices that the **KSeF** system successfully submitted, validated, and accepted. Printing of QR codes for **Offline24** mode isn't supported in this implementation.

Receive incoming electronic invoices

To receive electronic invoices, follow these steps:

1. In Finance, go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. Select **OK**, and then close the page.

During the import process, the system tries to automatically match incoming electronic vendor invoices with existing confirmed purchase orders.

If the system doesn't find a purchase order, it raises a warning but continues to import the products on invoice lines as **Non-stock** items. Therefore, it expects that the products belong to an item model group where the **Stocked product** checkbox is cleared in the inventory policy.

If no related **Non-stock** products exist, the system tries to import invoice lines by referring to a default item. You must configure the default item in the system as a released product where the code is defined exactly as **DEFAULT_ITEM**. In addition, the product must belong to an item model group where the **Stocked product** checkbox is cleared in the inventory policy. If you don't

configure a default item in the system, the import process fails, and a related error message is shown.

To view the receipt logs for electronic invoices, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**. For more information about how to receive electronic invoices, see Receive electronic documents.

To view successfully received invoices, go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.

Additional resources

- Electronic invoicing overview

Last updated on 08/06/2026

Electronic invoicing in Saudi Arabia

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to get started with phase two of electronic invoicing for Saudi Arabia in Microsoft Dynamics 365 Finance.

This article provides information that will help you get started with phase two of electronic invoicing for Saudi Arabia. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Dynamics 365 Supply Chain Management. These steps complement the steps that are described in Electronic invoicing setup.

Prerequisites

Before you begin the procedures in this article, complete the following prerequisites:

- Become familiar with electronic invoicing as it's described in Electronic Invoicing service overview.
- Do the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.
- Import the **Saudi Arabian Zatca submission (SA)** version **14** or later and **Saudi Arabian ZATCA compliance check (SA)** version **20** or later electronic invoicing features from the repository. For more information, see Import features from the repository.

Configure the application-specific parameters

Before you begin, make sure that the following Electronic reporting (ER) format configurations are imported:

- Sales e-invoice (SA)
- Project e-invoice (SA)

To configure the application-specific parameters, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic reporting** tile, and then select the **Reporting configurations** tile.
2. On the **Configurations** page, select the **Sales e-invoice (SA)** format configuration.
3. On the **Configurations** menu, in the **Application specific parameters** section, selected **Setup**.
4. In the **Lookups** section, make sure that the **PaymentMethodSubstitutionLookup** lookup is selected.
5. In the **Conditions** section, select **Add** to add a condition.
6. In the **Name** column for the new condition, select the method of payment that's defined in the application. Then, in the **Lookup result** column, select a standardized method of payment code according to UN/EDIFACT Code list 4461.
7. Add specific conditions for each method of payment that's defined in the system, and save your changes.

   > **Note**
   >
   > In the **Name** column, you can select the ***Blank*** or ***Not blank*** placeholder value instead of a specific method of payment.

8. In the **State** field, change the state of the application-specific parameters to **Completed**. Then select **Save**.
9. Repeat steps 2 through 8 for the **Project e-invoice (SA)** ER format configuration as required.

Configure electronic document parameters

To configure electronic document parameters, follow these steps:

1. Ensure that the country/region-specific ER configurations required for Saudi Arabia are imported. For more information, see Configure electronic invoicing parameters.
2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

3. On the **Electronic document** tab, add records for the **Customer Invoice journal** and, if required, for the **Project invoice**, **Journal lines**, and **Fiscal transaction document** table names.

4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.

5. For the **Customer Invoice journal** table name, select **Response types**.

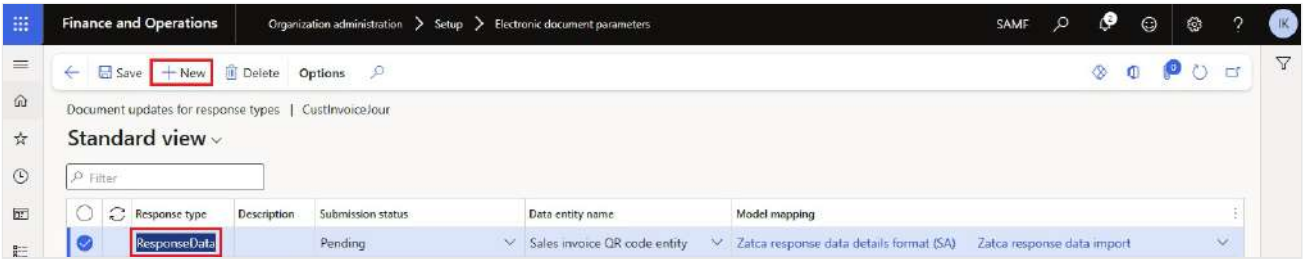
> **Note**
>
> When tax invoices are cleared, the Saudi Arabian tax authority (Zakat, Tax and Customs Authority [ZATCA]) generates a QR code that contains the digital signature for that clearance. This QR code must be imported back into Finance, together with other clearance information that the tax authority returns as a result of the submission of the tax invoices. To achieve this result, you must configure the response types in Finance.

6. Create a response type that has the same name as the related variable in the value of the **Configuration Integration point** parameter in the **Transform document (export), export QRCode** processing pipeline's action for the **Sales invoice** feature setup. The default value is **ResponseData**.

7. In the **Submission status** field, select **Pending**.

8. In the **Data entity name** field, select **Sales invoice QR code entity**.

9. In the **Model mapping** field, select **Zatca response data import**.



10. On the **Electronic invoicing** tab, on the **Number sequences** FastTab, add a record for the number sequence that should be used to count submitted electronic invoices.

Configure electronic invoicing features

> **Note**

This section describes the configuration steps for the **Saudi Arabian Zatca submission (SA)** electronic invoicing feature. These steps assume that the onboarding process is completed, and that you've obtained the Production Cryptographic Stamp Identifier (PCSID). If the onboarding process is still in progress, and you've obtained only the Compliance Cryptographic Stamp Identifier (CCSID), complete the same configuration steps for the **Saudi Arabian ZATCA compliance check (SA)** electronic invoicing feature.

Some of the parameters from the **Saudi Arabian Zatca submission (SA)** electronic invoicing feature are published with default values. Before you deploy the electronic invoicing feature to the service, review the default values, and update them as required so that they better reflect your business operations.

To configure electronic invoicing features, follow these steps:

1. In the **Globalization Studio** workspace, select the **Electronic invoicing** tile.
2. On the **Electronic invoicing features** page, verify that the **Saudi Arabian Zatca submission (SA)** electronic invoicing feature is selected.
3. On the **Versions** tab, verify that the **Draft** version is selected.
4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with ZATCA's API:

    - **ZatcaAPIEndPoint** – Enter the URL of the development portal or the production environment that ZATCA provided.
    - **ZatcaCertificateName** – Select **CCSID** or **PCSID**, depending on your place in the onboarding process.
    - **ZatcaSecretName** – Select **CCSIDSecret** or **PCSIDSecret**, depending on your place in the onboarding process.
    - **ZatcaNumberSequence** – Select the number sequence that you previously created on the **Electronic document parameters** page.

    The following illustration shows an example of a feature parameter setup that includes the URL for ZATCA's production endpoint. Update the values based on the configuration on the **Electronic document parameters** page.

5. On the **Setups** menu, select the **Sales invoice** feature setup, and then select **Edit**.

> **Note**
>
> For the **Saudi Arabian ZATCA compliance check (SA)** electronic invoicing feature, select the **Tax invoice compliance check** feature setup instead.

6. In the **Processing pipeline** section, select **Integrate with Saudi Arabia Zatca service**, and then follow these steps:

   a. Select the **API method name** name.

   b. In the **Value** field, select **Invoice clearance** for tax invoices or **Invoice reporting** for simplified invoices.

   > **Note**
   >
   > Depending on your place in the onboarding process, create either a CCSID or a PCSID, and configure it in advance in the Key Vault parameters.

7. Complete and deploy the **Saudi Arabian Zatca submission (SA)** feature to the service. For more information, see Complete and deploy a Globalization feature.

For more information about business data configuration and processing in Finance, see Customer electronic invoices in Saudi Arabia.

Electronic invoicing onboarding in Saudi Arabia

Onboarding is mandatory for all taxpayers who are subject to electronic invoicing in Saudi Arabia. Taxpayers and their software for e-invoicing must be onboarded with ZATCA. As a result of the onboarding process, taxpayers obtain Cryptographic Stamp Identifiers (CSIDs). CSIDs are required for integration with the electronic invoicing portal that ZATCA manages, and for further submission of electronic invoices.

Onboarding is an essential part of the electronic invoicing configuration. For more information about the onboarding process, see Onboarding for electronic invoicing in Saudi Arabia.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices. For more information about how to generate electronic invoices, see Submit electronic documents.

You can inquire about the results of a submission by going to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log** and selecting the required document type.

Additional resources

- Electronic Invoicing service overview
- Customer electronic invoices in Saudi Arabia

Last updated on 05/04/2026

Onboarding for electronic invoicing in Saudi Arabia

This article explains how to onboard taxpayers and their electronic invoicing software with the Saudi Arabian tax authority in Microsoft Dynamics 365 Finance.

Onboarding is mandatory for all taxpayers who are subject to electronic invoicing in Saudi Arabia. As a result of the onboarding process, taxpayers get Cryptographic Stamp Identifiers (CSIDs). You need CSIDs for integration with the electronic invoicing portal that the Saudi Arabian tax authority (Zakat, Tax and Customs Authority [ZATCA]) manages, and for further submission of electronic invoices.

Prerequisites

- The legal entity must be registered as a taxpayer in Saudi Arabia and must have a valid value-added tax (VAT) registration number.
- The legal entity must have access to the Saudi Arabian Taxation Portal (ERAD).

Onboarding process

The onboarding process consists of two steps:

1. Get a Compliance CSID (CCSID), which ZATCA assigns to perform compliance checks on electronic invoice generation solutions (EGSs).
2. Get a Production CSID (PCSID), which ZATCA assigns to compliant EGSs.

Obtain a CCSID

To obtain a CCSID, follow these steps:

1. In the Saudi Arabian Taxation Portal (ERAD), go to the Onboarding and Management Portal by selecting the relevant tile.

2. On the main landing page of the Onboarding and Management Portal, select the **Onboard new solution unit/device** tile, and then select **Generate OTP code**.

3. Select the number of one-time password (OTP) codes to generate. The number depends on the number of e-invoicing generation units (devices) that you'll use.

4. Save the generated OTP codes so you can use them in later steps.

> **ⓘ Important**
>
> OTP codes are valid for only one hour after they're generated. Ensure you use them within that time.

5. Prepare a configuration file for the certificate signing request (CSR). Create this configuration file as a plain text file that contains the following data.

```txt
oid_section = OIDs
[OIDs]
certificateTemplateName = 1.3.6.1.4.1.311.20.2
[req]
default_bits = 2048
emailAddress = MyEmail@email.com
req_extensions = v3_req
x509_extensions = v3_ca
prompt = no
default_md = sha256
req_extensions = req_ext
distinguished_name = dn
[dn]
C=SA
OU=Riyad Branch
O=Contoso
CN=PREZATCA-Code-Signing
[v3_req]
basicConstraints = CA:FALSE
keyUsage = digitalSignature, nonRepudiation, keyEncipherment
[req_ext]
certificateTemplateName = ASN1:PRINTABLESTRING:PREZATCA-Code-Signing
subjectAltName = dirName:alt_names
[alt_names]
SN=1-TST|2-TST|3-ed22f1d8-e6a2-1118-9b58-d9a8f11e445f
UID=310122393500003
title=1100
registeredAddress= MyAddress
businessCategory=Industry
```

6. Save the CSR configuration file to the same location as the onboarding script, and name it **csr_config.txt**.

7. In the CSR configuration file, update the **emailAddress** value and the following data.

Expand table

Code	Description	Specification
C	The country/region code.	A two-letter code (ISO 3166 Alpha-2).
OU	The name of the organization unit.	For normal taxpayers, the value is free text. For VAT groups, identify the value through the fact that the eleventh digit of the organization identifier is "1". Validate that the input is a 10-digit Tax Identification Number (TIN).
O	The name of the organization or taxpayer.	Free text.
CN	Unique Name or Asset Tracking Number of the Solution Unit provided by the Taxpayer for each Solution unit	Use • **PREZATCA-Code-Signing** value for simulation • your actual number for production
certificateTemplateName	The unique name of the solution or unit.	Use • **ASN1:PRINTABLESTRING:PREZATCA-Code-Signing** value for simulation • **ASN1:PRINTABLESTRING:ZATCA-Code-Signing** for production
SN	Manufacturer or Solution Provider Name, Model or Version and Serial Number. Automatically filled and not by the Taxpayer: Unique identification code for the EGS.	Free text.
UID	The VAT registration number of the taxpayer.	A 15-digit number that begins with "3" and ends with "3".
title	The document type that the taxpayer's solution unit will issue.	Four-digit numerical input that uses "0" and "1" mapped to "TSCZ": "0" = False/Not supported, and "1" = True/Supported. "T" = Tax invoice (standard),

Code	Description	Specification
		"S" = Simplified tax invoice, "C" = For future use, and "Z" = For future use.
registeredAddress	The address of the branch or location where the device or solution unit is primarily situated.	Free text.
businessCategory	The industry or sector that the device or solution will generate invoices for.	Free text.

8. Run the onboarding script that's provided later in this article. Specify the OTP code and the CSR configuration file as input parameters. Here's an example.

    - for CCSID in simulation environment: `.\OnboardingScript.ps1 -action getComplianceCSID -otp 123345 -csrconfig .\csr_config.txt -environment simulation`
    - for CCSID in production environment: `.\OnboardingScript.ps1 -action getComplianceCSID -password 123 -environment production`

    > **Note**
    >
    > The **password** parameter is optional and can be omitted. If you include it, the certificate you generate has the specified password.

9. You receive the CCSID as a certificate file, **CCSID.pfx**. Save the secret for the CCSID as a text file, **CCSIDSecret.txt**. Save the CCSID certificate file in the Microsoft Azure Key Vault certificate, and save the secret in the Key Vault secret. For more information, see Customer certificates and secrets.

10. Configure the related feature setup in the **Saudi Arabian ZATCA compliance check (SA)** Electronic invoicing feature, and reference the CCSID certificate that you saved in the key vault. Use the certificate for communication with the ZATCA electronic invoicing portal.

Compliance check

After you get a CCSID by using the PowerShell script, ZATCA requires that you complete specific compliance checks by submitting sample invoices. This step is a prerequisite for requesting a PCSID.

Ensure that you successfully submit to ZATCA all types of sample invoices that you configured in the CSR configuration file. Use the standard process for issuing electronic invoices. For more information, see Issue electronic invoices in Finance and Supply Chain Management.

Follow the steps in Electronic invoicing in Saudi Arabia - Phase two. Use the **Saudi Arabian ZATCA compliance check (SA)** Electronic invoicing feature and the CCSID that you get.

After you complete the compliance checks, use the PowerShell script to get a PCSID. (See the onboarding script that's provided later in this article.)

> **Note**
>
> If you set the document type in the **title** field of the CSR configuration file to **1000**, you must submit the following three sample invoices for the compliance check:
>
> - Standard Tax Invoice
> - Standard Debit Note
> - Standard Credit Note
>
> If you set the document type to **0100**, you must submit the following three sample invoices for the compliance check:
>
> - Simplified Tax Invoice
> - Simplified Debit Note
> - Simplified Credit Note
>
> If you set the document type to **1100**, you must submit all six sample invoices for the compliance check.

Obtain a PCSID

To obtain a PCSID, you must correctly configure the solution for electronic invoice generation and submission, and the solution must be fully functioning. To achieve this result, you must complete all the required preliminary configuration steps. For more information, see Electronic invoicing in Saudi Arabia - Phase two.

To obtain a PCSID, follow these steps:

1. Ensure that you successfully submit all electronic invoices to ZATCA.
2. Run the onboarding script that's provided later in this article. Specify the CCSID as an input parameter. Here's an example.

   ```
   .\OnboardingScript.ps1 -action getProductionCSID -password 123 -environment production
   ```

   > **Note**
   >
   > The **password** parameter is optional and can be omitted. If you include it, the generated certificate has the specified password.

3. Receive the PCSID as a certificate file in PFX format. Save this PCSID certificate and secret file in the key vault.
4. Configure the related feature setup in the **Saudi Arabian Zatca submission (SA)** Electronic invoicing feature. Include the PCSID certificate and secret in the Key Vault parameters.

After you complete all the configuration steps, the system is ready to use in production mode.

To review the obtained CSIDs on the ZATCA side, use the **Review Existing Cryptographic Stamp Identifier (CSID)** tile on the landing page of the Onboarding and Management Portal. You can access this portal from the main Saudi Arabian Taxation Portal (ERAD).

Onboarding script

> **Note**
>
> Microsoft doesn't support the sample scripts under any standard support program or service. The sample scripts are provided as-is without warranty of any kind. Microsoft disclaims all implied warranties, including without limitation, any implied warranties of merchantability or of fitness for a particular purpose. You assume all risk arising from the use or performance of the sample scripts and documentation. In no event is Microsoft, its authors, or anyone else involved in the creation, production, or delivery of the scripts liable for any damages whatsoever, including without limitation, damages for loss of business profits, business interruption, loss of business information, or other pecuniary loss, arising

from the use of or inability to use the sample scripts or documentation, even if Microsoft has been advised of the possibility of such damages.

1. Use the following Windows PowerShell script to get a CCSID and a PCSID.

PowerShell

```
#Saudi Arabian electronic invoice onboarding script
#Version 1.3 (adds -environment switch for Simulation vs Production, better error
output)
param(
    $action,
    $otp,
    $csrconfig,
    $password,
    [ValidateSet("simulation","production")]
    $environment = "production"
)
$env:path = $env:path + ";C:\Program Files\Git\usr\bin"

# Select the correct ZATCA base URL for the target environment.
# Simulation onboarding (Fatoora Simulation portal) and Production onboarding
# (Fatoora portal) use DIFFERENT endpoints on the same gateway host.
if ($environment -eq "simulation")
{
    $baseUri = 'https://gw-fatoora.zatca.gov.sa/e-invoicing/simulation'
}
else
{
    $baseUri = 'https://gw-fatoora.zatca.gov.sa/e-invoicing/core'
}
$complianceUri = "$baseUri/compliance"
$productionCsidUri = "$baseUri/production/csids"
Write-Host "Target environment: $environment"
Write-Host "Compliance endpoint: $complianceUri"

# Helper to print a meaningful error (HTTP status + response body when available)
function Write-ZatcaError($err, $contextMessage)
{
    Write-Host "`nZatca service communication error:"
    Write-Host $err.Exception.Message
    try {
        if ($err.Exception.Response -ne $null)
        {
            $statusCode = [int]$err.Exception.Response.StatusCode
            Write-Host "HTTP status code: $statusCode"
            $stream = $err.Exception.Response.GetResponseStream()
            if ($stream -ne $null)
            {
                $reader = New-Object System.IO.StreamReader($stream)
                $body = $reader.ReadToEnd()
                if ($body) { Write-Host "Response body: $body" }
```

```
            }
        }
    } catch { }
    Write-Host $contextMessage
    Write-Host "Also confirm you are using the correct environment endpoint
(simulation vs. core) and that the OTP was generated in the matching Fatoora portal
and used within 1 hour."
}

if ($action -eq "getComplianceCSID")
{
    if (-not (Test-Path -Path $csrconfig))
    {
        throw "CSR configuration file does not exist, please make sure to provide a
valid file path for the '-csrconfig' parameter."
    }

if ($otp -eq $null)
    {
        throw "OTP code is not provided, please carry correct parameters."
    }

#Generate private key
    openssl ecparam -name secp256k1 -genkey -noout -out privatekey.pem
    Write-Host "Private key generated."

#Generate public key
    openssl ec -in privatekey.pem -pubout -conv_form compressed -out publickey.pem
    Write-Host "Public key generated."

#Generate CSR(Certificate signing request)
    openssl base64 -d -in publickey.pem -out publickey.bin
    openssl req -new -sha256 -key privatekey.pem -extensions v3_req -config
$csrconfig -out .\taxpayer.csr
    openssl base64 -in taxpayer.csr -out taxpayerCSRbase64Encoded.txt
    $CSRbase64Encoded = Get-Content -path taxpayerCSRbase64Encoded.txt -Raw
    $CSRbase64Encoded = $CSRbase64Encoded -replace "`n",""
    $CSRbase64Encoded = $CSRbase64Encoded -replace "`r",""

#Init request for CCSID
    $postParams = @{"csr"=$CSRbase64Encoded} | ConvertTo-Json
    $postHeader = @{
           "Accept"="application/json"
           "OTP"=$otp
           "Content-Type"="application/json"
           "Accept-Version"="V2"}

try
    {
        $response = Invoke-WebRequest -Uri $complianceUri -Method POST -Body
$postParams -Headers $postHeader
    }
    catch
    {
        Write-ZatcaError $_ "Please make sure the OTP code in script parameter and
```

```
Serial Number (SN) in configuration file are valid."
        Write-Host "The process of obtaining a Compliance CSID (CCSID) is
interrupted."
    }

if ($response -ne $null)
    {
        $response = $response | ConvertFrom-Json
        $requestId = $response.requestID
        Write-Host "Request ID:"
        Write-Host $requestId
        $requestId | Out-File -FilePath .\requestId.txt -Encoding utf8 -NoNewline

$CCSIDbase64 = $response.binarySecurityToken
        Write-Host "`nCompliance CSID received from Zatca:"
        Write-Host $CCSIDbase64
        $CCSID =
[System.Text.Encoding]::UTF8.GetString([System.Convert]::FromBase64String($CCSIDbase64
))
        $CCSIDCertString = "-----BEGIN CERTIFICATE-----`n" + $CCSID + "`n" + "-----END
CERTIFICATE-----"

$CCSIDSecret = $response.secret
        Write-Host "`nCompliance CSID secret received from Zatca:"
        Write-Host $CCSIDSecret

$CCSIDStringFileName = "CCSIDString.txt"
        $CCSIDSecretFileName = "CCSIDSecret.txt"
        $CCSIDCertFileName = "CCSID.pem"
        $CCSIDFolderPath = Get-Location
        $CCSIDCertFilePath = Join-Path $CCSIDFolderPath $CCSIDCertFileName
        $CCSIDStringFilePath = Join-Path $CCSIDFolderPath $CCSIDStringFileName
        $CCSIDSecretFilePath = Join-Path $CCSIDFolderPath $CCSIDSecretFileName

$Utf8NoBomEncoding = New-Object System.Text.UTF8Encoding $False
        [System.IO.File]::WriteAllLines($CCSIDCertFilePath, $CCSIDCertString,
$Utf8NoBomEncoding)
        [System.IO.File]::WriteAllLines($CCSIDStringFilePath, $CCSIDbase64,
$Utf8NoBomEncoding)
        [System.IO.File]::WriteAllLines($CCSIDSecretFilePath, $CCSIDSecret,
$Utf8NoBomEncoding)

openssl pkcs12 -inkey privatekey.pem -in CCSID.pem -export -passout pass:$password -
out CCSID.pfx
        Write-Host "`nCertificate is saved to CCSID.pfx file and secret is saved to
CCSIDSecret.txt file."
        Write-Host "The process of obtaining a Compliance CSID (CCSID) is complete,
please process the compliance check and do not delete or move any created files before
getting PCSID."
    }
}

if ($action -eq "getProductionCSID")
{
    if (-not (Test-Path -Path requestId.txt))
```

```
    {
        throw "'requestId.txt' file is missing, please make sure you're running the
script in the same location where the results of getting the CCSID are stored."
    }
    if (-not (Test-Path -Path CCSIDString.txt))
    {
        throw "'CCSIDString.txt' file is missing, please make sure you're running the
script in the same location where the results of getting the CCSID are stored."
    }
    if (-not (Test-Path -Path CCSIDSecret.txt))
    {
        throw "'CCSIDSecret.txt' file is missing, please make sure you're running the
script in the same location where the results of getting the CCSID are stored."
    }

$requestId = Get-Content -path requestId.txt -Raw
    $requestId = $requestId -replace "`n",""
    $requestId = $requestId -replace "`r",""
    Write-Host "Request ID is:" $requestId
    $CCSID = Get-Content -path CCSIDString.txt -Raw
    $CCSID = $CCSID -replace "`n",""
    $CCSID = $CCSID -replace "`r",""
    Write-Host "`nCompliance CSID read locally:"
    Write-Host $CCSID
    $CCSIDSecretString = Get-Content -path CCSIDSecret.txt -Raw
    $CCSIDSecretString = $CCSIDSecretString -replace "`n",""
    $CCSIDSecretString = $CCSIDSecretString -replace "`r",""
    Write-Host "`nCompliance CSID secret read locally:"
    Write-Host $CCSIDSecretString
    $AuthTokenString = $CCSID + ":" + $CCSIDSecretString
    $BasicAuthToken = "Basic " +
[Convert]::ToBase64String([System.Text.Encoding]::UTF8.GetBytes($AuthTokenString))

#Init request for Production CSID (PCSID)
    $postParams = @{"compliance_request_id"=$requestId} | ConvertTo-Json
    $postHeader = @{
           "Accept"="application/json"
           "Authorization"=$BasicAuthToken
           "Content-Type"="application/json"
           "Accept-Version"="V2"}

try
    {
        $response = Invoke-WebRequest -Uri $productionCsidUri -Method POST -Body
$postParams -Headers $postHeader
    }
    catch
    {
        Write-ZatcaError $_ "Please make sure the compliance check process has been
done before obtaining a Production CSID (PCSID)."
        Write-Host "The process of obtaining a Production CSID (PCSID) is
interrupted."
    }

if ($response -ne $null)
```

```
    {
        $response = $response | ConvertFrom-Json
        $PCSIDbase64 = $response.binarySecurityToken
        Write-Host "`nProduction CSID received from Zatca:"
        Write-Host $PCSIDbase64

$PCSID =
[System.Text.Encoding]::UTF8.GetString([System.Convert]::FromBase64String($PCSIDbase64
))
        $PCSIDCertString = "-----BEGIN CERTIFICATE-----`n" + $PCSID + "`n" + "-----END
CERTIFICATE-----"

$PCSIDSecret = $response.secret
        Write-Host "`nProduction CSID secret received from Zatca:"
        Write-Host $PCSIDSecret

$PCSIDCertFileName = "PCSID.pem"
        $PCSIDSecretFileName = "PCSIDSecret.txt"
        $PCSIDFolderPath = Get-Location
        $PCSIDCertFilePath = Join-Path $PCSIDFolderPath $PCSIDCertFileName
        $PCSIDSecretFilePath = Join-Path $PCSIDFolderPath $PCSIDSecretFileName

$Utf8NoBomEncoding = New-Object System.Text.UTF8Encoding $False
        [System.IO.File]::WriteAllLines($PCSIDCertFilePath, $PCSIDCertString,
$Utf8NoBomEncoding)
        [System.IO.File]::WriteAllLines($PCSIDSecretFilePath, $PCSIDSecret,
$Utf8NoBomEncoding)

openssl pkcs12 -inkey privatekey.pem -in PCSID.pem -export -passout pass:$password -
out PCSID.pfx
        Write-Host "`nCertificate is saved to PCSID.pfx file and secret is saved to
PCSIDSecret.txt file."
        Write-Host "The process of obtaining a Production CSID (PCSID) is complete."
    }
}
```

1. Save the output .pfx certificate file that you receive in the key vault.

Additional resources

- Electronic invoicing overview
- Get started with Electronic invoicing service administration
- Get started with Electronic invoicing

Last updated on 09/01/2026

Generate and submit simplified electronic invoices for Saudi Arabia

> **Note**
>
> The functionality described in this article supports the Phase 2 requirements of the E-invoicing implementation in the Kingdom of Saudi Arabia.

This article provides an overview of the functionality for simplified electronic invoices (e-invoices) that are available for Saudi Arabia in Microsoft Dynamics 365 Commerce. It also provides guidelines for setting up the functionality.

The electronic invoicing functionality that's available for Saudi Arabia in Commerce provides the following capabilities:

- Generate an XML file of a simplified e-invoice when a sales transaction concludes in Commerce point of sale (POS).
- Generate a cryptographic stamp, or a digital signature, for the simplified e-invoice.
- Generate and print a QR code for the simplified e-invoice that includes the cryptographic stamp.
- Submit the simplified e-invoice from Commerce headquarters to Saudi Arabian tax authorities (Zakat, Tax and Customs Authority [ZATCA]) for reporting purposes.

For more information about the electronic invoicing requirements for Saudi Arabia, see the E-Invoicing portal by ZATCA.

The high-level, end-to-end process flow in Commerce for Saudi Arabia is as follows:

1. When the checkout process completes for a sales transaction in POS, POS uses Commerce Scale Unit (CSU) to send Commerce runtime (CRT) a request to generate and digitally sign a simplified e-invoice. The Fiscal registration framework and an internal connector are used to implement the generation and digital signing of simplified e-invoices.

   > **Note**

> If POS is in offline mode, the local copy of CRT on the POS machine generates and digitally signs the e-invoice.

2. CRT generates a simplified e-invoice in an XML format. Electronic reporting (ER) is used to implement the XML format of e-invoices for Saudi Arabia. A common format is used for simplified e-invoices in Commerce and regular tax e-invoices in Dynamics 365 Finance.
3. CRT sends Commerce headquarters a request to provide a digital certificate.
4. Commerce headquarters extracts the digital certificate from Azure Key Vault and sends it back to CRT. For more information about how Commerce handles digital certificates, see the Configure the digital signature parameters section of this article.

   > **Note**
   >
   > If POS is in offline mode, the local copy of CRT uses a digital certificate that's locally installed on the POS machine.

5. CRT calculates the invoice hash, digitally signs the e-invoice data, and generates a QR code that includes the invoice hash and digital signature data. CRT also updates the XML invoice with the invoice hash and digital signature data. The e-invoice, the invoice hash, the QR code, and other information are saved in the channel database in a fiscal transaction that's linked to the sales transaction.
6. POS requests a sales receipt from CRT. CRT builds the receipt, including the QR code, and sends it back to POS. POS sends the receipt to the receipt printer.
7. Commerce headquarters uses Commerce Data Exchange (CDX) to download the sales transaction data together with fiscal transactions from CSU. The data is stored in the headquarters database throughout the life of your production environment.
8. Commerce headquarters extracts the simplified e-invoice in the XML format from the fiscal transaction that's linked to the sales transaction. It then submits the e-invoice to ZATCA. The submission is done by integrating with the Electronic Invoicing service. For more information about the common electronic invoicing capabilities that are available to Saudi Arabia, see Customer electronic invoices in Saudi Arabia.

Set up Commerce for Saudi Arabia

This section describes the Commerce settings that are specific to and recommended for Saudi Arabia. For more information about common Commerce features and settings, see Commerce home page.

As a prerequisite, complete the setup of the electronic invoicing functionality for Saudi Arabia, including the configuration of the Electronic invoicing service. For more information, see Customer electronic invoices in Saudi Arabia.

To use the Commerce functionality that's specific to Saudi Arabia, complete the following tasks:

- Set the **Country/region** field to **SAU** (Saudi Arabia) in the primary address of the legal entity.
- Set the **ISO code** field to **SA** (Saudi Arabia) in the POS functionality profile of every store that's located in Saudi Arabia.

Specify the following settings for Saudi Arabia. Run appropriate distribution jobs after you complete the setup.

1. Enable Commerce features for Saudi Arabia in the **Feature management** workspace.
2. Set up value-added tax (VAT) per the Saudi Arabian VAT regulations.
3. Configure custom fields and receipt formats to print QR codes on receipts and comply with the local regulatory requirements.
4. Configure the fiscal registration functionality for Saudi Arabia to enable the generation and digital signing of simplified e-invoices.
5. Configure digital certificates and other parameters of digital signing for the Commerce channel side.
6. Specify the ER configurations that Commerce headquarters should use to generate simplified e-invoices in POS and submit them.
7. Configure e-invoice submission for simplified e-invoices that are generated in POS.
8. Enable the digital signature in offline mode.

Mix and match transactions aren't supported in Saudi Arabia

A transaction can't include both sales and return lines together. All lines in a transaction must be of the same type. To enforce this requirement, enable the **Prohibit mixing sales and returns in one receipt** setting in the POS functionality profile. To enable it, go to **Retail and Commerce** > **Channel setup** > **POS setup** > **POS profiles** > **Functionality profiles**, select the functionality profile that is linked to the store, and on the **Functions** FastTab, select the **Prohibit mixing sales and returns in one receipt** check box.

Enable features for Saudi Arabia

In the **Feature management** workspace, enable the following features:

- KSA Electronic-Invoicing capability for the fiscal integration framework
- (Saudi Arabia) Electronic invoicing integration

Set up VAT per Saudi Arabian requirements

Create sales tax codes, sales tax groups, and item sales tax groups. Set up sales tax information for products and services. For more information about how to set up and use sales tax, see Sales tax overview and Configure sales tax codes.

Specify sales tax groups and enable the **Prices include sales tax** option for stores that are located in Saudi Arabia.

Configure custom fields to use in receipt formats for sales receipts

You can configure the language text and custom fields that are used in the POS receipt formats. The default company of the user who creates the receipt setup should be the same legal entity where you create the language text setup. Alternatively, create the same language texts in both the user's default company and the legal entity of the store that the setup is created for.

On the **Language text** page, on the **POS** tab, add the following records for the labels of the custom fields for receipt layouts. The **Language ID**, **Text ID**, and **Text** values that are shown in the table are examples. You can change them to meet your requirements. However, the **Text ID** values that you use must be unique, and they must be equal to or higher than 900001.

Expand table

Language ID	Text ID	Text
en-US	900001	QR Code

On the **Custom fields** page, add the following records for the custom fields for receipt layouts. The **Caption text ID** values must correspond to the **Text ID** values that you specified on the **Language text** page.

Expand table

Name	Type	Caption text ID
INVOICEQRCODE_SA	Receipt	900001

On the **Commerce parameters** page, on the **Configuration parameters** tab, add the following records.

Expand table

Name	Value
RetailEInvoiceFeature_SA.QrCodeHeight	200
RetailEInvoiceFeature_SA.QrCodeWidth	200

Configure receipt formats

For each required receipt format, change the value of the **Print behavior** field to **Always print**. You must also configure hardware profiles to support receipt printers and to enable Hardware station. For more information about how to work with POS peripherals, see Peripherals.

In the Receipt format designer, add the following custom fields to the appropriate receipt sections. Field names correspond to the language texts that you defined in the previous section.

- **Footer:** Add the following field:
  - **QR Code** – This field prints the QR code for the receipt.

For more information about how to work with receipt formats, see Set up and design receipt formats.

Set up fiscal registration

Complete the fiscal registration setup steps that are described in Set up the fiscal integration for Commerce channels:

1. Set up a fiscal registration process. Note the settings of the fiscal registration process that are specific to Saudi Arabia.
2. Set error handling settings.
3. Enable manual execution of deferred fiscal registration.

Configure the fiscal registration process

To enable the fiscal registration process for Saudi Arabia in Commerce headquarters, follow these steps:

1. Download configuration files for the fiscal document provider and the fiscal connector from the Commerce software development kit (SDK):
   a. Open the Dynamics 365 Commerce Solutions repository.
   b. Open the latest available release branch.
   c. Open **src** > **FiscalIntegration** > **ElectronicInvoiceSaudiArabia**.
   d. Download the configuration file for the fiscal connector, **ConnectorSample.xml**.
   e. Download the configuration file for the fiscal document provider, **DocumentProviderSample.xml**.

2. Go to **Retail and Commerce** > **Headquarters setup** > **Parameters** > **Shared parameters**. On the **General** tab, set the **Enable fiscal integration** option to **Yes**.

3. Go to **Retail and Commerce** > **Channel setup** > **Fiscal integration** > **Fiscal connectors**, and load the fiscal connector configuration file that you downloaded earlier.

4. Go to **Retail and Commerce** > **Channel setup** > **Fiscal integration** > **Fiscal document providers**, and load the fiscal document provider configuration file that you downloaded earlier.

5. Go to **Retail and Commerce** > **Channel setup** > **Fiscal integration** > **Connector functional profiles**. Create a new connector functional profile, and select the document provider and the connector that you loaded earlier. Then update the data mapping settings as required.

6. Go to **Retail and Commerce** > **Channel setup** > **Fiscal integration** > **Connector technical profiles**. Create a new connector technical profile, and select the connector that you loaded earlier. Set the connector type to **Internal**, and update the other connection settings as required.

7. Go to **Retail and Commerce** > **Channel setup** > **Fiscal integration** > **Fiscal connector groups**, and create a new fiscal connector group for the connector functional profile that you created earlier.

8. Go to **Retail and Commerce** > **Channel setup** > **Fiscal integration** > **Fiscal registration processes**. Create a new fiscal registration process, create a fiscal registration process step, and then select the fiscal connector group that you created earlier.

9. Go to **Retail and Commerce** > **Channel setup** > **POS setup** > **POS profiles** > **Functionality profiles**, and select a functionality profile that's linked to the store where the registration

process should be activated. On the **Fiscal registration process** FastTab, select the fiscal registration process that you created earlier. On the **Fiscal services** FastTab, select the connector technical profile that you created earlier.

10. Go to **Retail and Commerce** > **Retail and Commerce IT** > **Distribution schedule**. Open the distribution schedule, and select jobs **1070** and **1090** to transfer data to the channel database.

Configure the digital signature parameters

To digitally sign and submit simplified e-invoices, you must obtain Cryptographic Stamp Identifiers (CSIDs) from ZATCA. CSIDs come as digital certificates. For more information about how to obtain CSIDs, see Electronic invoicing onboarding in Saudi Arabia. You must get a CSID for each POS register that you use, because sequential numbering and digital signing of simplified e-invoices happen per POS register.

Store the digital certificates that you use to digitally sign simplified e-invoices in Key Vault. For the offline mode of the Store Commerce app, you can also sign by using a digital certificate that's stored in the local storage of the machine where the Store Commerce app is installed. The User-defined certificate profiles for retail stores feature enables configuration of certificates that are stored in Key Vault. It also supports failover to offline mode when Key Vault or Commerce headquarters isn't available. This feature extends the Manage secrets for retail channels feature.

To configure certificates and certificate profiles that you can use for digital signing, follow the steps in Set up certificate profiles. You must configure a separate certificate profile for each CSID that you get from ZATCA.

After you configure certificate profiles, follow these steps:

1. Go to **Retail and Commerce** > **Channel setup** > **Fiscal integration** > **Connector technical profiles**, and select the connector technical profile that you created earlier.
2. To configure certificate profiles per POS register, select **Override**, and create records for all registers that you must specify CSIDs for.
3. In each record, on the **Device** FastTab, in the **Certificate profile** field, specify a corresponding certificate profile. For more information about how to override connector technical profile settings, see Create connector technical profiles.

Specify ER configurations

Download the latest versions of the following configurations from the Dataverse repository. For more information, see Import Electronic reporting (ER) configurations from Dataverse.

- E-invoice generation configurations:
  - **Invoice model** data model
  - **Invoice model mapping for commerce (SA)** data model mapping
  - **Sales e-invoice (SA)** format
- E-invoice submission configurations:
  - **Customer invoice context model** data model
  - **Retail channel data** data model
  - **Retail fiscal document mapping** data model mapping
  - **Retail fiscal document format** format

You must also configure legal entity–specific parameters of the e-invoice format.

1. In the **Electronic reporting** workspace, select **Reporting configurations**.
2. On the **Configurations** page, select the **Sales e-invoice (SA)** format that you imported earlier, and then select **Configurations** > **Application specific parameters** > **Setup**.
3. On the **Application specific parameters** page, select the version of the format configuration that you want to configure the parameters for.
4. On the **Lookups** FastTab, select the record for the **PaymentMethodSubstitutionLookup** lookup.
5. On the **Conditions** FastTab, create records to link methods of payment that you configured for customers in the **Accounts receivable** module to payment means that ZATCA defines for e-invoices.
6. In the **Lookup result** field, select a payment means code.
7. In the **Name** field, select a customer payment method that must correspond to the payment means code. If you want the payment means code to be selected when the customer payment method isn't specified, select ***Blank***. If you want the payment means code to be selected for any customer payment method, select ***Not blank***.

   The **Line** field is automatically set. It defines the order in which payment means codes are searched for a customer payment method. You can change the order by using the **Move up** and **Move down** buttons.

8. Complete the configuration by setting the **State** field to **Completed**.

Configure e-invoice submission

Before you set up e-invoice submission parameters for Commerce, configure the Electronic invoicing service so that it can be used for Saudi Arabia. For more information, see Get started with Electronic invoicing for Saudi Arabia. To support the configuration of CSIDs per POS register, follow these steps for each POS register in your Electronic Invoicing environment.

1. Add a corresponding CSID certificate to the Key Vault parameters of the environment.

2. Create a feature setup of the **Processing pipeline** type for the electronic invoicing feature. To complete this step, configure the first feature setup, and then create new feature setups by copying them from the first one and modifying the parameters.

3. In the feature setup, follow these steps:
   a. In the integration action of the processing pipeline, specify the name of the CSID certificate that you added earlier.
   b. In the applicability rule, specify the corresponding **Store** and **RegisterNumber** values.

In Commerce headquarters, follow these steps to set up e-invoice submission parameters that are specific to Commerce for Saudi Arabia.

1. On the **Electronic document parameters** page, on the **Electronic document** tab, create a new record, and specify the following parameters:
   a. In the **Table name** field, select **Fiscal transaction document**.
   b. In the **Document context** field, select **Retail fiscal document context**.
   c. In the **Electronic document model mapping** field, select **Retail fiscal document mapping**.
   d. Select number sequences for submitted file names and batch submission IDs.

2. On the **Features** tab, mark the **Saudi Arabia electronic invoice** feature as **Enabled**.

Enable the digital signature in offline mode

To enable the digital signature in offline mode, follow these steps after you activate POS on a new device.

1. Sign in to POS.
2. On the **Database connection status** page, ensure that the offline database is fully synchronized. When the value of the **Pending downloads** field is **0**, the database is fully

synchronized.
3. Sign out of POS.
4. Wait for the offline database to be fully synchronized.
5. Sign in to POS.
6. On the **Database connection status** page, ensure that the offline database is fully synchronized. When the value of the **Pending transactions in offline database** field is **0**, the database is fully synchronized.
7. Restart POS.

E-invoice submission

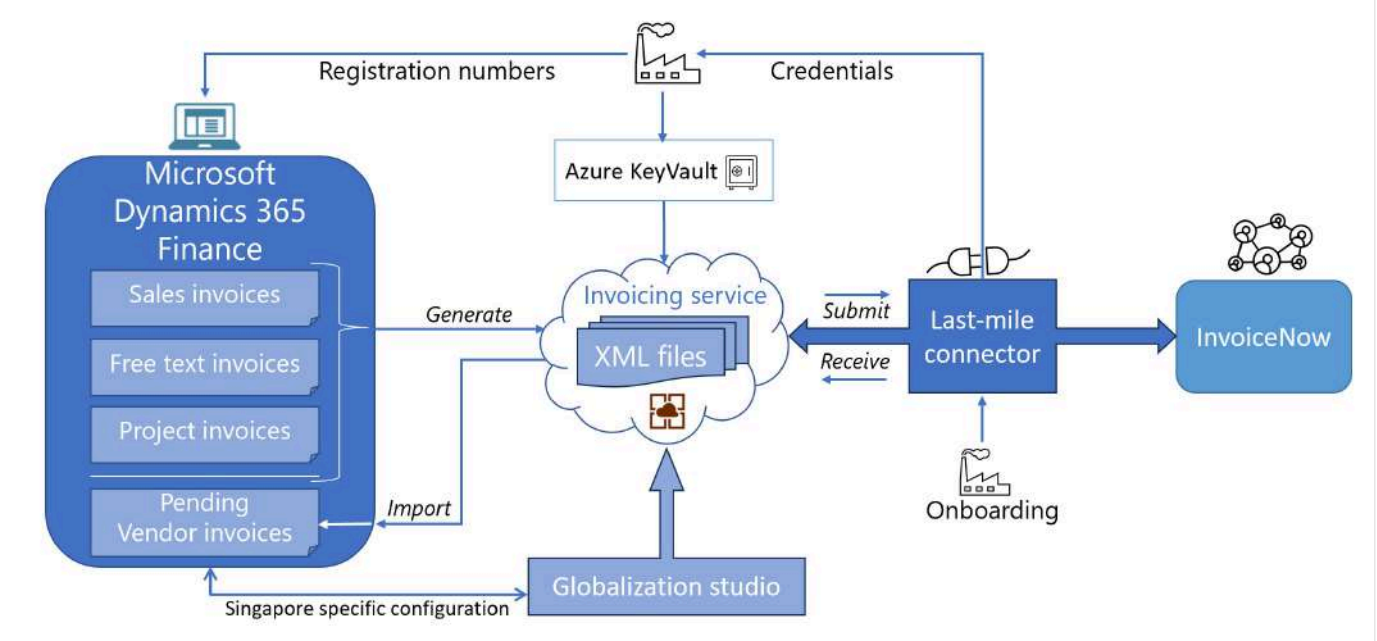
To initiate submission of simplified e-invoices to ZATCA, run the **Submit electronic documents** periodic operation in Commerce headquarters.

You can review the submission log on the **Electronic document submission log** page.

Last updated on 05/12/2026

Electronic invoicing for Singapore

This article helps you get started with electronic invoicing for Singapore. Set up the system to generate, submit, and receive electronic invoices in the Singaporean extension of the PEPPOL International (PINT) format in Microsoft Dynamics 365 Finance by using the last mile connector.



> **Note**
>
> This electronic invoicing approach uses an invoicing service that's applicable only to cloud deployments of Microsoft Dynamics 365 Finance.

Prerequisites

Before you start, make sure these prerequisites are in place:

- The company is a registered taxpayer in Singapore.
- The company has a signed agreement with the electronic document delivery service provider that supports document interchange in the PEPPOL (PINT) format.
- Register the following profile ID for Microsoft Dynamics 365 Finance electronic document interchange:
  - **urn:peppol:bis:billing** – Use this profile to interchange documents in PEPPOL (PINT) format.

- Get the credentials from the service provider to integrate the Electronic Invoicing service with the Electronic Invoicing service independent software vendor (ISV) last-mile connector.

  > **Note**
  >
  > This implementation assumes Edicom is the Electronic Invoicing ISV last-mile connection service provider. For more information, see Edicom integration with Microsoft Dynamics 365.

- Learn about Electronic Invoicing in the Electronic Invoicing service overview and Electronic invoicing components.
- Complete the common Electronic Invoicing service configuration in Electronic invoicing configuration.

Create the Azure Key Vault configuration

Create an Azure Key Vault to store required company secrets. For more information, see Configure Azure resources for Electronic invoicing.

Add the following element to the key vault:

- Add the secret for the **token** that authorizes access to Edicom services. Obtain it from Edicom as described in the prerequisites.

Set up electronic invoicing Key Vault parameters

Set up electronic invoicing Key Vault parameters.

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
2. On the **Electronic invoicing** tab, in **Key Vault settings**, in **Key Vault**, select the key vault reference you created earlier.
3. In **SAS token secret**, select the storage account secret URL used to authenticate access to the storage account.
4. Select **Key Vault parameters**.
5. On the **Key Vault parameters** page, in **Certificates**, select **Add** and create an element of the appropriate type for each secret described earlier.

- Add the **token** element of type **Secret**.

> ⓘ Note
>
> Match the value in the **Name** column to the secret name described earlier.

Import the electronic invoicing feature

1. Go to **Globalization Studio** and select the **Electronic invoicing** tile. Import the latest version of the **Singaporean electronic invoice (SG)** globalization feature as described in Import features from the repository.

2. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, confirm that the following Electronic reporting configurations are imported as a result of the **Singaporean electronic invoice (SG)** globalization feature import.

    - **Invoice model**
    - **Invoice model mapping**
    - **PINT Sales e-invoice**
    - **PINT Sales e-credit note**
    - **PINT Project e-invoice**
    - **PINT Project e-credit note**
    - **Customer invoice context model**
    - **Response message model**
    - **Edicom Response Processing**
    - **Error log import Json**

    > ⓘ Note
    >
    > If the configurations aren't imported, import them manually as described in Import Electronic reporting (ER) configurations from Dataverse.

3. In the **Electronic reporting** workspace, on the **Reporting configurations** tile, import the latest versions of the following Electronic reporting configurations to receive incoming vendor invoices.

    - **Vendor invoice import**
    - **Vendor invoice Mapping to destination**
    - **Import invoice context model**

Configure the electronic invoicing feature

The **Singaporean electronic invoice (SG)** feature publishes some parameters with default values. Before you deploy the feature, review the default values and update them so they reflect your business operations.

Review and update the **Singaporean electronic invoice (SG)** feature configuration:

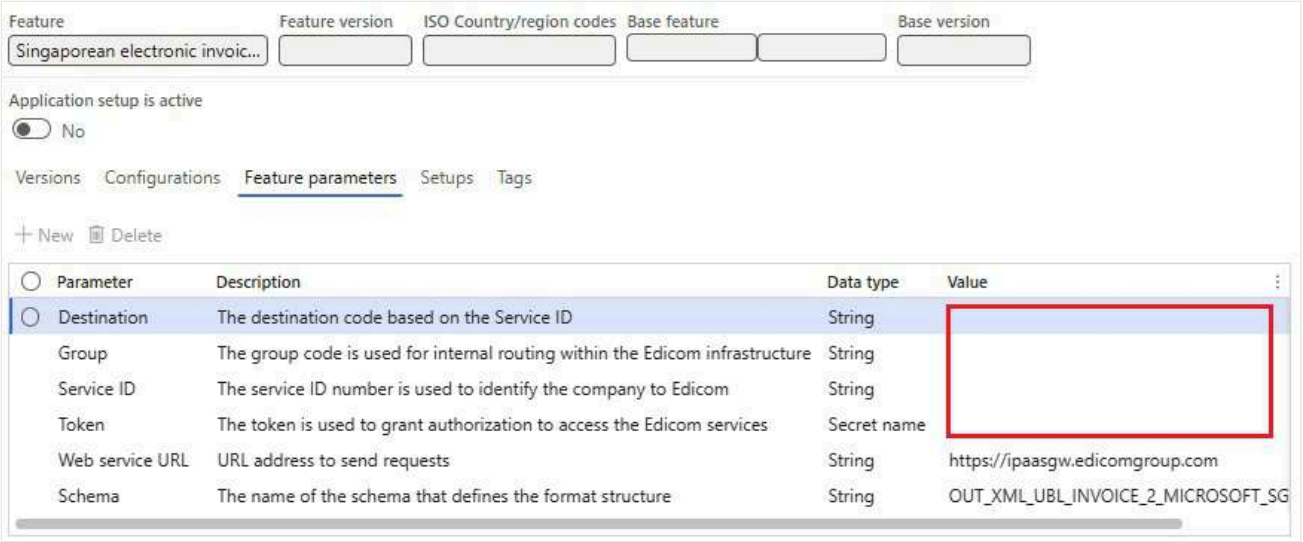
1. Go to **Globalization Studio** and select the **Electronic invoicing** tile. Import the globalization feature as described in Import features from the repository.

2. Copy the imported **Singaporean electronic invoice (SG)** globalization feature and select your configuration provider, as described in Create a Globalization feature.

3. On the **Versions** tab, check that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify these required **Edicom** connection and integration parameters:

    - **Service ID** – Use the service ID (Domain) number from Edicom to identify the company.
    - **Group** – Use the group code for internal routing within the Edicom infrastructure.
    - **Destination** – Construct the destination by appending **_EDIWIN** to the service ID number. For example, if the service ID number is **123456**, enter **123456_EDIWIN**.
    - **Token** – Select the name of the token you created earlier.



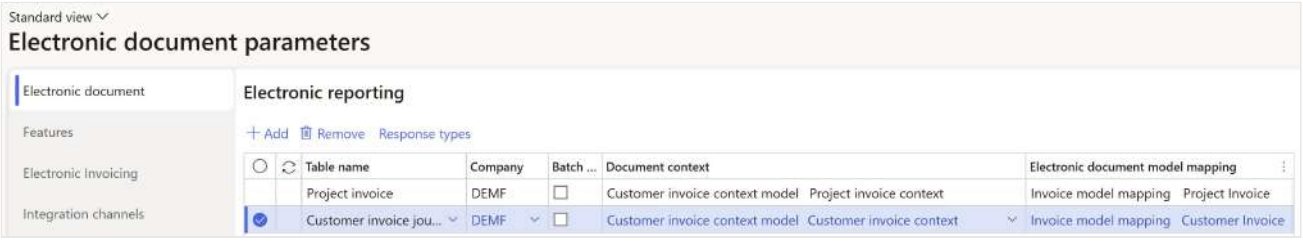
5. Each copy starts as a **Draft** version. Complete and deploy the feature as described in Complete and deploy a Globalization feature.

Configure electronic document parameters

1. Go to **Organization administration** > **Setup** > **Electronic document parameters**.

2. On the **Electronic document** tab, add records for the **Customer Invoice journal** and **Project invoice** tables.

3. For each table name, set the **Document context** and **Electronic document model mapping** fields as described in Set up electronic invoicing parameters.

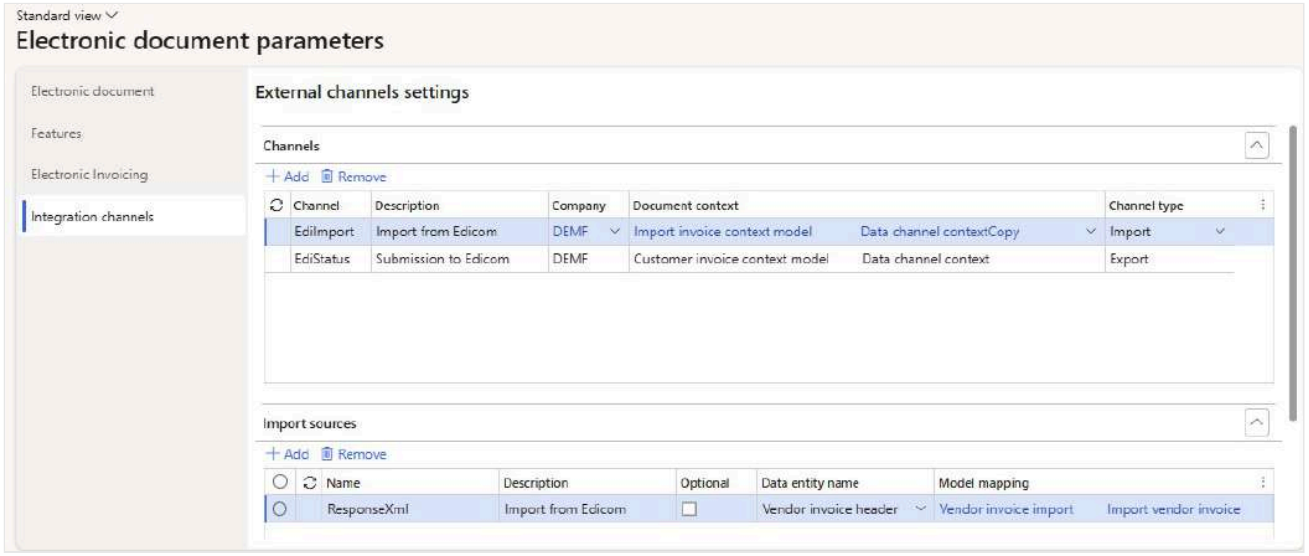


> **Note**
>
> If you created derived equivalents of the earlier Electronic Reporting configurations, use them instead of the standard configurations.

4. On the **Integration channels** tab, in the **Channels** section, select **Add** to create a new channel.

5. In the **Channel** field, enter **EdiStatus**. Enter the value exactly as shown. The system uses it to submit outgoing electronic invoices.

6. In the **Company** field, select the legal entity.
7. In the **Document context** field, select the **Data channel context** mapping from the **Customer invoice context model** configuration.
8. In the **Channel type** field, select **Export**.
9. In the **Channels** section, select **Add** to create another channel.
10. In the **Channel** field, enter **EdiImport**. Enter the value exactly as shown. The system uses it to import incoming electronic invoices.
11. In the **Company** field, select a required legal entity.
12. In the **Document context** field, select the **Data channel context** mapping from the **Import invoice context model** configuration.
13. In the **Channel type** field, select **Import**.
14. In the **Import sources** section, select **Add** to create an import source.
15. In the **Name** field, enter **ResponseXml**. Enter the value exactly as shown.
16. In the **Data entity name** field, select the **Vendor invoice header** entity.
17. In the **Model mapping** field, select the **Import vendor invoice** mapping from the **Vendor invoice import** configuration.
18. Select **Save**, and then close the page.



> ⓘ **Note**

If you use integration channels other than **EdiStatus** and **EdiImport**, do another configuration for the feature and related context configurations.

Set up registration numbers

If these registration types and categories already exist, skip this procedure.

Set up UEN number

To set up the Unique Entity Number (UEN) registration number, see UEN of the business user.

Set up GST number

To set up the Goods and Services Tax (GST) registration number, see GST Registration Number of the business user.

Set up address structure

Set up the postal address structure.

1. Go to **Organization administration** > **Global address book** > **Addresses** > **Address setup**.
2. Ensure the following required elements are configured:
    - Country code
    - Postal code
    - Street

Configure legal entity data

Enter the address

Add the primary address.

1. Go to **Organization administration** > **Organizations** > **Legal entities**, and select a legal entity.
2. On the **Addresses** FastTab, add the primary address for the legal entity.

Seller identification

Add the registration numbers.

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. On the Action Pane, select **Registration IDs**.
3. On the **Registration ID** FastTab, select **Add**, set **Registration type** to the Unique Entity Number (UEN) type you created earlier, and enter the UEN value in **Registration number**.
4. Select **Add**, set **Registration type** to the Goods and Services Tax (GST) type you created earlier, and enter the GST value in **Registration number**.

   The system puts the **UEN** number in the **Invoice\cac:AccountingSupplierParty\cac:Party\cbc:EndpointID** element and the **GST** number in the **Invoice\cac:AccountingSupplierParty\cac:Party\cac:PartyTaxScheme\cbc:CompanyID** element in the generated electronic invoice XML file. These values act as the seller's identification during submission.

Configure customer data

Enter the address

To enter the address, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer.
3. On the **Addresses** FastTab, add a valid address for the selected customer.

Buyer identification

To enter the registration numbers, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. On the Action Pane, on the **Customer** tab, in the **Registration** group, select **Registration IDs**.
3. On the **Registration ID** FastTab, select **Add** to create a registration ID.
4. In the **Registration type** field, select the Unique Entity Number (UEN) registration type that you created earlier.

5. In the **Registration number** field, enter a valid BRN registration number for the selected customer.
6. Select **Add** to create another registration ID, if necessary.
7. In the **Registration type** field, select the Goods and Services Tax (GST) registration type that you created earlier.
8. In the **Registration number** field, enter a valid SST registration number for the selected customer.

   The **UEN** number is entered in the **Invoice\cac:AccountingCustomerParty\cac:Party\cbc:EndpointID** element and the **GST** number is entered in the **Invoice\cac:AccountingCustomerParty\cac:Party\cac:PartyTaxScheme\cbc:CompanyID** element in the electronic invoice XML file that is generated. It's used as the buyer's identification during the submission process.

> **Note**
>
> For both, seller and buyer identification, the schema **0195** (Singapore UEN identifier) is used by default according to the Electronic Address Scheme (EAS). If you need to redefine the default schema, see the next section. Otherwise, you can ignore the next section.

Configure identification schemas

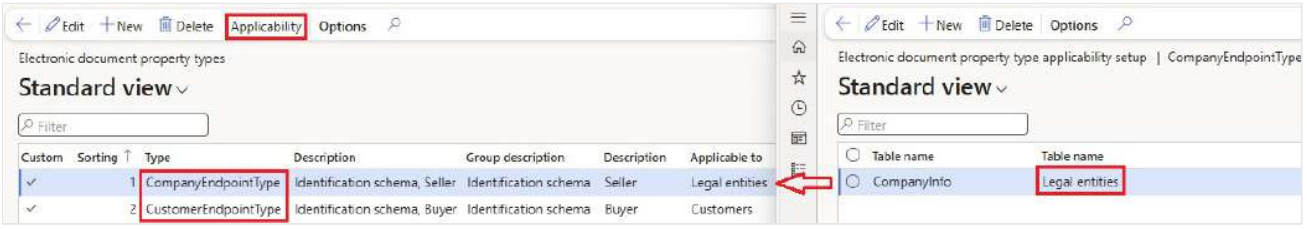
Follow these steps only if you need to redefine the default identification schemas for sellers or buyers.

Configure electronic document properties

Set up electronic document properties.

1. Go to **Accounts receivable** > **Setup** > **Electronic document property types**, and select **New**.
2. In the **Type** field, enter **CompanyEndpointType**. Enter the value exactly as shown. Use this type for the **Seller** identification schema definition.
3. Select **Applicability** to add an applicable table.

4. On the **Electronic document property type applicability setup** page, in the **Table name** field, select **Legal entities**.
5. Save your changes and return to the **Electronic document property types** page.
6. Select **New** to create another electronic document property type.
7. In the **Type** field, enter **CustomerEndpointType**. Enter the value exactly as shown. Use this type for the **Buyer** identification schema definition.
8. Select **Applicability** to add an applicable table.
9. On the **Electronic document property type applicability setup** page, in the **Table name** field, select **Customers**.
10. Save your changes, and return to the **Electronic document property types** page.
11. Save your changes, and close the page.



Enter the seller schema code

To enter the Seller schema code, follow these steps:

1. Go to **Organization administration** > **Organizations** > **Legal entities** and select a legal entity.
2. Select **Electronic document properties** from the Action Pane.
3. In the **Value** column, enter the required seller schema code.

Enter the buyer schema codes

To enter the buyer schema codes, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a specific customer in the list. On the Action Pane, select the **Customer** tab. In the **Properties** group, select **Electronic document properties**.
3. In the **Value** column, enter the required buyer schema code.

> **Note**
>
> Schema codes defined by electronic document property types take priority over the default **0195** schema code.

Set up units of measure

Set up units of measure.

1. Go to **Organization administration** > **Setup** > **Units** > **Units**.
2. Select a unit ID, and then select **External codes**.
3. On the **External codes** page, in **Overview**, in the **Code** column, enter the unit ID.
4. In the **Standard code** column, select the checkbox.
5. In the **Value** section, in the **Value** field, enter the external code from the UNECE Recommendation 20 code list.

   > **Note**
   >
   > If no specific unit of measure applies, the system uses the default value **EA**.

Configure sales tax codes

1. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax codes**.
2. Select a sales tax code. On the Action Pane, on the **Sales tax code** tab, in the **Sales tax code** group, select **External codes**.
3. In the **Overview** section, create a line for the selected unit. In the **External code** field, enter the sales tax code from step 2.
4. In the **Value** section, in the **Value** field, enter an external code that matches one of the Singapore-specific Duty or tax or fee category codes.

Issue electronic invoices

After you complete the required configuration steps, generate and submit electronic invoices for posted invoices: go to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. Learn more in Submit electronic documents.

> **Important**
>
> In current implementations, the standard submission procedure only generates electronic invoices and stores them in the service. It doesn't submit them. To submit invoices, complete these extra steps.

Follow these steps to submit the generated electronic invoices.

1. In the **Feature management** workspace, confirm that the **Export channels for electronic invoicing integration** feature is enabled. To learn more, see Feature management overview.
2. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Run submission process in export channels**.
3. In the **Channel** field, select the export channel you created, and then select **OK**.

Check the submission results at **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**. Learn more in Work with Electronic document submission log.

> **Note**
>
> You can also find submitted electronic invoices in the **Ediwin** portal in the **Outbound** folder and its subfolders. You can monitor further processing there.

According to the Singaporean electronic invoicing requirements, each generated e-invoice is assigned a unique identifier. This identifier is added to the XML files of related e-invoices in the **cbc:UUID** element. You can also see the identifier in the related Invoice Journals as the rightmost column of the invoices list.

Receive incoming electronic invoices

Set more parameters in Microsoft Dynamics 365 Finance to import incoming invoices.

Configure these master data types to match incoming electronic invoices:

- vendors
- products
- units

Follow the steps in Import vendor electronic invoices, starting at Configure vendor data.

Receive electronic invoices

After you complete the configuration, receive incoming electronic invoices in the **PEPPOL** format.

> **Note**
>
> Review incoming electronic invoices in the **Inbound** folder and its subfolders in your Ediwin portal.

Receive electronic invoices:

1. Go to **Organization administration** > **Periodic** > **Electronic documents** > **Receive electronic documents**.
2. Select **OK**, and then close the page.

To view receipt logs for processed electronic invoices, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document receipt log**.

To view successfully received invoices, go to **Accounts payable** > **Invoices** > **Pending vendor invoices**.

More information

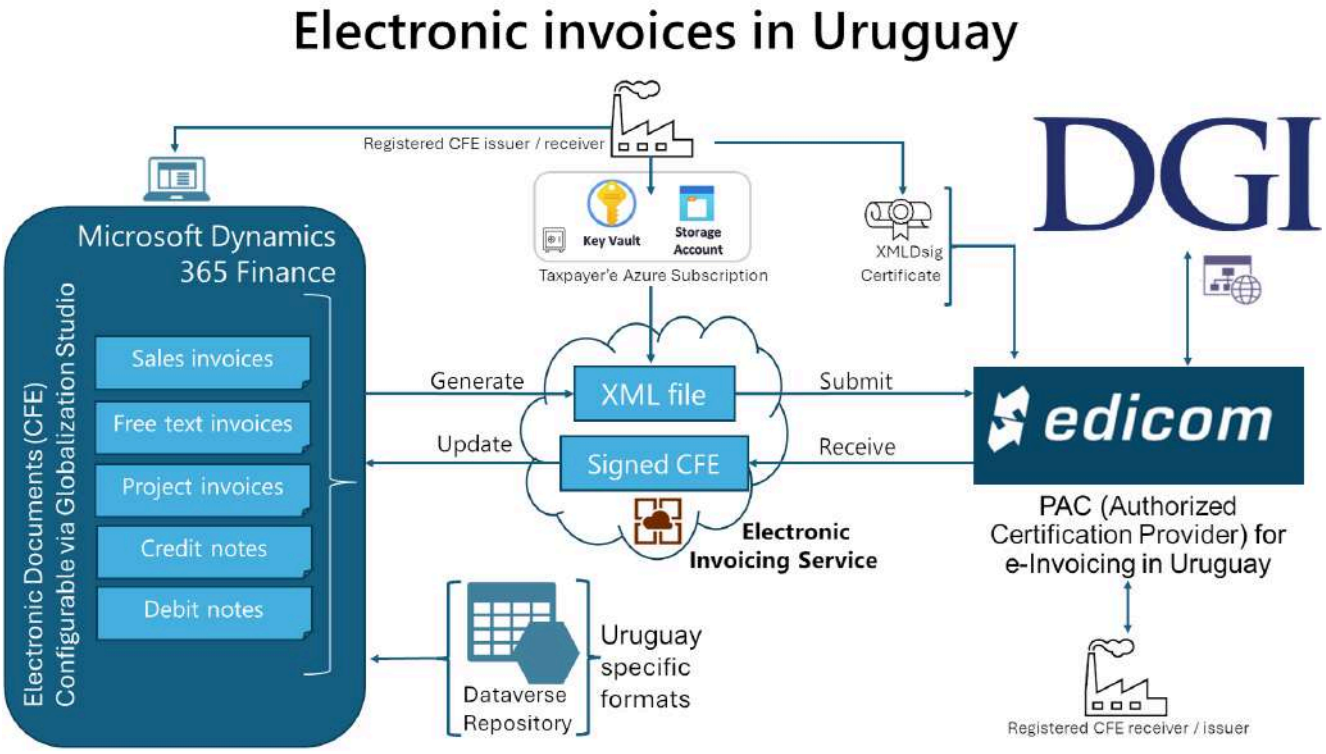
- Electronic invoicing coverage by country and region

Last updated on 03/11/2026

Get started with Electronic invoicing for Uruguay

This article provides information to help you get started with Electronic invoicing for Uruguay. It guides you through the configuration steps that are country/region-dependent in Microsoft Dynamics 365 Finance or Microsoft Dynamics 365 Supply Chain Management. These steps complement the steps that are described in Electronic invoicing setup. For the last-mile integration with the Uruguayan Tax Authorities, Microsoft is partnering with Edicom.

After you configure Electronic invoicing, you can generate, digitally sign, and submit the XML files of electronic invoices to the Edicom authorized certification provider (PAC) according to the regulatory requirements in Uruguay.



> **Note**
>
> The electronic invoicing approach that this article describes is implemented by using an invoicing service that is applicable only to cloud deployments of Finance or Supply Chain Management.

Prerequisites

Before you begin the procedures in this article, the following prerequisites must be met.

1. Ensure that the settings for the Uruguayan legal entity are in place. Learn more in Set up a legal entity and tax information for Uruguay.

2. Gain familiarity with and understanding of Electronic invoicing as it's described in Electronic invoicing overview.

3. Do the common part of Electronic Invoicing service configuration as described in Electronic invoicing configuration.

4. You must enable the following features in Feature management:

    - Electronic invoicing integration
    - Electronic invoicing integration resubmit document from failed action
    - Execute update actions for submitted documents

5. Ensure that the following Electronic reporting (ER) format configurations are imported. Learn more in Import Electronic reporting (ER) configurations from Dataverse, and Import Electronic reporting (ER) configurations.

    - Customer invoice context model
    - Edicom response error log import
    - Edicom Response Processing LATAM
    - Edicom source file response import format
    - Inventory e-Invoice (UY)
    - Inventory Export e-Invoice (UY)
    - e-Packing Slip (UY)
    - Export e-Packing Slip (UY)
    - Project e-Invoice (UY)
    - Project Export e-Invoice (UY)

    > **Note**
    >
    > These formats are based on the corresponding LATAM format configurations, which use the **Invoice model LATAM** and **Invoice model mapping LATAM** configurations. All other required configurations are imported automatically.

Configure the electronic invoicing feature

The "**Uruguay Electronic Invoice (UY) Edicom integration for Uruguay**" feature represents an outbound flow to issue the following sales documents.

Expand table

Name	Code	Original name
Electronic sales invoice	111	Factura electrónica de venta
Credit Note	112	Nota de Crédito
Debit Note	113	Nota de Débito
Packing Slip	181	Remito
Export invoice	121	Factura de exportación
Export Credit Note	122	Nota de Crédito de exportación
Export Debit Note	123	Nota de Débito de exportación
Export packing slip	124	Remito de exportación

Some parameters of the feature are published with default values. Before you deploy the Electronic invoicing feature to the service environment, add a feature that is based on the Microsoft-provided feature, and complete the common parameters on the **Feature parameters** tab. Review the default values, and update them as required, so that they better reflect your business operations.

Edicom Interactions for Uruguay

For Uruguay, there are at least four core interactions with Edicom in the base flow:

1. **Invoice submission**
2. **Signed XML retrieval**
3. **Status retrieval of the submitted invoice**
4. **Customer acceptance or rejection**

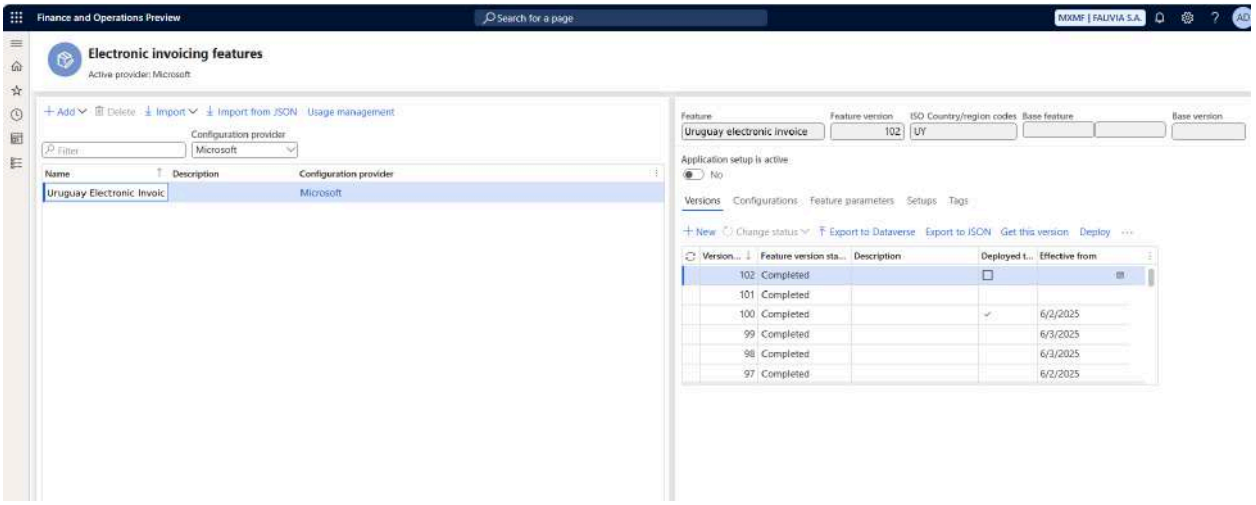
Each of these interactions requires common parameters, such as Edicom connection details and the authentication token provided by Edicom. These parameters are reused across the feature setup for all document types. Edicom supplies these values during the company's onboarding process.

Uruguay Electronic Invoice (UY) Edicom integration for Uruguay

> ⓘ **Important**

The **Uruguay Electronic Invoice (UY)** feature is provided by Microsoft. Before it can be used, it requires additional configuration, as described in this article. Learn how to configure invoicing features and apply changes in Work with feature setups. For example, in addition to the connection parameters, you can filter specific legal entities so that they are processed in applicability rules. By default, the feature is applicable to all legal entities that have a primary address in Uruguay.

1. Import the latest version of the **Uruguay Electronic Invoice (UY)** Globalization feature as described in Import features from the repository. The following illustrations show what the feature looks like after you import it from Dataverse.



If you go to the Configuration tab, as shown in the previous screenshot, there's a page displaying information similar to the following table:

Expand table

Name	Version	Status	Country/region code
Edicom Response Processing LATAM	13.2.5	Completed	
e-Packing Slip (UY)	296.140.89.125	Completed	UY
Export e-Packing Slip (UY)	296.140.89.41	Completed	UY
Inventory Export e-Invoice (UY)	296.140.89.52	Completed	UY
Inventory e-Invoice (UY)	296.140.89.64	Completed	UY
Project e-Invoice (UY)	296.139.40.39	Completed	UY

Note

The version numbers may change if the formats have been updated. The table is for illustrative purposes only.

2. Create a copy of the imported Globalization feature, and select your configuration provider. Learn more in Create a Globalization feature.

3. On the **Versions** tab, confirm that the **Draft** version is selected.

4. On the **Feature parameters** tab, specify values for the following connection and integration parameters that are required for interoperation with Edicom's API.
These parameters are general-purpose and are used across the different actions in the setups, so you only need to configure them once in this section.

> **Tip**
>
> The following table provides **example values** for each feature parameter, so you can understand the type of data that should be entered for each one. These values are **illustrative only** and were provided by Edicom to Microsoft for testing purposes. When you are onboarded, Edicom provides you with your own unique values to use in your configuration.

Expand table

Parameter	Description	Data type	Value
EdicomApplication	Application	String	<ID_PROVIDED_EDICOM>
EdicomDomain	Domain	String	<ID_PROVIDED_EDICOM>
EdicomGetStatusSchema	Get status schema name	String	IN_DOCUMENT_STATUS_MICROSOFT_UY
EdicomGroup	Group	String	<GROUP_ID>
EdicomSignedXMLSchema	Signed XML schema name	String	EDIWIN-SOURCE-FILE_XML_CFE_ADENDA_<ID_PROVIDED_EDICOM>
EdicomSubmitInvoiceDestination	Destination name	String	<ID_PROVIDED_EDICOM>_EDIWIN
EdicomSubmitInvoiceSchema	Submit invoice	String	OUTBOUND_DOCUMENT_MICROSOFT_UY

Parameter	Description	Data type	Value
	schema name		
EdicomToken	Auth token	Secret name	EdiwinUruguayToken
EdicomWebServiceURL	Web service URL	String	https://ipaasgw.edicomgroup.com

> **Tip**
>
> When configuring the feature parameters for Edicom integration, follow these steps:
>
> - Select **Application**, and enter the service ID number you obtained from Edicom.
> - Select **Domain**, and enter the same service ID number.
> - Select **Get status schema name**, and enter the provided schema name.
> - Select **Group**, and enter the group code you received.
> - Select **Signed XML schema name**, and enter the appropriate schema name.
> - Select **Destination name**, and enter the service ID number concatenated with "_EDIWIN". For example, if your service ID is 123456, enter **123456_EDIWIN**.
> - Select **Submit invoice schema name**, and enter the schema name.
> - Select **Auth token**, and choose the name of the secret you created for the token.
> - Select **Web service URL**, and verify the web address is correct.

5. The copy of the feature is always created as a **Draft** version. Regardless of whether you made changes, you must complete, publish, and deploy the feature as described in Complete and deploy a Globalization feature.

Outbound flow pipeline

There are six electronic invoice formats for Uruguay included in the **Uruguay Electronic Invoice (UY) feature**. Each format requires its own setup and configuration. The formats are:

- **Inventory e-Invoice (UY)**
- **Inventory Export e-Invoice (UY)**
- **e-Packing Slip (UY)**
- **Export e-Packing Slip (UY)**
- **Project e-Invoice (UY)**
- **Project Export e-Invoice (UY)**

The following sections have guidance on the general configuration steps that apply to all six formats. Specific considerations and requirements for each individual format are explained, so you can ensure that each one is configured correctly according to its unique needs.

Electronic invoice feature

Setups-View

After importing the feature, in the second screenshot, five tabs are available: **Versions**, **Configuration**, **Feature parameters**, **Setups**, and **Tags**. Select the **Setups** tab, and then select **View**. This opens the page **Feature version setup**:

Feature version setup

This page allows you to review and configure the details of the feature setup for Uruguay. On the left side of this page, there are four sections: **Processing pipeline**, **Applicability rules**, **Variables**, and **Parameters**.

Processing pipeline

The Setup Actions and Parameters section is divided into two distinct grids:

Top Right – Processing Pipeline

The **Processing pipeline** grid should is similar to the following table:

Expand table

Action	Action name	Description	Enable retry	Retry action	Export result	Update action
Transform document	Generate invoice XML file					
Integrate with Edicom	Submit invoice					
Get status from Edicom for an invoice	Get signed XML from Edicom					**Checked**
Get status from Edicom for an invoice	Get invoice status from Edicom					

Action	Action name	Description	Enable retry	Retry action	Export result	Update action
Process response	Process invoice status response					
Terminate pipeline	Terminate pipeline	Terminates the processing pipeline				

> **Note**
>
> The **Get status from Edicom for an invoice** action is crucial in the flow, as it is marked with the **Update action** checkbox. This designation means that this specific step—and all steps following it repeatedly execute in a loop. The system continues this process until the invoice reaches a final, terminal status, ensuring that the latest status from Edicom is always retrieved and processed.

Bottom Right – Parameters

The lower grid shows the Parameters associated with the selected processing action. These parameters allow for fine-tuning of the selected step's behavior. Each entry includes a parameter name, description, data type, and value, enabling users to configure processing logic according to their business requirements.

Transform document parameters

Expand table

Name	Description	Value
Input file	Source file provides to the action the data to be executed.	Variable: BusinessDocumentDataModel
Direction	Direction describes which format will be used: import or export	export
Configuration	Configuration describes format which will be executed	Inventory e-Invoice: Inventory e-Invoice
Configuration Integration point	Source file provides data to the reporting runtime	InvoiceCustomer
Custom file name	Custom file name from the client	

Linking the correct format in Transform document parameters

When configuring the **Transform document** action in the processing pipeline, ensure the following:

- **Configuration**: Set this to the exact name of the feature format you are configuring.
  - For example:
    - `Project e-Invoice (UY)` → Configuration: `Project e-Invoice (UY)`
    - `Inventory e-Invoice (UY)` → Configuration: `Inventory e-Invoice (UY)`
    - `Project Export e-Invoice (UY)` → Configuration: `Project Export e-Invoice (UY)`
    - `Inventory Export e-Invoice (UY)` → Configuration: `Inventory Export e-Invoice (UY)`
    - `e-Packing Slip (UY)` → Configuration: `e-Packing Slip (UY)`
    - `Export e-Packing Slip (UY)` → Configuration: `Export e-Packing Slip (UY)`
- **Configuration Integration Point**:
  - Use `InvoiceCustomer` for inventory-related formats (`Inventory e-Invoice (UY)`, `Inventory Export e-Invoice (UY)`, `e-Packing Slip (UY)`, `Export e-Packing Slip (UY)`).
  - Use `InvoiceProject` for project-related formats (`Project e-Invoice (UY)`, `Project Export e-Invoice (UY)`).

> **Tip**
>
> If the feature name contains "Project", use `InvoiceProject`. If it contains "Inventory", use `InvoiceCustomer`. Incorrect settings will result in empty XML or missing required data.

Integrate with Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
Http request body	Http request body (can be empty)	Generate invoice XML file: Output file
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	

Name	Description	Value
Auth token	Auth token secret name in the KeyVault	EdiwinUruguayToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Destination	Destination	FeatureParameter: EdicomSubmitInvoiceDestination
Schema	Schema	FeatureParameter: EdicomSubmitInvoiceSchema
Duplicates	Duplicates	0
Group	Group	FeatureParameter: EdicomGroup
Custom file name	Custom file name from the client	Generate invoice XML file: File name

Get signed XML from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinUruguayToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomSignedXMLSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference Id of the document submitted to Edicom	Submit invoice: ExternalID

Get invoice status from Edicom parameters

Expand table

Name	Description	Value
Web service URL	URL address to send request	FeatureParameter: EdicomWebServiceURL
User name	KeyVault secret name to store user name	
Password	KeyVault secret name to store user password	
Client ID	Client ID secret name in the KeyVault	
Client Secret	Client Secret secret name in the KeyVault	
Auth token	Auth token secret name in the KeyVault	EdiwinUruguayToken
Domain	Domain	FeatureParameter: EdicomDomain
Application	Application	FeatureParameter: EdicomApplication
Schema	Schema	FeatureParameter: EdicomGetStatusSchema
Group	Group	FeatureParameter: EdicomGroup
ERP UUID	Reference Id of the document submitted to Edicom	Submit invoice: ExternalID

Process response parameters

Expand table

Name	Description	Value
Input file	Response to analyze	Get invoice status from Edicom: Output file
Reporting configuration list	List of configurations used for response parsing	Edicom Response Processing LATAM: Edicom Response Processing LATAM
Custom file name	Custom file name from the client	
Reporting configuration list	List of configurations used for response parsing	Edicom response error log import: Edicom response error log import

Terminate pipeline parameters

Expand table

Name	Description	Value
Terminal state	Desired state of the pipeline when terminating	complete
Number of days	Number of days to wait before terminating	8

> **Note**
>
> - If the status response indicates a failure, the pipeline is terminated, and the submission is marked as failed.
> - If the response indicates successful submission to the Uruguayan Tax Authorities (DGI), the process will be completed after 8 days, provided no additional responses are received in the meantime.

Applicability Rules and Feature Setup Scope

Applicability rules must be correctly configured to provide context, so that the exact Electronic Invoicing Globalization feature that must run in the Electronic Invoicing service can be found. These applicability rules are provided out of the box by checking the legal entity's International Organization for Standardization (ISO) country/region code.

Applicability rules for Uruguayan electronic invoice formats

Below are the applicability rules for each electronic invoice format for Uruguay. Replace `<Legal Entity ID>` with the actual ID of your legal entity.

Inventory UY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	UY	string
		LegalEntityId	Equals	<Legal Entity ID>	string
Or		DocumentType	Equals	Customer invoice	string
		DocumentType	Equals	Customer credit note	string
		DocumentType	Equals	Customer debit note	string

Inventory Expo UY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	UY	string
		LegalEntityId	Equals	<Legal Entity ID>	string
		DocumentType	Equals	Export Customer invoice	string
		DocumentType	Equals	Export Customer debit note	string
		DocumentType	Equals	Export Customer credit note	string

e-Packing Slip UY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	UY	string
		LegalEntityId	Equals	<Legal Entity ID>	string
		DocumentType	Equals	Customer packing slip	string

Export e-Packing Slip UY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	UY	string
		LegalEntityId	Equals	<Legal Entity ID>	string
		DocumentType	Equals	Export Packing Slip	string

Project UY

Expand table

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	UY	string
		LegalEntityId	Equals	<Legal Entity ID>	string
Or		DocumentType	Equals	Project invoice	string

And/or	And/or	Field	Operator type	Value	Data type
		DocumentType	Equals	Project credit note	string
		DocumentType	Equals	Project debit note	string

Project Expo UY

And/or	And/or	Field	Operator type	Value	Data type
And		CountryISOCode	Equals	UY	string
		LegalEntityId	Equals	<Legal Entity ID>	string
		DocumentType	Equals	Export Project invoice	string
		DocumentType	Equals	Export Project credit note	string
		DocumentType	Equals	Export Project debit note	string

Variables

The following variables are used in the outbound data flow actions of the Uruguayan feature. (These variables are provided out of the box.)

- **BusinessDocumentDataModel** – The Business Document Data model variable that is received from Finance or Supply Chain Management and transformed into the format that is required for submission.
- **SignedXML** – The signed XML variable that is sent back to Finance or Supply Chain Management. This variable contains the base64-encoded response body from the **Get Signed XML from Edicom** step. It's used in the response types to save the signed XML that is obtained from Edicom as an attachment to the invoice journal. It's also used to generate printable reports through QR codes.

Name	Description	Type	Data type	Value
BusinessDocumentDataModel	Contains the electronic document from the client	From client	file	
SignedXML	Signed XML document in base64-encoded format	To client	file	Get signed XML from Edicom.Base64ResponseBody

After you import the **Electronic invoicing for Uruguay** feature that includes the out of the box default feature setup, follow the steps in the next section to configure electronic documents.

Setups-Application Setup

To configure the mapping between Dynamics 365 Finance source tables and electronic document formats, go to the **Setups** tab within the **Electronic invoice feature** section, and then select **Application Setup**. This step lets you define how data from Finance is linked to the correct electronic invoice formats for Uruguay.

> **Note**
>
> Although this configuration is not set at the general feature level, it acts globally for the selected feature setup. You don't need to repeat these steps for each individual format or feature—one configuration is sufficient for all supported document types within the feature.

The following grid defines the link between the source table in Dynamics 365 Finance, the document context model, and the mapping configuration used for electronic invoicing:

Table name	Context model / Context	Electronic document mapping	Mapping name	Description
Customer invoice journal	Customer invoice context model / Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice	Maps customer invoices from the CustInvoiceJour table to the LATAM invoice model for electronic submission.
Project invoice	Customer invoice context model / Project invoice context	Invoice Model mapping LATAM	Project E-Invoice	Maps project invoices to the LATAM invoice model for electronic submission.

> **Tip**
>
> Ensure that the Context and Mapping name match the document type. Incorrect mappings cause errors in XML generation or submission to DGI.

Configure electronic document parameters

To configure electronic document parameters, follow these steps:

1. Make sure that the country/region-specific ER configurations for the document context and electronic document model mapping that are required for Uruguay are imported. Learn more in Set up Electronic document parameters.
2. Go to **Organization administration** > **Setup** > **Electronic document parameters**.
3. In the **Electronic document** section, add records for the **Customer Invoice journal** and **Project invoice** table names.
4. For each table name, set the **Document context** and **Electronic document model mapping** fields in accordance with step 1.

The Electronic Document Parameters page in Dynamics 365 Finance. Shows on the left-hand navigation pane, we can see the following options:

- Electronic Document
- Features
- Electronic Invoicing
- Integration Channels

For the Electronic Document option on the right-hand side, the section labeled Electronic Reporting is displayed.

Within Electronic Reporting, a grid is presented which lists the electronic document configurations. This grid includes important associations between:

- The source table name
- The document context model and context
- The electronic document model mapping
- The final mapping name used for generating the electronic invoice.

This configuration is essential for determining how Dynamics 365 Finance maps internal data models to external electronic formats for submission to tax authorities or service providers.

Configure the grid as follows:

Table name	Microsoft-owned	Document context model	Document context	Electronic document model mapping	Mapping name
Customer invoice journal	No	Customer invoice context model	Customer invoice context	Invoice Model mapping LATAM	Customer E-Invoice

Table name	Microsoft-owned	Document context model	Document context	Electronic document model mapping	Mapping name
Project invoice	No	Customer invoice context model	Project invoice context	Invoice Model mapping LATAM	Project E-Invoice

1. Save your changes, and close the page.

2. For each table name, follow these steps:

   a. Select **Response types**, select **New** to create a response type that you will get from the back end, and enter the following values:

   - In the **Response type** field, enter **SignedXML** (the default value). (See the Variables section of this article.)
   - In the **Description** field, enter any meaningful name. Alternatively, leave the field blank.
   - In the **Submission status** field, select **Pending**.
   - In the **Model mapping** field, select **Edicom source file response format**.

   b. Repeat the preceding step, but set the **Submission status** field to **Pending update actions execution**. In this way, you can run a subset of the actions in the processing pipeline in a loop to continuously pull updated statuses and other information for the submitted documents from the tax authority. In the standard view, you should see a table similar to the following:

Response type	Description	Submission status	Data entity name	Model mapping
SignedXML	Signed XML obtained from EDICOM	Pending		Edicom source file response import format
SignedXML	Signed XML obtained from EDICOM	Pending update actions execution		Edicom source file response import format

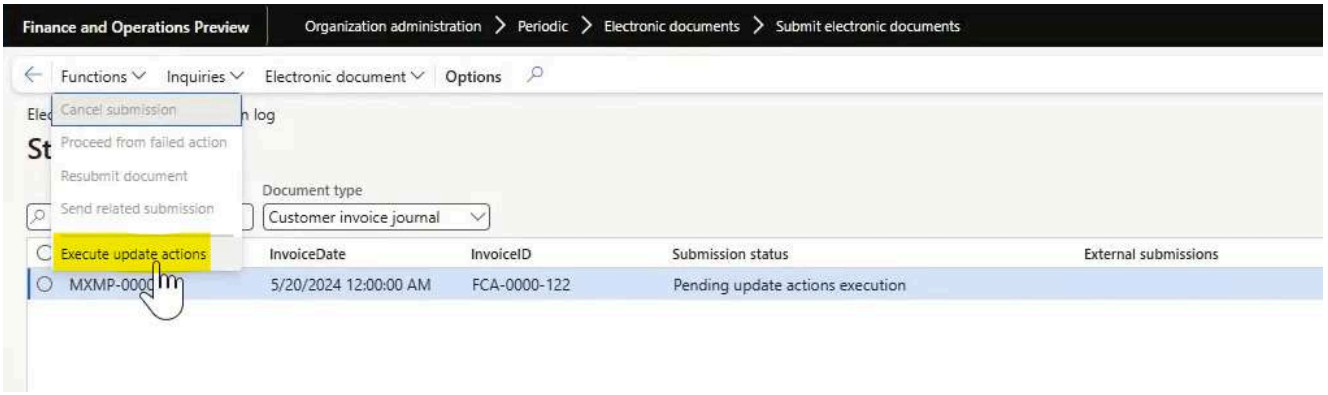
> **Note**
>
> The response includes the signed XML that is obtained from Edicom. This signed XML is stored in the system as an attachment to the corresponding invoice journal. It will eventually be used to generate printable invoices through QR codes.

Issue electronic invoices

After you complete all the required configuration steps, you can generate and submit electronic invoices for posted invoices by going to **Organization administration** > **Periodic** > **Electronic documents** > **Submit electronic documents**. For more information about how to generate electronic invoices, see Submit electronic documents.

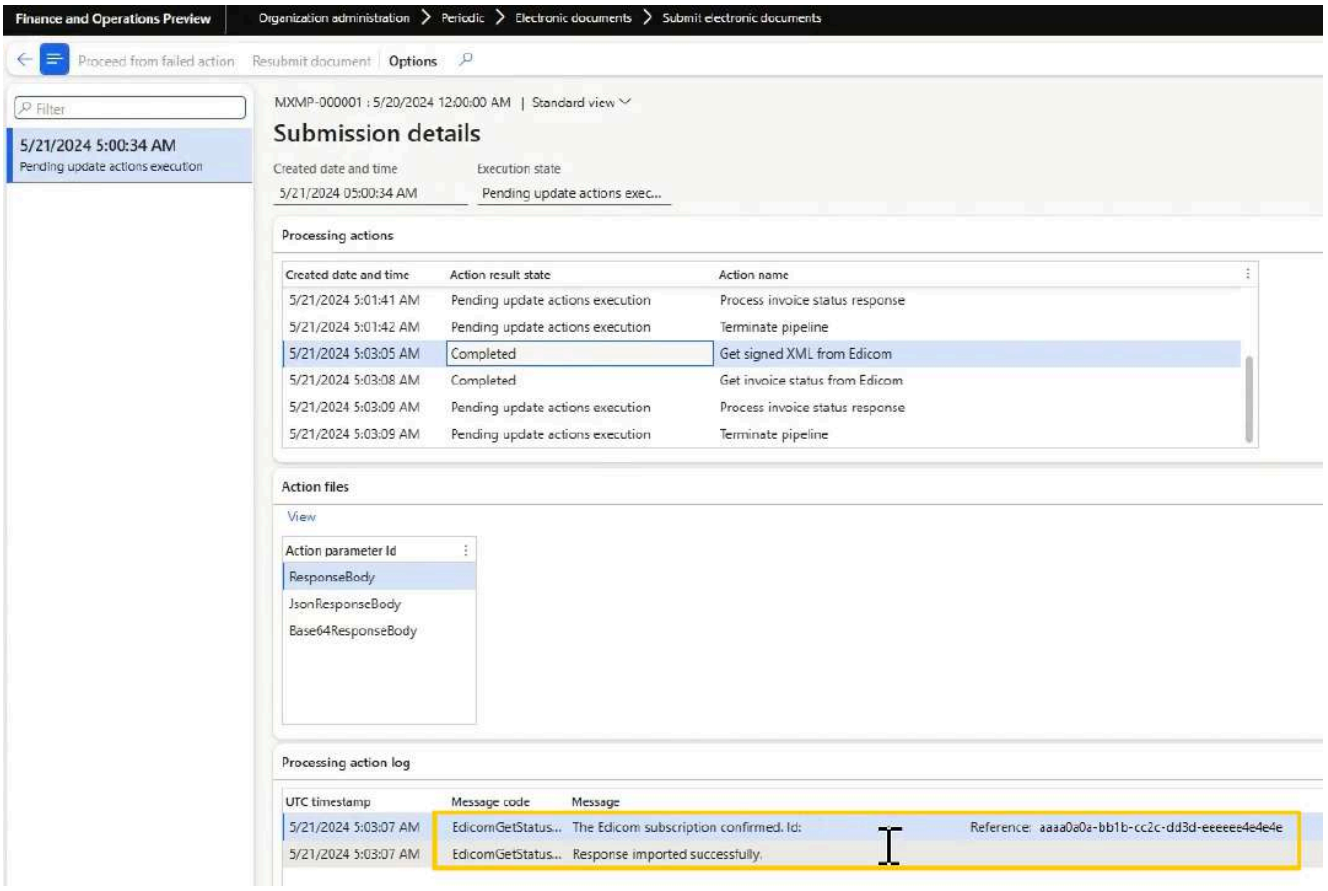
To inquire about the results of a submission, go to **Organization administration** > **Periodic** > **Electronic documents** > **Electronic document submission log**, and select the required document type. For more information, see Work with Electronic document submission log.

For Uruguay, after you submit the invoice, the submission status is set to **Pending update actions execution**. The response body is probably empty for the signed XML and the call to get the invoice status. An empty response body indicates that the XML wasn't available immediately after submission. To address the pending status, a function that is named **Execute update actions** is used. This function resumes the pipeline from the action that is marked as an update action. It then runs all subsequent actions in the pipeline in a loop. The status should change to **Executing** again. Then, after a few seconds, it reverts to **Pending update actions execution**.

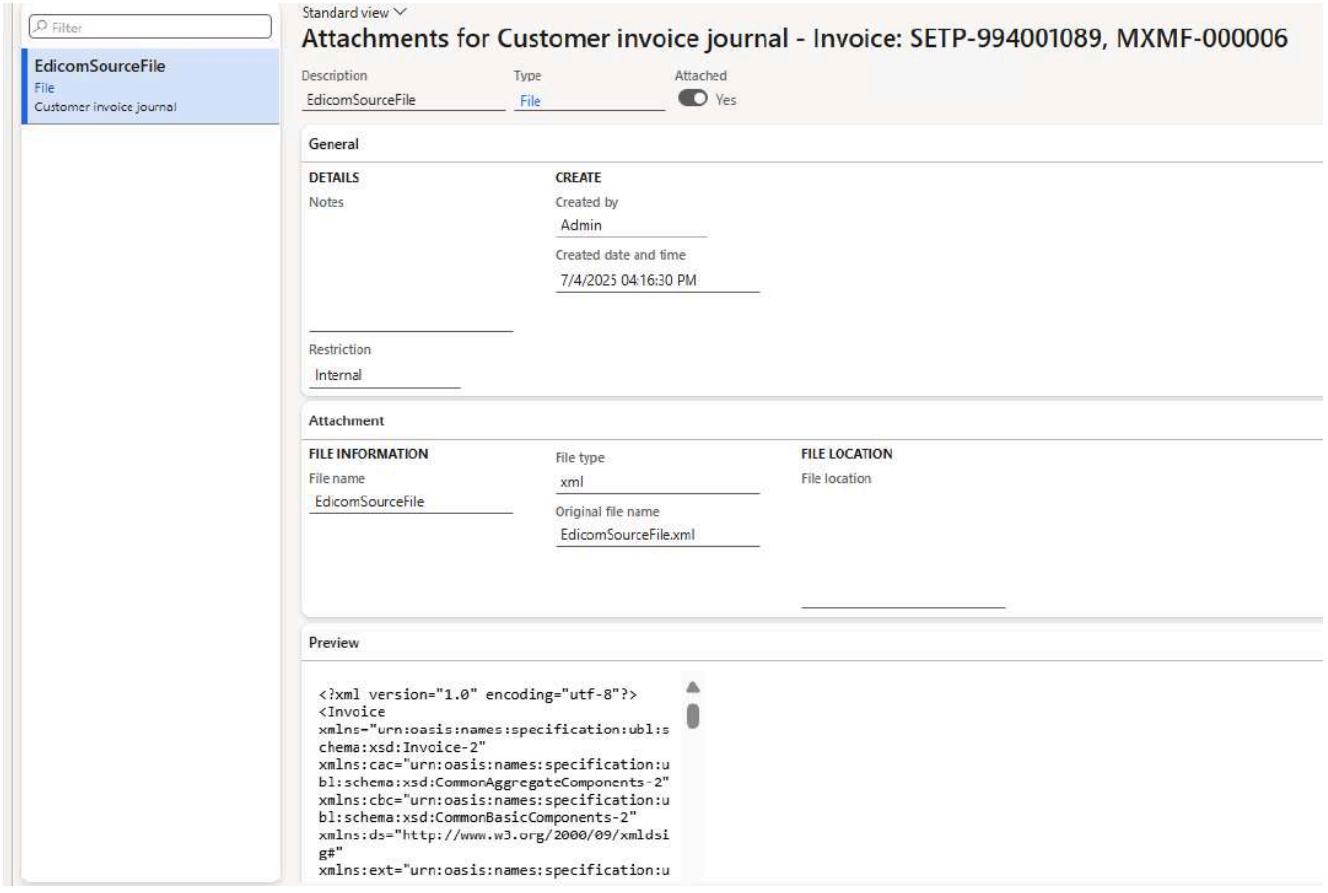


> **Note**
>
> By setting a recurrence schedule, you can configure the **Execute update actions** function to run in batch mode on a periodic basis.

When you review the submission details, you should notice that the steps run again. This time, the signed XML is received.



As a result of the completed outbound flow, the signed XML is attached to the invoice journal as **EdicomSourceFile**.



For Uruguay, the **Process response** action completes the pipeline after a few minutes.

More resources

- Electronic Invoicing service overview
- Get started with Electronic invoicing service administration
- Setting up Electronic Invoicing
- Electronic Invoicing service independent software vendor (ISV) last-mile connector
- Dynamics 365 Country/Region expansion: localizations for LATAM Countries/Regions | June 27, 2024
- Dynamics 365 Country/Region expansion: localizations for LATAM Countries/Regions | D365 FastTrack Tech Talks (youtube.com)
- Finance Localization for LATAM: Update on Additional Countries/Regions | Dynamics 365 Fast Track TechTalk | Jun 23, 2025
- Finance Localization for LATAM: Update on Additional Countries/Regions | Dynamics 365 Fast Track TechTalk (youtube.com)

Last updated on 09/02/2025

Electronic Invoicing service FAQ

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides answers to frequently asked questions about the Electronic Invoicing service. Electronic invoicing extends the electronic invoicing capabilities that exist in Microsoft Dynamics 365 Finance, Dynamics 365 Supply Chain Management, and Dynamics 365 Project Operations.

What's important about electronic invoicing, and why should it matter to my organization?

Operational complexity and risk intensify as organizations grow globally and expand their footprints across regions. Maintaining compliance and adapting to frequently changing regulations are growing challenges. These challenges are especially important for invoicing. Invoicing is traditionally expensive and prone to errors because companies rely on paper documents and manually intensive processes.

Organizations have begun to move away from paper invoices to help reduce cost and speed up the end-to-end process. Governments are increasingly turning to electronic invoicing as a key component of tax digitalization. By requiring that organizations digitally submit real-time tax information to tax authorities, governments can minimize tax evasion and manipulation, and reduce fraud.

The world is shifting to paperless document processing. Customers that don't implement electronic invoicing might risk compliance issues and unnecessary costs. They might also risk lagging behind their competitors.

Don't Finance, Supply Chain Management, and Project Operations already include electronic invoicing functionality? What value does this functionality provide to me as a customer?

Electronic invoicing extends the electronic invoicing capabilities that exist in Finance, Supply Chain Management, and Project Operations. Additionally, the functionality simplifies adherence to the newest electronic invoicing standards. It also provides coherent experiences for different geographies in electronic invoice processing and exchange. Electronic invoicing's capabilities unlock a range of benefits. Here are some examples:

- Faster and easier adoption of country/region-specific requirements
- Standardization of e-invoicing solution implementations
- Enhanced traceability of e-invoice processing
- Easier maintenance to comply with changing legal requirements and local business practices
- Simplified solution packaging
- Scaling capabilities for a large volume of submitted documents, to achieve faster turnaround
- The ability to share your solutions with other companies

Does the Electronic Invoicing service support on-premises installations of Finance, Supply Chain Management, and Project Operations?

No, the current platform doesn't support the on-premises versions, and there are no plans to support them in the future.

Does the Electronic Invoicing service interface with the vendor import automation feature?

No. There are plans for this interface, but there's no timeline. When Microsoft plans a timeline, it announces the dates in the Release plans.

How does the Electronic Invoicing service handle file attachments for an electronic invoice? Is a SharePoint server needed to embed PDF files into the XML file?

The Electronic Invoicing service handles files attached to an electronic invoice as embedded binary data in an XML element. You don't need a SharePoint server to embed PDF files.

However, you must store the attachment in the Finance, Supply Chain Management, and Project Operations document management system.

Is the Electronic Invoicing service available according to the regulations of my country/region?

The Electronic Invoicing service is a microservice platform that's globally available.

Microsoft plans to publish the electronic invoice formats and integrations for the countries/regions that are functionally localized by Microsoft. For more information, see Availability of electronic invoicing features.

If the electronic invoicing format for your country/region isn't listed, the platform aims to support this scenario in the future. You can still benefit from the configuration capabilities of Electronic invoicing and configure the electronic invoicing format by yourself. Alternatively, you can work with a partner or independent software vendor (ISV) to configure the electronic invoicing format for your organization. For more information, see Electronic Invoicing service ISV last-mile connector.

Do I have to sign up to get the Electronic Invoicing service, or do I just have to turn it on in Feature management?

If you have a license for Finance, Supply Chain Management, and Project Operations, see Get started with Electronic invoicing add-on service administration to sign up for Electronic invoicing.

Does the Electronic Invoicing service work with Tier 1 virtual machines? What's the minimal required environment?

Integration with the Electronic Invoicing service requires at least a Tier 2 virtual machine to host Finance or Supply Chain Management. For more information about environment planning, see Environment planning.

Does the Electronic Invoicing service have a test mode for testing invoice submission?

You can achieve testing through configuration. To test invoice submission, you can connect to Finance or Supply Chain Management from a user acceptance testing (UAT) environment and submit the test invoices. Electronic invoicing supports the configuration of test digital certificates. For e-invoices that require digital approval, it also supports the setup of a URL from test web services that the tax authorities publish.

Does the Electronic Invoicing service enable a legal entity in Finance or Supply Chain Management that's configured for a specific country/region to consume electronic invoicing features from different countries/regions?

Yes. You can use the electronic invoicing features to process business document submissions independently of the country/region of the legal entity. However, the following conditions must be met:

- The business document that's generated uses the appropriate document model mapping.
- There's a match between the business document and applicability rules that are configured in the electronic invoicing feature.

Does the Electronic Invoicing service use the same configuration and design experience that the Electronic reporting module in Finance and Supply Chain Management provides?

Yes. The same designer tools that the **Electronic reporting** (ER) module in Finance and Supply Chain Management uses are used to create and configure data model mapping and file format configurations.

Can I extend and configure the applicability rules so that they aren't tied to any specific parameter, such as a legal entity?

Yes. You can fully configure the applicability rules. You can add or remove clauses, build logic operations, and group and ungroup clauses.

Does the Electronic Invoicing service support adding other ER format configurations to generate other types of documents? Does it support electronically sending these documents to customers as delivery notes, for example?

You can use other ER format configurations to produce the desired output files. However, to generate business documents, the ER format configuration must be derived from the same ER invoice model mapping that's configured in Finance or Supply Chain Management. Out of the box, electronic invoicing doesn't support sending the output file directly to the customer as an electronic data interchange (EDI) transaction.

Does the Electronic Invoicing service support exchanging electronic invoices with customers? Does it support configuring different invoice formats for the same invoice?

The capability to receive and import vendor electronic invoices is on the roadmap. However, the capability to automatically send electronic invoices to customers isn't currently supported.

Does the Electronic Invoicing service support exchanging EDI messages with other companies?

The Electronic Invoicing service focuses on exchanging electronic invoice message types, driven by regulatory compliance requirements. The capability to receive and import vendor electronic invoices is on the roadmap. However, the capability to send electronic invoice files to customers isn't currently supported, except in scenarios where the electronic invoice must be sent to the customer as a regulatory requirement.

Does the Electronic Invoicing service support importing or merging customizations that are made in the ER format configurations from the legacy Electronic invoicing feature?

You can reuse ER format configurations in the Electronic Invoicing service. However, the ER format configuration must come from the same ER invoice model mapping that the Electronic invoicing feature was designed to use. Finance or Supply Chain Management uses this mapping to generate the business documents.

Does the Electronic Invoicing service support issuing electronic invoices from custom-made tables? If so, how do I create the ER data model configurations for these new tables and entities?

Yes, you can issue electronic invoices from custom-made tables. However, you must customize the invoice model mapping and add the required references to the custom-made tables. Alternatively, depending on the complexity, you must create a new invoice model mapping.

Does the Electronic Invoicing service support different web service endpoints?

Yes, Electronic Invoicing supports different web service endpoints. You can use configurable integration with REST web services or several parametrized country/region-specific web service integrations. You can configure different endpoints for the same web services and APIs by using different versions of configurations. For more information, see List of parameters by action.

Is Electronic invoicing integrated with the APIs of the various invoice operators from the Nordic countries/regions, or should this integration be handled on a case-by-case basis?

Microsoft continuously extends functional coverage to provide out-of-box integrations by using the Electronic invoicing features. For more information about the formats and integrations that are supported, see Availability of electronic invoicing features and Electronic Invoicing service ISV last-mile connector.

Last updated on 03/21/2026

Tax Calculation overview

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Tax Calculation helps automate and simplify the tax determination and calculation process. The tax engine is fully configurable. You can configure elements such as the taxable data model, tax code, tax applicability matrix, and tax calculation formula.

Tax Calculation is a highly scalable tax engine that's designed to support you in performing the following tasks:

- Automatically determine the correct sales tax group, item sales tax group, and tax codes through an enhanced determination mechanism.
- Support multiple tax registration numbers in one legal entity, and automatically determine the correct tax registration number on taxable transactions.
- Support tax determination, calculation, posting, and settlement for transfer orders.
- Define configurable tax calculation formulas and conditions for your specific business requirements.
- Share the tax determination and calculation solution across legal entities to save maintenance effort and avoid errors.
- Support customer and vendor tax registration number determination.
- Support list code determination.
- Support tax calculation parameters at the tax jurisdiction level.

For more information, see Get started with tax service.

Versions

Import and set up your Tax Calculation configuration by using the version that matches your Microsoft Dynamics 365 Finance or Microsoft Dynamics 365 Supply Chain Management version.

Expand table

Finance or Supply Chain Management version	Tax configuration version
10.0.47	Tax Calculation Configuration 50.78.269 This configuration is also compatible with all lower supported versions.
10.0.46	Tax Calculation Configuration 47.73.265
10.0.45	Tax Calculation Configuration 47.73.265

> **Important**
>
> To use **Tax Calculation Configuration 50.78.269**, you need to synchronize **Tax measure types**. To complete this step:
>
> 1. Open **Globalization studio** workspace, select **Electronic reporting** tile, and then select **Tax configurations** tile.
> 2. On **Configurations** page, select **Configurations**.
> 3. Under **Advanced settings** group, select **Tax measure types**.
> 4. On **Tax Measure types** page, select **Synchronize**.

Data flow

Here's an outline of the data flow process for Tax Calculation.

1. In the **Globalization Studio** workspace in Finance, view and import taxable document model configurations and model mapping configurations. If you need to extend configurations for an advanced scenario, see Add data fields in tax configurations.
2. In the **Globalization Studio** workspace in Finance, create or maintain tax features. Use tax features to maintain tax rates and tax applicability rules.
3. After you finalize the tax feature setup, change the **Feature version status** to *Completed*.
4. In **Tax calculation parameters**, select which tax feature setup version to use for a specific legal entity.
5. In Finance and Supply Chain Management, operate transactions as usual. When Tax Calculation is needed, the client collects information from the transaction, such as the sales order or purchase order, and packages the information as payload. The client then sends a request to calculate the tax.
6. The tax calculation service receives the tax calculation request from the client and completes the calculation. It returns the tax result to the client.

7. The Dynamics 365 client receives the tax result and presents the tax calculation result on a sales tax page.

Supported transactions

You can enable tax calculation by transaction.

The following table lists the transactions supported in each version.

Expand table

Version	Transactions
10.0.43	Customer advance invoice Vendor advance invoice Interest note Collection letter
10.0.42	Slip journal (Cash) [ITA] Customs declaration journal
10.0.41	Write-off journal Prepayment customer invoice (Sales order)
10.0.40	[POL] Single Administrative Document (SAD)
10.0.39	Prepayment handling
10.0.38	Project invoice proposal Journals (Hour/Expense/Item/Fee) Project quotations Intercompany customer invoice Microsoft Dynamics 365 Project Operations integration journal
10.0.36	Invoice register Invoice approval Invoice pool
10.0.29	Periodic journals
10.0.28	Vendor payment journal Customer payment journal
10.0.26	General journals Vendor invoice journal
10.0.23	Free text invoice
10.0.21	Sales

Version	Transactions
	• Sales quotation • Sales order • Confirmation • Picking list • Packing slip • Sales invoice • Credit note • Return order • Header miscellaneous charge • Line miscellaneous charge Purchase • Purchase order • Confirmation • Receipts list • Product receipt • Purchase invoice • Header miscellaneous charge • Line miscellaneous charge • Credit note • Return order • Purchase requisition • Purchase requisition line miscellaneous charge • Request for quotation • Request for quotation header miscellaneous charge • Request for quotation line miscellaneous charge Inventory • Transfer order – ship • Transfer order – receive

Supported countries/regions

You can run Tax Calculation with supported localization features. The following table lists the countries and regions for a legal entity's primary address.

Expand table

Version	Country or region
10.0.26	• China • Czech Republic

Version	Country or region
	• Spain
10.0.24	• Mexico
10.0.23	• Thailand • Japan • Malaysia • Singapore • Indonesia
10.0.22	• Australia • Bahrain • Canada • Egypt • Hong Kong SAR • Kuwait • New Zealand • Oman • Qatar • Saudi Arabic • South Africa • United Arab Emirates
10.0.21	• Austria • Belgium • Denmark • Estonia • Finland • France • Germany • Hungary • Iceland • Ireland • Italy • Latvia • Lithuania • Netherlands • Norway • Poland • Sweden • Switzerland • United Kingdom • United States

> **Note**
>
> For any country or region that Microsoft doesn't localize, you can enable and run Tax Calculation by using other global features.

Related resources

- Get started with tax service
- Multiple VAT registration number
- Tax feature support for transfer order
- How to build extension in tax service

Last updated on 02/17/2026

Get started with Tax Calculation

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about how to get started with Tax Calculation. The sections in this article guide you through the high-level design and configuration steps in Dynamics 365 Finance and Dynamics 365 Supply Chain Management.

High-level design

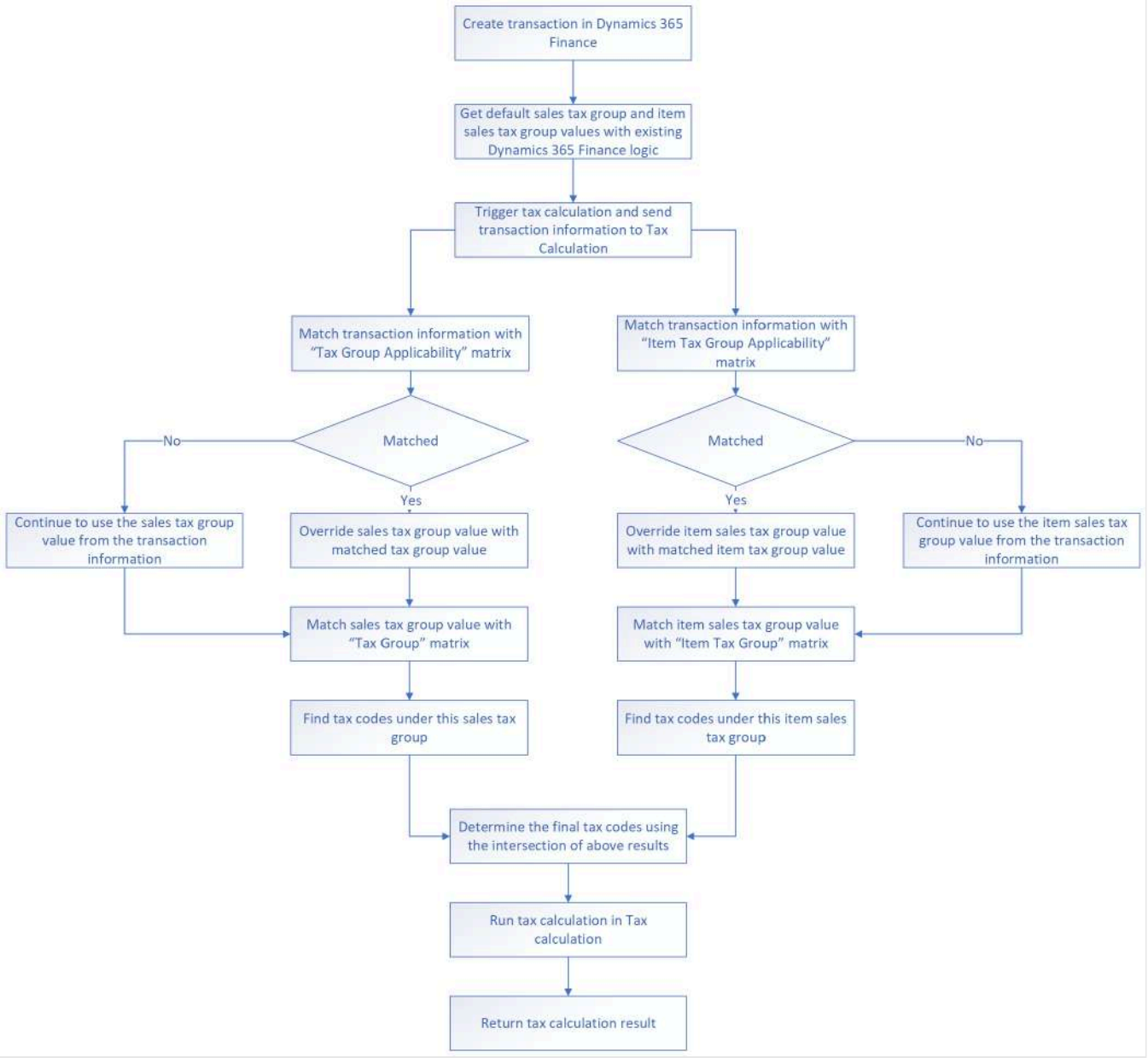
Runtime design

The following illustration shows the high-level runtime design of Tax Calculation.

1. A transaction, such as a sales order or purchase order, is created in Finance.
2. Finance automatically uses the default values of the sales tax group and the item sales tax group.
3. When the **Sales tax** button is selected on the transaction, the tax calculation is triggered. Finance then sends the transaction information to Tax Calculation.
4. Tax Calculation matches the transaction information with predefined rules in the tax feature to find a more accurate sales tax group and item sales tax group simultaneously.
    - If Tax Calculation matches the transaction information with the **Tax Group Applicability** matrix, it overrides the sales tax group value with the matched tax group value in the applicability rule. Otherwise, it continues to use the sales tax group value from the transaction information.
    - If Tax Calculation matches the transaction information with the **Item Tax Group Applicability** matrix, it overrides the item sales tax group value with the matched item tax group value in the applicability rule. Otherwise, it continues to use the item sales tax group value from the transaction information.
5. Tax Calculation determines the final tax codes by using the intersection of the sales tax group and the item sales tax group.

6. Tax Calculation calculates tax, based on the final tax codes that it determined.

7. Tax Calculation returns the tax calculation result. The **Sales tax** page shows the tax calculation result.



High-level configuration

The following steps provide a high-level overview of the configuration process for Tax Calculation.

1. In the **Globalization Studio** workspace, select the **Tax calculation** tile, and create the **Tax Calculation** feature.

2. Set up the **Tax Calculation** feature:
   a. Select the tax configuration version.
   b. Create tax codes.

c. Create a tax group.
d. Create an item tax group.
e. (Optional) Create tax group applicability if you want to override the default sales tax group that is entered from customer or vendor master data.
f. (Optional) Create item group applicability if you want to override the default item sales tax group that is entered from the item master data.

3. Complete the **Tax Calculation** feature.

4. In the **Tax calculation parameters**, select the **Tax Calculation** feature.

After you complete these steps, the following setups automatically sync from the Tax feature setup to Finance:

- Sales tax codes
- Sales tax groups
- Item sales tax groups

The remaining sections in this article provide more detailed configuration steps.

Configure the Tax Calculation feature

The steps in this section aren't related to a specific legal entity. Complete this procedure only once. You can complete it in any legal entity in Globalization Studio workspace in Finance (if you're using version 10.0.39 or later).

1. Import the correct tax configuration version, based on your Finance version. Follow the steps in Import Electronic reporting (ER) configurations from Dataverse.

2. Open the **Globalization Studio** workspace, select the **Tax Calculation** tile. On the **Tax calculation features** page, select the **Add** button, and select one of the following feature types:

   - **New feature** – Create a feature setup that has blank content.
   - **Based on existing feature** – Create a feature from an existing feature, and copy the content from the existing feature setup.

   > ⓘ **Note**
   >
   > You can import a demo feature for the **DEMF** demo legal entity. For more information, see **Import feature demo data**.

3. Enter a name and description for the feature, and then select **Create feature**.

After you create the feature, a draft version of it is automatically created. You can select **Get this version** to rebase the draft version on any completed version.

4. Select the draft version of the feature, and then select **Edit**. The **Tax Calculation setup** page is filled in.

5. Select **Configuration version**. You should see the configuration version that you imported.

   Microsoft provides a default tax configuration for tax calculation. This configuration covers most of the requirements for tax calculation behaviors. Microsoft updates it based on market feedback. If you must extend the configuration to meet specific requirements, see How to build extension in tax service for information about how to generate and select your own tax configuration.

6. After you select **Configuration version**, several additional tabs appear. Follow the order that is shown here to complete the mandatory tab setup.

   Mandatory setup

   - **Tax codes** – Maintain master data for tax codes. Finance automatically synchronizes all tax codes that you create on this tab when you enable the current version.
   - **Tax group** – Define the tax group master data and the tax codes under the group.
   - **Item tax group** – Define the item tax group master data and the tax codes under the group.

   Optional setup

   - **Tax group applicability** – Define a matrix that determines the tax group. If no applicability rules in this matrix match the taxable document from Dynamics 365, Tax Calculation uses the default value on the taxable document line.
   - **Item tax group applicability** – Define a matrix that determines the item tax group. If no applicability rules in this matrix match the taxable document from Dynamics 365, Tax Calculation uses the default value on the taxable document line.
   - **Customer tax registration number applicability** – If you have multiple tax registration numbers for one customer, Tax Calculation can automatically determine the correct tax registration number. In the matrix on this tab, define the rules that should be used to make the determination. Otherwise, Finance and Supply Chain Management continue to use the default tax registration number on taxable documents for sales transactions.
   - **Vendor tax registration number applicability** – If you have multiple tax registration numbers for one vendor, Tax Calculation can automatically determine the correct tax registration number. In the matrix on this tab, define the rules that should be used to make the determination. Otherwise, Finance and Supply Chain Management

continue to use the default tax registration number on taxable documents for purchase transactions.

- **List code applicability** – Automatically determine the value of the **List code** field through more flexible and configurable rules. In the matrix on this tab, define the rules that should be used to make the determination. Otherwise, Finance and Supply Chain Management continue to use the default code on taxable documents.

7. On the **Tax codes** tab, select **Add**, and enter the tax code and a description.

8. Select **Calculation origin**. The calculation origin defines how the tax base amount and the tax amount are calculated. The following values are available:

    - By net amount
    - By gross amount
    - By quantity
    - By margin
    - Tax on tax

9. Select **Save**. More fields become available, based on the **Calculation origin** that you selected.

10. Use the following options to identify the nature of the tax code:

    - Is exempt
    - Is use tax
    - Is reverse charge
    - Exclude from base amount calculation

    For a use tax scenario, set up a single tax code that has a positive tax rate, and mark it as **Is use tax**.

    For a reverse charge scenario, set up two tax codes, one of which has a positive tax rate, and the other of which has a negative tax rate but the same rate value. Mark the negative tax code as **Is reverse charge**. For more information about the reverse charge solution in Finance, see Reverse charge mechanism for VAT/GST scheme.

    For some tax types that should be excluded from the calculation of the tax base amount for price-inclusive transactions (for example, custom duty in some countries or regions), select the **Exclude from Base Amount Calculation** checkbox. For more information about this parameter, see Calculating tax on top of price when Prices include taxes is enabled.

    Maintain tax rates and the tax amount limits for this tax code.

11. Repeat steps 8 through 11 to add all other required tax codes.

12. On the **Tax group** tab, select the **Tax group** column, add it to the matrix as the input condition, and then add lines to maintain the tax group master data.

    Here's an example.

    Expand table

Tax group	Tax codes
DEU_Dom	DEU_VAT19; DEU_VAT7
DEU_EU	DEU_Exempt
BEL_Dom	BEL_VAT21; BEL_VAT6
BEL_EU	BEL_Exempt

13. On the **Item tax group** tab, select **Item tax group** column, add it to the matrix as the input condition, and then add lines to maintain the item tax group master data.

    Here's an example.

    Expand table

Item tax group	Tax codes
Full	DEU_VAT19; BEL_VAT21; DEU_Exempt; BEL_Exempt
Reduced	DEU_VAT7; BEL_VAT6; DEU_Exempt; BEL_Exempt

14. On the **Tax group applicability** tab, select the columns that are required to determine the correct tax group, and then select **Add**. Enter or select values for each column. The **Tax group** field is the output of this matrix. If you don't configure this tab, the sales tax group on the transaction line is used.

    Here's an example.

    Expand table

Business process	Ship from	Ship to	Tax group
Sales	DEU	DEU	DEU_Dom
Sales	DEU	FRA	DEU_EU
Sales	BEL	BEL	BEL_Dom
Sales	BEL	FRA	BEL_EU

> **Note**
>
> If the default sales tax group on your taxable document lines is correct, leave this matrix blank. For more information, see the Runtime design section in this article.

15. On the **Item tax group applicability** tab, select the columns that are required to determine the correct tax code, and then select **Add**. Enter or select values for each column. The **Item tax group** field is the output of this matrix. If you don't configure this tab, the item sales tax group on the transaction line is used.

    Here's an example.

    Expand table

Item code	Item tax group
D0001	Full
D0003	Reduced

    > **Note**
    >
    > If the default item sales tax group on your taxable document lines is correct, leave this matrix blank. For more information, see the Runtime design section in this article.

    For more information about how tax codes are determined in Tax Calculation, see Sales tax group and item sales tax group determination logic.

16. Set up the applicability of customer tax registration numbers, vendor tax registration numbers, and list codes, based on your business needs.

17. Select **Save**, and then close the page.

18. Select **Change status** > **Complete**. After the status is changed to **Complete**, you can't edit the version. Each legal entity can see this version of the tax feature setup on **Tax calculation parameters** page.

Set up Tax Calculation in Globalization studio Workspace

After you complete the setup in the Configure the Tax Calculation feature section, follow these steps to enable Tax Calculation and use the settings provided by the Tax Calculation feature.

Set up the configuration in this section by legal entity. You must configure it for each legal entity that you want to enable Tax Calculation for in Finance.

1. In Finance, go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**.
2. On the **General** tab, set the following fields:
    - **Enable advanced tax calculation** – Select this checkbox to enable Tax Calculation for the legal entity. If you don't enable it for the current legal entity, the legal entity continues to use the existing tax engine to determine and calculate tax.
    - **Feature name** – Select a completed tax feature setup and version for the legal entity. For more information about how to set up and complete a published tax feature, see the previous section of this article.
    - **Business Process** – Select the business processes to enable.
3. Use the **Sales tax rounding rule** group of fields to define the expected rounding rule for the legal entity. For more information about the rounding logic, see Tax calculation rounding rules.
4. On the **Error handling** tab, define the expected error handling method for the legal entity. Three options are available:
    - No
    - Warning
    - Error

    Set up an error handling method for each result code in the **Details** section.
5. On the **Multiple VAT registration** tab, you can turn on VAT declaration, EU Sales List, and Intrastat separately to work under a multiple VAT registrations scenario. For more information about tax reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.
6. Save the setup, and repeat the previous steps for each additional legal entity. When you complete a new version of Tax Calculation feature and want to apply it, set the **Feature name** field on the **General** tab of the **Tax calculation parameters** page (see step 2).

Last updated on 03/21/2026

Set up tax codes

This article explains how to set up tax codes in the Tax Calculation feature of Globalization studio workspace. It includes the setup for a simple scenario to make the tax code work and information about some advanced tax code functionality for complex scenarios.

> **Important**
>
> The setup of tax codes in the Tax Calculation of Globalization studio workspace is legal entity–agnostic. Tax codes are automatically synced to Sales tax codes when you enable the Enable advanced tax calculation in the Tax calculation parameters for a selected legal entity in Finance.

Simple setup

Follow these steps to use a tax code in a simple scenario, such as a scenario where there's only one tax rate.

1. Open the **Globalization Studio** workspace, select **Globalization services**, and then select the **Tax Calculation** tile.
2. On the **Tax calculation features** page, select the feature and version that you want to set up, and then select **Edit**.
3. On the **General** tab, select **Configuration version**.
4. On the **Tax codes** tab, select **Add**, and enter the tax code and a description.
5. Select **Calculation origin**. A calculation origin is a group of methods that are defined in the tax configuration version that you selected. For this simple scenario, select **By net amount**.
6. Select **Save**. More fields become available, based on the calculation origin that you selected.
7. On the **Rates** FastTab, select **Add** to add one tax rate for this tax code.
8. Select **Save**.

Calculation origin

The calculation origin defines how the system calculates the tax base amount and the tax amount. The **Calculation origin** field offers the following values:

- By net amount
- By gross amount

- By quantity
- By margin
- Tax on tax

By net amount

If you select **By net amount** in the **Calculation origin** field, the system calculates the tax amount as a percentage of the purchase or sales amount, excluding any other tax codes.

For example, the tax rate is 25 percent, the invoice line shows a quantity of 10 items at 1.00 each, and the customer is allowed a 10-percent line discount. In this case, the system calculates amounts as follows:

- **Net amount:** (10 × 1.00) – 10 percent = 9.00
- **Sales tax:** 9.00 × 25 percent = 2.25
- **Total invoice amount:** 9.00 + 2.25 = 11.25

By gross amount

If you select **By gross amount** in the **Calculation origin** field, the system calculates the tax amount as a percentage of the gross sales amount. The gross amount is the line's net amount plus all taxes and fees for the line except the one tax where the **Calculation origin** field is set to **By gross amount**.

For example, the tax authority imposes special duties on an item. The duty amounts are added to the net amount before tax is calculated. The following tax codes are used:

- **Duty 1** – The rate is 10 percent, and the **By net amount** calculation method is used.
- **Duty 2** – The rate is 20 percent, and the **By net amount** calculation method is used.
- **Tax rate** – The rate is 25 percent, and the **By gross amount** calculation method is used.

If the net amount is 10.00, the Duty 1 amount is 1.00 (10.00 × 10 percent), and the Duty 2 amount is 2.00 (10.00 × 20 percent). In this case, amounts are calculated in the following way:

- **Gross amount:** Net amount + Duty 1 amount + Duty 2 amount = 10.00 + 1.00 + 2.00 = 13.00
- **Tax amount:** 13.00 × 25 percent = 3.25
- **Total duties and tax:** 1.00 + 2.00 + 3.25 = 6.25
- **Total invoice amount:** 10.00 + 6.25 = 16.25

By quantity

If you select **By quantity** in the **Calculation origin** field, the tax amount is calculated as a fixed amount per unit and multiplied by the quantity that you enter on the document line. You specify the amount per unit on the **Rates** FastTab.

For example, you set up the tax code as 1.20 per unit. On a sales invoice line, you sell 25 units of an item. In this case, the tax amount is calculated as 25 × 1.20 = 30.00.

> ⓘ **Note**
>
> If you enter the transaction in a unit other than the unit that's specified on the tax code, the unit is automatically converted based on the unit conversions that are set up on the **Unit conversions** page. This function is available starting from the 10.0.40 update.

By quantity, additional option

You can choose whether to calculate a tax amount By quantity before other tax codes and add it to the net amount before other tax codes are calculated where **Calculation origin** = **By Net Amount**.

Examples

Assume two tax codes are calculated on a transaction:

- **Duty**: Calculation origin = By quantity, the value is set to 5.00 per unit = pcs
- **Tax**: Calculation origin = as shown in the examples below, the value is set to 25%

One piece of an item is sold at a unit price of 10.00.

Example 1

Tax: Calculation origin = By Gross Amount method. The **Calculate before sales tax** option has no effect, because **Tax** is calculated as a percentage of the gross amount. Duty: 1 × 5.00 = 5.00. Gross amount: 10.00 + 5.00 = 15.00. Tax: 15.00 × 25% = 3.75. Total tax: 5.00 + 3.75 = 8.75. Total amount: 10.00 + 8.75 = 18.75.

Example 2

Tax: Calculation origin = By Net Amount. The **Calculate before sales tax** option isn't selected for the **Duty** calculation. Net amount: 10.00. Duty: 1 × 5.00 = 5.00. Tax: 10.00 × 25% = 2.50. Total tax: 5.00 + 2.50 = 7.50. Total amount: 10.00 + 7.50 = 17.50.

Example 3

Tax: Calculation origin = By Net Amount. The **Calculate before sales tax option** is selected for the **Duty** calculation. Net amount: 10.00. Duty: 1 × 5.00 = 5.00. Tax: (10.00 + 5.00) × 25% = 3.75. Total tax: 5.00 + 3.75 = 8.75. Total amount: 10.00 + 8.75 = 18.75.

Example 4

The results of example 3 and example 1 are the same, because there's only one duty. Assume that you have two duties, and only one of them is included in the net amount for the sales tax calculation: Duty 1: 5.00, using the By Quantity method, and the **Calculate before sales tax** option is selected. Duty 2: 2.50, using the By Quantity method, and the **Calculate before sales tax** option isn't selected. Tax: 25%, using the By Net Amount method. Net amount: 10.00. Duty 1: 1 × 5.00 = 5.00. Duty 2: 1 × 2.50 = 2.50. Net amount subject to tax: 10.00 + 5.00 = 15.00. Tax: 15.00 × 25% = 3.75. Total taxes, including duties: 5.00 + 2.50 + 3.75 = 11.25. Total amount: 10.00 + 11.25 = 21.25. The 25% **Tax** is calculated for the sum of the net amount (10.00) + Duty 1 (5.00) = 15.00. Duty 2 is added to the tax amount after the sales tax is calculated.

By margin

If you select **By margin** in the **Calculation origin** field, the system calculates the tax amount as a percentage of the sales margin. The sales margin is the sales amount minus the cost amount. This calculation method applies only to sales transactions.

For example, the tax rate is 25 percent, the invoice line shows a quantity of 10 items at 10.00 each, and the cost per item is 6. In this case, amounts are calculated in the following way:

- **Sales margin:** 10 × (10.00 – 6.00) = 40.00
- **Tax amount:** 40.00 × 25 percent = 10.00
- **Total invoice amount:** 100.00 + 10.00 = 110.00

Tax on tax

If you select **Tax on tax** in the **Calculation origin** field, the system calculates the sales tax as a percentage of all the other tax amounts on the same document line.

For example, use the following tax codes:

- **Duty 1** – The rate is 10 percent, and the **By net amount** calculation method is used.
- **Duty 2** – The rate is 20 percent, and the **By net amount** calculation method is used.
- **Tax on duty** – The rate is 25 percent, and the **Tax on tax** calculation method is used.

In this case, amounts are calculated in the following way:

- **Net amount:** 10.00
- **Duty 1:** 10.00 × 10 percent = 1.00
- **Duty 2:** 10.00 × 20 percent = 2.00
- **Tax on duty:** 3.00 × 25 percent = 0.75
- **Total tax:** 1.00 + 2.00 + 0.75 = 3.75
- **Total invoice amount:** 10.00 + 3.75 = 13.75

Advanced functionality

This section explains some advanced functionality of the tax code setup for complex scenarios.

Tax exemption

If you set the **Is Exempt** option to **Yes** on the **General** FastTab, the tax amount is always 0, regardless of the actual tax rate.

Set the **Exempt Code** field to specify the reason for the exemption.

You can enable master data lookup for the **Exempt Code** field. By using this feature, you can select among the exempt code values that are defined in Finance. For information about how to enable master data lookup, see Enable master data lookup for tax calculation configuration.

For example, the tax rate is 25 percent, and the **Is Exempt** option is set to **Yes** on the tax code. The invoice line shows a quantity of 10 items at 1.00 each, and the customer is allowed a 10-percent line discount. In this case, amounts are calculated in the following way:

- **Net amount:** (10 × 1.00) – 10 percent = 9.00
- **Sales tax:** 0.00
- **Total invoice amount:** 9.00 + 0.00 = 9.00

Use tax

If you set the **Is Use Tax** option to **Yes** on the **General** FastTab, the tax amount posts to the **Use tax payable** account instead of the **Vendor summary** account.

For example, the tax rate is 25 percent, and you set the **Is Use Tax** option to **Yes** on the tax code. The invoice line shows a quantity of 10 items at 1.00 each, and the customer is allowed a 10-percent line discount. In this case, amounts are calculated in the following way:

- **Net amount:** (10 × 1.00) – 10 percent = 9.00
- **Use tax:** 9.00 × 25 percent = 2.25
- **Total invoice amount:** 9.00 + 0.00 = 9.00

> **Important**
>
> If you set both the **Is Exempt** option and the **Is Use Tax** option to **Yes** on a tax code, the code is recognized as tax-exempt for sales transactions and use tax for purchase transactions.

Reverse charges

If you set the **Is Reverse Charge** option to **Yes** on the **General** FastTab, you can configure the tax rate as negative. For a reverse charge scenario, set up two tax codes: one that has a positive tax rate and one that has a negative tax rate. Both tax codes should have the same rate value, and set the **Is Reverse Charge** option to **Yes** on the tax code that has a negative tax rate. For more information about the reverse charge solution in Finance, see Reverse charge mechanism for VAT/GST scheme.

For example, you determine two tax codes on one invoice line. One tax rate is 25 percent. The other tax rate is -25 percent, and you set the **Is Reverse Charge** option to **Yes** on the second tax code. The invoice line shows a quantity of 10 items at 1.00 each. In this case, the amounts are calculated in the following way:

- **Net amount:** (10 × 1.00) = 10.00
- **Tax code 1:** 10.00 × 25 percent = 2.50
- **Tax code 2:** 10 × -25 percent = -2.50
- **Total invoice amount:** 10.00 + 2.50 -2.50 = 10.00

Exclusion from base amount calculations

If you set the **Exclude from base amount calculation** option to **Yes** on the **General** FastTab, the calculated tax amount of the tax code is excluded from the tax base amount for other tax code calculations in the price-inclusive scenario.

For more information, see Calculate tax on prices when Prices include taxes is enabled.

Rates

On the **Rate** FastTab, you can define different tax rates for different ranges of tax base amounts. To calculate the tax amount, the Tax Calculation service always uses the rate that matches the tax base amount.

For example, you might configure tax rates as shown in the following table.

Expand table

Minimum amount	Maximum amount	Tax rate
0	1,000	10 percent
1,000	5,000	15 percent
5,000	10,000	20 percent
10,000	0	30 percent

In this case, calculate the tax amount as follows:

- If the tax base amount is 300.00, the tax rate is 10 percent, and the tax amount is 300.00 × 10 percent = 30.00.
- If the tax base amount is 3,000.00, the tax rate is 15 percent, and the tax amount is 3,000.00 × 15 percent = 450.00.
- If the tax base amount is 6,000.00, the tax rate is 20 percent, and the tax amount is 6,000.00 × 20 percent = 1,200.00.
- If the tax base amount is 20,000.00, the tax rate is 30 percent, and the tax amount is 20,000.00 × 30 percent = 6,000.00.

> **Note**
>
> If the tax base amount can match both the maximum amount on one line and the minimum amount on the other line, the base uses the tax rate that matches the minimum base amount. For example, if the tax base amount is 1,000.00, the tax rate is 15 percent, and the tax amount is 1,000.00 × 15 percent = 150.00.

Limits

On the **Limits** FastTab, you can define tax limits to override the calculated tax amount if the tax amount falls outside the minimum or maximum range.

- If the calculated tax amount is more than or equal to the maximum tax amount that you configure on the **Limits** FastTab, the final tax amount equals the maximum tax amount.
- If the calculated tax amount is less than the minimum tax amount that you configure on the **Limits** FastTab, the final tax amount is 0.

For example, the tax limits are configured in the following way:

- **Minimum tax amount:** 100
- **Maximum tax amount:** 1,000

If the calculated tax amount is 2,000, the final tax amount is 1,000.

If the calculated tax amount is 500, the final tax amount is 500.

If the calculated tax amount is 80, the final tax amount is 0.

Last updated on 03/21/2026

Set up tax groups

This article explains how to set up tax groups in the Tax Calculation feature. It also explains how to set up the tax group applicability rule matrix and configure lines in the matrix.

The concept of tax groups in the Tax Calculation feature resembles the concept of sales tax groups in Microsoft Dynamics 365 Finance. They're groups of tax codes. The Tax Calculation feature uses the intersection of a tax group and an item tax group to determine the tax codes.

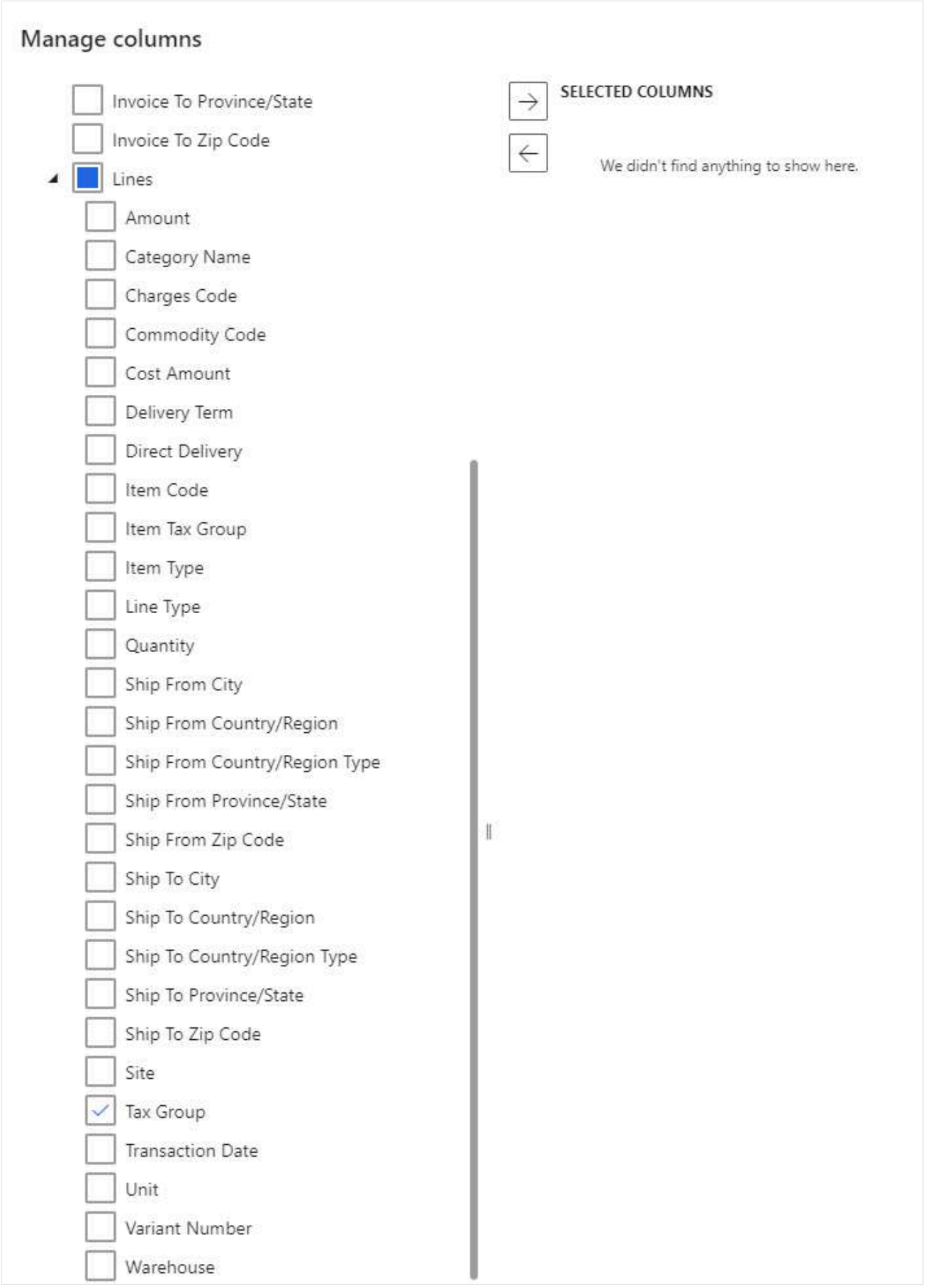
However, tax groups in the Tax Calculation feature differ from sales tax groups in Finance in that they don't have additional parameters, such as **Use tax** and **Exempt tax**. Instead, those parameters are available at the tax code level.

> **Important**
>
> The setup of tax groups in the Tax Calculation of Globalization studio workspace is legal entity–agnostic. When you enable the **Enable advanced tax calculation** in the Tax calculation parameters, the system automatically syncs tax groups to the Sales tax groups for the selected legal entity.

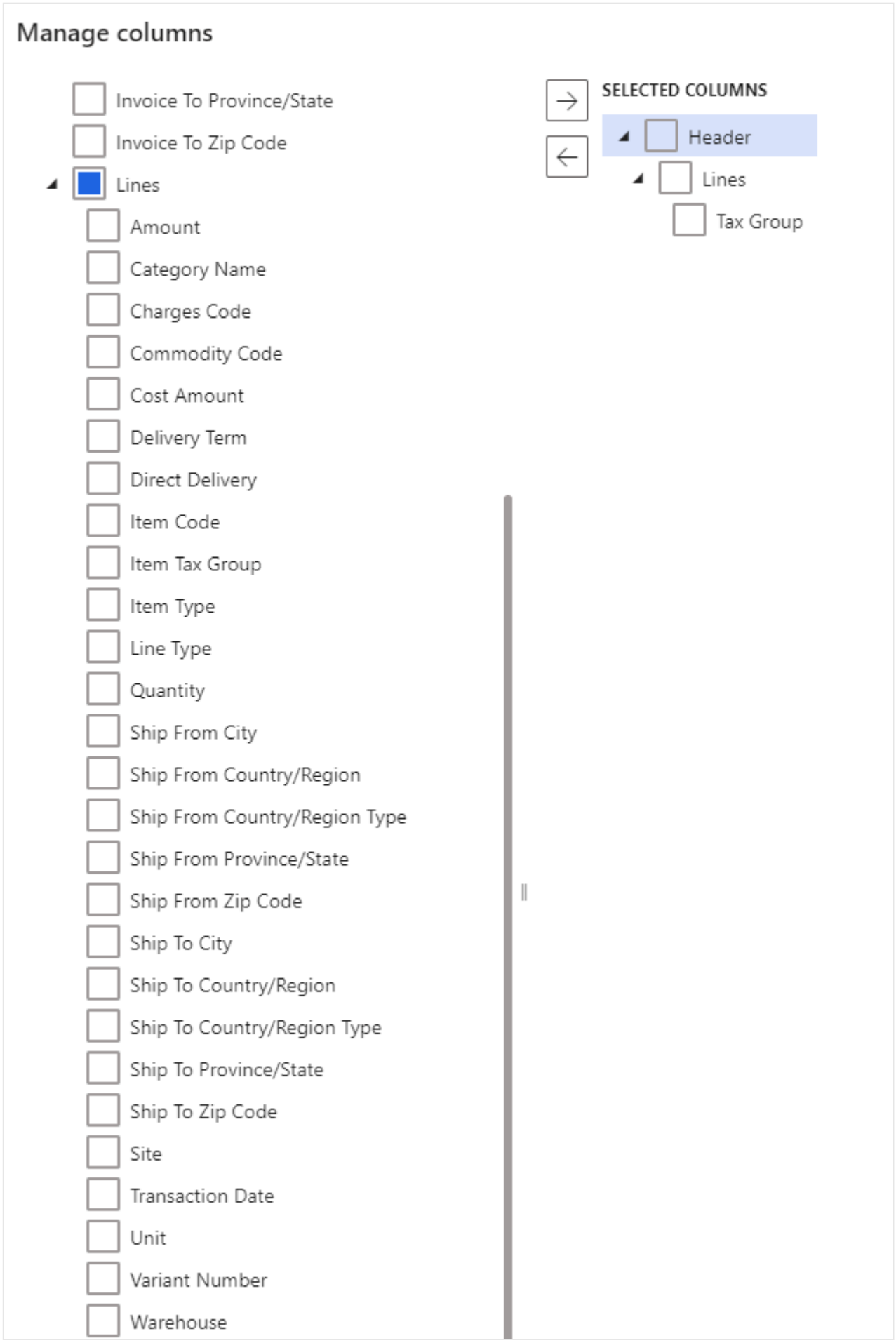
Set up a tax group

Follow these steps to set up a tax group.

1. In Finance, open the **Globalization Studio** workspace, select **Globalization services**, and then select the **Tax Calculation** tile.
2. On the **Tax calculation features** page, select the feature and version that you want to set up, and then select **Edit**.
3. On the **General** tab, select **Configuration version**.
4. On the **Tax group** tab, select **Manage Column**. If you're setting up a tax group for the first time, the fields in the **Manage column** dialog box are automatically set.
5. In the list on the left, expand the **Lines** node, and select the checkbox for **Tax Group**.



6. Select the right arrow button to add **Tax Group** to the **Selected columns** list on the right.


Manage columns
Invoice To Province/State
Invoice To Zip Code
Lines
Amount
Category Name
Charges Code
Commodity Code
Cost Amount
Delivery Term
Direct Delivery
Item Code
Item Tax Group
Item Type
Line Type
Quantity
Ship From City
Ship From Country/Region
Ship From Country/Region Type
Ship From Province/State
Ship From Zip Code
Ship To City
Ship To Country/Region
Ship To Country/Region Type
Ship To Province/State
Ship To Zip Code
Site
Transaction Date
Unit
Variant Number
Warehouse
SELECTED COLUMNS
Header
Lines
Tax Group

7. Select **OK**.

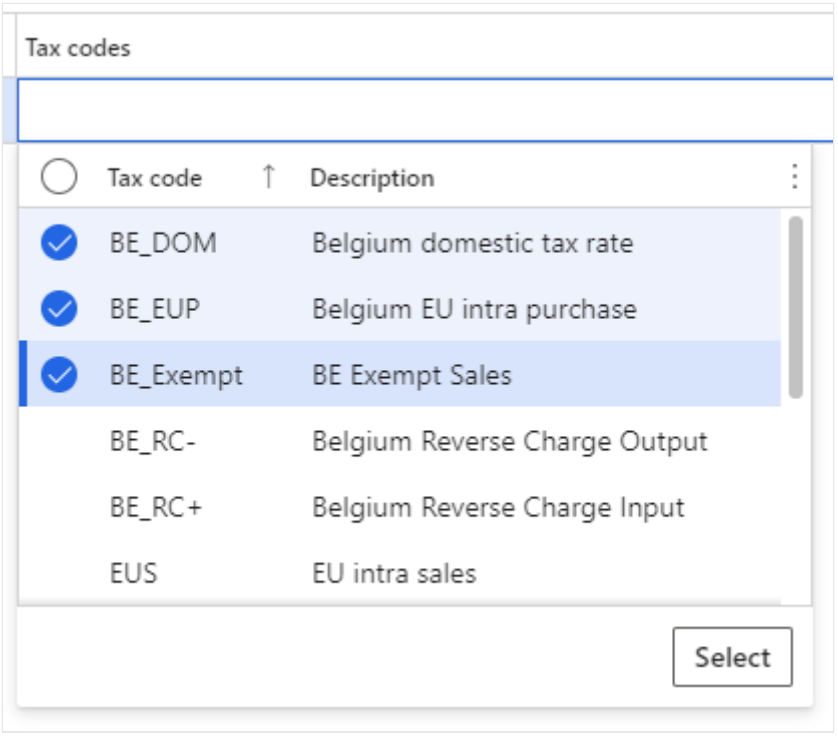
Configure a tax group

When you set up a tax group, you create the tax group applicability rule matrix. Add lines to the matrix to configure the tax group.

1. On the **Tax group** tab, select **Add**.
2. In the **Tax group** field, enter the name of the tax group.

   > **Important**
   >
   > Limit the name of the tax group to 10 characters. This name syncs with Finance, which has a limit of 10 characters for the names of sales tax groups.

3. In the **Tax codes** field, select the checkbox for each tax code that you want to include in the tax group. You can include multiple tax codes in one tax group.



4. Repeat steps 1 through 3 to add more tax groups.

Last updated on 03/21/2026

Set up item tax groups

This article explains how to set up item tax groups in the Tax Calculation feature. It also explains how to set up the item tax group applicability rule matrix and configure lines in the matrix.

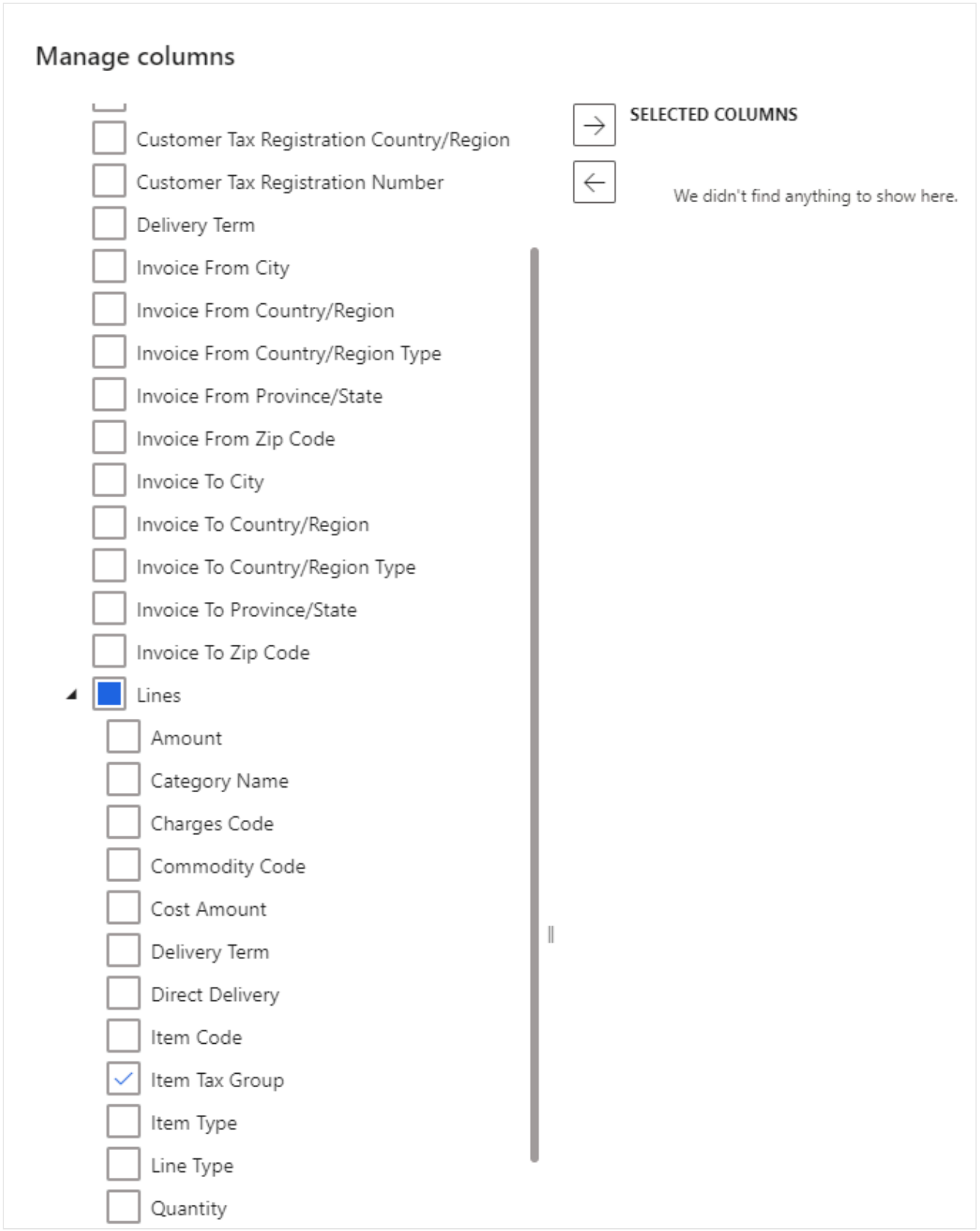
The concept of item tax groups in the Tax Calculation feature resembles the concept of item sales tax groups in Microsoft Dynamics 365 Finance. They're groups of tax codes. The Tax Calculation feature uses the intersection of a tax group and an item tax group to determine the tax codes.

> ⓘ **Important**
>
> The setup of item tax groups in the Tax Calculation feature of the Globalization studio workspace is legal entity–agnostic. When you enable the **Enable advanced tax calculation** in the Tax Calculation feature, the system automatically syncs item tax groups to the item sales tax groups for the selected legal entity.

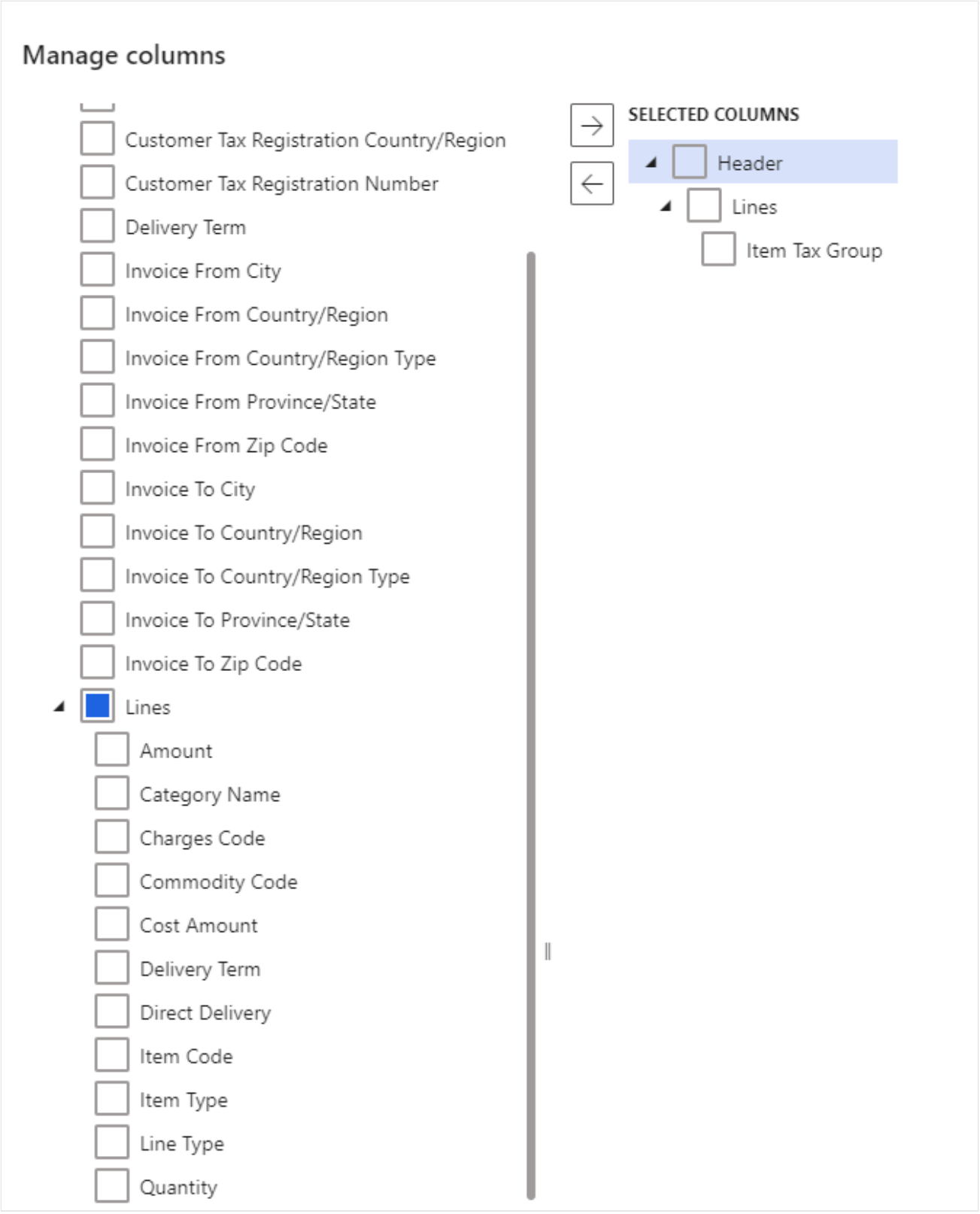
Set up an item tax group

Follow these steps to set up an item tax group.

1. In Finance, open the **Globalization Studio** workspace, select **Globalization services**, and then select the **Tax Calculation** tile.
2. On the **Tax calculation features** page, select the feature and version that you want to set up, and then select **Edit**.
3. On the **General** tab, select **Configuration version**.
4. On the **Item tax group** tab, select **Manage column**. If you're setting up an item tax group for the first time, the fields in the **Manage column** dialog box are automatically set.
5. In the list on the left, expand the **Lines** node, and select the checkbox for **Item Tax Group**.



6. Select the right arrow button to add **Item Tax Group** to the **Selected columns** list on the right.



7. Select **OK**.

Configure an item tax group

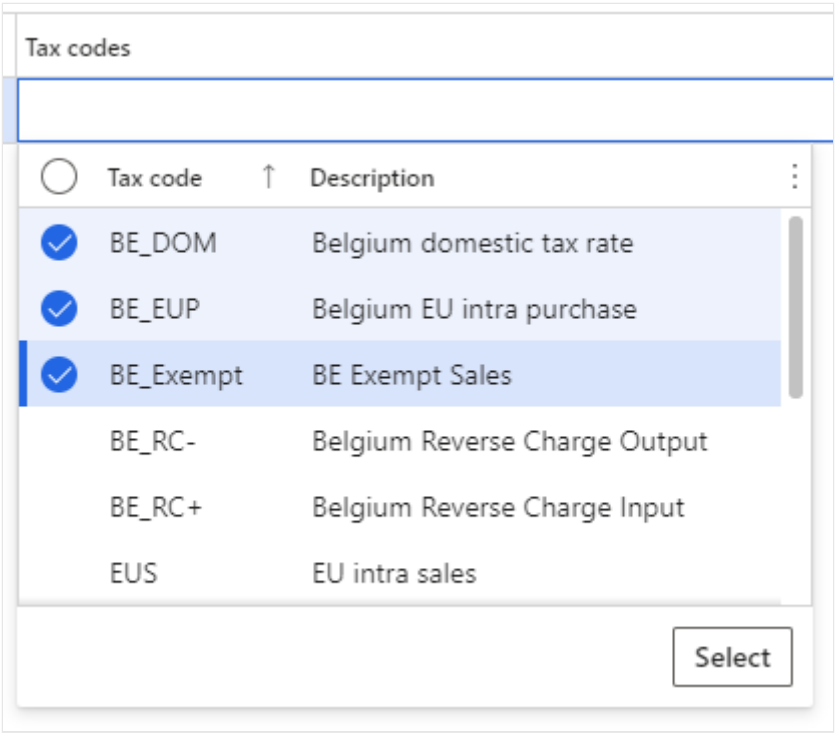
When you set up an item tax group, you create the applicability rule matrix. Add lines to the matrix to configure the item tax group.

1. On the **Item tax group** tab, select **Add**.

2. In the **Item tax group** field, enter the name of the item tax group.

> **Important**
>
> Limit the name of the item tax group to 10 characters. This name syncs with Finance, which has a limit of 10 characters for the names of item sales tax groups.

3. In the **Tax codes** field, select the checkbox for each tax code that you want to include in the item tax group. You can include multiple tax codes in one item tax group.



4. Repeat steps 1 through 3 to add more item tax groups.

Last updated on 03/21/2026

Import and export tax calculations

This article provides information about the import and export functionality of the tax calculation feature. This functionality helps ensure a flexible and efficient configuration experience.

Import and export tax codes

Export templates

1. Open the **Globalization Studio** workspace, select **Globalization services**, and then select the **Tax Calculation** tile.
2. On the **Tax calculation features** page, select an existing feature, or create a new feature.
3. In the **Versions** grid, select **Edit**.
4. On the **Tax calculation** feature page, set the configuration version.
5. On the **Tax codes** tab, select **Import**.
6. Select **Export a tax code template** to download the **TaxCodesTemplate.zip** file.
7. Unzip the **TaxCodesTemplate.zip** file. The tax code templates are compressed separately according to **Calculation origin** value.
8. Unzip the **By Net Amount.zip** file to get the following comma-separated values (CSV) files:
    - **TaxCode.csv** – Enter the tax codes.
    - **TaxLimit.csv** – Enter the tax limits for each tax code.
    - **TaxRate.csv** – Enter the tax rates for each tax code.

> ⓘ **Note**
>
> By default, the templates include entries for the **Sample** tax code. If you don't remove the **Sample** tax code from the CSV files, the import process adds it to the feature.

Import tax codes

Follow these steps to import the **Sample** tax code in the template into your tax calculation feature.

1. On the **Tax calculation features** page, select the feature. In the **Versions** grid, select **Edit**.
2. On the **Tax codes** tab, select **Import**.
3. Select **Browse**, find and select the **TaxCode.csv** file. In the **Target table** field, select **Tax code**.
4. Select **OK** to import the tax code.
5. Find and select the **TaxRate.csv** file. In the **Target table** field, select **Tax rate**.
6. Select **OK** to import the tax rate.
7. Find and select the **TaxLimit.csv** file. In the **Target table** field, select **Tax limit**.
8. Select **OK** to import the tax limit.

You can also directly import the ZIP file that includes all three CSV files. By using this method, you can quickly and easily complete the import.

1. On the **Tax calculation** feature page, on the **Tax codes** tab, select **Import**.
2. Select **Browse**, find and select the **By Net Amount.zip** file, and then select **OK**.

Export tax codes

1. On the **Tax calculation features** page, select the feature. In the **Versions** grid, select **Edit**.
2. On the **Tax codes** tab, select **Export**.

   The **Export** button is available when there's at least one tax code in the **Tax codes** grid.
3. Select **OK** to export all tax codes of the feature in a ZIP file.

> **Note**
>
> Use the exported file as a template to import tax codes into the corresponding feature.

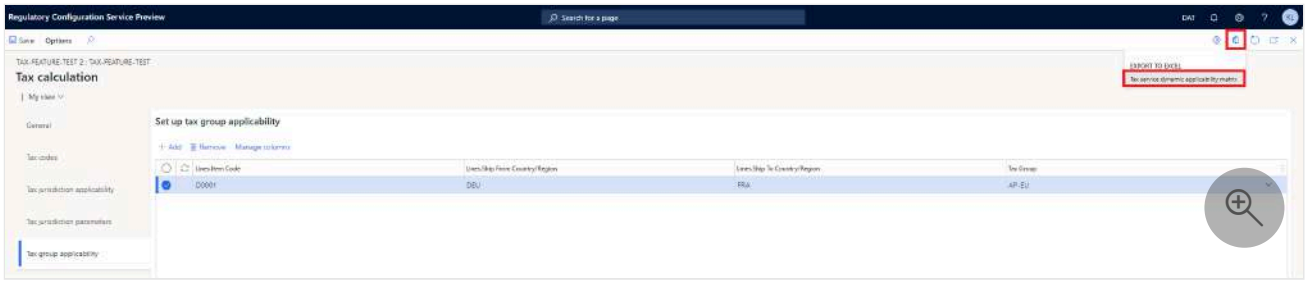
Import and export applicability rules

Export applicability rules

1. On the **Tax calculation features** page, select an existing feature, or create a new feature.
2. In the **Versions** grid, select **Edit**.
3. On the **Tax calculation** feature page, set the configuration version.

4. On the **Tax group applicability** tab, select the rows in the **Set up tax group applicability** grid.

5. Select the **Open in Microsoft Office** button, and then select **Tax service dynamic applicability matrix**.



6. Select **Download** to export the selected rows to a Microsoft Excel worksheet.

Import applicability rules

The Excel worksheet that you downloaded contains the structure of the **Set up tax group applicability** grid. You can add new rules directly in the worksheet. When you finish, follow these steps to import the new rules back into the **Set up tax group applicability** grid.

1. In the Excel worksheet, select and copy the rows to import.
2. On the **Tax group applicability** tab, select **Add** to insert an empty record at the bottom of the **Set up tax group applicability** grid.
3. Select **Ctrl+V** to paste the copied rows into the grid.
4. Select **Save**.

Import feature demo data

Follow these steps to import feature demo data.

1. On the **Tax calculation features** page, select **Import**. Then, on the **Import feature from Dataverse repository** page, select **Synchronize**.
2. In the table, select the **tax-calculation-feature-demo-data** feature, and then select **Import**.
3. Select **View** to review the tax codes, groups, and applicability rules that are defined in the imported feature.
4. Switch to the **DEMF** legal entity, and then go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**.
5. On the **General** tab, select **Enable advanced tax calculation**.
6. In the **Feature setup name** field, select **tax-calculation-feature-demo-data**.

7. Select a settlement period and a ledger posting group for the new demo tax codes, and then select **Confirm**.
8. Select **Save**.

> **Note**
>
> The **tax-calculation-feature-demo-data** demo feature is based on feature version **40.54.234** and is designed for the **DEMF** demo legal entity.

Last updated on 03/21/2026

Enable master data lookup for tax calculation configuration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up and enable the tax calculation master data lookup functionality. A dropdown list is available to select values in the tax calculation configuration for fields such as **Legal entity**, **Vendor account**, **Item code**, and **Delivery term**.

1. In the **Feature management** workspace, enable the **Enable applicability rules value lookup for Tax Calculation** feature.

   > **Note**
   >
   > Starting from version 10.0.43, you can't enable this feature through feature management. Instead, use the **Enable lookups in applicability rules** parameter in the **Tax feature setup**.

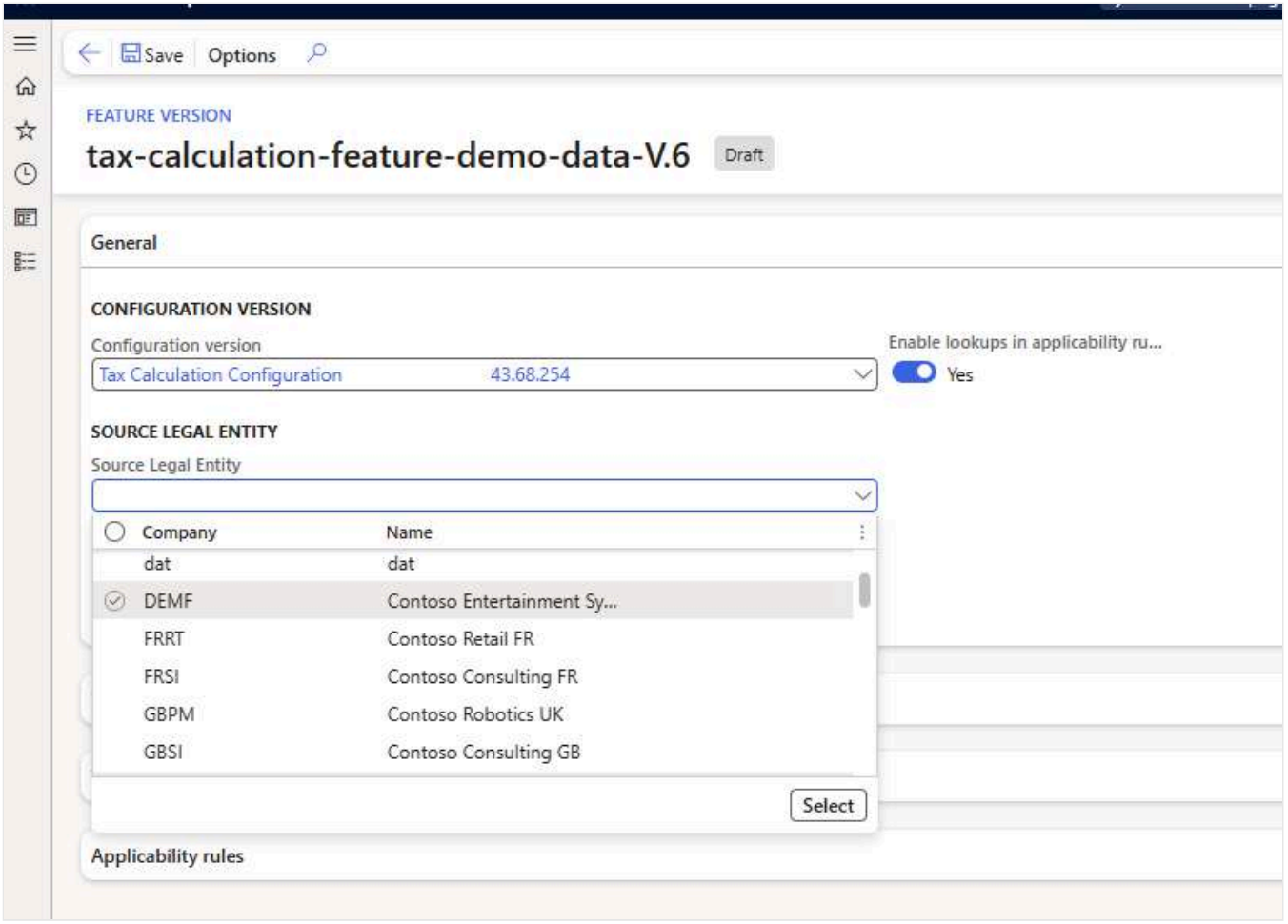
2. Make sure that you imported the latest configuration and model mapping versions. If you didn't, follow the steps in Import Electronic reporting (ER) configurations from Dataverse to import them.

   - Tax Data Model.version.40.xml
   - Tax Calculation Data Model 40.65.xml
   - FNO Model Mapping 40.65.35.xml (For this model mapping, set the **Default for model mapping** parameter to **Yes**.)
   - Tax Calculation Configuration 40.65.249.xml

3. On the tax **Feature version** setup page, select configuration version 40.65.249.

   > **Note**
   >
   > If you have a customized configuration, you must rebase the configuration and then use the customized configuration.

4. In the **Source legal entity** field, select legal entities. You can choose master data from these legal entities.



Last updated on 03/21/2026

Sync the tax setup from the Tax Calculation feature to Dynamics 365 Finance

This article explains how to sync tax setup master data from the Tax Calculation feature to Microsoft Dynamics 365 Finance.

After you complete the required setup steps in Get started with Tax Calculation, the Tax Calculation feature automatically syncs the following tax setup data to Finance.

Sales tax code

Expand table

Tax Calculation feature	Finance
Tax code	Sales tax code
Description	Name
User input	Settlement period
User input	Ledger posting group
User input	Sales tax currency
A negative tax rate is configured	Allow negative sales tax percentage

Tax value

Expand table

Tax Calculation feature	Finance
From transaction date	From date
To transaction date	To date
Minimum amount	Minimum limit
Maximum amount	Maximum limit
Tax rate	Value
Non-deductible rate	Non-deductible percentage

Tax limits

Expand table

Tax calculation feature	Finance
From transaction date	From date
To transaction date	To date
Minimum tax amount	Minimum sales tax
Maximum tax amount	Maximum sales tax

Sales tax group

Expand table

Tax calculation feature	Finance
Tax group	Sales tax group
Tax codes under this tax group	Sales tax codes under this sales tax group
The tax code is marked as **Is Exempt**	Exempt
The tax code is marked as **Is Exempt**	Exempt code
The tax code is marked as **Is Reverse Charge**	Reverse charge
The tax code is marked as **Is Use Tax**	Use tax

Item sales tax group

Expand table

Tax Calculation feature	Finance
Item tax group	Item sales tax group
Tax codes under this item tax group	Sales tax codes under this item sales tax group

After the synchronization finishes, continue to maintain the remaining parameters in Finance for posting and reporting purposes.

> ⓘ Note

Synchronization of the tax setup from Finance to the Tax Calculation feature isn't supported. For an existing Finance environment, first create the tax setup in the Tax Calculation feature. Then sync the setup data back to the Finance environment.

Last updated on 06/26/2026

Automate tax feature creation based on tax master data (Preview)

[This article is prerelease documentation and is subject to change.]

This article explains how to use the functionality to automatically create a Tax Calculation feature based on existing sales tax master data in Microsoft Dynamics 365 Finance.

This feature reduces time and effort for businesses by simplifying the transition from standard sales tax functionality to the advanced Tax Calculation functionality. An automated tax feature creation process enables you to adopt and realize the benefits of Tax Calculation faster and with less manual work. It reads existing sales tax master data in the legal entity and automatically creates the required records in the Tax feature setup, reusing existing tax codes, sales tax groups, and item sales tax groups. This approach minimizes setup complexity, preserves current configurations, and ensures a smoother transition to the advanced Tax Calculation experience.

> **Note**
>
> This functionality is supported globally, except for Brazil and India.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Prerequisites

Before you begin, ensure the following prerequisites are met:

- Import the latest Tax Configuration in **Electronic reporting** > **Tax configurations**. Learn more in Import Electronic reporting (ER) configurations from Dataverse or Import Electronic reporting (ER) configurations.

- Enable the **(Preview) Automate tax feature creation based on tax master data** feature in the **Feature management** workspace.

Enable the Automate tax feature

To enable the Automate tax feature, follow these steps:

1. In the **Feature management** workspace, find and enable the **(Preview) Automate tax feature creation based on tax master data** feature.
2. After you enable the Automate tax feature, the **Tax Data Migration (preview)** button becomes available in the header of the Tax Calculation workspace in Globalization Studio.

Run the autocreate tax feature process

To run the autocreate tax feature process, follow these steps:

1. Go to **Globalization Studio** > **Tax Calculation**.
2. In the header, select the **Tax Data Migration (preview)** button.

> ⓘ **Note**
>
> Although Globalization Studio settings are legal entity agnostic, this functionality works only for the currently selected legal entity. You must switch to the correct legal entity before starting the process.

When the process finishes, the system creates the tax feature version in a **Completed** state. The following message is displayed: **The tax feature 'xxxx' is created successfully. You can review the setup in the feature version by using the View button. If updates are needed, create a new version to edit the tax feature.**

To review the generated setup, use the **View** button. If you need to make changes, create a new version of the feature to make edits. Learn more in Configure the Tax Calculation feature.

Tax feature creation behavior overview

During the tax feature creation process, existing sales tax settings are mapped from the legal entity into the Tax Calculation feature setup. The following sections describe key behaviors.

Tax group level settings

In standard sales tax, define settings such as cash discount, use tax, reverse charge, and exempt at the tax group level. In Tax Calculation, define these settings at the line level in the tax feature. The functionality considers this setup mapping and creates the appropriate records during the tax feature creation.

Tax code variant handling

When you assign a tax code to multiple tax groups with different parameters, the system creates extra tax code variants during the tax feature creation process. For example:

- In Tax Group **Test1**, the same tax code **Test10** is marked as **Exempt**.
- In Tax Group **Test2**, tax code **Test10** has **Use tax** enabled.

The system creates multiple versions of the tax code (for example, Test10, Test10_1, Test10_2, and so on). Each version is assigned to the appropriate tax group with the corresponding parameters.

If a tax code name is already at the maximum length (10 characters) and a variant needs to be created, the name is truncated and a suffix is appended. For example, **Test123456** becomes **Test1234_1**. Despite truncation, the tax codes are correctly assigned to their respective tax groups.

Reverse sales tax on cash discount

The system sets the default tax jurisdiction parameters based on the General ledger parameter values. If the **Reverse sales tax on cash discount** value in a sales tax group differs from the General ledger parameter, the system displays the following warning: **The default Reverse tax on cash discount parameter differs from the one set in the group(s): 'xxxx', 'xxxx'. Once the process completes, please manually create rules to ensure proper configuration.**

After the tax feature is created, you must manually maintain the tax jurisdiction parameters to ensure the sales tax group-level parameter scenario works correctly. Learn more in Tax jurisdiction parameters setup.

Tax group character limit

If a tax group or item tax group contains tax codes whose total length exceeds 1,000 characters, the process stops and one of the following messages is displayed:

- **The tax group 'xxxx' contains too many or excessively long tax codes. The total length must not exceed 1,000 characters. The process has been canceled.**
- **The item tax group 'xxxx' contains too many or excessively long tax codes. The total length must not exceed 1,000 characters. The process has been canceled.**

Data mapping during tax feature creation

The following tables describe the mapping that applies during the tax feature creation process.

General ledger parameters to Tax jurisdiction parameters

Expand table

General ledger – Sales tax setting	Tax jurisdiction parameter after tax feature creation
Reverse sales tax on cash discount	Default Reverse Tax On Cash Discount
Deduct cash discount before sales tax calculation	Default Deduct Cash Discount Before Tax Calculation
Cash discount is calculated on amount including sales tax (Customer)	Default Calculate Cash Discount On Amount Including Tax – Customer
Cash discount is calculated on amount including sales tax (Vendor)	Default Calculate Cash Discount On Amount Including Tax – Vendor

Sales tax code origin to Tax code calculation origin

Expand table

Sales tax code origin	Tax Calculation feature origin
Percentage of gross amount	By Gross Amount
Percentage of net amount	By Net Amount
Percentage of margin	By Margin
Calculate percentage of net amount	By Percentage of Net
Amount per unit	By Quantity
Percentage of sales tax	Tax on Tax

Rounding precision behavior

In the Tax Calculation feature, the rounding precision behaves similarly to the configuration on the **Tax Configuration** page. However, the **Rounding Method** is initially set to **Ordinary**, which functions equivalently to **Normal** rounding.

After you create the tax feature

When the process finishes, review the following items:

1. **Tax jurisdiction parameters** – Verify that the mapped parameters from the general ledger match your business requirements. If the **Reverse sales tax on cash discount** value differs at the tax group level, manually create rules in the tax feature setup.
2. **Tax code variants** – Review any tax code variants that the process created with suffixes (_1, _2, and so on). Check the tax group assignments in the feature parameters to confirm they're correct.
3. **Tax code settings** – Verify that tax code origins, rates, and other settings are correct.
4. **Rounding method** – The default rounding method is set to **Ordinary**. Adjust if your business requires a different rounding method.
5. **No incremental updates** – The process always creates a tax feature based on the full current tax master data. If you modify tax master data after the tax feature is created (for example, add new tax codes, update tax groups, or change tax code settings), those changes aren't automatically reflected in the previously created tax feature. To incorporate the changes, you can either create a new version of the tax feature and update the setup manually, or run the **Tax Data Migration** process again to recreate the tax feature from scratch based on the current master data.

After you review and, if needed, update the created tax feature, assign it to the legal entity and start using it for tax calculation. To do so, follow the steps in Set up Tax Calculation in Globalization Studio workspace.

If you create new tax codes, sales tax groups, or item sales tax groups during the tax feature creation process (for example, tax code variants with suffixes), the process synchronizes those records back to the standard sales tax tables in Dynamics 365 Finance. Learn more in Sync the tax setup from the Tax Calculation feature to Finance.

Key considerations and known limitations

Expand table

Topic	Description
One legal entity at a time	The process can only create a tax feature for the legal entity that you're currently signed into. Switch to the correct legal entity before starting the process.
Manual adjustments for parameters	If tax code attributes differ from ledger parameters or tax groups, you might need to adjust parameters manually after the tax feature is created.

Topic	Description
Tax code name limited to 10 characters	Tax codes are limited to a maximum of 10 characters. Longer codes are truncated before a suffix is appended (for example, `Test123456` becomes `Test1234_1`).
Tax group character limit of 1,000	A tax group or item tax group that contains over 1,000 characters causes an error. The process stops and the following message is displayed: "The tax group 'xxxx' contains too many or excessively long tax codes. The total length must not exceed 1,000 characters. The process has been canceled."
Reverse Sales Tax on Cash Discount	If the value of **Reverse sales tax on cash discount** in a sales tax group differs from the default General ledger parameter, you must manually create rules after the tax feature is created to ensure proper configuration. Tax jurisdiction parameters setup
Tax code variant handling	If a tax code (for example, VAT0) exists in different tax groups with different settings (such as exempt or use tax), the process creates separate tax code variants with a suffix (for example, VAT0 and VAT0_1). The suffixes (_1, _2, and so on) are assigned in a nondeterministic order, but the parameter mappings remain correct. To identify which tax group a variant belongs to, check the tax group specified in the feature parameters.
Brazil and India not supported	The functionality isn't available for legal entities in Brazil and India.

Additional resources

- Tax Calculation overview
- Import Electronic reporting (ER) configurations from Dataverse
- Import Electronic reporting (ER) configurations

Last updated on 04/24/2026

Tax calculation data model

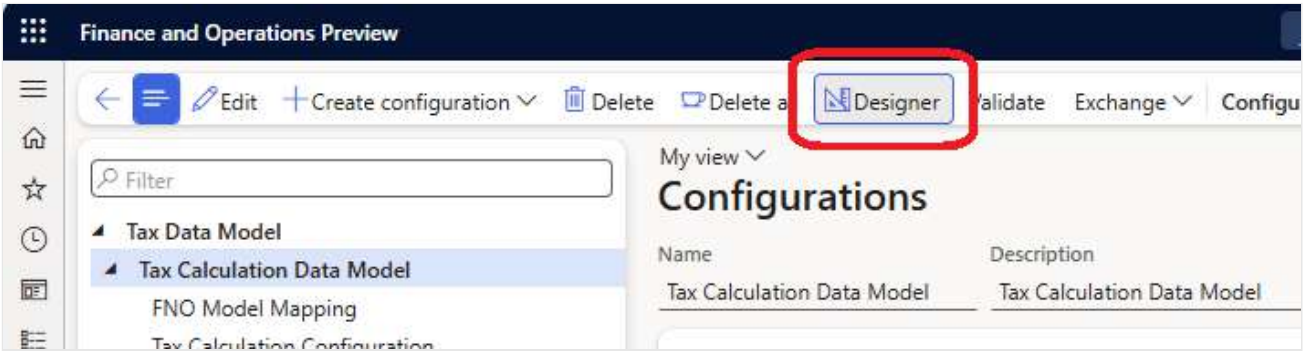
This article provides information about how the field values in the tax data model are determined in tax calculation transactions.

The *tax data model* consists of fields that are required for tax calculations. Microsoft provides the *tax calculation data model*, which includes the header fields and line fields of transaction documents in finance and operations apps. The fields that are defined in the tax calculation data model are the available columns of the applicability rules tables in the configuration of the tax calculation feature configuration.

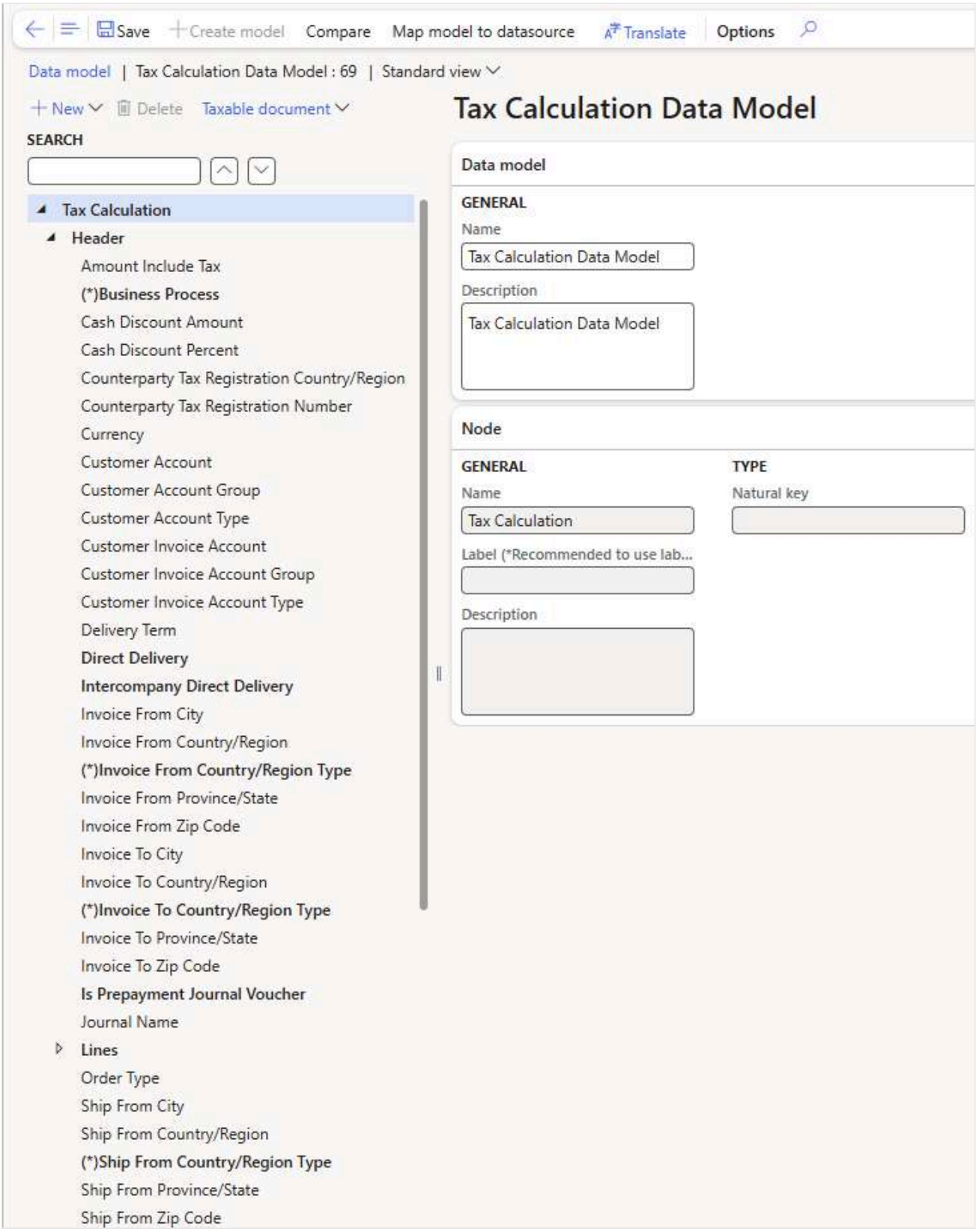
> **Note**
>
> Some nodes that are defined in the data model, such as **Record ID** and **Table ID**, are for technical purposes. They aren't available columns in the tax calculation feature configuration.

Follow these steps to view the tax calculation data model.

1. Open the **Globalization Studio** workspace, and select **Electronic reporting** > **Tax configurations**.
2. Select **Tax Data Model** > **Tax Calculation Data Model**.
3. On the **Versions** tab, select a version.
4. Select **Designer**.



5. Expand the **Tax Calculation** node.
6. Expand the **Header** node to view the header data models.
7. Expand the **Lines** node to view the line data models.



You can also view the tax calculation data model and the available columns in the applicability rules table.

1. Open the **Globalization Studio** workspace, select **Globalization services**, and then select the **Tax Calculation** tile.

2. On the **Tax calculation features** page, find a tax feature that has a status of **Draft**, and then select **Edit**.

3. On the **General** tab, select the configuration version, based on the version of the tax calculation data model. For example, select **40.46.212**.

4. On the **Tax group applicability** tab, select **Manage columns**.

The tax calculation data model is integrated with finance and operations apps. During each transaction that is related to tax calculation, the system collects the values of the fields that are defined in the applicability rules tables and sends them to the tax calculation service. In version 40.46, 57 predefined data models are available for the following transaction types: purchase orders, sales orders, transfer orders, purchase requisitions, requests for quotation, and sales quotations.

Version updates and restrictions

- **Free text invoice:** Supported as of version 40.50.221 and Finance version 10.0.23.
- **General journals:** Supported as of version 40.54.234 and Finance version 10.0.26.

  Restriction: One voucher and payment fee aren't supported.
- **Vendor invoice journal:** Supported as of version 40.54.234 and Finance version 10.0.26.
- **Periodic journals:** Supported as of version 40.55.236 and Finance version 10.0.29.

Field mapping

Thirty-two data models can be included as fields on the header of a transaction. However, not all fields are applicable to all transaction types. The following table shows which header fields are available for specific transaction types. In some cases, the field mapping must be determined by predefined logic. Where applicable, the logic is provided.

Expand table

Header field	Business transaction type and field mapping
Amount include tax	• **Sales order:** Prices include sales tax • **Purchase order:** Prices include sales tax • **Purchase requisition:** No • **Request for quotation:** Prices include sales tax • **Sales quotation:** Prices include sales tax • **Free text invoice:** Prices include sales tax • **Journal:** Amounts include sales tax
Business process	• **Sales order:** Sales • **Purchase order:** Purchase • **Transfer order - Ship:** Inventory • **Transfer order - Receive:** Inventory • **Purchase requisition:** Purchase • **Request for quotation:** Purchase • **Sales quotation:** Sales • **Free text invoice:** FTI • **Journal:** Journal
Currency	• **Sales order:** Transaction currency • **Purchase order:** Transaction currency • **Transfer order - Ship:** Accounting currency • **Transfer order - Receive:** Accounting currency • **Purchase requisition:** Transaction currency • **Free text invoice:** Transaction currency

Header field	Business transaction type and field mapping
	• **Journal:** Currency
Customer account	• **Sales order:** Customer account • **Sales quotation:** Customer account • **Free text invoice:** Customer account • **Journal (customer):** Customer account
Customer account group	• **Sales order:** Customer account group • **Sales quotation:** Customer account group • **Free text invoice:** Customer account group • **Journal (customer):** Customer account group
Customer invoice account	• **Sales order:** Invoice account • **Sales quotation:** Invoice account • **Free text invoice:** Invoice account • **Journal (customer):** 1. Customer master > Invoice account 2. Customer account
Customer invoice account group	• **Sales order:** Invoice account group • **Sales quotation:** Invoice account group • **Free text invoice:** Invoice account group • **Journal (customer):** 1. Customer master > Invoice account group 2. Customer account group
Customer tax registration country/region	• **Sales order:** Customer VAT ID country/region • **Sales quotation:** Customer VAT ID country/region • **Free text invoice:** Customer VAT ID country/region • **Journal (customer):** Customer VAT ID country/region
Customer tax registration number	• **Sales order:** Tax exempt number (registration category **VAT ID**) • **Sales quotation:** Tax exempt number (registration category **VAT ID**) • **Free text invoice:** Tax exempt number (registration category **VAT ID**) • **Journal (customer):** Tax exempt number (registration category **VAT ID**)
Delivery term	• **Sales order:** Delivery terms • **Purchase order:** Delivery terms • **Transfer order - Ship:** Delivery terms • **Transfer order - Receive:** Delivery terms • **Request for quotation:** Delivery terms • **Sales quotation:** Delivery terms

Header field	Business transaction type and field mapping
	• **Free text invoice:** Delivery terms
Direct delivery	• **Sales order:** Line details > Direct delivery • **Purchase order:** Direct delivery
Intercompany direct delivery	• **Sales order:** Header > Intercompany settings > Direct delivery
Invoice from city	• **Sales order:** 1. Legal entity > Default invoice address > City 2. Legal entity > Primary address > City • **Purchase order:** 1. Header > Invoice account > Default invoice address > City 2. Header > Invoice account > Primary address > City • **Sales quotation:** 1. Legal entity > Default invoice address > City 2. Legal entity > Primary address > City • **Free text invoice:** 1. Legal entity > Default invoice address > City 2. Legal entity > Primary address > City • **Journal (customer):** 1. Legal entity > Default invoice address > City 2. Legal entity > Primary address > City • **Journal (vendor):** 1. Vendor invoice account > Default invoice address > City 2. Vendor invoice account > Primary address > City 3. Vendor account > Default invoice address > City 4. Vendor account > Primary address > City
Invoice from country/region	• **Sales order:** 1. Legal entity > Default invoice address > Country/region 2. Legal entity > Primary address > Country/region • **Purchase order:** 1. Header > Invoice account > Default invoice address > Country/region 2. Header > Invoice account > Primary address > Country/region

Header field	Business transaction type and field mapping
	• **Sales quotation:** 1. Legal entity > Default invoice address > Country/region 2. Legal entity > Primary address > Country/region • **Free text invoice:** 1. Legal entity > Default invoice address > Country/region 2. Legal entity > Primary address > Country/region • **Journal (customer):** 1. Legal entity > Default invoice address > Country/region 2. Legal entity > Primary address > Country/region • **Journal (vendor):** 1. Vendor invoice account > Default invoice address > Country/region 2. Vendor invoice account > Primary address > Country/region 3. Vendor account > Default invoice address > Country/region 4. Vendor account > Primary address > Country/region
Invoice from country/region type	• If the **Invoice from country/region** and **Invoice to country/region** fields have the same value, the type is **Domestic**. • The mapping is determined by the configuration on the **Country/region properties** tab of the **Foreign trade parameters** page (**Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**), based on the value of the **Invoice from country/region** field. • Not applicable to **Purchase requisition** or **Request for quotation**.
Invoice from province/state	• **Sales order:** 1. Legal entity > Default invoice address > State 2. Legal entity > Primary address > State • **Purchase order:** 1. Header > Invoice account > Default invoice address > State 2. Header > Invoice account > Primary address > State • **Sales quotation:** 1. Legal entity > Default invoice address > State 2. Legal entity > Primary address > State • **Free text invoice:** 1. Legal entity > Default invoice address > State 2. Legal entity > Primary address > State

Header field	Business transaction type and field mapping
	• **Journal (customer):** 1. Legal entity > Default invoice address > State 2. Legal entity > Primary address > State • **Journal (vendor):** 1. Vendor invoice account > Default invoice address > State 2. Vendor invoice account > Primary address > State 3. Vendor account > Default invoice address > State 4. Vendor account > Primary address > State
Invoice from zip code	• **Sales order:** 1. Legal entity > Default invoice address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code • **Purchase order:** 1. Header > Invoice account > Default invoice address > ZIP/postal code 2. Header > Invoice account > Primary address > ZIP/postal code • **Sales quotation:** 1. Legal entity > Default invoice address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code • **Free text invoice:** 1. Legal entity > Default invoice address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code • **Journal (customer):** 1. Legal entity > Default invoice address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code • **Journal (vendor):** 1. Vendor invoice account > Default invoice address > ZIP/postal code 2. Vendor invoice account > Primary address > ZIP/postal code 3. Vendor account > Default invoice address > ZIP/postal code 4. Vendor account > Primary address > ZIP/postal code
Invoice to city	• **Sales order:** 1. Header > Invoice account > Default invoice address > City 2. Header > Invoice account > Primary address > City • **Purchase order:**

Header field	Business transaction type and field mapping
	1. Legal entity > Default invoice address > City 2. Legal entity > Primary address > City • **Sales quotation:** 1. Header > Invoice account > Default invoice address > City 2. Header > Invoice account > Primary address > City • **Free text invoice:** 1. Header > Invoice address > City 2. Header > Invoice account > Default invoice address > City 3. Header > Invoice account > Primary address > City • **Journal (customer):** 1. Customer invoice account > Default invoice address > City 2. Customer invoice account > Primary address > City 3. Customer account > Default invoice address > City 4. Customer account > Primary address > City • **Journal (vendor):** 1. Legal entity > Default invoice address > City 2. Legal entity > Primary address > City
Invoice to country/region	• **Sales order:** 1. Header > Invoice account > Default invoice address > Country/region 2. Header > Invoice account > Primary address > Country/region • **Purchase order:** 1. Legal entity > Default invoice address > Country/region 2. Legal entity > Primary address > Country/region • **Sales quotation:** 1. Header > Invoice account > Default invoice address > Country/region 2. Header > Invoice account > Primary address > Country/region • **Free text invoice:** 1. Header > Invoice address > Country/region 2. Header > Invoice account > Default invoice address > Country/region 3. Header > Invoice account > Primary address > Country/region • **Journal (customer):** 1. Customer invoice account > Default invoice address > Country/region 2. Customer invoice account > Primary address > Country/region

Header field	Business transaction type and field mapping
	3. Customer account > Default invoice address > Country/region 4. Customer account > Primary address > Country/region • **Journal (vendor):** 1. Legal entity > Default invoice address > Country/region 2. Legal entity > Primary address > Country/region
Invoice to country/region type	• If the **Invoice from country/region** and **Invoice to country/region** fields have the same value, the type is **Domestic**. • The mapping is determined by the configuration on the **Country/region properties** tab of the **Foreign trade parameters** page (**Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**), based on the value of the **Invoice to country/region** field. • Not applicable to **Purchase requisition** or **Request for quotation**.
Invoice to province/state	• **Sales order:** 1. Header > Invoice account > Default invoice address > State 2. Header > Invoice account > Primary address > State • **Purchase order:** 1. Legal entity > Default invoice address > State 2. Legal entity > Primary address > State • **Sales quotation:** 1. Header > Invoice account > Default invoice address > State 2. Header > Invoice account > Primary address > State • **Free text invoice:** 1. Header > Invoice address > State 2. Header > Invoice account > Default invoice address > State 3. Header > Invoice account > Primary address > State • **Journal (customer):** 1. Customer invoice account > Default invoice address > State 2. Customer invoice account > Primary address > State 3. Customer account > Default invoice address > State 4. Customer account > Primary address > State • **Journal (vendor):** 1. Legal entity > Default invoice address > State 2. Legal entity > Primary address > State

Header field	Business transaction type and field mapping
Invoice to zip code	• **Sales order:** 1. Header > Invoice account > Default invoice address > ZIP/postal code 2. Header > Invoice account > Primary address > ZIP/postal code • **Purchase order:** 1. Legal entity > Default invoice address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code • **Sales quotation:** 1. Header > Invoice account > Default invoice address > ZIP/postal code 2. Header > Invoice account > Primary address > ZIP/postal code • **Free text invoice:** 1. Header > Invoice address > ZIP/postal code 2. Header > Invoice account > Default invoice address > ZIP/postal code 3. Header > Invoice account > Primary address > ZIP/postal code • **Journal (customer):** 1. Customer invoice account > Default invoice address > ZIP/postal code 2. Customer invoice account > Primary address > ZIP/postal code 3. Customer account > Default invoice address > ZIP/postal code 4. Customer account > Primary address > ZIP/postal code • **Journal (vendor):** 1. Legal entity > Default invoice address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code
Is prepayment journal voucher	• **Journal:** From the **Prepayment journal voucher** field on the Customer or Vendor line
Journal name	• **Journal:** Name
Order type	• **Sales order:** Order type • **Purchase order:** Order type
Ship from city	• **Sales order:** 1. Header > Warehouse > Primary address > City 2. Header > Site > Primary address > City 3. Legal entity > Default delivery address > Country/region 4. Legal entity > Primary address > Country/region

Header field	Business transaction type and field mapping
	• **Purchase order:** 1. Transportation details > Loading address > City 2. Header > Vendor account > Default delivery address > City 3. Header > Vendor account > Primary address > City • **Transfer order - Ship:** 1. Header > From warehouse > Primary address > City 2. Header > From site > Primary address > City • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > City 2. Header > From site > Primary address > City • **Request for quotation:** 1. Header > Vendor account > Default delivery address > City 2. Header > Vendor account > Primary address > City • **Sales quotation:** 1. Header > Warehouse > Primary address > City 2. Header > Site > Primary address > City 3. Legal entity > Default delivery address > City 4. Legal entity > Primary address > City • **Free text invoice:** 1. Transportation details > Loading address > City 2. Legal entity > Default delivery address > City 3. Legal entity > Primary address > City • **Journal (customer):** 1. Legal entity > Default delivery address > City 2. Legal entity > Primary address > City • **Journal (vendor):** 1. Vendor account > Default delivery address > City 2. Vendor account > Primary address > City
Ship from country/region	• **Sales order:** 1. Header > Warehouse > Primary address > Country/region 2. Header > Site > Primary address > Country/region 3. Legal entity > Default delivery address > Country/region 4. Legal entity > Primary address > Country/region

Header field	Business transaction type and field mapping
	• **Purchase order:** 1. Transportation details > Loading address > Country/region 2. Header > Vendor account > Default delivery address > Country/region 3. Header > Vendor account > Primary address > Country/region • **Transfer order - Ship:** 1. Header > From warehouse > Primary address > Country/region 2. Header > From site > Primary address > Country/region • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > Country/region 2. Header > From site > Primary address > Country/region • **Request for quotation:** 1. Header > Vendor account > Default delivery address > Country/region 2. Header > Vendor account > Primary address > Country/region • **Sales quotation:** 1. Header > Warehouse > Primary address > Country/region 2. Header > Site > Primary address > Country/region 3. Legal entity > Default delivery address > Country/region 4. Legal entity > Primary address > Country/region • **Free text invoice:** 1. Transportation details > Loading address > Country/region 2. Legal entity > Default delivery address > Country/region 3. Legal entity > Primary address > Country/region • **Journal (customer):** 1. Legal entity > Default delivery address > Country/region 2. Legal entity > Primary address > Country/region • **Journal (vendor):** 1. Vendor account > Default delivery address > Country/region 2. Vendor account > Primary address > Country/region

Header field	Business transaction type and field mapping
Ship from country/region type	• If the **Ship from country/region** and **Ship to country/region** fields have the same value, the type is **Domestic**. • The mapping is determined by the configuration on the **Country/region properties** tab of the **Foreign trade parameters** page (**Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**), based on the value of the **Ship from country/region** field. • Not applicable to **Purchase requisition**.
Ship from province/state	• **Sales order:** 1. Header > Warehouse > Primary address > State 2. Header > Site > Primary address > State 3. Legal entity > Default delivery address > State 4. Legal entity > Primary address > State • **Purchase order:** 1. Transportation details > Loading address > State 2. Header > Vendor account > Default delivery address > State 3. Header > Vendor account > Primary address > State • **Transfer order - Ship:** 1. Header > From warehouse > Primary address > State 2. Header > From site > Primary address > state • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > State 2. Header > From site > Primary address > State • **Request for quotation:** 1. Header > Vendor account > Default delivery address > State 2. Header > Vendor account > Primary address > State • **Sales quotation:** 1. Header > Warehouse > Primary address > State 2. Header > Site > Primary address > State 3. Legal entity > Default delivery address > State 4. Legal entity > Primary address > State • **Free text invoice:** 1. Transportation details > Loading address > State 2. Legal entity > Default delivery address > State 3. Legal entity > Primary address > State • **Journal (customer):**

Header field	Business transaction type and field mapping
	1. Legal entity > Default delivery address > State 2. Legal entity > Primary address > State • **Journal (vendor):** 1. Vendor account > Default delivery address > State 2. Vendor account > Primary address > State
Ship from zip code	• **Sales order:** 1. Header > Warehouse > Primary address > ZIP/postal code 2. Header > Site > Primary address > ZIP/postal code 3. Legal entity > Default delivery address > ZIP/postal code 4. Legal entity > Primary address > ZIP/postal code • **Purchase order:** 1. Transportation details > Loading address > ZIP/postal code 2. Header > Vendor account > Default delivery address > ZIP/postal code 3. Header > Vendor account > Primary address > ZIP/postal code • **Transfer order - Ship:** 1. Header > From warehouse > Primary address > ZIP/postal code 2. Header > From site > Primary address > ZIP/postal code • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > ZIP/postal code 2. Header > From site > Primary address > ZIP/postal code • **Request for quotation:** 1. Header > Vendor account > Default delivery address > ZIP/postal code 2. Header > Vendor account > Primary address > ZIP/postal code • **Sales quotation:** 1. Header > Warehouse > Primary address > ZIP/postal code 2. Header > Site > Primary address > ZIP/postal code 3. Legal entity > Default delivery address > ZIP/postal code 4. Legal entity > Primary address > ZIP/postal code • **Free text invoice:** 1. Transportation details > Loading address > ZIP/postal code 2. Legal entity > Default delivery address > ZIP/postal code 3. Legal entity > Primary address > ZIP/postal code • **Journal (customer):**

Header field	Business transaction type and field mapping
	1. Legal entity > Default delivery address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code • **Journal (vendor):** 1. Vendor account > Default delivery address > ZIP/postal code 2. Vendor account > Primary address > ZIP/postal code
Ship to city	• **Sales order:** 1. Header > Delivery address > City • **Purchase order:** 1. Header > Delivery address > City • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > City 2. Header > To site > Primary address > City • **Transfer order - Receive:** 1. Header > To warehouse > Primary address > City 2. Header > To site > Primary address > City • **Request for quotation:** 1. Header > Delivery address > City • **Sales quotation:** 1. Header > Delivery address > City • **Free text invoice:** 1. Header > Customer account > Default delivery address > City 2. Header > Customer account > Primary address > City • **Journal (customer):** 1. Customer account > Default delivery address > City 2. Customer account > Primary address > City • **Journal (vendor):** 1. Legal entity > Default delivery address > City 2. Legal entity > Primary address > City
Ship to country/region	• **Sales order:** 1. Header > Delivery address > Country/region

Header field	Business transaction type and field mapping
	• **Purchase order:** 1. Header > Delivery address > Country/region • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > Country/region 2. Header > To site > Primary address > Country/region • **Transfer order - Receive:** 1. Header > To warehouse > Primary address > Country/region 2. Header > To site > Primary address > Country/region • **Request for quotation:** 1. Header > Delivery address > Country/region • **Sales quotation:** 1. Header > Delivery address > Country/region • **Free text invoice:** 1. Header > Customer account > Default delivery address > Country/region 2. Header > Customer account > Primary address > Country/region • **Journal (customer):** 1. Customer account > Default delivery address > Country/region 2. Customer account > Primary address > Country/region • **Journal (vendor):** 1. Legal entity > Default delivery address > Country/region 2. Legal entity > Primary address > Country/region
Ship to country/region type	• If the **Ship from country/region** and **Ship to country/region** fields have the same value, the type is **Domestic**. • The mapping is determined by the configuration on the **Country/region properties** tab of the **Foreign trade parameters** page (**Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**), based on the value of the **Ship to country/region** field. • Not applicable to **Purchase requisition**.
Ship to province/state	• **Sales order:** 1. Header > Delivery address > State

Header field	Business transaction type and field mapping
	• **Purchase order:** 1. Header > Delivery address > State • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > State 2. Header > To site > Primary address > State • **Transfer order - Receive:** 1. Header > To warehouse > Primary address > State 2. Header > To site > Primary address > State • **Request for quotation:** 1. Header > Delivery address > State • **Sales quotation:** 1. Header > Delivery address > state • **Free text invoice:** 1. Header > Customer account > Default delivery address > State 2. Header > Customer account > Primary address > State • **Journal (customer):** 1. Customer account > Default delivery address > State 2. Customer account > Primary address > State • **Journal (vendor):** 1. Legal entity > Default delivery address > State 2. Legal entity > Primary address > State
Ship to zip code	• **Sales order:** 1. Header > Delivery address > ZIP/postal code • **Purchase order:** 1. Header > Delivery address > ZIP/postal code • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > ZIP/postal code 2. Header > To site > Primary address > ZIP/postal code • **Transfer order - Receive:**

Header field	Business transaction type and field mapping
	1. Header > To warehouse > Primary address > ZIP/postal code 2. Header > To site > Primary address > ZIP/postal code • **Request for quotation:** 1. Header > Delivery address > ZIP/postal code • **Sales quotation:** 1. Header > Delivery address > ZIP/postal code • **Free text invoice:** 1. Header > Customer account > Default delivery address > ZIP/postal code 2. Header > Customer account > Primary address > ZIP/postal code • **Journal (customer):** 1. Customer account > Default delivery address > ZIP/postal code 2. Customer account > Primary address > ZIP/postal code • **Journal (vendor):** 1. Legal entity > Default delivery address > ZIP/postal code 2. Legal entity > Primary address > ZIP/postal code
Site	• **Sales order:** Site • **Purchase order:** Site • **Transfer order - Ship:** Site of the From warehouse • **Transfer order - Receive:** Site of the To warehouse • **Request for quotation:** Site • **Sales quotation:** Site
Tax direction	• **Sales order:** Output • **Purchase order:** Input • **Transfer order - Ship:** Output • **Transfer order - Receive:** Input • **Purchase requisition:** Input • **Request for quotation:** Input • **Sales quotation:** Output • **Free text invoice:** Output • **Journal:** Output
Vendor account	• **Purchase order:** Vendor account • **Journal (vendor):** Vendor account

Header field	Business transaction type and field mapping
Vendor account group	• **Purchase order:** Vendor account group • **Journal (vendor):** Vendor account group
Vendor invoice account	• **Purchase order:** Invoice account • **Journal (vendor):** 1. Vendor master > Invoice account 2. Vendor account
Vendor invoice account group	• **Purchase order:** Invoice account group • **Journal (vendor):** 1. Vendor master > Invoice account group 2. Vendor account group
Vendor tax registration country/region	• **Purchase order:** Vendor VAT ID country/region • **Journal (vendor):** Vendor VAT ID country/region
Vendor tax registration number	• **Purchase order:** Tax exempt number (registration category **VAT ID**) • **Journal (vendor):** Tax exempt number (registration category **VAT ID**)
Warehouse	• **Sales order:** Warehouse • **Purchase order:** Warehouse • **Transfer order - Ship:** From warehouse • **Transfer order - Receive:** To warehouse • **Request for quotation:** Warehouse • **Sales quotation:** Warehouse

Twenty-six data models are included as fields on the lines of a transaction. As for the header fields, not all fields are applicable to all transaction types. The following table shows which line fields are available for specific transaction types. In some cases, the field mapping must be determined by predefined logic. Where applicable, the logic is provided.

Expand table

Line field	Business transaction type and mapping
Amount	• **Sales order:** Net amount • **Purchase order:** Net amount • **Transfer order - Slip:** Amount • **Transfer order - Receive:** Amount • **Purchase requisition:** Net amount

Line field	Business transaction type and mapping
	• **Request for quotation:** Net amount • **Sales quotation:** Net amount • **Free text invoice:** Net amount • **Journal:** Debit/Credit
Category name	• **Sales order:** Sales category • **Purchase order:** Procurement category • **Purchase requisition:** Procurement category • **Request for quotation:** Procurement category • **Sales quotation:** Sales category
Charges code	• **Sales order:** Charges code • **Purchase order:** Charges code • **Purchase requisition:** Charges code • **Request for quotation:** Charges code • **Sales quotation:** Charges code • **Free text invoice:** Charges code
Commodity code	• **Sales order:** Line > Foreign trade > Commodity • **Purchase order:** Line > Foreign trade > Commodity • **Transfer order - Ship:** Line > Foreign trade > Commodity • **Transfer order - Receive:** Line > Foreign trade > Commodity • **Sales quotation:** Line > Foreign trade > Commodity • **Free text invoice:** Line > Foreign trade > Commodity
Cost amount	• **Sales order:** The cost price of the item in inventory • **Sales quotation:** The cost price of the item in inventory
Delivery term	• **Sales order:** Delivery terms • **Request for quotation:** Delivery terms • **Sales quotation:** Delivery terms
Direct delivery	• **Sales order:** Direct delivery • **Purchase order:** Direct delivery
Item code	• **Sales order:** Item number • **Purchase order:** Item number • **Transfer order - Ship:** Item number • **Transfer order - Receive:** Item number • **Purchase requisition:** Item number • **Request for quotation:** Item number • **Sales quotation:** Item number

Line field	Business transaction type and mapping
Item group	• **Sales order:** Item group of the item • **Purchase order:** Item group of the item • **Transfer order - Ship:** Item group of the item • **Transfer order - Receive:** Item group of the item • **Purchase requisition:** Item group of the item • **Request for quotation:** Item group of the item • **Sales quotation:** Item group of the item
Item type	• **Sales order:** The Product type of the Item code • **Purchase order:** The Product type of the Item code • **Transfer order - Ship:** The Product type of the Item code • **Transfer order - Receive:** The Product type of the Item code • **Purchase requisition:** The Product type of the Item code • **Request for quotation:** The Product type of the Item code • **Sales quotation:** The Product type of the Item code
Line type	• **Sales order:** The type of the current line. Options include header charge, line charge, and normal order line. • **Purchase order:** The type of the current line. Options include header charge, line charge, and normal order line. • **Transfer order - Ship:** Normal • **Transfer order - receive:** Normal • **Purchase requisition:** The type of the current line. Options include header charge, line charge, and normal order line. • **Request for quotation:** The type of the current line. Options include header charge, line charge, and normal order line. • **Sales quotation:** The type of the current line. Options include header charge, line charge, and normal order line. • **Free text invoice:** The type of the current line. Options include header charge, line charge, and normal order line. • **Journal:** Normal
Quantity	• **Sales order:** Quantity • **Purchase order:** Quantity • **Transfer order - Ship:** Quantity • **Transfer order - Receive:** Quantity • **Purchase requisition:** Quantity • **Request for quotation:** Quantity • **Sales quotation:** Quantity • **Free text invoice:** Quantity
Ship from city	• **Sales order:** 1. Header > Warehouse > Primary address > City 2. Header > Site > Primary address > City

Line field	Business transaction type and mapping
	3. Legal entity > Default delivery address > City 4. Legal entity > Primary address > City • **Purchase order:** 1. Transportation details > Loading address > City 2. Header > Vendor account > Default delivery address > City 3. Header > Vendor account > Primary address > City • **Transfer order - Ship:** 1. Header > From warehouse > Primary address > City 2. Header > From site > Primary address > City • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > City 2. Header > From site > Primary address > City • **Purchase requisition:** 1. Line > Vendor account > Default delivery address > City 2. Line > Vendor account > Primary address > City • **Request for quotation:** 1. Header > Vendor account > Default delivery address > City 2. Header > Vendor account > Primary address > City • **Sales quotation:** 1. Header > Warehouse > Primary address > City 2. Header > Site > Primary address > City 3. Legal entity > Default delivery address > City 4. Legal entity > Primary address > City • **Free text invoice:** 1. Transportation details > Loading address > City 2. Legal entity > Default delivery address > City 3. Legal entity > Primary address > City
Ship from country/region	• **Sales order:** 1. Line > Warehouse > Primary address > Country/region 2. Line > Site > Primary address > Country/region 3. Legal entity > Default delivery address > Country/region 4. Legal entity > Primary address > Country/region • **Purchase order:** 1. Transportation details > Loading address > Country/region

Line field	Business transaction type and mapping
	2. Header > Vendor account > Default delivery address > Country/region 3. Header > Vendor account > Primary address > Country/region • **Transfer order - Ship:** 1. Header > From warehouse > Primary address > Country/region 2. Header > From site > Primary address > Country/region • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > Country/region 2. Header > From site > Primary address > Country/region • **Purchase requisition:** 1. Line > Vendor account > Default delivery address > Country/region 2. Line > Vendor account > Primary address > Country/region • **Request for quotation:** 1. Header > Vendor account > Default delivery address > Country/region 2. Header > Vendor account > Primary address > Country/region • **Sales quotation:** 1. Line > Warehouse > Primary address > Country/region 2. Line > Site > Primary address > Country/region 3. Legal entity > Default delivery address > Country/region 4. Legal entity > Primary address > Country/region • **Free text invoice:** 1. Transportation details > Loading address > Country/region 2. Legal entity > Default delivery address > Country/region 3. Legal entity > Primary address > Country/region
Ship from country/region type	• If the **Ship from country/region** and **Ship to country/region** fields have the same value, the type is **Domestic**. • The mapping is determined by the configuration on the **Country/region properties** tab of the **Foreign trade parameters** page (**Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**), based on the value of the **Ship from country/region** field. • Not applicable to **Purchase requisition** or **Journal**.
Ship from province/state	• **Sales order:** 1. Line > Warehouse > Primary address > State 2. Line > Site > Primary address > State 3. Legal entity > Default delivery address > State 4. Legal entity > Primary address > State

Line field	Business transaction type and mapping
	• **Purchase order:** 1. Transportation details > Loading address > State 2. Header > Vendor account > Default delivery address > State 3. Header > Vendor account > Primary address > State • **Transfer order - Ship:** 1. Header > From warehouse > Primary address > State 2. Header > From site > Primary address > State • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > State 2. Header > From site > Primary address > State • **Purchase requisition:** 1. Line > Vendor account > Default delivery address > State 2. Header > Vendor account > Primary address > State • **Request for quotation:** 1. Header > Vendor account > Default delivery address > State 2. Header > Vendor account > Primary address > State • **Sales quotation:** 1. Line > Warehouse > Primary address > State 2. Line > Site > Primary address > State 3. Legal entity > Default delivery address > State 4. Legal entity > Primary address > State • **Free text invoice:** 1. Transportation details > Loading address > State 2. Legal entity > Default delivery address > State 3. Legal entity > Primary address > State
Ship from ZIP code	• **Sales order:** 1. Line > Warehouse > Primary address > ZIP/postal code 2. Line > Site > Primary address > ZIP/postal code 3. Legal entity > Default delivery address > ZIP/postal code 4. Legal entity > Primary address > ZIP/postal code • **Purchase order:** 1. Transportation details > Loading address > ZIP/postal code 2. Header > Vendor account > Default delivery address > ZIP/postal code 3. Header > Vendor account > Primary address > ZIP/postal code

Line field	Business transaction type and mapping
	• **Transfer order - Ship:** 1. Header > From warehouse > Primary address > ZIP/postal code 2. Header > From site > Primary address > ZIP/postal code • **Transfer order - Receive:** 1. Header > From warehouse > Primary address > ZIP/postal code 2. Header > From site > Primary address > ZIP/postal code • **Purchase requisition:** 1. Line > Vendor account > Default delivery address > ZIP/postal code 2. Header > Vendor account > Primary address > ZIP/postal code • **Request for quotation:** 1. Header > Vendor account > Default delivery address > ZIP/postal code 2. Header > Vendor account > Primary address > ZIP/postal code • **Sales quotation:** 1. Line > Warehouse > Primary address > ZIP/postal code 2. Line > Site > Primary address > ZIP/postal code 3. Legal entity > Default delivery address > ZIP/postal code 4. Legal entity > Primary address > ZIP/postal code • **Free text invoice:** 1. Transportation details > Loading address > ZIP/postal code 2. Legal entity > Default delivery address > ZIP/postal code 3. Legal entity > Primary address > ZIP/postal code
Ship to city	• **Sales order:** 1. Line > Delivery address > City • **Purchase order:** 1. Line > Delivery address > City 2. Header > Delivery address > City • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > City 2. Header > To site > Primary address > City • **Transfer order - Receive:** 1. Header > To warehouse > Primary address > City 2. Header > To site > Primary address > City

Line field	Business transaction type and mapping
	• **Purchase requisition:** 1. Line > Delivery address > City • **Request for quotation:** 1. Line > Delivery address > City • **Sales quotation:** 1. Line > Delivery address > City
Ship to country/region	• **Sales order:** 1. Line > Delivery address > Country/region • **Purchase order:** 1. Line > Delivery address > Country/region 2. Header > Delivery address > Country/region • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > Country/region 2. Header > To site > Primary address > Country/region • **Transfer order - Receive:** 1. Header > To warehouse > Primary address > Country/region 2. Header > To site > Primary address > Country/region • **Purchase requisition:** 1. Line > Delivery address > Country/region • **Request for quotation:** 1. Line > Delivery address > Country/region • **Sales quotation:** 1. Line > Delivery address > Country/region
Ship to country/region type	• If the **Ship from country/region** and **Ship to country/region** fields have the same value, the type is **Domestic**. • The mapping is determined by the configuration on the **Country/region properties** tab of the **Foreign trade parameters** page (**Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**), based on the value of the **Ship to country/region** field.

Line field	Business transaction type and mapping
	• Not applicable to **Purchase requisition** or **Journal**.
Ship to province/state	• **Sales order:** 1. Line > Delivery address > State • **Purchase order:** 1. Line > Delivery address > State 2. Header > Delivery address > State • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > ZIP/postal code 2. Header > To site > Primary address > ZIP/postal code • **Transfer order - Receive:** 1. Header > To warehouse > Primary address > State 2. Header > To site > Primary address > State • **Purchase requisition:** 1. Line > Delivery address > State • **Request for quotation:** 1. Line > Delivery address > State • **Sales quotation:** 1. Line > Delivery address > State
Ship to ZIP code	• **Sales order:** 1. Line > Delivery address > ZIP/postal code • **Purchase order:** 1. Line > Delivery address > ZIP/postal code 2. Header > Delivery address > ZIP/postal code • **Transfer order - Ship:** 1. Header > To warehouse > Primary address > ZIP/postal code 2. Header > To site > Primary address > ZIP/postal code • **Transfer order - Receive:** 1. Header > To warehouse > Primary address > ZIP/postal code 2. Header > To site > Primary address > ZIP/postal code

Line field	Business transaction type and mapping
	• **Purchase requisition:** 1. Line > Delivery address > ZIP/postal code • **Request for quotation:** 1. Line > Delivery address > ZIP/postal code • **Sales quotation:** 1. Line > Delivery address > ZIP/postal code
Site	• **Sales order:** 1. Line > Site 2. Product > Site • **Purchase order:** 1. Line > Site 2. Product > Site • **Transfer order - Ship:** The Site that is selected for the From warehouse on the line • **Transfer order - receive:** The Site that is selected for the From warehouse on the line • **Purchase requisition:** Line > Site • **Request for quotation:** Line > Site • **Sales quotation:** 1. Line > Site 2. Product > Site
Transaction date	• **Sales order:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page (**General ledger** > **Setup** > **General ledger parameters**). For example, if **Delivery date** is selected, the date is retrieved by using the following order: 1. Line confirmed ship date 2. Header confirmed ship date 3. Line requested ship date 4. Header requested ship date If the configured calculation date type isn't available in the transaction, the current session date is retrieved instead. • **Purchase order:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page. For example, if **Delivery date** is selected, the date is retrieved by using the following order:

Line field	Business transaction type and mapping
	1. Line confirmed delivery date 2. Line delivery date 3. Header delivery date If the configured calculation data type isn't available in the transaction, the current session date is retrieved instead. • **Transfer order - Ship:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page. For example, if **Delivery date** is selected, the date is retrieved by using the following order: 1. Ship date 2. Receipt date 3. Posting date If **Invoice date** or **Document date** is selected, the posting date is retrieved. • **Transfer order - Received:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page. For example, if **Delivery date** is selected, the date is retrieved by using the following order: 1. Ship date 2. Receipt date 3. Posting date If **Invoice date** or **Document date** is selected, the posting date is retrieved. • **Purchase requisition:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page. If the configured calculation date type isn't available in the transaction, the current session date is retrieved instead. • **Request for quotation:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page. If the configured calculation date type isn't available in the transaction, the current session date is retrieved instead. • **Sales quotation:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page. For example, if **Delivery date** is selected, the date is retrieved by using the following order: 1. Line > Attributes > Ship date 2. Line > Delivery > Requested ship date 3. Header requested ship date If the configured calculation date type isn't available in the transaction, the current session date is retrieved instead. • **Free text invoice:** The date that is used for tax calculation. The logic follows the configuration of the **Calculation date type** field on the **Sales tax** tab of the **General ledger parameters** page. For example, if **Delivery date** is selected, the date is retrieved by using the following order:

Line field	Business transaction type and mapping
	1. Line confirmed ship date 2. Header confirmed ship date 3. Line requested ship date 4. Header requested ship date If the configured calculation date type isn't available in the transaction, the current session date is retrieved instead. • **Journal:** Date
Unit	• **Sales order:** Unit • **Purchase order:** Unit • **Transfer order - Ship:** Unit • **Transfer order - Receive:** Unit • **Purchase requisition:** Unit • **Request for quotation:** Unit • **Sales quotation:** Unit
Variant number	• **Sales order:** Variant number • **Purchase order:** Variant number • **Sales quotation:** Variant number
Warehouse	• **Sales order:** 1. Line > Warehouse 2. Product > Warehouse • **Purchase order:** 1. Line > Warehouse 2. Product > Warehouse • **Transfer order - Ship:** From warehouse • **Transfer order - Receive:** To warehouse • **Purchase requisition:** Line > Warehouse • **Request for quotation:** Line > Warehouse • **Sales quotation:** 1. Line > Warehouse 2. Product > Warehouse

Last updated on 03/19/2026

Sales tax applicability and sales tax group determination logic

This article explains the logic for determining sales tax applicability and sales tax groups in the tax feature setup.

Matching logic

The applicability logic tries to match the condition that includes the most input fields. Each field that has a value has a *weight* of 10. A condition that has a higher total weight has a higher priority.

Example

The following example shows how the matching logic works.

The **Tax group applicability** tab is configured as shown in the following table.

Expand table

Business process	Currency	Item code	Tax group	Weight
Purchase	EUR		TG_A	20
Purchase	EUR	D0001	TG_B	30

For the first line in the table, two input fields are set. Therefore, the line has a weight of 20. For the second line, three fields are set. Therefore, the line has a weight of 30.

When sales tax is calculated for a purchase order that has the EUR currency and item D0001, Tax Calculation uses the condition that has a higher weight. Therefore, it uses the second line in the table.

Adjust execution sequence

You can adjust the execution sequence of the applicability rules that have equal weight.

Example

The following example shows how the **Adjust execution sequence** button works.

The **Tax group applicability** tab is configured as shown in the following table.

Expand table

Business process	Currency	Item code	Tax group	Weight
Purchase	EUR		TG_A	20
Purchase		D0001	TG_B	20

According to the matching logic, the rules in the table have equal weight. If you purchase item **D0001** with the transaction currency **EUR**, the first rule (Tax group **TG_A**) applies.

Complete the following steps to adjust the rule that applies.

1. Select **Adjust execution sequence**.
2. Select the second rule, and then select **Move up**.
3. Select **Complete**.

Now, the second rule (Tax group **TG_B**) is moved above the first one and applies first.

> **Note**
>
> You can't move a rule above or under another rule that has a different **Weight**.

Sales tax group and item sale tax group determination logic

When you create a document line on a business document, such as a sales order, purchase order, or vendor invoice, the system sets the **Sales tax group** and **Item sales tax group** fields to the values from the master data. Then, when tax is calculated, the system updates these fields according to the conditions that you configure in the Tax feature setup on the **Tax group applicability** and **Item tax group applicability** tabs.

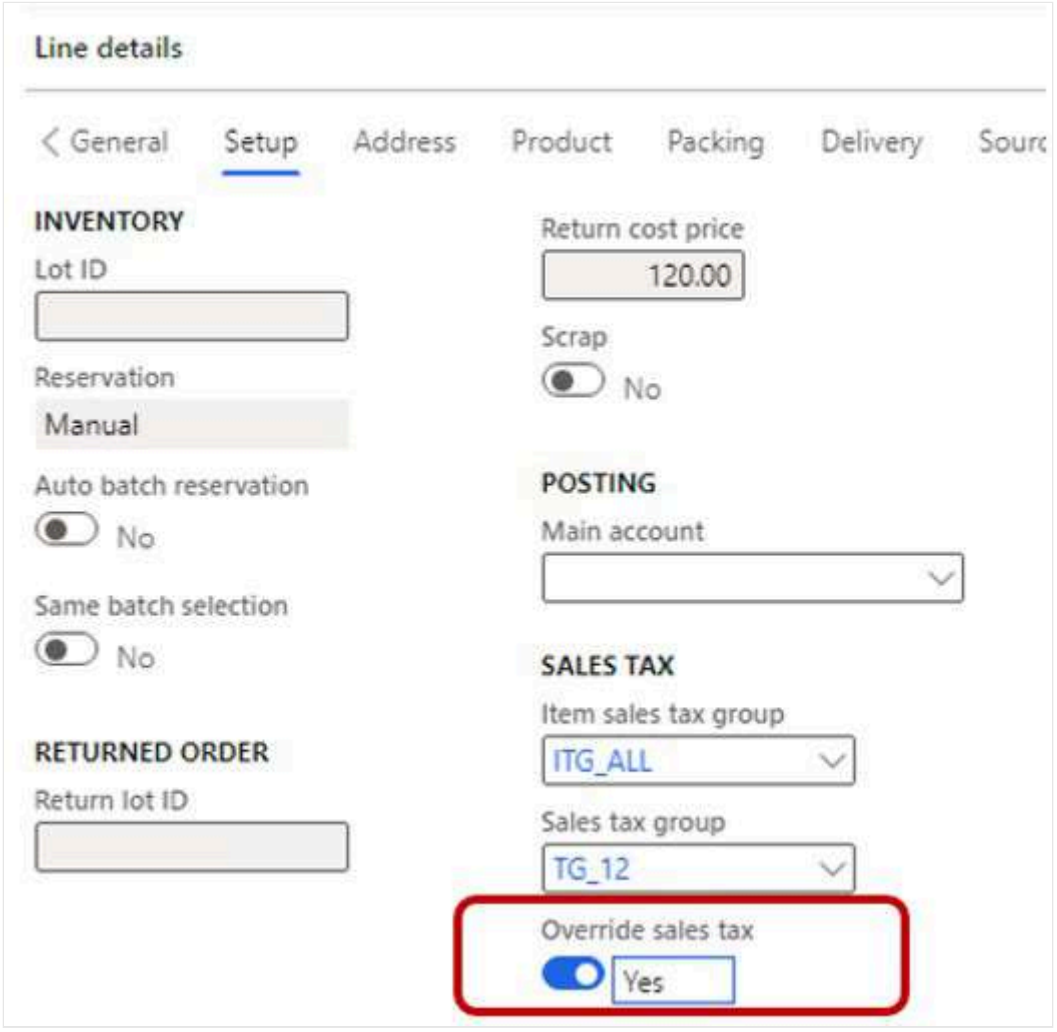
If you don't configure an applicability rule for the tax group or item tax group in the Tax feature setup, Tax Calculation uses the default rules from the master data in Microsoft Dynamics 365 Finance.

> **Note**

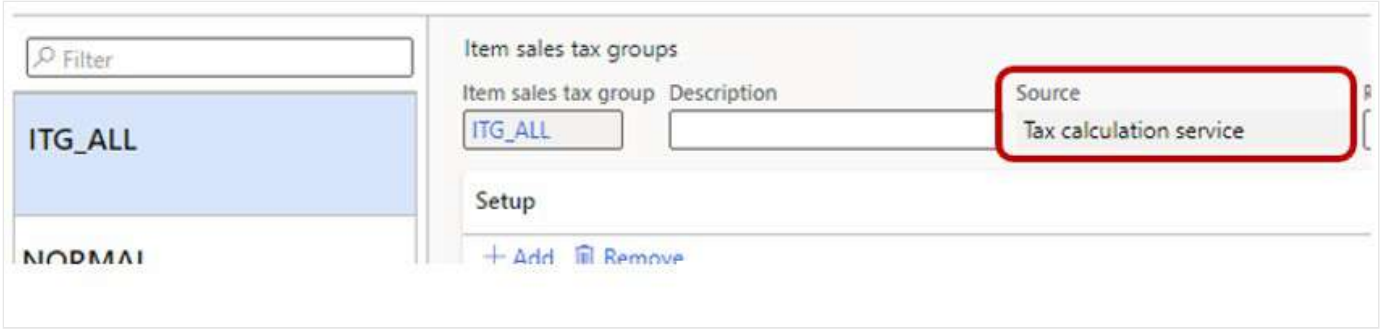
You must create and maintain the sales tax group and item sales tax group in the Tax calculation feature.

Override sales tax

You can update the tax groups that Tax Calculation determines without having to reconfigure the Tax feature. For example, you might need to update the tax groups because the rule is incorrectly set up, or because you need an exception to the rule. By setting the **Override sales tax** option to **Yes** on the line details page for a document line, you can select the required sales tax group and item sales tax group.



The system calculates the tax codes based on the intersection of the tax codes that you define on the **Tax group** and **Item tax group** tabs in the Tax feature setup. In Finance, the value of the **Source** field indicates that the system synchronizes the tax groups from the Tax feature setup.



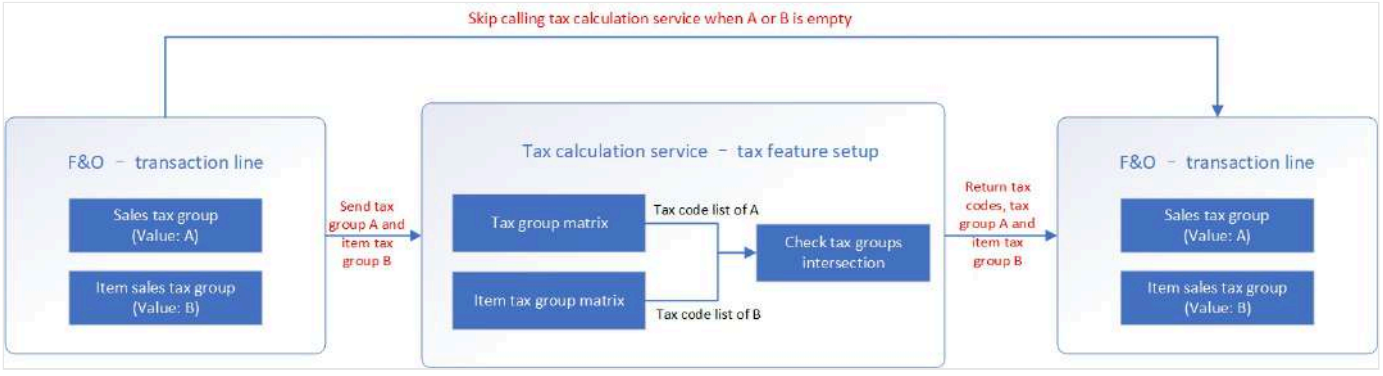
> **Note**
>
> If you leave the **Sales tax group** or **Item sales tax group** field blank, and set the **Override sales tax** option to **Yes**, the system doesn't send the line to the Tax calculation service for processing.

The system adds the **Override sales tax** checkbox to the **Customer** and **Vendor** master data on the **Invoice and delivery** FastTab. The system also adds the checkbox to the Sales order, Purchase order, Free text invoice header, and to the line level of those documents.

The following process combines tax code determination logic with override sales tax:

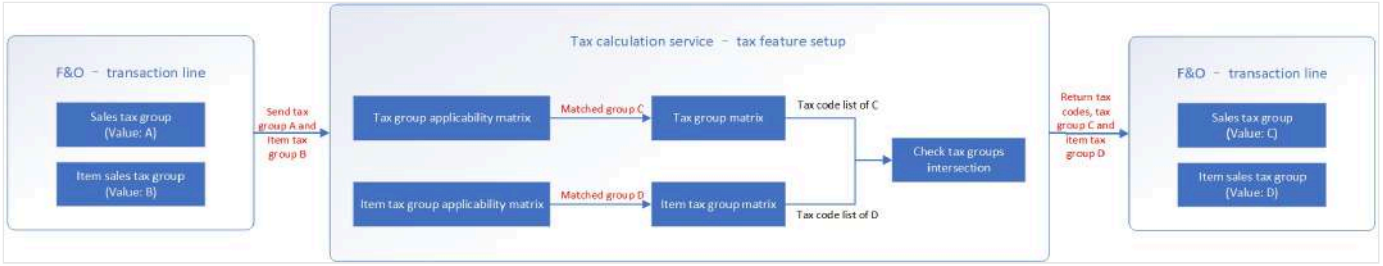
Scenario 1: Override sales tax = Yes

The flow that combines tax code determination logic with override sales tax yes.



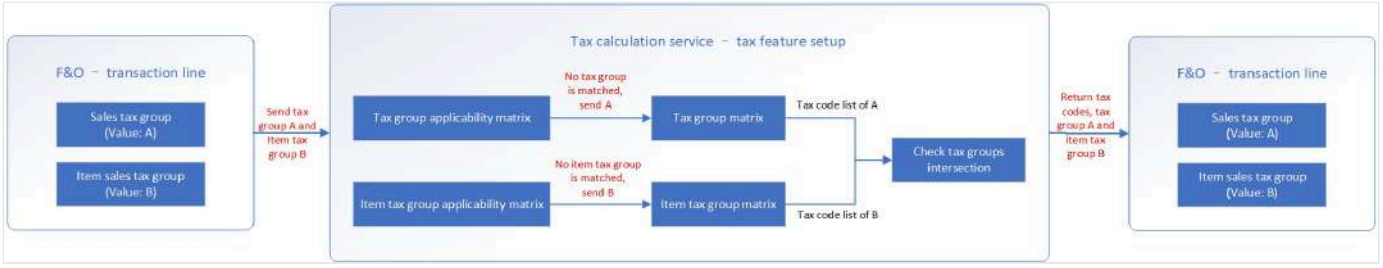
Scenario 2: Override sales tax = No and applicability rule can be matched

The flow that combines tax code determination logic with override sales tax no and matched applicability rules.



Scenario 3: Override sales tax = No and applicability rule can't be matched

The flow that combines tax code determination logic with override sales tax no and applicability rules mismatch.



Update order lines

You can bulk update lines if you change the **Override sales tax** checkbox on the header level. You can also add this checkbox to the **Update order lines** parameters pages for Sales order, Sales quotation, and Purchase order. For example, if you set the option to **Prompt**, when you change the **Override sales tax** checkbox on the header of the document, the dialog box opens. You can select whether the document lines should be updated.

> **Note**
>
> Header-level and line-level charges inherit the **Override sales tax** option from the header or line of the document, respectively.
>
> Current limitation: When you update the **Override sales tax** checkbox on a header or line level, the respective charges don't inherit this checkbox if you already created the charges.

Reverse charge applicability rules

For information about the configuration for reverse charge rules, see Set up reverse charge rules.

When the document lines meet the reverse charge rules that you configure in Finance, the **Override sales tax** checkbox is set to **Yes** for the affected lines. The default value for **Sales tax group** comes from the **Purchase order sales tax group** or **Sales order sales tax group** field. You

specify these fields on the **General ledger parameters** page, on the **Reverse charge** tab. For more information about these fields, see Set up default parameters.

> **Note**
>
> For reverse charges, you also need to create and maintain the sales tax groups and item sales tax groups in the Tax feature setup.

Last updated on 03/21/2026

Tax jurisdiction parameters for cash discount calculation setup

This article provides information about the setup of cash discount calculation options.

When a single legal entity has several value-added tax (VAT) registrations in different countries or regions, the requirements for calculating cash discounts and sales tax might vary by country or region. If you need different cash discount calculation options for different VAT registrations, set up the options on the **Tax jurisdiction parameters** and **Tax jurisdiction applicability** tabs of the **Tax calculation** page.

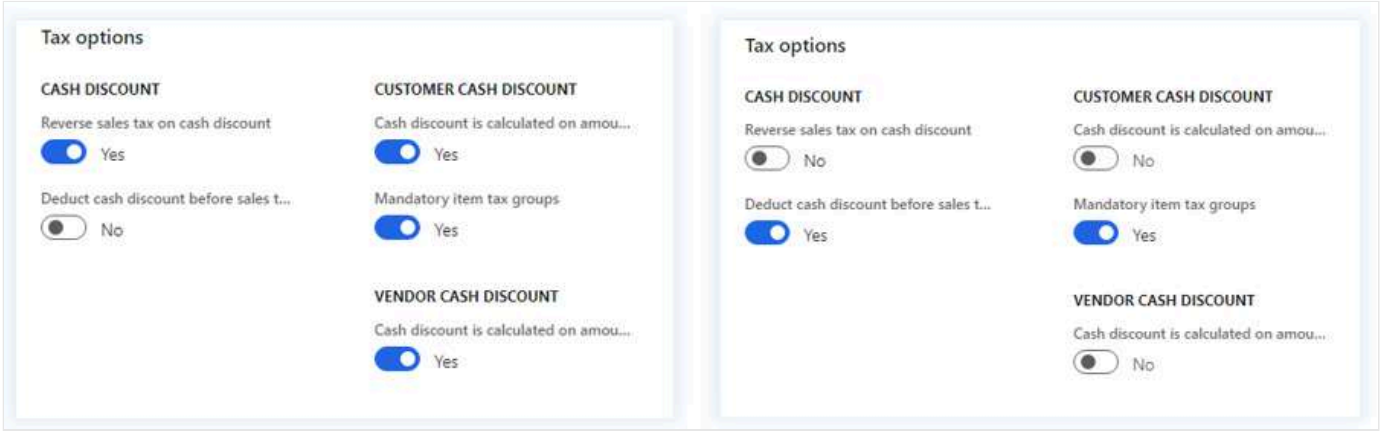
Tax jurisdiction parameters setup

Microsoft Dynamics 365 Finance supports the following generic combinations of cash discount calculation options.

Expand table

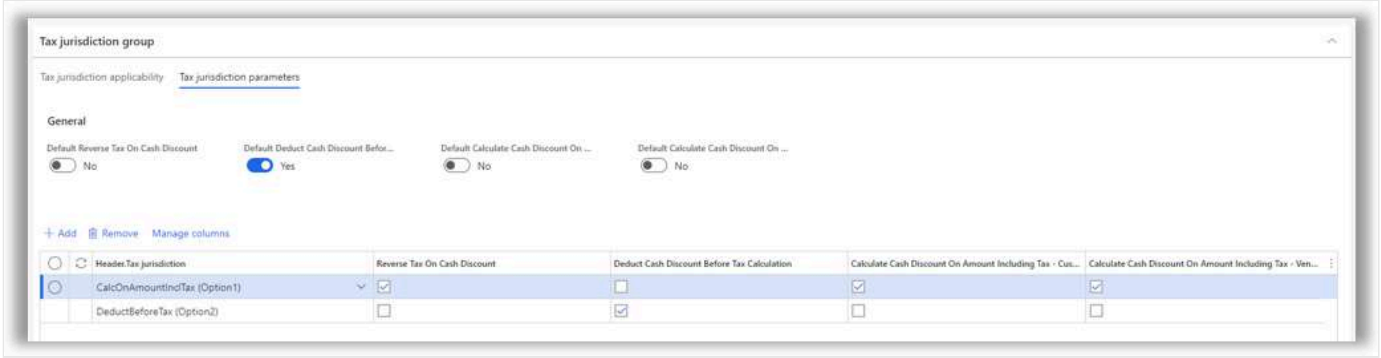
Parameter	Option 1	Option 2
Reverse sales tax on cash discount	Yes	No
Deduct cash discount before sales tax calculation	No	Yes
Cash discount is calculated on amount including sales tax (This parameter is available for both the customer side and the vendor side.)	Yes	No

The following illustration shows the parameter setup for both options.

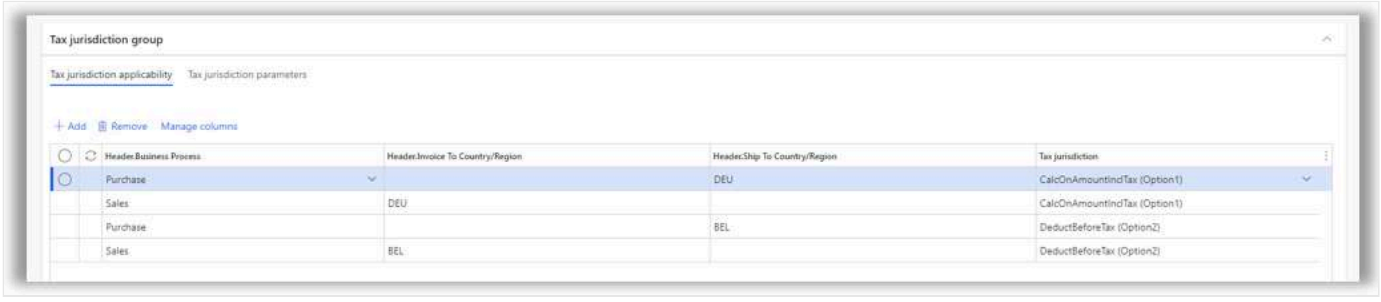


> **Note**
>
> - The **Deduct Cash Discount Before Tax Calculation** and **Calculate Cash Discount On Amount Including Tax** parameters are mutually exclusive.
> - Currently, the tax calculation feature setup doesn't support cross-measure validation. Finance performs the validation when it retrieves the parameters from the Tax calculation service.
> - For a tax jurisdiction code, don't set both the **Deduct Cash Discount Before Tax Calculation** parameter and the **Calculate Cash Discount On Amount Including Tax** parameter to **Yes** on the **Tax calculation** page.

To set up the cash discount calculation option in the tax calculation feature setup, create a tax jurisdiction code on the **Tax jurisdiction parameters** tab in the **Tax jurisdiction** group.



Use the fields on the **Tax jurisdiction applicability** tab in the **Tax jurisdiction** group to define rules that determine when a cash discount calculation option should be applied. You can select header-level fields for the determination.



If no matching rule is found on the **Tax jurisdiction applicability** tab, the default parameters on the **General** tab are used.

Example: Cash discount parameters setup

To determine a tax jurisdiction code, use **Ship To/Ship From Country/Region** or **Invoice To Country/Region**. When you create a sales order for a customer in Germany (DEU), **Option1** is selected. When you create a sales order for a customer in Belgium (BEL), **Option2** is selected.

The following table shows the results of the cash discount and sales tax calculation for a sales order that has a net amount of 1,000 and a 4-percent cash discount. For tax calculation purposes, the tax rate that is used for **Option1** is 19 percent. For **Option2**, it's 21 percent.

Expand table

Sales order or tax jurisdiction code	Option1	Option2
Net amount	1,000	1,000
Invoice amount	1,190	1,201.60
Sales tax amount	190 (= 1,000 × 19%)	201.60 (= [1000 – 40] × 21%)
Cash discount amount	47.60 (= 1,190 × 4%)	40 (= 1,000 × 4%)

> **Important**
>
> This functionality supports sales order, purchase order, vendor invoice, and free text invoice operations.

Last updated on 03/21/2026

Calculate tax on prices when prices include taxes is enabled

This article explains how to use the **Exclude from base amount calculation** parameter of a tax code.

Use the **Exclude from base amount calculation** parameter to calculate tax amounts in a flexible way. When you enable this parameter for a tax code and prices include sales tax, the Tax calculation service calculates tax amounts for the marked sales tax code as if it wasn't included in the base amount.

The following example shows how the tax calculation works.

A sales tax group and an item sales tax group contain two sales tax codes. The **Exclude from base amount calculation** parameter is enabled for one tax code.

Expand table

Tax code	Tax rate	Exclude from base amount calculation
TaxCodeA	20%	No
TaxCodeB	10%	Yes

If you create a sales order or a purchase order where you enable the **Prices include sales tax** option, and the net amount of a line equals 120, the tax is calculated in the following way.

Expand table

Tax code	Tax base amount	Amount
TaxCodeA	100	20
TaxCodeB	120	12

> **Note**

In this example, tax code TaxCodeA is configured as **By Net amount**, and tax code TaxCodeB is configured as **By Gross amount**. If tax code TaxCodeB is configured as **By Net Amount** instead, its tax base amount equals 100, and its tax amount equals 10.

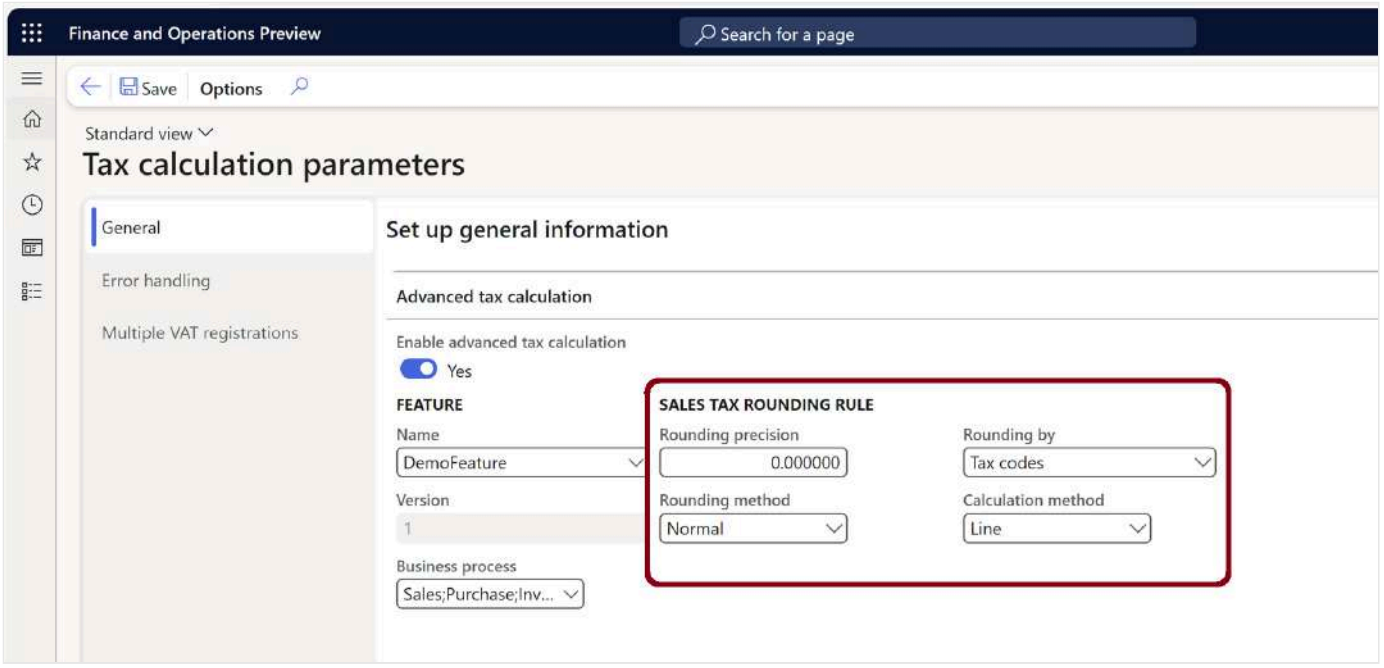
Last updated on 03/21/2026

Tax calculation rounding rules

This article provides information about how the rounding rules work on the **Tax calculation parameters** page.

> **Note**
>
> When the **Enable advanced tax calculation** parameter is enabled, the rounding rules on the **Sales tax code** and **Sales tax group** pages aren't effective.

You can view the configuration of rounding rules for the tax calculation service in the **Sales tax rounding rule** section on the **Calculation** FastTab on the **General** tab of the **Tax calculation parameters** page (**Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**).



The **Rounding precision** and **Rounding method** fields determine how calculated amounts in the payload from the tax calculation service are rounded.

Rounding precision

The **Rounding precision** field supports a value that has up to six decimal places. For example, if you set the **Rounding precision** field to **0.000000**, calculated amounts are rounded to six decimal places and then sent to Microsoft Dynamics 365 Finance. For example, if the **Normal** rounding method is used, the amount **987.1234567** is rounded to **987.123457**.

> **Note**
>
> Finance rounds amounts according to the currency rounding rules. Therefore, the tax amounts that are shown and recorded in transactions are affected by both tax calculation rounding rules and currency rounding rules.

Rounding method

You can configure the rounding method for tax calculation for each legal entity. In the **Rounding method** field, select from three options: **Normal**, **Downward**, and **Rounding-up**.

Rounding example

The following table provides an example that shows how the amount **987.345** is rounded for different combinations of rounding precisions and rounding methods.

Expand table

Rounding method	0.00	0.01	0.10	1.00	10.00	0.02	0.05	0.25
Normal	987.35	987.35	987.30	987.00	990.00	987.34	987.35	987.25
Downward	987.00	987.34	987.30	987.00	980.00	987.34	987.30	987.25
Rounding-up	988.00	987.35	987.40	988.00	990.00	987.36	987.35	987.50

You can configure the calculation and rounding logic of tax amounts according to taxation rules.

Rounding by

In the **Rounding by** field, select the rounding principle that applies to the taxes. The following options are available:

- **Tax codes** – Round the tax amount inside each tax code.
- **Tax code combinations** – Round the tax amount inside the tax code combination on the line.

Calculation method

In the **Calculation method** field, select whether to calculate taxes on invoices for each line or all lines. The following options are available:

- **Line** – Calculate the tax amount on a line-by-line basis. The tax amount on each line is unaffected by the tax amount on other lines.
- **Total** – Calculate the tax amount across all the lines on one document.

Rounding calculation example

This example shows the different calculations you can do for one invoice, for different combinations of **Rounding by** and **Calculation method** values. For this example, the following setup is in place:

- The invoice has four lines.
- There are two tax codes: **VAT1** (10 percent) and **VAT2** (10 percent).
- The rounding precision is set to **0.01**.
- The rounding method is set to **Rounding-up**.

Rounding by = Tax codes and Calculation method = Line

Expand table

Line number	Line net amount	Determined tax codes	Tax amount before rounding	Rounded tax amount
1	11.11	VAT1	1.111	1.12
2	22.22	VAT1; VAT2	2.222; 2.222	2.23; 2.23
3	33.33	VAT1	3.333	3.34
4	44.44	VAT1; VAT2	4.444; 4.444	4.45; 4.45

Rounding by = Tax code combinations and Calculation method = Line

Expand table

Line number	Line net amount	Determined tax codes	Tax amount before rounding	Rounded tax amount
1	11.11	VAT1	1.111	1.12
2*	22.22	VAT1; VAT2	2.222; 2.222	2.23; 2.22
3	33.33	VAT1	3.333	3.34
4**	44.44	VAT1; VAT2	4.444; 4.444	4.45; 4.44

* Line 2 = Round[22.22 × (10 percent + 10 percent)] = 2.23 + 2.22

** Line 4 = Round[44.44 × (10 percent + 10 percent)] = 4.45 + 4.44

Rounding by = Tax codes and Calculation method = Total

Expand table

Line number	Line net amount	Determined tax codes	Tax amount before rounding	Rounded tax amount
1	11.11	VAT1*	1.111	1.12
2	22.22	VAT1*; VAT2**	2.222; 2.222	2.22; 2.23
3	33.33	VAT1*	3.333	3.33
4	44.44	VAT1*; VAT2**	4.444; 4.444	4.44; 4.44

* VAT1(Line 1, Line 2, Line 3, Line 4) = Round[(11.11 + 22.22 + 33.33 + 44.44) × 10 percent] = 1.12 + 2.22 + 3.33 + 4.44

** VAT2(Line 2, Line 4) = Round[(22.22 + 44.44) × 10 percent] = 2.23 + 4.44

Rounding by = Tax code combinations and Calculation method = Total

Expand table

Line number	Line net amount	Determined tax codes	Tax amount before rounding	Rounded tax amount
1*	11.11	VAT1	1.111	1.12
2**	22.22	VAT1; VAT2	2.222; 2.222	2.23; 2.22
3*	33.33	VAT1	3.333	3.33
4**	44.44	VAT1; VAT2	4.444; 4.444	4.44; 4.45

* Line 1, Line 3 = Round[(11.11 + 33.33) × 10 percent] = 1.12 + 3.33

** Line 2, Line 4 = Round[(22.22 + 44.44) × (10 percent + 10 percent)] = 2.23 + 2.22 + 4.44 + 4.45

Last updated on 03/21/2026

Multiple VAT registration numbers

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about the functionality for multiple value-added tax (VAT) registration numbers. By using this functionality, you can set up the tax registration numbers of a legal entity and its customers and vendors in different countries or regions. You can then post and settle taxes according to registration in the appropriate country or region.

Here are the main steps to configure and use this functionality:

- Assign the registration type for VAT registration to the VAT ID registration category.
- Set up the legal entity's, customers', and vendors' VAT registration numbers on the **Registration ID** FastTab on the **Manage addresses** page.
- Specify the legal entity's VAT registration number for the sales tax authority, and specify the settlement period. The sales tax codes that are assigned to the settlement period identify the legal entity's VAT registration.

The tax calculation feature can identify the customer and vendor VAT registration numbers for transactions. The identified tax registration numbers are available in the sales tax transactions.

The sales tax settlement procedure uses the country or region code of the registration ID.

Prerequisites

Before you begin, Tax Calculation must be configured. For more information, see Tax Calculation.

Enable the feature

1. In the **Feature management** workspace, turn on the **Support multiple VAT registration numbers** feature.
2. Go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**, and turn on the **Enable advanced tax calculation** option.

> **Note**

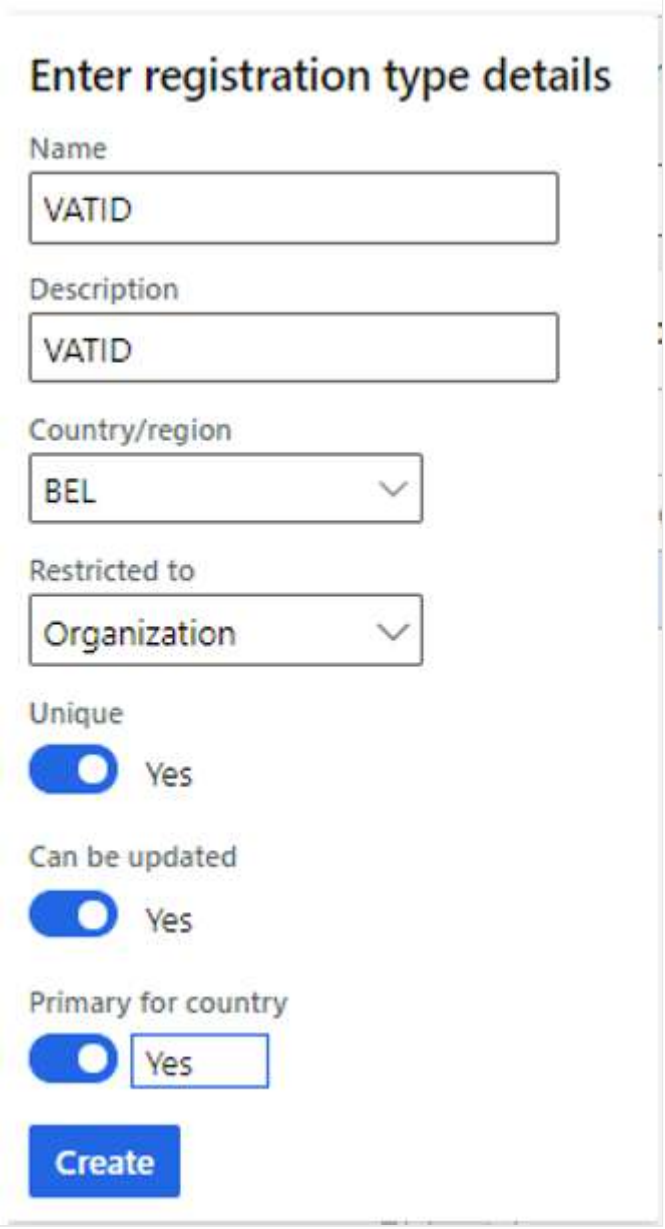
In the 10.0.39 update, you can no longer enable this feature through Feature management. It's now controlled by the **Support multiple VAT registration numbers** parameter on the **Tax calculation parameters** page.

Set up a VAT ID for a legal entity, customers, and vendors

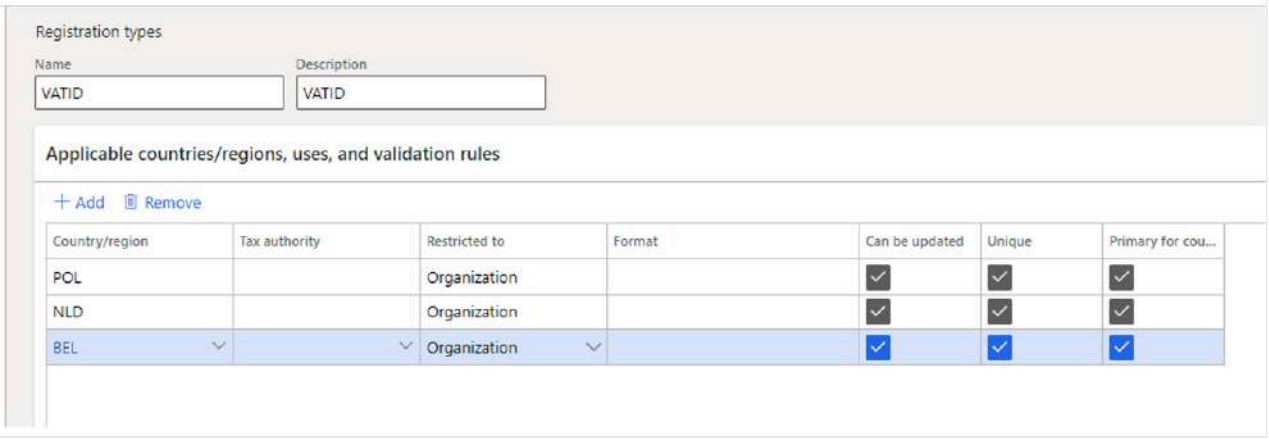
To set up the VAT registration numbers for a legal entity and its customers and vendors, use the registration ID framework to create these registration numbers. For more information, see Registration IDs.

Set up registration types and categories

1. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration types**, and create a new registration type, such as **VATID**.



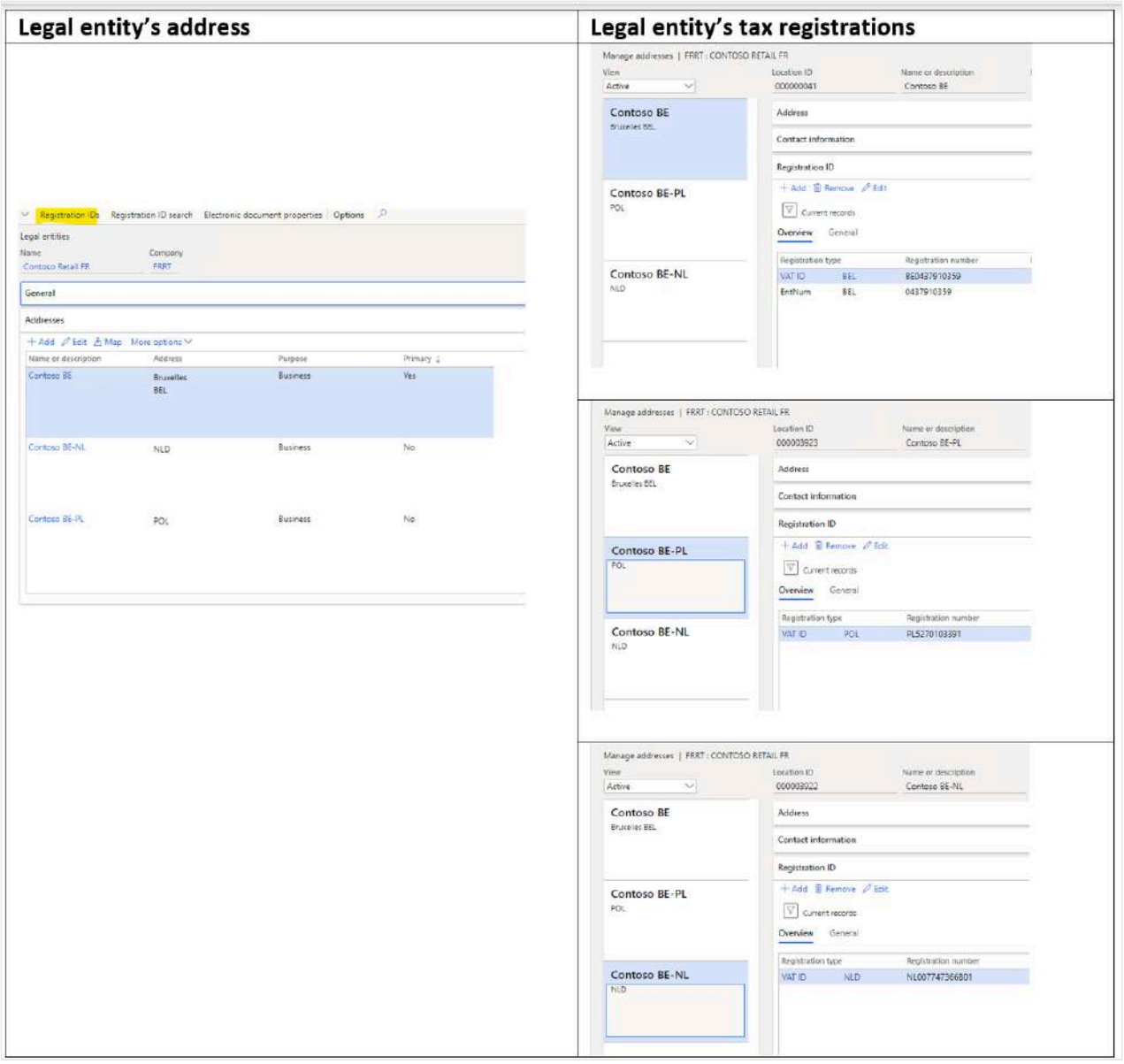
2. On the **Applicable countries/regions, uses, and validation rules** FastTab, create a registration type line for each country or region where the legal entity, customers, and vendors have registrations.



3. Go to **Organization administration** > **Global address book** > **Registration types** > **Registration categories**, and assign the registration types that you just created to the **VAT ID** registration category.

Create VAT registration numbers for a legal entity, customers, and vendors

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select **Registration IDs**, and assign VAT ID registration to each address where the legal entity has VAT registrations.

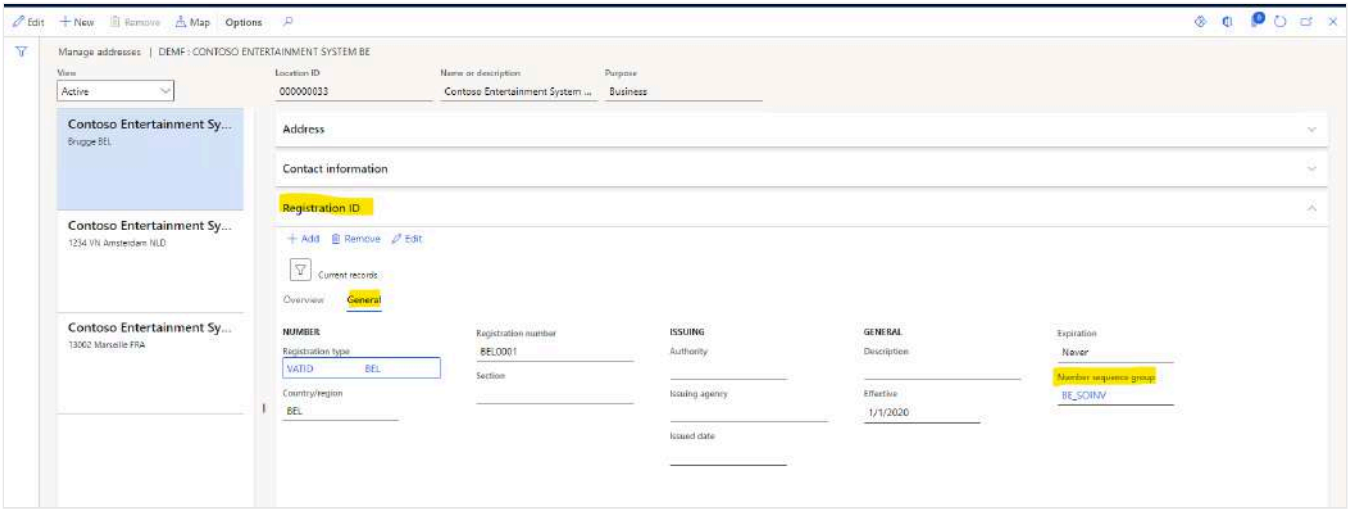


> **Note**
>
> To automatically identify customer and vendor VAT registration numbers for sales tax transactions, sales documents, and purchase documents, either create the numbers in the tax calculation feature on the **Customer tax registration number applicability** and **Vendor tax registration number applicability** tabs or enable the **Update tax exempt number from customer address** parameter on the **Multiple VAT registrations** tab of the **Tax calculation parameters** page.

Set up number sequences for a legal entity's registration numbers

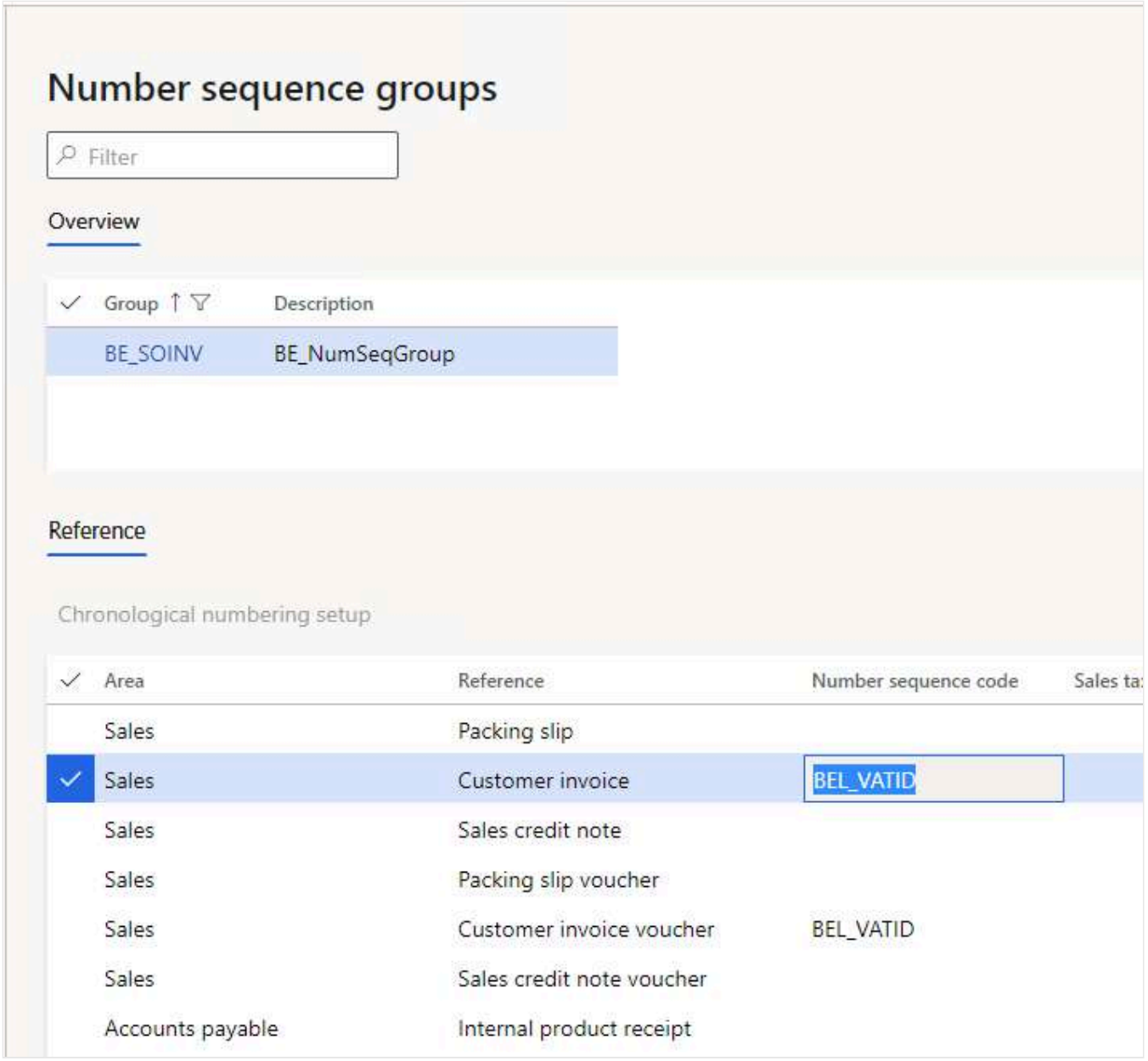
To generate separate number sequences for documents such as packing slips and invoices, create a number sequence group. Then, on the **Manage addresses** page, on the **Registration**

ID FastTab, on the **General** tab, assign the number sequence group to the VAT ID of the legal entity.



Next, on the **Number sequence groups** page, in the **Reference** section, set up the required number sequence codes for the supported references.

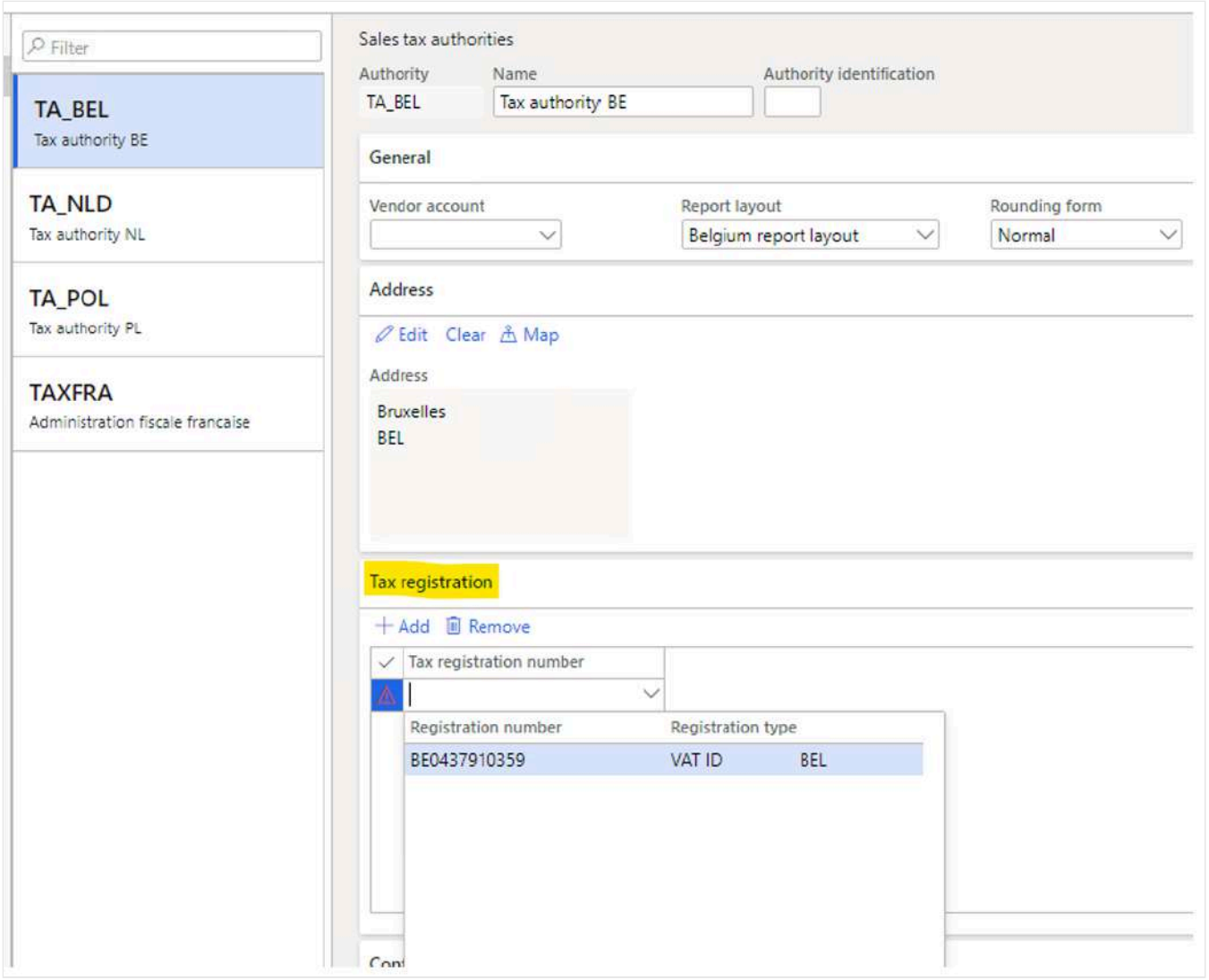
The code for the number sequence group is then entered by default on the sales order or purchase order header after the legal entity's tax registration is determined. The documents are numbered according to the number sequences that are assigned to the references.



> **Note**
>
> The default logic supports only sales orders, purchase orders, and free text invoices.

Set up tax authorities

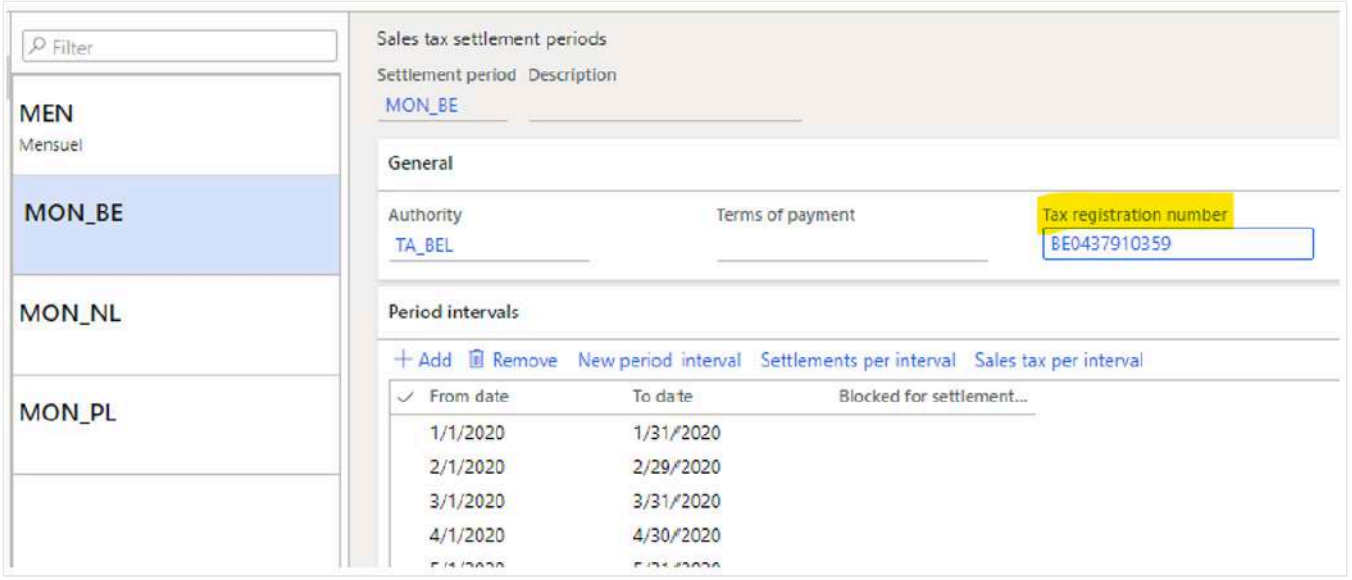
1. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax authorities**, and create all the tax authorities that the legal entity must report to.
2. On the **Tax registration** FastTab, add the appropriate VAT registration number.



> **Note**
>
> The lookup for the **Tax registration number** field on the **Tax registration** FastTab contains only the registration numbers of the legal entity that have a VAT ID registration category. The list of registration IDs is available for the corresponding country or region of the tax authority.
>
> Date effectivity isn't supported for assigned registration numbers. If you change or expire the registration number of the legal entity, you must manually update the tax registration in the setup of the tax authority and sales tax settlement period.

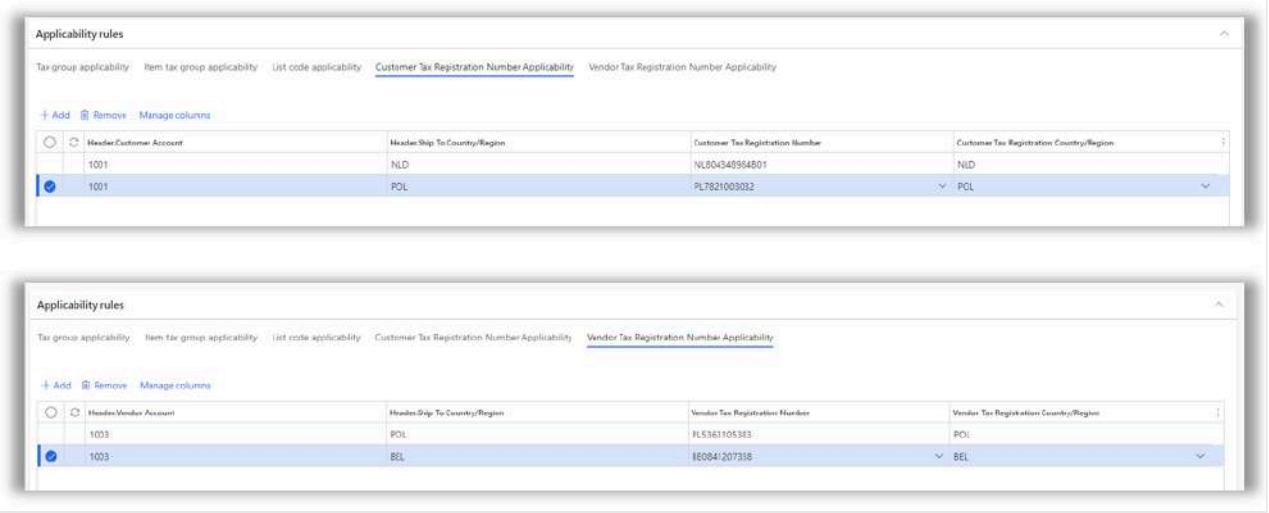
Set up the sales tax settlement period

On **Sales tax settlement periods**, create sales tax settlement periods. In the **Tax registration number** field, verify that the corresponding legal entity's VAT registration number is assigned.



Set up customer and vendor tax registration numbers in the Tax feature setup

1. Open the tax calculation feature setup.
2. On the **Customer Tax Registration Number Applicability** tab, under **Applicability rules**, verify that the registration IDs for customers are defined.
3. On the **Vendor Tax Registration Number Applicability** tab, under **Applicability rules**, verify that the registration IDs for vendors are defined.



> **Note**
>
> This step is optional unless the customer or vendor has multiple tax registration numbers, and you want the Tax Calculation feature to determine the ID. As the default value for the customer or vendor tax registration number, the Tax Calculation feature uses the value entered in the **Tax exempt number** field on the header of

sales and purchase documents. It replaces that default value with the value that you defined in customer or vendor tax registration number applicability rules.

During sales tax calculation and document posting, the tax feature returns the customer's or vendor's tax registration number to Dynamics 365 Finance and updates the **Tax exempt number** field on the sales order or purchase order. If the corresponding value isn't set up on the **Registration ID** FastTab on the **Manage addresses** page for the customer or vendor, the registration ID is left blank, and you receive the following message: "Customer tax registration 'xxx' is not found in the customer's Registration IDs setup. To add customer tax registration to sales tax transactions and posted documents, make sure the registration is defined in the Registration IDs setup."

Sales order and purchase order processing

On the **Tax calculation parameters** page, make sure that the **Enable advanced tax calculation** option is turned on, and that **Sales** and **Purchase** are selected in the **Business process** field.

If you assign sales tax codes on lines that you create for a sales order or purchase order to different sales tax settlement periods and tax registrations, the order has multiple registration numbers. To control the system behavior in this scenario, the **Check Tax registration number in document lines** option is available on the **Accounts receivable parameters** and **Accounts payable parameters** pages.

Before version 10.0.41, the system didn't support assigning sales tax codes to different registration numbers on a sales order or purchase order. During sales tax calculation and document posting, you receive an error message and can't continue the process. For information about the item ID, sales tax codes, settlement periods, and tax registration numbers that are identified for the order lines, view the message details.

Starting with version 10.0.41, you can create sales orders and sales quotations with lines that have different tax registration numbers. The **Check Tax registration number in document lines** option on the **Accounts receivable parameters** page still controls the level of validation: *None*, *Warning*, or *Error*.

> **Important**
>
> You can't assign multiple tax registrations to individual sales order document lines. If tax calculation determines that different tax registration numbers (VAT IDs) of the legal entity are assigned to an individual line, an error is displayed.

When you post, the system automatically splits sales orders into different sales confirmations, picking lists, packing slips, and invoices according to the VAT IDs it determines for document lines. Go to **Account receivable parameters** > **Summary update** > **Split based on** to select whether sales confirmation and/or picking list should be split based on the tax registration number. For packing slips and invoice postings, this option is mandatory and is set to **Yes**.

Temporary sales tax and posted sales tax

On the **Temporary sales tax** page, you can preview the identified VAT registration numbers of the legal entity, customers, and vendors. The following new fields are available on the page:

- **Tax registration number** – The VAT registration number of the legal entity.
- **Customer tax registration number** – The VAT registration number of the customer. This field is available only for sales orders.
- **Vendor tax registration number** – The VAT registration number of the vendor. This field is available only for purchase orders.

On the **Posted sales tax** page, the following new fields are available. You can sort and filter sales tax transactions by these fields.

- **Tax registration number** – The VAT registration number of the legal entity.
- **Counterparty tax registration number** – The VAT registration number of the counterparty.

Sales tax settlement procedure updates

The **Settle and post sales tax** periodic task now uses the country/region code of the legal entity's tax registration.

> **Note**
>
> If you don't set the tax registration number for a tax period, you receive the following error message: "Tax registration number is not set up for the tax settlement period xxx" and the settlement process stops.

After the settlement process finishes, the system doesn't print a sales tax payment report. Instead, you receive the following message: "The sales tax settlement and posting is completed. The voucher 'xxxx, m/d/yyyy' has been posted."

You can still manually run the sales tax payment report by going to **Tax** > **Inquiries and reports** > **Sales tax inquiries** > **Sales tax payments**.

> **Note**
>
> Even if you don't enable the feature on the **Tax calculation parameters** page, the system copies the tax registration ID from the original sales tax transactions to the offset sales tax transactions.

Last updated on 03/18/2026

Reporting for multiple VAT registrations

This article explains how to report for multiple value-added tax (VAT) registrations from a single legal entity. You can use this functionality in the following countries or regions:

- Austria
- Belgium
- Czechia
- Denmark
- Estonia
- Finland
- France
- Germany
- Italy
- Latvia
- Lithuania
- Netherlands
- Norway
- Poland
- Spain
- Sweden
- Switzerland
- United Kingdom

The scope of reporting is limited to generation of the following reports:

- Intrastat
- EU sales list
- VAT declaration

For an overview of the functionality, see the following TechTalk session from September 29, 2021: VAT Reporting for Multiple Tax Registrations in Single Legal Entity.

Prerequisites

Before you begin, configure the following prerequisites:

- **Tax Calculation service**. For more information, see Tax Calculation.

- **Support multiple VAT registration numbers**. For more information, see Multiple VAT registration numbers.

Enable features in feature management

1. Go to **Workspaces** > **Feature management**.
2. In the feature list, select and enable the following features:
    - Intrastat reporting for multiple VAT registrations
    - EU Sales list reporting for multiple VAT registrations
    - EU sales list transfer based on tax transactions only
    - Sales tax declaration for multiple VAT registrations

> ⓘ **Note**
>
> If you transfer between warehouses in different countries or regions that have a tax registration, you must also enable the **Tax in transfer order** feature and configure the Tax Calculation service for tax in transfer orders. For more information, see Tax feature support for transfer orders.

Activate feature for specific legal entity

1. Go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**.
2. On the **General** tab, set **Enable tax service** to **Yes**.
3. On the **Multiple VAT registrations** tab, set **VAT declaration**, **EU Sales List**, and **Intrastat** to **Yes** to activate VAT reporting, EU sales list reporting, or Intrastat reporting respectively, for the selected legal entity.

> ⓘ **Note**
>
> You can only access country or region-specific functionality that's applicable to the country or region of the primary address of the selected legal entity. Formats for the VAT declaration, EU sales list, and Intrastat are available for countries or regions of other VAT registrations. Other country or region-specific functionality that's applicable to the countries or regions of other VAT registrations isn't available in this legal entity.

Set up country/region properties for multiple VAT registrations

1. Go to **Tax** > **Set up** > **Foreign trade** > **Foreign trade parameters**.
2. On the **Country/region properties** tab, set up the following country/region types: **EU**, **EFTA**, or **Third country/region**.

   The following information applies to documents about movements of goods between European Union (EU) countries/regions, except movements within the same country/region (for example from Belgium to Belgium):

   - By default, the list code is **EU trade**, and it's transferred to the EU sales list.
   - Documents are transferred to Intrastat.

> ⓘ **Note**
>
> If you set the country/region of the legal entity's primary address as an EU member state, select **EU** (not **Domestic**) for the country/region type on the **Country/region properties** tab.

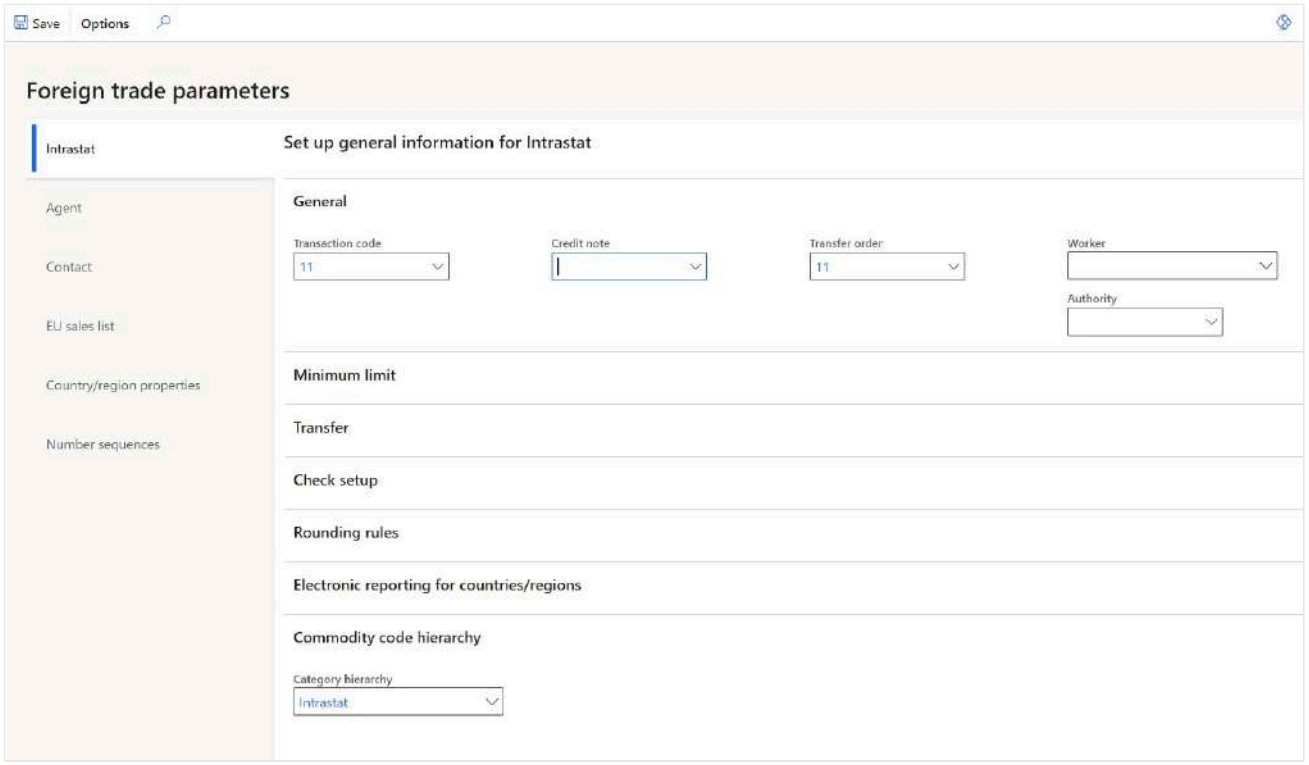
Intrastat reporting for multiple VAT registrations

Set up Intrastat for multiple VAT registrations

Intrastat parameters that you specify for a legal entity apply to all tax registrations that you set up in that legal entity.

1. Go to **Tax** > **Set up** > **Foreign trade** > **Foreign trade parameters**.
2. On the **Intrastat** tab, set up the parameters on the following FastTabs:

   - General: Transaction codes, worker, tax authority.
   - Minimum limit
   - Transfer
   - Check setup
   - Rounding rules
   - Electronic reporting for countries/regions
   - Commodity code hierarchy

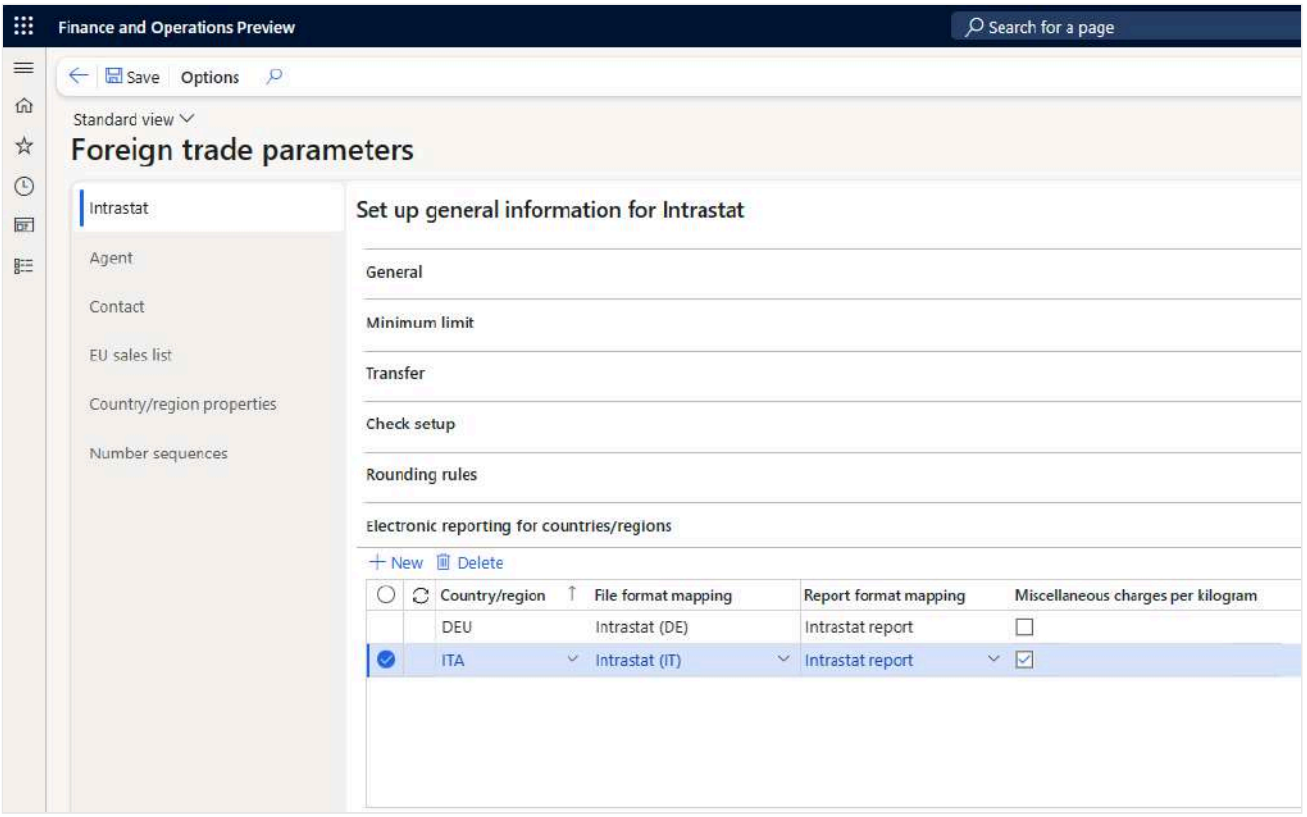


3. Go to **Tax** > **Setup** > **Foreign trade** > **Compression of Intrastat** to set up Intrastat compression rules. All tax registrations have the same setup for compression rules.

For more information about how to configure Intrastat, see Intrastat overview.

Set up Intrastat reporting formats

1. Go to **Tax** > **Set up** > **Foreign trade** > **Foreign trade parameters**.
2. On the **Intrastat** tab, on the **Electronic reporting for countries/regions** FastTab, select the Intrastat reporting format for each country/region of your tax registration.
3. In the **File format mapping** field, select the Electronic reporting (ER) format for Intrastat file export. For example, select **Intrastat (DE)** for Germany (DEU).
4. In the **Report format mapping** field, select the ER format for printable report layout, such as **Intrastat report**.
5. The **Miscellaneous charges per kilogram** checkbox is available when **Country/region** field is set to **ITA** (Italy). Select this checkbox to calculate miscellaneous charges by multiplying the cost of each kilogram and the weight of goods in kilograms. If you clear this checkbox, miscellaneous charges are calculated as a percentage of the invoice amount. For more information, see Miscellaneous charges per kilogram in an Intrastat declaration.



> **Note**
>
> You can select an ER format if the **ISO Country/region codes** field on the **ISO Country/region codes** FastTab is blank in this format.

The following table shows the earliest ER format versions that you can select the format for.

Expand table

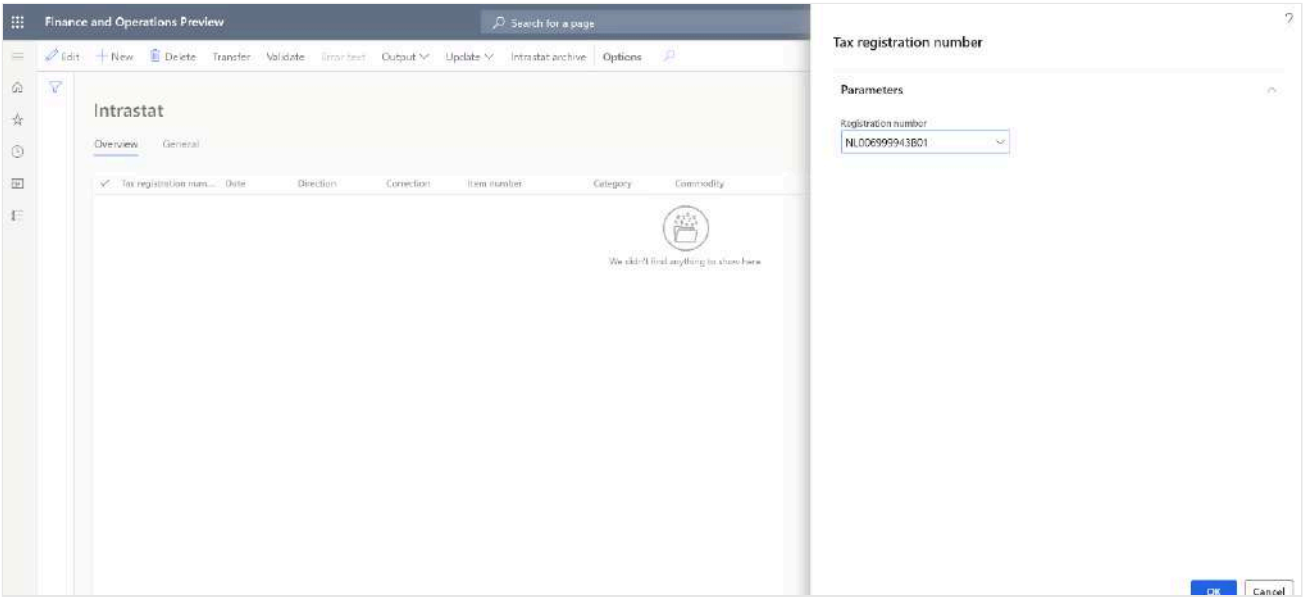
Release	Country/region	ER format
10.0.19	All	Intrastat model.version.16
10.0.19	Netherlands	Intrastat (NL).version.1.3
10.0.20	France	Intrastat INTRACOM (FR).version.13.5 Intrastat SAISUNIC (FR).version.1.3
10.0.20	United Kingdom (Northern Ireland)	Intrastat (UK).version.1.2
10.0.21	Austria	Intrastat (AT).version.16.3
10.0.21	Germany	INSTAT XML (DE).version.22.9.9
10.0.21	Spain	Intrastat (ES).version.16.7

Release	Country/region	ER format
10.0.21	Sweden	Intrastat (SE).version.16.4
10.0.23	Finland	Intrastat (FI).version.2.3
10.0.23	Denmark	Intrastat (DK).version.25.4
10.0.23	Poland	Intrastat (PL).version.25.5
10.0.25	Belgium	Intrastat (BE).version.2.9
10.0.34	The Czech Republic	Intrastat (CZ).version.24.6
10.0.38 10.0.37 - build 10.0.1725.25 10.0.36 - build 10.0.1695.56	Italy	Intrastat (IT).version.24.36

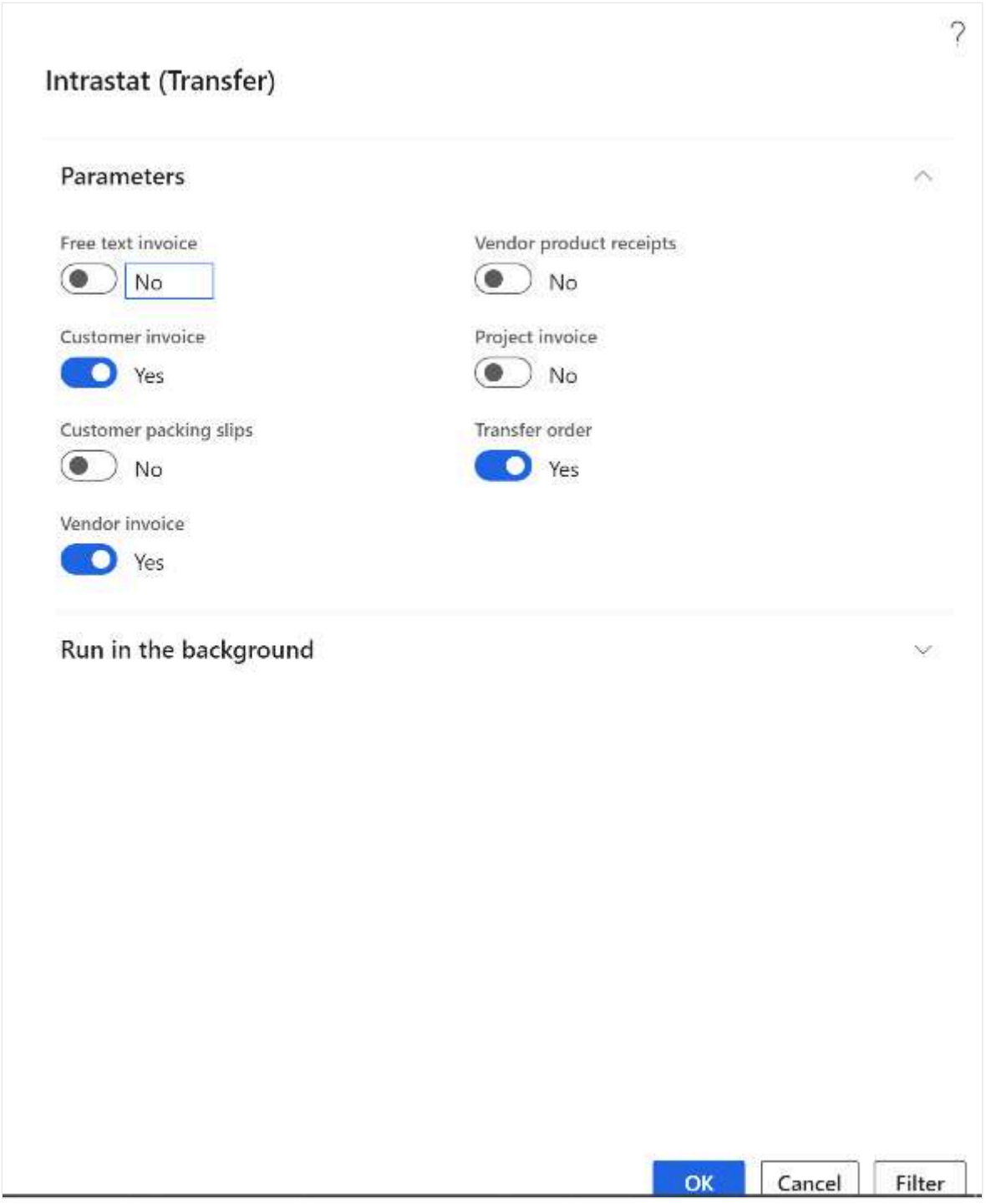
For more information, see Download ER configurations from the Global repository of Configuration service.

Transfer and report Intrastat

1. Go to **Tax** > **Declarations** > **Foreign trade** > **Intrastat**.
2. Select **Transfer**.
3. In the **Tax registration number** dialog box, select the tax registration number to transfer transactions for, and then select **OK**.



4. In the **Intrastat (Transfer)** dialog box, select the documents to transfer: **Free text invoice**, **Customer invoice**, **Customer packing slips**, **Vendor invoice**, **Vendor product receipts**, **Project invoice**, or **Transfer order**. Then select **OK**.

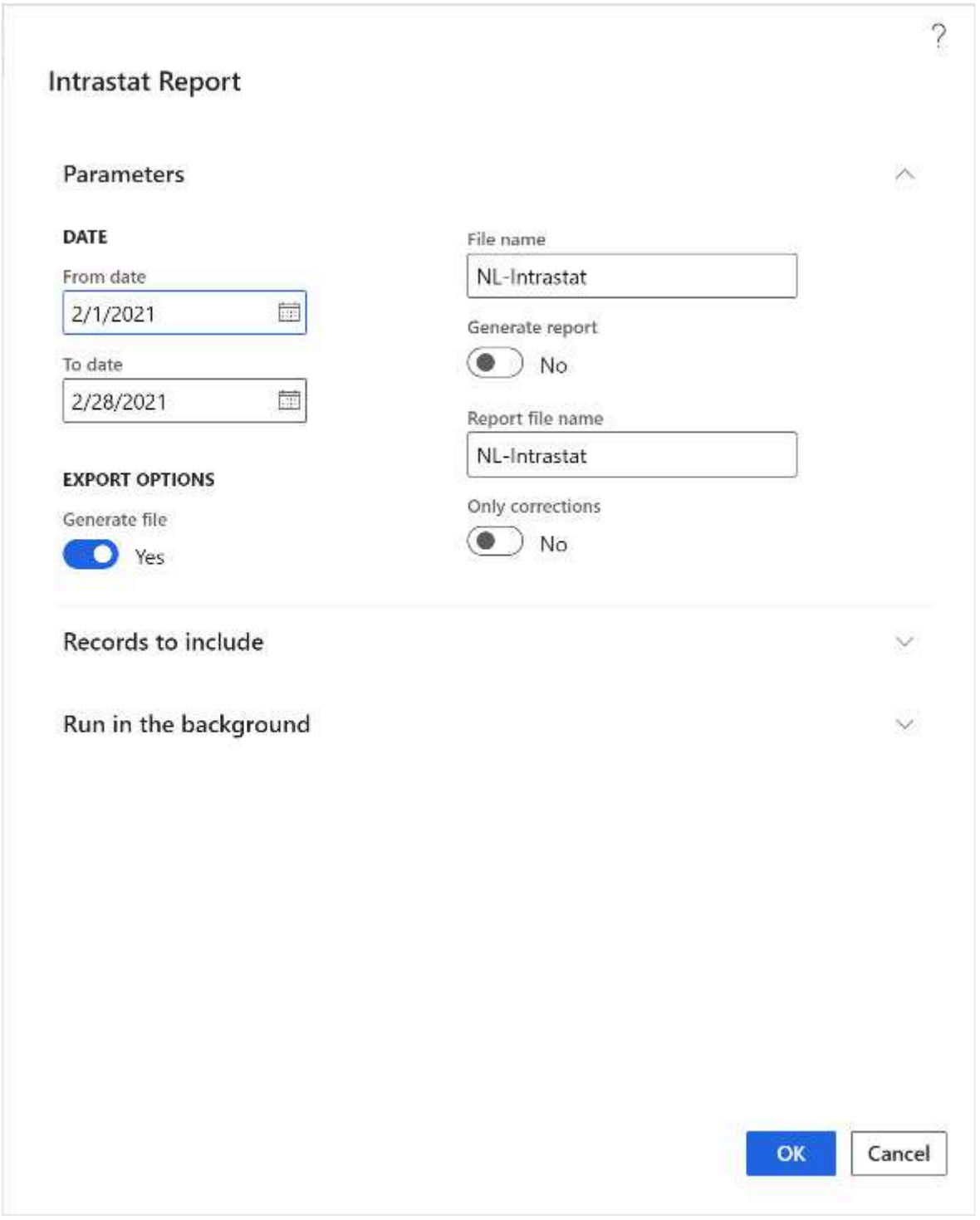


5. Transactions for the selected tax registration and documents are transferred. Review the transactions, and make any adjustments that are required.

> **Note**
>
> In the 10.0.19 and 10.0.20 releases, the Intrastat **Transfer** function has country/region-specific logic that isn't yet covered in the **Intrastat transfer for Multiple Tax ID** feature

for the following countries/regions: the Czech Republic, Finland, Germany, Hungary, Italy, Latvia, Lithuania, Poland, and Spain.

6. Select **Output** > **Report**.
7. In the **Tax registration number** dialog box, select the tax registration number to generate the Intrastat report for, and then select **OK**.
8. In the **Intrastat Report** dialog box, in the **From date** and **To date** fields, define the period to generate the Intrastat report for.
9. Set the **Generate file** option to **Yes** to generate an electronic reporting file. Then, in the **File name** field, enter the name of the output electronic file, if applicable.
10. Set the **Generate report** option to **Yes** to generate a Microsoft Excel report. Then, in the **Report file name** field, enter the name of the output Excel file, if applicable.



11. The dialog box also contains country/region-specific fields that are required in the country/region-specific Intrastat report. Set these fields as required.

12. Select **OK** to generate the report.

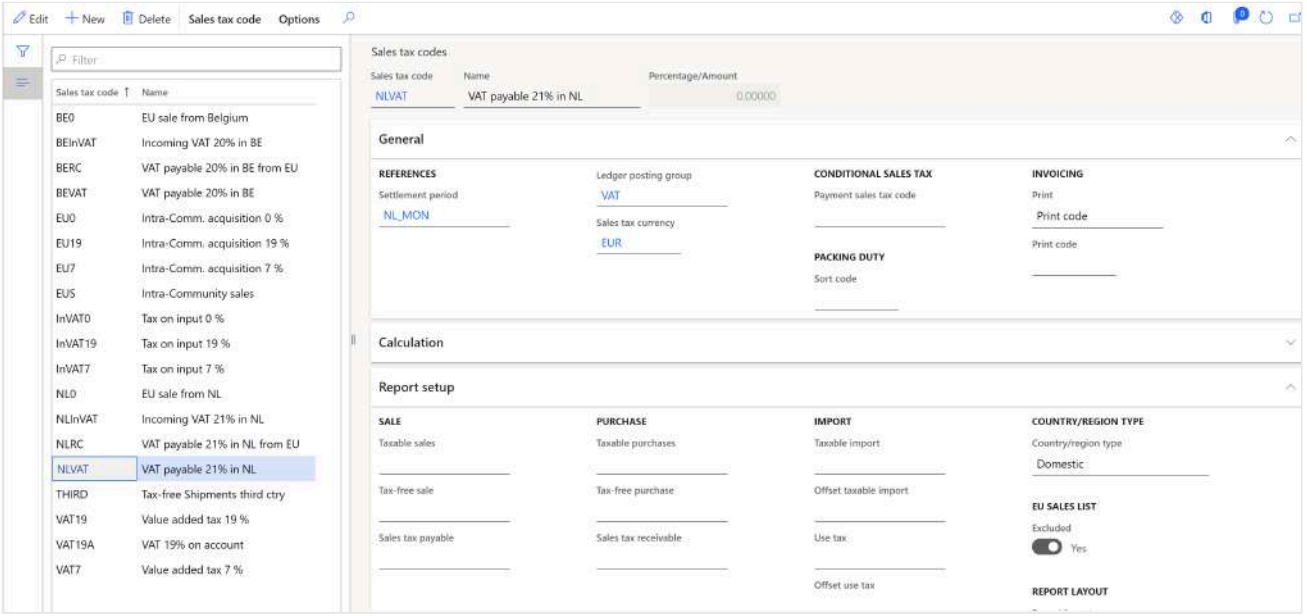
EU sales list reporting for multiple VAT registrations

Set up the EU sales list

Set up sales tax codes

1. Go to **Tax** > **Sales tax** > **Sales tax codes**.
2. Create unique sales tax codes for each of your tax registrations.
3. On the **Report setup** FastTab, in the **EU sales list** section, set the **Excluded** option to **Yes** if tax transactions that have a selected sales tax code shouldn't be transferred to the EU sales list.
4. In the **Country/region type** section, in the **Country/region type** field, define the tax transaction type for VAT reporting: **EU**, **Domestic**, or **Third country/region**.

   This setting is mandatory in some countries/regions, such as the United Kingdom.



Set up item sales tax groups

1. Go to **Tax** > **Sales tax** > **Item sales tax groups**.
2. In the **Reporting type** field, select the reporting type for EU sales list reporting: **Item**, **Service**, or **Investment** (for Belgium).

Set up EU sales list parameters

All tax registrations use the same settings for EU sales list parameters.

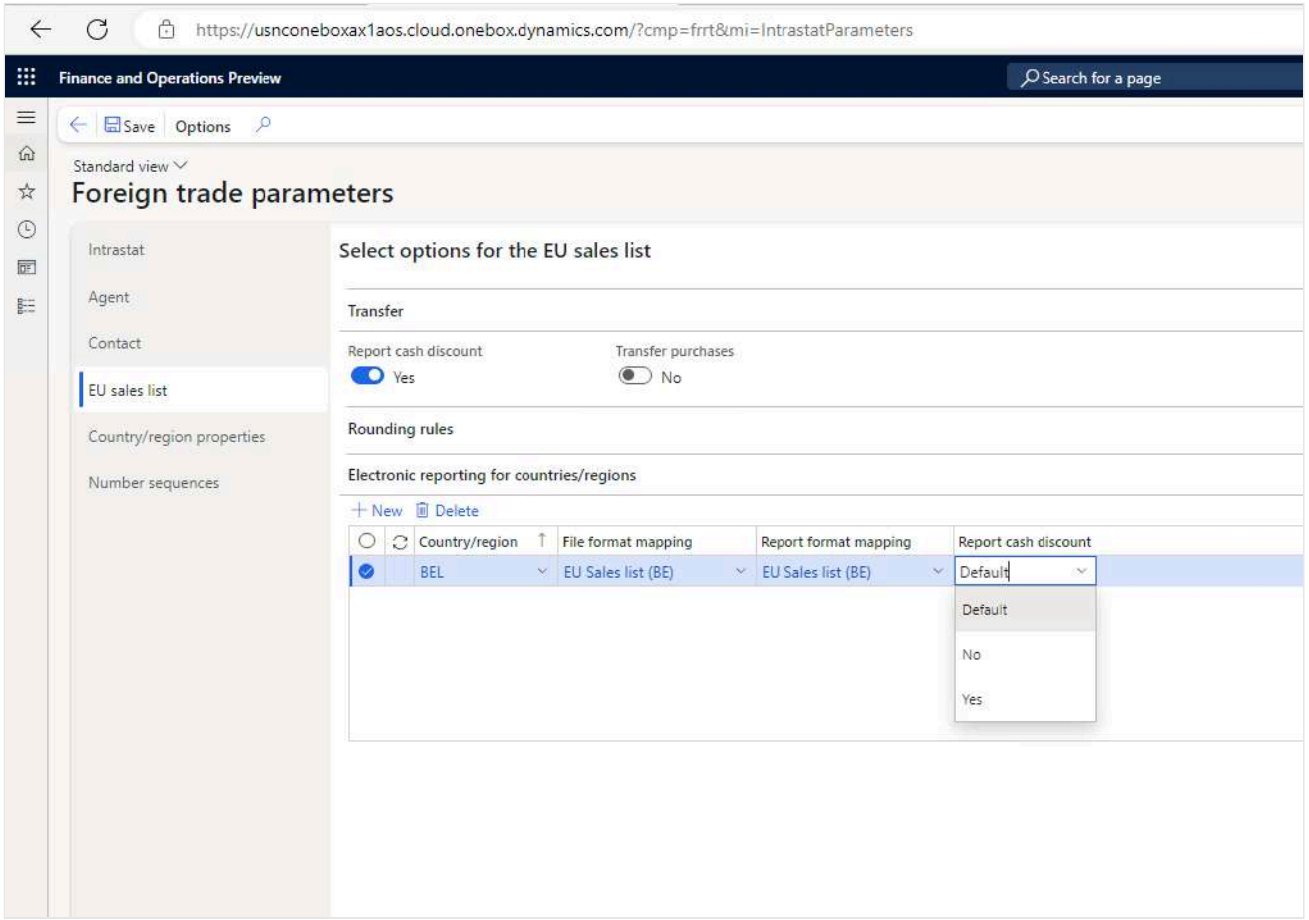
1. Go to **Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**.
2. On the **EU sales list** tab, set up the parameters on the **Transfer** and **Rounding rules** FastTabs.

> **Note**

If one of your tax registrations is in Poland or Hungary (countries/regions that report purchases in addition to sales), set the **Transfer purchases** option to **Yes**.

Set up EU sales list reporting formats

1. Go to **Tax** > **Set up** > **Foreign trade** > **Foreign trade parameters**.
2. On the **EU sales list** tab, on the **Electronic reporting for countries/regions** FastTab, select EU sales list reporting formats for each country/region of your tax registration.
3. In the **File format mapping** field, select the ER format for EU sales list report electronic format. For example, for Netherlands, select **EU Sales list (NL)**.
4. In the **Report format mapping** field, select the ER format for printable report layout. For example, select either **EU sales list by rows report** or **EU sales list by columns report**.
5. In the **Report cash discount** field, select **Yes** if the selected country/region cash discount must be transferred to the EU sales list. Select **No** if the selected country/region cash discount must not be transferred to EU sales list. Leave the value set to **Default** if the value specified in the **Report cash discount** field on the **Transfer** FastTab must be applied to the selected country/region.



> **Note**
>
> You can select an ER format if the **ISO Country/region codes** field on the **ISO Country/region codes** FastTab is blank in this format.

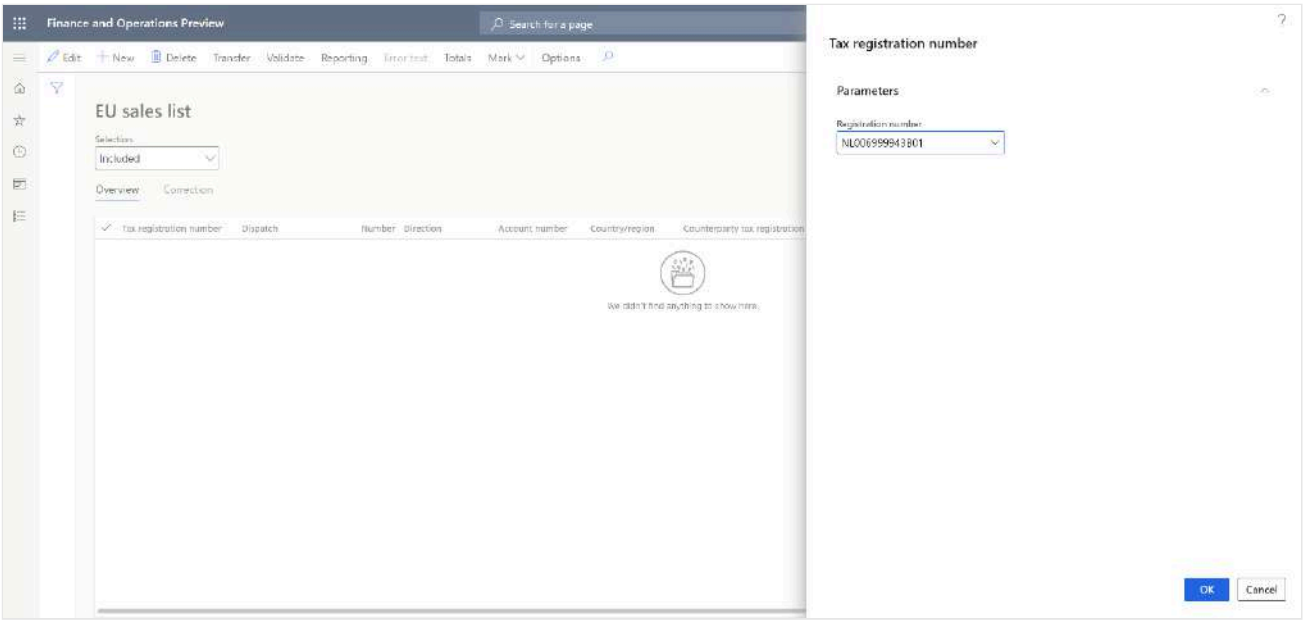
The following table shows the earliest ER format versions that you can select the format for.

Expand table

Release	Country/region	ER format
10.0.19	All	EU Sales list model.version.9
10.0.19	Netherlands	EU Sales list (NL).version.1.10
10.0.20	France	EU Sales list (FR).version.1.2
10.0.20	United Kingdom (Northern Ireland)	EU Sales list XML (UK).version.9.6 EU Sales list TXT (UK).version.9.7
10.0.21	Austria	EU Sales list (AT).version.9.5
10.0.21	Germany	EU Sales list (DE).version.9.5
10.0.21	Spain	EU Sales list (ES).version.9.2
10.0.21	Sweden	EU Sales list (SE).version.13.7
10.0.23	Finland	EU Sales list (FI).version.2.5
10.0.23	Denmark	EU Sales list (DK).version.13.4
10.0.24	Poland	EU Sales list (PL).version.14.7
10.0.25	Belgium	EU Sales list (BE).version.2.3
10.0.33	The Czech Republic	EU Sales list (CZ).version.15.9

Transfer and report the EU sales list

1. Go to **Tax** > **Declarations** > **Foreign trade** > **EU sales list**.
2. Select **Transfer**.
3. In the **Tax registration number** dialog, select the tax registration number to transfer transactions for, and then select **OK**.



4. In the **Transfer transactions for EU sales list** dialog box, select the documents and reporting types to transfer.

Transfer transactions for EU sales list

Select reporting types and invoice information to restrict the transactions that will be transferred to the EU sales list form and printed on the EU sales list report. You can make changes to the list before you print the report.

REPORTING TYPE (SALES)

Item
Yes

Service
No

Investment
No

Not assigned
No

REPORTING TYPE (PURCHASES)

Item
No

Service
No

Investment
No

Not assigned
No

REPORTING TYPE (TRANSFER ORDERS)

Item
Yes

Service
No

Investment
No

Not assigned
No

CUSTOMER INVOICE JOURNAL

Date
2/13/2021

List code
!Not included

PROJECT INVOICE

Date
2/13/2021

List code
!Not included

VENDOR INVOICE JOURNAL

Date
2/13/2021

List code
!Not included

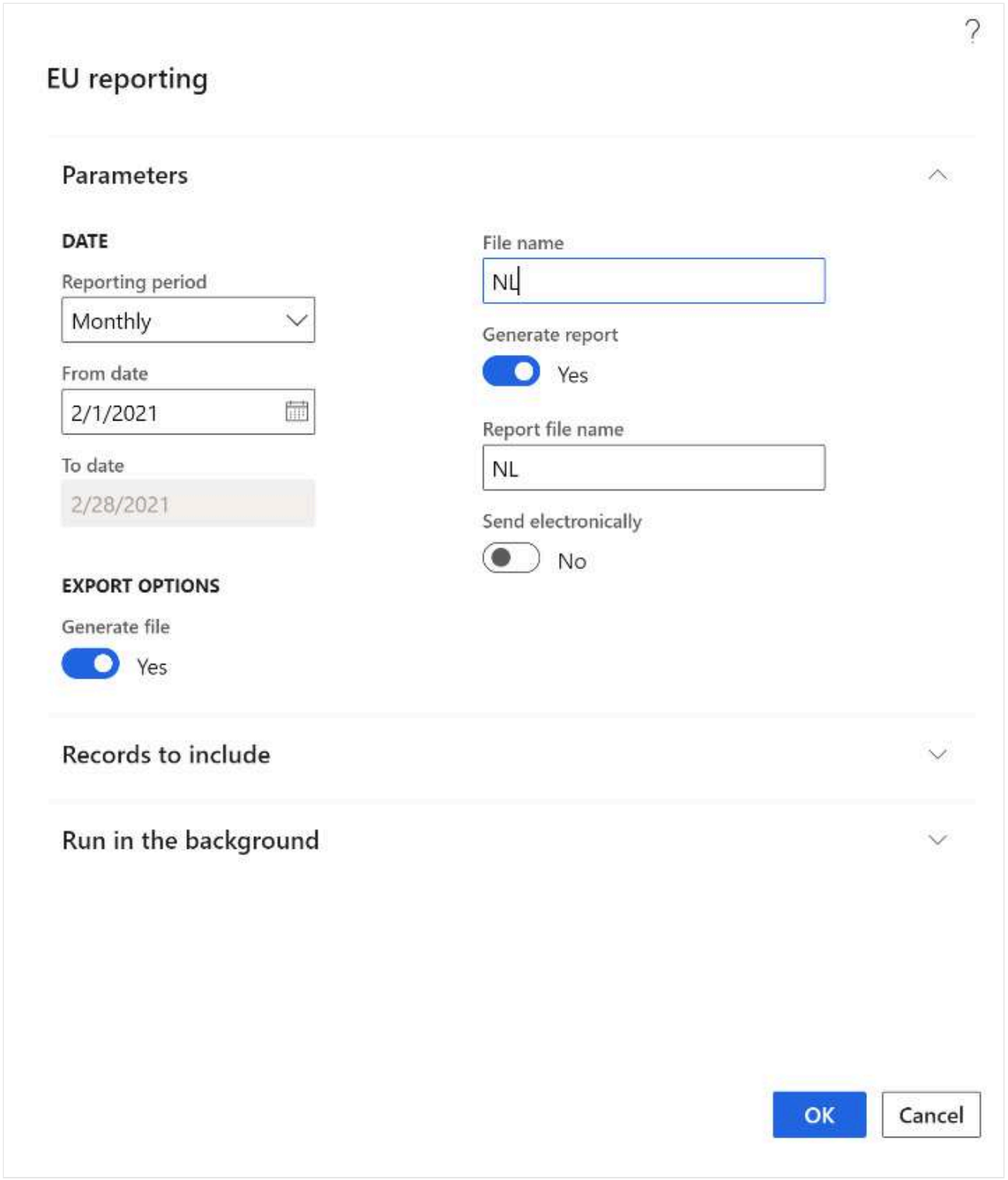
TRANSFER ORDER

Date
2/13/2021

List code
!Not included

Run in the background

5. Select **Select** to adjust the default filter for transactions that should be transferred, and then select **Transfer**.
6. The system transfers transactions for the selected tax registration, documents, and reporting types. Review the transactions, and make any adjustments that are required.
7. Select **Reporting**.
8. In the **Tax registration number** dialog box, select the tax registration number to generate the EU sales list report for, and then select **OK**.
9. In the **EU reporting** dialog box, in the **From date** field, specify the first date to generate the EU sales list report for.
10. Set the **Generate file** option to **Yes** to generate an electronic reporting file. Then, in the **File name** field, enter the name of the output electronic file.
11. Set the **Generate report** option to **Yes** to generate an Excel report. Then, in the **Report file name** field, enter the name of the output Excel file.



12. The dialog box also contains country/region-specific fields that are required in the country/region-specific EU sales list report. Set these fields as required.

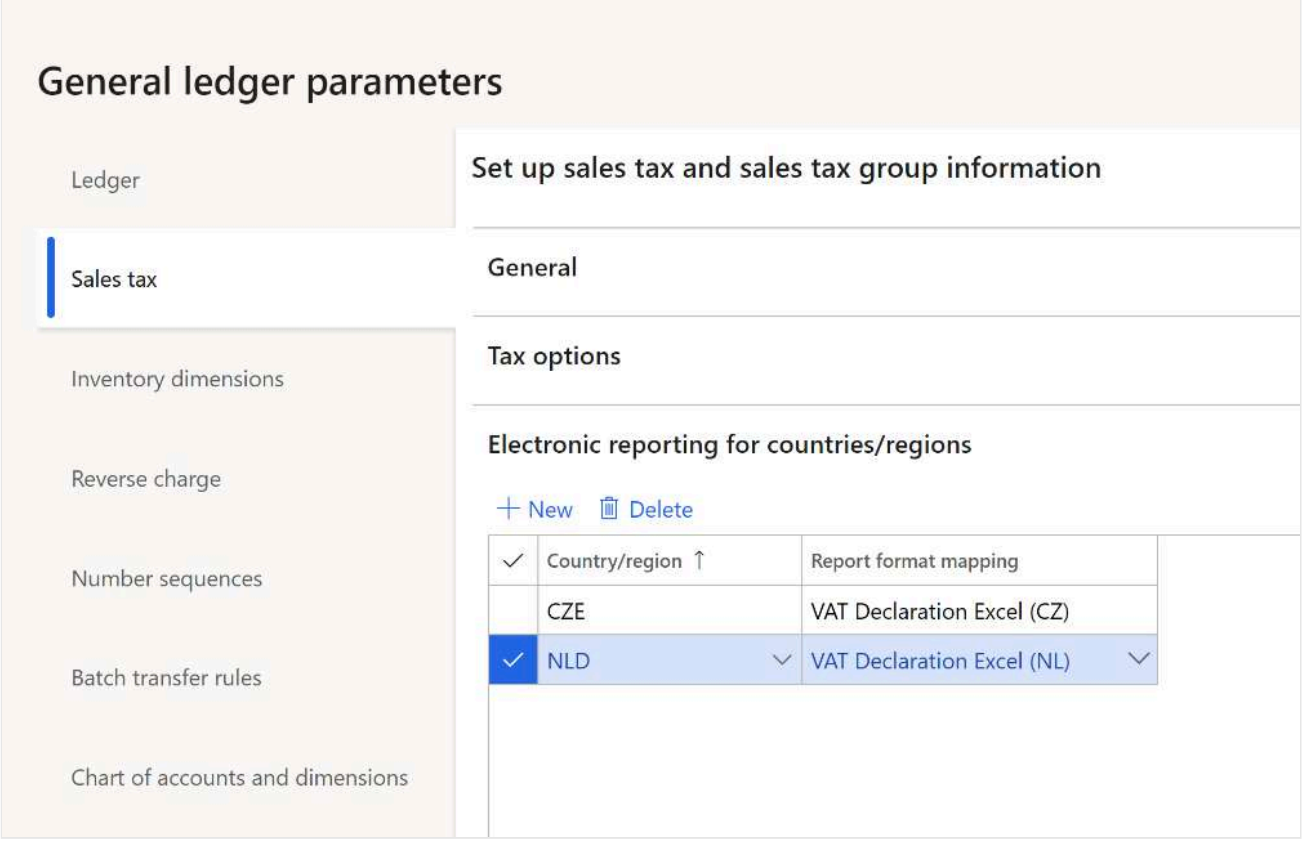
13. Select **OK** to generate the report.

VAT declaration reporting for multiple VAT registrations

Set up for VAT declaration reporting

1. Go to **Tax** > **Set up** > **Parameters** > **General ledger parameters**.

2. On the **Sales tax** tab, on the **Electronic reporting for countries/regions** FastTab, select the VAT reporting formats for each country or region of your tax registration.



> **Note**
>
> You can select an ER format if the **ISO Country/region codes** field on the **ISO Country/region codes** FastTab is blank in this format.

The following table shows the earliest ER format versions that you can select the format for. For more information about how to run VAT declaration for specific countries or regions, review the related article.

Expand table

Release	Country/region	ER format	Link to article
10.0.19	All	Tax declaration model.version.85 Tax declaration model mapping.version.85.138	
10.0.13	The Czech Republic	VAT declaration XML (CZ).version.64.66 VAT Control statement XML (CZ).version.64.66.65	VAT declaration (Czech Republic)

Release	Country/region	ER format	Link to article
		VAT Declaration Excel (CZ).version.64.66.25	
10.0.19	Netherlands	VAT Declaration XML (NL).version.85.14 VAT Declaration Excel (NL).version.85.14.17	VAT declaration (Netherlands)
10.0.20	France	VAT Declaration Excel (FR).version.85.15	VAT declaration (France)
10.0.21	United Kingdom	MTD VAT importing model mapping (UK).version.31.36 Tax declaration model mapping.version.95.158 VAT Declaration Excel (UK).version.32.30.16 VAT Declaration JSON (UK).version.32.31	Prepare for integration with MRD for VAT
10.0.21	Sweden	VAT Declaration XML (SE).version.95.11 VAT Declaration Excel (SE).version.95.11.13	VAT declaration (Sweden)
10.0.21	Switzerland	Tax declaration model.version.96 Tax declaration model mapping.version.96.164 VAT Declaration XML (CH).version.96.16 VAT Declaration Excel (CH).version.96.16.9	VAT declaration (Switzerland)
10.0.22	Austria	VAT Declaration XML (AT).version.101.23 VAT Declaration Excel (AT).version.101.23.17	VAT declaration (Austria)
10.0.23	Germany	VAT Declaration XML (DE).version.101.16 VAT Declaration Excel (DE).version.101.16.12	VAT declaration (Germany)
10.0.21	Norway	Tax declaration model.version.112 Tax declaration model mapping.version.112.192 VAT Declaration XML (NO).version.112.54	VAT return with direct submission to Altinn

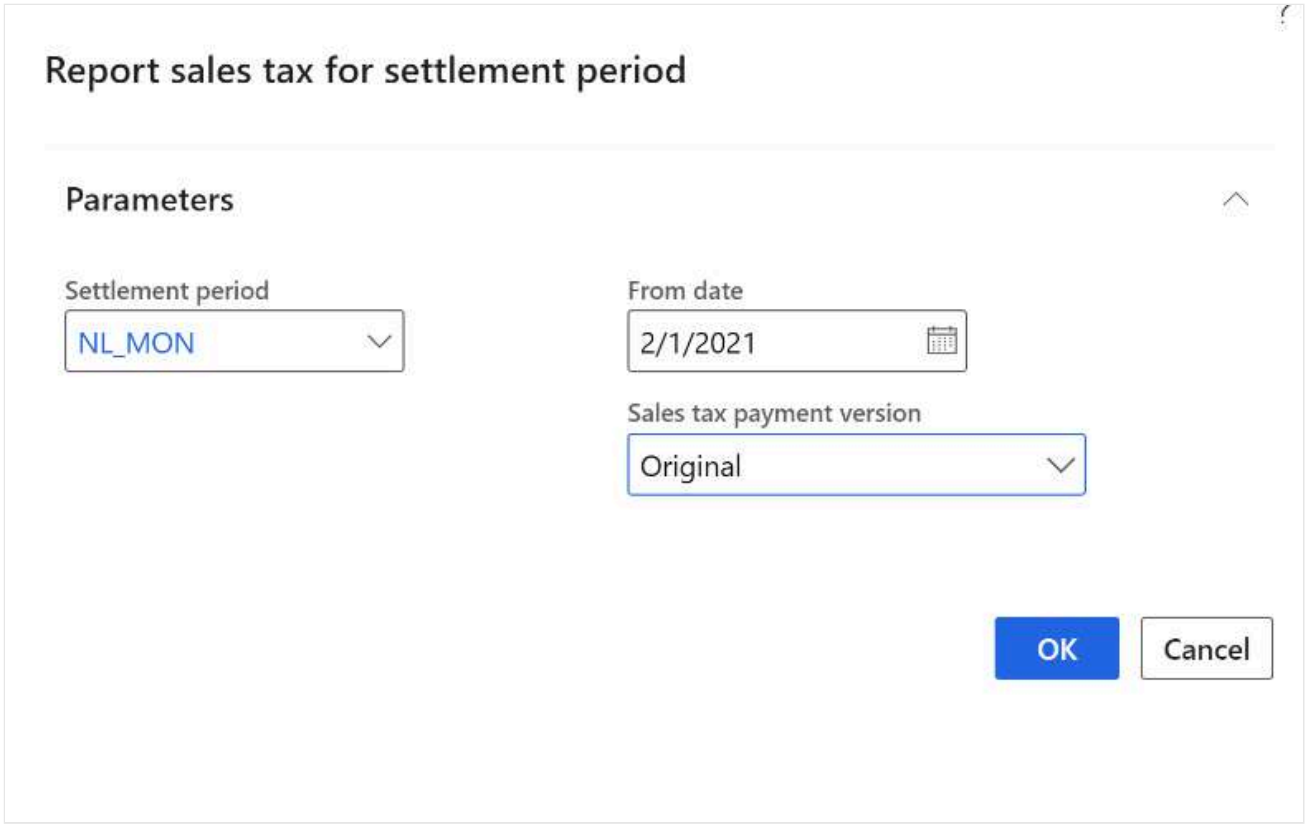
Release	Country/region	ER format	Link to article
		VAT Declaration Excel (NO).version.112.54.39	
10.0.23	Spain	VAT Declaration TXT(ES).version.101.28 VAT Declaration Excel (ES).version.101.28.17	VAT declaration (Spain)
10.0.25	Denmark	VAT Declaration Excel (DK).version.101.8 VAT Declaration XML (DK).version.101.15.7	VAT declaration (Denmark)
10.0.29	Poland	Standard Audit File (SAF-T).version.129 Standard Audit File model mapping.version.131.283 JPK-V7M XML format (PL).version.129.221 JPK-V7M Excel format (PL).version.129.221.70	VAT declaration (Poland)
10.0.36	Estonia	Tax declaration model.version.150 Tax declaration model mapping.version.150.317 VAT Declaration XML (EE).version.150.21 VAT Declaration Excel (EE).version.150.21.17	VAT declaration (Estonia)
10.0.39	Italy	Tax declaration model.version.153 Tax declaration model mapping (IT).version.153.320.4 VAT Declaration PDF (IT).version.153.19	Configure and report Italian sales tax books
10.0.41	Lithuania	Tax declaration model.version.156 Tax declaration model mapping.version.156.329 VAT Declaration XML (LT).version.156.10 VAT Declaration Excel (LT).version.156.10.9	VAT declaration for Lithuania (FR0600)
10.0.42 10.0.41 (build 10.0.2015.44) 10.0.40 (build 10.0.1935.114)	Latvia	Tax declaration model.version.167 Tax declaration model mapping.version.167.339 VAT Declaration XML (LV).version.167.19	VAT declaration for Latvia

Release	Country/region	ER format	Link to article
10.0.39 (build 10.0.1860.186)		VAT Declaration Excel (LV).version.167.19.10	

> **Note**
>
> The VAT declaration of Poland (JPK-V7) is supported by using the Electronic messaging functionality only. For more information, see **VAT declaration with registers (JPK-V7, VDEK)**.

Generate a VAT declaration for multiple VAT registrations

1. Go to **Tax** > **Declarations** > **Sales tax** > **Report sales tax for settlement period**.
2. In the **Report sales tax for settlement period** dialog box, in the **Settlement period** field, select a settlement period.
3. In the **From date** field, specify the first date to generate the VAT declaration for.
4. In the **Sales tax payment version** field, select one of the following values:
    - **Original:** Generate a report for sales tax transactions of the original sales tax payment or before the sales tax payment is generated.
    - **Corrections:** Generate a report for sales tax transactions of all subsequent sales tax payments for the period.
    - **Total list:** Generate a report for all sales tax transactions for the period, including the original and all corrections.



5. Select **OK** to generate the report.

Last updated on 03/18/2026

Support for multiple VAT registration numbers in reporting for Austria

This article provides information about the regulatory reports for Austria that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about the **Multiple VAT registration numbers** functionality, the prerequisites that must be met before you use it, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for Austria.

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.21	Intrastat (AT), version 16.3
EU sales list	10.0.21	EU Sales list (AT), version 9.5
VAT declaration	10.0.22	VAT Declaration XML (AT), version 101.23 VAT Declaration Excel (AT), version 101.23.17

Last updated on 03/04/2026

Support for multiple VAT registration numbers in reporting for Belgium

This article provides information about the regulatory reports for Belgium that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for Belgium.

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.25	Intrastat (BE), version 2.9
EU sales list	10.0.25	EU Sales list (BE), version 2.3
VAT declaration	10.0.27	VAT Declaration XML (BE), version 120.46 VAT Declaration Excel (BE), version 120.46.47
Annual VAT listing of domestic sales	10.0.44	Invoice turnover report model, version 13 Invoice turnover report (BE), version 13.13

Last updated on 03/04/2026

Support for multiple VAT registration numbers in reporting for Czechia

This article provides information about the regulatory reports for Czechia that are available for legal entities that use the multiple value-added tax (VAT) registration numbers functionality. For more information about the **Multiple VAT registration numbers** functionality, the prerequisites that must be met before you use it, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for Czechia:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.34	Intrastat (CZ), version 24.6
EU sales list	10.0.33	EU Sales list (CZ), version 15.9
VAT declaration	10.0.13	VAT declaration XML (CZ), version 64.66 VAT Control statement XML (CZ), version 64.66.65 VAT Declaration Excel (CZ), version 64.66.25

Last updated on 03/09/2026

Support for multiple VAT registration numbers in reporting for Denmark

This article provides information about the regulatory reports for Denmark that are available for legal entities that use the multiple value-added tax (VAT) registration numbers functionality. For more information about the **Multiple VAT registration numbers** functionality, the prerequisites that must be met before you use it, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for Denmark:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.23	Intrastat (DK), version 25.4
EU sales list	10.0.23	EU Sales list (DK), version 13.4
VAT declaration	10.0.25	VAT Declaration Excel (DK), version 101.8

Last updated on 03/09/2026

Support for multiple VAT registration numbers in reporting for Estonia

This article provides information about the regulatory reports for Estonia that legal entities can access when they use the functionality for multiple value-added tax (VAT) registration numbers. For more information about this functionality, the prerequisites that must be met before you use it, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, you can access the following reports for reporting for Estonia.

Expand table

Report name	Release	Electronic reporting (ER) format, version
VAT declaration	10.0.36	VAT Declaration XML (EE), version 150.21 VAT Declaration Excel (EE), version 150.21.17

Last updated on 03/09/2026

Support for multiple VAT registration numbers in reporting for Finland

This article provides information about the regulatory reports for Finland that are available for legal entities that use the multiple value-added tax (VAT) registration numbers functionality. For more information about the **Multiple VAT registration numbers** functionality, the prerequisites that must be met before you use it, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for Finland:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.23	Intrastat (FI), version 2.3
EU sales list	10.0.23	EU Sales list (FI), version 2.5
VAT declaration	10.0.26	VAT Declaration TXT (FI), version 112.9 VAT Declaration Excel (FI), version 112.9.6

Last updated on 03/14/2026

Support for multiple VAT registration numbers in reporting for France

This article provides information about the regulatory reports for France that are available for legal entities that use the multiple value-added tax (VAT) registration numbers functionality. For more information about the **Multiple VAT registration numbers** functionality, the prerequisites that must be met before you use it, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for France:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.20	Intrastat INTRACOM (FR), version 13.5 Intrastat SAISUNIC (FR), version 1.3
EU sales list	10.0.20	EU Sales list (FR), version 1.2
VAT declaration	10.0.20	VAT Declaration Excel (FR), version 85.15
France e-reporting	10.0.49	e-Reporting XML (FR)

Last updated on 06/08/2026

Support for multiple VAT registration numbers in reporting for Germany

This article provides information about the regulatory reports for Germany that are available for legal entities that use the multiple value-added tax (VAT) registration numbers functionality. For more information about the **Multiple VAT registration numbers** functionality, the prerequisites that must be met before you use it, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for Germany:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.21	INSTAT XML (DE), version 22.9.9
EU sales list	10.0.21	EU Sales list (DE), version 9.5
VAT declaration	10.0.23	VAT Declaration XML (DE), version 101.16 VAT Declaration Excel (DE), version 101.16.12

Last updated on 03/14/2026

Support for multiple VAT registration numbers in reporting for Italy

This article provides information about the regulatory reports for Italy that are available for legal entities that use the functionality for multiple value-added tax (VAT) registration numbers. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for Italy.

Expand table

Report name	Release	Electronic reporting (ER) format and version
Intrastat	10.0.38 10.0.37 - build 10.0.1725.25 10.0.36 - build 10.0.1695.56	Intrastat (IT), version 24.36
Yearly tax communication	10.0.38	Yearly tax communication (IT), version 1.5
Configure and report Italian sales tax books	10.0.39	VAT Declaration PDF (IT), version 153.19

Last updated on 05/02/2026

Support for multiple VAT registration numbers in reporting for Latvia

This article provides information about the regulatory reports for Latvia that legal entities can use with the functionality for multiple value-added tax (VAT) registration numbers. For more information about this functionality, the prerequisites, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, the following reports are available for reporting for Latvia.

Expand table

Report name	Release	Electronic reporting (ER) format
VAT declaration	10.0.42, 10.0.41HF, 10.0.40HF, 10.0.39HF	VAT Declaration XML (LV) VAT Declaration Excel (LV)

Last updated on 05/08/2026

Support for multiple VAT registration numbers in reporting for Lithuania

This article provides information about the regulatory reports for Lithuania that are available to legal entities that use the functionality for multiple value-added tax (VAT) registration numbers. For more information about this functionality, the prerequisites, and the required setup, see Multiple VAT registration numbers.

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When you configure your legal entity for reporting for multiple VAT registrations, you get the following reports for reporting for Lithuania.

Expand table

Report name	Release	Electronic reporting (ER) format
VAT declaration	10.0.41	VAT Declaration XML (LT) VAT Declaration Excel (LT)

Last updated on 05/08/2026

Support for multiple VAT registration numbers in reporting for Netherlands

This article explains how to support multiple value-added tax (VAT) registration numbers in reporting for Netherlands in Microsoft Dynamics 365 Finance, and provides information about the regulatory reports for Netherlands that are available for legal entities that use multiple VAT registration numbers functionality.

For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for Netherlands:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.19	Intrastat (NL), version 1.3
EU sales list	10.0.19	EU Sales list (NL), version 1.10
VAT declaration	10.0.19	VAT Declaration XML (NL), version 85.14 VAT Declaration Excel (NL), version 85.14.17

Last updated on 05/08/2026

Support for multiple VAT registration numbers in reporting for Norway

This article provides information about the regulatory reports for Norway that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for Norway:

Expand table

Report name	Release	Electronic reporting (ER) format, version
VAT return with direct submission to Altinn	10.0.21	VAT Declaration XML (NO), version 112.54 VAT Declaration Excel (NO), version 112.54.39

Last updated on 05/08/2026

Support for multiple VAT registration numbers in reporting for Poland

This article provides information about the regulatory reports for Poland that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for Poland:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.23	Intrastat (PL), version 25.5
EU sales list	10.0.24	EU Sales list (PL), version 14.7
VAT declaration	10.0.29	Standard Audit File (SAF-T), version 129 Standard Audit File model mapping, version 131.283 JPK-V7M XML format (PL), version 129.221 JPK-V7M Excel format (PL), version 129.221.70

Last updated on 05/08/2026

Support for multiple VAT registration numbers in reporting for Spain

This article provides information about the regulatory reports for Spain that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for Spain:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.21	Intrastat (ES), version 16.7
EU sales list	10.0.21	EU Sales list (ES), version 9.2
VAT declaration	10.0.23	VAT Declaration TXT(ES), version 101.28 VAT Declaration Excel (ES), version 101.28.17
Immediate Supply of Information on VAT (SII)(*)	10.0.22	SII Invoice Received Format (ES), version 107.45 SII Invoice Issued Format (ES), version 107.51

Note

(*) As of Finance version 10.0.22, if you're using the Tax Calculation service, and the Support multiple VAT registration numbers feature is enabled in the **Feature management** workspace, you can report the following reports to the SII system of Spain from a legal entity that has a primary address outside Spain:

- 'Libro de registro de facturas Expedidas': **Record book of issued invoices**
- 'Libro de registro de facturas Recibidas': **Record book of received invoices**

Last updated on 05/13/2026

Support for multiple VAT registration numbers in reporting for Sweden

This article provides information about the regulatory reports for Sweden that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for Sweden:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat	10.0.21	Intrastat (SE), version 16.4
EU sales list	10.0.21	EU Sales list (SE), version 13.7
VAT declaration	10.0.21	VAT Declaration XML (SE), version 95.11 VAT Declaration Excel (SE), version 95.11.13

Last updated on 05/13/2026

Support for multiple VAT registration numbers in reporting for Switzerland

This article provides information about the regulatory reports for Switzerland that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for Switzerland:

Expand table

Report name	Release	Electronic reporting (ER) format, version
VAT declaration	10.0.21	VAT Declaration XML (CH), version 96.16 VAT Declaration Excel (CH), version 96.16.9

Last updated on 05/13/2026

Support for multiple VAT registration numbers in reporting for the United Kingdom

This article provides information about the regulatory reports for the United Kingdom that are available for legal entities that use multiple value-added tax (VAT) registration numbers functionality. For more information about **Multiple VAT registration numbers** functionality, the prerequisites that must be met before it's used, and the required setup, see Multiple VAT registration numbers

For general information about reporting for multiple VAT registrations, see Reporting for multiple VAT registrations.

When your legal entity is configured for reporting for multiple VAT registrations, the following reports are available for reporting for the United Kingdom:

Expand table

Report name	Release	Electronic reporting (ER) format, version
Intrastat - United Kingdom (Northern Ireland)	10.0.20	Intrastat (UK), version 1.2
EU sales list - United Kingdom (Northern Ireland)	10.0.20	EU Sales list XML (UK), version 9.6 EU Sales list TXT (UK), version 9.7
VAT declaration - Making Tax Digital	10.0.21	VAT Declaration Excel (UK), version 32.30.16 VAT Declaration JSON (UK), version 32.31

Last updated on 05/13/2026

Tax feature support for transfer orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about tax calculation and posting integration in transfer orders. By using this functionality, you can set up tax calculation and posting in transfer orders for stock transfers. Under European Union (EU) value-added tax (VAT) regulations, stock transfers are considered intra-community supply and intra-community acquisitions.

Set up tax calculation for tax and transfer order transactions

Follow these steps to set up the tax that is involved in a transfer order. In the example that is shown here, the transfer order is from the Netherlands to Belgium.

1. On the **Tax calculation features** page, on the **Versions** tab, select the draft feature version, and then select **Edit**.

2. On the **Tax features setup** page, on the **Tax codes** tab, select **Add** to create new tax codes. For this example, create three tax codes: **NL-Exempt**, **BE-RC-21**, and **BE-RC+21**.

    - When a transfer order ships from a warehouse in the Netherlands, apply the Netherlands VAT exempted tax code (**NL-Exempt**).

      Create the tax code **NL-Exempt**.
      a. Select **Add**, enter **NL-Exempt** in the **Tax code** field.
      b. Select **By Net Amount** in the **Tax component** field.
      c. Select **Save**.
      d. Select **Add** in the **Rate** table.
      e. Set **Is Exempt** to **Yes** in the **General** section.
      f. In the **Exempt Code** field, enter **EC**.

    - When a transfer order arrives at a Belgium warehouse, apply the reverse charge mechanism by using the **BE-RC-21** and **BE-RC+21** tax codes.

      Create the tax code **BE-RC-21**.

a. Select **Add**, enter **BE-RC-21** in the **Tax code** field.
b. Select **By Net Amount** in the **Tax component** field.
c. Select **Save**.
d. Select **Add** in the **Rate** table.
e. Enter **-21** in the **Tax Rate** field.
f. Set **Is Reverse Charge** to **Yes** in the **General** section.
g. Select **Save**.

Create the tax code **BE-RC+21**.

a. Select **Add**, enter **BE-RC+21** in the **Tax code** field.
b. Select **By Net Amount** in the **Tax component** field.
c. Select **Save**.
d. Select **Add** in the **Rate** table.
e. Enter **21** in the **Tax Rate** field.
f. Select **Save**.

3. Define the tax group.
   a. Select **Manage columns**, and then select the line field **Tax Group**.
   b. Select ->, and then select **OK**.
   c. Select **Add** to add a tax group.
   d. In the **Tax Group** column, enter **AR-EU** and then select the **NL-Exempt** tax code.
   e. Select **Add** to add a tax group.
   f. In the **Tax Group** column, enter **RC-VAT** and then select the **BE-RC-21** and **BE-RC+21** tax codes.

4. Define the item tax group.
   a. Select **Manage columns**, and then select the line field **Item Tax Group**.
   b. Select -> and then select **OK**.
   c. Select **Add** to add an item tax group.
   d. Enter **FULL** in the column **Item Tax Group**. Select tax codes **BE-RC-21**, **BE-RC+21**, and **NL-Exempt**.

5. Define the applicability of the tax group.

   a. Select **Manage columns**, and then select columns that you want to use to build the applicability table.

   > ⓘ **Note**
   >
   > Be sure to add the **Business process** and **Tax directions** columns to the table. Both columns are essential to the functionality for tax in transfer orders.

b. Add applicability rules. Don't leave the **Tax group** field blank.

Add a new rule for transfer order shipment.

i. Select **Add** in the **Applicability rules** table.
ii. In the **Business process** field, select **Inventory** to make the rule applicable for a transfer order.
iii. In the **Ship From Country/Region** field, enter **NLD**.
iv. In the **Ship To Country/Region** field, enter **BEL**.
v. In the **Tax direction** field, select **Output** to make the rule applicable to transfer order shipment.
vi. In the **Tax Group** field, select **AR-EU**.

Add another rule for transfer order receipt.

i. Select **Add** in the **Applicability rules** table.
ii. In the **Business process** field, select **Inventory** to make the rule applicable for a transfer order.
iii. In the **Ship From Country/Region** field, enter **NLD**.
iv. In the **Ship To Country/Region** field, enter **BEL**.
v. In the **Tax direction** field, select **Input** to make the rule applicable to transfer order receipt.
vi. In the **Tax Group** field, select **RC-VAT**.

6. Define the applicability of the item tax group.

a. Select **Manage columns**, and then select columns that you want to use to build the applicability table.

b. Add applicability rules.

> ⓘ **Note**
>
> If the item sales tax group defaulted on your taxable document lines is already correct, leave this matrix blank.

Add a new rule for transfer order shipment and receipt.

i. On the **Applicability rules** page, select **Add**.
ii. In the **Business process** field, select **Inventory** to make the rule applicable for the transfer order.
iii. In the **Item Tax Group** field, select **FULL**.

7. Complete and publish the new tax feature version.

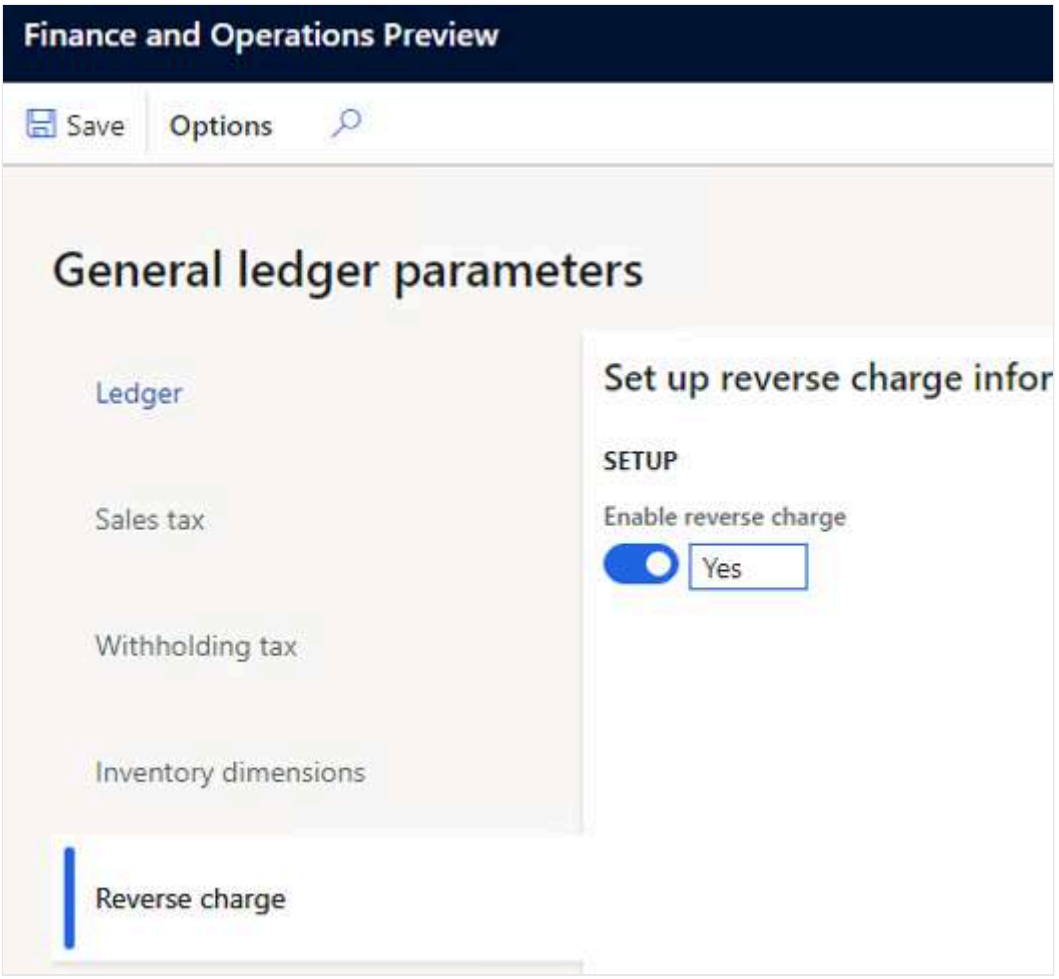
Set up Finance for transfer order transactions

1. Enable the tax calculation **Enable advanced tax calculation** parameter, and select the **Inventory** business process.

   > **Important**
   >
   > You must complete this step for each legal entity in Finance where you want the tax calculation feature and the functionality for tax in transfer orders to be available.

   a. Go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**.
   b. In the **Business process** field, select **Inventory**.

2. Verify that the reverse charge mechanism is set up. Go to **General ledger** > **Setup** > **Parameters**, and then, on the **Reverse charge** tab, verify that the **Enable reverse charge** option is set to **Yes**.

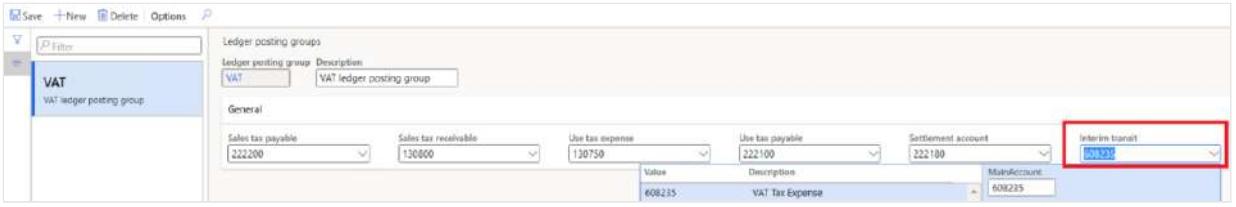


3. Verify that the related tax codes, tax groups, item tax groups, and VAT registration numbers are set up in Finance according to the tax calculation guidance.

4. Set up an interim transit account. This step is required only when the tax that you apply to a transfer order isn't applicable to a tax-exempted or reverse charge mechanism.

a. Go to **Tax** > **Setup** > **Sales tax** > **Ledger posting groups**.

b. In the **Interim transit** field, select a ledger account.

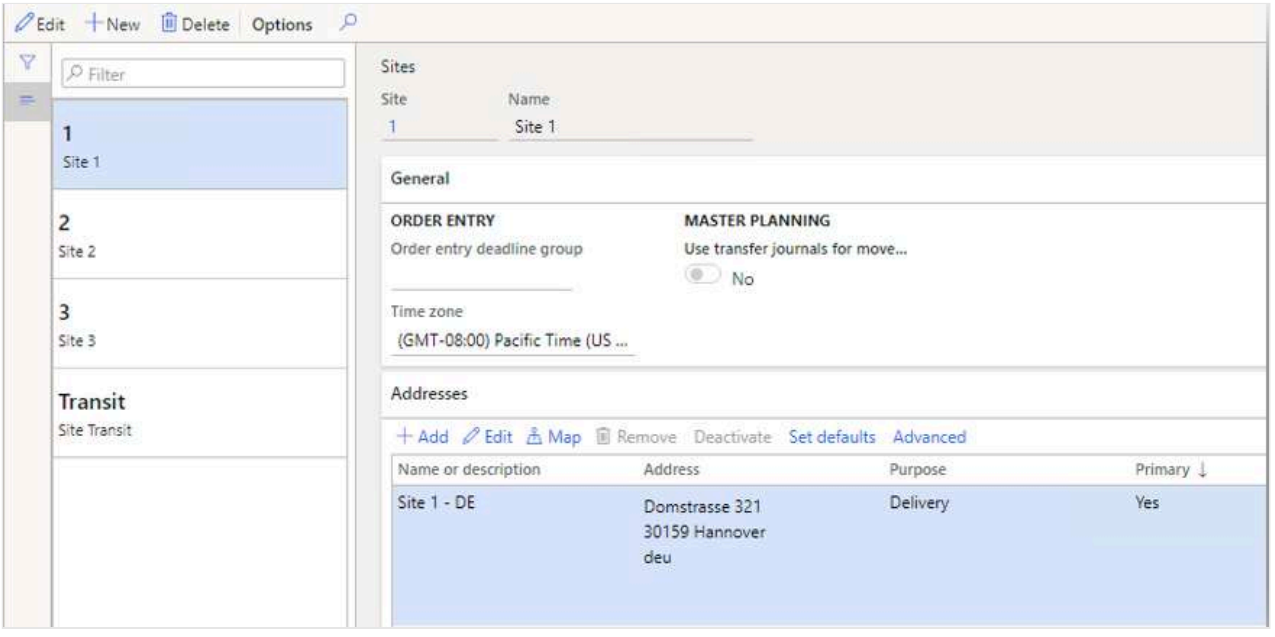


Set up basic inventory for transfer order transactions

Follow these steps to set up basic inventory to enable transfer order transactions.

1. Create ship-from and ship-to sites for your warehouses in different countries or regions, and add the primary address for each site.
   a. Go to **Warehouse management** > **Setup** > **Warehouse** > **Sites**.
   b. Select **New** to create the site that you assign to a warehouse later.
   c. Repeat step 2 for all the other sites that you must create.

> **Note**
>
> One of the sites that you create should be named **Transit**. In later steps of this procedure, you assign this site to the transit warehouse, so that tax-related inventory vouchers can be posted in "ship" and "receive" transactions for transfer orders. The address of the transit site is irrelevant to tax calculation. Therefore, you can leave it blank.

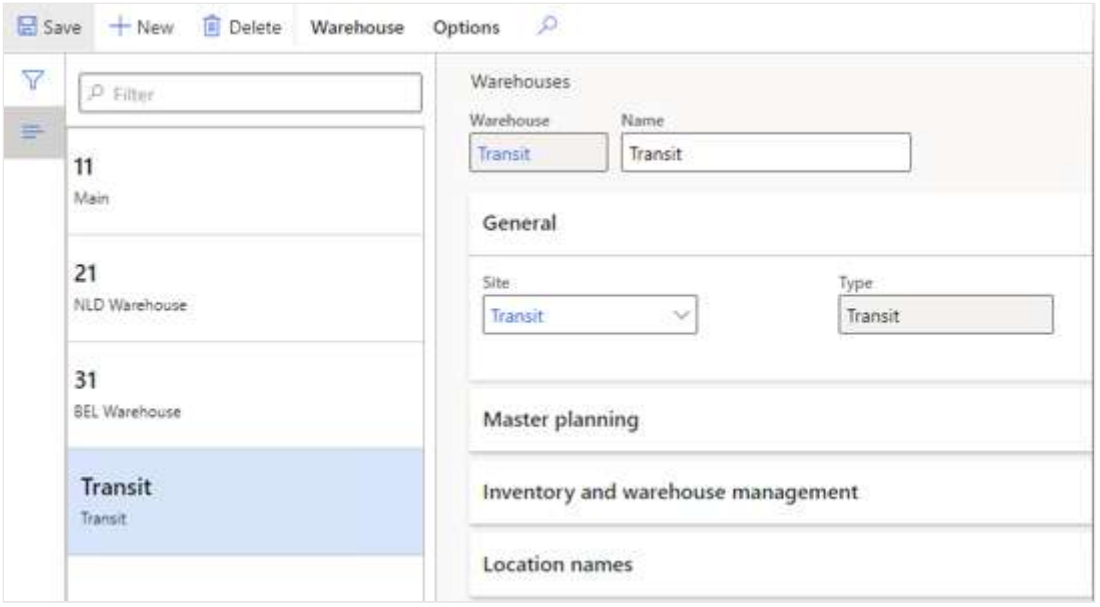


2. Create ship-from, transit, and ship-to warehouses. Any address information that you maintain in a warehouse overrides the site address during tax calculation.

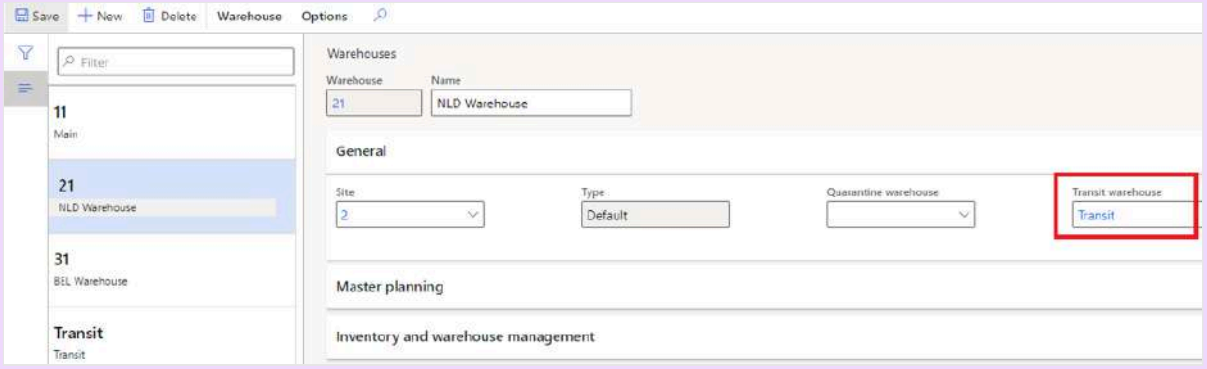
   a. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.

   b. Select **New** to create a warehouse, and assign it to the corresponding site.

   c. Repeat step 2 to create a warehouse for each site as required.



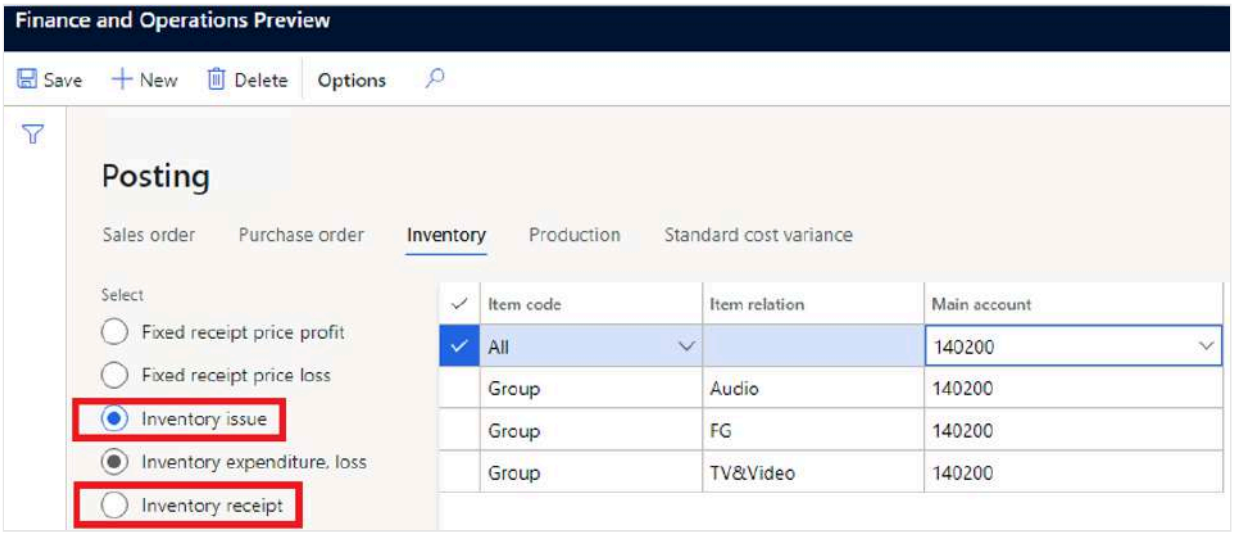
> ⓘ **Note**
>
> For a ship-from warehouse, you must select a transit warehouse in the **Transit warehouse** field for transfer order transactions.



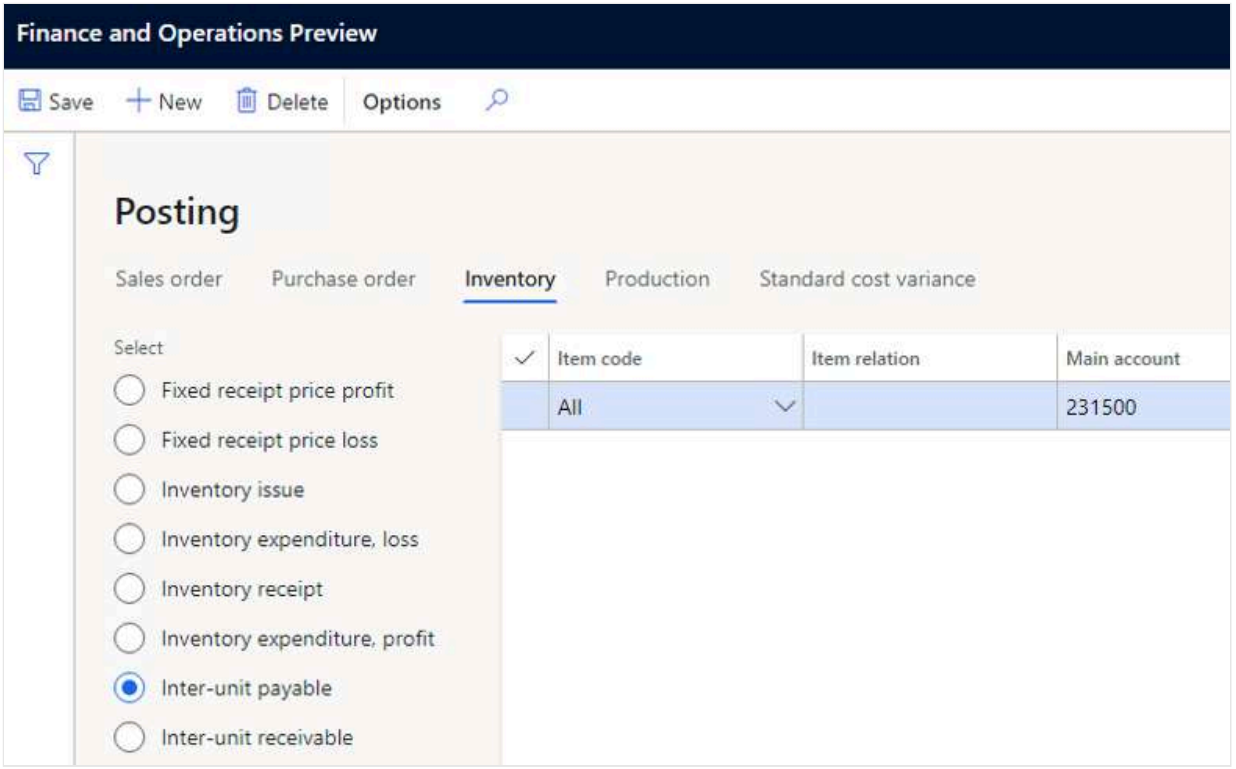
3. Verify that the inventory posting configuration is set up for transfer order transactions.

   a. Go to **Inventory management** > **Setup** > **Posting** > **Posting**.

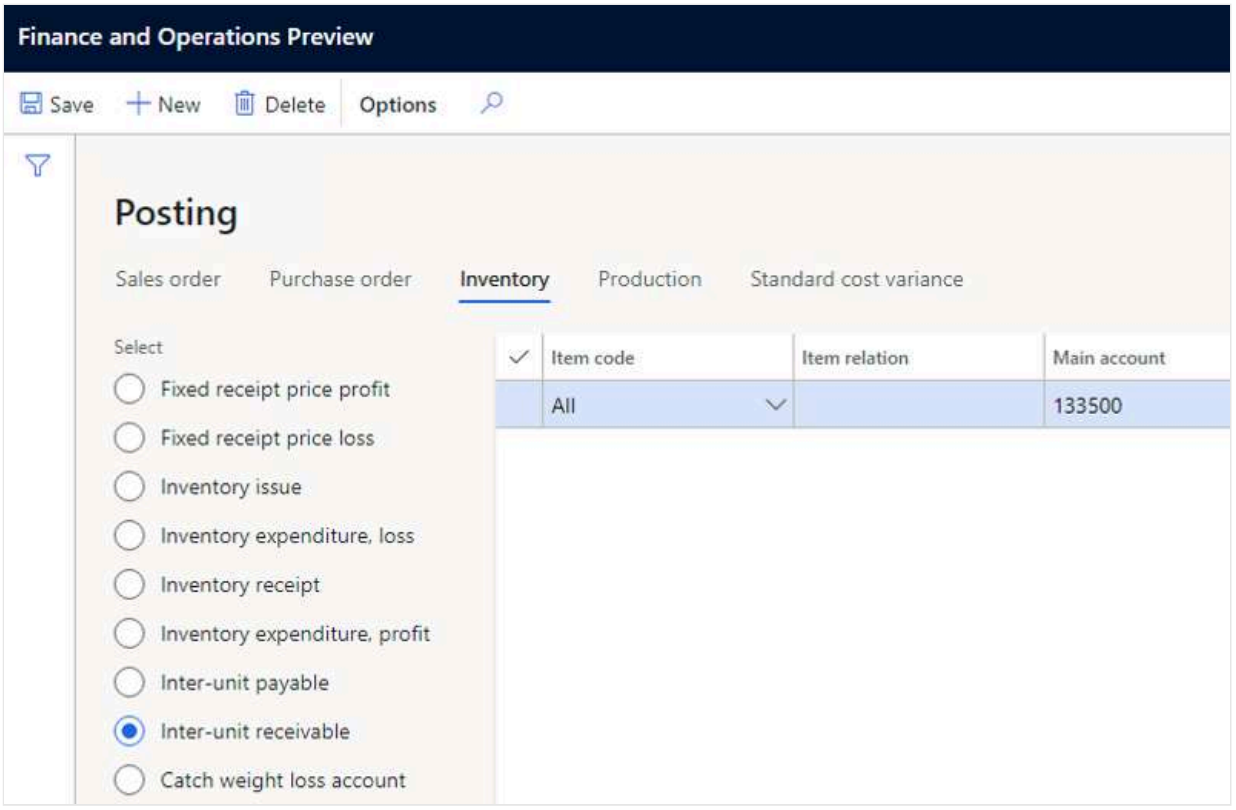
   b. On the **Inventory** tab, verify that a ledger account is set up for both **Inventory issue** and **Inventory receipt** posting.



c. Verify that a ledger account is set up for **Inter-unit payable** posting.



d. Verify that a ledger account is set up for **Inter-unit receivable** posting.



Last updated on 03/21/2026

Print tax information on transfer order documents

This article explains how to print tax information on transfer order documents. You can print the pro-forma invoice document of a transfer order for stock transfers that are considered intra-community supply and intra-community acquisitions under European Union (EU) value-added tax (VAT) regulations.

The following tax-relevant data is added to the transfer order documents:

- The names and the ship-from and ship-to addresses for the stock transfer
- The tax identification numbers of the supplier and the recipient
- The unit price and the taxable amount of the supplied goods
- The tax code, tax rate, tax amount, and tax directives

To configure this data, complete the following main steps.

1. Enable and set up the tax feature for transfer orders.

2. Create and set up multiple tax registration numbers for a legal entity.

3. Set up the exempt code, description, tax directives, and print code in tax codes. For this example, create and synchronize three tax codes in Microsoft Dynamics 365 Finance: **NL-Exempt**, **BE-RC-21**, and **BE-RC+21**.
   a. In Finance, go to **Tax** > **Setup** > **Sales tax** > **Sales tax exempt codes**.
   b. Select **Edit**, and enter a description for the **EC** exempt code. For example, enter **Tax-free EC shipments with tax registration number**.
   c. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax codes**.
   d. Select the **BE-RC-21** tax code, and then select **Tax directives**.
   e. Select **New**.
   f. In the **Language** field, select **EN-US**. Then, in the **Text** field, enter **Document laying down the transfer of goods in the sense of article 138. 2. c) of EU VAT Directive 2006/112/EC**.
   g. Close the page.
   h. On the **General** tab, in the **Print** field, select **Sales tax rate**.
   i. Select the **NL-Exempt** tax code, and then, on the **General** tab, in the **Print** field, select **Sales tax rate**.

> **Note**
>
> The tax code, tax rate, and tax-exempt description don't print on transfer order documents if you don't maintain an exempt code description or tax directives for the tax code.

Print the transfer order - shipment document

After you ship a transfer, follow these steps to print the transfer order - shipment document.

1. Go to **Inventory management** > **Inquiries and reports** > **Transfer orders** > **Shipment**.
2. On the **Parameters** tab, in the **Print** group, select **Tax information**.
3. On **Records to include** tab, select **Filter**, select the transfer order number, and then select **OK**.
4. Select **OK** to print the transfer order - shipment document.

Print the transfer order - receipt document

After you receive a transfer, follow these steps to print the transfer order - receipt document.

1. Go to **Inventory management** > **Inquiries and reports** > **Transfer orders** > **Receive**.
2. On the **Parameters** tab, in the **Print** group, select **Tax information**.
3. On the **Records to include** tab, select **Filter**, select the transfer order number, and then select **OK**.
4. Select **OK** to print the transfer order - receipt document.

Print the transfer overview document

Follow these steps to print the transfer overview document.

> **Note**
>
> Tax information is available only when the transfer order is shipped or received.

1. Go to **Inventory management** > **Inquiries and reports** > **Transfer orders** > **Transfer overview**.
2. On the **Parameters** tab, in the **Print** group, select **Tax information**.

3. On the **Records to include** tab, select **Filter**, select the transfer order number, and then select **OK**.
4. Select **OK** to print the transfer overview document.

Last updated on 03/21/2026

Exchange rate type for sales tax

This article explains the logic for exchange rate type setup and calculation for sales tax.

As of Microsoft Dynamics 365 Finance version 10.0.35, the functionality for value-added tax (VAT) exchange rates is extended for more countries and regions. In addition, the following capabilities are enhanced:

- Support for direct recalculation from the transaction currency to the tax currency
- Support for amount adjustments in the tax currency

Prerequisites and setup

This functionality is available only for legal entities that enable the Tax Calculation feature for the selected business processes. For more information about Tax Calculation, its capabilities, and how to set it up, see Tax Calculation overview.

In addition, you must enable the following features in the **Feature management** workspace:

- **Date of VAT register** (For more information about this feature, see Tax point date (Date of VAT register).)
- **Enable exchange rate types for sales tax**

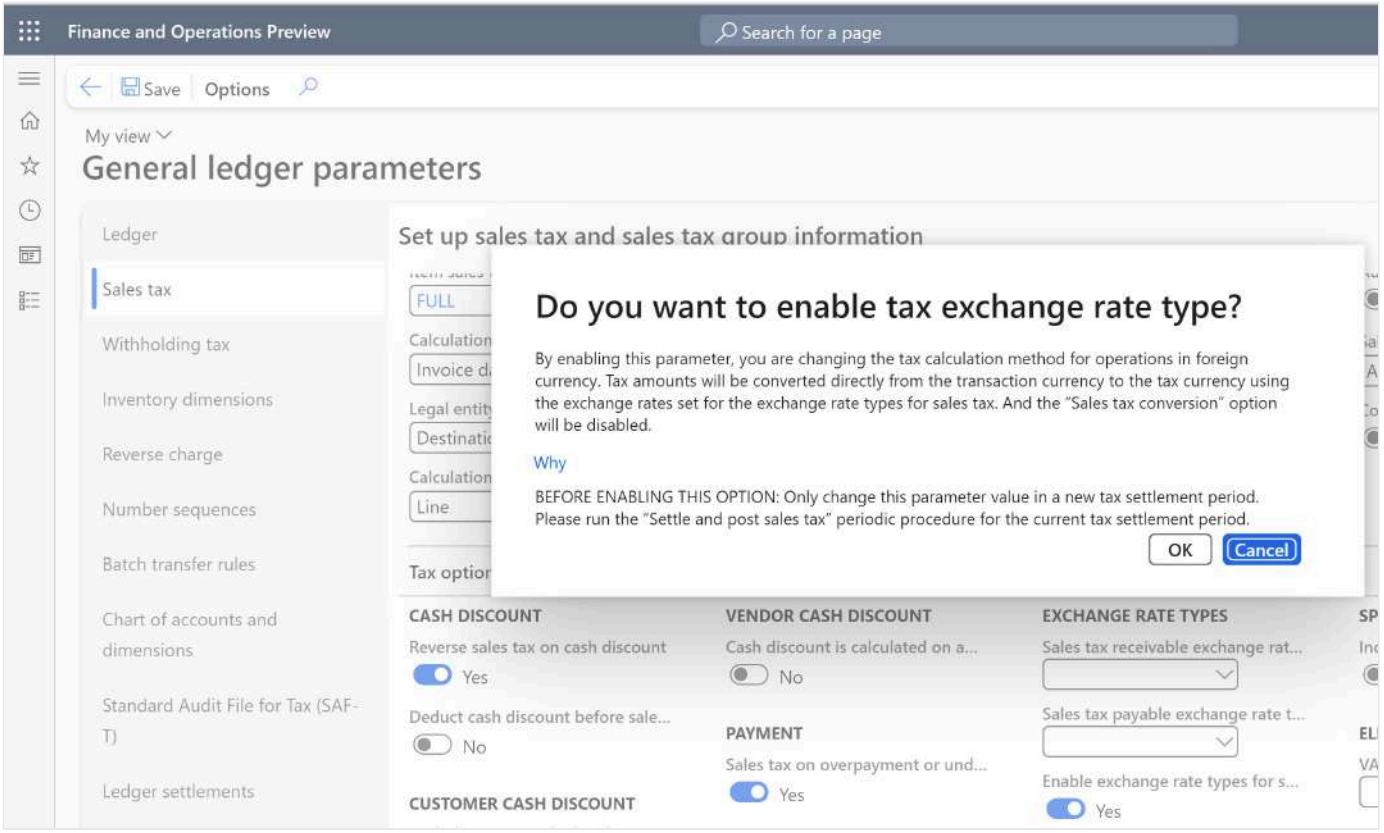
After you enable these features, you might have to refresh the page to update the user interface (UI).

> ⓘ **Note**
>
> If you didn't use the **Date of VAT register** functionality, follow the instructions in Sales tax transactions extension consistency check.

Next, on the **General ledger parameters** page, on the **Sales tax** tab, on the **Tax options** FastTab, set the **Enable exchange rate types for sales tax** option to **Yes**. Enable this parameter in a new tax settlement period. Be sure to run the **Settle and post sales tax** periodic procedure for the current tax settlement period.

When you enable the parameter, you receive the following message:

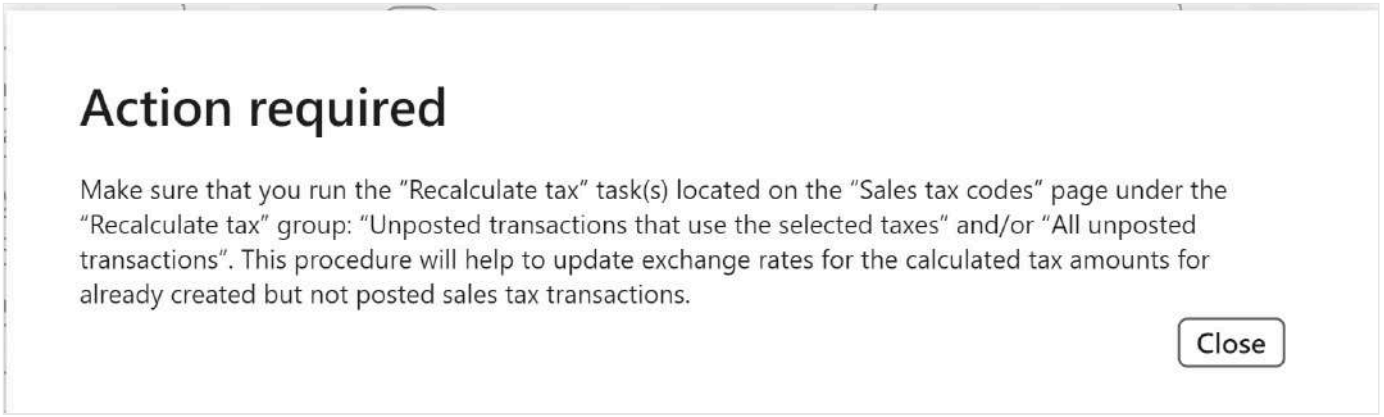
> By enabling this parameter, you change the tax calculation method for operations in foreign currency. Tax amounts are converted directly from the transaction currency to the tax currency by using the exchange rates set for the exchange rate types for sales tax. The **Sales tax conversion** option is disabled. This option controls how to convert tax amount from transaction currency to tax currency - via either Accounting currency or Reporting currency.



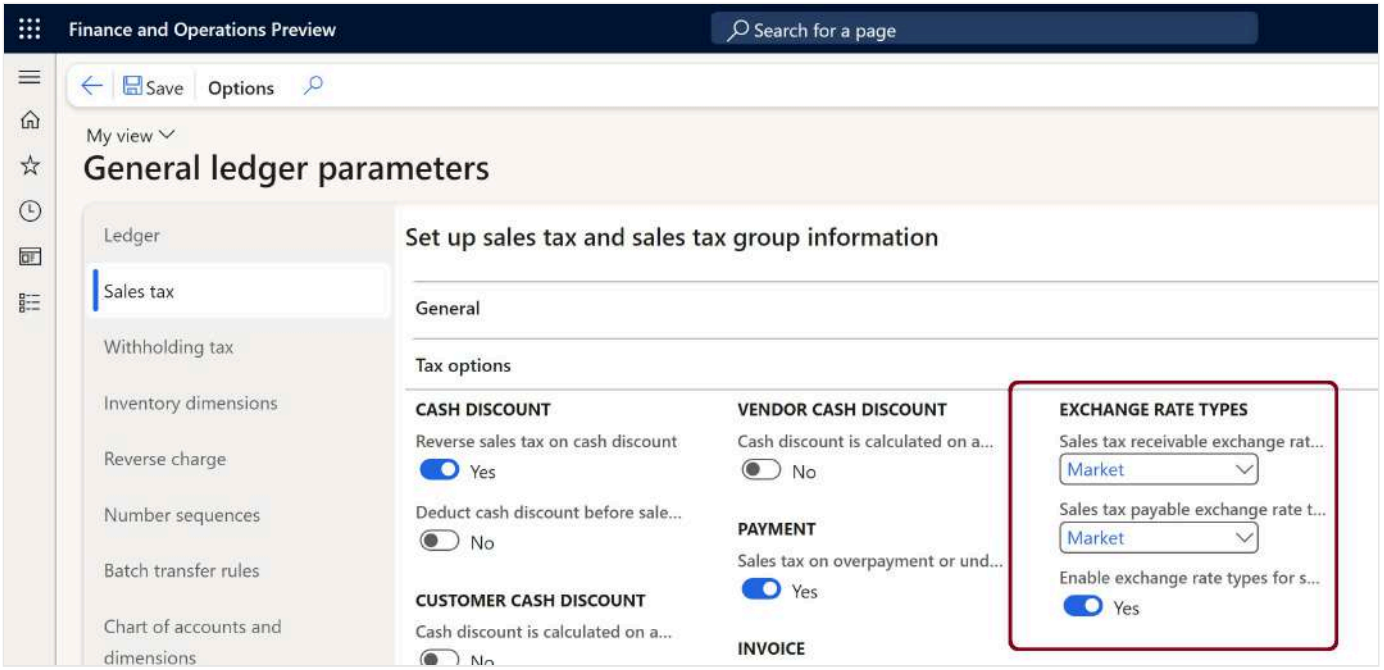
Another message prompts you to run the **Recalculate tax** tasks in the **Recalculate tax** group on the **Sales tax codes** page. Those tasks can include the following two tasks:

- Unposted transactions that use the selected taxes
- All unposted transactions

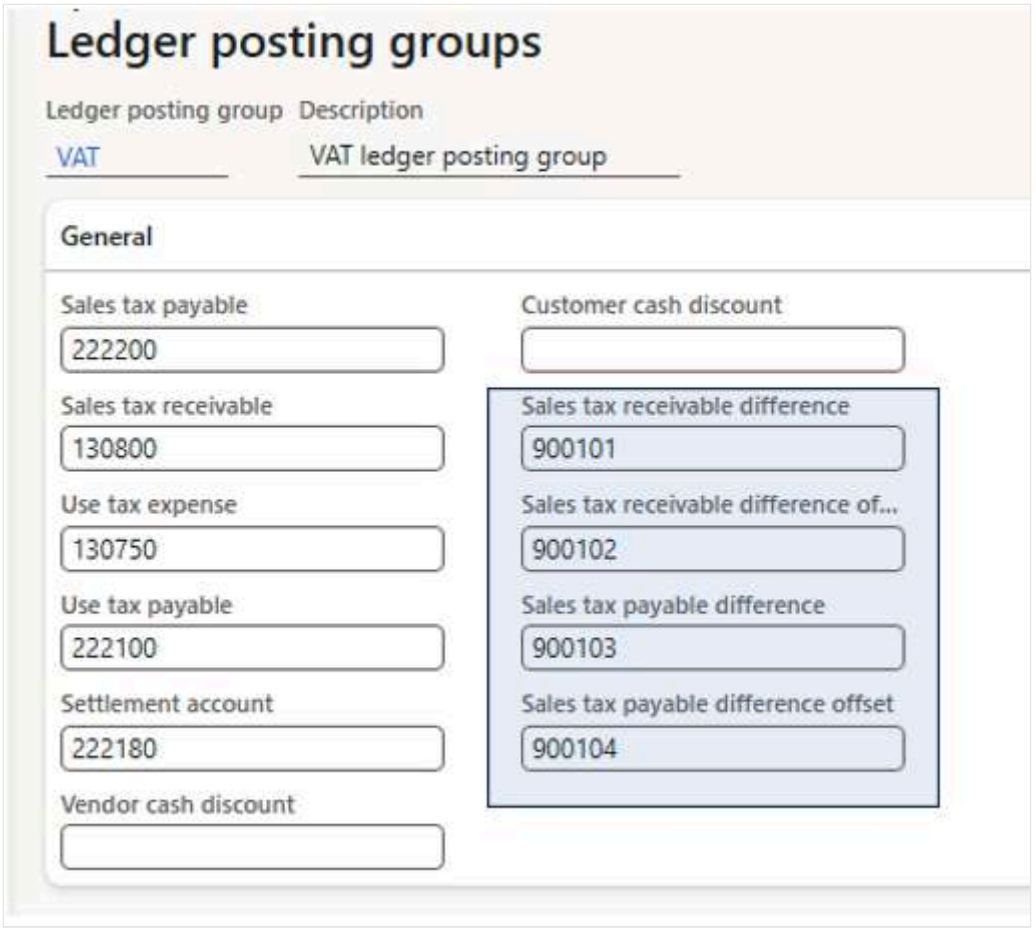
These tasks help you update exchange rates for the calculated tax amounts for sales tax transactions that you created but didn't post.



Next, on the **General ledger parameters** page, on the **Sales tax** tab, on the **Tax options** FastTab, in the **Sales tax receivable exchange rate type** and **Sales tax payable exchange rate type** fields, select the exchange rate types to use for purchase and sales operations. If you leave these fields blank, the exchange rates come from the exchange rate types that are set on the **Ledger** page.



Finally, on the **Ledger posting groups** page (**Tax** > **Setup** > **Sales tax** > **Ledger posting groups**), set up sales tax receivable and sales tax payable difference and difference offset accounts in the ledger posting groups.



Overview of the functionality

After you set up the functionality, you can post invoices that use foreign currency, and adjust the exchange rate and tax amounts on the **Sales tax transactions** page.

On the **Sales Tax transactions** page of the document you're processing, in the **Date of VAT register** field, select the date when the exchange rate for tax amount calculations should be applied. You can edit the **Sales tax exchange rate** and **Date of VAT register** fields.

You can also use the **Adjusted amount origin (VAT exchange rate)** or **Adjusted sales tax amount (VAT exchange rate)** field to enter actual VAT amounts in the local currency that's stated in an external document. When you review the accounting, you can view sales tax difference amounts on the **Subledger journal** page. This information is relevant to documents that follow the source document framework.

When you post a document, you can view any differences in sales tax amount for transactions that post to the general ledger accounts you configured. These differences happen because of the difference between the tax currency exchange rate and the accounting currency exchange rate for your organization. For the *Sales tax payable* sales tax direction, the tax difference posts with a *Sales tax payable difference* tax direction. For the *Sales tax receivable* sales tax direction, the

tax adjustment records with a *Sales tax receivable difference* tax direction. For use tax, the tax difference posts with a *Use tax* sales tax direction.

> **Note**
>
> After you post tax transactions, updating the **Date of VAT register** field on the **VAT register transactions** has no effect on the tax adjustment transaction posting.

Example

In this example, a legal entity issues a free text invoice in a foreign currency. The invoice amount is 1,100 US dollars (USD) and includes VAT at a rate of 10 percent. The VAT is calculated based on the special exchange rate.

Expand table

Currency	Currency code
Legal entity accounting currency	EUR
Legal entity reporting currency	USD
Tax code currency	PLN
Transaction currency	USD

Set up the following exchange rates for business accounting purposes on the **Ledger** page.

Expand table

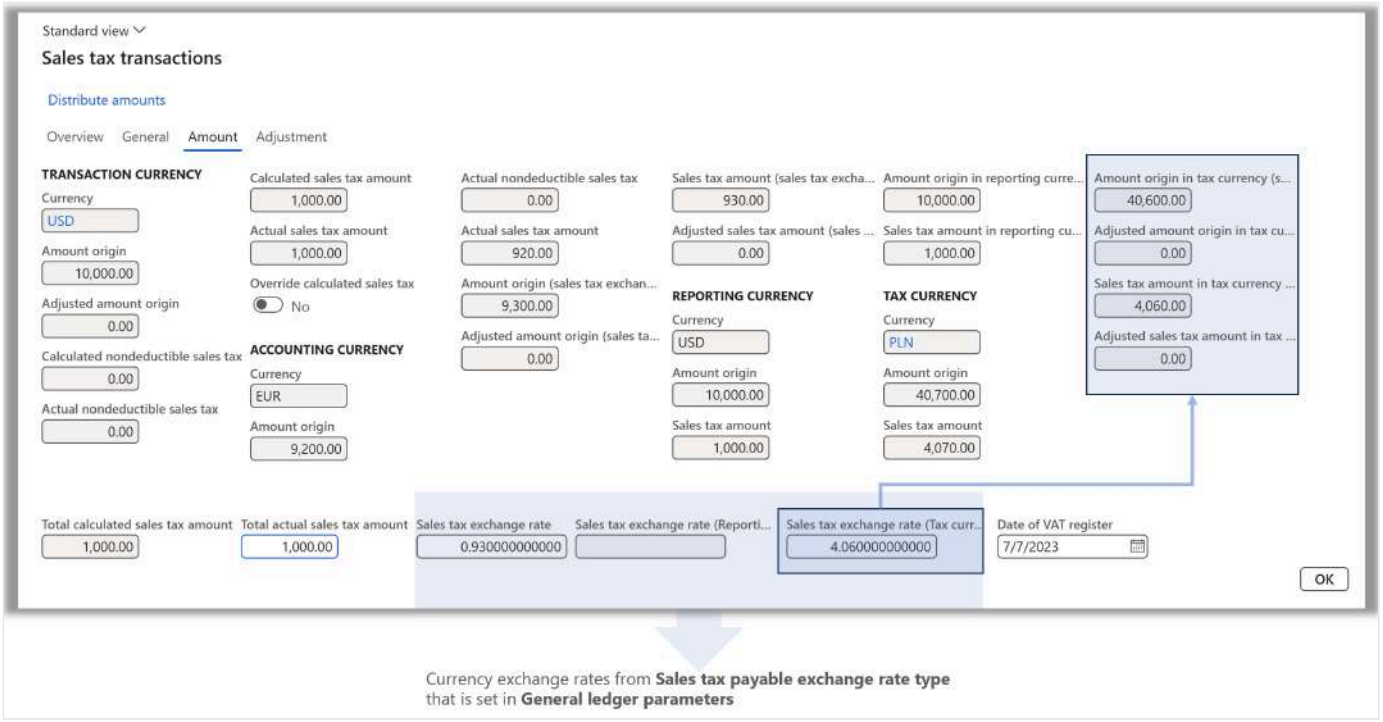
Currency code	Exchange rate
USD : EUR	1 : 0.92
EUR : PLN	1 : 4.44
USD : PLN	1 : 4.07

Set up the following exchange rates for tax calculation purposes on the **Sales tax** tab of the **General ledger parameters** page.

Expand table

Currency code	Exchange rate
USD : EUR	1 : 0.93
EUR : PLN	1 : 4.46
USD : PLN	1 : 4.06

On the **Sales tax transactions** page, on the **Amount** tab, review the calculated sales tax. Then, in the free text invoice, select **Sales tax**.



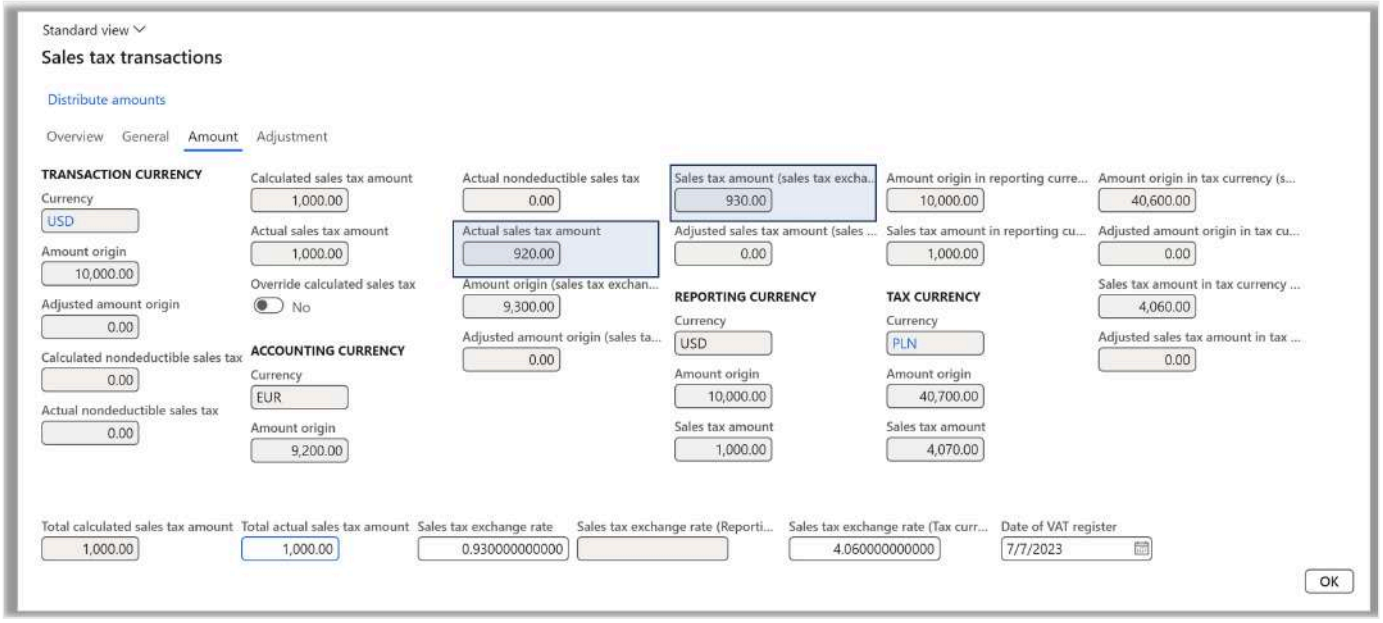
On the **Temporary sales tax transactions** page, you can review the exchange rates, amount origin, and tax amounts in the accounting currency, reporting currency, and tax currency. These amounts are shown for both exchange rate types: the one that's set on the **Ledger** page and the one that's set on the **General ledger parameters** page.

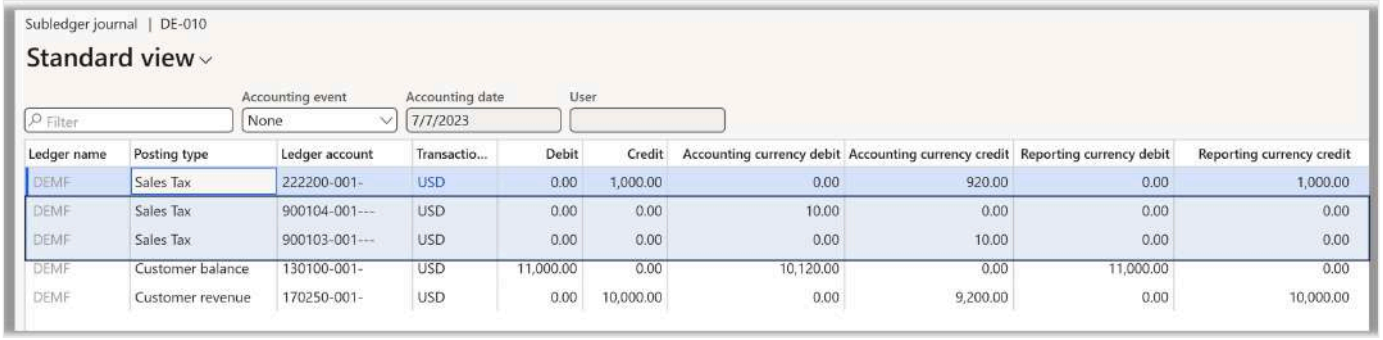
On the **Adjustment** tab, you can adjust tax amounts in the accounting currency and the tax currency.

> **Note**
>
> In this example, the reporting currency and transaction currency are the same (USD). Therefore, the exchange rate for the reporting currency isn't editable.

On the **Subledger journal** page, you can review amounts and accounts for the calculated sales tax difference. In this example, the exchange rate of the EUR-USD currency pair differs for the

accounting currency exchange rate type on the **Ledger** page and the sales tax payable exchange rate type on the **General ledger parameters** page.





After you post the document, review the posted sales tax transactions and voucher. The tax difference is posted with a *Sales tax payable difference* tax direction.

> **Note**
>
> For better user experience, add the following fields to the **Posted sales tax** page by using the *Insert columns* function and saving the *View*:
>
> - Base amount in accounting currency
> - Base amount in reporting currency
> - Base amount in tax currency
> - Sales tax amount in accounting currency
> - Sales tax amount in reporting currency
> - Sales tax amount in tax currency

Ledger account	Account name	Currency	Amount in transaction currency	Amount	Amount in reporting currency	Posting type
130100-001-	Accounts Receivable - Domestic	USD	11,000.00	10,120.00	11,000.00	Customer balance
170250-001-	Development	USD	-10,000.00	9,200.00	-10,000.00	Customer revenue
222200-001-	VAT Tax Payable	USD	-1,000.00	920.00	-1,000.00	Sales Tax
900103-001---	Sales tax payable difference	USD	0.00	10.00	0.00	Sales Tax
900104-001---	Sales tax payable difference offset	USD	0.00	10.00	0.00	Sales Tax

For this example, all the ledger accounts are different to demonstrate the postings.

To reflect the tax difference on the sales tax payable and sales tax receivable tax accounts, select the appropriate accounts in the **Sales tax payable difference** and **Sales tax receivable difference** fields.

> **Note**
>
> - If the tax currency differs from the accounting and reporting currencies, the sales tax transaction records the tax difference amounts in the tax currency.
> - In the voucher, the system records the tax difference in the accounting and reporting currencies on ledger accounts set in the sales tax code posting.

Last updated on 03/21/2026

Tax integration overview

Tax integration is a framework that replaces the legacy tax calculation engine and, as required, calls the Tax Calculation service. This article introduces the basic flow of tax integration.

Entrance to tax integration

The `TaxIntegrationFacade` class contains most of the code and functions for tax integration. However, some additional code is required to enter the tax integration flow before `TaxIntegrationFacade`. This code is the entrance to tax integration. The entrance code constructs a document object, `TaxIntegrationDocumentObject`, which it passes to `TaxIntegrationFacade`. The entrance code also prepares some basic information that's used in the following integration actions.

TaxIntegrationFacade

The `TaxIntegrationFacade` class implements and controls the real flow of tax integration by calling several classes, such as `TaxIntegration***ActivityOnDocument`. The class passes a `TaxIntegrationDocumentObject` from activity to activity. These activities perform their actions on the object in turn.

The following list shows all document-level activities that have the suffix "OnDocument." There are also line-level activities that work on `TaxIntegrationLineObject`. Those activities have the suffix "OnLine."

- **TaxIntegrationSettingRetrievalActivityOnDocument** – This activity retrieves settings that affect tax calculation and saves them to the `TaxIntegrationDocumentObject` object.
- **TaxIntegrationDataRetrievalActivityOnDocument** – This activity prepares the metadata that's required by tax calculation and then copies it to the `TaxIntegrationDocumentObject` object.
- **TaxIntegrationCalculationActivityOnDocument** – This activity transforms the `TaxIntegrationDocumentObject` object to a request, sends the request to the Tax Calculation service, and then parses the response.
- **TaxIntegrationErrorHandlingActivityOnDocument** – This activity handles the error message that's returned from the Tax Calculation service.
- **TaxIntegrationTaxIdActivityOnDocument** – This activity processes logic that's related to the **Multiple VAT ID** feature.

- **TaxIntegrationListCodeActivityOnDocument** – This activity processes logic that's related to the **List Code** feature.
- **TaxIntegrationCurrencyExchangeActivityOnDocument** – This activity processes logic that's related to the **Currency Exchange rate/Rounding/Penny difference adjustment** feature.
- **TaxIntegrationDataPersistenceActivityOnDocument** – This activity persists the tax calculation result and other information that must be saved to the database, including the value-added tax (VAT) ID and list code.

Last updated on 03/21/2026

Facade, sequence, and activities in tax integration

The **facade** controls the tax integration flow through a **sequence** that maintains a list of **activities**. This article introduces these three classes in tax integration.

Facade

X++

```
public final static class TaxIntegrationFacade
```

The facade (`TaxIntegrationFacade`) is the outermost interface of tax integration. It's a bridge to the Tax Calculation service.

The following example shows the most important method.

X++

```
[Replaceable]
public static void calculate(TaxIntegrationDocumentObject _document)
```

This method accepts a document (`TaxIntegrationDocumentObject`) as input, and it returns `void`.

The method modifies the document directly. The document also serves as the output.

> ⓘ **Note**
>
> The method is replaceable. In other words, it can be completely replaced (customized).

X++

```
TaxIntegrationSequence sequence = TaxIntegrationSequence::construct(LoggerName)

.appendActivityOnDocument(TaxIntegrationSettingRetrievalActivityOnDocument::construct(
))

.appendActivityOnDocument(TaxIntegrationDataRetrievalActivityOnDocument::construct())

.appendActivityOnDocument(TaxIntegrationCalculationActivityOnDocument::construct())
    // ...
```

```
.appendActivityOnDocument(TaxIntegrationDataPersistenceActivityOnDocument::construct()
);
sequence.act(_document);
if (!_document.isOK())
{
    throw _document.getException();
}
```

Inside the method, a sequence (`TaxIntegrationSequence`) is constructed and invoked. The method appends all activities to the sequence in order.

As required, the method explicitly throws an exception at the end.

Sequence

The sequence (`TaxIntegrationSquence`) is a special activity that maintains a sequence of activities. When you invoke its `act` method, it runs all activities in the order that you append them.

X++

```
public final class TaxIntegrationSequence
    extends TaxIntegrationAbstractActivity
{
    public final void act(TaxIntegrationDocumentObject _document);
    private void actInternal(TaxIntegrationDocumentObject _document);
}
```

To use a sequence (`TaxIntegrationSequence`), follow these steps:

1. Use the `construct` method to create the sequence.
2. Use the `appendActivitySequence`, `appendActivityOnDocument`, and `appendActivityOnLine` methods to build up the sequence.
3. Use the `act` method to play the sequence.

Internally, the sequence maintains an activity list (`List`). It then enumerates that list when you invoke `act`.

- The sequence calls the `shouldSkip` method of an activity to determine whether to skip the activity for the current document.
- For `TaxIntegrationSequence` and `TaxIntegrationAbstractActivityOnDocument`, if the activity isn't skipped, the sequence invokes the `act` method directly.

- For `TaxIntegrationAbstractActivityOnLine`, the sequence enumerates all `TaxIntegrationLineObject` objects in the document and invokes its `act` method. Charges are also implemented by the `TaxIntegrationLine` object. The sequence first invokes the `act` method on the charges of the document. Then, it invokes the `act` method on the lines of the document, together with the charges on each line. For example, there are two charges and two lines on a document. At the same time, there are two charges on each line. In this case, the complete sequence is as follows:

    1. The first charge of the document
    2. The second charge of the document
    3. The first line of the document
    4. The first charge of the first line
    5. The second charge of the first line
    6. The second line of the document
    7. The first charge of the second line
    8. The second charge of the second line

You can use `TaxIntegrationSequence` directly. It's a concrete class that has a public `act` method.

The following example shows the `actInternal` method for quick reference.

X++

```
private void actInternal(TaxIntegrationDocumentObject _document)
{
    var enumerator = list.getEnumerator();
    while (enumerator.moveNext())
    {
        anytype current = enumerator.current();
        if (current is TaxIntegrationSequence)
        {
            TaxIntegrationSequence sequence = current;
            sequence.act(_document);
            continue;
        }
        if (current is TaxIntegrationAbstractActivityOnDocument)
        {
            TaxIntegrationAbstractActivityOnDocument activity = current;
            if (!activity.shouldSkip(_document))
            {
                activity.act(_document);
            }
            continue;
        }
        if (current is TaxIntegrationAbstractActivityOnLine)
        {
            TaxIntegrationAbstractActivityOnLine activity = current;
```

```
            if (!activity.shouldSkip(_document))
            {
                SetEnumerator chargeEnumerator =
_document.getChargeSet().getEnumerator();
                while (chargeEnumerator.moveNext())
                {
                    TaxIntegrationLineObject charge = chargeEnumerator.current();
                    activity.act(charge);
                }
                SetEnumerator lineEnumerator = _document.getLineSet().getEnumerator();
                while (lineEnumerator.moveNext())
                {
                    TaxIntegrationLineObject line = lineEnumerator.current();
                    activity.act(line);
                    chargeEnumerator = line.getChargeSet().getEnumerator();
                    while (chargeEnumerator.moveNext())
                    {
                        TaxIntegrationLineObject charge = chargeEnumerator.current();
                        activity.act(charge);
                    }
                }
            }
            continue;
        }
    }
}
```

Activity

An activity does the real work in an integration flow. It acts as a segment in the integration flow. The name of an activity describes its purpose.

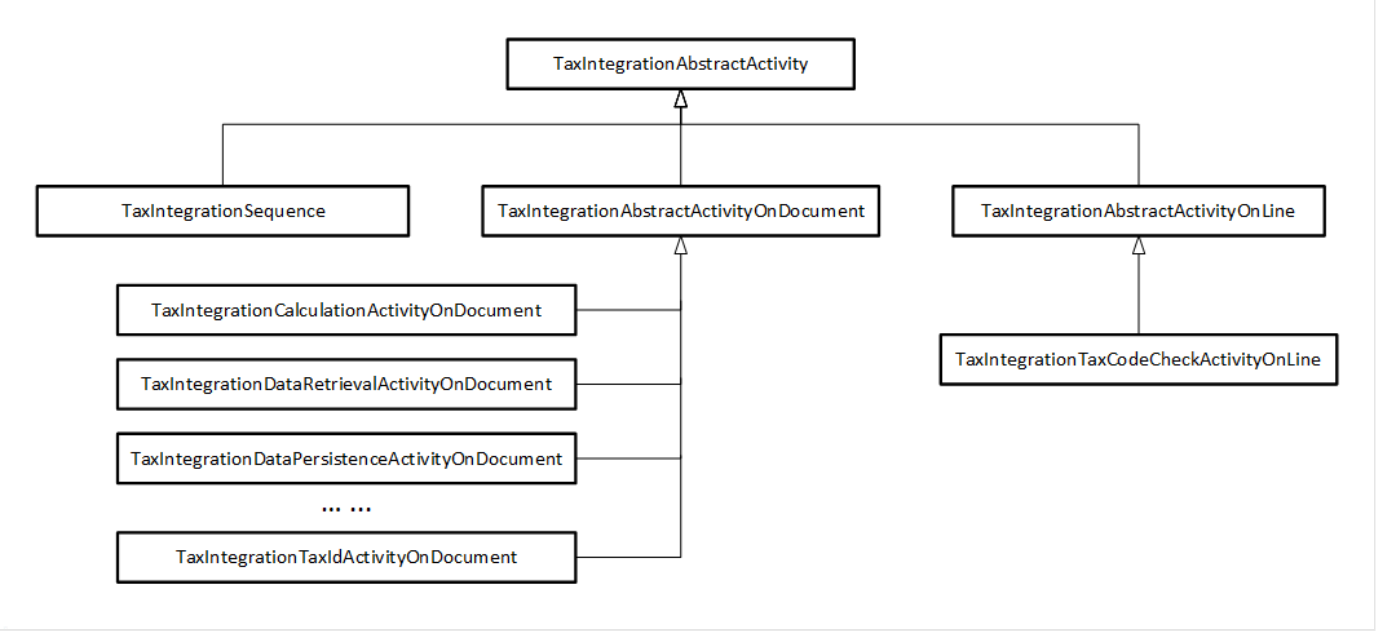
All activities inherit from the same class, `TaxIntegrationAbstractActivity`. There are three types of activities:

- `TaxIntegrationSequence`
- `TaxIntegrationAbstractActivityOnDocument`
- `TaxIntegrationAbstractActivityOnLine`

Each type of activity contains an `act` method and an `actInternal` method, but they accept different arguments. `TaxIntegrationAbstractActivityOnDocument` and `TaxIntegrationAbstractActivityOnLine` are abstract classes that are intended to be extended. They have an internal `act` method and an abstract `actInternal` method. Therefore, a new activity should extend one of these two abstract classes. The activity should implement its own `actInternal` method to hold the real logic of the activity. If the activity isn't applicable to all transactions and documents, it should implement its own `shouldSkip` method.

`TaxIntegrationSequence` is a concrete class that's intended to be used directly. It has a public `act` method.

All three types of activities log events during the `act` method as required after `TaxIntegrationLogLevelUtility` is checked. The logging context is inherited from `TaxIntegrationAbstractActivity`. The following illustration shows the hierarchy of activities.



For more information about how to add a new activity, see Add a new activity in tax integration.

Sequence

`TaxIntegrationSequence` is a special activity.

Document activity

The document activity acts on `TaxIntegrationDocumentObject` and extends `TaxIntegrationAbstractActivityOnDocument`.

X++

```
public abstract class TaxIntegrationAbstractActivityOnDocument
    extends TaxIntegrationAbstractActivity
{
    internal final void act(TaxIntegrationDocumentObject _document);
    protected abstract void actInternal(TaxIntegrationDocumentObject _document);
}
```

A new document activity should implement the `actInternal` method to hold the logic of this activity. The `actInternal` method accepts a `TaxIntegrationDocumentObject` object as input.

Therefore, unlike a line activity, it can manipulate data at the document level. However, a document activity can also act on `TaxIntegrationLineObject` by enumerating the `TaxIntegrationDocumentObject` object. Most document activities handle data at both the document level and the line level. If an activity isn't applicable to all transactions and documents, it should implement its own `shouldSkip` method.

Line activity

The line activity acts on `TaxIntegrationLineObject` and extends `TaxIntegrationAbstractActivityOnLine`.

X++

```
public abstract class TaxIntegrationAbstractActivityOnLine
    extends TaxIntegrationAbstractActivity
{
    internal final void act(TaxIntegrationLineObject _line);
    protected abstract void actInternal(TaxIntegrationLineObject _line);
}
```

A new line activity should implement the `actInternal` method to hold the logic for this activity. The `actInternal` method accepts a `TaxIntegrationLineObject` object as input and can manipulate data only on the given line object. However, because charges are also described as a `TaxIntegrationLineObject` object, the line activity also processes charges. If an activity isn't applicable to **all lines** in the document, it should implement its own `shouldSkip` method. The `shouldSkip` method accepts a `TaxIntegrationDocumentObject` object as input and indicates whether to skip the current activity for **all lines** (including charges) in the document.

Current activities

Here's the list of current activities, in the order of `TaxIntegrationFacade::calculate`:

1. TaxIntegrationSettingRetrievalActivityOnDocument
2. TaxIntegrationDataRetrievalActivityOnDocument
3. TaxIntegrationCalculationActivityOnDocument
4. TaxIntegrationErrorHandlingActivityOnDocument
5. TaxIntegrationTaxCodeCheckActivityOnLine
6. TaxIntegrationTaxIdActivityOnDocument
7. TaxIntegrationListCodeActivityOnDocument
8. TaxIntegrationCurrencyExchangeActivityOnDocument
9. TaxIntegrationDataPersistenceActivityOnDocument

Last updated on 03/21/2026

Entrance code to tax integration

This article explains how to integrate a new transaction into tax integration. Before you start to use tax integration, you need some "entrance code" so that you can bypass the legacy tax engine and use the new logic instead. This article describes how to introduce this entrance code.

Find a code point

The first step is to find a code point for the entrance code. This code point should meet the following conditions:

- The code point should be before tax is calculated by the legacy tax engine, so that tax calculation can be bypassed. Additionally, the code point should bypass only tax calculation. No other processes should be bypassed.
- The code should be in a context that contains the basic transaction information that you can use to initialize a `TaxIntegrationDocumentObject` object. You can initialize the object in two ways:
  - From the transaction header table record itself
  - By using the table ID and record ID to distinguish the header table record of the transaction

Usually, if the transaction that you must integrate is already supported by the legacy tax engine, it has a specific derived class of the `TaxCalculation` class, such as `TaxSales` for sales orders or `TaxPurch` for purchase orders. This class is an ideal code point. Currently, all transactions such as purchase orders and sales orders use this approach. You can refer to `TaxPurch.calculateTax()` and `TaxSales.calc()`.

If the transaction isn't supported by the legacy tax engine, you should create the subclass of `Tax`. This article doesn't cover this scenario.

Implement the entrance of tax integration

This example uses the entry point code of the sales order, `TaxSales.xpp`.

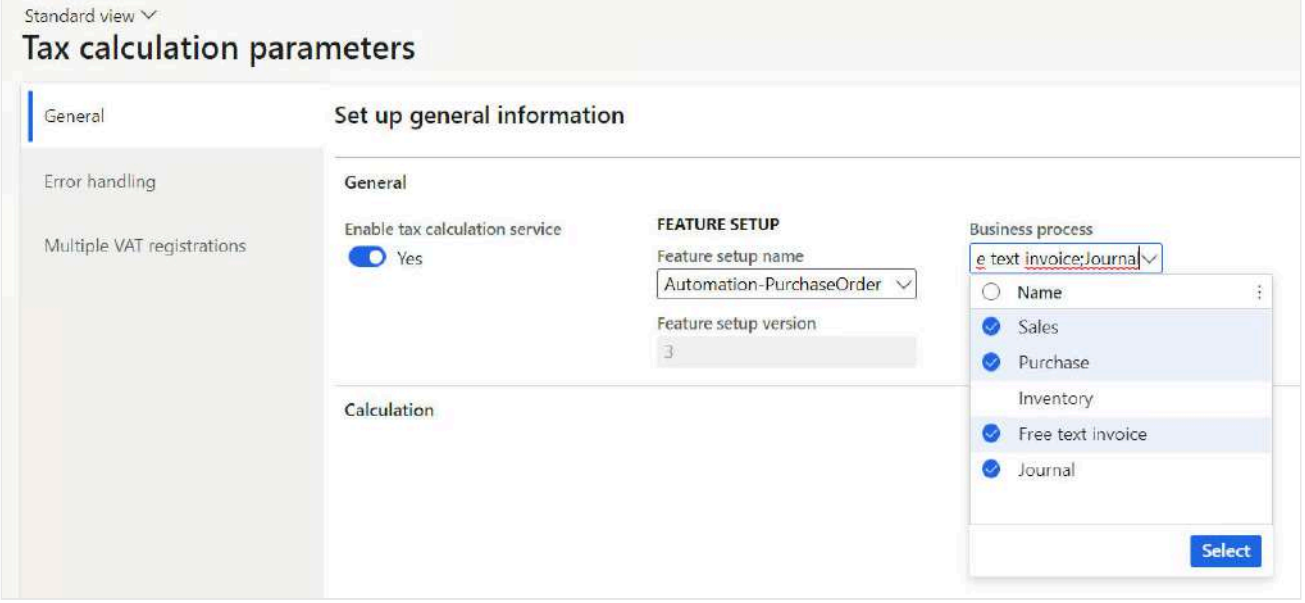
X++

```
TaxAmount calc()
{
```

```
    TaxAmount taxAmount;
    if
(Tax::isTaxIntegrationEnabledForBusinessProcess(TaxIntegrationBusinessProcess::Sales))
    {
        taxAmount = this.calcUsingTaxIntegration();
        return taxAmount;
    }
    boolean                 moreLines;
    TaxBaseCur              baseAmount;
    ...
}
```

- Use `Tax::isTaxIntegrationEnabledForBusinessProcess()` to check whether to use tax integration for a transaction. This line acts as a switch that you can use to enable tax integration for a transaction from the front end.

  You can control this switch by setting the **Business process** field in the **Feature setup** section of the **General** tab on the **Tax calculation parameters** page (**Tax** > **Setup** > **Tax configuration** > **Tax calculation**).



- `TaxIntegrationBusinessProcess` is an enumeration (enum) type that currently has five values: *Sales*, *Purchase*, *Inventory* (used for transfer orders), *Free text invoice*, and *Journal*. If your new transaction isn't covered by these five values, you can add a new business process enum value for it.

- All logic is wrapped in `this.calcUsingTaxIntegration()`.

The calcUsingTaxIntegration method

The `calcUsingTaxIntegration` method prepares a `TaxIntegrationDocumentObject` object and calls `TaxIntegrationFacade::calculate(document)`. The following example uses the purchase order.

X++

```
private TaxAmount calcUsingTaxIntegration()
{
    // code on lock
    ...
    TaxAmountCur taxAmount;
    try
    {
        TaxIntegrationDocumentObject document;
        document =
TaxIntegrationDocumentObject::constructWithRecord(purchCalcTax.getSource());
        this.setFieldsForLegacyTax();
        this.setFieldsForTaxIntegrationDocumentObject(document);
        TaxIntegrationFacade::calculate(document);
        amountInclTaxMap = document.getAmountIncludingTax();
        amountExclTaxMap = document.getAmountExcludingTax();
        ...
        taxAmount = this.finalizeCalculationForTaxIntegration(
            TaxParameters::isBankExchRateEnabled_W() &&
purchCalcTax.vendInvoiceInfoTable(),
            doIsolateTransactionScopeTrue);
    }
    // code on error handling
    ...
    return taxAmount;
}
```

This code completes the following steps.

1. Construct a `TaxIntegrationDocumentObject` object in one of two ways:
    - `TaxIntegrationDocumentObject::constructWithRecord(Common _record)` receives a transaction header table record as its parameter. This method is preferred because it avoids conflict when fields are updated in the transaction table in the data persistence activity.
    - `TaxIntegrationDocumentObject::construct(RefTableId _tableId, RefRecId _recId)` receives a table ID and a record ID as its parameters to distinguish the header table record of the transaction. However, because the header table record is reselected from the database, a conflict might occur.
2. Use the setter methods to set basic information for the `TaxIntegrationDocumentObject` object. In the preceding example, the code completes this step by calling the `this.setFieldsForTaxIntegrationDocumentObject(document)` method.
3. Call `TaxIntegrationFacade::calculate(document)` to trigger the real tax integration flow.
4. Get the `amountIncludingTax` map for compatibility with the existing tax adjustment function.

5. Call the `this.finalizeCalculationForTaxIntegration()` method to handle `TaxUncommitted`. This step is required only if the transaction is using `TaxUncommitted`.

6. Return the tax amount.

Set fields for the document object

The `this.setFieldsForTaxIntegrationDocumentObject()` method initializes basic data fields for a taxable document to use in later activities.

Retrieve most of the data fields in the data retrieval activity. However, set some of them here for the following reasons:

- The field is fundamental and is used before the data retrieval activity. Examples include `headingTableId` and `headingRecId`.
- The field can be accessed only in this context. For example, the tax object is set by `_document.setLegacyTax(this);`
- It's convenient to set some fields here because you can use local methods and methods in the subclass of `TradeCalcTax` to avoid duplicate code with the data retrieval class. Examples include the document date, invoice date, and delivery date.

Set the data fields according to the new transaction before you call the facade. Usually, you can retrieve data by using an object of `TradeCalcTax` in the context of the entrance point. For example, `SalesFormLetter` is an object of `SalesCalcTax_Sales` for sales orders, `purchCalcTax` for purchase orders, and `custInvoiceCalcTax` for free text invoices.

The following example is from `TaxPurch.xpp`.

X++

```
protected void setFieldsForTaxIntegrationDocumentObject(TaxIntegrationDocumentObject
_document)
{
    _document.setTransactionDate(this.taxDate);
    // This deliveryDateMarkup() is for header.
    _document.setDeliveryDate(salesFormLetter.deliveryDateMarkup());
    _document.setDocumentDate(salesFormLetter.documentDate());
    _document.setInvoiceDate(salesFormLetter.invoiceDate());
    _document.setTransactionCurrencyCode(salesFormLetter.currencyCode());
    _document.setCompany(this.getCompany());
    _document.setHeadingTableId(this.headingTableId());
    _document.setHeadingRecId(this.headingRecId());
    // this sign is used as sign in TmpTaxWorkTrans, 1 for purchase and -1 for sales.
    _document.setSign(-1);
    _document.setSource(TaxModuleType::Sales);
```

```
    _document.setBusinessProcess(TaxIntegrationBusinessProcess::Sales);
    _document.setPrepaid(this.isPrePayment());
    _document.setEUROTriangulation(this.getTriangulation());
    _document.setLegacyTax(this);
    _document.setShouldSkipDocumentCharge(skipTableMarkup);
    _document.setShouldSkipLineCharge(skipLineMarkup);
}
```

Last updated on 03/21/2026

Implement tax integration activities

This article provides information about how to implement the required activities for a new transaction.

Activities are the classes that complete the job in tax integration. The process passes a document object, `TaxIntegrationDocumentObject`, from activity to activity. Each activity performs an action on the metadata in `TaxIntegrationDocumentObject`. All tax integration activities extend the `TaxIntegrationAbstractActivity` class. You can append them to a sequence and run them by using `TaxIntegrationFacade::calculate()` in the order you appended them.

To integrate a new transaction, you don't need to implement or modify all the activities. In most cases, two specific activities are enough: `TaxIntegrationDataRetrievalActivityOnDocument` and `TaxIntegrationDataPersistenceActivityOnDocument`. If you enable more features for the new transaction, see Uptake features of tax integration.

Data retrieval activity

The data retrieval activity (`TaxIntegrationDataPersistenceActivityOnDocument`) copies all the metadata that's required for tax calculation from a database to a `TaxIntegrationDocumentObject` object.

This activity constructs and then calls a transaction-specific subclass of the `TaxIntegrationAbstractDataRetrievalTemplate` class, according to the `HeadingTableId` value of the current processed transaction in the factory pattern. Next, the subclass queries all the transaction lines, charges, and other useful information from the database and saves it to the document object.

To begin the data retrieval activity, create and implement a new subclass of `TaxIntegrationAbstractDataRetrievalTemplate`. Give the subclass a name, such as `TaxIntegration***DataRetrieval`. The following list shows some of the data retrieval classes for the integrated transactions:

- AxClass/TaxIntegrationPurchTableDataRetrieval
- AxClass/TaxIntegrationPurchParmTableDataRetrieval
- AxClass/TaxIntegrationPurchREQTableDataRetrieval
- AxClass/TaxIntegrationPurchRFQTableDataRetrieval
- AxClass/TaxIntegrationVendInvoiceInfoTableDataRetrieval

- AxClass/TaxIntegrationSalesTableDataRetrieval
- AxClass/TaxIntegrationSalesParmDataRetrieval

The names consist of the prefix *TaxIntegration*, the header table name of the transaction, and then *DataRetrieval*. If you need to integrate the free text invoice transaction, implement a `TaxIntegrationCustInvoiceTableRetrieval` class.

Implementing the data retrieval class

This section introduces the implementation of the data retrieval class. It uses `TaxIntegrationPurchTableDataRetrieval` as an example to provide an outline of this class.

X++

```
[TaxIntegrationDataRetrieval(tableStr(PurchTable))]     // Attribute for
TaxIntegrationDataRetrievalActivityOnDocument to call this class
public class TaxIntegrationPurchTableDataRetrieval extends
TaxIntegrationAbstractDataRetrievalTemplate
{
    protected PurchTable purchTable;        // Header table buffer
    protected PurchLine purchLine;          // Line table buffer
    // Query builder methods, for the SQL query for framework to execute
    [Replaceable]
    protected SysDaQueryObject getDocumentQueryObject()
    [Replaceable]
    protected SysDaQueryObject getLineQueryObject()
    [Replaceable]
    protected SysDaQueryObject getDocumentChargeQueryObject()
    [Replaceable]
    protected SysDaQueryObject getLineChargeQueryObject()
    // Construct a TaxIntegrationLineObject
    protected TaxIntegrationLineObject constructLine()
    // Data retrieval methods, copy the metadata from table buffer to Document object.
    protected void copyToDocument()
    protected void copyToDocumentFromHeaderTable()
    protected void copyToLine(TaxIntegrationLineObject _line)
    protected void copyToLineFromLineTable(TaxIntegrationLineObject _line)
    ...
}
```

Declaration

The following example is from `TaxIntegrationPurchTableDataRetrieval.xpp`.

X++

```
/// <summary>
/// The <c>TaxIntegrationPurchTableDataRetrieval</c> class retrieves data from
```

```
<c>PurchTable</c>, etc.
/// </summary>
[TaxIntegrationDataRetrieval(tableStr(PurchTable))]
public class TaxIntegrationPurchTableDataRetrieval
    extends TaxIntegrationAbstractDataRetrievalTemplate
{
    // ...
```

Declare the class with the `[TaxIntegrationDataRetrieval(tableStr(PurchTable))]` attribute. `TaxIntegrationDataRetrievalActivityOnDocument` uses this attribute to construct and call the class by using the factory pattern. Use the transaction header table as the table name. The table ID must match the `HeadingTableId` value that you initialize in the `TaxIntegrationDocumentObject` object.

Refer to the `actStatic` method of `TaxIntegrationDataRetrievalActivityOnDocument.xpp`, which is shown here.

X++

```
protected static void actStatic(TaxIntegrationDocumentObject _document)
{
    // ...
    TaxIntegrationAbstractDataRetrievalTemplate dataRetrievalInstance =
SysExtensionAppClassFactory::getClassFromSysAttributeWithInstantiationStrategy(
        classStr(TaxIntegrationAbstractDataRetrievalTemplate),
        new
TaxIntegrationDataRetrievalAttribute(tableId2Name(_document.getLocalTableId())),
        new SysExtensionGenericInstantiation(_document));
    /// ...
    dataRetrievalInstance.execute();
}
```

Declare at least two tables for each transaction. For this transaction, declare the header table and line table. (For example, for purchase orders, use the `PurchTable` and `PurchLine` tables.) If you use fields from other tables as input for the calculation, declare them here.

X++

```
[TaxIntegrationDataRetrieval(tableStr(PurchTable))]
public class TaxIntegrationPurchTableDataRetrieval
    extends TaxIntegrationAbstractDataRetrievalTemplate
{
    protected PurchTable purchTable;
    protected PurchLine purchLine;
    protected TransportationDocument transportationDocument;    // Other tables
through join relationship
```

```
    protected VendTable vendTable;
    private LedgerDimensionDefaultAccount ledgerAccount;
```

Create a query object

Tax integration queries use the `SysDaQuery` method. After you implement several `get***QueryObject()` methods, the tax integration framework calls those methods, retrieves the data from the database, and assigns it to the table buffer that you declared earlier.

Format four query objects so that you can perform the following queries:

- Query the header level tables by using the `HeadingTableId` and `HeadingRecId` values in `TaxIntegrationDocumentObject`. The header-level tables refer to tables that are related to the transaction header table (for example, `PurchTable` for purchase orders). If you construct the document object with a record by using `TaxIntegrationDocumentObject::constructWithRecord(Common _record)`, assign the header table to the buffer by using `this.document.getLocalRecord()` directly. You don't need to query the header table again.
- Query the line table by using the relationship between the header table and the line table, after you provide the header table buffer.
- Query the header charge (`MarkupTrans`).
- Query the line charge (`MarkupTrans`).

The following example is from `TaxIntegrationPurchTableDataRetrieval.xpp`.

X++

```
/// <summary>
/// Gets the query for the document.
/// </summary>
/// <returns>The query for the document.</returns>
[Replaceable]
protected SysDaQueryObject getDocumentQueryObject()
{
    this.purchTable = this.document.getLocalRecord();
    SysDaQueryObjectBuilder queryObjectBuilder =
SysDaQueryObjectBuilder::from(this.vendTable)
        .where(this.vendTable, fieldStr(VendTable, AccountNum))
            .isEqualTo(this.purchTable, fieldStr(PurchTable, OrderAccount))
        .outerJoin(this.transportationDocument) // primary key
            .where(this.transportationDocument, fieldStr(TransportationDocument,
RecId))
                .isEqualTo(this.purchTable, fieldStr(PurchTable,
TransportationDocument));
        .outerJoin(this.vendTableInvoice) // alternate key
            .where(this.vendTableInvoice, fieldStr(VendTable, AccountNum))
```

```
                .isEqualTo(this.purchTable, fieldStr(PurchTable, InvoiceAccount));
    return queryObjectBuilder.toSysDaQueryObject();
}
/// <summary>
/// Gets the query for the lines of the document.
/// </summary>
/// <returns>The query for the lines of the document</returns>
[Replaceable]
protected SysDaQueryObject getLineQueryObject()
{
    return SysDaQueryObjectBuilder::from(this.purchLine)
        .where(this.purchLine, fieldStr(PurchLine, PurchId))
            .isEqualToLiteral(this.purchTable.PurchId)
        .toSysDaQueryObject();
}
/// <summary>
/// Gets the query for the charges of the document.
/// </summary>
/// <returns>The query for the charges of the document</returns>
[Replaceable]
protected SysDaQueryObject getDocumentChargeQueryObject()
{
    return SysDaQueryObjectBuilder::from(this.markupTransOfDocument)
        .where(this.markupTransOfDocument, fieldStr(MarkupTrans, TransTableId))
            .isEqualToLiteral(this.purchTable.TableId)
        .where(this.markupTransOfDocument, fieldStr(MarkupTrans, TransRecId))
            .isEqualToLiteral(this.purchTable.RecId)
        .toSysDaQueryObject();
}
/// <summary>
/// Gets the query for the charges of each line.
/// </summary>
/// <returns>The query for the charges of each line.</returns>
[Replaceable]
protected SysDaQueryObject getLineChargeQueryObject()
{
    return SysDaQueryObjectBuilder::from(this.markupTransOfLine)
        .where(this.markupTransOfLine, fieldStr(MarkupTrans, TransTableId))
            .isEqualToLiteral(this.purchLine.TableId)
        .where(this.markupTransOfLine, fieldStr(MarkupTrans, TransRecId))
            .isEqualToLiteral(this.purchLine.RecId)
        .toSysDaQueryObject();
}
```

ⓘ Note

If you require fields from other tables, join them in the query object. In the preceding example, this approach is used for `transportationDocument` in the `getDocumentQueryObject()` method.

Construct the line object

```
X++
```

```
/// <summary>
/// Constructs a line of the document.
/// </summary>
/// <returns>The constructed line of the document.</returns>
protected TaxIntegrationLineObject constructLine()
{
    return TaxIntegrationLineObject::constructOnDocument(
        this.purchLine.TableId,
        this.purchLine.RecId,
        this.document);
}
```

Construct a `TaxIntegrationLineObject` object by using the line's table ID and record ID. The framework calls this method. Replace the table buffer for the new transaction.

Copy methods

Implement several methods that begin with *copy* to copy metadata from the table buffer to the document object. For more information, see Retrieve data from the database. Because you create new classes, you can disregard the information about extensions.

When you implement the methods, determine which fields are required for the Tax Calculation service, and copy only the required fields. For information about how to add new fields, see Add the data variable in the object class. For information about how to add them to the request, see Add data to the request.

Data persistence activity

When you run the data persistence activity (`TaxIntegrationDataPersistenceActivityOnDocument`), it calls the Tax Calculation service. It processes the response and saves it to `TaxIntegrationDocumentObject`. The data persistence activity updates the necessary metadata in the database, including tax calculation results, tax group, tax item group, VAT ID, and list code.

Like the data retrieval class, `TaxIntegrationDataPersistenceActivityOnDocument` builds and calls the data persistence class based on the `HeadingTableId` value of the current transaction. Your job is to implement a `TaxIntegration***DataPersistence` class. For example, you implement the `TaxIntegrationCustInvoiceTableDataPersistence` class for free text invoices.

However, unlike the data retrieval activity, `TaxIntegrationDataPersistenceActivityOnDocument` handles most of the persistence work, including the tax calculation result. This part is managed

by standard code, so no extra work is needed when you integrate a new transaction.

The new persistence class saves transaction-specific fields, such as the tax group, tax item group, VAT ID, number sequence group, and list code, to the database.

The following example shows the basic structure of `TaxIntegrationPurchTableDataPersistence`.

X++

```
/// <summary>
/// The <c>TaxIntegrationPurchTableDataPersistence</c> class persists data to
<c>PurchTable</c>, etc.
/// </summary>
[TaxIntegrationDataPersistence(tableStr(PurchTable))] // Attribute for
TaxIntegrationDataPersistenceActivityOnDocument to call
public class TaxIntegrationPurchTableDataPersistence
    extends TaxIntegrationAbstractDataPersistenceTemplate
{
    // Table buffer to be updated
    protected PurchTable purchTable;
    protected PurchLine purchLine;
    // Query builder methods
    [Replaceable]
    protected SysDaQueryObject getDocumentQueryObject()
    [Replaceable]
    protected SysDaQueryObject getLineQueryObject()
    [Replaceable]
    protected SysDaQueryObject getDocumentChargeQueryObject()
    [Replaceable]
    protected SysDaQueryObject getLineChargeQueryObject()
    // Get line object
    protected TaxIntegrationLineObject getLine()
    // Save to database
    protected boolean saveDocument()
    protected boolean saveLine(TaxIntegrationLineObject _line)
    ...
}
```

As the preceding example shows, the structure of the data persistence class resembles the structure of the data retrieval class. The only difference is that the data retrieval class copies data from the *database* to the *document object*, whereas the data persistence class copies data from the *document object* to the *database*. Here's a brief introduction to the implementation. Not all code examples are included.

- `[TaxIntegrationDataPersistence(tableStr(PurchTable))]` is an attribute for `TaxIntegrationDataPersistenceActivityOnDocument` to construct in the factory pattern.
- `QueryObject` is almost the same as for data retrieval, except for two points:

    - `.forUpdate()` is called for the persistence class to update data to the database.
    - The `Less` table is used because it must be updated.

- The `getLine()` method resembles the `constructLine()` method in the data retrieval class. However, it's enough to replace the table name with the new transaction table.

- Methods that begin with *save* are the main job in this class. For more information, see the next section.

Save methods

The following example is from `TaxIntegrationPurchTableDataPersistence`.

X++

```
/// <summary>
/// Saves the document.
/// </summary>
/// <returns>Always true.</returns>
protected boolean saveDocument()
{
    TaxIntegrationTaxIdUtility::saveTaxIDFromDocumentToTable(document, purchTable);
    TaxIntegrationListCodeUtility::saveListCodeFromDocumentToTable(document,
purchTable);
    this.saveNumberSequenceGroupToTable();
    return true;
}
/// <summary>
/// Saves the current line of the document.
/// </summary>
/// <param name = "_line">The current line of the document.</param>
/// <returns>true if the operation is successful; otherwise, false.</returns>
protected boolean saveLine(TaxIntegrationLineObject _line)
{
    if (!_line.isTaxable())
    {
        return true;
    }
    TaxGroup taxGroup = _line.getTaxGroup();
    TaxItemGroup taxItemGroup = _line.getTaxItemGroup();
    this.checkGroup(taxGroup, taxItemGroup);
    purchLine.TaxGroup = taxGroup;
    purchLine.TaxItemGroup = taxItemGroup;
    purchLine.doUpdate();
    return true;
}
```

The `saveDocument()` method updates fields in the header table, whereas the `saveLine()` method updates fields in the transaction line table.

Currently, the VAT ID, number sequence group, and list code are updated in the header table. In the `saveLine()` method, you must update the tax group and item tax group that the Tax Calculation service returns. This requirement applies to most transactions.

> **Note**
>
> To help avoid extra logic, use `doUpdate()` instead of `update()`.

Last updated on 03/21/2026

Uptake features and functions

This article introduces the features and functions you need to implement to integrate a new transaction.

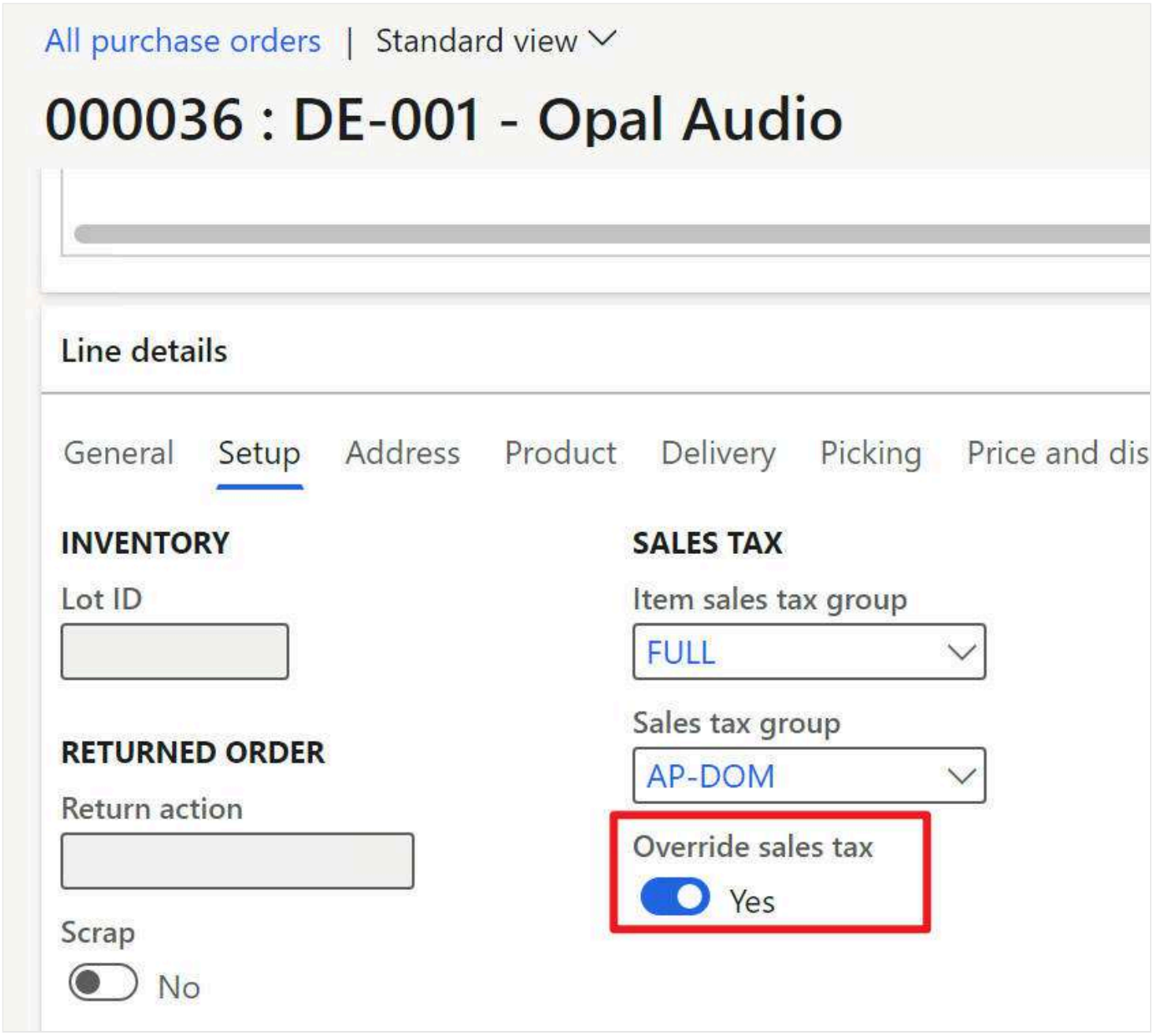
Tax integration supports the following features and functionalities:

- Override sales tax
- Multiple VAT ID
- List code
- Cash discount

Override sales tax

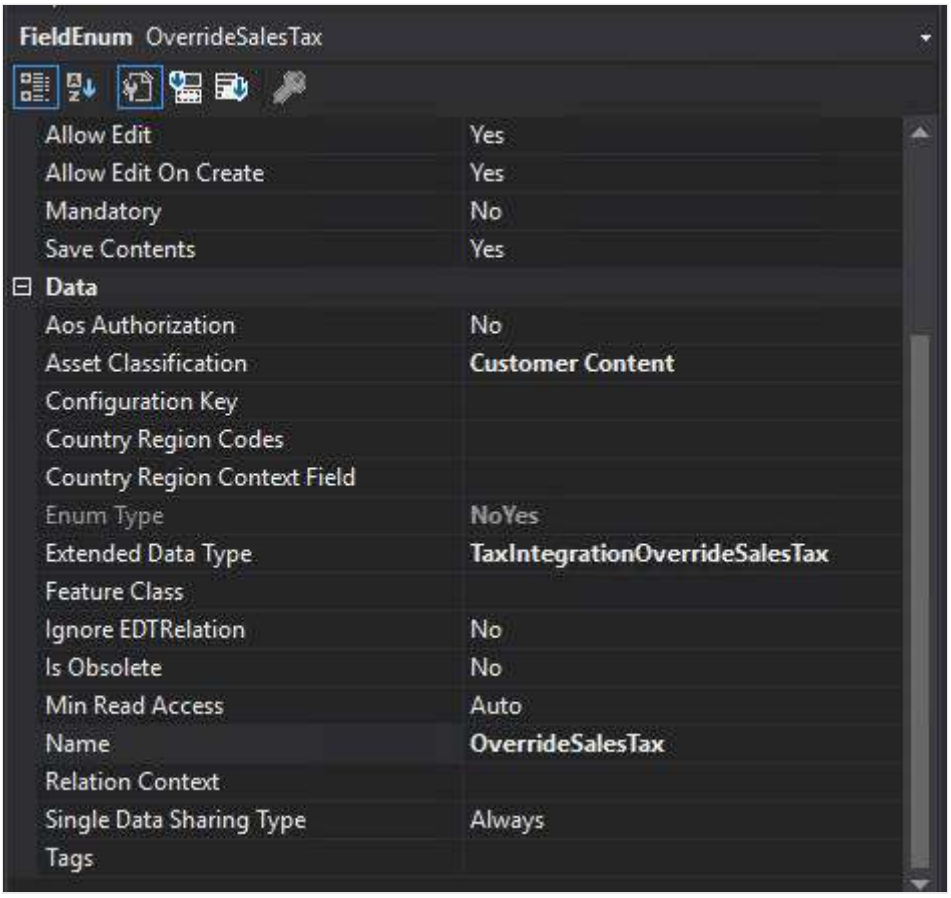
In tax integration, you can't edit the tax group and item tax group on a line, because the Tax Calculation engine determines taxes. By using the **Override sales tax group** functionality, you can change the tax group or item tax group on a line for calculating sales tax. This calculation overrides the tax groups that the Tax Calculation engine determines.

When you set the **Override sales tax** option to **Yes**, you can select a specific tax group and item tax group for tax calculation.



Follow these steps to enable the functionality.

1. Add the **Override sales tax** field to the transaction line table schema and to the **Sales tax** section, if it exists.



2. Set the default visibility to **false** on related transaction forms.

3. Map the **Override sales tax** field to the `SalesPurchJournalLine` map. This map is widely used in tax integration. If you don't map it, you might need extra code to realize the expected function.

4. Set the visibility and editability on the transaction form:

    - Set the default visibility to **false** on the transaction form data source.
    - Set the default visibility to **true** only if you enable tax integration for this transaction.

      X++

      ```
      if (isTaxIntegrationEnabledForPurchase) // Condition should be modified
      according to the business process
      {
          PurchTable_ds.object(fieldNum(PurchTable,
      OverrideSalesTax)).visible(true);
          PurchLine_ds.object(fieldNum(PurchLine,
      OverrideSalesTax)).visible(true);
      }
      ```

    - Add an `allowEdit` control to the transaction form to control the editability of the tax group and item tax group. Usually, add this control in three places:

- The `init()` method of the form
- The `active` method of line
- The `modified()` method of the **Override sales tax** field

```
X++
```

```
if (isTaxIntegrationEnabledForPurchase) // Condition should be modified
according to the business process
{
    PurchLine_ds.object(fieldNum(PurchLine,
TaxGroup)).allowEdit(purchLine.OverrideSalesTax == NoYes::Yes);
    PurchLine_ds.object(fieldNum(PurchLine,
TaxItemGroup)).allowEdit(purchLine.OverrideSalesTax == NoYes::Yes);
}
```

5. Set the value of the override sales tax in the data retrieval activity.

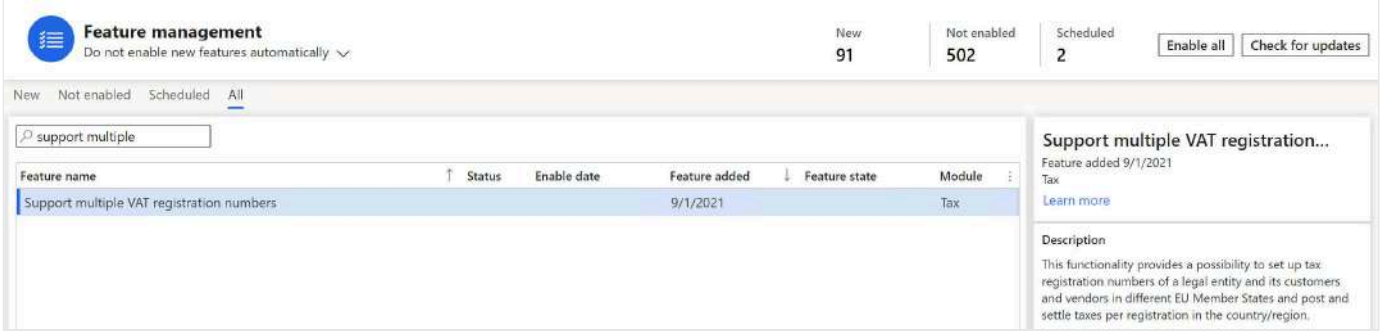
   `line.setOverrideSalesTax(this.vendInvoiceInfoLine.OverrideSalesTax);`

6. To keep the design consistent, the new charge enters the default override sales tax from its origin. Modify `MarkupTrans::getOverrideSalesTaxFromParentRecord(MarkupTransRefTableId _tableId, MarkupTransRefRecId _refRecId)` for new transaction support.

7. (Optional) Modify related entities to support the import and export of override sales tax.

Multiple VAT ID

The **Multiple VAT ID** feature enables the value-added tax (VAT) ID to be determined from the Tax Calculation engine. The **Support multiple VAT registration numbers** feature controls this feature. For more information, see Multiple VAT registration numbers.



You must enter and save two registration numbers in this feature: the registration number for the tax authority of the current legal entity and the registration number for the counterparty. If a transaction doesn't have a customer or vendor as a counterparty, the counterparty registration number doesn't work for it.

> **Note**
>
> There's only one legal entity registration number and one counterparty registration number for all lines in a transaction. Most of the code for this feature is in `TaxIntegrationTaxIdActivityOnDocument.xpp`.

Database schema

The **TaxId** and **PartyTaxId** data fields are used for the record IDs of registration numbers. Developers add these two fields to the following related tables to store the registration numbers of a transaction:

- TmpTaxWorkTrans
- TaxUncommitted
- TaxTrans
- CustPackingslipJour
- CustInvoiceJour
- VendPackingSlipJour
- VendInvoiceJour

The first three tables in the preceding list (tax-related tables) are line-level tables. The last four tables are header-level tables, because one transaction should have the same registration number.

Registration number for the legal entity

When you enable the **Support multiple VAT registration numbers** feature, you also enable the legal entity registration number. You don't need to enter anything here for a new transaction. If the system determines the tax code, it fills in the legal entity registration number with the registration number that's assigned to the settlement period of that tax code. The system determines this number at `TaxIntegrationTaxIdActivityOnDocument::populateTaxLineTaxId()` and saves it to the database together with the tax result.

The system also includes validation logic for the legal entity registration number in case it determines different registration numbers for different lines in a transaction. The system performs this validation at `TaxIntegrationTaxIdActivityOnDocument::checkTaxIdConsistency()`.

Counterparty registration number

The Tax Calculation engine determines the counterparty registration number. After the system receives the response, it validates the number and saves it to the database together with the legal entity registration number. However, if the Tax Calculation engine determines a number that isn't in the user's master data, the system writes the default value on the transaction header to the database instead of the returned value. The counterparty registration number doesn't apply to all transactions. However, the system includes extra logic to handle this approach.

Default logic

The system applies default logic from customer and vendor master data to the transaction header. You can set the **Tax-exempt number** value at the customer and vendor master data levels. When you create a new transaction, the system enters the default tax-exempt number on it. However, you can select a new number to override the default number. If the number that the tax engine returns isn't valid, the system uses the number on the header before tax calculation as the default number.

The tax-exempt number is a string field (**VATNum**) instead of a record ID, and it has two data sources. One data source is from the tax registration number, and the other is from the tax-exempt number. Tax integration supports only the tax registration number as its source. You should add two new fields, **VATNumRecId** and **VATNumTableType**, to the appropriate tables to distinguish the record.

When you add a new transaction, decide whether to add the **VATNum**, **VATNumRecId**, and **VATNumTableType** fields to the transaction header table. If you add these fields, also add the table to `TaxExemptVATNumMap`. The record ID and table type are then automatically saved to the header table when users select a number in the user interface (UI).

A developer must identify the default logic and call `copyPrimaryRegistrationNumberToVATMap` to copy the two fields to the newly created transaction header table. Use the following default code from `CustTable` to `SalesTable` as an example in `SalesTable.xpp`.

X++

```
private void initRegistrationNumbers(CustTable _custTable)
{
    this.vatNum =
_custTable.getPrimaryRegistrationNumber(TaxRegistrationTypesList::TAXID);
    _custTable.copyPrimaryRegistrationNumberToVATMap(this);     // This line is to
copy the new fields.
    this.EnterpriseNumber =
_custTable.getPrimaryRegistrationNumber(TaxRegistrationTypesList::UID);
}
```

Switch function

1. Modify the switch function to support the new transaction during journalization:

   `TaxIntegrationUtils::isMultipleTaxIdEnabledForJournalV3(RefTableId _journalHeaderTableId, RefRecId _journalHeaderRecId, Tax _tax = null)`

2. If there's no counterparty, or if the counterparty registration number doesn't apply to the new transaction, skip the counterparty logic at `TaxIntegrationTaxIdActivityOnDocument::shouldSetPartyTaxId()`.

   X++

```
protected static boolean shouldSetPartyTaxId(TaxIntegrationDocumentObject
_document)
{
    // Sales quotation for a prospect customer doesn't apply to the counterparty
VAT ID
    if (_document.getHeadingTableId() == tableNum(SalesQuotationTable) &&
_document.getCustVendAccount() == '')
        return false;
    // Currently for purchase order and sales order, the party tax ID is set only
if returned by the tax engine.
    // For Transfer Order, its party tax ID isn't determined by the tax engine
and must be set for the first calculation round.
    return _document.isPartyTaxIdReturned()
        || ((_document.getBusinessProcess() ==
TaxIntegrationBusinessProcess::Inventory)
            && !(TaxInventTransferCalcTaxContext::current() &&
TaxInventTransferCalcTaxContext::current().parmShouldSkipSetPartyTaxId()))
        // If a default exists, always go through the tax ID process.
        || _document.getPartyTaxIdRecIdDefault();
}
```

Number sequence group

The number sequence group is part of the **Multiple VAT ID** feature. When you update the VAT ID on the transaction header, update the corresponding number sequence group. However, this update doesn't apply to all transactions. If you need to integrate the number sequence group into the new transaction, use the following example in the data persistence activity.

X++

```
private void saveNumberSequenceGroupToTable()
{
    NumberSequenceGroupId numberSequenceGroupId = document.getNumberSequenceGroupId();
    if (numberSequenceGroupId && custInvoiceTable.numberSequenceGroup !=
```

```
numberSequenceGroupId)
    {
        ttsbegin;
        custInvoiceTable.numberSequenceGroup = document.getNumberSequenceGroupId();
        custInvoiceTable.doUpdate();
        ttscommit;
    }
}
```

The tax ID activity determines this value for all transactions. Add and call the logic in the data persistence activity to persist the number sequence group.

List code

The tax engine determines the list code, which resembles the counterparty VAT ID, and persists it in the database. Always enable the list code feature if you configure the list code applicability matrix in Tax calculation feature.

Data retrieval

Retrieve the default list code to `TaxIntegrationDocument`. Add a line in the `copyToDocumentFromHeaderTable` method of the newly created data retrieval class.

X++

```
protected void copyToDocumentFromHeaderTable()
{
    super();
    ...
    document.setListCode(this.purchTable.ListCode);
    ...
}
```

Map list code to map

The data persistence class uses a `SalesPurchJournalTable` map to reduce duplicate code. Map the list code from the header table to the map.

X++

```
<AxTableMapping>
<MappingTable>SalesPurchJournalTable</MappingTable>
    <Connections>
        ...
        <AxTableMappingConnection>
            <MapField>ListCode</MapField>
```

```
            <MapFieldTo>ListCode</MapFieldTo>
        </AxTableMappingConnection>
    </Connections>
</AxTableMapping>
```

If you don't map your transaction table to this `SalesPurchJournalTable` map, write your own logic in the data persistence activity to update the list code.

Data persistence

Call the `TaxIntegrationListCodeUtility::saveListCodeFromDocumentToTable()` method in the `saveDocument` method of the newly created data persistence class.

X++

```
/// <summary>
/// Saves the document.
/// </summary>
/// <returns>Always true.</returns>
protected boolean saveDocument()
{
    TaxIntegrationTaxIdUtility::saveTaxIDFromDocumentToTable(document, salesTable);
    TaxIntegrationListCodeUtility::saveListCodeFromDocumentToTable(document,
salesTable);
    return true;
}
```

Currency exchange, rounding, and penny difference adjustment

This functionality is enabled by default. Set the transaction currency, the fixed exchange rate for the reporting currency, and the fixed exchange rate for the accounting currency in the document object. The `TaxIntegrationCurrencyExchangeActivityOnDocument` activity performs this action. `TaxIntegrationFacade` calls this activity without any condition. You don't need to take any action to use this functionality.

Cash discount

If your finance and operations apps already support the cash discount function for the new transaction, tax integration also enables the Tax Calculation engine to determine the cash discount parameters. Here's a brief overview of the process for enabling this functionality.

1. In the code base, find all references to cash discount parameters.

2. Replace the references to cash discount parameters with parameters from the Tax Calculation engine. You can retrieve the parameters by using one of two methods:

    - `TaxIntegrationFacade::getTaxJurisdictionParameters(RefTableId _sourceHeadingTableId, RefRecId _sourceHeadingRecId)`
    - `TaxIntegrationFacade::getTaxJurisdictionParametersByTable(Common _sourceHeadingTable)`

3. Extend the switch functions to enable the new transaction:

    - `TaxIntegrationUtils::shouldRetrieveCashDiscParametersFromTaxService(RefTableId _refTableId, RefRecId _refRecId)`
    - `TaxIntegrationUtils::getBusinessProcessBySourceHeadingTable(RefTableId _sourceHeadingTableId, RefRecId _sourceHeadingRecId)`

Last updated on 03/21/2026

Modifications on the switch function and the front end

This article explains the modifications that are required on switch functions and the front end to integrate a new transaction.

Modification on switch functions to enable a new transaction

The switch function controls whether tax integration should be applied to a specific transaction, header table, and line table. As a developer, extend the switch function to enable the new transaction.

In `Tax.xpp`, add the header and line tables of new transactions according to the business process. Extend the method so that it returns *true* for the header and line tables of the new transaction.

X++

```
/// <summary>
/// Checks whether Tax Integration is enabled.
/// </summary>
/// <param name = "_refTableId">The table id.</param>
/// <returns>Whether Tax Integration is enabled.</returns>
public static boolean isTaxIntegrationEnabledForTable(RefTableId _refTableId)
{
    if (TaxIntegrationFlight::instance().isEnabled()
        && TaxIntegrationTaxServiceParameters::find().IsEnable)
    {
        switch (_refTableId)
        {
            case tableNum(SalesTable):
            case tableNum(SalesLine):
            case tableNum(SalesParmTable):
            case tableNum(SalesParmLine):
            case tableNum(SalesQuotationTable):
            case tableNum(SalesQuotationLine):
                return
TaxIntegrationBusinessProcessTable::exist(TaxIntegrationBusinessProcess::Sales);
            case tableNum(PurchTable):
            case tableNum(PurchLine):
            case tableNum(VendInvoiceInfoTable):
            case tableNum(VendInvoiceInfoSubTable):
            case tableNum(VendInvoiceInfoLine):
            case tableNum(PurchParmTable):
            case tableNum(PurchParmLine):
```

```
            case tableNum(PurchReqLine):
            case tableNum(PurchRFQCaseTable):
            case tableNum(PurchRFQCaseLine):
            case tableNum(PurchRFQTable):
            case tableNum(PurchRFQLine):
                return
TaxIntegrationBusinessProcessTable::exist(TaxIntegrationBusinessProcess::Purchase);
            default:
                return false;
        }
    }
    return false;
}
```

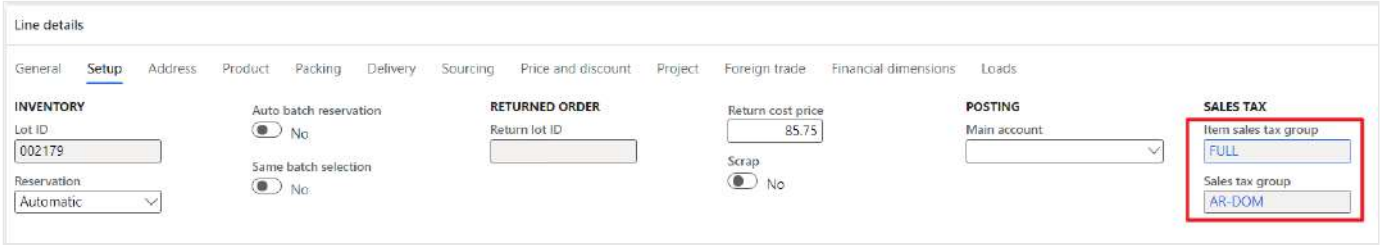
Modification on the front end

Uptake the front-end modification to adapt to the tax integration business logic.

In the transaction form, make the **Override sales tax** control visible. Don't allow users to edit the tax group or the item tax group. The following example uses a sales quotation in the `init()` method of the `SalesQuotationTable` form.

X++

```
if
(Tax::isTaxIntegrationEnabledForBusinessProcess(TaxIntegrationBusinessProcess::Sales))
{
    SalesQuotationTable_ds.object(fieldNum(SalesQuotationTable,
OverrideSalesTax)).visible(true);
    SalesQuotationLine_ds.object(fieldNum(SalesQuotationLine,
TaxGroup)).allowEdit(false);
    SalesQuotationLine_ds.object(fieldNum(SalesQuotationLine,
TaxItemGroup)).allowEdit(false);
    SalesQuotationLine_ds.object(fieldNum(SalesQuotationLine,
OverrideSalesTax)).visible(true);
}
```



Find all buttons that can trigger tax calculation, and reread the data source after tax calculation. Because several fields are updated during the calculation of tax integration, the data source of the form must also be refreshed to reflect the changes. The following example uses free text invoices. This code is from `CustFreeInvoice.xpp`.

X++

```
private void refreshDataSourceForTaxIntegration()
{
    if (isTaxIntegrationEnabledForFTI)
    {
        CustInvoiceTable_ds.reread();
        CustInvoiceLine_ds.research(true);
    }
}
```

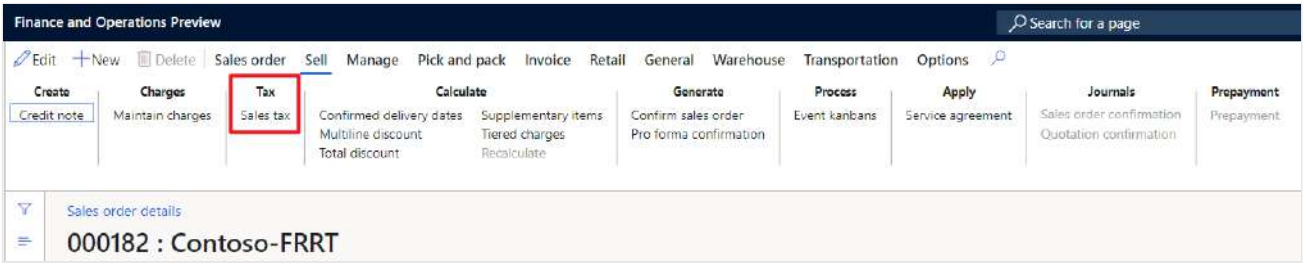
Make a method for the form, and then call that method after tax calculation is triggered by a user interface (UI) action. Here's an example.

X++

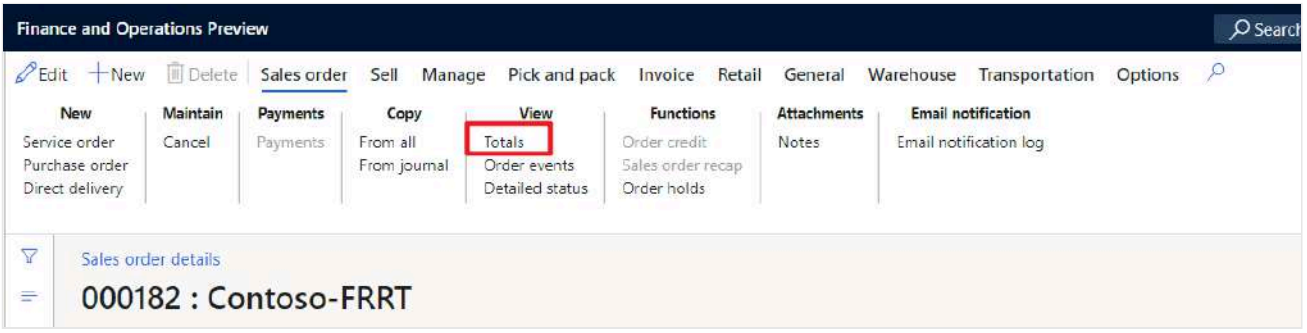
```
[Control("MenuFunctionButton")]
class CustInvoiceTableTotals
{
    public void clicked()
    {
        super();
        element.refreshDataSourceForTaxIntegration();
    }
}
```

Usually, there can be three buttons:

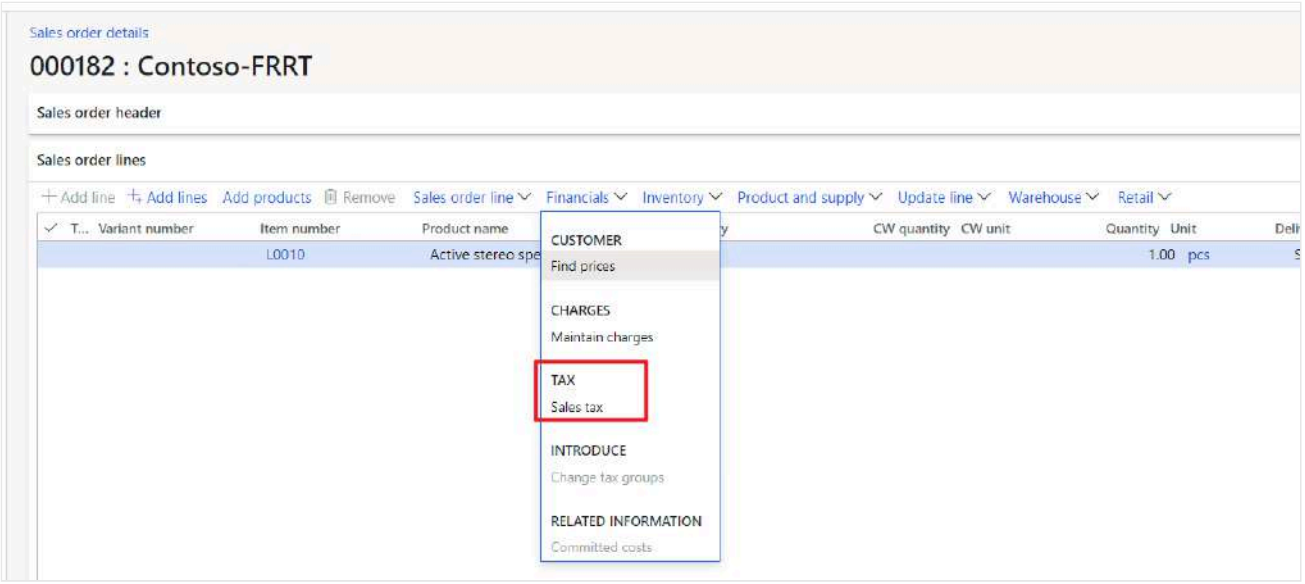
- **Sales tax** button on the Action Pane



- **Totals** button on the Action Pane



- **Sales tax** button on the line



If other buttons trigger tax calculation, call the refresh in the `clicked()` method of those buttons.

Last updated on 04/17/2026

Add a new activity in tax integration

This article provides information about how to add a new or customized activity to the tax integration framework. Before you begin, be sure to read Facade, sequence, and activities in tax integration.

Create a new activity

Document activity or line activity

Before you create the activity class, think about the kinds of activities that you need.

If all actions are at the line level, you don't need anything from the document level. Therefore, consider a line activity. Create a line activity, and extend `TaxIntegrationAbstractActivityOnLine`.

X++

```
public class TaxIntegrationTaxCodeCheckActivityOnLine
    extends TaxIntegrationAbstractActivityOnLine
{
}
```

If actions are at both the document level and the line level, you need a document object. Create a document activity, and extend `TaxIntegrationAbstractActivityOnDocument`.

X++

```
public class TaxIntegrationDataPersistenceActivityOnDocument
    extends TaxIntegrationAbstractActivityOnDocument
{
}
```

Implement shouldSkip

Implement the `shouldSkip` method to skip the activity in specific scenarios. This method is defined in `TaxIntegrationAbstractActivity`. The `shouldSkip` method of the document activity and the `shouldSkip` method of the line activity have the same signature, and they accept `TaxIntegrationDocumentActivity` as input.

For example, if the activity isn't applicable to purchase orders, add a business process check in its `shouldSkip` method.

X++

```
internal boolean shouldSkip(TaxIntegrationDocumentObject _document)
{
    // call super() for some common check
    if (super(_document))
    {
        return true;
    }
    // Skip specifically for purchase order
    if (_document.getBusinessProcess() == TaxIntegrationBusinessProcess::Purchase)
    {
        return true;
    }
}
```

> ⓘ **Note**
>
> In most cases, call `super()`, because `shouldSkip()` in `TaxIntegrationAbstractActivity` has some basic checks (for example, if the document has exceptions). For a line activity, `shouldSkip()` also takes a document object as input. Therefore, it determines whether **all lines (including charges)** in the current document should be skipped. There's no way to determine whether a specific line should be skipped in the framework. If such a determination is required, you can do it in the `actInternal` method of the line activity.

Implement actInternal

Write the real business logic of the activity in the `actInternal` method. The `actInternal` method of the document activity and the `actInternal` method of the line activity have different signatures.

For the line activity

The `actInternal` method of the line activity accepts a `TaxIntegrationLineObject` object as input and can manipulate data at the line level.

For the document activity

The `actInternal` method of the document activity accepts a `TaxIntegrationDocumentObject` object as input and can manipulate data in the whole document.

Most of the document activities manipulate data at the line level. Enumerate the lines in the document object, and then continue as required. The following example shows `TaxIntegrationTaxIdActivityOnDocument.xpp`.

X++

```
SetEnumerator chargeEnumerator = _document.getChargeSet().getEnumerator();
while (chargeEnumerator.moveNext())
{
    TaxIntegrationLineObject charge = chargeEnumerator.current();
    // process the document charge
    if (!TaxIntegrationTaxIdActivityOnDocument::populateLine(charge))
    {
        return;
    }
}
SetEnumerator lineEnumerator = _document.getLineSet().getEnumerator();
while (lineEnumerator.moveNext())
{
    TaxIntegrationLineObject line = lineEnumerator.current();
    // process the line
    if (!TaxIntegrationTaxIdActivityOnDocument::populateLine(line))
    {
        return;
    }
    chargeEnumerator = line.getChargeSet().getEnumerator();
    while (chargeEnumerator.moveNext())
    {
        TaxIntegrationLineObject charge = chargeEnumerator.current();
        // process the line charge
        if (!TaxIntegrationTaxIdActivityOnDocument::populateLine(charge))
        {
            return;
        }
    }
}
```

Append an activity to the sequence

The new activity can't take effect until you add it to a `TaxIntegrationSequence` object. Therefore, as for other activities, append it to the `TaxIntegrationSequence` object in the `TaxIntegrationFacade::calculate` method.

X++

```
TaxIntegrationSequence sequence = TaxIntegrationSequence::construct(LoggerName)

.appendActivityOnDocument(TaxIntegrationSettingRetrievalActivityOnDocument::construct(
))
```

```
.appendActivityOnDocument(TaxIntegrationDataRetrievalActivityOnDocument::construct())

.appendActivityOnDocument(TaxIntegrationCalculationActivityOnDocument::construct())
    // ...

.appendActivityOnDocument(TaxIntegrationDataPersistenceActivityOnDocument::construct()
);
sequence.act(_document);
```

> **Note**
>
> To append the activity by extension, replace the whole method, because the method is replaceable. Carefully arrange the order of the activities, because the activities usually depend on the activities that you append in front of them.

Last updated on 03/21/2026

Add data fields in tax configurations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to customize tax configurations by using data fields that are added in the tax integration.

Customize the tax data model

1. In Microsoft Dynamics 365 Finance, go to **Electronic Reporting** > **Tax configurations**.
2. In the configuration tree, select **Tax Calculation Data Model**. Then, on the action pane, select **Create configuration**.

> **Note**
>
> If there's no configuration provider available, create one and make it active for your tax configuration. For more information, see Create configuration providers and mark them as active.

1. In the drop-down dialog box, select **Taxable document model derived from Name: Tax Calculation Data Model, Microsoft**, enter a name for the new tax data model, and then select **Create configuration**.
2. Select the tax data model that you just created, and then, on the action pane, select **Designer**.
3. Expand the data model tree, select **Lines**, and then select **New**.
4. In the **Create node** dialog box, enter a name, specify the item type, and then select **Add**.
5. Add any required columns.
6. On the action pane, select **Save**, and then select **Complete**.
7. Close the page, and view the completed version of your tax data model.

Customize the tax configuration

1. In Finance, go to **Electronic reporting** > **Tax configurations**.

2. In the configuration tree, selectMarkdown, content, and Acrolinx fixes **Tax Calculation Configuration**. Then, on the action pane, select **Create configuration**.
3. In the drop-down dialog box, select **Tax service configuration derived from Name: Tax Calculation Configuration, Microsoft**, enter a name for the new tax configuration, and then select **Create configuration**.
4. Select the tax configuration that you just created. On the action pane, select **Designer**.
5. In the **Properties** section, in the **Data model** field, select the customized tax data model that you created earlier.
6. In the **Data model version** field, select the completed version of the tax data model.
7. Select **Add**, and add the required tax measures.
8. On the action pane, select **Save**, and then select **Complete**.
9. Close the page, and view the completed version of your tax configuration.

Implement tax features in the customized tax configuration

1. In Finance, go to **Globalization Features** > **Tax**.
2. Select **Add**, enter information about the new feature, and then select **Create feature**.
3. On the **Versions** tab, select the feature, and then select **Edit**.
4. On the **General** tab, in the **Configuration version** field, select the customized tax configuration and version.
5. In the **Manage columns** dialog box, select the header and line columns that you want to include in your customized tax measure, and then select the right arrow button to add them to the **Selected columns** list.

Last updated on 03/21/2026

Add data fields in the tax integration by using extension

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to use X++ extensions to add data fields in the tax integration. You can extend these fields to the tax data model of the tax service and use them to determine tax codes. For more information, see Add data fields in tax configurations.

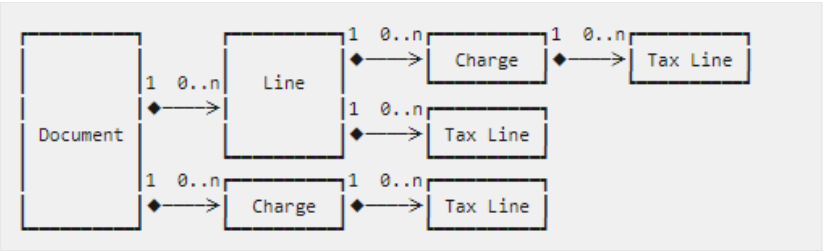
Data model

Objects carry the data in the data model, and classes implement these objects.

Here is a list of the major objects:

- AxClass/TaxIntegration**Document**Object
- AxClass/TaxIntegration**Line**Object
- AxClass/TaxIntegration**TaxLine**Object

The following illustration shows how these objects are related.



A **Document** object can contain many **Line** objects. Each object contains metadata for the tax service.

- `TaxIntegrationDocumentObject` has `originAddress` metadata, which contains information about the source address, and `includingTax` metadata, which indicates whether the line amount includes sales tax.
- `TaxIntegrationLineObject` has `itemId`, `quantity`, and `categoryId` metadata.

> **Note**
>
> `TaxIntegrationLineObject` also implements **Charge** objects.

Integration flow

Activities manipulate the data in the flow.

Key activities

- AxClass/TaxIntegration**Calculation**ActivityOnDocument

- AxClass/TaxIntegration**CurrencyExchange**ActivityOnDocument
- AxClass/TaxIntegration**DataPersistence**ActivityOnDocument
- AxClass/TaxIntegration**DataRetrieval**ActivityOnDocument
- AxClass/TaxIntegration**SettingRetrieval**ActivityOnDocument

Activities run in the following order:

1. Setting Retrieval
2. Data Retrieval
3. Calculation Service
4. Currency Exchange
5. Data Persistence

For example, extend **Data Retrieval** before **Calculation Service**.

Data Retrieval activities

Data Retrieval activities retrieve data from the database. Adapters for different transactions are available to retrieve data from different transaction tables:

- AxClass/TaxIntegration**PurchTable**DataRetrieval
- AxClass/TaxIntegration**PurchParmTable**DataRetrieval
- AxClass/TaxIntegration**PurchREQTable**DataRetrieval
- AxClass/TaxIntegration**PurchRFQTable**DataRetrieval
- AxClass/TaxIntegration**VendInvoiceInfoTable**DataRetrieval
- AxClass/TaxIntegration**SalesTable**DataRetrieval
- AxClass/TaxIntegration**SalesParm**DataRetrieval

In these **Data Retrieval** activities, you copy data from the database to `TaxIntegrationDocumentObject` and `TaxIntegrationLineObject`. Because all these activities extend the same abstract template class, they have common methods.

Calculation Service activities

The **Calculation Service** activity is the link between the tax service and the tax integration. This activity is responsible for the following functions:

1. Construct the request.
2. Post the request to the tax service.
3. Get the response from the tax service.
4. Parse the response.

A data field that you add to the request is posted together with other metadata.

Extension implementation

This section provides detailed steps that explain how to implement the extension. It uses the **Cost center** and **Project** financial dimensions as examples.

Step 1. Add the data variable in the object class

The object class contains the data variable and getter and setter methods for the data. Add the data field to either `TaxIntegrationDocumentObject` or `TaxIntegrationLineObject`, depending on the level of the field. The following example uses the line level, and the file name is `TaxIntegrationLineObject_Extension.xpp`.

> **Note**
>
> If the data field that you're adding is at the document level, change the file name to `TaxIntegrationDocumentObject_Extension.xpp`.

X++

```
[ExtensionOf(classStr(TaxIntegrationLineObject))]
final class TaxIntegrationLineObject_Extension
{
    private OMOperatingUnitNumber costCenter;
    private ProjId projectId;

    /// <summary>
    /// Gets a costCenter.
    /// </summary>
    /// <returns>The cost center.</returns>
    public final OMOperatingUnitNumber getCostCenter()
    {
        return this.costCenter;
    }

    /// <summary>
    /// Sets the cost center.
    /// </summary>
    /// <param name = "_value">The cost center.</param>
    public final void setCostCenter(OMOperatingUnitNumber _value)
    {
        this.costCenter = _value;
    }

    /// <summary>
    /// Gets a project ID.
    /// </summary>
    /// <returns>The project ID.</returns>
    public final ProjId getProjectId()
    {
        return this.projectId;
    }

    /// <summary>
    /// Sets the project ID.
    /// </summary>
    /// <param name = "_value">The project ID.</param>
    public final void setProjectId(ProjId _value)
    {
        this.projectId = _value;
    }
}
```

Add **Cost center** and **Project** as private variables. Create getter and setter methods for these data fields to manipulate the data.

Step 2. Retrieve data from the database

Specify the transaction, and extend the appropriate adapter classes to retrieve the data. For example, if you use a **Purchase order** transaction, extend `TaxIntegrationPurchTableDataRetrieval` and `TaxIntegrationVendInvoiceInfoTableDataRetrieval`.

> **Note**
>
> `TaxIntegrationPurchParmTableDataRetrieval` inherits from `TaxIntegrationPurchTableDataRetrieval`. Don't change `TaxIntegrationPurchParmTableDataRetrieval` unless the logic of the `purchTable` and `purchParmTable` tables differs.

If you need to add a data field for all transactions, extend all `DataRetrieval` classes.

Because all **Data Retrieval** activities extend the same template class, the class structures, variables, and methods are similar.

X++

```
protected TaxIntegrationDocumentObject document;

/// <summary>
/// Copies to the document.
/// </summary>
protected abstract void copyToDocument()
{
    // It is recommended to implement as:
    //
    // this.copyToDocumentByDefault();
    // this.copyToDocumentFromHeaderTable();
    // this.copyAddressToDocument();
}

/// <summary>
/// Copies to the current line of the document.
/// </summary>
/// <param name = "_line">The current line of the document.</param>
protected abstract void copyToLine(TaxIntegrationLineObject _line)
{
    // It is recommended to implement as:
    //
    // this.copyToLineByDefault(_line);
    // this.copyToLineFromLineTable(_line);
    // this.copyQuantityAndTransactionAmountToLine(_line);
    // this.copyAddressToLine(_line);
    // this.copyToLineFromHeaderTable(_line);
}
```

The following example shows the basic structure when the `PurchTable` table is used.

X++

```
public class TaxIntegrationPurchTableDataRetrieval extends TaxIntegrationAbstractDataRetrievalTemplate
{
    protected PurchTable purchTable;
    protected PurchLine purchLine;

    // Query builder methods
    [Replaceable]
    protected SysDaQueryObject getDocumentQueryObject()
    [Replaceable]
    protected SysDaQueryObject getLineQueryObject()
    [Replaceable]
    protected SysDaQueryObject getDocumentChargeQueryObject()
```

```
    [Replaceable]
    protected SysDaQueryObject getLineChargeQueryObject()

    // Data retrieval methods
    protected void copyToDocument()
    protected void copyToDocumentFromHeaderTable()
    protected void copyToLine(TaxIntegrationLineObject _line)
    protected void copyToLineFromLineTable(TaxIntegrationLineObject _line)
    ...
}
```

When the `CopyToDocument` method is called, the `this.purchTable` buffer already exists. The purpose of this method is to copy all the required data from `this.purchTable` to the `document` object by using the setter method that you created in the `DocumentObject` class.

Likewise, a `this.purchLine` buffer already exists in the `CopyToLine` method. The purpose of this method is to copy all the required data from `this.purchLine` to the `_line` object by using the setter method that you created in the `LineObject` class.

The most straightforward approach is to extend the `CopyToDocument` and `CopyToLine` methods. However, try the `copyToDocumentFromHeaderTable` and `copyToLineFromLineTable` methods first. If they don't work as you require, implement your own method, and call it in `CopyToDocument` and `CopyToLine`. There are three common situations where you might use this approach:

- The required field is in `PurchTable` or `PurchLine`. In this situation, extend `copyToDocumentFromHeaderTable` and `copyToLineFromLineTable`. Here's the sample code.

  X++

  ```
  /// <summary>
  /// Copies to the current line of the document from.
  /// </summary>
  /// <param name = "_line">The current line of the document.</param>
  protected void copyToLineFromLineTable(TaxIntegrationLineObject _line)
  {
      next copyToLineFromLineTable(_line);
      // if we already added XXX in TaxIntegrationLineObject
      _line.setXXX(this.purchLine.XXX);
  }
  ```

- The required data isn't in the default table of the transaction. However, there are some join relationships with the default table, and the field is required on most lines. In this situation, replace `getDocumentQueryObject` or `getLineObject` to query the table by join relationship. In the following example, the **Deliver Now** field is integrated with the sales order at the line level.

  X++

  ```
  public class TaxIntegrationSalesTableDataRetrieval
  {
      protected MCRSalesLineDropShipment mcrSalesLineDropShipment;

      /// <summary>
      /// Gets the query for the lines of the document.
      /// </summary>
      /// <returns>The query for the lines of the document</returns>
      [Replaceable]
      protected SysDaQueryObject getLineQueryObject()
      {
          return SysDaQueryObjectBuilder::from(this.salesLine)
  ```

```
            .where(this.salesLine, fieldStr(SalesLine,
SalesId)).isEqualToLiteral(this.salesTable.SalesId)
            .outerJoin(this.mcrSalesLineDropShipment)
            .where(this.mcrSalesLineDropShipment, fieldStr(MCRSalesLineDropShipment,
SalesLine)).isEqualTo(this.salesLine, fieldStr(SalesLine, RecId))
            .toSysDaQueryObject();
    }
}
```

In this example, a `mcrSalesLineDropShipment` buffer is declared, and the query is defined in `getLineQueryObject`. The query uses the relationship `MCRSalesLineDropShipment.SalesLine == SalesLine.RecId`. While you're extending in this situation, you can replace `getLineQueryObject` with your own constructed query object. However, note the following points:

- Because the return value of the `getLineQueryObject` method is `SysDaQueryObject`, you must construct this object by using the SysDa approach.
- You can't remove existing tables.

- The required data is related to the transaction table by a complicated join relationship, or the relation isn't one to one (1:1) but one to many (1:N). In this situation, things become a little complicated. This situation applies to the example of financial dimensions.

  In this situation, you can implement your own method to retrieve the data. Here's the sample code in the `TaxIntegrationPurchTableDataRetrieval_Extension.xpp` file.

X++

```
[ExtensionOf(classStr(TaxIntegrationPurchTableDataRetrieval))]
final class TaxIntegrationPurchTableDataRetrieval_Extension
{
    private const str CostCenterKey = 'CostCenter';
    private const str ProjectKey = 'Project';

    /// <summary>
    /// Copies to the current line of the document from.
    /// </summary>
    /// <param name = "_line">The current line of the document.</param>
    protected void copyToLineFromLineTable(TaxIntegrationLineObject _line)
    {
        Map dimensionAttributeMap =
this.getDimensionAttributeMapByDefaultDimension(this.purchline.DefaultDimension);
        if (dimensionAttributeMap.exists(CostCenterKey))
        {
            _line.setCostCenter(dimensionAttributeMap.lookup(CostCenterKey));
        }
        if (dimensionAttributeMap.exists(ProjectKey))
        {
            _line.setProjectId(dimensionAttributeMap.lookup(ProjectKey));
        }
        next copyToLineFromLineTable(_line);
    }
    private Map getDimensionAttributeMapByDefaultDimension(RefRecId _defaultDimension)
    {
        DimensionAttribute dimensionAttribute;
        DimensionAttributeValue dimensionAttributeValue;
        DimensionAttributeValueSetItem dimensionAttributeValueSetItem;
        Map ret = new Map(Types::String, Types::String);

        select Name, RecId from dimensionAttribute
            join dimensionAttributeValue
                where dimensionAttributeValue.dimensionAttribute == dimensionAttribute.RecId
            join DimensionAttributeValueSetItem
                where DimensionAttributeValueSetItem.DimensionAttributeValue ==
```

```
DimensionAttributeValue.RecId
                    && DimensionAttributeValueSetItem.DimensionAttributeValueSet ==
_defaultDimension;

        while(dimensionAttribute.RecId)
        {
            ret.insert(dimensionAttribute.Name, dimensionAttributeValue.DisplayValue);
            next dimensionAttribute;
        }
        return ret;
    }
}
```

Step 3. Add data to the request

Extend the `copyToTaxableDocumentHeaderWrapperFromTaxIntegrationDocumentObject` or `copyToTaxableDocumentLineWrapperFromTaxIntegrationLineObjectByLine` method to add data to the request. The following code shows an example in the `TaxIntegrationCalculationActivityOnDocument_CalculationService_Extension.xpp` file.

X++

```
[ExtensionOf(classStr(TaxIntegrationCalculationActivityOnDocument_CalculationService))]
final static class TaxIntegrationCalculationActivityOnDocument_CalculationService_Extension
{
    // Define the field name in the request
    private const str IOCostCenter = 'Cost Center';
    private const str IOProject = 'Project';
    // private const str IOEnumExample = 'Enum Example';

    /// <summary>
    /// Copies to <c>TaxableDocumentLineWrapper</c> from <c>TaxIntegrationLineObject</c> by line.
    /// </summary>
    /// <param name = "_destination"><c>TaxableDocumentLineWrapper</c>.</param>
    /// <param name = "_source"><c>TaxIntegrationLineObject</c>.</param>
    protected static void
copyToTaxableDocumentLineWrapperFromTaxIntegrationLineObjectByLine(Microsoft.Dynamics.TaxCalculation.ApiCo
ntracts.TaxableDocumentLineWrapper _destination, TaxIntegrationLineObject _source)
    {
        next copyToTaxableDocumentLineWrapperFromTaxIntegrationLineObjectByLine(_destination, _source);
        // Set the field we need to integrated for tax service
        _destination.SetField(IOCostCenter, _source.getCostCenter());
        _destination.SetField(IOProject, _source.getProjectId());

        // If the field to be extended is an enum type, use enum2Symbol to convert an enum variable
exampleEnum of ExampleEnumType to a string
        // _destination.SetField(IOEnumExample, enum2Symbol(enumNum(ExampleEnumType),
_source.getExampleEnum()));
    }
}
```

In this code, `_destination` is the wrapper object that generates the request, and `_source` is the `TaxIntegrationLineObject` object.

> **Note**
>
> Define the field name used in the request as **private const str**. The string should be exactly the same as the node name (not the label) added in the article Add data fields in tax configurations.

Set the field in the **copyToTaxableDocumentLineWrapperFromTaxIntegrationLineObjectByLine** method by using the **SetField** method. The data type of the second parameter should be **string**. If the data type isn't **string**, convert it to string. If the data type is X++ **enum type**, use the **enum2Symbol** method to convert the enum value to a string. The enum value added in the tax configuration should be exactly the same as the enum name. The following list describes the differences between enum value, label, and name.

- The name of enum is a symbolic name in code. **enum2Symbol()** converts the enum value to its name.
- The value of the enum is integer.
- The label of the enum can be different across preferred languages. **enum2Str()** converts the enum value to its label.

Model dependency

To successfully build the project, add the following reference models to the model dependencies:

- ApplicationPlatform
- ApplicationSuite
- Tax Engine
- Dimensions, if you use financial dimension
- Other necessary models referenced in the code

Validation

After you complete the previous steps, you can validate your changes.

1. In Finance, go to **Accounts payable** and add **&debug=vs%2CconfirmExit&** to the URL. For example, `https://usnconeboxax1aos.cloud.onebox.dynamics.com/?cmp=DEMF&mi=PurchTableListPage&debug=vs%2CconfirmExit&`. The final **&** is essential.
2. Open the **Purchase order** page and select **New** to create a purchase order.
3. Set the value for the customized field, and then select **Sales tax**. A troubleshooting file with prefix, **TaxServiceTroubleshootingLog** is downloaded automatically. This file contains the transaction information posted to the Tax Calculation Service.
4. Check if the customized field added is present in the **Tax service calculation input JSON** section and if its value is correct. If the value isn't correct, double check the steps in this document.

File example:

```
===Tax service calculation input JSON:===
{
  "TaxableDocument": {
    "Header": [
      {
        "Lines": [
          {
            "Line Type": "Normal",
            "Item Code": "",
            "Item Type": "Item",
            "Quantity": 0.0,
            "Amount": 1000.0,
            "Currency": "EUR",
            "Transaction Date": "2022-1-26T00:00:00",
            ...
```

```
            /// The new fields added at line level
            "Cost Center": "003",
            "Project": "Proj-123"
          }
        ],
        "Amount include tax": true,
        "Business Process": "Journal",
        "Currency": "",
        "Vendor Account": "DE-001",
        "Vendor Invoice Account": "DE-001",
        ...
        // The new fields added at header level, no new fields in this example
        ...
      }
    ]
  },
}
...
```

Appendix

This appendix shows the complete sample code for the integration of the financial dimensions, **Cost center** and **Project** at the line level.

TaxIntegrationLineObject_Extension.xpp

X++

```
[ExtensionOf(classStr(TaxIntegrationLineObject))]
final class TaxIntegrationLineObject_Extension
{
    private OMOperatingUnitNumber costCenter;
    private ProjId projectId;

    /// <summary>
    /// Gets a costCenter.
    /// </summary>
    /// <returns>The cost center.</returns>
    public final OMOperatingUnitNumber getCostCenter()
    {
        return this.costCenter;
    }

    /// <summary>
    /// Sets the cost center.
    /// </summary>
    /// <param name = "_value">The cost center.</param>
    public final void setCostCenter(OMOperatingUnitNumber _value)
    {
        this.costCenter = _value;
    }

    /// <summary>
    /// Gets a project ID.
    /// </summary>
    /// <returns>The project ID.</returns>
    public final ProjId getProjectId()
    {
        return this.projectId;
    }

    /// <summary>
    /// Sets the project ID.
```

```
    /// </summary>
    /// <param name = "_value">The project ID.</param>
    public final void setProjectId(ProjId _value)
    {
        this.projectId = _value;
    }
}
```

TaxIntegrationPurchTableDataRetrieval_Extension.xpp

X++

```
[ExtensionOf(classStr(TaxIntegrationPurchTableDataRetrieval))]
final class TaxIntegrationPurchTableDataRetrieval_Extension
{
    private const str CostCenterKey = 'CostCenter';
    private const str ProjectKey = 'Project';

    /// <summary>
    /// Copies to the current line of the document from.
    /// </summary>
    /// <param name = "_line">The current line of the document.</param>
    protected void copyToLineFromLineTable(TaxIntegrationLineObject _line)
    {
        Map dimensionAttributeMap =
this.getDimensionAttributeMapByDefaultDimension(this.purchline.DefaultDimension);
        if (dimensionAttributeMap.exists(CostCenterKey))
        {
            _line.setCostCenter(dimensionAttributeMap.lookup(CostCenterKey));
        }
        if (dimensionAttributeMap.exists(ProjectKey))
        {
            _line.setProjectId(dimensionAttributeMap.lookup(ProjectKey));
        }
        next copyToLineFromLineTable(_line);
    }
    private Map getDimensionAttributeMapByDefaultDimension(RefRecId _defaultDimension)
    {
        DimensionAttribute dimensionAttribute;
        DimensionAttributeValue dimensionAttributeValue;
        DimensionAttributeValueSetItem dimensionAttributeValueSetItem;
        Map ret = new Map(Types::String, Types::String);
        select Name, RecId from dimensionAttribute
            join dimensionAttributeValue
                where dimensionAttributeValue.dimensionAttribute == dimensionAttribute.RecId
            join DimensionAttributeValueSetItem
                where DimensionAttributeValueSetItem.DimensionAttributeValue ==
DimensionAttributeValue.RecId
                    && DimensionAttributeValueSetItem.DimensionAttributeValueSet == _defaultDimension;
        while(dimensionAttribute.RecId)
        {
            ret.insert(dimensionAttribute.Name, dimensionAttributeValue.DisplayValue);
            next dimensionAttribute;
        }
        return ret;
    }
}
```

TaxIntegrationCalculationActivityOnDocument_CalculationService_Extensic

X++

```
[ExtensionOf(classStr(TaxIntegrationCalculationActivityOnDocument_CalculationService))]
final static class TaxIntegrationCalculationActivityOnDocument_CalculationService_Extension
{
    // Define the field name in the request
    private const str IOCostCenter = 'Cost Center';
    private const str IOProject = 'Project';

    /// <summary>
    /// Copies to <c>TaxableDocumentLineWrapper</c> from <c>TaxIntegrationLineObject</c> by line.
    /// </summary>
    /// <param name = "_destination"><c>TaxableDocumentLineWrapper</c>.</param>
    /// <param name = "_source"><c>TaxIntegrationLineObject</c>.</param>
    protected static void
copyToTaxableDocumentLineWrapperFromTaxIntegrationLineObjectByLine(Microsoft.Dynamics.TaxCalculation.ApiCo
ntracts.TaxableDocumentLineWrapper _destination, TaxIntegrationLineObject _source)
    {
        next copyToTaxableDocumentLineWrapperFromTaxIntegrationLineObjectByLine(_destination, _source);
        // Set the field we need to integrated for tax service
        _destination.SetField(IOCostCenter, _source.getCostCenter());
        _destination.SetField(IOProject, _source.getProjectId());
    }
}
```

Last updated on 03/21/2026

Customize tax configurations for master data lookup

Follow the steps in this article to customize tax configurations to extend master data lookup functionality.

Import a tax configuration provided by Microsoft

Follow the steps in Import Electronic reporting (ER) configurations from Dataverse and import the correct tax configuration version, based on your Finance version.

> **Note**
>
> By default, the Dataverse model mapping is imported. If you receive warning messages during the configuration import process, enable the virtual entities in Dataverse. For more information, see Enable Dataverse virtual entities.

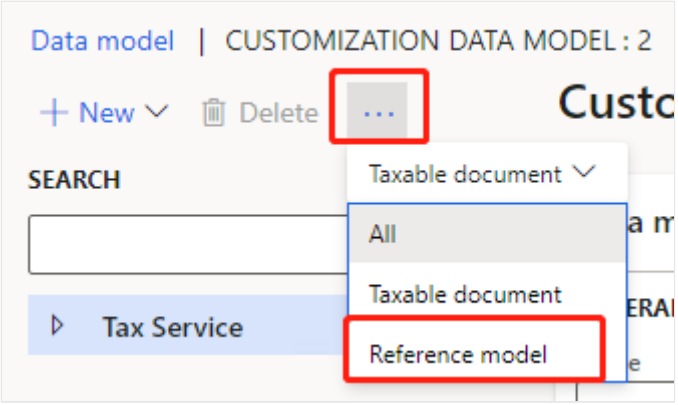
Create a customized data model configuration

1. In the **Electronic reporting** workspace, select **Tax configurations**, and then select the data model configuration to extend. For example, select **Tax Calculation Data Model**.
2. Select **Create configuration**.
3. Select **Taxable document model derived from Name: Tax Calculation Data Model, Microsoft**.
4. In the **Name** field, enter **Customization data model**.
5. Select **Create configuration**.

Create customized reference models

1. On the **Tax configurations** page, select **Customization data model**, and then select **Designer**.

2. Select the ellipsis button (**...**), and then select the **Reference model** view.



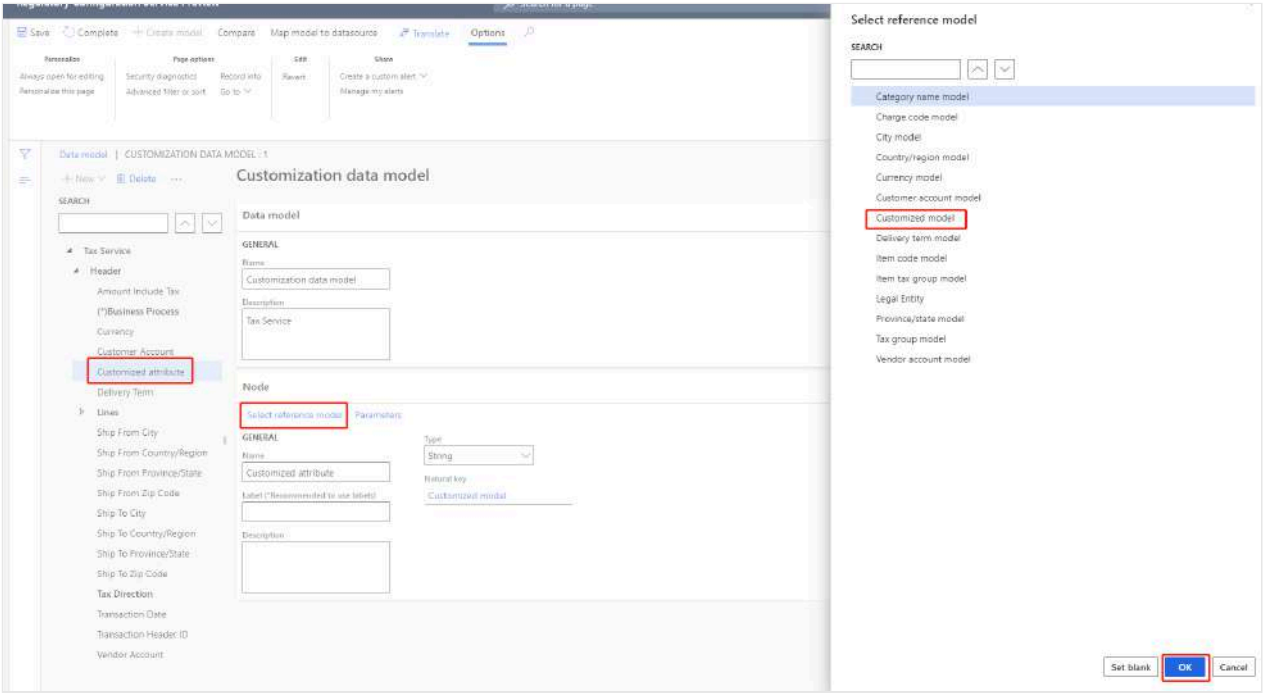
3. Create the customized reference model. The customized model is a root model. The customized entity is a record list. The customized field is a string field that you want to use in the lookup. You can add more fields as you require.

4. Select the ellipsis button (...), and then select the **Taxable document** view.

5. Select the attribute to bind to the customized reference model. For example, select **Customized attribute**, and then follow these steps:

   a. Select **Select reference model**.

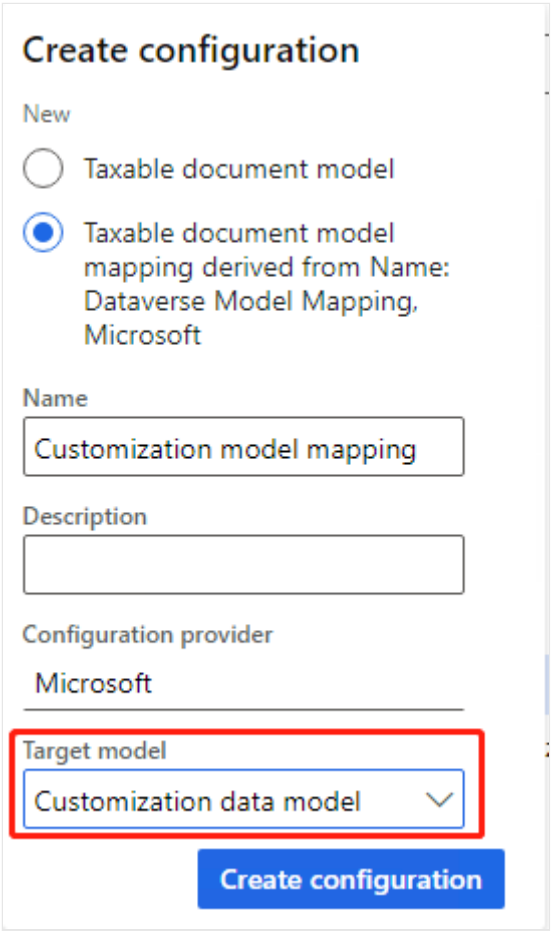
   b. Select **Customized model**, and then select **OK**. The reference model name is updated to the value of the **Natural key** field.



   c. Select **Save**, and then select **Complete**.

Create a customized model mapping configuration

1. In the **Electronic reporting** workspace, select **Tax configurations**, and then select the **Dataverse model mapping** configuration.
2. In the **Default for model mapping** field, select **No**.
3. Select **Create configuration**.
4. Select **Taxable document model mapping derived from Name: Dataverse Model Mapping, Microsoft**.
5. In the **Name** field, enter **Customization model mapping**.
6. In the **Target model** field, select the **Customization data model** data model.
7. Select **Create configuration**.

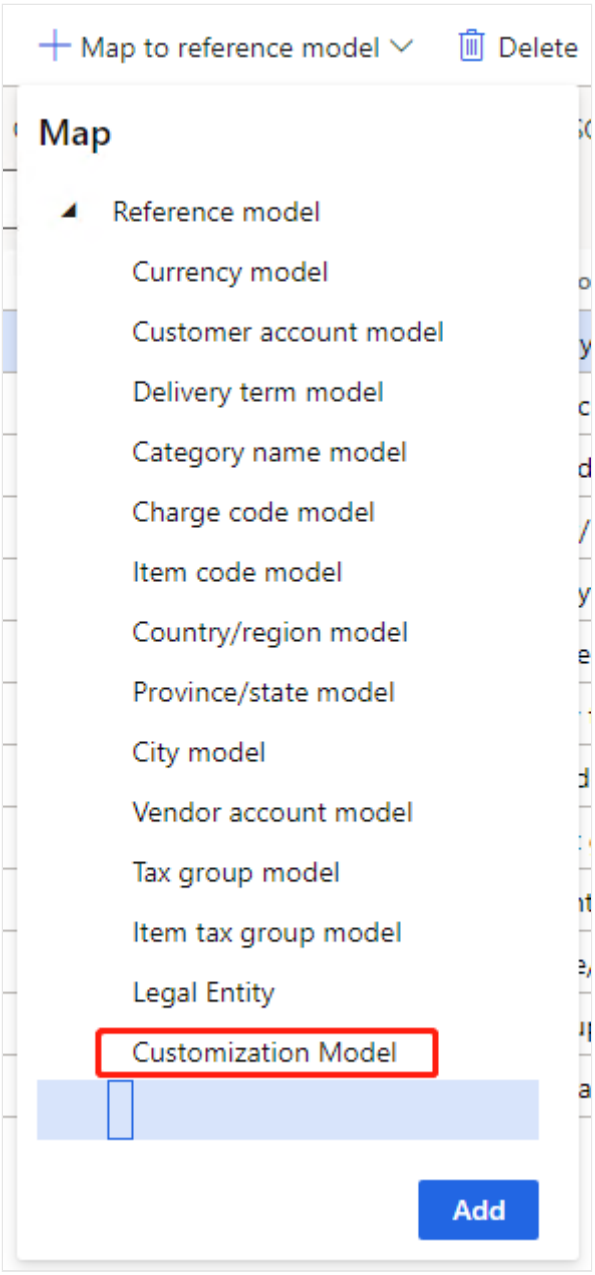


8. Select **Customization model mapping**, and set the **Connected application** field to the connection that you created in step 8 of Set up an environment for master data lookup.
9. Set the **Default for model mapping** field to **Yes**.

Create customized model mappings

1. On **Tax configurations**, select **Customization model mapping**.

2. Select **Designer**, and then select **Customization Model**.



Map a model mapping to a Dataverse entity

1. On the **Model mapping designer** page, select **Customization Model**, and then select **Designer**.

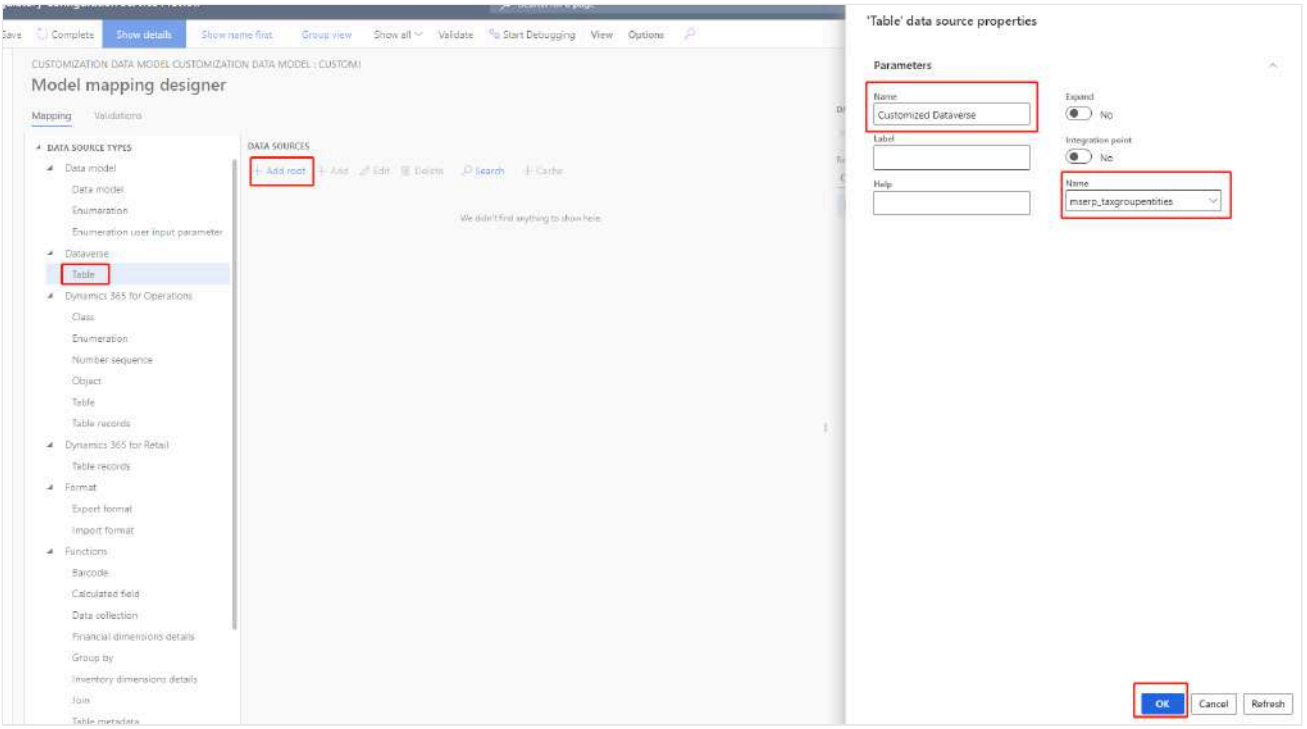
2. In the **Data source types** tree, select **Dataverse Table**.

3. On the **Data sources** tab, select **Add root**.

4. In the **Name** field, enter **Customized Dataverse**.

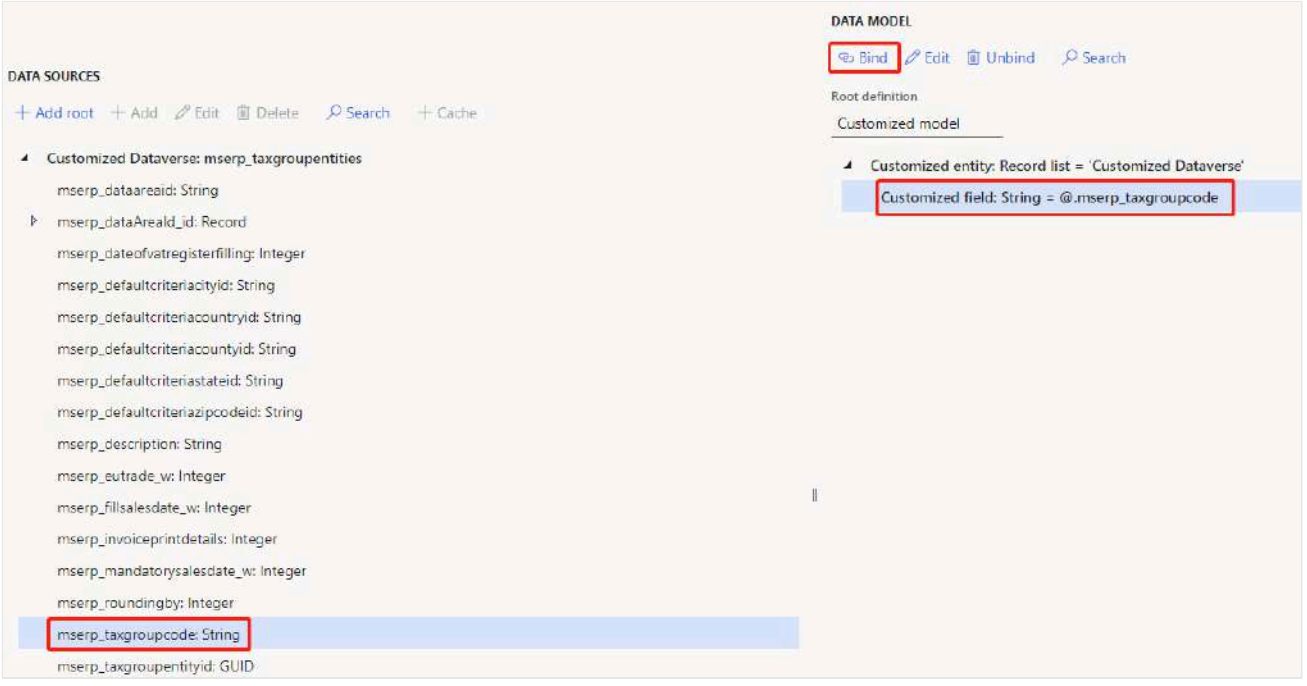
5. In the second **Name** field, select an entity.

6. Select **OK**.



7. Select **Customized Dataverse** and **Customized entity**, and then select **Bind**.

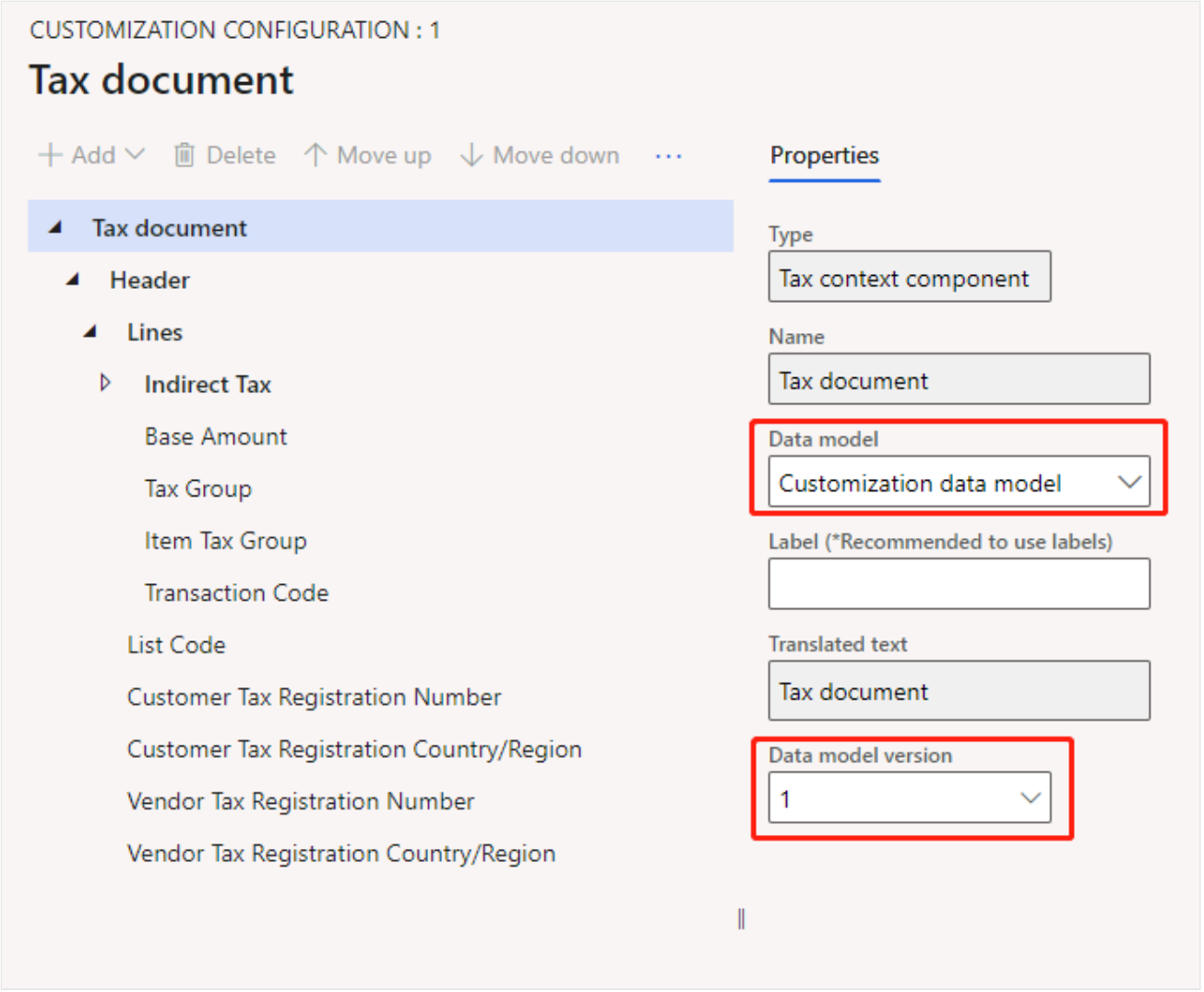
8. Under **Customized Dataverse** and **Customized field**, select a field, and then select **Bind**.



9. Select **Save**, and then select **Complete**.

Create a customized tax configuration

1. In the **Electronic reporting** workspace, select **Tax configurations**, and then select **Tax Calculation Configuration**.
2. Select **Create configuration**.
3. Select **Tax service configuration derived from Name: Tax Calculation Configuration, Microsoft**.
4. In the **Name** field, enter **Customization configuration**.
5. Select **Create configuration**.
6. Select **Customization configuration**, and then select **Designer**.
7. In the **Data model** field, select **Customization data model**.
8. In the **Data model version** field, select the corresponding data model version.



9. Select **Complete**.

Last updated on 03/14/2026

The tax features list is empty on the Tax calculation parameters page

This article provides a resolution for an issue where the list of tax features on the **Tax calculation parameters** page is empty in Dynamics 365 Finance.

Symptoms

In Dynamics 365 Finance, when you try to select a value in the **Name** field under **FEATURE** on the **Tax calculation parameters** page, the list of values is empty.

Cause

The tax feature isn't created in Globalization Studio.

Resolution

To solve this issue, create the tax feature in Globalization Studio.

For more information, see Configure the Tax Calculation feature.

Last updated on 11/22/2024

"Tax code cannot be determined" error in Tax Calculation

This article explains the troubleshooting steps that you can take if you receive a "Tax code cannot be determined" error in Tax Calculation in Dynamics 365 Finance.

Symptoms

You receive the following error message:

> Header/Lines - 1, tax code cannot be determined.

Alternatively, you find the error message in the troubleshooting file, as shown in the following example. For more information, see How to enable debug mode for troubleshooting.

jsonc

```
======================Tax calculation result JSON:==========================
{
    "taxDocument": {
        "Header": [
            {
                "Lines": [
                    {
                        ...
                        "Errors": [
                            {
                                "Code": "TaxSetup20001",
                                "Message": "Header/Lines - 1, tax code cannot be
determined."
                            }
                        ],
                        "Adjustment": null
                    }
                ],
                "Measures": {
                    ...
                },
                ...
            }
        ]
    },
    ...
}
```

Cause

The issue is probably occurring because the tax group and the item tax group don't intersect.

Resolution

To solve the issue:

1. In the troubleshooting file, verify that tax group and item tax group are determined. If the values for `Tax Group` and `Item Tax Group` are blank, the tax group and item tax group aren't determined. If they're determined, the results might be incorrect.

   Here's an example of a troubleshooting file:

   jsonc

   ```
   ======================Tax calculation result JSON:============================
   {
       "taxDocument": {
           "Header": [
               {
                   "Lines": [
                       {
                           "Tax Codes": {},
                           "Measures": {
                               "Tax Group": "Group A",
                               "Item Tax Group": "Group B"
                           },
                           "Adjustment": null
                       }
                   ],
                   "Measures": {
                       ...
                   },
                   ...
               }
           ]
       },
       ...
   }
   ```

2. Verify that the **Override sales tax** option on the **Setup** tab of the sales order line details is enabled.

   - If it's enabled, tax codes are determined by the `Tax group` and `Item tax group` values that you enter on the transaction line. Verify that these values are entered correctly.

- If it isn't enabled, verify that correct values are set for the **Tax group applicability** and **Item tax group applicability** fields. For more information, see "No matching result could be found" error in the Tax Calculation.

3. If the tax group and item tax group are determined correctly, determine whether there's any intersection for them.

   a. In Globalization Studio, go to **Tax features** > **Tax codes and groups** > **Tax group**.

   Expand table

Line.Tax Group	Tax Codes
Group A	A

   b. Go to **Tax features** > **Tax codes and groups** > **Item tax group**.

   Expand table

Line.Item Tax Group	Tax Codes
Group B	B

   If there's no intersection for the tax group and the item tax group, the tax code won't be determined.

Mitigation

1. Go through each step in the Resolution section of this article, and fix the setup as required. If the tax group and item tax group aren't determined correctly, see "No matching result could be found" error in Tax Calculation.

2. If there's no intersection for the tax group and the item tax group, create a new feature version in Globalization Studio, and then fix the setup.

   - Go to **Tax features** > **Tax codes and groups** > **Item tax group**.

   Expand table

Line.Item Tax Group	Tax codes
Group B	A;B

The tax code will be determined as **A**.

Last updated on 11/22/2024

"No matching result could be found" error in the Tax Calculation

This article explains the troubleshooting steps that you can take if you receive the "No matching result could be found" error in Tax Calculation in Dynamics 365 Finance.

Symptoms

You receive the following error message:

> Header/Lines - 1, Tax group applicability, no matching result could be found.

jsonc

```
======================Tax calculation result JSON:==========================
{
    "taxDocument": {
        "Header": [
            {
                "Lines": [
                    {
                        ...
                        "Errors": [
                            {
                                "Code": "TaxSetup20000",
                                "Message": "Header/Lines - 1, Tax group applicability,
no matching result could be found."
                            }
                        ],
                        "Adjustment": null
                    }
                ],
                "Measures": {
                    ...
                },
                ...
            }
        ]
    },
    ...
}
```

Cause

The issue occurs when the feature setup in Globalization Studio is incorrect.

Resolution

1. Download the troubleshooting file. For more information, see How to enable debug mode for troubleshooting.

2. Compare the Tax calculation input with the feature setup to fix the setup issue.

   The following example shows the Tax calculation input.

   jsonc

```
==============================Tax calculation input
JSON:====================================
{
    "TaxableDocument": {
        "Header": [
            {
                "Lines": [
                    {
                        ...
                    }
                ],
                "Business Process": "Sales",
                "Ship From Zip Code": "30159",
            }
        ]
    },
    "Parameter": {
        ...
    },
    "Adjustment": {
        "Lines": {}
    }
}
```

   The following table lists the tax group applicability in Globalization Studio.

   Expand table

Header.Business process	Lines.Business Unit	Header.Ship From Zip Code	Tax Group
Journal			Group A
Sales		30160	Group B

According to the Tax calculation input, the **Business Process** value on the header is **Sales**, and the **Ship From Zip Code** value on the header is **30159**. This input is based on the setup of the applicability rules in Globalization Studio. Because there's no matching line, the error message occurs.

> **Note**
>
> If the value in the applicability rule is blank, the rule is applicable to any value.

Mitigation

Follow these steps to mitigate the error.

1. In Globalization Studio, go to **Globalization features** > **Tax calculation**.
2. Create a new version of the feature.
3. Add a line for the corresponding information.

Expand table

Header.Business process	Lines.Business Unit	Header.Ship From Zip Code	Tax Group
Journal			Group A
Sales		30160	Group B
Sales		30159	Group B

4. Publish the feature setup version.
5. In Dynamics 365 Finance, go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**, and then select the new version.

Last updated on 11/22/2024

How to enable debug mode in Tax Calculation

To enable the debug mode in Tax Calculation in Dynamics 365 Finance, take the following steps:

1. Add *&debug=vs%2CconfirmExit&* to the URL of the Application Object Server (AOS), and then refresh the page.
2. When you select **Sales tax** to calculate the sale tax, a text file that is named *TaxServiceTroubleshootingLog.txt* is opened. The *TaxServiceTroubleshootingLog.txt* file contains `TaxableDocument` and the calculation parameter. These results are returned from the tax calculation and exception information for troubleshooting.

Sample

jsonc

```
==============================Tax calculation input
JSON:================================
{
    "TaxableDocument": {
    "Header": [
        {
        "Lines": [
            {
            "Transaction Line ID": "5816,68719677391",
            ...
            }
        ],
        "Transaction Header ID": "2022,68719506302",
        ...
        }
    ]
    },
    "Parameter": {
    "ContinueOnErrors": true,
    ...
    },
    "Adjustment": {
    "Lines": {}
    }
}
==========================Tax calculation result
JSON:================================
{
    "taxDocument": {
    "Header": [
```

```
        {
        "Lines": [
            {
            "Tax Codes": {
                ...
            }
        ],
        "Measures": {
            "List Code": "EUTrade"
        },
        "Tax Registration ID": null,
        "Transaction Header ID": "2022,68719506302",
        "Errors": null
        }
    ]
    },
    "solutionId": "b51e0025-cbbe-4d37-bf0b-90d7be4f214d|1",
    "isPartialSuccess": false
}
==============================CorrelationId:==============================
"21f3a8a1-ee9a-477b-b44e-b8ec79e74d16"
```

Last updated on 11/22/2024

Tax calculation FAQ

This article provides answers to frequently asked questions about the functionality of the Tax Calculation.

Will the Tax Calculation be available on tier-1 machines and on-premises machines?

Yes.

Do I have to continue to maintain tax codes and sales tax groups in finance and operations and Globalization studio, or can I maintain them in only one place?

You can create and maintain a tax setup in the Tax calculation feature in Globalization studio. When you select the Tax feature on the **Tax calculation parameters** page in finance and operations, the setting synchronizes from Globalization studio to finance and operations.

Do you have any plans to support Russia, Brazil, and India?

Microsoft doesn't currently plan to support Russia, Brazil, or India. For more information about country/region support, see Release plans on Dynamics 365 and Microsoft Power Platform release plans.

Brazil tax reform taxes are supported with Tax calculation, details: Tax Reform Overview.

What is the recommendation for global implementations that involve countries or regions such as India? Can I use the Goods and Services Tax (GST) engine that Microsoft provides and the Tax Calculation on the same instance of Dynamics 365 Finance or Dynamics 365 Supply Chain Management, or do I have to have separate instances?

You can use different tax engines for different legal entities. For example, for a legal entity in India, you can use GST that is based on Global Tax Engine (GTE). Then, for other legal entities, you can enable the Tax Calculation.

If I post a transaction with incorrect tax, how can I correct it?

The correction process is the same as it currently is. If you discover that the tax is incorrect before you post the transaction, you can either make a tax amount adjustment or adjust the applicability rule by changing tax groups. If you discover that the tax is incorrect after you post the transaction, you can either reverse the posted document or post a tax adjustment journal.

What are the triggers for calling the Tax Calculation?

The triggers for the Tax Calculation in finance and operations apps are the same as the current triggers for the existing tax engine.

On the Tax calculation parameters page, why is the list of values in the Feature setup name field blank?

You need to complete and publish the Tax calculation feature in Globalization studio.

Why can't I turn on the Enable advanced tax calculation parameter?

The **Enable advanced tax calculation** parameter isn't yet supported in some countries and regions. Make sure that the primary address of the legal entity is in one of the supported countries/regions.

Last updated on 06/25/2026

Connect to an external tax solution provider via the Universal Tax Rate API

A connection to an external tax solution provider helps simplify and reduce the effort of maintaining the tax rates and tax applicability rules for Tax calculation. This result is especially critical when you implement Tax calculation for countries or regions where a significant number of tax jurisdictions must be covered.

The Universal Tax Rate API is a set of standard application programming interfaces that Microsoft has defined in Tax calculation, based on the tax calculation data model. It's an extended feature of Tax calculation that enables external tax services to be connected under the same framework.

Through this extension, external tax solution providers can provide tax rate determination, tax calculations, and extended tax functionality via standardized APIs. This extension introduces the following benefits:

- **Seamless integration with Microsoft Dynamics 365 applications** – The external tax solution is seamlessly integrated into Dynamics 365 applications. This integration ensures a smooth and unified experience across Dynamics 365 applications.
- **Improved scalability** – By connecting to an external tax solution provider, you can take advantage of its infrastructure and expertise to efficiently handle large-scale tax calculations.
- **Reduced maintenance overhead** – External tax solution providers are responsible for maintaining up-to-date tax rates and tax applicability rules. Therefore, they can help save you time and effort.
- **Enhanced accuracy** – External tax services often have access to comprehensive tax databases and advanced algorithms. Therefore, tax calculations are more accurate.
- **Flexibility and customization** – The standardized APIs let you integrate with various external tax solution providers. Therefore, you can select the one that best fits your business needs.
- **Compliance with local regulations** – External tax solution providers are well-versed in local tax laws and regulations. Therefore, they can ensure that your tax calculations comply with the specific requirements of each jurisdiction.

> ⓘ **Note**

The Universal Tax Rate API shares the same integration of Tax calculation. Different tax solution providers might support different regions, scopes, scenarios, and extended functionality. For more information, contact the tax solution providers.

The following functionality is available via the Universal Tax Rate API that the tax solution providers support:

- **Encrypted connection** – The Universal Tax Rate API supports an encrypted connection via Azure Key Vault. This connection helps ensure the security and privacy of data that's transmitted to external tax solution providers.
- **Address validation** – This functionality lets you validate addresses that are maintained in the global address book. It helps ensure accurate tax calculations by confirming that addresses are valid and complete.
- **Sales tax calculation** – External tax solution providers can use the Universal Tax Rate API to calculate sales tax, based on factors such as the location and the product type.
- **Posted tax transactions** – This functionality enables tax transaction postings in the external tax solution system of the tax solution providers. Tax reporting is generated after the posting in the external tax solution system.
- **Accrual of use tax** – The Universal Tax Rate API enables external tax solution providers to accrue use tax. Use tax is a type of tax that's imposed on the use, storage, or consumption of taxable items or services that weren't taxed at the time of purchase in the United States.

Availability

The Universal Tax Rate API is available in Dynamics 365 Finance or Dynamics 365 Supply Chain Management version 10.0.38 and later.

Versions

We recommend that you import and set up your Tax calculation configuration with the version that matches your Finance or Supply Chain Management version.

Expand table

Finance or Supply Chain Management version	Tax configuration version
10.0.48	Tax Calculation Data Model for ISV Integration 50.78.13
10.0.39	Tax Calculation Data Model for ISV Integration 40.65.5

Enable a tax solution provider

To enable a tax solution provider, follow these steps:

1. Enable the following features in Feature management:

    - Tax calculation service
    - Enable external tax solution providers for Tax calculation service
    - Globalization features
    - Enable Globalization feature setup for Tax calculation service
    - Electronic reporting globalization feature JSON import/export

2. Engage and select one tax solution provider. Install the independent software vendor (ISV) application when the tax solution provider requires it for the complete and extended functionality.

3. Set up a client ID and client secret for the application access credentials that your tax solution provider provides.

4. Import the tax calculation data models provided by Microsoft from the global repository or the Dataverse repository via Globalization Studio.

    - **Configuration providers** > **Microsoft** > **Repositories** > **Global** > **Open** > **Import**
        - Tax Calculation Data Model for ISV integration.xml (version 40.65.5)

    > **Note**
    >
    > If you want to import the tax data model from the Dataverse repository, make sure you have enabled the option **Enable Finance and Operations User Impersonation in Dataverse** under **Power Platform admin center** > **Environments** > **your environment id** > **settings** > **Features**.
    >
    >
    >

5. Load the ISV provided tax configuration and tax features via Globalization Studio. Follow the implementation guidance that your tax solution provider provides.

    - **Tax configurations** > **Exchange** > **Load from XML**
        - ISV tax configuration.xml (version 40.65.5.x)
    - **Tax calculation** > **Import from JSON**
        - ISV tax solution.json (version x)

    > **Note**
    >
    > The ISV distributes the preceding .xml and .json files for load and import into the Tax calculation setup. The name of the files can be vary per different ISV naming rules.

6. In Globalization Studio, select **Tax calculation** > **Add**, and create and configure the customer ISV tax feature for your credentials, parameter values, mapping rules, and so on. Follow the implementation guidance that your tax solution provider provides.
7. Go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**.
8. On the **General** tab, in the **Tax solution provider** section, enable the **Enable tax solution provider** option.
9. Configure the tax solution provider.
10. You can now perform transactions as usual. When Tax calculation is required, the client collects information from the transaction, such as the sales order or purchase order, and it packages that information as a payload. A request is then sent to your tax solution provider to calculate the tax.
11. The calculation result is returned from your tax solution provider to your Finance system and presented on the sales tax transaction page.
12. When you post a sales tax–related voucher in Finance, a Posted tax transaction request is sent to your tax solution provider to record the necessary tax information in its tax solution system for tax reporting.

Supported countries/regions

The Universal Tax Rate API can be enabled for for any country/region.

For more information about supported countries/regions, contact your tax solution providers.

Available tax solution providers

- Vertex (available)
- Avalara (planned)
- Wolters Kluwer (available)
- Thomson Reuters (available)

For additional details on release information, available solutions, and supported countries/regions, we recommend reaching out to your tax solution providers.

Last updated on 03/20/2026

Configure a tax solution provider

This article explains how to configure a tax solution provider for Tax calculation. After you complete the steps according to the implementation guidance that your tax rate provider provides, select your tax feature setup in Tax calculation parameters for the corresponding legal entity.

To configure a tax solution provider, follow these steps:

1. Go to **Tax** > **Setup** > **Tax configuration** > **Tax calculation parameters**.
2. On the **General** tab, in the **Tax solution provider** section, select **Enable tax solution provider**.
3. In the **Feature** section, in the **Name** field, select your tax feature setup.
4. If the feature setup defines new tax codes, the **Maintain mandatory fields for new tax codes** page appears. For each tax code, select a settlement period and a ledger posting group. To bulk update the mandatory fields, select all tax codes and then select **Bulk update**.
5. Select **Confirm**. A log file is downloaded in your browser. This file contains the information for the imported and updated tax codes, tax groups, and item tax groups for the current legal entity.
6. In the **Business process** field, select the processes that access the external tax service for tax calculation.

   > ⓘ **Note**
   >
   > To learn about the supported business processes that you should select, contact your tax solution provider.

7. Select **Enable address validation**, and select the country/region keys that your tax rate provider supports.

   > ⓘ **Note**

To ensure an accurate location-based tax calculation, your tax rate provider validates and corrects the addresses in the master data. After you enable address validation, a **Validate** button is available when you add or maintain the addresses in the master data.

For batch mode validation, go to **Organization administration** > **Global address book** > **Global address book**, and then, on the **Registration** tab, select **Address validation**.

8. For US legal entities, enable the **Accrue use tax** option, and maintain the fields in the **Overcharge tolerance** section to activate the assessment and processing of vendor-charged sales taxes for Accounts Payable (AP) invoicing scenarios. For more information, see Use tax assessment.

Last updated on 03/21/2026

Set up a client ID and a client secret

This article explains how to set up a client ID and a client secret for the Universal Tax Rate API.

> **Note**
>
> You can create a support request to ask Microsoft support to connect you with the Tax Calculation - Universal Tax Rate API product team for detailed guidance on setting up the client ID and client secret in your environment.

Prerequisites

Before you can access a tax provider's service, you must set up your client ID and client secret for the Universal Tax Rate API. Your tax provider provides these values. When you receive them, store them in a Microsoft Azure Key Vault, and configure Key Vault parameters for Dynamics 365 finance and operations apps. After you set up the client ID and client secret in the key vault, you can select the corresponding Key Vault secret name in the tax feature setup.

Create a key vault in the Azure portal

You must store all the secrets and certificates that are used in the Tax Calculation service in an Azure Key Vault. This approach helps ensure that you don't work directly with the secrets, and that the secrets are securely stored. When you must use digital signing or secure a connection to external web services, set the reference to the Key Vault secrets instead of using the secrets and certificates directly. For more information, see About Azure Key Vault.

Create a key vault

To create a key vault, follow these steps:

1. From the Azure portal, open the menu or **Home**, and select **Create a resource**.
2. Enter **Key vault** in the search field.
3. Select **Key vault** in the list of results.
4. In the **Key Vault** section, select **Create**.

5. In the **Create key vault** section, set the following fields:

    - **Subscription** – Select a subscription.
    - **Resource group** – Select a resource group, or select **Create new**.
    - **Name** – Enter a unique name. For this example, use **TaxProvider-vault**.
    - **Location** – Select a location.

    Leave the other fields set to their default values.

    > ⓘ **Note**
    >
    > In the Set up Key Vault parameters in finance and operations apps section of this article, the name that you enter in the **Name** field is referred to as **<KeyVaultName>**.

6. Select **Next**, and set the **Permission model** field to **Vault access policy**.

7. Select **Review + Create**.

8. After the deployment finishes, expand the **Deployment details** section, and select the **Key vault** detail.

    > ⓘ **Note**
    >
    > Copy and save the **Vault URI** value. You must enter it in the **Key Vault URL** field when you complete the procedure in the Set up Key Vault parameters in finance and operations apps section.

9. Select **Next**, and set the **Permission model** field to **Vault access policy**.

10. Select **Secrets**, and then select **Generate/Import**.

11. Set the following fields:

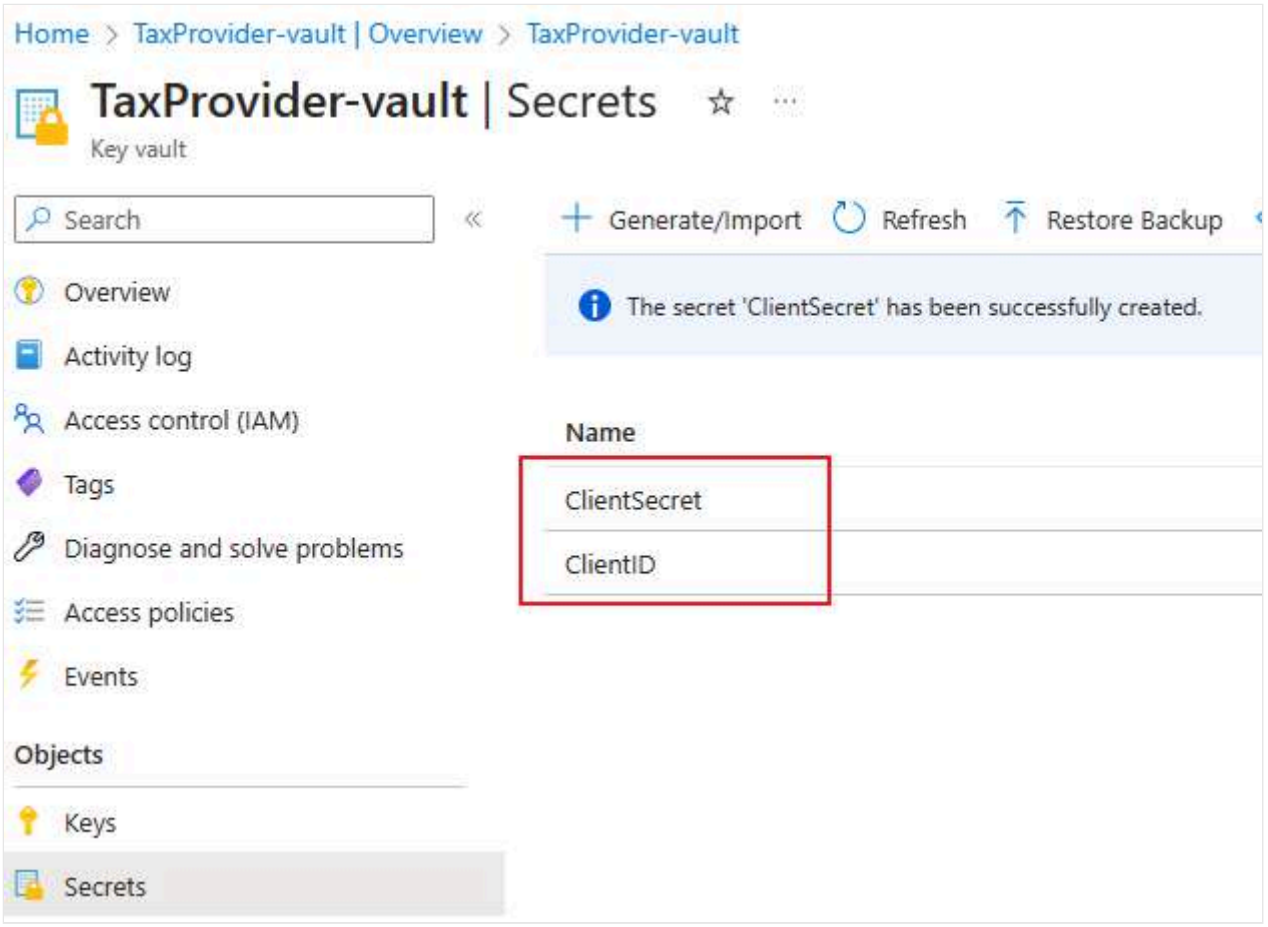
    - **Upload options** – Specify **Manual**.
    - **Name** – Enter a name for the secret. For example, enter **ClientID**.
    - **Secret value** – Enter the client ID that you get from the tax provider.

12. Select **Create**.

    > ⓘ **Note**

> The secret name is a mandatory parameter for integration with the key vault. Therefore, specify it in the application. In the Set up Key Vault parameters in finance and operations apps section, it's referred to as the **<SecretName>** parameter.

13. Repeat step 10 for the client secret.



Set up permissions

This section explains how to complete the following procedures:

- Create an app registration.
- Set up the access policy of the key vault.
- Set up Key Vault parameters in finance and operations apps.

Create an app registration

To access Key Vault, you must create an app registration in Azure Active Directory (Azure AD).

To create an app registration, follow these steps:

1. In Azure AD, search for **register**, and then select **App registrations**.

2. Set the **Name** and **Supported account types** fields, and then select **Register**.

3. Copy and save the **Application (client) ID** value. Specify it in the application. In the Set up Key Vault parameters in finance and operations apps section, it's referred to as the **Key Vault client** parameter.

4. Create a new client secret.

> **Note**
>
> The client secret is a mandatory parameter for integration with the key vault. Copy it and specify it in the application. In the Set up Key Vault parameters in finance and operations apps section, it's referred to as the **Key Vault secret key** parameter.

Set up the access policy of the key vault

Set up the access policy to grant the app registration the correct level of secure access to the secret that you created.

1. Open the Key Vault storage that you created earlier.
2. Go to **Settings** > **Access policies**, and select **Create**.
3. In the **Secret permissions** section, select the **Get** and **List** operations.
4. Select **Next**.
5. In the **Select principal** step, select the app registration that you created earlier.
6. Select **Next** until you reach the **Review + create** step, and then select **Create**.

Set up Key Vault parameters in finance and operations apps

After you finish the prerequisite steps, set up the Key Vault parameters to link to the key vault in finance and operations apps.

1. Go to **System administration** > **Setup** > **Key vault parameters**.

2. Select **New** to create a new instance.

3. Enter a name and description. On the **General** tab, set the following fields that are required for integration with Key Vault storage:

- **Key Vault URL** – Enter the key vault URI that you saved in the Create a key vault section.
- **Key Vault client** – Enter the interactive client ID of the Microsoft Entra ID application that's associated with the Key Vault storage for authentication.
- **Key Vault secret key** – Enter the secret key that's associated with the Microsoft Entra ID application that's used for authentication with the Key Vault storage.

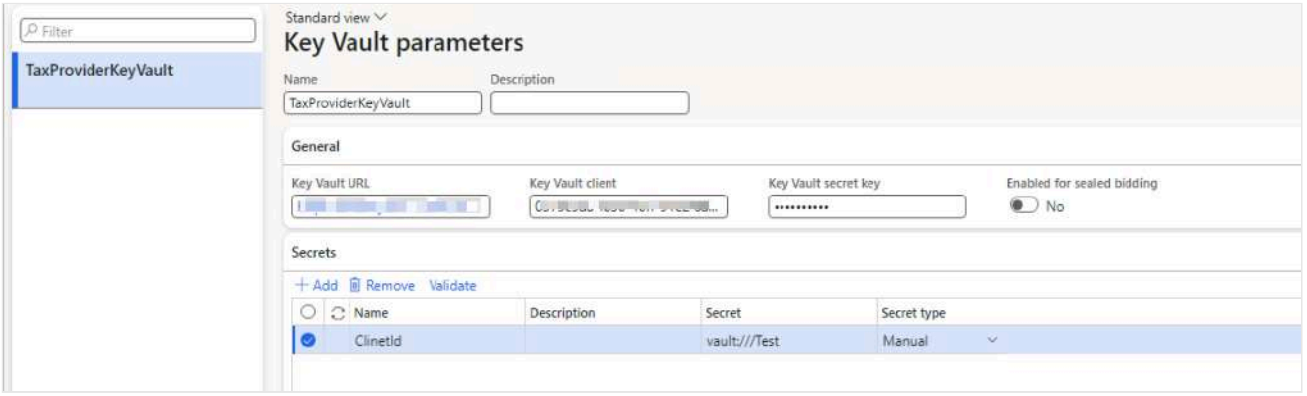
4. On the **Secrets** tab, select **Add** to add your secret. For each secret, set the following fields:

   - **Name**
   - **Description**
   - **Secret** – Enter a secret reference to the Key Vault secret.

   The format of a secret must resemble the following example: **vault://<KeyVaultName>/<SecretName>/<SecretVersion>**.

   The **<KeyVaultName>** and **<SecretVersion>** attributes are **optional**. However, the **<SecretName>** attribute is **required**. In most cases, you can define a Key Vault secret key in the following format: **vault:///<SecretName>**.

   If the secret version isn't defined in the Key Vault secret key, the system retrieves the active secret that has the latest expiration date.



> **Note**
>
> You must create the Key vault parameters in each legal entity that connects to the external tax solution provider for tax calculation.

Last updated on 03/19/2026

Use tax assessment

Use tax assessment is a feature that US legal entities can use to calculate the tax liability for purchases that are subject to use tax. Use tax is a type of sales tax that applies when a taxable item is purchased from an out-of-state vendor or an online retailer that doesn't collect sales tax from the buyer. The buyer is responsible for paying the use tax to the state where the item is used, stored, or consumed.

By using the **Accrue use tax** option in Tax calculation parameters, US legal entities can comply with the tax laws and regulations of their states by automatically generating use tax transactions, based on the purchase transactions and the tax rates of the states.

Enable the **Accrue use tax** option to activate the use tax assessment–related fields and functionality in Accounts Payable (AP) invoicing scenarios. After you enable it, the following fields and parameters are activated for use tax assessment.

- The **Vendor charged sales tax** field is available on the header of a vendor invoice. Use it to record the total sales tax amount that the vendor charges on the invoice. If you import the vendor invoice, the amount in the **Imported sales tax** field automatically carries over to the **Vendor charged sales tax** field.

- The **Accrue sales tax type** field is available in the vendor master and on the header of the vendor invoice. It controls the tax assessment process. Select one option for each vendor in the vendor master to set the default behavior for processing of the **Vendor charged sales tax** value. On the header of the vendor invoice, you can select a different behavior for each invoice transaction.
  - **Default** – Use the default behavior, which ignores the amount in the **Vendor charged sales tax** field. The system calculates and populates the sales tax amount and sales tax transactions through sales tax logic. No other assessment is done.
  - Adjust sales tax
  - **Accrue use tax** – This option is available if you enabled the **Accrue use tax** option according to the guidance in Configure a tax solution provider. The system populates use tax transactions when the **Vendor charged sale tax** value is less than the tax liability that your tax solution provider calculates. For more details, contact your tax solution provider.
  - **Advanced** – Select this option when your tax solution provider supports advanced use tax assessment processes that consider the fields in the **Overcharge tolerance** section. For more details, contact your tax solution provider.

- The fields in the **Overcharge tolerance** section are available in both Tax calculation parameters and the vendor master. Use them to perform advanced tax transaction handling when the **Vendor charged sales tax** value differs from the tax calculation result. For more details, contact your tax solution provider.

Adjust sales tax

The **Adjust sales tax** option is available after you enable the **Adjust sales tax amount per vendor charged sales tax** feature (in version 10.0.39 and later). Select this option if you want to fully accept the sales tax amount that the vendor charges. In this case, the system generates a sales tax adjustment transaction to override the tax calculation result.

> **Note**
>
> The **Adjust sales tax amount per vendor charged sales tax** feature is a standalone core feature. It doesn't require that you enable a tax solution provider. You can enable the feature to activate the **Vendor charged sales tax** and **Accrue sales tax type** fields without connecting to an external tax solution provider.

Last updated on 03/21/2026

Stock transfer orders for India

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use transfer orders to process inventory transfers between warehouses. In India, if the shipping and receiving branches of the organization have different tax registration numbers, calculate and post the India Goods and Services Tax (GST) for the transfer order. You can define the tax base as the current cost price of the item being transferred or a special transfer price. Post the tax amount as GST payable for the transfer shipment and as GST recoverable upon the transfer receipt. Use an **Interim transit** account as an offset account for the posting, and nullify it when the transfer order is fully received.

The **Stock transfer** functionality that's available for India supports this process.

> **Note**
>
> Unrealized profit or loss isn't posted during the shipment or receipt of a stock transfer line if the unit price differs from the inventory cost price of the item. If mutual settlements and profit or loss transactions are required for internal transfers between your own company's warehouses, use other functionality. For example, you can configure purchase orders and sales orders invoices or intercompany invoices for intracompany transfers.
>
> For more information, see Set up intercompany trade.

Set up stock transfers

Enable stock transfer functionality

In the **Feature management** workspace, enable the following features:

- *(India) Improvements in unit price and cost price handling in Stock transfer orders* (As of Supply Chain Management version 10.0.43, this feature is turned on by default. As of version 10.0.45, it's mandatory and can't be turned off.)
- *Enable uniform tax amount and GST transaction ID for both shipment and receipt transaction of a stock transfer order* (As of version 10.0.43, this feature is mandatory and can't be turned off.)

For more information, see Feature management overview.

You also need to configure the India GST functionality to enable GST calculation for stock transfer orders. For more information, see India Goods and Services Tax (GST) overview.

Set up default transfer type and price type for transfer orders

You can define a default transfer order type and a default price type for transfer orders that you create manually. On the **Inventory and warehouse management parameters** page, on the **Transfer orders** tab, select **Stock transfer** in the **Transfer type** field to enable the **Stock transfer** functionality for all newly created transfer orders. In the **Price type** field, select a default price type for newly created stock transfer orders:

- **Cost price** – The cost price, or the on-hand price, of the item is used for stock transfer orders.
- **Transfer price** – The transfer price that you set up for the item is used for stock transfer orders.

> ⓘ **Note**
>
> For the **Cost price** price type, the system calculates a running average cost price, based on the inventory dimensions that you specify for an item line. The system stores this running average price in the **Cost price** field when you create or update the line. The system never recalculates the saved cost price when you post shipping and receiving transactions.

Configure Master planning parameters

You can define a default transfer order type and a default price type for transfer orders that you create when confirming planned orders in the **Master planning** module. On the **Master planning parameters** page, on the **Standard update** tab, select **Transfer type** and **Price type** in the **Firm - Transfer** group.

Set up item master parameters

To calculate GST for an item in a stock transfer order, you need to configure certain parameters on the **Released product details** page, such as **HSN codes** and **Tax rate type**. For more information, see Assign HSN codes and SACs to products.

Configure transfer pricing

You can configure the prices that the system uses when processing stock transfers for an item. To do this, open the **Transfer price** page from the **Released product details** page.

Configure stock transfer posting

You must configure the **Interim transit** account. Complete this configuration on the following pages:

- In the **Inventory management** module, on the **Posting** page, on the **Transfer order** tab, specify the **Interim transit** account.
- On the **Tax setup** page, select **Setup**, and on the **Setup** page specify the **Interim transit for stock transfer** account for all GST components that you can create for GST on stock transfer. For more information, see Define main accounts.

> ⓘ **Note**
>
> The accounts that you use to post interim transit amounts should have **Interim transit** specified in the **Posting type** field on the **Posting validation** tab of the **Main accounts** page.

You must also set up main accounts to post inventory cost for transfer orders. Specify the following accounts from the **Inventory** tab on the **Posting** page in **Inventory management**:

- **Inventory issue**
- **Inventory receipt**
- **Inter-unit payable**
- **Inter-unit receivable**
- **Inventory expenditure, loss**

> ⓘ **Note**

The **Inventory expenditure, loss** account is required to post the scrap amount when receiving a transfer order with scrap.

Configure posting to General ledger based on financial dimension links to sites

In the standard finance and operations functionality, you can post transfer orders to General ledger only if *both* of the following conditions are met:

- You enable a dimension link, so there's a link between the **Site inventory** dimension and some financial dimension.
- The transfer order movement occurs between warehouses that belong to different sites. This condition applies to the **From**, **Transit**, and **To** warehouses. If the **From** warehouse and **Transit** warehouse belong to the same site, the transfer order shipment isn't posted to General ledger. Similarly, if the **Transit** warehouse and **To** warehouse belong to the same site, the transfer order receipt isn't posted to General ledger.

If you must post stock transfers to General ledger and track inventory in transit warehouses, enable a dimension link and create separate sites for transit warehouses.

For more information, see Configure and manage financial dimension links to sites.

> **Note**
>
> Financial dimensions come from product master data and the dimension link. You can't change financial dimensions in transfer orders.

Create and post a stock transfer order

The following example scenario shows frequently performed actions that are associated with transfer orders where the **Transfer type** field is set to **Stock transfer** and the dimension link is enabled.

1. Go to **Inventory management** > **Outbound orders** > **Transfer order** and create a new transfer order.

2. In the **From warehouse** field, select the supply warehouse that the items are dispatched from.

3. In the **To warehouse** field, select the receiving warehouse that the items are delivered to.

> ⓘ Note
>
> If the **From** warehouse and **Transit** warehouse belong to the same site, the transfer order shipment isn't posted to General ledger. Similarly, if the **Transit** warehouse and **To** warehouse belong to the same site, the transfer order receipt isn't posted to General ledger.

4. In the **Transfer type** field, select **Stock transfer** to apply GST to the transfer of items.

> ⓘ Note
>
> If you select **Transfer order** in the **Transfer type** field, the transfer order is posted based on the standard transfer order process.

5. In the **Price type** field, select a default price type for transfer order lines.

6. On the **Transfer order lines** tab, create a new line. In the **Item number** field, select the item to transfer.

7. In the **Transfer quantity** field, enter the quantity of the items to transfer. In the **Unit** field, change the default unit of measurement, if required.

> ⓘ Note
>
> Changes to the unit of measure of stock on a transfer order line are restricted.

8. In the **Price type** field, select the price type for the transfer order line, from the following options:

   - **Cost price** – The cost price, or the on-hand price, of the item is used for the transfer order line.
   - **Transfer price** – The transfer price that is set up for the item is used for the transfer order line.

> ⓘ Note

If the **Price type** field is set to **Transfer price**, the **Transfer quantity** field shows the quantity of the items that are defined for the combination of an item and a dimension on the **Transfer price** page. This value can be modified.

9. In the **Unit price** field, enter the cost price or the transfer price for one unit of the item.

> ⓘ **Note**
>
> The unit price on a stock transfer order line is recalculated based on the inventory dimensions that are specified on the line until the first shipment is posted for the line.
>
> Shipment of a stock transfer order line that has a unit price of 0 (zero) can be controlled through the **Prevent shipment with zero unit price** field in Inventory and warehouse management parameters.

10. Select **Tax information** to review the tax configuration for the transfer order.
    - At the line level, on the **Invent transfer order from location** tab, you find tax information that is linked to the shipment (from warehouse).
    - At the line level, on the **Invent transfer order to location** tab, you find tax information that is linked to the receipt (to warehouse).
11. Go to **Inquiries** > **Tax document**, and verify that the tax is calculated correctly.
12. Select **Ship** > **Ship transfer order**. Then, on the **Shipment** page, post the transfer order shipment:
    a. Select the **Explode lines** checkbox, and then, in the **Update** field, select **All**.
    b. Select **Setup** > **Tax document**, and review the tax settings and amounts. Make any adjustments that are required.
    c. Close the **Tax document** page, and then select **OK** on the **Ship** page to post the operation.
13. Select **Receive** > **Receive**. Then, on the **Receive** page, post the transfer order receipt:
    a. Select the **Explode lines** checkbox, and then, in the **Update** field, select **All**.
    b. Select **Setup** > **Tax document**, and review the tax settings and amounts. Make any adjustments that are required.
    c. Close the **Tax document** page, and then select **OK** on the **Receive** page to post the operation.

> **Note**
>
> If the "Enable uniform tax amount and GST transaction ID for both shipment and receipt transaction of a stock transfer order" feature is enabled in the **Feature management** workspace, only a previously posted shipment can be received. You must select **Shipment** in the **Update** field when you post a receipt, and you must select a previously posted shipment in the **Shipment voucher** field.

You can also cancel a previously posted stock transfer order shipment if no receipts are posted for this order. On the **Transfer orders** page, select **Transfer order** > **Transfer order history**. On the **Transfer order history** page, select a previously posted shipment. Select **Cancel**, and confirm the cancellation of the shipment. The shipment is canceled, and all inventory movements and GST that was posted for the shipment are reversed. The "Transfer Order Cancellation" feature in the **Feature management** workspace must be enabled to cancel transfer order shipments.

The rest of this article provides several examples that show how to post stock transfer orders under different conditions.

Example 1: Standard transfer order posting

This example uses the following preconditions:

- Current inventory cost price at the moment when the shipment is posted = 100 Indian rupees (Rs.)
- From warehouse = Site 1
- Transit warehouse = Site 2
- To warehouse = Site 3

Transfer order shipment posting

Expand table

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory issue	Site 1		100
Inventory inter-unit receivable	Site 1	100	
Inventory inter-unit payable	Site 2		100

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory receipt	Site 2	100	

Transfer order receipt posting

Expand table

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory issue	Site 2		100
Inventory inter-unit receivable	Site 2	100	
Inventory inter-unit payable	Site 3		100
Inventory receipt	Site 3	100	

You can calculate the balance on the inventory in transit as *InventoryIssue-Site2* – *InventoryReceipt-Site2*. The receipt nullifies this balance.

Example 2: Standard transfer order posting without a transit site

This example uses the following preconditions:

- Current inventory cost price when you post the shipment = 100 Rs.
- From warehouse = Site 1
- Transit warehouse = Site 1
- To warehouse = Site 3

Transfer order shipment posting

No posting to the general ledger occurs.

Transfer order receipt posting

Expand table

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory issue	Site 1		100
Inventory inter-unit receivable	Site 1	100	
Inventory inter-unit payable	Site 3		100
Inventory receipt	Site 3	100	

In this case, the system **doesn't track an inventory in transit balance**.

Example 3: Stock transfer order that has tax on the transfer price (GST)

This example uses the following preconditions:

- Current inventory cost price when you post the shipment = 100 Rs.
- Unit price that is specified on the line = 120 Rs. This price can be the inventory cost price at the moment when the line was created or updated, or it can be the transfer price of the item.
- Taxable value (base amount) = 120.00.
- GST rate = 10 percent.
- From warehouse = Site 1.
- Transit warehouse = Site 2.
- To warehouse = Site 3.

Transfer order shipment posting

Expand table

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory issue	Site 1		100
Inventory inter-unit receivable	Site 1	100	
Inventory inter-unit payable	Site 2		100
Inventory receipt	Site 2	100	

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
GST payable	Site 1		12
Interim account	Site 1	12	

Transfer order receipt posting

Expand table

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory issue	Site 2		100
Inventory inter-unit receivable	Site 2	100	
Inventory inter-unit payable	Site 3		100
Inventory receipt	Site 3	100	
GST receivable	Site 3		12
Interim account	Site 3	12	

The balance on the inventory in transit can be calculated as *InventoryIssue-Site2* – *InventoryReceipt-Site2*. The balance is nullified upon the receipt.

> **Note**
>
> - The **Inventory issue** and **Inventory receipt** accounts are probably the same balance account, such as **Finished goods**.
> - The settings for dimension links and sites (that is, a separate site for the transit warehouse) are recommended if you must track the in-transit inventory in General ledger.

Example 4: Stock transfer order with standard cost

This example uses the following preconditions:

- Create an item that uses the **Standard cost** inventory model.

- Create and activate different item prices for the item, where **Costing type** = **Standard cost** and **Price type** = **Cost**:
  - Price = 115 for Site 1
  - Price = 112 for Site 2
  - Price = 111 for Site 3
- The inventory cost price is defined by the standard cost on the **From warehouse** site and equals 115 Rs. in this case.
- From warehouse = Site 1.
- Transit warehouse = Site 2.
- To warehouse = Site 3.

Transfer order shipment posting

Expand table

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory issue	Site 1		115
Inventory inter-unit receivable	Site 1	115	
Inventory inter-unit payable	Site 2		115
Inventory receipt	Site 2	112	
Cost change variance	Site 2	3	

Transfer order receipt posting

Expand table

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory issue	Site 2		112
Inventory inter-unit receivable	Site 2	112	
Inventory inter-unit payable	Site 3		112

Ledger account name	Financial dimension linked to site	Debit amount (Rs.)	Credit amount (Rs.)
Inventory receipt	Site 3	111	
Cost change variance	Site 3	1	

You can calculate the balance on the inventory in transit as *InventoryIssue-Site2* – *InventoryReceipt-Site2*. The balance is nullified upon receipt.

Last updated on 05/02/2026

Russia overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about functionality that supports Russian regulations.

General ledger

- Correspondence of accounts
- Preview ledger transactions
- Set up exchange rates for currency transactions
- Import currency exchange rates
- Ledger translation
- Turnover balance statement reports
- Account activity reports
- White paper: Dynamics 365 Finance and Dynamics 365 Supply Chain Management-Federal Accounting Standard 27-2021 in Russia
- Enable Electronic reporting destinations to store Russian accounting documents

Advance holders

- Russia advance holders overview
- Create an advance holder
- Daily operations for advance holders
- Advance reports with budget control
- Revaluate foreign currency for advance holders
- Advance holder reports

Cash and Bank

- Set up bank accounts

- Set up and process payment orders
- Exchange adjustment
- Foreign currency sales, purchases, and transfers
- Petty cash
- Cash - Local primary forms and unified reports
- Client-Bank interface and reconciliation procedure

VAT

- Local specific posting of sales VAT
- Invoice factures processing and printing
- Revision factures and corrective factures
- Value-added tax (VAT) for tax agents
- VAT declaration
- Sales books, purchase books, and invoice-factures journals
- Prepayments management
- Incoming VAT restoring
- Process invoice factures for export trade

Fixed assets (Russia)

- Fixed assets overview

Set up Fixed assets

- Set up fixed assets
- Set up depreciation
- Depreciation methods
- Depreciation bonuses
- Set up fixed asset locations and numbering

Working with fixed assets

- Create fixed assets (Russia)
- Register fixed assets acquisitions
- Acquiring fixed assets and putting them into operation
- Calculate depreciation
- Partial fixed asset disassembly (liquidation)
- Sell, dispose, and write-off assets

- Create fixed asset lease and return from lease
- Maintain fixed assets
- Fixed asset counting
- Fixed asset currency revaluation
- Revaluate fixed asset cost and depreciation
- Create and post budget journals for fixed asset acquisitions
- Transfer fixed assets

Not valuable fixed assets, and working clothes

- Not valuable fixed assets
- Working clothes/special rigging

Printing forms

- Printing forms for fixed assets

Accounts payable and receivable

- Credit note corrections
- Invoices for payment
- Set up dimension control for settlements
- Register goods shipped to customers
- Cash flow management
- Allocation of miscellaneous charges in proportion to weight and volume
- Third party miscellaneous charges
- Agreements
  - Set up and create agreements
  - Register transactions with reference to agreements
  - Inquiries and reports with agreements
- Transactions through intermediary
  - Purchases on commission
  - Sales on commission
- Local reports and primary forms in Accounts receivable and Accounts payable

Inventory

- Goods in transit from vendor
- Customs declaration numbers

- Inventory cost structure
- Inventory adjustment wizard
- Inventory reports
- Inventory profile
  - Set up an inventory profile
  - Use an inventory profile in documents and queries
- Customs clearance
  - Work with customs clearance

Customs

- Set up customs clearance
- Work with customs clearance

Reporting

- Financial reporting
- Configure financial reports in Excel
- Accounting reporting in electronic format
- Assessed tax declaration
- Transport tax declaration
- Land tax declaration
- Profit tax declaration

Organization administration

- Registration IDs
- Set up officials
- Company divisions
- Russian address formats and import from FIAS

Deferrals

- Set up deferrals
- Create or generate referrals
- Receipt of deferrals
- Write off deferrals
- Dispose of deferrals

- Deferrals counting
- Reporting for deferrals

Tax accounting

- Create tax registers and the tax register journal
- Expense and income codes
- Assets tax registers
- Deferrals tax register
- Cash flow tax registers
- Rated expenses registers
- Accounts payable debt tax registers and debt write-offs
- Bad debt reserves in Accounts receivable
- Calculation of tax asset and tax liability registers
- Goods movement tax registers
- Other tax registers

Additional resources

- Electronic reporting overview
- Download Electronic reporting configurations from Lifecycle Services

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Last updated on 05/12/2026

Goods in transit from vendor (Russia)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to work with the Counting act for goods in transit (INV-6) report for Russia in Microsoft Dynamics 365 Finance.

The item counting process (INV-6) is used to identify the quantity and value of inventory items that are in transit when inventory is counted.

The item counting process is done in the context of nomenclatures, shipping documents, and vendors.

Set up an inventory profile for a transferable item

To reflect the goods and materials that are in transit, you can set the kind of inventory for the **Basic** kind of activity. The **Kind of inventory** field can have the following values:

- Common
- Purchased items in route

To set up an inventory profile for a transferable item, follow these steps:

1. In Dynamics 365 Finance, go to **Inventory management** > **Setup** > **Dimensions** > **Inventory profiles**.
2. Select **New** to create an inventory profile for an item.
3. In the **Inventory profile** field, enter the name of the inventory profile.
4. In the **Name** field, enter a description of the inventory profile.
5. On the **Setup** FastTab, in the **Kind of activity** field, select **Basic**.

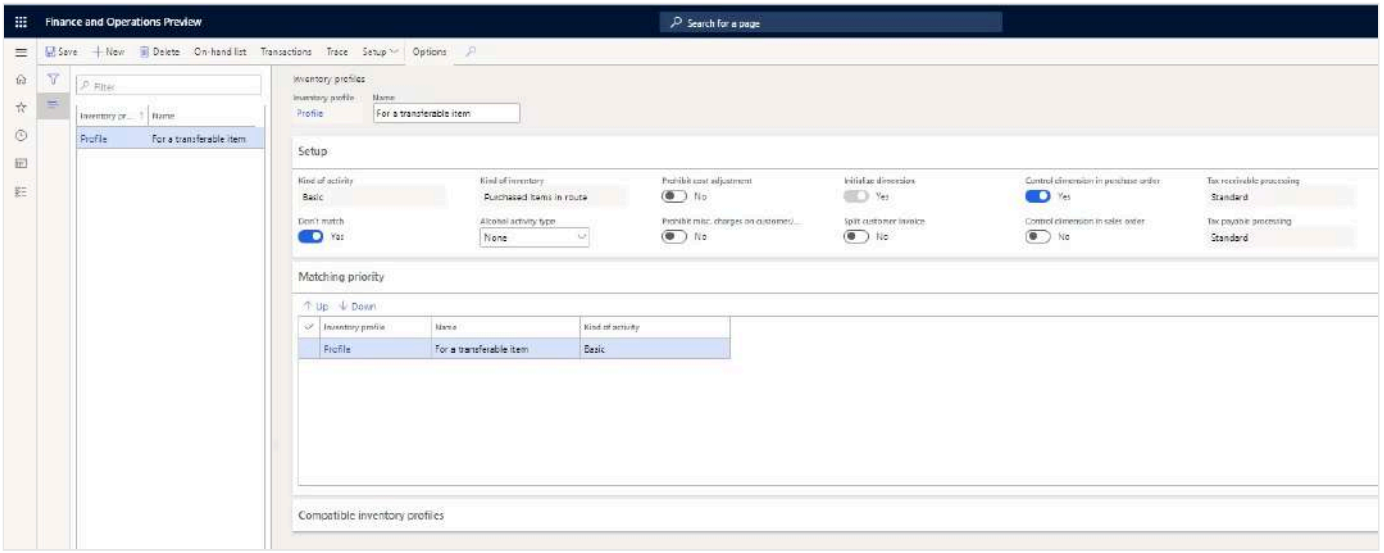
6. Set the **Don't match option** to **Yes** to prevent on-hand inventory that uses this inventory profile from being automatically matched with a compatible inventory profile.

7. In the **Kind of inventory** field, select **Purchased items in route**. The **Initialize dimension** and **Control dimension in purchase order** options are automatically set to **Yes**.

   > ⓘ **Note**
   >
   > The **Kind of inventory** field is available only if you select **Basic** in the **Kind of activity** field.

8. On the **Matching priority** FastTab, select **Up** or **Down** to change the order of the inventory profile.

9. Select **Save**, and close the page.



Set up a number sequence for the Counting act (INV-6) report

To set up a number sequence for the Counting act (INV-6) report, follow these steps:

1. In Dynamics 365 Finance, go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. On the **Number sequences** tab, in the **Number sequence code** field, select a number sequence for the **Counting act (INV-6)** reference.
3. Select **Save**, and close the page.

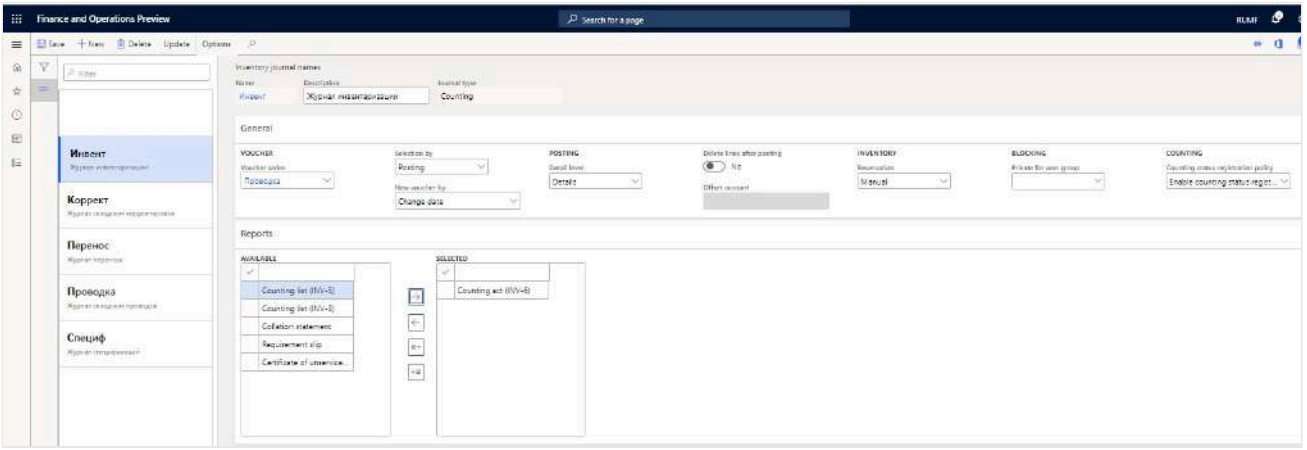
Set up an inventory journal name for the Counting act (INV-6) report

To set up an inventory journal name for the Counting act (INV-6) report, follow these steps:

1. In Dynamics 365 Finance, go to **Inventory management** > **Setup** > **Journal names** > **Inventory**.
2. Create a journal name.
3. In the **Name** field, enter the inventory journal name.
4. In the **Description** field, enter a description.
5. In the **Journal type** field, select **Counting**.
6. On the **Reports** FastTab, in the **Available** field, select **Counting act (INV-6)**, and then select the left arrow button to move the report to the **Selected** field.

   > **Note**
   >
   > If the **Available** field is blank, select **Update** on the Action Pane to update the list of available reports.

7. Select **Save**, and close the page.



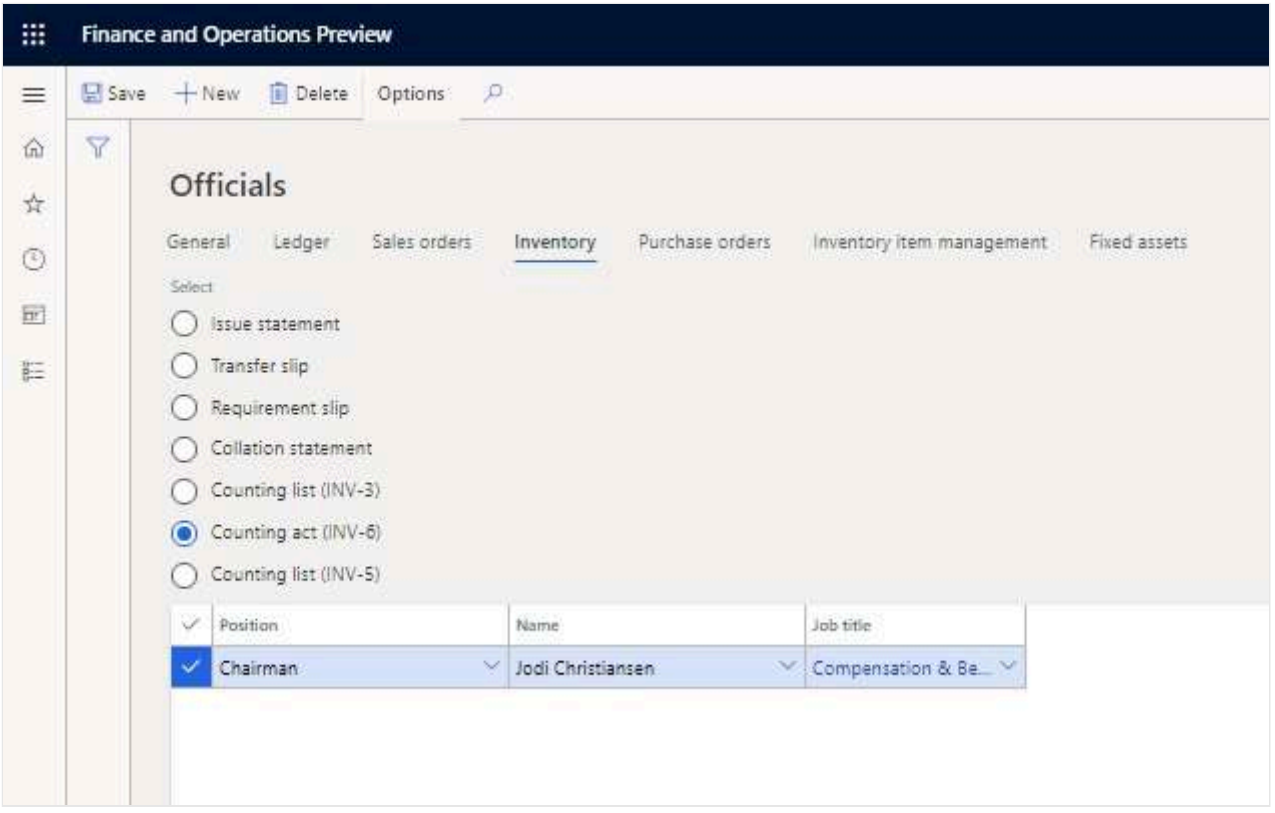
Add officials to the Counting act (INV-6) report

You can specify the company officials who are involved in the item counting process for the **Counting act (INV-6)** report.

To add officials to the Counting act (INV-6) report, follow these steps:

1. In Dynamics 365 Finance, go to **Organization administration** > **Setup** > **Contacts** > **Officials**.
2. On the **Inventory** tab, select the **Counting act (INV-6)** option.
3. Select **New** to create an official for the **Counting act (INV-6)** report.
4. In the **Position** field, select the designation of the official.
5. In the **Name** field, enter the name of the official.
6. In the **Job title** field, select the job title of the official.

> **Note**
>
> By default, the job title is copied from the selected employee's record.



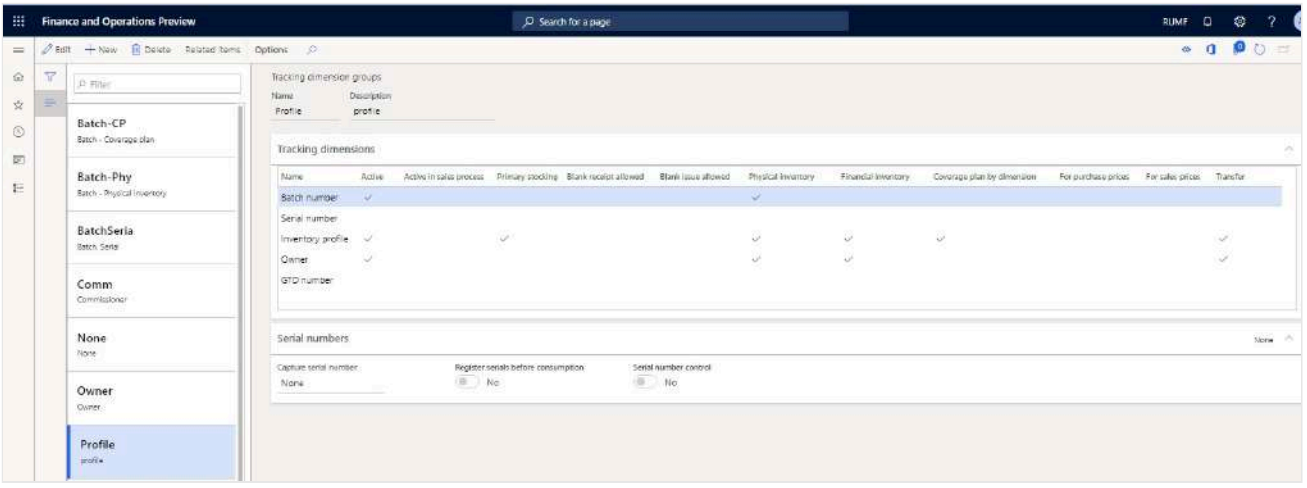
7. Select **Save**, and close the page.

Create an inventory tracking dimension group for the Counting act (INV-6) report

To create an inventory tracking dimension group for the Counting act (INV-6) report, follow these steps:

1. In Dynamics 365 Finance, go to **Product information management** > **Setup** > **Dimension and variant groups** > **Tracking dimension groups**.
2. Select **New** to create a dimension group.
3. In the **Name** field, enter the name of the dimension group.
4. In the **Description** field, enter a description.
5. On the **Tracking dimensions** FastTab, on the **Inventory profile** line, select the **Active**, **Primary stocking**, **Physical inventory**, **Financial inventory**, **Coverage plan by dimension**, and **Transfer** checkboxes.
6. On the **Owner** line, select the **Active**, **Physical inventory**, **Financial inventory**, and **Transfer** checkboxes.

   > **Note**
   >
   > To make the report reflect the details of shipping and payment documents, on the **Batch number** line, select the **Active** and **Physical inventory** checkboxes.

7. Select **Save**.

Create a purchase agreement

To create a purchase agreement, follow these steps:

1. In Dynamics 365 Finance, go to **Accounts payable** > **Purchase orders** > **Purchase agreements**.

2. Select **New** to open the **Create purchase agreement** dialog.

3. On the **Vendor** FastTab, in the **Vendor account** field, select the vendor account.

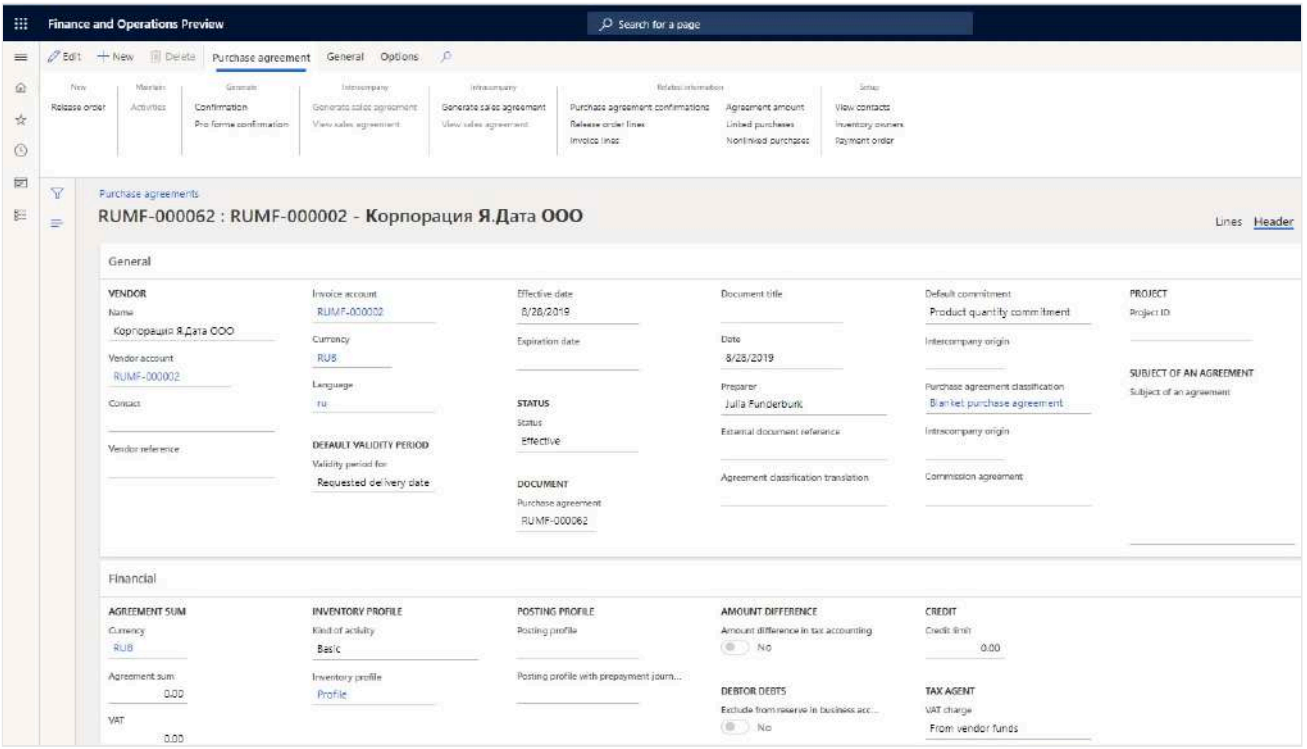
4. In the **Purchase agreement classification** field, select **Blanket purchase agreement**.

5. On the **General** FastTab, in the **Document** section, in the **Purchase agreement** field, enter the ID of the purchase agreement.

6. Specify other details, and then select **OK**.

7. On the **Purchase agreements** page, switch to the **Header** view, and then, on the **Financial** FastTab, in the **Inventory profile** section, set the following fields:

    - In the **Kind of activity** field, select **Basic**.
    - In the **Inventory profile** field, select the inventory profile that you created earlier.

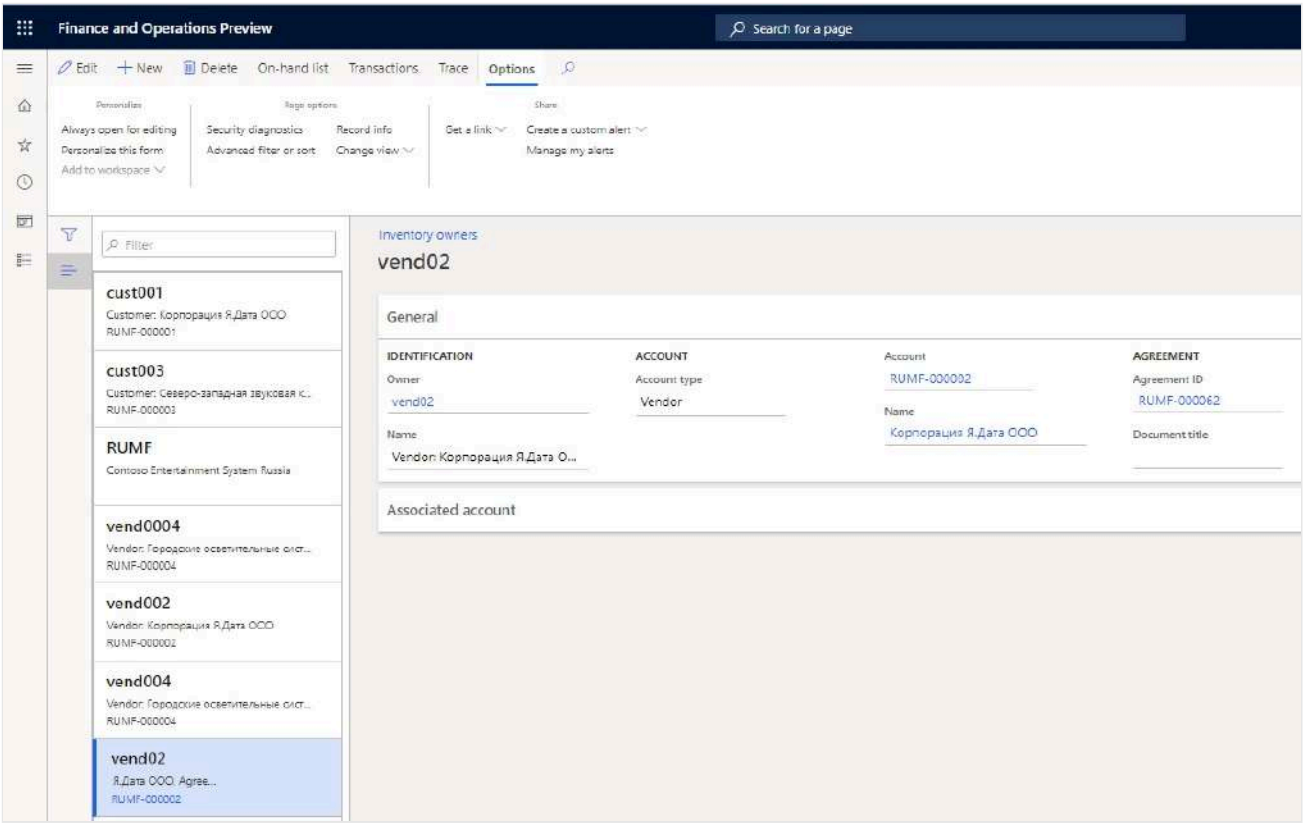


8. On the Action Pane, on the **Purchase agreement** tab, in the **Generate** group, select **Confirmation** to update the status of the purchase agreement to **Effective**.

Add an inventory owner to the Counting act (INV-6) report

To add an inventory owner to the Counting act (INV-6) report, follow these steps:

1. In Dynamics 365 Finance, go to **Inventory management** > **Setup** > **Dimensions** > **Inventory owners**.
2. Select **New** to add an inventory owner.
3. In the **Owner** field, enter the owner code.
4. In the **Account type** field, select **Vendor**.
5. In the **Account** field, select the principal code. The **Name** field is filled in automatically.
6. In the **Agreement ID** field, select the purchase agreement that you created earlier. In this way, you associate the new owner with the agreement.



7. Select **Save**.

Register goods and materials that are in transit

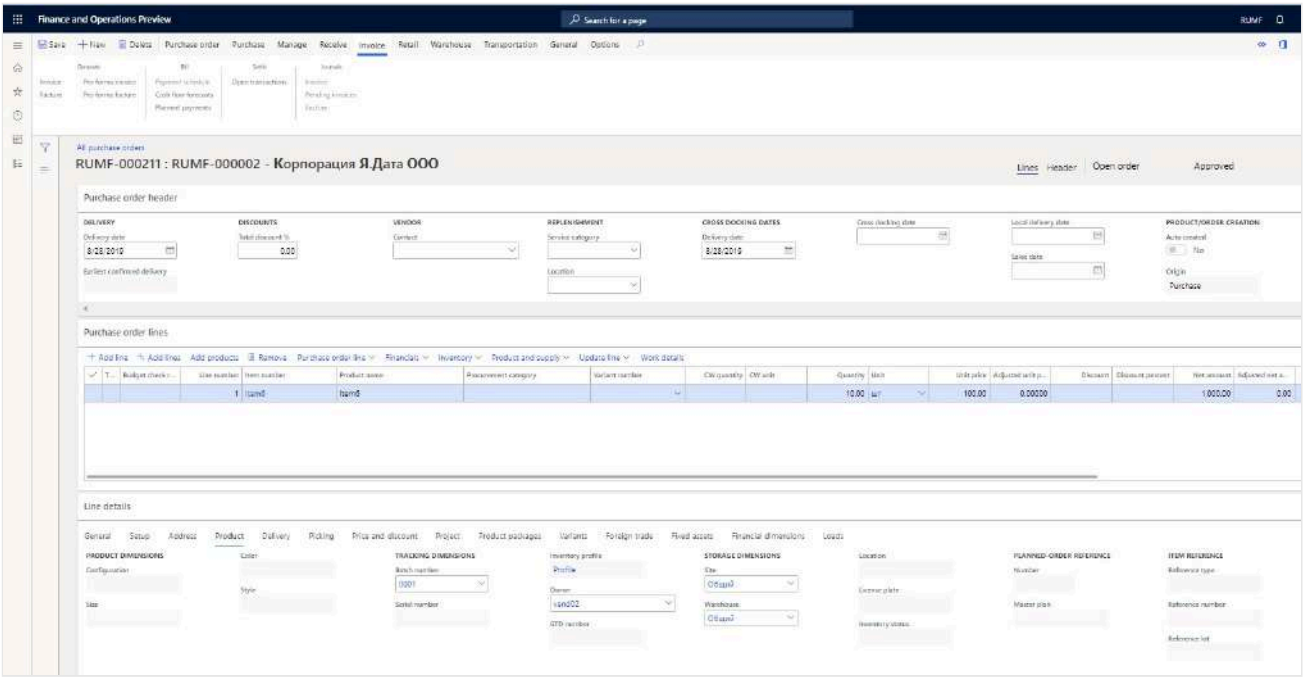
To register goods and materials that are in transit, follow these steps:

1. Create a new purchase order. In the **Purchase agreement** field, select the purchase agreement that you created earlier, and then, on the purchase order line, in the **Item number** field, select the item number.

> **Note**
>
> The **Tracking dimension** field for the item should be set to the inventory profile that you created earlier.

2. On the **Line details** FastTab, on the **Product** tab, in the **Tracking dimension** section, validate that the **Inventory profile** field is set to the inventory profile that you created earlier.

3. In the **Owner** field, select the record that you created earlier.



4. Post the invoice in the usual way.

Generate a Counting act (INV-6) report

You can use the **Print of counting act (INV-6)** dialog to generate a **Counting act (INV-6)** report as a Microsoft Excel file. You must generate this report to track items that are transferred between warehouses and to create a counting list of the items in the transfer that have been purchased. The counting list can contain on-hand inventory holdings.

To generate a Counting act (INV-6) report, follow these steps:

1. In Dynamics 365 Finance, go to **Inventory management** > **Journal entries** > **Item counting** > **Counting**.

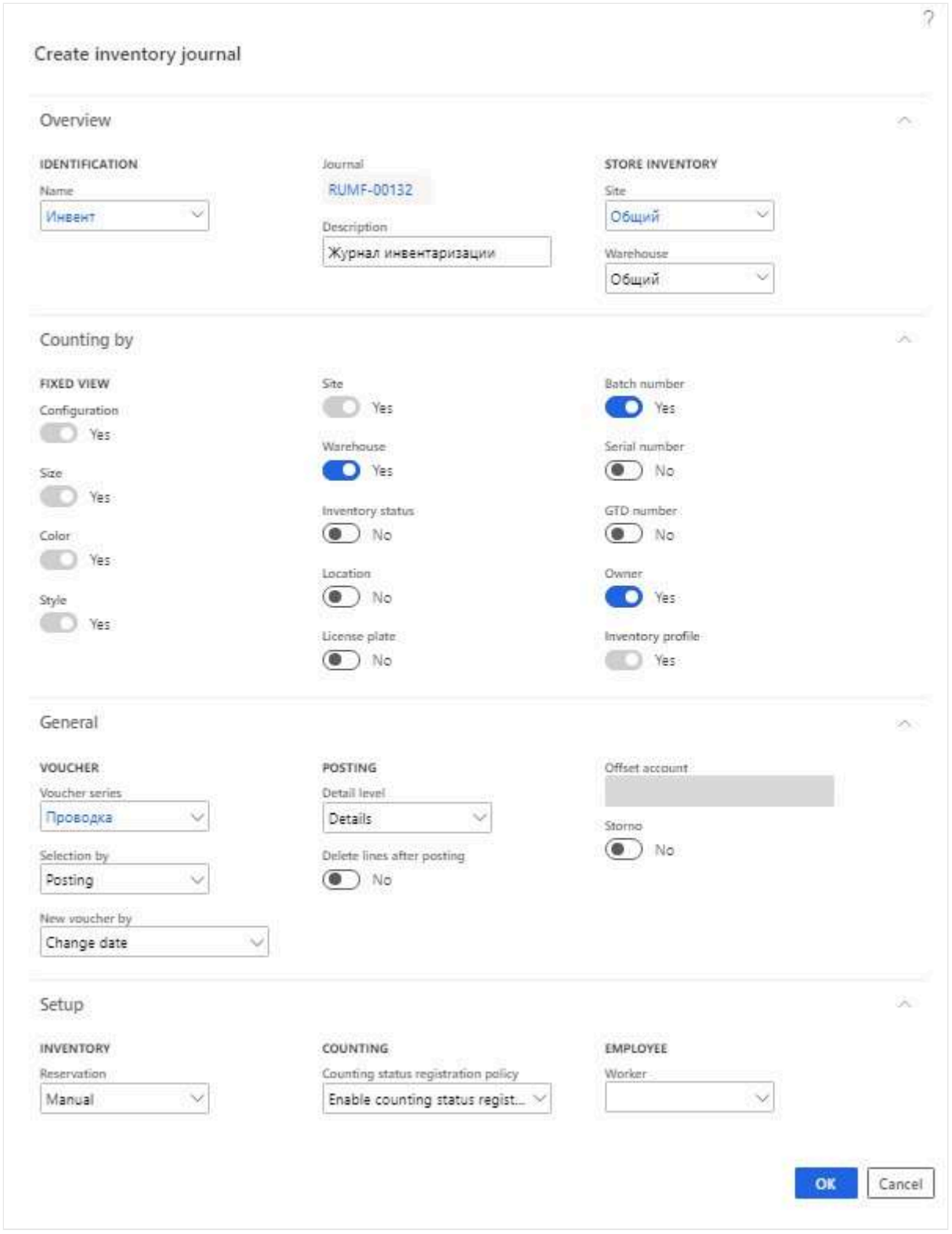
2. Select **New** to open the **Dimensions display** dialog.

3. On the **Overview** FastTab, validate that the **Name** field is set to the inventory journal name that you created earlier.

4. In the **Store inventory** section, set the **Site** and **Warehouse** fields.

5. On the **Counting by** FastTab, validate that the **Inventory profile** and **Owner** options are set to **Yes**.

> **Note**
>
> You can also set the **Batch number** and **Warehouse** options to **Yes**, if those dimensions are applicable.

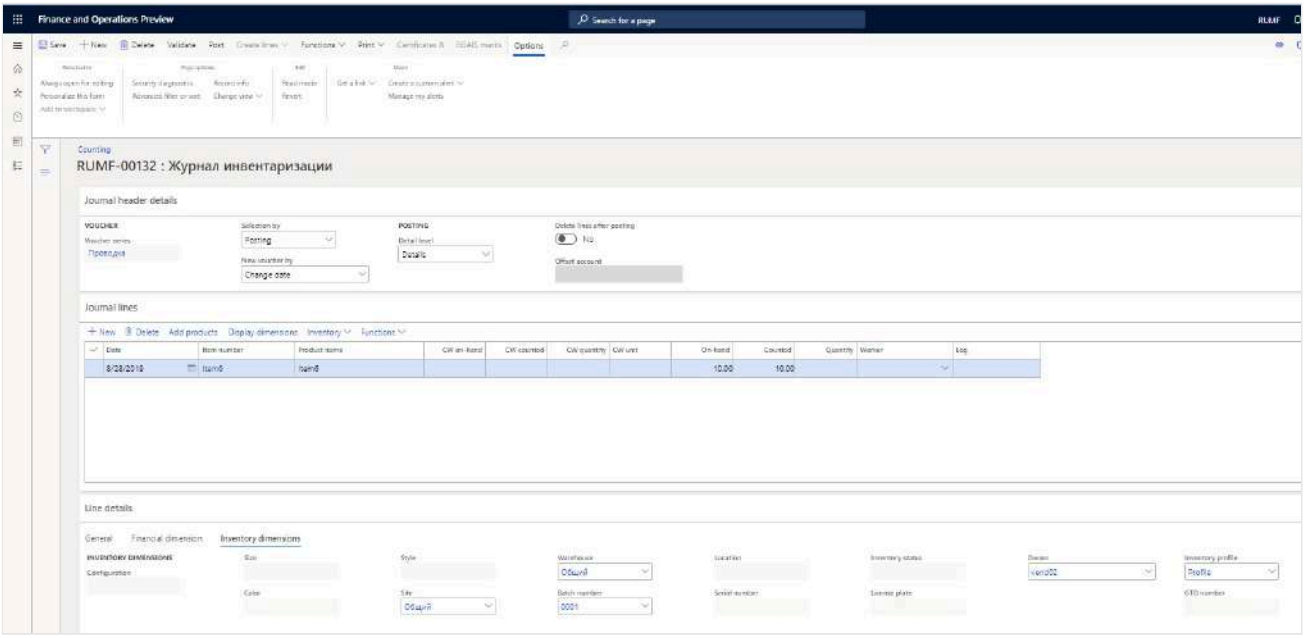


6. Select **OK** to create an inventory counting journal for the items in the transfer.

7. On the **Counting** page, in the **Journal lines** FastTab, create a line, and select the item number that you created earlier.

8. On the **Line details** FastTab, on the **Inventory dimensions** tab, in the **Inventory profile** and **Owner** fields, select the records that you created earlier. The **On-hand** field on the journal line is automatically updated.

9. On the **Journal lines** FastTab, select **Functions** > **Create counting list** to create a counting list. For each line, the **Counted** field is updated with the value that is specified in the **On-hand** field.



10. On the Action Pane, select **Print** > **Counting act (INV-6)** to open the **Print of counting act (INV-6)** dialog.

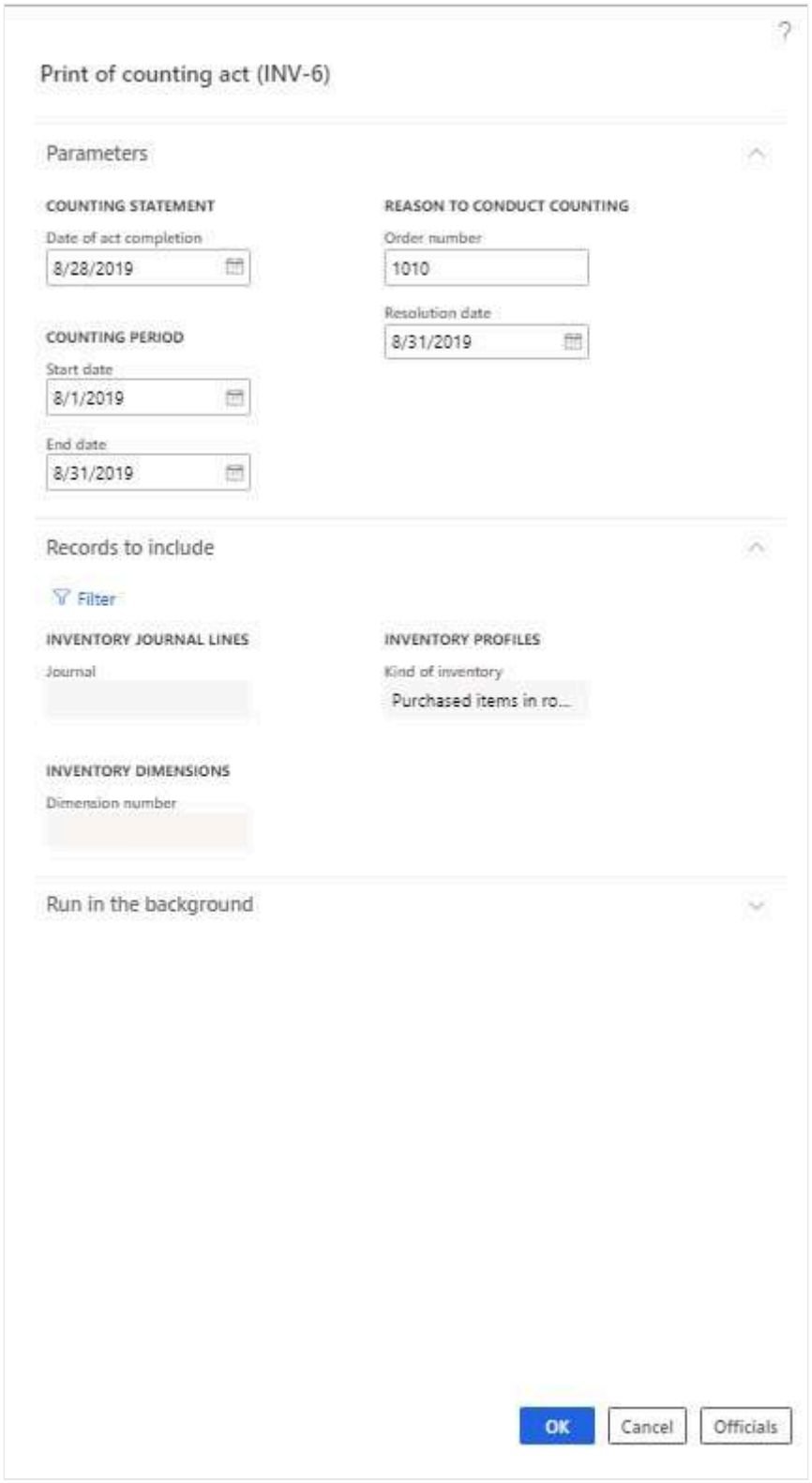
11. In the **Date of act completion** field, select the date when the counting process is scheduled to be completed.

12. In the **Start date** and **End date** fields, select the start and end dates of the inventory counting period.

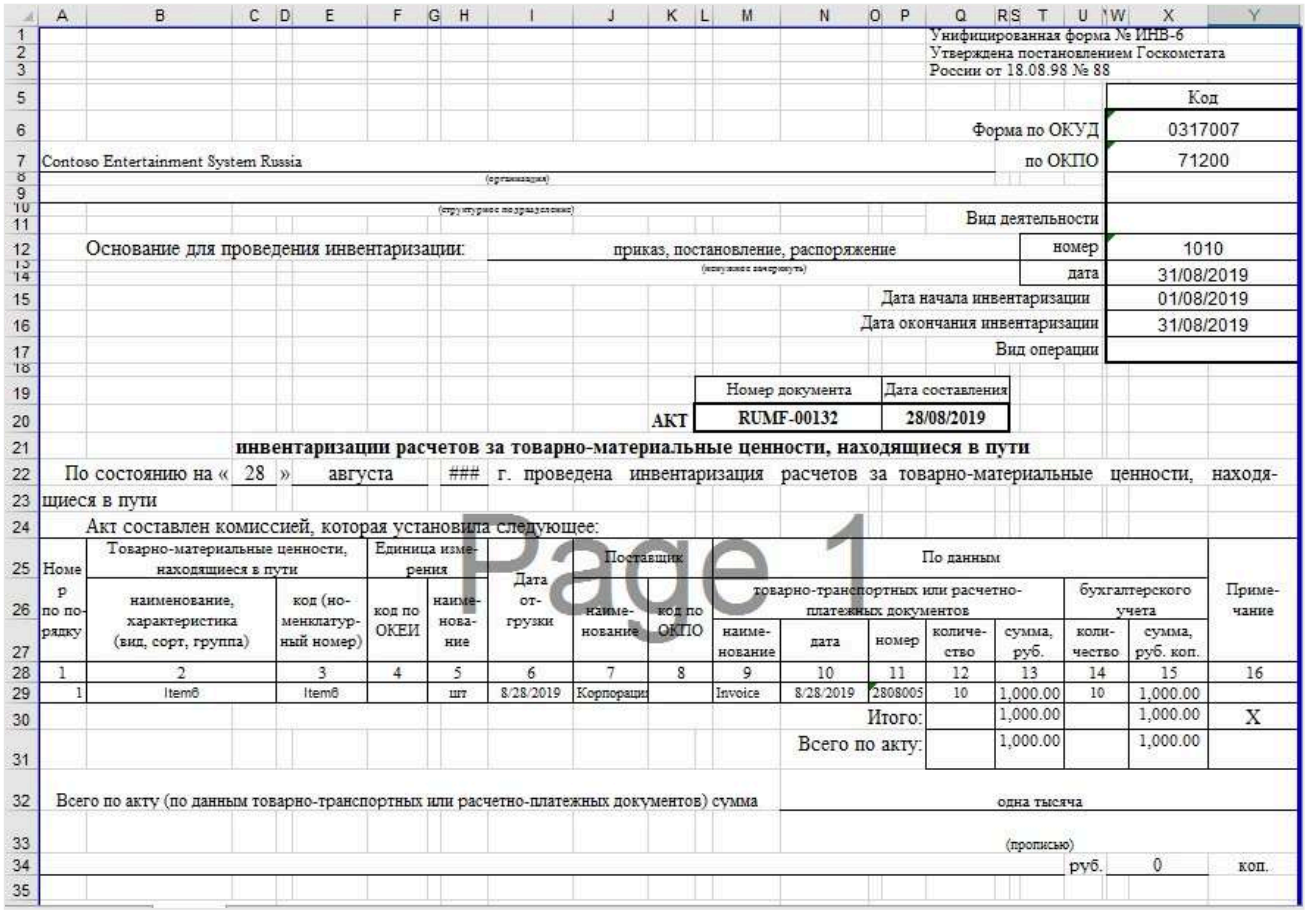
13. In the **Order number** field, enter the order number that you're generating the **Counting act INV-6** report for.

14. In the **Resolution date** field, select the transaction date of the order that you're generating the **Counting act INV-6** report for.

> **Note**
>
> The values in the **Journal**, **Dimension number**, and **Kind of inventory** fields are based on the inventory counting transactions. To change the values of these fields, select **Filter**. For example, you can modify these values to generate the report for a different journal or a different dimension.



15. Select **OK** to generate the **Counting act (INV-6)** report.



Унифицированная форма № ИНВ-6
Утверждена постановлением Госкомстата России от 18.08.98 № 88

	Код
Форма по ОКУД	0317007
по ОКПО	71200
Вид деятельности	
номер	1010
дата	31/08/2019
Дата начала инвентаризации	01/08/2019
Дата окончания инвентаризации	31/08/2019
Вид операции	

Contoso Entertainment System Russia
(организация)
(структурное подразделение)

Основание для проведения инвентаризации: приказ, постановление, распоряжение
(документ-основание)

Номер документа	Дата составления
RUMF-00132	28/08/2019

АКТ

инвентаризации расчетов за товарно-материальные ценности, находящиеся в пути

По состоянию на « 28 » августа ### г. проведена инвентаризация расчетов за товарно-материальные ценности, находящиеся в пути

Акт составлен комиссией, которая установила следующее:

Номер по порядку	Товарно-материальные ценности, находящиеся в пути		Единица измерения		Дата отгрузки	Поставщик		По данным товарно-транспортных или расчетно-платежных документов					По данным бухгалтерского учета		Примечание
	наименование, характеристика (вид, сорт, группа)	код (номенклатурный номер)	код по ОКЕИ	наименование		наименование	код по ОКПО	наименование	дата	номер	количество	сумма, руб.	количество	сумма, руб. коп.	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Item6	Item6		шт	8/28/2019	Корпорации		Invoice	8/28/2019	2808005	10	1,000.00	10	1,000.00	
										Итого:		1,000.00		1,000.00	X
										Всего по акту:		1,000.00		1,000.00	

Всего по акту (по данным товарно-транспортных или расчетно-платежных документов) сумма одна тысяча
(прописью)
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Page 1

Last updated on 05/12/2026

Approvals Management mobile app overview

Many businesses use approval workflows to ensure that new requests (such as proposed purchase requisitions and purchase orders) are valid, within budget, and in line with all business practices and standards. Unfortunately, this extra layer of security can sometimes add delays to the purchasing process. Therefore, Dynamics 365 Supply Chain Management now provides a mobile approval experience that adds flexibility, efficiency, resiliency, and responsiveness to the approval process.

The Approvals Management mobile app provides the following capabilities for managing approval requests:

- View requests that are assigned to you.
- Approve or deny requests.
- Delegate requests to another approver.
- Ask requesters to make changes.

The following illustration shows an example where the Approvals Management mobile app is being used to manage multiple purchase order requests.

> **Note**
>
> Use of the Approvals Management mobile app requires a license. You can find complete and up-to-date licensing information in the Dynamics 365 Licensing Guide. Licensing terms are subject to change without notice.

The Approvals Management mobile app replaces the now-deprecated Purchase order approval mobile workspace.

Next steps

- Onboard the Approvals Management mobile app
- Manage approvals using the Approvals Management mobile app

Last updated on 11/07/2025

Onboard the Approvals Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how administrators can prepare your Microsoft Dynamics 365 Supply Chain Management and Dataverse environments to support the Approvals Management mobile app. It also explains how to install the app on your mobile devices.

Prerequisites

Before you can start to onboard the Approvals Management mobile app, you must meet the requirements that are outlined in this section.

License requirements

Use of the Approvals Management mobile app requires a license. You can find complete and up-to-date licensing information in the Dynamics 365 Licensing Guide. Licensing terms are subject to change without notice.

System requirements

To run the Approvals Management mobile app, you must be using Supply Chain Management version 10.0.41 or later.

Region requirements

The Approvals Management mobile app is available in all regions where Supply Chain Management and Power Apps are supported. However, it isn't available for Microsoft Cloud for Sovereignty.

Set up Dataverse for your Supply Chain Management environment

The Approvals Management mobile app uses Dataverse virtual tables that are connected to Supply Chain Management. Therefore, a Dataverse environment must be integrated with your Supply Chain Management environment. If Dataverse isn't already set up for your environment, follow the instructions in Microsoft Power Platform integration with finance and operations apps.

When you create the Dataverse environment where you want to install the app, be sure to enable Dynamics 365 apps.

The Power Apps component framework feature must be enabled for your environment.

Allow publishing of canvas apps in Dataverse

The Approvals Management mobile app is a canvas app. Therefore, before you can install it in Dataverse, you must enable the **Allow publishing of canvas apps with code components** option for your environment.

1. Sign in to the Power Platform admin center.
2. On the navigation pane, select **Environments**, and then select the environment that is integrated with Supply Chain Management.
3. On the toolbar, select **Settings**.
4. On the **Settings** page, expand the **Product** section, and select **Features**.
5. Set the **Allow publishing of canvas apps with code components** option to *On*.

Install the mobile app in Dataverse

To make the Approvals Management mobile app available to users when they sign in using the Power Apps mobile app, you must install it using the Power Platform admin center. The installation process also sets up the required user roles and other dependencies in Dataverse.

Follow these steps to install the Approvals Management mobile app.

1. Sign in to Power Platform admin center.
2. On the left navigation pane, select **Resources** > **Dynamics 365 apps**.
3. Search for and select the app named *Dynamics 365 Approvals Management*.
4. On the top toolbar, select **Install**.
5. Select the environment where you want to install the app, review the packages that will be installed, and select the **I agree to the terms of service** checkbox.
6. Select **Install**.

Grant access to the mobile app in Dataverse

After the mobile app solution is installed in your Dataverse environment, you must share it with your users. The Approvals Management mobile app is a canvas app. To share it, follow the instructions in Share an app.

Each relevant user must also have the *Finance and Operations Basic User* role.

- To assign a role to a group team, follow the instructions in Manage the security roles of a team. We recommend that you use group teams if you must assign the role to multiple users. Learn more about how to manage team members in Manage team members.
- To assign a role directly to a user, follow the instructions in Assign a security role to a user.

Set up user accounts to manage approvals in Supply Chain Management

Each Supply Chain Management user who should be able to use the Approvals Management mobile app must have one or all of the following standard security roles:

- *Buying agent* – This role allows a Supply Chain Management user account to use the app to manage purchase orders.
- *Purchasing agent* – This role allows a Supply Chain Management user account to use the app to manage purchase requisitions.
- *Purchasing manager* – This role allows a Supply Chain Management user account to use the app to manage both purchase orders and purchase requisitions.

Users who have the *System administrator* role can use the app to manage all types of approval requests.

Follow these steps to add the required security roles to a Supply Chain Management user.

1. Sign in to your Supply Chain Management environment by using an administrator account.
2. Go to **System administration** > **Users** > **Users**.
3. Use the filter to find a user who should be able to use the app.
4. Open the details page for the selected user by selecting the link in the **User ID** column.
5. On the **User's roles** FastTab, on the toolbar, select **Assign roles** to open the **Assign roles to user** dialog box.
6. In the **Role name** column, find and select the *Buying agent*, *Purchasing agent* and/or *Purchasing manager* roles, as needed.
7. Select **OK** to apply your settings and close the dialog box.

> ⓘ **Note**

User role organization assignments aren't supported. If one or more organizations that differ from a selected user's **default company** have been assigned to any security role that is set for that user, app installation fails.

Learn more about how to set up roles and security in Supply Chain Management in Security roles.

Set up approval workflows

To enable the approval process for a document type, you must set up approval workflows for it in Supply Chain Management. Learn more in Workflow system overview.

After your approval workflows are set up, approval requests appear in the Approvals Management mobile app just as they do when the same user signs in to Supply Chain Management to manage approvals in the web client.

Install and open the mobile app

Follow these steps to install and use the Approvals Management mobile app on a mobile device.

1. Install the Power Apps mobile app by following the instructions in Install the Power Apps mobile app.
2. Open the Power Apps mobile app, and sign in by using the same corporate account that you use to sign in to Supply Chain Management.
3. Use the **Search** field to search for *Approvals Management*. Because the Approvals Management mobile app is a canvas app, you can add it to your favorites list in Power Apps by swiping left on it after you find it.
4. Open the Approvals Management mobile app, and start to use it.

Clear the Power Apps cache on a mobile device after an update of the back-end setup

If the back-end setup is updated in Supply Chain Management after you install the Power Apps mobile app on a mobile device (for example, if the legal entity setup is changed), you might have to clear the cache in Power Apps to update the local setup data. Power Apps can then reload the settings that otherwise might not take effect until the cache is cleared.

Follow these steps to clear the Power Apps cache on a mobile device.

1. Sign in to the Power Apps mobile app.
2. Select the image for your user account in the upper left of the page to open the settings menu.
3. Select **Clear cache** to open the **Clear cache** dialog box.
4. Select **Confirm**.

Last updated on 12/09/2025

Manage approvals using the Approvals Management mobile app

This article explains how to use the Approvals Management mobile app to approve, reject, or delegate approval requests.

User requirements

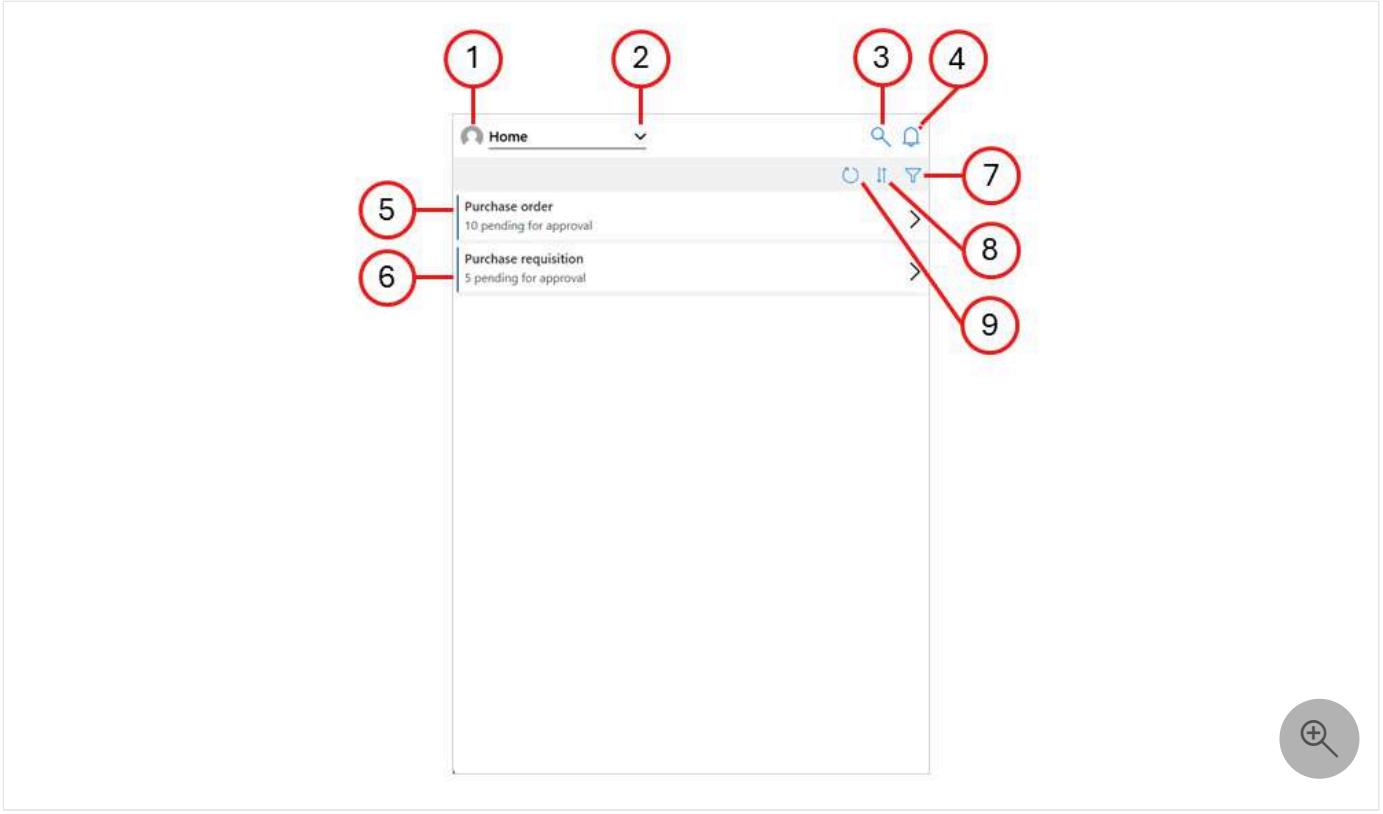
To manage purchasing requests from the Approvals Management mobile app, you must meet the following requirements:

- You must sign in to Microsoft Power Apps by using a domain account, and that domain account must match a user account in Dynamics 365 Supply Chain Management that has the same Microsoft Entra ID user name.
- Your user account in Supply Chain Management must be assigned the required security roles for the types of approvals that you must manage. For more information, see Set up user accounts to manage approvals in Supply Chain Management.

Home page

When you open the Approvals Management mobile app and sign in, the home page appears. This page provides an overview of requests that are awaiting your approval.

The following illustration highlights the user interface (UI) elements on the home page. Many of these elements are common to all pages in the app.



Legend:

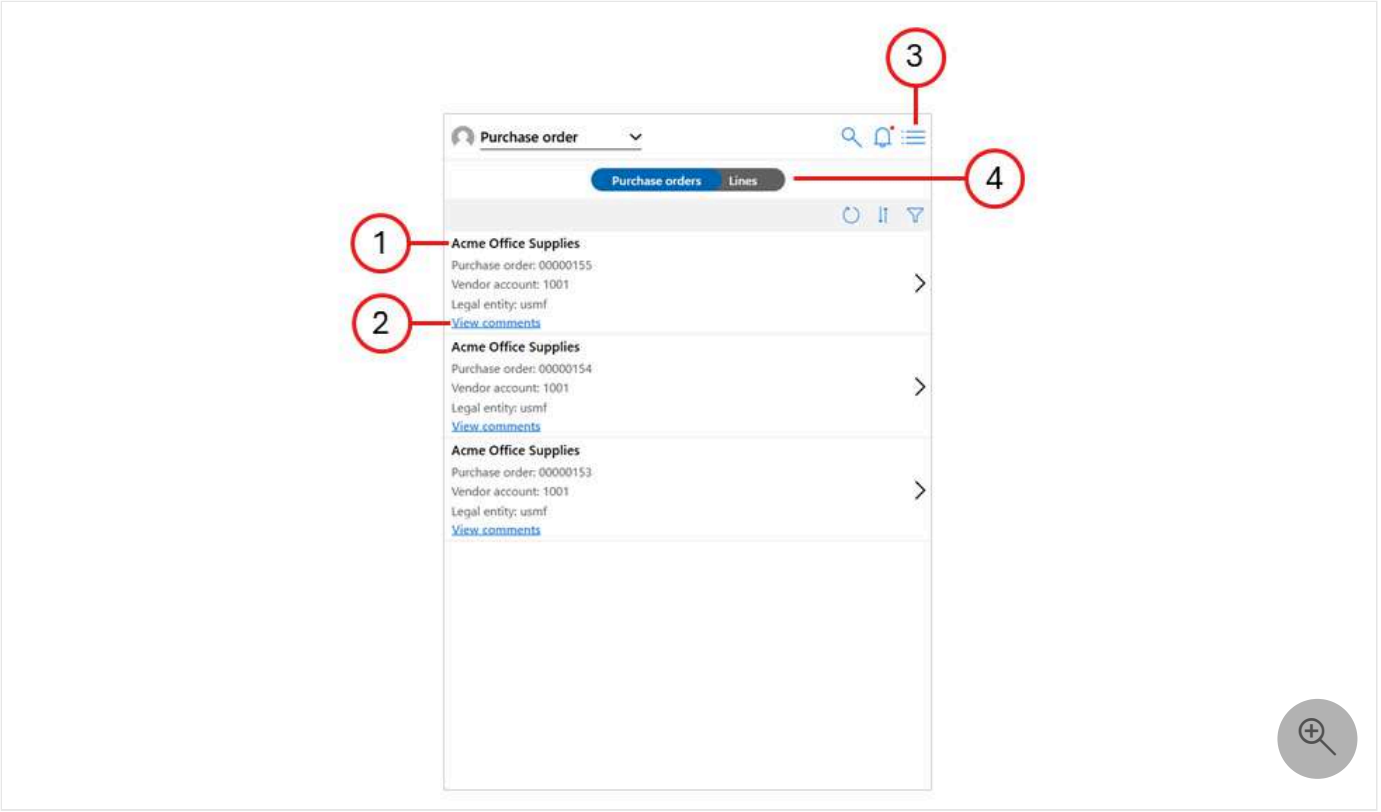
1. **About button** – View information about the app, such as terms and conditions, and the current version.
2. **Navigator** – Move between the main pages of the app (such as **Home**, **Purchase orders**, and **Purchase requisitions**).
3. **Search** – Search for specific documents to approve. For example, you can search for a specific order number, vendor account, vendor name, legal entity, or item number.
4. **Alerts** – Read your notifications. A red dot indicates that you have new notifications.
5. **Purchase orders** – If you can manage purchase orders, this item shows the number of purchase orders that are awaiting your approval. Select it to view the list of purchase orders.
6. **Purchase requisitions** – If you can manage purchase requisitions, this item shows the number of purchase requisitions that are awaiting your approval. Select it to view the list of purchase requisitions.
7. **Filter** – Filter the list by defining specific values to look for. This capability is useful on pages that show longer lists.
8. **Sort** – Select how items are sorted in the list.
9. **Refresh** – Manually refresh the list. (The list is automatically refreshed every 30 seconds.)

List pages

List pages show a list of orders, requisitions, or line items that are awaiting your approval. The type of item that is shown depends on how you opened the list page. From a list page, you can select any item in the list to view more details about it. If you turn on multiselect, you can select one or more items, and then approve, reject, delegate, or request changes for all of them directly from the list page.

List pages without multiselect

List pages where multiselect is turned off show a list of items that you can navigate and select to view more details. The following illustration highlights the UI elements on a list page where multiselect is turned off.

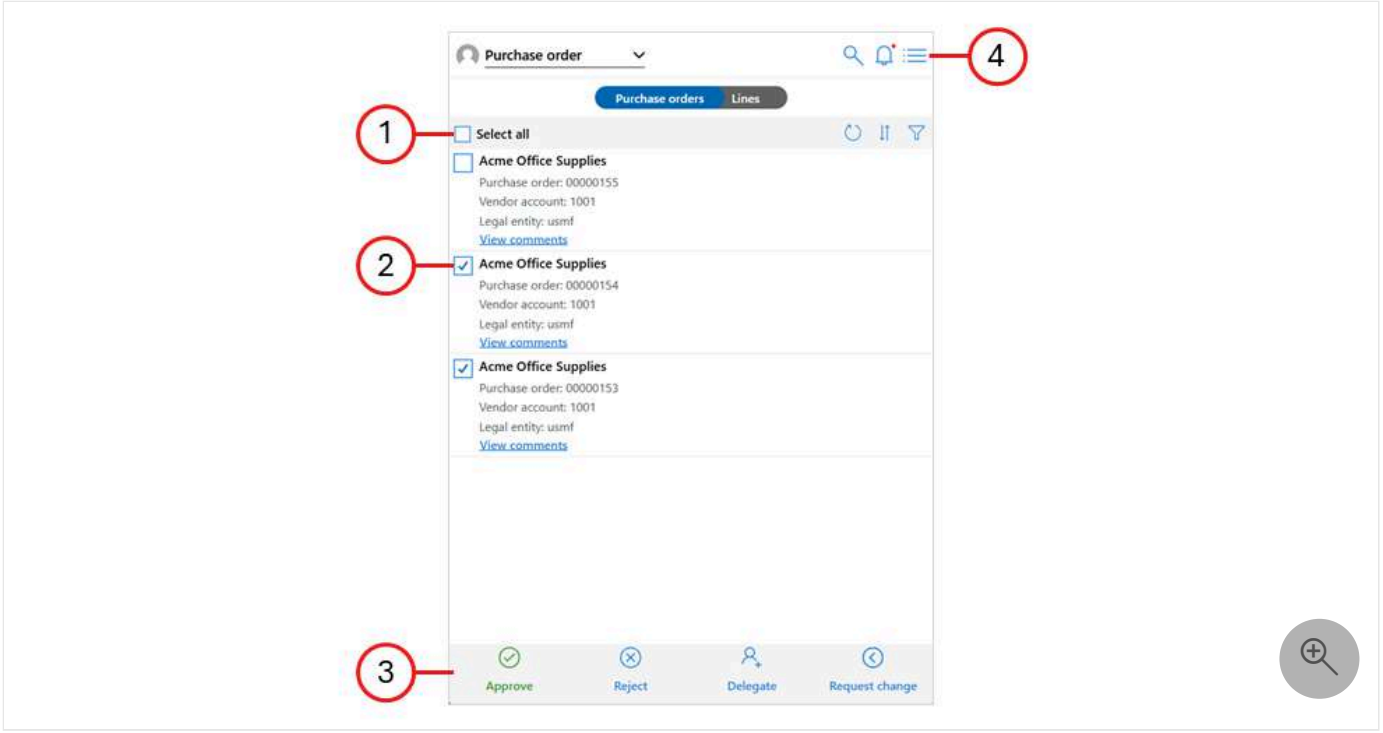


Legend:

1. **List item** – Each item in the list represents an individual document header or line. A summary of the item is shown. Select the item to view details about it.
2. **View comments** – View comments about the related item. These comments are usually comments that the requester wrote for the reviewer.
3. **Multiselect** – Turn multiselect on or off. (In this illustration, multiselect is turned off.)
4. **Header/Lines selector** – Select whether to view a list of document headers or individual lines.

List pages with multiselect

On list pages where multiselect is turned on, you can select one or more items in the list at the same time. You can then approve, reject, delegate, or request changes for all of those items directly from the list page. The following illustration highlights the UI elements on a list page where multiselect is turned on.



Legend:

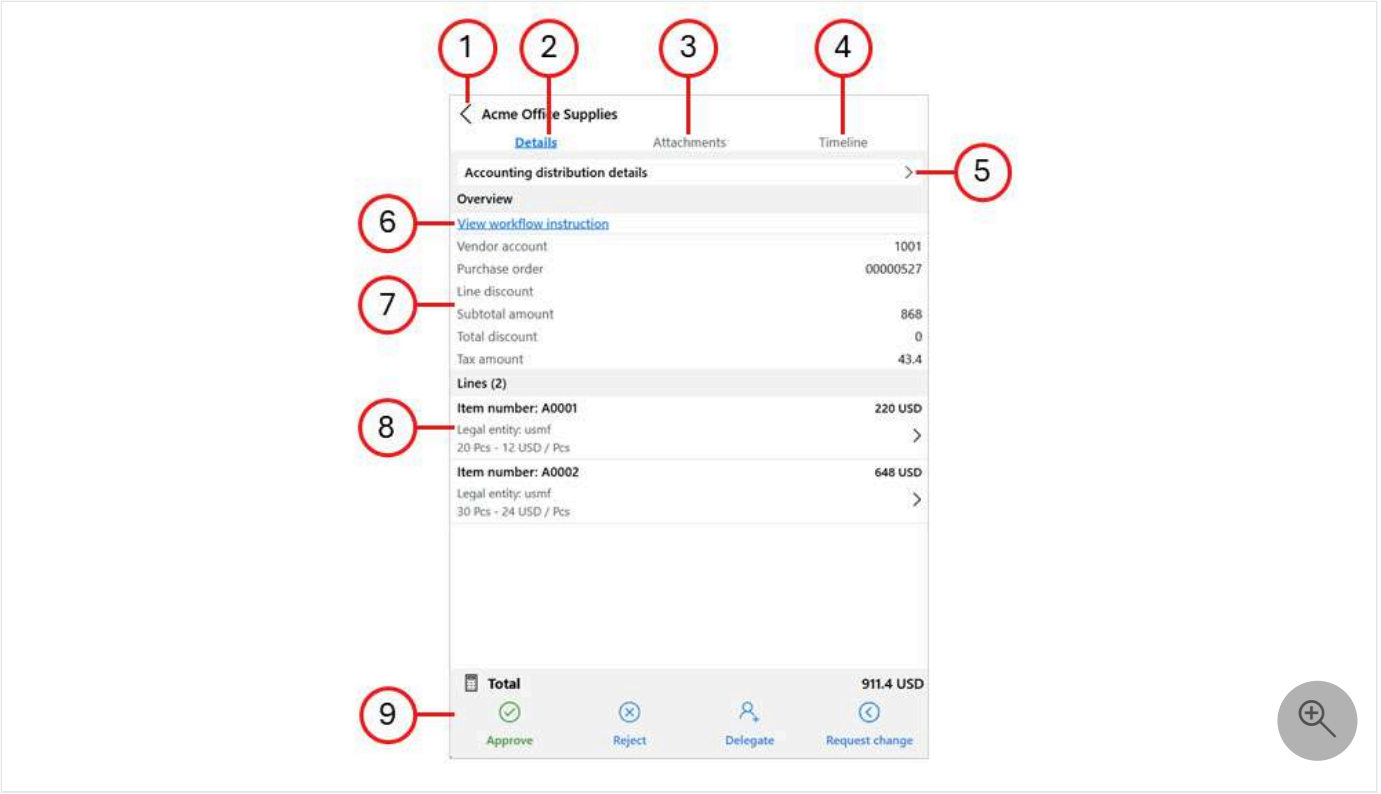
1. **Select all** – Select or clear the selection of all items in the list.
2. **Checkbox** – Select or clear the selection of the related item.
3. **Action buttons** – Approve, reject, delegate, or request changes for all the selected items.
4. **Multiselect** – Turn multiselect on or off. (In this illustration, multiselect is turned on.)

Detail pages

Detail pages show detailed information about a specific document header or line item. The details that are provided vary slightly, depending on whether you view header details or line details.

Header details

Header details pages show information about the selected document as a whole. The following illustration highlights the UI elements on a header details page.

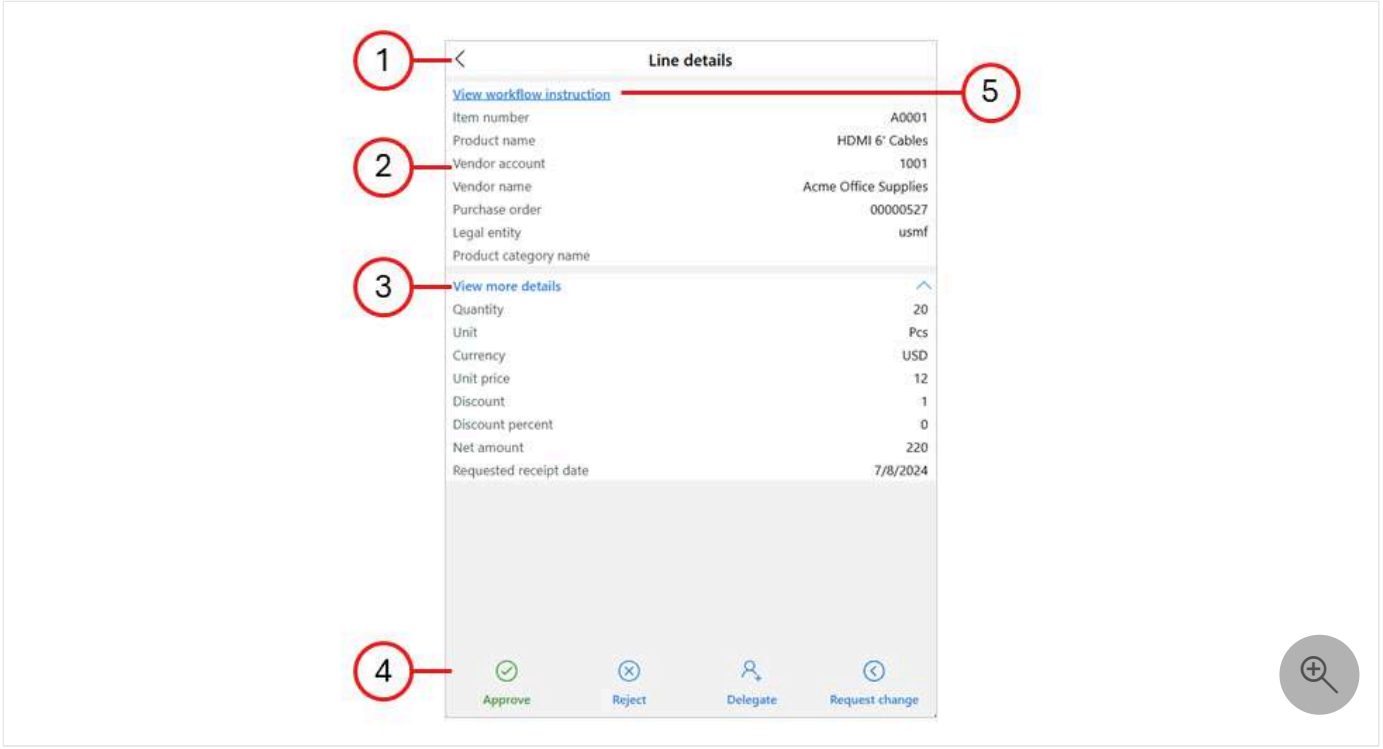


Legend:

1. **Back** – Return to the page that you came from.
2. **Details tab** – View general information about the document. (In this illustration, this tab is selected.)
3. **Attachments tab** – View and open any attachments that are associated with the document.
4. **Timeline tab** – View a timeline of events that are related to the document.
5. **Accounting distribution details** – View accounting distribution details that are linked to the document.
6. **View workflow instructions** – View instructions for the workflow.
7. **Header details** – Shows detailed information about the document.
8. **Lines** – Lines are listed under the header details. Each line shows a summary of the line item. Select a line to view more details about it.
9. **Action buttons** – Approve, reject, delegate, or request changes for the document.

Line details

Line details pages show information about a specific document line. The following illustration highlights the UI elements on a line detail page.



Legend:

1. **Back** – Return to the page that you came from.
2. **Standard details** – View general information about the line item.
3. **Extra details** – Expand or collapse additional information about the line item.
4. **Action buttons** – Approve, reject, delegate, or request changes for the line.

   > **Note**
   >
   > Not all line details pages include these buttons.

5. **View workflow instructions** – View instructions for the workflow.

Last updated on 05/22/2026

Approvals Management mobile app FAQ and known issues

06/30/2025

This article provides answers to several frequently asked questions (FAQ) about the Approvals Management mobile app. It also describes some known issues that affect the app and explains how to work around them.

What platforms and mobile devices are supported?

The Approvals Management mobile app runs within the Power Apps mobile app. All platforms that the Power Apps mobile app supports can also run the Approvals Management mobile app. Learn more in System requirements, limits, and configuration values for Power Apps.

Why isn't anything shown when I sign in to the mobile app?

If no features are visible in the Approvals Management mobile app, you probably lack the permissions that are required to access them. Access to each feature is controlled by security roles that are assigned to your user account in Dynamics 365 Supply Chain Management. Learn more in Onboard the Approvals Management mobile app.

What versions of Supply Chain Management support the mobile app?

The Approvals Management mobile app requires Supply Chain Management version 10.0.41 or later.

Does the mobile app support Microsoft Intune deployment?

Yes, the Approvals Management mobile app supports deployment through Intune. Learn more in Manage your mobile app with Microsoft Intune.

Can I customize and extend the mobile app?

No, you can't currently customize or extend the Approvals Management mobile app.

Does the mobile app support offline mode?

No, the Approvals Management mobile app doesn't currently support offline mode.

Is there multi-language support?

Yes, the Approvals Management mobile app is available in all the same languages as Supply Chain Management, except right-to-left (RTL) languages. We plan to add RTL support in a future release, after it's added to the canvas app platform.

Where can I go to discuss the mobile app with the community and submit suggestions to Microsoft?

The Dynamics 365 Procurement and Sourcing group on Viva Engage is a great place to go if you want to exchange tips, ask questions, or submit suggestions for improvement. Viva Engage group participants include Microsoft partners, customers, experts, and employees.

I receive the following error: "Error when trying to retrieve data from the network." How can I diagnose the issue?

If you receive an error message that states, "Error when trying to retrieve data from the network," follow these steps to diagnose and fix the issue.

1. Run the Approvals Management mobile app in a browser as described in Run an app in a web browser.
2. If you receive the same error while you run the app in a browser, set up your browser to capture an HTTP Archive file (HAR) file. Then reload the tab where the app is running.
3. Use a tool such as the Google HAR analyzer to redact all secrets from the HAR file.
4. Contact Microsoft Support, and share the redacted HAR file with them.
5. Microsoft Support will help you diagnose and fix the issue.

The content of the HAR file might reveal an error that resembles the following example:

```
text
```

```
SecLib::CheckPrivilege failed. User: <EntraObjectID>, PrivilegeName:
prvReadRelationship, PrivilegeId: <PrivilegeID>, Required Depth: Basic,
BusinessUnitId: <BusinessUnitID>, MetadataCache Privileges Count: <CacheCount>,
User Privileges Count: <UserPrivilegeCount>
```

This error can occur because the *Approvals Management app* role doesn't have the privileges that are required to read the `Relationship` table. Instead, that privilege is provided by the *Finance and Operations Basic User* role in Dataverse. To fix this issue, add the *Finance and Operations Basic User* role to each user or team that uses the app.

When I update the mobile app, I receive the following error: "Could not delete solution...because it has been modified." How can I fix the error?

The Approvals Management mobile app is a canvas app. The update process fails with an error if it detects that the canvas app has been customized. This behavior helps prevent the updater from overwriting your customizations. To fix the error, remove any customization layers by restoring the oldest version of the app. If the oldest version is more than six months old, you must manually delete the `msdyn_ApprovalsManagementAnchor` and `msdyn_ApprovalsManagement` solutions, and then reinstall the app from Dynamics 365 apps or AppSource.

Consider cloning the canvas app before you remove your customizations, just in case you want to copy the customizations to the updated app.

To delete the solutions, follow these steps:

1. Open the Power Apps Maker portal for your environment.
2. On the left navigation pane, select **Solutions**.
3. Open the **Managed** tab to view managed solutions.
4. Find and delete each of the following solutions. The **Search** field in the top-right corner of the page can help you find the solutions quickly. The **Delete** button is available in the **Commands** menu (three dots) provided for each solution.

    - Dynamics 365 Approvals Management Anchor Solution
    - Dynamics 365 Approvals Management Solution

> **Note**

Microsoft doesn't support **customizing or extending** the Approvals Management mobile app.

Are there any known issues that affect the release of the mobile app?

Yes, the Approvals Management mobile app has a few known issues that we expect to correct in an upcoming release. The following subsections describe these issues.

Limited file type support for attachments

The current release provides only limited support for viewing file attachments on the job details page. We plan to add support for more file types in the final release.

Supported attachment types:

- URL (link)
- PNG/JPEG/JPG/BMP/GIF (image)
- PDF (document)

Unsupported attachment types:

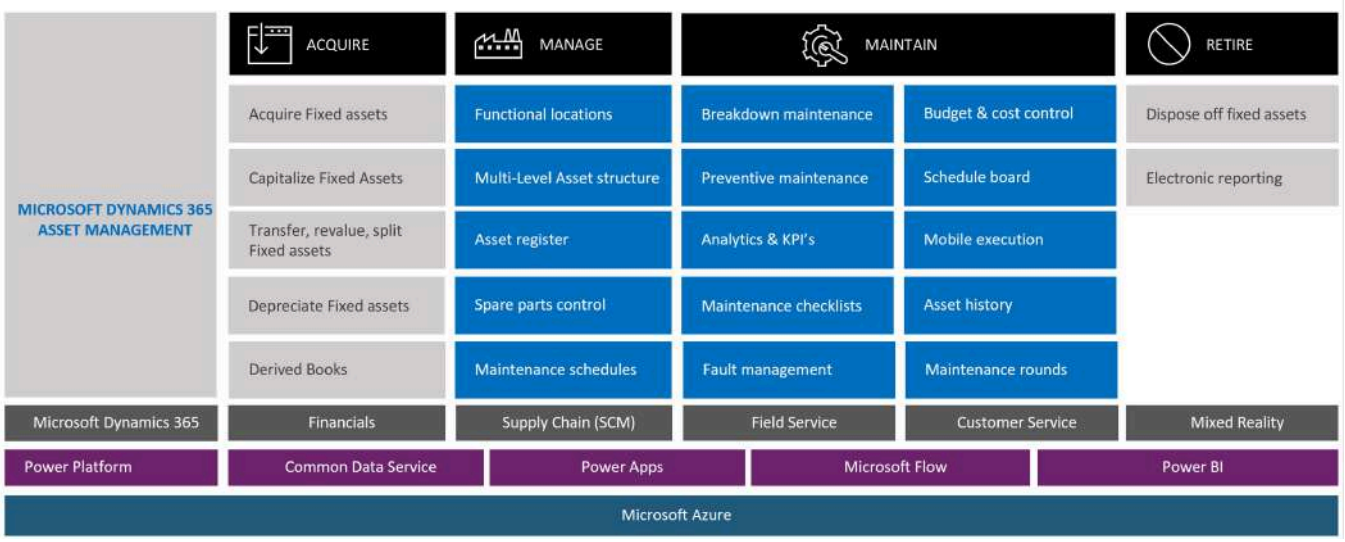
- Note
- All other file types

Asset management overview

07/10/2025

Asset Management is an advanced module for managing assets and maintenance jobs in Dynamics 365 Supply Chain Management. Asset Management integrates seamlessly with several other finance and operations apps.

The following illustration shows the interfaces between Asset Management and other finance and operations apps.



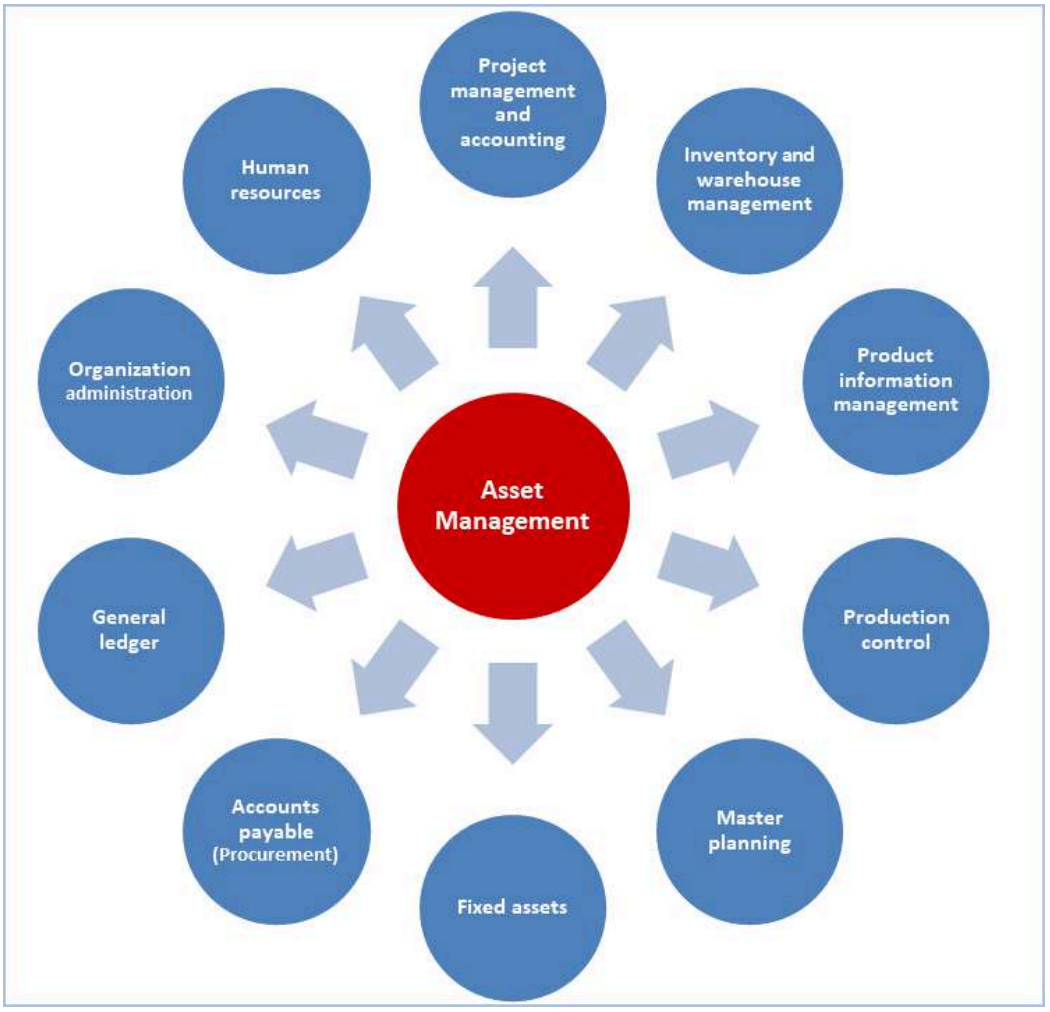
Asset Management enables you to efficiently manage and service many types of equipment in your company (such as machines, production equipment, and vehicles). Asset Management supports solutions across numerous industries.

Functional locations and assets

07/10/2025

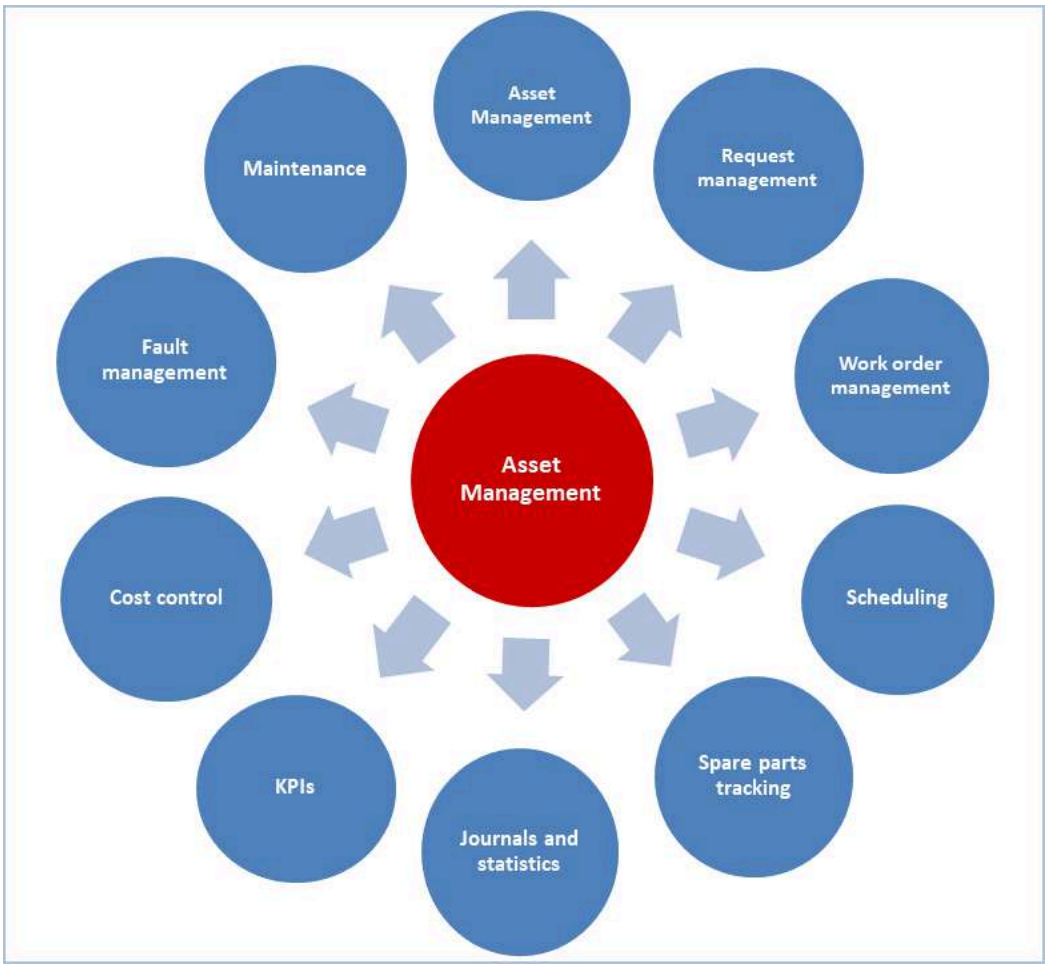
This article describes functional locations and assets in Asset Management.

Asset Management is integrated seamlessly with several modules for Supply Chain Management and other finance and operations apps. The following illustration shows the interfaces with other modules.



Asset Management lets you efficiently manage and perform all tasks that are related to managing and servicing many types of equipment in your company. This equipment includes machines, production equipment, and vehicles. Asset Management also supports solutions across numerous industries.

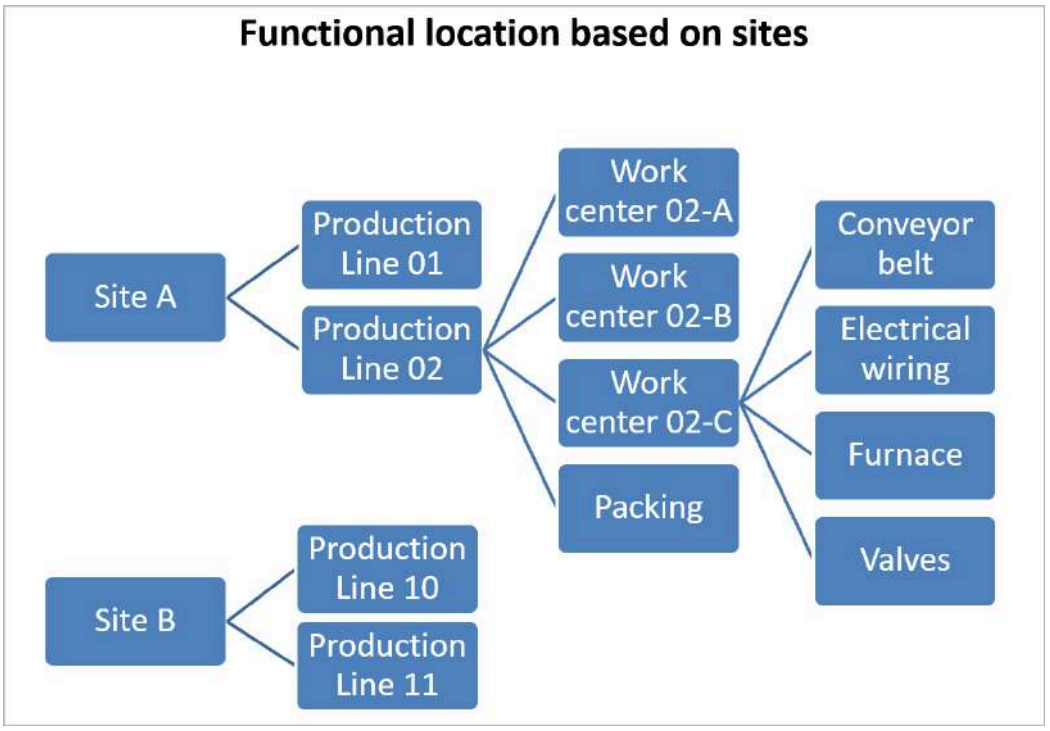
The following illustration shows an overview of the main functionality that is covered by Asset Management.



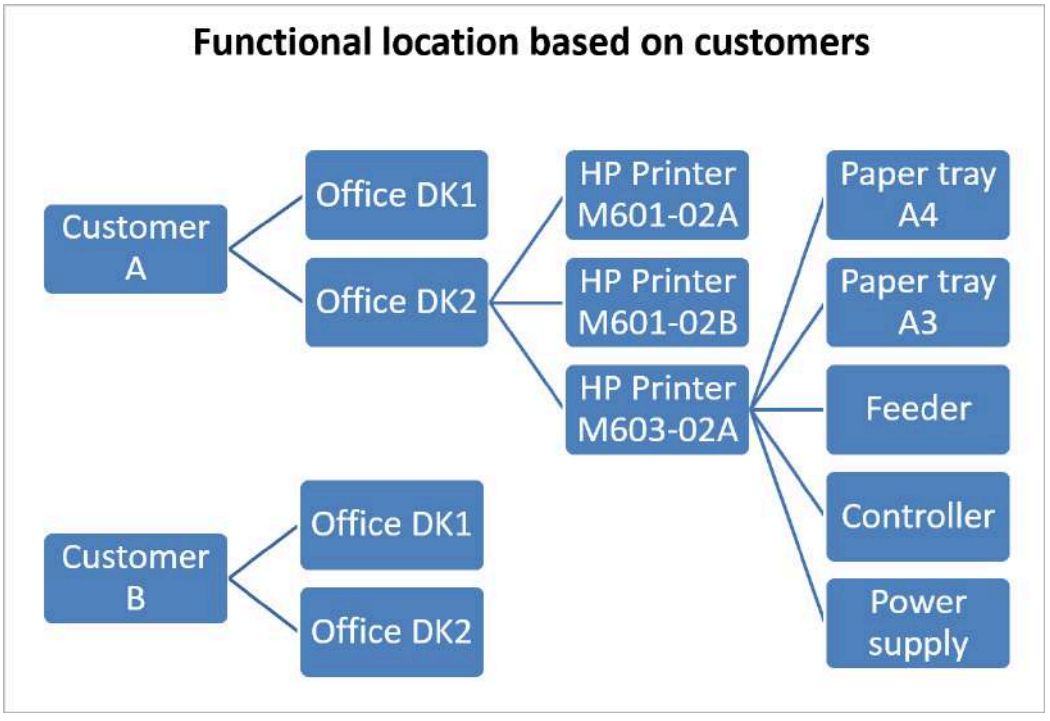
Functional locations are used to manage assets on locations. This management includes tracking of asset costs on functional locations. Functional locations are structured hierarchically, and locations can have sublocations. The structure of functional locations is static. In other words, locations can't change place. Assets can be installed on functional locations and, as required, can be installed on other functional locations later.

Asset costs always follow the location of the asset. In other words, if you install an asset on a new functional location, the asset automatically uses the financial dimensions that are related to the new functional location. Therefore, asset costs are always related to the functional location that the asset is currently installed on. This automatic handling of financial dimensions helps guarantee complete tracking of costs when your company does project controlling and reporting on functional locations.

The way that you build your hierarchy of functional locations depends on your company's requirements for maintaining internal equipment or servicing customer equipment. The following figure shows an example of functional locations that are based on geographical locations.



The following figure shows an example of functional locations that are based on customers.



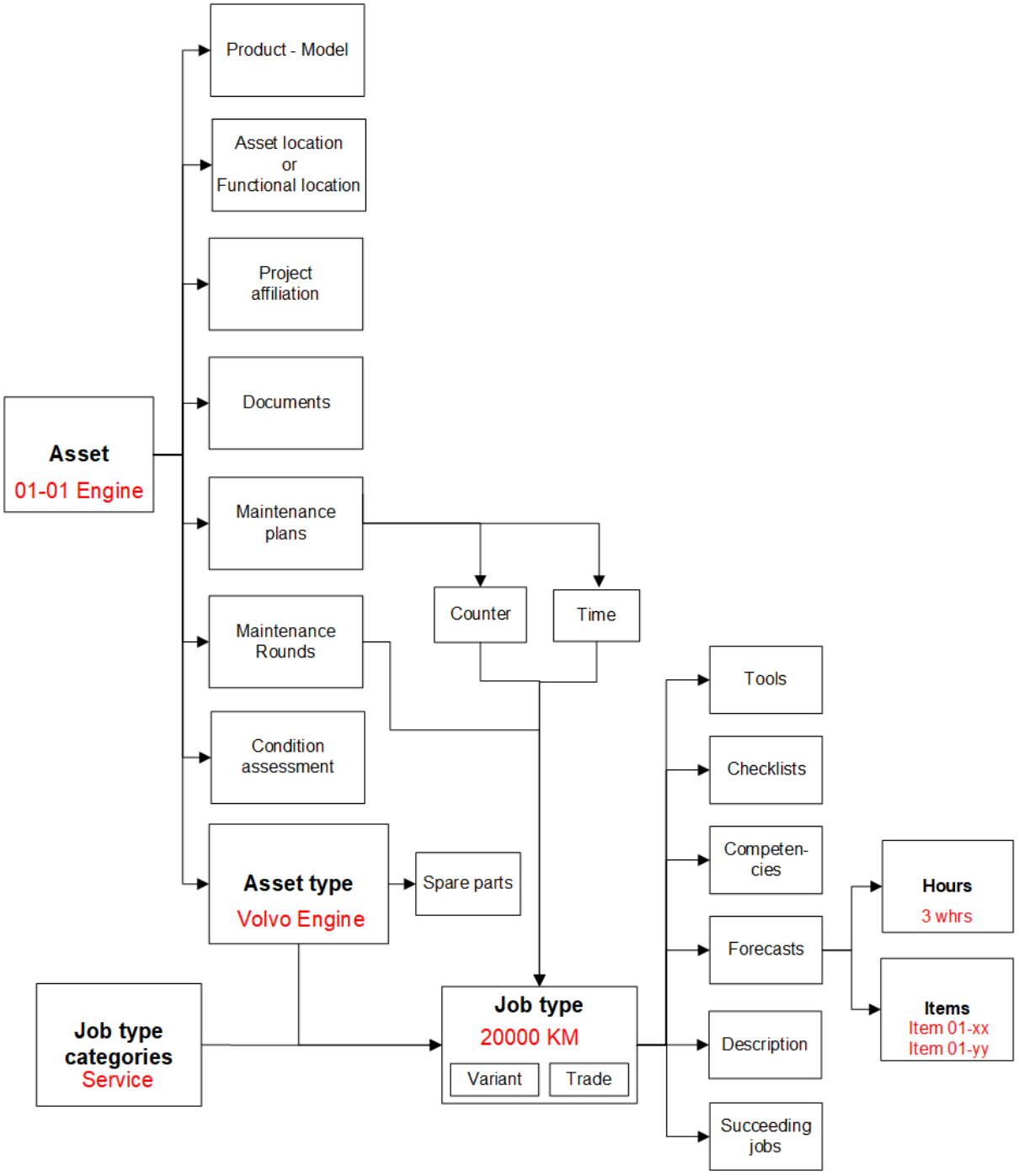
Assets and work orders

07/10/2025

This article describes assets and work orders in Asset Management. Assets and work orders are the central parts of Asset Management. An *asset* is a machine or machine part that requires continuous maintenance and service. Assets can be created in a hierarchical structure, and they can be related to functional locations. Maintenance jobs can be planned at all levels in the asset structure.

Asset data and job types

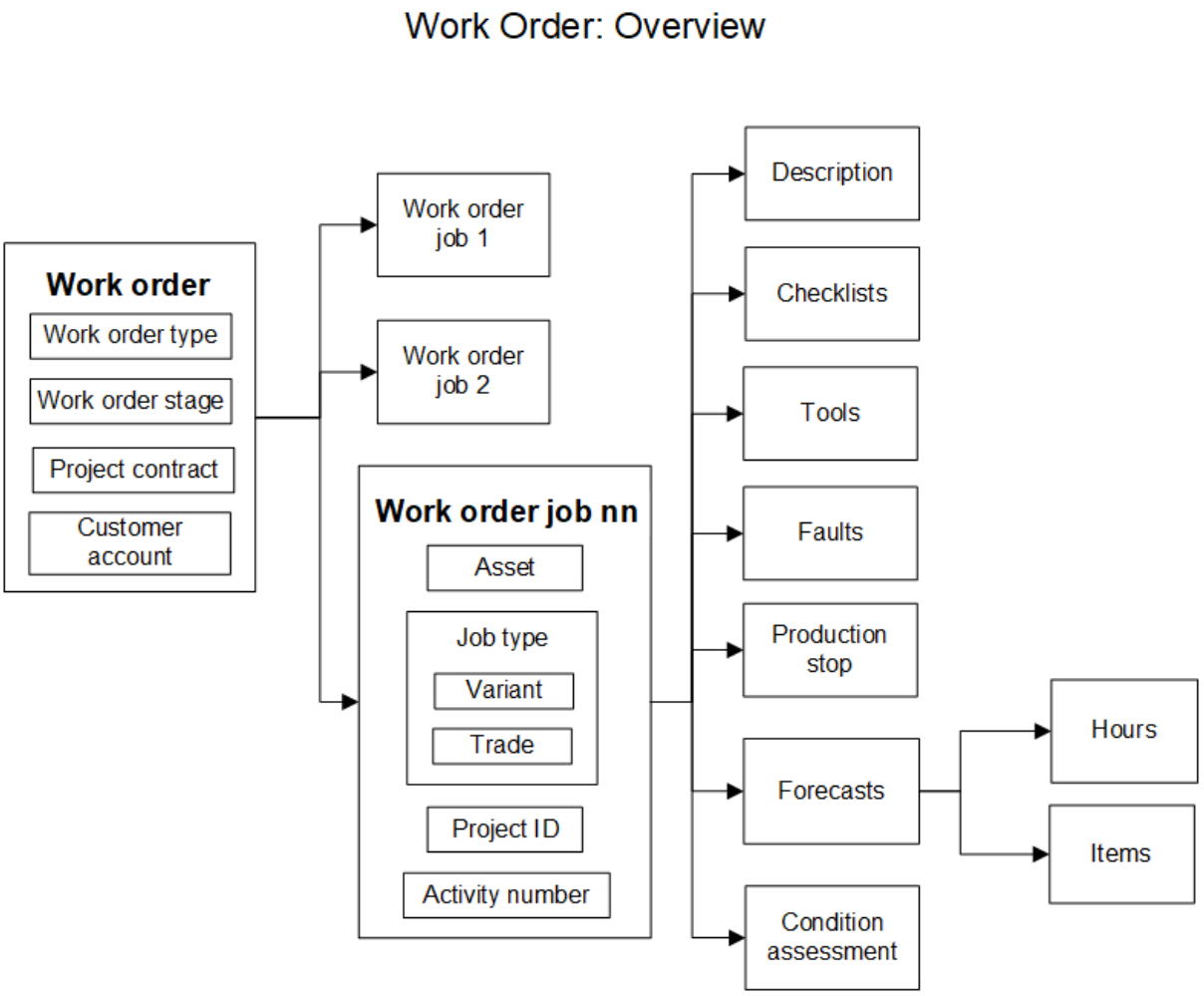
Various data, such as product information and asset specification, and required maintenance plans are set up on each asset. The following illustration shows an overview of asset data and the affiliation of assets to job types. Red text is used for examples that show inheritance and dependencies.



Key data in a work order

Every work order has a work order type, such preventive maintenance, corrective maintenance, or inspection. The work order contains one or more work order jobs. Every work order job defines a job that must be performed on an asset and a related job type. Examples of related job types include 10,000 km, 50,000 km, 1-year overhaul, and safety inspection. One work order can be related to multiple assets.

The following illustration shows an overview of the key data in a work order.

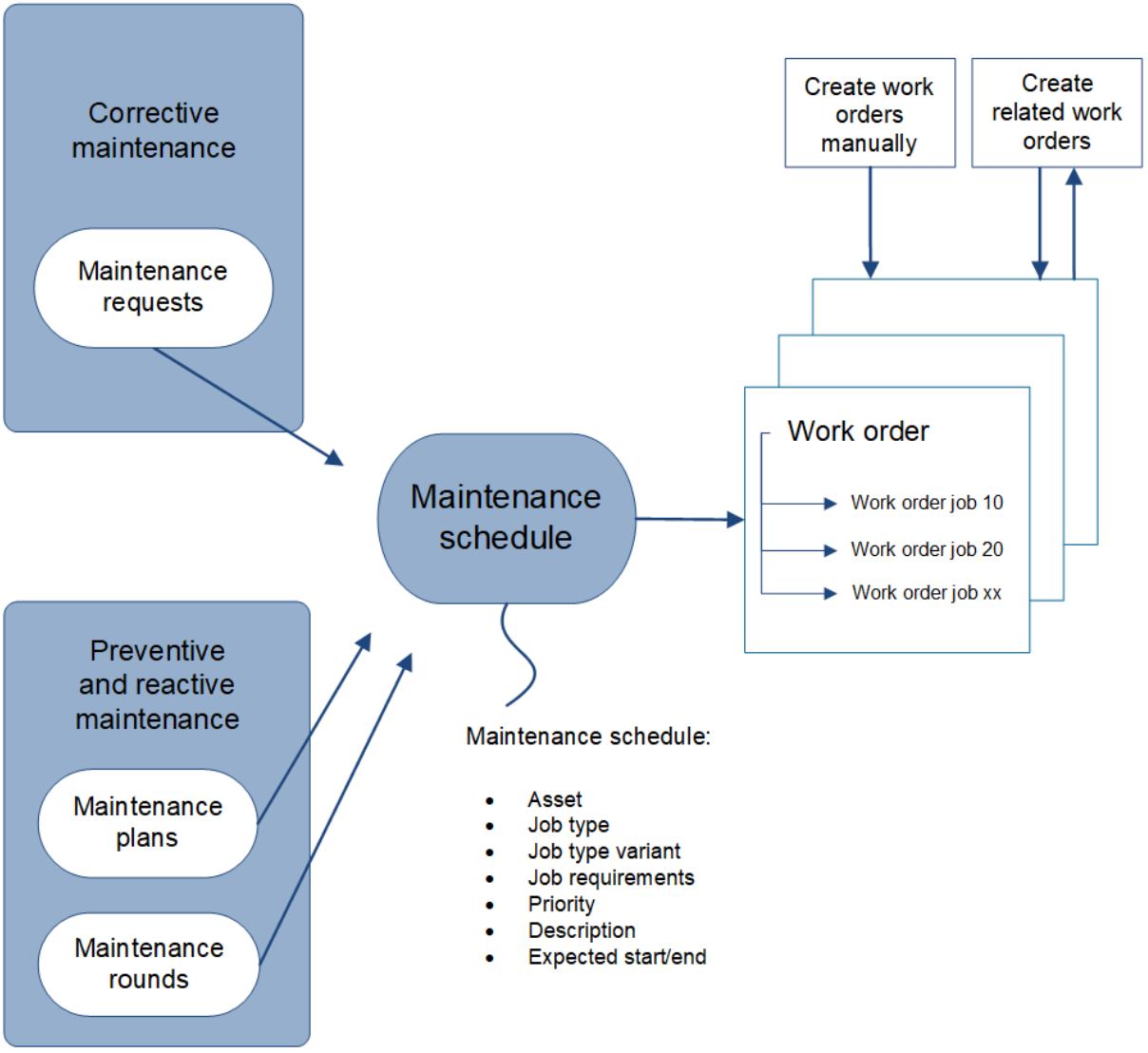


A work order can be related to another work order, and job types can contain succeeding jobs that create a work order. In general, there are no dependencies between work orders. Therefore, they can change their work order lifecycle state and can be scheduled independently of each other.

Creating work orders automatically or manually

Work orders can be created in various ways that are related to corrective, preventive, or reactive maintenance. You can also create work orders manually. The following illustration shows an overview of the process for automatic or manual creation of work orders.

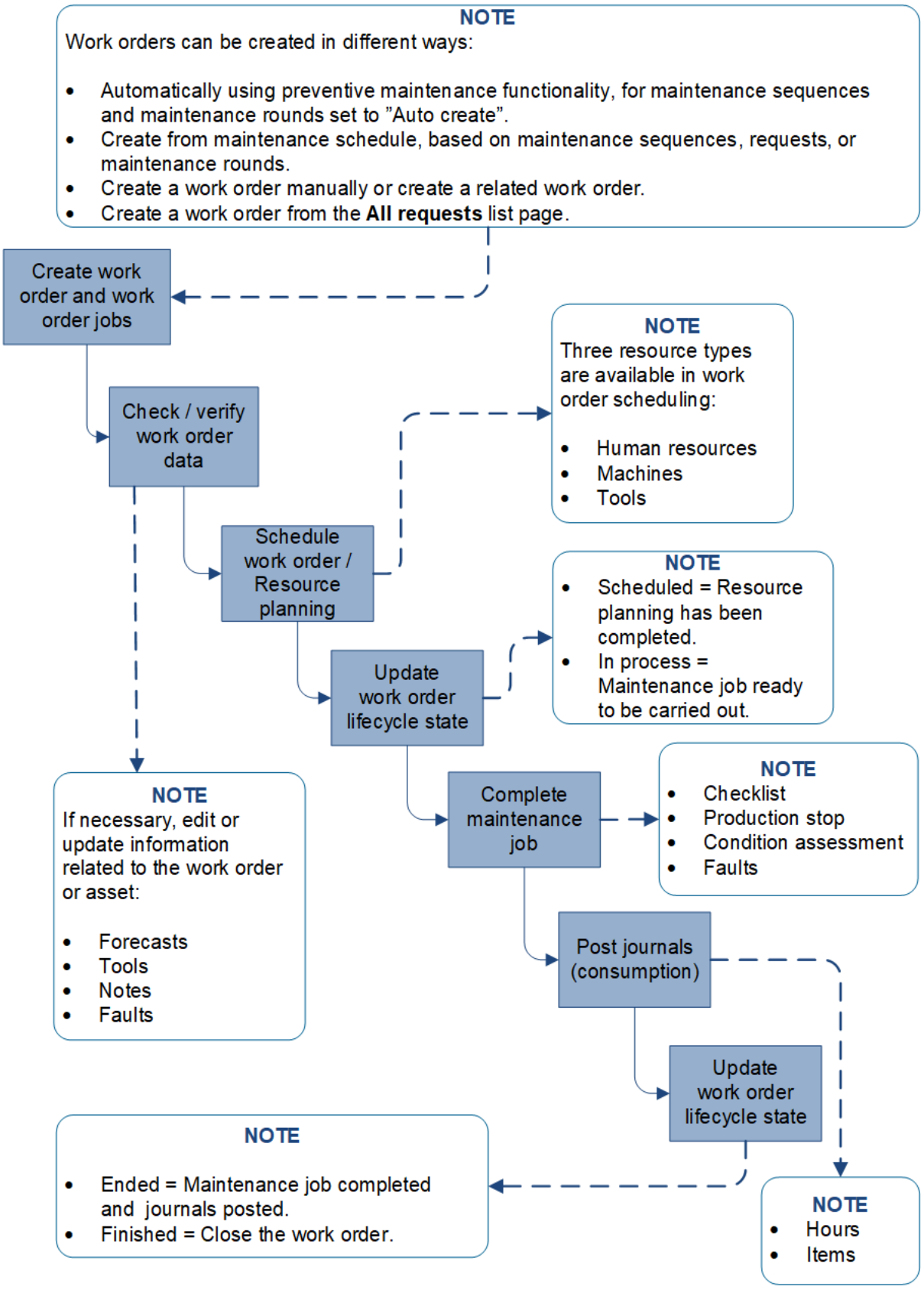
Process flow – Creating work orders



Processing a work order

Several steps must be completed when you want to schedule and run a maintenance job on a work order. The following illustration shows an overview of the processing for a work order.

Work Order - Flow Diagram



> ⓘ Note
>
> In general, when you work in Dynamics 365 Supply Chain Management and the **Asset Management** module, you select **New** to create a new record, you select **Edit** to update an existing record, and you select **Save** to save new or edited data.

Next steps

- Introduction to assets
- Introduction to work orders

Functional location lifecycle states

Article • 01/06/2025

This article describes how to set up functional location lifecycle states and lifecycle models in Asset Management. Functional location lifecycle states define the states that a functional location can go through, for example, created, active, and ended. You're able to view all functional locations, regardless of their lifecycle state, in the **All functional locations** list page. You can change the state of a functional location by selecting it in the **All functional locations** list page and selecting **Update functional location state**.

Set up functional location lifecycle states

1. Select **Asset management** > **Setup** > **Functional locations** > **Lifecycle states**.
2. Select **New** to create a new functional location state.
3. Insert the state ID in the **Lifecycle state** field and a name for the functional location state in the **Name** field. In the **Lifecycle models** field, you can see the number of functional location lifecycle models that uses the functional location state.
4. On the **General** FastTab, select *Yes* on the **Active** toggle button if the functional location should be active at this state.
5. Select *Yes* on the **Create assets** toggle button if it should be possible to automatically create an asset with the same name as the functional location and install it on the functional location at this state.

   > **Note**
   >
   > This toggle button relates to the **Asset type** field on the **General** FastTab on the **Functional location types** page (**Asset management** > **Setup** > **Functional locations** > **Functional location types**).

6. Set **Rename location** to *Yes* to make it possible to change the name of the functional location at this state.
7. Set **New sub locations** to *Yes* to make it possible to add new sub locations to the functional location at this state.
8. Set **Install assets** to *Yes* to make it possible to install assets on the functional location at this state.

9. Set **Install assets** to *Yes* to make it possible to delete the functional location at this state.

10. Select an asset state in the **Lifecycle state** field if you want the asset lifecycle state for all assets installed on the functional location to be automatically updated at this state. For example, if you close down a functional location and set the functional location lifecycle state to *Ended*, you might want to automatically change the lifecycle state of the assets installed on that functional location to *Not in use*.

> **Note**
>
> Functional location lifecycle states, lifecycle models, and types are related and used in the same way as work order lifecycle states, work order lifecycle models, and work order types.

Set up functional location lifecycle models

When you have created the lifecycle states required for your functional locations, they can be divided into groups. This is done to create the lifecycle model flow that could be used for different types of functional locations. As a minimum, one standard functional location lifecycle model should be created.

1. Select **Asset management** > **Setup** > **Functional locations** > **Lifecycle models**.
2. Select **New** to create a new lifecycle model.
3. Insert the lifecycle model ID in the **Lifecycle model** field and a name for the lifecycle model in the **Name** field. In the **Functional location types** and **Lifecycle states** fields, you can see the number of functional location types that uses the lifecycle model and the number of states selected in the lifecycle model.
4. On the **Lifecycle states** FastTab, select the states that should be included in the model. This is done by selecting a state in the **Lifecycle states remaining** section and selecting the button.
5. If you want to select all the available states for a model, select the button. All states are transferred to the **Lifecycle states selected** section.
6. If you want to remove a selected state from the model, select the state in the **Lifecycle states selected** section and then select the button.
7. Select **Lifecycle state updates** to define which lifecycle states can follow a selected state.

Feedback

Was this page helpful? Yes No

Provide product feedback

Maintenance attribute types

Article • 01/09/2023

This article explains how to create attribute types in Asset Management. Attributes are used to describe the properties of various elements. You can set up attributes on the following elements:

- Functional location types
- Create functional locations
- Asset types
- Assets

The attributes that you can set up vary, depending on the element. For example, for a functional location, you can set up attributes for the configuration and physical size of the location. For an asset type or an asset, you can set up attributes for engine volume, power consumption, and maximum load capacity under different conditions.

Create attribute types

You can create your own attribute types. Additionally, you can transfer product dimensions to the **Attribute types** page.

1. Select **Asset management** > **Setup** > **Attribute types**.
2. The first time that you set up attribute types, select **Create product dimensions** to automatically transfer standard product dimensions.
3. Select **New** to create a new attribute type.
4. In the **Attribute type** field, enter a name for the attribute type.
5. In the **Description** field, enter a description.
6. In the **Unit** field, select the relevant attribute unit, as required.
7. In the **Data type** field, select a data type for the unit.
8. If you selected **String** as the data type, follow these steps to create values for the attribute type:
   a. Select the attribute type, and then select **Values**.
   b. In the **Attribute values** field, select **New**.
   c. In the **Attribute type** field, select an attribute type (dimension).
   d. In the **Value** field, enter a related value.

e. In the **Description** field, enter a description.
f. Save the record.
g. Return to the **Attribute types** page.

9. Save the record.

   The **Functional location types** field shows the number of functional locations that are using the attribute type. The **Asset types** field shows the number of asset types that are using it.

Feedback

Was this page helpful? Yes No

Provide product feedback

Functional location types

06/17/2025

This article describes how to create functional location types in Asset Management. Functional location types are used to manage requirements for functional locations, including how assets are installed on a functional location. You can set up asset types, maintenance plans, functional location attributes, and asset attribute requirements to be used on a functional location that uses the specific functional location type. When you create a functional location, the functional location type is mandatory.

> **Note**
>
> To work with functional locations, you must create a default functional location to be used only for the purpose of creating new assets. For that default functional location, you should create a default functional location type that is simple and allows multiple assets to be installed on the default functional location. To learn more about how to set up functional locations, go to Create functional locations.

Create a default functional location type

This procedure shows how to create a default functional location type to be used for a default functional location.

1. Select **Asset management** > **Setup** > **Functional locations** > **Functional location types**.
2. Select **New** to create a functional location type.
3. Insert a functional location type ID in the **Functional location type** field, for example, *Default*, and a name in the **Name** field.
4. Select a lifecycle model in the **Functional location lifecycle model** field.
5. Select *Yes* on the **Multiple assets** toggle button to allow more assets to be installed on a functional location (the default functional location) using this type.

Now, the default functional location type to be used only on a default functional location is created. You shouldn't add any more requirements or restrictions to this default functional location type.

Create Functional Location Types

1. Select **Asset Management** > **Setup** > **Functional locations** > **Functional location types**.
2. Select **New** to create a functional location type.

3. Insert a functional location type ID in the **Functional location type** field and a name in the **Name** field.
4. Select a lifecycle model in the **Functional location lifecycle model** field. To learn more about functional location lifecycle states and lifecycle models, go to Functional location lifecycle states.
5. Select *Yes* on the **Multiple assets** toggle button if it should be possible to install several assets on a functional location using this functional location type. If you select *No*, you can only install *one* asset on a functional location using this functional location type.
6. Select *Yes* on the **Update asset dimension** toggle button if you want assets installed on a functional location of this type to automatically use the financial dimensions related to the functional location. This means that if you change financial dimensions in the Create functional locations form, and the functional location uses a functional location type with this toggle button set to *Yes*, financial dimensions are automatically updated on all assets installed on that functional location.
7. The **Asset type** field is used if you want to automatically create *one* asset for the functional location with the same ID and name as the functional location you're creating. For example, this might be relevant if you create a static functional location, such as a building or a pipeline. In that case, select the asset type you want to use for the automatically created asset. Remember that if you make a selection in this field, the **Multiple assets** toggle button must be set to *No*.
8. On the **Asset types** FastTab, select the asset types to be related to the functional location type. Select **Add line** and select the asset types. If you add asset types here, only assets using those asset types can be installed on a functional location using this functional location type. If no asset types are selected on the **Asset types** FastTab, all asset types might be installed.
9. On the **Maintenance plans** FastTab, select the maintenance plans that should automatically be set up on new functional locations using this functional location type. Select **Add line** and select the maintenance plans. If you add maintenance plans here, only those plans can be used on a functional location using this functional location type.
10. On the **Asset attribute requirements** FastTab, set up the asset attributes that should automatically be set up on new functional locations using this functional location type. Select **Add line** and select the attribute. These attribute requirements function as guidelines. They aren't validated against attributes set up on an asset (to see the attributes, go to **Asset management** > **Assets** > **All assets**, select an asset in the list page, and on the Action Pane, open the **General** tab and select **Attributes**). The attribute requirements are shown when you install assets on functional locations.
11. On the **Permitted types** FastTab, select the functional location types that should be valid for sub functional location types related to a parent functional location type, which uses the selected functional location type.
12. On the **Attributes** FastTab, select the functional location attributes that should automatically be set up on functional locations using this functional location type. Select

Add line and select the attribute.

> **Note**
>
> On the **General** FastTab, you can get an overview of the number of asset types, maintenance plans, asset attribute requirements, permitted types, attributes, and functional locations set up on the functional location type. The **Functional locations** field shows the number of functional locations that use the functional location type. You can use the **Copy** button to copy settings from a functional location type to the selected functional location type.

Introduction to functional locations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an overview of functional locations in Asset Management. Functional locations are elements of a technical structure, such as the functional units in a system. Functional locations are created hierarchically, and you install assets on them. The setup of functional locations in your company depends on the company's requirements.

Here are some examples of how you can use functional locations:

- **Functional** – The functional locations can be user-oriented and used to manage assets that have similar behavior.
- **Process-related** – The functional locations can be workflow-oriented.
- **Spatial** – The functional locations can represent geographical locations or sites.

Each functional location is managed independently in Asset Management. Here are some of the useful features of functional locations:

- Set up functional location specifications.
- Set up asset specification requirements.
- Set up maintenance sequences for preventive and reactive maintenance.
- Manage installed assets.
- Track active requests and work orders that are related to installed assets.
- Track faults that are registered on assets.
- Track maintenance costs on the assets that are related to a functional location at any given time.

Functional locations provide traceability of assets in relation to requests, work orders, fault registrations, condition assessments, production stop registrations, and asset counter registrations.

> **Note**

Even if an asset is installed on various functional locations during its lifetime, the costs can be related to each location. In other words, asset costs are always related to the functional location that the asset was installed on at a given time.

Functional locations are **not** flexible. Therefore, after you set up a functional location hierarchy, you can't move locations around in it.

After you create a functional location hierarchy, the next step is to install assets on it. Learn more in Install assets on functional locations.

All functional locations

Select **Asset management** > **Functional locations** > **All functional locations** to open the **All functional locations** list page. This page shows all functional locations and some of the information that is related to each. To view only active functional locations, select **Active functional locations**. To view only the functional locations that you're related to as a worker, select **My active functional locations**. (This relation is set up on the **Workers** page. Learn more in Maintenance workers and worker groups.)

On the **All functional locations** list page, select a link in the **Functional location** column to view the details of the selected record. To edit the functional location, select the **Edit** button. The details view shows detailed information that is related to the location. It also includes a **Related information** pane on the right. This pane shows the functional location hierarchy. You can expand and collapse the **Related information** pane.

The buttons on the Action Pane are organized on tabs. The following table briefly describes the buttons that are related to Asset Management.

Expand table

Button name	Description
Edit	Switch between edit mode and view mode for the page.
New	Create a new functional location.
Delete	Delete the selected functional location.
Rename	Rename the selected functional location.
Copy functional location structure	Copy the functional location hierarchy.
Install asset	Install an asset, including child assets, on the functional location.

Button name	Description
Replace asset	Replace the asset hierarchy with another asset hierarchy on the functional location.
Cost control	Open the **Functional location cost control** page, where you can do a cost calculation for the selected functional location.
Hour control	Open the **Functional location hour control** page, where you can do a cost calculation for the selected functional location.
Assets	Open the **All assets** page, where you can view a list of assets that are related to the selected functional location.
Requests	Open the **Active requests** page, where you can view a list of requests that are related to the selected functional location.
Work orders	Open the **Active work orders** page, where you can view a list of work orders that are related to the selected functional location.
Faults	Open the **Asset faults** page, where you can view a list of asset fault registrations that are related to the selected functional location.
Update functional location state	Update the stage of the selected functional location.
Lifecycle state log	View a log that shows the stages of the selected functional location.

Last updated on 05/07/2025

Create functional locations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to create a functional location in Asset Management.

When you create a functional location structure, be aware once you have created a functional location, you cannot move it from the original location. This means that you should carefully consider the structure of your functional locations before you start creating them in Asset Management. If you regret a functional location, you can delete it, provided that it has not yet been taken into use.

To be able to work with functional locations, you start by creating two "categories" of functional locations:

- Create *one* default functional location with not sub locations. This functional location is used only as the standard location for assets when you create new assets.
- Create the functional location structures required for managing maintenance jobs in your company.

Create a default functional location

When you use functional locations, start by creating one default location to be used when you create new assets. This functional location is the one you select in **Asset management** > **Setup** > **Asset management parameters** > **Assets** link > **Default functional location** field. The default functional location can be used when you create new assets, and you have not yet set up a functional location structure for those assets.

1. Select **Asset management** > **Functional locations** > **All Functional locations**.
2. In **All functional locations**, select **New**.
3. Insert an ID in the **Functional location** field, for example, "0000" or "Default", to indicate that this is a special functional location.
4. Insert name for the default functional location in the **Name** field.
5. Do *not* select a parent in the **Parent** field – leave this field blank.
6. In the **Functional location type** field, select the functional location type to be used for the default functional location. See Functional location types for more information on how to

set up functional location types.

7. Select **OK**. You should not add further data to this functional location as it is only used as a temporary location for new assets until you install the assets on the functional locations used by your company.

Create functional locations

The following procedure describes how you create the functional locations required for maintenance management in your company.

1. Select **Asset management** > **Functional locations** > **All Functional locations**. You can create a functional location from grid view or details view.
2. Select the **New** button.
3. Insert an ID in the **Functional location** field.
4. Insert a name for the functional location in the **Name** field.
5. If the functional location is a sub location in a structure, select the parent location in the **Parent** field.
6. Select a type in the **Functional location type** field.
7. Select **OK**.
8. Select the functional location and click the **Edit** button to add further information.

> ⓘ **Note**
>
> Depending on your setup of functional location lifecycle states, you may have to create all sub locations for a functional location, and then change the functional location lifecycle state before you can start installing assets. See Install assets on functional locations for more information on asset installation. See Functional location lifecycle states to learn more about setup of functional location lifecycle states.

In Details view, you will see FastTabs on which you can add and edit information about the functional location.

General Information

This section provides an overview of parent and child information in the functional location structure. In the **Details** section, you can see the number of asset attributes, maintenance plans, and assets related to the functional location. In the **Inventory** section, you can select the site and warehouse to which the functional location is related. Site and warehouse is used in connection with work order item forecasts. When creating an item forecast, site and warehouse

information from the functional location of the asset is automatically used. In the **Lifecycle state** section, information about the functional location lifecycle state is displayed.

Installed assets

Refer to Install assets on functional locations for more information on asset installation. You can use the **View** button on this FastTab to show more fields on the FastTab. The **Valid from** and **Sub asset** fields can be shown in the grid.

Asset attribute requirements

On this FastTab you can add specific attribute requirements for the assets that you install on the functional location. These requirements are for information purposes only. They do not prevent you from installing assets with other attribute requirements. Select **Add line** and select the attribute type. Then you insert the relevant **Value**, select a threshold in the **Threshold criteria** field and save the record.

Maintenance plans and Maintenance rounds

Here you can add maintenance plans and maintenance rounds to the functional location, including a start date. The assets installed on a functional location may have other maintenance plans set up. All maintenance plans and maintenance rounds can be used for scheduling asset calendar entries for a functional location and its currently installed assets.

> **Note**
>
> If you update the setup of asset types, asset brands, and asset models on maintenance plans in **All functional locations** detail view > **Maintenance plans** FastTab after you have scheduled maintenance plans, existing maintenance schedule entries related to that functional location are automatically deleted. In order to create new schedule entries, which correspond with the updated maintenance plan setup on the functional location, you must run a new maintenance plan schedule for that functional location.

Address

Insert the functional location address on the **Address** FastTab. Addresses on functional locations are inherited, meaning if a sub location has no address defined, the address of the parent location is used.

Workers

On this FastTab, you can add workers affiliated with the functional location, and you can select a functional location as primary for the worker.

Attributes

On this FastTab, you can set values for functional location attributes. These attributes can be used to describe properties or characteristics pertinent to the functional location, for example, structural properties, building type, area descriptions, or location above or under ground.

Select **Add line** and select the attribute type. Next, insert the **Value** related to the attribute type and save the record.

Financial dimensions

You can select financial dimensions for the functional location. Functional location types can be set up to allow for automatic update of financial dimensions from a functional location. This means that assets installed on a financial dimension automatically get the financial dimensions for the functional location. This is useful if you want different cost centers, depending on locations.

When data regarding **Site**, **Warehouse**, **Address**, and **Financial dimensions** are updated on a parent functional location, the related sub functional locations can be updated accordingly if you make that selection during the update. A dialog opens, providing you with the update options.

Copy a functional location structure

If your company has several functional locations with similar location structures, you can use the copy function in Asset Management to quickly create a number of similar location hierarchies. When you copy a specific functional location or an entire structure, the new location or structure has the same name as the one you copied. After the copy procedure is done, you can easily change the name or other settings on the new functional location, provided that the functional location lifecycle state selected for the new functional location allows it.

1. In **All functional locations**, select the functional location you want to copy. For example, you select a top location (parent) if you want to copy the entire functional location structure including sub locations.

2. Select the **Copy functional location structure** button. The location you selected in the list page is shown in the **Copy from** field.
3. Insert the name of the new location in the **New functional location** field.
4. In the **Parent to paste under** field, you should only insert a parent ID if the location you are creating should be part of an existing functional location structure.
5. Click **OK**. The new functional location structure is shown in **All functional locations**.

> **Note**
>
> When you copy a functional location structure, functional location lifecycle states in the new structure are set to the "first state" that you have created for functional locations. Whether you can rename or delete a functional location using the **Rename** and **Delete** buttons in **All functional locations**, depends on the current lifecycle state of the functional location.

Delete a Functional Location

A functional location with related sub locations can be deleted if no assets have been installed on any of the functional locations you are trying to delete, and if the current functional location lifecycle state allows it.

1. In **All functional locations**, select the functional location you want to delete.
2. If required, update the functional location to a functional location lifecycle state that allows deletion of a functional location.
3. Select **Delete**.

> **Note**
>
> If you cannot delete a functional location, instead you can handle deletion by setting up a functional location lifecycle state for this purpose. For example, you can set up a "Scrapped" or "Deleted" stage, which should not be an active stage, in the **Functional location lifecycle states** form.

Last updated on 05/07/2025

Change the type of existing functional locations

This article explains how to change the functional location type of an existing functional location.

Prerequisites

Before you can use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management 10.0.38 or later.
- The feature that's named *Change types on assets and functional locations* must be turned on in Feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default. As of version 10.0.45, it's mandatory and can't be turned off.

Change the functional location type of one or more existing functional locations

To change the functional location type of one or more existing functional locations, follow these steps:

1. Go to **Asset management** > **Functional locations** > **All functional locations**.

2. On the **All functional locations** list page, select one or more functional locations that you want to change the type for. If you select multiple functional locations, all of them must be of the same type.

3. On the Action Pane, on the **Functional locations** tab, in the **Maintain** group, select **Change functional location type**.

4. In the **Change functional location type** dialog box, on the **Parameters** FastTab, set the following fields:

    - **New functional location type** – Select the new type to assign to the selected functional locations.
    - **Attributes** – Set this option to *Yes* to append the attribute types that are associated with the new functional location type to the selected functional locations.
    - **Asset attribute requirements** – Set this option to *Yes* to append the asset attribute requirements that are associated with the new functional location type to the selected

functional locations.

- **Maintenance plans** – Set this option to *Yes* to append the maintenance plans that are associated with the new functional location type to the selected functional locations.

5. On the **Run in background** FastTab, set the **Batch processing** option to *Yes* to run the process of changing functional location types as a batch job.

6. Select **OK** to apply your settings.

> **Note**
>
> Each functional location type has settings that define which supporting data applies to functional locations of that type. This supporting data includes information such as asset types, maintenance plans, and asset attribute requirements. When you change the functional location type, the supporting data for existing functional locations might not match the supporting data for the new functional location type.

Last updated on 12/30/2025

Install assets on functional locations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

After you've created functional location structures, the next step is to install assets on the relevant functional locations. This article explains how to install assets on those functional locations in Asset Management. For information about how to create assets, see Introduction to assets.

If you've created an asset structure, the whole asset structure must be installed on a functional location. Therefore, only parent assets (top-level assets that have no parent asset) can be selected on a functional location. All related child assets (sub-assets) will also be installed on the functional location. When you install assets on a functional location, the financial dimensions of the functional location might be automatically transferred to them, depending on the setup on the functional location type that is selected for the functional location. For more information about how to set up functional location types, see Functional location types.

> **Note**
>
> You can set up asset types on the functional location type that is used for a functional location. In this case, when you install assets on the functional location, only parent assets that have the same asset type are shown in the list of assets that can be installed on the functional location.

After you've installed assets on a functional location, you can replace a parent asset or an asset structure as you require. As when you install assets, you select a parent asset to replace. All related child assets will also be replaced.

Install an asset structure on a functional location

1. Select **Asset management** > **Functional locations** > **All Functional locations** or **Active functional locations**.
2. Select the functional location to install an asset on.
3. Select **Install asset**.

The **Attributes** section shows a list of the asset attribute requirements that are set up on the functional location type that is selected for the functional location. The attributes are for informational purposes only. The system doesn't validate the attributes against the asset attributes that are set up on the asset that you're installing. You must do that validation after you select an asset in the **Asset** field.

4. In the **Asset** field, select the parent asset to install. All related child assets are automatically included in the installation.

   The **Asset attributes** section to the right of the asset list shows the asset attributes that are related to the selected asset.

5. In the **Effective** field, select the date and time that the asset installation is valid from. After that date and time, costs for the asset and related sub-assets will be related to the functional location.

   > ⓘ **Note**
   >
   > The asset attributes that are set up on the asset are added to the **Attributes** section. For example, the **Weight** attribute requirement has been added as a requirement on both the asset and the functional location. If the asset has attribute requirements of the same type as the functional location, the values of the asset's attribute requirements are entered in the **Value** fields. Therefore, you can validate the asset values against the attribute requirements that are set up on the functional location. The attribute requirements that are set up on the functional location are marked with a check mark.

6. Select **OK**.

   > ⓘ **Note**
   >
   > To change the installation of an asset by installing it on a new functional location, follow steps 1 through 6 of this procedure. When you install an asset on a new functional location, the asset is automatically uninstalled from the previous functional location. Any active maintenance requests or work orders that were created on the asset before you installed it on a new functional location are **not** automatically transferred to the new functional location. If those maintenance requests and work orders are still required for the asset, you must manually re-create them after the asset is installed on the new location.

7. To view a list of all the assets, including sub-assets, that are installed on the functional location, select the functional location on the **All Functional locations** page, and then select **Assets**.

8. To view a list of active maintenance requests, active work orders, or fault registrations that are related to the assets that are installed on a functional location, select the functional location on the **All Functional locations** page, and then select **Requests**, **Work orders**, or **Faults**.

> **Note**
>
> When asset-related data is changed, it's automatically updated on the functional location that the asset is installed on. This automatic update pertains to changes to maintenance requests, work orders, asset fault registrations, maintenance downtime registrations, and asset measure registrations.

Automatically create one asset on a functional location

You can set up functional location stages and functional location types to handle the automatic creation of *one* asset on a functional location. The asset gets the same ID and name as the functional location. This functionality might be useful when you're handling maintenance on a large, static asset, such as a building.

Before you can automatically create an asset on a functional location, the following setup data must be available:

- Create a functional location type to handle the automatic creation of an asset. In the **Asset type** field, select an asset type. Learn more in Functional location types.
- Create a functional location lifecycle state to handle the automatic creation of an asset. Set the **Create asset** option to **Yes**. Learn more in Functional location lifecycle states.

After the setup data is available, you're ready to create an asset.

1. On the **All Functional locations** page, make sure that the functional location where you want the asset to be automatically created uses the functional location type that you created for this purpose.
2. Select the functional location in the list.
3. Select **Update functional location state**, and then select the lifecycle state that you created for this purpose. One asset is now automatically installed on the functional location. This asset has the same name as the functional location.

Last updated on 05/07/2025

Asset types

Article • 01/06/2025

This article explains how to create asset types. It also describes the elements that are related to asset types. Asset types are used as general categories for assets. Examples include CNC machines, measuring equipment, and truck engines. Asset types are used to manage the maintenance job types (maintenance tasks), asset lifecycle states, counters, asset attributes, condition assessment templates, and asset models that can be selected for an asset. When you create an asset, you must specify the asset type.

For each asset type, variations of the asset type setup can be created. For example, if you have an asset type that is named *Trucks*, you can create variations of that asset type for different asset manufacturers and asset models. To each asset type setup, you can add the required spare parts and maintenance plans.

First, you set up the required asset types. Next, you create the asset models that should be related to the asset types. Finally, on the **Asset type defaults** page, you create all the variations of asset types that are required for your equipment.

Create an asset type

1. Select **Asset management** > **Setup** > **Asset types** > **Asset types**.
2. Select **New** to create an asset type.
3. In the **Asset type** field, enter an asset type ID.
4. In the **Name** field, enter a name.
5. In the **Asset lifecycle model** field, select an asset lifecycle model. For more information about asset lifecycle states and asset lifecycle models, see Asset lifecycle states.
6. Set the **Total** option to *Yes* if summarized key performance indicator (KPI) values should be calculated for assets that have this asset type.
7. Select **Save**.
8. On the **Maintenance job types** FastTab, select the maintenance job types that should be related to the asset type:
    - To select a maintenance job type, select it in the **Maintenance job types remaining** field, and then select the right arrow button → to move it to the

Maintenance job types selected section.

- To select all available maintenance job types, select the button. All maintenance job types are transferred from the **Maintenance job types remaining** field to the **Maintenance job types selected** field.
- To cancel the selection of a maintenance job type, select it in the **Maintenance job types selected** field, and then select the left arrow button to move it to the **Maintenance job types remaining** field.

9. You can also select the counters that should be related to the asset type. On the **Counters** FastTab, make your selections by using the methods that are described for maintenance job types in step 8. For more information about the setup of counters, see Counters.

10. You can also select the attribute types that should be related to the asset type. On the **Attribute types** FastTab, make your selections by using the methods that are described for maintenance job types in step 8. Then, to create the preferred sequence of attribute types, select an attribute type in the **Attribute types selected** field, and use the up arrow and down arrow buttons to move it. The sequence of attribute types will be shown on assets that use this asset type. For more information about asset attributes, see Maintenance attribute types.

    > ⓘ **Note**
    >
    > When you add new attribute types on the **Attribute types** FastTab, existing assets are automatically updated with that information.

11. You can also select the condition assessment templates that should be related to the asset type. On the **Condition assessments** FastTab, make your selections by using the methods that are described for maintenance job types in step 8. For more information about condition assessment templates and registrations, see Condition assessment.

12. The **Asset model** FastTab shows all the combinations of asset manufacturers and models that are set up on the selected asset type. To see the combinations divided according to manufacturer, select **Asset model** to open the **Asset model** page.

    On the **Asset model** page, you can add asset model–asset type relations. Additionally, on the **Asset types** page, you can add asset manufacturer–asset model relations directly to an asset type. Finally, on the **Asset model** page (**Asset management** > **Setup** > **Assets** > **Asset model**), you can create new asset manufacturer–asset model–asset type relations. Therefore, there are three ways to set up and edit asset manufacturer–asset model–asset type relations. All the

available combinations are shown from different perspectives, and you can select your preferred point of entry when you work with the setup.

> ⓘ Note
>
> - If you select counters on an asset type, the selections are automatically updated on the **Counters** page (**Asset management** > **Setup** > **Assets** > **Asset types** > **Counters**).
> - The fields in the **Details** section on the **General** FastTab show the number of maintenance job types, counters, attributes, and so on, that are set up on the selected asset type.

Typically, work orders that are manually created are related to corrective maintenance, whereas work orders that are automatically created are related to preventive maintenance. When you manually create work orders, only the maintenance job types that are selected on the **maintenance job types** FastTab of the **Asset types** page can be used. However, automatically created work orders can use all the maintenance job types you create on the **Maintenance job types** page (**Asset management** > **Setup** > **Jobs** > **Maintenance job types**).

Create asset type setup lines

1. Select **Asset management** > **Setup** > **Assets** > **Asset types** > **Asset type setup**. Alternatively, select **Asset management** > **Setup** > **Assets** > **Asset types** > **Asset types**, select an asset type, and then select **Asset type setup**.

2. The first time that you use the **Asset type setup** page, you might find the **Create combinations** button useful. You can use this button to quickly create all combinations of an asset model on an asset type. Select **Create combinations**, select the asset type to create combinations for, and then select **OK**.

   > ⓘ Note
   >
   > If you won't use all the asset type setup combinations that were automatically created, you can delete a setup by selecting it and then selecting **Delete**.

3. Select **New** to manually create an asset type setup.

4. Depending on how specific the asset type setup should be, make selections in the **Asset type**, **Manufacturer**, and **Model** fields.

5. If a warranty agreement is related to the asset type, select the agreement in the **Vendor warranty** and **Customer warranty** fields.

6. On the **Spare parts** FastTab, select **Add** to add spare parts to the selected asset type setup.

7. To approve a spare part, select the spare part line, and then select **Approve**. You can select multiple lines for approval.

8. To see whether a spare part is used somewhere else in Asset Management (for example, in relation to assets and work orders), select the spare part line, and then select **Item where used** to open the **Item where used** page. To see all active spare parts in the list, select the **Active** check box. To see only approved spare parts, select the **Approved** check box.

9. On the **Maintenance plans** FastTab, select **Add** to add maintenance plans to the selected asset type setup.

10. To copy an asset type setup to another setup, you can use the Copy function. Select the asset type setup to copy a setup to, select **Copy setup**, and select the asset type setup to copy the setup from. The settings of the various options determine how much information is included. When you've finished, select **OK** to copy the setup.

> ⓘ **Note**
>
> If you have many spare part lines and maintenance plan lines that you'll reuse, the Copy function lets you quickly and easily set up data for many asset type setup combinations.

Spare parts on the asset type setup

As was described in the "Create asset type setup lines" section, spare parts are set up on asset models on the **Asset type setup** page. Therefore, when you open the **Asset type setup** page, you see only the spare parts that are related to the selected combination of an asset type, asset manufacturer, and asset model. To see a list of all spare part records, open the **Spare parts** page (**Asset management** > **Inquiries** > **Spare parts**).

On the **Spare parts** page, you can also create new spare parts for existing combinations of an asset type, asset manufacturer, and asset model. You can decide whether you prefer to create spare part records on the **Asset type setup** page or the **Spare parts** page. The **Asset type setup** page provides an overview of data on the selected

combination of an asset type, asset manufacturer, and asset model, whereas the **Spare parts** page provides a complete overview of all asset type setup lines. If the **Spare parts** page contains many records, the **Asset type setup** page might give you a better overview.

To see whether the spare part on the selected line is used anywhere else in Asset Management (for example, in relation to assets and work orders), select **Item where used** to open **Item where used** page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Condition assessment

> **Note**
>
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This article explains how to create a condition assessment template and registration on an asset in Asset Management. Condition assessment is performed at regular intervals, and the primary objective is to create and maintain condition data on assets. Seen from a preventive maintenance perspective, it is important to monitor key information such as current condition, and remaining life span. Furthermore, if you carry out condition assessment at regular intervals, you will be able to monitor and compare conditions on the machinery in your factory.

Condition assessment can be used to measure and monitor many conditions on your equipment. Example: You could measure vibrations on your machinery. After you have registered vibration measurements in Asset Management on various types of equipment, you can search for the latest registered assessment and view vibration measurements.

Condition assessment is created on assets. You set up a condition assessment template on an asset type before you carry out the condition assessment procedure. The reason for using templates for condition assessment is to avoid variation of condition data on similar assets. The sequence for setting up and using condition assessment in Asset Management is: First you set up the required condition assessment templates. Next, you associate templates with asset types in the **Asset types** form. Finally, you can create condition assessment registrations on an asset in the **Asset** form.

Create a condition assessment template

1. Select **Asset management** > **Setup** > **Asset types** > **Condition assessment**.
2. Select **New** to create a new template.
3. Insert and ID for the template in the **Template** field.
4. Insert a name for the template in the **Name** field.
5. On the **Condition assessment lines** FastFab, add the lines required for the condition assessment, including selection of the appropriate condition type and measurement unit.
6. On the **Asset types** FastTab, add the asset types that should use the condition assessment template.

7. In the **Lines** and **Asset types** fields in the **Details** group at the top of the screen, you will see the number of assessment lines and asset types related to the selected condition assessment template.

Create condition assessment registration on an asset

1. Select **Asset management** > **Assets** > **All Assets**.
2. In the list, select the asset for which you want to create a condition assessment registration.
3. On the **General** tab, click **Condition assessment**.
4. Click **New** to make a new registration.
5. Select the date for the condition assessment in the **Date** field.
6. Select the name of the worker who carried out the assessment registration in the **Worker** field.
7. In the **Lines** field, you see the number of assessment lines set up on the condition assessment.
8. Select a template for the condition assessment in the **Template** field. The name of the template is automatically inserted in the **Name** field, and the related registration lines are inserted on the **Condition assessment lines** FastTab.
9. You can insert notes relating to the selected condition assessment on the **Notes** FastTab.
10. For each condition assessment line, insert measurement data in the **Value** field.
11. You can insert a comment relating to the selected registration line on the **Condition assessment lines** FastTab > **Comments** field. If you add a comment on a line, the **Comment** check box is automatically selected.

After you have made a condition assessment registration on an asset, you can print a condition assessment report.

> ⓘ **Note**
>
> You can also register condition assessment on a work order (**Asset management** > **Work orders** > **All Work orders** > **Condition assessment** button.)

Last updated on 05/07/2025

Asset manufacturers and models

Article • 08/15/2024

This article explains how to set up asset manufacturers and related models in Asset Management. Models can be related to asset types.

Set up product-model relations

1. Select **Asset management** > **Setup** > **Assets** > **Manufacturer and model**.
2. Select **New** to create a new product.
3. In the **Manufacturer** field, enter a name for the asset manufacturer.
4. In the **Description** field, enter a description.
5. On the **Models** FastTab, select **Add** to create an asset model that should be related to the asset manufacturer.
6. In the **Model** field, enter a name for the asset model.
7. In the **Description** field, enter a description.
8. In the **Asset type** field, select the asset type that the manufacturer model should be related to.

   > ⓘ **Note**
   >
   > You can also set up relations for asset types, manufacturers, and models in the **Asset types** lookup. Learn more in Asset types.

   In the **Details** FastTab, the **Models** field shows the number of asset models that are set up on the selected asset manufacturer. The **Assets** field shows the number of assets that are using the selected manufacturer.

   The **Assets** field shows the number of objects that are using the manufacturer model.

> ⓘ **Note**
>
> An asset type can have no asset manufacturer model relations, it can be related to one asset manufacturer model, or it can be related multiple asset manufacturer

models. If an asset type is related to at least one manufacturer model, only the combinations that are set up in the **Manufacturer model** lookup can be selected on those Asset Management pages where a combination of an asset type, manufacturer, and model can be set up. These pages include **All assets**, **Asset service levels**, **Job type defaults**, and **Maintenance budget lines**. If some asset types aren't related to any manufacturer model, only those asset types, and manufacturer models that also have no relation to asset types, are shown on the pages.

Select a manufacturer and model on an object

1. Select **Asset management** > ***Assets** > **All assets**.
2. In the **Asset** column, select the link for the asset. The **Details** page appears.
3. Select **Edit**.
4. On the **General** FastTab, select values in the **Manufacturer** and **Model** fields.

Feedback

Was this page helpful? Yes No

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Counters

Article • 01/06/2025

The article explains how to create counter types in Asset Management. Counter types are used to make counter registrations on assets, for example, regarding number of production hours, or quantity produced on the asset. Asset types are related to the counter types. This means that a counter can only be used on an asset if the counter is set up on the asset type used on the asset.

Before you can make counter registrations on assets, you first create the counter types you want to use in **Counters**. Next, you can create counter registrations on assets in **Counters**.

Counters can be used on maintenance plans. A maintenance plan line can be of type *Counter*, for example, relating to number of production hours or quantity produced.

A counter registration can be updated manually or automatically based on production hours or quantity produced. A counter can be set up to use one of three update methods (selected in the **Update** field in **Counters**):

- Manual - you must manually register counter values.
- Production hours - the counter is automatically updated based on number of production hours.
- Production quantity - the counter is automatically updated based on quantity produced.

> ⓘ **Note**
>
> If quantity produced is used, *all* registered items are included in the counter registration, good quantity and error quantity. It's always possible to make manual counter registrations, if necessary.

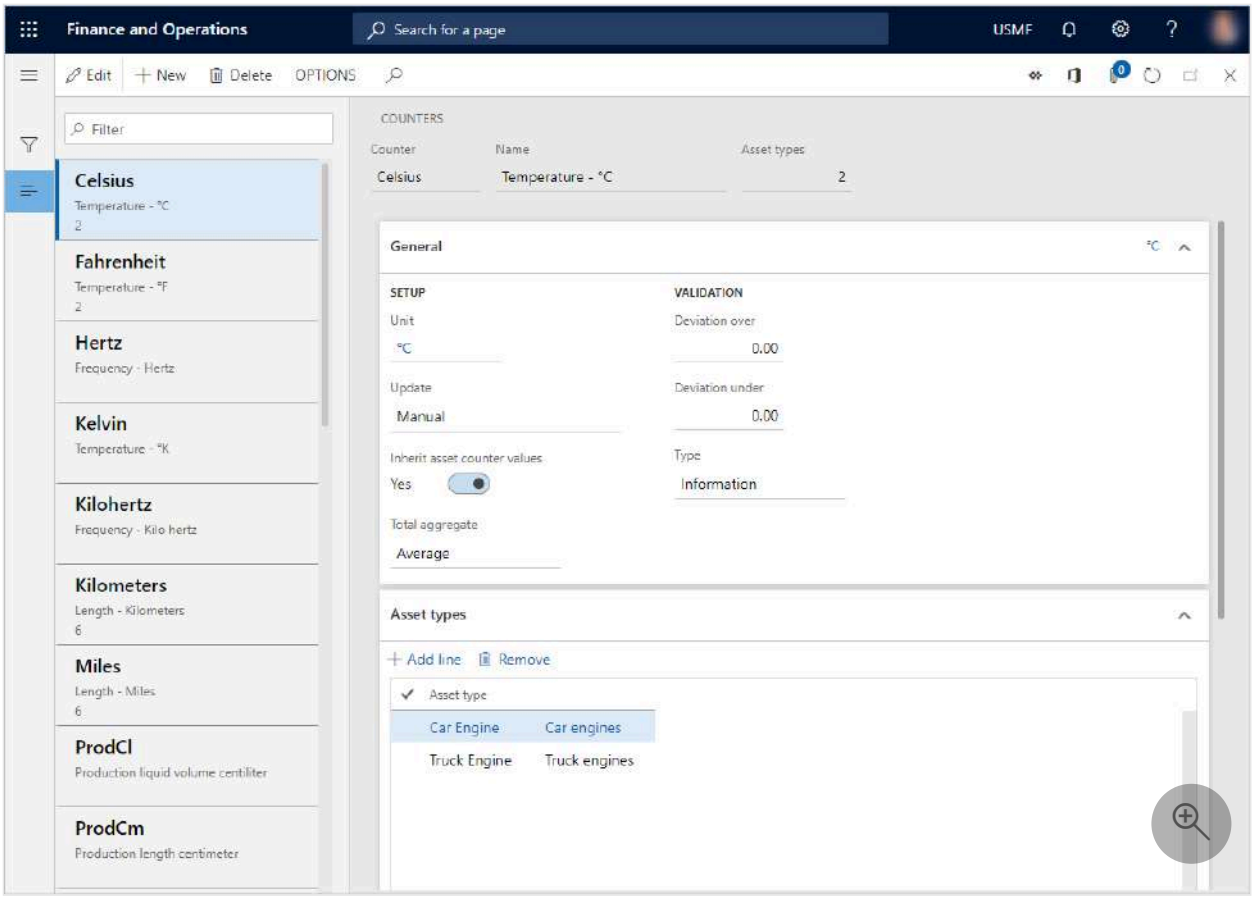
Create counter types for asset counter registrations

1. Go to **Asset management** > **Setup** > **Asset types** > **Counters**.
2. Select **New** to create a new counter type.
3. Insert an ID in the **Counter** field, and a counter name in the **Name** field.
4. On the **General** FastTab, select a counter unit in the **Unit** field.
5. In the **Update** field, select the update method to be used for the counter.

6. Select *Yes* on the **Inherit counter values** toggle button if child assets in an asset structure should automatically inherit counter registrations made on the parent asset.
7. In the **Total aggregate** field, select the summation method to be used for a counter using this counter type. *Sum* is the standard selection used to continuously add registered values to the total value. *Average* can be used if a counter is set up to monitor a threshold, for example, regarding temperature, vibrations, or wear and tear on an asset.
8. In the **Deviation over** field, insert the upper level in percent for validating if manual counter registrations are within an expected range. The validation is based on a linear increase in existing counter registrations.
9. In the **Deviation under** field, insert the lower level in percent for validating if manual counter registrations are within an expected range. The validation is based on a linear decrease in existing counter registrations.
10. In the **Type** field, select the type of message (information, warning, error) to be shown if deviations outside the defined range occur when you make manual counter registrations.
11. On the **Asset types** FastTab, add the asset types that should be able to use the counter.
12. On the **Related asset counters** FastTab, add the counter that you want to be automatically updated when this counter is updated.

> **Note**
>
> A related counter is automatically updated only if the related counter has the asset type, to which it's related, in the counter setup. For example, suppose you set up a counter for *Production hours* and add the asset type *Truck Engine*. When that counter is updated, a related counter *Oil* is also updated with the same counter values. The setup in **Counters** includes the setup on *Hours*. Also, on the *Oil* counter, the asset type *Truck Engine* should be added to the **Asset types** FastTab to ensure the counter relation.

When asset types are added to a counter type in **Counters**, that counter is automatically added to the asset types on the **Counters** FastTab in Asset types.



Feedback

Was this page helpful? Yes No

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Asset Management parameters

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to set up the general parameters for Asset Management, which relate to assets, work orders, and work order scheduling. Select **Asset management** > **Setup** > **Asset management parameters** to open the page.

> **Note**
>
> If you'd like to set up a system that includes demo data for testing Asset Management features, see Deploy a demo environment for instructions.

The Assets tab

The **Assets** tab provides the following settings:

- **Default functional location** is the standard functional location that's automatically selected on assets when you create new assets.
- In the **Standard calendar** field, select a calendar to be used for calculating asset KPIs if no resource is selected on an asset.
- In the **View** field, select the standard view that's shown when you open **Asset view** (**Asset management** > **Assets** > **Asset view**).
- **Default request type** is the standard maintenance request type that's automatically selected when you create a new request.
- The project selected in the **Forecast project** field stores forecasts on job types. For each job type, the system automatically creates a new activity on the forecast project. The system then saves forecasts on the job type to the forecast project.
- In the **Model** field, select the forecast model used on job type and work order forecasts.

The Work orders tab

The **Work orders** tab provides the following settings:

- **Default work order type** defines standard settings when creating a work order.
- **Preventive work order type** defines the work order type used when creating work orders from maintenance plans. If you leave this field blank, the work order type in the **Default work order type** field is used.
- In the **Related work order mask** field, you define the maximum number of work orders that can be related to a work order. For example, `##` allows you to have up to 99 work orders related. If you define a mask as described here, related work orders are numbered `[work order ID of the work order to which a work order is related] -01`, `-02`, `-03`, and so on. If you don't define a mask in this field, a related work order gets the next sequential work order ID.
- Select *Yes* for **Copy faults** if you want to automatically copy faults registered on maintenance requests to related work orders.
- In the **Level** field, you define the functional location level that is automatically inserted on a work order if all related work order jobs refer to the same functional location. If the work order jobs don't all relate to the same functional location on the defined level, the **Functional location** field is left blank on the work order. For example, if you insert the number *1* in this field, that number is the top level in a functional location structure. If you insert the number *0* in this field, you haven't defined a specific functional location level, only that all work order jobs on a work order must be related to the same functional location for that functional location to be added to the work order.
- Select the journals used when posting consumption on a work order on the **General** FastTab in the **Hour**, **Item**, and **Expense** fields.
- In the **Product language source** field, select which language to use for product names in Asset management reports. You can select the language set up on the company account, or the language set up for the user currently logged in.
- Select *Yes* for **Real time update** if you want to automatically update changes to job type defaults, maintenance plans, and maintenance rounds.
  - If you select *No*, changes to job type defaults, maintenance plans, and maintenance rounds aren't updated automatically in Asset Management.
  - Select *No* if you have large amounts of data being synchronized, for example, many assets or functional locations set up on maintenance plans or maintenance rounds, or a large number of maintenance plans or rounds.

    - If you make changes to job type defaults or maintenance plans or maintenance rounds, and you have selected *No* to real time update, a warning might not be shown if the changes influence:
        - Functional locations set up on maintenance plans or rounds
        - Objects set up on maintenance plans or rounds
        - Maintenance plans setup
    - Maintenance rounds setup
- On the **Category** FastTab, define default categories relating to consumption on work orders.

The Work order scheduling tab

The **Work order scheduling** tab provides the following settings:

- **Schedule time fence** defines the period in days, calculated from the expected start date of the work order, during which work order jobs are planned.

- **Default scheduled start date** - Controls how the system pre-fills the **Scheduled start** field on the **Schedule work order** dialog when you schedule one or more work orders or work order lines. Choose one of the following options:
    - *Expected start date from header* – Use the earliest **Expected start** date assigned among the selected work order headers. This option is the default.
    - *Scheduled start date from lines* – Use the earliest **Scheduled start** date already set on the selected work order lines. This option is useful when line schedules are planned in advance and you want the dialog to honor those dates instead of the header's expected start.

- **Service level** - A rating score calculated together with the rating score in the **Criticality** and **Start date** fields. The number in this field is divided by the number in the **Service level** field on a work order. For example, if you insert the value *5.00* in this field, and a work order has **Service level** of *20*, the rating score for **Service level** is *0.25*.

- **Criticality** - A rating score calculated together with the rating score in the **Service level** and **Start date** fields. The number in this field is multiplied by the number in the **Criticality** field on a work order. For example, if you insert the value *10.00* in this field, and a work order has criticality *5*, the rating score for criticality is 50.

- **Start date** - A rating score calculated together with the rating score in the **Service level** and **Criticality** fields. This field indicates daily score as a negative value and is compared to the **Expected start** field on a work order. For example, if you insert the value *10.00* in this field,

and the expected start date of a work order is three days from now, the rating result is minus 30.00. Adding the results of 0.25 and 50 from the examples in the **Service level** and **Criticality** fields described earlier provides a total of plus 20.25. That number is compared to all other work order rating scores during work order scheduling, and the highest rating scores are then planned first.

- **Responsible worker** - A rating score calculated together with the **Responsible maintenance worker group**, **Preferred maintenance worker**, **Preferred maintenance worker group**, **Asset location**, and **Start date** rating score values. If you insert the value *50.00* in this field, and a responsible worker is selected on a work order, the worker gets 50 points in the overall worker calculation during work order scheduling.

- **Responsible maintenance worker group** - A rating score calculated together with the **Responsible worker**, **Preferred maintenance worker**, **Preferred maintenance worker group**, **Asset location**, and **Start date** rating score values. If you insert the value *50.00* in this field, and a responsible worker is selected on a work order, the worker gets 50 points in the overall worker calculation during work order scheduling.

- **Limit to responsible** - This setting limits the number of workers available for work order scheduling. Select *No* if you want to calculate a score for all workers, regardless if they're set up as responsible workers or part of a responsible maintenance worker group. Select *Yes* if you want to calculate a score for workers who are set up as responsible worker on the work order, and/or included in a responsible maintenance worker group selected on the work order.

- **Preferred maintenance worker** - A rating score calculated together with the **Responsible worker**, **Preferred maintenance worker**, **Preferred maintenance worker group**, **Asset location**, and **Start date** rating score values. The four rating scores are calculated and added together to provide a score used for selecting which worker should be assigned to which work order during work order scheduling. If you insert the value *10.00* in this field, and a worker is selected as preferred maintenance worker on a work order, that worker gets 10 points in the overall worker calculation during work order scheduling.

- **Preferred maintenance worker group** - A rating score calculated together with the **Responsible worker**, **Preferred maintenance worker**, **Preferred maintenance worker group**, **Asset location**, and **Start date** rating score values. If you insert the value *10.00* in this field, and a worker is assigned to a preferred maintenance worker group selected on a work order, that worker gets 10 points in the overall worker calculation during work order scheduling.

- **Limit to preferred** - This setting limits the number of workers available for work order scheduling. Select *No* if you want to calculate a score for all workers, whether they are set up as preferred maintenance workers or part of a preferred maintenance worker group. Select *Yes* if you only want to calculate a score for workers who are set up as preferred maintenance workers, and/or included in a preferred maintenance worker group.

- **Location** - A rating score calculated together with the **Responsible worker**, **Preferred maintenance worker**, **Preferred maintenance worker group**, **Asset location**, and **Start date** rating score values. If you insert the value *3,000.00* in this field, a worker gets 3,000 points in the calculation if the worker is located in the same factory or facility as the asset on which a job is to be scheduled.
  - If your company uses functional locations, workers get full score if they're located on the functional location related to the asset. If the functional location of the asset has a parent asset, workers on that functional location get half score. If that location also has a parent, workers on that location get one-third score. If that location also has a parent, workers on that location get one-fourth score, and so on.
  - If your company uses asset location, which isn't recommended, location, area, and zone are used to calculate location scores. Workers get full score if they're located in the location and area and zone related to the asset. If worker location only matches location and area, the rating score for the worker is two-thirds of the full score. If worker location only matches location, the rating score for the worker is one-third of the full score.

- **Limit to location** - This setting limits the number of workers available for work order scheduling. Select *No* if you want to calculate a score for all workers across all functional locations. Select *Yes* if you only want to calculate a score for workers who are associated with the work order's functional location.

  > ⓘ **Note**
  >
  > The three *Limit to* fields increase the speed of work order scheduling by limiting the number of scores calculated for workers.

- **Worker's start date** - A rating score calculated together with the **Responsible worker**, **Preferred maintenance worker**, **Preferred maintenance worker group**, **Asset location**, and **Start date** rating score values. This field indicates daily score as a negative value and is compared to the **Expected start** field on a work order. If you insert the value *10.00* in this field, and the expected start date of a work order is tomorrow, the rating result is minus 10.00.

- Assuming that no responsible worker and responsible maintenance worker group are selected on a work order to be scheduled - you add and subtract the rating score values in the examples in the **Preferred maintenance worker**, **Preferred maintenance worker group**, **Asset location**, and **Start date** fields described earlier, you get a total of 3,010.00. This total means a high score for the worker who is already selected as preferred maintenance worker and included in the preferred maintenance worker group on the work order, and the worker is also located in the same facility as the asset for which a job needs to be scheduled. This means there's a good chance that the worker in question is selected to complete the job during work order scheduling.
- If you insert the value *0.00* in one of the eight fields described earlier, that rating score isn't used during work order scheduling.

The Document types tab

Select the document types that you want to print attachments for when they're related to a work order report. Select a document type in the **Available** section and select . To remove a selected document type, select the document type in the **Selected** section and select .

The Number sequences tab

Select the required number sequences in this section. Two number sequences exist for assets: one for manually created assets and one for assets created through pending assets.

Last updated on 05/07/2026

Asset lifecycle states

09/24/2025

This article explains asset lifecycle states and lifecycle models in Asset Management. Asset lifecycle states are used to define whether an asset is active or inactive. For example, you can set up asset lifecycle states such as **Created**, **Active**, and **Terminated**.

Asset lifecycle states can be set up in asset lifecycle models, where you can define the required lifecycle states for various types of assets. You first set up lifecycle states. You then create a lifecycle model and select lifecycle states for it.

1. Select **Asset management** > **Setup** > **Assets** > **Lifecycle states**.
2. Select **New** to create a new asset lifecycle state.
3. In the **Lifecycle state** field, enter the lifecycle state ID.
4. In the **Name** field, enter a description.

   The **Lifecycle models** field shows the number of asset lifecycle models that use the asset lifecycle state.
5. Set the **Active** option to **Yes** if this lifecycle state should be an active lifecycle state (in other words, if work orders can be created for assets that are in this lifecycle state).
6. Set the **Delete open calendar lines** option to **Yes** if open asset calendar lines that have an asset lifecycle state of **Created** should be deleted when they are in this lifecycle state. This setting is useful if you want to clean up any open maintenance schedules that are no longer relevant for the asset (for example, if the asset is no longer active).

> **Note**
>
> Asset lifecycle states, asset lifecycle models, and asset types are related. They're used in the same way as work order lifecycle states, work order lifecycle models, and work order types.

After you've created the required asset lifecycle states, you can set up lifecycle states in asset lifecycle models.

1. Select **Asset management** > **Setup** > **assets** > **lifecycle models**.
2. Select **New** to create a new asset lifecycle model.
3. In the **Lifecycle model** field, enter the lifecycle model ID.

4. In the **Name** field, enter a description.

   The **Lifecycle states** field shows the number of asset lifecycle states that are selected in the asset lifecycle model. The **Asset types** field shows the number of asset types that use the lifecycle model.

5. On the **Lifecycle states** FastTab, select the asset lifecycle states that should be included in the asset lifecycle model:

   - To use a lifecycle state for the model, select it in the **Lifecycle states remaining** section, and then select the right arrow button to move it to the **Lifecycle states selected** section.
   - To use all the available lifecycle states for the model, select the **All available lifecycle states** button. All lifecycle states are transferred to the **Lifecycle states selected** section.
   - To remove a lifecycle state from the model, select it in the **lifecycle states selected** section, and then select the left arrow button to move it to the **Lifecycle states remaining** section.

6. Select **Lifecycle state updates** to define the asset lifecycle states that can follow a selected lifecycle state.

7. You use the **Asset state** FastTab if you handle assets that you receive for repair. In the **Inbound/outbound** section, you can select asset lifecycle states to indicate the workflow of an asset that you receive for repair. If you offer loan assets to customers or departments, in the **Loan** section, you can select lifecycle states for loan assets.

> ⓘ **Note**
>
> You can link work order lifecycle states to asset and maintenance request lifecycle states. If you configure this link, changing the work order's lifecycle state also changes the linked asset and maintenance request to a new lifecycle state.

Asset service levels

Article • 01/09/2023

This article explains asset service levels in Asset Management. Asset service levels are related to assets, and are transferred to maintenance requests and work orders. They are used to calculate the priority of work orders during work order scheduling. Asset service levels can be changed, if changes are required.

For more information about the setup that is related to the calculation of rating scores for work order scheduling, see Asset Management parameters. You must set up at least one default record for the asset service level. This default record is used if no other match is found during work order scheduling.

Example 1: The default service level that is used if no other match is found. In this record, you select only a service level.

Example 2: A high service level that is used to schedule jobs for a Volvo truck engine. In this record, you select a relevant asset type (such as **Truck Engine**), a manufacturer (**Volvo**), and a service level.

Set up asset service levels

1. Select **Asset management** > **Setup** > **Asset service levels**.
2. Select **New** to create a record.
3. Depending on the detail level that is required for the asset service level, make relevant selections in the **Functional location**, **Asset type**, **Manufacturer**, **Model**, **Asset**, **Work order type**, and **Service level** fields.

> **Note**
>
> When the asset service level is used for maintenance requests and work orders, Asset Management goes through all asset service level records to check for a possible match. It always checks the most specific combination first. In other words, Asset Management first checks for a match for the **Work order type** field. If no match is found, it checks for a match for the **Asset** field, and so on. As you can see in the layout of the **Asset service levels** page, this behavior means that, to find the most specific combination, Asset Management checks each record from right to left for a match. If no match is found, the default record that has no selections in those fields is used.

4. In the **Service level** field, select a number that indicates the service level (priority).

> **Note**
>
> If you change an asset service level record on the **Asset service levels** page after you've already used it on a work order, the service level on maintenance requests and work orders isn't updated accordingly.

Feedback

Was this page helpful? Yes No

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Asset criticality types

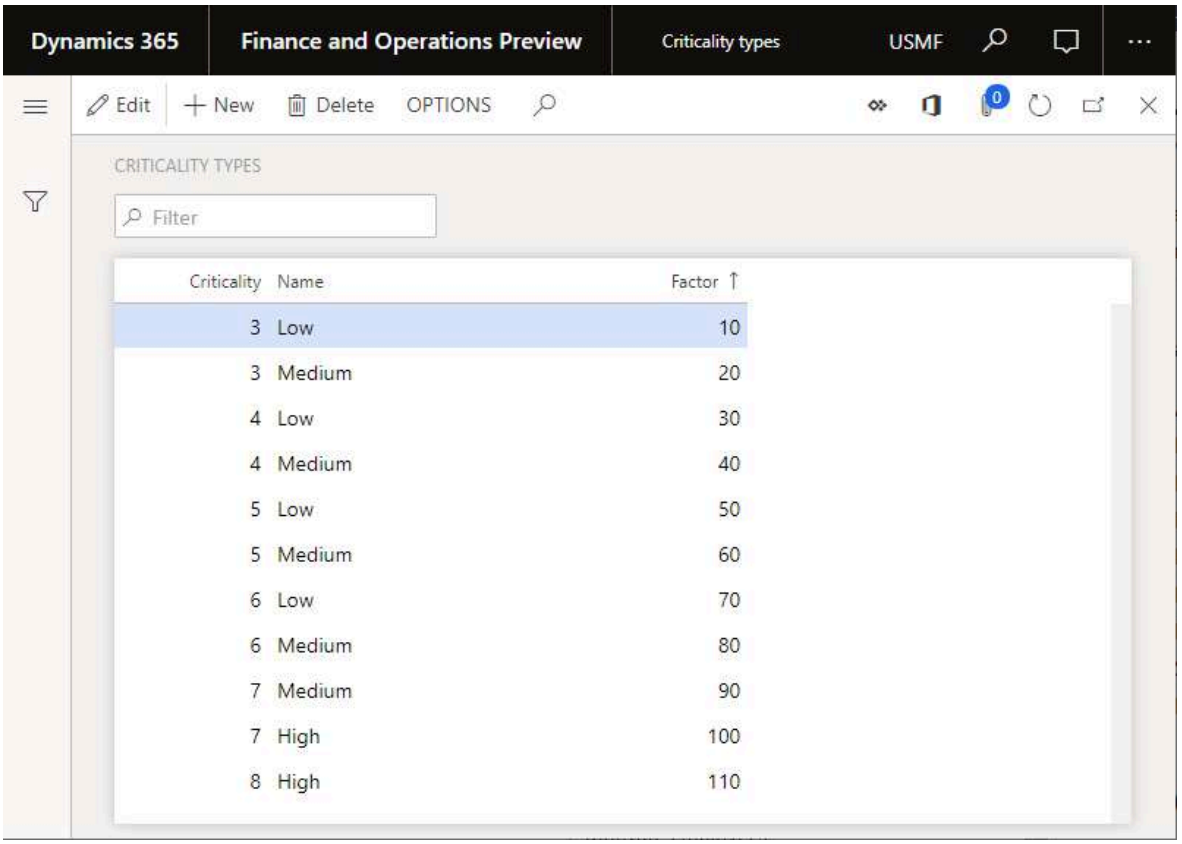
Article • 01/09/2023

The article explains asset criticality types in Asset Management. Asset criticality is related to assets and is transferred to work orders. It can't be changed on a work order. Asset criticality is used to calculate work order criticality during work order scheduling. In other words, it's used to calculate the extent to which a maintenance job on an asset affects the production schedule and productivity in your company. For more information about the setup that is related to the calculation of rating scores for work order scheduling, see Asset Management parameters.

To set up criticality, you first create the criticality types that should be used in the asset setup. You then set up asset criticalities.

Set up criticality types

1. Select **Asset management** > **Setup** > **Assets** > **Criticality types**.
2. Select **New** to create a record.
3. In the **Criticality** field, enter a number that indicates the criticality.
4. In the **Name** field, enter a name for the criticality type.
5. In the **Factor** field, enter a factor. This factor is used during the calculation of work order scheduling to determine the criticality record that should be used. (The record that has the highest factor is always used.) This setting is relevant if, as shown in the following illustration, criticality lines are created that have the same criticality value.



Set up asset criticalities

1. Select **Asset management** > **Setup** > **Asset criticalities**.

2. Select **New** to create a record.

3. Depending on the required level of detail for asset criticality, make relevant selections in the **Functional location**, **Asset type**, **Manufacturer**, **Model**, **Asset**, **Job type category**, **Job type**, **Job type variant**, and **Job requirement** fields.

   > **Note**
   >
   > When an asset criticality is selected, Asset Management goes through all asset criticality records to check for a possible match. It always checks the most specific combination first. In other words, Asset Management first checks **Job requirement**. If no match is found, it checks **Job type variant**. If no match is found, it checks **Job type**, and so on. As you can see in the layout of the page, this behavior means that, to find the most specific combination, Asset Management checks each record from right to left for a match. If no match is found, the "default" record that has no selections is used.

4. In the **Criticality** field, select one of the criticality values that you created in the previous procedure.

Notes about criticality setup

- If you change an asset criticality in this setup after you've already used it on a work order, the criticality on the work order isn't updated accordingly.
- The criticality on a work order is recalculated every time that a work order line is added to or deleted from the work order.
- If a work order contains several work order jobs, the highest criticality, according to the **Factor** field on the **Criticality types** page, is always used on the work order.
- Generally, asset criticality can change over a period. Criticality can be affected by the purchase of new equipment, refurbishments, and so on. Consider reevaluating your asset criticalities at regular intervals (for example, once per year or every other year) to make sure that your criticality definitions match your current production setup.

Feedback

Was this page helpful? Yes No

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Asset documents

Article • 01/09/2023

This article explains asset documents in Asset Management.

In Asset Management, you can set up documents so that they are automatically related to job types, asset manufacturers, asset types, or assets, for example. This functionality is useful when updated document versions are released. In that case, you just have to put the updated document in the standard location that you use for your Supply Chain Management documents, and attach the document to the asset document record that you've created. The updated document can then be accessed from the **All assets**, **Active assets**, **My active assets**, **All work orders**, and **Active work order jobs** menu items. The process for attaching documents to an asset document record uses the standard document handling system.

Example 1: A document that is related to a job type might describe a procedure for that job type.

Example 2: A document that is related to a combination of an asset type, manufacturer, and model might be the standard manual for the selected asset manufacturer model.

Create asset document relation

1. Select **Asset management** > **Setup** > **Asset documents**.
2. Select **New** to create an asset document record.
3. Depending on how specific you want the document relation to be, make relevant selections in one or more of the following fields: **Asset type**, **Manufacturer**, **Model**, **Asset**, **Job type category**, **Job type**, **Job type variant**, and **Job requirement**. The options that are available in the **Job type variant** and **Job requirement** fields depend on your selection in the **Job type** field.

   > ⓘ **Note**
   >
   > When the system searches for documents that should be related to an asset or a work order, Asset Management goes through all asset document records to check for a possible match. It always checks the most specific combination first. In other words, Asset Management first checks for a match for the **Job requirement** field. If no match is found, it checks for a match for the **Job type variant** field. If no match is found, it checks for a match for the **Job type** field,

and so on. As you can see in the layout of the **Asset documents** page, this behavior means that, to find the most specific combination, Asset Management checks each record from right to left for a match. Several documents might be related to an asset or a work order. You can edit the service level on a maintenance request or a work order as you require.

4. Select **Attachments**. The standard **Document handling** page appears.

5. Set up the documents or notes that should be attached to the asset document record. After you attach documents, the **Attachments** field shows the number of documents that are related to the record.

Feedback

Was this page helpful? Yes No

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Maintenance workers and worker groups

08/14/2025

This article explains maintenance workers and worker groups in Asset Management. In Asset Management, you can connect maintenance workers to functional locations. (For more information about functional locations, see Create functional locations.) This functionality might be useful if, for example, you're scheduling a maintenance job on a machine that is located in functional location 01, and you want to allocate maintenance workers from the same location to perform the job.

You can also create maintenance worker groups and associate maintenance workers with them. This functionality is useful when you do simple work order scheduling, and you want to schedule a group of maintenance workers on a work order. You can use maintenance workers and maintenance worker groups to set up preferred maintenance workers and responsible maintenance workers.

Create workers

1. Select **Asset management** > **Setup** > **Workers** > **Workers**.
2. Select **New** to add a worker to the list.
3. In the **Worker** field, select the worker.
4. Set the **Active** option to *Yes* to schedule the worker on work orders.

   On the **General** FastTab, the **Resource** and **Description** fields are automatically filled in if a resource is selected for the worker. The **Calendar** field is also automatically filled in if the worker is set up as a resource and a calendar is allocated to that resource on the **Resources** page (**Organization administration** > **Resources** > **Resources**).
5. On the **Groups** FastTab, select **Add**, and then select a maintenance worker group for the worker. A worker can be affiliated with more than one group.
6. In the standard setup, a worker's affiliation with a group is effective from the date when you add the group, and it never expires. This date is shown in the **Effective** field. To see the **Effective** field, select **View** > **All**. If the worker's affiliation with a group should be limited to a specific period, use the **Effective** and **Expiration** fields to define the period.
7. On the **Functional locations** FastTab, select **Add**, and then select a functional location for the maintenance worker. Also specify which location is the primary functional location for the worker.

> **Note**
>
> When you add functional locations to a worker, all active assets that are related to those functional locations appear on various menu items, such as **My active assets** and **My active functional locations**. They also appear in the asset lookups that are shown when you create a new asset, maintenance request, or work order.

The fields on the **Details** FastTab show the number of maintenance worker groups and functional locations that the selected maintenance worker is related to.

Create worker groups

1. Select **Asset management** > **Setup** > **Workers** > **Maintenance worker groups**.
2. Select **New** to add a worker group to the list.
3. In the **Maintenance worker group** field, enter a group ID.
4. In the **Name** field, enter a name.
5. On the **Workers** FastTab, select **Add**, and then select a maintenance worker for the worker group. For information about the **Effective** and **Expiration** fields, see step 6 in the previous procedure.
6. If a resource group should be related to the selected maintenance worker group, select **Copy from resource group**. In the **Group** field, select the resource group to copy calendar settings from. Then, in the **Worker group** field, select the worker group to copy the resource group's calendar settings to. This step is relevant only if you want maintenance workers to use the calendar that is related to a resource (work center) during work order scheduling.

   The field on the **Details** FastTab shows the number of maintenance workers that are set up on the selected maintenance worker group.

Introduction to assets

This article provides an overview of assets in Asset Management. An *asset* is any type of equipment, such as a machine or a machine part, that requires maintenance, service, or repair.

An asset is automatically updated with related information. For example, this related information might be about new or updated work orders. You can create assets by using either the **All assets** menu item or the **Pending assets** menu item. This article explains how to create assets by using the **All assets** menu item. For information about how to create assets by using the **Pending assets** menu item, see Create assets based on purchase orders.

All assets

Select **Asset management** > **Assets** > **All assets**. The **All assets** list page shows all assets and some of the information that is related to them. To view only active assets, select **Active assets**. To view only assets that are installed on the functional locations that you're related to as a maintenance worker, select **My active assets**. (This relation is set up on the **Workers** page. Learn more in Maintenance workers and worker groups.)

In the **All assets** grid view, select a link in the **Asset** column to view the details of the selected record. To edit the record, select the **Edit** button. The details view shows detailed information that is related to the asset. A **Related information** pane on the right contains additional asset-related information. Expand the pane to show the related information for the selected asset.

The buttons on the Action Pane are organized on tabs. The following table briefly describes the buttons that are related to Asset Management.

Expand table

Button name	Description
Edit	Edit the selected asset.
New	Create a new asset.
Delete	Delete the selected asset.
Move asset	Move assets to another asset structure, or to another location in the same asset structure.
Replace asset	Replace one child asset in an asset hierarchy with another asset.

Button name	Description
Install asset	Install an asset on a functional location.
Copy asset	Copy an asset hierarchy to another asset.
Requests	Open the **Active requests** list page, where you can view maintenance requests that have been created for the selected asset.
Event history	View an overview of the various registrations that have been made on the asset.
Asset BOM	View a list of all items (spare parts and other items) that are used on an asset.
Work orders	Open the **Active work orders** list page, where you can view work orders for the asset.
Checklist	View an overview of maintenance checklists and measurements that have been registered on the asset.
Maintenance downtime	Create or view maintenance downtime registrations on the asset.
Project transactions	View all posted transactions that are related to work orders that have been created for the asset.
Asset measures	Create or view asset measures on the asset.
Maintenance schedule	Open the **Open maintenance schedule** list page, where you can view maintenance plans, maintenance requests, and maintenance rounds that are associated with the asset, and that have a status of **Created**.
Update asset state	Update the asset lifecycle state. You can select multiple assets on the **All assets** list page and then update the asset lifecycle state for all of them at the same time. You can add remarks, which are visible in the lifecycle state log. Remarks require Supply Chain Management version 10.0.46 or higher with *Remarks can be associated with each change in lifecycle states* enabled in Feature management (As of Supply Chain Management version 10.0.49, this feature is turned on by default).
Lifecycle state log	Open a log that shows the lifecycle states of the selected asset. The log includes any remarks that were added by users during manual lifecycle changes. Remarks require Supply Chain Management version 10.0.46 or higher with *Remarks can be associated with each change in lifecycle states* enabled in Feature management (As of Supply Chain Management version 10.0.49, this feature is turned on by default).
Asset documents	View a list of the documents that are attached to an asset. These documents are set up at **Asset management** > **Setup** > **Asset documents**.
Attributes	Create or view asset attributes.
Image	Select an image for the asset.

Button name	Description
Parent assets	View parent asset history for the selected asset.
Functional locations	View functional location history for the selected asset.
Condition assessment	Register condition assessment measurements on the asset.
Faults	Open the **Asset faults** list page, where you can view faults that have been registered on the asset.
Cost control	Compare budget costs and actual costs on the asset.
Hour control	Compare forecast hours and actual hours on the asset.
Asset KPIs	Calculate and view key performance indicators (KPIs) for the asset.
Job types	View an overview of the current default job types for the asset.
Criticality types	View or update asset criticality types.
Spare parts	View a list of approved and alternative spare parts that can be used on the asset.
Asset consumption	Print a report that shows consumption registrations on the asset.
Asset fault	Print a report that shows fault registrations on the asset.

Last updated on 07/23/2026

Create an asset

Article • 01/06/2025

This article describes how to create an asset in Asset Management.

1. Go to **Asset management** > **Assets** > **All assets** or **Active assets**.
2. Select **New**.
3. In the **Create assets** dialog, insert data regarding **Asset** (the asset ID) and the asset name. Select date and time for the asset in the **Effective** field. From that date, you're able to install the asset on a functional location and move and replace the asset in an asset structure.
4. In the **Asset type** field, select the asset type for the asset (mandatory field). If necessary, select **Asset manufacturer** and **Asset model** for the asset. If only one product is up, that product is automatically selected in the **Asset manufacturer** field. The selections available in the **Asset manufacturer** and **Asset model** fields depend on the setup in Asset manufacturers and models.
5. In the **Parent asset** group, the **Asset** field is blank as default. If you select a parent asset, then all fields in the **Parent asset** group are automatically filled out.

   > ⓘ Note
   >
   > When you select a parent asset, two or three tabs are available: The **My assets** tab contains assets related to the functional locations to which you (the maintenance worker who is logged on the system) can be allocated. If no functional locations are set up on a maintenance worker on the Maintenance workers and worker groups page, the **My assets** tab isn't shown. The **Active assets** tab contains a list of all assets with asset lifecycle state *Active*. The **Asset view** tab displays a tree view of functional locations and assets installed on those locations.

6. The default functional location you have set up is suggested for the asset in the **Asset** group **Functional location** field. Select another functional location if necessary.

   > ⓘ Note

After you have created an asset, you can install it on another functional location, if required. Only top-level assets (assets without a current parent asset) can be installed on a functional location. This means that you install the top level and any child assets on the selected functional location. Read more about installing assets on functional locations in Introduction to functional locations.

7. Select **OK**.
8. Select the asset in the **All Assets** list and select the **Edit** button to add further information to the asset.

General information

The functional location to which the asset is related is shown in the **Functional location** field. If the asset is a parent asset, the number of children related to the asset is shown in the **Children** field. If the asset is a sub asset to an existing asset, the ID of the parent asset is shown in the **Parent** field.

You can edit **Asset manufacturer** and **Asset model** information on the asset, which is used to manage spare parts, alternative spare parts, and job type defaults. Learn more in Asset manufacturers and models. You can also add information about **Model year** and **Serial number**, if necessary.

Current lifecycle state is used to define if the asset is active or inactive. When creating an asset, the stage is always set to the first stage in the asset stage group. When you're ready to activate an asset, select **Update asset state**, and select the lifecycle state that you have defined as *Active*, and select **OK**.

> **Note**
>
> When an asset is set to *Inactive*, it's no longer possible to create work orders for the asset. Also, you can't schedule preventive maintenance jobs for an inactive asset.

The **Service level** and **Criticality** fields relate to work orders created for the asset. The fields show the **Service level** and **Criticality** numbers calculated for the current setup for the asset. Refer to Asset service levels and Asset criticality types regarding setup of those values.

Asset

You can select a **Resource** for the asset. The resource selection determines which calendar is used for work order scheduling. Resource selection is often used for fixed assets. Resources and resource groups are set up in **Organization administration** > **Resources** > **Resource groups** or **Resources**.

In the **Fixed assets number** field, you can select a fixed asset to be related to the asset. This is relevant if your asset is related to an investment project.

- If the asset is related to a fixed asset, you can create a work order type to be used for work orders related to an investment project.
- Information about fixed assets for an asset is related to the **Fixed assets** module in Dynamics 365 Supply Chain Management. This means that in **Fixed assets** > **Fixed assets** > **Fixed assets**, you can get an overview of the Asset Management projects that are related to a fixed asset by selecting the asset in the list and viewing the contents on the **Related information** pane, **Associated projects** section.

Details

In the **Active from** field, the date on which you updated the asset lifecycle state to an active state (refer to Asset lifecycle states regarding setup of asset lifecycle states), is shown. If the asset is no longer active, and you have updated the asset lifecycle state to an inactive lifecycle state, the date from which the asset is inactive is shown in the **Active to** field. If necessary, you can manually change those dates.

If necessary, you can insert an expected date for replacement of the asset in the **Replacement date** field. An estimated value for replacing the asset can be inserted in the **Replacement value** field. For example, you can use replacement information to compare it with the costs of maintaining an asset, and then make a decision for purchasing a new asset if maintenance costs on the existing asset increase rapidly.

Notes

You can add notes related to the asset on the **Notes** FastTab. Select the **Add timestamp** button before you write the note, if you want to add user information and a date/timestamp to the note.

Attributes

On this FastTab, you can set values for asset attributes. These attributes can be used to describe properties or characteristics pertinent to the asset, for example, size, weight, or machine configuration.

Select **Add line** and select the attribute type. Next, insert the **Value** related to the attribute type and save the record.

> ⓘ Note
>
> You can get an overview of asset attribute types and their relation to the assets in **Asset attribute** and **Asset attribute overview**. Refer to Asset attribute overview for more information.

Vendor

On the **Vendor** FastTab, select a vendor account for the asset. Also, if a vendor warranty has been granted, you can insert warranty information here.

Address

On the **Address** FastTab, you can insert the address of the equipment. If no address is inserted on the asset, the asset uses the address of a parent asset, if the parent asset has an address. If no address is related to the asset or any parents in the asset hierarchy, the address of the functional location on which the asset is installed can be used. If that functional location doesn't have an address related to it, the address of the parent functional location is used on the asset.

Asset management plans

Maintenance plans are used for scheduling preventive maintenance jobs at regular intervals on the asset. On this FastTab, you can set up maintenance plan lines for the selected asset. Maintenance rounds can be set up for various assets, on which you need to carry out a similar task at regular intervals. On the **Functional location maintenance plans** tab, you see the maintenance plans related to the functional location on which the asset is installed.

> ⓘ Note
>
> If you delete a maintenance plan line or a maintenance round related to an asset in **All Asset**, you also automatically delete all maintenance schedules with status *Created* that have been created based on that maintenance plan or maintenance round.

Functional location maintenance plans

The **Functional location maintenance plans** FastTab provides an overview of the maintenance plans related to the functional location on which the asset is installed.

Maintenance rounds

The **Maintenance rounds** FastTab lets you add or remove maintenance rounds, which are related to the asset.

Financial dimensions

The **Financial dimensions** FastTab lets you select financial dimensions for the asset.

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Change the asset type of existing assets

This article explains how to change the asset type of an existing asset.

Prerequisites

Before you can use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management 10.0.38 or later.
- The feature that's named *Change types on assets and functional locations* must be turned on in Feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default. As of version 10.0.45, it's mandatory and can't be turned off.

Change the asset type of one or more existing assets

To change the asset type of one or more existing assets, follow these steps:

1. Go to **Asset management** > **Assets** > **All assets**.
2. On the **All assets** list page, select one or more assets that you want to change asset type for. If you select multiple assets, all of them must be of the same asset type.
3. On the Action Pane, on the **Asset** tab, in the **Maintain** group, select **Change asset type**.
4. In the **Change asset type** dialog box, on the **Parameters** FastTab, set the following fields:
    - **New asset type** – Select the new asset type to assign to the selected assets.
    - **Attributes** – Set this option to *Yes* to append the attribute types that are associated with the new asset type to the selected assets.
    - **Maintenance plans** – Set this option to *Yes* to append the maintenance plans that are associated with the new asset type to the selected assets.
5. On the **Run in background** FastTab, set the **Batch processing** option to *Yes* to run the process of changing asset types as a batch job.
6. Select **OK** to apply your settings.

> ⓘ **Note**

Each asset type has settings that define which supporting data can be used for assets of that type. This supporting data includes information such as maintenance job types, counters, and attribute types. When you change the asset type, the supporting data for existing assets might not match the supporting data for the new asset type.

For example, asset *A1* has an asset type of *AT1*. Two maintenance job types, *JT1* and *JT2*, are selected for asset type *AT1*. Work orders have been created for asset *A1*, and maintenance job types *JT1* and *JT2* are selected for those work orders. Later, the asset type of asset *A1* is changed to *AT2*. Maintenance job types *JT3* and *JT4* are selected for asset type *AT2*. Therefore, after the asset type is changed, the work orders have maintenance job types that aren't associated with the asset type of the asset for those work orders.

Last updated on 12/30/2025

Multi-level assets

08/13/2025

This article explains how to create and delete multi-level assets. You can create assets and related sub-assets in a hierarchical tree structure. In this way, you can show relations and dependencies among assets. Maintenance jobs can be related to all levels of the tree structure. Statistics can also be created for an individual level or as a sum of all sub-asset levels.

On the **All Assets** list page (**Asset management** > **Assets** > **All assets**), the **Asset** column lists assets in hierarchical order. The **Parent** column shows the related parent. Additionally, if assets and sub-assets have already been created, the **Asset tree** section in the **Related information** pane shows the assets in a tree structure.

For information about how to create an asset, see Create an asset. To create a sub-asset, select the parent asset in the **Parent** field on the **General** FastTab.

Copy an asset or asset structure

If your company has several similar asset structures, you can use the Copy function in Asset Management to quickly create them.

1. Select **Asset management** > **Assets** > **All assets**.
2. On the **All assets** list page, select the asset to copy. For example, if you want to copy the whole asset structure, including sub-assets, select a parent asset.
3. Select **Copy asset**. In the **Copy from** section, the **Asset** field is set to the asset that you selected on the list page.
4. In the **Copy to** section, in the **Asset** field, enter the name of the new asset.
5. If the asset that you're creating should be part of an existing asset structure, in the **Parent asset** section, in the **Asset** field, select a parent ID.
6. Select **OK**. The new asset structure is shown on the **All assets** list page. All asset attributes, maintenance plans, and maintenance rounds that are related to the asset that you copied are transferred to the new asset or asset structure.

When you copy an asset structure, the sub-assets in the new structure have the same name as the sub-assets that you copied. After the copy procedure is completed, you can easily change the name and other settings for an asset. Select the asset on the **All assets** list page, and then select the **Edit** button.

> **Note**

When you copy an asset or asset structure, the lifecycle state of the new assets is reset to whatever state you defined as the initial lifecycle state for assets. The functional location is reset to the default functional location.

Delete an asset or asset structure

If an asset has related sub-assets, you can delete it only if no maintenance requests, work order jobs, fault registrations, or condition assessments are registered on any of the assets.

1. On the **All assets** list page, select the asset to delete.
2. Select **Delete**.

> **Note**
>
> If you can't delete an asset by using this procedure, another way to handle deletion is to set up an asset lifecycle state for this purpose. For example, you can set up a **Scrapped** or **Deleted** lifecycle state on the **Asset lifecycle states** page.

Move, replace, and install assets

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to move, replace, and install assets in Asset Management. You can create individual assets that have no relations to other assets, or you can create an asset structure that includes a parent asset (top-level asset) and related child assets (sub-assets). In Asset Management, there are three approaches to moving and changing the location of an asset:

- **Move** – Move an asset either to another asset structure or to another location in the same asset structure.
- **Replace** – Temporarily remove an asset from an asset structure so that it can be repaired or refurbished, and then add the refurbished asset back to the asset structure later. Alternatively, permanently replace a used asset with a new asset.
- **Install** – Install a parent asset and any related child assets on a functional location.

> **Note**
>
> The asset that you move, replace, or install might be related to another functional location. In that case, the asset might use financial dimensions of the functional location. On the **Functional location types** page, you set up the handling of financial dimensions on functional locations.

Move asset

Use the **Move asset** function to move an asset either to another asset structure or to another location in the same asset structure. You can also move an asset out of an asset structure so that it becomes a standalone asset that has no structure relations.

> **Note**
>
> Don't use this function if assets are being repaired or temporarily replaced. Instead, use the **Replace asset** functionality that is described later in this article.

1. Select **Asset management** > **Assets** > **All assets** or **Active assets**.
2. In the list, select the asset to move. If the asset has child assets, you also move those assets.
3. Select **Move asset**.
4. To move the asset so that it becomes part of an asset structure, select the new parent asset in the **Parent asset** field. If you're moving a child asset, and you want to make it a standalone asset that has no structure relations, leave the **Parent asset** field blank.
5. By default, the **Effective** field is set to the current date and time. However, you can select a different date and time that the asset move is valid from.
6. Select **OK**.

Replace asset

Use the **Replace asset** function in connection with repairs, refurbishment, or permanent replacement of a worn-out asset by a new asset. This function is used to replace child assets in an asset structure. For parent assets (that is, assets that don't currently have a parent asset), this replacement is done on a functional location. For more information about how to replace parent assets on a functional location, see Install assets on functional locations.

> **Note**
>
> If a repair shop is related to your production department, you can create functional locations such as **Repair**, **Scrap**, and **Storage** to handle the repair and replacement of assets.

1. Select **Asset management** > **Assets** > **All assets** or **Active assets**.
2. In the list, select the child asset to replace. If the asset has child assets, you also replace those assets.
3. Select **Replace asset**.

   The **Structure change** field shows the last date and time that the selected asset and related child assets were moved in the asset structure.
4. In the **Selected asset** section, in the **Target functional location** field, select the functional location to move the asset to. For example, select the **Repair** or **Scrap** location.

   In the **Parent asset** section, the **Effective** field shows the last date and time that the parent asset and related child assets were installed or moved on the current functional location.

5. In the **New asset** section, in the **Asset** field, select the asset to insert as a temporary or permanent replacement for the selected asset. This asset might currently be located on another functional location, such as **Storage**.
6. In the **Effective from** section, the **Effective** field is set to the current date and time by default. However, you can select a different date and time that the asset replacement is valid from.
7. Select **OK**.

Install asset

Use the **Install asset** function to install an asset structure on a functional location.

> **Note**
>
> Always select a parent asset. The parent asset and related child assets will be moved to the selected functional location.

1. Select **Asset management** > **Assets** > **All assets** or **Active assets**.
2. In the list, select the parent asset to install on another functional location.
3. Select **Install asset**.

   The **Attributes** section shows the asset attributes that are set up on the parent asset.
4. In the **Functional location** field, select the new location.
5. By default, the **Effective** field is set to the current date and time. However, you can select a different date and time that the installation on the asset structure is valid from.
6. Select **OK**.

Last updated on 05/07/2025

Create assets based on purchase orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how you can create a list of asset items that can be used as a basis for creating assets for maintenance jobs in Asset Management. Based on the asset items, you are able to view a list of the purchase order lines that have been created on those items. The purpose of this functionality is to easily create an asset in Asset Management based on a purchase order.

First, you set up the items to be used for creating assets from a purchase order in **Asset items**. After creating a purchase order line, you create the assets in **Pending assets**. It is possible to decide at which stage of the purchase order the asset should be created.

Select asset items

1. Click **Asset management** > **Setup** > **Assets** > **Items**.
2. Click **New** to create a new asset item.
3. Select the item in the **Item number** field. When you leave that field, the item name is automatically inserted in the **Product name** field.
4. On the **General** FastTab, select an asset type for the item.
5. Select asset manufacturer and model for the item.
6. Save the item.

Create assets from pending assets

1. Click **Asset management** > **Assets** > **Pending Assets**.
2. You will see an updated list of the purchase orders based on the items selected in **Asset items**.
3. You can filter the status of purchase orders to select at which lifecycle state the asset should be created. For example, you may only want to create assets when a product receipt has been posted on a purchase order.
4. Select the **Reference number** link on a purchase order line to view detailed information about the item.

5. If you want to edit which dimensions are displayed on the **Inventory dimensions** FastTab, click the **Display Dimensions** button, and make your selections.
6. If you want to create an asset based on a purchase order line, select the check box in the **Mark** column for that line on the list page, and click **Create assets**. A message will be displayed informing you of the asset ID.
7. Select the asset in the **All assets** list and click **Edit** to add more information to the asset.
8. In **Pending assets**, if you want to delete a line because you don't want to create an asset, select the check box in the **Mark** column for that line, and click **Discard pending assets**. A message will be displayed informing you of the asset ID. You are not deleting the purchase order or sales order, just the record from which you could have created an asset in **Asset Management**.

> ⓘ **Note**
>
> All product dimensions (size, color, configuration etc.) are automatically transferred to the asset attributes. Tracking dimensions (serial number) are stored directly on the asset when the asset is created.

Find pending assets

You can run a **Pending asset count** to check for pending assets. For example, this function can be used for receiving a notification each time a pending asset is ready to be created as an asset.

1. Click **Asset management** > **Periodic** > **Assets** > **Pending asset count**.
2. Click **OK** to run the job and update the list in **Pending assets**.
3. You can set up this job to run as a batch job, for example, once each day.

Caution: If data is changed on a purchase order *after* you have created an asset based on the appertaining item, those changes will not be reflected on the asset.

Last updated on 05/07/2025

Asset BOMs

Article • 01/06/2025

This article describes asset bills of materials (BOMs) in Asset Management. The **Asset BOM** page shows a list of all items (spare parts and other items) that are used on an asset during its whole lifetime. When you create a new asset, you should consider setting up an asset BOM as a part of the setup procedure. In this way, you can track item history for the asset from the creation date.

After you've completed a maintenance job, and item consumption has been registered on a work order, you can track consumption of spare parts and other items that are used on the asset. This functionality lets you keep a complete item consumption record for all your assets. For example, you can use the record to monitor whether a specific spare part is often replaced, or to keep track of the spare parts or other items that are currently used on an asset.

> **Note**
>
> On a work order, item consumption might include both spare parts and other, additional items, such as lubricants, bolts, and gaskets.

An asset BOM can be automatically updated based on the setup in Asset Management. If a work order lifecycle state has been created to handle finished or completed work orders, and if the **Update asset BOM** option for that work order lifecycle state is set to *Yes* on the **Work order lifecycle states** page, all items that are used on the work order will be automatically updated on the **Asset BOM** page when the work order is updated to that lifecycle state.

You can also manually update an asset BOM by creating new item lines on the **Asset BOM** page.

On the **Asset BOM** page, you can track spare parts history for assets after item consumption is registered on a work order. To use this functionality, you must select the item groups that should be used for spare parts registration on the **Spare parts item groups** page.

To use asset BOM functionality, you must first set up the required spare parts items groups. Then, if you want the asset BOM to be automatically updated when a work order is completed, you can set up a work order lifecycle state to handle that update.

Set up spare parts item groups

The setup of spare parts history is based on item groups that are created in the **Inventory and warehouse management** module. In Asset Management, you set up item groups so that you can track spare parts history for the items in the selected item groups.

1. Select **Asset management** > **Setup** > **Asset** > **Spare parts item groups**.
2. Select **New** to set up an item group.
3. In the **Item group** field, select the group. The name of the group is automatically entered in the **Name** field.

View and update asset BOMs

After you post item consumption on a work order, you can view the registered item consumption on the **Asset BOM** page.

1. Select **Asset management** > **Assets** > **Active assets**. Select the asset in the list, and then select **Asset BOM**.

   > ⓘ **Note**
   >
   > To view all item consumption registrations on all assets, select **Asset management** > **Inquiries** > **Assets** > **Asset BOM**.

2. Select **Update asset BOM**. You can add assets, asset types, and resources to the update as you require by selecting **Select**. Select **OK** to start the update. You can also set up the Update function as a batch job.

3. If you want to see more information that is related to the items, you can add inventory dimensions. Select **Inventory** > **Dimensions display**, and then select the check boxes for the dimensions that you want to see. To keep this setup for all assets on the **Asset BOM** page, set the **Save setup** option to *Yes*.

4. To get an overview of where in Asset Management the item on the selected line is used, in relation to assets, job type defaults, spare parts, and work orders, select **Item where used**.

5. If you want to see only active item lines, select **View** > **Current**. To see all item lines, including lines where the expiration date is earlier than the current date, select **View** > **All**.

> **Note**
>
> When you've completed a work order, if some items or spare parts that are related to the related asset have expired or have been replaced by other items or spare parts, we recommend that you update the asset BOM accordingly.

Create a new item line in an asset BOM

You can manually create item lines for assets.

1. On the **Asset BOM** page, select **New**.
2. In the **Asset** field, select the asset to add an item line for.
3. In the **Line number** field, enter a sequential line number.
4. In the **Effective** field, enter a start date for the item.
5. If the item will expire, in the **Expiration** field, enter an end date.
6. In the **Item number** field, select the item. The name is automatically entered in the **Product name** field.
7. In the **Quantity** field, enter the quantity that is used. The **Unit** field is automatically updated.

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Asset event history

Article • 01/09/2023

This article explains how to access asset event history in Asset Management. The **Asset event history** page shows the registration history that has been made during the lifetime of an asset. You can access this page from the **All assets**, **Active assets**, and **My active assets** menu items. Select an asset, and then select **Event history**.

On the **Details** FastTab of the **Asset event history** page, you can do a search on all the available information. For example, you can use the asset event history to find the following information:

- When a job type was last used on an asset
- When a specific worker entered a remark on a work order job

The timeline is updated every time that the page is opened. It contains the following information:

- Changes that have been made on the asset: asset location, asset ID, name, and vendor warranty
- Asset parent
- Asset bill of materials (BOM)
- Asset lifecycle state log
- Functional location
- Maintenance requests
- Work orders, including posted item and notes
- Faults
- Condition assessments

Feedback

Was this page helpful? Yes No

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Asset view

Article • 01/09/2023

This article describes the asset view in Asset Management. The **Asset view** page shows active assets and functional locations in a tree view. Therefore, you can easily get an overview of asset relations to functional locations. Additionally, you can view detailed information about functional locations, assets, and related bills of materials (BOMs). You can also get a quick overview of active maintenance requests and work orders that are related to an asset.

1. Select **Asset management** > **Assets** > **Asset view**.
2. To change the view that is shown on the page, select a new value in the **View** field.

> **Note**
>
> The default view that is shown when you open the **Asset view** page depends on the value that is selected in the **View** field on the **Assets** tab of the **Asset management parameters** page (**Asset management** > **Setup** > **Asset management parameters**).

On the right side of the page, FastTabs shows details of the selected view.

The breadcrumb trail that appears above the tree view shows the current selection in the tree view. This breadcrumb trail uses the following format:

Functional location ID / Functional location ID (if there is more than one functional location) > Asset ID / Asset ID (if there is more than one asset) - Item number.

If you've selected an asset in the tree view, you can select **Active requests** or **Active work orders** to view the maintenance requests or work orders that are related to the asset. You can also select **Open** > **Functional location**, **Asset**, or **Asset BOM** to open the related view.

The **Asset functional locations** option that you can select in the **View** field is also available in any asset lookup where you can select an asset. The tree view is shown on an **Asset view** tab, for example, where you create an asset, create a maintenance request, or create a work order.

Feedback

Was this page helpful? Yes No

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Asset attribute overview

Article • 01/09/2023

This article provides an overview of asset attributes in Asset Management. Asset attributes are properties that are related to an asset type or an asset. If you've set up attribute types and used them on assets, you can get an overview of the attribute values that are set up on the assets. This overview is shown on two pages in Asset Management: **Asset attribute overview** and **Asset attribute**.

The **Asset attribute overview** page shows a separate line for each asset. The line shows all attribute types that are related to that asset. Follow these steps to open and use the **Asset attribute overview** page.

1. Select **Asset management** > **Inquiries** > **Assets** > **Asset attribute overview**.
2. In the **Asset type** field, select an asset type.
3. Select **OK**. A list shows all the assets that use the selected asset type. For each asset, all related asset attribute types are shown on the same line.

The **Asset attribute** page shows a separate line for each attribute type that is used on an asset. To open this page, select **Asset management** > **Inquiries** > **Assets** > **Asset attribute**.

Feedback

Was this page helpful? Yes No

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Maintenance request lifecycle states

07/30/2025

Maintenance request lifecycle states define the stages that a request can go through. Examples include *Created*, *Active*, and *Ended*. When a maintenance request is converted to a work order, the maintenance request lifecycle state should be updated to *Ended* or *Closed* to indicate that the maintenance request is no longer active. On the **All maintenance requests** list page, you can view all maintenance requests, regardless of their lifecycle state.

Turn the depot repair option on or off

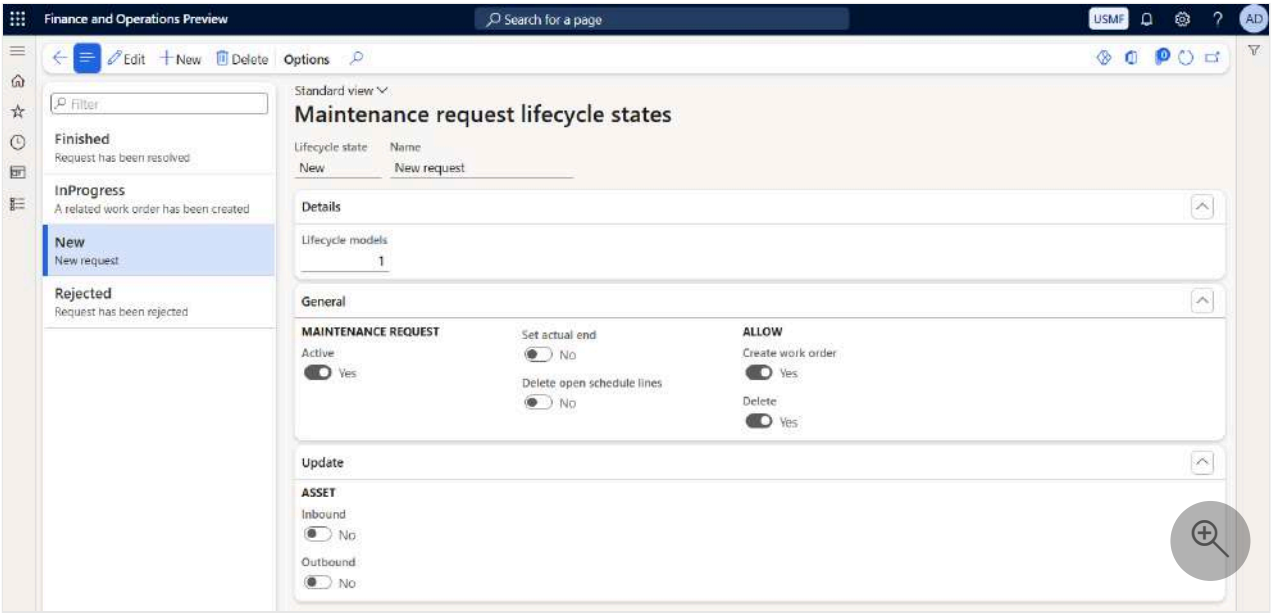
Depot repair is a maintenance request type that lets you receive assets from another location so that you can do a maintenance or repair job, and then return the asset after the job is completed. To use depot repair, you must enable its configuration key by following these steps:

1. Put your system into maintenance mode, as described in Maintenance mode.
2. Go to **System administration** > **Setup** > **License configuration**.
3. On the **Configuration keys** tab, expand the **Asset management** node and then select the check box for **Inbound/outbound**.
4. Turn off maintenance mode, as described in Maintenance mode.

Manage maintenance request lifecycle states

To manage the maintenance request lifecycle states that you need, follow these steps.

1. Go to **Asset management** > **Setup** > **Maintenance requests** > **Lifecycle states**.



2. Use the buttons on the Action Pane to add, remove, or edit a lifecycle state.

3. Make the following settings in the header of your new or selected lifecycle state:

    - **Lifecycle state** – Enter an ID for the lifecycle state.
    - **Name** – Enter a name for the lifecycle state.

4. For existing lifecycle states, the **Lifecycle models** field on the **Details** FastTab shows the number of maintenance request lifecycle models that use this lifecycle state.

5. Set the following options on the **General** FastTab:

    - **Active** – Choose whether a maintenance request should be active while it's in this lifecycle state.
    - **Set actual end** – Choose whether an actual end date and time should automatically be entered on a maintenance request that's in this lifecycle state.
    - **Delete open schedule lines** – Choose whether all maintenance schedule lines without a work order reference should be deleted automatically on a maintenance request that's in this lifecycle state.
    - **Create work order** – Choose whether a work order can be created from a maintenance request that's in this lifecycle state.
    - **Delete** – Choose whether a maintenance request can be deleted while it's in this lifecycle state.

6. If you use depot repair, then the **Update** FastTab is shown. Make the following settings here:

    - **Inbound** – Choose whether the asset lifecycle state of assets that are selected on a maintenance request should automatically be updated to **Inbound** when the maintenance request lifecycle state of that maintenance request is set to **Inbound**.
    - **Outbound** – Choose whether the asset lifecycle state of assets that are selected on a maintenance request should automatically be updated to **Outbound** when the maintenance request lifecycle state of that maintenance request is set to **Outbound**.

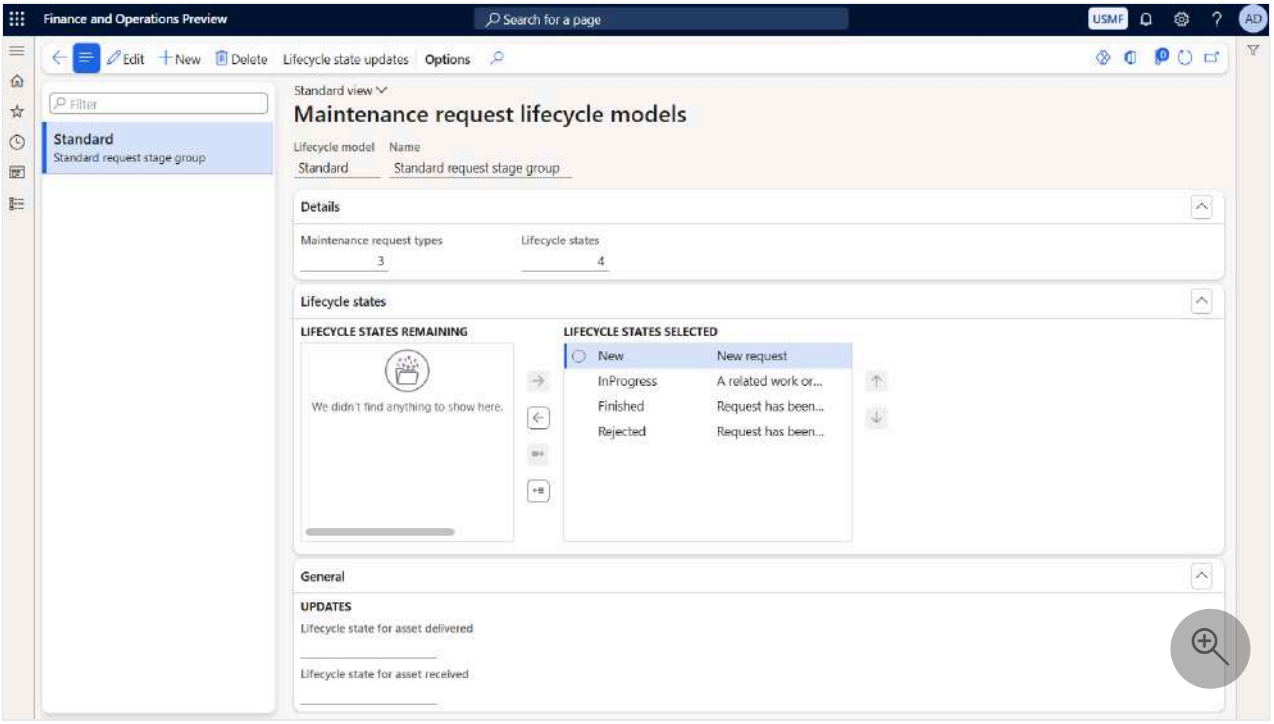
> ⓘ **Note**
>
> Maintenance request lifecycle states, lifecycle state groups, and lifecycle state types are related to, and used in the same way as, work order lifecycle states, lifecycle state groups, and lifecycle state types.

Set up maintenance request lifecycle models

After you've created the lifecycle states that are required for your maintenance requests, they can be divided into lifecycle state groups, or lifecycle models. Maintenance request lifecycle

models are used to create the flow that can be used for different types of maintenance requests. At a minimum, one standard maintenance request lifecycle model should be created.

1. Go to **Asset management** > **Setup** > **Maintenance requests** > **Lifecycle models**.



2. Use the buttons on the Action Pane to add, remove, or edit a lifecycle model.

3. Make the following settings in the header of your new or selected lifecycle model:

    - **Lifecycle model** – Enter an ID for the lifecycle model.
    - **Name** – Enter a name for the lifecycle model.

4. For existing lifecycle models, the **Details** FastTab displays the following information:

    - **Lifecycle states** – Shows the number of lifecycle states that are selected in this lifecycle model.
    - **Maintenance request types** – Shows the number of maintenance request types that use this lifecycle model.

5. On the **Lifecycle states** FastTab, move all of the lifecycle states that should be included in this lifecycle model into the **Lifecycle states selected** column.

    - Use the buttons between the columns to move one or more selected lifecycle states from one column to the other.
    - When a user changes the lifecycle state of a maintenance request, the states are displayed in the order shown in the **Lifecycle states selected** column, which should indicate the way maintenance requests typically progress from start to finish. Use the up and down buttons next to the **Lifecycle states selected** column to adjust the order of the lifecycle states.

6. If you use depot repair, then the **General** FastTab is shown. Make the following settings here:

    - **Lifecycle state for asset received** – Select the asset lifecycle state that assets selected on a maintenance request should automatically be updated to when they're received for depot repair.
    - **Lifecycle state for asset delivered** – Select the lifecycle state that assets selected on a maintenance request should automatically be updated to when they're returned after depot repair.

Maintenance request types

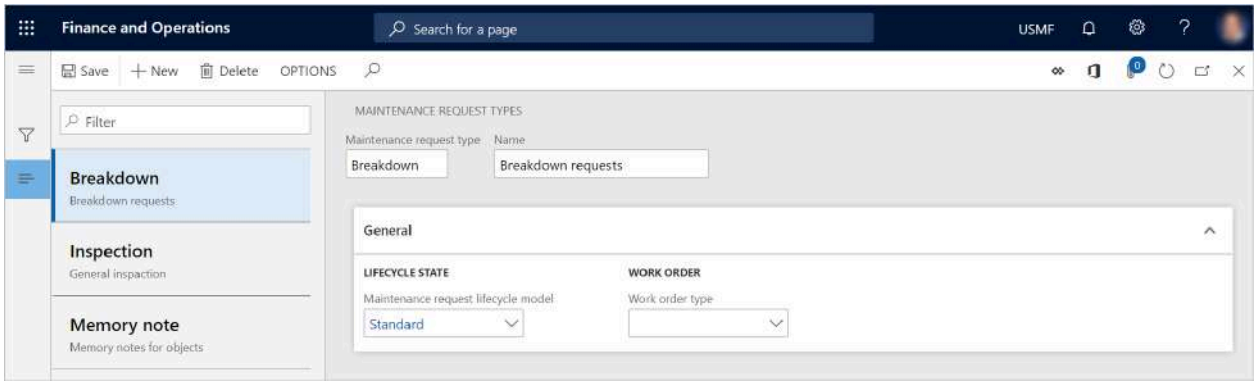
Article • 06/03/2022

Maintenance request types are used to categorize maintenance requests. For example, you might have maintenance request types that are related to preventive maintenance and corrective maintenance. Or you might have a special maintenance request type that is used to manage repair of assets (depot repair).

A maintenance request type defines the affiliation with a maintenance request lifecycle state group (maintenance lifecycle model). Maintenance request lifecycle models define the lifecycle states that can be set for a maintenance request. (Examples of maintenance request lifecycle states include **Created**, **Active**, and **Ended**.)

1. Select **Asset management** > **Setup** > **Maintenance requests** > **Maintenance request types**.
2. Select **New** to create a maintenance request type.
3. In the **Maintenance request type** field, enter an ID for the maintenance request type.
4. In the **Name** field, enter a name.
5. On the **General** FastTab, in the **Maintenance request lifecycle model** field, select a maintenance request lifecycle model.
6. In the **Work order type** field, select a work order type. When a maintenance request is converted to a work order, the work order automatically gets the work order type that is related to the maintenance request type.

The following illustration shows an example of the **Maintenance request types** page.



Feedback

Was this page helpful? Yes No

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Responsible maintenance workers

Article • 06/03/2022

Responsible maintenance workers can be related to asset types, assets, functional locations, maintenance job type categories, maintenance job types, maintenance job type variants, and trades. They can be used on work orders and maintenance requests to indicate a preference about the maintenance workers who should be responsible for a work order. (However, those maintenance workers aren't necessarily the same workers who are scheduled to carry out the work order.) Use of this functionality is optional. For example, it can be used to select responsible workers or worker groups for specific work types or work areas.

During a work order lifecycle or a maintenance request lifecycle, responsible maintenance workers can be changed or updated. This change or update might be related to, for example, a change in the maintenance request lifecycle state or the work order lifecycle state.

The setup on the **Responsible maintenance workers** page is *not* used during work order scheduling.

> ⓘ **Note**
>
> In Asset Management, you can also set up *preferred* maintenance workers who might be allocated to work orders during work order scheduling.

Before you can set up responsible maintenance workers, you must set up the workers and maintenance worker groups that should be available for selection. For information about how to set up workers and maintenance worker groups, see Maintenance workers and worker groups.

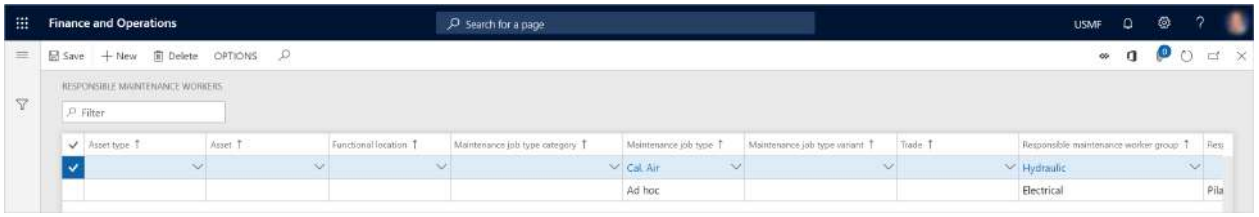
Set up responsible maintenance workers

1. Select **Asset management** > **Setup** > **Workers** > **Responsible maintenance workers**.
2. Select **New** to create a record.
3. First create a default responsible maintenance worker or responsible maintenance worker group setup, where you set only the **Responsible maintenance worker group** field and/or the **Responsible worker** field. Leave the remaining fields blank.

This default setup will be used during work order scheduling if no other, more specific combination matches the contents of the work order.

> **Note**
>
> During the creation or update of a maintenance request, Asset Management goes through all responsible maintenance worker records to check for a possible match. It always checks the most specific combination first. In other words, Asset Management first checks for a match for the **Trade** field. If no match is found, it checks for a match for the **Maintenance job type variant** field. If no match is found, it checks for a match for the **Maintenance job type** field, and so on. As you can see in the layout of the page, this behavior means that, to find the most specific combination, Asset Management checks each record from right to left for a match (first **Trade**, then **Maintenance job type variant**, then **Maintenance job type**, then **Maintenance job type category**, then **Functional location**, then **Asset**, and finally **Asset type**). If no match is found, the default record that has no selections in those seven fields is used.

The following illustration shows an example of the **Responsible maintenance workers** page.



Feedback

Was this page helpful? Yes No

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Maintenance requests

Article • 01/06/2025

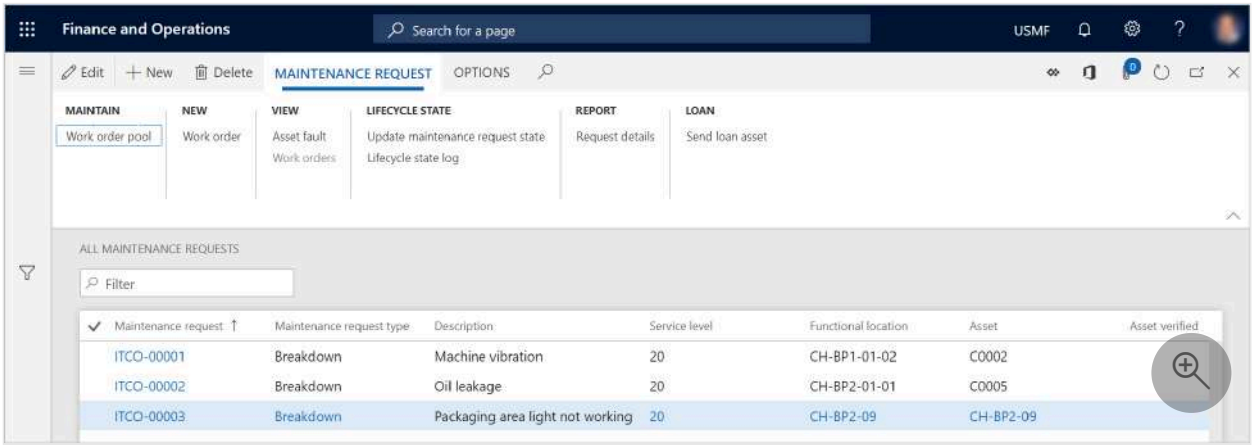
Maintenance requests are notes or declarations that are created to notify a manager or planner that an asset might require a maintenance or repair job, but without creating a work order. If the contents of a maintenance request are considered valid, a work order can then be created based on the maintenance request.

Maintenance requests can be created for any asset in Asset Management. Various types of maintenance requests can be created, depending on how your company uses maintenance requests. Here are some examples:

- Maintenance requests
- Notes
- Corrections or enhancements
- Investments
- Depot repair (This type is used when you receive assets from another location so that you can do a maintenance or repair job, and you then return the asset after the job is completed.)

View maintenance requests

To view maintenance requests, select **Asset management** > **Maintenance requests** > **All maintenance requests**, **Active maintenance requests**, or **My functional location maintenance requests**. Each list page shows some of the information that is related to a maintenance request.

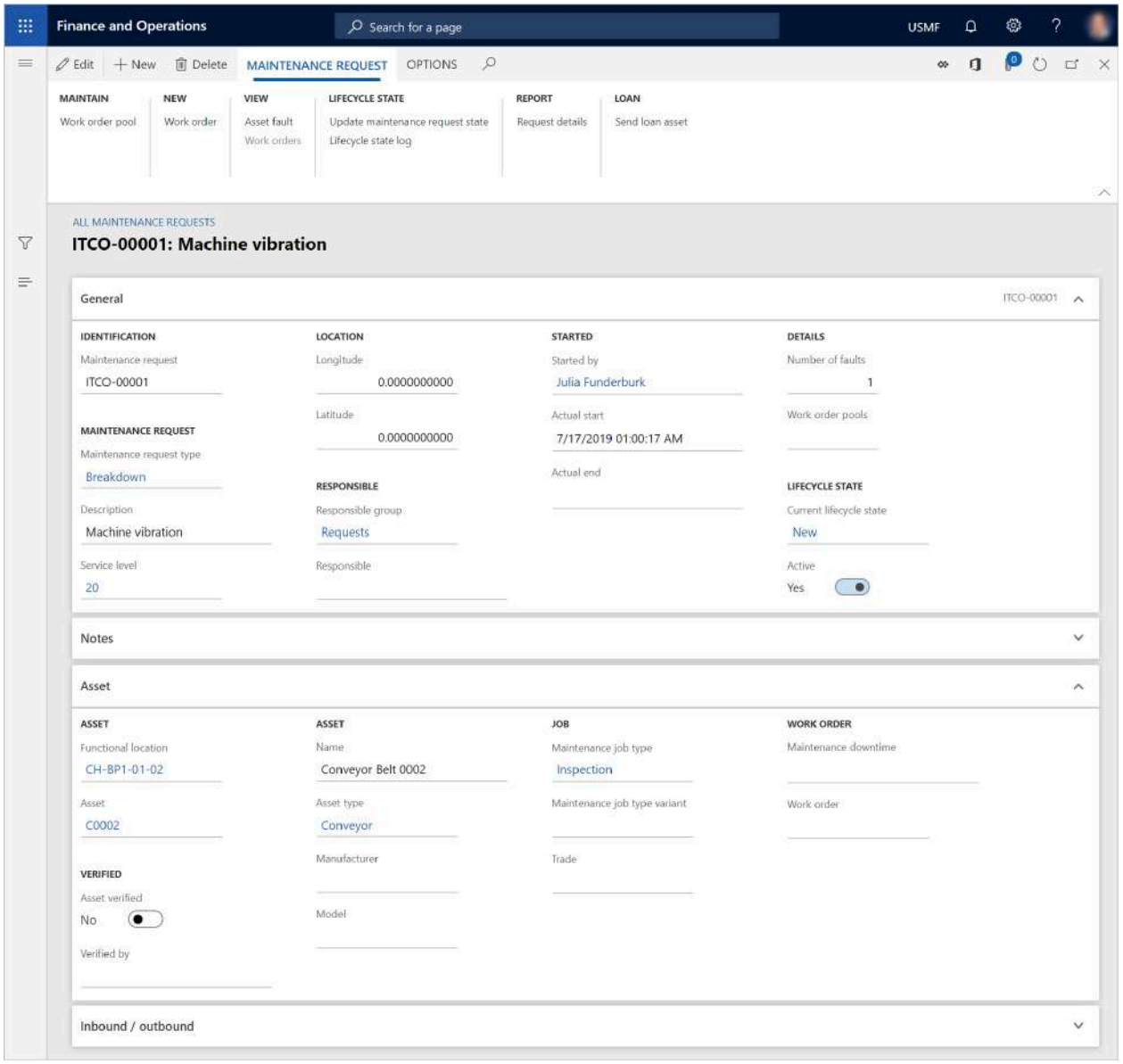


> **Note**
>
> Use the **My functional location maintenance requests** list page to view a list of maintenance requests that contain either functional locations that you're related to

as a worker or assets that are installed on functional locations that you're related to as a worker. (Read Maintenance workers and worker groups to learn more about how to set up functional locations on maintenance workers.)

Although customer account information is available in Asset Service Management (external maintenance), it isn't available in Asset Management (internal maintenance).

To open the details view of a record, on the **All maintenance requests** list page, in the grid view, select a link in the **Maintenance request** column.



The buttons on the Action Pane are organized on tabs. The following table briefly describes the buttons that are related to Asset Management.

Expand table

Button name	Description
Edit	Edit the selected maintenance request.
New	Create a new maintenance request.
Delete	Delete the selected maintenance request.
Work order pool	Connect the selected maintenance request to a work order pool.
Work order	Create a work order, based on the selected maintenance request.
Asset fault	Select **Asset faults** to create a fault registration on the selected maintenance request.
Work orders	Show a list of all work orders that are connected to the selected maintenance request.
Update maintenance request state	Update the maintenance request state.
Lifecycle state log	View a log that shows the lifecycle states of the selected maintenance request.
Maintenance request details	Print a report that shows details of the selected maintenance request.
Send loan asset	Select a loan asset that should be a temporary replacement for the asset that is selected on the selected maintenance request.
Return loan asset	Register the loan asset as returned.

Feedback

Was this page helpful? Yes No

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Create maintenance requests

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Maintenance requests can be used if maintenance workers or production workers discover that equipment requires repair, but the repair job can't be done right away.

Example: While a maintenance worker is making a repair, they discover that another asset at the same location must be serviced. However, the maintenance worker doesn't have the time or the required spare parts to do the repair job. Therefore, they create a maintenance request on the asset and enters a short description of the issue.

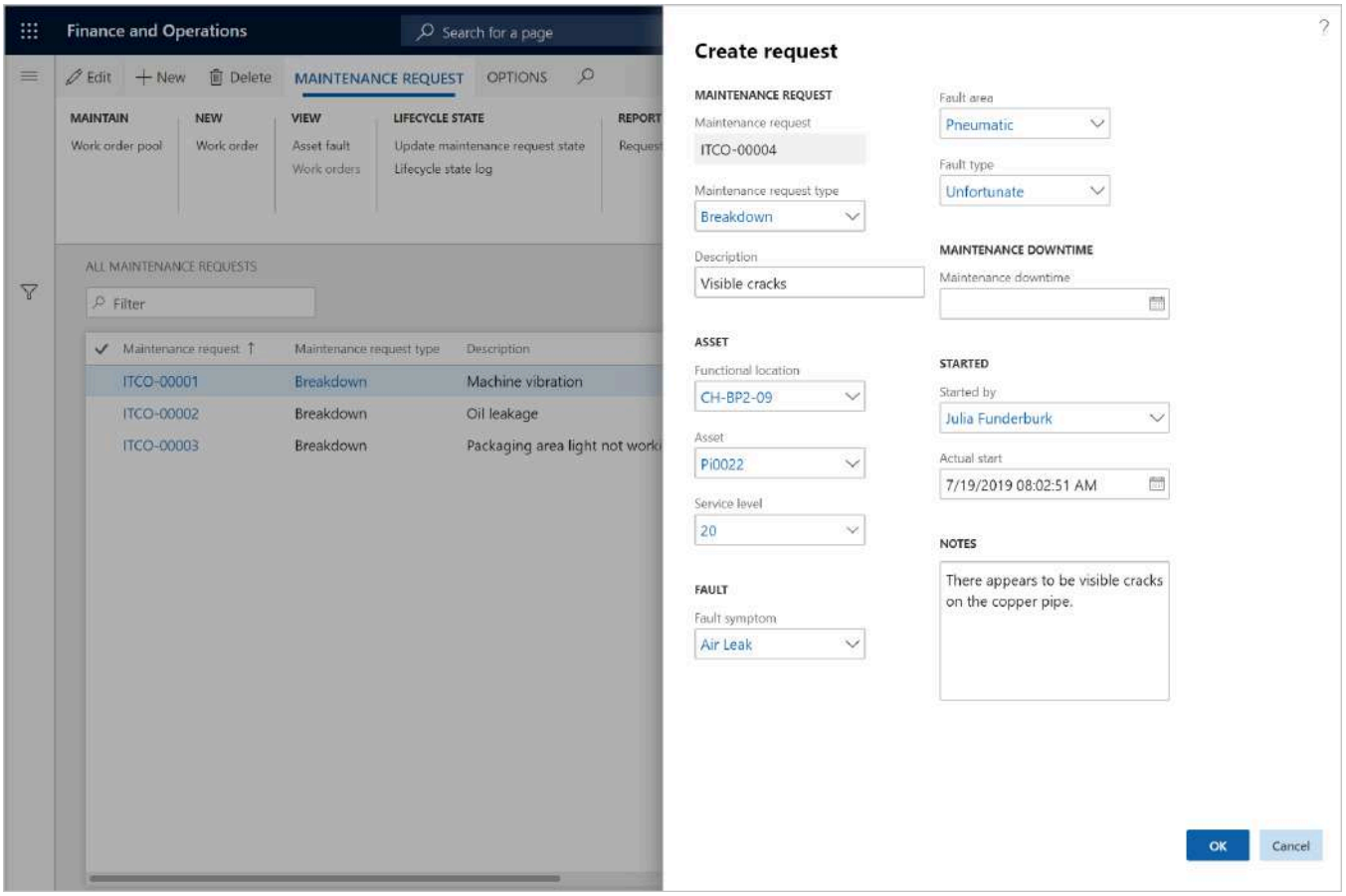
The **Active maintenance requests** section of the **Related information** pane on the right side of the **All assets** or **Active assets** page (**Asset management** > **Assets** > **All assets** or **Active assets**) shows the active maintenance requests that are attached to the selected asset.

1. Select **Asset management** > **Maintenance requests** > **All maintenance requests** or **Active maintenance requests**.
2. Select **New**.
3. In the **Create request** dialog box, in the **Maintenance request type** field, select the type of maintenance request. A default type is suggested.
4. In the **Description** field, enter a name or title that briefly describes the maintenance request.
5. In the **Functional location** and **Asset** fields, select a functional location or an asset, or a combination of a functional location and an asset, as you require. You can create a maintenance request without selecting an asset, and the asset can be added to the maintenance request later. If the maintenance worker who is signed in is related to a resource that is related to an asset, the **Asset** field is automatically set.

   If a maintenance request is already attached to the selected asset, a message bar appears at the top of the **Create request** dialog box to notify you about the ID of the existing maintenance request. A message bar also notifies you if the asset is covered by a warranty agreement.

6. In the **Service level** field, select a service level that indicates the urgency of the request.
7. If you selected an asset in step 5, you can use the **Fault symptom**, **Fault area**, and **Fault type** fields to create a fault registration.
8. If the maintenance request has caused maintenance downtime, enter the start date and time of the downtime.

   The **Started by** field is automatically set to your name.
9. The **Actual start** field is automatically set to the current date and time. However, you can change the value as you require.
10. In the **Notes** field, enter any additional notes that are required.
11. Select **OK**.



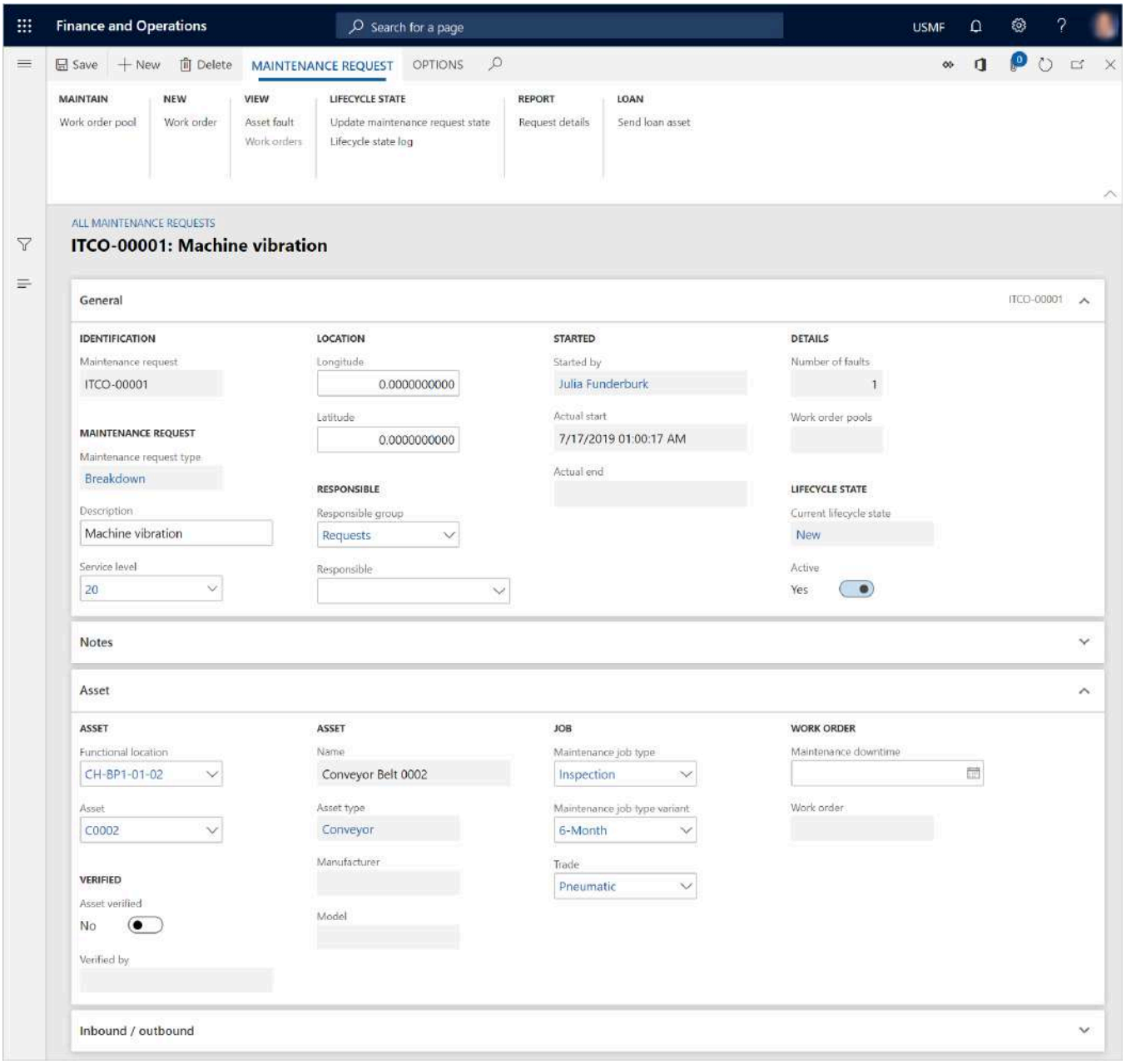
Subsequent processing of maintenance requests

After a maintenance request is created, but before it's converted to a work order, various information should be updated on it. Typically, a planner or another administrative employee completes this task.

- On the **All maintenance requests** or **Active maintenance requests** page, select the request to work with, and then select **Edit**.

In the details view, you can update various information. Here are some examples:

- Select and verify the asset. If you must select a different asset later, you can set the **Asset verified** option to **No**.
- Select a responsible maintenance worker group and/or a responsible maintenance worker. For more information about the required setup, see Responsible maintenance workers.
- Select a maintenance job type and, if this information is relevant, a related maintenance job variant and a job trade.
- In the **Latitude** and **Longitude** fields, enter geographic coordinates. Any coordinates that are added to a maintenance request are automatically transferred to a related work order.



> **Note**
>
> If you select an asset when you create a maintenance request, you can add one fault to the asset. After the maintenance request has been created, you can add more faults, as you require. To add faults, select **Asset fault** on the **All maintenance requests** page.

Last updated on 05/07/2025

Create work orders from maintenance requests

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

After you've created maintenance requests, you can easily convert them to work orders. This article describes the quickest way to work with maintenance requests, update several maintenance requests at the same time, and then create a work order for several maintenance requests at the same time. On the **Active maintenance requests** or **My functional location maintenance requests** page, you can also work with one maintenance request at a time and convert one maintenance request to a work order.

> **Note**
>
> Every maintenance request can be related to only one work order. However, multiple maintenance requests can be included in one work order, even if the maintenance requests have different assets.

1. Select **Asset management** > **Maintenance requests** > **All maintenance requests**.
2. Before you can create a work order from maintenance requests, you must select, at a minimum, a maintenance job type for the maintenance requests, and also a maintenance job type variant and trade, if this information is relevant. In the grid view, you can easily update information for a maintenance request.
3. When you're ready to create a work order, select the maintenance requests to include in it.
    - If you select several maintenance requests to convert to a work order, both the **Asset** field and the **Maintenance job type** field must be set before you create the work order.
    - If you select one maintenance request to convert to a work order, only the **Asset** field must be set before you create the work order. However, when you create the work order, you can select a maintenance job type (and a related maintenance job

type variant and trade, if this information is relevant) in the **Create work order** dialog box.

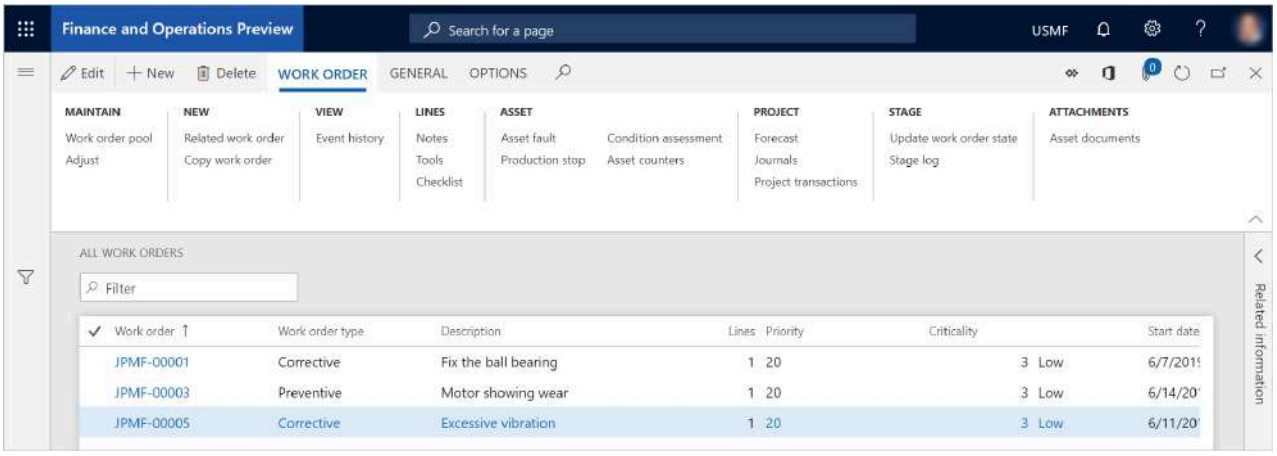
4. Select **Work order**.

5. In the **Create work order** dialog box, set the fields, and then select **OK**.

   A message bar might notify you that a new work order has been created.

   Additionally, when you create a work order that is based on a maintenance request, if the asset that is related to the maintenance request is included in a warranty agreement, a message bar notifies you about the warranty agreement.

6. Select **Asset management** > **Work orders** > **All work orders**, and open the new work order.



Last updated on 05/07/2025

Asset loans

Article • 06/15/2022

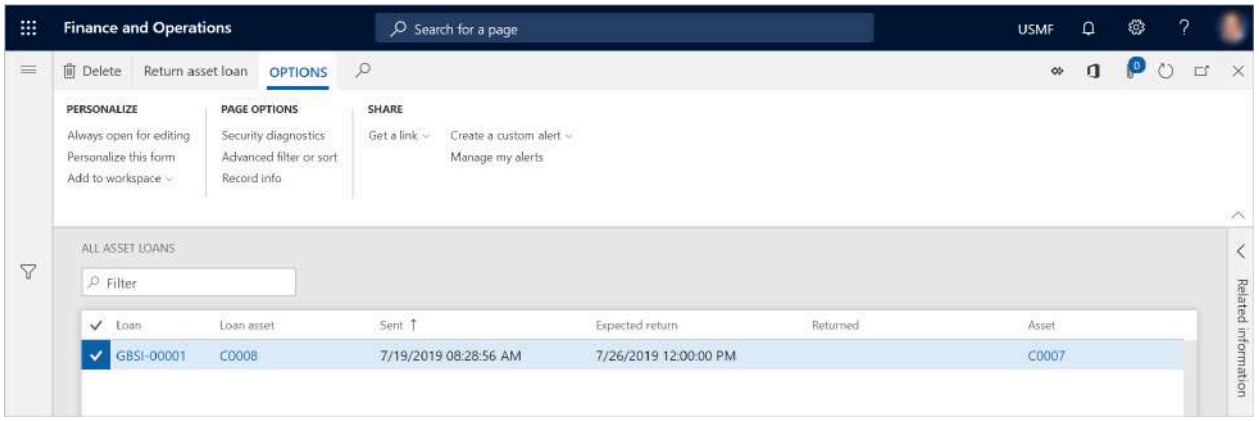
If your company receives assets for repair or maintenance jobs from either internal locations or customers, and if you temporarily loan assets to those locations or customers, Asset Management can track the asset loans.

Register asset loans on a maintenance request

1. Select **Asset management** > **Maintenance requests** > **All maintenance requests** or **Active maintenance requests**.
2. Select the maintenance request to register an asset loan on, and then select **Edit**.
3. On the **Request** page, select **Send loan asset**.
4. Select the asset, and enter the expected return date.
5. Select **OK**.

> **Note**
>
> - You can send a loan asset only if an asset of the same asset type is available.
> - The asset that you loan must have an asset lifecycle state that allows it to be used as a loan asset, such as **InStorage**. When the asset loan is registered, the asset lifecycle state of the asset is automatically updated to, for example, **OnLoan**.

To view a list of all the assets that you've loaned to other locations or customers, select **Asset management** > **Asset loan** > **All asset loans**. If the **Ended** check box is selected for an asset, the asset has been registered as returned to your company.



On the **Active asset loans** page, you can view a list of all the loan assets that haven't yet been returned to your company.

Register loan assets as returned

1. Select **Asset management** > **Asset loan** > **Active asset loans**.
2. Select the asset loan to register as returned, and then select **Return asset loan**.
3. In the **Returned** field, enter the date and time.
4. Select **OK**.
5. Refresh the **Active asset loans** list page, and notice that the asset loan no longer appears in the list.

Feedback

Was this page helpful? Yes No

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Inbound and outbound assets

Article • 06/15/2022

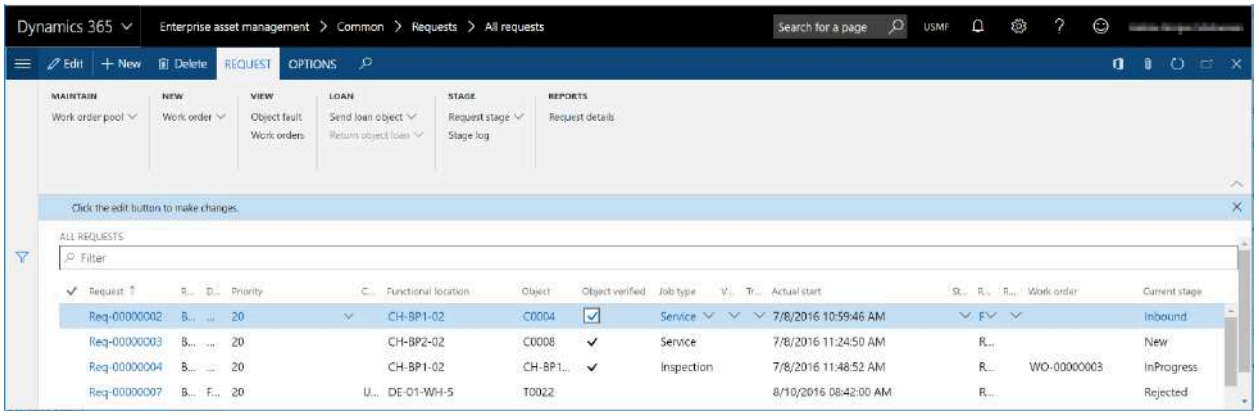
If your company does repair jobs or maintenance jobs on assets that are received from other locations or customers, Asset Management can track both inbound assets that are on their way to your company and outbound assets that are being returned.

> **Note**
>
> If you want to use inbound and outbound lifecycle states to manage assets that are coming in and being returned, you must set up maintenance request lifecycle states and lifecycle models to support these actions. Learn more in Maintenance requests.

The setup of Asset Management determines whether you can work with inbound and outbound assets.

Register assets as inbound

1. Select **Asset management** > **Maintenance requests** > **Active maintenance requests**.
2. Select the maintenance request.
3. Select **Update maintenance request state**.
4. Select **Inbound** (or another lifecycle state that you've created for inbound assets), and then select **OK**.



Register inbound assets as received

1. Select **Asset management** > **Inbound/outbound** > **Inbound assets**.
2. Select the asset or maintenance request.

3. Select **Receive assets**.
4. In the **Received** field, enter the date and time. Then select **OK**. The record is removed from the **Inbound assets** list page.

Register assets as outbound

When you've completed the maintenance or repair job, you can register the asset as returned.

1. Select **Asset management** > **Maintenance requests** > **Active maintenance requests**.
2. Select the maintenance request.
3. Select **Update maintenance request state**.
4. Select **Outbound** (or another lifecycle state that you've created for outbound assets), and then select **OK**.

Register outbound assets as delivered

1. Select **Asset management** > **Inbound/outbound** > **Outbound assets**.
2. Select the asset or maintenance request.
3. Select **Deliver assets**.
4. In the **Delivered** field, enter the date and time. Then select **OK**. The record is removed from the **Outbound assets** list page.

Feedback

Was this page helpful? Yes No

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Maintenance request reports

Article • 06/03/2022

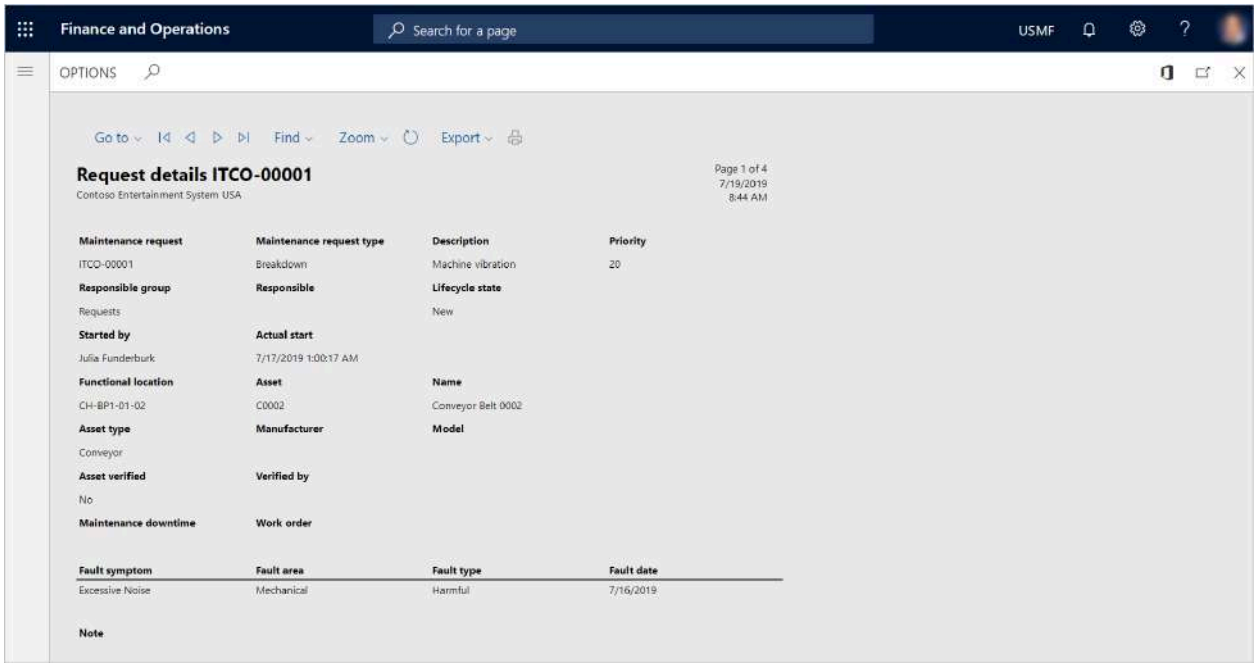
In Asset Management, you can generate two reports that are related to maintenance requests. One report shows details, and the other report provides a list that can be used for planning and follow-up.

Create a Maintenance request details report

The **Maintenance request details** report shows various information that is related to maintenance requests.

1. Select **Asset management** > **Reports** > **Maintenance requests** > **Maintenance request details**.
2. On the **Records to include** FastTab, you can select specific maintenance requests to include on the report.
3. On the **Run in the background** FastTab, you can set up report generation as a batch job, as you require.
4. Select **OK** to generate the report.

The following illustration shows an example of the **Maintenance request details** report.

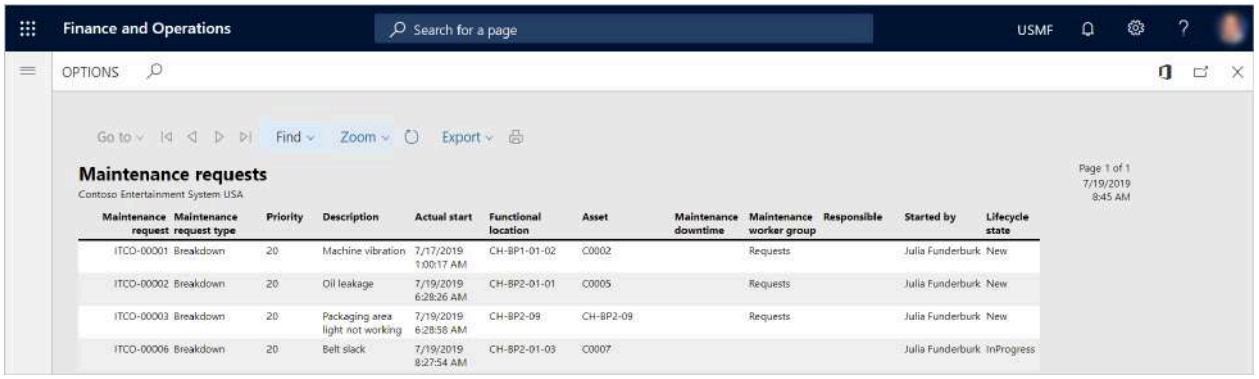


Create a Maintenance request list report

The **Maintenance request list** report shows a list of all maintenance requests of the same request type.

1. Select **Asset management** > **Reports** > **Maintenance requests** > **Maintenance request list**.
2. On the **Records to include** FastTab, you can make selections to define which maintenance requests are included on the report.
3. On the **Run in the background** FastTab, you can set up report generation as a batch job, as you require.
4. Select **OK** to generate the report.

The following illustration shows an example of the **Maintenance request list** report for all active maintenance requests.



Feedback

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Maintenance job types, categories, variants, trades, and checklists

08/13/2025

An asset type is attached to every asset. Asset types define the maintenance job types (and therefore, the maintenance jobs) that can be performed on assets. When you create a work order, you must select a maintenance job type. You can select only the maintenance job types that are related to the setup of the asset type that is used for the asset.

For a graphical overview of assets and maintenance job types, and their connection to work orders, see Functional locations and assets.

Maintenance job type variants can be set up on a maintenance job type. Maintenance job type variants define variations of a job type, such as sizes (small, medium, or large), periods (weekly, biweekly, one month, or three months), and configurations (low standard, flexible, or high performance).

Maintenance job trades provide information about professional trades, such as mechanical, electrical, and hydraulic trades. Competency requirements can be set up on a maintenance job trade. All maintenance job trades can be used in relation to all maintenance job types. Selection of a maintenance job type variant and/or maintenance job trade on a work order is optional.

For each maintenance job type, variations of the maintenance job type setup can be created. For example, if you have a maintenance job type that is named *Service*, you can create the following variations for that maintenance job type: *Trucks 30,000 km*, *Cars 30,000 km*, and *Vans 30,000 km*.

Maintenance job type categories are used to collect a group of maintenance job types for overview purposes. Examples of maintenance job type categories might include *Calibration*, *Inspection*, *Overhaul*, and *Instrumentation*.

Maintenance checklist templates and maintenance checklist variables are used to set up maintenance checklists. Maintenance checklists are set up on maintenance job types and used on work orders.

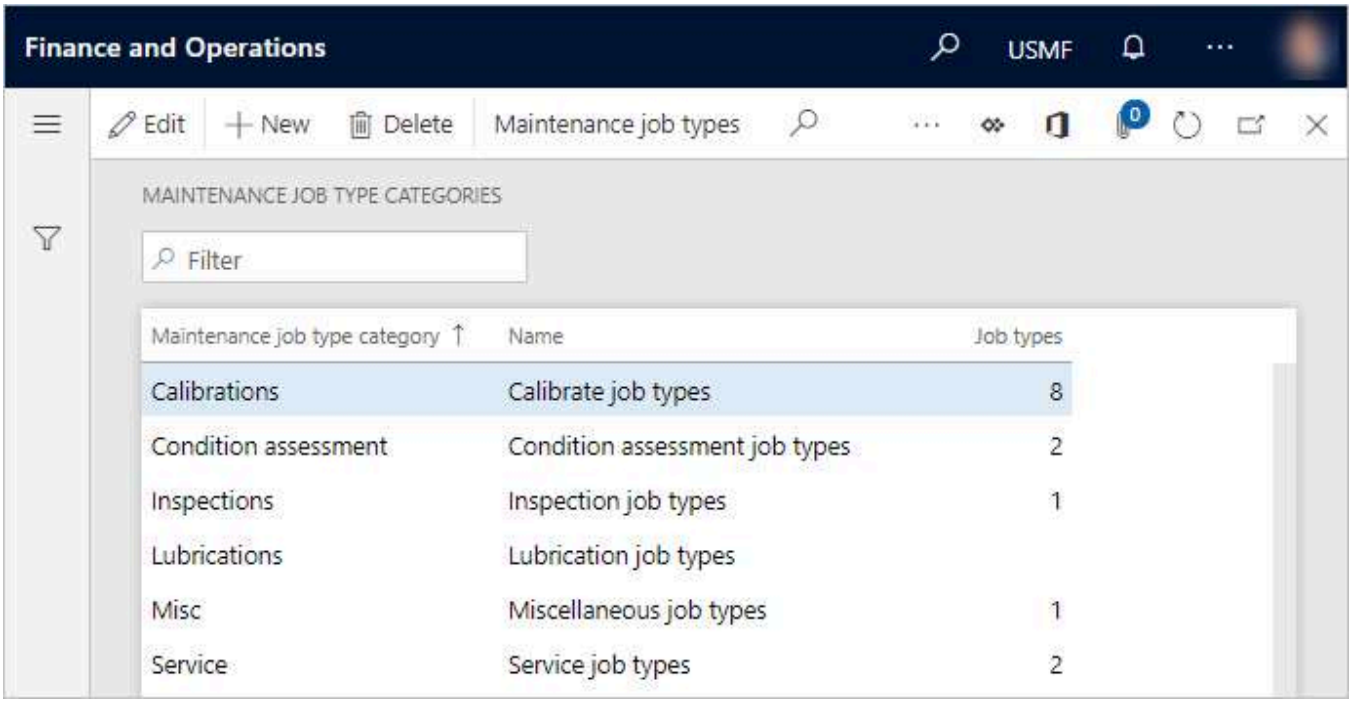
You first set up the required maintenance job type categories, maintenance job type variants, and maintenance job trades. You then create maintenance job types. Finally, on the **Maintenance job type defaults** page, you create all the variations of maintenance job types that are required for your equipment. On that page, you can also set up forecasts, maintenance checklists, and tools for a combination of maintenance job types.

> **Note**
>
> A maintenance job type can be related to only one maintenance job type category.

Create a maintenance job type category

1. Select **Asset management** > **Setup** > **Jobs** > **Maintenance job type categories**.
2. Select **New**.
3. In the **Maintenance job type category** field, enter an ID for the maintenance job type category.
4. In the **Name** field, enter a name.

   After you relate maintenance job type categories to maintenance job types, the **Job types** field shows the number of maintenance job types that are related to this maintenance job type category.

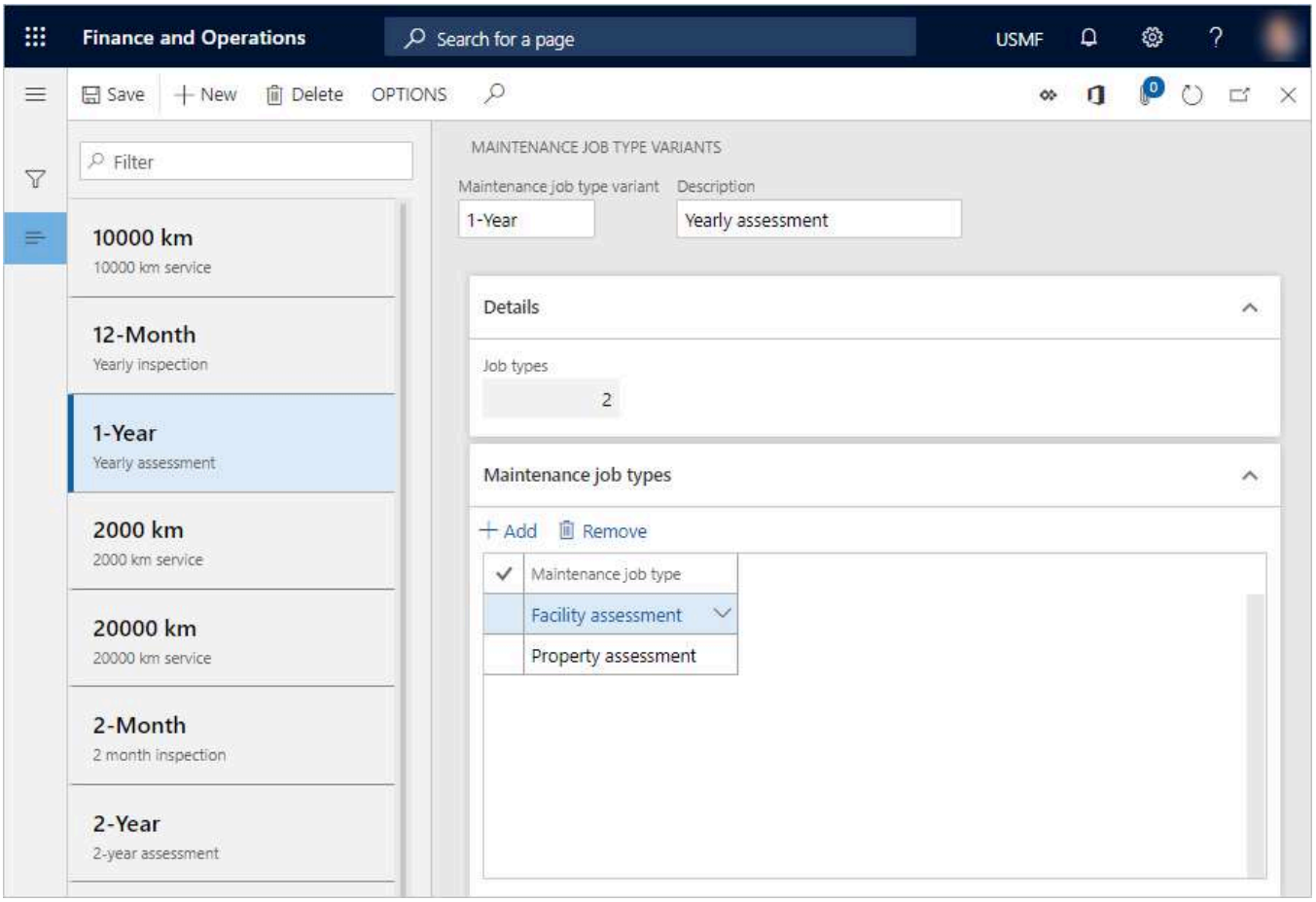


Create a maintenance job type variant

1. Select **Asset management** > **Setup** > **Jobs** > **Maintenance job type variants**.
2. Select **New**.
3. In the **Maintenance job type variant** field, enter an ID for the maintenance job type variant.

4. In the **Description** field, enter a description.
5. On the **Maintenance job types** FastTab, select **Add** to add a maintenance job type.
6. In the **Maintenance job type** field, select the maintenance job type.
7. Repeat steps 5 through 6 to add more maintenance job types to the maintenance job type variant.

   On the **Details** FastTab, the **Job types** field shows the number of maintenance job types that have been added to this maintenance job type variant.



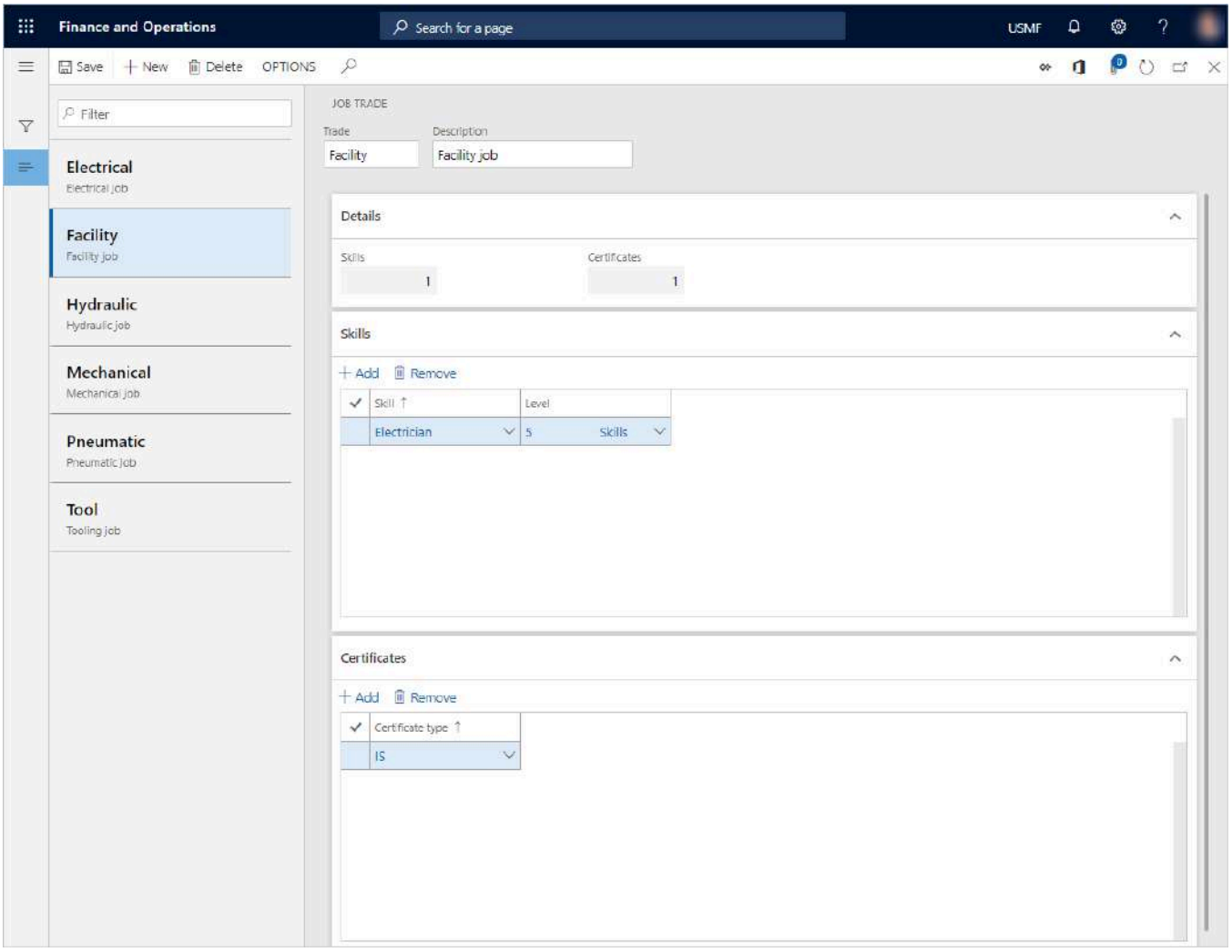
Create a maintenance job trade

1. Select **Asset management** > **Setup** > **Jobs** > **Maintenance job trade**.
2. Select **New**.
3. In the **Trade** field, enter an ID for the maintenance job trade.
4. In the **Description** field, enter a description.
5. On the **Skills** FastTab, select **Add** to add a new skill that should be related to the maintenance job trade.

6. In the **Skill** field, select the skill.
7. In the **Level** field, select the skill level.
8. Repeat steps 5 through 7 to add more skills to the maintenance job trade.

   On the **Details** FastTab, the **Skills** field shows the number of skills that have been added to this maintenance job trade.
9. On the **Certificates** FastTab, select **Add** to add a certificate to the maintenance job trade.
10. In the **Certificate type** field, select the certificate.
11. Repeat steps 9 through 10 to add more certificates to the maintenance job trade.

    On the **Details** FastTab, the **Certificates** field shows the number of certificates that have been added to this maintenance job trade.



Create a maintenance checklist variable

When you create maintenance checklist lines in the maintenance job type default, you must select a maintenance checklist type. *Variable* is one maintenance checklist type. It's used to

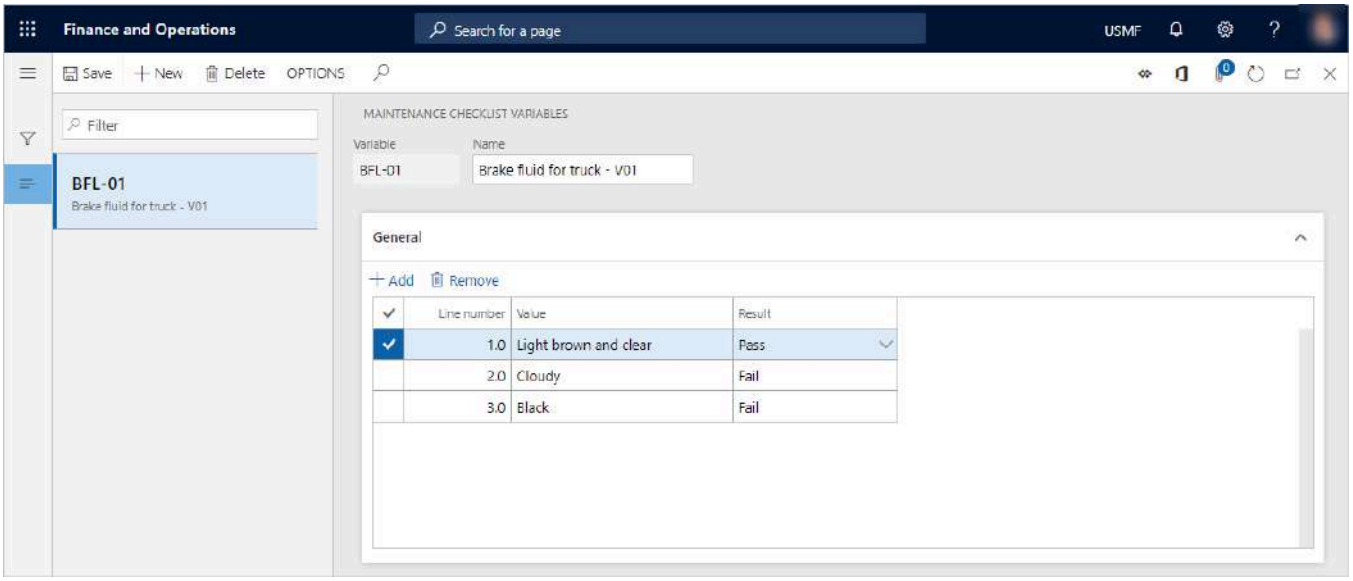
define a possible result in a range on a maintenance checklist line that is related to a work order line. A variable lets you create a set of predefined outcomes without having to make an exact measurement.

Example 1: You can measure oil level by defining three values: *Level too high*, *Level too low*, and *Level within range*. For each value, you define whether the value result is *Pass*, *Fail*, or *None*.

Example 2: You make a visual inspection of a piece of equipment to assess wear and tear.

1. Select **Asset management** > **Setup** > **Maintenance checklists** > **Maintenance checklist variables**.
2. Select **New**.
3. In the **Variable** field, enter an ID for the maintenance checklist variable.
4. In the **Name** field, enter a name.
5. On the **General** FastTab, select **Add** to add a line for a variable.

   A sequential line number is automatically entered in the **Line number** field. After you've added all the lines, you can change the line numbers as you require. When you select line and then press the **Down arrow** key, the next number in the sequence is automatically entered on the next line.
6. In the **Value** field, enter a value description.
7. In the **Result** field, select a result for the line.



Create a maintenance checklist template

Maintenance checklist templates can be used as a common set of tasks that a worker must perform to complete a work order correctly. The templates are referenced from maintenance checklist lines on the maintenance job type default. Templates can be referenced across multiple maintenance job type default lines. Therefore, you can easily reuse a set of common maintenance checklist tasks. Examples of maintenance checklist templates include general safety instructions, and a list of items and conditions that must be checked on a specific pump or similar models of a conveyor belt.

1. Select **Asset management** > **Setup** > **Maintenance checklists** > **Maintenance checklist templates**.
2. Select **New**.

   A template ID is automatically entered in the **Maintenance checklist template** field.
3. In the **Name** field, enter a name for the maintenance checklist template.
4. On the **Maintenance checklist lines** FastTab, select **Add** to add a template line.

   A sequential line number is automatically entered in the **Line number** field. After you've added all the lines, you can change the line numbers as you require. When you select line and then press the **Down arrow** key, the next number in the sequence is automatically entered on the next line.
5. In the **Type** field, select a type for the maintenance checklist line. For each maintenance checklist type, the **Line details** FastTab shows related fields. The following values are available:

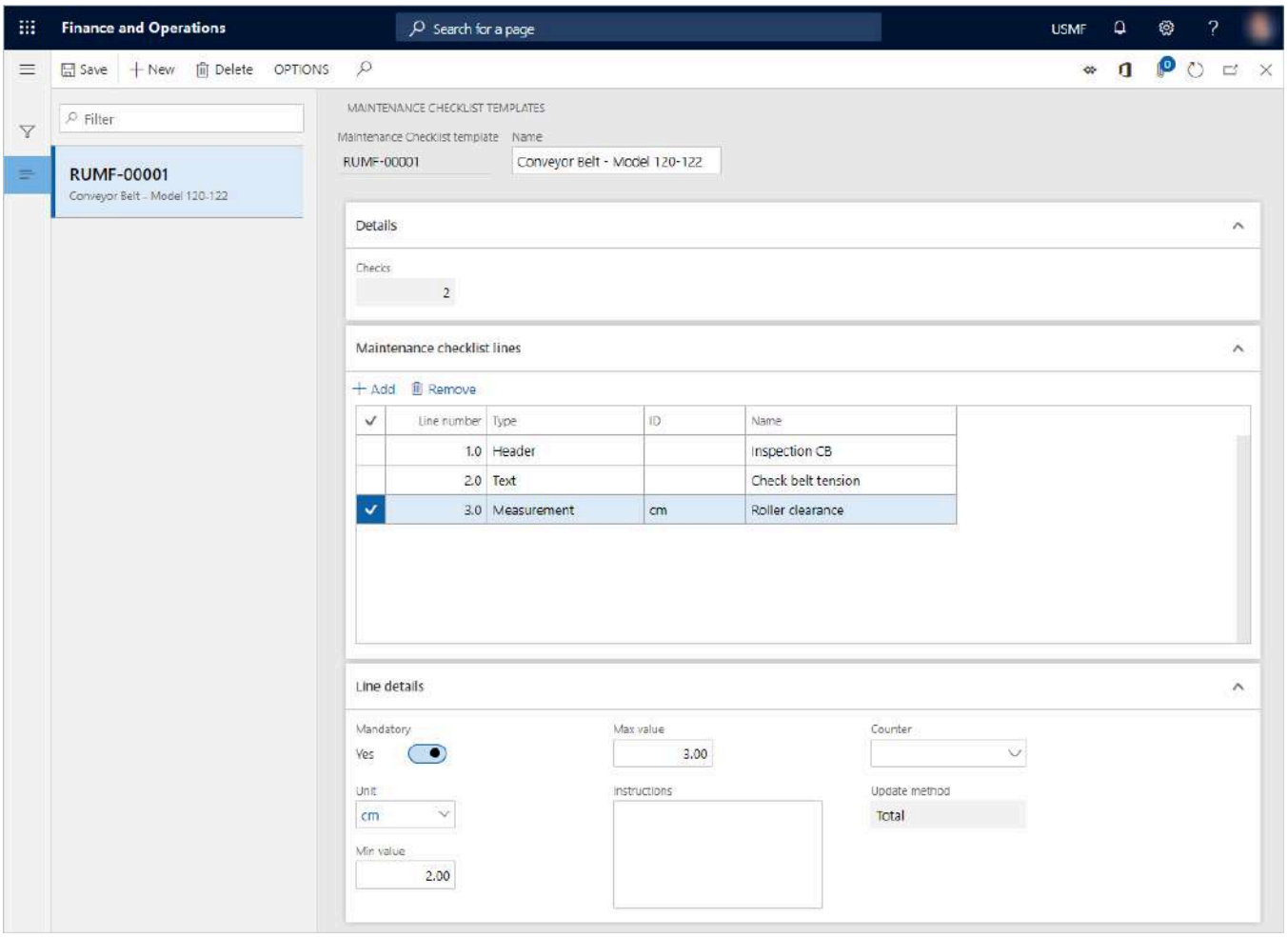
   - **Text** – The line has text that describes what to do. Use this maintenance checklist type if you want a worker to check or inspect something, but you don't expect a specific (measurable) result. After you select this type, enter a name or heading in the **Name** field. In the **Instructions** field, enter a description of what must be done. If the step is mandatory for the maintenance checklist, set the **Mandatory** option to *Yes*.
   - **Header** – The line is used as a heading to group the maintenance checklist lines that appear below it. This type is useful if you have several maintenance checklist lines that can be divided into specific areas. Headers provide an overview for the worker who will complete a maintenance checklist that has many maintenance checklist lines. After you select this type, enter a descriptive name in the **Name** field.
   - **Template** – The line is used to make a reference to an existing template. After you select this type, enter a name for the template in the **Name** field. In the **Template** field, select the template.

- **Variable** – The line is used to define a possible result in a range. For information about how to set up maintenance checklist variables, see the Create a maintenance checklist variable section. After you select this type, enter a descriptive name for the variable in the **Name** field. In the **Variable** field, select the variable. In the **Instructions** field, enter a description of what must be done. If the step is mandatory for the maintenance checklist, set the **Mandatory** option to *Yes*.

- **Measurement** – The line is used to record a specific measurement. You can set up the measurement that should be related to a predefined counter. After you select this type, enter a name for the template in the **Name** field. If this step is mandatory for the maintenance checklist, set the **Mandatory** option to *Yes*. If you want to use the measurement line as a counter registration, select the counter in the **Counter** field. The related **Unit** field is then automatically updated. If you've selected a counter, select the update method in the **Value** field. In the **Min. value** and **Max. value** fields, enter the allowed value range. In the **Instructions** field, enter a description of what must be done.

  > **Note**
  >
  > Any line of the **Measurement** type that doesn't have a counter setup is treated as an independent measurement registration that there's no automatic follow-up for in Asset Management. Likewise, if the selected counter type isn't present on the asset that is related to the work order, the maintenance checklist task is treated as an independent measurement. The counter value can be changed multiple times. It isn't posted until the **work order lifecycle state** is changed to a state where the **Process maintenance checklist** option is set to *Yes*.

On the **Details** FastTab, the **Checks** field shows the total number of checklist lines in your template. This number includes the nested lines in any existing template that you've referenced in your template.



Create a maintenance job type

1. Select **Asset management** > **Setup** > **Jobs** > **Maintenance job types**.
2. Select **New**.
3. In the **Maintenance job type** field, enter an ID for the maintenance job type.
4. In the **Name** field, enter a name.

   The **Details** FastTab shows an overview of the number of maintenance job type variants, skills, certificates, succeeding jobs, and asset types that have been created on this maintenance job type. The **Setup lines** field shows the number of maintenance job type default lines that have been set up on this maintenance job type. The **Assets** field shows the number of active assets that currently use this maintenance job type.
5. On the **General** FastTab, in the **Maintenance job type category** field, select a maintenance job type category.
6. Set the **Maintenance downtime activities** option to *Yes* if the maintenance job type requires a maintenance stop of the equipment before the job can be performed.
7. On the **Description** FastTab, enter a description of the maintenance job type.

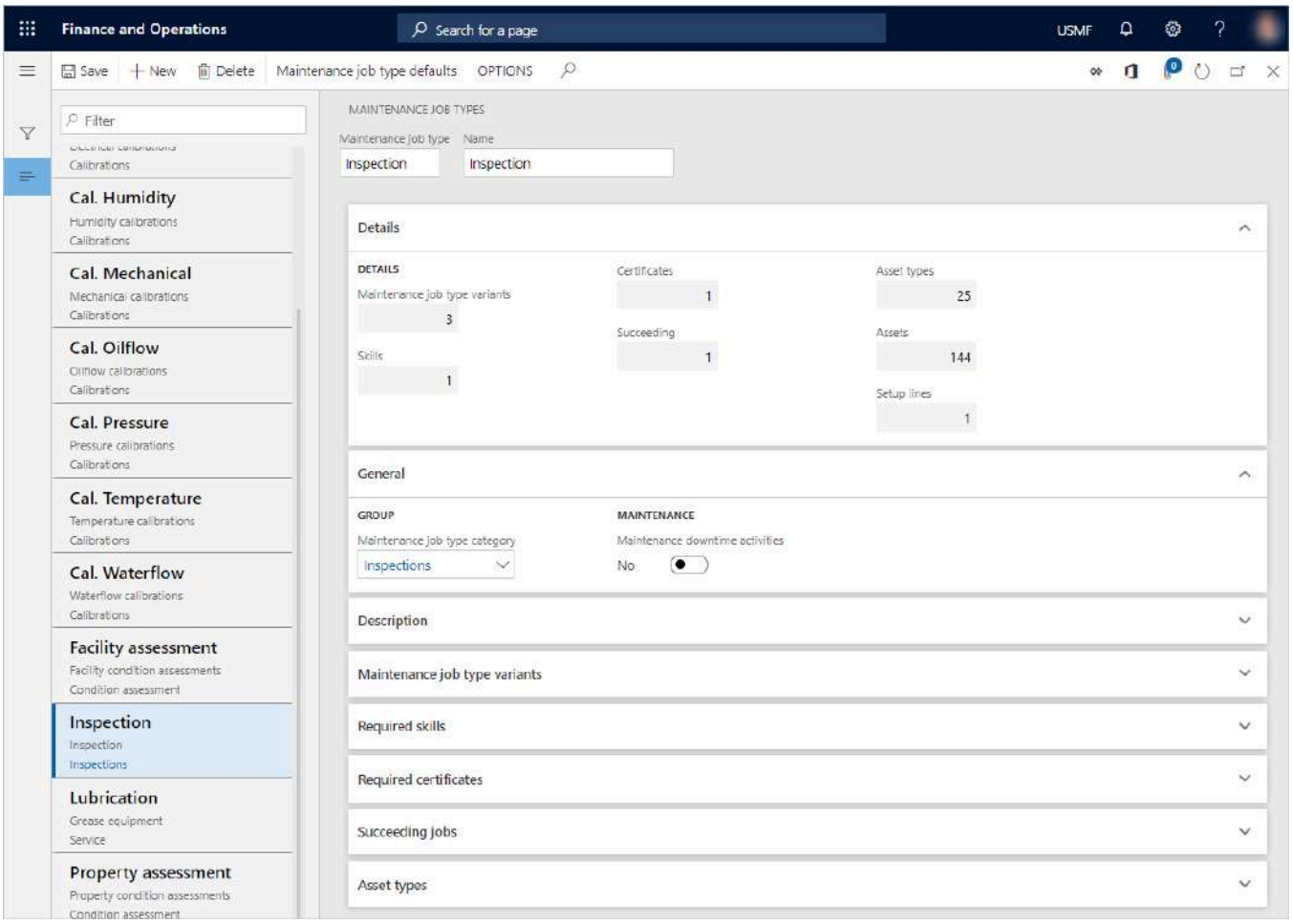
8. On the **Maintenance job type variants** FastTab, you can add variants to the maintenance job type.

9. On the **Required skills** and **Required certificates** FastTabs, you can add skills and certificate requirements to the maintenance job type.

10. If a specific maintenance job type must be performed next, add it on the **Succeeding jobs** FastTab. You can also set up a maintenance job type variant and trade that are related to the maintenance job type. If the succeeding job should start a specific number of days before or after the job that uses this maintenance job type has started, enter the number of days in the **Delay by days** field. Positive numbers represent days after the start of the related job, and negative numbers represent days before the scheduled start of the related job. For example, if you enter *5*, the succeeding job will start five days after the start of the job that is related to the maintenance job type. If you enter *-3*, the succeeding job will start three days before the scheduled start of the job that is related to the maintenance job type.

    > **Note**
    >
    > If you add more than one maintenance job type line, the sequence of the lines indicates the order that they should be performed in. The sequence starts at the top of the list.

11. On the **Asset types** FastTab, you can add asset types to the maintenance job type.



Create maintenance job type default lines and related forecasts, maintenance checklists, tools, description, and attachments

1. Select **Asset management** > **Setup** > **Jobs** > **Maintenance job type defaults**.

   –or–

   Select **Asset management** > **Setup** > **Jobs** > **Maintenance job types**, select a maintenance job type, and then select **Maintenance job type defaults**.

2. Select **New**.

3. In the **Functional location**, **Asset type**, **Manufacturer**, **Model**, and **Asset** fields, select appropriate values, depending on how specific the maintenance job type default should be.

4. In the **Maintenance job type** field, select a maintenance job type if it wasn't automatically selected.

5. In the **Maintenance job type variant** and **Trade** fields, select a maintenance job type variant and a maintenance job trade as you require.

6. Select **Forecast**.

7. On the **Maintenance job type default forecast** page, you can make forecasts on hours, items, and expenses. On the relevant tabs, select **Add**, and make selections to create the required forecasts for the maintenance job type.

8. On the **Item forecast** tab, you can select inventory dimensions that should be shown on the item line. Select **Inventory** > **Dimensions**, select the dimensions to show, set the **Save setup** option to *Yes*, and then select **OK**.

9. On the **Item forecast** tab, select **Item where used** to see an overview of where the item on the selected line is used in Asset Management in relation to assets, maintenance job type default, spare parts, and work orders.

   The **Maintenance forecast totals** FastTab shows an overview of forecast totals. This overview includes the total number of hours and forecast lines that have been created.

   > ⓘ **Note**
   >
   > To copy the forecast setup from another maintenance job type, select **Copy forecast**, and then select the maintenance job type to copy the setup from.

10. Select **Save** to save your changes.

11. Close the **Maintenance job type default forecast** page to return to the **Maintenance job type defaults** page.

12. Select **Maintenance checklist**.

13. On the **Maintenance job type defaults checklist** page, you can add maintenance checklist lines to the selected maintenance job type default. On the **Maintenance check lines** FastTab, select **New** to add a maintenance checklist line.

    Line numbers are automatically entered in the **Line number** field to indicate the sequence of the maintenance checklist lines. You can edit line numbers as you require. After you've created the first maintenance checklist line, select the line, and then press the **Down arrow** key to add a line below it. Alternatively, you can select a maintenance checklist line and then select **New**. In this case, a new line is added above the selected maintenance checklist line.

14. In the **Type** field, select the line type, and then add information that is related to the maintenance checklist type. For a description of the available types and related fields, see the Create a maintenance checklist template section.

> **Note**
>
> To copy the maintenance checklist setup from another maintenance job type, select **Copy maintenance checklist**, and then select the maintenance job type to copy the setup from.
>
> You can easily create a template from an existing maintenance checklist. You can then reuse the template across multiple maintenance checklists. The new template will be an exact copy of the active maintenance checklist. Select **Create template**, and then enter a name for the template. To replace the existing maintenance checklist with a single line that references the new template, set the **Replace** option to *Yes*. You can view the contents of the template in the **Maintenance checklist templates** details page.

15. Select **Save** to save your changes.
16. Return to the **Maintenance job type defaults** page.
17. Select **Tools**.
18. On the **Maintenance job type default tools** page, you can add the tools (resources) that should be used for the maintenance job type. Select **New**, and then select the tool in the **Resource** field.

    > **Note**
    >
    > To copy the tool setup from another maintenance job type, select **Copy tools**, and then select the maintenance job type to copy the setup from.

19. Select **Save** to save your changes.
20. Return to the **Maintenance job type defaults** page.
21. Select **Work description**.
22. On the **Work description** page, select **Edit**, and then add a description that is related to the selected maintenance job type default, as you require.
23. Select **Save** to save the description.

    If you add a work description here, it overrides any description that is set up for the maintenance job type on the **Maintenance job types** page. If you don't add a work description here, any description that is set up for the maintenance job type is used.

Descriptions are automatically transferred to work orders that use the maintenance job type or maintenance job type default.

24. Return to the **Maintenance job type defaults** page.

25. To set up attachments on a selected maintenance job type default line, select **Attach documents**. Attachments that are set up on a maintenance job type default line are automatically included on work order lines that use that maintenance job type default line.

26. Select **New**, and then select a document type.

27. Upload the document or file.

28. Set the fields on the **Attachments** page. The attachment setup uses standard document setup functionality.

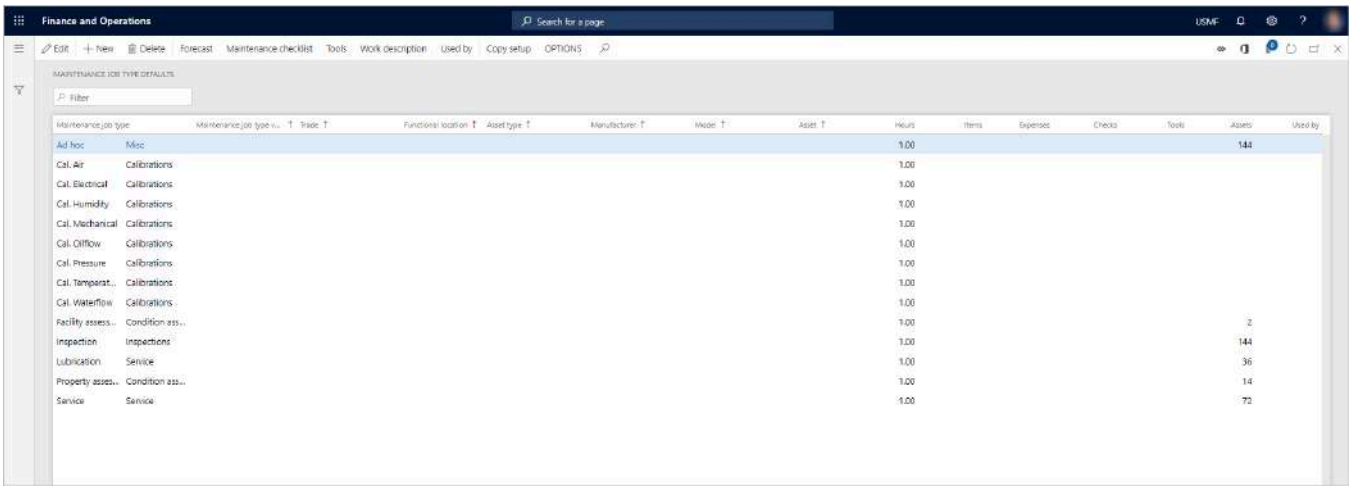
29. Select **Save** to save the attachment.

    > ⓘ **Note**
    >
    > Attachments on a maintenance job type default line are printed together with a work order report only if the document types of the attachments are selected on the **Document types** tab of the **Asset management parameters** page (**Asset management** > **Setup** > **Asset management parameters**). Examples of attachments include guidelines that explain how to complete a specific job or a predefined maintenance checklist (if you don't use the maintenance checklist functionality for maintenance job type default lines).

    On the **Maintenance job type defaults** page, each line shows the number of forecasted hours, and also the number of lines that have been created for items, expenses, maintenance checklists, and tools. The **Assets** field shows the number of active assets that are related to the maintenance job type default line.

30. To copy a maintenance job type default to another maintenance job type default, select the maintenance job type default line to copy another setup to, select **Copy setup**, and then select the maintenance job type default to copy.

31. To view a list of the maintenance plans or maintenance rounds that currently use a maintenance job type default line, select the line, and then select **Used by**.



When the system selects the available maintenance job type default that should be used on a work order line, the selection is based on the asset and the related asset type setup. Asset Management goes through all maintenance job type default records that are related to the maintenance job type that is related to the asset type to check for a possible match. It always checks the most specific combination first. In other words, to find the most specific combination, Asset Management first checks for a possible match for the **Trade** field. If no match is found, it checks for a match for the **Maintenance job type variant** field. If no match is found, it checks for a match for the **Maintenance job type** field, and so on (**Trade**, then **Maintenance job type variant**, then **Maintenance job type**, then **Asset**, then **Model**, then **Manufacturer**, and then **Asset type**). If no match is found, the default record where only the maintenance job type is selected is used.

A project activity ID is automatically related to each maintenance job type default line that you create. The project activity is created on the forecast project that is selected in the **Maintenance forecast project** field on the **Assets** tab of the **Asset management parameters** page. The purpose of the project activity is to manage forecasts on hours, items, and expenses in relation to work orders. Maintenance job type forecasts are automatically transferred to the work order line, and they're copied from the forecast project to the work order project that is created for the work order line. The purpose of the project activity is to manage forecasts in relation to work orders.

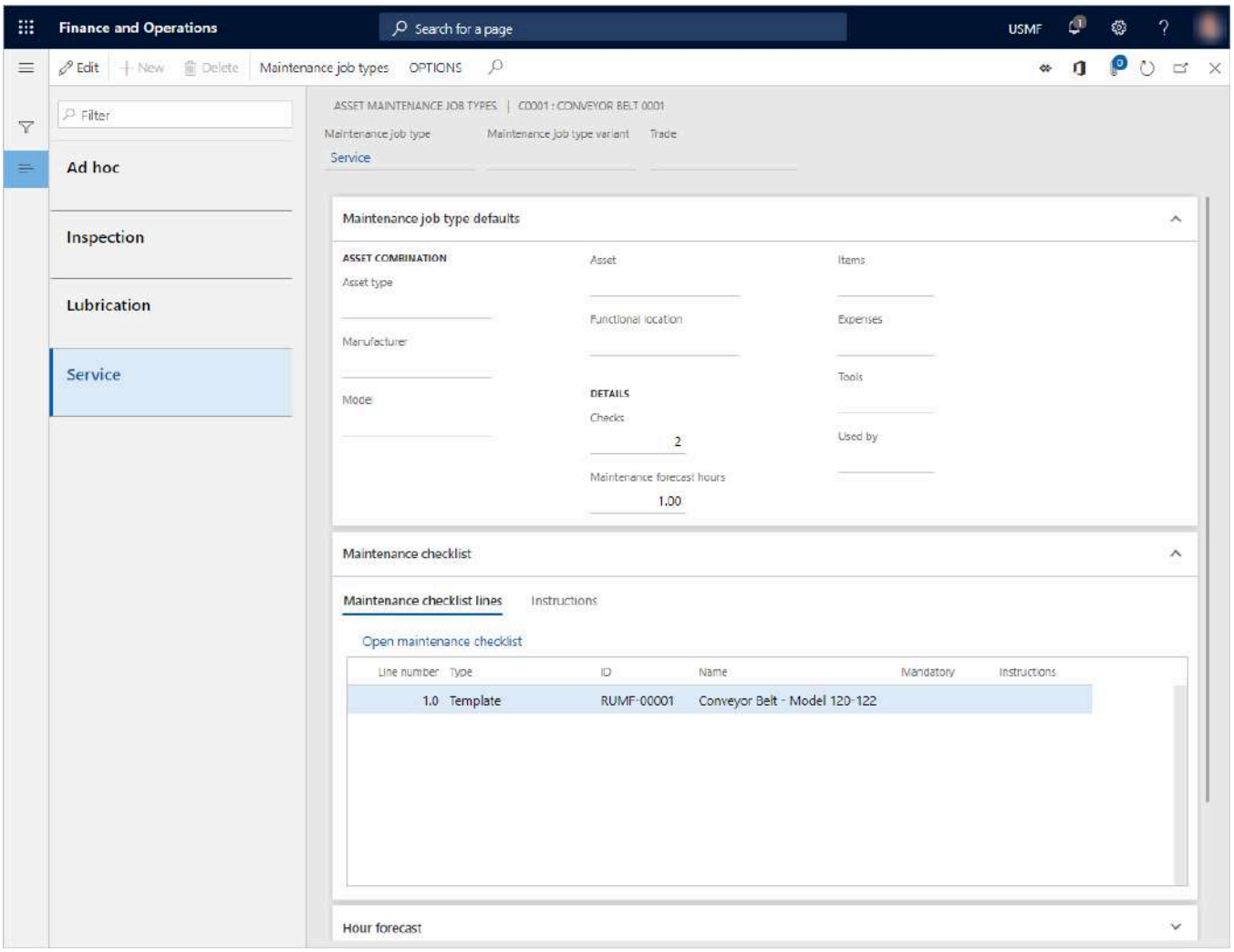
You can set up a batch job to update maintenance job type default references at regular intervals, or you can manually run the job. To create a batch job or run a manual update, select **Asset management** > **Periodic** > **Preventive maintenance** > **Update maintenance job type default references**.

Overview of maintenance job types that are related to assets

After you've created the required maintenance job type default combinations, you can use the **All assets** page to get an overview of the current maintenance job type default that is related to a specific asset. The overview shows all maintenance job type default combinations that can be used on the asset type that is selected for the asset. These combinations include combinations that have variations of maintenance job type variants and maintenance job trades.

1. Select **Asset management** > **Assets** > **All assets** or **Active assets**.
2. In the list, select the asset to see an overview of maintenance job type combinations for.
3. On the Action Pane, on the **General** tab, in the **Related information** group, select **Maintenance job types**.

   The left pane of the **Asset maintenance job types** page shows a list of all the maintenance job type combinations that are related to the selected asset.
4. Select a maintenance job type combination to see the related setup for maintenance checklists, forecasts, and tools. The **Details** section on the **Maintenance job type defaults** FastTab shows the number of related maintenance checklists, forecasted hours, items, and so on, that are related to the selected maintenance job type combination.
5. To view details for the selected maintenance job type, select **Maintenance job types**.



Automatic update of maintenance job type forecasts

In Asset Management, you can automatically update any changes to maintenance job type forecasts for hour costs, item costs, and expenses that have been updated in other modules. In this way, you help guarantee that your maintenance job type forecasts always use the latest cost prices.

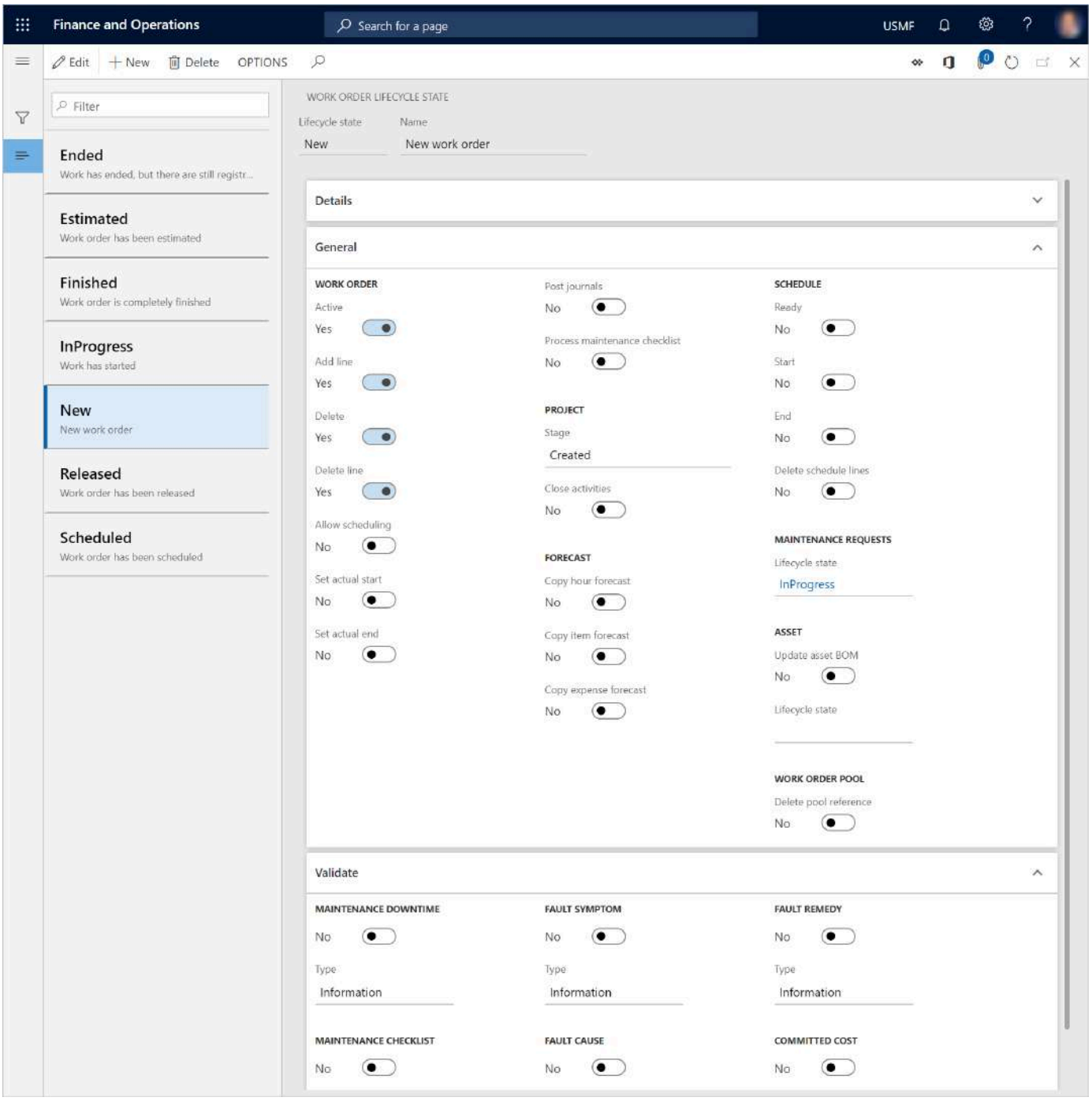
1. Select **Asset management** > **Periodic** > **Forecast** > **Update maintenance job type forecast**.
2. In the **Update maintenance job type forecast** dialog box, on the **Records to include** FastTab, you can add selections for specific maintenance job types as you require. Select **Filter**, and then select **Select** to make the selections.
3. On the **Run in the background** FastTab, you can set up the automatic update as a batch job, as you require.
4. Select **OK** to start the forecast update.

Work order lifecycle states

Work order lifecycle states define the states that a work order can go through. Examples include **Created**, **Scheduled**, **In progress**, and **Ended**. Work order lifecycle states can be manually updated on a work order, or they can be automatically updated (for example, during work order scheduling).

The work order lifecycle states that are required for your work orders must be attached to matching project stages on the **Project management and accounting parameters** page (**Project management and accounting** > **Project management and accounting parameters**). You first set up project stages in Project management and accounting. You then set up work order lifecycle states and work order lifecycle models in Asset Management.

The following table describes the options in the **Work order** and **Schedule** sections on the **General** FastTab of the **Work order lifecycle state** page (**Asset management** > **Setup** > **Work orders** > **Lifecycle states**).



Expand table

Option name	Description
Active	Set this option to *Yes* if the work order should be active while it's in this lifecycle state.
Add line	Set this option to *Yes* if work order jobs can be added to a work order that is in this lifecycle state.
Delete	Set this option to *Yes* if a work order can be deleted while it's in this lifecycle state.
Delete line	Set this option to *Yes* if work order jobs can be deleted from a work order that is in this lifecycle state.
Allow	Set this option to *Yes* if a work order can be scheduled while it's in this lifecycle state.

Option name	Description
scheduling	
Set actual start	Set this option to *Yes* if the user should be prompted to select an actual start date and time for a work order when it's updated to this lifecycle state.
Set actual end	Set this option to *Yes* if the user should be prompted to select an actual end date and time for a work order when it's updated to this lifecycle state.
Post journals	Set this option to *Yes* if work order journals should be automatically posted when a work order is updated to this lifecycle state. If an error occurs during journal posting, a message is shown, and the update of the work order lifecycle state is canceled. To view the journal lines for a work order, select **Asset management** > **Work orders** > **All work orders**, **Active work orders**, or **My active work orders**, select the work order in the list, and then select **Journals**. This setup of automatic work order journal posting at a specific lifecycle state is the same as when you select **Post journals** on the **Work order journals** page. **Note:** If you set this option to *Yes*, journals are automatically posted if no approval workflow has been set up. If your company uses the journal approval setup that is configured on the **Journal approval** page (**Project management and accounting** > **Setup** > **Journals** > **Journal approval**), a manager or clerk must validate and post consumption registrations.
Process maintenance checklist	Set this option to *Yes* if all attached maintenance checklists should be processed when a work order is updated to this lifecycle state. As part of this processing, any counter registrations that were made on a maintenance checklist are posted, and the result of the whole maintenance checklist is evaluated. Maintenance checklist lines that have *Pass* and *Fail* results are evaluated, and if at least one line fails, the whole maintenance checklist is marked as *Failed* in Asset Management.
Ready	Set this option to *Yes* if the work order job schedule status for all work order jobs that are created on a work order should automatically be updated to **Ready** when the work order is updated to this lifecycle state.
Start	Set this option to *Yes* if the work order job schedule status for all work order jobs that are created on a work order should automatically be updated to **Started** when the work order is updated to this lifecycle state.
End	Set this option to *Yes* if the work order job schedule status for all work order jobs that are created on a work order should automatically be updated to **Ended** when the work order is updated to this lifecycle state.
Delete schedule lines	Set this option to *Yes* if scheduling on all work order jobs that are created on a work order that has already been scheduled should be deleted when the work order is updated to this lifecycle state. In other words, capacity reservations on the asset, the related maintenance worker, and related tools will be deleted. For example, you set this option to *Yes* on a work order lifecycle state that is named **Estimated**. Then, when a work order is rolled back to this lifecycle state because rescheduling is required, scheduling can be deleted on that work order.

Set up project stages and work order lifecycle states

1. Select **Project management and accounting** > **Setup** > **Project management and accounting parameters**.

2. On the **Project stage** tab, for each stage that you want to use, select the check box for every project type that is required for your work orders. The project types define rules about the financial tasks that are allowed. Examples of financial tasks include creating a forecast, creating estimates, and creating beginning balances.

   > **Important**
   >
   > If a project stage isn't selected for a project type, but the project stage is used for a work order lifecycle state, the work order projects won't be updated in the appropriate manner.

3. Close the **Project management and accounting parameters** page.

4. Select **Asset management** > **Setup** > **Work orders** > **Lifecycle states**.

5. Select **New** to create a work order lifecycle state.

6. In the **Lifecycle state** field, enter an ID for the lifecycle state.

7. In the **Name** field, enter a name.

   On the **Details** FastTab, the **Lifecycle models** field shows the number of work order lifecycle models that use this lifecycle state.

8. On the **General** FastTab, in the **Work order** section, select the functions that should be available for this lifecycle state by setting the relevant options to *Yes*. For descriptions of the options, see the table earlier in this article.

9. In the **Project** section, in the **Stage** field, select the project stage that should be related to this lifecycle state.

10. In the **Project** section, set the **Close activities** option to *Yes* if project activities that are related to each work order job should be automatically closed when the work order is in this lifecycle state.

    > **Note**

To find the number of the project activity that is related to a work order job, select **Asset management** > **Work orders** > **All work orders**, **Active work orders**, or **My active work orders**. Open the work order, and then select the work order job. The activity number is shown in the **Activity number** field in the **Project** section on the **General** tab of the **Line details** FastTab.

11. In the **Forecast** section, set the **Copy hour forecast**, **Copy item forecast**, and/or **Copy expense forecast** option to *Yes* if work order project forecasts should automatically be copied to work order journals when the work order is in this lifecycle state.

12. In the **Schedule** section, set one of the options to *Yes* if the schedule status for work order jobs should be updated when the work order is in this lifecycle state. For descriptions of the **Ready**, **Start**, **End**, and **Delete schedule lines** options, see the table earlier in this article.

    > ⓘ **Note**
    >
    > To view schedule lines that are related to work order jobs, select **Asset management** > **Work orders** > **All work orders**, **Active work orders**, or **My active work orders**. Open the work order, select the work order job on the **Work order jobs** FastTab, and view related information on the **Line details** FastTab. The **Status** field on the **Schedule** tab shows the status of the work order job. The **Status** field can be set to the following values: *Scheduled*, *Ready*, *Started*, *Stopped*, and *Ended*.

13. In the **Maintenance requests** section, in the **Lifecycle state** field, select the maintenance request lifecycle state that related maintenance requests should be updated to. This update occurs automatically. It can be done only if the maintenance request lifecycle state is selected in the maintenance request lifecycle model that is used for the maintenance request.

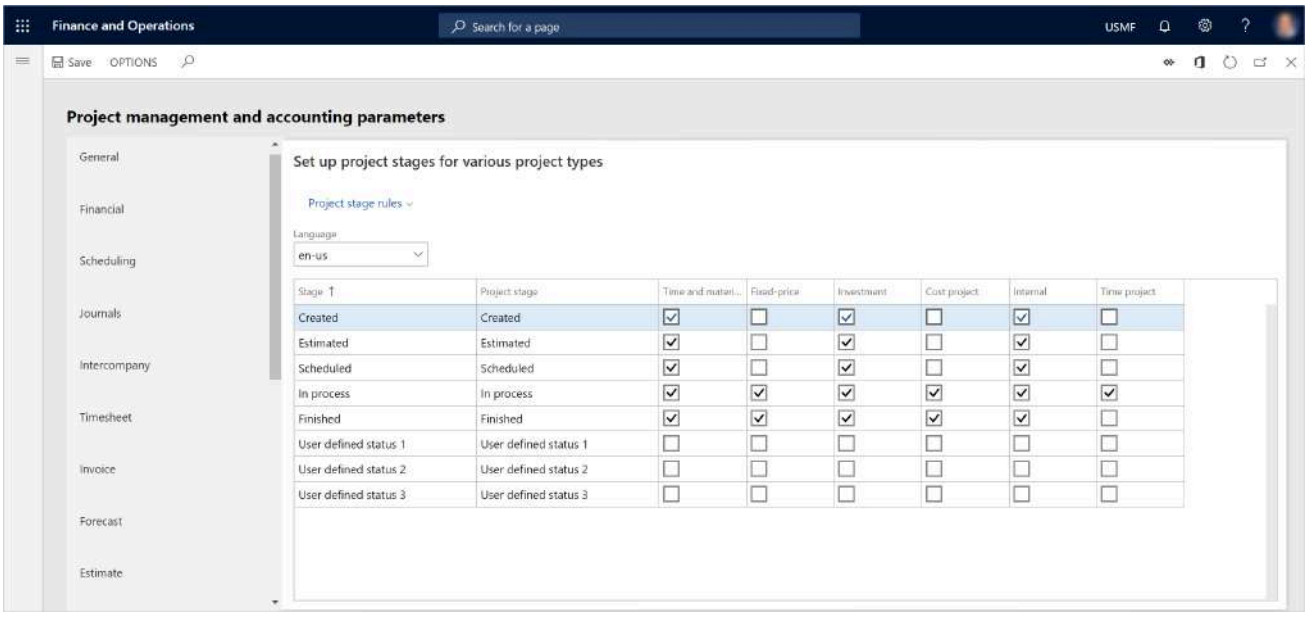
14. In the **Asset** section, set the **Update asset BOM** option to *Yes* if all items that are used on a work order should automatically be updated on the **Asset BOM** page when the work order is updated to this lifecycle state. This setting might be relevant if, for example, the work order lifecycle state defines the work order as completed or ended.

15. In the **Work order pool** section, set the **Delete pool reference** option to *Yes* if work orders that are in this lifecycle state should automatically be deleted from work order pools.

16. On the **Validate** FastTab, select the validation types that should be activated in this lifecycle state by setting the relevant options to *Yes*. Then, in the **Type** field for each validation type

that you select, select the type of message that should be shown if mandatory fields that are related to the validation type haven't been validated. If you select **Error**, the update of the work order lifecycle state is canceled until the related mandatory fields have been updated on the work order.

- The **Maintenance downtime**, **Maintenance checklist**, **Fault symptom**, **Fault cause**, and **Fault remedy** options are related to the options in the **Mandatory** section on the **Work order types** page (**Asset management** > **Setup** > **Work orders** > **Work order types**). To activate these validations, the related options must also be set to *Yes* on the work order type that is used for the work order.
- If the **Maintenance checklist** option is set to *Yes* for the lifecycle state that a work order is updated to, validation is done to verify that maintenance checklist lines that are marked as **Mandatory** have been registered as either **Checked** or **Not applicable**. If neither of those registrations has been made on the mandatory lines, an informational, error, or warning message is shown when the work order is updated to this lifecycle state.
- If the **Committed cost** option is set to *Yes* for the lifecycle state that a work order is updated to, total amount of committed costs (that is, the total amount of expenses that the legal entity has committed to pay) are calculated for each work order job. A message is shown if the committed cost amount is more than 0 (zero). You select the types of cost commitment that are included on the **Cost commitments** FastTab on the **Cost control** tab of the **Project management and accounting parameters** page (**Project management and accounting** > **Setup** > **Project management and accounting parameters**).
- If the **Maintenance downtime** option is set to *Yes* for the lifecycle state that a work order is updated to, maintenance downtime validation is done on the asset that is related to the work order. If a maintenance downtime registration has been made, but there's no **Ended** registration, a message is shown when the work order is updated to this lifecycle state.
- If the standard project setup doesn't include all the stages that you require for your Asset Management setup, you can set up user-defined project stages on the **Project stage** tab of the **Project management and accounting parameters** page. The following illustration shows the **Project stage** tab on the **Project management and accounting parameters** page.



17. On the **Security roles** FastTab, select the security roles that are allowed to change to this lifecycle state. If no roles are selected, then all users that have access to work order lifecycle states are able to change to this lifecycle state.

> **Note**
>
> If the lifecycle state that you update a work order to is inactive, journals that are related to the work order but that haven't yet been posted are automatically deleted. This behavior helps guarantee automatic cleanup of unused data. (A lifecycle state is inactive if the **Active** option for it is set to *No* on the **General** FastTab of the **Work order lifecycle state** page.)
>
> However, if you manually set a work order so that it's inactive, journals that are related to the work order but that haven't yet been posted are *not* automatically deleted. (To manually deactivate a work order, select **Asset management** > **Work orders** > **All work orders** or **Active work orders**. Open the work order, and switch to the **Header** view. On the **General** FastTab, select **Edit**, and then set the **Active** option to *No*.)

Relations among work order lifecycle models, work order types, and work order lifecycle states

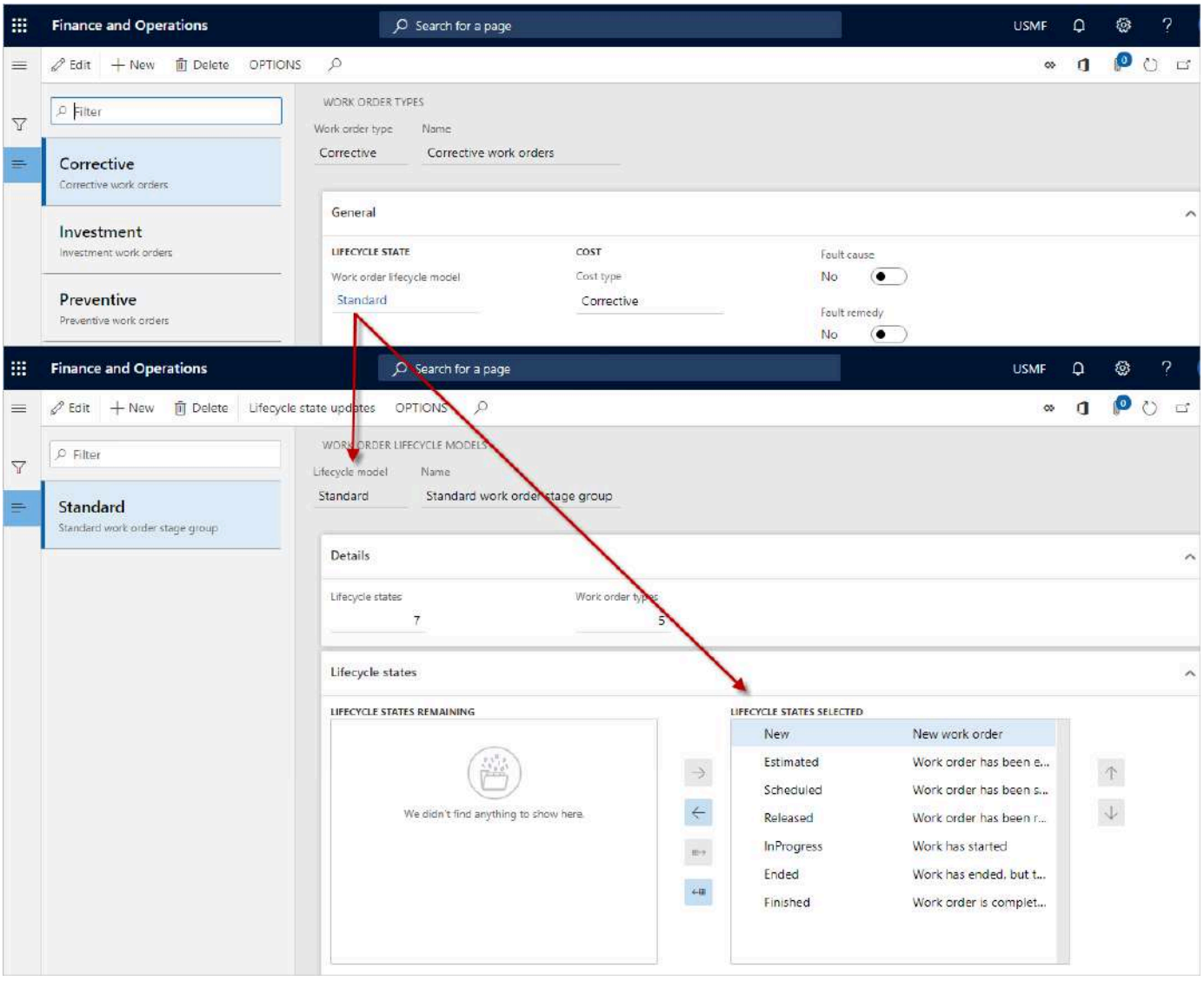
Lifecycle models refer to workflows, and lifecycle states are selected in a lifecycle model in sequential order. Lifecycle models are set up on work order types. Work order types determine the size or extent of workflows and work processes. For example, *Maintenance*, which is a standard work order type, might be related to a work order lifecycle model that has many lifecycle states. By contrast, you might have a *Corrective* work order type that is used for work

orders that haven't been scheduled, or for work orders where the job is completed before the work order is made because of an urgent situation. This work order type might be related to a work order lifecycle model that has only a few lifecycle states.

The reason for using types is that when a type is defined on, for example, a work order or an asset, the related work processes (lifecycle states) are automatically defined. For more information about how to set up work order types, see Work order types.

> **Note**
>
> Lifecycle states, lifecycle models, and types apply to functional locations, assets, and maintenance requests, in addition to work orders.

The following illustration shows the relation between work order types, lifecycle models, and lifecycle states.



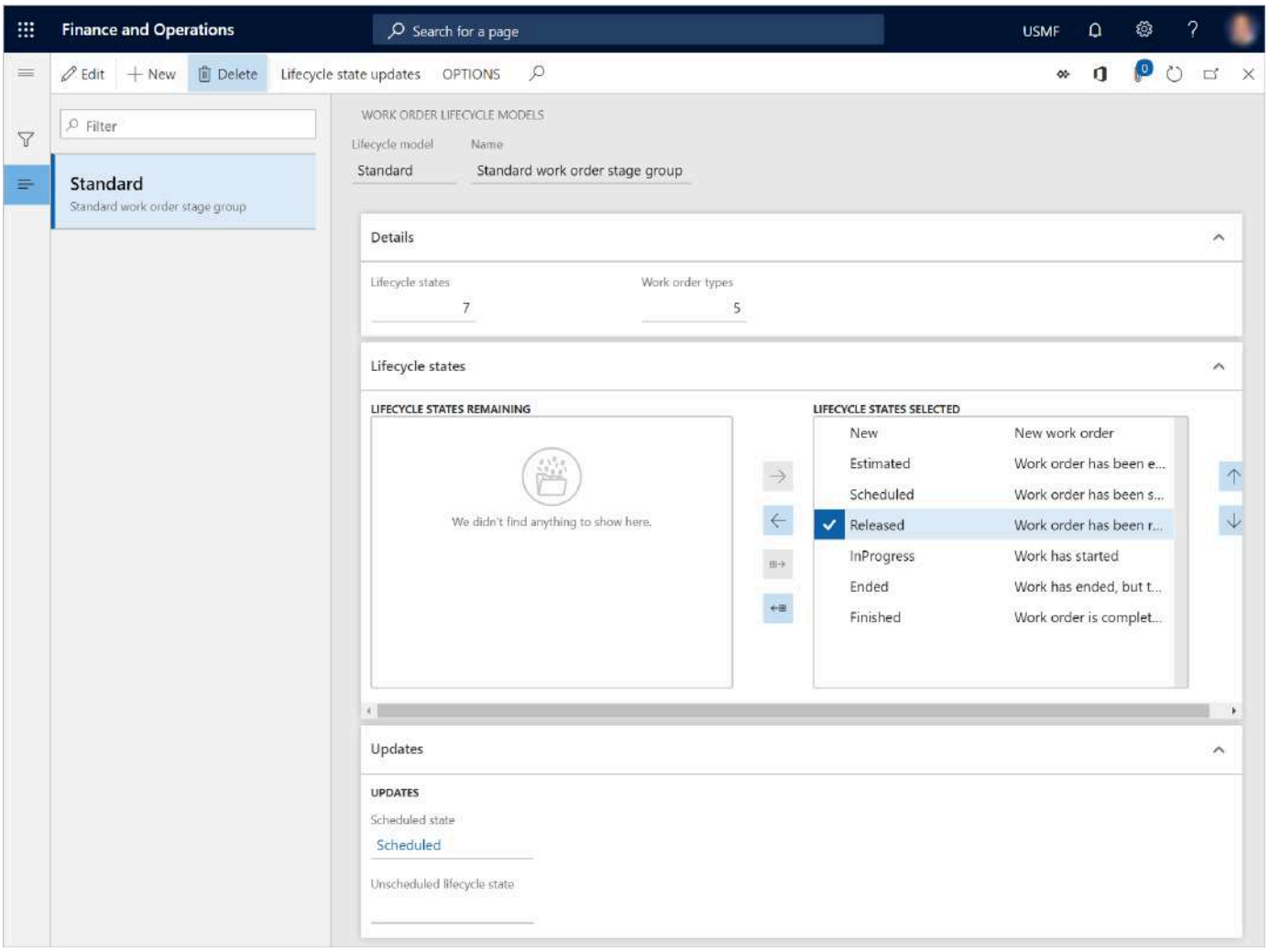
Work order lifecycle models

After you've created the work order lifecycle states that are required for your work orders, they can be divided into work order lifecycle models. At a minimum, you should create one standard lifecycle model.

1. Select **Asset management** > **Setup** > **Work orders** > **Lifecycle models**.
2. Select **New** to create a work order lifecycle model.
3. In the **Lifecycle model** field, enter an ID for the lifecycle model.
4. In the **Name** field, enter a name.

   On the **Details** FastTab, the **Lifecycle states** field shows the number of lifecycle states that are selected in this lifecycle model. The **Work order types** field shows the number of work order types that use this lifecycle model.
5. On the **Lifecycle states** FastTab, select the lifecycle states that should be included in the lifecycle model:

   - To include a lifecycle state in the lifecycle model, select it in the **Lifecycle states remaining** section, and then select the right arrow button to move it to the **Lifecycle states selected** section.
   - To include all the available lifecycle states in the lifecycle model, select the **Select all available stages** button . All lifecycle states are moved to the **Lifecycle states selected** section.
   - To remove a lifecycle state from the lifecycle model, select it in the **Lifecycle states selected** section, and then select the left arrow button to move it to the **Lifecycle states remaining** section.
6. Select **Lifecycle state updates** to define the lifecycle states that can follow a selected lifecycle state.
7. On the **Updates** FastTab, in the **Scheduled state** field, select the lifecycle state that should always be selected for a work order that you've completed work order scheduling for, regardless of the previous lifecycle state of the work order.
8. In the **Unscheduled lifecycle state** field, select the lifecycle state that should always be selected for a work order if work order scheduling is deleted.
9. Save the work order lifecycle model.



Last updated on 07/27/2026

Work order types

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Work order types are used to categorize work orders. For example, you might have work orders that are related to preventive maintenance or corrective maintenance.

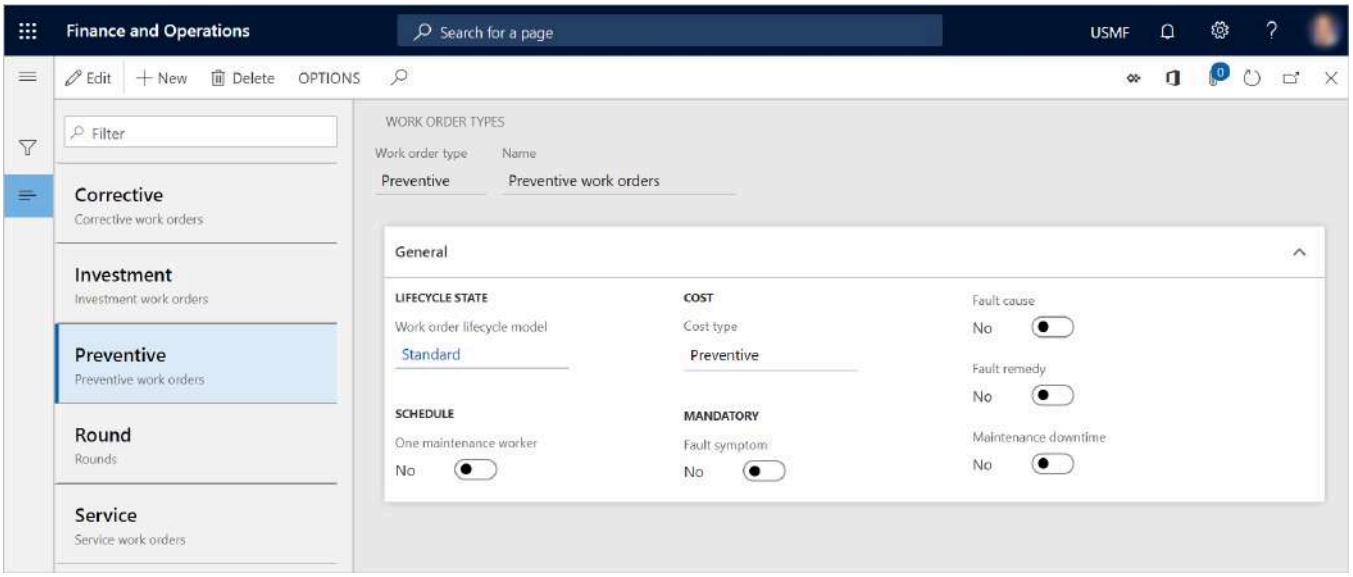
A work order type defines an affiliation with a work order lifecycle model. A work order lifecycle model defines the work order lifecycle states that can be set on a work order. (Examples of work order lifecycle states include **Created**, **In Process**, and **Finished**.)

For more information about work order lifecycle states and project stages, see Work order lifecycle states.

1. Select **Asset management** > **Setup** > **Work orders** > **Work order types**.
2. Select **New** to create a work order type.
3. In the **Work order type** field, enter an ID for the work order type.
4. In the **Name** field, enter a name.
5. In the **Work order lifecycle model** field, select a lifecycle model.
6. Set the **One maintenance worker** option to **Yes** if all work order jobs that are related to a work order of this type should be scheduled to the same maintenance worker.
7. In the **Cost type** field, select **Corrective**, **Preventive**, or **Investment**, as appropriate. All work order jobs on a work order must have the same cost type.
8. In the **Mandatory** section, set the relevant options to **Yes** to specify which fault-related or maintenance downtime–related information is added to a work order of this type.

   > **Note**
   >
   > The options in the **Mandatory** section are related to the options on the **Validate** FastTab of the **Work order lifecycle states** page (**Asset management** > **Setup** > **Work orders** > **Lifecycle states**).

9. Select **Save**.



Last updated on 05/07/2025

Work order project setup

06/17/2025

In the **Asset management** module, a project relation is required for every work order job. The project that is associated with a work order job lets you track costs on various projects that are related to Asset Management, such as internal maintenance projects, service management projects, and investment projects.

Project setup for a work order job

When you create a work order job on a work order, the project setup in the **Project management and accounting** module and the work order project setup in the **Asset management** module determine how projects can be used for cost control on the asset that is selected on that work order job. This section describes the following parts of the project setup that is used for a work order: the parent project (project ID), project type, project activities, and financial dimensions:

- When you create a work order job on a work order, a unique project ID and a related project activity are automatically created for it. However, if several work order jobs on a work order include the same asset, the same project ID is used for them. In other words, one project ID is created for every asset on a work order.
  - The parent project (project ID) for a work order job is found in the parent project setup. (For more information about the parent project setup, see the next section.) For example, if you associate a customer or a functional location with a specific parent project, the parent project is used every time that you create work orders for that customer or that functional location. If you don't relate a specific project ID to, for example, a functional location, the next relevant parent project in the work order project setup is used.
  - A project type is required for every project ID. The project type is found in the setup of the project group setup. (For more information about the project group setup, see the next section.) If no match is found in the project group setup, the project group setup on the parent project is used.
  - The setup for requiring project activities on forecasts and journals is copied from the parent project to the work order project. If the **Hour**, **Expense**, and **Item** options are set to *Yes* for the project that is used as a parent project, a project activity is mandatory on forecasts and journals. (To access these options, select **Project management and accounting** > **Projects** > **All Projects**, and then select the project that is used as a parent project. The options are in the **Require activity on journals** section on the **Setup** FastTab.)

- Financial dimensions are copied from the asset and merged with the parent project.

The next section explains how to set up parent projects and project groups. Parent project and parent groups are used to control work orders. They're also used for reporting about work orders.

Set up work order projects

Before you start to create work orders, you must set up work order projects. The **Work order project setup** page (**Asset management** > **Setup** > **Work orders** > **Project setup**) contains two tabs: **Parent project** and **Project group**.

On the **Parent project** tab, you can set up project relations that can be used if no project is set up on the asset that is selected on the work order job. A parent project setup isn't required if your company uses asset projects. It's relevant only if you want to use work order projects instead of asset projects. In that case, you must set up at least one parent project.

On the **Project group** tab, you can set up project groups that can be associated with work order types, asset types, and assets.

Project groups can be used to create specific categories (groups) that are used for cost control. For example, by creating project groups for specific asset types or work order types, you can do detailed tracking of maintenance costs by type.

Project groups aren't mandatory. If you don't set up project groups, the parent project is used to determine the project group, and a child project is created from the parent project's project group.

The setup allows for complete integration with the **Project management and accounting** module. Therefore, you can track the costs that are related to work orders in the related projects. The following procedure describes the setup for work order projects.

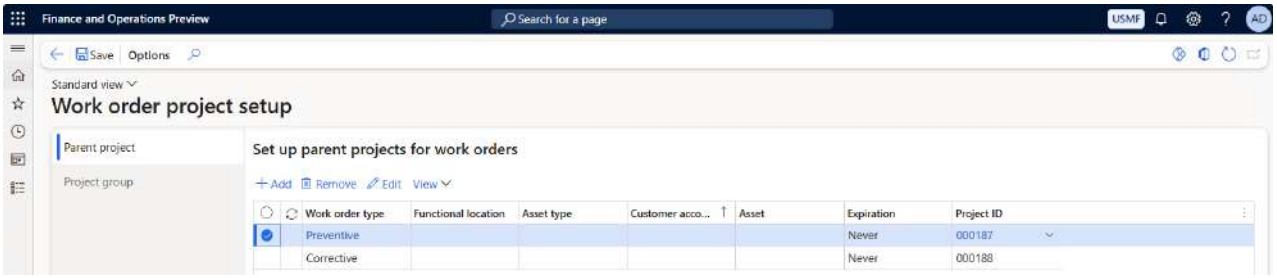
1. Select **Asset management** > **Setup** > **Work orders** > **Project setup**.
2. On the **Parent project** tab, select **Add**.
3. In the **Work order type**, **Functional location**, **Asset type**, and **Asset** fields, select values as you require. For each line that you add, you can set just one field or multiple fields. The number of fields that you set determines the combination that is used when a project ID is selected in Asset Management.

   If you select a functional location, the related child locations are automatically included. If you select an asset, you can create more work order project setup lines for the same asset, but you can select different projects for that asset.

4. In the **Project ID** field, select the project that should be related to the setup that you created in step 3.

5. If the project setup should be valid for only a limited period, select an end date in the **Expiration** field. Otherwise, select *Never*.

   By default, the start date is the date when you add the work order project to the page. It's controlled by the **Effective** field, which is hidden by default. To show the **Effective** field, select **View** > **All**. You can then use the **Effective** field together with the **Expiration** field to set up a limited period of validity for the work order project.



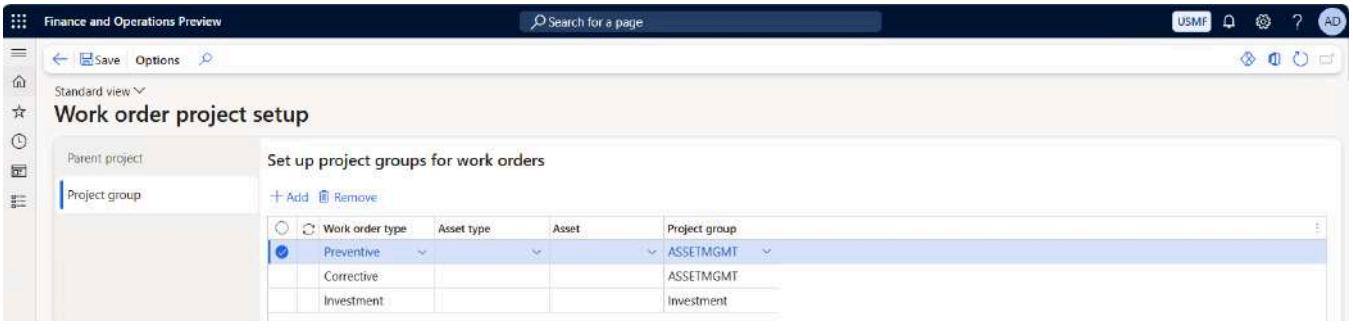
6. On the **Project group** tab, select **Add**.

7. In the **Work order type** field, select a work order type.

8. If you want the project group association to be more specific, select an asset type in the **Asset type** field or an asset in the **Asset** field.

9. In the **Project group** field, select the project group that should be related to the work order type. For example, a work order type that is named *Preventive maintenance* might be associated with a project group that is named *Prev Maint* or *Internal*. Alternatively, an *Investment* work order type that is used for work orders that are related to investments and fixed assets might be associated with a project group that is named *Invest* or *Investment*.

10. Select **Save**.



> **Note**

Every time that a work order line is created, Asset Management searches for a project group that should be related to the work order job project. The search is based on the setup that is described in this article. Every project group has a related project type. Project groups that have the *Time and material* or *Fixed-price* project type are valid only for assets that are related to a customer account.

For parent projects and project groups, when the system selects the available work order project or project group, the selection is based on the records that you created by using the preceding procedure. Asset Management goes through records that are related to the work order project to check for a possible match. It always checks the most specific combination first. In other words, for the work order parent project, Asset Management first checks for a possible match for the **Asset** field. If no match is found, it checks for a match for the **Asset type** field. If no match is found, it checks for a match for the **Functional location** field, and so on. As you can see in the layout of the **Work order project setup** page, this behavior means that, to find the most specific combination, Asset Management checks each record from right to left for a match. If no match is found, the default record where only a project ID is selected is used. The process for finding the related project group is similar. Asset Management first checks for a possible match for the **Asset** field, then the **Asset type** field, and then the **Work order type** field. If no match is found, the default record where only a project group is selected is used.

Service level and description

> **Note**
>
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When you create a work order, you might want to define the service levels for it and add a general description to it. You can create work order service levels on the **Work order service levels** page and descriptions on the **Work order description** page.

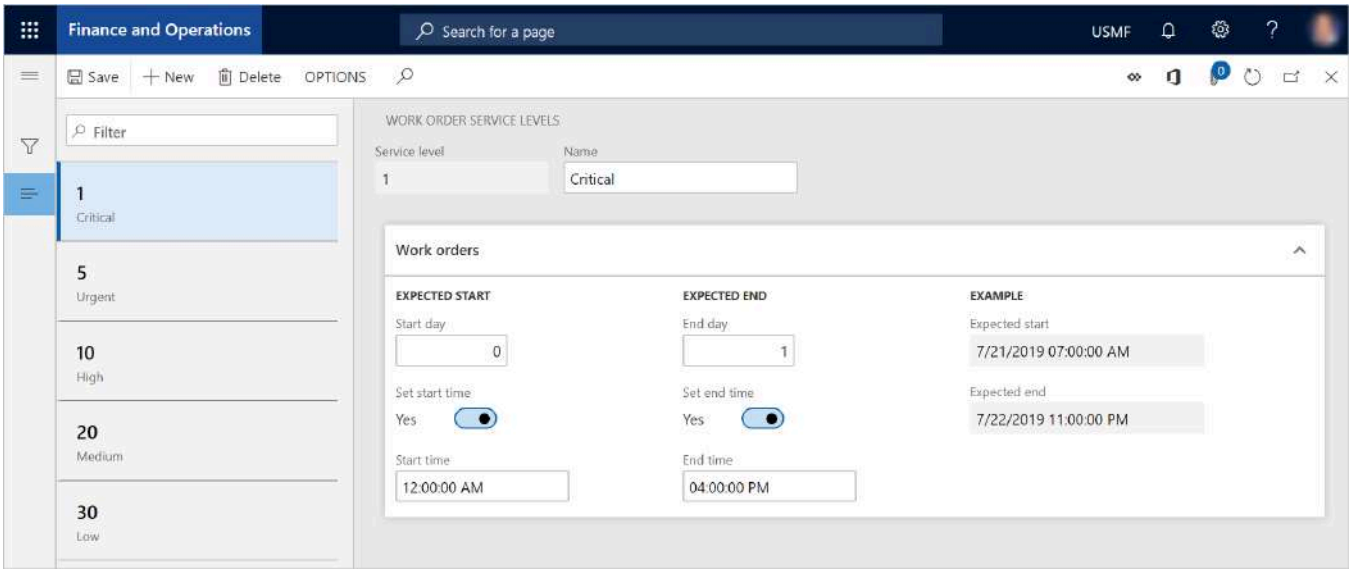
Create a service level

1. Select **Asset management** > **Setup** > **Work orders** > **Service level**.
2. Select **New**.
3. In the **Service level** field, enter the service level (for example, a number).
4. In the **Name** field, enter a name.

   On the **Work orders** FastTab, you can define expected start and end dates and times. The fields on this FastTab define the period that work orders should be started and ended during. They are used for work orders that are manually created and work orders that are created based on maintenance requests.
5. In the **Start day** field, enter a number of days to define the period that the work order should start during. The number of days is calculated from the creation date of the work order. For example, if the work order should start when it's created, enter **0**. If the work order should start within one week after it's created, enter **7**.
6. To set a start time for the work order, in addition to a start date, set the **Set start time** option to **Yes**. Then enter the start time in the **Start time** field. If you set the option to **No**, the current time of day is used.
7. In the **End day** field, enter a number of days to define the period that the work order should end during. The number of days is calculated from the start date of the work order. For example, if the work order should end within one week of its start date, enter **7**.
8. To set an end time for the work order, in addition to an end date, set the **Set end time** option to **Yes**. Then enter the end time in the **End time** field. If you set the option to **No**,

the current time of day is used.

9. Select **Save**.



Create a description

1. Select **Asset management** > **Setup** > **Work orders** > **Descriptions**.
2. Select **New**.
3. In the **Description** field, enter the description.
4. Select **Save**.

Last updated on 05/07/2025

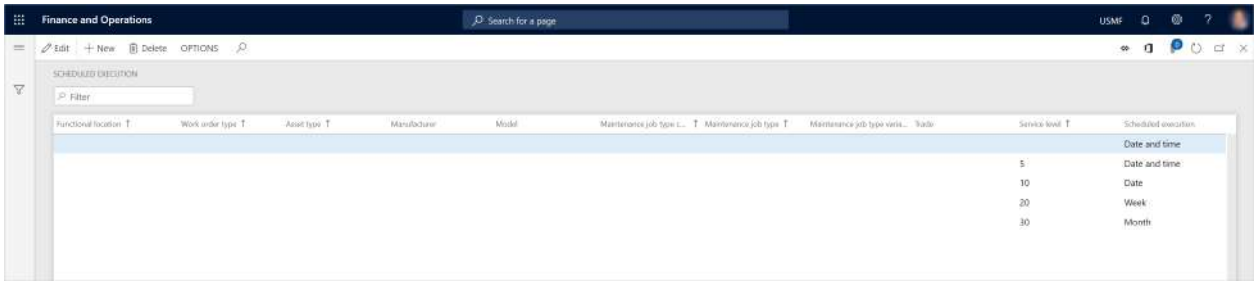
Scheduled execution

Article • 06/03/2022

You can use work order service levels to set up scheduled execution. (For more information about work order service levels, see Service level and description.) Scheduled execution provides flexibility in work planning for maintenance workers, because you can set up more detailed or less detailed requirements for the interval that a work order should be completed during. For example, a maintenance worker who completes a job faster than expected in a production facility might be able to move on to another nearby job that was planned for the current week but not necessarily for the current day. This approach allows for optimization of worker planning and job completion.

Scheduled execution setup, which is related to work orders, can be generic or specific. You can set up generic lines that aren't limited to specific work order types, asset types, and so on. Alternatively, you can create scheduled execution lines that apply to a specific work order type, asset type, maintenance job type, and so on.

1. Select **Asset management** > **Setup** > **Work orders** > **Scheduled execution**.
2. Select **New** to create a scheduled execution line.
3. In the **Functional location**, **Work order type**, **Asset type**, **Manufacturer**, **Model**, **Maintenance job type category**, **Maintenance job type**, **Maintenance job type variant**, and **Trade** fields, select values as you require.
4. In the **Service level** field, select a work order service level. If you leave this field blank, you make the most generic type of scheduled execution line. For an example of a generic line, see the first record in the illustration that follows. That line enables all work orders that have no work order service level to be scheduled for a specific date and time.
5. In the **Scheduled execution** field, select the time interval.
6. Select **Save**.



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Fault management

> **Note**
>
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In Asset Management, you can use the fault designer to set up fault symptoms, fault areas, and fault types on asset types. In this way, you can manage faults that are detected on assets. Additionally, fault causes and suggestions for fixing faults can be registered on a work order.

The process for fault registration and management consists of these steps.

1. Create a list of fault symptoms, fault areas, and fault types that might occur on your asset types.
2. In the fault designer, set up symptoms, fault areas, and fault types.

Here are some examples to help you understand the difference between fault symptoms, fault areas, and fault types.

Fault symptoms:

- Unbalanced voltages
- Short circuit
- Noise
- Leak
- Vibrations

Fault areas:

- Electrical
- Mechanical
- Hydraulic
- Pneumatic

Fault types:

- Faulty main stator winding
- Faulty diode
- Dirty windings
- Defective generator

- Defective sensor

Create fault symptoms

Follow these steps to create a list of symptoms that can be used in the fault designer.

1. Select **Asset management** > **Setup** > **Fault** > **Fault symptoms**.
2. Select **New** to create a record.
3. In the **Fault symptom** field, enter a name for the fault symptom.
4. In the **Description** field, enter a description.
5. Select **Save**.

Create fault areas

Follow these steps to create a list of areas or locations that can be used in the fault designer.

1. Select **Asset management** > **Setup** > **Fault** > **Fault areas**.
2. Select **New** to create a record.
3. In the **Fault area** field, enter a name for the fault area.
4. In the **Description** field, enter a description.
5. Select **Save**.

Create fault types

Follow these steps to create a list of fault types that can be used in the fault designer.

1. Select **Asset management** > **Setup** > **Fault** > **Fault types**.
2. Select **New** to create a record.
3. In the **Fault type** field, enter a name for the fault type.
4. In the **Description** field, enter a description.
5. Select **Save**.

Set up the fault designer

In the fault designer, you set up fault data on asset types.

1. Select **Asset management** > **Setup** > **Fault** > **Fault designer**.

2. In the left pane, select the type of asset to set up a fault record for.

3. On the **Fault symptom** FastTab, select **Add line**, and then, in the **Fault symptom** field, select a fault symptom.

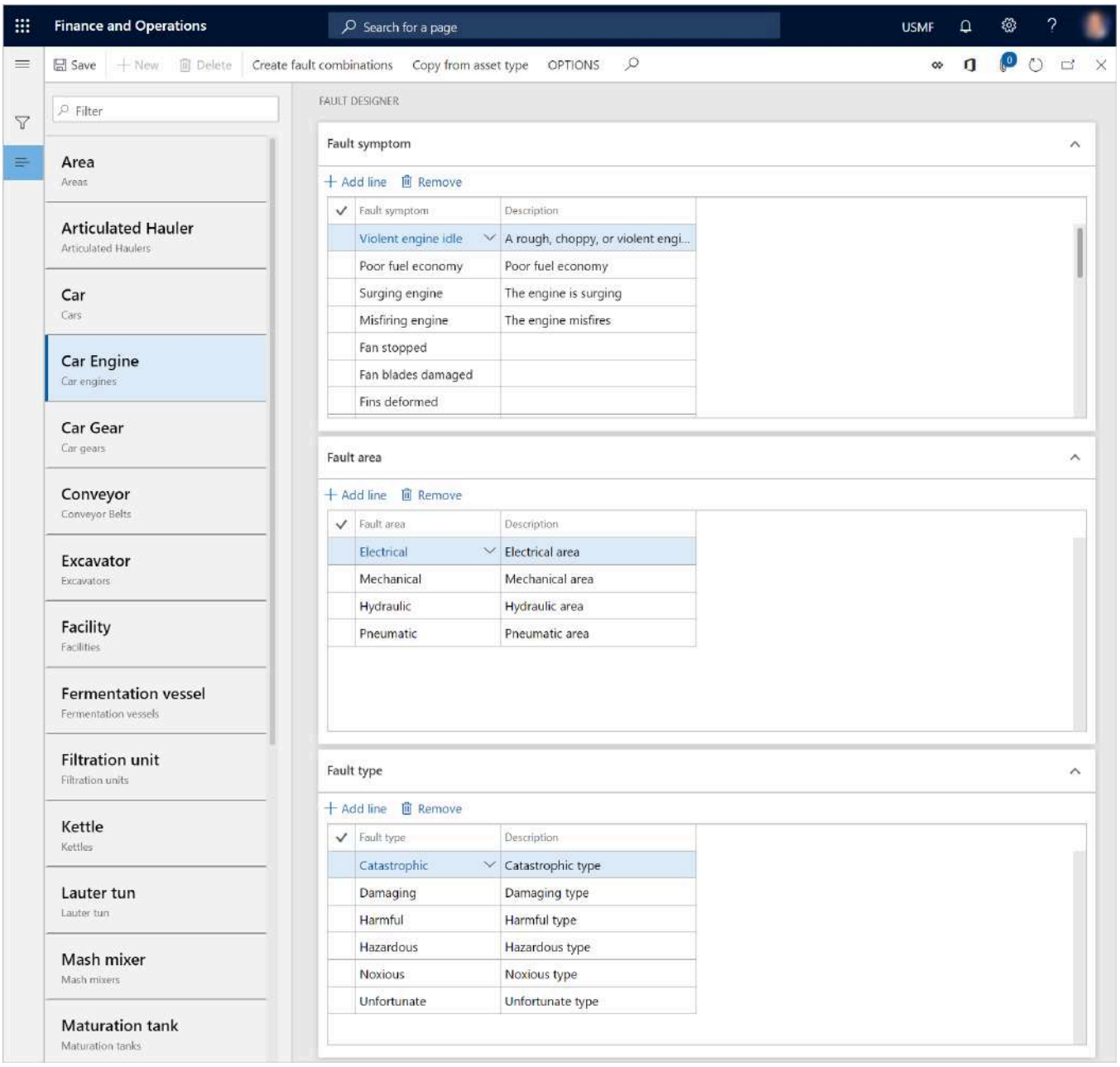
4. On the **Fault area** FastTab, select **Add line**, and then, in the **Fault area** field select a fault area.

5. On the **Fault type** FastTab, select **Add line**, and then, in the **Fault type** field, select a fault type.

6. To quickly create combinations of all existing fault symptoms, areas, and types for the selected asset type, select **Create fault combinations**. This function is useful if you've added many fault symptoms, areas, and types. It's easier to delete the lines for any combinations that aren't relevant to the asset type than to create new lines manually.

   > ⓘ **Note**
   >
   > To copy the setup of fault symptoms, areas, and types from one asset type to the selected asset type, select **Copy from asset type**.

7. Select **Save** to save your changes.



Create fault causes

Follow these steps to create a list of known fault causes that can be added to a work order or a maintenance request.

1. Select **Asset management** > **Setup** > **Fault** > **Fault causes**.
2. Select **New** to create a record.
3. In the **Fault cause** field, enter a name for the fault cause.
4. In the **Description** field, enter a description.
5. Select **Save**.

Create fault remedies

Follow these steps to create a list of suggestions for remedy and repair that can be added to a work order or a maintenance request.

1. Select **Asset management** > **Setup** > **Fault** > **Fault remedies**.
2. Select **New** to create a record.
3. In the **Fault remedy** field, enter a name for the fault remedy.
4. In the **Description** field, enter a description.
5. Select **Save**.

> **Note**
>
> You can change the names of your fault symptoms, areas, types, causes, and remedies as you require. The name changes are automatically reflected in the related fault registrations.

Last updated on 05/07/2025

Introduction to work orders

Work orders are used to manage maintenance jobs, provide required information for them, and register consumption on them. Each work order can contain one or more work order jobs, and one or more assets can be connected to each work order. Each work order job defines a maintenance job that is scheduled on the asset.

Work orders can be created in the following ways:

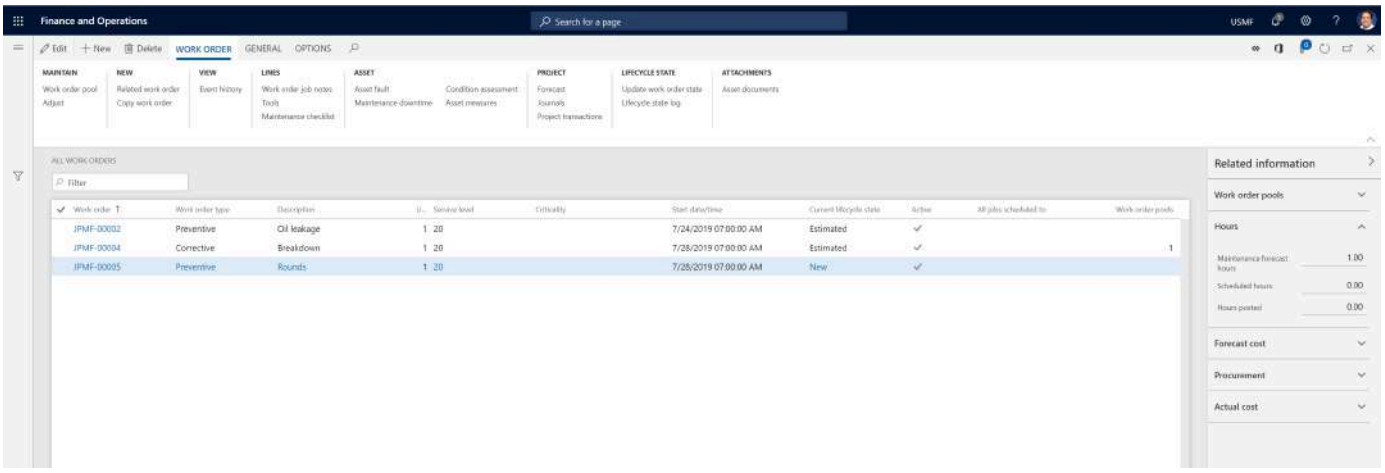
- For calendar-based maintenance plans where the "Auto create" setting is turned on, automatically by using Schedule maintenance plans.
- For maintenance rounds where the "Auto create" setting is turned on, automatically by using Schedule maintenance rounds.
- For preventive maintenance jobs or maintenance requests, from Maintenance schedule.
- Manually
- From the **All maintenance requests**, **Active maintenance requests**, or **My functional location maintenance requests** page.

> **Note**
>
> Work order jobs that are related to the same asset are related to the same project ID.

All work orders

Select **Asset management** > **Work orders** > **All work orders** to open the **All work orders** list page. This page shows all work orders and some of the information that is related to each.

The illustration below shows an example of the **All work orders** list page.



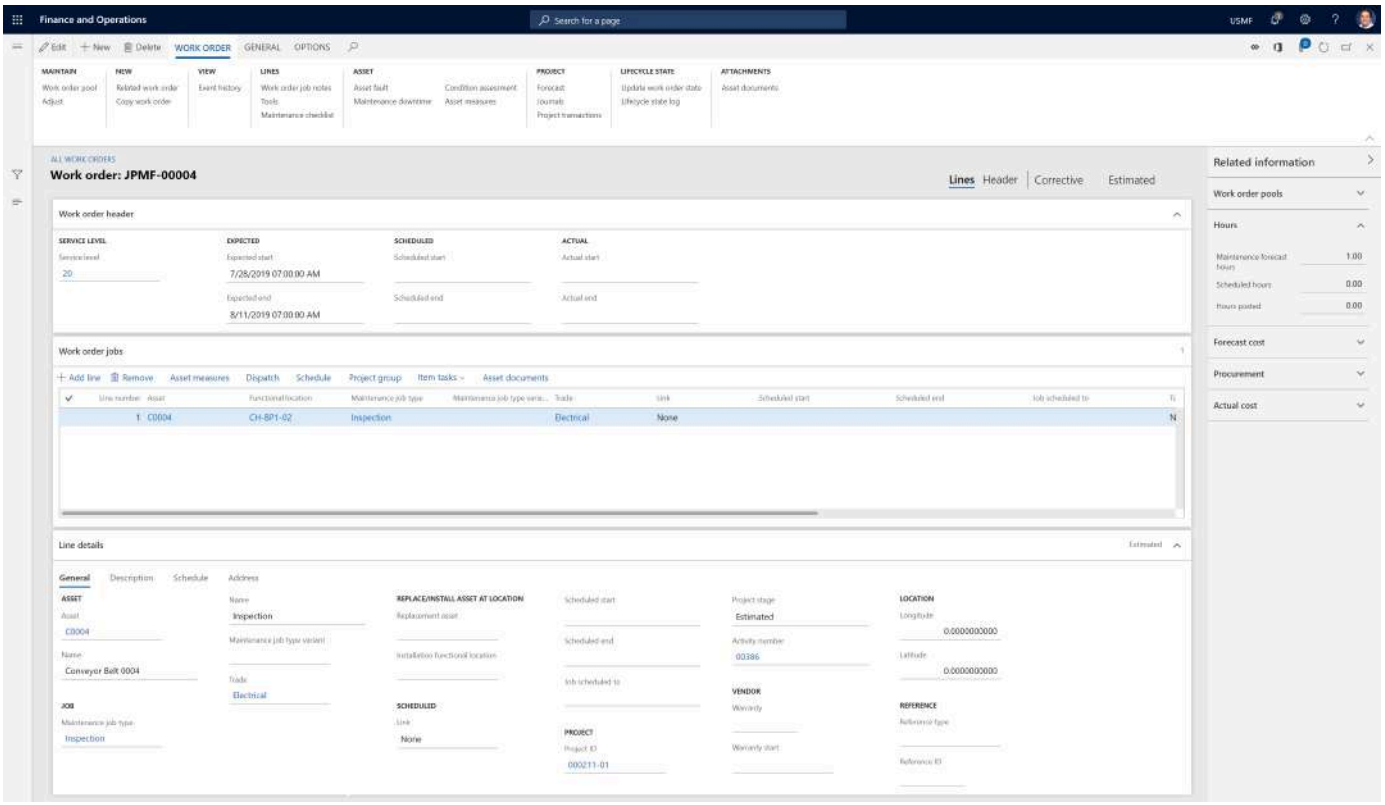
To view a list of only active work orders, select **Asset management** > **Work orders** > **Active work orders**.

To view a list of work order jobs that contain assets that are installed on functional locations that you're related to as a worker, select **Asset management** > **Work orders** > **My functional location work order maintenance jobs**. (The relation between workers and functional locations is set up on the **Workers** page. Learn more in Maintenance workers and worker groups.)

Here are some ways that you can use the **All work orders** page:

- In the grid view, select a link in the **Work order** column to show the details view for the selected record. You can then select **Edit** to open the record for editing.
- In the details view, you view detailed information that is related to the work order.
- In the details view, select the **Lines** tab to view details of the work order job, or select the **Header** tab to view details of the work order.
- Expand the **Related information** pane on the right side of the page to view additional information that is related to the selected work order.

The illustration below shows an example of the **All work orders** details view.



The buttons on the Action Pane are organized on tabs. The following table briefly describes the buttons that are related to Asset Management:

Expand table

Button name	Description
Edit	Edit the selected work order.
New	Create new work order.
Delete	Delete the selected work order.
Work order pool	Add the selected work order to a work order pool, or remove it from work order pool.
Adjust	Adjust information about expected start and end, service level, responsible maintenance worker, or responsible maintenance worker group on selected work orders.
Related work order	Create a new work order related to the selected work order job. This is useful if you want to register primary and secondary work orders.
Copy work order	Create a new work order that is based on an existing work order.
Event history	View the registration history for the work order.
Work order maintenance job notes	Create a description on a work order, or insert notes or remarks about it. First, select **Add timestamp** to add your user name and a timestamp to the note. Notes are shown on the **Description** tab on the **Line details** FastTab of the **Work order** page.

Button name	Description
Tools	Create a list of required tools on a work order. Tools are set up as resources in **Organization administration** > **Resources** > **Resources**.
Maintenance checklist	View the checklist for the asset that is connected to the work order.
Asset fault	View or register fault information on an asset. This information is used for fault management.
Maintenance downtime	Specify maintenance downtime for a work order.
Condition assessment	Register condition assessment measurements on a work order.
Asset counters	Create or view counter registrations on the asset.
Forecast	View or create forecasts on a work order.
Journals	View or create work order journals. Journal lines can be copied from forecasts.
Project transactions	View all posted transactions that are related to work orders created for the asset.
Update Work order state	Update the work order lifecycle state. You can add remarks, which are visible in the lifecycle state log. Remarks require Supply Chain Management version 10.0.46 or higher with *Remarks can be associated with each change in lifecycle states* enabled in Feature management (As of Supply Chain Management version 10.0.49, this feature is turned on by default).
Lifecycle state log	View a log that shows the lifecycle states of the selected work order. The log includes any remarks that were added by users during manual lifecycle changes. Remarks require Supply Chain Management version 10.0.46 or higher with *Remarks can be associated with each change in lifecycle states* enabled in Feature management (As of Supply Chain Management version 10.0.49, this feature is turned on by default).
Asset documents	View the list of documents attached to assets that are related to a work order. These documents are set up in **Asset management** > **Setup** > **Asset documents**.
Schedule	Schedule the selected work orders.
Dispatch	Schedule the selected work order for one worker.
Delete schedule	Delete the scheduling for the selected work order.
Scheduled work order maintenance	Open the **Scheduled work order maintenance jobs** list page.

Button name	Description
jobs	
Work order purchase requisition	Open the **Work order purchase requisition** list page.
Work order purchase	Open the **Work order purchase** list page.
Cost control	Compare budget costs and actual costs on the work order.
Hour control	Compare forecasted hours and actual hours on the work order.
Work order report	Print a work order report.
Work order consumption	Print a consumption report.

The buttons in the **Project** group on the **Work order** tab of the Action Pane are related to the functionality for forecasts, journals, and invoicing in the **Project management and accounting** module.

> **Note**
>
> To include forecasts that have been created on a work order when you run master scheduling, use the forecast model that is selected on the **Asset management parameters** page.

Last updated on 07/27/2026

Manually created work orders

08/13/2025

You can create work orders manually in two places:

- On the **All work orders** or **Active work orders** page.
- On the **All maintenance requests** or **Active maintenance requests** or **My functional location maintenance requests** page.

Create work order

1. Go to **Asset management** > **Work orders** > **All work orders** or **Active work orders**.
2. Select **New**.
3. In the **Create work order** dialog, select a work order type in the **Work order type** field.
4. If necessary, select a **Description**.
5. In the **Asset** field, select the asset.

   > ⓘ Note
   >
   > When you select an asset, three tabs might be available in the **Asset** drop-down:
   >
   > - **Active assets** – This tab contains a list of all assets that have an "Active" asset lifecycle state.
   > - **Asset view** – This tab displays a tree view of functional locations and the assets installed on them.
   > - **My assets** – This tab contains assets that are related to the functional locations that you (the worker who is signed in to the system) might be allocated to. (For information about the setup, see Maintenance workers and worker groups.) If no functional locations are set up on a worker in Maintenance workers and worker groups, the **My assets** tab isn't available.

6. In the **Maintenance job type** field, select a maintenance job type for the work order.
7. If necessary, select **Maintenance job type variant** and **Trade**.
8. If necessary, you can change the work order service level in the **Service level** field.
9. Select **Expected start** and **Expected end** dates in the related fields.

10. Select **OK** to create the work order.

11. On the **All work orders** list page, you can edit the work order as you require.

Note the following points:

- In the details view on the **All work orders** list page, you can add several assets to a work order by adding lines on the **Work order maintenance jobs** FastTab. On an asset, you can select only the maintenance job types that are defined on the asset type that is selected on the asset.
- If you change an asset service level or an asset criticality after you've used the asset on a work order, the service level or criticality on the work order isn't updated accordingly. For more information about service levels and criticalities, see Asset service levels and Asset criticality types.
- Criticality on a work order is recalculated every time a work order job is added to or deleted from the work order.
- In the **All work orders** details view > **Header** tab > **Schedule** FastTab, in the **Responsible group** or **Responsible** field, you can select a responsible maintenance worker group or a responsible maintenance worker. These settings can be changed while the work order is active. For example, they can be changed when the work order lifecycle state changes. The automatic selection that is made during work order creation is based on the setup on the **Responsible maintenance workers** page. If you add or remove work order jobs after you've created a work order, and if the **Responsible group** and **Responsible** fields are blank when you update the work order, Asset Management tries to find a responsible maintenance worker group or a responsible maintenance worker for a possible match on the setup page. If the **Responsible group** or **Responsible** field is already set when you update the work order, no changes are made. For more information about responsible maintenance workers and worker groups, see Responsible maintenance workers.
- From the Maintenance status page, you can do a calculation to get an overview of the workload for incoming and completed work orders.
- In the details view of the **All work orders** page, on the **Line details** FastTab, you can use the **Latitude** and **Longitude** fields to add geographic coordinates for the asset that is selected on the work order job.

Create related work order

You can create a work order that is related to an existing work order. This capability is useful if, for example, you want to work with primary and secondary work orders. A new work order is based on a work order job from an existing work order.

1. Select **Asset management** > **Work orders** > **All work orders** or **Active work orders**.

2. Select the work order to create a related work order for.

3. On the Action Pane, on the **Work order** tab, in the **New** group, select **Related work order**.

4. In the **Create related work order** dialog, in the **Work order job** field, select the work order job to create a related work order for.

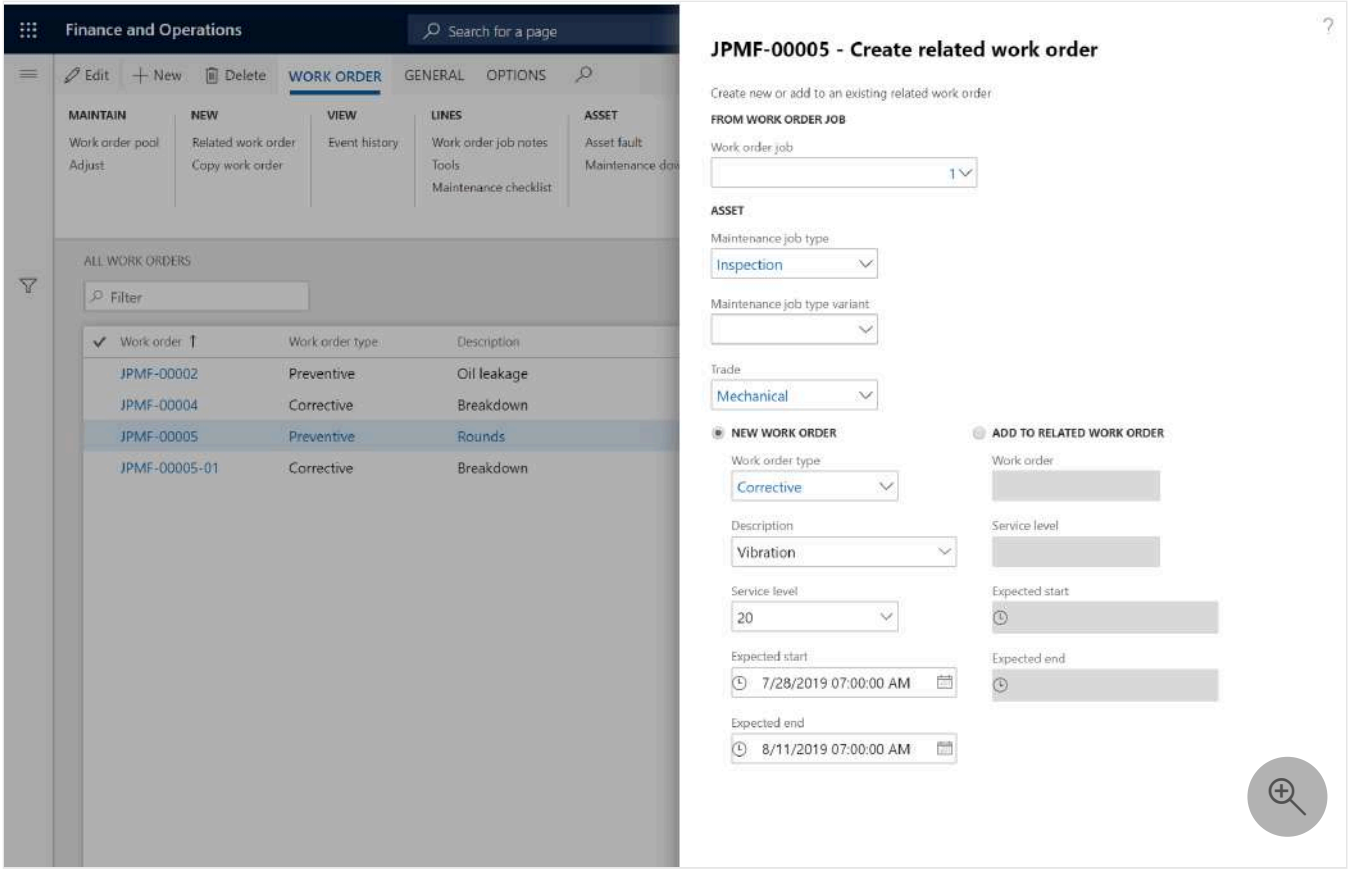
5. Select a maintenance job type in the **Maintenance job type** field.

6. Select a related maintenance job type variant and trade in the **Maintenance job type variant** and **Trade** fields, as you require.

7. If this work order is the first related work order that has been created for the selected work order, follow these steps:
   a. Select the **New work order** option.
   b. In the **Work order type** field, select a work order type.
   c. In the **Description**, enter a description.
   d. In the **Service level** field, change the work order service level as you require.
   e. In the **Expected start** and **Expected end** fields, select the expected start and end dates.
   f. Select **OK**. The new related work order is shown on the **All work orders** list page.

8. If the work order that you're creating this related work order for already has related work orders, follow these steps to add a new work order job to an existing related work order:
   a. Select the **Add to related work order** option.
   b. In the **Work order** field, select the related work order to add a new work order job to.
   c. In the **Service level** field, change the work order service level as you require.
   d. In the **Expected start** and **Expected end** fields, change the expected start and end dates as you require.
   e. Select **OK**. The work order job is added to the existing related work order.

The following illustration shows an example of the **Create related work order** dialog.



> **Note**
>
> If you've set up a related work order mask in **Asset management parameters** > **Work orders** tab > **Related work order mask** field, work order IDs are created according to the mask setup. If no related work order mask is set up, the next available work order ID is used for related work orders.

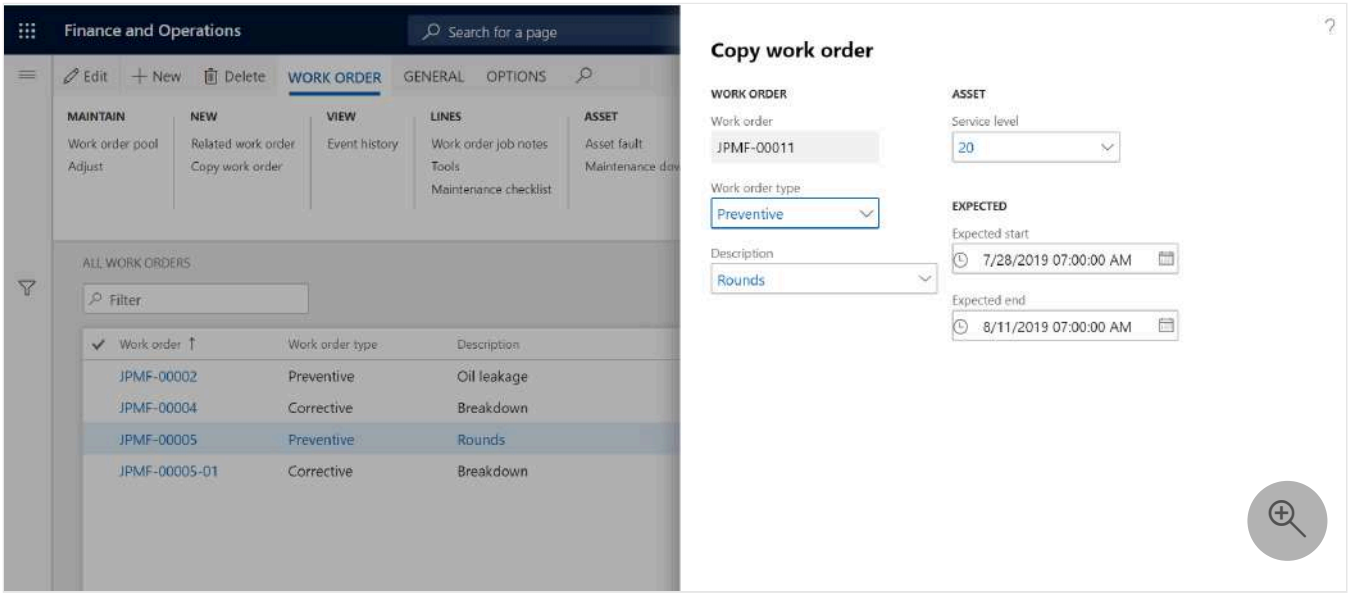
Copy a work order

You can quickly create a new work order from an existing work order. This way of working with work orders differs from the creation of work orders based on maintenance plans. It's useful if, for example, a work order contains many work order jobs, and the various jobs should be completed on different assets at regular intervals.

1. Select **Asset management** > **Work orders** > **All work orders** or **Active work orders**.
2. Select the work order to copy content from.
3. On the Action Pane, open the **Work order** tab and, from the **New** group, select **Copy work order**.
4. The work order setup from the selected work order is shown. You can edit some of the fields as you require.
5. Select **OK** to create the new work order.
6. On the **All work orders** list page, you can edit the work order as you require.

> **Note**
>
> When the new work order is created, some information is copied directly from the existing work order. Information about forecasts, tools, maintenance checklists, functional location, addresses, and scheduling isn't copied. Instead, it's initialized from the current setup in Asset Management. Therefore, if that information was changed between the time when the first work order was created and the time when you made a copy of the work order, the changes are included in the new work order. Examples include changes to forecasts and updates to maintenance checklists.

The following illustration shows an example of the **Copy work order** dialog.

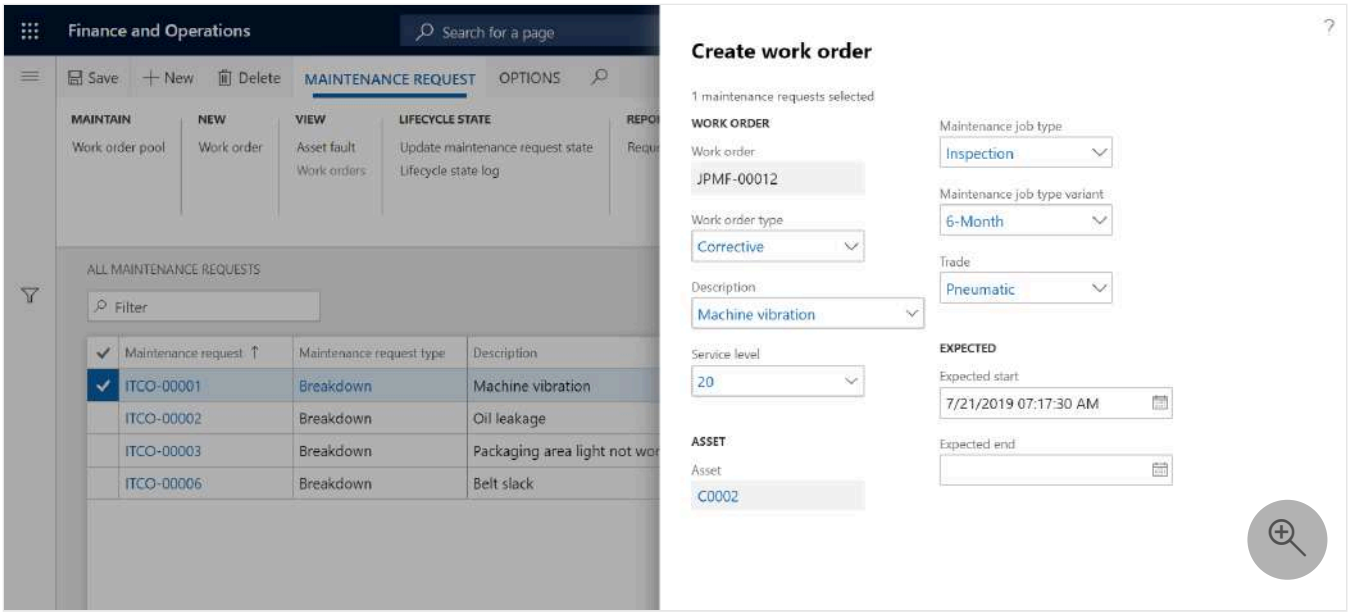


Create a work order based on a maintenance request

1. Select **Asset management** > **Maintenance requests** > **All maintenance requests** or **Active maintenance requests**.
2. Select the maintenance request to create a work order for, and select **Edit**.
3. On the Action Pane, open the **Maintenance request** tab and, from the **New** group, select **Work order**.
4. In the **Work order** dialog, set the fields. If a maintenance job type has been selected in the maintenance request, you can select a different maintenance job type when you create the work order, as you require.
5. Select **OK**. A message notifies you that the work order has been created.
6. To view the work orders that are connected to a maintenance request, on the **All maintenance requests** or **Active maintenance requests** list page, select the maintenance

request. Then, on the Action Pane, on the **Maintenance request** tab, in the **View** group, select **Work orders**.

The following illustration shows an example of the **Create work order** dialog.



> **Note**
>
> If you want work orders to be created automatically, you can schedule maintenance plan jobs, or you can set up "Auto create" maintenance plans or maintenance rounds on an asset. Work orders that are created from maintenance requests on the **All maintenance schedule** list page have the maintenance job types that are selected on the maintenance requests.

Maintenance forecasts

Article • 01/29/2025

When you create a work order, you create work order jobs that have related assets and maintenance job types. When you select a maintenance job type that contains maintenance forecasts, the forecasts are automatically copied to the work order.

You might be able to add forecast lines to a work order or delete them from a work order. The setup of the work order lifecycle state, the related project type, and the stage rules that are related to the project type determine whether you can add or edit forecast lines. For more information about work order lifecycle states and related project stages, see Forecasts, work orders, and projects.

1. Go to **Asset management** > **Work orders** > **All work orders** or **Active work orders**.
2. Select the work order in the list, and then, on the Action Pane, on the **Work order** tab, in the **Project** group, select **Forecast**. The **Work order maintenance forecast** page shows forecast lines from the maintenance job type that is selected on the work order job.

Add an hours forecast to a work order

1. On the **Work order maintenance forecast** page, select the work order job to add a forecast to.
2. On the **Hours** FastTab, select **Add** to create a new line.
3. In the **Category** field, select a category.
4. In the **Hours** field, enter the number of forecasted hours.
5. In the **Line property** field, select the type of charge that should be used on the line.

Add an items forecast to a work order

There are three ways to add items to a work order maintenance forecast. You can create lines for items (spare parts) that aren't included on the spare parts list or the asset bill of materials (BOM), you can select spare parts from the approved spare parts list, or you can select items from the asset BOM.

- On the **Work order maintenance forecast** page, select the work order job to add a forecast to.
- On the **Items** FastTab, add items to the maintenance forecast by using the appropriate method.

To create a line for a spare part that isn't on the spare parts list or the asset BOM, follow these steps:

1. Select **Add**.
2. In the **Item number** field, select the item.
3. In the **Sales quantity** field, enter the quantity.
4. In the **Unit** field, select the unit of measure for the quantity.
5. In the **Cost price** and **Currency** fields, enter appropriate values.
6. In the **Line property** field, select a line property.
7. To change the list of dimensions that is shown on the item lines, select **Inventory** > **Display dimensions**, select the dimensions, and then set the **Save setup** option to *Yes*.

To add a spare part from an approved spare parts list, follow these steps:

1. Select **Add spare parts**.
2. Select the spare part, and edit the related information as you require.
3. Select **OK**.

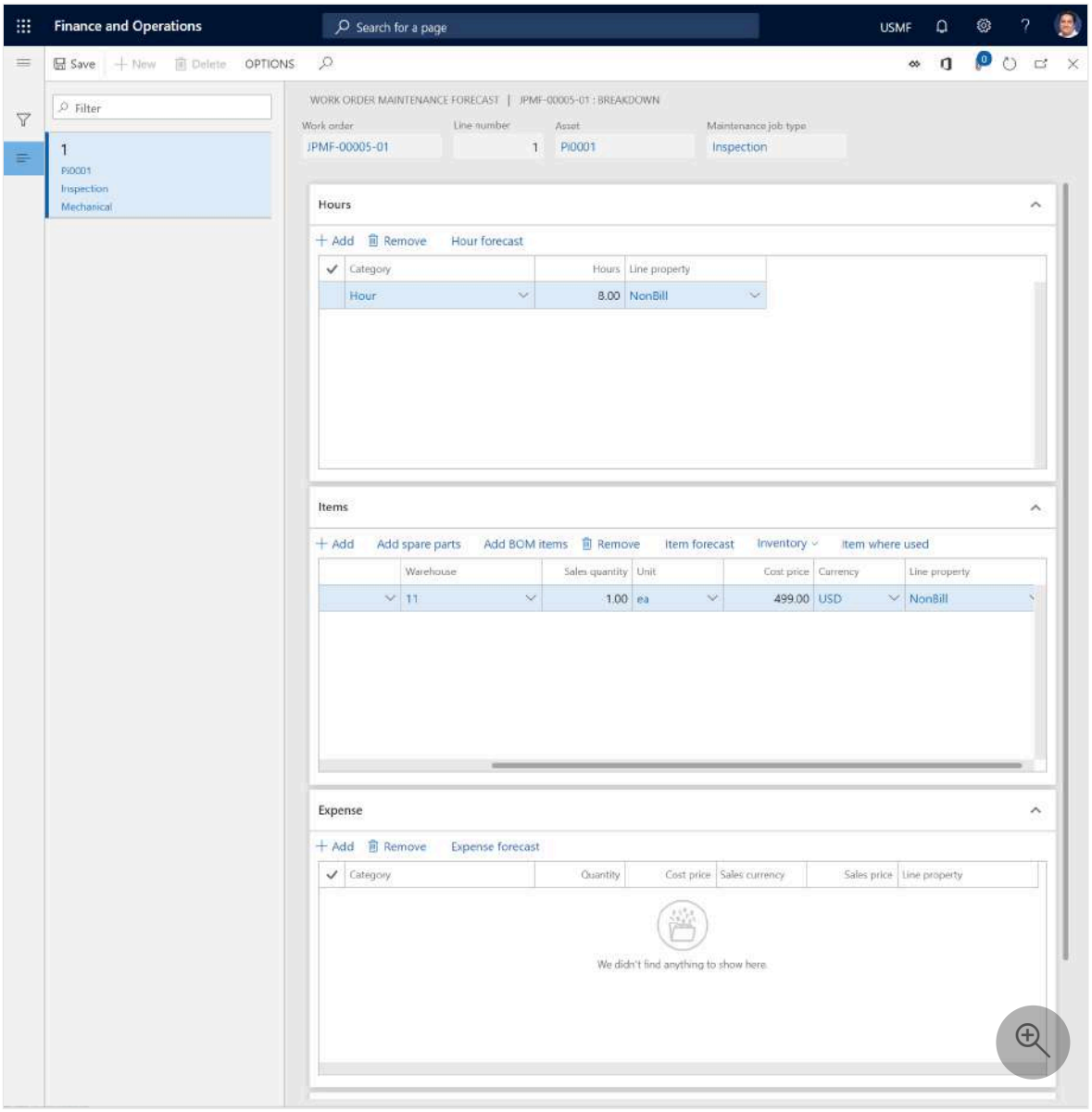
To add an item from the asset BOM, follow these steps:

1. Select **Add BOM items**.
2. Select the item, and edit the related information as you require.
3. Select **OK**.

To get an overview that shows where the item on the selected line is used in relation to assets, maintenance job type defaults, spare parts, and work orders in Asset Management, select **Item where used**. For more information about this overview, see Item where used.

Add an expense forecast to a work order

1. On the **Work order maintenance forecast** page, select the work order job to add a forecast to.



2. On the **Expense** FastTab, select **Add** to create a line.
3. In the **Category** field, select a category.
4. In the **Quantity** field, enter the quantity.
5. In the **Cost price**, **Sales currency**, and **Sales price** fields, enter appropriate values.
6. In the **Line property** field, select the type of charge that should be used on the line.

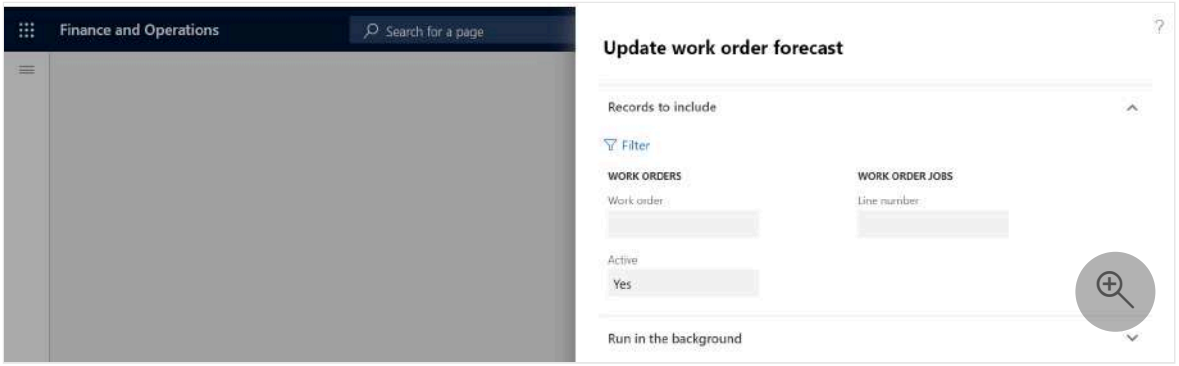
Note

The **Maintenance forecast totals** FastTab shows an overview of the number of lines that were created for the selected work order job and the work order on each FastTab. It also shows the total forecasted work hours for the work order job and the work order.

Automatic update of work order forecasts

If hour costs, item costs, and expenses are updated in other modules, work order forecasts in Asset Management can automatically be updated to reflect those changes. This capability helps guarantee that the latest cost prices are always used in your work order forecasts. You can also do similar updates for maintenance job type forecasts.

1. Go to **Asset management** > **Periodic** > **Forecast** > **Update work order forecast**.



2. In the **Update work order forecast** dialog box, on the **Records to include** FastTab, you can select **Filter** to add selections that are related to specific work orders or work order jobs, as you require.

3. On the **Run in the background** FastTab, you can set up the automatic update as a batch job, as you require.

4. Select **OK** to start the forecast update.

Feedback

Was this page helpful? Yes No

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Procurement

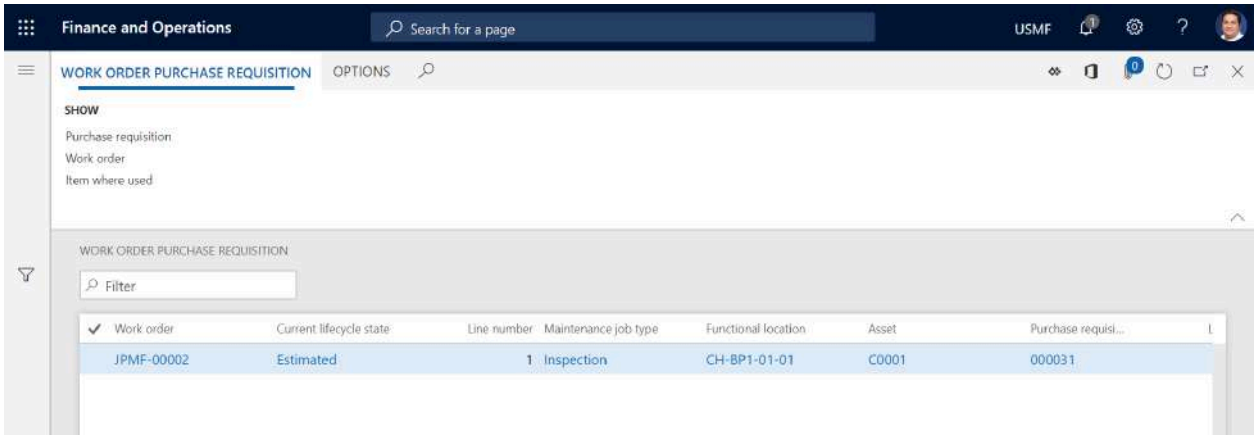
Article • 06/15/2022

In Asset Management, you can get an overview of purchase requisitions and purchase orders that are related to work orders. You can also create a purchase order or a purchase requisition from a work order.

The **Work order purchase requisition** list page (**Asset management** > **Procurement** > **Work order purchase requisition**) shows a list of purchase requisitions that are related to work orders. When you select a work order job on this page, you can use the buttons in the **Show** group on the **Work order purchase requisition** Action Pane tab to perform various actions:

- To open the related purchase requisition, select **Purchase requisition**.
- To open the related work order, select **Work order**.
- To get an overview that shows where the item on the selected line is used in relation to assets, maintenance job type defaults, spare parts, and work orders in Asset Management, select **Item where used**. For more information about this overview, see Item where used.

The illustration below shows an example of the **Work order purchase requisition** list page.

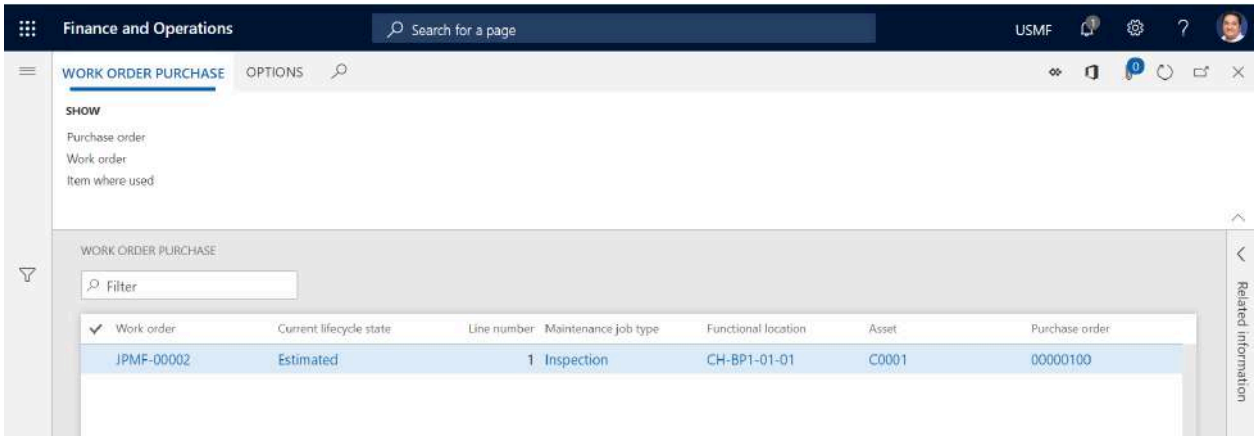


The **Work order purchase** list page (**Asset management** > **Procurement** > **Work order purchase**) shows a list of purchase orders that are related to work orders. When you select a work order job on this page, you can use the buttons in the **Show** group on the **Work order purchase** tab of the Action Pane to perform various actions:

- To open the related purchase order, select **Purchase order**.
- To open the related work order, select **Work order**.
- To get an overview that shows where the item on the selected line is used in relation to assets, maintenance job type defaults, spare parts, and work orders in

Asset Management, select **Item where used**. For more information about this overview, see Item where used.

The illustration below shows an example of the **Work order purchase** list page.



On both the **Work order purchase** list page and the **Work order purchase requisition** list page, a symbol that is related to delivery date control appears on the right side of each line. If the symbol is an exclamation point in a red circle, delivery of the related purchase order or purchase requisition might be delayed.

For a purchase order, the date that is related to the purchase order line is used to calculate a possible delay. To view this date, on the **Purchase order** page, select the purchase order line. The date is shown in the **Confirmed delivery date** field on the **Setup** tab of the **Line details** FastTab. If the **Confirmed delivery date** field isn't set, the date in the **Delivery date** field on the **Purchase order header** FastTab is used for the calculation. One of those dates is compared to the available date on the work order or work order job, in the following order:

1. Actual start date on the work order
2. Scheduled start date on the related work order job
3. Scheduled start date on the work order
4. Expected start date on the work order

For a purchase requisition, the date in the **Requested date** field on the **Purchase requisition header** FastTab of the **Purchase requisitions** page is used to calculate a possible delay. The date in that field is compared to the available date on the work order or work order job, in the same order that is used for a purchase order.

Create a purchase order from a work order

On the **All work orders** list page, you can select a work order job, and then create a related purchase order or a related purchase requisition. In this way, you help guarantee that project relations exist between the purchase order or purchase requisition and the work order.

1. Select **Asset management** > **Work orders** > **All work orders** or **Active work orders**.
2. Select the work order to create a purchase order for, and then select **Edit**.
3. On the **Work order maintenance jobs** FastTab, select the work order job to create the purchase order for.
4. Select **Item tasks** > **Purchase order from work order job**.
5. On the **Project purchase orders** list page, click **New**.
6. Create the purchase order.

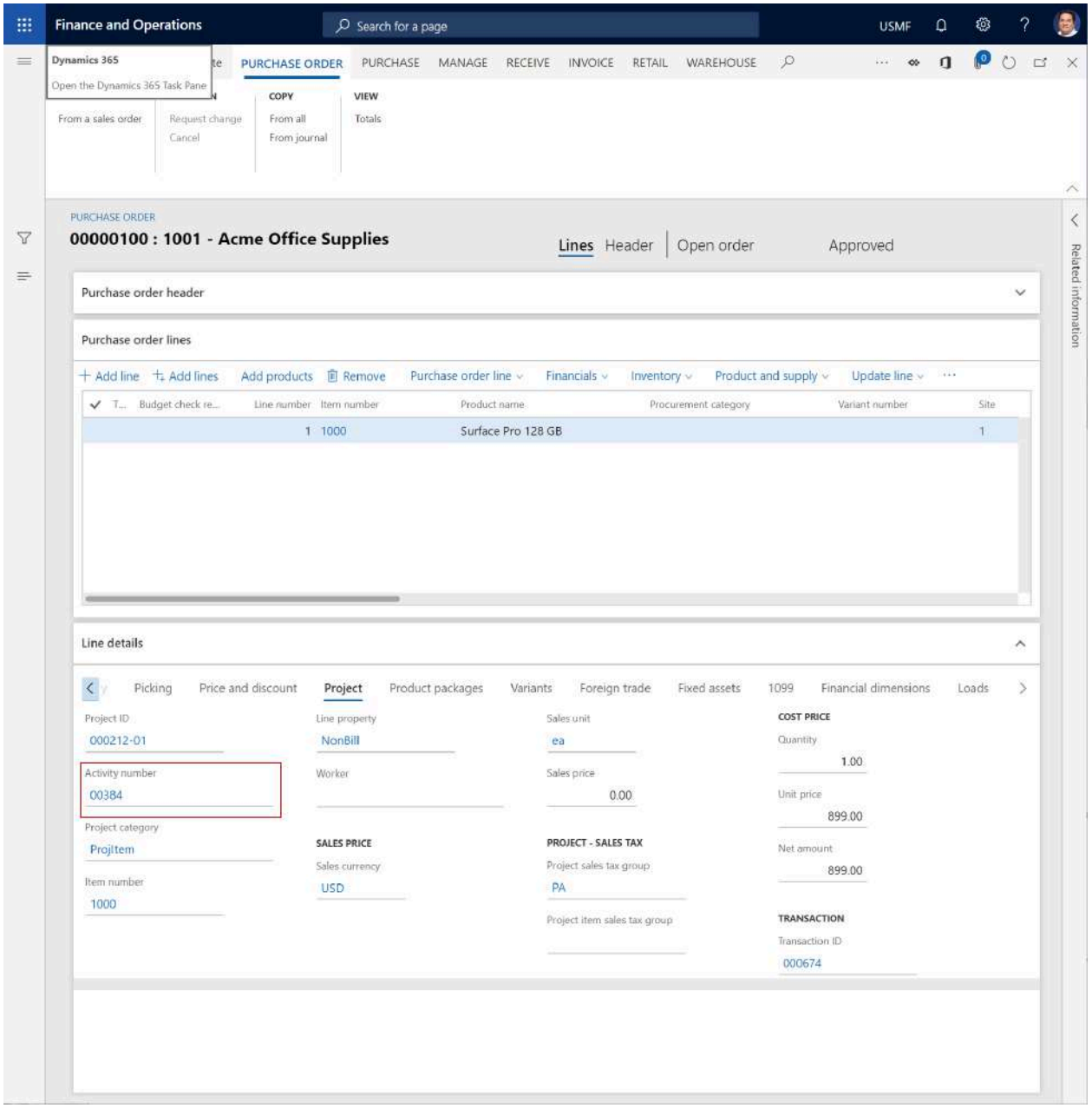
> ⓘ **Note**
>
> To create a related purchase requisition, follow the same steps. However, select **Item tasks** > **Purchase requisition from work order job** in step 4.

Project relation between work order and purchase order or purchase requisition

A purchase order line or purchase requisition line is related to a work order job via the work order project and the related project activity number. When you create a purchase order or purchase requisition from a work order job, the related project activity number is mandatory. If all the work order jobs in the related work order have the same maintenance job type, the project activity number is automatically entered on the purchase order or purchase requisition. If the work order jobs have different maintenance job types, you must manually enter the project activity number on the purchase order or purchase requisition.

To view or enter the activity number that is related to a purchase order line, on the **Work order purchase** list page, select the purchase order record, and then, in the **Purchase order** column, select the link for the purchase order. You can find the **Activity number** field on the **Project** tab of the **Line details** FastTab.

The illustration below shows an example of the **Purchase order** page, with focus on the **Activity number**.



Likewise, to view or enter the activity number that is related to a work order purchase requisition line, on the **Work order purchase requisition** list page, select the purchase requisition record, and then, in the **Purchase requisition** column, select the link for the purchase requisition. You can find the **Activity number** field on the **Project** tab of the **Line details** FastTab.

Feedback

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Material availability check for work orders

Supervisors can do a material availability check for work orders. This process is relevant for organizations that want to confirm that all the required spare parts are available before workers start maintenance work. For example, a supervisor is responsible for all corrective maintenance at a manufacturing site. As a standard procedure, the supervisor checks whether all the materials that are required for each work order are available. After availability is confirmed, the supervisor releases the materials to maintenance workers for processing.

Prerequisites

Before you can use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.37 or later.
- The feature that's named *Material availability check on maintenance work orders* must be turned on in feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default. As of version 10.0.45, it's mandatory and can't be turned off.

How the material availability check works

The material availability check for work orders uses a standard master planning run to calculate the inventory availability of all required materials. Material demand for work orders is generated by journal lines in the project item journal that's related to each work order. A journal line can be created, or it can be copied from the work order's item forecast. Learn more in Maintenance forecasts.

Supervisors use a dedicated page to run the material availability check for a selected range of work orders. From this page, supervisors can run the master plan and then view the material availability status for each of the selected work orders. The page also provides information about material availability of the individual spare parts that are required for the selected work orders.

Check material availability for work orders

Follow these steps to check material availability for work orders.

1. Go to **Asset management** > **Work orders** > **All work orders**.

2. Use the **Filter** field and the column header filters to select the range of work orders that you want to check material availability for. All work orders that are shown on the page will be checked.

3. On the Action Pane, on the **Work Order** tab, select **Material availability check**.

   The **Work order material availability check** page that appears is divided into two sections. The **Work orders** section in the upper part of the page shows the work orders. The **On-hand inventory** section in the lower part of the page shows journal lines that are related to the work order that's selected in the **Work orders** section.

4. In the **Work orders** section, select a work order. Confirm that there's a check mark in the **Journal lines created** column for the order. If there's no check mark, use one of the following methods to create the journal lines:

   - **Create the journal lines from a forecast:**

     a. Select **Forecast** on the toolbar.

     b. On the **Work order maintenance forecast** page, on the **Items** FastTab, confirm that there are item forecast lines for the order. If there are no lines, you can add them by selecting **Add**, **Add spare parts**, or **Add BOM items** on the toolbar.

     > **Tip**
     >
     > The system will automatically generate item forecast lines for work orders if you set up default forecasts on the **Maintenance job type defaults** page.

     c. Close the **Work order maintenance forecast** page to return to the **Work order material availability check** page.

     d. Select **Copy journals from forecast** on the toolbar.

     e. In the **Copy from forecast** dialog box, set the **Item** field to **Yes**.

     f. Select **OK**.

     g. Select **Refresh** at the top of the page.

     h. Confirm that there's now a check mark in the **Journal lines created** column.

   - **Create the journal from the Work order journals page:**

a. Select **Journals** on the toolbar.
b. On the **Work order journals** page, on the **Items** FastTab, add a line to the grid.
c. Close the **Work order journals** page to return to the **Work order material availability check** page.
d. Select **Refresh** at the top of the page.
e. Confirm that there's now a check mark in the **Journal lines created** column.

5. Run the material availability check by selecting **Material availability check** on the toolbar.

> ⓘ **Note**
>
> You can select multiple work orders for the material availability check.

6. In the **Material availability check** dialog box, set the following fields:

- **Run mode** – Select one of the following values to specify the type of check that you want to do:
  - *Master planning + Availability check* – First, master planning is run for the products that are related to the work orders. A material availability check for the orders is then run based on the master planning result.
  - *Availability check only* – A material availability check is run based on data from the latest version of the master planning.

  > 💡 **Tip**
  >
  > If you're currently just exploring the feature (for example, by using demo data), select *Master planning + Availability check*.

- **Plan** – Select the master plan to use when master planning is run. Typically, the master plan that represents the dynamic master plan should be used.

7. Select **OK** to run the check.

In the **Work orders** section of the **Work order material availability** page, a symbol in the **Ready to be released** column indicates the material availability status of each selected work order.

Expand table

Symbol	Description
⃠	The check couldn't be run for this work order because there's no material demand for it. Make sure that there's a journal line for the order.
✓	All materials for the work order are fully available.
⚠	Some or all of the materials for the work order aren't available.

8. In the **Work orders** section, select a work order. In the **On-hand inventory** section, a symbol in the **Material availability** column indicates the availability of each material for the selected work order.

Expand table

Symbol	Description
✓	The required material is fully available. The **Requested quantity** field shows the quantity that's requested by the work order. For material in this status, the quantity in the **Requested quantity** field equals the quantity in the **On-hand settled** field.
⚠	The required material is unavailable or only partially available. For material in this status, the quantity in the **Requested quantity** field isn't fully covered by the quantity in the **On-hand settled** field. Instead, it's covered by a combination of the quantities in the **On-hand settled**, **Planned order settled**, and/or **Order settled** fields.

> **Tip**
>
> To get more information about how the demand for the requested quantity is covered, select **Inventory** > **Explosion** on the toolbar in the **On-hand inventory** section.

The material availability check is now completed. If you want, you can now update the state of orders that are ready to be worked on by maintenance workers, as described in the next section.

Update a work order state

After you've confirmed that all materials are available for a work order, you can update the work order state. In this way, you can indicate that the order is now ready to be worked on by maintenance workers.

1. If the **Work order material availability check** page isn't already open, open it as described in the previous section.

2. If you need help finding the work orders that you're looking for, use the **Material availability** field to filter the work orders in the **Work orders** section. Select one of the following options:

    - *All* – Show all work orders.
    - *Upcoming* – Show only work orders where some or all materials aren't available.
    - *Available* – Show only work orders where all materials are available. These orders are ready to be worked on by maintenance workers.

    For this procedure, select *Available* to show the work orders where all materials are available. You can now change the work order state of these orders to indicate that they're ready to be worked on by maintenance workers.

3. In the **Work orders** section, select one or more orders that you want to update.

4. Select **Update work order state** on the toolbar.

5. In the **Update work order state** dialog box, select the next work order state for the selected work orders.

6. Select **OK** to confirm the change.

Aggregated material availability check

The *aggregated material availability check* feature aggregates supply and demand per item to provide a comprehensive view of potential material shortages. You can use this feature to view inventory levels and the total material requirements across selected work orders. This consolidated view makes it easy to identify and assess any deficiencies in your inventory. It helps you gain a comprehensive understanding of your material needs, so that you can take proactive measures to address any shortages.

Prerequisites for this feature

Before you can use the aggregated material availability check feature, your system must meet the following requirements:

- Your system must meet the prerequisites that are listed at the beginning of this article.
- You must be running Supply Chain Management version 10.0.44 or later.

- The feature named *Aggregated material availability check* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

View information about aggregated material availability

To gain an overview of the total material requirements by spare part and warehouse dimension across all selected work orders, follow these steps:

1. If the **Work order material availability check** page isn't already open, open it as described earlier in this article.
2. In the bottom section of the page, on the **Aggregated on-hand inventory** tab, review the information that is shown.

Last updated on 07/27/2026

Manual update of asset counters

Article • 06/15/2022

Counters are used to create registrations on an asset, such as registrations about the number of hours that the asset has been in operation or the quantity that has been produced.

The counter type that is selected for a counter might be set to inherit counter values. In other words, the **Inherit asset counter values** option is set to **Yes** on the **General** FastTab of the **Counters** page (**Asset management** > **Setup** > **Asset types** > **Counters**). In this case, when you create a new counter line of that type, every child asset that uses the same counter type is automatically updated.

On the **All assets** page, you create hours or quantity counter registrations on an asset, based on your readings on the asset.

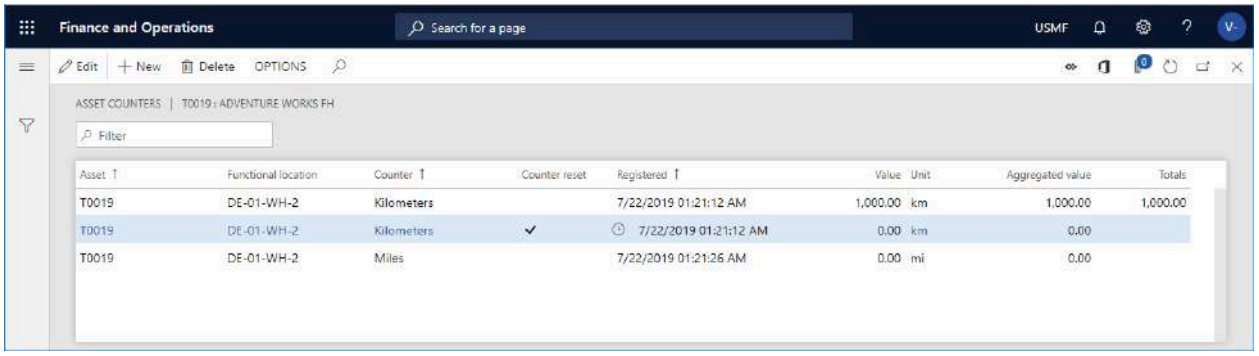
1. Select **Asset management** > **Assets** > **All assets**.
2. Select the asset, and then, on the Action Pane, on the **Asset** tab, in the **Preventive** group, select **Counters**. The **Asset counters** page shows a list of all previous counter registrations that have been made on the selected asset.
3. Select **New** to create a registration. The asset ID is automatically entered in the **Asset** field.
4. In the **Counter** field, select the relevant counter. Only counters that are related to the asset type selected on the asset are available for selection. The related unit is automatically entered in the **Unit** field.
5. In the **Registered** field, select the date and time for the counter registration.
6. In the **Value** field, enter the number since the last counter registration. Alternatively, in the **Aggregated value** field, enter the total count number.

Note the following points:

- If you physically install a new counter on an asset, you must register the change on the asset on the **Asset counters** page. Next, you must create two registration lines that have identical timestamps. The first line must be for the counter that you're replacing. On the line that is related to the new counter, select the **Counter reset** check box. In the **Totals** field, the total count number is the sum of the counter totals of all registered values on that counter type.

- If the **Counter reset** check box is selected on an asset using a maintenance plan that has a "Once from..." or "Once reached..." interval type, the counter is still active on the new counter line, because you create a separate counter line and start over with a new counter.

The illustration below shows an example of the **Asset counters** page.



On the **Asset counters** page (**Asset management** > **Inquiries** > **Assets** > **Asset counters**), you can make counter registrations on several assets at the same time, as you require.

> **Note**
>
> You can set up a range to define deviations in manual counter registrations. You can also specify the type of message that is shown if registrations are outside the defined range. For more information about how to set up counters, see Counters.

Feedback

Was this page helpful? Yes No

Provide product feedback

Automatic update of asset counters

Article • 06/15/2022

For information about manual registration of asset counters, see Manual update of asset counters. For information on how to set up asset counters, see Counters.

Counter values can also be automatically updated from production registrations, based on the production hours or production quantity (that is, the quantity that is produced). This update is done on the **Update asset counters** page. You can update one or several assets by setting one parameter, **From date**. This parameter specifies the start date for production registrations (production hours or production quantities). In other words, it specifies the date that counter values should be updated from.

All assets that are related to a resource, *and* that have asset counters that are set up to be updated based on the production hours or production quantity, will be included in an automatic update. New counter values will be created.

For counters that are based on the production quantity, the count includes both the good quantity and the error quantity that are registered. If the unit that is used for production quantity registration differs from the unit that is used for the counter, the quantity is converted so that it corresponds to the counter unit.

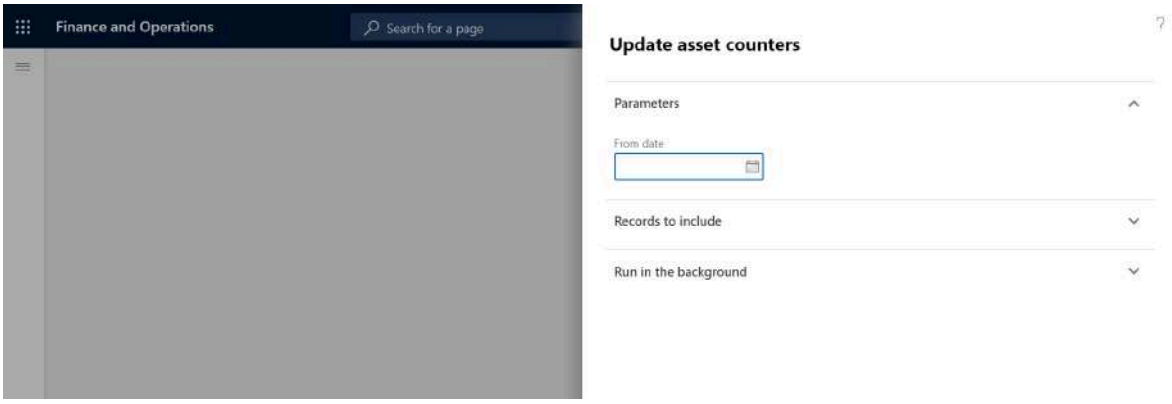
As mentioned above, automatic counters can be updated from production registrations. Therefore, the asset for which you want to automatically update counters must be related to a resource (machine). When produced quantities or production hours have been registered on the resource, you can update the related asset counters.

1. Select **Asset management** > **Periodic** > **Assets** > **Update asset counters**.
2. In the **From date** field, select the start date of the automatic update.

   > ⓘ **Note**
   >
   > The date in this field is the "work in progress" date from **Route transactions** (**Production control** > **Inquiries and reports** > **Production** > **Route transactions** > **Physical date** field).

3. On the **Records to include** FastTab, you can select specific assets, asset types, or resources for the automatic update. Select **Filter**, and make the relevant selections.
4. On the **Run in the background** FastTab, you can set up the automatic update as a batch job, as you require.

The illustration below shows an example of the **Update asset counters** dialog.

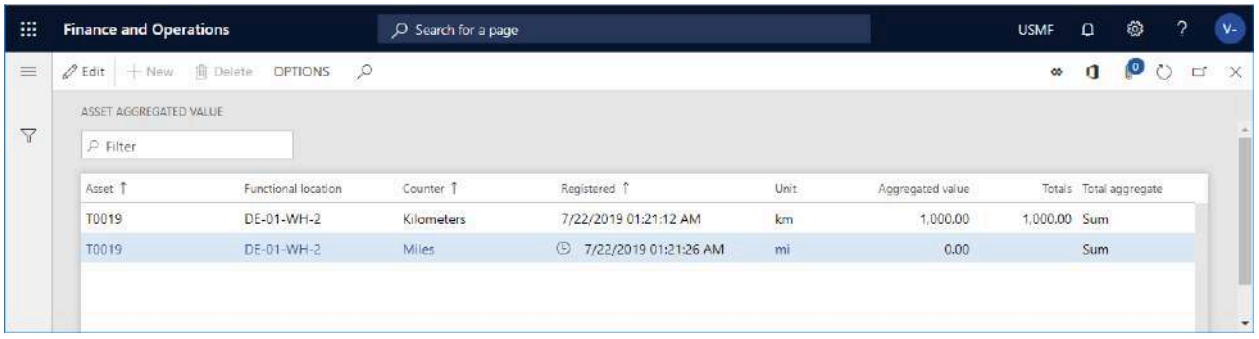


5. Select **OK**.

After the automatic asset counter update is done, you can view the counter registrations that are related to the asset on the **Asset counters** page. Select **Asset management** > **Assets** > **All assets**, select the asset, and then, on the Action Pane, on the **Asset** tab, in the **Preventive** group, select **Counters**.

On the **Asset aggregated value** page, you can get an overview of the latest registration that have been made on all counter types on all assets. Select **Asset management** > **Inquiries** > **Assets** > **Asset aggregated value**. This page resembles the **Asset counters** page, but you can't add or edit registrations. It's for overview only.

The illustration below shows an example of the **Asset aggregated value** page.



Note the following points:

- You can still create manual counter value registrations for counter types that are automatically updated. Learn more in Manual update of asset counters.
- You can set up counters that are related to another counter. In this case, when a counter is updated, related counters are automatically updated at the same time. For more information about how to set up related counters, see Counters.

Feedback

Was this page helpful? Yes No

Provide product feedback

Maintenance checklists

Set up maintenance checklists on maintenance job types. Fill in maintenance checklists as part of the process of completing a work order. For more information about how to set up maintenance checklists on maintenance job types, see Maintenance job type categories and maintenance job types, maintenance job type variants, maintenance job trades, and maintenance checklists.

When you work with maintenance checklists on a work order, you can fill in the predefined maintenance checklists that are related to maintenance job types. You can also add more maintenance checklist lines manually or from a maintenance checklist template.

Fill in a maintenance checklist

1. Go to **Asset management** > **Work orders** > **All work orders** or **Active work orders**.
2. Select the work order. On the Action Pane, open the **Work order** tab and, in the **Lines** group, select **Maintenance checklist**.
3. The **Work order maintenance job checklist** page shows the checklists for all work order jobs. If the work order jobs have different maintenance job types, the maintenance checklists might differ for each work order job. Select a work order job to work with the related maintenance checklist. The **Line details** FastTab shows details of the selected maintenance checklist line.
4. Complete all the maintenance checklist lines, one at a time, in the order that they appear. You complete a maintenance checklist line by filling in the fields on the **Line details** FastTab. The information that is required to complete a line can vary, depending on the line type. For example, on a line of the *Text* type, you add a note that explains the result of your check. On a line of the *Measurement* type, you enter the counter value that you read on the equipment, and you can also add a note as you require. A maintenance checklist line of the *Header* type is used as a heading to group the maintenance checklist lines that appear below it. You don't have to fill in a header. As for all other types of maintenance checklist lines, you can add a note to a line of the *Header* type.
5. If instructions are related to a maintenance checklist line, the **Instructions** check box is selected. Read instructions for the selected maintenance checklist line in the **Instructions** field on the **Line details** FastTab.

6. When you complete a maintenance checklist line, select the **Checked** check box on that line to mark it as completed. To discard a maintenance checklist line because it isn't relevant to the work order job, select the **N/A** check box on the line. If the **Mandatory** check box is selected on a maintenance checklist line, you must select either the **Checked** check box or the **N/A** check box.

> ⓘ **Note**
>
> You can only update maintenance checklist registrations if the work order is in an Active lifecycle state.

Add a maintenance checklist line

Create maintenance checklists from the definition on the maintenance job type default. The system transfers these checklists to a work order job. Add maintenance checklist lines to a work order job as needed. Maintenance checklist lines that you manually add get the **Manual** reference.

1. On the **Work order maintenance job checklist** page, select the work order job to add a maintenance checklist for.

2. On the **Maintenance checklist lines** FastTab, select a maintenance checklist line. To insert a new line after the selected line, press the **Down arrow** key. The next number in the sequence is automatically entered in the **Line number** field. To insert a new line before the selected line, select **Add line**.

3. On the **Name** field, enter a name for the maintenance checklist line.

4. In the **Type** field, select a type for the maintenance checklist line. The **Line details** FastTab contains related fields for each maintenance checklist type.

    - *Text* – Use this type to add a maintenance checklist line that has text that describes what must be done. For example, use this type if you want a worker to check or inspect something, but you're not expecting a specific (measurable) result. After you select this type, on the **Lines details** FastTab, in the **Instructions** field, enter text that describes what must be done.
    - *Header* – Use this type as a heading to group the maintenance checklist lines that appear below it. This type is useful if you have several maintenance checklist lines that

you can divide into specific areas. After you select this type, in the **Name** field, enter a descriptive name.

- *Template* – This type isn't applicable when you manually add a maintenance checklist line on a work order job.
- *Variable* – Use this type to define a possible result in a range on the maintenance checklist line. For information about how to set up maintenance checklist variables, see Maintenance job type categories and maintenance job types, maintenance job type variants, maintenance job trades, and maintenance checklists. After you select this type, in the **Name** field, enter a name to describe the variable. On the **Line details** FastTab, in the **Variable** field, select the variable. In the **Instructions** field, enter text that describes what must be done.
- *Measurement* – Use this type to record a specific measurement on the maintenance checklist line. After you select this type, in the **Name** field, enter a name for the measurement. On the **Line details** FastTab, in the **Counter** and **Unit** fields, select appropriate values. In the **Instructions** field, enter text that describes what must be done.

5. When you finish manually adding maintenance checklist lines, fill in the lines as described in the previous section.

Add maintenance checklist lines from a template

In addition to the maintenance checklist lines that the maintenance job type provides, you can add maintenance checklist lines directly to a work order job from a maintenance checklist template. This capability is useful when a work order requires a standard set of tasks that the maintenance job type doesn't already include. Learn more about how to create maintenance checklist templates in Maintenance job type categories and maintenance job types, maintenance job type variants, maintenance job trades, and maintenance checklists.

1. On the **Work order maintenance job checklist** page, select the work order job to add maintenance checklist lines to.

2. On the **Maintenance checklist lines** FastTab toolbar, select **Add lines from template**.

3. In the dialog box, in the **Maintenance checklist template** field, select the template that contains the lines you want to add.

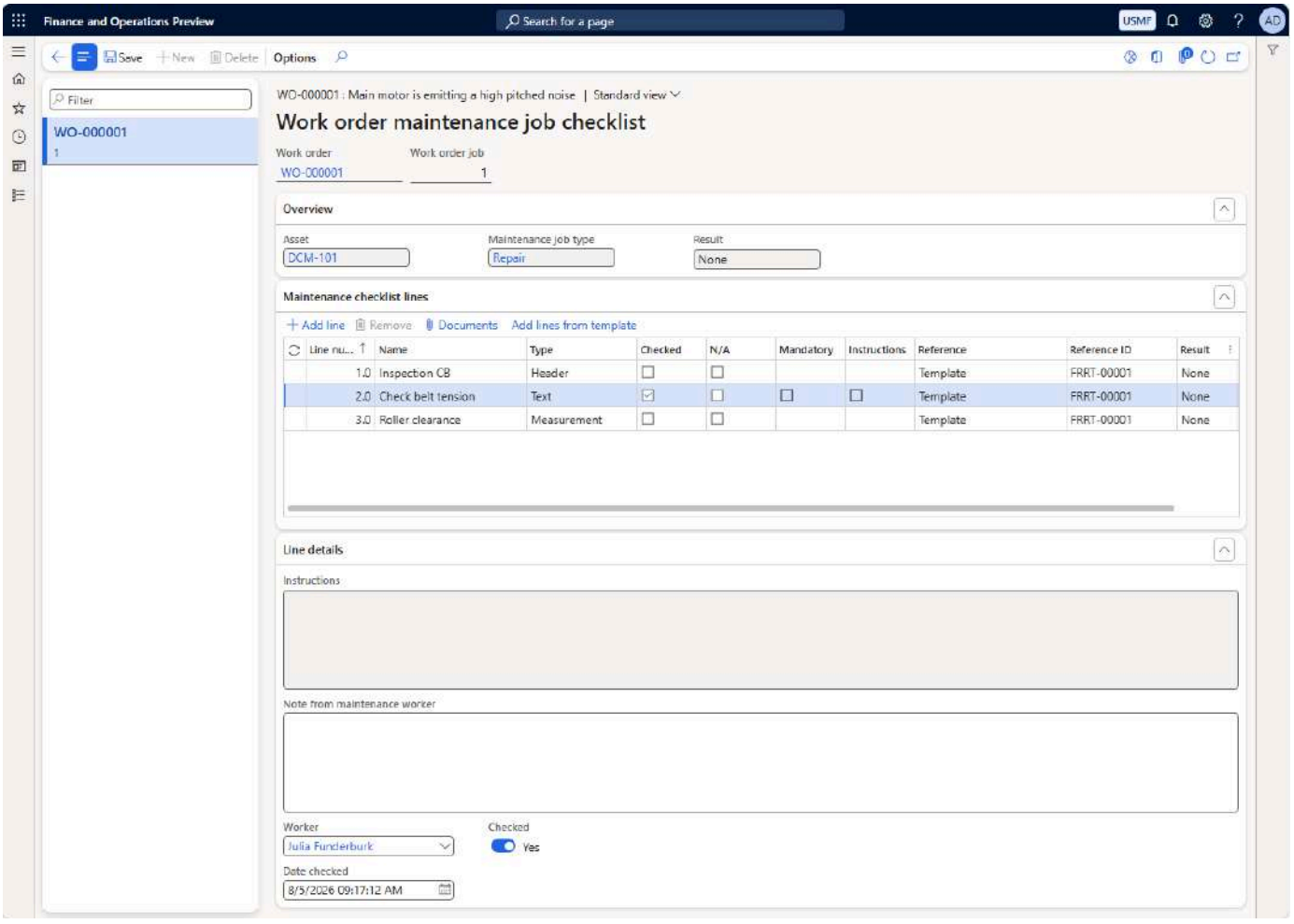
4. Select **OK**. The lines from the template are appended after any existing maintenance checklist lines on the work order job, and they're assigned a **Reference** of *Template*. A

message shows how many lines were added from the template.

You can add lines from more than one template, and you can combine template lines with manually added lines. The template governs maintenance checklist lines that you add from a template. As with lines that have the *Job type* reference, you can't delete template lines individually on the work order job. If a template line doesn't apply to the current work order job, select the **N/A** check box on the line to exclude it.

> **Note**
>
> On the **Work order maintenance job checklist** page, you can only delete maintenance checklist lines that you or other maintenance workers created manually, and that have the *Manual* reference. You can't delete maintenance checklist lines that have the *Job type* reference or the *Template* reference. If one of those lines doesn't apply to the work order job, select the **N/A** check box on the line to exclude it. When you select **N/A**, the line remains on the work order for traceability, but it's excluded from the mandatory maintenance checklist validation.

The following illustration shows an example of a maintenance checklist.



Last updated on 08/10/2026

Maintenance downtime for work orders

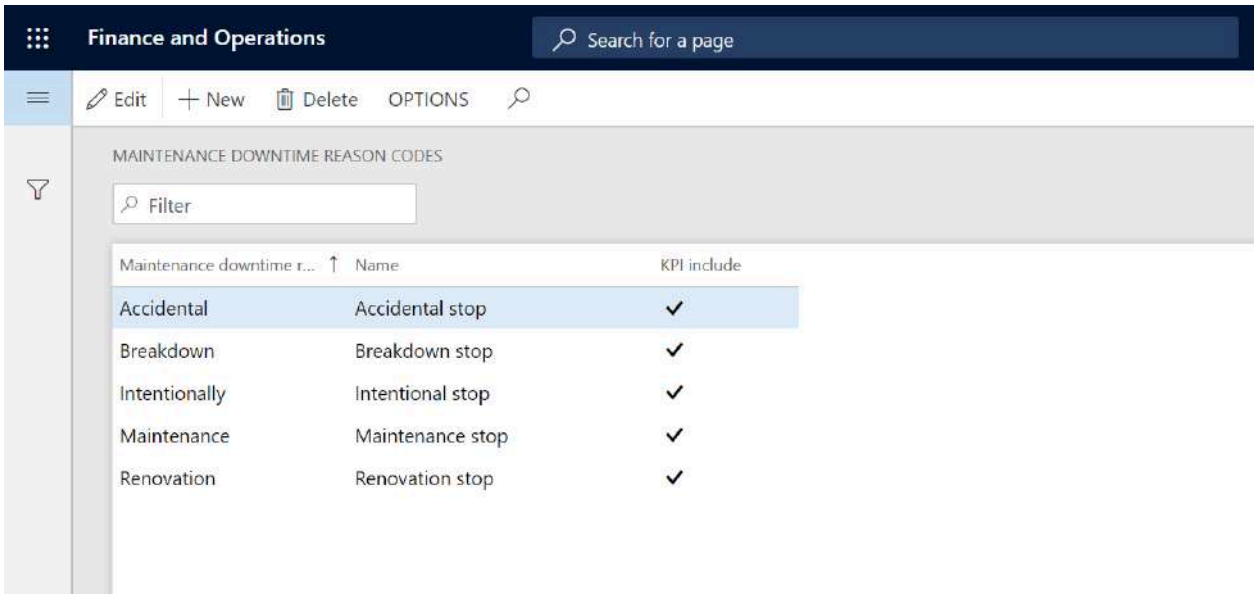
Article • 06/15/2022

You can create maintenance downtime registrations on the asset that is selected on a work order. This capability is useful if you want to register maintenance downtime on one or more machines in the production area. You first create the maintenance downtime reason codes that you want to use, such as **Breakdown** and **Planned stop**. This step is done on the **Maintenance downtime reason codes** page. You can then create maintenance downtime registrations on the **Maintenance downtime** page and add the relevant maintenance downtime reason codes.

Create maintenance downtime reason codes

1. Select **Asset management** > **Setup** > **Work orders** > **Maintenance downtime reason codes**.
2. Select **New**.
3. In the **Maintenance downtime reason code** field, enter an ID for the maintenance downtime reason code.
4. In the **Name** field, enter a name.
5. Select the **KPI include** check box if the reason code should be included in calculations of key performance indicators (KPIs) for the asset. In general, planned production stops should not be included in KPI calculations, because they don't affect expected performance.
6. Select **Save**.

The illustration below shows an example of the **Maintenance downtime reason codes** page.



After you've created the maintenance downtime reason codes that you want to use, you can create maintenance downtime registrations for work orders and assets.

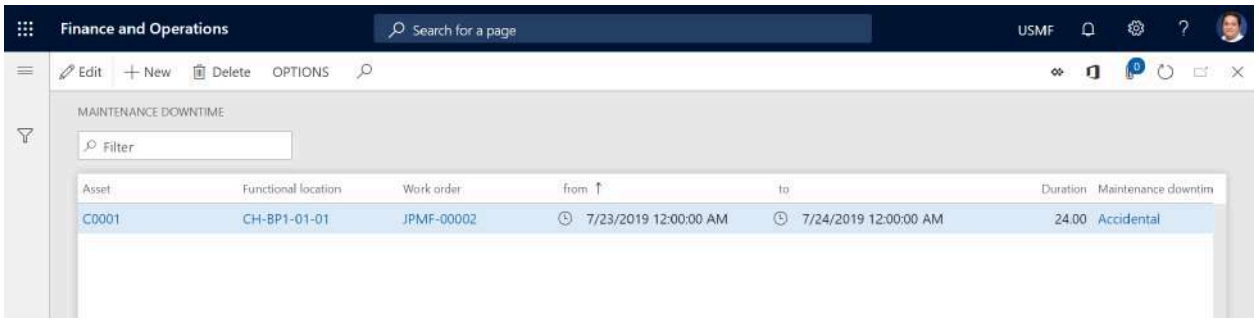
Create maintenance downtime registrations

1. Click **Asset management** > **Work orders** > **All work orders** or **Active work orders**.
2. Select the work order, and then, on the **Work order** tab, in the **Asset** group, select **Maintenance downtime**.
3. Select **New**.
4. In the **From** and **To** fields, define the date and time interval for the maintenance downtime registration.

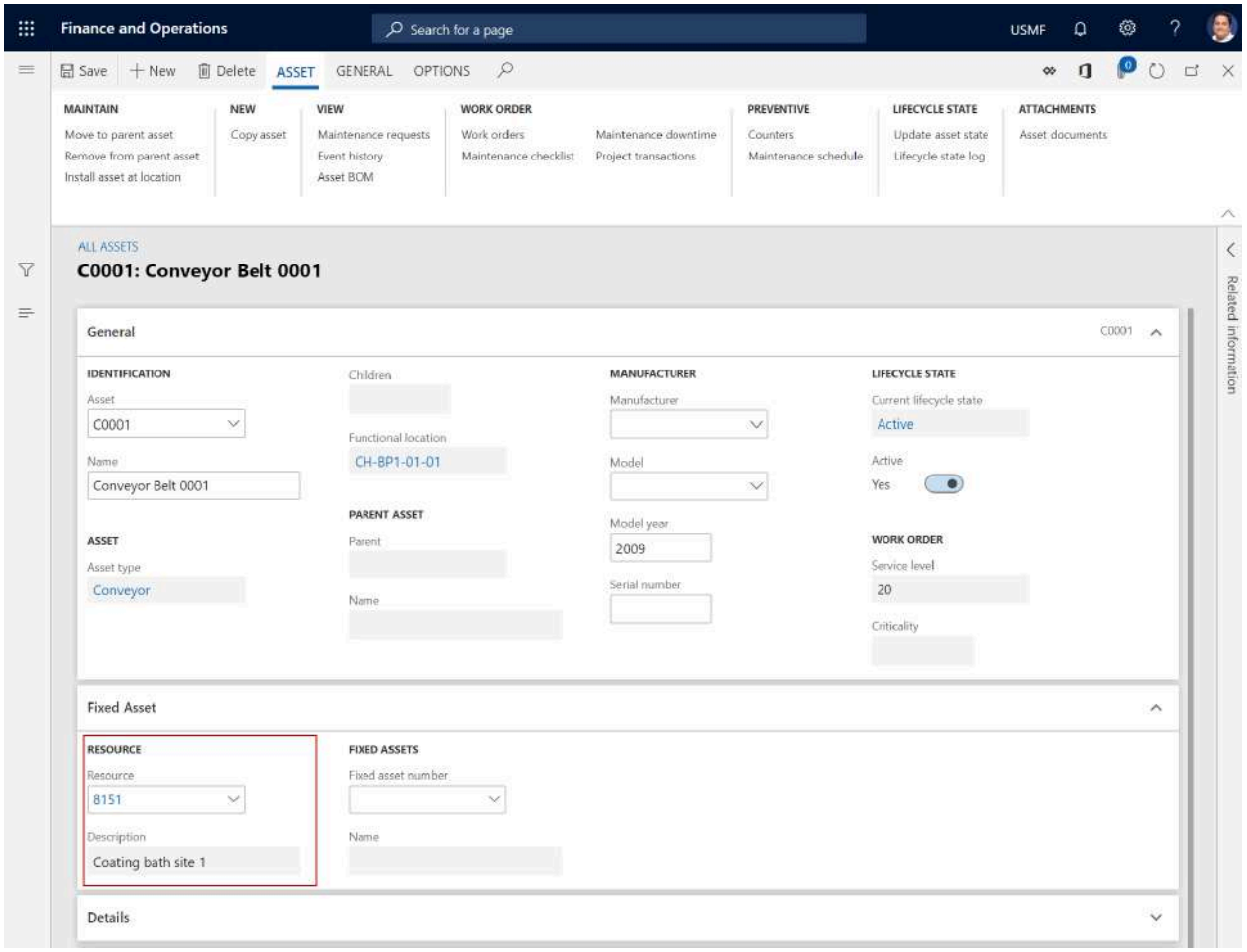
> **Note**
>
> When you leave the **To** field, the duration in hours is automatically inserted in the **Duration** field.

5. In the **maintenance downtime reason code** field, select a reason code.
6. Repeat steps 3 through 5 to add more registrations.
7. Select **Save**.

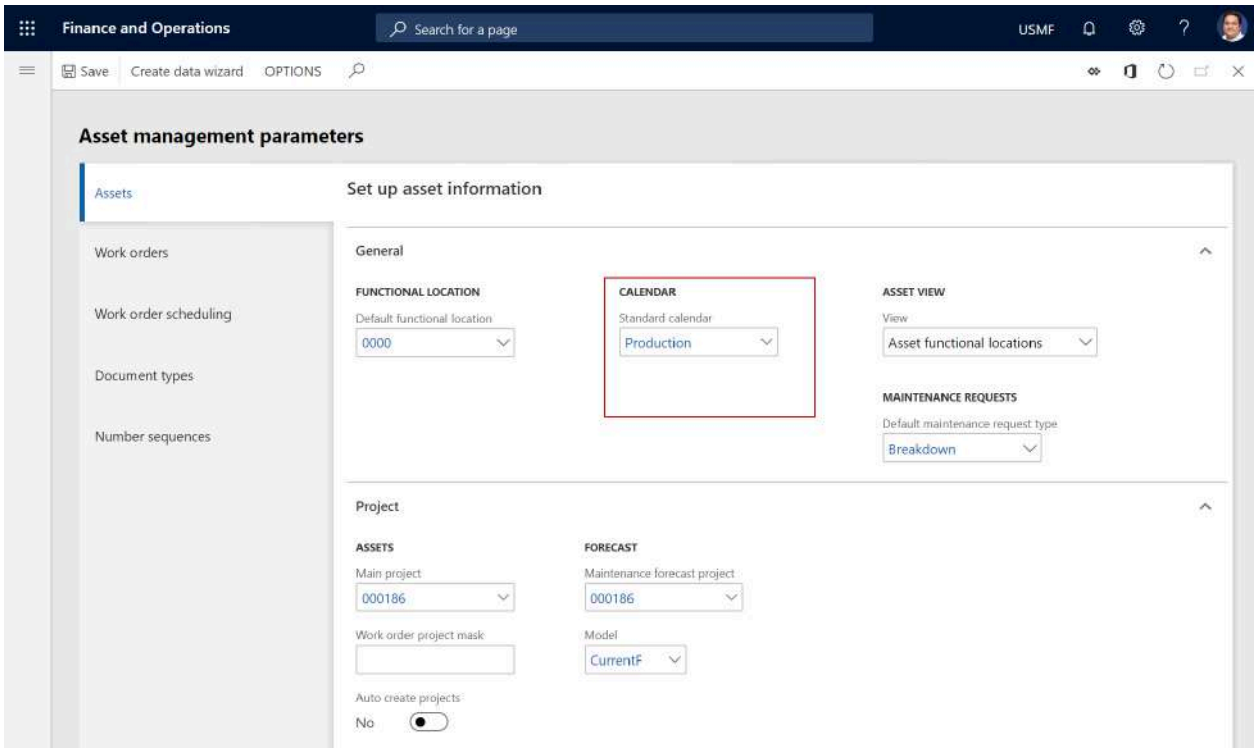
The illustration below shows an example of maintenance downtime registration.



The calendar that is used to calculate a maintenance downtime registration depends on your selection in the setup of assets and parameters. If a resource is selected on an asset in the **Resource** field on the **Fixed asset** FastTab of the **All assets** page, the calendar that is set up for the associated resource group is used, as shown in the following illustration.



If no resource is selected on the asset, the standard calendar that is selected on the **Asset management parameters** page is used, as shown in the following illustration.



To see an overview of all maintenance downtime registrations, click **Asset management** > **Inquiries** > **Maintenance downtime**.

> **Note**
>
> All calendars that are used in the **Asset Management** module are set up in **Organization administration** > **Setup** > **Calendars** > **Calendars**.

Feedback

Was this page helpful? Yes No

Provide product feedback

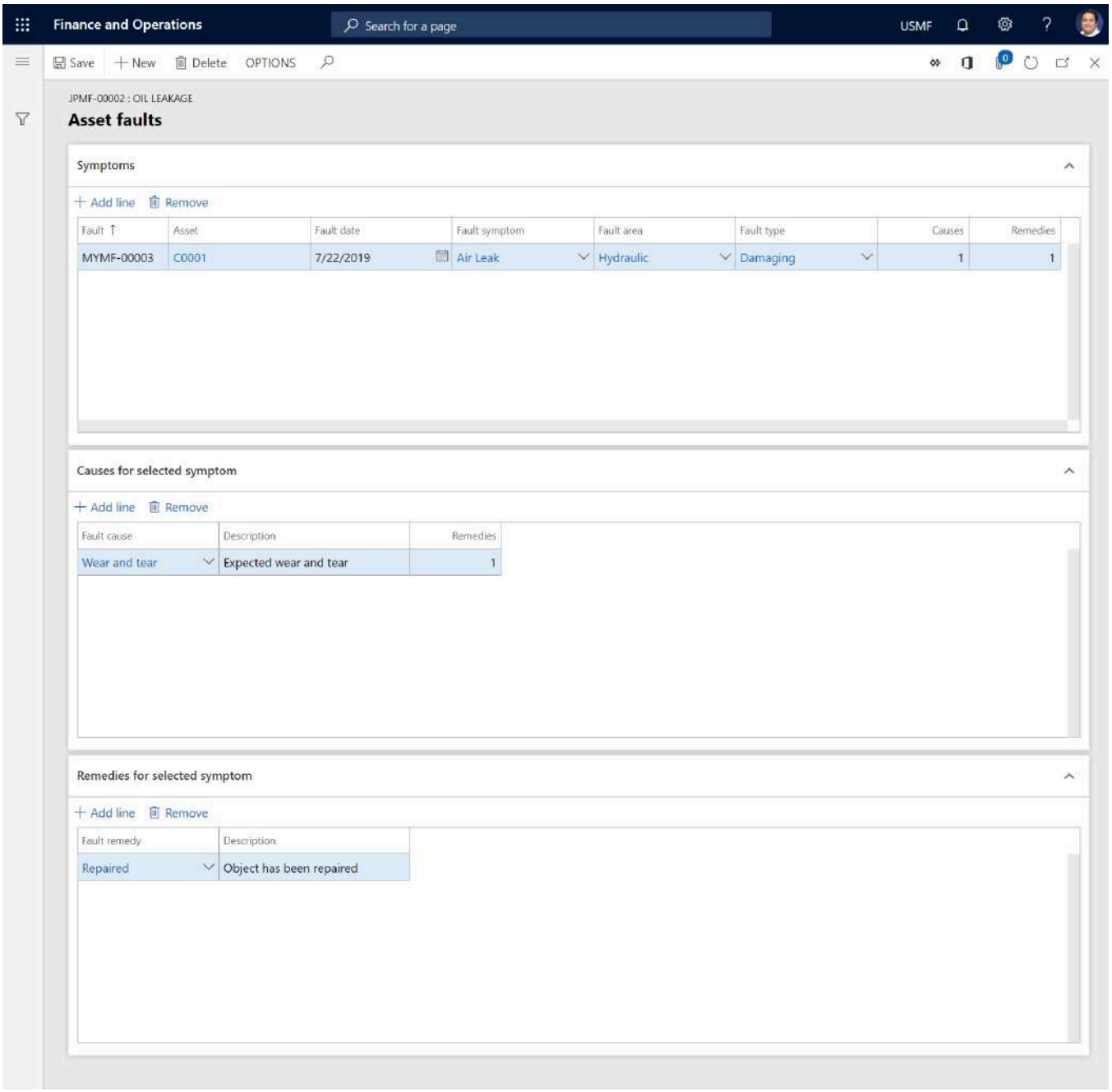
Add fault to work order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can add faults that were set up in the fault designer to a work order. One or more fault records must be connected to the asset types that are used for the asset that is selected in the work order. For more information about the setup, see Fault management.

1. Select **Asset management** > **Work orders** > **All Work orders** or **Active work orders**.
2. Select the work order to make a fault registration on, and then, on the Action Pane, on the **Work order** tab, in the **Asset** group, select **Asset fault**.
3. On the **Symptoms** FastTab, select **Add line**. A sequential fault number is automatically entered in the **Fault** field.
4. In the **Fault symptom** field, select the relevant symptom.
5. In the **Fault area** and **Fault type** fields, select the appropriate values.
6. In the **Fault date** field, the current date is automatically inserted. You can select a different date as you require.
7. On the **Causes for selected symptom** FastTab, add a line to describe the cause of the issue.
8. On the **Remedies for selected symptom** FastTab, add a line to describe a possible solution to the issue.
9. Select **Save**.

The illustration below shows an example of a fault registration.



View asset faults

In the **Asset faults** list, you can get an overview of all faults registered on assets.

On the **Asset faults** list page, you can get an overview of all faults that have been registered on assets. To open the page, select **Asset management** > **Inquiries** > **Asset fault** > **Asset faults**.

Print asset fault report

From the **All assets** list page, you can print an asset fault report that shows all fault registrations and a graphical overview of fault statistics.

1. Select **Asset management** > **Assets** > **All assets**.

2. Select the asset to print a fault report for.

3. On the Action Pane, on the **General** tab, in the **Reports** group, select **Asset fault**.

4. Enter a specific period, or select a fault type.

5. Select **OK** to print the report.

> **Note**
>
> To print a fault report for several assets or asset types, select **Asset management** > **Reports** > **Assets** > **Asset fault**.

Last updated on 05/07/2025

Work order report

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can generate a work order report that shows detailed information about one or more work orders that you select.

1. Select **Asset management** > **Reports** > **Work orders** > **Work order report**.
2. Select "Yes"/"No" on the toggle buttons as you require, to determine the details to be included in the report.
    - In the **Print settings** section, you can select whether to include attachments from the setup of the related maintenance job type.
3. On the **Records to include** FastTab, you can filter the contents of the report by **Work order**.
4. On the **Run in the background** FastTab, you can set up generation of the work order report as a batch job, as you require.
5. Select **OK** to generate the report.

The following illustration shows an example of how the report parameters can be set up.

Work order report

Parameters

WORK ORDER

Work order pool

Yes

ASSET

Asset image

Yes

Asset Address

No

Asset counters

No

Asset Notes

Yes

Asset attributes

Yes

WORK ORDER JOB NOTES

Notes: Description

Yes

Notes: Maintenance worker

Yes

Notes: Internal

Yes

Blank note lines

5

MAINTENANCE CHECKLIST

Maintenance checklist

Yes

Maintenance checklist N/A

Yes

FORECAST

Forecast

Yes

Blank maintenance forecast lines

5

WORK ORDER JOB

Schedule

No

Blank fault lines

1

Tools

Yes

Unposted Journal lines

No

Purchase

Yes

PRINT SETTINGS

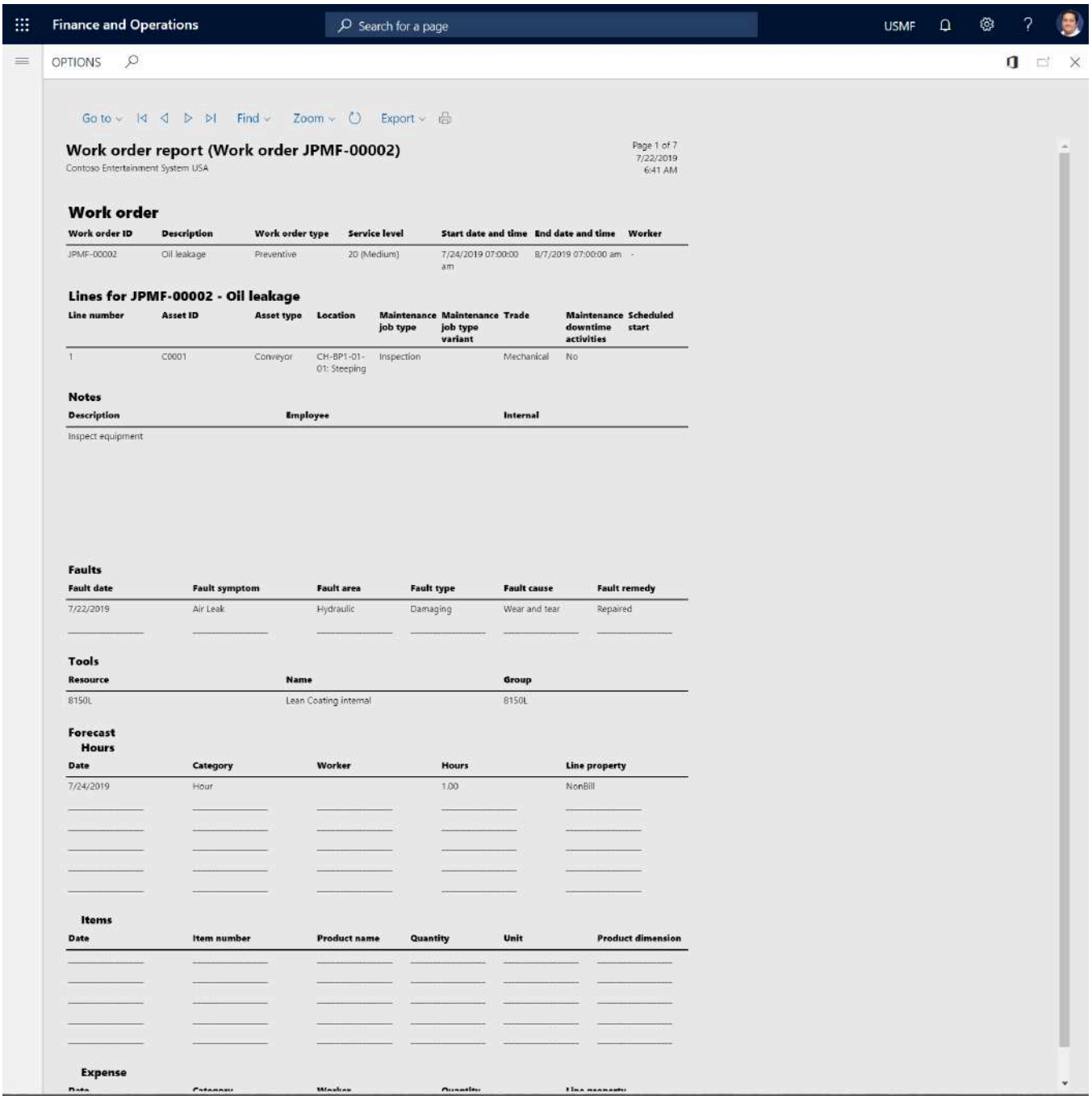
Print attached documents

No

Asset attributes

No

The following illustration shows an example of a work order report that is generated based on that setup.



Last updated on 05/07/2025

Work order pools

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can use work order pools to group work orders that have something in common. Here are some examples of things that you can create work order pools for:

- Work crews, for example, Maintenance Crew A or Maintenance Crew B
- Professional skills, such as electricians or plumbers
- Physical locations
- Time schedules, such as weeks or other periods

As you require, you can put one work order in multiple work order pools.

Create a work order pool

On the **All work order pools** or **Active work order pools** list page, you can get an overview of your work order pools and create new pools.

1. Select **Asset management** > **Work order pools** > **All work order pools** or **Active work order pools**.
2. Select **New**.
3. In the **Pool** field, enter an ID for the work order pool.
4. the **Name** field, enter a name.
5. Set the **Active** option to **Yes** to indicate that the work order pool is active.
6. Set the **Delete work order relations** option to **Yes** if work orders should automatically be removed from the work order pool.
7. In the **Delete lifecycle state** field, select the work order lifecycle state. For example, the work order lifecycle state for completing a work order could be set to automatically delete relations to work order pools.

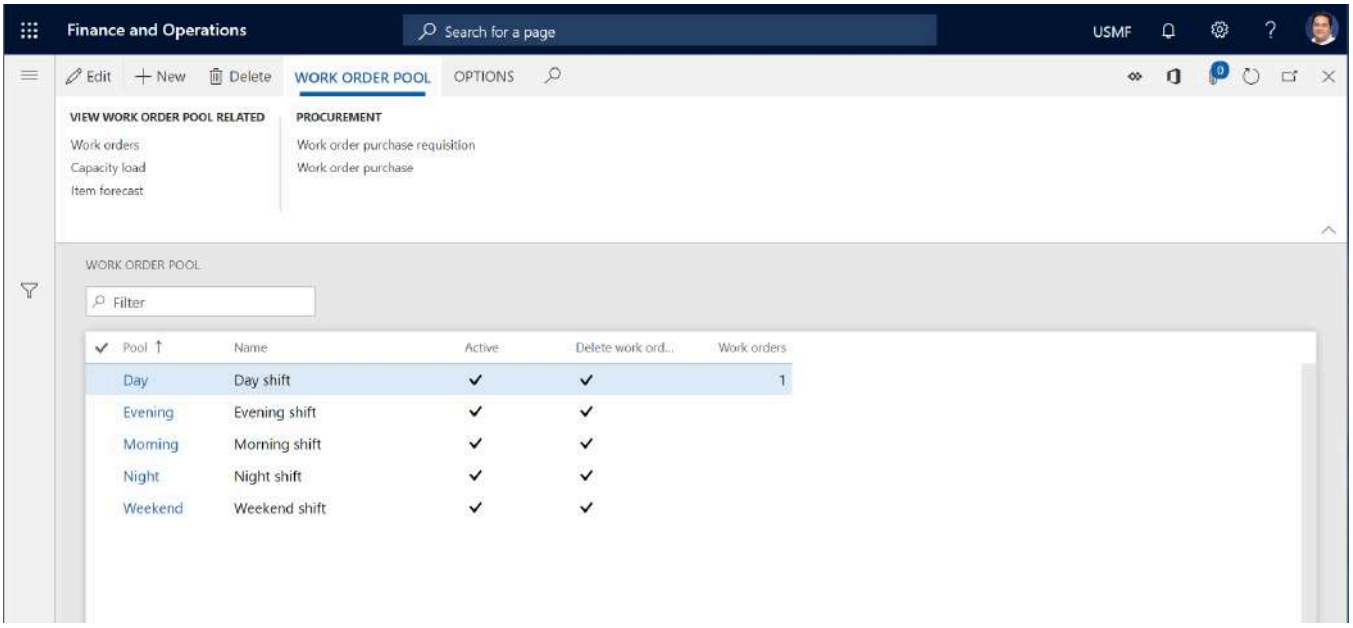
You can start adding work orders to your work order pool right away.

8. On the **Work orders** FastTab, select **Add line**.
9. In the **Work order** field, select a work order. The related fields are automatically updated.
10. Repeat steps 8 through 9 to add more work orders.
11. If the work orders that you added should be done in a specific order, in the **Sort order** field, you can enter the numbers **1**, **2**, **3**, and so on, to specify that order.
12. To view a list of all the work orders that are included in the work order pool, on the Action Pane, on the **Work order pool** tab, in the **View work order pool related** group, select **Work orders** to open the **All work orders** list page.
13. To calculate and view capacity load for the maintenance schedule, unscheduled work orders, and scheduled work orders, on the Action Pane, on the **Work order pool** tab, in the **View work order pool related** group, select **Capacity load** to open the **Calculate capacity load** dialog.
14. To calculate and view forecasts for items (spare parts and other required items) that are related to maintenance schedule, unscheduled work orders, and scheduled work orders, on the Action Pane, on the **Work order pool** tab, in the **View work order pool related** group, select **Item forecast** to open the **Calculate item forecast** dialog.
15. To view a list of purchase requisitions that are related to the work orders in the work order pool, on the Action Pane, on the **Work order pool** tab, in the **Procurement** group, select **Work order purchase requisition** to open the **Work order purchase requisition** list page.
16. To view a list of purchase orders that are related to the work orders in the work order pool, on the Action Pane, on the **Work order pool** tab, in the **Procurement** group, select **Work order purchase** to open the **Work order purchase** list page.

> ⓘ **Note**
>
> When a work order pool is no longer relevant to your work planning, set the **Active** option for that pool to **No** in the list view of the **Work order pool** page.

To delete all worker order lines, set the **Delete work order relations** option to **Yes**. This option is useful if, for example, you want to create an empty pool that you can use later for other work orders. When you're ready to use the work order pool to create new work order relations later, remember to set the **Delete work order relations** option to **No**.

The illustration below shows an example of the **Work order pool** list page.



Add a work order to a work order pool

As described in the previous section, you can add work orders to a work order pool when you create that pool. You can also add work orders to a work order pool on the **All work orders** or **Active work orders** list page.

1. Select the work order, and then, on the Action Pane, on the **Work order** tab, in the **Maintain** group, select **Work order pool**.
2. Select the work order in the list, and click **Work order pool**.
3. In the **Maintain work order pool** dialog, in the **Add/remove** field, select **Add**.
4. In the **Pool** field, select the work order pool.
5. Select **OK**.
6. To put the work order that you added in a specific order in the work order pool, on the **All work order pools** or **Active work order pools** list page, select the pool, and then select **Edit**. Then, on the **Work order pool** page, on the **Work orders** FastTab, use the **Sort order** field to adjust the sort order of the work orders that are included in pool.

To remove a work order from a work order pool, repeat these steps, but select **Remove** in step 3.

Last updated on 05/07/2025

Active work order maintenance jobs overview

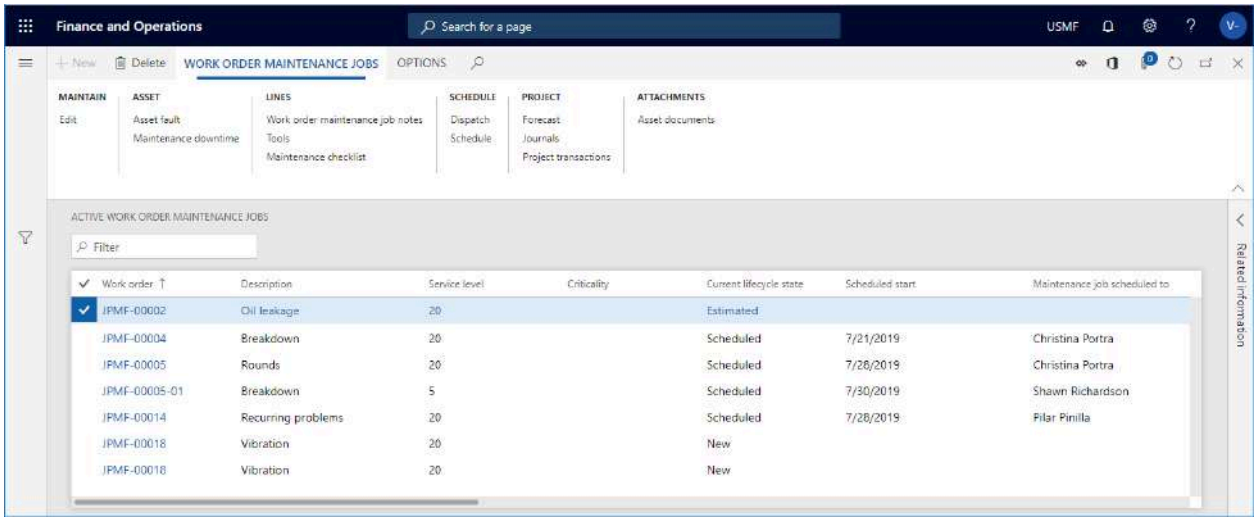
Article • 06/15/2022

On the **Active work order maintenance jobs** list page, you can get an overview of work orders, in terms of the number of work orders that have been created on specific assets, asset types, manufacturers, models, maintenance job types, and so on. By selecting a work order job and then selecting **Edit**, you can open the related work order. By selecting a work order job and then selecting one of the buttons on the **Work order maintenance jobs** tab of the Action Pane, you can view data for the work order that the work order job is related to.

To open the list page, select **Asset management** > **Work orders** > **Active work order maintenance jobs**. The page lists all active work order jobs, and it also shows some of the information that is related to the work order or work order job.

In the % column, a number shows completion of the work order as a percentage. Completion is based on two calculations: posted hours compared to forecasted hours, and the number of maintenance checklists that have been completed.

The illustration below shows an example of the **Active work order maintenance jobs** list page.



For a short description of the buttons on the **Active work order maintenance jobs** list page, refer to Introduction to work orders where identical buttons are described.

Feedback

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Work orders and fixed assets

> **Note**
>
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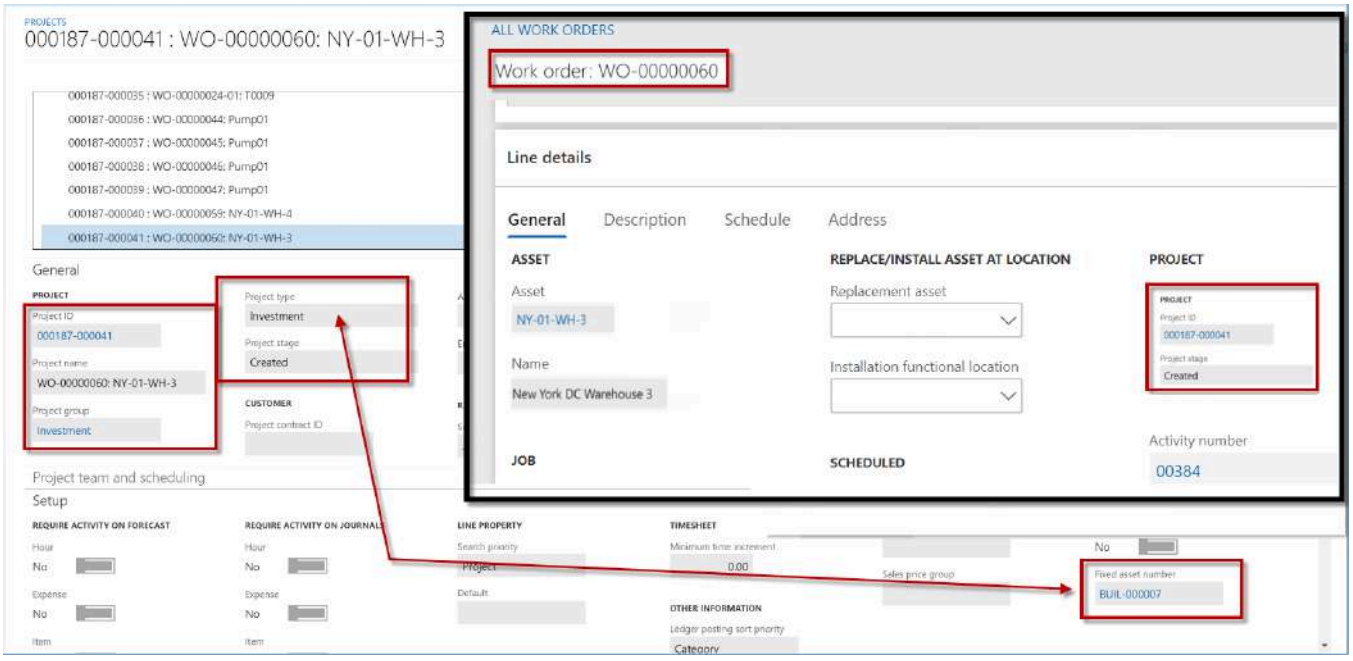
In Asset Management, assets can be related to fixed assets, and you can create work orders for those assets. If you use this functionality, you can get a complete overview of fixed assets, related investment projects, and the costs that are registered on the investment projects in the **Project management and accounting** and **Fixed assets** modules in the finance and operations apps.

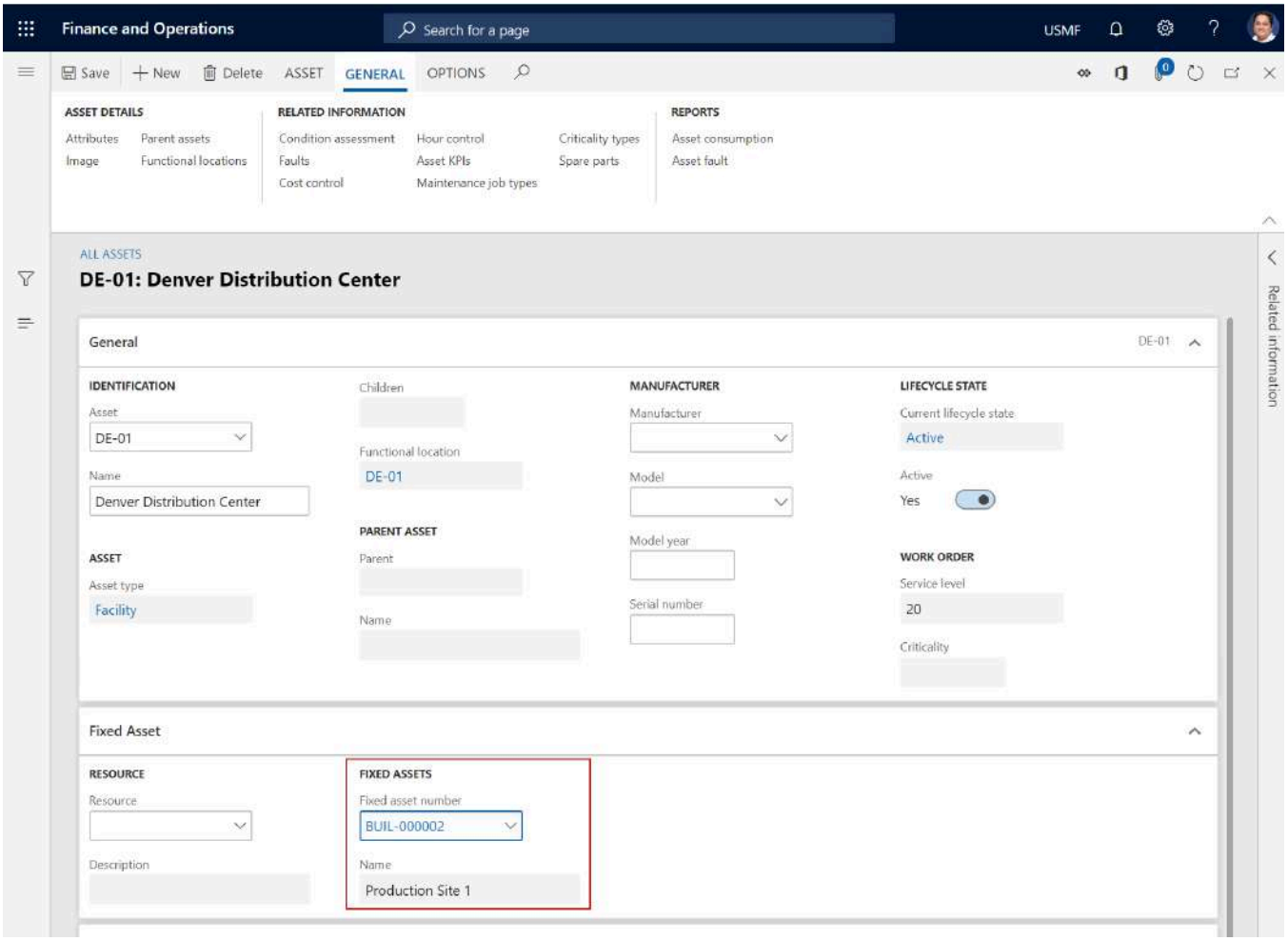
> **Note**
>
> The **Fixed asset number** field on the work order job project is set only if **Investment** is selected as the project type on the work order job project.

The illustration below shows the relation between an investment project in the **Project management and accounting** module and a work order job project.

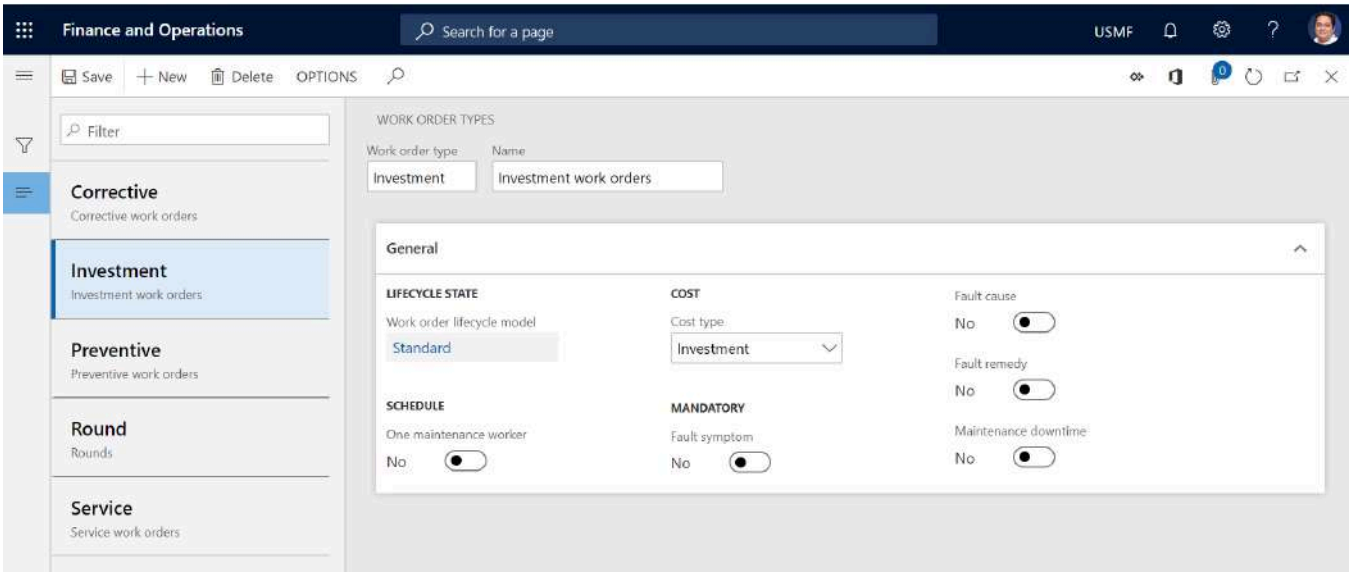


The following procedure describes the relation between assets, work orders, work order job projects, and fixed assets.

1. You create an asset that you relate to a fixed asset.

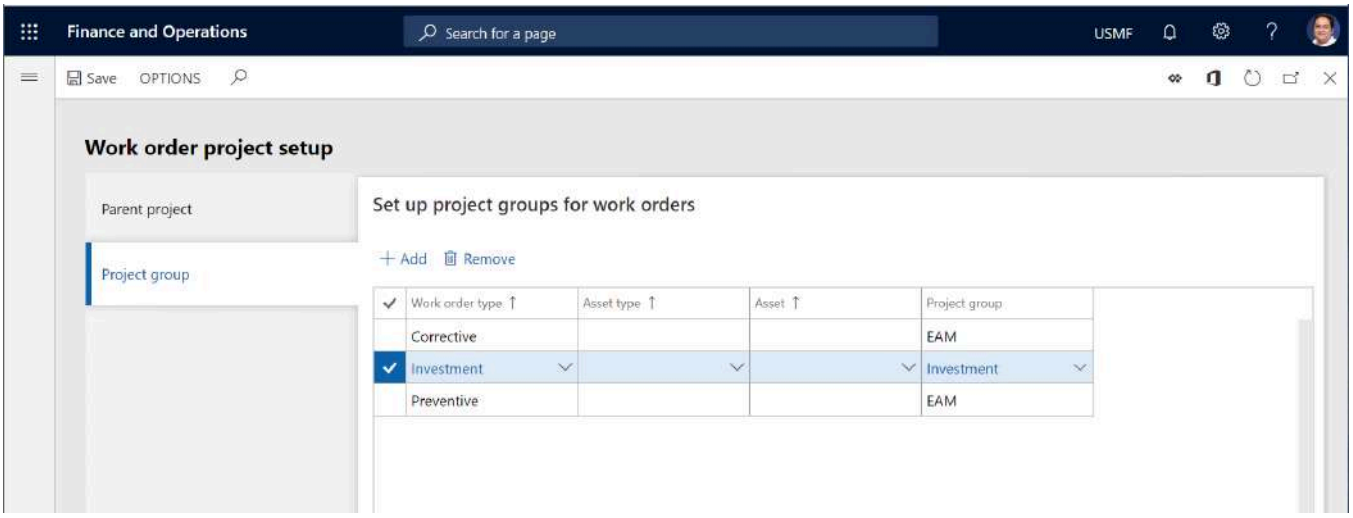


2. When you set up work order types on the **Work order types** page (**Asset management** > **Setup** > **Work orders** > **Work order types**), you create a work order type for handling investments. Learn more in Work order types.



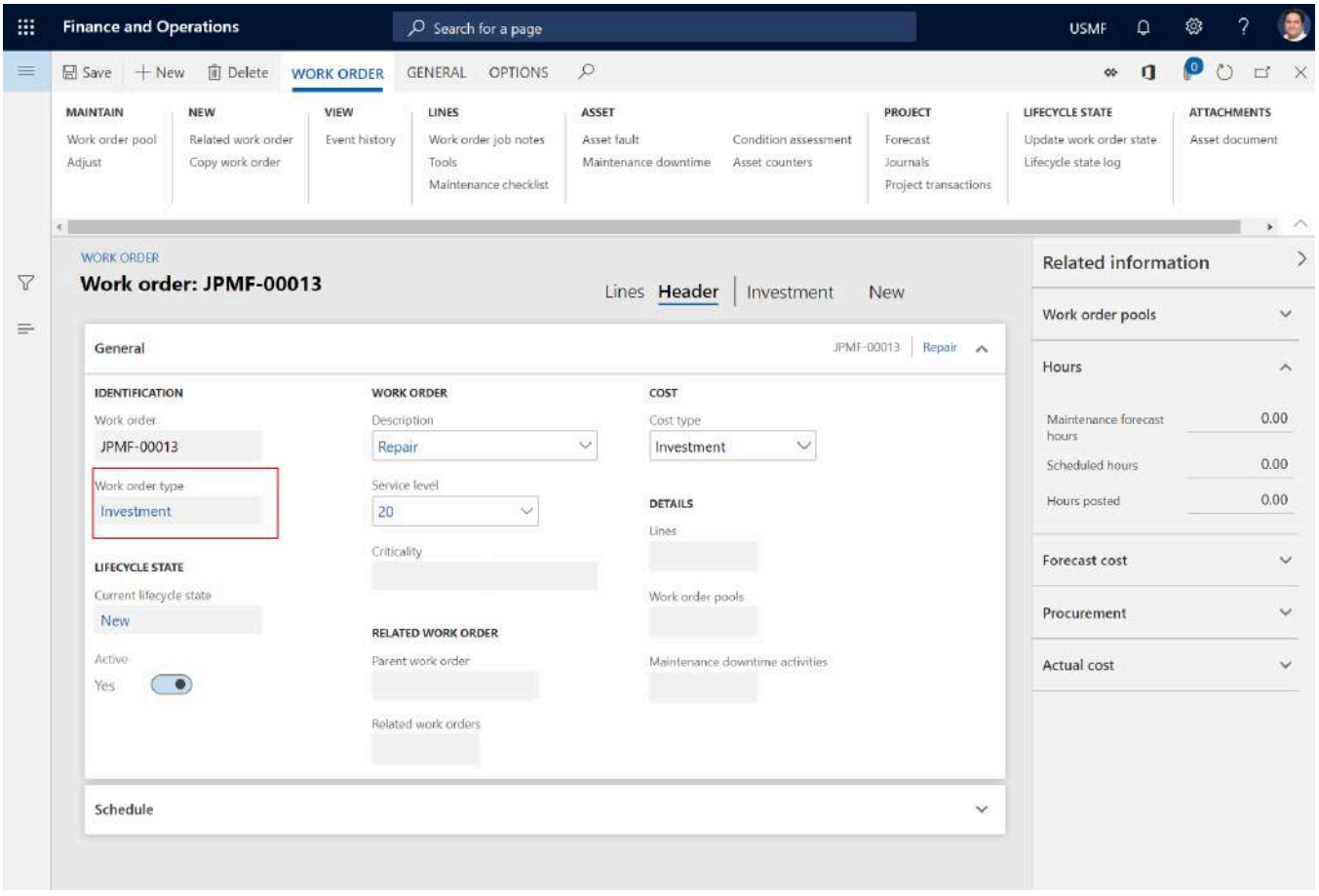
3. When you set up work order project groups on the **Project group** tab of the **Work order project setup** page (**Asset management** > **Setup** > **Work orders** > **Project setup**), you create a relation between the work order type that is used for investments and the project group that was created for investments on the **Project groups** page in the **Project**

management and accounting module (**Project management and accounting** > **Setup** > **Posting** > **Project groups**).



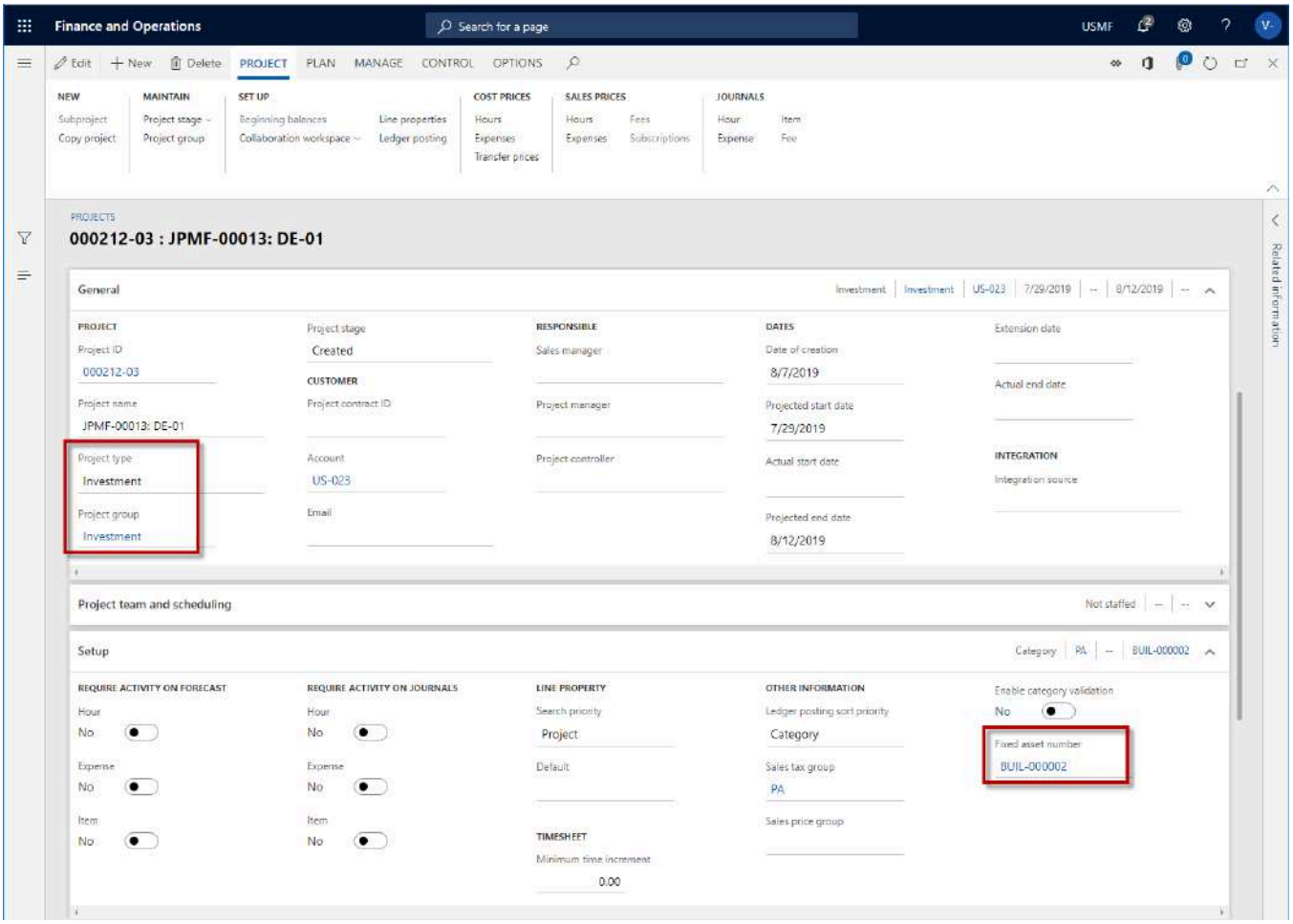
4. When you create a work order that is related to a fixed asset, you select the work order type that is used to handle investments, such as **Investment**.

5. When the work order is created, the related work order type is shown on the **All work orders** page.



6. When the work order is created, the project that is related to the work order is created on the **All projects** page in the **Project management and accounting** module (**Project management and accounting** > **Projects** > **All projects**). To view project-related

information, select the link in the **Project ID** field on the **General** tab on the **Line details** FastTab in the details view of the **All work orders** page in the **Asset management** module (**Asset management** > **Commom** > **Work orders** > **All work orders**).



7. To see an overview of the projects associated with a fixed asset, select **Fixed assets** > **Fixed assets** > **Fixed assets**, and then, in the **Fixed asset number** field, select the link for the fixed asset to open the details view. Expand the **Related information** pane on the right side of the page, and select the **Associated projects** FastTab.

Last updated on 05/07/2025

Forecasts, work orders, and projects

08/13/2025

In Asset Management, integration with the **Project management and accounting** module helps optimize cost control, so that users can track costs on maintenance job type forecasts and work order jobs.

Tracking of maintenance job type forecasts requires two settings:

1. Select a project in **Asset management** > **Setup** > **Asset management parameters**, and then, on the **Assets** tab, on the **Project** FastTab, in the **Maintenance forecast project** field, select a project.
2. When you create a maintenance job type default line, an activity number is automatically created for the line on the **Maintenance job type defaults** page (**Asset management** > **Setup** > **Jobs** > **Maintenance job type defaults**).

Maintenance job type forecasts serve two purposes:

- You can track costs on maintenance job type forecasts in the **Project management and accounting** module.
- Forecasts are automatically transferred to a work order job project when you select a maintenance job type on a work order job.

To track costs on work order jobs, you must first set up work order projects. Learn more in Work order project setup.

Work order job projects

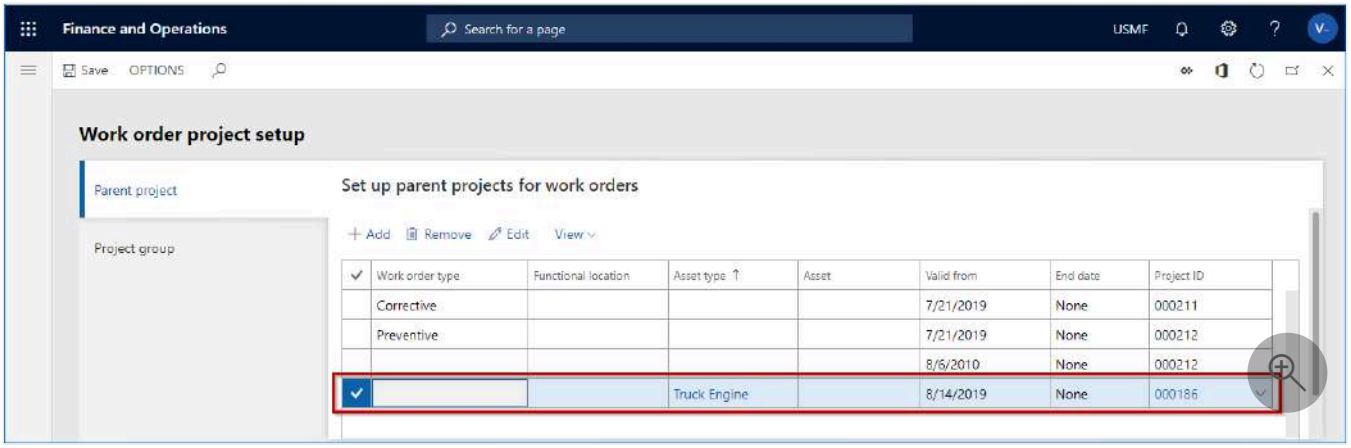
When you create a work order job on a work order, the work order project is determined by the setup of the parent project for work orders on the **Work order project setup** page (**Asset management** > **Setup** > **Work orders** > **Project setup**).

Work order job projects are created by using a combination of the following work order information:

- The work order type selected on the work order
- The functional location related to the asset on the work order job
- The asset type that is related to the asset on the work order job
- The expected start and end times that are set on the work order

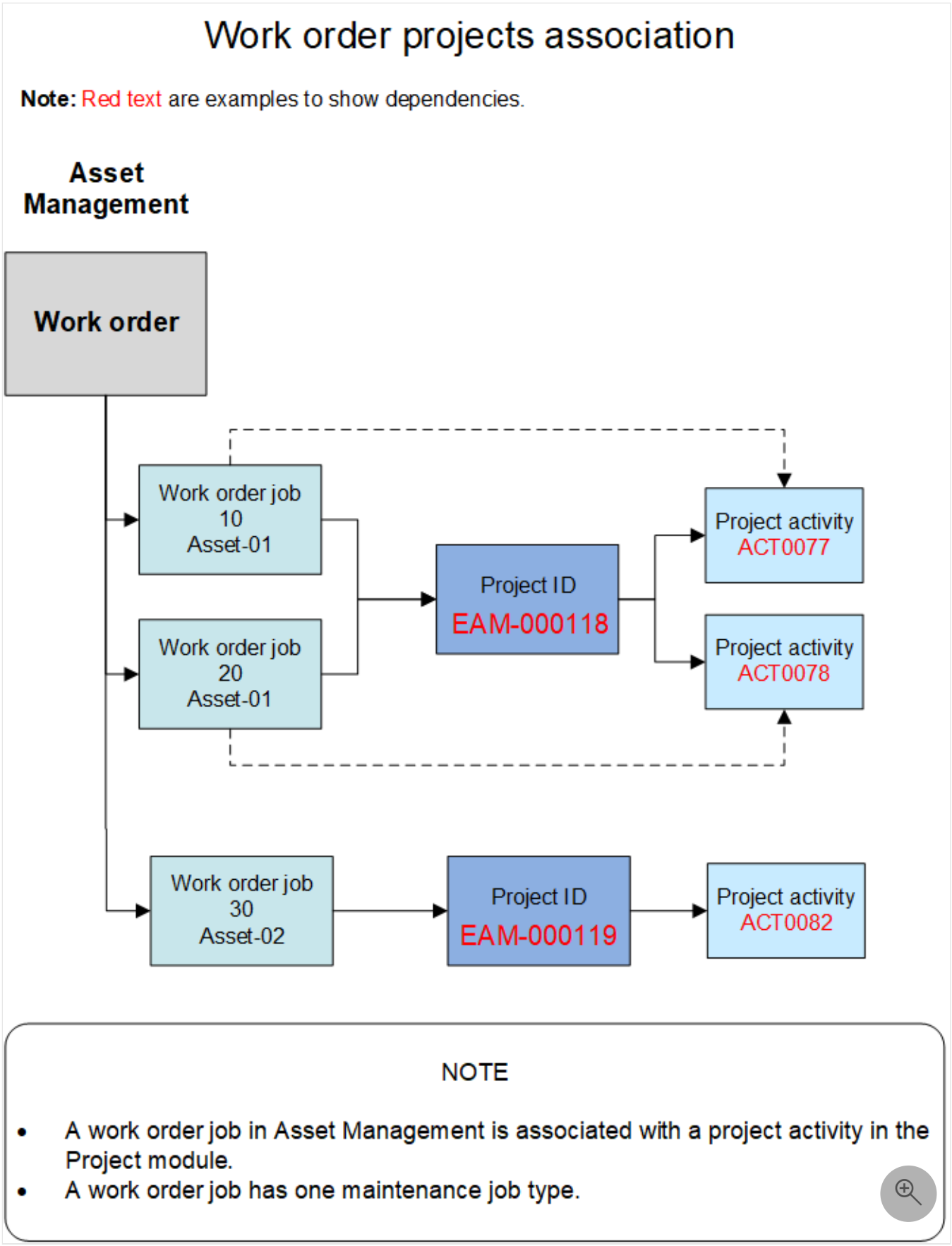
Some of this information might not be found on a work order. Therefore, the search for a work order parent project is done by using the available combination of data and selecting the project ID that corresponds to work order data.

For example, in the following illustration, because of the way that the *Truck Engine* asset type is set up, every work order job that is created with the *Truck Engine* asset type will be a sub-project of project ID 000186.



The purpose of the project ID on the work order job, and the related activity number, is to track costs that are related to the work order job, and the asset that is selected on it, in the **Project management and accounting** module. (To view the project ID and activity number, select **Asset management** > **Work orders** > **All work orders**, and then select the work order. On the **Line details** FastTab, the **Project ID** field shows the project ID, and the **Activity number** field shows the activity number.) For more information about cost control in Asset Management, see Cost and date control.

The following illustration shows a graphical overview of work order projects and related project activities.



When a new work order job is created on a work order, a work order project is automatically created for the job. The financial dimensions for the asset that is related to the work order job are automatically transferred to the work order project.

The project activity that is created for a work order job has related information attached to it. This information is about the maintenance job type, maintenance job type variant, and trade.

It's useful if, for example, you create a purchase order from a work order (see Procurement), or if you use the **Project management and accounting** module for time registration.

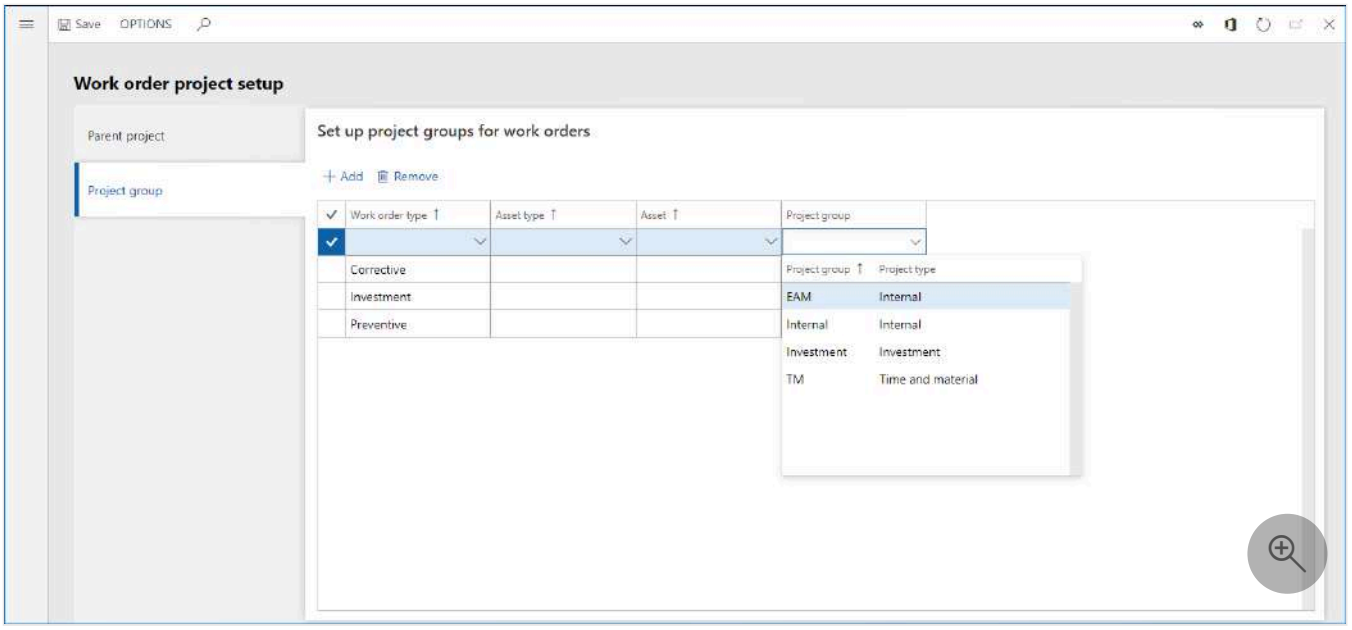
If the asset was installed on a functional location but is later installed on a different functional location, the financial dimensions that are related to the new functional location are automatically updated on the asset. Then, when you create a work order job for the asset, the work order project for the work order job automatically gets the financial dimensions that are now related to the asset. Therefore, when you use functional locations, costs can always be tracked on the functional locations that an asset was installed on at any given time. The automatic update of financial dimensions helps guarantee complete traceability of costs for project control and reporting.

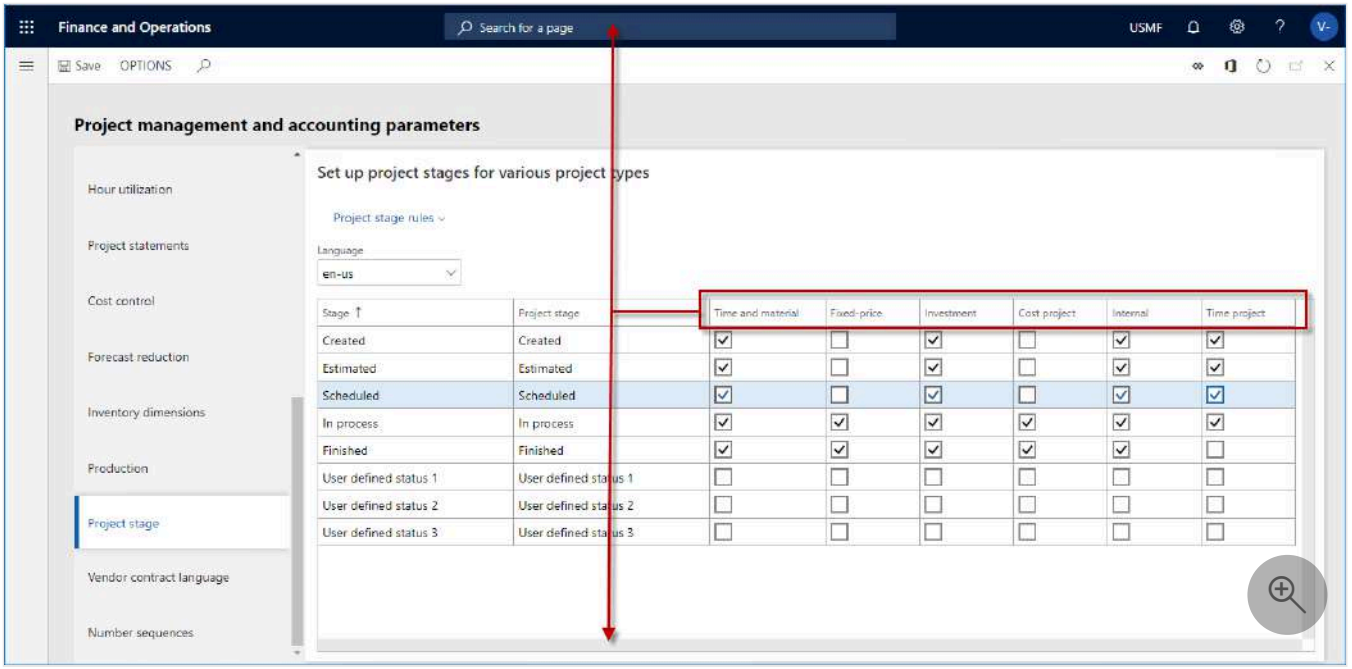
Work order projects, work order lifecycle states, project stages, and project types

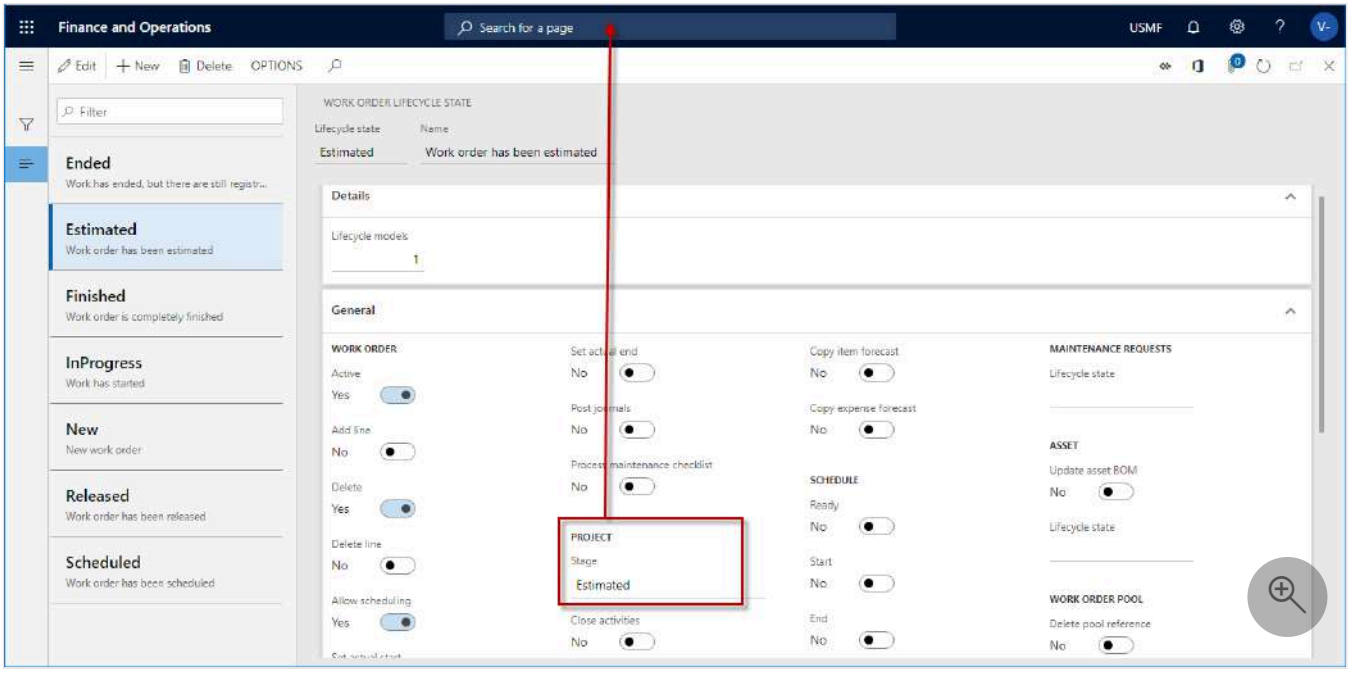
To help guarantee that work order lifecycle states and related project stages on work orders are used correctly, consider the dependencies in relation to the **Project management and accounting** module:

- In the **Project management and accounting** module, project stages are set up on project types on the **Project management and accounting parameters** page.
- On the **Project management and accounting parameters** page, use the check boxes to select relevant project stages for all the project types that you'll use. In the following illustrations, five stages (**Created**, **Estimated**, **Scheduled**, **In process**, and **Finished**) have been selected for the **Time and material** and **Internal** project types. Those five stages are relevant to both internal maintenance jobs and service maintenance jobs.
- In the **Asset Management** module, project types are defined by the project groups that you set up on the **Work order project setup** page > **Project group** tab (**Asset management** > **Setup** > **Work orders** > **Project setup**).
- The project groups that are set up on the **Work order project setup** page are used when you create work orders. When a work order is created, a work order project is automatically created for the work order.
- Every work order lifecycle state must have a related project stage.
- The project stage that is related to a work order lifecycle state must be defined as an active stage for the project group that is defined in the work order project. The work order project is automatically created on a work order.
- When you create a new work order, the automatic allocation of a work order project is based on the setup on the **Work order project setup** page.

The following illustrations show the associations between work order project groups, related project types, project stages, and work order lifecycle states.


Save
OPTIONS
Work order project setup
Parent project
Project group
Set up project groups for work orders
Add
Remove
Work order type
Asset type
Asset
Project group
Corrective
Investment
Preventive
Project group
Project type
EAM
Internal
Internal
Internal
Investment
Investment
TM
Time and material

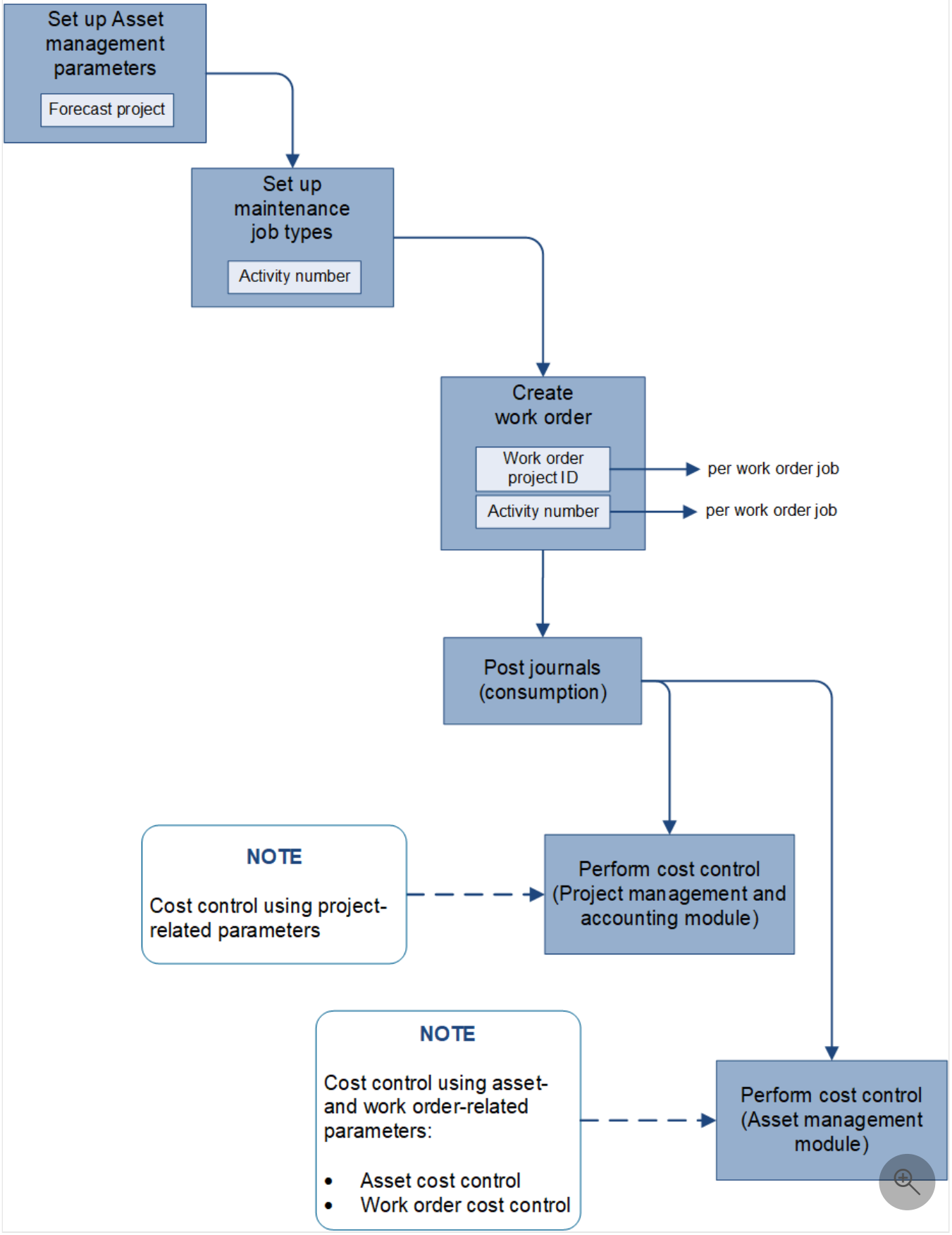

Finance and Operations
Search for a page
USMF
Save
OPTIONS
Project management and accounting parameters
Hour utilization
Project statements
Cost control
Forecast reduction
Inventory dimensions
Production
Project stage
Vendor contract language
Number sequences
Set up project stages for various project types
Project stage rules
Language
en-us
Stage
Project stage
Time and material
Fixed-price
Investment
Cost project
Internal
Time project
Created
Estimated
Scheduled
In process
Finished
User defined status 1
User defined status 2
User defined status 3


Finance and Operations
Search for a page
USMF
Edit
New
Delete
OPTIONS
Filter
Ended
Work has ended, but there are still registr...
Estimated
Work order has been estimated
Finished
Work order is completely finished
InProgress
Work has started
New
New work order
Released
Work order has been released
Scheduled
Work order has been scheduled
WORK ORDER LIFECYCLE STATE
Lifecycle state
Name
Estimated
Work order has been estimated
Details
Lifecycle models
1
General
WORK ORDER
Active
Yes
Add line
No
Delete
Yes
Delete line
No
Allow scheduling
Yes
Set actual end
No
Post journals
No
Process maintenance checklist
No
PROJECT
Stage
Estimated
Close activities
No
Copy item forecast
No
Copy expense forecast
No
SCHEDULE
Ready
No
Start
No
End
No
MAINTENANCE REQUESTS
Lifecycle state
ASSET
Update asset BOM
No
Lifecycle state
WORK ORDER POOL
Delete pool reference
No

For information about how to set up work order projects, see Work order project setup. For information about how to create work order lifecycle states, see Work order lifecycle states.

The following illustration shows a graphical overview of the various projects that are created in the **Asset management** module to enable integration with the **Project management and accounting** module. It also shows the work processes that the projects are related to.

Project Integration - Flow Diagram



Bill for maintenance on customer-owned assets

The *Work order billing* feature lets you create, process, and bill maintenance work that is done on assets that your customers own. This feature lets you perform the following tasks:

- Connect customers to the assets that they own.
- Select a customer and view the assets that customer owns when you create a work order.
- Set up a parent project for each customer.
- Register hours, items, expenses, and fees against the work order, and then create an invoice proposal for the customer later.
- Check customer credit limit before dispatching a work order.

In addition, the feature provides the following functionality:

- The project contract from a customer's parent project is automatically copied to the relevant work order project.
- Asset management can now use the *fee* project transaction type on both work order forecasts and work order journals.

Turn the work order billing feature on or off

To use the work order billing feature, it must be turned on for your system. As of Supply Chain Management version 10.0.32, this feature is mandatory and can't be turned off. If you're running a version older than 10.0.32, then admins can turn this functionality on or off by searching for the *Work order billing* feature in the **Feature management** workspace.

Example scenario

To learn how this feature works, work through the following example scenario.

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. You must select the **USMF** legal entity before you begin.

You can also use this scenario as guidance for using the feature when you work on a production system. However, in that case, you must substitute your own values, and you might be missing some types of required records that the standard demo data provides.

Step 1: Create a new project contract for the customer

1. Go to **Project management and accounting** > **Projects** > **Project contracts**.
2. On the Action Pane, select **New**.
3. In the **New project contract** dialog box, set the following values:
    - **Name:** *Pelican Wholesales*
    - **Funding type:** *Customer*
    - **Funding source:** *US-013* (*Pelican Wholesales*)
4. Select **OK**.

Step 2: Create a new parent project for the customer

The parent project that you create here will be used when work orders for the customer are created.

1. Go to **Project management and accounting** > **Projects** > **All projects**.
2. On the Action Pane, select **New**.
3. In the **New project** dialog box, set the following values:
    - **Project type:** *Time and material*
    - **Project name:** *Pelican Wholesales work orders*
    - **Project group:** *TM*
    - **Project contract ID:** *Pelican Wholesales* (the contract that you created earlier)
    - **Start date:** Select today's date.
4. Select **Create project**.
5. The new project is opened. Make a note of the **Project ID** value. You will have to enter it later.

Step 3: Create a new work order type in Asset management

1. Go to **Asset management** > **Setup** > **Work order** > **Work order types**.
2. On the Action Pane, select **New**.
3. A new record is added to the list. Set the following values for it:

- **Work order type:** *Service*
- **Name:** *Service work orders*
- **Work order lifecycle state:** *Standard*

Step 4: Link the customer account to the parent project

You must now link the customer account to the parent project in the project setup in Asset management.

1. Go to **Asset management** > **Setup** > **Work orders** > **Project setup**.
2. On the **Parent project** tab, select **Add** to add a line.
3. On the new line, set the following values:
    - **Customer account:** *US-013* (*Pelican Wholesales*)
    - **Project ID:** Enter the project ID that you made a note of earlier.

Step 5: Link the project group and type to the work order project

You should still be on the **Project setup** page (**Asset management** > **Setup** > **Work orders** > **Project setup**).

1. On the **Project group** tab, select **Add** to add a line.
2. On the new line, set the following values:
    - **Work order type:** *Service* (the work order type that you created earlier)
    - **Project group:** *TM*

> ⓘ **Note**
>
> The project contract on the work order project is always inherited from the parent project.

Step 6: Link an asset to the customer ID

1. Go to **Asset management** > **Assets** > **Active assets**.
2. In the **Filter** field, enter *VE-102*, and select to filter by **Asset**.
3. Select the asset to open its settings.

4. On the **Customer** FastTab, set the **Customer account** field to *US-013* (*Pelican Wholesales*).

   The **Name** field is automatically updated to *Pelican Wholesales*.

Step 7: Create a new work order on the asset

1. Go to **Asset management** > **Work orders** > **Active work orders**.
2. On the Action Pane, select **New**.
3. In the **Create work order** dialog box, set the following values:
   - **Work order type:** *Service*
   - **Description:** *Repair Truck*
   - **Customer account:** *US-013* (*Pelican Wholesales*)
   - **Asset:** *VE-102*

     > ⓘ **Note**
     >
     > The lookup shows only assets that are linked to the selected customer account.

   - **Maintenance job type:** *Repair*
   - **Trade:** *Mechanic*
   - **Service level:** *4*
4. Select **OK**.

Step 8: Review the work order and start to work on it

In this section, you will work on the work order that you created in the previous section.

1. Follow these steps to verify that the parent project is the *Pelican wholesales Work order* project:
   a. In the **Work order maintenance jobs** section, select a line.
   b. On the **Line details** FastTab, inspect the **Project ID** value. It should be a hyphenated number in the form <*Parent-Project-ID*>-<*Project-ID*>. This value is a link.
   c. Select the project ID link to open a page where you can view the parent project and project names.

2. On the Action Pane, on the **Work order** tab, in the **Lifecycle state** group, select **Update work order state**.
3. In the **Update work order state** dialog box, in the **Select** column, select the check box for the row where the **Lifecycle state** field is set to *In progress*.
4. Select **OK**.
5. In the **Lifecycle state: InProgress** dialog box, select **OK**.

Step 9: Post the hours that were spent on the work order and create a new invoice proposal

In this section, you will continue to work on the work order that you worked on in the previous section.

1. On the Action Pane, on the **Work order** tab, in the **Project** group, select **Journals**.

   The **Work order journals** page appears. Here, you can register the time that you spent on the work order.
2. On the **Hours** FastTab, select **Add line**.
3. On the new, line, set the **Hours** field to *4*.
4. On the Action Pane, select **Post journals**.
5. Close the **Work order journals** page to return to the work order.
6. On the Action Pane, on the **Invoicing** tab, select **New invoice proposal**.
7. In the **Create invoice proposal** dialog box, in the **Project transactions** section, select the **Select** check box for every line that you want to invoice.
8. Select **OK** to close the dialog box and view the new invoice proposal.

Credit limit check

The credit limit check feature checks work order dispatches to confirm whether a forecasted work order will cause the customer to exceed their credit limit. If the credit limit is exceeded, a warning or error is triggered based on the feature's configuration.

Each customer's total outstanding credit is calculated as the sum of outstanding sales orders and project credit (work orders accrue toward project credit).

Prerequisites

To use the credit limit feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature named *Credit limit check on work order dispatch* must be turned on in feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Configure the credit limit check

To configure the credit limit check, follow these steps:

1. Go to **Asset Management** > **Setup** > **Asset management parameters**.
2. Open the **Work orders** tab.
3. Make the following settings on the **Credit limits** FastTab.

    - **Credit limit check on work order dispatch** – Set to *Yes* to turn on the credit limit check.
    - **Message when exceeding credit limit** – Choose what to do to if the limit is exceeded. Select one of the following values.
        - *Warning* – Display a warning but allow the user to override the warning and proceed with dispatching the work order.
        - *Error* – Display an error and block the user from dispatching the order.

Set customer credit limits

To set the credit limit for a customer, follow these steps:

1. Go to **Credit and collections** > **Customers** > **All customers**.
2. Select and open a customer record (if you're following the examples provided previously, you could choose *US-013* (*Pelican Wholesales*)).
3. On the **Credit and collections** FastTab, enter the customer's credit limit in the **Credit limit** field.
4. On the Action Pane, select **Save**.

Last updated on 07/27/2026

Integrate asset management with fixed assets

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

By integrating the **Asset management** and **Fixed assets** modules, you can link fixed assets with maintenance assets. Fixed assets users can then create a maintenance asset from a new or existing fixed asset, and Asset management users can associate a maintenance asset with an existing fixed asset. This feature also makes it easy for Fixed assets users to view the costs that were posted from work orders for related maintenance assets.

> **Note**
>
> In this article, *maintenance assets* refer to assets from the **Asset management** module, and *fixed assets* refer to assets from the **Fixed assets** module.

Set a default location for new maintenance assets that are created from fixed assets (optional)

This optional configuration lets you set a default functional location for new maintenance assets that are created from fixed assets. You typically complete this configuration this just one time, if you complete it at all.

To complete the configuration, follow these steps:

1. Go to **Asset management** > **Setup** > **Asset management parameters**.
2. On the **Fixed assets** tab, in the **Functional location** field, select the default location.
3. On the Action Pane, select **Save**.

Work with integrated maintenance assets and fixed assets

This section provides a collection of procedures that show various ways that you can work with the integrated Asset management and Fixed assets features.

Associate an existing maintenance asset with a fixed asset

To associate an existing maintenance asset with a fixed asset, follow these steps:

1. Go to **Asset management** > **Assets** > **All assets** (or **Active assets**).
2. Select an asset.
3. On the **Fixed asset** FastTab, in the **Fixed asset number** field, select an existing fixed asset.
4. On the Action Pane, select **Save**.

View the fixed asset that is associated with a selected maintenance asset

To view the fixed asset that is associated with a selected maintenance asset, follow these steps:

1. Go to **Asset management** > **Assets** > **All assets** (or **Active assets**).
2. Select an asset.
3. On the **Fixed asset** FastTab, in the **Fixed asset number** field, select the link.

   The **Fixed assets** page for the selected asset is opened. (Typically, this page is found at **Fixed assets** > **Fixed assets** > **Fixed assets**.)

View the maintenance asset that is associated with a selected fixed asset

To view the maintenance asset that is associated with a selected fixed asset, follow these steps:

1. Go to **Fixed assets** > **Fixed assets** > **Fixed assets**.
2. Select an asset.
3. On the Action Pane, on the **Asset management** tab, in the **View** group, select **Maintenance asset**.

   The **Maintenance asset** page for the asset that is associated with the selected fixed asset is opened. (Typically, this page is found at **Asset management** > **Assets** > **All assets**.)

View maintenance costs that are associated with a fixed asset

Asset management work orders can be posted for maintenance assets, and each of those work orders typically has a cost. When a fixed asset is associated with a maintenance asset, you can go directly from the fixed asset to view the related costs.

To view the maintenance costs that are associated with a fixed asset, follow these steps:

1. Go to **Fixed assets** > **Fixed assets** > **Fixed assets**.
2. Select an asset.
3. On the Action Pane, on the **Asset management** tab, in the **View** group, select **Maintenance cost**.

   The **Maintenance cost** page is opened and shows the related costs.

Create a new maintenance asset for an existing fixed asset

To create a new maintenance asset for an existing fixed asset, follow these steps:

1. Go to **Fixed assets** > **Fixed assets** > **Fixed assets**.
2. Select an asset.
3. On the Action Pane, on the **Asset management** tab, in the **New** group, select **Create maintenance asset**. (If this option is unavailable, a maintenance asset might already be associated with the selected fixed asset.)
4. Finish creating the asset as described in Create an asset.

Create a new fixed asset and add a new maintenance asset for it

To create a new fixed asset and add a new maintenance asset for it, follow these steps:

1. Go to **Fixed assets** > **Fixed assets** > **Fixed assets**.
2. On the Action Pane, select **New**.
3. Finish creating the fixed asset as described in Create a fixed asset.
4. On the Action Pane, on the **Asset management** tab, in the **New** group, select **Create maintenance asset**.
5. Finish creating the asset as described in Create an asset.

Remove the association between two assets

In some cases, you might have to disassociate a maintenance asset from its fixed asset. For example, if you want to create a new maintenance asset from a fixed asset, you must first remove any existing association.

To remove an existing association between a maintenance asset and a fixed asset, follow these steps:

1. Go to **Asset management** > **Assets** > **All assets** (or **Active assets**).
2. Find and open the maintenance asset.
3. On the **Fixed asset** FastTab, clear the value from the **Fixed asset number** field.
4. On the Action Pane, select **Save**.

Last updated on 12/30/2025

Preventive maintenance overview

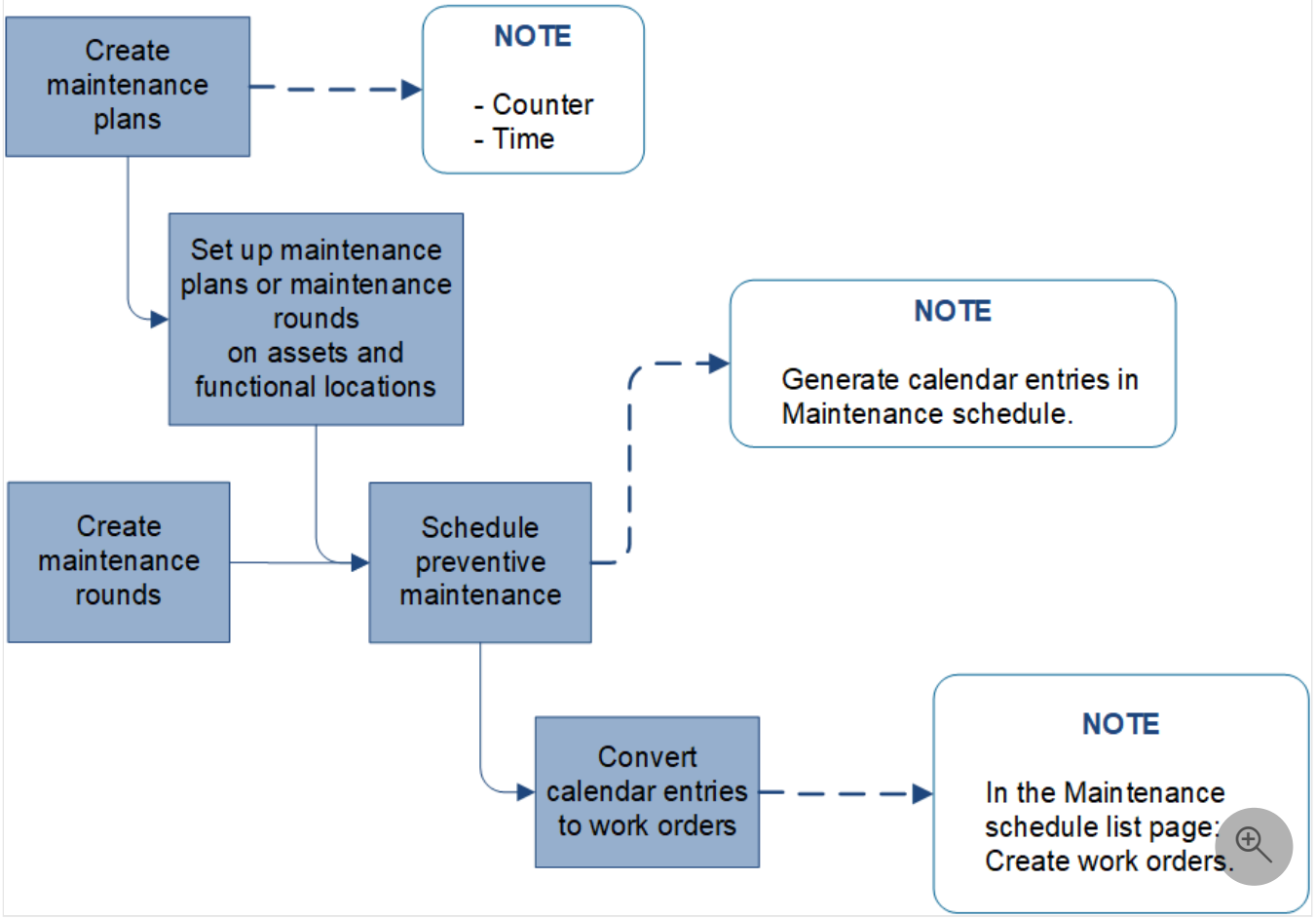
08/12/2025

This article explains preventive maintenance in Asset Management. Preventive maintenance is a discipline involving planned maintenance jobs, for example, regular service, calibration, and inspections. In Asset Management, you can create maintenance plans and set them up on assets and functional locations. You can also set up maintenance rounds on functional locations. Maintenance plans on assets are active regardless of where the asset is installed. Maintenance plans and maintenance rounds on functional location are active for the assets currently installed at the location. Instead of setting up maintenance plans on assets, or setting up maintenance rounds on functional locations, you can create maintenance rounds that include multiple assets on which you need to perform related types of maintenance jobs in the same work routine. Maintenance rounds created from assets (instead of on functional locations) let you select several assets for one maintenance round, even if they aren't installed on the same functional location.

Maintenance plans are used for preventive maintenance on individual assets. Maintenance rounds are used for preventive maintenance on a group or a set of assets. Maintenance plans and maintenance rounds are used for generating work order proposals. The work order proposals are saved as maintenance schedule lines, which can be bundled and converted into work orders.

The following illustration provides an overview of the work flow from creating maintenance plans and maintenance rounds to creating work orders for assets, based on those maintenance plans and maintenance rounds.

Flow Diagram: Schedule Preventive Maintenance Work Orders



Maintenance plans

07/10/2025

A maintenance plan defines when a pre-planned preventive maintenance job is to be carried out on an asset. Maintenance plans can be related to assets, asset types, functional locations, or functional location types, but first you create the maintenance plans to be used in your company.

A maintenance plan can have multiple maintenance plan lines. Maintenance job type and interval are specified on the maintenance plan line. There are two types of maintenance plan lines:

- Time
- Counter

Maintenance plan lines of type "Time" are used for recurring planned maintenance based on a fixed time interval. Maintenance plan lines of type "Counter" are used for planned maintenance or reactive maintenance based on asset counter registrations. A maintenance plan may include several maintenance plan lines of both types.

> ⓘ **Note**
>
> If no counter values have been registered for a counter type on an asset, the maintenance plan lines are omitted.

First, you create the maintenance plans you require for your preventive maintenance jobs and select the asset types, assets, functional location types, and functional locations that should be related to each maintenance plan. Afterwards, if required, you can also add maintenance plans to an asset or a functional location, which is done in **All assets** > select asset > **Asset maintenance plans** FastTab, or **All functional locations** > select functional location > **Maintenance plans** FastTab.

If you add a maintenance plan to asset types or functional location types, it means that when you create new assets or functional locations with those asset types or functional location types, the asset or functional location will automatically be added to the maintenance plan. The start date of the relation to a maintenance plan will be the current date, which may need to be adjusted.

Set up maintenance plans

This section describes how to set up maintenance plan lines and provides examples of how they can be used.

1. Go to **Asset management** > **Setup** > **Preventive maintenance** > **Maintenance plans**.
2. Select **New** to create a new sequence.
3. Insert an ID in the **Maintenance sequence** field, and a name in the **Name** field.
4. In the **Plan date** field, insert the start date from which planning can be done on the maintenance plan. Note that time-based maintenance plan lines may have other plan dates.
5. Select "Yes" in the **Active** toggle button to activate the maintenance plan.

   > ⓘ **Note**
   >
   > If you deactivate a maintenance plan, no schedule posts will be created in the maintenance schedule when you run a schedule maintenance plan job.

6. The **Tolerance days before** and **Tolerance days after** fields relate to maintenance plan lines in which the **Suppress overlapping maintenance jobs** check box is selected (refer to step 17). The "Tolerance" fields are used to extend the interval in days in which, if several maintenance plans overlap, the most comprehensive / largest job is created as a maintenance schedule line during maintenance plan scheduling, while more frequent, overlapping jobs are omitted during maintenance plan scheduling. Insert number of days in the **Tolerance days before** field, for example "2".
7. If you have inserted a value in **Tolerance days before**, also insert number of days in the **Tolerance days after** field, for example "2".

   > ⓘ **Note**
   >
   > The example described in this and the previous step means that if several maintenance plan lines overlap, and **Suppress overlapping maintenance jobs** is selected for one or more lines, the period of omitting maintenance schedule lines is extended to a total of five days (the expected start date on the maintenance schedule line *and* two days before *and* two days after that date).

8. The fields in the **Details** group on the **Details** FastTab show number of maintenance plan lines set up on the maintenance plan, and number of assets and functional locations related to the maintenance plan.

9. On the **Lines** FastTab, select **Add time line** or **Add asset counter line** to create a new maintenance plan line.

10. Insert a description for the line in the **Work order description** field. The description is transferred to related work orders.

11. In the **Maintenance job type** field, select the job type to which the maintenance plan line is related.

12. In the **Maintenance job type variant** and **Trade** fields, select the variant and trade related to the maintenance job type.

13. In the **Finish within days** and **Finish within hours** fields, you can insert expected end date in days or hours. The expected end date is inserted relative to the expected start date, which is calculated when maintenance schedule lines are created. For example, you can insert "7" in the **Finish within days** field to indicate that the related job should be completed within a week from the expected start date.

14. In the **Interval type** field, select the type of interval to be used on the maintenance plan line, for example, "Repeated..." or "Once...". Refer to the Interval types overview table below for a description of the relation between interval types and line types.

15. The **Period** field only relates to time-based line types. Select the period type related to the period frequency.

16. In the **Period frequency** field, insert the number of times the line should be used for planning preventive maintenance jobs. Example: If you have created a line of type "Counter", and your counter is production quantity, and you insert the number "20000" in this field, new maintenance sequence lines are created during preventive maintenance scheduling every time you are expected to produce 20,000 more items.

17. The **Suppress overlapping maintenance jobs** check box relates to time-based as well as counter-based line types. Select the check box to delete maintenance schedule entries that are created on the same date. This is relevant if, for example, you have created a 1-month inspection line, a 6-month inspection line, and a 1-year inspection line. For the 1-year inspection you only want that inspection to be done, not the other two inspections, which would also fit in the time frame. In order to set up this example correctly, you set up the 1-year inspection line as the first line, the 6-month line as the second line, and the 1-month line as the third line, and you select the **Suppress overlapping maintenance jobs** check box for the 1-month and 6-month lines. That way you ensure that when you reach the 1-year mark, the inspections for one month and six months are omitted, and a maintenance schedule line is only created for the 1-year inspection line.

> **Note**
>
> The example described in this step shows that the most comprehensive job, which contains the largest number of tasks, and which is not done so often, should always be inserted as the first line. The more frequent jobs are then inserted as separate lines in the order of frequency, placing the most frequent job at the bottom of the list.

18. The **Counter** field only relates to counter-based line types. Select the counter type to be used on the line. If a counter type isn't active on a related asset, the maintenance plan line is omitted.

19. The **Asset counter time fence in days** field only relates to counter-based line types. Insert a number that defines how many days back counter registrations are checked when maintenance plan scheduling is done. This means how far back are data (existing counter registrations) used as basis for calculating the trend that determines how many maintenance schedule lines are created.

    > *Example:* If counter registrations are expected to be made once a month, you may insert the number '365' in this field because maintenance plan scheduling will always be based on the last 12 months and therefore create maintenance schedule lines based on the trend of the past year. On the other hand, if you insert the number '10' in this field, you expect counter registrations to be made more often, for example, on a daily basis. This means that when you schedule maintenance plan, counter registrations for the last 10 days are used as the basis for the scheduling of maintenance schedule lines.

20. The **Plan date** field only relates to time-based line types. If the maintenance plan line has another planning date than the entire maintenance plan, select a date in the **Plan date** field on the line.

21. In the **Service level** field, you can select a work order service level as a further delimitation on the maintenance plan line - to be used as a service level on work orders.

22. Select the **Auto create** check box if you want a work order to be automatically created according to the selected maintenance plan line when scheduling maintenance plans.

23. If you have selected the **Auto create** check box, you can select a work order type for the autocreated work order in the **Work order type** field. If you have selected the **Auto create** check box, and you don't select a work order type in this field, the work order type

selected in **Asset management** > **Setup** > **Asset management parameters** > **Work orders** link > **Preventive work order type** field is used.

24. Use the **Season from** and **Season to** fields to create a repeated time-based maintenance plan line within a 12-month period. *Example:* Equipment used for maintaining green areas requires service each spring within a predefined period. Insert the start date of the period to be repeated in the **Season from** field.

25. Insert the end date of the period to be repeated in the **Season to** field.

26. In the **Resulting period** field, the current period to be repeated is shown. When the current period has passed, and you start a new year, the period shown in this field will be updated to reflect the next period in the repeat sequence.

27. On the **Assets** FastTab, select the assets that should be related to the maintenance plan.

28. On the **Asset types** FastTab, select the asset types that should be related to the maintenance plan.

29. On the **Functional locations** FastTab, select the functional locations that should be related to the maintenance plan. If necessary, you can make the setup more specific by selecting a related asset type, manufacturer, and model.

30. On the **Functional location types** FastTab, select the functional location types that should be related to the maintenance plan.

> **Note**
>
> When work orders are manually created on assets that are covered by a vendor warranty, a dialog box is shown to make the user aware of the warranty. The creation of the work order can then be canceled. The check for a warranty relation is omitted for work orders that are automatically created.

Interval types overview

Expand table

Interval type and description	Line Type: Time	Line Type: Counter
Interval type: Repeated from plan date The count starts from the plan date used. When you schedule maintenance plans, maintenance	The **Plan date** on the maintenance plan line is used. If no plan date is selected on the line, the	The **Plan date** for the maintenance plan is used. If the counter has been replaced, the latest replacement date is used as the plan date.

Interval type and description	Line Type: Time	Line Type: Counter
schedule lines are created when the interval is reached.	**Plan date** for the maintenance plan is used. Example: If the number "3" is inserted in the **Period frequency** field, and "Year" is selected in the **Period** field, a new maintenance schedule line will be created once every 3 years.	
Interval type: Repeated from start date The count starts from the start date on the asset relation. The date is selected in the **All asset** details view > **Asset maintenance plans** FastTab > **Start date** field, or in the **All functional locations** details view > **Maintenance plans** FastTab > **Start date** field. When you schedule maintenance plans, a maintenance schedule line is created when the interval is reached.	The start date of the maintenance plan line on asset or functional location is used. If that field is blank, the **Plan date** for the maintenance plan is used.	The start date of the maintenance plan line on asset or functional location is used. If that field is blank, the **Plan date** for the maintenance plan is used.
Interval type: Repeated from last work order The count starts from the actual end date and time of the latest work order that was completed on the asset with that specific maintenance job type / maintenance job type variant / trade combination. That date and time are shown in the **Actual end** field in the **All work order** details view.	The actual end date and time of the work order completed on the asset with that specific maintenance job type / maintenance job type variant / trade combination. If no completed work order is found, one of the dates used in the "Repeated from start date" interval type is used instead.	The actual end date and time of the work order completed on the asset *and* the maintenance job type / maintenance job type variant / trade combination. is used. If the end date and time was left blank on the work order, one of the dates used in the "Repeated from start date" interval type is used instead.
Interval type: Once from plan date See description for the "Repeated from plan date" interval type. Only difference is that this interval type is to be used only once.	See description for "Repeated from plan date" interval type. This interval is typically used for a one-time maintenance or service job.	See description for "Repeated from plan date" interval type above. This interval is typically used for a one-time maintenance or service job. **Note 1:** This interval type is only relevant if the counter is replaced every time you carry out a maintenance or service job. If, for some reason, a counter has been replaced before the end of the

Interval type and description	Line Type: Time	Line Type: Counter
		planned interval, a new time is calculated for the job from the time of the counter replacement. **Note 2:** If the counter is replaced when completing the maintenance or service job, this interval type functions as the "Repeated from plan date" interval type.
Interval type: Once from start date See description for the "Repeated from start date" interval type. Only difference is that this interval type is to be used only once.	See description for "Repeated from start date" interval type. This interval is typically used for a one-time maintenance or service job.	See description for "Repeated from start date" interval type. This interval is typically used for a one-time maintenance or service job. **Note 1** above also applies to this interval type. **Note 3:** If the counter is replaced when completing the maintenance or service job, this interval type functions as the "Repeated from start date" interval type.
Interval type: Once reached above This interval type only relates to counters and is used for indicating an upper limit set up on the maintenance plan line. Maintenance schedule entries will have the expected start date and time of the counter registration, meaning these entries will be created with an expected start date equal to or earlier than the system date.	Not applicable	The counter interval indicates an upper limit. If that limit is exceeded when you create a counter registration, a maintenance schedule line is created when you schedule preventive maintenance.
Interval type: Once reached below This interval type only relates to counters and is used for indicating a lower limit set up on the maintenance plan line. Maintenance schedule entries will have the expected start date and time of the counter registration, meaning these entries will be created with an expected start date equal to or earlier than the system date.	Not applicable	The counter interval indicates a lower limit. If that limit is passed when you create a counter registration, a maintenance schedule line is created when you schedule preventive maintenance.
Interval type: Linked from start date This interval type only creates a maintenance schedule line once. A	See description for "Once from start date" above. Example: You create two	See description for "Once from start date" above. Example: You create two lines in a maintenance plan for a

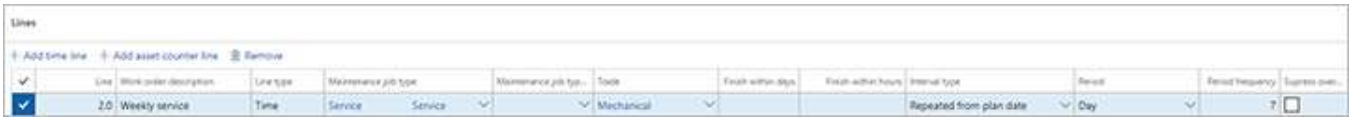
Interval type and description	Line Type: Time	Line Type: Counter
maintenance plan can contain more maintenance plan lines using this interval type, and those lines are linked. Typically, you'll create a maintenance plan that contains lines of only this interval type. Maintenance schedule lines are created by identifying the maintenance plan line that has the first expected start date and time.	lines in a maintenance plan for a service job on a car: one time-based line with a 1-year period, and one counter-based line with a 25,000 km limit. A maintenance schedule line is created for the limit that is reached first. For this line type you create the line with the 1-year period.	service job on a car: one time-based line with a 1-year period, and one counter-based line with a 25,000 km limit. A maintenance schedule line is created for the limit that is reached first. For this line type, you create the line with the 25,000 km limit. Example creating two counter lines: You can also set up a maintenance plan with two linked, counter-based lines in which the first line has a limit of 10,000 items quantity produced, and the second line relates to the machine or work center requiring service after running 3,000 hours.
Interval type: Linked from last work order This interval type only creates a maintenance schedule line once, based on the latest work order completed on the asset that matches the maintenance job type, maintenance job type variant, or trade combination. A maintenance plan can contain more lines using this interval type, and those lines are linked. Typically, you'll create a maintenance plan that contains maintenance plan lines of only this interval type. Maintenance schedule lines are created by identifying the maintenance plan line that has the first expected start date and time.	This interval type basically works as "Linked from start date" described above. Only difference is the date on which the interval type is based. The date used is the actual date and time on the latest work order completed on the asset *and* the maintenance job type / maintenance job type variant / trade combination.	This interval type basically works as "Linked from start date" described above. Only difference is the date on which the interval type is based. The date used is the actual date and time on the latest work order completed on the asset *and* the maintenance job type / maintenance job type variant / trade combination.
Interval type: Repeated on aggregated value (Counter only) When the maintenance plan is run, a scheduled maintenance line will be created each time that the accumulated value for an asset counter reaches the period frequency or an even multiple of the period frequency. (The period frequency is defined on the maintenance plan line.) For more information about how to enable and use this functionality,	Not applicable	**Example:** An hour counter is set up for asset AK-101. An asset plan line is also set up for the asset. The interval type of this line is *Repeated on aggregated value (Counter only)*, and the period frequency is *1000*. When the maintenance plan is run, a scheduled maintenance line will be generated when the aggregated value for the counter exceeds 1,000 hours. Then, when the aggregated value for the counter exceeds 2,000 hours, another scheduled maintenance line will be generated,

Interval type and description	Line Type: Time	Line Type: Counter
see the Counter-based maintenance enhancements section later in this article.		and so on for every additional 1,000 hours.
Interval type: Once on aggregated value (Counter only) When the maintenance plan is run, a scheduled maintenance line will be created when the accumulated value for an asset counter reaches the period frequency that is defined on the maintenance plan line. For more information about how to enable and use this functionality, see the Counter-based maintenance enhancements section.	Not applicable	**Example:** An hour counter is set up for asset AK-101. An asset plan line is also set up for the asset. The interval type of this line is *Once on aggregated value (Counter only)*, and the period frequency is *1000*. When the maintenance plan is run, a scheduled maintenance line will be generated when the aggregated value for the counter exceeds 1,000 hours.

> **Note**
>
> When maintenance schedule lines are created for time-based maintenance plan lines, expected time is always at the start of the day. For counter-based maintenance plan lines, expected time can be anytime during the day.

Here are some examples of how to set up time-based and counter-based maintenance plan lines:

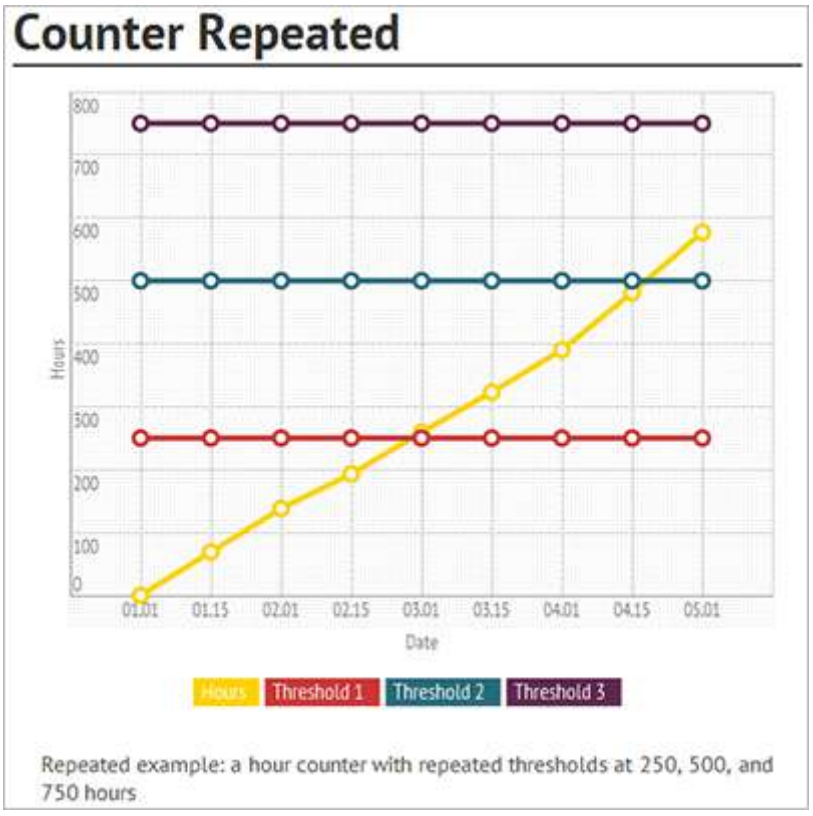
Example 1 - Time-based maintenance plan line: A lubrication job is set up in a fixed interval, occurring once a week. For that purpose, select "Repeated from plan date" in the **Interval type** field. See an example in the following illustration.



Example 2 - Time-based maintenance plan line: An inspection job is set up to be carried out approximately once a week. For that purpose, select "Repeated from last work order" in the **Interval type** field. See an example in the following illustration.

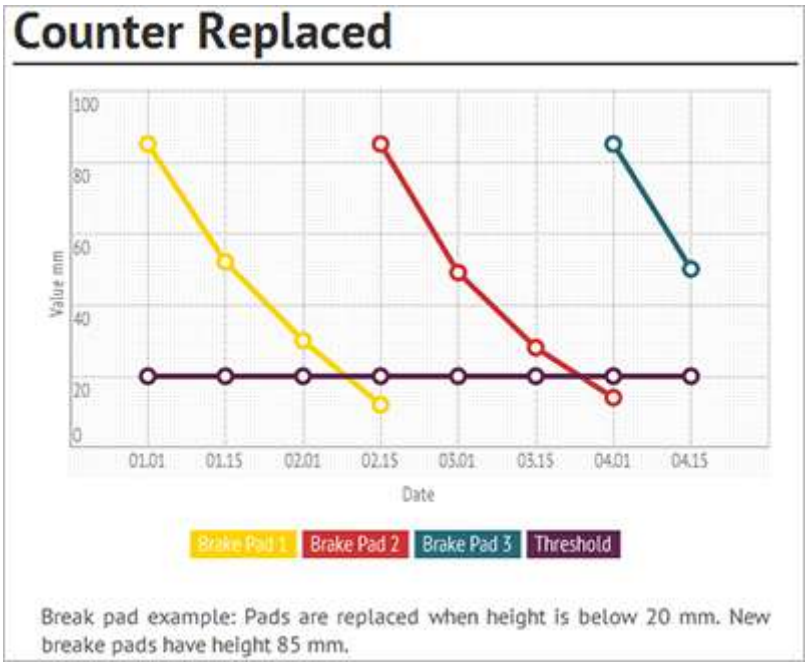
Example 3 - Counter-based maintenance plan line: The following graphic illustration shows an hour counter for which a new maintenance schedule line is created each time 250 hours have

passed. The interval type for this counter-based line is "Repeated from start date". The start date is the start date of the related assets in the **All assets** details view > **Asset maintenance plans** FastTab > **Start date** field, or in the **Functional location** details view > **Maintenance plans** FastTab > **Start date** field. This is an example of a *preventive* maintenance plan because the maintenance schedule line is automatically created each time the threshold (+ 250) is reached.

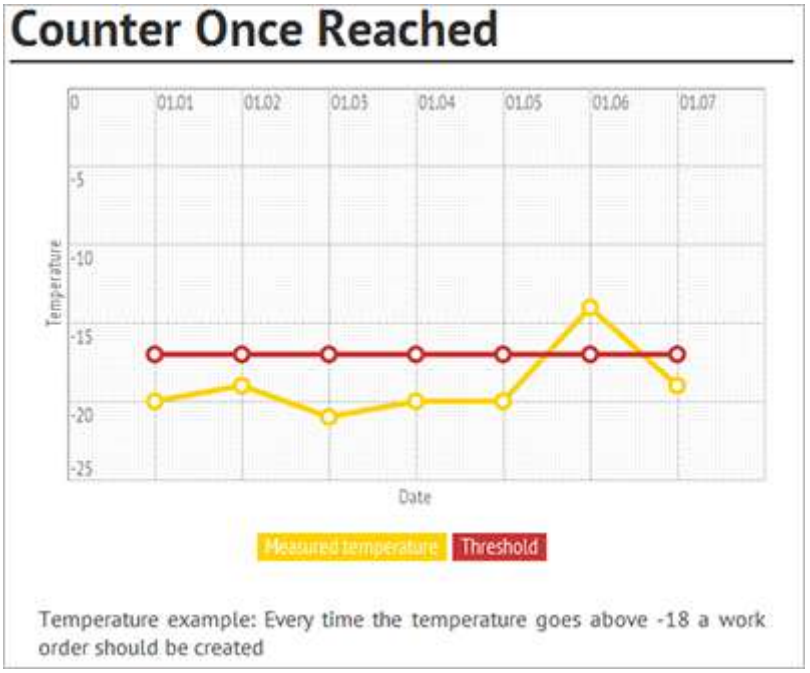


Repeated example: a hour counter with repeated thresholds at 250, 500, and 750 hours

Example 4 - Counter-based maintenance plan line: The following graphic illustration shows a decrease in counter value, measuring brake pad wear. A maintenance schedule line is created when a counter registration below 20 mm is created on the brake pad. The interval type for this counter-based line is "Once reached below" or "Once from last start date". This is an example of a *reactive* maintenance plan because the maintenance schedule line isn't created until a measurement below 20 mm is registered.



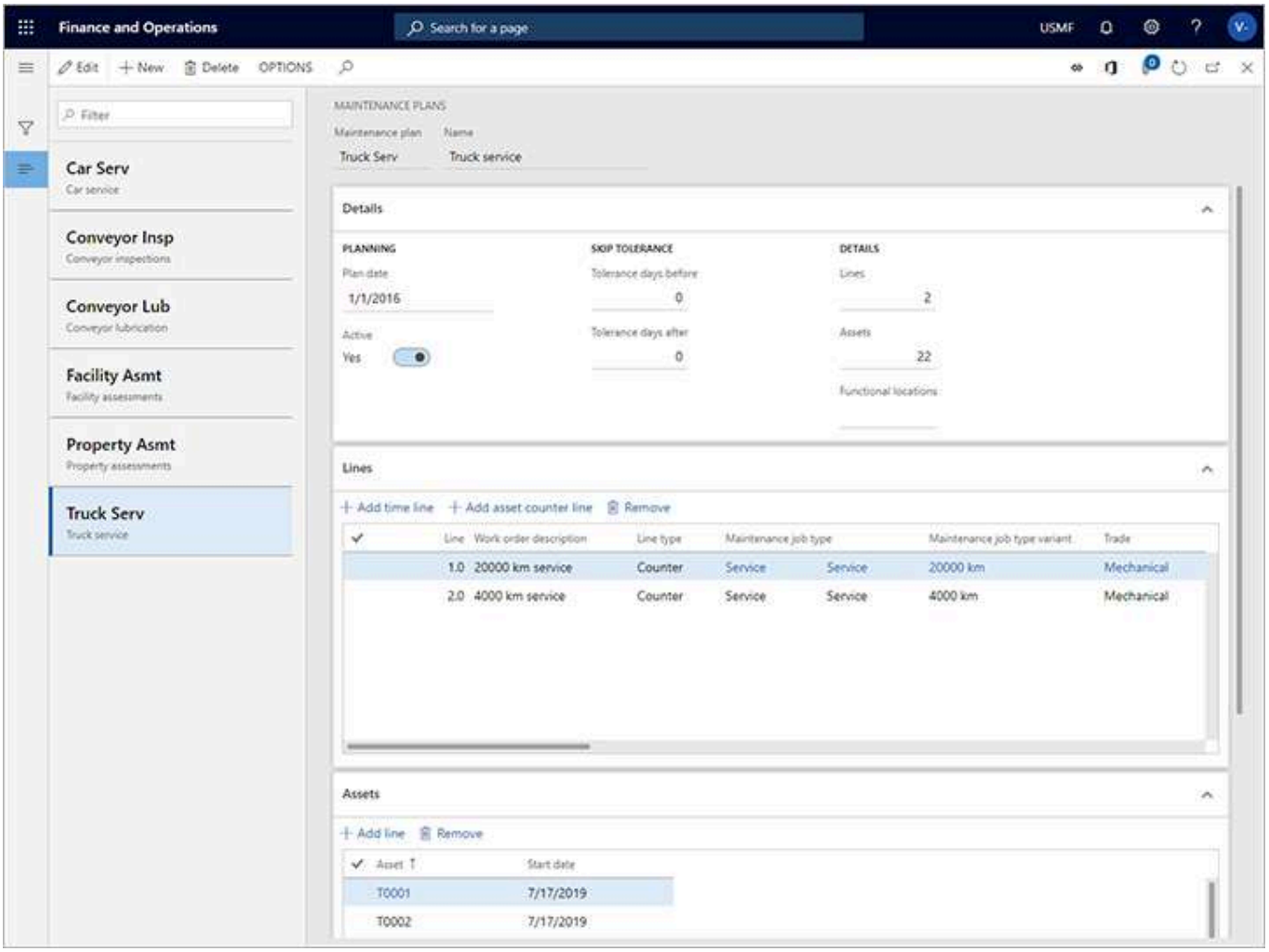
Example 5 - Counter-based maintenance plan line: The following graphic illustration shows a counter with a threshold of -18° Celsius. A maintenance schedule line is created when a counter registration above -18° Celsius is made. The interval type for this counter-based line is "Once reached above". This is an example of a *reactive* maintenance plan because the maintenance schedule line isn't created until a measurement higher than -18° Celsius is registered.



- When you create a new asset, and that asset uses an asset type related to a maintenance plan, the maintenance plan is automatically inserted in **All objects** > **Asset maintenance plans** FastTab. Also, in the **Asset type defaults**, on the **Maintenance plans** FastTab, the related maintenance plans will automatically be inserted.
- If you add or remove asset types or functional location types in **Maintenance plans**, that change will only reflect on new assets created after you made the change.

- If you add or remove assets or functional locations in **Maintenance plans**, that change will automatically be updated in **All assets** > **Asset maintenance plans** FastTab, or in **All functional locations** > **Maintenance plans** FastTab.

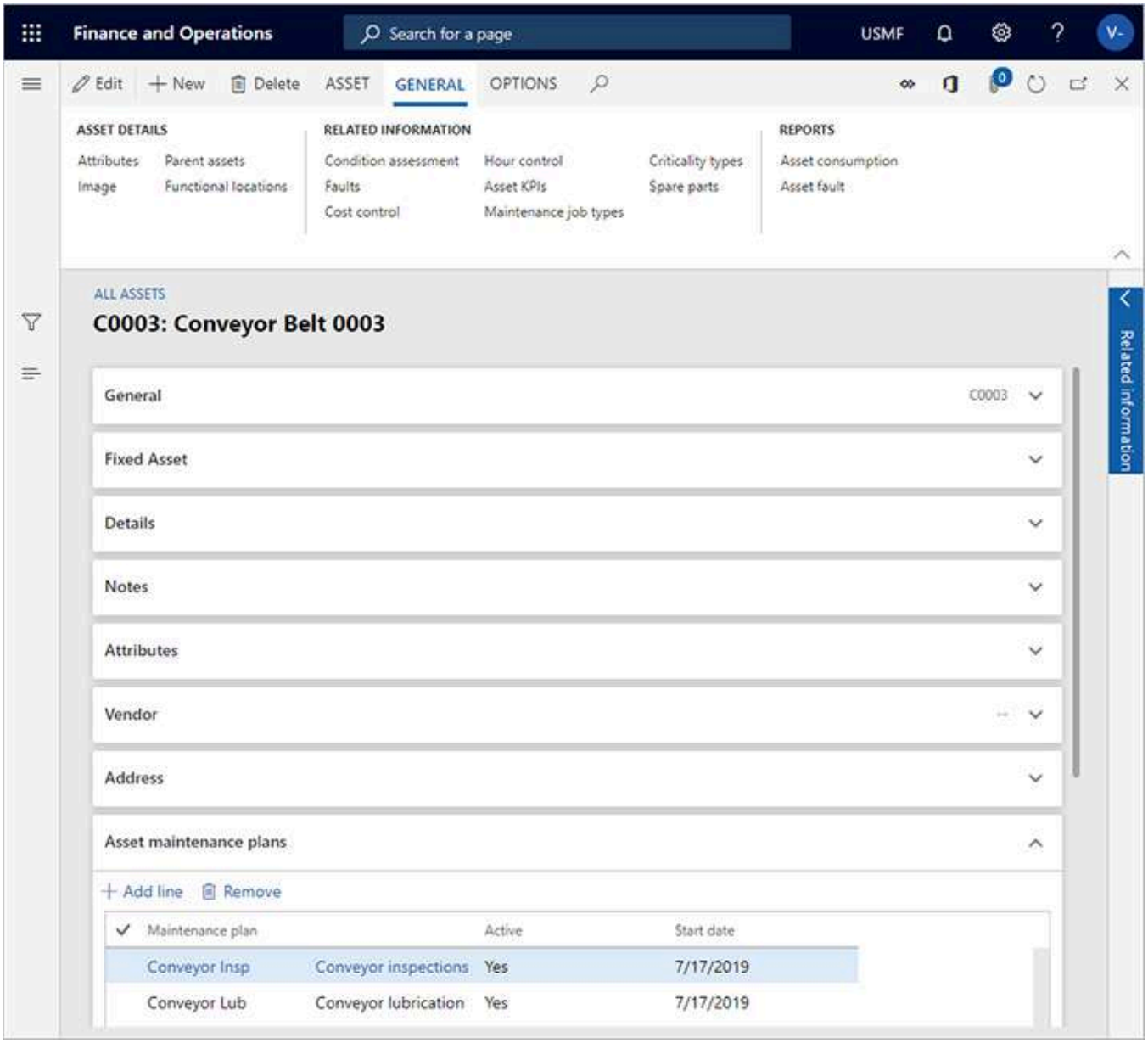
The following illustration shows an example of a "Truck service" maintenance plan on the **Maintenance plans** page.



Add a maintenance plan to an asset

1. Go to **Asset management** > **Common** > **Assets** > **All assets** or **Active assets**.
2. Select the asset on which you want to set up a maintenance plan and select **Edit**.
3. On the **Asset maintenance plans** FastTab, select **Add line** to add a maintenance plan to the asset.
4. In the **Maintenance plan** field, select the relevant maintenance plan.
5. In the **Start date** field, select the date from which planning of preventive maintenance jobs can be done.
6. Select **Save**. The **Active** field is automatically updated.

The following illustration shows an example of maintenance plans set up on an asset on the **All assets** page.



Counter-based maintenance enhancements

The *Counter-based maintenance enhancements* feature introduces the following functionality:

- The option to automatically insert a counter that has a value of *0* (zero) when an asset is created. This option can be useful when you use predictive maintenance that is based on counters. When the *Counter-based maintenance enhancements* feature isn't used, counters that have a value of *0* (zero) must be manually inserted.
- The ability to configure a counter so that it's automatically reset when a work order is completed. This functionality is useful when you want to schedule maintenance based on the aggregated value since the last work order was completed.
- A new type of maintenance plan interval that is named *Repeated on aggregated value (Counter only)*. This type triggers maintenance each time that an aggregated counter

reaches a multiple of a specific value. For example, maintenance can be triggered every 10,000 hours. Learn more in the Interval types overview section earlier in this article.

- Another new type of maintenance plan interval that is named *Once on aggregated value (Counter only)*. This type triggers maintenance when an aggregated counter reaches a specific value, such as 8,000 hours. Learn more in the Interval types overview section.

Create and initialize counters when an asset is created

Each time that you create an asset, related asset counters that are initialized to a value of *0* (zero) can automatically be created, provided that you set up your system and create the asset correctly.

1. Go to **Asset management** > **Setup** > **Asset types**.
2. Make sure that you have an asset type that is applicable to your planned new asset. Create an asset type as required. Make sure that all the relevant counters are selected on the **Counters** FastTab.
3. Go to **Asset management** > **Setup** > **Asset types** > **Counters**.
4. For each relevant counter, make sure that the **Total aggregate** field is set to *Sum*.
5. On the **All assets** page, create the asset.
6. Set the **Asset type** field to the asset type that you identified or created in step 2.
7. Finish setting up the new asset as required.
8. Go to **Asset management** > **Inquiries** > **Assets** > **Counters**. You should be able to find the initialized counters that are set up for your new asset.

> **Note**
>
> When initialized asset counters are created, the assumption is that the asset had never been used before it was added to the system. When the maintenance schedule is run for the first time, the calculation uses the date and the *0* (zero) counter value as a baseline for calculating future maintenance. If the asset wasn't new when it was added to the system, you can manually adjust the counter value so that it matches the actual counter value. To adjust a counter value, open the relevant asset on the **All assets** page, and then, on the Action Pane, on the **Asset** tab, in the **Preventative** group, select **Counters**. On the **Asset counters** page for the selected asset, adjust the value in the **Value** column for the initial counter record as required.

Automatically reset a counter value

You can configure the system to automatically reset a counter each time that a relevant work order reaches a selected status value.

1. Go to **Asset management** > **Setup** > **Preventive maintenance** > **Maintenance plans**.
2. In the list pane, select a maintenance plan. The counter reset will apply to all assets that use this plan.
3. In the **Lines** section, find an asset counter line that you want to reset a counter for, and select the **Reset counter** check box for that line. (Asset counter lines have a value in the **Counter** column. The counter that is specified in that column is the counter that will be reset for the relevant asset.)
4. Go to **Asset management** > **Setup** > **Work orders** > **Lifecycle states**.
5. In the list pane, select the work order lifecycle state that the relevant counter should be reset at.
6. On the **General** FastTab, set the **Reset counter** option to *Yes*.

Schedule maintenance plans

08/13/2025

Preventive maintenance scheduling generates calendar entries on assets based on the maintenance plans set up on the assets. You can schedule calendar entries based on selected maintenance plans, asset types, and assets.

1. Go to **Asset management** > **Periodic** > **Preventive maintenance** > **Schedule maintenance plans**.

2. You can select a time interval in the **Period** and **Period frequency** fields.

   > ⓘ **Note**
   >
   > The **Period** and **Period frequency** fields indicate how far ahead in time you want maintenance schedule lines to be created, based on the maintenance plans you have created (time-based or counter-based). In the figure shown later in this article, maintenance schedule lines (= work order proposals) are created from the current date and three months onwards.

3. Select *Yes* on the **Auto create if specified in the line** toggle button if work orders should automatically be created according to the maintenance plan line.

   > ⓘ **Note**
   >
   > If this toggle button is set to *Yes*, and the **Auto create** check box is also selected on maintenance plan lines in **Maintenance plans**, work orders are created based on the maintenance plan lines, and maintenance schedule lines with status *Work order created* are also created. If only one option is selected (**Auto create if specified in the line** toggle button in this dialog or **Auto create** check box in **Maintenance plans** form), only maintenance schedule lines are created with status *Created*. In that case, no work orders are created.

4. It's possible to generate calendar entries based on maintenance plans (time or counter), assets, asset types, functional locations, and functional location types. If necessary, select the **Filter** button and make your selections as needed.

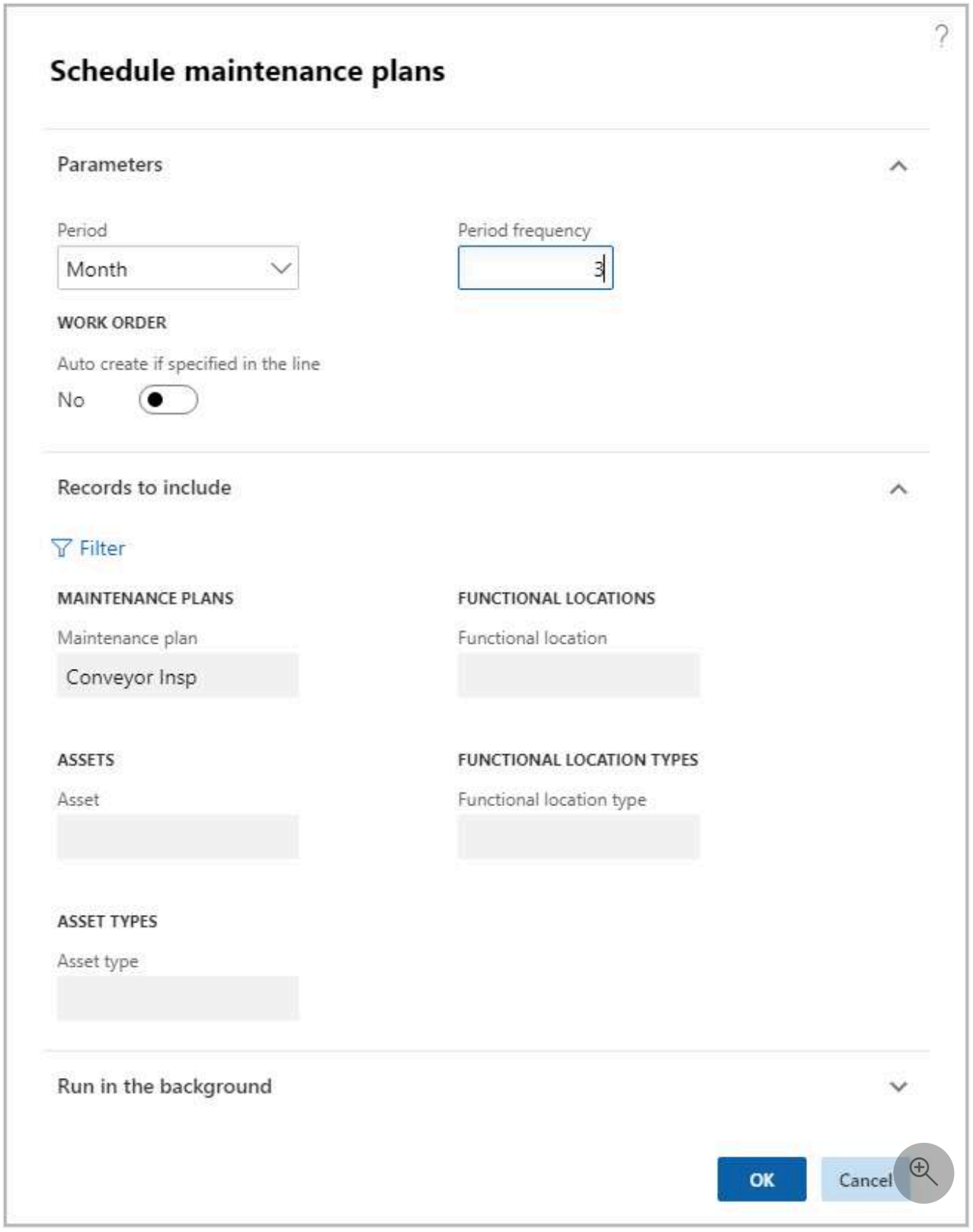
   Regarding scheduling of maintenance plans on functional locations: If you update the setup of asset types, manufacturers, and models on maintenance plans in **All functional locations** > **Maintenance plans** FastTab after you have scheduled maintenance plans, existing maintenance schedule entries related to that functional location are automatically

deleted. In order to create new calendar entries, which correspond with the updated maintenance plan setup on the functional location, you must run a new maintenance plan schedule for that functional location. Read more about the setup of asset types, manufacturers, and models on functional locations in Create functional locations.

Example: You want to create a maintenance plan for a specific functional location, meaning all assets set up on that functional location at any given time will be included when you schedule the maintenance plan. In that case, you create a maintenance plan and select the specific functional location, but you do NOT add any assets in the maintenance plan. The result is that when you schedule that maintenance plan, maintenance schedule lines will be created for all the assets related to the functional location at that time.

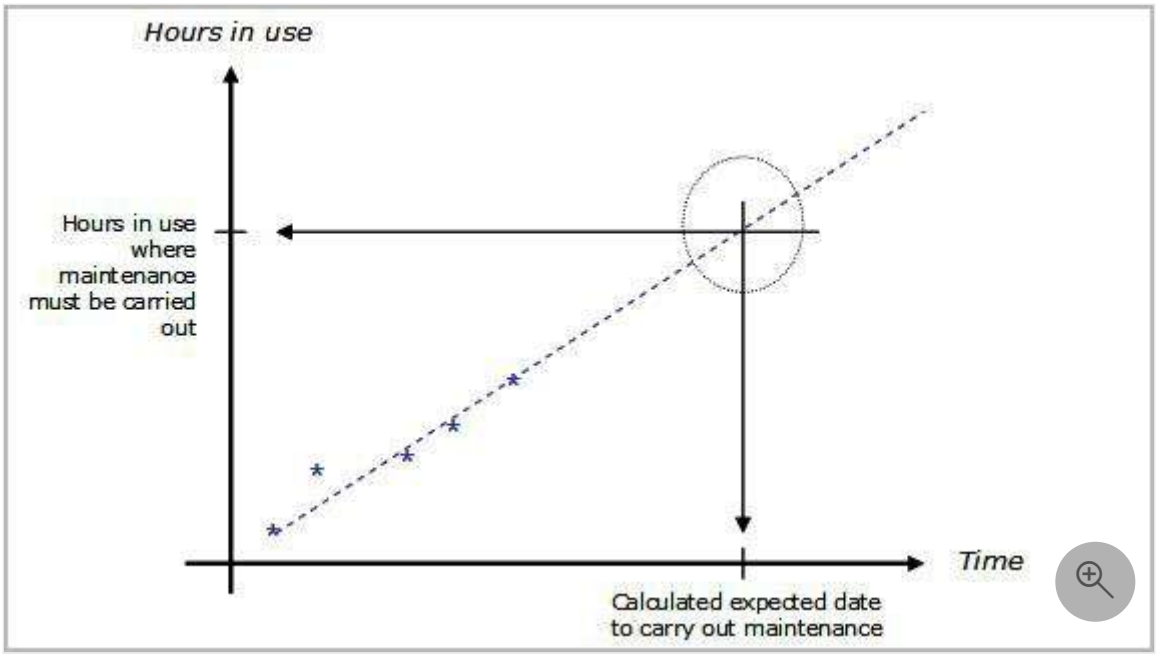
If you make changes to asset types, manufacturers, and models in **Asset types**, those changes only affect new assets that use the updated asset type. Read more about asset type setup in Asset types.

5. Select **OK** to start the generation of maintenance schedule entries on assets. The generated entries will be shown in the **All maintenance schedule** list page. The following illustration shows an example of the **Schedule maintenance plans** dialog.



- In the **Schedule maintenance plans** dialog, you can set up batch jobs on the **Run in the background** FastTab to automatically generate calendar entries at regular intervals.
- When you schedule preventive maintenance, maintenance schedule lines with expected start date and time earlier than the system date and time won't be created.

The following figure provides a graphic illustration of a time-based maintenance plan calculation.

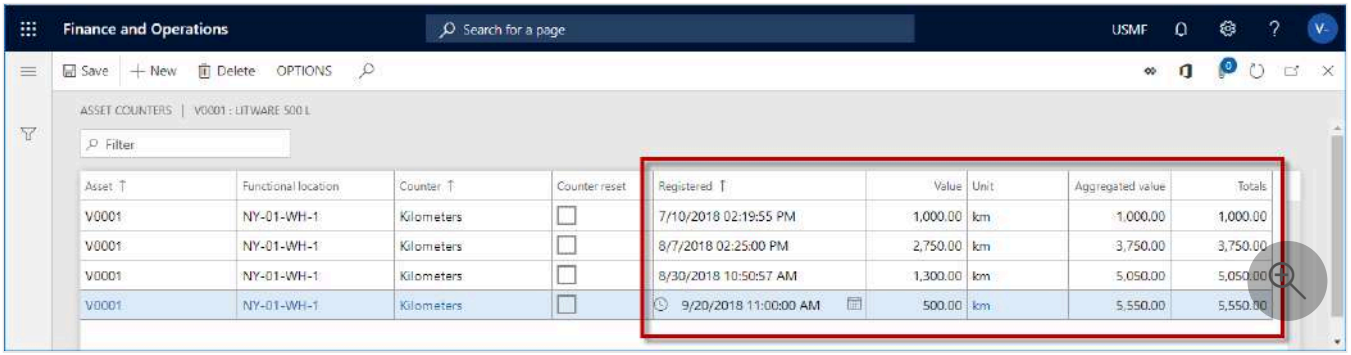


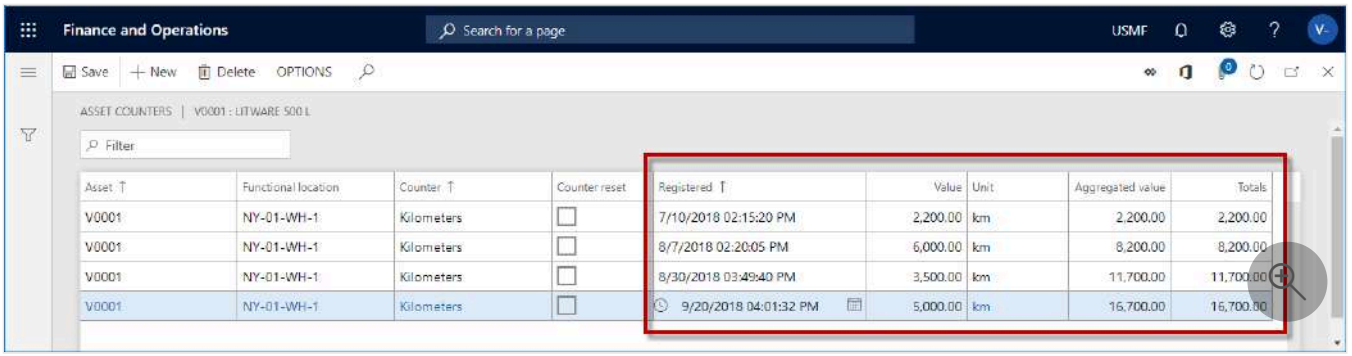
Regarding counter-based maintenance plans: In the following figures, two different counter registration cycles are shown. They're based on a maintenance plan set up for asset *V0001*, expecting the asset (a car) to run approx. 2,000 km every month.

- In the first example, the expected 2,000 km aren't reached every month. According to the counter-based maintenance plan, the threshold is 2,000 km, meaning when you run a maintenance plan scheduling, a maintenance schedule line should be created each time the 2,000-kilometer threshold is reached. In example 1, there are four registration lines, but the 2,000-kilometer threshold is only reached once. This means that when you run schedule maintenance plans this asset, for example for a three-month period, only one maintenance schedule line will be created.

- In the second example, 2,000 km or more are registered every month. Therefore, three maintenance lines would be created if you schedule maintenance plans for this asset for a three-month period.

The examples described here show that all counter registrations made on an asset show a trend describing wear and tear on the asset. That trend is used as calculation basis at the time of maintenance plan scheduling.




Finance and Operations
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ASSET COUNTERS | V0001 : LITWARE 500 L
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Counter
Counter reset
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Aggregated value
Totals
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V0001 NY-01-WH-1 Kilometers 8/7/2018 02:20:05 PM 6,000.00 km 8,200.00 8,200.00
V0001 NY-01-WH-1 Kilometers 8/30/2018 03:49:40 PM 3,500.00 km 11,700.00 11,700.00
V0001 NY-01-WH-1 Kilometers 9/20/2018 04:01:32 PM 5,000.00 km 16,700.00 16,700.00

Maintenance rounds

Article • 08/28/2024

In Asset Management, you can create maintenance rounds for various assets, on which you need to carry out a similar task at regular intervals. For example, lubrication jobs or safety inspection jobs that need to be carried out on several machines within the same intervals. First step is to create a maintenance round, including assets that require the same form of maintenance job. Next, you schedule the maintenance rounds. When you have completed the maintenance rounds schedule, you can see all the job records relating to the round in the **All maintenance schedule** and **Open maintenance schedule lines**.

> **Note**
>
> Maintenance rounds can also be set up on functional locations to be completed on the assets installed on the functional location at the time of creation of the round-based work order. For more information on the setup of maintenance rounds on functional locations, see Create functional locations.

Set up a maintenance round

1. Go to **Asset management** > **Setup** > **Preventive maintenance** > **Maintenance rounds**.
2. Select **New** to create a new maintenance round.
3. Make the following settings in the heading of the new record:
    - **Maintenance round** – Enter a unique ID for the new record.
    - **Name** – Enter a name for the maintenance round.
4. Make the following settings on the **Header** FastTab:
    - **Start date** – Select a start date for the round.
    - **Finish within days** – Enter the number of days the job should take to complete. The expected end date is calculated relative to the start date, which is calculated when maintenance schedule lines are created. For example, you can enter *7* in this field to indicate that the related job should be completed within a week of the start date.
    - **Finish within hours** – Enter the number of hours the job should take to complete.

- **Auto create** – Set to *Yes* if work orders should automatically be created from maintenance schedule lines that are created from this maintenance round.
- **Work order type** – Select the work order type to be used on work orders created from this maintenance round.
- **Service level** – Select the work order service level to be used on work orders created from this maintenance round.
- **Active** – Set to *Yes* to activate the maintenance round.
- **Assets** – This read-only value shows the number of assets related to the maintenance round.
- **Lines** – This read-only value shows the number of lines related to the maintenance round.

> **Important**
>
> If you set **Active** to *No*, the system won't add any lines for this round to the maintenance schedule when the *Schedule maintenance rounds* job runs.

5. On the **Asset lines** FastTab, select **Add** to add an asset to the maintenance round. Make the following settings for the new line:

   - **Line number** – A line number is automatically inserted in this field to indicate the sequence of the assets in the maintenance round.
   - **Asset** – Select the asset to be included in the maintenance round.
   - **Maintenance job type** – Select the maintenance job type for the asset.
   - **Maintenance job type variant** – If necessary, select a maintenance job type variant related to the maintenance job type.
   - **Trade** – If necessary, select a trade related to the maintenance job type.
   - **Period type** – Select the recurrence (such as *Day*, *Week*, or *Month*) for the maintenance round.
   - **Period frequency** – Enter the number of recurrences for the maintenance round. For example, if you have selected *Day* in the **Period type** field, and you enter the number *7* in this field, new maintenance round lines are created during preventive maintenance scheduling once a week.
   - **Start date** – Select a start date for the asset to be included in the maintenance round. This date might differ from the start date set on the maintenance round.

6. Repeat the previous step to add more assets to the maintenance round.

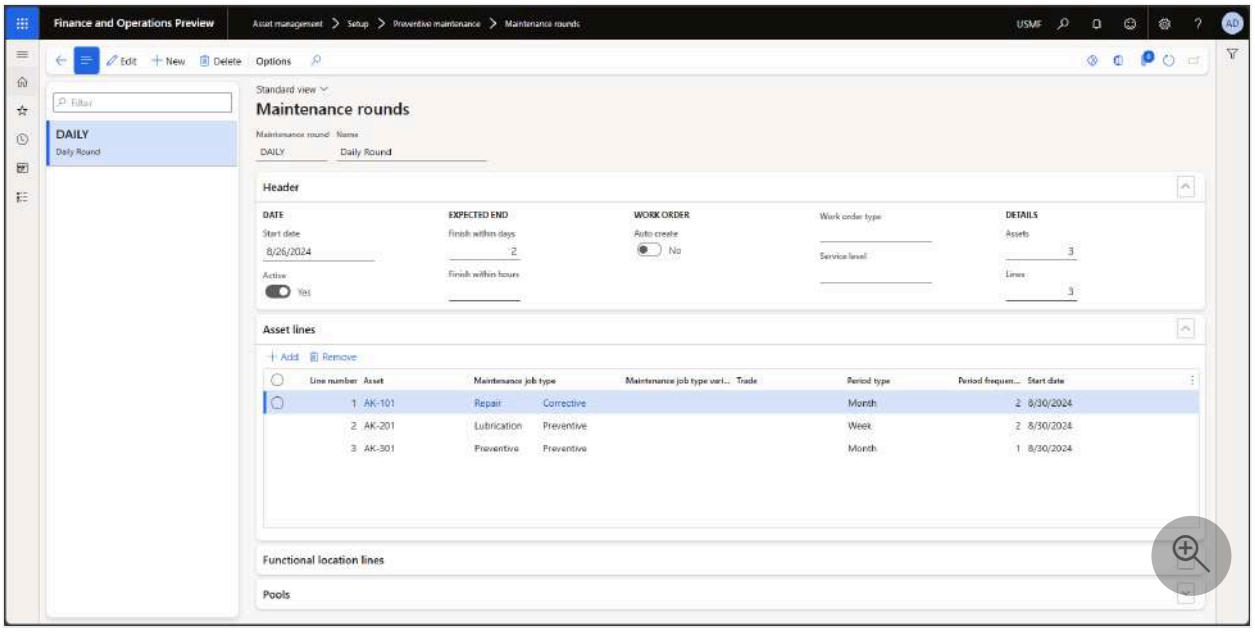
7. On the **Functional location lines** FastTab, select **Add** to add a functional location to the maintenance round. Refer to the descriptions of the related fields provided

previously. The same fields are available as for creating an asset line, but you can also select **Manufacturer** and **Model** for a functional location, if necessary. If you only select a functional location on a line, but make no selections in **Asset type**, **Manufacturer**, **Model**, **Maintenance job type**, **Maintenance job type variant**, and **Trade**, all assets related to that functional location at the time of maintenance scheduling will be included in the maintenance round.

8. On the **Pools** FastTab, select **Add** to select a work order pool to be included in the maintenance round. Several work order pools can be connected to one maintenance round.

9. On the Action Pane, select **Save**.

The following illustration shows an example of a maintenance round containing three assets.



Schedule maintenance rounds

When you've set up a maintenance round, you run a schedule job to schedule all the jobs related to the maintenance round.

1. Do one of the following steps:

    - Go to **Asset management** > **Periodic** > **Preventive maintenance** > **Schedule maintenance rounds**.

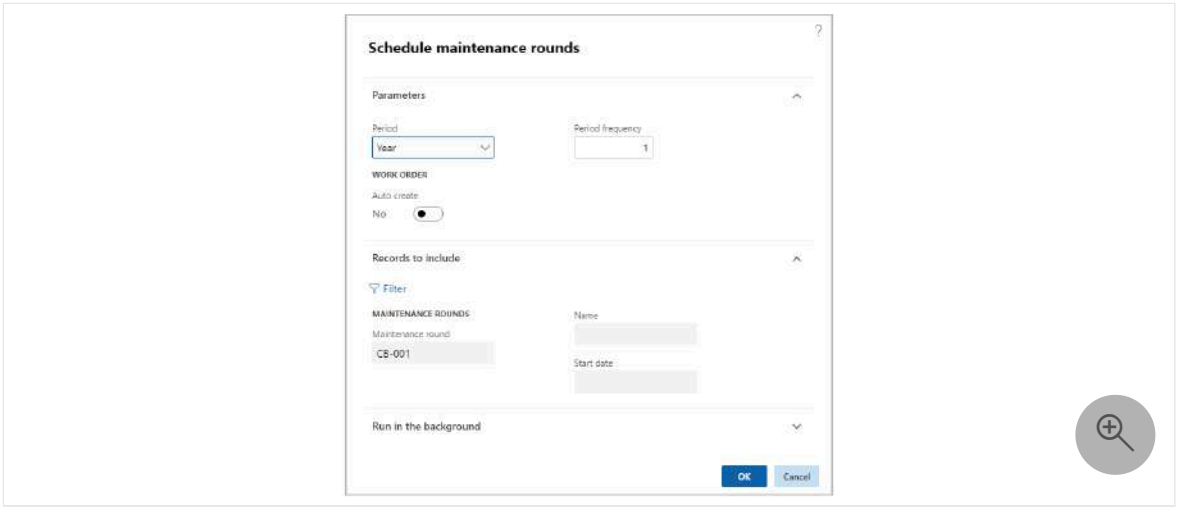
    - Go to **Asset management** > **Maintenance schedule** > **All maintenance schedule**. Select a maintenance schedule line in the grid. On the Action Pane,

open the **Maintenance schedule** tab and, from the **Schedule** group, select **Maintenance rounds**.

- Go to **Asset management** > **Maintenance schedule** > **Open maintenance schedule lines**. Select a maintenance schedule line in the grid. On the Action Pane, open the **Maintenance schedule** tab and, from the **Schedule** group, select **Maintenance rounds**.

- Go to **Asset management** > **Maintenance schedule** > **Open maintenance schedule pools**. Select a maintenance schedule line in the grid. On the Action Pane, open the **Maintenance schedule** tab and, from the **Schedule** group, select **Maintenance rounds**.

2. The **Schedule maintenance rounds** dialog opens.



Make the following settings:

- **Period** – Select the period type to use for the scheduling job.
- **Period frequency** – Enter the number of periods to be included in the scheduling job. The start of the scheduling is the current date.
- **Auto create** – Set to *Yes* if a work order should automatically be created based on a maintenance round.

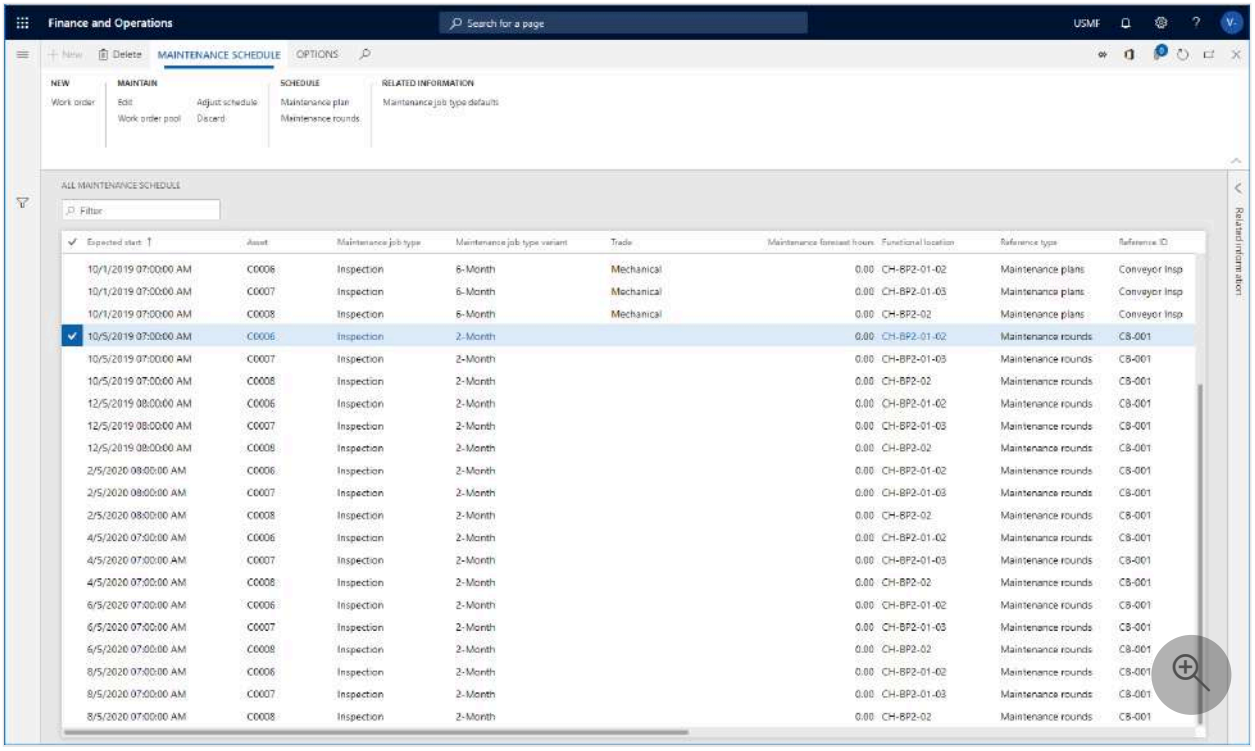
> **Note**
>
> If **Auto create** is set to *Yes* on the **Schedule maintenance rounds** dialog, and **Auto create** is also set to *Yes* on the **Maintenance rounds** page, the system creates work orders based on the maintenance round lines and also creates maintenance schedule lines with status *Work order created*. If only one of the **Auto create** toggle buttons is set to *Yes*, the system only creates maintenance schedule lines with status *Created*. In that case, no work orders are created.

3. If necessary, you can select specific rounds or another start date for the schedule job. To do so, select **Filter** and add the rounds to be included.

4. Select **OK**.

5. You're now able to see the maintenance rounds jobs in **Asset management** > **Maintenance schedule** > **All maintenance schedule** or **Open maintenance schedule lines**. If the scheduled rounds are connected to a work order pool, you also see maintenance schedule lines in **Open maintenance schedule pools**. Maintenance schedule lines created from a round have the reference type *Maintenance rounds*.

The following illustration shows the maintenance schedule lines listed on the **All maintenance schedule** page.



- When work orders are manually created on assets that are covered by a vendor warranty, a dialog box is shown to make the user aware of the warranty. The creation of the work order can then be canceled. The check for a warranty relation is omitted for work orders that are automatically created.
- You can set up a batch job on the **Run in the background** FastTab to schedule rounds at regular intervals.
- If a round is included in several work order pools (refer to Work order pools), one record is shown for each pool in **Open maintenance schedule pools**. This is done to optimize the filtering options for work order pools.

Feedback

Was this page helpful? Yes No

Provide product feedback

Maintenance schedule

Article • 01/06/2025

The maintenance schedule contains a list of all the expected preventive maintenance plans, maintenance requests, and maintenance rounds to be carried out. Some schedule lines might have been converted to work orders.

The four maintenance schedule views are slightly different, depending on which maintenance schedule lines you want to see.

Expand table

Menu item	Description of contents
All maintenance schedule	All maintenance schedule lines are shown.
My asset schedule	All maintenance schedule lines containing assets installed on functional locations to which you're related as a worker (set up in Maintenance workers and worker groups) are shown in this list.
Open maintenance schedule lines	Maintenance schedule lines with status *Created* - meaning they haven't yet been converted to a work order or discarded - are shown in this list.
Open maintenance schedule pools	Maintenance schedule lines related to a work order pool are shown in this list.

> **Note**
>
> If a maintenance schedule line is included in several work order pools (refer to Work order pools), one record is shown for each pool in **Open maintenance schedule pools**. This is done to optimize filtering options on work order pools.

To open a maintenance schedule:

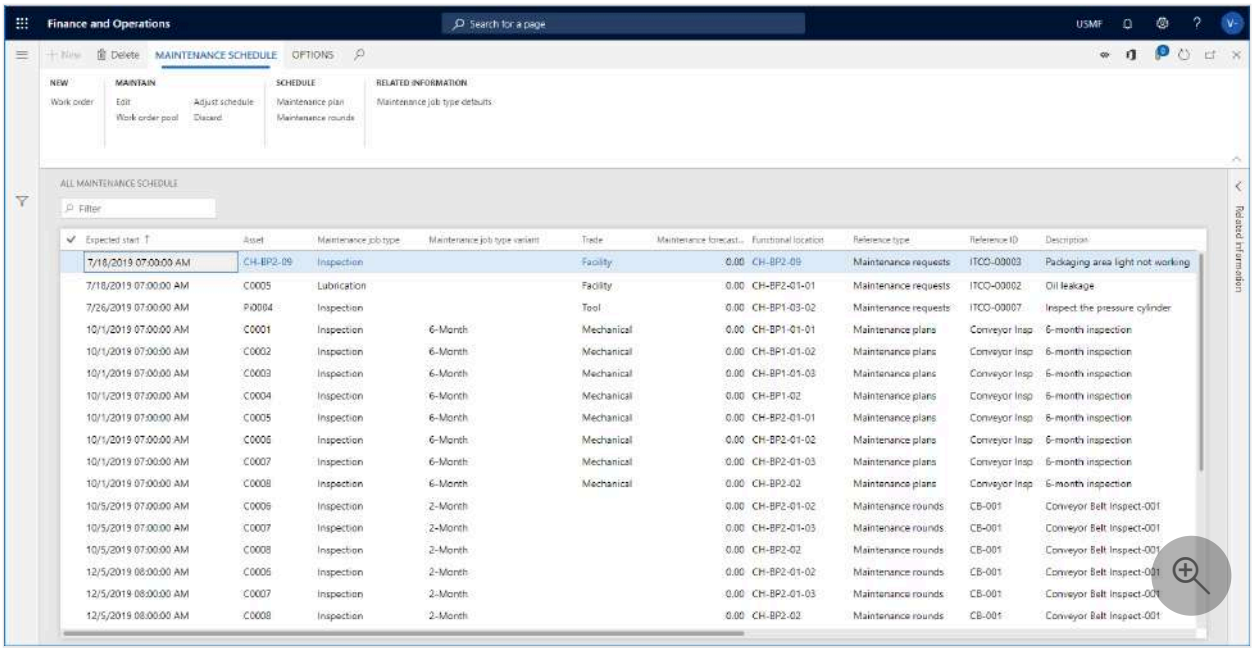
1. Select **Asset management** > **Maintenance schedule** > **All maintenance schedule** or **Open maintenance schedule lines** or **Open maintenance schedule pools**.
2. To update the maintenance schedule, select **Maintenance plan** or **Maintenance rounds**.
3. You can edit a maintenance schedule line by selecting it and selecting **Edit**. For example, you can easily update the service level or the worker responsible for the job. You can only edit maintenance schedule lines that haven't yet been connected to a work order.

4. You can delete a maintenance schedule line by selecting it in the list page and selecting **Delete**.
5. You can discard a maintenance schedule line by selecting it in the list page and selecting **Discard**. This function is useful if, for example, an asset has a 2,000-km maintenance plan and a 10,000-km maintenance plan. Then, you might want to discard the line created from the 2,000-km maintenance plan when it coincides with 10,000 km, 20,000 km, 30,000 km, and so on. If you discard a maintenance schedule line related to a maintenance plan, that line will never again appear when that maintenance plan is scheduled.
6. You can select several maintenance schedule lines in **All maintenance schedule** and select **Work order pool**, if you want all selected lines to be included in the same work order pool.
7. You can select several maintenance schedule lines in **All maintenance schedule** or **Open maintenance schedule lines** or **Open maintenance schedule pools** and select **Adjust schedule** if you want to make the same adjustments on multiple lines. You can change expected start and end dates, service level, and the responsible maintenance worker group or responsible maintenance worker related to the selected maintenance schedule lines.

When a maintenance schedule line has been related to a work order, the work order ID will be displayed in the **Work order** field.

In the **All assets** details view, on the **Asset maintenance plans** FastTab, you can select maintenance plans for the asset. Later, if you delete a maintenance plan line related to an asset in **All assets**, you also automatically delete all maintenance schedule lines with status *Created* that have been created based on that maintenance plan. Learn more in Create an asset.

The following below shows the **All maintenance schedule** list page.



Feedback

Was this page helpful? Yes No

Provide product feedback

Maintenance schedule cost

Article • 06/03/2022

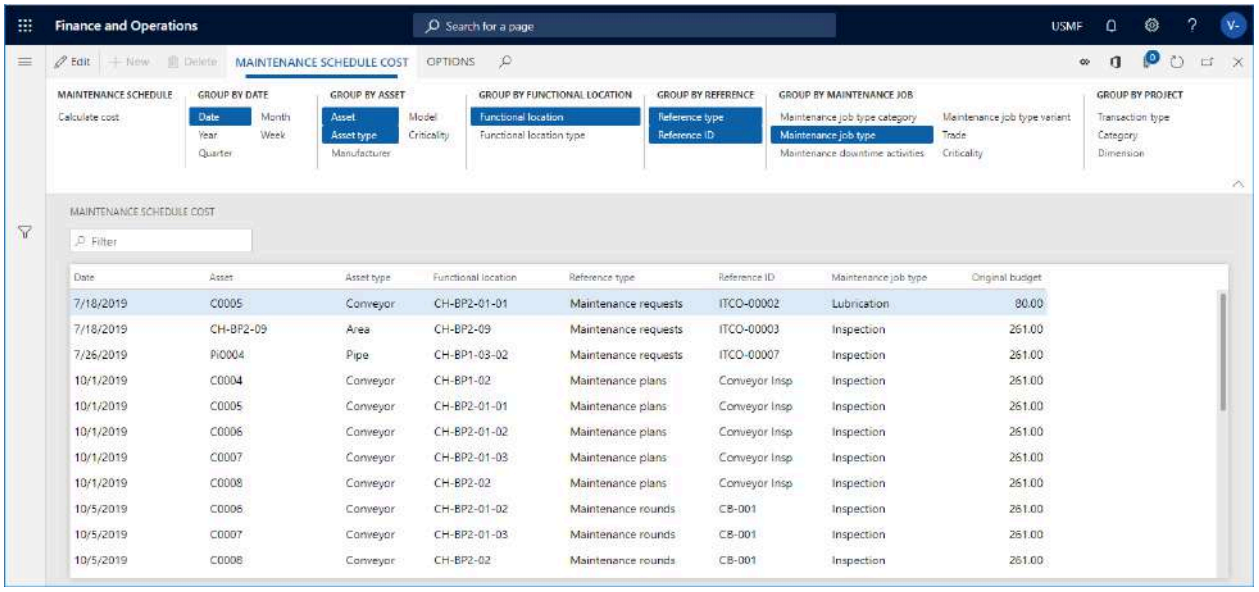
In Asset Management, you can calculate budget costs on maintenance schedule lines. This is useful if you want to get an overview of expected costs, for example, costs relating to planned preventive maintenance jobs for the next year. The calculations are based on existing maintenance schedule lines of type "Maintenance plans" and "Maintenance rounds" and "Maintenance requests".

1. Click **Asset management** > **Inquiries** > **Assets** > **Maintenance schedule cost**.
2. In the **Maintenance schedule cost** dialog, you can select a **Financial dimension set** if you want to see costs grouped in financial dimensions.

> ⓘ **Note**
>
> Financial dimension sets are set up in **General ledger** > **Chart of accounts** > **Dimensions** > **Financial dimension sets**.

3. You can use the **Level** field to indicate how detailed you want the maintenance schedule lines to be regarding functional locations. For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all maintenance schedule lines for a functional location will be shown on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level. If you insert the number "0" in the **Level** field, you will see a detailed result showing all maintenance schedule lines on all the functional location levels to which they are related.
4. If you want to make a calculation for specific assets, click **Filter** on the **Records to include** FastTab, and select the relevant assets. If required, you can also specify an **Expected start** date for the cost calculation or select a different **Status** for the cost calculation
5. Click **OK** to start the cost calculation.
6. On the **Maintenance schedule cost** tab > in the **Group by...** Action Pane groups, click the relevant buttons to show the required detail level of the cost calculation. The selected Action Pane group buttons are highlighted. Click on a button to activate or deactivate it.
7. Click the **Calculate cost** button if you want to make a new cost calculation.

The illustration below shows the results of a maintenance schedule cost calculation.



Feedback

Was this page helpful? Yes No

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Creating work orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

After you've scheduled preventive maintenance jobs, the next step is to create work orders for them. You can complete this step by using one of the maintenance schedules. The scheduled jobs in a maintenance schedule can have different reference types, as described in the following table.

Expand table

Reference type	Description
Maintenance plans	Preventive maintenance jobs that are based on the *Time* or *Counter* maintenance plan type.
Maintenance rounds	Preventive maintenance jobs that contain several assets that require a similar type of maintenance.
Maintenance request	A manually created request for maintenance or repair of an asset. This request can be converted to a work order.

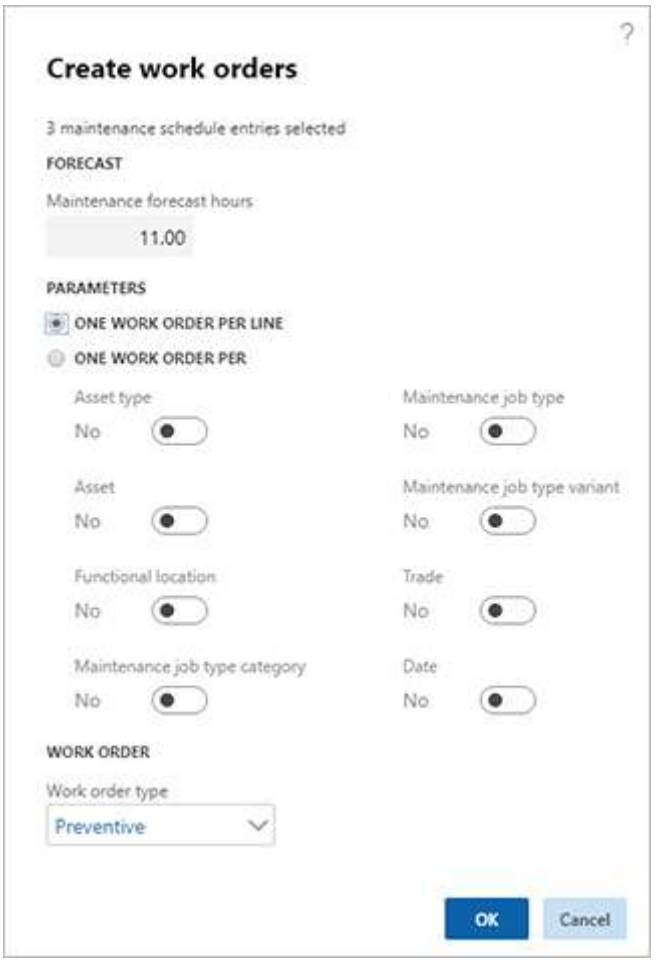
Create work orders based on your maintenance schedule

To create work orders that are based on your maintenance schedule, follow these steps:

1. Open one of the following pages, depending on how you want to select schedule items for your work orders:
    - All maintenance schedule (**Asset management** > **Management schedule** > **All maintenance schedule**)
    - Open maintenance schedule lines (**Asset management** > **Management schedule** > **Open maintenance schedule lines**)
    - Open maintenance schedule pools (**Asset management** > **Management schedule** > **Open maintenance schedule pools**)

2. In the grid, select the check box for every scheduled maintenance job that you want to create a work order for. Then, on the Action Pane, select **Work order**.

   The **Create work orders** dialog box appears. The **Maintenance forecast hours** field shows the total number of forecast hours for the selected lines.



3. In the **Parameters** section, specify the number of work orders that should be created. Select one of the following options:

   - **One work order per line** – Create one work order per maintenance schedule line.
   - **One work order per** – Create work orders that are grouped according to the settings of the other options that become available when you select this option.

4. In the **Work order type** field, select the work order type to use for all the work orders that you create.

5. Select **OK** to create the work orders according to your settings.

Group work order lines that are automatically created while a maintenance plan runs

This feature lets you define rules for grouping work order lines under a single work order when the system is set up to generate work orders automatically, based on a maintenance plan. Previously, automatically generated work orders could contain only one line. However, you can now group work orders by, for example, asset, asset type, or functional location. (Manually generated work orders could already be grouped in this way, as described in the previous section of this article.)

To set up grouping for automatically generated work orders, follow these steps:

1. Go to **Asset management** > **Setup** > **Preventative maintenance** > **Maintenance plans**.
2. For each plan where you want to generate grouped work orders, follow these steps:
   a. Select the plan in the list pane.
   b. On the **Lines** FastTab, make sure that the **Auto create** check box is selected on every line.
3. Go to **Asset management** > **Periodic** > **Preventive maintenance** > **Schedule maintenance plans**.
4. In the **Schedule maintenance plans** dialog box, in the **Period** section, specify the time horizon for the plan (how far to look ahead when finding scheduled maintenance jobs to generate work for).
5. Set the **Automatically create work order from schedule** option to *Yes*.
6. In the **Work order** section, select one of the following options:
   - **One work order per line** – Create one work order per maintenance schedule line.
   - **One work order per** – Create work orders that are grouped according to the settings of the other options that become available when you select this option.
7. If you want the options to apply when you run only some of your maintenance plans, on the **Records to include** FastTab, add filters as you require, just as you might do for other batch jobs in Microsoft Dynamics 365 Supply Chain Management.
8. On the **Run in the background** FastTab, set up batch and scheduling options as you require, just as you might do for other batch jobs in Supply Chain Management.
9. Select **OK** to run and/or schedule the selected maintenance plans.

Last updated on 12/30/2025

Maintenance downtime activities

Article • 06/15/2022

Maintenance downtime is used to get an overview of the capacity required to carry out maintenance jobs on specific assets during a specific period. For example, you can create a maintenance downtime registration for Production line 10 in Production Hall 29-A on production site 02. The maintenance downtime registration has a start and end time indicating the period in which the assets related to the maintenance stop are not available for production.

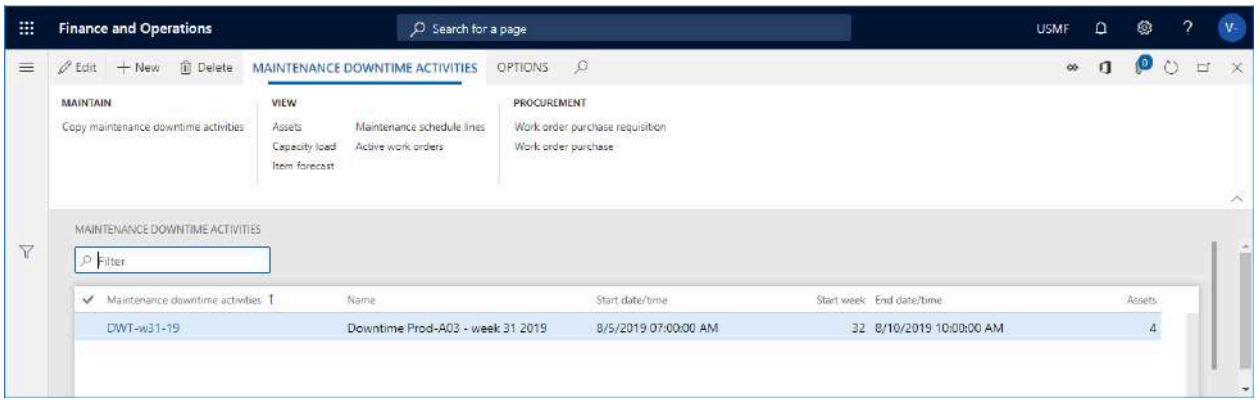
Maintenance downtime activities is an overview of maintenance schedule lines and work order maintenance jobs on related assets during a specified period. The lines related to work order maintenance jobs all have an expected start date within the maintenance stop period. You can extract useful information and make adjustments to planned maintenance jobs:

- Get an overview of required shut-down periods of production equipment (assets).
- Get an overview of planned maintenance (hours), grouped by competencies (responsible maintenance worker groups or maintenance workers), for example capacity load on electricians, smiths, or other maintenance work groups required to do the planned maintenance jobs.
- Make adjustments to maintenance schedule lines or work order maintenance jobs related to the assets, for example, change expected start and end times on a line, or select other maintenance workers to optimize the workflow for maintenance workers and maintenance worker groups.

When assets have been selected on a maintenance downtime registration, all open maintenance schedule lines and work order maintenance jobs relating to active work orders are included in the maintenance downtime registration.

Maintenance downtime activities

Click **Asset management** > **Maintenance downtime activities** > **All maintenance downtime activities** to open a list of all maintenance downtime activities and see some of the information related to the activities. Click on a link in the **Maintenance downtime activities** column to open the details view. The illustration below shows an example of the **Maintenance downtime activities** list.



Create a maintenance downtime activity

1. Click **Asset management** > **Maintenance downtime activities** > **All maintenance downtime activities** or **Active maintenance downtime activities**.

2. Click **New**.

3. Insert an ID in the **Maintenance downtime activities** field and a name in the **Name** field.

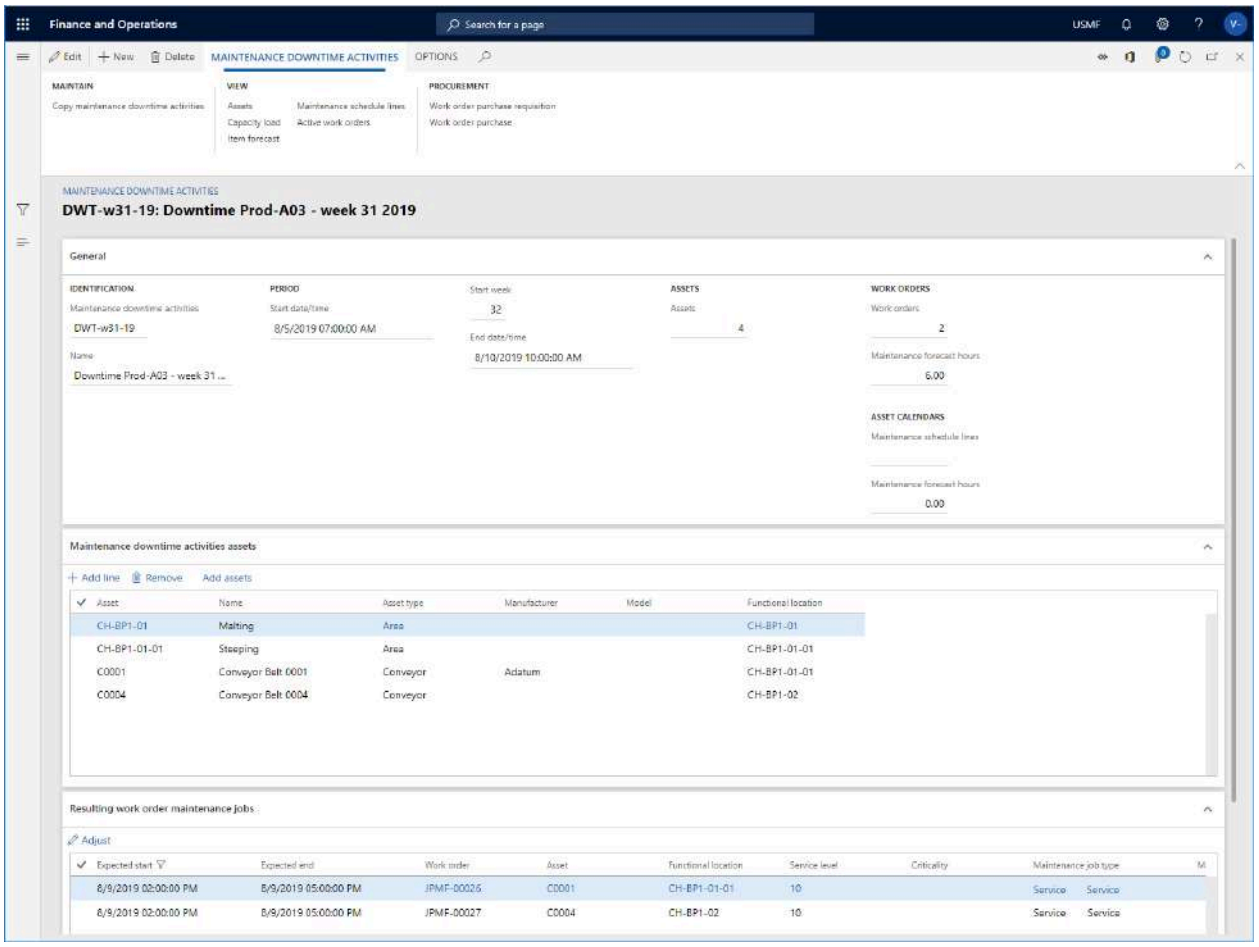
4. Insert the period for the maintenance stop in the **Start date/time** and **End date/time** fields.

5. On the **Maintenance downtime activities assets** FastTab> click **Add line** to add assets, one at a time, to the maintenance downtime activity.

6. Click **Save** when all assets have been added. The illustration below shows an example of a maintenance downtime activity with related assets and maintenance jobs.

7. The work order maintenance jobs and open maintenance schedule lines related to the selected assets are shown on the **Resulting work order maintenance jobs** and **Maintenance schedule lines** FastTabs. On the **General** FastTab > **Work orders** group > **Maintenance forecast hours** field and **General** FastTab > **Maintenance schedule** group > **Maintenance forecast hours** field , you see the total number of hours forecasted for work order maintenance jobs and maintenance schedule lines.

The illustration below shows an example of the **Maintenance downtime activities** details view.



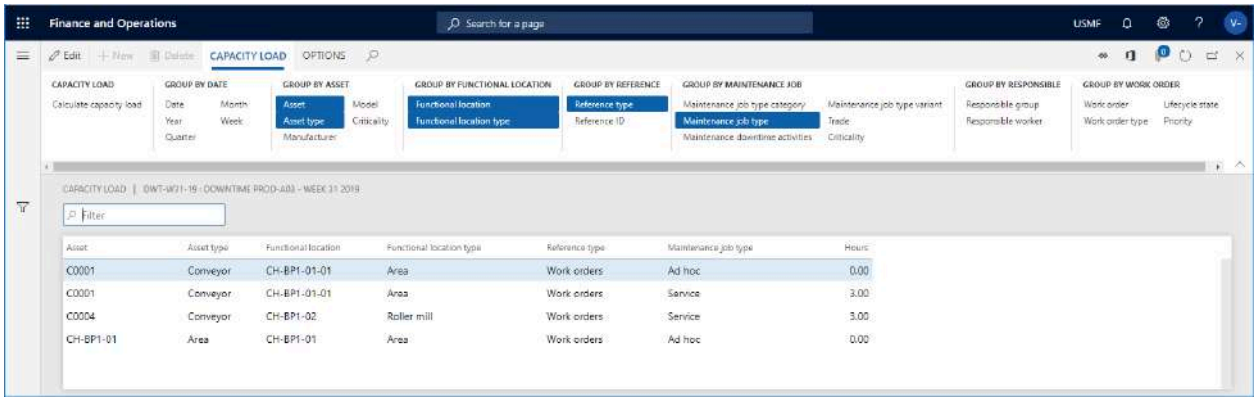
> **Note**
>
> The work order maintenance jobs and maintenance schedule lines related to the selected assets are automatically updated if new work orders or maintenance schedule lines are created after you created the maintenance downtime activity. For example, if you schedule maintenance plans or maintenance rounds on the related assets two days after the maintenance downtime activity was created, new maintenance schedule lines are automatically added to the maintenance downtime activity.

8. In **All maintenance downtime activities** > **Maintenance downtime activities** > select a maintenance downtime activity in the list and click **Capacity load** to open the **Calculate capacity load** dialog. Use this dialog to get an overview of capacity load on, for example, dates, assets, asset types, and maintenance job types. Note that the dates shown in the dialog are the start and end dates selected in **Maintenance downtime activities**. This calculation includes the assets related to the maintenance downtime activity.

9. In the **Calculate capacity load** dialog, edit start and end times if required, and select if you want to include work orders and maintenance schedules in the calculation. You can use the **Level** field to indicate how detailed you want the

capacity load calculation to be regarding functional locations. For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all assets for a functional location, which are selected on the maintenance downtime activity, will be shown on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level. If you insert the number "0" in the **Level** field, you will see a detailed result showing all capacity load lines on all the functional location levels to which they are related.

10. Click **OK** to start the calculation. The total number of hours is shown in the **Capacity load** overview. On the **Capacity load** tab > the **Group by...** Action Pane groups, click the relevant buttons to get a more detailed overview of the allocation of forecasted hours. The illustration below shows the results of a **Capacity load** calculation.



11. After you get an overview of the capacity load, if you want to make adjustments on work order maintenance jobs or maintenance schedule lines, return to the **Maintenance downtime activities** details view and select the lines you want to adjust on the **Resulting work order maintenance jobs** and **Maintenance schedule lines** FastTabs.

12. Click the **Adjust** button and update expected start/end dates, service level, or responsible maintenance workers for the selected work order maintenance jobs or maintenance schedule lines.

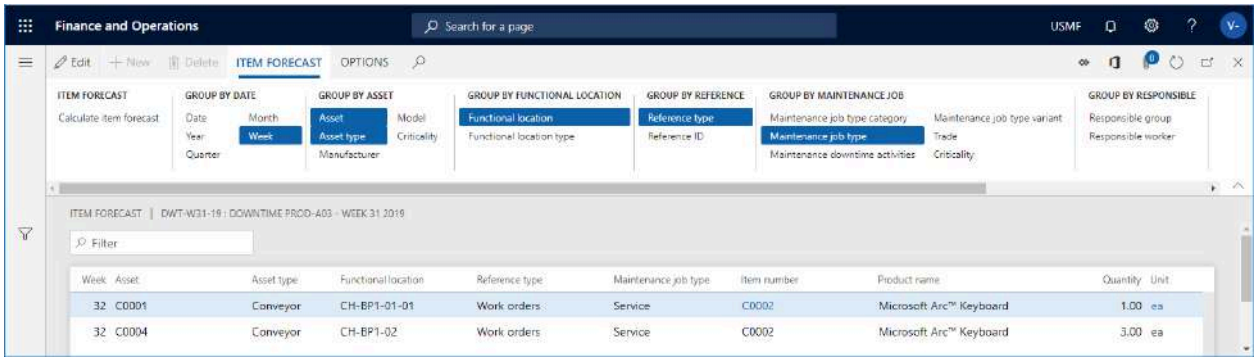
13. Click **OK** when you have made the required adjustments.

> **Note**
>
> Work order maintenance jobs and maintenance schedule lines that are not included in the maintenance downtime period after you have made adjustments are automatically removed from **Maintenance downtime activities**.

14. In **All maintenance downtime activities** > **Maintenance downtime activities** > select a maintenance downtime activity in the list and click **Item forecast** to open the **Calculate item forecast** dialog. Use this dialog to calculate forecasts for items (spare parts and other required items) and group them to get an overview, for example, by date, asset, asset type, and maintenance job type. Note that the dates shown in the dialog are the start and end dates selected in **Maintenance downtime activities**. This calculation includes spare parts and items related to the assets that are selected on the maintenance downtime activity.

15. In the **Calculate item forecast** dialog, edit start and end times if required, and select if you want to include work orders and maintenance schedules in the calculation. You can use the **Level** field to indicate how detailed you want the capacity load calculation to be regarding functional locations. For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all assets for a functional location, which are selected on the maintenance downtime activity, will be shown on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level. If you insert the number "0" in the **Level** field, you will see a detailed result showing all capacity load lines on all the functional location levels to which they are related.

16. Click **OK** to start the calculation. The total number of item forecasts is shown in the **Item forecast** overview. On the **Item forecast** tab > the **Group by...** Action Pane groups, click the relevant buttons to get a more detailed overview of the allocation of forecasted items.The illustration below shows the results of an **Item forecast** calculation.



- You can copy assets from one maintenance downtime activity to another. In **All maintenance downtime activities**, select the **Copy maintenance downtime activities** button, and make your selections in the **From maintenance downtime activities** and **To maintenance downtime activities** fields, and click **OK**.
- In **All maintenance downtime activities**, click the **Maintenance schedule lines** button or the **Active work orders** button to open the related lists and view the lines related to the selected maintenance downtime activity.

Feedback

Was this page helpful? Yes No

Provide product feedback

Manage test instrument calibration with Asset Management (preview)

[This article is prerelease documentation and is subject to change.]

Calibration ensures that test instruments provide accurate and reliable measurements. This accuracy is critical for maintaining product quality, meeting regulatory requirements, and avoiding costly errors. Asset Management in Dynamics 365 Supply Chain Management provides tools to schedule, track, and document calibration activities for all your test instruments.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Maintaining accurate and reliable test instruments is essential for ensuring product quality, meeting regulatory requirements, and reducing operational risks. Supply Chain Management provides a connected framework that brings together quality control and asset management to achieve these goals.

- **Standardized instrument management** – Define test instrument types to guide quality teams in selecting the right tools for inspections and calibration. This approach ensures consistency, reduces errors, and supports compliance across all quality processes.
- **Full traceability and digital records** – Use test instrument tags as digital twins of physical instruments. These tags store calibration history, specifications, and usage data, enabling complete traceability and simplifying audits.
- **Automated quality tracking** – Quality orders not only capture inspection results but can also track instrument usage automatically. This usage data drives usage-based calibration schedules, ensuring instruments remain accurate without manual intervention.
- **Lifecycle control through asset management** – Model each test instrument as an asset, giving you full visibility into its lifecycle, from acquisition to calibration and retirement, while aligning with maintenance best practices.

- **Proactive calibration scheduling** – Implement maintenance plans to automate calibration scheduling. Plans can be time-based or usage-based, reducing the risk of missed calibrations and ensuring compliance with industry standards.
- **Efficient execution and documentation** – Generate maintenance work orders to perform and document calibration events. These work orders include all necessary details—tasks, resources, and compliance notes. These features help streamline execution and record keeping.

Prerequisites

To use Asset Management to maintain calibration of test instruments, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- The following features must be turned on in feature management:
  - *Advanced quality management* (As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.)
  - *Optional linking of test instruments to maintainable assets*

Set up calibration in Asset Management

To set up your system to manage test instrument calibration with Asset Management, complete the following steps. Follow the links to learn more about each step.

1. **Create assets for your test instruments** – Create assets for the test instruments that require scheduled calibration in Asset Management. When defining an asset, you must assign an **Asset type**, which categorizes the asset and helps standardize its management and maintenance processes. Learn more about defining assets in Create an asset.
2. **Set up test instrument types** – Test instrument types serve as categories for grouping similar test instruments. Each test instrument type must be linked to an **Asset type** to ensure consistency between quality management and asset management processes. Learn more about test instrument types in Quality management test instruments.
3. **Create test instrument tags** – Each tag represents a physical test instrument and links to the test instrument asset in Asset Management. For instructions, see the next section.
4. **Create a maintenance plan** – Create or edit a maintenance plan for the test instrument asset. A maintenance plan is a predefined schedule that automatically creates work orders

for recurring maintenance tasks such as calibrating test instruments based on time intervals or usage thresholds. Learn more in Maintenance plans.

Set up test instrument tags for test instruments managed with Asset Management

Each test instrument tag represents a physical test instrument and also links to the test instrument asset in Asset Management. Follow these steps to create a test instrument tag with an asset assigned:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test instruments** to open the test instruments page.

2. Select or create a test instrument type record where an **Asset type** is assigned, as described in Quality management test instruments.

3. On the Action Pane, select **Test instrument tags**.

4. On the Action Pane, select **New** to add a new tag. Then make the following settings in the header for the new tag:

    - **Tag number** – Enter a unique name, number, or code for the instrument.

    - **Test instrument type** – Specify the test instrument type that this tag belongs to. This field is prefilled to show the test instrument type that was selected when you opened the **Test instrument tags** page.

    - **Instrument usage status** – The current usage status of the test instrument. Although the field is read-only, you can manually change the status to *Out of service* or *Available* by using the buttons on the **Update usage status** tab of the Action Pane. The following status values are used:
        - *Available* – The test instrument is ready to be assigned to a new quality order.
        - *Calibration* – The test instrument is currently being calibrated. This status is automatically set when an open (in-progress) calibration record exists for this test instrument tag.
        - *Out of service* – The test instrument is out of service and can't be used for quality order tests. This status is automatically set when the lifetime usage maximum is reached.

5. On the **General** FastTab, make the following settings:

- **Asset** – Select the asset that represents the test instrument that requires scheduled calibration in Asset Management.
- **Usage increment** – If the field Fixed increment counter type is set on the related instrument type record, then this field is incremented each time a quality order is validated with the increment specified in the field Usage increment, on the related instrument type record.
- **Skip check for test instrument on open quality order** – Set this option to *No* to prevent test instrument tags that are already assigned to open quality orders from being assigned to new quality orders. Set it to *Yes* to skip this check (in this case, a test instrument tag might be assigned to more than one open quality order at a time).
- **Check if test instrument is being calibrated** – Select the action that is taken on a quality order when the **Instrument usage status** value of the instrument is *Calibration*. The default value for new tags is defined in Inventory and warehouse management parameters, but you can change it for individual test instrument tags. The following values are available:
  - *No check* – Skip this check.
  - *Warning only* – If the instrument is being calibrated, show a warning message, but allow the instrument to be used.
  - *Not allowed* – If the instrument is being calibrated, show an error message that states that the instrument can't be used.

6. On the Action Pane, select **Save** to save the new test instrument tag.

Last updated on 07/27/2026

Calculate capacity load

06/17/2025

In Asset Management, you can calculate capacity load on:

- Maintenance schedule lines
- Work orders that haven't yet been scheduled
- Scheduled work orders

This is useful if you want to get an overview of expected capacity load for a specific period. Calculation of capacity load can be done on all assets or selected assets. You can also make a calculation on maintenance downtime activities or work order pools.

1. Go to one of the following pages, depending on what you want to calculate capacity load for:

   - **Asset management** > **Inquiries** > **Capacity load**
   - **Asset management** > **Work order pools** > **All work order pools**
   - **Asset management** > **Work order pools** > **Active work order pools**
   - **Asset management** > **Maintenance downtime activities** > **All maintenance downtime activities**
   - **Asset management** > **Maintenance downtime activities** > **Active maintenance downtime activities**

2. Select the record you want to work with and then, on the Action Pane, select **Capacity load**.

3. In the **Calculate capacity load** dialog, select a period for the calculation in the **Start date/time** and **End date/time** fields.

4. Select *Yes* on the **Include maintenance schedule** toggle button if you want to include maintenance schedule lines in the calculation.

5. Select *Yes* on the **Include work order** toggle button if you want to include work order jobs in the calculation.

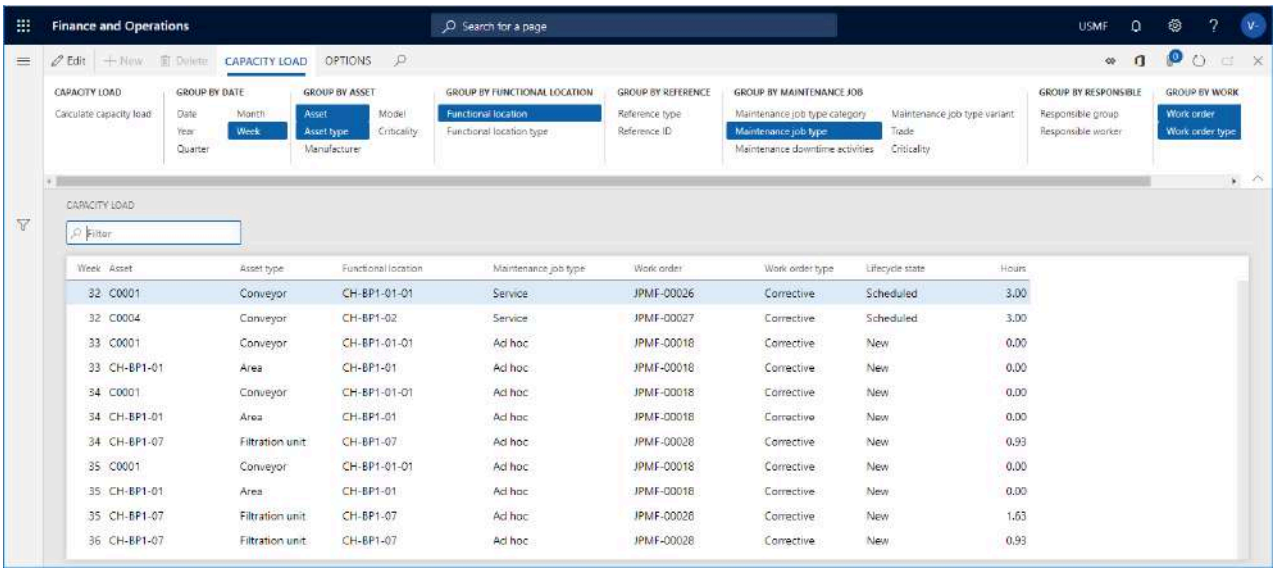
6. You can use the **Level** field to indicate how detailed you want the capacity load lines to be regarding functional locations.

   For example, if you insert the number *1* in the field, and you have a multi-level functional location structure, all maintenance schedule lines and work orders for a functional location will be shown on the top level, and therefore the hours on a line might be added up from functional locations located at a lower level.

If you insert the number *0* in the **Level** field, you'll see a detailed result showing all maintenance schedule lines and all work orders on all the functional location levels to which they're related.

7. Select **OK** to start the calculation.

8. In the **Group by...** groups, select the relevant buttons to show the required detail level of the calculation. In the following screenshot, the selected **Group by** buttons are highlighted in blue color. Select on a button to activate or deactivate it.



> **Note**
>
> If you want to focus only on capacity planning regarding scheduled work orders, see Calculate capacity load on scheduled work orders.

Calculate item forecast

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Just as you can make capacity load calculations, which are described in the previous section, you can also make item forecast calculations on:

- maintenance schedule lines
- work orders that have not yet been scheduled
- scheduled work orders

This is useful if you want to get an overview of expected item consumption (spare parts as well as other items required for completing work orders) for a specific period. Calculation of item forecast can be done on all assets or selected assets. You can also make a calculation on a maintenance downtime activity (**All maintenance downtime activities** or **Active maintenance downtime activities**), or on a work order pool (**All work order pools** or **Active work order pools**).

1. Click **Asset management** > **Inquiries** > **Item forecast**, or **Asset management** > **Work order pools** > **All work order pools** / **Active work order pools** > select work order pool in the list > **Item forecast** button, or **Asset management** > **Maintenance downtime activities** > **All maintenance downtime activities** / **Active maintenance downtime activities** > select maintenance downtime activity in the list > **Item forecast** button.

2. In the **Calculate item forecast** dialog, select a period for the calculation in the **Start date/time** and **End date/time** fields.

3. Select "Yes" on the **Include maintenance schedule** toggle button if you want to include maintenance schedule lines in the forecast calculation.

4. Select "Yes" on the **Include work order** toggle button if you want to include work order jobs in the forecast calculation.

5. You can use the **Level** field to indicate how detailed you want the item forecast lines to be regarding functional locations.

   For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all maintenance schedule lines and work orders for a

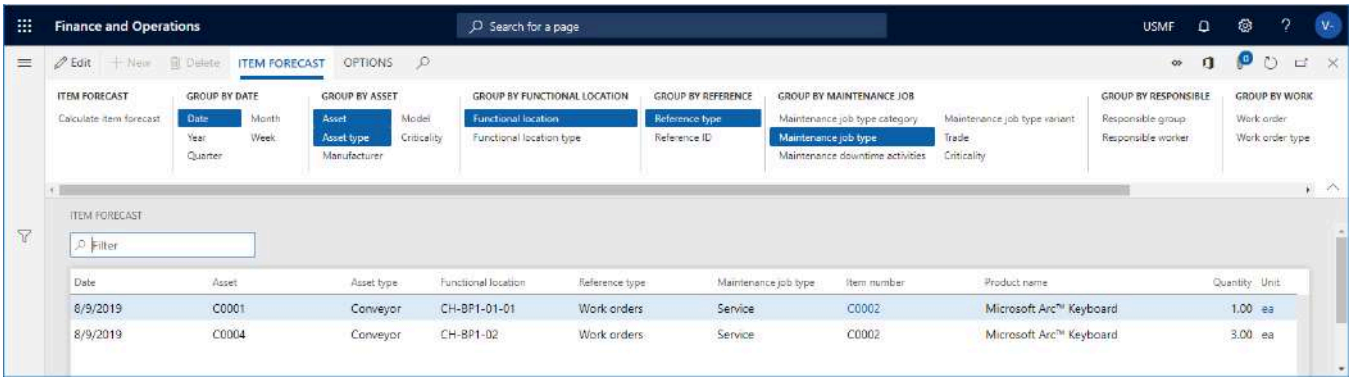
functional location will be shown on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level.

If you insert the number "0" in the **Level** field, you will see a detailed result showing all maintenance schedule lines and all work orders on all the functional location level to which they are related.

6. Click **OK** to start the calculation.

7. In the **Group by...** groups, click the relevant buttons to show the required detail level of the calculation. In the screenshot below, the selected **Group by** buttons are highlighted in blue color. Click on a button to activate or deactivate it.

8. Click the **Display dimensions** button if you want to see the product, storage, or tracking dimensions related to the items. Select the relevant check boxes, and click **OK**.



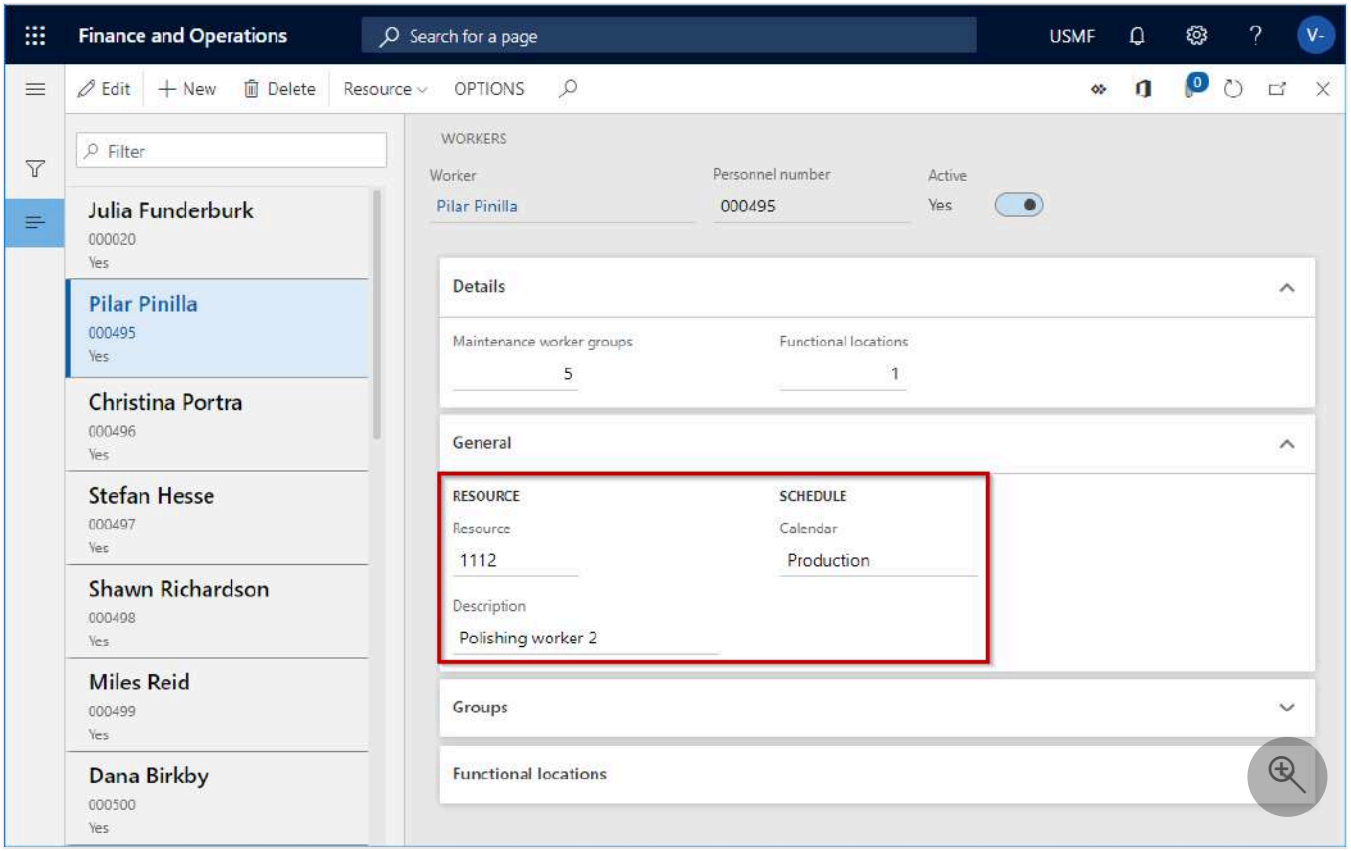
Last updated on 05/07/2025

Maintenance worker calendar and scheduling

08/14/2025

When you schedule work orders, you create a schedule for maintenance workers, tools, and assets. In order to schedule maintenance workers, a calendar must be set up for each maintenance worker. Maintenance workers are related to a resource, and working time calendars are set up for resources. You set up the resource and calendar in **Asset management** > **Setup** > **Workers** > **Workers**, which is described in Maintenance workers and worker groups.

The following screenshot shows an example of a maintenance worker who is related to a resource that uses the working time calendar *Production*.



Calendar setup for tools and assets isn't needed in relation to work order scheduling. The assumption is that tools and assets are available 24 hours a day for maintenance.

Set up preferred maintenance workers

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

During work order scheduling, you can make a preference regarding which maintenance worker or worker group is allocated to complete the work order. The use of this functionality is optional, but it can help you make a choice for the most qualified maintenance worker to complete a job, based on worker skills and competencies. Only maintenance workers that are available at scheduling time will be scheduled. If a preferred maintenance worker setup matches a work order during scheduling, but the maintenance worker is allocated to other jobs, the work order will be scheduled to another available maintenance worker.

Before you can set up preferred maintenance workers, you must first set up the maintenance workers and worker groups. For a description about how to set up maintenance workers and worker groups, see to Maintenance workers and worker groups.

Set up preferred workers

A preferred maintenance worker or worker group can be related to one or more of the following:

- trade
- maintenance job type variant
- maintenance job type
- maintenance job type category
- asset
- asset type

The more selections you make for the same record, the more specific your setup will be.

1. Click **Asset management** > **Setup** > **Workers** > **Preferred maintenance workers**.
2. Click **New** to create a new record.
3. Start by creating a "default" maintenance worker or worker group. This means that you only make a selection in the **Preferred maintenance worker group** field or the **Preferred**

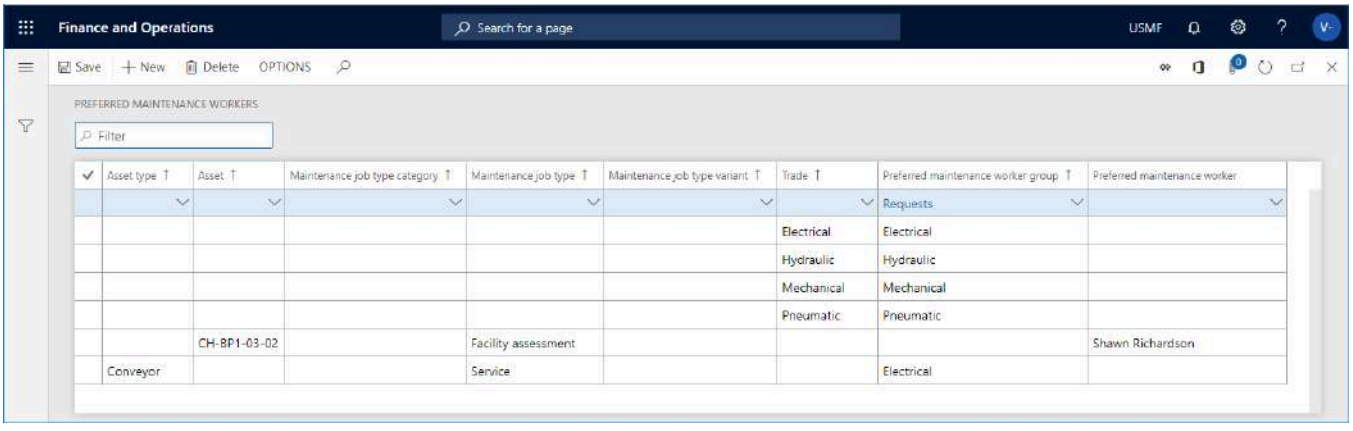
maintenance worker field. In the screenshot below, you see an example in the first record in which "Requests" is selected as **Preferred maintenance worker group**.

> **Note**
>
> The default setup will be used during work order scheduling if no other, more specific, combination matches the contents of the work order.

4. Repeat step 2 to create a new record. Make the required selections, depending on the detail level for the preferred worker or worker group.

   Example: In the screenshot below, in the sixth record, the maintenance worker Shawn Richardson is selected as preferred worker. Shawn will automatically be selected during scheduling of a work order that includes the asset "CH-BP1-03-02 and the maintenance job type "Facility assessment", if Shawn is available at the scheduled time.

   > **Note**
   >
   > Generally, when a preferred maintenance worker is selected during work order scheduling, Asset Management goes through all **Preferred maintenance workers** records to check for a possible match, always checking the most specific combination first. If no match is found, the "default" record with a selection in either the **Preferred maintenance worker group** field or the **Preferred maintenance worker** field is used.



You can also set up *responsible* maintenance workers who can be selected when a maintenance request or a work order is created. You can edit the selection in **All work orders** and **All maintenance requests**, if required. Learn more in Responsible maintenance workers.

During work order scheduling, different scores are calculated to determine which workers should complete the jobs related to a work order (those scores are set up in **Asset management parameters** > **Work order scheduling** link). If two or more preferred

maintenance workers or responsible maintenance workers get the same score during work order scheduling, one worker is randomly selected. Otherwise, it is always the worker with the highest score who is allocated to complete a work order.

Last updated on 05/07/2025

Schedule work orders

Article • 01/29/2025

This article explains how to schedule work orders in Asset Management.

The required number of hours for a work order is defined by the sum of forecasted hours minus posted hours. If more time is required, the forecast must be adjusted accordingly. On the **All work orders** page (**Asset management** > **Work orders** > **All work orders**) or the **Active work orders** page (**Asset management** > **Work orders** > **Active work orders**), you can view or edit forecasts on a work order. Select the work order, and then, on the **Work order** tab, select **Forecast**. After work orders are created and estimated, the next step is to allocate the required maintenance workers and tools.

> **Important**
>
> Work orders can be scheduled only if their work order lifecycle state allows for scheduling. To set up a lifecycle state so that it allows for scheduling, go to **Asset management** > **Setup** > **Work orders** > **Lifecycle states**, and then, on **General** FastTab, set the **Allow scheduling** option to *Yes*.

1. Go to **Asset management** > **Work orders** > **All work orders**.
2. In the list, select the work orders that you want to schedule. To find the correct work orders, you can sort the list by **Current lifecycle state** value, for example.
3. On the **General** tab, select **Schedule**.
4. In the **Schedule work orders** dialog box, you can add selections that are related to the expected start date and service level, as you require. If the scheduling process should observe capacity limitations for resources that are already scheduled for other jobs, make sure that the **Asset**, **Tool**, and **Worker** options are set to *Yes*.

   > **Note**
   >
   > If you set the **Asset**, **Tool**, and **Worker** options to *No*, existing reservations are ignored. The Action center shows a list of overlapping work order schedules. As you require, you can select a message link to open a work order and reschedule it.

5. To view detailed information about the scheduling process, set the **Verbose** option to *Yes*. In this case, the Action center shows detailed information about the

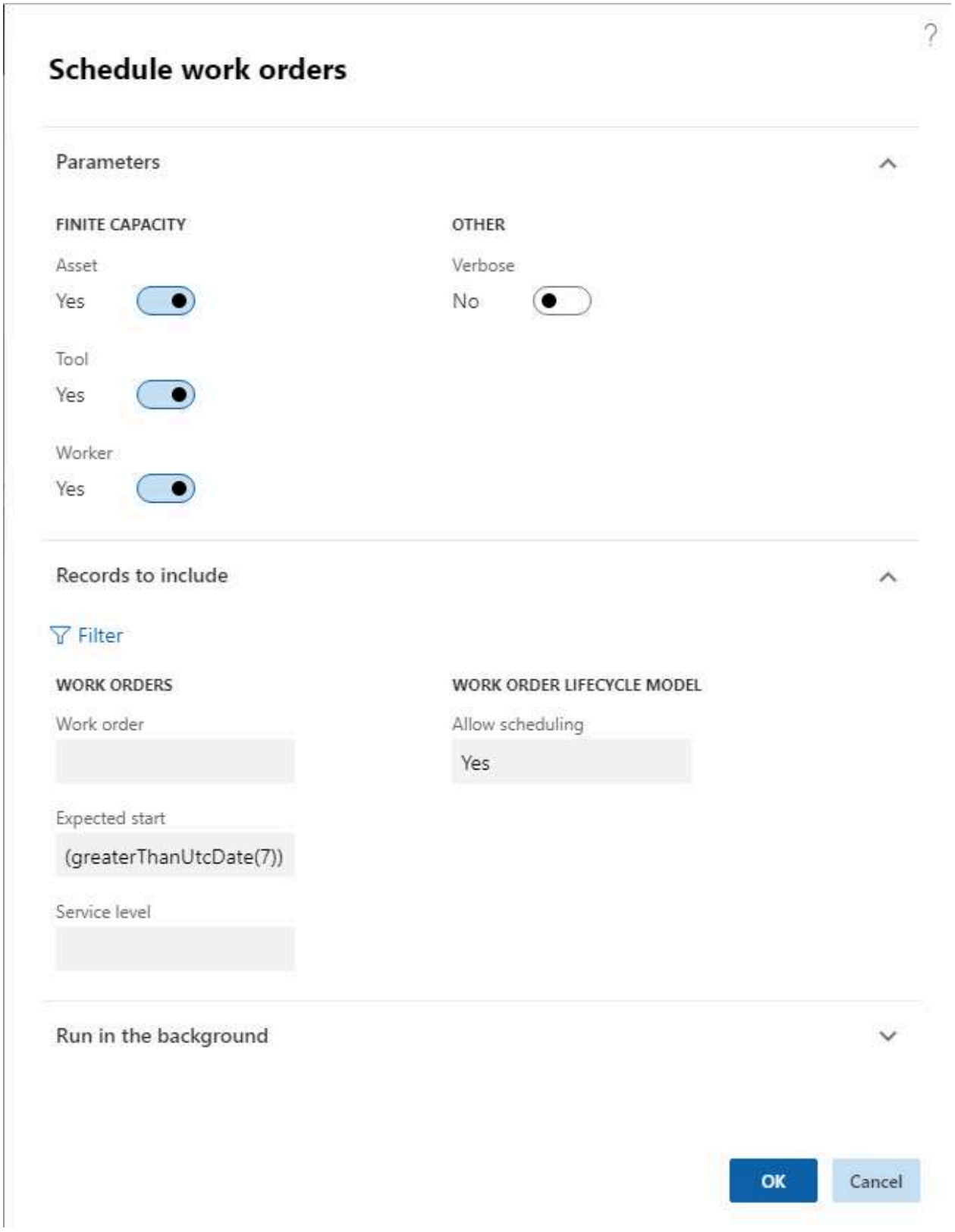
calculated scores on the work orders and maintenance workers.

6. Select **OK** to start the work order scheduling process.
7. When scheduling is completed, a message in the Action center shows the number of work orders that were scheduled. If you set the **Verbose** option to *Yes*, the Action center also shows more detailed information.

> **Note**
>
> Work orders are scheduled in one cycle per work order, not per work order job.
>
> You can also open the **Schedule work orders** dialog box by going to **Asset management** > **Periodic** > **Work orders** > **Schedule work orders**. Make your selections, and then select **OK** to start the work order scheduling process.
>
> On the **Run in the background** FastTab of the **Schedule work orders** dialog box, you can set up work order scheduling as a batch job.

Example

In the following illustration, the **Expected start** field in the **Schedule work orders** dialog box is set to the formula `(greaterThanUtcDate(7))`. This formula generates work order scheduling for all work orders where the expected start date is one week from the current date and later. This formula can be useful when you run work order scheduling on an ongoing basis, but you want to ensure that work orders that are scheduled for the next five to six days aren't rescheduled.



On the **Work order types** page (**Asset management** > **Setup** > **Work orders** > **Work order types**), if the **One maintenance worker** option is set to *Yes* for a work order type, that work order type sets up scheduling for a single maintenance worker. If that work order type is used on a work order, the **One maintenance worker** option is automatically set to *Yes* on the **Schedule** FastTab in the **Header** view of the **All work orders** details page. Then, during work order scheduling, all work order jobs that are created on the work order are scheduled to the same maintenance worker. As you require, you can change the setting of the **One maintenance worker** option on the **All**

work orders details page to enable either multiple workers or one worker to be scheduled on the work order jobs.

The scheduling process in Asset Management includes several factors in the scheduling calculation:

- Calculating scores for both work orders and maintenance workers. Scores for work orders and maintenance workers are set up in Asset management parameters.
- Checking for matching competencies (that is, skills and certificates) that are required to perform the job. Skills and certificates are set up on maintenance workers in the **Human resources** module. (Go to **Human resources** > **Workers** > **Workers**, select a worker in the list, and then, on the **Worker** tab, in the **Competencies** section, select the **Skills** and **Certificates** buttons.) Skills and certificates can also be added to maintenance job types and maintenance job trades. Learn more about competencies and maintenance job types in Maintenance job type categories and maintenance job types, maintenance job type variants, maintenance job trades, and maintenance checklists.

Scores used in work order scheduling

Calculating scores for a work order job is based on the expected start date and service level of the work order.

- **Start date calculation** – For every future date that is calculated from the expected start date, the start date score is subtracted and multiplied by the score. (In the examples that follow, the score is *10*.)
- **Criticality calculation** – The criticality score is multiplied by the criticality on the work order.
- **Service level calculation** – The service level score is divided by the service level on the work order.

In the following examples, the criticality score is *2*, and the service level scores are *5* and *100*.

Example 1

Expand table

Work order ID	Expected start date	Work order criticality	Work order service level	Calculation	Score
WO-00010816	Tomorrow	2	20	(-1 × 10) + (2 × 2) + (5 ÷ 20)	-5.75

Work order ID	Expected start date	Work order criticality	Work order service level	Calculation	Score
WO-00010817	Two days from now	2	20	(-2 × 10) + (2 × 2) + (5 ÷ 20)	-15.75
WO-00010818	Two days from now	3	5	(-2 × 10) + (2 × 3) + (5 ÷ 5)	-13

The work orders will be scheduled in the following order: WO-00010**816**, WO-00010**818**, WO-00010**817**.

Example 2

Expand table

Work order ID	Expected start date	Work order criticality	Work order service level	Calculation	Score
WO-00010816	Tomorrow	2	20	(-1 × 10) + (2 × 2) + (100 ÷ 20)	-1
WO-00010817	Two days from now	2	20	(-2 × 10) + (2 × 2) + (100 ÷ 20)	-11
WO-00010818	Two days from now	3	5	(-2 × 10) + (2 × 3) + (100 ÷ 5)	6

If the service level score is increased from *5* to *100*, the scheduling order will be WO-00010**818**, WO-00010**816**, WO-00010**817**.

All the rating scores that are related to calculating which maintenance workers should work on the work orders are set up as numbers. Those numbers are then added to each maintenance worker calculation during work order scheduling. The maintenance worker who has the highest score is selected for the work order. The following table provides a short description of the maintenance worker scores.

Expand table

Maintenance worker score	Description
Responsible worker	If the maintenance worker is selected as the responsible worker on the work order, the score is added.
Responsible maintenance	If the maintenance worker is part of the responsible maintenance worker group on the work order, the score is added.

Maintenance worker score	Description
worker group	
Preferred maintenance worker	If the worker is selected as the preferred maintenance worker on the asset, the score is added.
Preferred maintenance worker group	If the worker is part of the preferred maintenance worker group on the asset, the score is added.
Location	If your company uses functional locations, maintenance workers get the full score if they are in the functional location that is related to the asset. If the functional location of the asset has a parent location, maintenance workers in that functional location get one-half of the full score. If that location also has a parent, maintenance workers in that location get one-third of the full score. If that location also has a parent, maintenance workers in that location get one-quarter of the full score, and so on. If your company uses asset locations, which we don't recommend, the location, area, and zone are used to calculate location scores. Maintenance workers get the full score if they are in the location, area, and zone that are related to the asset. If the maintenance worker location matches only the location and area, the rating score for the maintenance worker is two-thirds of the full score. If the maintenance worker location matches only the location, the rating score for the maintenance worker is one-third of the full score.
Worker's start date	For every date that the scheduled start date is later than the expected start date, the score is subtracted.

Note

If a score is set to *0*, that score isn't calculated. This behavior is useful if, for example, you don't want to include a responsible worker in your scheduling.

Competencies used in work order scheduling

Skills and certificate requirements can be set up on maintenance job types (**Asset management** > **Setup** > **Jobs** > **Maintenance job types**) and maintenance job trades (**Asset management** > **Setup** > **Jobs** > **Maintenance job trade**).

Maintenance job types and maintenance job trades are selected on work order jobs. If skills or certificates have been selected on a maintenance job type or maintenance job trade, and that maintenance job type or maintenance job trade is used on a work order

job, only maintenance workers with matching skills and certificates are scheduled to work on the work order.

Work with scheduled work orders using a Gantt chart

The **Scheduled work orders gantt chart** provides a graphical interface where you can view and reschedule your work orders.

To view and work with the Gantt chart:

1. Go to **Asset management** > **Work orders** > **Scheduled work orders gantt chart**.
2. Use the drop-down lists and fields in the **Settings** section to set up which functional location, time span, and time scale to show in the Gantt chart.
3. Select **Apply**.
    - The Gantt chart updates to show the scheduled work orders that match your settings. Each work order is represented by a blue rectangle.
    - To reschedule a displayed work order, select and then drag it to the appropriate new date and time.
4. If you made any changes, select **Save** on the Action Pane to save them.

Feedback

Was this page helpful? Yes No

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Dispatch work orders

The dispatch functionality in Asset Management schedules work orders and work order jobs.

Dispatch a work order

To dispatch a work order, follow these steps:

1. Go to **Asset management** > **Work orders** > **All work orders** or **Active work orders**.
2. In the grid, select the work order that you want to dispatch.
3. On the Action Pane, on the **General** tab, select **Dispatch**.
4. In the **Schedule work order** dialog box, set the following fields:
    - **Worker** – Select the worker.
    - **Schedule hours** – If the expected work hours differ from the forecast hours, enter the number of expected work hours.
    - **Scheduled start** – Specify the start date and time.
    - **Asset**, **Tool**, and **Worker** – Set one or more of these options to *Yes* if the scheduling process should observe capacity limitations for resources of the specified type that are already scheduled on other jobs.
    - **Ignore schedule** – Set this option to *Yes* to ignore closed days in the calendar. (This option applies to assets, workers, and tools.)
    - **Ignore scheduled execution** – Set this option to *Yes* to ignore any scheduling-related limitations that were selected on the work order. Learn more in Scheduled execution.
    - **Verbose** – Set this option to *Yes* if you want the system to show detailed information about the scheduling process. In this case, the Action center shows detailed information about the calculated scores on the work order.
    - **Update forecast** – Set this option to *Yes* to update the forecast so that it reflects any scheduling change that you made in this dialog box.
5. Select **OK**. The work order lifecycle state is automatically updated to the **Scheduled state** value that is specified on the **Lifecycle models** page (**Asset management** > **Setup** > **Work orders** > **Lifecycle models**).

> ⓘ **Note**

To delete the schedule on a work order, select it, and then, on the Action Pane, on the **General** tab, select **Delete schedule**. If you delete the schedule on a work order, remember to manually update the work order lifecycle state.

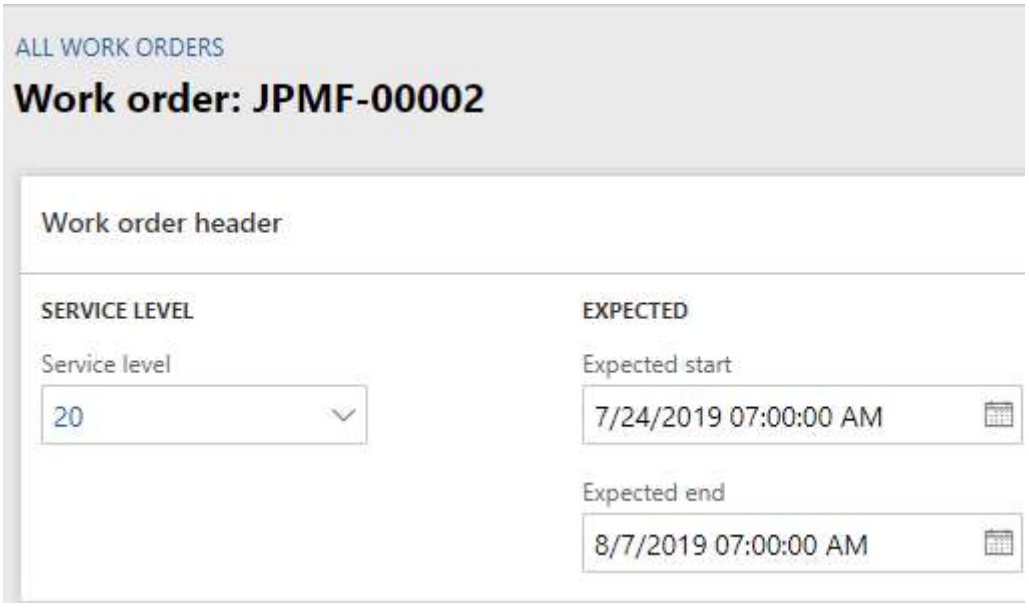
Last updated on 07/23/2026

Schedule work order on specific date and time

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

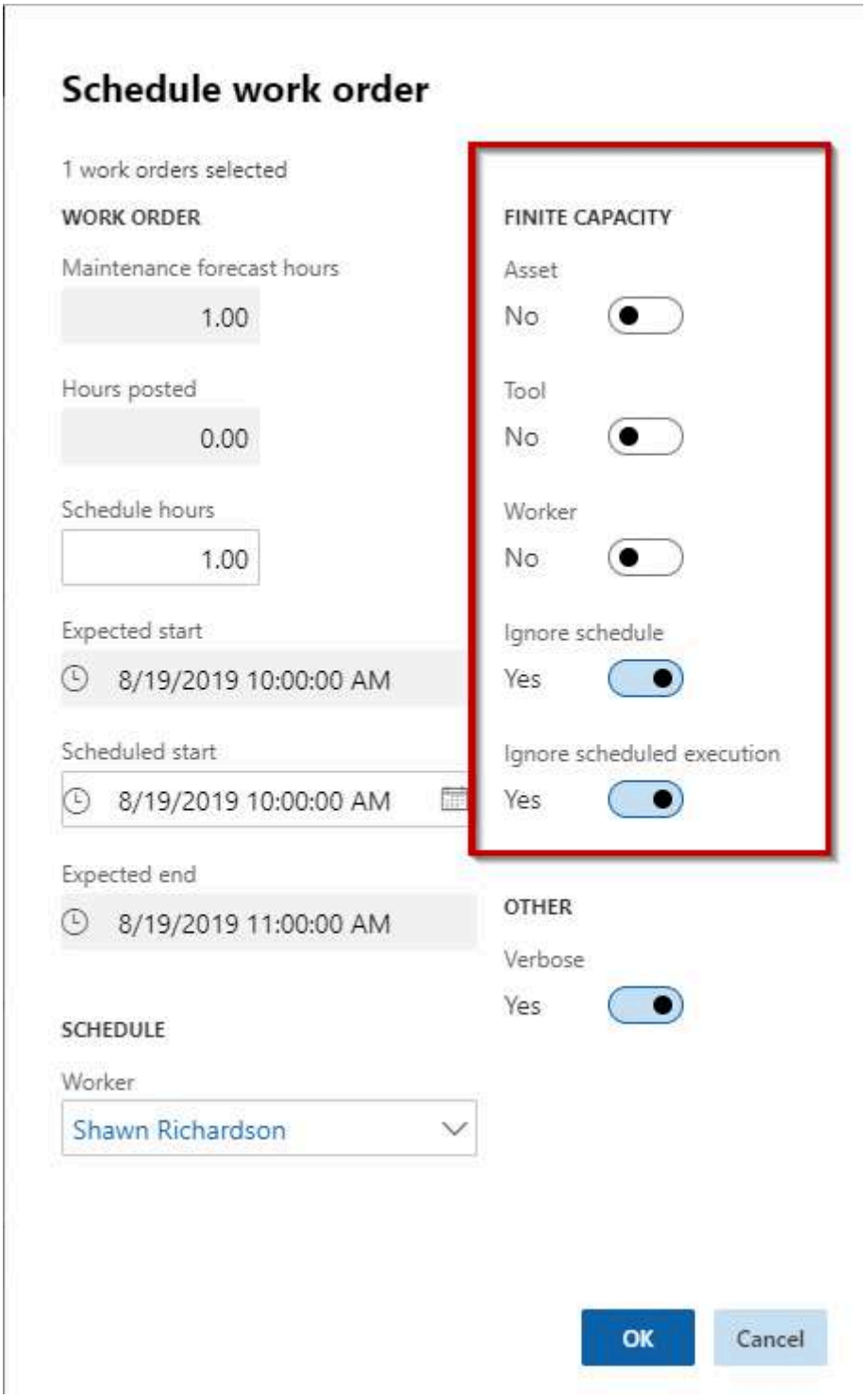
If a work order must be scheduled on a specific date *and* time, you can override the standard scheduling process in Asset Management and create a specific schedule for a work order.

1. Click **Asset management** > **Work orders** > **All Work orders** or **Active work orders**.
2. In the work order list, click on the Work order ID in the **Work order** column.
3. Click **Edit**.
4. On the **Work order header** FastTab, insert start and end dates and times in the **Expected start** and **Expected end** fields.



5. On the **General** tab, click **Schedule** to use the standard scheduling process, or click **Dispatch** if you want to assign the work order to a specific worker.
6. In order to override any existing capacity reservations to ensure that the work order is scheduled in the expected period, make the selections as shown in the figure below in the **Schedule work order** dialog > **Finite capacity** section. This means that the scheduling

process will ignore existing capacity reservations because the work order must start on the expected start time.



7. Click **OK** to start scheduling.
8. If the scheduling process results in double bookings, you will see a message on the screen, and you can adjust the related work orders.

> ⓘ **Note**
>
> In order to schedule a maintenance worker for the work order, that maintenance worker must be available at the expected start date and time. Worker availability is set up in the

worker calendar.

Last updated on 05/07/2025

Scheduled work order maintenance jobs

Article • 06/15/2022

The **Scheduled work order maintenance jobs** page shows an overview of the work orders allocated to a resource. Work orders using resource types "Human resources", "Tool", and "Machine" are shown. For example, if a maintenance worker calls in sick, you can use this page to quickly find work orders allocated to the worker, and then allocate another maintenance worker to the job.

View scheduled work order maintenance jobs

1. Click **Asset management** > **Work orders** > **Scheduled work order maintenance jobs**. You see a list of all work orders set to work order lifecycle state "Scheduled" or "In progress".

2. You can sort the list, for example, by maintenance worker. You can also use the filter to limit the list to display work orders allocated to a specific resource or maintenance worker.

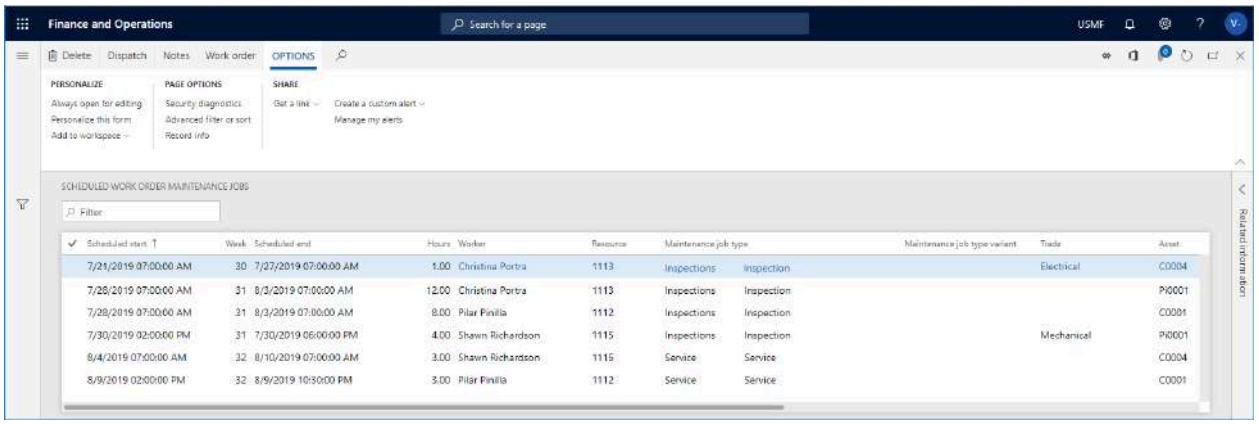
3. You can see notes on the work order and, if required, add new notes by selecting the work order job, and then click **Notes**.

4. If you want to allocate one maintenance worker to a work order, select the work order, and then click **Work order**.

5. The **Work order** page opens. Click **Dispatch** to schedule the work order to a specific maintenance worker.

> ⓘ **Note**
>
> Read more about scheduling several work orders or one work order in Schedule work orders and Dispatch work order.

The following screenshot shows an example of the **Scheduled work order maintenance jobs** page.



Feedback

Was this page helpful? Yes No

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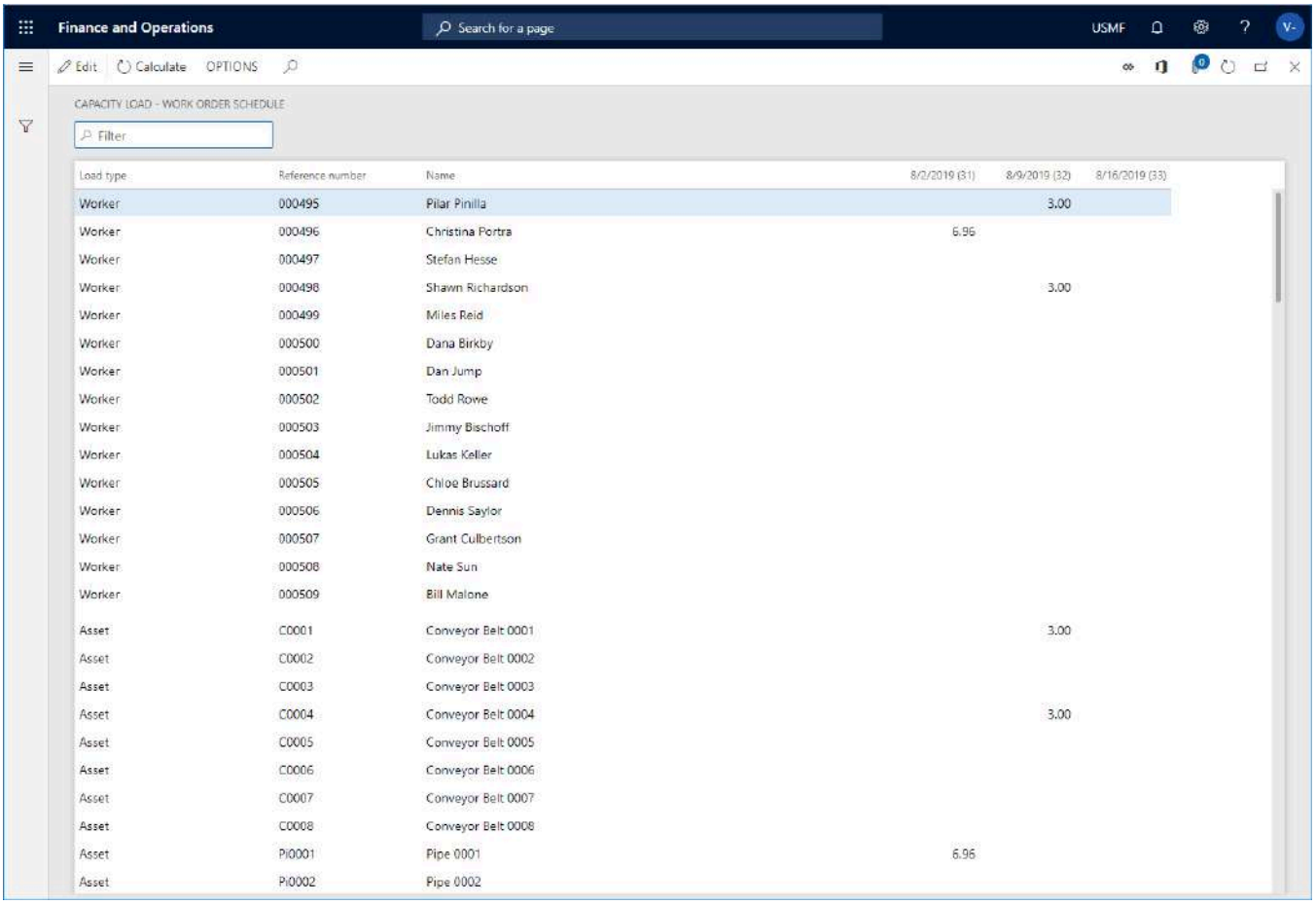
Calculate capacity load on scheduled work orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can calculate capacity load on scheduled work orders to get an overview of the work load on resources for a specific period. Calculations can be made for the following resources: Maintenance workers, worker groups, tools, and assets.

1. Click **Asset management** > **Inquiries** > **Schedule** > **Capacity load**.
2. In the **Calculate capacity load** dialog > **Show** field, select which load type you want to calculate: **Capacity**, **Reserved**, or **Remainder**.
3. Select **Yes** on the **Skip zero** toggle button if you do not want to show results containing zero.
4. Select the resource types for which you want to calculate capacity load by selecting **Yes** on the relevant toggle buttons: **Worker**, **Maintenance worker group**, **Tool**, and **Asset**.
5. Select the start date for the calculation in the **From date** field.
6. In the **Interval type** field, select the interval for the calculation: **Day**, **Week**, **Month**, or **Quarter**.
7. In the **Period frequency** field, insert the number of intervals you want to calculate. For example, if you have selected **Day** as the interval type, and you enter the number "5" in this field, a calculation of five days from the start date will be made.
8. Click **OK** to start the calculation.

The figure below shows the result of a calculation covering three weeks for the load type **Reserved**.



Load type	Reference number	Name	8/2/2019 (31)	8/9/2019 (32)	8/16/2019 (33)
Worker	000495	Pilar Pinilla		3.00	
Worker	000496	Christina Portra	6.96		
Worker	000497	Stefan Hesse			
Worker	000498	Shawn Richardson		3.00	
Worker	000499	Miles Reid			
Worker	000500	Dana Birkby			
Worker	000501	Dan Jump			
Worker	000502	Todd Rowe			
Worker	000503	Jimmy Bischoff			
Worker	000504	Lukas Keller			
Worker	000505	Chloe Brussard			
Worker	000506	Dennis Saylor			
Worker	000507	Grant Culbertson			
Worker	000508	Nate Sun			
Worker	000509	Bill Malone			
Asset	C0001	Conveyor Belt 0001		3.00	
Asset	C0002	Conveyor Belt 0002			
Asset	C0003	Conveyor Belt 0003			
Asset	C0004	Conveyor Belt 0004		3.00	
Asset	C0005	Conveyor Belt 0005			
Asset	C0006	Conveyor Belt 0006			
Asset	C0007	Conveyor Belt 0007			
Asset	C0008	Conveyor Belt 0008			
Asset	PI0001	Pipe 0001	6.96		
Asset	PI0002	Pipe 0002			

[!NOTE] If you select the load types **Capacity** or **Remainder** for your calculation, the same result will be displayed if no reservations have been made for the resources in the selected period.

For information about how to calculate capacity load on maintenance schedule lines and not scheduled work orders, refer to Calculate capacity load.

Last updated on 05/07/2025

Register consumption

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When a maintenance job on a work order is complete, the next step is to register consumption and post the journals. Register consumption for the following types: hours, items, and expenses. Register and post the different consumption types on the **Work order journals** page. The journal setup in **Asset Management** is used for creating and posting separate journals for hours, items, and expenses in the **Project management and accounting** module.

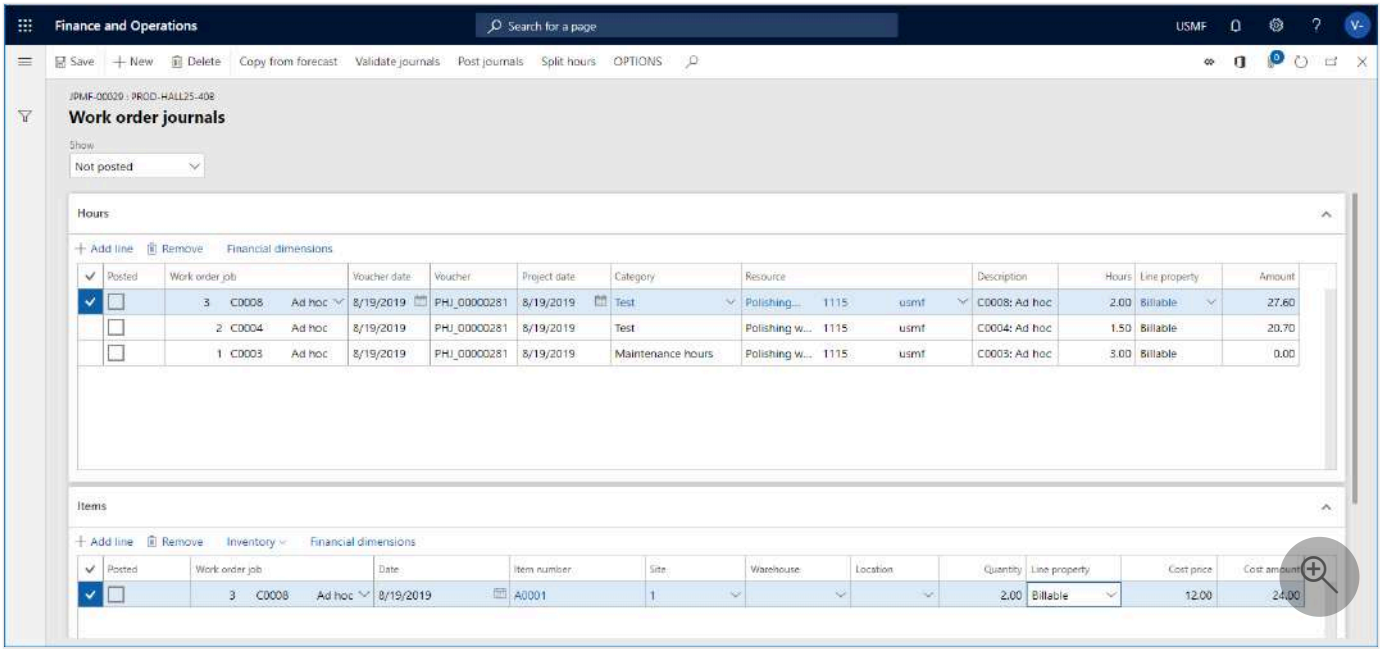
In some cases, you can add or delete forecast lines on a work order. The setup of a work order lifecycle state, the related project type, and the stage rules related to the project type determine if you can add or edit journal lines. Learn more about work order lifecycle states and related project stages in Forecasts, work orders, and projects.

> **Note**
>
> You can set up automatic posting of journals on a work order lifecycle state. Learn more in Work order lifecycle states.

1. Select **Asset management** > **Work orders** > **All Work orders** or **Active work orders**.
2. Select the work order, and select **Journals**.
3. Select **Copy from forecast** to transfer any forecast lines that might be connected to the work order. You can select which consumption types you want to transfer.
4. If necessary, add more consumption lines on the relevant FastTab by selecting **Add line** and filling out data on the line.
5. Select **Validate journals** to validate the journal lines before posting.
6. Select **Post journals** to post the journal lines.
7. After you post the consumption journals, you can update the work order lifecycle state. For example, to indicate that the work order is complete, update the lifecycle state to **Ended**.

- In the **Show** field at the top of the **Work order journals** page, select which journal lines you want to see: **All**, **Not posted**, or **Posted**. Posted journals have a check mark in the **Posted** check box.
- When you create item lines in the work order journal, product dimensions and tracking dimensions related to the item automatically transfer to the journal line.

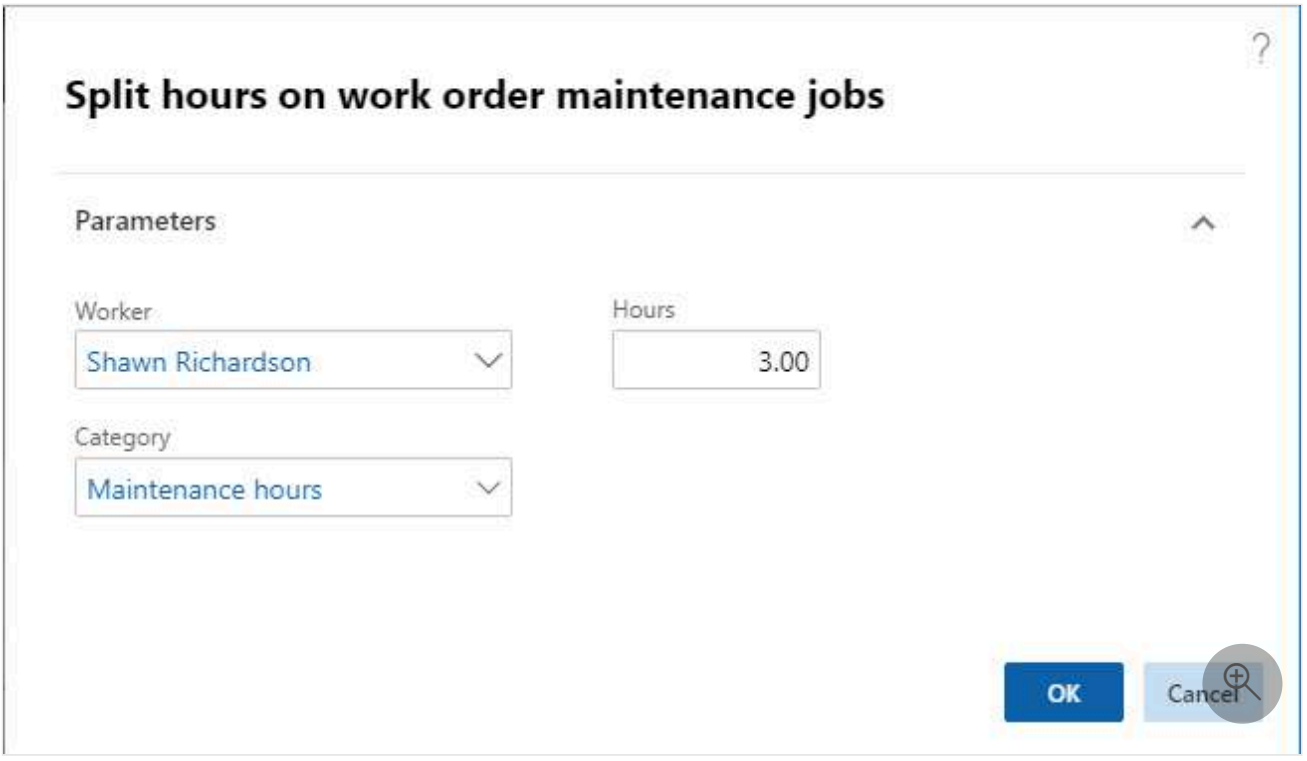
The following screenshot shows an example of hour and item registrations on a work order in **Work order journals**.



Split hours on work orders with several work order jobs

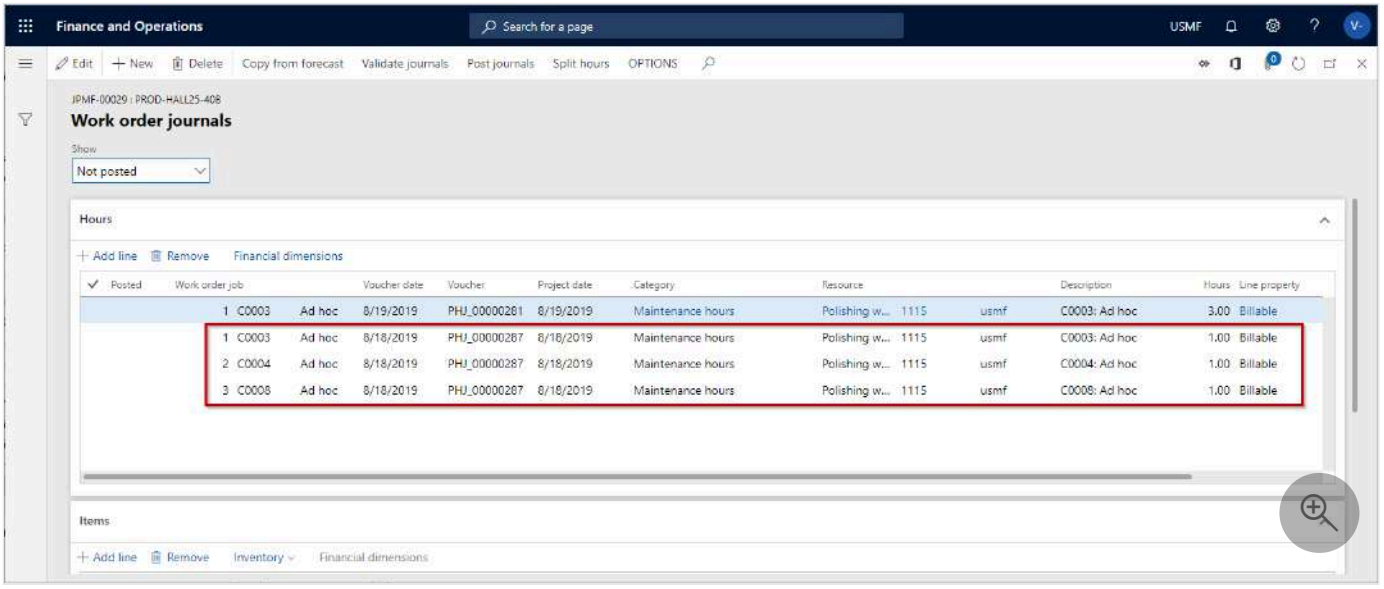
If a work order contains several work order jobs, use the **Split hours** functionality to register work hours. You can distribute one hour registration line evenly across each work order job.

1. Select **Asset management** > **Work orders** > **All Work orders** or **Active work orders**.
2. Select the work order and select **Journals**.
3. Select the hour registration line you want to split, and select **Split hours**.
4. In the **Split hours on work order maintenance jobs** dialog, the name of the worker who is signed in automatically appears in the **Worker** field. If required, select another worker.
5. Select a category for the hour registration in the **Category** field.
6. Enter the number of work hours to split in the **Hours** field.



7. Select **OK**.

Example: The following screenshot shows journal lines for a work order containing three work order jobs. The first line, which contains three work hours, is split, and one work hour is registered on each work order job. After the three hour registration lines are created, you decide what to do with the original hour registration line (the first line in the example). You can keep it as-is or delete it.



Financial dimensions on consumption registrations

When you make consumption registrations, the system adds financial dimensions related to the different registration types to the registrations in a specific sequence.

- *Hour and Expense registrations:* First, the system adds financial dimensions from the journal header, if any. Next, it adds financial dimensions from the related work order project. Finally, it adds financial dimensions from the resource (worker).
- *Item registrations:* First, the system adds financial dimensions from the related work order project. Next, it adds financial dimensions from the site. Finally, it adds financial dimensions from the item.

> **Important**
>
> If the dimension link feature is active (on **Inventory management** > **Setup** > **Posting** > **Dimension link**) and a financial dimension is linked to the inventory site, the linked dimension values from the site take precedence on item registrations. In this case, the values from the dimension attributes override the values that came from the related work order project.

> **Note**
>
> For all three registration types, the system validates the financial dimension combination and blanks invalid combinations.

Note: The author created this article with assistance from AI. Learn more

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Last updated on 05/13/2026

Create consumption reports

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you've created and posted consumption registrations on work orders in Asset Management, two reports are available to display consumption details.

Asset consumption report

When you have posted consumption on work orders, you can print an asset consumption report. The report displays hours, hour costs, item costs, and expenses posted on assets.

1. Click **Asset management** > **Reports** > **Assets** > **Asset consumption**.
2. In the **Asset consumption** dialog, select the parameters and detail level you want to see by selecting **Yes** on the relevant toggle buttons, and inserting a functional location level in the **Show** section.
    - You can use the **Levels** field to indicate how detailed you want the asset lines to be regarding functional locations.

      For example, if you enter the number "1" in the field, and you have a multi-level functional location structure, all assets for a functional location will be shown on the top level, and therefore a line may be added up from functional locations located at a lower level.

      If you enter the number "0" in the **Levels** field, you will see a detailed result showing all assets on all the functional location levels to which they are related.
    - Select **Yes** on the **Sum on all sub assets** toggle button to see sums for each sub asset in the report.
3. Select a date interval in the **Dates** section.
4. On the **Destination** FastTab, select if you want to display the report on screen, print it, or save it as a file or email.

5. If required, you can select specific assets to be displayed in the report. On the **Records to include** FastTab, click **Filter**, and add the assets you want to include in the report.
6. Click **OK** to generate the report.

Work order consumption report

When you have posted consumption on work orders, you can print a work order consumption report. The report displays hours, hour costs, item costs, and expenses posted on work orders.

1. Click **Asset management** > **Reports** > **Work orders** > **Work order consumption**.
2. In the **Work order consumption** dialog, select the parameters you want to include in the report by selecting **Yes** on the relevant toggle buttons in the **Show** section.
3. Select a date interval in the **Dates** section.
4. On the **Destination** FastTab, select if you want to display the report on screen, print it, or save it as a file or email.
5. If required, you can select specific work orders to be displayed in the report. On the **Records to include** FastTab, click **Filter**, and add the work orders you want to include in the report.
6. Click **OK** to generate the report.

> **Note**
>
> You can also generate a work order report, which contains more work order details.

Last updated on 05/07/2025

Warranty agreements

Article • 06/03/2022

In Asset Management, you can set up warranty terms that can be connected to an asset or an asset type. Warranty terms are created for a specific period. Warranty can be set up to provide full coverage or partial coverage, and you can set up terms that are related to hours, expenses, and items.

The first step is to create any vendor warranty agreements that you have for your equipment. You then attach warranty agreements to assets or asset types. Vendor warranty agreements are used only for informational purposes. If vendor warranty is set up on an asset, you can see the warranty coverage period on the asset.

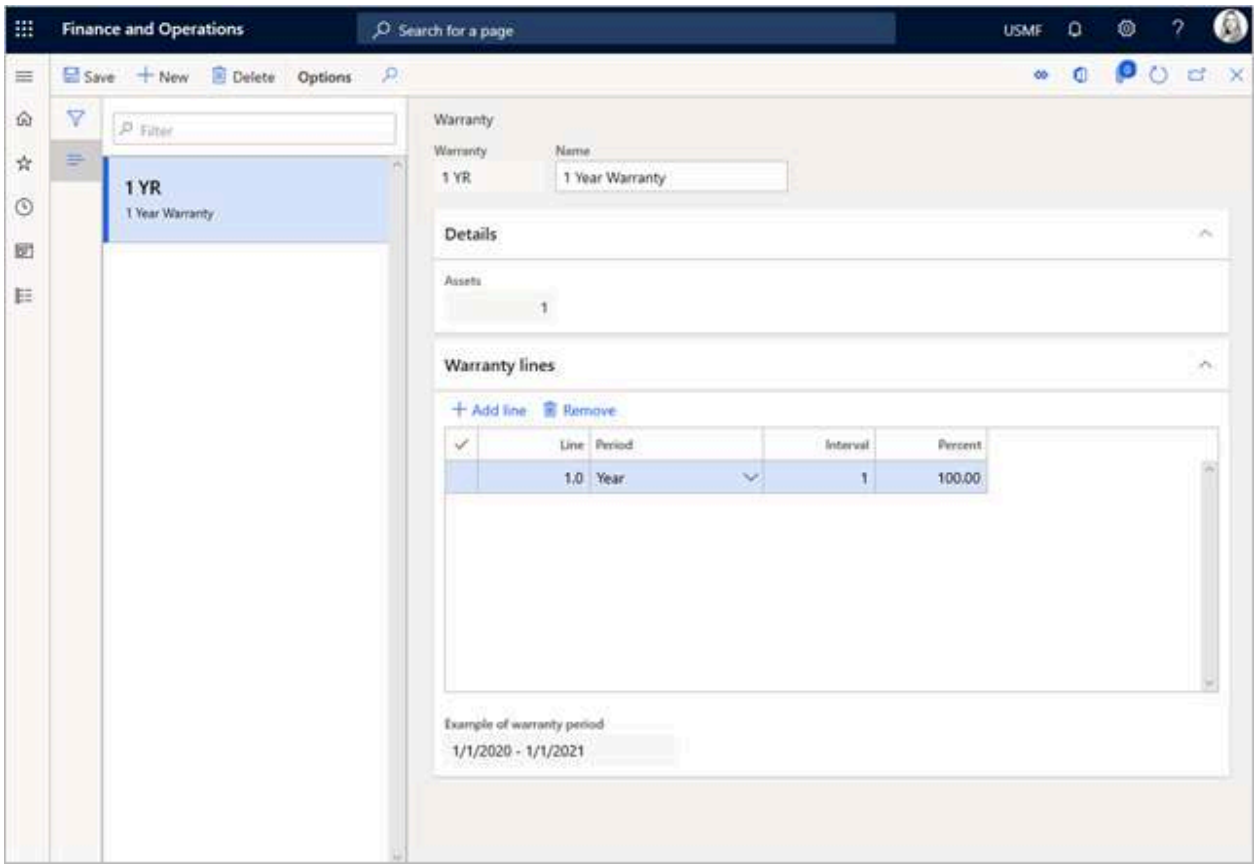
Create a warranty agreement

A warranty agreement can include several agreement lines to cover the warranty for work hours, expenses, and items.

1. Select **Asset management** > **Setup** > **Assets** > **Warranty**.
2. Select **New** to create a product.
3. In the **Warranty** field, enter a warranty ID.
4. In the **Name** field, enter a description.

   On the **Details** FastTab, the **Assets** field shows the number of active assets that use the warranty agreement.

5. On the **Warranty lines** FastTab, follow these steps to add lines that should be included in a warranty agreement:
   a. Select **Add line** to add a new condition to the warranty. A sequential line number is automatically entered in the **Line** field.
   b. In the **Period** field, select the type of warranty period.
   c. In the **Interval** field, enter a number. This field defines the number of periods that the warranty should be valid for.
   d. In the **Percent** field, enter the coverage percentage for the warranty line. The percentage indicates how much is covered by your company.



Feedback

Was this page helpful? Yes No

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Warranties on assets and asset types

Article • 06/15/2022

This article explains how to set up warranties on assets and asset types in Asset Management.

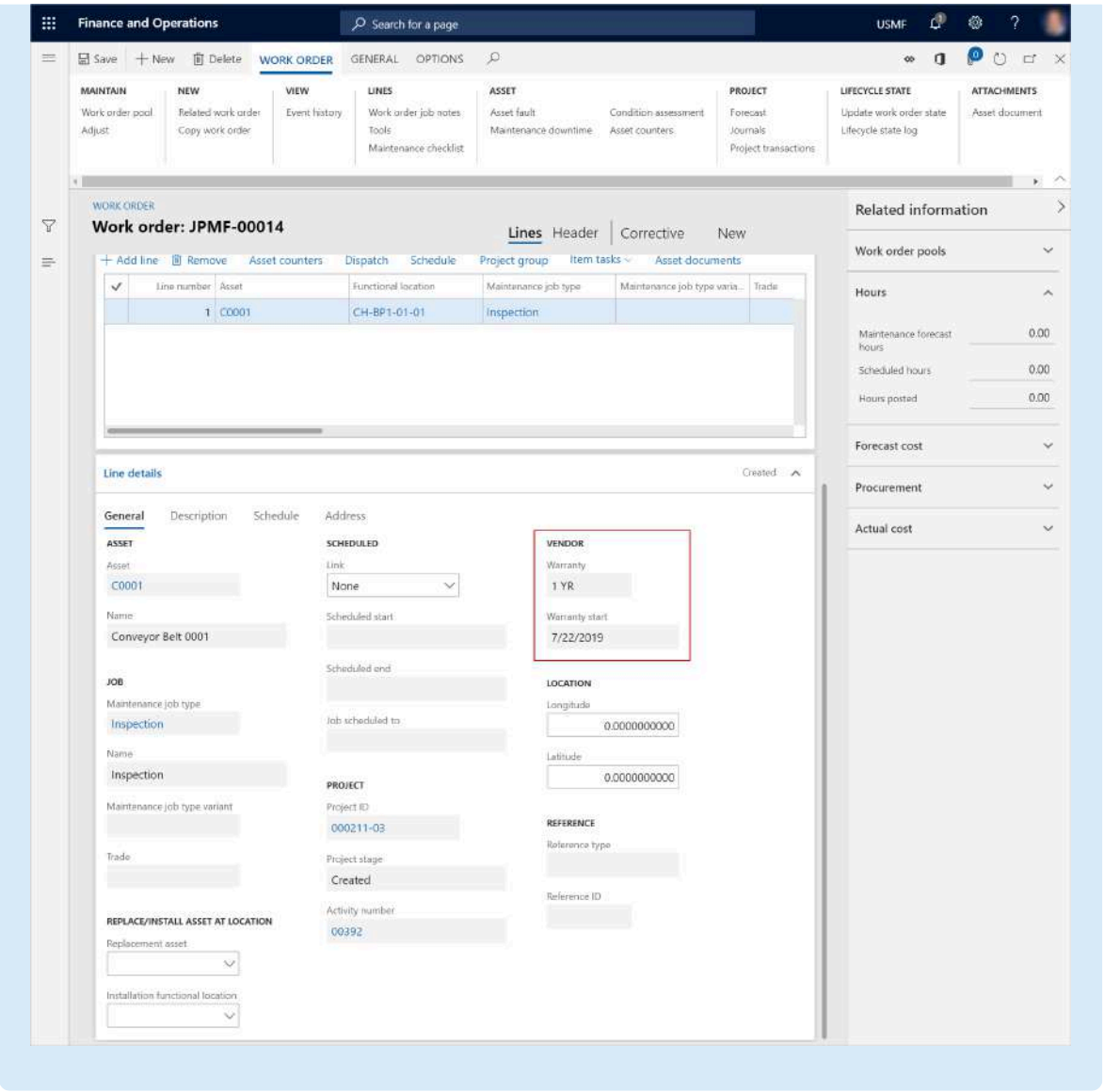
Set up a warranty on an asset type

1. Select **Asset management** > **Setup** > **Asset types** > **Asset types**.
2. In the left pane, select the asset type to attach a vendor warranty agreement to, and then select **Asset type defaults**.
3. On the **General** FastTab, in the **Vendor warranty** field, select the agreement.

Set up a warranty on an asset

1. Select **Asset management** > **Assets** > **All assets**.
2. Select the asset, and then select **Edit**.
3. On the **Vendor** FastTab, in the **Vendor warranty** section, in the **Warranty** field, select the warranty agreement.
4. In the **Warranty start** and **Warranty end** fields, select the start and end dates.

> **Important**
>
> If a date is selected in the **Warranty start** field on a work order, the warranty becomes valid for the work order on that date. When you create a work order, the **Warranty start** field is automatically set to the date of creation. However, you can change the date so that it corresponds to, for example, the start date of a warranty agreement.



> **Note**
>
> When you create a work order for an asset that is covered by a vendor warranty, if the work order has an expected start date during the warranty period, you receive a notification about the warranty agreement. You can then cancel the work order, as you require.

Feedback

Was this page helpful? Yes No

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Create maintenance budgets

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Maintenance budgets are used to provide an overview of expected costs for preventive maintenance. Budget lines are calculated based on maintenance schedule lines that have an expected start date during the budget period.

Maintenance budgets are based on the cost types that are used in Asset Management: **Preventive**, **Corrective**, and **Investment**. Budget costs for investment maintenance are included for active assets that have a replacement date during the budget period and a related replacement value. Budget costs for corrective maintenance are included if a past corrective date is included in the budget calculation. In that case, corrective costs from an earlier period are calculated for the same future period that you calculate the maintenance budget for.

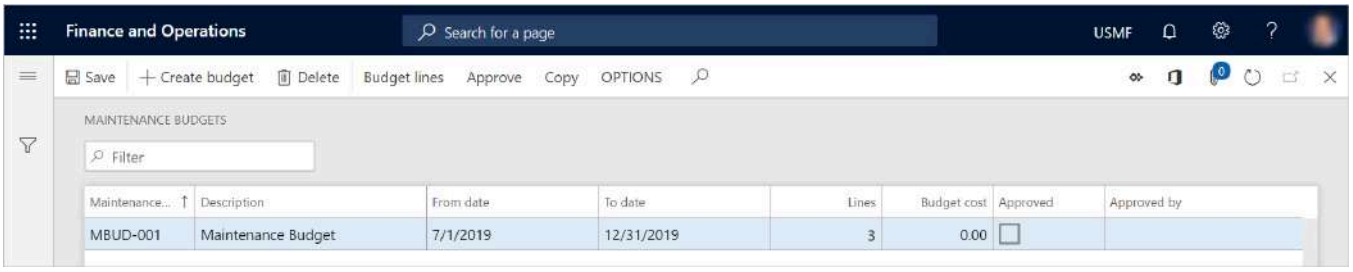
Create a maintenance budget

1. Select **Asset management** > **Inquiries** > **Maintenance budget** > **Budget**.
2. Select **Create budget**.
3. In the **Maintenance budget** field, enter a budget ID.
4. In the **Description** field, enter a description.
5. On the **Period** FastTab, in the **From date** and **To date** fields, enter the start and end dates of the budget period.
6. To include corrective budget costs that are calculated on the basis of actual costs from a previous period, in the **Corrective from date** field, enter the start date of the period that those costs should be included from.
7. Depending on the level of detail that is required in the budget, set the relevant options on the five **Group by** FastTabs.
8. Select **OK**.

9. Select **Budget lines** to open **Maintenance budget lines** page, where you can view all the budget lines that have been created for the period.

10. To approve the budget, select it on the **Maintenance budgets** page, and then select **Approve**. Then, in the **Approve budget** dialog box, select **OK**. Your name is entered in the **Approved by** field on the **Maintenance budgets** page.

> **Note**
>
> After you've approved a maintenance budget, you can't recalculate or adjust the related lines on the **Maintenance budget lines** page unless you first remove the approval. To remove the approval of a maintenance budget, select it on the **Maintenance budgets** page, and then select **Approve**. Then, in the **Approve budget** dialog box, select **OK**.



You can also create a new maintenance budget by copying an existing budget. On the **Maintenance budgets** page, select the budget to copy, and then select **Copy**. This approach is useful if, for example, you've created a budget for one month and want to copy it to other months.

> **Note**
>
> The maintenance budget calculates only budget costs based on maintenance schedule lines. To calculate actual costs for the same period, you can do that calculation on the **Asset cost control** page.

Last updated on 05/07/2025

Update maintenance budgets

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Maintenance budget lines** page shows all the budget lines that have been created for the budget that is selected on the **Maintenance budgets** page. (Learn more in Create maintenance budgets.) You can recalculate and adjust maintenance budget lines until the maintenance budget is approved. After the budget period has passed, and costs have been posted in Asset Management, you can also update the budget lines with actual costs.

Recalculate a maintenance budget

You can recalculate a maintenance budget on the **Maintenance budget lines** page. When you recalculate a maintenance budget, the existing budget lines are deleted, and a new calculation is done.

1. On the **Maintenance budget lines** page, select **Recalculate**.
2. In the **Recalculate budget** dialog box, make the required changes for the new calculation, and then select **OK**.

New budget lines are created according to the values that you set.

Adjust budget lines

Instead of recalculating the whole maintenance budget, you can adjust selected lines that are related to budget costs.

1. On the **Maintenance budget lines** page, select the lines to update the budget cost for.
2. Select **Adjust**.
3. To add an amount to the selected lines, select the **Add cost** check box, and then enter the value in the **Add value** field.
4. To multiply the budget cost on the selected budget lines by a factor, select the **Multiply cost** check box, and enter the factor in the **Multiply value** field.

For example, if you enter **1.2** in the **Multiply value** field, you increase the budget cost on the selected lines by 20 percent. If you enter **0.7**, you reduce the budget cost on the selected lines by 30 percent.

5. Select **OK**.

The selected budget lines are updated according to the values that you set in step 3 or 4.

Update actual costs

After the dates on the budget lines have passed, and actual costs have been posted in Asset Management, you can update the actual costs on the maintenance budget.

1. On the **Maintenance budget lines** page, select **Update cost**.
2. In the **Calculate actual cost** dialog box, select **OK**.

The **Actual cost** fields on the budget lines are updated if actual costs have been posted. New budget lines might be generated if new asset types have been created since you created the budget, and if those asset types have been used on assets that work orders have been created for and related costs have been posted for. New budget lines show only actual costs, because no budget costs were calculated for them.

> ⓘ **Note**
>
> To see an overview of actual costs divided into preventive, corrective, and investment costs, you can do a calculation for the same period on the **Asset cost control** page.

Manually add budget lines

On the **Maintenance budget lines** page, you can manually add a new budget line by selecting **New** and then making selections on the line. Here are some examples of situations where this approach might be useful:

- You know that refurbishment of some assets is currently in the planning phase, but related jobs haven't yet been created in Asset Management. However, you want budget costs for those jobs to be included in the maintenance budget.
- New assets or asset types have been created since you made the maintenance budget, but maintenance plans haven't yet been set up on those assets or asset types. However, you want budget costs for those asset types to be included in the maintenance budget.

Last updated on 05/07/2025

Cost and date control

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In Asset Management, you can calculate costs to get an overview of actual costs compared to budget costs on assets, functional locations, and work orders. Actual costs are based on posted transactions.

You can also make a date calculation if you want to compare scheduled start and end dates to actual start and end dates on work orders.

Cost control for assets, functional locations, and work orders

The calculations made for assets, functional locations, and work orders are almost identical. The only difference is that for assets and functional locations, you can also include sub assets and sub locations in your calculation. The date is the transaction date when the registration was recorded.

1. Click **Asset management** > **Inquiries** > **Assets** > **Asset cost control** or **Functional location cost control**, or **Asset management** > **Inquiries** > **Work orders** > **Work order cost control**.
2. In the **Asset cost control** / **Functional location cost control** / **Work order cost control** dialog, select a time range to be calculated.
3. If required, select a financial dimension set to be included in the calculation.
4. Select "Yes" on the **Skip zero** toggle button if you don't want to show results with a cost of zero.
5. You can use the **Level** field to indicate how detailed you want the cost control lines to be regarding functional locations.

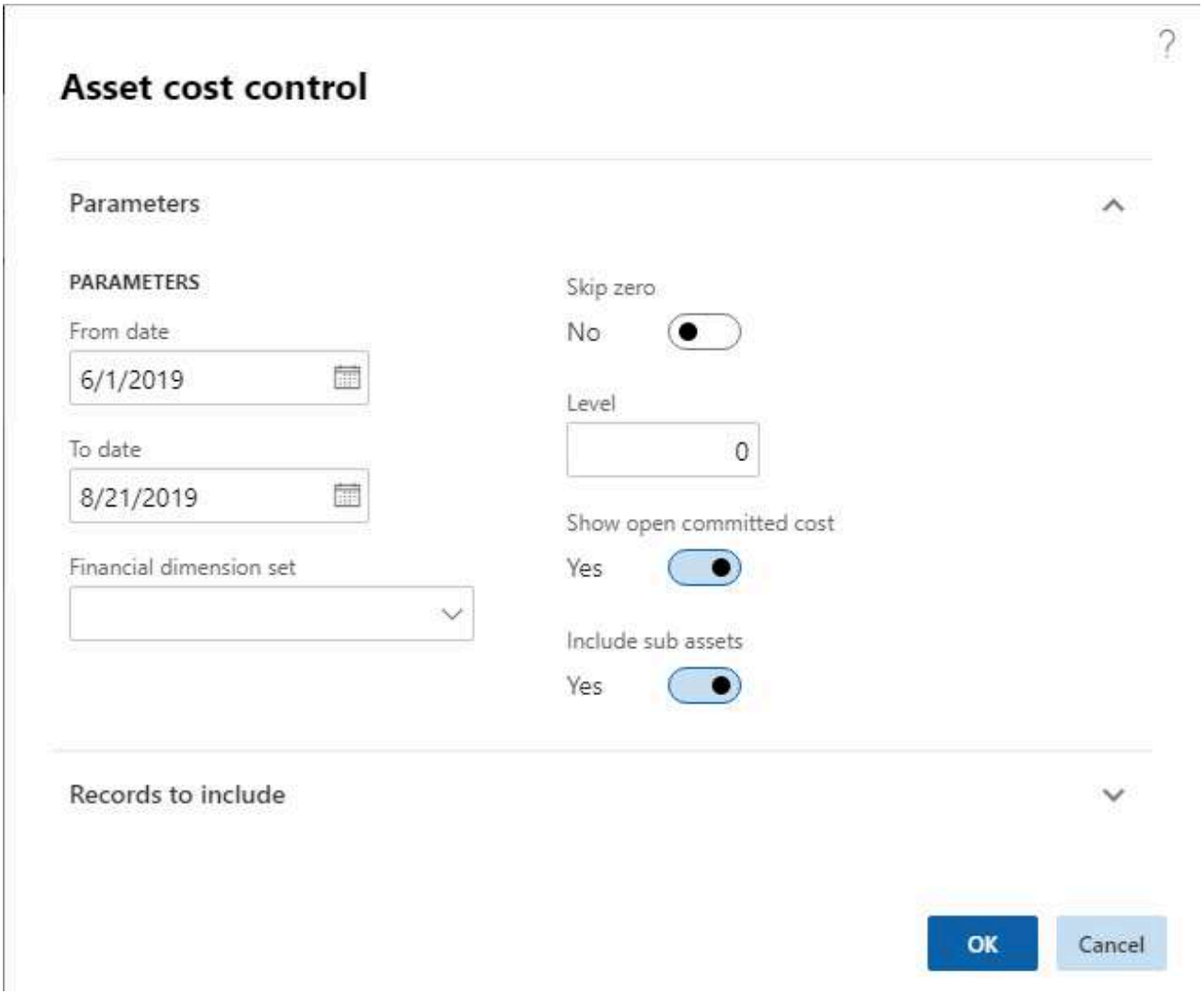
   For example, if you insert the number "1" in the field, and you have a multi-level functional location hierarchy, all cost control lines for a functional location will be shown

on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level.

If you insert the number "0" in the **Level** field, you will see a detailed result showing all cost control lines on all the functional location level to which they are related.

6. Select "Yes" on the **Show open committed cost** toggle button if you want to include that column in the calculation.
7. Select "Yes" on the **Include sub assets** toggle button to show costs related to sub assets as separate lines.
8. If you want to limit the search, you can select specific assets / functional locations / work orders on the **Records to include** FastTab.
9. Click **OK** to start the calculation.

   The figure below shows an example of the **Asset cost control** dialog.

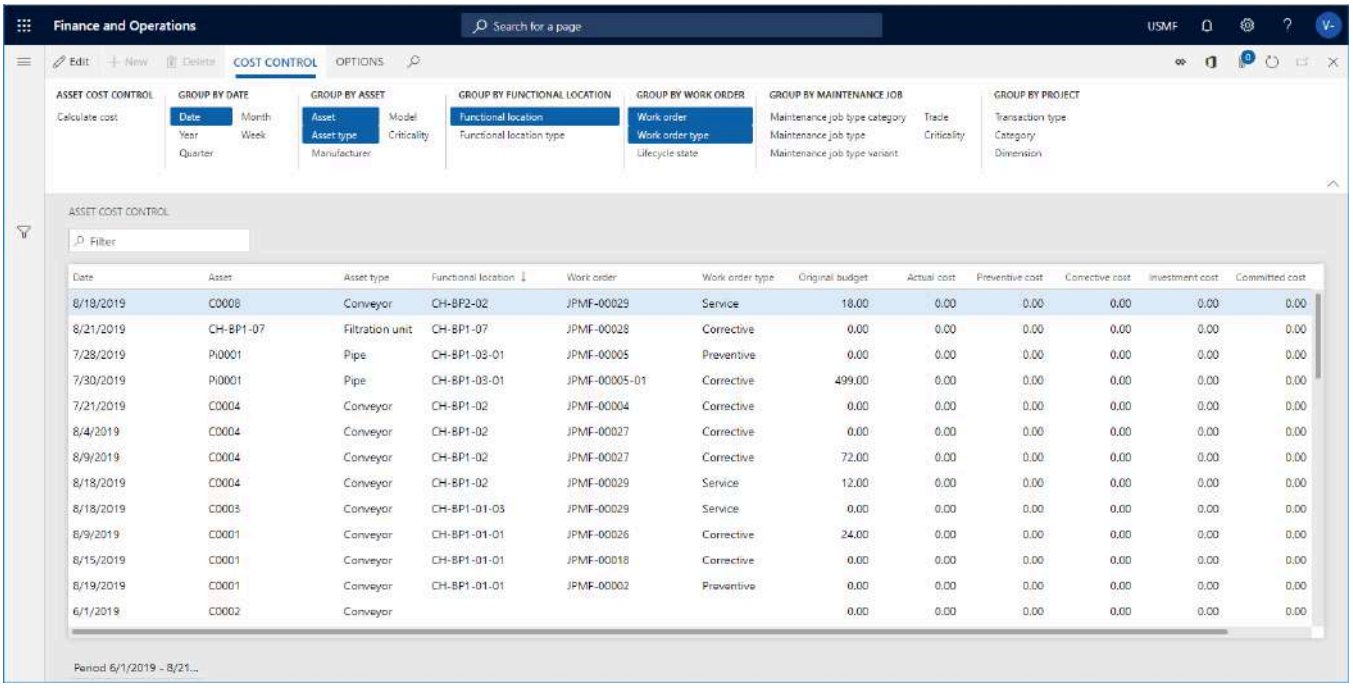


10. On the **Asset cost control** page, click the **Group by** buttons to show the required detail level of the calculation. The selected **Group by** buttons are highlighted. Click on a button to activate or deactivate it.

Example of calculation results in asset cost control

The screenshot below shows an example of calculation results in **Asset cost control**.

- The **Original budget** field shows budget costs from the work order forecast.
- The **Committed cost** field shows the total amount of expenses that a legal entity has committed itself to pay.
- The **Open committed cost** field shows commitments to pay for items, hours, and services you have ordered or received but not yet paid for.
- The **Actual cost** field shows related costs after all consumption registrations have been posted.



Another way of making a cost calculation is to multi-select assets in **All assets** or **Active assets**. Then, you click the **Cost control** button on the **General** tab. In the **Asset cost control** dialog, the selected assets are automatically inserted in the **Asset** field on the **Records to include** FastTab. Click **OK**, and a cost calculation for the selected assets is shown. The same procedure can be done for functional locations in **All functional locations** or **Active functional locations**, and for work orders in **All work orders** or **Active work orders**.

Work order date control

Use this page to get an overview of expected start and end dates compared to actual start and end dates on work orders.

1. Click **Asset management** > **Inquiries** > **Work orders** > **Work order date control**.

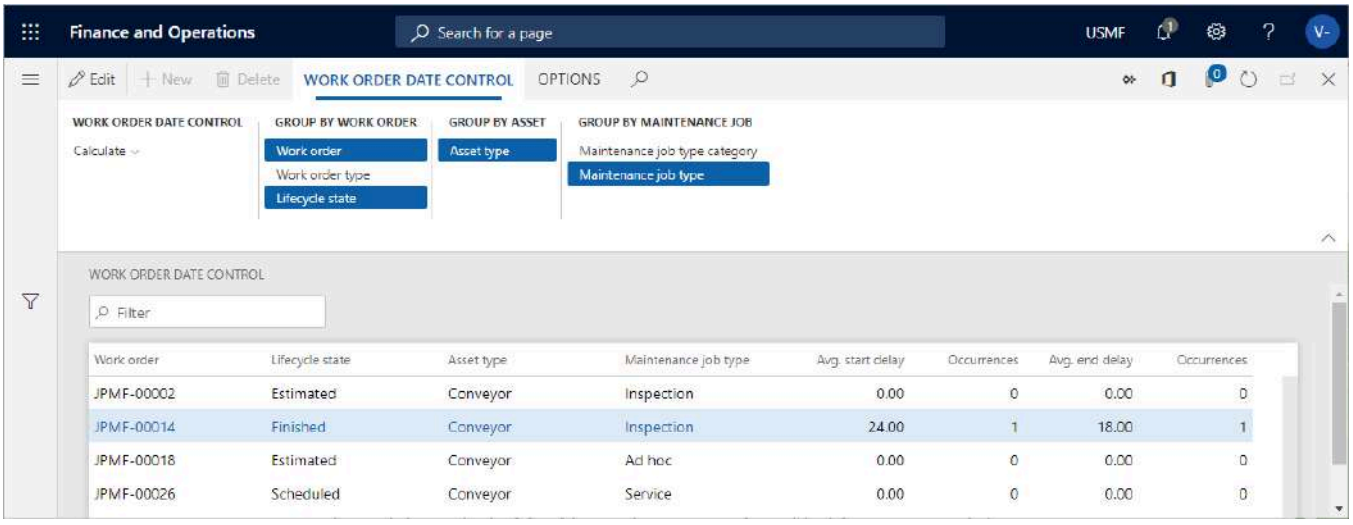
2. Click **Calculate**.

3. Select a functional location in the **Functional location** field.
4. Insert the range for which you want to make the calculation in the **From date** and **To date** fields. All work orders with expected start date within the range will be included.
5. Click **OK**.
6. Click the **Group by** buttons to show the required detail level of the calculation. The selected **Group by** buttons are highlighted. Click on a button to activate or deactivate it.

Example of calculation results in work order date control

The screenshot below shows an example of calculation results in **Work order date control**.

- The **Avg. start delay** field shows the difference in days between scheduled start date for a work order compared to actual start date. If, for example, the actual start date was two days before the scheduled start date, "-2" will be displayed in this field.
- The **Avg. end delay** field shows the difference in days between scheduled end date for a work order compared to actual end date. If, for example, the actual end date was three days after the scheduled end date, "3" will be displayed in this field.
- The **Occurrences** fields show the number of times deviations occur in relation to scheduled and actual start date, and scheduled and actual end date on the work order.



Last updated on 05/07/2025

Work hour control

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In Asset Management, you can calculate hours to get an overview of actual hours compared to budget hours on assets, functional locations, or work orders. Actual hours are based on posted transactions.

Work hour control for assets, functional locations, and work orders

The calculations made for assets, functional locations, and work orders are almost identical. Only difference is that for assets and functional locations, you can also include sub assets and sub locations in your calculation. The date is the transaction date when the registration was recorded.

1. Click **Asset management** > **Inquiries** > **Assets** > **Asset hour control** or **Functional location hour control**, or **Asset management** > **Inquiries** > **Work orders** > **Work order hour control**.

2. In the **Asset hour control** dialog, .

3. In the **Asset hour control** / **Functional location hour control** / **Work order hour control** dialog, select a period to be calculated in the **From date** and **To date** fields.

4. If required, select a **Financial dimension set** to be included in the calculation.

5. Select "Yes" on the **Skip zero** toggle button if you don't want to show results containing zero hours.

6. You can use the **Level** field to indicate how detailed you want the hour control lines to be regarding functional locations.

   For example, if you insert the number "1" in the field, and you have a multi-level functional location hierarchy, all hour control lines for a functional location will be shown on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level.

If you insert the number "0" in the **Level** field, you will see a detailed result showing all hour control lines on all the functional location levels to which they are related.

7. Select "Yes" on the **Include sub assets** toggle button to show costs related to sub assets as separate lines.

8. If you want to limit the search, you can select specific assets / functional locations / work orders on the **Records to include** FastTab.

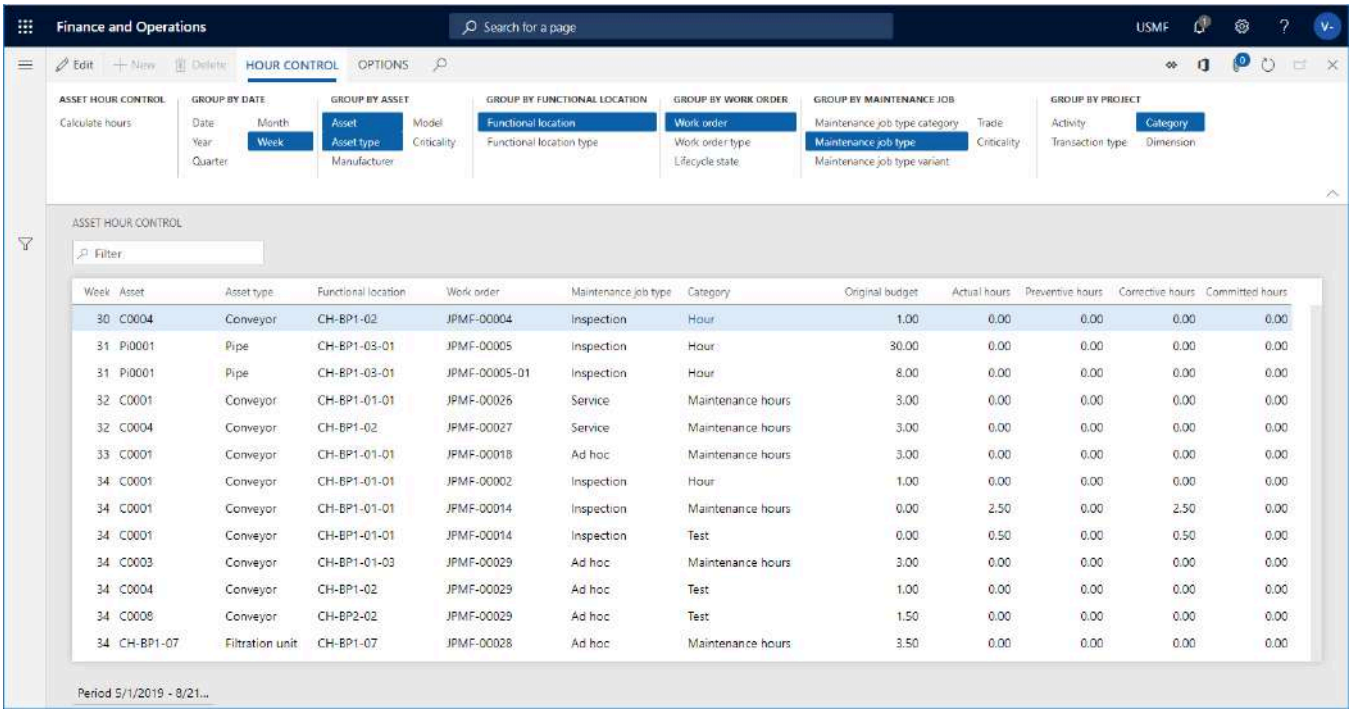
9. Click **OK** to start the calculation.

10. On the **Asset hour control** page, click the **Group by** buttons to show the required detail level of the calculation. The selected **Group by** buttons are highlighted. Click on a button to activate or deactivate it.

Example

The screenshot below shows an example of an **Asset hour control** calculation.

- The **Original budget** field shows budget hours from the work order forecast.
- The **Actual hours** field shows posted hours on work orders.
- The **Committed hours** field shows total amount of hours that your company is committed to in relation to work orders.



Another way of making an hour calculation is to multi-select assets in **All assets** or **Active assets**. Then you click the **Hour control** button on the **General** FastTab. The selected assets are automatically inserted in the **Asset** field on the **Records to include** FastTab. Click **OK** in the **Asset hour control** dialog, and the calculation for the selected assets is shown. The same

procedure can be done for functional locations in **All functional locations** or **Active functional locations**, and for work orders in **All work orders** or **Active work orders**.

Last updated on 05/07/2025

Asset fault cost control

Article • 06/03/2022

In Asset Management, you can calculate costs on asset fault registrations to get an overview of actual costs compared to budget costs. Actual costs are based on posted transactions. The date is the fault date on which the symptom was recorded.

1. Click **Asset management** > **Inquiries** > **Asset fault** > **Asset fault cost control**.
2. In the **Asset fault cost control** dialog, select a financial dimension set to be included in the calculation, if required.
3. Select "Yes" on the **Skip zero** toggle button if you don't want to show results with a cost of zero.
4. You can use the **Level** field to indicate how detailed you want the cost control lines to be regarding functional locations.

   For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all asset fault cost control lines for a functional location will be shown on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level.

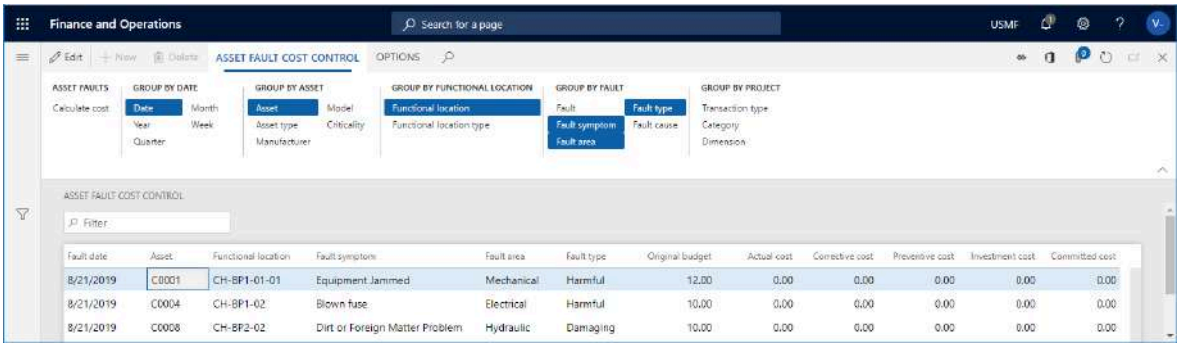
   If you insert the number "0" in the **Level** field, you will see a detailed result showing all asset fault cost control lines on all the functional location levels to which they are related.
5. If you want to limit the search, you can select specific assets, fault dates, and fault causes on the **Records to include** FastTab.
6. Click **OK** to start the calculation.
7. Click the **Group by** buttons to show the required detail level of the calculation. The selected **Group by** buttons are highlighted. Click on a button to activate or deactivate it.

Example

This example shows an asset fault cost control calculation.

- The **Original budget** field shows budget costs from the work order forecast.
- The **Actual cost** field shows posted costs on work orders.

- The **Committed cost** field shows total costs that your company is committed to in relation to work orders.



For information about how to set up faults, see the Fault management article.

Feedback

Was this page helpful? Yes No

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Asset fault analysis

Article • 06/03/2022

In Asset Management, you can analyze asset fault registrations to get an overview of the total number of faults registered during a specific period. Fault registrations can be analyzed from different perspectives, for example with focus on assets, asset types, functional locations, fault symptoms, or fault types.

1. Click **Asset management** > **Inquiries** > **Asset fault** > **Asset fault analysis**.
2. In the **Asset fault analysis calculation** dialog, you can use the **Level** field to indicate how detailed you want the asset fault lines to be regarding functional locations.

   For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all asset fault lines for a functional location will be shown on the top level, and therefore the hours on a line may be added up from functional locations located at a lower level.

   If you insert the number "0" in the **Level** field, you will see a detailed result showing all asset fault lines on all the functional location level to which they are related.
3. If you want to limit the search, you can select specific assets, fault dates, fault causes, and fault remedies on the **Records to include** FastTab.
4. Click **OK** to start the calculation.
5. On the **Asset fault analysis** tab, click one or more **Group by** buttons to display the detail level you want to see. Activated buttons are highlighted. Activate or deactivate buttons by clicking on them.
6. Click **Update calculations** to show your selections on the screen.

> ⓘ **Note**
>
> Every time you activate or deactivate a **Group by** button, remember to click the **Update calculations** button. This is required because a large amount of data is processed as you are recalculating fault probability.

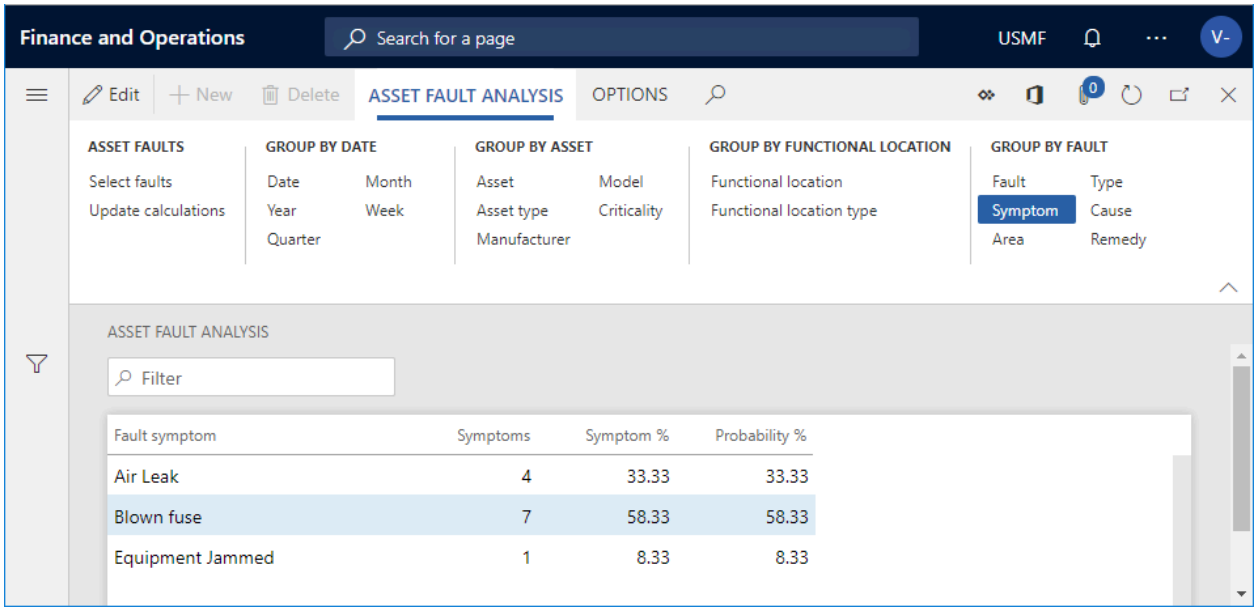
Examples

There are many ways to analyze fault registrations. This section has five examples of how different data selections can provide more insight and detail when analyzing asset fault registrations.

Group by symptoms

In the screenshot below, only the **Symptom** button is selected.

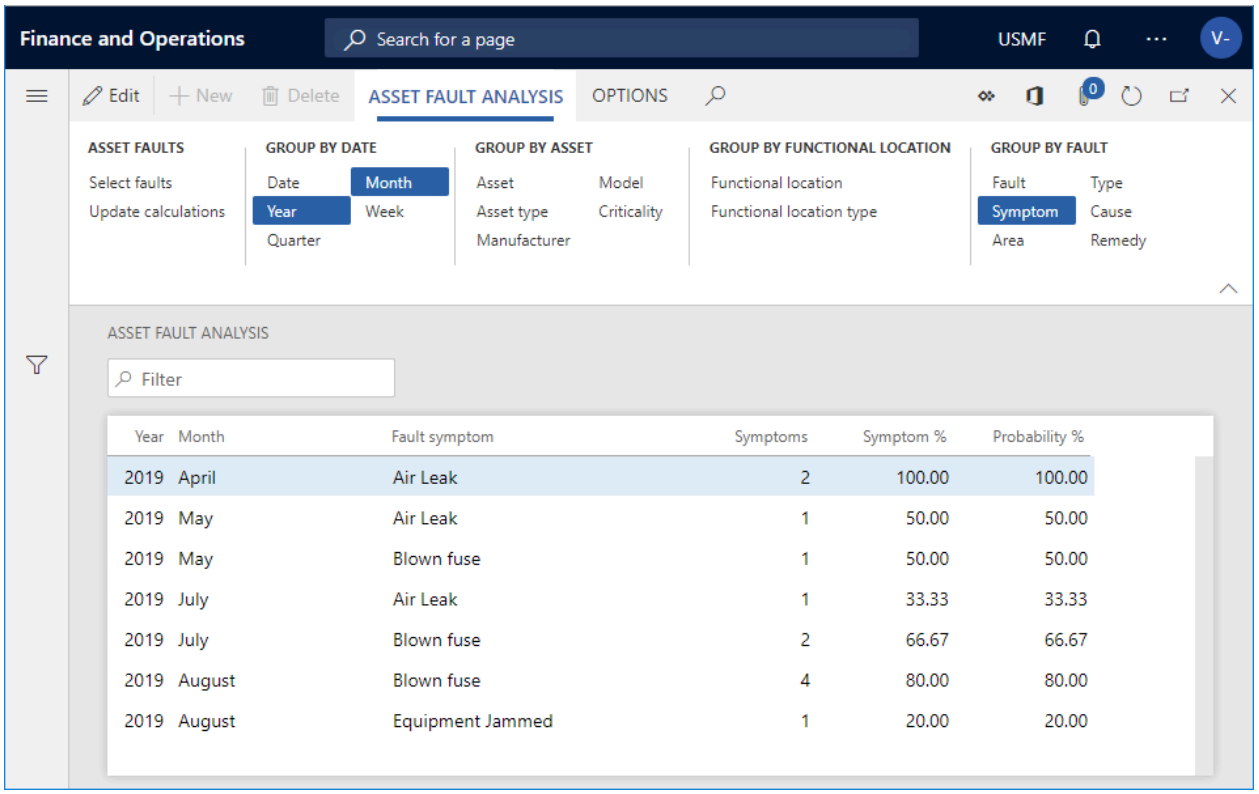
- Fault registrations have been made on three fault symptoms: "Air leak", "Blown fuse", and "Equipment jammed".
- In the **Probability %** column, all percentages add up to 100%. Probability is based on all **Symptom** registrations in this fault analysis.



Group by symptoms and time period

In the screenshot below, **Year** and **Month** are added to show how you can view fault registrations during a selected period.

- The fault symptoms are now shown as registrations per year/month.
- In the **Probability %** column, if you add all percentages for each month, they add up to 100%. Probability is based on the **Symptom** registrations in this fault analysis. If you have a large number of lines on an asset, but a large percentage stands out on a line, that would be an indication of a fault symptom to examine more closely to find a way to limit the number of registrations for that fault symptom.



Group by multiple symptoms and assets

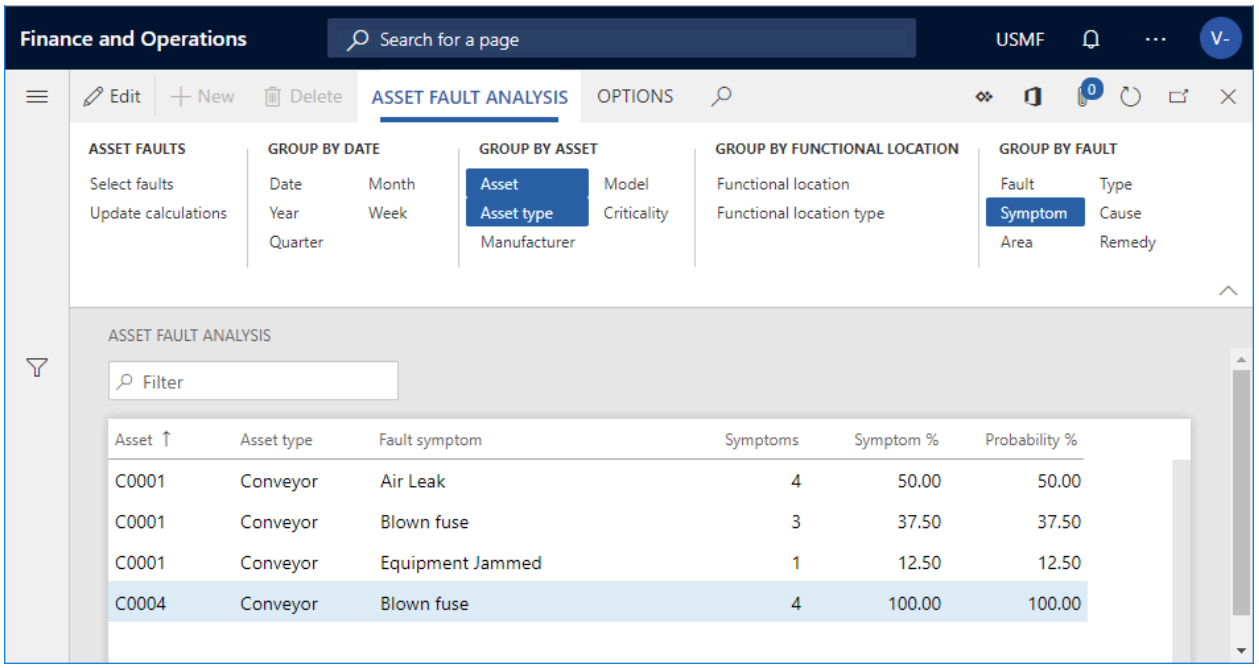
The combination of assets and an asset type is used as a basis for the calculations shown in the three screenshots below, which will increase in detail level.

Generally, the buttons in the **Group by date**, **Group by asset**, **Group by functional location** Action Pane groups, as well as the **Fault** button (Fault ID), contain periods or asset relations. The **Symptom**, **Area**, **Type**, **Cause**, and **Remedy** buttons are categorizations used in fault management to analyze asset fault registrations and pinpoint problem areas.

Group by symptom, asset, and asset type

In the screenshot below, **Asset** and **Asset type** were added to provide more detail regarding fault registrations.

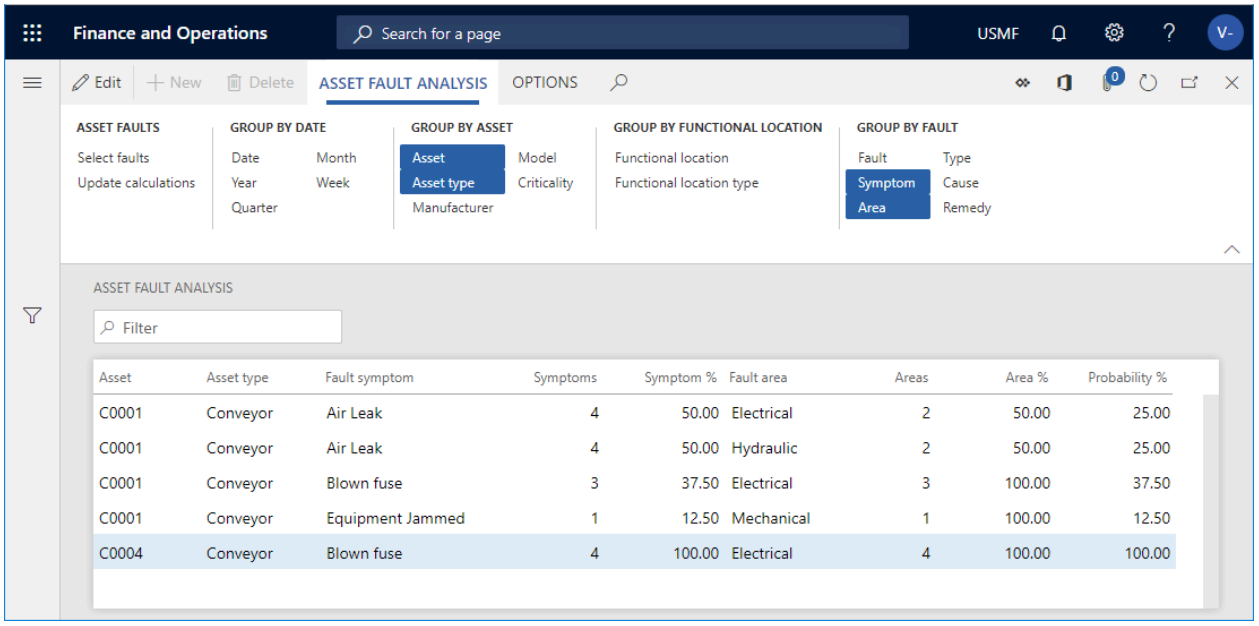
- The fault symptoms are now split up in **Asset** / **Asset type** / **Symptom** combinations.
- In the **Probability %** column, if you add all percentages for the combination of **Asset** / **Asset type** / **Symptom** respectively, they each add up to 100%. Probability is based on **Symptom** registrations in this fault analysis. If you have a large number of lines on an asset, but a large percentage stands out on a line, that would be an indication of a fault symptom to examine more closely to find a way to limit the number of registrations for that fault symptom.



Group by two symptoms, asset, and asset type

In the screenshot below, **Area** was added to **Symptom**, **Asset**, and **Asset type** to provide more detail regarding fault registrations.

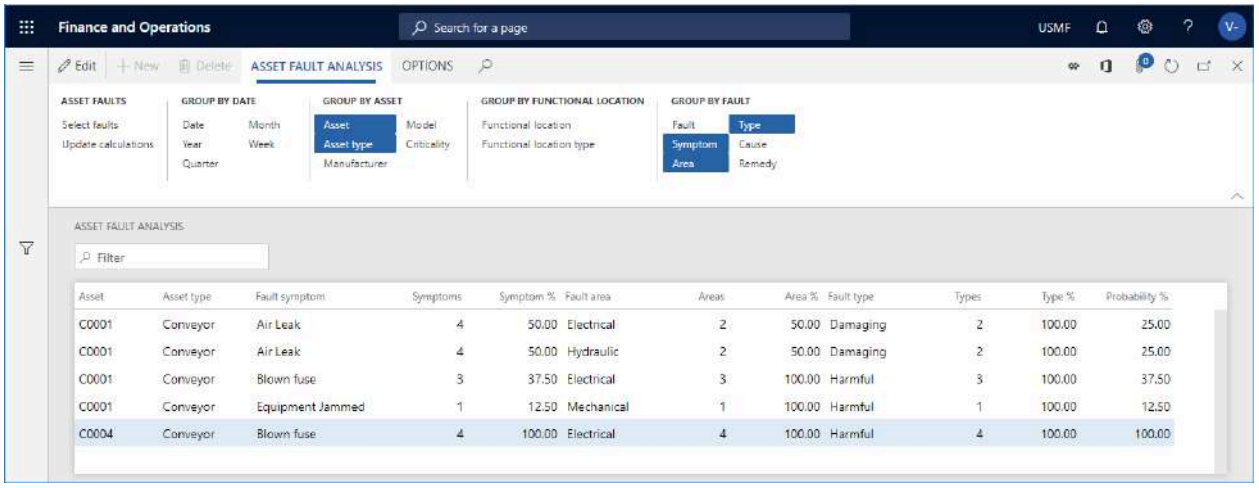
- In the **Probability %** column, if you add all percentages for the combination of **Asset** / **Asset type** / **Symptom** on an asset, they each add up to 100%. Probability is based on the combination of **Symptom** and **Area** in this fault analysis. If you have a large number of lines on an asset, but a large percentage stands out on a line, that would be an indication of a fault area to examine more closely to find a way to limit the number of registrations for that fault area.



Group by three symptom, asset, and asset type

In the screenshot below, **Type** was added, and the most detailed calculation in this example is shown.

- In the **Probability %** column, if you add all percentages for the combination of **Asset** / **Asset type** / **Symptom** on an asset, they each add up to 100%. Probability is based on the combination of **Symptom**, **Area**, and **Type** in this fault analysis. If you have a large number of lines on an asset, but a large percentage stands out on a line, that would be an indication of a fault type to examine more closely to find a way to limit the number of registrations on that fault type.



> **Note**
>
> For an overview of all fault registrations created on work orders and maintenance requests, click **Asset management** > **Inquiries** > **Asset fault** > **Asset faults**. On the **Asset faults** page, select an asset fault registration and expand the **Related information** pane to see information regarding the related work order or maintenance request.

Feedback

Was this page helpful? Yes No

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Asset KPIs

Article • 06/03/2022

In Asset Management, you can calculate various Key Performance Indicators (KPIs) for assets and asset types. You use KPIs to get an overview of performance on assets in relation to, for example, uptime, downtime, repair time, and Mean Time Between Failure (MTBF).

1. Click **Asset management** > **Inquiries** > **Assets** > **Asset KPIs**.
2. In the **Calculate asset KPIs** dialog, you select the **Time scale** to be used in the calcualtion, and a period in the **From date** and **To date** fields.
3. On the **Records to include** FastTab, you can select specific assets and asset types to be included in the calculation, if required.
4. Click **OK** to start the calculation.
5. On the **Overview** FastTab, the results of the calculation are displayed in grid view. Each asset is displayed on a separate line.
6. On the **KPIs for selected line** FastTab, you see calculations for the asset selected on the **Overview** FastTab. The KPI values are categorized regarding **Time**, **Availability**, **Work orders**, **Maintenance**, **Faults**, **Maintenance downtime**, and **Cost**.

In the table below, you'll find a description of the fields on the **Asset KPIs** page.

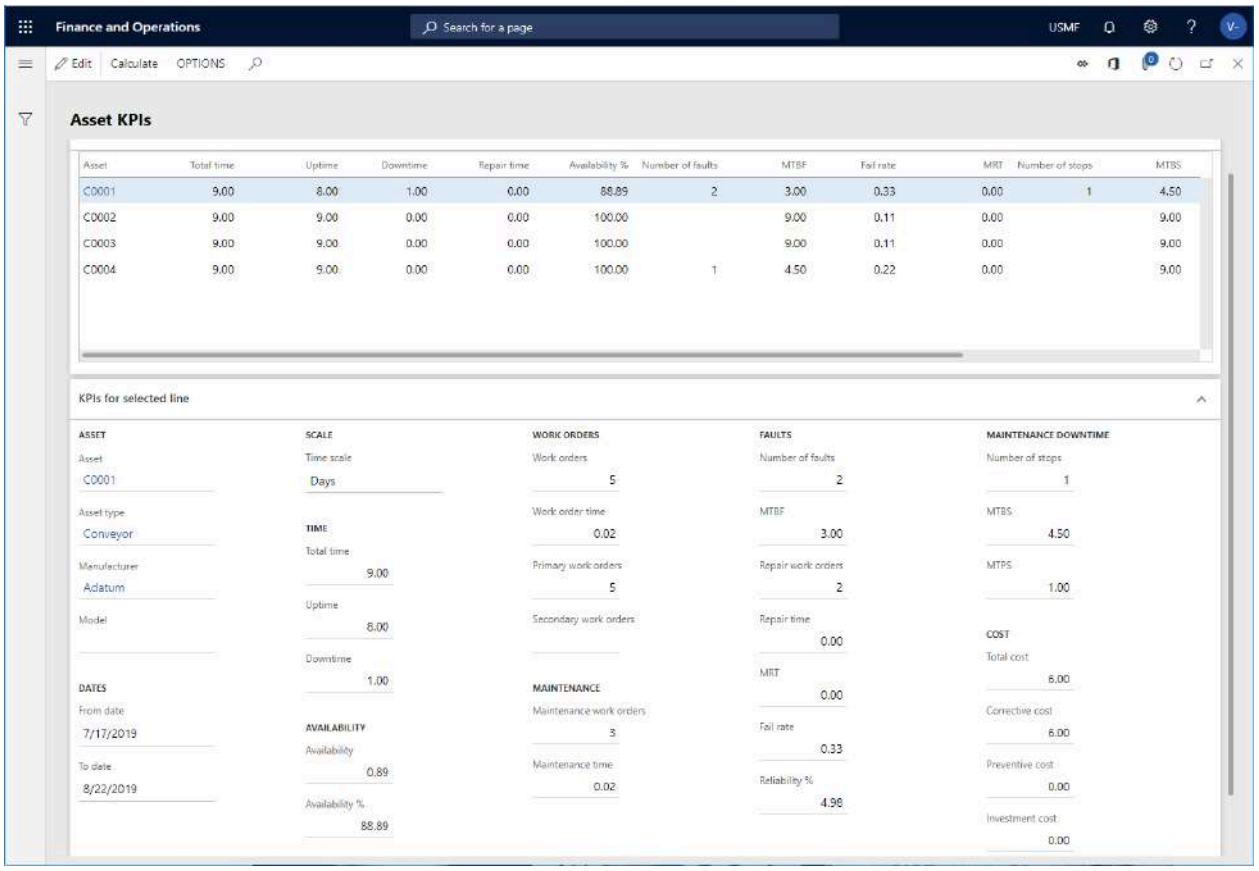
Expand table

Field	Description
Asset	Asset ID.
Total time	Total time set up in the calendar used in the calculation. If the asset is related to a resource, the related resource calendar is used. If the asset isn't related to a resource, the calendar selected in the **Standard calendar** field in **Asset management parameters** is used.
Uptime	Total time minus downtime.
Downtime	Maintenance downtime registrations made on the asset in the selected period.
Repair time	Total number of work hours consumed on repair work orders.
Availability %	Uptime divided by total time and multiplied by 100.

Field	Description
Number of faults	Number of fault causes registered on fault symptoms on the asset in the selected period.
MTBF	Mean Time Between Failure, which is total time divided by number of faults registered on the asset in the selected period. If number of faults is zero, MTBF is set to total time.
Fail rate	1 divided by MTBF.
MRT	Mean Repair Time, which is repair time divided by number of faults registered on the asset in the selected period. If number of faults is zero, MRT is set to repair time.
Number of stops	Number of maintenance downtime registrations created on the asset.
MTBS	Mean Time Between Stops, which is total time divided by number of maintenance downtime registrations. If number of maintenance downtime registrations is zero in the selected period, the MTBS value is set to total time.
Preventive cost	Costs posted on the asset related to cost type "Preventive" in the selected period. Cost types are set up on work order types.
Corrective cost	Costs posted on the asset related to cost type "Corrective" in the selected period. Cost types are set up on work order types.
Replacement value	Value defined on the asset as the replacement cost.
Asset type	Identification of the asset type selected on the asset.
Manufacturer	Identification of the manufacturer selected on the asset.
Model	Identification of the manufacturer model selected on the asset.
From date	Start date of the KPI calculation. If the asset was created after the start date selected for the calculation, the start date of the asset is shown in this field.
To date	End date of the KPI calculation. If the asset was registered as inactive before the end date selected for the calculation, the date from which the asset was no longer active is shown in this field.
Time scale	During calculation of the KPIs, you select a time scale to be used: Hours, Days, or Weeks.
Availability	Uptime divided by total time.
Work orders	Total number of work orders included in the KPI calculation.
Work order time	Total number of work hours consumed on the work orders.

Field	Description
Primary work orders	Number of primary work orders included in the KPI calculation.
Secondary work orders	Number of secondary work orders included in the KPI calculation.
Maintenance work orders	Number of maintenance work orders included in the KPI calculation. A maintenance work order is a work order with no related fault causes.
Maintenance time	Total number of work hours consumed on maintenance work orders.
Repair work orders	Number of repair work orders included in the KPI calculation. A repair work order is a work order with a related fault cause.
Reliability %	Calculation based on the expected exponential development in fault registrations on an asset meaning that, over time, the asset becomes less reliable due to wear and tear. The calculation of this KPI is based on MTBF and total time.
MTPS	Mean Time Production Stops, which is maintenance downtime divided by number of maintenance downtime registrations. If number of maintenance downtime registrations is zero in the selected period, the MTPS value is set to zero.
Total cost	Total costs posted on the asset in the selected period.
Investment cost	Costs posted on the asset related to cost type "Investment" in the selected period. Cost types are set up on work order types.

The figure below shows a screenshot of a KPI calculation for four assets.



- You can multi-select several assets in **All assets** and click the **Asset KPIs** button on the **General** tab. Then click **OK** in the **Calculate asset KPIs** dialog to calculate KPIs for the selected assets.
- Results from a KPI calculation may or may not include maintenance downtime registrations, depending on the setup and use of maintenance downtime reason codes.

Feedback

Was this page helpful? Yes No

Provide product feedback

Item where used

Article • 06/03/2022

You can make a calculation for a specific item to get an overview of where in Asset Management the item has been used. The results show the context in which the item has been used during its lifetime. The **Item where used** page can be opened from the main Asset Management menu, and it can also be accessed from the following pages:

- Asset BOMs
- Spare parts on asset type defaults
- Maintenance job type categories and maintenance job types, maintenance job type variants, maintenance job trades, and maintenance checklists
- Maintenance forecast
- Procurement
- Work order purchase

Make an item-where-used calculation

1. Click **Asset management** > **Inquiries** > **Item where used**, or select the **Item where used** button on one of the pages mentioned above.
2. In the **Item where used** dialog, select the item for which you want to make the calculation in the **Item number** field.
3. You can use the **Level** field to indicate how detailed you want the item lines to be regarding functional locations.

   For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all item lines for a functional location will be shown on the top level. Therefore, relation/quantity on a line may be added up from functional locations located at a lower level.

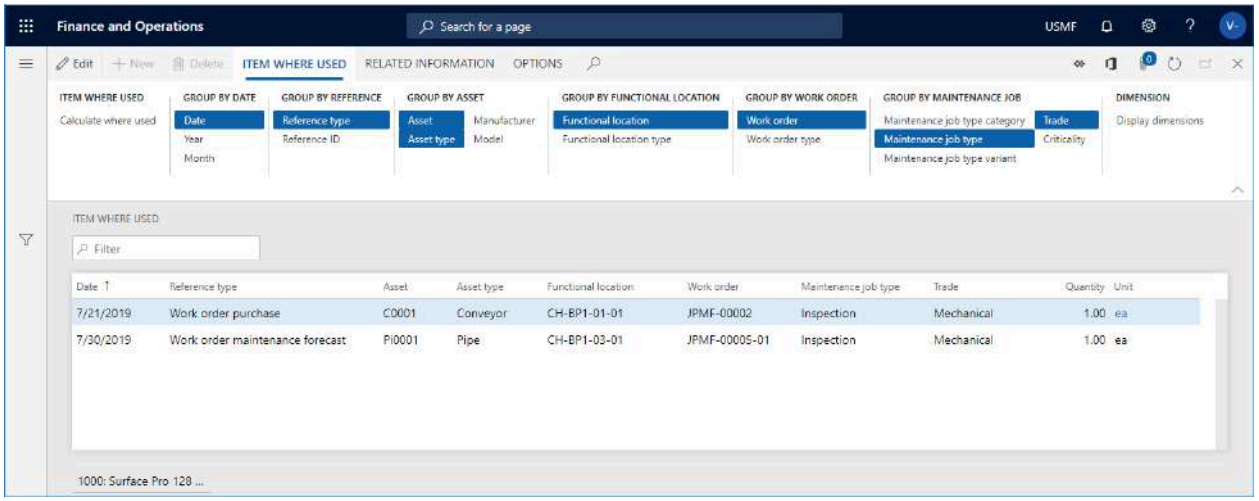
   If you insert the number "0" in the **Level** field, you will see a detailed result showing all item lines on all the functional location levels to which they are related.
4. In the **Include** section, select "Yes" on the toggle buttons that you want to include in the calculation.
5. Click **OK** to start the calculation.

6. On the **Item where used** tab, select the **Group by** buttons to show the required detail level of the calculation. The selected **Group by** buttons are highlighted. Click on a button to activate or deactivate it.

7. If you want to show dimensions related to the item, click **Display dimensions**, and select the dimensions to be shown.

Example

In the screenshot below, you see an example of an item-where-used calculation for item number "1000".



Feedback

Was this page helpful? Yes No

Provide product feedback

Maintenance status

Article • 06/03/2022

In Asset Management, you can make an overview calculation for a specific period for new, active, and completed maintenance requests, work orders, and maintenance downtime activities. You can also see the number of completed condition assessments for the same period. Use this calculation to get an overview of workload for incoming and completed maintenance requests and work orders.

Make a maintenance status calculation

1. Click **Asset management** > **Inquiries** > **Maintenance status**.
2. In the **Calculate status** dialog, select the time range that you want to make the calculation in the **From date** and **To date** fields.
3. You can use the **Level** field to indicate how detailed you want the maintenance lines to be regarding functional locations.

For example, if you insert the number "1" in the field, and you have a multi-level functional location structure, all maintenance lines for a functional location will be shown on the top level, and therefore the status on a line may be added up from functional locations located at a lower level.

If you insert the number "0" in the **Level** field, you see a detailed result showing all maintenance lines on all the functional location levels to which they are related.

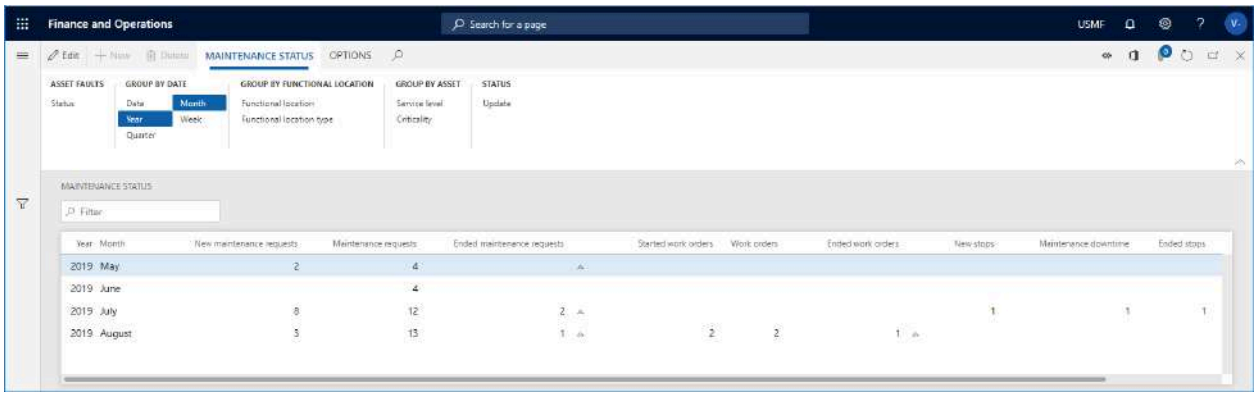
4. Click **OK** to start the calculation.
5. Click the **Group by** buttons to show the required detail level of the calculation. The selected **Group by** buttons are highlighted. Click on a button to activate or deactivate it.
6. Remember to click the **Update** button to update the calculation each time you make changes by activating or deactivating **Group by** buttons, or by making a calculation for a new period.
7. Click **Status** if you want to create a new maintenance status calculation.

> **Note**

The results shown in **Maintenance status** only include maintenance requests and work orders that have an actual start date and time. End date and time may be blank.

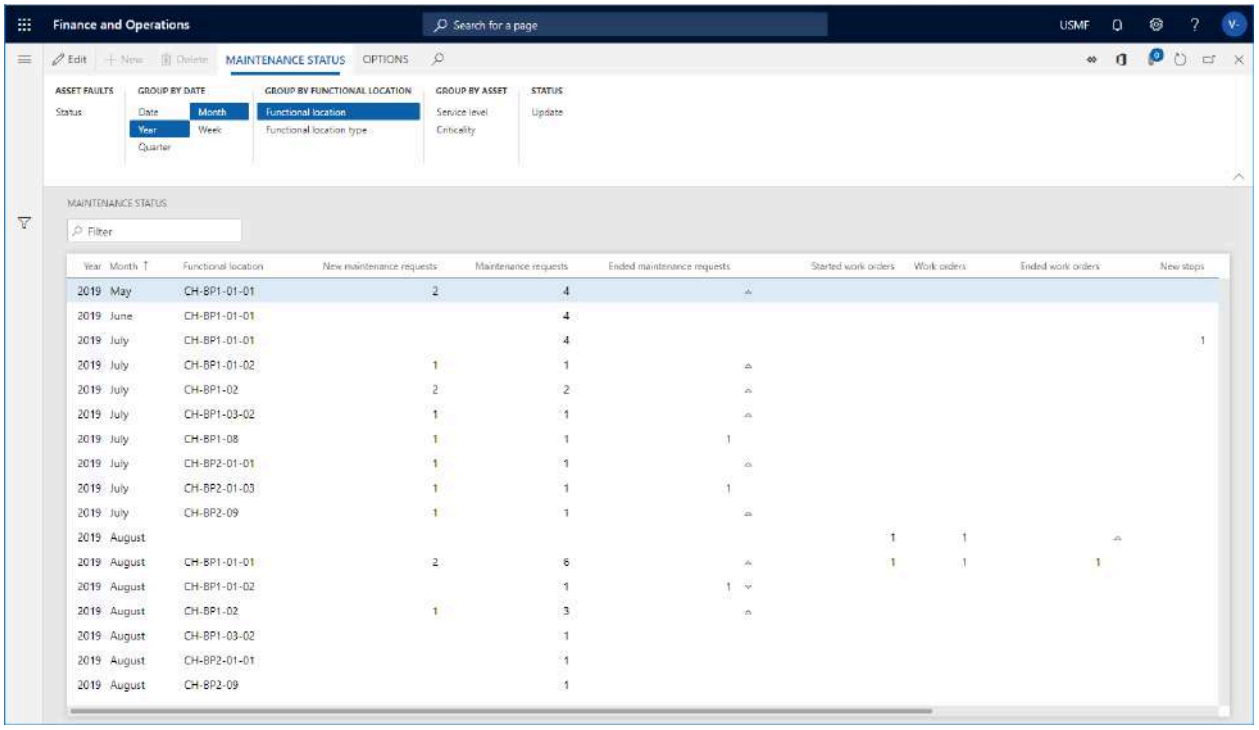
Example 1

In the screenshot below, the **Year** and **Month** buttons have been activated. With these **Group by** options selected, you get a general overview on a monthly basis of workload and throughput related to maintenance requests and work orders.



Example 2

In the screenshot below, information about functional locations has been added. Now it is possible to compare workload and throughput across functional locations, which may represent geographical locations, factories, or work areas.



Feedback

Was this page helpful? Yes No

Provide product feedback

Asset Management mobile app overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> The Asset Management mobile app is now provided on GitHub as an open-source sample at microsoft/scmsamples-EnterpriseAssetManagement. It provides the same functionality as the original app and is freely customizable. For timeline and transition guidance, see Asset Management mobile app open-source release.

The Asset Management mobile app provides the following capabilities for workers and organizations that use Asset Management for Microsoft Dynamics 365 Supply Chain Management:

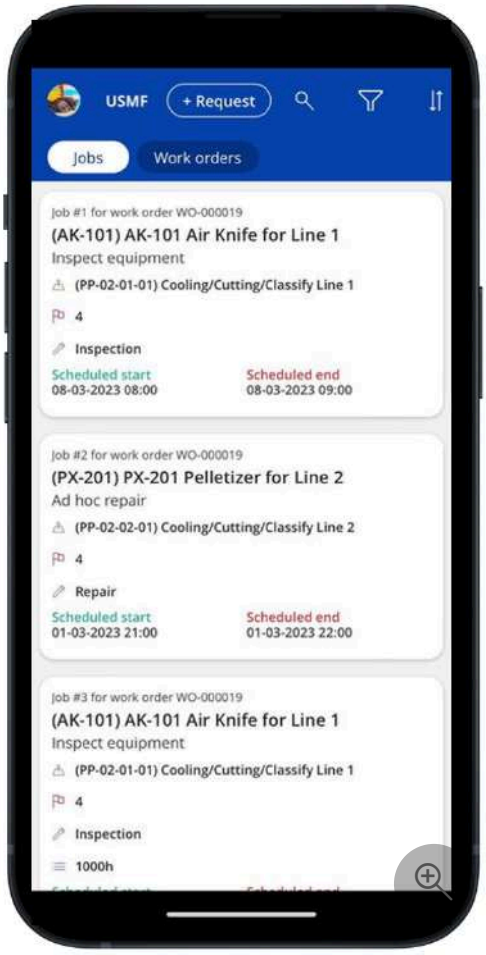
- **Manage work orders.** Maintenance workers use work orders as a daily to-do list that provides the information that they need to complete their planned maintenance tasks. In the app, workers can search for work orders that have been assigned to them. They then use the app to record their progress, so that the system can track labor, materials, and services for the work that's done. Workers can process work orders and perform tasks such as updating checklist items, registering time and materials, and viewing and adding notes.
- **Create maintenance requests.** Administrators can use the role-based security setup in Supply Chain Management to grant workers permission to create new maintenance requests. Workers who receive this permission will then be able to use the mobile app to proactively request maintenance of assets.
- **Create work orders.** Maintenance workers can create new work orders from scratch, or from existing work orders that need further work in the future.

To use the app, you must have a fully implemented setup of Asset Management in your Supply Chain Management environment.

The following video shows how the Asset Management mobile app supports various business roles in common asset management scenarios.

https://learn-video.azurefd.net/vod/player?id=0781126d-d706-42af-b4f5-e4f727a55e8e&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fasset-management%2Fasset-management-mobile-app%2Fasset-management-mobile-app-overview

The following illustration shows an example of a list of jobs for a work order in the Asset Management mobile app.



> **Note**
>
> Use of the Asset Management mobile app is covered by the *Dynamics 365 Supply Chain Management* user licenses. Using the app for maintenance requests requires a *Dynamics 365 Team Members* license. Using the app for processing work orders requires a *Dynamics 365 Operations – Activity* license. Licensing terms are subject to change without notice. For complete and up-to-date licensing information, see the Dynamics 365 Licensing Guide.

Next steps

- Asset Management mobile app open-source release
- Onboard the Asset Management mobile app
- Manage work orders using the Asset Management mobile app
- File maintenance requests using the Asset Management mobile app

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Last updated on 05/29/2026

Asset Management mobile app open-source release

> **Important**
>
> The Asset Management mobile app is now available on GitHub as an open-source sample at microsoft/scmsamples-EnterpriseAssetManagement. It provides the same functionality as the original app and is freely customizable. The existing app remains available for installation through Dynamics 365 apps in the Power Platform admin center until November 30, 2026, with critical fixes from Microsoft during this transition period. After that date, the app is no longer distributed through the admin center. Existing installations continue to work in customer tenants.

Until now, canvas-app customizations made in Power Apps Studio weren't preserved across official updates. Releasing the source code as open source removes that limitation: you can build from the source, customize it freely, and deploy under your own publisher without your changes being overwritten by Microsoft updates.

Open-source release timeline

The following table summarizes the key milestones in the open-source release of the Asset Management mobile app.

Expand table

Date	Milestone
June 1, 2026	Source code released as open source. The app enters a 6-month transition period during which it remains available for installation through the Power Platform admin center, with critical fixes from Microsoft.
November 30, 2026	End of the transition period. Microsoft maintenance concludes and the app is no longer distributed through the Power Platform admin center. Existing installations continue to function in customer tenants.

Guidance for existing installations

Continue using the installed app

Existing installations require no action. Microsoft delivers any critical fixes during the maintenance window through the same Power Platform admin center channel as previous versions. After November 30, 2026, the installed app continues to function in the customer tenant. Customers who plan to evolve the app beyond that date should follow the Build from the open-source code guidance.

Customize the app for in-tenant use

For additive customizations to the data model (such as adding new fields, tables, or security roles), use solution layering in the Power Apps maker portal. These customizations live in the customer tenant's unmanaged solution layer and are preserved when Microsoft delivers managed-solution updates during the maintenance window.

For customizations to the canvas app itself (such as screens, formulas, or UI changes), we recommend that you build from the open-source code so your changes persist. Customizations made directly in Power Apps Studio aren't preserved across Microsoft updates during the maintenance window.

Build from the open-source code

To customize the app and make it your own, build from the open-source code provided on GitHub at microsoft/scmsamples-EnterpriseAssetManagement. The source code is licensed under the MIT license, which permits free use, modification, and redistribution.

By default, builds from the open-source code use a Microsoft-branded sample publisher (`AssetManagementMobileSample`, display name "Microsoft Asset Management Mobile Sample") in Microsoft Dataverse. This publisher is distinct from the first-party Microsoft publisher used by in-market Dynamics 365 products. The default publisher is appropriate when you customize and run the app in your own organization's tenant. If you plan to distribute your custom build to other organizations, change the publisher and customization prefix to your own brand before building. The repository README documents this process.

After producing a custom build under your own publisher, transitioning an existing tenant requires that you delete the existing solutions (`msdyn_EnterpriseAssetManagementMobileV2` and `msdyn_EnterpriseAssetManagementMobileAnchor`) and import the custom-built solution. For details on deleting solutions, see Work with solutions in Power Apps.

Guidance for new installations

For Supply Chain Management customers who need field-level work order management, we recommend that you build a custom solution from the open-source code provided on GitHub at microsoft/scmsamples-EnterpriseAssetManagement. The MIT-licensed source can be customized freely and deployed under your own publisher to fit your needs. Alternatively, evaluate Microsoft AppSource for partner-built alternatives.

The existing app remains available for installation through the Power Platform admin center until November 30, 2026. For new deployments, build from the open-source code instead.

Support

For product issues during the maintenance window, use your existing Microsoft support agreements and Dynamics 365 support channels. After November 30, 2026, Microsoft doesn't provide product support for the app. Issues filed against the open-source repository aren't triaged.

Last updated on 06/01/2026

Onboard the Asset Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> The Asset Management mobile app is now provided on GitHub as an open-source sample at microsoft/scmsamples-EnterpriseAssetManagement. It provides the same functionality as the original app and is freely customizable. For timeline and transition guidance, see Asset Management mobile app open-source release.

This article describes how administrators can prepare your Microsoft Dynamics 365 Supply Chain Management and Dataverse environments to support the Asset Management mobile app. It also describes how to install the app on your mobile devices.

Prerequisites

Before you can start to onboard the Asset Management mobile app, you must meet the requirements that are outlined in this section.

System requirements

To run the latest Asset Management mobile app, you must be using Supply Chain Management version 10.0.36 or later.

Set up Dataverse for your Supply Chain Management environment

The mobile app uses Dataverse virtual tables that are connected to Supply Chain Management. Therefore, a Dataverse environment must be integrated with your Supply Chain Management

environment. If Dataverse isn't already set up for your environment, follow the instructions in Microsoft Power Platform integration with finance and operations apps.

When you create the Dataverse environment where you want to install the app, be sure to enable Dynamics 365 apps.

The Power Apps component framework feature must be enabled for your environment.

Licensing requirements

To gain access to the app, users must have the appropriate security roles assigned to them in Supply Chain Management. The roles that are available for each user in Supply Chain Management depend on the licenses assigned to them in Microsoft Entra ID. For details about which security roles grant access to which functionality in the Asset Management mobile app, see Configure users and workers in Supply Chain Management. For details about which licenses grant access to which security roles, see the Dynamics 365 Licensing Guide.

Install the mobile app in Dataverse

You must install the Asset Management mobile app in your Dataverse environment to enable users to access it when they sign in by using the Power Apps mobile app. The installation process also sets up the required user roles and other dependencies in Dataverse.

Follow these steps to install the Asset Management mobile app in Dataverse.

1. Sign in to Power Platform admin center.
2. On the left navigation pane, select **Resources** > **Dynamics 365 apps**.
3. Search for and select the app named *Dynamics 365 Asset Management Mobile Application*.
4. On the top toolbar, select **Install**.
5. Select the environment where you want to install the app, review the packages that will be installed, and select the **I agree to the terms of service** checkbox.
6. Select **Install**.

Grant access to the mobile app in Dataverse

After the mobile app solution is installed in your Dataverse environment, you must share it with your users. The Asset Management mobile app is a canvas app. To share it, follow the instructions in Share an app.

Each relevant user must be assigned a role that lets them work with the Dataverse Supply Chain Management virtual tables that the solution installs. The solution installs a role named *Asset Management Mobile Application User Role* that grants access privileges to all the required virtual tables. You can assign this role to a Dataverse group team. Any user who's a member of that team will then have the role too. Alternatively, you can assign the role directly to a user.

Each relevant user must also have the *Finance and Operations Basic User* role.

- To assign a role to a group team, follow the instructions in Manage the security roles of a team. We recommend that you use group teams if you must assign the role to multiple users. For information about how to manage team members, see Manage team members.
- To assign a role directly to a user, follow the instructions in Assign a security role to a user.

Configure users and workers in Supply Chain Management

Each user who will use the Asset Management mobile app must be set up as described in this section.

Set up user accounts to maintain assets and/or request maintenance

Each Supply Chain Management user who should be able to operate the Asset Management mobile app must have one or both of the following standard security roles:

- *Maintenance requester* – This role allows a Supply Chain Management user account to use the app to create maintenance requests.
- *Maintenance worker* – This role allows a Supply Chain Management user account to use the app to manage maintenance work orders.

Users who have the *System administrator* role can use the app both to create maintenance requests and to manage work orders.

Follow these steps to add the required security roles to a Supply Chain Management user and review that user's **Person** setting.

1. Sign in to your Supply Chain Management environment by using an administrator account.
2. Go to **System administration** > **Users** > **Users**.
3. Use the filter to find a user who should be able to use the app.

4. Open the details page for the selected user by selecting the hyperlink in the **User ID** column.
5. Review the value of the **Person** field to determine which person is assigned to the current user account. If a person is assigned, write down their name. If no person is assigned, you'll assign one in the next section.
6. On the **User's roles** FastTab, on the toolbar, select **Assign roles** to open the **Assign roles to user** dialog box.
7. In the **Role name** column, find and select the *Maintenance requester* and *Maintenance worker* roles.
8. Select **OK** to apply your settings and close the dialog box.

> ⓘ **Note**
>
> User role organization assignments aren't supported. If one or more organizations that differ from a selected user's default company have been assigned to any security role that's set for that user, app installation will fail.

For more information about how to set up roles and security in Supply Chain Management, see Security roles.

Set up a maintenance worker for each user

Follow these steps to confirm or create a maintenance worker for a selected user account.

1. Go to **Asset Management** > **Setup** > **Workers** > **Workers**.
2. Follow one of these steps, depending on whether a person was assigned to the user account that you set up in the previous section:
   - If a person was assigned, confirm that the same person is listed here as a maintenance worker. If they aren't, set up a maintenance worker for that person now. If you must set up a worker, follow the instructions in Maintenance workers and worker groups.
   - If no person was assigned, either select a maintenance worker in the list here, or add the person that you want to use as a maintenance worker by following the instructions in Maintenance workers and worker groups. Then go back to **System administration** > **Users** > **Users**, and open the user account that you're setting up. In the **Person** field, assign the maintenance worker that you just selected or added.

> ⊗ **Caution**

The **Person** setting for user accounts is permanent. You won't be able to edit it after you save your changes. Therefore, make sure that you select the correct person when you set up the user account.

Install and open the Asset Management mobile app

Follow these steps to install and use the Asset Management mobile app on a mobile device.

1. Install the Power Apps mobile app by following the instructions in Install the Power Apps mobile app.
2. Open the Power Apps mobile app, and sign in by using the same corporate account that you use to sign in to Supply Chain Management.
3. Use the **Search** field to search for *Asset Management V2*. Because it's a canvas app, you can add it to your favorites list in Power Apps by swiping left on it after you find it.
4. Open the Asset Management mobile app, and start to use it.

Clear the Power Apps cache on a mobile device after an update of the back-end setup

If the back-end setup is updated in Supply Chain Management after you install the Power Apps mobile app on a mobile device (for example, if the legal entity setup is changed), you might have to clear the cache in Power Apps to update the local setup data. Power Apps will then be able to reload the settings that otherwise might not take effect until the cache is cleared.

Follow these steps to clear the Power Apps cache on a mobile device.

1. Sign in to the Power Apps mobile app.
2. Select the image for your user account in the upper left of the page to open the settings menu.
3. Select **Clear cache** to open the **Clear cache** dialog box.
4. Select **Confirm**.

Last updated on 05/29/2026

Manage work orders using the Asset Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> The Asset Management mobile app is now provided on GitHub as an open-source sample at microsoft/scmsamples-EnterpriseAssetManagement. It provides the same functionality as the original app and is freely customizable. For timeline and transition guidance, see Asset Management mobile app open-source release.

Maintenance workers can use the Asset Management mobile app to manage and process maintenance work orders. The app supports maintenance workers through the following main capabilities:

- It lists maintenance jobs and work orders that are assigned to a worker, including all information that the worker needs to process each job.
- Workers can register the time and spare parts that are consumed for each job.
- Workers can view and update the maintenance checklist that's associated with a job.

For more information about maintenance work orders in Supply Chain Management, see Introduction to work orders.

User requirements

To view and process work orders by using the Asset Management mobile app, you must meet the following requirements:

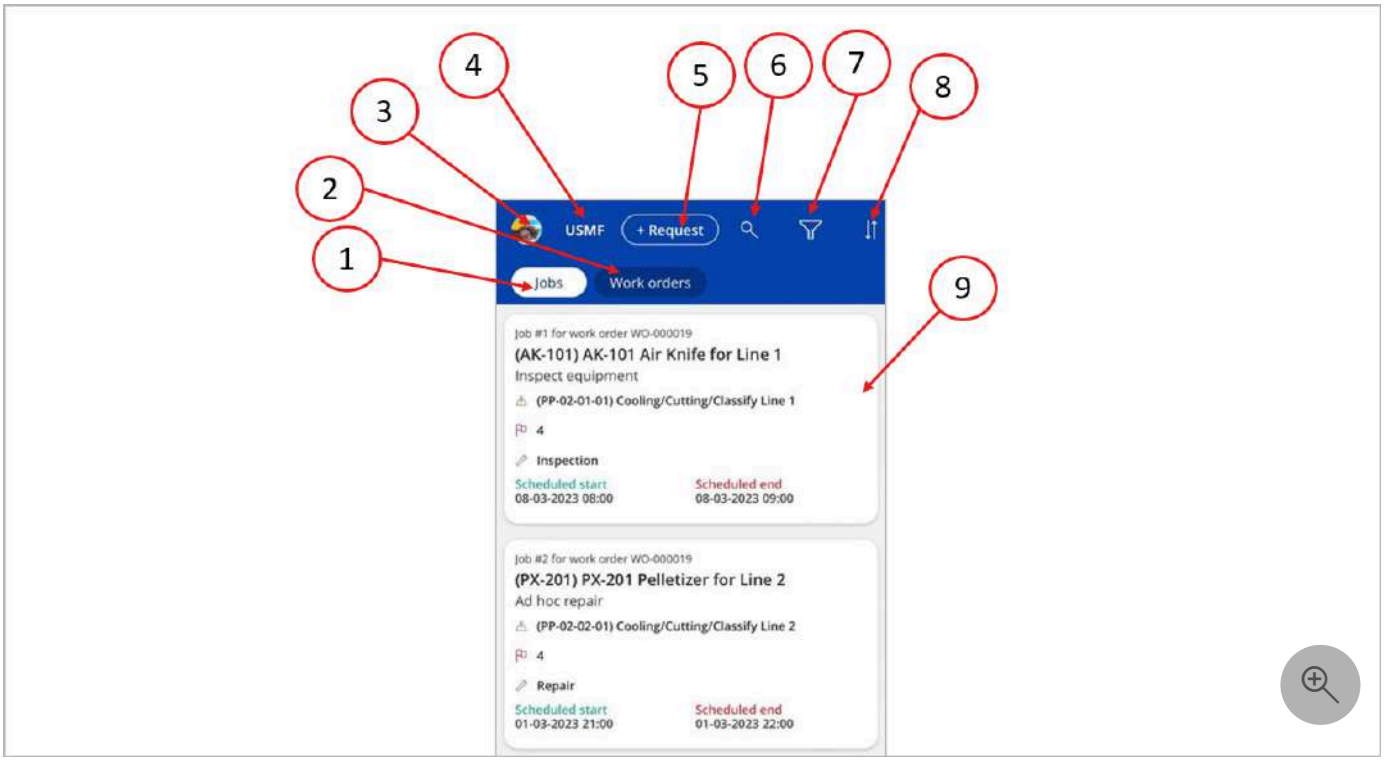
- Your user account in Supply Chain Management must be assigned the *Maintenance worker* security role. Learn more in Onboard the Asset Management mobile app.

- Your user account in Supply Chain Management must be associated with a human resources *Worker* record that's also set up as an Asset Management worker. Learn more in Onboard the Asset Management mobile app.
- You must sign in to Power Apps by using a domain account that matches a user account in Supply Chain Management that has the same Microsoft Entra ID username.

View the jobs and work orders assigned to you

When you open the Asset Maintenance mobile app and sign in as a user who has the *Maintenance worker* security role, the list of jobs and work orders is shown.

The following illustration highlights the different user interface (UI) elements for working with maintenance work orders.



The job and work order list includes the UI following elements. The numbers correspond to the numbers in the previous illustration.

1. **Jobs tab** – Select this tab to view the job list. The list shows only maintenance jobs that both are assigned to you and belong to a maintenance work order that's *Active*.

2. **Work orders tab** – Select this tab to view the work order list. The list shows only active work orders that have jobs that are assigned to you.

3. **User image** – Select this image to view information about the app, such as terms and conditions and the current version of the app.

4. **Legal entity identifier** – This label shows the legal entity (company) that you're currently working in. The lists show only jobs and work orders that are associated with this legal entity. If your Supply Chain Management user account is set up as an Asset Management worker in more than one legal entity, you can select this label to switch between legal entities.

5. **Request button** – If your Supply Chain Management user account is assigned the *Maintenance requester* security role, you can use this button to create a maintenance request. For more information about this functionality, see Maintenance requests.

6. **Search button** – When you select this button, a search field appears. There, you can enter text to search for the ID of the work order, asset, or functional location that you're looking for.

7. **Filter button** – Select this button to filter the jobs or work orders in the list, based on the following criteria:

    - *Today* – Show only jobs that are scheduled for today.
    - *This week* – Show only jobs that are scheduled to start during the current week.
    - *All time* – Show all jobs.

8. **Sort order button** – Select this button to specify how the list is sorted. You can sort by work order service level, scheduled start date, or work order ID.

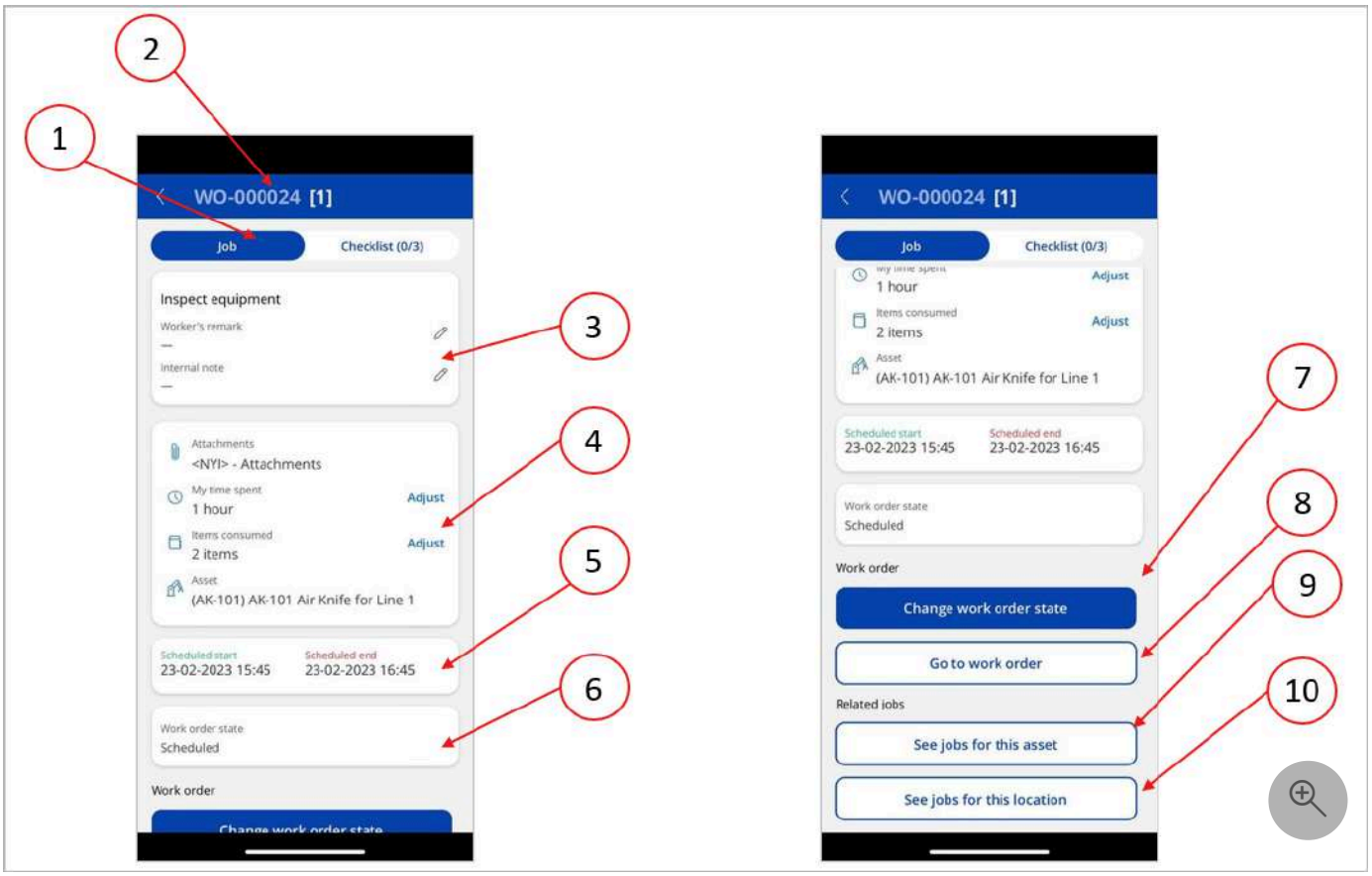
9. **Job or work order cards** – Each job or work order in the list is presented as a *card* that summarizes the item. Tap a card to open its details page, which provides more information about the selected job or work order.

Job details page

In the job lists, each job is presented as a *card* that shows summary information. Tap a card to open its details page, which provides more information about the selected job. The job details page provides access to both job information and the maintenance checklist.

Job information

To view the job information, select the **Jobs** tab at the top of the page. The following illustration highlights the different UI elements on the **Jobs** tab.



The **Jobs** tab of the job details page includes the following UI elements. The numbers correspond to the numbers in the previous illustration.

1. **Job and Checklist tabs** – Select a tab to switch between the job details and the checklist. The numbers in parentheses on the **Checklist** tab indicate the number of completed tasks and the total number of tasks on the checklist.

2. **Work order identifier** – This heading shows the ID of the work order that the current job belongs to. The number in square brackets indicates the total number of jobs in the work order.

3. **Job information** – This section shows the job description, and lets you view and edit a worker's remark and an internal note.

4. **Attachment, time spent, and materials consumed information** – This section lets you view and open documents that are attached to the job. You can also adjust the time that's spent and the materials that are consumed while you work on the job. In Supply Chain Management, the hours that are spent on a maintenance job are accounted for in a project journal. For more information about how materials and time are accounted for in journals, see Register consumption.

    - To adjust the number of hours that are spent working on the job, select **Adjust** next to the **My time spent** heading.

- To adjust the materials that are consumed while you work on the job, select **Adjust** next to the **Items consumed** heading. You can adjust quantities in the list of items that are expected to be consumed for the job. You can also add new items. When you add new items, you can select from lists of released products, items on the asset bill of materials (BOM), and spare parts for the current asset. When you select a product that will be consumed, you can specify storage dimensions (site, warehouse, and location) and tracking dimensions (batch and serial number) as required. Items that have product variants (such as configuration, color, and size) aren't listed.

5. **Scheduled start and end dates and times** – This section shows the dates and times when the current job was expected to be done.

6. **Work order state** – This section shows the current state of the work order.

7. **Change work order state button** – Select this button to change the state of the parent work order for the current job. You'll typically use this button to mark the work order as completed or to note a problem that prevents it from being completed.

8. **Go to work order button** – Select this button to open the work order that the current job belongs to.

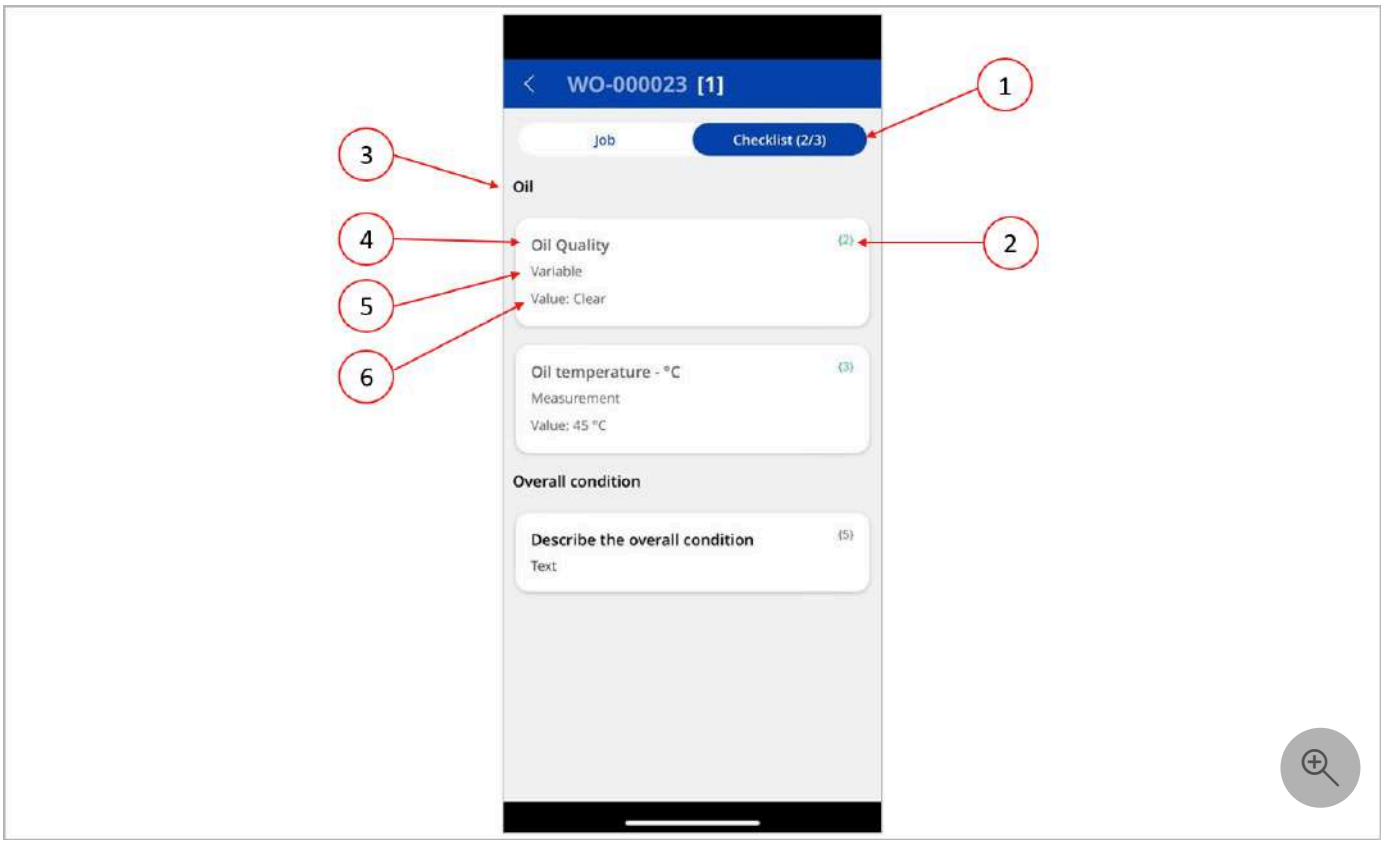
9. **See jobs for this asset button** – Select this button to view a list of all open maintenance jobs that are associated with the same asset as the current job. This list includes jobs that are assigned to you.

10. **See jobs for this location** – Select this button to view a list of all open maintenance jobs that are associated with the same location as the current job. This list includes jobs that are assigned to you.

Maintenance checklist

A maintenance checklist is a set of tasks that the maintenance worker must complete to close the maintenance job. For more information about how to define a checklist for a job, including how to use item types and create groups, see Maintenance checklists. For more information about how to set up default checklists that can be assigned to various maintenance jobs or asset types, see Maintenance job types, categories, variants, trades, and checklists.

To view the maintenance checklist, select the **Checklist** tab at the top of the page. The following illustration highlights the different UI elements on the **Checklist** tab.

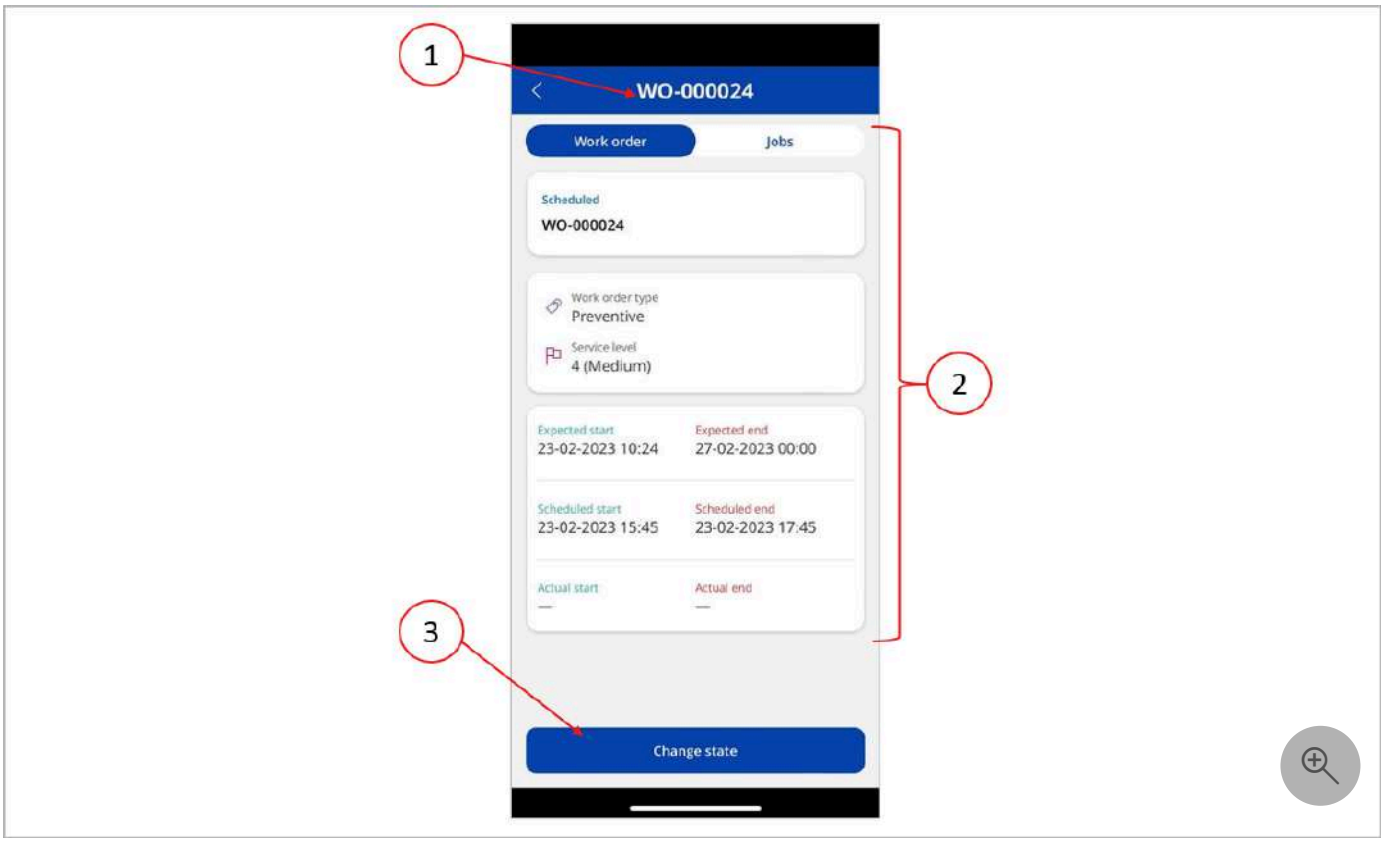


The **Checklist** tab of the job details page includes the following UI elements. The numbers correspond to the numbers in the previous illustration.

1. **Job and Checklist tabs** – Select a tab to switch between the job details and the checklist. The numbers in parentheses on the label of the **Checklist** tab indicate the number of completed items and the total number of items on the checklist.
2. **Checklist line number** – Each checklist item has a system-generated line number. You can use this number to refer to the checklist item when you communicate with other workers. The text color changes to green when the item is marked as completed by the worker.
3. **Group name** – Checklist items can be grouped under a heading. (You set up group names in Supply Chain Management by creating a checklist item of the *Header* type.)
4. **Checklist item title** – This text shows the title of the checklist item.
5. **Checklist item type** – This text shows the type of checklist item (*Variable*, *Text*, or *Measurement*).
6. **Value** – This field shows the value that has been entered for checklist items of the *Measurement* or *Variable* type. To enter these values, select the checklist item card.

Work order details page

The work order details page shows more information about a selected work order. To open it, select a card in the work order list. The following illustration highlights the different UI elements on a work order details page.



The work order details page includes the following UI elements. The numbers correspond to the numbers in the previous illustration.

1. **Work order and Jobs tabs** – Select a tab to switch between the work order details and the list of jobs that belong to the work order. The job list shows a list of job cards. Select a card to open the details page for that job.
2. **Work order information** – Information about the work order is shown on various cards.
3. **Change state button** – Select this button to change the lifecycle state of the work order. For more information about work order states, see Work order lifecycle states.

Last updated on 05/29/2026

File maintenance requests using the Asset Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> The Asset Management mobile app is now provided on GitHub as an open-source sample at microsoft/scmsamples-EnterpriseAssetManagement. It provides the same functionality as the original app and is freely customizable. For timeline and transition guidance, see Asset Management mobile app open-source release.

You can use the Asset Management mobile app to file maintenance requests.

For more information about maintenance requests in Microsoft Dynamics 365 Supply Chain Management, see Maintenance requests.

User requirements

To submit maintenance requests from the Asset Management mobile app, you must meet the following requirements:

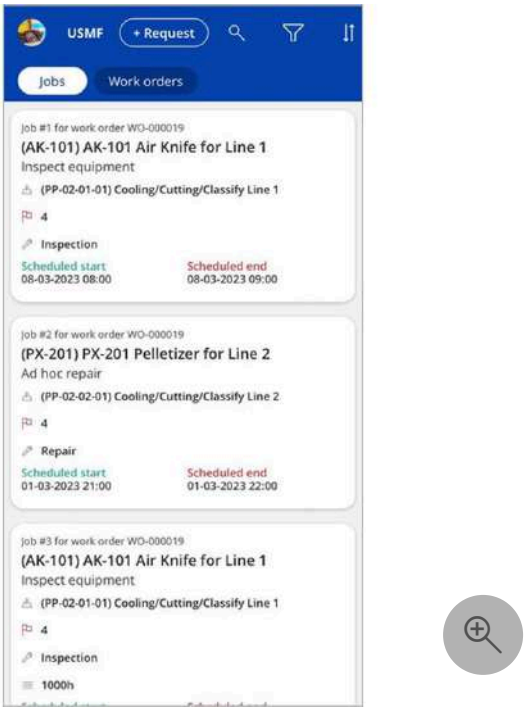
- You must sign in to Power Apps by using a domain account that matches a user account in Supply Chain Management that has the same Microsoft Entra ID username.
- Your user account in Supply Chain Management must be assigned the *Maintenance requester* security role.

For more information about how to set up roles and security in Supply Chain Management, see Security roles.

Identify the asset or functional location that you want to request maintenance for

The procedure for creating maintenance requests varies slightly, depending on the security roles that are assigned to your user account in Supply Chain Management. The following combinations are possible:

- If you have the *Maintenance requester* role but not the *Maintenance worker* role, the **Search** and **Scan** buttons are shown when you first open the app. Select one of these button to start to create a maintenance request as described later in this article.
- If you have both the *Maintenance requester* role and the *Maintenance worker* role, a list of jobs that are assigned to you is shown when you first open the app. To start to create a maintenance request, select **Request** at the top of the page to open a dialog box that includes the **Search** and **Scan** buttons.
- If you have the *System administrator* role, the app behaves as though you have both the *Maintenance requester* role and the *Maintenance worker* role.



Start page for users who have both the *Maintenance requester* role and the *Maintenance worker* role (or the *System administrator* role)

Start page for users who have the *Maintenance requester* role but not the *Maintenance worker* role

Search

Select **Search** to manually search for the asset or functional location that you want to create a maintenance request for. If you want to search for assets, select **Asset** under the search field. If you want to search for functional locations, select **Functional locations**. The system will search for assets and functional locations across all legal entities where the user's *Maintenance requester* security role is granted access in Supply Chain Management.

In the search field, enter the name or ID of the asset or functional location that you want to submit a maintenance request for. Then select an asset or functional location to continue.

The following illustration shows an example of search results for an asset.

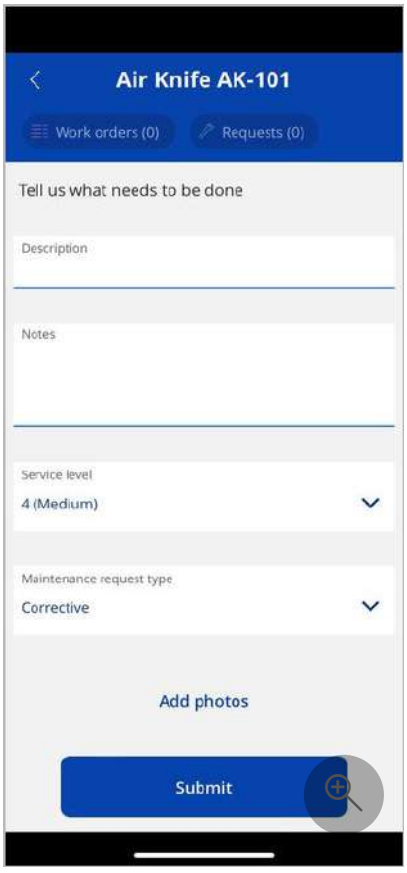
Scan

If your organization uses bar codes or QR codes to uniquely identify each asset, you can use the **Scan** button to identify the asset that you want to submit a maintenance request for. The bar code or QR must contain the ID of the asset or functional location.

When you scan the bar code or QR code on a tag, the system automatically searches for matching assets or functional locations, and then shows a list of results. Select an asset or functional location to continue.

Document the maintenance request

When you've identified the target asset or functional location, you're directed to a documentation page where you can add details about the maintenance request.



Before you file the request, look at the **Work orders** and **Requests** buttons at the top of the page to determine whether any open maintenance requests or work orders already exist for the selected item. If a button shows a value that's more than 0 (zero), select the button to view a list of relevant records and get more information.

Set the following fields as required:

- **Description** – Enter a description of the request.
- **Notes** – Add details about the request.
- **Service level** – Select a service level for the request. The service level is typically used to identify how critical the maintenance request is. For more information about how to set up service levels and set related default values, see Asset service levels.

- **Maintenance request type** – Select the request type. The request type is typically used to categorize the type of incident that's being reported. For more information about how to set up request types and set related default values, see Maintenance request types.

Add photos

You can use the camera on your mobile device to take photos that document the incident that you're filing. The photos will be attached to your maintenance request. If the maintenance request is later converted into a maintenance work order, the system will copy these photos to the work order.

To add photos, select **Add photos** at the bottom of the documentation page.

Submit the maintenance request

When you've finished filling in the documentation page, select **Submit** to create the maintenance request. You'll receive a confirmation message when the maintenance request is created in the system.

Last updated on 05/29/2026

Asset Management mobile app FAQs and known issues

> **Note**
>
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> **Important**
>
> The Asset Management mobile app is now provided on GitHub as an open-source sample at microsoft/scmsamples-EnterpriseAssetManagement. It provides the same functionality as the original app and is freely customizable. For timeline and transition guidance, see Asset Management mobile app open-source release.

This article provides answers to several frequently asked questions (FAQs) about the Asset Management mobile app. It also describes a few known issues that affect the app and how to work around them.

What platforms and mobile devices are supported?

The Asset Management mobile app runs within the Power Apps mobile app. All platforms supported by the Power Apps mobile app can also run the Asset Management mobile app. For details, see System requirements, limits, and configuration values for Power Apps.

Why don't I see anything when I sign in to the app?

If you don't see any features in the app, you probably lack the permissions required to access them. Access to each feature is controlled by security roles assigned to your user account in Supply Chain Management. Learn more in Configure users and workers in Supply Chain Management

What versions of Supply Chain Management support the app?

The Asset Management mobile app requires Supply Chain Management version 10.0.36 or newer.

Can I customize and extend the app?

Yes — the source code is now available as an open-source sample on GitHub at microsoft/scmsamples-EnterpriseAssetManagement and can be customized freely. See Asset Management mobile app open-source release for guidance.

Does the app support offline mode?

No, the Asset Management mobile app doesn't currently support offline mode.

Is there multi-language support?

Yes, the Asset Management mobile app is available in all the same languages as Supply Chain Management, except for right-to-left (RTL) languages. We're awaiting RTL support from the canvas app platform and plan to add RTL support in a future release.

Where can I go to discuss the app with the community and submit suggestions to Microsoft?

The Asset Management community on Microsoft Viva Engage is a great place to go if you want to exchange tips, ask questions, or submit suggestions for improvement. Viva Engage community participants include Microsoft partners, customers, experts, and employees.

How can I diagnose "Error when trying to retrieve data from the network" issues?

If you see the error message "Error when trying to retrieve data from the network," follow these steps to diagnose and solve the issue:

1. Run the Asset Management app in a browser as described in Run an app in a web browser.
2. If you see the same error while running in a browser, set up your browser to capture a HAR file and then reload the tab where the app is running.
3. Use a tool such as the Google HAR analyzer to redact all secrets from the HAR file.

4. Contact Microsoft Support and share the redacted HAR file with them.
5. Microsoft Support will help you diagnose and solve the issue.

The content of the HAR file might reveal an error similar to the following example:

text

```
SecLib::CheckPrivilege failed. User: <EntraObjectID>, PrivilegeName:
prvReadRelationship, PrivilegeId: <PrivilegeID>, Required Depth: Basic,
BusinessUnitId: <BusinessUnitID>, MetadataCache Privileges Count: <CacheCount>, User
Privileges Count: <UserPrivilegeCount>
```

This error can occur because the *Asset Management app* role doesn't have the privileges required to read the `Relationship` table. Instead, that privilege is provided by the *Finance and Operations Basic User* role in Dataverse. To resolve this issue, add the *Finance and Operations Basic User* role to each user or team that's using the app.

When updating the app, how can I solve the error: "Could not delete solution ... because it has been modified"?

The update process fails with an error if it detects that the Asset Management mobile application canvas app has been customized. This helps prevent the updater from overwriting your customizations. To solve this issue, remove any customization layers by restoring the oldest version of the app. If the oldest version is over six months old, you'll need to manually delete the `msdyn_EnterpriseAssetManagementMobileV2` solution and then reinstall the app from *Dynamics 365 apps* or *Marketplace*.

Consider cloning the canvas app before removing your customizations, just in case you want to copy the customizations to the updated app.

> **Note**
>
> For canvas-app customizations that need to be preserved across official updates, use the open-source release rather than editing in Power Apps Studio.

Are there any known issues affecting the release of the Asset Management mobile app?

Yes, the Asset Management mobile app has a few known issues that we expect to correct in an upcoming release. The following subsections describe them.

Limited file-type support for attachments

The current release provides only limited support for viewing file attachments on the job details page. We plan to add support for more file types in the final release.

Supported attachment types:

- URL (link)
- PNG/JPEG/JPG/BMP/GIF (image)
- PDF (document)

Unsupported attachment types:

- Note
- All other file types

Last updated on 05/29/2026

Integrate Dynamics 365 Supply Chain Management (Asset management) with Dynamics 365 Guides

You can integrate the **Asset management** module in Microsoft Dynamics 365 Supply Chain Management with Dynamics 365 Guides to take advantage of mixed-reality guides in your day-to-day service and maintenance workflows. If a guide is associated with an Asset Management work order, a worker who opens the work order's maintenance checklist in the Supply Chain Management (Dynamics 365) mobile app sees that a guide is available. The worker can then find and open the guide in the Dynamics 365 Guides HoloLens app.

Prerequisites

Before you can attach guides to Asset management work orders, you must complete these prerequisites:

- Set up Dynamics 365 Supply Chain Management version 10.0.9 or later.
- Turn on dual-write for Supply Chain Management apps.
- Turn on the following configuration keys on the **License configuration** page:
  - Asset management > Asset management mixed reality
  - Mixed reality > Mixed reality guide
- Set up Dynamics 365 Guides version 200.0.0.96 or later.

Use Dynamics 365 Guides with Asset management

To associate a guide, you use a maintenance checklist line in Asset management. You can create the association through a maintenance checklist template, a maintenance job type, or a work order, because all three contain maintenance checklist lines. You can save time by using a template, because a template can be associated with all the maintenance job types that use it. For example, a guide that is associated with a maintenance job type is automatically associated with all work orders that specify that job type. On the other hand, a guide that is associated directly with a work order exists only for that work order.

Associate a guide with a maintenance checklist template

To associate a guide with a maintenance checklist template, follow these steps:

1. Create a guide by using the Dynamics 365 Guides PC and HoloLens apps. For information about how to create a guide, see the following articles:

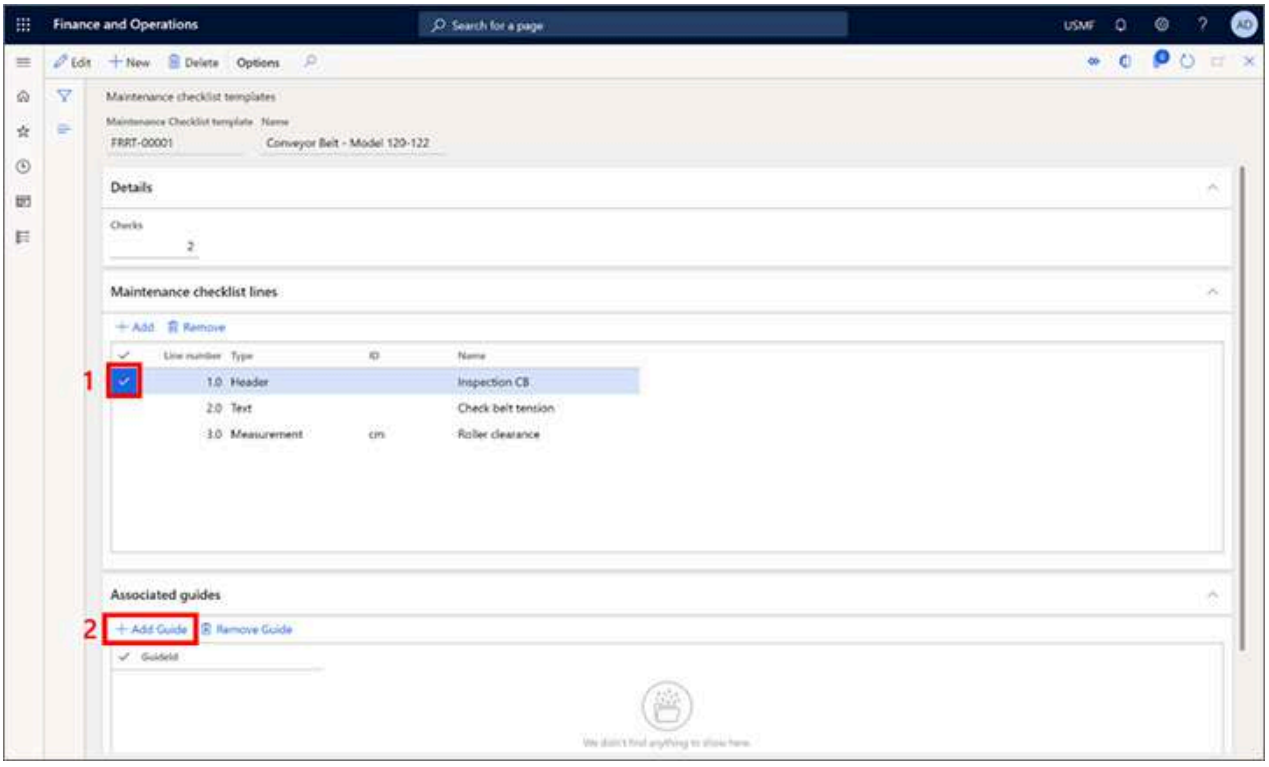
    - Use the PC app to create a guide
    - Use the HoloLens app to place your holograms

2. In Supply Chain Management, create a maintenance checklist template.

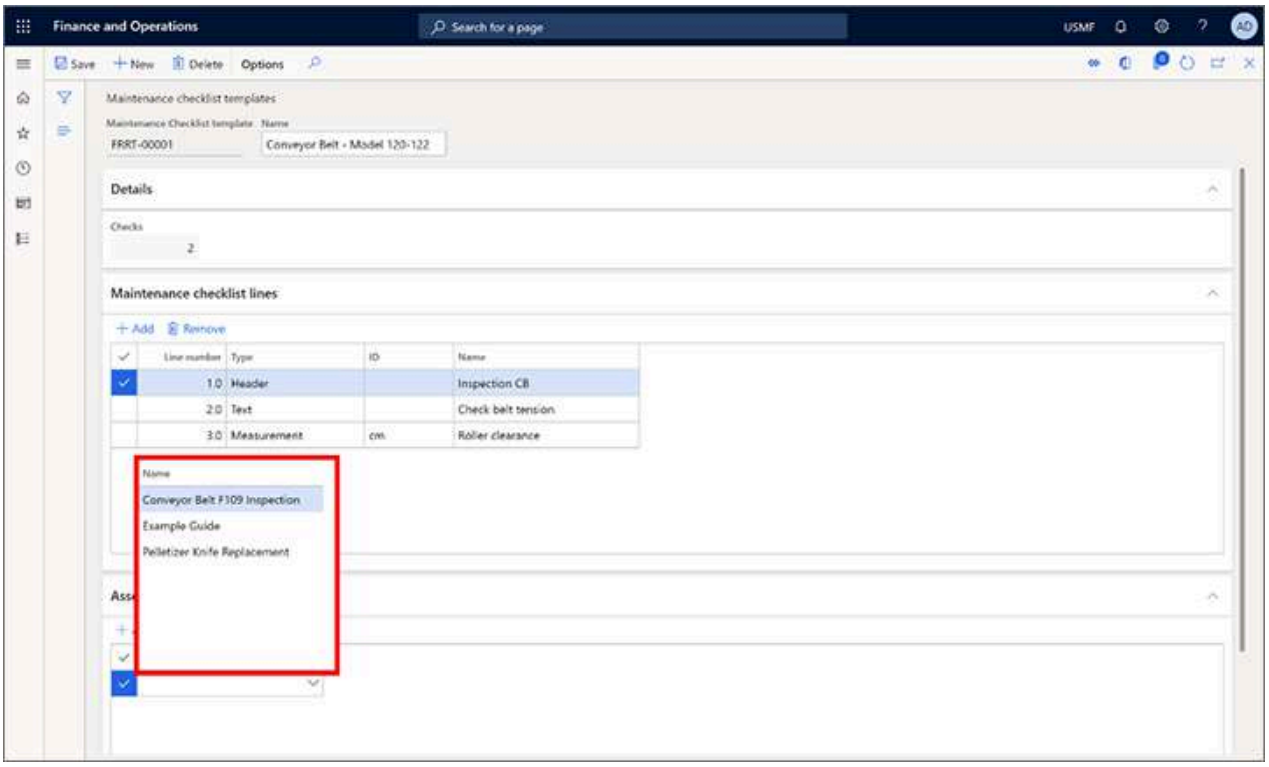
3. Associate the guide that you created with a maintenance checklist line in the new maintenance checklist template:

    a. On the **Maintenance checklist lines** FastTab, select the line that you want to associate the guide with.

    b. On the **Associated guides** FastTab, select **Add Guide**.



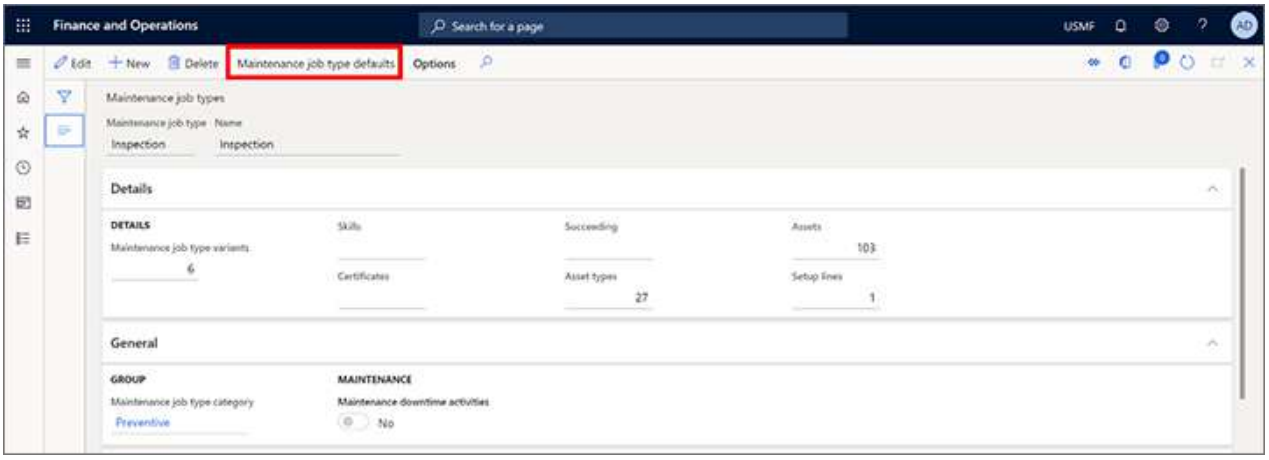
    c. In the **Name** field, select a guide, and then select **Save**.



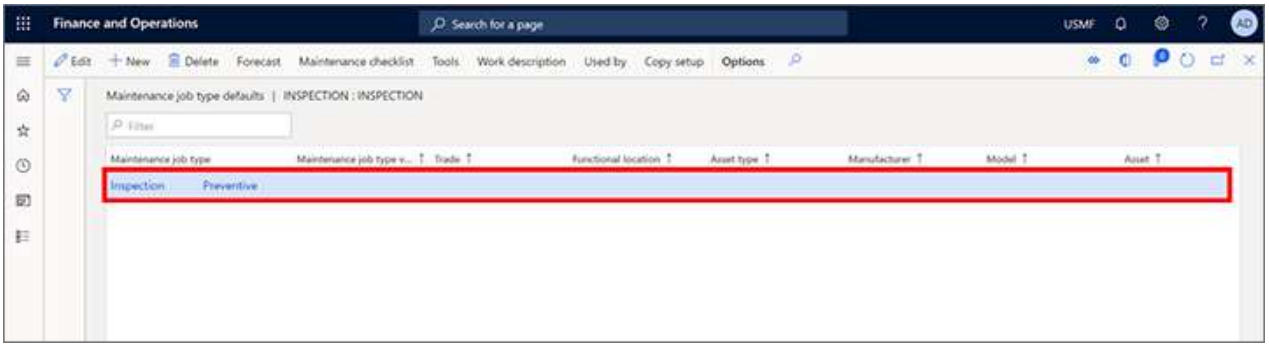
4. Associate the maintenance checklist template with a job type:

   a. Create a maintenance job type, or select an existing maintenance job type.

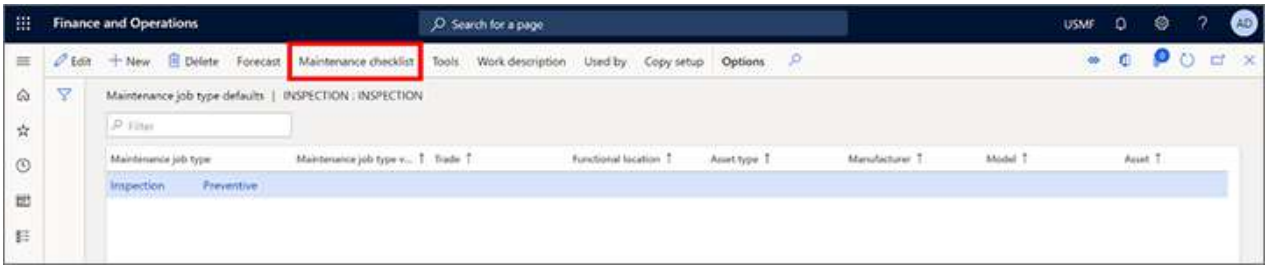
   b. On the Action Pane, select **Maintenance job type defaults**.



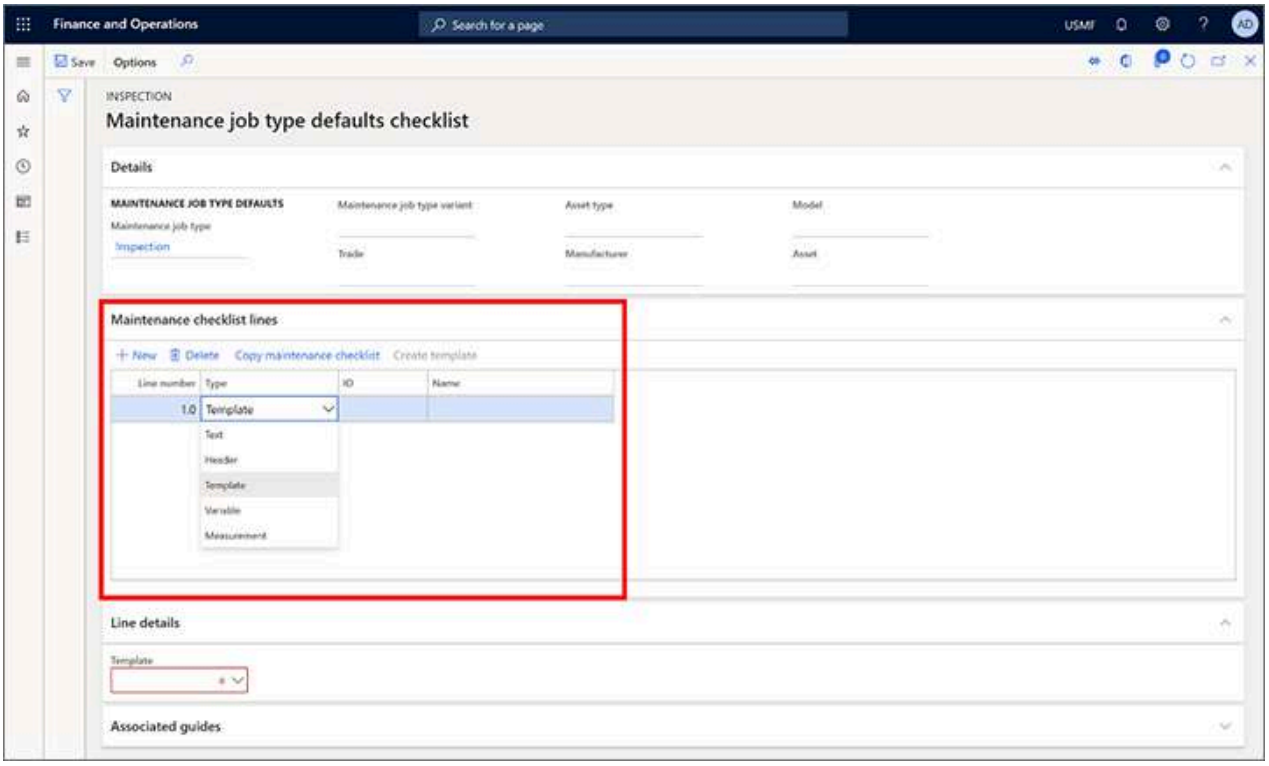
   c. Create a line, and then select **Save**.



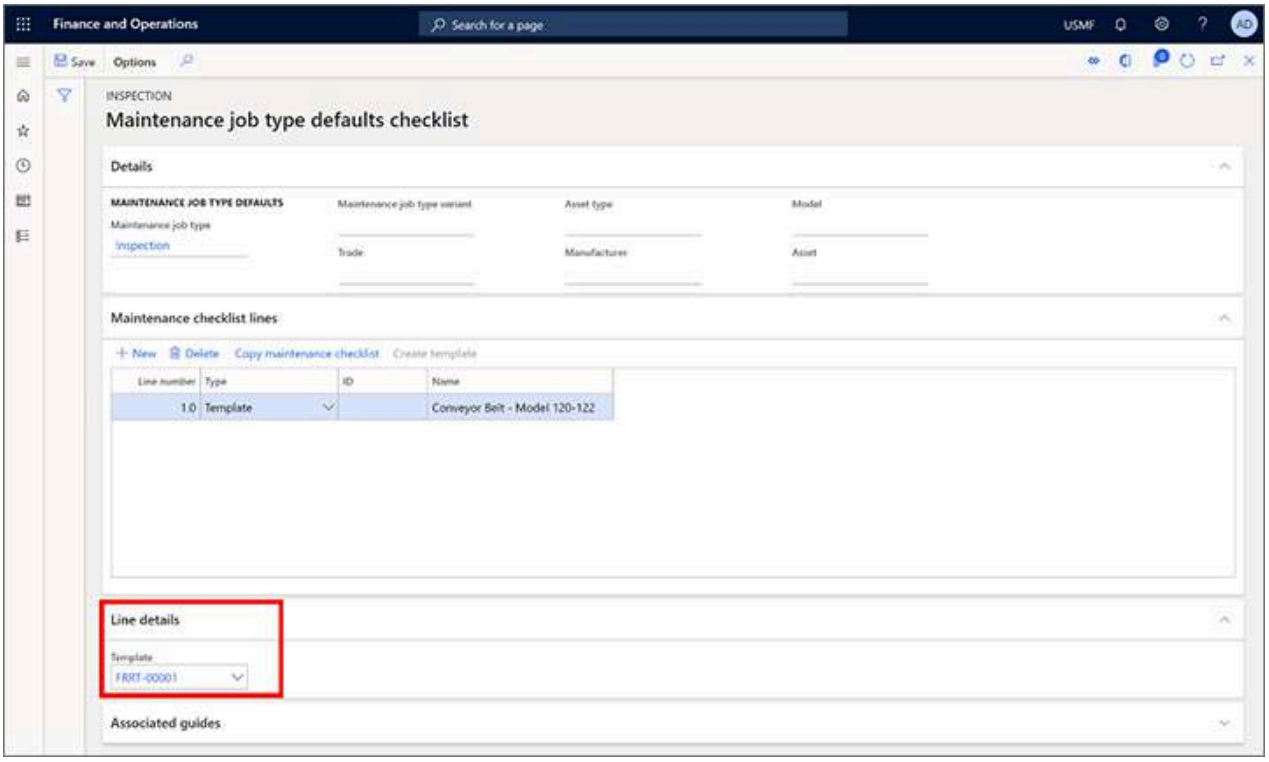
   d. On the Action Pane, select **Maintenance checklist**.



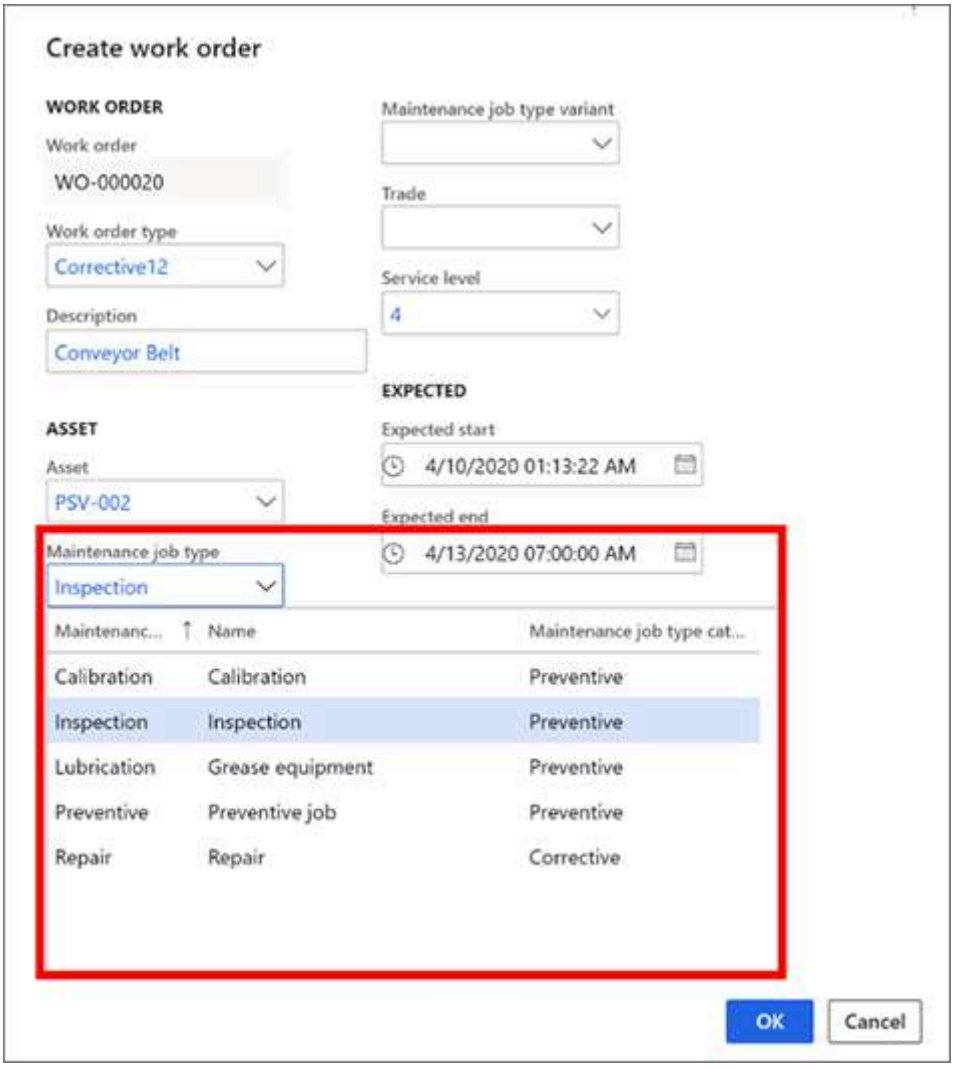
e. On the **Maintenance checklist lines** FastTab, add a line, and then change the value of the **Type** field to **Template**.



f. On the **Line details** FastTab, in the **Template** field, select the template that you associated the guide with, and then select **Save**.



5. Create a work order, and then select the maintenance job type that uses the maintenance checklist template that you associated the guide with. The guide is automatically associated with the work order.

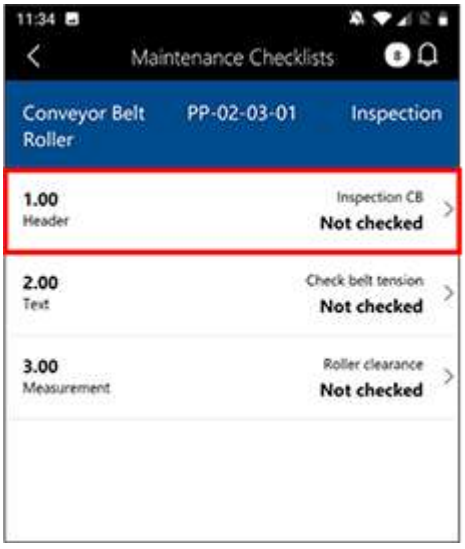


6. View the guide that is associated with the work order and workers:

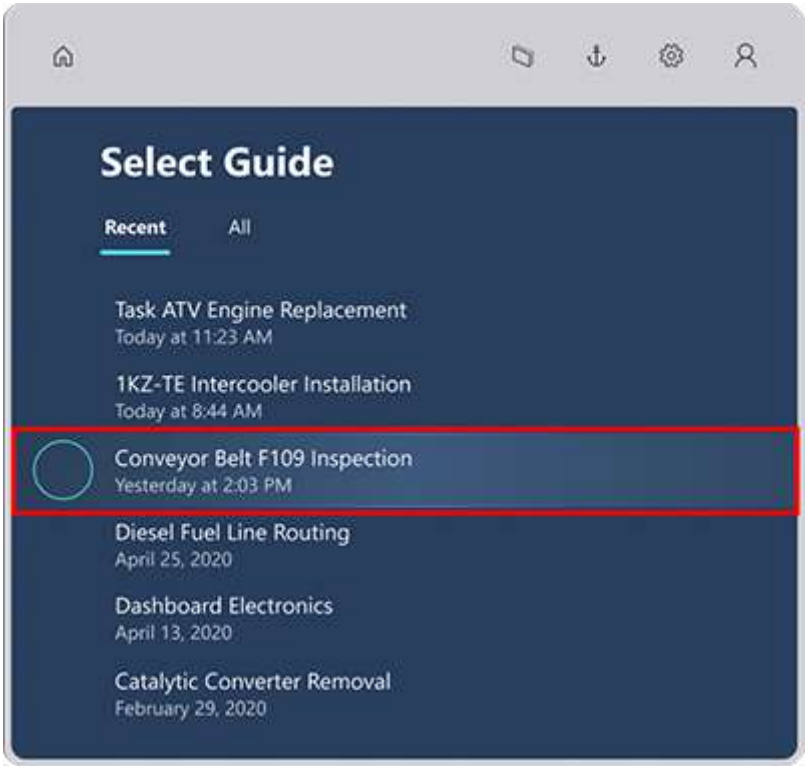
   a. Open the Asset management mobile workspace to access the work order.

   b. Open the maintenance checklist for the work order.

   c. Select a checklist line to see the associated guide.



   d. Open the guide on HoloLens.



> **Note**

You can also associate a guide directly in the maintenance checklist of a work order or a job type.

Important

There is a known issue where, when you associate a maintenance checklist template with a default maintenance job type, the guide that is linked to the template doesn't appear on the **Associated guides** FastTab of the **Maintenance job type defaults** page. However, the guide will appear after that job type is applied to a work order on the **Associated guides** FastTab.

Related information

- Dual-write overview
- Asset management overview

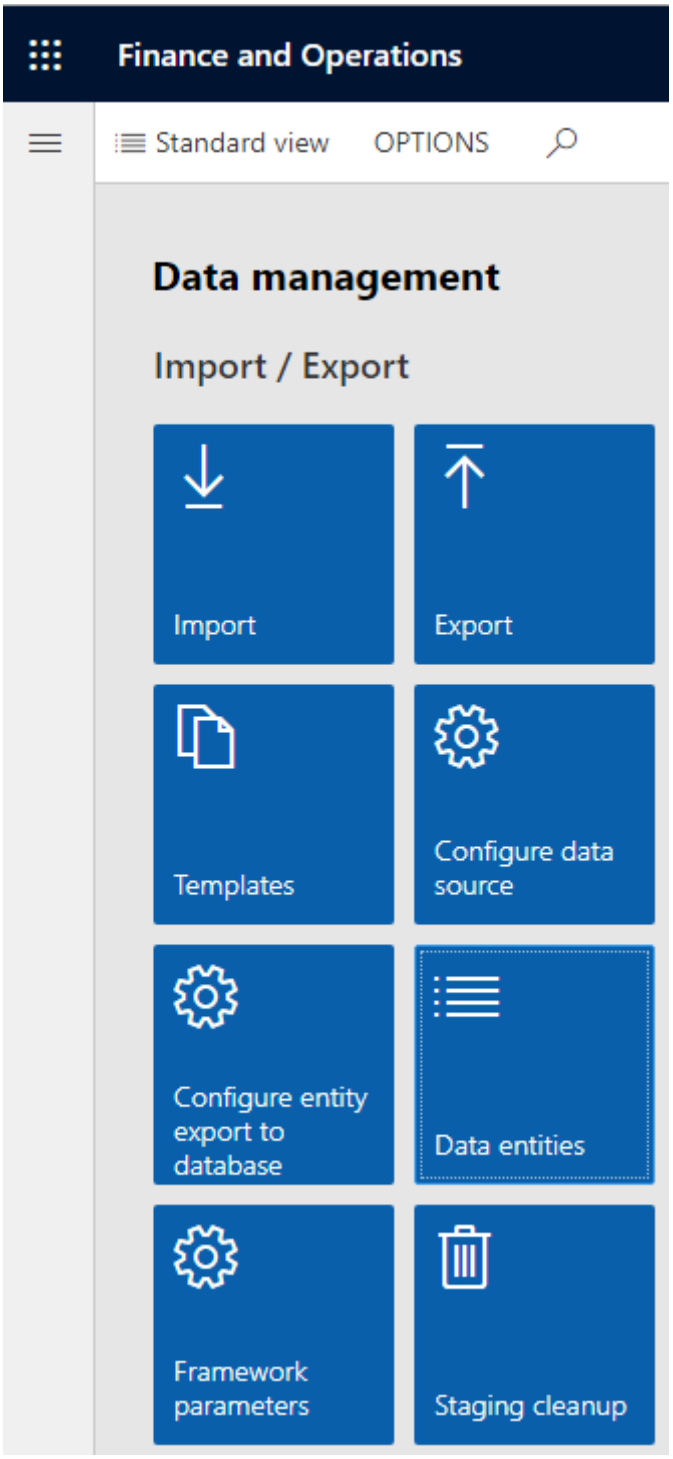
Last updated on 12/30/2025

Target entities

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

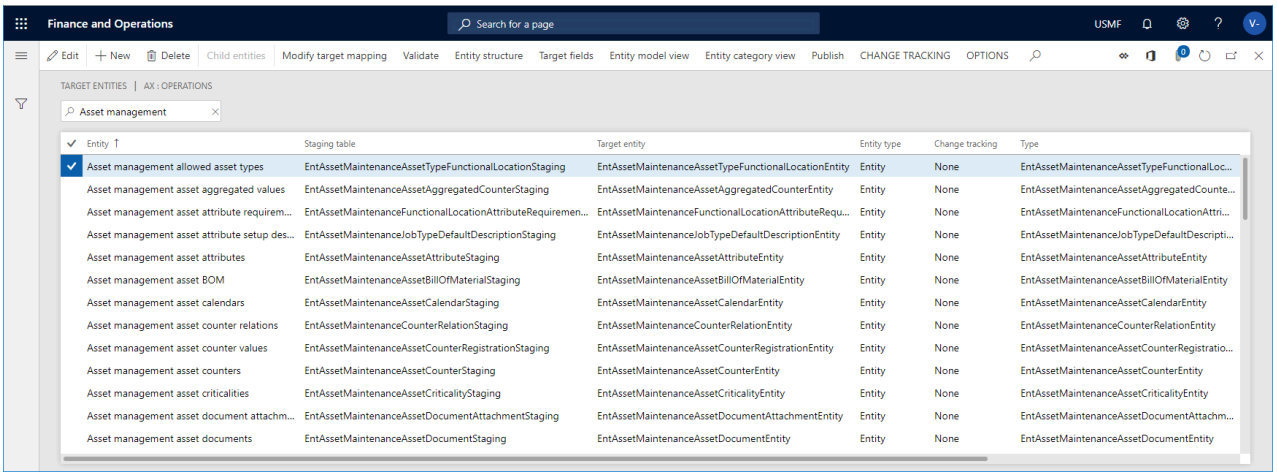
In the **Data management** workspace, you can get an overview of target entities, related entity types and staging tables related to the **Asset management** module.

1. Click the **Finance and operations** button in the upper-left corner of the screen, and click **Data management**.
2. In the **Import / Export** section, click **Data entities**.



3. On the **Target entities** page, search for "asset management" in the filter, and press Enter to see a list of the entities related to Asset management.

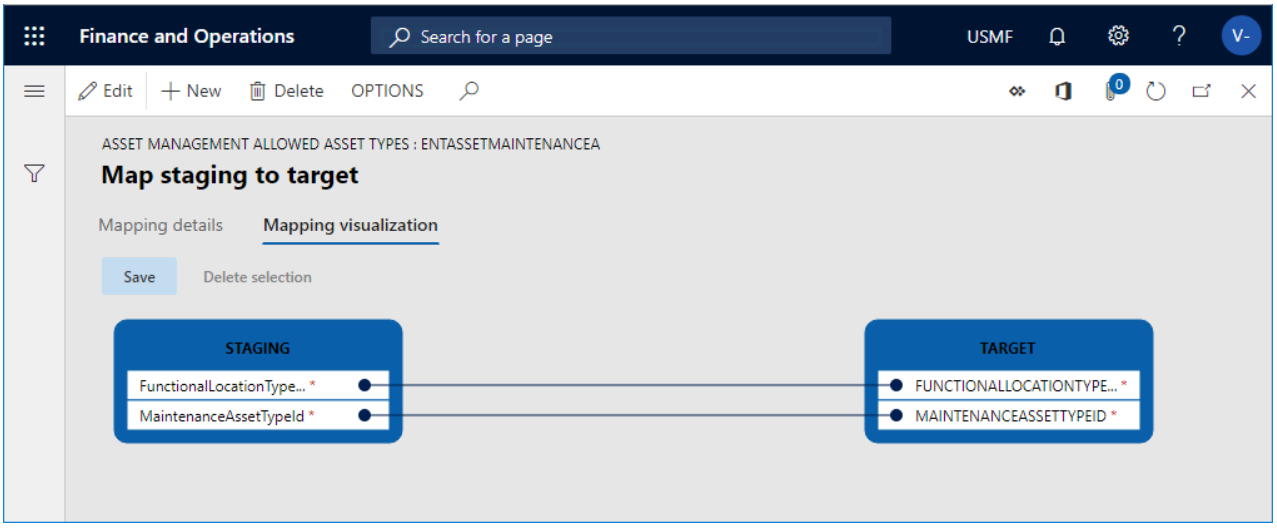
   The figure below shows some of the Asset management entities.



4. Select an entity and click **Modify target mapping**.

5. On the **Map staging to target** page, you see a list of the staging fields related to the selected entity. Click **Mapping visualization** to see a graphic overview of how staging data and target data are related.

   In the figure below, a visualization of the fields related to the asset types entity is shown.



Last updated on 05/07/2025

Cost accounting home page

> **Note**
>
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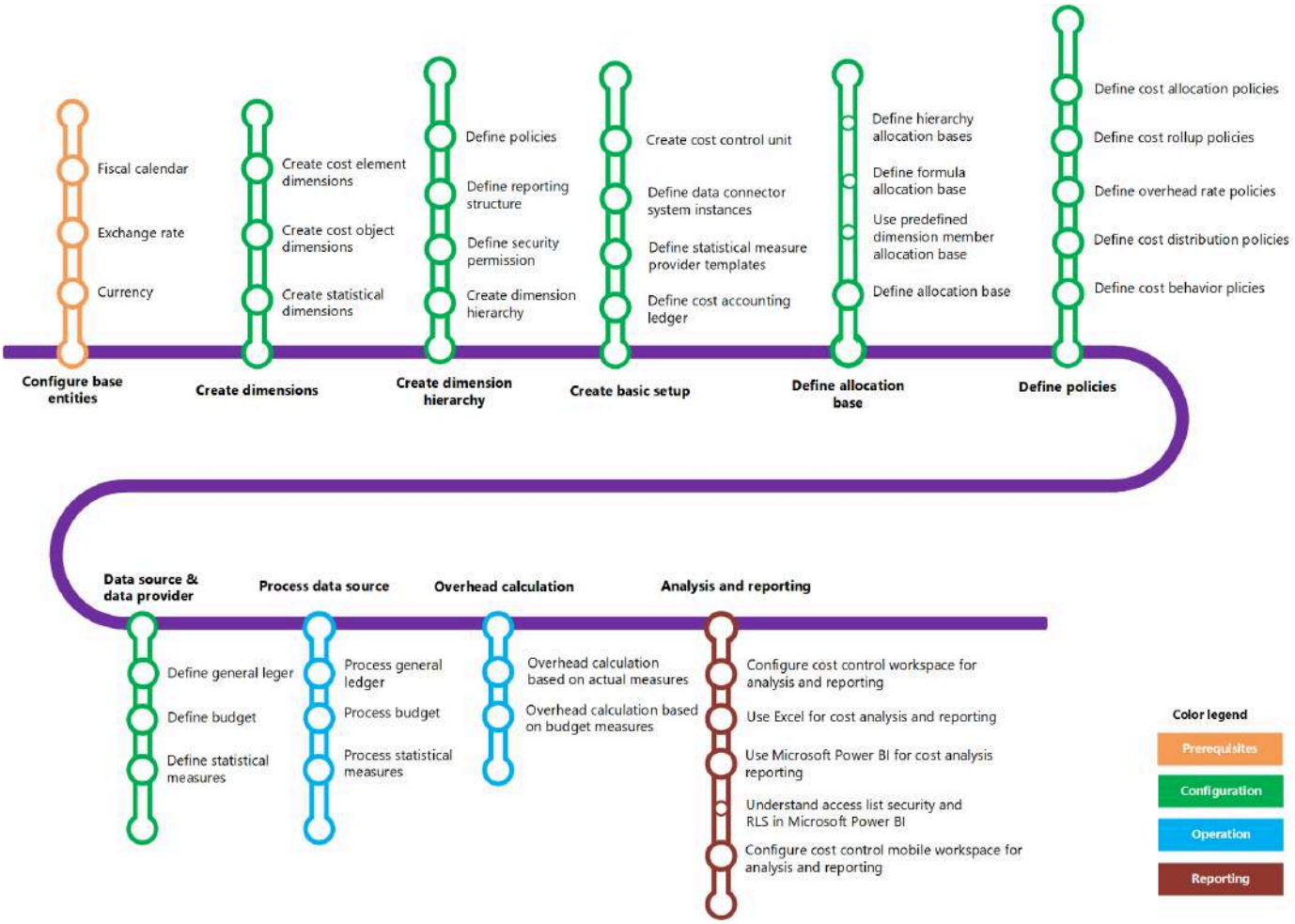
Learning map

The following learning map shows the major concepts and tasks that make up the framework of the Cost accounting module. Select the links in the Quick links section to learn how to use the module.



Quick links

Expand table

Get started with cost accounting (Learn)	Cost element dimensions
Dimension hierarchy	Define cost control units
Allocation bases	Create and assign a cost allocation policy to a cost control unit
Manage a data source for the cost accounting ledger	Process and trace source data
Overhead calculation	Cost control workspace

Get started videos

Expand table

Get started with cost accounting	Cost accounting mobile workspace	Use Excel for cost accounting analysis

Last updated on 06/23/2026

Cost accounting terminology

> **Note**
>
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This article defines the key terms that are used in Cost accounting.

Allocation base

The allocation base is used to measure and quantify activities, such as machine hours that are used, kilowatt hours that are consumed, or square footage that is occupied. It's used as basis for allocating costs to one or more cost objects

Cost accounting

Cost accounting lets you collect data from various sources, such as the general ledger, subledgers, budgets, and statistical information. You can then analyze, summarize, and evaluate cost data, so that management can make the best possible decisions for price updates, budgets, cost control, and so on. The source data that is used for cost analysis is treated independently in Cost accounting. Therefore, updates in Cost accounting don’t affect the source data. However, when you collect cost data from various sources, and especially when you import the main accounts from General ledger as cost elements, there's data redundancy, because the same data exists in both General ledger and Cost accounting. This redundancy is required, because you use financial management for external reporting and Cost accounting for internal reporting.

Cost accounting ledger

Defined by calendar, currency, and cost element dimension, it controls processes and policies for measuring costs.

Cost entry

Cost entries result from a transfer via data connectors from general ledger entries, cost allocations, and posted cost entries in cost journals.

Cost object

Any type of object that is selected for cost control. Costs or revenues are either directly posted on or allocated to cost objects. Some typical cost objects are:

- Products
- Projects
- Departments
- Cost centers

Management uses cost objects to quantify costs, but also to drive profitability analysis.

Cost element

Use cost elements as a function to track and categorize costs. There are two types of cost elements: primary and secondary.

Primary cost elements represent the cost flow from financial accounting to Cost accounting. The structure typically corresponds to the profit and loss account structure in the general ledger where a cost element can correspond to a main account. Not all main accounts must be represented as cost elements, depending on business requirements.

Secondary cost elements represent the internal cost flow because these costs are only used in Cost accounting. They're used in cost roll-up rules to aggregate costs into meaningful buckets used by overhead calculation.

Cost classification

Cost classification groups cost according to their shared characteristics. For example, costs can be grouped by elements, traceability, and behavior.

- **By elements** – Materials, labor, and expenses.
- **By traceability** – Direct costs and indirect costs. Direct costs are assigned directly to cost objects. Indirect costs aren't directly traceable to cost objects. Indirect costs are allocated to cost objects.
- **By behavior** – Fixed, variable, and semi-variable.

Cost behavior

Cost behavior classifies costs according to their behavior in relation to changes in key business activities. To control costs effectively, management must understand the cost behavior. There are three types of cost behavior pattern: fixed, variable, and semi-variable.

- **Fixed cost** - A fixed cost is a cost that doesn't vary in the short term, regardless of changes in activity level. For example, a fixed cost can be a basic operating expense of a business, such as rent, that isn't affected even if the activity level increases or decreases.
- **Variable cost** - A variable cost changes according to changes in activity level. For example, a specific direct materials cost is associated with each product that is sold. The more products that are sold, the more direct materials costs there are.
- **Semi-variable cost** - Semi-variable costs are partly fixed and partly variable costs. For example, an Internet access fee includes a standard monthly access fee and a broadband usage fee. The standard monthly access fee is a fixed cost, whereas the broadband usage fee is a variable cost.

Cost control unit

The cost control unit represents the cost structure. The structure determines how cost flows in a hierarchical order between cost object dimensions and their respective cost objects.

Cost distribution

Distribute cost from one cost object to one or more other cost objects by applying a relevant allocation base. Cost distribution and cost allocation differ in that cost distribution always occurs at the level of the primary cost element of the original cost and involves no reciprocal processing.

Cost allocation

Is used to allocate the balance of a cost object to other cost objects by applying an allocation base. Finance supports the reciprocal allocation method. In the reciprocal allocation method, the mutual services that auxiliary cost objects exchange is fully recognized. The system automatically determines the order to perform the allocations in and iterate over it. The balance of a cost object is allocated by a single allocation base. Allocations across cost object dimensions and their respective members are supported. The allocation order is controlled by the cost control unit.

Cost allocation policy

A cost allocation policy defines the amounts and quantities that you must allocate. Allocation rules include allocation source rules, which determine the costs that are allocated, and allocation target rules, which determine where the costs are allocated. For example, all costs for facility services are an allocation source that you can allocate to various departments in an organization (that is, to allocation targets).

Cost roll-up

The purpose of cost roll-up rules is to aggregate costs into meaningful buckets. The level of aggregation is user-defined and involves the assignment of a secondary cost element. If cost roll-up isn't used, every element of a cost is allocated from one cost object to another

Cost rate policy

Use the cost rate to calculate price per cost object. To understand the elements of the price, define cost rate policies. There are two types of cost rate: historical cost rate and planned cost rate. A historical cost rate is a calculated rate that is used as a multiplier for the allocation base of a cost object. The rate is calculated based on the cost allocations in the previous period. A planned rate is a user-defined rate.

Dimension hierarchy

There are two dimension hierarchies: categorization hierarchy and classification hierarchy. Use the dimension categorization hierarchy type for reporting purposes. It supports only the cost element dimensions. Use the dimension classification hierarchy type to define policies and for reporting purposes. It supports all dimensions, such as cost objects, cost elements, and statistical dimensions.

Data connector

Cost accounting supports integration of data from source systems via a set of data connectors. The following data connectors are available:

- Imported transactions (preconfigured)
- Dynamics 365 Finance (preconfigured)
- Dynamics AX (configuration required)

> **Note**
>
> The data connector Imported transactions are based on data entities.

Data provider

Most source systems can provide data that matches one or more data sources in Cost accounting. To align data from the source systems with the data source in Cost accounting, you need to configure a data provider. The following table lists the availability of data providers per data connector and data source.

Expand table

Data sources	Imported transactions data connector	Dynamics 365 Finance data connector	Dynamics AX data connector
Cost element dimension members	Yes	Yes	Yes
Cost object dimension members	Yes	Yes	Yes
Statistical dimension members	Yes	No	No
General ledger	Yes	Yes	Yes
Budget entries	Yes	Yes	Yes
Statistical measures	Yes	Yes	Yes

Formula

Formula allocation bases let you define advanced formulas to achieve the correct allocation basis. You can manually create formula allocation bases. You can use the following operators to define your formula.

Expand table

Symbols	Text
( )	Parentheses
<	Smaller than

Symbols	Text
>	Larger than
+	Addition
–	Subtraction
*	Multiplication

Traditional IF statements aren't supported. However, you can create statements and validate whether they're true.

Expand table

Statement Validation	Result
a > b	True
a > b	False

Overhead cost

Overhead costs refer to the ongoing expenses of operating a business. They're costs that you can't link directly to specific business activities. Here are some examples of overhead costs:

- Accounting fees
- Depreciations
- Insurance
- Interest
- Legal fees
- Taxes
- Utilities costs

Overhead rate

Define rates per cost object and cost element. Two types of rates exist: fiscal period and user-specified. The overhead calculation determines fiscal period rates. A user-specific rate is user-defined and can allocate cost between cost objects at a predetermined rate in the overhead calculation.

Published

If you set this field to Yes, a user assigned one of the following roles can view the report in the Cost control workspace:

- Cost accounting manager
- Cost accountant
- Cost accountant clerk
- Cost object controller

If you set this field to No, only users who are assigned one of the following roles can view the report in the Cost control workspace:

- Cost accounting manager
- Cost accountant
- Cost accountant clerk

Statistical dimension

A statistical dimension and its members are used to register and control nonmonetary entries in Cost accounting. Statistical dimension members can be used for two purposes:

- As an allocation base in policies such as cost distribution or cost allocation.
- For reporting of nonmonetary consumption.

A statistical dimension is the expression of a count or sum of an activity that can be used as the basis for allocations or rate calculations. It's either created manually or imported from source systems. Examples of statistical dimensions include the number of employees, the count of licensed software on each device, power consumption of each machine, or square meters for a cost center.

Statement

Statements are views for the managers who are responsible for controlling costs. Statements are defined by a cost controller, and they provide a quick overview of actual costs, budgeted costs, and deviations. A manager can drill further into details if necessary. To help ensure that managers view only data that they're accountable for, data that appears in the statements is subject to access rules.

Version

Use versions to simulate, view, and compare various outcomes. By default, you view all actual costs in one base version that's known as *actual*. For budgets and calculations, you can work with as many versions as you require. For example, you can import budget data into an original version and then revise the budget in a revised version. For calculations, you can create multiple versions. In these various versions, you can then create calculations by using different calculation rules that apply for cost allocation.

Last updated on 06/23/2026

Fiscal calendars, fiscal years, and periods

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article discusses fiscal calendars, fiscal years, and periods, and how to use them for legal entities, fixed assets, and budgeting.

Fiscal calendars provide a framework for the financial activity of an organization. Each fiscal calendar contains one or more fiscal years, and each fiscal year contains multiple periods. Fiscal calendars can be based on a January 1 to December 31 calendar year, or on any dates that you select. For example, some organizations select a fiscal calendar that starts on July 1 of one year and ends on June 30 of the following year.

You can create unlimited fiscal calendars and fiscal years for a fiscal calendar. Each fiscal calendar is independent of your organization, and multiple legal entities in the organization can use it. For example, an organization has eight departments and each department is a separate legal entity. Five of them share the same fiscal calendar and three use different fiscal calendars. You can create one fiscal calendar for the five legal entities that share the same fiscal calendar, and then create separate fiscal calendars for the other legal entities.

Create fiscal calendars, fiscal years, and periods

You can create and delete fiscal calendars, fiscal years, and periods on the **Fiscal calendars** page. You can also divide existing periods and create closing periods that you can use to close a fiscal year.

A closing period is used to separate general ledger transactions that are generated when a fiscal year is closed. When the closing transactions are in one fiscal period, it's easier to create financial statements that either include or exclude different types of closing entries. If a fiscal year is divided into 12 fiscal periods, the closing period is usually the 13th period. However, a closing period can be created from any period that has a status of Open.

When you create a closing period, select a period that has a status of **Open** and that has the dates that you want to use. The new closing period copies the starting and ending dates from the existing period. The original period continues to exist. For example, you select Period 12, which is the last period in the fiscal year, and that has dates of August 1 through August 31. You enter a name for the closing period, such as **Close**. After you create the new closing period, you now have the original period and the closing period. Both have dates that start on August 1 and end on August 31.

Select fiscal calendars for ledgers, fixed assets, and budget cycles

Fiscal calendars are used with fixed asset depreciation, financial transactions, and budget cycles. When you create a fiscal calendar, you can use it for multiple purposes. You can select a fiscal calendar for a fixed asset book to make it a fixed asset calendar. You can select a fiscal calendar for a ledger to make it a ledger calendar. And you can select a fiscal calendar for a budget cycle to make it a budget calendar. You can use the same fiscal calendar for all of these.

Select a fiscal calendar for your legal entity

Select the fiscal calendar that you want to use for the ledger for your legal entity in the **Ledger** page. A fiscal calendar must be selected on the **Ledger** page for every legal entity. After a fiscal calendar is selected, you can set up period statuses and permissions on the **Ledger calendar** page for any of the periods that are part of a fiscal year.

Select a fiscal calendar for fixed assets

You can select a fiscal calendar for a fixed asset book, and that fiscal calendar will be used by the fixed assets that use the selected book. You can select from any fiscal calendar that is defined on the **Fiscal calendars** page.

Define budget cycle time spans

Budget cycles are the length of time during which a budget is used. Budget cycles can include part of a fiscal year or multiple fiscal years, such as a biennial budget cycle of two years or a triennial budget cycle of three years. The budget cycle time span defines the number of periods that are included in the budget cycle. To specify the budget cycle time span, use the **Budget cycle time spans** page.

Maintain periods for your organization

Use the **Ledger calendar** page to view the details of the fiscal calendar, fiscal years, and periods used by your organization. You can also change the status of the periods and select which users can post accounting transactions to periods. For example, at the start of a new period, you might want a group of users to finish posting financial transactions in the previous period, while other groups work only in the new period.

Last updated on 05/28/2026

Ledger and subledger accounting overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

What is a ledger?

A *ledger* captures the collection of main accounts in a defined structure. It represents the general ledger for a legal entity, where all the financial transactions are summarized and posted. Therefore, it serves as the basis for the preparation of statutory finance reports (balance sheets and profit and loss statements). Each legal entity can have only one ledger. However, up to 10 posting layers can be used to track the deltas for additional accounting standards.

Set up a ledger

For each legal entity in Microsoft Dynamics 365 Finance, details about the ledger must be configured. To configure ledger settings, go to **General ledger** > **Ledger setup** > **Ledger**.

A ledger has the following main components:

- **Chart of accounts** – Determines the main accounts that are used for all accounting entries that are posted to the ledger.

  Account structures determine which financial dimensions are used with the main accounts. (Financial dimensions can be different for each main account or group of accounts.) Account structures also determine which combination of financial dimension values is valid when entries are posted to the ledger.

- **Calendar** – Defines the fiscal year and periods of the ledger.

- **Currency** – Defines the accounting currency, or home currency, of the ledger. The reporting currency is considered a second accounting currency and is optional. The exchange rate types determine which exchange rate tables are used for transaction entry.

Learn more about how to configure a ledger in Configure ledgers.

What is a subledger?

Most documents that are posted affect a *subledger* or a *module*. Examples of subledgers in Dynamics 365 finance and operations apps include Accounts payable, Accounts receivable, Inventory, Fixed assets, Tax, and Production. Each subledger captures the document details that are relevant to it. For example, when a customer invoice is posted, the customer's balance is captured in the **Accounts receivable** module, the tax codes and amounts are captured in the **Tax** module, and the cost of inventory that was sold is captured in the **Inventory/Costing** modules.

What are subledger journal account entries?

Each source document maintains the subledger journal account entries, which represent the detailed accounting entries of the document. The subledger journal account entries are ultimately transferred to the ledger.

The subledger journal account entries are created through accounting rules that are defined through the posting profile for each module. Posting profiles use the main accounts from the chart of accounts that is defined in the ledger setup.

To set up posting profiles in finance and operations apps, follow these steps:

- **Accounts payable:** Go to **Accounts payable** > **Setup** > **Vendor posting profile**.
- **Accounts receivable:** Go to **Accounts receivable** > **Setup** > **Customer posting profile**.
- **Inventory:** Go to **Inventory management** > **Setup** > **Posting** > **Posting**.
- **Production groups:** Go to **Production control** > **Setup** > **Production** > **Production groups**.

Additional posting profiles exist for other modules.

Learn more about posting profiles in Posting profiles overview.

Link between a subledger and a ledger

When a source document is posted, the subledger journal account entry captures the accounting impact of that document. This subledger accounting contains the affected ledger accounts and the corresponding debit and credit amounts. The ledger accounts in the subledger journal account entries are created by using the posting profiles for the main accounts and the financial dimensions from the document. The resulting ledger account (main account + financial dimension values) adheres to the chart of accounts and account structures that are defined in the ledger.

The total debit and credit amounts must be balanced. According to the requirements of the double-entry bookkeeping principle, every transaction that is posted to an account must have an equal and opposite entry in another account. For example, a product receipt for $100 in purchases results in a $100 debit to the Inventory account and an equivalent credit of $100 to the Purchase accrual account.

The balanced accounting entry in the subledger journal account entry is then transferred to the ledger, either in detail or summary, depending on the setup in General ledger parameters.

To view transactions that are posted to the main account, go to **General ledger** > **Inquiries and reports** > **Voucher transactions**. Alternatively, open the **Trial balance** list page by going to **General ledger** > **Inquiries and reports** > **Trial balance**. Drill down to the transaction details.

The voucher is used as the shared reference between the subledger document and the accounting entries in the ledger.

Example: Product receipt subledger and ledger postings

After a product receipt is posted for a purchase order, you can view the product receipt journal that was created during posting. From the **Product receipt** journal, select **Vouchers** to view the ledger accounting entry for the product receipt.

In the **Voucher transactions** view, select the **View subledger journal** tab to view the subledger details. Because summarization can occur during the transfer to the ledger, more account entries might be shown in the subledger journal account entry.

Last updated on 01/21/2026

Financial dimensions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

Use financial dimensions to further categorize financial transactions. Financial dimension values become segments within the ledger account. Use them for further analysis, such as generating a profit and loss financial statement by a dimension or a trial balance by dimension.

Financial tags (tags) are an alternative to financial dimensions. An organization can create up to 20 user-defined financial tags and enter values for them on transactions. For more information, see Financial tags. To understand the differences between the two, see Differences between financial tags and financial dimensions.

This article explains the various types of financial dimensions and their key properties. For step-by-step creation instructions, see Define financial dimensions.

Financial dimension types

Use the **Financial dimensions** page to create financial dimensions that you can use as account segments for charts of accounts. Two types of financial dimensions exist: custom dimensions and entity-backed dimensions.

Custom dimensions

A custom dimension is a financial dimension that reflects your organization's specific business needs - for example, an internal cost center classification, fleet identifier, or expense purpose. Unlike entity-backed dimensions, you create and maintain custom dimension values manually on the **Financial dimension values** page. In addition, custom dimensions are always shared across legal entities.

Entity-backed dimensions

An entity-backed financial dimension is based on a system-defined entity that you select in the **Use values from** field. For entity-backed dimensions, the values are defined somewhere else in the system, such as in Customers or Stores entities. Some entity-backed dimensions are shared across legal entities, whereas other entity-backed dimensions are company-specific. Dimensions that are company-specific (also known as company-striped dimensions) are only visible when accessed as a user from the associated company.

For example, to create dimension values for projects, select **Projects**. This selection allows any value from the project table to be used as a dimension value directly. You create a dimension value for each project name. The **Financial dimension values** page shows the values for the entity. If those values are company-specific, the page also shows the company.

Entity-backed dimension values aren't available in the dimension framework until you use the value in a transaction, posting profile, journal, or similar context. A record that exists in the source entity (for example, a new customer) doesn't appear as a selectable dimension value until it's referenced in one of these areas.

If you want to rename or delete dimension values from entity-backed dimensions, make these changes from the source entity rather than from the **Financial dimension values** page. For more information, see Modifying and deleting financial dimensions.

Financial dimension values

After you create a financial dimension, use the **Financial dimension values** page to create, view, or assign additional properties to each financial dimension value.

For a custom financial dimension, use this page to create and edit dimension values. You can only enter or edit the **Dimension value** and **Description** fields for custom dimensions. Dimension values have a maximum length of 30 characters.

For an entity-backed financial dimension, you can't create dimension values from the **Financial dimension values** page. You also can't edit the dimension value and descriptions from within the

page. For example, let's say you created the Project financial dimension previously described. On the **Financial dimension values** page, you can't edit the project **Dimension value** or **Description**. This information is taken directly from the project setup. If a new project value is necessary, you must create it from the **Project** page.

To validate whether a ledger account or default dimension combination is structurally valid, use the **Dimension data validation** page. For more information, see Dimension data validation.

Legal entity overrides

All custom dimensions and some entity-backed dimensions, such as dimensions created from an operating unit, are global. When you use global values, all dimension values for that dimension are active for all legal entities that include that dimension in their account structure.

But, not all dimensions or main accounts are valid for all legal entities. Additionally, some might be relevant only for a specific period. In these cases, use the **Legal entity overrides** section to specify the companies that the dimension or main account should be suspended for, the owner, and the period when the dimension is active. The overrides at the shared level can't be more restrictive than the overrides at the legal entity level.

Example

Department has dimension values of 100, 200, and 300. USMF, USSI, and DEMF all use the Department dimension. DEMF is only permitted to use department 100 and 200. To restrict DEMF from using Department 300, create a legal entity override where you mark the dimension value as Suspended. USMF and USSI still have full access to departments 100, 200, and 300.

Financial dimensions as legal entities

You can use financial dimensions to represent legal entities. You don't need to create the legal entities in Dynamics 365 Finance. However, financial dimensions aren't designed to address the operational or business requirements of legal entities. The intercompany accounting functionality in Finance is designed to address only the accounting entries that each transaction creates.

Before you set up financial dimensions as legal entities, evaluate your business processes in the following areas to determine if this setup works for your organization:

- Inventory
- Sales and purchases between financial dimensions and legal entities
- Sales tax calculation and reporting

- Operational reporting

Here are some of the limitations:

- You can use sales tax functionality only with legal entities, not with financial dimensions.
- Some reports don't include financial dimensions. Therefore, to report by financial dimension, you might have to modify the reports.

Financial dimension service

> **Important**
>
> Starting September 11, 2026, Microsoft deprecates the Financial dimension service with the release of Dynamics 365 Finance version 10.0.49 and aligns future investment with Optimized Financial dimensions. The Financial dimension service was introduced as an optional performance optimization for specific journal import scenarios, mainly large general journal imports through the Data Management Framework where ledger account values are provided on journal lines. The service applies to a limited set of scenarios and isn't used for Customer, Vendor, Bank account types, or when derived dimensions are configured.

You can access the Financial dimension service add-in in your Microsoft Dynamics Lifecycle Services environment. It provides improved performance when you use the Data management framework to import a journal that has a large number of lines. To use the service, you must enable it on the **Financial dimension service parameters** page. Currently, the service works only on imported journals. In addition, it can currently process only general journals where the **Ledger** account type is set on the journal lines. Other account types on journal lines, such as **Customer**, **Vendor**, and **Bank**, aren't currently supported. This service isn't invoked when derived dimensions are set up in the system.

The Financial dimension service provides improved performance when you import journals by using a new service that runs in parallel to the data import. It runs only on the main account and financial dimension data in the journal, and it generates the dimension combinations that are specified in the ledger account string field on the journal lines. The processing converts this string into the structured data storage that the Financial dimension framework uses throughout the rest of the product for validation, summary reporting, and inquiries. For more information about summary reporting of financial dimension data, see Financial dimension sets.

For more information, see the following topics:

- Define financial dimensions
- Maintain financial dimension default templates

Last updated on 05/27/2026

Financial dimensions and posting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you plan and set up your chart of accounts, consider how the various components work together when you post a document or journal. These components include account structures, advanced rules, and balancing and fixed dimensions. This article explains what each component is and how the components work together.

Chart of accounts and financial dimension components

A rich, rule-based system defines valid combinations of main accounts and financial dimension values. This section gives a brief overview of the functionality of each component and explains where you can find the component.

Account structures

You need an account structure when you set up your ledger. You must define and activate at least one account structure, and you must assign it to the ledger. The account structure must include the main account. You can define the order of segments that works best for the business. After you define the main account, the system can determine the account structure that is used. By putting the main account first or near the front of a structure, you can help limit the values and also help the system apply the last known valid value as a default value. You can include up to 10 additional financial dimensions in the account structure. The account structure defines which dimension values are valid in combination with other values. It also defines whether you must enter dimension values.

Advanced rules

Advanced rules are an optional component when you set up the chart of accounts. You can add as many advanced rules as you want to an account structure. Use advanced rules to handle

scenarios where you must track extra financial dimensions when specific criteria are met. For example, if you use a Travel expense account, you might want to track extra information, such as the event that the employee is traveling for. If there are multiple advanced rules, the system applies them in alphabetical order, based on the names of the rules. The segments that a rule adds can be applied only after the segments of the account structure.

Balancing dimension

You can optionally define a balancing financial dimension. On the **Ledger** page, you can define the financial dimension that should be balanced. Then, whenever you post transactions to that financial dimension, the system automatically creates and posts entries to make the financial dimension balanced.

Default and fixed financial dimensions on the main account

Default dimensions come from various places, such as master records (for example, customer or vendor records), document headers, and the main account. This article focuses on default dimensions on the main account by legal entity. You can define whether a main account has a **Not fixed** or **Fixed** value for each financial dimension that is used across all account structures for the ledger. If a financial dimension is **Not fixed**, it uses a default value, but you can overwrite that value. This behavior applies to all default values in the system, even default values that come from master records. If you set a financial dimension to a **Fixed** value, the system always applies that value, regardless of whether it came from somewhere as a default value or the user entered it.

> ⓘ **Note**
>
> A related concept is derived dimensions. Derived dimensions let the value of one financial dimension automatically set the values of other dimensions, based on rules that you define. By default, derived values are applied only when the target dimensions are blank, but you can enable the **Replace existing dimension values with derived values** option to overwrite values that are already populated. For more information, see Derived dimensions.

Order in which the system applies default dimensions during posting

You might have questions about the order that the various components run in. It's very important that you understand the order that the system applies default dimensions, because this behavior

affects the approach that you take to set up.

> **Note**
>
> This information applies only to the application of default dimensions in the application. If you import data by using Microsoft Excel or the Data Management Framework, the behavior differs.

Example 1

Account structure

Expand table

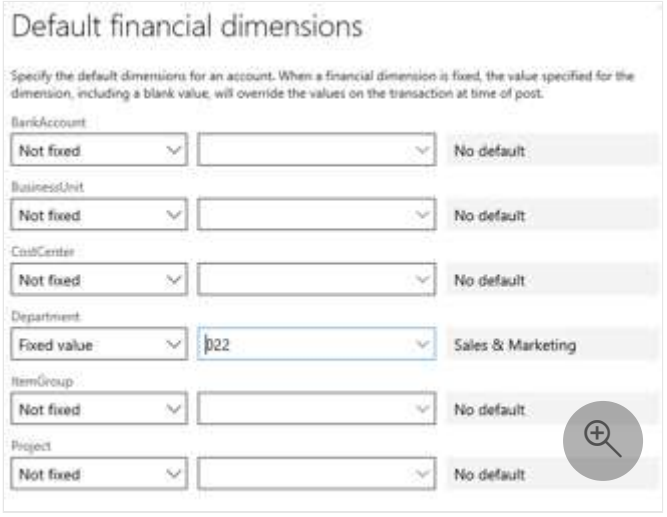
Main account	Business unit	Department	Cost center
All values are allowed	All values are allowed	All values are allowed	All values are allowed

Main account

Expand table

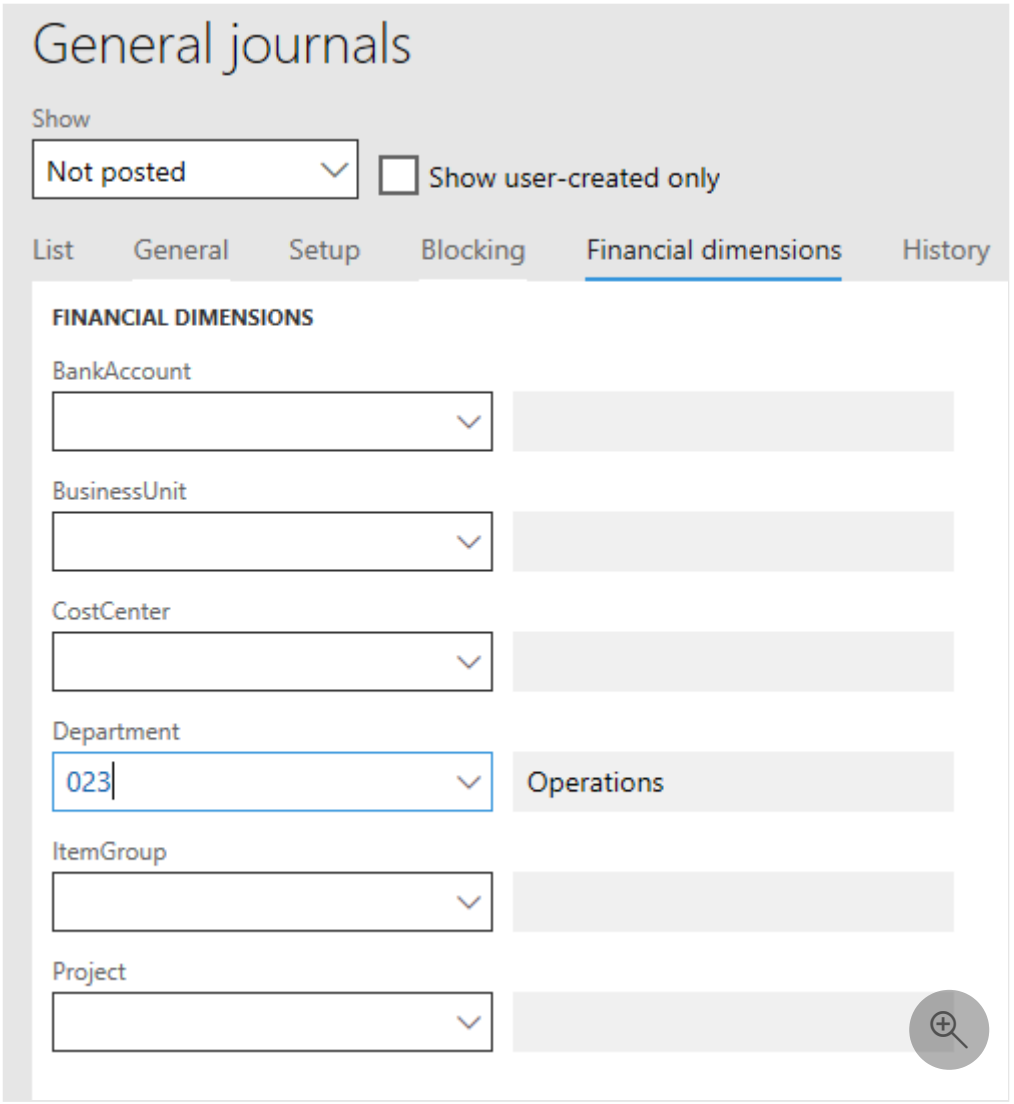
Main account	Name	Legal entity	Department
401100	Product Sales	USMF	Fixed – 022 Sales and Marketing department

The following illustration shows the fixed default dimension that is set on main account 401100.

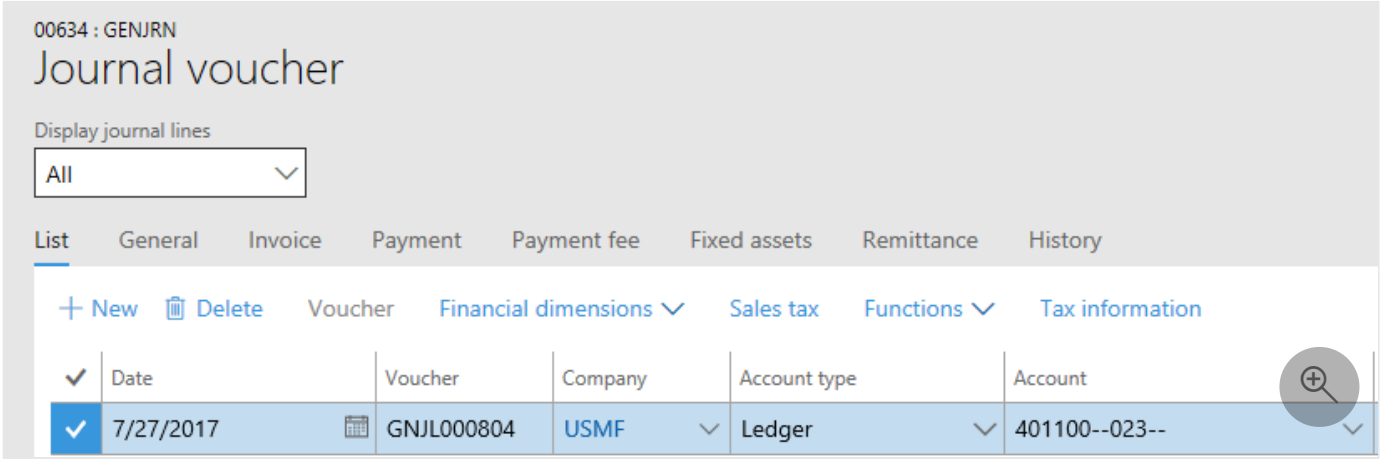


For this basic example, you enter a general journal where the Department dimension is set to use the default value **023** (Operations). You enter and post a ledger account. The following illustration

shows the default financial dimension on the general ledger header.

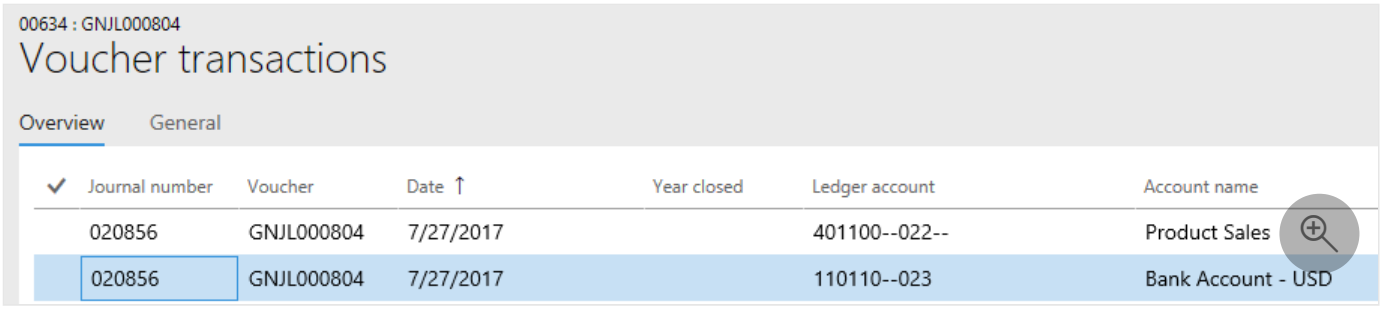


The default dimension on the journal header causes department 023 to be applied by default on the sales account line. The following illustration shows the general journal line, where the **023** default dimension value from the header is applied.



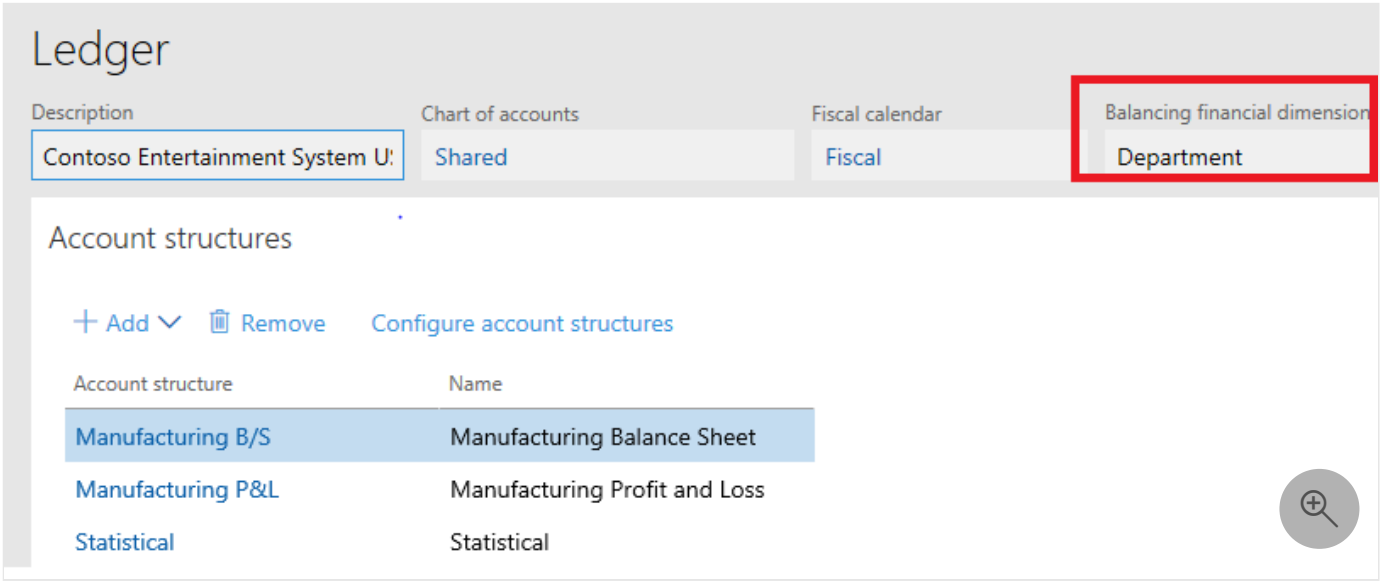
However, when you post the line, the fixed dimension is applied, and the line is posted to department 022. The following illustration shows the posted voucher, where the fixed dimension

is applied for the sales account.

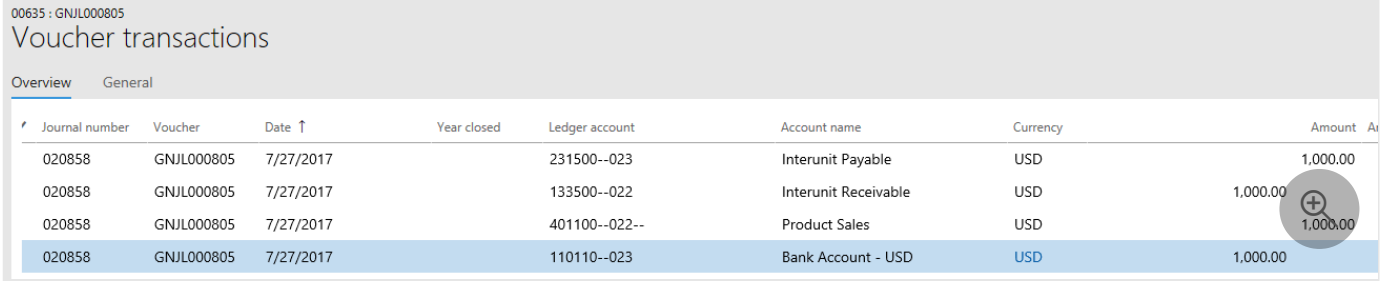


Example 2

This example uses the same setup as the first example. However, you add a second component and use the Department dimension as a balancing dimension. In the following illustration, **Department** is the balancing financial dimension for the USMF ledger.



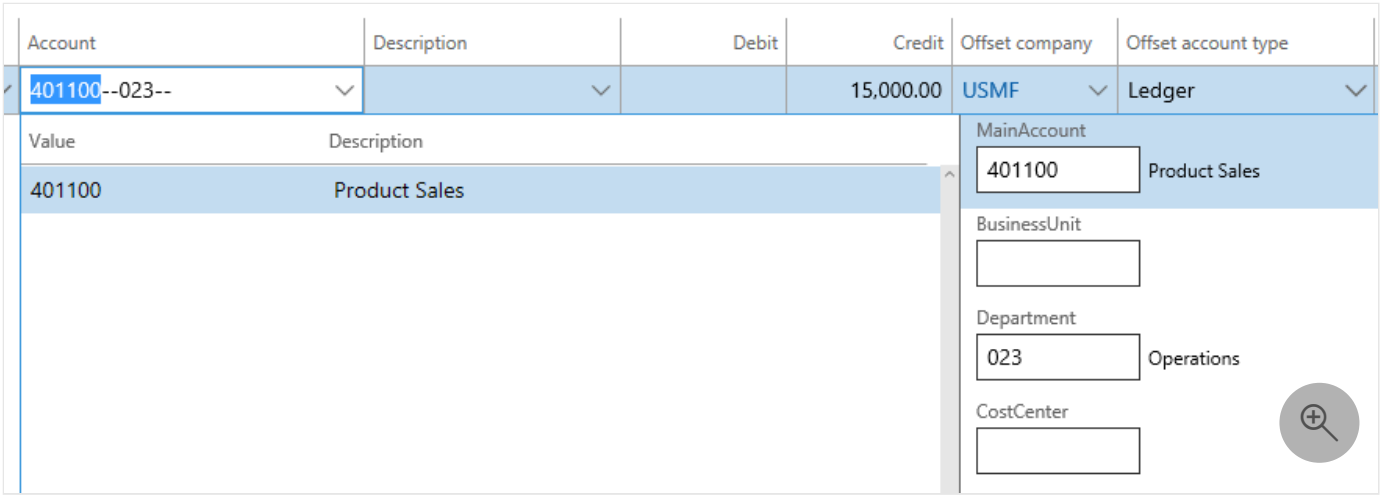
When you use the same journal header setup and post the same transaction, the fixed dimension is applied first. Then the balancing logic is applied to help guarantee that every department has a balanced entry. The following illustration shows voucher transactions that include the balancing entry after the fixed dimension is applied.



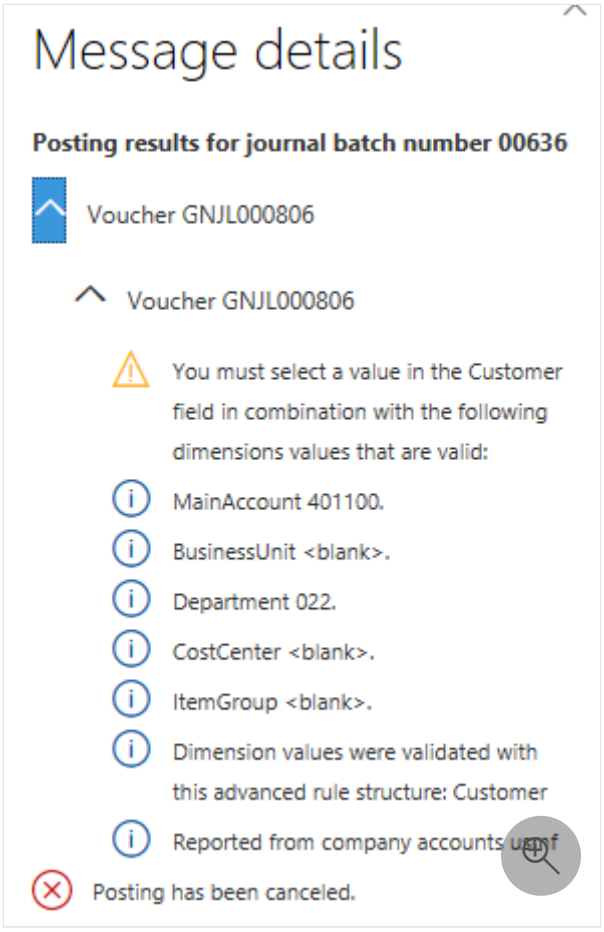
Example 3

In this example, you add an advanced rule. The advanced rule specifies that if sales account 401100 and department 022 (Sales and Marketing) are used, the system should track an additional segment that is named Customer.

This example is important because of the order. The account structure is determined after the main account is entered. If you refer to the account structure setup, the system can determine that the main account, business unit, department, and cost center are relevant. At this point, the advanced rule isn't triggered, because fixed dimensions aren't applied until default dimensions are applied for the journal voucher during posting. In the following illustration, the Customer segment isn't present, because the criteria for the advanced rule aren't met.



The posting isn't successful, because the fixed dimension was applied at the end of the process. Dimension validation determines that the Customer segment is required if the main account is 401100 and the department is 022. Posting can't occur because of the validation error. The following illustration shows the message that appears after dimension validation determines that Customer is a required segment.



In this example, you must overwrite the default value so that the advanced rule is triggered and you can enter the Customer segment. However, this solution isn't always possible, and some users aren't even aware of the posting rules. Therefore, it's important that you understand the order that default dimensions are applied in when you set up your chart of accounts.

To achieve what you want in this example, you can change the configuration in several ways. For example, you can create a new account structure for sales accounts and include the Customer segment in the structure. You can also add more rows in an existing account structure, and specify the main account and valid department values. Then, in the additional customer structure, you might find it useful to have a separate account structure of sales accounts where the Customer segment is present.

Additional resources

Some of the following resources refer to an earlier version. However, much of the information about the application of default dimensions and many of the concepts are the same in the earlier version, so the references are still valid.

Balanced journals for interunit accounting

Plan your chart of accounts

Planning your chart of accounts in AX 2012 blog – This link goes to part one of a seven-part series.

Dimension defaulting in accounting distributions

Dimension defaulting in Dimensions framework

Last updated on 05/29/2026

Add financial dimensions to the CFO workspace

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

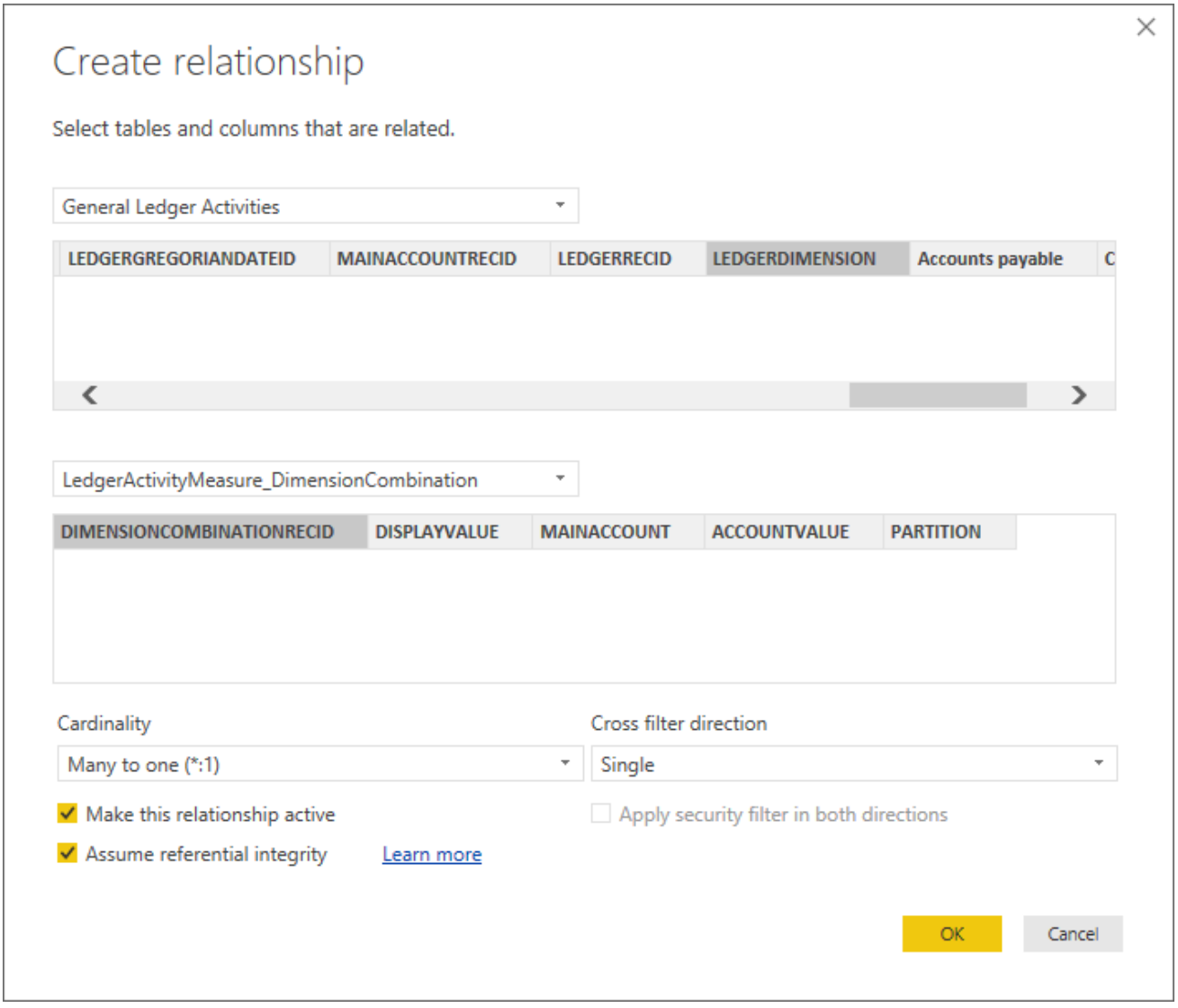
This article explains how to add financial dimensions to the Chief Financial Officer (CFO) workspace, so that they can be used for the ledger and budget reports. The CFO workspace has an **Overview** tab and a **Financial** tab. The reports on these two tabs are backed by two measures: LedgerActivityMeasure and BudgetActivityMeasure. There's a relation between those two measures and the DimensionCombinationEntity entity. Therefore, you can select dimensions.

1. In Microsoft Dynamics 365 Finance, on the **Entity Store** page, update the **LedgerActivityMeasure** and the **BudgetActivityMeasure** measures.
2. In Microsoft Visual Studio, open Application Explorer, and search for **LedgerCFO**.
3. Under **Resources**, open **LedgerCFOWorkspacePBIX**.
4. When the resource opens in Microsoft Power BI Desktop, select **Get Data**, select **SQL Server database**, and then select **Connect**.
5. Enter the server name, and enter **AxDW** as the database. Select **DirectQuery**, and then select **OK**.
6. Search for and select **LedgerActivityMeasure_DimensionCombination**, and then select **Load**.

   > **Tip**
   >
   > In the **Fields** list, rename the table **Financial dimensions**, so that it's easy to identify.

7. Select **Manage Relationships**, and then select **New**.

8. In the first field, select **General Ledger Activities**, and then select **LedgerDimension**.
9. In the second field, select **LedgerActivityMeasure_DimensionCombination** (or **Financial dimensions** if you renamed the table). Select the **DimensionCombinationRECID** header.
10. In the **Cardinality** field, select **Many to One**.
11. Change the **Cross filter direction** value to **Single**.
12. Select both **Make this relationship active** and **Assume referential integrity**, select **OK**, and then select **Close**.



13. In the **Fields** list, you see the table and the available financial dimensions. Drag the financial dimensions that you want to the report-level filters.
14. Save your changes.
15. In the Application Object Tree (AOT), right-click your project, and then select **Synchronize**.
16. Build your project, and then open the application to view the results.

Last updated on 06/24/2026

Creating, modifying, and deleting account structures

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article walks through the process of creating, activating, and deleting account structures. The steps use the USMF demo data company.

Creating account structures

1. Go to **General ledger** > **Chart of accounts** > **Structures** > **Configure account structures**.
2. On the **Action pane**, click **New** to open the drop dialog.
3. In the **Account structure** field, enter a name to describe the purpose of the account structure.
4. In the **Description** field, enter a description to specify the purpose of the account structure.
5. Click **Create**.
6. In the **Segments and allowed values**, click **Add segment**.
7. In the dimensions list, select the dimension to add to the account structure.
8. At the end of the list, click **Add segment**.
9. Repeat step 6 to 9 as needed.
10. In the **Allowed value details** section, select the segment to edit the allowed values. For example, click the **Main Account** field.
11. In the **Operator** field, select an option, such as is between and includes.
12. In the **Value** field, type a value. For example, 600000.
13. In the **through** field, type a value. For example, 699999.
14. In the **Allowed value details** section, click **Apply**.
15. Repeat step 10 to 15 as needed.
16. In the **Allowed value details** section, click **Add new criteria**.
17. In the **Operator** field, select an option, such as is between and includes.

18. In the **Value** field, type a value. For example, 033.
19. In the **through** field, type a value. For example, 034.
20. Click **Apply**.
21. In the grid, select the segment to edit the allowed values. For example, Cost Center.
22. In the **CostCenter** field, type a value. For example, 007..021.
23. In the **Segments and allowed values**, click **Add**.
24. In the **MainAccount** field, type a value. For example, 600000..699999
25. In the grid, select the segment to edit the allowed values. For example, Department.
26. In the **Department** field, type a value. For example, 032.
27. In the **CostCenter** field, type a value. For example, 086.
28. On the **Action pane**, click **Validate**.
29. On the **Action pane**, click **Activate**.
30. Click **Activate**.

Activating account structures

When you're satisfied with your new setup or a change to an account structure, you must activate it before it takes effect. During activation, the system validates the structure and then synchronizes all unposted transactions to match the new structure. Posted transactions aren't affected by account structure changes.

> ⓘ **Note**
>
> Account structure activation is not the same as dimension activation. Account structure activation synchronizes unposted transactions to match your updated structure and runs as a batch job during normal operations. It can't be performed in maintenance mode.

The activation process runs as a batch job. Depending on the volume of unposted transactions, activation can take anywhere from a few seconds to several hours. While activation is running, the structure is marked as **Activating** on the **Account structures** page.

> ⓘ **Important**
>
> Don't cancel a running activation. Canceling resets the process, and resubmitting starts the work over from the beginning — it doesn't resume from where it left off.

Monitoring activation status

When you activate an account structure, the system validates the structure and marks it as active. Unposted transaction processing then continues in the background, with multiple transaction updates running at the same time.

To monitor activation progress:

1. On the **Account structures** page, select **View activation status** above the grid, or select **View** on the Action Pane and then **Activation status**.
2. A dialog box appears showing the individual tasks running as part of the activation process, along with each task's status and completion time.

For more information, see Account structure activation.

Account structure activation failure

If an activation fails, the structure reverts to **Draft** status. The error is typically not displayed on the account structure form itself — check the **batch job error log** for the specific error message.

The most common reasons for activation failure are overlapping criteria, either within the structure or across other structures on the same ledger. For more information, see Account structures overview.

Modifying an existing account structure

In many cases, modifying an existing account structure is preferable to deleting it and creating a new one. Modification avoids the overhead of reassigning the structure across legal entities and reconfiguring dependent processes.

Modification is suitable for scenarios such as:

- **Broadening a range** — For example, changing from 1000-2000 to 1000-3000.
- **Restricting a range** — For example, specifying a **Department** segment range of 012-014.
- **Subdividing a main account range** — For example, splitting a **BalanceSheet** structure with range 1000-3000 into two structures: **BalanceSheet_Small** with range 1000-2000 and **BalanceSheet_Large** with range 2000-3000.
- **Adding or removing dimensions** — For example, adding a new segment or removing one that is no longer needed.

To modify an account structure:

1. Go to **General ledger** > **Chart of accounts** > **Structures** > **Configure account structures** and select the desired structure.
2. Select **Edit** on the Action Pane. The system creates a draft copy while keeping the original active for ongoing transactions.
3. Make your changes — update criteria, add or remove segments, or adjust ranges as needed.
4. Select **Validate** to check your changes.
5. Select **Activate** to apply your changes and wait for successful activation.

Deleting account structures

If an account structure is no longer needed and has no blocking unposted transactions, it can be deleted. Before deleting, consider whether modifying the structure would better serve your needs.

Resolving blocking transactions

You can't delete an account structure if unposted transactions reference it. To find and resolve these transactions:

1. Navigate to the journal which has the transactions (e.g. **General ledger** > **Journal entries** > **General journals**).
2. Filter by **Show** > **Not posted**.
3. Select a journal and select **Lines** on the Action Pane to view its transactions.
4. Transactions that reference the account structure can be removed with the **Delete** button or updated to reference a different ledger account or offset account.

Deleting the structure

After all blocking transactions are resolved:

1. Go to **General ledger** > **Chart of accounts** > **Structures** > **Configure account structures**.
2. Select the account structure you want to delete.
3. Select **Delete** on the Action Pane.

If you can't locate the blocking transactions, contact Microsoft Support for assistance.

Last updated on 04/13/2026

Budgeting overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Almost every company that uses Financials functionality in Microsoft Dynamics 365 Finance needs to create reports of budget versus actuals. This article explains the minimum configuration that you need to create budgets in finance and operations or load them from a third-party program.

Overview

You maintain the approved budget for a legal entity in a document known as a *budget register entry*. The lines in a budget register entry document are known as *budget account* entries. They contain financial dimension information, dates, and the amounts of the approved budget. The budget register entry document integrates with basic financial reports and inquiry pages where ledger actual amounts are compared to budget amounts.

You can create budget register entries in several ways:

- Manually enter the document information on the **Budget register entries** page.
- Use the Microsoft Excel template that you can open by selecting the **Open in Excel** button on the **Budget register entries** page.
- Use the **Budget Account Entries** data entity in Data management to import budget register entries. Consider using this method and turning on the **Set based processing** parameter when you must import many budget account entries.
- If the company uses Budget planning functionality to prepare budget data, you can use the **Generate budget register entry** periodic process.

The budget register entry is considered completed when you update the budget balances. On the **Budget register entries** page, select **Update budget balances** for a selected budget register entry or multiple entries. After you update the budget balances, the status of the budget register entry changes to **Completed**. You can't reopen a completed budget register entry for edits.

Therefore, if you must adjust the budget data, you must create a new budget register entry instead of correcting data in the completed budget register entry.

Configuration

When you configure budgeting, start on the **Budgeting parameters** page. On this page, you must define the budget journal, the number sequence for budget register entries, and the default behavior in the workspaces.

On the **Budgeting parameters** page, the following fields are available:

- **Allocation key** – This field determines how the amount for a budget account entry is allocated in a cash flow forecast.
- **Use rules for budget transfers** – Select this option to enable budget transfers.
- **Display legacy budget analysis inquiry** – Select this option to show the legacy budget analysis inquiry.
- **Default account structure** – Select this checkbox to automatically populate the account structure in the budget register, based on the main account. This checkbox is available if the **Default the account structure in the budget register entry** feature is enabled in Feature management.
- **Budget journal** – Select a journal name for budget appropriations. This field is available if the **Enable budget appropriation** field is selected on the **General Ledger parameters** page.

Next, if policies govern the approval of budget register entries based on budget type (for example, transfers or revisions), create budget register entry workflows on the **Budgeting workflows** page. If there are scenarios where transfers might be allowed without workflow approval, you can define budget transfer rules to support those scenarios.

On the **Budgeting dimensions** page, select the financial dimensions that are used for budgeting, based on the dimensions that are used in the chart of accounts. You can select all financial dimensions or a subset of them for budgeting.

Define a *budget model* that corresponds to all or some of the budgets. You can use a single budget model for all budget register entries. Alternatively, you can create separate models that are based on the budget type, the geographical location, or some other way that a budget can be classified.

> ⓘ **Note**

If you use budget control, you can associate only one budget model with a specific budget cycle time span.

Create *budget codes* that identify the type of budget transactions to record and any related workflow. Budget codes can support the following budget types:

- Original budget
- Transfer
- Revision
- Encumbrance
- Preencumbrance
- Carry-forward budget

Budget codes provide an audit trail of approved budget modifications throughout the course of the budget cycle. If you associate a workflow with a budget code, the workflow is enabled for all budget register entries that use that budget code, and workflow steps must be completed before the budget register entry can reach the **Completed** stage.

You can also optionally set up *budget transfer rules*. To use budget transfer rules, select **Use rules for budget transfers** on the **Budget parameters** page. When budget transfer rules are used, if a user creates a document by using a budget code of the **Transfer** type, budget balances aren't updated if the budget transfer rules are violated. For example, you can allow budget transfer documents where the expense budget is transferred between the main accounts for the Sales and Marketing department, but can prohibit budget from being transferred from or to that department unless workflow approval is granted for that type of budget account entry.

Functionality that was introduced in Microsoft Dynamics 365 Finance version 10.0.7 (January 2020) adds capability and flexibility for budget register entries. To enable these enhancements, go to the **Feature management** workspace and select **Budget register entries for quantity only** and/or **Budget register entries defaulting of amount type**.

The **Budget register entries for quantity** only feature lets you post a budget register entry with quantity-only amounts. For example, you could post a budget entry with a quantity of 32 and a price of zero, which results in an amount of zero. You can then use this quantity within the context of a financial report to determine a price per quantity. No inquiries or reports were updated as part of this feature; the feature just enables you to post an amount of zero.

The **Budget register entries defaulting of amount type** feature allows the default amount type within a budget register entry to be an amount type other than expense. The budget register

entry line now defaults to expense when the main account type is expense; it defaults to revenue when the main account type is revenue; and it defaults to expense for all other account types.

Using workspaces and inquiry pages to track budget vs. actuals

The budget manager can review the current state of a budget in the **Ledger budgets and forecasts** workspace. The **Expense over budget** and **Revenue under budget** tabs provide a quick view of the financial dimension combinations where budget targets aren't being met or are approaching the threshold. You can personalize the budget threshold percentage and financial dimension sets that are used on those tabs by clicking **Configure my workspace**. You can select **Unit managers** to see the workers who are responsible for specific financial dimension combinations that are selected on those tabs. For example, if you see that the expense budget of the Operations department is going over the budget threshold, you can easily find and contact the Operations department manager to discuss the issue.

> **Note**
>
> The **Department manager** field on the **Organization Units** page determines which managers support specific financial dimension combinations. Select **See more** at the bottom of the tab to open the **Budget vs actuals** inquiry page for more details about budget amounts versus actual amounts.

The **Actual vs budget** inquiry page lets you drill into the details of the budget versus actual amounts. Select a line on the inquiry page, and then select **Period balances** to see budget and actual amounts spread across fiscal periods. The **Budget account entries** page provides drill-through to the details of the budget amount in budget register entries. The **General journal entries** page opens the ledger transactions that are included in the calculated **Actuals** amount.

A company that's using Budget planning functionality can create and use *budget forecasts* in the **Ledger budgets and forecasts** workspace.

Last updated on 05/28/2026

Create dimensions and import dimension members

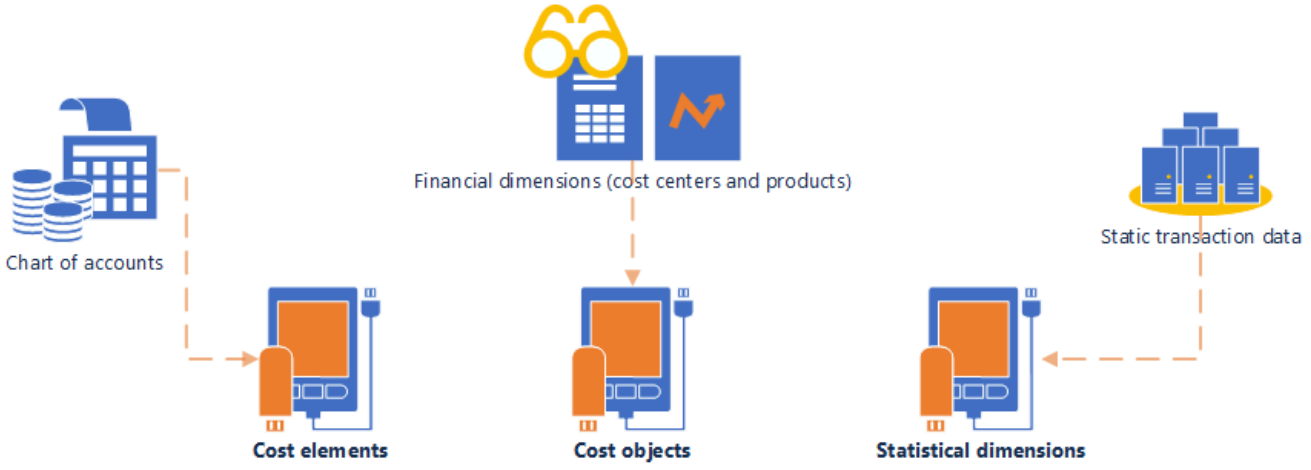
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Cost accounting is an independent module that requires data from other modules. Categorize this data into the following categories:

- Cost elements
- Cost objects
- Statistical dimensions

A **Cost element** corresponds to a cost-relevant item in the chart of accounts. A **Cost object** corresponds to any type of financial dimension, such as products, cost centers, and projects that you want to estimate, allocate costs to, or measure directly. A **Statistical dimension** and its members are used to register nonmonetary entries. Statistical dimension members can be used as an allocation base in cost distribution and allocation

The following diagram illustrates the dimensions that are used in Cost accounting.



After you import the data into Cost accounting, use it to build various perspectives that provide insights to managers at all levels of the organization. The following articles provide information

about creating dimensions and importing dimension members.

- Cost element dimensions
- Create cost elements
- Cost object dimensions
- Map cost element dimension members to a common set of dimension members
- Map a cost element dimension
- Statistical dimension members and statistical measure provider templates

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Last updated on 06/25/2026

Cost element dimensions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

As a core pillar in cost accounting, use cost element dimensions to categorize and track where costs flow.

A cost element corresponds to a cost-relevant item in the chart of accounts. It can be any type of element at the lowest level in a business where costs can flow. The concept of cost elements includes ledger accounts and all cost-relevant resources. Currently, Cost accounting supports ledger accounts.

Two types of cost elements

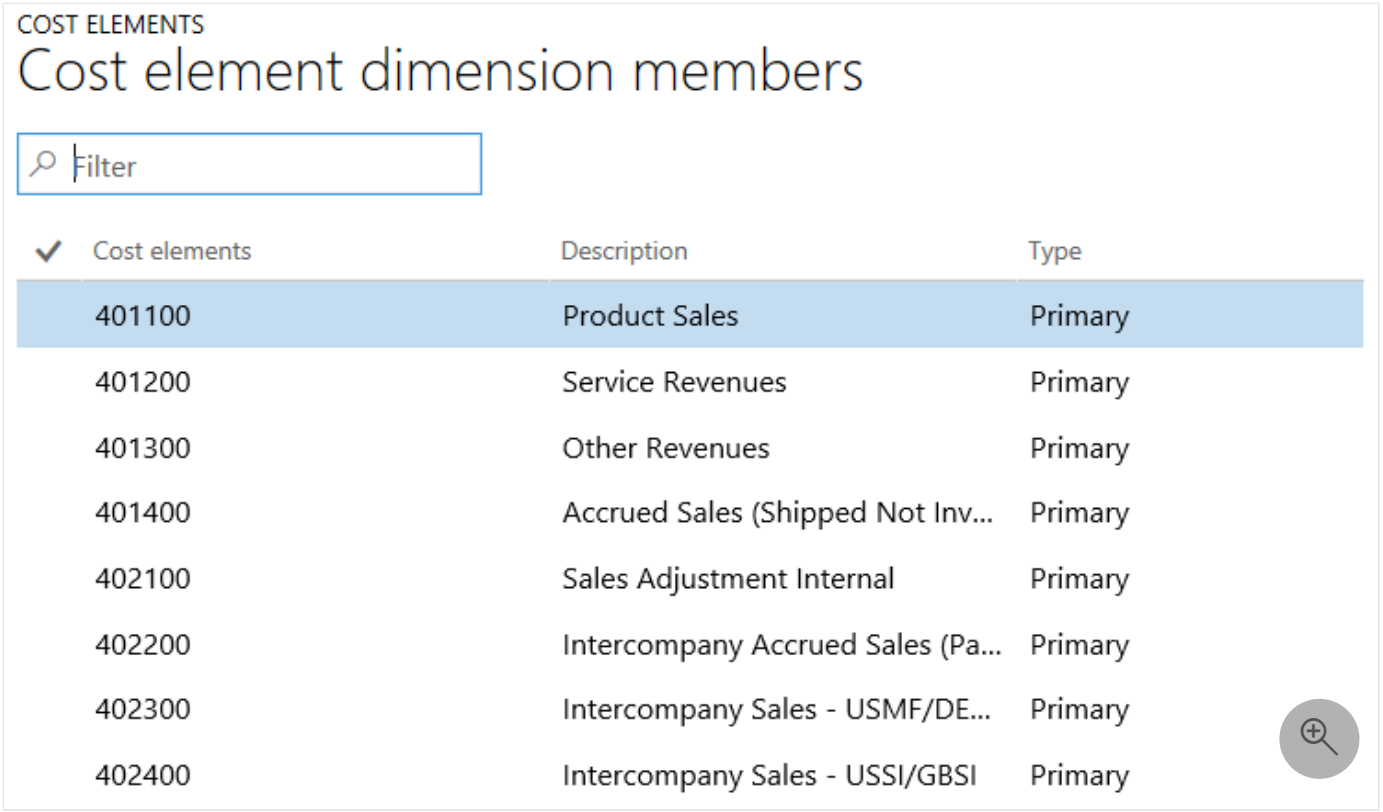
There are two types of cost elements: primary cost elements and secondary cost elements. The following table describes the difference between the two types.

Expand table

Primary cost elements	Secondary cost elements
The primary cost elements represent the flow of costs from financial accounting to cost accounting. The cost element structure corresponds to the profit and loss account structure in the general ledger, where a cost element can correspond to a main account. Not all main accounts need to be represented as cost elements depending on the business needs. Examples of primary cost elements include: • Costs of goods sold (COGs) • Indirect material costs • Personnel costs • Energy costs	The secondary cost elements represent the flow of costs internally because these costs are created and used only in Cost accounting. Use them to ensure that you can trace the source of costs. Use these cost elements in cost allocations and overhead calculations. Examples of secondary cost elements include: • Production costs • Production, material, and marketing overheads

Cost element dimensions and cost element dimension members

Cost elements are *cost element dimensions*. The individual dimension values are *cost element dimension members*. For example, you have a US chart of accounts structure (COA) that serves as the base for your statutory reporting. Use this COA as the cost element dimension. The accounts, which are primary cost elements, appear as the cost element dimension members in Cost accounting. The following screenshot shows an example of Main Accounts as the cost element dimension with its actual main accounts as the cost element dimension members.



Import cost element dimension members through data connectors

To simplify the setup of cost element dimension members in Cost accounting, use data connectors that are either prebuilt or custom built to retrieve the primary cost elements from one or more source systems.

Implementation considerations

Because cost elements represent the lowest level of cost details, make sure you include all the cost elements required for managerial reporting when you implement the cost elements structure. It can be a challenge to find an appropriate number of cost elements for cost control.

Having thousands of cost elements can make it difficult to control each cost element. As an alternative, group cost elements and manage cost control at an aggregated level.

Last updated on 05/28/2026

Create cost elements

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

There are several ways to create cost elements in Cost accounting. This procedure shows how to create cost elements by importing main accounts via a data connector. The USMF demo company was used to create this procedure. This procedure is for a Cost accounting feature that was added in Dynamics 365 for Operations, version 1611.

Create new cost elements

1. Go to **Cost accounting** > **Dimensions** > **Cost element dimensions**.
2. Click **New**.
3. In the **Name** field, type a value.
4. In the **Data connector for dimension members** field, enter or select a value.
5. In the **Description** field, type a value.
6. Click **Save**.

Configure the data connector

1. Click **Configure dimension member provider**.
2. In the **Chart of accounts** field, enter or select a value.
    - Select **Shared** to use the shared chart of accounts.
3. Click **New**.
4. In the list, mark the selected row.
    - You can apply filters to accounts to meet your criteria.
5. In the **From main account** field, enter or select a value.
6. In the **To main account** field, enter or select a value.
7. Click **OK**.

Import main accounts

1. Click **Import dimension members**.
    - Main accounts will be imported into Cost accounting and used as cost elements.
2. Click **OK**.

View the imported accounts as cost elements

1. Click **View dimension members**.
    - View the imported ledger accounts as cost elements in your business that costs can flow to.

Last updated on 10/02/2025

Cost object dimensions

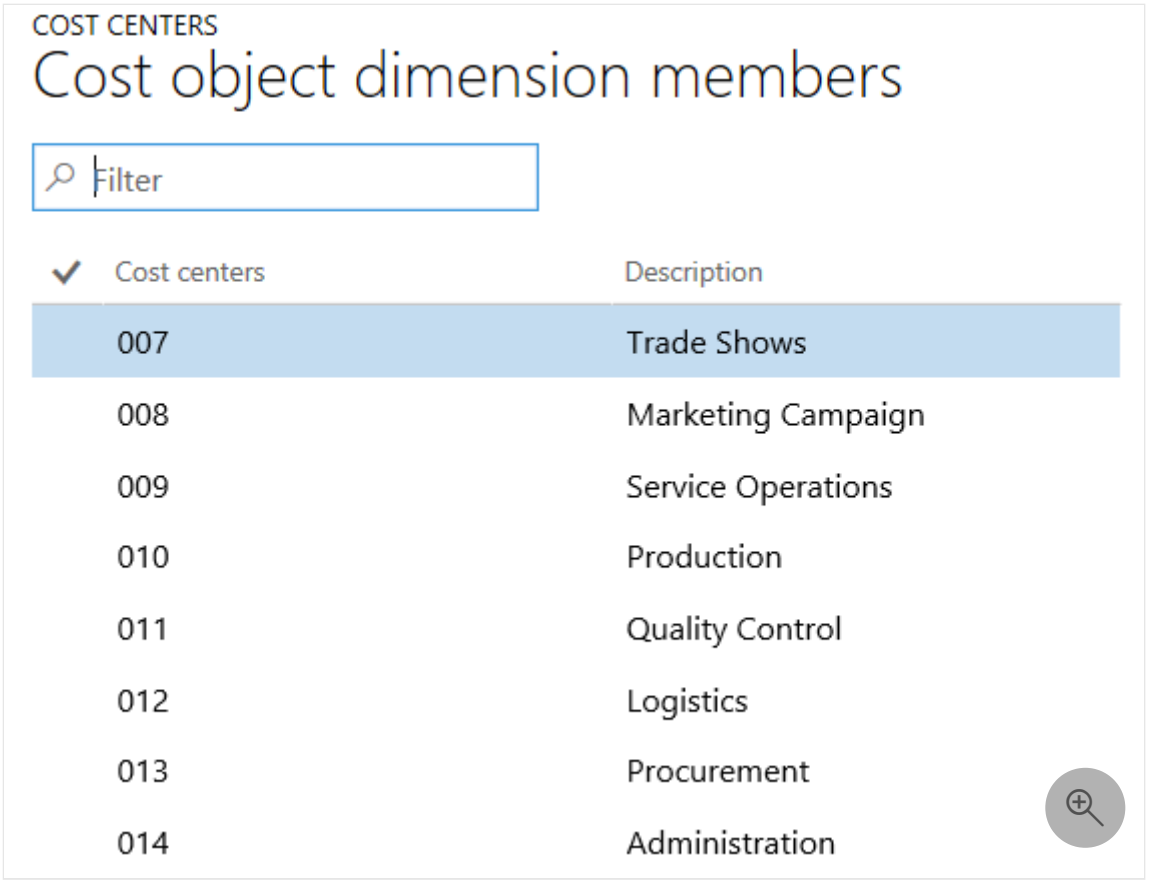
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you analyze costs, use cost element dimensions to determine where costs flow. Use cost object dimensions to determine where you should assign costs. This article provides information about cost object dimensions.

A cost object can be any type of object that you want to estimate, allocate costs to, or measure directly. Typical cost objects include products, projects, resources, departments, cost centers, and geographical regions. Management uses cost objects to quantify costs and also to drive profitability analysis.

Cost object dimensions and cost object dimension members

Cost objects are known as *cost object dimensions*. After you decide which entity the cost object dimension should refer to, specify the individual dimension values or import them into Cost accounting from other source systems. These individual dimension values are known as *cost object dimension members*. For example, you want to use the financial dimension that is named Cost center as the cost object dimension. To see how costs flow to the individual cost centers, you must import the cost object dimension members. In this case, the cost object dimension members are the actual cost centers, such as Sales, Production, Administration, and Geographic locations. The following screenshot shows an example of Cost Centers as the cost object dimension with its actual cost centers as cost object dimension members.



Import cost object dimension members through data connectors

To make the import of cost object dimension members easier, use data connectors to retrieve the values from the entities that you want to use as cost object dimensions. You can use either the prebuilt data connectors or custom data connectors that you build.

Last updated on 05/27/2026

Create cost objects

Article • 03/28/2023

This procedure shows how to create cost objects by importing the cost center financial dimension into Cost accounting via a data connector. The USMF demo company was used to create this procedure.

Create new cost objects

1. Go to **Cost accounting** > **Dimensions** > **Cost object dimensions**.
2. Click **New**.
3. In the **Name** field, type a value.
4. In the **Data connector for dimension members** field, enter or select a value.
5. In the **Description** field, type a value.
6. Click **Save**.

Configure the data connector

1. Click **Configure dimension member provider**.

    - Select **CostCenter** to import the CostCenter dimension into Cost accounting.

2. In the **Dimension name** field, select Cost center.
3. Click **OK**.

Import cost centers

1. Click **Import dimension members**.
2. Click **OK**.

View the imported cost centers

1. Click **View dimension members**.

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Map cost element dimension members to a common set of dimension members

> **Note**
>
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By mapping different cost element dimension members to a common set of cost element dimension members, you merge data into a common format for analysis purposes.

If you're a global company and comply with statutory accounting requirements, you might use multiple charts of accounts. When you import cost element dimension members from different charts of accounts, you can end up with a mix of accounts. However, these accounts might actually have the same nature, and you might want to analyze and allocate costs for them by using a common format.

Map cost element dimension members to a common format

The following example shows how you, as a cost controller, can create a new cost element dimension in Cost accounting that maps cost element dimension members from the US chart of accounts structure and the French chart of accounts structure to a common set of cost element dimension members. You can then use the common set of cost element dimension members to analyze cost data from the two legal entities in a cost accounting ledger.

Expand table

Source: US chart of accounts	Source: French chart of accounts	New common set of cost element dimension members
Imported cost element dimension members from the US chart of accounts	Imported cost element dimension members from the French chart of accounts	Mapping of US and French cost element dimension members to a common set

Source: US chart of accounts	Source: French chart of accounts	New common set of cost element dimension members
5001: Sales	5001: Sales and advertising	5000: Sales and advertising
5030: Advertising	6390: Stock purchase*	7000: Cleaning expenses
7001: Cleaning expenses	7001: Travel expense	7001: Travel expenses

*The Stock purchase French cost element dimension member isn't mapped.

Currency conversion

The various charts of accounts that you use might be set up to use different currencies. In this case, be sure to specify a currency conversion, so that cost data is processed by using the correct currency, as defined in the cost accounting ledger where the cost element dimension members are used. In the preceding example, if US dollars (USD) are used in the cost accounting ledger, you must create a currency conversion from USD to euros (EUR) to process transactions for the mapped cost element dimension members.

Update mappings at any time

You can update the mapping definitions for a cost element dimension at any time. Because mappings aren't date-effective, changes apply the next time you process cost transactions or run cost calculations.

Last updated on 05/27/2026

Map a cost element dimension

Article • 03/28/2023

A cost controller can use this procedure to map a cost element dimension to a cost element dimension in the MXMF legal entity. This recording uses the USP2 demo data company.

1. Go to **Cost accounting** > **Dimensions** > **Cost element dimensions**.
2. In the list, find and select the desired record.
    - For this example, select **Cost elements**.
3. Click **Dimension mappings**.
4. Click **Configure mappings** from this dimension.
5. Click **New**.
6. In the **To dimension** field, enter or select a value.
    - For this example, select MXMF Cost elements.
7. Click **New**.
8. In the list, mark the selected row.
9. In the **From dimension member** field, enter or select a value.
    - For this example, select dimension member 606400 Telephone & Fax Expense.
10. In the **To dimension member** field, enter or select a value.
    - For this example, select dimension member 6001004 Telefono.
11. Click **Save**.

Feedback

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Dimension hierarchy

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

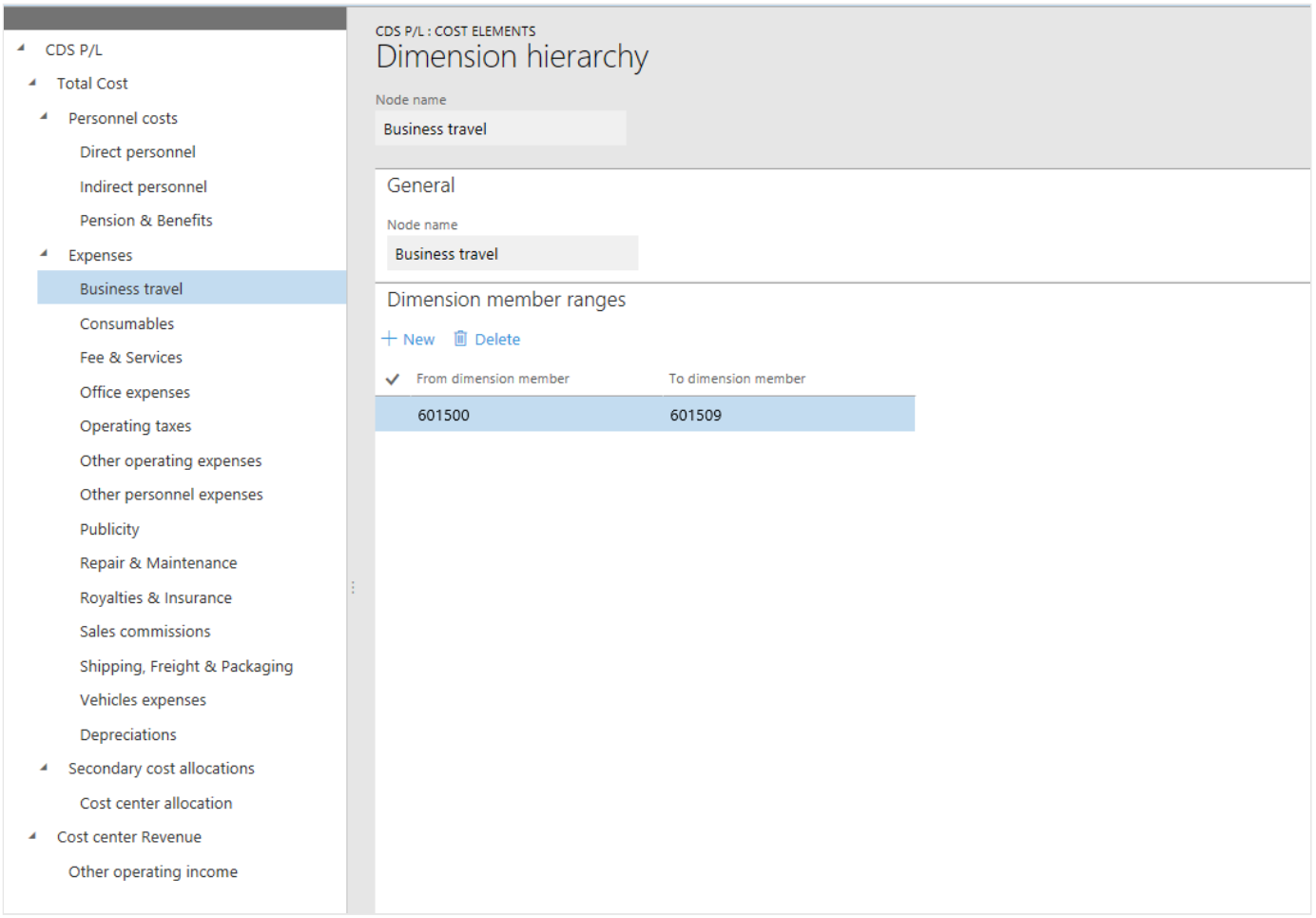
This article provides information about dimension hierarchies. Use a dimension hierarchy to define the reporting structure, cost policies, and security setup in Cost accounting.

Overview

Cost accounting uses dimension hierarchies in various places. A dimension hierarchy lets you define the following information:

- The reporting structure that fits into the organization's requirements
- Cost policies
- The security setup

Here's an example of a dimension hierarchy.



You can create a dimension hierarchy for the following types of dimensions:

- Cost element dimensions
- Cost object dimensions
- Statistical dimensions

> **Note**
>
> - You can create multiple dimension hierarchies for the same dimension if you need different perspectives.
> - You can associate a dimension hierarchy with only one dimension.
> - A dimension hierarchy can have unlimited levels in its structure. All the levels appear in the **Cost control** workspace. When you use Microsoft Excel or Microsoft Power BI for reporting purposes, only the first 15 levels of the dimension hierarchy are exported. This limitation exists because both Excel and Power BI require a fixed schema.
> - A dimension hierarchy isn't date-effective. Therefore, any change to a dimension hierarchy is immediately saved to the record, and you can't compare the before date and after date.

Dimension hierarchy type

When you create a new dimension hierarchy, you must select a hierarchy type. Go to **Cost accounting** > **Dimensions** > **Dimension hierarchies**. Select **New**, and choose a dimension hierarchy type. You can select either **Dimension categorization hierarchy** or **Dimension classification hierarchy**.

Dimension categorization hierarchy

Use the **Dimension categorization hierarchy** type for reporting. It supports only the cost element dimensions. When you select this type, the following rules apply:

- You can associate a dimension member more than once in the hierarchy structure.
- You can put a cost element dimension member in different nodes by assigning a cost behavior to the leaf node.

Dimension classification hierarchy

Use the **Dimension classification hierarchy** type for defining rules and reporting purposes. It supports all dimensions, such as cost objects, cost elements, and statistical dimensions. When you select this type, you can associate a dimension member only once in the hierarchy structure.

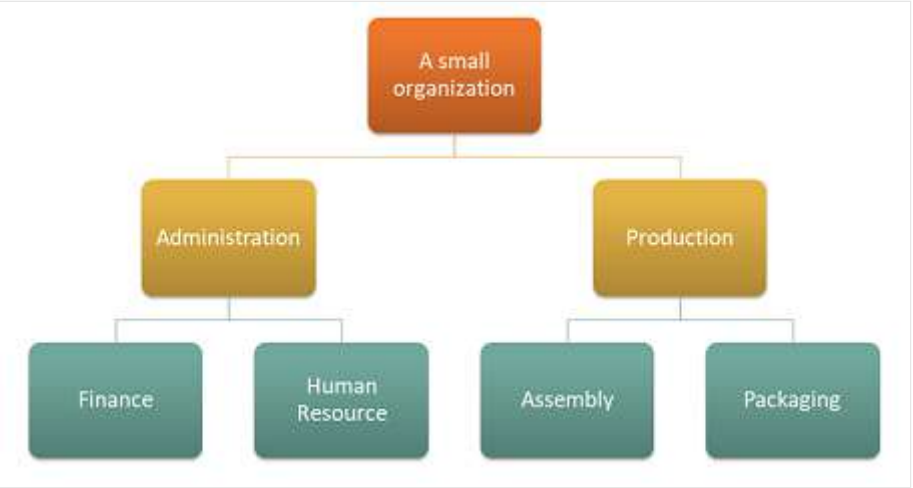
Create and maintain a dimension hierarchy

Create a dimension hierarchy as a tree structure that has node and leaf node relationships.

- A node can have 1:*n* subnodes.
- A node can't have both subnodes and leaf nodes assigned to it.
- You can assign a leaf node only at the lowest level in the hierarchy.

Example

A small company has the following organization structure, where Finance and Human resources are departments under Admin, and Assembly and Packaging are departments under Production.



A cost object dimension represents all the cost centers in the organization.

- Cost object dimension
  - Cost centers

Set up the cost object dimension that represents all the cost centers as shown in the following table.

Expand table

Cost centers	Description
CC001	HR
CC002	Finance
CC003	Tax
CC007	AR/AP
CC005	Assembly
CC006	Packaging

A cost element dimension represents all the cost elements in the organization.

- Cost element dimension
  - Cost elements

Set up the cost element dimension that represents all the cost elements as shown in the following table.

Expand table

Cost elements	Description
10001	Electricity
10010	Cleaning
10011	Heating
40001	COGS

Set up a dimension hierarchy that meets the organizational reporting requirements as shown in the following table.

Dimension hierarchy details

Expand table

Dimension hierarchy name	Dimension	Dimension hierarchy type name	Access list hierarchy
Organization	Cost centers	Dimension classification hierarchy	No

Set up the dimension hierarchy for reporting as shown in the following table.

Dimension member ranges

Expand table

Nodes	From dimension member	To dimension member
Organization		
Admin		
Finance	CC002	CC003
	CC007	CC007
HR	CC001	CC001
Production		
Packaging	CC005	CC005
Assembly	CC006	CC006

Set up a dimension hierarchy that meets the policy requirement as shown in the following table.

Dimension hierarchy details

Expand table

Dimension hierarchy name	Dimension	Dimension hierarchy type name
Cost behavior	Cost elements	Dimension classification hierarchy

Set up the dimension hierarchy for the policy as shown in the following table.

Dimension member ranges

Expand table

Nodes	From dimension member	To dimension member
Cost behavior		
Fixed cost	10001	10011
Variable cost	40001	40010

> **Note**
>
> Under **Dimension member ranges**, a node can contain one-to-many dimension member ranges. You can insert dimension member IDs that don't yet exist as dimension members. This approach makes the hierarchy resilient for the future.

Copy a hierarchy

You can copy a current dimension hierarchy as the starting point for a new dimension hierarchy. This approach is useful if you want to compare the previous dimension hierarchy to the new dimension hierarchy.

Rearrange nodes in a hierarchy

You can move a node up and down within its current level in the structure. Use this method to rearrange the order of nodes for reporting in the **Cost control** workspace.

To move a node to a new location in the hierarchy, select the target node. Use one of two options to move a node:

- **Move below** – Move the selected node from its current position in the hierarchy, and insert it **under** the selected target node.

- **Move after** – Move the selected node from its current position in the hierarchy, and insert it **after** the selected target node at its level of the hierarchy.

> ⓘ **Note**
>
> The order of the nodes isn't maintained when you export data to Excel or Power BI, because those tools use an alphanumeric sort order by default. You should manually rearrange the order.

Define dimension hierarchies for reporting

Dimension hierarchies are important for reporting. They let you define the specific structure that fits into the individual organization. The aggregations that you perform at the node level of the dimension hierarchy let stakeholders at any level of the organization see data at any level.

You can find dimension hierarchies in the following reporting tools. This approach helps guarantee consistency in the reporting structure.

- **Cost control** workspace (Client):
  - Controlled by configuration.
- **Cost control** workspace (Mobile application):
  - Controlled by configuration.
- Excel
  - Provides the option to select specific dimension hierarchies per export definition:
    - One cost element dimension hierarchy (mandatory)
    - One cost object dimension hierarchy (optional)
    - One statistical dimension hierarchy (optional)
- Power BI:
  - All dimension hierarchies are available.

If you create reports by using Excel or Power BI, you can export only the first 15 levels of the dimension hierarchies. This limitation exists because Excel and Power BI require a fixed schema. If a hierarchy has more than 15 levels, the extra levels aren't exported. The normalized table contains a record for each dimension member in the hierarchy. Therefore, automated aggregation occurs. This behavior helps guarantee that the balances at any of the 15 available levels in the hierarchy are still correct.

The following example shows what a dimension hierarchy might look like in the reporting structure.

Expand table

Cost object dimension hierarchy – Level 1	Cost object dimension hierarchy – Level 2	Cost object dimension hierarchy – Level 3	Cost object dimension hierarchy – Level 4	Cost object dimension hierarchy – Level 15
Organization	Admin	Finance	CC002	
Organization	Admin	Finance	CC003	
Organization	Admin	Finance	CC007	
Organization	Admin	HR	CC001	
Organization	Production	Packaging	CC005	
Organization	Production	Assembly	CC006	

Update the dimension hierarchies that you use for reporting

Over time, you need to update the dimension hierarchies that you use in the previously mentioned reporting tools. You can update dimension hierarchies by refreshing the client.

- **Cost control** workspace (Client)
- **Cost control** workspace (Mobile application)

Updates to dimension hierarchies are picked up every 24 hours by a precached job. After the exported data is updated, the updated dimension hierarchies are available in the following tools:

- Excel
- Power BI

> **Note**
>
> To manually trigger an update of the dimension hierarchy cache, create a new export to Excel for the dimension hierarchy or hierarchies that you need to update.

Define dimension hierarchies for cost policies

Cost accounting consists of multiple policies where you define detailed rules. You must define one or more dimension hierarchies for the following policies:

- Cost behavior
- Cost distribution
- Cost allocation
- Cost rollup

Dimension hierarchies make it easy to create rules. To avoid having to create rules for every dimension member, you can take advantage of the aggregations of dimension members that are provided by dimension hierarchy levels. If you have overlapping rules, you must define specific rules that the system will consider when it does the overhead calculation.

Example: Define a cost behavior policy

Create a new cost behavior policy and assign the appropriate dimension hierarchies to the policy, as shown in the following table.

Cost behavior policy

Expand table

Policy name	Cost element dimension hierarchy	Cost object dimension hierarchy	Accounting currency
Cost behavior	Cost behavior	Organization	USD

Rules

Expand table

Cost element dimension hierarchy node	Cost object dimension hierarchy node	Fixed percentage	Fixed amount	Valid from	Valid to
Fixed cost	Organization	100.00	0.00	1/1/2025	Never
10001	Organization	0.00	150.00	1/1/2025	Never
10001 (*)	Finance	50.00	1/1/2025	Never	
Cost behavior or Variable cost (**)	Organization	0.00	0.00	1/1/2025	Never

* The variable cost node isn't required. If a cost isn't classified as a fixed cost, it must be a variable cost.

** A detailed rule is created for the combination of cost element member 10001 and all cost object members that are aggregated under the Finance hierarchy level (CC002, CC003, CC007).

The preceding rules show the flexibility that dimension hierarchies provide. By defining high-level rules, you can help minimize maintenance. You can then define detailed rules to fit into a specific business objective.

If the dimension hierarchies that are used in rules are updated, the system automatically brings the updates forward.

If a level of granularity in the rules is no longer required, you can expire the rule.

For example, a specific cost behavior rule for the Finance cost object dimension hierarchy node is no longer required. In this case, select **Expire rule** to expire the rule.

Expand table

Cost element dimension hierarchy node	Cost object dimension hierarchy node	Fixed percentage	Fixed amount	Valid from	Valid to
Fixed cost	Organization	100.00	0.00	1/1/2025	Never
10001	Organization	0.00	150.00	1/1/2025	Never
10001	Finance		50.00	1/1/2025	20/1/2025
Cost behavior or Variable cost	Organization	0.00	0.00	1/1/2025	Never

Any overhead calculation that runs after January 20, 2025, no longer considers this rule.

> **Note**
>
> The **Valid from** and **Valid to** fields are date-effective and time-effective. You can expire the rule and run a new overhead calculation on the same day.

Define dimension hierarchies for security setup

Cost accounting data should be made available to all managers who are responsible for a reporting unit. In Cost accounting terminology, a reporting unit is represented as a cost object or a set of cost objects.

Potentially, all managers are able to access highly sensitive business data, such revenues and margins. Therefore, it's important that you set up security, so that managers see only the data that's relevant to them. To help control data security, you define dimension hierarchies.

- The use of dimension hierarchies applies only when the dimension value that is selected in the dimension hierarchy reference is a cost object dimension.
- You can enable only one dimension hierarchy per cost object dimension in the access list hierarchy.

Dimension hierarchy details

Expand table

Dimension hierarchy name	Dimension	Dimension hierarchy type name	Access list hierarchy
Organization	Cost centers	Dimension classification hierarchy	**Yes**

A new **Users** FastTab is available in the hierarchy designer. Here, you can insert one or more user IDs at each node in the hierarchy.

Users and dimension member ranges

Expand table

Nodes	User ID	From dimension member	To dimension member
Organization	Benjamin, Claire		
Admin	April		
Finance	Alicia	CC002	CC003
		CC007	CC007
HR	Arnie	CC001	CC001
Production	David		
Packaging	Ellen	CC005	CC005
Assembly	Chris	CC006	CC006

> **Note**
>
> Cost accountants should be assigned to the top level of the hierarchy, so that they can see all entries in Cost accounting.

To enable the access list hierarchy and its security settings, go to **Cost accounting** > **Setup** > **Parameters** > **General**. Select the **Enable view access for cost object dimension members** parameter.

The settings for the access list hierarchy control the data that is shown in the following areas:

- **Cost control** workspace (Client):
  - Data in pages that are used to drill through scenarios
- **Cost control** workspace (Mobile application):
  - Balances in cards
- Power BI:
  - Data that is shown in Power BI visualizations
  - Data Power BI visualizations that are embedded in the Dynamics 365 Finance client

> **Note**
>
> - Before the access list hierarchy can affect data in Power BI, access list hierarchy and row-level security in Power BI must be paired. For more information, see Set up security for Cost accounting content pack.
> - The access list hierarchy doesn't help secure the export of data to Excel. Therefore, that reporting tool should be used only by cost accountants and managers who must have full access to view the data.

Last updated on 05/27/2026

Create an organization report hierarchy

Use this procedure to create a report hierarchy for organization reporting. The purpose of this recording is to guide you through the dimension hierarchy so that you can continue until the whole organization reporting structure is created. This recording uses the USP2 demo data company.

1. Go to **Cost accounting** > **Dimensions** > **Dimension hierarchies**.
2. Select **New**.
3. In the **HierarchyTypeComboBox** field, select **Dimension classification hierarchy**.
    - The **Dimension classification hierarchy** type defines rules and supports reporting for all dimensions, such as cost objects, cost elements, and statistical dimensions.
4. Select **Create**.
5. In the **Dimension hierarchy name** field, enter **Oganization USP2**.
6. In the **Dimension** field, enter or select a value.
    - Select **Cost centers**.
7. Select **Save**.
8. Select **View hierarchy**.
9. Select **New**.
10. In the **Node name** field, enter **CEO**.
11. Select **Save**.
12. Select **New**.
13. In the **Node name** field, enter **CEO cost centers**.
14. Select **Save**.
15. Select **New**.
16. In the **Node name** field, enter **Region East**.
17. Select **Save**.
18. Select **New**.
19. In the list, select the row.
20. In the **From dimension member** field, enter or select a value.
    - Select the dimension member that corresponds to the node.
21. Select **Save**.

22. In the tree, select **Oganization USP2\CEO\CEO cost centers**.
23. Select **New**.
24. In the **Node name** field, enter **Region West**.
25. Select **Save**.
26. Select **New**.
27. In the list, select the row.
28. In the **From dimension member** field, enter or select a value.

    - Select the dimension member that corresponds to the node.

29. Select **Save**.
30. In the tree, select **Oganization USP2\CEO**.
31. Select **New**.
32. In the **Node name** field, enter **CFO cost centers**.
33. Select **Save**.
34. Select **New**.
35. In the **Node name** field, enter **Marketing campa**.
36. In the **Node name** field, enter **Marketing campaign**.
37. Select **Save**.
38. Select **New**.
39. In the list, select the row.
40. In the **From dimension member** field, enter or select a value.

    - Select the dimension member that corresponds to the node.

41. Select **Save**.
42. In the tree, select **Organization USP2\CEO\CFO cost centers**.
43. Select **New**.
44. In the **Node name** field, enter **Trade shows**.
45. Select **Save**.
46. Select **New**.
47. In the list, select the row.
48. In the **From dimension member** field, enter or select a value.

    - Select the dimension member that corresponds to the node.

49. Click **Save**.
50. In the tree, select `Oganization USP2\CEO`.
51. In the **Node name** field, type `CIO cost centers`.
52. Click **Save**.

53. Click **New**.
54. In the **Node name** field, type `Call centers`.
55. Click **Save**.
56. Click **New**.
57. In the list, mark the selected row.
58. In the **From dimension member** field, enter or select a value.
    - Select the dimension member that corresponds to the node.
59. Click **Save**.

Last updated on 08/11/2026

Access rights for cost object controllers

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Cost control** workspace is a central point where managers can view the performance of their cost objects. This workspace lets managers consume Cost accounting data even though they aren't cost accountants. For security reasons, managers should only see the Cost accounting data that relates to the specific cost objects that they're responsible for.

Cost accounting has four unique roles.

Expand table

Role name	License
Cost accounting manager	Activity
Cost accountant	Operations
Cost accountant clerk	Operations
Cost object controller	Team members

This article explains how to assign the **Cost object controller** role to a manager.

When you assign the **Cost object controller** role to a manager, the manager can perform the following tasks:

- Access the **Cost control** workspace (in the client).
  - Drill through and have view access to the pages that support the drill-through experience.
- Access the **Cost control** workspace (in the mobile application).

> **Note**

The **Cost object controller** role doesn't control which cost objects the user can access and view data for. Row-level security is provided via dimension hierarchies and the Access list hierarchy.

Grant access rights

The following example shows what a dimension hierarchy can look like.

Dimension hierarchy details

Expand table

Dimension hierarchy name	Dimension	Dimension hierarchy type name	Access list hierarchy
Organization	Cost centers	Dimension classification hierarchy	**Yes**

Use the **Users** FastTab in the hierarchy designer to add one or more user IDs on each node.

Expand table

Nodes	Users	From dimension member	To dimension member
Organization	Benjamin, Claire		
Admin	April		
Finance	Alicia	CC002	CC003
		CC007	CC007
HR	Arnie	CC001	CC001
Production	David		
Packaging	Ellen	CC005	CC005
Assembly	Chris	CC006	CC006

Note

Assign cost accountants to the top level of the hierarchy, so they can see all entries in Cost accounting.

Before you can apply the Access list hierarchy and its security settings, set the **Enable view access for cost object dimension members** option to **Yes** on the **General** tab of the **Cost accounting parameters** page (**Cost accounting** > **Setup** > **Parameters**).

Use the settings for the Access list hierarchy to control the data that is shown in the following areas:

- **Cost control** workspace (in the client):
  - Data on the pages that are used for drill-through
- **Cost control** workspace (in the mobile application):
  - Balances in cards
- Microsoft Power BI:
  - Data that is shown in Power BI visualizations
  - Data in Power BI visualizations that are embedded in the Dynamics 365 Finance client

> **Important**
>
> - Before the Access list hierarchy can affect data in Power BI, you must pair the Access list hierarchy with row-level security in Power BI. For more information, see Set up security for Cost accounting content pack.
> - This article shows the prerequisites that must be in place before you can use the **Cost control** workspace.

Additional resources

- Cost control workspace
- Dimension hierarchy
- Set up security for Cost accounting content pack

Last updated on 06/23/2026

Configure access rights for a cost object controller

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use this procedure to configure access rights for a cost object controller. This recording uses the USP2 demo data company.

Assign the cost object controller role

1. Go to **System administration** > **Users** > **Users**.
2. Use the Quick Filter to find records. For example, filter on the **User name** field with a value of 'alicia'.
3. In the list, click the link in the selected row.
4. Click **Assign roles**.
5. In the list, find and select the desired record.
6. Click **OK**.

Enable access list security

1. Go to **Cost accounting** > **Dimensions** > **Dimension hierarchies**.
2. In the list, find and select the desired record.

    - Select **Organization**.

3. Click **Edit**.
4. Select **Yes** in the **Access list hierarchy** field.
5. Click **View hierarchy**.

Assign access rights to user

1. Click **New**.
2. In the list, mark the selected row.
3. In the **User ID** field, enter or select a value.

- Select **Admin**.

4. In the tree, select 'Organization\CEO\CFO\FIM'.
5. Click **New**.
6. In the list, mark the selected row.
7. In the **User ID** field, enter or select a value.

    - Select Alicia.

8. Click **Save**.

Enable access rights in Cost accounting

1. Go to **Cost accounting** > **Setup** > **Parameters**.
2. Click the **General** tab.
3. Select **Yes** in the **Enable view access for cost object dimension members** field.
4. Click **Save**.
5. Close the page.
6. Go to **Cost accounting** > **Setup** > **Cost control workspace configuration**.
7. Click **Edit**.
8. Select **Yes** in the **Published** field.

    - If you select **Yes**, a user who is assigned one of the following four roles can see the reports in the **Cost control** workspace: cost accounting manager, cost accountant, cost accountant clerk, and cost object controller. If you select No, only a user who is assigned one of the following roles can see the reports: cost accounting manager, cost accountant, and cost accountant clerk.

9. Click **Save**.

Last updated on 10/02/2025

Define cost control units

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to define cost control units in the cost accounting ledger. You select the cost object dimension for which you want to control costs in the ledger. The cost object dimension members are transformed into cost objects of the cost control units. The demo data company used to create this procedure is USP2.

1. Go to **Cost accounting** > **Ledger setup** > **Cost accounting ledgers**.
2. In the list, find and select the desired record.
3. Click **Control units**.
4. Click **New**.
5. In the **Name** field, type a value.
6. In the **Dimension** field, enter or select a value.
7. Click **Save**.
8. Click **New**.
9. In the **Name** field, type a value.
10. In the **Dimension** field, enter or select a value.
11. Click **Save**.

Last updated on 10/02/2025

Create a cost accounting ledger

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

A cost accounting ledger represents the overall reporting unit. It is defined by a cost element dimension, statistical dimension, fiscal calendar, and currency. It is agnostic to the concept of legal entities. A legal entity and its data can be associated with many cost accounting ledgers. This recording uses the USP2 demo data company.

1. Go to **Cost accounting** > **Ledger setup** > **Cost accounting ledgers**.
2. Click **New**.
3. In the **Name** field, type a value.
4. In the **Cost element dimension** field, enter or select a value.
5. In the **Fiscal calendar** field, enter or select a value.
6. In the **Accounting currency** field, enter or select a value.
7. In the **Exchange rate type** field, enter or select a value.
8. In the **Statistical dimension** field, enter or select a value.
9. Click **Save**.

Last updated on 10/02/2025

Provider templates for statistical dimension members and measure providers

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use a statistical dimension and its members to register and control nonmonetary entries in Cost accounting. Use statistical dimension members for two purposes:

- As an allocation base in policies such as cost distribution or cost allocation
- For reporting of nonmonetary consumption

Statistical dimension

A statistical dimension has a unique name and a set of unique dimension members. Assign the statistical dimension to a Cost accounting ledger ID. This relationship ties all corresponding statistical dimension members to the Cost accounting ledger. Therefore, you create all statistical entries in the context of the Cost accounting ledger.

> **Note**
>
> You can assign a statistical dimension to more than one Cost accounting ledger.

Here's an example of a statistical dimension.

Expand table

Name	Data connector for dimension members
Shared Statistical elements	Imported dimension members

Here's an example of a statistical dimension that you assigned to a Cost accounting ledger.

Expand table

Name	Accounting currency	Exchange rate type	Fiscal calendar	Cost element dimension	Statistical dimension
Managerial accounting	USD	Constant currency	Fiscal period	Shared Cost elements	Shared Statistical elements

Statistical dimension members

A statistical dimension member represents an entity that you want to register nonmonetary measures for. Use these measures as an allocation base or just to report nonmonetary values.

Create statistical dimension members manually. Alternatively, import them from a file by using the Data management import/export tool.

A statistical dimension member automatically becomes a predefined allocation base. Use it as an allocation base in policies or as input in other types of allocation bases.

Here are some examples of typical statistical dimension members.

Expand table

Statistical dimension name	Statistical elements	Description	Unit
Shared Statistical elements	FTE	Full time employees	Ea.
Shared Statistical elements	Electricity	Electricity consumption	kWh
Shared Statistical elements	Pack CC	Packaging Cost center	Hrs.

Statistical measure provider template

Statistical measures can come from many kinds of sources. Dynamics 365 Finance is a great source for extracting statistical measures. Use a statistical measure provider template to easily configure the statistical measures you want to extract.

The definition of a statistical measure provider template is generic and can be reused in multiple statistical dimension members.

> **Note**
>
> You can use all tables that contain financial dimensions as sources for statistical measures.

Example: Count of employees per cost center

The count of employees per cost center is a statistical measure that you can use for various purposes that provide managerial insight:

- A statistical reporting measure by cost center
- An allocation base for various types of expenses
- Internal cost rates by cost center:
  - Cost by employee
  - Revenue by employee

The HcmEmployment table holds a list of all employees in the instance. This table is a global table. Therefore, the records aren't related to a specific data area ID.

Here's an example of employees in the HcmEmployment table.

Expand table

Name	Cost center	Description	Worker type
Employee 1	CC001	HR	Employee
Employee 2	CC002	FI	Employee
Employee 3	CC002	FI	Employee
Employee 4	CC003	IT	Employee
Employee 5	CC003	IT	Employee
Employee 6	CC002	FI	Contractor

When you create a **Statistical measure provider template** record, decide which function to use:

- **Count** – A count of records per cost object is transferred.
- **Sum** – A sum for records per cost object is transferred. (The **Sum** field and **Date** field are required.)

Using the Count function

For example, set up a statistical measure provider template as follows.

Expand table

Name	Function	Source table	Sum field	Date field
FTEs	Count	HcmEmployment	Not applicable	Not applicable

You can also add one or more ranges to narrow the measures from the source table.

In this example, if you want a count of all full-time employees (FTEs), add a range in the **Worker type** field. In the **Criteria** field, select **Employee** to limit the output range as follows.

Ranges

Expand table

Source table	Field	Criteria
HcmEmployment	Worker type	Employee

Before you can get statistical measures into Cost accounting, you must establish the relation between the statistical measure provider template and the statistical dimension member. Create this relation per Cost accounting ledger and version. The relation consists of a data connector and a data provider. You can have several data connectors and data providers per statistical dimension member.

> **Note**
>
> In this example, you create a relation only for the **Actual version**.

Go to **Cost accounting ledger** > **Actual version** > **Manage** > **Statistical measures** to establish the relation. For this scenario, select the **Dynamics 365 Finance – Statistical measures** data connector, because you want to extract data from Finance.

Data source

Expand table

Name	Data connector	Statistical dimension member
FTEs D365FO	Dynamics 365 Finance – Statistical measures	FTEs

Data provider configuration

Expand table

Statistical template name
FTEs

After the source data for the statistical measure is processed, Cost accounting creates the following statistical entries.

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Year	Period	Version	Data connector source entries
00001	Statistical entry transfer journal	Fiscal	2017	Period 1	CA ledger USMF	FTEs

Statistical entry transfer journal entries

Expand table

Accounting date	Magnitude	Statistical element	Description	Cost center
31-01-2017	1.00	FTEs	Full time employees	CC001
31-01-2017	2.00	FTEs	Full time employees	CC002
31-01-2017	2.00	FTEs	Full time employees	CC003

Statistical entries

Expand table

Cost object	Description	Accounting date	Statistical dimension member	Description	Magnitude
CC001	HR	31-01-2017	FTEs	Full time employees	1.00
CC002	FI	31-01-2017	FTEs	Full time employees	2.00
CC003	IT	31-01-2017	FTEs	Full time employees	2.00

If you assign the FTEs predefined dimension member allocation basis as an allocation base in a cost distribution rule, the cost distributes by using the following allocation factor.

Expand table

Cost object	Description	Magnitude	Allocation factor
CC001	HR	1.00	(1/5) × Amount
CC002	FI	2.00	(2/5) × Amount
CC003	IT	2.00	(2/5) × Amount

Using the Sum function

A production cost center, CC010 (Packaging), is responsible for packaging the products before they're shipped to customers. Add the direct labor cost to the products through the bill of materials (BOM) and route. You must also allocate the indirect cost of running the cost center to the produced products. Often, the best statistical measure for such an allocation is the number of registered production hours per product within the given period.

The ProdRouteTrans table holds all production labor transactions per legal entity DataAreadID.

Here's an example of the ProdRouteTrans table.

Expand table

Reference	Number	Operation	Type	Time	Physical date	Product group (Financial dimension)	Legal entity
Production order	P0001	Packaging	Time	8.00	01-01-2017	Orange juice B2B	USMF
Production order	P0001	Packaging	Time	8.00	02-01-2017	Orange juice B2B	USMF
Production order	P0002	Packaging	Time	4.00	03-01-2017	Orange juice Consumer	USMF
Production order	P0003	Packaging	Time	4.00	03-01-2017	Orange juice Consumer	USMF
Production order	P0004	Reconst.	Time	10.00	03-01-2017	Orange juice Consumer	USMF

When you create a **Statistical measure provider template** record, decide which function to use:

- **Count** – A count of records per cost object is transferred.
- **Sum** – A sum for records per cost object is transferred. (The **Sum** field and **Date** field are required.)

The statistical measure provider template can be set up as follows.

Expand table

Name	Function	Source table	Sum field	Date field
Pack CC	Sum	ProdRouteTrans	Hours	Physical date

You can also add ranges to narrow the measures from the source table.

In this example, if you want only the sum of hours that relate to the CC010 Packaging cost center, add a range in the **Operation** field. In the **Criteria** field, select **Packaging** to limit the output range.

Ranges

Expand table

Source table	Field	Criteria
ProdRouteTrans	Operation	Packaging

Before you can get statistical measures into Cost accounting, you must establish the relation between the statistical measure provider template and the statistical dimension member. Create this relation per Cost accounting ledger and version. The relation consists of a data connector and a data provider. You can have several data connectors and data providers per statistical dimension member.

> **Note**
>
> In this example, you create a relation only for the **Actual version**.

Go to **Cost accounting ledger** > **Actual version** > **Manage** > **Statistical measures** to establish the relation. For this scenario, select the **Dynamics 365 Finance – Statistical measures** data connector, because you want to extract data from Finance.

Data source

Expand table

Name	Data connector	Statistical dimension member
Pack CC D365FO	Dynamics 365 Finance – Statistical measures	Pack CC

The system recognizes that ProdRouteTrans is a table where each record belongs to a separate legal entity. Therefore, you're asked to select the legal entity that transactions should be imported from.

Data provider configuration

Expand table

Statistical template name	Legal entity
Pack CC	USMF

After the source data for the statistical measure is processed, the following statistical entries are created in Cost accounting.

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Year	Period	Version	Data connector source entries
00002	Statistical entry transfer journal	Fiscal	2017	Period 1	CA ledger USMF	Pack CC

Statistical entry transfer journal entries

Expand table

Accounting date	Magnitude	Statistical element	Description	Product group
31-01-2017	16.00	Pack CC	Packaging Cost center	Orange juice B2B
31-01-2017	8.00	Pack CC	Packaging Cost center	Orange juice Consumer

Statistical entries

Expand table

Cost object	Accounting date	Statistical dimension member	Description	Magnitude
Orange juice B2B	31-01-2017	Pack CC	Packaging Cost center	16.00
Orange juice Consumer	31-01-2017	Pack CC	Packaging Cost center	8.00

If you assign the Pack CC predefined dimension member allocation basis as an allocation base in a cost distribution rule, the system distributes the cost by using the following allocation factor.

Expand table

Cost object	Magnitude	Allocation factor
Orange juice B2B	16.00	(16 ÷ 24) × Amount
Orange juice Consumer	8.00	(8 ÷ 24) × Amount

Imported statistical measures

You can import statistical measures into Cost accounting by using the Data management import/export tool.

The data entity that is used for the import is named **Imported statistical measures**.

> **Note**
>
> This data entity is designed to allow a maximum of five unique dimension values per entry.

You can record the consumption of electricity in Microsoft Excel by using the predefined format of the data entity. Here's an example.

Expand table

Accounting date	Dimension member name1	Dimension member name2	Dimension member name5	Magnitude	Source identifier
31-01-2017				2,450.00	Electricity
31-01-2017				4,100.00	Electricity
31-01-2017	CC003			15,000.00	Electricity

When you've imported your data via Data management, the data is stored in a Cost accounting staging table. Therefore, the imported data can be used in multiple Cost accounting ledgers. A reload of data isn't required.

To import the data, go to **Imported data** > **Data entity** > **Imported statistical measures**.

Expand table

Source identifier	Accounting date	Magnitude	Dimension member name1	Dimension member name2	Dimension member name5
Electricity	31-01-2017	2,450.00	CC001		
Electricity	31-01-2017	4,100.00	CC002		
Electricity	31-01-2017	15,000.00	CC003		

Before you can get statistical measures into Cost accounting, you must establish the relation between the source identifier and the statistical dimension member. This relation is created per Cost accounting ledger and version. The relation consists of a data connector and a data provider. You can have several data connectors and data providers per statistical dimension member.

> **Note**
>
> In this example, you create a relation only for the **Actual version**.

Go to **Cost accounting ledger** > **Actual version** > **Manage** > **Statistical measures** to establish the relation. For this scenario, select the **Imported statistical measures** data connector, because you imported data from a third-party system into Cost accounting via Excel.

Data source

Expand table

Name	Data connector	Statistical dimension member
Electricity	Imported statistical measures	Electricity

Data provider configuration

Expand table

Import source identifier	Function	Dimension1	Dimension2	Dimension5
Electricity	Sum	Cost centers		

> **Note**
>
> When you define the data provider configuration, you must specify which cost object dimensions to match against the imported transactions. After the system processes the source data for the statistical measure, it creates the following statistical entries in Cost accounting.

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Year	Perid	Version	Data connector source entries
00002	Statistical entry transfer journal	Fiscal	2017	Period 1	CA ledger USMF	Electricity

Statistical entry transfer journal entries

Expand table

Accounting date	Magnitude	Cost element	Description	Cost center
31-01-2017	2,450.00	Electricity	Electricity consumption	CC001
31-01-2017	4,100.00	Electricity	Electricity consumption	CC002
31-01-2017	15,000.00	Electricity	Electricity consumption	CC003

Statistical entries

Expand table

Cost object	Description	Accounting date	Statistical dimension member	Description	Magnitude
CC001	HR	31-01-2017	Electricity	Electricity consumption	2,450.00
CC002	FI	31-01-2017	Electricity	Electricity	4,100.00

Cost object	Description	Accounting date	Statistical dimension member	Description	Magnitude
				consumption	
CC003	IT	31-01-2017	Electricity	Electricity consumption	15,000.00

If you assign the Electricity predefined dimension member allocation basis as an allocation base in a cost distribution rule, the system distributes the cost by using the following allocation factor.

Expand table

Cost object	Description	Magnitude	Allocation factor
CC001	HR	2,450.00	(2,450 ÷ 21,550) × Amount
CC002	FI	4,100.00	(4,100 ÷ 21,550) × Amount
CC003	IT	15,000.00	(15,000 ÷ 21,550) × Amount

Additional resources

Allocation bases

Last updated on 05/27/2026

Allocation bases

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

An allocation base is the basis on which Cost accounting allocates overhead costs. An allocation base can be a quantity, such as machine hours that are used, kilowatt hours (kWh) that are consumed, or square footage that is occupied. Allocation bases are mostly used to assign overhead costs to inventory that is produced. For example, an IT department allocates its expenses according to the number of computers that each department uses.

Cost accounting has three types of allocation bases:

- Predefined dimension member allocation bases
- Hierarchy allocation bases
- Formula allocation bases

Predefined dimension member allocation bases

The system automatically creates predefined dimension member allocation bases when you create a dimension member of one of the following types:

- Statistical dimension members
- Cost element dimension members

> **Note**
>
> The predefined dimension member allocation bases that are based on a cost element dimension member consider only the values from the data source provider, such as the general ledger or budget.

Example 1: Use a cost element dimension member as the allocation base

This example shows how to create a cost allocation rule to allocate cost element 10002 (Employee insurance) to the balance that is recorded on cost element 10001 (Salaries). The allocation rule is based on the ratio of each department's salaries to total salaries.

In the general ledger, the chart of accounts is defined as follows.

Expand table

Chart of accounts	Main account	Description	Main account type
Shared	10001	Salaries	Expense
Shared	10002	Employee insurance	Expense

Define a cost element dimension, and configure the data connector. After you import the data, Cost accounting creates the following entries.

Cost element dimension members

Expand table

Cost element dimension name	Cost element	Description	Type
Cost elements	10001	Salaries	Primary
Cost elements	10002	Employee insurance	Primary

Predefined dimension member allocation bases

Expand table

Name	Description	Cost element dimension
10001	Salaries	Cost elements
10002	Employee insurance	Cost elements

In the general ledger, you post the following entries:

- The entries that show the Salaries main account come from the Payroll system and are posted to cost centers.
- You manually post the expense for employee insurance to a default cost center.

Expand table

Accounting date	Cost center	Description	Main account	Description	Amount in accounting currency
03-01-2017	CC001	HR	10001	Salaries	2,000.00
03-01-2017	CC002	FI	10001	Salaries	5,000.00
03-01-2017	CC003	IT	10001	Salaries	3,000.00
03-01-2017	CC099	Default cost center	10002	Employee insurance	1,000.00

After the general ledger source data is processed, Cost accounting creates the following entries.

Cost entries

Expand table

Cost object	Description	Cost element	Description	Cost behavior	Amount	Accounting date
CC001	HR	10001	Salaries	Unclassified	2,000.00	03-01-2017
CC002	FI	10001	Salaries	Unclassified	5,000.00	03-01-2017
CC003	IT	10001	Salaries	Unclassified	3,000.00	03-01-2017
CC099	Default cost center	10002	Employee insurance	Unclassified	1,000.00	03-01-2017

In this simplified example, you create a cost allocation rule to allocate cost element 10002 (Employee insurance) relative to the balance that is recorded on cost element 10001 (Salaries).

Cost distribution rule

Expand table

Cost object dimension hierarchy node	Cost element dimension hierarchy node	Cost behavior	Allocation base
CC099	10002	Unclassified	10001

Perform overhead calculation

After you apply cost element 10001 (Salaries) as the allocation base, the result of the overhead calculation is as follows.

Expand table

Cost object	Description	Magnitude	Allocation factor	Amount
CC001	HR	2,000	(2,000 ÷ 10,000) × 1,000.00	200.00
CC002	FI	5,000	(5,000 ÷ 10,000) × 1,000.00	500.00
CC003	IT	3,000	(3,000 ÷ 10,000) × 1,000.00	300.00

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Year	Period	Version
00001	Cost distribution calculation journal	Fiscal	2017	Period 1	Overhead calculation / 01-02-2017 11:51:00 PM / Ledger /2017 / Period 1

Cost object balance journal entries

Expand table

Accounting date	Cost object	Description	Cost element	Description	Cost behavior	Amount
31-01-2017	CC099	Default cost center	10002	Employee insurance	Unclassified	1,000.00

Cost entries

Expand table

Cost object	Description	Cost element	Description	Cost behavior	Amount	Accounting date
CC099	Default cost center	10002	Employee insurance	Unclassified	-1,000.00	31-01-2017
CC001	HR	10002	Employee insurance	Unclassified	200.00	31-01-2017

Cost object	Description	Cost element	Description	Cost behavior	Amount	Accounting date
CC002	FI	10002	Employee insurance	Unclassified	500.00	31-01-2017
CC099	IT	10002	Employee insurance	Unclassified	300.00	31-01-2017

Example 2: Use a statistical dimension member as the allocation base

You can use statistical dimension members as allocation bases to define policies or report nonmonetary consumption by cost objects. You can manually create statistical dimension members or import them from a file by using the Data management import/export tool.

Statistical dimension members

Expand table

Statistical dimension name	Statistical element	Description	Unit
Statistical elements	FTE	Full time employees	Ea
Statistical elements	Electricity	Electricity consumption	kWh

When a statistical dimension member is saved, a corresponding record is created in the predefined dimension member allocation bases.

Predefined dimension member allocation bases

Expand table

Name	Description	Statistical element dimension
FTE	Full time employees	Statistical elements
Electricity	Electricity consumption	Statistical elements

Statistical measures can come from various sources:

- Electricity consumption can be measured by meters that are installed in different areas of the company.

- Tables hold statistical measures. For example, the HcmEmployment table holds a list of all employees and the cost centers that they work for.

Expand table

Name	Cost center	Description	Worker type
Employee A	CC001	HR	Employee
Employee B	CC002	FI	Employee
Employee C	CC002	FI	Employee
Employee D	CC003	IT	Employee
Employee F	CC003	IT	Employee

> **Note**
>
> You can use all the tables that contain financial dimensions as sources for statistical measures.

Cost accounting supports a collection of statistical measures by using the following data connections:

- Data management import/export tool
- Statistical measures

To pull statistical measures from the system, a statistical measure provider template is required. For more information, see Statistical measure provider template. (Will add a link once this article is written.)

Statistical measure provider templates

Expand table

Name	Function	Source table	Sum field	Date field
FTE's	Count	HcmEmployment	Not applicable	Not applicable

After the system processes the statistical measure source data, it creates the following entries in Cost accounting.

Statistical entries

Expand table

Cost object	Description	Accounting date	Statistical dimension member	Description	Magnitude
CC001	HR	31-01-2017	FTE's	Full time employees	1.00
CC002	FI	31-01-2017	FTE's	Full time employees	2.00
CC003	IT	31-01-2017	FTE's	Full time employees	2.00

Here is an example of a cost distribution rule if the FTE's predefined dimension member allocation basis is assigned as the allocation base in it.

Expand table

Cost object	Description	Magnitude	Allocation factor
CC001	HR	1.00	(1/5) × Amount
CC002	FI	2.00	(2/5) × Amount
CC003	IT	2.00	(2/5) × Amount

Use the **Imported statistical measures** data entity to import statistical measures into Cost accounting. You can also use the Data management import/export tool. In Excel, the consumption of electricity is recorded as follows.

Expand table

Accounting date	Dimension member	Magnitude	Source identifier
31-01-2017	CC001	2,450.00	Electricity
31-01-2017	CC002	4,100.00	Electricity
31-01-2017	CC003	15,000.00	Electricity

After the system processes the statistical measure source data, it creates the following entries in Cost accounting.

Statistical entries

Expand table

Cost object	Name	Accounting date	Statistical dimension member	Description	Magnitude
CC001	HR	31-01-2017	Electricity	Electricity consumption	2,450.00
CC002	FI	31-01-2017	Electricity	Electricity consumption	4,100.00
CC003	IT	31-01-2017	Electricity	Electricity consumption	15,000.00

Here's an example of a cost distribution rule if you assign the **Electricity** predefined dimension member allocation basis as the allocation base.

Expand table

Cost object	Description	Magnitude	Allocation factor
CC001	HR	2,450.00	(2,450 ÷ 21,550) × Amount
CC002	FI	4,100.00	(4,100 ÷ 21,550) × Amount
CC003	IT	15,000.00	(15,000 ÷ 21,550) × Amount

Hierarchy allocation bases

Cost accountants can manually create hierarchy allocation bases by applying a cost object dimension hierarchy node to an existing allocation base. By using this method, you can limit the range of the original predefined dimension member allocation basis. You can use one predefined dimension member allocation basis to create several hierarchy allocation bases. You can maintain ranges in the cost object dimension hierarchy that you associate with the hierarchy allocation bases.

Example: Hierarchy allocation bases that are based on full-time employees in the organization

Here's an example of a cost object dimension hierarchy that you can create to describe a simplified organization.

Expand table

Hierarchy name	Node level 0	Node level 1	Node level 2	Dimension members
Organization	CEO	CFO	FICO	CC001
Organization	CEO	CFO	HR	CC002
Organization	CEO	CIO	IT	CC003

The FTE's predefined dimension member allocation basis that was created in the previous section holds the following entries.

Statistical entries

Expand table

Cost object	Description	Accounting date	Statistical dimension member	Description	Magnitude
CC001	HR	31-01-2017	FTE's	Full time employees	1.00
CC002	FI	31-01-2017	FTE's	Full time employees	2.00
CC003	IT	31-01-2017	FTE's	Full time employees	2.00

A cost must be distributed between cost centers that report to the organization's chief financial officer (CFO). It's acknowledged that the correct allocation ratio is the number of full-time employees (FTEs) by cost center.

Hierarchy allocation bases

Expand table

Name	Allocation base	Cost object dimension hierarchy	Cost object dimension hierarchy node
Number of FTEs in CFO	FTE's	Organization	CFO

A Preview function lets you validate the hierarchy allocation basis that you create, based on statistical entries in the system.

Allocation base details

Expand table

Cost object	Description	Magnitude
CC001	HR	1.00
CC002	FI	2.00

Here's an example of a cost distribution rule if the Number of FTEs in CFO hierarchy allocation basis is assigned as the allocation base in it.

Expand table

Cost object	Description	Magnitude	Allocation factor
CC001	HR	1.00	(1/3) × Amount
CC002	FI	2.00	(2/3) × Amount

Formula allocation bases

Formula allocation bases let you define advanced formulas to achieve the correct allocation basis. You can manually create formula allocation bases.

When you create a formula allocation base, you select which statistical dimension and cost element dimension should be the basis for the formula. You can use all allocation bases that come from the previously selected dimensions in a formula allocation base.

> **Note**
>
> You can use previously defined formula allocation bases to define a new formula allocation base.

In formula allocation base factors, you create an alias and associate it with either an allocation base or a constant. Use the aliases to define the formula.

To define your formula, use the following operators.

Expand table

Symbols	Text
( )	Parentheses

Symbols	Text
<	Smaller than
>	Larger than
+	Addition
–	Subtraction
*	Multiplication

Traditional **IF** statements aren't supported. However, you can create statements and validate whether they're true.

Expand table

Statement	Validation	Result
a > b	True	1
a > b	False	0

Example 1: A simple formula

Electricity bills often consist of two parts:

- A fixed fee for being connected to the grid
- A cost that is associated with consumption per kWh

The Electricity predefined dimension member allocation basis already defined holds these values.

Statistical entries

Expand table

Cost object	Name	Accounting date	Statistical dimension member	Description	Magnitude
CC001	HR	31-01-2017	Electricity	Electricity consumption	2,450.00
CC002	FI	31-01-2017	Electricity	Electricity consumption	4,100.00
CC003	IT	31-01-2017	Electricity	Electricity consumption	15,000.00

If you need to spread the fixed fee evenly over cost objects that consume electricity, you have two options for allocating the costs:

- Create a new predefined allocation base, Electricity fixed, and then apply a statistical measure of 1.00 for each cost object that consumed electricity.
- Create a formula allocation base, Electricity fixed, that takes advantage of the Electricity predefined allocation base that's already defined in the system. The benefit of this option is that you only need to load data into Cost accounting for one Electricity statistical dimension member.

Formula allocation base

Expand table

Name	Cost element dimension	Statistical dimension	Formula
Electricity fixed		Statistical elements	

Before you can fill the **Formula** field, you must specify the alias that should be used in the formula.

Formula allocation base factors

Expand table

Alias	Constant	Allocation base
a		Electricity
b	0.01	

Note that 0 (zero) isn't supported as a constant.

Formula allocation base

Expand table

Name	Cost element dimension	Statistical dimension	Formula
Electricity fixed		Statistical elements	a > b

A Preview function lets you validate the formula allocation base that you created, based on statistical entries in the system.

Allocation base details

Expand table

Cost object	Description	Formula	Magnitude
CC001	HR	2,450.00 > 0.01	1.00
CC002	FI	4,100.00 > 0.01	1.00
CC003	IT	15,000.00 > 0.01	1.00

Here's an example of a cost distribution rule if you assign the Electricity formula allocation base as the allocation base in it.

Cost object magnitude allocation factor

Expand table

Cost object	Name	Magnitude	Allocation factor
CC001	HR	1.00	(1/3) × Amount
CC002	FI	1.00	(1/3) × Amount
CC003	IT	1.00	(1/3) × Amount

Example 2: An advanced formula

For this example, the cost of electricity shouldn't just follow the actual electricity that is consumed in kWh. Management wants to incorporate an incentive for lowering electricity usage.

Expand table

Rule	Rate
a <= 10,000.00 kWh	0.75
a > 10,000.00 kWh	1.15

Create a new formula allocation base named **Electricity usage**.

Formula allocation base

Expand table

Name	Cost element dimension	Statistical dimension	Formula
Electricity usage		Statistical elements	

Before you can fill the **Formula** field, you must specify the alias that should be used in the formula.

Formula allocation base factors

Expand table

Alias	Constant	Allocation base
a		Electricity
b	10,000.00	
c	0.75	
d	1.15	

Formula allocation base

Expand table

Name	Cost element dimension	Statistical dimension	Formula
Electricity fixed		Statistical elements	((a > b) × ((b × c) + (a – b) × d)) + ((a <= b] × a × c)

A Preview function lets you validate the formula allocation base that you created, based on statistical entries in the system.

Allocation base details

Expand table

Cost object	Name	Formula	Magnitude
CC001	HR	((2,450.00 > 10.000.00) × ((10,000.00 × 0.75) + (2,450.00 – 10,000.00) × 1.15)) + ((2,450.00 <= 10,000.00) × 2,450.00 × 0.75)	1,837.50
CC002	FI	((4,100.00 > 10.000.00) × ((10,000.00 × 0.75) + (4,100.00 – 10,000.00) × 1.15)) + ((4,100.00 <= 10,000.00) × 4,100.00 × 0.75)	3,075.00

Cost object	Name	Formula	Magnitude
CC003	IT	((15,000.00 > 10.000.00) × ((10,000.00 × 0.75) + (15,000.00 – 10,000.00) × 1.15)) + ((15,000.00 <= 10,000.00) × 15,000.00 × 0.75)	1,3250.00

Here's a closer look at the formula for CC003 (IT):

((15,000.00 > 10,000.00) × ((10,000.00 × 0.75) + (15,000.00 – 10,000.00) × 1.15)) + ((15,000.00 <= 10,000.00) × 15,000.00 × 0.75) = 13,250.00

(1 × (7,500.00 + 5,000.00 × 1.15)) + (0 × 15,000.00 × 0.75)

1 × 7,500.00 + 5,750.00 + 0

Here's an example of a cost distribution rule if you assign the **Electricity fixed** formula allocation base as the allocation base.

Expand table

Cost object	Description	Magnitude	Allocation factor
CC001	HR	1,837.50	(1,837.50 ÷ 18,162.50) × Amount
CC002	FI	3,075.00	(3,075.00 ÷ 18,162.50) × Amount
CC003	IT	13,250.00	(13,250.00 ÷ 18,162.50) × Amount

Last updated on 05/28/2026

Create and assign a cost allocation policy to a cost control unit

Article • 05/09/2022

Use this procedure to create and assign a cost allocation policy and the corresponding rules to a cost control unit. This recording uses the USP2 demo data company.

Create a policy

1. Go to **Cost accounting** > **Policies** > **Cost allocation policies**.
2. Click **New**.
3. In the **Policy name** field, type a value.
4. In the **Cost object dimension hierarchy** field, enter or select a value.

    - Select Organization.

5. In the **Statistical dimension** field, enter or select a value.
6. Click **Save**.

Create allocation rules

1. Click **New**.
2. In the list, mark the selected row.
3. In the **Cost object dimension hierarchy node** field, enter or select a value.
4. In the **Cost behavior** field, select 'Total'.
5. In the **Allocation base** field, enter or select a value.
6. Click **New**.
7. In the list, mark the selected row.
8. In the **Cost object dimension hierarchy node** field, enter or select a value.
9. In the **Cost behavior** field, select 'Total'.
10. In the **Allocation base** field, enter or select a value.
11. Click **New**.
12. In the list, mark the selected row.
13. In the **Cost object dimension hierarchy node** field, enter or select a value.
14. In the **Cost behavior** field, select 'Total'.
15. In the **Allocation base** field, enter or select a value.

    - Continue until you've created all the rules.

16. Click **Save**.

Assign the policy to a cost control unit

1. Click **Policy assignments for cost control unit**.
2. Click **New**.
3. In the list, mark the selected row.
4. In the **Valid from accounting date** field, enter a date.

    - The rules are date-effective. A user or the system can expire the rules if a newer version is created.

5. In the **Cost control unit** field, enter or select a value.
6. Click **Save**.

Feedback

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Create and assign a cost behavior policy to a cost control unit

Article • 05/09/2022

Cost behavior is the classification of costs as either fixed or variable. A policy and the corresponding rules have to be assigned to a cost control unit for the policy to become effective. Use this procedure to create a policy and then assign the policy to a cost control unit.

Create a cost behavior hierarchy

1. Go to **Cost accounting** > **Dimensions** > **Dimension hierarchies**.
2. Click **New**.
3. Click **Create**.
4. In the **Dimension hierarchy name** field, type 'Cost behavior hierarchy'.
5. In the **Dimension** field, enter or select a value.

    - Select Cost elements.

6. Click **Save**.
7. Click **View hierarchy**.
8. Click **New**.
9. In the **Node name** field, type a value.

    - Enter Fixed cost.

10. In the tree, select 'Cost behavior hierarchy'.
11. Click **New**.
12. In the **Node name** field, type a value.

    - Enter Variable cost.

13. Click **Save**.
14. In the tree, select 'Cost behavior hierarchy\Fixed cost'.
15. Click **New**.
16. In the list, mark the selected row.
17. In the **From dimension member** field, enter or select a value.

    - The range of dimension members can contain gaps, but the members can't overlap.

18. In the **To dimension member** field, enter or select a value.
    - The range of dimension members can contain gaps, but the members can't overlap.
19. In the tree, select 'Cost behavior hierarchy\\Variable cost'.
20. Click **New**.
21. In the list, mark the selected row.
22. In the **From dimension member** field, enter or select a value.
    - The range of dimension members can contain gaps, but the members can't overlap.
23. In the **To dimension member** field, enter or select a value.
    - The range of dimension members can contain gaps, but the members can't overlap.
24. Click **Save**.

Create the policy and rules

1. Go to **Cost accounting** > **Policies** > **Cost behavior policies**.
2. Click **New**.
3. In the **Policy name** field, type a value.
4. In the **Cost element dimension hierarchy** field, enter or select a value.
    - Select the policy hierarchy that you just created.
5. In the **Cost object dimension hierarchy** field, enter or select a value.
    - Select Organization.
6. Click **Save**.
7. Click **New**.
8. In the list, mark the selected row.
9. In the **Cost element dimension hierarchy node** field, enter or select a value.
    - Expand the hierarchy to select Variable cost.
10. In the **Cost object dimension hierarchy node** field, enter or select a value.
    - By default, the variable percentage is 100 percent.
11. Click **Policy assignments for cost control unit**.

12. Click **New**.
13. In the list, mark the selected row.
14. In the **Valid from accounting date** field, enter a date.

    - The rules are date-effective, and a user or the system can expire a rule if a newer version is created.

15. In the **Cost control unit** field, enter or select a value.
16. Click **Save**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create and assign a cost distribution policy to a cost control unit

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Cost distribution rules are used to distribute costs that have been financially counted on a collective cost center. The cost accountant makes sure that the cost is distributed to the cost centers, based on the selected allocation base. A policy and the corresponding rules are assigned to a cost control unit. This task guide uses an example to show how to create a cost distribution policy and the corresponding rules.

Create a policy

1. Go to **Cost accounting** > **Policies** > **Cost distribution policies**.
2. Click **New**.
3. In the **Policy name** field, type a value.
4. In the **Description** field, type a value.
5. In the **Cost object dimension hierarchy** field, enter or select a value.
    - Select **Organization**.
6. In the **Cost element dimension hierarchy** field, enter or select a value.
    - Select **CDS P/L**.
7. In the **Statistical dimension** field, enter or select a value.
    - Select Statistical elements.
8. Click **Save**.

Create rules for the policy

1. Click **New**.
2. In the list, mark the selected row.
3. In the **Cost object dimension hierarchy node** field, enter or select a value.

- Expand the hierarchy to select 094.

4. In the **Cost element dimension hierarchy node** field, enter or select a value.
    - Select **Other operating expenses** and then select 605110 Cleaning.
5. In the **Cost behavior** field, select an option.
    - Select **Fixed cost**.
6. In the **Allocation base** field, enter or select a value.
7. Click **New**.
8. In the list, mark the selected row.
9. In the **Cost object dimension hierarchy node** field, enter or select a value.
    - Expand the hierarchy to select 094.
10. In the **Cost element dimension hierarchy node** field, enter or select a value.
    - Select **Other operating expenses** and then select 605150 Rent.
11. In the **Cost behavior** field, select an option.
    - Select **Fixed cost**.
12. In the **Allocation base** field, enter or select a value.
13. Click **Save**.

Assign rules to a cost control unit

1. Click **Policy assignments** for cost control unit.
2. Click **New**.
3. In the list, mark the selected row.
4. In the **Valid from accounting date** field, enter a date.
    - Select September 1 in the valid fiscal year.
5. In the **Cost control unit** field, enter or select a value.
6. Click **Save**.

Last updated on 10/02/2025

Create a cost rollup policy

Article • 03/28/2023

This procedure shows how to create a cost rollup policy and create rules for the policy. The demo data used to create this procedure is USP2.

Create a policy

1. Go to **Cost accounting** > **Policies** > **Cost rollup policies**.
2. Click **New**.
3. In the **Policy name** field, type a value.
4. In the **Description** field, type a value.
5. In the **Cost object dimension hierarchy** field, enter or select a value.
    - Select Cost rollup CC.
6. In the **Cost element dimension hierarchy** field, enter or select a value.
    - Select Cost rollup CC.
7. Click **Save**.

Create rules for the cost rollup policy

1. Click **New**.
2. In the list, mark the selected row.
3. In the **Cost object dimension hierarchy node** field, enter or select a value.
    - Select 007.
4. In the **Cost element dimension hierarchy node** field, enter or select a value.
    - Select Cost rollup CE.
5. In the **Secondary cost element** field, enter or select a value.
    - For this example, map the secondary cost element CC-007 to the cost center.
6. Click **New**.
7. In the list, mark the selected row.
8. In the **Cost object dimension hierarchy node** field, enter or select a value.
    - Select 008.

9. In the **Cost element dimension hierarchy node** field, enter or select a value.
    - Select Cost rollup CE.
10. In the **Secondary cost element** field, enter or select a value.
    - For this example, map the secondary cost element CC-008 to the cost center.
11. Click **New**.
12. In the list, mark the selected row.
13. In the **Cost object dimension hierarchy node** field, enter or select a value.
    - Select 009.
14. In the **Cost element dimension hierarchy node** field, enter or select a value.
    - Select Cost rollup CE.
15. In the **Secondary cost element** field, enter or select a value.
    - For this example, map the secondary cost element CC-009 to the cost center.
    - Continue until all cost centers are mapped to their corresponding secondary cost elements.
16. Click **Save**.

Feedback

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Manage a data source for the cost accounting ledger

Use this procedure to manage the general ledger data source for a cost accounting ledger. Before you complete this task, make sure that you complete the "Create a cost accounting ledger" and "Define cost control units" task guides. This recording uses the USP2 demo data company.

1. Go to **Cost accounting** > **Ledger setup** > **Cost accounting ledgers**.
2. In the list, find and select the desired record.
3. Select **Actual versions**.
4. On the Action Pane, select **Manage**.
5. Select **General ledger**.
6. Select **New**.
7. In the **Name** field, type a value.
8. In the **Data provider** field, enter or select a value.
    - For this example, select **Dynamics 365 Finance - General ledger entries**.
9. In the **Cost element dimension** field, enter or select a value.
    - For this example, select **Cost elements**.
10. Select **Save**.
11. Select **Configure data provider**.
12. In the **Legal entity** field, enter or select a value.
    - For this example, select **USP2**.
13. Select **New**.
14. In the **Posting layer** field, select **Current**.
15. Select **OK**.

Last updated on 08/11/2026

Process and trace source data

Article • 03/27/2023

All data processing is run by jobs. For each job and data provider, a journal is created to document that the process has been run, and that the entries were processed in the current job. Use this procedure to set up a data source and then trace the origin of a specific cost entry. This recording uses the USP2 demo data company USP2. Before you complete this task, make sure that you play the following task guides: "Create a cost accounting ledger," "Define cost control units," and "Manage data source for the cost accounting ledger."

1. Go to **Cost accounting** > **Ledger setup** > **Cost accounting ledgers**.
2. In the list, find and select the desired record.
    - Select the cost accounting ledger that you created earlier.
3. Click **Actual versions**.
4. On the Action Pane, click **Source data processing**.
5. Click **General ledger entry transfer journals**.
6. In the list, find and select the desired record.
7. Click **Journal entries**.
8. In the list, mark the selected row.
9. Click **Cost entries**.
10. Click **Source entry**.
11. On the Action Pane, click **Source data processing**.
12. Click **General ledger**.
13. In the **Fiscal calendar period** field, enter or select a value.
    - For this example, select Fiscal 2017 Period 9.
14. Click **OK**.

Feedback

Was this page helpful? Yes No

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Overhead calculation

This article describes the typical processes for calculating and allocating overhead costs.

Term definition

Overhead costs are the costs that you incur to run a business but can't directly attribute to any specific business activity, product, or service. Overhead costs provide critical support for the generation of profit-making activities. Here are some examples of overhead costs:

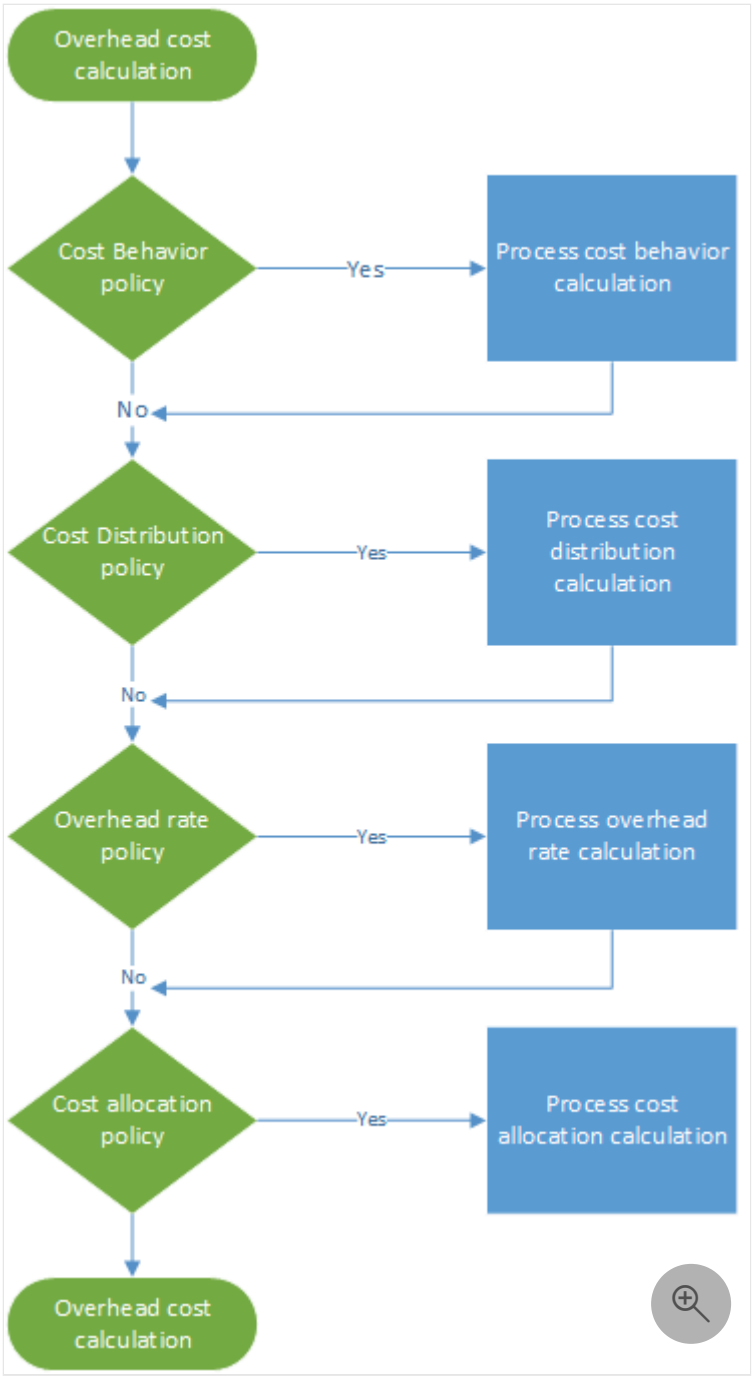
- Rent
- Electricity
- Administrative salaries

Overhead calculation overview

Overhead calculation runs the cost accounting policies in the correct order. You can run overhead calculation multiple times for the same fiscal period if you change cost accounting policies or detect specific errors. Each run of the overhead calculation is stored and receives a unique version ID that lets you compare the calculations in various versions. The cost entries that the overhead calculation generates receive an accounting date. This accounting date matches the end date of the fiscal period that is used in the calculation. The unique version ID consists of the following elements:

- Version type
- Date and time
- Cost accounting ledger
- Fiscal year
- Fiscal period

You run overhead calculation independently of the version. Therefore, you can calculate the Budget version before the Actual version. Overhead calculation consists of four steps, as shown in the following illustration. In each step, the process creates a journal header that has journal entries. This journal header keeps the input data for each calculation step. The process applies policies and rules to each journal line, and generates cost entries as output. Therefore, you always have full traceability.



Calculate and allocate the electricity overhead cost

In Financial accounting, some costs, such as electricity, are registered as a lump sum. Therefore, detailed managerial insight isn't provided for Cost accounting. In Cost accounting, to provide correct managerial insight across all organizational units and levels, costs must flow through the organizational units. This flow must be based on either an accurate record of the consumption or a fair assessment. In the general ledger, an electricity cost can be posted as shown in the following table.

Expand table

Accounting date	Cost center	Main account	Amount in the accounting currency
January 3, 2023	CC099 - Default cost center	10001 - Electricity	10,000.00

Step 1: Process the cost behavior calculation

By default, when you import cost entries from the source data, they receive the **Unclassified** cost behavior classification in Cost accounting. By applying cost behavior policy rules, you can reclassify cost entries as either **Fixed cost** or **Variable cost**.

Define the cost behavior rule

In some cases, part of the cost is a fixed fee, and the remaining cost is based on consumption. Electricity bills often match this definition. After you pay a specific fixed fee, you pay for consumption per kilowatt hour (Kwh). For example, if the fixed cost fee is 1,000.00, here's how you define the cost behavior rule:

- Fixed amount 1,000.00
  - 0 <= 1,000.00 = Fixed
  - 1000,01 < N = Variable

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Version
00001	Cost behavior calculation journal	Fiscal 2023 Period 1	Overhead calculation / 01-02-2023 11:51:00 PM / Ledger /2023 / Period 1

Journal entries (Cost object balance journal entries)

Expand table

Accounting date	Cost object	Cost element	Cost behavior	Amount
January 3, 2023	CC099 - Default cost center	10001 - Electricity	Unclassified	10,000.00

Cost entries

Expand table

Cost object	Cost element	Cost behavior	Amount	Accounting date
CC099 - Default cost center	10001 - Electricity	Unclassified	10,000.00	January 3, 2023
CC099 - Default cost center	10001 - Electricity	Unclassified	-10,000.00	January 31, 2023
CC099 - Default cost center	10001 - Electricity	Fixed cost	1,000.00	January 31, 2023
CC099 - Default cost center	10001 - Electricity	Variable cost	9,000.00	January 31, 2023

For more information, see Create and assign a cost behavior policy to a cost control unit.

Step 2: Process the cost distribution calculation

Use cost distribution to redistribute costs from one cost object to one or more other cost objects by applying a relevant allocation base. Cost distribution and cost allocation differ in that cost distribution always occurs at the level of the primary cost element of the original cost.

Define the cost distribution rule

In financial accounting, register electricity costs as a lump sum. In cost accounting, this approach isn't detailed enough. Distribute the variable cost to the individual cost objects on a fair basis. The most logical allocation basis is the consumption of electricity (Kwh). Create a statistical dimension member named **Electricity**, and record electricity consumption. By default, all statistical dimension members become available as allocation bases.

Expand table

Cost object	Kwh
CC001 - HR	1,000
CC002 - Finance	6,000
CC003 - Assembly	0

The following table shows the result when you apply electricity consumption as an allocation base for variable costs.

Expand table

Cost object	Magnitude	Allocation factor	Amount
CC001 - HR	1,000	(1,000 ÷ 7,000) × 9,000.00	1,285.71

Cost object	Magnitude	Allocation factor	Amount
CC002 - Finance	6,000	(6,000 ÷ 7,000) × 9,000.00	7,714.29
CC003 - Assembly	0	(0 ÷ 7,000) × 9,000.00	0.00

Distribute the fixed cost evenly to the individual cost objects that consume electricity. To achieve this result, use the Electricity statistical dimension member in a formula allocation base: (Electricity > 0.00). The following table shows the result when you apply electricity consumption as an allocation base for variable costs.

Expand table

Cost object	Formula	Magnitude	Allocation factor	Amount
CC001 - HR	(1,000 > 0.00)	1	(1 ÷ 2) × 1,000.00	500.00
CC002 - Finance	(6,000 > 0.00)	1	(1 ÷ 2) × 1,000.00	500.00
CC003 - Assembly	(0 > 0.00)	0	(0 ÷ 2) × 1,000.00	0.00

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Version
00002	Cost distribution calculation journal	Fiscal - 2023 Period 1	Overhead calculation / 01-02-2023 11:51:00 PM / Ledger /2023 / Period 1

Journal entries (Cost object balance journal entries)

Expand table

Accounting date	Cost object	Cost element	Cost behavior	Amount
January 31, 2023	CC099 - Default cost center	10001 - Electricity	Fixed cost	1,000.00
January 31, 2023	CC099 - Default cost center	10001 - Electricity	Variable cost	9,000.00

Cost entries

Expand table

Cost object	Cost element	Cost behavior	Amount	Accounting date
CC099 - Default cost center	10001 - Electricity	Fixed cost	-1,000.00	January 31, 2023
CC001 - HR	10001 - Electricity	Fixed cost	500.00	January 31, 2023
CC002 - Finance	10001 - Electricity	Fixed cost	500.00	January 31, 2023
CC099 - Default cost center	10001 - Electricity	Variable cost	-9,000.00	January 31, 2023
CC001 - HR	10001 - Electricity	Variable cost	1,285.71	January 31, 2023
CC002 - Finance	10001 - Electricity	Variable cost	7,714.29	January 31, 2023

For more information, see Create and assign a cost distribution policy to a cost control unit.

Step 3: Process the overhead rate calculation

Use the overhead rate to charge one or more specific cost objects. The charge is based on a predetermined cost rate and the magnitude from the assigned allocation base.

Define the overhead rate

Cost object CC001 HR contributes to a set of internal projects. Create a statistical dimension member named **HR projects** to measure the consumed magnitude.

Expand table

Cost object	Hours
Proj 1 - Project 1	3
Proj 2 - Project 2	1

Define a predetermined cost rate for the cost projects contribution.

Expand table

Cost object	Cost element	Cost behavior	Units	Rate
CC001 - HR	10001	Variable cost	1	10

The following table shows the result when the HR projects are applied as an allocation base.

Expand table

Cost object	Magnitude	Cost element	Allocation factor	Amount
Proj 1 - Project 1	3	10001	(3 ÷ 1) × 10.00	30.00
Proj 2 - Project 2	1	10001	(1 ÷ 1) × 10.00	10.00

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Version
00003	Overhead rate calculation journal	Fiscal 2023 - Period 1	Overhead calculation / 01-02-2023 11:51:00 PM / Ledger /2023 / Period 1

Journal entries (Journal entries for overhead rate calculation)

Expand table

Accounting date	Cost object	Magnitude
January 31, 2023	Proj 1 - Internal Proj 1	3.00
January 31, 2023	Proj 2 - Internal Proj 2	1.00

Cost entries

Expand table

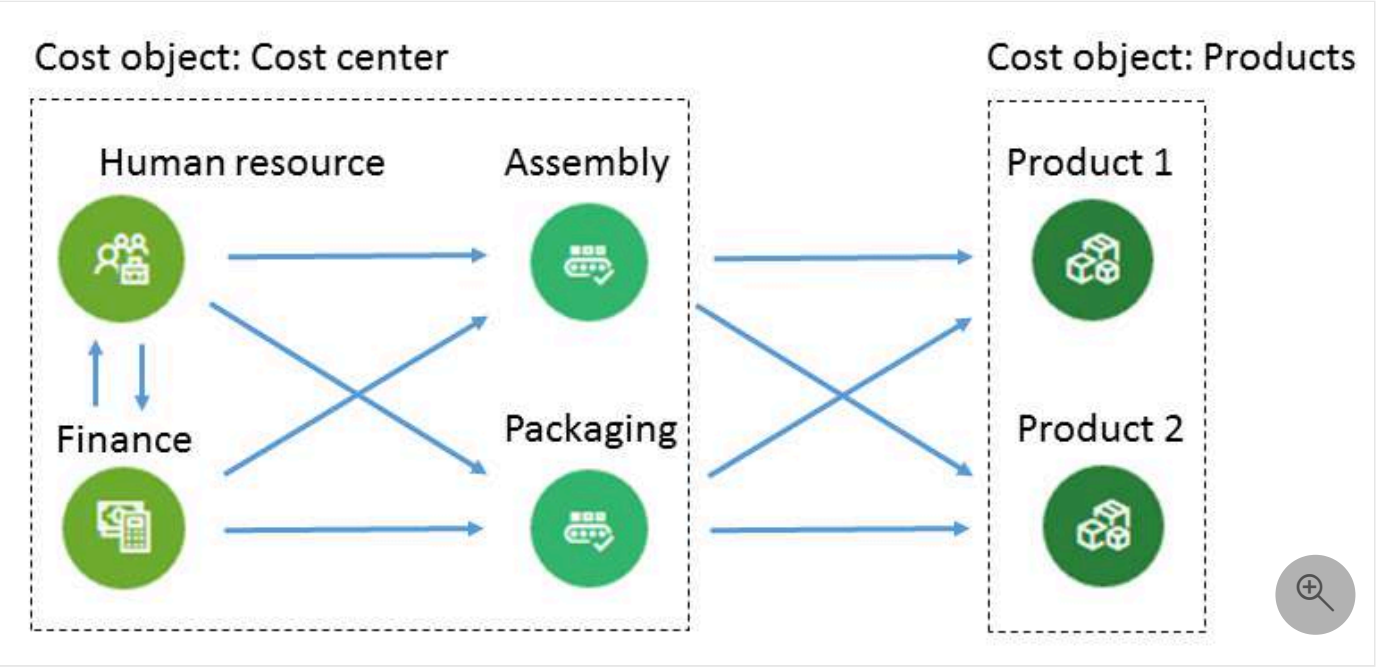
Cost object	Cost element	Cost behavior	Amount	Accounting date
CC0001 - HR	10001 - Electricity	Variable cost	-30.00	January 31, 2023
Proj 1 - Internal Proj 1	10001 - Electricity	Variable cost	30.00	January 31, 2023
CC001 - HR	10001	Electricity - Variable cost	-10.00	January 31, 2023
Proj 2 - Internal Proj 2	10001 - Electricity	Variable cost	10.00	January 31, 2023

For more information, see Perform overhead calculation.

Step 4: Process the cost allocation calculation

Allocation is used to allocate the balance of a cost object to other cost objects by applying an allocation base. Finance supports the reciprocal allocation method. In the reciprocal allocation

method, the mutual services that auxiliary cost objects exchange are fully recognized. The system automatically determines the correct order to perform the allocations in. The balance of a cost object is allocated by a single allocation base. Allocations across cost objects dimensions and their respective members are supported. The allocation order is controlled by the cost control unit.



Define the cost allocation

Here's a simple example that explains how you can trace the flow of cost. Cost object CC001 HR contributes to several cost objects. You create a statistical dimension member named HR services to measure the consumed magnitude.

Expand table

Cost object	HR services
CC002 - Finance	35
CC003 - Assembly	55
CC004 - Packaging	10

Cost object CC002 Finance contributes to several cost objects. You create a statistical dimension member named Finance services to measure the consumed magnitude.

Expand table

Cost object	Finance services
CC003 - Assembly	65
CC004 - Packaging	35

Cost object CC003 Assembly contributes to several cost objects. You create a statistical dimension member named Assembly services to measure the consumed magnitude.

Expand table

Cost object	Assembly services (hours)
Prod 1 - Product 1	60
Prod 2 - Product 2	20

Cost object CC004 Packaging contributes to several cost objects. You create a statistical dimension member named Packaging services to measure the consumed magnitude.

Expand table

Cost object	Packaging services (hours)
Prod 1 - Product 1	80
Prod 2 - Product 2	15

> **Note**
>
> Statistical measures such as the production hours that a product consumes can be derived from source data. For more information, see Statistical measure provider template. The following table shows the result when the HR services are applied as an allocation base for total cost (fixed cost and variable cost).

Expand table

Cost object	Magnitude	Allocation factor	Amount	Cost behavior
CC002 - Finance	35	(35 ÷ 100) × 500.00	175.00	Fixed cost
CC003 - Assembly	55	(55 ÷ 100) × 500.00	275.00	Fixed cost
CC004 - Packaging	10	(10 ÷ 100) × 500.00	50.00	Fixed cost

Cost object	Magnitude	Allocation factor	Amount	Cost behavior
CC002 - Finance	35	(35 ÷ 100) × 1,245.71	436.00	Variable cost
CC003 - Assembly	55	(55 ÷ 100) × 1,245.71	685.14	Variable cost
CC004 - Packaging	10	(10 ÷ 100) × 1,245.71	124.57	Variable cost

The following table shows the result when the Finance services are applied as an allocation base for total cost (fixed cost and variable cost).

Expand table

Cost object	Magnitude	Allocation factor	Amount	Cost behavior
CC003 - Assembly	65	(65 ÷ 100) × (500.00 + 175.00)	438.75	Fixed cost
CC004 - Packaging	35	(35 ÷ 100) × (500.00 + 175.00)	236.25	Fixed cost
CC003 - Assembly	65	(65 ÷ 100) × (7,714.29 + 436.00)	5,297.69	Variable cost
CC004 - Packaging	35	(35 ÷ 100) × (7,714.29 + 436.00)	2,852.60	Variable cost

The following table shows the result when the Assembly services are applied as an allocation base for total cost (fixed cost and variable cost).

Expand table

Cost object	Magnitude	Allocation factor	Amount	Cost behavior
Prod 1 - Product 1	60	(60 ÷ 80) × (275.00 + 438.75)	535.31	Fixed cost
Prod 2 - Product 2	20	(20 ÷ 80) × (275.00 + 438.75)	178.44	Fixed cost
Prod 1 - Product 1	60	(60 ÷ 80) × (5,297.69 + 685.14)	4,487.12	Variable cost
Prod 2 - Product 2	20	(20 ÷ 80) × (5,297.69 + 685.14)	1,495.71	Variable cost

The following table shows the result when the Packaging services are applied as an allocation base for total cost (fixed cost and variable cost).

Expand table

Cost object	Magnitude	Allocation factor	Amount	Cost behavior
Prod 1 - Product 1	80	(80 ÷ 95) × (50.00 + 236.25)	241.05	Fixed cost

Cost object	Magnitude	Allocation factor	Amount	Cost behavior
Prod 2 - Product 2	15	(15 ÷ 95) × (50.00 + 236.25)	45.20	Fixed cost
Prod 1 - Product 1	80	(80 ÷ 95) × (2,852.60 + 124.57)	2,507.09	Variable cost
Prod 2 - Product 2	15	(15 ÷ 95) × (2,852.60 + 124.57)	470.08	Variable cost

Journal entries (cost object balance journal entries)

Expand table

Journal	Journal type	Fiscal calendar period	Version
00004	Cost allocation journal	Fiscal 2023 - Period 1	Overhead calculation / 01-02-2023 11:51:00 PM / Ledger /2023 / Period 1

Journal lines

Expand table

Accounting date	Cost object	Cost element	Cost behavior	Amount
January 31, 2023	CC001 - HR	10001 - Electricity	Fixed cost	500.00
January 31, 2023	CC001 - HR	10001 - Electricity	Variable cost	1,245.71
January 31, 2023	CC002 - Finance	10001 - Electricity	Fixed cost	675.00
January 31, 2023	CC002 - Finance	10001 - Electricity	Variable cost	8,150.29
January 31, 2023	CC003 - Assembly	10001 - Electricity	Fixed cost	713.75
January 31, 2023	CC003 - Assembly	10001 - Electricity	Variable cost	5,982.83
January 31, 2023	CC003 - Packaging	10001 - Electricity	Fixed cost	286.25
January 31, 2023	CC003 - Packaging	10001 - Electricity	Variable cost	2,977.17
January 31, 2023	Prod 1 - Product 1	10001 - Electricity	Fixed cost	776.36
January 31, 2023	Prod 1 - Product 1	10001 - Electricity	Variable cost	6,994.21
January 31, 2023	Prod 2 - Product 1	10001 - Electricity	Fixed cost	223.64
January 31, 2023	Prod 2 - Product 1	10001 - Electricity	Variable cost	1,965.79

Cost entries

Expand table

Cost object	Cost element	Cost behavior	Amount	Accounting date
CC001 - HR	10001 - Electricity	Fixed cost	-500.00	January 31, 2023
CC002 - Finance	10001 - Electricity	Fixed cost	175.00	January 31, 2023
CC003 - Assembly	10001 - Electricity	Fixed cost	275.00	January 31, 2023
CC004 - Packaging	10001 - Electricity	Fixed cost	50.00	January 31, 2023
CC001 - HR	10001 - Electricity	Variable cost	-1,245.71	January 31, 2023
CC002 - Finance	10001 - Electricity	Variable cost	436.00	January 31, 2023
CC003 - Assembly	10001 - Electricity	Variable cost	685.14	January 31, 2023
CC004 - Packaging	10001 - Electricity	Variable cost	124.57	January 31, 2023
CC002 - Finance	10001 - Electricity	Fixed cost	-675.00	January 31, 2023
CC003 - Assembly	10001 - Electricity	Fixed cost	438.75	January 31, 2023
CC004 - Packaging	10001 - Electricity	Fixed cost	236.25	January 31, 2023
CC002 - Finance	10001 - Electricity	Variable cost	-8,150.29	January 31, 2023
CC003 - Assembly	10001 - Electricity	Variable cost	5,297.69	January 31, 2023
CC004 - Packaging	10001 - Electricity	Variable cost	2,852.60	January 31, 2023
CC003 - Assembly	10001 - Electricity	Fixed cost	-713.75	January 31, 2023
Prod 1 - Product 1	10001 - Electricity	Fixed cost	535.31	January 31, 2023
Prod 2 - Product 2	10001 - Electricity	Fixed cost	178.44	January 31, 2023
CC003 - Assembly	10001 - Electricity	Variable cost	-5,982.83	January 31, 2023
Prod 1 - Product 1	10001 - Electricity	Variable cost	4,487.12	January 31, 2023
Prod 2 - Product 2	10001 - Electricity	Variable cost	1,495.71	January 31, 2023
CC003 - Assembly	10001 - Electricity	Fixed cost	-286.25	January 31, 2023
Prod 1 - Product 1	10001 - Electricity	Fixed cost	241.05	January 31, 2023
Prod 2 - Product 2	10001 - Electricity	Fixed cost	45.20	January 31, 2023
CC003 - Assembly	10001 - Electricity	Variable cost	-2,977.17	January 31, 2023
Prod 1 - Product 1	10001 - Electricity	Variable cost	2,507.09	January 31, 2023

Cost object	Cost element	Cost behavior	Amount	Accounting date
Prod 2 - Product 2	10001 - Electricity	Variable cost	470.08	January 31, 2023

Conclusion

In financial accounting, you post a cost of 10,000.00 for electricity to a dummy cost center ID. This posting alerts cost accountants that they need to allocate this cost. In cost accounting, costs flow across organizational units and levels based on the policies and rules that you apply. Each cost associates with an allocation base that provides the best assessment for the allocation of costs.

Expand table

Cost element	Cost object CC099	Cost object CC001	Cost object CC002	Cost object CC003	Cost object CC004	Cost object Proj 1	Cost object Proj 2	Cost object Prod 1	Cost object Prod 2	Total
10001 Electricity	0.00	0.00	0.00	0.00		30.00	10.00	7,770.57	2,189.43	10,000.00
Unclassified	0.00									
Fixed cost	0.00	0.00	0.00	0.00	0.00			776.36	223.64	1,000.00
Variable cost	000	0.00	0.00	0.00	0.00	30.00	10.00	6,994.21	1,965.79	9,000.00

> **Note**
>
> This article shows how a primary cost element, 10001 Electricity, flows through the cost objects. Therefore, you allocate this overhead cost to the lowest level in the organization. In other words, the cost objects at the lowest level bear the cost. If you require a visual flow of the cost between the cost objects, you can use the cost roll-up policy rules to visualize the flow of the cost. For more information, see Cost rollup policy and overhead calculation.

Last updated on 08/04/2026

Cost rollup policy and overhead calculation

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Cost accounting helps you understand how costs flow to the products and services that an organization delivers. To achieve cost transparency, you need to allocate costs between cost objects using an appropriate allocation basis. By default, the system allocates costs for the primary cost element. This default allocation works well in some situations, but it has a few implications that you should consider.

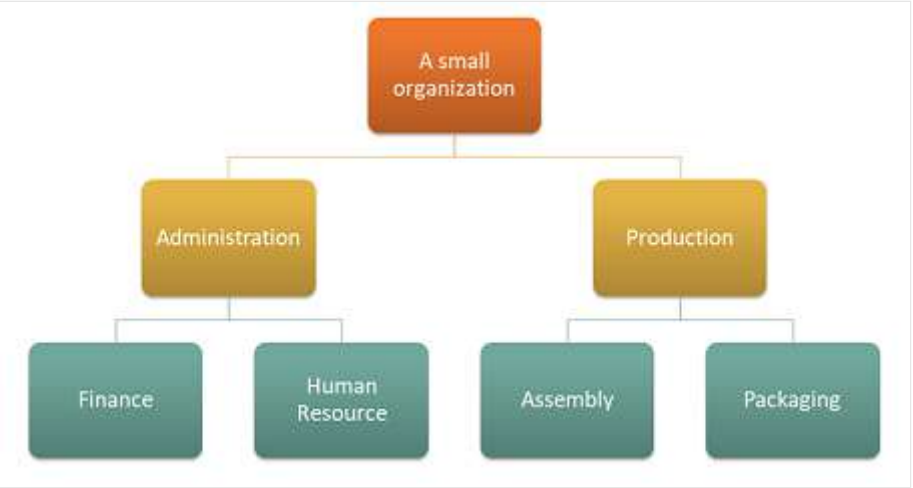
- After overhead calculation, auxiliary cost objects have a zero balance for the primary cost element.
- The volume of cost entries generated by overhead calculation can be very high.
- You can't track the cost flow between cost objects.

To avoid these implications, use Cost accounting to configure cost allocation to fit your organization's managerial reporting requirements. This article discusses how you can determine the correct level of secondary cost elements and create cost rollup rules that fit organization reporting and cost traceability.

> **Note**
>
> You can change configurations if reporting requirements change.

Example of cost rollup policy setup

Imagine that an organization has the following structure with four cost centers.



Cost object dimension

Expand table

Cost centers	Description
CC001	HR
CC002	Finance
CC003	Assembly
CC004	Packaging

Cost element dimension

Expand table

Cost elements	Description	Type
1001	Electricity	Primary
1002	Salaries	Primary
1003	Advertising	Primary

You can set up a dimension hierarchy that fulfills the organizational reporting requirements as follows.

Dimension hierarchy details

Expand table

Dimension hierarchy name	Dimension	Dimension hierarchy type name	Access list hierarchy
Organization	Cost centers	Dimension classification hierarchy	No

Dimension hierarchy

Expand table

	Dimension member ranges	
Nodes	From dimension member	To dimension member
Organization		
Admin		
Finance	CC001	CC001
HR	CC002	CC002
Production		
Packaging	CC003	CC003
Assembly	CC004	C004

You can set up a dimension hierarchy that fulfills the policy requirement as follows.

Dimension hierarchy details

Expand table

Dimension hierarchy name	Dimension	Dimension hierarchy type name
Profit & Loss statement	Cost elements	Dimension classification hierarchy

Dimension hierarchy

Expand table

	Dimension member ranges	
Nodes	From dimension member	To dimension member
Profit & Loss statement		
Primary cost	10001	10003

After processing the general ledger entries, the cost entry balance by cost object looks like this.

Expand table

	Cost object				Total
Cost element	**CC001**	**CC002**	**CC003**	*CC004**	
1001 Electricity	100.00	200.00	6,000.00	2,000.00	**8,300.00**
1002 Salaries	10,000.00	10,000.00	8,000.00	6,500.00	**34,500.00**
1003 Advertising	0.00	4,000.00	0.00	0.00	**4,000.00**
	10.100,00	14.200,00	14.000,00	8.500,00	**46.800,00**

Statistical dimension

Expand table

Statistical elements	**Description**
SE-1	HR services
SE-2	Finance services

Cost object CC001 HR contributes HR services to several cost objects.

The following table shows how HR services are consumed by the distribution of magnitude.

Expand table

Cost object	**Description**	**HR services**
CC002	Finance	35
CC003	Assembly	55
CC004	Packaging	10

Cost object CC002 Finance contributes to several cost objects.

The following table shows how finance services are consumed by the distribution of magnitude.

Expand table

Cost object	Description	Finance services
CC003	Assembly	65
CC004	Packaging	35

Set up cost allocation policies as follows.

Expand table

Policy name	Description	Cost object dimension hierarchy	Statistical dimension	Cost element dimension
2017	Cost allocation	Organization	Statistical elements	Cost elements

Set up cost allocation rules as follows.

Expand table

Cost object dimension hierarchy node	Cost behavior	Allocation base
CC001	Total	**HR services**
CC002	Total	**Financial services**

How cost flows between cost centers

If you want to learn how cost flows between the cost centers in the organization, you can create cost elements of the type **Secondary** for each cost center. These cost elements will then be used to transfer balances between the cost centers during the overhead calculation.

Set up cost element dimension members as follows.

Expand table

Cost elements	Type	
1001	Electricity	Primary
1002	Salaries	Primary
1003	Advertising	Primary
SC-CC001	**HR**	**Secondary**

Cost elements	Type	
SC-CC002	Finance	Secondary
SC-CC003	Assembly	Secondary
SC-CC004	Packaging	Secondary

Update the dimension hierarchy **Profit & Loss statement** with the new dimension members so that the dimension hierarchy contains the correct data for defining reporting and policies.

Dimension hierarchy details

Expand table

Dimension hierarchy name	Dimension	Dimension hierarchy type name
Profit & Loss statement	Cost elements	Dimension classification hierarchy

Dimension hierarchy

Expand table

	Dimension member ranges	
Nodes	From dimension member	To dimension member
Profit & Loss statement		
Primary cost	10001	10003
Secondary cost	**SC-CC001**	**SC-CC004**

Create a **Cost rollup policy** where you map each cost center to a corresponding cost element of the type **Secondary**.

Cost rollup polices

Expand table

Policy name	Description	Cost object dimension hierarchy	Cost element dimension hierarchy
2017	Cost flow	Organization	Profit & Loss statement

Cost rollup rules

Expand table

Cost object dimension hierarchy node	Cost element dimension hierarchy node	Secondary cost element
CC001	Profit & Loss statement	**SC-CC001**
CC002	Profit & Loss statement	**SC-CC002**
CC003	Profit & Loss statement	**SC-CC003**
CC004	Profit & Loss statement	**SC-CC004**

Perform overhead calculation

Journal

Expand table

Journal	Journal type	Fiscal calendar period	Year	Period	Version
00002	Cost allocation journal	Fiscal	2017	Period 1	Overhead calculation / 01-02-2017 11:51:00 PM / Ledger /2017 / Period 1

The system applies the **Cost rollup policy** when it creates the **Cost object balance journal entries**.

Cost object balance journal entries

Expand table

Accounting date	Cost object	Description	Cost element	Description	Amount
31-01-2017	CC001	HR	SC-CC001	HR	10,100.00
31-01-2017	CC002	Finance	SC-CC002	Finance	17,735.00
31-01-2017	CC003	Assembly	SC-CC003	Assembly	31,082.75
31-01-2017	CC004	Packaging	SC-CC004	Packaging	15,717.25

> **Note**

The system creates the journal entries based on the rules in the **Cost rollup policy** if a policy exists. The balance displayed is the balance of the overhead calculation.

The **Cost object cost balance journal entry details** page that you access from the journal entries displays how the balance is obtained.

Example: The journal entry for Cost object CC002 Finance

Cost object cost balance journal entry details

Expand table

Cost element dimension member	Description	Amount
1001	Electricity	200.00
1002	Salaries	10,000.00
1003	Advertising	4,000.00
SC-CC001	HR	3,535.00

Cost entries generated by the Overhead calculation

Expand table

Cost object	Description	Cost element	Description	Amount	Accounting date
CC001	HR	SC-CC001	HR	-10,100.00	31-01-2017
CC002	Finance	SC-CC001	HR	3,535.00	31-01-2017
CC003	Assembly	SC-CC001	HR	5,555.00	31-01-2017
CC004	Packaging	SC-CC001	HR	1,010.00	31-01-2017
CC002	Finance	SC-CC002	Finance	-17,735.00	31-01-2017
CC003	Assembly	SC-CC002	Finance	11,527.75	31-01-2017
CC004	Packaging	SC-CC002	Finance	6,207.25	31-01-2017

After the **Overhead calculation** finishes, you can report the results by using tools such as Microsoft SharePoint Workspace, Excel, or Power BI.

View reporting in Excel

The dimension hierarchies allow you to view data at different aggregation levels.

Here's an example of Power Pivot reporting in Excel.

Expand table

Profit & Loss statement	Cost object				Total
	CC001	**CC002**	**CC003**	CC004**	
Primary cost	**10,100.00**	**14,200.00**	**14,000.00**	**8,500.0**	**46,800.00**
1001	100.00	200.00	6,000.00	2,000.00	**8,300.00**
1002	10,000.00	10,000.00	8,000.00	6,500.00	**34,500.00**
1003	0.00	4,000.00	0.00	0.00	**4,000.00**
Secondary cost	**-10,100.00**	**-14,200.00**	**17,082.75**	**7,217.25**	**0.00**
SC-CC001	-10,100.00	3,535.00	5,555.00	1,010.00	**0.00**
SC-CC002	0.00	-17,735.00	11,527.75	6,207.25	**0.00**
SC-CC003	0.00	0.00	0.00	0.00	0.00
SC-CC004	0.00	0.00	0.00	0.00	0.00
Total	**0.00**	**0.00**	**31,082.75**	**15,717.25**	**46,800.00**

By using **Cost rollup policy** and **Cost elements of the type secondary**, you can keep the primary cost per cost object for internal reporting as the primary cost that remains after **Overhead calculation**.

If you perform the same example without creating the **Cost rollup policy**, the reporting result is as shown in the following table. The cost flows correctly but the traceability and insight into how cost flows between the cost centers are lost.

Expand table

Profit & Loss statement	Cost object				Total
	CC001	**CC002**	**CC003**	**CC004**	
Primary cost	**0.00**	**0.00**	**31,082.75**	**15,717.25**	**46,800.00**
1001	0.00	0.00	6,207.75	2,092.25	**8,300.00**
1002	0.00	0.00	22,275.00	12,225.00	**34,500.00**

Profit & Loss statement	Cost object				Total
1003	0.00	0.00	2,600.00	1,400.00	**4,000.00**
Secondary cost	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
SC-CC001	0.00	0.00	0.00	0.00	**0.00**
SC-CC002	0.00	0.00	0.00	0.00	**0.00**
SC-CC003	0.00	0.00	0.00	0.00	0.00
SC-CC004	0.00	0.00	0.00	0.00	0.00
Total	**0.00**	**0.00**	**31,082.75**	**15,717.25**	**46,800.00**

Depending on the reporting and traceability requirements in your organization, you can determine the correct level of secondary cost elements and create cost rollup rules that fit your needs.

The clear separation between **Cost allocation** and **Cost rollup policies** provides the flexibility to make continuous updates without affecting each other.

Additional resources

- Cost object dimensions
- Cost element dimensions
- Dimension hierarchy
- Overhead calculation

Last updated on 05/27/2026

Cost control workspace

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Cost control** workspace is a central point where managers who are responsible for controlling a cost object or a set of cost objects within a dimension or across dimensions (for example, cost centers and product groups) can access reports. Cost accountants fully manage the reports in the workspace, so the layout and data used for reporting are consistent across the whole organization.

Cost control workspace configuration

Cost accountants can define as many report configurations as they require for the desired data composition or layout. A report configuration consists of six sections, each of which contributes to either the selection of the targeted data composition or the layout.

To configure a cost control workspace, select **Cost accounting** > **Setup** > **Cost control workspace configuration**.

General

On the **General** FastTab, create a unique report layout. The name of the report is a unique identifier that users recognize in the **Cost control** workspace. You can also specify whether to share the report or keep it internal for cost accountants.

Expand table

Field	Description
Name	Enter a unique name for the layout.
Description	Enter a detailed description.

Field	Description
Published	If you set this field to **Yes**, a user assigned one of the following roles can see the report in the **Cost control** workspace: • Cost accounting manager • Cost accountant • Cost accountant clerk • Cost object controller If you set this field to **No**, only users assigned one of the following roles can see the report in the **Cost control** workspace: • Cost accounting manager • Cost accountant • Cost accountant clerk

Data filtering

On the **Data filtering** FastTab, you define the data foundation for the report. Users of this report see values on the report after the system processes the source data.

Expand table

Field	Description
Cost accounting ledger	The **Cost accounting ledger** that the report is based on. The value comes from the **Cost control unit** field.
Cost control unit	The value that you select determines the cost accounting ledger and cost objects that this report is based on.
Statistical dimension hierarchy, Cost element dimension hierarchy	A **Cost control** workspace configuration record can report either nonmonetary or monetary values, but not in the same layout. Select a value in the **Cost element dimension hierarchy** field to report monetary values. Select a value in the **Statistical dimension hierarchy** field to report nonmonetary values. The dimension hierarchy record that you select determines the structure of the reporting and aggregation levels.**NOTE:** To view nonmonetary and monetary values side by side, you can export data to Microsoft Excel for the Microsoft Power BI content pack.
Cost object dimension hierarchy	Select the dimension hierarchy of the cost object dimension that suits the purpose of the reporting that you're defining.
Budget original version	Select the budget version ID that acts as the original budget in the context of this report.

Field	Description
Budget revised version	Select the budget version ID that acts as the revised budget in the context of this report.

Assign calculation records

The overhead calculation performs several calculation steps on the source data, such as cost behavior classification, cost distribution, and cost allocation. You can perform multiple overhead calculations for the same fiscal period in case missing source data is discovered or rules must be updated. Each overhead calculation is saved with a unique ID. The cost accountant selects a specific overhead calculation ID. Users of the report, such as managers, see the results of the overhead calculation in the **Cost control** workspace.

Expand table

Field	Description
Fiscal calendar period	Select the fiscal calendar period to assign an overhead calculation ID to.**NOTE:** The fiscal periods that are listed in the field come from the fiscal calendar that is associated with the cost accounting ledger.
Actual version	Select the appropriate overhead calculation ID.
Budget version	Select the appropriate overhead calculation ID.
Revised budget version	Select the appropriate overhead calculation ID.

Fiscal periods per column

On the **Fiscal periods per column** FastTab, the cost accountant decides which fiscal period shows in the report layout.

The values in the selected columns multiply by the selected values on the **Fiscal periods per column** FastTab.

Expand table

Field	Description
Current period	The balance of the current fiscal period shows.**NOTE:** By default, the session date determines the current period. In the **Cost control** workspace, you can select a specific fiscal period. The selected value then represents the current period.
Previous period	The balance of the previous fiscal period shows. The following formula is used: Current fiscal period – 1**NOTE:** By default, the session date determines the previous period. In the **Cost control** workspace, you can select a specific fiscal period as the current period. The system recalculates **Previous period** accordingly.
Year to date	The value for the year to date shows. The following formula is used: YearToDate (Current fiscal period)**NOTE:** By default, the session date determines the current period. In the **Cost control** workspace, you can select a specific fiscal period. The selected value then represents the current period, and the **Year to date** value updates accordingly.
Year to date average	The average for the year to date shows. The following formula is used: (YearToDate [Current fiscal period]) ÷ (Count [Current fiscal period]) **Example** - **Statistical dimension member:** Full time employees - **Current date:** 3-21-2017 - **Period:** Fiscal period 1, Fiscal period 2, Fiscal period 3 - **Magnitude:** 10, 10, 12 In this case, **Year to date average** = (10 + 10 + 12) ÷ 3 = 10.67 The **Year to date average** value can be calculated for cost element dimension members and statistical dimension members. **NOTE:** By default, the session date determines the current period. In the **Cost control** workspace, you can select a specific fiscal period. The selected value then represents the current period, and the **Year to date** and **Year to date average** values update accordingly.

Columns to display for costs

On the **Columns to display for costs** FastTab, the cost accountant decides which columns the report layout should contain. There are three categories: Fixed cost, Variable cost, and Unclassified cost.

Expand table

Field	Description
Fixed cost	This column type shows the fixed cost, based on the selected overhead calculation ID.**NOTE:** This column type shows a balance only when you select an overhead calculation ID for the fiscal period.

Field	Description
Variable cost	This column type shows the variable cost, based on the selected overhead calculation ID.**NOTE:** This column type shows a balance only when you select an overhead calculation ID for the fiscal period.
Fixed + variable cost	This column type shows the fixed cost and variable cost, based on the selected overhead calculation ID.**NOTE:** This column type shows a balance only when you select an overhead calculation ID for the fiscal period.
Total cost	This column type shows the total cost (unclassified cost, fixed cost, and variable cost).**NOTE:** The column type always shows the balance.
Unclassified cost	This column type shows the unclassified cost.**NOTE:** You can use this column to validate whether all costs are correctly classified by the overhead calculation, or whether the cost behavior rules must be adjusted.

Columns to display for budgeted costs

On the **Columns to display for budgeted costs** FastTab, the cost accountant decides which columns to show for the selected budget versions. You can make individual selections for the original and revised budget.

> **Note**
>
> Because the following fields behave in the same manner for original budget and revised budget, the descriptions apply to both the original budget and the revised budget.

Expand table

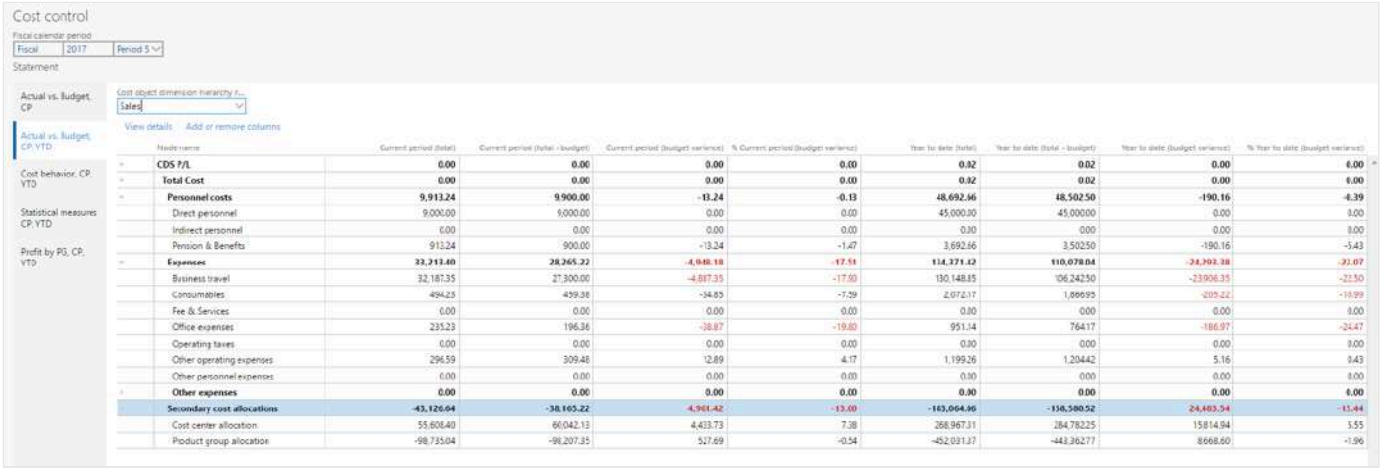
Field	Description
Budget	Shows budget balances per the selected columns.**NOTE:** The balances are based on the budget versions that you select on the **Data filtering** FastTab.
Budget variance	Calculates and shows the difference between budget and actual. The following formula is used: Budget balance – Actual balance
Budget variance in %	Calculates and shows the difference in percentage between budget and actual. The following formula is used: (Budget balance – Actual balance) ÷ Budget balance

Field	Description
Variance period threshold	Sets a threshold for the variance in monetary amount for the current period. If the threshold is exceeded, the line is highlighted in red in the **Cost control** workspace.**NOTE:** This field applies only to the cost elements that represent expenditures.
Variance year threshold	Sets a threshold for the variance in monetary amount for the year. If the threshold is exceeded, the line is highlighted in red in the **Cost control** workspace.
Variance threshold %	Sets a threshold for the variance in percentage. If the threshold is exceeded, the line is highlighted in red in the **Cost control** workspace.**NOTE:** The same percentage threshold applies to the current period and year.

Cost control workspace

The **Cost control** workspace is designed as a web report. Therefore, you can grant access to all managers who are responsible for a cost object as described in Define access rights for cost object controllers.

The list of reports that are available for users, such as managers, is controlled by the setting of the **Published** option on the **Cost control workspace configurations** page.



A manager can select the fiscal calendar period to view. The session date determines the default current period.

The values in the fiscal calendar period are determined by the report name and the fiscal calendar that you select for the cost accounting ledger that is associated with the report name on the **Cost control workspace configurations** page.

In the cost object dimension hierarchy, users can select the aggregation level at which balances should be shown. By enabling access-level security, you control the permissions, so that users can select only the hierarchy levels that they have access to. Therefore, they can see only the aggregated balances that they have access to.

Use the Cost control workspace

Add or remove columns

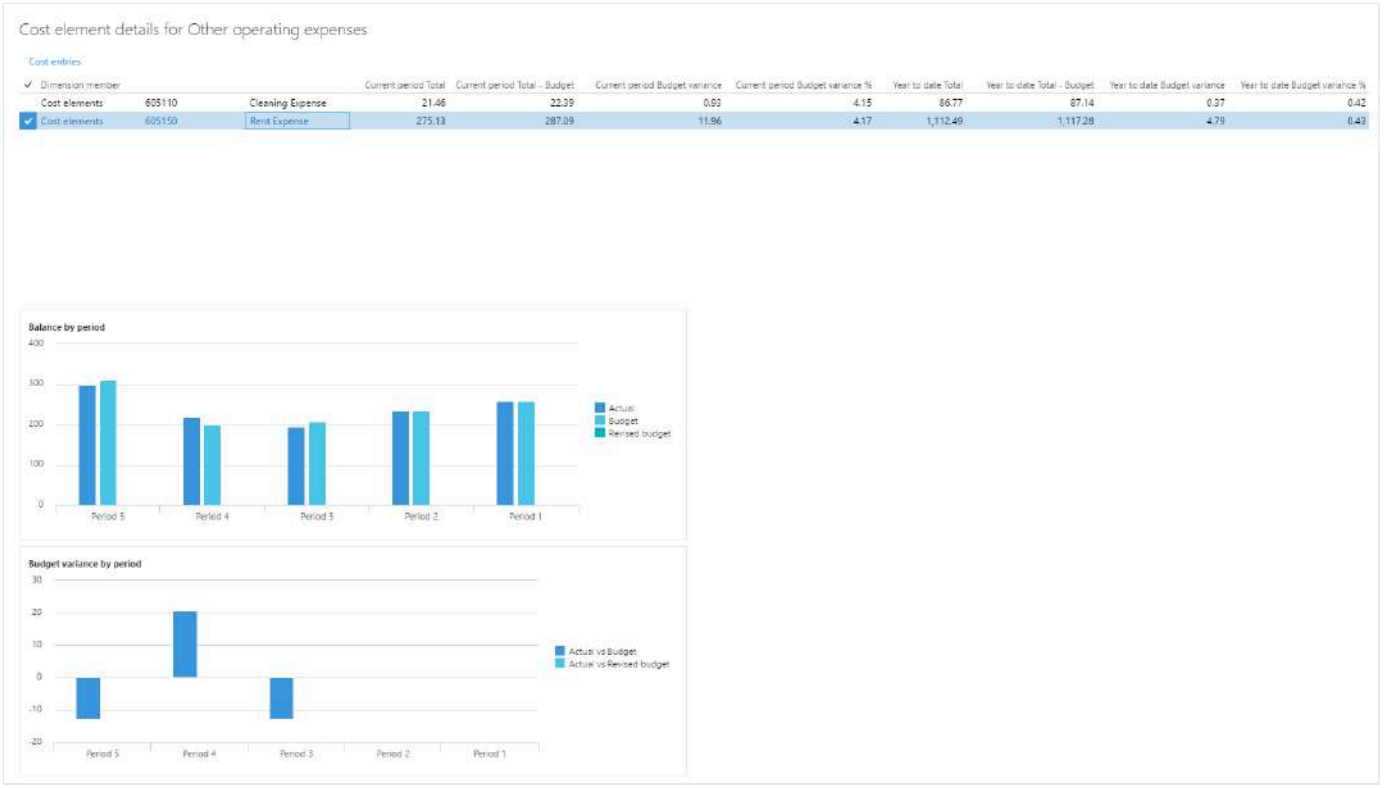
Users can customize the columns on a report to fit their requirements.

View details

You can view the details behind the balances that appear in the workspace. If you select a cost element dimension hierarchy node and then select **View details**, the **Cost element details** dialog box shows detailed information for the node.

A grid shows each cost element that is associated with the cost element dimension hierarchy node, and its values. The columns that appear in the grid match the workspace settings.

Two charts show a summary of actual versus budget and budget variance by period.

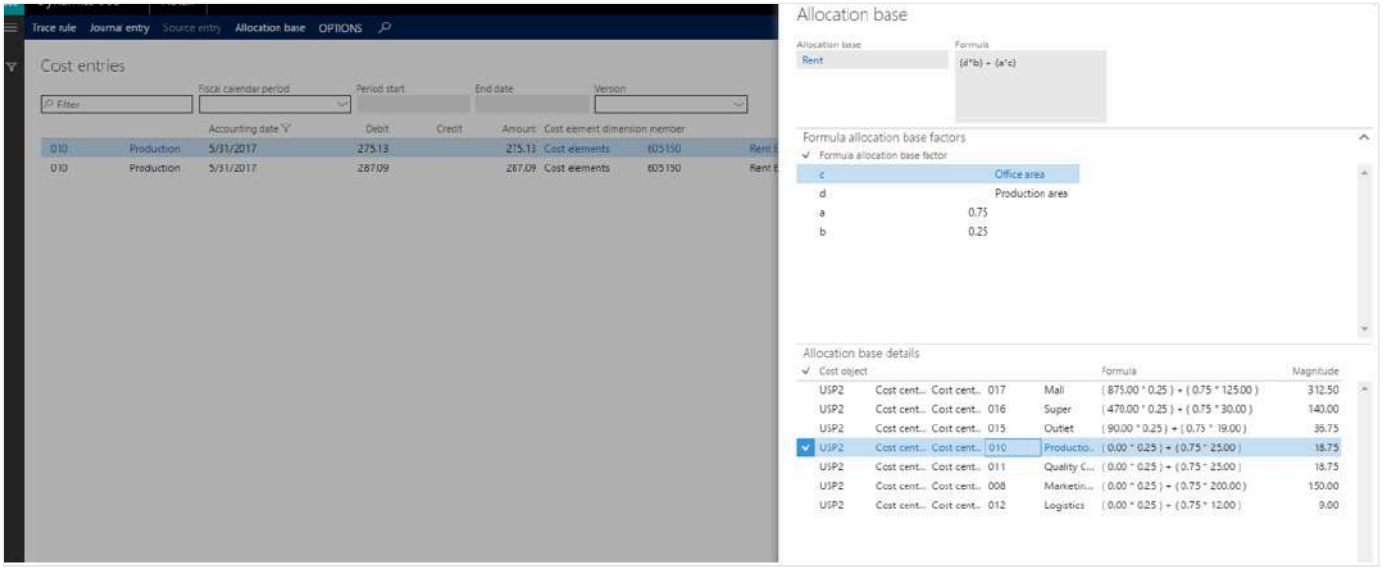


Select **Cost entries** to drill down into the entry details as required.

For example, rent is an expenditure that you distribute to cost centers. If you want to understand the rent cost that your cost center must carry, drill down to see how rent is calculated.

If you select **Allocation base** on the **Cost entries** page, a dialog box appears. You can then assign the allocation base to the rule and view the corresponding statistical measures that are registered for the period.

In the following example, the allocation base is of the **Formula allocation base** type, and the formula is shown. The factors that define the formula are listed. Additionally, a grid shows the calculation that is done per cost object.



Additional resources

Define access rights for cost object controllers

Last updated on 05/27/2026

Configure cost control workspace parameters

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use this procedure to configure the Cost control workspace so that managers at various levels in an organization can gain insight into their cost objects, such as cost centers and product groups. The USP2 demo company was used to create this recording.

1. Go to **Cost accounting** > **Setup** > **Cost control workspace configuration**.
2. Click **New**.
3. In the **Name** field, type a value.
4. In the **Description** field, type a value.
5. Select **Yes** in the **Published** field.

    - If you set this option to **Yes**, users who are assigned one of these roles can see the report in the **Cost control workspace**: cost accounting manager, cost accountant, cost accountant clerk, or cost object controller. If you set this option to **No**, only users who are assigned one of these roles can see the report in the **Cost control workspace**: cost accounting manager, cost accountant, or cost accountant clerk.

6. Expand the **Data filtering** section.
7. In the **Cost control unit** field, enter or select a value.
8. In the **Budget original version** field, enter or select a value.
9. In the **Cost element dimension hierarchy** field, enter or select a value.
10. In the **Cost object dimension hierarchy** field, enter or select a value.
11. Expand the **Assign calculation records** section.
12. Click **New**.
13. In the list, mark the selected row.
14. In the **Fiscal calendar period** field, enter or select a value.
15. In the **Actual version** field, enter or select a value.
16. Expand the **Fiscal periods per column** section.
17. Select **Yes** in the **Current period** field.
18. Expand the **Columns** to display for costs section.
19. Select **Yes** in the **Fixed cost** field.

20. Select **Yes** in the **Variable cost** field.
21. Select **Yes** in the **Total cost** field.
22. Click **Save**.
23. Close the page.
24. Go to **Cost accounting** > **Workspaces** > **Cost control**.
25. Select a statement to see fixed, variable, total, and unclassified costs for the selected fiscal periods.
26. In the **Fiscal calendar period** field, enter or select a value.
27. In the **Cost object dimension hierarchy node** field, enter or select a value.

    - After you've selected a **Cost object dimension hierarchy**, expand the **Cost element dimension hierarchy** to see the desired cost values. For example, you can expand the hierarchy to Manufacturing overhead to see the value.

Last updated on 10/02/2025

Set up security for the Cost accounting analysis Power BI content

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how you can propagate the access-level security in Cost accounting to row-level security in Microsoft Power BI. This functionality helps guarantee that users see only Power BI data that they're granted access to.

Overview

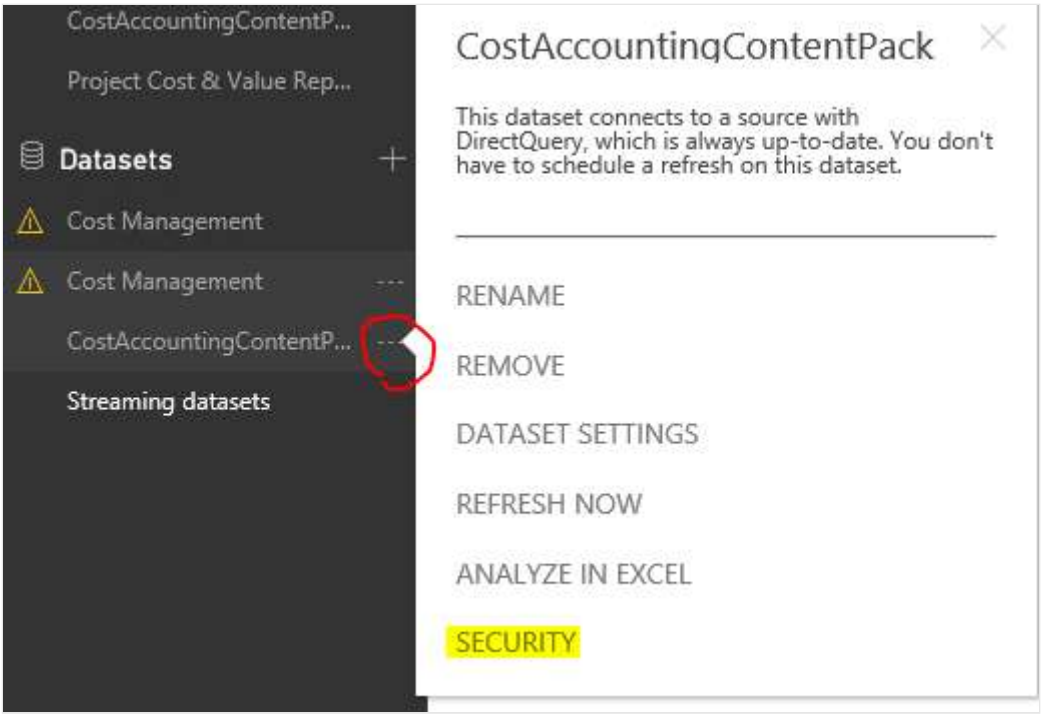
The **Cost accounting analysis** Microsoft Power BI content uses Power BI row-level security to limit a user's access. Set up security based on the access-level organizational hierarchy in the Cost accounting parameters. For more information about the **Cost accounting analysis** Power BI content, see Cost accounting analysis Power BI content.

Setup

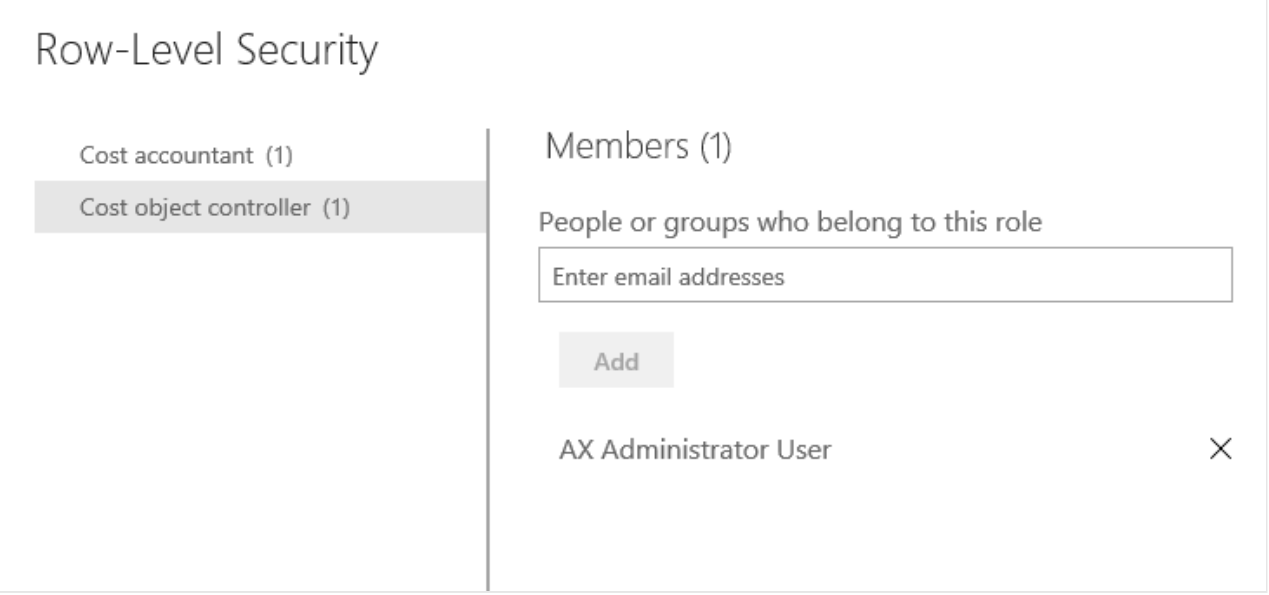
To propagate access-level security to Power BI, the owner of the Power BI content must follow these steps:

> **Note**
>
> The user who publishes the **Cost accounting analysis** Power BI content automatically becomes the owner. Only an owner can set up security in Power BI. Additionally, until an owner adds other users on PowerBI.com, no one except the owner can see any data in the **Cost accounting analysis** Power BI content.

1. Publish the definition file to Power BI.
2. Sign in to PowerBI.com.
3. Find the dataset for the **Cost accounting analysis** Power BI content.
4. Open the security page.



5. The **Cost object controller** role is already created. Add other members who are part of the Cost accounting access-level organizational hierarchy.



Users who are added to the **Cost object controller** role see only the data that they're allowed to see, according to the definition in the Cost accounting access-level organizational hierarchy.

> ⓘ **Note**
>
> Row-level security applies to tiles and reports that are embedded from Power BI.

Updating security

If you make updates to access-level security in Cost accounting, and you want Power BI to reflect those updates, you must update the entity store for the **Cost accounting analysis** Power BI content. After you complete the entity store update, you must update the artifacts on PowerBI.com. For more information about how to do an entity store update, see Power BI integration with Entity store. The owner of the **Cost accounting analysis** Power BI content must also do an entity store update if new users are granted access to the organizational hierarchy. Additionally, the owner must add the new users to the **Cost object controller** role on PowerBI.com, so that row-level security is applied for them.

Disabling security

This article assumes that your organization wants to restrict data access. If you disable the security parameters when you run Cost accounting, the owner must add users to the **Cost accountant** role in Power BI instead. If you change security from an enabled state to a disabled state, remove users from the **Cost object controller** role. And vice versa if you re-enable security. Users can belong to both roles. Joint access is the union of both roles. For the **Cost accounting analysis** Power BI content, users who have joint access have unrestricted data access. If your goal is to apply restricted access, assign users only to the **Cost object controller** role. These row-level security updates take effect immediately. Affected users should refresh their browsers.

Additional resources

To learn more about Power BI row-level security, see Manage security on your model in Power BI .

Last updated on 01/22/2026

Cost accounting analysis Power BI content

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes what the **Cost accounting analysis** Microsoft Power BI content includes. It explains how to access the Power BI reports, and provides information about the data model and entities that were used to build the content.

Overview

The **Cost accounting analysis** Power BI content is intended for cost controllers or anyone who's responsible for performing cost control of an organization. It includes the key metrics, such as cost, magnitude, and cost rate by actual cost, budget cost, and flexible budget cost. It uses transaction data from the **Cost accounting** module and provides an aggregate view of costs for the whole organization in one reporting currency. Managers can filter the data by cost objects to perform cost control of their organizational units, even if the organization has several legal entities.

Because the **Cost accounting analysis** content highlights variances between the actual costs and budgeted costs, it notifies managers about positive and negative trends for their operational units. Managers can drill down to the cost element hierarchies or individual cost elements. In this way, managers can gain detailed insight into how cost variances occurred, and then take effective action.

The **Cost accounting analysis** content lets cost accountants analyze how cost flows through the cost objects of the whole organization.

To learn more about Cost accounting, see Cost accounting home page.

By defining access-level security in Cost accounting and combining it with row-level security in Power BI, you can grant all cost object owners access to the **Cost accounting analysis** Power BI content. All data in the visualizations is then filtered based on the access level that's controlled in Cost accounting. To learn more about access-level security and row-level security, see Set up security for the Cost accounting analysis Power BI content.

Accessing the Power BI content

You can find the **Cost accounting analysis** Power BI content in the Shared assets library in Microsoft Dynamics Lifecycle Services (LCS). For more information about how to download the content and implement it in your organization, see Power BI content in LCS from Microsoft and your partners.

Be sure to download the **Cost accounting analysis** content that applies to the version of Microsoft Dynamics 365 that you're using.

> **Note**
>
> KB 4011327 is a prerequisite for this Power BI content. After you sign in to LCS, you can access the KB at https://fix.lcs.dynamics.com/issue/results/?q=kb4011327.

Metrics included in the Power BI content

The content includes a set of report pages. Each page consists of a set of metrics that the report visualizes as charts, tiles, and tables. The following table provides an overview of the visualizations in the **Cost accounting analysis** Power BI content.

Expand table

Report page	Chart	Tile
Cost control by fiscal period	Actual cost and Budget cost by Cost element hierarchy level	Actual cost vs Budget cost
	Budget variance by Cost element hierarchy level	Actual cost rate vs Budget cost rate
	Top 10 Budget variance in percentage by Cost element	Actual magnitude vs Budget magnitude
Cost control by Year to date	Actual cost and Budget cost by Calendar Year Period	Actual cost vs Budget cost
	Budget variance by Calendar Year Period	Actual cost rate vs Budget cost rate
	Top 10 Budget variance in percentage by Cost element	Actual magnitude vs Budget magnitude
Cost rate by fiscal year	Actual cost rate by Cost behavior	Actual cost rate vs Budget cost rate

Report page	Chart	Tile
	Actual cost rate, Budget cost rate variance, Budget cost rate percentage, and Budget cost rate by Cost element hierarchy level	Actual magnitude vs Budget magnitude
	Budget variance by Cost element hierarchy level	
	Top 10 Budget variance in percentage by Cost element	
Flexible budget by fiscal period	Actual cost, Budget cost, and Flexible budget cost by Cost element hierarchy level	Actual magnitude vs Budget magnitude
	Budget variance and Flexible budget variance by Cost element hierarchy level	Actual cost vs Flexible budget cost
	Actual cost, Budget cost, and Flexible cost by Cost behavior and Cost element hierarchy level	Actual cost rate vs Flexible budget cost rate
Cost statement by fiscal period	Actual cost by Cost element hierarchy level and Cost object dimension member name	
	Actual cost by Cost object dimension member name and Cost element dimension member name	

Understanding the data model and entities

The report pages in the **Cost accounting analysis** Power BI content use the following data. This data is represented as aggregate measurements staged in the Entity store. The Entity store is a Microsoft SQL Server database that's optimized for analytics. For more information, see Power BI integration with Entity store.

The content uses the following key aggregate measurements.

Expand table

Entity	Key aggregate measurement	Data source for Dynamics 365	Field	Description
Cost accounting entries	SUM(Amount)	CAMDATAAggregatedCostEntry	Amount	The amount in the Cost accounting ledger currency.
Statistical entries	SUM(Magnitude)	CAMDATAAggregatedStatisctialEntry	Magnitude	

The following table shows how the key aggregate measurements are used to create several calculated measures in the content's dataset.

Expand table

Measure	How the measure is calculated
Actual cost	CALCULATE('Cost accounting entries'[Measure], 'Transaction versions'[ISSOURCEVERSIONBUDGET_VALUE] = 0)
Budget cost	CALCULATE('Cost accounting entries'[Measure], 'Transaction versions'[ISSOURCEVERSIONBUDGET_VALUE] = 1)
Budget cost variance	[Budget cost] - [Actual cost]
Budget variance percentage	IF([Budget cost] = 0, blank(), [Budget variance] / [Budget cost])
Actual magnitude	CALCULATE('Statistical entries'[FullMagnitude], 'Transaction versions'[ISSOURCEVERSIONBUDGET_VALUE] = 0)
Budget magnitude	CALCULATE([FullMagnitude], 'Transaction versions'[ISSOURCEVERSIONBUDGET_VALUE] = 1)
Statistical budget variance	[Budget magnitude] - [Actual magnitude]
Statistical budget variance percentage	IF([Budget magnitude] = 0, blank(), [Statistical budget variance] / [Budget magnitude])
Actual cost rate	IF([Actual magnitude] = 0, BLANK(), [Actual cost] / [Actual magnitude])
Budget cost rate	IF([Budget magnitude] = 0, BLANK(), [Budget cost] / [Budget magnitude])
Budget cost rate variance	[Budget cost rate] - [Actual cost rate]
Budget cost rate variance percentage	IF([Budget cost] = 0, blank(), [Budget cost rate variance] / [Budget cost rate])
Fixed budget cost	CALCULATE([Budget cost], 'Cost accounting entries'[COSTBEHAVIOR] = 1)
Variable budget cost	CALCULATE([Budget cost], 'Cost accounting entries'[COSTBEHAVIOR] = 2)
Fixed flexible budget cost	[Fixed budget cost]
Variable flexible budget cost	IF([Budget magnitude] = 0, BLANK(), ([Variable budget cost] / [Budget magnitude]) * [Actual magnitude])
Flexible budget cost	[Fixed flexible budget cost] + [Variable flexible budget cost]
Flexible budget variance	[Flexible budget cost] - [Actual cost]

Measure	How the measure is calculated
Flexible budget variance percentage	IF([Flexible budget cost] = 0, BLANK(), [Flexible budget variance] / [Flexible budget cost])
Flexible budget cost rate	IF([Actual magnitude] = 0, BLANK(), [Flexible budget cost] / [Actual magnitude])
Flexible budget cost rate variance	[Flexible budget cost rate] - [Actual cost rate]
Flexible budget cost rate variance percentage	IF([Flexible budget cost rate] = 0, BLANK(), [Flexible budget cost rate variance] / [Flexible budget cost rate])

The following key dimensions are used as filters to slice the aggregate measurements. By using these filters, you can achieve greater granularity and provide deeper analytical insights.

Expand table

Entity	Examples of attributes
Cost accounting ledgers	Cost accounting ledger
Cost control units	Cost control unit name
Cost element dimensions	Cost elements dimension name, Cost element dimension member name, Cost element dimension member description
Cost object dimensions	Cost object dimension name, Cost object dimension member name, Cost object dimension member description
Statistical dimensions	Statistical dimension name, Statistical dimension member name, Statistical dimension member description
Cost object dimension hierarchies	Cost object dimension hierarchy name, Cost object dimension hierarchy level, Cost object dimension hierarchy tree
Cost element dimension hierarchies	Cost element dimension hierarchy name, Cost element dimension hierarchy level, Cost element dimension hierarchy tree
Statistical dimension hierarchies	Statistical dimension hierarchy name, Statistical dimension hierarchy level, Statistical dimension hierarchy tree
Transaction versions	Version name
Fiscal calendars	Calendar, Calendar description
Fiscal years	Calendar year
Fiscal periods	Calendar year period

Last updated on 03/09/2026

Cost management home page

Cost management (video) helps you work with the valuation and accounting of raw materials, semi-finished goods, finished goods, and work-in-progress assets. It's the process of defining, managing, and reporting Inventory accounting and Manufacturing accounting.

You can define cost policies in the following areas:

- Predetermined cost
- Inventory accounting
- Manufacturing accounting
- Indirect cost accounting
- Ledger integration

For example, you can define which inventory valuation methods, such as FIFO, Weighted average, Standard cost, or Moving average that you want to apply to products in the Item model group in Inventory accounting.

You can access Inventory accounting and Manufacturing accounting from the **Cost administration** and **Cost analysis** workspaces. These workspaces provide a comprehensive overview of the current status, key performance indicators (KPIs), and detection of deviation.

> **Tip**
>
> Are you new to inventory costing? Check out the following resources:
>
> - Glossary of terms in Dynamics 365 business processes
> - Define service costing overview and relation to other business processes
> - Business process for product costing and how it relates to other processes with Dynamics 365

Manufacturing accounting lets you handle Job order costing in production orders and batch orders, as well as Backflush costing in lean manufacturing.

The Cost management Power BI content provides managerial insight into inventory and work-in-progress (WIP) inventory, and how cost flows through them by category over time. The information can also be used as a detailed supplement to the financial statement.

Related information

What's new and in development

To see what new features are released and what new features are in development, go to the Dynamics 365 Release Planner.

White paper

BOM calculation by using a costing sheet describes how to set up a costing sheet that includes material and manufacturing, and how the setup affects the BOM calculation results. To better explain the articles, it provides concrete scenarios and data that demonstrate the effect of the various settings and configurations.

Blogs

You can find opinions, news, and other information about Cost management on the Dynamics AX Manufacturing R&D Team blog and Supply Chain Management in Dynamics AX R&D Team blog. Although some of these posts were written for the previous version of Cost management, the same concepts still apply, and the procedures are also similar in the current version.

Last updated on 07/01/2026

Inventory costing FAQ

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article answers some frequently asked questions about inventory costing in Microsoft Dynamics 365 Supply Chain Management.

Inventory closing, adjustments, recalculation, pre-closing, and reverse

Why is inventory closing required?

The inventory close process matches and settles inventory issue transactions (resulting from sales, production order issue, transfer journals, and other transactions) to their receipt transactions (such as purchase, counting journals, production receipts, and other transactions). The process uses the inventory valuation method that you define in the item's *item model group* and considers the item's active financial *product*, *storage*, and *tracking* dimensions.

You perform inventory closing for periodic inventory valuation methods (FIFO, LIFO, weighted average, and other methods). It ensures that the financial statements (income statement, balance sheet, trial balance, and other statements) reflect appropriate inventory values and cost of goods sold (COGS) based on all receipts and issues up to the period end.

You should run inventory closing regularly for a specific period. Once you close the inventory for that period, you can't post any transactions in that period.

The frequency of inventory close runs varies by company. However, transaction volume should determine how often you decide to run inventory close. In general, most companies run inventory closing as part of their month-end closing and reconciliation procedures, and before any audit activities.

To balance out the transactional load, run inventory closing regularly, once inventory transactions are finalized for a period, typically at a monthly frequency.

You must run inventory closing if you plan to change the valuation model in *item model group* of an item from periodic to perpetual, or if you plan to archive old inventory transactions.

Why is inventory recalculation required?

If you need to adjust inventory and the general ledger during a month or other inventory period, run inventory recalculation instead of inventory close. Inventory recalculation adjusts the inventory transactions according to the applicable valuation method but doesn't settle inventory transactions. Like inventory closing, you can only run inventory recalculation for periodic inventory valuation methods.

During inventory recalculation, the system adjusts on-hand and inventory transactions. These tasks affect any ledger account that's linked to the original inventory transaction.

Inventory recalculation doesn't close any period, unlike inventory closing. You can still post transactions in the period for which you already ran inventory recalculation.

The frequency of inventory recalculation runs depends on your business requirements. You can run recalculations daily, weekly, or monthly to adjust the on-hand and inventory transactions and reflect the accurate cost of the items based on their valuation method. However, running inventory recalculation is an optional procedure and not a prerequisite to inventory closing. Run inventory closing monthly. Once you run closing, the system automatically cancels all the recalculations you ran for dates after the closing date. Running many recalculations or running scheduled recalculations has a performance impact.

What are the differences between inventory closing and inventory recalculation?

These are the key differences between inventory closing and inventory recalculation:

- Inventory closing matches and settles transactions, whereas inventory recalculation adjusts transactions.
- Inventory closing closes the period for which you run it, so you can't post any further transactions in that period. Inventory recalculation doesn't restrict inventory postings for any period.
- You must run inventory closing before making any modifications in an item's *item model group* and before archiving old transactions. Besides, you should regularly run inventory closing at periodic intervals to reduce the system load for high transactional volume. Inventory recalculation isn't a mandatory process.

Is it mandatory to run inventory closing or recalculation?

You must run inventory closing before making any modifications in an item's *item model group* and before archiving old inventory transactions. Besides, you should regularly run inventory closing at periodic intervals to reduce the system load for high transactional volume. Inventory

closing closes the settled inventory transactions, thereby excluding the closed transactions from further inventory processes. Inventory recalculation isn't a mandatory process. You can run it according to your business requirements—either daily, weekly, or monthly—to adjust the inventory transactions for reporting the accurate cost of the items based on the valuation method. Run inventory closing monthly for best performance.

What is inventory pre-closing?

Pre-closing replaces the *activate closing of non-financial transfer* functionality that was available in older versions of Supply Chain Management. Pre-closing finalizes non-financial transfers; this finalization allows the inventory closing process to safely ignore them, since closing and recalculation only affect financial transactions. *Non-financial transfers* are inventory transactions that have no effect on inventory costing, like inventory movement journal between two warehouses in the same site if warehouse isn't a financial dimension. You can explicitly run pre-closing if required as per the business requirements. Moreover, the closing and recalculation procedure always implicitly runs pre-closing for the non-financial transfers (among the transfer orders) against which markings are present.

What is inventory reversal? How is it different from canceling a closing?

If you need to revert the settlements or adjustments made by a closing or recalculation voucher, select the respective voucher and then select **Reverse**. This action posts equal but opposite settlements or adjustments in the ledger, balancing the changes made by the respective closing or recalculation voucher.

Use inventory reversal when you need to post any transaction in an already closed inventory period. In that case, the closing voucher is reversed, which opens the period and allows you to post transactions.

Inventory reversal and cancellation refer to the same operation and can be used interchangeably.

What is inventory revaluation?

For perpetual valuation models like moving average or standard cost, use inventory revaluation to revalue the on-hand inventory. Inventory revaluation works similarly to inventory recalculation. Use inventory revaluation to adjust the recorded value of inventory items to reflect changes in cost, market conditions, accounting corrections, or policy updates. This adjustment can either increase (appreciate) or decrease (depreciate) the value of inventory on hand in the system and general ledger.

What is the typical duration required to complete the inventory closing and recalculation process, and what factors impact the run time?

The duration required to complete the inventory closing and recalculation process can vary widely depending on several factors. It can range from a few minutes to several hours for most environments. Overall run time depends on:

- Volume of inventory transactions (issues versus receipts)
- Type of inventory transactions (source documents)
- Length of the period
- Number of items and inventory dimensions (site, warehouse, location, and so on)
- Real-time resource utilization and availability
- Parallel processes being run at the same time, number of active sessions, and so on

Because the closing process can run from few minutes to several hours, schedule the process during off-peak hours to avoid any business disruptions.

Why does my inventory closing or recalculation take so long? Can I speed it up?

As discussed previously, there's no fixed duration for inventory closing or recalculation. The overall run depends on the count of open or unsettled transactions, the nature of source documents, resource availability, parallel processing, parallel user sessions, and other factors. The following recommendations help you analyze the closing process:

- **Track progress of the run** – You can always track the progress of the closing or recalculation voucher from the **Closing and adjustment** page. Select the specific voucher and go to *Progress details*. You can find the bundles that have completed running and the ones that are still pending.

- **Run inventory closing regularly** – If you have a high volume of transactions and don't run inventory closing regularly at periodic intervals, you might experience performance issues as the count of open or unsettled transactions gradually increases in the system. Hence, run inventory closing regularly (preferably at monthly intervals) for customers with a high volume of transactions for better performance. You can check the time stamp for the most recent closing run. You can view this information from the **Closing and adjustment** page by filtering the **Type** as *Closing*. If you didn't run closing recently and have items that use any periodic valuation methods in your inventory, proceed with closing the inventory for the period nearest to your recalculation period.

- **Count of issue and receipt transactions** – During the adjustment process, issue transactions are adjusted with the receipt transactions. A large difference in the count of issue transactions and receipt transactions can impact the overall performance because the system might not find the required number of receipts to settle the large quantity of issues.

- **Circularity in inventory transactions** – Overall run time largely depends on your inventory transactions and the nature of your source documents. BOM journals are one of the source documents that consume a considerable amount of time in calculating the cost of the final product when the BOM structure is quite complicated, with lots of depth, consumption of scrap products, circular links in production and consumption, and other factors. If you have inventory transactions that pertain to any BOM journal, you can perform a basic BOM circularity check and eliminate any circular link. BOM circularity doesn't throw errors in the closing or recalculation process because the process has a specific design to handle such scenarios. However, BOM circularity increases the overall run time to a large extent. The same situation applies to transfer order circularity. Consider the scenario when there's a transfer from site 1 to site 2 and then again from site 2 to site 1. To improve performance, reduce and eliminate the possibility of any such circularity in BOM journals or transfer orders. For more information about BOM circularity check strategies, refer to the other FAQs in this article.

- **Review inventory performance parameters** – You can configure a few parameters to control the performance of closing and recalculation. Here's how to optimize these parameters:

  Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters** and open the **Inventory accounting** tab. In the **Closing** section, make the following settings:
  - **Extra Batch Helpers** – Set the number of extra threads to spawn based on thread pool availability to aid the running of the respective batch job. The default value for this parameter is *8*. You can increase this value based on the availability of your system resources. Learn more in batch capacity.
  - **Number of items per bundle** – Set the maximum number of items to keep in a bundle. The default value is *40*. An item is always present in a single bundle, and a single thread or task always processes a bundle. If your transactions have many items and a relatively comparable number of transactions per item, keep this value at *40*. If your inventory transactions don't have many items or instead have a large number of transactions for only a few items, set this value to *10*. Modify these values before starting the closing or recalculation process, not during the run.

  In the **Closing** or **Recalculation** dialog, make the following settings:

- **Maximum number of iterations allowed per item** – Pertains to cost calculations for circular links and is a trade-off between performance and accuracy. It determines the number of iterations for which the closing or recalculation process runs. Every iteration adjusts the cost of the item involved in the circular BOM. Hence, the more iterations, the better the cost accuracy for those items in the circular BOM. But you can't practically run this process infinitely, so the default iteration value is *100*. If your business transactions don't have many circular inventory transactions, or if you're fine with some approximate cost adjustments, consider reducing this value to reduce the overall run time.
- **Minimum amount allowed to be expensed** – Determines the minimum adjustment amount to be expensed while processing inventory transactions in every iteration. This parameter is again a tradeoff, like the previous one, but instead, performance is reversely proportional to the value. A lower value contributes to poorer performance but more accurate results. The default value is *1.00* in the accounting currency. Most companies don't modify this value, but it's up to your business requirements to determine an optimized value.

To improve the closing or recalculation performance, eliminate any circular links from your inventory transactions.

If you feel that the operation is running for an unusual amount of time, contact Microsoft Support or your partner to monitor the real-time run status and resource utilization.

How do I check circularity in BOM journals?

To check circularity in BOM journals, go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters** > **Bill of materials**. Enter the required values in **Level of circularity** and **Circularity check strategy**. The **Circularity check strategy** option works in conjunction with the **Level of circularity** option, so start by looking at the **Level of circularity** setting.

The **Level of circularity** setting determines when the system checks for circularity in BOM structures. A circular BOM exists when an item is defined as a component of itself within a BOM structure. A BOM is considered circular regardless of whether it's a first-level component or a lower-level component in the BOM structure that is referenced. Set it to one of the following values:

- *Line* – The circularity check occurs when you save the BOM line. This setting prevents you from adding a BOM line when the component item causes BOM circularity.
- *Never* – The system doesn't check for circularity. You can manually check for circularity errors by using the **Check** function on the **BOM line** page. If circularity exists, the error is also reported when you perform cost rollup and planning calculations.

- *BOM* – The circularity check occurs when you do any of the following activities: close the **Bills of materials** page, use the **Check** function on the **Bills of materials** page, or associate a BOM with a BOM version.

The controls that prevent a circular BOM don't apply to a production BOM or a production order. You can manually add a component or consume a component that creates BOM circularity. When the component consumption reflects BOM circularity, the inventory closing process might not calculate a sensible cost for a manufactured item that has an inventory valuation method using actual cost. In such an occurrence, the use of a manufacturing indirect cost of the surcharge type can also cause problems in inventory closing, because the calculated cost can grow exponentially.

For the **Circularity check** field, choose one of the following values:

- *Optimize for low complexity* – Requests data from storage as it's needed for the circularity check. For single-level or simple BOM structures, performance is better when you select this option.
- *Optimize for high complexity* – The system caches the BOM structure and all its dependencies before processing the circularity check. When processing a BOM with many lines, it might take a moment for the processing to start since it's loading the information into cache. However, the *Optimize for high complexity* option can significantly improve overall performance of the circularity check. This improvement is especially true if you're running a task that performs a circularity check on multiple BOMs or if you're doing a consistency check on the bill of materials.

You can use the following SQL query to fetch the items with circular links directly from the database. You could also convert this query to an X++ script and run it.

SQL

```
select p.itemid, c.itemid, p.REFERENCECATEGORY, c.REFERENCECATEGORY, c.REFERENCEID,
p.REFERENCEID, *
from INVENTTRANSORIGINASSEMBLYCOMPONENT a, INVENTTRANSORIGIN p, INVENTTRANSORIGIN c
where a.ASSEMBLYINVENTTRANSORIGIN=p.RECID
  and a.COMPONENTINVENTTRANSORIGIN=c.RECID
  and p.ITEMID=c.ITEMID
  and p.DATAAREAID=c.DATAAREAID
  and p.DATAAREAID = 'abc';
```

What are the best practices for running inventory closings and recalculations?

Here are some best practices for running inventory closings and recalculations:

- Run inventory closings and recalculations during off-peak business hours when the overall system load is lower. This schedule increases the availability of resources.
- Always run inventory closings, recalculations, and resume calculations as batch processes for better performance and error tracing.
- Verify the inventory closing parameters to ensure they fit your business operations.
- If you have a high transactional volume for a period that might prolong the closing run, divide the period into smaller time intervals and close those periods consecutively.
- If recalculation takes a long time, check when you last ran closing. You can view this information on the **Closing and adjustment** page by filtering the **Type** as *Closing*. If you didn't run closing recently and your inventory includes items that use periodic valuation methods, run inventory closing for the period nearest to your recalculation period. Closing the inventory closes the transactions and reduces the scope and volume for further inventory operations.

How can I analyze the issue and receipt transactions as part of the inventory closing and recalculation process?

Inventory closings and recalculations settle and adjust issue transactions to receipt transactions. If there aren't enough receipt transactions, the process temporarily stores many issue transactions in memory during the adjustment process. Run inventory closings or recalculations when the number of issue and receipt transactions is comparable. The inventory value report can provide a summarized view of issue versus receipt transactions.

You can also get this information directly from the database by running the following SQL query, which you could also convert to an X++ script.

SQL

```
select ITEMID, STATUSISSUE, count(*) countOfOpenTrans
from InventTrans
where PARTITION = 5637144576
  and DATAAREAID = 'abc'
  and STATUSISSUE <= 1
  and STATUSRECEIPT <= 1
  and VALUEOPEN = 1
  and DATEFINANCIAL <= 'closing_trans_date'
group by ITEMID, STATUSISSUE
order by ITEMID, STATUSISSUE, countOfOpenTrans desc;
```

What are the best practices for running inventory reversals?

Here are some best practices for running inventory reversals:

- Run inventory reversals during off-peak business hours to make better use of system resources.
- Always run inventory reversals one at a time. Running multiple reversals at the same time might lead to errors (such as *duplicate reverse is not allowed*) or even ledger data corruption.
- If your reversal run is very slow, check the number of batch helpers you're using. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters** and open the **Inventory accounting** tab. The **Extra batch helpers** field in the **Closing** section decides the number of parallel threads to use for the reversal task, based on thread pool availability. The recommended and default number of extra batch helpers is *8*. If your reversal run is very slow, consider increasing this value based on the real-time system resources availability.

Why did my inventory closing or recalculation end in an error? What should I do?

Several issues can cause inventory closings, recalculations, or reversals to end in an error. To find the exact error message, check the **Batch jobs** page for the related batch job and the **Closing and adjustment** page after you select the specific voucher, then view *log* details. If you can't fix these issues, contact Microsoft Support or your partner.

Possible reasons for errors include:

- **Over utilization of system resources** – Because these operations consume a considerable amount of resources, try running these operations during off-peak business hours and minimizing other parallel and background processes and user sessions. Also check whether the closing parameters match your business requirements and resource availability. For more information, see the other sections of this article.

- **SQL deadlock or availability issues** – Most of the SQL issues are transient. Retry the operation after selecting the specific voucher and selecting **Resume Calculation**. If the issue persists, check for the exact error from the **Batch jobs** or **Closing and adjustment** page and share them with your partner.

- **Business data corruption and inconsistencies such as rounding or currency issues** – Run a consistency check for the relevant item. Go to **System administration** > **Periodic tasks** > **Database** > **Consistency check**. In the **Consistency check** dialog, set **Module** to *Inventory management*, and set **Check/Fix** to *Check* or *Fix error* (depending or your business requirements). Then expand the **Item** tree and select the checkboxes for **Inventory transactions** and **On-hand**. Then open the **More** menu (three dots) and select *Dialog*. Use the dialog to filter for the exact item you want to check. If required, you can run the check as a batch process in the background. You can view the final check/fix logs

from the batch job logs or in the notification panel. If the consistency check doesn't highlight or fix any issues, use the **Potential conflicts - inventory and general ledger** and **Inventory value report storage** reports to drill down further into the discrepancies and inventory postings respectively.

- **Incorrect business practices**
- **Configuration issues like ledger, account, project, or warehouse setup**

When should I use the on-hand inventory adjustment on the Closing and adjustments page?

You can run the on-hand inventory adjustment only after you complete an inventory close. Typically, you run it for the date after the last close. The on-hand adjustment can adjust only inventory that's still on hand as of the date when you run the adjustment.

When should I use the inventory transaction adjustment on the Closing and adjustments page?

Use the inventory transaction adjustment before you run an inventory close. Typically, use it to correct an incorrect receipt. You can't post the inventory transaction adjustment after you run an inventory close and settle the transaction.

Are purchase order returns treated like other issues during the inventory close?

Yes. A purchase order receipt is an issue that the heuristic model you select for the item settles to a receipt. If you use a periodic costing model, you can use marking to override the heuristic cost.

What happens to sales order returns during the inventory close?

When you run the inventory close, the system treats a sales order return as a receipt into inventory. The system settles issues against the inventory based on the heuristic model that you select for the item.

What cost price is used on a sales order return?

When you create a return that relates to a sales order, the system copies the value of the **Unit price** field from the original sales order. The system sets the **Return cost price** field on the return to the adjusted cost price from the original inventory transaction for the sales order line that you're returning. If you set the **Value open** option for the related inventory transaction to *Yes*, the inventory close can update the issue cost on the original sales order. The **Return cost price** field isn't updated in this scenario. However, when you post a return order packing slip, the system checks the cost price and uses the updated cost on the return inventory transaction.

For a return of a standard cost item that relates to a sales order, the system uses the standard cost from the time of the original sales order, even if a new standard cost is active for the item.

When you create a return that doesn't relate to a sales order, the system sets the **Return cost price** field to the active item price for the item in the site that you're creating the return order for. If you don't have an active cost price in a costing version for the item, the value is 0. If you leave the value as 0, you receive a warning that states that the return lot ID or return cost price isn't specified.

Why are my inventory transactions posted at a different cost price?

Issue transactions are always posted at the running average cost. Receipt transactions are posted at the cost specified during posting the document line, or the latest activated cost price.

The system estimates this running average cost price for an item by using the following formula:

Estimated price = (Physical amount + Financial amount) ÷ (Physical quantity + Financial quantity)

If the **Include physical value** option isn't selected for an item in its item model group, the system uses 0 for both the physical amount and the physical quantity.

Learn more in Running average cost price

Why is item cost inflated?

The running average cost can be inflated when you price several inventory issue transactions that don't have enough receipt transactions. This issue can occur when you enable both **Include physical value** and **Physical negative inventory** for the item model group. If you disable **Include physical value** but still see this issue, make sure that you have also disabled **Physical negative inventory** for the item model group.

Learn more in Avoiding pricing amplification.

Why does the cost change after running inventory closing or recalculation?

Inventory closing or recalculation adjusts on-hand inventory and inventory transactions. After these operations, a cost adjustment or settlement modifies the resulting cost of the inventory transactions.

Why does the system reopen and adjust already closed inventory transactions?

This behavior is by design and might occur for items that use *Weighted average (date)* as the valuation model.

When the system settles or adjusts inventory through inventory closing or recalculation, it creates a summarized transaction for all issues and receipts separately. These summarized transactions settle or adjust issues to receipts. If you post a manual adjustment to any inventory transaction involved in previous summarized transactions and then run a new closing or recalculation, the system reopens and resettles or readjusts all inventory transactions that make up the summarized transaction.

I'm getting abnormal inventory costs reported for an item. How can I use the system to check or fix the issue?

Run a consistency check for the relevant item by following these steps:

1. Go to **System administration** > **Periodic tasks** > **Database** > **Consistency check**.
2. In the **Consistency check** dialog, set **Module** to *Inventory management*, and set **Check/Fix** to *Check* or *Fix error* (depending on your business requirements).
3. Expand the **Item** tree and select the checkboxes for **Inventory transactions** and **On-hand**.
4. Open the **More** menu (three dots) and select **Dialog**.
5. Use the dialog to filter for the exact item you want to check. If required, you can run the check as a batch process in the background.
6. View the final check or fix logs from the batch job logs or in the notification panel. If the consistency check doesn't highlight or fix any issues, use the *Potential conflicts - inventory and general ledger* and *Inventory value report storage* reports to drill down further into the discrepancies and inventory postings.

Once this process is completed and if the valuation method for that item is periodic, execute closing or recalculation to adjust the valuation and on-hand inventory as per the valuation

method.

Why does my ledger trial balance differ from the inventory value after I run closing, recalculation, or reverse?

Normally, you resolve this issue by rebuilding the trial balance for the corresponding main accounts from **General Ledger** > **Inquiries and reports** > **Trial Balance**.

Verify whether the inventory closing, recalculation, or reverse execution completed successfully and whether voucher postings are present. Cross check the status of batch job with the voucher execution status. Check the logs in the Closing and Adjustment form for the corresponding voucher. Once verified, rebuild the trial balance for the specific account where you observe the differences.

If the issue persists, then there's probably an inconsistency between the general ledger and inventory postings. Try generating a potential conflicts report to trace the items causing conflicts.

Next, generate an inventory value report for those specific items to trace the inconsistent inventory transactions. Run a consistency check for those items for on-hand inventory and inventory transactions to check whether it resolves the data inconsistencies. If the consistency check fix makes any changes to the inventory postings, execute inventory closing again and rebuild the trial balances to check whether the balance matches.

If these steps don't resolve the issue, contact Microsoft Support or your partner.

Costing sheet and indirect costs

Which costing models support the costing sheet?

Although organizations that use standard costing typically use the costing sheet, you can use it with any costing model that's available in Supply Chain Management.

Can I have multiple costing sheets for various parts of my organization?

No. You can have only one costing sheet per legal entity.

Can I have different costing sheets for each site?

No, you can't have a different costing sheet for each site. You can create only one costing sheet for each legal entity. However, you can configure total nodes, cost groups, or indirect cost nodes per site. This configuration is a manual process, and you must maintain the hierarchy and item assignments on the costing sheet when organizational or operational changes occur. If a single item is produced or purchased in more than one site, there's no mechanism that lets you treat the item differently on the costing sheet for each site. All indirect cost codes have a rate or surcharge that is defined in your costing version. The costs that you define are always site specific.

Can I deactivate and activate versions of the costing sheet?

Although the system doesn't keep detailed version history for the costing sheet, you can make changes to the costing sheet and then, when you're ready, save and validate it. There's no mechanism that lets you go back to an older version of the costing sheet or view the changes that you made to the costing sheet. If you start changes and don't want them to take effect, you can close the page without saving and validating the changes. You're prompted to discard the changes.

Can I create indirect costs for each item?

On your costing sheet, you can create indirect cost codes where the **Node type** field is set to *Surcharge*, *Rate*, or *Output unit based*. You can then define the rate or surcharge so that it's specific to an item number. On the **Rate** or **Surcharge** FastTab, add a row to the grid. Set the **Valid for** field to *Table* and the **Relation** field to the specific item number.

Can I create an indirect cost that isn't related to a specific item?

Yes. You can create an indirect cost code where the **Node type** field is set to *Output unit based*. You can then set the **Subtype** field to *Quantity*, *Weight*, or *Volume* to specify the quantity, weight, or volume of the item that you're producing. The rate that you specify on the **Rate** FastTab is applied to the subtype that you selected. For example, you set the **Subtype** field to *Quantity* and the **Rate** field to *1.00 USD*, and then create a production or batch order for a quantity of 10. In this case, the indirect cost that is added to the finished good is 10.00 USD.

Can I use the costing sheet to split my production costs by hours and materials?

Yes. You can create **Total** nodes on your costing sheet to separate the costs by any grouping that you choose. For example, you can create one **Total** node that is named *Hours* and another

that is named *Materials*. Under the **Hours** node, add each code group that is related to your hours. Under the **Materials** node, add each cost group that is related to your materials.

Dimension groups

Can I manage cost at the batch or serial number level?

Yes, if you're using a periodic costing model such as FIFO, LIFO, LIFO date, weighted average, or weighted average date, you can enable the **Financial inventory** option for the **Batch** or **Serial number** dimension in the tracking dimension group to track costs at a detailed level.

Can I manage costs at the location level?

No, you can't enable the **Financial inventory** option for the **Location** dimension in the storage dimension group. If your organization must track costs at a more detailed level, consider whether you can create virtual warehouses, and then select the **Financial inventory** option for the **Warehouse** dimension in your storage dimension group.

Should I enable the Use warehouse management processes option for the storage dimension group?

If you think that you might want to use the warehouse management processes (WMS) features in the future, you should enable the **Use warehouse management processes** option. After you save a storage dimension group, you can no longer change the setting of the **Use warehouse management processes** option for it. If you decide to use warehouse management processes later, you'll have to create a new warehouse where the option is enabled. There's no automated process that you can use to move all the inventory from one warehouse to another warehouse, or to copy related configurations to a new warehouse.

Can I enable the Use warehouse management processes for the storage dimension group even if I'm not planning to use warehouse management processes (WMS)?

Yes, even if you don't plan to use the warehouse management processes (WMS) features, you can enable the **Use warehouse management processes** option for the storage dimension group. To create and process transactions, you need to complete the minimum configuration, such as reservation hierarchies and unit sequence groups. However, the settings for WMS are generally ignored when you manually process picking lists, packing slips, and product receipts (for example, on the sales order and purchase order pages).

When should I enable the Physical inventory option for a storage or tracking dimension group?

Enable the **Physical inventory** option for storage and tracking dimension groups when you need to keep detailed inventory records based on that dimension. Typically, any dimension that's active is also physically tracked. Therefore, the system tracks any receipt, issue, or movement of the inventory by the selected dimension. If a dimension isn't mandatory (for example, the license plate), you can enable the **Blank receipt allowed** and **Blank issue allowed** options to let users receive, issue, or move inventory even when the dimension isn't specified.

When should I enable the Financial inventory option for a storage or tracking dimension group?

Enable the **Financial inventory** option for storage and tracking dimension groups when you need to keep detailed financial records based on that dimension. The **Site** dimensions are always financially tracked, whereas other dimensions are optional for financial tracking. If you use a periodic costing model such as FIFO, LIFO, or weighted average, enabling the **Financial inventory** option for a dimension indicates that you make settlements only in cases where the receipt and issue have the same dimension values. For example, if you enable the **Financial inventory** option for the **Warehouse** dimension, you have a different cost in each warehouse, and receipts and issues from different warehouses can't be settled.

Can I make changes to a product, storage, or tracking dimension group after transactions exist?

After you create an item model group, you can change the setting of the **Coverage plan by dimension**, **For purchase prices**, and **For sales prices** fields if you select the **Active** checkbox for a dimension in the item model group. You can't make any other changes. For example, you can't enable or disable the **Active**, **Blank issue allowed**, **Blank receipt allowed**, **Physical inventory**, and **Financial inventory** options.

Can I change the product, storage, or tracking dimension group for a released product?

If the on-hand inventory for a product is 0 (zero), and you set the **Value open** option to *No* for all the inventory transactions, you can change the storage and tracking dimension groups on the released production page. You can't change the product dimension group after the record is created.

Item model groups

When should I enable the Stocked product option?

Enable the **Stocked product** option for any item that you want to track in your inventory. When you enable this option, the system keeps detailed inventory transactions that track the receipt and issue of the item. Typically, you enable this option for any tangible item that you keep in your warehouse. You should also enable this option if you plan to add a non-tangible item such as a service item to your bills of materials (BOMs) or formulas. If you don't enable the **Stocked product** option, the system doesn't track any inventory transactions in the inventory subledger, and the cost of the items is typically expensed into your general ledger. For example, you might use this option for shop supplies or services that aren't included in your BOMs or formulas.

When should I enable the Post physical inventory option?

Typically, you enable the **Post physical inventory** option when you also enable the **Stocked product** option. When you enable this option, the system tracks the physical update to the item on the inventory transaction. Use this option in coordination with parameters in Accounts payable, Accounts receivable, and Production control that specify that the physical update should create a voucher. You typically enable this option when you want the ledger to be updated whenever you physically update the inventory. If Supply Chain Management isn't your system of record for financials, you might want to disable this option.

When should I enable the post Financial inventory option?

Typically, you enable the **Post financial inventory** option when you also enable the **Stocked product** option. Use this option in coordination with parameters in Accounts payable, Accounts receivable, and Production control that specify that the financial update should create a voucher. You typically enable this option when you want the ledger to be updated whenever you financially update the inventory (for example, by invoicing sales orders and purchase orders, or ending a production order). If Supply Chain Management isn't your system of record for financials, you might want to disable this option.

When should I enable the Post to Deferred Revenue Account on Sales Delivery option?

Use the **Post to Deferred Revenue Account on Sales Delivery** option to indicate whether you want to recognize revenue in the general ledger when you post a packing slip for a sales order. Use this option when there's a long delay between the packing slip and invoice on a sales

order, or when it's not possible to invoice all sales orders in the period. Typically, disable this option if you post your sales order invoices immediately after you post the packing slip. In this way, you avoid generating extra general ledger postings that are immediately reversed when you invoice a sales order.

When should I enable the Accrue liability on product receipt option?

In most organizations, enable the **Accrue liability on product receipt** option for all item model groups, regardless of whether you have a stocked product or a not-stocked product. Use this option to post an accrual to the general ledger, based on product receipt price. Because there's typically a delay between posting of the product receipt for a purchase order and posting of the invoice, most organizations must recognize the liability on the balance sheet to comply with local regulations such as Generally Accepted Accounting Practices (GAAP). If Supply Chain Management isn't your system of record for financials, you might want to disable this option. If your organization uses procurement categories on purchase orders, you can control the accrual requirement by enabling the **Accrue purchase expense on receipt** option for the category policy rule on the **Purchasing policies** page.

How can I prevent a user from posting a purchase order product receipt if a receipt registration isn't yet posted?

You can prevent a user from posting a purchase order product receipt if a purchase order registration isn't yet posted by enabling the **Registration requirements** option for the item model group. Use this option in organizations that use a two-step receiving process, or in scenarios where you must register a batch or serial number, for example, on the items that you're receiving. The **Registration requirement** option applies to *all* inventory receipts for an item, not just to purchase orders. For example, it applies to an inventory journal that has a positive quantity and a production order report as finished journal.

How can I prevent a user from posting a purchase order invoice if a product receipt isn't yet posted?

You can prevent a user from posting a purchase order product invoice if a purchase order product receipt isn't yet posted by enabling the **Receiving requirements** option for the item model group. Use this option in organizations that require that receipts be physically recognized in the general ledger for accruals. The **Receiving requirement** option applies to *all* inventory receipts for an item, not just to purchase orders. For example, it applies to an inventory journal that has a positive quantity and a production order report as finished journal. If your organization uses procurement categories on purchase orders, you can control the

requirement for a receipt by enabling the **Receiving requirements** option for the category policy rule on the **Purchasing policies** page.

How can I prevent a user from posting a sales order packing slip if a sales order picking list isn't yet posted?

You can prevent a user from posting a sales order packing slip or invoice if a sales order picking list isn't yet posted by enabling the **Picking requirements** option for the item model group. Use this option in organizations that perform a physical picking process in the warehouse (for example, by using the warehouse mobile device to do picking). The **Picking requirement** option applies to *all* inventory issues for an item, not just to sales orders. For example, it applies to an inventory journal that has a negative quantity and a production order picking list journal.

How can I prevent a user from posting a sales order invoice if the sales order packing slip isn't yet posted?

You can prevent a user from posting a sales order invoice if a sales order packing slip isn't yet posted by enabling the **Deduction requirements** option for the item model group. Use this option in organizations that perform a physical packing process in the warehouse (for example, by using the Warehouse Management mobile app to do packing or by generating a document that will be included with the items that are shipped). The **Deduction requirement** option applies to *all* inventory issues for an item, not just to sales orders. For example, it applies to an inventory journal that has a negative quantity and a production order picking list journal.

Can I prevent items that are registered from being sold?

When you register an item in your inventory in Supply Chain Management, the quantity is physically available to be issued in the system. If items that you registered but didn't yet receive shouldn't be available to be issued to sales orders or production orders, for example, consider using the inventory status, inventory blocking, quality orders, quarantine orders, or reservations features to manage the business process.

Production costing

Can I use one costing model for raw materials and a different costing model for finished goods?

Yes, you can use different costing models for each item. It's common for manufacturers to use a periodic costing model for raw materials and standard cost for semi-finished and finished goods.

How can I drive overhead off machine costs?

To drive overhead off your machine costs, you must create resources and resource groups for each machine, according to your business requirements. You can assign each resource or resource group to cost categories to control the cost of the machine. You can link each cost category to a cost group, and use each cost group as the basis for calculating indirect costs on the costing sheet.

How do I recognize costs related to energy consumption (for example, water, energy, or gas consumption) on the costing sheet?

You can generally recognize costs related to energy consumption in one of two ways:

- Create a line in your BOM or formula. Typically, you create this line as a service item, and you can specify the unit of measure that you're consuming in relationship to the quantity you're producing. In this way, the user can consume a different amount during the production process. You can automatically consume the items based on the value that you select in the **Flushing principle** field.
- Create an indirect cost on your costing sheet. Typically, you use this approach to allocate the total cost of your energy consumption across your production process. You can use the cost group and the absorption basis to scale the consumption based on materials or labor in your route, for example.

Select the best option based on your reporting, reconciliation, and operational requirements.

Can I capture resource details in the BOM or formula?

You can capture resource details only in a route operation. Although you can create a service item to represent a resource and assign a cost to increase the cost calculation for a finished good, we don't typically recommend this approach. Instead, create a simple route that has one line to track the resource costs, and configure the operation so that it's automatically consumed at either the start or the end of the production order.

Can I view the calculation details if I manually enter the cost?

No. If you manually enter the price on the **Item price** page, the **View calculation details** and **Report calculation details** buttons aren't available. If you select the **Cost rollup by cost group** button for a cost price that you manually enter, a summarized line is shown, and all costs roll up to the finished good item.

Does the system calculate variances on a production order when I manually enter the cost?

Yes, the system calculates variances when you manually enter a standard cost. However, when you manually enter a standard cost instead of calculating it, all material, route, and indirect cost consumptions in the production order are considered a substitution variance. If there are additional variances, such as consumption of extra materials or labor, the system also records them as variances from the production BOM. Therefore, always run a calculation for items that have a BOM, route, or indirect costs.

How can I carry the variances from a subproduction order to the parent production order?

When you use a periodic costing model such as FIFO, LIFO, or weighted average, the costs from a subproduction are recognized in the heuristic model that you select for the items. If you require actual costing, use the marking principle to indicate which subproduction is issued to a parent production order. Alternatively, use the **Financial inventory** option for the **Batch** or **Serial number** dimension in the tracking dimension group.

How does the flushing principle affect consumption?

The flushing principle on a BOM, formula, or route line controls the timing and technique that the system uses to consume the item or labor. If you select *Start*, the system automatically consumes the item or labor when you start the production order. If you select *Finish*, the system automatically consumes the item or labor when you report the production order as finished. If you select *Manual*, a user must manually pick materials or record the time in a job or route card journal. BOMs and formulas also have an *Available on location* option. If you select this option, the system automatically consumes the items after it transfers them to the production floor location.

How should I run cost calculations if I have multilevel BOMs or formulas?

In general, start at the lowest level of your BOMs or formulas for the calculation. To make filtering easier when you mass-run cost calculations, use calculation groups to help segregate

products. Run the *Recalculate BOM levels* periodic job before you start to mass-run cost calculations. Each organization should consider the mix of products and define a strategy that meets the specific needs of your product and BOM or formula structures.

Transfer order costing

Is there a way to allocate freight to a transfer order cost?

Add charges to a transfer order to add costs. The charges code defines the debit and credit for the charge that you're adding. First, create charges codes in the **Inventory management** module. To add a charge to a transfer order, select **Charges** on the transfer order line that you want to add a charge to.

Variances

Can I treat variances differently, based on the site or warehouse?

There's no option to configure variance accounts by site. When you use standard costing for a released product, you can select the main account that is used for standard cost variance postings on the **Posting** page. You can select to configure the accounts for one item, a group of items, or all items. You can also configure one cost group, a group of cost groups, or all cost groups.

Can I separate variances that are the result of currency exchange rates from other types of variances?

If a variance results from a currency exchange rate difference between the purchase order price and the standard cost for an item, you can't separate the exchange rate difference from other variances.

Reporting

How many inventory value report configurations can I create and use?

You can create unlimited inventory value report configurations. Evaluate your specific reporting requirements and create the number of configurations that you need to meet those requirements. You need at least one inventory value report configuration to run the report or the report storage option.

Can I use the inventory value report to analyze the cost of an item in each warehouse?

Yes. You can enable the **View** or **Total** option for each **Warehouse** dimension in the inventory value report configuration. However, the report shows values only for dimensions where the **Financial inventory** option is enabled for the storage dimension group. For other dimensions, it shows blank columns. Learn more in Inventory value report examples and logic.

How can I view the inventory quantity as of a specific date with the weighted average?

You can use the **Inventory aging** report, which includes an **Average unit cost** column that shows the value of inventory as of a specific date. Learn more in Inventory aging report examples and logic.

How can I view which receipt transactions are settled against an issue transaction?

You can view the settlements for an inventory transaction by selecting **Settlements** or **Cost explorer** on the **Inventory** tab on the Action Pane of the **Inventory transaction** or **Inventory transaction details** page. If you select a receipt to view the issues that are related to the transaction, the cost explorer doesn't show the details. Details are shown only if you select an issue transaction.

How does the inventory value report show items that have a positive physical quantity and a negative financial value?

The inventory value report separates the physical and financial amounts and quantities into their own columns. The values that are shown on the report are as of the date range that you selected when you ran the report. The level of summarization that is shown depends on the settings that you selected. If an item has transactions that are physically received and financially issued, the quantities and values are summed independently. If you select to view the detail level as transactions, rows for each receipt and issue are shown separately, and the

physical and financial quantities and amounts, respectively, are shown. Learn more in Inventory value report examples and logic.

What is the impact of storage and tracking dimension groups on the inventory value report?

If you enable the **Financial value** option for a dimension in a storage or tracking dimension group, you can select the **View** or **Total** option for the dimension in the inventory value report configuration. If you select the **View** or **Total** option for a dimension where the **Financial value** option isn't selected, the column is blank in the report output. If you enable the **Financial value** option for a dimension in a storage or tracking dimension group, and you don't select the **View** or **Total** option on the inventory value report configuration, the system summarizes the values for the selected dimensions where they're financially tracked.

Can I customize the Power BI embedded reports for costing?

Yes, users who have the correct security permissions can update the report design canvas for any Power BI embedded report in Supply Chain Management. Learn more in Customize embedded reports in analytical workspaces.

Where can I view the variance analysis statement?

You can access the variance analysis statement by going to **Cost management** > **Inquiries and reports** > **Inventory accounting – analysis reports** or **Cost management** > **Inquiries and reports** > **Manufacturing accounting – analysis reports**. Both options open the same report, and the report has the same behavior.

Item prices and default costs

Can I maintain the cost for each product variant?

Yes. You can enable the **Use cost price by variant** option on the **Manage cost** FastTab of the **Released product** page to enable pricing by product variant. (This option is available only for product masters.) Then, on the **Item price** page (which you can open from either the **Costing version** page or the **Released product** page), you can select **Dimensions display** to specify whether you want to show the **Configuration**, **Size**, **Color**, or **Style** dimension. To save the setup and use the selected dimensions every time that you open the page, enable the **Save setup** option, and then select **OK**. You can enter the dimensions only for items where the dimensions are active in the product dimension group.

Can I maintain the cost for each warehouse?

No, you can't maintain the item price by warehouse. However, you can maintain trade agreements for purchase prices or sales prices by warehouse. First, select the **For purchase prices** or **For sales prices** option for the dimension on the **Storage dimension group** page. Then, when you create the trade agreement journal and open the lines for the dimensions, select **Inventory** > **Display dimensions** to select which dimension should be shown in the grid. To save the setup and use the selected dimensions every time that you open the page, enable the **Save setup** option, and then select **OK**. You can enter the dimensions only for items where the dimensions are active in the product dimension group.

What are price charges?

Price charges provide a way to add a fixed amount to the unit price of the item price or trade agreement. When you enter an amount in the **Price charges** field, you can also enter a value in the **Charges quantity** field. The price charges are amortized over the charges quantity that you specify. Enable the **Incl. in unit price** option if you want to include the price charges in the unit price for the item. This option is always enabled for a standard cost.

How should I configure prices for items that are procured in multiple currencies?

If you enter a default price in the **Purchase price** field on the **Released product** page, the system assumes the price is in the accounting currency of the ledger for the legal entity that you're in. Likewise, if you enter a purchase price in a costing version where the **Price type** field is set to *Purchase*, you assume the price is in the accounting currency of your legal entity. When you create a purchase order for a vendor that has a different currency, the system automatically converts the currency from the accounting currency amount to the transaction currency by using the exchange rate that you specify in the **Accounting currency exchange rate type** field in your ledger.

When you create a trade agreement journal, you can specify the currency that you're expressing the price in on each line. You can create trade agreements for multiple currencies, specific vendors, and many other combinations of factors. If you create a purchase order where a trade agreement exists for the currency that you selected, the system uses the trade agreement that has the matching currency. When you post transactions such as product receipts or invoices, the amount is converted to the ledger's accounting currency by using the accounting currency exchange rate that you specify in the ledger.

How should I configure costs for items that are procured in multiple currencies?

You can't configure costs in more than one currency. The cost that you specify for the item price or default costs that you enter in the **Price** field on the **Manage cost** FastTab of the **Released product** page are always expressed in the accounting currency of the ledger for the legal entity that you select.

If your organization uses standard costing, you should define a strategy for defining the costs when there are multiple currencies. You should also define a process for regularly updating the cost to help reduce the number of variances that are posted.

Can I use the Profit setting on the Cost group page to calculate sales prices?

Yes, you can use the **Profit** setting on the **Cost group** page to add a percentage when sales prices are calculated by using a cost calculation. Learn more in BOM calculations.

Marking

How does marking affect periodic costing models?

If you use a periodic costing model such as FIFO, LIFO, or weighted average, and you mark a receipt transaction against an issue transaction, the inventory close process ignores the heuristic model of the item. Instead, the system uses the actual cost of the receipt for the cost of the issue.

What happens during the inventory close when I use marking?

When you mark receipts and issues, the inventory close settles the marked transactions together. When marking is used to create the settlement, the **Principle** field on the **Settlement** page is set to *Marking*. If you mark a transaction before physically or financially updating it, the issue uses the marked receipt's cost instead of the running average cost. If you mark the transactions after the financial update, the inventory close and adjustment process adjusts the issue cost so that it matches the receipt cost.

Can I manually mark transactions when I use standard costing or moving average?

No, you can't manually mark receipts or issues when you use standard costing or moving average. If transactions (for example, direct delivery or intercompany orders) are automatically marked, the marking record remains and acts as a hard reservation. However, when you use standard costing or moving average, the marking of records has no effect on the cost of the items.

How does marking affect the profit and loss statement?

When you mark an issue transaction against a receipt, the cost for the issue matches the selected receipt. When you physically and financially post the issue, the posting affects the **Cost of goods sold, delivered** and **Cost of goods sold, invoiced** accounts that you specify in the inventory posting profile. If you mark a transaction after the physical or financial update, the *Inventory close and adjustment* process creates an adjustment that has a matching voucher that adjusts the **Cost of goods sold, invoiced** account and offsets to the account that you specify for **Cost of units, invoiced** (inventory).

How does marking affect master planning?

The **Standard update** tab on the **Master planning parameters** page includes a field named **Update marking**. The option that you select is used when you firm a planned order that master planning generates. The following options are available:

- *No* – The system doesn't perform any marking.
- *Standard* – The system marks receipts against issues according to the pegging. A requirement order is marked against a fulfillment order. If some quantity remains on the fulfillment, the fulfillment order isn't marked.
- *Extended* – The system marks both the requirement orders and fulfillment orders, regardless of whether any quantity remains open on the fulfillment order.

Negative inventory

When should I allow physical negative inventory?

In general, don't allow physical negative inventory, because you can't have less than zero of a tangible item in your warehouse. However, some industries and business scenarios require physical negative inventory. For example, retailers might not want to prevent the sale of an item that a customer brings to the register, even when the system indicates that no items are available. Process manufacturers provide another example. For these manufacturers, the amount that is consumed can exceed what is recommended in the formula. Alternatively, the

consumption might be estimated instead of precise, so that the consumption exceeds the amount in a specific location, such as a tank.

Whenever possible, evaluate your business process and try to improve it to ensure that inventory can't be negative. If you must allow negative inventory, establish a clear business process for correcting the negative inventory, because it can adversely affect costing.

When should I allow financial negative inventory?

In general, allow financial negative inventory. (In most cases, the system doesn't process purchase order invoices before you can ship the items.) Enable this option when you have specific business processes that require the sales price to reflect the final cost of a purchase order. For example, this requirement might apply in a make-to-order industry or in specific regions where it's dictated by law.

What happens to the cost of my issues when the inventory is negative?

When the inventory for your item is negative, and you issue more items than you physically have, the system uses the default item price to calculate the running average if you use a periodic costing model such as FIFO, LIFO, or weighted average. If you don't specify a default price for the item, the system issues the inventory with a value of zero. This behavior can cause future calculations of your running average or moving average to be inaccurate.

Can I prevent items from being picked, packed, or sold on sales orders and production orders if there isn't enough on-hand inventory?

Yes. Disable the **Physical negative** option for the item model group to prevent items from being picked, packed, or sold on sales orders and production orders.

Can I prevent items from being invoiced on a sales order if no purchase orders have been invoiced for the same item?

Yes. Disable the **Financial negative** option for the item model group to prevent items from being invoiced on a sales order if no purchase orders are invoiced for the same item.

How does negative inventory affect financial ratios such as gross profit margin?

When you allow your inventory to go negative, the inventory value in your balance sheet and the cost of goods sold in your profit and loss statement can be understated, especially if you don't configure a default price for your items. Therefore, financial reporting and ratios such as gross profit margins can be overstated, because the cost is incorrect. If you use a periodic costing model such as FIFO, LIFO, or weighted average, you can adjust the value of the issues when you run the inventory close and adjustment process after the negative inventory quantities are corrected. However, if you use moving average, there's no way to revalue the individual transactions.

How do I correct negative inventory?

Frequently monitor and correct negative inventory when your organization or business requirements dictate that you allow your inventory to go negative. You can correct the inventory values by performing a cycle count, posting an adjustment, or posting a movement journal. If you must recognize the unexpected gains in inventory in a specific general ledger account, plan to use a movement journal. Otherwise, when you use the cycle counting or inventory adjustment process, the system posts the inventory adjustment to the **Inventory receipt** account and offsets to the **Inventory expenditure, profit** accounts that you specify on the inventory posting profile.

Do I have to create a new item if my inventory goes negative and I use moving average?

No. If your organization allows inventory to go physically negative, and you're using moving average as your inventory model, the system uses the fallback cost sequence that you assign on the **Inventory and warehouse management parameters** page to determine how the system assigns cost to your issues. In general, avoid allowing your inventory to go physically negative. For more information, see the other questions in the Negative inventory section of this article.

Not-stocked products

Can I post sales order picking lists for not-stocked products?

When you generate a picking list for a sales order that includes items in an item model group that isn't stocked, you don't see any lines for those not-stocked items. You can't use the Warehouse Management mobile app to process not-stocked items.

When you generate a packing slip for a sales order that includes items in an item model group that isn't stocked, set the **Quantity** field to *Picked quantity and not stocked products* to include

the not-stocked items in the document generation. If you select *All*, only stocked items are included in the packing slip.

Should I use a not-stocked product or a category (sales category or procurement category)?

The choice between a not-stocked product and a category depends on your specific business requirements. Not-stocked products generally offer more control over default values, such as quantities and prices on purchase orders and sales orders. Therefore, use not-stocked products in scenarios where you purchase or sell the same product or service more than once. Categories are useful in scenarios where the price, items, descriptions, and so on are inconsistent from transaction to transaction. You can also use categories on any product to help classify the type of product that you're selling or purchasing.

Service items

What is the difference between a service item and a not-stocked product?

A service item is a product type. When you create a newly released product, you can set the **Product type** field to either *Item* or *Service*. Typically, select *Item* to indicate that the item is tangible, and select *Service* to indicate that the item is intangible.

Any released product, regardless of whether it's an item or service product, can be stocked or not-stocked. The stocked or not-stocked setting is controlled by the item model group that you select for the released product. When you select an item group that isn't stocked, the system doesn't create inventory transactions for the related sales order or purchase order, for example.

Can I include not-stocked items in a BOM?

No, you can't include a released product that is linked to an item model group where the **Stocked** option is disabled in a BOM or formula. It doesn't matter whether the **Product type** field is set to *Item* or *Service*. You can only include stocked items on BOM or formula lines.

Costing model–specific questions

Which costing model should I use?

The costing model you select depends on your business requirements. Before you select a costing model to use in Supply Chain Management, validate that your local regulations allow the model. We recommend that you validate your selection with a certified accountant in your region.

Can I use more than one costing model in my organization?

Yes. You can select as many item model groups or costing models as you need in Supply Chain Management.

Can I use more than one costing model for each item?

No. You can select only one costing model for each released product. The item model group controls this behavior. If you need to use more than one costing model to report on inventory values, consider using the Global Inventory Accounting add-in.

When I use manufacturing execution, which costing methodology should I use?

The costing model you select depends on your business requirements. There's no specific advantage or disadvantage to using any costing model when your organization also uses manufacturing execution.

When is FEFO used?

First expired, first out (FEFO) isn't a costing methodology. Instead, it's a reservation technique that's often used in manufacturing organizations. You can enable FEFO reservations for an item model group by enabling the **FEFO date-controlled** option and then selecting a value in the **Pick criteria** field.

Are there performance benefits to selecting one costing model over another?

In general, the costing model that you select for an item has minimal impact on the overall performance of the system. However, the inventory costing model that you select might affect the amount of time required to run the inventory close or recalculation process. For example, if you use standard cost or moving average, the inventory close process has minimal impact on the system, because it doesn't update each inventory transaction and create settlements. However, a periodic costing model such as FIFO, LIFO, or weighted average can increase the amount of time required to complete the inventory close process. Specifically, the close

process can be longer when you enter a high number in the **Maximum number of iterations allowed per item** field or a low number in the **Minimum amount allowed** field for the *Close inventory* process.

When should I use the Fixed receipt price option?

When you select the **Fixed receipt price** checkbox on the **Item model group** page for an item, the system uses the receipt price as the standard cost for the inventory receipt. If there's a difference between the purchase price and the default item cost price that you configure for a product, the system posts the difference to the **Fixed receipt price profit** or **Fixed receipt price loss** account, and offsets it to the **Fixed receipt price offset** account. You specify all these accounts on the **Posting** page.

Select the **Fixed receipt price** checkbox when you use a periodic costing model such as FIFO, LIFO, or weighted average, and you want to track a purchase price variance in the ledger if the price of a receipt differs from the default item cost.

Moving average

Is moving average the same as a floating average?

The terms moving average, floating average, and running average are often used synonymously. Of these terms, moving average is the only official costing model that's available in Supply Chain Management. Although running average isn't an official costing model, it's the technique that is used when you use a periodic costing model such as FIFO, LIFO, or weighted average. The term floating average isn't used in Supply Chain Management and might have different connotations in other systems.

How can I accommodate the difference between the product receipt price and invoice price when I use moving average?

When you use moving average, the physical cost (receipt price) is used to calculate the moving average for all issue transactions. If there's a difference between the physical cost (receipt price) and the financial cost (invoice price), the system automatically posts the difference to the main account that you specify for the **Price difference for moving average** posting type on the **Inventory** tab of the **Inventory posting profile** page.

Where is the price difference for moving average presented in the general ledger?

When there's a price difference between the posting of a physical update and a financial update for a receipt, the system posts the difference to the main account that you specify for the **Price difference for moving average** posting type on the **Inventory** tab of the **Inventory posting profile** page. Learn more in Moving average.

When I use moving average, what happens if there's an issue before the receipt?

Typically, there might be an issue before the receipt either because you allow physical negative inventory for the item model group or because the issue is backdated. Learn more in the Negative inventory section of this article.

If you backdate transactions, carefully consider your business process and operations to determine whether there's a way to avoid this scenario. If you backdate a transaction for an item that uses moving average, the system assigns the current moving average to the transaction. Later issues aren't adjusted. Learn more about moving average with backdated transactions in Moving average.

Standard costing

What is the difference between standard costing and fixed receipt price?

Standard costing requires that you define an item price and activate the cost in a costing version. The system uses that cost for all receipts and issues. The system records variances for the receipts to inventory in the general ledger by using the standard cost variance accounts that you specify on the **Standard cost** tab of the **Inventory posting profile** page.

However, when you use fixed receipt price, only the cost for receipts is fixed. The system uses the cost that you specify on the **Manage costs** FastTab of the **Released product** page. Differences between the default cost and the purchase order price cause the purchase price variance to be posted to the fixed receipt price profit and loss accounts. Issues don't use fixed receipt price. Instead, issues are valued at the running average at the time of posting (unless you use marking), and they're revalued to the heuristic model that you select when you run the inventory close.

Can I use FEFO reservations with standard costing?

Yes, you can use FEFO reservations on an item model group when you select standard costing. FEFO reservations control the reservation logic that's used for the physical handling of the

items, whereas standard costing controls the physical and financial cost for an item.

Can I upload pending prices?

Yes, you can use the Excel add-in or the Data Management Framework to upload a pending price. Use the following entities:

- Pending item prices (V2)
- Pending route cost category unit costs
- Costing sheet node calculation factors (V2)

How often can I update the standard cost for an item?

You can activate a new standard cost as often as you want. If you activate a new cost for an item on the same day that you activated the last cost, the system uses the most recent activated cost on new transactions or updates (such as updates to existing transactions).

Can I deactivate or delete an activated cost?

No, you can't deactivate or delete an activated cost. If you incorrectly activate a cost, you can activate a new cost that has the correct cost.

Are calculation groups used with standard costing?

You can use calculation groups with any item, regardless of the item model group that you choose. The cost rollup or cost calculation process uses the calculation groups to determine the settings for the items that you're running the calculation for. Learn more about calculation groups in BOM calculations groups.

When should I use a planned costing version?

Costing versions can have a type of *Standard cost* or *Planned cost*. Use the *Standard cost* type for costs that are active in the system and that you post in transactions. Use the *Planned cost* type for running simulations on costs. This type doesn't affect the cost of transactions.

Can I transfer the total cost from one entity to another entity as the selling cost?

There's no automated way to copy costs from one company to another. Additionally, there's no automated way to copy costs from a purchase price to a sales price. If your organization must

complete one of these tasks, consider whether you can use the Data Management Framework to export the data out of your costing version and upload it into a different company, either as a sales price in the costing version or as a trade agreement. You might need to manually manipulate the files.

What's the best way to copy planned costs to a standard costing version?

Use the **Copy** button on the **Costing versions** page to copy item prices, cost category prices, or indirect costs from one costing version to another.

Last updated on 11/26/2025

Prerequisites for standard costs

This article describes the basic steps for using standard costs. Subsequent steps depend on the company's operations. For example, the steps differ for a non-manufacturing environment, a manufacturing environment that doesn't use routings, and a manufacturing environment that uses routings.

To set up standard costs, follow these steps:

1. Create an item model group for standard costs.

   Use the **Item model groups** page to create a new group for standard costs, and assign an inventory model of *Standard cost*. The identifier for the item model group should be meaningful, such as *Std Cost*. Select the check boxes to indicate that the group should allow financial negative inventory, and that it should post physical inventory and financial inventory. This standard cost group is assigned to items.

2. Define ledger accounts that are related to standard cost variances.

   Use the **Chart of accounts** page to define ledger accounts that are related to standard cost variances. These ledger accounts must be defined before they can be assigned on the **Posting** page. The ledger accounts can reflect item groups and cost groups.

3. Assign ledger accounts to item postings that are related to standard cost variances.

   Use the **Posting** page to assign the ledger accounts that are related to standard cost variances. You can specify a variance's ledger account by item (or item group) and by cost group (or cost group type), or you can specify that the ledger account applies to all items and all cost groups. These options correspond to cost relations for tables, groups, and all.

   Before you define the item posting rules, use the **Transaction combinations** page to enable cost relations (for tables, groups, and all).

4. Define inventory parameters that are related to standard costs.

   a. Use the **Inventory accounting** tab on the **Inventory accounting policies setup** > **Parameters** page to define two cost control parameters that are related to standard costs.

- In the **Cost breakdown** field, select *None* or *Sub ledger*. If you select *Sub ledger*, the cost breakdown is an *active* cost breakdown. An active cost breakdown is critical for calculating, retaining, and viewing cost group segmentation across a multilevel product structure for standard cost items. When the cost breakdown is active, you can report and analyze inventory, work in process (WIP), and cost of goods sold (COGS) per cost group in a single-level, multilevel, or total format. When the cost breakdown is active, if you activate a manufactured item's cost, the cost group segmentation is stored in the item's cost record.

- If you select *None*, cost group segmentation won't be maintained for standard cost items. In other words, a manufactured item's standard cost is calculated as a single amount, without cost group segmentation. The cost contributions of manufactured components are aggregated into the single amount.

b. In the **Variances to standard** field, select *Summarized* or *Per cost group*. If you select *Per cost group*, you can identify purchase price variances and production variances by cost group. You can also identify the four types of production variances: the lot size, quantity, price, and substitution variances. If you select *Summarized*, you can't identify variances by cost group, and you can't identify the four types of production variances. You can just view a summarized production variance.

   The policy about variance to standard works independently of the cost breakdown policy. In other words, you can select a cost breakdown policy of *None* and select variances per cost group, so that production variances by cost group will still be captured.

5. Create costing versions for standard costs.

   Use the **Costing version setup** page to create one or more costing versions for standard costs. Each costing version must be designated by a costing type of *Standard cost* and must allow content to include cost data.

6. Prepare an existing customer to use standard costs.

   Customers who want to change their existing items to a standard cost inventory model must use the **Standard cost conversions** page.

Related information

Help articles

- Standard cost conversion overview

Community blogs

- How to set up standard costs for direct materials in finance and operations
- Standard direct labor costs in finance and operations

Last updated on 12/30/2025

Standard cost conversion overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides a process overview to help you set up and run a standard cost conversion. Complete the steps after you complete the prerequisites for a standard cost conversion.

Use the **Standard cost conversions** page to convert the inventory model for a batch of selected items from an actual costing approach to a standard costing approach. The conversion process involves a prerequisite inventory close, several steps during a transition period, which is defined by a transition start date and a planned conversion date, and then the conversion and an associated inventory close.

- Inventory close before the transition period – An inventory close is a prerequisite step because it settles an item's open transactions by using the old valuation method. During the transition period, you can enter and post backdated transactions, such as invoices, so that you can close the previous period. The inventory close date must be one day before the transition start date to ensure a clean break from the old valuation method.
- Conversion steps during the transition period – Use the **Standard cost conversions** page to create a conversion record that also contains the user-defined identifier for a new costing version. You identify the items that require conversion, and then enter the items' pending standard costs in the new costing version. You perform a check of the selected items to identify issues that might prevent conversion, resolve the issues, and then perform another check. After the items pass the checks, you change the status of the conversion record to *Ready*. On the planned conversion date, you perform the conversion and optionally include an inventory close. An item's inventory movements during the transition period are posted and valued according to the old inventory model. Then, after the conversion is successfully completed, the inventory movements are revalued to standard cost.
- Inventory close before the conversion – You can include the inventory close as part of the conversion on the planned conversion date, or you can perform it as a separate step before the conversion.

After the conversion process is successfully completed, the inventory model for each item is based on standard cost, and the item's standard cost is enabled. Subsequent inventory transactions are valued at the item's standard cost. In addition, the system converts the item's physical inventory transactions for receipts and issues to standard cost based on the conversion date. The system also converts the item's financial on-hand inventory to standard cost, and posts the difference in value as an inventory revaluation. Any transactions that occur after the conversion are valued at the item's standard cost. You can't enter backdated transactions before the conversion date, because an inventory close must be performed one day before the conversion date. A conversion can be performed only if an inventory close was performed one day earlier. You can't cancel this inventory close.

Step 1: Define a standard cost conversion record and the associated costing version

Use the **Standard cost conversions** page to create a conversion record. You can create a new conversion record only when existing conversion records have been completed. The duration of the planned transition period is defined by the transition start date and the planned conversion date. A planned transition period can be as short as one day. A planned transition period helps to ensure that the conversion process has enough time to complete all its steps. An inventory close must be performed on a date that is one day before the transition start date, to help ensure that settlements are completed before you start the conversion process. To make sure that the transition start date and inventory close date are correctly aligned, you can either change the transition start date to one day after an existing inventory close or perform an inventory close. When you enter a conversion record, also enter a user-defined identifier for a new costing version that contains the standard costs for converted items. The costing version is generated automatically when you save the conversion record.

Step 2: Review and change the new costing version for the conversion record

The new costing version is dedicated to the conversion record, as the *Conversion* costing type indicates. The dedicated costing version is similar to a costing version for standard costs and contains the item cost records for items that are associated with the conversion record. The dedicated costing version for a conversion record has the following settings, which you should review and modify on the various tabs as required:

- **Costing type:** Set this field to *Standard cost*.

- **Version:** The identifier reflects the information entered on the conversion record for the costing version ID.
- **Name:** By default, the name is blank. You can optionally enter a name.
- **Block:** Set this field to *No*. You can enter cost records into the costing version until you change the status of the conversion record to *Ready*. A status of *Ready* indicates that the selected items are checked, and that changes to cost records aren't permitted.
- **Block activation:** Set this field to *Yes*. You can't manually activate a pending cost record in the dedicated costing version. Activation occurs when you perform the conversion.
- **From date:** The from date reflects the planned conversion date that is entered on the conversion record.
- **Site:** Leave this field blank, so that cost records can be entered for any site.
- **Allow price type field group:** Set this field to *Yes*, so that only cost price records can be entered.
- **Fallback principle:** Set this field to *None*. Change the fallback principle to *Active* if you require cost information that is activated in other costing versions. For example, cost information about components, cost categories, and indirect costs might be required in order to calculate the costs of manufactured items.
- **Fallback costing version:** Leave this field blank.

You can maintain item cost information in the dedicated costing version only from the **Standard cost conversions** page. You can't use the **Costing version setup** page or the **Costing version maintenance** page to calculate costs for the costing version during conversion. However, you can use these pages to maintain the dedicated costing version after you complete the conversion process.

Step 3: Identify the items to convert to standard cost

Use the **Standard cost conversions** page to identify the individual items that you want to convert to standard cost. You can add multiple items by using the **Add items to standard cost conversion** page. In general, include all manufactured items in a single conversion record to help ensure that costs are calculated correctly.

Step 4: Enter or calculate the pending standard cost for each item that is being converted

Use the **Item price** page to enter pending standard costs in the dedicated costing version for purchased items and transfer items. Cost records are site-specific, and an item's pending costs must be entered for every site. Use the **Item price** page to calculate pending standard costs for

manufactured items. Calculate a manufactured item's pending costs for every manufacturing site, unless the site represents a transfer site. In this case, enter the pending costs manually. Some items might have color, size, or configuration product dimensions. On the **Standard cost conversions** page, the **Use cost price by variant** check box shows the standard cost for every combination of product dimensions. When this check box is cleared, you must enter only a pending cost for the item.

Step 5: Check and resolve any issues for the items that are being converted

Use the **Standard cost conversion checks** report to identify issues for the items that are converted. If an item doesn't have any issues, its status in the conversion record changes to *Checked*. If an item has issues, resolve the issues and then run the report again until the item's status changes to *Checked*. If you can't resolve an item's issues in a timely manner, you can optionally delete the item from the conversion record and then convert the item later.

Step 6: Change the status of the conversion record to Ready

When the status of the conversion record is changed to *Ready*, the system performs a final check before it runs a standard cost conversion. The status is changed to *Ready* only if the following conditions are met:

- Every item in the conversion record has a status of *Checked*.
- An inventory close was performed on a date that is one day before the transition start date. To make sure that the transition start date and inventory close date are correctly aligned, you can either change the transition start date to one day after an existing inventory close or perform an inventory close.

Step 7: Back up the database before conversion

The backup lets you restore the database if errors are encountered during the conversion.

Step 8: Perform the conversion when the conversion record has a Ready status

The conversion process requires that an inventory close be performed on a date that is one day before the planned conversion date. This requirement helps ensure that back-dated transactions

can't be entered during the transition period. If an inventory close isn't yet performed, you're asked if you want to perform it as part of the conversion process. The conversion process handles one item at a time. It starts with the lowest items in a product structure, based on the item's low-level code. When an item is successfully converted, its status in the conversion record changes to *Converted*. If the conversion process is interrupted, any items that aren't successfully converted still have a status of *Checked*. Successful completion of the conversion process has the following effects:

- The status of the conversion record changes from *Ready* to *Completed*, and the status of each selected item changes from *Checked* to *Converted*.
- The item model group for converted items changes so that it reflects a new group that has a standard cost inventory model.
- The standard costs for the converted items are enabled in the dedicated costing version.
- The costing type of the costing version changes from *Conversion* to *Standard cost*, and the costing version is now like other costing versions for standard costs.

Step 9: Validate and reconcile the inventory values for the converted items

The **Variance analysis statement** report lets you analyze revaluation variance and the **Inventory value** report lets you view inventory value on a specific date.

- Analyze revaluation variances. Use the **Variance analysis statement** report to view inventory revaluation variances for the converted items. You can also use the **Standard cost transactions** page to view the inventory revaluation transactions for the converted items that have inventory.
- Analyze the inventory value before the transition start date. Use the **Inventory value** report to view inventory values for the converted items. For the To date for the report, use a date that is one day before the transition start date.
- Analyze the inventory value before the conversion date. Use the **Inventory value** report to view inventory values for the converted items. As the To date for the report, use a date that reflects one day before the conversion date.
- Analyze the inventory value on the conversion date. Use the **Inventory value** report to view inventory values as of the conversion date. Both the From date and the To date for the report should match the conversion date. The report selection criteria should reflect the converted items.
- Analyze back-dated inventory movements. Use the **Inventory value** report to view back-dated inventory movements that were entered after the conversion. The "From date and the

To date for the report should correspond to the transition start date and the conversion date, minus one day. The report selection criteria should reflect the converted items. The report shows inventory movements that were made at standard cost during the transition period.

Related information

- Prerequisites for a standard cost conversion

Last updated on 05/20/2026

Prerequisites for a standard cost conversion

This article discusses tasks to perform before you run a standard cost conversion.

Before you run a standard cost conversion, follow these steps:

1. Define an **item model group** for standard costs. Use the Item model groups page to create an item model group that uses a standard cost inventory model. When setting up a standard cost conversion, you assign this item model group to the items that are being converted.

2. Assign a **cost group** to each item.

    - The cost group that is assigned to a purchased item can act as the basis for assigning ledger accounts that are related to standard costs. This can include assigning ledger accounts for purchase price variances. In a manufacturing environment, the cost group that is assigned to purchased components also provides cost segmentation in the calculated costs of a manufactured item.
    - The cost group that is assigned to a manufactured item can act as the basis for assigning ledger accounts that are related to standard costs, such as assigning ledger accounts for production variances.

3. Assign a *standard order quantity* to a manufactured item when it has constant costs. The standard order quantity for a manufactured item acts as an accounting lot size for amortizing, or prorating, constant costs. These can include setup times in routing operations or a constant component quantity in a bill of material (BOM).

4. Assign *general ledger accounts* that are related to standard costs, especially the revaluation variance. Use the **Posting** page (**Inventory management** > **Setup**) to assign general ledger accounts that are related to standard costs. As a minimum for the standard cost conversion process, you must assign the account for the revaluation variance for all items and all cost groups. Use the **Chart of accounts** page to define the general ledger accounts that will be needed for standard costs. Use the **Transaction combinations** page to enable cost relations (for tables, groups, and all) before you define the item posting rules.

5. Define the inventory parameters that are related to standard costs. Use the **Number sequences** tab on the **Inventory and warehouse management parameters** page to assign a number sequence to revaluation vouchers. A revaluation voucher is generated when the

standard cost conversion creates a change of an item's inventory value. Use the **Inventory and warehouse management parameters** page to define Cost Control parameters (on the **Inventory accounting** tab) to define two parameters that are related to standard costs.

- Use the **Cost breakdown** field to select No or subledger. The selection of subledger is termed an active cost breakdown. An active cost breakdown is very important for calculating, retaining, and viewing cost group segmentation across a multilevel product structure for standard cost items. When the cost breakdown is active, you can report and analyze the following in a single level, multi-level, or total format:
  - Inventory
  - Work in process (WIP)
  - Cost of goods sold (COGS) per cost group

  An active cost breakdown means that if you enable a manufactured item's cost, the result is stored in the cost group segmentation in the item's cost record. If you put no value in the **Cost breakdown** field, the cost group segmentation isn't maintained for standard cost items. That is, a manufactured item's standard cost is calculated and maintained as a single amount without cost group segmentation, and the cost contributions of manufactured components are aggregated into the single amount.

- Use the **Variances to standard** field to select summarized or per cost group. The selection of per cost group enables you to identify purchase price variances and production variances by cost group. This also enables you to identify the four types of production variances (the lot size, quantity, price, and substitution variances). If you select summarized, you can't identify variances by cost group, and you can't identify the four types of production variances. You can only view a summarized production variance. The policy about variance to standard is independent of the cost breakdown policy. That is, you can select a cost breakdown policy of none, and select variances per cost group, so that production variances by cost group is still captured.

Last updated on 11/26/2025

Manage standard cost updates

Updates to standard cost data can be managed by using two different approaches—the one-version approach or the two-version approach.

The one-version approach uses a single costing version that contains all cost records. These records include the original costs and all cost updates.

The two-version approach uses one version that contains records of the original costs and a second version that contains records of all cost updates. A primary advantage of the two-version approach is the clear delineation and tracking of cost updates in a separate costing version, without affecting the original costing version. The two-version approach can be used to identify multiple incremental updates, where each incremental update has a separate costing version that contains the incremental cost records.

Example

The following example illustrates how the one-version and two-version approaches can be used for updating standard costs in a manufacturing environment. For example, updates that reflect new items or error corrections. Assume that a single costing version represents the standard costs for the current year. The identifier for this version is 2020-STD. Version 2020-STD contains the current active costs for all items. Additionally, it contains all routing-related cost categories and overhead calculation formulas that were known at the start of the year 2020. 2020-STD is the original costing version.

- **One-version approach to cost data updates** – In the one-version approach, the original costing version 2020-STD contains all cost records. Cost updates are recorded in 2020-STD and set to a status of Pending. The pending costs can be manually entered for new purchased items, or they can be calculated for a manufactured item to reflect corrections. When the one-version approach is used, the bill of material (BOM) calculations don't require a fallback data source because all active costs are contained in the costing version. After the pending costs become active, the original costing version 2020-STD will again contain all the current active costs.
- **Two-version approach to cost data updates** – The two-version approach requires an additional costing version that contains only the cost updates. The identifier for this version is 2020-STD-CHANGES. Cost updates are recorded in 2020-STD-CHANGES and are set to a status of Pending. With the two-version approach, the BOM calculations of pending costs

for manufactured items require a fallback data source. This is because the additional costing version 2020-STD-CHANGES contains only a subset of cost data. The fallback can be expressed as the active costs or as the costing version 2020-STD, because both identify the source of cost data when it isn't included in 2020-STD-CHANGES. After the pending costs become active, the costing version 2020-STD-CHANGES will contain the current active costs that reflect the updates, whereas the original costing version 2020-STD won't be affected. When the two-version approach is used, blocking policies for the original costing version should be set up to prevent updates. Identical blocking policies should be set up for the additional costing version, except for the specified from-date and the selective use of blocking policies to allow for updates. The specified from-date should be updated with each batch of changes to reflect the scheduled activation date.

This example used one additional costing version for managing updates throughout the year 2020. More than one additional costing version can be used, such as a separate version for each batch of updates. When more than one additional costing is used, the fallback must be expressed as the active costs, because the active costs are spread over multiple costing versions.

Financial dimensions for the standard cost revaluation

Activating a new standard price will typically revaluate the on-hand inventory value by standard cost revaluation transactions. Usually, the financial dimensions of the item are then posted on the transactions. However, if you would like to control whether and how the financial dimensions are posted, use feature management to turn on the feature named *Options of defaulting financial dimensions for inventory standard cost revaluation*. After enabling this feature, go to **Cost management** > **Inventory accounting policies setup** > **Parameters** and set the new **Origin of financial dimension** drop-down list to one of the following values:

- *None* – No financial dimensions are posted on the revaluation transactions. If your account structure includes a required financial dimension, the revaluation process will still run, but it will create accounting entries that have no financial dimensions. In this case, users will receive a warning message first, so they can cancel the revaluation if necessary.
- *Table* – The financial dimensions of the item are posted on the revaluation transactions. This is the default setting and is consistent with the original system behavior without turning on the feature *Options of defaulting financial dimensions for inventory standard cost revaluation*.
- *Posting* – The financial dimensions of the transaction that is being revalued are posted on the revaluation transactions. By default, the financial dimensions from the original transaction's inventory account will be used for both the inventory account and the revaluation account.

Last updated on 11/26/2025

Restrictions on costing versions for standard costs

This article describes the restrictions that apply to a costing version for standard costs.

The following restrictions help guarantee adherence to standard costing principles:

- Charges must be included in an item's cost. The charges for a manufactured item represent the amortized constant costs in the bill of materials (BOM) and route information. Therefore, the charges must be included in the unit cost. The charges for a purchased item are also included in the item's unit cost.
- Calculation of standard costs for manufactured items must be based on the cost records in a costing version for standard costs. Alternative sources of cost data can be used only with a costing version for planned costs, such as purchase price trade agreements for purchased items. Alternative sources of cost data are defined by the BOM calculation group.
- BOM calculations must be performed in a single-level explosion mode.

The item cost data for standard costs can be copied to another costing version that contains standard costs or planned costs. However, the item cost data for planned costs can't be copied to a cost version that contains standard costs, because the restrictions that are listed earlier in this article don't apply to planned costs.

Related information

- Costing versions overview
- Update standard costs in a non-manufacturing environment
- Prepare to maintain standard costs for manufactured items

Last updated on 11/26/2025

Information used in BOM calculations with standard costs

Article • 02/11/2025

Bills of material (BOM) calculations use data from several sources to calculate the standard costs of a manufactured item. The sources include information about items, bills routings, indirect cost calculation formulas, and the costing version.

The purchased item information that is used in a standard cost BOM calculation includes the following:

- Cost – A purchased item's costs are maintained as site-specific cost records in a costing version for standard costs. Each cost record has an effective date, and the BOM calculation date determines which cost record will be used. For example, a BOM calculation with a future calculation date might use a cost record with a pending status and a future effective date.
- Cost group – The cost group that is assigned to a purchased item provides the basis for cost segmentation in the calculated costs of a manufactured item.
- Warning conditions that are embedded in the item's BOM calculation group enable the BOM calculation to identify potential problems. This can be when the purchased item has a zero cost, a zero quantity in a BOM, or an out-of-date cost record. The applicable warning conditions can be overridden when initiating a BOM calculation.

The manufactured item information that is used in a standard cost BOM calculation includes the following:

- Standard order quantity for inventory – The item's standard order quantity for inventory, acts as the default accounting lot size for amortizing constant costs in a BOM calculation. The accounting lot size will also reflect the order quantity multiple if it's specified.
- Warning conditions that are embedded in the item's BOM calculation group enable the BOM calculation to identify potential problems. One example could be that the manufactured item doesn't have a BOM or route. The applicable warning conditions can be overridden when initiating a BOM calculation.

The bill of material information that is used in a standard cost BOM calculation includes the following:

- BOM version – The BOM version that is assigned to the manufactured item has effective from and to dates, and a status for approved and active. The bill version

can be company-wide or site-specific, and it can optionally reflect quantity breakpoints.

- BOM line item quantity − A component typically has a variable quantity required, but it can be a constant. The component quantity is typically expressed for producing one parent item, but it could be expressed per 100 or per 1000 to handle decimal precision issues. The component quantity can also be calculated based on measurements.
- BOM line item scrap − A component can have a variable or constant quantity for planned scrap.
- BOM line item valid dates − A component can have valid from and to dates.
- BOM line item type of production − The costing lot size for amortizing constant costs reflect the BOM calculation quantity and a make-to-order explosion mode, because the BOM calculation assumes that the manufactured component will be produced to the exact quantity instead of its standard order quantity.
- Sub-BOM for a manufactured component − A manufactured component can optionally have a specified BOM version, which would be used instead of the item's current active BOM version in a BOM calculation.
- Sub-route for a manufactured component − A manufactured component can optionally have a specified route version, which would be used instead of the item's current active route version in a BOM calculation.
- Operation number and the impact of operation scrap percentages − The operation number links a component to a specific operation, and operations can have a scrap percentage. The linkage enables BOM calculations to account for scrap percentages and cumulative scrap percentages across multiple operations on the component's required quantity.
- Ignore BOM line item in cost calculations − A component can be ignored for BOM calculation purposes.

The operations resource information that is used in a standard cost BOM calculation includes:

- Hourly costs − The hourly costs that are associated with an operations resource are expressed as cost categories that are assigned to set up time and run time. These cost categories should be identified for resource groups and operations resources. The hourly costs that are associated with a cost category can be site-specific or company-wide.
- Per unit costs − Some manufacturing environments assign operations resource costs per unit of output, which would be expressed as a different cost category for quantity. For example, the quantity-related costs can reflect piece rates for labor, or a paint manufacturer might assign operations resource costs per gallon of output.

- Overriding operations resource information on routing operations − The resource information about cost categories will be inherited by operations, where it can be overridden. BOM calculations use the cost category information that is specified on the routing operations.
- Cost group for a cost category − The cost group that is assigned to a cost category provides cost segmentation in the calculated costs of a manufactured item. For example, different cost groups might be used to segment the calculated costs that are associated with machines and labor or with setup and run time.

The route information that is used in a standard cost BOM calculation includes:

- Route version − The route version that is assigned to the manufactured item has effective from and to dates, and a status for approved and active. The route version must be site-specific, and it can optionally reflect quantity breakpoints.
- Routing operation time − The time can be specified for runtime and setup time. The hourly time is typically expressed for producing one parent item, but it might be expressed per 100 or per 1000 to handle decimal precision issues.
- Routing operation time for secondary resources − A secondary resource has the same operation number as the primary resource, and the same routing operation time. For example, an operation might require a machine as the primary resource and labor and tools as secondary resources.
- Routing operation scrap percentage − BOM calculations account for scrap percentages and cumulative scrap percentages across multiple operations. This applies to the required time for routing operations and the required quantity for components that are linked to operation numbers.
- Cost categories for a routing operation − Operations resource information about cost categories will be inherited by operations, where it can be overridden. BOM calculations use the cost category information that is specified on the routing operations.

The manufacturing overhead information that is used in a standard cost BOM calculation includes:

- Surcharge − A surcharge calculation formula reflects a percentage of value, such as 100 percent, that is tied to a specific cost group, such as labor.
- Rate − A rate calculation formula reflects an amount per unit, such as USD 10.00 per hour, that is tied to a specific cost group, such as labor.
- Time-based versus material-based overhead − The manufacturing overhead can be tied to routing operations or material components.

The costing version information that is used in a standard cost BOM calculation includes:

- Costing type – The costing version must contain standard costs. Several restrictions apply to BOM calculations that use standard costs. For example, these restrictions specify that miscellaneous charges must be included in unit costs and that the BOM calculation explosion mode must be single level.
- Mandated fallback principle – The costing version can mandate the use of a fallback principle, such as using a specified costing version or the active cost records. The mandated fallback principle typically applies to a costing version that represents the incremental updates in a two-version approach for cost updates.
- Blocking flag for pending costs – A blocking flag can prevent BOM calculations of the pending cost for manufactured items.
- Specified from-date – The specified from-date acts as the default calculation date for all BOM calculations that involve the costing version.
- Specified site – A specified site limits BOM calculations to the single site.
- Content of the costing version must include costs – The content must include costs. It can optionally include sales prices in order to calculate suggested sales prices for manufactured items.

Several sources of information can be specified when initiating a BOM calculation. This includes the site, the calculation date, and the costing version.

Feedback

Was this page helpful? Yes No

Provide product feedback

Update standard costs in a manufacturing environment

This article provides guidance about how to update standard costs in a manufacturing environment.

Updates can reflect new items, cost categories, or indirect cost calculation formulas. They can also reflect corrections and cost changes. The type of update affects the steps that you must complete to update standard costs, as illustrated in the following cases:

- Enter expected standard cost changes for purchased items, and then change the status of the item cost records to **Active** on the appropriate date. However, don't recalculate the costs of manufactured items that use the purchased items.
- Enter standard costs for a new purchased item, but don't recalculate the costs of manufactured items that have a bill of materials (BOM) version that contains the new purchased item as a component.
- Correct or change the cost of a purchased item, or change the cost group that is assigned to a purchased item, and calculate the cost for all manufactured items that have a BOM version that contains the purchased item as a component.
- Change the cost for a cost category, and calculate the cost for all manufactured items that have a route version that contains routing operations that use the cost category.
- Change the cost categories that are assigned to routing operations or the cost group that is assigned to cost categories. Then calculate the cost for all manufactured items that have a route version that contains routing operations that use the cost category.
- Change an indirect cost calculation formula, and calculate the cost for all manufactured items that are affected by the change.
- Change or add a manufacturing site for a manufactured item, and calculate the item's manufactured cost for the site.
- Calculate, or recalculate, the cost for a manufactured item, and recalculate the cost for all manufactured items that have a BOM version that contains the manufactured item as a component.
- Calculate costs for a new manufactured item, based on its defined, approved, and active BOM and route information.

Each case requires careful consideration about how to update standard costs.

Last updated on 11/26/2025

Update standard costs for a new manufactured item

This article provides guidance for updating standard costs for a new manufactured item.

The following guidelines assume that you use a two-version approach to update standard costs. In this approach, one costing version contains the standard costs that were originally defined for the frozen period, and the second costing version contains the incremental updates that pertain to the new manufactured items. The incremental updates are entered as cost records in the second costing version, and eventually they're enabled. The two-version approach requires that you define a second costing version. Here are the guidelines for defining this costing version:

- Assign a costing type of **Standard cost**.
- Assign a significant identifier that indicates the contents of the costing version, such as **2016-UPDATES**.
- In the **Allow price types** field group, make sure that **Cost price** is set to **Yes**.
- Allow cost records to be entered for all sites (that is, leave the **Site** field blank). If you enter a site, cost records can be entered only for that site.
- Use a fallback principle of **Active**.

To add new manufacturing items throughout the frozen period, follow these steps:

1. Use the **Costing version setup** page to enable cost records to be entered into the second costing version that contains the incremental updates. Prevent the activation of pending costs, where activation is allowed after pending costs have been completely and accurately defined. Indicate a blank from-date as a policy in the costing version, and then enter the from-date when you enter each cost record. The from-date should represent a date before the new items are purchased or manufactured.
2. Use the **Item price** page to enter cost records for new purchased items. For each cost record, enter the costing version that contains incremental updates, and use a from-date that comes before the expected manufacturing date for new manufactured items.
3. Calculate the cost of new manufactured items by using the **Calculation** page. Open the **Calculation** page from the **Costing version maintenance** page, and then select the costing version that contains the incremental updates. Use the selection criteria to specify the new manufactured item and any one of its manufactured components. In a multilevel product structure, you might also have to specify any parent item that contains the new manufactured items as a component. Enter a from-date for the bill of materials (BOM)

calculation that corresponds to the start of manufacturing for the new manufactured items. The from-date should be in the effective dates for the item's BOM version and route version. **Note:** A missing cost record can indicate a new manufactured item. BOM calculations can be limited to items that have missing costs.

4. Verify the completeness and accuracy of the calculated costs for new manufactured items. Use the **Complete** page to review the calculated costs for each item cost record, and also review any Action center warning messages. Alternatively, use the **Calculation** page to review the calculated costs.
5. Use the **Costing version setup** page to change the blocking flag to allow activation of the pending cost records in the second costing version.
6. Use the **Activate prices** page (which you open from the **Costing version maintenance** page) to enable all pending cost records in the second costing version. You can also enable the pending cost records for individual items by clicking the **Activate** button on the **Item price** page.
7. Use the **Costing version setup** page to change the blocking flags in the second costing version to prevent additional data maintenance. The blocking policies prevent the entry of new pending costs and the activation of pending costs.

Last updated on 12/30/2025

Update standard costs in a non-manufacturing environment

This article provides guidance for updating standard costs in a non-manufacturing environment.

The following guidelines assume that you use a two-version approach to update standard cost. In this approach, one costing version contains the standard costs that were originally defined for the frozen period, and the second costing version contains the incremental updates. Each update is entered as a cost record that is enclosed in the second costing version, and eventually it's enabled. An alternative, one-version approach uses the set of standard costs that was originally defined. The two-version approach requires that you define a second costing version. Here are the guidelines for defining this costing version:

- Assign a costing type of **Standard costs**.
- Assign a meaningful identifier that indicates the contents of the costing version, such as *2025-UPDATES*.
- Make sure that the content includes cost records.
- Allow cost records to be entered for all sites. If you specify a site, cost records can be entered only for that site.
- Indicate a fallback principle of *None*. The fallback principle applies only to cost calculations for manufactured items.

To correct, adjust, or update standard costs for new items, follow these steps:

1. Use the **Costing version setup** page to enable cost data to be entered into the second costing version. Optionally, prevent the activation of pending costs, so that activation will be allowed after pending costs have been completely and accurately defined. You can also optionally enter a date in the **From date** field. As a general guideline, use the date when you intend to enable the incremental updates. Alternatively, leave the **From date** field blank for the costing version, and then enter a date in the **From date** field for each cost record.
2. Use the **Item price** page to enter updates as item cost records that are enclosed in the second costing version.
3. Use the **Item prices** report to verify the completeness and accuracy of the item cost records that are enclosed in the second costing version.
4. Use the **Costing version maintenance** page to change the blocking flag to allow activation of the pending cost records that are enclosed in the second costing version.

5. Use the **Activate prices** page (which you open from the **Costing version maintenance** page) to activate all pending item cost records that are enclosed in the second costing version. You can also activate the pending cost records for individual items by clicking the **Activate pending price** button on the **Item price** page.
6. To prevent additional data maintenance, use the **Costing version setup** page to change the blocking flags that are enclosed in the second costing version. The blocking policies will prevent the entry of new pending costs and the activation of pending costs.

Last updated on 12/30/2025

Prepare to maintain standard costs for manufactured items

This article describes the steps for preparing to maintain costs for manufactured items. The steps for manufactured items differ slightly from the steps for purchased items.

Policies that are assigned to manufactured items can affect the cost calculations for the parent manufactured items. To prepare to maintain costs for manufactured items, follow these steps:

1. Assign an item model group to the manufactured item.

   The item model group identifies that standard costs will be used.

2. Assign an item group to the manufactured item.

   The item group contains ledger accounts that apply to the manufactured item. The ledger accounts for a manufactured item that has a standard cost inventory model include several production variances, the cost change variance, and the inventory cost revaluation.

3. Assign the inventory unit of measure to the item.

   The item's costs are always expressed in the item's inventory unit of measure.

4. Don't assign a cost group to the manufactured item unless the item will be treated as a purchased item.

5. Assign a bill of materials (BOM) calculation group to the manufactured item.

   The item's BOM calculation group defines warning conditions that apply. In that way, when a BOM calculation is done, warning messages can be generated about possible sources of calculation errors. For example, a warning message can identify when an active BOM or route doesn't exist. The BOM calculation group contains a stop explosion policy that indicates when a manufactured item should be treated as a purchased item.

6. If the manufactured item has constant costs, assign a standard order quantity to it.

   The standard order quantity is an accounting lot size for amortizing constant costs. Examples of constant costs include setup times in routing operations and a constant component quantity in the BOM.

7. Define the BOM for the manufactured item.

   One or more BOM versions can be defined for the manufactured item. Verify that the versions that you want have been marked as approved and active, and that they have the effective dates that you want. The BOM version can be company-wide or site-specific.

8. Define the routing for the manufactured item.

   One or more route versions can be defined for the manufactured item. Verify that the versions that you want have been marked as approved and active, and that they have the effective dates that you want. The route version must be site-specific.

If you want to use routing information for costing purposes, additional preparation steps are required. For example, the cost categories that are assigned to routing operations must be correct and completed.

Related information

- Amortize constant costs for a manufactured item
- Set up products that can be produced or procured

Last updated on 12/30/2025

Costing versions overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about costing versions, how to maintain them, and the types of data that you can include in them.

A costing version can serve one or more purposes, depending on the data that the costing version contains. The primary purpose of a costing version is to contain cost records about items, cost categories, and calculation formulas for indirect costs. A costing version can contain a set of standard cost records or a set of planned cost records that are based on the costing type that is assigned to the costing version.

Costing versions for standard or planned costs

Standard costs

A costing version can support a standard cost inventory model for items, where the costing version contains a set of standard cost records about items and manufacturing processes. Cost data about manufacturing processes is expressed in terms of the cost categories for routing operations and the calculation formulas for manufacturing overheads.

Planned costs

A costing version can contain a set of planned cost records about items and manufacturing processes. A costing version that contains planned costs is often used to support cost calculation simulations, such as simulations of the effect that cost changes to purchased materials or manufacturing processes have on the calculated costs of manufactured items. The item cost records for planned costs can also be used to support an actual cost inventory model by providing the initial values for item costs. These values include the calculation of planned costs for manufactured items.

Entering costs

Data maintenance for cost records in a costing version involves entering costs for purchased items and for items that are transferred between sites. Additional data maintenance for manufacturers involves entering costs for cost categories that are associated with routing operations, entering calculation formulas for the indirect costs that reflect manufacturing overhead, and calculating costs for manufactured items.

The item cost data in a costing version consists of one or more cost records for each item. When an item cost record is first entered, it has *Pending* status and an intended effective date. When you activate the item cost record, the status is updated to *Active*, and the effective date is updated to the activation date. Different item cost records can reflect different sites, effective dates, or statuses. When you calculate costs for manufactured items for a future date, the bill of materials (BOM) calculation uses cost records that have the relevant effective date, regardless of whether the status is *Pending* or *Active*. An item's current active cost record is used to estimate production order costs and to value inventory transactions under a standard costing inventory model. The maintenance of cost records for cost categories and indirect cost calculation formulas resembles the maintenance of item cost records.

Two blocking policies for a costing version determine whether pending costs can be maintained and whether the pending cost can be activated. Use the blocking policies to permit data maintenance, and then use them to prevent data maintenance for cost records in a costing version. The following blocking policies can prevent pending costs from being updated or modified until they are finalized:

- *Block* – Prevents pending costs from being edited in or added to a cost version.
- *Block activation* – Prevents pending costs from being activated.

A costing version can also contain data about item sales prices or purchase prices for BOM calculation purposes.

Item sales prices for BOM calculations

The main reason for including data about sales prices is to retain a manufactured item’s calculated sales price. The calculated sales price can then be analyzed to determine how components, routing operations, and overhead contribute to the cost and sales price.

A secondary reason for including data about sales prices is to define the sales price records for component items. These records can then be used to calculate the sales price of manufactured

items. You first define the sales price model that is embedded in a BOM calculation group, and assign the BOM calculation group to purchased items. Then, when you perform BOM calculations that use planned costs, you select the cost price model of the BOM calculation group.

Otherwise, the sales price records for items are used only as reference information, regardless of whether the records are manually entered or calculated. By activating an item's sales price record, you can update the item's base sales price. However, the base sales price isn't site-specific and can be manually overridden. The item's base sales price is used as a default sales price on sales orders and sales quotations.

Item purchase prices for BOM calculations

The main reason for enabling purchase price data is to define purchase price records for component items to calculate the costs of manufactured items. Enter the item purchase price records manually.

To enable purchase price content, you first define a BOM calculation group that contains a cost price model for the item's purchase price, and assign the BOM calculation group to purchased items. You then use a cost price model for the BOM calculation group when you perform BOM calculations that use planned costs to calculate the sales price of manufactured items.

The purchase price records for items are also used as reference information. By changing the status of an item's purchase price record from *Pending* to *Active*, you can update the item's base purchase price. However, the base purchase price isn't site-specific and can be manually overridden. The item's base purchase price is used as a default purchase price on purchase orders.

Related information

- Glossary of terms in Dynamics 365 business processes
- Define service costing overview and relation to other business processes
- Business process for product costing and how it relates to other processes with Dynamics 365

Last updated on 06/22/2026

Costing sheets

Setting up the costing sheet involves two objectives. As the first objective, you define the format for displaying cost of goods sold information about a manufactured item or production order. The formatted display is termed a costing sheet. As the second objective, you define the basis for calculating indirect costs. The costing sheet setup builds on the cost group feature for displaying information and for the indirect cost calculation formulas. The two objectives of costing sheet setup are described in this article.

The following table lists the out-of-box security roles that can access costing sheets, including the level of access granted to each role by the default settings.

Expand table

Role	Access
Accounting manager	Edit
Inventory accountant clerk	View
Inventory accountant	View

A costing sheet is the formatted display of information about the cost of goods that are sold for a manufactured item or a production order. When you set up a costing sheet, you define the format for the information and also define the basis for calculating indirect costs. The costing sheet setup builds on the cost group features for displaying information and for the formulas that are used to calculated indirect cost. Here's more information about the two objectives of costing sheet setup:

- **Define the format for the costing sheet.** The user-defined format for a costing sheet identifies the segmentation of costs that contain a manufactured item's cost of goods sold. For example, the information about an item's cost of goods sold can be segmented into material, labor, and overhead, based on cost groups. These cost groups are assigned to items, cost categories for routing operations, and indirect cost calculation formulas. The format for the costing sheet typically requires intermediate totals when multiple cost groups are defined. For example, multiple cost groups that are related to material can be aggregated. The definition of a costing sheet format is optional, but a costing sheet format must be defined if indirect costs will be calculated.

- **Define the basis for calculating indirect costs.** Indirect costs reflect manufacturing overhead that is associated with the production of a manufactured item. An indirect cost calculation formula can be expressed as either a surcharge or a rate. A surcharge represents a percentage of value, whereas a rate represents an amount per hour for a routing operation. A cost group defines the basis for the calculation formula, such as a 100-percent surcharge for a labor cost group or a USD 50.00 hourly rate for a machine cost group. If you want to define a calculation formula and its cost group basis, the costing sheet setup requires that you identify the cost group that represents the overhead, and select whether a surcharge or rate approach is used.

Each calculation formula must be entered as a cost record. The cost record consists of a specified costing version, a surcharge percentage or a rate amount, the cost group basis, a status, and an effective date. When a cost record is first entered, it has *Pending* status and an effective date. When you activate the cost record, the status is updated to so that the record is the current active record, and the effective date is updated to the activation date. The cost record can also specify a site for a site-specific calculation formula. Alternatively, you can leave the **Site** field blank to indicate that the calculation formula is a company-wide formula. The cost record can optionally consist of a specified item or item group when the calculation formula is marked as a per-item formula.

The current active cost records for indirect cost calculation formulas are used to estimate production order costs. They're also used to calculate actual costs that are related to actual consumption of time and material. Pending cost records are used in bill of materials (BOM) calculations for a future date.

Two blocking policies for a costing version determine whether pending costs can be maintained, and whether the pending cost can be started. Use the blocking policies to permit data maintenance, and then to prevent data maintenance for the cost data in a costing version.

After you define the costing sheet format and calculations for indirect costs, you must perform a separate step to validate and save the information. The costing sheet represents a company-wide format for consistently displaying information about the costs of goods sold.

The costing sheet is displayed as part of the **Calculate item cost** page. The costing sheet can be displayed for a manufactured item's calculated cost record on the **Item price** page or for an order-specific calculation record on the **BOM calculation results** page. It can also be displayed as part of the **Price calculation** page for a production order.

Related information

- The white paper BOM calculation by using a costing sheet describes how to set up a costing sheet that includes material and manufacturing, and how the setup affects the BOM calculation results. To better explain the articles, it provides concrete scenarios and data that demonstrates the effect of the various settings and configurations.

Last updated on 11/26/2025

Simulate cost changes by using a costing version for planned costs

This article explains how you can simulate the effects of cost changes on a manufactured item's calculated costs by using a separate costing version for planned costs.

You can simulate the effects of cost changes on the calculated costs of a manufactured item by using a separate costing version for planned costs. Use this separate costing version to enter pending cost records that reflect incremental cost changes, and to calculate the cost impact on manufactured items. Because the Active costs fallback principle will be used in the bill of materials (BOM) calculations, only the incremental cost changes must be entered.

Guidelines for defining the simulation costing version

Use the following guidelines when you define the costing version for simulations:

- Assign a costing type of *Planned costs*.
- Assign a meaningful identifier for the costing version, such as *Simulation*.
- Make sure that the content includes cost records.
- Allow the entry of cost records. Don't block the entry of pending costs.
- Allow the entry of cost records for all sites. If you specify a site, you'll limit the entry of cost records to that site.
- Prevent the activation of pending costs. Only pending costs must be entered for cost records in the simulation costing version.
- Don't enter a "from" date. A calculation date will be entered for each BOM calculation that uses the simulation costing version.
- Specify a fallback principle of *Current active*. The fallback principle enables incremental cost changes to be entered for simulation purposes but uses the current active costs as the source for all other cost records. We assume that all current active costs exist for all other cost records.
- Specify a cost price model of *Version cost price*, but don't restrict calculations. For example, the simulations can also use the *BOM calculation group* cost price model to indicate the source of cost contribution data for purchased items.
- Specify an explosion mode of *Multilevel*, but don't restrict calculations. The simulations can use other explosion modes.

Costing versions

The following scenarios illustrate how the simulation costing version is used to simulate the impact of cost changes. Before you enter a simulated cost change, delete the pending cost records in the simulation costing version, so that you have a clean starting point. Use the **Item price** page to view and delete the pending cost records that are related to item prices and cost category prices.

- Simulate the cost change for a purchased item. For example, the cost change might reflect an expected increase or decrease in the cost of critical purchased materials. To define the different cost for a purchased item, use the **Item price** page to enter a pending cost record in the simulation costing version.
- Simulate the cost change for a cost category. For example, the cost change might reflect an expected increase or decrease in labor rates, or in the hourly rates for operations resources. To define the different cost for a cost category, use the **Cost category price** page to enter a pending cost record in the simulation costing version.
- Simulate the cost change in an indirect cost calculation formula. For example, the cost change might reflect an expected increase or decrease in manufacturing overhead. To define the change in an indirect cost calculation formula, use the **Costing sheet setup** page to enter a pending cost record in the simulation of costing version, and to validate and save the change.

After you enter the simulated cost changes, calculate the costs for manufactured items that are affected by the cost changes. Use the **Calculation** page for the simulation costing version, and identify the selected manufactured items that will be affected by the cost changes. The BOM calculations apply to all manufactured items unless you select specific items. Alternatively, you can use the BOM calculation option for where-used updates. View the item cost records in the simulation costing version to analyze how the simulated cost changes affected the costs of the selected manufactured items. Use the **Item price** page and the **Calculate item cost** page to view and analyze the costs.

Last updated on 11/26/2025

Compare item prices storage report

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to run a **Compare item prices storage** report and make the output available digitally, either as an interactive page in Dynamics 365 Supply Chain Management, or as an exported document in any of several formats.

When you view the report in your browser, columns and aggregate balances are dynamically adjusted, depending on your configured layout. You can sort the results, filter them, drill down into the data, and more.

Report results are stored in the **Compare item prices** data entity, which lets you filter and export the results to a format such as CSV or Microsoft Excel.

The **Compare item prices storage** report is helpful in cases where the output contains many lines. For example, the output will contain many lines if you have more than 40,000 items holding a pending item price in the costing version.

Generate a Compare item prices storage report

Follow these steps to generate and store a **Compare item prices storage** report:

1. Go to **Cost management** > **Inquiries and reports** > **Predetermined cost reports** > **Compare item prices storage**.

2. Select **New** to open the **Compare item prices** pane. Set the following options to define which prices to compare in your report:

    - On the **Parameters** FastTab, give the report a unique **Name** and use the fields in the **Pending prices to compare** and **Prices used for comparison** sections to define which prices and dates to compare.
    - On the **Records to include** FastTab, set up filters and constraints to define which data to include in the report.
    - On the **Run in the background** FastTab, set up how, when, and the frequency at which you want to generate the report.

> **Note**
>
> This report is always executed as part of a batch job.

3. Select **OK** to apply your settings and close the pane.
4. After the batch job is completed, it will be listed on the **Compare item prices storage** page. You might need to refresh the page to see the report.

Explore the Compare item prices storage report

After generating a report, you can view and explore it at any time as follows:

1. Go to **Cost management** > **Inquiries and reports** > **Predetermined cost reports** > **Compare item prices storage**.
2. Select a report from the list.
3. Do one of the following:
   - Select **Overview** to get an overview of your report results.
   - Select **View details** to get a more detailed view of your report
4. After the selected view opens, you can do the following:
   - Select almost any column heading to sort or filter the table by values in that column, just as with most standard forms in Supply Chain Management. Note, you can't sort or filter the **Net change price %** column because it's a calculated field.
   - Select **Dimension display** to open a pane where you can choose which dimension column to include on the form. Set **Save setup** to *Yes* if you'd like to save these settings so they'll be preserved the next time you open the report. Select **OK** to apply your settings and close.
   - Select any row in the form and then select **View details** to see more information about the selected item. You'll be able to drill down into the data from here.
   - Select any row in the form and then select **View comparison chart** to see an interactive graphical representation of your results as they relate to your selected item. You can explore these results by selecting various graphical elements in the chart and chart legend.
   - Select any row in the form and then select **View calculation details** to see more information about calculations related to the selected item. You'll be able to drill down into the data from here.

Export the Compare item prices storage report

Each report that you generate is stored in the **Compare item prices** data entity. You can use the standard data management features of Supply Chain Management to export data from this entity to any supported data format, including CSV or Microsoft Excel.

The following is an example of how to export a **Compare item prices storage** report:

1. Go to **System administration** > **Workspaces** > **Data management**.
2. Select the **Export** button in the **Data management** section.
3. The **Export** page opens, which you'll use to set up your export job. Start by giving your job a **Group name**.
4. In the **Selected entities** section, select **Add entity** to open a dialog box where you can set the following options:
    - **Entity name** - Select **Compare item prices**.
    - **Target data format** - Choose the format that you want to export to.
5. Select **Add** to add the new row and then select **Close** to close the dialog box.
6. Usually you'll export one report at a time. To do this, set up a **Filter** for the row you just added to the **Inquiry** pane. This will let you define which reports from the **Compare item prices** entity that you want to include in your export. Set the following filter options to export a single report:
    - On the **Range** tab, select **Add** to add a new row.
    - Set **Table** to *Compare item prices*.
    - Set **Derived table** to *Compare item prices*.
    - Set **Field** to the field that you want to filter by. Usually you'll use *Execution name* or *Execution time*.
    - Set **Criteria** to the value from your selected field that you want to look for (either the name of the report or the time the report was generated).
    - If necessary, add more rows to the **Range** table until you have uniquely identified the report that you're looking for.
7. Select **OK** to save your settings and close.
8. Select **Save** to save your export setup.
9. Open the **Export options** tab and select **Export now** to generate the export file.

10. The **Execution summary** page opens, where you can see the status of your export job and a list of entities that were exported. Select the **Compare item prices** entity listed in the **Entity processing status** area and then select **Download file** to download the data exported from that entity.

For more information about how to use data management to export data, see Data import and export jobs overview.

Last updated on 10/03/2025

Running average cost price

The inventory close process settles issue transactions to receipt transactions, based on the inventory valuation method that is selected in the item's item model group. However, before inventory close is run, the system calculates a running average cost price that is typically used when issue transactions are posted.

The system estimates this running average cost price for an item by using the following formula:

Estimated price = (Physical amount + Financial amount) ÷ (Physical quantity + Financial quantity)

Default item cost

The default item cost for a released product can be configured in one of two ways on the **Released product details** page:

- Create a standard cost by selecting **Item price** in the **Set up** group on the **Manage costs** tab of the Action Pane. If you use this method, you must use a standard cost costing version, and the cost must be activated.
- Define a default item cost price for a released product by entering a value in the **Price** field on the **Manage costs** FastTab.

In addition to entering or creating a price, you can select the **Use latest cost price** checkbox on the **Manage costs** FastTab of the **Released product details** page. In this case, the system automatically updates the **Price** field when you post a financial update. For example, if you post a purchase order invoice, the field is set to the purchase price from that invoice.

If you have a cost price in an active costing version, and you enter a price on the **Manage costs** FastTab, the system uses the price from the active costing version before it uses the price that is defined on the **Manage costs** FastTab.

Using the running average cost price

The following table shows when the system posts inventory transactions by using the running average cost price, and when it uses the cost price that is defined on the item master record instead.

Expand table

Condition	The system uses the estimated running average cost price	The system uses the default item cost price
Both the numerator* and denominator** are positive.	Yes	No
The numerator*, denominator**, or both are negative.	No	Yes
The denominator** is 0 (zero).	No	Yes

* Numerator = (Physical amount + Financial amount)

** Denominator = (Physical quantity + Financial quantity)

> **Note**
>
> If the **Include physical value** option isn't selected for an item, the system uses 0 (zero) for both the physical amount and the physical quantity. For information about this option, see Options for including physical value in cost calculations.

Avoiding pricing amplification

On rare occasions, the system prices several issues before it has enough receipts to base the price on. This scenario can cause estimates of the running average cost price to be overly inflated. However, there are steps that you can take to avoid pricing amplification, or to reduce its impact when it does occur.

Scenario: The following transactions occur for an item where the **Include physical value** option is selected:

1. You financially receive a quantity of 100 at USD 100.00.
2. You financially issue a quantity of 200.
3. You physically receive a quantity of 101 at USD 202.00.

When you examine the estimated running average cost price for the item, you expect a cost price of USD 1.51. Instead, you find an estimated running average of USD 102.00, which is based on the following formula:

Estimated price = [202 + (-100)] ÷ [101 + (-100)] = 102 ÷ 1 = 102

This pricing amplification occurs because, when 200 items are financially issued in step 2, the system must price 100 of the items before it has any corresponding receipts. This situation causes

negative inventory. The system then estimates a unit price of USD 1.00, as you might expect. However, when the corresponding 100 receipts arrive, they are at a unit price of USD 2.00 each.

> **Note**
>
> Although the issues create negative inventory, inventory is positive when the issue price is calculated. Therefore, the running average cost price is used instead of the price on the item master. At this point, the system has an inventory value offset of USD 100.00. Although that offset was built up over 100 pieces, where there was a unit offset of USD 1.00 each, you now have only one piece in inventory. Therefore, the offset of USD 100.00 is allocated to this one piece. The result is the overly inflated estimated cost price.
>
> For comparison, notice that if steps 2 and 3 are reversed in the scenario, 200 items will be issued at a unit price of USD 1.51, and one piece will remain at a unit price of USD 1.51. Because this pricing amplification scenario can occur when negative inventory is involved, it's difficult to avoid in the following cases:
>
> - You must estimate issue prices on the on-hand value and quantity.
> - You must adjust the on-hand value and quantity on issues and receipts.
> - Your business model allows you to send out, or price, more pieces than you have.
> - You must accept any receipt value and quantity that are submitted to you.

However, if your business model allows for the following practices, they can help you avoid the negative quantities that make the pricing amplification scenario possible:

- If you select the **Include physical value** option for an item, clear the **Physical negative inventory** checkbox on the **Item model groups** page.
- If you do **not** select the **Include physical value** option for an item, clear the **Financial negative inventory** option on the **Item model groups** page.

Additionally, consider that the maximum offset in your physical inventory value is limited by the number of physical transactions, and the difference between physical and financial prices. If all physical transactions are eventually updated financially, the physical value can't rise to extreme levels. The amplification effect decreases significantly when the accumulated offset is spread out over several on-hand pieces instead of just one.

Avoid a zero cost price on issues

When the **Include physical value** option isn't selected on the **Item model group** page, an issue from inventory can have a zero cost price if there are no financially updated receipts in inventory. To help avoid this scenario, consider the following options:

- Select the **Include physical value** option on the **Item model group** page. This option prevents a zero cost price on an issue, provided that the receipt is physically updated. If you don't allow negative physical inventory, issues calculate the running average from the physically updated transactions.
- Create a default item cost price by activating a costing version that has a standard cost, or enter the price on the **Manage costs** FastTab of the **Released product details** page. This option is best when the **Include physical value** option isn't selected on the **Item model group** page, because the system will always have a fallback price to use.
- Select the **Use latest cost price** option on the **Manage costs** FastTab of the **Released product details** page. This option keeps the **Price** field up to date each time that you financially update a receipt. If you select this option, but you don't enter a default price or activate a cost in a standard costing version, you can still have a zero cost on an issue.

If you have issue transactions that have zero cost, the *Inventory close and adjustment* process will correct the cost price by creating an adjustment after the receipts are financially updated. Keep in mind that the financial period when this update occurs might differ from the financial period when the items were physically received or issued.

Last updated on 11/26/2025

Track running average cost per inventory dimension

An inventory dimension group is attached to every inventory item. Therefore, the running average cost price of an item is calculated based on the selected inventory dimensions that are being tracked financially.

There are three types of inventory dimensions: product, storage, and tracking. Product dimensions include configuration, size, and color. Product dimensions are always tracked financially. Storage and tracking dimensions include site, warehouse, location, inventory status, license plate, batch number, and serial number. You can decide which storage and tracking dimensions are tracked financially.

Example 1

If the storage dimension group that is attached to the item is financially tracked by warehouse, the running average cost price is calculated per warehouse. The following purchase orders have been invoiced:

- A purchase order for a quantity of 2 at a cost price of USD 10.00 has been invoiced for warehouse GW.
- A purchase order for a quantity of 3 at a cost price of USD 12.00 has been invoiced for warehouse GW.
- A purchase order for a quantity of 5 at a cost price of USD 15.00 has been invoiced for warehouse MW.

The running average cost price for warehouse GW is USD 11.20, and the running average cost price for warehouse MW is USD 15.00. A sales order invoice is posted for warehouse GW. The value of the inventory and the cost of goods sold (before inventory close is run and without marking) is USD 11.20. Another sales order is posted for warehouse MW. The value of the inventory and the cost of goods sold (before inventory close is run and without marking) is USD 15.00.

Example 2

If the storage dimension group that is attached to the item is financially tracked by warehouse, and the tracking dimension group is financially tracked by batch number, the running average

cost price is calculated for each batch. The following purchase orders have been invoiced:

- A purchase order for a quantity of 2 at a cost price of USD 10.00 has been invoiced for warehouse GW and batch AAA.
- A purchase order for a quantity of 3 at a cost price of USD 12.00 has been invoiced for warehouse GW and batch AAA.
- A purchase order for a quantity of 2 at a cost price of USD 15.00 has been invoiced for warehouse GW and batch BBB.

The running average cost price for warehouse GW and batch AAA is USD 11.20, and the running average cost price for warehouse GW and batch BBB is USD 15.00.

> **Note**
>
> We recommend that you always view the cost price for all financial dimensions that are being tracked.

Last updated on 11/26/2025

FIFO with physical value and marking

07/21/2025

First in, first out (FIFO) is an inventory management and valuation method where inventory that is produced or acquired first is sold, used, or disposed of first. During the inventory close process in Microsoft Dynamics 365 Supply Chain Management, the system creates settlements where the first receipt is matched against the first issue, and so on.

The settlements and matching principle are based on the financial date of the inventory transactions. A preliminary assessment of the settlements and adjustments can be performed by running the inventory recalculation process.

You can override the FIFO principle by marking inventory transactions so that a specific item receipt is settled against a specific issue. A periodic inventory close is required when you use the FIFO inventory model to create settlements and adjust the value of issues according to the FIFO principle. Until you run the inventory close process, issue transactions are valued at the running average when the physical and financial updates occurred. Unless you're using marking, the running average is calculated when the physical or financial update is performed. The following examples show the effect of using FIFO in three configurations:

- FIFO without the **Include physical value** option
- FIFO with the **Include physical value** option
- FIFO with marking

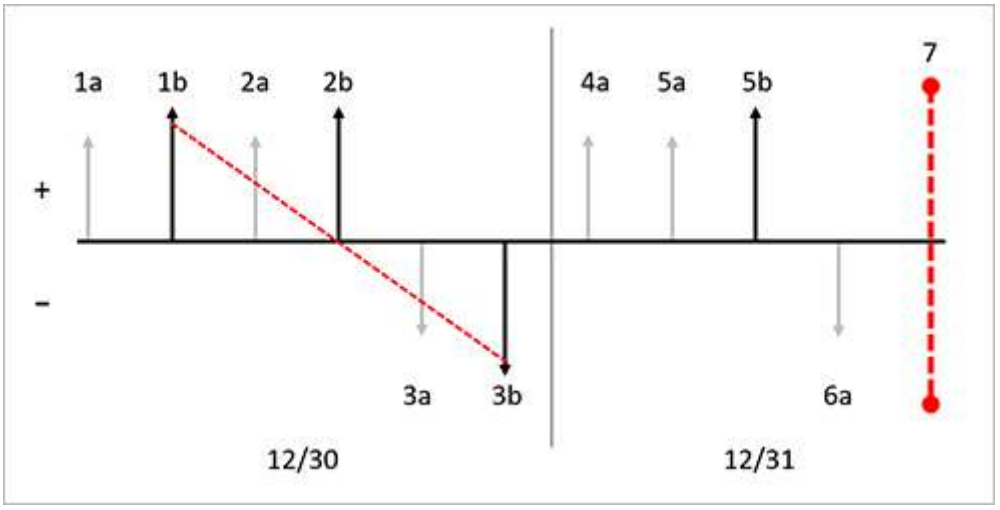
FIFO without the Include physical value option

In this example, the **Include physical value** checkbox is cleared on the item model group for the released product. The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.

- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions)
- 7. Inventory close is performed. Based on the FIFO method, the first financially updated issue is settled against the first financially updated receipt, and so on. In this example, one settlement is created between 1b and 3b. An adjustment of USD –6.00 will be made to 3b, and the resulting final cost will be USD 10.00.

The new running average cost price reflects the average of the financially updated transactions. The following illustration shows the effects of the FIFO inventory model on this series of transactions when the **Include physical value** option isn't used.



Key to the diagram

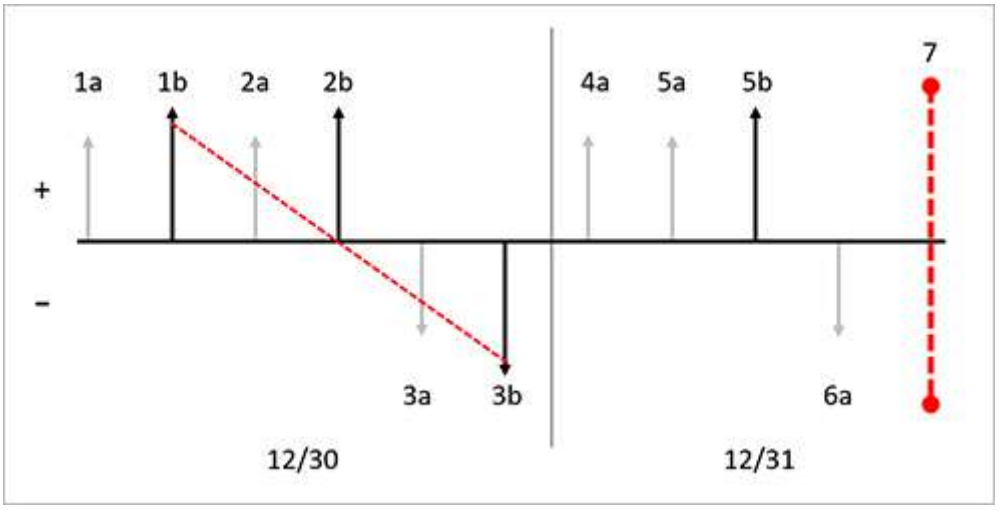
- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the timeline.
- Issues out of inventory are represented by vertical arrows below the timeline.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

FIFO with the Include physical value option

If the **Include physical value** checkbox is selected for an item on the **Item model group** page, the system uses both physical and financial receipt transactions to calculate the running

average cost price. Where applicable, the system also adjusts the physically updated issue transaction. Inventory close that uses the FIFO inventory model makes settlements only to transactions that are financially updated. The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.67 (running average of physically and financially posted transactions).
- 7. Inventory close is performed. Based on the FIFO method, the first financially updated issue is settled against the first financially updated receipt, and so on. In this example, one settlement is created between 1b and 3b. An adjustment of USD –6.00 will be made to 3b, and the resulting final cost will be USD 10.00. Additionally, transaction 6a will be adjusted to the receipt transaction cost of 2b. The system won't settle these transactions because the receipt is updated physically but not financially. Instead, only an adjustment of USD -1.67 will be posted to the physical issue transaction, and the resulting adjusted cost will be USD 22.00.



Notice that the result of running the inventory close process is the same, regardless of whether you select the **Include physical value** option on the **Item model group** page. The **Include physical value** option affects only the running average.

Key to the diagram

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- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

FIFO with marking

Marking is a process that lets you link, or mark, an issue transaction to a receipt transaction. Marking can occur either before or after a transaction is posted. You can use marking when you want to be sure of the exact cost of inventory when the transaction is posted or the inventory close is performed.

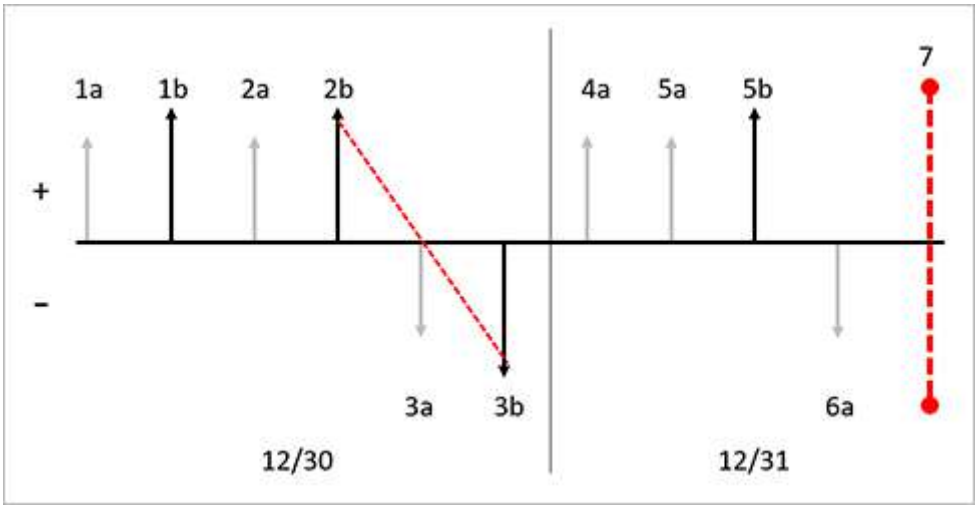
For example, the Customer Service department accepted a rush order from an important customer. Because this order is a rush order, you must pay more for this item to fulfill your customer's request. You must make sure that the cost of the inventory item is reflected in the margin, or cost of goods sold (COGS), for the sales order invoice. When the purchase order is posted, the inventory is received at a cost of USD 120.00. If the sales order document is marked to the purchase order before the packing slip or invoice is posted, the COGS will be USD 120.00, not the current running average cost for the item. If the sales order packing slip or invoice is posted before the marking occurs, the COGS is posted at the running average cost price. Before inventory close is performed, these two transactions can still be marked to each other.

When a receipt transaction is marked to an issue transaction, the valuation method that is defined in the item model group is disregarded, and the system settles these transactions to each other. You can manually mark a transaction against a sales order line on the **Sales order details** page by selecting **Inventory** > **Marking** on the **Sales order lines** FastTab. You can view the open receipt transactions on the **Marking** page. You can match or mark an issue transaction to an open receipt transaction for an inventoried item from a posted inventory adjustment journal. The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.

- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3c. Inventory financial issue for 3b is marked to inventory financial issue for 2b.
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions)
- 7. Inventory close is performed. Based on the marking principle that uses the FIFO method, the marked transactions are settled against each other. In this example, 3b is settled against 2b, and an adjustment for USD 6.00 is posted to 3b to bring the value to USD 22.00. In this example, no additional settlements are made, because the close creates settlements only for financially updated transactions.

The following illustration shows the effects of the FIFO inventory model on this series of transactions when marking between issues and receipts is used.



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- Each new receipt or issue transaction is designated by a new label.

- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
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- Settlements and markings that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

LIFO date with physical value and marking

07/21/2025

Last in, first out date (LIFO date) is an inventory management and valuation method where inventory that was produced or acquired last is sold, used, or disposed of first. During the inventory close process in Microsoft Dynamics 365 Supply Chain Management, the system creates settlements where the last receipt is matched against the first issue for each given date starting with the oldest date first, and so on. When you use the last in, first out date (LIFO date) inventory model, if there's no receipt before the issue, the issue is settled against any receipts that occur after the date of the issue. The settlements and matching principle are based on the financial date of the inventory transactions. When there are several issues on the same date, they're settled in the order of last issue, last receipt. A preliminary assessment of the settlements and adjustments can be performed by running the inventory recalculation process.

You can override the LIFO date principle by marking inventory transactions so that a specific item receipt is settled against a specific issue. A periodic inventory close is required when you use the LIFO date inventory model to create settlements and adjust the value of issues according to the LIFO principle. Until you run the inventory close process, issue transactions are valued at the running average when the physical and financial updates occurred. Unless you're using marking, the running average is calculated when the physical or financial update is performed.

We recommend a periodic inventory closing when you use the LIFO date inventory model.

The following examples show the effect of using LIFO date in three configurations:

- LIFO date without the **Include physical value** option
- LIFO date with the **Include physical value** option
- LIFO date with marking

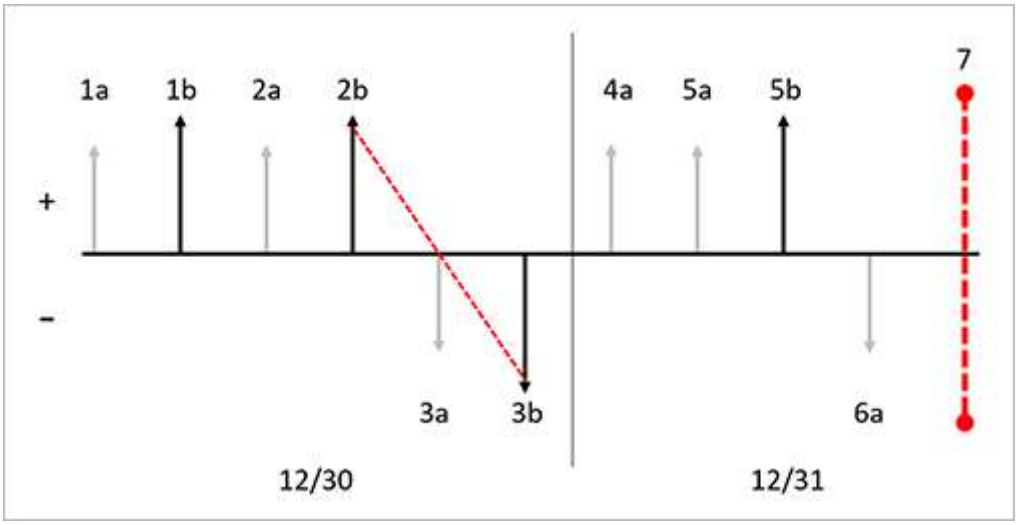
LIFO date without the Include physical value option

In this example, the item model group isn't marked to include physical value. The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.

- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions)
- 7. Inventory close is performed. Based on the LIFO date method, the first financially updated issue is settled against the last financially updated receipt starting on the first date, and so on. In this example, one settlement is created between 3b and 2b. An adjustment of USD 6.00 is made to 3b, and the resulting final cost is USD 22.00.

The following illustration shows the effects of the LIFO date inventory model when the **Include physical value** option isn't used.



Key to the diagram

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- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.

- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

LIFO date with the Include physical value option

If the **Include physical value** checkbox is selected for an item on the **Item model groups** page, the system uses both physical and financial receipt transactions to calculate the running average cost price. Where applicable, the system also adjusts the physically updated issue transaction. When the **Include physical value** checkbox is cleared, inventory close that uses the LIFO date inventory model makes settlements only to transactions that are financially updated.

In this example, the item model group is marked to include physical value.

The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
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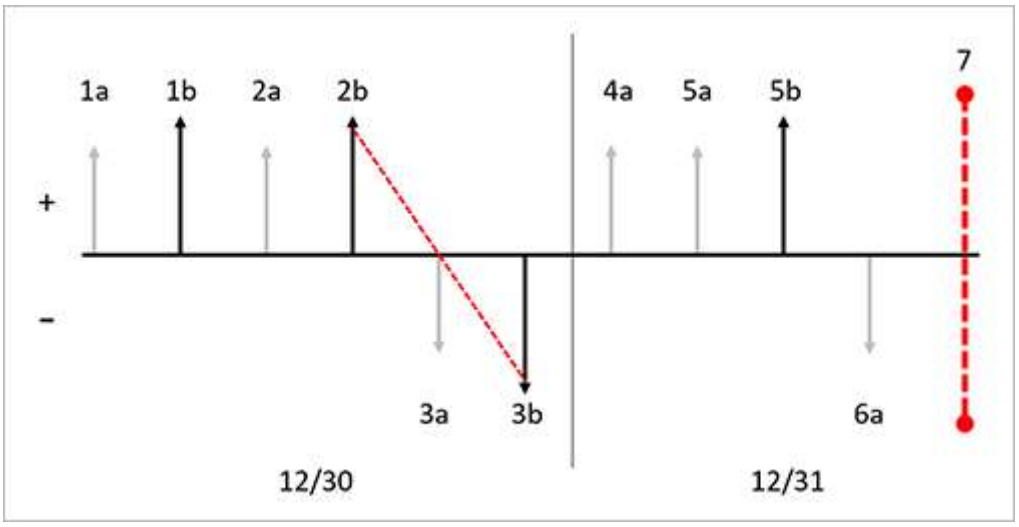
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.67 (running average of physically and financially posted transactions).
- 7. Inventory close is performed. Based on the LIFO date method, the first financially updated issue is settled against the last financially updated receipt for each date starting on the first date, and so on. In this example, one settlement is created between 2b and 3b. An adjustment of USD 6.00 is made to 3b, and the resulting final cost is USD 22.00. Additionally, transaction 6a is adjusted to the receipt transaction cost of 5b. The system doesn't settle these transactions because the receipt is updated physically but not financially. Instead, only an adjustment of USD 6.33 is posted to the physical issue transaction, and the resulting adjusted cost is USD 30.00.

The following illustration shows the effects of the LIFO inventory model when the **Include physical value** option is used.



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- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

LIFO date with marking

Marking is a process that lets you link, or mark, an issue transaction to a receipt transaction. Marking can occur either before or after a transaction is posted. You can use marking when you want to be sure of the exact cost of inventory when the transaction is posted or the inventory close is performed. For example, the Customer Service department accepted a rush order from an important customer. Because this order is a rush order, you must pay more for the item to fulfill the customer's request.

You must make sure that the cost of the inventory item is reflected in the margin, or cost of goods sold (COGS), for the sales order invoice. When the purchase order is posted, the inventory is received at a cost of USD 120.00. If this sales order document is marked to the purchase order before the packing slip or invoice is posted, the COGS will be USD 120.00, not

the current running average cost for the item. If the sales order packing slip or invoice is posted before the marking occurs, the COGS will be posted at the running average cost price.

Before inventory close is performed, these two transactions can still be marked to each other.

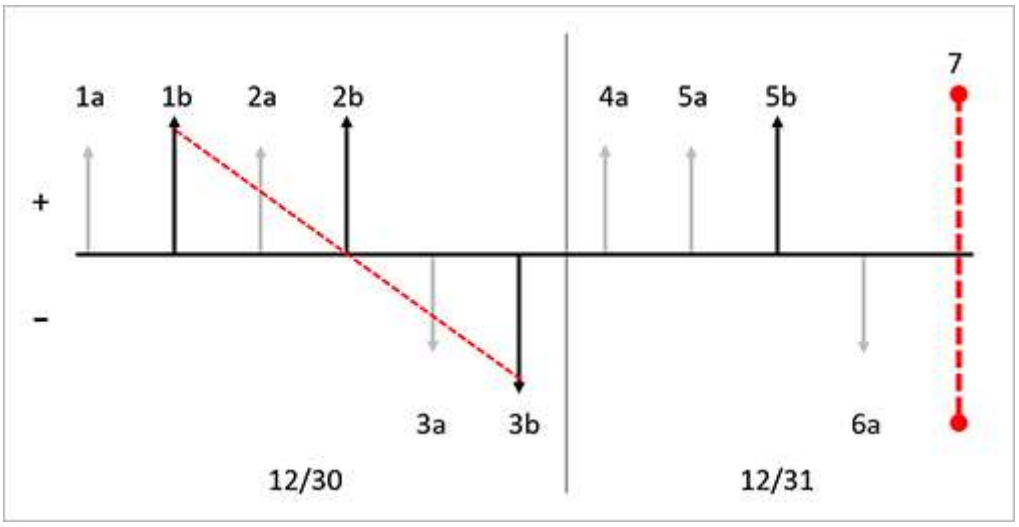
You can mark an issue transaction to a receipt before the transaction is posted. You can do this marking from a sales order line on the **Sales order details** page by selecting **Inventory** > **Marking** on the **Sales order lines** FastTab. You can view the open receipt transactions on the **Marking** page.

You can also mark an issue transaction to a receipt after the transaction is posted. You can match or mark an issue transaction for an open receipt transaction for an inventoried item from a posted inventory adjustment journal.

The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3c. Inventory financial issue for 3b is marked to inventory financial issue for 1b.
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions)
- 7. Inventory close is performed. Based on the marking principle that uses the LIFO date method, the marked transactions are settled against each other. In this example, 3b is settled against 1b, and an adjustment for USD -6.00 is posted to 3b to bring the value to USD 10.00. In this example, no additional settlements are made, because the close creates settlements only for financially updated transactions.

The following illustration shows the effects of the LIFO date inventory model when marking between issues and receipts is used.



Key to the diagram

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- Inventory closings are represented by a red vertical dashed line.
- Settlements and markings that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

LIFO with physical value and marking

Article • 02/11/2025

Last in, first out (LIFO) is an inventory management and valuation method where inventory that was produced or acquired last is sold, used, or disposed of first. During the inventory close process in Microsoft Dynamics 365 Supply Chain Management, the system creates settlements where the last receipt is matched against the first issue, and so on. The settlements and matching principle are based on the financial date of the inventory transactions. A preliminary assessment of the settlements and adjustments can be performed by running the inventory recalculation process.

You can override the LIFO principle by marking inventory transactions so that a specific item receipt is settled against a specific issue. A periodic inventory close is required when you use the LIFO inventory model to create settlements and adjust the value of issues according to the LIFO principle. Until you run the inventory close process, issue transactions are valued at the running average when the physical and financial updates occurred. Unless you're using marking, the running average is calculated when the physical or financial update is performed.

The following examples show the effect of using LIFO in three configurations:

- LIFO without the **Include physical value** option
- LIFO with the **Include physical value** option
- LIFO with marking

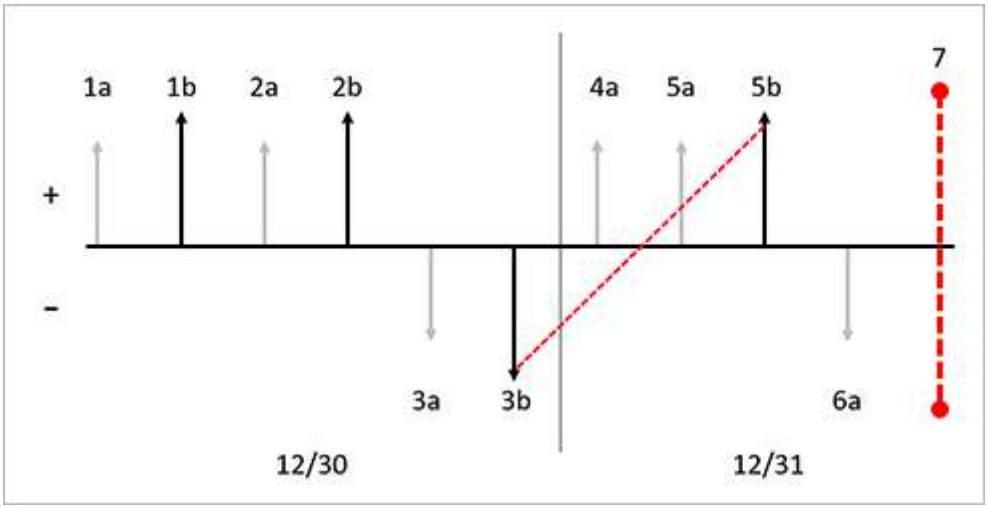
LIFO without the Include physical value option

In this example, the **Include physical value** checkbox is cleared on the item model group for the released product. The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.

- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions)
- 7. Inventory close is performed. Based on the LIFO method, the first financially updated issue is settled against the last financially updated receipt, and so on. In this example, one settlement is created between 5b and 3b. An adjustment of USD 14.00 is made to 3b, and the resulting final cost is USD 30.00.

The following illustration shows the effects of the first in, first out (FIFO) inventory model on this series of transactions when the **Include physical value** option isn't used.



Key to the diagram

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

LIFO with the Include physical value option

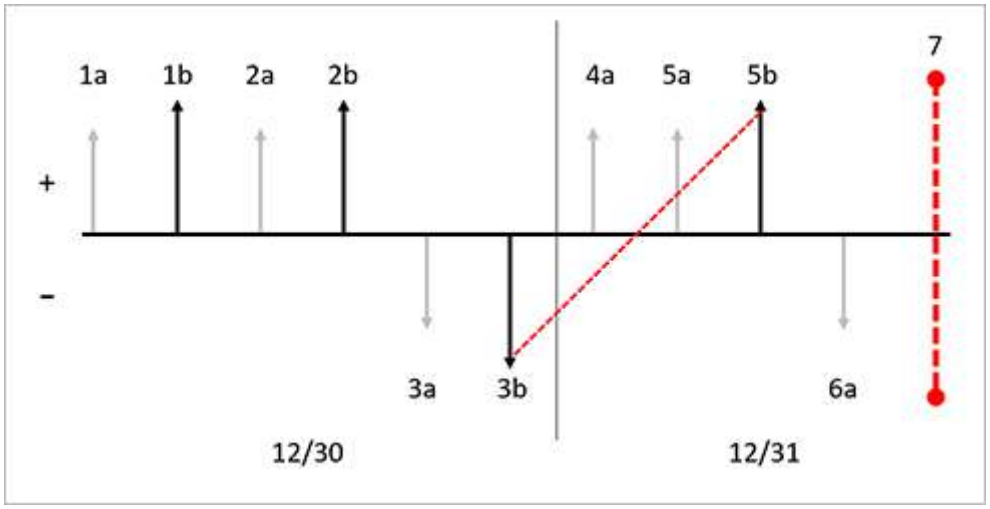
If the **Include physical value** checkbox is selected for an item on the **Item model groups** page, the system uses both physical and financial receipt transactions to calculate the running average cost price. Where applicable, the system also adjusts the physically

updated issue transaction. When the **Include physical value** checkbox is cleared, inventory close that uses the LIFO inventory model makes settlements only to transactions that are financially updated.

The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.67 (running average of physically and financially posted transactions).
- 7. Inventory close is performed. Based on the LIFO method, the first financially updated issue is settled against the last financially updated receipt, and so on. In this example, one settlement is created between 3b and 5b. An adjustment of USD 14.00 is made to 3b, and the resulting final cost is USD 30.00. Additionally, transaction 6a is adjusted to the receipt transaction cost of 4a. The system won't settle these transactions, because the receipt is updated physically but not financially. Instead, only an adjustment of USD 1.33 is posted to the physical issue transaction, and the resulting adjusted cost will be USD 25.00.

The following illustration shows the effects of the LIFO inventory model on this series of transactions when the **Include physical value** option is used.



Key to the diagram

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

LIFO with marking

Marking is a process that lets you link, or mark, an issue transaction to a receipt transaction. Marking can occur either before or after a transaction is posted. You can use marking when you want to be sure of the exact cost of inventory when the transaction is posted or the inventory close is performed. For example, the Customer Service department accepted a rush order from an important customer. Because this order is a rush order, you must pay more for the item to fulfill the customer's request.

You must make sure that the cost of the inventory item is reflected in the margin, or cost of goods sold (COGS), for the sales order invoice. When the purchase order is posted, the inventory is received at a cost of USD 120.00. If this sales order document is marked to the purchase order before the packing slip or invoice is posted, the COGS is USD 120.00, not the current running average cost for the item. If the sales order packing slip or invoice is posted before the marking occurs, the COGS is posted at the running average cost price.

Before inventory close is performed, these two transactions can still be marked to each other.

You can mark an issue transaction to a receipt before the transaction is posted. You can do this marking from a sales order line on the **Sales order details** page by selecting **Inventory** > **Marking** on the **Sales order lines** FastTab. You can view the open receipt transactions on the **Marking** page.

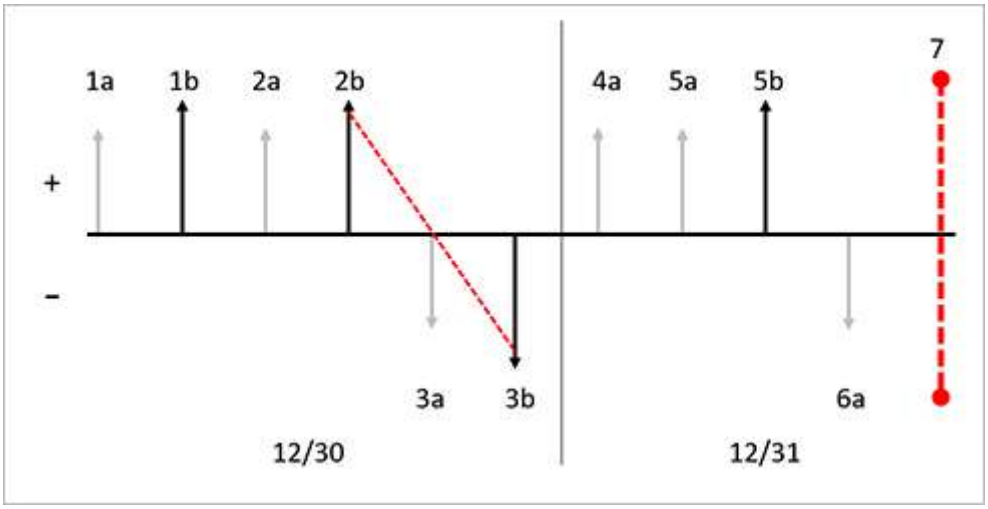
You can also mark an issue transaction to a receipt after the transaction is posted. You can match or mark an issue transaction for an open receipt transaction for an inventoried item from a posted inventory adjustment journal.

The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3c. Inventory financial issue for 3b is marked to inventory financial issue for 2b.
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions)
- 7. Inventory close is performed. Based on the marking principle that uses the LIFO method, the marked transactions are settled against each other. In this example, 3b is settled against 2b, and an adjustment for USD 6.00 is posted to 3b to bring the value to USD 22.00. In this example, no additional settlements are made, because the close creates settlements only for financially updated transactions.

The new running average cost price reflects the average of the financially and physically updated transactions, USD 17.50.

The following illustration shows the effects of the LIFO inventory model on this series of transactions when marking between issues and receipts is used.



Key to diagram

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements and markings that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Feedback

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Weighted average with physical value and marking

07/21/2025

Weighted average is an inventory model based on an average that results from the multiplication of each component (item transaction) by a factor (cost price) reflecting its importance (quantity). Another way to say this is that weighted average is an inventory model that assigns the cost of issue transactions based on the mean value of all inventory received during the period, plus any on-hand inventory from the previous period.

When you run an inventory closing using the weighted average inventory model, there are two ways a settlement can be created. Typically, all receipts are settled against a virtual issue, which holds the total received quantity and value. This virtual issue has a corresponding virtual receipt from which the issues are settled. In this manner, all issues get the same average cost. The virtual issue and receipt can be seen as a virtual transfer, which is named the *weighted average inventory closing transfer*. This settlement method is called a *weighted average summarized settlement*. If there's only one receipt, all issues can be settled from it and the virtual transfer won't be created. This settlement method is referred to as a *direct settlement*. Any inventory that is on hand after the inventory close is performed is valued at the weighted average from the previous period and included in the weighted average calculation in the next period.

You can override the weighted average principle by marking inventory transactions so that a specific item receipt is settled against a specific issue. A periodic inventory close is required when you use the weighted average inventory model to create settlements and adjust the value of issues according to the weighted average principle. Until you run the inventory close process, issue transactions are valued at the running average when the physical and financial updates occurred. Unless you're using marking, the running average is calculated when the physical or financial update is performed.

The weighted average inventory costing method is calculated by the following formula:

- Weighted average = ([Q1 × P1] + [Q2 × P2] + [Q*n* × P*n*]) ÷ (Q1 + Q2 + Q*n*)

Q = quantity of the transaction
P = price of the transaction

Settlements are inventory close postings that adjust the issues to the correct weighted average as of the closing date. The following examples illustrate the effect of using weighted average with five different configurations:

- Weighted average direct settlement without the **Include physical value** option

- Weighted average summarized settlement without the **Include physical value** option
- Weighted average direct settlement with the **Include physical value** option
- Weighted average summarized settlement with the **Include physical value** option
- Weighted average with marking

Weighted average direct settlement without Include physical value

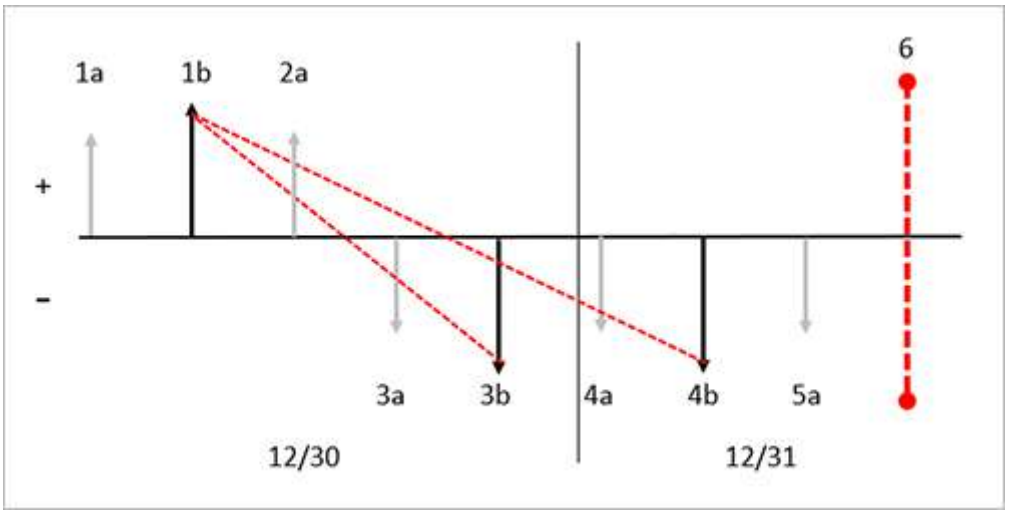
The direct settlement principle creates settlements directly between receipts and issues without creating additional inventory transactions. The system uses this direct settlement principle in the following situations:

- One receipt and one or more issues were posted in the period.
- Only issues were posted in the period and the inventory contains on-hand items from a previous closing.

In this example, the **Include physical value** checkbox is cleared on the **Item model group** for the released product. The illustration that follows shows these transactions:

- 1a. Inventory physical receipt for a quantity of 10 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 10 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 10 at a cost of USD 20.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 10.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 10.00 (running average of financially posted transactions).
- 4a. Inventory physical issue for a quantity of 1 at a cost of USD 10.00 each (running average of financially posted transactions).
- 4b. Inventory financial issue for a quantity of 1 at a cost of USD 10.00 each (running average of financially posted transactions).
- 5a. Inventory physical issue for a quantity of 1 at a cost of USD 10.00 each (running average of financially posted transactions).
- 6. Inventory close is performed. Based on the weighted average method, the system uses the direct settlement method because only one receipt is financially updated in the period. In this example, one settlement is created between 1b and 3b, and another between 1b and 4b. No adjustment is made because the running average is the same as the weighted average.

The following diagram illustrates this series of transactions with the effects of choosing the weighted average inventory model and the direct settlement principle without the **Include physical value** option.



Key to diagram

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average summarized settlement without the Include physical value option

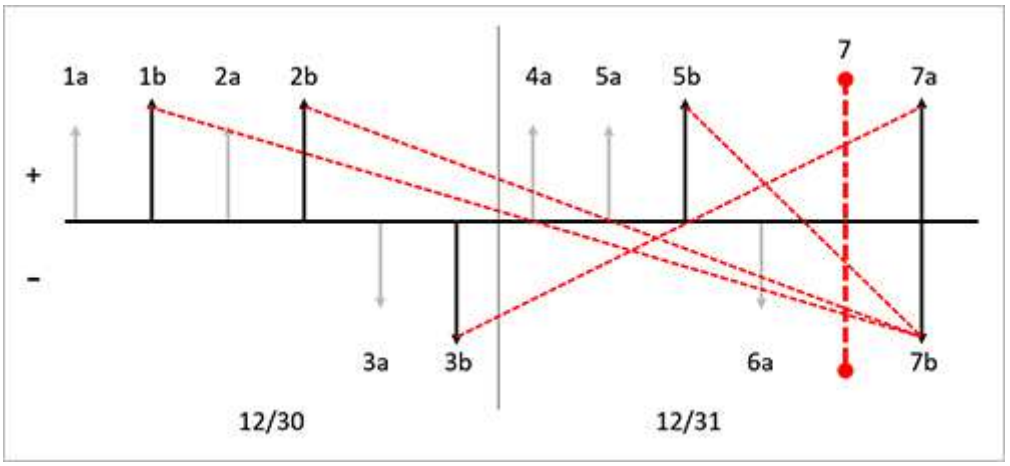
When there are multiple receipts in a period, weighted average uses the summarized settlement principle where all receipts within in a closing period are summarized into a transaction called *weighted average inventory closing*. All the receipts for the period will be settled against the issue of the newly created inventory transaction. All issues for the period will be settled against the receipt of the new inventory transaction. If there's remaining on-hand inventory after the inventory close, the on-hand inventory value is included in the receipt transaction of the weighted average inventory closing transactions.

The following transactions are illustrated in the graphic that follows:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.

- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions).
- 7. Inventory close is performed.
- 7a. Weighted average inventory close transaction financial issue is created to sum the settlements of all the inventory financial receipts.
    - Transaction 1b is settled for a quantity of 1 with an amount settled of USD 10.00.
    - Transaction 2b is settled for a quantity of 1 with an amount settled of USD 22.00.
    - Transaction 5b is settled for a quantity of 1 with an amount settled of USD 30.00.
    - Transaction 7a. is created for a quantity of 3 with an amount settled of USD 62.00. This transaction offsets the sum of the three receipt transactions that are financially updated in the period.
- 7b. Weighted average inventory close transaction financial receipt is created as the offset to financially posted issues.
    - Transaction 3b is settled for a quantity of 1 with an amount settled of USD 20.67. This transaction is adjusted by USD 4.67 to bring the original value of USD 16.00 to 20.67, which is the weighted average of financially posted transactions for the period.
    - Transaction 7b. is created for a quantity of 1 with an amount settled of USD 20.67 to offset 3b. This transaction offsets the sum of the one issue transaction that is financially updated in the period.

The following diagram illustrates this series of transactions with the effects of choosing the weighted average inventory model and the summarized settlement principle without the **Include physical value** option.



Key to diagram

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average direct settlement with the Include physical value option

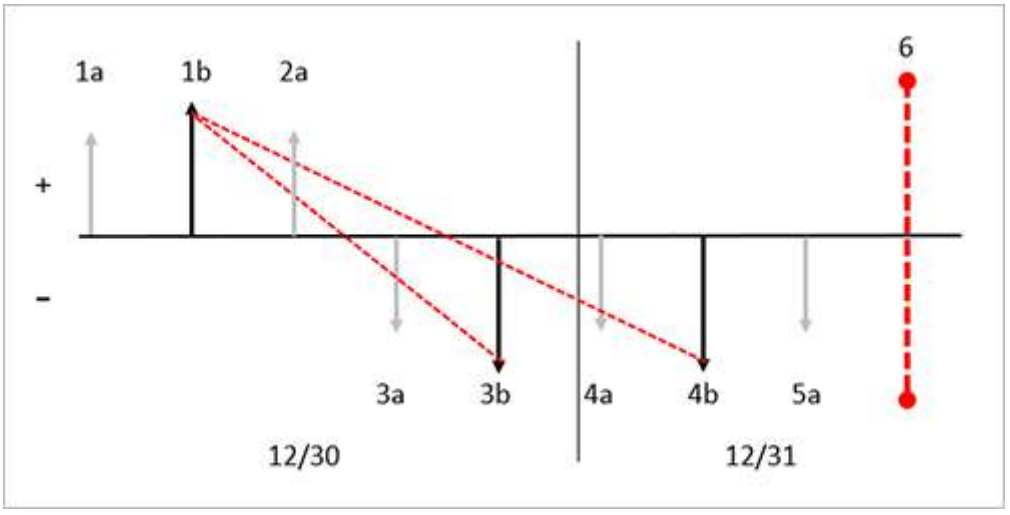
The parameter **Include physical value** works differently with the weighted average inventory model than in earlier versions of the product. When you select the **Include physical value** option for an item in the **Item model group** form, the system uses physically updated receipts when calculating the estimated issue cost price, or running average. Issues are posted based on this estimated cost price during the period. During the inventory close, financially updated receipts are only considered in the weighted average calculation.

The following transactions are illustrated in the graphic that follows:

- 1a. Inventory physical receipt for a quantity of 10 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 10 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 10 at a cost of USD 20.00 each.

- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 15.00 (running average of physically and financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 15.00 (running average of physically and financially posted transactions).
- 4a. Inventory physical issue for a quantity of 1 at a cost of USD 15.00 each (running average of physically and financially posted transactions).
- 4b. Inventory financial issue for a quantity of 1 at a cost of USD 15.00 each (running average of physically and financially posted transactions).
- 5a. Inventory physical issue for a quantity of 1 at a cost of USD 15.00 each (running average of physically and financially posted transactions).
- 6. Inventory close is performed. Based on the weighted average method, the system uses the direct settlement method because only one receipt is financially updated in the period. In this example, one settlement is created between 1b and 3b, and another between 1b and 4b. Transaction 3b and 4b are each adjusted by USD -5.00 to bring the value to USD 10.00.

The following diagram illustrates this series of transactions with the effects of choosing the weighted average inventory model and the direct settlement principle with the **Include physical value** option.



Key to diagram

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.

- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average summarized settlement with the Include physical value option

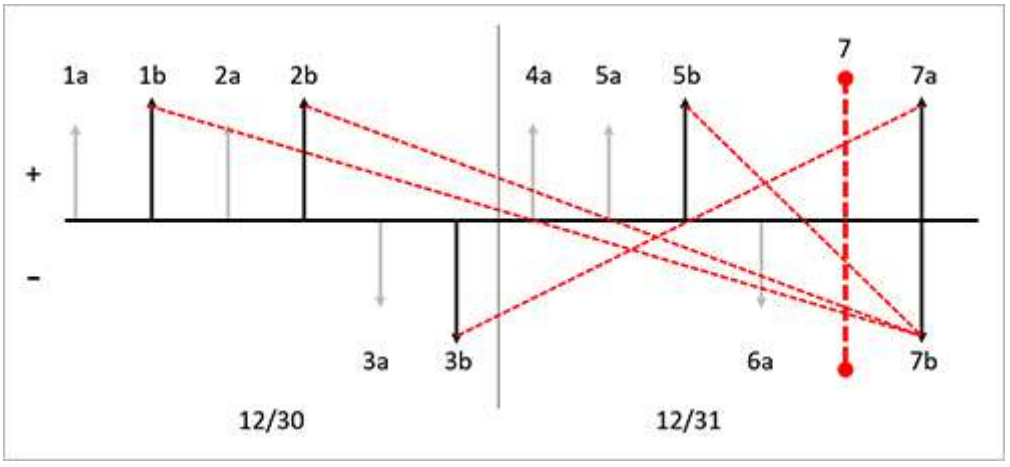
The **Include physical value** parameter works differently with weighted average than in earlier versions. Select the **Include physical value** checkbox for an item on the **Item model group** page. Then the system uses physically updated receipts in the calculation of estimated cost price, or running average. Issues are posted based on this estimated cost price during the period. During the inventory close, financially updated receipts are only considered in the weighted average calculation. We recommend a monthly inventory close when you use the weighted average inventory model. In this weighted average summarized settlement example, the inventory model is marked to include physical value.

The following transactions are illustrated in the graphic that follows:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.67 (running average of physically and financially posted transactions).
- 7. Inventory close is performed.
- 7a. Weighted average inventory close transaction financial issue is created to sum the settlements of all the inventory financial receipts.
    - Transaction 1b is settled for a quantity of 1 with an amount settled of USD 10.00.
    - Transaction 2b is settled for a quantity of 1 with an amount settled of USD 22.00.
    - Transaction 5b is settled for a quantity of 1 with an amount settled of USD 30.00.
    - Transaction 7a. is created for a quantity of 3 with an amount settled of USD 62.00.

- 7b. Weighted average inventory close transaction financial receipt is created as the offset the financially closed issue transactions.
    - Transaction 3b is settled for a quantity of 1 with an amount settled of USD 20.67. This transaction is adjusted by USD 4.67 to bring the original value of USD 16.00 to 20.67, which is the weighted average of financially posted transactions for the period.
    - Transaction 7b. is created for a quantity of 1 with an amount settled of USD 20.67 to offset 3b.

The following diagram illustrates this series of transactions with the effects of choosing the weighted average inventory model and the summarized settlement principle without the **Include physical value** option.



Key to diagram

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- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average with marking

Marking is a process that lets you link, or mark, an issue transaction to a receipt transaction. Marking can occur either before or after a transaction is posted. You can use marking when you

want to make sure of the exact cost of the inventory when the transaction is posted or when the inventory close is performed.

For example, your Customer Service department accepted a rush order from an important customer. Because this is a rush order, you'll have to pay more for this item to service your customer's request. You must be certain the cost of this inventory item is reflected in the margin, or cost of goods sold (COGS), for this sales order invoice.

When the purchase order is posted, the inventory is received at a cost of USD 120.00. For example, this sales order document is marked to the purchase order before the packing slip or invoice is posted. Then COGS will be USD 120.00 instead of the current running average cost for the item. If the sales order packing slip or invoice is posted before the marking occurs, the COGS will be posted at the running average cost price.

Before inventory close is performed, these two transactions can still be marked to each other.

A receipt transaction is marked to an issue transaction. Then, the valuation method selected for the item's item model group will be disregarded and the system will settle these transactions to each other.

You can mark an issue transaction to a receipt before the transaction is posted. You can do this from a sales order line on the **Sales order details** page. The open receipt transactions are shown on the **Marking** page.

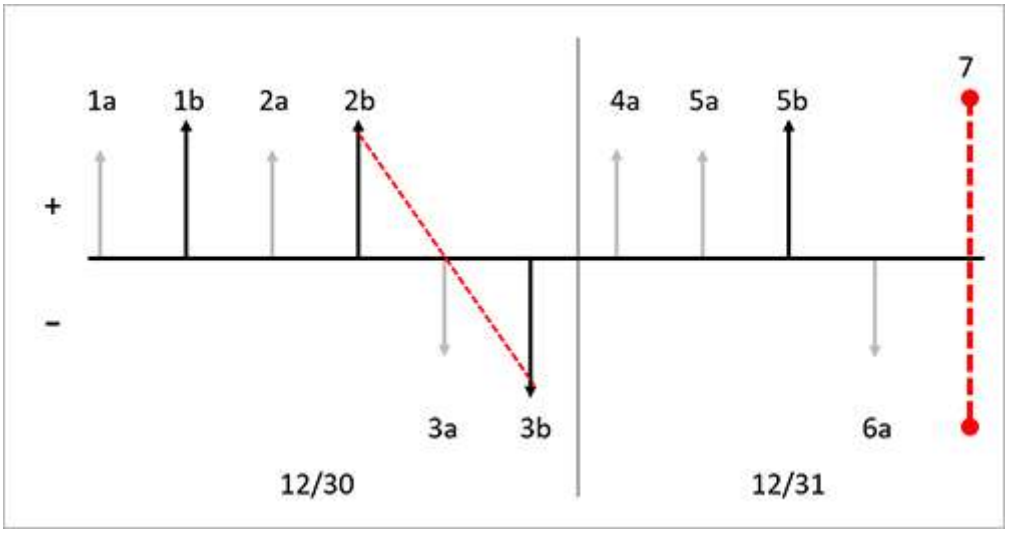
You can mark an issue transaction to a receipt after the transaction has been posted. You can match or mark an issue transaction for an open receipt transaction for an inventoried item from a posted inventory adjustment journal.

The following transactions are illustrated in the graphic that follows:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3c. Inventory financial issue for 3b is marked to inventory financial issue for 2b.
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions).

- 7. Inventory close is performed. Based on the marking principle that uses the weighted average method, the marked transactions are settled against each other. In this example, 3b is settled against 2b, and an adjustment for USD 6.00 is posted to 3b to bring the value to USD 22.00. In this example, no additional settlements are made because the close creates settlements only for financially updated transactions.

The following diagram illustrates this series of transactions with the effects of choosing the weighted average inventory model with marking.



Key to diagram

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.
- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Each date in the diagram is separated by a thin black vertical line. The date is noted at the bottom of the diagram.
- Inventory closings are represented by a red vertical dashed line.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average date with Include physical value and marking

Article • 01/24/2025

Weighted average date is an inventory model that is based on an average that is calculated by multiplying each component (item transaction) by a factor (cost price) that reflects its importance (quantity) on each day in the period. In other words, this inventory model assigns the cost of issue transactions based on the mean value of all inventory that is received each day, plus any on-hand inventory from the previous day.

When you run an inventory closing by using the weighted average date inventory model, two methods can be used to create a settlement. Typically, all receipts are settled against a virtual issue that holds the total received quantity and value. This virtual issue has a corresponding virtual receipt that the issues are settled from. In this way, all issues get the same average cost each day. The virtual issue and the virtual receipt can be considered a virtual transfer. This virtual transfer is referred to as *weighted average inventory closing transfer*. Therefore, this settlement method is known as a *weighted average summarized settlement*. If there's only one receipt, all issues can be settled from it, and no virtual transfer is created. This settlement method is referred to as *direct settlement*. Any inventory that is on hand after the inventory close is performed is valued at the weighted average from the last day of the previous period and included in the weighted average date calculation in the next period.

You can override the weighted average date principle by marking inventory transactions so that a specific item receipt is settled against a specific issue. A periodic inventory close is required when you use the weighted average date inventory model to create settlements and adjust the value of issues according to the weighted average date principle. Until you run the inventory close, issue transactions are valued at the running average when the physical and financial updates occurred. Unless you're using marking, the running average is calculated when the physical or financial update is performed.

The weighted average date inventory costing method is calculated by using the following formula each day:

$$Cost = [(Q_b \times P_b) + \sum_n (Q_n \times P_n)] \div (Q_b + \sum_n Q_n)$$

- Q = Quantity of the transaction
- P = Price of the transaction

In other words, the weighted average cost equals the beginning quantity times the beginning price, plus the sum of each receipt quantity times its receipt price, all divided

by the beginning quantity plus the sum of the receipt quantities.

During inventory close, the calculation is performed every day during the closing period.

> **Note**
>
> For more information about settlements, see Inventory close.

The following examples show the effect of using weighted average date with five configurations:

- Weighted average date direct settlement when the **Include physical value** option isn't used
- Weighted average date summarized settlement when the **Include physical value** option isn't used
- Weighted average date direct settlement when the **Include physical value** option is used
- Weighted average date summarized settlement when the **Include physical value** option is used
- Weighted average date when marking is used

Weighted average date direct settlement when the Include physical value option isn't used

The direct settlement principle creates settlements directly between receipts and issues, without creating additional inventory transactions. The system uses this direct settlement principle in the following situations:

- One receipt and one or more issues have been posted in the period.
- Only issues have been posted in the period, and the inventory contains on-hand items from a previous closing.

In this example, the **Include physical value** checkbox is cleared on the **Item model group** page for the released product. The diagram that follows shows these transactions:

Day 1:

- 1a. Inventory physical receipt for a quantity of 10 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 10 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 10 at a cost of USD 20.00 each.

- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 10.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 10.00 (running average of financially posted transactions).

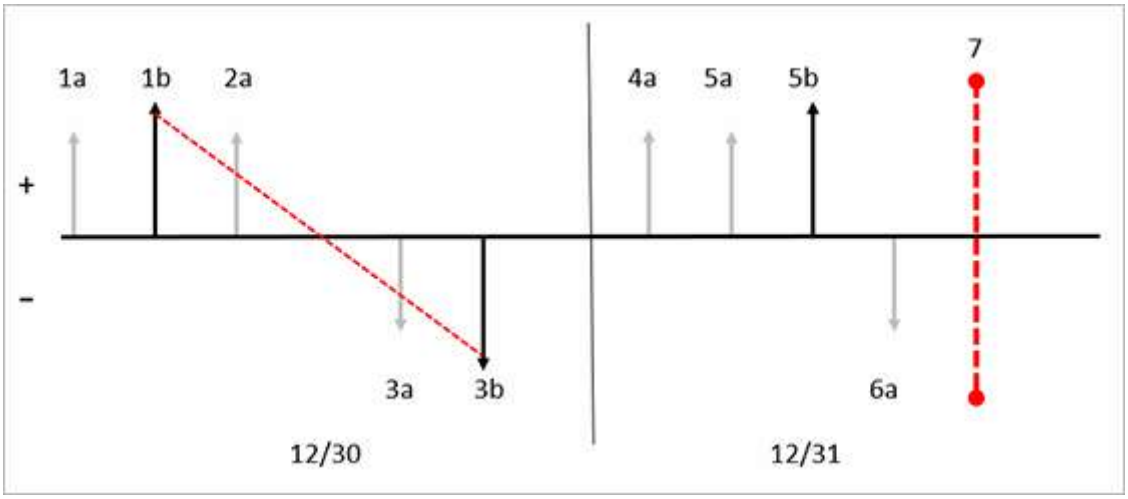
Day 2:

- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory financial issue for a quantity of 1 at a cost of USD 20.00 each (running average of financially posted transactions).

Day 3:

- 7. Inventory close is performed. Based on the weighted average date method, the system uses the direct settlement method for December 30 (12/30) because only one receipt is financially updated on 12/30. In this example, one settlement is created between transactions 1b and 3b. An adjustment of USD 10.00 is made to bring the value of transaction 3b up to 20.00. No adjustment or settlement is made on December 31 (12/31) because there are no financially updated issues on 12/31.

The following diagram shows this series of transactions and the effects of using the weighted average inventory model and the direct settlement principle without the **Include physical value** option.



Key to the diagram:

- Inventory transactions are represented by vertical arrows.
- Physical transactions are represented by shorter light gray arrows.
- Financial transactions are represented by longer black arrows.
- Receipts into inventory are represented by vertical arrows above the axis.
- Issues out of inventory are represented by vertical arrows below the axis.

- Each new receipt or issue transaction is designated by a new label.
- Each vertical arrow is labeled with a sequential identifier, such as *1a*. The identifiers indicate the order of inventory transaction postings in the timeline.
- Dates are separated by thin black vertical lines. The dates are noted at the bottom of the diagram.
- Inventory closings are represented by red vertical dashed lines.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average date summarized settlement when the Include physical value option isn't used

When there are multiple receipts in a period, weighted average date uses the summarized settlement principle, where all receipts in a single day are summarized into a transaction that is named *weighted average inventory closing*. All the receipts for the day will be settled against the issue of the newly created inventory transaction. All issues for the day will be settled against the receipt of the new inventory transaction. If any on-hand inventory value remains after the inventory close, the on-hand inventory value is included in the receipt transaction of the weighted average inventory closing transactions.

The diagram that follows shows these transactions:

Day 1:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).

Day 2:

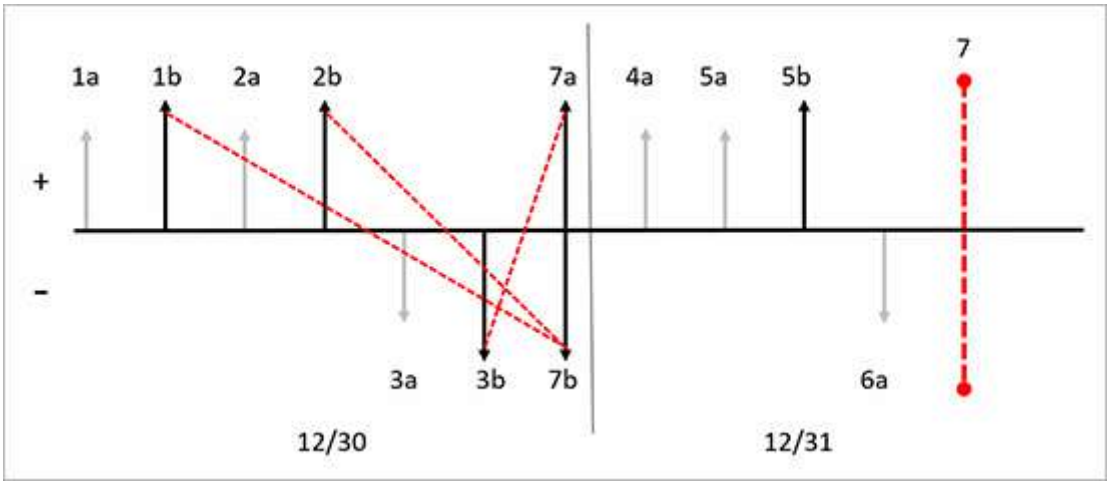
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.

- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions).

Day 3:

- 7. Inventory close is performed.
- 7a. Weighted average inventory close transaction financial issue is created to sum the settlements of all the inventory financial receipts.
  - Transaction 1b is settled for a quantity of 1 with a settled amount of USD 10.00.
  - Transaction 2b is settled for a quantity of 1 with a settled amount of USD 22.00.
  - Transaction 7a is created for a quantity of 2 with a settled amount of USD 32.00. This transaction offsets the sum of the two receipt transactions that are financially updated in the period.
- 7b. Weighted average inventory close transaction financial receipt is created as the offset for financially posted issues.
  - Transaction 3b is settled for a quantity of 1 with a settled amount of USD 16.00. This transaction isn't adjusted because it's the weighted average of financially posted transactions on December 1 (12/1).
  - Transaction 7b is created for a quantity of 2 with a financial amount of USD 32.00 and a settled amount of USD 16.00 to offset transaction 3b. This transaction offsets the sum of the one issue transaction that is financially updated in the period. The transaction remains open because there's still one on hand.

The following diagram shows this series of transactions and the effects of using the weighted average inventory model and the summarized settlement principle, but without using the **Include physical value** option.



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- Inventory closings are represented by red vertical dashed lines.
- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average date direct settlement when the Include physical value option is used

In the current version of the product, the **Include physical value** option works differently with the weighted average date inventory model than it worked in earlier versions. When you select the **Include physical value** checkbox for an item on the **Item model group** page, the system will use physically updated receipts when it calculates the estimated issue cost price, or running average. Issues will be posted based on this estimated cost price during the period. During the inventory close, only financially updated receipts will be considered in the weighted average calculation.

The diagram that follows shows these transactions:

Day 1:

- 1a. Inventory physical receipt for a quantity of 10 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 10 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 10 at a cost of USD 20.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 15.00 (running average of physically and financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 15.00 (running average of physically and financially posted transactions).

Day 2:

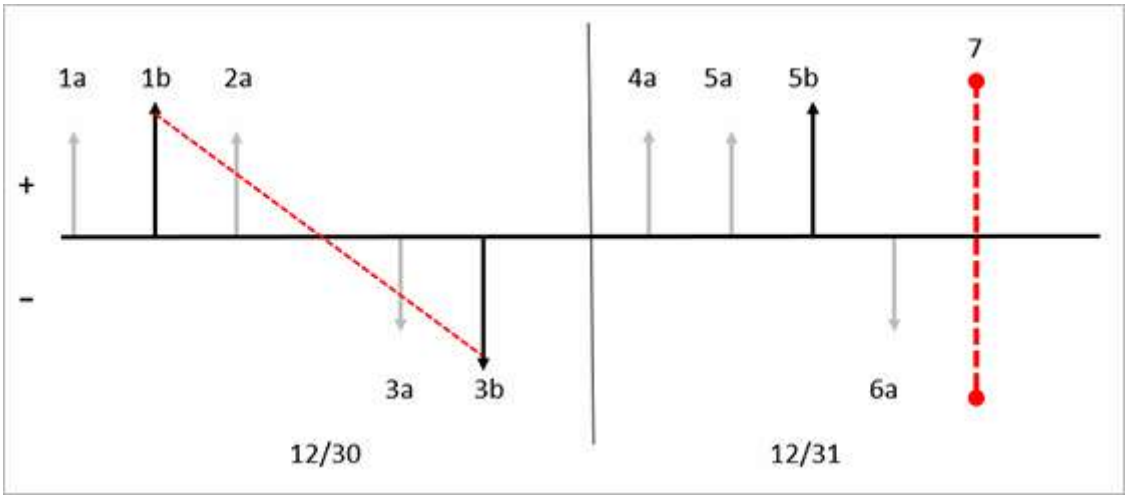
- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.

- 6a. Inventory physical issue for a quantity of 1 at a cost of USD 21.25 each (running average of physically and financially posted transactions).

Day 3:

- 7. Inventory close is performed. Based on the weighted average date method, the system uses the direct settlement method for December 30 (12/30) because only one receipt is financially updated on 12/30. In this example, one settlement is created between transactions 1b and 3b. No adjustment is made to the issue on 12/30. Additionally, no adjustment or settlement is made on December 31 (12/31) because there are no financially updated issues on 12/31.

The following diagram shows this series of transactions and the effects of using the weighted average inventory model and the direct settlement principle with the **Include physical value** option.



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- Settlements that are performed by inventory close are represented by red diagonal dashed arrows that go from a receipt to an issue.

Weighted average date summarized settlement when the Include physical value option is used

In the current version of the product, the **Include physical value** option works differently with weighted average than it worked in earlier versions. When you select the **Include physical value** checkbox for an item on the **Item model group** page, the system will use physically updated receipts when it calculates the estimated cost price, or running average. Issues will be posted based on this estimated cost price during the period. During the inventory close, only financially updated receipts will be considered in the weighted average calculation. We recommend that you perform a monthly inventory close when you use the weighted average inventory model. In this example of weighted average date summarized settlement, the inventory model is marked to include physical value.

The diagram that follows shows these transactions:

Day 1:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.
- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of physically and financially posted transactions).

Day 2:

- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.67 (running average of physically and financially posted transactions).

Day 3:

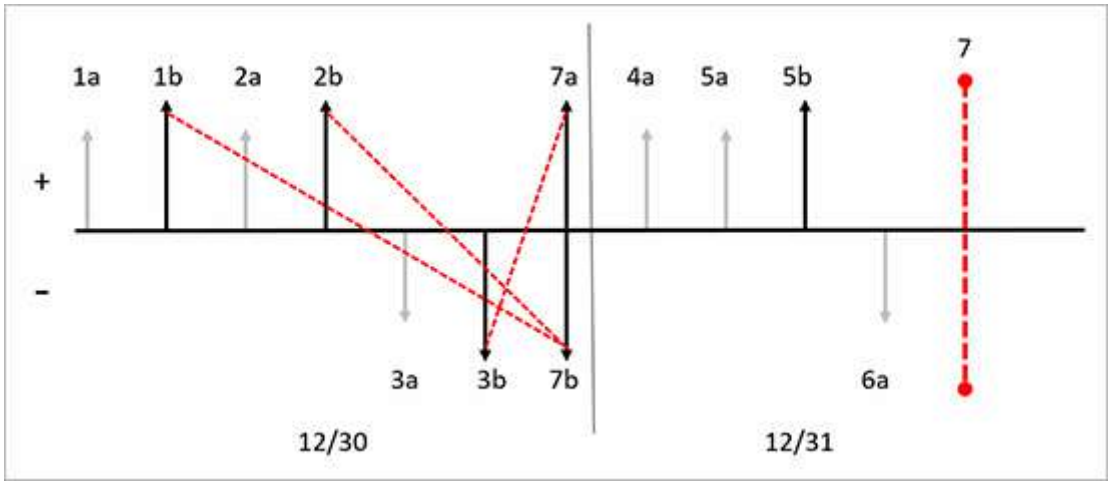
- 7. Inventory close is performed.

- 7a. Weighted average inventory close transaction financial issue is created to sum the settlements of all the inventory financial receipts.
  - Transaction 1b is settled for a quantity of 1 with a settled amount of USD 10.00.
  - Transaction 2b is settled for a quantity of 1 with a settled amount of USD 22.00.

    - Transaction 7a is created for a quantity of 2 with a settled amount of USD 32.00. This transaction offsets the sum of the two receipt transactions that are financially updated in the period.

- 7b. Weighted average inventory close transaction financial receipt is created as the offset for financially posted issues.
    - Transaction 3b is settled for a quantity of 1 with a settled amount of USD 16.00. This transaction isn't adjusted because it's the weighted average of financially posted transactions on December 1 (12/1).
    - Transaction 7b is created for a quantity of 2 with a financial amount of USD 32.00 and a settled amount of USD 16.00 to offset transaction 3b. This transaction offsets the sum of the one issue transaction that is financially updated in the period. The transaction remains open because there's still one on hand.

The following diagram shows this series of transactions and the effects of using the weighted average inventory model and the summarized settlement principle without the **Include physical value** option.



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Weighted average date when marking is used

Marking is a process that lets you link, or mark, an issue transaction to a receipt transaction. Marking can occur either before or after a transaction is posted. You can use marking when you want to make sure of the exact cost of the inventory when the transaction is posted or the inventory close is performed.

For example, your Customer Service department accepted a rush order from an important customer. Because it's a rush order, you'll have to pay more for this item to service your customer's request. You must be sure that the cost of this inventory item is reflected in the margin, or cost of goods sold (COGS), for this sales order invoice.

When the purchase order is posted, the inventory is received at a cost of USD 120.00. If the sales order document is marked to the purchase order before the packing slip or invoice is posted, COGS will be USD 120.00 instead of the current running average cost for the item. If the marking occurs after sales order packing slip or invoice is posted, the COGS will be posted at the running average cost price.

Before inventory close is performed, these two transactions can still be marked to each other.

If a receipt transaction is marked to an issue transaction, the valuation method that is selected for the item's item model group will be disregarded, and the system will settle these transactions to each other.

You can mark an issue transaction to a receipt before the transaction is posted. You can do this marking from a sales order line on the **Sales order details** page. The open receipt transactions are shown on the **Marking** page.

You can also mark an issue transaction to a receipt after the transaction is posted. You can match or mark an issue transaction for an open receipt transaction for an inventoried item from a posted inventory adjustment journal.

The diagram that follows shows these transactions:

Day 1:

- 1a. Inventory physical receipt for a quantity of 1 at a cost of USD 10.00 each.
- 1b. Inventory financial receipt for a quantity of 1 at a cost of USD 10.00 each.
- 2a. Inventory physical receipt for a quantity of 1 at a cost of USD 20.00 each.
- 2b. Inventory financial receipt for a quantity of 1 at a cost of USD 22.00 each.

- 3a. Inventory physical issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3b. Inventory financial issue for a quantity of 1 at a cost price of USD 16.00 (running average of financially posted transactions).
- 3c. Inventory financial issue for transaction 3b is marked to inventory financial issue for transaction 2b.

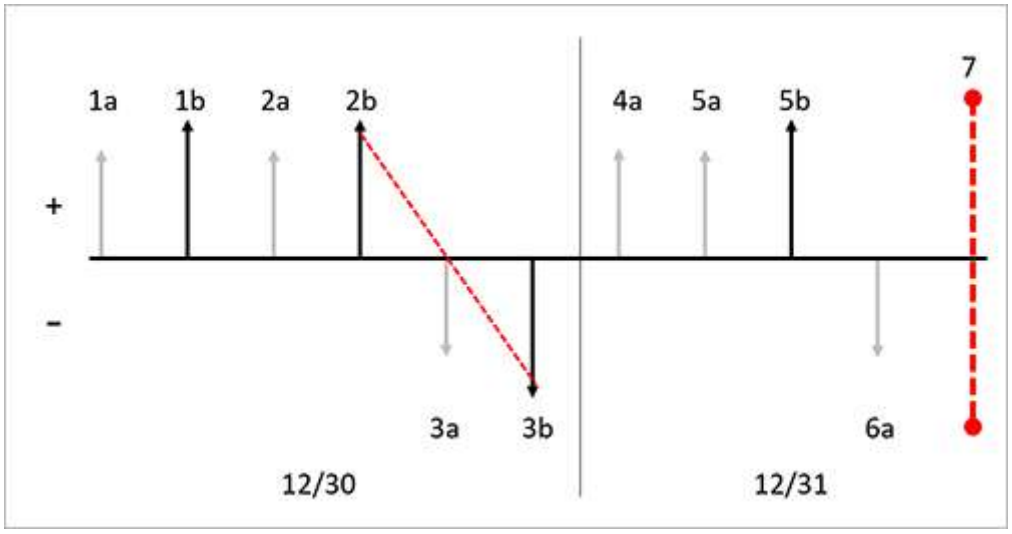
Day 2:

- 4a. Inventory physical receipt for a quantity of 1 at a cost of USD 25.00 each.
- 5a. Inventory physical receipt for a quantity of 1 at a cost of USD 30.00 each.
- 5b. Inventory financial receipt for a quantity of 1 at a cost of USD 30.00 each.
- 6a. Inventory physical issue for a quantity of 1 at a cost price of USD 23.00 (running average of financially posted transactions).

Day 3:

- 7. Inventory close is performed. Based on the marking principle that uses the weighted average method, the marked transactions are settled against each other. In this example, transaction 3b is settled against transaction 2b, and an adjustment for USD 6.00 is posted to transaction 3b to bring the value to USD 22.00. In this example, no additional settlements are made because the close creates settlements only for financially updated transactions.

The following diagram shows this series of transactions and the effects of using the weighted average inventory model with marking.



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Feedback

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Moving average

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Moving average is a perpetual costing method based on the average principle, where the costs on inventory issues don't change when the purchase cost does. The difference is capitalized and is based on a proportional calculation. The amount that remains is expensed.

When you use moving average, inventory settlements and inventory marking aren't supported. Inventory close doesn't affect products that have moving average as the inventory model group, and it doesn't generate any settlements between the transactions.

The following are prerequisites when you use moving average cost as a costing method.

1. On the **Item model groups** page, set up an item model group that has **Moving average** selected in the **Inventory model** field.

   > **Note**
   >
   > By default, when **Moving average** is selected, the **Post physical inventory** and **Post financial inventory** fields are also selected.

2. On the **Posting** page, assign accounts to the **Price difference for moving average**. You use the **Price difference for moving average** account when cost has to be proportionally expensed. This occurs in following two scenarios:

   - There's a difference in cost between a purchase receipt and the purchase invoice because there's a difference between the original inventory quantity and the current on-hand quantity.
   - The transactions bring the inventory from negative to zero, and there's a difference between the transaction cost and the current moving average cost.

3. On the **Posting** page, assign accounts to the **Cost revaluation for moving average** accounts on the **Inventory** tab. You use the **Cost revaluation for moving average** account when you want to adjust the moving average cost for a product to a new unit price.

4. On the **Released products** page, assign the moving average item model group to the product.

   > ⓘ **Note**
   >
   > The inventory close process only closes the accounting period. It doesn't affect products that have moving average assigned to them as an item model group.

Convert to the moving average costing method

You can convert products to use the moving average inventory valuation method. You usually do this type of conversion at the end of the year, after the last month of the current year is closed, by using the product's current costing model. You can change your inventory costing method from a costing method that is based on average cost or standard cost to a method that is based on moving average.

If you're changing your costing method from a standard costing method to a moving average method, you have to complete the following tasks:

1. Make adjustments to get inventory quantities and values down to 0 (zero).
2. After the inventory value and quantity are 0 (zero), change the item model group to moving average.
3. Make adjustments to get the quantity and value back into inventory.

You can't change your inventory costing method from a moving average method to a First in, First out (FIFO) method, a Last in, First out (LIFO) method, or a weighted average method.

> ⓘ **Note**
>
> Converting from standard cost to moving weighted average is a manual process.

The following examples illustrate the effect of using the moving average costing method. There are four configurations:

- Purchase order and proportionally expensed cost difference

- Moving average product and inventory adjustment
- Moving average with production
- Moving average with a backdated transaction

Purchase order and proportionally expensed cost difference

With moving average, the purchase receipt determines the product's cost. When the purchase invoice is posted, if there's a difference in cost between the purchase receipt and the purchase invoice, the difference is proportionally adjusted to the current products in stock, and any remaining amount is expensed.

In this example, a purchase order is created and received at one cost, and the purchase invoice is posted with a different cost.

1. Create a purchase order for a quantity of 2 and a unit price of 10.00.
2. Create a purchase receipt of the product.
3. Create a sales order for a quantity of 1 and a unit price of 10.00.
4. Create a purchase invoice for a quantity of 2 and a unit price of 12.00.

The difference in unit price, 2.00, is posted to the Price difference for moving average account when the purchase invoice is posted. The reason is that two products were purchased for a cost of 20.00. One of the products was sold for a unit price of 10.00. The purchase invoice was posted at a unit price of 12.00 with a quantity of 2. The unit price of the product can't be posted at 14.00.

Moving average product and inventory adjustment

If you need to adjust the moving average cost of a product, inventory adjustments are allowed as of today's date. You can't backdate an inventory adjustment to correct the moving average cost of a product. You can't have the cost flow through subsequent transactions.

In this example, the moving average cost is adjusted for a product.

1. Select the product that you want to adjust the moving average cost for.

> **Note**

The **Revaluation for moving average** page examines the inventory available for a product. The product selected has a posted quantity of 1, a posted a value of 12.00, a posted unit cost of 12.00, and a unit cost of 12.00.

2. Update the **Unit cost** field to 16.00. The system calculates the remaining fields.

3. The adjustment is posted.

> **Note**
>
> You can only adjust the moving average cost as of today's date.

On the **Settlements for voucher** page, you can see an adjustment of 4.00 posted to the Cost revaluation for moving average account.

Moving average with production

Moving average supports produced items. If you plan to use moving average in a production environment, select **Use estimated cost price** on the **Production control parameters** page. This means that the cost price that is calculated during estimation is used instead of the actual bill of material (BOM) calculation cost price.

Moving average with a backdated transaction

Backdated transactions are assigned the current moving average cost, and the product's physical quantity is updated, but the product's moving average cost isn't affected. In this moving average example, a backdated transaction for a moving average product is posted.

1. Create an inventory adjustment for the moving average product for a quantity of 1 and a cost of 20.00.
2. The inventory transaction history for the product resembles the following:

    - An inventory transaction of 1, a cost of 16.00, a posting date of January 15, and a transaction date of January 15.
    - An inventory adjustment of 1, a cost of 20.00, a posting date of January 1, and a transaction date of January 15.

3. Post the adjustment.

On the **Inventory transactions** page, you can see that 4.00 is expensed as the current moving average for the product is 16.00. You can post in the past, but the difference in cost is expensed, so the moving average cost isn't affected.

Negative inventory balances

Transactions are handled differently depending on whether the new on-hand quantity after the transaction is negative, zero, or positive.

New balance is negative or zero

If the new on-hand quantity is negative or zero, the transaction is costed at the current average costs. If there's a difference between the purchase price and the current average costs, it's posted to **Price difference for moving average**.

New balance is positive

If the new on-hand quantity is positive after the transaction, the transaction is split into two parts and costed differently, as summarized in the following table.

Expand table

Part	Description
Quantity from negative to zero	Inventory uses the current moving average cost of the item rather than the transaction cost for that portion of the receipt quantity that increases the on-hand balance from negative to zero. The difference between the transaction cost and the current moving average cost is posted to **Price difference for moving average**.
Quantity from zero to positive	Inventory uses the transaction cost for that portion of the receipt quantity that increases the on-hand balance from zero to positive.

Inventory value report

In this moving average example, the inventory value report is printed to support the current moving average calculation for a product. The Inventory value report can print the transactions in chronological order, together with the cost to support the moving average cost calculation of a product. The report displays the moving average cost for the product. In the **Inventory value reports** dialog box, a date interval allows you to select the **Transaction time** or the **Posting date** to sort the report by. The **Posting date** option is how the report is traditionally printed. The

Transaction time option is the actual date that the transaction is reported and the moving average cost for the product is updated. You can print the Inventory value report by using the **Transaction time sorting** option if you want to see the moving average cost calculation over time. The following table displays the transactions for the product that the report covers when using the **Transaction time sorting** option.

Expand table

Transaction time	Date	Transaction type	Quantity	Amount	Average unit cost
	October 1	Beginning balance	0	0.00	0.00
October 8	September 28	Backdated receipt	1	16.00	16.00
October 3	October 3	Purchase receipt	2	20.00	12.00
October 5	October 5	Sales order	-1	-10.00	13.00
October 7	October 7	Purchase invoice		2.00	14.00
October 8	October 8	Moving average revaluation		4.00	16.00
	October 31	Total	2	32.00	16.00

Note

You can't reconcile the general ledger with inventory by using the **Transaction time sorting** option. The report must be printed by using the **Posting date** option.

As of the release of the Inventory value report storage report in Supply Chain Management version 10.0.29, the older Inventory value report no longer supports transaction-level calculations. As a result, the **Average unit cost** is hidden by default for the older report. However, an extension point is provided that makes it possible to customize the system to add support for calculating cost prices on a transaction level and to make the **Average unit cost** field visible in the Inventory value report.

Last updated on 06/22/2026

Moving average fallback cost sequence

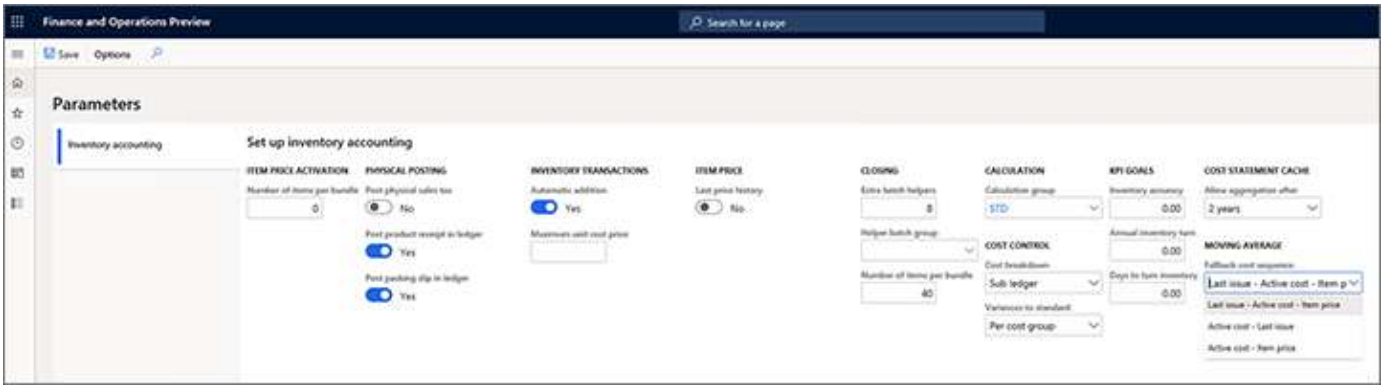
One way that you can calculate the cost of your inventory is by using a *moving average*. Up to three cost values can be associated with each inventory item:

- **Last issue** – The last issue cost that was assigned before inventory went negative
- **Active cost** – The latest cost that was activated in a costing version
- **Item price** – The cost that is specified for the released product

To determine which of these cost values should be used in moving average calculations, the system uses a *fallback cost sequence* to establish the order of preference for the values. If the preferred cost value isn't available, the system uses the next-preferred value, and so on.

To select the fallback cost sequence for moving average calculations, follow these steps:

1. Go to **Cost management** > **Inventory accounting policies setup** > **Parameters**.
2. On the **Inventory accounting** tab, in the **Moving average** section, set the **Fallback cost sequence** field to one of the following values:
    - *Last issue – Active cost – Item price* – This is the default sequence.
    - *Active cost – Last issue*
    - *Active cost – Item price* – Organizations might experience performance issues if they use business processes where inventory regularly goes negative and, at the same time, the transaction volume is high. This setting can help mitigate those performance issues.



Last updated on 12/30/2025

Physical and financial updates

This article provides an overview of which types of transactions increase or decrease inventory quantities.

You can physically update and financially update inventory transactions in Dynamics 365 Supply Chain Management. Some types of physical and financial transactions increase inventory quantities, whereas others decrease the quantity.

Physical increases

When a physical transaction is posted, the status of the transaction record is *Received*. The following transactions are considered physical increases:

- Purchase order receipt
- Sales order packing slip return
- Reporting a production order as finished
- By-product on a production order picking list

Financial increases

When a financial receipt transaction is posted, the status of the transaction record that increases the quantity is *Purchased*. The following transactions are considered financial increases:

- Vendor invoice
- Sales order invoice for a return
- Production order costing
- Positive quantity inventory journals, such as movement, profit and loss, counting, bill of materials, and transfer

Transactions that increase quantity

Transactions that increase quantity are posted at the running average cost price. The calculated running average cost price is based on the cost of each of these transactions for each inventory dimension that is tracked financially. For information about running average cost prices, see Running average cost price.

Transactions that decrease quantity

The calculated running average cost price is used when a transaction that decreases quantity is posted, regardless of the inventory model associated with that inventory. The transaction that decreases quantity shouldn't marked to another transaction before it's posted. If the physical on-hand inventory becomes negative, the inventory cost defined for the item on the **Item** page is used.

> ⓘ **Note**
>
> If multisite functionality is enabled, this cost is the inventory cost defined for a site on the **Default order settings** page.

Physical issues vs. financial issues

When a physical issue transaction is posted, the status of the transaction record is *Deducted*. The following transactions are considered physical issues:

- Production order picking list journal
- Sales order packing slip
- Purchase order packing slip return

When a financial transaction is posted, the status of the transaction record is *Sold*. The following transactions are considered financial issues:

- Ending a production order
- Sales order invoice
- Vendor invoice return
- Negative quantity inventory journals, such as movement, profit and loss, counting, bill of materials, and transfer

Transactions that decrease quantity are posted at the running average cost price. Therefore, you need to run the inventory close procedure to settle issue transactions to receipt transactions, based on the inventory model assigned to each item.

Last updated on 07/01/2026

Options for including physical value in cost calculations

Supply Chain Management provides two options for including physical value in cost calculations:

- Include physical value
- Include physical value in weighted average recalculation

You can also choose not to include physical value in cost calculations.

This article explains these options and how to set them up.

Include physical value

You can choose whether the system should consider physically updated transactions when calculating the average cost price for items belonging to a given item model group. To set this option, follow these steps:

1. Go to **Cost management** > **Inventory accounting policies setup** > **Item model groups**.
2. On the list pane, select the item model group that you want to configure (or create a new one).
3. Expand the **Costing method & cost recognition** FastTab.
4. Set the **Include physical value** check box to one of the following values:

    - *Selected* – Both physically updated transactions and financially updated transactions are used to calculate the running average cost price.
    - *Cleared* – Only financially updated transactions are used to calculate the running average cost price.

The check box has slightly different effects, depending on the inventory model that you use.

- If you select the **Include physical value** check box when using the *FIFO* (first in, first out), *LIFO* (last in, first out), or *LIFO date* inventory model, inventory close also makes adjustments to physically updated transactions.
- If you don't select the **Include physical value** check box when using the *FIFO* (first in, first out), *LIFO* (last in, first out), or *LIFO date* inventory model, inventory close makes settlements only to financially updated transactions.

- When you use the *weighted average* or *weighted average date* inventory model, inventory close settles only financially updated transactions, regardless of whether you select the **Include physical value** check box.

Include physical value example 1

The **Include physical value** check box is selected and you receive the following purchase orders:

- A purchase order for a quantity of 2 and a cost price of USD 10.00 that has been packing slip–updated.
- A purchase order for a quantity of 3 and a cost price of USD 12.00 that has been invoice-updated.

In this case, the running average cost price is USD 11.20 = (2x10+3x12)/(2+3), because both physically updated transactions and financially updated transactions are used to calculate the cost price.

Include physical value example 2

The **Include physical value** check box isn't selected, and the cost price on the item setup is USD 10.00.

- You receive a purchase order for a quantity of 20 and a cost price of USD 12.00 that has been packing slip–updated.

When a sales order is posted, the posted cost amount is USD 10.00, because the running average cost price doesn't include physically posted transactions.

> **Note**
>
> For comparison, if you select the **Include physical value** check box for this item, when a sales order is posted, the posted cost amount will be USD 12.00.

Include physical value in weighted average recalculation

This feature lets you choose whether to include physically updated transactions that haven't yet been financially updated when recalculating weighted average.

> **Important**
>
> This capability is only available during recalculation and isn't used during inventory close. If you would like to check the results including physical value after inventory closing, you should run a new recalculation.

Prerequisites

To use the feature described in this section, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.41 or later.
- The feature named *Recalculation for Weighted Average Cost including Physical Transaction* must be turned on in feature management.

Choose whether to include physical value in weighted average recalculation

To choose whether the system should include physically updated transactions that haven't yet been financially updated when recalculating weighted average, follow these steps:

1. Go to **Cost management** > **Inventory accounting policies setup** > **Item model groups**.
2. On the list pane, select the item model group that you want to configure (or create a new one).
3. Expand the **Costing method & cost recognition** FastTab.
4. Set the **Include physical value in weighted average recalculation** check box to one of the following values:

    - *Selected* – Both physically updated transactions and financially updated transactions are used to calculate the running average cost price.
    - *Cleared* – Only financially updated transactions are used to calculate the running average cost price.

> **Note**
>
> - This check box is only visible for item model groups where **Inventory model** is set to *Weighted avg.* or *Weighted avg. date*.

- This option is only available during recalculation and isn't used during inventory close. To ensure accurate results, we recommend that you perform the recalculation immediately after inventory closing.

Last updated on 03/25/2026

Fixed receipt price

Fixed receipt price is an option that you can select on an item model group when you use an inventory model other than *Standard cost* or *Moving weighted average*. In early versions of Microsoft Dynamics AX, this option was named **Standard cost**. It was renamed **Fixed receipt price** when the new standard cost inventory model was introduced in Dynamics AX 2012. This article explains how you can configure and use fixed receipt prices in Dynamics 365 Supply Chain Management.

About fixed receipt prices

When you select the **Fixed receipt price** checkbox on the **Item model group** page for an item, the system uses the receipt price as a standard cost for the inventory receipt. If the purchase price and the default item cost price that is configured for a product differ, the difference is posted to the **Fixed receipt price profit** or **Fixed receipt price loss** account and offset to the **Fixed receipt price offset** account that you specify on the **Inventory posting profile** page.

Because the **Fixed receipt price** option is used with different inventory models (such as first in, first out (FIFO); last in, first out (LIFO); and weighted average), the *Inventory close and adjustment* process will still create settlements and adjustments according to the inventory principle that you select on the **Item model group** page. However, adjustments are made only to issues from inventory.

For example, you've configured a product with an item model group where the **Inventory model** field is set to *FIFO* and the **Fixed receipt price** option is selected. When you receive inventory through a purchase order, the inventory value is set to the value of the item's default cost price at the time of product receipt. When you invoice the purchase order, if the invoice price is different, the system posts the default cost price on the released product to the inventory account. It posts the difference to the fixed receipt price profit or loss account. Later, when you sell or consume the product, the system uses the running average for the item at the time when the sales order invoice was posted (unless you use marking).

When you run the *Inventory close and adjustment* process, the system creates a settlement with the first issue against the first receipt. The difference between the receipt price that was used on the receipt and the running average that was used on the issue is posted in a new voucher.

Enable fixed receipt prices

To enable fixed receipt prices for an item, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory** > **Item model groups**.
2. Create a new item model group, or select an existing model group.
3. On the **Costing method & cost recognition** FastTab, select the **Fixed receipt price** checkbox.

   > ⓘ **Note**
   >
   > You can't select the **Fixed receipt price** checkbox if the **Inventory model** field is set to *Standard cost* or *Moving average*.

4. Close the page.

After you enable the **Fixed receipt price** option, you must update your inventory posting profile by following these steps.

1. Go to **Inventory management** > **Setup** > **Posting** > **Posting**.
2. On the **Purchase order** tab, in the **Select** column, select **Fixed receipt price profit**.
3. On the Action Pane, select **New** to add a row to the grid.
4. Set up the new row so that it applies to the item model group that you enabled fixed receipt pricing for, and specify the main account that is used to record fixed receipt price profits for purchase orders. Configure other settings as you require.
5. In the **Select** column, select **Fixed receipt price loss**. Add a new row that applies to the item model group that you enabled fixed receipt pricing for, and specify the main account that is used to record fixed receipt price losses for purchase orders. Configure other settings as you require.
6. In the **Select** column, select **Fixed receipt price offset**. Add a new row that applies to the item model group that you enabled fixed receipt pricing for, and specify the main account that is used to record fixed receipt price offsets for purchase orders. Configure other settings as you require.
7. On the **Inventory** tab, in the **Select** column, select **Fixed receipt price profit**. Add a new row that applies to the item model group that you enabled fixed receipt pricing for, and specify the main account that is used to record fixed receipt price profits for inventory. Configure other settings as you require.
8. In the **Select** column, select **Fixed receipt price loss**. Add a new row that applies to the item model group that you enabled fixed receipt pricing for, and specify the main account that is

used to record fixed receipt price losses for inventory. Configure other settings as you require.

> **Note**
>
> We usually recommend that the **Fixed receipt price profit** and **Fixed receipt price loss** fields use the same main account for both purchase orders and inventory.

> **Important**
>
> When you change the value in the **Price** field on the **Manage costs** FastTab of the **Released products** page, the system doesn't automatically revalue the inventory that is on hand. As a manual workaround, open the **Closing and adjustment** page, and select **Adjustment** > **On-hand** on the Action Pane. Then select the items that are on hand, and adjust them to the current price. One clear advantage of using the standard cost inventory model is that it causes the system to automatically revalue the on-hand inventory.

Last updated on 12/30/2025

Post to charge account accounting principle

07/10/2025

The *post to charge account* accounting principle lets you account for and more easily reconcile any differences that occur in the unit price between a physical posting and financial posting, indirect costs on purchased items, or charges on a purchase order.

Two configurations for Accounts payable charges codes on the **Charges code** page (**Accounts payable** > **Charges setup** > **Charges code**) can cause a purchase order to affect the valuation of inventory assets:

- For charges codes where the **Debit type** field is set to *Item* and the **Credit type** field is set to *Ledger account*, the ledger account that is selected as the absorption account acts as a stock variation account.
- For charges codes where the **Debit type** field is set to *Item* and the **Credit type** field is set to *Customer/Vendor*, the charge will be accounted as material cost, and the stock variation account of the item will be used.

European special accounting rule

In Europe, the *post to charge account* accounting principle is often used to incorporate a special accounting rule. For example, one of the following methods is typically used to account for inventory changes:

- **Cost of sales method** – The standard inventory posting profile configuration supports this method. The *post to charge account* accounting principle isn't required.
- **Nature of expense method** – Smaller organizations often use this method. It typically involves the following steps:

    1. Goods or services are fully expensed at the time of receipt.
    2. At the end of the period, a cycle count is performed.
    3. A manual adjustment for the quantity and value is posted into inventory. (The offset account is a stock variation account that offsets the expense that was posted in step 1. Therefore, you see the variation in the stock value only on this account.)

The *post to charge account* principle lets you fully automate the two postings. In this way, you can eliminate a manual period-end closing adjustment posting.

Enable the post to charge account accounting principle

To enable the *post to charge account* accounting principle, follow these steps.

1. Go to **Accounts payable** > **Setup** > **Accounts payable parameters**.
2. On the **Invoice** tab, on the **Invoice** FastTab, set the **Post to charge account in ledger** option to *Yes*.
3. Close the page.

Prerequisites and recommended parameters for the post to charge account accounting principle

If you plan to account for purchase charges and stock variations, the following prerequisites must be in place:

- On the **Invoice** tab of the **Accounts payable parameters** page (**Accounts payable** > **Setup** > **Accounts payable parameters**), the **Post to charge account in ledger** option must be set to *Yes*.

- On the **Item model groups** page (**Cost management** > **Inventory accounting policies setup** > **Item model groups**), all the following options must be set to *Yes* for every relevant model group that contains items that are included on a purchase order:
  - Post physical inventory
  - Post financial inventory
  - Accrue liability on product receipt

- On the **Delivery** tab of the **Procurement and sourcing parameters** page (**Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**), the **Generate charges on product receipt** option must be set to *Yes*.

- On the **Inventory accounting** tab of the **Inventory and warehouse management parameters** page (**Inventory management** > **Setup** > **Inventory and warehouse management parameters**), the **Post packing slip in ledger** option must be set to *Yes*.

- On the **Purchase order** tab of the **Posting** page (**Inventory management** > **Setup** > **Posting** > **Posting**), main accounts that apply to every relevant item must be specified for each of the following posting types:
  - Purchase expenditure, un-invoiced
  - Purchase expenditure for product
  - Stock variation

Example scenario 1: Change in unit cost price

This example scenario has the following prerequisites:

- First in, first out (FIFO) costing model

The following procedure provides an example that shows what happens when you change the unit cost price on a purchase order.

1. Create a purchase order for a quantity of 1 of an item that has a unit price of 100.00.
2. Confirm the purchase order.
3. Post the product receipt for the purchase order.
4. Validate the voucher on the product receipt. The following table shows a sample voucher.

Expand table

Posting type	Main account	Main account name	Account type	Debit/Credit?	Clearing account	Physical/Financial?	Amount
Purchase, stock variation	600170	Stock variation materials	Expense	Credit	No	Physical	-100.00
Cost of purchased materials received	140100	Materials Inventory	Asset	Debit	Yes	Physical	100.00
Purchase expenditure, un-invoiced	600180	Material Receipts	Expense	Debit	Yes	Physical	100.00
Purchase, accrual	200140	Accrued Purchases	Liability	Credit	Yes	Physical	-100.00

5. Post the invoice for the purchase order that has an updated unit price of 110.00.
6. Validate the voucher on the invoice. The following table shows a sample voucher.

Expand table

Posting type	Main account	Main account name	Account type	Debit/Credit?	Clearing account	Physical/Financial?	Amount
Purchase, stock variation	600170	Stock variation materials	Expense	Credit	No	Financial	-10.00
Cost of purchased materials received	140100	Materials Inventory	Asset	Debit	Yes	Financial	-100.00

Posting type	Main account	Main account name	Account type	Debit/Credit?	Clearing account	Physical/Financial?	Amount
Purchase expenditure, un-invoiced	600180	Material Receipts	Expense	Debit	Yes	Financial	-100.00
Purchase, accrual	200140	Accrued Purchases	Liability	Credit	Yes	Financial	100.00
Cost of purchased materials invoiced	140100	Materials Inventory	Asset	Debit	No	Financial	110.00
Purchase expenditure for product	600180	Materials Receipt	Expense	Credit	No	Financial	110.00
Vendor balance	211000	Accounts Payable Trade	Liability	Credit	No	Financial	-110.00

Example 2: Charges and indirect costs on the purchase order

This example scenario has the following prerequisites:

- FIFO costing model
- **Charges code 1:** Debit item and credit ledger account for 10%
- **Charges code 2:** Debit item and credit customer/vendor for 10.00 proportional
- **Indirect cost:** 2.00% added to the purchase price

The following procedure provides an example that shows what happens when you include charges and indirect costs on a purchase order.

1. Create a purchase order for a quantity of 1 of an item that has a unit price of 100.00.
2. Add one charges code for 10 percent that debits the item and credits the ledger.
3. Add one charges code for 10.00 that debits the item and credits the customer/vendor.
4. Allocate the charges to the purchase order lines.
5. Confirm the purchase order.
6. Post the product receipt for the purchase order.

7. Validate the voucher on the product receipt. The following table shows a sample voucher.

Expand table

Posting type	Main account	Main account name	Account type	Debit/ Credit?	Clearing account	Physical or Financial	Amount
Purchase, stock variation	600170	Stock variation materials	Expense	Credit	No	Physical	-110.00
Estimated indirect cost absorbed	600520	Absorbed Indirect Costs	Expense	Credit	Yes	Physical	-2.40
Purchase freight	600120	Freight/ Transportation Costs	Expense	Credit	No	Physical	-10.00
Cost of purchased materials received	140100	Materials Inventory	Asset	Debit	Yes	Physical	122.40
Purchase expenditure, un-invoiced	600180	Material Receipts	Expense	Debit	Yes	Physical	110.00
Purchase, accrual	200140	Accrued Purchases	Liability	Credit	Yes	Physical	-110.00

8. Post the invoice for the purchase order.

9. Validate the voucher on the invoice. The following table shows a sample voucher.

Expand table

Posting type	Main account	Main account name	Account type	Debit/Credit?	Clearing account	Physical/Financial?	Amount
Purchase, stock variation	600170	Stock variation materials	Expense	Credit	No	Financial	0.00
Estimated indirect cost absorbed	600520	Absorbed Indirect Costs	Expense	Debit	Yes	Financial	2.40
Indirect cost absorbed	600520	Absorbed Indirect Costs	Expense	Debit	No	Financial	-2.40

Posting type	Main account	Main account name	Account type	Debit/Credit?	Clearing account	Physical/Financial?	Amount
Cost of purchased materials received	140100	Materials Inventory	Asset	Credit	Yes	Financial	-110.00
Purchase expenditure, un-invoiced	600180	Material Receipts	Expense	Credit	Yes	Financial	-110.00
Purchase, accrual	200140	Accrued Purchases	Liability	Debit	Yes	Financial	110.00
Cost of purchased materials invoiced	140100	Materials Inventory	Asset	Debit	No	Financial	110.00
Purchase expenditure for product	600180	Materials Receipt	Expense	Credit	No	Financial	110.00
Vendor balance	211000	Accounts Payable Trade	Liability	Credit	No	Financial	-110.00

Cost objects

This article provides information about costs objects, and explains how costs and quantities are accumulated. A cost object is an entity that costs and quantities are accumulated for. A cost object entity can be either a product or product variants, such as variants for style and color.

The **Cost objects** page lists all cost objects that are registered on a product. The cost objects are defined by data from the following sources:

- Product
- Product dimension group
- Storage dimension group
- Tracking dimension group

A cost object represents a cost element of the *Direct material* type only. A cost object and an inventory object differ in the way that a cost object is defined by the inventory dimensions that are selected for financial inventory.

For example, suppose you have an item with following configuration:

- **Site:** Physical inventory = Yes, Financial inventory = Yes
- **Warehouse:** Physical inventory = Yes, Financial inventory = No
- **Batch No.:** Physical inventory = Yes, Financial inventory = No

For this example, the following table shows what is a cost object and what is an inventory object.

Expand table

Object type	Item number	Site	Warehouse	Batch No.
Cost object	x	x		
Inventory object	x	x	x	x

Accumulation of costs and quantities

- The value in the **Value** field is a sum of the following values:
  - Physical cost amount
  - Financial cost amount
  - Adjustments

- The value in the **Quantity** field is a sum of the following values:
  - Received
  - Deducted
  - Posted quantity
- The **Average unit cost** field is a calculated field. The value is calculated by dividing the **Value** value by the **Quantity** value.

> **Note**
>
> The **Include physical value** parameter has no effect on the preceding calculations.

Related information

- Product dimension group
- Storage dimension group
- Tracking dimension group
- What's new or changed
- Cost entries

Last updated on 11/26/2025

View cost object balance

Follow these steps to view your cost object balance. This procedure is intended for the cost controller.

1. Go to **Cost management** > **Workspaces** > **Cost administration**.
2. On the Action Pane, select **Released products**.
3. Find and select the product you're looking for. If necessary, you can use the **Filter** field to help find it.
4. On the Action Pane, open the **Manage costs** tab and select **Cost objects**.
5. Find and select the cost object you're looking for.
6. On the Action Pane, select **Physical quantity**.
7. If you'd like to change the set of dimensions displayed, select **Dimensions display** from the Action Pane and then use the dialog to choose which dimensions to show. The **Physical quantity** grid shows a row for each unique combination of displayed dimensions, so if you show fewer dimensions, you'll probably see fewer rows, but the total quantity for all rows remains the same.

Last updated on 11/26/2025

View inventory statement and KPI by cost object

Follow these steps to view inventory statement and key performance indicators (KPIs) by cost object. This procedure is intended for the cost controller.

1. Go to **Cost management** > **Workspaces** > **Cost administration**.
2. On the Action Pane, select **Released products**.
3. Find and select the product you're looking for. If necessary, you can use the **Filter** field to help find it.
4. On the Action Pane, open the **Manage costs** tab and select **Cost objects**.
5. Find and select the cost object you're looking for.
6. On the Action Pane, select **Inventory statement and KPIs**.
7. Use the **From date** and **To date** fields to specify the date range you're looking for.

Last updated on 11/26/2025

Cost entries

This article provides information about cost entries and when they're created. A cost entry is a record that registers the quantity and cost of a given event.

Cost entries are aggregations of inventory transactions that are recorded on active financial inventory dimensions.

Examples

Example 1: No cost entries are created

A transfer journal event is registered. The event transfers one piece of item A from location A to location B. The Location inventory dimension isn't considered part of the cost object. Therefore, the event creates two inventory transactions and no cost entries.

Example 2: Cost entries are created

A transfer journal event is registered. The event transfers one piece of item A from site 1 to site 2. The Site inventory dimension is considered part of the cost object. Therefore, the event creates two inventory transactions and two cost entries.

Example 3: One cost entry is created

A product receipt event is registered for a purchase order. The event registers 100 pieces of item A at a unit cost of 10.00 U.S. dollars (USD). Because item A uses a serial number to track the purpose of inventory management, a unique serial number is created for each item that is received. Therefore, the event creates 100 inventory transactions and one cost entry.

Cost entries page

The new **Cost entries** page lets you view and control registrations of quantities and costs. This page complements the **Inventory transaction** and **Inventory settlement** pages. Records are registered in chronological order for an event. Therefore, you can quickly find and control the accumulated costs of a specific event or all events that are related to a document. Here's an example:

- A product receipt event is registered for item A. One hundred pieces are received at a unit cost of 10.00 USD.
- A few days after the invoice event is registered, the cost increases to 11.00 USD. Therefore, the total amount is 1,100 USD. A second voucher is created to account for the difference of 100 USD.
- A few days later, a miscellaneous charge of 15.00 USD to cover the transportation cost is registered on the purchase order.

Expand table

Voucher	Date	Reference	Number	Lot ID	Quantity	Amount
00001	01-01-2015	Purchase order	100001	0000101	100.00	1000.00
00002	20-01-2015	Purchase order	100001	0000101		100.00
00003	31-01-2015	Adjustment	100001	0000101		15.00

The **Cost entries** page enables filtering by document ID and document date.

> **Note**
>
> Cost entries are available only for cost objects or released products.

Related information

- Cost objects

Last updated on 11/26/2025

View cost entries for a cost object

This procedure shows how to view cost entries for a cost object. This procedure is intended for the cost controller.

1. Go to **Cost management** > **Workspaces** > **Cost administration**.
2. On the Action Pane, select **Released products**.
3. Find and select the released product you're looking for. If necessary, you can use the **Filter** field to help find it.
4. On the Action Pane, open the **Manage costs** tab and select **Cost objects**.
5. Find and select the cost object you're looking for.
6. On the Action Pane, select **Cost entries**.
7. View the entries shown in the grid. If necessary, you can use the **Filter** field to help find a particular entry.

Last updated on 11/26/2025

Cost groups

Cost groups provide the basis for segmenting and analyzing cost contributions in a manufactured item's calculated cost, such as the cost contributions for material, labor, and overhead. Cost group segmentation has several synonyms within manufacturing environments, such as cost breakdown, cost decomposition, or cost classification.

Cost group segmentation has several synonyms within manufacturing environments, such as cost breakdown, cost decomposition, or cost classification. Cost group segmentation can serve several purposes. Here are some examples:

- It can segment costs for different types of material, such as ingredients and packaging material for a canned goods product, based on the cost groups that are assigned to items.
- It can segment costs for different types of operations resources, such as different types of labor or machines, based on the cost groups that are assigned to cost categories that are related to operations resources and routing operations.
- It can segment costs for setup time and run time in routing operations, based on the cost groups that are assigned to cost categories that are related to the routing operations.
- It can segment costs for different types of overhead, such as labor-related and machine-related overhead, based on cost groups that are assigned to indirect costs in the costing sheet setup.
- It can act as the basis for calculating various types of manufacturing overhead in the costing sheet setup. This overhead can include different types of overhead that are related to routing information about operations resources, or information about setup time and run time. Manufacturing overhead can also be related to cost contributions of component material, reflecting a lean manufacturing philosophy that eliminates the need for routing information.

Cost group segmentation applies to a manufactured item's calculated cost, regardless of whether that cost was based on standard costs or planned costs. The **Summary** page displays this segmentation by cost group, by level, or by both cost group and level.

The ability to retain cost group segmentation across multiple levels in a product structure is known as *splitting*. Splitting applies only to manufactured items that use a standard cost inventory model. If splitting isn't used, the cost of a manufactured component is treated as a material cost contribution. The inventory parameter for cost breakdown by subledger indicates that cost group segmentation will be retained across multiple levels in standard cost calculations.

By contrast, if a policy has no levels, cost group segmentation applies only to a single-level calculation. For example, the **Cost rollup by cost group** page displays the cost group segmentation across multiple levels for standard cost items.

Cost group segmentation can also apply to variances for a standard cost item. A second inventory parameter defines whether variances are identified by cost group or just summarized.

A cost group can be assigned a cost group type and a behavior for supplemental segmentation purposes.

- **Cost group type** – Each cost group must be assigned a cost group type to indicate that the cost group applies to direct material, direct manufacturing, or direct outsourcing, or to designate it as indirect or undefined. A cost group that is designated as direct material can be assigned to items. A direct manufacturing cost group can be assigned to cost categories. A direct outsourcing cost group can be assigned to a product type of service, so that you can classify costs that are associated with the service purchase to subcontracting activities. An indirect cost group can be assigned to indirect costs for surcharges or rates. A cost group that is designated as undefined can be assigned to items, cost categories, or indirect costs. The assignment of a cost group type serves several purposes. First, it constrains the ability to assign a cost group and to view a list of applicable cost groups. Second, it provides supplemental segmentation for reporting purposes. Third, it can be used to assign ledger accounts for variances.
- **Behavior** – Each cost group can optionally be assigned a behavior to indicate that the cost group applies to fixed costs or variable costs. A cost group that has a null value for behavior is treated as a variable cost. The assignment of a behavior serves only a reporting purpose. For example, costs can be displayed with segmentation of fixed and variable costs on the costing sheet and on the **Cost rollup by cost group** page. If you assign a profit setting percentage to each cost group, the bill of materials (BOM) calculation provides a suggested sales price, based on a cost-plus-markup approach.

Last updated on 11/26/2025

Inventory object values

This article provides information about how the values of an inventory object are calculated.

A new functionality that is named **physical quantity** lets you see the values of a specific inventory object.

A cost object represents the entity level where inventory accounting is performed. For more information about cost objects, see Cost objects.

To see the values of a specific inventory object, click **Physical quantity** on the **Cost object** page. Here is how the value of an inventory object is calculated:

Inventory object.Value = Cost object.Average unit cost × Inventory object.Quantity

The following example shows how the values of an inventory object and a cost object are calculated. Two product receipt events are registered on item A:

- Product receipt 1: Quantity = 100 pcs., Amount = $1,000.00, Site = 1, Warehouse =11, Batch No. = B1
- Product receipt 2: Quantity = 50 pcs., Amount = $800.00, Site = 1, Warehouse =11, Batch No. = B2

The following table shows the calculation result for a cost object. You can view the result on the **Cost object** page.

Expand table

Object type	Item number	Site	Quantity	Inventory unit	Value	Average unit cost
Cost object	A	1	150	Pcs.	$1800.00	$12.00

The following table shows the calculation result for an inventory object. You can view the result by clicking **Physical quantity** on the **Cost object** page.

Expand table

Object type	Item number	Site	Warehouse	Batch No.	Quantity	Inventory unit	Value	Average unit cost
Inventory object	A	1	11	B1	100	Pcs.	$1200.00	$12.00
Inventory object	A	1	11	B2	50	Pcs.	$600.00	$12.00

Related information

- Cost objects
- Cost entries
- What's new and changed

Last updated on 11/26/2025

Inventory close

As part of the process to settle issue transactions with receipt transactions, you can also choose to have the general ledger updated to reflect the adjustments that have been made.

The inventory close process settles issue transactions to receipt transactions, based on the inventory valuation method that is selected in the item's item model group. As part of the settlement process, you can specify that the general ledger should be updated, so that it reflects the adjustments that have been made. However, until inventory close or recalculation has been run, issue transactions are posted at the calculated running average cost price.

After inventory close, you can no longer post in periods that are before the inventory closing date that you set, unless you reverse a completed inventory close process. For example, if inventory close is run for the period that ends on January 31, you can't post transactions that have a date that is earlier than January 31.

Items in inventory are assigned to one of two inventory types: item or service. Inventory close performs the same functions for both types. However, for service items, inventory close still settles issues to receipts.

How often the inventory close process is run varies by company. However, transaction volume should help determine how often you decide to run inventory close. In general, most companies run inventory close as part of their month-end close and reconciliation procedures.

Inventory recalculation and the general ledger

If adjustments to inventory and the general ledger are required during a month or other inventory period, you can run inventory recalculation instead of inventory close. Inventory recalculation makes adjustments but doesn't make settlements to inventory transactions.

During inventory recalculation, on-hand inventory is adjusted, inventory transactions are adjusted, and inventory recalculations and inventory closes are run. These tasks affect any ledger account that is linked to the original inventory transaction.

Example

When you create a purchase order from a sales order, the general ledger accounts that are used for the original sales order are updated. Even if the ledger accounts for the item group that is

assigned to the item were changed after the sales order was posted, and an inventory recalculation created an adjustment amount, the adjustment amount is posted to the original ledger accounts. The adjusted amount isn't posted to the new ledger accounts that are assigned to the item.

After the update is completed, you can review the ledger voucher that is posted because of one of these tasks.

1. On the **Closing and adjustment** page, on the **Overview** tab, select the update to review.
2. Select **Details**, and then select **Voucher**.

Effects of the inventory close process on the general ledger

Several of the tasks that you can perform on the **Closing and adjustment** page cause an update to general ledger. For example, the general ledger is updated when you make inventory on-hand adjustments, make inventory transaction adjustments, run inventory recalculation, and run inventory close.

The ledger accounts that are updated because of these tasks are linked to the original inventory transaction. For example, if a sales order is settled to a purchase order, the general ledger accounts that were used for the original sales order are adjusted. This behavior occurs even if the ledger accounts for the item group that is assigned to the item have changed since the sales order was posted. After inventory close creates a settlement amount, the settlement amount is still posted to the original ledger accounts, not to the new ledger accounts that are assigned to the item. The general ledger might also be updated if you reverse an inventory close.

> ⓘ **Note**
>
> - Inventory close is a required step in the month-end closing procedure for all inventory models except moving average. You'll be warned if you try to close a financial period without first performing the inventory close as of the period end date.
> - Before you run the closing procedure, you can view a list of items that can't be settled during the update.
> - We recommend that you run inventory close during off-peak hours, to distribute computing resources more evenly.

The inventory close log

After the inventory close process has been completed, a message in the message center might inform you that a unit cost price might be incorrect because a transaction couldn't be fully settled.

Before this message is shown, the system reports the item number and the affected transaction. The message informs you that the cost amount that is used for this transaction wasn't updated because of the inventory close. This message appears when a transaction of the issue type can't be settled.

During the inventory close process, the system checks each financial dimension to see whether there are more issues than receipts up to the specified closing date. This type of imbalance can occur when an inventory transaction from a purchase order isn't fully posted financially, either because the vendor invoice hasn't been received, or because bill of materials (BOM) components that are included in a production on a higher level aren't financially posted. (The subproduction isn't cost-calculated.) In this case, the subsequent close won't adjust all issues to the correct cost price, because not enough receipt information is available.

For each run of the closing procedure, the system indicates whether a log that contains the warnings is stored and can be viewed.

If you receive many warnings in the message, we recommend that you perform the following actions:

- Update receipts financially.
- Advance the closing date.
- Reevaluate the business procedures.

In some circumstances, you might not be able to do anything about the warnings. For example, if marking is used, and the marked purchase order has a financial date that is after the closing date, the closing date can't be changed.

Reversing a completed inventory close

Occasionally, you might have to reverse a completed inventory close to return settlements to the state that they had before adjustments were made. When you reverse a completed inventory close, inventory is reopened to enable posting in the period that the inventory close covers. Related changes might also be made in the general ledger. After you've finished making adjustments, you can run inventory close again for the period that you're working with.

> ⓘ **Note**

Only the last inventory period that was closed can be reopened. To reverse an earlier inventory close, you must reverse each subsequent inventory close one at a time, starting with the most recent close.

Last updated on 11/26/2025

Adjust on-hand inventory cost values

Use the **Adjustment of on-hand inventory** page to adjust the cost value of the on-hand inventory quantities after running an inventory close process. To open the **Adjustment of on-hand inventory** page, start on the **Closing and adjustment** page, select the record of a completed inventory close process, and then select **Adjustment** > **On-hand**.

For example, suppose you have the following transactions in February:

- February 1: An inventory financial receipt for a quantity of 2 at a cost of USD 10.00
- February 5: An inventory financial receipt for a quantity of 1 at a cost of USD 13.00
- February 19: An inventory financial issue for a quantity of 1 at a running average cost of USD 11.00

This item was set up with the first in, first out (FIFO) inventory model, and inventory close was performed as of February 28. The financial issue transaction of USD 11.00 is settled against the financial receipt that is dated February 1, and an adjustment of USD 1.00 is made. The following inventory receipts then contain open inventory quantities:

- February 1: A quantity of 1 at a cost of USD 10.00
- February 5: A quantity of 1 at a cost of USD 13.00

To set the cost of these two items to USD 15.00, use the on-hand adjustment option to adjust the open on-hand quantities as of the last inventory close period.

> **Note**
>
> The posting date of the on-hand adjustment transaction is the date of the last inventory close. You can't modify this date.

Last updated on 07/01/2026

Inventory value reports

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Inventory value reports provide details about your inventory physical and financial quantities and amounts. You can view the reports in many different ways. For example, you can show totals or transactions, or filter by items or a time range. You can view cost of goods sold (COGS) values or work in process (WIP) values, and set other options.

Inventory value reports let you perform the following tasks:

- Reconcile the general ledger and inventory.
- Consult on-hand inventory and values for a specific period.
- Create report configurations that are tailored to a specific purpose.
- Save report configurations so that you can use them multiple times.
- Add ranges to filter data when you run a report.

Types of inventory value report

There are two types of inventory value report: **Inventory value** (the standard report) and **Inventory value report storage**.

Standard inventory value report

The standard **Inventory value** report is a simple report that lets you select the information that is included and then shows that information on the screen. It doesn't save the results. It also doesn't provide interactive features for filtering, drilling down, browsing, or exporting. For these reasons, use the **Inventory value report storage** report instead in most cases.

Inventory value report storage report

The **Inventory value report storage** report provides output either as an interactive page in Microsoft Dynamics 365 Supply Chain Management or as an exported document in one of several formats.

When you view the report in your browser, columns and aggregate balances dynamically adjust, depending on the layout that you configure. You can sort the results, filter them, drill down into the data, and more.

Report results are stored in the **Inventory value** data entity. Therefore, you can filter and export the results to a format such as comma-separated values (CSV) or Microsoft Excel format.

The **Inventory value report storage** report is helpful when the output contains many lines. For example, you have 50,000 items, and 300 stores are created as warehouses. In this case, if you request inventory ending balances by item, site, and warehouse, the output contains many lines.

> ⓘ **Note**
>
> The **Inventory value report storage** report doesn't include subtotals that are defined in the report layout. It also doesn't include general ledger balances, even if those balances are defined in the report layout. Reconciliation to the general ledger must be done by using trial balances. However, the standard **Inventory value** report does include these subtotals and balances.

Define inventory value report configurations

Use the **Inventory value reports** page to set up the content that is included in the different types of inventory value report. You can define any number of report types. Each time that you generate either type of inventory value report, select a report type.

1. Go to **Cost management** > **Inventory accounting policies setup** > **Inventory value reports**.
2. Follow one of these steps:
    - To edit an existing report, select it in the list pane, and then select **Edit** on the Action Pane.
    - To create a new report, select **New** on the Action Pane.
3. In the header of the new or selected report, set the following fields:

- **ID** – Enter a short identifier for the report. This value must be unique among all inventory value report configurations. You can't edit it after you save a new configuration.
- **Name** – Enter a descriptive name for the report.

4. If you're creating a new report configuration, select **Save** on the Action Pane to make the remaining fields available.

5. On the **General** FastTab, set the following fields:

    - **Date interval** – Select a predefined date interval. You can override this date interval when you run the report.
    - **Range** – Select either *Posting date* or *Transaction Time*, depending on the date and time that should be used when retrieving records for the report.
    - **Dimension set** – Select the set of dimensions to run the data for. (The dimensions are defined in the general ledger.) For example, you might run the data for *Main account* or for *Main account* + *Business unit*. The dimension set that you select must have no more than two dimensions. For more information, see Financial dimension sets.

6. On the **Columns** FastTab, set the following fields. These fields control the columns that your report includes and the types of data that those columns contain.

    - **Inventory** – Set this option to *Yes* to show the inventory values. You can then reconcile those values with the general ledger account balances.

    - **WIP** – Set this option to *Yes* to show the WIP values. You can then reconcile those values with WIP account balances in the general ledger. When you set this option to *Yes*, the report shows only the physical quantities and amounts of inventory that have WIP status. Production orders that have WIP status are picked or reported as finished, but they aren't ended.

    - **Deferred COGS** – Set this option to *Yes* to display a column that shows the physical quantities and amounts of inventory for deferred COGS. Deferred COGS is shown by using physical quantities and amounts, because it offsets packing slip quantities and amounts.

    - **COGS** – Set this option to *Yes* to display a column that shows the financial quantities and amounts for COGS. COGS is shown by using financial quantities and amounts, because it offsets the invoice quantities and amounts.

- **Profit and loss** – Set this option to *Yes* to display a column that shows the financial amount that was posted to the profit and loss accounts for inventory.

- **Print cumulative account values for comparison** – Set this option to *Yes* to display a column that shows the general ledger account balance. In this way, you don't have to check the trial balance. This option works only with the standard **Inventory value** report, not with the **Inventory value report storage** report. After you set this option to *Yes*, use the following fields to specify each general ledger account that you want to list, depending on the **Financial position** options that you enabled.

  > ⓘ **Note**
  >
  > If you select a *total* account for any of these fields, both the amount of each inventory account that is included in the total account and the amount of the total account are shown.

  - **Inventory account** – Specify the general ledger account to show inventory information for. (Both the **Inventory** option and the **Print cumulative account values for comparison** option must be set to *Yes*.)
  - **WIP account** – Specify the general ledger account to show WIP information for. (Both the **WIP** option and the **Print cumulative account values for comparison** option must be set to *Yes*.)
  - **Deferred COGS account** – Specify the general ledger account to show deferred COGS information for. (Both the **Deferred COGS** option and the **Print cumulative account values for comparison** option must be set to *Yes*.)
  - **COGS account** – Specify the general ledger account to show COGS information for. (Both the **COGS** option and the **Print cumulative account values for comparison** option must be set to *Yes*.)

- **Summarize physical and financial values** – Set this option to *Yes* to display a column that shows the total inventory quantity and inventory amount (a summary of both physical and financial inventory values). If this option is set to *No*, the report shows both physical and financial inventory values.

- **Include not posted to ledger** Set this option to *Yes* to display a column that shows the transactions that were never posted to the general ledger. Transactions for the following types of items might not be posted to the general ledger:
  - Received and not yet invoiced items when the **Post physical inventory** option is cleared for the relevant item model group.

    - Received and not yet invoiced items when the **Post product receipt in ledger** option is cleared on the **Product receipt** FastTab on the **General** tab of the **Accounts payable parameters** page (**Accounts payable** > **Setup** > **Accounts payable parameters**).

- **Calculate average unit cost** – Set this option to *Yes* to display a column that shows the average unit cost. The average unit cost is the total amount divided by the total quantity.

- **Total quantity and value** – Set this option to *Yes* to display columns that show the total quantity of physical inventory (and financial quantities) and the total amount of physical inventory (and financial amounts). You can set this option to *Yes* only if the **Summarize physical and financial values** option is set to *No*.

- **Inventory dimensions** – In this grid, select the **View** checkbox for each dimension that you want to show on the report. Only dimensions where the **Financial inventory** option is enabled show values on the report. Other dimensions show only blank columns. For the dimensions that you select to show, you can select the **Total** checkbox to include totals too.

- **Resource ID** – Set the **View** option to *Yes* to display a column that identifies the item for each row. Set the **Total** option to *Yes* to include totals too. Depending on the type of item that is listed in each row, the column shows one of the following types of information:
    - **Material** – The column shows the `ItemID` field value for the relevant material record.
    - **Labor** – The column shows the `WorkCenterID` field value for the relevant labor record.
    - **Indirect cost** – The column shows the `CodeID` field value for the relevant cost record.

  If the **View** option is set to *No* for both the **Resource ID** field and the **Resource Group** field, you see only a total inventory value that is based on the inventory dimensions that you selected.

- **Resource Group** – Set the **View** option to *Yes* to display a column that identifies the resource group for each row. Set the **Total** option to *Yes* to include totals too. Depending on the type of item that is listed in each row, the column shows one of the following types of information:

- **Material** – The column shows the `ItemGroup` field value for the relevant material record.
- **Labor** – The column shows the `WorkcenterGroup` field value for the relevant labor record.
- **Indirect cost** – The column shows the `CostGroup` field value for the relevant cost record. (The `CostGroupType` value must be *Indirect*.)

If the **View** option is set to *No* for both the **Resource ID** field and the **Resource Group** field, you see only a total inventory value that is based on the inventory dimensions that you selected.

7. On the **Rows** FastTab, set the following fields. These fields let you add corresponding WIP-related subsections to the report or remove them.

- **Material** – Set this option to *Yes* to show information about materials. *Material* is a default resource type because you must include materials in all report configurations to create reliable output.
- **Labor** – Set this option to *Yes* to show labor costs of WIP.
- **Indirect cost** – Set this option to *Yes* to show indirect costs of WIP.
- **Direct outsourcing** – Set this option to *Yes* to show direct outsourcing costs of WIP. This information is useful for subcontracting.
- **Detail Level** – Select a view option for the report:
- *Transactions* – View all relevant transactions on the report. You might experience performance issues when you view reports that include a large volume of transactions. Therefore, if you want to use this view option, use the **Inventory value report storage** report.
- *Totals* – View the total result.
- **Include beginning balance** – Set this option to *Yes* to show the beginning balance. This option is available only when the **Detail level** field is set to *Transactions*.

Generate an Inventory value report storage report

Follow these steps to generate and store an **Inventory value report storage** report.

1. Go to **Cost management** > **Inquiries and reports** > **Inventory value report storage**.

2. On the Action Pane, select **New**.

3. In the **Inventory value** dialog box, on the **Parameters** FastTab, set the following fields:

    - **Name** – Enter a unique name for the report.
    - **Large item volume** – To use the large-volume report generation feature, set this option to *Yes*. Only enable this option when generating reports for a significantly large number of items (for example, 50,000 or more) because this option takes longer to run than the standard-volume option. Learn more in Generate an inventory value report storage report for a large number of items.
    - **ID** – Select the inventory value report configuration to use for the report. The configuration establishes options for the columns and rows that the report includes.
    - **Date interval** – Use the fields in this section to define which records are included on the report. To define the date interval, select either a preset range (relative to the report generation date) in the **Date interval code** field, or specific dates in the **From date** and **To date** fields.

4. On the **Records to include** FastTab, set up filters and constraints to define which data is included on the report. Select **Filter** to open a standard query editor dialog, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Supply Chain Management. All these filters apply to the inventory transactions but not to the general ledger balance. Keep this behavior in mind when you set up your filters. Otherwise, you might see a discrepancy between inventory and the general ledger.

5. On the **Run in the background** FastTab, specify how, when, and how often the report is generated. The fields work just as they do for other types of background jobs in Supply Chain Management.

    > ⓘ **Note**
    >
    > This report is always run as part of a batch job.

6. Select **OK** to apply your settings and close the dialog box.

After the batch job completes, the report is listed on the **Inventory value report storage** page. You might have to refresh the page to see the report.

> ⓘ **Important**

In the selected inventory value report configuration, you might get an incorrect beginning balance if you select the same date in both the **From date** field and the **To date** field, and if you also set the **Include beginning balance** option to *Yes*.

Explore an Inventory value report storage report

After you generate a report, you can view and explore it at any time by following these steps.

1. Go to **Cost management** > **Inquiries and reports** > **Inventory value report storage**.
2. Select a report in the list. The page shows the details of the inventory value report configuration that was used to generate the selected report.
3. On the Action Pane, select **View details** to show the report content.
4. Explore the report by following any of these steps:
    - As with most standard pages in Supply Chain Management, select almost any column heading to sort or filter the grid by the values in that column.
    - Use the **Filter** field to filter the report by any value in any of several available columns.
    - Use the view menu (above the **Filter** field) to save and load your favorite combinations of sort and filter options.

Export an Inventory value report storage report

Every report that you generate is stored in the **Inventory value** data entity. You can use the standard data management features of Supply Chain Management to export data from this entity to any supported data format, such as CSV or Excel format.

The following example shows how to export an **Inventory value report storage** report.

1. Go to **System administration** > **Workspaces** > **Data management**.
2. In the **Import / Export** section, select the **Export** tile.
3. On the **Export** page, enter a **Group name** for the job.
4. On the **Selected entities** FastTab, select **Add entity**.
5. In the **Add entity** drop-down dialog, set the following fields:
    - **Entity name** – Select *Inventory value report storage*.

- **Target data format** – Select the format to export data to.

6. Select **Add** to add the new row, and then select **Close** to close the dialog box.

7. Usually, you export one report at a time. To export a single report, select the button in the **Filter** column for the row that you just added to **Selected entities** FastTab. Then use the query designer to set up the following filter:

   - **Table** – Select *Inventory value report storage*.
   - **Field** – Select *Execution name*.
   - **Criteria** – Enter the name of the report that you want to export.

8. On the Action Pane, select **Save**.

9. On the Action Pane, open the **Export options** tab and select **Export now**.

10. The **Execution summary** page opens. Here you can view the status of your export job and a list of the entities that were exported. In the **Entity processing status** section, select the **Inventory value** entity in the list, and then select **Download file** to download the data that was exported from that entity.

For more information about how to use data management to export data, see Data import and export jobs overview.

Delete stored inventory value reports

As the number of stored inventory value reports increases, they can take up too much space in your database. This situation can affect system performance and cause higher data-storage costs. To avoid these issues, clean up the reports by deleting older reports.

> **Important**
>
> Before you delete any previously generated inventory value reports, export the reports and store them externally. You might not be able to regenerate the reports later. This limitation exists because when you generate an inventory value report, the system works backwards from today and processes each inventory transaction record in reverse order as it goes. If you try to look too far back when you generate a report, the volume of transactions to process might become so large that the system times out before it can finish generating the report. The distance into the past that you can generate new reports for depends on the number of inventory transactions that you have in your system for the relevant time span.

Delete one report at a time

Follow these steps to delete one stored report at a time.

1. Export the report that you plan to delete, and store it in an external location for future reference.
2. Go to **Cost management** > **Inquiries and reports** > **Inventory value report storage**.
3. In the list pane, select the report to delete.
4. On the Action Pane, select **Delete**.
5. A warning message reminds you to back up generated reports. Select **Yes** if you're ready to proceed with the deletion.

Delete several reports at the same time

Follow these steps to delete several stored reports at the same time.

1. Export all the reports that you plan to delete, and store them in an external location for future reference.
2. Go to **Cost management** > **Inventory accounting** > **Clean up** > **Inventory value report data clean up**.
3. In the **Inventory value report data clean up** dialog box, in the **Delete inventory value report executed before** field, select the date before which all inventory value reports should be deleted.
4. On the **Records to include** FastTab, set additional filter conditions to limit the set of reports that are deleted. Select **Filter** to open a standard query editor where you can define the properties of the reports to delete.
5. On the **Run in the background** FastTab, specify how, when, and how often the reports should be deleted. The fields work just as they do for other types of background jobs in Supply Chain Management. However, you typically run this job manually each time that it's required.
6. Select **OK** to delete the specified reports.

Generate a standard Inventory value report

Use the following procedure to generate a standard **Inventory value** report.

1. Go to **Cost management** > **Inquiries and reports** > **Inventory accounting - status reports** > **Inventory value**.

2. In the **Inventory value report** dialog box, on the **Parameters** FastTab, set the following fields:

    - **Name** – Enter a unique name for the report.
    - **ID** – Select the inventory value report configuration to use for the report. The configuration establishes options for the columns and rows that the report includes.
    - **Date interval** – Use the fields in this section to define which records are included on the report. To define the date interval, select either a preset range (relative to the report generation date) in the **Date interval code** field, or specific dates in the **From date** and **To date** fields.

3. On the **Records to include** FastTab, set up filters and constraints to define which data is included on the report. Select **Filter** to open a standard query editor dialog, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Supply Chain Management.

4. On the **Run in the background** FastTab, specify how, when, and how often the report is generated. The fields work just as they do for other types of background jobs in Supply Chain Management.

5. Select **OK** to apply your settings and close the dialog box. The report appears.

Reading inventory value reports

This section provides guidance on how to read and understand an inventory value report.

Supply Chain Management supports the following two important concepts that are related to inventory status:

- **Financial updated** – This concept indicates that the inventory transactions are already invoiced. For production orders, it indicates the end of a production order.
- **Physical updated** – This concept indicates that the inventory transactions aren't yet invoiced, but they're received or shipped. For production orders, it indicates that material is picked or the production order is reported as finished.

When you understand these two concepts, it should be easy to understand the following columns in the report output:

- **Inventory: Financial Quantity** – The quantity that is financially updated.
- **Inventory: Financial Amount** – The amount value of inventory that is financially updated.
- **Inventory: Physical Quantity Posted** – The quantity that is physically updated.

- **Inventory: Physical Amount Posted** – The amount value of inventory that is physically updated.
- **Inventory: Physical Quantity Not Posted** – The quantity that has inventory transactions but isn't posted to the general ledger. For example, you have an item model group where the **Post physical inventory** and **Post financial inventory** options are cleared, and you have an item that is linked to that group. You then create a purchase order, receive it, and invoice it. At this point, if you review the inventory value report for the item, you see that the quantity and the value on the purchase order are shown in the **Inventory: Physical Quantity Not Posted** and **Inventory: Physical Amount Not Posted** columns.
- **Inventory: Physical Amount Not Posted** – You can set up your reports so that they show unposted amounts. However, if you're using the report for inventory reconciliation, don't use this value. Otherwise, the amount isn't posted to the general ledger.
- **Inventory: Quantity** – The total quantity of all the quantity columns on the report.
- **Inventory: Amount** – The total quantity of all the amount columns on the report. When you do inventory reconciliation, don't use this column if your report includes the **Inventory: Physical Amount Not Posted** column. In this case, you must exclude the **Inventory: Physical Amount Not Posted** amount from the total amount.
- **Average unit cost** – The total amount divided by the total quantity.

Typically, you use an inventory value report to view the inventory value and quantity. However, sometimes the report doesn't show all the relevant inventory dimensions. If you don't see the dimensions that you expect, validate the following settings:

- Review the item storage and tracking dimension groups. Only dimensions where the **Financial inventory** option is enabled can be shown on the report.
- Go to **Cost management** > **Inventory accounting policies setup** > **Inventory value reports**, select the report configuration that you used to generate the report, and make sure that the required inventory dimensions are selected in the **View** column.

For example, you have an item that has the item number *A0001*. In the storage dimensions group, only the site is enabled for financial inventory. The site and warehouse are both enabled for physical inventory. In the tracking dimension group, the batch number is enabled for physical inventory but not for financial inventory. You then use a report configuration where site, warehouse, and batch number are all selected. When you view the report, you see a value only for the site. The columns for the warehouse and batch number are blank. As this example shows, inventory value reports can show only inventory dimensions that are enabled for financial inventory.

Generate an inventory value report storage report for a large number of items

This feature helps you handle extensive datasets while maintaining control over the process of generating **Inventory value report storage** reports. It ensures smooth and uninterrupted operation. It improves performance by dividing large datasets into manageable bundles. The system processes each bundle sequentially, which prevents timeout errors. You can also pause and resume the process to support incremental report generation.

Prerequisites

To use this feature, your system must meet the following requirements:

- You're running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature named *Large Volume Historical Inventory Value Report Generation* is turned on in feature management.

Configure the large-volume option

To configure this feature, follow these steps:

1. Go to **Cost management** > **Inventory accounting policies setup** > **Parameters**.
2. In the **Inventory value report storage** field group, set the **Number of items per bundle** field to the number of items that you want to process in one bundle when generating large-volume reports. For example, if you often deal with reports that cover 800,000 items, a bundle size of 30,000 might be appropriate.

> **Important**
>
> Proper configuration of the bundle size is crucial for balancing performance and reliability. If the bundle size is too small, the system takes longer to process the report. If the bundle size is too large, the system might time out before it can finish processing the report. Start with a bundle size of 30,000 items and adjust it as needed based on your experience.

Use the large-volume option when generating a report

When you generate a report, choose whether to use standard-volume processing or large-volume processing.

> **Important**
>
> Use the large-volume option only when generating reports for a significantly large number of items (for example, 50,000 or more) because it takes longer to run than the standard-volume option.

To generate a large-volume historical inventory value report, follow these steps:

1. Go to **Cost management** > **Inquiries and reports** > **Inventory value report storage**.
2. On the Action Pane, select **New**.
3. In the **Inventory value** dialog, expand the **Parameters** FastTab and set **Large item volume** to *Yes*.
4. Fill out the rest of the dialog as described in Generate an Inventory value report storage report.
5. Select **OK** to start the batch job.

When the batch job finishes, the report appears on the **Inventory value report storage** page. Refresh the page if needed to view the newly generated report.

Pause and resume functionality

To pause and resume report generation:

1. Go to **Cost management** > **Inquiries and reports** > **Inventory value report storage**.
2. Select the report in progress from the list. The page displays the report configuration details.
3. Select one of the following buttons on the Action Pane:

    - **Pause** – Temporarily stop generating a running report.
    - **Resume** – Resume a paused process. The system picks up from where it left off.

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Inventory value report examples and logic

Inventory value reports provide details about your inventory physical and financial quantities and amounts. This article provides some examples of results that are presented on each type of inventory value report.

For more information about how to generate and use each type of inventory value report, see Inventory value reports.

Sample data that is used in these examples

The examples in this article are based on the sample inventory transaction data that is described in this section.

Storage dimension setup

The example system contains the following setup of storage dimensions.

Expand table

Name	Active	Physical inventory	Financial inventory
Site	Yes	Yes	Yes
Warehouse	Yes	Yes	No

Inventory model

For the example system, the inventory model for the released products is *FIFO*, and the **Cost price** field for the inventory model is set to *Include physical value*.

Inventory transactions

The example system contains the following inventory transactions for a released product that has the item number *B0001*.

Expand table

Reference	Site	Warehouse	Receipt	Issue	Physical date	Financial date	Quantity	Cost amount	Physical cost amount
Purchase order	1	11	Purchased		March 15	March 15	10	1,000	1,000
Purchase order	2	21	Purchased		March 15	March 15	10	2,000	2,000
Purchase order	1	11	Received		April 15		5		375
Transfer order	1	11		Sold	May 2	May 2	-5	-458.33	-458.33

Reference	Site	Warehouse	Receipt	Issue	Physical date	Financial date	Quantity	Cost amount	Physical cost amount
Transfer order	1	12	Purchased		May 2	May 2	5	458.33	458.33
Sales order	1	12		Sold	May 3	May 3	-1	-91.67	-91.67

Inventory value report configuration

The example system includes an inventory value report configuration that has the following settings:

- **Range:** *Posting date*
- **Inventory:** *Yes*
- **Calculate average unit cost:** *Yes*
- **Total quantity and value:** *Yes*
- **Site, View:** *Selected*
- **Resource ID, View:** *Yes*
- **Level:** *Totals*

Inventory value report example 1

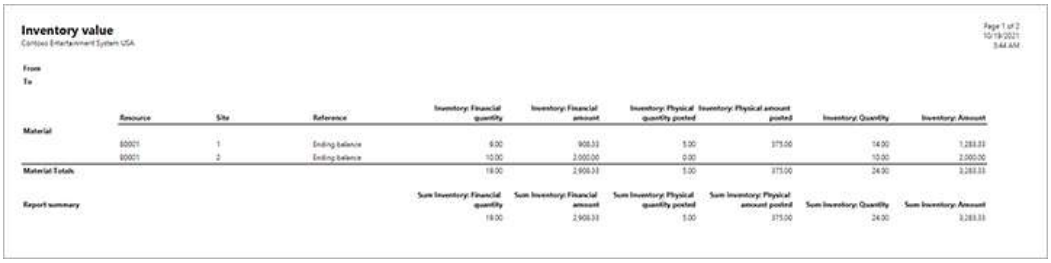
The following table and illustrations show the results when you use the sample data and report configuration described earlier in this article.

Expand table

Resource type	Resource	Site	Reference	Inventory: Financial quantity	Inventory: Financial amount	Inventory: Physical quantity posted	Inventory: Physical amount posted	Inventory: Quantity	Inventory: Amount	Avera unit cost
Material	B0001	1	Ending balance	9.00	908.33	5.00	375.00	14.00	1,283.33	91.67
Material	B0001	2	Ending balance	10.00	2,000.00	0.00	0.00	10.00	2,000.00	200.0

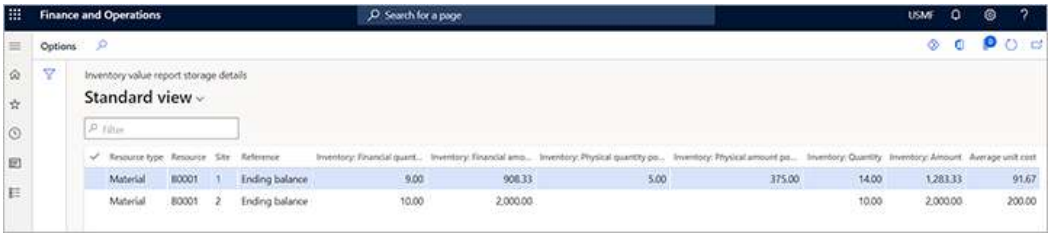
Standard Inventory value report for example 1

The following illustration shows the standard **Inventory value** report for example 1. (Select the illustration to open a larger version.)



Inventory value report storage report for example 1

The following illustration shows the **Inventory value report storage** report for example 1. (Select the illustration to open a larger version.)



Inventory value report example 2

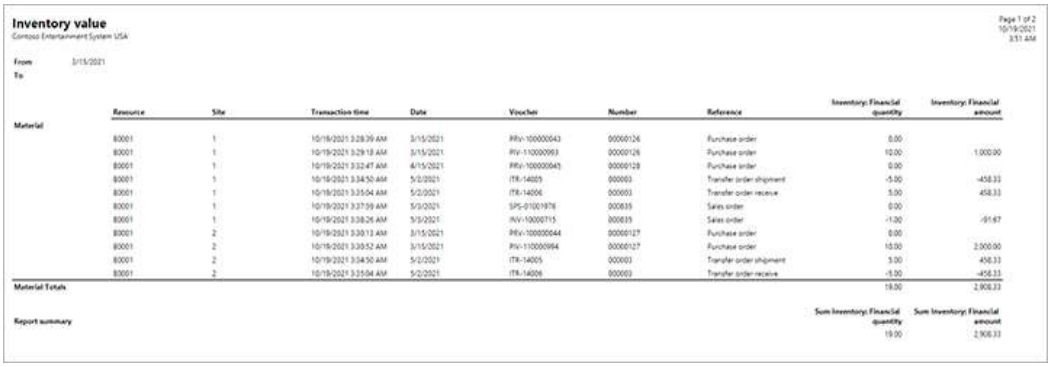
The following table and illustrations show the results when you use the sample data described earlier in this article, but you change the value of the **Level** field to *Transactions* in the report configuration, and you set the **From date** field to *March 15* when you run the report.

Expand table

Resource type	Resource	Site	Date	Number	Reference	Inventory: Financial quantity	Inventory: Financial amount	Inventory: Physical quantity posted	Inventory: Physical amount posted	Invento Quanti
Material	B0001	1	3/15/2021	00000126	Purchase order	0.00	0.00	10.00	1,000.00	**10.00**
Material	B0001	1	3/15/2021	00000126	Purchase order	10.00	1,000.00	-10.00	-1,000.00	**0.00**
Material	B0001	1	4/15/2021	00000128	Purchase order	0.00	0.00	5.00	375.00	**5.00**
Material	B0001	1	5/2/2021	000003	Transfer order shipment	-5.00	-458.33	0.00	0.00	**-5.00**
Material	B0001	1	5/2/2021	000003	Transfer order receive	5.00	458.33	0.00	0.00	**5.00**
Material	B0001	1	5/3/2021	000835	Sales order	0.00	0.00	-1.00	-91.67	**-1.00**
Material	B0001	1	5/3/2021	000835	Sales order	-1.00	-91.67	1.00	91.67	**0.00**
Material	B0001	2	3/15/2021	00000127	Purchase order	0.00	0.00	10.00	2,000.00	**10.00**
Material	B0001	2	3/15/2021	00000127	Purchase order	10.00	2,000.00	-10.00	-2,000.00	**0.00**
Material	B0001	2	5/2/2021	000003	Transfer order shipment	5.00	458.33	0.00	0.00	**5.00**
Material	B0001	2	5/2/2021	000003	Transfer order receive	-5.00	-458.33	0.00	0.00	**-5.00**

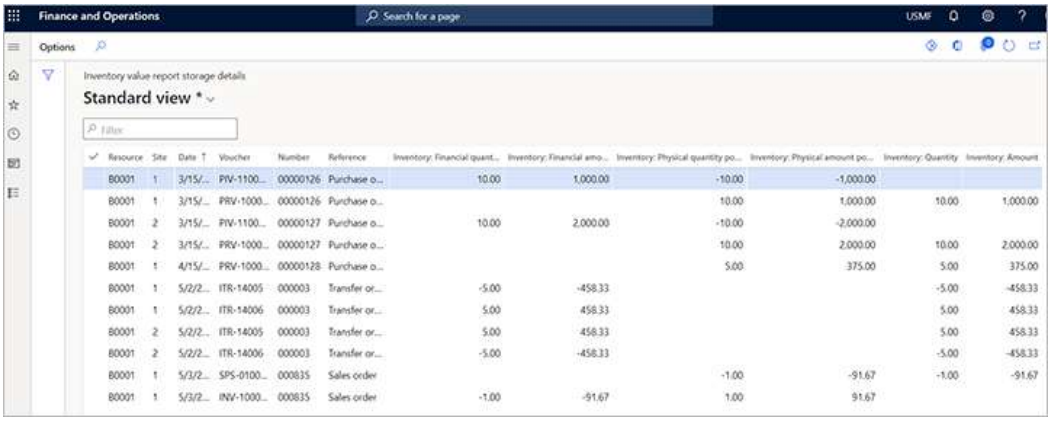
Standard Inventory value report for example 2

The following illustration shows the standard **Inventory value** report for example 2. (Select the illustration to open a larger version.)



Inventory value report storage report for example 2

The following illustration shows the **Inventory value report storage** report for example 2. (Select the illustration to open a larger version.)



Inventory value report example 3

Run inventory value reports after recalculation or inventory closing, so that you have the transactions and amounts affected by the recalculation or inventory closing.

The following subsections show the inventory value reports that are generated after you close inventory until May 30.

Example 3 when the Totals level is used

The following table shows the results when you use the sample data and report configuration that are described earlier in this article. (In that report configuration, the **Level** field is set to *Totals*.)

Expand table

Resource type	Resource	Site	Reference	Inventory: Financial quantity	Inventory: Financial amount	Inventory: Physical quantity posted	Inventory: Physical amount posted	Inventory: Quantity	Inventory: Amount	Avera unit cost
Material	B0001	1	Ending balance	9.00	900.00	5.00	375.00	14.00	1,275.00	91.07
Material	B0001	2	Ending	10.00	2,000.00	0.00	0.00	10.00	2,000.00	200.0

Resource type	Resource	Site	Reference	Inventory: Financial quantity	Inventory: Financial amount	Inventory: Physical quantity posted	Inventory: Physical amount posted	Inventory: Quantity	Inventory: Amount	Avera unit cost
			balance							

Example 3 when the Transactions level is used

The following table shows the results when you use the sample data described earlier in this article, but you change the value of the **Level** field to *Transactions* in the report configuration.

Expand table

Resource type	Resource	Site	Date	Number	Reference	Inventory: Financial quantity	Inventory: Financial amount	Inventory: Physical quantity posted	Inventory: Physical amount posted	Invento Quanti
Material	B0001	1	3/15/2021	00000126	Purchase order	0.00	0.00	10.00	1,000.00	10.00
Material	B0001	1	3/15/2021	00000126	Purchase order	10.00	1,000.00	-10.00	-1,000.00	0.00
Material	B0001	1	4/15/2021	00000128	Purchase order	0.00	0.00	5.00	375.00	5.00
Material	B0001	1	5/2/2021	000003	Transfer order shipment	-5.00	-458.33	0.00	0.00	-5.00
Material	B0001	1	5/2/2021	000003	Transfer order receive	5.00	458.33	0.00	0.00	5.00
Material	B0001	1	5/3/2021	000835	Sales order	0.00	0.00	-1.00	-91.67	-1.00
Material	B0001	1	5/3/2021	000835	Sales order	-1.00	-91.67	1.00	91.67	0.00
Material	B0001	1	5/31/2021	000835	Sales order	0.00	-8.33	0.00	0.00	0.00
Material	B0001	1	5/31/2021	000003	Transfer order shipment	0.00	-41.67	0.00	0.00	0.00
Material	B0001	1	5/31/2021	000003	Transfer order receive	0.00	41.67	0.00	0.00	0.00
Material	B0001	2	3/15/2021	00000127	Purchase order	0.00	0.00	10.00	2,000.00	10.00
Material	B0001	2	3/15/2021	00000127	Purchase order	10.00	2,000.00	-10.00	-2,000.00	0.00
Material	B0001	2	5/2/2021	000003	Transfer order shipment	5.00	458.33	0.00	0.00	5.00
Material	B0001	2	5/2/2021	000003	Transfer order receive	-5.00	-458.33	0.00	0.00	-5.00

Resource type	Resource	Site	Date	Number	Reference	Inventory: Financial quantity	Inventory: Financial amount	Inventory: Physical quantity posted	Inventory: Physical amount posted	Invento Quanti
Material	B0001	2	5/31/2021	000003	Transfer order shipment	0.00	41.67	0.00	0.00	0.00
Material	B0001	2	5/31/2021	000003	Transfer order receive	0.00	-41.67	0.00	0.00	0.00

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Inventory aging report storage

In Microsoft Dynamics 365 Supply Chain Management, you can run an *inventory aging report storage* report and make the output available as a form and a chart. In the form, columns and aggregate balances dynamically adjust based on the configured layout. The chart provides a visual overview that supports filtering and lets you drill down into details. Additionally, a data entity named *Inventory aging report* lets you export the results of an *inventory aging report storage* report run to a format such as a Microsoft Excel file or a PDF file.

This method of running an inventory aging report is helpful when the output contains many lines. For example, the output contains many lines if you have 50,000 items and 300 stores that are created as warehouses, and you request inventory aging by item, site, and warehouse.

Set up and generate reports

To set up and generate *inventory aging report storage* reports, follow these steps:

1. Go to **Cost management** > **Inquiries and reports** > **Inventory aging report storage**.

2. Select **New**. The **Inventory aging** dialog opens.

3. On the **Parameters** FastTab, make the following settings:

    - **Name** – Enter a unique name for the report.
    - **As of date** – Select the date to base the values of the report on. Leave blank to use the date on which the report runs, which is useful if you choose to set up a repeating schedule.
    - **View** – Choose which fields to include in your report. Tooltips are provided for each of them, so you can hover your mouse pointer for more information. Select one of the following values for each field:
        - *No* – Don't include the field in the report.
        - *View* – Include the field in the report.
        - *Total* – Display the field as a subtotal (subtotals aren't shown in the online reports).
    - **Unit of aging period** – Choose how you want to define your aging periods. Select one of the following values:
        - *Days* – Specify each period as a number of days previous to the **As of date**.
        - *Dates* – Select a specific calendar date for each period.

    - *Date intervals* – Select from a list of predefined date intervals related to the **As of date**.
  - **Aging periods** – You can specify up to four aging periods using the unit you selected for **Unit of aging period**

4. If you want to limit the records included in your report, expand the **Records to include** FastTab and add the filters you require, just as you might do for other batch jobs in Supply Chain Management.

5. On the **Run in the background** FastTab, set up batch, scheduling, and recurrence options as you require, just as you might do for other batch jobs in Supply Chain Management. The report always runs in a batch job.

6. Select **OK** to run and/or schedule the report.

7. After the batch job completes, the output appears on the **Inventory aging report storage** page. If you set up a recurring report, then a new report is added to the list each time the report runs.

View a report

To view an inventory aging report storage, follow these steps:

1. Go to **Cost management** > **Inquiries and reports** > **Inventory aging report storage**.
2. On the list pane, select the report that you want to view.
3. On the Action Pane, select one of the following actions:

   - **View details** – View the report as a form that has a traditional grid layout.
   - **View chart** – View the report as an aggregated chart.

> ⓘ **Note**
>
> The views don't include the subtotals that are defined in the report layout.

Export a report

Every report that you generate is stored in the *Inventory value report storage* data entity. You can use the standard data management features of Supply Chain Management to export data from this entity to any supported data format, such as CSV or Excel format.

The following example shows how to export an inventory aging report storage report.

1. Go to **System administration** > **Workspaces** > **Data management**.
2. In the **Import / Export** section, select the **Export** tile.
3. On the **Export** page, enter a **Group name** for the job.
4. On the **Selected entities** FastTab, select **Add entity**.
5. In the **Add entity** drop-down dialog, set the following fields:
    - **Entity name** – Select *Inventory aging report storage*.
    - **Target data format** – Select the format to export data to.
6. Select **Add** to add the new row, then select **Close** to close the dialog box.
7. Usually, you export one report at a time. To export a single report, select the button in the **Filter** column for the row that you just added to **Selected entities** FastTab. Then use the query designer to set up the following filter:
    - **Table** – Select *Inventory aging report storage*.
    - **Field** – Select *Execution name*.
    - **Criteria** – Enter the name of the report that you want to export.
8. On the Action Pane, select **Save**.
9. On the Action Pane, open the **Export options** tab and select **Export now**.
10. The **Execution summary** page opens. Here you can view the status of your export job and a list of the entities that were exported. In the **Entity processing status** section, select the *Inventory aging report storage* entity in the list. Then select **Download file** to download the data that was exported from that entity.

For more information about how to use data management to export data, see Data import and export jobs overview

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Last updated on 12/30/2025

Inventory aging report examples and logic

This article presents some examples that show how to interpret the results of an **Inventory aging** report. This report categorizes on-hand quantity and inventory values for a selected item or item group into several period buckets. This article also shows the internal logic of the report.

The examples in this article show results that are presented on a standard **Inventory aging** report. However, in general, we recommend that you use the Inventory aging report storage version of this report, especially when you have many items and warehouses that must be processed. Inventory aging report storage saves each report that you generate, shows the results as an interactive page and a chart, and lets you export any saved report.

Sample data that is used in these examples

The examples in this article are based on the sample inventory transaction data that is described in this section.

Storage dimension setup

The example system contains the following setup of storage dimensions.

Expand table

Name	Active	Physical inventory	Financial inventory
Site	Yes	Yes	Yes
Warehouse	Yes	Yes	No

Inventory model

For the example system, the inventory model for the released products is *FIFO*, and the **Cost price** setting for the inventory model is *Include physical value*.

Inventory transactions

The example system contains the following inventory transactions for a released product that has the item number *1000*.

Expand table

Reference	Site	Warehouse	Receipt	Issue	Physical date	Financial date	Quantity	Cost amount	Physical cost amount
Purchase order	1	11	Purchased		March 15	March 15	10	1,000	1,000
Purchase order	2	21	Purchased		March 15	March 15	10	2,000	2,000
Purchase order	1	11	Received		April 15		5		375
Transfer order	1	11		Sold	May 2	May 2	-5	-458.33	-458.33
Transfer order	1	12	Purchased		May 2	May 2	5	458.33	458.33
Sales	1	12		Sold	May 3	May 3	-1	-91.67	-91.67

Reference	Site	Warehouse	Receipt	Issue	Physical date	Financial date	Quantity	Cost amount	Physical cost amount
order									

How quantities and amounts in each period bucket are calculated

By using the sample data that is described in the previous sections, you can run an **Inventory aging** report that has the following settings:

- **As of date:** *May 9, 2020*
- **Site:** *View*
- **Warehouse:** *No*
- **Item number:** *Total*
- **Aging period:** Set this field to generate monthly buckets.

In this case, the content of the report that is generated will resemble the following example.

Expand table

Item number	Site	On-hand quantity	On-hand value	Inventory value quantity	Inventory value	Average unit cost	P1:Quantity (5/8/2020 - 5/1/2020)	P1:Amount (5/8/2020 - 5/1/2020)	P2:Quantity (4/30/2020 - 4/1/2020)	P2:Amount (4/30/2020 - 4/1/2020)	P3:Q (3/3 - 3/1
1000	1	14	1,283.33	14	1,283.33	91.67			5.00	458.33	9.00
1000	2	10	2,000.00	10	2,000.00	200.00					10.0
1000 Totals		**24.00**	**3,283.33**						**5.00**	**458.33**	**19**

Note the following details in this example report:

- The **Inventory value quantity**, **Inventory value**, and **Average unit cost** values that are shown on the report are values for the financial inventory dimension (**Site**, in this case).

  For example, for site 1, the report shows the following information:
  - The **Inventory value quantity** value is *14* (= 10 + 5 – 5 + 5 – 1).
  - The **Inventory value** value is *1,283.33* (= 1,000 + 375 – 458.33 + 458.33 – 91.67).
  - The **Average unit cost** value is *91.67*.
  - The **On-hand value** value and the **Amount** value in each period bucket are calculated by using the **Average unit cost** value.

- The report determines the on-hand quantity for each period bucket by summarizing the total received inventory quantity for each period bucket. It then applies the first in, first out (FIFO) principle to deduct the total issued quantity, regardless of the inventory model that the items use.

If you run the same report again, but this time you set both the **Site** and **Warehouse** fields to *View*, the new report will resemble the following example.

Expand table

Item number	Site	Warehouse	On-hand quantity	On-hand value	Inventory value quantity	Inventory value	Average unit cost	P1:Quantity (5/8/2020 - 5/1/2020)	P1:Amount (5/8/2020 - 5/1/2020)	P2:Quantity (4/30/2020 - 4/1/2020)	P2:A (4/3 - 4/1/
1000	1	11	10	916.67	14	1,283.33	91.67			5.00	458.3
1000	1	12	4	366.67	14	1,283.33	91.67	4.00	366.67		
1000	2		10	2,000.00	10	2,000.00	200.00				

Item number	Site	Warehouse	On-hand quantity	On-hand value	Inventory value quantity	Inventory value	Average unit cost	P1:Quantity (5/8/2020 - 5/1/2020)	P1:Amount (5/8/2020 - 5/1/2020)	P2:Quantity (4/30/2020 - 4/1/2020)	P2:A (4/3 - 4/1/
1000 Totals			**24.00**	**3,283.33**				**4.00**	**366.67**	**5.00**	**458.**

This time, site 1 is split into two rows, one for warehouse 11 and one for warehouse 12. However, the **Inventory value quantity**, **Inventory value**, and **Average unit cost** values are the same, because **Warehouse** isn't a financial inventory dimension.

Additionally, notice that the quantity distribution of site 1 is different. In the first report that you ran, the system ignored the transfer order that occurred in the same site and deducted the quantity of the sales invoice from the **3/31/2020 - 3/1/2020** period bucket in site 1. However, in the new report, the system deducts the quantity of the sales invoice from the **5/8/2020 - 5/1/2020** period bucket in warehouse 12.

Effects of inventory closing

If you run the inventory closing for May and then run the previous report again, but you set the **As of date** field to *May 31, 2020*, you will notice the following results:

- The **Inventory value** and **Average unit cost** values are updated.
- The **On-hand value** value and all the **Amount** values in every period bucket are updated accordingly.

The new report will resemble the following example.

Expand table

Item number	Site	Warehouse	On-hand quantity	On-hand value	Inventory value quantity	Inventory value	Average unit cost	P1:Quantity (5/31/2020 - 5/1/2020)	P1:Amount (5/31/2020 - 5/1/2020)	P2:Quantity (4/30/2020 - 4/1/2020)	P2:A (4/3 - 4/1/
1000	1	11	10	910.70	14	1,275.00	91.07	0.00		5.00	455.3
1000	1	12	4	364.29	14	1,275.00	91.07	4.00	364.29		
1000	2		10	2,000.00	10	2,000.00	200.00				
1000 Totals			**24.00**	**3,275.00**				**4.00**	**364.29**	**5.00**	**455.3**

Last updated on 07/01/2026

BOM calculations

07/21/2025

The cost roll-up and sales price calculations are known as bill of materials (BOM) calculations, and you initiate them from the Calculations page. This article provides information about BOM calculations.

The cost roll-up and sales price calculations are known as bill of materials (BOM) calculations, and you initiate them from the **Calculations** page. You use the **Calculations** page to perform the following tasks:

- Calculate the cost of a manufactured item, and generate an associated item cost record within a costing version.
- Calculate the sales price of a manufactured item, and generate an associated item sales price record within a costing version.

The way that you use the **Calculations** page varies slightly, depending on how you initiate the BOM calculations. The way that you use the **Calculations** page also depends on whether the BOM calculations involve a costing version for standard costs or planned costs, and on several policies that are defined in the costing version that is used in the BOM calculations. A variation of the **Calculations** page is used in the context of a sales order, sales quotation, or service order line item. These calculations are known as order-specific BOM calculations. An order-specific BOM calculation doesn't generate an item cost record within a costing version. Instead, it generates a calculation record that appears on the **BOM calculation details** page. The calculation record includes a calculated cost and a calculated sales price. The **Calculations** page can be opened for a single manufactured item or for a costing version.

- To calculate costs for a single manufactured item, you initiate BOM calculations from the **Item price** page. The **Calculations** page inherits the item identifier. The costing version, BOM version, route version, calculation quantity, calculation date, and site must be specified.
    - By default, the BOM version and route version are set to the active versions for the item, site, date, and calculation quantity. However, you can override the default values with approved versions.
    - By default, the calculation quantity is set to the item's standard order quantity. However, you can override the default value.
    - The calculation date or site can be mandated by the costing version, or user-specified values can be set when the date or site isn't mandated in the costing version. A future calculation date determines how pending cost records are used. BOM calculations use a pending cost record that has the nearest from-date that is on or before the calculation date.

- To calculate costs for all manufactured items or selected items, or to update items on a where-used basis, you initiate BOM calculations from the **Costing version setup** page or the **Costing version maintenance** page. The **Calculations** page inherits the costing version.
    - For the calculations, it's assumed that the active BOM version and route version are used for a manufactured item (and for the related site, date, and quantity), unless a manufactured component has a specified sub-BOM or subroute.
    - For the calculations, it's assumed that the standard order quantity is used for manufactured items. The standard order quantity provides the basis for calculating component quantities, determining the relevant BOM versions and route versions (when you use quantity-sensitive BOMs and routes), and amortizing constant costs. However, when a manufactured component has a BOM line type of *Production* or *Vendor*, or when you use a make-to-order explosion mode for the BOM calculations, this assumption doesn't apply, because quantities reflect the specified calculation quantity.
    - The calculation date or site can be mandated by the costing version, or user-specified values can be set when the date or site isn't mandated in the costing version.

Other variations in BOM calculations reflect the costing type and restrictions of the costing version:

- BOM calculations that use standard costs must be restricted by costing version policies, because the restrictions help guarantee that standard costing principles are used. Standard costing principles require the enforcement of restrictions about the use of standard costs for purchased items, a single-level explosion mode, and the inclusion of miscellaneous charges in unit costs.
- BOM calculations that use planned costs don't have to follow standard costing principles. These BOM calculations can use different explosion modes, alternative sources of cost data for purchased items, and optional enforcement of restrictions within the costing version.

BOM calculations that use standard costs

Policies within the costing version (for standard costs) can mandate enforcement of five BOM calculation policies. The *Recording restriction* option in the costing version mandates one of these policies, where miscellaneous charges must be included in the unit price. Miscellaneous charges for purchased items can be entered manually, whereas miscellaneous charges for manufactured items reflect the calculated amortization of constant costs. The *Calculation restriction* option in the costing version mandates the other four BOM calculation policies:

- The source of cost contributions for purchased items must be based on standard costs. In other words, BOM calculations must use the item cost records within the specified costing version, or within the fallback that contains standard costs.
- To help guarantee accurate and consistent calculation of standard costs, the explosion mode must be single-level.
- To help guarantee consistent results when the sales price of the items is calculated, the profit setting must be mandated. The profit setting can be used, and the item sales price records can be generated, only if the costing version allows for content of sales prices.
- The fallback principle must be mandated, and can be set to *None*, *Active* (for active cost records), or *Costing version* (for a specified costing version).

BOM calculations that use planned costs

Policies within the costing version (for planned costs) can optionally mandate enforcement of five BOM calculation policies. Alternatively, the policies can just provide default values. The *Recording restriction* option in the costing version determines whether the BOM calculation policy about miscellaneous charges will be mandated or act as a default value. Miscellaneous charges can optionally be included in the unit price. The *Calculation restriction* option in the costing version determines whether the other four BOM calculation policies will be mandated or act as default values:

- The source of cost contributions for a purchased item can be the item cost records within a costing version. Alternatively, the source can be defined by the BOM calculation group that is assigned to the item. For example, the BOM calculation group can define purchase price trade agreements as the source of cost contribution data.
- The explosion mode can be single-level, multilevel, or make-to-order, or it can be based on the BOM line item. The explosion mode for the BOM line type replicates the cost calculation logic for production order estimates.
- The profit setting can be mandated, or it can be a default value. The profit setting can be used, and the item sales price records can be generated, only if the costing version allows for content of sales prices.
- The fallback principle can be mandated, or it can be a default value. The fallback principle can be set to *None*, *Active* (for active cost records), or *Costing version* (for a specified costing version).

BOM calculations generate warning messages and other types of messages. Several BOM calculation policies determine the types of messages. The warning conditions are defined in the BOM calculation group that is assigned to items. However, you can override these warning conditions when you initiate a BOM calculation. When the fallback principle is used, it's often helpful if the fallback is shown as an information message. When you're trying to update

calculated costs for items that have missing cost records, it's also helpful if the information message identifies items that weren't updated.

BOM calculations that use the fallback principle

The following situations illustrate two uses of the fallback principle:

- **Two-version approach to standard cost updates** - A costing version can contain the incremental changes to standard costs, such as pending cost records that represent new items or cost changes. In this situation, the fallback principle can identify the use of the active standard costs that are contained in other costing versions.
- **Simulation of the effect of cost changes by using planned costs** - A costing version for planned costs can contain incremental changes for simulation purposes. This costing version will include pending cost records that represent the simulated cost changes to items, cost categories, and calculation formulas for indirect cost. In this situation, the fallback principle can identify the use of the active standard costs that are contained in other costing versions.

BOM calculation of a suggested sales price

When you use a cost-plus-markup approach, the calculated sales price for an item reflects the set of profit-setting percentages that is specified for the BOM calculation, and the costs that are associated with the item's component items, routing operations, and applicable manufacturing overheads. The markup reflects profit-setting percentages that are assigned to cost groups, and the cost groups that are assigned to items, cost categories for routing operations, and the indirect cost calculation formulas for manufacturing overheads. The sets of profit-setting percentages are labeled *Standard*, *Profit 1*, *Profit 2*, and *Profit 3*. Within the Profit 1 set, for example, a profit-setting percentage of 50 percent can be defined for a cost group that is assigned to purchased material, and a profit-setting percentage of 80 percent can be defined for a cost group that is assigned to cost categories for routing operations. The context of the BOM calculation determines how the results of a calculated sales price are handled:

- **BOM calculation for an item and specified costing version** - The BOM calculation generates a pending sales price record within the costing version. This sales price record provides the starting point for viewing the calculation details (for example, on the **Calculate item cost** page). The sales price record acts mainly as reference information and isn't used as the basis for a sales price on sales orders.
- **Order-specific BOM calculation** - A variation of the **BOM calculation** page is used in the context of a sales order, sales quotation, or service order line item. An order-specific BOM calculation doesn't generate a record within a costing version. Instead, it generates a calculation record that appears on the **BOM calculation results** page. This calculation

record provides the starting point for viewing the calculation details (for example, on the **Calculate item cost** page). Information about a selected calculation record can be transferred to the originating line item. For example, the calculated sales price can be transferred to a sales order line item.

Order-specific BOM calculations

An order-specific BOM calculation represents a variation of a BOM calculation for a manufactured item. The order-specific BOM calculation is performed in the context of a sales order, sales quotation, or service order line item. An order-specific BOM calculation generates a calculation record that appears on the **BOM calculation results** page. The calculation record includes a calculated weight, a calculated cost that is based on active cost records, and a calculated sales price. The calculation record that each order-specific BOM calculation for an item generates on the **BOM calculation results** page is uniquely identified by a calculation number. The results of a calculation record can be optionally transferred to the originating line item. An order-specific BOM calculation differs from a BOM calculation for a manufactured item in two ways:

- An order-specific BOM calculation doesn't generate an item cost record within a costing version. Therefore, the BOM calculation policies aren't applied when an item cost record is created, or when a cost record is overwritten.
- An order-specific BOM calculation always uses the active cost records for components, cost categories, and indirect cost calculation formulas.

BOM calculation groups

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about calculation groups for bills of materials (BOMs) and how to set them up. To run a BOM calculation, you must either set up calculation groups and assign them to individual items, or set a default calculation group. The calculation settings from the calculation group are then used as default values on the BOM calculation page at the time of BOM calculation.

You must set a default calculation group on the **Inventory and warehouse management parameters** page, or set a product-specific calculation group on the **Released product details** page. The system first looks for the calculation group setup on the **Released product details** page. If it doesn't find a calculation group there, it looks on the **Inventory and warehouse management parameters** page. If the system can't find a calculation group, the user receives an error message during calculation. A calculation group contains policies for the cost price model, the sales price model, and the warnings checklist. The calculation settings from the calculation group are used as default values on the **BOM calculation** page at the time of BOM calculation.

Purposes of BOM calculation groups

Assign a BOM calculation group to items for several reasons:

- By setting the **Cost price model** field, you indicate the source of a purchased component's cost contribution data during the calculation of the planned cost of a manufactured item. Some manufacturers calculate planned costs by using the purchase price trade agreements for purchased components or another basis, such as the purchase price records in a costing version.
- By setting the **Sales price model** field, you indicate how the item's data is used to calculate a suggested sales price. You can specify either the item sales price or the cost group. Some manufacturers want to calculate a suggested sales price for manufactured items. The

calculated sales price can reflect a rolled-price approach that is based on the component's sales price record. Alternatively, the calculated sales price can reflect a cost-plus-markup approach that is based on the component's cost and applicable profit percentage, which is associated with the item's cost group.

- By using the **Stop explosion** field, you indicate that a manufactured item should be treated as a purchased item for cost roll-up purposes during BOM calculation. Typical scenarios include a purchased item that is occasionally manufactured or a manufactured item that is now being purchased. The item is first designated as a manufactured item to define BOM and route information, and to support production orders for the item. However, the **Stop explosion** flag prevents cost calculations from using the item's BOM and route. Instead, the BOM calculation uses the item's specified costs.

Calculation groups

Define calculation groups under **Predetermined cost policies setup** in Cost management. Calculation groups assigned to items let you specify how the cost or sales price of components, as outlined by the calculation group, is sourced for the calculation. On the **Calculation groups** page, you can define a cost price model, an alternative cost price model, a sales price model, and warnings.

Cost price model

There are four options for the **Cost price model** field:

- *Item cost price* – The cost price from the **Released product** table is used, or the combination of item dimensions is used as the cost price.
- *Item purchase price* – The purchase price from the **Cost price** field on the **Purchase** tab of the **Released product list** page is used.
- *Trade agreements* – You can configure trade agreements for specific combinations of items and vendors, or for specific sites. Then, when you select the *Trade agreements* option here, the trade agreement that you created for the purchase price together with the item and site is used.
- *Inventory price* – The current inventory value for the item is used to calculate the unit cost in the BOM calculation. A unit cost price is calculated only if the posted quantity and the physical quantity are more than 0 (zero).

Alternative cost price

The **Alternate cost price** field has the same options as the **Cost price model** field. However, this field is used only when a price can't be found in the primary cost price model.

Sales price model

For the **Sales prices** field, choose from two options for calculation:

- **Cost group** – When you select this option, the sales price is calculated based on the cost price and the profit setting percentage from the cost group.
- **Item sales price** – When you select this option, the sales price on the **Sell** FastTab from the Released product table is used.

Stop explosion

Use the **Stop explosion** check box to indicate when a manufactured item should be treated as a purchase item. Typically, you leave the **Stop explosion** check box cleared. By selecting this check box, you indicate that a manufactured item must be treated as a purchase component instead of a manufactured component for the purpose of BOM calculations. Depending on the site, the item's cost can still be calculated by using BOM calculations. Explosion of planned purchase orders and production orders stops at the BOM whose components are associated with the calculation group that the **Stop explosion** check box is selected for. Master scheduling generates the planned orders on the BOM itself, not on the items that are included in the BOM. By selecting this check box, you specify that a cost isn't added into the BOM calculation for items that have this calculation group.

Warnings

On the **Warnings** FastTab, select the options for any warning messages that users should receive when they do BOM calculations.

For example, if you select the **No BOM** check box, the user receives a warning if no active BOM version is found for one of the components or the parent item that the BOM calculation is run for. If you select the **No route** check box, the user receives a warning if no active route version is found. If you're using resources on your routes and operations, you can instruct the system to check for those resources. Then, if a resource isn't found on every line in the active route, the user receives a warning.

You can also verify and check for consumption. Consumption is the quantity in a particular route. Typically, it represents the amount of time that is required to perform a specific operation for a

production process. You can check whether an item has no cost price. If there's no active cost price for an item, no cost is added into the BOM calculation.

You can also check and verify the age of the cost price. For example, enter *60* to indicate that the unit cost price must be reevaluated after 60 days. If this limit is reached, the system generates a warning. For example, a cost price was entered for an item in January of this year. If it's now August, which is more than 60 days after the cost price was entered, the user receives a warning when the BOM calculation is run. You can enter a percentage in the **Minimum contribution margin** field. This value indicates the point at which the minimum contribution margin isn't being met. If the contribution margin for a particular component isn't met, the user receives a warning. Therefore, this field helps guarantee that you don't undercut the costs and the additional carrying costs that might be required for your items.

Default setup in Inventory and warehouse management parameters

Because calculation groups are required to run calculations, set up a default calculation group in the Inventory management parameters. This setup enables companies to have a standard cost group and profit setting for all items. If a particular item has special calculation requirements, assign a different calculation group to that item. Typically, you set calculation groups on BOM component items instead of BOM items. However, when warning messages are shown, you can apply calculation groups. A calculation group that you assign to items overrides the default value that you set up in the Inventory management parameters.

Set up the default parameter at **Cost management** > **Inventory accounting policies setup** > **Parameters** > **Inventory accounting** > **Calculation group**. By setting up a default configuration group, you can also configure the warning conditions that prompt users during the BOM calculation process, if the selected components might cause calculation errors.

View warning messages on the Complete page

A BOM calculation generates warning messages. You can view the warnings about a selected item. For example, in Sales and marketing, create a new sales order for item D0001. Then, on the sales order line, on the **Update line** menu, select **Calculate Based on BOM/Formula** to view the calculation details and warnings. You can also view BOM calculation results on the **Complete** page. For the warning messages, only two warning conditions apply to manufactured items, whereas four warning conditions apply to any item:

- Identify when a manufactured item doesn't have an active BOM.

- Identify when a manufactured item doesn't have an active route.
- Identify when the item on a BOM line has a quantity of 0 (zero).
- Identify when the item on a BOM line has a cost of 0 (zero), or when it doesn't have a cost record.
- Identify when the item on a BOM line has an out-of-date cost. The warning reflects a comparison of the calculation date to the specified days for a maximum age of cost.
- Identify when the item on a BOM line has a profitability percentage that is less than you want.

You can define multiple BOM calculation groups, depending on your requirements for variations in warning messages. For example, one BOM calculation group that has warning conditions about an active BOM, a component quantity of 0 (zero), and component cost of 0 (zero) might be enough. When you start a BOM calculation, you can override the warning conditions that are associated with the BOM calculation group. You can also add or remove warning conditions. For example, if the current situation doesn't involve routing data, you can remove the warning condition about an active route.

Time and attendance includes a **Calculation groups** page, but that page has no relationship to BOM calculation groups. In Time and attendance, workers can be assigned to calculation groups that reflect the grouping of workers who are associated with the same supervisor or manager. Calculation of worker registrations can be done either automatically or manually by a supervisor or manager.

Last updated on 05/05/2026

Cost calculation level

The bill of materials (BOM) level that is named **Cost calculation level** excludes production orders and batch orders from its calculations. The system uses this level when it runs cost calculations in costing versions. In processes such as recalculation and inventory close, the system uses the **Costing level** BOM level instead.

Example scenario

The following simple scenario shows the differences between the **Cost calculation level** BOM level and the **Costing level** BOM level.

You have three products: A, B, and C. Product C is the component of product B, and product B is the component of product A.

- **Costing level** assigns the following BOM levels:
  - **Product A:** 0
  - **Product B:** 1
  - **Product C:** 2

- **Cost calculation level** assigns the following BOM levels:
  - **Product A:** 0
  - **Product B:** 1
  - **Product C:** 2

A production order for product C is then created, and product A is added to the production order BOM. After the order is estimated, BOM levels are updated in the following way:

- **Costing level** assigns the following BOM levels:
  - **Product B:** 1
  - **Product C:** 2
  - **Product A:** 3

- **Cost calculation level** assigns the following BOM levels:
  - **Product A:** 0
  - **Product B:** 1
  - **Product C:** 2

This behavior ensures that changes to production order BOMs don't affect subsequent cost calculations.

Last updated on 11/26/2025

Backflush costing

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This article introduces the concept of backflush costing that is used for Lean manufacturing.

Costing for Lean manufacturing enables the production flow to use the cost accumulation method that is known as backflush costing. In the backflush costing method, the direct materials that are consumed are accumulated in the production flow's work in progress (WIP) cost account. The standard cost inventory model group is used. The products that are received from the production flow are deducted from WIP at their standard cost. The main difference between backflush costing and standard cost is that, for backflush costing, variances aren't calculated per kanban or finished product. Instead, variances are calculated per production flow over a period. This method introduces a truly lean concept for reporting material consumption. Dedicated picked quantities of material aren't reported to a kanban or production order. Instead, full batches or handling units are staged to the production flow. After the batches or handling units are registered as empty, they're declared consumed. Advanced consumption might be used, depending on the configuration of the production flow. Before advanced consumption can be used, organizations must allow themselves to make material vanish in the WIP of the production flow. The periodic backflush costing determines the effective value of WIP to the end of the period. This determination is based on the kanban handling units and the kanban job status. Deviations between the effective values and the actual WIP values per cost group and item are accounted and shown as variances.

Configuring backflush costing

To enable costing, you must complete the following setup:

- **Set up WIP accounts for the production group and production flow.** The WIP accounts for the production flow are specified in the production group. The backflush costing production flow calculates variances as the difference in the WIP value before and after backflush costing is run for each production flow. Therefore, we recommend that you create a WIP account for every production flow.
- **Assign a cost category to the resource group.** You must assign a cost category to the run-time category of the work cell. To determine variances by activity, you should create a cost category for every work cell. The cost categories for setup and quantity aren't considered in costing for lean manufacturing. The WIP accounts per resource group are ignored in backflush costing. For subcontracted activities, no

cost category is required. The cost group that is assigned to the active service is used instead.

- **Assign cost groups.** To enable segmentation of the cost contribution in a production flow, you must assign cost groups by cost group type:
  - *Direct material cost group* – Direct material requires a cost group that identifies the material category for costing. This cost group enables an aggregated view of cost, WIP, and variances by direct material.
  - *Direct manufacturing cost group* – The direct manufacturing cost group captures the direct cost contribution of operational resources to the production flow. This cost group enables an aggregated view of cost, WIP, and variances by direct manufacturing cost.
  - *Indirect cost group* – The indirect cost group is required in order to calculate the indirect cost contribution to the production flow. This cost group enables an aggregated view of cost, WIP, and variances by indirect cost.
  - *Direct outsourcing cost group* – The cost group for the services enables an aggregated view of assigned cost and WIP, and determines the cost variances of the subcontracted services.
  - *Cost group for a finished product* – Finished products require a cost group that identifies the product category for costing. This cost group enables an aggregated view of cost, WIP, and variances by product category. The standard cost for products is calculated by using the cost calculation that is based on the bill of materials (BOM), and either the production flow and kanban rules or the route.

Costing sheet

The costing sheet models the cost structure for the company and is built by the cost groups to classify the cost. The costing sheet has various forms. It shows cost information according to the structure that is designed in it. In the costing sheet, you also define the formula that is used to calculate the indirect cost. The calculation formula can be based on quantities, weight, volume, or value.

- **Define a costing version.** In the costing version, the company defines how the cost should be maintained. A costing version can contain a set of standard cost records or a set of planned cost records, depending on the costing type that is assigned to the costing version. The costing version that is used for costing for Lean manufacturing must be based on standard cost.
- **Assign an inventory model group for released products.** All products that are related to the production flow must be assigned to an inventory model group that uses the *Standard cost* inventory model group. Standard cost is maintained per site

and activation date. For product masters, the inventory model group can be selected if the cost is maintained per variant or product master.

- **By definition, subcontracted services are non-inventoried services.** Subcontracted activities have no inventory model group. To cost a subcontracted activity correctly, you must make sure that the service activity belongs to an inventory model group where the inventory policy is set to *Stocked product = False*.

For output products, cost calculation that is based on the production flow requires that a standard cost be maintained for the services that are related to subcontracted activities. The cost group that is assigned to the services is used to determine the cost variances of the subcontracted activity.

Cost calculation for Lean manufacturing

For products that are supplied out of a production flow, the BOM calculation must be based on either a route version or a production flow. The BOM calculation calculates the cost of a product and the related breakdown to the resources and material that are required in order to build the product. The deduction from the WIP account for the production flow is done by using the breakdown of a product by item and cost group.

Calculation that is based on the production flow

Lean manufacturing for Dynamics 365 Supply Chain Management is independent of routes. The cost calculation for products that are supplied from a production flow can be based on the production flow itself. Before the calculation can be done, a kanban rule must be created that supplies the product out of the production flow. If a product can be supplied from multiple production flows at the same site on the calculation date, you can select the production flow for the BOM calculation. On the **Default production flow** page, you can configure a default production flow for each item. If multiple kanban rules exist for the same product in the same production flow that is active on the calculation date, the calculation selects the first kanban rule that is active for the calculation.

Calculation that is based on the route

Calculation that is based on a route is as valid as calculation that is based on a production flow. However, calculation that is based on a route doesn't use the costing for Lean manufacturing functionality. The route should use resource requirements for resource groups. To avoid systematic variances, it should also use the same work cells, or at least the same cost categories. Again, you should avoid cost categories for setup

and quantity. They don't help calculate the cost in a more granular breakdown than Lean manufacturing cost backflushing. To determine which option (production flow or route) you should use to calculate the cost, consider the results of the cost breakdown. The version that comes closer to reality and produces fewer variances overall is the better option. In a Lean manufacturing environment where a product is supplied by a single production flow and a single kanban rule, the calculation that is based on the production flow is probably more accurate. For a product that can be supplied by Lean manufacturing and production orders on the same site, or that can have multiple production flows or multiple kanban rules in the same flow, a calculation might be more accurate if it's based on a route version that is built specifically for the cost calculation, not for the production. The production flow calculation must be used to calculate products that involve subcontracting. The cost models for subcontracting via production orders and subcontracting in Lean manufacturing are using two different approaches. Lean manufacturing introduces a new cost group type, *Direct outsourcing*, to calculate subcontracted services.

Material consumption

When material is consumed from inventory to WIP, the cost of material is added to WIP at its actual standard cost for a cost group. This operation occurs under the following conditions:

- Kanban issues are posted for kanban picking lines that update inventory.
- Transfer jobs are completed that update inventory on pick but not receipt (Transfer of material from inventory to WIP).

Receiving products from the production flow

Products are received from the production flow under the following conditions:

- Process jobs are completed that have **Update inventory on receipt** set to *Yes*.
- Transfer jobs are completed that update inventory on receipt, but that have **Update inventory on pick set** to *No* (Transfer from WIP to inventory). This option lets you receive any products out of a production flow independently of the BOM and route configuration. The process just follows the physical flows. This option is especially suitable for receiving by-products, co-products, or scrap out of a production flow, and for correcting the cost balance of the production flow WIP accordingly.

Products that are received from the production flow are deducted from WIP.

Products in WIP

The WIP model of Lean manufacturing lets you use the kanban handling unit status to manage the material, semi-finished products, and finished products that are part of WIP.

- **Assigned** - The kanban can have consumed material that is accounted in WIP.
- **Received** - If the kanban refers to a last activity where **Update inventory on receipt** is set to *No*, it represents a full handling unit of a product or a semi-finished product that isn't registered to inventory.

Material in WIP isn't visible in inventory on-hand overviews. However, it's visible in the kanban quantity overviews.

Consuming products in WIP

Products in WIP are consumed when the corresponding kanban handling units are emptied. A kanban empty signal doesn't produce an active costing transaction but will take effect when the next backflush costing is run. Emptied kanban handling units are no longer accounted as on-hand and therefore calculated as consumed within the period.

Automatic empty registration

Scheduled or event kanbans can be set to automatic empty registration in the kanban rule:

- **When handling units are received** - By default, for scheduled kanbans, the material is declared as consumed when the last job of the kanban is completed. For fixed quantity kanbans, this option is recommended only for single-bin kanbans, because it returns the card to the source of demand whenever a kanban is received at the final destination.
- **When source requirement is registered** - This option is available only for event kanbans and is the default option for them. In connection to WIP, this option is useful for keeping WIP clean, because kanbans for components in WIP are automatically emptied, and therefore consumed from WIP, when the parent kanban is prepared.

In conclusion, kanban handling units can be assigned (= in process), received (=full), or emptied. There's no partial emptying. Therefore, to enable accurate registration of consumption, it's important that you limit the product quantities of a kanban so that they're less than the consumption per period. Products that are moved to the shop floor

in large batches that cover days or weeks of demand shouldn't be consumed to WIP. Instead, these products should be kept in inventory.

Backflush costing steps

You should run backflush costing to periodically value the WIP and produce an end-of-period status that calculates the variances of material, labor, and indirect costs. The calculated variances are posted to the variance accounts. In the backflush costing process, all production flows of the legal entity are used in the same batch run. When backflush costing is run in a batch, the processing might be multi-threaded by production flow. The backflush period is defined by an end date. You can't post new transactions to a date when a backflush costing calculation has been run. You should never run backflush costing for the current date before the day is actually over. Backflush costing performs the following steps.

1. Determine the unused quantities in the production flow as of the period's end date. After the backflush costing is run, you can view the unused quantities on the date of the costing run in the **Unused quantities** dialog box.

2. Calculate the production flow's net realized usage over the period.

3. Clear the WIP from the realized resource consumption and products.

4. Calculate production variances to standard cost for the period.

   For consumed components for the period:

   - Financially update the net realized quantities of material that the production flow consumed over the period. The system processes in first in, first out (FIFO) order on the individual inventory transactions as to financially update the physically updated transactions for the production flow, until the net realized quantities for the period are reached.
   - Transactions are financially split to map the exact consumed quantities.
   - Unused quantities in the production flow WIP remain in physically updated status.

   For production completed quantities of the period:

   - Financially update the inventory transactions for the completed quantities for the period.

   For the conversion cost:

- The applied conversion cost transactions (route transactions) that were recorded for the period are financially updated.
- All direct manufacturing cost is financially updated. All kanban process jobs that are completed during the period are accumulated.
- All indirect cost calculated for the consumed material within the period is calculated and deducted from WIP. The remaining indirect cost is posted as a variance.

5. Calculate the production variances to standard cost. The variance is calculated per cost group.

Feedback

Was this page helpful? Yes No

Provide product feedback

Common sources of production variances

This article explains various typical sources of each type of production variance.

Here are some typical sources of a *lot size* variance:

- The good quantity for a production order differs from the calculation quantity that is used in the standard cost calculation. The quantity provides the basis for amortizing constant costs.
- The value of constant costs on the production order differs from the constant costs that are used in the standard cost calculation. The constant costs on the production order can differ for several reasons. For example, the constant costs might reflect the following factors:
  - Manual changes to the production bill of materials (BOM) or route
  - The selection of a different BOM version or route version when you create the production order
  - Planned engineering changes to the BOM version or route version that is assigned to the item

Here are some typical sources of a *production price* variance:

- The cost category (and cost category price) for the reported consumption of a routing operation differs from the cost category that is used in standard cost calculation.
- The active cost for the cost category price differs from the cost category price that is used in standard cost calculation.

Here are some typical sources of a *production quantity* variance:

- You over-issue or under-issue a material component.
- You over-report or under-report the time for a routing operation.
- You over-receive or under-receive the good quantity of the parent item, relative to the order quantity. However, you issue components and report operations completely, based on the order quantity for the production order.

Here are some typical sources of a *production substitution* variance:

- You issue a material component that isn't on the production BOM.
- You manually add a component to the production BOM and report that component as consumed.
- You report an item as consumed but don't manually add it to the production BOM.

- You manually add an operation to the production route and report that operation as consumed.
- When you create the production order, you select a BOM version that differs from the BOM version that is used in the standard cost calculation.
- When you create the production order, you select a route version that differs from the route version that is used in the standard cost calculation.

Last updated on 11/26/2025

Cost categories used in production routing

This article provides information about cost categories that apply to manufacturing environments that use routing.

Cost categories apply to manufacturing environments that use routing. They're assigned to operations resources and routing operations to define hourly costs and to segment cost contributions in a manufactured item’s calculated costs. The cost groups assigned to cost categories classify manufacturing cost contributions based on the operation resources and the type of activity, such as setup time and run time. The specificity of cost group assignments enables manufacturing overhead to be calculated based on routing information.

> **Note**
>
> In manufacturing environments, cost categories are known by several other names, such as labor rate codes or machine rate codes.

Each cost category has associated cost records and an assigned cost group. Different cost categories are required for different production purposes.

- Assign different hourly costs, depending on the operations resource. For example, the costs can differ for various types of labor skills, machines, or manufacturing cells.
- Assign different hourly costs for the setup time or run time that is associated with a routing operation.
- Assign operations resource costs based on the output quantity instead of hourly costs, such as the piece rates for paying labor.
- Provide cost group segmentation of cost contributions to a manufactured item’s calculated cost. For example, you can segment labor and machine costs.
- Provide the cost group basis for overhead calculation formulas, such as formulas for labor-related and machine-related overhead, or overhead that is related to setup and run time.

A cost category can be assigned to the setup time, the process time, and the quantity for a routing operation. When, for example, costs or cost group segmentation differs between the setup time and the process time, different cost categories should be defined and assigned to the setup time and the process time. The selective use of cost categories for setup time, process time, and quantity is determined by the route group that is assigned to an operation. The assignment

of cost categories to time and quantity can be mandated by company-wide policies that are defined on the **Production control parameters** page.

Each cost category has associated costs that are based on the definition of cost records in a costing version. Use the **Cost category price** page to define the cost records for a specified costing version and site. When the cost record for a cost category is first entered, it has a status of *Pending* and an intended effective date. When you activate the cost record, the status is updated to *Current active*, and the effective date is updated to the activation date. Different cost records might reflect different sites, effective dates, or statuses. Bills of materials (BOM) calculations for a future or historical date use cost records that have the relevant effective date. The current active cost record can be used to estimate production order costs and to value reported time against a production order.

The cost record for a cost category can be site-specific or company-wide. To make a cost record site-specific, assign a site to it. Otherwise, a blank value indicates that the cost record applies to all sites in the company. Because costs can differ between sites, for example, the cost records must be defined as site-specific.

A routing operation generally inherits the cost categories that are assigned to the operations resource or master operation. When a production order is created, the routing operations in the production route reflect the selected route version. You can override the cost categories that are assigned to the operations in the production route.

Some types of production work can apply to project time estimates and reporting. In this case, a cost category is required for production and project purposes. You must define additional project-related information when a cost category is flagged for use in projects.

Last updated on 07/01/2026

Cost categories used in Production control and Project management accounting

Some types of production work can apply to project time estimates and reporting. This article provides information about the cost categories that you must define for these types of production work for production and project purposes.

Some types of production work can apply to project time estimates and reporting. In this case, a cost category is required for production and project purposes. When a cost category is used in production and projects, additional project-related information must be defined. For example, the hourly costs that are associated with projects can differ from the hourly costs that are associated with production. You can use the **Cost categories** page to define a cost category that is used in Production control and Project management accounting.

> **Note**
>
> Cost accounting has a **Project categories** page, but this page has no relationship to the functionality that is described in this article. When you use a cost category in projects, the **Cost categories** page has additional tabs that show additional project-related information. This information includes the category group, a line property, and ledger accounts that are assigned to the cost category.

- The cost category must be assigned to a category group that supports a transaction type of **Hours**.
- The line property indicates default information about how reported time can be charged to a project.
- Typically, the ledger accounts that are related to costs and sales are defined for the category group that is assigned to the cost category. However, specific accounts can be defined for an individual cost category.

Additional buttons on the **Cost categories** page let you access project-related information about a selected cost category. For example, you can view project-related transactions, define employees or projects, define hourly costs and sales prices, and view reports.

Last updated on 11/26/2025

Display charges for a manufactured item

The constant costs of a manufactured item reflect the operation setup times and the components that have a constant quantity or a constant scrap amount.

The calculated amount of an item's charges can be displayed with the item's unit costs. However, the charges are sometimes displayed as separate fields, and they aren't included in the item's unit costs. When the charges appear as separate fields, one field displays the total charge amount, and another field displays the costing lot size that is used to amortize the amount. The **Item price** page, for example, displays the charges as two separate fields. However, the **Complete** page displays the item's total cost per unit, and the amortized costs are included in the unit costs.

The charges for a manufactured item are always included in the item's unit cost for standard cost purposes. They can optionally be included for planned cost purposes. A policy in the costing version enforces the inclusion of charges in the cost of a manufactured item. When you activate an item's cost record, you update the charges for the item's base cost information, which is displayed on the **Item price** page. The charges are displayed as two separate fields, and they aren't included in the item's unit cost. Each activation updates the item's base cost information, even if the activation reflects different sites. Therefore, you should view the base cost information as reference information.

Last updated on 11/26/2025

Production order cost analysis

This article provides information about the cost analysis that you can do for completed and current production orders. You can analyze the estimated costs and actual costs by using the **Price calculation** page or the **Cost estimates and costings** report. You can view information about the estimated and actual costs (and quantity) for each component item, the routing operation, and the indirect cost.

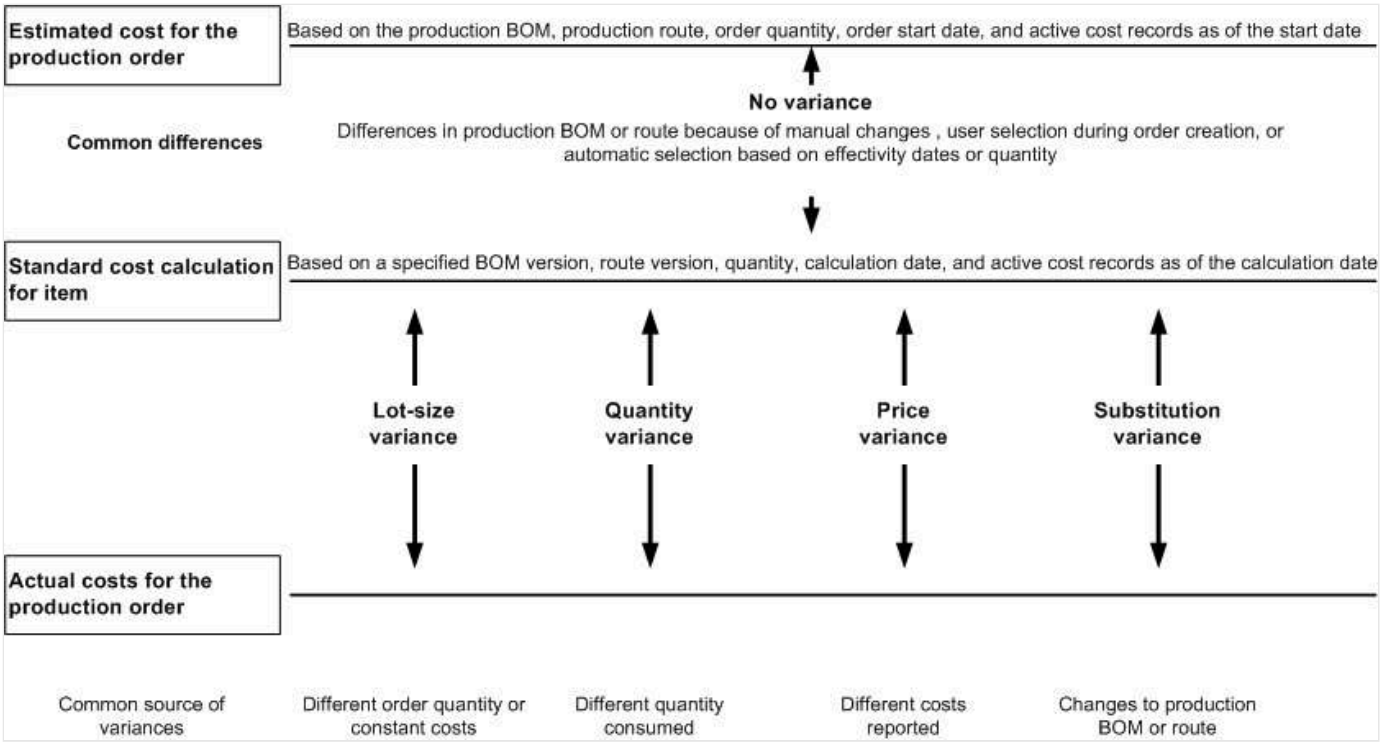
The actual costs for a production order are based on the reported consumption of material and routing operations. You can access detailed transactions about the reported consumption of material, routing operations, and indirect costs for a production order on the **Production posting** page.

Variance analysis for a completed production order

The variances reflect a comparison of the reported production activities and the calculation of standard costs for the production item. The variances don't reflect a comparison to the production order's estimated costs. The production activities that are reported include the consumption of material and routing operations, together with the associated indirect costs, and the quantity that is reported as finished. The following four types of variance are calculated:

- Lot size variance
- Production quantity variance
- Production price variance
- Production substitution variance

The following diagram shows the four variances that account for the difference between a production order's actual costs and the calculated costs within the item's cost record when the production order is ended.



You can analyze the production variances by using the **Variance** page or the **Production variance** report. Use the display options to view detailed variances by item and operations resource, or by cost group. The policy for cost breakdown in the inventory parameters determines whether the variances are tracked by cost group. You can also use the **single**, **multi**, and **total** display options to view summarized variances. The information about detailed variances can help you understand the source of each variance. To predict variances before you end a production order, analyze the detailed information that the **Cost estimates and costings** report provides.

Cost analysis for current production orders

Separate reports provide information about each type of transaction. Use these reports to analyze costs for reported production activities. Information is displayed only for current production orders that have a status of *Started* or *Reported as finished*.

- **Materials in process** – This report lists the picking list transactions that are reported against the current production orders as of a specified transaction date. The report indicates the quantity of a component that's issued and the cost amount for each transaction. Use the selection criteria for a single component item. For example, you can print information about the component's issued quantity against applicable production orders. The issued quantity isn't updated by the quantities reported as finished for the parent item. Therefore, the actual quantity of raw materials in process might be overstated.
- **Work in process** – This report lists route transactions (or job transactions) reported against the current production orders as of a specified transaction date. The report indicates the hours, amount, and quantity (both good quantity and error quantity) that are reported for

each transaction. It also includes information such as the operation number, operation ID, and operations resource. Additionally, this report shows the total time and amount for all transactions against the production order, and the quantity reported as finished.

- **Indirect costs in process** – This report lists the indirect costs incurred against production orders. This data is based on reported consumption of routing operations and components as of a specified transaction date. The report indicates the type of indirect cost (surcharge or rate), the costing sheet code for the indirect cost, and the cost amount for each transaction. This report doesn't provide information about the route card or pick list transaction that generated the indirect cost.

Related information

- Common sources of production variances

Last updated on 07/01/2026

Production order cost estimation

This article provides information about production cost estimation. Production cost estimation provides the projected material and capacity consumption costs of producing an item in the planned production order quantity.

After you create a production order, you must estimate production costs. The purpose is to estimate item and route consumption for the production process, because these estimates are used as the basis for subsequent scheduling and production processes.

Production cost estimation

Estimates of production costs are based on the following information:

- The quantity on the production order
- The components on the production bills of materials (BOMs)
- The routing operations in the production route
- The indirect costs that apply to the components and operations
- The active cost data as of the calculation date

If there's a phantom line item on the production BOMs, the calculations reflect the phantom's components and route operations. You can use the estimation task to recalculate estimated costs so that they reflect updated information. For example, the updated information might be changes to the quantity on the production order, the components on the production BOMs, the routing operations in the production route, the indirect costs that apply to these components and operations, or the active cost data as of the recalculation date. The calculations of estimated cost also suggest a sales price for the production item, based on a cost-plus-markup approach. The calculations of estimated cost can optionally apply to reference orders that reflect other production orders that are linked to the production order.

View the estimated costs

After you run estimation, you can view the results on the **Price calculation** page. The estimation calculates the following values:

- **Production cost** – The production cost is the top line of the estimate. It shows the complete cost of running the production order and the total sales price for the production. It's the sum of all the cost lines on the estimate.

- **Route or resource costs** – Route or resource costs are the costs for the production operations. They include the cost of elements such as setup time, run time, and overhead.
- **Material costs** – Material costs are the costs and prices of the BOM components that are required in order to produce the item. These costs were previously established and entered into the system.

Other uses of cost estimation

A cost estimate also provides the following information:

- Meaningful price quotations
- Estimates of the profitability of the order
- Estimates of raw material usage
- Comparisons of cost information from previous productions
- Budget and forecasting information
- Estimates of the production size that is required in order to maintain a particular cost

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Last updated on 11/26/2025

View current WIP status on a production order

This procedure shows how to view a work in progress (WIP) statement for a production order. This procedure is intended for the cost controller.

1. Go to **Cost management** > **Workspaces** > **Cost administration**.
2. On the Action Pane, select **Production orders**.
3. Find and select the production order you're looking for. If necessary, you can use the **Filter** field to help find it.
4. On the Action Pane, open the **Manage costs** tab and select **Production WIP statement**.
5. Use the **From date** and **To date** fields to specify the date range you're looking for.

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Compare active, estimated, and realized costs on a production order

This procedure shows how to view reasons for high production variance for a production order. The demo data company used to create this procedure is USMF. This procedure is intended for the cost controller.

1. Select **Cost administration**.
2. In the **Date** field, enter or select a value. (This procedure uses the fiscal year 2012. You can set **From date** to *January 1, 2012* and **To date** to *December 31, 2012*.)
3. Open the **High production variances** tab.
4. Select the relevant link in the **Production** field. (If you're working with USMF sample data, select *P000116*).
5. On the Action Pane, select **Manage costs**.
6. Select **View cost comparison**.
7. Select **Close**.

Last updated on 11/26/2025

Production posting

This article provides information about different types of postings in the production process.

Production posting activities follow production processes that are described in the sections below.

Material consumption

Materials are registered as consumed during production when the production picking list journal is posted. This process generates issue transactions that deduct the on-hand inventory. In the production parameters, you can specify whether the value of raw materials that are in progress (work in process [WIP]) should be posted in the ledger. The value of raw materials that are in progress (WIP) is then posted to a dedicated Picking list account and a dedicated Picking list offset account.

Time consumption

The time that workers spend on production jobs is recorded in the Route card journal or the Job card journal. When these journals are posted, ledger posting to a dedicated account for resources that are in progress (WIP) is processed. This posting represents the value of the time that is spent on the production order. After the production order is registered as ended, the WIP accounts are settled.

Reporting finished goods and error quantities

When a production order is reported as finished, the quantity of the finished goods that have been completed is updated in Inventory management through the Report as finished journal. If you're using WIP accounting, which can be set up in the production parameters, a ledger journal is made to reduce the WIP accounts and increase the inventory of the finished goods. The journal uses the standard cost that is defined for the product.

Ending the production order

Before a production order is ended, actual costs are calculated for the quantity that was produced. All estimated costs of material, labor, and overhead are reversed and replaced with actual costs. The overall cost of the finished item is debited from the inventory Receipt account

and credited to the inventory Issues account. If you select the **End job** check box when you run the cost calculation, the status of the production order is changed to *Ended*. This status prevents any additional costs from being unintentionally posted to a completed production order. You can specify that the value of error quantities that are reported during reporting as finished should be allocated to the good quantities that are reported as finished. Alternatively, you can specify that the value of error quantities should be posted to a dedicated scrap account.

Controlling the level of ledger posting

In the **Production control parameters**, you can use the **Ledger posting** field to set the level of ledger posting for production processes. The following values are available:

- *Item and resource* – Use the ledger accounts that are set up on the item groups for raw materials and finished goods. WIP for time consumption is taken from resource or resource group from the route operations.
- *Item and category* – Use the ledger accounts that are set up on the item groups for raw materials and finished goods. WIP for time consumption is taken from the cost categories that are associated with the route operations.
- *Production groups* – Use the ledger accounts that are set up on the production groups for both material and time consumption. The production groups are associated with the released products and copied to the production orders when those orders are created. The posting on the production orders will then follow the production groups that are associated with the production order.

> **Note**
>
> If the standard method for calculating the cost of the finished item was used, the final transactions reflect this fact. If actual costs and the costs that are calculated by using the standard method differ, the difference is posted to the account that shows profit or loss.

Last updated on 11/26/2025

Set up products that can be produced or procured

Products can be sourced in various ways. They can be produced (manufactured) or procured (purchased). This article describes some typical points to consider when you configure products to support multi-sourcing.

Multi-sourcing is typically used for a purchased item that is occasionally manufactured, or when an item that was primarily a manufactured item is changed so that it's now primarily a purchased item. The item is first designated as a manufactured item, so that bill of materials (BOM) and route information can be defined, and to support production orders for the item. The production type should be set to *BOM* (or, for process manufacturing, *Formula* or *Co-Product*).

When you use standard cost, the item cost record can be calculated for the manufactured item. However, the item cost record might not match the standard cost that you want for purchasing purposes. In this case, the standard cost must be manually entered and activated for the item cost record. For the cost calculation, consider using a special BOM and route that represent the supply mix of the product over the course of a fiscal period, to minimize the variances over time. Additionally, a manufactured item at one site can be transferred to another site. Therefore, the item's cost must be manually entered and activated for the site that the item is transferred to. When you use the manufactured item as a component in higher-level products, the component's costs should be treated as a purchased item. This guideline applies, regardless of whether the component’s costs were calculated or manually entered. In other words, a BOM calculation should treat the item's costs as a purchased component instead of using the item's BOM and route information to calculate costs.

To prevent the calculation from occurring, select the **Stop explosion** flag that is embedded in the BOM calculation group that is assigned to the item. To prevent master scheduling calculations from exploding requirements through the item, set the explosion fence to 0 (zero) days in item coverage or in the coverage group. The master scheduling calculation will then treat the item as a purchased item and won't perform more calculations for the item's BOM and route information.

Last updated on 11/26/2025

Total cost allocation method

Article • 01/28/2025

Total cost allocation (TCA) is a method of calculating the cost between the main formula item for a batch order and the co-products that are defined for the formula. This method is dynamic. It calculates the cost as a weighted average between the quantities that are reported as finished for the formula item and the co-products. When TCA is used, you don't have to review cost allocations for every batch order. If TCA isn't used, the formula calculation uses existing functionality.

Using TCA for co-products

Here are some of the guidelines for using TCA for co-products:

- If you set **Total Cost Allocation** to *Yes* for a formula version, co-products must have a cost price that is more than 0 (zero). The value can be retrieved from the active cost version for the same site, or for the first site for a formula that isn't site-specific. This condition is validated when the formula is approved.
  - You don’t need to manually enter cost allocation percentages for co-products. Instead, the system automatically creates the cost allocation percentage as the average of active cost prices of co-products.
  - You don’t need to enter standard cost for non-standard cost items that are co-products. There are two types of costing versions in the system: standard cost and planned cost
  - If an item isn’t valuated by the standard cost valuation method, we recommend you use an active cost price in the planned cost version. This price is used for cost estimation, for example, bill of material (BOM) calculation, production cost estimation, and fallback price in the inventory valuation process.

- If you set **Total Cost Allocation** to *Yes* for the formula version and the following conditions are true, the method of cost allocation is **TCA**, and the percentage of cost allocation is unchanged:
  - You added co-products.
  - You used a different method of cost allocation for the co-products.

- If you set **Total Cost Allocation** to *No* for the formula version and the following conditions are true, the method of cost allocation is changed to **Manual**, and the percentage of cost allocation is unchanged:
  - You added co-products.
  - The percentage of cost allocation is more than 0 (zero).

- Before you can successfully perform a formula calculation, you must estimate the percentages of cost allocation. You can complete this step either manually or by using the **Estimate cost** option on the **Co-products** page. The **Estimate cost** option is available only when **Total Cost Allocation** is set to *Yes* for the formula version. You can view the expected allocation if the batch order quantities that are reported as finished match quantities that appear on the formula.

- When a batch order is created manually or a planned batch order is firmed, the value of the **Total Cost Allocation** slider for the formula version is copied to the batch order. However, you can change this setting on the batch order. If **Total Cost Allocation** is set to *No* for the formula version and then changed to *Yes* for the batch order, the method of cost allocation for each line previously set to *Manual* is changed to *TCA*. A cost allocation of *None* is unchanged. If **Total Cost Allocation** is set to *Yes* for the formula version and then changed to *No* for the batch order, the method of cost allocation for each co-product of the *Production* type is changed to *Manual*. Any estimated percentage of cost allocation is unchanged.

- The **Co-product cost allocation** page shows the calculated cost allocation percentage. You can open this page from the **Batch order** page. This information is useful when the products and quantities that are reported differ from the scheduled or started quantities on the batch order. When the cost is complete, these new percentage allocations from TCA are shown on the **Co-product cost allocation** page.

Calculating the burden for byproducts

The **By-product cost allocation** field on the **Co-products** page is an enumerator field that is used only for by-products. For co-products, the value of this field is always *None*. For by-product lines, this field determines how the cost amount for the by-product line is added to the total cost of the production. The following options are available:

- *None* — No amount is added to the total cost of the production for this by-product line.
- *Percent* — The cost amount is calculated as a percentage of the total cost of raw materials that are consumed in the production. The percentage that is used for the calculation is entered in the field.
- *Per series* — The cost amount is calculated as an amount per standard batch size of the production order. This amount is independent of the reported quantity in the production. The amount that is used for the calculation is entered in the field.
- *Per quantity* — The cost amount is calculated as an amount per reported quantity of the formula item in the production. The amount that is used for the calculation

is entered in the field.

Feedback

Was this page helpful? Yes No

Provide product feedback

Amortize constant costs for a manufactured item

A manufactured item's constant costs reflect the operation setup times and the components that have a constant quantity or a constant scrap amount.

The concept of a costing lot size is used to amortize these constant costs in the calculated cost of a manufactured item. This concept has several synonyms, one of which is accounting lot size. The concept of amortizing constant costs also has several synonyms, one of which is proportional constant costs.

The quantity of a costing lot size for a manufactured item is used in a bill of material (BOM) calculation. The quantity depends on how you initiate and perform the BOM calculation, as illustrated by the following:

- Default calculation quantity in an item's BOM calculation – The item's standard order quantity for inventory acts as the costing lot size, but the default value might be a greater quantity to reflect the item's order quantity multiple. The item's standard order quantity and multiple can be defined within its default order settings or site-specific order settings.
- Specified calculation quantity in an item's BOM calculation – The specified calculation quantity acts as the costing lot size for the item. The calculation quantity initially uses the item's standard order quantity, but the default value can be manually overridden. The specified calculation quantity represents the costing lot size for the specified item, and for manufactured components that have a BOM line type of production. This is because it is assumed that the component will be produced to the exact quantity. The costing lot size for other manufactured components that have a BOM line type of item will reflect their standard order quantity.
- Specified make-to-order calculation quantity in an item's BOM calculation – The specified calculation quantity acts as the costing lot size for the item and its manufactured components when BOM calculations use a make-to-order explosion mode. It is assumed that the manufactured components will be produced to the exact quantity, just like a manufactured component that has a BOM line type of production.
- Specified calculation quantity in an order-specific BOM calculation – An order-specific BOM calculation can be performed for a line item on a sales order, sales quotation, or service order. The specified calculation quantity uses the quantity on the originating line item, but

the default quantity can be overridden. You can select whether the order-specific BOM calculation uses a make-to-order or multilevel explosion mode.

The calculated amount of a manufactured item's amortized constant costs is termed charges.

Last updated on 11/26/2025

Cost management Power BI content

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Overview

The **Cost management** Microsoft Power BI content is intended for inventory accountants or individuals in the organization who are responsible for or interested in the status of inventory or work in progress (WIP), or who are responsible for or interested in analyzing standard cost variances.

This Power BI content provides a categorized format that helps you monitor the performance of inventories and visualize how cost flows through them. You can gain managerial insights such as the turnover ratio, number of days that inventory is on hand, accuracy, and "ABC classification" at your preferred aggregated level (company, item, item group, or site). The information that is made available can also be used as a detailed supplement to the financial statement.

The Power BI content is built on the **CostObjectStatementCacheMonthly** aggregated measurement, which has the **CostObjectStatementCache** table as its primary data source. This table is managed by the Data set cache framework. By default, the table is updated every 24 hours, but you can change the update frequency or enable manual updates in the configuration of the data set cache. Manual updates can be run in either the **Cost administration** workspace or the **Cost analysis** workspace.

After every update of the **CostObjectStatementCache** table, the **CostObjectStatementCacheMonthly** aggregated measurement must be updated before data in the Power BI visualizations is updated.

Accessing the Power BI content

The **Cost management** Power BI content is shown in the **Cost administration** and **Cost analysis** workspaces.

The **Cost administration** workspace contains the following tabs:

- **Overview** – This tab shows application data.
- **Inventory accounting status** – This tab shows Power BI content.
- **Manufacturing accounting status** – This tab shows Power BI content.

The **Cost analysis** workspace contains the following tabs:

- **Overview** – This tab shows application data.
- **Inventory accounting analysis** – This tab shows Power BI content.
- **Manufacturing accounting analysis** – This tab shows Power BI content.
- **Std. cost variance analysis** – This tab shows Power BI content.

Report pages that are included in the Power BI content

The **Cost management** Power BI content includes a set of report pages that consist of a set of metrics. These metrics are visualized as charts, tiles, and tables.

The following tables provide an overview of the visualizations in the **Cost management** Power BI content.

Inventory accounting status

Expand table

Report page	Visualization
Inventory overview	Beginning balance
	Net change
	Net change %
	Ending balance
	Inventory accuracy
	Inventory turnover ratio
	Days inventory on-hand
	Active product in period
	Active cost objects in period
	Balance by item group
	Balance by site
	Statement by category
	Net change by quarter
Inventory overview by site and item group	Inventory accuracy by site
	Inventory turnover ratio by site
	Inventory ending balance by site

Report page	Visualization
	Inventory accuracy by item group
	Inventory turnover ratio by item group
	Inventory ending balance by site and item group
Inventory statement	Inventory statement
Inventory statement by site	Inventory statement by site
Inventory statement by product hierarchy	Inventory statement
Inventory statement by product hierarchy	Inventory statement by site

Manufacturing accounting status

Expand table

Report page	Visualization
WIP overview YTD	Beginning balance
	Net change
	Net change %
	Ending balance
	WIP turnover ratio
	Days WIP on-hand
	Active cost object in period
	Net change by resource group
	Balance by site
	Statement by category
	Net change by quarter
WIP statement	Beginning balance
	Ending balance
	WIP statement by category
WIP statement by site	Beginning balance
	Ending balance
	WIP statement by category and site

Report page	Visualization
WIP statement by hierarchy	Beginning balance
	Ending balance
	WIP statement by category hierarchy

Inventory accounting analysis

Expand table

Report page	Visualization
Inventory details	Top 10 resources by ending balance
	Top 10 resources by net change increase
	Top 10 resources by net change decrease
	Top 10 resources by inventory turnover ratio
	Resources by low inventory turnover ratio and ending balance above threshold
	Top 10 resources by low accuracy
ABC classification	Inventory ending balance
	Consumed material
	Sold (COGS)
Inventory trends	Inventory ending balance
	Inventory net change
	Inventory turnover ratio
	Inventory accuracy

Manufacturing accounting analysis

Expand table

Report page	Visualization
WIP trends	WIP ending balance
	WIP net change
	WIP turnover ratio

Std. cost variance analysis

Expand table

Report page	Visualization
Purchase price variance (Std. cost) YTD	Procured balance
	Purchase price variance
	Purchase price variance ratio
	Variance by item group
	Variance by site
	Purchase price by quarter
	Purchase price by quarter and Item group
	Top 10 resources by unfavorable purchase price ratio
	Top 10 resources by favorable purchase price ratio
Production variance (Std. cost) YTD	Manufactured cost
	Production variance
	Production variance ratio
	Variance by item group
	Variance by site
	Production variance by quarter
	Production variance by quarter and variance type
	Top 10 resources by unfavorable production variance
	Top 10 resources by favorable production variance

Understanding the data model and entities

Data from the application is used to fill the report pages in the **Cost management** Power BI content. This data is represented as aggregate measurements that are staged in the entity store, which is a Microsoft SQL Server database that is optimized for analytics. For more information, see Power BI integration with Entity store.

The key aggregate measurements of the following objects are used as the basis of the Power BI content.

Expand table

Object	Key aggregate measurements	Data source for finance and operations	Field
CostObjectStatementCacheMonthly	Amount	CostObjectStatementCache	Amount
CostObjectStatementCacheMonthly	Quantity	CostObjectStatementCache	Qty
CostInventoryAccountingKPIGoal	AnnualInventoryTurn	CostInventoryAccountingKPIGoal	AnnualInventoryTurn
CostInventoryAccountingKPIGoal	InventoryAccuracy	CostInventoryAccountingKPIGoal	InventoryAccuracy

The following table shows the key calculated measurements in the Power BI content.

Expand table

Measure	Calculation
Beginning balance	Beginning balance = [Ending balance]-[Net change]
Beginning balance qty.	Beginning balance qty. = [Ending balance qty.]-[Net change qty.]
Ending balance	Ending balance = (CALCULATE(SUM([Amount]), FILTER(ALL(FiscalCalendar) ,FiscalCalendar[MONTHSTARTDATE] <= MAX(FiscalCalendar[MONTHSTARTDATE]))))
Ending balance qty.	Ending balance qty. = CALCULATE(SUM([QTY]), FILTER(ALL(FiscalCalendar),FiscalCalendar[MONTHSTARTDATE] <= MAX(FiscalCalendar[MONTHSTARTDATE])))
Net change	Net change = SUM([AMOUNT])
Net change qty.	Net change qty. = SUM([QTY])
Inventory turnover ratio by amount	Inventory turnover ratio by amount = if(OR([Inventory average balance] <= 0, [Inventory sold or consumed issues] >= 0), 0, ABS([Inventory sold or consumed issues])/[Inventory average balance])
Inventory average balance	Inventory average balance = (([Ending balance] + [Beginning balance]) / 2)
Days inventory on-hand	Days inventory onhand = 365 / CostObjectStatementEntries[Inventory turnover ratio by amount]
Inventory accuracy	Inventory accuracy by amount = IF([Ending balance] <= 0, IF(OR([Inventory counted amount] <> 0, [Ending balance] < 0), 0, 1), MAX(0, ([Ending balance] - ABS([Inventory counted amount]))/[Ending balance]))

The following key dimensions are used as filters to slice the aggregate measurements, so that you can achieve greater granularity and gain deeper analytical insights.

Expand table

Entity	Examples of attributes
Products	Product number, Product name, Unit, Item groups
Category hierarchies (Assigned to role Cost management)	Category hierarchy, Category level
Legal entities	Legal entity names
Fiscal calendars	Fiscal calendar, Year, Quarter, Period, Month
Site	ID, Name, Address, State, Country/Region

Last updated on 03/05/2026

Reporting and analytics with Power BI home page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article points you to resources that you can use to learn more about the business intelligence (BI) and reporting tools that are available.

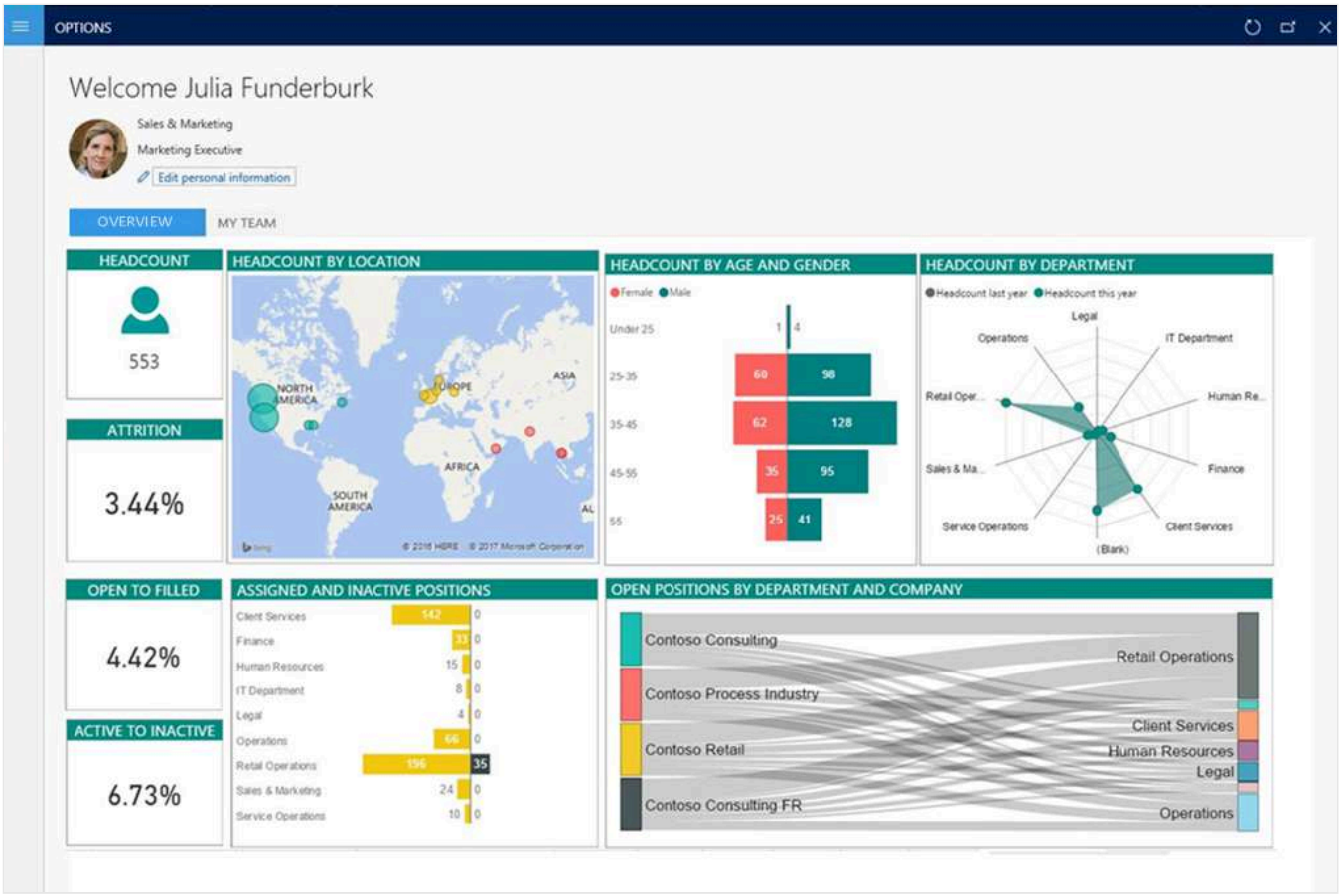
Get started

- Information access and reporting
- Tech Talk: Reporting options (video)
- finance and operations: Power BI Analytics & Reporting Services blog (blog)

Analytical workspaces

Workspaces can use rich infographics and visuals that Microsoft Power BI supports. These infographics and visuals include many controls that third parties provide. Therefore, workspaces can provide a highly visual, interactive experience for users.

Users can interact with data by clicking or touching visuals on the page. They can see cause and effect, and do simple what-if operations without leaving the workspace. Thanks to stunning, interactive visuals, your users have fun exploring data and discovering hidden trends.

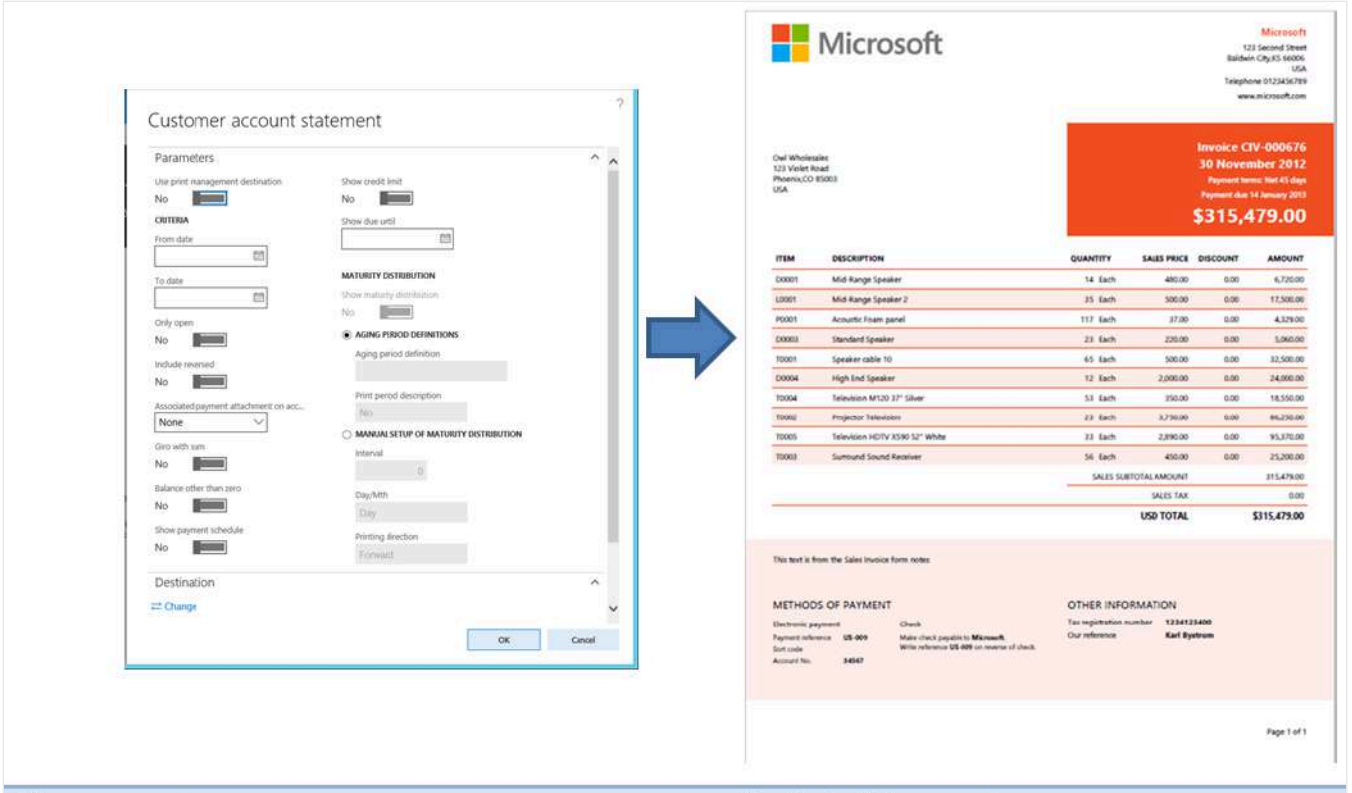


Learn more in the following articles:

- Embedded Power BI in workspaces
- Power BI Embedded integration
- Add analytics to workspaces by using Power BI Embedded
- Help secure analytical workspaces and reports by using Power BI Embedded
- Power BI content home page

Business documents and printing

Reporting solutions often capture and communicate the details of business transactions. Therefore, a reporting solution must produce physical manifestations of business data by using existing devices, such as network printers. Examples of business documents include sales invoices, customer statements, and checks.



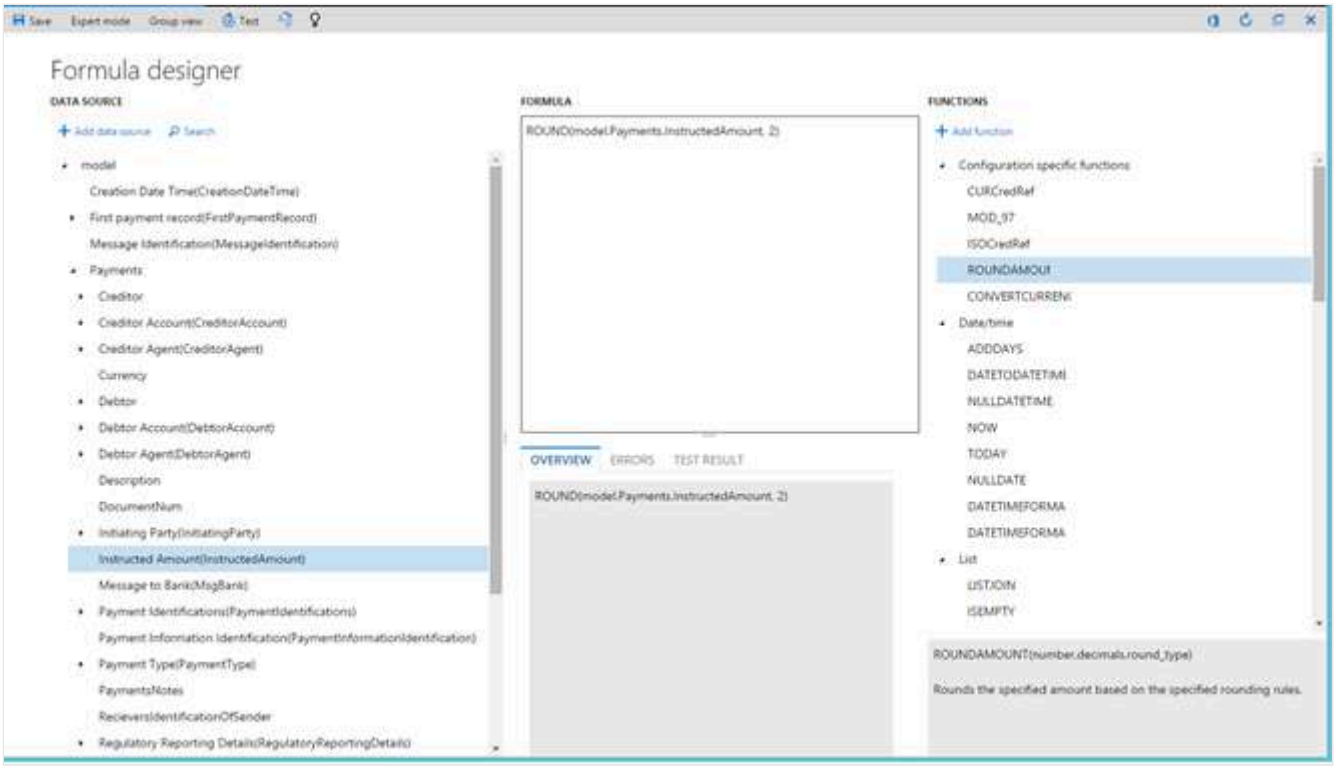
User parameters	Paginated document

Learn more in the following articles:

- Document Reporting Services
- Document printing overview
- Install the Document Routing Agent to enable network printing

Electronic reporting

Electronic reporting (ER) is the tool that you use to configure electronic document formats that comply with the legal requirements of various countries or regions. You can use electronic reporting for financial auditing, tax reporting, and electronic invoicing.

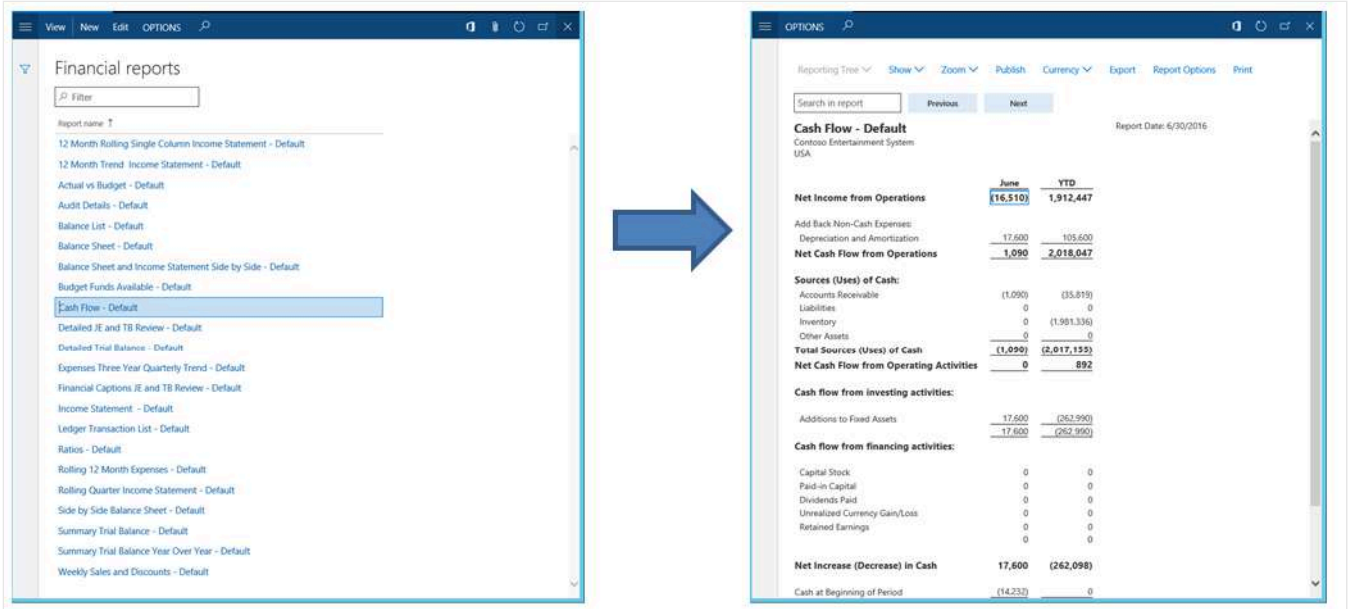


Learn more in the following articles:

- Electronic reporting (ER) overview
- Manage the Electronic reporting (ER) configuration lifecycle
- Create Electronic reporting (ER) configurations

Financial reporting

Standard financial reports use the default main account categories. Use the report designer to create or modify traditional financial statements, such as income statements and balance sheets. You can then share the results with other members of your organization. Examples of financial reporting include balance sheets, cash flow, and summary trial balance year over year.



Standard templates (GL > Inquiries > Fiancial reports)	Interactive financial reports

Learn more in the following articles:

- Financial reporting
- Generate financial reports
- Financial report components

Technical reference reports

The following reports provide reference information about the objects:

- Find information about standard data entities
- License codes and configuration keys report
- SQL Server Reporting Services (SSRS) reports that are available
- Workflow types report

Last updated on 11/26/2025

Information access and reporting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains the various reporting options available as part of the platform.

Why information access is important

Information access is an essential part of an ERP solution. It represents a significant portion of the user engagement with the system. Consider the numerous methods of capturing information related to daily activities and the level of investments required to manage the data. Employees depend on logical interpretations of massive amounts of data to stay on top of their daily tasks. Out of the box, the platform provides a collection of reporting solutions to address the various information access needs of an ERP solution. In an increasingly competitive environment, mergers and acquisitions seem to occur as often as the change in seasons. More than ever before, businesses today are finding ways to expand their global reach to attract more customers. To be successful, they must adapt legacy solutions used to communicate with customers and prepare for the enforcement of regional regulatory requirements associated with new markets. Beyond the primitive functions provided by Microsoft Azure – compute, networking, storage, and authentication – the platform provides tools to manage resources for organizations that span in size from small businesses to global enterprise conglomerates. These tools are designed with flexibility in mind, to accommodate a dynamic world of business.

What is a "report" in the application?

A *report* is a structured presentation of data. Under ideal circumstances, reports present data in such a way that it allows the user to make informed decisions. The application supports a broad spectrum of information access scenarios: cross company all-up financial reporting; analytical dashboards and tiles; electronically transferable funds; customer sales invoices; checks and tax documents; and many more. Examples of integrated report scenarios that involve the consumption of business data include:

- **Native controls** including list pages, grid controls, and chart visualizations.
- **Dashboards and workspaces** containing predefined and personalized views.
- **Financial reporting** providing all-up views across legal entities.

- **Structured documents** distributed internally to employees or externally to customers and vendors.

Although each of these scenarios at its core involves the presentation of structured business data, the process of accessing these reports and how the results are then consumed varies greatly. Flexibility in the user tooling is essential for scenarios that involve data exploration. By contrast, layout precision is required for compliance with most regulatory documents. Given the diversity of information access scenarios, it's understandable that not all reports are created equal. This article is intended to help familiarize you with the various reporting options available as part of the platform.

Common myths of reporting

To become a proficient *report maker*, it's often useful to let go of past inhibitions. The following section seeks to rebuke three common myths about reporting.

- **Myth #1: Operational reports require "real-time" data** To the contrary, relatively few reporting scenarios require *real-time* results. And, in the grand scheme of things, taking a critical stance on a request for *real-time* views is recommended given the high development costs and potential heavy burden these solutions may incur on production environments.
- **Myth #2: The best tool is the one the developer is most comfortable using** Consider a customer request for a report that allows them to monitor company's assets. In the past, a developer would build a static report displaying a list of inventory items with complex calculations relying entirely on the user to provide filters to sufficiently reduce the result set. This solution might function perfectly in developer environments with a reduced data set. However, this approach can unnecessarily consume significant amounts of compute resources when utilized in production.
- **Myth #3: Developers are good at creating visually compelling designs** In reality, developers are often the worst offenders when it comes to producing elegant design layouts that appeal to the customer's aesthetic preferences. When it comes to analytical reports, you're better off empowering users to both explore the data directly and share personalized views.

Understanding report requirements

The best reporting solutions are designed with the expertise, daily functions, and information access needs of the target user in mind. The platform offers several tools designed to meet the functional requirements that are common across various reporting experiences. Selecting the right tool that most effectively addresses the *need* requires a clear understanding of the customer experience. You can drastically increase your chances of delivering a complete and

robust solution that fully satisfies customer requirements by simply asking the right questions. Here are some leading questions to ask when evaluating customer requirements for reporting solutions:

- **Get to know the user**
  - What is the proficiency of the target persona? Are they familiar with analytical tools like Microsoft Excel?
  - Does the user require a guided parameter experience to refine the dataset?
  - How frequently is the report accessed?

- **Familiarize yourself with the data**
  - Are they looking for transactional, analytical, and predictive information?
  - Does the shape of the data change and if so, how often?
  - Does the report include data from external sources?

- **Determine how the results are used**
  - Are you going to explore the data to gain insights?
  - How are the results shared with others?
  - Is there a fixed document structure for the target output?

It's understandable that customers *want* a solution that aligns with the existing processes they're comfortable using. However, you can learn a lot through these leading questions used to discover what the customer actually *needs* to be successful in their task. Delight your customers by providing them with solutions that empower them to be more productive.

Reporting experiences

Applications support information access scenarios that break down into five distinct reporting experiences. Specialized tools meet the complex and diverse reporting needs of various functions throughout the organization.

- **Operational views** – Designed to address the specific needs of a given business persona.
- **Business documents** – Static documents used to capture and exchange processed business data.
- **Analytical tools and visualizations** – Personalized presentations of logical calculations that allow the user to explore their data.
- **Electronic reporting** – Tool used to configure formats for electronic documents.
- **Financial reporting** – Designed to provide in-depth accounting management tools based on standard views of financial activities across legal entities.

Scorecard

Use the following table as a guide when choosing the right tool for your reporting solution.

Expand table

MAKER	Operational views	Business documents	Analytical tools & visualizations	Electronic reporting	Financial reporting
Persona	Developer	Developer	Power user	Power user	Power user
Authoring tool	Visual Studio	Visual Studio	PowerBI.com PowerBI app	Excel	Management Reporter Designer
Time to market	Weeks	Weeks	Hours	Hours	Hours
Data sources	Entity DB OLTP	OLTP	Entity DB Azure Catalog	OLTP	OLTP
Effort	Days	Days	Minutes	Hours	Hours

Expand table

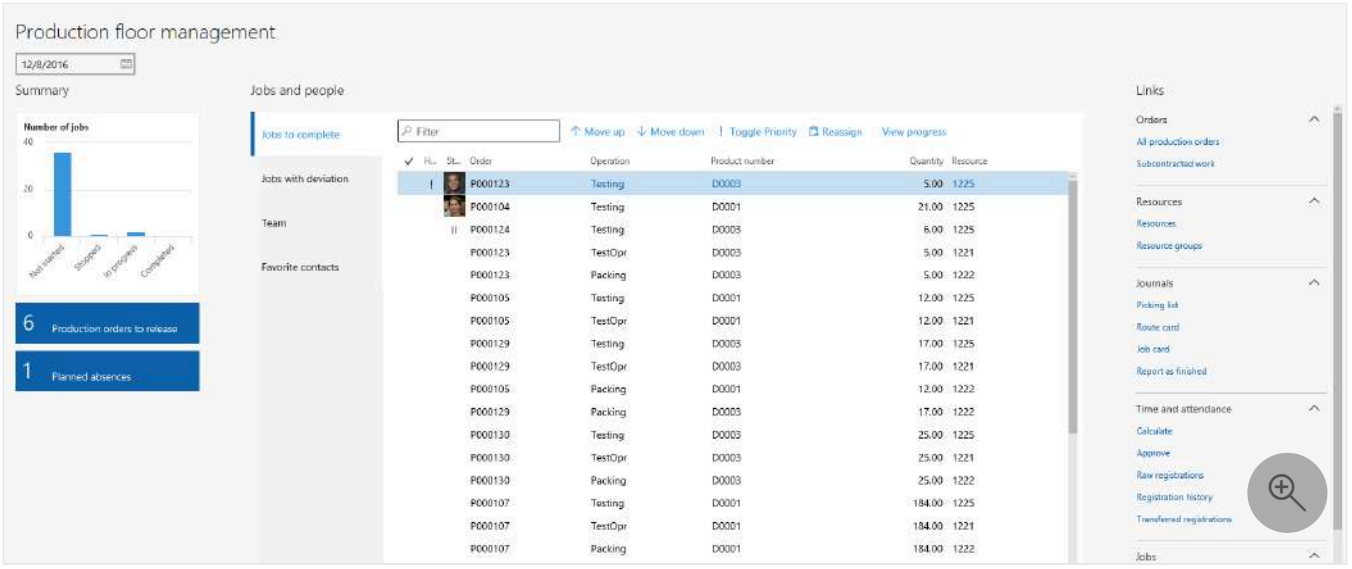
VIEWER	Operational views	Business views	Analytical tools & visualizations	Electronic reporting	Financial reporting
Target	Organization	Back Office	Power user	Power user	Finance officers
Data accuracy	Near real-time	Real-time	Near real-time	Real-time	Cached views
Personalization	Medium-Modeled	None	High - Free form designer	Low - Expressions	Medium - Modeled
Sharing	None	PDF export 0365 export Email	Dashboards Reports Tiles	Excel export	Excel export
Printing	Screen captures	Local printer Network devices	Screen captures	Excel	Local printer
Automation	Autorefresh	Batch integration	Scheduled refresh	Batch jobs	None
Scenarios	Monitoring	Transactions	Exploratory	Transactions	Accounting

> **Note**

"Near real-time" denotes processed data that's slightly slower than real-time.

Operational views

Operational views play an essential role in the daily work of most employees. As important as a brush is to a painter, operational views empower people to be productive. These views contain logical presentations of data to help users discover patterns, highlight anomalies, and act on the most important tasks. Targeted experiences satisfy the unique information access requirements for a given persona. These views provide actionable controls that help maximize efficiency for common user actions. For more information about constructing custom operational workspaces, see Build operational workspaces. Example applications of operational views include controller operations, production floor management, and customer collections monitoring.



What are the characteristics and capabilities?

- A fully integrated experience with responsive visualizations that understand user context and selections.
- Views that users can personalize to meet their unique and changing needs.
- Actionable controls that let users efficiently transact and monitor activities.
- A combination of analytical data that answers general questions and transactional views that provide access to record details.

What distinguishes "operational views" from other types of visualizations?

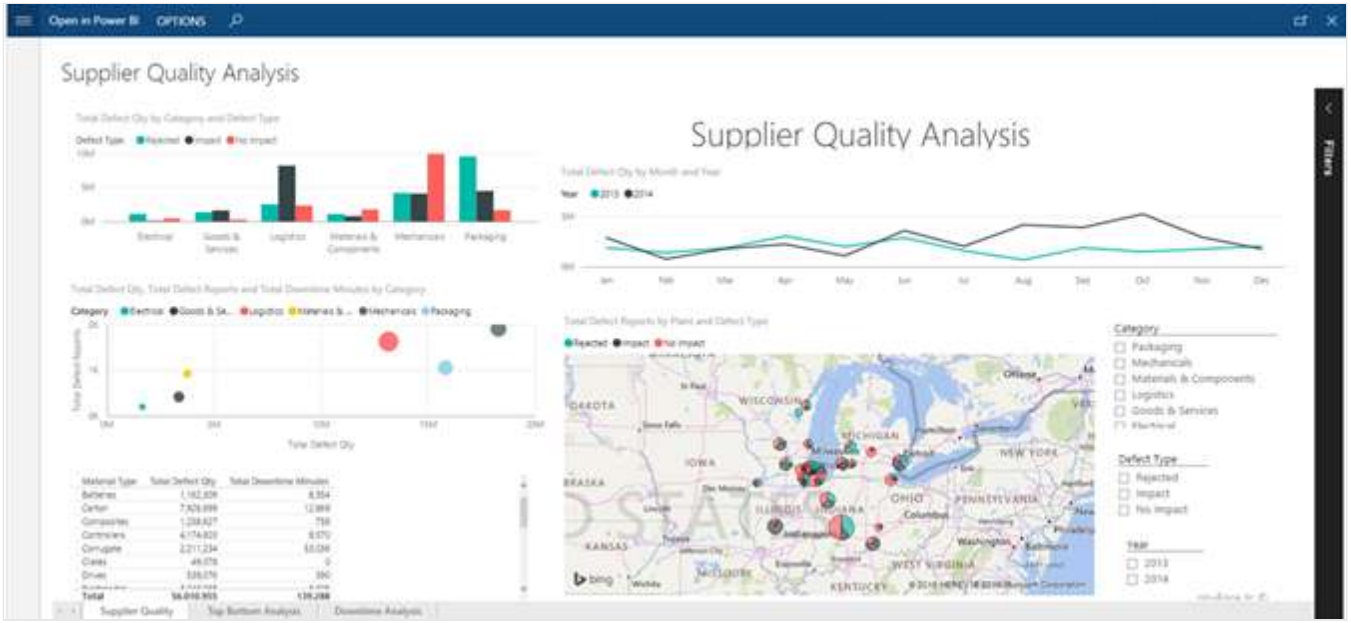
- General purpose tools designed for use at all levels of the organization.
- Predefined views based on common information access requirements associated with specific roles within the organization.
- Highly responsive to user interactions and changes made to the transactional database.

What should you consider when selecting this tool?

- The platform allows users to embed Power BI tiles and links to reports directly in workspaces.
- Users can introduce personalized workspaces to create their own custom operational views.
- Form data sources now support aggregate queries for analytical views using native controls.

Analytical tools and visualizations

Embedded visuals based on analytical data let users navigate between aggregate views and transactional details. Power BI service integration delivers world-class analytical tools with built-in support for accessing data. These tools empower "citizen developers" to author the reports they need and share the reports with others within the organization. Use the Power BI content packs available in Lifecycle Services to get started. For more information, see Features and services available through Power BI integration. Example applications of analytical tools and visualizations include customer sales per quarter, total revenue by region, and inventory turn-over.



What are the characteristics and capabilities?

- Near real-time results that provide macro level insights based on micro level activities.
- Common applications include charts, KPIs, and more complex visuals.
- Offer a deep exploratory experience with interactive controls that provide drill-through navigations.

What distinguishes "analytical tools and visualizations" from other types of reports?

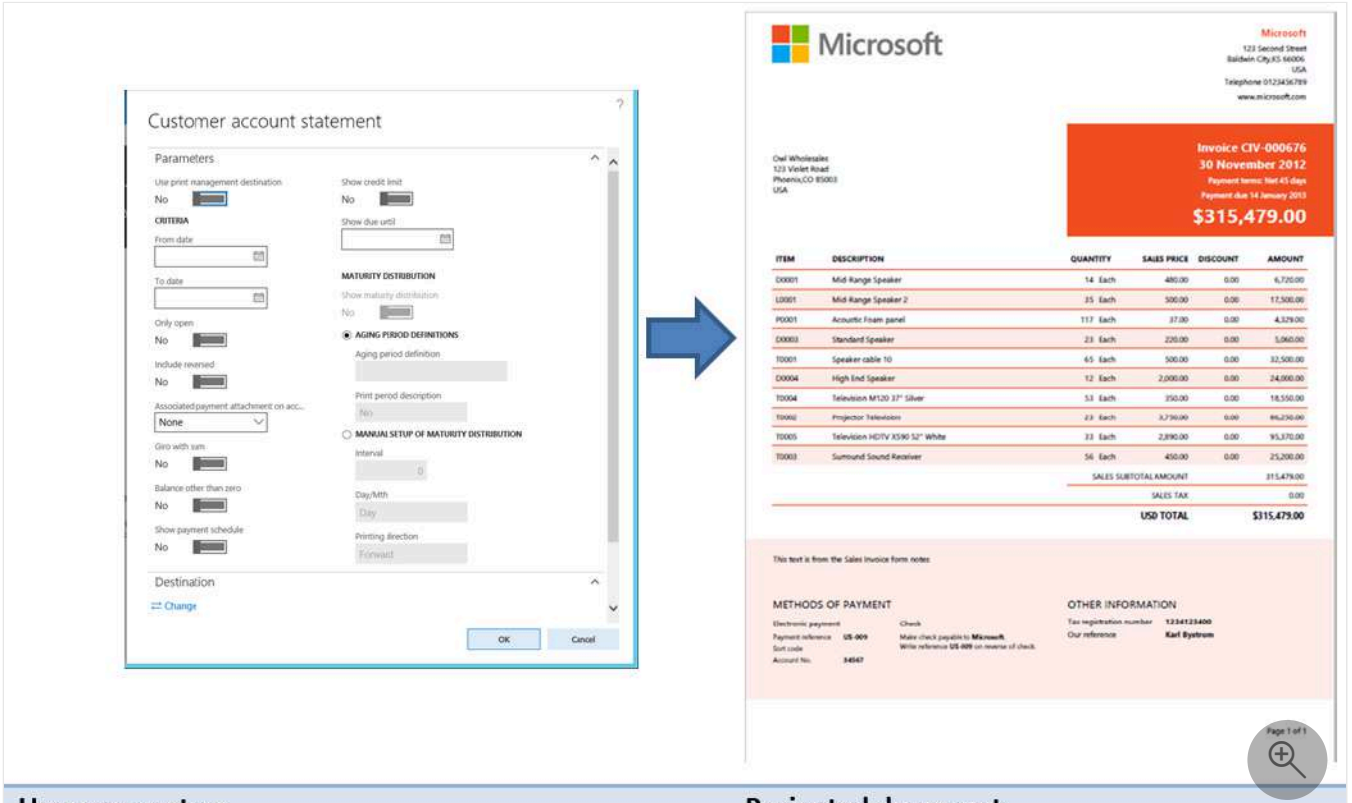
- Highly graphical nature, these presentations help you find the hidden meaning behind the data.
- Free form web designer that supports rich visualizations with built-in user interactions.
- Utilized by power users to explore data and gain insight through analysis.
- Personal nature by allowing the user to choose which information to include.
- Built-in sharing capabilities and user controlled access management.

What should you consider when selecting this tool?

- Developers are responsible for publishing data entities that Power BI can consume.
- Power users can produce mash-up views based on application data combined with external data sources.
- Visuals are highly responsive to user interactions and provide near real-time results when using Direct Query access to the data source.

Business documents

Use these reporting solutions to capture and communicate the details of business transactions. As such, you need a reporting solution that can produce physical manifestations of business data by using existing devices like network printers. For more information about the enhancements to the Document reporting service, see Document Reporting Services. Example applications of business documents include sales invoices, customer statements, and checks.



User parameters **Paginated document**

What are the characteristics and capabilities?

- Paginated documents that you print on paper or distribute through email.
- Heavy dependence on parameters to filter and produce the desired result set.
- Business documents capture a snapshot of customer and vendor activity that you can archive for future reference.
- Complex solutions that you develop in Visual Studio and deploy as part of the application.

What distinguishes "business documents" from other types of visualizations?

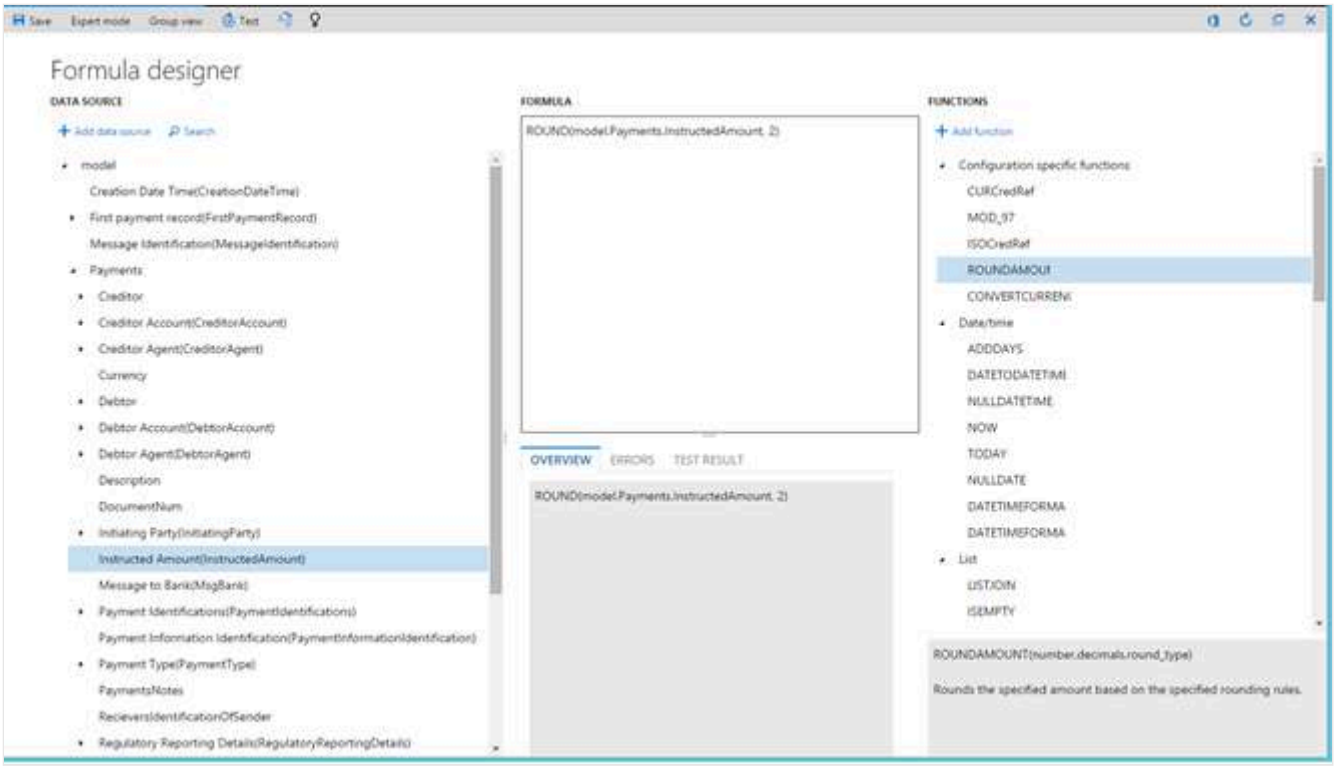
- Asynchronous data access and rendering solution designed to handle relatively large data sets.
- Dedicated reporting services that offer distributed resource utilization.
- Ideal solution for automated processes that involve bulk generation of business documents.
- Built-in support for document archive and data extraction via file export to PDF in addition to Word, Excel, and CSV.

What should you consider when selecting this tool?

- Use application suite reports as a starting point for custom solutions.
- Solutions that heavily depend on metadata changes and don't offer personalization.
- You must manage modifications to out-of-box solutions as a metadata change.

Electronic reporting

Electronic reporting (ER) is the tool to use to configure electronic document formats in accordance with the legal requirements of various countries and regions. For more information about the Electronic reporting tool, see Electronic reporting (ER) overview. Example applications of electronic reporting include financial auditing, tax reporting, and electronic invoicing.



What are the characteristics and capabilities?

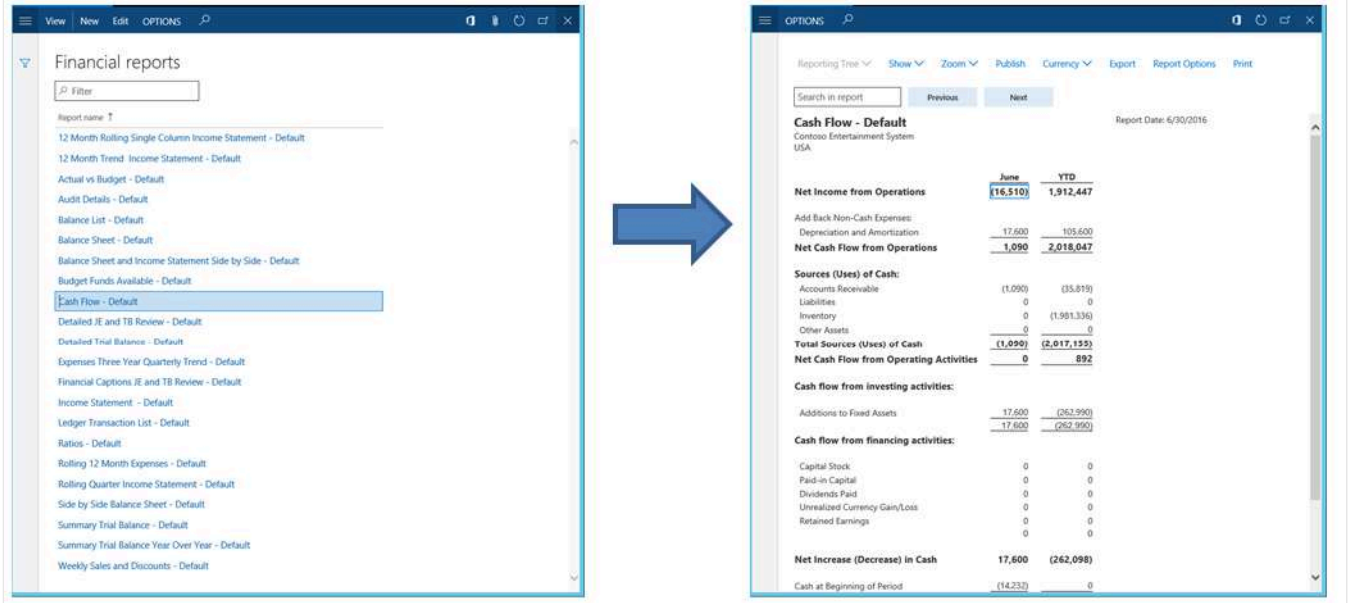
- Perfect tool for producing TEXT, XML, and OPENXML worksheet formats.
- Tooling is designed for business users familiar with Excel-based formulas.
- Highly adaptable to adhere to changes in regulatory requirements.
- Component versioning is available to manage draft definitions.

What distinguishes "electronic reports" from other types of visualizations?

- Designed for electronic submission to banks, governments, and other external entities.
- Use formulas to define data transformations into groups containing summary data and logical calculations.

Financial reporting

Standard financial reports use the default main account categories. Use the report designer to create or modify traditional financial statements, such as Income statement and Balance sheet, and share the results with other members of your organization. For detailed information about the Financial reporting tooling, see General ledger and Financial reporting overview. Example applications of financial reporting include balance sheets, cash flow, and summary trial balance year over year.



Standard templates (GL > Inquiries > Fiancial reports)	Interactive financial reports

What are the characteristics and capabilities?

- Built-in flexible financial reporting solution designed to handle complex organizational structures.
- Fully integrated with General ledger.
- Create custom financial reports using the default solutions as a starting point.
- Interactive reports with drill-down capabilities to navigate down to transaction details.

What distinguishes "financial reports" from other types of visualizations?

- User controls tailored for the specialized needs of financial reporting.
- Create roll-up reports containing data across companies or business units.
- Utilizes a financial data mart for optimized performance.

Last updated on 11/26/2025

Features available through the Power BI integration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft Power BI is a suite of business analytics tools that you can use to analyze data and share insights. By using Power BI tools, you can explore data and quickly create rich reports and dashboards. You and your colleagues can then use the reports interactively on many devices. The application uses Power BI for data exploration.

Data exploration through Power BI

There are various types of reporting.

Use chart controls to build embedded experiences that require visuals.

Microsoft SQL Server Reporting Services (SSRS) is an engine that's designed for pixel-perfect, formatted reports that often require printing. Use SSRS for document-style reports, such as invoices and purchase orders. Investments in SSRS integration focus on document generation and printing scenarios.

For all non-document reports or reports that don't require printing, use Power BI.

In the past, the terms *self-service reports* and *ad-hoc reports* referred to Power BI. Now, use the term *data exploration*. This change in terminology reflects a subtle shift in paradigms. Self-service reports were reports that users created. (Alternatively, a power user created the reports and shared them with other people. These people then continued to adjust the reports according to their requirements.) Often, users had to change the shape of a chart, add a new column, change the grouping, or just create a new view of data. Although you might think of the results as reports, users are just trying to understand the data by exploring it, pivoting it around columns, and changing the shape of charts. Technologies of the past didn't let users interactively explore large volumes of data. Therefore, users had to create "reports," or many views of the same data.

In Power BI, thanks to the in-memory database technologies in Microsoft SQL Server, a report is just the starting point for interactive data exploration. Charts in a Power BI report invite users to click them, visuals change shape interactively, and data can easily be filtered. Users can easily adjust existing reports and create their own views of the data. The reports can be shared, and teams can collaborate over data.

Therefore, although you might use the term *report* to refer to Power BI artifacts, think about the larger scenario that is involved. Your users are exploring data! Power BI is the tool of choice when you must address data exploration and visualization requirements.

For a detailed discussion of reporting concepts, see Information access and reporting.

Ready-made Power BI content

You can use ready-made Power BI reports right away. Two types of Power BI content are available:

- Power BI content that's available in Microsoft Dynamics Lifecycle Service
- Power BI content packs that are distributed in the PowerBI.com marketplace

Depending on your version, you can use one or both types of content.

Power BI content that's available in Lifecycle Services

Lifecycle Services is a service that can manage your environments. Microsoft operates Lifecycle Services. Developers create Power BI reports by using Entity store, and then distribute them in Lifecycle Services as implementation assets. In Lifecycle Services, you find not only content that Microsoft develops, but also content that independent software vendors (ISVs) and partners develop.

Microsoft continues to release Power BI content that's based on Entity store. For information, see the Roadmap.

Power BI content packs that are distributed in the PowerBI.com marketplace

Several Power BI content packs are available in the PowerBI.com marketplace. Although Microsoft continues to support these content packs until further notice, future investments in content packs are based on Entity store, and content is released via Lifecycle Services.

For more information about the content, see Power BI content.

Extending, creating, and distributing Power BI reports

Use the ready-made Power BI content as a first step. You can modify ready-made reports or extend them by using capabilities that are built into PowerBI.com. In addition to modifying ready-made reports, you can extend them by using Power BI authoring tools such as Power BI Desktop. You can also create new reports. You can use several approaches to create new Power BI reports.

Create high-volume, near-real-time "operational Power BI reports" by using Entity store

Entity store is an operational data store that's built specifically for Power BI integration. To create high-volume, near-real-time Power BI reports that use Entity store, a business analyst or a developer can use Power BI Desktop, which is the authoring tool for Power BI reports. Like other artifacts that developers create, you must distribute these reports to your users via Lifecycle Services.

Reports that you create by using Entity store take advantage of DirectQuery technology. This technology enables reports to be created over large volumes of data. Reports that are created by using DirectQuery technology don't cache data in the PowerBI.com service. Instead, data is always stored in application.

For an overview of Power BI integration with Entity store, see Power BI integration with Entity store.

If you're upgrading from Microsoft Dynamics AX 2012, you can upgrade cubes to aggregate measurements that use Entity store. You can then create Power BI reports by using Entity store. For more information, see Migrate upgraded AX 2012 R3 sales cubes to the entity store.

Create Power BI reports by using Excel

In addition to using the Power BI desktop authoring tool, you can use "Power tools" that are incorporated into Excel to create visualizations. Your organization might have many users who already use Excel every day. For a quick "one-off" report, Excel might be the best option for these users.

There are several scenarios where you can use Excel:

- Export data from a page in Dynamics 365 into Excel. You can then use the Power View add-in that is built into Excel to visualize the data. The Excel workbook can be used as a stand-alone visualization. In addition, you can import the report into the PowerBI.com service.
- Use the Power Query extension in Excel to combine the data in another worksheet (or data that is imported from OData endpoints) with external data. You can visualize the resulting data by using Power View.
- Use the PowerPivot extension in Excel to ingest a larger amount of data into Excel.

Consider using Export to Excel functionality for ad-hoc "one-off" reports. If you want to share the reports with a group of users, consider using Entity store to create them.

Sharing and using reports in PowerBI.com

PowerBI.com is a service that's offered by Microsoft. It lets you create dashboards and reports, and also enables collaboration with a group of users. Regardless of how you create your reports, you can share reports with users by uploading them into the PowerBI.com service. This process is also known as *publishing*.

After you upload your reports, your users can view, adjust, and explore them either on the web (when they're connected to the Internet at home or in the office) or by using apps on a device.

For more information about Power BI concepts, see the Power BI documentation.

Pin Power BI content to the client

You can use PowerBI.com on its own as a reporting and dashboard solution for your organization or business unit. However, users can also pin tiles and reports from their own PowerBI.com accounts to workspaces. Power BI content in Dynamics 365 provides contextual insights that are related to business operations.

You can pin two types of objects from PowerBI.com: tiles in PowerBI.com dashboards, and reports.

Before you can pin Power BI content, you must configure Power BI in your Dynamics 365 environment.

One-time configuration of PowerBI.com integration

Before you can pin tiles or reports, an administrator must configure integration with PowerBI.com in your environment. This configuration must be done only one time per environment. For instructions, see Configure Power BI integration for workspaces.

Pin PowerBI.com tiles

Power BI tiles that you pin to the Dynamics 365 client provide insightful visuals at a glance. They also let users open PowerBI.com for interactive analysis. For more information, see Configure Power BI integration for workspaces.

Pin PowerBI.com reports to workspaces

As a power user, a business analyst, or a developer, you can use Power BI desktop to create reports that use Entity store. These reports are rich and interactive, and your users can make changes without having to rely on another person.

Although the reports in PowerBI.com are powerful and interactive on their own, you can also pin them into workspaces. Your users can pin reports to workspaces themselves. For more information about how to pin reports to workspaces, see Pin Power BI reports to workspaces.

Last updated on 04/13/2026

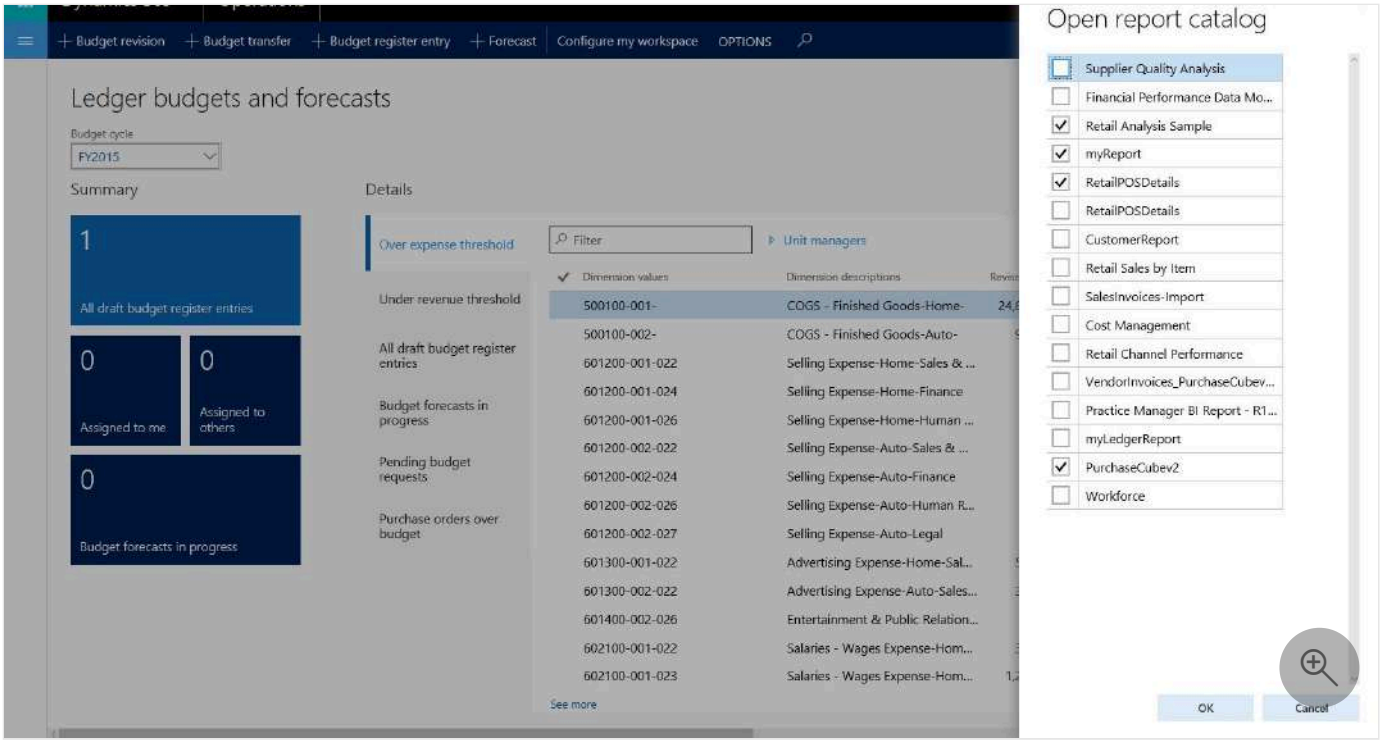
Pin Power BI content

Microsoft Dynamics 365 Finance uses Power BI for data exploration. This article explains how you can pin full-page Power BI reports to workspaces to give your users an interactive data exploration experience.

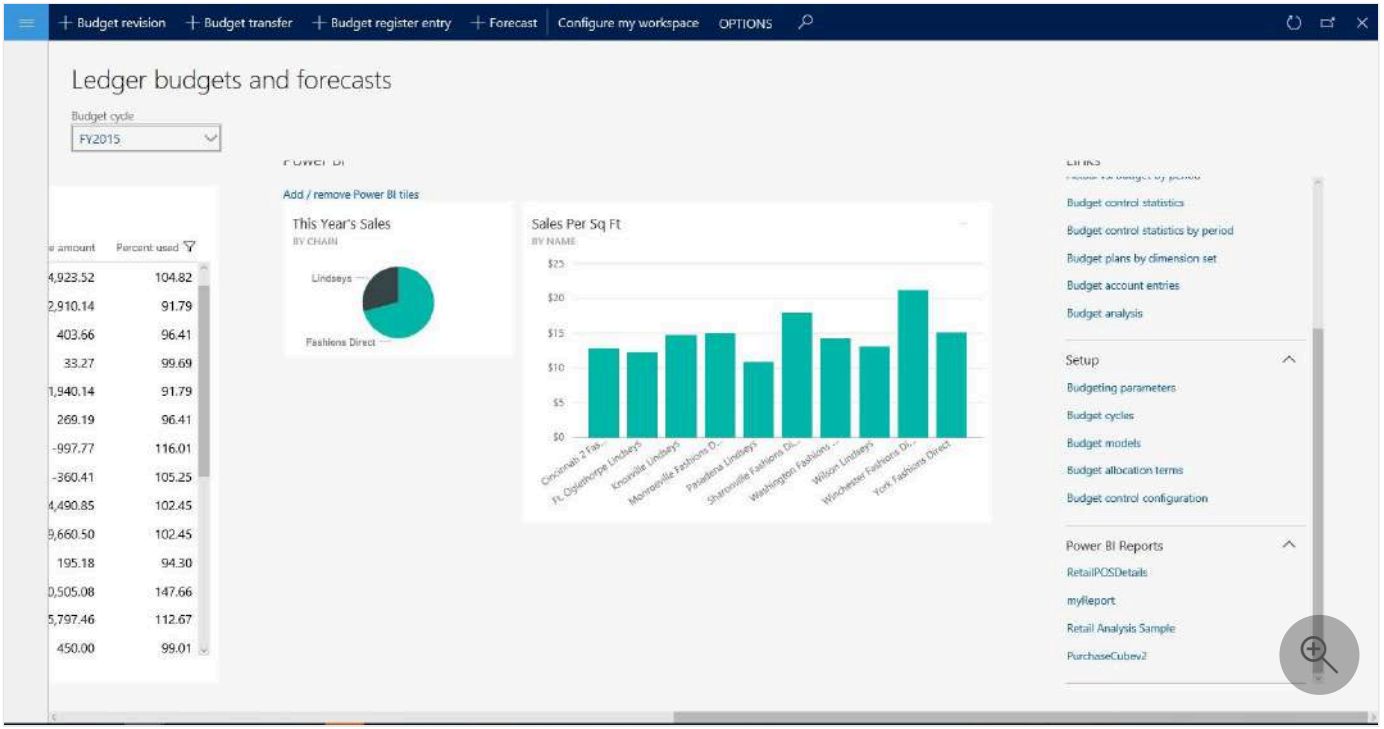
This article assumes that you're familiar with the feature that lets you pin Microsoft Power BI tiles to a workspace. For more information, see Features and services available through Power BI integration. If you're a developer who is creating a workspace, to let users pin Power BI tiles to the workspace, embed the Power BI tile control.

Pin Power BI reports to workspaces

Microsoft Dynamics AX platform update 1 (May 2016) introduced the capability to pin Power BI reports to workspaces. Power BI reports can be added to any workspace that contains a **Links** section. In other words, the reports can be added to most of the out-of-box workspaces that are included in the product. To enable Power BI reports and tiles, you must configure Power BI to work with the application. This one-time operation must be completed by an administrator of the environment. For instructions, see Configure Power BI integration for workspaces. After you've configured Power BI to work with the application, open the **Ledger budgets and forecasts** workspace in the client. In the workspace, select the **Options** tab. Notice that this tab contains buttons to open the (Power BI) tile catalog and the (Power BI) report catalog. Select **Open report catalog**. A dialog box that contains a list of reports appears. The list of reports comes from the reports that you have in your Power BI account. If you open PowerBI.com in a browser, you'll see that the same list of reports is used across your Power BI dashboards. Select some reports, as shown in the following illustration, and then select **OK** to continue.



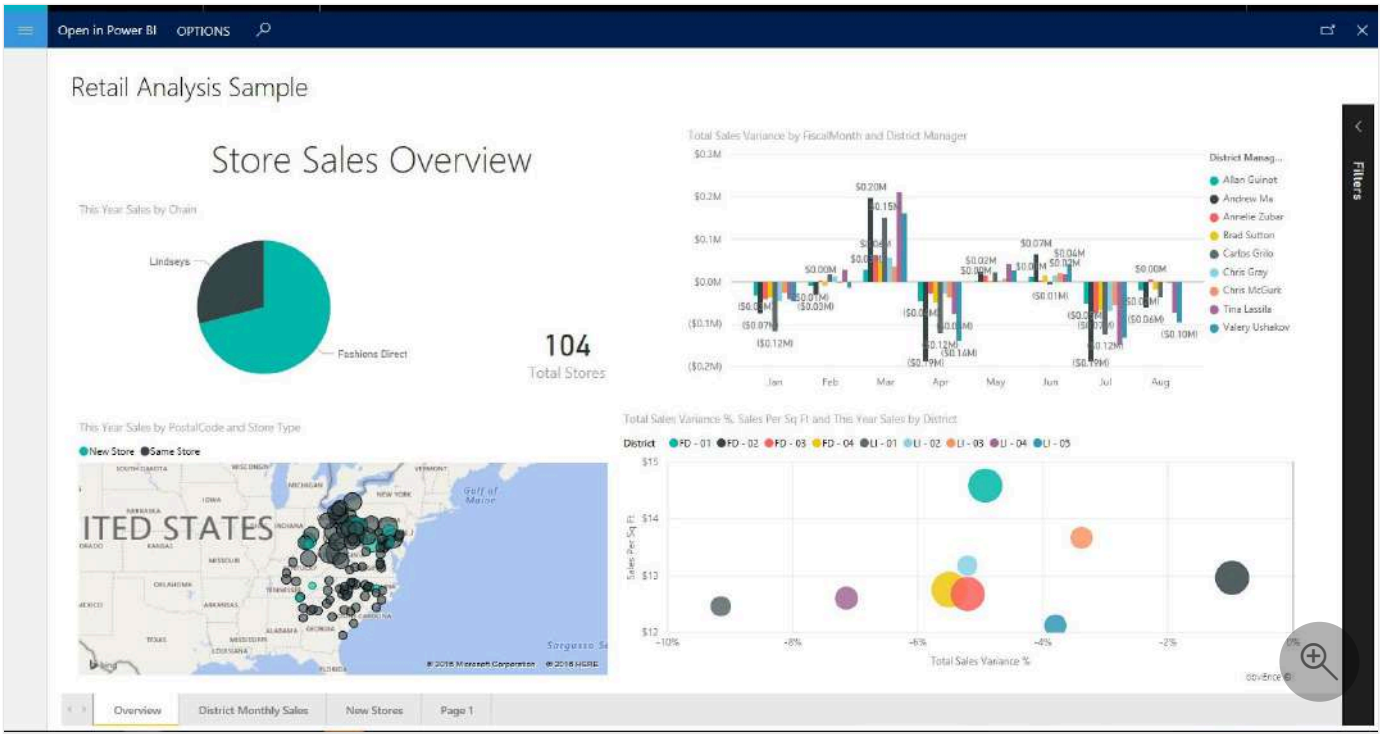
Next, scroll to the bottom of the **Links** section in the workspace. A new section for Power BI reports is added to your links.



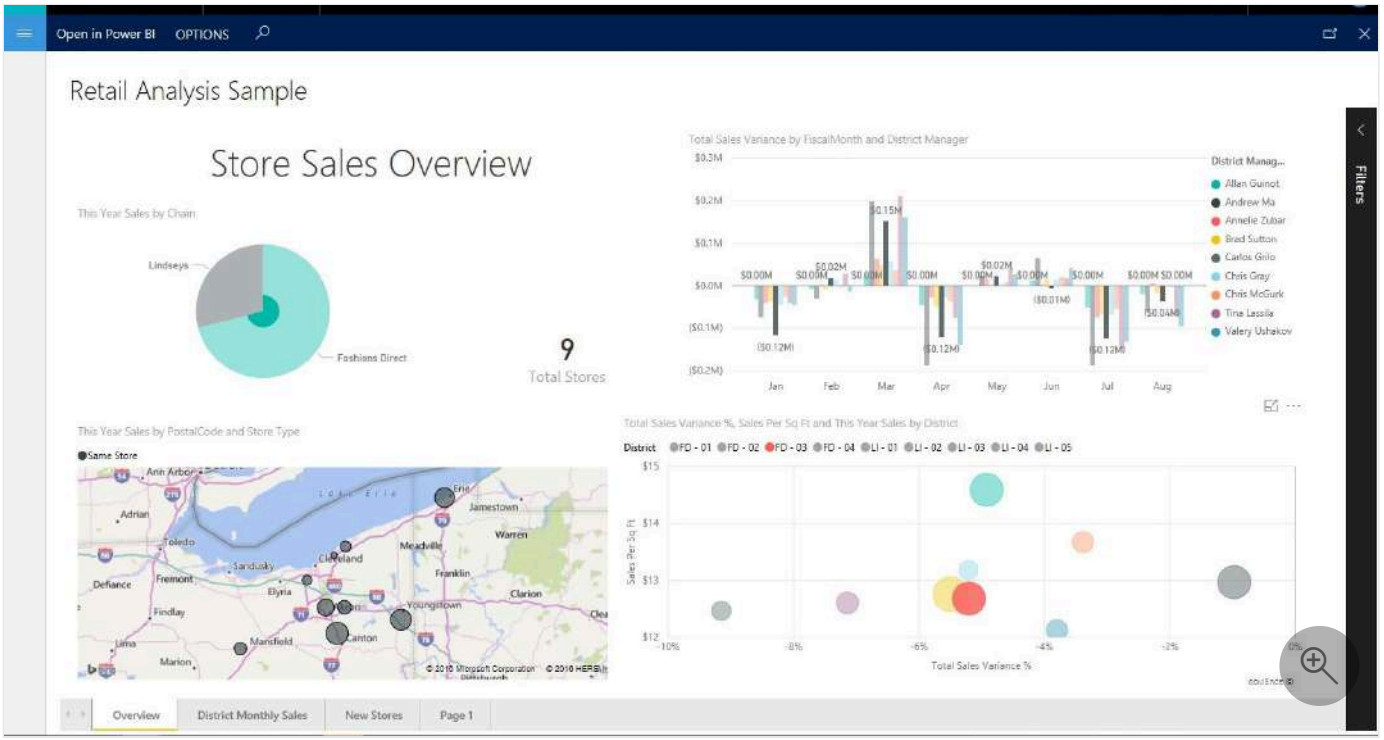
Full-page Power BI reports in the client

You can open and run Power BI reports in the client. The functionality resembles the functionality for running Microsoft SQL Server Reporting Services (SSRS) reports. To run a Power BI report, in the **Links** section, select the link for one of the Power BI reports. For this example, select the **Retail Analysis Sample** link. The Power BI report opens in the client in a full-page view, as shown

in the following illustration. This report is interactive. As you select regions of the report, the remaining visuals react to your selection.



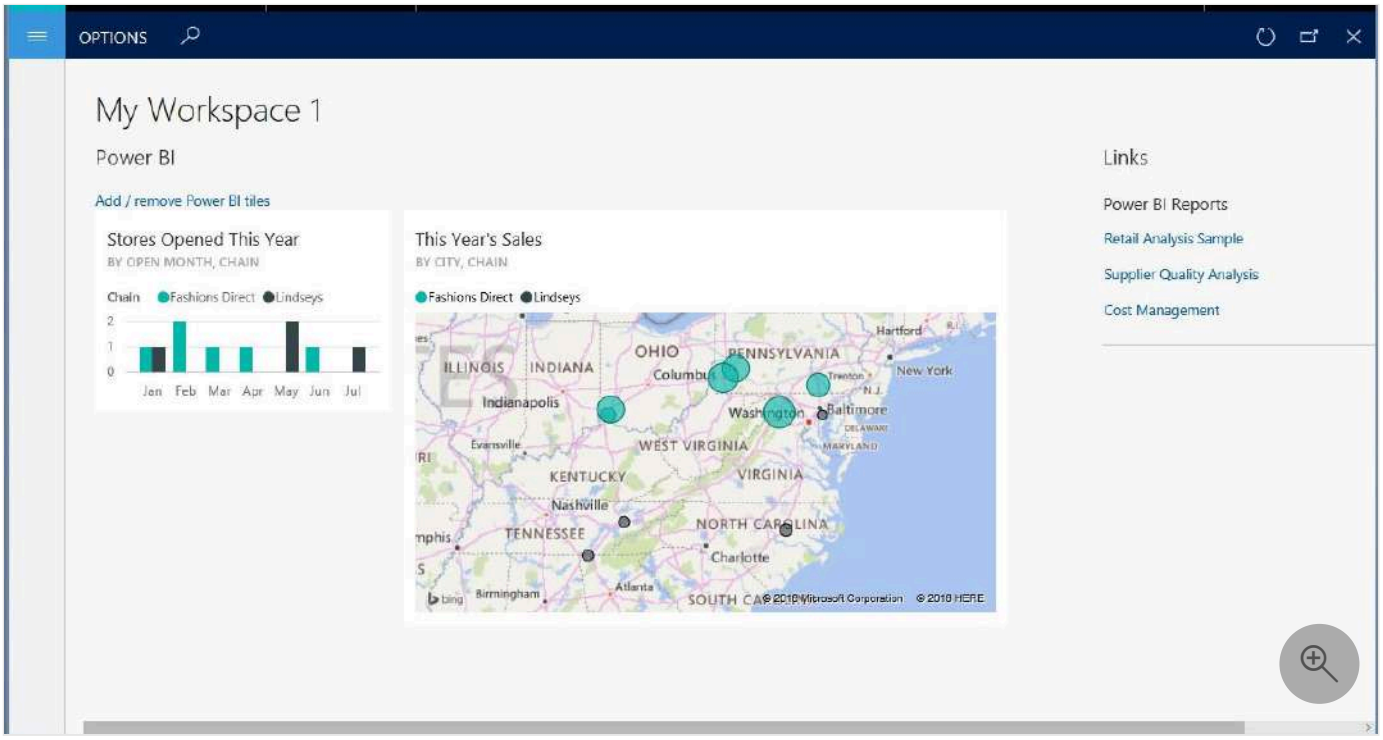
You can filter the data on the report by using the filter pane. The following illustration shows the report after filters have been applied.



You can also open this report on PowerBI.com and make changes. You can then save the modified report as another copy that has a different name, and even pin the new report to the workspace.

Power BI in user-created workspaces

So far, you learned how to add Power BI tiles and reports to **developer-created** workspaces. Developer-created workspaces are workspaces that Microsoft (that is, they're built into the product), your independent software vendor (ISV) or partner, or in-house developers create. However, in Microsoft Dynamics AX platform update 1 (May 2016), users can create new workspaces by using the personalization capabilities of the client. To create a new workspace, on the home page (or the dashboard), right-click the tile for a workspace, and then select **Add a workspace**. A new workspace is created. New workspaces are named **My Workspace 1**, **My Workspace 2**, and so on. You can change the name later. Select the workspace that you created. You can now add Power BI tiles and reports by using the same options that we discussed earlier. The following illustration shows an example.



Last updated on 06/26/2026

Select analytical workspaces from Power BI

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Analytical workspaces

The analytical workspaces that are bundled with the application suite offer users relevant insights into their business data. However, in some cases, it makes sense to replace standard reports with custom reports that are designed specifically for the users in your organization.

The tooling that PowerBI.com provides lets you produce analytical reports that contain mashup views that use data from external sources. In Platform update 26 for finance and operations, power users can replace the standard embedded reports with those that are hosted on PowerBI.com.

> **Important**
>
> The functionality that this article describes isn't a personalization. The customization of analytical workspaces applies to all users in the active legal entity.

Motivations for embedding PowerBI.com reports

Although standard reports deliver insights that are tailored for a given business persona, an organization can prefer a custom report in some cases. The application lets power users promote custom reports that are hosted on PowerBI.com and shared with members of the organization.

Here are some of the top motivations for selecting reports that are hosted on PowerBI.com:

- PowerBI.com reports support data mashups that use external data sources and can be accessed outside the application.
- The reports are appropriate for demonstrating custom solutions that are hosted on PowerBI.com and embedded in the application in one box deployments.
- Organizations that have Microsoft Power BI Premium services want to augment the standard reports.

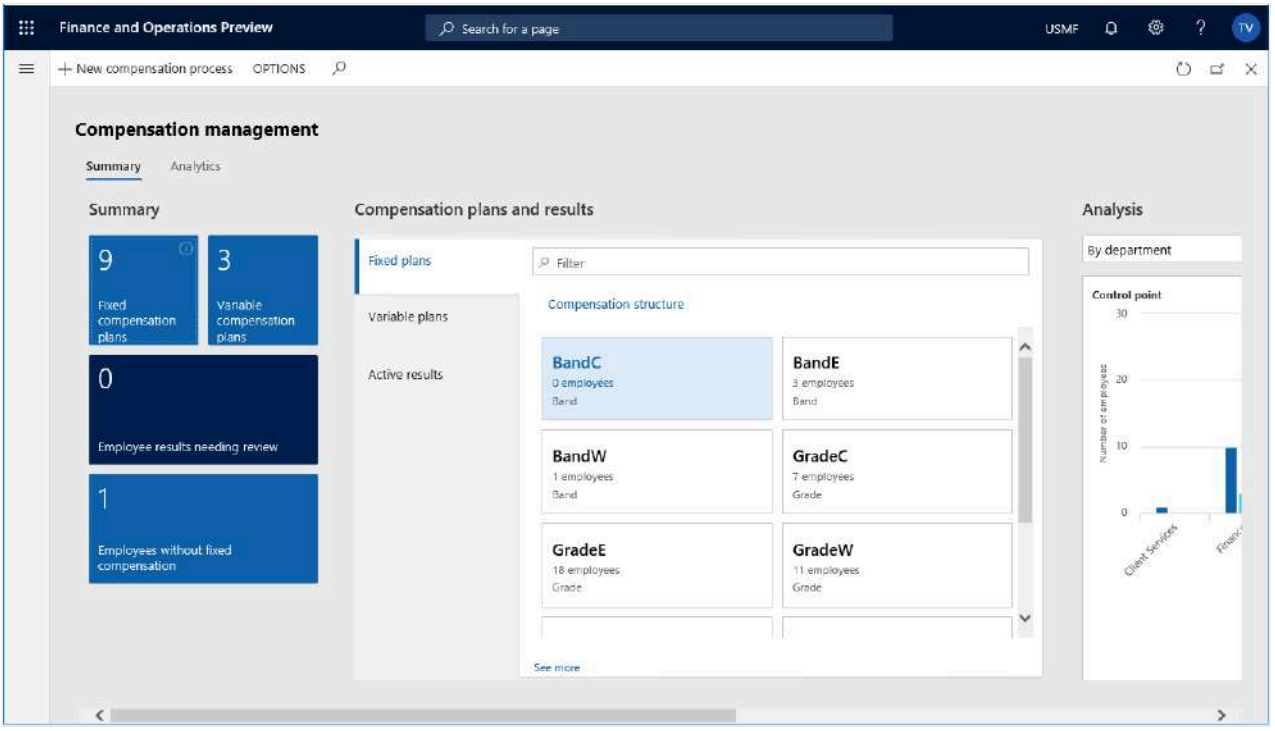
Embed a PowerBI.com report in an analytical workspace

https://learn-video.azurefd.net/vod/player?id=ecad8101-d1a3-4045-bf49-8703338f355a&locale=en-us&embedUrl=%2Fdynamics365%2Ffin-ops-core%2Fdev-itpro%2Fanalytics%2Fselect-analytical-workspace

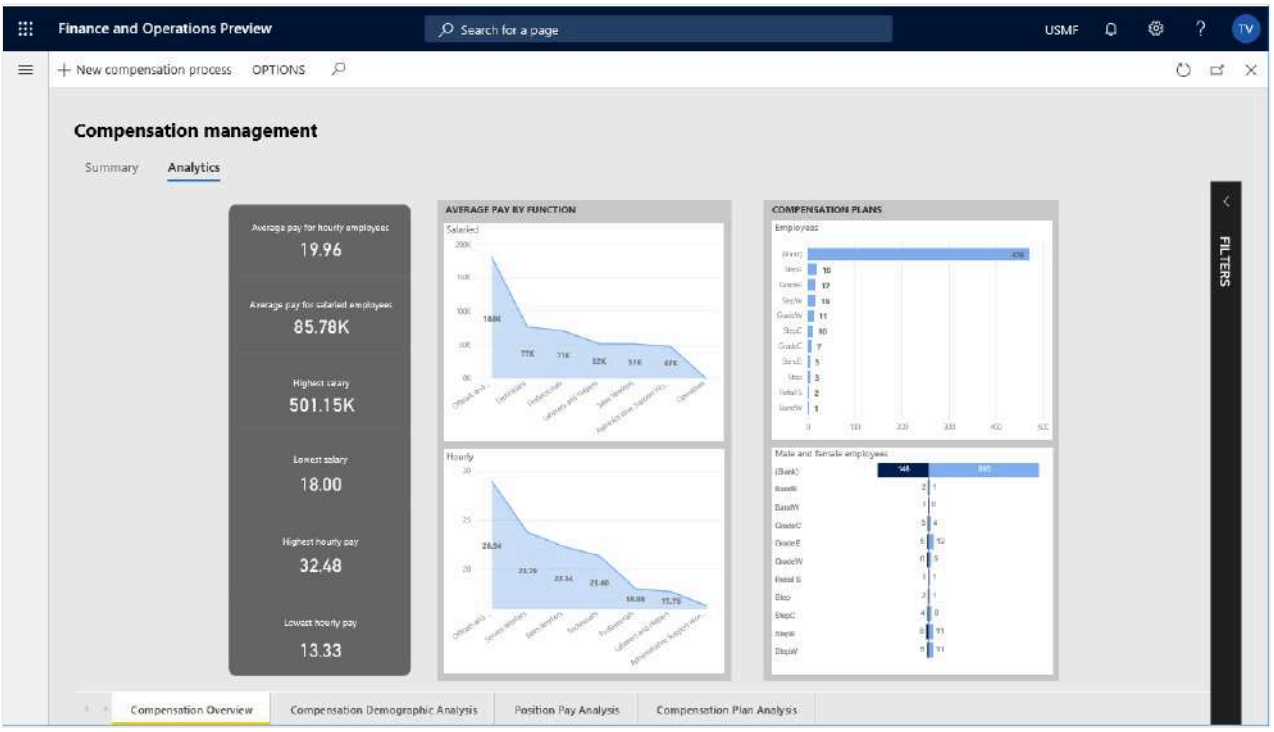
The How to embed PowerBI.com reports video (shown above) is included in the finance and operations playlist available on YouTube.

To replace the standard reports, you need to be a member of the System Report Editors security group. Members of this security group can access the options in application workspaces that let them customize the standard reports. This example shows how to replace the standard analytical report with a customized report that is hosted on PowerBI.com.

1. Sign in and open the application report that you want to customize. In this example, you'll replace the standard analytical report that is embedded in the **Compensation management** workspace.



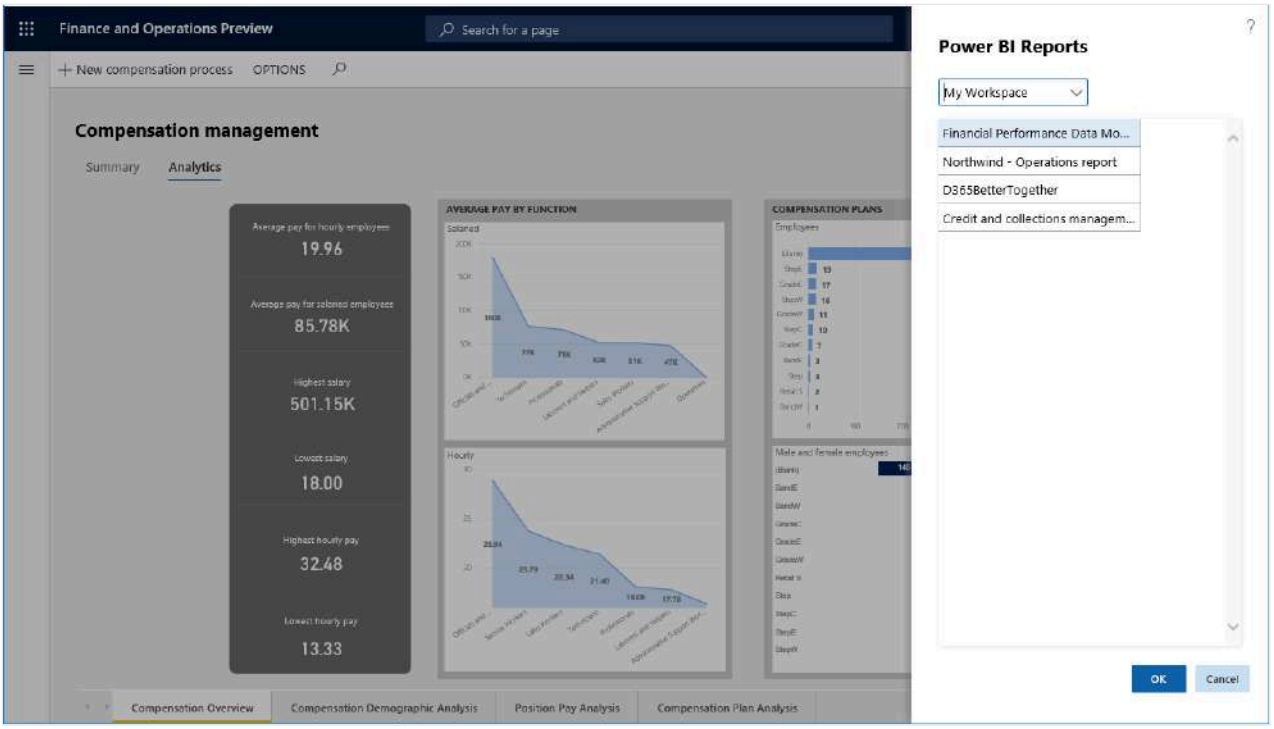
2. Select the **Analytics** tab to access the workspace's embedded analytical report.



By default, you see the standard analytical workspace solution that is included with your application. The reports in this solution are automatically deployed and configured for your environment during the provisioning process.

> **Note**
>
> The analytical workspaces require a hosted Power BI service that's available only for dedicated environments. For more information, see the blog post, Accessing Analytical Workspaces and Reports on 1-Box Environments.

3. On the Action Pane, on the **Options** tab, in the **Power BI** group, select **Select Analytics** to open the **Power BI Reports** dialog box.



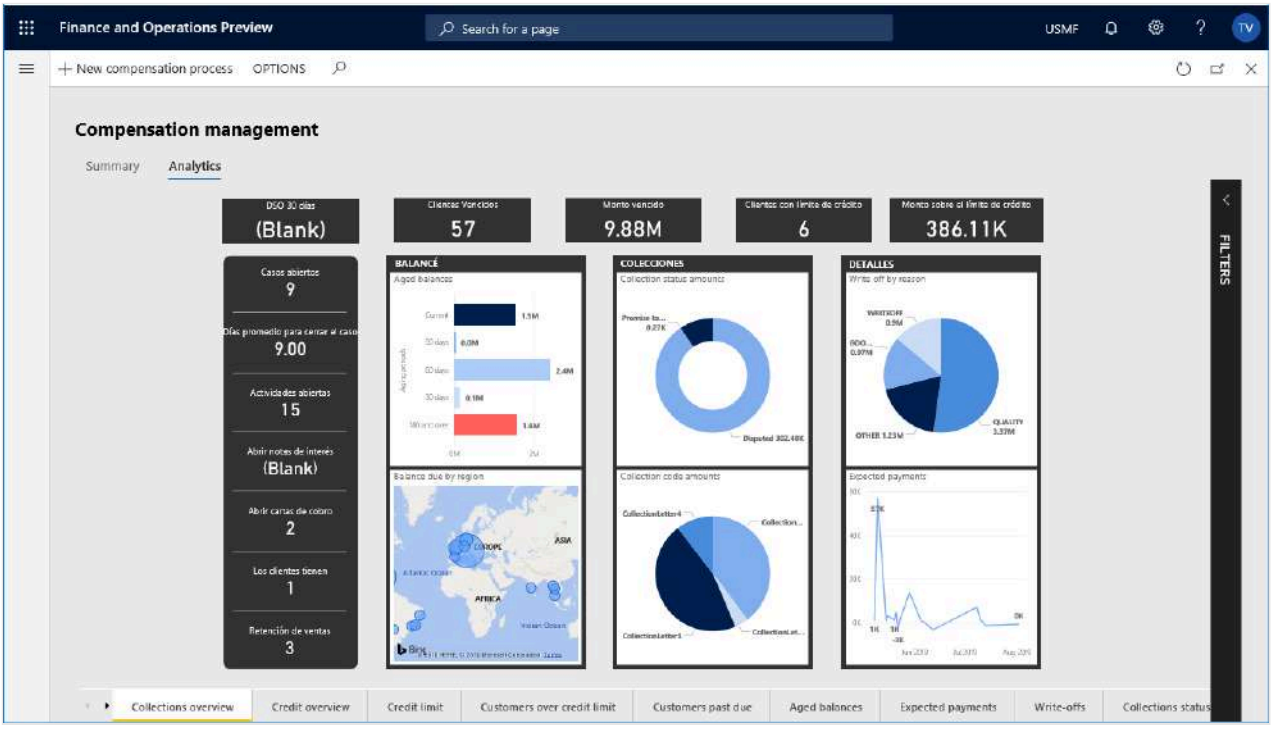
This dialog box lets you select among the reports that are shared on the PowerBI.com service. The reports are organized by workspace.

4. In the drop-down list, select the workspace that contains the report.

5. Select the report to embed in the application workspace, and then select **OK**.

6. To view the updates to the workspace, you need to reload the page. Either move away from the workspace and then return, or refresh your browser.

7. In the **Compensation management** workspace, select the **Analytics** tab to access the PowerBI.com report that is now embedded in the analytical workspace.



Revert to the standard solution

After a PowerBI.com report has been embedded in an application workspace, updates to the report are reflected immediately for users. However, to replace the report with another PowerBI.com solution, a power user needs to first revert to the standard application solution. Follow these steps to revert to the standard application solution.

1. On the Action Pane, on the **Options** tab, in the **Power BI** group, select **Restore Analytics**.

2. To view the updates to the workspace, you must reload the page. Either move away from the workspace and then return, or refresh your browser.

3. In the **Compensation management** workspace, select the **Analytics** tab to access the standard solution that is now embedded in the analytical workspace.

Last updated on 11/26/2025

Power BI content home page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following PowerBI.com solutions are available for application environments.

PowerBI.com solutions available from Lifecycle Services

The Shared assets library of LCS provides the following PowerBI.com solutions. Check the version information in the Shared assets library to download the correct version for your environment.

> **Note**
>
> Most of the PowerBI.com solutions in the following list are embedded in analytical workspaces in recent versions. By using analytical workspaces, you don't need to download these solutions from the Shared assets library in LCS. The topics in this list indicate where these PowerBI.com solutions are embedded where applicable.

- Actual vs budget
- Cash overview
- Compensation and benefits
- Cost accounting analysis
- Credit and collections management
- Employee competencies and development
- Financial performance
- Fixed asset management
- Organizational training
- Practice manager
- Production performance
- Purchase spend analysis
- Recruiting
- Sales and profitability performance

- Vendor payments
- Warehouse performance
- Workforce metrics

PowerBI.com solutions available from Marketplace

The following PowerBI.com solutions are available from Microsoft Marketplace.

> **Note**
>
> These solutions are deprecated as documented in Power BI content packs available on Marketplace.

- Cost management
- Financial performance
- Retail channel performance

Last updated on 01/14/2026

Cost management Power BI content

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Overview

The **Cost management** Microsoft Power BI content is intended for inventory accountants or individuals in the organization who are responsible for or interested in the status of inventory or work in progress (WIP), or who are responsible for or interested in analyzing standard cost variances.

This Power BI content provides a categorized format that helps you monitor the performance of inventories and visualize how cost flows through them. You can gain managerial insights such as the turnover ratio, number of days that inventory is on hand, accuracy, and "ABC classification" at your preferred aggregated level (company, item, item group, or site). The information that is made available can also be used as a detailed supplement to the financial statement.

The Power BI content is built on the **CostObjectStatementCacheMonthly** aggregated measurement, which has the **CostObjectStatementCache** table as its primary data source. This table is managed by the Data set cache framework. By default, the table is updated every 24 hours, but you can change the update frequency or enable manual updates in the configuration of the data set cache. Manual updates can be run in either the **Cost administration** workspace or the **Cost analysis** workspace.

After every update of the **CostObjectStatementCache** table, the **CostObjectStatementCacheMonthly** aggregated measurement must be updated before data in the Power BI visualizations is updated.

Accessing the Power BI content

The **Cost management** Power BI content is shown in the **Cost administration** and **Cost analysis** workspaces.

The **Cost administration** workspace contains the following tabs:

- **Overview** – This tab shows application data.
- **Inventory accounting status** – This tab shows Power BI content.
- **Manufacturing accounting status** – This tab shows Power BI content.

The **Cost analysis** workspace contains the following tabs:

- **Overview** – This tab shows application data.
- **Inventory accounting analysis** – This tab shows Power BI content.
- **Manufacturing accounting analysis** – This tab shows Power BI content.
- **Std. cost variance analysis** – This tab shows Power BI content.

Report pages that are included in the Power BI content

The **Cost management** Power BI content includes a set of report pages that consist of a set of metrics. These metrics are visualized as charts, tiles, and tables.

The following tables provide an overview of the visualizations in the **Cost management** Power BI content.

Inventory accounting status

Expand table

Report page	Visualization
Inventory overview	Beginning balance
	Net change
	Net change %
	Ending balance
	Inventory accuracy
	Inventory turnover ratio
	Days inventory on-hand
	Active product in period
	Active cost objects in period
	Balance by item group
	Balance by site
	Statement by category
	Net change by quarter
Inventory overview by site and item group	Inventory accuracy by site
	Inventory turnover ratio by site
	Inventory ending balance by site

Report page	Visualization
	Inventory accuracy by item group
	Inventory turnover ratio by item group
	Inventory ending balance by site and item group
Inventory statement	Inventory statement
Inventory statement by site	Inventory statement by site
Inventory statement by product hierarchy	Inventory statement
Inventory statement by product hierarchy	Inventory statement by site

Manufacturing accounting status

Expand table

Report page	Visualization
WIP overview YTD	Beginning balance
	Net change
	Net change %
	Ending balance
	WIP turnover ratio
	Days WIP on-hand
	Active cost object in period
	Net change by resource group
	Balance by site
	Statement by category
	Net change by quarter
WIP statement	Beginning balance
	Ending balance
	WIP statement by category
WIP statement by site	Beginning balance
	Ending balance
	WIP statement by category and site

Report page	Visualization
WIP statement by hierarchy	Beginning balance
	Ending balance
	WIP statement by category hierarchy

Inventory accounting analysis

Expand table

Report page	Visualization
Inventory details	Top 10 resources by ending balance
	Top 10 resources by net change increase
	Top 10 resources by net change decrease
	Top 10 resources by inventory turnover ratio
	Resources by low inventory turnover ratio and ending balance above threshold
	Top 10 resources by low accuracy
ABC classification	Inventory ending balance
	Consumed material
	Sold (COGS)
Inventory trends	Inventory ending balance
	Inventory net change
	Inventory turnover ratio
	Inventory accuracy

Manufacturing accounting analysis

Expand table

Report page	Visualization
WIP trends	WIP ending balance
	WIP net change
	WIP turnover ratio

Std. cost variance analysis

Expand table

Report page	Visualization
Purchase price variance (Std. cost) YTD	Procured balance
	Purchase price variance
	Purchase price variance ratio
	Variance by item group
	Variance by site
	Purchase price by quarter
	Purchase price by quarter and Item group
	Top 10 resources by unfavorable purchase price ratio
	Top 10 resources by favorable purchase price ratio
Production variance (Std. cost) YTD	Manufactured cost
	Production variance
	Production variance ratio
	Variance by item group
	Variance by site
	Production variance by quarter
	Production variance by quarter and variance type
	Top 10 resources by unfavorable production variance
	Top 10 resources by favorable production variance

Understanding the data model and entities

Data from the application is used to fill the report pages in the **Cost management** Power BI content. This data is represented as aggregate measurements that are staged in the entity store, which is a Microsoft SQL Server database that is optimized for analytics. For more information, see Power BI integration with Entity store.

The key aggregate measurements of the following objects are used as the basis of the Power BI content.

Expand table

Object	Key aggregate measurements	Data source for finance and operations	Field
CostObjectStatementCacheMonthly	Amount	CostObjectStatementCache	Amount
CostObjectStatementCacheMonthly	Quantity	CostObjectStatementCache	Qty
CostInventoryAccountingKPIGoal	AnnualInventoryTurn	CostInventoryAccountingKPIGoal	AnnualInventoryTurn
CostInventoryAccountingKPIGoal	InventoryAccuracy	CostInventoryAccountingKPIGoal	InventoryAccuracy

The following table shows the key calculated measurements in the Power BI content.

Expand table

Measure	Calculation
Beginning balance	Beginning balance = [Ending balance]-[Net change]
Beginning balance qty.	Beginning balance qty. = [Ending balance qty.]-[Net change qty.]
Ending balance	Ending balance = (CALCULATE(SUM([Amount]), FILTER(ALL(FiscalCalendar) ,FiscalCalendar[MONTHSTARTDATE] <= MAX(FiscalCalendar[MONTHSTARTDATE]))))
Ending balance qty.	Ending balance qty. = CALCULATE(SUM([QTY]), FILTER(ALL(FiscalCalendar),FiscalCalendar[MONTHSTARTDATE] <= MAX(FiscalCalendar[MONTHSTARTDATE])))
Net change	Net change = SUM([AMOUNT])
Net change qty.	Net change qty. = SUM([QTY])
Inventory turnover ratio by amount	Inventory turnover ratio by amount = if(OR([Inventory average balance] <= 0, [Inventory sold or consumed issues] >= 0), 0, ABS([Inventory sold or consumed issues])/[Inventory average balance])
Inventory average balance	Inventory average balance = (([Ending balance] + [Beginning balance]) / 2)
Days inventory on-hand	Days inventory onhand = 365 / CostObjectStatementEntries[Inventory turnover ratio by amount]
Inventory accuracy	Inventory accuracy by amount = IF([Ending balance] <= 0, IF(OR([Inventory counted amount] <> 0, [Ending balance] < 0), 0, 1), MAX(0, ([Ending balance] - ABS([Inventory counted amount]))/[Ending balance]))

The following key dimensions are used as filters to slice the aggregate measurements, so that you can achieve greater granularity and gain deeper analytical insights.

Expand table

Entity	Examples of attributes
Products	Product number, Product name, Unit, Item groups
Category hierarchies (Assigned to role Cost management)	Category hierarchy, Category level
Legal entities	Legal entity names
Fiscal calendars	Fiscal calendar, Year, Quarter, Period, Month
Site	ID, Name, Address, State, Country/Region

Last updated on 03/05/2026

Learning Power BI content

This article describes the **Learning** Microsoft Power BI content.

Reports that are included in the Power BI content

The reports that are included in the **Learning** Power BI content have both charts and tables that contain additional information. The following table describes the reports.

Expand table

Report	Contents
Learning overview	Summary of other reports
Course analysis	Registration by location, attendee by status, courses by type per company, and course attendance by job
Registration analysis	Registration list
Course types	Course types by skill
Instructor analysis	Ratio of courses to instructors, number of instructors, courses by instructor, courses per instructor, and course agenda by instructor
Courses offered	List of courses
Courses design	Course agenda

You can filter the charts and tiles on these reports, and pin the charts and tiles to the dashboard. For more information about how to filter and pin in Power BI, see Create and Configure A Dashboard.

Understanding the data model and entities

The **Learning** Power BI content uses the following data to fill the reports. This table shows the entities that the content is based on.

Expand table

Entity	Contents	Relationships with other entities
Calendar Offset	Calendar offsets to slice reports	Course Agenda, Course Attendees
Company	Companies to filter reports by	Course Agenda, Course Attendees
Course	Course, description, instructor name, location, room, and status	Course Agenda, Course Attendees, Course Skill
Course Agenda	Agenda, course, and start and end times	Company, Calendar Offset, Date, Course
Course Attendees	Name, status, job, and registration date	Company, Calendar Offset, Date, Course, Demographics, Employment, Course, Employee Name, Employee Title, Job, Position
Course Skill	Skill, skill type, and level	Course
Date	Days, weeks, months, and years	Course Agenda, Course Attendees
Demographics	Date of birth, gender, ethnic origin, and marital status	Course Agenda, Course Attendees
Employment	Start date, end date, and transition date	Course Agenda, Course Attendees
Job	Function, type, and title	Course Agenda, Course Attendees
Position	Position, title, and full-time equivalent (FTE)	Course Agenda, Course Attendees
Employee Name	First name, last name, and full name	Course Attendees
Employee Title	Title and seniority date	Course Attendees

Last updated on 08/04/2026

Organizational training Power BI content

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the finance and operations - organizational training Power BI content.

Reports that are included in the content pack

After you connect the content pack to your data, the reports show your organization's data. If you never used Microsoft Power BI before, you can learn more about it on the Guided Learning page for Power BI. The reports that are included in the content pack have both charts and tables that contain additional information. The following table describes the reports.

Expand table

Report	Contents
Course Analysis	Registration by location, course attendees by status, and registration list
Course Types	Course types by skill

You can filter the charts and tiles on these reports, and pin the charts and tiles to the dashboard. For more information about how to filter and pin in Power BI, see Create and Configure A Dashboard.

Understanding the data model and entities

Application data populates the reports in the Organizational training content pack. The following table shows the entities that the content pack is based on.

Expand table

Entity	Contents	Relationships with other entities
Training_CalendarOffset	Calendar offsets to slice reports	Training_CourseAgenda, Training_CourseAttendees
Training_Company	Companies to filter reports by	Training_CourseAgenda, Training_CourseAttendees
Training_Course	Course, description, instructor name, location, room, and status	Training_CourseAgenda, Training_CourseAttendees, Training_CourseSkill
Training_CourseAgenda	Agenda, course, and start and end times	Training_Company, Training_CalendarOffset, Training_Date, Training_Course
Training_CourseAttendees	Name, status, job, and registration date	Training_Company, Training_CalendarOffset, Training_Date, Training_Demographics, Training_Employment, Training_Course, Training_WorkerName, Training_WorkerTitle, Training_Job, Training_Position
Training_CourseSkill	Skill, skill type, and level	Training_Course
Training_Date	Days, weeks, months, and years	Training_CourseAgenda, Training_CourseAttendees
Training_Demographics	Date of birth, gender, ethnic origin, and marital status	Training_CourseAgenda, Training_CourseAttendees
Training_Employment	Start date, end date, and transition date	Training_CourseAgenda, Training_CourseAttendees
Training_Job	Function, type, and title	Training_CourseAgenda, Training_CourseAttendees
Training_Position	Position, title, and full-time equivalent (FTE)	Training_CourseAgenda, Training_CourseAttendees
Training_WorkerName	First name, last name, and full name	Training_CourseAttendees
Training_WorkerTitle	Title and seniority date	Training_CourseAttendees

Last updated on 04/08/2026

Production performance Power BI content

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes what is included in the **Production performance** Microsoft Power BI content. It explains how to access the Power BI reports, and provides information about the data model and entities that were used to build the content.

Overview

The **Production performance** Power BI content is for production managers or individuals in the organization who are responsible for production control.

The reports that are included let you use Power BI to monitor the performance of manufacturing operations with regard to timely execution, quality, and cost. The reports use transactional data from production orders and batch orders, and provide both an aggregate view of company-wide production metrics and a breakdown of metrics by product and resource.

The Power BI content highlights the organization's ability to complete production on time and in full. Future performance is projected based on the production plans. Comprehensive reports provide detailed insights into product defects that are caused by production, and also the defect rates for resources and operations.

This Power BI content also lets you analyze production variances. Production variances are calculated as the difference between estimated cost and realized cost. Production variances are calculated when production orders or batch orders reach **Ended** status.

The **Production performance** Power BI content includes data that originates from production orders and batch orders. The reports don't include data that is related to kanban productions.

Accessing the Power BI content

The **Production performance** Power BI content is shown on the **Production performance** page (**Production control** > **Inquiries and reports** > **Production performance analysis** > **Production performance**).

Metrics that are included in the Power BI content

The **Production performance** Power BI content includes a set of report pages. Each page consists of a set of metrics that are visualized as charts, tiles, and tables.

The following table provides an overview of the visualizations that are included.

Expand table

Report page	Charts	Tiles
Production performance	• Number of production by date • Number of productions by product and item group • Number of planned productions by date • Bottom 10 products by on-time & in-full	• Total orders • On-time & in full % • Incomplete % • Early % • Late %
Defects by product	• Defective rate (ppm) by date • Defective rate (ppm) by product and item group • Quantity produced by date • Top 10 products by effective rate	• Defective rate (ppm) • Defective quantity • Total quantity
Defects trend by product	Defect rate (ppm) by quantity produced	Defect rate (ppm)
Defects by resource	• Defect rate (ppm) by date • Defect rate (ppm) by resource and Site • Defect rate (ppm) by operation • Top 10 resources by defect rate	Defective quantity
Defects trend by resource	Defect rate (ppm) by quantity processed	
Production variances for job order costing	• Production variance by date and cost group type • Production variance by site and cost group type • Top 10 products with unfavorable production variance • Top 10 unfavorable production variance by resource	• Realized cost • Production variance • Production variance %

Understanding the data model and entities

The following data is used for the report pages in the **Production performance** Power BI content. This data is represented as aggregate measurements that are staged in the Entity store. The Entity store is

a Microsoft SQL Server database that is optimized for analytics. To learn more about the entity store, see Power BI integration with Entity store.

The following table shows the key aggregate measurements that are used as the basis of the Power BI content.

Expand table

Entity	Key aggregate measurements	Data source for finance and operations apps	Field
CostCalculation	CostAmount	ProdCalcTransExpanded	CostAmount
CostCalculation	CostMarkup	ProdCalcTransExpanded	CostMarkup
CostCalculation	ActualCostAmount	ProdCalcTransExpanded	RealCostAmount
CostCalculation	RealCostAdjustment	ProdCalcTransExpanded	RealCostAdjustment
RouteTransactions	ErrorQuantity	ProdRouteTransExpanded	QtyError
RouteTransactions	GoodQuantity	ProdRouteTransExpanded	QtyGood
ProductionOrder	DaysDelayed	ProdTableExpanded	DaysDelayed
ProductionOrder	LeadTime	ProdTableExpanded	LeadTime
ProductionOrder	PlannedLeadTime	ProdTableExpanded	PlannedLeadTime
ProductionOrder	ProductionOrderCount	ProdTableExpanded	
CoproductCostCalculation	CoproductCostAmount	PmfCoByProdCalcTransExpanded	CostAmount
CoproductCostCalculation	CoproductCostMarkup	PmfCoByProdCalcTransExpanded	CostMarkup
CoproductCostCalculation	CoproductRealCostAdjustment	PmfCoByProdCalcTransExpanded	RealCostAdjustment
CoproductCostCalculation	CoproductActualCostAmount	PmfCoByProdCalcTransExpanded	RealCostAmount

The following table shows how the key aggregate measurements are used to create several calculated measures in the content's dataset.

Expand table

Measure	How the measure is calculated
Production variance, %	SUM('Production variance'[Production variance]) / SUM('Production variance'[Estimated cost])
All planned orders	COUNTROWS('Planned production order')
Early	COUNTROWS(FILTER('Planned production order', 'Planned production order'[Scheduled end date] < 'Planned production order'[Requirement date]))

Measure	How the measure is calculated
Late	COUNTROWS(FILTER('Planned production order', 'Planned production order'[Scheduled end date] > 'Planned production order'[Requirement date]))
On-time	COUNTROWS(FILTER('Planned production order', 'Planned production order'[Scheduled end date] = 'Planned production order'[Requirement date]))
On-time %	IF ('Planned production order'[On-time] <> 0, 'Planned production order'[On-time], IF ('Planned production order'[All planned orders] <> 0, 0, BLANK())) / 'Planned production order'[All planned orders]
Completed	COUNTROWS(FILTER('Production order', 'Production order'[Is RAF'ed] = TRUE))
Defective rate (ppm)	IF('Production order'[Total quantity] = 0, BLANK(), (SUM('Production order'[Defective quantity]) / 'Production order'[Total quantity]) * 1000000)
Delayed production rate	'Production order'[Late #] / 'Production order'[Completed]
Early & in full	COUNTROWS(FILTER('Production order', 'Production order'[Is in full] = TRUE && 'Production order'[Is early] = TRUE))
Early #	COUNTROWS(FILTER('Production order', 'Production order'[Is early] = TRUE))
Early %	IFERROR(IF('Production order'[Early #] <> 0, 'Production order'[Early #], IF('Production order'[Total orders] = 0, BLANK(), 0)) / 'Production order'[Total orders], BLANK())
Incomplete	COUNTROWS(FILTER('Production order', 'Production order'[Is in full] = FALSE && 'Production order'[Is RAF'ed] = TRUE))
Incomplete %	IFERROR(IF('Production order'[Incomplete] <> 0, 'Production order'[Incomplete], IF('Production order'[Total orders] = 0, BLANK(), 0)) / 'Production order'[Total orders], BLANK())
Is delayed	'Production order'[Is RAF'ed] = TRUE && 'Production order'[Delayed value] = 1
Is early	'Production order'[Is RAF'ed] = TRUE && 'Production order'[Days delayed] < 0
Is in full	'Production order'[Good quantity] >= 'Production order'[Scheduled quantity]
Is RAF'ed	'Production order'[Production status value] = 5 || 'Production order'[Production status value] = 7
Late & in full	COUNTROWS(FILTER('Production order', 'Production order'[Is in full] = TRUE && 'Production order'[Is delayed] = TRUE))
Late #	COUNTROWS(FILTER('Production order', 'Production order'[Is delayed] = TRUE))
Late %	IFERROR(IF('Production order'[Late #] <> 0, 'Production order'[Late #], IF('Production order'[Total orders] = 0, BLANK(), 0)) / 'Production order'[Total orders], BLANK())
On-time & in full	COUNTROWS(FILTER('Production order', 'Production order'[Is in full] = TRUE && 'Production order'[Is delayed] = FALSE && 'Production order'[Is early] = FALSE))
On-time & in full %	IFERROR(IF('Production order'[On-time & in full] <> 0, 'Production order'[On-time & in full], IF('Production order'[Completed] = 0, BLANK(), 0)) / 'Production order'[Completed], BLANK())

Measure	How the measure is calculated
Total orders	COUNTROWS('Production order')
Total quantity	SUM('Production order'[Good quantity]) + SUM('Production order'[Defective quantity])
Defect rate (ppm)	IF('Route transactions'[Processed quantity] = 0, BLANK(), (SUM('Route transactions'[Defective quantity]) / 'Route transactions'[Processed quantity]) * 1000000)
Defect ratio mixed (ppm)	IF('Route transactions'[Total mixed quantity] = 0, BLANK(), (SUM('Route transactions'[Defective quantity]) / 'Route transactions'[Total mixed quantity]) * 1000000)
Processed quantity	SUM('Route transactions'[Good quantity]) + SUM('Route transactions'[Defective quantity])
Total mixed quantity	SUM('Production order'[Good quantity]) + SUM('Route transactions'[Defective quantity])

The following table shows the key dimensions that are used as filters to slice the aggregate measurements, so that you can achieve greater granularity and gain deeper analytical insights.

Expand table

Entity	Examples of attributes
Reported as finished date	Completion (RAF) date, Month, and Year offset
Ended date	Ended month offset and Month
Requirement date	Requirement date month offset and Requirement date
Route transaction date	Route transaction month offset and Date
Sites	Sites ID, Site name, State, and City
Entities	ID and Name
Resources	Resource ID, Resource name, Resource type, and Resource group
Products	Product number, Product name, Item ID, and Item group

Last updated on 03/05/2026

Warehouse performance Power BI content

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes what's included in the **Warehouse performance** Microsoft Power BI content. It explains how to access the Power BI reports, and provides information about the data model and entities that are used to build the content.

Overview

The **Warehouse performance** Power BI content was created so that warehouse and operations managers can monitor important inbound, outbound, and inventory metrics. It uses Warehouse management, product, and other transactional data from your system, and provides both an aggregate view of warehouse performance and a breakdown for vendors, product groups and products, and site and warehouses.

Warehouse managers can use the **Warehouse performance** Power BI content to measure the following three areas:

- **Inbound performance** – Measure how well a vendor is performing against customer requirements (in other words, measure delivery performance), and measure put-away performance, so that you can identify issues that involve workers or items over a period. It's important that you know whether vendors are delivering on time, early, or late, so that you can determine how vendor performance is affecting overall put-away performance. A vendor that delivers outside the dates that were agreed on can put extra pressure on the warehouse because of unexpected work, and can increase the average put-away time.
- **Shipping performance** – Measure whether your warehouse is shipping in full and on time to customers (in other words, measure outbound shipping and delivery performance), so that you can identify any issues that involve products, sites or warehouses, or dedicated customers. If you find that you're shipping late to specific areas or towns, you might have to pay more attention to transportation or account management.

- **Location inventory accuracy** – Inventory accuracy is important internal warehouse business intelligence (BI). It's very important that you determine how accurately you're counting in general. However, it's also important that you determine how accurate you are at storing items in the correct locations, and that you highlight discrepancy data, so that you can find better positions for items or initiate total counting on specific items. (Currently, the new item-based counting functionality is delivered as a hotfix.) If you're using this Power BI content to determine the correctness of on-hand inventory data per location, you can also identify theft in your shops. You can also determine whether any locations have on-hand quantities that differ from enterprise resource planning (ERP) data. Those locations might be too large, or they might be impossible to count. Alternatively, some of the physical positioning might be bad, so that it's difficult to keep a single type of item in sync with on-hand data.

Prerequisites

The **Warehouse performance** Power BI content only shows data for the dates defined on the **Date dimensions** page (**Organization administration** > **Setup** > **Calendars** > **Date dimensions**).

Accessing the Power BI content pack

The **Warehouse performance** Power BI content is shown on the **Warehouse performance** page (**Warehouse management** > **Inquiries and reports** > **Warehouse performance analysis** > **Warehouse performance**).

Metrics that are included in the Power BI content

The **Warehouse performance** Power BI content includes a report. This report consists of a set of metrics that are visualized as charts, tiles, and tables. The following table provides an overview of the visualizations in the **Warehouse performance** Power BI content.

Expand table

Report page	Charts	Description
Inbound Performance	Total put aways	The number of put-away work lines that are processed during a given time.
Inbound Performance	Put away average time	The average time, in hours, for all purchase order put-away lines that are processed, from registration of the items until the last put is

Report page	Charts	Description
		processed.
Inbound Performance	Received early	The number of purchase order lines that are received early.
Inbound Performance	Received on time	The number of purchase order lines that are received on time.
Inbound Performance	Received late	The number of purchase order lines that are received late.
Inbound Performance	On time by vendor	A view of the percentage of purchase order lines that are received from a vendor or vendor group early, on time, and late.
Inbound Performance	Dock to stock average put away (hours)	The average dock-to-stock put-away time in hours over time.
Inbound Performance	Average put away by worker	The average time, in hours, that a worker has been taking for put-away processing of purchase order lines.
Inbound Performance	Average put away hour by vendor	The average put-away time in hours by vendor or vendor group.
Inbound Performance	Average put away hour by product	The average put-away time in hours by item or item group.
Location inventory accuracy	Total count	The number of counting work lines that are processed for a given period.
Location inventory accuracy	Discrepancy rate	The total discrepancy rate as a percentage of all lines that are counted for a given period.
Location inventory accuracy	Count without discrepancy	Of the total number of counting work lines that are processed, the number of lines that are processed without any discrepancy.
Location inventory accuracy	Items counted over time	The percentage of items that are counted with and without discrepancy over time.
Location inventory accuracy	Item quantity discrepancy greater than %	A table view of counting lines that have discrepancies that exceed a specified percentage. The table includes information about locations, items, the average discrepancy, the total counting work lines that are counted, the number of counting lines that have discrepancies for the location, the average quantity that is counted, the expected total

Report page	Charts	Description
		quantity that will be counted, and the actual item quantity that is counted.
Location inventory accuracy	Item count by worker	The counting performance of workers. Performance is expressed as a percentage of counting work lines that have and don't have discrepancies.
Location inventory accuracy	Item count by site / warehouse	Counting performance by the number of processed counting work lines by site or warehouse that have and don't have discrepancies.
Location inventory accuracy	Discrepancy rate by product	The discrepancy rate for counting performance. The rate is expressed as a percentage of processed counting work lines by item or item group.
Shipping performance	Lines shipped	The total number of shipment lines that are shipped over a given period.
Shipping performance	Early	The percentage of shipment lines that are shipped early.
Shipping performance	On time	The percentage of shipment lines that are shipped on time.
Shipping performance	Late	The percentage of shipment lines that are shipped late.
Shipping performance	Shipments over time	The percentage of shipment lines that are shipped on time, early, or late over a given period. A trend line shows the trend for lines that are shipped on time.
Shipping performance	Shipped late by city	A map visualization of cities and areas that are affected by late shipments.
Shipping performance	Shipped by product	The percentage that is shipped early, on time, or late by item or item group.
Shipping performance	Shipped by customer	The percentage that is shipped early, on time, or late by customer or customer group.
Shipping performance	Shipped by site / warehouse	The percentage that is shipped early, on time, or late by site or warehouse.

Understanding the data model and calculations

The following data is used to fill the report pages in the **Warehouse performance** Power BI content. This data is represented as aggregate measurements that are staged in the Entity

store. The Entity store is a Microsoft SQL Server database that is optimized for analytics. For more information, see Power BI integration with Entity store.

The following key aggregate measurements are used as the basis of the content.

Expand table

Report page	Charts	Tables	Calculation descriptions
Inbound Performance	Total put aways	WHSWorkLine	The count of work lines where the work type is **put**.
Inbound Performance	Put away average time	WHSWorkLine	The sum of work lines max. time divided by the count of work lines max. time, where work lines max. time is the maximum gap between the work created date and the closed date.
Inbound Performance	Received early	WHSWorkLine	The count of work lines where the work created date is earlier than the delivery date that is used. If the delivery confirmed date isn't set, use the default delivery date.
Inbound Performance	Received on time	WHSWorkLine	The count of work lines where the work created date is equal to the delivery date that is used. If the delivery confirmed date isn't set, use the default delivery date.
Inbound Performance	Received late	WHSWorkLine	The count of work lines where the work created date is later than the delivery date that is used. If the delivery confirmed date isn't set, use the default delivery date.
Inbound Performance	On time by vendor	WHSWorkLine	Received early, Received on time, and Received late (see the descriptions earlier in this table).
Inbound Performance	Dock to stock average put away (hours)	WHSWorkLine	Put away average time (see the description earlier in this table).
Inbound Performance	Average put away by worker	WHSWorkLine	Put away average time (see the description earlier in this table).
Inbound Performance	Average put away hour by	WHSWorkLine	Put away average time (see the description earlier in this table).

Report page	Charts	Tables	Calculation descriptions
	vendor		
Inbound Performance	Average put away hour by product	WHSWorkLine	Put away average time (see the description earlier in this table).
Location inventory accuracy	Total count	WHSWorkLineCycleCount	Cycle counts where the absolute discrepancy quantity is equal to or more than 0 (zero).
Location inventory accuracy	Discrepancy rate	WHSWorkLineCycleCount	Cycle counts that have discrepancies, divided by the total count. A cycle count is considered to have discrepancies if the absolute discrepancy quantity is more than 0 (zero).
Location inventory accuracy	Count without discrepancy	WHSWorkLineCycleCount	Cycle counts where the absolute discrepancy quantity is more than 0 (zero).
Location inventory accuracy	Count with discrepancy	WHSWorkLineCycleCount	Cycle counts where the absolute discrepancy quantity is equal to 0 (zero).
Location inventory accuracy	Items counted over time	WHSWorkLineCycleCount	Count without discrepancy and Count with discrepancy (See the descriptions earlier in this table.)
Location inventory accuracy	Item quantity discrepancy greater than %	WHSWorkLineCycleCount	The average discrepancy is the absolute discrepancy quantity divided by the expected quantity where the absolute discrepancy quantity is more than 0 (zero). The average discrepancy quantity is the average absolute discrepancy quantity where the absolute discrepancy quantity is more than 0 (zero). Count with discrepancy, Total count, Expected quantity, and Counted quantity where the whole quantity is grouped by item and location (see the descriptions earlier in this table).
Location inventory accuracy	Item count by worker	WHSWorkLineCycleCount	Count without discrepancy and Count with discrepancy (see the descriptions earlier in this table).
Location inventory accuracy	Item count by site / warehouse	WHSWorkLineCycleCount	Count without discrepancy and Count with discrepancy (see the descriptions earlier in this table).

Report page	Charts	Tables	Calculation descriptions
Location inventory accuracy	Discrepancy rate by product	WHSWorkLineCycleCount	Discrepancy rate (see the description earlier in this table).
Shipping performance	Lines shipped	SalesLine	The count of sales lines where sales lines are shipped.
Shipping performance	Early	CustPackingSlipOnTimeStatus	Sales lines where the ship date status is **1 (Early)**. Early means that the ship date of the packing slip is earlier than the confirmed ship date of the sales line. If the confirmed ship date isn't set, use the requested ship date.
Shipping performance	On time	CustPackingSlipOnTimeStatus	Sales lines where the ship date status is **2 (On time)**. On time means that the ship date of the packing slip is equal to the confirmed ship date of the sales line. If the confirmed ship date isn't set, use the requested ship date.
Shipping performance	Late	CustPackingSlipOnTimeStatus	Sales lines where the ship date status is **3 (Late)**. Late means that the ship date of the packing slip is later than the confirmed ship date of the sales line. If the confirmed ship date isn't set, use the requested ship date.
Shipping performance	Shipments over time	SalesLine CustPackingSlipOnTimeStatus	Orders that are fully shipped, where the whole quantity of the sales line is shipped, plus Early, On time, and Late (see the descriptions earlier in this table).
Shipping performance	Shipped late by city	CustPackingSlipOnTimeStatus	Late (see the descriptions earlier in this table).
Shipping performance	Shipped by product	CustPackingSlipOnTimeStatus	Early, On time, and Late (see the descriptions earlier in this table).
Shipping performance	Shipped by customer	CustPackingSlipOnTimeStatus	Early, On time, and Late (see the descriptions earlier in this table).
Shipping performance	Shipped by site / warehouse	CustPackingSlipOnTimeStatus	Early, On time, and Late (see the descriptions earlier in this table).

Last updated on 04/08/2026

Analyze warehouse material movement through process mining

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Microsoft Power Automate Process Mining feature offers *task mining* and *process mining* capabilities to help you better understand your business processes, so that you can optimize them. Dynamics 365 Supply Chain Management offers a *Warehouse material movement analysis* template for the Power Automate Process Mining. Use this template to create processes that help warehouse and operations managers gain insights into the material flow in the warehouse. It helps find inefficiencies that you can remove to improve the performance of the warehouse.

The *Warehouse material movement analysis* template uses warehousing, product, and transactional data that's stored in your system. It lets you visualize and analyze material movements on the warehouse floor, based on closed warehouse work records. In process mining terminology, each warehouse work record corresponds to a *case*, and each pick or put work line corresponds to an *event*. Depending on your configuration, an *activity* can be any one of the following items:

- Warehouse location
- Warehouse location profile
- Warehouse zone

For more information about Power Automate Process Mining, see Overview of Power Automate Process Mining.

This article describes what's included in the *Warehouse material movement analysis* template, and explains how to deploy and access Power Automate Process Mining processes.

Prerequisites

The *Warehouse material movement analysis* template for Power Automate Process Mining has the following prerequisites:

- You're running Supply Chain Management version 10.0.35 or later.
- Your Supply Chain Management environment must be linked to a Dataverse environment.
- The user who deploys the process in Power Automate Process Mining must have the *Warehouse manager*, *Warehouse planner*, and/or *System administrator* role.
- The Supply Chain Manager user who deploys the process in Power Automate Process Mining must also be configured as a user in the linked Dataverse environment. To deploy the process, a user must have the *Environment maker* and/or *System administrator* role. After the process is deployed, it can be shared with other users. Those users automatically get the permissions required to use the process.

Deploy a new Power Automate Process Mining process based on the template

Set up as many processes as you need, each with a different name and filters. Follow these steps to deploy a *Warehouse material movement analysis* process.

1. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**.

2. On the Action Pane, select **Deploy process** to open the deployment wizard.

3. Follow the on-screen instructions. Provide the following information:

    - **Process name** – Enter a name for the process. Include the legal entity (company) and other filter information in the name, so you can easily identify the process later.
    - **Company** – Select the legal entity that you want to use this process to analyze. Only warehouse work records in the selected legal entity are included in Power Automate Process Mining process.
    - **Number of months to load** – Specify the number of months to load at a time. This filter helps prevent excessive or irrelevant data from being loaded into Power Automate Process Mining. Only warehouse work records that were closed after the specified time threshold are included in Power Automate Process Mining process.
    - **Activity** – Specify which column Power Automate Process Mining should treat as an *activity*. To learn more about process mining terminology, see Glossary in Power Automate Process Mining.

> **Note**
>
> Try to limit the data by optimizing the **Number of months to load** filter value. Power Query Online includes several threshold checks. Therefore, if you try to load too much data, time-out or quota violation exceptions might occur, and the system might fail. Load less than 25 months of data into the *Warehouse material movement analysis* process and ensure that the total number of events in the process is less than 4 million.

When you complete the deployment wizard, the following actions occur:

- **Initial data is loaded.** The system loads 200 *cases* into the staging table. In the context of the *Warehouse material movement analysis* process template, each case corresponds to a single closed warehouse work record. Most likely, many more than 200 records meet the filtering criteria set in the deployment wizard. By loading only 200 cases, the system enhances performance and the user experience, so that you can access the visualization more quickly.
- **You're prompted to complete the deployment process.** Power Automate Process Mining user interface (UI) opens on a new tab in your browser. You might be asked to sign in. Power Automate Process Mining uses your credentials to read data from Supply Chain Management. Don't close the page right away. Otherwise, you interrupt the deployment. You don't have to do anything else on the page. Power Automate Process Mining lets you know when deployment is completed.

Refresh data in the staging table and Power Automate Process Mining

As mentioned earlier, after you deploy a new process, the system loads an initial set of 200 cases into the staging table and loads that data into Power Automate Process Mining. This behavior lets you get started quickly. However, to work with your full data set, you must refresh the staging table and Power Automate Process Mining. This section describes how to manually trigger this process.

> **Tip**
>
> Although it can be useful to refresh the data on demand from time to time, you probably want to set up each process so that it automatically refreshes its data on a regular basis. For

instructions, see the Configure the warehouse material movement process section of this article.

Follow these steps to manually refresh data in the staging table and Power Automate Process Mining.

1. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**.
2. In the list pane, select the process to refresh the data for.
3. On the Action Pane, select **Refresh staging table**. The batch job that's responsible for refreshing the staging table runs every five minutes. The next time that it runs, it picks up the request to refresh the table. It then loads all the data that matches the filters specified during deployment.
4. On the **General** FastTab, the **Last staging table refresh** field is updated to indicate that the table is refreshed. However, the new data isn't available in Power Automate Process Mining until you refresh it there. Select **View process details** on the Action Pane, and then select **Refresh data** in Power Automate Process Mining.

A staging table cleanup batch job runs daily and removes any records from the staging table that no longer satisfy the filter criteria. For example, it removes any records that are now too old.

Configure warehouse material movement processes

After you deploy a process, you can configure it at any time by following these steps.

1. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**.

2. In the list pane, select the process to refresh the data for.

3. On the **General** FastTab, set the following fields:

    - **Automatic refresh recurrence** – Select how often the system should update the data table (*None*, *Once per day*, or *Once per week*). Although this update refreshes data in the staging table, it doesn't automatically load that data into Power Automate Process Mining. To set Power Automate Process Mining to periodically load this data from the staging table, select **View process details** on the Action Pane. Then, in Power Automate Process Mining, select **Schedule refresh**, and select a refresh schedule. You can also manually refresh the data in Power Automate Process Mining by selecting **Refresh data** in Power Automate Process Mining.

- **Last staging table refresh** – This read-only field shows the date and time when the staging table was last refreshed.
- **Enforce offline hours** – Set this option to *Yes* to prevent the system from refreshing the staging table during the specified hours. This option can be useful if you want to avoid refreshing the table during peak hours.

4. On the **Process parameters** FastTab, review the following fields:

    - **Legal entity** – This read-only field shows the legal entity specified during deployment.
    - **Number of months to load** – This read-only field shows the number of months to load, as specified during deployment.
    - **Activity** – This read-only field shows the activity specified during deployment.

    The **Processing log** FastTab shows a history of data refresh and cleanup operations. This information can be useful during troubleshooting.

Access the warehouse material movement analysis process

You can access the warehouse material movement analysis process from both Supply Chain Management and Power Automate Process Mining UI.

To access the process from Supply Chain Management, follow these steps:

1. Go to **Warehouse management** > **Enquiries and reports** > **Warehouse performance analysis** > **Warehouse material movement analysis**.

2. In the **Process name** field, select the name of the process to view. The list includes only processes that meet both the following conditions:

    - The process is ready for analysis. (In other words, it has been deployed and analyzed in Power Automate Process Mining.)
    - The current user can access the process. (In other words, the current user created or shared the process.)

3. The Power BI report for your selected process opens. You can interact with the report to view the process and analyze it in various ways.

4. To view the selected process in Power Automate Process Mining, select **View process report** on the Action Pane. Power Automate Process Mining UI opens on a new tab in your browser.

5. To perform advanced process analysis, assess process behavior with business rules, detect rework, and build your own business metrics, select **Open Power Automate Process Mining** to open process in Power Automate Process Mining desktop app. For more information about its capabilities, see Power Automate Process Mining overview.

FAQ

Why don't I see all the events (work lines) and cases (work headers) that I expect in Power Automate Process Mining process?

Several reasons might prevent you from seeing all the events and cases that you expect. Check the following items:

- Confirm that the staging table is recently refreshed. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**, and review the **Last staging table refresh** field on the **General** tab and the information on the **Processing log** FastTab.
- Confirm that the warehouse work that you expect to see in the process is closed. The warehouse work closed date and legal entity (company) should meet the process filtering criteria that are specified during process deployment. You can check these details on the **Warehouse material movement process configuration** page.
- Confirm that Power Automate Process Mining process is recently refreshed. For information about how to view refresh information, and how to set up schedules to automatically refresh the staging table and Power Automate Process Mining process, see the Configure the warehouse material movement process section of this article.

Why don't I see the process that I'm looking for on the Warehouse material movement analysis page?

Several reasons might prevent you from seeing a process on the **Warehouse material movement analysis** page. Check the following items:

- Confirm that you have access to the process. Usually, to view a process, you must have created it yourself, or the creator must share it with you. To confirm that you have access, sign in to Power Automate Process Mining in Power Automate by using the same user account, and determine whether you can access the process.
- Confirm that the process is deployed and ready for analysis. The **Warehouse material movement analysis** page shows only processes that are ready for analysis. If the process is

still being deployed, it might not yet appear on the page. To confirm that the process is ready for analysis, try to open the Power BI report for the process in Power Automate Process Mining UI.

Why do I receive an error during process deployment?

The system performs several checks while it's provisioning a process. Most failures occur either because the Supply Chain Management user account that's used isn't registered in Dataverse or because the account doesn't have the required privileges. If deployment takes a long time (about four hours), you're probably exceeding the threshold limits for Power Automate Process Mining integration. Consider changing the filters to reduce the number of events that are loaded into Power Automate Process Mining. You can typically lower the **Number of months to load** filter value.

Last updated on 05/29/2026

Analytical Workspaces (using Power BI Embedded)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Dynamics finance and operations apps now deliver rich, interactive reports that seamlessly integrate into application workspaces. By using graphics and visuals supported by Power BI, workspaces provide a highly visual, yet interactive experiences for users.

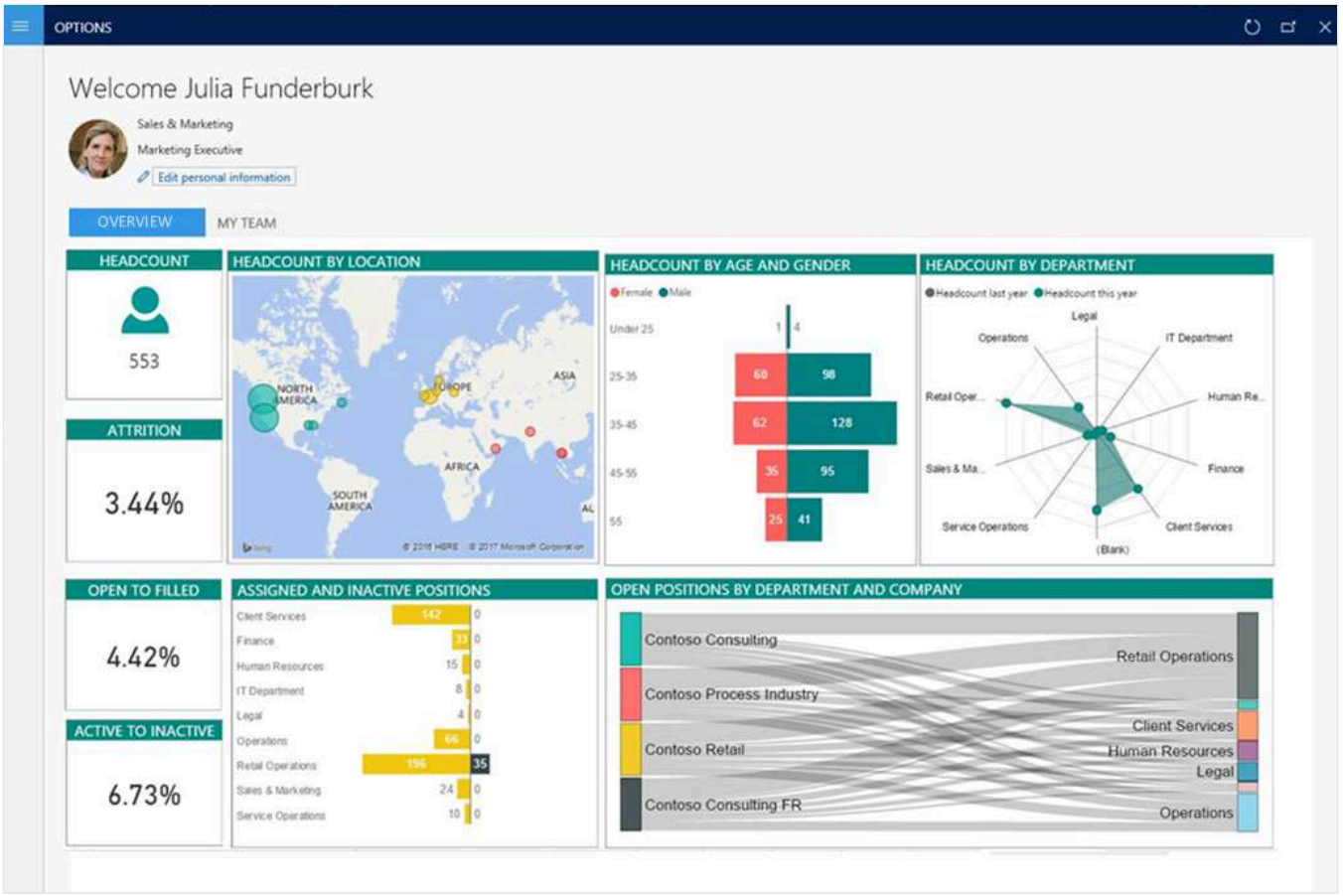
Overview

Workspaces in the application provide an overview of business processes or business units. With rich workspaces, users get a bird's-eye view of the state of business before diving into details and taking action. Workspaces contain visuals, count tiles, and KPIs as well as quick links to reports and pages. Within a workspace, all controls are tightly integrated to provide a highly productive and engaging work environment to the user. Power BI Embedded is a Microsoft Azure service that enables ISVs and app developers to surface Power BI data experiences within their applications. With Power BI Embedded, developers can deliver always-up-to-date views with Direct Query. To learn more about how the Power BI Embedded service integrates with the application, see Power BI Embedded integration.

Power BI in workspaces

The application now delivers interactive reports that seamlessly integrate into application workspaces. By using rich infographics and visuals supported by Power BI (including the large number of controls provided by third parties), workspaces provide a highly visual, yet interactive experience for users. Using infographics in the overview page, users get a quick glance of the state of the business. They can interact with data by simply clicking or touching visuals on the page. They can see the cause and effect, perform simple what-if operations without leaving the workspace. With interactive visuals, your users have fun exploring data and discovering hidden trends.

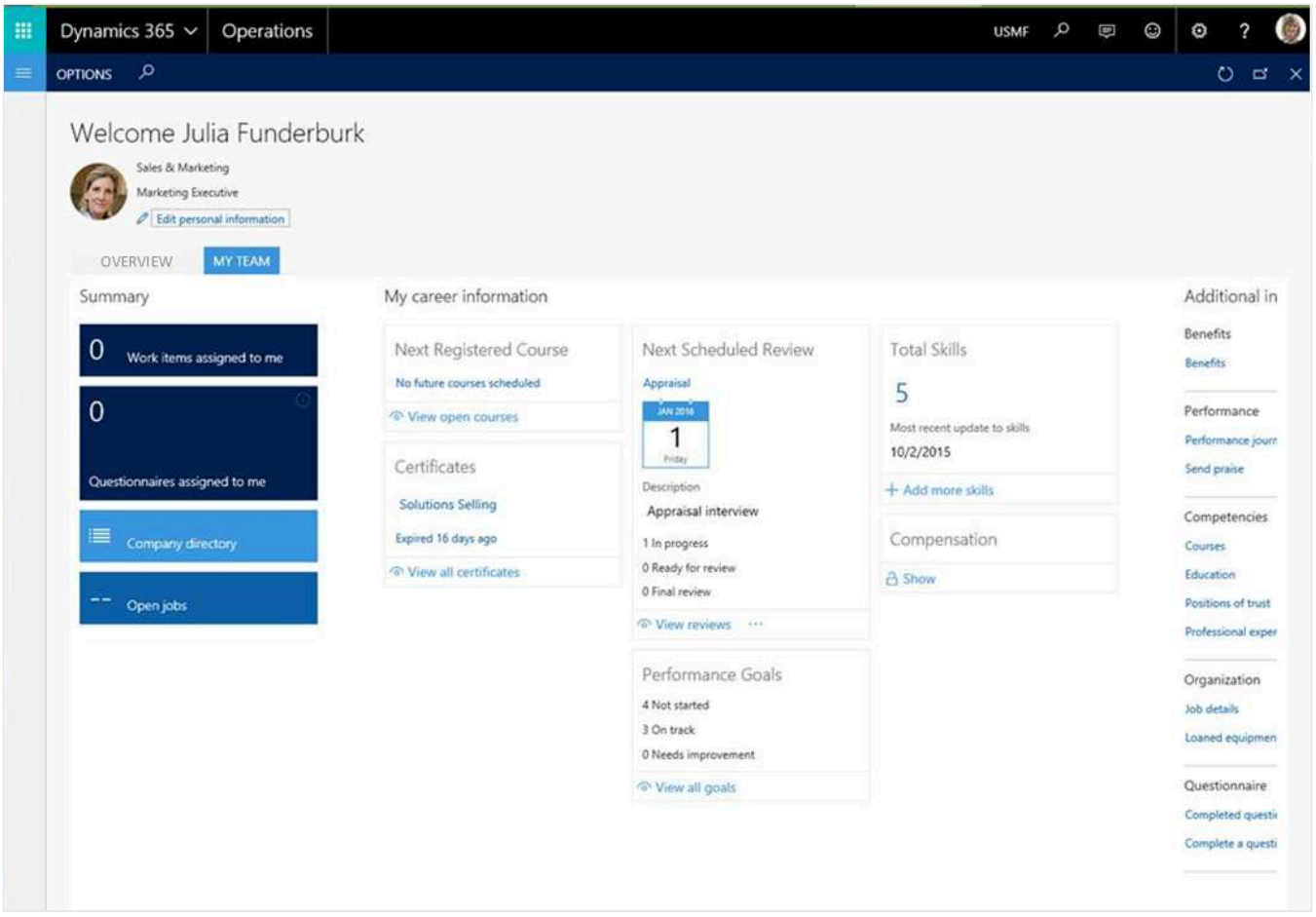
The following screenshot shows Power BI in a workspace.



Power BI vs operational workspaces

Power BI workspaces complement operational views with analytical insights based on near real-time information. The following section offers a visual comparison of a Power BI workspace and an operational workspace.

The following screenshot shows an operational workspace.



Edit embedded reports in analytical workspaces

https://learn-video.azurefd.net/vod/player?id=ea74f876-1a5f-49cc-b692-6a7bacf12ea3&locale=en-us&embedUrl=%2Fdynamics365%2Ffin-ops-core%2Fdev-itpro%2Fanalytics%2Fembed-power-bi-workspaces

The How to edit an embedded report in an analytical workspace video (shown preceding) is part of the Playlist available on YouTube.

What's next?

Going forward, new cloud deployments come bundled with the Power BI Embedded service. Other documentation describing the Developer ALM process is available to help partners and ISVs create new solutions that take advantage of the Power BI Embedded service integration options.

Last updated on 11/26/2025

Power BI Embedded integration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can embed Microsoft Power BI content that partners and independent software vendors (ISVs) develop directly into Microsoft Dynamics 365 Finance. This article describes some of the ways that you can use the Microsoft Power BI Embedded integration.

Overview

The integration of the application and Power BI enables data mash-up scenarios that require access to external data sources that Microsoft Power Query for Excel supports. Users can personalize workspaces by embedding tiles that PowerBI.com hosts. Users can also add direct links to reports that PowerBI.com hosts. In this way, users can access and interact with the reports without leaving the application. Partners and ISVs can embed Power BI content (PBIX files) directly into the application. PBIX files that are associated with a model file are automatically published in Power BI Embedded as part of the application deployment process. Additionally, you can add X++ extensions for embedded reporting scenarios that require the following functionality:

- Drill-down navigation into detailed pages in response to user interactions
- Report filters that are based on user and session context information, such as company or date range
- The ability to navigate directly to a specific tab on a Power BI report via menu items

For more information about customizations that use extensions, see Customize through extension and overlayering.

Advantages of Power BI Embedded

- **Deliver Power BI workspaces and reports in the application.** If you're a power user or a business analyst, you can tweak ready-made reports or create new reports by using Power BI tools. As a developer, you can use the reports that your users create to provide rich navigation experiences in the product through workspaces. If you're in the partner and

ISV community, you can build rich workspaces that include Power BI experiences, and then release those workspaces as part of your solution.

- **The Power BI Embedded service license is bundled with the application.** If you're an ISV or a systems integrator, you can package workspaces that are enabled for Power BI (and navigational experiences that those workspaces provide) as part of a Microsoft Dynamics Lifecycle Services (LCS) solution. Your customers get the same experience without needing a PowerBI.com subscription. The workspaces just work with finance and operations applications.
- **Enable drill-down into detailed pages from Power BI.** The visuals are the starting point for action. Your users can drill down to business processes and pages to act immediately on issues that they uncover. The visuals let users filter data and uncover trends. Action pages reflect just the set of data that requires attention.
- **Help secure access to Power BI reports by using menu items.** As a developer, you can use familiar programming concepts that are available in finance and operations apps, because we extended the same concepts to workspaces that are based on Power BI. You can create new workspaces or extend existing workspaces by adding an overview page that Power BI drives. Developers can associate menu items with Power BI reports and include them as links in workspaces. The role-based and task-based security in finance and operations apps can help secure these menu items.
- **Filter reports based on application context.** You can build navigation experiences by passing one or more filters to Power BI reports. For example, depending on a user's actions or context, you can filter the Power BI report to reflect data from one business unit or a specific product. The user doesn't have to filter the data. You can define drill-through links to finance and operations pages, so that users can go directly to the transactional details pages.

For more information about the Power BI Embedded service, see the Power BI FAQ.

Service availability

The Power BI Embedded service is automatically deployed and configured for all cloud-hosted, multibox deployments. Because the service relies on Microsoft Azure services, application analytical workspaces and reports are unavailable in one-box environments. The Power BI Embedded service is already available in most Azure datacenters. You can check the latest availability on the Azure status page.

Frequently asked questions

Can I customize the Power BI embedded reports?

Yes. To customize the Power BI embedded reports, install Power BI Desktop in a one-box environment, and follow the steps in Create analytical reports by using Power BI Desktop.

Do customers have to purchase a separate Power BI license to use the new embedded analytics?

No, customers don't have to purchase a separate Power BI license to use the new embedded analytics. However, a Power BI Pro license is required to connect to Entity Store from PowerBI.com by using DirectQuery.

Can I help secure data to only those companies that I have access to?

Yes, the single company view prevents users from accessing data from companies that they don't have access to. For more information about how to help secure custom solutions, see Help secure analytical workspaces and reports by using Power BI Embedded.

How is currency shown across multiple companies?

Currency is shown as a system currency. The system currency is defined on the **System parameters** page.

Can I drill from summary balances back into finance and operations?

Yes, you can drill into the details on a Power BI report. However, there's limited support for drill-down into finance and operations apps.

What languages are currently supported?

Currently, only English is supported. However, the Power BI team plans to add support for other languages.

Can I access analytical workspaces and reports in the on-premises version of finance and operations?

No, you can't currently access analytical workspaces and reports in Dynamics 365 Finance + Operations (on-premises). Systems of Intelligence functions rely on cloud-hosted solutions.

Last updated on 11/26/2025

Customize embedded reports in analytical workspaces

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Analytical workspaces

Analytical workspaces come with the application suite. Through reporting, they give users insights into data based on standard business operations. The reports are generic reports that business professionals define. They include metrics that interest a wide range of users from any industry.

However, in some cases, the standard reports include data that isn't relevant to all customers. More often, customers want to access data points or calculations that the standard reports leave out.

Power users can use web-friendly design tools to customize the analytical reports embedded in the application. By using the free-form canvas designer, users who are familiar with the relevant business insights that are required can help make the organization successful.

> **Important**
>
> The service automatically deploys customizations that you make to the embedded analytical reports and makes them available to other users of the system.

Edit embedded reports in analytical workspaces

https://learn-video.azurefd.net/vod/player?id=ea74f876-1a5f-49cc-b692-6a7bacf12ea3&locale=en-us&embedUrl=%2Fdynamics365%2Ffin-ops-core%2Fdev-itpro%2Fanalytics%2Fcustomize-analytical-workspace

The How to edit an embedded report in an analytical workspace video (shown above) is included in the Playlist available on YouTube.

Important points about embedded analytical reports

Although the standard reports deliver insights that are tailored to a given business persona, customizations can often maximize the value of these standard reports.

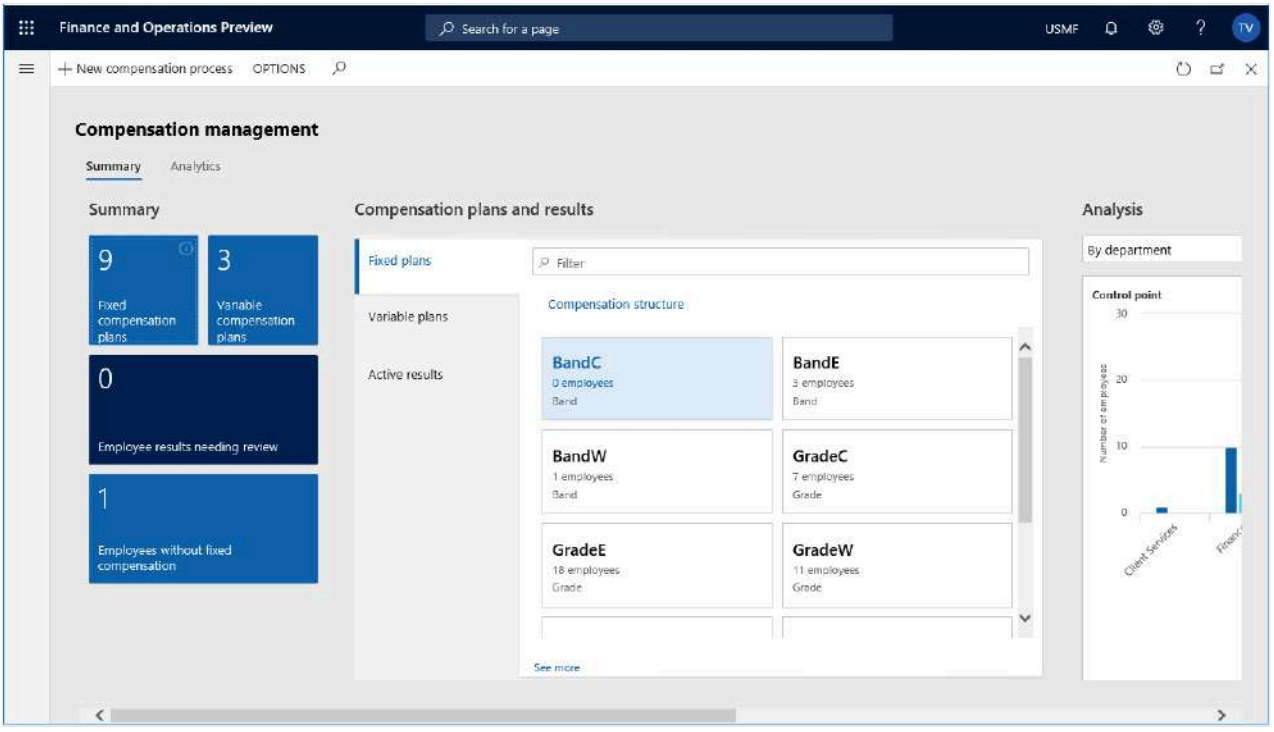
Here are some important points to note about this service capability:

- You can only customize the report design canvas. You can't change the definitions of report data sets.
- When you customize reports in the analytical workspace, the changes apply to all users in the environment.
- The service automatically preserves report customizations during product upgrades.
- The service doesn't support the export of customizations that you make to analytical workspaces.

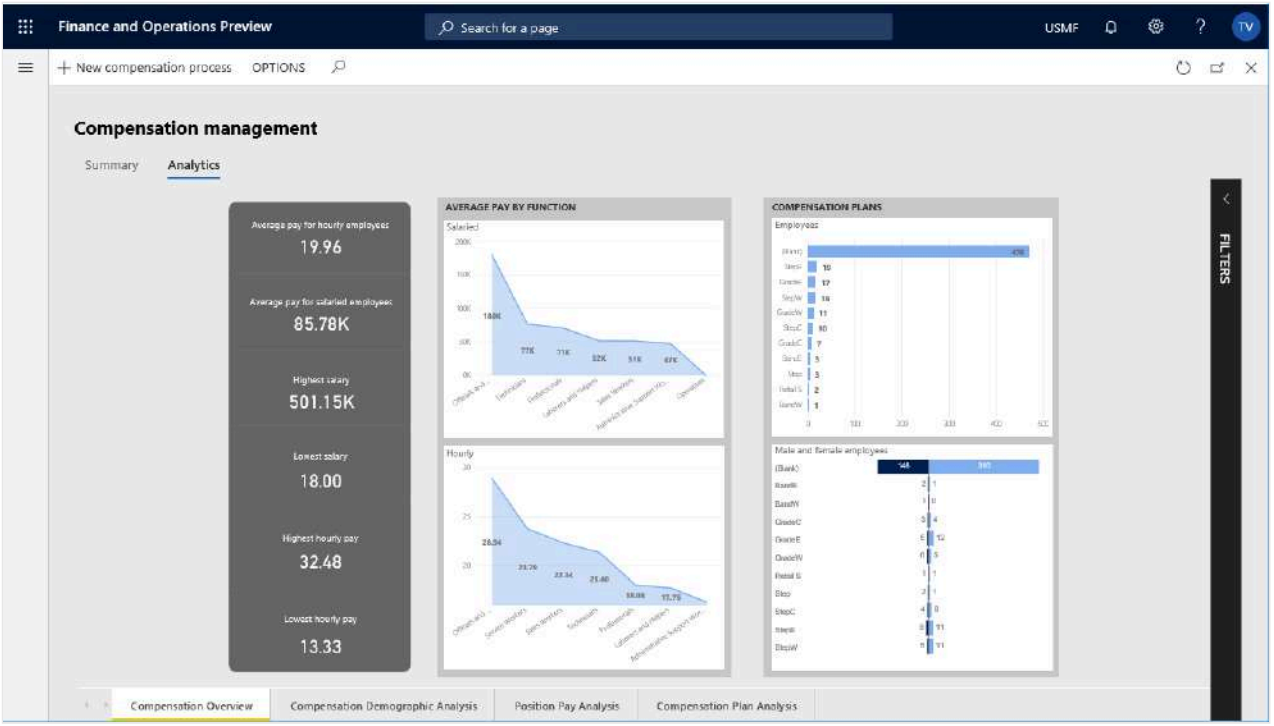
Customize an analytical workspace

To customize the embedded application solutions, assign the **System report editor** security role to a user. Users with this security role can use the buttons on the **Options** tab on the Action Pane of the application workspaces to make customizations. This example shows how to customize one of the standard analytical workspaces that are bundled with the application suite.

1. Sign in and open the application workspace that you want to customize. In this example, you replace the standard analytical report that's embedded in the **Compensation management** workspace.



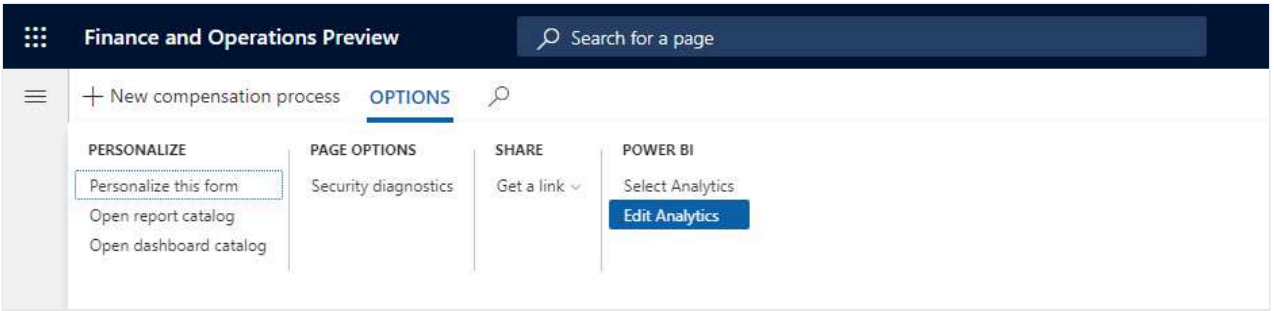
2. Select the **Analytics** tab to access the workspace's embedded analytical report.



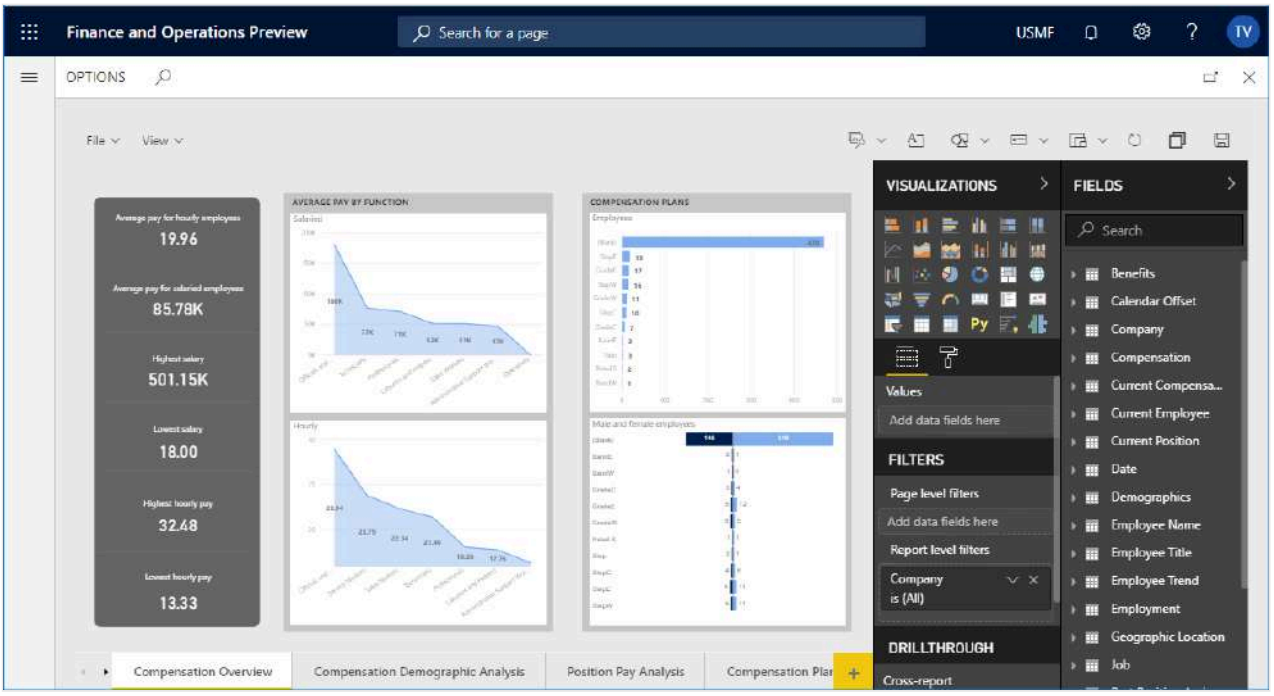
By default, you see the standard analytical workspace solution that's packaged with your application. The provisioning process automatically deploys and configures the reports in this solution for your environment.

> **Note**
>
> The analytical workspaces require a hosted Microsoft Power BI service that's available only for dedicated environments. For more information, see Accessing Analytical Workspaces and Reports on 1Box environment.

3. On the Action Pane, on the **Options** tab, in the **Power BI** group, select **Edit Analytics**.



The analytical workspace opens in edit mode, and you have direct access to the Power BI web designer tools.



4. Use the Power BI web designer tools to customize the report canvas. The intuitive web controls let you perform typical actions such as adding and removing visuals, changing visual types, and formatting the content. You can also inspect the source of the report visualizations to make sure that decisions are based on the most relevant data that's available in the system. For more information, see Add visualizations to a Power BI report.

5. After you complete your report customizations, select the **Save** button to promote the report edits. Customizations to the report are reflected immediately in the service. Therefore, users in your organization have access to the latest innovations.

> **Note**
>
> Customizations made in the web editor aren't saved back to the underlying PBIX report and aren't retrievable if the PBIX report is later customized on a development environment.

Restore the standard application solution

Follow these steps to restore the analytical workspaces that are bundled with the application solution.

1. In the analytical workspace, on the Action Pane, on the **Options** tab, in the **Power BI** group, select **Restore Analytics**.
2. To view the updates to the workspace, reload the page. Either move away from the workspace and then return, or refresh your browser.

3. In the **Compensation management** workspace, select the **Analytics** tab to access the original analytical workspace that was packaged with the application.

Troubleshooting

Follow these steps to address common issues encountered while attempting to use analytical workspaces.

Error message: *Please log into Power BI to access its resource*

The Power BI service requires explicit permission from the user to allow access to hosted content. Use the following steps to ensure the current user is able to connect to reports hosted on PowerBI.com from the application suite.

1. Open any application workspace containing a section titled **Link**. For example, "Bank management."
2. Select **Options**, and then select **Open report catalog** on the top left.
3. Follow the steps in the dialog box to **Authorize to Power BI** to access finance and operations apps for the current user.

Last updated on 11/26/2025

Document Reporting Services

This article describes the integrated reporting solutions that are available. This solution simplifies service administration, increases developer productivity, and provides an enhanced report viewing experience for users.

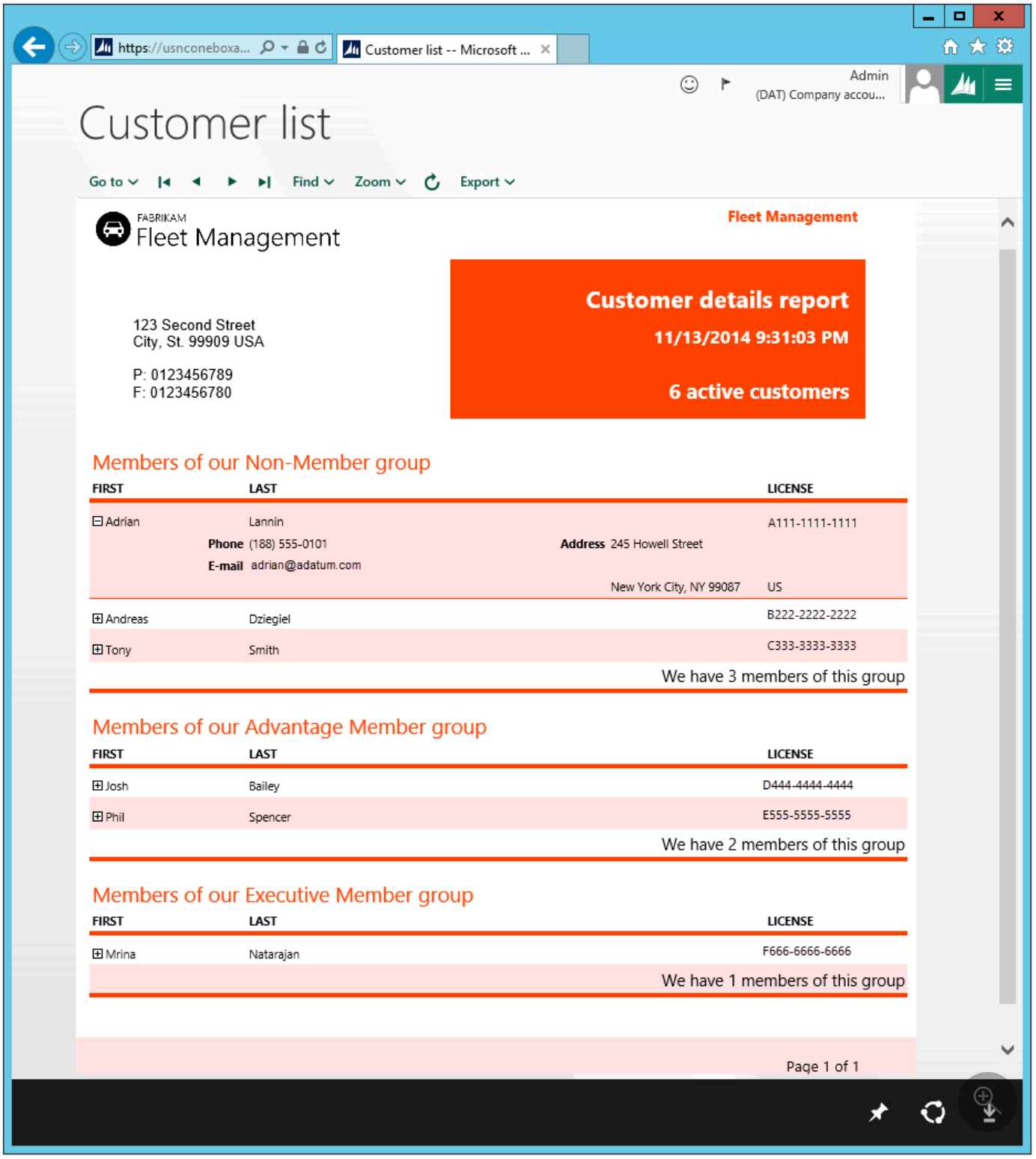
Document Reporting Services are based on Microsoft SQL Server Reporting Services (SSRS). In the current version of the application, these services are hosted in the Microsoft Azure Compute service. If you're developing in a one-box environment, the services also run locally in the Azure Compute Emulator.

Service deployment – Local vs. cloud

In a one-box environment, developers can create, modify, and preview reports, from end to end, by using Microsoft Visual Studio. You don't need a separate process to add reports to the application's metadata store. Changes to reports are packaged together with other solution updates and then deployed to the cloud after development is completed in the local environment.

Viewing reports

The enhanced report viewing experience that the solution provides for end users is the same as the report preview experience in Microsoft Visual Studio. You no longer use a separate design preview in Visual Studio. Instead, just press Ctrl+F5 to build and preview the report in a Microsoft Edge window. The report appears exactly as it would appear in the client. Even the user's parameter experience is the same. The following image shows an example of a report preview that is opened from Visual Studio.



Service administration prerequisites

The following table compares the service administration prerequisites for Microsoft Dynamics AX 2012 and the current version of the application.

Expand table

AX 2012	The current version of the application
A report development environment has the following prerequisites: • SSRS must be installed. • SSRS must be configured by using Reporting Services Configuration Manager. • SSRS extensions for the application must be installed.	Reporting services run in the Azure compute emulator, together with the application server. Therefore, there are no SSRS service administration prerequisites. After the reports are deployed to the local reporting services, you can access them from the client.

Developing application reports

Developing a report in the current version is easier than it is in AX 2012, because you can create and validate a reporting solution entirely in Visual Studio. The following table describes how the application simplifies the basic procedure for adding an automatic design report that's based on a query.

Expand table

AX 2012	The current version of the application
1. In the application, create a query in the Application Object Tree (AOT). 1. In Visual Studio, create a reporting project, and add the query to it. 1. Edit the report in the Visual Studio model editor. 1. Preview the report design in Visual Studio by using the model editor toolbar. 1. Use Visual Studio to add the report to the AOT. 1. Use the AOT in the client to create a menu item for the report, and add the menu item to a menu. 1. Use the AOT to deploy the report to the report server. 1. Verify the report in the client.	1. In Visual Studio, create a reporting project and the query. 1. Edit the report in Visual Studio. 1. In Visual Studio, add the report to a menu item, and set the menu item as a startup object. 1. Use the AOT to deploy the report to the report server. 1. Press Ctrl+F5 to verify the report in the application. **NOTE:** There's no longer a separate preview of the report design from the model editor. 1. When the whole solution is completed, deploy it to the cloud in one package.

Last updated on 06/26/2026

Supported fonts

> **Note**
>
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Finance and operations apps include access to hundreds of standard, business-ready fonts for documents rendered by the cloud-hosted service.

This portfolio continues to grow as the service expands into new regions and industries. The service doesn't support installing custom fonts in customer environments.

The following list captures the portfolio of font families available for documents produced using SQL Server Reporting Services (SSRS) services in finance and operations apps.

Supported fonts for application version 8.3 with Platform update 32 or later

Expand table

Font family	Source
Agency FB	Office font
Algerian	Office font
Arial	Finance and operations font
Arial Black	Finance and operations font
Arial Narrow	Finance and operations font
Arial Rounded MT Bold	Office font
Baskerville Old Face	Office font
Bauhaus 93	Office font
BC C128 HD Medium	Finance and operations font
BC C128 HD Narrow	Finance and operations font

Font family	Source
BC C128 HD Wide	Finance and operations font
BC C128 Medium	Finance and operations font
BC C128 Narrow	Finance and operations font
BC C128 Wide	Finance and operations font
BC C39 2 to 1 HD Medium	Finance and operations font
BC C39 2 to 1 HD Narrow	Finance and operations font
BC C39 2 to 1 HD Wide	Finance and operations font
BC C39 2 to 1 Medium	Finance and operations font
BC C39 2 to 1 Narrow	Finance and operations font
BC C39 2 to 1 Wide	Finance and operations font
BC C39 3 to 1 HD Medium	Finance and operations font
BC C39 3 to 1 HD Narrow	Finance and operations font
BC C39 3 to 1 HD Wide	Finance and operations font
BC C39 3 to 1 Medium	Finance and operations font
BC C39 3 to 1 Narrow	Finance and operations font
BC C39 3 to 1 Wide	Finance and operations font
BC CBar 2 to 1 HD Medium	Finance and operations font
BC CBar 2 to 1 HD Narrow	Finance and operations font
BC CBar 2 to 1 HD Wide	Finance and operations font
BC CBar 2 to 1 Medium	Finance and operations font
BC CBar 2 to 1 Narrow	Finance and operations font
BC CBar 2 to 1 Wide	Finance and operations font
BC CBar 3 to 1 HD Medium	Finance and operations font
BC CBar 3 to 1 HD Narrow	Finance and operations font
BC CBar 3 to 1 HD Wide	Finance and operations font
BC CBar 3 to 1 Medium	Finance and operations font

Font family	Source
BC CBar 3 to 1 Narrow	Finance and operations font
BC CBar 3 to 1 Wide	Finance and operations font
BC I25 HD Medium	Finance and operations font
BC I25 HD Narrow	Finance and operations font
BC I25 HD Wide	Finance and operations font
BC I25 Medium	Finance and operations font
BC I25 Narrow	Finance and operations font
BC I25 Wide	Finance and operations font
BC Postnet	Finance and operations font
BC UPC HD Medium	Finance and operations font
BC UPC HD Narrow	Finance and operations font
BC UPC HD Wide	Finance and operations font
BC UPC Medium	Finance and operations font
BC UPC Narrow	Finance and operations font
BC UPC Wide	Finance and operations font
Bell MT	Office font
Berlin Sans FB	Office font
Berlin Sans FB Demi	Office font
Bernard MT Condensed	Office font
Blackadder ITC	Office font
Bodoni MT	Office font
Bodoni MT Black	Office font
Bodoni MT Condensed	Office font
Bodoni MT Poster Compres	Office font
Book Antiqua	Office font
Bookman Old Style	Office font

Font family	Source
Bookshelf Symbol 7	Office font
Bradley Hand ITC	Office font
Britannic Bold	Office font
Broadway	Office font
Brush Script MT	Office font
Buxton Sketch	Office font
Calibri	Finance and operations font
Calibri Light	Finance and operations font
Californian FB	Office font
Calisto MT	Office font
Cambria	Finance and operations font
Cambria Math	Finance and operations font
Candara	Finance and operations font
Castellar	Office font
Centaur	Office font
Century	Office font
Century Gothic	Office font
Century Schoolbook	Office font
Chiller	Office font
Colonna MT	Office font
Comic Sans MS	Finance and operations font
Consolas	Finance and operations font
Constantia	Finance and operations font
Cooper Black	Office font
Copperplate Gothic Bold	Office font
Copperplate Gothic Light	Office font

Font family	Source
Corbel	Finance and operations font
Courier New	Finance and operations font
Curlz MT	Office font
DengXian	Office font
Dotum	Finance and operations font
DotumChe	Finance and operations font
Ebrima	Finance and operations font
Edwardian Script ITC	Office font
Elephant	Office font
Engravers MT	Office font
Eras Bold ITC	Office font
Eras Demi ITC	Office font
Eras Light ITC	Office font
Eras Medium ITC	Office font
Felix Titling	Office font
Footlight MT Light	Office font
Forte	Office font
Franklin Gothic Book	Office font
Franklin Gothic Demi	Office font
Franklin Gothic Demi Con	Office font
Franklin Gothic Heavy	Office font
Franklin Gothic Medium	Finance and operations font
Franklin Gothic Medium C	Office font
Freestyle Script	Office font
French Script MT	Office font
Gabriola	Finance and operations font

Font family	Source
Gadugi	Finance and operations font
Garamond	Office font
Georgia	Finance and operations font
Gigi	Office font
Gill Sans MT	Office font
Gill Sans MT Condensed	Office font
Gill Sans MT Ext Condens	Office font
Gill Sans Ultra Bold	Office font
Gill Sans Ultra Bold Con	Office font
Gloucester MT Extra Cond	Office font
Goudy Old Style	Office font
Goudy Stout	Office font
Gulim	Finance and operations font
GulimChe	Finance and operations font
Haettenschweiler	Office font
Harlow Solid Italic	Office font
Harrington	Office font
High Tower Text	Office font
IDAutomationMICR	Finance and operations font
Impact	Finance and operations font
Imprint MT Shadow	Office font
Informal Roman	Office font
Javanese Text	Finance and operations font
Jokerman	Office font
Juice ITC	Office font
Kristen ITC	Office font

Font family	Source
Kunstler Script	Office font
Leelawadee	Office font
Leelawadee UI	Finance and operations font
Leelawadee UI Semilight	Finance and operations font
Lucida Bright	Office font
Lucida Calligraphy	Office font
Lucida Console	Finance and operations font
Lucida Fax	Office font
Lucida Handwriting	Office font
Lucida Sans	Office font
Lucida Sans Typewriter	Office font
Lucida Sans Unicode	Finance and operations font
Magneto	Office font
Maiandra GD	Office font
Malgun Gothic	Finance and operations font
Malgun Gothic Semilight	Finance and operations font
Marlett	Finance and operations font
Matura MT Script Capital	Office font
MICR E13B 2.1	Finance and operations font
Microsoft Himalaya	Finance and operations font
Microsoft JhengHei	Finance and operations font
Microsoft JhengHei Light	Finance and operations font
Microsoft JhengHei UI	Finance and operations font
Microsoft JhengHei UI Li	Office font
Microsoft MHei	Office font
Microsoft NeoGothic	Office font

Font family	Source
Microsoft New Tai Lue	Finance and operations font
Microsoft PhagsPa	Finance and operations font
Microsoft Sans Serif	Finance and operations font
Microsoft Tai Le	Finance and operations font
Microsoft Uighur	Office font
Microsoft YaHei	Finance and operations font
Microsoft YaHei Light	Finance and operations font
Microsoft YaHei UI	Finance and operations font
Microsoft YaHei UI Light	Finance and operations font
Microsoft Yi Baiti	Finance and operations font
MingLiU	Finance and operations font
MingLiU-ExtB	Finance and operations font
MingLiU_HKSCS	Finance and operations font
MingLiU_HKSCS-ExtB	Finance and operations font
Mistral	Office font
Modern No. 20	Office font
Mongolian Baiti	Finance and operations font
Monotype Corsiva	Office font
MS Gothic	Finance and operations font
MS Outlook	Office font
MS PGothic	Finance and operations font
MS Reference Sans Serif	Office font
MS Reference Specialty	Office font
MS UI Gothic	Finance and operations font
MT Extra	Office font
MV Boli	Finance and operations font

Font family	Source
Myanmar Text	Finance and operations font
Niagara Engraved	Office font
Niagara Solid	Office font
Nirmala UI	Finance and operations font
Nirmala UI Semilight	Finance and operations font
NSimSun	Finance and operations font
OCR A Extended	Office font
OCRB	Finance and operations font
Old English Text MT	Office font
Onyx	Office font
Palace Script MT	Office font
Palatino Linotype	Finance and operations font
Papyrus	Office font
Parchment	Office font
Perpetua	Office font
Perpetua Titling MT	Office font
Playbill	Office font
PMingLiU	Finance and operations font
PMingLiU-ExtB	Finance and operations font
Poor Richard	Office font
Pristina	Office font
Rage Italic	Office font
Ravie	Office font
Rockwell	Office font
Rockwell Condensed	Office font
Rockwell Extra Bold	Office font

Font family	Source
Script MT Bold	Office font
Segoe Marker	Office font
Segoe MDL2 Assets	Finance and operations font
Segoe Print	Finance and operations font
Segoe Script	Finance and operations font
Segoe UI	Finance and operations font
Segoe UI Black	Finance and operations font
Segoe UI Emoji	Finance and operations font
Segoe UI Historic	Finance and operations font
Segoe UI Light	Finance and operations font
Segoe UI Semibold	Finance and operations font
Segoe UI Semilight	Finance and operations font
Segoe UI Symbol	Finance and operations font
Segoe WP	Office font
Segoe WP Black	Office font
Segoe WP Light	Office font
Segoe WP Semibold	Office font
Segoe WP SemiLight	Office font
Showcard Gothic	Office font
SimSun	Finance and operations font
SimSun-ExtB	Finance and operations font
Sitka Banner	Finance and operations font
Sitka Display	Finance and operations font
Sitka Heading	Finance and operations font
Sitka Small	Finance and operations font
Sitka Subheading	Finance and operations font

Font family	Source
Sitka Text	Finance and operations font
SketchFlow Print	Office font
Snap ITC	Office font
Stencil	Office font
Sylfaen	Finance and operations font
Symbol	Finance and operations font
Tahoma	Finance and operations font
Tempus Sans ITC	Office font
Times New Roman	Finance and operations font
Trebuchet MS	Finance and operations font
Tw Cen MT	Office font
Tw Cen MT Condensed	Office font
Tw Cen MT Condensed Extr	Office font
Verdana	Finance and operations font
Viner Hand ITC	Office font
Vivaldi	Office font
Vladimir Script	Office font
Webdings	Finance and operations font
Wide Latin	Office font
Wingdings	Finance and operations font
Wingdings 2	Office font
Wingdings 3	Office font
Yu Gothic	Finance and operations font
Yu Gothic Light	Finance and operations font
Yu Gothic Medium	Finance and operations font
Yu Gothic UI	Finance and operations font

Font family	Source
Yu Gothic UI Light	Finance and operations font
Yu Gothic UI Semibold	Finance and operations font
Yu Gothic UI Semilight	Finance and operations font

Supported fonts for application version 8.0 with Platform Update 23

Expand table

Number	Font family
1	Agency FB
2	Algerian
3	Arial
4	Arial Black
5	Arial Narrow
6	Arial Rounded MT Bold
7	Baskerville Old Face
8	Bauhaus 93
9	BC C128 HD Medium
10	BC C128 HD Narrow
11	BC C128 HD Wide
12	BC C128 Medium
13	BC C128 Narrow
14	BC C128 Wide
15	BC C39 2 to 1 HD Medium
16	BC C39 2 to 1 HD Narrow
17	BC C39 2 to 1 HD Wide
18	BC C39 2 to 1 Medium

Number	Font family
19	BC C39 2 to 1 Narrow
20	BC C39 2 to 1 Wide
21	BC C39 3 to 1 HD Medium
22	BC C39 3 to 1 HD Narrow
23	BC C39 3 to 1 HD Wide
24	BC C39 3 to 1 Medium
25	BC C39 3 to 1 Narrow
26	BC C39 3 to 1 Wide
27	BC CBar 2 to 1 HD Medium
28	BC CBar 2 to 1 HD Narrow
29	BC CBar 2 to 1 HD Wide
30	BC CBar 2 to 1 Medium
31	BC CBar 2 to 1 Narrow
32	BC CBar 2 to 1 Wide
33	BC CBar 3 to 1 HD Medium
34	BC CBar 3 to 1 HD Narrow
35	BC CBar 3 to 1 HD Wide
36	BC CBar 3 to 1 Medium
37	BC CBar 3 to 1 Narrow
38	BC CBar 3 to 1 Wide
39	BC I25 HD Medium
40	BC I25 HD Narrow
41	BC I25 HD Wide
42	BC I25 Medium
43	BC I25 Narrow
44	BC I25 Wide

Number	Font family
45	BC Postnet
46	BC UPC HD Medium
47	BC UPC HD Narrow
48	BC UPC HD Wide
49	BC UPC Medium
50	BC UPC Narrow
51	BC UPC Wide
52	Bell MT
53	Berlin Sans FB
54	Berlin Sans FB Demi
55	Bernard MT Condensed
56	Blackadder ITC
57	Bodoni MT
58	Bodoni MT Black
59	Bodoni MT Condensed
60	Bodoni MT Poster Compres
61	Book Antiqua
62	Bookman Old Style
63	Bookshelf Symbol 7
64	Bradley Hand ITC
65	Britannic Bold
66	Broadway
67	Brush Script MT
68	Buxton Sketch
69	Calibri
70	Calibri Light

Number	Font family
71	Californian FB
72	Calisto MT
73	Cambria
74	Cambria Math
75	Candara
76	Castellar
77	Centaur
78	Century
79	Century Gothic
80	Century Schoolbook
81	Chiller
82	Colonna MT
83	Comic Sans MS
84	Consolas
85	Constantia
86	Cooper Black
87	Copperplate Gothic Bold
88	Copperplate Gothic Light
89	Corbel
90	Courier New
91	Curlz MT
92	DengXian
93	Dotum
94	DotumChe
95	Ebrima
96	Edwardian Script ITC

Number	Font family
97	Elephant
98	Engravers MT
99	Eras Bold ITC
100	Eras Demi ITC
101	Eras Light ITC
102	Eras Medium ITC
103	Felix Titling
104	Footlight MT Light
105	Forte
106	Franklin Gothic Book
107	Franklin Gothic Demi
108	Franklin Gothic Demi Con
109	Franklin Gothic Heavy
110	Franklin Gothic Medium
111	Franklin Gothic Medium C
112	Freestyle Script
113	French Script MT
114	Gabriola
115	Gadugi
116	Garamond
117	Georgia
118	Gigi
119	Gill Sans MT
120	Gill Sans MT Condensed
121	Gill Sans MT Ext Condens
122	Gill Sans Ultra Bold

Number	Font family
123	Gill Sans Ultra Bold Con
124	Gloucester MT Extra Cond
125	Goudy Old Style
126	Goudy Stout
127	Gulim
128	GulimChe
129	Haettenschweiler
130	Harlow Solid Italic
131	Harrington
132	High Tower Text
133	IDAutomationMICR
134	Impact
135	Imprint MT Shadow
136	Informal Roman
137	Javanese Text
138	Jokerman
139	Juice ITC
140	Kristen ITC
141	Kunstler Script
142	Leelawadee
143	Leelawadee UI
144	Leelawadee UI Semilight
145	Lucida Bright
146	Lucida Calligraphy
147	Lucida Console
148	Lucida Fax

Number	Font family
149	Lucida Handwriting
150	Lucida Sans
151	Lucida Sans Typewriter
152	Lucida Sans Unicode
153	Magneto
154	Maiandra GD
155	Malgun Gothic
156	Malgun Gothic Semilight
157	Marlett
158	Matura MT Script Capital
159	MICR E13B 2.1
160	Microsoft Himalaya
161	Microsoft JhengHei
162	Microsoft JhengHei Light
163	Microsoft JhengHei UI
164	Microsoft JhengHei UI Li
165	Microsoft MHei
166	Microsoft NeoGothic
167	Microsoft New Tai Lue
168	Microsoft PhagsPa
169	Microsoft Sans Serif
170	Microsoft Tai Le
171	Microsoft Uighur
172	Microsoft YaHei
173	Microsoft YaHei Light
174	Microsoft YaHei UI

Number	Font family
175	Microsoft YaHei UI Light
176	Microsoft Yi Baiti
177	MingLiU
178	MingLiU-ExtB
179	MingLiU_HKSCS
180	MingLiU_HKSCS-ExtB
181	Mistral
182	Modern No. 20
183	Mongolian Baiti
184	Monotype Corsiva
185	MS Gothic
186	MS Outlook
187	MS PGothic
188	MS Reference Sans Serif
189	MS Reference Specialty
190	MS UI Gothic
191	MT Extra
192	MV Boli
193	Myanmar Text
194	Niagara Engraved
195	Niagara Solid
196	Nirmala UI
197	Nirmala UI Semilight
198	NSimSun
199	OCR A Extended
200	OCRB

Number	Font family
201	Old English Text MT
202	Onyx
203	Palace Script MT
204	Palatino Linotype
205	Papyrus
206	Parchment
207	Perpetua
208	Perpetua Titling MT
209	Playbill
210	PMingLiU
211	PMingLiU-ExtB
212	Poor Richard
213	Pristina
214	Rage Italic
215	Ravie
216	Rockwell
217	Rockwell Condensed
218	Rockwell Extra Bold
219	Script MT Bold
220	Segoe Marker
221	Segoe MDL2 Assets
222	Segoe Print
223	Segoe Script
224	Segoe UI
225	Segoe UI Black
226	Segoe UI Emoji

Number	Font family
227	Segoe UI Historic
228	Segoe UI Light
229	Segoe UI Semibold
230	Segoe UI Semilight
231	Segoe UI Symbol
232	Segoe WP
233	Segoe WP Black
234	Segoe WP Light
235	Segoe WP Semibold
236	Segoe WP SemiLight
237	Showcard Gothic
238	SimSun
239	SimSun-ExtB
240	Sitka Banner
241	Sitka Display
242	Sitka Heading
243	Sitka Small
244	Sitka Subheading
245	Sitka Text
246	SketchFlow Print
247	Snap ITC
248	Stencil
249	Sylfaen
250	Symbol
251	Tahoma
252	Tempus Sans ITC

Number	Font family
253	Times New Roman
254	Trebuchet MS
255	Tw Cen MT
256	Tw Cen MT Condensed
257	Tw Cen MT Condensed Extr
258	Verdana
259	Viner Hand ITC
260	Vivaldi
261	Vladimir Script
262	Webdings
263	Wide Latin
264	Wingdings
265	Wingdings 2
266	Wingdings 3
267	Yu Gothic
268	Yu Gothic Light
269	Yu Gothic Medium
270	Yu Gothic UI
271	Yu Gothic UI Light
272	Yu Gothic UI Semibold
273	Yu Gothic UI Semilight

Last updated on 10/29/2025

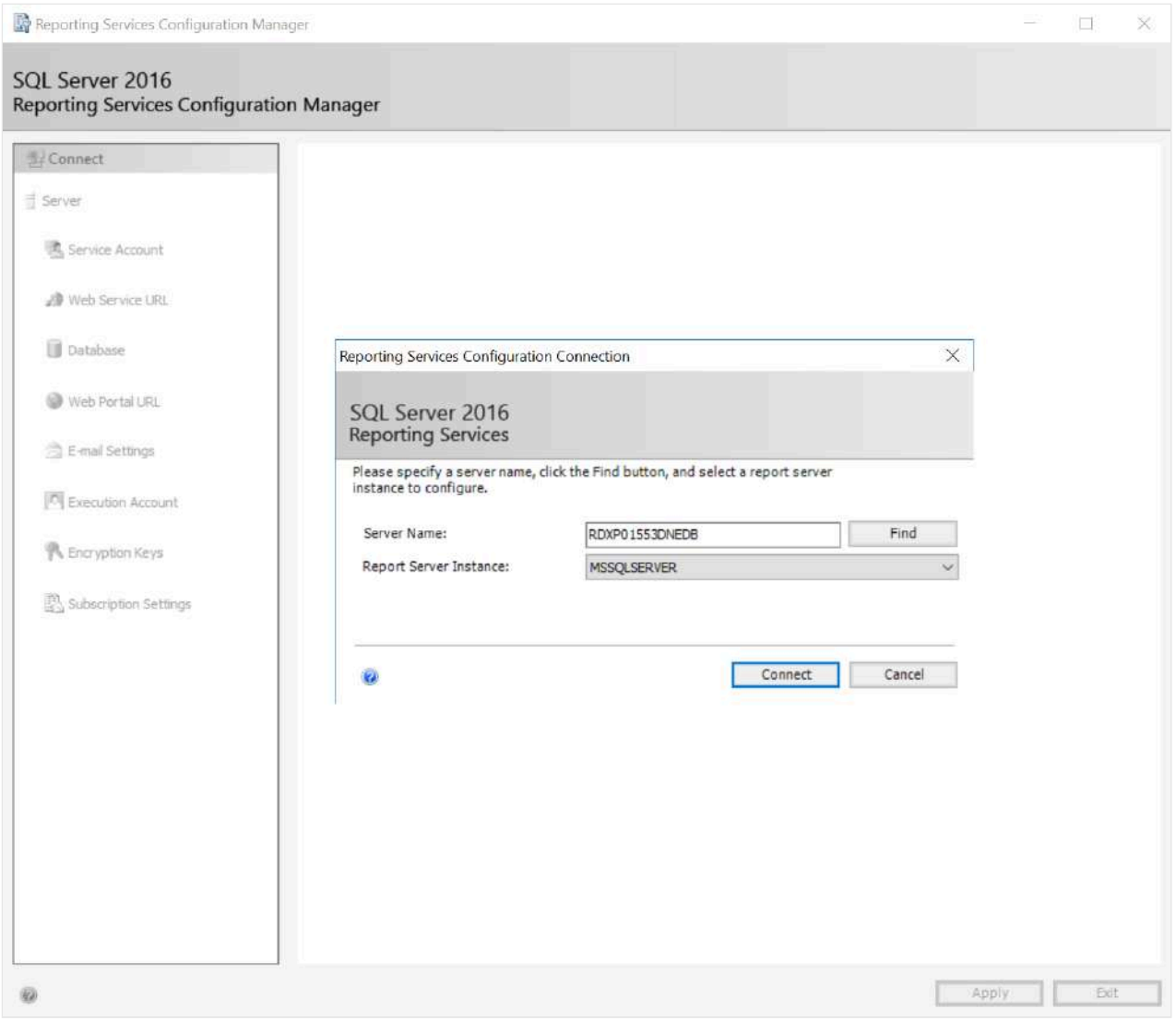
Configure SQL Server Reporting Services for on-premises deployments

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the steps in this article to configure SQL Server Reporting Services (SSRS) for your Microsoft Dynamics 365 Finance + Operations (on-premises) deployment.

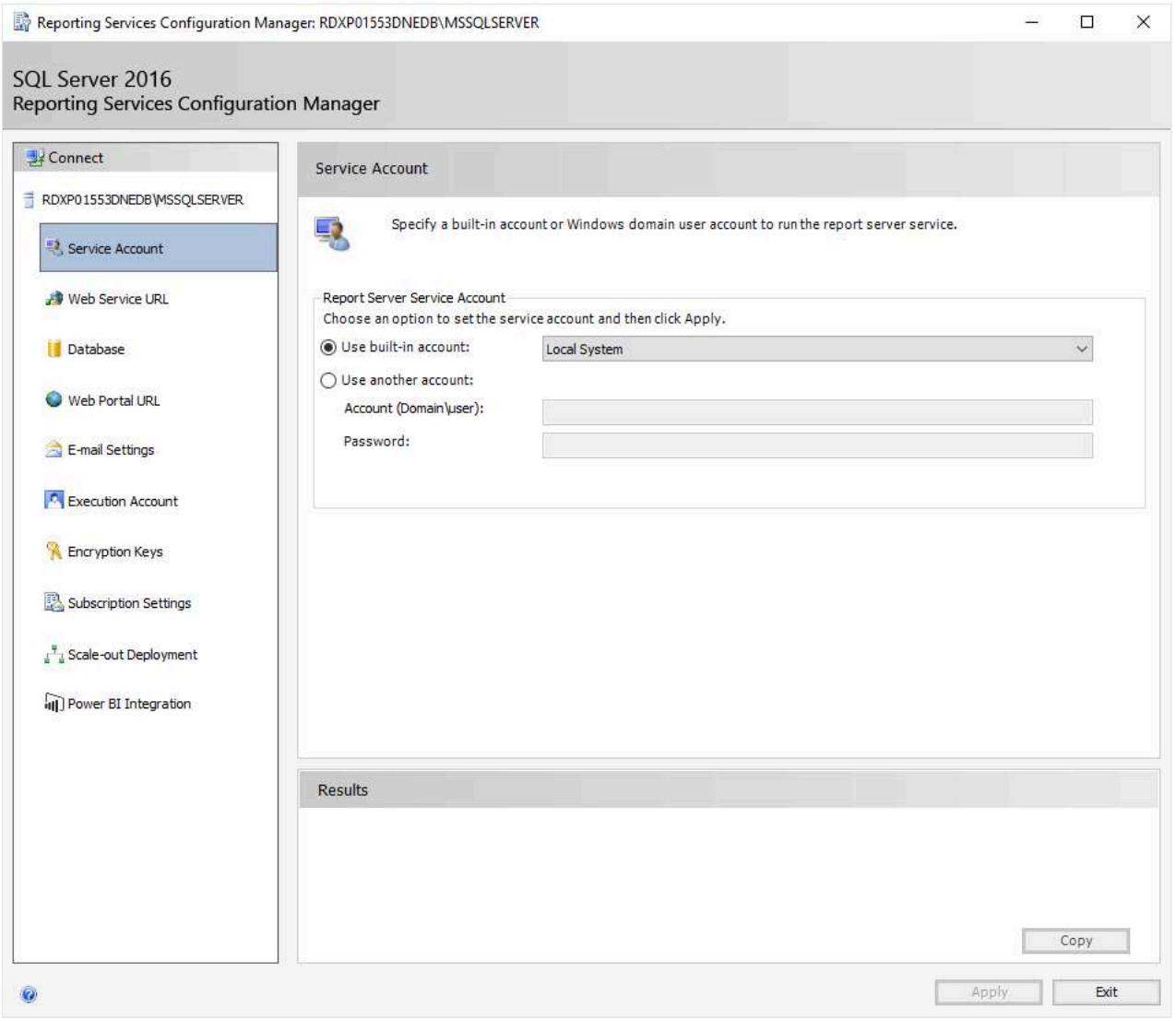
> **Important**
>
> Only follow this guide if you deployed your environment by using a base deployment older than Application version 10.0.17 (with Platform update 41).

1. Open the Reporting Services Configuration Manager application.
2. Leave the default **Server name**, which should be the name of the current machine, and the **Report Server Instance**, **MSSQLSERVER**.
3. Select **Connect**.

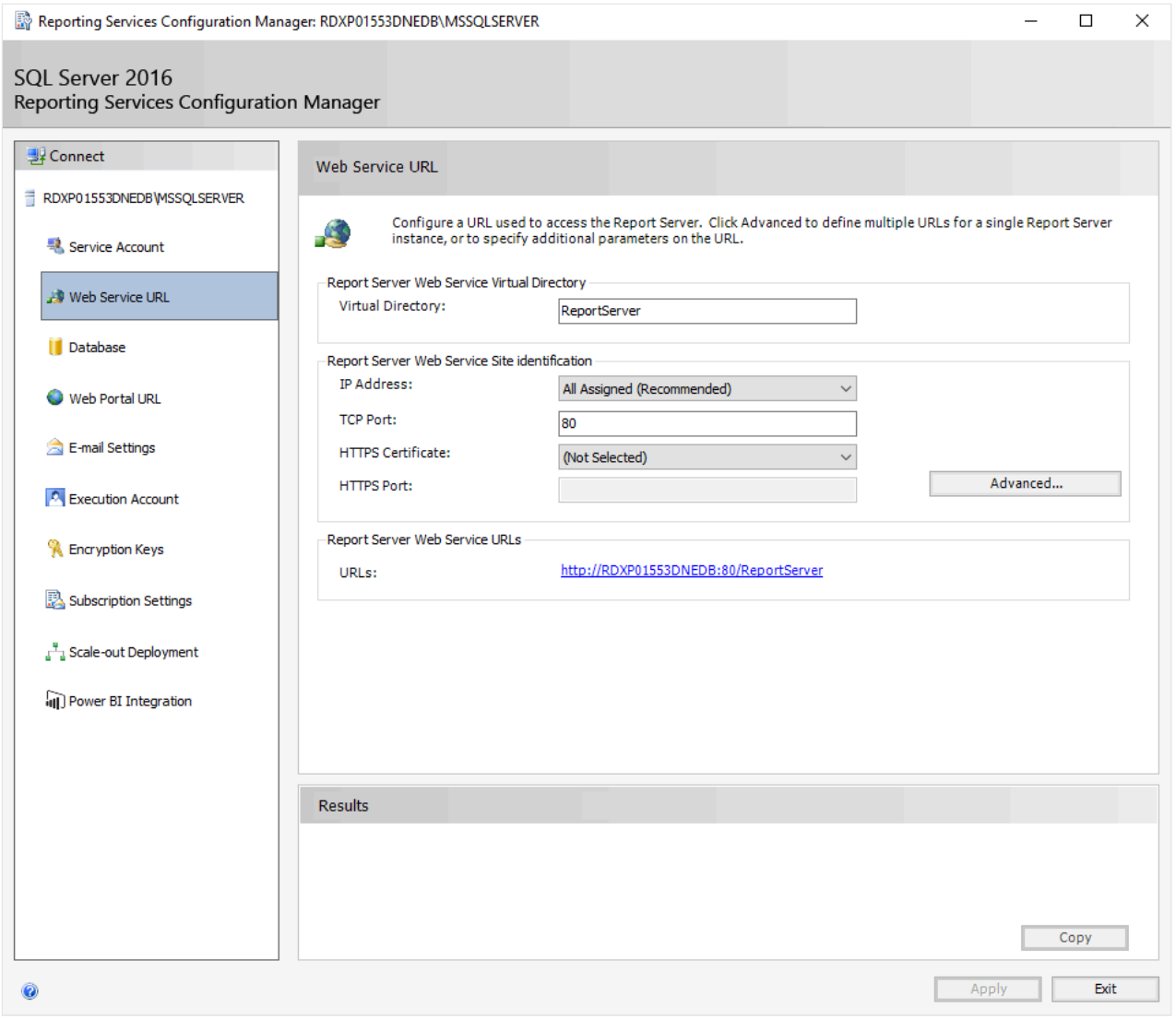


4. Select the **Service Account** tab and verify that the settings match the following graphic.

> ⓘ **Note**
>
> SQL Server Reporting Services 2019 no longer supports choosing the **Local System** account. Instead, use the **NETWORK SERVICE** account.

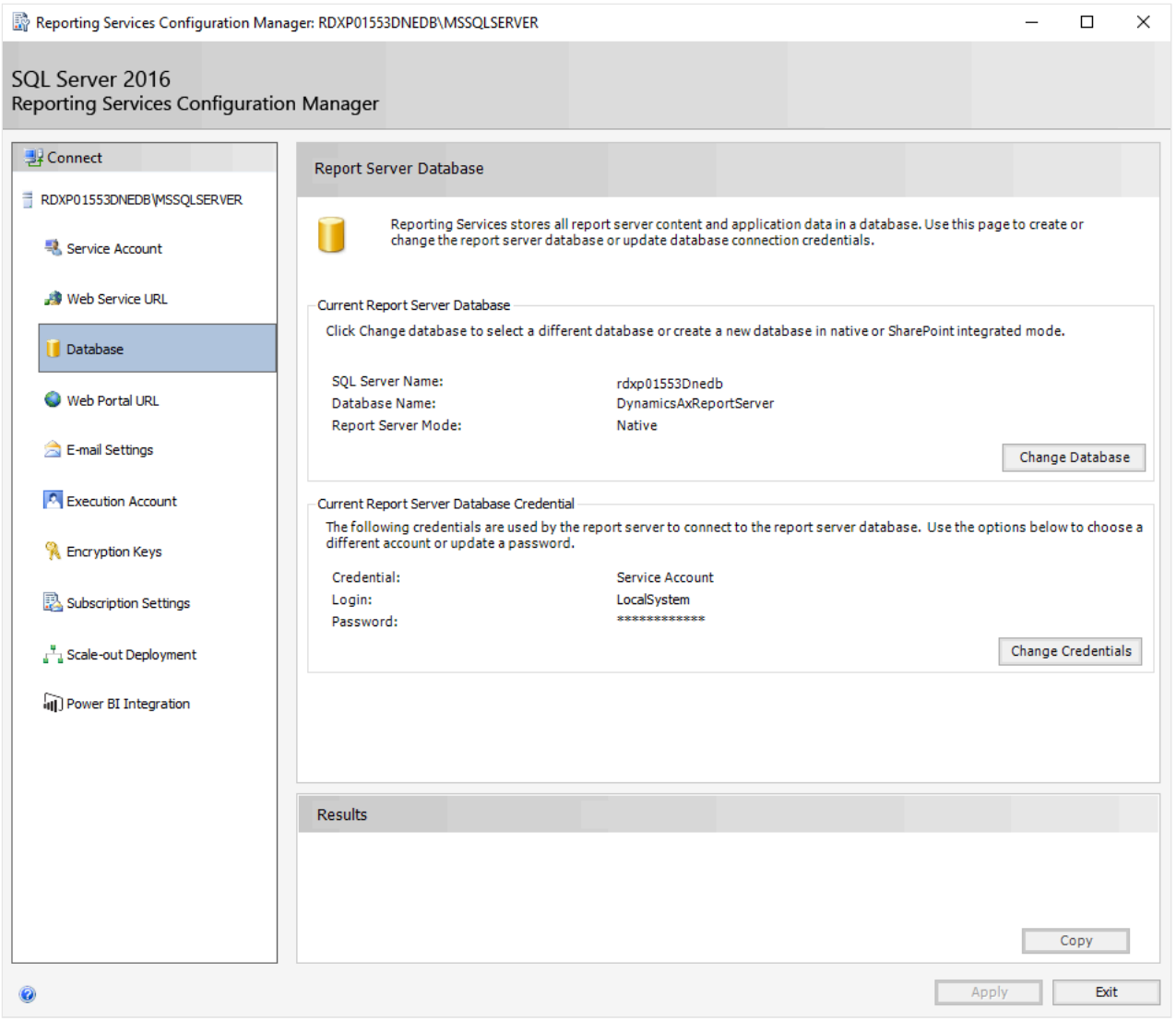


5. Select the **Web Service URL** tab and verify that the settings match the following graphic.



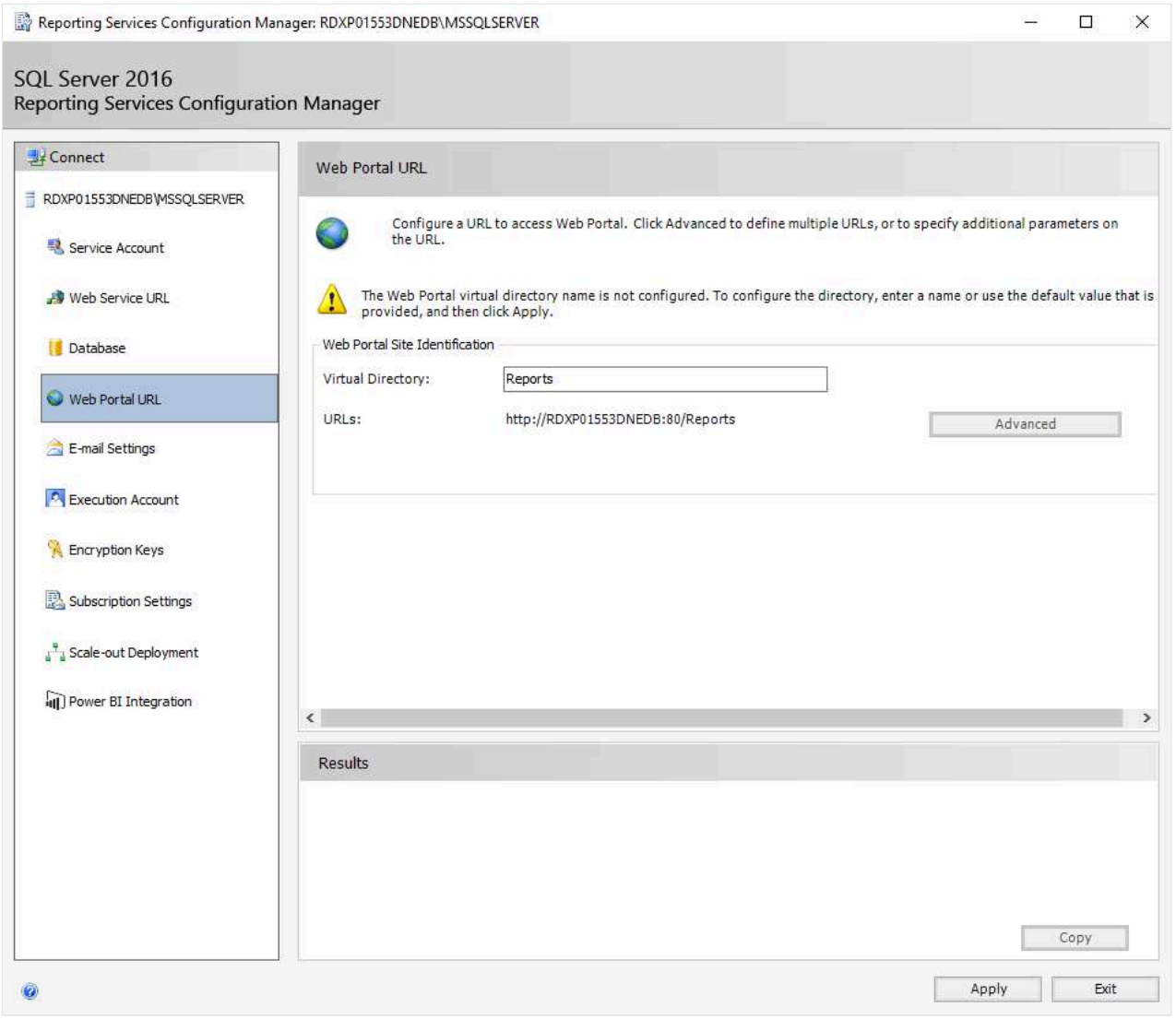
6. Select the **Database** tab and verify that the **Database Name** and **Credential settings** match the following graphic.

> **Note**
>
> You need to create a new database. To do this, select **Change Database**, and then verify that the new database name is: **DynamicsAxReportServer**.

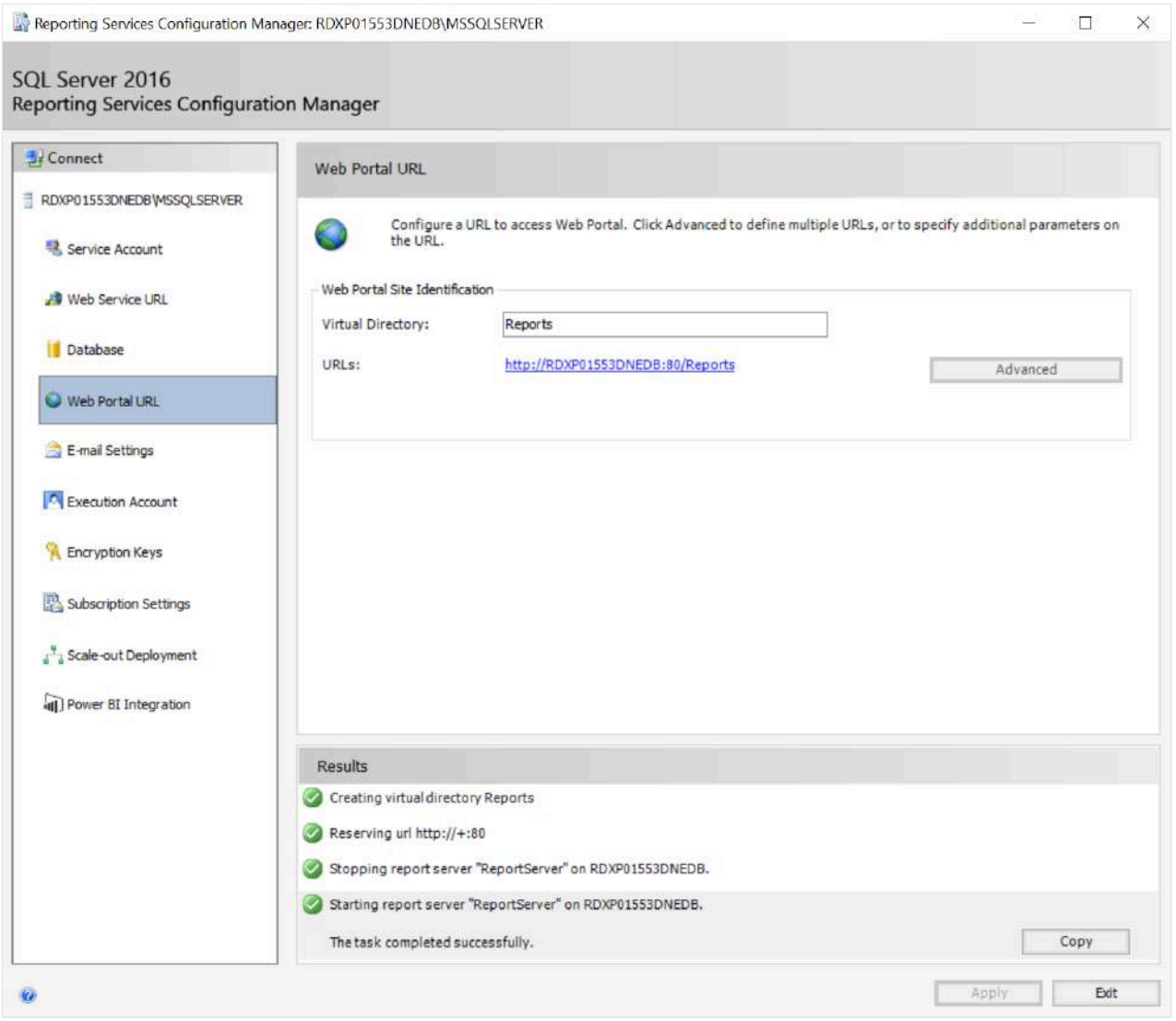


7. Select the **Web Portal URL** tab and verify that the settings match the following graphic.

> ⓘ **Note**
>
> You must select **Apply** to create and properly configure the portal.

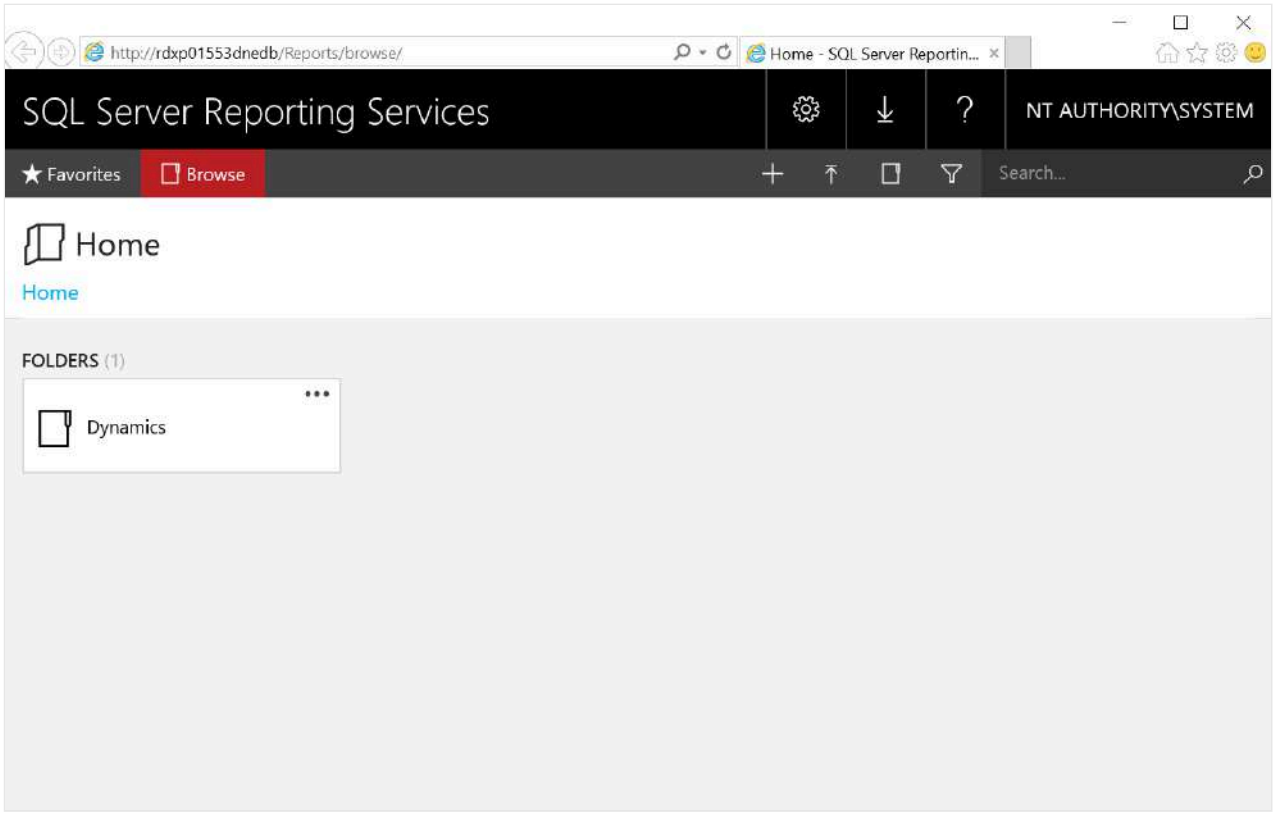


After the portal is configured, the **Web Portal** tab matches the following graphic.

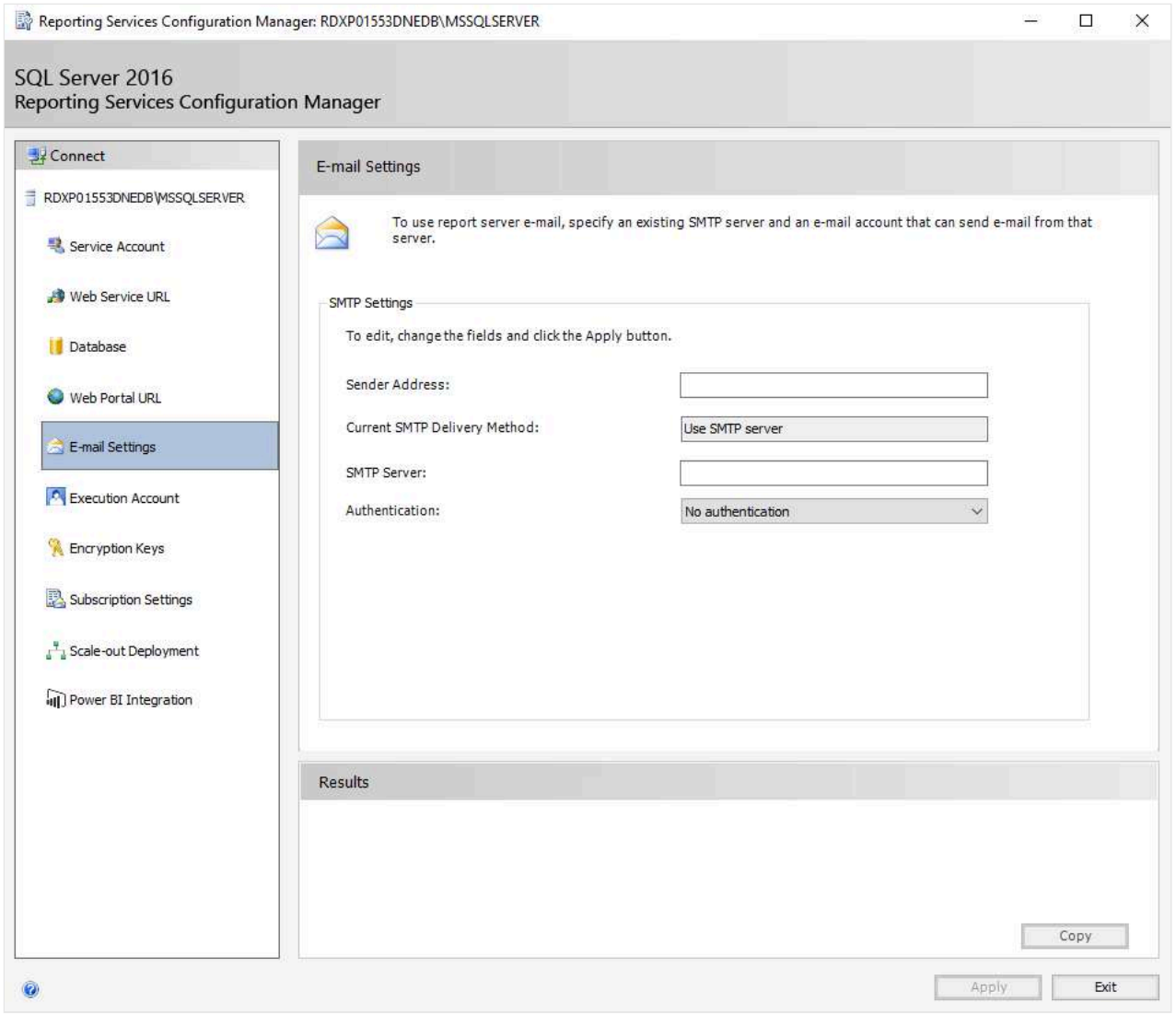


8. Select the reports URL to view the SQL Server Reporting Services web portal.

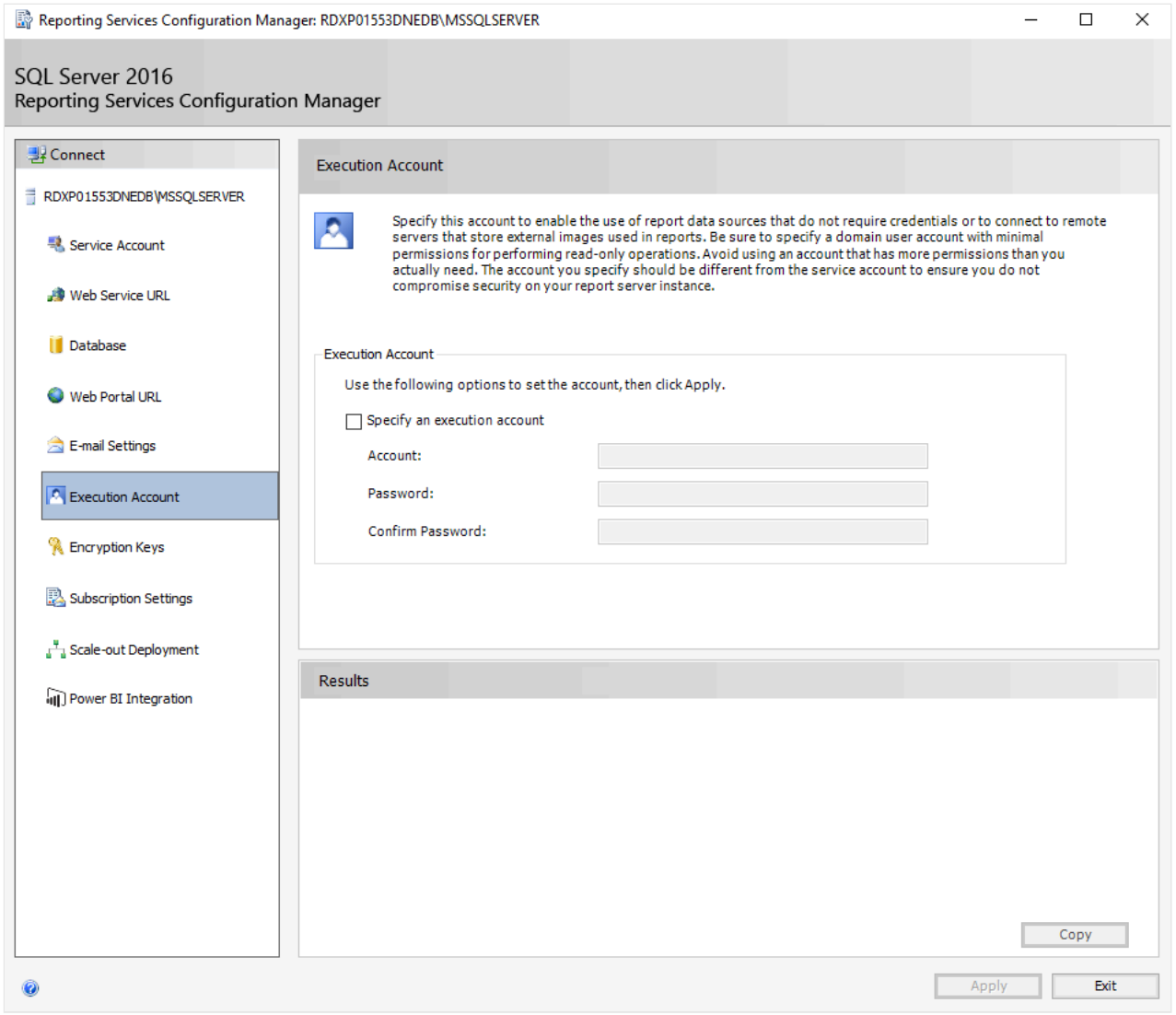
9. When you're in the portal, create a new folder named **Dynamics**.



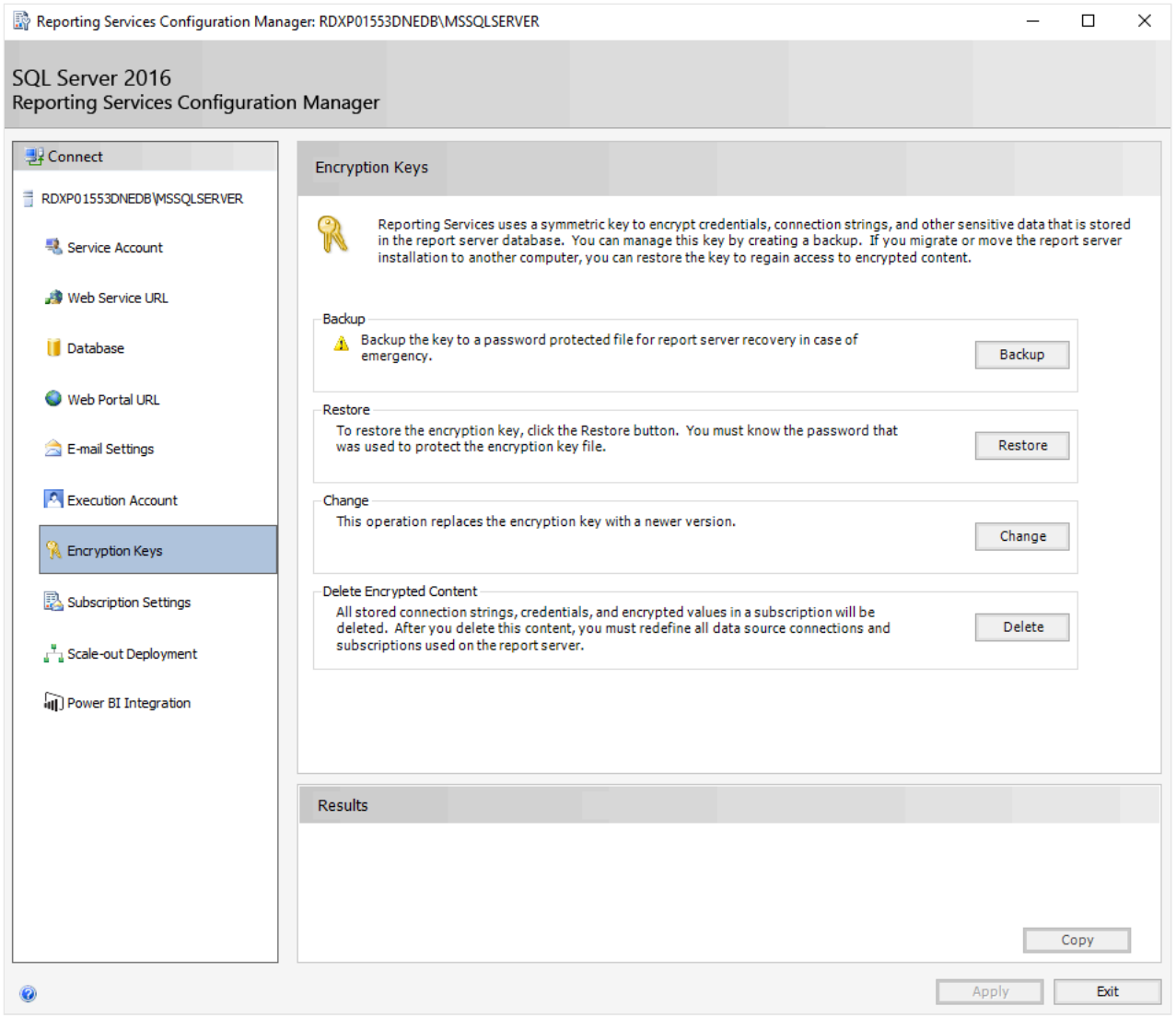
10. In the **Reporting Services Configuration Manager**, select the **E-mail Settings** tab and verify that the settings match the following graphic.



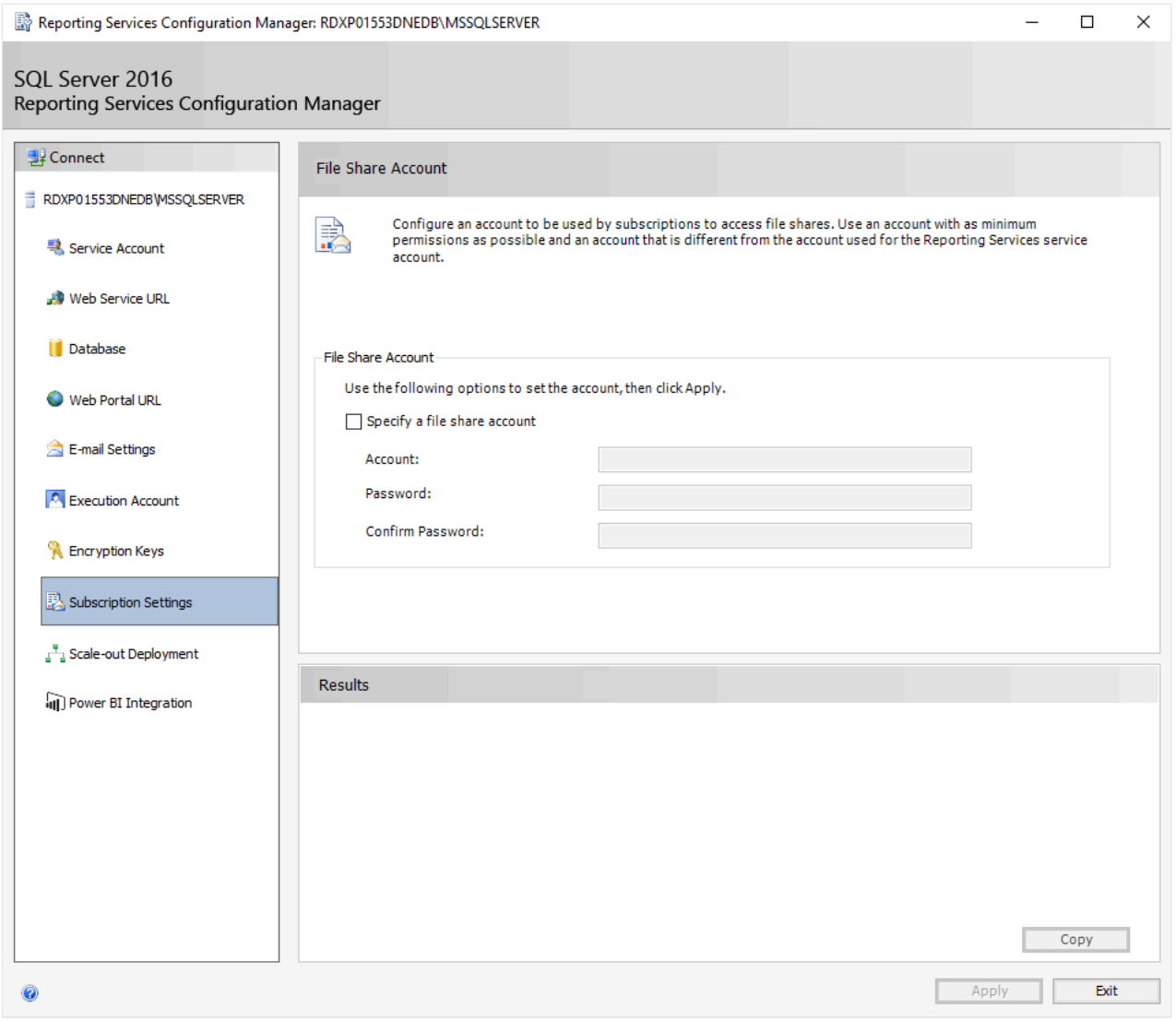
11. Select the **Execution Account** tab and verify that the settings match the following graphic.



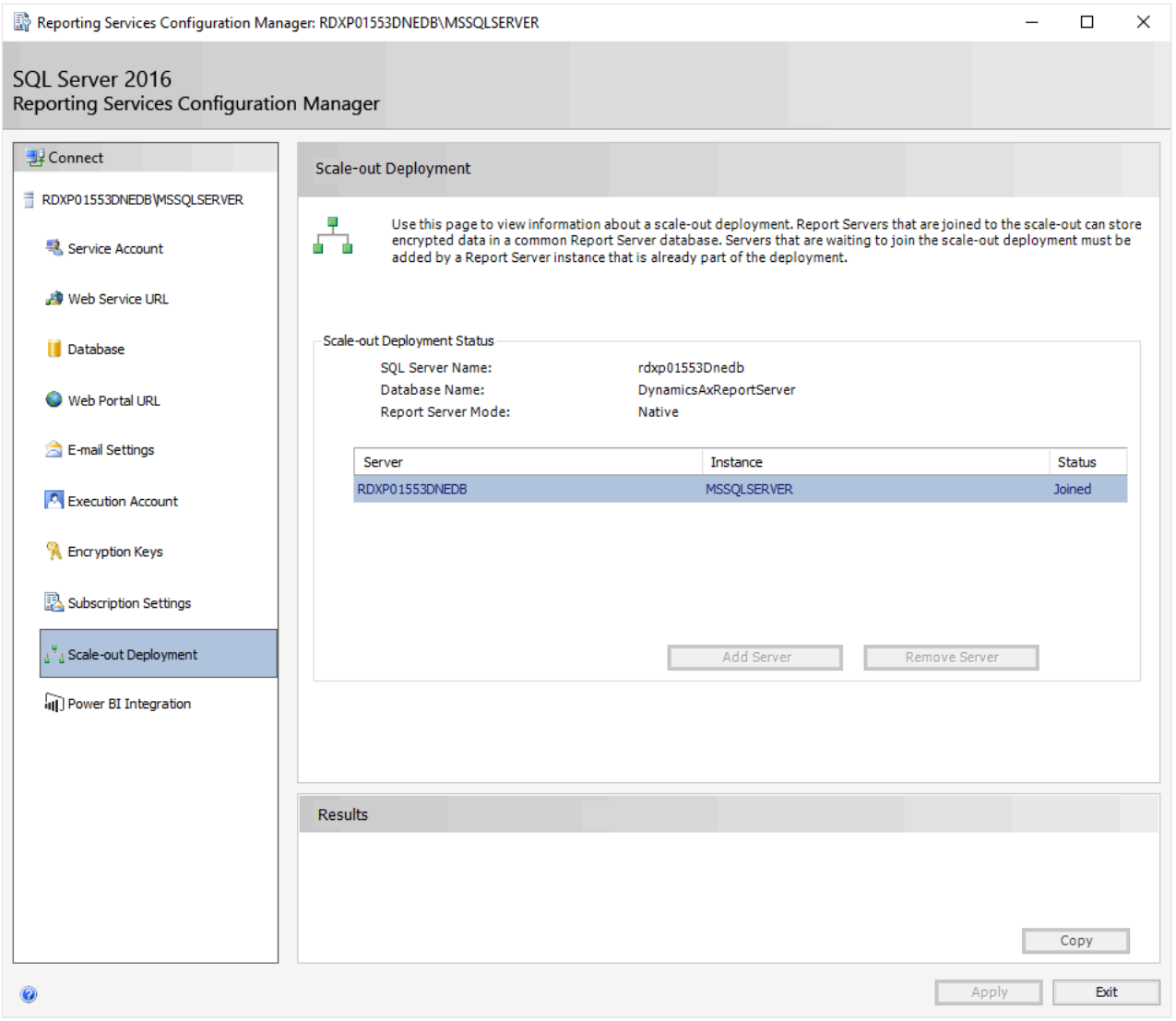
Don't change the default settings on the **Encryption Keys** tab.



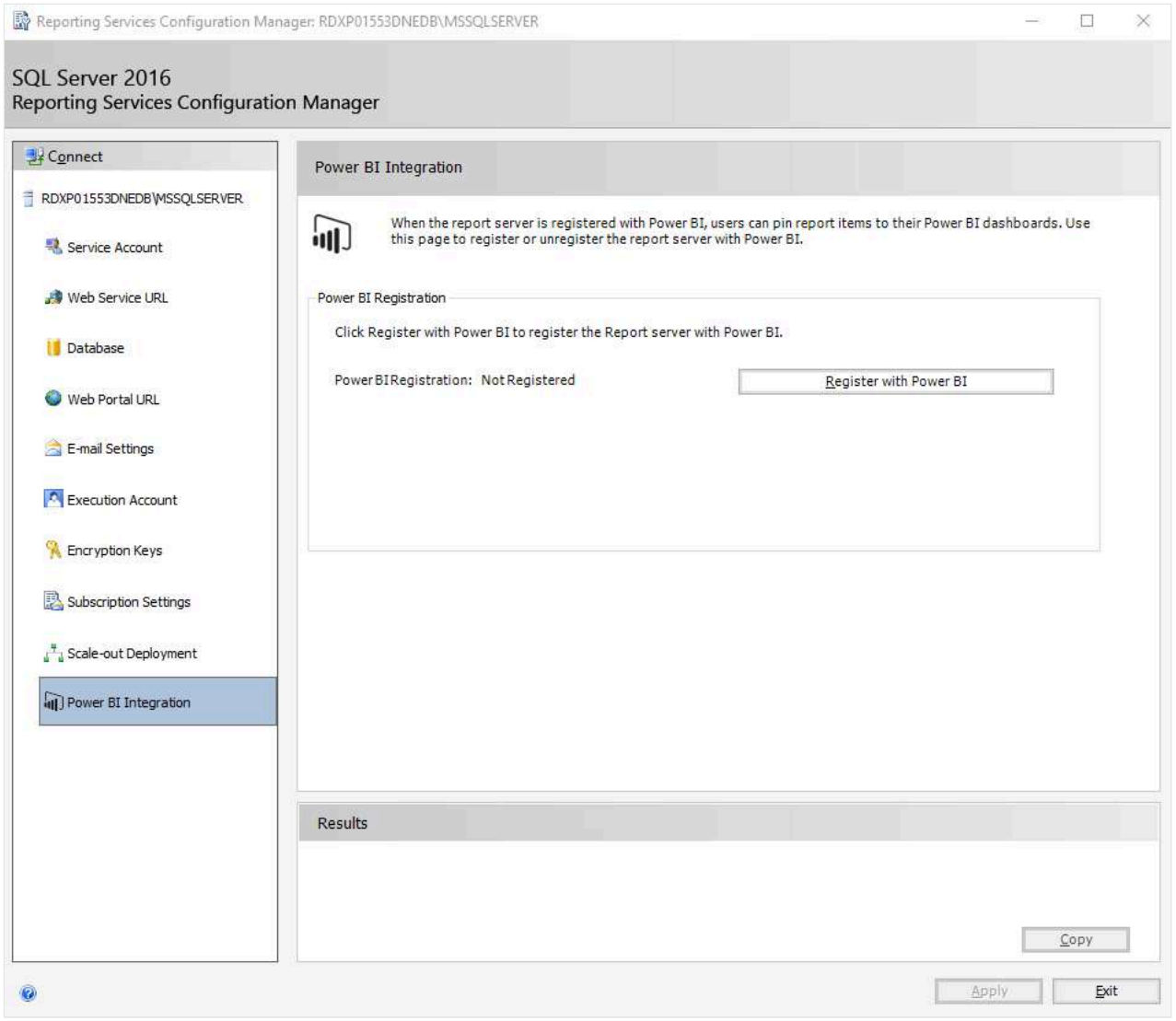
12. Select the **Subscription Settings** tab, and verify that the settings match the following graphic.



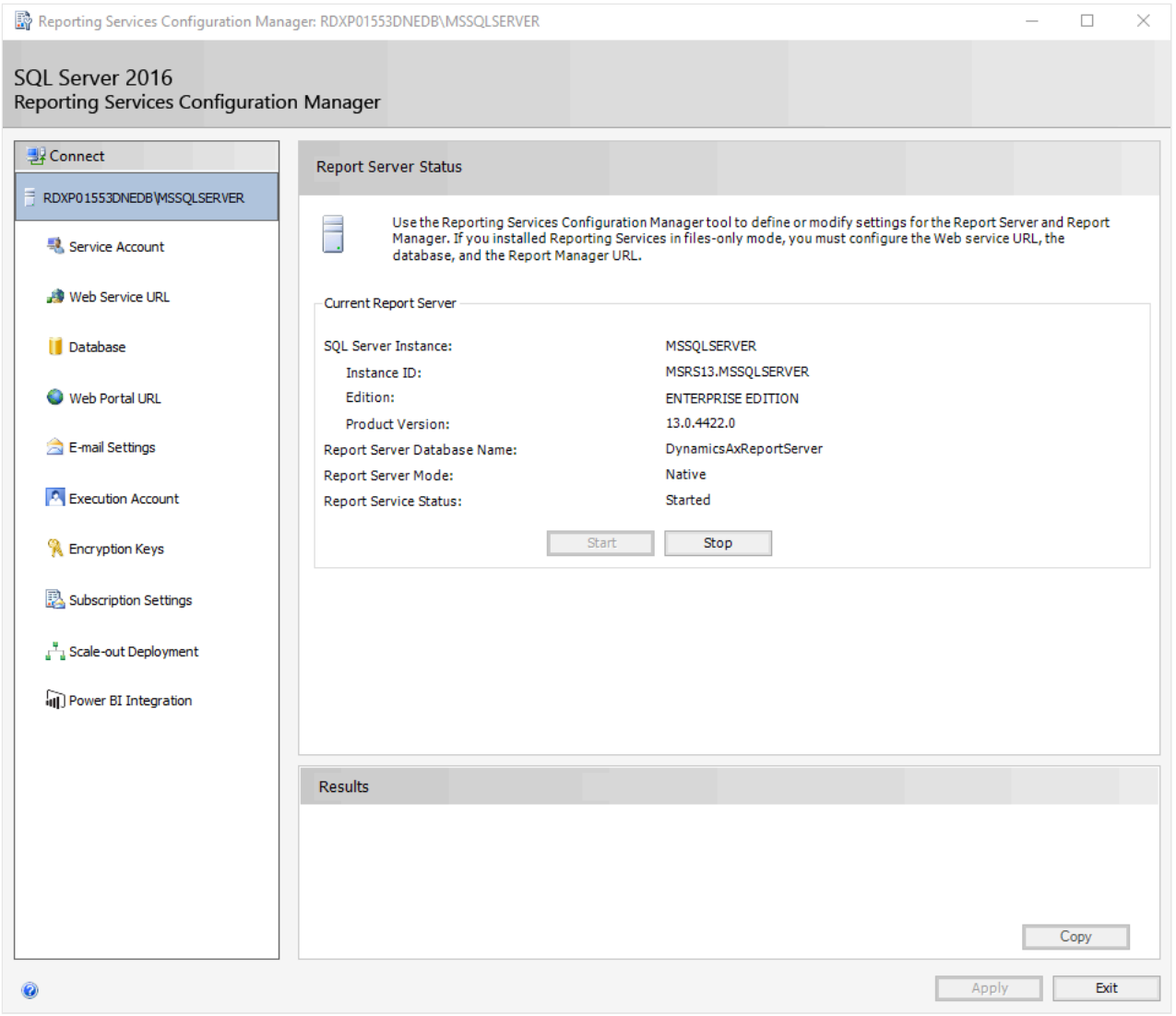
Don't change the default settings on the **Scale-out Deployment** tab.



Don't change the default settings on the **Power BI Integration** tab.



13. Select **Exit** to close the **Reporting Services Configuration Manager**.



14. For SQL Server 2019 only, you need to manually grant **NETWORK SERVICE** the permissions to read the private keys of these certificates:

    - DataEncryption
    - DataSigning
    - ReportingService

Last updated on 04/13/2026

Reduce long-running transactions in TempDB preprocessing reports

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to reduce long-running transactions and tempDB transaction log pressure caused by preprocessing SSRS reports that use the TempDB pattern, by opting out of the reporting framework's transaction scope.

This article is for anyone who maintains custom SSRS reports and whose Report Data Provider (RDP) class extends `SrsReportDataProviderPreProcessTempDB`.

Applies to

Platform Update 55 and later. Every currently supported environment includes the fix.

Background

Reports that use the TempDB preprocessing pattern stage their results in temporary database tables during a preprocessing step. By default, the reporting framework wraps the entire preprocessing step in a single transaction scope and commits only after preprocessing finishes.

On cloud deployments backed by Azure SQL, the tempDB transaction log is a shared resource. While a long-lived transaction stays open, log space can't be released. For reports that process large data sets, this behavior can hold a transaction open for the full duration of preprocessing, delay log truncation, and in extreme cases, fill the tempDB log and impact other workloads on the same database.

What this capability changes

The framework exposes an opt-in switch (`parmSkipReportTransaction`) that lets a report data provider skip the framework's transaction scope. When a report opts in:

- The framework no longer opens transactions around your `processReport` method.
- Each statement you execute commits independently.
- Transaction log space is released as work progresses, instead of being held until the end of preprocessing.

This change isn't a runtime optimization for the report itself. The report does the same amount of work in roughly the same amount of time. What changes is the transaction boundary, which improves the health of the shared environment.

When to use it

For any report that extends `SrsReportDataProviderPreProcessTempDB` and processes meaningful data volumes, **opting in is the recommended default**. The legacy behavior (a framework-managed transaction wrapping the entire preprocessing step) exists for historical reasons, not because it's the better design. Holding a single transaction open during preprocessing increases tempDB log pressure and can contend with other operations running in your environment at the same time: batch jobs, other reports, and interactive users.

The only valid reason to keep the legacy behavior is when your preprocessing step has a genuine business requirement for atomic rollback of the entire step, and the work can't be restructured into explicit transaction boundaries. This situation is uncommon. "It's easier to leave it as-is" isn't a valid reason. The transaction management you take on is straightforward and is fully covered in the next section.

How to opt in

Override `parmSkipReportTransaction` in your report data provider so it returns `true`. The recommended pattern mirrors the platform team's reference implementation:

X++

```
public class MyLargeReportDP extends SrsReportDataProviderPreProcessTempDB
{
    public boolean parmSkipReportTransaction(boolean _skipReportTransaction =
skipReportTransaction)
    {
        skipReportTransaction = true;
        return skipReportTransaction;
    }
```

```
    public void processReport()
    {
        // Your preprocessing logic.
        // See "Manage your own transactions" below.
    }
}
```

The override is read at run time, so the change is scoped to the report data providers you explicitly opt in to. All other reports continue to behave as before.

Important: manage your own transactions

When you opt in, the framework removes **two** transactions that previously surrounded your `processReport` call:

Expand table

Transaction	What it covered	What replaces it when you opt in
Ambient transaction (`ttsbegin` / `ttscommit`)	Regular table operations in `processReport`	You wrap them yourself with `ttsbegin;` / `ttscommit;` where needed.
User connection transaction (`userConnection.ttsBegin()` / `ttsCommit()`)	Operations against tables attached to the report's user connection (the temp tables)	You wrap them yourself with `parmUserConnection().ttsBegin()` / `ttsCommit()`.

The most common mistake is updating or deleting a user-connection-bound record without an open transaction on that connection. You see the following error message:

> Cannot select a record for update when the transaction is not started on the user connection attached. You need to begin transaction on the user connection first.

A plain `ttsbegin;` doesn't resolve this issue. The data tables are on the user connection, not the ambient connection. Begin the transaction on the user connection explicitly:

X++

```
public void processReport()
{
    UserConnection userConnection = this.parmUserConnection();

    // Wrap operations that update or delete user-connection-bound records.
    userConnection.ttsBegin();
    // selectForUpdate / update() / delete() against temp tables
```

```
    userConnection.ttsCommit();

    // Wrap operations against regular (non-temp) tables if they need atomicity.
    ttsbegin;
    // updates against regular tables
    ttscommit;
}
```

Guidelines:

- Prefer set-based operations (`insert_recordset`, `update_recordset`, `delete_from`). They minimize the time any transaction stays open.
- Watch for silent row-by-row downgrades. A set-based statement that is internally downgraded to row-by-row execution (for example, because a data event handler is registered on the table) reintroduces the same long-running pattern you were trying to avoid. Where possible, restructure the code or remove the cause of the downgrade.
- Keep each explicit transaction as small as possible. Wrapping the whole `processReport` body defeats the purpose of opting in.

How to opt out

Opt-out is the default. To revert a report you previously opted in, change the override to return `false` or remove the override entirely, and redeploy.

Things to keep in mind

- **No automatic rollback.** A failure partway through preprocessing no longer rolls back the whole step. If a section needs all-or-nothing behavior, define an explicit transaction boundary around it (on the right connection).
- **Review customizations.** Code paths that update or delete records during preprocessing are the most common source of the user-connection error. Audit them before enabling the switch.
- **Validate report output.** Confirm the report produces the same output as before, particularly in failure and cancellation scenarios.

Recommended rollout

Roll out the change gradually so you can isolate and revert any individual report quickly.

1. **Development / test.** Opt in one report at a time. Exercise the update/delete paths and any failure paths. Confirm the report produces correct output.

2. **Sandbox (UAT) with production-like data.** Validate correctness at realistic scale and confirm that the long-running transaction is gone.
3. **Production.** Enable for a small set of high-impact reports first, then expand.

Common issues

Expand table

Symptom	Likely cause	Action
"Can't select a record for update when the transaction isn't started on the user connection attached."	An update or delete on a user-connection-bound table is no longer inside a transaction.	Wrap the affected statements in `parmUserConnection().ttsBegin()` / `ttsCommit()`. A plain `ttsbegin` doesn't resolve this issue.
Partial or inconsistent data after a failure	No explicit transaction boundary around logic that needs atomicity.	Add an explicit transaction block around the section that must be all-or-nothing, on the correct connection (ambient or user connection).
Report compiles but `parmSkipReportTransaction` is reported as unknown.	Environment is on a platform build older than PU55.	Update the environment to a current platform release.

Last updated on 06/25/2026

Document Reporting Service in Dynamics 365 applications

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Reporting scenarios in ERP

The Document Reporting Service bundled with finance and operations apps provides user tooling to facilitate two basic user functions, or generic scenarios, in enterprise resource planning (ERP): **document generation** and **business intelligence (BI) and analytics**. These generic scenarios represent the two most frequently used methods for accessing business data.

- **Document generation** – The production of structured documents, where the primary purposes are often printing and bulk email distribution.
- **BI and analytics** – Interactive visualizations that are based on aggregations and that contain embedded links to referenced data. Users rely on these visualizations to identify patterns and trends in large amounts of data.

Although the Document Reporting Service continues to be the ideal tool for bulk production of business documents, it lacks the extensibility options that power users require so that they can adapt analytical views and account for frequent changes in the business environment. At the same time, the Microsoft Power BI service is recognized as an industry leader in the delivery of powerful analytics for businesses of all sizes. Integration options for finance and operations apps and the Power BI service are available to accommodate seamless exploration of customer business data.

> **Note**
>
> When the platform updates for version 10.0.12 are released, the Dynamics 365 service discontinues support for the Report Viewer control that facilitates BI and analytics

interactions with paginated reports. Instead, reports that the Document Reporting service renders on the screen use the embedded PDF viewer. For more information about the embedded PDF viewer, see Preview PDF documents with an embedded viewer.

Comparison of report scenarios

In document generation scenarios, a simple business activity, either automated or manual, can trigger the production of between one and hundreds of documents. These documents often include graphics with commercial branding. You can distribute the documents to both internal and external recipients by using predefined delivery routing instructions.

The following table compares the fundamentals of the two reporting experiences.

Expand table

Item	Document generation	BI and analytics
Display format	PDF and Office documents	HTML
Data volume	One to hundreds of rows	Tens of thousands of rows or more
Interactive features	Printing, text search, and sharing	Drill-through, drill-down, and nested regions
Data source	Transactional database	Data warehouse
Report author	Developer	Power user
Layout	Structured, predefined layout	Adaptive, flexible layout
Recommended tooling	Document Reporting Service/configurable Office documents	Power BI.com/analytical workspaces

Enhancements in paginated reporting

Using PDFs to interact with documents that the Document Reporting Service renders and bundles with finance and operations apps provides several advantages:

- **Reports appear on the screen more quickly.** Users benefit from improved performance when reports appear on the screen. This performance improvement includes a reduction in the number of progress bars, so that reports are faster.
- **Service reliability is enhanced.** The latest Document Reporting Service architecture delivers the power of the cloud scale to customer environments. The service includes automatic

scaling to help maximize resource utilization.

- **Printed output has higher fidelity.** Documents appear in PDF formats to provide consistency with printer output.
- **Local devices can print documents.** You can maintain user identity when you send documents directly from your browser to a printer. You can use the built-in user options on the control toolbar to help secure print jobs when you print documents.

For more information about how to preview the embedded PDF viewer in environments, see Preview PDF documents with an embedded viewer.

Turn off BI and analytics in one-box environments

In the platform updates for version 10.0.11, developers can preview existing solutions by forcing the local Document Reporting Service to render RDL while the BI and analytics features are turned off. This viewing mode is referred to as **RDL sandboxing**. You can turn it on or off by using Windows PowerShell commands.

Follow these steps to turn on RDL sandboxing in your local one-box environment.

1. Sign in to your one-box environment that uses version 10.0.11 or later.
2. Open Windows PowerShell, and go to `C:\AOSService\PackagesLocalDirectory\Plugins\AxReportVmRoleStartupTask\`.
3. Run the **UpdateRDLSandboxRule.ps1** command. Here's a list of the parameters:
    - **ConfigPath** – If SQL Server Reporting Services (SSRS) isn't deployed in the default location, specify the configuration location.
    - **RemoveRules** – Set this parameter to remove the rules.
    - **ForceUpdate** – Set this parameter to force the rules to be re-created.
    - **LogDir** – Specify a location to write the log to. The default location is the location of the **UpdateRDLSandboxRule.ps1** command.

Last updated on 11/26/2025

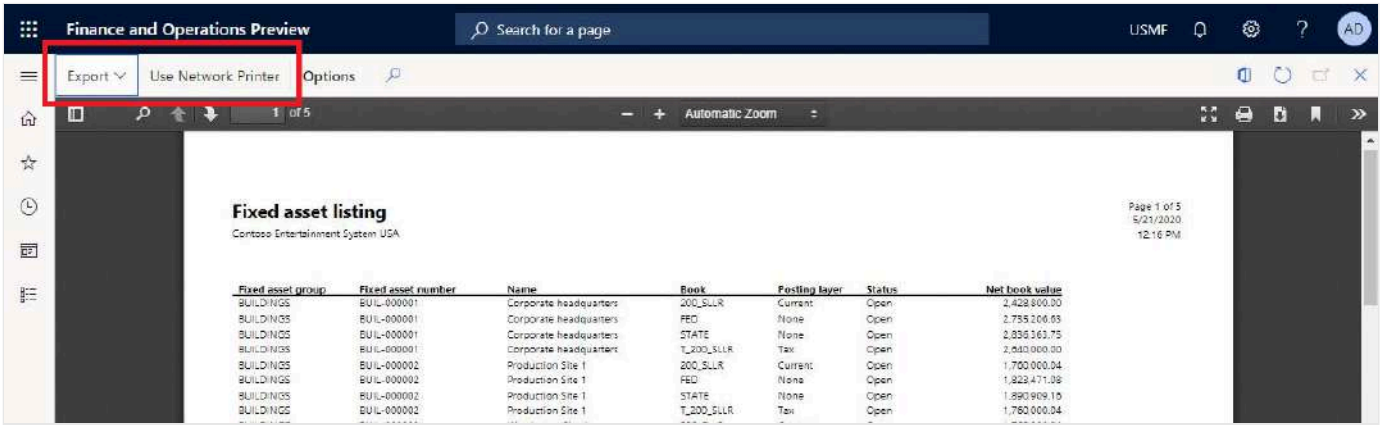
Document reporting preview options

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Finance and operations apps offer an expanded collection of options to use while previewing documents within the embedded PDF viewer. This article describes the **Export** and **Network Printer** options.

Overview

The efficient publication and distribution of documents is not only a key component to many operations, but it's also critical to daily business functions. Use documents to capture transaction details and represent agreements between two parties. Business documents come in many forms, from quotations to sales invoices used in the packing slip process, which ultimately produces receipts and customer statements. The document reporting preview options empower you to quickly take the appropriate action on documents.



File export

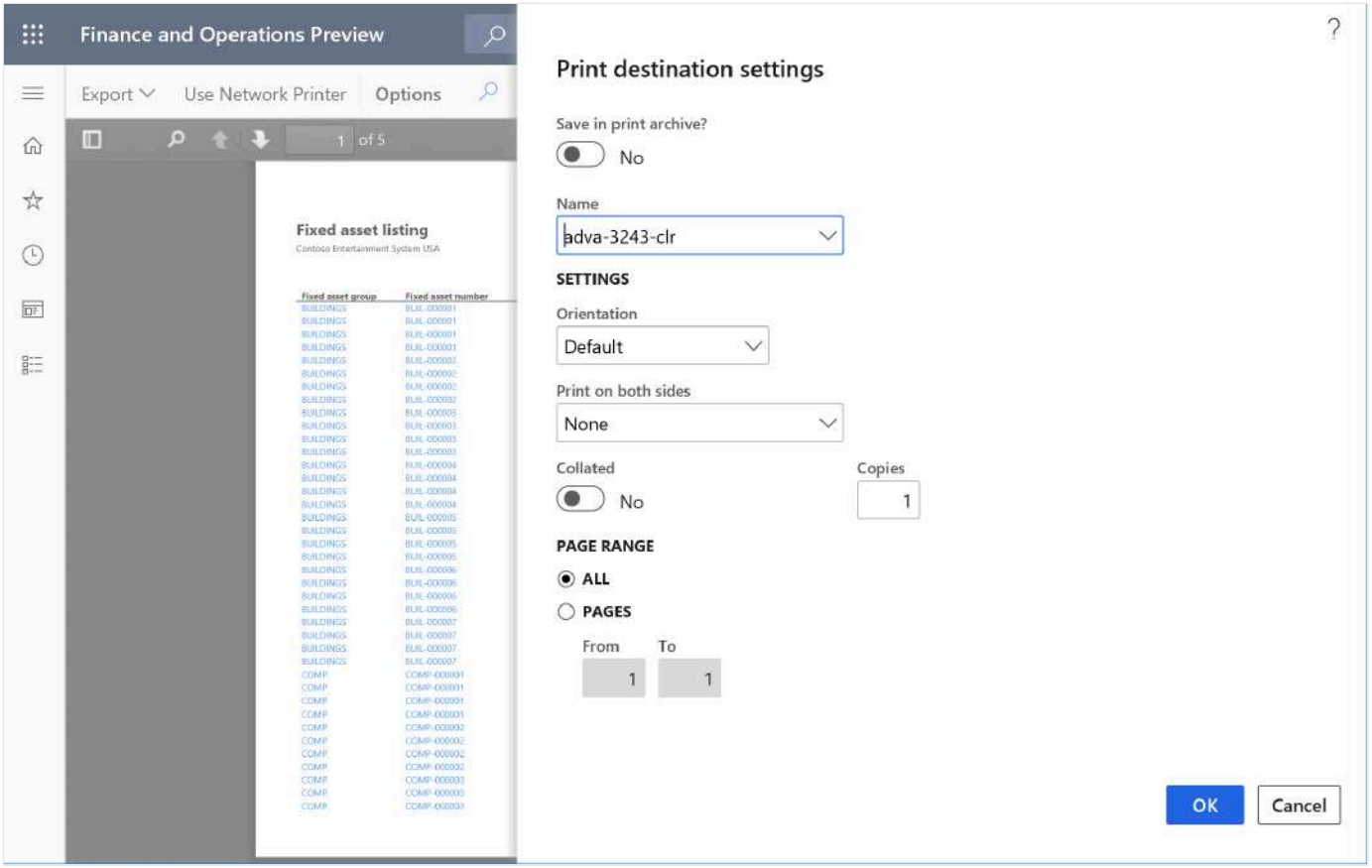
Additional user options replace functions previously supplied by the HTML-based **Report viewer** control. Use these toolbar options to export files to the various supported formats. This list includes Microsoft Word, Excel, HTML, CSV, and more. To ensure a consistent layout with the

printer output, the new modern experience displays documents by using the PDF pagination rules. For more information, see Pagination in reports.

Network printing

You can choose to print on locally connected devices by using the **Print** button or send documents directly to network printers by using the toolbar option. With the network print option, you can initiate print instructions to network connected printers while working remotely from nearly any internet connected device, including mobile phones.

Use the toolbar option to choose the network printer and establish print settings.



Access preview options

Use the Feature Management area to access the document preview options. The two additional features are:

- **Enable Export on Report PDF viewer**
- **Enable Network Printing on Report PDF viewer**

Select these features, then select **Enable now** to start using the new user options. For more information, see Feature management overview.

Last updated on 11/26/2025

Document printing overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can print documents by using either a local printer or a network-connected device. This article provides an overview of how documents are printed.

Printing overview

The application provides integrated services and client applications that make it easy to generate, store, and distribute documents that support business activity. You can print documents by using either a local printer or a network-connected device. In addition, you can export pages and reports directly from the client as PDF files or Microsoft Office documents. Finally, the distributed workload lets you print business documents directly from a mobile device by using network resources. Although printing requirements might vary, all industries typically must create hard copies of business documents by using the application. Printing documents on network devices from hosted applications presents a unique set of challenges. Here are some examples:

- Print drivers might not be available on the user's device.
- The user's device might not be connected to the corporate network.

By using a dedicated host and following a few easy steps, system administrators can configure deployments so that users can print directly from business applications on network devices.

Application printing scenarios

The following table describes the three primary printing scenarios.

Expand table

Scenario	Goal	Solution
1. Printing what you see	Print what is currently shown in the browser.	Generate a "print-friendly" version of the webpage for the browser.

Scenario	Goal	Solution
2. Interactive printing	Print a precision document on a locally connected device.	Export a PDF version of the report and download it to the browser.
3. Printing on a network device	Send a precision document to a domain printer device.	Send a precision document to a client application that runs on a server that is hosted in the customer's domain.

Because the solution varies depending on the scenario, applications provide built-in services and tooling to help users accomplish their goals:

- **Scenario 1** is supported by the browser's rendering of the HTML5 client.
- **Scenario 2** uses client applications and Microsoft 365 services.
- **Scenario 3** requires support from client applications and from services that are hosted in Microsoft Azure.

In addition to the platform that you deploy to the Azure subscription, finance and operations applications provide customers with an integrated, first-party Azure application that helps them more easily use domain-hosted devices to print documents.

> ⓘ **Note**
>
> When using the Document Routing Agent to print a report that includes barcodes on a network printer, some printouts might show plain text (for example, *LoadNumber*) instead of the barcodes. This issue can occur because the required font isn't available to Microsoft SQL Server Reporting Services (SSRS). To work around this issue, print the report to the screen and then print the preview document.

Service overview

While documents that the hosted applications produce wait to be printed on a network-connected device, the service stores them in Azure blob storage. The Install the Document Routing Agent to enable network printing uses Azure authentication to establish a secure channel to the Azure services.

Here's the execution sequence:

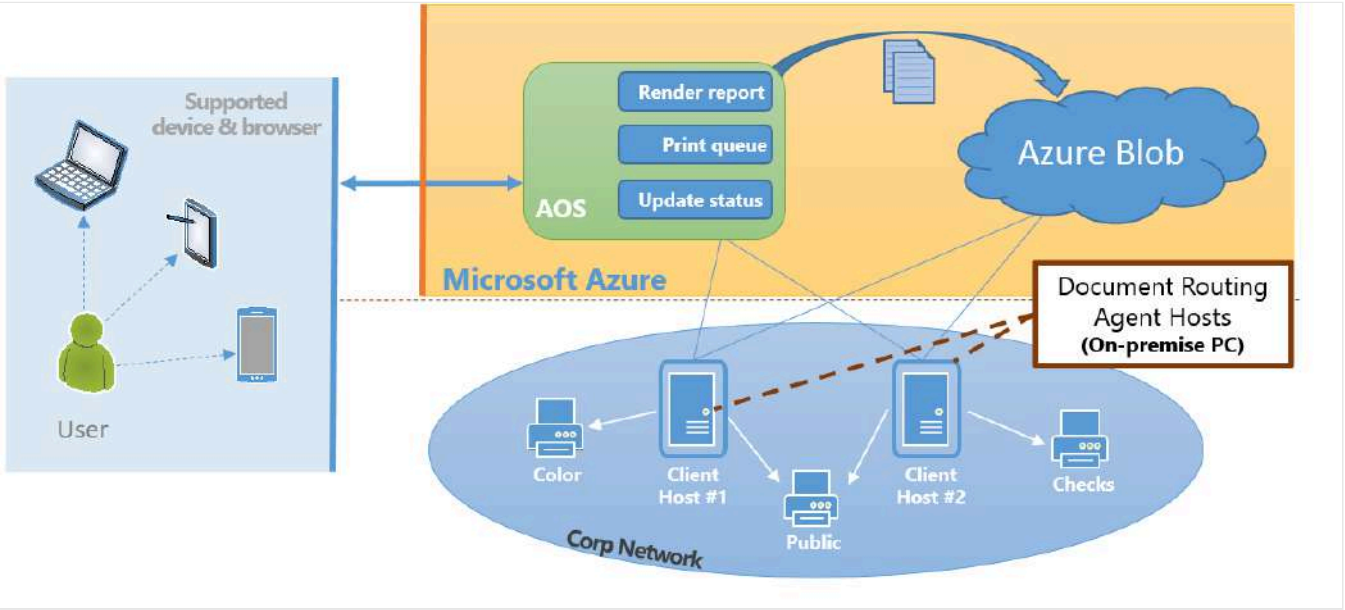
1. Microsoft SQL Server Reporting Services (SSRS) generates the report and stores it in Azure blob storage. The service stores the attached printer settings with the document.
2. The Document Routing Agent queries the Azure Service Bus queue for active jobs.

3. The Document Routing Agent downloads the document and spools it to the network printer.

This client-based solution lets customers manage the scale of their printing needs. Customers who have heavy-volume printing workloads can install many Document Routing Agents to increase the number of concurrent printing operations. Alternatively, some customers require very few installations of the Document Routing Agent to handle their anticipated printing needs.

Service components for network printing

The following diagram shows the basic components that help support network printing operations.



A single printer can be registered with multiple Document Routing Agents. To resolve the printer preferences, the hosted service uses the network path that uniquely identifies every network printer. As a result, even when multiple clients register a printer, it appears as a single selection in the list of printers available in applications.

Last updated on 11/26/2025

Install the Document Routing Agent to enable network printing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to install and configure the Document Routing Agent. The Document Routing Agent is a downloadable application that you can use to enable network printing scenarios. You can enable network printers for specific companies by using in-client administrative pages.

Preparing to install the Document Routing Agent

- Supported on Windows 8.1, Windows 10, Microsoft Windows Server 2012 R2, Microsoft Windows Server 2016, Microsoft Windows Server 2019, Microsoft Windows Server 2022, or Microsoft Windows Server 2025.
- Access to network printing resources requires Active Directory Domain Services (AD DS) authentication.
- When installing the Document Routing Agent, make sure you're signed in as the Admin user.
- The Microsoft Entra ID (Azure AD) account that you use to configure the Document Routing Agent must share the same domain as the Azure tenant.
- The Document Routing Agent requires .NET 4.7.2 or later and Adobe Acrobat Reader 32-bit or 64-bit on the client.
- Configure Adobe client print settings to prevent document scaling.
- The person installing the Document Routing Agent and configuring printers needs to have the "Document routing client" security role.

Network printers that are registered for applications can be used by all legal entities (also known as companies) that are defined in the environment. Network printer settings are company-specific. Therefore, administrators can restrict access, based on the user's active company. For

example, users in the active company might have access to all the network printers that are registered by the Document Routing Agent. However, users in another company don't have access to those printers until access is explicitly enabled for that company.

Key concepts

This article helps you with the following tasks:

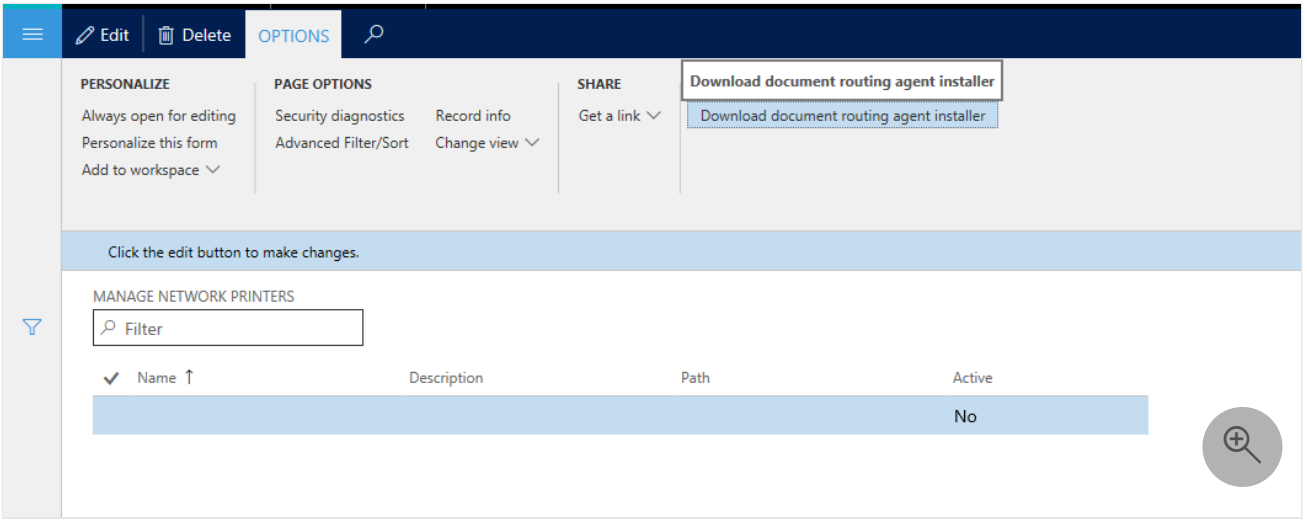
- Identify the key components that are involved in the support for network printing in applications.
- Learn about the function of the Document Routing Agent.
- Configure the Document Routing Agent to work against an existing application.
- Use administration pages to manage access to network printers.

Install the Document Routing Agent

Applications use the Document Routing Agent to manage the spooling of documents to network printer devices. You can get the client by using direct links that are embedded in the web application. Use the following procedure to download the application to your local computer. You can then access both local and network printers that are connected to your computer, from a single deployment. Ensure the user installing the agent has the **Document routing client** security role.

1. Open the **Manage network printers** page (**Organization administration** > **Setup** > **Network printers**).

2. On the **Options** tab, in the **Application** group, select **Download document routing agent installer**.



3. Run the downloaded file to begin the installation process.

4. Complete the setup process.

After you install the application, you can register local printers as network printers for the applications.

> ⓘ **Note**
>
> It's recommended to update the Document Routing Agent to the most current version whenever the application release on the environment has been updated, othgerwise you might experience issues such as:
>
> Observable performance degradation in applications, Document loss that is associated with orphaned print jobs, Inconsistent handling of printed documents that have custom margins
>
> For more information, see **Update the Document Routing Agent**.

If you don't update existing installations of the DRA the most current version, you might experience issues such as:

Observable performance degradation in applications Document loss that is associated with orphaned print jobs Inconsistent handling of printed documents that have custom margins

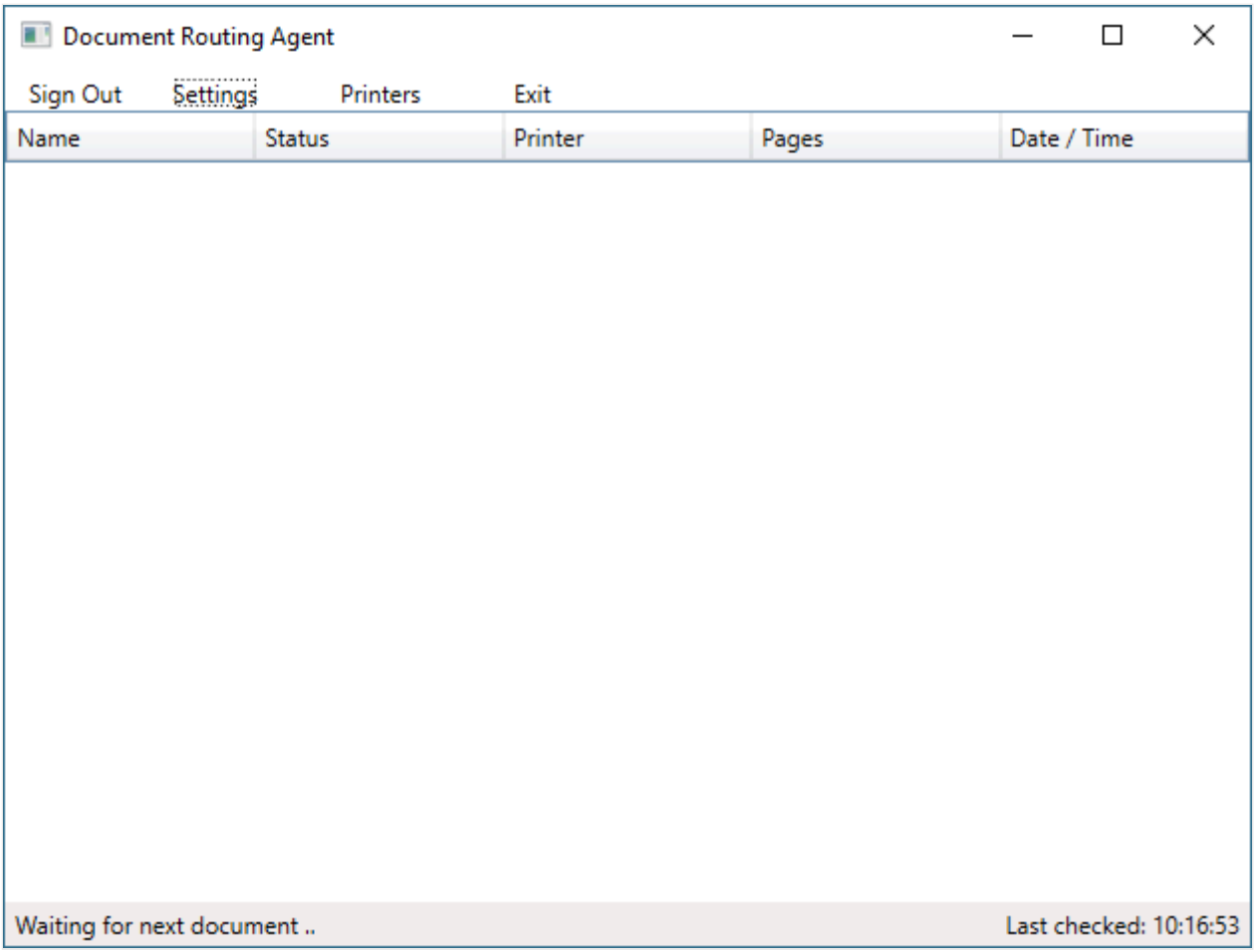
Configure the Document Routing Agent

Use the following procedure to configure the client application so that it can communicate with the Azure services that host the documents that are in-flight.

1. Close all browser instances that are running the application. This action resets the local Azure authentication tokens.

2. On your desktop, run the Document Routing Agent.

3. On the toolbar, select **Settings**.



4. Add the following settings:

    - **Application ID** – The ID that's unique to the application and should be entered automatically.
    - **finance and operations URL** – The base URL of the application.
    - **Azure AD tenant** – The domain name of the Azure AD.

5. Select **OK**.

6. Select **Sign In** to sign in to your account.

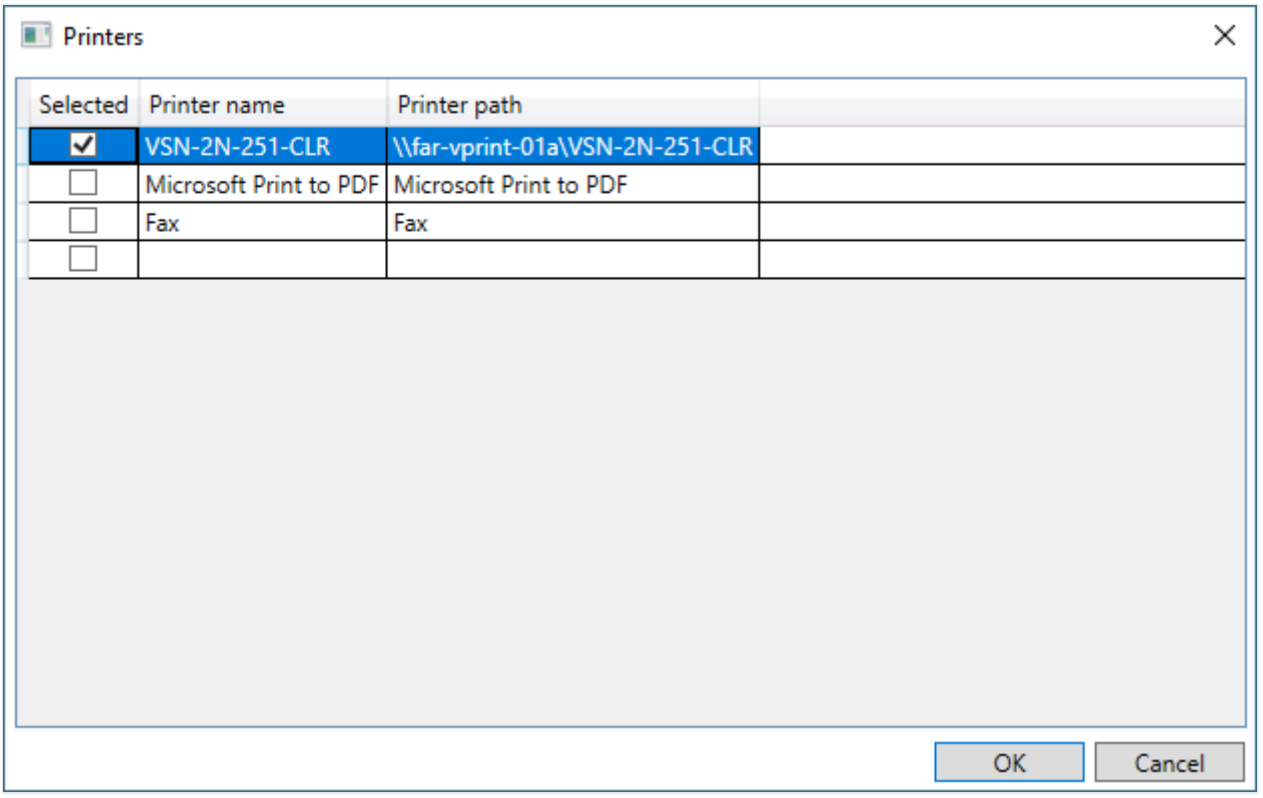
> ⓘ **Note**
>
> The account must share the same domain as the Azure AD that is associated with the application. The Document Routing Agent is now ready to process documents.

After you sign in, the **Printers** button becomes available on the toolbar.

Register network printers

Before you complete this procedure, make sure that you installed all the network printers on the local host computer. All the printer devices that you install are available for service registration. Be sure to select only the printers that you want to expose in the applications.

1. On the toolbar, select **Printers**.
2. Select the printers to make available in the applications.



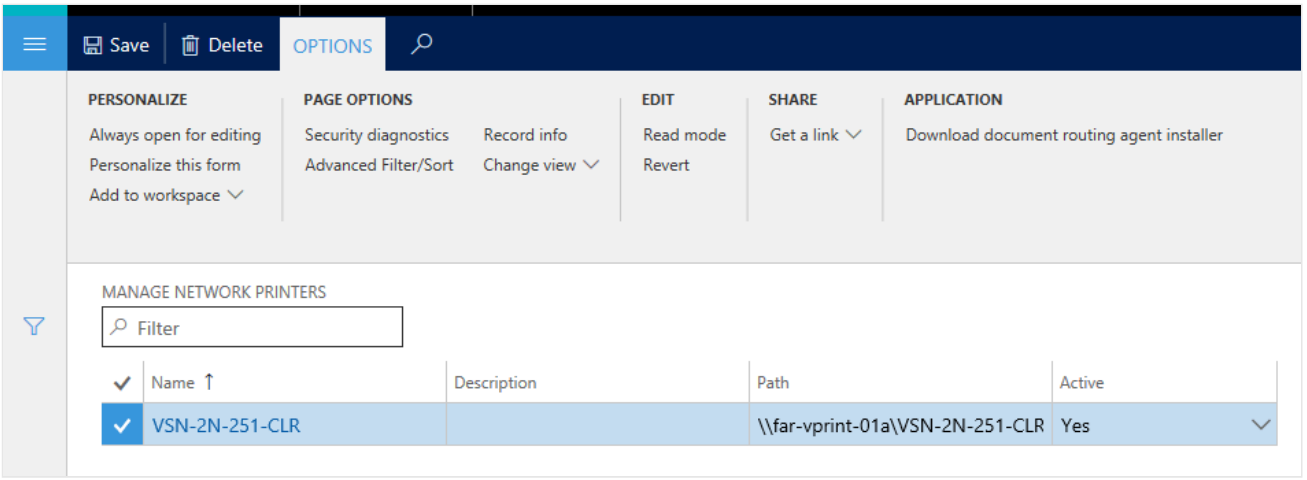
3. Specify a default name for the printer.
4. Select **OK**.

After you complete this procedure, the selected printer devices are registered in the application's network printer catalog. System administrators can now enable the printers for access from within the application.

Administer network printers

Use client pages to manage access to the network printers that one or more Document Routing Agents register. Network printers are uniquely identified by their path. Therefore, printers are listed one time, even if more than one Document Routing Agent registers them. Use the following procedure to activate the Application Object Server (AOS) network printers.

1. Open the **Manage network printers** page (**Organization administration** > **Setup** > **Network printers**).



2. Edit the existing entries that are mapped to each network printer. As part of your changes, edit the connection path.

3. To include a printer as an option in the **Print Destinations** field, set the **Active** field to **Yes**.

The network printers can now be used in the application.

> **Note**
>
> Ensure that your network printer destinations are kept up to date, and that document routing is properly configured with printers that are registered against Document Routing Agents. If you send documents to a printer that no longer exists, the print queue continues to grow and slow down queries that poll the print queue.

Adjust the document routing history cleanup batch job

A cleanup batch job for document routing history is enabled by default and runs daily. This batch job purges document routing history that's older than seven days. Use this history for troubleshooting or traceability if there are issues with printing. Depending on how you intend to access this historical data, you can reduce the retention period from the default value of seven days, which is considered an upper limit. Having fewer records in this table ensures that printing has optimal performance. You can configure this setting at `https://[host_adress]/?mi=DocumentRoutingHistoryCleanupConfig`. Set the value for **JobHistoryHours** (number of hours to retain history).

As part of the Document Routing Agent polling, a query runs against this table. This query should execute quickly, but if there are many records in this table, a large print job can be slow. Ensure that this batch job runs daily, and configure this setting to reduce how much print history you retain.

Excluding printers with stuck print jobs

The **Enable excluded printers** setting handles problematic printers and drivers. When you enable this setting, if a print job is sent to the printer spool and doesn't return with a **Pending** status, the Document Routing Agent adds the printer to an excluded list after the time specified in the **Abort a stuck print job at** field. (The default time is five minutes). The **Reset this printer every x minutes** field, which has a default value of 30 minutes, adds the printer back after the specified time and attempts sending print jobs.

The administrator can also see any excluded printers in the **Network printers** section in the **Spooler status** column. The administrator can reset any excluded printer by selecting the **Reset** icon in the **Reset** column. In addition, the administrator can send a test page to the printer by using the **Print test page** button.

Frequently asked questions

Does the Document Routing Agent need to be installed on each computer where a user connects by using a browser?

No. Individuals who access the provisioned environment can share client installations of the Document Routing Agent. We recommend that you install agents on one or more print servers or other domain-hosted clients that have access to network printers.

If the Document Routing Agent belongs on a network print server, why doesn't the client run as a service?

The Document Routing Agent now supports running in the background as a service. Ensure that you download the latest version of the client. For more information, see Run the Document Routing Agent as a Windows service.

Do I need to update credentials or refresh Microsoft Entra authentication tokens on a recurring basis?

Yes. The Microsoft Entra token must be refreshed every 90 days. If you don't refresh the token, the Document Routing Agent can't authenticate and retrieve printing instructions applications.

Is the Document Routing Agent supported on Microsoft Windows Server 2019?

Yes. Microsoft Windows Server 2019 supports the Document Routing Agent.

> ⓘ **Note**
>
> If you configure the server to prevent background service, the Document Routing Agent client can't run as a service. For more information, see **Run the Document Routing Agent as a Windows service**.

Will Microsoft add support for Microsoft Windows Server 2008 servers?

No, not at this time. The Document Routing Agent has several dependencies on Azure capabilities that are available only in Microsoft Windows Server 2012 R2 and Microsoft Windows Server 2016.

Does the user who installs the Document Routing Agent have to be part of a finance and operations apps security group?

Yes. To access the agent installation links, the user must be part of the **Document routing client** security role.

How many network printers can the Document Routing Agent support?

The number of supported network printers depends on the number of legal entities and the number of network printers deployed. If you have 50 printers and one legal entity, a single Document Routing Agent can handle the load (although you'd want more than one to ensure high availability). If you have a large number of printers and legal entities, you should do some performance testing to determine the number of Document Routing Agents that you need.

How many Document Routing Agents should be configured per printer?

You should configure multiple Document Routing Agents for your printers to ensure high availability. However, limit the number of agents per printer to no more than three agents. Each Document Routing Agent poll needs to query the queue to pick up documents sent to printers registered in that Document Routing Agent client. The more printers that are associated with a Document Routing Agent, the slower the query is. This issue is especially true when there are a

large number of pending jobs in the queue. It's better to have a smaller number of printers across two to three Document Routing Agents, than a large number of printers across three or more Document Routing Agents.

Last updated on 05/05/2026

Update the Document Routing Agent

The solution for managing the print job queue helps you scale Dynamics 365 finance and operations apps to meet high-volume printing needs. Although the public service endpoints that manage print jobs are backward-compatible, update **all** existing Document Routing Agent (DRA) clients.

If you don't update existing installations of the DRA to the most current version, you might encounter problems such as:

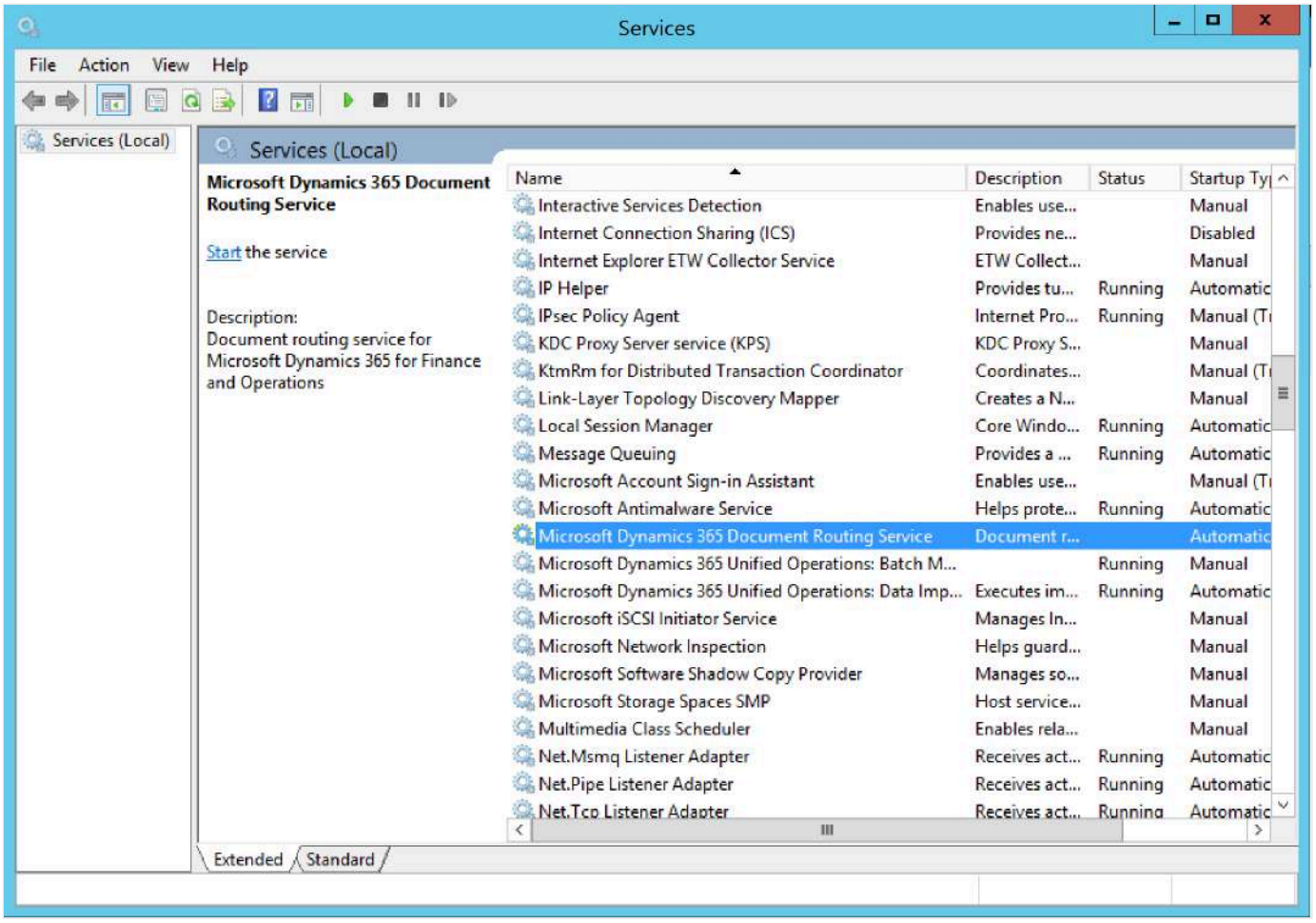
- Observable performance degradation in applications
- Document loss associated with orphaned print jobs
- Inconsistent handling of printed documents that have custom margins

IT administrators must perform the following procedures on each domain resource that hosts a DRA.

> ⓘ **Note**
>
> When you complete a DRA update, register any printers that connect through the host server. For network printers that are identified by their network paths, if the paths didn't change, updates aren't required.

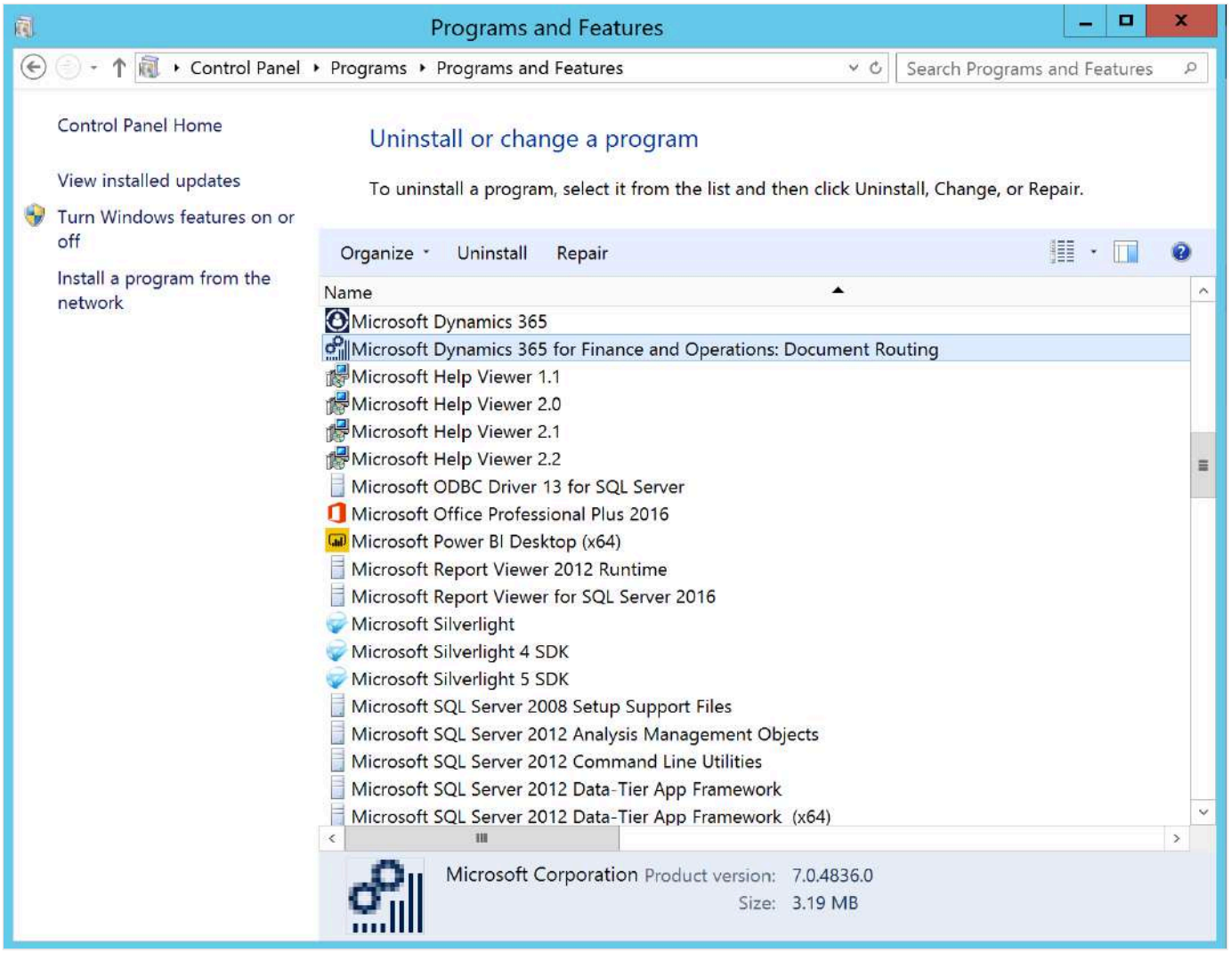
Get started

To keep running the DRA as a Microsoft Windows service, you need both the user name and password of the domain account that runs the service. You must have this information after the update finishes. To find the information for the active service account, start the Microsoft Management Console (MMC) Services snap-in and select **Microsoft Dynamics 365 Document Routing Service** in the list.

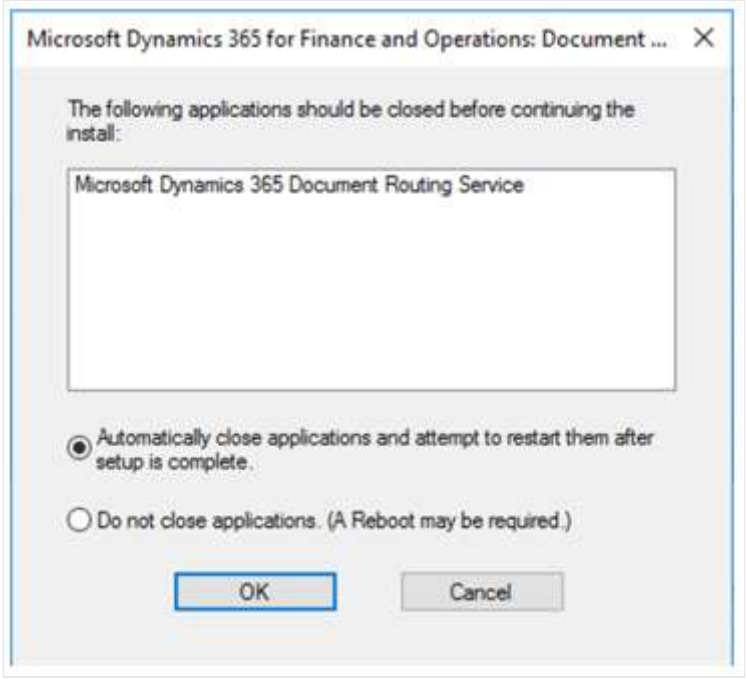


Uninstall an existing Document Routing Agent

Open **Programs and Features**, and then find and uninstall **Microsoft Dynamics 365 Finance: Document Routing**.



During the uninstallation process, if you're prompted to close the Microsoft Dynamics 365 Document Routing Service application, select **Automatically close applications and attempt to restart them after setup is complete**.



Install the latest Document Routing Agent

For information about how to install the latest DRA that's available with your subscription, see Install the Document Routing Agent to enable network printing.

> **Note**
>
> Open the DRA client after upgrading to refresh network user credentials.

Best practices

When you create a support ticket specific to the Document Routing Agent, consider exporting event logs that contain events at the time the issue occurred from the Document Routing Agent machine.

The default event log size is only 1 MB, so if the issue is intermittent, such as a performance issue, consider increasing the event log size. Then wait some time before you export the event logs.

- \Applications and Services Logs\Microsoft\Dynamics\Ax-DocumentRouting\Operational
- \Applications and Services Logs\Microsoft\Dynamics\Ax-DocumentRouting\Admin

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Last updated on 06/26/2026

Install network printer devices in on-premises environments

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to connect an on-premises deployment of Microsoft Dynamics 365 Finance + Operations (on-premises) to existing network printer devices. The Print and Document Services feature in Microsoft Windows Server 2016 supports network printing in the on-premises application. This feature centralizes tasks that are related to printer management. To install and configure Print and Document Services, you must have administrative access to the server that hosts the primary instance of Application Object Server (AOS).

Two roles are associated with the configuration of network printing services:

- **Service Administrator** – The person responsible for installing and configuring components of the platform infrastructure. Traditionally, this role is an Active Directory account that has elevated domain privileges. It has enough privileges to install components of Microsoft Windows Server.
- **Organization Administrator** – The person who manages application security privileges. This Active Directory account is assigned to the **System Administrator** role.

Before the organization administrator can add network printers, the service administrator must install and configure Print and Document Services on the server that hosts the primary AOS instance. The organization administrator can then use built-in tools to configure network printer devices.

Install and configure Print and Document Services

The environment administrator uses the information in this section to enable network printing services.

1. Install Print and Document Services by following the instructions in Install Print and Document Services.
2. Configure Print and Document Services by following the instructions in Configure Print and Document Services.
3. Follow these steps for each server that hosts the AXService application:
   a. On the local server, start the **Local Users and Groups** manager.
   b. Select the **Groups** node.
   c. Right-click **Print Operators**, and then select **Add to Group**.
   d. Add the network Active Directory account that runs the AXService application to the group.

> ⓘ **Note**
>
> Starting with version 2.9.0 of the infrastructure scripts, the accounts are automatically added to the appropriate group.

Install printers on nodes where AXService runs under a domain account

To install printers on nodes where AXService runs under a domain account, follow these steps:

1. Install network printers by using the AXService user account. This step ensures that the printer driver is available to the AXService user account.
2. Print a test page on the installed printers to make sure that all the connections are correctly configured.
3. Restart the AXService application to help ensure that the user's profile is correctly loaded so that it can locate the printer driver.

Install printers on nodes where AXService runs under a group Managed Service Account (gMSA)

To install printers on nodes where AXService runs under a gMSA, follow these steps:

> ⓘ **Important**
>
> This section requires at least version 2.9.0 of the infrastructure scripts. Additionally, you need to download the SysInternals Suite.

1. Update the `Printers.json` file by adding the network location of each printer that should be available to the AOS. Ensure that you remove the example entries.
2. Run the following command from the Infrastructure Scripts folder.

PowerShell

```
# Exports the script files to be executed on each VM into a directory VMs\<VMName>.
.\Export-Scripts.ps1 -ConfigurationFilePath .\ConfigTemplate.xml
```

1. Copy the contents of each `infrastructure\VMs\<VMName>` folder to the corresponding VM. If you use the remoting scripts, they automatically copy the contents to the target VMs. Then run the following command as an administrator.

> **Note**
>
> - The following procedure requires execution on multiple VMs. However, to simplify the process, you can use the remoting scripts that are provided. These scripts let you run the required scripts from a single machine, such as the same machine that you use to run the **.\Export-Scripts.ps1** command. When the remoting scripts are available, they are declared after a **# If Remoting** comment in the Windows PowerShell sections.
> - Remoting uses WinRM. In some cases, it requires that CredSSP be enabled. The remoting module enables and disables CredSSP on an execution-by-execution basis. Disable CredSSP when it isn't used. Otherwise, there's a risk of credential theft. When you complete the setup, run the following command.
>
> PowerShell
>
> ```
> .\Disable-CredSSP-AllVMs.ps1 -ConfigurationFilePath .\ConfigTemplate.xml.
> ```

PowerShell

```
# If Remoting, execute
# .\Install-PrintersOnGmsa-AllVMs.ps1 -ConfigurationFilePath .\ConfigTemplate.xml -
SysInternalsFolderLocation \\networkshare\SysInternalsSuite -ForcePushLBDScripts

.\Install-PrintersOnGmsa.ps1 -PrintersJsonFilePath .\Printers.json -
SysInternalsFolderLocation \\networkshare\SysInternalsSuite
```

Manage network printers

Use the information in this section to define network printers.

1. Go to **Organization administration** > **Setup** > **Network printers**.
2. On **Network printers**, add new printers. For each printer, specify a name, description, path, and status. Make sure that the printer path matches the network path of the installed printer.

When you mark a printer **Active**, application users can immediately use it to print document-style reports on network printer devices.

Last updated on 04/13/2026

Document generation, publishing, and printing in on-premises deployments

This article describes the capabilities for generating, publishing, and printing documents in on-premises deployments of Microsoft Dynamics 365 Finance + Operations (on-premises). The application provides a fully integrated experience for enterprise document generation that's powered by Microsoft SQL Server Reporting Services (SSRS). From any supported device, users can produce standard industry documents that are linked to business processes. These documents include sales invoices, checks, and packing slips. Built-in tools let administrators configure the service so that users can securely connect to network printers.

You can upgrade solutions that are built on the Microsoft Dynamics AX 2012 SQL Reporting Services framework, or you can take advantage of the modern solutions that are available in Microsoft Dynamics Lifecycle Services.

Document publishing services: secure, reliable, and convenient

Employees spend lots of time on the go. Therefore, businesses depend on their employees' ability to stay productive while they work remotely. However, even today, documents remain critical for business transactions and record keeping.

From their mobile devices, users can print documents on network printers. Users can also automate the creation of business documents and use built-in tools to configure instructions for routing documents to multiple recipients.

The following options are available for document publishing:

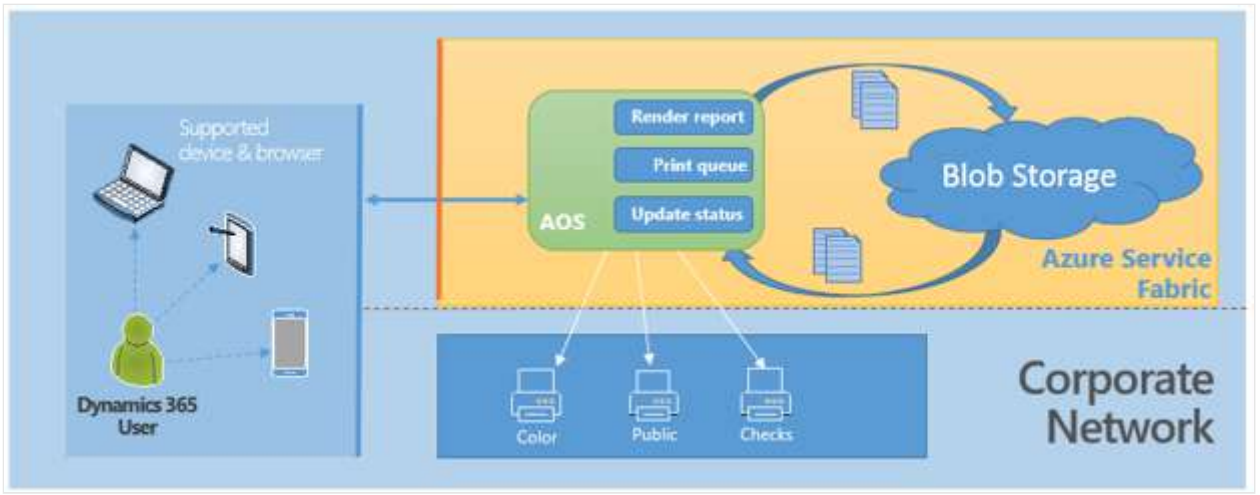
- **Email** – Distribute mail via a server, and link reports as attachments.
- **Archive** – Store reports for future reference and regulatory compliance.
- **File** – Produce a PDF file that is downloaded directly to the browser for local printing.
- **Print** – Send documents directly to network printers from all supported platforms. These platforms include mobile devices.

For a high-level summary of the options for information access that are available in the cloud-hosted solution, see Document printing overview.

Comparing the cloud-hosted and on-premises services

Unlike the cloud-hosted service, the on-premises publishing service produces documents as PDF files that the browser automatically downloads. Therefore, users can save documents or print hard copies by using locally connected devices. Administrators can manage access to network printers directly from the application by using built-in administrative pages. Users can interact with reports on demand, or they can schedule automatic jobs to securely generate and distribute documents on a recurring basis.

The following illustration shows the components that are involved in document printing.



For information about how to use extensions to manage availability of the embedded drill-through links in application reports, see the Appendix.

Managing access to network printers

Administrators can use built-in administrative pages to manage access to network printers. Network printers are secured per company and shared by users of the application. Documents are then printed by using a privileged domain account, based on settings that the user provides. In on-premises deployments, you don't have to install an adapter to connect to domain resources such as printers and fax machines.

The following illustration shows the page that is used to manage network printers.

Appendix

Turning on embedded links in business documents

Here's the code that you can use to make embedded drill-through links available in PDF documents.

```
X++
```

```
class Controller extends SrsReportRunController
{
    protected void preRunModifyContract()
    {
        this.parmReportContract().parmRdlContract().parmEnableFileDrillThrough(true);
        super();
    }
    static void main(Args _args)
    {
        ...
    }
}
```

Last updated on 06/26/2026

Run the Document Routing Agent as a Windows service

The Document Routing Agent includes an option that lets you select the mode of execution. The process can run as either a desktop application or a Microsoft Windows service. When the application runs as a Windows service, it can start automatically after a computer restart. You can also configure it to run under the security context of a specific user account. This enhancement lets you host the Document Routing Agent on secured domain resources such as network print servers.

This article provides important information that helps you select the correct execution mode.

Service applications

An application is a program that a user interacts with on the desktop. A service is a process that runs in the background and doesn't have an active window. The Document Routing Agent now supports both execution modes. It's important that you understand why you might select one mode instead of the other and the steps that are involved in running the process as a service. For more information about Windows services, see Introduction to Windows Service Applications. Here are some of the main benefits of running the Document Routing Agent as a background service:

- You can configure the service to start automatically after a computer restart. No user intervention is required.
- The service runs in the background. No active application runs in the notification area.
- The service routes documents without requiring that a user sign in by using cached credentials.

Although there are many benefits of running the Document Routing Agent as a Windows service, there are also limitations. The next section discusses an issue that affects the handling of document reports, such as checks, that require custom page margins.

Documents that require custom margins

When the Document Routing Agent runs as a Windows service, document reports, such as checks, that require custom margins can't be printed directly to network printers. Instead, the Document Routing Agent automatically routes those documents to a target folder. New

configuration properties in the application's **Settings** dialog box let you define the target location for document reports that require custom margins.

When the Document Routing Agent runs as a desktop application, it continues to take advantage of Adobe Reader to spool the document to the shared printer device that is selected in finance and operations. To handle scenarios where documents that have custom margins must be printed, install the Document Routing Agent in multiple locations. Then install the printers that handle those documents only on the Document Routing Agents that run in desktop application mode. Alternatively, use a post-execution process to pick up the files in the target directory and direct them in the appropriate manner.

Install the latest build

1. Save a copy of the current Document Routing Agent configuration file. This file is located at `C:\Users\<UserID>\AppData\Local\Microsoft\Microsoft Dynamics 365 Finance Document Routing\Microsoft.Dynamics.AX.Framework.DocumentRouting.config`. In this path, `<UserID>` is the Active Directory Domain Services (AD DS) user name that you used to install the Document Routing Agent.

2. Uninstall the current version of the Document Routing Agent.

3. Install the latest version of the Document Routing Agent by following the instructions in Install the Document Routing Agent to enable network printing.

   > **Note**
   >
   > Although you install the application at this point, don't run it yet.

4. Copy the configuration file from step 1, and paste it into the following directory: `C:\ProgramData\Microsoft\Microsoft Dynamics 365 Finance Document Routing`. This step helps guarantee that all your previous configuration settings are used for the new version of the Document Routing Agent application.

5. Run the Document Routing Agent.

6. Sign in by using your Microsoft Entra ID or your credentials for your finance and operations apps.

7. View and verify your settings and printers by clicking the appropriate menu items.

The next section provides detailed instructions for selecting the Windows service execution mode.

Change the default execution mode

By default, the Document Routing Agent runs as a desktop application. To run the process as a Windows service, make sure that you're familiar with the process for installing the Document Routing Agent to enable network printer devices. Then complete the following tasks.

Update the execution mode for the Document Routing Agent

1. Start the Document Routing Agent by using the desktop icon.
2. Select the **Sign In** option, and sign in by using your Microsoft Entra ID credentials.
3. On the ribbon, select **Settings**.
4. Enable the **Run as a Windows Service** option.
5. Provide a target folder for PDF files that are produced for documents that have custom margins.
6. Select **OK**, and close the Document Routing Agent window.

Configure and start the Windows service

1. In Windows, start Service Manager.
2. In the list, select **Microsoft Dynamics 365 Finance Document Routing Service**.
3. Right-click the name, then select **Properties**.
4. On the **Log On** tab, select the **This account** option, then supply the AD DS credentials that run the service.

   > ⓘ **Note**
   >
   > The selected account must have access to the shared network devices. The Windows domain account or local machine account that runs the Windows service must be the same as the account that starts the Document Routing Agent desktop app.

5. Select **OK**.
6. Start the service.

The Document Routing Agent now runs as a Windows service.

Troubleshooting tips

Verify the network printer connection

- Verify that the active account has enough access rights to the network printer.
- Verify that the user account can successfully print to the network device by using Notepad or another local application.

Verify security roles

- To access the application links that are used to install the Document Routing Agent, the user must be part of the **Document routing client** security role.

Review the service account's access rights

- Verify that the **DocumentRoutingService** service runs as a domain account that has access to the network devices.

Refresh the Azure service token

- Azure authentication tokens must be **refreshed every 90 days** while the Document Routing Agent runs as a Windows service. To refresh the service token, start the client, then sign out and sign back in by using the menu items.

Disable shared printers for remote access

- When you connect to the host machine by using Microsoft Remote Desktop, make sure that you clear the **Printers** option in the **Local devices and resources** section on the **Local Resources** tab.

Review the event logs

1. On the host machine, start Event Viewer.
2. Review the logs at **Application and Services Logs** > **Microsoft** > **Dynamics** > **AX-DocumentRouting**.

Last updated on 08/03/2026

SQL Server Reporting Services (SSRS) reports that are available

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article points you to a report that lists the SQL Server Reporting Services (SSRS) reports available.

You can define reports as any visualization of a structured data set. This definition includes transactional data presented in a tabular layout and advanced graphical views of aggregate information. To account for this broad definition, the application offers several tools to produce reports that satisfy complex business requirements. One of these tools is SSRS. SSRS reports provide the following advantages:

- Back office document management capabilities, including email support, scheduled executions via batches, and print archive functionality.
- Parameterized views with drill-through navigation to application pages and other reports.
- Precision documents for compliance with local regulatory business practices.

For more information, see Create reporting solutions.

To view the report

The **SQL Server Reporting Services Reports report**, included with the Technical reference reports, lists each available SSRS report. The report indicates the data set used for each report, as well as the filters and fields available on each report.

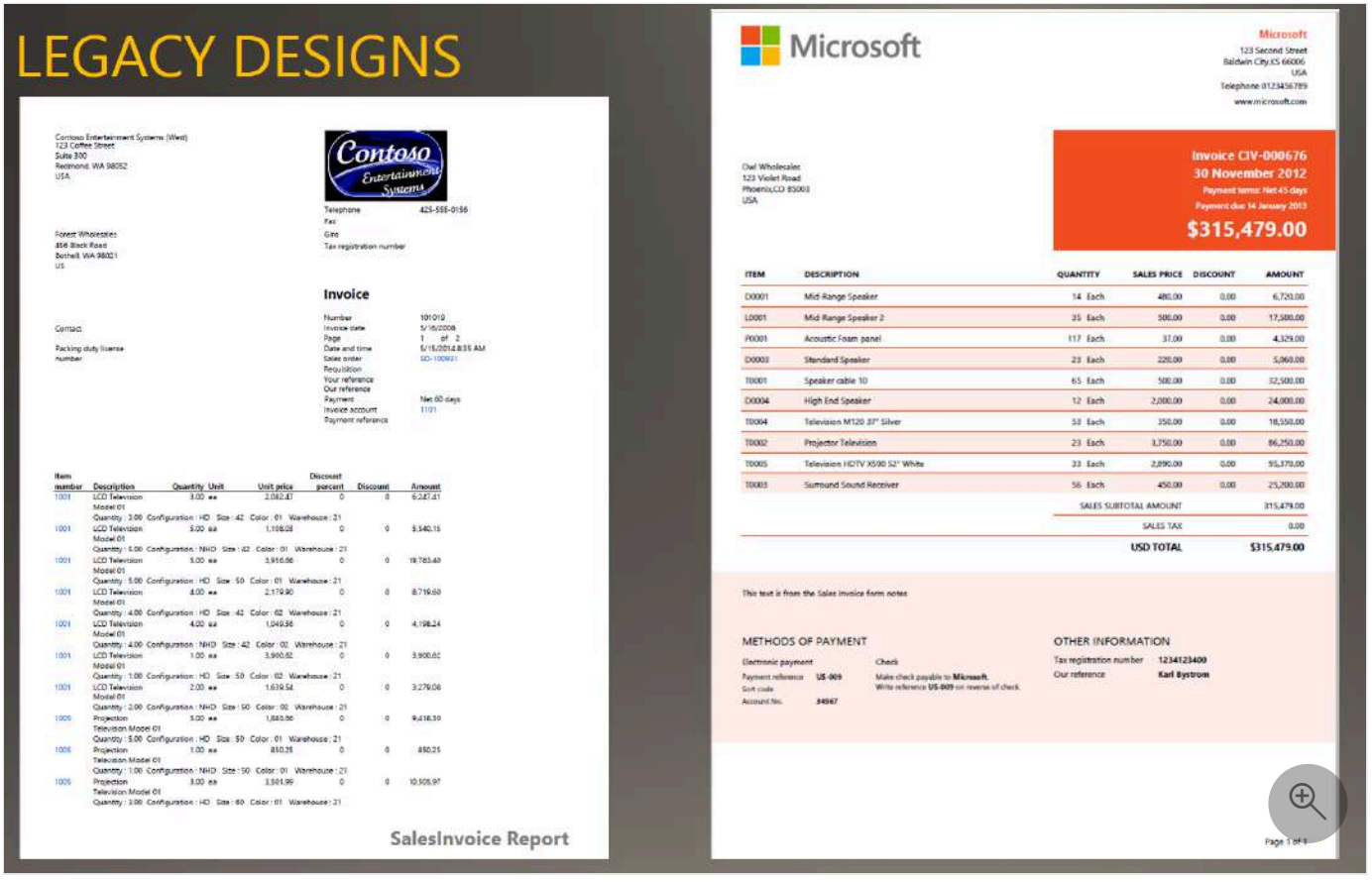
Last updated on 11/26/2025

Install modern report design templates

This article explains how to install the modern report design templates in the application suite. You can use these samples to create graphically rich business documents that have flexible branding in the header and footer.

Introduction

A new set of developer tools is available that takes the form of report designs for several core business documents in the application suite. These report designs have been reimagined so that flexible branding appears in the header and footer of public-facing documents when transactions are generated in the application. The following illustration shows how an earlier design for a sales invoice differs from a modern sales invoice design.



After you complete the installation, you can use the built-in brand management tools to define brand settings that should be applied to the modern designs for application business documents. The brand management tools are available at **Organization administration** > **Setup** > **Document branding** > **Branding details**.

Why aren't these designs the default designs for the application suite reports?

We're maintaining the legacy solutions for two primary reasons:

- **Modern designs don't include code.** Although the legacy solutions use embedded Microsoft Visual Basic (VB) code to recognize configuration keys and honor regulatory requirements that vary by region, the modern report designs offer much less flexibility. The benefit of a simple design that has minimal code behind it comes at the expense of reusability across regions.
- **Modern designs aren't available for all business documents.** There's a gap between the supported business documents and the availability of modern report designs. Although the legacy designs aren't as aesthetically pleasing, they provide a sense of consistency.

> **Important**
>
> The simple modern designs aren't recommended for all types of deployments. They're intended for cases where the customer doesn't require runtime control over the layout of the document through existing application configuration settings.

Apply the modern designs

The modern report designs are bundled into a model file and posted to Microsoft Dynamics Lifecycle Services. You can easily access them from your existing subscription. Use the following procedure to obtain the modern report design solutions and install them in your local development environment. You must then apply some customizations to incorporate the modern report designs into the appropriate scenarios.

Follow these steps to install the modern report designs for the application suite.

1. Sign in to Lifecycle Services to access the deployment dashboard. Then, on the **Shared asset library** page, select the **Model** asset type, and download the **ApplicationSuiteModernDesigns** model file. Save the model file to a location that's accessible from the development environment.

   > **Note**

Be sure to select the appropriate model file for the version of the application that you're using.

2. Import the model file into your local development environment. To install a model file in a development environment, use the ModelUtil.exe tool and the **-import** directive. Here's an example.

   Console

   ```
   ModelUtil.exe -import -metadatastorepath=[path of the metadata store] -file=[full path of the file to import]
   ```

3. Navigate to the **<AOSServiceDrive>:\AOSService\PackagesLocalDirectory\bin** folder.

4. Run the following command.

   Console

   ```
   ModelUtil.exe -import -metadatastorepath=J:\AOSService\PackagesLocalDirectory -file="E:\Test\AppSuiteModernDesigns.axmodel"
   ```

   For more information about how to import model files, see Export and import models. After you import the model file, start Microsoft Visual Studio. In Application Explorer, verify that the **Application Suite - Modern Designs** collection appears under the **AOT** node. For more information about how to use the Application Explorer, see Development tools tutorial.

Now that you successfully imported the Application Suite Modern Designs model, you must rebuild the application suite to update the metadata elements.

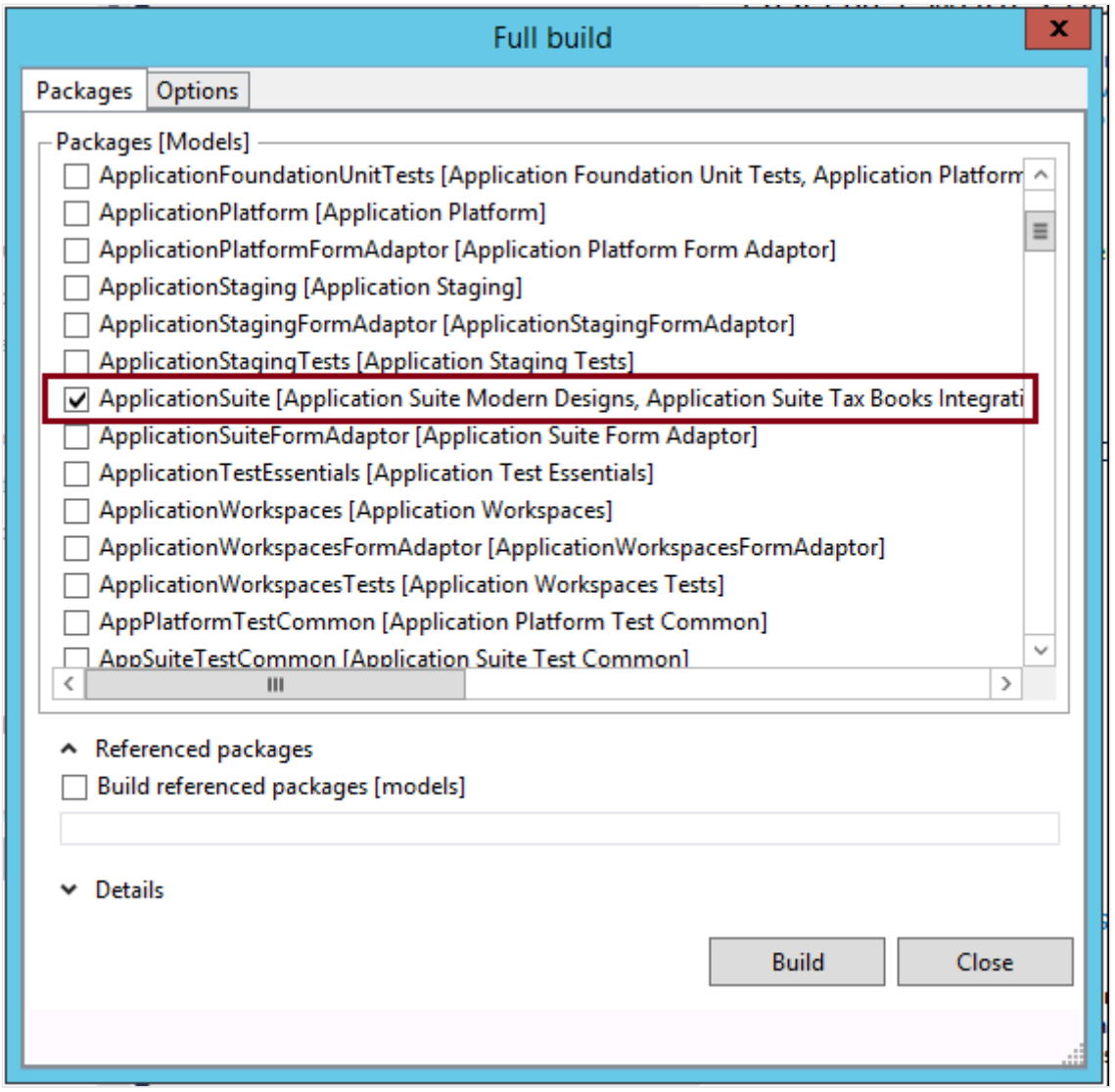
Build the application suite

The Application Suite Modern Designs model extends the Application Suite model. To update all application references so they target the model extensions, build the Application Suite model by using Microsoft Visual Studio.

1. Start Visual Studio, or use the existing instance.

2. On the **Dynamics 365** menu, select **Build models**.

3. In the list, select the check box for the **ApplicationSuite** package.



> ⓘ **Note**
>
> The Application Suite Modern Designs model is included in the package definition.

4. Select **Build** to do a full build of the application suite.

This process might take up to 20 minutes, depending on the size of your machine.

Deploy the modern designs (one-box environments)

After you compile the application suite that includes the modern report design templates, verify the changes locally. To verify the changes, deploy the new modern report design solutions to the instance of Microsoft SQL Server Reporting Services (SSRS) that's running locally.

Follow these steps to incorporate the modern report design into an existing application suite report.

1. Create a project that contains the application suite report. In Application Explorer, under the **Application Suite Modern Designs** model, expand the **Reports** node, and then expand the **Reports** subnode. Select all the items in the folder, right-click, and then select **Add to new project**.
2. Complete the **New Project** wizard. Accept all default values.
3. In Solution Explorer, select the project, right-click, and then select **Deploy reports** to deploy the build and deploy the reports locally.

When you add the modern report design to an existing report, you can reuse both the parameter handling and the data provider that the out-of-box solution uses.

Update Print management settings

At this point, you should be able to access the modern report designs from the application. Make sure that you thoroughly test the modern report design templates before you deploy them to production environments. To test the templates, you must activate the modern report designs for the application business process.

Follow these steps to update the Print management settings for customer sales orders by selecting the modern report design solution as the default report design.

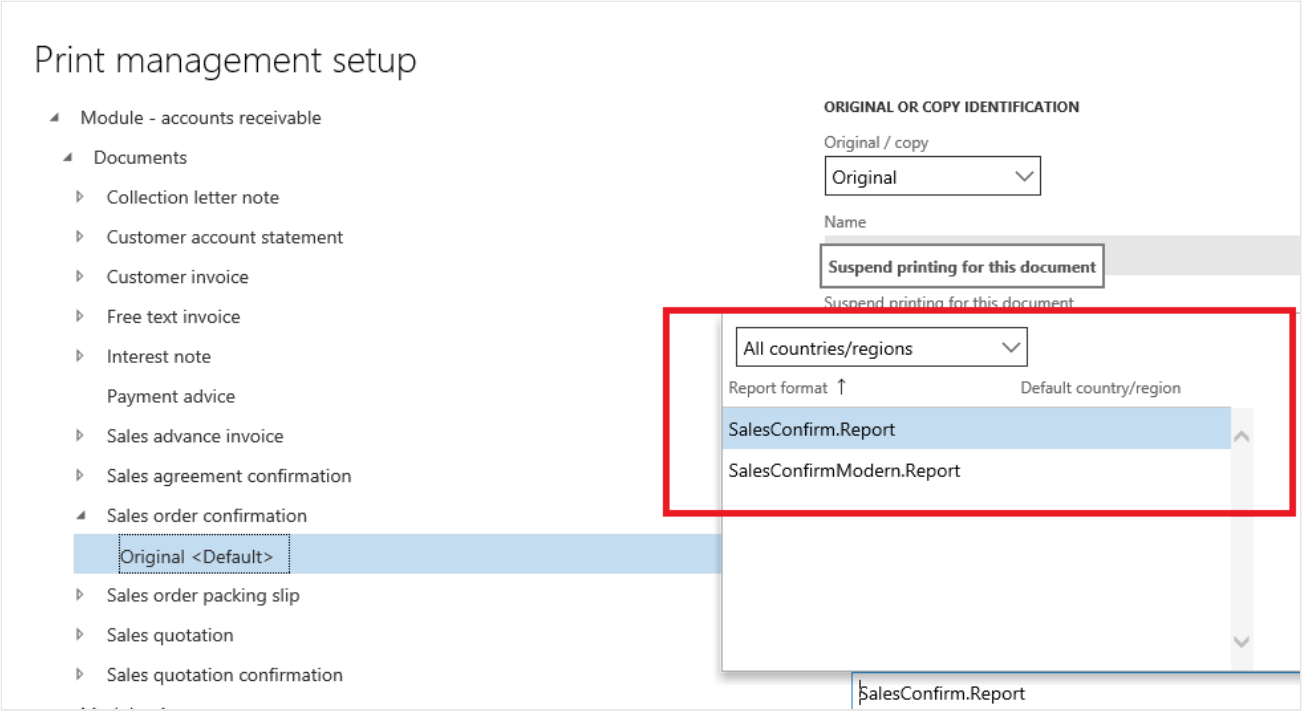
1. Open the **Form setup** page for the module. For example, for Accounts receivable, select **Accounts receivable** > **Setup** > **Forms** > **Form Setup**.

2. Select **Print management** to open the **Print management setup** page.

3. Expand the tree, and find the settings for the **Sales order confirmation** document.

4. Select **Original <Default>** to begin to modify the default document routing.

5. In the **Report format** list, select **SalesConfirmModern.Report** to enable the modern report design solution.



6. Open another page. This step forces a save operation to occur.

7. Post a sales order to view the modern design in the application.

Last updated on 04/13/2026

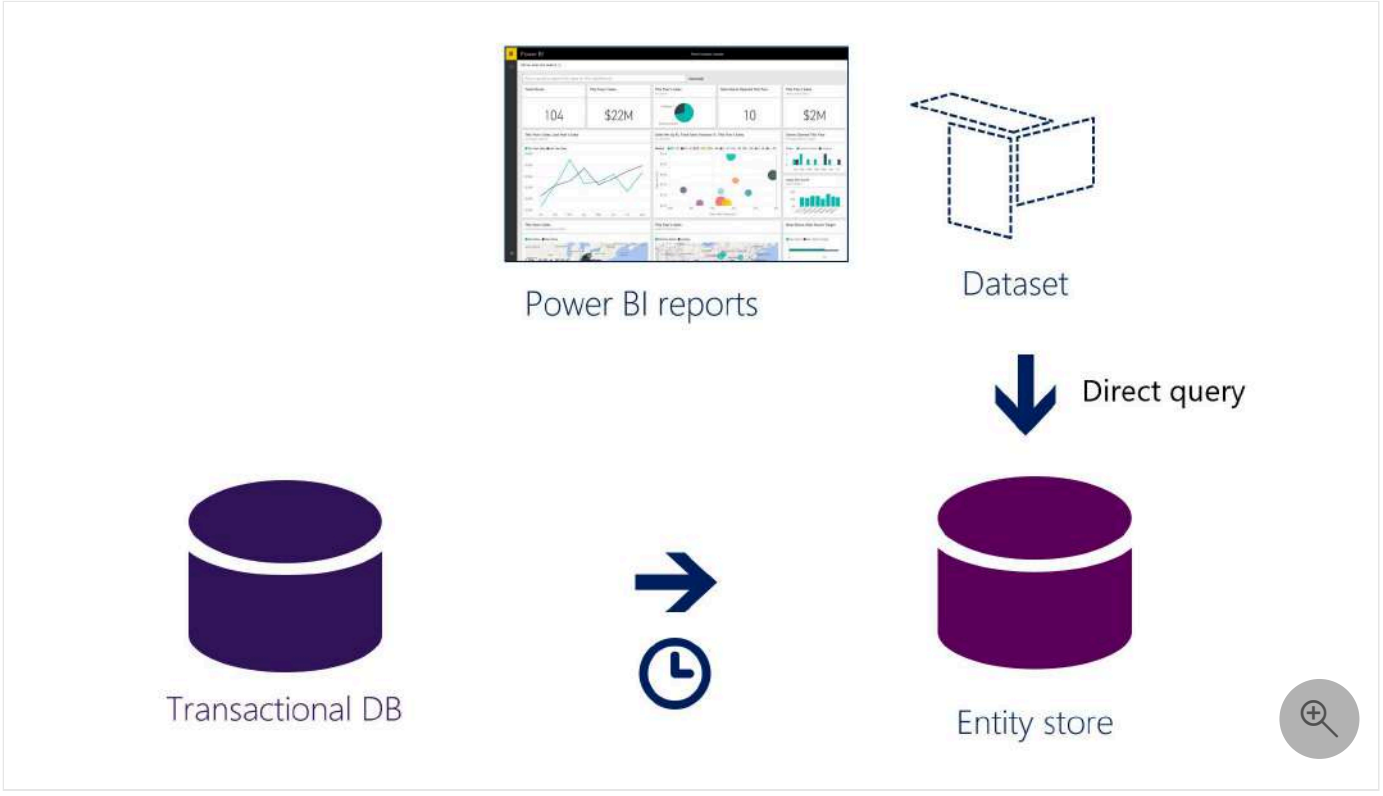
Power BI integration with Entity store

Entity store is an operational data store that's included with Microsoft Dynamics 365 Finance. This article describes how Entity store enables Power BI integration.

Entity store is an operational data store that's included with the application. The Microsoft Dynamics AX platform update 1 (May 2016) release introduced the Entity store feature. This feature lets an administrator or power user stage aggregate measurements in a dedicated data store for reporting and analytics. (Aggregate measurements are a star schema that is modeled by using entities.) This data store is called Entity store. It's a database that's optimized for reporting purposes. Entity store uses the in-memory, clustered columnstore index (CCI) functionality that is built into Microsoft SQL Server to optimize reporting and queries. Customers can use Microsoft Power BI DirectQuery models together with Entity store to enable high-volume, near-real-time analytical reporting over large volumes of data.

Power BI DirectQuery mode

In the February 2016 release of Microsoft Dynamics AX, you could create Power BI reports by using OData endpoints that data entities expose. Both aggregate data entities and detailed or regular data entities expose these OData endpoints. Although this approach is still supported, Entity store also lets power users create Power BI DirectQuery reports.



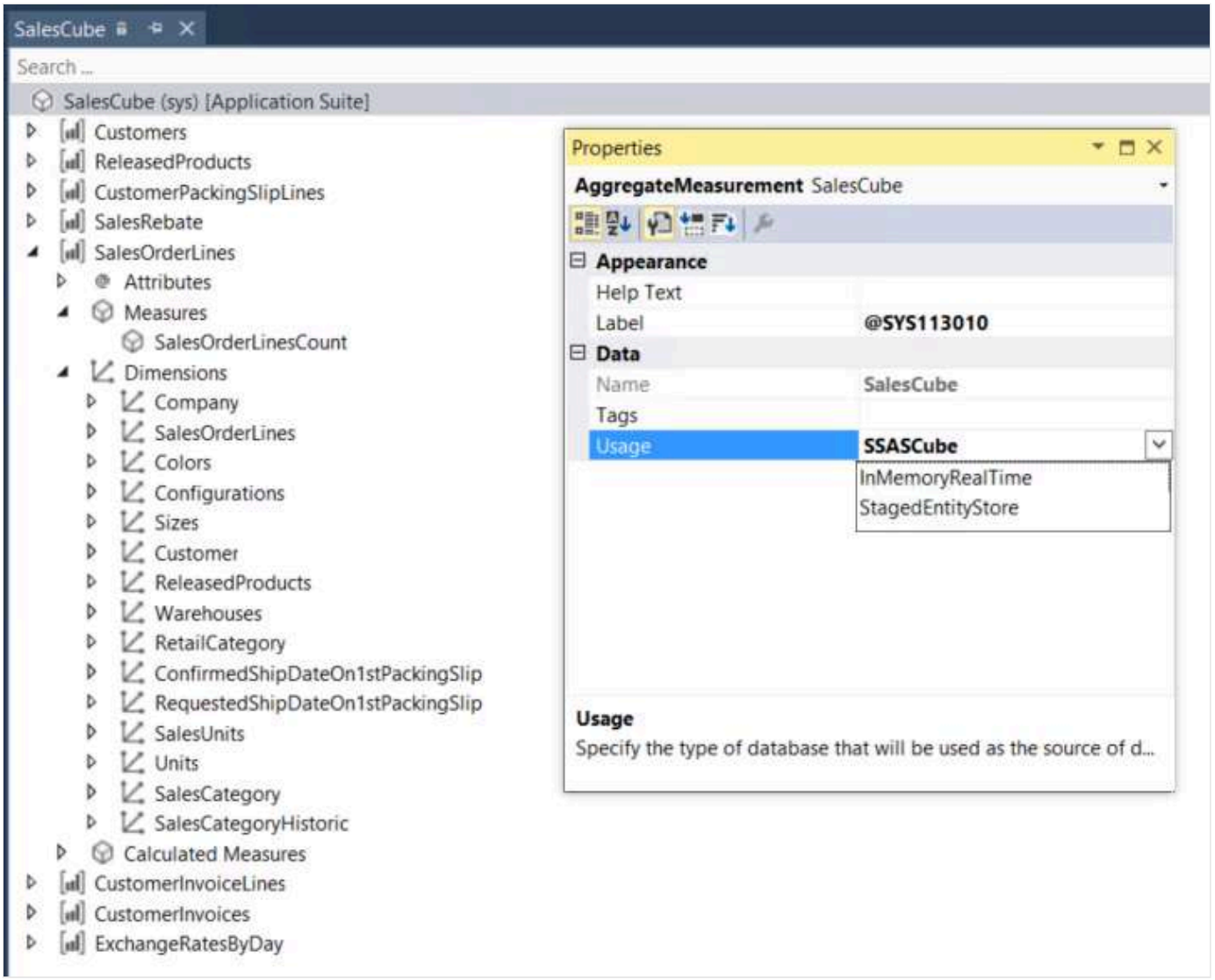
As the preceding illustration shows, DirectQuery is a reporting mode that runs reports directly on Entity store. In this reporting mode, data isn't staged in Power BI caches. This mode provides two immediate benefits:

- You can create Power BI reports over large data volumes.
- Reports don't have to be updated by using Power BI. When Entity store is updated, reports reflect the latest data.

Additionally, data doesn't leave your environment, because no data is cached in the Power BI service.

Stage aggregate measurements in Entity store

Aggregate measurements use a star schema that's designed for analytical scenarios. In the February 2016 release, Microsoft introduced real-time, in-memory aggregate measurements. By using real-time aggregate measurements, you can enable embedded charts and key performance indicators (KPIs) that react to real-time operations on data. For more information, see Transition from Analysis Services cubes to aggregate models. Real-time aggregate measurements take advantage of the in-memory, nonclustered columnstore index (NCCI) technology. Visuals and aggregate calculations that are built over real-time aggregate measurements reflect transactions within seconds. In the platform update 1 (May 2016) release, Microsoft introduced aggregate measurements that you can stage in Entity store. You can use aggregate measurements that are staged in Entity store for near-real-time analytical scenarios where you need to explore large volumes of data by using Power BI. As a developer, you learned how to model an aggregate measurement for real-time analytics in Model aggregate data. In the platform update 1 (May 2016) release, Microsoft added the capability to model aggregate measurements that you can stage in Entity store. In Microsoft Visual Studio, you can specify **StagedEntityStore** as the usage property of an aggregate measurement. This new property was added in May 2016. Previously, **InMemoryRealTime** was available as the usage property.



However, you might wonder why you would model an aggregate measurement for staging. Why wouldn't you use in-memory real-time aggregate measurements all the time? Several reasons exist for using the **StagedEntityStore** pattern:

- You need to explore and analyze large amounts of data.
- You migrated Analysis projects (that is, cubes) from Microsoft Dynamics AX 2012 R3 as part of the code upgrade process. Because of the complex views and joins that are present in the schema, query response times might not be acceptable for embedded visuals. However, you might not want to refactor the visuals to take advantage of NCCI technology immediately.
- Unlike the operational database schema, the schema in Entity store is modeled specifically for reporting. Therefore, it's easier to build new reports from the schema in Entity store.
- Your scenario doesn't require that analytical data be updated within seconds of an operation. Most Power BI reports that are built to enable data exploration fall into this category. If data freshness of approximately 10 minutes is acceptable for your scenario, you might want to use the staged pattern.

If one of the preceding reasons covers your situation, stage your aggregate measurement in Entity store and use it for Power BI integration.

Update Entity store

The system automates and manages Entity store refresh. In the client, you can find the **Entity Store** page at **Systems administration** > **Setup** > **Entity Store**. For more information, see Automated Entity store refresh.

Connecting to the Entity store database

For troubleshooting and diagnostics, you can connect to the Entity store database directly from a related sandbox environment. To connect:

1. Use Remote Desktop to access the sandbox. You can download the RDP file from the **Environment Details** page after you include your IP address in a safe list.
2. Open SQL Server Management Studio, and connect to the server specified on the **Environment Details** page.
    - Find the section titled **Database Accounts** Locate the entry for the user with the name **axdwadmin**.
    - The server name is the first portion of the **SQL Server\Database Name** field. Use the format **SQLServerName.database.windows.net**, where SQLServerName is the value from LCS.
    - Change the authentication type from Windows Authentication to SQL Server Authentication.
    - Use **axdwadmin** as the sign-in name and use the password from LCS.
3. Use the **Options** button or browse to the **Connection Properties** tab. Change the **Connect to database** property from the default value to your **Database Name** value from LCS.
4. Select **Connect** to access the database.

Last updated on 06/26/2026

Resolve problems after entity store maintenance

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you perform maintenance on the entity store, it affects the following components:

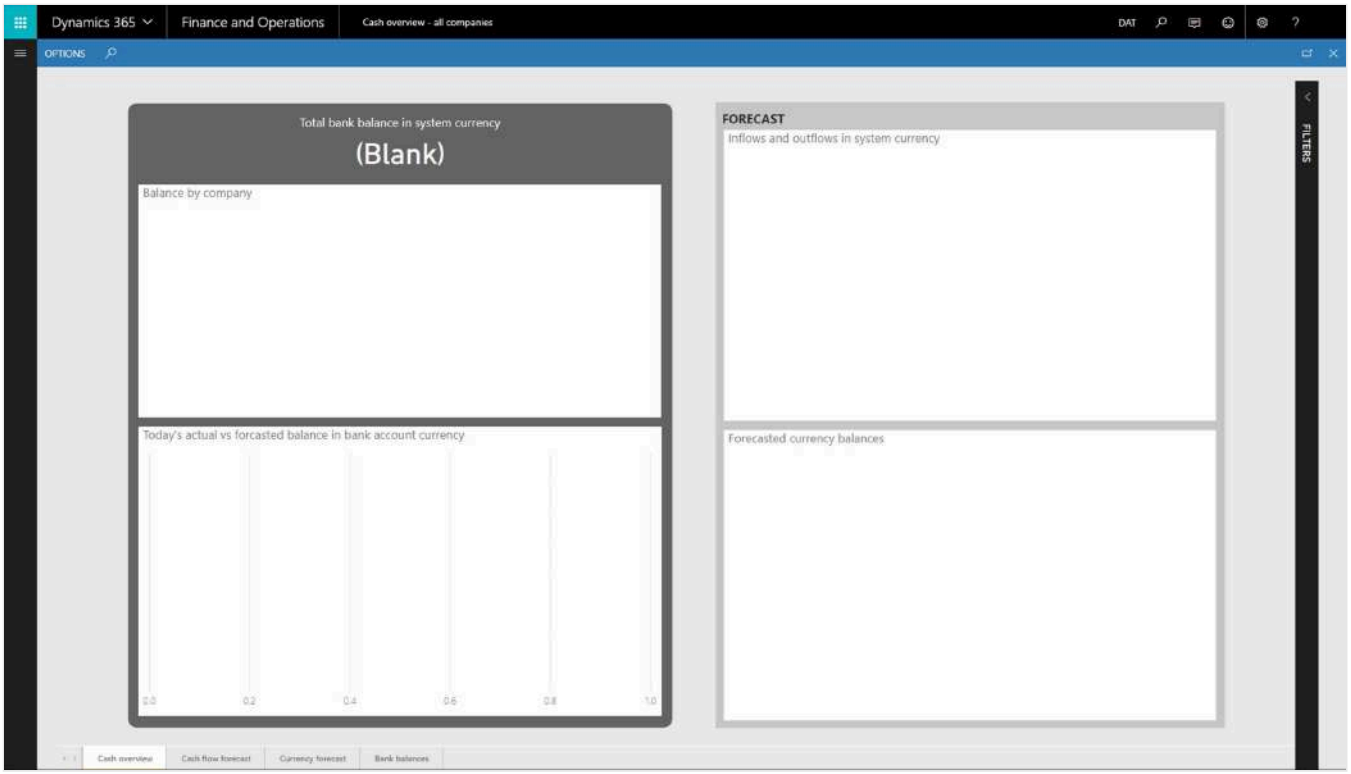
- Application analytical workspaces.
- Entity store-based reports deployed to PowerBI.com.

To resolve problems with these components, follow the procedures in this article.

> **Note**
>
> There's no impact to the normal operation of your application.

If you're using application analytical workspaces

Application analytical workspaces and reports might not show data after certain maintenance operations. The following screenshot shows an example of this problem.



To fix this problem:

1. Sign in to the application.
2. Go to **System administration** > **Inquiries** > **Batch jobs**.
3. On the **Batch jobs** page, delete all pending batch jobs that are associated with the entity store. These batch jobs:
   - Have a status of **Waiting**.
   - Typically have a description of **Deploy measurement**, **Full reset**, or **Incremental update**.

   > ⓘ **Note**
   >
   > The default description is **Deploy measurement** in older versions and **Full reset** in newer versions. If you enable Data Lake integration by using the **Trickle update Data Lake** option, an **Incremental update** batch job is created. You can verify whether a batch job is associated with the entity store by looking at the class name if the description is customized. Batch jobs associated with the entity store have a class name of **BIMeasurementDeployManagementEntityBatchJob**, **BIMeasurementProcessorFull**, or **BIMeasurementProcessorIncremental**.
4. Go to the **Entity store** page (**System Administration** > **Setup** > **Entity Store**).
5. Select all measurements that need to be refreshed.

6. Select **Refresh**, and then select **OK**.

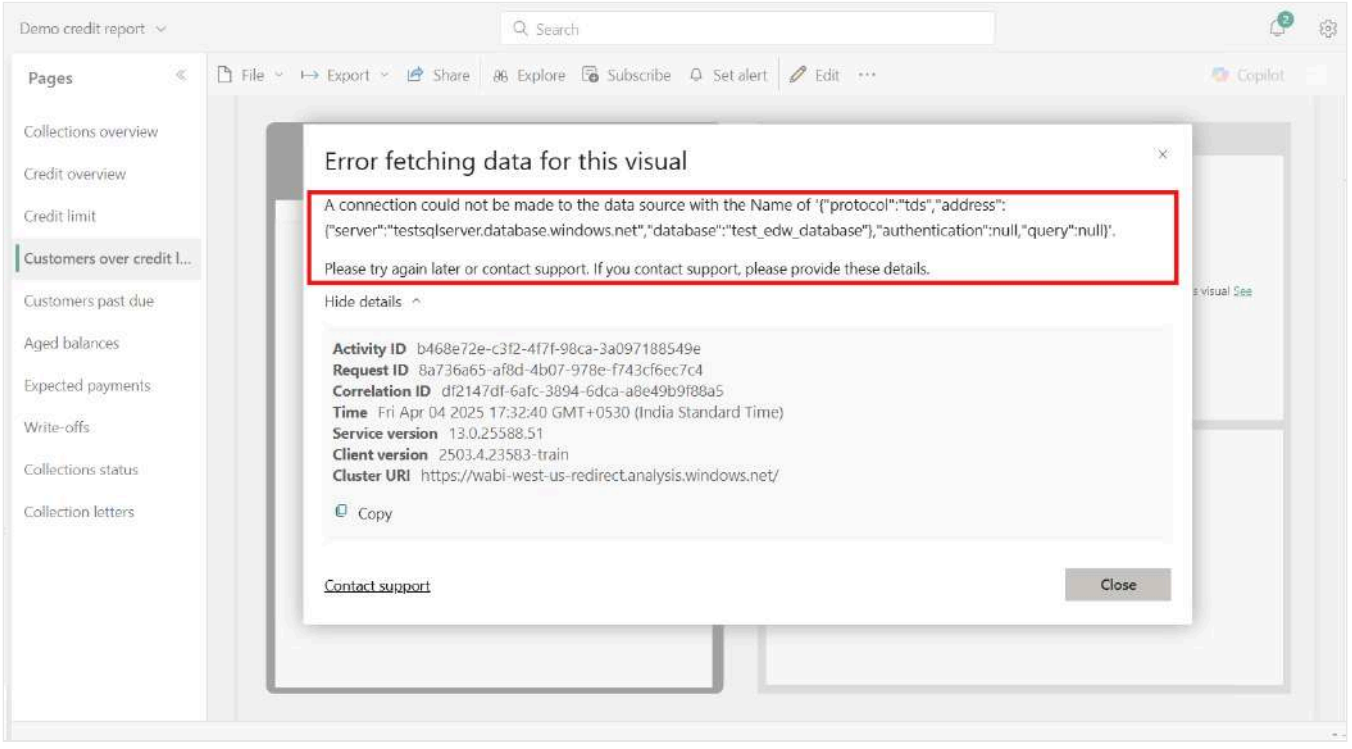
After the refresh finishes, the application analytical workspaces and reports show data.

Deployed entity store-based reports to PowerBI.com and using the PowerBI.com reports

Refresh the Entity store measurements as described above.

The reports deployed to PowerBI.com may receive the following errors:

- The credentials provided for the SQL source are invalid.
- Login failed for user <user_name>.
- A connection couldn't be made to the data source with the Name of '{"protocol":"tds","address":{"server":"testsqlserver.database.windows.net","database":"test_edw_database"},"authentication":null,"query":null}'. Try again later or contact support. When you contact support, provide these details:



These errors can occur if the credentials of Entity store database are rotated and PowerBI.com reports are configured with old credentials.

To resolve this issue, follow one of the following solutions:

1. Redeploy the reports where the error occurred. Go to **System Administration** > **Setup** > **Deploy Power BI files**.

- This action overwrites the current report and any customizations not in Microsoft Dynamics Lifecycle Services as PBIX are lost. If there are customizations in PowerBI.com, export the report as a PBIX file and upload it to Microsoft Dynamics Lifecycle Services for redeployment.

2. Deploy a sample report Go to **System Administration** > **Setup** > **Deploy Power BI files**. This fixes the above errors for any Entity store based reports in PowerBI.com

    - Users encountering this issue need to perform this action individually.

> **Note**
>
> For errors with the PowerBI.com reports after maintenance activity or if the above solution didn't resolve the issue, you may need to delete the report and the related dataset, and redeploy the report. To redeploy the report, go to **System Administration** > **Setup** > **Deploy Power BI files**. Any customizations not in Microsoft Dynamics Lifecycle Services as PBIX are lost.

Last updated on 01/22/2026

Configure PowerBI.com integration

Overview

Users can pin tiles, dashboards, and reports from their own PowerBI.com account to application workspace.

This functionality requires a one-time configuration of your environment. An administrator must complete this step to enable Microsoft Power BI to communicate and authenticate correctly.

For a workspace to show a Power BI tile, the server must contact the Power BI service on behalf of a user and access the visualization. It must then redraw the visualization in the application workspace. The fact that the server contacts the Power BI service "on behalf of a user" is important. When a user, such as `D365User@contoso.com`, contacts the PowerBI.com service, Power BI shows only tiles and reports from the user's PowerBI.com subscription.

By completing this configuration step, you enable them to contact the PowerBI.com service.

Things you must know before you start

- You must be a system administrator in the application. This option is available on the **System administration** menu.
- You must have a PowerBI.com account. You can create a trial account if you don't have an account. (A Pro license isn't required for this configuration step.)
- You must have at least one dashboard and one report in your Power BI account. Although the dashboard and report aren't required for this configuration step, you might not be able to validate the configuration if you don't have any content in your PowerBI.com account.
- You must be an administrator for your Microsoft Entra account. If you aren't the administrator, an administrative user must perform this configuration step for you.
- The Microsoft Entra domain that is configured must be the same domain that you used for your PowerBI.com account. For example, if you provisioned the application in the Contoso.com domain, you must have Power BI accounts in that domain, such as `Tim@ContosoAX7.onmicrosoft.com`.
- Users navigating to these pinned tiles must have a valid Power BI.com license.

Registration process

1. Sign in to https://portal.azure.com/ with an Azure tenant admin account.

   > **Note**
   >
   > The user who completes this procedure must have admin rights for the tenant to register applications.

2. Go to **Microsoft Entra** > **App registrations** > **New application registration**.

3. Enter the following values:

   - **Name** – Enter the name of your app.
   - **Supported account types** – Enter only accounts that are directly in this organization (single tenant).
   - **Application type** – Enter **Web**.
   - **Redirect URI** – Enter the base URL of your client with **oauth** appended to the end, such as `https://contosoax7.cloud.dynamics.com/oauth`.

4. Select **Register**.

5. Copy the **Application (client) ID** value. You use this value to connect to the PowerBI.com service.

   > **Important**
   >
   > Be sure to capture the **Application (client) ID** value.

6. Select **Manage** > **API permissions** > **Add a permission** > **Microsoft APIs** > **Power BI Service**.

   > **Note**
   >
   > For Government Community Cloud, instead of selecting **Power BI Service**, search for and select **Microsoft Power BI Government Community Cloud** in the **APIs my organization uses** tab.

7. Select **Select Delegated permissions**.

8. Enable the following permissions:

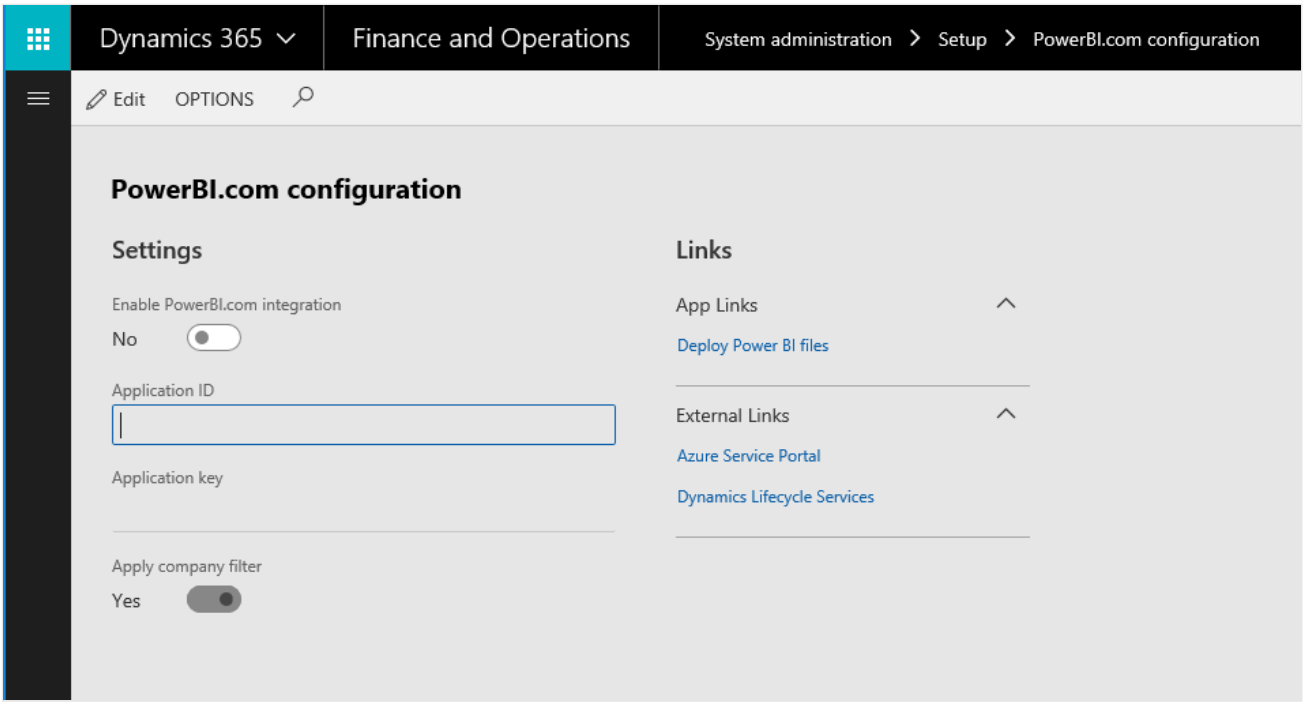
- **Content**
  - Content.Create
- **Dashboard**
  - Dashboard.Read.All
- **Dataset**
  - Dataset.Read.All
  - Dataset.ReadWrite.All
- **Report**
  - Report.Read.All
- **Workspace**
  - Workspace.Read.All

9. Select **Add permissions**. You must grant admin consent for the permissions to allow reports to be published to Power BI.
10. Select **Manage** > **Certificates & secrets**.
11. In the **Client secrets** field, select **New client secret**.
12. Enter a value in the **Description** and **Expires** fields, and then select **Add**.
13. Make a note of the **Application ID** and **Secret Value** values. You use these values in the next procedure.

    > ⓘ **Important**
    >
    > Be sure to capture the **Application (client) ID** and **Secret Value** values before you continue.

Specify Power BI settings in finance and operations

1. In the finance and operations client, open the **PowerBI.com configuration** page.



2. Select **Edit**.
3. Set the **Enabled** option to **Yes**.
4. In the **Application ID** field, enter the **Application ID** value that you got from Power BI in the previous procedure.
5. In the **Application key** field, enter the **Secret Value** value that you captured in Microsoft Entra ID in the previous procedure.

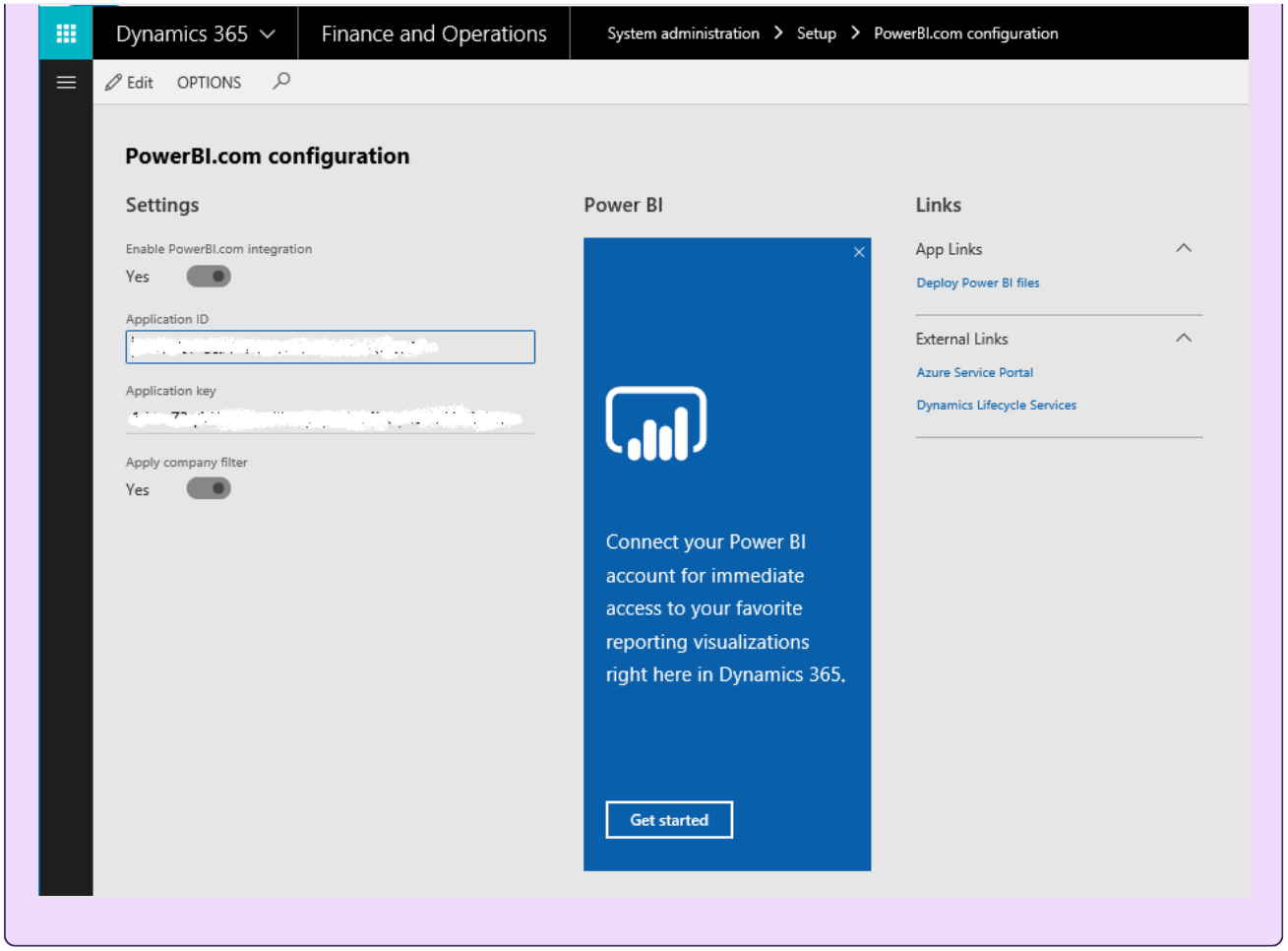
   You can apply the company filter only if your Power BI content has a table named **Company** and a column named **ID**. Ready-made Power BI content that is released uses this convention.
6. Select **Save**.

To verify the changes and enable PowerBI.com integrations, complete the steps in the next section.

Pin tiles to a workspace

1. To validate the PowerBI.com configuration, select **Get started**.

> **Note**
>
> You might need to refresh the browser to apply the changes.



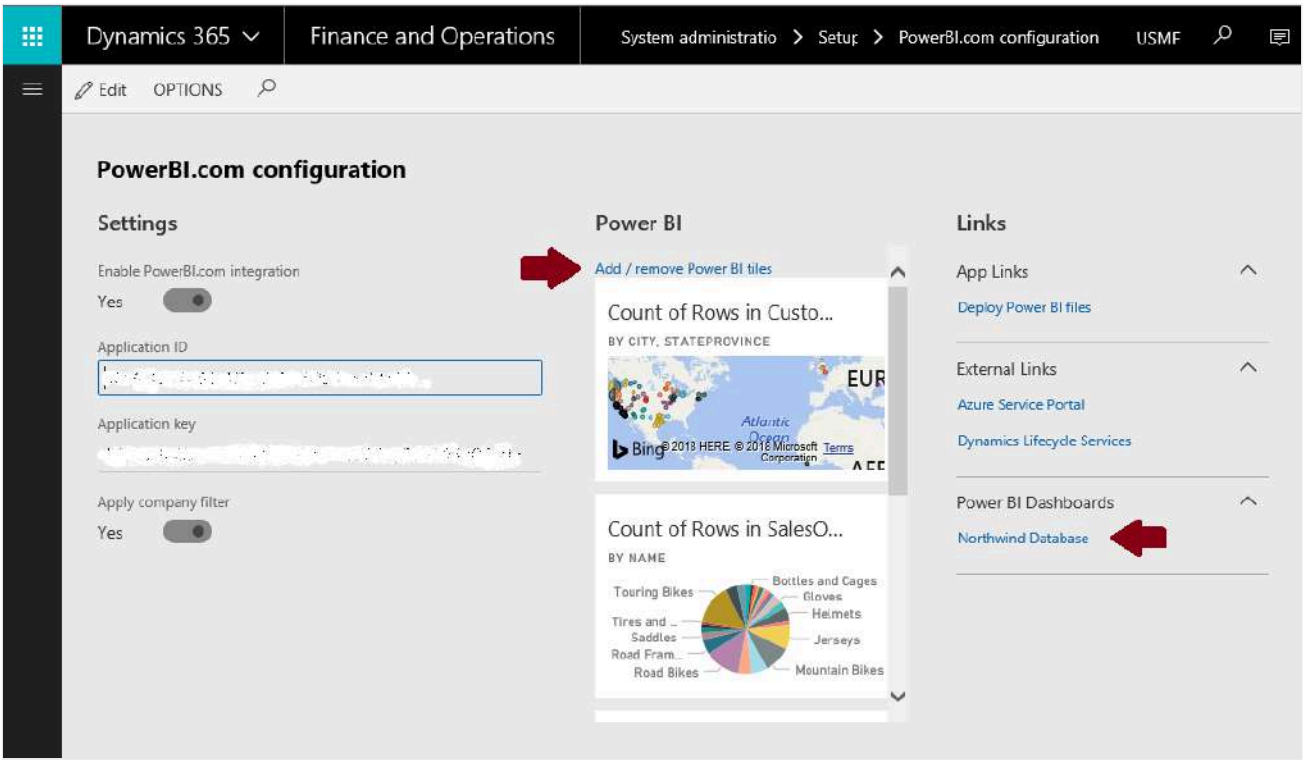
If you start Power BI from the application for the first time, you're prompted to authorize sign-in to Power BI from the client. Select **Click here to provide authorization to Power BI**.

Users must complete this step the first time they pin Power BI content.

2. The Microsoft Entra consent page asks for your consent. User consent is required for the application to access PowerBI.com on behalf of the user. Select **Accept**.

3. Because you're already signed in to Microsoft Entra ID, you don't have to enter your credentials again. A new tab appears, where you're prompted to authorize the connection between the application and Power BI. Authorize the connection, then return to the original tab.

4. A list of tiles from your PowerBI.com account appears. Select one or more tiles to pin to the selected workspace.



Troubleshooting common errors

In the previous procedure, after you select **Accept**, you might receive the following error message if the process isn't successful. The details of the error appear at the bottom of the message. More technical information provides clues that can help you determine what went wrong.

Common issues and resolution steps

Expand table

Error	Resolution
The Power BI service is unavailable.	This issue doesn't occur often, but the Power BI service might sometimes be unreachable. You don't need to re-register. Try to pin a tile to a workspace later.
You can't access the application.	You probably didn't select all the check boxes under **Step 8 in the Registration process** during the registration process. Start Power BI, and rerun the registration process.
The Power BI tiles page is empty (no content is shown).	Your PowerBI.com account might not have a dashboard or any tiles. Add a dashboard, such as a sample dashboard, and try to pin a tile again.
Error when authorizing Power BI	On the Azure Admin dashboard, under **Users and Groups** > **User settings**, make sure that the **Users can consent to apps accessing company data on their behalf** option is set to **Yes**.

Error	Resolution
Sorry, something went wrong. The authentication process wasn't successful. Contact your system administrator.	This error might occur when service keys are expired. Perform the registration process previously described, starting with step 3. When complete, be sure to update the PowerBI.com settings at **AX client** > **System administration** > **Set up** > **PowerBI.com integration**.

Last updated on 11/26/2025

Bring your own database (BYOD)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how administrators can export data entities from the application into their own Microsoft Azure SQL database. This feature is also known as *bring your own database* (BYOD).

The BYOD feature lets administrators configure their own database, and then export one or more data entities that are available in the application into the database. (Currently, more than 1,700 data entities are available.) Specifically, this feature lets you complete these tasks:

- Define one or more SQL databases that you can export entity data into.
- Export either all the records (*full push*) or only the records that changed or deleted (*incremental push*).
- Use the rich scheduling capabilities of the batch framework to enable periodic exports.
- Access the entity database by using Transact-SQL (T-SQL), and even extend the database by adding more tables.

Entity store or BYOD?

If you followed the series of blog posts about Microsoft Power BI integration, you're familiar with Entity store. Entity store is the operational data warehouse. Entity store provides built-in integration of operational reports with Power BI. Ready-made reports and analytical workspaces use Entity store. If you write Power BI reports by using data in your application environment, you should use Entity store.

However, the BYOD feature is recommended for the following scenarios:

- You must export data into your own data warehouse.
- You use analytical tools other than Power BI, and those tools require T-SQL access to data.
- You must perform batch integration with other systems.

> **Note**

The application doesn't allow T-SQL connections to the production database. If you're upgrading from a previous version of finance and operations, and you have integration solutions that require direct T-SQL access to the database, BYOD is the recommended upgrade path.

You can use either Entity store or BYOD. The default operational reports take advantage of embedded Power BI and Entity store. Use the default operational reports as your first choice. You can also extend the ready-made operational reports to meet your requirements. Consider BYOD a complementary option that you use as you require.

Creating a SQL database

Before you can configure the entity export option and use the BYOD feature, you must create a SQL database by using Azure portal.

For one-box development environments, you can create a database in the local Microsoft SQL Server database. However, use this database only for development and testing purposes. For production environments, you must create an Azure SQL database.

You should also create a SQL user account for sign-in to the database. Write down the server name, database name, and the SQL user ID and password. You use this information when you configure the entity export option in the next section.

If you're using the BYOD feature for integration for analytical purposes, consider using clustered columnstore indexes as described in Columnstore indexes: Overview.

Note

Finance and operations apps must access your BYOD database. If you encounter problems where you can't access BYOD, make sure firewall rules in your BYOD are configured appropriately. For more information, see Self-service deployment FAQ.

Selecting the correct service tier and compute size is critical to secure expected performance. While doing this, consider the total, targeted workload, and not just the load based on the finance and operations export. For production environments, use at least the minimum tier specified in the following table:

Expand table

Edition	Minimum Tier
Premium	P4 or higher
Standard	S6 or higher
Business Critical	BC_Gen5_8 or higher
Hyperscale	HS_Gen5_24 or higher
General Purpose	GP_Gen5_8 or higher

Your specific BYOD usage might require a service tier greater than the minimum. For more information about tiers and compute sizes, see SQL Azure service tiers and Detailed resource limits. To determine DTU needs or utilization, see Determine number of DTUs needed by a workload.

Configuring the entity export option

1. Start the client, and then, in the **Data management** workspace, select the **Configure Entity export to database** tile.

2. If you configured any databases, you see a list. Otherwise, you must configure a new database. In this case, select **New**, and then enter a unique name and a description for the new database. You can export entities into multiple databases.

3. Enter the connection string in the following format:

   Data Source=<logical server name>,1433; Initial Catalog=<your DB name>; Integrated Security=False; User ID=<SQL user ID>; Password=<password>

   In this connection string, the logical server name should resemble **nnnn.database.windows.net**. You find the logical server name in Azure portal. The following illustration shows an example of a connection string.

> ⓘ **Note**
>
> The default extension field shown in the previous image doesn't apply to BYOD.

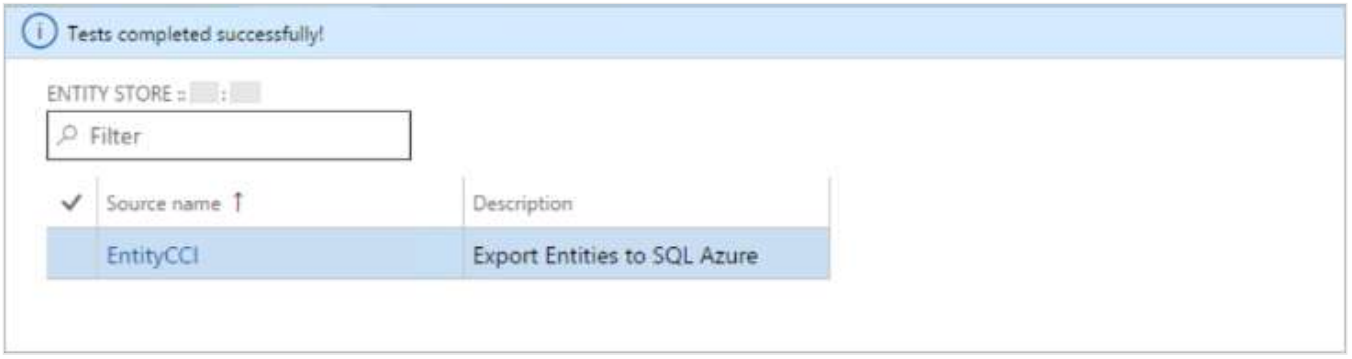
1. Select **Validate**, and make sure that the connection is successful.

   - The **Create clustered column store indexes** option optimizes the destination database for selected queries by defining columnstore indexes for entities that you copy.

- The **Enable triggers in target database** option sets jobs to enable SQL triggers in the target database. This option lets you hook downstream processes into the trigger to orchestrate actions that must be started after records are inserted. One trigger is supported per bulk insert operation. The size of the bulk insert is determined by the **Maximum insert commit size** parameter in the Data management framework.

For scenarios in which analytical applications read data from BYOD, there's always the challenge of ensuring that the reporting systems get consistent data from BYOD while the sync is in progress. You can achieve this result by not reading the analytical data apps directly from the staging tables that the BYOD process creates. The staging tables hold the data while data is being synced from the instance and hence constantly changes. Use the SQL trigger feature to determine when the data sync is completed, and then transform and fill data to the downstream analytical data scenarios.

When the validation passes, the database that you configured for entity appears in lists of databases, as shown in the following illustration.



You can now publish one or more entities to the new database by selecting the **Publish** option on the menu.

Publishing the entity schema to the database

The **Publish** page enables several scenarios:

- Publish new entities to the database.
- Delete previously published entities from the database. For example, you might want to re-create the schema.
- Compare published entities with the entity schema. For example, if you add new fields later, you can compare the fields with your database schema.
- Configure change tracking functionality that enables incremental updates of your data.

The following sections discuss each option.

Publish

The **Publish** option defines the entity database schema on the destination database. When you select one or more entities, and then select the **Publish** option, a batch job starts. This job creates the entities in the destination database. When the database definition job finishes, you receive a message, which you can access by using the bell symbol in the upper right.

The actual data update occurs when you export data. At this point, you're just creating the schema.

Drop entity

The **Drop entity** option deletes the data and the entity definition from the destination database.

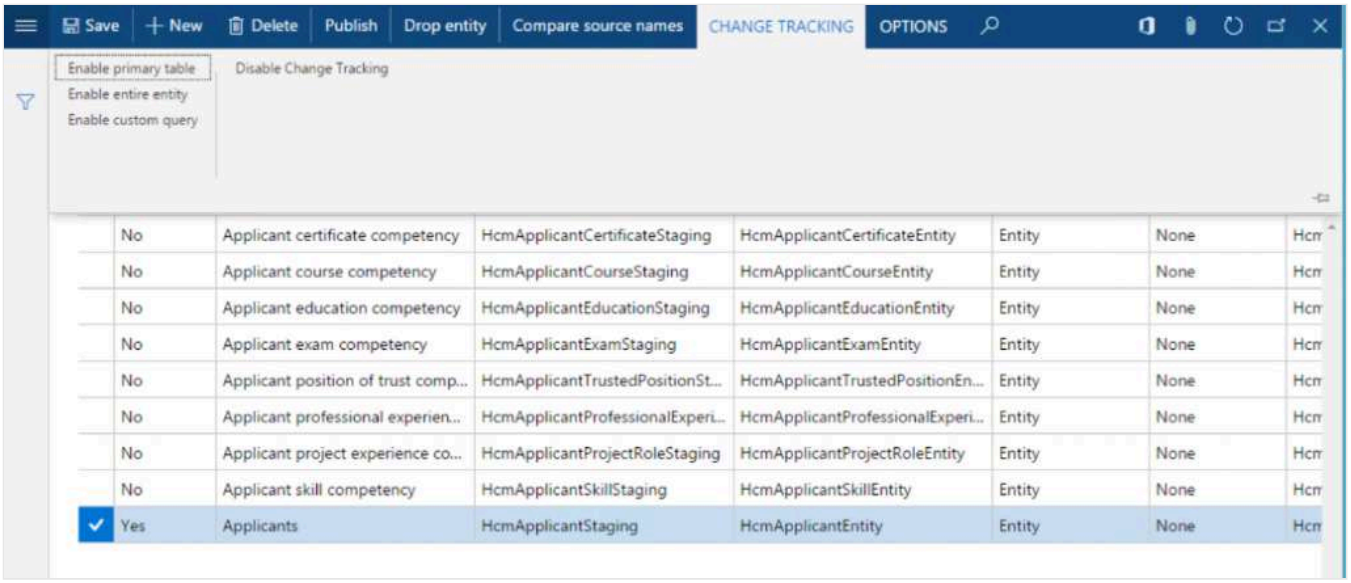
Compare source names

The **Compare source names** option lets you compare the entity schema in the destination with the entity schema in the application. Use this option for version management. You can also use this option to remove any unwanted columns from the destination table.

Configure change tracking

Change tracking is a feature that's provided in SQL Server and SQL Database. Change tracking enables the database to track changes, including deletes, that are made on tables. The system uses change tracking to identify changes that are made to tables as transactions. However, because the application must track changes at the data entity level, there's more logic on top of SQL change tracking to make this functionality work. The steps to enable change tracking are explained later in this section.

The **Change tracking** option on the **Publish** page lets you configure how changes are tracked on the underlying entity.



The following table describes the change tracking options that are available.

Expand table

Option	Description
Enable primary table	An entity consists of several tables. Select this option to track all changes that are made to the primary table of the entity. When you change the primary table, the system inserts or updates the corresponding record in the destination database. Although data from the whole entity is written to the destination table, the system triggers the insert or update option only when the primary table is modified.
Enable entire entity	Select this option to track all changes to the entity. These changes include changes to all the tables that make up the entity. When you change the entity, you make corresponding updates to the destination.
Enable custom query	This option lets a developer provide a custom query that the system runs to evaluate changes. This option is useful when you have a complex requirement to track changes from only a selected set of fields. You can also select this option when the exported entities were built by using a hierarchy of nested views. For more information, see Enable change tracking for entities.

To use change tracking, you must enable the **Change tracking** option as shown earlier in data management. This action is available on the **Data entities** list page, by going to **Data management** > **Data entities**. You need to select an entity and select from one of the options listed earlier to enable change tracking on the data entity.

If you republish an entity that exists in the destination database, the system warns you that existing data is deleted because of the new operation.

When you confirm the publish operation, the system publishes the schema to the database, and you're notified when the operation is completed.

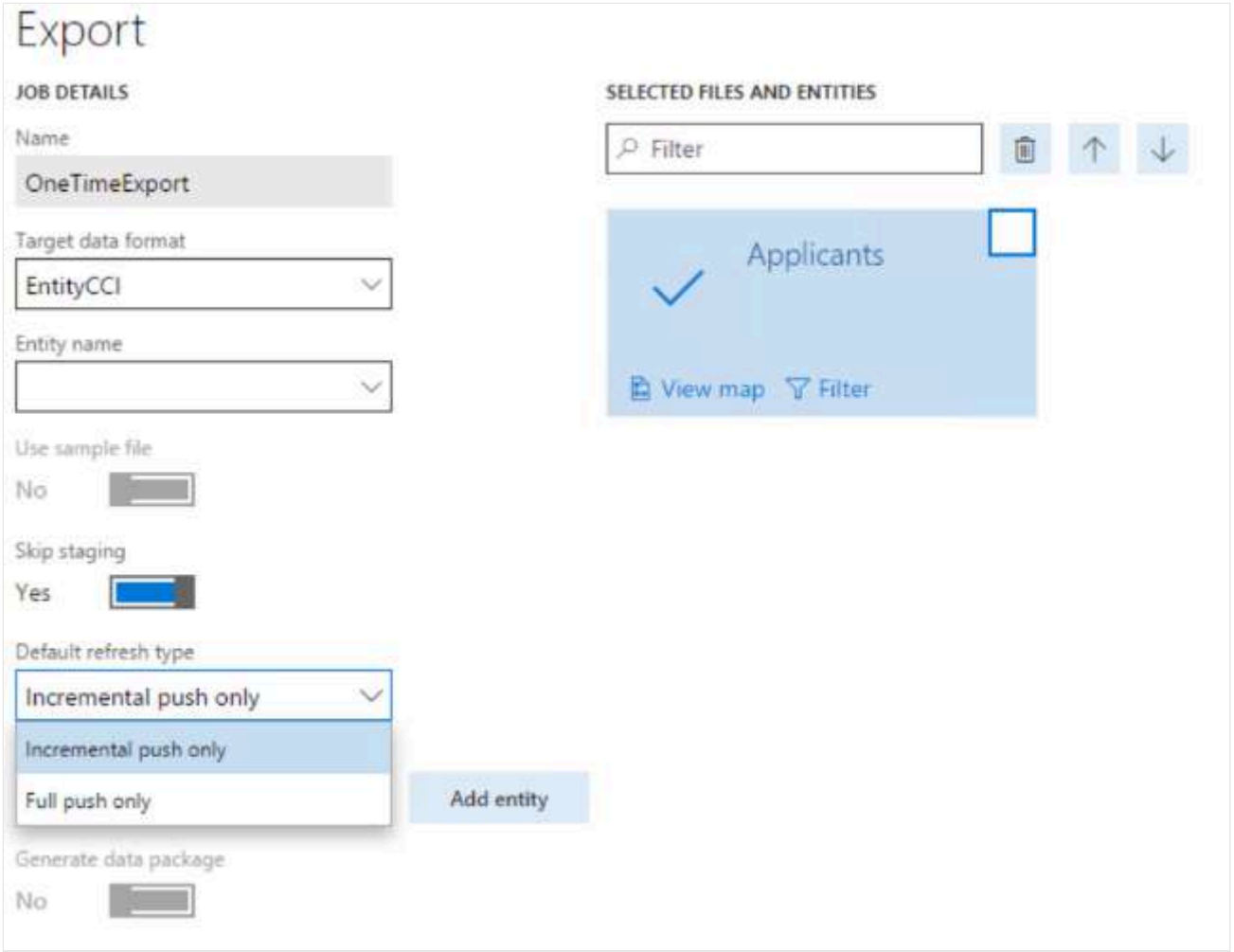
By selecting the **Show published only** option on the **Publish** page, you can show only the entities that you published to a given destination database. The Publish function creates the entity schema in the database. You can go to the database and see the table schemas that were created, together with corresponding indexes.

> **Note**
>
> Currently, you can't use BYOD to export composite entities into a database. You must export each entity in the composite entity.

Export data into your database

After you publish entities to the destination database, use the Export function in the **Data management** workspace to move data. The Export function lets you define a Data movement job that contains one or more entities.

Use the **Export** page to export data into many target data formats, such as a comma-separated values (CSV) file. This page also supports SQL databases as another destination.



You can create a data project that has multiple entities. Schedule this data project to run by using the batch framework. You can also schedule the data export job to run on a periodic basis by selecting the **Export in batch** option.

Export data across companies

Use jobs that run in batch to export data across companies. This batch export requires that the **Enable all company export** option is enabled under **Data import/export framework parameters** > **Bring your own database**. Concurrent exports for the same entity to the BYOD database might cause high DTU usage and can result in data loss for incremental exports. To avoid this risk, starting in version 10.0.16, all execution across companies is sequential per company. This execution means that jobs with a high number of entities and companies take longer to run.

To reduce the overall export times, consider the following practices:

- Make sure that the same data entity isn't in multiple projects so that the projects can run without conflicting with each other.
- Put entities that take a long time to run in separate projects. This separation allows the other projects to run faster.
- Use **Full push only** exports instead of **Incremental only** exports for smaller data sizes. For example, you might want to use this approach if some of the entities have record counts of around 1,000 or less.
- Create cross-company entities if you don't need to export data per company, individually.

To create a cross-company entity:

1. Copy the current per company entity.
2. Change the name of **PublicCollectionName** and **PubliceEntityName** of the copied entity.
3. Add a column for **DataAreaId**, if needed.
4. Remove the value for the **PrimaryCompanyContext** property in order to not filter by the primary company during export.
5. Generate the staging table, and then build and deploy the new entity.
6. Test the new entity to make sure that it works properly and performs adequately at required scale.
7. Schedule an export job in batch for the new entity, without selecting the **Export across all companies** option.

> **Note**
>
> Add multiple entities to an export project for BYOD carefully to ensure that the overall reliability of the BYOD export isn't compromised. Consider different parameters when

deciding the number of entities to add to the same project. Consider the degree of complexity of the entities, data volume per entity that you expect, and the overall time for export to complete at the job level. Avoid adding hundreds of entities. Create multiple jobs with a smaller number of entities.

Use of recurring exports in **Manage** > **Manage recurring data jobs** for BYOD isn't supported. You must use the **Export in batch** option.

Incremental export

When you add an entity for data export, you can select an incremental export (also known as incremental push) or a full push. For incremental push to work, you must enable the **Change tracking** option in the database and specify an appropriate change tracking option, as described earlier in this article.

> ⓘ **Note**
>
> A full push deletes all existing records from an entity and then inserts the current set of records from the selected entity.

If you select an incremental push, the first push always uses a full push. This push uses a full push because SQL needs to know which records are tracked in order to track subsequent changes. Whenever you insert a new record or add or delete a record, the corresponding changes are reflected in the destination entity.

Because the first push always uses a full push, don't perform an explicit full push before you enable change tracking.

Enable change tracking and schedule an export job with incremental push. This change takes care of the first full push and the subsequent incremental exports.

> ⓘ **Note**
>
> If you notice that the incremental push setting acts like a full push in certain scenarios, two possible reasons exist:
>
> 1. The entity is being exported for the first time by using incremental push, so it behaves as a full push.
> 2. The incremental push export doesn't run within the retention period. To prevent data loss, the export automatically falls back to the full push setting.

Timeouts

The default timeouts for BYOD exports are set to 10 minutes for truncation operations and one hour for actual bulk insert operations. When volumes are high, these timeout settings might not be sufficient and you must update them. You can update the timeout settings by going to **Data management** > **Framework parameters** > **Bring your own database**. These timeouts are company specific and you must set them separately for each company.

Known limitations

The BYOD feature has the following limitations.

Don't have active locks on your database during synchronization

Because BYOD is your own database, you must ensure that there are no active locks on your Azure SQL database when data is syncing. Active locks on your database during synchronization can result in slow writes or even failure to export to your Azure SQL database.

You can't export composite entities into your own database

Currently, composite entities aren't supported. You must export individual entities that make up the composite entity. You can do this export in the same data project.

You can't export entities that don't have unique keys by using incremental push

You might face this limitation especially when you try to incrementally export records from a few ready-made entities. Because these entities were designed to enable the import of data, they don't have a unique key. However, you can enable change tracking only for entities that have a unique key. Therefore, there's a limitation on incremental push. One workaround is to extend the required entity and define a unique key.

Troubleshooting

Incremental push isn't working correctly

Problem - When a full push occurs for some entity, you can see a large set of records in BYOD by using a select statement. However, an incremental push results in only a few records in BYOD. It seems as if the incremental push deleted all the records and added only the changed records in BYOD.

Solution - In cases like this, disable and re-enable change tracking for the entity in question. The state of the SQL change tracking tables might not be in the expected state. Also verify that there are no other incremental exports that cover the same tables (DMF, MR, Retail).

SSIS error code DTS_E_OLEDBERROR. An OLE DB error occurred. Error code: 0x80004005

Problem - Export to BYOD fails with an SSIS exception.

Console

```
An OLE DB error has occurred. Error code: 0x80004005.

An OLE DB record is available. Source: "Microsoft SQL Server Native Client 11.0"
Hresult: 0x80004005  Description: "Communication link failure".

An OLE DB record is available. Source: "Microsoft SQL Server Native Client 11.0"
Hresult: 0x80004005  Description: "TCP Provider: An existing connection was forcibly
closed by the remote host.

Failed to open a fastload rowset for <entityStaging>. Check that the object exists
in the database.

OLE DB Destination failed the pre-execute phase and returned error code 0xC0202040.
```

Solution - This problem occurs if the connection policy on the Azure SQL BYOD server is set to Proxy. Change this policy to **Redirect**. For more information, see SQL DB Connectivity Architecture.

Last updated on 01/22/2026

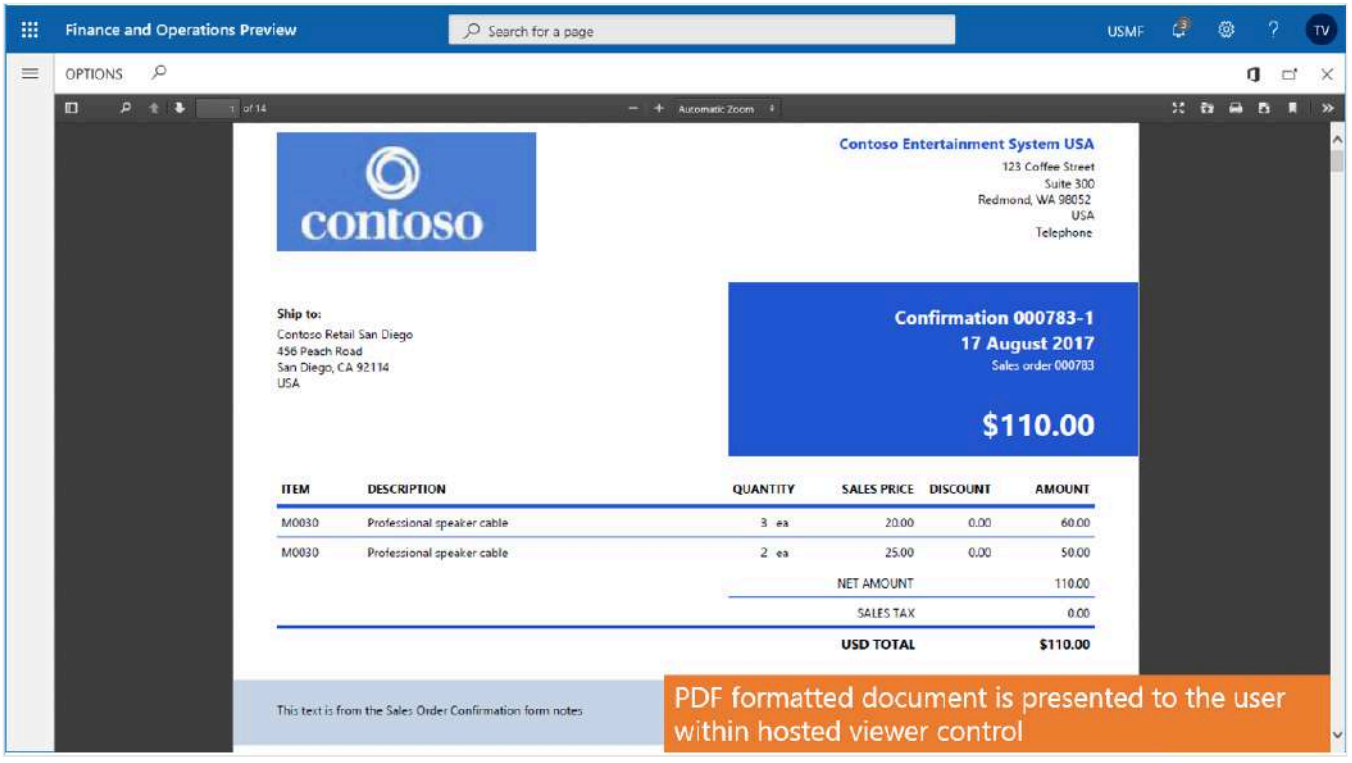
Preview PDF documents by using a PDF viewer

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Streamline application experiences that result in the production of business documents by taking advantage of the embedded PDF Preview option. Finance and operations applications deliver a modern experience to preview business documents that the service produces. You can use the built-in toolbar to navigate and download the document or to print to locally connected devices.

The embedded viewer offers consistency between the screen presentation and the printed output. In addition, report viewing times are drastically reduced when compared to the legacy experience. The Preview option is available on all supported devices and doesn't require any additional third-party software. You can easily download and navigate documents by using the built-in viewer toolbar options.

The following illustration shows a preview of the experience with a modern business document.



The legacy HTML-based preview experience is being replaced by a true document preview experience. The modern PDF preview experience offers several key advantages. These advantages include:

- Fidelity between the screen presentation and the printed output.
- A consistent document report preview experience across devices and platforms, including on-premises deployments.
- Server-side rendering that improves the performance when producing the document.
- Built-in tooling that allows users to quickly navigate the contents of the business document.

Accessing the PDF preview experience (Platform update 36 or later)

The PDF preview experience is enabled by default in Self-Service deployments and in environments hosted on Platform update 39 or later. To use the PDF preview experience in cloud-hosted and Microsoft-managed environments running Platform updates 36 through 38, use Feature Management to enable the **Report PDF viewer** feature.

Additional feature information

- Expandable and collapsible sections are available by default. These interactive operations don't function after the PDF document is created.
- The printer drop-down menu allows users to choose from locally connected devices. This list doesn't include network printers connected through the service.
- The built-in toolbar actions download documents to the local device.
- Use the **Print destination** options to produce documents in formats other than PDF.
- Use the **Enable accessibility feature on report PDF Files** feature flag to render PDF files as an accessible and tagged PDF, which is larger in size but easier for screen readers and other assistive technologies to read and navigate. The default is off.

Feature limitations

The Embedded PDF viewer experience delivers a closed document that exactly matches the printed output of the document. Recipients can't modify these documents, making the format ideal for business operations. However, as a closed format, the documents are far less interactive on the screen when compared to HTML presentations. The following end-user capabilities aren't supported when previewing documents by using the embedded PDF viewer.

- By default, embedded drill-thru navigation links are only available while previewing PDF documents.
- PDF documents don't support expandable and collapsible sections.

Last updated on 11/26/2025

Analytics, aggregate measurements, and KPI modeling

This article discusses the embedded business intelligence (BI), aggregate measurements, dimensions, data entities, and aggregate programming model.

Embedded business intelligence

The term embedded business intelligence (BI) refers to experiences that use highly intuitive and fluid visualizations to provide insights that are relevant to a task, so that the user is more informed and can make better choices. For example, a rental clerk in a car rental organization views the previous rental preferences of a customer when the customer makes a reservation. In this case, the clerk can see the vehicles and colors that the customer selected in the past, and therefore can offer options that are likely to please the customer. Use embedded BI throughout the user interface. At a technical level, building rich visualizations requires a powerful charting framework, and also an efficient way to access aggregated data that enables the display of fluid visualizations. Microsoft Dynamics 365 finance and operations meets both of these requirements, so that application developers can build rich and deep embedded BI–enabled scenarios.

Where are the perspectives?

Perspectives were designed to present data from a reporting point of view. Perspectives have evolved over the past three releases for analytical scenarios.

Expand table

Version	Description
Microsoft Dynamics AX 4.0	Perspectives provided the ability to model ad-hoc reporting models.
Microsoft Dynamics AX 2009	Perspectives added support for modeling analysis cubes.
Microsoft Dynamics AX 2012	Perspective modeling capabilities were improved through richer modeling support and deeper integration with the Application Object Tree (AOT).
Finance and operations apps	Perspectives are a first-class citizen in the data access framework. You can consume them via X++ or C# code, and also in a model-driven way.

Perspectives reside within the analytics collection in the Application Explorer. Perspectives underwent a major upgrade and now incorporate the following improvements:

- You can model new aggregate models and customize existing aggregate models as a star schema within Application Explorer.
- Modeling for key performance indicators (KPIs) in Application Explorer is supported.
- You can model data entities by referencing aggregate models. You can expose data entities to external reporting tools, such as PowerBI, as OData endpoints. You can also consume data entities.
- You can consume aggregate models directly in the programming model by using X++ or C# code. You no longer have to write MDX code to consume aggregate data.
- Aggregate data is a first-class citizen within application data access. Its behavior is similar to the behavior of detailed data. For example, you can enrich aggregate data with extended data types (EDTs) and enumerations, and you can help secure them by using application security concepts.
- The aggregate data infrastructure is maintained completely within the environment. By default, aggregate measurements are real-time. As a system administrator, you can manage the latency of aggregate data and controls based on available resources and business needs, without having to deal with the complexity of scheduling and external tools.
- Developers can reuse existing business models, making the modeling process quicker and easier.

You can upgrade projects that were generated by using perspectives from Dynamics AX 2012 and later to finance and operations metadata equivalents.

Aggregate measurements and aggregate dimensions

An aggregate measurement is a model that contains a collection of measures together with their corresponding dimensions. Measures are aggregate numbers, such as Total Sales or Number of Orders. Dimensions are slicers, such as Product, Vendor, or Customer, that help you analyze the measure. For example, the measure of Total Sales isn't useful unless it can be sliced by Product, Region, and Customer. Aggregate measurements are the evolution of AX 2012 analysis cubes. Whereas a cube was based on a multidimensional online analytical processing (OLAP) technology, an aggregate measurement abstracts the underlying technology. Therefore, you no longer have to know a lot about the underlying implementation technology. Additionally, the underlying technology infrastructure can take advantage of improvements in in-memory real-time technology without requiring the developer to rewrite the program. Aggregate dimensions are shared across an implementation. Aggregate measurements associate themselves with

relevant aggregate dimensions. For example, Total Sales can be associated with Customer, Product, and Sales Region dimensions. However, Total Sales can't be associated with Vendor and Warehouse dimensions. You model aggregate dimensions and aggregate measurements by using Visual Studio tools. The Upgrade tool lets customers and partners migrate existing Dynamics AX 2012 cubes to newer versions of the product.

Aggregate data entities

By using the model-driven approach, you can create data entities by directly referencing aggregate measurements and aggregate dimensions. These data entities are known as aggregate data entities. Aggregate data entities are read-only data entities that you use for reporting purposes. To consume aggregate data when you build charts and other client controls, add the aggregate data to a form as a data source. You can also consume aggregate data entities programmatically in C# or X++ code.

Aggregate programming model

The Aggregate programming model lets a developer consume aggregate data programmatically by using either X++ or C# code. Data that you retrieve by using the Aggregate programming model can be used as a data source in forms and reports. A developer can add aggregate data that is modeled in perspectives to an **AXQuery** object. The developer can also use an existing aggregate data entity to create a query that can be extended by adding filters and additional columns that aren't present in the aggregate data entity. Bulk Move is a capability that is associated with the Aggregate programming model. When a query is run by the kernel, the developer can move all the records to a temporary or regular table without iterating row by row. Bulk Move provides a very efficient way to move data from aggregate models to temporary tables.

Method expressions

Method expressions are a programming model for constructing rich calculations that are used to define fields in a data entity. Method expressions enhance the computed column capability that was introduced in AX 2012 views. Method expressions let you build expressions by using the C# or X++ programming language. You can also create calculations on aggregate data that was previously coded by using MDX. Method expressions can be shared across the program.

In-memory, near-real-time aggregate measurements

You deploy aggregate measurements to Microsoft SQL Server nonclustered columnstore indexes (NCCI). Therefore, they take advantage of the in-memory computing (IMC) engine that is built into Microsoft SQL Server 2016 as Azure DB. Aggregate measurements that have the IMC engine as their destination are referred to as in-memory, near-real-time (IM-NRT) aggregate measurements. These aggregate measurements don't require that a Microsoft SQL Server Analysis Services (SSAS) server be used. Because these models don't involve data updates, the queries that are sent to them reflect the latest state of data in the operational database. That is why it's referred to as near-real-time.

KPI modeling and customization

In AX 2012 and earlier versions, you had to model KPIs and business indicators by using native SQL Server development tools. Although users could pin a KPI or business indicator to a Role Center by using the **Business Overview** Web Part, they couldn't modify a KPI definition, such as the goal. Users can use a rich client form to modify a KPI definition that a developer built and shipped. Users can also define new KPIs by using the aggregate data that is contained in aggregate measurements. A developer can model a KPI definition in Microsoft Visual Studio and ship it to a customer, either as a project or together with an independent software vendor (ISV) solution. After a KPI is defined, users can customize it at run time.

Last updated on 06/26/2026

Model aggregate data

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This tutorial shows you how to model aggregate data.

Prerequisites

This tutorial requires you to access the environment by using Remote Desktop and be provisioned as an administrator. For more information, see Deploy and access development environments.

Key concepts

- **Aggregate measurements**, similar to **perspectives** and **Cubes** from earlier versions, enable you to model and consume aggregate data.
- **Key Performance Indicators (KPIs)** are a form of analytic controls that track organizational performance against the current status. KPIs are represented as tiles in a **workspace** or the **dashboard**. In this tutorial, you model a KPI in Visual Studio.
- You can modify KPIs that you model in Visual Studio in the client. A user can also model new KPIs by using the client.
- A **workspace** is an overview page that's specific to a particular subject area. All users share workspaces.
- The **dashboard** is the default home page for each user.
- **Tiles** are securable objects that you can pin to a workspace or the dashboard. You can secure KPIs and aggregate data that show on the dashboard or a workspace by using menu items.
- You can consume aggregate data in building charts and other controls. By using the model driven approach, you can create data entities by directly referencing aggregate measurements and aggregate dimensions. These entities are referred to as **Aggregate data entities**. Aggregate data entities are read-only data entities used for reporting purposes.
- **The aggregate programming model** enables a developer to consume aggregate data programmatically by using either X++ or C# code. You can use data retrieved by using

the aggregate programming model as a data source in forms.

- **Method expressions** enable a developer to build rich expressions by using aggregate data. Method expressions are an X++ class. By using method expressions, you can model KPIs, thereby eliminating the need to build MDX expressions.
- **Contextual BI** refers to providing required insights as part of the user experience such that the user has relevant insights to not only achieve the task at hand, but be highly-productive during the day.
- **Embedded BI** refers to analytic content being embedded within the user experience. Contextual BI and embedded BI teams are closely related. Contextual BI implies the added notion that the context of analytic context revolves around the data or the task.
- **Self-service BI** refers to enabling a user to tweak existing and/or create new analytic content such as reports, KPIs, and dashboards.

Set up

If this tutorial is the first tutorial you're working on, make sure you configure the administrator user if you're running on a local VM.

Import the tutorial project

If you already imported the Fleet management tutorial project, go to the next section. In Visual Studio, on the **Dynamics 365** menu, select **Import Project**.

1. Download the Fleet Management sample from https://github.com/Microsoft/FMLab, save it to **c:**, and unzip it.
2. In the **Import Project** window, next to the **Filename** text box, select the ellipsis button.
3. In the **Select the file to import** window, browse to the location of the **FMLab** folder, select **FMTutorialDataModel.axpp**, and then select **Open**.
4. In the **Project file location** text box, enter **C:\FMLab**.
5. Select the **Overwrite Elements** check box, and then select **OK**.

Open the tutorial project

1. In Visual Studio, open the **FMTutorial** project. On the **File** menu, point to **Open**, and then select **Project/Solution**.
2. In the **Open Project** dialog box, browse to C:\FMLab\FMTutorial, and then select FMTutorial. Select **Open**. The **FMTutorial** project appears in **Solution Explorer**.
3. Use the FMTDataHelper class to load data for the Fleet Management tutorial. In **Solution Explorer**, in the FMTutorial project, expand **Classes**. Right-click **FMTDataHelper**, and then select **Set as Startup Object**.

4. On the **Build** menu, select **Rebuild Solution**. Use the rebuild to update the timestamps of the imported artifacts. You can view the build progress in the **Output** window.
5. Press Ctrl+F5 to run the project and load the data.

Model an aggregate measurement for rental charges

Often, when a user asks for additional information, you get a request for one or more new reports. Imagine that the manager of a rental car company calls and asks for a report. The manager is interested in finding out how the rental business is performing. The manager wants a report that shows rental revenue by month. You soon find out that the manager is interested in a breakdown of rental revenues. The manager wants to know whether the rental revenue is high in cases where they sold additional services, such as car seats, GPS, refueling, as opposed to the base rental charge. As it turns out, the manager suspects that specific customer groups are driving up revenue, and this is why the manager wanted the report in the first place. The manager insists on adding Customer group to the report. Because the revenue must be considered in relation to the number of rentals, the manager doesn't want a few large corporate rentals to skew the analysis. You both agree that the number of rentals needs to be shown along with revenue. You could represent this requirement as a set of business questions using a matrix. Rows indicate the **measures** (or numbers) and the columns indicate the **dimensions** (or slicers). An "X" in the intersection between a measure and a dimension indicates that the measure needs to be "grouped by" the dimension.

Expand table

Item	Rental date	Customer group	Rental charge type
Revenue	X	X	X
Number of rentals	X	X	X

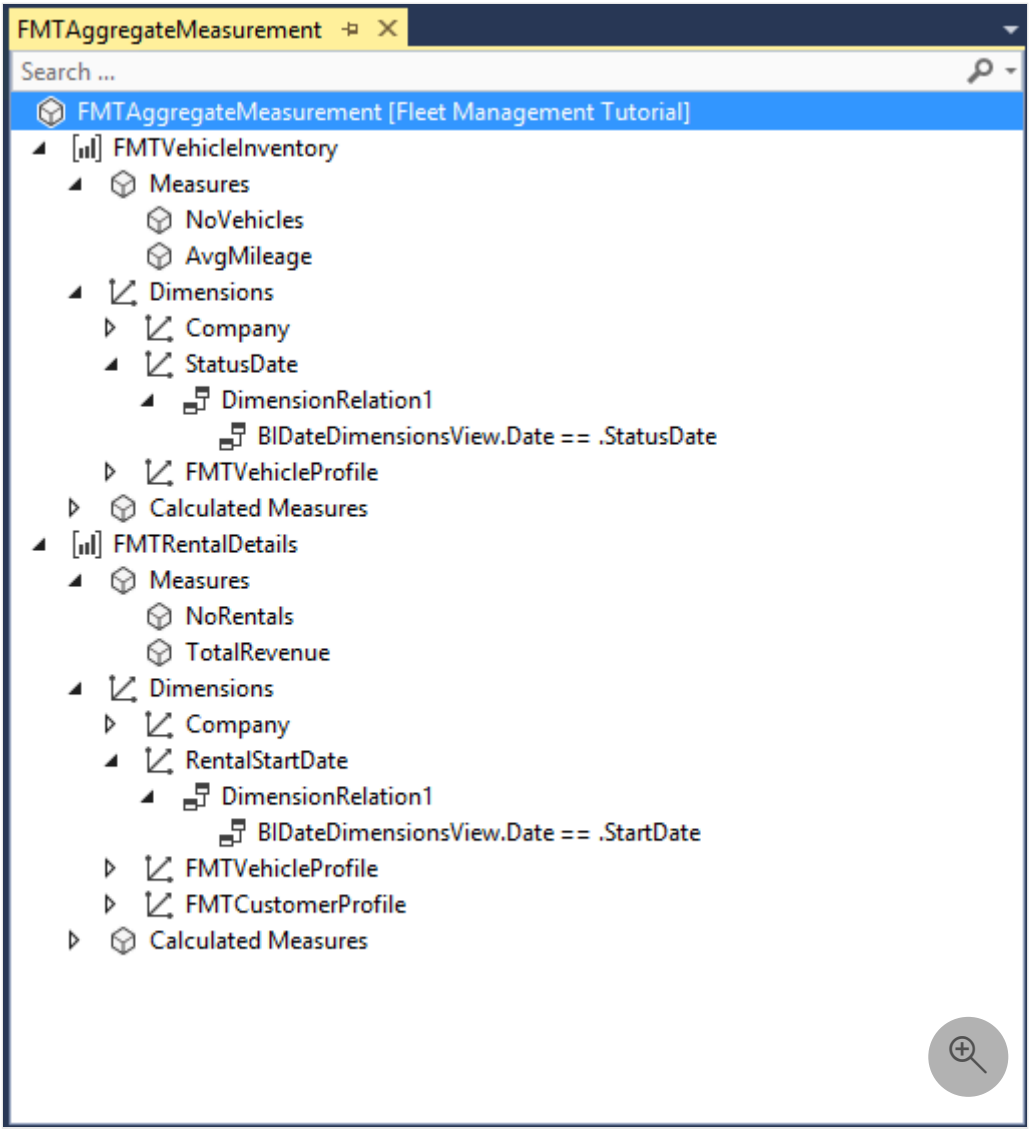
Next, model an aggregate measurement to answer this business question.

Add a measure group for rental charges by using an existing view

In this section, you add a new measure group to an existing aggregate measurement.

1. In **Solution Explorer**, expand the **Analytics** folder of the project, and then double-click the aggregate measurement, **FMTAggregateMeasurement**. The aggregate measurement opens in the designer. Notice that the existing aggregate measurement contains two

measure groups related to vehicle inventory and rental details. You create a new measure group related to rental charges.



2. In **Solution Explorer**, expand the **Views** folder of the project, and then select the **FMTRentalChargeExtendedView** view.

3. Drag and drop the **FMTRentalChargeExtendedView** into the root node of the **FMTAggregateMeasurement** aggregate measurement in the designer. A new measure group is created and the values of properties are applied as follows.

Expand table

Property	Value
Name	FMRentalChargeExtendedView
Table	FMRentalChargeExtendedView

4. In **Solution Explorer**, double-click the **FMRentalChargeExtendedView** view. When the designer form opens, expand the **Fields** node.

5. Select the **ExtendedAmount** and **RentalID** fields, and then drag and drop the two fields onto the **Measures** node of the newly created **FMAggregateMeasurement** measure group called **FMRentalChargeExtendedView**. As you drag the fields, hover your cursor over the **FMAggregateMeasurement** tab to access the **Measures** node. By default, when you drag and drop the fields, the system assumes that you want counts of the measures. In this case, you need to modify default properties for the **ExtendedAmount** and **RentalId** measures as follows:

Expand table

Property (ExtendedAmount)	Value
Default Aggregate	**Sum**
Field	ExtendedAmount (no change required)
Name	Revenue

Expand table

Property (RentalId)	Value
Default Aggregate	**DistinctCount**
Field	RentalId (no change required)
Name	NumRentals

6. Next, create another measure that calculates **Revenue per Rental** by copying an existing measure. Select the **TotalRevenue** measure. Right-click and select **Copy** from the **Context** menu.

7. Select the **FMRentalDetails** measure group. Right-click and select paste. Rename the newly created measure to **RevenuePerRental**. Modify the aggregation function to **Average**.

Model an aggregate dimension charge type

To analyze rental revenue by the different charge types, you need to be able to slice revenue by the charge type. For this purpose, model a charge type dimension.

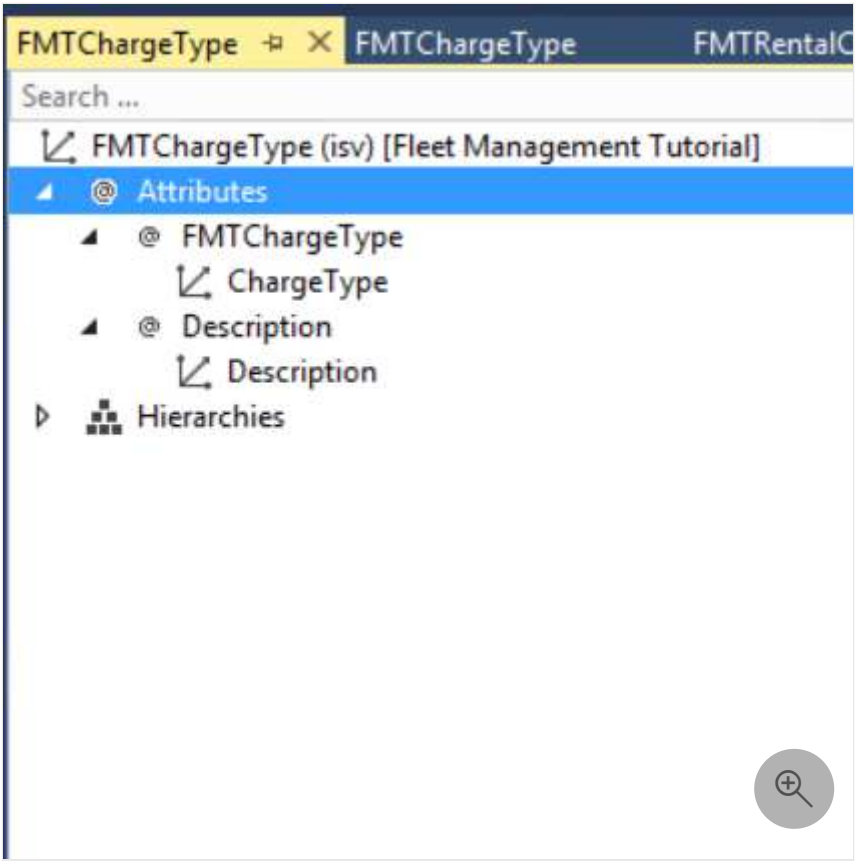
1. In **Solution Explorer**, under **FMTutorial**, right-click **Analytics**, point to **Add**, and then select **New Item**.

2. Select **Dynamics 365 Artifacts** > **Analytics** > **Aggregate Dimension** from the list of items.

3. In the **Name** property, enter **FMTChargeType**. This name is the aggregate dimension that you create. It must be unique. Select **Add**. The new dimension appears in Visual Studio.

4. In **Application Explorer**, expand the **AOT** and select **Data Model** > **Tables**. Drag and drop the **FMTChargeType** table from **Application Explorer** onto the root node of the newly created **FMTChargeType** dimension in the designer. The **AutoReport** field group of the table adds dimension attributes and corresponding keys.

5. Expand the **Attributes** node of the new dimension. Several attributes are created by default. The system also creates a dimension key based on unique indexes of the table. You can add extra fields by dragging and dropping them into the **Fields** node.



6. Save the new dimension.

7. You might get a warning that asks you to rename the field name **Description** to avoid the MDX reserved word. Even though you might not deploy the aggregate dimension to SSAS in this aggregate measurement, it's possible that this dimension might be used by an aggregate measurement deployed to SSAS in the future. To avoid potential problems, rename the field name **Description** to **ChargeDescription**.

Model dimension references for customer profile and charge type dimensions

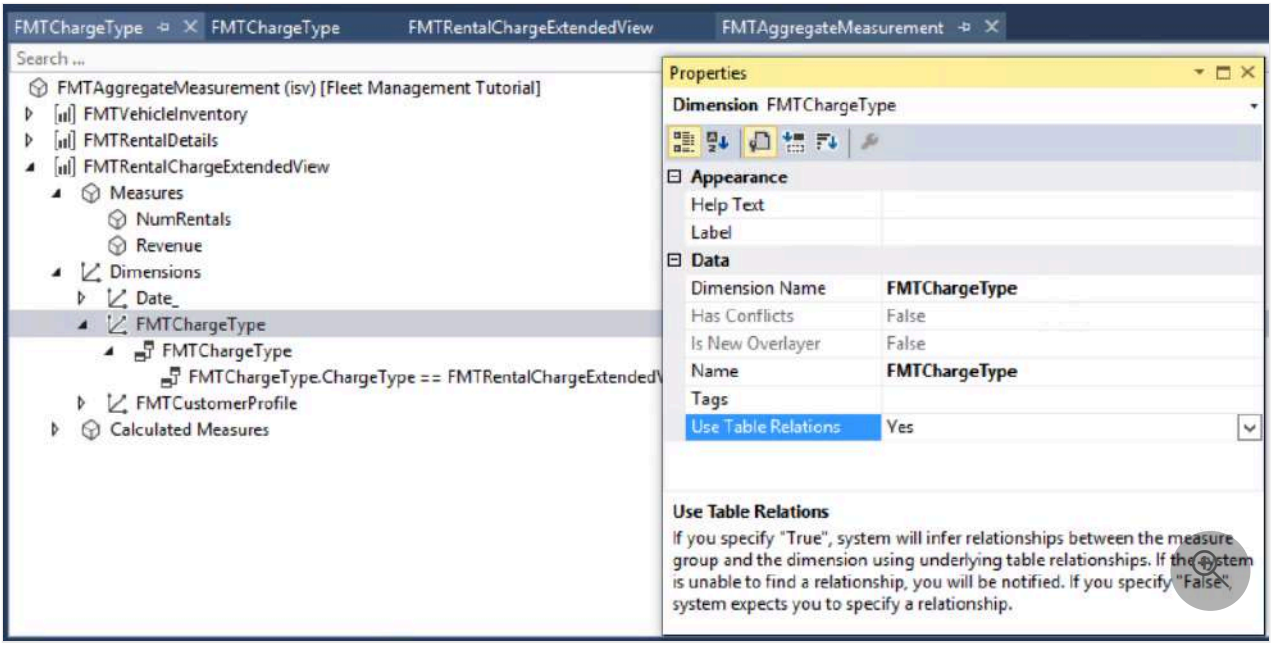
Next, create dimension references to new and existing dimensions so that you can slice revenue by customer and charge type.

1. In **Solution Explorer**, double-click **FMTAggregateMeasurement** or, if you have it open, navigate to it in the designer.

2. In **Solution Explorer**, select the dimensions **FMTChargeType** and **FMTCustomerProfile**.

3. Drag and drop these dimensions into the **Dimensions** node of the **FMTRentalChargeExtendedView** measure group. Notice that you create dimension references along with relations.

4. Save changes to **FMTAggregateMeasurement**. Review the property sheet for the dimension relation and notice that the **Use Table relations** property is set to **Yes**. Notice that the drag-and-drop operation created relationships between the measure group dimensions **FMTRentalChargeExtendedView** and **FMTChargeType**, **FMTCustomerProfile**. Review the property sheet for the dimension relation and notice that the **Use Table relations** property is set to **Yes**.



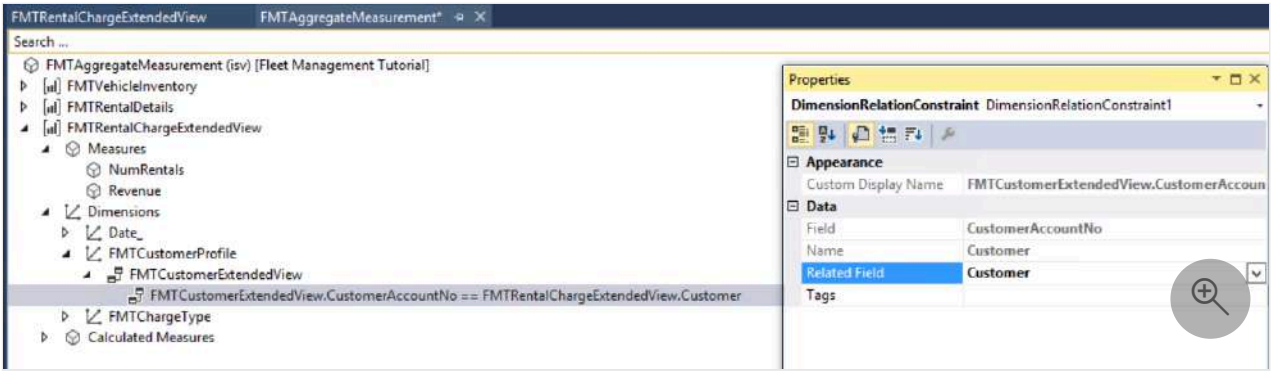
> **Note**
>
> In platform update 1611 and later, the **UseTableRelations** property is removed. When you create a new dimension reference, the system defaults existing relationships. You can continue to provide an explicit relationship by changing the relationship field that was defaulted. Providing an explicit relationship is equal to setting **UseTableRelationship** to **No**.

5. Expand the Dimension relations node for the **FMTCustomerProfile** dimension. Notice that the **UseTableRelations** property is set to **No**. In this case, the system can't find a suitable relationship between the measure group and dimension. You need to specify one manually.

6. Expand the **FMTCustomerProfile** dimension reference if you didn't already. Select the node **FMTCustomerExtendedView**. Right-click and see the property sheet.

7. Select **CustomerID** as the value for property **DimensionAttribute**. Select the relationship shown in the following image. Select **Customer** for the property value **RelatedField**.

8. Save changes to **FMTAggregateMeasurement**.



9. In this scenario, you specify a relationship because the system was unable to find one. You can also specify a different relationship if you want to override the system choice by setting **Use Table Relations property** to **No**.

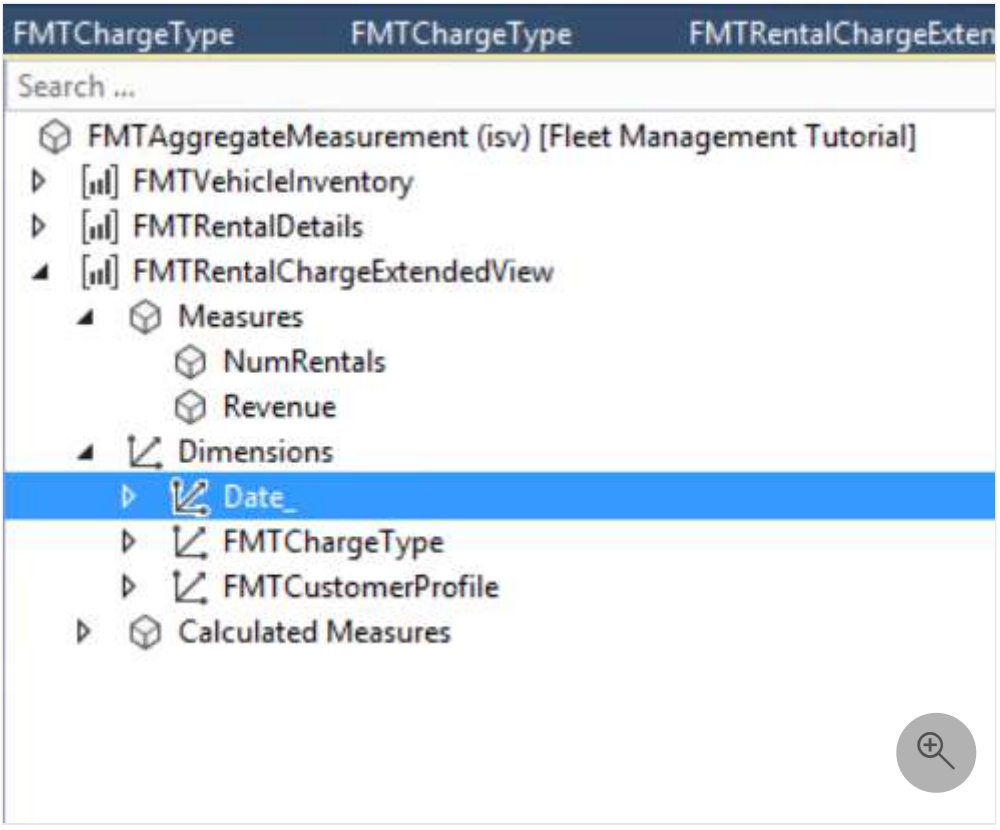
Model dimension references for the date dimension's rental start date and transaction date

Assume that for analysis purposes, you want to enable slicing by the start date of the rental. But for accounting purposes, you want to enable slicing by the transaction date for each of the charges. To do this, you need to associate the rental charges measure group with two date dimensions. In the BI world, this pattern is known as **Role Playing dimensions**. By default, a date dimension is added to the measure group. You can rename the default name appropriately and add new date dimensions as required.

1. Expand the **Dimensions** node of the **FMTRentalChargeExtendedView** measure group. Notice that the **Date_** dimension is already included as dimension slicers.

> ⓘ **Note**
>
> If the table or the view used to model the measure group is a Company-specific table, for example it contains DATAAREAID as part of the key, a **Company** dimension

relation is created by default. In this case, the view you used isn't a company specific one.



2. Select the **Date** dimension and specify the following properties.

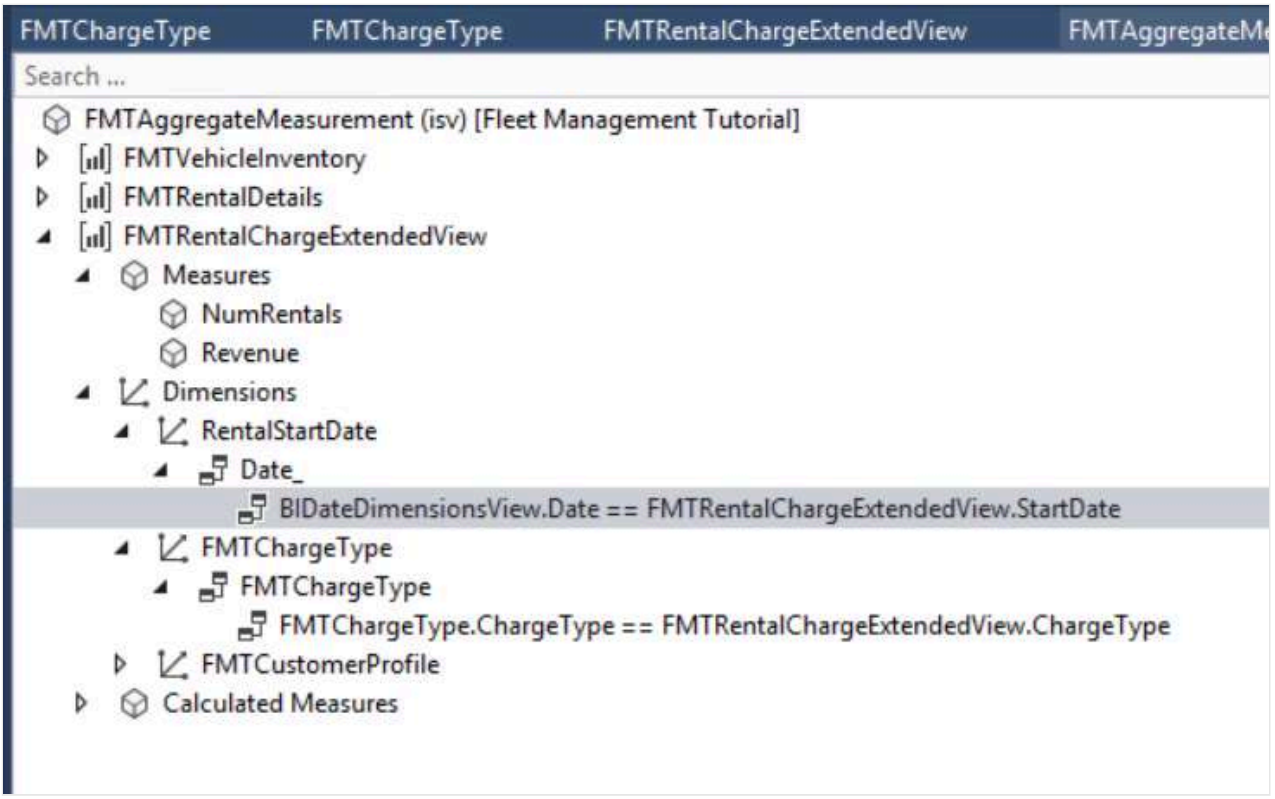
Expand table

Property	Value
Name	RentalStartDate
Use Table Relations	No (This is the default – no need to change)

3. Define the table relationship. Expand **RentalStartDate**, and then expand the **Date** node.

4. Select the Relationship shown. Right-click and select the property sheet. Select **StartDate** for the value of **Related field** property.

5. The relationship you defined should look like the following screenshot.



6. Next, enable slicing of measures by the TransactionDate dimension. TransactionDate is also a date dimension, so add another reference to the date dimension and associate that reference with the corresponding field that contains the transaction date. When you use more than one date dimension as a slicer, each date dimension is known as a **Role playing date dimension**.

7. Under the **FMTRentalChargeExtendedView** measure group, right-click the **Dimensions** node, and then select **New Dimension**. A new dimension is added to the list of dimension references.

8. Specify the following properties for the new dimension reference.

Expand table

Property	Value
Dimension Name	Date_ **NOTE:** If you select the Dimension Name in the wrong order, it resets the other values you already set.
Name	TransactionDate
Use table relations	No
Tags	RolePlayingDate; Fleet

Notice the new property called **Tags**. This property enables the discovery of patterns within code and metadata from within the Visual Studio environment. You can enter any

number of tags and they can be searched by using the hot keys or the **Dynamics 365** menu in Visual Studio.

9. Define the table relationship. Right-click **TransactionDate**, and then select **New Relation**. You don't need to specify any properties in the DimensionsRelation at this point.

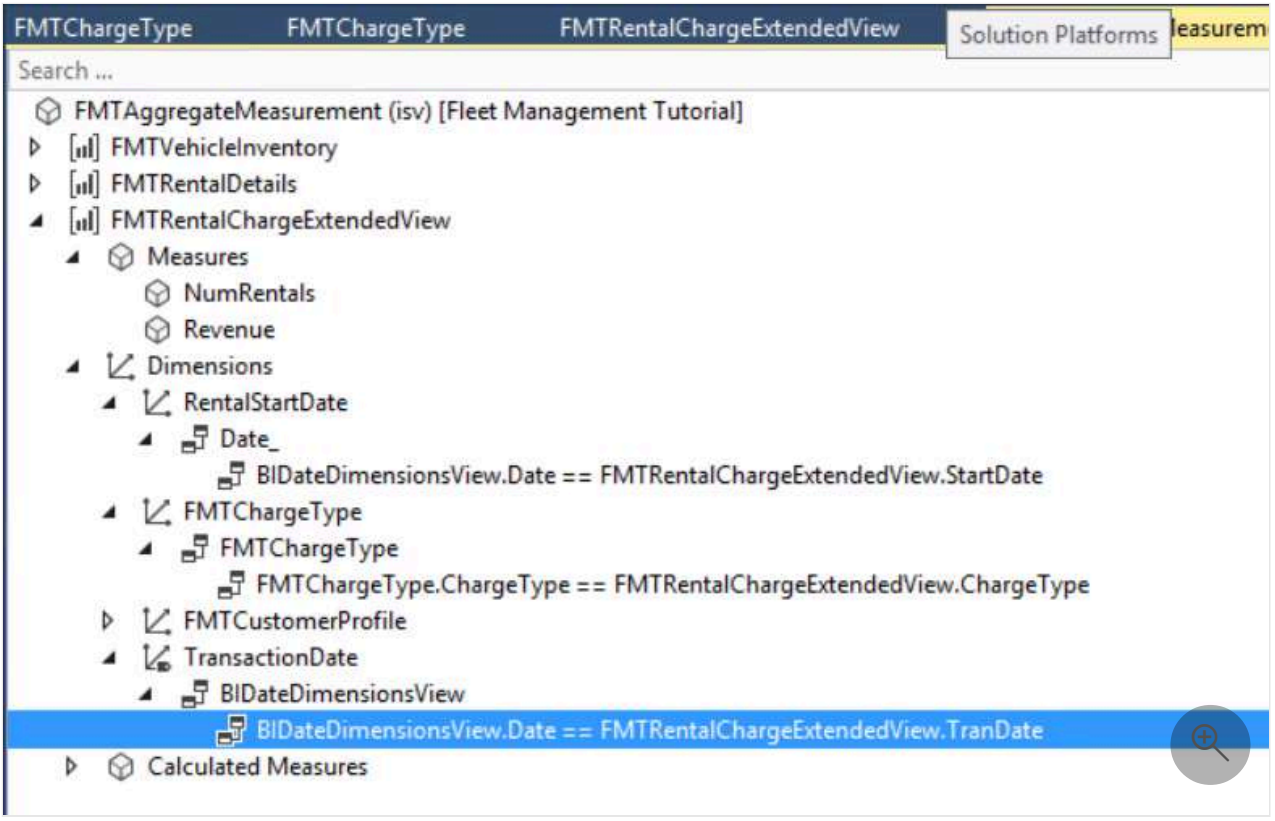
10. Expand **BIDateDimensionValue**, and then select the **Relationship Constraint**. Right-click and select the property sheet.

11. Specify the following properties for the BIDateDimensionsView relation constraint.

Expand table

Property	Value
Field	Date
Related Field	TranDate

The relationship you defined should look like the following screenshot.



12. Save the aggregate measurement.

Deploy the newly generated aggregate measurement

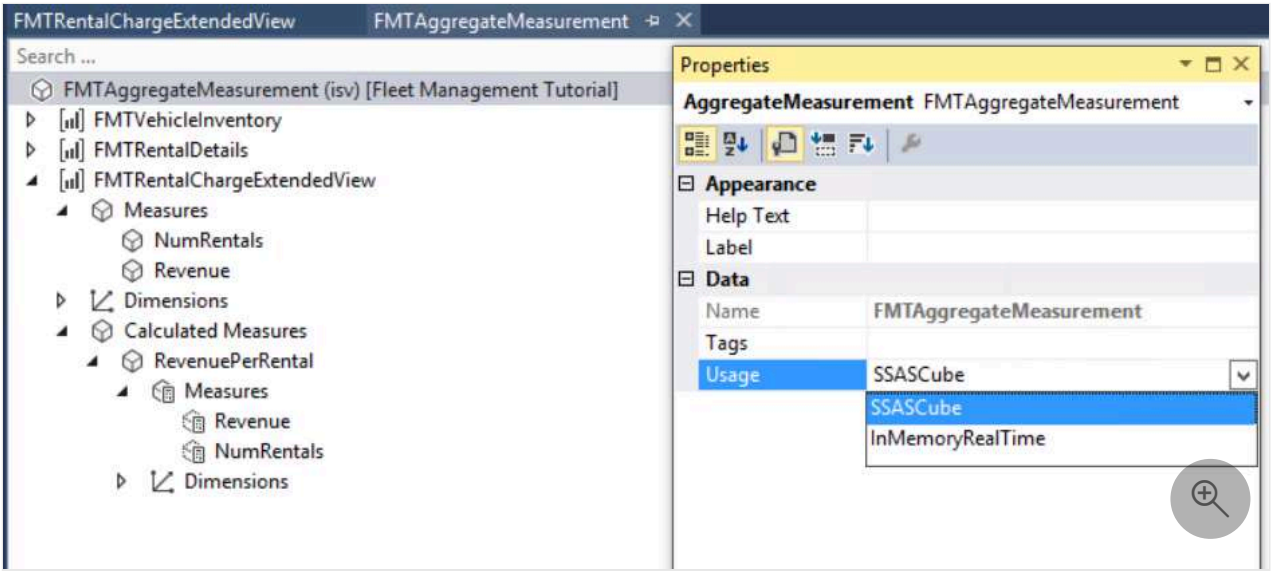
After modeling the aggregate measurement, deploy it so you can continue building KPIs and visualizations. You have two deployment choices:

Expand table

Option	Considerations
In-memory real-time	Use this option to apply the in-memory column store indexes of the SQL Server database for deploying aggregate measurements. Use this option when the aggregate measurement supports embedded analytics within the client and you need **real-time analytics**. For an overview of concepts on real-time analytics, see Analytics, aggregate measurements, and KPI modeling.
Stage in Entity Store	Use this option to apply the entity store, the operational data store that enables **near real-time PowerBI reporting**. If you choose this option, you can deploy the aggregate measurement to the entity store and schedule the data to be refreshed periodically. For an overview of this approach, see Power BI integration with Entity Store in Dynamics AX 7 May Update.

> **Note**
>
> The **SSAS Cube** option is no longer supported when modeling aggregate measurements.

1. Select the **FMTAggregateMeasurement** node. Right-click and select **Properties**. Select **InMemoryRealTime** as the value for the property **Usage**.



2. Deploy InMemoryRealTime aggregate models to SQL Server by using Non-Clustered Column Store Index (NCCI) technology. NCCIs is an in-memory technology that enables analytical and operational workloads to be served from SQL server database. Define NCCI indexes on tables similar to any other index. While you can define NCCI indexes manually, the framework can analyze index requirements and add them to underlying tables where necessary.

3. Right-click **FMAggregateMeasurement** in Solution Explorer, and then select **Add Column store indices option**. You add several new indexes.

4. Save and build the project.

5. InMemoryRealTime aggregate models don't require data processing because the models are queried real-time. If you didn't enable database synchronization along with the build, manually synchronize the database.

Model a KPI to show revenue per rental

Model a KPI in Visual Studio

Model a KPI definition in Visual Studio by using the aggregate measurement you defined earlier.

1. In **Solution Explorer**, right-click **FMTutorial**, point to **Add**, and then select **New Item**.

2. Select **Dynamics 365 Artifacts** > **Analytics** > **Key Performance Indicator**. Enter `FMTRevenuePerRental` as the name of the KPI, and then select **Add**. The name must be unique across KPIs. The KPI is created.

3. Select **FMTRevenuePerRental**, and specify the **Measurement**. Leave the default values for the other properties.

Expand table

Property	Value
Measurement	FMTAggregateMeasurement
Bad Threshold	0
Good Threshold	0
Scoring Pattern	MoreIsBetter
Show Goal	Yes
Show Status and Trend	Yes
Threshold Type	Value

4. Define the expression for the KPI value. Under **FMTRevenuePerRental**, select **Value**, and specify the following properties.

> **Note**
>
> Enter the values in the order they appear in the table:

Expand table

Property	Value
Value Type	BasedOnMeasure
Measure Group	FMTRentalChargeExtendedView
Measure	RevenuePerRental

5. Define the expression for the KPI Goal. Select **Goal**, and specify the following properties.

Expand table

Property	Value
Goal Type	BasedOnValue
Value	250

> **Note**
>
> You can define a goal based on an aggregate measure as well. In this case, define a number as the goal.

6. Save the KPI definition.

Preview KPI in client

Next, preview the KPI definition in the client.

1. Right-click **FMTutorial**, and then select **Re-Build**. When the build finishes, select the **Synchronize ... database** option.
2. Open Microsoft Edge, and go to your Rainier instance base URL.
3. Go to the **Reservation Management workspace** under **App links** > **Fleet Management** > **Workspaces** > **Reservation Management**.
4. Select the KPI tile **Total Revenue**. The KPI details page for the **Total Revenue** KPI appears.

5. To go to the newly defined KPI, select the **Show list** icon on the top left. From the list of KPIs, select **FMTRevenuePerRental**.

   The KPI details page for the new KPI, **FMTRevenuePerRental** appears. Even though you didn't define trend charts, the system created a set of charts based on the limited metadata you defined. Users can modify KPI definitions and create new ones in the client. Next, modify the newly defined KPI.

6. To demonstrate this capability, change the KPI goal. Select the **Edit** button on the top left, and enter 900 as the goal value.

7. Modify the threshold properties as follows:

Expand table

Property	Value
Threshold Type	Percentage
Red if less than	90
Green if more than	110

8. Select **Save** on the bottom left to save changes. The KPI status color changes in the KPI tile.

Last updated on 01/22/2026

Add financial dimensions to aggregate measurements

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This feature lets power users include financial dimensions in ready-made Microsoft Power BI reports. Power users can also create new Power BI reports that use financial dimensions.

Financial dimensions are user-defined configuration data that enables the ledger chart of accounts to retain extra information. By adding financial dimensions, you can configure extra data fields that appear with the chart of accounts and financial reports. You can slice and dice the ledger chart of accounts based on those fields. This feature provides powerful financial reporting. For example, you can explore ledger data by using the new financial dimensions that you add. Because the ready-made Power BI reports don't include financial dimensions when they're released, you must add these extra fields to the ready-made reports.

After you add financial dimension fields to the report, you can share the report with other users. Those other users can modify existing visuals, such as charts, by including all or some of the financial dimension fields.

> **Note**
>
> This feature is available in Microsoft Dynamics 365 Finance, Enterprise edition (July 2017). Platform update 8 or later is also required.

How does this feature work?

The Entity store is an operational data warehouse that power users can use to create reports. Whenever the Entity store updates, it includes all available financial dimensions. Here's an example from the **Ledger Activity** aggregate measurement that contains General ledger journal-level details.

The following list shows some of the tables that the Entity store fills when it updates. Tables that changed are bold.

- LedgerActivityMeasure_LedgerActivityMeasureGroup
- LedgerActivityMeasure_TransactionDate
- LedgerActivityMeasure_Currency
- LedgerActivityMeasure_FiscalPeriodDateAggregtateDimension
- LedgerActivityMeasure_LedgerFactDimension
- LedgerActivityMeasure_FiscalYearOffsetDimension
- LedgerActivityMeasure_MainAccount
- **LedgerActivityMeasure_DimensionCombination**
- LedgerActivityMeasure_Ledger
- LedgerActivityMeasure_MainAccountCategory
- LedgerActivityMeasure_Company
- LedgerActivityMeasure_MainAccountLegalEntity
- **LedgerActivityMeasure_Agreement**
- **LedgerActivityMeasure_BankAccount**
- **LedgerActivityMeasure_BusinessUnit**
- **LedgerActivityMeasure_Campaign**
- **LedgerActivityMeasure_Cargo**
- **LedgerActivityMeasure_Cargo_CN**

For each financial dimension that you define in your system, you might see a corresponding table in the Entity store.

If you examine the LedgerActivityMeasure_DimensionCombination table, you see that the list of fields has been expanded and now includes additional fields. Each new field corresponds to a financial dimension. For example, here are some of the additional fields:

- Agreement_FK
- Agreement_Description
- Agreement_Value
- BankAccount_FK
- BankAccount_Description
- BankAccount_Value
- BusinessUnit_FK
- BusinessUnit_Description
- BusinessUnit_Value

If a user defines a new financial dimension named **vendor**, the Entity store adds a new table named LedgerActivityMeasure_Vendor when it updates.

The LedgerActivityMeasure_DimensionCombination table also contains the following new set of fields:

- Vendor_FK

- Vendor_Description
- Vendor_Value

How a Power BI report author can create reports that use financial dimensions

A business user can create a new report that uses financial dimensions by using Power BI desktop. Existing reports that use financial dimensions can be updated so that they include extra fields.

In this example, use Power BI desktop to create a report that uses the Ledger Activity measure group.

1. In a development environment, connect to the Entity store database by using Power BI desktop.

2. Select the following tables:

    - LedgerActivityMeasure_LedgerActivityMeasureGroup
    - LedgerActivityMeasure_FiscalPeriodDateAggregtateDimension
    - LedgerActivityMeasure_DimensionCombination

3. Use the **Manage relationships** option in Power BI desktop to define the following relationships between table fields:

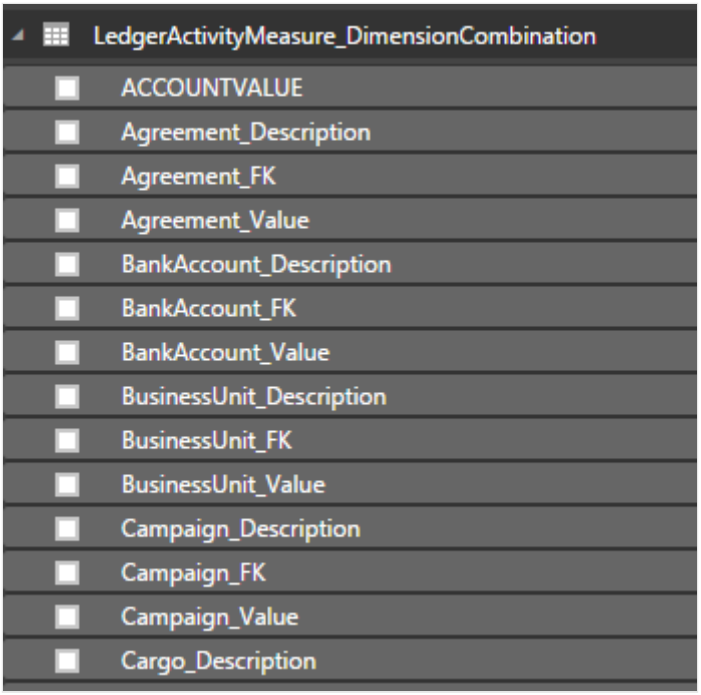
    - Define a join between **LedgerActivityMeasure_LedgerActivityMeasureGroup.LEDGERDIMENSION** and **LedgerActivityMeasure_DimensionCombination.DIMENTIONCOMBINATIONRECID**.
    - Define a join between **LedgerActivityMeasure_LedgerActivityMeasureGroup.LEDGERGREGORIANDATEID** and **LedgerActivityMeasure_FiscalPeriodDateAggregateDimension.LEDGERPERIODGREGORIANDATEID**.

4. Create a matrix report that uses the **Sales** and **YearName** fields. The report should resemble the following example.

YEARNAME	SALES
2015	355,958,240.86
2016	478,064,122.48
2017	10,712,838.30
Total	**844,735,201.64**

Next, add financial dimension values.

5. In the list of fields in Power BI desktop, expand the **LedgerActivityMeasure_DimensionCombination** table. You see that the list of dimension fields is expanded into table fields, as shown here.



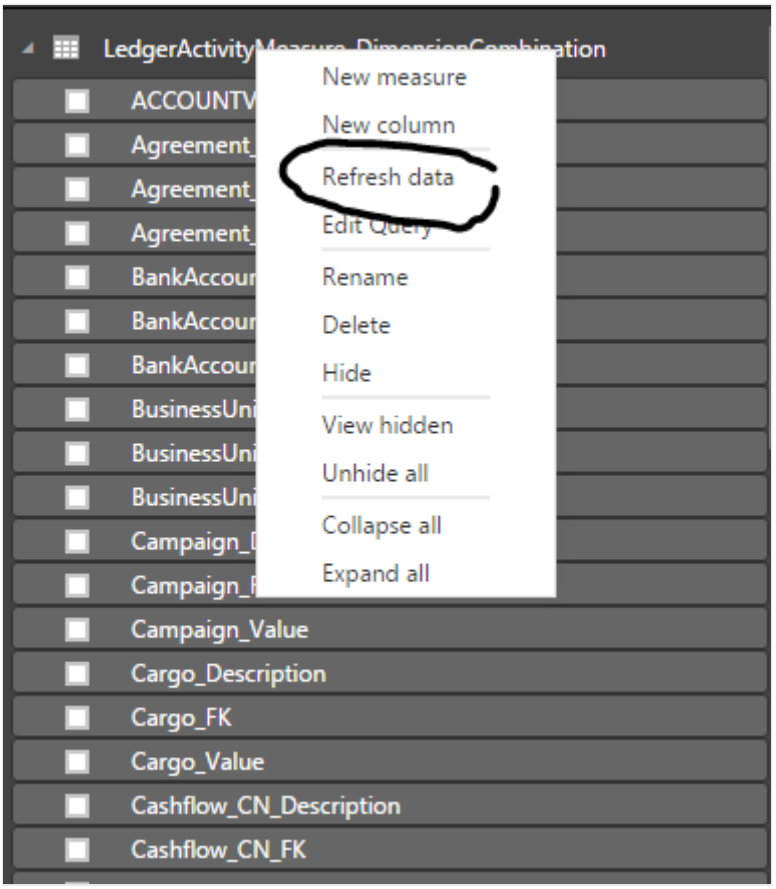
6. Include the **BusinessUnit_Description** field in the report. Your report now shows sales by business unit, as shown in the following example.

YEARNAME	Electronics	Home	IT Consulting Practice	Management Consulting Practice	Sporting	**Total**
2015	7,362,985.20	152,524,507.23	100,045,909.46	699,928.66	7,937,635.30	**355,958,240.86**
2016	6,374,315.40	167,483,885.95	115,009,475.39	769,787.36	7,622,420.20	**478,064,122.48**
2017	100,000.00	10,390,021.77	0.00	26,571.26		**10,712,838.30**
Total	**13,837,300.60**	**330,398,414.95**	**215,055,384.85**	**1,496,287.28**	**15,560,055.50**	**844,735,201.64**

Notice that you have the **BusinessUnit_description** field and the **BusinessUnit_Value** field. The value field lets you get a numerical value that you can use to sort columns.

7. Define a new financial dimension, and update the Entity store.

8. When the update finishes, right-click the **LedgerActivityMeasure_DimensionCombination** table, and then select **Refresh data**. Notice that the new financial dimension that you defined is reflected in the list of fields that are available for reporting.



9. You can include the new financial dimension fields in the report.

Creating reports that use expanded financial dimension tables

As discussed earlier, new tables are created in the Entity store. Each financial dimension field is added to a new table. The new tables contain individual fields for the name, value, and description.

The new tables contain a key that you can use to join them with the **LedgerActivityMeasure_DimensionCombination** table that you use to create the report. If you want to use fields from these additional tables, include them in the report and relate them by using the keys.

1. Open the report that you created earlier.

   Now, add a financial dimension table to the report.

2. In Power BI Desktop, select **Recent Sources** on the menu, and then select the **AXDW** data connection. The table navigator is shown.

3. Select the **LedgerActivityMeasure_LegalEntity** table for the report.

   > **Note**
   >
   > If you're using a development or demo environment, you see this table because **LegalEntity** is a financial dimension that's defined in demo data. If you're working with your own data, the tables that you see correspond to financial dimensions that you define.

   Now, relate the selected table to existing tables in the report.

4. In the report, open the **Manage relationships** dialog box.

5. Join **LedgerActivityMeasure_DimensionCombination.LegalEntity_FK** to **LedgerActivityMeasure_LegalEntity.KEY_**.

If you select a different dimension table in step 3, you must relate the corresponding **FieldName_FK** field from the combination table to the **KEY_** field in the dimension table.

How a developer can enable financial dimensions

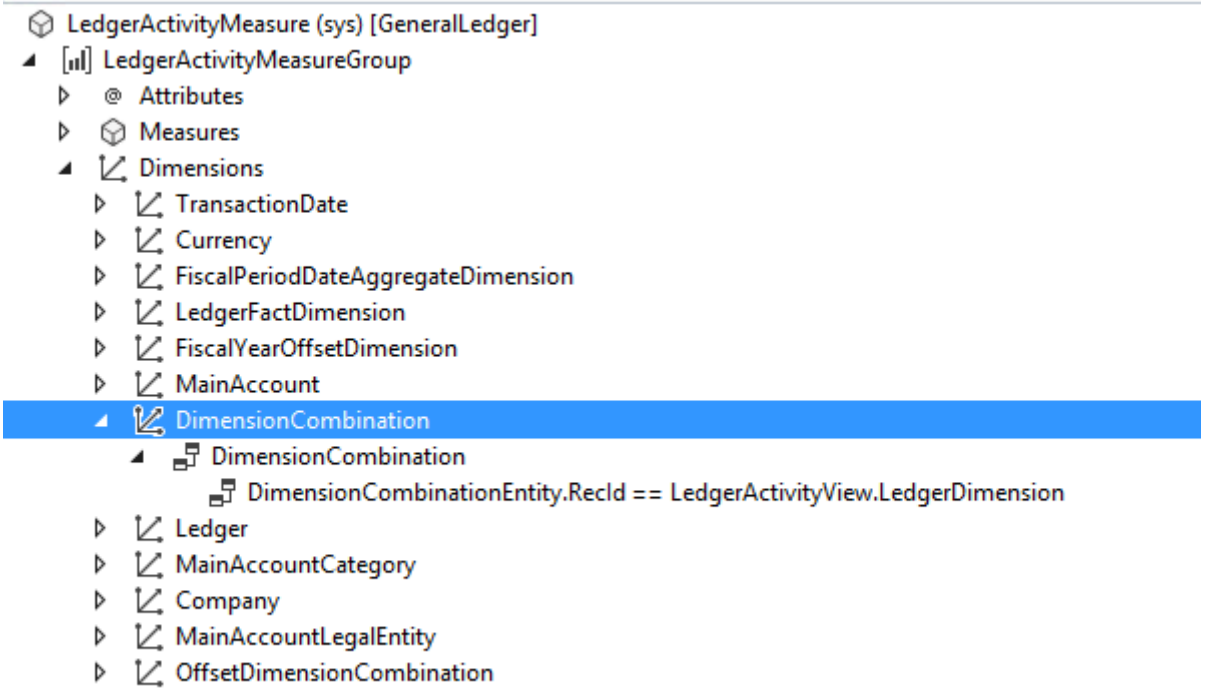
Before users can see expanded financial dimensions in the Entity store, a developer must enable the use of financial dimensions in aggregate measurements. As a best practice, include financial dimensions when you model any aggregate measurement that involves financial data. To include financial dimensions, create an aggregate dimension that is based on "financial dimension tables." As shown earlier, an aggregate dimension that you create by using financial dimension tables expands at runtime.

What are financial dimension tables?

Financial dimension tables are base tables that contain financial dimension data. Examples include **DimensionAttributeValueCombination** and **DimensionAttributeValueSet**.

If you use a table, a view, or an entity that's based on those two tables, you see the fields of the aggregate dimension expanded at runtime.

Consider the following example from the LedgerActivityMeasure aggregate measurement.



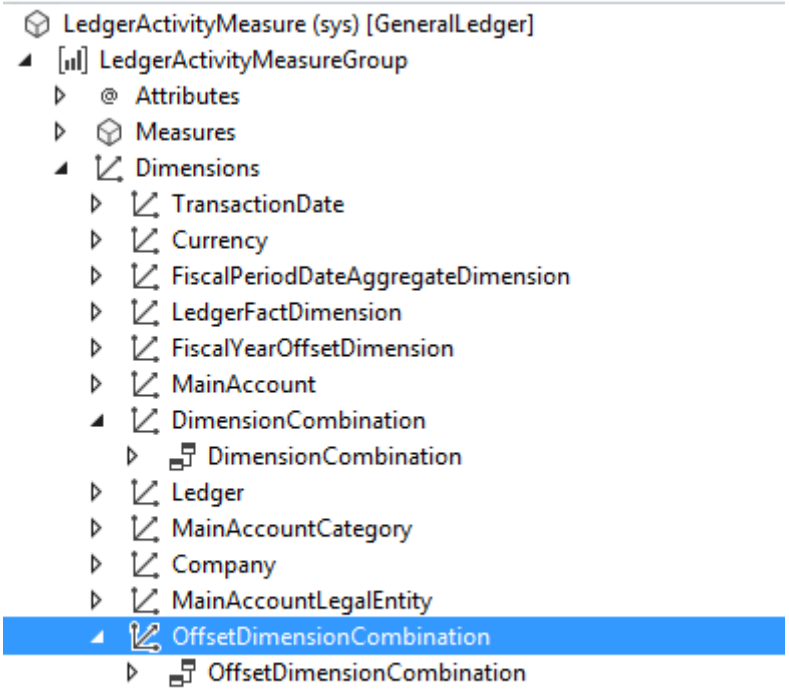
DimensionCombination is an aggregate dimension that the DimensionAttributeValueCombination base table models. In this case, the developer references the aggregate dimension by using the LedgerActivityMeasureGroup measure group.

At runtime, new dimension fields are added (that is, the DimensionCombination table is expanded) as you define new financial dimensions in the system.

Creating role-playing financial dimensions

When you report by using ledger data, you might need to report on primary accounts and offset accounts. For example, if a ledger transaction involves the transfer of an amount from one account to another account, the primary account is the "from" account, whereas the offset account is the "to" account. This pattern is known as the *role-playing dimensions* pattern.

You must associate both primary accounts and offset accounts with transaction data. Therefore, you must expand financial dimension fields of both primary accounts and offset accounts. The following example shows how you can meet this requirement.



You model two dimension references for `LedgerActivityMeaureGroup`. The first reference, `DimensionCombination`, is joined by using the **LedgerDimension** field. You saw this pattern earlier in this article.

The second reference, `OffsetDimensionCombination`, is another reference to the same dimension. You reuse the `DimensionCombination` aggregate dimension and give it a new name. In the second case, you can join by using the **OffsetLedgerDimension** field.

At runtime, the system expands both these dimensions with extra fields. Therefore, you can report on primary and offset dimension fields.

Last updated on 01/22/2026

Create analytical reports by using Power BI Desktop

If you're a power user or a business analyst, you probably create many reports for your organization. You might create these reports in Microsoft Excel by formatting and relating data before you share it with other people. People in your organization might even come to you when they require modifications to the report. This solution offers an easy way to create rich, interactive reports. As a report writer, you can use Microsoft Power BI Desktop as the reporting tool. The reports that you create can then be published to PowerBI.com. For more information about Power BI Desktop, see Create stunning reports and visualizations with Power BI Desktop.

Accessing the local Entity Store by using DirectQuery

You can create Microsoft Power BI reports by using Open Data Protocol (OData) endpoints that are exposed via data entities. Despite the limitations of this approach, the Entity Store still supports it for legacy solutions. However, DirectQuery is now the preferred method for sourcing data for analytical solutions. For more information about the benefits and limitations of DirectQuery, see Use DirectQuery in Power BI Desktop.

When you use Power BI Desktop, you can create a report in your development or test environment by connecting directly to the local Entity Store database. When you're satisfied with the report, your administrator can help you migrate it to your production environment. The rest of this section walks you through this process.

> **Note**
>
> To develop or extend analytical workspaces and reports in the application suite, customers must use a development environment running in their own subscription or on local machines. You won't be able to develop or extend embedded analytical reports in Microsoft-provided Tier-1 environments. You need administrator rights to install Power BI Desktop.

> Power BI embedded integration has upgraded the infrastructure as part of the 10.0.22 release. This was an infrastructure change and will happen automatically when an environment is upgraded to release 10.0.22. With this change, users are now able to use the *latest version* of Power BI desktop when editing workspace reports.

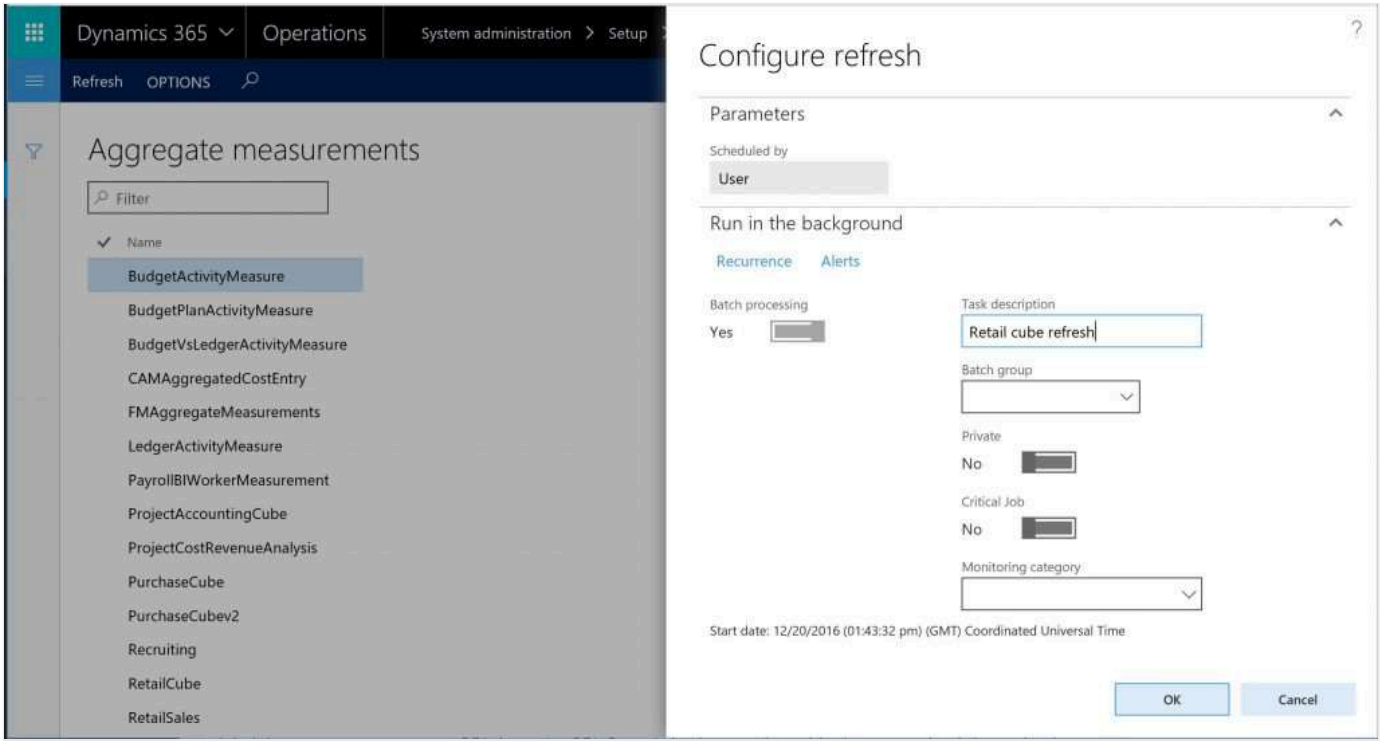
> For environments running version 10.0.21 or earlier, you must still use the previous build of Power BI Desktop pre-installed on the development environment. Alternatively, you can use the *compatible* Power BI Desktop August 2020 release of Power BI Desktop with preview features turned off to author analytical reports for finance and operations apps. You can download the Power BI Desktop August 2020 release from the LCS Shared asset library.

Step 1: Populate the local Entity Store database

For this example, we'll stage the aggregate models that the Commerce analytical solution consumes in the local Entity Store. The models that the application uses are defined in the RetailCube aggregate measurement.

1. In the client, open the **Entity Store** page. (Select **System administration** > **Setup** > **Entity Store**.)
2. Select the **RetailCube** aggregate measurement, and then select **Refresh**.
3. Enter a name for the job that will be run in the background, and then select **OK**.

The following illustration shows the administrator dialog box that is used to configure the frequency of updates for the aggregate model.



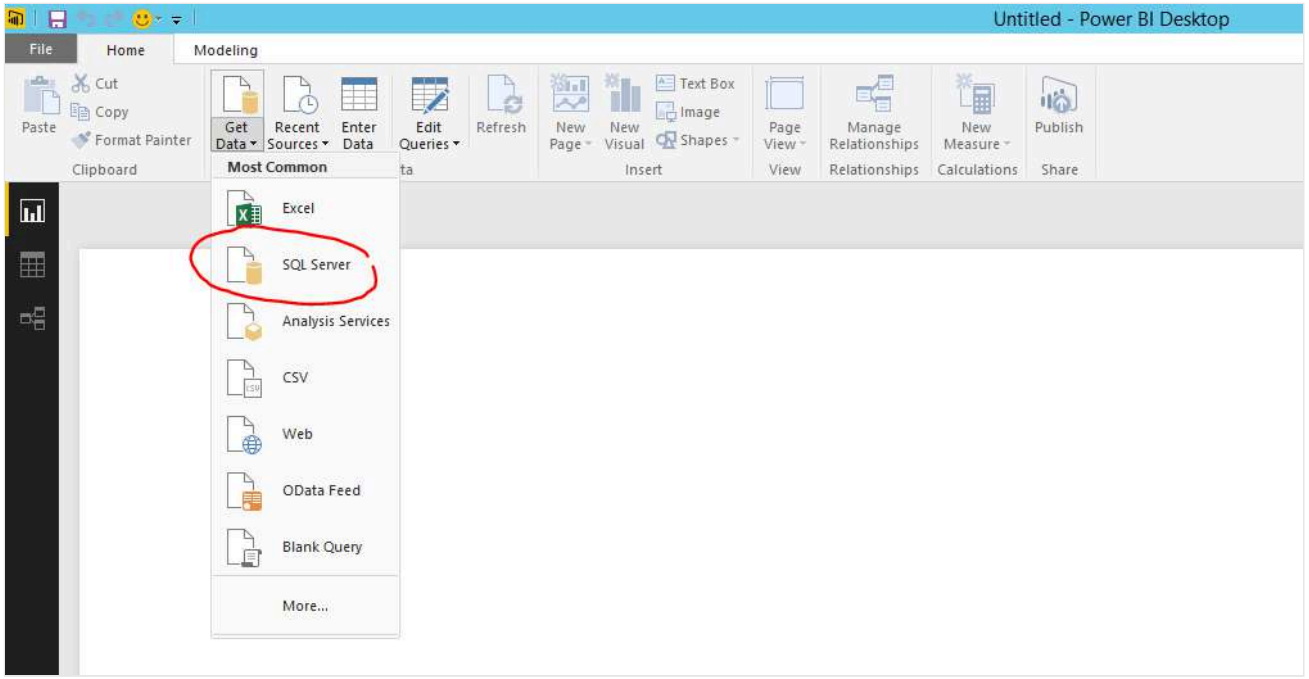
To monitor the progress of the job that stages the data, you can use the batch job monitoring page. (Select **System administration** > **Database** > **Batch jobs**.) If you're using demo data, the job should take about a minute. After the data is in the Entity Store, you can write reports.

Step 2: Connect to the local Entity Store database

1. Start Power BI Desktop. If any updates are available for Power BI Desktop, you might have to download and apply them.

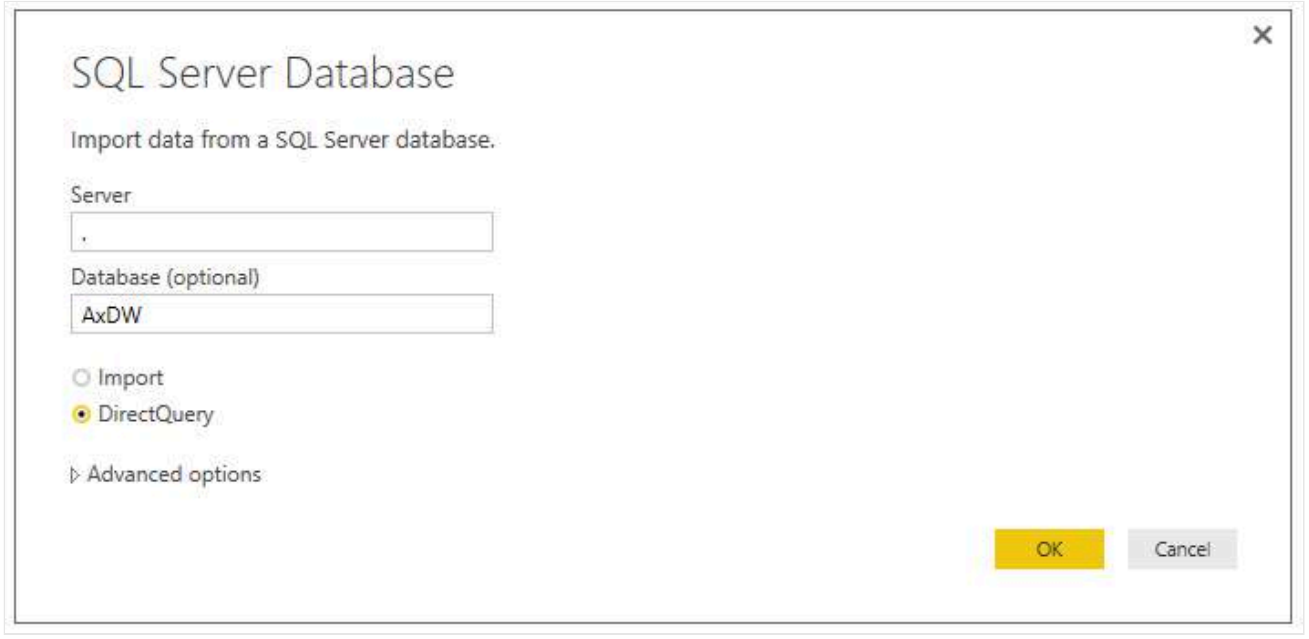
2. On the Power BI **Welcome** page, select **Get data**.

   Alternatively, when Power BI Desktop starts, you can select **Get Data** > **SQL Server**.



3. In the **SQL Server Database** dialog box, enter.** as the server name and **AxDW** as the database name. Then select the **DirectQuery** option.

   The following illustration shows the settings that enable Power BI Desktop to access the local Entity Store database.



> **Note**
>
> The **Import** option isn't currently supported.

4. Select **OK**.

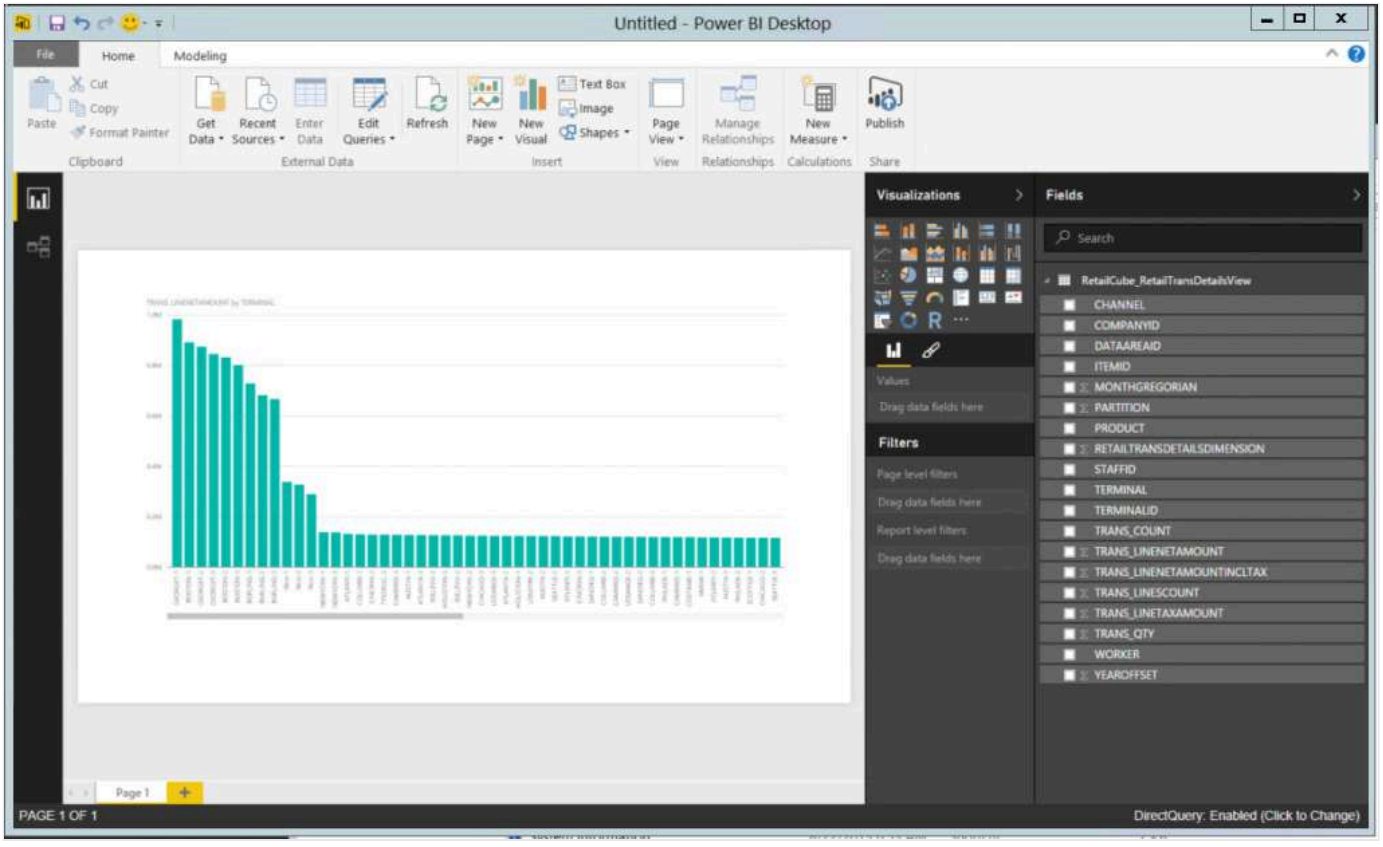
   The **Navigator** dialog box appears. You use this dialog box to select which tables and views from the Entity Store you want to report on.

5. In the search box, enter **Retail** to filter for entities that are related to the RetailCube aggregate measurement.

6. Select the **RetailCube_RetailTransDetailsView** table that is shown in the navigator, and then select **Load**.

You can now create a report. You can drag measures and fields to the canvas, and can explore data and trends interactively.

The following illustration shows a basic report that uses the local Entity Store database as its source.



Power BI Desktop also supports the creation of calculations and lets you combine data from multiple aggregate measurements. Within minutes, you can create analytical reports by using

data in the local development environment. When you're satisfied with the report, you can migrate it to the production environment, so that users can use the report to interact with production data.

Validating reports in a demo environment

The report shows the demo or test data in your developer environment. If you want to integrate the report into a demo environment, you can continue to publish this report to your PowerBI.com account and pin it to the client.

Last updated on 11/26/2025

Add analytics to workspaces by using Microsoft Power BI Embedded

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Note**
>
> This feature is supported in finance and operations (version 7.2 and later).

Introduction

This article shows how to embed a Microsoft Power BI (PBI) report on the **Analytics** tab of a workspace. In this example, you extend the **Reservation management** workspace in the Fleet Management application to embed an analytical workspace on an **Analytics** tab.

Prerequisites

- Access to a developer environment that runs Platform update 8 or later.
- An analytical report (.pbix file) that you created by using Microsoft Power BI Desktop, with a data model sourced from the Entity store database.

Overview

Whether you extend an existing application workspace or introduce a new workspace, you can use embedded analytical views to deliver insightful and interactive views of your business data. The process for adding an analytical workspace tab has four steps.

1. Add a .pbix file as a Microsoft Dynamics 365 resource.
2. Define an analytical workspace tab.
3. Embed the .pbix resource on the workspace tab.
4. Optional: Add extensions to customize the view.

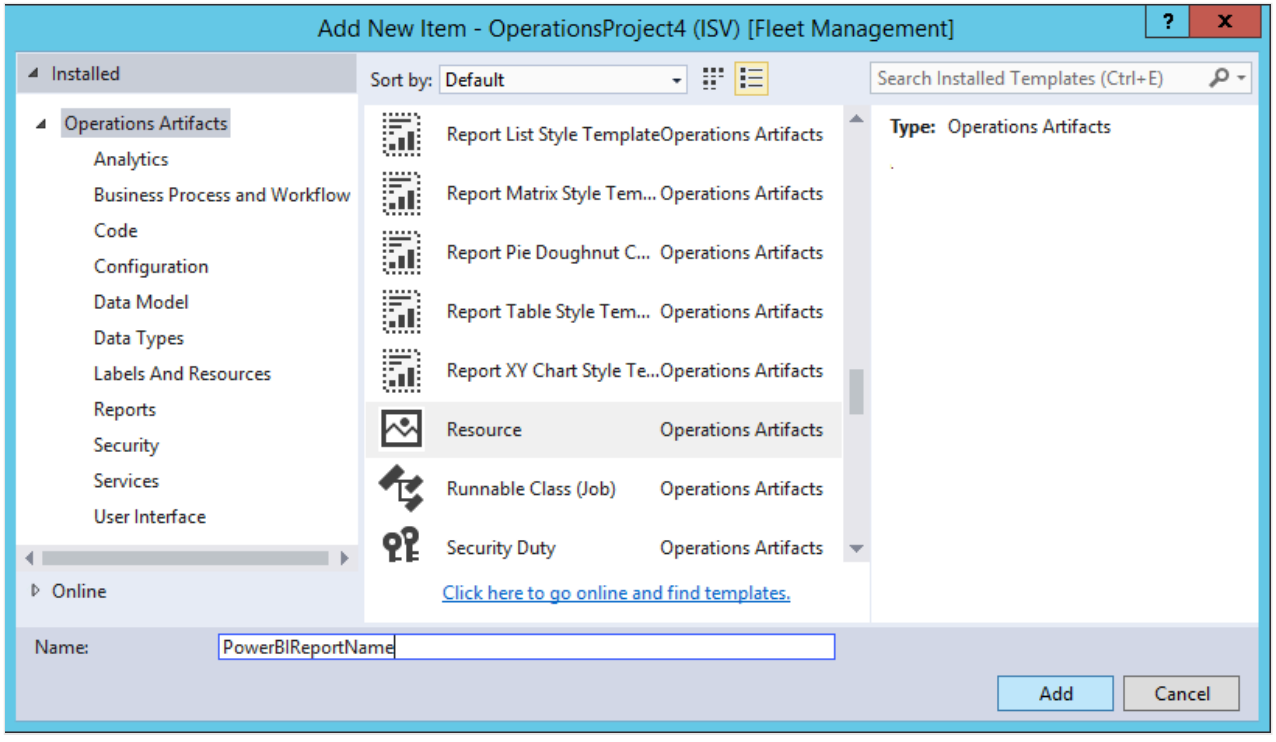
> **Note**
>
> For more information about how to create analytical reports, see Getting started with Power BI Desktop. This page is a great source for insights that can help you create compelling analytical reporting solutions.

Add a .pbix file as a resource

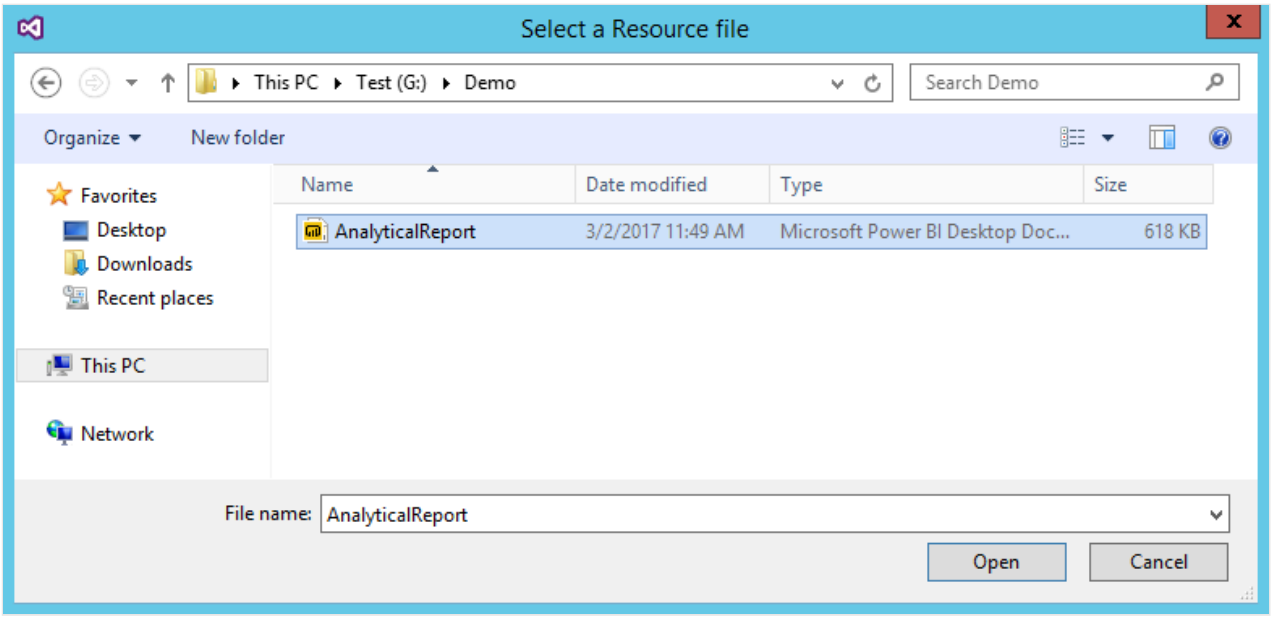
Before you begin, you must create or obtain the PBI report that you want to embed in the workspace. For more information about how to create analytical reports, see Getting started with Power BI Desktop.

Follow these steps to add a .pbix file as a Microsoft Visual Studio (VS) project artifact.

1. Create a new project in the appropriate model.
2. In Solution Explorer, select the project, right-click, and then select **Add** > **New Item**.
3. In the **Add New Item** dialog box, under **Operations Artifacts**, select the **Resource** template.
4. Enter a name that you use to reference the report in X++ metadata, then select **Add**.



5. Find the .pbix file that contains the definition of the analytical report, then select **Open**.

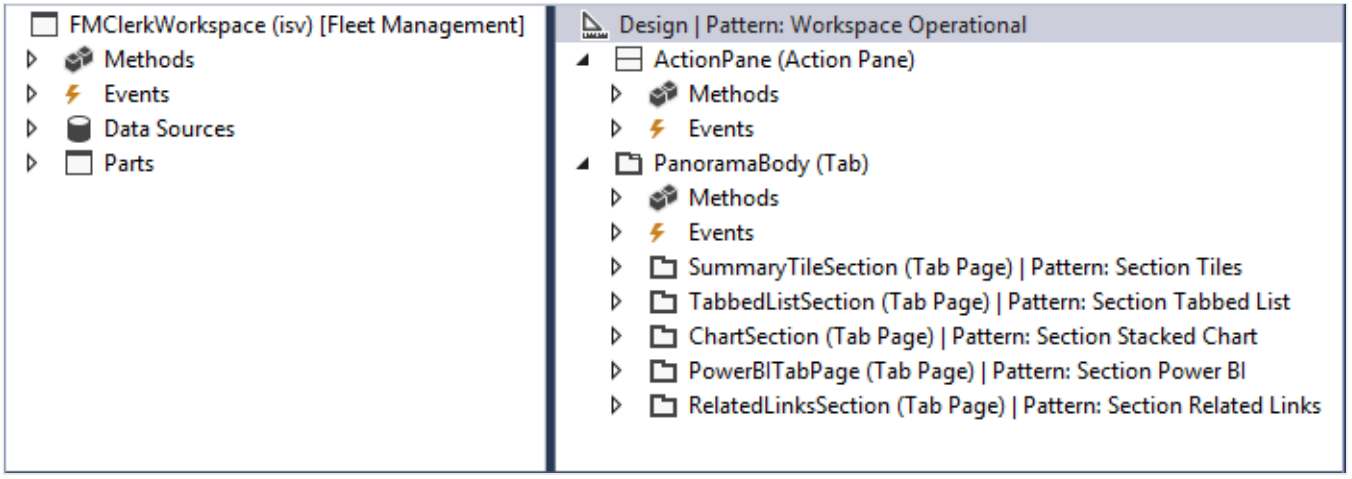


After you add the .pbix file as a Microsoft Dynamics 365 resource, you can embed the reports in workspaces and add direct links by using menu items.

Add a tab control to an application workspace

In this example, you extend the **Reservation management** workspace in the Fleet Management model by adding the **Analytics** tab to the definition of the **FMClerkWorkspace** form.

The following illustration shows what the **FMClerkWorkspace** form looks like in the designer in Microsoft Visual Studio.

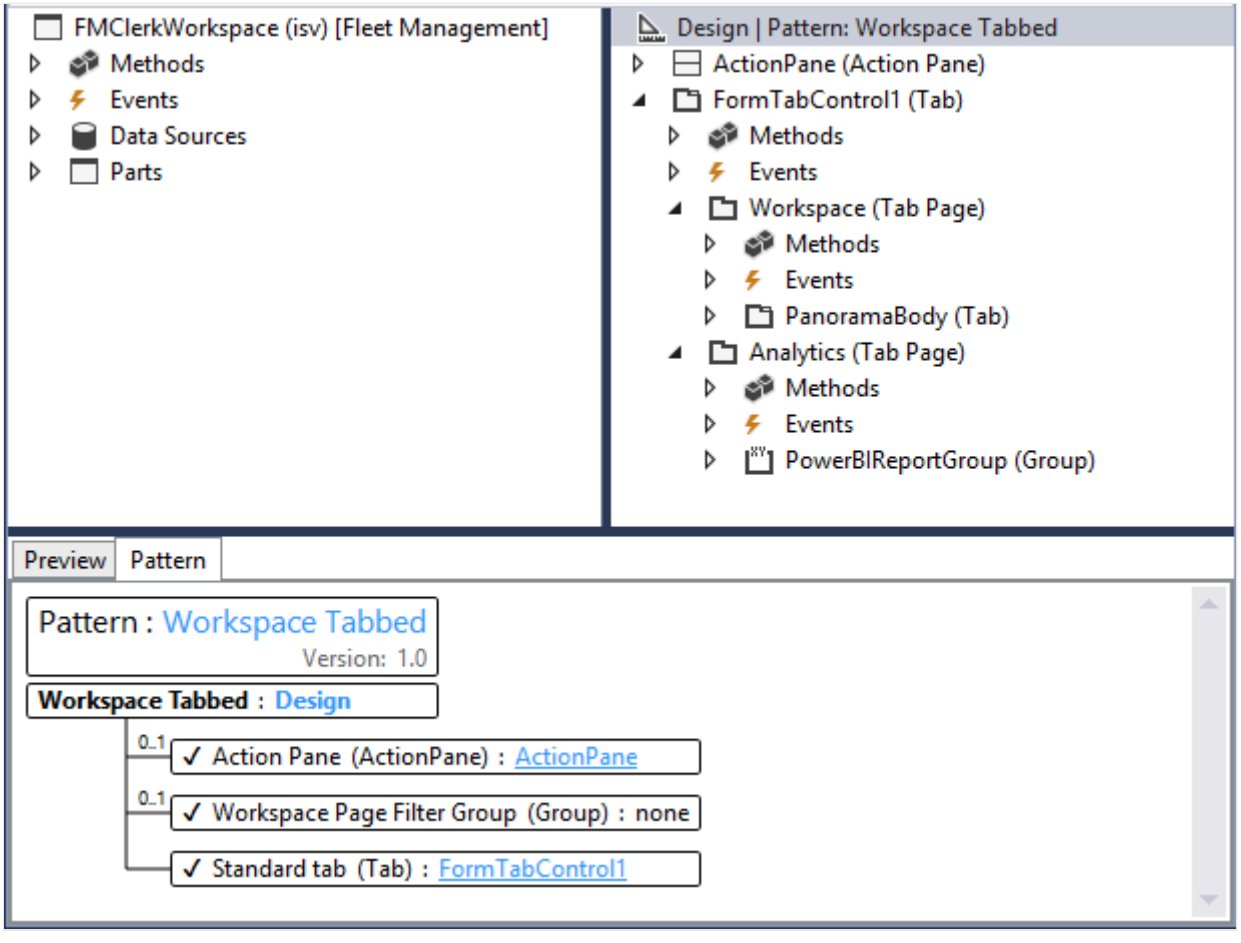


Follow these steps to extend the form definition for the **Reservation management** workspace.

1. Open the form designer to extend the design definition.
2. In the design definition, select the top element that is labeled **Design | Pattern: Workspace Operational**.
3. Right-click, and then select **New** > **Tab** to add a new control that is named **FormTabControl1**.

4. In the form designer, select **FormTabControl1**.
5. Right-click, and then select **New Tab Page** to add a new tab page.
6. Rename the tab page to something meaningful, such as **Workspace**.
7. In the form designer, select **FormTabControl1**.
8. Right-click, and then select **New Tab Page**.
9. Rename the tab page to something meaningful, such as **Analytics**.
10. In the form designer, select **Analytics (Tab Page)**.
11. Set the **Caption** property to **Analytics**, and set the **Auto Declaration** property to **Yes**.
12. Right-click the control, and then select **New** > **Group** to add a new form group control.
13. Rename the form group to something meaningful, such as **powerBIReportGroup**.
14. In the form designer, select **PanoramaBody (Tab)**, and then drag the control onto the **Workspace** tab.
15. In the design definition, select the top element that is labeled **Design | Pattern: Workspace Operational**.
16. Right-click, and then select **Remove pattern**.
17. Right-click again, and then select **Add pattern** > **Workspace Tabbed**.
18. Build the project to verify your changes.

The following illustration shows what the design looks like after these changes are applied.



Now that you've added the form controls that are used to embed the workspace report, you must define the size of the parent control so that it accommodates the layout. By default, both

the **Filters Pane** page and the **Tab** page are visible on the report. However, you can change the visibility of these controls as appropriate for the target consumer of the report.

> ⓘ Note
>
> For embedded workspaces, use extensions to hide both the **Filters Pane** and **Tab** pages, for consistency.

You've now completed the task of extending the application form definition. For more information about how to use extensions to do customizations, see Customize through extension and overlayering.

Add X++ business logic to embed a viewer control

Follow these steps to add business logic that initializes the Microsoft Report Viewer control embedded in the **Reservation management** workspace.

1. Open the **FMClerkWorkspace** form designer to extend the design definition.
2. Press F7 to access the code behind the code definition.
3. Add the following X++ code.

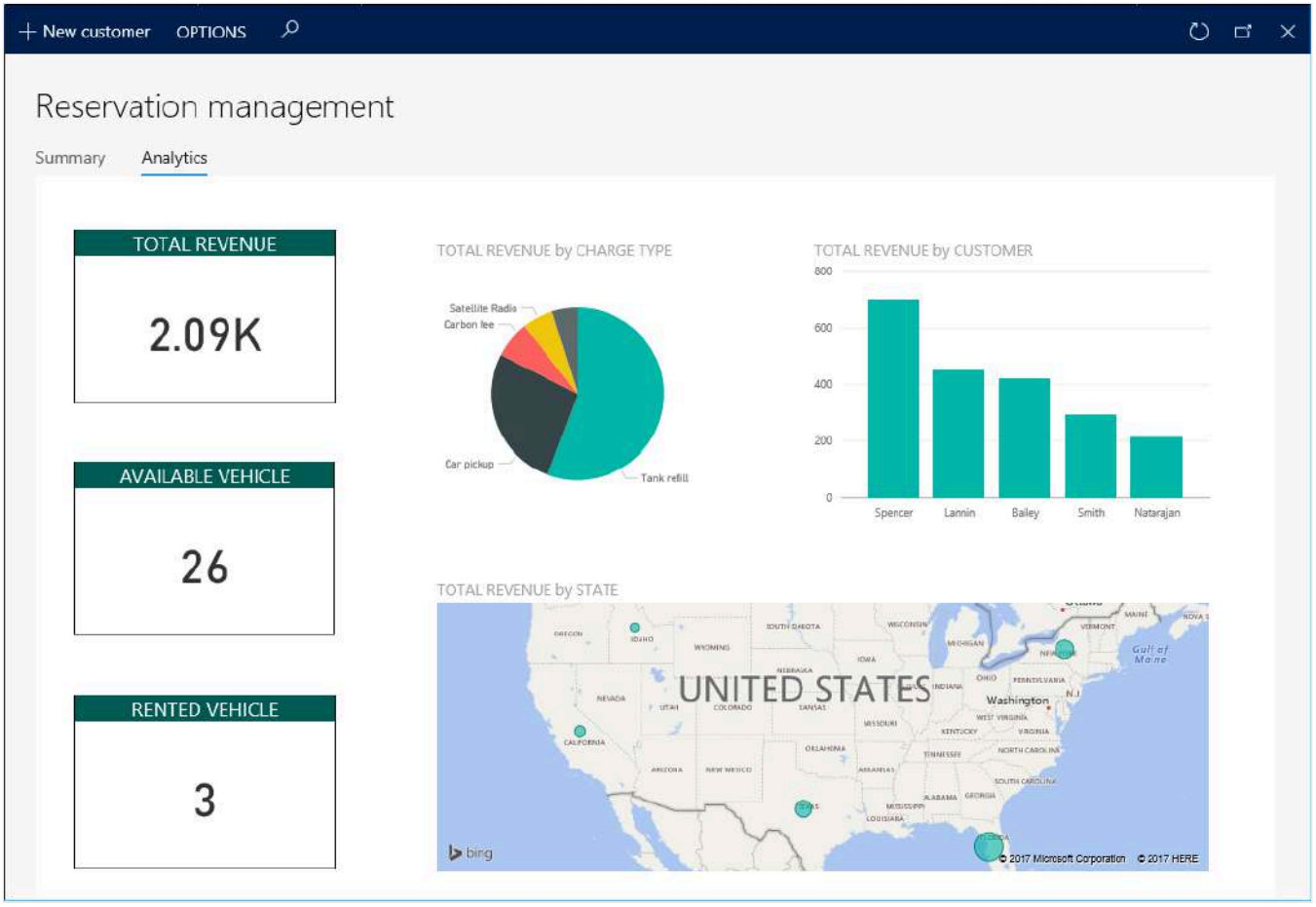
X++

```
[Form]
public class FMClerkWorkspace extends FormRun
{
    private boolean initReportControl = true;
    protected void initAnalyticalReport()
    {
        if (!initReportControl)
        {
            return;
        }
        // Note: secure entry point into the Workspace's Analytics report
        if (Global::hasMenuItemAccess(menuItemDisplayStr(FMClerkWorkspace),
MenuItemType::Display))
        {
            // initialize the PBI report control using shared helper
            PBIReportHelper::initializeReportControl('FMPBIWorkspaces',
powerBIReportGroup);
        }
        initReportControl = false;
    }
    /// <summary>
    /// Initializes the form.
```

```
    /// </summary>
    public void init()
    {
        super();
        this.initAnalyticalReport();
    }
}
```

4. Build the project to verify your changes.

You've now completed the task of adding business logic to initialize the embedded Microsoft Report Viewer control. The following illustration shows what the workspace looks like after these changes are applied.



> **Note**
>
> You can access the existing operational view by using the workspace tabs that appear below the page title.

Reference

PBIReportHelper.initializeReportControl method

This section provides information about the helper class that is used to embed a Power BI report (.pbix resource) in a form group control.

Syntax

X++

```
public static void initializeReportControl(
    str                 _resourceName,
    FormGroupControl    _formGroupControl,
    str                 _defaultPageName = '',
    boolean             _showFilterPane = false,
    boolean             _showNavPane = false,
    List                _defaultFilters = new List(Types::Class))
```

Parameters

Expand table

Name	Description
resourceName	The name of the .pbix resource.
formGroupControl	The form group control to apply the Power BI report control to.
defaultPageName	The default page name.
showFilterPane	A Boolean value that indicates whether the filter pane should be shown (**true**) or hidden (**false**).
showNavPane	A Boolean value that indicates whether the navigation pane should be shown (**true**) or hidden (**false**).
defaultFilters	The default filters for the Power BI report.

Last updated on 11/26/2025

Secure analytical workspaces and reports by using Power BI Embedded

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Note**
>
> This feature is supported in Microsoft Dynamics 365 Finance, Enterprise edition (July 2017) (version 7.2), and later releases.

Introduction

This article provides a walkthrough for application developers who want to help secure analytical workspaces and reports that are delivered by using Microsoft Power BI Embedded. It describes the recommended strategies for securing access to both the reports and the dataset, based on viewer access rights. By using the techniques described in this article, you can hide reports from users and filter reports to show the dataset that's appropriate for a specific user, based on the active company context.

Prerequisites

- Access to a developer environment that runs Platform update 8 or later
- An analytical report (.pbix file) that was created by using Microsoft Power BI Desktop, and that has a data model that's sourced from the Entity store database

Overview

Whether you're extending an existing application workspace or adding your own workspace, you can use embedded analytical views to deliver insightful and interactive views of your business data. Before you add new analytical workspaces and reports, it's important that you establish a strategy to secure the content.

Consider several factors when you develop analytical solutions by using Power BI Embedded. Analytical reports are secured by using menu items. After they have access to a report, all viewers can access the underlying data model that is defined in the report. Although service options are available that automatically hide the fields that back a report data set, all viewers of the report have effective access to the fields in the data model. Additionally, X++ extensions are available that influence how the report is presented in the client. Both the **Filter** pane and the **Report** tabs can be hidden. However, Microsoft Power BI filters can be modified by client-side script injections.

Recommendation

To deliver analytical views, create scenario-specific .pbix files:

- Area overviews that are delivered by workspaces
- Subject matter–specific analytical reports

> **Note**
>
> These analytical reports are often used to deliver reports that contain medium-business-impact and high-business-impact data.

For more information about how to create analytical reports, see Getting started with Power BI Desktop. This page is a great source for insights that can help you create compelling analytical reporting solutions.

Help secure analytical views that are provided through embedded Power BI reports

Power BI report filters and the **Filter** pane serve as a mechanism for passing session context into the report embedded on the **Analytics** tab. The capability to turn the visibility of the **Filter** pane on and off isn't a security feature. Power BI report filters and the capability to hide and show the **Filter** pane are user experience (UX) decisions that the application designer makes.

Row-level security defined isn't inherited by Power BI reports. Instead, application developers can help secure the workspace or the report.

Help secure analytical workspaces on the Analytics tab

Analytical workspaces are embedded Power BI reports that are shown in a form control. Unless you complete the following procedure, anyone who has access to the workspace can see the **Analytics** tab and access the Power BI reports.

1. Add a menu item for the analytical workspace.
2. Verify that the form initialization uses the **hasMenuItemAccess** application programming interface (API) to verify that the user has access to the menu item.

```
X++
```

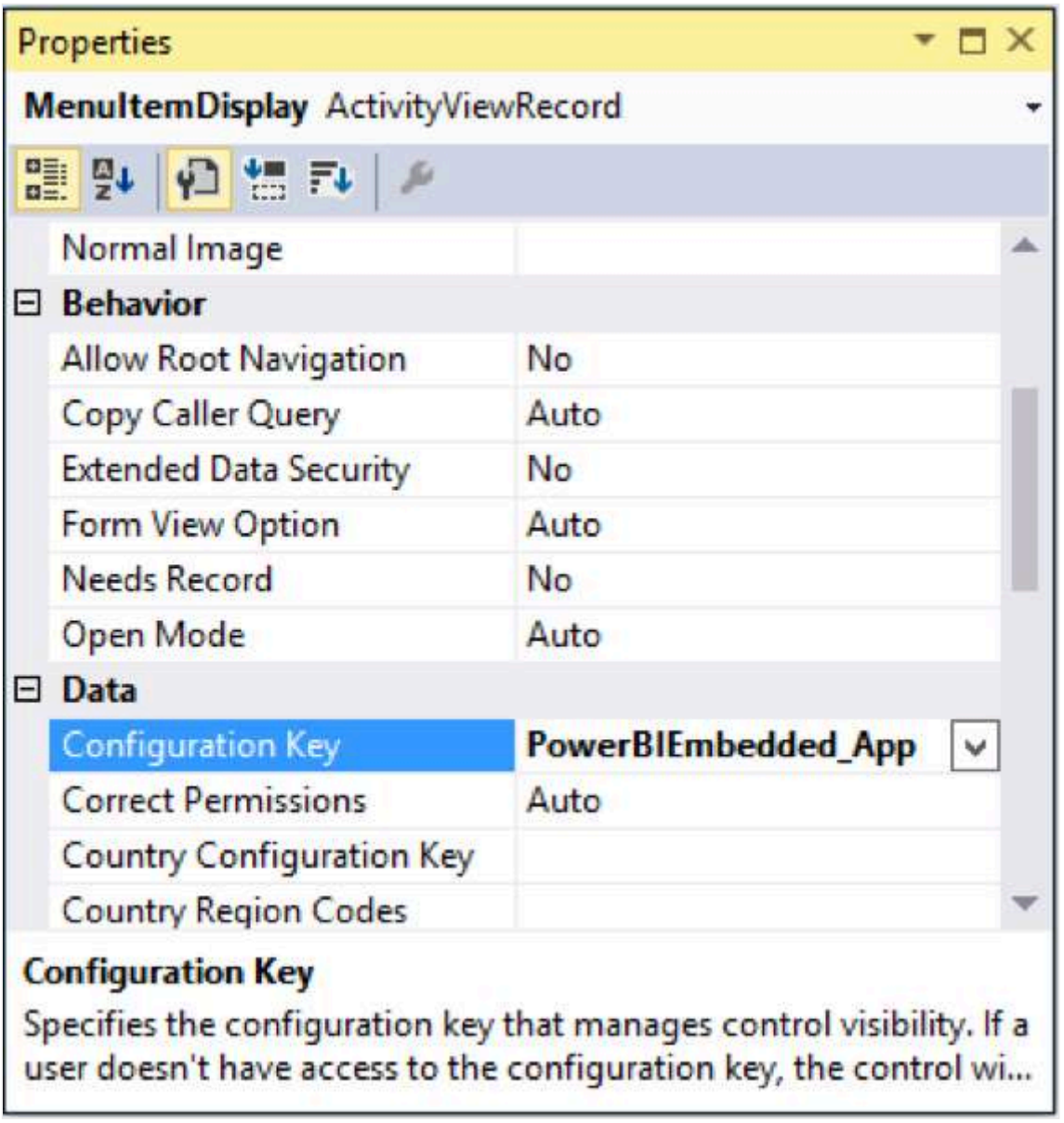
```
// Note: secure entry point into the Workspace's Analytics report
if (Global::hasMenuItemAccess(menuItemDisplayStr(FMClerkWorkspace),
MenuItemType::Display))
{
    FMPBIWorkspaceController controller = new FMPBIWorkspaceController();
    PBIReportHelper::initializeReportControl('FMPBIWorkspaces',
powerBIReportGroup);
}
```

The preceding logic prevents the Power BI Viewer control from being initialized. Therefore, an empty tab appears on the page. By default, the framework automatically hides empty tabs. Therefore, the **Analytics** tab is hidden and can't be access if the user doesn't have access to the menu item that is associated with the analytical workspace.

Help secure analytical reports

Embedded Power BI reports in the application are secured by using menu items. Users who try to access a Power BI report directly, by using a menu item in application, will receive an error. Follow these steps to help secure the analytical reports.

1. Add a menu item for the report or the appropriate tab. By default, the first tab of the report is shown if no other tab is selected.
2. Link the menu item to the **PowerBIEmbedded_App** configuration key.



The menu item is now associated with the availability of the Power BI Embedded service. If the service is unavailable, the links for the menu items are removed from the application.

Help secure analytical workspaces and reports by company

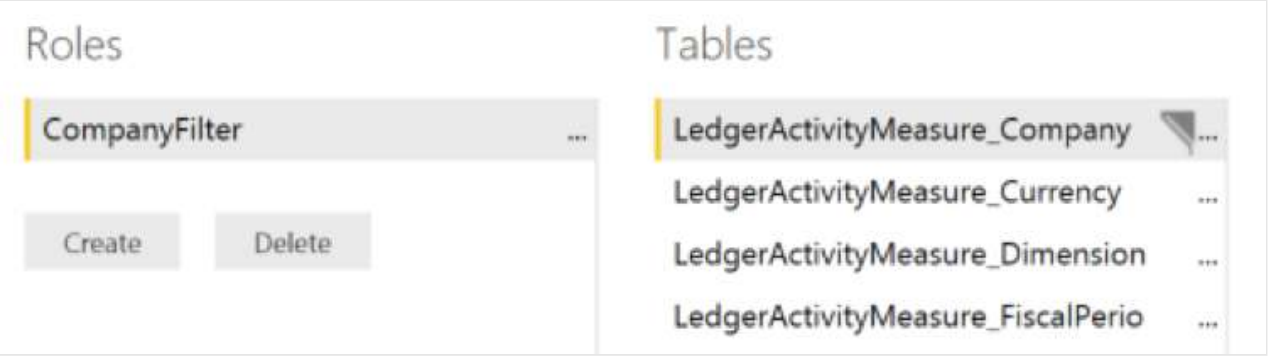
You can secure Power BI workspaces and reports by company (for example, **DataAreaID** value). Application solutions apply the following steps for company-level security in analytical workspaces and reports.

In this scenario, the workspaces and reports that the sales manager from Contoso USA sees are limited to data that is related to Contoso USA. The report viewer doesn't have access to data associated with any other company, unless the company context is changed.

1. Open the analytical report in Power BI Desktop by double-clicking the resource in a Microsoft Visual Studio project.

2. On the **Modeling** tab, select **Manage Roles**.

3. Create a new role against a column in the data model that contains the **Company** field. Name the new role **CompanyFilter**. A **COMPANY** field is present in the data model to restrict access by company.



4. In the **Table filter DAX expression** field, enter **[COMPANY]=username()**.

5. To make sure that the rules work, on the **Modeling** tab, select **View as Roles**. In the dialog box, set the following fields:
   a. Clear the **None** check box.
   b. Select the **Other user** check box, and then enter **USMF** in the text box.
   c. Select the **CompanyFilter** check box.

The reports show data as if you're running the USMF company.

Last updated on 11/26/2025

Create reporting solutions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This tutorial shows how to export data and create a report, expand predefined views and add navigation to charts, use the free-form report designer, and customize the parameter experience.

Prerequisites

For this tutorial, you must access the application environment, and you must be provisioned as an administrator on the instance.

Key concepts

- Describe the various ways reports can be created and consumed
- Add interactivity to embedded aggregate reports in forms and workspaces
- Use framework extensions to customize the parameter experience for SSRS based business documents
- Export List Page data to create reports with external tools including Microsoft Excel
- Author modern report designs using enhanced developer tooling in Visual Studio

What's new in Reporting?

- Embedded report drill-through navigation to AX forms and reports
- Navigate between reports using embedded report drill-through links
- Several Document Routing Service enhancements including support for custom print settings
- Introduced Document Brand Management administrative tools
- More visualizations available through the Embedded Charting Control
- Dates and Amounts in the report body are formatted based on the "Date, time, and number format" user setting
- Network Print monitoring form
- Table extensions need to be supported through the VS Query picker

What is a report in Dynamics 365 finance and operations apps?

Reports are any visualization of a structured data set. This definition includes transactional data presented in a tabular layout and advanced graphical views of aggregate information. To account for this broad definition, the application offers several tools to produce reports that satisfy complex business requirements. Some common applications of reports in an ERP include:

- Creating and archiving transactional documents as part of a posting process
- Producing packing slips for tracking orders from Manufacturing to Warehousing to Sales
- Monitoring key performance metrics and surfacing trends in data
- Navigating the client through filtered searches
- Distributing heavily branded documents to customers and employees
- Extracting data in such a way that it articulates the health of a business

The most difficult task for developers is selecting the *right* Business Intelligence visualization tool for the job, given a customer's requirements. To accomplish this task, it's important to understand the capabilities offered through the tools that are available for creating reports. We offer tooling to support the following basic reporting requirements:

- **Excel Integration** – Allows data management and analysis using Microsoft Excel
- **Embedded Analytics** – Add aggregate data to Workspaces by using native controls like charts and grids
- **Reporting Services** – Create business documents that require precision by using SSRS-based solutions
- **Power BI Integration** – Author and share reports that can be accessed anywhere
- **Management Reporter** – Designed to help users create financial reports

The following table can be used to compare the basic characteristics of these reporting tools:

Expand table

Tool	Author	Target	Data	Consumption	Designer
Excel	Power User	Power User	Transactional	External	Free form
Embedded BI	Developer	All users	Aggregates	Internal	Modeled
SSRS Report	Developer	All users	Transactional and Aggregates	Internal/External	Free form

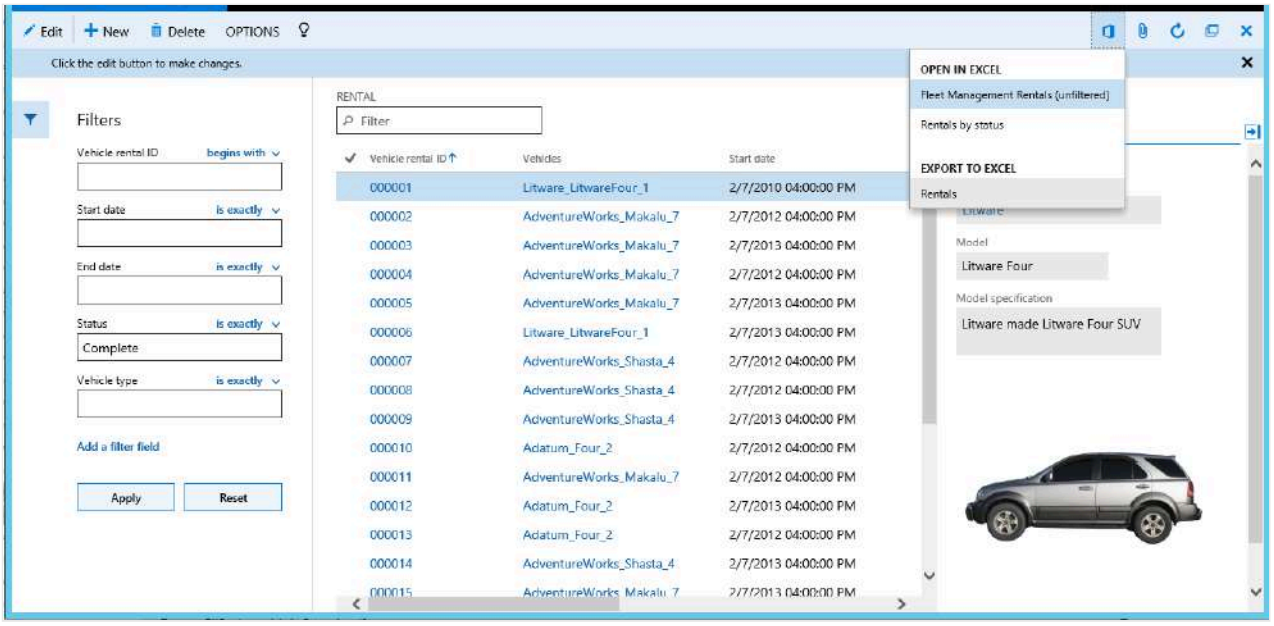
Tool	Author	Target	Data	Consumption	Designer
Power BI	Power User	All users	Aggregates	Internal/External	Free form
Management Reporter	Power User	Power User	Transactional	External	Modeled

Create a report using List Pages

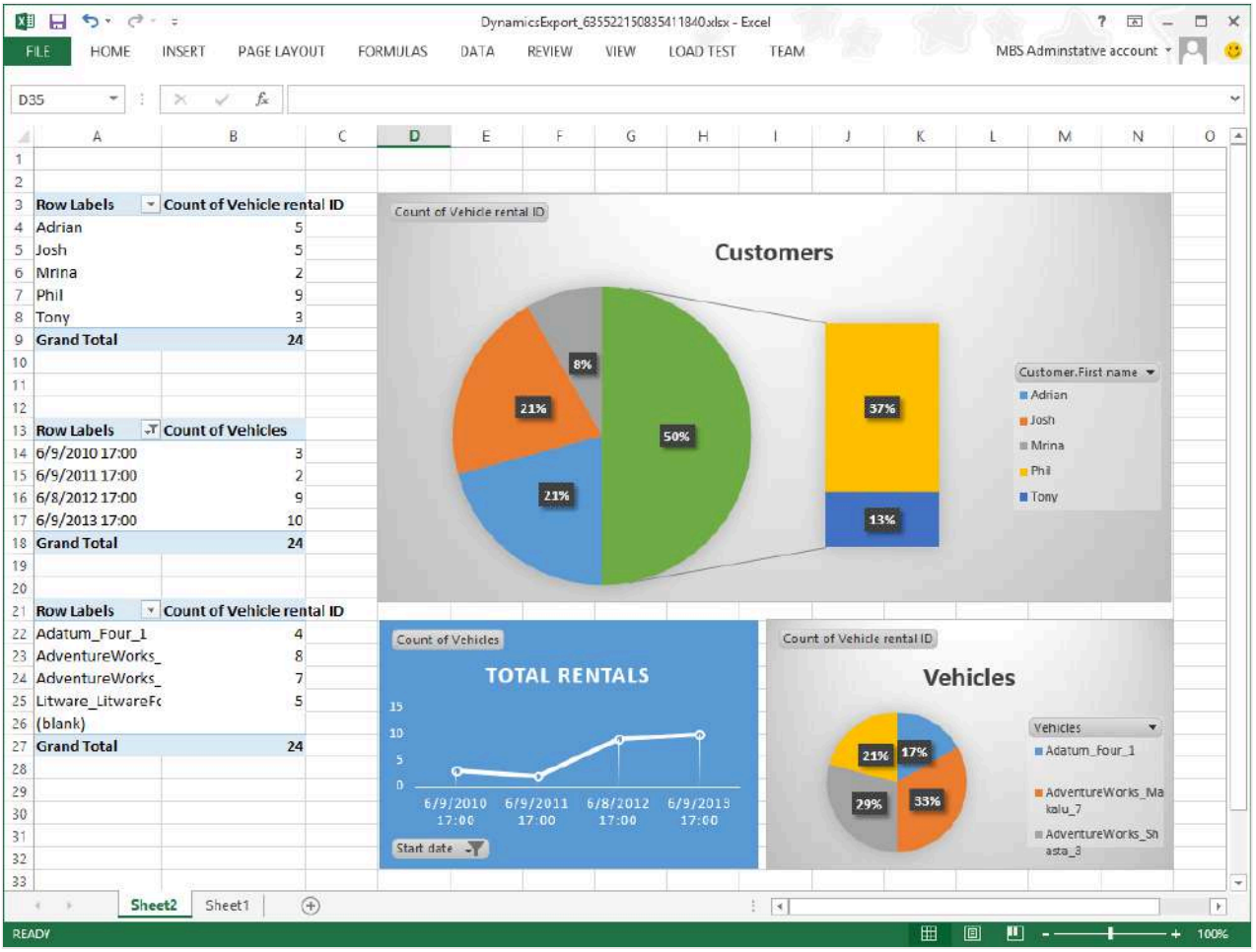
This section walks you through the process of exporting data displayed in an entity details form. It demonstrates how every form can be viewed as a source of data for management and analysis by using the power of Excel.

Create an analysis report based on all rentals in Fleet Management that are marked as complete

1. Open Microsoft Edge, go to your instance base URL, and sign in.
   a. On the cloud environment, you get the base URL from LCS.
   b. On a local VM, the base URL is `https://usnconeboxax1aos.cloud.onebox.dynamics.com`.
2. On the Dashboard, scroll to the far right, and select the **Reservation management** tile.
3. Under the **Summary** section, select on the **All rentals** tile.
4. Expand the **Filters** pane by selecting on the Filters icon next to the main grid.
5. Select **Add a filter field**, then select **Status** in the drop-down list.
6. Enter **Complete** in the Status filter field, then select **Apply**.
7. Expand the page action bar, select **Open in Office**, then select **Rentals**.



After the data is exported, you can use the power and flexibility of the Excel tools to create reports for presentation or other analysis. The following image is an example of a self-service report authored with Excel.



8. Close the browser session and the exported Excel file.

Advantages of list pages

- Flexible source for accessing transactional business data by using external reporting and analytical tools like Excel
- Allows end users to model the data sets they need without requiring a developer
- Direct access to real-time business information

Expand predefined views and add navigation to charts

Business Intelligence can be useful at every level of an organization. Use embedded controls to elevate the most relevant information based on the target persona. Native controls offer users an intuitive and convenient way of interacting with aggregate data, allowing for informed decision making.

Create development project

1. On the desktop, right-click the **Visual Studio** shortcut, and select **Run as administrator** to open the development environment.

2. On the toolbar, select **View**, then select **Application Explorer**.

3. Use the **Application Explorer** to search for the **FMReservationsReport** form in the **Fleet Management** module.

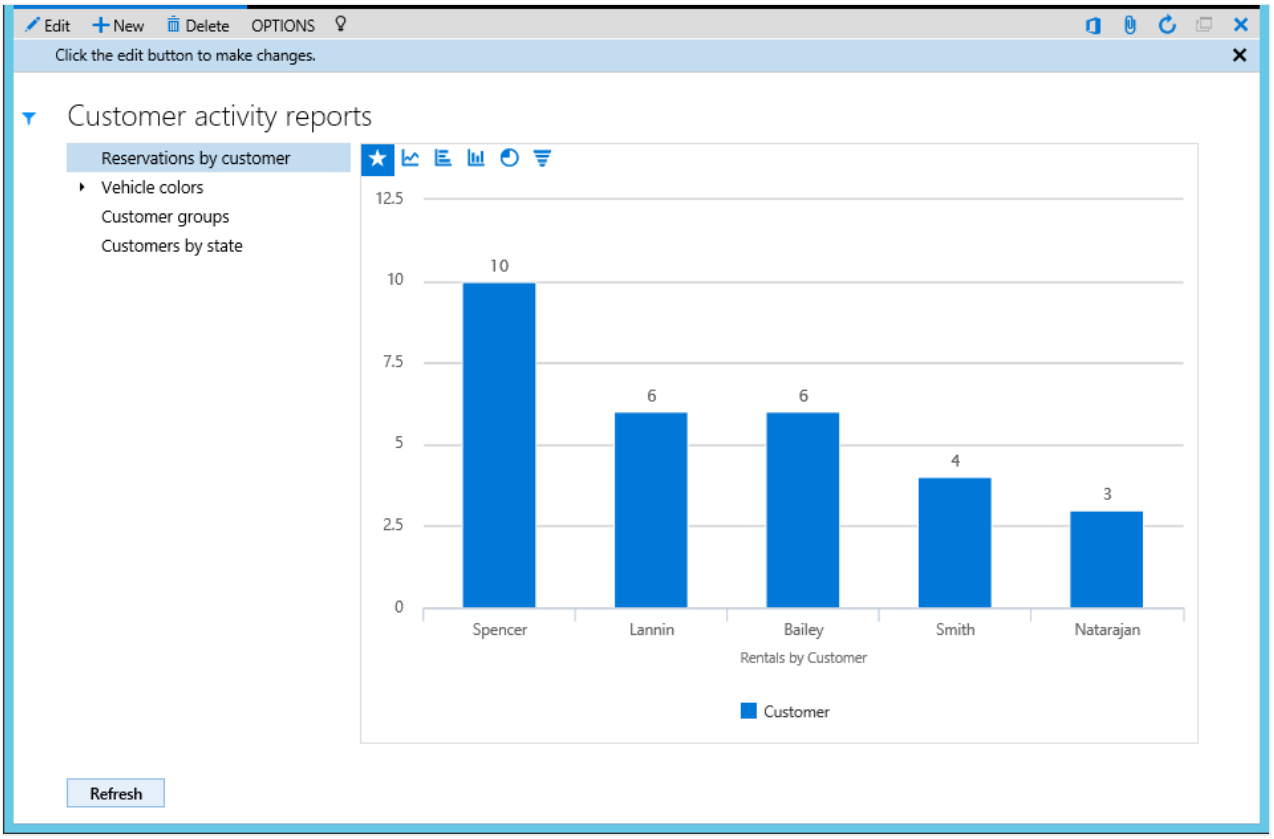
4. In the **Application Explorer's** search results, right-click **FMReservationsReport** form, then select **Add to new project**.

5. Select the **finance and operations Project** template, then select **OK** to create the project.

6. In **Solution Explorer**, right-click **FMReservationsReport** menu item, then select **Set as startup object.**

7. Press Ctrl+Shift+B to save and build the project.

8. Press Ctrl+F5 to load the form containing the report.

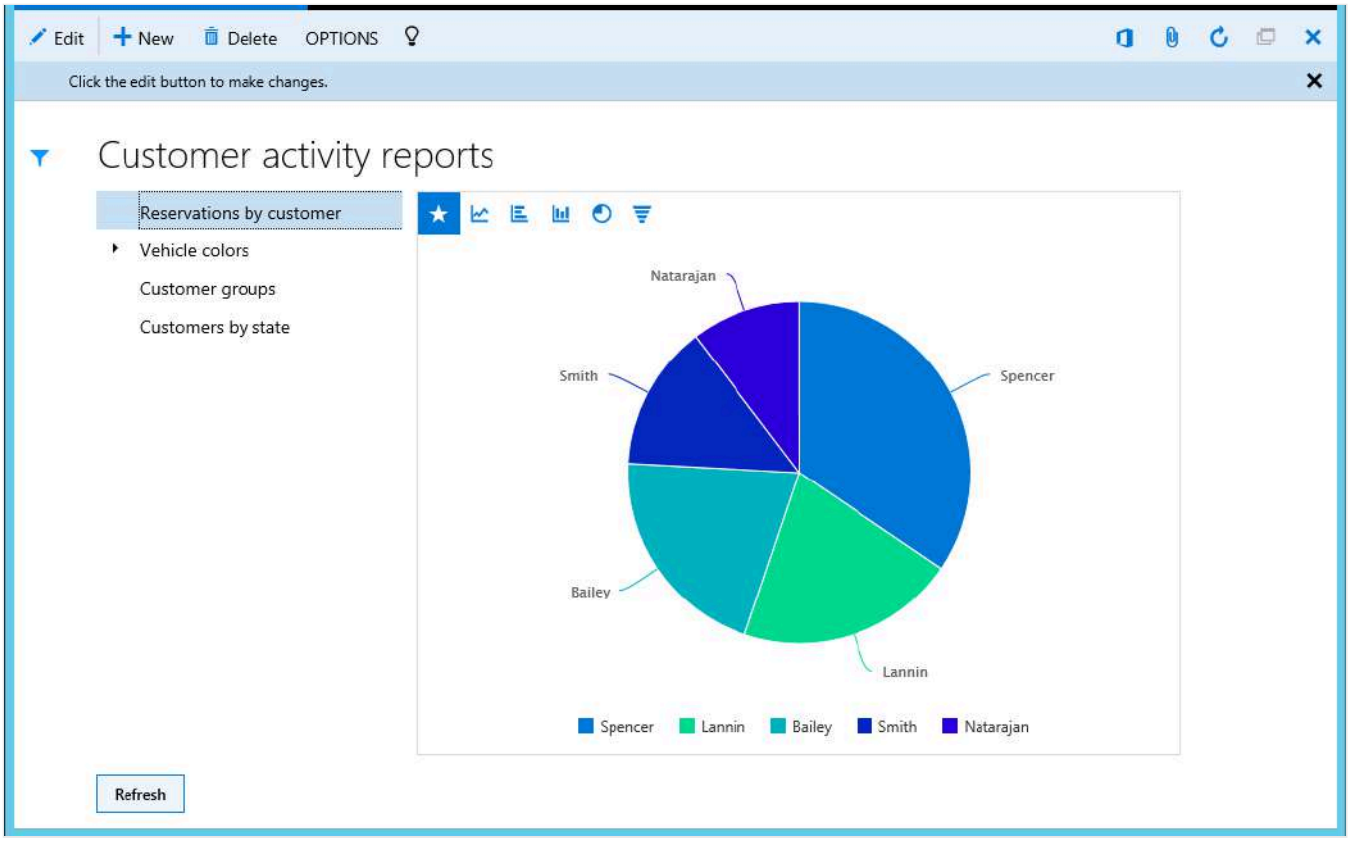


At this point, you have a collection of predefined chart visualizations of aggregate data. The controls offer basic interactive features like hover text, series filtering, and touch expansion. However, it's often appropriate that a greater degree of user interactivity is required.

Change the default chart type for the visualization

1. Close the browser session and return to the Visual Studio project.
2. Open the **Application Explorer**, locate the **FMReservationsByCustPart** form, right-click, and then select **Add to project**.
3. Open the **FMReservationsByCustPart** form designer.
4. Access the **Series** collection in the **ReservationsByCust** chart definition.
5. Select the **SysBuildChatMeasure1** series definition.
6. Locate the **Appearance** section in the **Properties** window.
7. Update the **Chart type** value and select **Pie**.
8. Press Ctrl+Shift+B to save and rebuild the project.
9. To run the report, press Ctrl+F5.

The **Reservations by customer** view now visualizes the aggregate data set by using a Pie chart. This chart type simplifies the task of comparing rental activity across customers.



Other functions include:

- Define contextual drill-through navigation by using modeled properties
- Manage drill-through links by using X++ form logic

Advantages of embedded aggregate visualizations

- Designed to be simple and intuitive to promote informed decisions at every level of the organization
- Predefined views tailored to meet the needs of the consumer
- Augments the user's ability to recognize key performance trends in the data

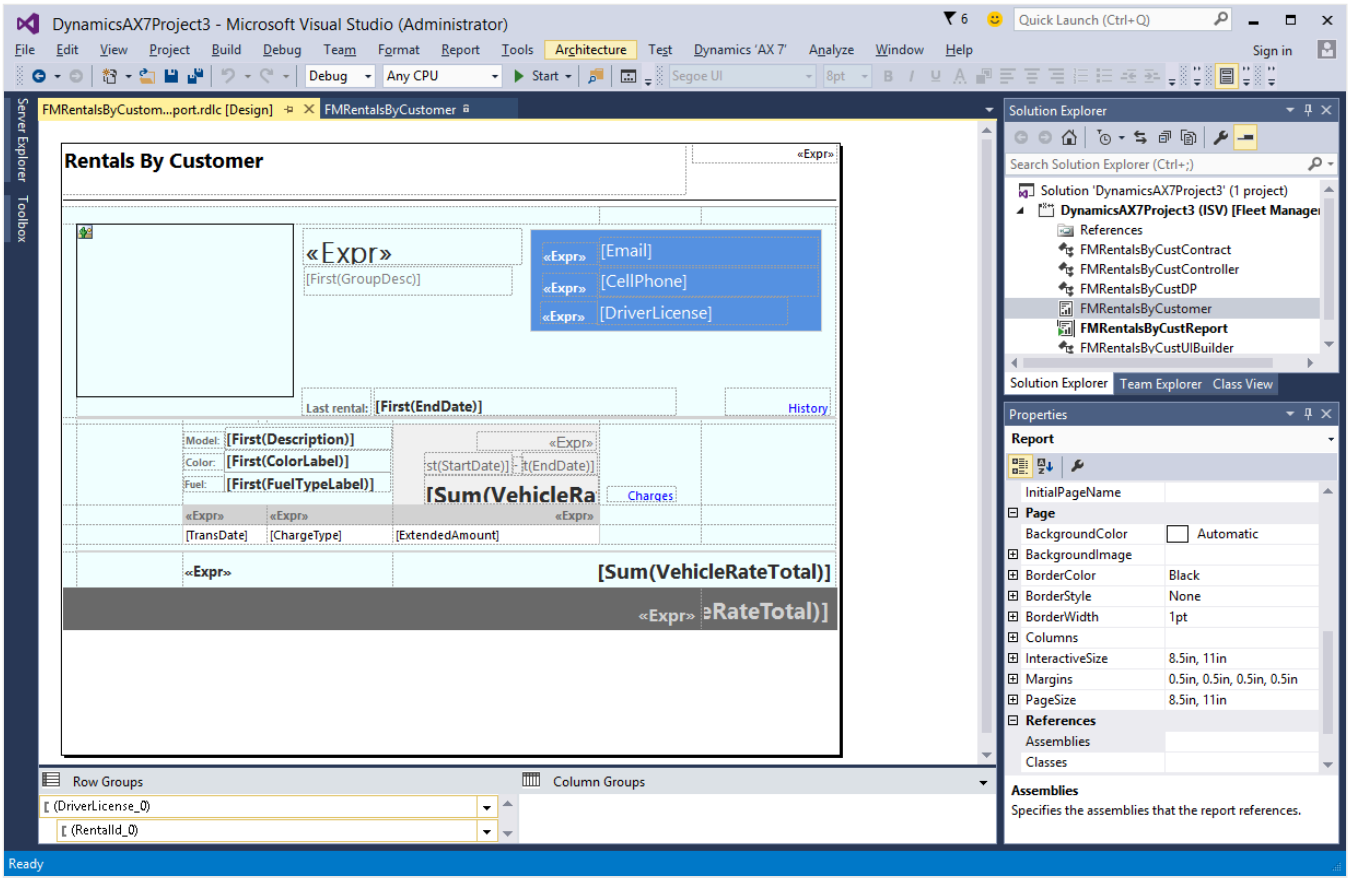
Using the freeform report designer

SSRS continues to be the platform for producing advanced Business Document solutions for your ERP application. The hosted service is designed for high volume and heavily transactional reports with an integrated parameter experience. Backed by the Document Printing & Distribution Services, these solutions are ideal for producing precision documents that are intended for email, printing, archive, and bulk distribution.

Open the Precision Designer

1. Close the browser session and return to the Visual Studio project.

2. Select **File** > **Close Solution** to close the active project.
3. Use the **Application Explorer** to search for objects with **FMRentalsByCust** in the name.
4. In the **Application Explorer's** search results, select all **Classes**, the **Output Menu Item**, and the **FMRentalsByCustomer** report.
5. Right-click and then select **Add to new project**.
6. Select the **Dynamics 365** template, then select **OK** to create the project.
7. In **Solution Explorer**, double-click on the **FMRentalsByCustomer** report to open the designer.
8. Expand the **Designs** collection in the designer to view the list of design definitions.
9. Double-click on the **Report** design to view the report definition using the Precision Designer.



The preceding screenshot shows the FMRentalsByCustomer report design definition as viewed by using the Visual Studio Precision Designer. The Precision Designer offers a free-form design surface with built-in tools that you can use to customize the content and layout of the report. You can also take advantage of embedded VB code to create run-time design manipulations and support user interactions. As an integrated tool, developers can reference AX labels and public APIs to format data in the report body based on AX EDTs. MSDN offers a rich collection of developer documentation related to SSRS formatting capabilities. For a good primer on designing effective SSRS reports, see the article Reporting Services Reports (SSRS).

Customizing the parameter experience

The Reporting Framework offers flexibility through service extensions to facilitate advanced solutions with requirements that can't be addressed by using a modeled solution. Use the VS designer to add basic parameter formatting, grouping, and input validation. X++ based data contract validation is available for more advanced scenarios. Consider adding User Interface (UI) Builder Classes to customize the parameter pane used to prompt for session inputs before running a report. These custom extensions are effective for addressing the following functions:

- Overriding dialog field events
- Validating report parameters inline
- Adding customized lookups to dialog fields
- Changing the layout of the dialog controls
- Adding custom controls including images

Defining parameters defaulting using code

1. In **Solution Explorer**, double-click on the **FMRentalsByCustUIBuilder** class to open the designer.

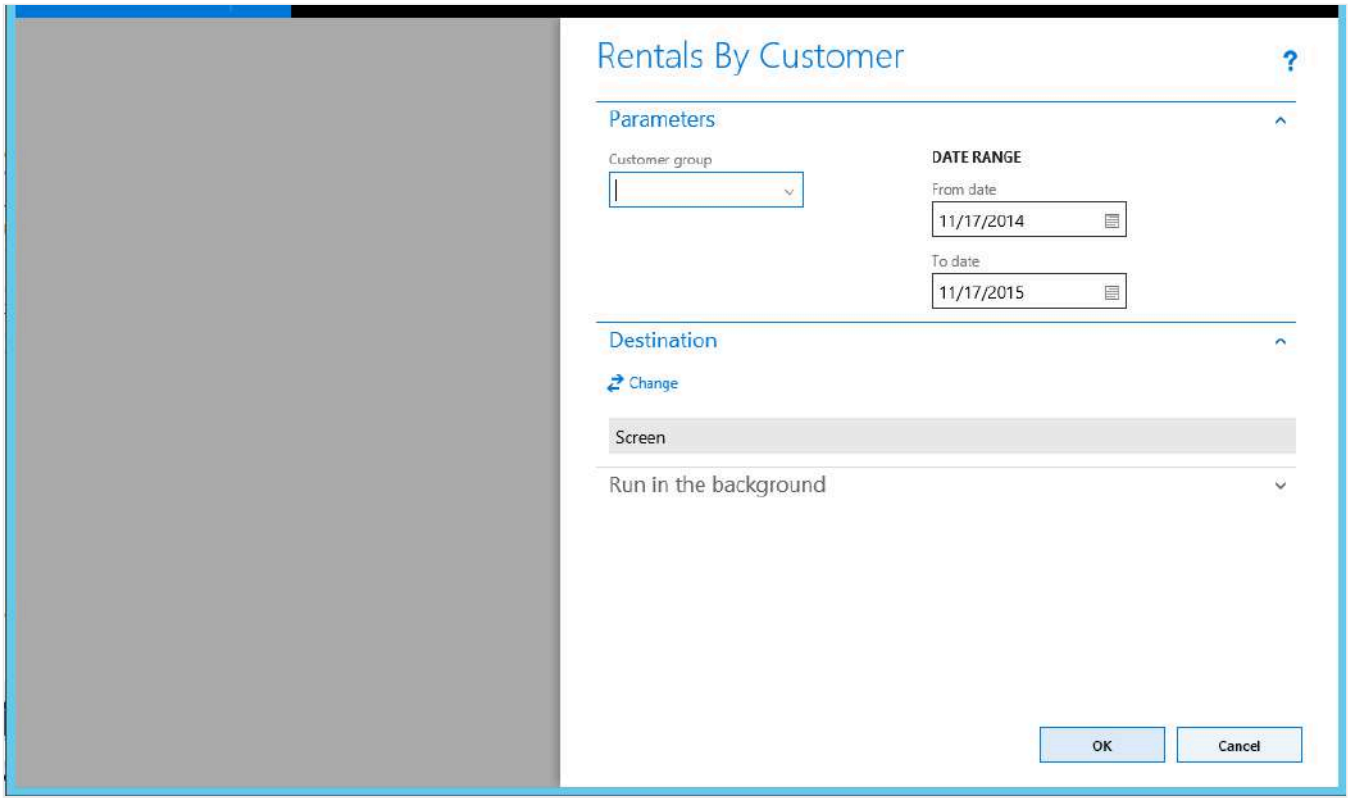
2. Locate the class **build** method and update the initialization code as follows

   X++

   ```
   public void build()
   {
       Dialog dialogLocal = this.dialog();
       contract = this.dataContractObject() as FMRentalsByCustContract;//
   associate dialog field with data contract method
       this.addDialogField(methodStr(FMRentalsByCustContract, parmCustGroupId),
   contract);dialogLocal.addGroup("@SYS41297");
       fromDateField = this.addDialogField(methodStr(FMRentalsByCustContract,
   parmFromDate), contract);
       toDateField = this.addDialogField(methodStr(FMRentalsByCustContract,
   parmToDate), contract);// set the default date range values
       fromDateField.value(today() - 365);
       toDateField.value(today());
   }
   ```

3. Press Ctrl+Shift+B to save and rebuild the project.

4. Press Ctrl+F5 to run the report.



The parameter initialization code sets the default values of the report execution relative to today's date. Use the classes UIBuilder to override the framework's default handling of report parameters. Other extension scenarios supported:

- Automatically set query ranges based on session context using Controller classes
- Select report designs at runtime using a shared menu item
- Modify report data contract values using business logic

> **Note**
>
> This example demonstrates a UIBuilder extension with an RDP-based report. For a Query based example that includes a UIBuilder extension, see the **FMCustomerList** report.

Using VB code to manage running balances

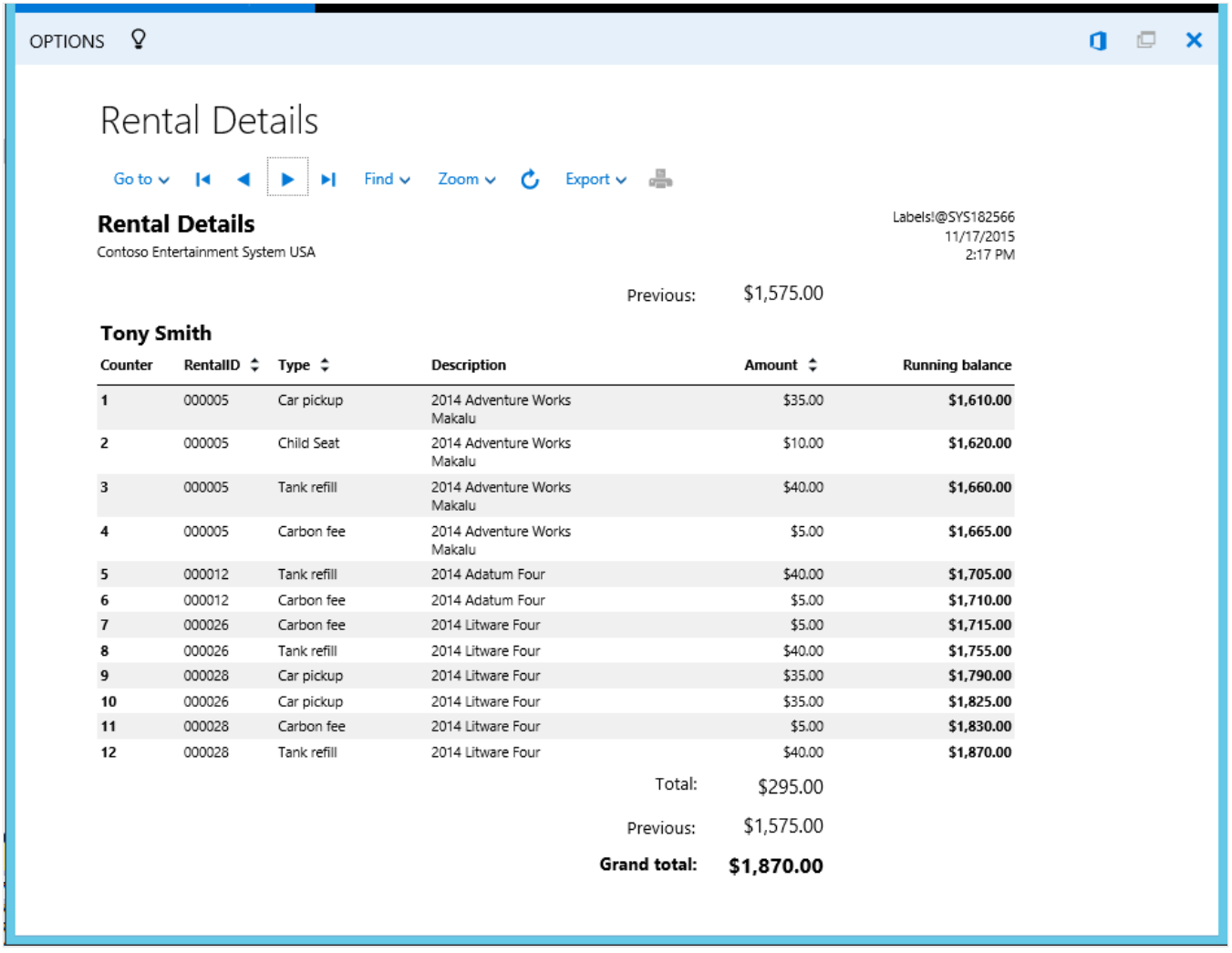
The Reporting Framework offers flexibility through service extensions to facilitate advanced solutions with requirements that can't be addressed by using a modeled solution. Use the VS designer to add basic parameter formatting, grouping, and input validation. X++ based data contract validation is available for more advanced scenarios. Consider adding User Interface (UI) Builder Classes to customize the parameter pane used to prompt for session inputs before running a report. These custom extensions are effective for addressing the following functions:

- Overriding dialog field events
- Validating report parameters inline

- Adding customized lookups to dialog fields
- Changing the layout of the dialog controls
- Adding custom controls including images

Import the section resources

1. Close the browser session and return to the Visual Studio project.
2. On the toolbar, select **DYNAMICS AX**, then select **Import Project...**
3. In the File name window, browse to `C:\FmLab\Lab10-3`, select **FMRentalDetailsReport.axpp**, then select **Open**.
4. In the Project file location text box, enter `C:\FmLab\Lab10-3`.
5. Select the **Current solution** radio button.
6. Select **OK** to close. Wait for the project to be imported and opened.
7. In the **Solution Explorer**, select the new project, right-click, then select **Set as startup project.**
8. In the **Solution Explorer**, right-click **FMRentalDetails** menu item, then select **Set as startup object.**
9. Select the project in Solution Explorer, right-click, then select **Deploy Reports**
10. Press Ctrl+F5 to view the report.



This report uses embedded VB script to keep track of running totals so that the balances from the previous page can be referenced on the active page. To inspect the report's VB code, load the report design in the Precision Designer, access the **Report Properties**, then select the **Code** section. Here you see several functions referenced by the report designs to surface running balances within the report headers and footers.

Advantages of SSRS reports

- Built-in back office document management capabilities including email support, scheduled executions via Batch, and Print Archive
- Parameterized views with drill-through navigation to forms and other reports
- Used to produce precision documents for compliance with local regulatory business practices

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Last updated on 11/26/2025

Customize App Suite reports by using extensions

This article discusses a series of scenarios for customizing App Suite reports.

Finance and operations offer an expanded set of tools to support custom solutions. Customizations to reporting solutions in the standard application are fully supported using a pure extension model. This article contains guidance about how to add the most common customizations to standard application reports without overlayering Application Suite artifacts. Here are some of the key benefits of using an extension-based approach when customizing the application:

- It reduces the footprint of your application solutions by minimizing code duplication.
- Custom reports benefit from enhancements made to standard solutions including updates to business logic in Report Data Provider (RDP), data contracts, and UI Builder classes.
- Standard application solutions are unaffected and continue to be available in concert with custom reports.

Report extensions don't break or prevent access to standard application reports. Instead, the platform supports run-time selection of the target report allowing you to choose the appropriate report design based on the context of the user session. For more information about customizations using extensions, see Customize through extension and overlayering

Scenarios

There are four key scenarios that demonstrate the flexibility available. The first two scenarios involve extending existing RDP classes for custom reporting solutions.

- Expand Application Suite report data sets – Use table extensions and integrate custom business logic to add custom columns to an existing dataset.
- Composing custom datasets – Add more data to application reports by extending an existing RDP class to return a custom dataset.

The other two scenarios offer insights on how to use extensions to redirect application navigation to your custom solutions.

- Extend report menu items to redirect user navigation – Customize application menu items to redirect references to a custom report design.

- Create custom designs for business documents – Delegate handlers allow you to add custom report designs to an existing Print Management document instance.

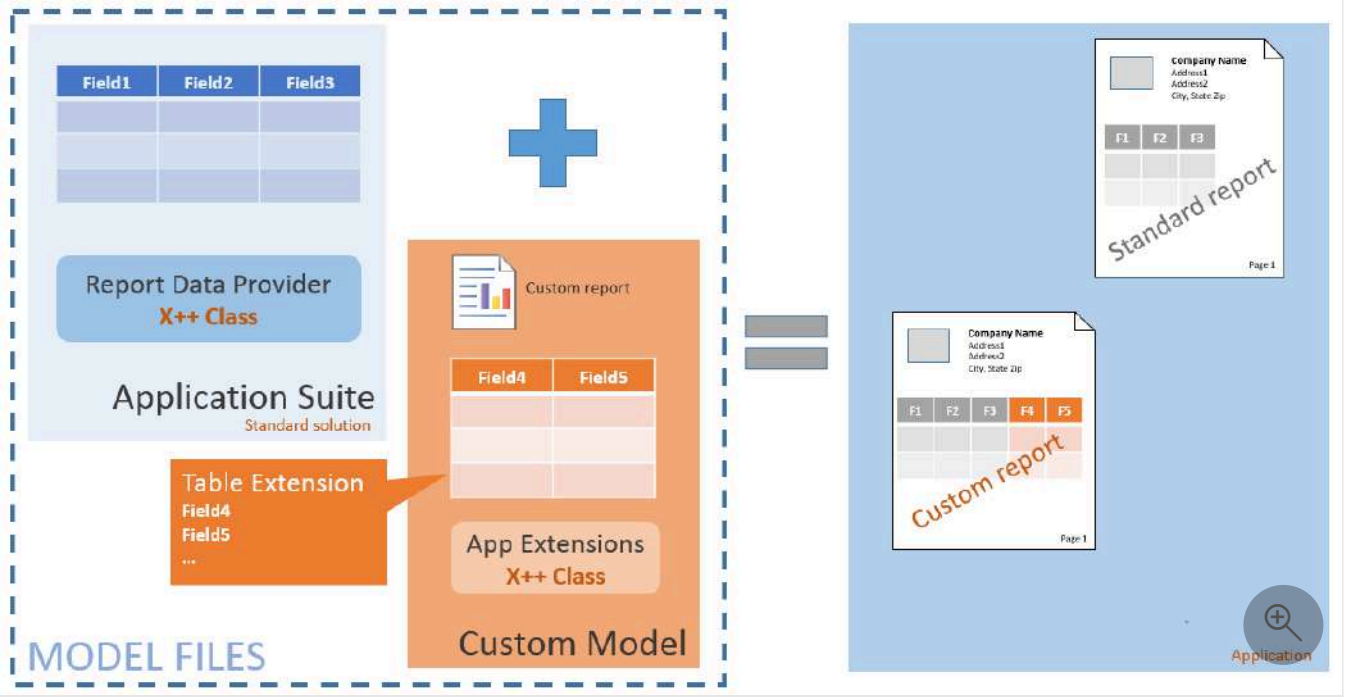
Last updated on 06/26/2026

Expand Application Suite report data sets

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows how to expand an existing report data set that uses X++ business logic in a report data provider (RDP) class.

This article focuses on expanding an existing report data set that uses X++ business logic in a report data provider (RDP) class. You use custom delegate handlers and table extensions to include more field data and calculations. You don't have to over-layer the Application Suite. You then create custom designs that replace the standard application solutions and present the data to users. The following illustration shows a typical application customization, as described in this article.



Important considerations

Before you apply this solution, be aware of the following considerations.

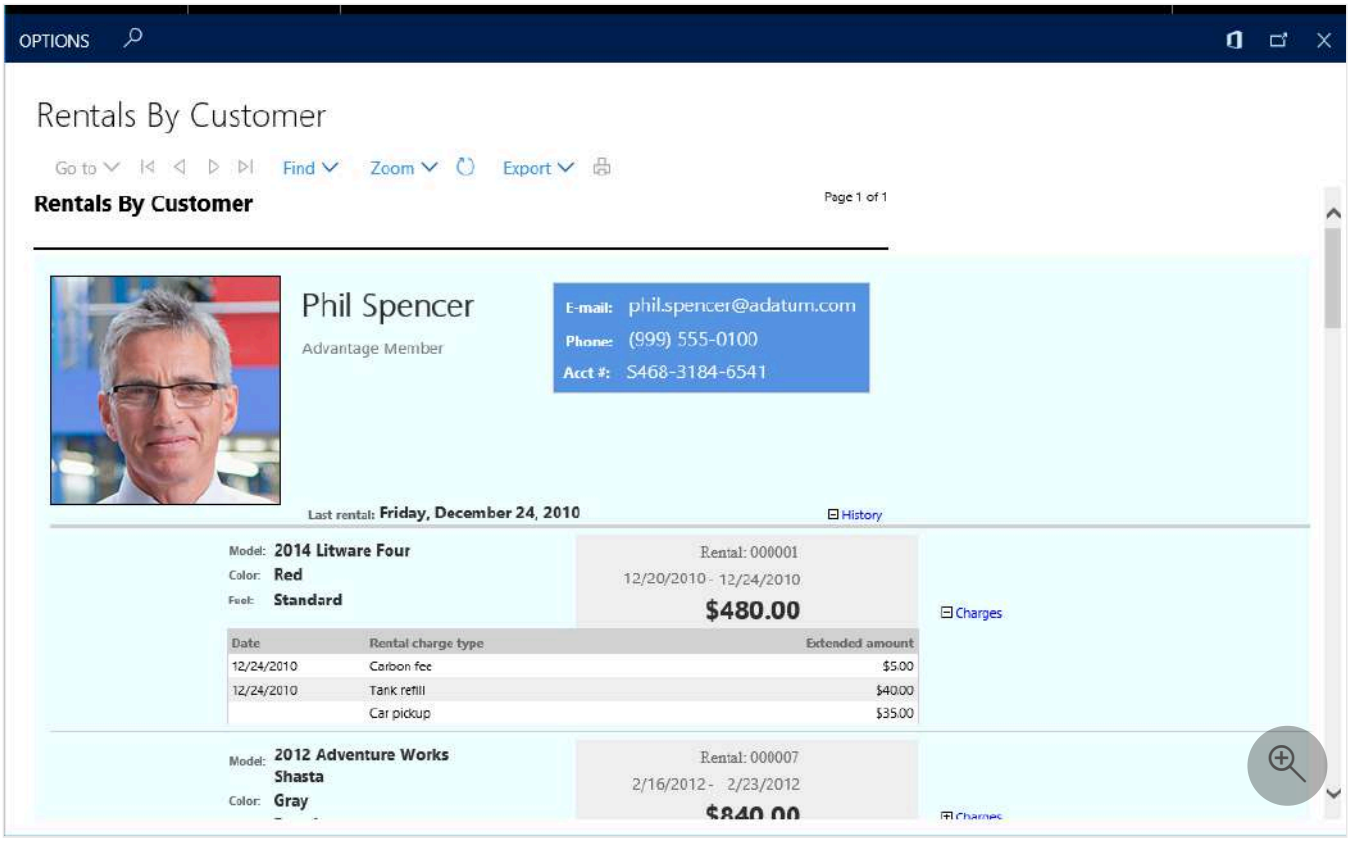
- **You can't directly extend RDP classes.** However, the platform provides extension points that enable data set expansion without duplicating business logic in the standard application.

- **Two methods are available to expand report data sets.** Use the strategy that's appropriate for your solution:
  - **Data processing post-handler** – This method is called only one time, after the **ProcessReport** method completes and before the data set is returned to the report server. Register for this post-handler to perform bulk updates on the temporary data set that the standard application solution produces.
  - **Temp table inserting event** – This method is called for each row that is added to the temporary table. It's more suitable for calculations and inline evaluations. Avoid expensive queries that have many joins and look-up operations.
- **Use event handlers to redirect menu items to your new report design.** You can customize all aspects of an application reporting solution by using event handlers. Add a **PostHandler** event for the controller class to reroute user navigation to a custom report design.

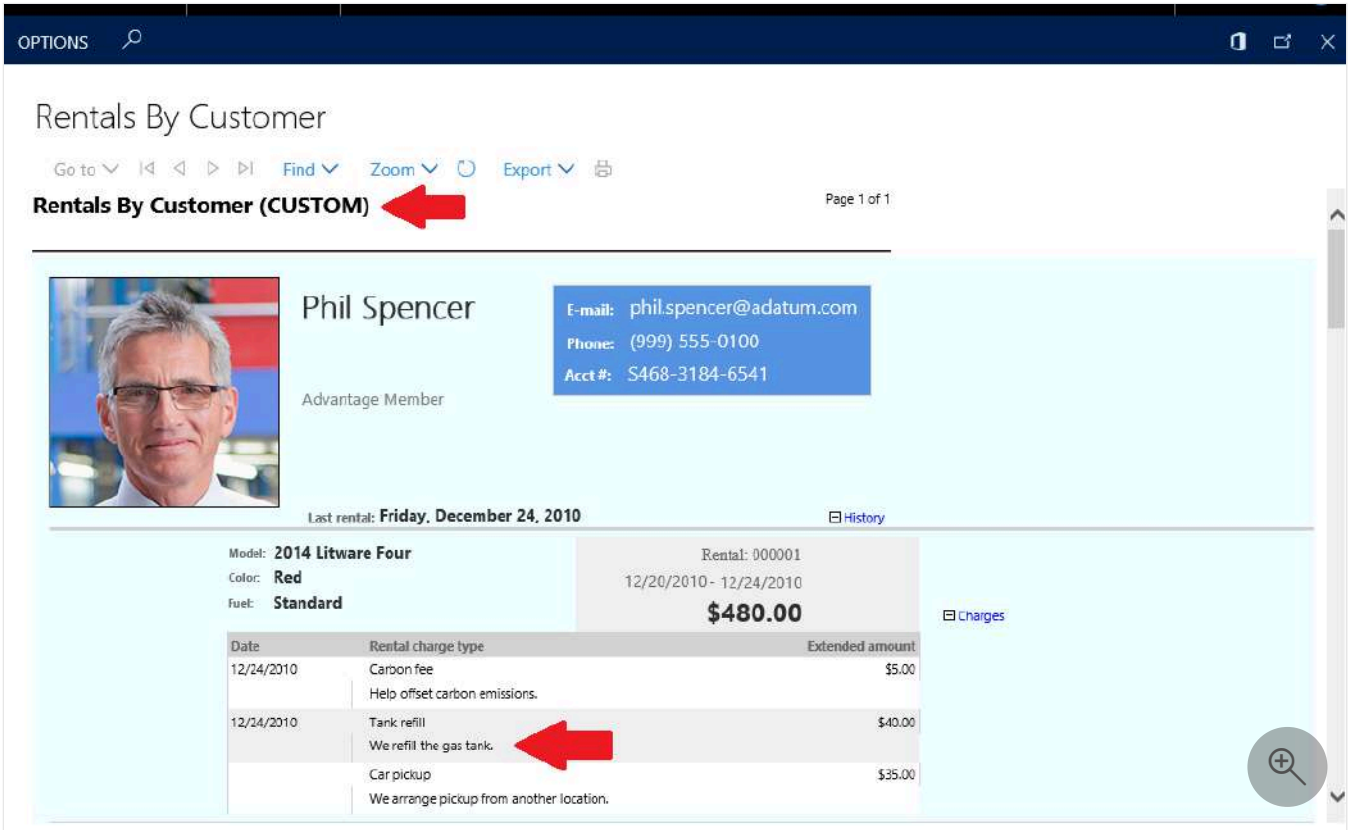
Expand a report data set

The following walkthrough shows the process of expanding an existing application data set by using a "pure" extension-based solution. The solution includes a custom **Rentals list** report for the Fleet Management application. The new report includes more rental charge data in the rental details. The application customizations are defined in an extension model. The following illustrations show the standard design and the custom solution.

Before (standard design)

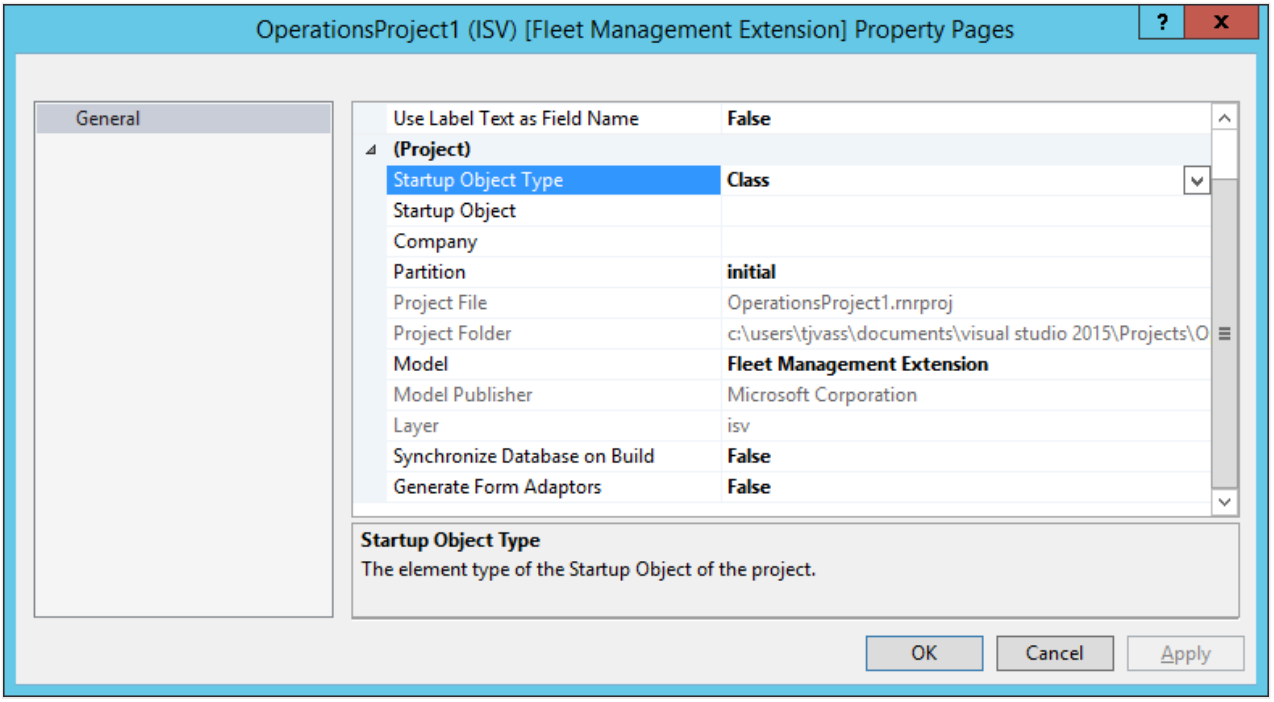


After (custom solution)

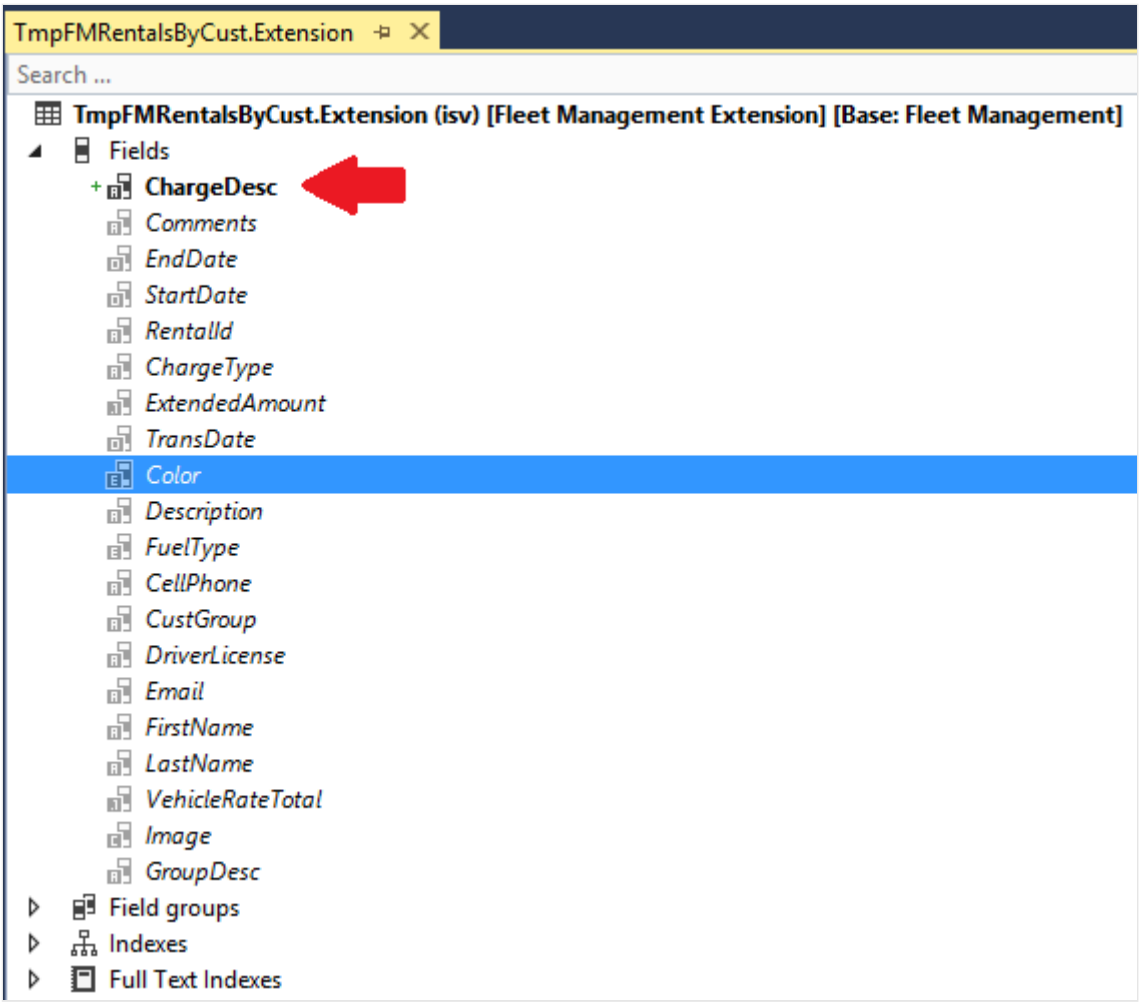


1. **Create a new model for your application customizations.** For more information about extension models, see Customize through extension and overlayering. For this example, add a custom report to the **Fleet Management Extensions** model.

2. **Create a new project in Microsoft Visual Studio.** Make sure that the project is associated with your extension model. The following illustration shows the project settings.



3. **Add a table extension to store the custom report data.** Find the temporary cache for the **TmpFMRentalsByCust** data set that the RDP class populates, and create an extension in your model. Define the fields that store the data for the report server, and then select **Save** to save your changes. The following illustration shows the table extension that is required for this example.

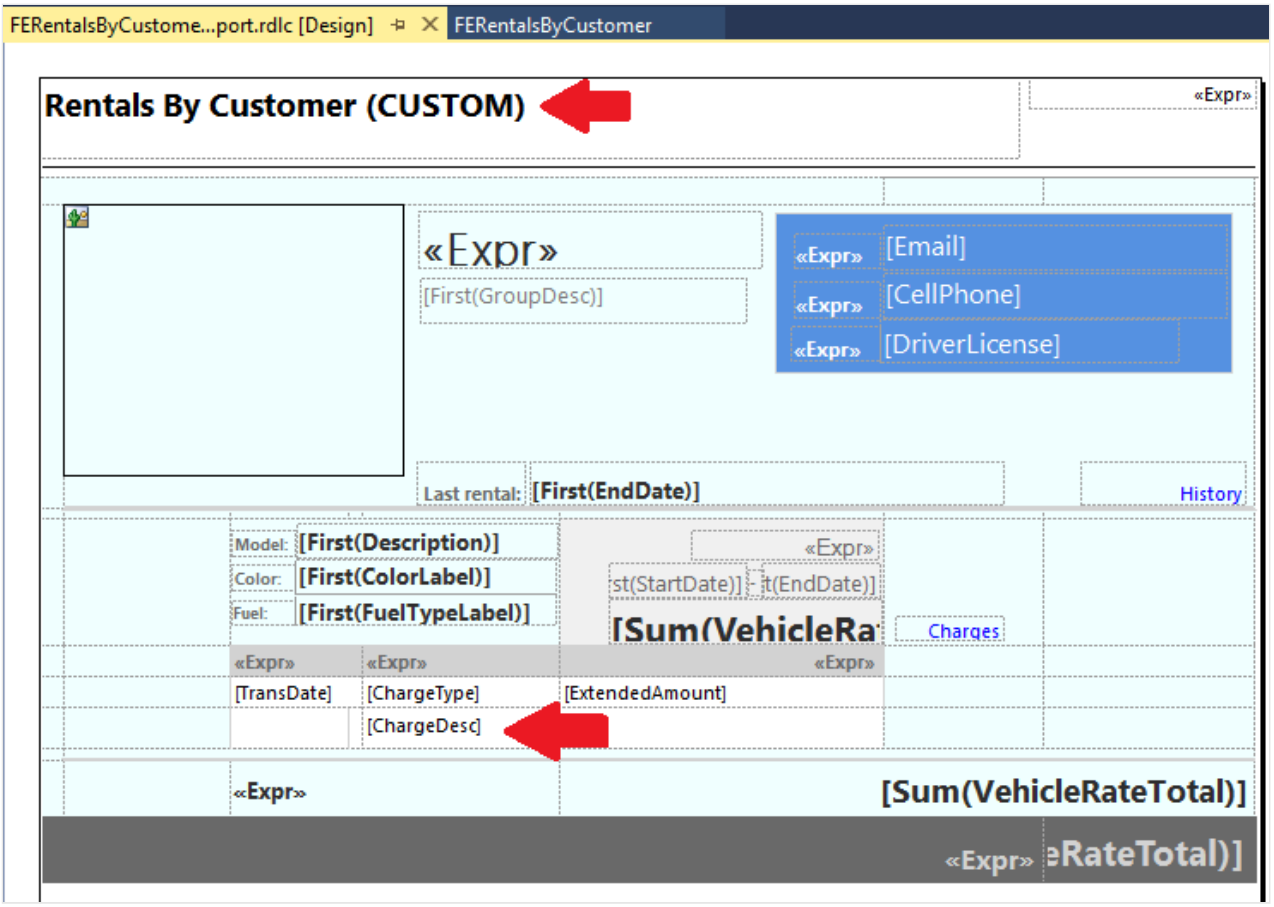


4. **Add your custom report to the project.** The custom design closely resembles the standard solution. Therefore, you can just duplicate the existing application report in the **Fleet Management Extension** model, and then update the report design so that it includes the custom title and another text box in the Rental Charges container.

5. **Rename the report so that it has a meaningful name.** For this example, rename the custom report **FERentalsByCustomer** to distinguish it from the standard solution.

6. **Restore the report data set references.** Open the report designer, expand the **Datasets** collection, right-click the data set named **FMRentalsByCustDS**, and then select **Restore**. The data set expands so that it includes the newly introduced columns. Therefore, these columns are now available in the report designer.

7. **Customize the report design.** The designer offers a free-form design surface that you can use to create the custom solution. The following illustration shows the custom design that is used for this example.



8. **Add a new report handler (X++) class to the project.** Give the class a name that appropriately describes that it's a handler for an existing application report. For this example, rename the class **FERentalsByCustomerHandler** to distinguish it from other report handlers.

9. **Add a PostHandler method to begin to use your custom report.** In this example, extend the controller class in the standard solution, **FMRentalsByCustController**, by using the following code.

X++

```
class FERentalsByCustomerHandler
{
    [PostHandlerFor(classStr(FMRentalsByCustController),
staticMethodStr(FMRentalsByCustController, construct))]
    public static void ReportNamePostHandler(XppPrePostArgs arguments)
    {
        FMRentalsByCustController controller = arguments.getReturnValue();
        controller.parmReportName(ssrsreportstr(FERentalsByCustomer, Report));
    }
}
```

User navigation in the application now reroutes to the custom reporting solution. Take some time to deploy the custom report to the report server and verify that the application is using it. At this point, you just need to add the business logic that

populates the custom fields that you introduced in step 3. In the next step, you must select the method of data set expansion that's appropriate for your solution.

10. **Add X++ business logic to populate the custom field data.** Select the data processing technique that makes sense for the type of transformation that you require for the solution.

    - **Option 1: Add a data processing post-handler.** Apply this technique for bulk insert operations that use a single pass over the result set of the standard solution. Here's the code that expands the data set by using a table lookup.

      X++

      ```
      class FERentalsByCustomerHandler
      {
          [PostHandlerFor(classStr(FMRentalsByCustDP),
      methodstr(FMRentalsByCustDP, processReport))]
          public static void TmpTablePostHandler(XppPrePostArgs arguments)
          {
              FMRentalsByCustDP dpInstance = arguments.getThis() as
      FMRentalsByCustDP;
              TmpFMRentalsByCust tmpTable = dpInstance.getTmpFMRentalsByCust();
              FMRentalCharge chargeTable;
              ttsbegin;
              while select forUpdate tmpTable
              {
                  select * from chargeTable where chargeTable.RentalId ==
      tmpTable.RentalId;
                  tmpTable.ChargeDesc = chargeTable.Description;
                  tmpTable.update();
              }
              ttscommit;
          }
      }
      ```

    - **Option 2: Add a temp table Inserting event.** Apply this technique for row-by-row calculations. Here's the code that expands the data set by using a table lookup.

      X++

      ```
      class FERentalsByCustomerHandler
      {
          [DataEventHandlerAttribute(tableStr(TmpFMRentalsByCust),
      DataEventType::Inserting)]
          public static void TmpFMRentalsByCustInsertEvent(Common c,
      DataEventArgs e)
          {
              TmpFMRentalsByCust tempTable = c;
              FMRentalCharge chargeTable;
              // update the value of the 'ChargeDesc' column during 'insert'
      ```

```
operation
        select * from chargeTable where chargeTable.RentalId ==
tempTable.RentalId
        && chargeTable.ChargeType == tempTable.ChargeType;
        tempTable.ChargeDesc = chargeTable.Description;
    }
}
```

You finished expanding the report data set. After you compile the application, it reroutes user navigation to the new report design by using the custom X++ business logic that you defined in the report class handler that is defined in the extension model.

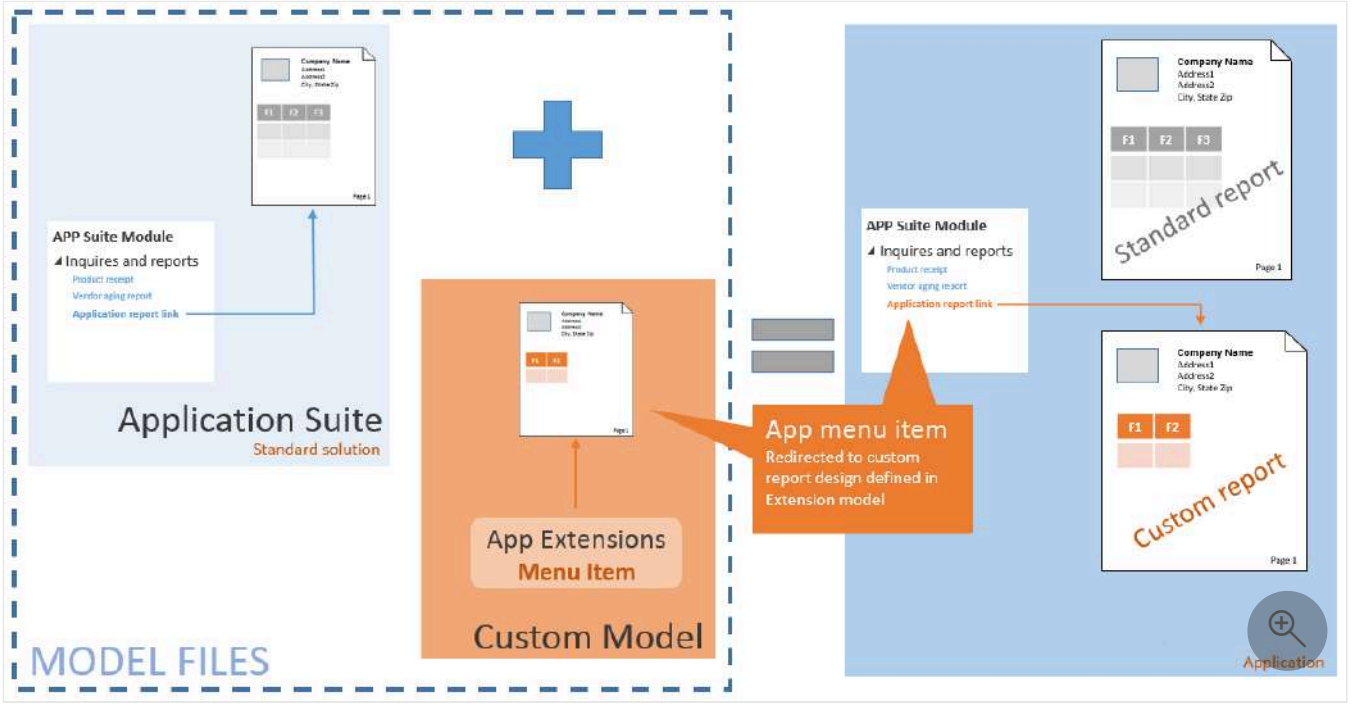
Last updated on 11/26/2025

Extend report menu items to redirect user navigation

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows how to extend existing application menu items so that, with only minimal code changes, you can redirect navigation to a custom reporting solution.

This article focuses on the process of extending existing application menu items so that, with only minimal code changes, you can redirect navigation to a custom reporting solution. By using this technique, you avoid the inconvenience of tracking down and replacing all references to an existing application report. Just extend an existing application menu item to redirect application navigations to reports that you define in an extension model. The following illustration shows a typical application customization.



What's important to know?

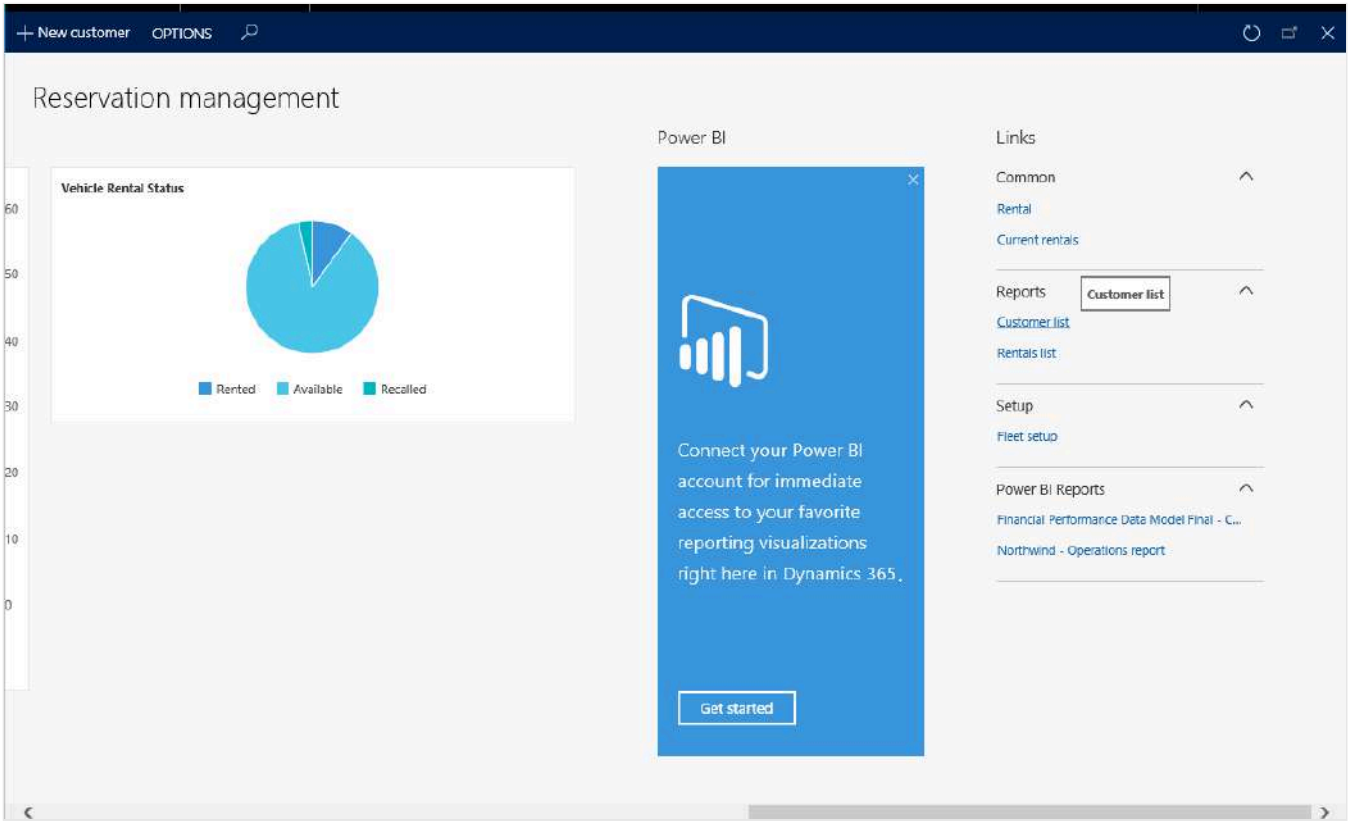
Before you apply this solution, be aware of the following basic assumptions.

- Extended menu items let you override both the display string and the target.

- You can use this technique for all types of reports, from simple query-based reports to complex report data provider (RDP)–based reports.
- Extended menu items are available for direct references to reports and solutions that orchestrate the reporting session by using a controller class.

Extend report menu items

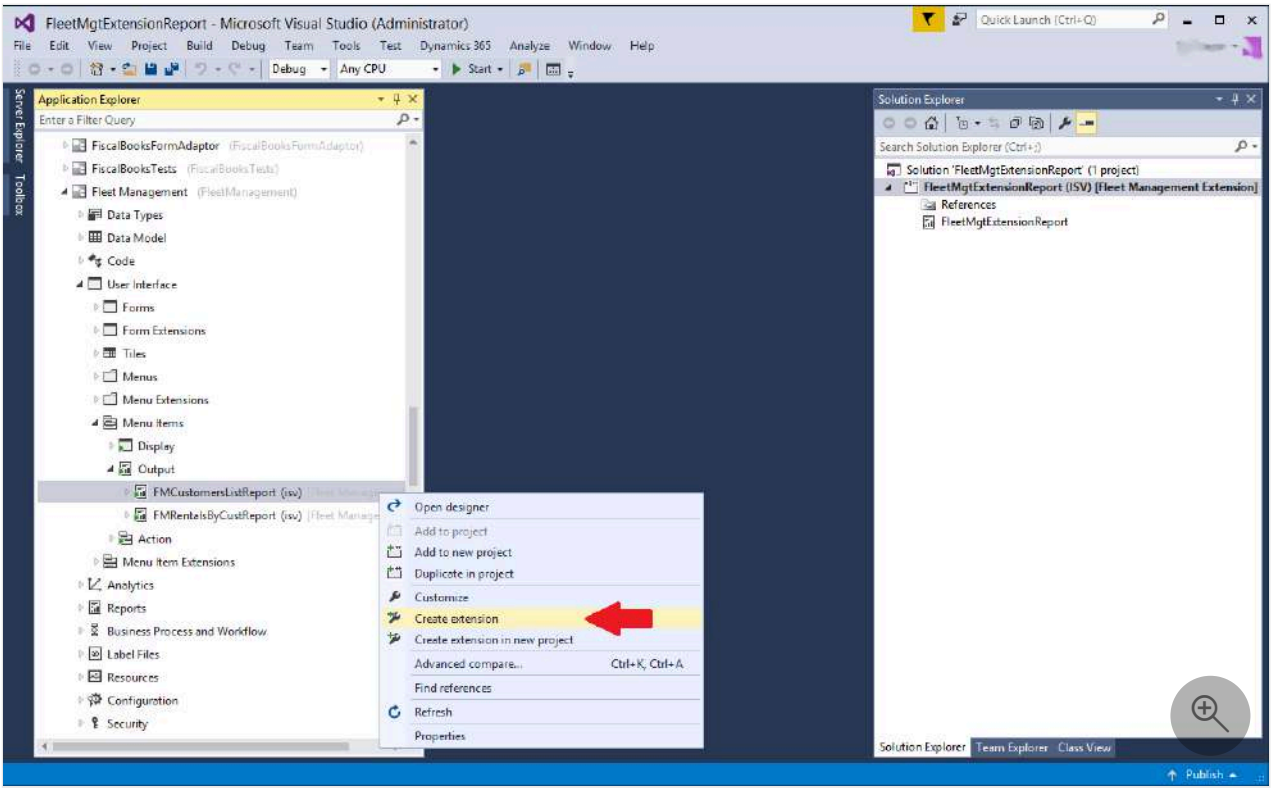
The following walkthrough shows how to use menu item extensions to redirect user navigations in the application to a custom solution. The solution includes a custom **Customer list** report for the Fleet Management application and defines all the application customizations in a pure extension model. The following illustration shows the menu item that you use to access the custom **Customer list** report.



1. **Create a new model for your application customizations.** For more information about extension models, see Customize through extension and overlayering.

2. **Create a new project in Microsoft Visual Studio, and add your custom report.** Additionally, add all the solution artifacts. These artifacts include the RDP class or source query, the controller class, and UI builders, if they're present.

3. **Create an extension of the menu item that is used to access the report.** In this example, the output menu item is named **FMCustomerListReport**. Use the menu item structure to find the menu item name that the application exposes. The following illustration shows the action in Application Explorer.



4. **Modify the properties of the menu item extension.** Update the report design or controller reference in the menu item to direct navigations to your custom solution.

> ⓘ **Note**
>
> The property changes that you can make on the object depend on the original application solution. If the application report manages the solution by using a controller, a controller class is required for the report.

5. **Rebuild the solution, and deploy the custom report.**

You finished extending the report menu item. Navigations to the standard menu item now redirect to your custom reporting solution.

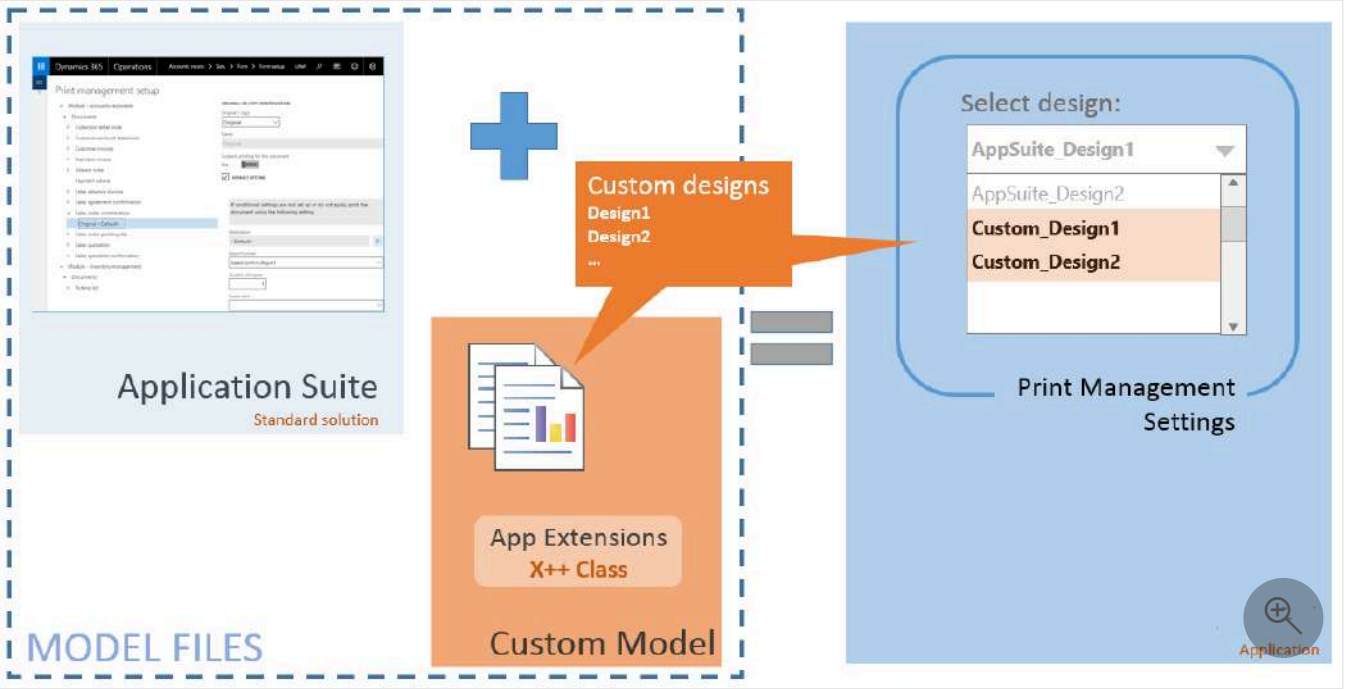
Last updated on 11/26/2025

Create custom designs for business documents

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows how to create a custom report design for an existing application business document by using a pure extension model.

Microsoft Dynamics 365 Finance includes an expanded set of tools to support custom solutions. This article focuses on the steps for creating a custom report design for an existing application business document by using a pure extension model. Follow the steps later in this article to associate a custom report design with an instance of an application document. When you finish, users can configure Print management settings to select the custom design whenever it's appropriate, based on transaction details. The following illustration shows a typical application customization.



What's important to know?

Before you apply this solution, be aware of the following points:

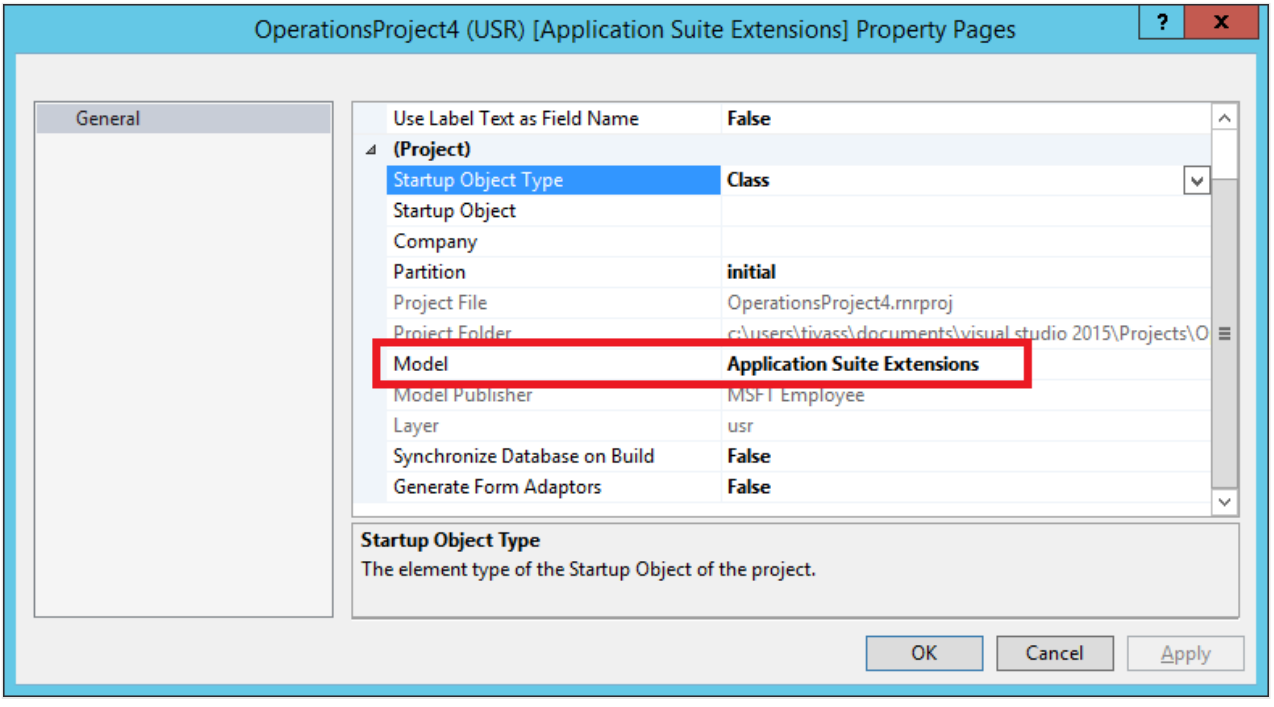
- Print management settings are scoped to the active legal entity. You can associate custom designs with one or more Print management settings.
- Standard report designs continue to be available alongside custom solutions. Use Print management settings to select the appropriate design, based on transaction details.
- If you introduce a business document for a custom business process, you need to do more work. For more information about how to create a custom business document solution, see the Print Management Integration Guide.

Customize a business document

The following walkthrough shows the process of introducing a custom report design for an existing application business document and then using Print management to select the new design. The solution includes a custom design definition for the **Sales confirmation** report that is provided in the standard application as part of the Application Suite model. You define the application customizations in an extension model.

1. **Create a new model for your application customizations.** For more information about extension models, see Customize through extension and overlayering. For this example, add a model named **Application Suite Extensions** that references the Application Suite, Application Platform, and Application Foundation packages.

2. **Create a new project in Microsoft Visual Studio.** Make sure that the project is associated with your extension model. The following illustration shows the project settings.

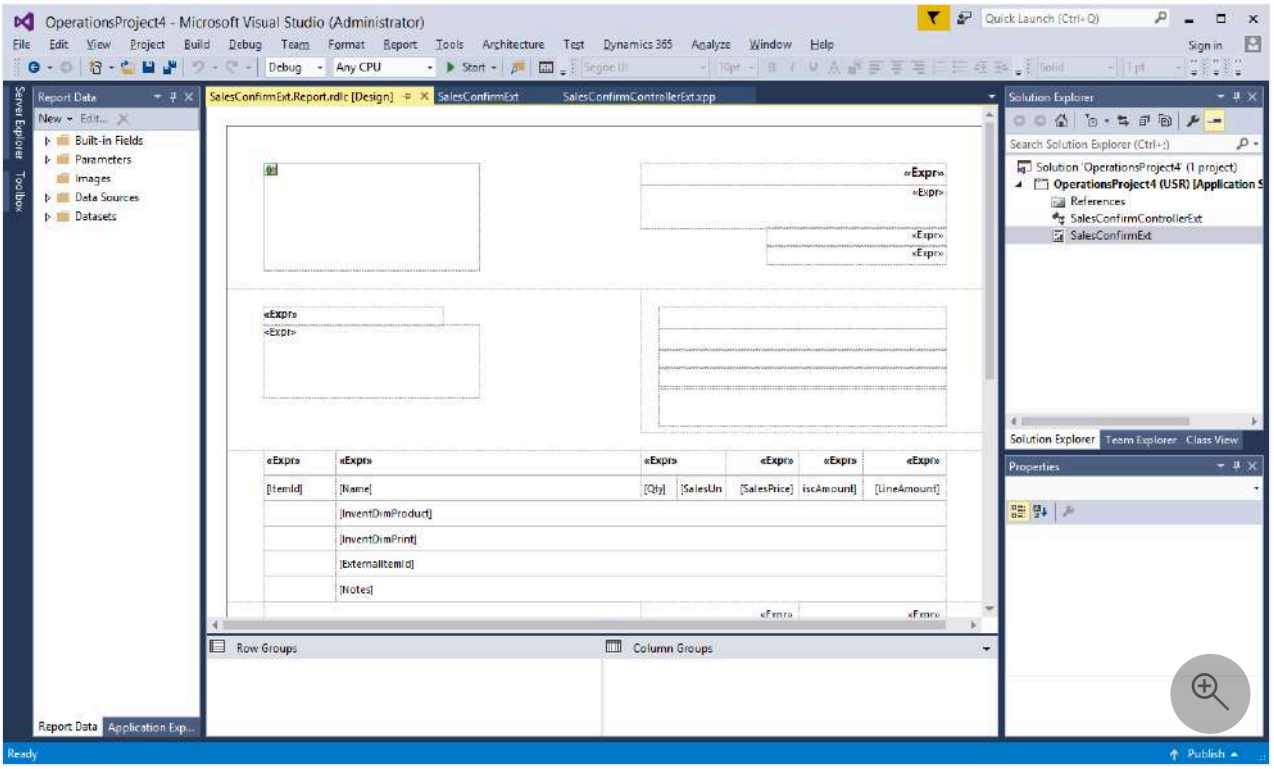


3. **Create a custom report design for the business document.** Make sure that your custom solution consumes the correct report data contract. Find the existing Application Suite

report in Application Explorer. This report is named **SalesConfirm**. Right-click it, then select **Duplicate in project** to create the custom solution.

4. **Rename the report so that it has a meaningful name.** For this example, name the custom report **SalesConfirmExt** to distinguish it from the standard solution. Compile the project, and deploy the report to verify that the changes have no errors.

5. **Use the free-form designer to customize the report design.** Select the report design named **Report**, right-click it, and open the precision designer. Customize the design to satisfy the organization's business requirements. The following illustration shows a custom design definition for the **Sales confirmation** report.



6. **Add a new X++ class that extends the standard report controller.** Give the class a name that appropriately describes that it's a handler for an existing application report. For this example, rename the class **SalesConfirmControllerExt** to distinguish it from other report controllers.

7. **Use the extended class to load the custom design.** Add a **main** method that refers to the custom report design. (You can just copy the **main** method from the standard solution and add references to the new **Controller** class.) Here's the code that extends the standard solution.

X++

```
class SalesConfirmControllerExt extends SalesConfirmController
{
    public static SalesConfirmControllerExt construct()
    {
```

```
        return new SalesConfirmControllerExt();
    }
    public static void main(Args _args)
    {
        SrsReportRunController formLetterController =
SalesConfirmControllerExt::construct();
        SalesConfirmControllerExt controller =
formLetterController;controller.initArgs(_args, ssrsReportStr(SalesConfirmExt,
Report));
        if (classIdGet(_args.caller()) == classNum(SalesConfirmJournalPrint))
        {
            formLetterController.renderingCompleted +=
eventhandler(SalesConfirmJournalPrint::renderingCompleted);
        }
        formLetterController.startOperation();
    }
}
```

8. **Add a new report handler (X++) class to the project.** Give the class a name that appropriately describes that it's a handler for Print management–based documents. For this example, rename the class **PrintMgtDocTypeHandlerExt** to distinguish it from other object handlers.

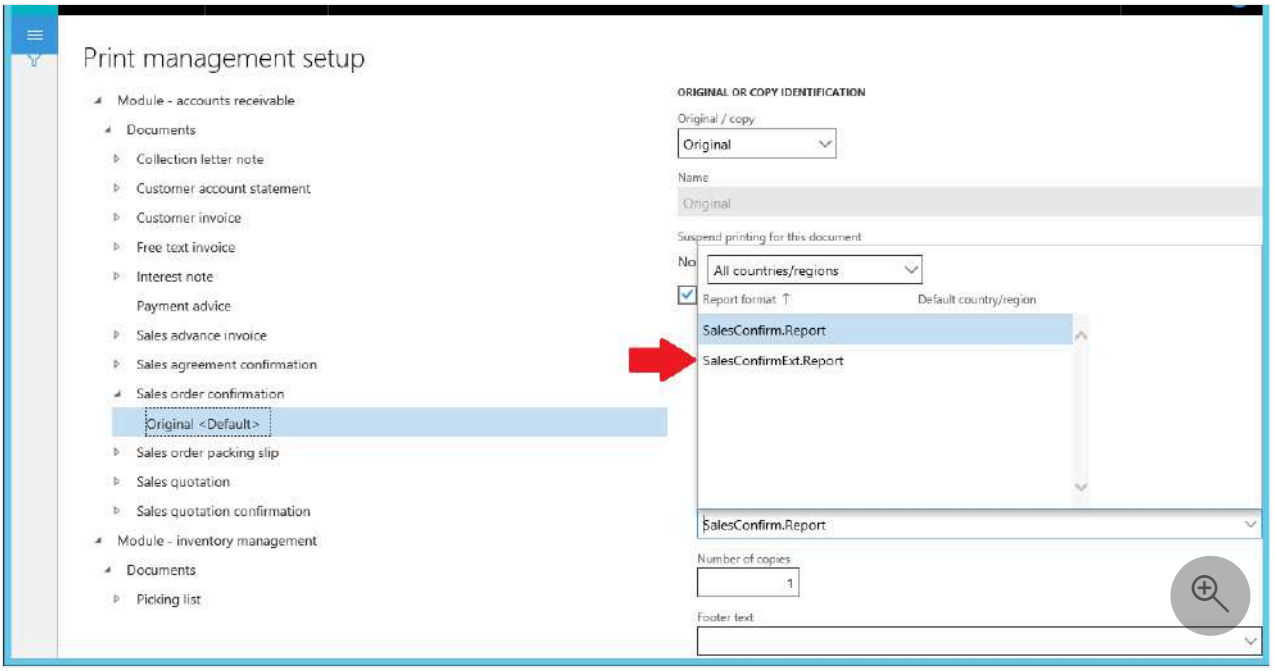
9. **Add a delegate handler method to start to use your custom report.** In this example, extend the **getDefaultReportFormatDelegate** method in the **PrintMgtDocTypeHandlerExt** class by using the following code.

X++

```
class PrintMgtDocTypeHandlersExt
{
    [SubscribesTo(classstr(PrintMgmtDocType), delegatestr(PrintMgmtDocType,
getDefaultReportFormatDelegate))]
    public static void getDefaultReportFormatDelegate(PrintMgmtDocumentType
_docType, EventHandlerResult _result)
    {
        switch (_docType)
        {
            case PrintMgmtDocumentType::SalesOrderConfirmation:
                _result.result(ssrsReportStr(SalesConfirmExt, Report));
                break;
        }
    }
}
```

10. **Extend the menu item for the application report.** Find the existing Application Suite menu item in Application Explorer. This menu item is named **SalesConfirmation**. Right-click it, then select **Create extension**. Open the new extension object in the designer, and set the value of the **Object** property to **SalesConfirmControllerExt** to redirect user navigation to the extended solution.

11. **Update the Print management settings to use the custom business document.** For this example, go to **Accounts receivable** > **Setup** > **Forms** > **Form setup**. Select **Print Management**, find the document configuration settings, then select the custom design. The following illustration shows the Print management settings after the changes have been compiled.



You've now finished customizing the business document. Users will now be presented with the custom report design for the business document when they process transactions in the application.

Last updated on 11/26/2025

Help prevent long-running reports from timing out

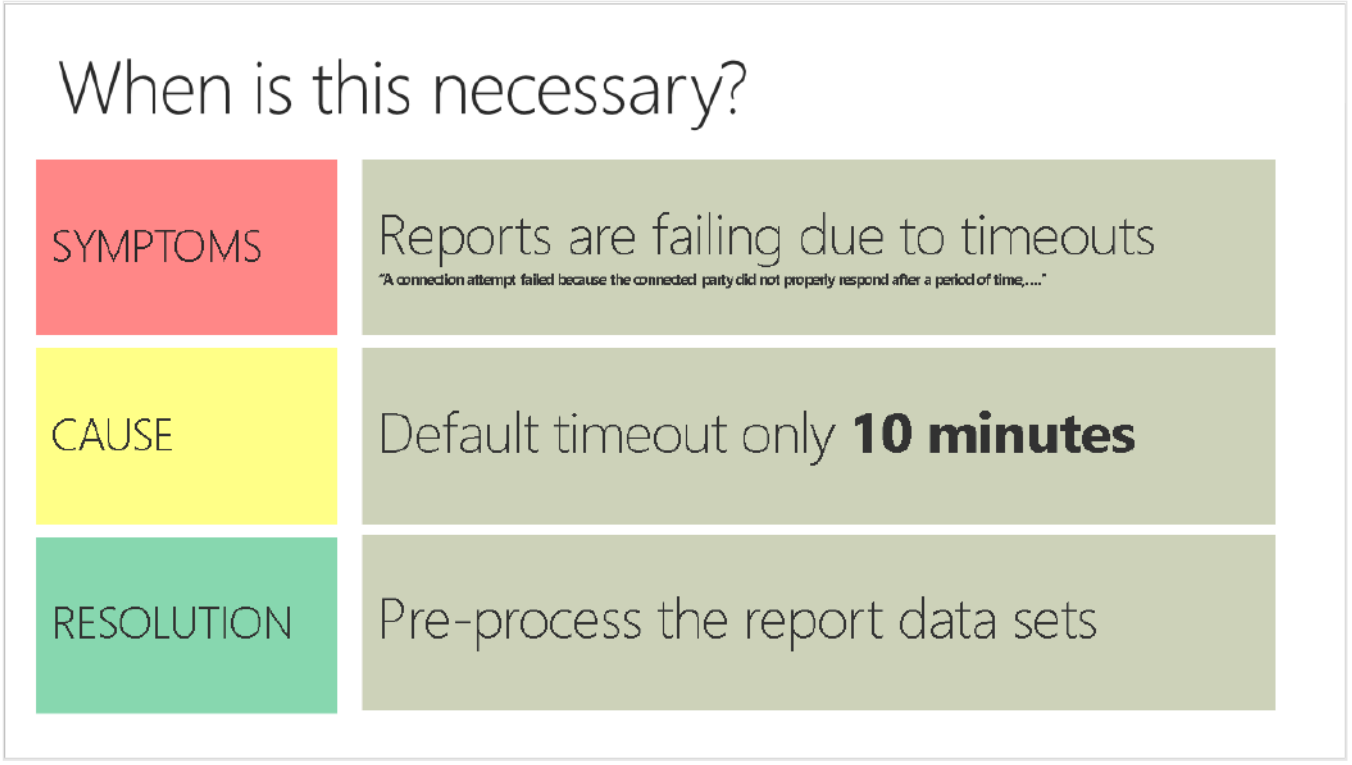
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides tips that can help you prevent reports that run for a long time from timing out.

Paginated reports and documents are generated by using Microsoft SQL Server Reporting Services. Reporting Services retrieves report data from Application Object Server (AOS) by using a custom extension that uses Windows Communication Foundation (WCF) to communicate with AOS. The size of the data set and the complexity of the report that is generated can affect the time that is required to display the report. Additionally, if various time-outs and other thresholds are reached, report generation might fail. Service time-outs in deployments are fixed and limit interactive connections to 10 minutes. Any data set generation process that exceeds this service time-out limit isn't completed. This article describes the extensions that support long-running reports. These extensions help guarantee that long-running reports can be generated even if the process exceeds the 10-minute service time-out limit.

Preprocess the data source

If the report uses the Report Data Provider (RDP) to retrieve data, modify the report to use a pre-processed RDP class as the data source. In this way, processing logic is invoked before a call is made to Reporting Services. For more information about RDP classes, see Using Report Data Provider Classes to Access Report Data and Report Programming Guide.



How do I migrate a regular RDP to a pre-process data access solution by using TempDB?

1. Change the RDP base class from **SRSReportDataProviderBase** to **SRSReportDataProviderPreProcessTempDB**.
2. Update the table type from **InMemory** to **TempDB**.
3. Rebuild the report's RDP class.
4. Restore the data source that is linked to the RDP class in the report designer.
5. Redeploy the report.
6. Introduce a **Controller** class to run the report.
7. Update the Output Menu Item so that it points to the **Controller** class instead of the report.

Use batch processing

To improve performance when you print statements or reports that include large amounts of data, use batch processing. When you use batch processing, you can run specific tasks as batch jobs and then schedule those batch jobs to run on a different computer (a batch server). By moving the processing of these tasks to a batch server, you can improve the report performance on the client computer. You can also apply range restrictions to limit the size of each batch.

To further improve performance, don't submit one large batch. Instead, submit multiple smaller batches for processing at the same time on different servers. You can run many tasks as part of

batch jobs. For more information, see Batch processing overview.

Last updated on 11/26/2025

Electronic reporting (ER) overview

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>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an overview of the Electronic reporting (ER) tool. It includes information about key concepts, the scenarios that ER supports, and a list of formats that are designed and released as part of the solution.

ER is a configurable tool that helps you create and maintain regulatory electronic reporting and payments. It's based on the following three concepts:

- Configuration instead of coding:
  - A business user can configure the solution without needing a developer.
  - The data model is defined in business terms.
  - You use visual editors to create all components of the ER configuration.
  - The language used for data transformation resembles the language used in Microsoft Excel.

- One configuration for multiple Dynamics 365 Finance releases:
  - You manage one domain-specific data model that's defined in business terms.
  - You isolate application release details in release-dependent data model mappings.
  - You maintain one format configuration for multiple releases of the current version, based on the data model.

- Easy or automatic upgrade:
  - The solution supports versioning of ER configurations.
  - You can use the Dataverse as a repository for ER configurations, for version exchange.
  - You can introduce localizations that are based on original ER configurations as child versions.
  - An ER configuration tree is provided as a tool that helps control dependencies for versions.

    - The solution records differences in localization, or the delta configuration, to enable automatic upgrade to a new version of the original ER configuration.
    - You can easily manually resolve conflicts that are discovered during automatic upgrade of localization versions.

ER lets you define electronic format structures and then describe how the structures should be filled by using data and algorithms. You can use a formula language that resembles the Excel language for data transformation. To make the database-to-format mapping more manageable, reusable, and independent of format changes, the solution introduces an intermediate data model concept. This concept enables implementation details to be hidden from the format mapping and enables a single data model to be reused for multiple format mappings.

You can use ER to configure formats for both incoming and outgoing electronic documents in accordance with the legal requirements of various countries and regions. ER lets you manage these formats during their lifecycle. For example, you can adopt new regulatory requirements and generate business documents in the required format to electronically exchange information with government bodies, banks, and other parties.

The ER engine is targeted at business users instead of developers. Because you configure formats instead of code, the processes for creating and adjusting formats for electronic documents are faster and easier.

ER currently supports the TEXT, XML, JSON, PDF, Microsoft Word, Microsoft Excel, and OPENXML worksheet formats.

Capabilities

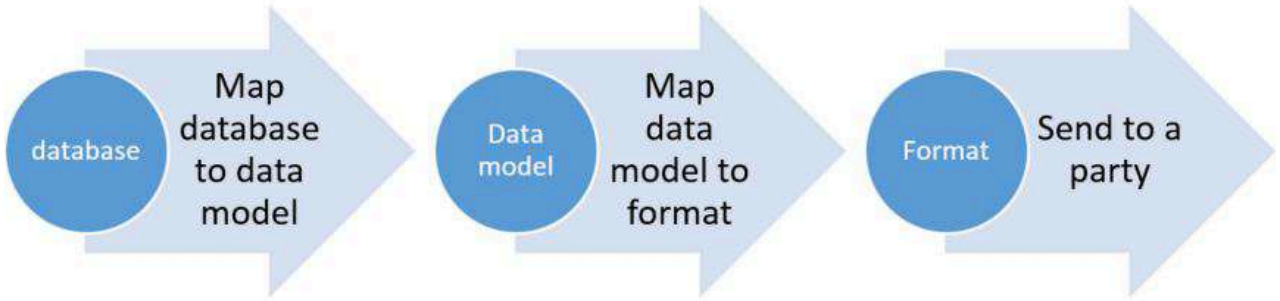
The ER engine has the following capabilities:

- It serves as a single shared tool for electronic reporting in different domains. It replaces more than 20 different engines that handle some type of electronic reporting for finance and operations.
- It insulates a report's format from the current implementation. In other words, the format works for different versions.
- It supports the creation of a custom format that's based on an original format. It also includes capabilities for automatically upgrading the customized format when the original format changes due to localization or customization requirements.

- It becomes the primary standard tool to support localization requirements in electronic reporting, both for Microsoft and for Microsoft partners.
- It supports the ability to distribute formats to partners and customers through Dataverse.

Key concepts

Main data flow



Component

ER supports the following types of components:

- Data model
- Model mapping
- Format
- Metadata

For more information, see Electronic reporting components.

Configuration

An ER configuration is the wrapper for a particular ER component. That component can be either a data model component or a format component. A configuration can include different versions of an ER component. A specific configuration provider owns each configuration. You can edit the **Draft** version of a component of a configuration when you select the owner of the configuration as an active provider in the ER settings in the application.

Each model configuration contains a data model component. You can derive a new format configuration from a specific data model configuration. In the configuration tree, the format configuration that you create appears as a child of the original data model configuration.

The format configuration that you create contains a format component. The data model component of the original model configuration is automatically inserted into the format component of the child format configuration as a default data source.

You share an ER configuration for application companies.

Provider

The ER provider is the party identifier that indicates the author (owner) of each ER configuration. ER lets you manage the list of configuration providers. Format configurations that are released for electronic documents as part of the finance and operations solution are marked as owned by the **Microsoft** configuration provider.

To learn how to register a new ER provider, see the task guide **ER Create a configuration provider and mark it as active** (part of the **7.5.4.3 Acquire/Develop IT service/solution components (10677)** business process).

Repository

An ER repository stores ER configurations. The following types of ER repositories are currently supported:

- File system
- Operations resources
- Dataverse

A **File system** repository provides access to the list of configurations that are located as XML files in the specific folder of the local file system of the machine where the AOS service is hosted. You select the required folder at the repository registration stage. You can import configurations from a **File system** repository into the current instance.

You can access this repository type in the following environments:

- Cloud-hosted environments deployed for development purposes (containing test models of enclosed suites)
- Locally deployed environments (on-premises)

For more information, see Import Electronic reporting (ER) configurations.

An **Operations resources** repository provides access to the list of configurations that Microsoft, as an ER configuration provider, initially releases as part of the application solution. You can import these configurations into the current instance and use them for electronic reporting or playing sample task guides. You can also use them for additional localizations and customizations. You must import the latest versions provided by Microsoft ER configurations from the Dataverse by using the corresponding ER repository.

A **Dataverse** repository provides access to the list of configurations that Microsoft, as an ER configuration provider. You can import these configurations into the current instance and use them for electronic reporting. Dataverse serves as the central source for configurations. You can import configurations from Dataverse into your Finance, and likewise, export configurations into Dataverse. For details, see Import Electronic reporting (ER) configurations from Dataverse.

Register required repositories individually for each configuration provider of the current instance. Each repository can be dedicated to a specific configuration provider.

Supported scenarios

Building a data model

ER provides a model designer that you can use to build a data model for a particular business domain. You can present all domain-specific business entities and the relations between them in a data model as a hierarchical structure.

To become familiar with the details of this scenario, review the article, Design domain specific data model.

Translating data model content

You can translate data model content (labels and descriptions) into other languages that the applications support. Translate data model content for the following reasons:

- At design time, to make the content more intelligible for format designers who speak other languages and who use the data model for data mapping of format components.
- At run time, to make the content more user-friendly by presenting prompts and help for run-time parameters, and configured validation messages (errors and warnings), in the language that the currently signed-in user prefers.

Configuring data model mappings for outgoing documents

ER provides a model mapping designer that users can use to map data models they design to specific application data sources. Based on the mapping, the data imports at run time from selected data sources into the data model. The data model is then used as an abstract data source of ER formats that generate outgoing electronic documents.

To become familiar with the details of this scenario, review the articles, Define ER model mappings and select data sources for them and Map data model to selected data sources.

Configuring data model mappings for incoming documents

ER provides a model mapping designer that users can use to map data models they design to specific destinations. For example, you can map data models to the updatable data components (tables, data entities, and views). Based on the mapping, the data updates at run time by using the data from the data model. As abstract storage of the ER format, the data model fills with data that imports from an incoming electronic document.

Storing a designed model component as a model configuration

ER can store a designed data model, together with associated data mappings, as a model configuration of the current instance.

To become familiar with the details of this scenario, review the article Map data model to selected data sources.

Building a format that uses a data model as a base

ER supports a format designer that you can use to build the format of an electronic document for a selected business domain by selecting the model component as a base. The same ER format designer lets you map a format that you create to a selected domain's data model mapping as a data source.

Building a configuration to generate electronic documents in OPENXML worksheet format

Use the ER format designer to build an electronic document in OPENXML worksheet format.

To become familiar with the details of this scenario, review the Design a configuration for generating reports in OPENXML format. As part of importing a template, use the Template of Payment Report (SampleVendPaymWsReport.xlsx) Excel file as a template.

Building a configuration to generate electronic documents in a Word document format

Use the ER format designer to build an electronic document in a Word document format. This format reuses the existing ER configuration that was originally designed to generate the report output in OPENXML format.

To become familiar with the details of this scenario, review the article, Design a new ER configuration to generate reports in Microsoft WORD format. As part of the task guide step for importing a template, use the following Word files as templates for the ER format:

- Template of Payment Report (SampleVendPaymDocReport.docx)
- Bounded template of Payment Report (SampleVendPaymDocReportBounded.docx)

Building a configuration to import data from incoming electronic documents

Use the ER format designer to describe an electronic document that you plan to import data from in either XML or text format. The designed format is used to parse an incoming document. Use the ER format mapping designer to define the binding of the elements of the designed format to the data model.

To become familiar with the details of this scenario, see [Create required configurations to import data from an external file](Create required configurations to import data from an external file.md). Use the following files to complete the steps in this article:

- ER data model configuration (1099model.xml)
- ER format configuration (1099format.xml)
- Sample of the incoming document in XML format (1099entries.xml)
- Sample of the workbook to manage data of incoming document (1099entries.xlsx)

Storing a designed format component in a format configuration

Electronic Reporting (ER) can store a designed format together with the configured data mappings as a format configuration of the current instance. The preceding illustration shows

an example of this type of format configuration (**BACS (UK)**, which is a child of the **Payment model** configuration).

Configuring Finance to start using a created format internally

You can configure the application to start using a created format to generate electronic reports. Define the reference to the created format configuration in the settings of a specific domain. For example, to start using an ER format configuration for electronic vendor payments in BACS format, reference the format configuration in specific methods of payment.

For more information, see Generate electronic documents for payments using a format configuration.

Handling ER components

Building a format by selecting another format as a base (customization)

ER lets you create (derive) a new component from the current version of a component (base) that you imported from a repository. For example, you might want to derive a new format to implement some special requirements for an electronic document, such as an additional field or an extensive description, to support a customization scenario. For more information, see ER Upgrade your format by adopting a new, base version of that format.

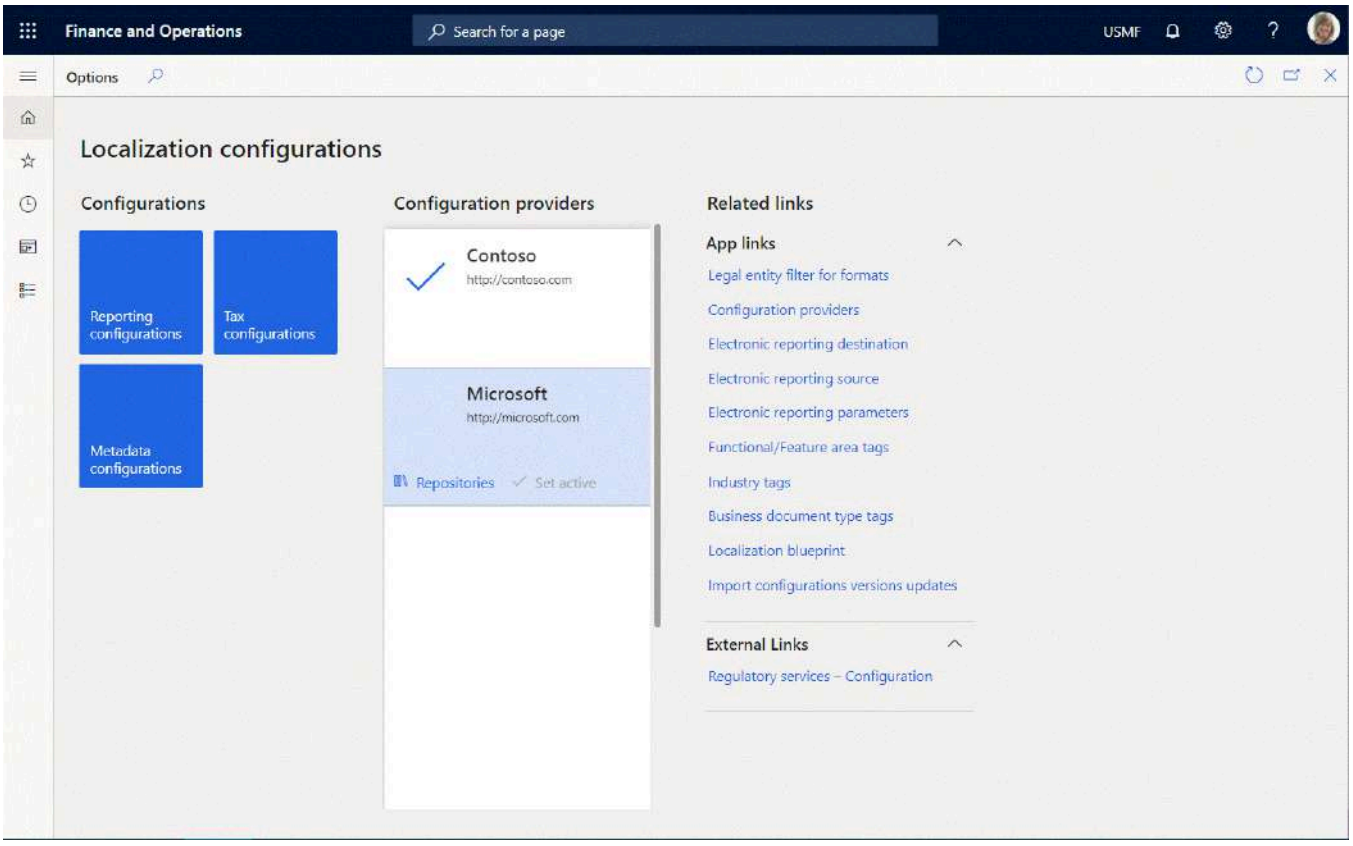
Upgrading a format by selecting a new version of base format (rebase)

ER automatically adopts changes from the latest version of the base component in the current draft version of the derived component. This process is known as *rebasing*. For example, a new regulatory change that is introduced in the latest version of the format that you imported from Dataverse can be automatically merged into the customized version of this format of the electronic document. Any changes that can't be merged automatically are considered conflicts. The designer tool presents these conflicts for manual resolution. For more information, see ER Upgrade your format by adopting a new, base version of that format.

List of ER configurations that are released in Finance

The list of ER configurations for Finance is constantly updated. Open the **Global repository** to review the list of ER configurations that are currently supported. On the **Discontinuation**

details FastTab, you can review the information about configurations that are discontinued or that are no longer maintained.



Additional resources

- Electronic reporting components
- Create Electronic reporting (ER) configurations
- Manage the Electronic reporting (ER) configuration lifecycle

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Last updated on 04/13/2026

Electronic reporting components

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>
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Electronic reporting (ER) supports the following types of components:

- Data model
- Model mapping
- Format
- Metadata

Data model component

A data model component is an abstract representation of a data structure. It describes a specific business domain area in enough detail to satisfy the reporting requirements for that domain. A data model component consists of the following parts:

- **Data model** – A set of domain-specific business entities and a hierarchically structured definition of relations between those entities.
- **Model mapping** – Selected application data sources are linked to individual elements of a data model that specifies, at runtime, the data flow and rules for entering business data in a data model component.

The business entity of a data model is represented as a container, or record. Business entity properties are represented as data items, or fields. Each data item has a unique name, label, description, and value. The value of each data item can be designed so that it's recognized as a string, integer, real, date, enumeration (enum), or Boolean value. Additionally, the data item can be another record or record list.

A single data model component can contain several hierarchies of domain-specific business entities. It can also contain model mappings that support a report-specific data flow at runtime. The hierarchies are differentiated by a single record that is selected as a root for model

mapping. For example, the data model of the payment domain area might support the following mappings:

- Company > Vendor > Payment transactions of the AP domain
- Customer > Company > Payment transactions of the AR domain

Business entities, such as company and payment transactions, are designed only one time. Different mappings can reuse them as required.

Model mapping component

Model mapping links application data sources to individual elements of a data model. It specifies the data flow and rules for entering business data in a data model component at runtime.

A model mapping that supports outgoing electronic documents has the following capabilities:

- It uses different data types as data sources for a data model. These data types include tables, data entities, methods, and enums.
- It supports user input parameters that you can define as data sources for a data model when some data must be specified at runtime.
- It supports data transformation into required groups. You can also filter, sort, and sum data, and append logical calculated fields that are designed through formulas that resemble Microsoft Excel formulas. For more information, see Formula designer in Electronic reporting (ER).

A model mapping that supports incoming electronic documents has the following capabilities:

- It uses different updatable data elements as targets. These data elements include tables, data entities, and views. The data can be updated by incoming data from electronic documents. You can use multiple targets in a single model mapping.
- It supports user input parameters that you can define as data sources for a data model when some data must be specified at runtime.

You design a data model component for each business domain. Use it as a unified data source for reporting. The unified data source isolates reporting from the physical implementation of data sources. The component represents domain-specific business concepts and functionalities in a form that makes a reporting format's initial design and further maintenance more efficient.

Format component

Format components for outgoing electronic documents

A format component is the scheme of the reporting output that the system generates at runtime. A scheme consists of the following elements:

- A format that defines the structure and content of the outgoing electronic document.
- Data sources, as a set of user input parameters and a domain-specific data model that uses a selected model mapping.
- A format mapping, as a set of bindings of format data sources that have individual elements of a format that specify, at runtime, the data flow and rules for generating format output.
- A format validation, as a set of configurable rules that control report generation at runtime, depending on the running context. For example, there might be a rule that stops output generation of a vendor's payments and throws an exception when specific attributes of the selected vendor are missing, such as the bank account number.

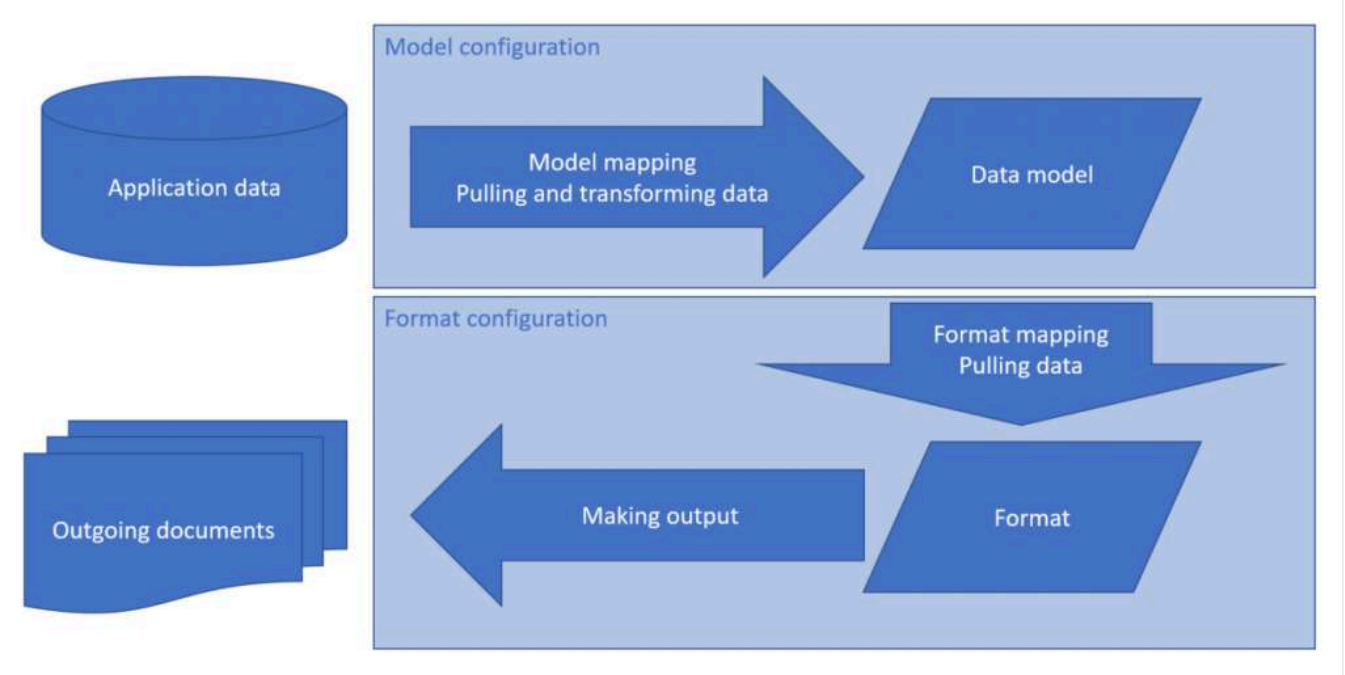
A format component supports the following functions:

- Creating a reporting output as individual files in various formats, such as text, XML, Microsoft Word document, or worksheet.
- Creating multiple files separately and encapsulating them into zip files.

A format component lets you attach specific files that you can use in the reporting output:

- Excel workbooks that contain a worksheet that you can use as a template for output in the OPENXML worksheet format.
- Word files that contain a document that you can use as a template for output in the Microsoft Word document format.
- Other files that you can incorporate into the format's output as predefined files.

The following illustration shows how the data flows for these formats.



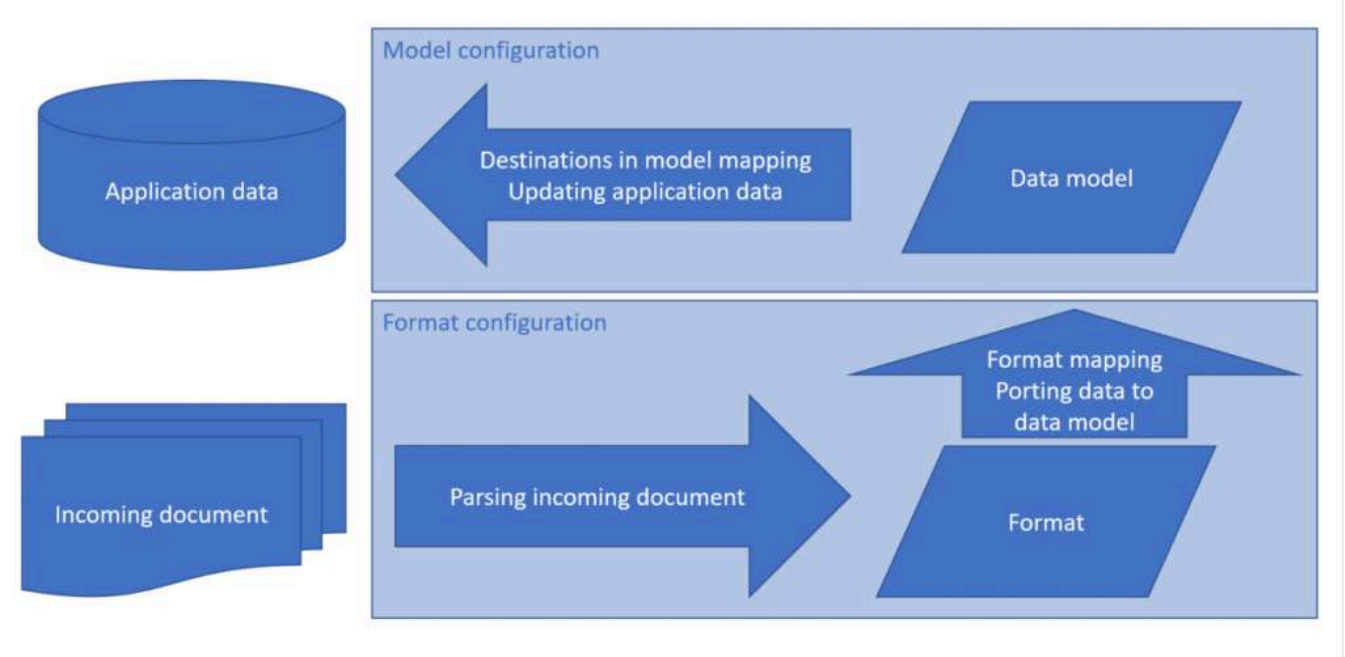
To run a single ER format configuration and generate an outgoing electronic document, you must identify the mapping of the format configuration.

Format components for incoming electronic documents

A format component is the scheme of the incoming document that the system imports at run time. A scheme consists of the following elements:

- A format that defines the structure and content of the incoming electronic document. The document contains data that the system imports at run time. Use a format component to parse an incoming document in various formats, such as text and XML.
- A format mapping that binds individual format elements to elements of a domain-specific data model. At run time, the elements in the data model specify the data flow and the rules for importing data from an incoming document, and then store the data in a data model.
- A format validation, as a set of configurable rules that control data import at run time, depending on the running context. For example, there might be a rule that stops data import of a bank statement that has a vendor's payments and throws an exception when a specific vendor's attributes are missing, such as the vendor identification code.

The following illustration shows how the data flows for these formats.



To run a single ER format configuration to import data from an incoming electronic document, you must identify the desired mapping of a format configuration, and also the integration point of a model mapping. You can use the same model mapping and destinations together with different formats for different types of incoming documents.

Component versioning

ER components support versioning. Use the following workflow to manage changes in ER components:

1. Mark the originally created version as a **Draft** version. You can edit this version, and it's available for test runs.
2. Convert the **Draft** version to a **Completed** version. Use this version in local reporting processes.
3. Convert the **Completed** version to a **Shared** version. Publish this version in Microsoft Dynamics Lifecycle Services and use it in global reporting processes.
4. Convert the **Shared** version to a **Discontinued** version. You can delete this version.

Other data interchange can access versions with **Completed** or **Shared** status. You can perform the following actions on a component with these statuses:

- Serialize the component in XML format and export it as an XML file.
- Reserialize the component from an XML file and import it into the application as a new version of an ER component.

For more information, see Import a new data model configuration and Export completed version of a derived format.

Draft versions at runtime

In your personal user parameters for the ER framework, you can enable the option that lets you specify whether the runtime should use the draft version of an ER configuration. For information about how to make the **Run Draft** option available for your ER configurations, see Mark a custom format as runnable.

> ⓘ **Note**
>
> The ER user parameters are company-specific and user-specific.

Draft format versions at runtime

By default, when you run an ER solution, the runtime ignores the draft versions of its format components. Instead, it uses only the relevant version that has a status other than **Draft**. Sometimes, you might want to force ER to use the draft version of your ER format configuration at runtime. For example, after you introduce necessary changes in your draft version, you can use that draft version to do the test run. In this way, you can validate the correctness of your changes. To start using the draft format version, set the **Run Draft** option of the relevant ER configuration to **Yes**.

Draft model mapping versions at runtime

By default, when you run an ER solution, it uses the draft versions of its model mapping components. Sometimes, you might want to force ER to ignore the draft version of your ER model mapping configuration at runtime. In **version 10.0.29 and later**, you can enable the **Always take into consideration the "Run draft" option for ER model mappings** feature to control the model mapping version that is used at runtime. When you enable this feature, the following behavior occurs:

- When you set the **Run Draft** option to **No** for a model mapping configuration, the highest non-draft version of that configuration is used at runtime. An exception is thrown if the configuration isn't available in the current Finance instance.
- When you set the **Run Draft** option to **Yes** for a model mapping configuration, the draft version of that configuration is used at runtime.

Component date effectivity

ER format component versions are date-effective. You can set the "effective from" date for an ER format component to specify the date when the component becomes effective for reporting processes. The application session date is used to define whether a component is valid for execution. If more than one version is valid for a specific date, the latest version is used for reporting processes.

Component access

Access to ER format and model mapping components at runtime depends on the setting for the International Organization for Standardization (ISO) country/region code. If this setting is blank for a selected version of a format or model mapping configuration, you can access a format or model mapping component from any company at runtime. If the setting contains ISO country/region codes, a format or model mapping component is available only from companies that have a primary address defined for one of a format component's ISO country/region codes.

Different versions of a format or a model mapping component can have different settings for ISO country/region codes.

For more information, see Configure country/region context dependent ER model mappings.

Last updated on 04/13/2026

Create Electronic reporting (ER) configurations

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>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

As part of the requirements for Microsoft Dynamics Lifecycle Services solutions for localization and translation, localization ISV solution providers must implement features specific to a country/region or solutions by using the Electronic reporting tool. This article provides background information that helps you start using Electronic reporting for creating configurations. This article isn't meant to replace any available and upcoming Electronic reporting documentation, but is intended as a supplemental view from the perspective of localization requirements.

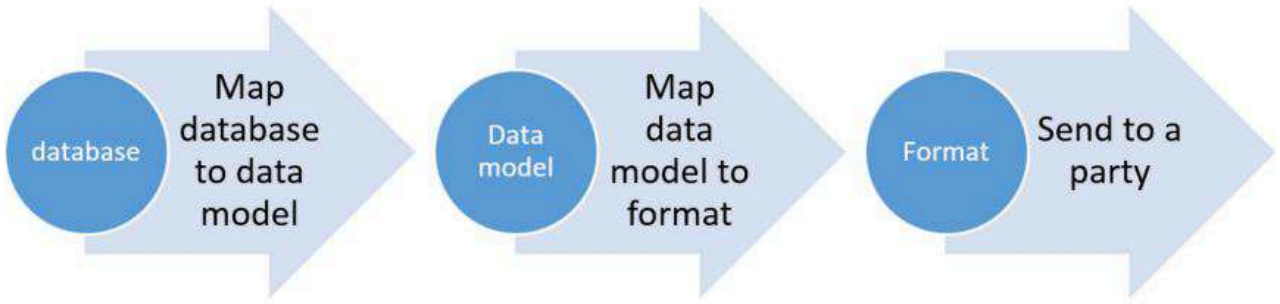
Electronic reporting (ER) is a configurable tool that helps you create and maintain regulatory electronic reporting and payments, based on the following three concepts:

- Configuration instead of coding
- One configuration for multiple Dynamics 365 Finance releases
- Easy or automatic upgrade

By using Lifecycle Services, ER provides one common way for Microsoft and partners to distribute electronic document configurations to other partners and customers. ER also makes it easier for partners and customers to customize, upgrade, and distribute electronic document formats for their specific business requirements.

You can use ER to set up data models that are domain-specific and independent of the database as data sources for document formats. You can configure formats based on these domain-specific data models by using simple visual tools that are similar to Excel. Data models and formats support versioning, and formats can be date-effective.

Main data flow



Data model configuration creation

Reuse and customize data models that Microsoft releases whenever you can. Or, create a business domain area–specific data model that introduces the abstract model of required entities and their relations. By using this approach, you stay aligned with future updates that Microsoft releases. You can also reuse your model for the design and maintenance of multiple domain–specific electronic documents that have different formats that are required in different scenarios or countries/regions.

Design the data model of the created model configuration

Design a data model to recognize and describe the required business entities and the relations between them in the selected domain. A data model consists of descriptors that express entities by using data containers (records). Use data items to express properties of entities. A record definition is an entity that contains fields (the data items). Each data item has a unique name, label, description, and value. You can desigh the value of each data item so that it's recognized as string, integer, real, date, enumerate type, and so on. Additionally, the value can be another record or record list. You can select a single record definition as a root of the data model. A root is the starting point of the entire model for data source mapping. In this case, the model is used as a data source that delivers data according to the single predefined data flow. If no record definition is selected as a root of the data model, the data model contains record definitions that can be assigned as a root at the format mapping stage. You can define the data flow of such a model as a data source in multiple ways, depending on the nature of the format. For example, you can design a single data model for the payments domain area. This data model can include data record definitions for the company as a legal entity, for vendors and customers, and also for payments. However, according to the nature of the

format, the data must be presented in the following way: payer > payee > payments. Therefore, a single data model can offer data according to the following alternative paths:

- Company > vendor > payment for the Accounts payable domain when you select the company record definition as a root.
- Customer > company > payment for the Accounts receivable domain when you select the customer record definition as a root.

Format configuration creation

Use the data model configuration created to hold abstract data for a new electronic format that you want to design. If you intend to consume a data model prepared earlier when you create a new format, make sure that you select the **Format based on data model** option for **Create configuration**. After you have a format configuration, you must define a format structure. You can create the structure manually or automatically by importing an example of an XML file or an Excel template. The data model of the parent configuration is automatically offered for format mapping, together with a proper root container. Nevertheless, a format might require that data be represented in a specific way. Therefore, you can use formula designer to define expressions as virtual data items (calculated fields) for your data containers.

> **Note**
>
> Although you can map format components directly to database and application artifacts, such as tables or data entities, enumerations, or classes and objects in ER, don't use this approach. It's likely that multiple formats are maintained in some business domain areas that use the same data sources. Whenever the structure of the database artifacts changes, you must also change the format mapping to those database artifacts. The cost of these changes multiplies by the number of maintained formats. Therefore, work through the data model as the abstract description of the domain-specific data structure. Use the direct binding of format elements to database components only for simplification and for coverage for specific customizations. For example, use direct binding to refer to custom tables when those references are required in a limited number of maintained formats.

> **Tip**

If you still have to add ER data sources to configure a format component to access database and application artifacts at runtime, select **Show details** on the Action pane of the **Format designer** page.

Version control

One of the principles behind the design of Electronic reporting is that you can easily distribute a data model and formats together with an enhanced maintenance model for their customizations. All the configurations are versioned, and you can "clone" an existing customization to derive a new configuration for localization or customization implementation. For example, you represent a company named Proseware, Inc. You subscribe to the service of a company named Litware Inc., which provides you with the Intrastat returns configuration and supports all legal requirements in it. You receive specific configurations from Litware Inc., together with data model and formats, and deploy them. Your company works in a district where, in addition to the federal requirements, you must support the following regional requirements:

- As part of Intrastat transactions details, your XML file must show the statistical procedure code that isn't required anywhere else.
- You must limit the length of the company name that is presented in the Intrastat returns header block to 200 characters.

To support these requirements and comply with local district authorities, you must implement this localization as a localized configuration. However, you must keep the link with the origin configuration, so that you can adopt any future changes that are introduced at the federal level as new versions of the origin configuration. Therefore, you import the Litware Inc. origin configuration from Lifecycle Services, derive it as a new localized configuration, introduce the required changes, complete this work by introducing a first version of the localized format, and start to use it internally. Whenever Litware Inc. offers you a new version of the origin configuration, you import it from Lifecycle Services, rebase your localized configuration to this version, adopt changes to support new federal requirements, complete this work by introducing a next version of the localized format, and continue to use it internally.

Note

The draft version of any configuration must be "completed" before it can become available locally for further action, such as the following:

- Make it available so that it can be referenced as a data source from a new format.
- Enable configuration exchange between companies or instances via configuration import/export, and so on.

Electronic reporting domain coverage

You can use several out-of-the-box configurations to meet electronic reporting requirements for specific countries/regions. The following list shows some examples of format configurations that are grouped into business domains. To get a complete, up-to-date list of available and supported configurations, open a configuration repository setup to show the configurations that are available for import from either resources or an Lifecycle Services Assets library.

- Audit file
  - FEC
  - GDPdU...
- Payments (ISO20022)
  - SEPA CT
  - SEPA DD
  - JBA
  - BACS...
- Statistical reports
  - EU Intrastat...
- Tax reports
  - CIS
  - BAS
  - ELSTER
  - EU Sales list...
- Customer e-Invoice
  - OIOUBL...

Your solution uptake

You can choose how to move your electronic reporting functionality into ER. However, consider the following high-level steps when you plan that move.

1. Review the electronic reporting functionality that your solution currently provides.

2. Identify domain areas that your solution covers, such as Payments and E-Invoices.
3. Review the configurations that Microsoft provides. It's likely that you find a configuration that you can use as a base. For example, if your solution customizes the SEPA CT payment format, extend the SEPA CT configuration.
4. Create new configurations that are based on either an existing model or format, or a new model or format.
5. Define input parameters that users must select when they run the report, and validations for the content of the report.
6. Define mappings with the model by using arithmetic, string, date, or other available Excel-like functions.
7. Define labels and translation to different languages, where applicable.
8. Define templates that include named ranges, and set links from the configuration for the Excel report, if applicable.

Terminology

Expand table

Term	Definition
ER	Electronic reporting is an engine that simplifies the creation of electronic reports for information interchange with governments, banks, and other parties. Currently, Electronic reporting supports text, XML, and OpenXML spreadsheet formats, and provides an extension interface to support more formats.
Transformation	If you have a typical action that must be done on the source of data before it's sent as output to a format, you can introduce a transformation and attach it to format components. A transformation is an ER formula that takes one value as a parameter and returns another value. For example, you have many format fields that contain spaces, and the spaces should be replaced by spaces when the fields are exported. In this case, you can create a transformation that takes a string argument and uses the REPLACE function to do the job. You can then create string components and associate them with that transformation.
Data model	A data model provides a structure for data. This structure is used to abstractly describe certain business domain areas at sufficient detail to satisfy the reporting requirements in this domain.
Configuration	A container for either a data model or a format, together with its mappings to data sources, that can be maintained and executed, and that supports versioning. The configuration is the entity that's imported or exported to organize electronic document format exchange between finance and operations instances.
Derive action	An operation that uses an existing configuration as a basis to create a new configuration.

Term	Definition
Rebase action	An operation that updates a derived configuration with changes that were introduced in a new version of the base configuration. The version number is selected at the rebase initialization stage.
Update conflict	A conflict discovered during the rebase action, where the new base version contains adjustments of a format/mapping element (name, property, and so on) that's also adjusted in the derived version.
Relocation conflict	A conflict discovered during the rebase action, where the new base version contains a new position (parent element) of a format element (name, property, and so on) that's also relocated to a different position in the derived version.
Duplication conflict	A conflict discovered during the rebase action, where the new base version introduced a new format element that is the same as an element (it has the same name and child components) that's also entered in the derived version.

Additional resources

Electronic reporting (ER) overview

Manage the Electronic reporting (ER) configuration lifecycle

Last updated on 04/13/2026

Define the dependency of ER configurations on other components

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete these steps, you must first complete the steps in the task guide, **ER Manage model mapping configurations**, and you must have access to Microsoft Dynamics Lifecycle Services.

This procedure shows how to design an Electronic reporting (ER) configuration and specify its dependency on other software components. By specifying these dependencies, you help guarantee that the configuration is correctly downloaded to a specific version of finance and operations. In this example, you create required ER configurations for the sample company Litware, Inc.

This procedure is intended for users who have the **System administrator** or **Electronic reporting developer** role assigned to them. You can perform the steps in any company, because ER configurations are shared among companies.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
    - Make sure that the configurations tree contains the **Sample data model** configuration and subordinate items. Otherwise, complete the steps in the task guide, **ER Manage model mapping configurations**, and then start this guide again.

Define the dependency of ER configurations on other components

1. In the tree, expand **Sample data model**.
2. In the tree, select **Sample data model\Sample mapping**.

- You selected the draft version of the **Sample mapping** model mapping configuration. You can now define its dependency on other software components. This step is a prerequisite for controlling the download of this configuration version from an ER repository and any further use of this version.

3. Expand the Prerequisites section.

   - The **Implementations** prerequisites group is added automatically at this stage. This group contains the prerequisite component that refers to the data model configuration and has the Implementation flag turned on. This flag indicates that the **Sample mapping** mapping configuration is considered the implementation of the **Sample data model** data model. This component forces ER to download the **Sample mapping** mapping configuration from an ER repository whenever the **Sample data model** model configuration is downloaded.

4. Select **Edit**.

   - You can specify a single dependency of the current version of a configuration on a software component by using the definition of the component's type, and either the component version or a range of component versions.
   - You can group desired dependencies together. When you select the **All of** grouping type, the dependency condition of this group is considered satisfied when each dependency condition from this group and subordinate group is satisfied. When you select the **One of** grouping type, the dependency condition of this group is considered satisfied when at least one dependency condition from this group is satisfied.

5. Select **New**.
6. Select **Product prerequisite component**.
7. Select **Microsoft Dynamics 365 for Operations (1611)**.
8. In the **Version** field, type `[7.1.1541.3036,8)`.

   - [7.1.1541.3036,8)
   - Dependencies that you enter are evaluated when this configuration is downloaded from any ER repository. This configuration version is downloaded from the ER repository when version 1 of the **Sample data model** configuration is either already in place or downloaded in advance. If it's downloaded in advance, it must be completed in finance and operations version 7.1.1541.3036 or later, but must not exceed major version 8.

9. Select **Save**.

10. Close the page.
11. Select **Change status**.
12. Select **Complete**.
13. Select **OK**.
14. In the tree, select **Sample data model\Sample mapping (alternative)**.
15. Select **Edit**.
16. Select **New**.
17. Select **Product prerequisite component**.
18. Select **Microsoft Dynamics AX 7.0 RTW**.
19. In the **Version** field, type `[7.0.1265.3015,7.1)`.

    - [7.0.1265.3015,7.1)
    - Dependencies are evaluated when the configuration is downloaded from any ER repository. This configuration version is downloaded from the ER repository when version 1 of the **Sample data model** configuration is either already in place or downloaded in advance. If it's downloaded in advance, it must be completed in Microsoft Dynamics 365 Finance, Enterprise edition, the version of which must be 7.0.1265.3015 or later, but must not exceed minor version 1.

20. Select **Save**.
21. Close the page.
22. Select **Change status**.
23. Select **Complete**.
24. Select **OK**.

Configure the ER repository

1. Close the page.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

    - Open the list of ER repositories for the current ER provider, Litware, Inc.

3. In the list, select the row.
4. Select **Repositories**.
5. Select **Show filters**.
6. Enter a filter value of **LCS** in the **Type name** field by using the **contains** filter operator.

    - If the Lifecycle Services repository is already registered for the current ER provider, you can skip the remaining steps in this sub-task. If the Lifecycle Services repository

isn't already registered, complete the remaining steps.

7. Select **Add** to open the drop dialog.
8. In the **Configuration repository type** field, enter **LCS**.
9. Select **Create repository**.
10. In the **Project** field, enter or select a value.

    - Select the desired Lifecycle Services project from the lookup of the **Project** field.

11. Select **OK**.
12. Close the page.

Upload configurations to Lifecycle Services

1. Select **Reporting configurations**.
2. In the tree, select **Sample data model**.
3. Select the completed version of this configuration.
4. Select **Change status**.
5. Select **Share**.
6. Select **OK**.

    - Version 1 of this model configuration is uploaded to Lifecycle Services by using the Lifecycle Services project for the ER repository that you previously configured.

7. In the tree, expand **Sample data model**.
8. In the tree, select **Sample data model\Sample mapping**.
9. Select the completed version of this configuration.
10. Select **Change status**.
11. Select **Share**.
12. Select **OK**.

    - Version 1.1 of this model mapping configuration is uploaded to Lifecycle Services by using the Lifecycle Services project for the ER repository that you previously configured.

13. In the tree, select **Sample data model\Sample mapping (alternative)**.
14. Select the completed version of this configuration.
15. Select **Change status**.
16. Select **Share**.
17. Select **OK**.

- Version 1.1 of this model mapping configuration is uploaded to Lifecycle Services by using the Lifecycle Services project for the ER repository that you previously configured.

Evaluate ER configuration dependencies

You delete created configurations from the system and download them back from the Lifecycle Services repository.

1. In the tree, select **Sample data model\Sample mapping**.
2. Select **Delete**.
3. Select **Yes**.
4. In the tree, select **Sample data model\Sample mapping (alternative)**.
5. Select **Delete**.
6. Select **Yes**.
7. In the tree, select **Sample data model\Sample format**.
8. Select **Delete**.
9. Select **Yes**.
10. In the tree, select **Sample data model**.
11. Select **Delete**.
12. Select **Yes**.
13. Close the page.

    - Open the list of ER repositories for the current ER provider, Litware, Inc.

14. Select **Repositories**.
15. Select **Show filters**.
16. Enter a filter value of **LCS** in the **Type name** field by using the **contains** filter operator.
17. Select **Open**.
18. In the tree, select **Sample data model**.

    - You can view an evaluation of whether prerequisite conditions are satisfied for each version of the ER configurations for the current repository. To view this evaluation, select **Check prerequisites**.

19. Select **Check prerequisites**.
20. Select **Import**.
21. Select **Yes**.
22. Close the page.

23. Close the page.
24. Close the page.
25. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
26. In the tree, expand **Sample data model**.

    - The model **Sample mapping** mapping configuration is downloaded together with the selected data model configuration. You download the two files together because **Sample mapping** is defined as implementing the selected data model, and because it's applicable for the application. The **Sample mapping (alternative)** configuration isn't downloaded because the condition for the required application version isn't satisfied.
    - If you sign in to finance and operations, register the same provider, access the same Lifecycle Services project, and download the same data model configuration, the **Sample mapping (alternative)** configuration downloads, whereas the **Sample mapping** configuration is skipped.

Additional resources

Manage the Electronic reporting (ER) configuration lifecycle

Last updated on 04/13/2026

ER design domain specific data model

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can create a new Electronic reporting (ER) configuration that contains a data model for electronic payment documents. You use this data model as a data source when you create the format of the payment documents.

In this example, you create a configuration for sample company, Litware, Inc. You can perform these steps in any company because ER configurations are shared among companies. To complete these steps, you must first complete the steps in the "Create a configuration provider and mark it as active" procedure.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

   Select the configuration provider for sample company, Litware, Inc. If you don't see this configuration provider, you must first complete the steps in the "Create a configuration provider and mark it as active" procedure.

2. Select **Reporting configurations**.

   You create a configuration that contains a data model for electronic payment documents. You use this data model as a data source when you create the format for the payment documents.

Create a new data model configuration

1. Select **Create configuration** to open the dialog.

2. In the **Name** field, type *Payments (simplified model)*.

3. In the **Description** field, type *Payment model configuration*.

The active configuration provider is automatically entered here. This provider can maintain this configuration. Other providers can use this configuration, but can't maintain it.

4. Select **Create configuration** to complete the configuration creation task.

Create a data model

Create a new data model for the selected configuration. This configuration version has a status of Draft.

1. Select **Designer**.

Define the structure of a party participating in a payment process

1. Select **New** to open the drop dialog.
2. In the **Name** field, type `Party`.
3. Select **Add**.
4. Select **New** to open the drop dialog.
5. In the **Name** field, type `Name`.
6. In the **Item type** field, select `String`.
7. Select **Add**.
8. In the **Find** field, type `Party`.
9. Select **Find previous**.

Define the bank structure for this model

1. Select **New** to open the drop dialog.
2. In the **Name** field, type `Agent`.
3. In the **Item type** field, select `Record`.
4. Select **Add**.
5. In the **Description** field, enter `Financial institution (for instance, a bank) servicing an account for the party (debtor/creditor).`.

   Financial institution (for instance, a bank) servicing an account for the party (debtor/creditor).

6. Select **New** to open the drop dialog.
7. In the **Name** field, type `Name`.
8. In the **Item type** field, select `String`.
9. Select **Add**.
10. Select **New** to open the drop dialog.
11. In the **Name** field, type `SWIFT`.
12. Select **Add**.
13. In the **Description** field, enter `Bank identification code`.
14. Select **New** to open the drop dialog.
15. In the **Name** field, type `RoutingNumber`.
16. Select **Add**.
17. In the **Description** field, enter `Routing number`.
18. Select **Find previous**.

Define the bank account structure for this model

1. Select **New** to open the drop dialog.
2. In the **Name** field, type `Account`.
3. In the **Item type** field, select `Record`.
4. Select **Add**.
5. In the **Description** field, enter `Identification of an account of a party in a financial institution (for instance, a bank).`.

   Identification of an account of a party in a financial institution (for instance, a bank).
6. Select **New** to open the drop dialog.
7. In the **Name** field, type `Currency`.
8. In the **Item type** field, select `String`.

9. Select **Add**.
10. In the **Description** field, enter `Currency code`.
11. Select **New** to open the drop dialog.
12. In the **Name** field, type `Number`.
13. Select **Add**.
14. Select **New** to open the drop dialog.
15. In the **Name** field, type `IBAN`.
16. Select **Add**.
17. In the **Description** field, enter `International bank account number`.

Define the payment message structure for credit transfer payment type

1. Select **New** to open the drop dialog.
2. In **New node as a field**, enter *Model root*.
3. In **Name**, type *CustomerCreditTransferInitiation*.
4. Select **Add**.
5. In **Find**, type *CustomerCreditTransferInitiation*.
6. Select **Find previous**.
7. Select **New** to open the drop dialog.
8. In **Name**, type *MessageIdentification*.
9. Select **Add**.
10. In **Description**, enter *The point-to-point reference assigned by the instructing party (and sent to the next party) to identify a message.*.

    The point-to-point reference assigned by the instructing party (and sent to the next party) to identify a message.

11. Select **New** to open the drop dialog.
12. In **Name**, type *ProcessingDateTime*.
13. In **Item type**, select *DateTime*.
14. Select **Add**.
15. In **Description**, enter *Date and time at which the payment message was created.*.
16. Select **New** to open the drop dialog.

    Define the payment transaction structure for this model.

17. In **Name**, type *Payments*.
18. In **Item type**, select *Record list*.
19. Select **Add**.
20. In **Description**, enter *Payment lines of the current message*.
21. Select **New** to open the drop dialog.
22. In **Name**, type *Creditor*.
23. In the **Item type** field, select `Record`.
24. Select **Add**.
25. In **Description**, enter *Party to which an amount of money is due.*.
26. Select **Switch item reference**.
27. In the **Find** field, type `Party`.
28. Select **Find next**.
29. Select **OK**.
30. In **Find**, type *Payments*.
31. Select **Find next**.
32. Select **New** to open the drop dialog.
33. In **Name**, type *Debtor*.

34. Select **Add**.

35. In **Description**, enter *Party that owes an amount of money to the (ultimate) creditor.*.

36. Select **Switch item reference**.

37. In the **Find** field, type `Party`.

38. Select **Find next**.

39. Select **OK**.

40. Select **Find next**.

41. Select **New** to open the drop dialog.

42. In **Name**, type *Description*.

43. In the **Item type** field, select `String`.

44. Select **Add**.

45. Select **New** to open the drop dialog.

46. In the **Name** field, type `Currency`.

47. Select **Add**.

48. In the **Description** field, enter `Currency code`.

49. Select **New** to open the drop dialog.

50. In **Name**, type *TransactionDate*.

51. In **Item type**, select *Date*.

52. Select **Add**.

53. In the **Description** field, enter *Transaction date*.

54. Select **New** to open the drop dialog.

55. In the **Name** field, type *InstructedAmount*.

56. In the **Item type** field, select *Real*.

57. Select **Add**.

58. In the **Description** field, enter *The amount of money to be moved between the debtor and creditor, before deduction of charges. The amount should be expressed in the currency as ordered by the initiating party.*

    The amount of money to be moved between the debtor and creditor, before deduction of charges. The amount should be expressed in the currency as ordered by the initiating party.

59. Select **New** to open the drop dialog.

60. In the **Name** field, type *End2EndID*.

61. In the **Item type** field, select `String`.

62. Select **Add**.

63. In the **Description** field, enter *The unique identification assigned by the initiating party. This identification is passed on, unchanged, throughout the entire end-to-end chain.*

64. In the **Name** field, type *PaymentModel*.

    The **PaymentModel** name aligns with predefined interfaces of payment forms.

65. Select **Save**.

66. Close the page.

Last updated on 04/13/2026

Prepare application metadata to use in RCS

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can create a new Electronic reporting (ER) configuration that contains application metadata for designing ER model mapping configurations in Regulatory configuration service (RCS). Use this configuration to design a sample ER model mapping configuration to access foreign trade transactions. In this example, you create a configuration for sample company, Litware, Inc. You can perform these steps in any company. To complete these steps, you must first complete the steps in the article, Create configuration providers and mark them as active.

Prerequisites

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the procedure Create configuration providers and mark them as active.
3. Select **Metadata configurations**.
4. Assume that RCS is used to design an ER solution for a Finance and Operation application that generates electronic documents that contain information from the foreign trade business domain. To specify the mapping between the ER data model and sources of required data, RCS needs access to metadata of the Finance and Operation application. Therefore, as part of designing the ER solution, you configure a new ER metadata configuration containing all metadata that is currently required for generating ER reports for the selected business domain.

Add metadata configuration

1. Select **Create configuration** to open the form.

2. In the **Name** field, enter *Foreign trade metadata*.
3. Select **Create configuration**.
4. Select **Designer**.
5. Select **Add**.

> **Note**
>
> You can select all metadata for the entire application or selected models or selected modules. When you choose this option, the following metadata is automatically added: tables of records, enumerations, and extended data types. If you need other types of metadata, add them manually.

You can add foreign trade transaction-related metadata by selecting metadata items manually.

1. Select **Add data source**.
2. Select **Table records**.
3. Use the Quick Filter to filter on the **Name** field with a value of *Intrastat*.
4. Select the **Intrastat** table record.
5. Select **OK**.

You added metadata information about the Intrastat table of records.

1. In the tree, expand **Table records Intrastat** > **Relations**.
2. In the tree, select **Table records Intrastat** > **Relations** > **IntrastatCommodity (Table records EcoResCategory)**.
3. Select **Add metadata**.

> **Note**
>
> You must add metadata about required relations for selected table of records manually.

1. Select **Add data source**.
2. Select **Enumeration**.
3. Use the Quick Filter to filter on the **Name** field with a value of *IntrastatDirection*.
4. Select the **IntrastatDirection enumeration** record.
5. Select **OK**.
6. Select **Save**.
7. Close the page.

Complete the draft version of metadata configuration

1. Select **Change status**.
2. Select **Complete**.
3. Select **OK**.
4. Select version **1**.

Export the completed version of metadata configuration from application as XML file

1. Select **Exchange**.
2. Select **Export as XML file**.
3. Select **OK**.

The created ER metadata configuration is saved as an XML file. You can import this file to RCS and use it as the source of information about metadata for the foreign trade business domain. Based on this information, you can specify the mapping between application metadata and ER data model.

Last updated on 04/13/2026

Access application metadata by using ER configuration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a Regulatory configuration service (RCS) user in the System Administrator or Electronic Reporting Developer role can design a new Electronic reporting (ER) model mapping by using the application metadata. You access application metadata by using an ER metadata configuration that contains a sample set of metadata to access foreign trade transactions. To complete these steps, in RCS you must first complete the steps in the article, Create configuration providers and mark them as active procedure. Then complete the steps in the article, Prepare application metadata to be used in RCS.

Prerequisites

1. Go to **All workspaces** > **Electronic reporting**.
2. Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the procedure Create configuration providers and mark them as active.

Import metadata configuration

1. Select **Metadata configurations**.
2. Import the ER metadata configuration that contains metadata that is configured to generate electronic documents for foreign trade business. This ER metadata configuration is exported as XML file while the steps in the Prepare application metadata to be used in RCS procedure are completed.
3. Select **Exchange**.
4. Select **Load from XML file**.
5. Select **Browse** and select the **Foreign trade metadata.xml** file.

6. Select **OK**.
7. Close the page.

Create data model configuration

1. Select **Reporting configurations**.
2. Select **Create configuration** to open the form.
3. In the **Name** field, enter *Foreign trade model*.
4. Select **Create configuration**.
5. Select **Designer**.
6. Select **New** to open the form.
7. In the **Name** field, enter *Root*.
8. Select **Add**.
9. Select **New** to open the form.
10. In the **Name** field, enter *Transaction*.
11. In the **Item type** field, select **Record list**.
12. Select **Add**.
13. Select **New** to open the form.
14. In the **Name** field, enter *Commodity code*.
15. In the **Item type** field, select **String**.
16. Select **Add**.
17. Select **New** to open the form.
18. In the **Name** field, enter *Invoiced amount*.
19. In the **Item type** field, select **Real**.
20. Select **Add**.
21. Select **New** to open the form.
22. In the **Name** field, enter *Date*.
23. In the **Item type** field, select **Date**.
24. Select **Add**.
25. Select **Root reference**.
26. Select **OK**.
27. Select **Save**.
28. Close the page.
29. Select **Change status**.
30. Select **Complete**.
31. Select **OK**.

Create model mapping configuration

1. Select **Create configuration** to open the form.
2. In **New**, enter *Model Mapping based on data model Foreign trade model*.
3. In **Name**, type *Foreign trade mapping*.
4. Select **Create configuration**.
5. Expand the **Prerequisites** section.
6. Select **Edit**.
7. Select **New**.
8. In the list, select the row.
9. In **Prerequisite component type**, select **Configuration**.
10. Select **Foreign trade metadata** configuration.
11. Select **Save**.
12. You add the reference to version 1 of the **Foreign trade metadata** configuration. Application metadata from this configuration is offered while you design this model mapping.
13. Close the page.
14. Select **Designer**.
15. Select **Designer**.
16. In the tree, select **Dynamics 365 for Operations\Table records**.
17. Select **Add root**.
18. In **Name**, type *Intrastat*.
19. Select **Intrastat** table records.
20. Select **OK**.

> ⓘ **Note**
>
> You see only two tables because you added only two tables into the set of metadata that you're currently using.

1. Select **Bind**.
2. In the tree, expand **Intrastat**.
3. In the tree, select **Intrastat\AmountMST**.
4. In the tree, expand **Transaction = Intrastat**.
5. In the tree, select **Transaction = Intrastat\Invoiced amount**.
6. Select **Bind**.
7. In the tree, select **Intrastat\TransDate**.

8. In the tree, select **Transaction = Intrastat\Date**.
9. Select **Bind**.
10. In the tree, expand **Intrastat>Relations**.
11. In the tree, expand **Intrastat>Relations\IntrastatCommodity**.
12. In the tree, select **Intrastat>Relations\IntrastatCommodity\Code**.
13. In the tree, select **Transaction = Intrastat\Commodity code**.
14. Select **Bind**.
15. Select **Validate**.

> ⓘ **Note**
>
> You successfully bind elements of data model with items of data sources that are described by using details of application metadata from the referred ER metadata configuration.

1. Select **Save**.
2. Close the page.
3. Close the page.
4. When needed, you can extend the existing set of metadata and then export the new completed version of ER metadata configuration. You can then import it to RCS, and update the prerequisites of the configured model mapping configuration referring to a new version of imported metadata configuration.

> ⓘ **Note**
>
> This way of getting information about application metadata is the only one available for locally deployed applications (when local business data (LBD), or on-premises, deployment model is used).

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Access application metadata by using connected applications

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a Regulatory configuration service (RCS) user in the System Administrator or Electronic Reporting Developer role can design a new Electronic reporting (ER) model mapping by using metadata in finance and operations. The RCS connected application accesses application metadata online. You configure a sample ER model mapping to access foreign trade transactions. To complete these steps, in RCS you must first complete the steps in the article, Create configuration providers and mark them as active. If you didn't complete the steps in the article, Access application metadata by using ER configuration, download the Electronic reporting examples and save the following ER configurations: Foreign trade metadata.xml, Foreign trade model.xml, and Foreign trade mapping.xml, and then complete the steps in the procedure.

Prerequisites

1. Go to **All workspaces** > **Electronic reporting**.
2. Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the procedure Create configuration providers and mark them as active.

Get required ER configurations

1. Select **Reporting configurations**.
2. If you already completed the steps in the Access application metadata by using ER configuration procedure, you already have all necessary ER configurations (foreign trade metadata, model, and mapping configurations) in the current RCS instance. You can skip all the remaining steps of this sub-task.

3. Select **Exchange**.
4. Select **Load from XML file**.
5. Select **Browse** and select the **Foreign trade metadata.xml** file.
6. Select **OK**.
7. Select **Exchange**.
8. Select **Load from XML file**.
9. Select **Browse** and select the **Foreign trade model.xml** file.
10. Select **OK**.
11. Select **Exchange**.
12. Select **Load from XML file**.
13. Select **Browse** and select the **Foreign trade mapping.xml** file.
14. Select **OK**.

Register a connected application

1. Close the page.
2. Close the page.
3. Go to **All workspaces** > **Electronic reporting**.
4. Select **Connected applications**.
5. Make sure that the configured application is Azure based and accessible for the current RCS user. The current RCS user must also have access to the selected application and be registered as a user of this application with a role that gives them privileges to access the application's metadata.
6. Select **New**.
7. In the **Name** field, type `MyConnectedApp`.
8. In the **Application** field, type `https:// mycompany.operations.dynamics.com`.
9. In the **Tenant** field, type `mycompany.onmicrosoft.com`.
10. Select **Save**.
11. When you check connection to configured application, on the **Connect to remote application** page select **Click here to connect to selected remote application** link.
12. Select **Check connection**.
13. Select **Close**.
14. When the connection validation succeeds, version and tenant details are updated for the configured application in the current grid.

Review existing model mapping configuration

1. Close the page.

2. Select **Reporting configurations**.
3. In the tree, expand **Foreign trade model**.
4. In the tree, select **Foreign trade model\Foreign trade mapping**.
5. Expand the **Prerequisites** section.

> ⓘ **Note**
>
> Currently, this mapping refers to the metadata configuration. Application metadata from this configuration is offered while you design this model mapping.

6. Select **Designer**.
7. Select **Designer**.
8. In the tree, select **Dynamics 365 for Operations\Table records**.
9. Select **Add root**.
10. In the **Table** field, enter or select a value.

> ⓘ **Note**
>
> Currently, this mapping refers to the metadata configuration. Application metadata from this configuration is offered while you design this model mapping.

11. Select **Cancel**.
12. Close the page.
13. Close the page.

Assign connected application to model mapping

1. Select **Edit**.
2. Select **MyConnectedApp** application.

> ⓘ **Note**

Currently, this mapping refers to the metadata of the selected connected application. When the same mapping refers to metadata configuration and connected application at the same time, the metadata of the connected application is used.

3. Select **Designer**.
4. Select **Designer**.
5. In the tree, select **Dynamics 365 for Operations\Table records**.
6. Select **Add root**.
7. In the **Table** field, enter or select a value.

   > ⓘ **Note**
   >
   > You see more than two application tables because this mapping uses all the metadata of the connected application that you assigned to it.

8. Select **Cancel**.
9. Select **Validate**.

   > ⓘ **Note**
   >
   > You successfully bound elements of data model with items of data sources that are described by using details of metadata of the connected application that you assigned to this mapping.

10. Close the page.
11. Close the page.

To evaluate this model mapping by using metadata of a different version application, register another connected application, assign it to this model mapping, and validate it against new metadata.

Last updated on 04/13/2026

Define ER model mappings and select data sources for them

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can select data sources for an Electronic reporting (ER) data model. Bind the data sources to individual components of the selected data model at design time. The data sources populate business data to that data model at runtime. In this example, you select data sources for an existing data model created for the sample company Litware, Inc.

Open the Electronic Reporting configurations tree

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.

Insert a new model mapping

1. In the tree, select **Payments (simplified model)**.
2. Select **Designer**.
3. Select **Map model to datasource**.
4. Select **New**.

    - This action creates a new record that maps the data model to data sources. In this example, you map the data model to data sources for the desired payment type: credit transfer. You can design more than one mapping for a particular data model. For example, you could create a mapping for different types of payments, such as for direct debit or for credit transfers. In this example, you create a mapping for credit transfers.

5. In the **Name** field, type **CT mapping**.

6. In the **Description** field, type **Payment model mapping CT**.
7. In the **Definition** field, type **CustomerCreditTransferInitiation**.
8. Resolve changes the Definition.
9. Select **Save**.

Define required data sources for the current model mapping

1. Select **Designer**.
2. In the tree view, select **Dynamics 365 for Operations\Table records**.
3. Select **Add root**.

    - Enter this data source to access payment transactions.

4. In the **Name** field, type **Transactions**.
5. In the **Label** field, enter **Transactions**.
6. In the **Help** field, enter **Ledger journal lines**.
7. In the **Ask for query** field, select **Yes**.
8. In the **Table** field, type **LedgerJournalTrans**.
9. Select **OK**.

    - Select the LedgerJournalTrans table as a data source for the current data model.

10. In the tree view, select **Functions\Calculated field**.
11. Select **Add** to add a new calculated field.
12. In the **Name** field, type **$EndToEndID**.
13. Select **Edit formula**.
14. In the tree view, select **String\CONCATENATE**.
15. Select **Add function**.
16. In the tree view, expand **Transactions**.
17. In the tree view, select **Transactions\Voucher**.
18. Select **Add data source**.
19. In the **Formula** field, enter **CONCATENATE(Transactions.Voucher, "-",**.

    - Type `, "-",` at the end of the formula.

20. In the tree view, select **String\TEXT**.
21. Select **Add function**.
22. In the tree view, select **Transactions\Record-ID(RecId)**.
23. Select **Add data source**.

24. In the **Formula** field, enter **CONCATENATE(Transactions.Voucher, "-", TEXT(Transactions.RecId))**.
    - Type `))` at the end of the formula.
25. Select **Save**.
    - Make sure that no errors exist for the created formula. See the **ERRORS** tab below the formula editor control.
26. Close the page.
27. Select **OK**.
    - Add the calculated field to this data source.
28. Select **Add** to add a new calculated field.
29. In the **Name** field, type **$Amount**.
30. Select **Edit formula**.
31. In the tree view, expand **Transactions**.
32. In the tree view, select **Transactions\Debit(AmountCurDebit)**.
33. Select **Add data source**.
34. In the **Formula field**, enter **Transactions.AmountCurDebit -**.
    - Type `-` at the end of the formula.
35. In the tree view, select **Transactions\Credit(AmountCurCredit)**.
36. Select **Add data source**.
37. Select **Save**.
38. Close the page.
39. Select **OK**.
    - This step adds the **$Amount** calculated field to the selected data source for the current data model.
40. In the tree view, select **Transactions$Amount**.
41. In the tree view, expand **Transactions**.
42. In the tree view, expand or collapse **Transactions$Amount**.
43. In the tree view, expand or collapse **Transactions**.
44. In the tree view, select **Dynamics 365 for Operations\Table records**.
45. Select **Add root**.
    - Enter this data source to access the company's bank account details.

46. In the **Name** field, type **BankAccount**.
47. In the **Label** field, enter **Bank Account**.
48. In the **Help** field, enter **Bank Account**.
49. Select **Yes** in the **Ask for query** field.
50. In the **Table** field, type **BankAccountTable**.
51. Select **OK**.

    - Select the **BankAccountTable** table as a data source for the current data model.

52. Select **Add root**.

    - Enter this data source to access the company's requisites.

53. In the **Name** field, type **Company**.
54. In the **Label** field, type a value.
55. In the **Help** field, enter **Company information**.
56. Select **Yes** in the **Ask for query** field.
57. In the **Table** field, type **CompanyInfo**.
58. Select **OK**.

    - Select the **CompanyInfo** table as a data source for the current data model.

59. In the tree view, select **Functions\Calculated field**.
60. Select **Add root**.

    - Insert a calculated field as a new data source.

61. In the **Name** field, type **ProcessingDateTime**.
62. In the **Label** field, enter **Processing date & time**.
63. Select **Edit formula**.
64. In the tree, select `Date/time\SESSIONNOW`.
65. Select **Add function**.
66. Select **Save**.
67. Close the page.
68. Select **OK**.

    - Add the **ProcessingDateTime** calculated field as a data source for the current data model.

69. Select **Save**.
70. Close the page.

Last updated on 04/13/2026

ER Map data model to selected data sources

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can map an Electronic reporting (ER) data model to selected data sources. You use this model mapping as a data source in a format configuration that manages electronic payment documents. In this example, you map a data model for sample company, Litware, Inc. to data sources. To complete these steps, you must first complete the steps in the "Select data sources for model mapping" procedure.

Open ER configurations tree

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Configurations**.

Select created model mapping

1. In the tree, select **Payments (simplified model)**.

    - Make sure that you created the model configuration **Payments (simplified model)**. Otherwise, stop now and return after completion of the task guide **Create a new configuration with data model of the selected domain**.

2. Select **Model designer**.
3. Select **Map model to datasource**.
4. Select the **CT mapping** record.

    - CT mapping

Bind created data sources to data model elements

1. Click Designer.
2. In the tree, select 'Processing date & time(ProcessingDateTime)'.
3. In the tree, select 'Processing date(ProcessingDateTime)'.
4. Click Bind.
5. In the tree, select 'Payment message identification(MessageIdentification)'.
6. In the tree, expand 'Transactions'.
7. In the tree, select 'Transactions\Journal batch number(JournalNum)'.
8. Click Bind.
9. In the tree, select 'Payments'.
10. In the tree, select 'Transactions'.
11. Click Bind.
12. In the tree, expand 'Payments= Transactions'.
13. In the tree, expand 'Payments= Transactions\Creditor'.
14. In the tree, expand 'Payments= Transactions\Creditor\Account'.
15. In the tree, select 'Payments= Transactions\Creditor\Account\Currency code(Currency)'.
16. In the tree, expand 'Transactions\vendBankAccountInTransactionCompany()'.
17. In the tree, select 'Transactions\vendBankAccountInTransactionCompany()\Currency(CurrencyCode)'.
18. Click Bind.
19. In the tree, select 'Payments= Transactions\Creditor\Account\IBAN code(IBAN)'.
20. In the tree, select 'Transactions\vendBankAccountInTransactionCompany()\IBAN(BankIBAN)'.
21. Click Bind.
22. In the tree, select 'Payments= Transactions\Creditor\Account\Number'.
23. In the tree, select 'Transactions\vendBankAccountInTransactionCompany()\Bank account number(AccountNum)'.
24. Click Bind.
25. In the tree, expand 'Payments= Transactions\Creditor\Agent'.
26. In the tree, select 'Payments= Transactions\Creditor\Agent\Name'.
27. In the tree, select 'Transactions\vendBankAccountInTransactionCompany()\Name'.
28. Click Bind.
29. In the tree, select 'Payments= Transactions\Creditor\Agent\Routing number(RoutingNumber)'.
30. In the tree, select 'Transactions\vendBankAccountInTransactionCompany()\Routing number(RegistrationNum)'.

31. Click Bind.
32. In the tree, select 'Payments= Transactions\Creditor\Agent\SWIFT code(SWIFT)'.
33. In the tree, select 'Transactions\vendBankAccountInTransactionCompany()\SWIFT code(SWIFTNo)'.
34. Click Bind.
35. In the tree, select 'Payments= Transactions\Creditor\Name'.
36. In the tree, expand 'Transactions\findVendTable()'.
37. In the tree, select 'Transactions\findVendTable()\name()'.
38. Click Bind.
39. In the tree, select 'Payments= Transactions\Currency code(Currency)'.
40. In the tree, expand 'Transactions>Relations'.
41. In the tree, expand 'Transactions>Relations\Currency table(Currency)'.
42. In the tree, select 'Transactions>Relations\Currency table(Currency)\Currency code(CurrencyCodeISO)'.
43. Click Bind.
44. In the tree, expand 'Payments= Transactions\Debtor'.
45. In the tree, expand 'Payments= Transactions\Debtor\Account'.
46. In the tree, select 'Payments= Transactions\Debtor\Account\Currency code(Currency)'.
47. In the tree, select 'Bank Account(BankAccount)'.
48. In the tree, expand 'Bank Account(BankAccount)'.
49. In the tree, select 'Bank Account(BankAccount)\Currency(CurrencyCode)'.
50. Click Bind.
51. In the tree, select 'Bank Account(BankAccount)\IBAN'.
52. In the tree, select 'Payments= Transactions\Debtor\Account\IBAN code(IBAN)'.
53. Click Bind.
54. In the tree, select 'Payments= Transactions\Debtor\Account\Number'.
55. In the tree, select 'Bank Account(BankAccount)\Bank account number(AccountNum)'.
56. Click Bind.
57. In the tree, expand 'Payments= Transactions\Debtor\Agent'.
58. In the tree, select 'Payments= Transactions\Debtor\Agent\Name'.
59. In the tree, select 'Bank Account(BankAccount)\Name'.
60. Click Bind.
61. In the tree, select 'Payments= Transactions\Debtor\Agent\Routing number(RoutingNumber)'.
62. In the tree, select 'Bank Account(BankAccount)\Routing number(RegistrationNum)'.
63. Click Bind.
64. In the tree, select 'Payments= Transactions\Debtor\Agent\SWIFT code(SWIFT)'.

65. In the tree, select 'Bank Account(BankAccount)\SWIFT code(SWIFTNo)'.
66. Click Bind.
67. In the tree, select 'Payments= Transactions\Debtor\Name'.
68. In the tree, select 'Company information(Company)'.
69. In the tree, expand 'Company information(Company)'.
70. In the tree, select 'Company information(Company)\Name'.
71. Click Bind.
72. In the tree, select 'Payments= Transactions\Description'.
73. In the tree, select 'Transactions\Description(Txt)'.
74. Click Bind.
75. In the tree, select 'Payments= Transactions\End to end identification code(End2EndID)'.
76. In the tree, select 'Transactions$EndToEndID'.
77. Click Bind.
78. In the tree, select 'Payments= Transactions\Instructed amount(InstructedAmount)'.
79. In the tree, select 'Transactions$Amount'.
80. Click Bind.
81. In the tree, select 'Payments= Transactions\Transaction date(TransactionDate)'.
82. In the tree, select 'Transactions\Date(TransDate)'.
83. Click Bind.

Validate created mapping

1. Click Validate.

    - Validate the new mapping to ensure that all bindings are okay.

2. Click the arrow to expand or collapse the Error List section.
3. Click Save.
4. Close the page.
5. Close the page.
6. Close the page.

Change the status of the current version of model configuration

1. Click Change status.

    - Change the status of designed model configuration – from Draft to Completed to make it available for payment format design.

2. Click Complete.
    - Select Complete.
3. In the Description field, type a value.
    - For example, 'version 1'.
4. Click OK.
5. Select the completed version of the current configuration.
    - Note that the created configuration is saved as completed version 1.

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ER Create a format configuration (November 2016)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user in the System Administrator or Electronic Reporting Developer role can create a format configuration for Electronic reporting (ER). This format configuration defines the format of electronic documents that are used for processing payments. In this example, you create a format configuration for sample company, Litware, Inc. To complete these steps, you must first complete the steps in the "Map model to selected datasources" procedure.

Create a new format configuration

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the tree, select **Payments (simplified model)**.
4. Select **Create configuration** to open the dialog.

> **Note**
>
> If you don't see **Create configuration**, you must enable design mode on the **Electronic reporting parameters** page.

1. In **New**, enter **Format based on data model PaymentModel**.
2. In **Name**, type **BACS (UK fictitious)**.
3. In **Description**, type **BACS vendor payment format (UK fictitious)**.

    - The active configuration provider is automatically entered here. This provider can maintain this configuration. Other providers can use this configuration, but can't

maintain it.
    - You can define a particular format of electronic document. Leave this field blank if you want to select a format at run-time.
4. In **Data model definition**, enter or select a value.
5. Select **Create configuration**. You created a new configuration. Use the draft version to store the design format for managing electronic documents.

Design the format of an electronic document

1. Select **Designer**.
2. Select **Add root** to open the drop dialog.
3. In the tree, select **Common\File**.
4. In the **Name** field, type **Xml**.
5. In the **Encoding** field, type **UTF-8**.
6. Select **OK**.
7. Select **Add**.
8. In the tree, select **XML\Element**.
9. In the **Name** field, type **Message**.
10. Select **OK**.
11. In the tree, select **Xml\Message**.
12. Select **Add Element**.
13. In the **Name** field, type **ProcessingDate**.
14. Select **OK**.
15. Select **Add Element**.
16. In the **Name** field, type **MessageId**.
17. Select **OK**.
18. Select **Add Element**.
19. In the **Name** field, type **Payments**.
20. Select **OK**.
21. In the tree, select **Xml\Message\Payments**.
22. Select **Add Element**.
23. In the **Name** field, type **Item**.
24. Select **OK**.
25. In the tree, select **Xml\Message\Payments\Item**.
26. Select **Add**.
27. In the tree, select **XML\Attribute**.
28. In the **Name** field, type **Id**.

29. Select **OK**.
30. Select **Add**.
31. In the tree, select **XML\Element**.
32. In the **Name** field, type **Vendor**.
33. Select **OK**.
34. In the tree, select **Xml\Message\Payments\Item\Vendor**.
35. Select **Add Element**.
36. In the **Name** field, type **Name**.
37. Select **OK**.
38. Select **Add Element**.
39. In the **Name** field, type **Bank**.
40. Select **OK**.
41. In the tree, select **Xml\Message\Payments\Item\Vendor\Bank**.
42. Select **Add Element**.
43. In the **Name** field, type **RoutingNumber**.
44. Select **OK**.
45. Select **Add Element**.
46. In the **Name** field, type **AccountNumber**.
47. Select **OK**.
48. In the tree, select **Xml\Message\Payments\Item\Vendor**.
49. Select **Copy**.
50. In the tree, select **Xml\Message\Payments\Item**.
51. Select **Paste**.
52. In the **Name** field, type **Payer**.
53. In the tree, select **Xml\Message\Payments\Item**.
54. Select **Add Element**.
55. In the **Name** field, type **Currency**.
56. Select **OK**.
57. Select **Add Element**.
58. In the **Name** field, type **Description**.
59. Select **OK**.
60. Select **Add Element**.
61. Enter **TransDate** in the Name field.
62. Select **OK**.
63. Select **Add Element**.
64. Enter **Amount** in the Name field.
65. Select **OK**.

Prepare format components for mapping to data model elements

1. In the tree, select **Xml\Message\ProcessingDate**.
2. Click **Add** to open the drop dialog.
3. In the tree, select **Text\DateTime**.
4. In the **Format** field, type **yyyy-MM-dd**.
5. Select **OK**.
6. In the tree, select **Xml\Message\Payments\Item\TransDate**.
7. Click **Add DateTime**.
8. In the **Format** field, type **yyyy-MM-dd**.
9. In the **DateTime** type field, select **Date**.
10. Select **OK**.
11. In the tree, select **Xml\Message\MessageId**.
12. Click **Add** to open the drop dialog.
13. In the tree, select **Text\String**.
14. Select **OK**.
15. In the tree, select **Xml\Message\Payments\Item\Vendor\Name**.
16. Click **Add String**.
17. Select **OK**.
18. In the tree, select **Xml\Message\Payments\Item\Vendor\Bank\RoutingNumber**.
19. Click **Add String**.
20. Select **OK**.
21. In the tree, select **Xml\Message\Payments\Item\Vendor\Bank\AccountNumber**.
22. Click **Add String**.
23. Select **OK**.
24. In the tree, select **Xml\Message\Payments\Item\Payer\Name**.
25. Click **Add String**.
26. Select **OK**.
27. In the tree, select **Xml\Message\Payments\Item\Payer\Bank\RoutingNumber**.
28. Click **Add String**.
29. Select **OK**.
30. In the tree, select **Xml\Message\Payments\Item\Payer\Bank\AccountNumber**.
31. Click **Add String**.
32. Select **OK**.
33. In the tree, select **Xml\Message\Payments\Item\Currency**.
34. Click **Add String**.

35. Select **OK**.
36. In the tree, select **Xml\Message\Payments\Item\Description**.
37. Click **Add String**.
38. Select **OK**.
39. In the tree, select **Xml\Message\Payments\Item\Amount**.
40. Click **Add String**.
41. Select **OK**.
42. Click **Save**.
43. Close the page.

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ER map components of the created format to data model elements (November 2016)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following procedure shows how a user in either the System administrator or Electronic reporting developer role can map data model elements to components of the created Electronic reporting (ER) configuration. This configuration defines an electronic document format for the payments business domain. You use this format later to generate electronic documents for processing payments. In this example, you create a format configuration for the sample company, Litware, Inc. You can perform these steps in any company because all companies share ER configurations. To complete these steps, you must first complete the steps in the "Create a format configuration" task guide.

Select a format configuration

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the tree, expand **Payments (simplified model)**.
4. In the tree, select **Payments (simplified model)\BACS (UK fictitious)**.
5. Select **Designer**.

Map format components to data model elements

1. Select **Expand/collapse**.
2. Select the **Mapping** tab.
3. In the tree, expand `model`.
4. In the tree, select `Xml\Message\ProcessingDate\DateTime`.
5. In the tree, select `model\ProcessingDateTime`.
6. Select **Bind**.

7. In the tree, select `Xml\Message\MessageId\String`.
8. In the tree, select `model\MessageIdentification`.
9. Select **Bind**.
10. In the tree, expand `model\Payments`.
11. In the tree, select `Xml\Message\Payments\Item\Amount\String`.
12. In the tree, select `model\Payments\InstructedAmount`.
13. Select **Bind**.
14. In the tree, select `Xml\Message\Payments\Item\TransDate\DateTime`.
15. In the tree, select `model\Payments\TransactionDate`.
16. Select **Bind**.
17. In the tree, select `Xml\Message\Payments\Item\Description\String`.
18. In the tree, select `model\Payments\Description`.
19. Select **Bind**.
20. In the tree, select `Xml\Message\Payments\Item\Currency\String`.
21. In the tree, select `model\Payments\Currency`.
22. Select **Bind**.
23. In the tree, select `Xml\Message\Payments\Item\Id`.
24. In the tree, select `model\Payments\End2EndID`.
25. Select **Bind**.
26. In the tree, expand `model\Payments\Creditor`.
27. In the tree, expand `model\Payments\Creditor\Account`.
28. In the tree, expand `model\Payments\Creditor\Agent`.
29. In the tree, select `Xml\Message\Payments\Item\Vendor\Name\String`.
30. In the tree, select `model\Payments\Creditor\Name`.
31. Select **Bind**.
32. In the tree, select `Xml\Message\Payments\Item\Vendor\Bank\RoutingNumber\String`.
33. In the tree, select `model\Payments\Creditor\Agent\RoutingNumber`.
34. Select **Bind**.
35. In the tree, select `Xml\Message\Payments\Item\Vendor\Bank\AccountNumber\String`.
36. In the tree, select `model\Payments\Creditor\Account\Number`.
37. Select **Bind**.
38. In the tree, select `Xml\Message\Payments\Item\Payer\Name\String`.
39. In the tree, expand `model\Payments\Debtor`.
40. In the tree, expand `model\Payments\Debtor\Account`.
41. In the tree, expand `model\Payments\Debtor\Agent`.
42. In the tree, select `model\Payments\Debtor\Name`.
43. Select **Bind**.

44. In the tree, select `Xml\Message\Payments\Item\Payer\Bank\RoutingNumber\String`.
45. In the tree, select `model\Payments\Debtor\Agent\RoutingNumber`.
46. Select **Bind**.
47. In the tree, select `Xml\Message\Payments\Item\Payer\Bank\AccountNumber\String`.
48. In the tree, select `model\Payments\Debtor\Account\Number`.
49. Select **Bind**.
50. In the tree, select `Xml\Message\Payments\Item`.
51. In the tree, select `model\Payments`.
52. Select **Bind**.
53. Select **Save**.

Validate format mapping

1. Select **Validate**.

    - Validate the new mapping to ensure that all bindings are correct.

2. Close the page.

Change status of the current version of format configuration

In the next steps, change the status of the format configuration from **Draft** to **Completed** to make it available for payment document generation.

1. Select **Change status**.
2. Select **Complete**.
3. In the **Description** field, enter a value.

    - For example, 'version 1'.

4. Select **OK**.
5. Select the completed version of the current configuration.

    - The configuration is saved as completed version 1.1: version 1 of the format based on version 1 of the data model.

Define effective date for completed version of format

You can configure each format version as available for usage starting from a certain date. When more than one format version is active on a certain date, the latest format (based on version number) is selected for usage. The session date value is used for proper version selection.

Restrict access to created format from companies

1. Expand the **ISO Country/region codes** section.

    - You can restrict each format access by identifying particular countries or regions in which a format is applicable. When the list of countries or regions for a particular format is empty, any company can use this format. When you add some ISO country or region codes to the list, companies can only use the format if the primary address is in the country or region.

Last updated on 04/13/2026

Globalization Studio workspace features

> **Important**
>
> Regulatory Configuration Service (RCS) will be deprecated. All new RCS provisioning is stopped as of 10.0.39 GA. If provisioning is required, please register a support ticket. We will be providing tools and required support for migration from RCS to the **Globalization Studio** workspace. We plan to fully shut down RCS by August 1, 2024. For more information about migration to **Globalization Studio** workspace, see Regulatory Configuration Service merge to the Globalization Studio workspace

By using Globalization Studio, you can create Globalization features for use in the following Globalization services: Electronic Invoicing and Tax Calculation. You can:

- Define related components of a feature.
- Manage the feature lifecycle through a feature's status.

To use a Globalization feature, you must first import it from a Dataverse repository and create your own version of it. You can add Globalization features by:

- Adding a derived feature based on an existing feature that's published or shared.
- Adding a new feature that you created from scratch.

Feature component overview

Globalization features include several components:

- **Version** – This component supports feature lifecycle management. It lets you manage the status of different versions of the feature.
- **Configurations** – This component lets you manage, view, and edit related Electronic reporting (ER) formats and format mappings.
- **Setups** – This component lets users of Globalization services, such as an e-invoicing service, manage the setup of the related feature version. Therefore, it supports the flexible construction of communication and responses rules.

Last updated on 03/21/2026

ER Generate electronic documents for payments using a format configuration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can use a new Electronic reporting (ER) format configuration to generate electronic documents for processing payments. You can perform these steps in the GBSI sample company.

To complete these steps, you must first complete the steps in the "Create a configuration with format of payment document" procedure.

Change the configuration of the electronic payment method

1. Go to **Accounts payable** > **Payment setup** > **Methods of payment**.
2. Toggle the **File format** section to expand it, if needed.
3. Use the Quick Filter to find records. For example, filter on the **Method of payment** field with a value of **Electronic**.
4. Select **Edit**.
5. Set the **General electronic reporting** field to **Yes**.

    - Select **Yes** to use the General electronic reporting pattern for payment files generation.

6. In the **Name** field, select the drop-down button to open the lookup.
7. Select **BACS (UK fictitious)** format configuration.
8. Select **Save**.
9. Close the page.

Test the format of generated payment files

1. Go to **Accounts payable** > **Payments** > **Payment journal**.
2. Select **New**.
3. In the list, select the row.
4. In the **Name** field, select the drop-down button to open the lookup.
5. In the list, select the link in the selected row.

    - Select **VendPay**.

6. Select **Save**.
7. Select **Lines**.
8. In the **Company** field, enter `DEMF`.

    - Enter **DEMF**

9. In the **Account** field, enter `DE-01001`.

    - Enter **DE-01001**

10. In the **Description** field, enter `Payment`.

    - Enter **Payment**

11. In the **Debit** field, enter a number.

    - 1000

12. Select the **Payment** tab.
13. In the **Method of payment** field, select the dropdown button to open the lookup.
14. In the list, find and select the desired record.

    - Select the **Electronic** value.

15. In the list, select the link in the selected row.
16. Select **Save**.
17. Select **Generate payments**.
18. In the **Method of payment** field, select the dropdown button to open the lookup.
19. In the list, find and select the desired record.

    - Select the **Electronic** value.

20. In the list, select the link in the selected row.

- Select the **Electronic** value.

21. In the **File name** field, enter a value.
    - For example, enter `payments`.
22. In the **Bank account** field, select the dropdown button to open the lookup.
23. In the list, select the link in the selected row.
    - Select the value **GBSI OPER**.
24. Select **OK**.
25. Select **OK**.
    - Analyze the created payment file in XML format. Compare it with the designed document layout and defined payment transaction attributes.

Last updated on 04/13/2026

ER Upgrade your format by adopting a new, base version of that format

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can maintain an Electronic reporting (ER) format configuration. This procedure explains how to create a custom version of a format based on the format received from a configuration provider (CP). It also explains how to adopt a new, base version of that format.

To complete these steps, you must first complete the steps in the "Create a configuration provider and mark it as active" and "Use created format to generate electronic documents for payments" procedures. You can perform these steps in the GBSI company.

Select format configuration for customization

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

   In this example, sample company Litware, Inc. (https://www.litware.com) acts as a configuration provider that supports format configurations for electronic payments for a particular country/region. Sample company Proseware, Inc. (https://www.proseware.com) acts as a consumer of the format configuration that Litware, Inc. provided. Proseware, Inc. uses formats in certain regions of that country/region.

2. Select **Reporting configurations**.

3. Select **Show filters**.

4. Apply the following filters: Enter a filter value of **BACS (UK fictitious)** on the **Name** field by using the **begins with** filter operator.

The selected format configuration **BACS (UK fictitious)** is owned by provider Litware, Inc.

5. Select **Show filters**.
6. In the list, find and select the desired record.

   Proseware, Inc. uses the version of the format with the status of **Completed** for customization.

Create a new configuration for your custom format of electronic document

Proseware, Inc. receives version 1.1 of BACS (UK fictitious) configuration that contains the initial format to generate electronic payment documents from Litware, Inc. in accordance with their service subscription. Proseware, Inc. wants to start using this format as a standard for their country/region but some customization is required to support specific regional requirements. Proseware, Inc. also wants to keep the ability to upgrade a custom format as soon as a new version of it (with changes to support new country/region-specific requirements) comes from Litware, Inc. They want to perform this upgrade with the lowest cost.

To do this upgrade, Proseware, Inc. needs to create a configuration by using the Litware, Inc. configuration BACS (UK fictitious) as a base.

1. Close the page.
2. Select Proseware, Inc. to make it an active provider.
3. Select **Set active**.
4. Select **Reporting configurations**.
5. In the tree, expand **Payments (simplified model)**.
6. In the tree, select **Payments (simplified model)\BACS (UK fictitious)**.

   Select the BACS (UK fictitious) configuration from Litware, Inc. Proseware, Inc. uses version 1.1 as a base for the custom version.

7. Select **Create configuration** to open the drop dialog.

   This step lets you create a new configuration for a custom payment format.

8. In the **New** field, enter `Derive from Name: BACS (UK fictitious), Litware, Inc.`.

Select the **Derive** option to confirm the usage of BACS (UK fictitious) as the base for creating the custom version.

9. In the **Name** field, type `BACS (UK fictious custom)`.
10. In the **Description** field, type `BACS vendor payment (UK fictitious custom)`.

    The active configuration provider (Proseware, Inc.) is automatically entered here. This provider can maintain this configuration. Other providers can use this configuration, but can't maintain it.

11. Select **Create configuration**.

Customize your format for the electronic document

1. Select **Designer**.
2. Select **Expand/collapse**.
3. Select **Expand/collapse**.
4. In the tree, select `Xml\Message\Payments\Item\Vendor\Bank`.
5. Select **Add** to open the drop dialog.
6. In the tree, select `XML\Element`.
7. In the **Name** field, type `IBAN`.
8. Select **OK**.
9. In the tree, select `Xml\Message\Payments\Item\Vendor\Bank\IBAN`.
10. Select **Add** to open the drop dialog.
11. In the tree, select `Text\String`.
12. Select **OK**.
13. In the tree, select `Xml\Message\Payments\Item\Vendor\Name\String`.
14. In the **Maximum length** field, enter `60`.
15. Select the **Mapping** tab.
16. In the tree, expand `model`.
17. In the tree, expand `model\Payments`.
18. In the tree, expand `model\Payments\Creditor`.
19. In the tree, expand `model\Payments\Creditor\Account`.
20. In the tree, select `model\Payments\Creditor\Account\IBAN`.
21. In the tree, select `Xml\Message\Payments\Item = model.Payments\Vendor\Bank\IBAN\String`.
22. Select **Bind**.
23. Select **Save**.

Validate the customized format

1. Select **Validate**.

   Validate the customized format layout and data mapping changes to make sure that all bindings are correct.

2. Close the page.

Change the status of the current version of the custom format configuration

Change the status of the designed format configuration from **Draft** to **Completed** to make it available for payment document generation.

1. Select **Change status**.

   The current version of the selected configuration is in **Draft** status.

2. Select **Complete**.

3. In the **Description** field, enter a value.

4. Select **OK**.

5. In the list, find and select the desired record.

   The created configuration is saved as **completed version 1.1.1**. This version means it's version 1 of the custom BACS (UK fictitious custom) format, which is based on version 1 of the BACS (UK fictitious) format, which is based on version 1 of the Payments (simplified model) data model.

Test the customized format to generate payment files

Complete the steps in the "Use created format to generate electronic documents for payments" procedure in a parallel finance and operations session. Select the BACS (UK fictitious custom) format in electronic payment method parameters. Make sure that the created payment file contains the recently introduced XML node presenting IBAN code in accordance to regional requirements.

Update the existing country/region-specific configuration

Litware, Inc. needs to update the BACS (UK fictitious) configuration and adopt new country/region requirements for managing the format of the electronic document. Later, Litware, Inc. encloses this configuration in a new version that it offers to service subscribers, including Proseware, Inc.

In real service provision related processes, Proseware, Inc. can import each new version of BACS (UK fictitious) from Litware, Inc. configurations' LCS repository. In this procedure, you simulate this process by updating BACS (UK fictitious) on behalf of a service provider.

1. Close the page.
2. Select Litware, inc. provider.
3. Select **Set active**.
4. Select **Reporting configurations**.
5. In the tree, expand **Payments (simplified model)**.
6. In the tree, select **Payments (simplified model)\BACS (UK fictitious)**.

   The draft version owned by Litware, Inc. provider BACS (UK fictitious) is selected to bring in changes to support new country/region-specific requirements.

Localize the base format of the electronic document

Assume that Litware, Inc. needs to support new country or region-specific requirements:

- A value for the creditor's bank SWIFT code in each payment transaction.
- A limit of 100 characters for the length of text for the vendor's name in a generating file.
- New country or region-specific requirements.
- Select the draft version of the desired configuration to introduce required changes.

1. Select **Designer**.
2. Select **Expand/collapse**.
3. Select **Expand/collapse**.
4. In the tree, select `Xml\Message\Payments\Item\Vendor\Bank`.
5. Select **Add** to open the drop dialog.

6. In the tree, select `XML\Element`.
7. In the **Name** field, type `SWIFT`.
8. Select **OK**.
9. In the tree, select `Xml\Message\Payments\Item\Vendor\Bank\SWIFT`.
10. Select **Add** to open the drop dialog.
11. In the tree, select `Text\String`.
12. Select **OK**.
13. In the tree, select `Xml\Message\Payments\Item\Vendor\Name\String`.
14. In the **Maximum length** field, enter `100`.
15. Select the **Mapping** tab.
16. In the tree, expand `model`.
17. In the tree, expand `model\Payments`.
18. In the tree, expand `model\Payments\Creditor`.
19. In the tree, expand `model\Payments\Creditor\Agent`.
20. In the tree, select `model\Payments\Creditor\Agent\SWIFT`.
21. In the tree, select `Xml\Message\Payments\Item = model.Payments\Vendor\Bank\SWIFT\String`.
22. Select **Bind**.
23. Select **Save**.

Validate the localized format

1. Select **Validate**.
2. Close the page.

Change the status of the current version of the base format configuration

Change the status of the updated base format configuration from **Draft** to **Completed** to make it available for generating payment documents and updating format configurations derived from it.

1. Select **Change status**.

   The current version of the selected configuration is in **Draft** status.

2. Select **Complete**.

3. In the **Description** field, enter a value.

4. Select **OK**.

5. In the list, find and select the desired record.

Change the base version for the custom format configuration

Proseware, Inc. is informed that a new version 1.2 of BACS (UK fictitious) configuration is available to generate electronic payment documents in accordance with recently announced country/region-specific requirements. Proseware, Inc. wants to start using it as a standard for the country/region.

To do this, Proseware, Inc. needs to change the base configuration version for the custom configuration BACS (UK fictitious custom). Instead of version 1.1 of BACS (UK fictitious), use new version 1.2.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

2. Select the Proseware, Inc. provider to mark it as active.

3. Select **Set active**.

4. Select **Reporting configurations**.

5. In the tree, expand **Payments (simplified model)**.

6. In the tree, expand **Payments (simplified model)\BACS (UK fictitious)**.

7. In the tree, select **Payments (simplified model)\BACS (UK fictitious)\BACS (UK fictitious custom)**.

   Select the BACS (UK fictitious custom) configuration, which Proseware, Inc. owns.

   Use the draft version of the selected configuration to introduce required changes.

8. Select **Rebase**.

   Select the new version 1.2 of the base configuration to apply as a new base for updating the configuration.

9. Select **OK**.

The system discovers some conflicts between merging the custom version and a new base version. These conflicts represent some format changes that can't be merged automatically.

Resolve rebase conflicts

1. Select **Designer**.

   The system can't automatically resolve changes to the vendor's name text length limit. Therefore, the system presents this change in a conflicts list. For each conflict of type **Update**, you see the following options:

   - **Apply a prior base value** (button on top of the grid): Bring in the previous base version value (0 in this case).

- **Apply a base value** (button on top of the grid): Bring in the new base version value (100 in this case).
  - **Keep your own (custom) value** (60 in this case).

1. Select **Save**.

   Saving the format will remove resolved conflicts from the conflicts list.

2. Close the page.

Change the status of the new version of the custom format configuration

1. Select **Change status**.

   Change the status of the updated custom format configuration from **Draft** to **Completed**. This change makes the format configuration available for generating payment documents. The current version of the selected configuration is in **Draft** status.

2. Select **Complete**.

3. In the **Description** field, enter a value.

4. Select **OK**.

   The created configuration is saved as completed version 1.2.2: version 2 of base BACS (UK fictitious custom) format, which is based on version 2 of base BACS (UK fictitious) format,

which is based on version 1 of Payments (simplified model) data model.

Test the customized format for payment files generation

Complete the steps in the "Use created format to generate electronic documents for payments" procedure in parallel finance and operations session. Select the created **BACS (UK fictitious custom)** format in electronic payment method parameters. Make sure that the created payment file contains recently introduced by Proseware, Inc. XML node presenting IBAN account code in accordance to regional requirements. The file also should contain the recently introduced by Litware, Inc. XML node presenting SWIFT bank code in accordance to country/region requirements.

Last updated on 04/13/2026

Supported data source types for ER components

You can use the Electronic reporting (ER) framework to design an ER solution that you can run in the Microsoft Dynamics 365 Finance application to generate an outbound business document or parse an inbound business document. To fill in a generated document with application data or update the application database by using data from an inbound business document, configure your ER solution to specify how the data should be fetched from the appropriate data storage, transformed in accordance with the layout of the processed document, and formatted based on specific requirements. To specify data fetching, transformation, and formatting, use different types of ER data sources. The following ER data source types are currently supported for Finance:

- Data model
  - Data model
  - Enumeration
  - Enumeration user input parameter
  - Lookup
- Dynamics 365 application (Finance)
  - Class
  - Enumeration
  - Lookup
  - Number sequence
  - Object
  - Table
  - Table records
- Format
  - Export format
  - Import format
- Format enumeration
  - Enumeration
  - Enumeration user input parameter
  - Lookup
- Functions

    - Barcode
    - Calculated field
    - Data collection
    - Financial dimensions details
    - Group by
    - Inventory dimensions details
    - Join
    - Table metadata

- General
    - Empty container
    - Endless list
    - User input parameter

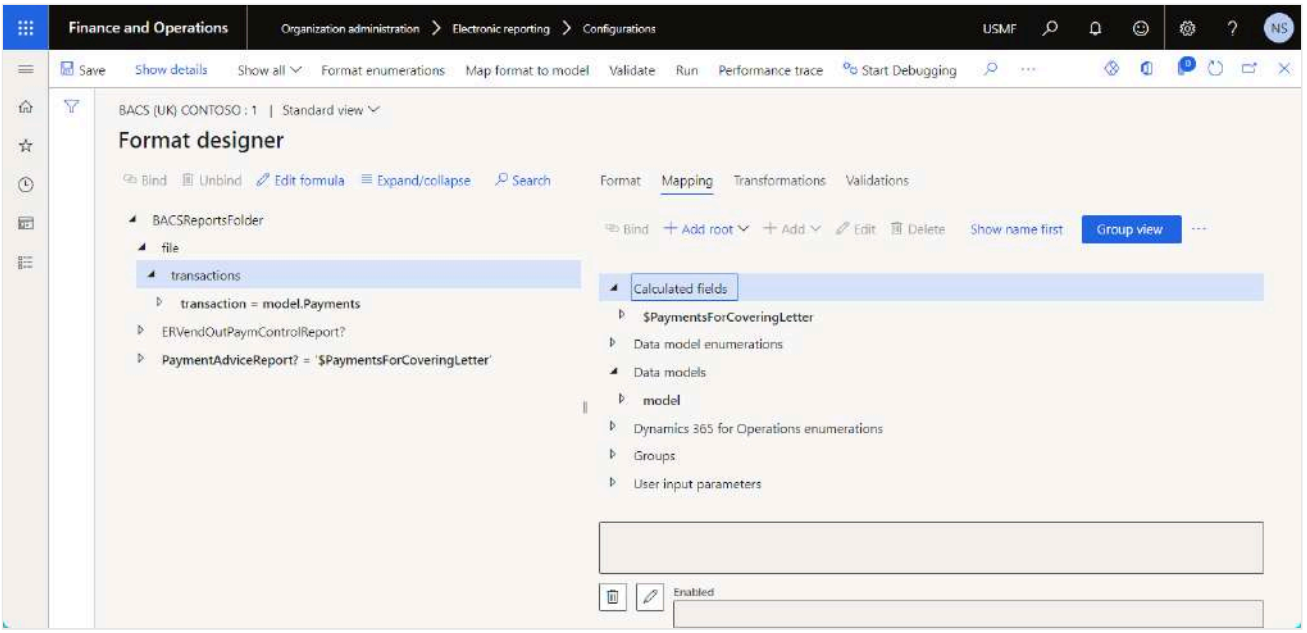
This article provides a summary of the ER data source types.

Add a new data source

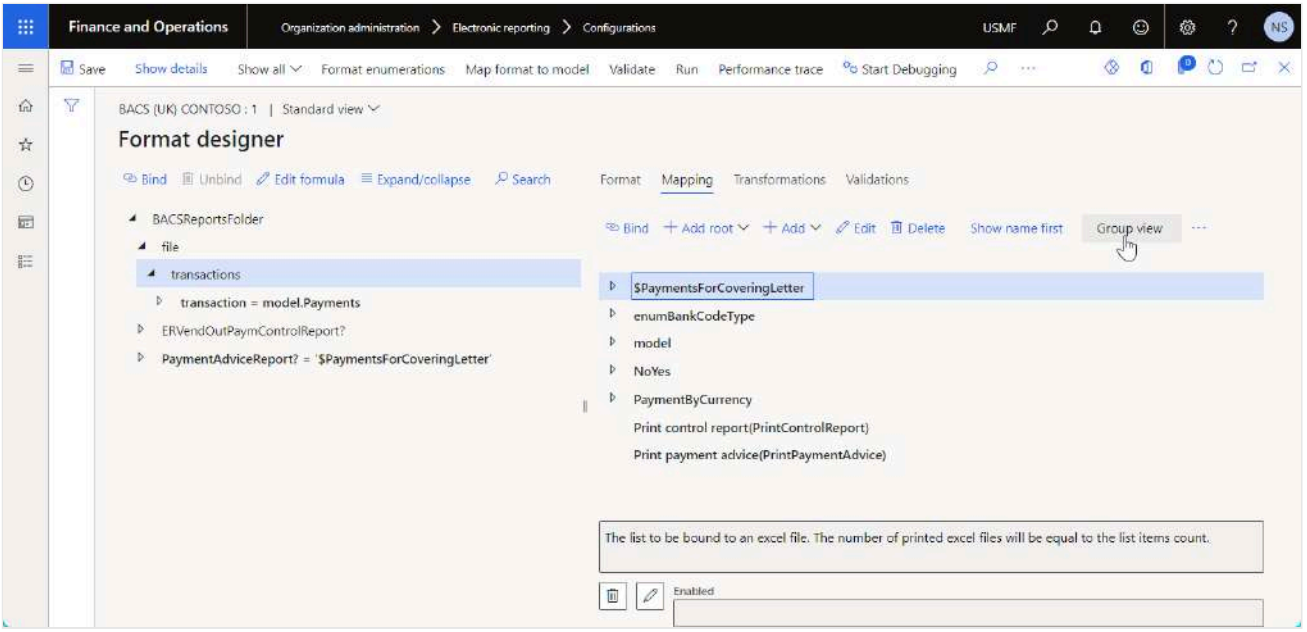
You can configure ER data sources in the ER model mapping and ER format components of your ER solution for data fetching, transformation, and formatting. You can then bind configured data sources with data consumers. Data consumers include a data model of the designed ER model mapping or format elements of the designed ER format.

Select a more convenient grouping mode to quickly discover a data source that you want to review or modify.

- When you turn on the **Group view** toggle button on the **Model mapping designer** or **Format designer** page, the configured data sources are grouped by the data source type.

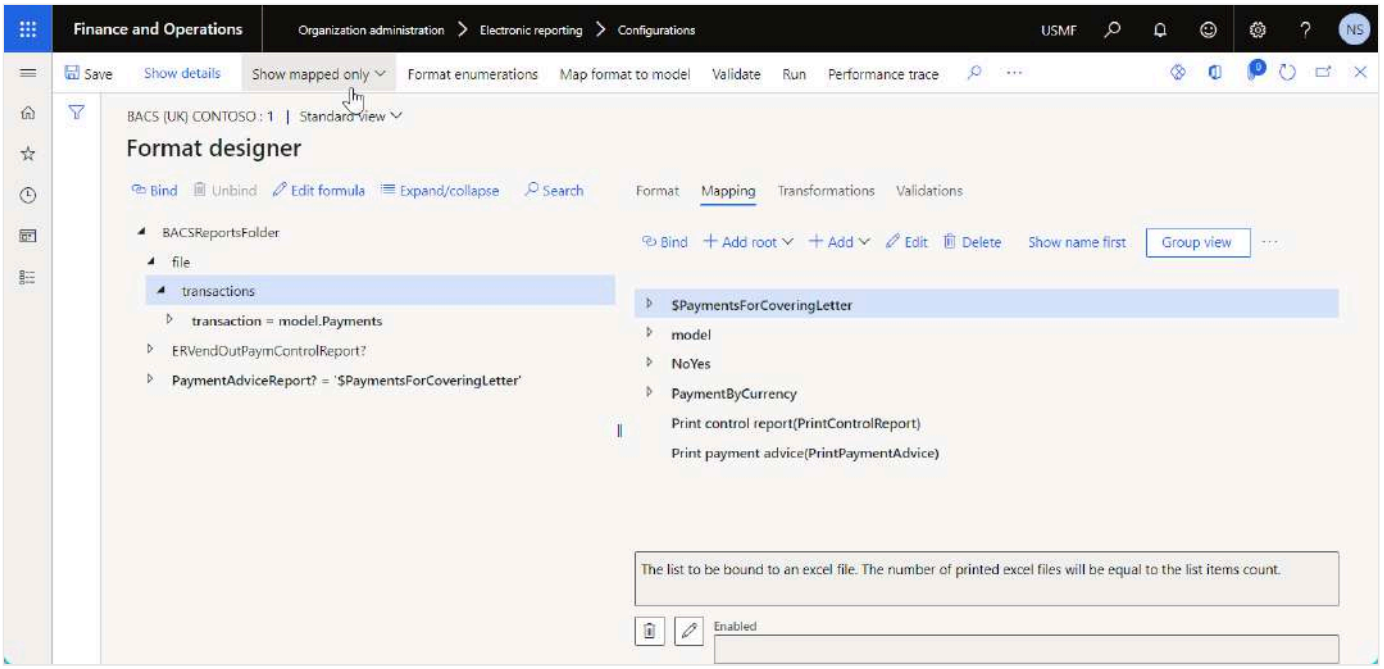


- When you turn off the **Group view** toggle button, the data sources are sorted in alphabetical order.



You can filter the list of configured data sources by selecting the more appropriate view mode on the **Model mapping designer** or **Format designer** page:

- Select **Show all** to list all configured data sources.
- Select **Show mapped only** to list only the data sources that you explicitly use in configured bindings.
- Select **Show not-mapped only** to list only the data sources that you don't use in configured bindings.



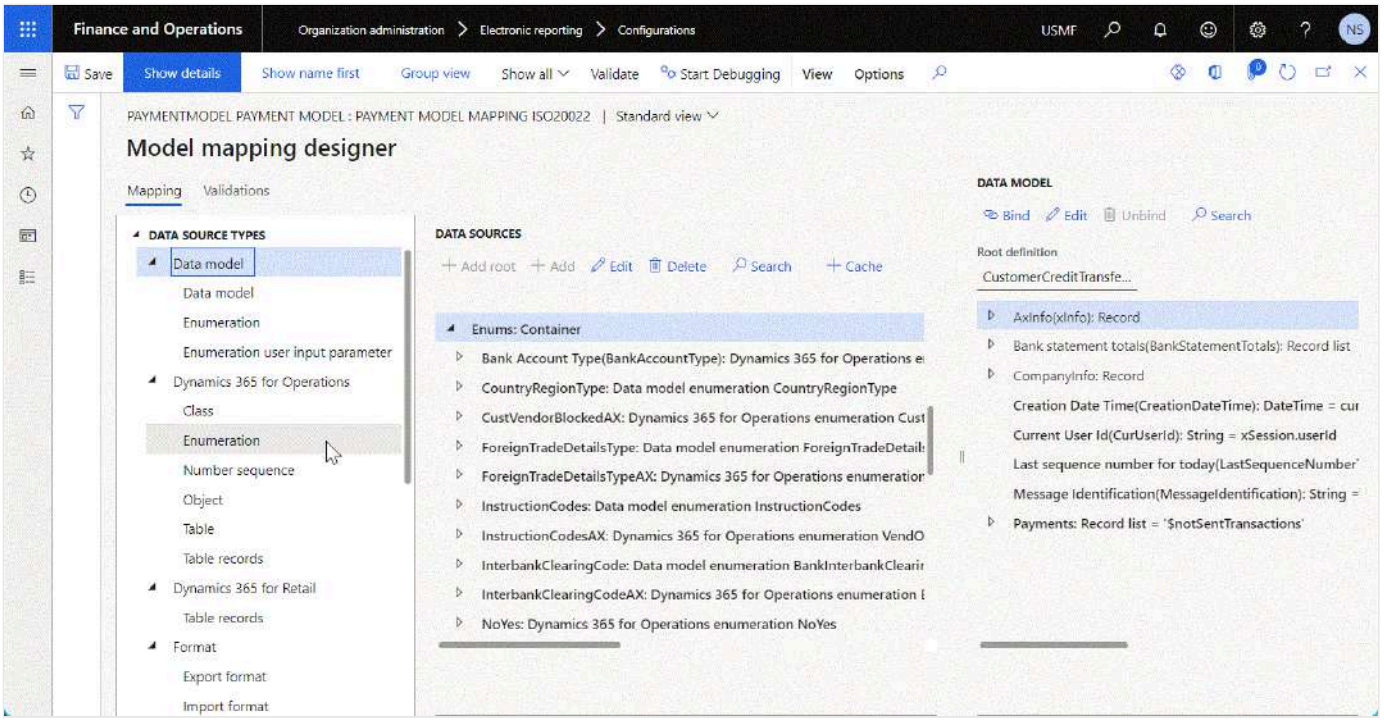
Model mapping designer

To add a new data source, select the data source type in the **Data source types** pane on the left side of the **Model mapping designer** page. This pane shows the tree of supported ER data sources.

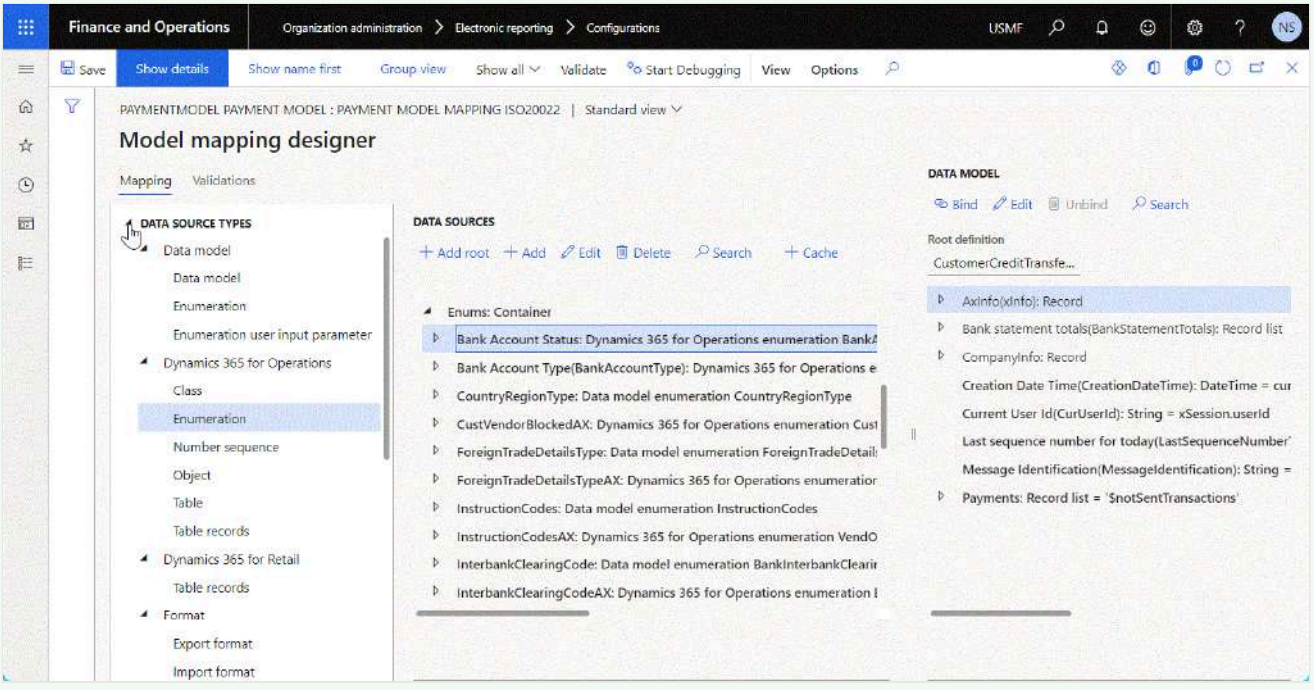
The **Data sources** pane in the middle of the page shows the data sources that you already added to the current model mapping. Use the **Add root** and **Add** buttons to add a new data source entry.

- To add a new top-level data source, select **Add root**.
- To add a new child data source, in the **Data sources** pane, select an existing data source as the parent node to add the new data source to. Then select **Add**.

The following illustration shows the addition of a new child data source of the *Enumeration* type.



> **Tip**
>
> To free up space so that you can enlarge the **Data sources** and **Data model** panes, select the arrow next to the **Data source types** heading in the left pane to collapse the tree. The whole pane is then collapsed.

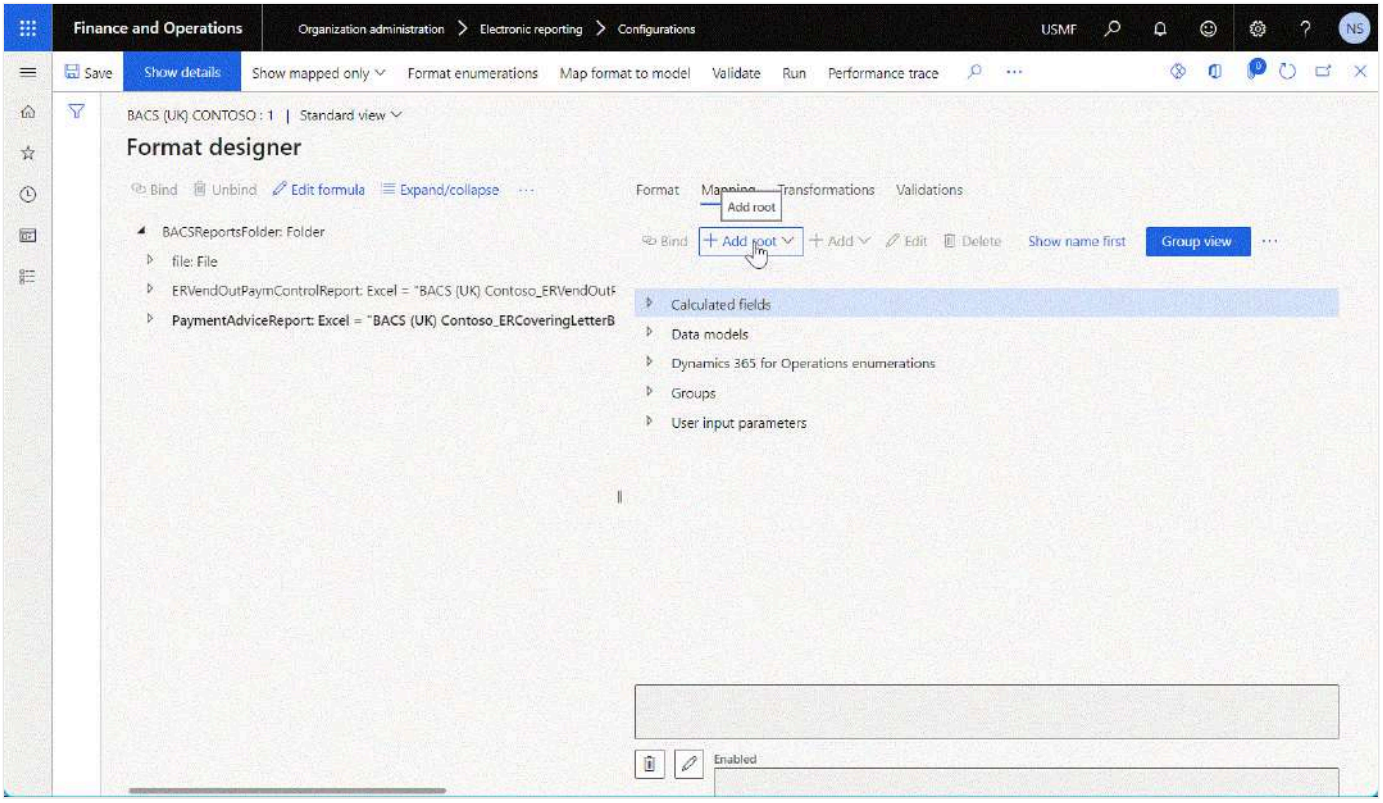


Format designer

The **Mapping** tab of the **Format designer** page shows the data sources that you already added to the current format. Use the **Add root** and **Add** buttons to add a new data source entry.

- To add a new top-level data source, select **Add root**. Then, in the **Add data source** dialog box, select the desired data source type.
- To add a new child data source, select an existing data source as the parent node to add the new data source to, and select **Add**. Then, in the **Add data source** dialog box, select the desired data source type.

The following illustration shows the addition of a new top-level data source of the *Data model* > *Enumeration* type.



> **Note**
>
> The following data source types are available for selection only when you turn on the **Show details** toggle button on the **Format designer** page:
>
> - Class
> - Object
> - Table
> - Table records

Common properties of ER data sources

You can specify the properties described in this section for any data source type.

Name property

Because data source names must be unique in an ER component, you must provide the **Name** property. Use this property to uniquely identify the added data source in the configured ER component.

Label property

The **Label** property is optional. Use it to specify an alternative name for the data source. To specify language-dependent text for the label of the editable data source, select **Translate**. When you translate the label, the **Model mapping designer** or **Format designer** page presents it in the preferred language of the current user.

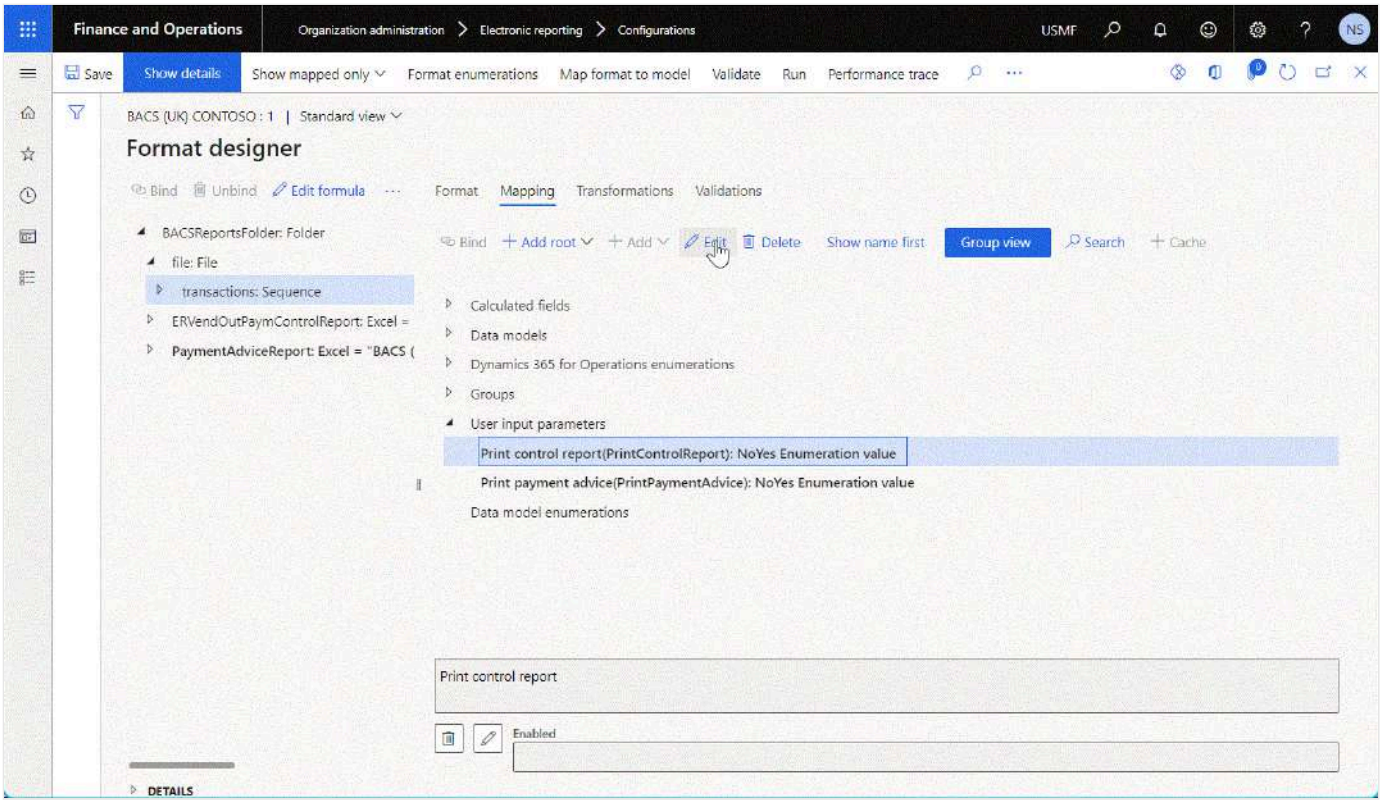
> **Tip**
>
> Turn the **Show name first** toggle button on the **Model mapping designer** or **Format designer** page on or off to sort data sources by name or label.

When you specify this property for a data source that lets you enter information in a data entry field, the specified label is the prompt text for that data entry field. When you translate the label, the prompt text for the data entry field appears in your preferred language, or the preferred language of the current user.

Help property

The **Help** property is optional. Use it to specify a description of the data source. To specify language-dependent text for the description of the editable data source, select **Translate**. When you translate the description, the **Model mapping designer** or **Format designer** page presents it in the preferred language.

The following illustration shows the entry of data source properties on the **Format designer** page.



Data model group

You can configure data sources of different types that represent a data model that you select as the basis for the editable ER model mapping or ER format component. This section provides a summary of the properties that are common to all types of ER data sources that belong to the **Data model** group.

Model property

The mandatory **Model** property references the ER data model component that specifies the structure of this data source.

Definition property

The optional **Definition** property points to a single root element of the selected ER data model component to specify what part of the referenced ER data model is used in this data source.

> **Tip**
>
> You can leave the **Definition** property blank if the default root definition is specified in the referenced data model.

Version property

The optional **Version** property specifies the revision of the data model component for this data source. To review the list of data model revisions that are available in the current Finance instance, expand the **Versions** FastTab on the **Configurations** page for the appropriate data model configurations.

> **Tip**
>
> Leave this property blank to use a data model component from the configuration **version** that's in the **Draft** status. In this way, you can simultaneously edit your ER model mapping or ER format and its base data model.

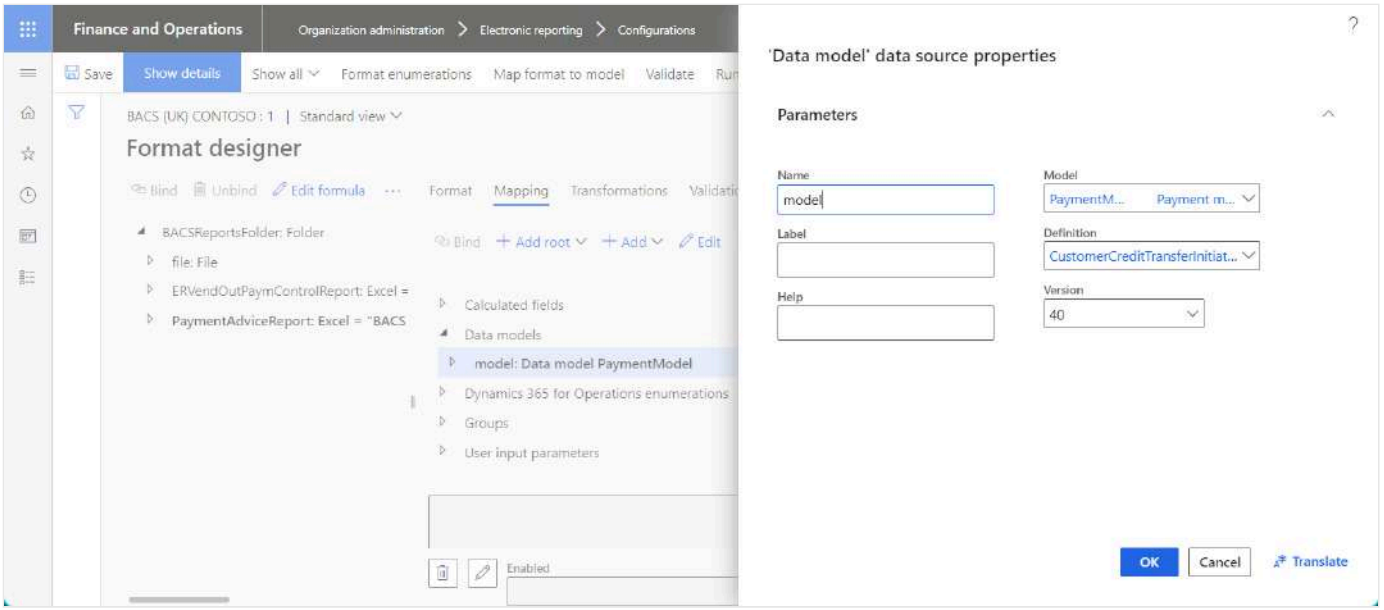
> **Important**
>
> You must specify this property when you change the status of the editable configuration from **Draft** to **Completed**. If you change the structure of the draft data model and use your changes in the edited ER model mapping or ER format, change the status of the edited base data model configuration version from **Draft** to **Completed** before you make the same change for the edited configuration version of the ER model mapping or ER format. Otherwise, a Path not found exception is thrown when you try to complete the draft version of the edited ER model mapping or ER format configuration.

Data model

The **Data model** data source represents the ER data model component that's filled in by actual data at runtime when a data consumer request occurs.

> **Tip**
>
> This data source is automatically added for any ER format component that you create to generate an outbound document.

The following illustration shows the properties for a data source of the *Data model* type in the **Data source properties** dialog box.

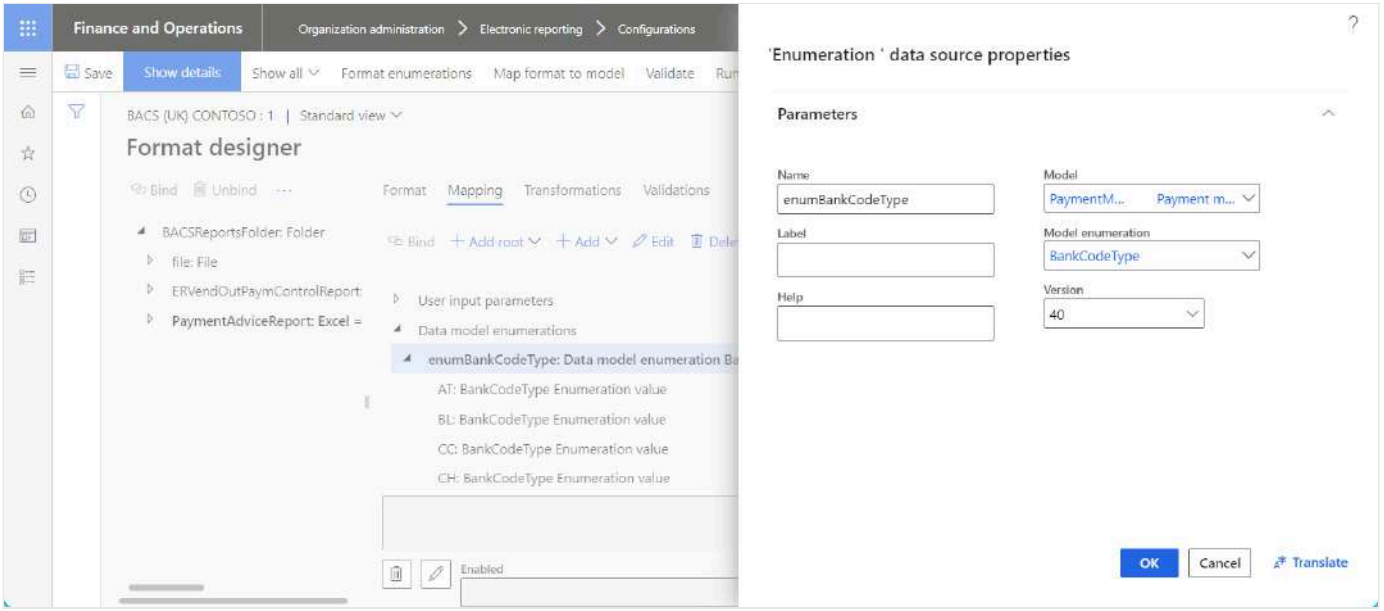


Model enumeration

The **Data model enumeration** data source represents the enumeration of the base data model. You usually need this data source when you configure an ER expression to compare the value of a data model field with a specific enumeration value.

The mandatory **Model enumeration** property points to a data model enumeration of the selected ER data model component. It specifies which values of the referenced ER data model enumeration are used in this data source.

The following illustration shows the properties for a data source of the *Model enumeration* type in the **Data source properties** dialog box.



Model enumeration user input parameter

The **Model enumeration user input parameter** data source represents the enumeration of the base data model. You usually need this data source when you want the value of the referenced enumeration that's specified in the related data entry field in the dialog box at runtime, before the ER format runs.

The mandatory **Model enumeration** property points to a data model enumeration of the selected ER data model component. It specifies what values of the referenced ER data model enumeration this data source returns.

The **Allow multiple selection** property identifies whether you can specify a single value or multiple values of the referenced enumeration.

Use the **Read only** property to configure an ER expression and specify whether the appropriate data entry field in the dialog box is editable at runtime.

Use the **Default value** property to configure an ER expression that specifies the default value for the appropriate data entry field in the dialog box at runtime.

The **Always reset to default value** property identifies whether the value that's returned by the **Default value** ER expression should always be used as the default value, regardless of the previously used value.

Use the **Edit visibility** function to configure an ER expression that specifies, at runtime, whether the appropriate data entry field should be shown in the dialog box at runtime.

- If this expression is configured and returns *False* at runtime, the appropriate data entry field in the dialog box is hidden.
- Otherwise, the appropriate data entry field is visible in the dialog box at runtime.

For more information, see Use USER INPUT PARAMETER data sources to specify parameters for a report.

Model enumeration lookup

Use the **Model enumeration lookup** data source to configure an ER data source of the *Lookup* type that returns the values of an ER data model enumeration.

> ⓘ **Note**
>
> You can only configure the **Model enumeration lookup** data source for an ER format component.

Configure a data source of the *Model enumeration lookup* type to filter data in an ER format so that it's based on a set of abstract rules. You can then specify real rules outside the ER component designers by using the user interface (UI) that's automatically generated based on the settings of the **Lookup** data source of the corresponding ER format and the current legal entity data. When you run this ER format, the ER format's **Lookup** data source accesses the specified rules.

The mandatory **Model enumeration** property of the **Model enumeration lookup** data source points to a data model enumeration of the base ER data model component to specify what values of the referenced ER data model enumeration this data source returns.

The **Cross-company** property identifies whether you plan to use the configured data source to evaluate the data of multiple companies at runtime.

The **Extended** property reveals more methods for accessing values of the configured data source.

For more information, see Configure Lookup data sources to use ER application-specific parameters.

Format enumeration group

You can configure different data source types that represent the editable format component and return values of an enumeration of that format. This section provides a summary of the properties that are common to all types of ER data sources that belong to the **Format enumeration** group.

> ⓘ **Note**
>
> You can only configure data sources of the **Format enumeration** group for an ER format component.

The mandatory **Enumeration** property references an ER format enumeration whose value the configured data source returns.

Format enumeration

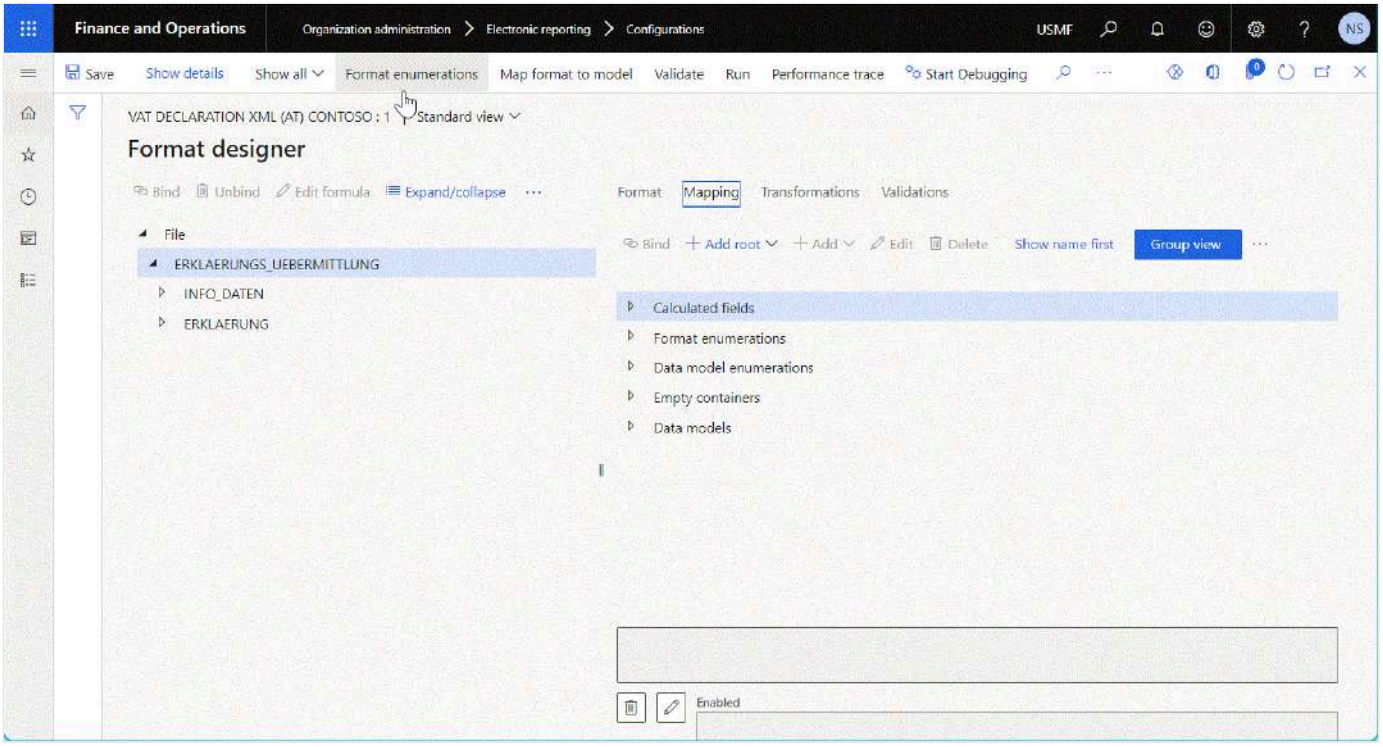
The **Format enumeration** data source represents the enumeration of the editable format.

- On the **Format enumeration** FastTab, select **Add** to add a new format enumeration.

- On the **Format enumeration values** FastTab, select **Add** to add a new value for the selected format enumeration.

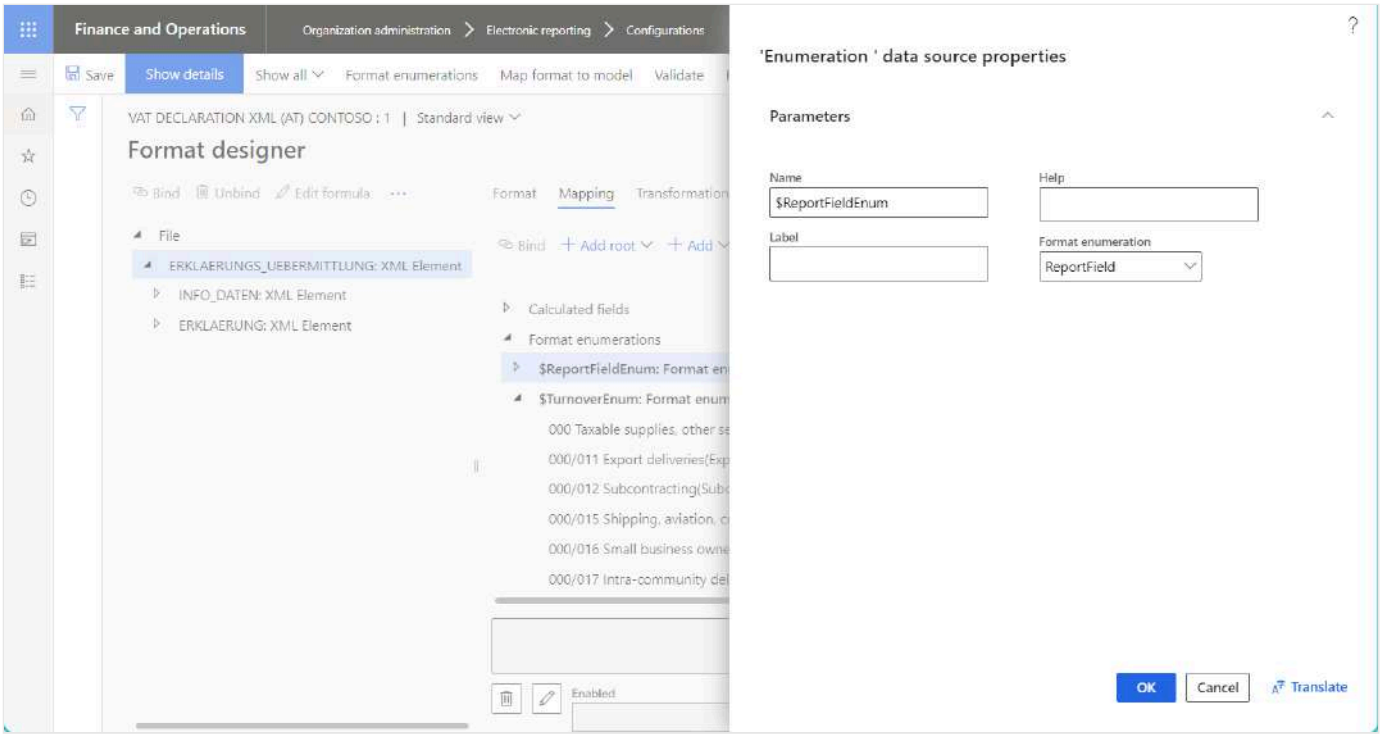
> **Tip**
>
> The **Label** and **Help** properties of format enumerations and format enumeration values are translatable.

The following illustration shows a format enumeration that you configured for the editable format component.



You usually need this data source when you configure an ER expression to compare a specific format enumeration value with a value of the relevant enumeration that another data source returns.

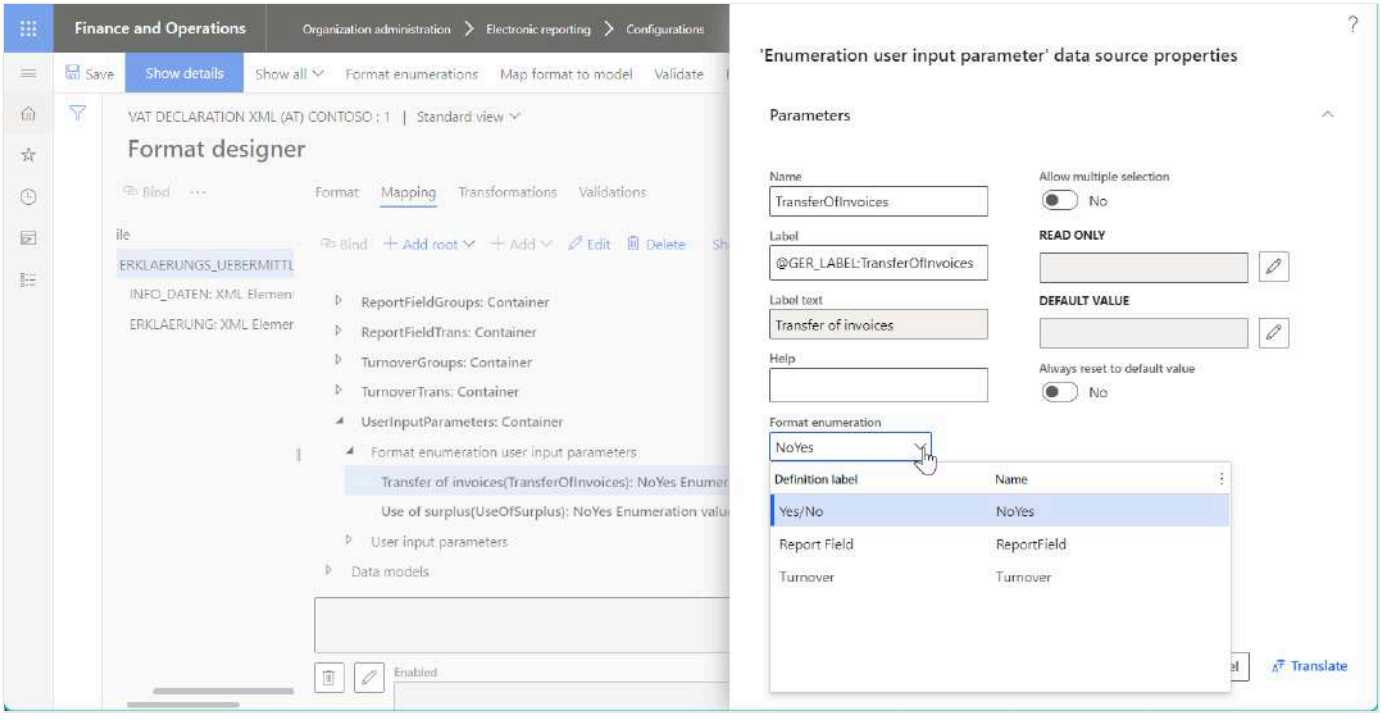
The following illustration shows the properties for a data source of the *Format enumeration* type in the **Data source properties** dialog box.



Format enumeration user input parameter

The **Format enumeration user input parameter** data source represents the enumeration of the editable format. You usually need this data source when you want to specify a value of the referenced enumeration in the related data entry field in the dialog box at runtime, before an ER format runs.

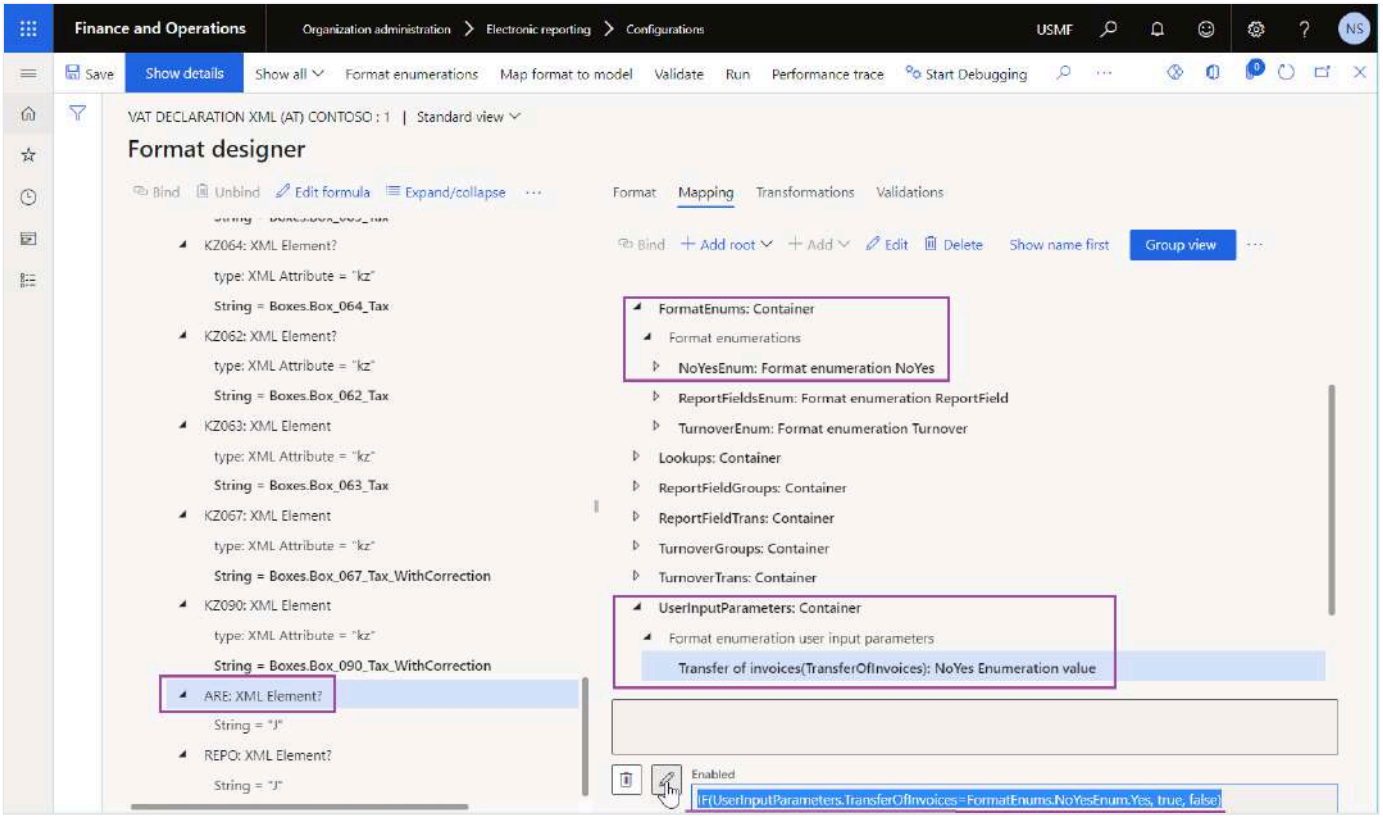
The mandatory **Format enumeration** property of the **Format enumeration user input parameter** data source points to an enumeration of the editable format to specify what values of the referred format enumeration this data source returns.



Set the **Allow multiple selection**, **Read only**, **Default value**, and **Always reset to default value** properties as described earlier for the *Model enumeration user input parameter* data source type.

> **Tip**
>
> If you leave the **Label** or **Help** property blank for a data source of the *Format enumeration user input parameter* type, the appropriate property of the referenced format enumeration is used, if it's configured.

The following illustration shows an ER expression that's configured to enable an ER format element, depending on the result of comparison of values of data sources of the *Format enumeration user input parameter* and *Format enumeration* types.



Format enumeration lookup

Use the **Format enumeration lookup** data source to configure an ER data source of the *Lookup* type to return the values of an ER format enumeration.

Configure a data source of the *Format enumeration lookup* type to filter data in an ER format so that it's based on a set of abstract rules. You can then specify real rules outside the ER component designers by using the UI that's automatically generated based on the settings of the **Lookup** data source of the corresponding ER format and the current legal entity data. When you run this ER format, the specified rules are accessed by the ER format's **Lookup** data source.

The mandatory **Format enumeration** property of the **Format enumeration lookup** data source points to a format enumeration of the editable format to specify what values of the referenced format enumeration this data source returns.

Set the **Cross-company** and **Extended** properties as described earlier for the *Model enumeration lookup* data source type.

Format group

Export format

Use the **Export format** data source to run another ER format during the execution of the editable format. The execution results of the nested ER format serve as a data source for the execution of the editable ER format.

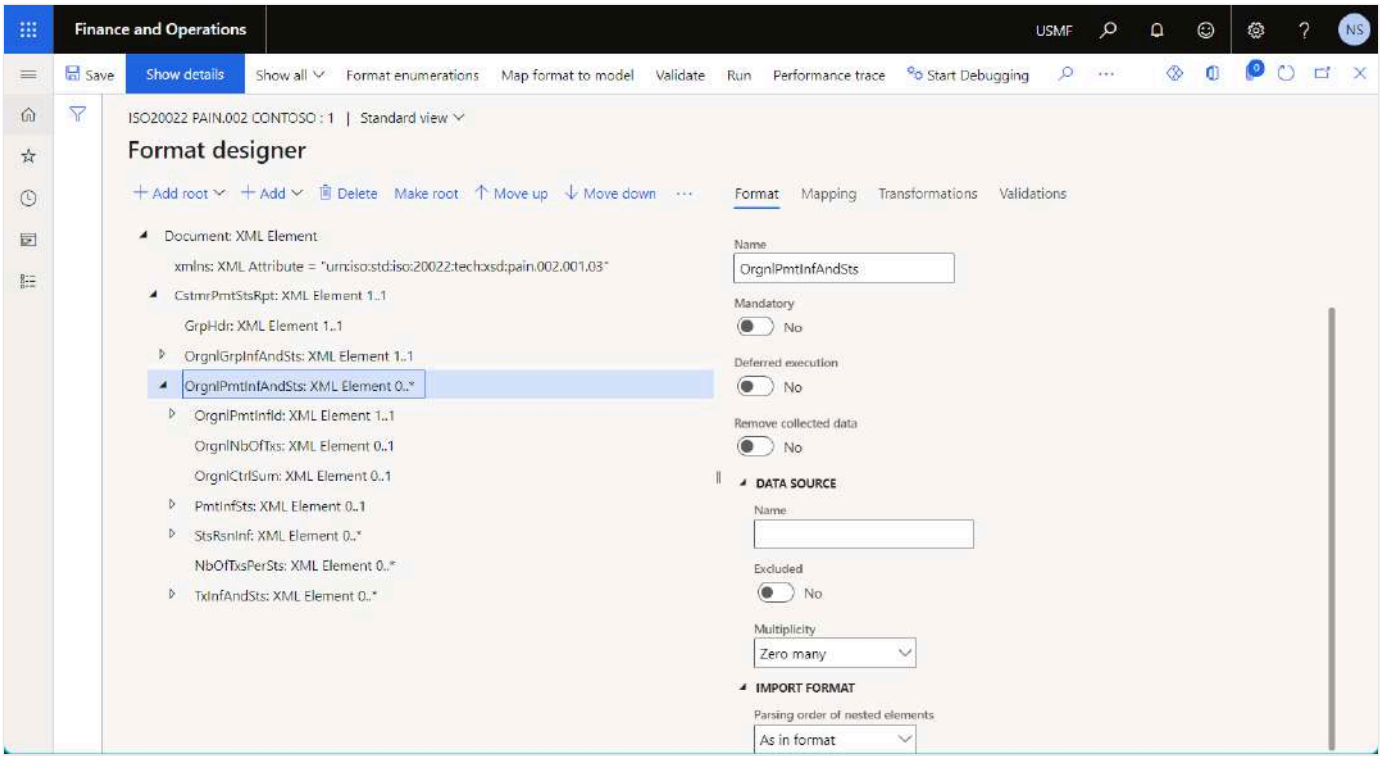
Use the mandatory **Format** property to specify an ER format that must run during the execution of the editable format.

Import format

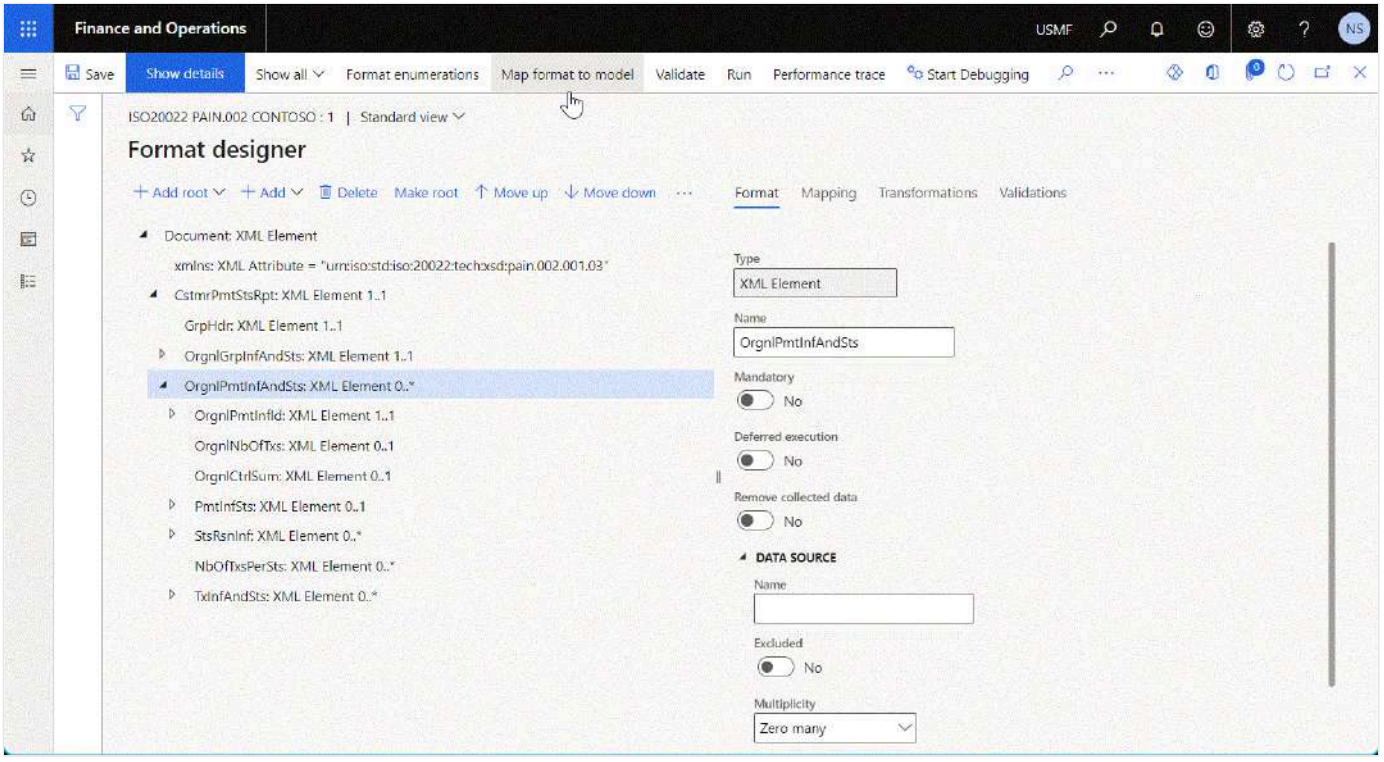
Use the **Import format** data source to parse an inbound business document. The results of the inbound document parsing can then be used in the format mapping to fill a data model.

Use the mandatory **Format** property to specify an ER format that must be used for inbound document parsing.

The following illustration shows the ER format that's configured to parse an inbound document in XML format.



The following illustration shows the model mapping that contains a data source of the *Import format* type to expose data of a parsed inbound document and then use that data to fill a data model.



For more information, see Parse incoming documents.

Dynamics 365 Finance group

Class

Use the **Class** data source to call a static method of a public class from your ER solution. You can dynamically define argument values for calling classes at runtime. To invoke a class, configure a data source of the *Calculated field* type with an expression that references the desired class of application source code. For more information, see Design ER expressions to call application class methods.

You can add a data source of the *Class* type in the editable **Model mapping** component. If you identify the editable model mapping as the preferred one for execution, set the **Integration point** option of the **Class** data source to **Yes**. For more information about this option, see API to run a format mapping for the generation of outbound documents.

Object

Use the **Object** data source to call a public method of a class instance from your ER solution. You can dynamically define argument values for calling classes at runtime. To invoke a class, configure a data source of the *Calculated field* type with an expression that references the desired class of application source code.

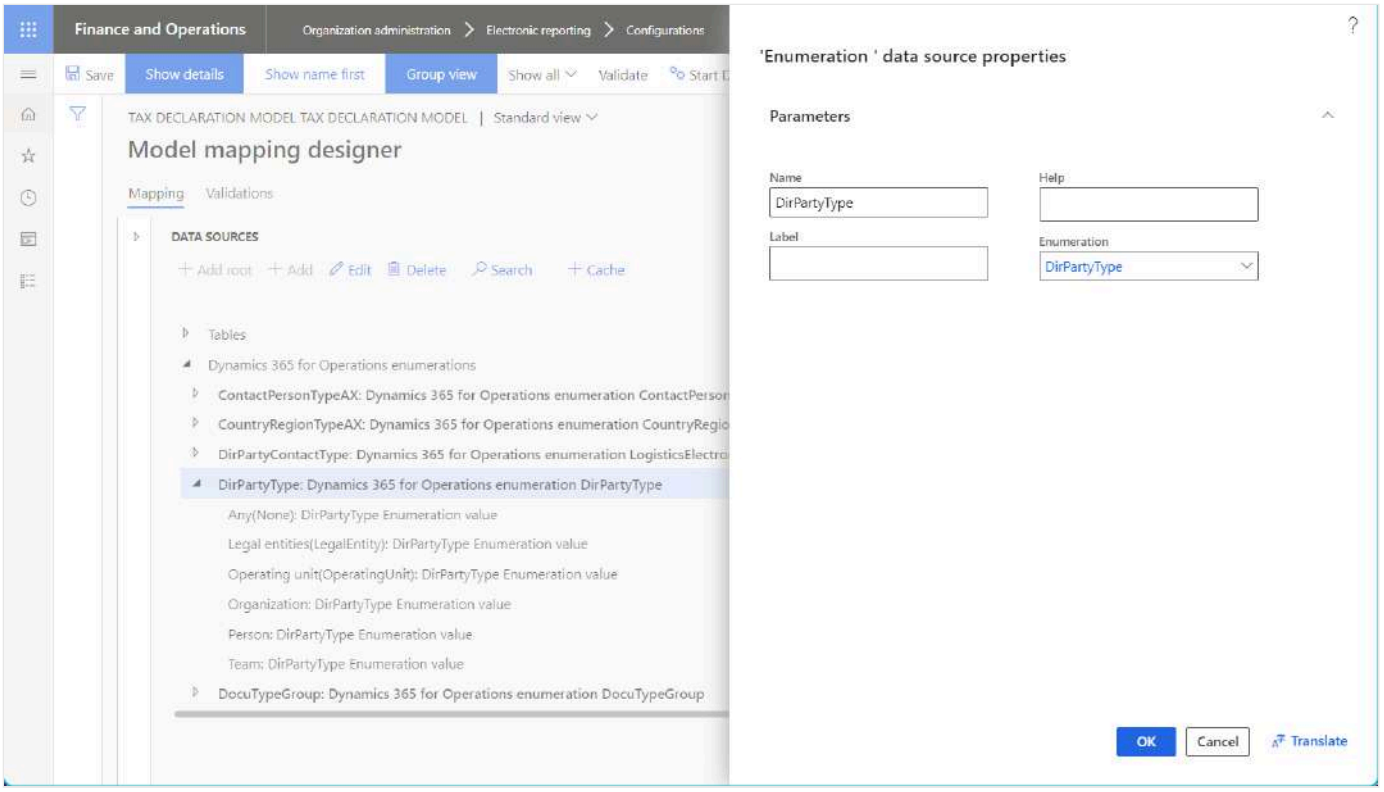
Add a data source of the *Object* type in the editable **Model mapping** component. If you identify the editable model mapping as the preferred one for execution at runtime, set the **Integration point** option of this **Object** data source to **Yes**.

Enumeration

The **Enumeration** data source represents the application enumeration. You usually need this data source when you configure an ER expression to compare a specific application enumeration value with a value of the relevant enumeration that another data source returns.

Use the mandatory **Enumeration** property to specify an application enumeration.

The following illustration shows the properties for a data source of the *Application enumeration* type in the **Data source properties** dialog box.



Application enumeration lookup

Use the **Application enumeration lookup** data source to configure an ER data source of the *Lookup* type to return the values of an application enumeration.

Configure a data source of the *Application enumeration lookup* type to filter data in an ER format so that it's based on a set of abstract rules. You can then specify real rules outside the ER component designers by using the UI that's automatically generated based on the settings of the **Lookup** data source of the corresponding ER format and the current legal entity data. When you run this ER format, the specified rules are accessed by the ER format's **Lookup** data source.

The mandatory **Operations data type name (EDT, enum)** property of the **Application enumeration lookup** data source points to an application enumeration or an application extended data type (EDT) to specify what values are returned by this data source.

Set the **Cross-company** and **Extended** properties as described earlier for the *Model enumeration lookup* data source type.

Number sequence

Use the **Number sequence** data source to return a value from the referenced number sequence for the current application instance. When a data consumer calls this data source, the referenced number sequence increments and returns the received value returned.

You must specify the mandatory **Number sequence code** property to identify the required number sequence.

In addition to **Number sequence** data sources, you can use the built-in ER **NUMSEQVALUE** function.

Table

Use the **Table** data source to call static methods of an application table, entity, or view. You call these static methods in the same way that you call static methods of an application class, as described earlier.

You must specify the mandatory **Table** property to identify the required table, entity, or view.

Table records

Use the **Table records** data source to fetch records from an application table, entity, or view.

You must specify the mandatory **Table** property to identify the required table, entity, or view.

If the table, entity, or view that you reference in the **Table** property keeps records company-specific, specify the **Cross-company** property.

- Set this property to **No** to fetch records only for the company that an ER solution runs in scope of.
- Set this property to **Yes** to fetch records for all companies of the current application instance.

Set the **Ask for query** option to **Yes** to enable the system query dialog box at runtime, so that users can filter data for the configured data source. For more information, see Cross-company data sources in Electronic reporting (ER).

If you identify the editable model mapping as the preferred one for execution, set the **Integration point** option to **Yes**.

Use the **Select fields** function to limit the list of fields whose values are fetched at runtime. For more information, see Improve performance of ER solutions by reducing the number of table fields that are fetched at runtime.

Use the **Refill table** function whenever you add or update Application Object Tree (AOT) artifacts. This function resembles the **Rebuild table references** function.

General group

Empty container

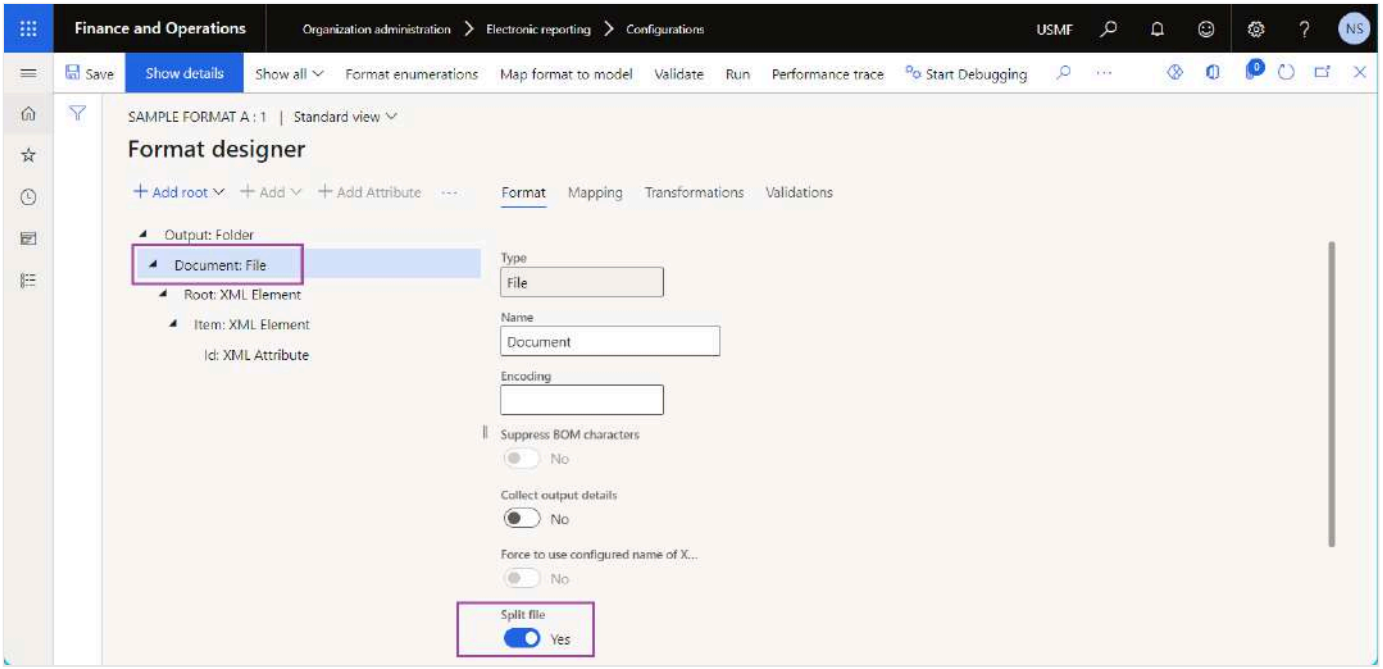
Use the **Empty container** data source as the container for other data sources. Use it to organize your data sources or to make groups of data sources of different types for specific purposes.

In addition, you can use an **Empty container** data source as the mockup of the desired record. Then, by using the LIST function, you can limit the list of fields for the final list of records.

Endless list

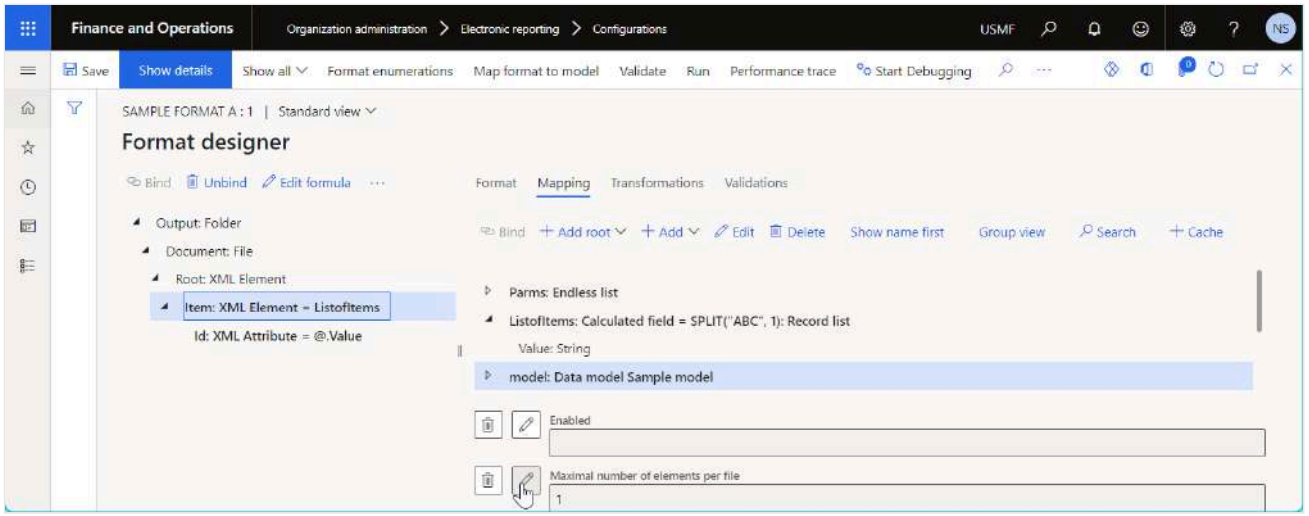
You can configure an ER format to split generated XML output into multiple files whenever specific limits are exceeded. Use the **Endless list** data source to uniquely name those outbound files.

The following illustration shows a sample ER format that's configured to generate zipped XML output that must be split into multiple files.

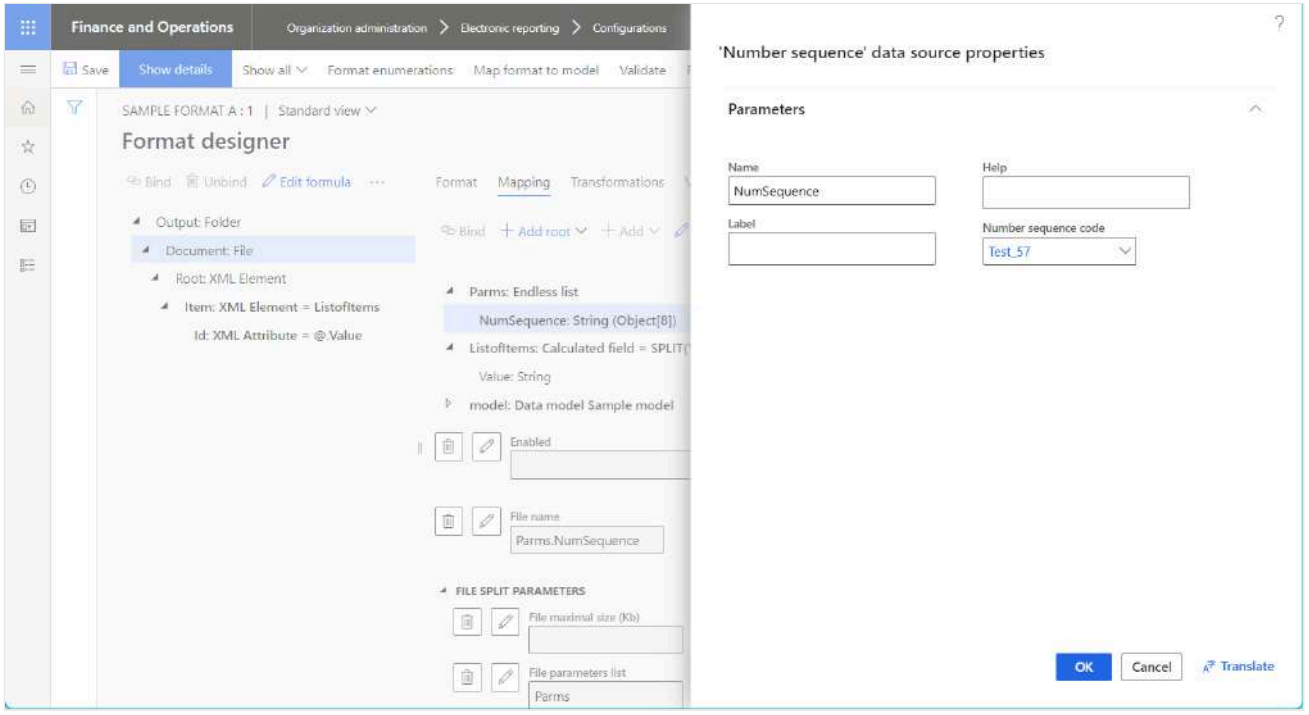


The following illustrations show the bindings of the sample ER format:

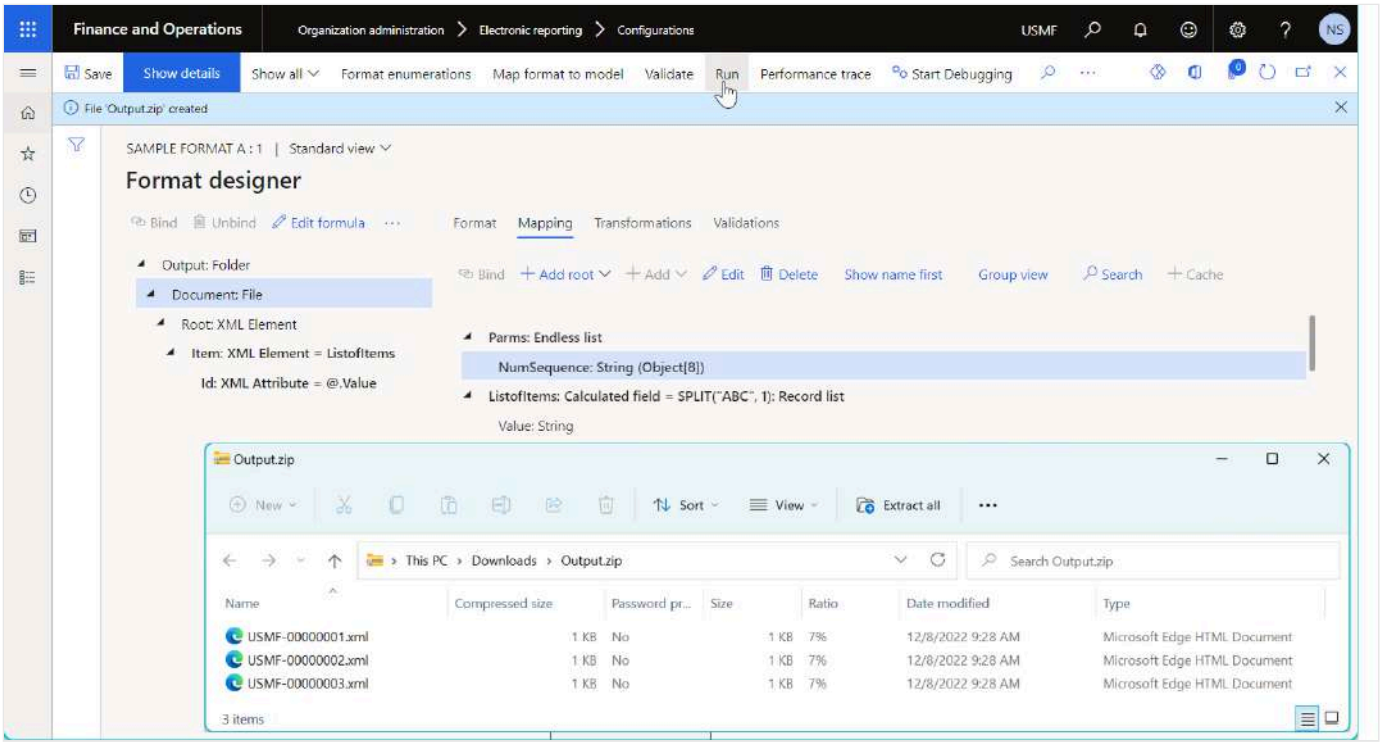
- The **Item** format element is bound to the `ListofItems` data source that returns three records. The **Maximal number of elements per file** property of this format element is configured to split generated output into multiple XML files, each of which must contain only one `Item` XML node.



- The **File parameters list** property of the **File** format element references the `Parms` data source of the *Endless list* type to ensure that this data source is invoked whenever the output split occurs. Because the `Parms` data source contains the nested `NumSequence` data source of the *Number sequence* type, the application number sequence that's referenced in `NumSequence` is also iterated for each split. Therefore, the `Parms.NumSequence` ER expression that's configured in the **File name** property of the **File** component generates a unique name for each output file, based on the current value of the referenced number sequence.



The following illustration shows the result of running the previously described sample ER format. The result represents a zip file that contains XML output that's split into three uniquely named XML files.



For more information, see Split generated XML files based on file size and content quantity.

User input parameter

Use a data source of the *User input parameter* type to let users specify a value of the referenced application enumeration or application EDT in the related data entry field in the dialog box at runtime, before an ER format runs.

The mandatory **Operations data type name (EDT, enum)** property of the **User input parameter** data source points to an application enumeration or an application EDT to specify what values this data source returns.

Set the **Allow multiple selection**, **Read only**, **Default value**, and **Always reset to default value** properties as described earlier for the *Model enumeration user input parameter* data source type.

Use the **Edit visibility** function for the *Model enumeration user input parameter* data source type.

Functions group

Barcode

Use a data source of the *Barcode* type to generate an image that represents the bar code for specified text.

The mandatory **Barcode format** property of the **Barcode** data source specifies the bar code type. The following bar code formats are currently supported:

- One-dimensional bar codes:
  - Codabar
  - Code 39
  - Code 93
  - Code 128
  - EAN-8
  - EAN-13
  - ITF-14
  - Intelligent Mail
  - MSI
  - Plessey
  - PDF417
  - UPC-A
  - UPC-E
- Two-dimensional bar codes:
  - Aztec
  - Data Matrix
  - QR Code

The optional **Width** property specifies the bar code's width in pixels. A value of **0** (zero) indicates that the default width is used. The meaning can vary for different formats.

The optional **Height** property specifies the bar code's height in pixels. A value of **0** (zero) indicates that the default height is used. The meaning can vary for different formats.

The optional **Margin** property specifies the size of the bar code's margin in pixels. The margin is the area on each side of a bar code that must be kept clear (quiet zone). A value of **0** (zero) indicates that the default margin is used. The meaning can vary for different formats.

The **Output content** option specifies whether a bar code image contains the encoded information as text. Set this option to **Yes** to generate a bar code image that contains the encoded information as text. The default value is **No**.

The mandatory **Encoding** property specifies the type of characters that are encoded in the generated bar code image. By default, the **UTF-8** encoding is used.

> **Important**

When you add a new **Barcode** data source, place it under another item (container) as a nested element. The **Barcode** data source appears in the configured ER component as a function that has the **Value** parameter of the ***String*** type. Provide the bar code text as the argument of this parameter.

To learn more, see Use Barcode data sources to generate bar code images.

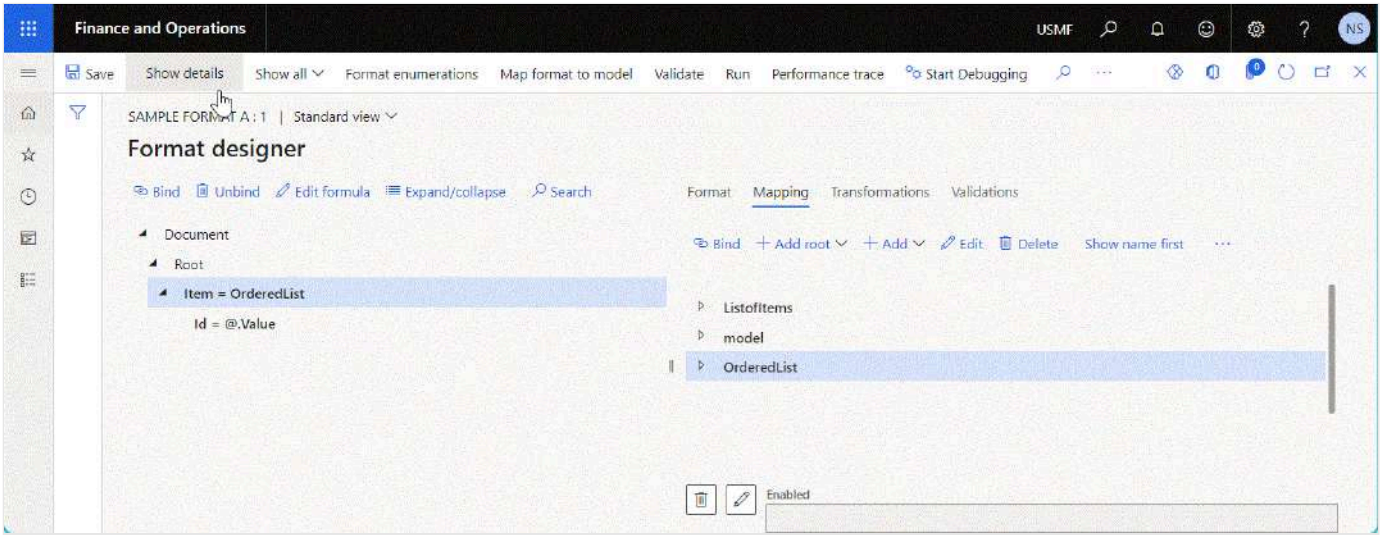
Calculated field

Use a data source of the *Calculated field* type to configure an ER expression for transforming values of other data sources.

Use the **Edit formula** function to configure an ER expression of the editable data source of the *Calculated field* type.

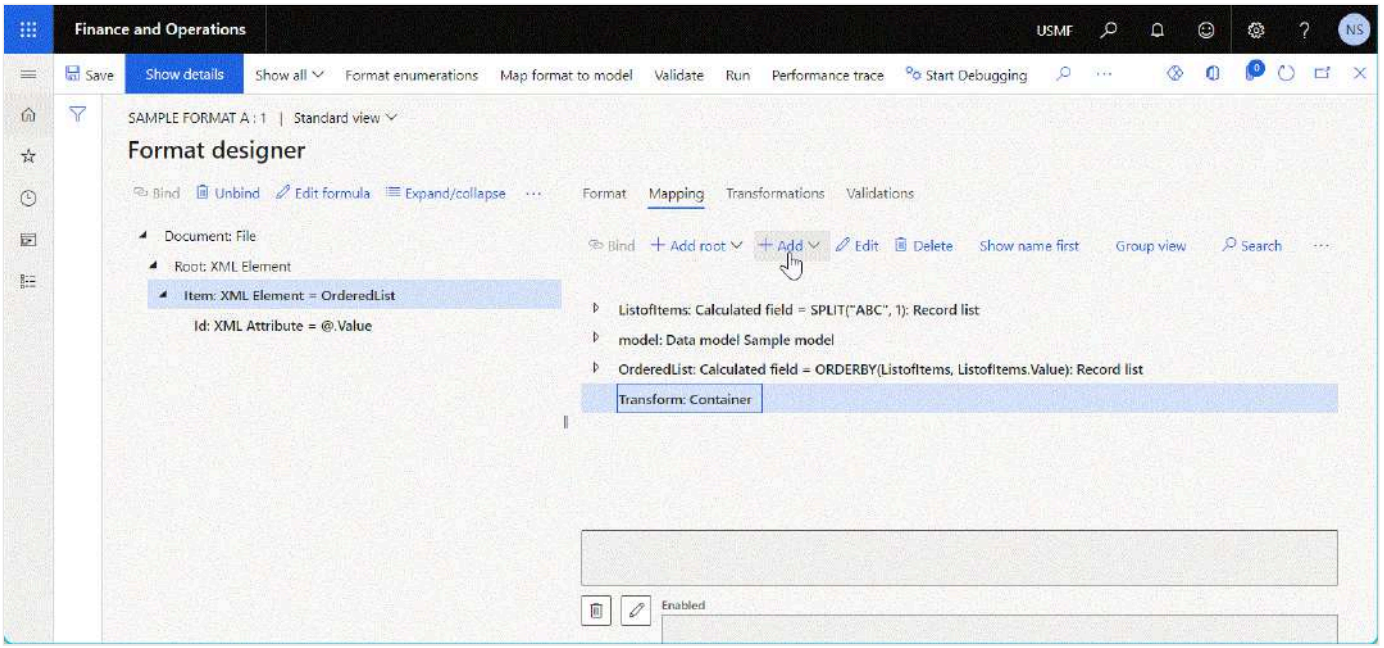
> **Tip**
>
> Turn on the **Show details** toggle button to preview configured expressions for data sources of the *Calculated field* type in the **Data sources** pane of the **Model mapping designer** page and on the **Mapping** tab of the **Format designer** page.

The following illustration shows the **Calculated field** data source that sorts the initially provided list of records in alphabetic order.

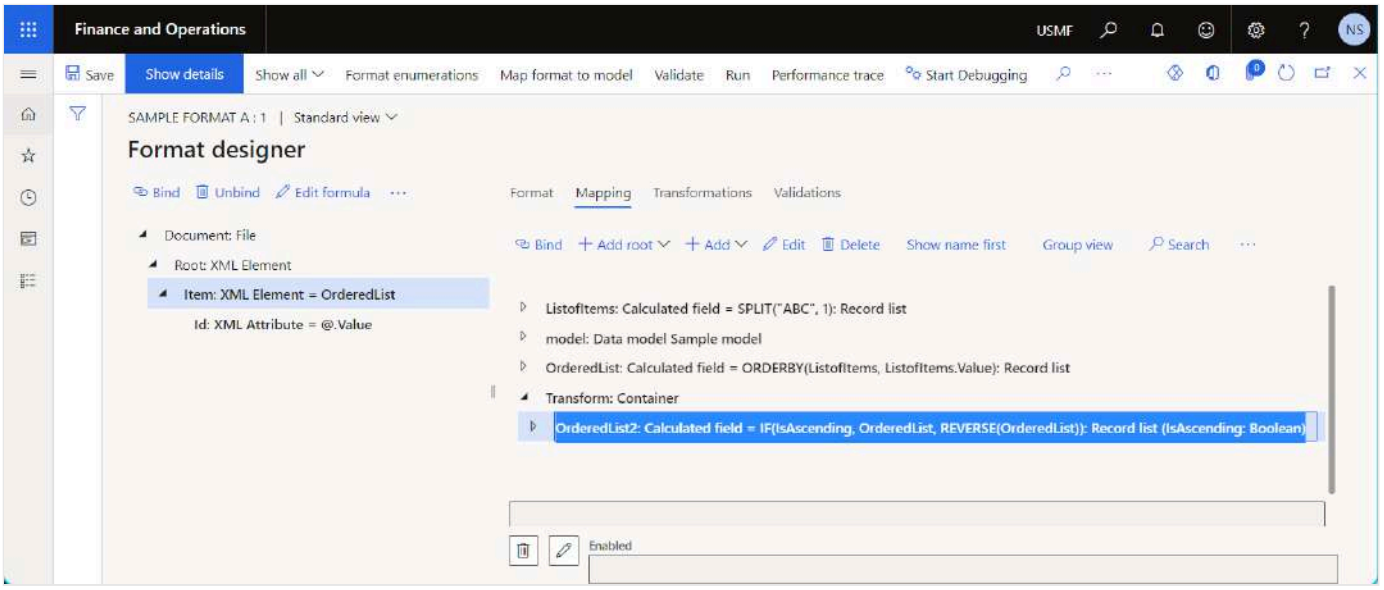


When you add a new **Calculated field** data source by placing it under another item as a nested element, you can specify multiple parameters that you can then reference in the configured ER expression. You can also configure other formulas to call this data source. In this way, you can control the values of arguments of those parameters. Therefore, you can reuse a single data

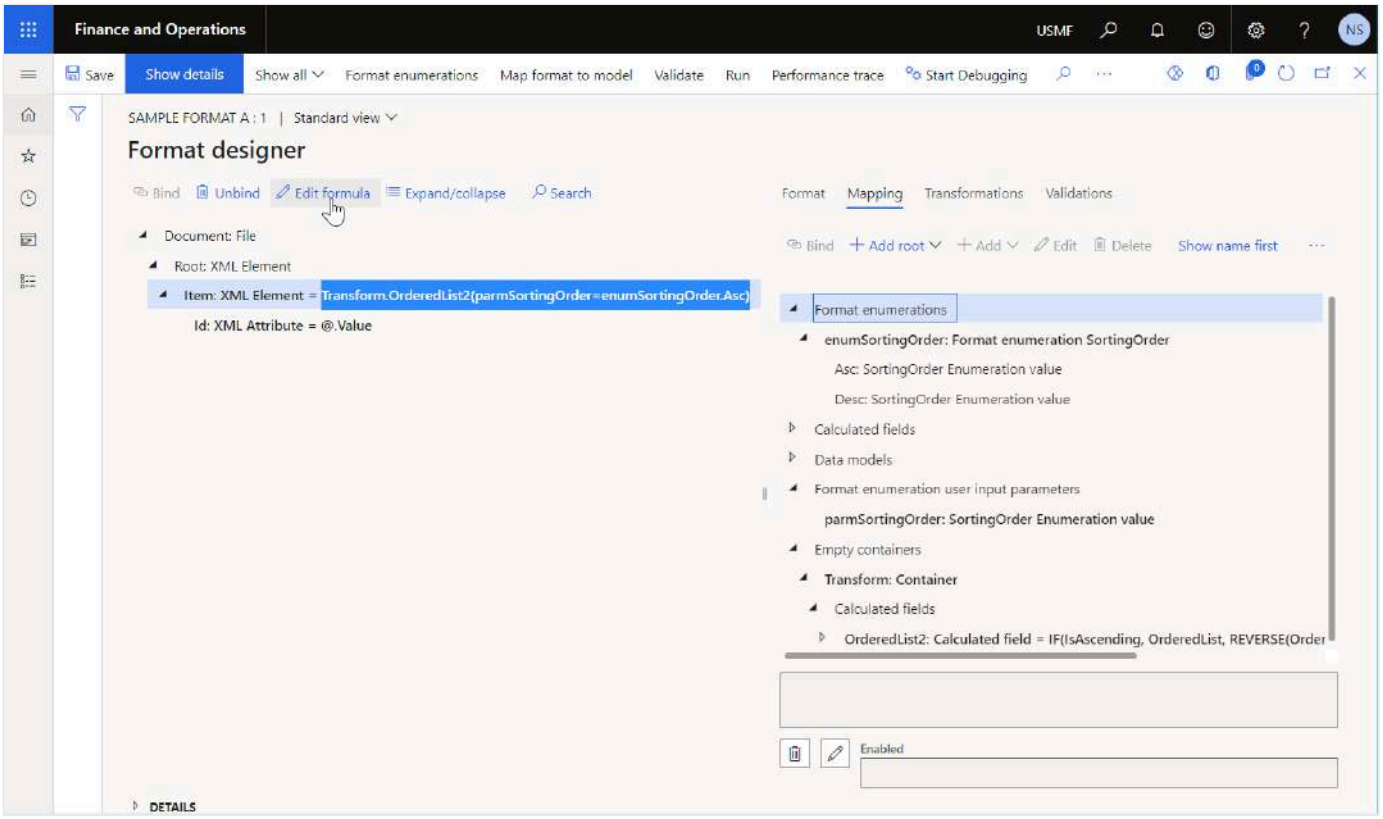
source in many bindings and reduce the total number of data sources that you must configure in an ER model mapping or ER format component.



The following illustration shows the **Calculated field** data source that sorts the initially provided list of records according to the sort direction that's specified at runtime.



The sort order of this data source is specified by using the value of the argument of the data source parameter.



For more information, see Support parameterized calls of ER data sources of the Calculated field type.

Data collection

You can configure an ER model mapping or an ER format to process specific elements, such as records of a model mapping data source or format elements, in a specific order. Additionally, you can configure a data source of the *Data collection* type to collect, during element processing at runtime, values of data model fields or values of bindings of the processed format elements, and then sum those values. The data source then uses the collected values to fill in a generated document or perform other calculations.

The mandatory **Item type** property of the **Data collection** data source specifies the type of collected data. The following data types are currently supported:

- *Boolean*
- *Date*
- *DateTime*
- *GUID*
- *Int64*
- *Integer*
- *Real*
- *String*

- *Time*

By default, a **Data collection** data source collects only unique values. To access the list of collected values, use the **Result** property of a **Data collection** data source. This property returns a record list. The records of the record list contain the **Value** field that you can use to access collected values.

To force a data source of the *Data collection* type to collect **all** values, follow these steps:

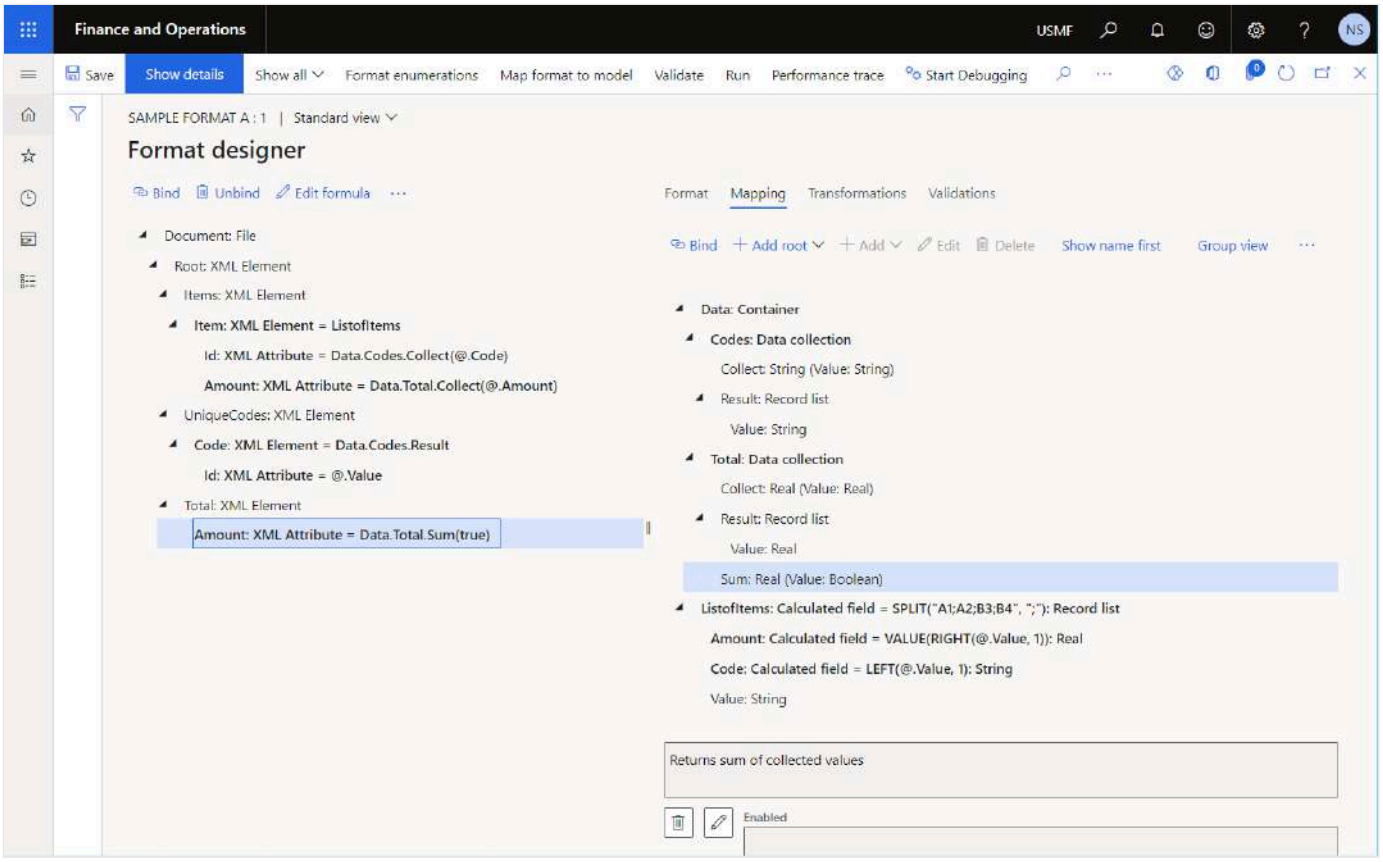
1. Select one of the following data types for the **Item type** property: *Int64*, *Integer*, or *Real*.
2. Set the **Collect all values** option of the configured **Data collection** data source to **Yes**.

When you set the **Collect all values** option to **Yes**, the **Sum(Flag)** parameterized property becomes available. Use this property to get the total amount of all currently collected values. In this property, the **Flag** argument is a *Boolean* value that indicates whether the total value must be reset.

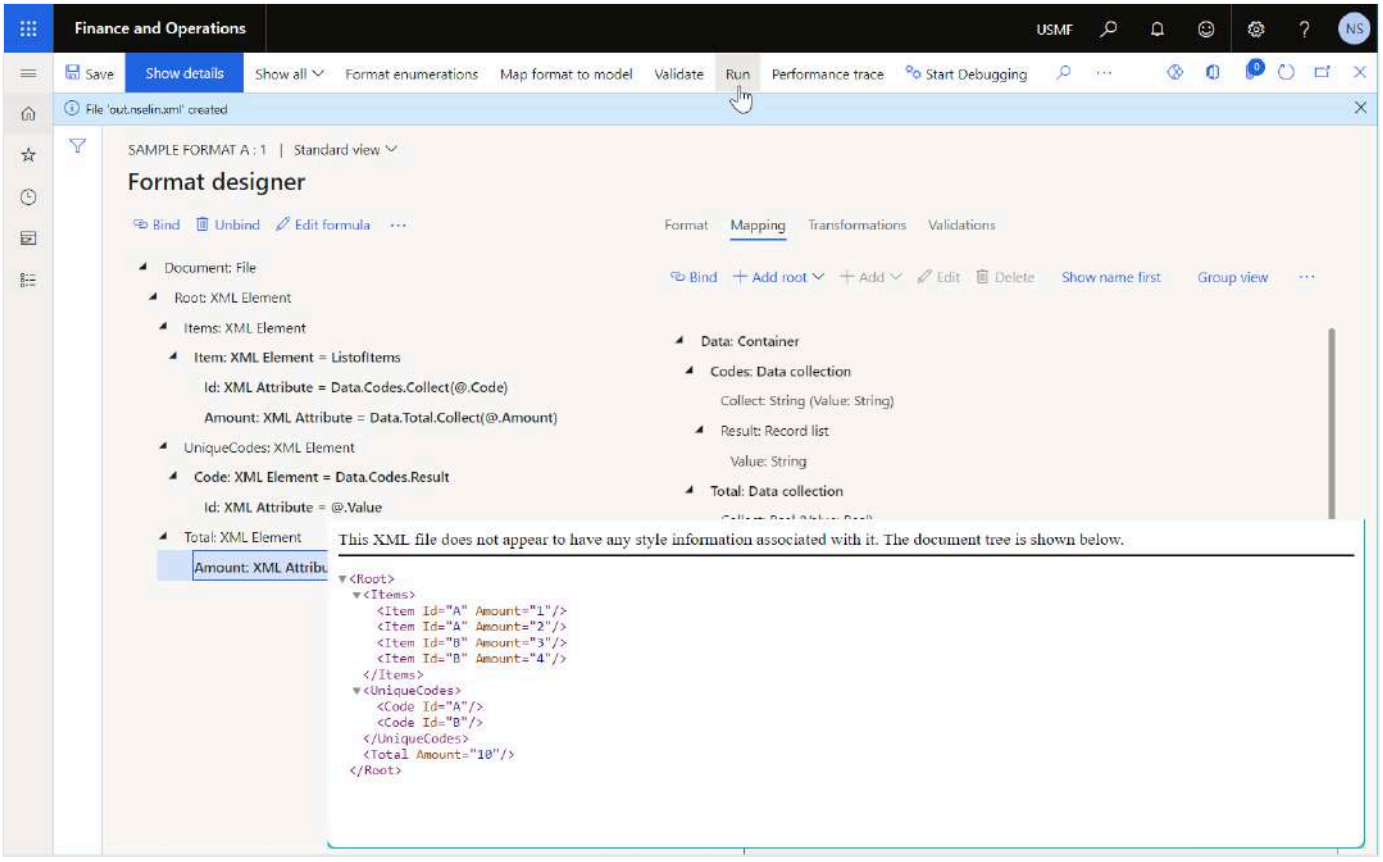
- If the value *False* is provided, summation continues from the previously collected amount.
- If the value *True* is provided, a new summation starts.

The following illustration shows the sample ER format that has two data sources of the *Data collection* type:

- The `Data.Codes` data source is configured to collect unique codes that are put in a generated document. The configured `Data.Codes.Result` binding is then used to put the list of unique codes in the footer of the generated document.
- The `Data.Total` data source is configured to calculate the total of values that are put in a generated document. The configured `Data.Total.Sum(true)` binding is then used to put this total value in the footer of the generated document.



The following illustration shows the result of running the previously described sample ER format that has two data sources of the *Data collection* type.



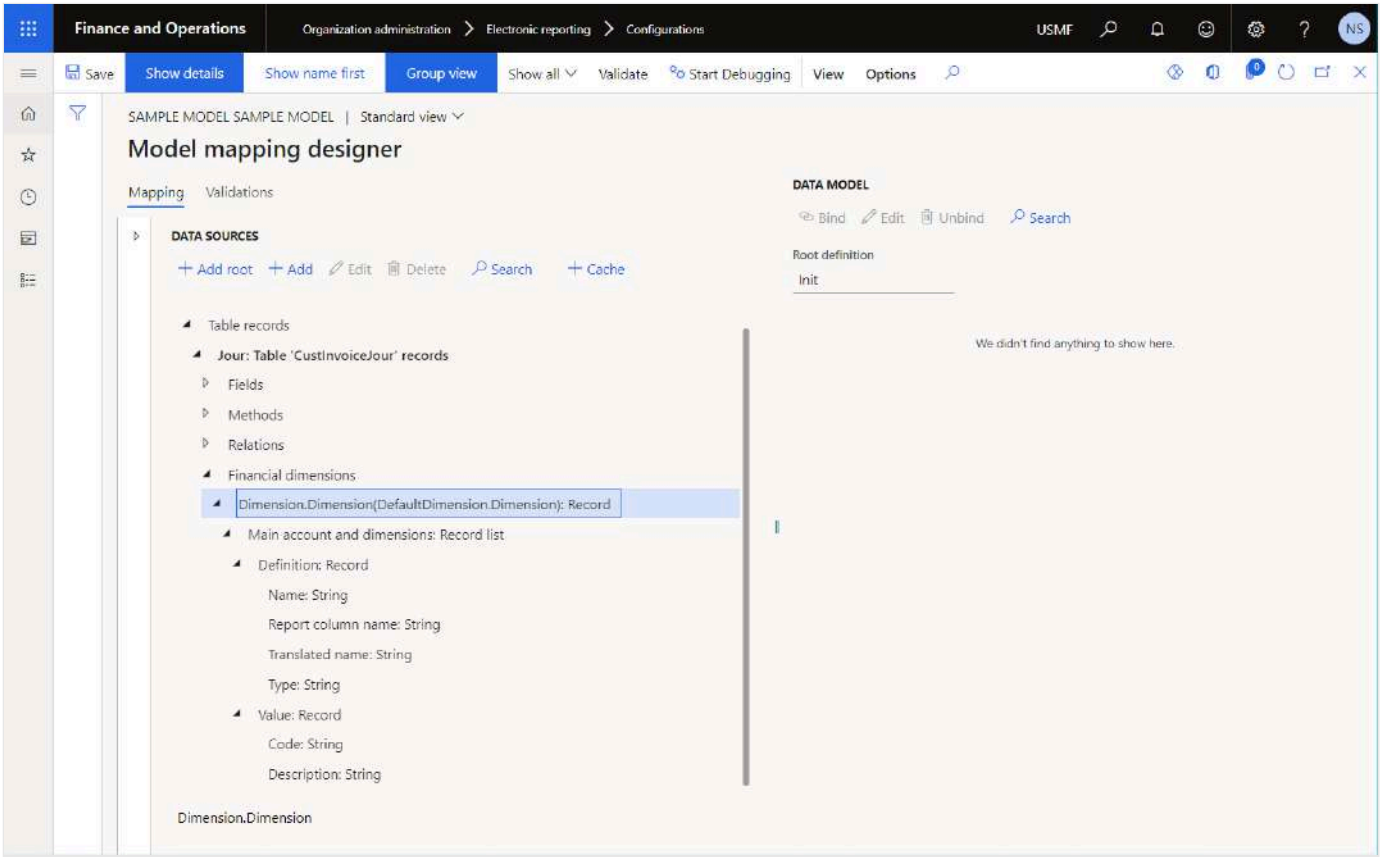
For more information, see Use DATA COLLECTION data sources in Electronic reporting formats.

Financial dimensions details

When you configure a data source of the *Table record* type that references an application table that supports the relation to financial dimensions, you see that this ER data source offers a field of the *Financial dimensions* type for every financial dimension relation. This field exposes the **Main account and dimensions** record list that enables access to financial dimensions as the list of records. Every record in this list represents a single dimension.

- The **Definition** record lets you access financial dimension metadata.
  - The **Name** field exposes the name of a financial dimension.
  - The **Report column name** field exposes the report alias of a financial dimension.
  - The **Translated name** field exposes the name of a financial dimension in the language that's specified by the language context of the ER solution that's running.
- The **Value** record lets you access a financial dimension value.
  - The **Code** field returns a financial dimension code.
  - The **Description** field returns a financial dimension name.

The following illustration shows a field of the *Financial dimensions* type of the **Table record** data source that you can use to access default financial dimensions for every record in the **CustInvoiceJour** application table.



Use the **Financial dimensions details** data source in the configured ER component to specify how the scope of financial dimensions is defined for any report that uses this component when a data consumer accesses a field of the *Financial dimensions* type. If you don't configure the **Financial dimensions details** data source in the editable ER component, the report uses all financial dimensions of the current application instance.

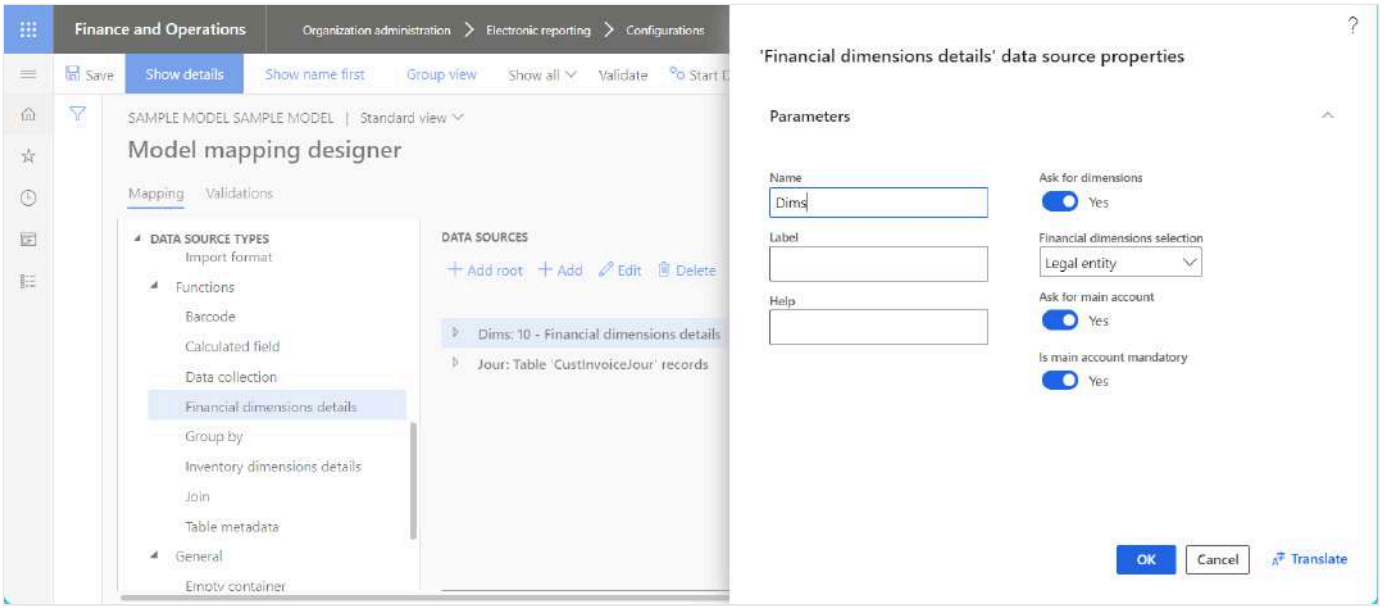
Set the **Ask for dimensions** option of the configured **Financial dimensions details** data source to **Yes** to let the user select dimensions in the dialog box at runtime. If you set the option to **No**, the report uses all financial dimensions of the current application instance by default.

When you set the **Ask for dimensions** option to **Yes**, the **Financial dimensions selection** option is enabled, and you can specify the scope of financial dimensions. The following options are available:

- Select **All** to use all financial dimensions of the current application instance.
- Select **Legal entity** to use only the financial dimensions of the company that an ER solution runs in the scope of.
- Select **Dimension set** to use only the financial dimensions of the dimension set that the user specifies in the dialog box at runtime.

Set the **Ask for main account** option to **Yes** to let the user select the main account as part of the list of dimensions in the dialog box at runtime. If you set the option to **No**, the main account isn't included in the list of dimensions, and the **Is main account mandatory** option is enabled. If you set the **Is main account mandatory** option to **Yes**, the main account is included in the list of dimensions, regardless of the user's selection.

The following illustration shows the data source of the *Financial dimensions details* type that you can use to specify the scope of financial dimensions for fields of the *Financial dimensions* type when the configured model mapping is used at runtime.



To learn more, complete the procedure in the following "ER Use financial dimensions as a data source" articles:

- Part 1 - Design data model
- Part 2 - Model mapping
- Part 3 - Design the report
- Part 4 - Run the report

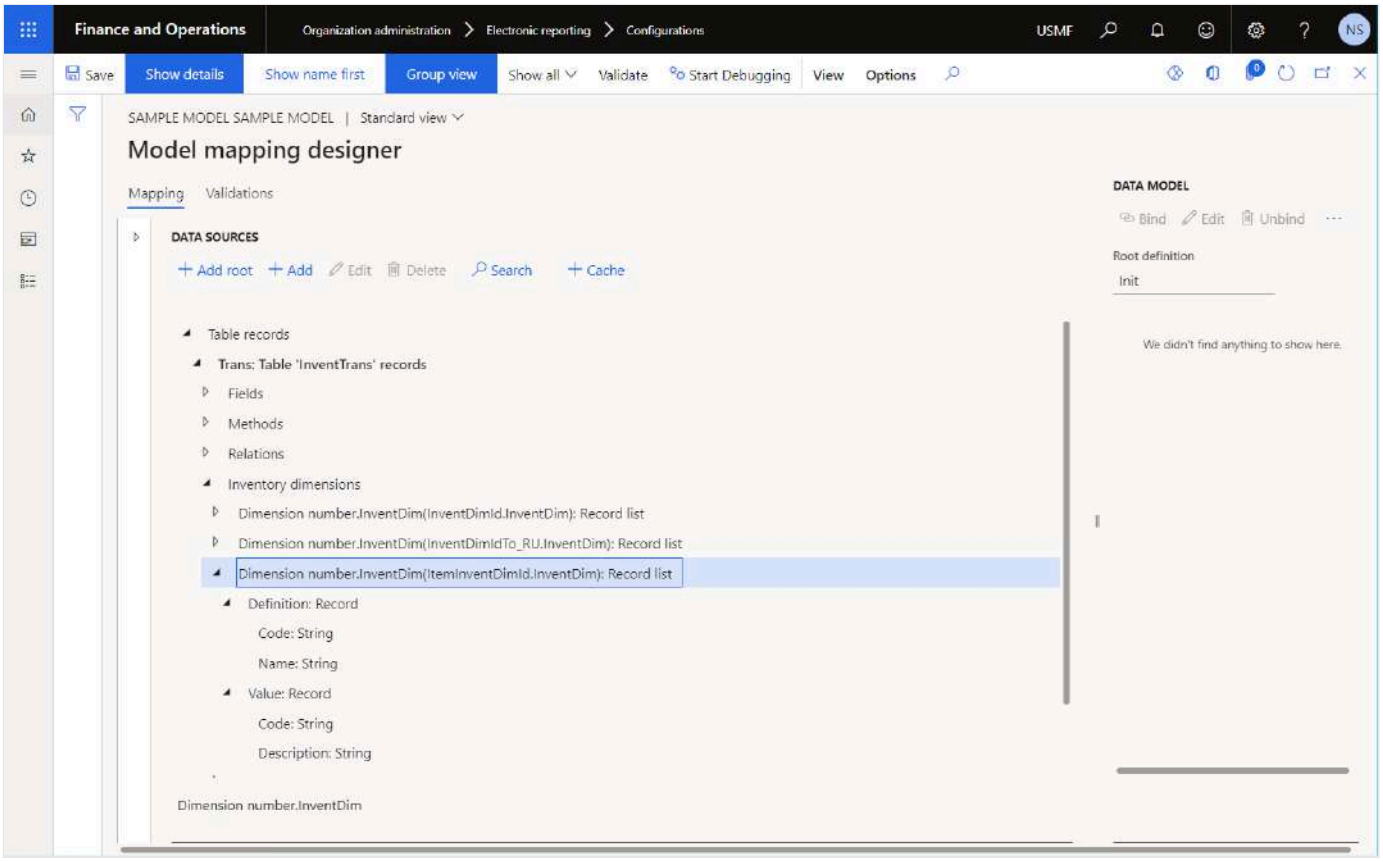
Inventory dimensions details

When you configure a data source of the *Table record* type that references an application table that supports the relation to inventory dimensions, you see that this ER data source offers a field of the *Inventory dimensions* type for every inventory dimension relation. This field exposes the **InventDim** record list that enables access to inventory dimensions as the list of records. Every record in this list represents a single dimension.

- The **Definition** record lets you access inventory dimension metadata.
  - The **Code** field exposes the code of an inventory dimension.
  - The **Name** field exposes the name of an inventory dimension.

- The **Value** record lets you access an inventory dimension value.
  - The **Code** field returns an inventory dimension code.
  - The **Description** field returns an inventory dimension name.

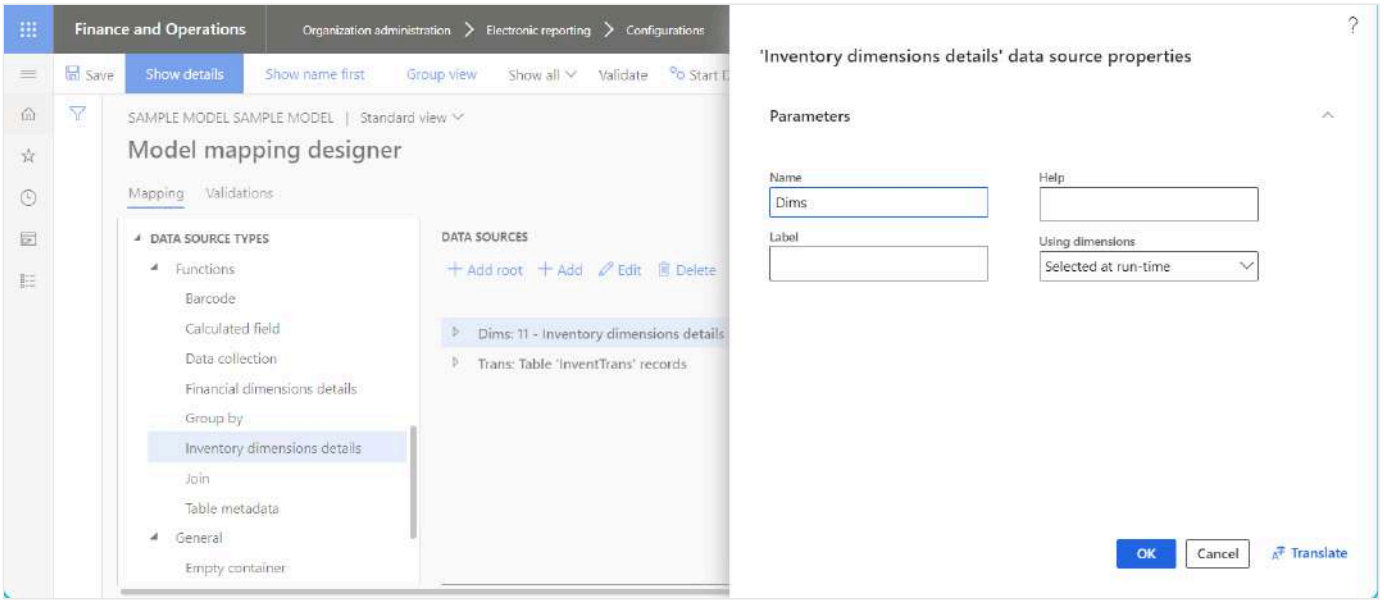
The following illustration shows a field of the *Inventory dimensions* type of the **Table record** data source that you can use to access several sets of inventory dimensions for every record in the **InventTrans** application table.



Use the **Inventory dimensions details** data source in the configured ER component to specify how the scope of inventory dimensions is defined for any report that uses this component when a field of the *Inventory dimensions* type is accessed by a data consumer. Set the **Using dimensions** property to specify the scope of inventory dimensions. The following options are available:

- Select the **All supported** option to use all inventory dimensions of the current application instance.
- Select the **Selected at run-time** option to use only the inventory dimensions that the user specifies in the dialog box at runtime.

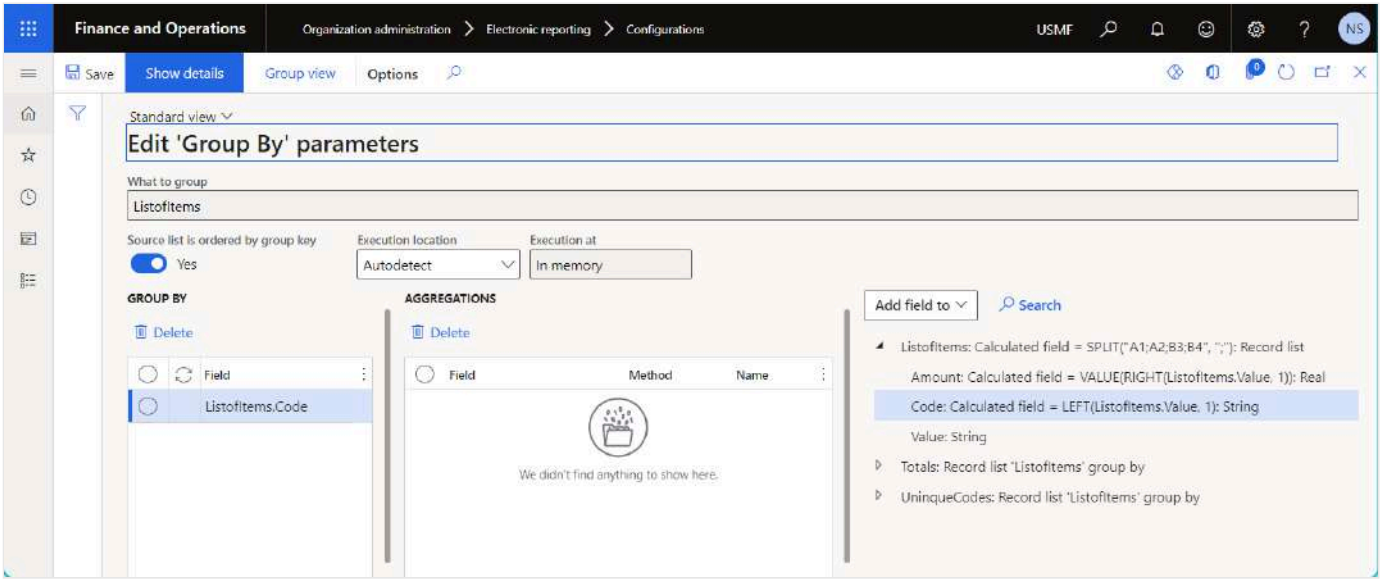
The following illustration shows the data source of the *Inventory dimensions details* type that you can use to specify the scope of inventory dimensions for the **Inventory dimensions* fields when the configured model mapping is used at runtime.



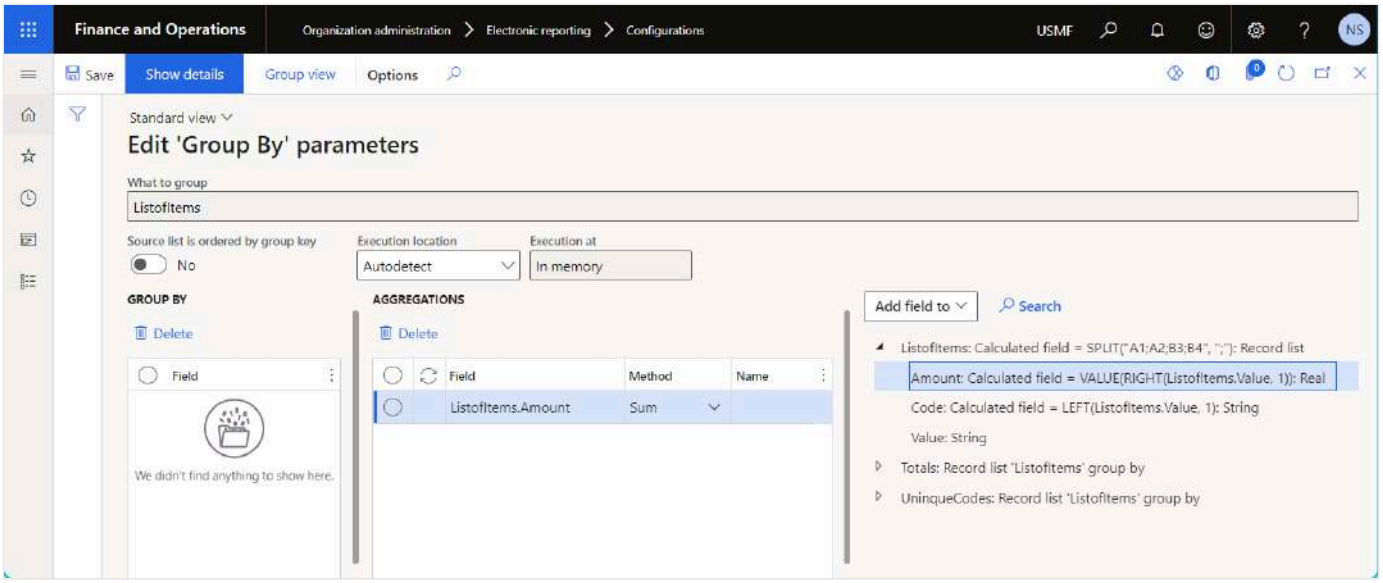
Group by

You can configure the **Group by** data source to group records of the base data source of the *Record list* type. For every generated group of records, this data source can calculate different aggregate functions.

The following illustration shows a data source of the *Group by* type that's set up to group records by code value.



The following illustration shows a data source of the *Group by* type that's set up to calculate the total amount for the initial list of records.



Follow these steps to configure grouping parameters on the **Edit Group By parameters** page.

- To specify the base data source for grouping, select the desired data source on the right side of the page, then select **Add field** > **What to group**.
- To specify a field for grouping, select the desired field of the specified base data source on the right side of the page, then select **Add field to** > **Grouped fields**.
- To specify a field for calculating the aggregated value, select the desired field of the specified base data source on the right side of the page, then select **Add field to** > **Aggregation fields**.
- Set the **Source list is ordered by group key** option to **Yes** to sort the group of records based on the values of fields that you select for grouping.

The following aggregate functions are currently supported.

Expand table

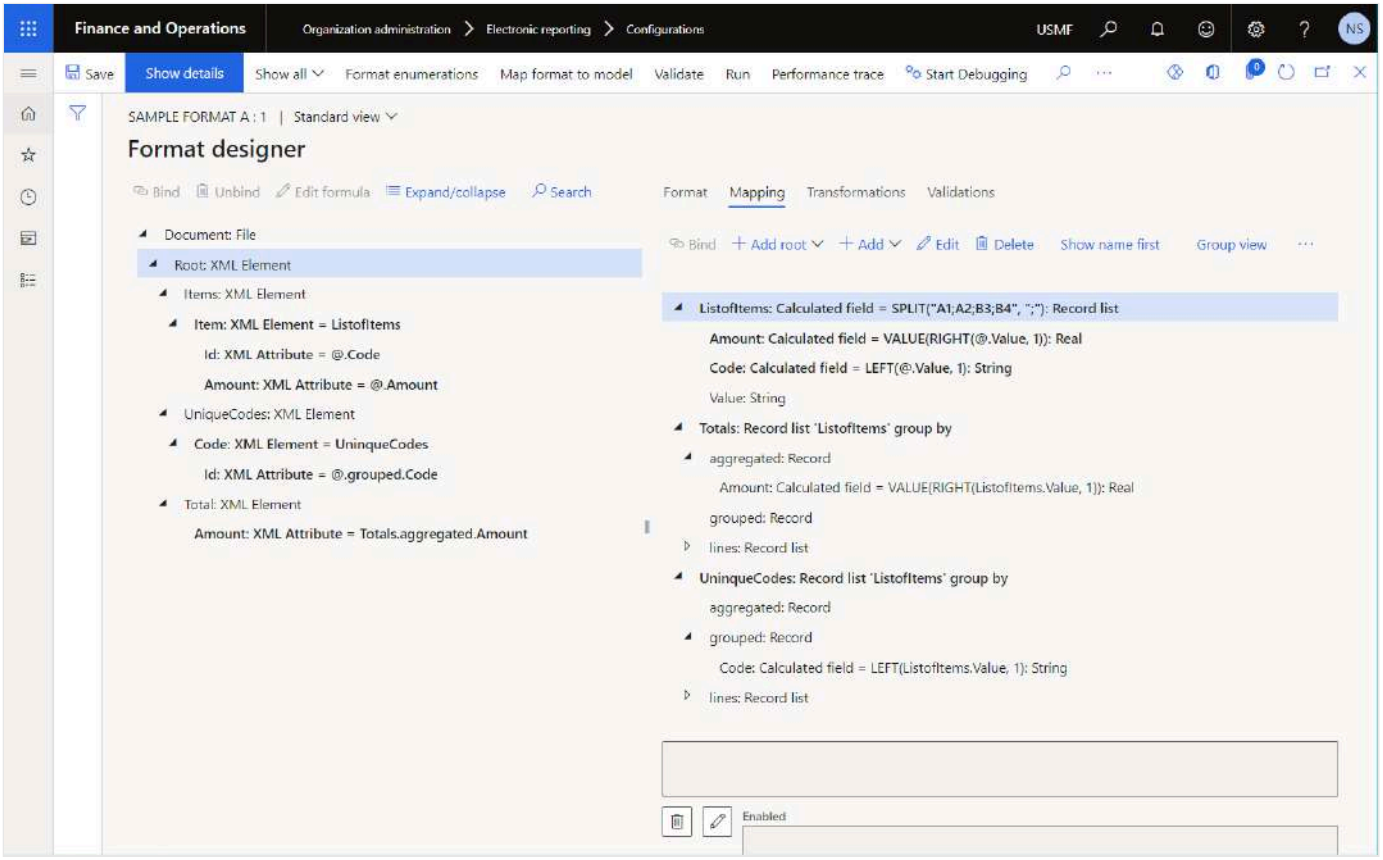
Name	Description	Applicability
AVG	This function returns the average of the values in a group.	This function can be used only with numeric fields.
COUNT	This function returns the number of items that were found in a group.	
MIN	This function returns the minimum value among the values in a group.	
MAX	This function returns the maximum value among the values in a group.	
SUM	This function returns the sum of all the values in a group.	This function can be used only with numeric fields.

Depending on the nature of the base data source, the grouping and aggregation can be done either at the database level or in memory. To specify the execution location, select the desired option for the **Execution location** property of the configured data source:

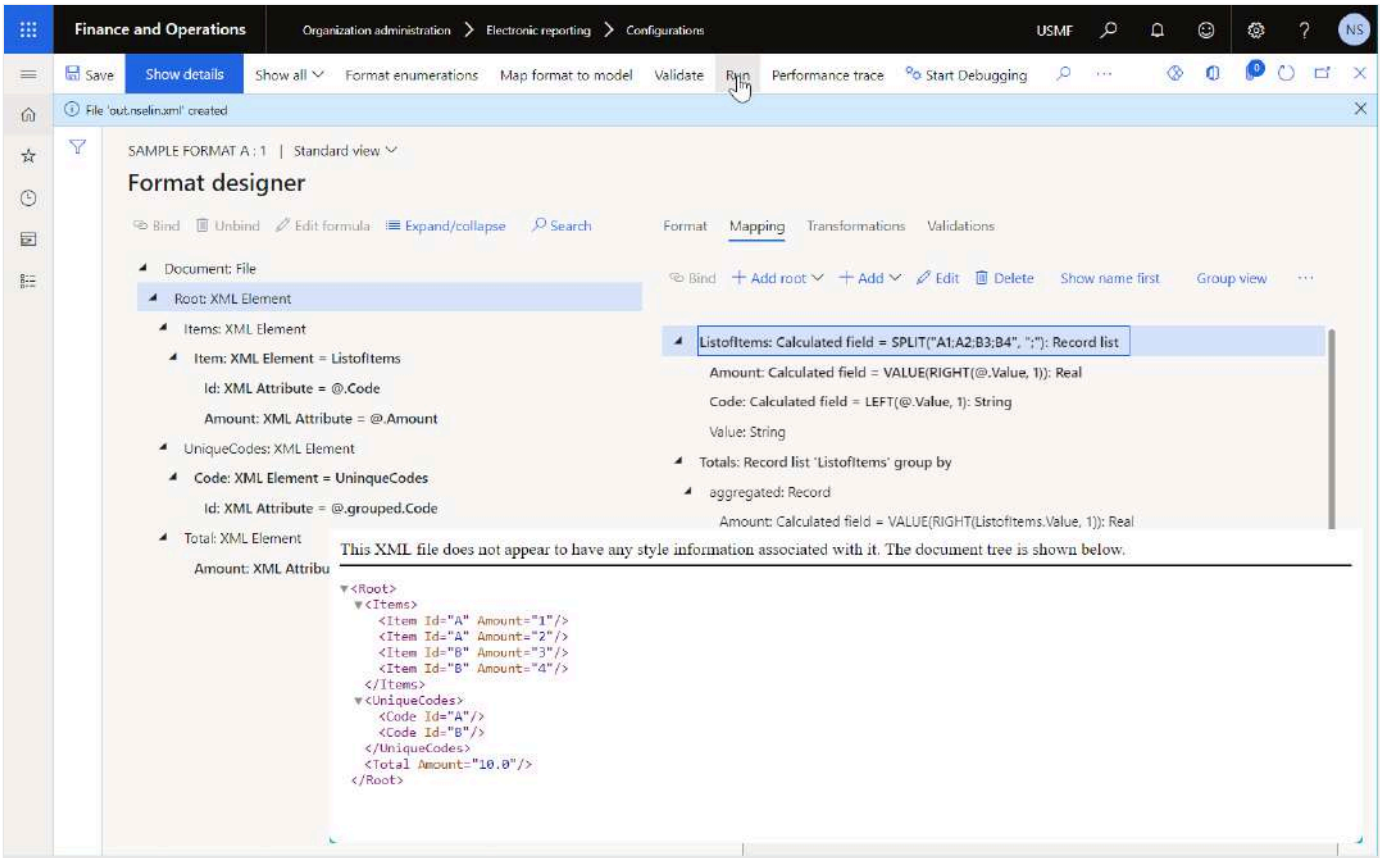
- Select the **In memory** option to do grouping and aggregation in application memory.
- Select the **Query** option to do grouping and aggregation in the application database when applicable.
- Leave the default **Autodetect** option to let ER identify the most appropriate execution location. If the base data source is queryable, the **Query** option is applied when you save changes to the configured data source. Otherwise, the **In memory** option is applied.

The configured **Group by** data source exposes a single record for every group of records of the base data source. This **Group by** data source is structured to expose the results of data grouping and aggregation for each group of records at runtime.

- The **aggregated** record exposes the value of each configured aggregation.
- The **grouped** record exposes the value of each grouping field.
- The **lines** record list exposes records of the base data source that belongs to the current group.



The following illustration shows the result of running the previously described sample ER format that has two data sources of the *Group by* type.



For more information, see Group records and aggregate calculations by using GROUPBY data sources.

Join

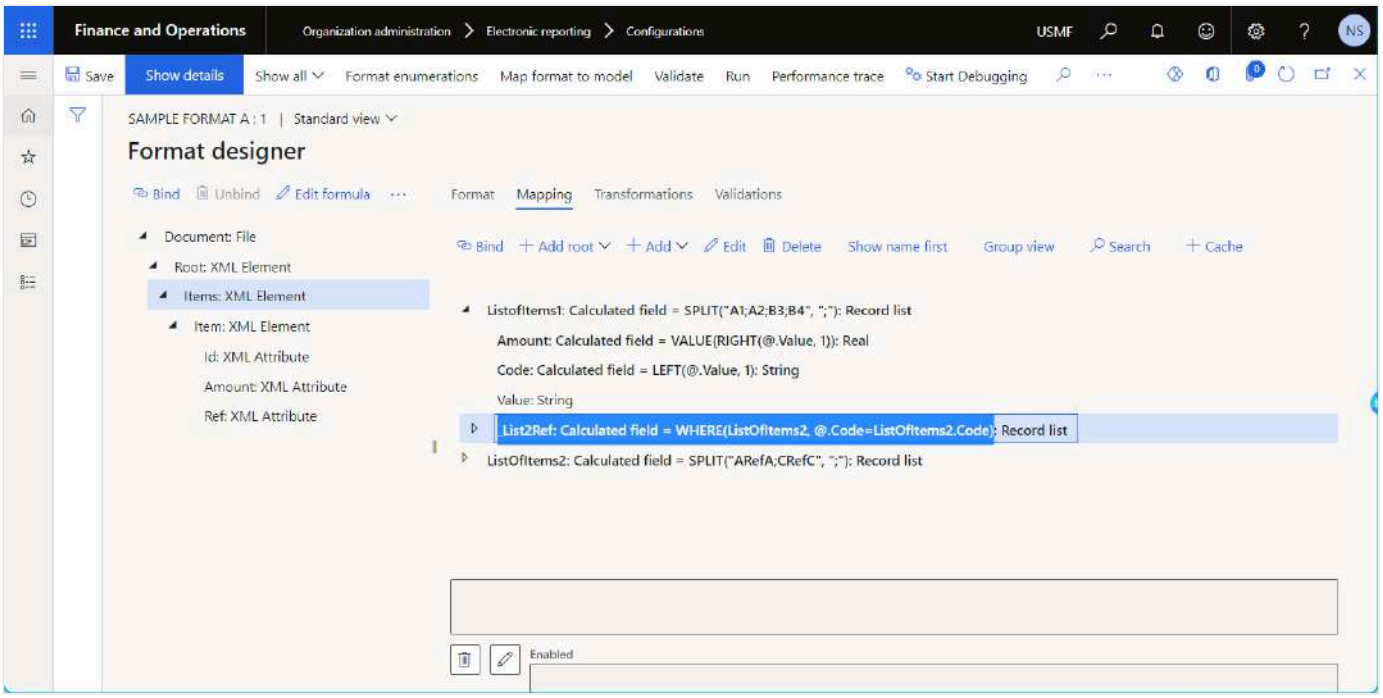
You can configure the **Join** data source to fetch records from several base data sources and return a joined list of records that contains fields from the records of the base data sources.

> **Tip**
>
> Because you can join many data sources that have multiple fields, use the **Select fields** function for any base data source of the configured **Join** data source to limit the list of fields whose values are fetched at runtime. For more information, see **Improve performance of ER solutions by reducing the number of table fields that are fetched at runtime**.

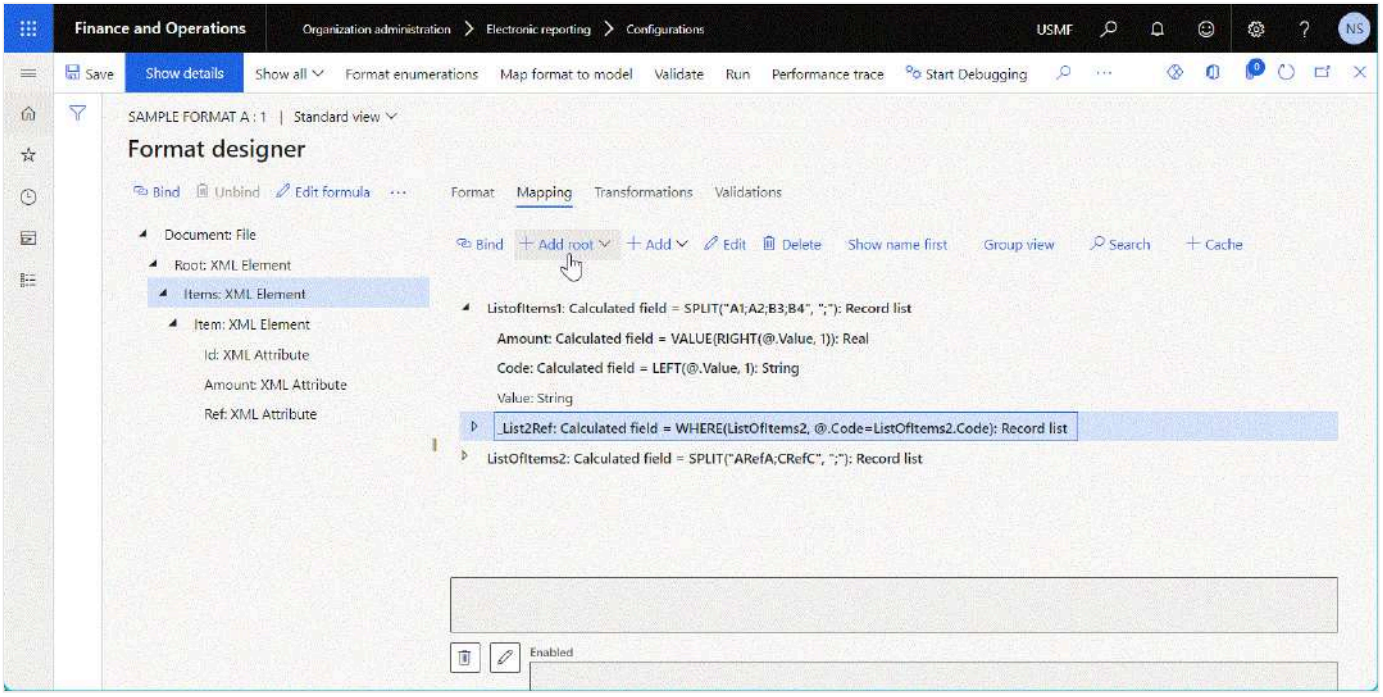
The following types of joins are currently supported:

- **Outer (left)** – Join all records of the first (leftmost) data source and then any records of the second (rightmost) data source that match in accordance with the configured conditions.
- **Inner (right)** – Join only those records of the first (leftmost) and second (rightmost) data sources that match each other in accordance with the configured conditions.

> **Note**
>
> You must configure the data sources that you want to join as nested fields of the *Calculated field* type, and you must specify the join criteria for every configured base data source.



The following illustration shows the configuration of a data source of the *Join* type on the **Join designer** page. To add a data source to the joined list, select the data source on the right side of the page, then use the **Add** button above the **Joined lists** grid.

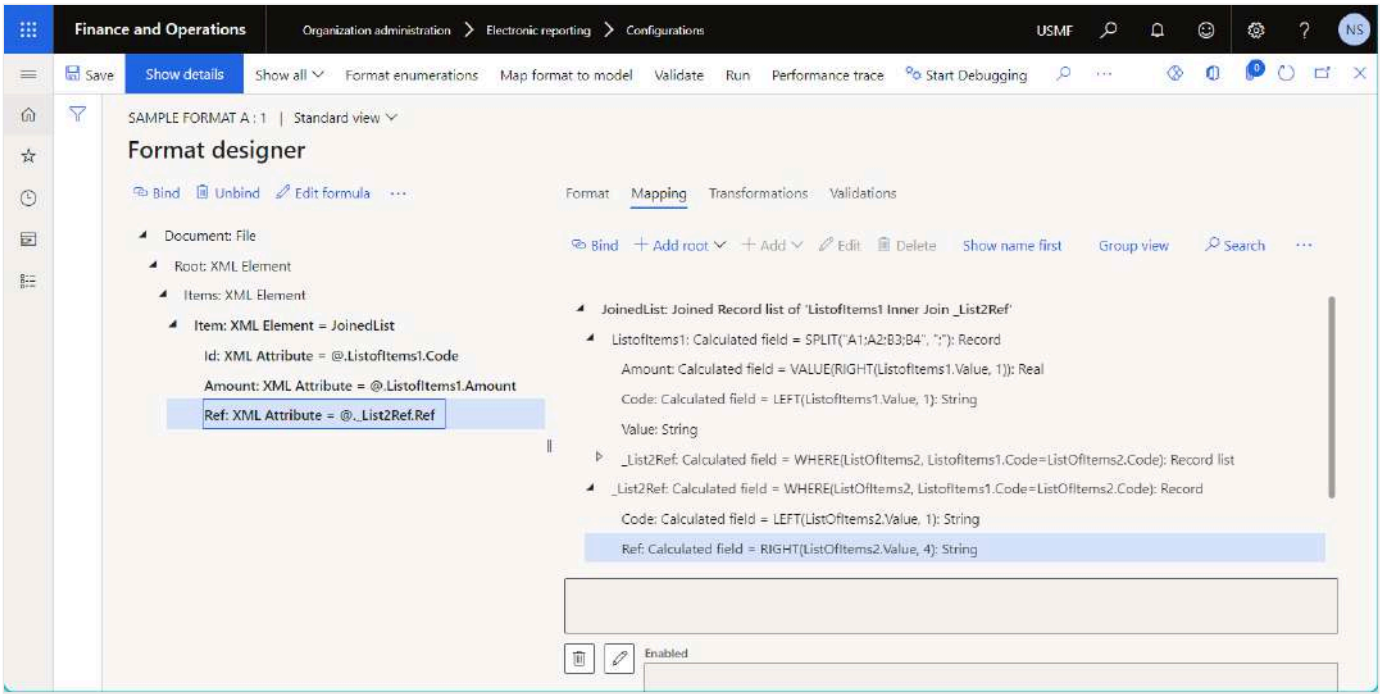


Depending on the nature of the base data sources, the join can be done either at the database level or in memory. To specify the execution location, select the desired option for the **Execute**

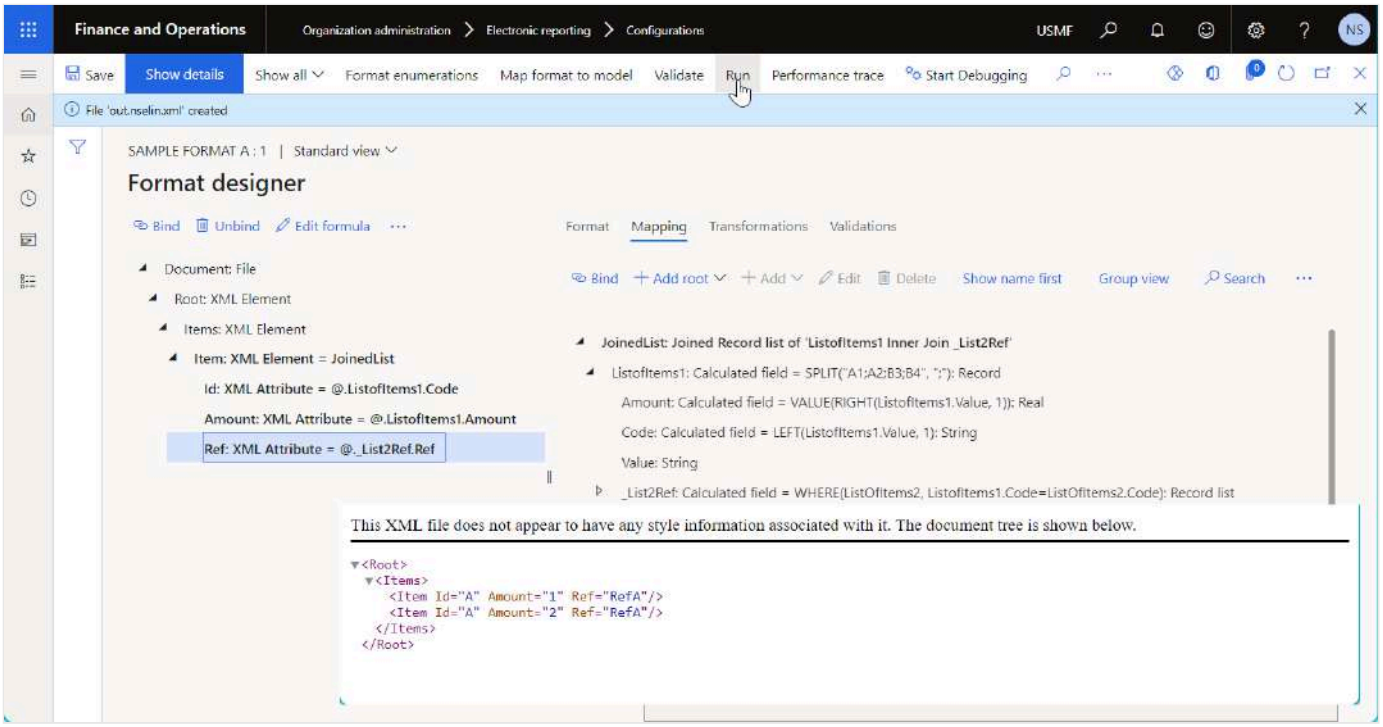
property of the configured **Join** data source:

- Select the **In memory** option to do the join in application memory.
- Select the **Query** option to do the join in the application database when applicable.

This **Join** data source includes several base data sources that you configure under each other as nested fields of the *Calculated field* type. The expression of every configured calculated field includes the condition that's used to associate records of the base data source with records of the nested data source.



The following illustration shows the result of running the previously described sample ER format that has a data source of the *Join* type.

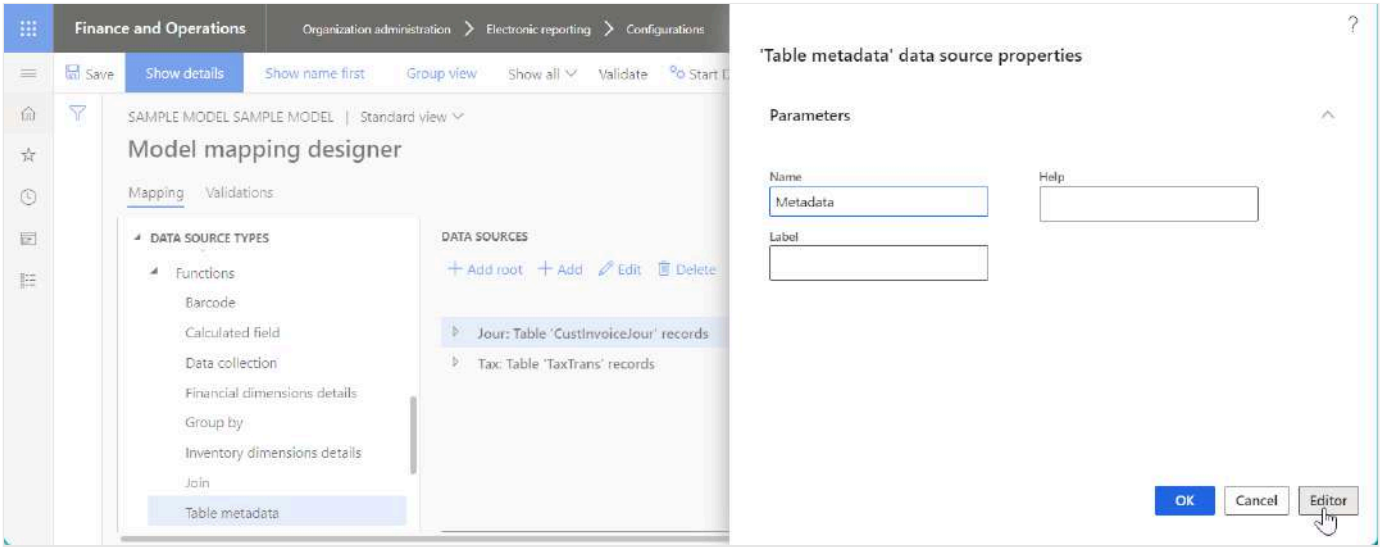


For more information, see Use JOIN data sources to get data from multiple application tables in Electronic reporting (ER) model mappings.

Table metadata

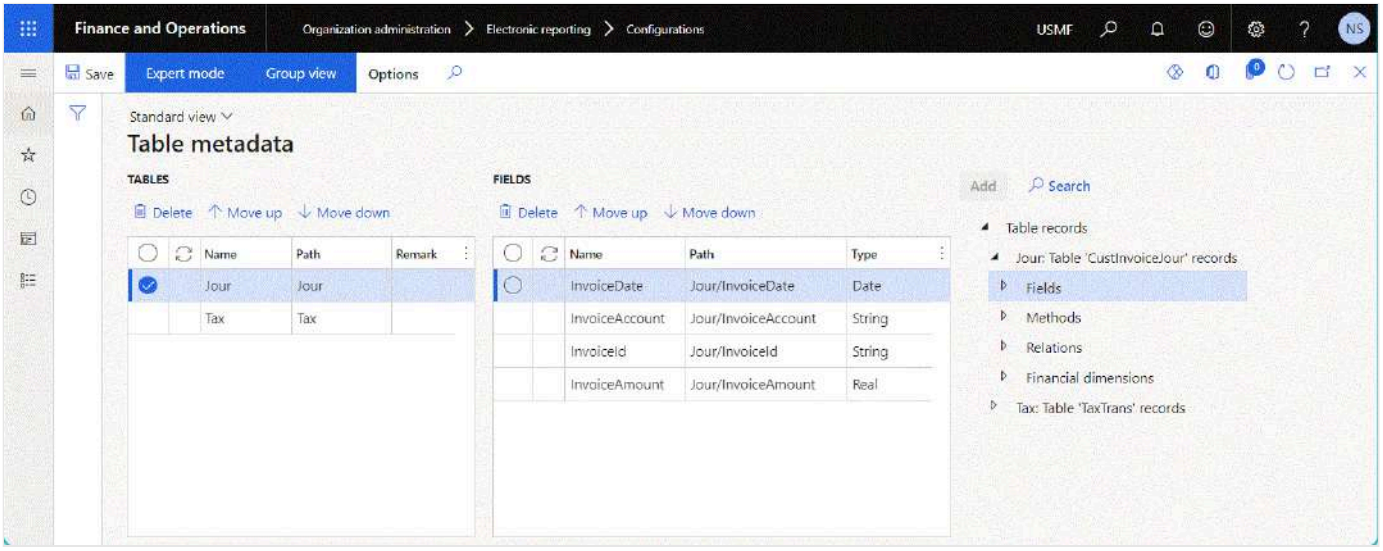
You can configure a **Table metadata** data source to simultaneously access the metadata and data of one or many ER data sources that you configured to return data of the *Record list* type.

When you add a new **Table metadata** data source, use the **Editor** button in the **Data source properties** dialog box to access the **Table metadata** page, where you can configure the added data source.

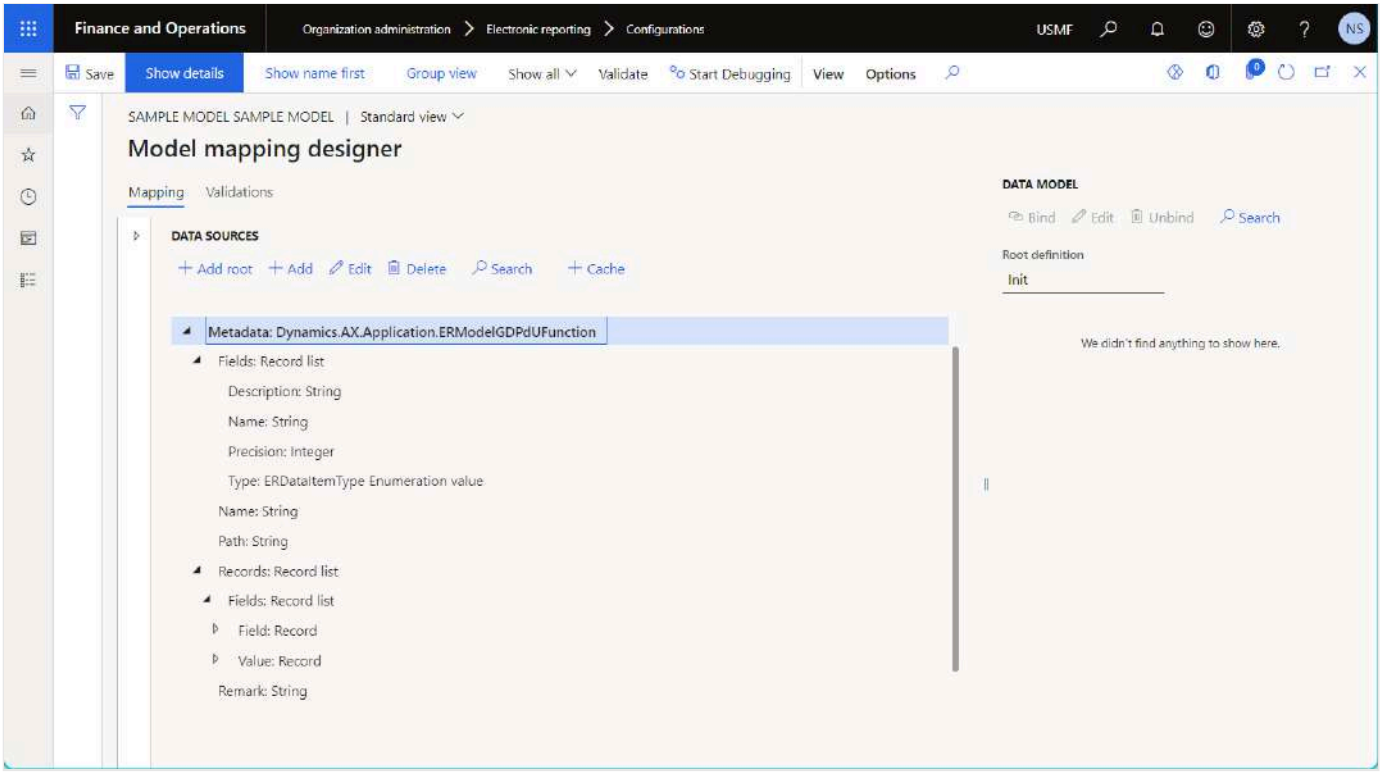


You can add the following items to the editable **Table metadata** data source:

- A data source that returns *Record list* data. Select the desired data source on the right side of the **Table metadata** page, then select **Add**.
- A field of the added data source. Select the desired field on the right side of the **Table metadata** page, then select **Add**.



The following illustration shows the configured data source of the *Table metadata* type.

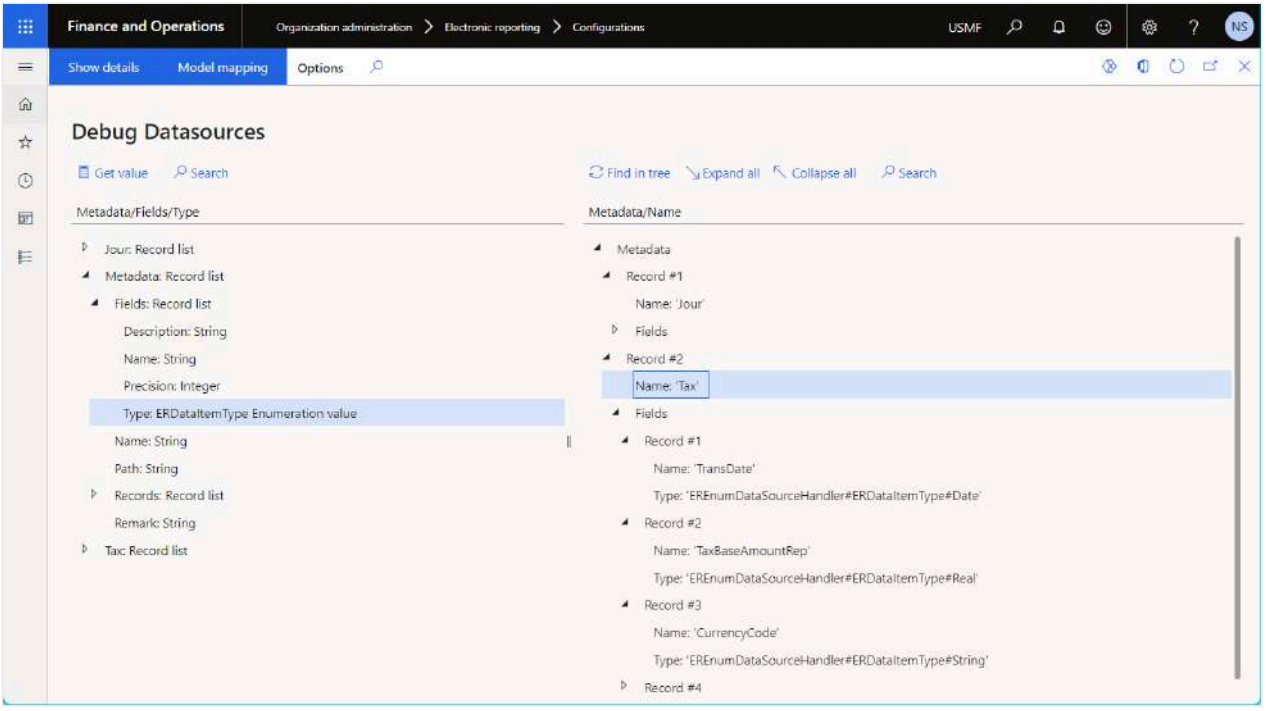


The configured data source of the *Table metadata* type returns a single record for each added ER data source that returns *Record list* data.

- **Name** returns the title of the appropriate data source.
- **Path** returns the path of the appropriate data source in the editable ER component (model mapping).

- **Remark** returns the description of the appropriate data source.
- **Fields** returns the list of fields of the current data source.
    - **Fields.Name** returns the title of the appropriate field.
    - **Fields.Description** returns the description of the appropriate field.
    - **Fields.Type** returns the type of the appropriate field as a value of the **ERDataItemType** enumeration.
    - **Fields.Precision** returns the precision of the appropriate field if this field returns a numeric value.

      The following illustration shows results of metadata debugging for the configured data source of the *Table metadata* type.



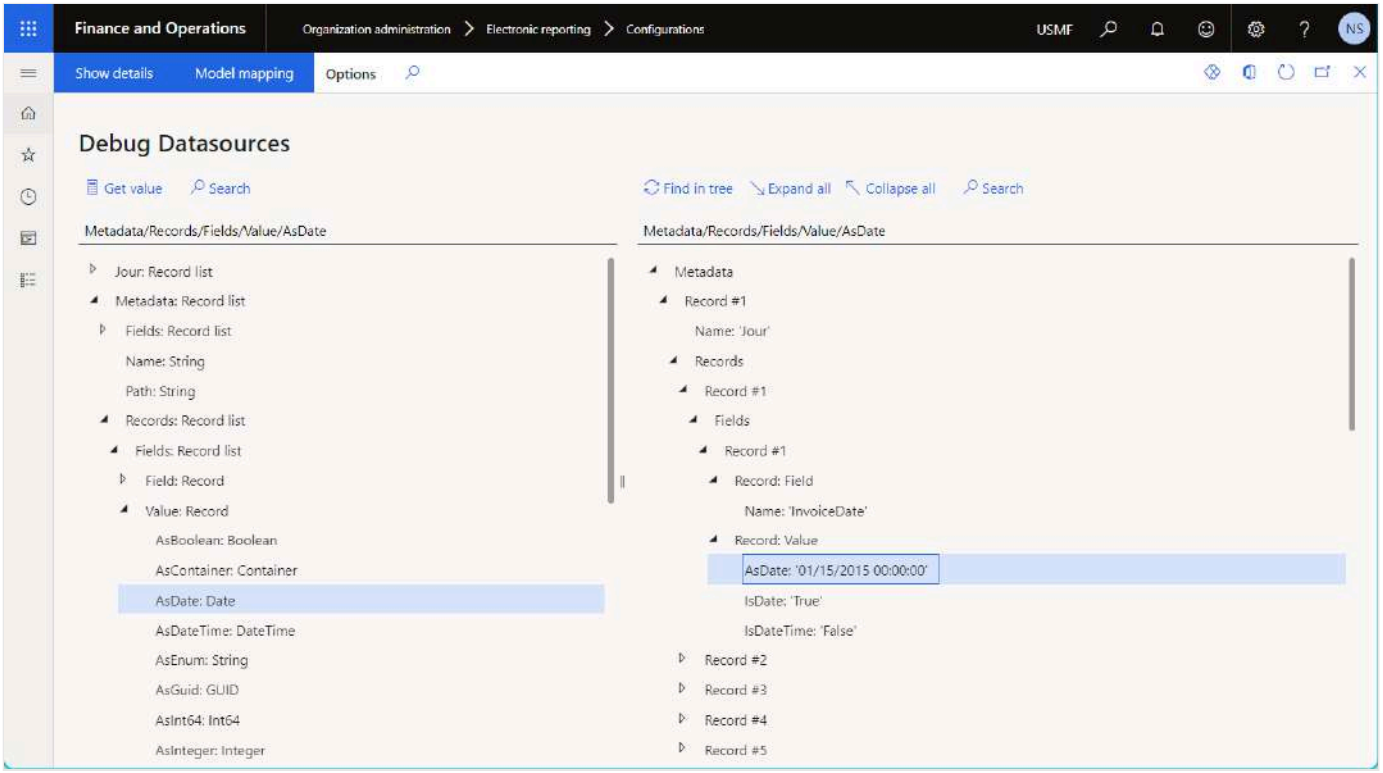
- **Records** returns the records of the appropriate data source.
    - **Records.Fields** represents fields of the current record.
        - **Records.Fields.Field.Name** returns the title of the appropriate field.
        - **Records.Fields.Field.Description** returns the description of the appropriate field.
        - **Records.Fields.Field.Type** returns the type of the appropriate field as a value of the **ERDataItemType** enumeration.

- **Records.Fields.Field.Precision** returns the precision of the appropriate field if this field returns a numeric value.

- **Records.Fields.Value** represents the value of the current field of the current record.
  - **Records.Fields.Value.AsBoolean** returns the value of the current field when the type of this field is *Boolean*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Boolean* type.
  - **Records.Fields.Value.AsContainer** returns the value of the current field when the type of this field is *Container*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Container* type.
  - **Records.Fields.Value.AsDate** returns the value of the current field when the type of this field is *Date*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Date* type.
  - **Records.Fields.Value.AsDateTime** returns the value of the current field when the type of this field is *DateTime*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *DateTime* type.
  - **Records.Fields.Value.AsEnum** returns the value of the current field when the type of this field is *Enumeration*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Enumeration* type.
  - **Records.Fields.Value.AsGuid** returns the value of the current field when the type of this field is *Guid*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Guid* type.
  - **Records.Fields.Value.AsInt64** returns the value of the current field when the type of this field is *Int64*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Int64* type.
  - **Records.Fields.Value.AsInteger** returns the value of the current field when the type of this field is *Integer*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Integer* type.
  - **Records.Fields.Value.AsReal** returns the value of the current field when the type of this field is *Real*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Real* type.
  - **Records.Fields.Value.AsString** returns the value of the current field when the type of this field is *String*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *String* type.
  - **Records.Fields.Value.AsTime** returns the value of the current field when the type of this field is *Time*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Time* type.

- **Records.Fields.Value.AsVoid** returns the value of the current field when the type of this field is *Void*. An exception is thrown at runtime if you try to fetch this property for a field that isn't of the *Void* type.
- **IsBoolean** returns **True** for a field of the *Boolean* type. Otherwise, it returns **False**.
- **IsContainer** returns **True** for a field of the *Container* type. Otherwise returns **False**.
- **IsDate** returns **True** for a field of the *Date* type. Otherwise, it returns **False**.
- **IsDateTime** returns **True** for a field of the *DateTime* type. Otherwise, it returns **False**.
- **IsEnum** returns **True** for a field of the *Enumeration* type. Otherwise, it returns **False**.
- **IsGuid** returns **True** for a field of the *Guid* type. Otherwise, it returns **False**.
- **IsInt64** returns **True** for a field of the *Int64* type. Otherwise, it returns **False**.
- **IsInteger** returns **True** for a field of the *Integer* type. Otherwise, it returns **False**.
- **IsReal** returns **True** for a field of the *Real* type. Otherwise, it returns **False**.
- **IsString** returns **True** for a field of the *String* type. Otherwise, it returns **False**.
- **IsTime** returns **True** for a field of the *Time* type. Otherwise, it returns **False**.
- **IsVoid** returns **True** for a field of the *Void* type. Otherwise, it returns **False**.

The following illustration shows the results of data debugging for the configured data source of the *Table metadata* type.



To learn more about data sources of the *Table metadata* type, download and review the **German audit file output** ER format configuration from the Global repository. Notice that the **Data export model** configuration that you download with the selected ER format contains the **German audit file - default groups** model mapping that's configured to use the *Table metadata* data source to

access different application tables. The downloaded **German audit file output** ER format configuration consumes this model mapping to generate country/region-specific audit reports.

Frequently asked questions

Question: I'm trying to add a new data source of the *Table records* type that references a custom table. However, I can't find that table in the lookup of the **Table** field in the **Data source properties** dialog box.

Answer: The ER metadata of your application instance is obsolete. Use either the **Refill table** function or the **Rebuild table references** function to refresh the ER metadata and align it with your source code base.

Additional resources

Debug data sources of an executed ER format to analyze data flow and transformation

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Last updated on 07/17/2026

Formula designer in Electronic reporting (ER)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to use the formula designer in Electronic reporting (ER). When you design a format for a specific electronic document in ER, use formulas to transform data so that it meets the requirements for the document's fulfillment and formatting. These formulas resemble formulas in Microsoft Excel. The formulas support various types of functions, including text, date and time, mathematical, logical, information, and data type conversion functions, as well as other business domain–specific functions.

Formula designer overview

ER supports the formula designer. At design time, you can configure expressions that can be used for the following tasks at runtime:

- Transform data that you receive from an application database and that you enter in an ER data model designed to be a data source for ER formats. For example, these transformations might include filtering, grouping, and data type conversion.
- Format data that you must send to a generating electronic document in accordance with the layout and conditions of a specific ER format. For example, the formatting might be done in accordance with the requested language or culture, or the encoding.
- Control the process of creating electronic documents. For example, the expressions can enable or disable the output of specific elements of the format, depending on processing data. They can also interrupt the document creation process or throw messages to users.

Open the **Formula designer** page when you perform any of the following actions:

- Bind data source items to data model components.
- Bind data source items to format components.

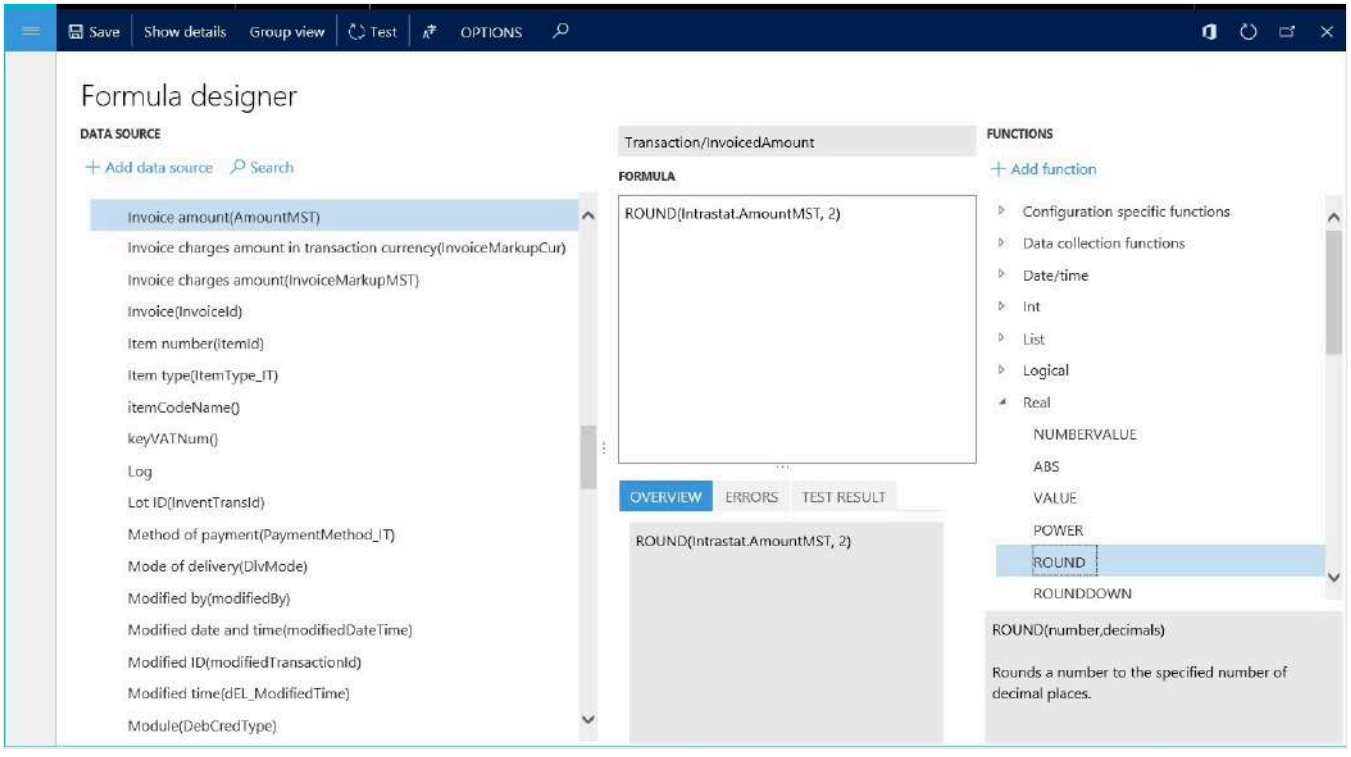
- Complete maintenance of calculated fields that are part of data sources.
- Define the visibility and editability conditions for user input parameters.
- Define the default values for user input parameters.
- Design a format's transformations.
- Define the enabling conditions for the format's components.
- Define the file names for the format's FILE components.
- Define the conditions for process control validations.
- Define the message text for process control validations.

Data binding

Use the ER formula designer to define an expression that transforms data received from data sources. At runtime, enter the data in the data consumer in the following ways:

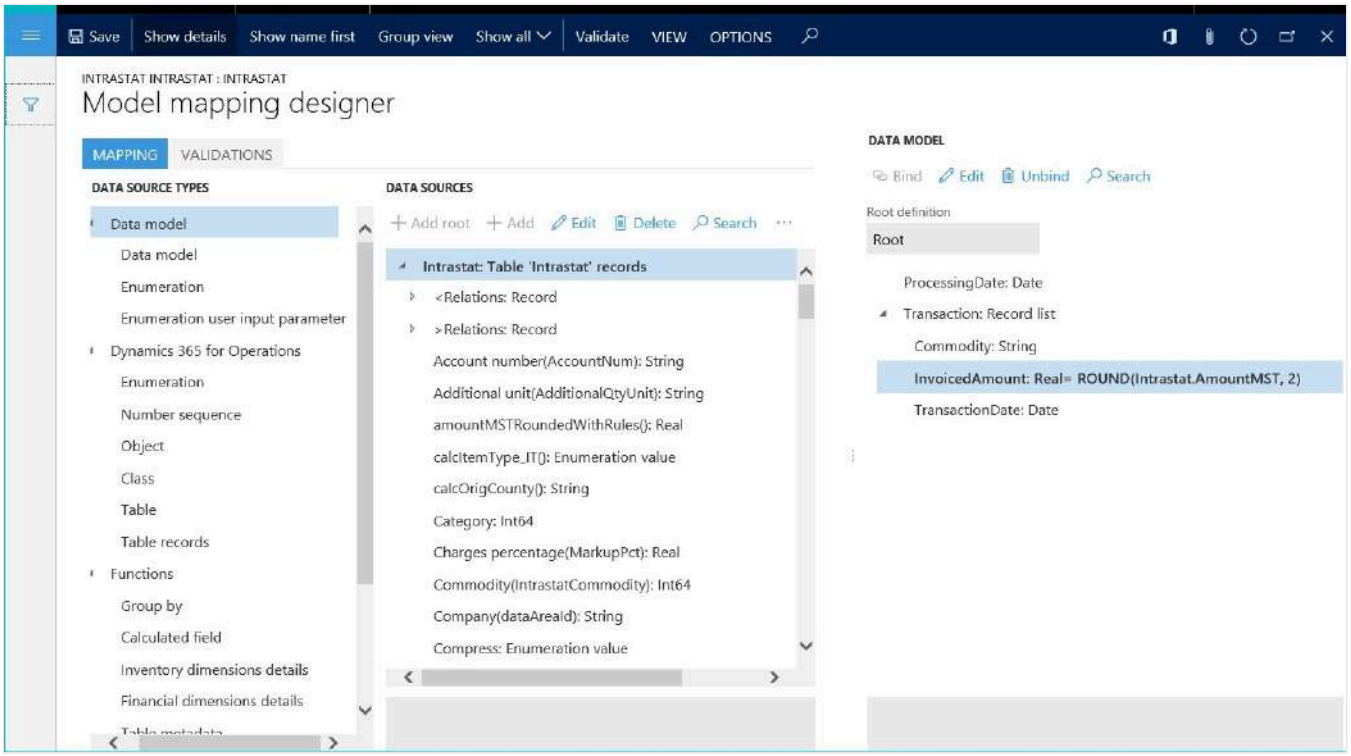
- From application data sources and runtime parameters to an ER data model
- From an ER data model to an ER format
- From application data sources and runtime parameters to an ER format

The following illustration shows the design of an expression of this type. In this example, the expression rounds the value of the **Intrastat.AmountMST** field in the Intrastat table to two decimal places and then returns the rounded value.



The following illustration shows how you can use an expression of this type. In this example, the result of the designed expression is entered in the **Transaction.InvoicedAmount**

component of the **Tax reporting model** data model.

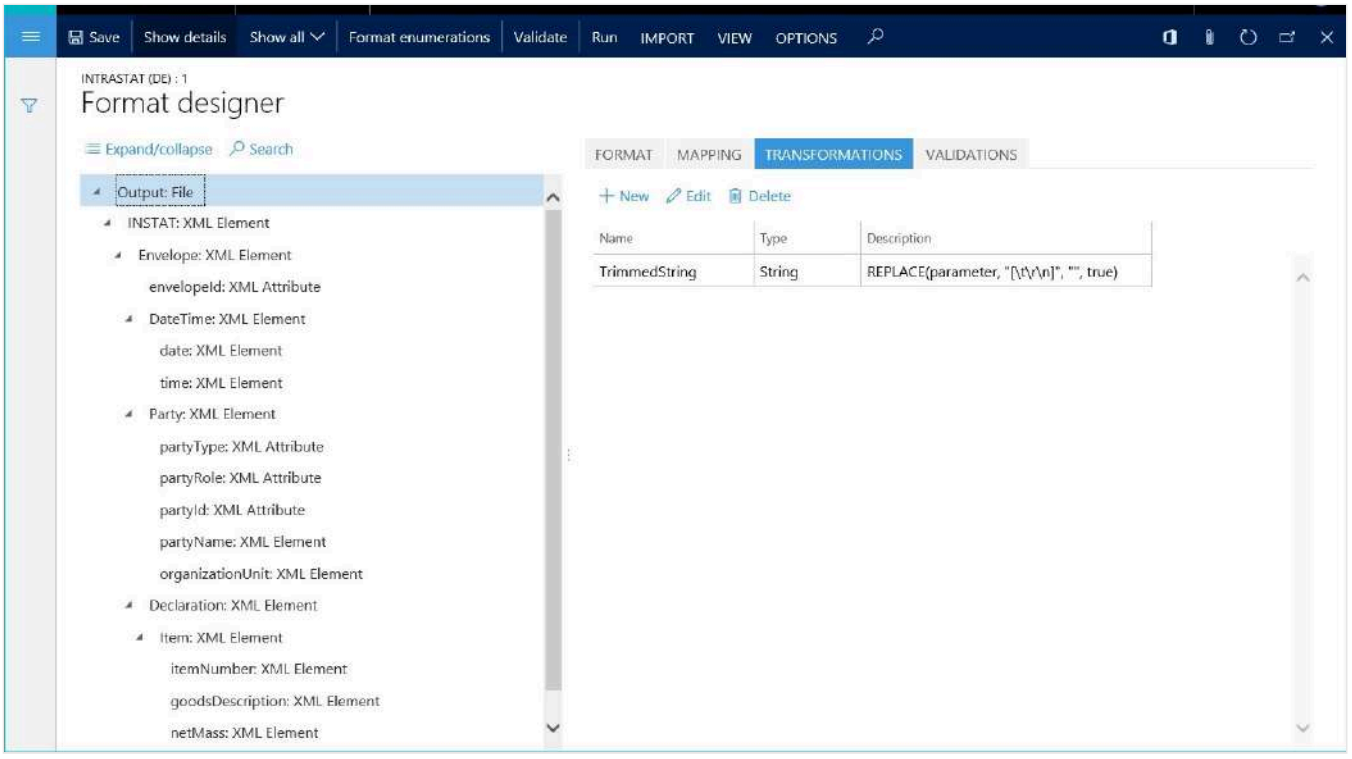


At runtime, the designed formula, `ROUND (Intrastat.AmountMST, 2)`, rounds the value of the **AmountMST** field for each record in the Intrastat table to two decimal places. It then enters the rounded value in the **Transaction.InvoicedAmount** component of the **Tax reporting** data model.

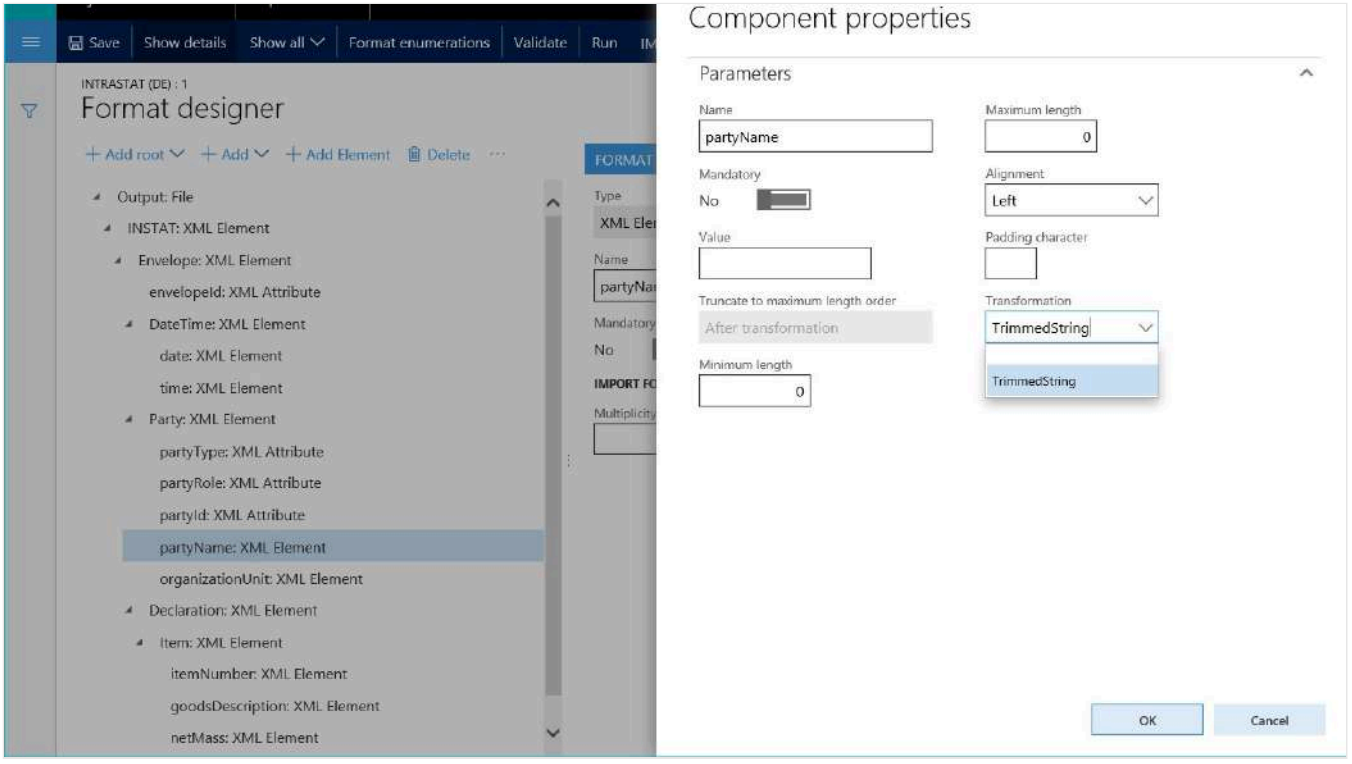
Data formatting

Use the ER formula designer to define an expression that formats data received from data sources. You can send the formatted data as part of the generating electronic document. If you have formatting that you want to apply as a typical rule and reuse for a format, introduce that formatting one time in the format configuration as a named transformation that has a formatting expression. You can link this named transformation to many format components where the output must be formatted according to the formatting expression that you created.

The following illustration shows the design of a transformation of this type. In this example, the **TrimmedString** transformation truncates incoming data of the *String* data type by removing leading and trailing spaces. It then returns the truncated string value.

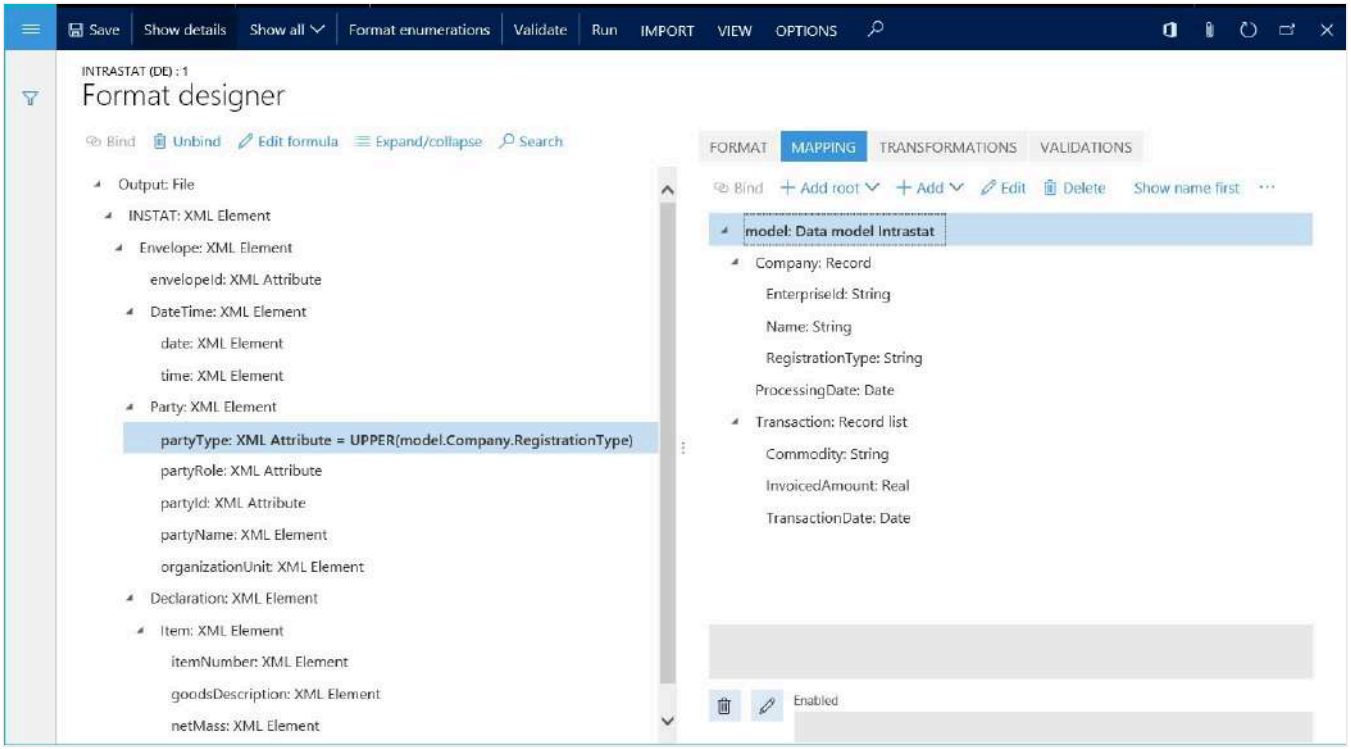


The following illustration shows how you can use a transformation of this type. In this example, several format components send text as output to the generating electronic document at runtime. All these format components refer to the **TrimmedString** transformation by name.



When format components, such as the **partyName** component in the preceding illustration, refer to the **TrimmedString** transformation, the transformation sends text as output to the generating electronic document. This text doesn't include leading and trailing spaces.

If you have formatting that you must apply individually, introduce that formatting as an individual expression of a binding of a specific format component. The following illustration shows an expression of this type. In this example, the **partyType** format component is bound to the data source via an expression that converts incoming data from the **Model.Company.RegistrationType** field in the data source to uppercase text. The expression then sends that text as output to the electronic document.



Process flow control

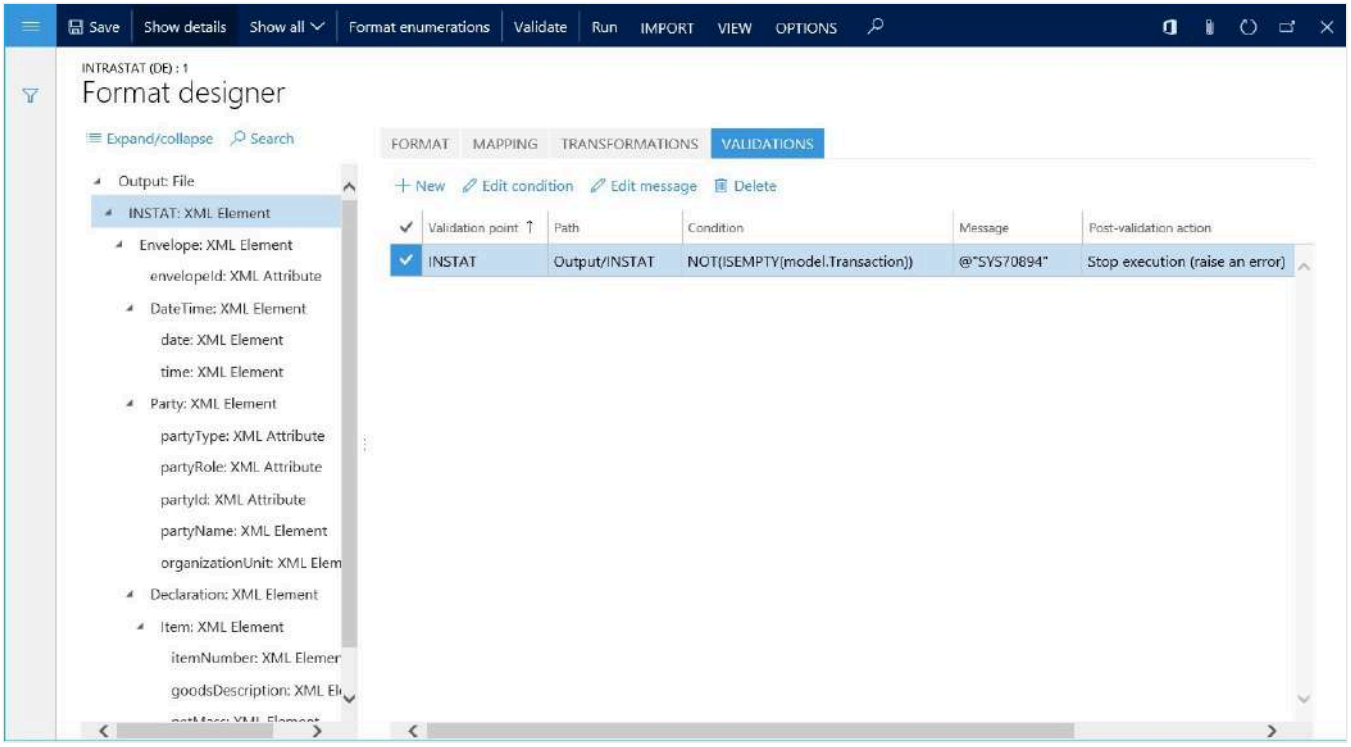
Use the ER formula designer to define expressions that control the process flow of generating electronic documents. You can perform the following tasks:

- Define conditions that determine when to stop a document creation process.
- Specify expressions that either create messages for the user about stopped processes or throw execution log messages about the continuing process of report generation.
- Specify the file names of generating electronic documents, and control the conditions of their creation.

Each rule of the process flow control is designed as an individual validation. The following illustration shows a validation of this type. Here's an explanation of the configuration in this example:

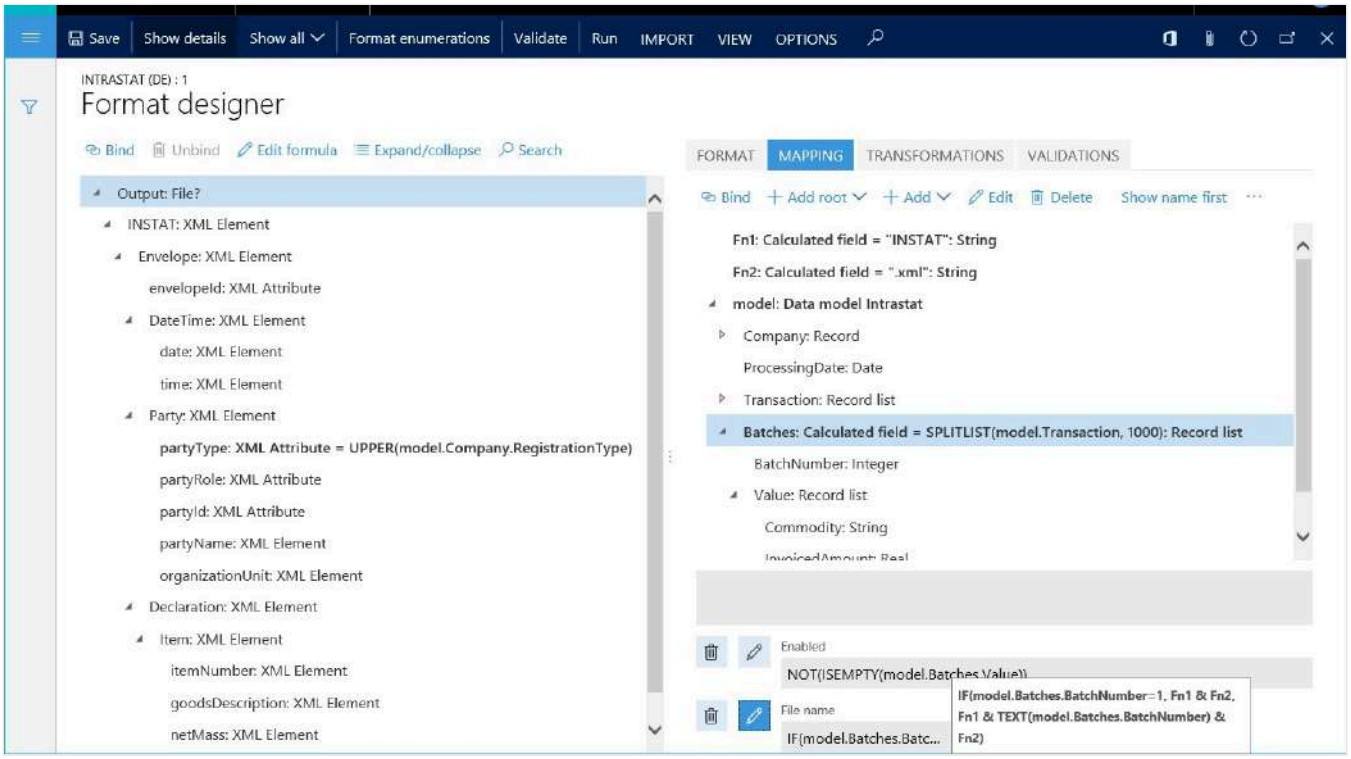
- The validation is evaluated when the **INSTAT** node is created during generation of the XML file.

- If the list of transactions is empty, the validation stops the execution process and returns **FALSE**.
- The validation returns an error message that includes the text of label SYS70894 in the user's preferred language.



Use the ER formula designer to generate a file name for a generating electronic document and to control the file creation process. The following illustration shows the design of a process flow control of this type. Here's an explanation of the configuration in this example:

- The list of records from the **model.Intrastat** data source is divided into batches. Each batch contains up to 1,000 records.
- The output creates a zip file that contains one file in XML format for every batch that was created.
- An expression returns a file name for generating electronic documents by concatenating the file name and the file name extension. For the second batch and all subsequent batches, the file name contains the batch ID as a suffix.
- An expression enables (by returning **TRUE**) the file creation process for batches that contain at least one record.



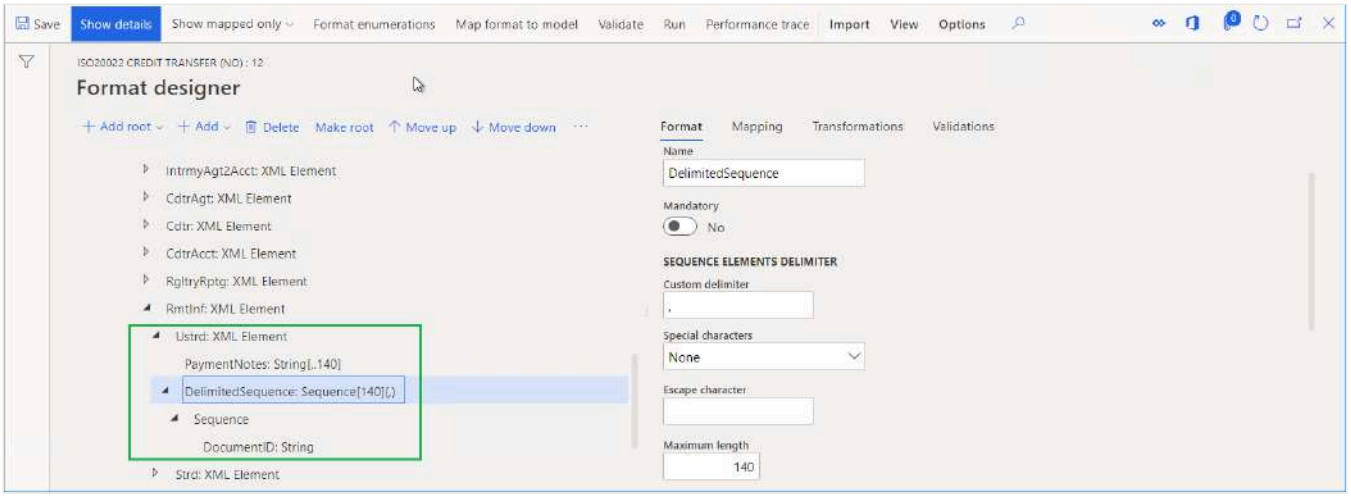
Document content control

Use the ER formula designer to configure expressions that control what data goes into generated electronic documents at runtime. These expressions can enable or disable the output of specific elements of the format, depending on processing data and configured logic. Enter these expressions for a single format element in the **Enabled** field on the **Mapping** tab of the **Operations designer** page. Enter the expressions as a logic condition that returns a *Boolean* value:

- If the condition returns **True**, the current format element runs.
- If the condition returns **False**, the current format element is skipped.

The following illustration shows expressions of this type. (Version 11.12.11 of the **ISO20022 Credit transfer (NO)** format configuration that Microsoft provides is used as an example.) The **XMLHeader** format component describes the structure of the credit transfer message according to the ISO 20022 XML message standards. The **XMLHeader/Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/RmtInf/Ustrd** format component adds the **Ustrd** XML element to the generated message and puts the remittance information in an unstructured format as text of the following XML elements:

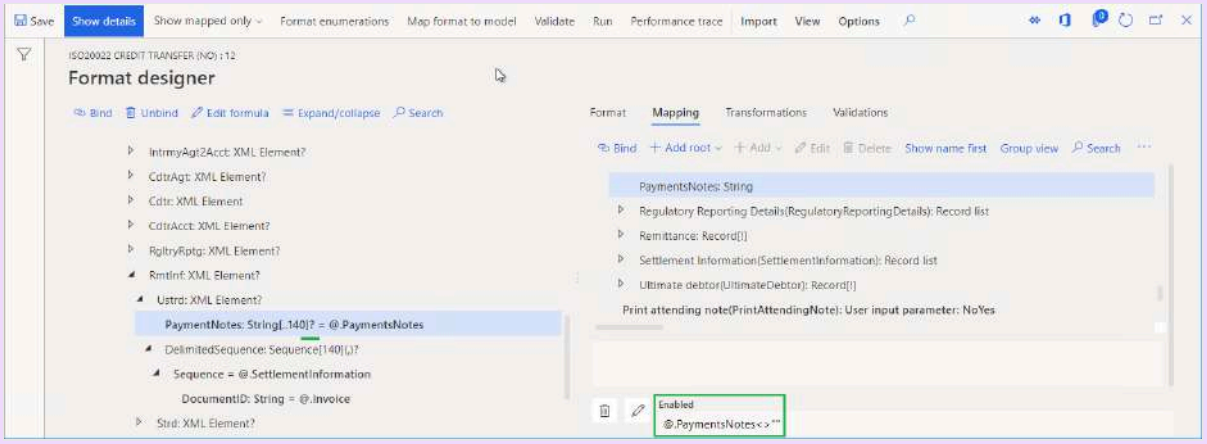
- The **PaymentNotes** component generates the text of payment notes.
- The **DelimitedSequence** component generates comma-separated invoice numbers that settle the current credit transfer.



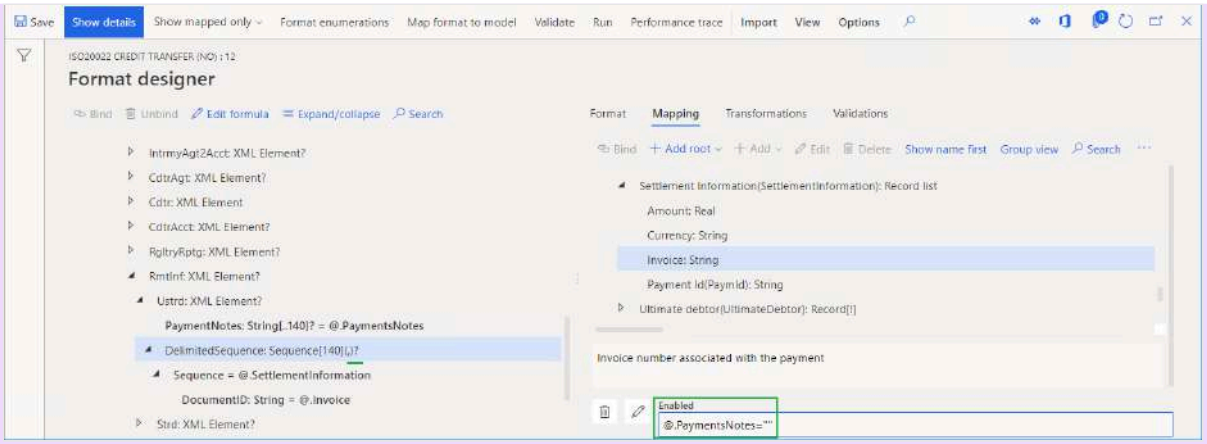
ⓘ Note

The **PaymentNotes** and **DelimitedSequence** components are labeled by using a question mark. A question mark indicates that the use of a component is conditional. In this case, use of the components is based on the following criteria:

- The `@.PaymentsNotes <> ""` expression that you define for the **PaymentNotes** component enables (by returning **TRUE**) the **Ustrd** XML element to be filled with the text of payment notes, if that text isn't blank for the current credit transfer.



- The `@.PaymentsNotes = ""` expression that you define for the **DelimitedSequence** component enables (by returning **TRUE**) the **Ustrd** XML element to be filled with a comma-separated list of the invoice numbers that settle the current credit transfer, if the text of payment notes for that credit transfer is blank.

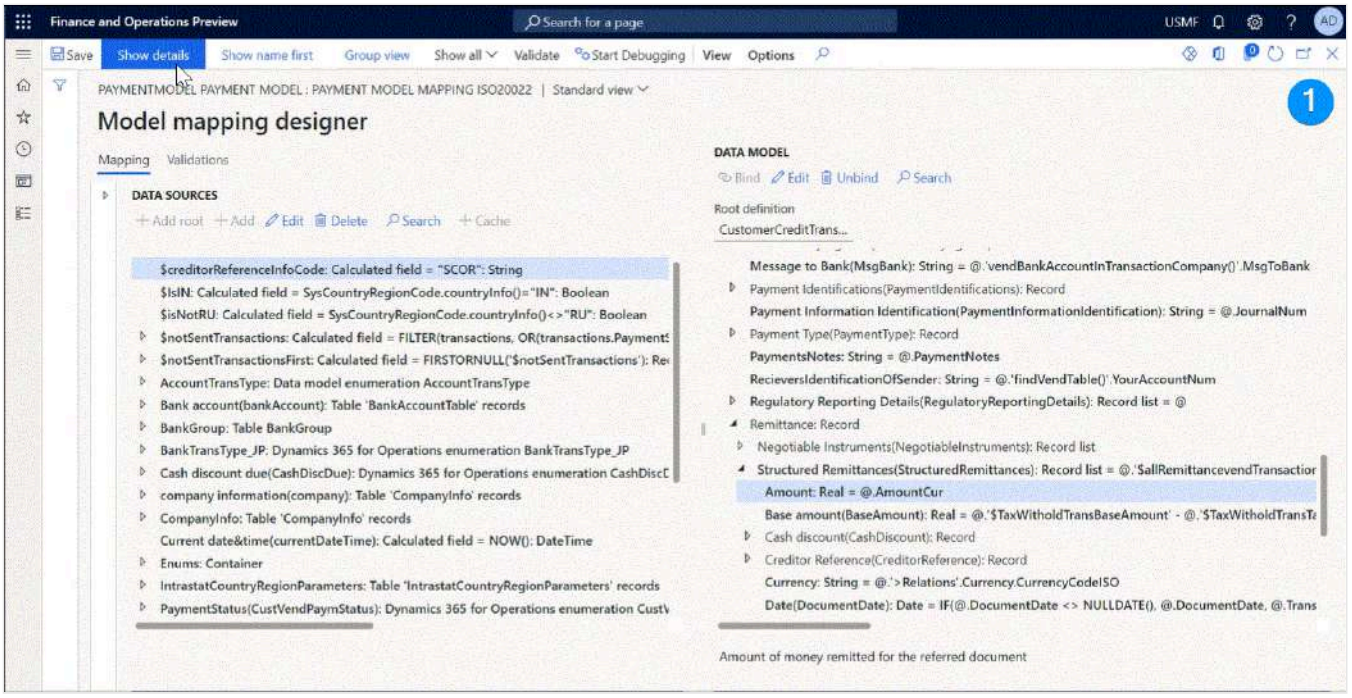


Based on this setup, the message that is generated for each debtor payment, the **Ustrd** XML element, contains either the text of payment notes or, when that text is blank, a comma-separated list of the invoice numbers that settle the payment.

Assistance in writing formulas

Data sources navigator

You can edit a formula that represents an element of a structured data source. When you configure your ER parameters to present the path to an element of a structured data source as the relative path, the formula shows the "at" (@) sign instead of the remaining part of the absolute path of the hierarchical tree structure. This remaining part of the absolute path points to a parent element of the editable one. In Finance version **10.0.30 and later**, on the **Formula designer** page, in the **Data sources** pane, you can select the **Go to @** option to position the cursor of the data sources tree to an element that is the parent of the editable one. The structure of all collapsed ascending elements is automatically and recursively expanded when required. This expansion helps you quickly visualize the base element of the editable one, observe siblings of the editable element in the data sources tree, and use each of them in the editable formula if needed.



Data sources picker

On the **Formula designer** page, in the **Data sources** pane on the left, select an element of a data source that you want to bring into the editable formula. Then select **Add data source**. The selected element is added to the text of the editable formula.

> **Tip**
>
> When you use the **Add data source** option in the default formula editor, the selected element is always added to the end of the formula text. When you use the same option in the Advanced formula editor, the selected element is inserted to the formula text at the current cursor position.

Built-in functions picker

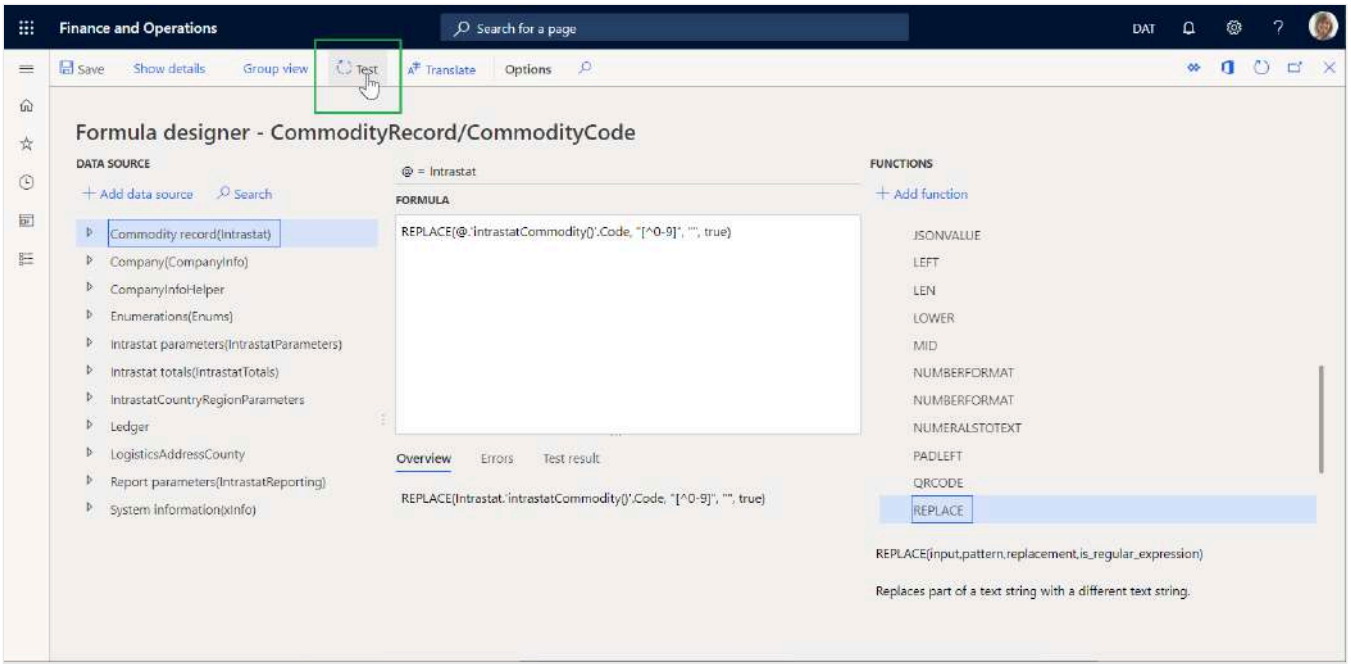
On the **Formula designer** page, in the **Functions** pane on the right, select an ER built-in function that you want to bring into the editable formula. Then, select **Add function**. The selected function is added to the text of the editable formula.

> **Tip**
>
> When you use the **Add function** option in the default formula editor, the selected function is always added to the end of the formula text. When you use the same option in the

Advanced formula editor, the selected function is inserted to the formula text at the current cursor position.

Validation of configured formulas

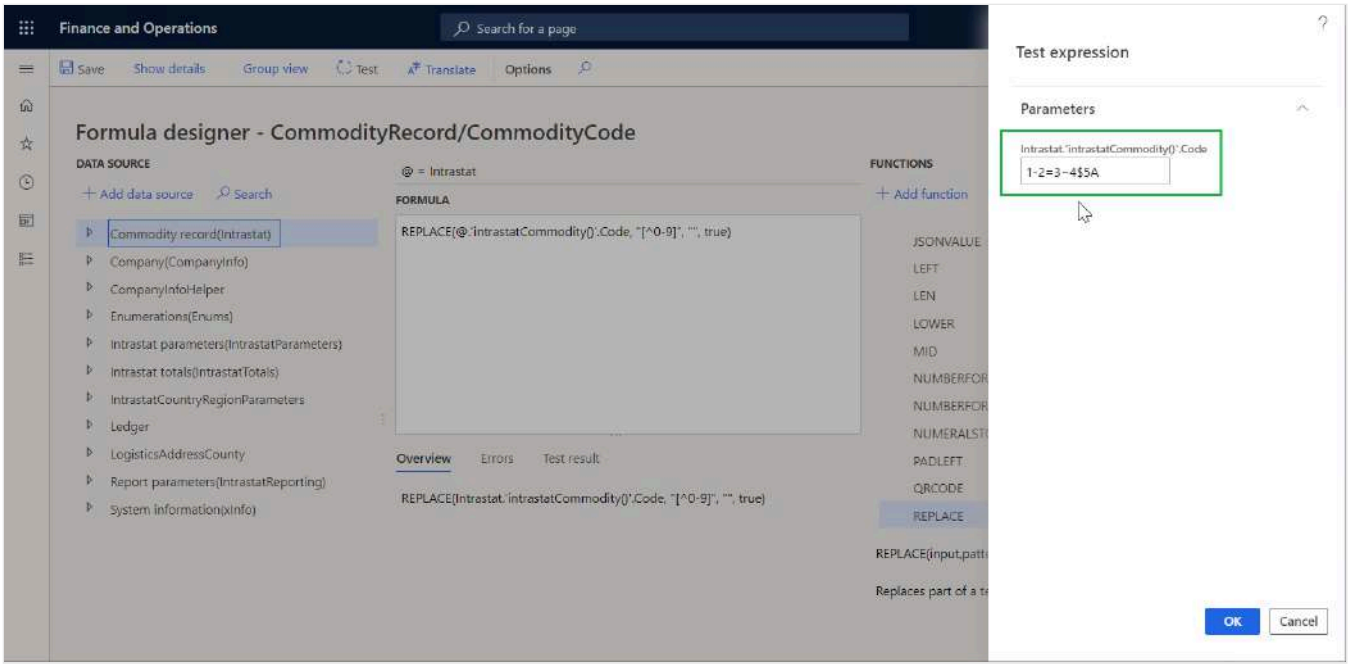
On the **Formula designer** page, select **Test** to validate how the configured formula works.



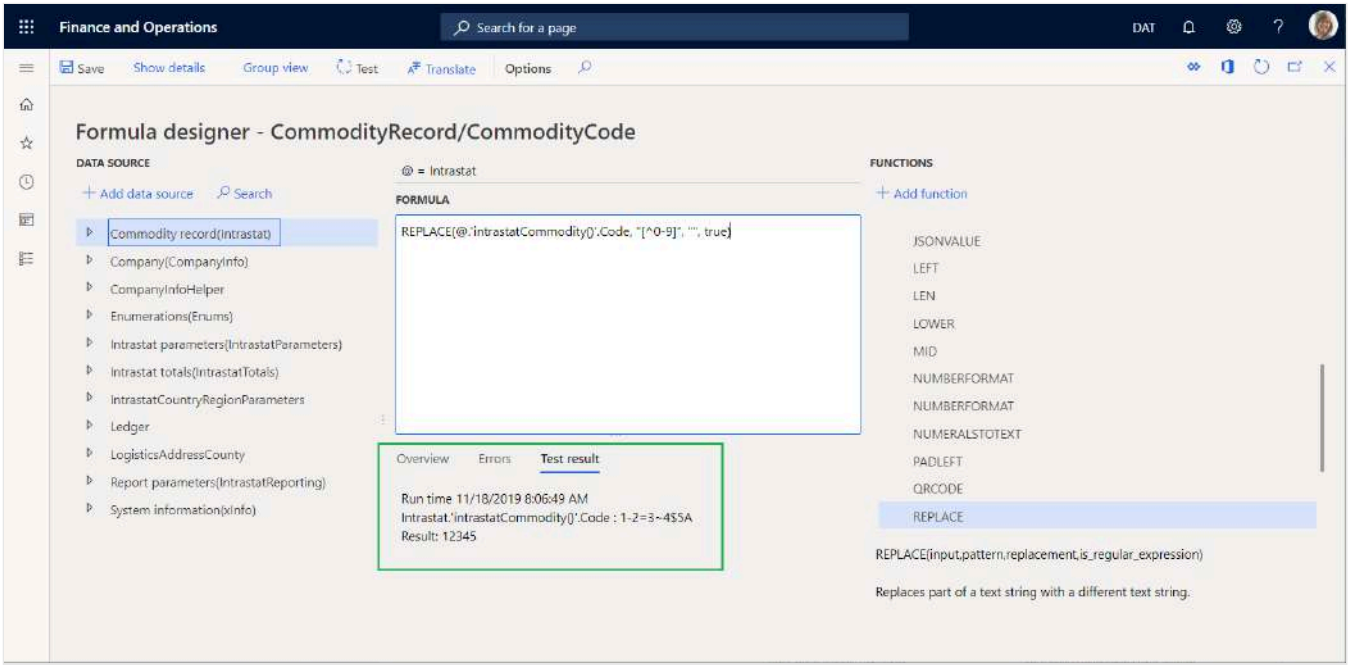
When the formula needs values for arguments, open the **Test expression** dialog box from the **Formula designer** page. In most cases, you must manually define these arguments because the configured bindings don't run at design time. The **Test result** tab on the **Formula designer** page shows the result from execution of the configured formula.

The following example shows how you can test the formula that you configured for the foreign trade domain to make sure that the Intrastat commodity code contains only digits.

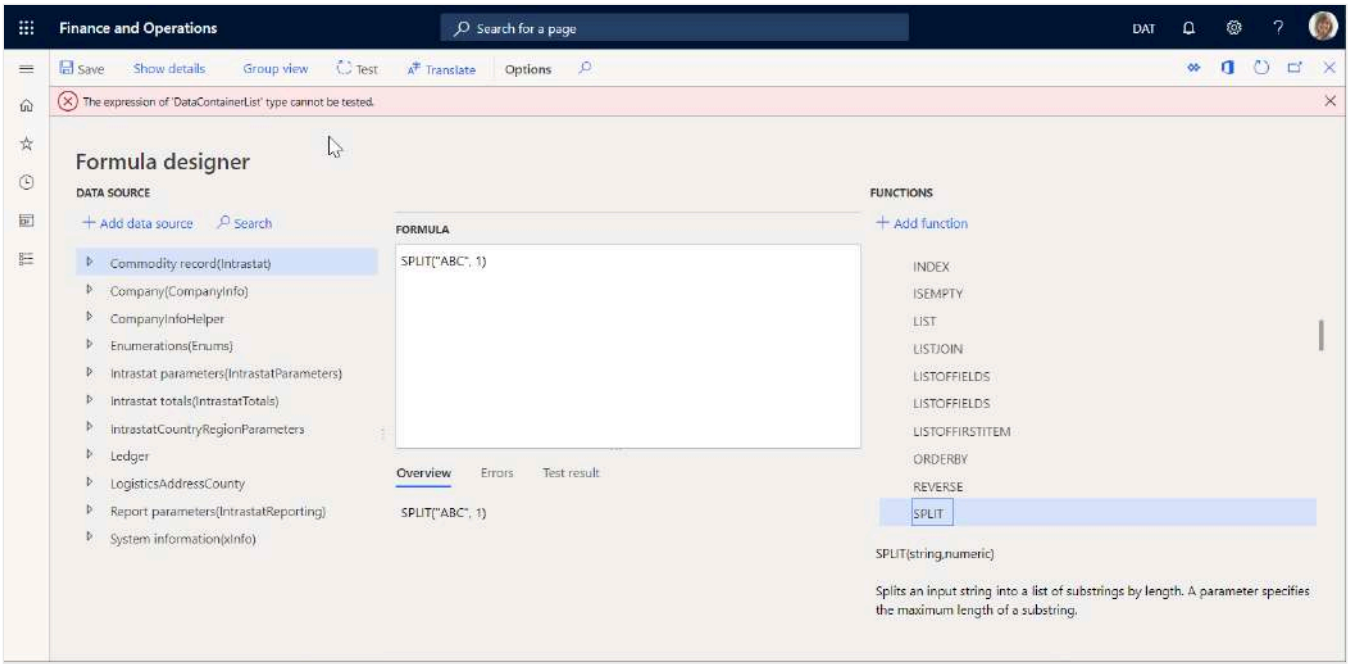
When you test this formula, use the **Test expression** dialog box to specify the value of the Intrastat commodity code for testing.



After you specify the Intrastat commodity code and select **OK**, the **Test result** tab on the **Formula designer** page shows the result of execution of the configured formula. You can then evaluate whether the result is acceptable. If the result isn't acceptable, you can update the formula and test it again.



Some formulas can't be tested at design time. For example, a formula might return a result of a data type that can't be shown on the **Test result** tab. In this case, you receive an error message that states that the formula can't be tested.



Additional resources

- Electronic Reporting overview
- Electronic reporting formula language

Last updated on 04/13/2026

Electronic reporting advanced formula editor

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

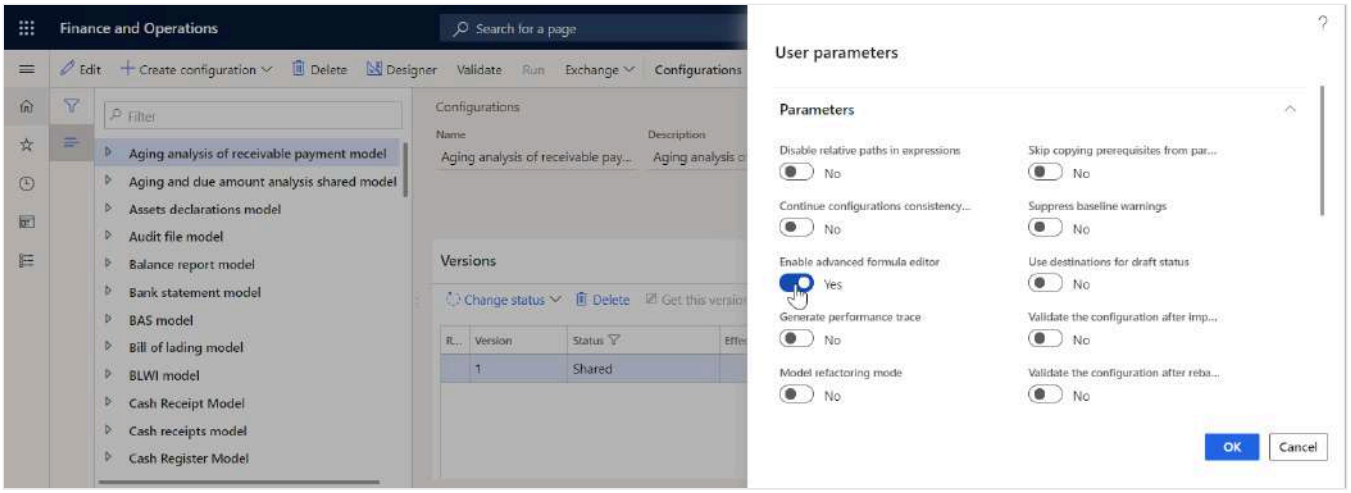
In addition to the Electronic reporting formula editor, you can use the advanced Electronic reporting formula editor to improve the experience of configuring Electronic reporting (ER) expressions. The advanced editor is browser-based and powered by the Monaco editor. This article describes the most commonly used advanced editor features:

- Code autoformatting
- IntelliSense
- Code completion
- Code navigation
- Code structuring
- Find and replace
- Data pasting
- Syntax colorization

Activate the advanced formula editor

Complete the following steps to start using the advanced formula editor in your instance of Microsoft Dynamics 365 Finance.

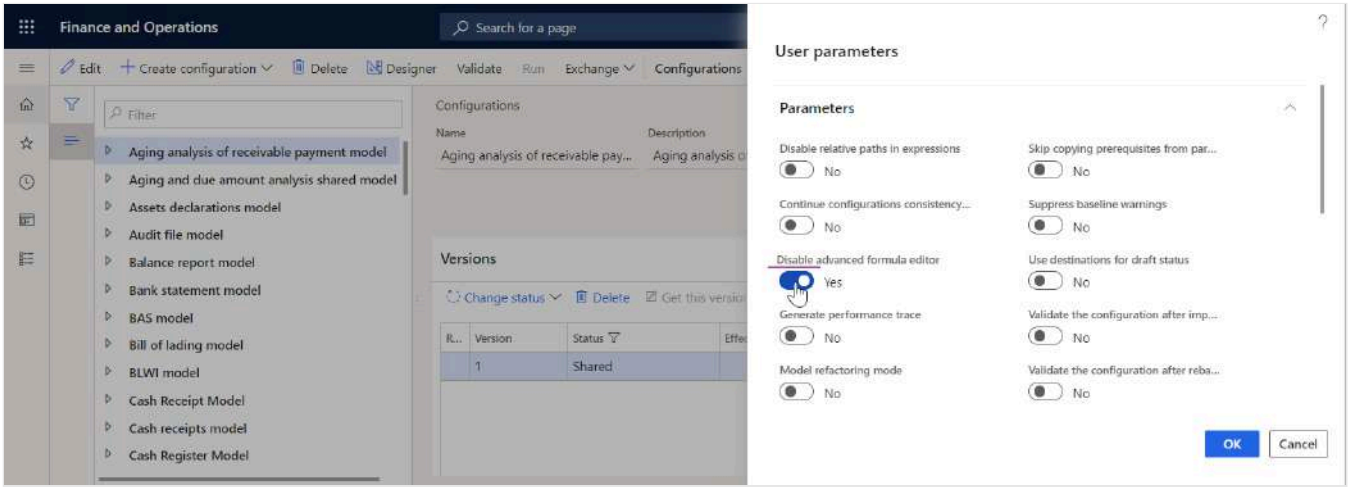
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In **User parameters**, in the **Execution tracing** section, set the **Enable advanced formula editor** parameter to **Yes**.



> **Note**
>
> This parameter is user specific and company specific.

Starting in Microsoft Dynamics 365 Finance version 10.0.19, you can control which ER formula editor is offered by default. Complete the following steps to enable the advanced formula editor for all users and companies of the current Finance instance.

1. Open the **Feature management** workspace.
2. Find and select the feature **Set the ER advanced formula editor as the default one for all users** in the list, and then select **Enable now**.
3. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
4. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
5. In **User parameters**, find the **Disable advanced formula editor** parameter and verify that it's set to **No**.

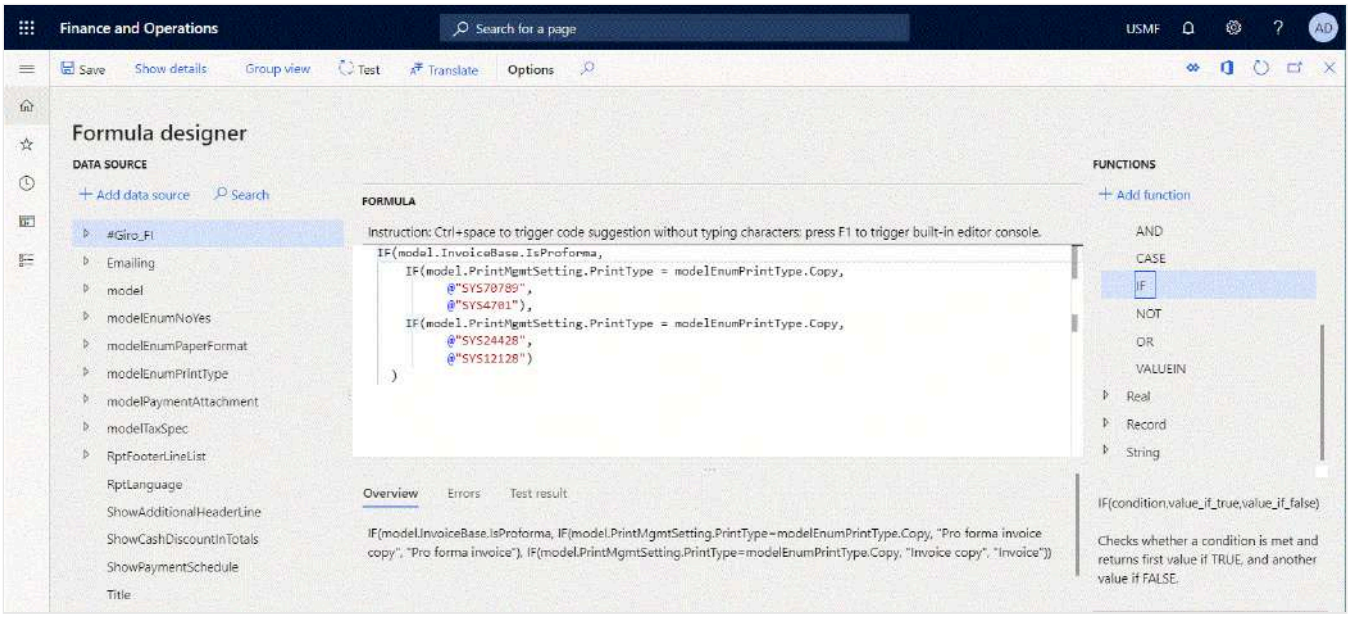


> **Note**

The values of the parameters **Enable advanced formula editor** and **Disable advanced formula editor** are kept separate for each user. The **User parameters** dialog box displays these parameters depending on the status of the **Set the ER advanced formula editor as the default one for all users** feature.

Code autoformatting

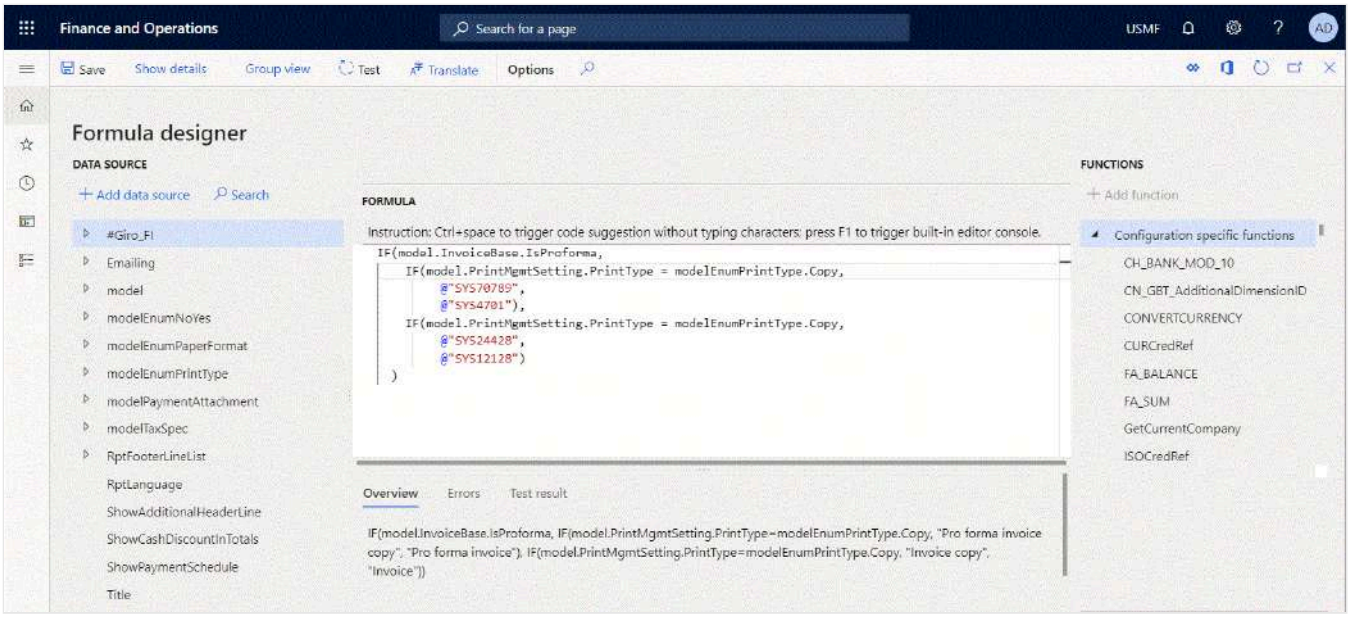
When you write a complex expression that consists of multiple rows of code, the editor automatically indents a new line based on the indentation of the previous row. You can select lines and change their indentation by typing **Tab** or **Shift+Tab**.



Autoformatting helps you keep the entire expression well formatted, which makes further maintenance easier and simplifies understanding of the configured logic.

IntelliSense

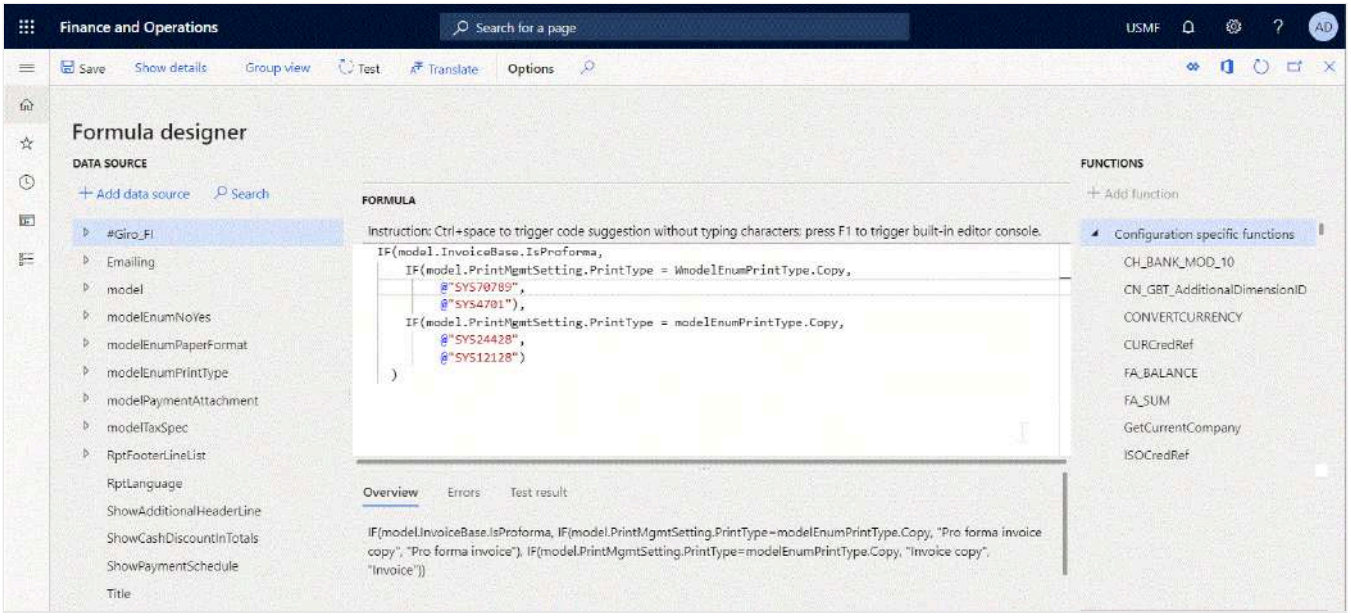
The editor provides word completion to help you write expressions faster and avoid typos. When you start adding new text, the editor automatically offers a list of functions supported in ER functions that contain the characters you entered. You can also trigger IntelliSense in any place of a configured expression by typing **Ctrl+Space**.



Code completion

The editor automatically provides code completion by:

- Inserting a closing bracket when you enter an opening bracket, and keeping the cursor inside the brackets.
- Inserting the second quotation mark when you enter the first one, and keeping the cursor inside the quotations.
- Inserting the second double quotation mark when you enter the first one, and keeping the cursor inside the quotations.



When you point to a typed bracket, the editor automatically highlights the second bracket of this pair to show the construct that they support.

Code navigation

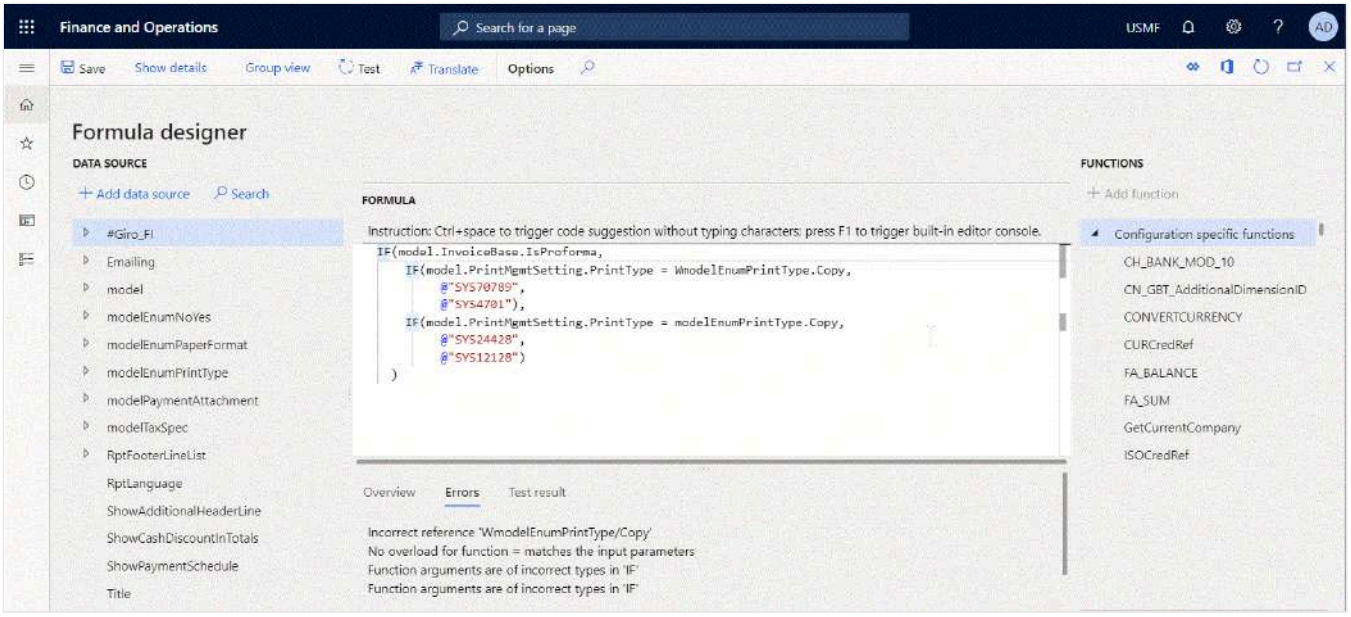
You can find required symbols or lines in your expression by using the **Go to** command from the command palette or the context menu.

For example, to jump to line **8**, use one of the following methods:

- Press **Ctrl+G**, enter the value **8**, and then press **Enter**.

  -or-

- Press **F1**, type **G**, select **Go to line**, enter the value **8**, and then press **Enter**.



Code structuring

The code for some functions, such as IF or CASE, is automatically structured. You can expand and collapse any or all of the folding regions of this code to reduce the editable part of an expression so you can focus on only the piece of code that requires your attention. Use the toggle fold and unfold commands for that task.

For example, to fold all regions, use one of the following methods:

- Press **Ctrl+K**

  -or-

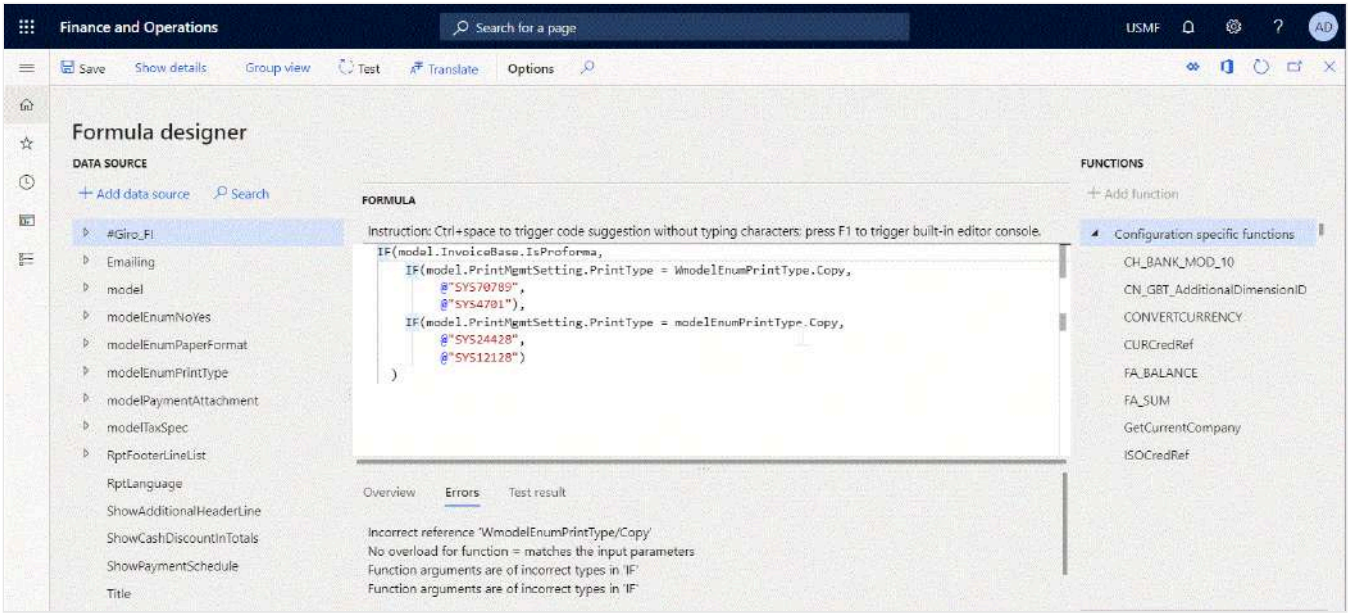
- Press **F1**, press **FO**, select **Fold all**, and then press **Enter**.

To unfold all regions, use one of the following methods:

- Press **Ctrl+J**

-or-

- Press **F1**, type **UN**, select **Unfold all**, and then press **Enter**.



Find and replace

To find occurrences of certain text, select the text in your expression, and do the following steps:

- Press **Ctrl+F** and then press **F3** to find the next occurrence of the selected text, or press **Shift+F3** to find the previous occurrence.

  -or-

- Press **F1**, type **F**, and then select the required option to find the selected text.

To replace occurrences of certain text, select the text in your expression, and do the following steps:

- Press **Ctrl+H**. Enter the alternative text and select the replacement option to replace either the selected text or all occurrences of this text in the current expression.

  -or-

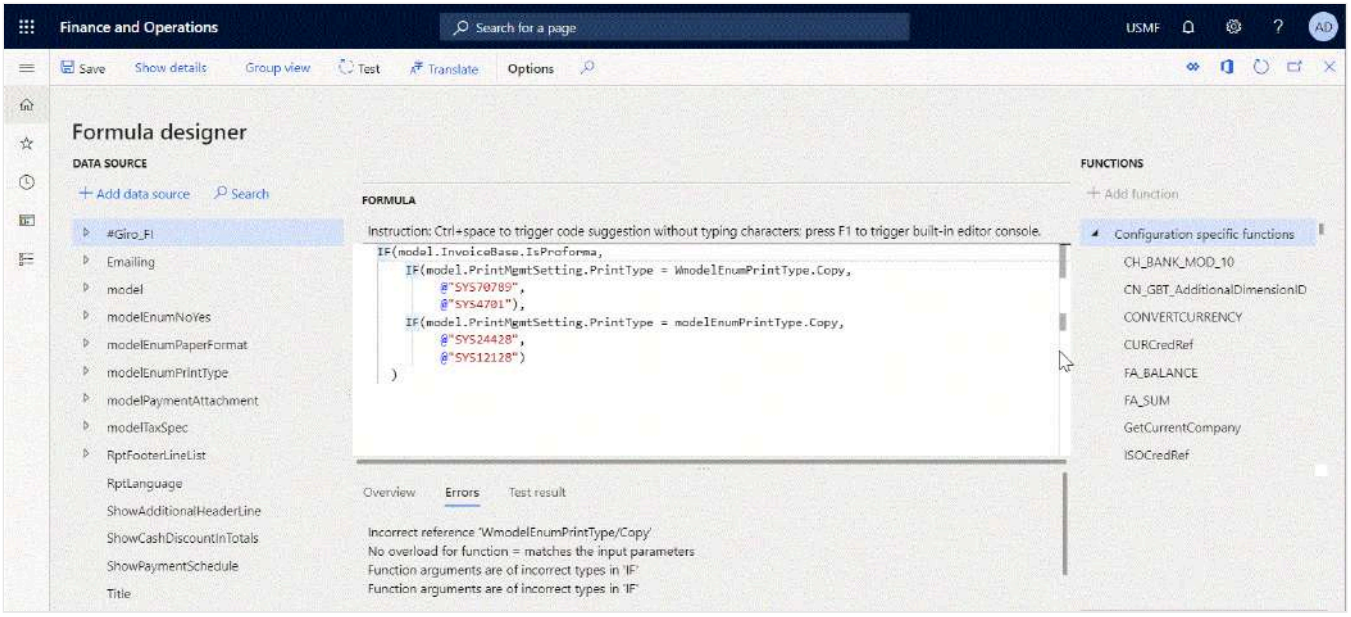
- Press **F1**, type **R**, and then select the required option to replace the selected text. Enter the alternative text and select the replacement option to replace either the selected text or all occurrences of this text in the current expression.

To change all occurrences of certain text, select the text in your expression, and do the following steps:

- Press **Ctrl+F2** and then enter the alternative text.

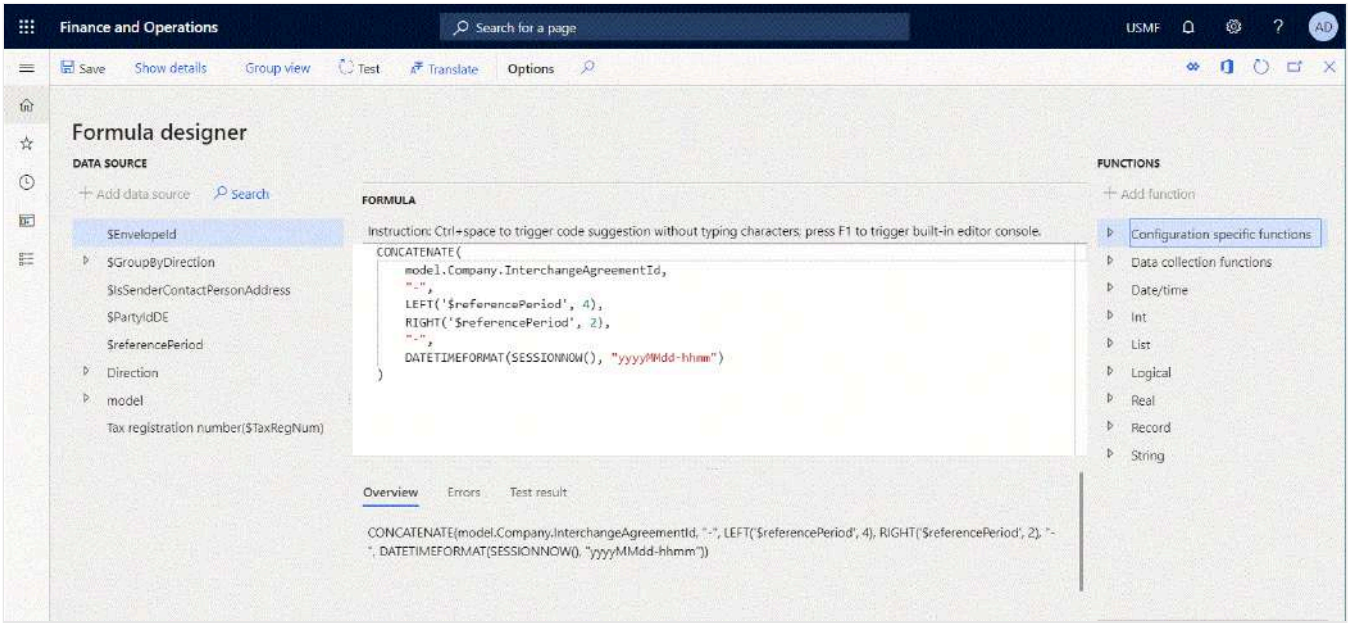
  -or-

- Press **F1**, type **C**, and then select the required option to change the selected text. Enter the alternative text.



Data sources and functions pasting

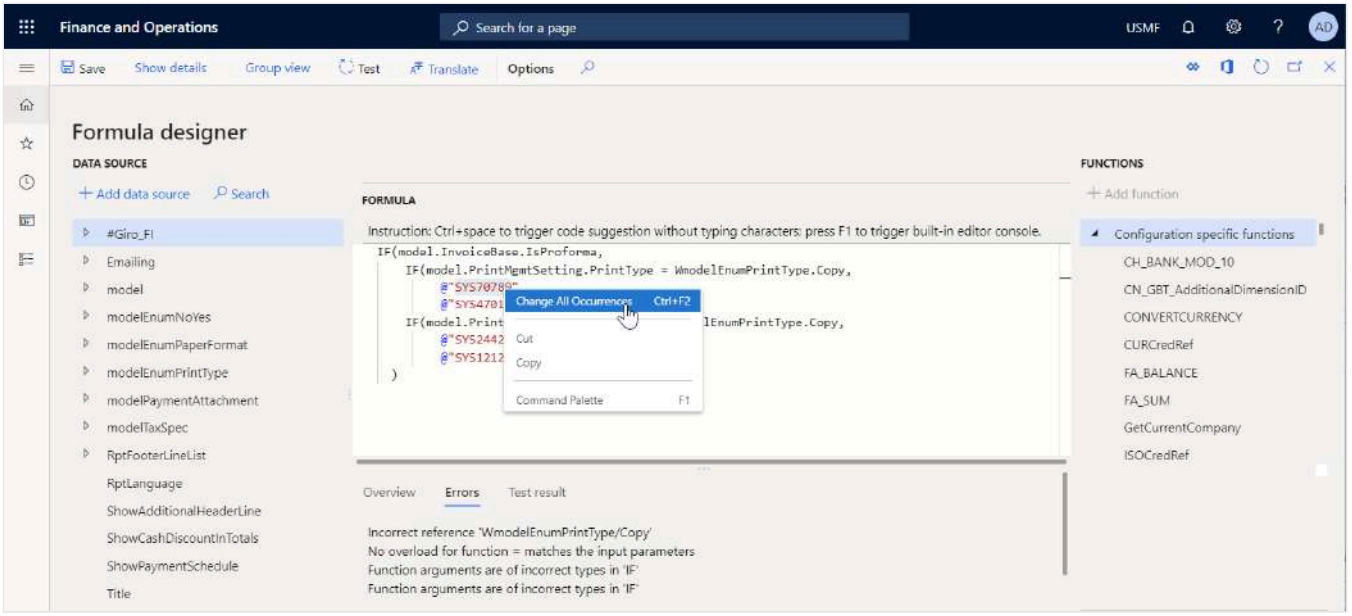
Select **Add data source** to paste a data source from the **Data source** left panel into the current expression. Select **Add function** to paste a function from the **Functions** right panel into the current expression. If you use the ER formula editor, the selected function or data source is always added to the end of the configured expression. When you use the advanced ER formula editor, you can add the selected function or data source to any part of the configured expression. Use the cursor to specify where you want to paste the data.



Syntax colorization

Currently, the editor uses different colors to highlight the following parts of expressions:

- The text in double brackets that can represent a label ID of a text constant.



Limitations

The editor currently supports the following web browsers:

- Chrome
- Edge
- Firefox
- Opera
- Safari

Additional resources

- Electronic reporting (ER) overview
- Formula designer in Electronic reporting
- Monaco editor

Last updated on 03/17/2026

Electronic reporting formula language

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) provides a powerful data transformation experience. The language that you use to express the required data manipulations in the ER formula designer resembles the formula language in Microsoft Excel.

Basic syntax

ER expressions can contain any or all of the following elements:

- Constants
- Operators
- References
- Paths
- Functions

Constants

When you design expressions, use text and numeric constants (that is, values that aren't calculated). For example, the expression `VALUE ("100") + 20` uses the numeric constant **20** and the string constant **"100"**, and it returns the numeric value **120**.

The ER formula designer supports escape sequences. Therefore, you can specify an expression string that should be handled differently. For example, the expression `"Leo Tolstoy ""War and Peace"" Volume 1"` returns the text string **Leo Tolstoy "War and Peace" Volume 1**.

Operators

The following table shows the arithmetic operators that you can use to do basic mathematical operations, such as addition, subtraction, multiplication, and division.

Expand table

Operator	Meaning	Example
+	Addition	`1+2`
-	Subtraction, negation	`5-2`, `-1`
*	Multiplication	`7\*8`
/	Division	`9/3`

The following table shows the comparison operators that are supported. You can use these operators to compare two values.

Expand table

Operator	Meaning	Example
=	Equal	`X=Y`
>	Greater than	`X>Y`
<	Less than	`X<Y`
>=	Greater than or equal to	`X>=Y`
<=	Less than or equal to	`X<=Y`
<>	Not equal to	`X<>Y`

Additionally, you can use an ampersand (&) as a text concatenation operator. In this way, you can join, or concatenate, one or more text strings into a single piece of text.

Expand table

Operator	Meaning	Example
&	Concatenate	`"Nothing to print:" & " " & "no records found"`

Operator precedence

The order in which the parts of a compound expression are evaluated is important. For example, the result of the expression `1 + 4 / 2` varies, depending on whether the addition or division operation is done first. You can use parentheses to explicitly define how an expression is evaluated. For example, to indicate that the addition operation should be done first, you can

change the preceding expression to `(1 + 4) / 2`. If you don't explicitly indicate the order of operations in an expression, the order is based on the default precedence that is assigned to the supported operators. The following table shows the precedence that is assigned to each operator. Operators that have a higher precedence (for example, 7) are evaluated before operators that have a lower precedence (for example, 1).

Expand table

Precedence	Operators	Syntax
7	Grouping	(...)
6	Member access	... .
5	Function call	... (...)
4	Multiplicative	... * ... ... / ...
3	Additive	... + ... - ...
2	Comparison	... < ... ... <= ... ... => ... ... > ... ... = ... <> ...
1	Separation	... , ...

If an expression includes multiple consecutive operators that have the same precedence, those operations are evaluated from left to right. For example, the expression `1 + 6 / 2 \* 3 > 5` returns **true**. Use parentheses to explicitly indicate the desired order of operations in expressions, so that the expressions are easier to read and maintain.

References

You can use all data sources of the current ER component that are available during the design of an expression as named references. The current ER component can be either a model mapping or a format. For example, the current ER model mapping contains the **ReportingDate** data source, which returns a value of the *DateTime* data type. To correctly format that value in the generating document, reference the data source in the expression as `DATETIMEFORMAT (ReportingDate, "dd-MM-yyyy")`.

Precede all characters in the name of a referencing data source that don't represent a letter of the alphabet with a single quotation mark ('). If the name of a referencing data source contains at least one symbol that doesn't represent a letter of the alphabet, enclose the name in single quotation marks. For example, these non-alphabetic symbols can be punctuation marks or other written symbols. Here are some examples:

- Refer to the **Today's date & time** data source in an ER expression as `'Today''s date & time'`.
- Refer to the **name()** method of the **Customers** data source in an ER expression as `Customers.'name()'`.

If the methods of application data sources have parameters, use the following syntax to call those methods:

- If the **isLanguageRTL** method of the **System** data source has an **EN-US** parameter of the *String* data type, refer to this method in an ER expression as `System.isLanguageRTL("EN-US")`.
- Quotation marks aren't required when a method name contains only alphanumeric symbols. However, they're required for a method of a table if the name includes brackets.

When you add the **System** data source to an ER mapping that refers to the **Global** application class, the expression `System.isLanguageRTL("EN-US ")` returns the *Boolean* value **FALSE**. The modified expression `System.isLanguageRTL("AR")` returns the *Boolean* value **TRUE**.

You can limit the way that values are passed to the parameters of this type of method:

- Only constants can be passed to methods of this type. Define the values of the constants at design time.
- Only primitive (basic) data types are supported for parameters of this type. The primitive data types include *Integer*, *Real*, *Boolean*, and *String*.

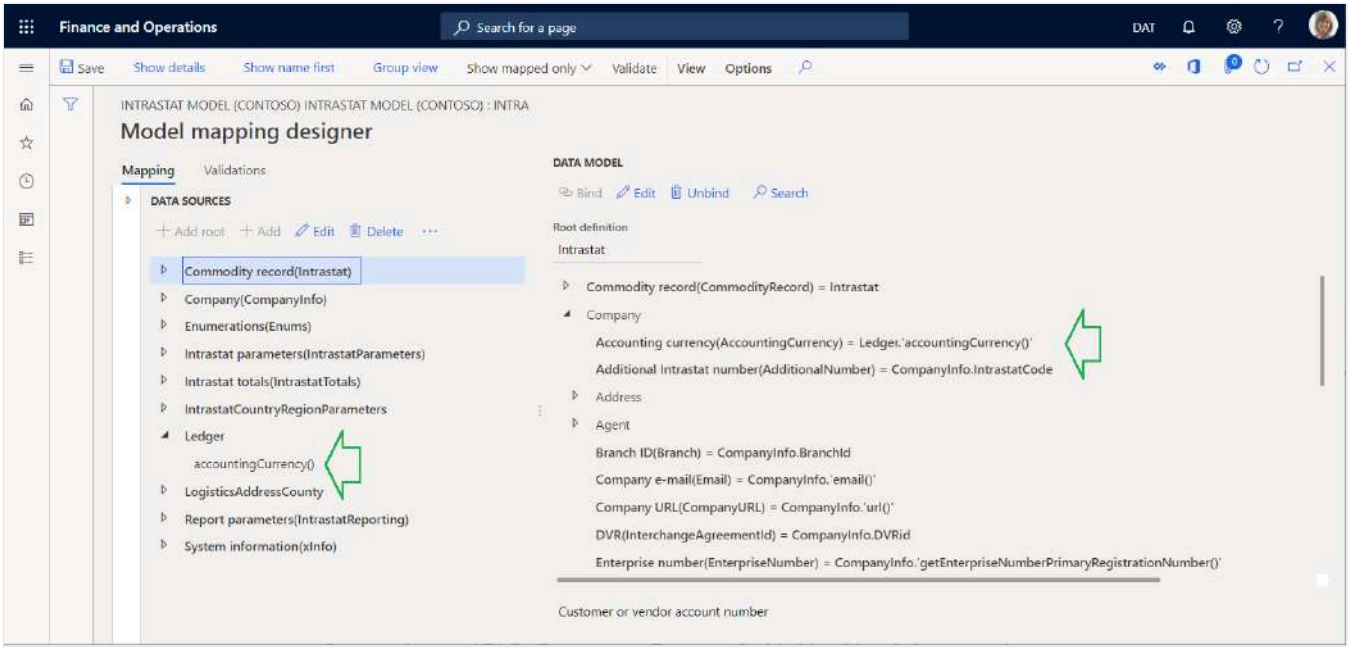
Paths

When an expression references a structured data source, use the path definition to select a specific primitive element of that data source. Use a dot character (.) to separate individual elements of a structured data source. For example, the current ER model mapping contains the **InvoiceTransactions** data source, and this data source returns a list of records. The **InvoiceTransactions** record structure contains the **AmountDebit** and **AmountCredit** fields, and both these fields return numeric values. Therefore, you can design the following expression to calculate the invoiced amount: `InvoiceTransactions.AmountDebit -`

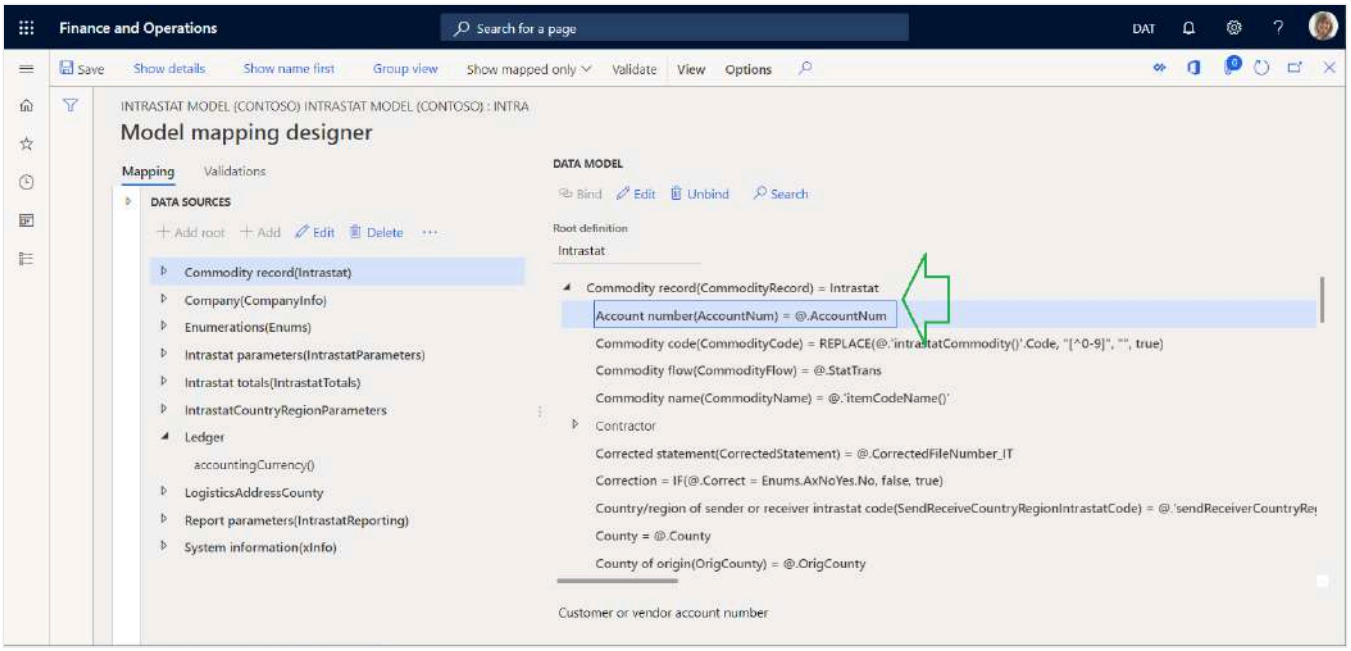
`InvoiceTransactions.AmountCredit`. The `InvoiceTransactions.AmountDebit` construction in this expression is the path that is used to access the **AmountDebit** field of the **InvoiceTransactions** data source of the *Record list* type.

Relative path

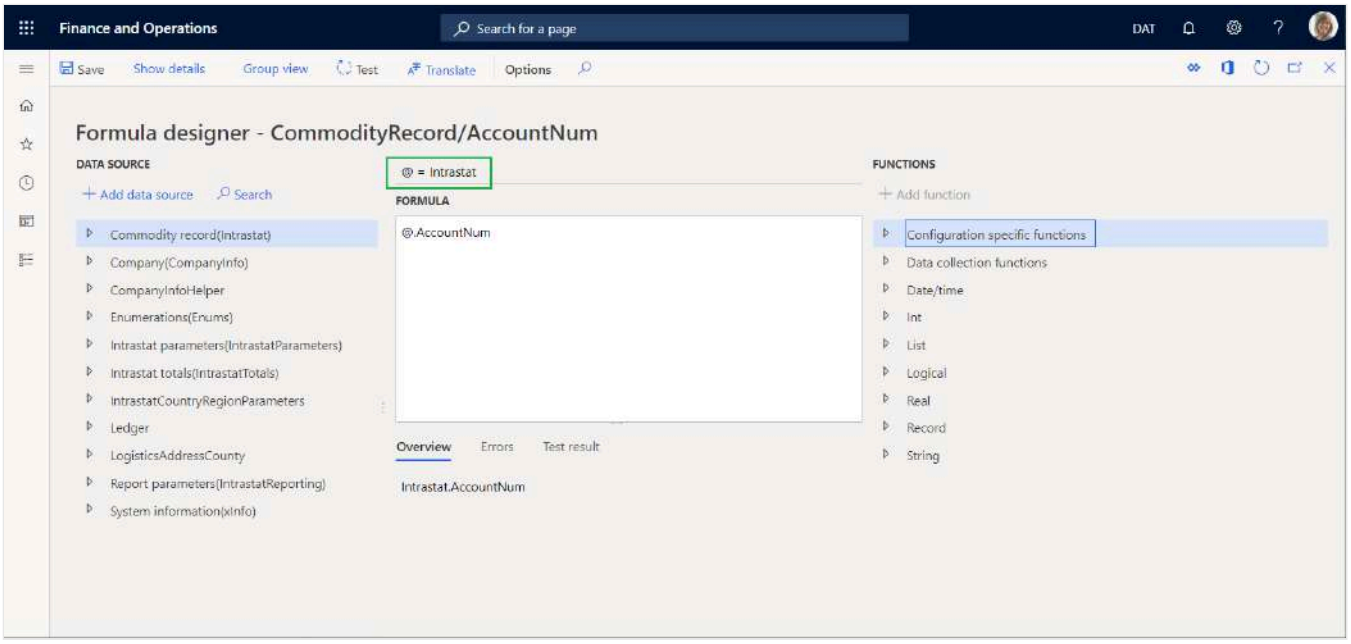
If the path of a structured data source starts with an `@` sign, it's a relative path. The `@` sign represents the remaining part of the absolute path of the hierarchical tree structure that the relative path uses. The following illustration shows an example. In this example, the absolute path `Ledger.'accountingCurrency()'` indicates that the accounting currency value from the **Ledger** data source is entered in the **AccountingCurrency** field of the data model.



The example in the following illustration shows how a relative path is used. The relative path `@.AccountNum` indicates that the **AccountNum** field of the **Intrastat** data source (which appears one level above the **AccountNum** field in the data model's hierarchical tree) is used to enter the customer or vendor account number in the data model's **AccountNum** field.



The ER formula editor also shows the remaining part of the absolute path.



For more information, see Use a relative path in data bindings of ER models and formats.

Functions

Use ER built-in functions in ER expressions. You can use all data sources of the expression context (that is, the current ER model mapping or ER format) as parameters when calling functions, as long as they follow the list of arguments for calling functions. You can also use constants as parameters. For example, the current ER model mapping contains the **InvoiceTransactions** data source, and this data source returns a list of records. The **InvoiceTransactions** record structure contains the **AmountDebit** and **AmountCredit** fields, and both these fields return numeric values. To calculate the invoiced amount, you can design the

following expression that uses the built-in ER rounding function: `ROUND(InvoiceTransactions.AmountDebit - InvoiceTransactions.AmountCredit, 2)`.

When you design ER model mappings and ER reports, use ER functions from the following categories:

- Date and time functions
- List functions
- Logical functions
- Mathematical functions
- Record functions
- Text functions
- Data collection functions
- Other (business domain–specific) functions
- Type conversion functions

Functions list extension

ER lets you extend the list of functions that you use in ER expressions. Some engineering effort is required. For detailed information, see Extend the list of Electronic reporting (ER) functions.

Compound expressions

You can create compound expressions that use functions from different categories, as long as the data types match. When you use functions together, match the data type of the output from one function to the input data type that another function requires. For example, to avoid a possible "list-is-empty" error in a binding of a field to an ER format element, combine functions from the List category with a function from the Logical category, as the following example shows. Here, the formula uses the IF function to test whether the **IntrastatTotals** list is empty before it returns the value of the required aggregation from that list. If the **IntrastatTotals** list is empty, the formula returns **0** (zero).

VB

```
IF(ISEMPTY(IntrastatTotals), 0.0, IntrastatTotals.aggregated.'$AmountMSTRounded')
```

Multiple solutions

Often, you can get the same data transformation result in multiple ways, by using functions from different categories or different functions from the same category. For example, the previous expression can also be configured by using the COUNT function from the List category.

VB

```
IF(COUNT (IntrastatTotals)=0, 0.0, IntrastatTotals.aggregated.'$AmountMSTRounded')
```

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Extend the list of Electronic reporting functions

Supported primitive data types

Supported composite data types

Last updated on 04/13/2026

Supported primitive data types for Electronic reporting formulas

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about the primitive data types that are supported in Electronic reporting (ER) expressions. Here is a list of the primitive data types:

- boolean
- date
- datetime
- enumeration
- guid
- integer
- int64
- real
- string

Boolean

The *boolean* primitive data type contains a value that is either *true* or *false*. Use the reserved literal keywords **True** and **False** wherever a *boolean* expression is expected. The default value is *false*.

The internal representation of a *boolean* is an *integer*. The *integer* value 0 (zero) represents *false*, and all other *integer* values represent *true*. When you validate a configured expression that returns a *boolean* in the ER formula designer, the test result pane shows *0* (zero) when an expression returns *false*. Otherwise, the test result pane shows *1*.

A *boolean* has no implicit conversions. However, you can use the TEXT function to explicitly convert a *boolean* to a *string*:

- The *false* value converts to the text string **False**.
- The *true* value converts to the text string **True**.

> ⓘ **Note**
>
> This conversion doesn't depend on the provided language and culture **context**.

Comparison operators are the only type of operator that you can use with the *boolean* data type. The following operators can compare two *boolean* values: <> and =.

Date

The *date* primitive data type contains the day, month, and year. Use the following functions to create dates:

- DATEVALUE
- NULLDATE
- SESSIONTODAY
- TODAY

The *date* data type holds dates between January 1, 1900, and December 31, 2154. The default value is **null**, and the internal representation is the date January 1, 1900.

A *date* has no implicit conversions. However, you can use the following explicit conversion functions:

- DATEFORMAT
- DATETODATETIME
- TEXT

The ADDDAYS function adds and subtracts days from dates. By using this function, you can move the date a specific number of days into the future or the past. The DAYS function subtracts dates from each other and calculates the difference in days. For more information about the transformation of *date* values, see List of ER functions in the Date and time category.

Comparison operators are the only type of operator that you can use with the *date* data type. Use the following operators to compare two *date* values: <>, <, <=, =, >, and >=.

Datetime

The *datetime* primitive data type combines the *date* type and a value that represents the time that has passed since midnight. The time is expressed in hours, minutes, seconds, and fractions of a second. A *datetime* value also holds information about the time zone.

The *datetime* data type can hold dates between January 1, 1900 (1900-01-01T00:00:00.0000000+00:00 in the round-trip format) and December 31, 2154 (2154/12/31T11:59:59.9999999+00:00 in the round-trip format). The smallest unit of time in a *datetime* is one ten millionth of a second.

> ⓘ **Note**
>
> When you use the **hh** specifier for hours, time values that go beyond 12:59:59.9999999 aren't valid times.
>
> When you use the **HH** specifier for hours, time values that go beyond 23:59:59.9999999 aren't valid times.

The default value is **null**, and the internal representation is the date January 1, 1900 (1900-01-01T00:00:00.0000000+00:00 in the round-trip format).

Initiate datetimes by using the following functions:

- DATETIMEVALUE
- NULNULLDATETIMELDATE
- SESSIONNOW
- NOW

A *datetime* has no implicit conversions. However, you can use the following explicit conversion functions:

- DATETIMEFORMAT
- TEXT

For more information about the transformation of *datetime* values, see List of ER functions in the Date and time category.

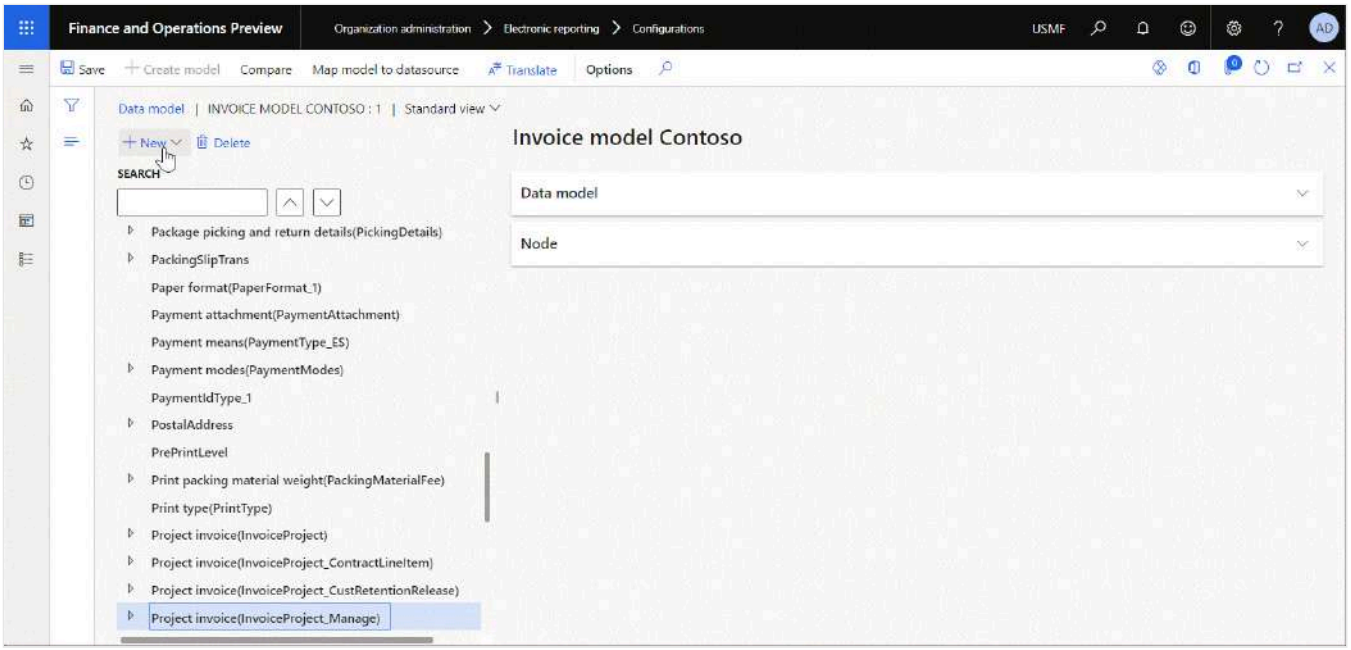
Comparison operators are the only type of operator that you can use with the *datetime* data type. The following operators can compare two *datetime* values: <>, <, <=, =, >, and >=.

Enumeration

The *enumeration* primitive data type is a list of literals. You can use enumerations that are defined in the application source code. You can also introduce your own enumerations in the ER data model and ER format components.

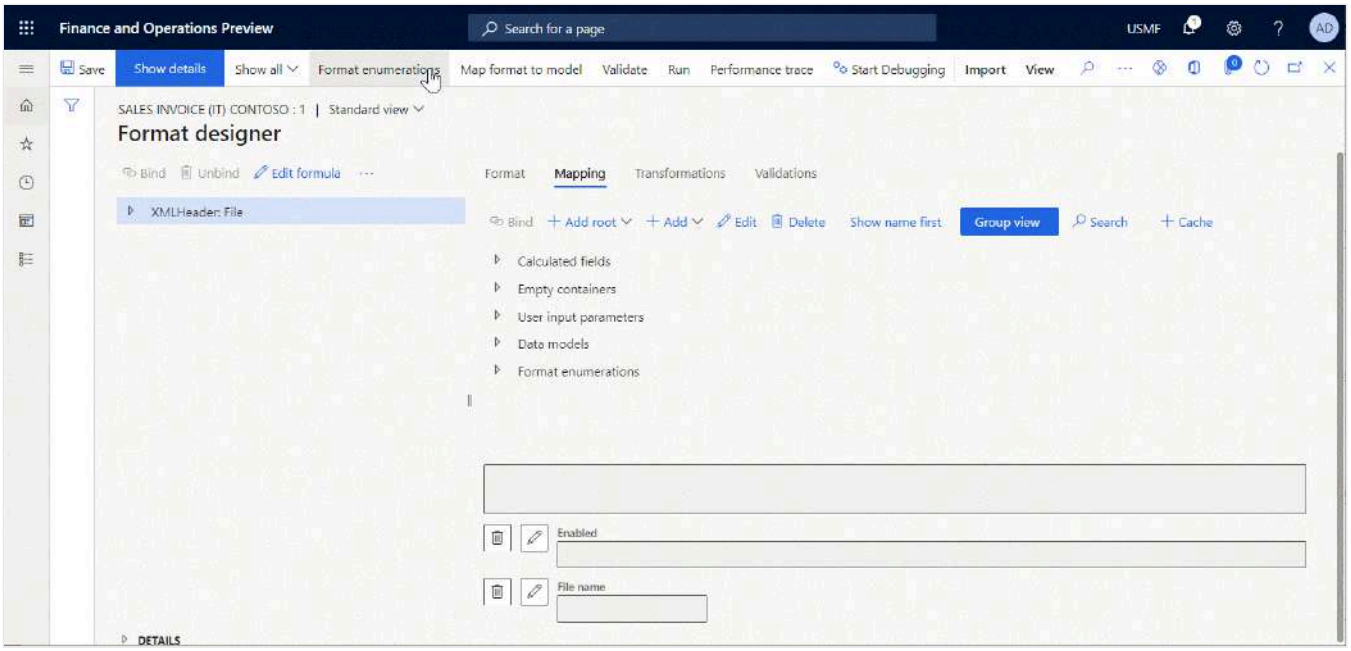
You can use an application *enumeration* in expressions of any ER model mapping and ER format.

The following illustration shows how you can add the **CustVendCorrectiveReasonCode** model enumeration to the editable ER data model.



You can use a model *enumeration* in expressions of any ER model mapping and ER format that you create under a data model where you introduced the *enumeration*.

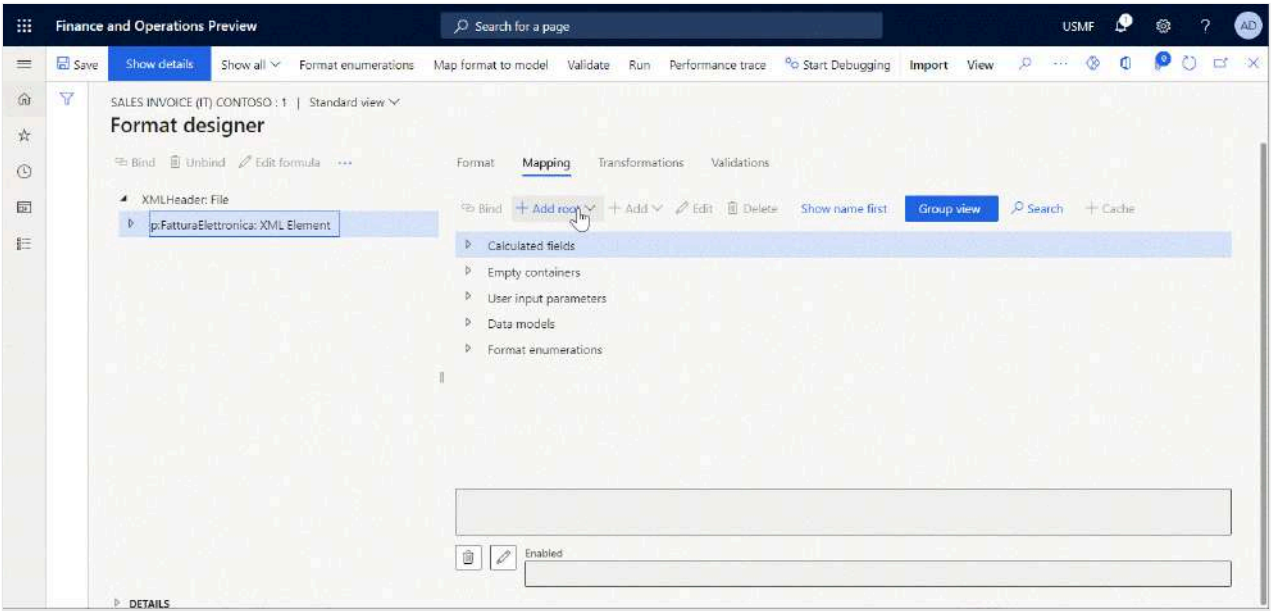
The following illustration shows how you can add the **List of Natura reverse charge subcategories** format enumeration to the editable ER format.



You can use a format *enumeration* only in expressions of the ER format where you introduced the *enumeration*.

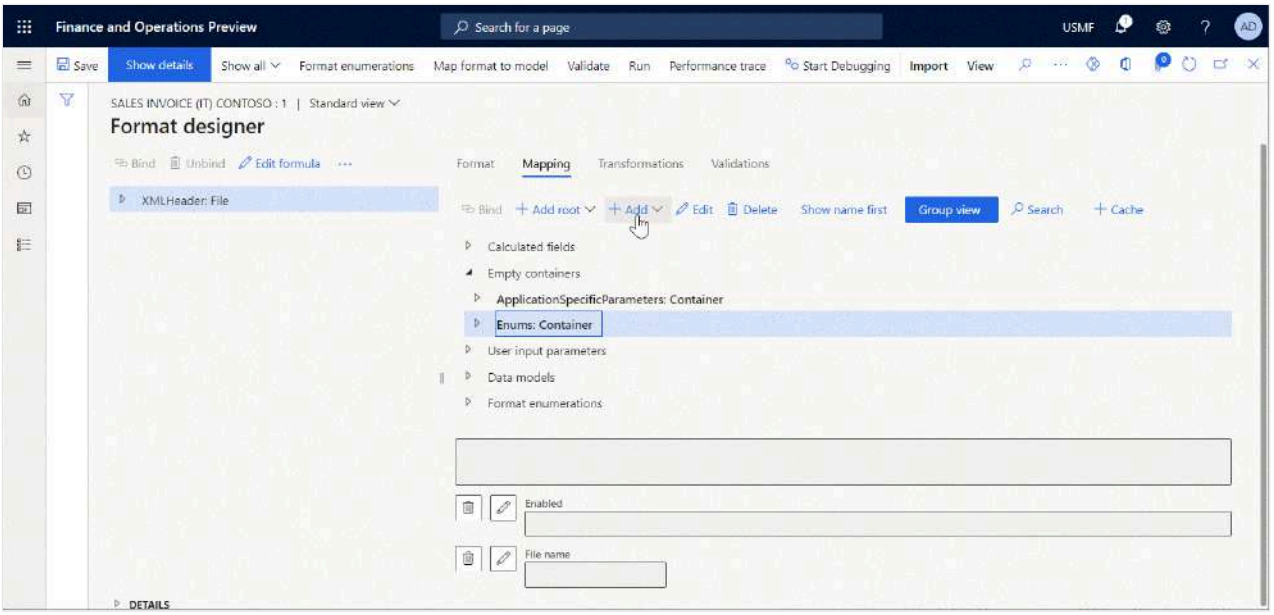
Use the appropriate type of ER data sources to bring a specific enumeration to a configured ER component as a constant or as a value that the user running an ER solution defined in the dialog box at runtime.

- Access application enumerations by using the **Dynamics 365 for Operations \ Enumeration** and **General \ User input parameters** data sources. The following illustration shows how you can add to the editable ER format the **appenumNoYes** and **uipNoYes** data sources that refer to the **NoYes** application enumeration.

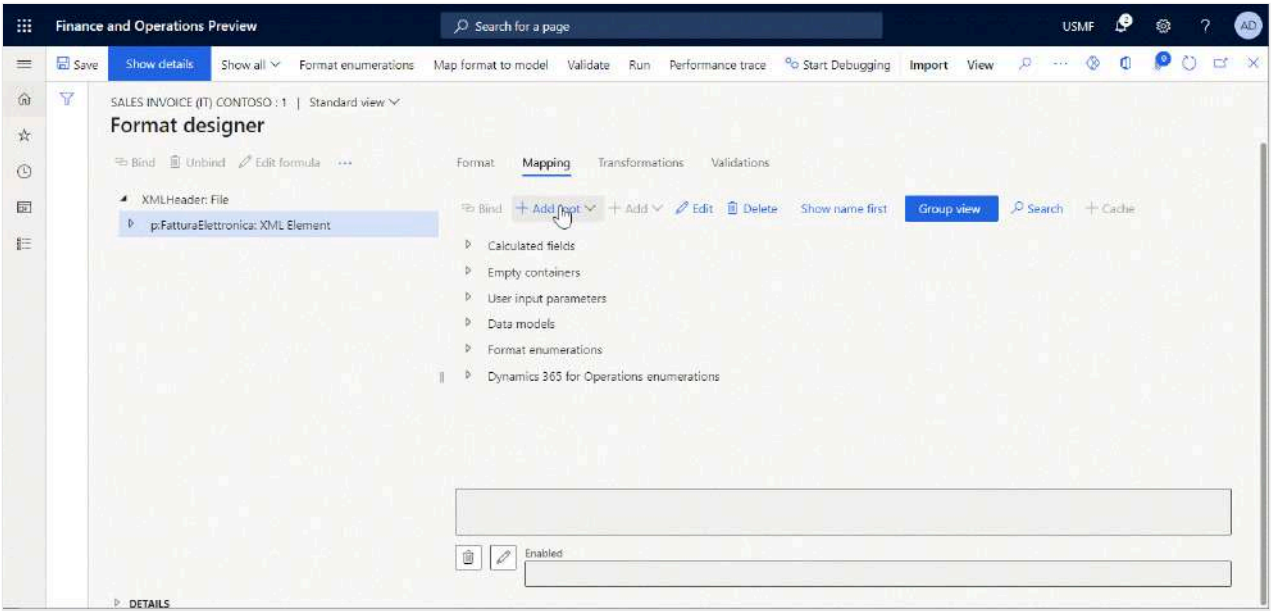


- Access data model enumerations by using the **Data model \ Enumeration** and **Data model \ Enumeration user input parameters** data sources. The following illustration

shows how you can add to the editable ER format the **CustVendCorrectiveReasonCode** data source that refers to the **CustVendCorrectiveReasonCode** data model enumeration.



- Access format enumerations by using the **Format \ Enumeration** and **Format \ Enumeration user input parameters** data sources. The following illustration shows how you can add to the editable ER format the **NaturaReverseCharge** data source that refers to the **Natura reverse charge subcategories** format enumeration.



An *enumeration* has no implicit conversions. However, you can use the TEXT conversion function to convert an *enumeration* to a text string. This conversion isn't language dependent. To learn how you can associate an *enumeration* value with the appropriate language-specific labels, see the usage examples for the LISTOFFIELDS and GETENUMVALUEBYNAME functions.

Comparison operators are the only type of operator that you can use with the *enumeration* data type. The following operators can be used to compare two *enumeration* values: <> and =.

Guid

The *guid* primitive data type holds a globally unique identifier (GUID) value. A GUID is a value that you can use across all computers and networks, wherever a unique identifier is required. It's unlikely that the number is duplicated. A valid GUID meets all the following specifications:

- It has 32 hexadecimal digits.
- It has four dash characters that are embedded at the following locations: 8-4-4-4-12.
- You can add optional braces {} at the beginning and end of the string. For example, both **{2CDB0FE7-D7B3-4938-A0F0-FE28FB8FE212}** and **2CDB0FE7-D7B3-4938-A0F0-FE28FB8FE212** are valid GUID strings.
- It has a total of 36 or 38 characters, depending on whether you add braces.
- The letters that are used as hexadecimal digits can be uppercase (A–F), lowercase (a–f), or mixed.

Use the following explicit conversion functions:

- GUIDVALUE
- TEXT

Comparison operators are the only type of operator that you can use with the *guid* data type. Use the following operators to compare two *guid* values: <> and =.

Integer

The *integer* primitive data type represents a number that has no decimal places. Use integers as control variables in repetitive statements or as indexes in record lists.

An *integer* literal is the integer as it's entered directly in an ER expression (formula), such as **12345**. An *integer* is 32 bits wide. The default value is **0**, and the internal representation is a long number. An *integer* is automatically converted to a *real*.

Additionally, use the following explicit conversion functions:

- INTVALUE
- NUMBERFORMAT
- TEXT

The range of an *integer* is [-2,147,483,647 : 2,147,483,647]. You can use all integers in this range as literals.

Use all comparison and mathematical operators with the *integer* data type.

Int64

The *int64* primitive data type represents a number that has no decimal places. Use *Int64* values as control variables in repetitive statements or as record identifiers.

An *int64* is 64 bits wide. The default value is **0**, and the internal representation is a long number. An *int64* automatically converts to a *real*.

Additionally, use the following explicit conversion functions:

- INT64VALUE
- NUMBERFORMAT
- TEXT

The range of an *int64* is [-9,223,372,036,854,775,807 : 9,223,372,036,854,775,807].

You can use all comparison and mathematical operators with the *int64* data type.

Real

The *real* primitive data type can hold decimal values in addition to integers. You can use decimal literals anywhere that a *real* is expected. A decimal literal is the decimal as it's entered directly in code, such as **2.19**.

> ⓘ **Note**
>
> In ER expressions, a period (.) is always used as the decimal separator.

You can use reals in all expressions, and you can use them with both comparison and arithmetic operators. A *real* has a precision of 16 significant digits. The default value for a *real* is **0.0**, and the internal representation is a binary-coded digital (BCD) number. The BCD encoding enables exact representations of values that are multiples of 0.1. The range of a *real* variable is $-(10)^{127}$ through $(10)^{127}$. All reals in this range can be used as literals in ER expressions.

A *real* has no implicit conversions. However, you can use the following functions to explicitly convert a *real* to other data types and other data types to a *real*:

- INTVALUE

- INT64VALUE
- NUMBERFORMAT
- TEXT
- VALUE

You can use all comparison and mathematical operators with the *real* data type.

String

The *string* primitive data type represents a sequence of characters that are used as texts, account numbers, addresses, and telephone numbers.

String literals are characters that are enclosed in quotation marks (""). *String* literals can be used wherever *string* values are expected in ER expressions. You can use strings in logical expressions, such as comparisons. You can also concatenate *string* values by using the **&** operator or the CONCATENATE function.

> ⓘ **Note**
>
> If you concatenate two *string* values, and you want the resulting *string* to span more than one line, use the line break separator between the values. For the TEXT output, this separator can be a character that is generated by using the CHAR(10) or CHAR(13) expression. For HTML, it can be the **
** tag.

The default value for a *string* is a blank text string that has no characters, and the internal representation is a list of characters.

There are no automatic conversions for strings. However, you can use the following explicit conversion functions:

- CHAR
- FORMAT
- LEFT
- LEN
- MID
- PADLEFT
- REPLACE
- RIGHT
- TEXT

- TRANSLATE
- TRIM
- UPPER

For more information about the transformation of *string* values, see List of ER functions of the text category.

A *string* can hold an indefinite number of characters.

All comparison operators can be used with the *string* data type.

Additional resources

- Electronic Reporting overview
- Electronic reporting formula language
- Supported composite data types

Last updated on 04/13/2026

Supported composite data types for Electronic reporting formulas

This article provides information about the composite data types that are supported in Electronic reporting (ER) expressions. The composite data types are class, container, record, record list, and object.

Class

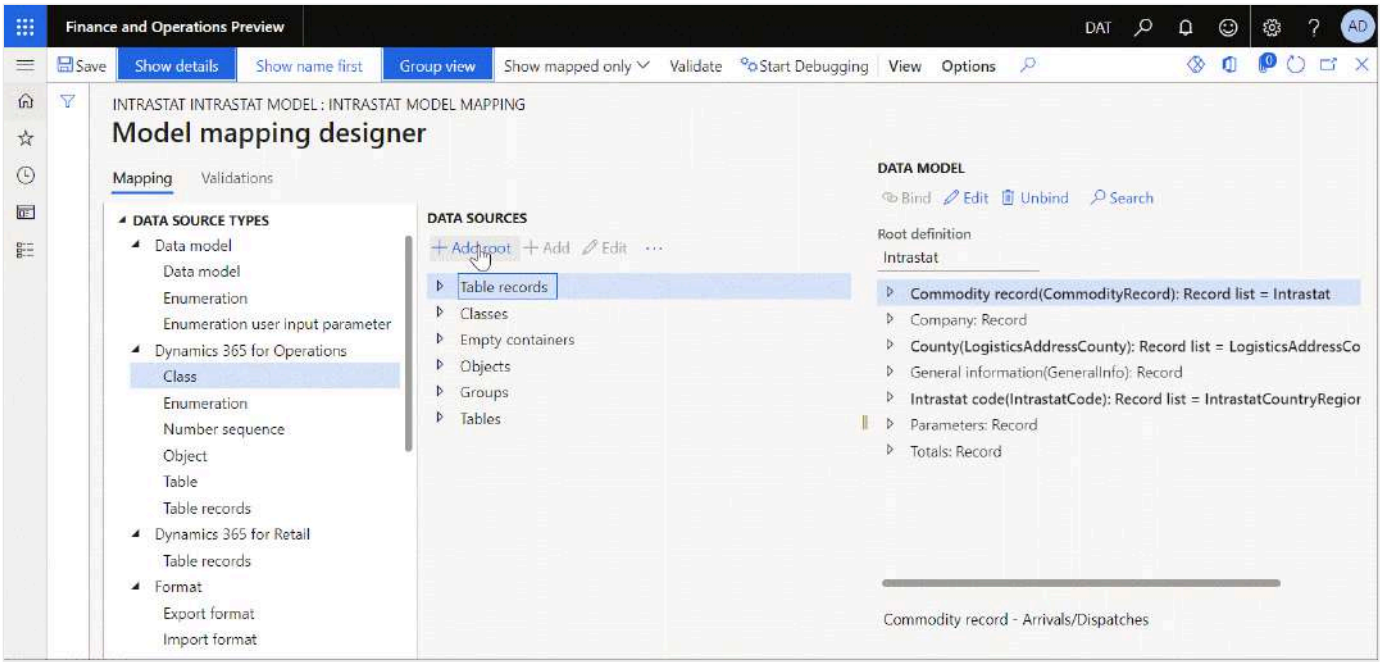
The *class* data type refers to a public application class. In ER, it's represented as a *record* that contains a separate field for every public method of the referenced class. When the call of the method is parameterized, you must also specify the required arguments of the appropriate types in an ER expression that is configured to call the method.

In ER mapping and format components, you can add the **Class** data source that is presented as a data source and that returns a value of the *class* type. This data source exposes public methods of the class that you can call at runtime.

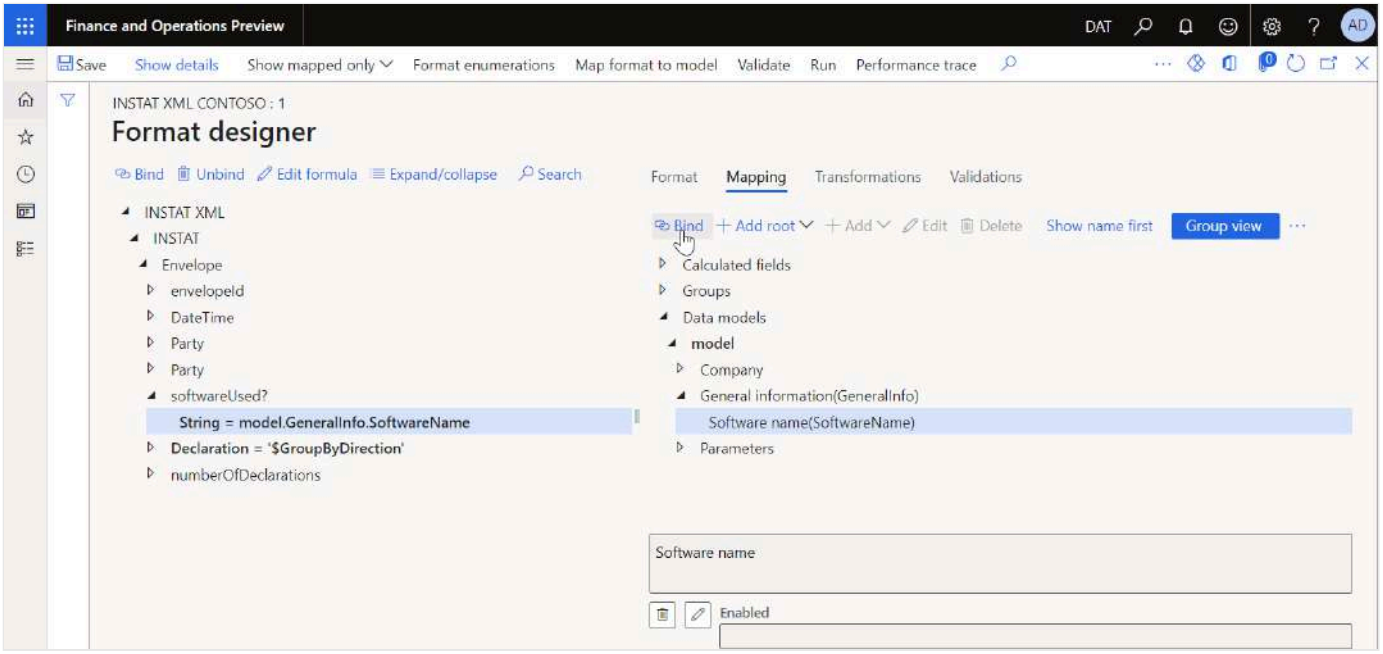
> ⓘ **Note**
>
> Only methods that return a value can be called from ER expressions.
>
> Only methods that have a range of zero to eight arguments can be called from ER expressions.

The default value of a *class* is **null**.

The following illustration shows how the **System information(xInfo)** data source of the **Class** type is added to make the instance of the **xInfo** application class and call its **productName()** method to receive the name of the current application. The name of the current application is fetched at runtime by execution of the `xInfo.productName` binding that you configured for the **Software name(SoftwareName)** field of the ER data model. This binding calls the `productName()` method of the **xInfo** application class that is represented in the current model mapping as the **System information(xInfo)** data source.



The following illustration shows how the ER format is configured to put the provided application name in generated documents. The **Software name(SoftwareName)** field of the used data model is bound to the **String** component that is nested under the **softwareUsed** XML element of the ER format. So, the name of the current application is placed at runtime to the **softwareUsed** XML element of a generated document in XML format.



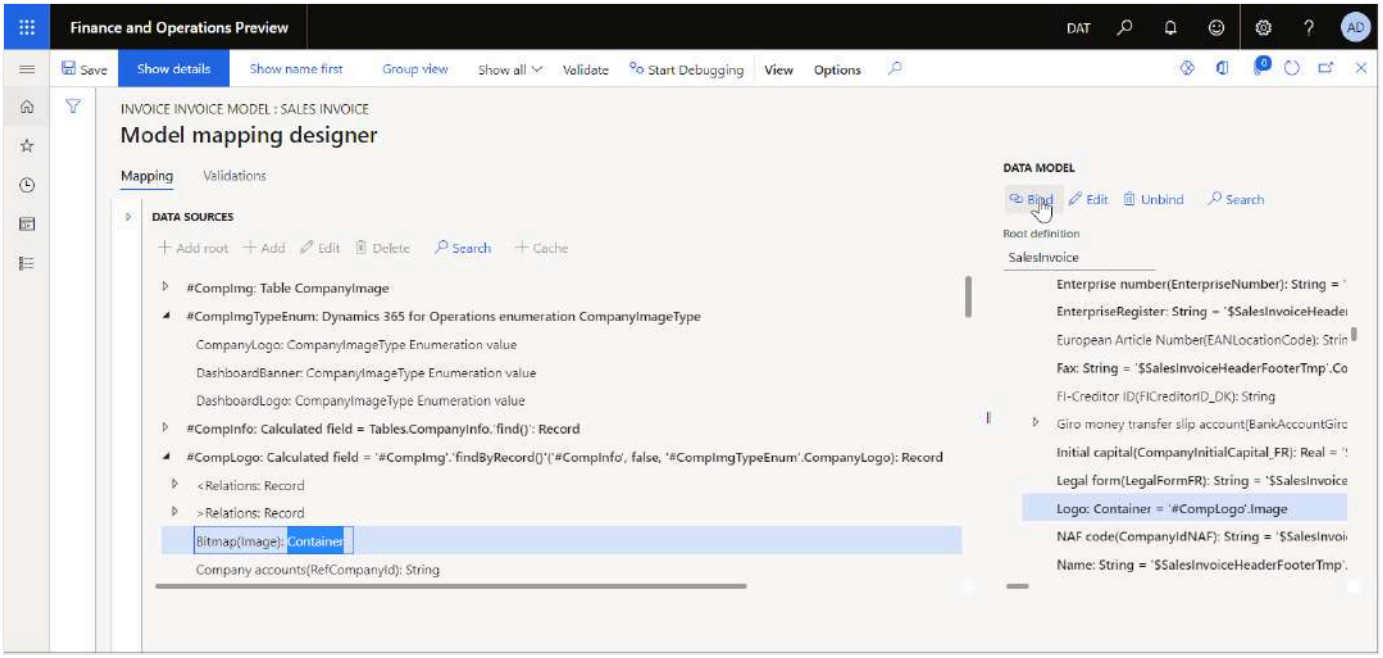
Container

The *container* data type holds binary content. Use a *container* value to pass specific information from storage to a generated document. In the ER framework, use this data type to include media content such as a company logo in generated documents.

> **Note**
>
> Although you can represent every media item as a *container* value, not every *container* value represents a media item. Therefore, if you configure an ER format to use a *container* to include an image in generated documents, but the referenced *container* doesn't return media content, an exception might be thrown. The exception resembles the following example: "Error executing code: Binary (object), method constructFromContainer called with invalid parameters."

The default value of a *container* is **null**.

The following illustration shows how the **Bitmap(Image)** field of the *Container* type is bound to the data model **Logo** field of the **Container** type in the **Sales invoice** model mapping. This binding makes the company logo available to any ER format that is designed for the **SalesInvoice** root definition and that uses this model mapping at runtime.



Record

A *record* is a collection of named fields, each associated with a value of either a primitive data type or a composite data type. Usually, you use a *record* to represent a single record in a record list. In this case, every item represents individual fields, methods, and relations.

The default value of a *record* is **empty**.

> **Note**

When you get the value of a field in an empty *record*, the default value of the appropriate data type is returned.

You can get a *record* by using the following functions:

- FIRST
- FIRSTORNULL
- EMPTYRECORD
- NULLCONTAINER

For more information about the transformation of *record* values, see List of ER functions in the list category.

Record list

A *record list* is a list of items of the *record* type. Usually, you use a *record list* to represent the list of records that you fetch from a database table.

By default, you access records in a *record list* sequentially. To access a specific record, use the INDEX function and specify the *integer* index.

The default value of a *record list* is **empty**. Use the ISEMPTY function to check whether a *record list* is empty.

> ⓘ **Note**
>
> If a *record list* is empty, any attempt to get a field value for a *record* in it causes an exception to be thrown at runtime. To learn how you can help prevent runtime exceptions of this type, see **Consideration of empty list cases**.

Initiate a *record list* by using the following functions:

- ALLITEMS
- ALLITEMSQUERY
- EMPTYLIST
- LIST
- LISTDISTINCT

For more information about the transformation of *record list* values, see List of ER functions in the list category. To learn how to introduce *record list* items, fill them with application data, and then

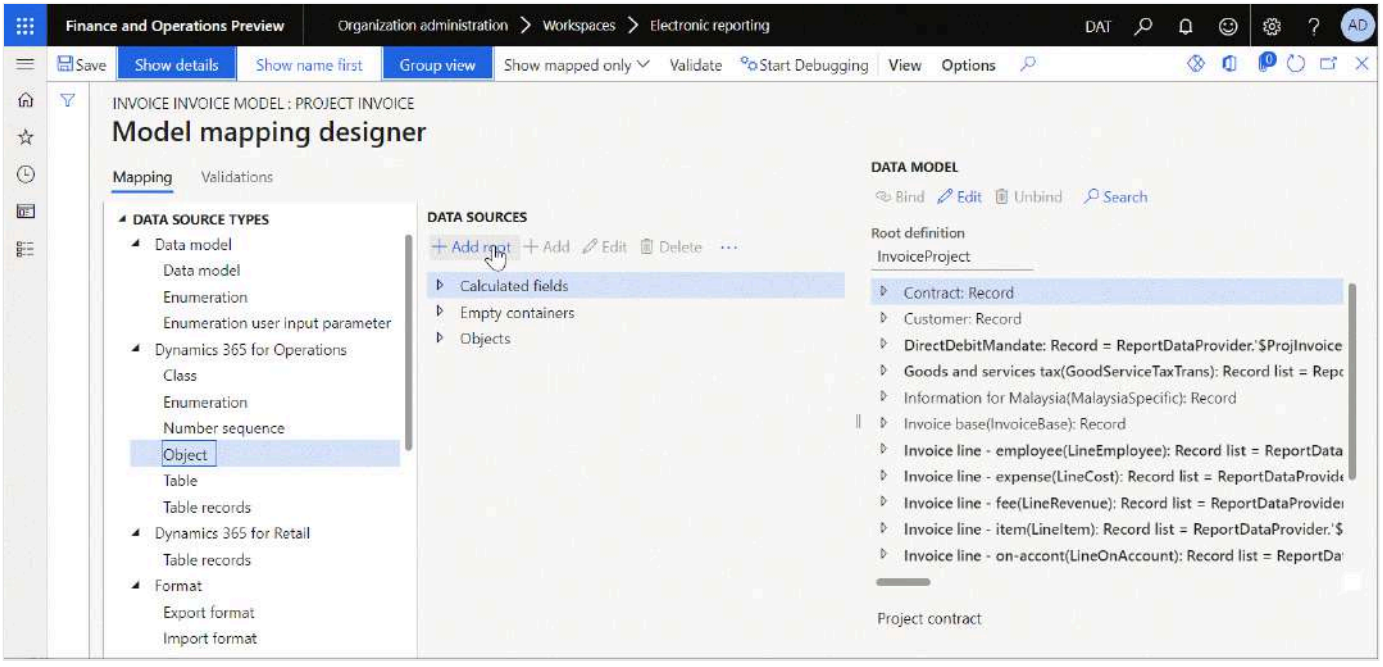
use the data to generate business documents, see Design a new ER solution to print a custom report.

Object

An *object* is a stateful instance of a *class*. You usually create an *object* in source code. You then pass the *object* to an ER model mapping, where it provides details about the execution context.

The default value of an *object* is **null**.

The following illustration shows how you add the **ReportDataContract** data source of the *Object* type to pass information about a generated invoice from source code to the **Project invoice** model mapping. For example, the invoice instance text is part of the execution context. The execution context text comes from source code at runtime by execution of the `ReportDataContract.parmInvoiceInstanceText` binding that you configured for the **Note** field of the ER data model. This binding calls the `parmInvoiceInstanceText()` method of the **PSAProjInvoiceContract** application class that the current model mapping represents as the **ReportDataContract** data source.



To learn how to pass details of the execution context from source code to the running ER solution, see Develop application artefacts to call the designed report.

Additional resources

- Electronic Reporting overview
- Electronic reporting formula language

- Supported primitive data types

Last updated on 07/17/2026

List of ER functions in the container category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) container functions can be used to perform operations that involve data sources of the *Container* data type. These operations occur when the processing data represents a collection of binary data in binary large object (BLOB) format. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
Base64StringToContainer	This function returns a *Container* value that consists of binary content that is decoded from the specified ASCII string that represents a Base64 group of binary-to-text encoding schemes.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

Base64StringToContainer ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `BASE64STRINGTOCONTAINER` function converts the specified input of the *String* type to a data item of the *Container* type.

Syntax

VB

```
BASE64STRINGTOCONTAINER (input)
```

Arguments

`input`: *String*

The valid path of a data source of the *String* type.

Return values

Container

The resulting binary value in binary large object (BLOB) format.

Usage notes

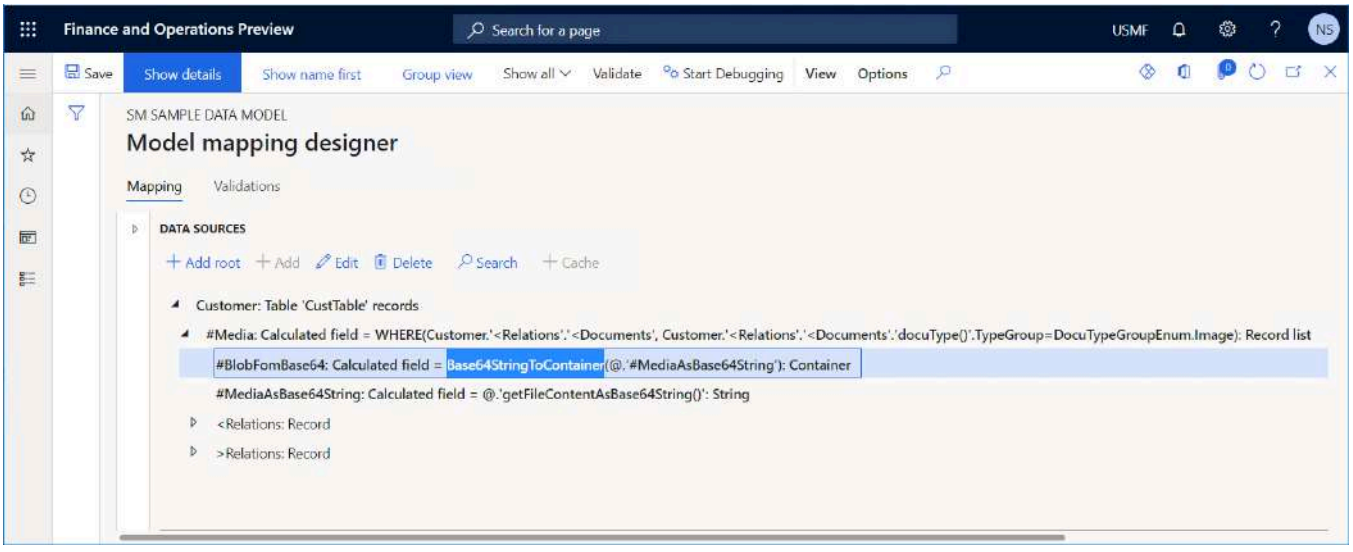
The "Parameter is not valid" exception is thrown if the input string doesn't provide a correct Base64 group of binary-to-text encoding schemes.

Example

You define the following data sources in your model mapping:

- The root **DocuTypeGroupEnum** data source of the *Dynamics 365 for Operations / Enumeration* type that refers to the **DocuTypeGroup** application enumeration
- The root **Customer** data source of the *Dynamics 365 for Operations / Table records* type that refers to the **CustTable** application table
- The **#Media** data source of the *Calculated field* type that is configured in the following way:
  - It's added as a child data source of the **Customer** data source.
  - It contains the expression `WHERE(@.'<Relations'.'<Documents', @.'<Relations'.'<Documents'.'docuType()'.TypeGroup = DocuTypeGroupEnum.Image)`.
- The **#MediaAsBase64String** data source of the *Calculated field* type that is configured in the following way:
  - It's added as a child data source of the **Customer.'#Media'** data source.
  - It contains the expression `Customer.'#Media'.'getFileContentAsBase64String()'`.
- The **#BlobFomBase64** data source of the *Calculated field* type that is configured in the following way:
  - It's added as a child data source of the **Customer.'#Media'** data source.
  - It contains the expression `Base64StringToContainer(Customer.'#Media'.'#MediaAsBase64String')'`.

In this example, the **#MediaAsBase64String** data source encodes the binary content of the current media attachment as text that represents a Base64 group of binary-to-text encoding schemes. The **#BlobFomBase64** data source decodes the Base64 string and returns a binary value in BLOB format.



Additional resources

Container functions

Last updated on 04/08/2026

List of ER functions in the Date and time category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) date and time functions can be used to extract information from date and time values, and to perform operations on them. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
AddDays	This function returns a *DateTime* value that is the specified number of days before or after a specified start date.
ChangeTimeZone	This function returns a *DateTime* value that is converted from a given date/time value in one time zone to a date/time value in another time zone.
DateFormat	This function returns a *String* value that presents a given date value as text in the specified format and in an optionally specified culture.
DateTimeFormat	This function returns a *String* value that presents a given date/time value as text in the specified format and in an optionally specified culture.
DateTimeValue	This function returns a *DateTime* value that is converted from a given text value in the specified format and in an optionally specified culture to a date/time value.
DateToDateTime	This function returns a *DateTime* value that is converted from a given date value to a date/time value in Coordinated Universal Time (Greenwich Mean Time [GMT]).
DateValue	This function returns a *Date* value that is converted from a given text value in the specified format and in an optionally specified culture to a date value.

Function	Description
DayOfYear	This function returns an *Integer* value that represents the number of days between January 1 and the specified date.
Days	This function returns an *Integer* value that represents the number of days between one specified date and a second specified date.
Now	This function returns a *DateTime* value that represents the current application server date and time.
NullDate	This function returns a *Date* value that represents the **null** date (January 1, 1900).
NullDateTime	This function returns a *DateTime* value that represents the **null** date/time value (January 1, 1900) in Coordinated Universal Time.
SessionNow	This function returns a *DateTime* value that represents the current application session date and time.
SessionToday	This function returns a *Date* value that represents the current application session date.
Today	This function returns a *Date* value that represents the current application server date.
WeekNum	This function returns an *Integer* value that represents the week of the year.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

ADDDAYS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ADDDAYS` function calculates a *DateTime* value that is the specified number of days before or after a specified start date.

Syntax

```
VB

ADDDAYS (datetime, days)
```

Arguments

`datetime`: *DateTime*

A date/time value that represents the start date.

`days`: *Integer*

The number of days before or after `datetime`.

Return values

DateTime

The resulting date/time value.

Usage notes

A positive value for `days` yields a future date. A negative value yields a past date.

Example 1

`ADDDAYS (NOW(), 7)` returns the date and time seven days in the future.

Example 2

`ADDDAYS (NOW(), -3)` returns the date and time three days in the past.

Additional resources

Date and time functions

Last updated on 04/08/2026

CHANGETIMEZONE ER function

The `CHANGETIMEZONE` function returns a *DateTime* value in Coordinated Universal Time (Greenwich Mean Time [GMT]) that is converted from a given date/time value in one time zone to a date/time value in another time zone.

Syntax

VB

```
CHANGETIMEZONE (datetime, base time zone, target time zone)
```

Arguments

`datetime`: *DateTime*

A date/time value in the Coordinated Universal Time time zone that represents the date/time value to convert.

`base time zone`: *String*

The name of the time zone that a given date/time value is shifted to before conversion.

`target time zone`: *String*

The name of the time zone that a converted date/time value is shifted to during conversion.

Return values

DateTime

The resulting date/time value in the Coordinated Universal Time time zone.

Usage notes

To specify source and target time zones, you can use time zone names that are provided by the Internet Assigned Numbers Authority (IANA) or that are supported by Microsoft Windows.

At runtime, the exception "Time zone '<time zone name>' does not exist" is thrown if the provided name isn't found in the IANA list or in the Windows registry.

For time zones where daylight saving time is observed, the conversion considers the Coordinated Universal Time daylight saving time offset. The latest available information about this offset is used during conversion.

Example 1

In this example, the time zone names for Windows are used.

You configure the **DSX** data source of the **Calculated field** type. It contains the following expression.

VB

```
CONCATENATE(
    DATETIMEFORMAT( DSY, "O"),
    " -> ",
    DATETIMEFORMAT( CHANGETIMEZONE(DSY, "E. Europe Standard Time", "Hawaiian Standard Time"), "O")
)
```

If you configure the expression of the **DSY** data source of the **Calculated field** type as `DATETIMEVALUE ("01-Jun-2021 12:55:00", "dd-MMM-yyyy HH:mm:ss", "EN")`, the **DSX** data source returns the text **2021-06-01T12:55:00.0000000+00:00 -> 2021-05-31T23:55:00.0000000+00:00**. This text shows that the time difference between the two provided time zones on June 1 is more than 24 hours. Therefore, the converted date/time value is one day earlier than the given date/time value, because the base time zone is ahead of the target time zone.

If you configure the expression of the **DSY** data source of the **Calculated field** type as `DATETIMEVALUE ("01-Dec-2021 12:55:00", "dd-MMM-yyyy HH:mm:ss", "EN")`, the **DSX** data source returns the text **2021-12-01T12:55:00.0000000+00:00 -> 2021-12-01T00:55:00.0000000+00:00**. This text shows that the time difference between the two provided time zones on December 1 is less than 24 hours. Therefore, the converted date/time value equals the given date/time value.

> ⓘ **Note**
>
> The same expression returns a different variance between the provided and converted date/time values for the same pair of time zones because a different Coordinated Universal Time daylight saving time offset is observed for the provided time zones on a given date/time.

Example 2

In this example, the IANA time zone names are used.

You configure the **DSX** data source of the **Calculated field** type. It contains the following expression.

VB

```
CONCATENATE(
    DATETIMEFORMAT( DSY, "O"),
    " -> ",
    DATETIMEFORMAT( CHANGETIMEZONE(DSY, "Europe/Athens", "US/Hawaii"), "O")
)
```

If you configure the expression of the **DSY** data source of the **Calculated field** type as `DATETIMEVALUE ("01-Jun-2021 12:55:00", "dd-MMM-yyyy HH:mm:ss", "EN")`, the **DSX** data source returns the text **2021-06-01T12:55:00.0000000+00:00 -> 2021-05-31T23:55:00.0000000+00:00**. This text shows that the time difference between the two provided time zones on June 1 is more than 24 hours. Therefore, the converted date/time value is one day earlier than the given date/time value, because the base time zone is ahead of the target time zone.

If you configure the expression of the **DSY** data source of the **Calculated field** type as `DATETIMEVALUE ("01-Dec-2021 12:55:00", "dd-MMM-yyyy HH:mm:ss", "EN")`, the **DSX** data source returns the text **2021-12-01T12:55:00.0000000+00:00 -> 2021-12-01T00:55:00.0000000+00:00**. This text shows that the time difference between the two provided time zones on December 1 is less than 24 hours. Therefore, the converted date/time value equals the given date/time value.

Example 3

You configure the **DSX** data source of the **Calculated field** type. It contains the following expression.

VB

```
CONCATENATE(
    DATETIMEFORMAT( DSY, "O"),
    " -> ",
    DATETIMEFORMAT( CHANGETIMEZONE(DSY, "US/Hawaii", "Europe/Athens"), "O")
)
```

If you configure the expression of the **DSY** data source of the **Calculated field** type as `DATETIMEVALUE ("01-Jun-2021 12:55:00", "dd-MMM-yyyy HH:mm:ss", "EN")`, the **DSX** data source returns the text **2021-06-01T12:55:00.0000000+00:00 -> 2021-06-02T01:55:00.0000000+00:00**'. This text shows that the time difference between the two provided time zones on June 1 is more than 24 hours. Therefore, the converted date/time value is one day later than the given date/time value, because the target time zone is ahead of the base time zone.

Example 4

You might receive a date and time stamp from an external source as text that contains no time zone information. However, you might know the time zone that the source operates in. For example, you receive the date and time stamp **01/12/2025 12:55:00** from a service that operates in Spain. To correctly save this date and time value to the database, complete the following conversion steps:

- Configure the **DSY** data source of the **Calculated field** type to convert a date/time stamp from text to the Coordinated Universal Time date/time value.

  ```
  DATETIMEVALUE ("01/12/2025 12:55:00", "dd/MM/yyyy HH:mm:ss", "ES")
  ```

- Configure the **DSX** data source of the **Calculated field** type to shift the converted date/time value to Coordinated Universal Time as the date/time value of the time zone of the external source.

  ```
  CHANGETIMEZONE(DSY, "Romance Standard Time", "GMT Standard Time")
  ```

> **Note**
>
> When you use the `CHANGETIMEZONE` function for date/time conversion, be aware that any date/time value is stored in the database as the value in the Coordinated Universal Time time zone. Before this value can be presented on application pages, it's transformed. The transformation considers the time zone that is **set** as a preferred zone for the currently signed-in application user.

Additional resources

Date and time functions

Last updated on 04/13/2026

DATEFORMAT ER function

The `DATEFORMAT` function returns a *String* value that presents a given date value as text in the specified format and in an optionally specified culture. For information about the supported formats, see standard and custom.

Syntax 1

VB

```
DATEFORMAT (date, format)
```

Syntax 2

VB

```
DATEFORMAT (date, format, culture)
```

Arguments

`date`: *Date*

A date value that represents the date to format.

`format`: *String*

The format of the output string. For information about the supported formats, see standard and custom.

> ⓘ Note
>
> The format string is case-sensitive when you use either a standard format or a custom format. For example, the **standard** "d" format specifier returns the date by using the short date pattern, whereas the standard "D" format specifier returns the date by using the long date pattern. Additionally, the **custom** "M" format specifier returns the month from 1 through 12, whereas the custom "m" format specifier returns the minute from 0 through 59.

`culture`: *String*

The culture to use for formatting. For information about the supported cultures, see culture.

Return values

String

The resulting string value.

Usage notes

If the culture isn't defined as an argument of the called function, the value of `culture` is defined by the calling context. For example, if the `DATEFORMAT` function is called by using syntax 1 in an Electronic reporting (ER) format for a **FILE** element that is configured to use the German culture, the conversion will be done by using the German culture. The default `culture` value is **EN-US**.

Example 1

`DATEFORMAT (TODAY (), "dd-MM-yyyy")` returns the current application server date, December 24, 2025, as the string "**24-12-2025**", based on the specified custom format.

Example 2

`DATEFORMAT (SESSIONTODAY (), "d", "DE")` returns the current application session date, December 24, 2025, as the string "**24-12-2025**", based on the selected German culture and the specified format.

Additional resources

Date and time functions

Last updated on 08/04/2026

DATETIMEFORMAT ER function

The `DATETIMEFORMAT` function returns a *String* value that presents a given date/time value as text in the specified format and in an optionally specified culture. For information about the supported formats, see standard and custom.

Syntax 1

VB

```
DATETIMEFORMAT (datetime, format)
```

Syntax 2

VB

```
DATETIMEFORMAT (datetime, format, culture)
```

Arguments

`datetime`: *DateTime*

A date/time value that represents the date and time to format.

`format`: *String*

The format of the output string. For information about the supported formats, see standard and custom.

> ⓘ Note
>
> The format string is case-sensitive when you use either a standard format or a custom format. For example, the **standard** "d" format specifier returns the date by using the short date pattern, whereas the standard "D" format specifier returns the date by using the long date pattern. Additionally, the **custom** "M" format specifier returns the month from 1 through 12, whereas the custom "m" format specifier returns the minute from 0 through 59.

`culture`: *String*

The culture to use for formatting. For information about the supported cultures, see culture.

Return values

String

The resulting string value.

Usage notes

If the culture isn't defined as an argument of the called function, the value of `culture` is defined by the calling context. For example, if the `DATETIMEFORMAT` function is called by using syntax 1 in an Electronic reporting (ER) format for a **FILE** element that is configured to use the German culture, the conversion will be done by using the German culture. The default `culture` value is **EN-US**.

When the `DATETIMEFORMAT` function converts a given date/time value, it considers the time zone setting of the application user who is running the ER format that the function is called in the context of.

Example 1

`DATETIMEFORMAT (NOW(), "dd-MM-yyyy")` returns the current application server date/time value, December 24, 2025, as "**24-12-2025**", based on the specified custom format.

Example 2

`DATETIMEFORMAT (SESSIONNOW(), "d", "DE")` returns the current application session date/time value, December 24, 2025, as "**24.12.2025**", based on the selected German culture and the specified format.

Example 3

`DATETIMEFORMAT (DATETIMEVALUE( "2019-11-12T09:00:00.0000000-07:00", "O"), "O")` returns the string value **2024-11-12T08:00:00.0000000-08:00** when the function is called during a process that an application user initiates who has the time zone value **(GMT-08:00) Pacific Time (US & Canada)** in the **Language and country/region preferences** section.

Additional resources

Date and time functions

Last updated on 08/04/2026

DATETIMEVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `DATETIMEVALUE` function returns a *DateTime* value that is converted from a given text value in the specified format and in an optionally specified culture to a date/time value. For information about the supported formats, see standard and custom.

Syntax 1

```
VB

DATETIMEVALUE (text, format)
```

Syntax 2

```
VB

DATETIMEVALUE (text, format, culture)
```

Arguments

`text`: *String*

Text that represents the value to format.

`format`: *String*

The format of the given text. For information about the supported formats, see standard and custom.

`culture`: *String*

The culture that is used for formatting of the given text. For information about the supported cultures, see culture.

Return values

DateTime

The resulting date/time value.

Usage notes

When the culture isn't defined as an argument of the called function, the value of `culture` is defined by the calling context. For example, if the `DATETIMEVALUE` function is called by using syntax 1 in an Electronic reporting (ER) format for a **FILE** element that is configured to use the German culture, the conversion will be done by using the German culture. The default `culture` value is **EN-US**.

Example 1

`DATETIMEVALUE ("21-Dec-2016 02:55:00", "dd-MMM-yyyy hh:mm:ss")` returns **2:55:00 AM on December 21, 2016**, based on the specified custom format and the default application's **EN-US** culture.

Example 2

`DATETIMEVALUE ("21-Gen-2016 02:55:00", "dd-MMM-yyyy hh:mm:ss", "IT")` returns **2:55:00 AM on December 21, 2016**, based on the specified custom format and culture.

However, `DATETIMEVALUE ("21-Gen-2016 02:55:00", "dd-MMM-yyyy hh:mm:ss", "EN-US")` throws an exception to inform the user that the specified string isn't recognized as a valid date/time value for the specified culture.

Additional resources

Date and time functions

Last updated on 04/08/2026

DATETODATETIME ER function

The `DATETODATETIME` function returns a *DateTime* value that is converted from a given date value to a date/time value in Coordinated Universal Time (Greenwich Mean Time [GMT]).

Syntax

VB

```
DATETODATETIME (date)
```

Arguments

`date`: *Date*

A date value that represents the date to convert.

Return values

DateTime

The resulting date/time value.

Example 1

`DATETODATETIME (CompInfo. 'getCurrentDate()')` returns the date of the current Microsoft Dynamics 365 Finance session, December 24, 2025, as **12/24/2025 12:00:00 AM**. In this example, **CompInfo** is an Electronic reporting (ER) data source of the **finance and operations/Table** type, and it refers to the CompanyInfo table.

Example 2

`DATETODATETIME (DATEVALUE ("2024-11-12T16:00:00.0000000-07:00", "O"))` returns the date and time value **11/12/2024 12:00:00 AM**.

Additional resources

Date and time functions

Last updated on 08/04/2026

DATEVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `DATEVALUE` function returns a *Date* value that is converted from a given text value in the specified format and in an optionally specified culture to a date value. For information about the supported formats, see standard and custom.

Syntax 1

```
VB

DATEVALUE (text, format)
```

Syntax 2

```
VB

DATEVALUE (text, format, culture)
```

Arguments

`text`: *String*

Text that represents the value to format.

`format`: *String*

The format of the given text. For information about the supported formats, see standard and custom.

`culture`: *String*

The culture that is used for formatting of the given text. For information about the supported cultures, see culture.

Return values

Date

The resulting date value.

Usage notes

When the culture isn't defined as an argument of the called function, the value of `culture` is defined by the calling context. For example, if the `DATEVALUE` function is called by using syntax 1 in an Electronic reporting (ER) format for a **FILE** element that is configured to use the German culture, the conversion will be done by using the German culture. The default `culture` value is **EN-US**.

Example 1

`DATEVALUE ("21-Dec-2016", "dd-MMM-yyyy")` returns the date value **December 21, 2016**, based on the specified custom format and the default application's **EN-US** culture.

Example 2

`DATEVALUE ("21-Gen-2016", "dd-MMM-yyyy", "IT")` returns the date value **January 21, 2016**, based on the specified custom format and culture.

However, `DATEVALUE ("21-Gen-2016", "dd-MMM-yyyy", "EN-US")` throws an exception to inform the user that the specified string isn't recognized as a valid date for the specified culture.

Additional resources

Date and time functions

Last updated on 04/08/2026

DAYOFYEAR ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `DAYOFYEAR` function returns an *Integer* value that represents the number of days between January 1 and the specified date.

Syntax

VB

```
DAYOFYEAR (date) as Integer
```

Arguments

`date`: *Date*

A date value that represents the date to use for the calculation of the number of days.

Return values

Integer

The resulting numeric value.

Example 1

`DAYOFYEAR (DATEVALUE ("01-03-2016", "dd-MM-yyyy"))` returns **61**.

Example 2

`DAYOFYEAR (DATEVALUE ("01-01-2016", "dd-MM-yyyy"))` returns **1**.

Additional resources

Date and time functions

Last updated on 04/08/2026

DAYS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `DAYS` function returns an *Integer* value that represents the number of days between one specified date and a second specified date.

Syntax

VB

```
DAYS (date 1, date 2) as Integer
```

Arguments

`date 1`: *Date*

A date value that represents the start date for the calculation of the number of days.

`date 2`: *Date*

A date value that represents the end date for the calculation of the number of days.

Return values

Integer

The resulting numeric value.

Usage notes

The `DAYS` function returns a positive value when the first date is later than the second date, it returns **0** (zero) when the first date equals the second date, and it returns a negative value

when the first date is earlier than the second date.

Example

```
DAYS (TODAY (), DATEVALUE( DATETIMEFORMAT( ADDDAYS ( NOW(), 1), "yyyyMMdd"), "yyyyMMdd"))
```

returns **-1**.

Additional resources

Date and time functions

Last updated on 04/08/2026

NOW ER function

The `NOW` function returns a *DateTime* value that represents the current application server date and time.

Syntax

```VB
NOW ()
```

Return values

DateTime

The resulting date/time value.

Example

`DATETIMEFORMAT (NOW(), "dd-MM-yyyy")` returns the current application server date/time value, December 24, 2025, as "**24-12-2025**", based on the specified custom format.

Additional resources

Date and time functions

Last updated on 08/04/2026

NULLDATE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NULLDATE` function returns a *Date* value that represents the **null** date (January 1, 1900).

Syntax

VB

```
NULLDATE () as
```

Return values

Date

The resulting date value.

Example 1

`DATEFORMAT (NULLDATE(), "yyyy-MM-dd")` returns the **null** date, January 1, 1900, as **"1900-01-01"**, based on the specified custom format.

Example 2

The expression `IF( Invoice.DocumentDate = NULLDATE(), true, false)` returns **True** when the value of the **DocumentDate** field equals the **null** date. In this example, **Invoice** is an Electronic reporting (ER) data source of the **Finance/Table records** type, and it refers to the CustInvoiceJour table.

Additional resources

Date and time functions

Last updated on 04/08/2026

NULLDATETIME ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NULLDATETIME` function returns a *DateTime* value that represents the **null** date/time value (January 1, 1900) in Coordinated Universal Time (Greenwich Mean Time [GMT]).

Syntax

VB

```
NULLDATETIME ()
```

Return values

DateTime

The resulting date/time value.

Example

`DATETIMEFORMAT( NULLDATETIME(), "O")` returns the string value **1900-01-01T00:00:00.0000000+00:00** when it's called during a process that was initiated by an application user who has the time zone value **(GMT) Coordinated Universal Time** in the **Language and country/region preferences** section.

Additional resources

Date and time functions

Last updated on 04/08/2026

SESSIONNOW ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `SESSIONNOW` function returns a *DateTime* value that represents the current application session date and time.

Syntax

VB

```
SESSIONNOW ()
```

Return values

DateTime

The resulting date/time value.

Example

`DATETIMEFORMAT (SESSIONNOW(), "d", "DE")` returns the current application session date/time value, December 24, 2015, as "**24.12.2015**", based on the selected German culture and the specified format.

Additional resources

Date and time functions

Last updated on 04/08/2026

SESSIONTODAY ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `SESSIONTODAY` function returns a *Date* value that represents the current application session date.

Syntax

VB

```
SESSIONTODAY ()
```

Return values

Date

The resulting date value.

Example

`DATEFORMAT (SESSIONTODAY (), "d", "DE")` returns the current application session date, December 24, 2015, as the string **"24-12-2015"**, based on the selected German culture and the specified format.

Additional resources

Date and time functions

Last updated on 04/08/2026

TODAY ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `TODAY` function returns a *Date* value that represents the current application server date.

Syntax

```
X++
TODAY ()
```

Return values

Date

The resulting date value.

Example

`DATEFORMAT (TODAY (), "dd-MM-yyyy")` returns the current application server date, December 24, 2015, as the string **"24-12-2015"**, based on the specified custom format.

Additional resources

Date and time functions

Last updated on 04/08/2026

WEEKNUM ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `WEEKNUM` function returns an *Integer* value that represents the week of the year that includes a specified *Date* value. The calculation is based on culture-dependent rules that define a calendar week and the first day of the week.

Syntax

VB

```
WEEKNUM (date, culture) as Integer
```

Arguments

`date`: *Date*

A date value that represents the date to use to calculate the week of the year.

`culture`: *String*

The culture to use for the calculation. You can use culture codes that are supported in accordance with .NET standards.

Return values

Integer

The resulting numeric value.

Usage notes

The week of the year is calculated based on the ISO 8601 standard, if this standard has been adopted by a country or region that the locale is provided for at runtime. Otherwise, the calculation is based on country/region-specific national standards.

If an unsupported culture code is provided as an argument of the `WEEKNUM` function at runtime, an exception is thrown. If the blank string is provided as a culture code, the English country/region-neutral calendar is used to calculate the week number.

Examples

`WEEKNUM (DATEVALUE ("01-01-2020", "de"))` returns **1**.

`WEEKNUM (DATEVALUE ("01-01-2021", "de"))` returns **53**.

`WEEKNUM (DATEVALUE ("01-01-2022", "de"))` returns **52**.

`WEEKNUM (DATEVALUE ("01-01-2022", "ar"))` returns **21**.

Additional resources

Date and time functions

Last updated on 04/08/2026

List of ER functions in the list category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) list functions can be used to extract information from, and perform operations on, data sources of the *Record list* and *Container (record)* data types. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
AllItems	This function runs as an in-memory selection. It returns a new flattened *Record list* value that consists of a list of records that represents all items that match the specified path.
AllItemsQuery	This function runs as a joined SQL query. It returns a new flattened *Record list* value that consists of a list of records that represents all items that match the specified path.
Count	This function returns an *Integer* value that represents the number of records in the specified list, if the list isn't empty. If the list is empty, this function returns **0** (zero).
EmptyList	This function returns an empty *Record list* value by using the specified list as a source for the list structure.
Enumerate	This function returns a new *Record list* value that consists of enumerated records of the specified list.
Filter	This function returns the specified list as a *Record list* value after the query has been changed so that it filters for the specified condition.
First	This function returns the first record of the specified list as a *Container (record)* value, if that list isn't empty. If the list is empty, this function throws an exception.
FirstOrNull	This function returns the first record of the specified list as a *Container (record)* value, if that record isn't empty. If the record is empty, this function returns a null *Container (record)* value.

Function	Description
Index	This function returns a *Container (record)* value that is selected by using the specified numeric index in the specified list. If the index is out of range for the records in the specified list, this function throws an exception.
IsEmpty	This function returns a *Boolean* value of **TRUE** if the specified list contains no records. Otherwise, it returns a *Boolean* value of **FALSE**.
List	This function returns a *Record list* value that consists of a new list that is created from the specified arguments.
ListDistinct	This function calculates the specified expression as a selector for every record of the specified list. It returns a new *Record list* value that contains a single record for each unique selector value.
ListJoin	This function returns a *Record list* value that represents a new joined list that is created from the specified arguments.
ListOfFields	This function returns a *Record list* value that is created based on the structure of the specified argument of the *Enumeration* or *Container (record)* type.
ListOfFirstItem	This function returns a *Record list* value that consists of only the first record of the specified list.
OrderBy	This function returns the specified list as a *Record list* value after it has been sorted according to the specified arguments. These arguments can be defined as expressions.
Repeat	This function builds a record that contains the field that has a value that matches the specified input. It then returns a new *Record list* of a record that is repeated a specified number of times.
Reverse	This function returns the specified list as a *Record list* value in reversed sort order.
Split	This function splits the specified input string into substrings and returns the result as a new *Record list* value.
SplitList	This function splits the specified list into sublists (or batches), each of which contains the specified number of records. It then returns the result as a new *Record list* value that consists of the batches.
SplitListByLimit	This function splits the specified list into a new list of sublists (batches). The number of records in each batch is dynamically calculated. The function then returns the result as a new *Record list* value that consists of the batches.
StringJoin	This function returns a *String* value that consists of concatenated values of the specified field from the specified list. The values can be separated by the specified delimiter.
Where	This function returns the specified list as a *Record list* value after it has been filtered according to the specified condition.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

ALLITEMS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ALLITEMS` function runs as an in-memory selection and returns a new flattened *Record list* value as a list of records that represents all items that match the specified path.

Syntax

VB

```
ALLITEMS (path)
```

Arguments

`path`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Record list

The resulting list of records.

Usage notes

The path must be defined as a valid data source path of a data source element of the *Record list* data type. Data elements such as the path string and date should raise an error in the Electronic reporting (ER) expression builder at design time.

We don't recommend that you use this function for transactional data sources that might contain a large volume of data. Instead, consider using the ALLTEMSQUERY function.

Example 1

If you enter `SPLIT("abcdef" , 2)` as data source **DS**, the expression `COUNT( ALLITEMS (DS))` returns **3**.

Example 2

If you enter **Vend** as the data source of the *Record list* data type that refers to the VendTable application table, the expression `ALLITEMS (Vend.'<Relations'.ContactPerson)` returns a flattened list of records that has the **ContactPerson** table structure and contains all contact persons that can be accessed by using the **ContactPerson.ContactForParty == VendTable.Party** relation, and that is available for all vendors from the referenced vendor table.

Additional resources

List functions

Last updated on 04/08/2026

ALLITEMSQUERY ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ALLITEMSQUERY` function runs as a joined SQL query. It returns a new flattened *Record list* value that consists of a list of records that represent all items that match the specified path.

Syntax

VB

```
ALLITEMSQUERY (path)
```

Arguments

`path`: *Record list*

The valid path of a data source of the *Record list* data type. It must contain at least one relation.

Return values

Record list

The resulting list of records.

Usage notes

The specified path must be defined as a valid data source path of a data source element of the *Record list* data type. It must also contain at least one relation. Data elements such as the path *String* and *Date* should raise an error in the Electronic reporting (ER) expression builder at design time.

When this function is applied to data sources of the *Record list* data type that refer to an application object that can be directly called by using SQL (for example, an table, entity, or query), it runs as a joined SQL query. Otherwise, it runs in memory as the ALLITEMS function.

Example

You define the following data sources in your model mapping:

- A **CustInv** data source of the *Table records* type that refers to the CustInvoiceTable table
- A **FilteredInv** data source of the *Calculated field* type that contains the expression `FILTER (CustInv, CustInv.InvoiceAccount = "US-001")`
- A **JourLines** of the *Calculated field* type that contains the expression `ALLITEMSQUERY ( FilteredInv.'<Relations'.CustInvoiceJour.'<Relations'.CustInvoiceTrans)`

When you run the model mapping to call the **JourLines** data source, the following SQL statement is run:

SQL

```
SELECT ... FROM CUSTINVOICETABLE T1 CROSS JOIN CUSTINVOICEJOUR T2 CROSS JOIN
CUSTINVOICETRANS T3 WHERE...
```

Additional resources

List functions

Last updated on 04/08/2026

COUNT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `COUNT` function returns an *Integer* value that represents the number of records in the specified list, if the list isn't empty. If the list is empty, this function returns **0** (zero).

Syntax

VB

```
COUNT (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Integer

The resulting numeric value.

Example

`COUNT (SPLIT("abcd" , 3))` returns **2**, because the `SPLIT` function that is used in this example creates a list that consists of two records.

Additional resources

List functions

Last updated on 04/08/2026

EMPTYLIST ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `EMPTYLIST` function returns an empty *Record list* value by using the specified list as a source for the list structure.

Syntax

```
VB

EMPTYLIST (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Record list

The resulting list of records.

Example

`EMPTYLIST (SPLIT ("abc", 1))` returns a new empty list that has the same structure as the list that is returned by the `SPLIT` function that is used.

Additional resources

List functions

Last updated on 04/08/2026

ENUMERATE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ENUMERATE` function returns a new *Record list* value that consists of enumerated records of the specified list.

Syntax

VB

```
ENUMERATE (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Record list

The resulting list of records.

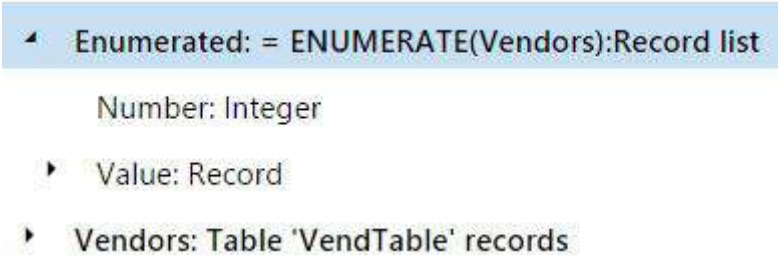
Usage notes

The list of enumerated records that is returned exposes the following additional elements:

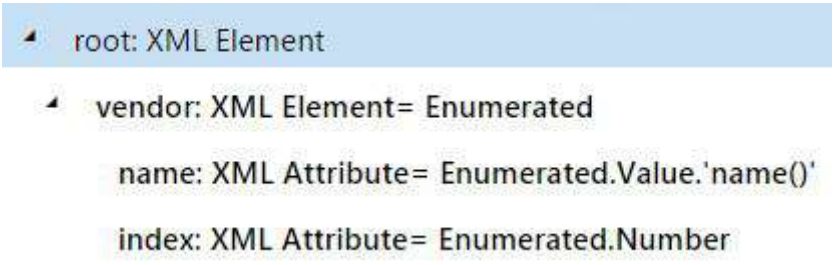
- The record of fields (**Value** component)
- The current record index (**Number** component)

Example

In the following illustration, an **Enumerated** data source is created as an enumerated list of vendor records from the **Vendors** data source that refers to the VendTable table.



The following illustration shows the Electronic reporting (ER) format. In this format, data bindings are created to generate output in XML format. This output presents individual vendors as enumerated nodes.



The following illustration shows the result when the designed format is run.

Additional resources

List functions

Last updated on 04/08/2026

FILTER ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `FILTER` function returns the specified list as a *Record list* value after the query has been changed so that it filters for the specified condition.

Syntax

```
VB

FILTER (list, condition)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`condition`: *Boolean*

A valid conditional expression that is used to filter records of the specified list.

Return values

Record list

The resulting list of records.

Usage notes

This function differs from the WHERE function, because the specified condition is applied to any Electronic reporting (ER) data source of the *Table records* type at the database level. The list

and condition can be defined by using tables and relations.

If one or both arguments that are configured for this function (`list` and `condition`) don't allow this request to be translated to the direct SQL call, an exception is thrown at design time. This exception informs the user that either `list` or `condition` can't be used to query the database.

> ⓘ **Note**
>
> The `FILTER` function behaves differently than the `WHERE` function when the **VALUEIN** function is used to specify the selection criteria.
>
> - If the `VALUEIN` function is used in scope of the `WHERE` function, and the second argument of `VALUEIN` refers to a data source that returns no records, the Boolean ***False*** value that `VALUEIN` returns is considered. Therefore, the expression `WHERE(Vendors, VALUEIN(Vendors.VendGroup, VendGroups, VendGroups.VendGroup))` returns no vendor records if the **VendGroups** data source returns no vendor group records.
> - If the `VALUEIN` function is used in scope of the `FILTER` function, and the second argument of `VALUEIN` refers to a data source that returns no records, the Boolean ***False*** value that `VALUEIN` returns is ignored. Therefore, the expression `FILTER(Vendors, VALUEIN(Vendors.VendGroup, VendGroups, VendGroups.VendGroup))` returns all vendor records of the **Vendors** data source, even if the **VendGroups** data source returns no vendor group records.

Example 1

If **Vendor** is configured as an ER data source that refers to the VendTable table, the expression `FILTER (Vendors, Vendors.VendGroup = "40")` returns a list of only vendors that belong to vendor group 40.

Example 2

If **Vendor** is configured as an ER data source that refers to the VendTable table, and if **parmVendorBankGroup** is configured as an ER data source that returns a value of the *String* data type, the expression `FILTER ( Vendor.'<Relations'.VendBankAccount, Vendor.'<Relations'.VendBankAccount.BankGroupID = parmVendorBankGroup)` returns a list of only vendor accounts that belong to a specific bank group.

Example 3

You enter data source **DS** of the *Calculated field* type, and it contains the expression `SPLIT ("A,B,C", ",")`. You then enter another expression, `FILTER( DS, DS.Value = "B")`. When you try to save this expression in the ER formula designer, the following exception is thrown: "Validation error: The list expression of FILTER function is not queryable."

Additional resources

List functions

Last updated on 04/08/2026

FIRST ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `FIRST` function returns the first record of the specified list as a *Container (record)* value, if that list isn't empty. If the list is empty, this function throws an exception.

Syntax

VB

```
FIRST (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Container (record)

The resulting record value.

Example 1

The expression `FIRST(SPLIT("ABC",1)).Value` returns the text value "**A**".

Example 2

The expression `FIRST(SPLIT("",1)).Value` throws an exception at runtime.

Additional resources

List functions

Last updated on 04/08/2026

FIRSTORNULL ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `FIRSTORNULL` function returns the first record of the specified list as a *Container (record)* value, if that record isn't empty. If the record is empty, this function returns a null *Container (record)* value.

Syntax

```
VB

FIRSTORNULL (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Container (record)

The resulting record value.

Example

The expression `FIRSTORNULL(SPLIT("",1)).Value` returns an empty string ("").

Additional resources

List functions

Last updated on 04/08/2026

INDEX ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `INDEX` function returns a *Container (record)* value that is selected by using the specified numeric index in the specified list. If the index is out of range for the records in the specified list, an exception is thrown.

Syntax

```
VB

INDEX (list, index)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`index`: *Integer*

A numeric index that indicates the position of the desired record in the specified list.

> **Note**
>
> Because one-based numbering is used for this function, specify the value **1** to return the first record of the specified list.

Return values

Container (record)

The resulting record value.

Example 1

If you enter data source **DS** of the *Calculated field* type, and it contains the expression `SPLIT ("A|B|C", "|")`, the expression `DS.Value` returns the text value "**B**" for the second record of this record list. The expression `INDEX (SPLIT ("A|B|C", "|"), 2).Value` also returns the text value "**B**".

Example 2

If you enter data source **DS** of the *Calculated field* type, and it contains the expression `SPLIT ("A|B|C", "|")`, the expression `INDEX (SPLIT ("A|B|C", "|"), 4).Value` throws an exception at runtime.

Additional resources

List functions

Last updated on 04/08/2026

ISEMPTY ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ISEMPTY` function returns a *Boolean* value of **TRUE** if the specified list contains no records. Otherwise, it returns a *Boolean* value of **FALSE**.

Syntax

VB

```
ISEMPTY (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Boolean

The resulting *Boolean* value.

Example 1

If you enter data source **DS** of the *Calculated field* type, and it contains the expression `SPLIT ("A|B|C", "|")`, the expression `ISEMPTY(DS)` returns **FALSE**.

Example 2

The expression `ISEMPTY (SPLIT ("",1))` returns **TRUE**.

Additional resources

List functions

Last updated on 04/08/2026

LIST ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LIST` function returns a *Record list* value that consists of a new list of records that is created from the specified arguments.

Syntax

VB

```
LIST (record 1 [, record 2, …, record N])
```

Arguments

`record 1`: *Container (record)*

A reference to a data source of the *Record* data type. This argument is required.

`record N`: *Container (record)*

A reference to a data source of the *Record* data type. These additional arguments are optional.

Return values

Record list

The resulting list of records.

Usage notes

The structure of the list that is created contains only the fields that are presented in the structure of every record that is mentioned in the arguments.

Example

You enter data source **Record 1** of the *Container* type. This data source contains the following nested fields of the *Calculated field* type:

- **Code:** This field contains an expression that returns a value of the *String* type.
- **Amount:** This field contains an expression that returns a value of the *Real* type.

You then enter data source **Record 2** of the *Container* type. This data source contains the following nested fields of the *Calculated field* type:

- **Amount:** This field contains an expression that returns a value of the *Real* type.
- **IsValid:** This field contains an expression that returns a value of the *Boolean* type.

In this case, the expression `LIST('Record 1', 'Record 2')` returns a new list that contains two records. The structure of this list consists of a single **Amount** field of the *Real* type, because this field is the only field that is presented in every argument of the called function.

Additional resources

List functions

Last updated on 04/08/2026

LISTDISTINCT ER Function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LISTDISTINCT` function calculates the specified expression as a selector for every record of the specified list. It returns a new *Record list* value that contains a single record for each unique selector value.

Syntax

```
LISTDISTINCT (list, selector)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`selector`: *Primitive data type*

A valid expression that is used to calculate a selector value for every record in the specified list.

The following data types are supported for this parameter:

- Boolean
- Date
- DateTime
- GUID
- Integer
- Int64
- Real

- String

Return values

Record list

The resulting list of records.

Usage notes

The structure of the list that is created matches the structure of the specified list.

The same selector value might be calculated for multiple records in the specified list. In this case, field values of the corresponding record in the created list equal the values of the first record from the specified list that the selector value is calculated for.

The execution of this function is done on any Electronic reporting (ER) data source of the *Record list* type that is present in memory.

The **GROUPBY** data source can also be used to generate the list of records that the selector that has distinct values is calculated for. However, from a performance and memory consumption perspective, it's better to use the `LISTDISTINCT` function than the **GROUPBY** data source, because the execution of the function is done in memory.

Example

The following example shows how you can get the list of unique customer account numbers that at least one sales invoice or project invoice has been issued to during a specific period.

1. Enter the **SalesInvoice** data source of the `Record list` type that refers to the **CustInvoiceJour** application table and filters sales invoices for specific periods.

   The `InvoiceAccount` field of this data source returns the account number of an invoiced customer.

2. Enter the **ProjectInvoice** data source of the `Record list` type that refers to the **ProjInvoiceJour** application table and filters project invoices for specific periods.

   The `InvoiceAccount` field of this data source returns the account number of an invoiced customer.

3. Configure the **AllInvoices** data source of the `Calculated field` type that contains the expression `LISTJOIN(SalesInvoice, ProjectInvoice)`.

   This data source returns the joined list of sales invoices and project invoices.

4. Configure the **InvoicedCustomer** data source of the `Record list` type that contains the expression `LISTDISTINCT(AllInvoices, AllInvoices.InvoiceAccount)`.

   This data source returns a new list that contains a single record for every unique customer that has been invoiced during the defined period. The `InvoiceAccount` field of this list contains a customer account number.

Additional resources

List functions

Last updated on 04/08/2026

LISTJOIN ER function

Article • 08/12/2022

The `LISTJOIN` function returns a *Record list* value that represents a new joined list of records that is created from the specified arguments.

Syntax

```
VB

LISTJOIN (list 1 [, list 2, …, list N])
```

Arguments

`list 1`: *Record list*

A reference to a data source of the *Record list* data type. This argument is mandatory.

`list N`: *Record list*

A reference to a data source of the *Record list* data type. These additional arguments are optional.

Return values

Record list

The resulting list of records.

Usage notes

The structure of the list that is created contains only the fields that are present in the structure of every record list that is referenced in the arguments.

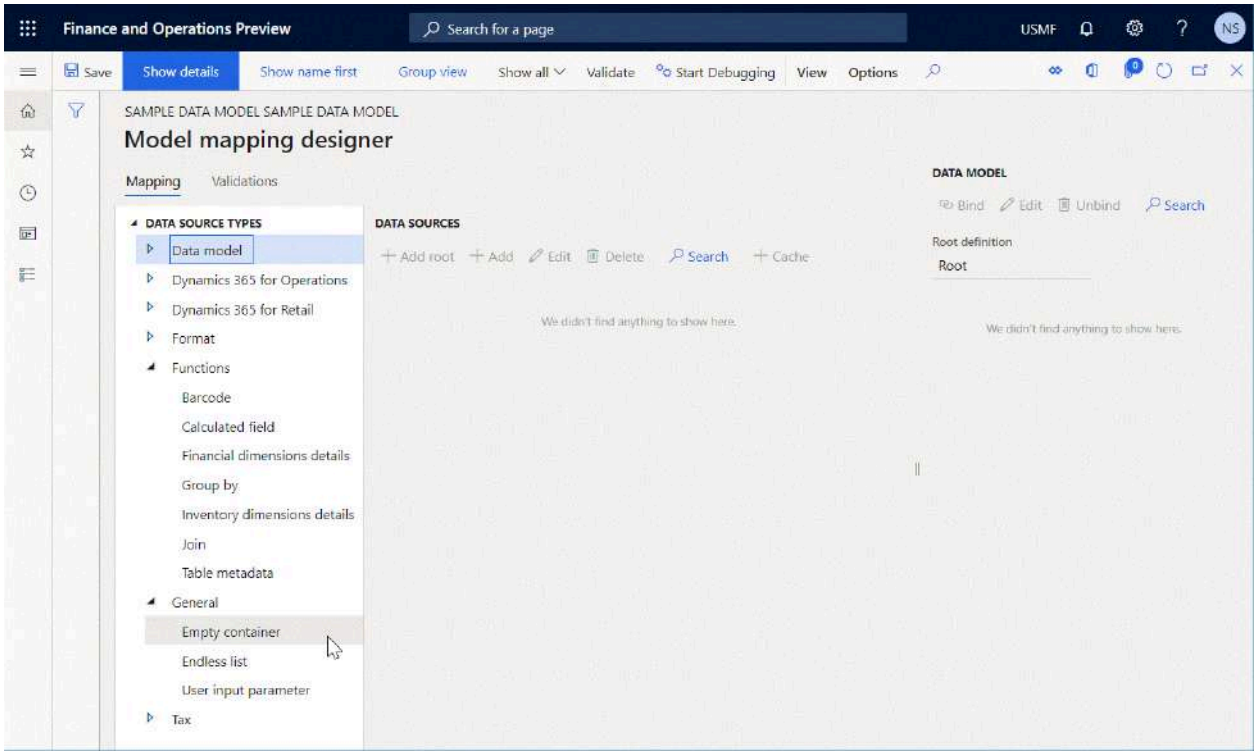
Example

You enter data source **Record 1** of the `Container` type. This data source contains the following nested fields of the `Calculated field` type:

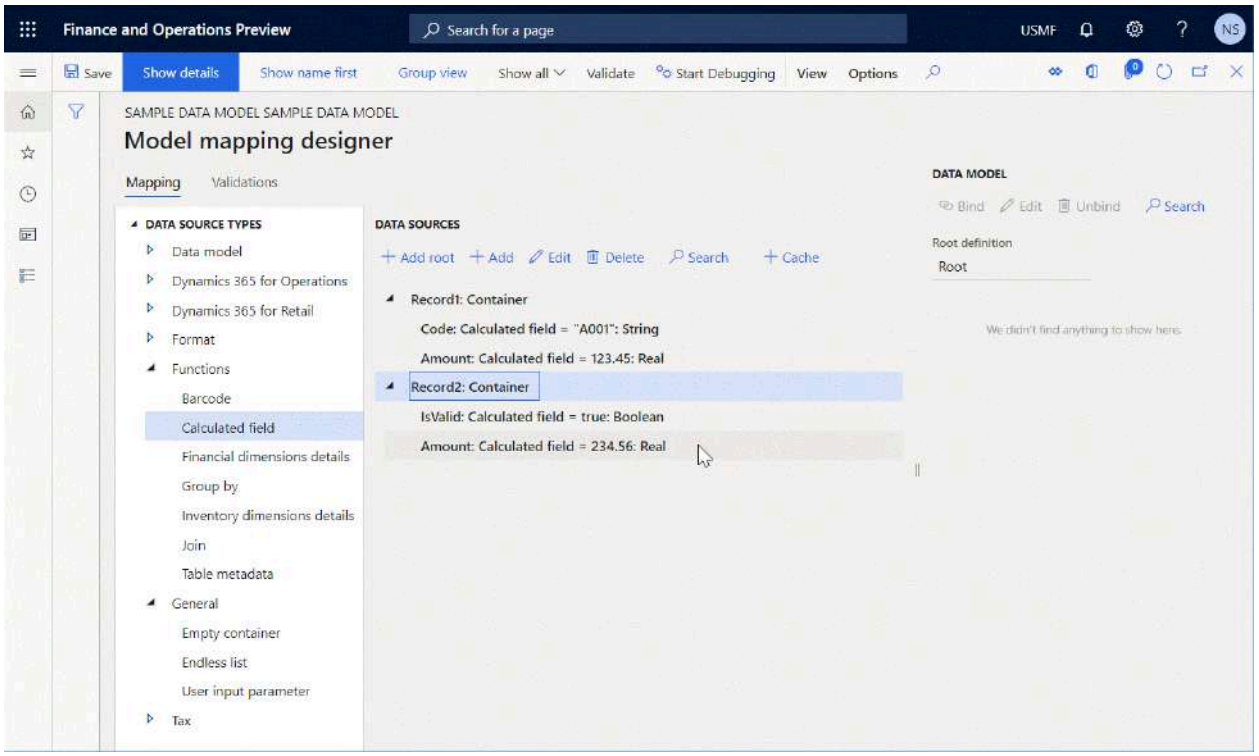
- **Code**: This field contains an expression that returns a value of the `String` type.
- **Amount**: This field contains an expression that returns a value of the `Real` type.

You then enter data source **Record 2** of the `Container` type. This data source contains the following nested fields of the `Calculated field` type:

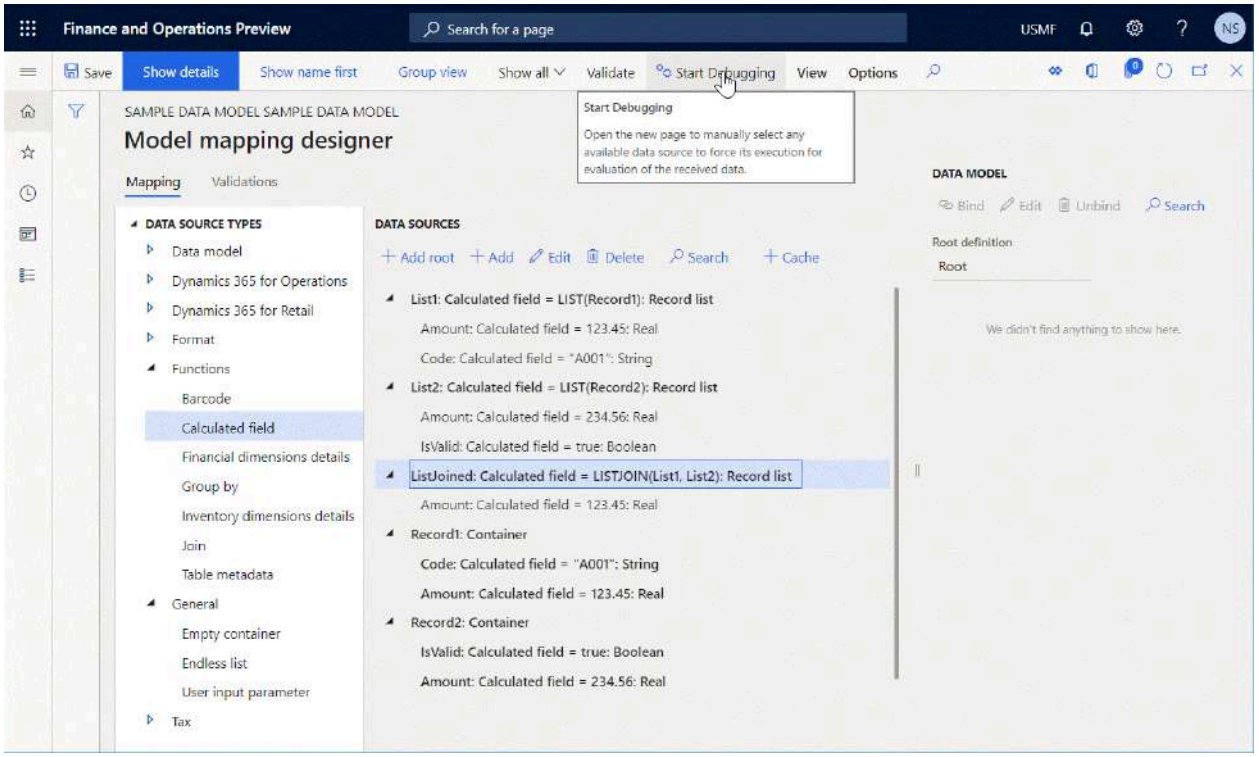
- **Amount**: This field contains an expression that returns a value of the `Real` type.
- **IsValid**: This field contains an expression that returns a value of the `Boolean` type.



In this case, the expression `LISTJOIN(LIST('Record 1'), LIST('Record 2'))` returns a new list that contains two records.



The structure of this list consists of a single **Amount** field of the `Real` type, because this field is the only field that is presented in every argument of the called function.



Additional resources

List functions

Debug data sources of an executed ER format to analyze data flow and transformation

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LISTOFFIELDS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LISTOFFIELDS` function returns a *Record list* value that is created based on the structure of the specified argument of the *Enumeration* or *Container (record)* type.

Syntax 1

```
VB

LISTOFFIELDS (path)
```

Syntax 2

```
VB

LISTOFFIELDS (path, language)
```

Arguments

`path`: Data source reference

The valid reference path of a data source of one of the following data types:

- Model enumeration
- Format enumeration
- Application enumeration
- Container (record)

`language`: *String*

Text that represents a language code.

Return values

Record list

The resulting list of records.

Usage notes

The list that is created consists of records that have the following fields:

- **Name** (*String* data type)
- **Label** (*String* data type)
- **Description** (*String* data type)
- **IsTranslated** (*Boolean* data type)

If the `path` argument refers to a data source of the *Container (Record)* type, for every field of the referenced container record, a new record is added to the list that is created. For every record that is created, the **Name** field returns the name of the field of the referenced container record that the current record was created for.

If the `path` argument refers to a data source of one of the *Enumeration* types, for every enumeration value of the referenced enumeration, a new record is added to the list that is created. For every record that is created, the **Name** field returns the value of the referenced enumeration that the current record was created for, the **Description** field returns the description of that enumeration, and the **Label** field returns the label of that enumeration.

At runtime, when syntax 1 is used, the **Label** and **Description** fields must return values that are based on the language settings of the Electronic reporting (ER) format that is running:

- If the labels and descriptions for the requested language are available, the **Label** and **Description** fields return values that are based on that language, and the **IsTranslated** field returns **True**.
- If the labels and descriptions for the requested language aren't available, the **Label** and **Description** fields return values that are based on the default **EN-US** language, and the **IsTranslated** field returns **False**.

At runtime, when syntax 2 is used, the **Label** and **Description** fields must return values that are based on the language that is defined as the second argument of the called function:

- If the labels and descriptions for the requested language are available, the **Label** and **Description** fields return values that are based on that language, and the **IsTranslated**

field returns **True**.

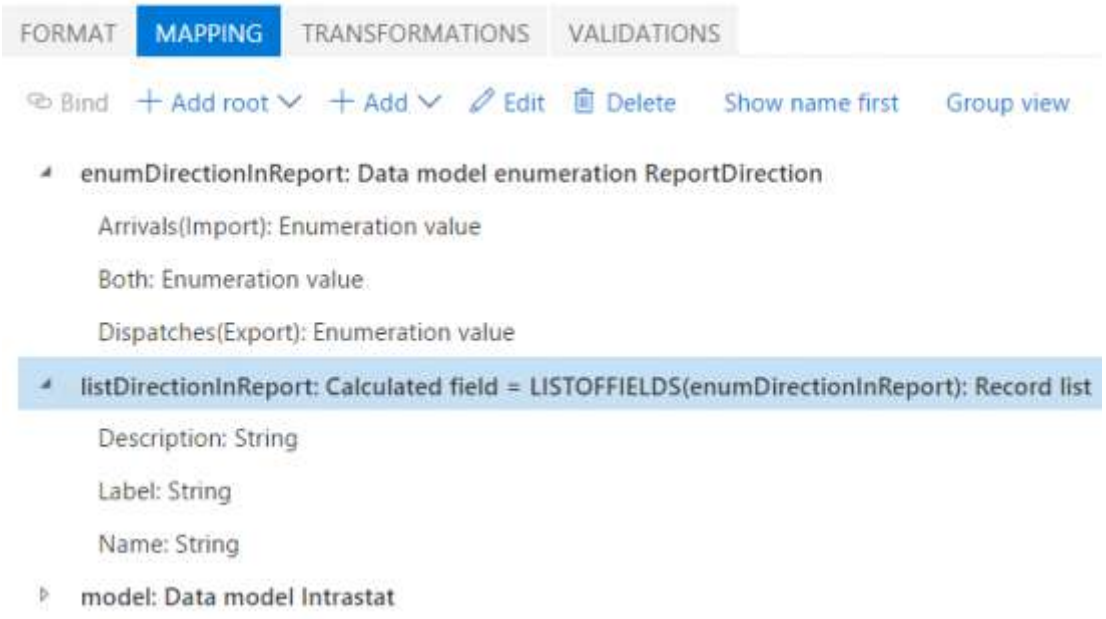
- If the labels and descriptions for the requested language aren't available, the **Label** and **Description** fields return values that are based on the **EN-US** language, and the **IsTranslated** field returns **False**.

Example 1

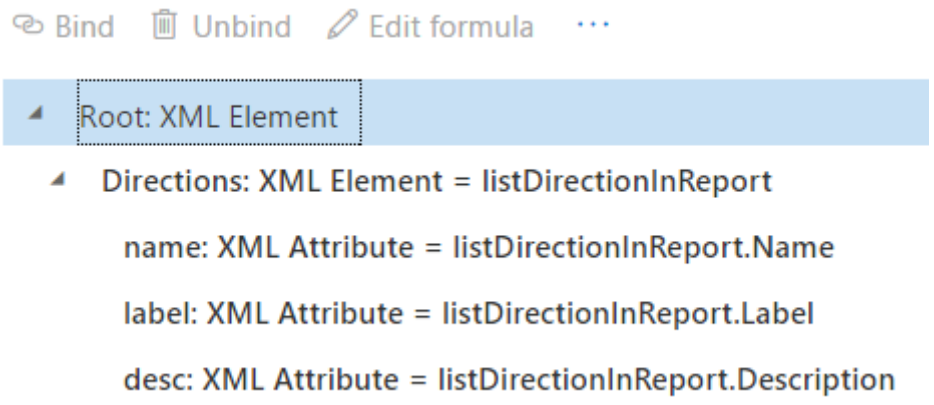
In the following illustration, an enumeration is introduced in an ER data model.

The following illustration shows these details:

- The model enumeration is inserted into a report as a data source.
- An ER expression uses the model enumeration as a parameter of the `LISTOFFIELDS` function.
- A data source of the *Record list* type is inserted into a report by using the ER expression that is created.



The following example shows the ER format elements that are bound to the data source of the *Record list* type that was created by using the `LISTOFFIELDS` function.



The following illustration shows the result when the designed format is run.

```
<?xml version="1.0" encoding="UTF-8"?>
- <Root>
    <Directions desc="" label="Both" name="Both"/>
    <Directions desc="" label="Dispatches" name="Export"/>
    <Directions desc="" label="Arrivals" name="Import"/>
  </Root>
```

> ⓘ Note
>
> Based on the language settings of the parent **FILE** and **FOLDER** format elements, translated text for labels and descriptions is entered in the output of the ER format.

Example 2

You use the *Calculated field* data source type to configure **enumType_de** and **enumType_deCH** data sources for the **enumType** data model enumeration:

- **enumType_de** = `LISTOFFIELDS (enumType, "de")`
- **enumType_deCH** = `LISTOFFIELDS (enumType, "de-CH")`

In this case, you can use the following expression to get the label of the enumeration value in Swiss German, if that translation is available. If the Swiss German translation isn't available, the label is in German.

VB

```
IF (NOT (enumType_deCH.IsTranslated), enumType_de.Label, enumType_deCH.Label)
```

Additional resources

List functions

Last updated on 04/08/2026

LISTOFFIRSTITEM ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LISTOFFIRSTITEM` function returns a *Record list* value that consists of only the first record of the specified list.

Syntax

VB

```
LISTOFFIRSTITEM (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Record list

The resulting list of records.

Example

The expression `FIRST( LISTOFFIRSTITEM ( SPLIT ("ABC",1))).Value` returns the text value "**A**".

Additional resources

List functions

Last updated on 04/08/2026

ORDERBY ER function

The `ORDERBY` function returns the specified list as a *Record list* value after it has been sorted according to the specified arguments. These arguments can be defined as expressions.

Syntax 1

```
VB
ORDERBY (list, expression 1[, expression 2, …, expression N])
```

Syntax 2

```
VB
ORDERBY (location, list, expression 1[, expression 2, …, expression N])
```

> **Note**
>
> This syntax is supported for Microsoft Dynamics 365 Finance version 10.0.25 and later.

Arguments

`location`: *String*

The location where sorting should be run. The following options are valid:

- "Query"
- "InMemory"

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`expression 1`: *Field*

The valid path of a field of the data source that is referenced by the `list` argument of the called function. The referenced field must be a field of the primitive data type. This argument is required.

`expression N`: *Field*

The valid path of a field of the data source that is referenced by the `list` argument of the called function. The referenced field must be a field of the primitive data type. These additional arguments are optional.

Return values

Record list

The resulting list of records.

Usage notes

Syntax 1

Data sorting is always done in the memory of the application server. For more details, see example 1.

Syntax 2

Sorting in memory

When the `location` argument is specified as **InMemory**, data sorting is done in the memory of an application server. For more details, see example 2.

Sorting in database

When the `location` argument is specified as **Query**, data sorting is done at the database level. In this case, the `list` argument must point to one of the following Electronic reporting (ER) data sources that specifies the application source that a direct database query can be established for:

- Data source of the *Table records* type
- Relation of a data source of the *Table records* type
- Data source of the *Calculation field* type

The `expression 1` and `expression N` arguments must point to fields of an ER data source that specifies the relevant fields of the application source that a direct database query can also be established for.

If a direct database query can't be established, a validation error occurs in the ER model mapping designer. The message that you receive states that the ER expression that includes the `ORDERBY` function can't be run at runtime.

For better performance, we recommend that you use the **Query** option when the sorting is configured for application data sources that might contain the large number of records (for example, for transactional application tables).

> **Note**
>
> The `ORDEBY` function itself can't be translated to a direct database query. Therefore, an ER data source that contains this function isn't queryable. It also can't be used in the scope of ER functions such as **FILTER** and **ALLITEMSQUERY**, where only queryable data sources can be used.

For more details, see example 3 and example 4.

Comparability

Because the SQL database engine and Finance application server can use a different ranking value for a single character, the sorting result of the same list of records can differ when a String field is used for sorting. For more details, see example 5.

Example 1: In-memory default execution

If you enter data source **DS** of the *Calculated field* type, and it contains the expression `SPLIT ("C|B|A", "|")`, the expression `FIRST( ORDERBY( DS, DS. Value)).Value` returns the text value "**A**".

Example 2: In-memory explicit execution

If **Vendor** is configured as an ER data source of the *Table records* type that refers to the **VendTable** table, both the expression `ORDERBY (Vendor, Vendor.'name()')` and the expression `ORDERBY ("InMemory", Vendor, Vendor.'name()')` return a list of vendors that is sorted by name in ascending order.

When you configure the expression `ORDERBY ("Query", Vendor, Vendor.'name()')` in the ER model mapping designer, a validation error occurs at design time, because the `Vendor.'name()'`

path refers to an application method that has logic that can't be translated to a direct database query.

Example 3: Database query

If **TaxTransaction** is configured as an ER data source of the *Table records* type that refers to the **TaxTrans** table, the expression `ORDERBY ("Query", TaxTransaction, TaxTransaction.TaxCode)` sorts records at the application database level and returns a list of tax transactions that is sorted by tax code in ascending order.

Example 4: Queryable data sources

If **TaxTransaction** is configured as an ER data source of the *Table records* type that refers to the **TaxTrans** table, the **TaxTransactionFiltered** ER data source can be configured so that it contains the expression `FILTER(TaxTransaction, TaxCode="VAT19")` that will fetch transactions for a specified tax code. Because the configured **TaxTransactionFiltered** ER data source is queryable, the expression `ORDERBY ("Query", TaxTransactionFiltered, TaxTransactionFiltered.TransDate)` can be configured to return the list of filtered tax transactions that is sorted by transaction date in ascending order.

If you configure **TaxTransactionOrdered** as an ER data source of the *Calculated field* type that contains the expression `ORDERBY ("Query", TaxTransaction, TaxTransaction.TransDate)` and an ER data source of the *Calculated field* type that contains the expression `FILTER(TaxTransactionOrdered, TaxCode="VAT19")`, a validation error occurs at design time in the ER model mapping designer. This error occurs because the first argument of the FILTER function must refer a queryable ER data source, but the **TaxTransactionOrdered** data source that contains the `ORDERBY` function isn't queryable.

Example 5: Comparability

Prerequisites

1. Enter data source **DS1** of the *Calculated field* type that contains the expression `SPLIT ("D1|_D2|D3", "|")`.
2. Open the **Financial dimension values** page, and select the **CostCenter** dimension.
3. Enter the following dimension values: **D1**, **_D2**, and **D3**.

Sorting in memory

1. Configure data source **DS2** of the *Calculated field* type that contains the expression `ORDERBY("InMemory", DS1, DS1.Value)`.
2. Notice that the expression `FIRST(DS2).Value` returns the text value "**D1**", the expression `INDEX(DS2, COUNT(DS2)).Value` returns the text value "**_D2**", and the expression `STRINGJOIN(DS2, DS2.Value, "|")` returns the text value "**D1|D3|_D2**".

Sorting in database

1. Enter data source **DS3** of the *Table records* type that refers to the **FinancialDimensionValueEntity** entity.
2. Configure data source **DS4** of the *Calculated field* type that contains the expression `FILTER(DS3, DS3.FinancialDimension="CostCenter")`.
3. Configure data source **DS5** of the *Calculated field* type that contains the expression `ORDERBY(DS4, DS4.DimensionValue)`.
4. Notice that the expression `FIRST(DS5).Value` returns the text value "**_D2**", the expression `INDEX(DS5, COUNT(DS5)).Value` returns the text value "**D3**", and the expression `STRINGJOIN(DS5, DS5.Value, "|")` returns the text value "**_D2|D1|D3**".

Additional resources

List functions

Last updated on 07/17/2026

REPEAT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `REPEAT` function builds a record that contains the field with a value that matches the specified input. It then returns a new *Record list* of a record that repeats a specified number of times.

Syntax

```
VB
REPEAT (item, number)
```

Arguments

`item`: Any supported primitive or composite data type

The value to repeat.

`number`: *Integer*

The number of repeats.

Return values

Record list

The resulting list of records.

Usage notes

The list of repeated records that the function returns exposes the following fields:

- The specified value (`Item` field)

- The current record index (`Number` field)

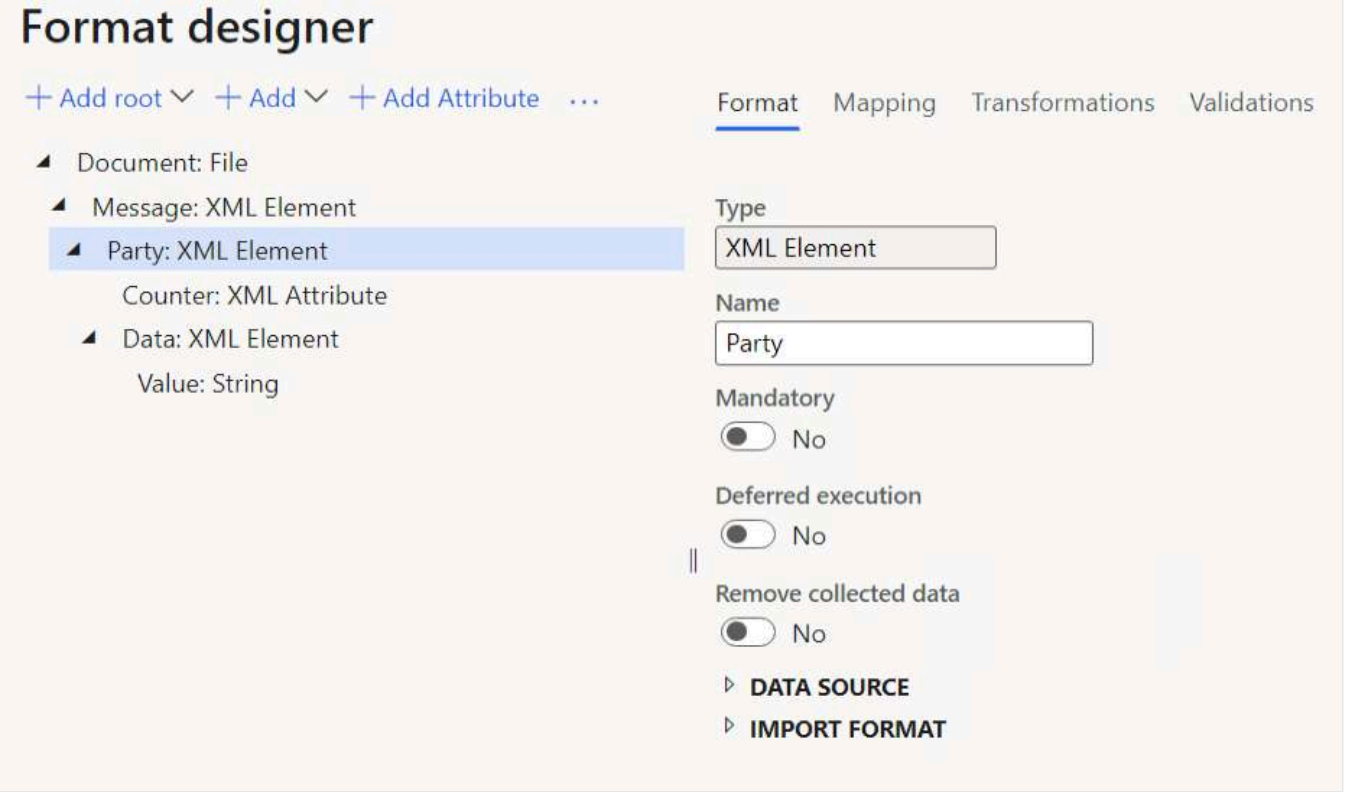
> **Note**
>
> Because this function uses one-based numbering, the `Number` field has the value **1** for the first record of the resulting list.

Use this function to multiply existing data so that you can do performance and volume testing of Electronic reporting (ER) solutions by using the Regression suite automation tool (RSAT).

Example

You want to generate a document in XML format that contains as many `Party` XML elements as you specify in a data entry field in the dialog box at runtime, before the execution of an ER format begins.

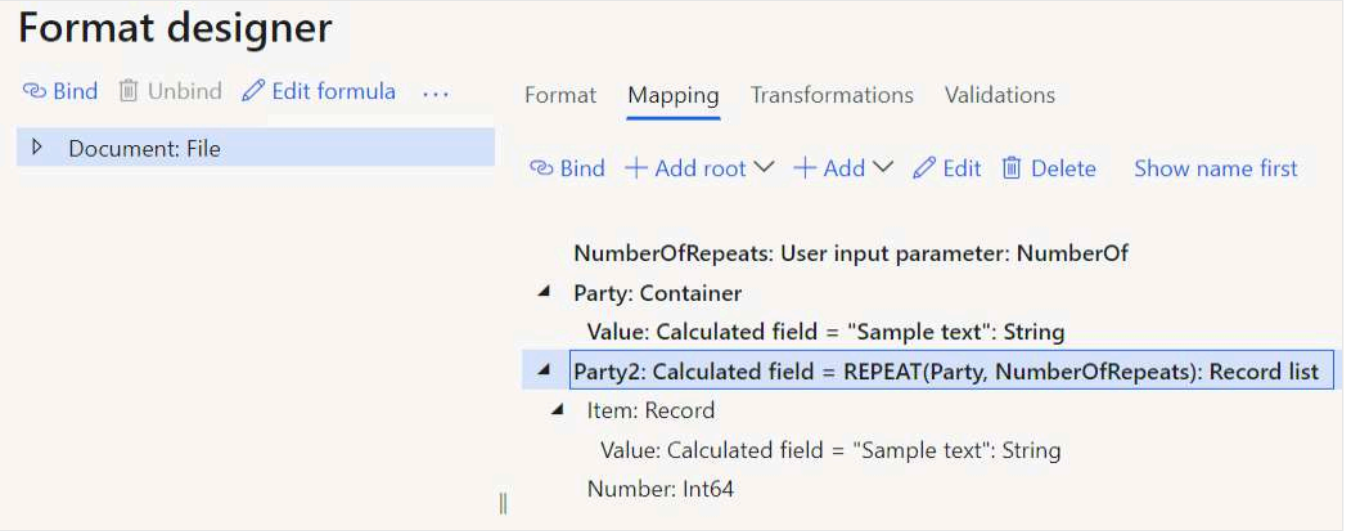
The following illustration shows the ER format. In this format, the single `Party` XML element is added to expose the properties of a single party.



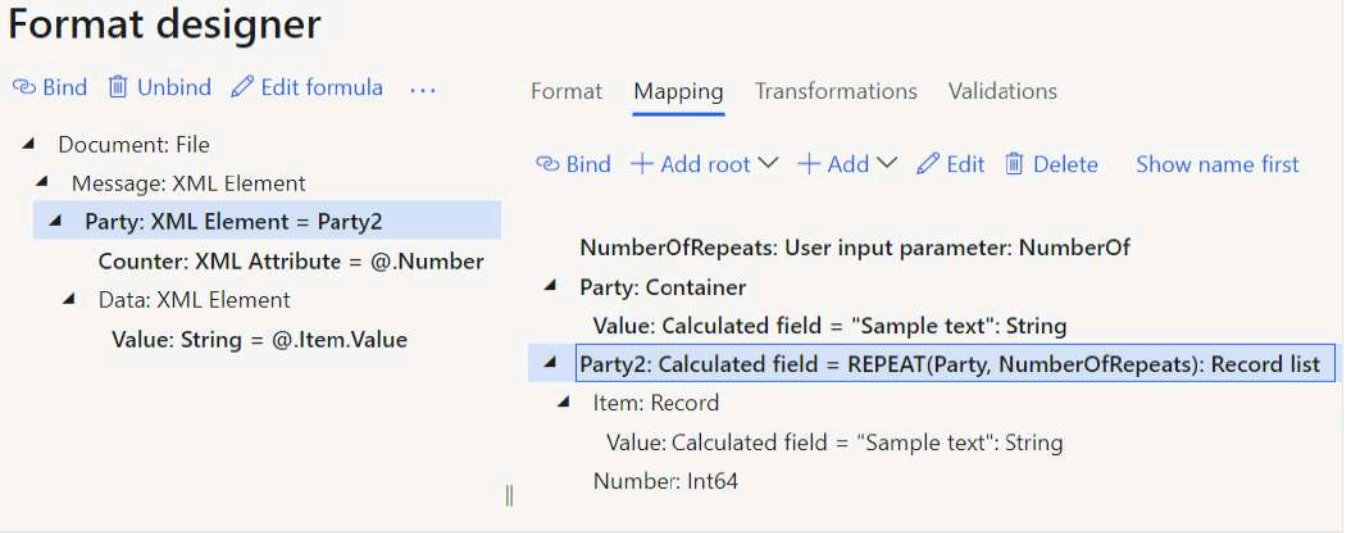
The next illustration shows the following configured data sources:

- The `Party` data source that represents a single party. The `Party.Value` field exposes a single text value.

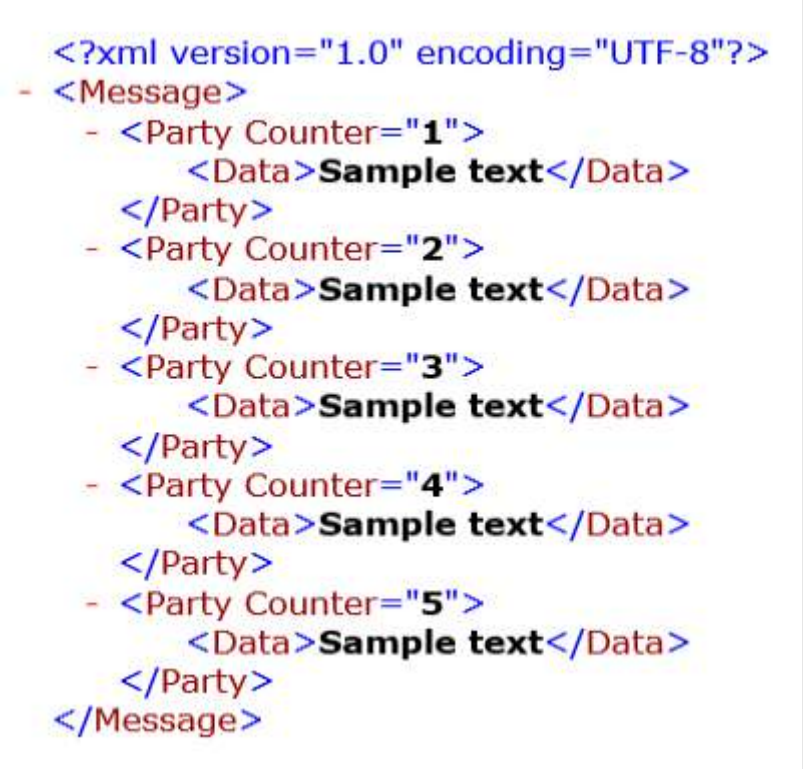
- The `NumberOfRepeats` data source that offers a data entry field in the dialog box at runtime, so that you can specify the number of parties that should be entered in the generated document.
- The `Party2` data source that repeats the `Party` record the number of times that you specified in the `NumberOfRepeats` data source.



The next illustration shows data bindings of the editable ER format that are created to generate output in XML format. This output presents individual parties as enumerated nodes.



The following illustration shows the result when you run the designed format and specify the value of the `NumberOfRepeats` data source as **5**.



Additional resources

List functions

Last updated on 12/09/2025

REVERSE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `REVERSE` function returns the specified list as a *Record list* value in reversed sort order.

Syntax

```
VB

REVERSE (list)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

Return values

Record list

The resulting list of records.

Example 1

If you enter data source **DS** of the *Calculated field* type, and it contains the expression `SPLIT ("C|B|A", "|")`, the expression `FIRST( REVERSE( ORDERBY( DS, DS. Value))).Value` returns the text value "**C**".

Example 2

If **Vendor** is configured as an Electronic reporting (ER) data source that refers to the VendTable table, the expression `REVERSE (ORDERBY (Vendors, Vendors.'name()'))` returns a list of vendors that is sorted by name in descending order.

Additional resources

List functions

Last updated on 04/08/2026

SPLIT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `SPLIT` function splits the specified input string into substrings and returns the result as a new *Record list* value.

Syntax 1

```
VB

SPLIT (input, length)
```

This syntax is used to split the specified input string into substrings, each of which has the specified length.

Syntax 2

```
VB

SPLIT (input, delimiter)
```

This syntax is used to split the specified input string into substrings, based on the specified delimiter.

Arguments

`input`: *String*

The text to split.

`length`: *Integer*

The maximum length of a single substring.

`delimiter`: *String*

A delimiter that is used to separate substrings.

Return values

Record list

The resulting list of records.

Usage notes

The record structure of the list that is returned consists of the **Value** field of the *String* type. Every record of the list that is returned contains generated substrings in this field.

If the `delimiter` argument is empty, the new list that is returned consists of one record that has the **Value** field of the *String* type. This field contains the input text.

If the `input` argument is empty, a new empty list is returned. If either the `input` or `delimiter` argument is unspecified (null), an application exception is thrown.

Example 1

`SPLIT ("abcd", 3)` returns a new list that consists of two records that have the **Value** field of the *String* type. The **Value** field in the first record contains the text "**abc**", and the **Value** field in the second record contains the text "**d**".

Example 2

`SPLIT ("XAb aBy", "aB")` returns a new list that consists of three records that have the **Value** field of the *String* type. The **Value** field in the first record contains the text "**X**", the **Value** field in the second record contains the text " ", and the **Value** field in the third record contains the text "**y**".

Example 3

Yo can use the INDEX function to access individual elements of the specified input string. If you enter the **MyList** data source of the **Calculated field** type and configure for it the `SPLIT("abc", 1)` expression, the expression `INDEX(MyList,2).Value` returns the text "**b**".

Example 4

The ENUMERATE function can also help you access individual elements of the specified input string. If you first enter the **MyList** data source of the **Calculated field** type and configure for it the `SPLIT("abc", 1)` expression, and then enter the **EnumeratedList** data source of the **Calculated field** type and configure for it the `ENUMERATE(MyList)` expression, the expression `FIRSTORNULL(WHERE(EnumeratedList, EnumeratedList.Number=2)).Value` returns the text "**b**".

Additional resources

List functions

Last updated on 04/08/2026

SPLITLIST ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `SPLITLIST` function splits the specified list into sublists (or batches), each of which contains the specified number of records. It then returns the result as a new *Record list* value that consists of the batches.

Syntax 1

```
VB

SPLITLIST (list, number)
```

Syntax 2

```
VB

SPLITLIST (list, number, on-demand reading flag)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`number`: *Integer*

The maximum number of records per batch.

`on-demand reading flag`: *Boolean*

A *Boolean* value that specifies whether elements of sublists should be generated on demand.

Return values

Record list

The resulting list of records.

Usage notes

The list of batches that is returned contains the following elements:

- **Value:** *List*

  The list of records that belong to the current batch.

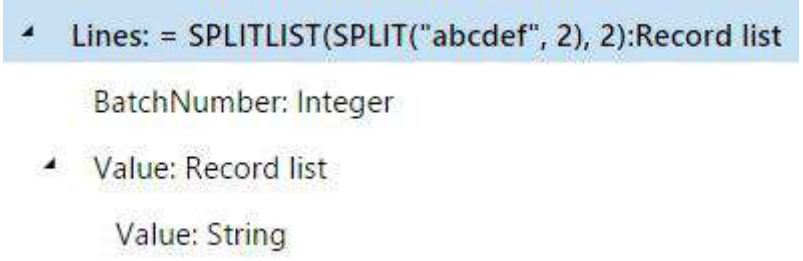
- **BatchNumber:** *Integer*

  The number of the current batch in the returned list.

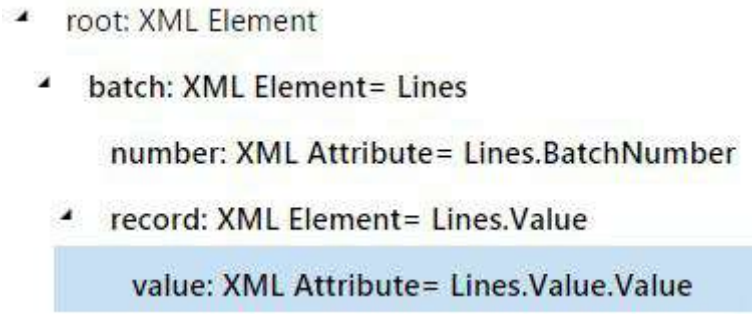
When the on-demand reading flag is set to **True**, sublists are generated upon request which allows for a reduction in memory consumption but may cause performance degradation if elements aren't used sequentially.

Example

In the following illustration, a **Lines** data source is created as a record list that has three records. This list is divided into batches, each of which contains up to two records.



The following illustration shows the designed format layout. In this format layout, bindings to the **Lines** data source are created to generate output in XML format. This output presents individual nodes for each batch and the records in it.



The following illustration shows the result when the designed format is run.

```
<?xml version="1.0" encoding="UTF-8"?>
- <root>
  - <batch number="1">
        <record value="ab"/>
        <record value="cd"/>
    </batch>
  - <batch number="2">
        <record value="ef"/>
    </batch>
</root>
```

Additional resources

List functions

Last updated on 04/08/2026

SPLITLISTBYLIMIT ER function

Article • 08/12/2022

The `SPLITLISTBYLIMIT` function splits the specified list into a new list of sublists (batches). The number of records in each batch is dynamically calculated. The function then returns the result as a new *Record list* value that consists of the batches.

Syntax

```
VB
```

```
SPLITLISTBYLIMIT (list, limit value, limit source)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`limit value`: *Integer* or *Real*

The maximum value of the limit that is used to split the original list into batches.

`limit source`: *Field*

The valid path of a field of the *Integer* or *Real* type in the specified list. The value of this field defines the step that the total sum is increased on.

Return values

Record list

The resulting list of records.

Usage notes

The list of batches that is returned contains the following elements:

- **Value**: *List*

  The list of records that belong to the current batch.

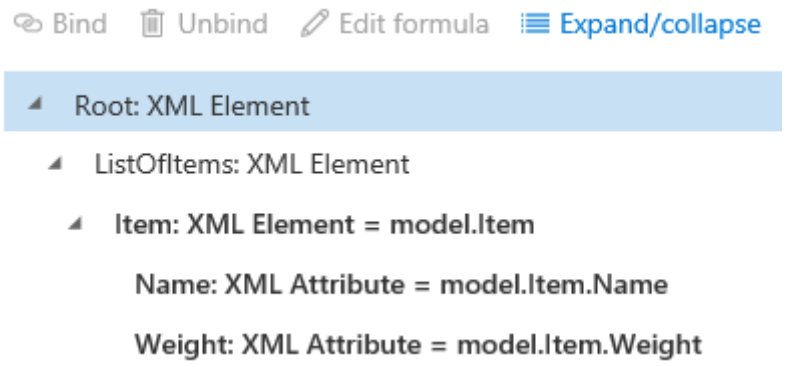
- **BatchNumber**: *Integer*

  The number of the current batch in the returned list.

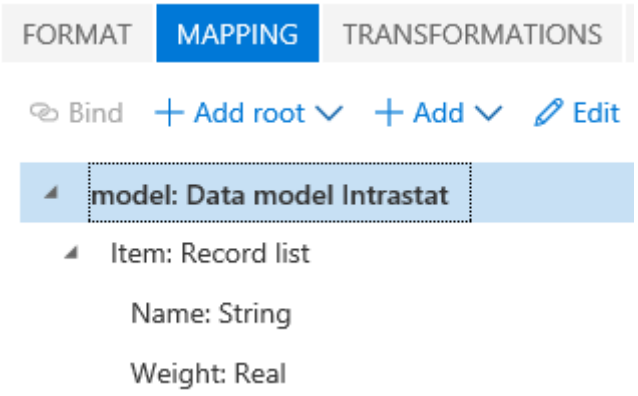
The limit isn't applied to a single item of the original list if the limit source exceeds the defined limit.

Example

The following illustration shows an Electronic reporting (ER) format.



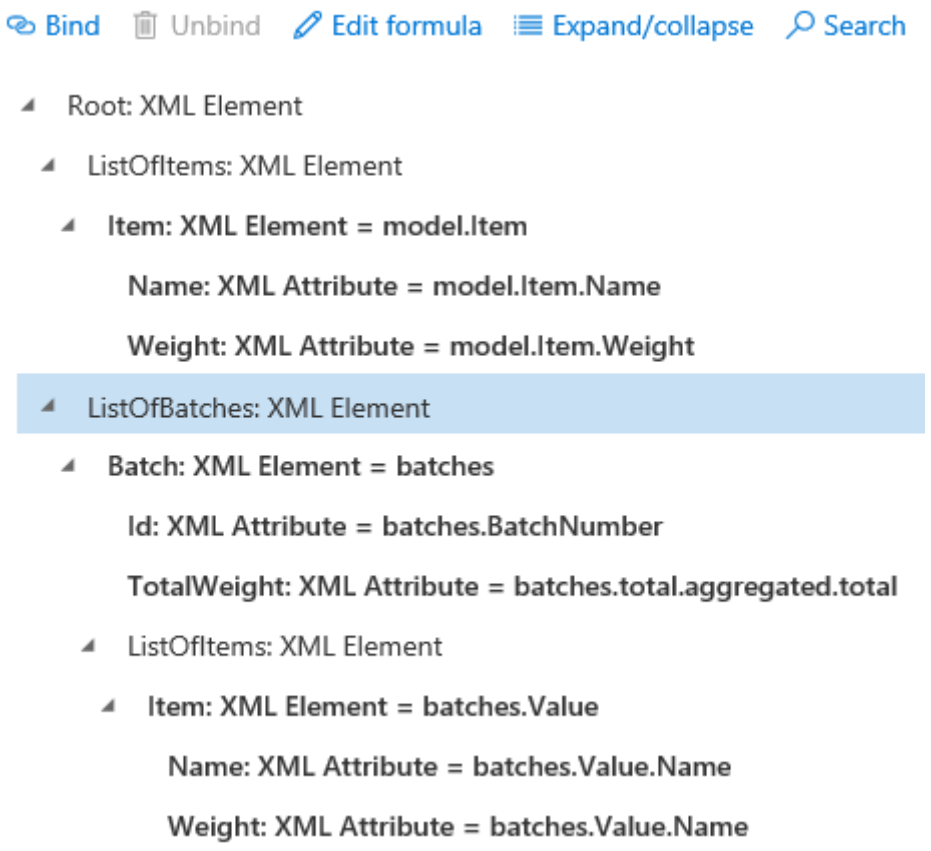
The following illustration shows the data sources that are used for the format.

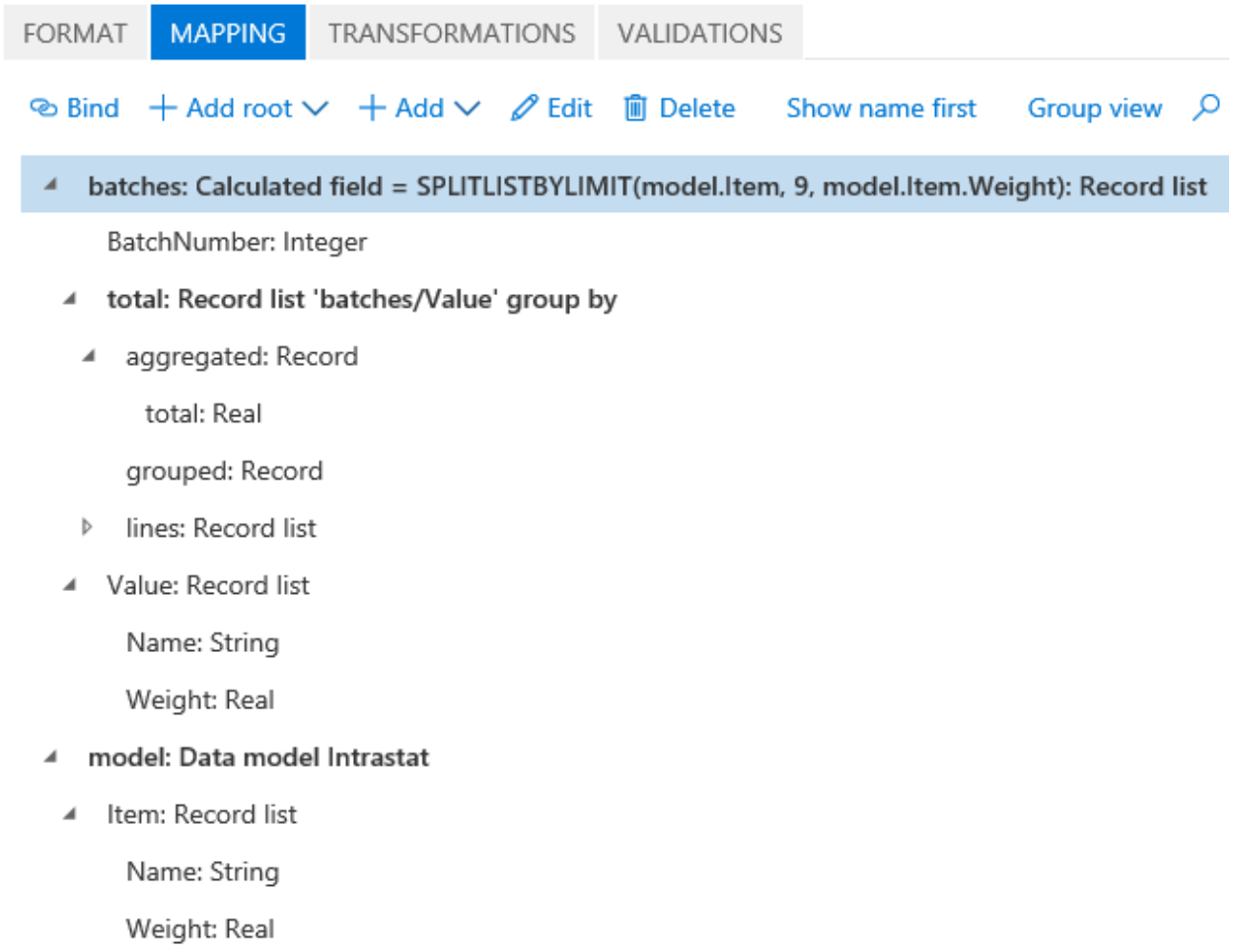


The following illustration shows the result when the format is run. In this case, the output is a flat list of commodity items.

```
<?xml version="1.0" encoding="UTF-8"?>
- <Root>
  - <ListOfItems>
      <Item Weight="1" Name="Speaker"/>
      <Item Weight="2" Name="Projector"/>
      <Item Weight="3" Name="Radio receiver"/>
      <Item Weight="4" Name="Monitor"/>
      <Item Weight="5" Name="Transmitter or receiver"/>
      <Item Weight="6" Name="Parts and components"/>
      <Item Weight="11" Name="Hardware"/>
    </ListOfItems>
</Root>
```

In the following illustrations, the same format has been adjusted so that it presents the list of commodity items in batches if a single batch must include commodities and the total weight should not exceed a limit of 9.





The following illustration shows the result when the adjusted format is run.

```
<?xml version="1.0" encoding="UTF-8"?>
- <Root>
  - <ListOfItems>
      <Item Weight="1" Name="Speaker"/>
      <Item Weight="2" Name="Projector"/>
      <Item Weight="3" Name="Radio receiver"/>
      <Item Weight="4" Name="Monitor"/>
      <Item Weight="5" Name="Transmitter or receiver"/>
      <Item Weight="6" Name="Parts and components"/>
      <Item Weight="11" Name="Hardware"/>
    </ListOfItems>
  - <ListOfBatches>
    - <Batch TotalWeight="6.0" Id="1">
      - <ListOfItems>
          <Item Weight="Speaker" Name="Speaker"/>
          <Item Weight="Projector" Name="Projector"/>
          <Item Weight="Radio receiver" Name="Radio receiver"/>
        </ListOfItems>
      </Batch>
    - <Batch TotalWeight="9.0" Id="2">
      - <ListOfItems>
          <Item Weight="Monitor" Name="Monitor"/>
          <Item Weight="Transmitter or receiver" Name="Transmitter or receiver"/>
        </ListOfItems>
      </Batch>
    - <Batch TotalWeight="6.0" Id="3">
      - <ListOfItems>
          <Item Weight="Parts and components" Name="Parts and components"/>
        </ListOfItems>
      </Batch>
    - <Batch TotalWeight="11.0" Id="4">
      - <ListOfItems>
          <Item Weight="Hardware" Name="Hardware"/>
        </ListOfItems>
      </Batch>
    </ListOfBatches>
</Root>
```

> ⓘ Note
>
> The limit isn't applied to the last item of the original list, because the value (**11**) of the limit source (**weight**) exceeds the defined limit (**9**). To ignore sublists during report generation, use either the `WHERE` function or the **Enabled** expression of the corresponding format element, as you require.

Additional resources

List functions

Feedback

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STRINGJOIN ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `STRINGJOIN` function returns a *String* value that consists of concatenated values of the specified field from the specified list. The values can be separated by the specified delimiter.

Syntax

```
VB

STRINGJOIN (list, field, delimiter)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`field`: *Field*

The valid path of a field of the *String* data type in the specified list.

`delimiter`: *String*

A delimiter that is used to separate substrings.

Return values

String

The resulting text value.

Example

If you enter `SPLIT("abc" , 1)` as data source **DS**, the expression `STRINGJOIN (DS, DS.Value, "-")` returns "**a-b-c**".

Additional resources

List functions

Last updated on 04/08/2026

WHERE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `WHERE` function returns the specified list as a *Record list* value after it has been filtered according to the specified condition.

Syntax

```VB
WHERE (list, condition)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`condition`: *Boolean*

A valid conditional expression that is used to filter records of the specified list.

Return values

Record list

The resulting list of records.

Usage notes

This function differs from the FILTER function, because the specified condition is applied to any Electronic reporting (ER) data source of the *Record list* type that is present in memory.

If the arguments that are configured for this function (`list` and `condition`) allow this request to be translated to the direct SQL call, a warning message is thrown at design time. This message informs the user that performance might be improved if the FILTER function is used instead of `WHERE`.

Example 1

If **Vendor** is configured as an ER data source that refers to the VendTable table, the expression `WHERE (Vendors, Vendors.VendGroup = "40")` returns a list of only vendors that belong to vendor group 40.

Example 2

If you enter data source **DS** of the *Calculated field* type, and it contains the expression `SPLIT ("A|B|C", "|")`, the expression `WHERE( DS, DS.Value = "B")` returns a list of only one record that contains the text **"B"** in the **Value** field.

Additional resources

List functions

Last updated on 04/08/2026

List of ER functions in the logical category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) logical functions can be used to work with logical values to perform more than one comparison in a single expression or test multiple conditions. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
And	This function returns a *Boolean* value of **TRUE** if all the specified conditions are true. Otherwise, it returns a *Boolean* value of **FALSE**.
Case	This function evaluates the value of the specified expression against the specified alternative options and returns the result of the first option that equals the value of the specified expression. Otherwise, it returns an optional default result, if a default result is specified as the last argument of the called function that isn't preceded by an option. The value that is returned can be a value of any of the supported data types.
Contains	This function determines whether the specified input contains the specified text. It returns a *Boolean* value of **TRUE** if the specified input contains the specified text. Otherwise, it returns a *Boolean* value of **FALSE**.
EndsWith	This function determines whether the specified input ends with the specified text. It returns a *Boolean* value of **TRUE** if the specified input ends with the specified text. Otherwise, it returns a *Boolean* value of **FALSE**.
If	This function returns the first specified value if the specified condition is met. Otherwise, it returns the second specified value. The value that is returned can be a value of any of the supported data types.
Not	This function returns the reversed logical value of the specified condition as a *Boolean* value.

Function	Description
Or	This function returns a *Boolean* value of **FALSE** if all the specified conditions are false. If any specified condition is true, the function returns a *Boolean* value of **TRUE**.
StartsWith	This function determines whether the specified input starts with the specified text. It returns a *Boolean* value of **TRUE** if the specified input starts with the specified text. Otherwise, it returns a *Boolean* value of **FALSE**.
ValueIn	This function determines whether the specified input matches any value of a specified item in the specified list. It returns a *Boolean* value of **TRUE** if the specified input matches the result of running the specified expression for at least one record of the specified list. Otherwise, it returns a *Boolean* value of **FALSE**.
ValueInLarge	This function determines whether the specified input of the *Int64* or *Integer* type matches any value of a specified item in the specified list. It returns a *Boolean* value of **TRUE** if the specified input matches the result of running the specified expression for at least one record of the specified list. Otherwise, it returns a *Boolean* value of **FALSE**.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

AND ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `AND` function returns a *Boolean* value of **TRUE** if all the specified conditions are true. Otherwise, it returns a *Boolean* value of **FALSE**.

Syntax

```
VB
AND (condition 1[, condition 2, …, condition N])
```

Arguments

`condition 1`: *Boolean*

A valid conditional expression that must be tested. This argument is required.

`condition N`: *Boolean*

A valid conditional expression that must be tested. These additional arguments are optional.

Return values

Boolean

The resulting *Boolean* value.

Usage notes

In the arguments of logical functions, you can use data source references, numeric and text values, Boolean values, comparison operators, and other Electronic reporting (ER) functions.

However, all the arguments must be evaluated to a *Boolean* value of **TRUE** or **FALSE**.

Example

`AND (1=1, "a"="a")` returns **TRUE**.

`AND (1=2, "a"="a")` returns **FALSE**.

Additional resources

Logical functions

Last updated on 04/08/2026

CASE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CASE` function evaluates the value of the specified expression against the specified alternative options and returns the result of the first option that equals the value of the specified expression. Otherwise, it returns the optional default result, if a default result is specified as the last argument of the called function that isn't preceded by an option. The value that is returned can be a value of any of the supported data types.

Syntax

VB

```
CASE (expression, option 1, result 1[, option 2, result 2, …, option N, result N,
default result])
```

Arguments

`expression`: *Primitive data type* (Boolean, numeric, or text)

A valid expression that returns a value of the primitive data type.

`option 1`: *Primitive data type* (Boolean, numeric, or text)

A valid expression that returns a value of the same primitive data type as the `expression` argument of the called function. This argument is required.

`result 1`: *Any of the supported data types*

The returned result that corresponds to the preceding option. This argument is required.

`option N`: *Primitive data type* (Boolean, numeric, or text)

A valid expression that returns a value of the same primitive data type as the `expression` argument of the called function. This argument is optional.

`result N`: *Any of the supported data types*

The returned result that corresponds to the preceding option. This argument is optional.

`default result`: *Any of the supported data types*

The result that should be returned if there is no match. This argument is optional.

Return values

Any of the supported data types

The resulting value of any of the supported data types.

Usage notes

An exception is thrown at runtime if there is no match and an optional default result isn't defined.

All results must be specified by using the same data type. An exception is thrown at design time if the data types of the configured results don't match.

If the first result value and the *N*th result value are values of the *Container (record)* or *Record list* data type, the result has only the fields that exist in both values.

Example

`CASE( DATETIMEFORMAT( NOW(), "MM"), "10", "WINTER", "11", "WINTER", "12", "WINTER", "")` returns the string "**WINTER**" if the current application session date is between October and December. Otherwise, it returns a blank string.

Additional resources

Logical functions

Last updated on 04/08/2026

CONTAINS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CONTAINS` function determines whether the specified input contains the specified text. It returns a *Boolean* value of **TRUE** if the specified input contains the specified text. Otherwise, it returns a *Boolean* value of **FALSE**.

Syntax

VB

```
CONTAINS (input, search text)
```

Arguments

`input`: *String*

The valid path of an item of a data source of the *String* type or a string constant, the value of which might contain the second argument.

`search text`: *String*

The valid path of a data source of the *String* data type or a string constant, the value of which might be contained in the first argument.

Return values

Boolean

The resulting *Boolean* value.

Usage notes

This function can be used to specify a condition expression of the FILTER function only when the relevant mapping is configured in Regulatory Configuration Services to access Microsoft Dataverse. Otherwise, an exception is thrown at design time. The message that you receive recommends that you use the WHERE function instead of the `FILTER` function.

Example 1

The expression `CONTAINS ("abc", "d")` returns **FALSE**, whereas the expression `CONTAINS ("abc", "C")` returns **TRUE**.

Example 2

The expression `CONTAINS (model.InvoiceBase.Customer.PostalAddress.Address, "DEU")` returns **TRUE** if the value of the **Address** field of the **model** data source contains the string **DEU**. Otherwise, it returns **FALSE**.

Additional resources

Logical functions

Last updated on 04/08/2026

ENDSWITH ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ENDSWITH` function determines whether the specified input ends with the specified text. It returns a *Boolean* value of **TRUE** if the specified input ends with the specified text. Otherwise, it returns a *Boolean* value of **FALSE**.

Syntax

```
VB

ENDSWITH (input, end text)
```

Arguments

`input`: *String*

The valid path of an item of a data source of the *String* type or a string constant, the value of which might end with the second argument.

`start text`: *String*

The valid path of a data source of the *String* data type or a string constant, the value of which might be at the end of the first argument.

Return values

Boolean

The resulting *Boolean* value.

Usage notes

This function can be used to specify a condition expression of the FILTER function only when the relevant mapping is configured in Regulatory Configuration Services to access Microsoft Dataverse. Otherwise, an exception is thrown at design time. The message that you receive recommends that you use the WHERE function instead of the `FILTER` function.

Example 1

The expression `ENDSWITH ("abc", "d")` returns **FALSE**, whereas the expression `ENDSWITH ("abc", "C")` returns **TRUE**.

Example 2

The expression `ENDSWITH (model.InvoiceBase.Customer.PostalAddress.Address, "USA")` returns **TRUE** if the value of the **Address** field of the **model** data source ends with the string **USA**. Otherwise, it returns **FALSE**.

Additional resources

Logical functions

Last updated on 04/08/2026

IF ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `IF` function returns the first specified value if the specified condition is met. Otherwise, it returns the second specified value. The value that is returned can be a value of any of the supported data types.

Syntax

```VB
IF (condition, first value, second value) as any of the supported data types
```

Arguments

`condition`: *Boolean*

A valid conditional expression that must be tested.

`first value`: *Any of the supported data types*

The result that is returned if the condition is met.

`second value`: *Any of the supported data types*

The result that is returned if the condition isn't met.

Return values

Any of the supported data types

The resulting value of any of the supported data types.

Usage notes

The `first value` and `second value` arguments must be specified by using the same data type. An exception is thrown at design time if the data types of the configured values don't match.

If the first value and the second value are values of the *Container (record)* or *Record list* data type, the result has only the fields that exist in both values.

Example

`IF (1=2, "condition is met", "condition is not met")` returns the string "**condition is not met**".

Additional resources

Logical functions

Last updated on 04/08/2026

NOT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NOT` function returns the reversed logical value of the specified condition as a *Boolean* value.

Syntax

VB

```
NOT (condition)
```

Arguments

`condition`: *Boolean*

A valid conditional expression that must be reversed.

Return values

Boolean

The resulting *Boolean* value.

Example

`NOT (TRUE)` returns **FALSE**.

Additional resources

Logical functions

Last updated on 04/08/2026

OR ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `OR` function returns a *Boolean* value of **FALSE** if all the specified conditions are false. If any specified condition is true, the function returns a *Boolean* value of **TRUE**.

Syntax

VB

```
OR (condition 1[, condition 2, …, condition N])
```

Arguments

`condition 1`: *Boolean*

A valid conditional expression that must be tested. This argument is required.

`condition N`: *Boolean*

A valid conditional expression that must be tested. These additional arguments are optional.

Return values

Boolean

The resulting *Boolean* value.

Example

`OR (1=2, "a"="a")` returns **TRUE**.

Additional resources

Logical functions

Last updated on 04/08/2026

STARTSWITH ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `STARTSWITH` function determines whether the specified input starts with the specified text. It returns a *Boolean* value of **TRUE** if the specified input starts with the specified text. Otherwise, it returns a *Boolean* value of **FALSE**.

Syntax

VB

```
STARTSWITH (input, start text)
```

Arguments

`input`: *String*

The valid path of an item of a data source of the *String* type or a string constant, the value of which might start with the second argument.

`start text`: *String*

The valid path of a data source of the *String* data type or a string constant, the value of which might be at the beginning of the first argument.

Return values

Boolean

The resulting *Boolean* value.

Usage notes

This function can be used to specify a condition expression of the FILTER function only when the relevant mapping is configured in Regulatory Configuration Services to access Microsoft Dataverse. Otherwise, an exception is thrown at design time. The message that you receive recommends that you use the WHERE function instead of the `FILTER` function.

Example 1

The expression `STARTSWITH ("abc", "d")` returns **FALSE**, whereas the expression `STARTSWITH ("abc", "A")` returns **TRUE**.

Example 2

The expression `STARTSWITH (model.InvoiceBase.Customer.PostalAddress.Address, "123 Coffee Street")` returns **TRUE** if the value of the **Address** field of the **model** data source starts with the string **123 Coffee Street**. Otherwise, it returns **FALSE**.

Additional resources

Logical functions

Last updated on 04/08/2026

VALUEIN ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `VALUEIN` function checks whether the specified input matches any value of a specified item in the specified list. It returns **TRUE** if the specified input matches the result of running the specified expression for at least one record of the specified list. Otherwise, it returns **FALSE**.

Syntax

VB

```
VALUEIN (input, list, list item expression)
```

Arguments

`input`: *Field*

The valid path of an item of a data source of the *Record list* type. The value of this item is matched.

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`list item expression`: *Boolean*

A valid conditional expression that either points to or contains a single field of the specified list that you use for the matching.

Return values

Boolean

The resulting *Boolean* value.

Usage notes

In general, the `VALUEIN` function translates to a set of **OR** conditions. If the list of **OR** conditions is large and the maximum total length of an SQL statement might be exceeded, consider using the VALUEINLARGE function.

VB

```
(input = list.item1.value) OR (input = list.item2.value) OR …
```

In some cases, the function translates to a database SQL statement by using the `EXISTS JOIN` operator.

> ⓘ **Note**
>
> Depending on whether you use the `VALUEIN` function to specify the selection criteria for the **FILTER** function or the **WHERE** function, the value that the `VALUEIN` function returns is **used differently**.

Example 1

In your model mapping, you define the **List** data source of the *Calculated field* type. This data source contains the expression `SPLIT ("a,b,c", ",")`.

When you call a data source, if you configure it as the `VALUEIN ("B", List, List.Value)` expression, it returns **TRUE**. In this case, the `VALUEIN` function translates to the following set of conditions: `(("B" = "a") or ("B" = "b") or ("B" = "c"))`, where `("B" = "b")` equals **TRUE**.

When you call a data source, if you configure it as the `VALUEIN ("B", List, LEFT(List.Value, 0))` expression, it returns **FALSE**. In this case, the `VALUEIN` function translates to the following condition: `("B" = "")`, which doesn't equal **TRUE**.

The upper limit for the number of characters in the text of such a condition is 32,768 characters. Therefore, don't create data sources that might exceed this limit at runtime. If the limit is exceeded, the application stops running, and an exception is thrown. For example, this situation can occur if the data source is configured as `WHERE (List1, VALUEIN (List1.ID, List2, List2.ID)`, and the **List1** and **List2** lists contain a large volume of records.

In some cases, the `VALUEIN` function translates to a database statement by using the `EXISTS JOIN` operator. This behavior occurs when the FILTER function is used and the following conditions are met:

- The **ASK FOR QUERY** option is turned off for the data source of the `VALUEIN` function that refers to the list of records. No additional conditions are applied to this data source at runtime.
- No nested expressions are configured for the data source of the `VALUEIN` function that refers to the list of records.
- A list item of the `VALUEIN` function refers to a field of the specified data source, not to an expression or method of that data source.

Consider using this option instead of the WHERE function that is described earlier in this example.

Example 2

You define the following data sources in your model mapping:

- The **In** data source of the *Table records* type. This data source refers to the Intrastat table.
- The **Port** data source of the *Table records* type. This data source refers to the IntrastatPort table.

When you call a data source that you configured as the `FILTER (In, VALUEIN(In.Port, Port, Port.PortId))` expression, the system generates the following SQL statement to return filtered records of the Intrastat table.

VB

```
select … from Intrastat
exists join TableId from IntrastatPort
where IntrastatPort.PortId = Intrastat.Port
```

For **dataAreaId** fields, the final SQL statement uses the `IN` operator.

Example 3

You define the following data sources in your model mapping:

- The **Le** data source of the *Calculated field* type. This data source contains the expression `SPLIT ("DEMF,GBSI,USMF", ",")`.

- The **In** data source of the *Table records* type. This data source refers to the Intrastat table, and the **Cross-company** option is turned on for it.

When you call a data source that you configured as the `FILTER (In, VALUEIN (In.dataAreaId, Le, Le.Value))` expression, the final SQL statement contains the following condition.

VB

```
Intrastat.dataAreaId IN ('DEMF', 'GBSI', 'USMF')
```

Additional resources

Logical functions

VALUEINLARGE functions

Last updated on 04/13/2026

VALUEINLARGE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `VALUEINLARGE` function determines whether the specified input of the *Int64* or *Integer* type matches any value of a specified item in the specified list. The function returns a *Boolean* value of **TRUE** if the specified input matches the result of running the specified expression for at least one record of the specified list. Otherwise, it returns a *Boolean* value of **FALSE**. To understand the difference with the `VALUEIN` function, see the Usage note section later in this article.

Syntax

VB

```
VALUEINLARGE (input, list, list item expression)
```

Arguments

`input`: *Field*

The valid path of a data source item of the *Record list* type. The value of this item will be matched.

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`list item expression`: *Expression*

A valid conditional expression that either points to or contains a single field of the specified list that should be used for the matching.

Return values

Boolean

The resulting *Boolean* value.

Usage notes

When the specified input represents an *Int64* or *Integer* type of a data source item, the call to which is translatable to a direct SQL statement, the specified list is converted to a temporary SQL table and matching is performed in the database by executing a single `EXISTS JOIN` query. Otherwise, this function works as the VALUEIN function.

When the specified input represents a data source item that is designed as an item other than *Int64* and *Integer* type, an error occurs at design time informing you that the `VALUEINLARGE` function is not applicable for the configured ER expression.

When the `VALUEINLARGE` function expression is executed and more than one temporary table is used in scope of this execution, a runtime error occurs.

Example

You define the following data sources in your model mapping:

- The **In** data source of the *Table records* type.
  - This data source refers to the **Intrastat** table.
  - The **Cross-company** option is set to **No**.
- The **InMemory** data source of the *Calculated field* type.
  - This data source contains the expression `WHERE (In, In.Port <> "")`.
- The **InFiltered** data source of the *Calculated field* type.
  - This data source contains the expression `FILTER (In, VALUEINLARGE(In.RecId, InMemory, InMemory.RecId)`.

When the data source **InFiltered** is called under the context of the company **DEMF**, a new temporary table is created in the application database, the collected in memory list of record identification codes are inserted to this table, and the following SQL statement is generated to return filtered records of the **Intrastat** table.

X++

```
SELECT … from Intrastat T1
WHERE ((T1.PARTITION=?) AND (T1.DATAAREAID IN (N'DEMF'))) AND
```

```
EXISTS (SELECT 'x' FROM tempdb."DBO".? T2 WHERE ((T2.PARTITION=?) AND
(T1.RecId=T2.RecId)))
```

Additional resources

Logical functions

VALUEIN functions

Last updated on 04/08/2026

List of ER functions in the mathematical category

Electronic reporting (ER) mathematical functions can be used to do many common mathematical calculations. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
Abs	This function returns the absolute value (modulus) of the specified number as a *Real* value. In other words, it returns the number without its sign.
DIV	This function returns the integer division of two integers. The result is the quotient of the division of `dividend` by `divisor`, rounded down to the nearest integer.
MOD	This function returns the remainder after a number is divided by another number. In other words, it returns the modulus of the specified numbers.
Power	This function returns a *Real* value that represents the result of raising the specified positive number to the specified power.
Round	This function returns the specified number as a *Real* value after it has been rounded to the specified number of decimal places.
RoundDown	This function returns the specified number as a *Real* value after it has been rounded down to the specified number of decimal places.
RoundUp	This function returns the specified number as a *Real* value after it has been rounded up to the specified number of decimal places.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 07/22/2026

ABS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ABS` function returns the absolute value (modulus) of the specified number as a *Real* value. In other words, it returns the number without its sign.

Syntax

VB

```
ABS (number)
```

Arguments

`number`: *Real*

A numeric value that you want the modulus of.

Return values

Real

The resulting numeric value.

Example

`ABS (-1)` returns **1**.

Additional resources

Mathematical functions

Last updated on 04/08/2026

DIV ER function

The `DIV` function is available in Dynamics 365 Finance **10.0.49** and later.

The `DIV` function returns the integer division of two integers. The result is the quotient of the division of `dividend` by `divisor`, rounded down to the nearest integer.

Syntax

VB

```
DIV (dividend, divisor)
```

Arguments

`dividend`: *Integer*

The dividend, or the number to be divided.

`divisor`: *Integer*

The divisor, or the number by which `dividend` is divided.

Return values

Integer

The resulting numeric value.

Example

`DIV (5, 2)` returns **2**.

Additional resources

Mathematical functions

Last updated on 07/22/2026

MOD ER function

The `MOD` function is available for Finance and Operations **10.0.49** and later.

The `MOD` function returns the remainder after a number is divided by another number. In other words, it returns the modulus of the specified numbers.

Syntax

VB

```
MOD (dividend, divisor)
```

Arguments

`dividend`: *Real* or *Integer*

The dividend, or the number to be divided.

`divisor`: *Real* or *Integer*

The divisor, or the number by which `dividend` is divided.

Return values

Real or *Integer*

The resulting numeric value.

Example

`MOD (5, 2)` returns **1**.

Example 2

`MOD (6.6, 3.1)` returns **0.4**.

Additional resources

Mathematical functions

Last updated on 07/22/2026

POWER ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `POWER` function returns a *Real* value that represents the result of raising the specified positive number to the specified power.

Syntax

VB

```
POWER (number, power)
```

Arguments

`number`: *Real* or *Integer*

A numeric value that must be raised to the specified power.

`power`: *Real* or *Integer*

A numeric value that represents the specific power.

Return values

Real

The resulting numeric value.

Example 1

`POWER (10, 2)` returns **100**.

Example 2

`POWER (4, 0.5)` returns 2.

Additional resources

Mathematical functions

Last updated on 04/08/2026

ROUND ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ROUND` function returns the specified number as a *Real* value after it has been rounded to the specified number of decimal places.

Syntax

VB

```
ROUND (number, decimals)
```

Arguments

`number`: *Real*

A numeric value that must be rounded.

`decimals`: *Integer*

A numeric value that represents the number of decimal places.

Return values

Real

The resulting numeric value.

Usage notes

If the value of the `decimals` argument is more than 0 (zero), the specified number is rounded to that many decimal places.

If the value of the `decimals` argument is **0** (zero), the specified number is rounded to the nearest even integer.

If the value of the `decimals` argument is less than 0 (zero), the specified number is rounded to the left of the decimal point.

Example 1

`ROUND (1200.767, 2)` rounds to two decimal places and returns **1200.77**.

Example 2

`ROUND (1200.767, -3)` rounds to the nearest multiple of 1,000 and returns **1000**.

Example 3

`ROUND (1200.5, 0)` rounds to the nearest even integer and returns **1200**, while `ROUND (1201.5, 0)` does the same and returns **1202**.

Additional resources

Mathematical functions

Last updated on 04/08/2026

ROUNDDOWN ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ROUNDDOWN` function returns the specified number as a *Real* value after it has been rounded down to the specified number of decimal places.

Syntax

VB

```
ROUNDDOWN (number, decimals)
```

Arguments

`number`: *Real*

A numeric value that must be rounded down.

`decimals`: *Integer*

A numeric value that represents the number of decimal places.

Return values

Real

The resulting numeric value.

Usage notes

This function behaves like ROUND, but it always rounds the specified number down (toward zero).

Example 1

`ROUNDDOWN (1200.767, 2)` rounds down to two decimal places and returns **1200.76**.

Example 2

`ROUNDDOWN (1700.767, -3)` rounds down to the nearest multiple of 1,000 and returns **1000**.

Additional resources

Mathematical functions

Last updated on 04/08/2026

ROUNDUP ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ROUNDUP` function returns the specified number as a *Real* value after it has been rounded up to the specified number of decimal places.

Syntax

VB

```
ROUNDDOWN (number, decimals)
```

Arguments

`number`: *Real*

A numeric value that must be rounded up.

`decimals`: *Integer*

A numeric value that represents the number of decimal places.

Return values

Real

The resulting numeric value.

Usage notes

This function behaves like ROUND, but it always rounds the specified number up (away from zero).

Example 1

`ROUNDUP (1200.763, 2)` rounds up to two decimal places and returns **1200.77**.

Example 2

`ROUNDUP (1200.767, -3)` rounds up to the nearest multiple of 1,000 and returns **2000**.

Additional resources

Mathematical functions

Last updated on 04/08/2026

List of ER functions in the record category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) record functions can be used to extract information from, and perform operations on, data sources of the *Container (record)* data type. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
NullContainer	This function returns a null *Container (record)* value that has the same structure as the specified record list or record.
EmptyRecord	This function returns a null *Container (record)* value that has the same structure as the specified record list or record.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

EMPTYRECORD ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `EMPTYRECORD` function returns a null *Container (record)* value that has the same structure as the specified record list or record.

Syntax

VB

```
EMPTYRECORD (list)
```

Arguments

`list`: *Record list* or *Container (record)*

The valid path of a data source of either the *Record list* or *Container (record)* type.

Return values

Container (record)

The resulting record value.

Usage notes

> **Note**
>
> A null record is a record where all fields have an empty value. An empty value is **0** (zero) for numbers, an empty string for strings, and so on.

Example

`EMPTYRECORD (SPLIT ("abc", 1))` returns a new empty record that has the same structure as the list that is returned by the `SPLIT` function. For more information, see SPLIT.

Additional resources

Record functions

Last updated on 04/08/2026

NULLCONTAINER ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NULLCONTAINER` function returns a null *Container (record)* value that has the same structure as the specified record list or record.

Syntax

VB

```
NULLCONTAINER (list)
```

Arguments

`list`: *Record list* or *Container (record)*

The valid path of a data source of either the *Record list* or *Container (record)* type.

Return values

Container (record)

The resulting record value.

Usage notes

> **Note**
>
> This function is obsolete. Use the `EMPTYRECORD` function instead. For more information, see EMPTYRECORD.

Example

`NULLCONTAINER (SPLIT ("abc", 1))` returns a new empty record that has the same structure as the list that is returned by the `SPLIT` function. For more information, see SPLIT.

Additional resources

Record functions

Last updated on 04/08/2026

List of ER functions of the text category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) text functions can be used to perform operations on data sources of the *String* data type. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
Char	This function returns a *String* value that presents a single character that is referenced by the specified Unicode number.
Concatenate	This function returns all the specified text strings as a *String* value after they have been joined into one string.
Format	This function returns the specified string a *String* value after it has been formatted by substituting any occurrences of **%N** with the *N*th argument.
GetEnumValueByName	This function searches for a specific *Enum* value in the specified enumeration data source by using the enumeration name that is specified as a *String* value. If the *Enum* value is found, the function returns it.
GetLabelText	This function searches for a specific label to return a *String* value that represents the translation of the specified label in the specified language.
GuidValue	This function converts the specified input of the *String* type to a data item of the *GUID* type.
JsonValue	This function parses data in JavaScript Object Notation (JSON) format that is accessed at the specified path, and it extracts a scalar value that is based on the specified ID. It then returns the extracted scalar value as a *String* value.
Left	This function returns a *String* value that presents the specified number of characters from the start of the specified string.

Function	Description
Len	This function returns an *Integer* value that presents the number of characters in the specified string.
Lower	This function returns the specified text string as a *String* value after it has been converted to lowercase letters.
Mid	This function returns a *String* value that presents the specified number of characters from the specified string, starting at the specified position.
NewGUID	This function returns a newly generated *GUID* value.
NumberFormat	This function returns a *String* value that presents the specified number in the specified format and in an optionally specified culture.
NumeralsToText	This function returns the specified number as a *String* value after it has been spelled out (that is, converted to text strings) in the specified language.
PadLeft	This function returns a *String* value of the specified length, where the start of the specified string is padded with one or more instances of the specified characters.
QrCode	This function returns a *Container* value that presents the Quick Response code (QR code) image for the specified string in binary format.
Replace	This function returns the specified text string as a *String* value after all or part of it has been replaced with another string.
Right	This function returns a *String* value that presents the specified number of characters from the end of the specified string.
Text	This function returns the specified number as a *String* value after it has been converted to a text string that is formatted according to the server locale settings of the current application instance.
Translate	This function returns a *String* value that contains the result of the replacement the specified text in characters for another provided set of characters.
Trim	This function returns the specified text string as a *String* value after tab, carriage return, line feed, and form feed characters have been replaced by a single space character, after leading and trailing spaces have been truncated, and after multiple spaces between words have been removed.
Upper	This function returns the specified text string as a *String* value after it has been converted to uppercase letters.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

CHAR ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CHAR` function returns a *String* value that presents a single character that is referenced by the specified Unicode number.

Syntax

VB

```
CHAR (number)
```

Arguments

`number`: *Integer*

A number that corresponds to an expected single character.

Return values

String

The resulting text value.

Usage notes

The string that this function returns depends on the encoding that is selected in the parent **FILE** format element. For a list of the supported encodings, see Encoding class.

Example

`CHAR (255)` returns "ÿ".

Additional resources

Text functions

Last updated on 04/08/2026

CONCATENATE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CONCATENATE` function returns all the specified text strings as a *String* value after they have been joined into one string.

Syntax

VB

```
CONCATENATE (text 1[, text 2, …, text N])
```

Arguments

`text 1`: *String*

A reference to a data source of the *String* data type. This argument is required.

`text N`: *String*

A reference to a data source of the *String* data type. These additional arguments are optional.

Return values

String

The resulting text value.

Example

`CONCATENATE ("abc", "def")` returns "**abcdef**".

Usage notes

The expression `"abc" & "def"` also returns "**abcdef**".

Additional resources

Text functions

Last updated on 04/08/2026

FORMAT ER function

The `FORMAT` function returns the specified string as a *String* value after it formats the string by substituting any occurrences of **%N** with the *N*th argument.

Syntax

```VB
FORMAT (string, argument 1[, argument 2, …, argument N])
```

Arguments

`string`: *String*

A reference to a data source of the *String* type that you want to format. This argument is required.

`argument 1`: *String*

The first argument, which replaces occurrences of **%1**. This argument is required.

`argument N`: *String*

The *N*th argument, which replaces occurrences of **%2**, **%3**, and so on. These additional arguments are optional.

Return values

String

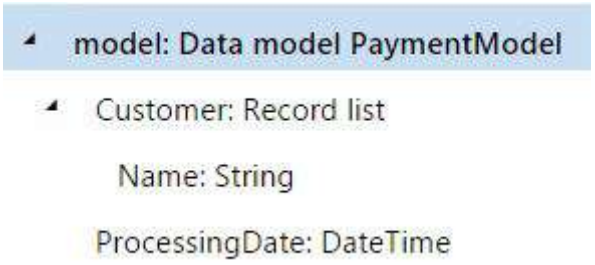
The resulting text value.

Usage notes

If you don't provide an argument for a parameter, the string shows the parameter as **"%N"**. For *Real* type values, the default string conversion rounds the number to two decimal places.

Example

In the following illustration, the **PaymentModel** data source returns a list of customer records by using the **Customer** component. It returns the processing date value by using the **ProcessingDate** field.



In the Electronic reporting (ER) format that is designed to generate an electronic file for selected customers, select **PaymentModel** as a data source. It controls the process flow. If a selected customer is stopped for the date when the report is processed, an exception is thrown to notify the user. The formula that is designed for this type of processing control can use the following resources:

- Label SYS70894, which has the following text:
  - **For the EN-US language:** "Nothing to print"
  - **For the DE language:** "Nichts zu drucken"
- Label SYS18389, which has the following text:
  - **For the EN-US language:** "Customer %1 is stopped for %2."
  - **For the DE language:** "Debitor '%1' wird für %2 gesperrt."

Here's the expression that you can design.

VB

```
FORMAT (CONCATENATE (@"SYS70894", ". ", @"SYS18389"), model.Customer.Name,
DATETIMEFORMAT (model.ProcessingDate, "d"))
```

If you process a report for the **Litware Retail** customer on December 17, 2025, in the **EN-US** culture and the **EN-US** language, this formula returns the following text, which you can present to the user as an exception message:

Nothing to print. Customer Litware Retail is stopped for 12/17/2025.

If you process the same report for the **Litware Retail** customer on December 17, 2025, in the **DE** culture and the **DE** language, the formula returns the following text, which uses a different date format:

Nichts zu drucken. Debitor 'Litware Retail' wird für 17.12.2025 gesperrt.

> **Note**
>
> The following syntax is applied in ER formulas for labels:
>
> - **For labels from resources in the Microsoft Dynamics 365 Finance app: @X**, where **X** is the label ID in the Application Object Tree (AOT)
> - **For labels that reside in ER configurations: @"GER_LABEL:X"**, where **X** is the label ID in the ER configuration

Additional resources

Text functions

Last updated on 08/04/2026

GETENUMVALUEBYNAME ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `GETENUMVALUEBYNAME` function searches for a specific *Enum* value in the specified enumeration data source by using the enumeration name that you specify as a *String* value. If the function finds the *Enum* value, it returns it. Otherwise, the function returns the **null** enumeration value.

Syntax

VB

```
GETENUMVALUEBYNAME (enumeration data source path, enumeration value text)
```

Arguments

`enumeration data source path`: *Enumeration*

The valid path of a data source for one of the following enumeration types:

- Electronic reporting (ER) model enumeration
- ER format enumeration
- Microsoft Dynamics 365 Finance enumeration

`enumeration value text`: *String*

A string value that represents the name of a single enumeration value.

Return values

Nullable *Enum*

The resulting enumeration value.

Usage notes

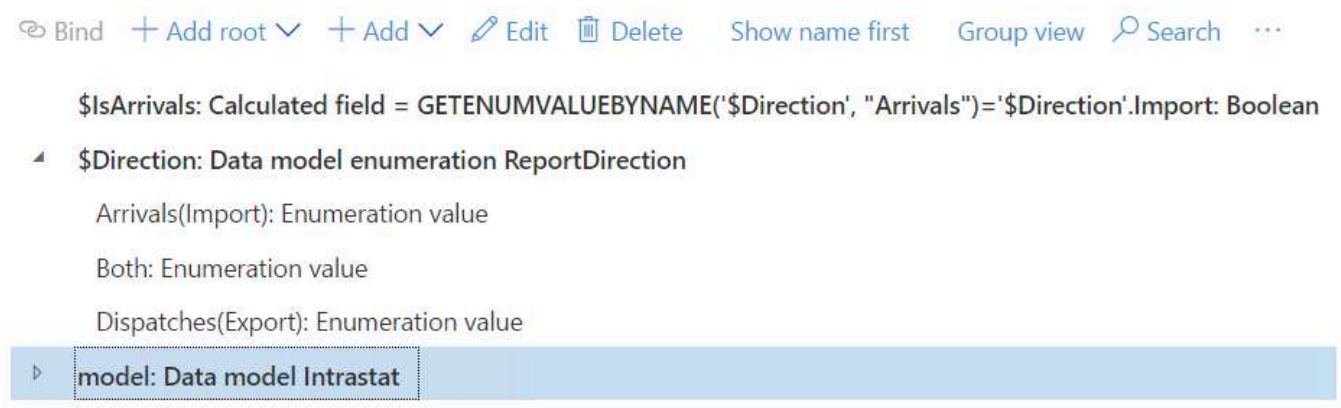
No exception is thrown if an *Enum* value isn't found by using the name of the enumeration value that you specify as a *String* value.

Example 1

In the following illustration, the **ReportDirection** enumeration is introduced in a data model. The labels for the enumeration values are defined.

The following illustration shows these details:

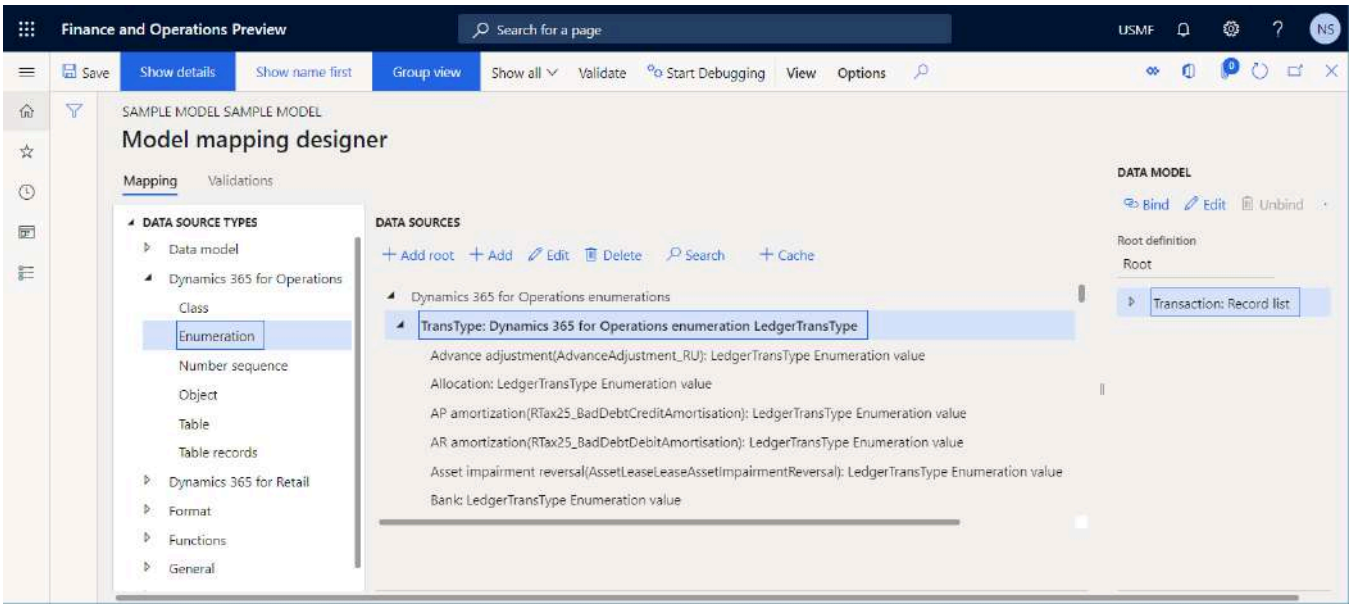
- The **$Direction** data source is configured in an ER report. This data source is configured based on the **ReportDirection** model enumeration.
- The `$IsArrivals` expression uses the model enumeration–based **$Direction** data source as a parameter of this function.
- The value of this comparison expression is **TRUE**.



Example 2

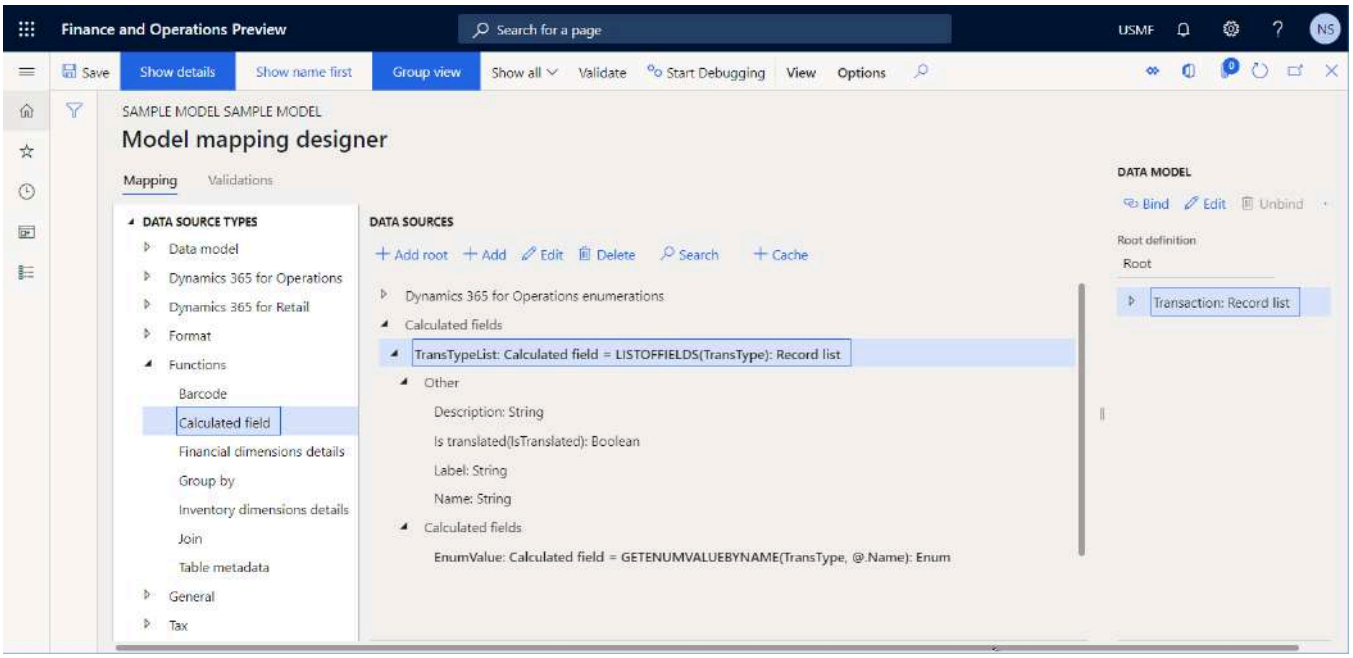
Use the `GETENUMVALUEBYNAME` and LISTOFFIELDS functions to get the values and labels of supported enumerations as text values. Supported enumerations include application enumerations, data model enumerations, and format enumerations.

In the following illustration, you add the **TransType** data source in a model mapping. This data source refers to the **LedgerTransType** application enumeration.



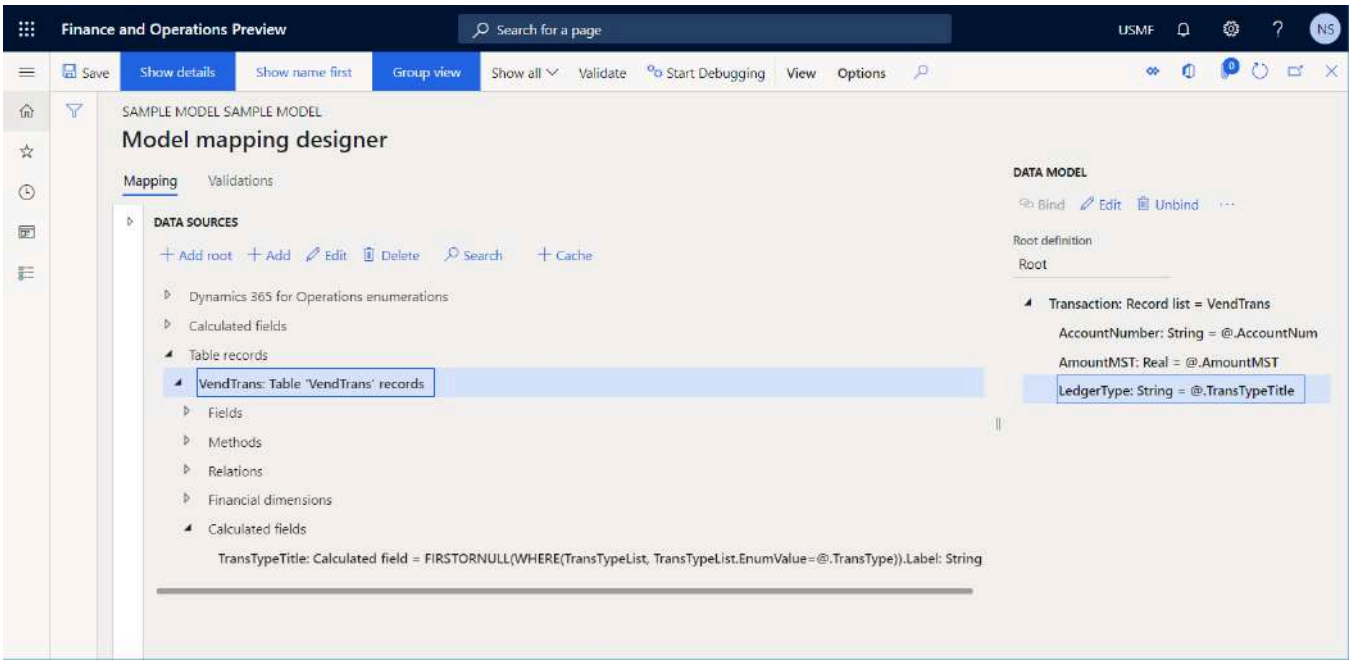
The following illustration shows the **TransTypeList** data source that you configure in a model mapping. You base this data source on the **TransType** application enumeration. Use the `LISTOFFIELDS` function to return all enumeration values as a list of records that contain fields. This approach exposes the details of every enumeration value.

> **Note**
>
> Use the `GETENUMVALUEBYNAME(TransType, TransTypeList.Name)` expression to configure the **EnumValue** field for the **TransTypeList** data source. This field returns an enumeration value for every record in this list.

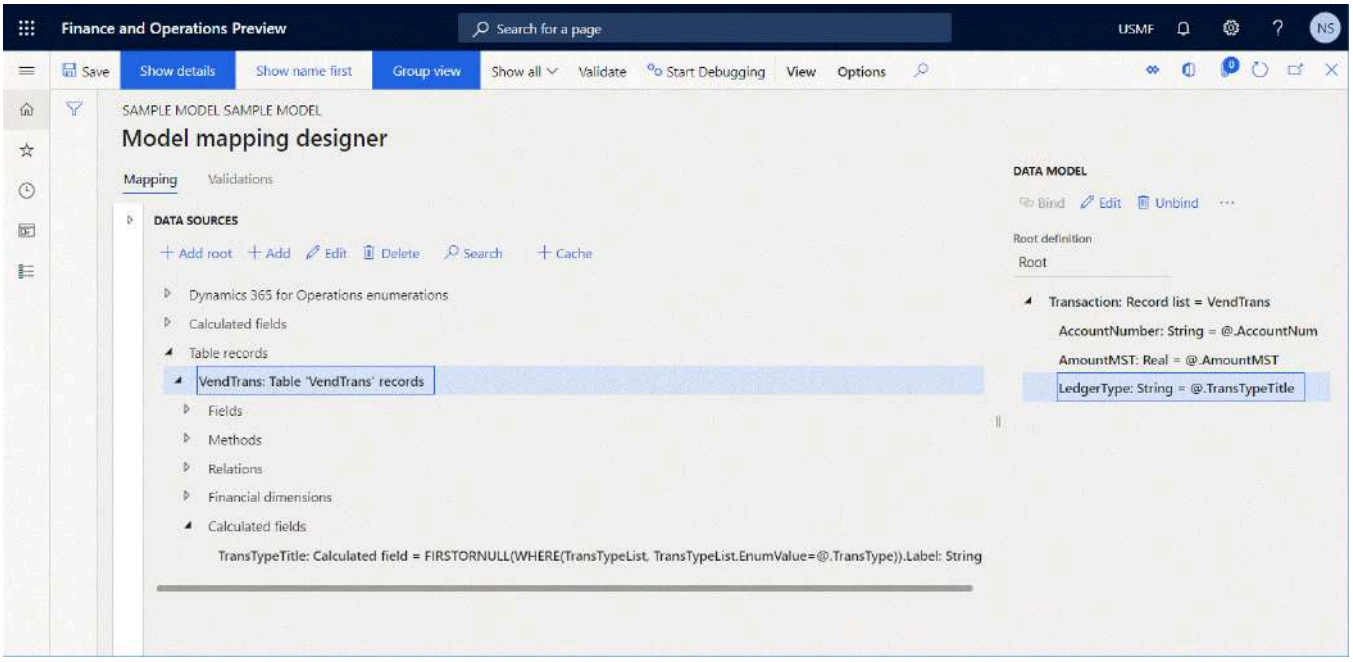


The following illustration shows the **VendTrans** data source that you configure in a model mapping. This data source returns vendor transaction records from the **VendTrans** application table. The ledger type of every transaction is defined by the value of the **TransType** field.

> **Note**
>
> Use the `FIRSTORNULL(WHERE(TransTypeList, TransTypeList.EnumValue = @.TransType)).Label` expression to configure the **TransTypeTitle** field for the **VendTrans** data source. This field returns the label of an enumeration value of the current transaction as text, if this enumeration value is available. Otherwise, it returns a blank string value.
>
> Bind the **TransTypeTitle** field to the **LedgerType** field of a data model. This binding enables you to use the information in every ER format that uses the data model as a source of data.



The following illustration shows how you can use the data source debugger to test the configured model mapping.



The **LedgerType** field of a data model exposes labels of transaction types as expected.

If you plan to use this approach for a large amount of transactional data, consider execution performance. For more information, see Trace the execution of ER formats to troubleshoot performance issues.

Additional resources

Text functions

Trace the execution of ER formats to troubleshoot performance issues

LISTOFFIELDS ER function

FIRSTORNULL ER function

WHERE ER function

Last updated on 04/13/2026

GETLABELTEXT ER function

The `GETLABELTEXT` function searches for a specific label to return a *String* value that represents the translation of the specified label in the specified language.

Syntax

```
VB
GETLABELTEXT (label id, language)
```

Arguments

Label ID

`label id`: *String* or *Label Id*

The valid ID of one of the following label types:

- Electronic reporting (ER) label
- Microsoft Dynamics 365 Finance label

Usage notes

This argument can be defined only as a constant, by using one of the following supported patterns:

- For ER labels:
  - `@"GER_LABEL:<LABEL ID>"`
  - `"GER_LABEL:<LABEL ID>"`
- For Finance labels:
  - `@"<LABEL ID>"`
  - `"<LABEL ID>"`

> **Note**

At design time, a validation error message is shown on the Formula designer page if no label can be found by using the provided label ID.

Language

`language`: *String*

A string that represents a language code.

Usage notes

This argument can be defined either as a text constant or as the path of a data source field that returns a *String* value.

> ⓘ Note
>
> At design time, a validation error message is shown if no language code can be found by using the provided `language` argument when it has been defined as a text constant.
>
> At runtime, the translation for the `EN-US` system language is returned for a specified label if no language code has been found by using the provided `language` argument.

Return values

String

The resulting text value.

Example 1: System label

The expressions `GETLABELTEXT (@"SYS70894", "en-us")` and `GETLABELTEXT ("SYS70894", "en-us")` return the English translation "Nothing to print" for the `@SYS70894` application label.

Example 2: ER label

You start to edit an ER configuration that has been derived from the **ISO20022 Credit transfer (DE)** configuration, enter a new data source of the *Calculated field* type, and configure the expression `GETLABELTEXT(@"GER_LABEL:VendorName", "de")` for this data source. In this case, at runtime, the data source returns the German translation "Kreditorenname" for the

`@GER_LABEL:VendorName` ER label that was initially configured in the base **ISO20022 Credit transfer (DE)** ER configuration.

Additional resources

Text functions

Design multilingual reports in Electronic reporting

Last updated on 04/13/2026

GUIDVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `GUIDVALUE` function converts the specified input of the *String* type to a data item of the *GUID* type.

Syntax

```
VB

GUIDVALUE (input)
```

Arguments

`input`: *String*

The valid path of a data source of the *String* type.

Return values

GUID

The resulting globally unique identifier (GUID) value.

Usage notes

To do a conversion in the opposite direction (that is, to convert specified input of the *GUID* data type to a data item of the *String* data type), you can use the TEXT function.

Example

You define the following data sources in your model mapping:

- A **myID** data source of the *Calculated field* type that contains the expression `GUIDVALUE ("AF5CCDAC-F728-4609-8C8B- A4B30B0C0AA0")`
- A **Users** data source of the *Table records* type that refers to the UserInfo table

You can then use an expression such as `FILTER (Users, Users.objectId = myID)` to filter the UserInfo table by the **objectId** field of the *GUID* data type.

Additional resources

Text functions

Last updated on 04/08/2026

JSONVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `JSONVALUE` function parses data in JavaScript Object Notation (JSON) format that is accessed at the specified path, and it extracts a scalar value that has the specified ID. It then returns the extracted scalar value as a *String* value.

Syntax

VB

```
JSONVALUE (input, path)
```

Arguments

`input`: *String*

The valid path of a data source of the *String* type that contains JSON data.

`path`: *String*

The identifier of a scalar value of JSON data. Use a forward slash (/) to separate the names of related JSON nodes. Use the bracket ([]) notation to specify the index of a particular value in a JSON array. Note that zero-based numbering is used for this index.

Return values

String

The resulting text value.

Example 1

The **JsonField** data source contains the following data in JSON format: **{"BuildNumber":"7.3.1234.1", "KeyThumbprint":"7366E"}**. In this case, the expression `JSONVALUE (JsonField, "BuildNumber")` returns the following value of the *String* data type: **"7.3.1234.1"**.

Example 2

The **JsonField** data source of the *Calculated field* type contains the following expression: `"{""workers"": [ {""name"": ""Adam"", ""age"": 30, ""emails"": [""AdamS@Contoso.com"", ""AdamS@Hotmail.com"" ]}, { ""name"": ""John"", ""age"": 21, ""emails"": [""JohnS@Contoso.com"", ""JohnS@Aol.com""]}]}"`

This expression configured to return a *String* value that represents the following data in JSON format.

JSON

```
{
    "workers": [
        {
            "name": "Adam",
            "age": 30,
            "emails": [ "AdamS@Contoso.com", "AdamS@Hotmail.com" ]
        },
        {
            "name": "John",
            "age": 21,
            "emails": [ "JohnS@Contoso.com", "JohnS@Aol.com" ]
        }
    ]
}
```

In this case, the expression `JSONVALUE(json, "workers/[1]/emails/[0]")` returns the following value of the *String* data type: `JohnS@Contoso.com`.

Additional resources

Text functions

Last updated on 04/08/2026

LEFT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LEFT` function returns a *String* value that presents the specified number of characters from the start of the specified string.

Syntax

VB

```
LEFT (text, number)
```

Arguments

`text`: *String*

A *String* value that represents the original text.

`number`: *Integer*

The number of characters that must be returned from the start of the original text.

Return values

String

The resulting text value.

Example

`LEFT ("Sample", 3)` returns "**Sam**".

Additional resources

Text functions

Last updated on 04/08/2026

LEN ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LEN` function returns the number of characters in the specified string as an *Integer* value.

Syntax

VB

```
LEN (text)
```

Arguments

`text`: *String*

A *String* value that specifies the text.

Return values

Integer

The resulting numeric value.

Example

`LEN ("Sample")` returns **6**.

Additional resources

Text functions

Last updated on 04/08/2026

LOWER ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LOWER` function returns the specified text string as a *String* value after it has been converted to lowercase letters.

Syntax

VB

```
LOWER (text)
```

Arguments

`text`: *String*

A *String* value that specifies the text.

Return values

String

The resulting text value.

Example

`LOWER ("Sample")` returns **"sample"**.

Additional resources

Text functions

Last updated on 04/08/2026

MID ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `MID` function returns a *String* value that presents the specified number of characters from the specified string, starting at the specified position.

Syntax

```
VB

MID (text, starting position, number of characters)
```

Arguments

`text`: *String*

A *String* value that specifies the text to return characters from.

`starting position`: *Integer*

An *Integer* value that specifies the position of the first character that must be returned from the specified text.

`number of characters`: *Integer*

An *Integer* value that specifies the number of characters that must be returned, starting at the specified starting position.

Return values

String

The resulting text value.

Usage notes

If the value of the `starting position` argument is less than 0 (zero), the characters that are returned are counted from the first position in the specified string.

If the value of the `starting position` argument exceeds length of the specified string, an empty string is returned.

Example

`MID ("Sample", 2, 3)` returns "**amp**".

Additional resources

Text functions

Last updated on 04/08/2026

NEWGUID ER function

The `NEWGUID` function generates a new globally unique identifier (GUID) and returns it as a *GUID* value.

Syntax

```
VB

NEWGUID ()
```

Return values

GUID

The resulting GUID value.

Example

`NEWGUID()` always returns a new *GUID* value, such as **3afcf7ff-af10-ec11-b6e6-000d3a13209e**.

Additional resources

Text functions

Last updated on 04/13/2026

NUMBERFORMAT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NUMBERFORMAT` function returns a *String* value that presents the specified number in the specified format and in an optionally specified culture. For information about the supported formats, see standard and custom.

Syntax 1

```
VB

NUMBERFORMAT (number, format)
```

Syntax 2

```
VB

NUMBERFORMAT (number, format, culture)
```

Arguments

`number`: *Integer* or *Real*

A numeric value that specifies the number that must be formatted.

`format` : *String*

A *String* value that represents the format.

`culture`: *String*

A *String* value that represents the culture to use for formatting.

Return values

String

The resulting text value.

Usage notes

If the culture isn't defined as an argument of the called function, the context that this function is run in determines the culture that is used to format numbers.

Example 1

For the **EN-US** culture, `NUMBERFORMAT (0.45, "p")` returns "**45.00 %**", and `NUMBERFORMAT (10.45, "#")` returns "**10**".

Example 2

`NUMBERFORMAT (10/3, "F2", "de")` returns **3,33**, whereas `NUMBERFORMAT (10/3, "F2", "en-us")` returns **3.33**.

Additional resources

Text functions

Last updated on 04/08/2026

NUMERALSTOTEXT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NUMERALSTOTEXT` function returns the specified number as a *String* value after it spells out the number in the specified language.

Syntax

VB

```
NUMERALSTOTEXT (number, language, currency, print currency name flag, decimal points)
```

Arguments

`number`: *Integer* or *Real*

A numeric value that specifies the number to spell out.

`language`: *String*

A *String* value that represents the language code.

`currency`: *String*

A *String* value that represents the currency code.

`print currency name flag`: *Boolean*

A *Boolean* value that indicates whether to add a currency name to the spelled-out text.

`decimal points`: *Integer*

An *Integer* value that indicates the number of decimal places that the spelled-out text should have.

Return values

String

The resulting text value.

Usage notes

The language code is optional. If you define it as an empty string, the language code for the running context is used. The default language code is **EN-US**. The language code for the running context is defined in a **Folder** or **File** element of the Electronic reporting (ER) format that is running.

The currency code is optional. If you define it as an empty string, the company currency for the running context is used.

> **Note**
>
> The `print currency name flag` and `decimal points` arguments are analyzed only for the following language codes: **CS**, **ET**, **HU**, **LT**, **LV**, **PL**, and **RU**. Additionally, the `print currency name flag` argument is analyzed only for companies where the country's or region's context supports declension of currency names.

Example 1

`NUMERALSTOTEXT (1234.56, "EN-US", "", false, 2)` returns "**One Thousand Two Hundred Thirty Four and 56**".

Example 2

`NUMERALSTOTEXT (120, "PL", "", false, 0)` returns "**Sto dwadzieścia**".

Example 3

`NUMERALSTOTEXT (120.21, "RU", "EUR", true, 2)` returns "**Сто двадцать евро 21 евроцент**".

Additional resources

Text functions

Last updated on 04/13/2026

PADLEFT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `PADLEFT` function returns a *String* value of the specified length, where the start of the specified string is padded with the specified characters.

Syntax

VB

```
PADLEFT (text, length, padding chars)
```

Arguments

`text`: *String*

A *String* value that represents the original text.

`length`: *Integer*

An *Integer* value that represents the final number of characters in the padded string.

`padding chars`: *String*

The characters to use for padding.

Return values

String

The resulting text value.

Example

`PADLEFT ("1234", 10, " ")` returns the text string "   1234".

Additional resources

Text functions

Last updated on 04/08/2026

QRCODE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `QRCODE` function returns a *Container* value that presents the Quick Response code (QR code) image for the specified string in binary format.

Syntax

VB

```
QRCODE (text)
```

Arguments

`text`: *String*

A *String* value that represents the original text.

Return values

Container

The resulting binary stream.

Example

You can configure an Electronic reporting (ER) format to generate an outbound document in Microsoft Office format (Excel workbooks or Word documents) by using a predefined template. This template might contain a **Picture** object (Excel workbook) or a **Picture Content Control** (Word document) as a placeholder for a QR code image. Add a **Cell** element to the configured ER format that fills this placeholder. To specify what information the QR code stores, define a

binding for this **Cell** element. For example, you can configure the binding to contain the following expression:

```
VB

QRCODE (model.ListOfShelfLabels.LabelText)`
```

When you run the configured ER format, the text value of the **LabelText** field of the **model.ListOfShelfLabels** data source generates a QR code image. This image replaces a QR code image placeholder in the document template that you use to generate an outbound document. When you scan this image of the generated document, it returns the text that was taken from the **LabelText** field of the **model.ListOfShelfLabels** data source. For more information, see Embed images and shapes in documents that you generate by using ER.

Additional resources

Text functions

Last updated on 04/13/2026

REPLACE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `REPLACE` function returns the specified text string as a *String* value after all or part of it is replaced with another string.

Syntax

VB

```
REPLACE (text, pattern, replacement, regular expression flag)
```

Arguments

`text`: *String*

The valid path of a data source of the *String* type.

`pattern`: *String*

If the `regular expression flag` argument is **FALSE**, this argument contains the text that must be replaced.

If the `regular expression flag` argument is **TRUE**, this argument contains a regular expression that defines both a search pattern and the replacement text.

`replacement`: *String*

If the `regular expression flag` argument is **FALSE**, this argument contains the text to use as a replacement.

If the `regular expression flag` argument is **TRUE**, this argument isn't used.

`regular expression flag`: *Boolean*

A *Boolean* value that indicates whether a regular expression is used to do the replacement.

Return values

String

The resulting text value.

Usage notes

If the `regular expression flag` argument is **TRUE**, this function returns the specified string after it has been changed by applying the regular expression that is specified by the `pattern` argument. The regular expression is used to find the characters that must be replaced.

If the `regular expression flag` argument is **FALSE**, this function returns the specified string after the set of characters that are defined in the `pattern` argument have been replaced by characters of the `replacement` argument.

Example 1

`REPLACE ("+1 923 456 4971", "[^0-9]", "", true)` applies a regular expression that removes all non-numeric symbols, and it returns "**19234564971**".

Example 2

`REPLACE ("abcdef", "cd", "GH", false)` replaces the pattern "**cd**" with the string "**GH**" and returns "**abGHef**".

Additional resources

Text functions

Last updated on 04/13/2026

RIGHT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `RIGHT` function returns a *String* value that presents the specified number of characters from the end of the specified string.

Syntax

VB

```
RIGHT (text, number)
```

Arguments

`text`: *String*

A *String* value that represents the original text.

`number`: *Integer*

The number of characters that you want to return from the end of the original text.

Return values

String

The resulting text value.

Example

`RIGHT ("Sample", 3)` returns "**ple**".

Additional resources

Text functions

Last updated on 04/13/2026

TEXT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `TEXT` function returns the specified number as a *String* value after it converts the number to a text string that's formatted according to the server locale settings of the current application instance.

Syntax

```
VB

TEXT (number)
```

Arguments

`number`: *Integer* or *Real*

The number to convert to a text string.

Return values

String

The resulting text value.

Usage notes

For *Real* type values, the string conversion includes up to two decimal places.

Example

If the server locale of the Microsoft Dynamics 365 Finance instance is defined as **EN-US**, `TEXT (NOW ())` returns the current Finance session date, December 17, 2015, as the text string "**12/17/2015 07:59:23 AM**". `TEXT (1/3)` returns "**0.33**".

Additional resources

Text functions

Last updated on 04/13/2026

TRANSLATE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `TRANSLATE` function returns a *String* value that contains the result of the character replacement of specified text in characters of another provided set.

Syntax

```
VB

TRANSLATE (text , pattern, replacement)
```

Arguments

`text`: *String*

The valid path of a data source of the *String* type.

`pattern`: *String*

The text that you want to replace.

`replacement`: *String*

The text to use as a replacement.

Return values

String

The resulting text value.

Usage notes

The `TRANSLATE` function replaces one character at a time. The function replaces the first character of the `text` argument with the first character of the `pattern` argument and then the second character and follows the same flow until finished. When a character from the `text` and `pattern` arguments match, the function replaces it with a character from the `replacement` argument that is in the same position as the character from the `pattern` argument. If a character appears multiple times in the `pattern` argument, the function uses the `replacement` argument mapping that corresponds to the first occurrence of this character.

Example 1

`TRANSLATE ("abcdef", "cd", "GH")` replaces the "**c**" character of the specified "**abcdef**" text with the "**G**" character of the `replacement` text due to the following reasons:

- The "**c**" character is presented in the `pattern` text in the first position.
- The first position of the `replacement` text contains the "**G**" character.

Example 2

`TRANSLATE ("abcdef", "ccd", "GH")` returns "**abGdef**".

Example 3

`TRANSLATE ("abccba", "abc", "123")` returns "**123321**".

Additional resources

Text functions

Last updated on 04/13/2026

TRIM ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `TRIM` function returns the specified text string as a *String* value after it replaces tab, carriage return, line feed, and form feed characters with a single space character. It also removes leading and trailing spaces and removes multiple spaces between words.

Syntax

```
VB

TRIM (text)
```

Arguments

`text`: *String*

The valid path of a data source of the *String* type.

Return values

String

The resulting text value.

Usage notes

In some cases, you might want to remove leading and trailing spaces but prefer to keep formatting for the specified text. For example, when this text represents an address that you enter in the multi-line text box and could contain line feed and carriage return formatting. In

this case, use the following expression: `REPLACE(text,"^[ \t]+|[ \t]+$","", true)` where `text` is the argument that refers to the specified text string.

Example 1

`TRIM ("  Sample  text  ")` returns "**Sample text**".

Example 2

`TRIM (CONCATENATE (CHAR(10), "  Sample  ", CHAR(9),"  text  ", CHAR(13)))` returns "**Sample text**".

Additional resources

Text functions

REPLACE ER function

Last updated on 04/13/2026

UPPER ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `UPPER` function returns the specified text string as a *String* value after it converts the text to uppercase letters.

Syntax

VB

```
UPPER (text )
```

Arguments

`text`: *String*

The valid path of a data source of the *String* type.

Return values

String

The resulting text value.

Example

`UPPER ("Sample")` returns **"SAMPLE"**.

Additional resources

Text functions

Last updated on 04/13/2026

List of ER functions in the data collection category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) data collection functions are used to do counting and summing in an ER format that is being run, based on data of the output that has already been generated in **Text** or **Xml** format. This approach is used to help improve performance of an ER format that is run, to enter values of running totals in generated documents, and for other purposes. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
CollectedList	This function returns a *Record list* value that contains the list of values that were returned by the **Collected data key value** property of format elements and collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified conditions. Each condition consists of a key range and a key value.
CountIF	This function returns an *Integer* value that represents the number of format elements that was collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified condition. The condition consists of a key range and a key value.
CountIFs	This function returns an *Integer* value that represents the number of format elements that was collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified conditions. Each condition consists of a key range and a key value.
FormatElementName	This function returns a *String* value that represents the name of the current ER format's element.

Function	Description
SumIF	This function returns a *Real* value that represents the sum of values that were returned by bindings of format elements and collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified condition. The condition consists of a key range and a key value.
SumIFs	This function returns a *Real* value that represents the sum of values that were returned by bindings of format elements and collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified conditions. Each condition consists of a key range and a key value.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

COLLECTEDLIST ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `COLLECTEDLIST` function a *Record list* value that contains the list of values that were returned by the **Collected data key value** property of format elements and collected when the format elements were used to generate outbound documents during the format run, and that satisfies the specified conditions. Each condition consists of a key range and a key value.

Syntax

VB

```
COLLECTEDLIST (condition 1 range, condition 1 value[, condition 2 range, condition 2 value, …, condition N range, condition N value])
```

Arguments

`condition 1 range`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of an Electronic reporting (ER) format component. This argument is mandatory.

`condition 1 value`: *String*

A value that is returned by the expression that has been configured in the **Collected data key value** property of an ER format component. This argument is mandatory.

`condition N range`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of an ER format component. These additional arguments are optional.

`condition N value`: *String*

A value that is returned by the expression that has been configured in the **Collected data key value** property of an ER format component. These additional arguments are optional.

Return values

Record list

The resulting list of records.

Usage notes

The **Collected data key name** and **Collected data key value** properties can be configured for either the **Sequence** component or the **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

This function returns an empty list when the **Collect output details** option of the current **Common\File** component is turned off.

In `condition range` arguments, the wildcard character "*" can be used to represent any multiple characters.

In `condition value` arguments, the wildcard character "*" can be used to represent any multiple characters.

Example

For more information about how to use this function, see the ER Use data of format output for counting and summing task guide, which is part of the **Acquire/Develop IT service/solution components** business process.

Additional resources

Data collection functions

Last updated on 04/08/2026

COUNTIF ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `COUNTIF` function returns an *Integer* value that represents the number of format elements that was collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified condition. The condition consists of a key range and a key value.

Syntax

```
VB

COUNTIF (condition range, condition value)
```

Arguments

`condition range`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of an Electronic reporting (ER) format component.

`condition value`: *String*

A value that is returned by the expression that has been configured in the **Collected data key value** property of an ER format component.

Return values

Integer

The resulting numeric value.

Usage notes

The **Collected data key name** and **Collected data key value** properties can be configured for either the **Sequence** component or the **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

This function returns a **0** (zero) value when the **Collect output details** option of the current **Common\File** component is turned off.

In the `condition range` argument, the wildcard character "*" can be used to represent any multiple characters.

In the `condition value` argument, the wildcard character "*" can be used to represent any multiple characters.

Example

For more information about how to use this function, see the ER Use data of format output for counting and summing task guide, which is part of the **Acquire/Develop IT service/solution components** business process.

Additional resources

Data collection functions

Last updated on 04/08/2026

COUNTIFS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `COUNTIFS` function returns an *Integer* value that represents the number of format elements that was collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified conditions. Each condition consists of a key range and a key value.

Syntax

VB

```
COUNTIFS (condition 1 range, condition 1 value[, condition 2 range, condition 2 value, …, condition N range, condition N value])
```

Arguments

`condition 1 range`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of an Electronic reporting (ER) format component. This argument is mandatory.

`condition 1 value`: *String*

A value that is returned by the expression that has been configured in the **Collected data key value** property of an ER format component. This argument is mandatory.

`condition N range`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of an ER format component. These additional arguments are optional.

`condition N value`: *String*

A value that is returned by the expression that has been configured in the **Collected data key value** property of an ER format component. These additional arguments are optional.

Return values

Integer

The resulting numeric value.

Usage notes

The **Collected data key name** and **Collected data key value** properties can be configured for either the **Sequence** component or the **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

This function returns a **0** (zero) value when the **Collect output details** option of the current **Common\File** component is turned off.

In `condition range` arguments, the wildcard character "*" can be used to represent any multiple characters.

In `condition value` arguments, the wildcard character "*" can be used to represent any multiple characters.

Example

For more information about how to use this function, see the ER Use data of format output for counting and summing task guide, which is part of the **Acquire/Develop IT service/solution components** business process.

Additional resources

Data collection functions

Last updated on 04/08/2026

FORMATELEMENTNAME ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `FORMATELEMENTNAME` function returns a *String* value that represents the name of the current Electronic reporting (ER) format's element.

Syntax

VB

```
FORMATELEMENTNAME ()
```

Return values

String

The resulting text value.

Usage notes

This function can be called in ER expressions that were configured for the **Collected data key name** and **Collected data key value** properties of an ER format component from the **Text** group that resides under the **Common\File** component where the **Collect output details** option is turned on.

Example

For more information about how to use this function, see the ER Use data of format output for counting and summing task guide, which is part of the **Acquire/Develop IT service/solution components** business process.

Additional resources

Data collection functions

Last updated on 04/08/2026

SUMIF ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `SUMIF` function returns a *Real* value that represents the sum of values that were returned by bindings of format elements and collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified condition. The condition consists of a key range and a key value.

Syntax

VB

```
SUMIF (key name for summing, condition range, condition value)
```

Arguments

`key name for summing`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of the Electronic reporting (ER) format component for which the value of the binding must be used for summing purposes.

The **Collected data key value** property can be configured for either a **Sequence** component or an **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

Return values

Real

The resulting numeric value.

Usage notes

This function returns a **0** (zero) value when the **Collect output details** option of the current **Common\File** component is turned off.

In the `condition range` argument, the wildcard character "*" can be used to represent any multiple characters.

In the `condition value` argument, the wildcard character "*" can be used to represent any multiple characters.

Example

For more information about how to use this function, see the ER Use data of format output for counting and summing task guide, which is part of the **Acquire/Develop IT service/solution components** business process.

For more information and examples about using this function, see Defer the execution of sequence elements in ER formats and Defer the execution of XML elements in ER formats.

Additional resources

Data collection functions

Last updated on 04/08/2026

SUMIFS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `SUMIFS` function returns a *Real* value that represents the sum of values that were returned by bindings of format elements and collected when the format elements were used to generate an outbound document during the format run, and that satisfies the specified conditions. Each condition consists of a key range and a key value.

Syntax

VB

```
SUMIFS (key name for summing, condition 1 range, condition 1 value[, condition 2 range, condition 2 value, …, condition N range, condition N value])
```

Arguments

`key name for summing`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of the Electronic reporting (ER) format component for which the value of the binding must be used for summing purposes.

The **Collected data key name** property can be configured for either a **Numeric** component or a **String** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

`condition 1 range`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of an ER format component. This argument is mandatory.

The **Collected data key name** property can be configured for either a **Sequence** component or an **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

`condition 1 value`: *String*

A value that is returned by the expression that has been configured in the **Collected data key value** property of an ER format component. This argument is mandatory.

The **Collected data key value** property can be configured for either a **Sequence** component or an **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

`condition N range`: *String*

A value that is returned by the expression that has been configured in the **Collected data key name** property of an ER format component. These additional arguments are optional.

The **Collected data key name** property can be configured for either a **Sequence** component or an **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

`condition N value`: *String*

A value that is returned by the expression that has been configured in the **Collected data key value** property of an ER format component. These additional arguments are optional.

The **Collected data key value** property can be configured for either a **Sequence** component or an **XML Element** component of an ER format that resides under the **Common\File** component where the **Collect output details** option is turned on.

Return values

Real

The resulting numeric value.

Usage notes

This function returns a **0** (zero) value when the **Collect output details** option of the current **Common\File** component is turned off.

In the `condition range` arguments, the wildcard character "*" can be used to represent any multiple characters.

In the `condition value` arguments, the wildcard character "*" can be used to represent any multiple characters.

Example

For more information about how to use this function, see the ER Use data of format output for counting and summing task guide, which is part of the **Acquire/Develop IT service/solution components** business process.

Additional resources

Data collection functions

Last updated on 04/08/2026

List of ER functions in the business domain–specific category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) domain-specific functions can be used to perform calculations and data access requests that are specific to the implementation of Microsoft Dynamics 365 Finance. This article provides a summary of these functions.

List of supported functions

Expand table

Function	Description
CH_Bank_Mod_10	This function returns a *String* value that represents a creditor reference as an MOD10 expression, based on the digits of the specified invoice number.
CN_GBT_AdditionalDimensionID	This function returns a *String* value that represents a single financial dimension ID that is taken from the specified string. The specified string presents all dimensions as a comma-separated list of IDs.
ConvertCurrency	This function returns a *Real* value that represents the result of converting the specified monetary amount from the specified source currency to the specified target currency by using the settings of the specified company on the specified date.
CurCredRef	This function returns a *String* value that represents a creditor reference, based on the digits of the specified invoice number.
FA_Balance	This function returns a *Container (record)* value that consists of data for the fixed asset balance for the specified fixed asset item, value model code, reporting year, and reporting date.
FA_Sum	This function returns a *Container (record)* value that consists of data for the fixed asset amounts for the specified fixed asset item, value model

Function	Description
	code, and period of dates.
GetCurrentCompany	This function returns a *String* value that represents the code for the legal entity (company) that a user is currently signed in to.
ISOCredRef	This function returns a *String* value that represents an International Organization for Standardization (ISO) creditor reference, based on the digits and alphabetic symbols of the specified invoice number.
IsValidCharacterISO7064	This function returns a *Boolean* value of **TRUE** if the specified string represents a valid international bank account number (IBAN). Otherwise, it returns a *Boolean* value of **FALSE**.
Mod_97	This function returns a *String* value that represents a creditor reference as a MOD97 expression, based on the digits of the specified invoice number.
NumSeqValue	This function returns a *String* value that represents the new generated value of a number sequence, based on the specified number sequence, scope, and scope ID. The scope ID equals the company code that is supplied by the context that the ER format is run under.
RoundAmount	This function returns a *Real* value that represents the result of rounding the specified amount to the specified number of decimal places according to the specified rounding rule.
TableName2ID	This function returns a numeric representation of the table ID for the specified table name as an *Integer* value.
TaxUnitConversion	This function returns a *Real* value that represents the quantity result of converting the specified unit from a transaction to the specified unit in a tax configuration. This function requires context that's from an original sourcing system. The method `ConverUnit` in interface `IExternalAPIProvider` must be implemented in a sourcing ERP environment.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

CH_BANK_MOD_10 ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CH_BANK_MOD_10` function returns a *String* value that represents a creditor reference as an MOD10 expression, based on the digits of the specified invoice number.

Syntax

VB

```
CH_BANK_MOD_10 (invoice number digits)
```

Arguments

`invoice number digits`: *String*

A text value that represents the digits of an invoice number.

Return values

String

The resulting text value.

Example

`CH_BANK_MOD_10 ("VEND-200002")` returns **3**.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

CN_GBT_ADDITIONALDIMENSIONID ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CN_GBT_ADDITIONALDIMENSIONID` function returns a *String* value that represents a single financial dimension ID that is taken from the specified string. The specified string presents all dimensions as a comma-separated list of IDs.

Syntax

VB

```
CN_GBT_ADDITIONALDIMENSIONID (text, number)
```

Arguments

`text`: *String*

A *String* value that presents all dimensions as a comma-separated list of IDs.

`number`: Integer

An *Integer* value that defines the sequence code of the requested dimension in the specified string.

Return values

String

The resulting text value.

Example

`CN_GBT_AdditionalDimensionID ("AA,BB,CC,DD,EE,FF,GG,HH", 3)` returns "**CC**".

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

CONVERTCURRENCY ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CONVERTCURRENCY` function returns a *Real* value that represents the result of converting the specified monetary amount from the specified source currency to the specified target currency by using the settings of the specified company on the specified date.

Syntax

VB

```
CONVERTCURRENCY (amount, source currency, target currency, date, company)
```

Arguments

`amount`: *Integer* or *Real*

A numeric value that represents the monetary amount that must be converted.

`source currency`: *String*

The code of the source currency.

`target currency`: *String*

The code of the target currency.

`date`: *Date*

A *Date* value that represents the date that is used to determine the exchange rate for the conversion.

`company`: *String*

A *String* value that represents the code of a company that supplies the settings that are used for the conversion.

Return values

Real

The resulting numeric value.

Example

`CONVERTCURRENCY (1, "EUR", "USD", TODAY(), "DEMF")` returns the equivalent of one euro in US dollars on the current session date, based on settings for the **DEMF** company.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

CURCREDREF ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CURCREDREF` function returns a *String* value that represents a creditor reference, based on the digits of the specified invoice number.

Syntax

```
VB

CURCREDREF (invoice number digits)
```

Arguments

`invoice number digits`: *String*

A text value that represents the digits of an invoice number.

Return values

String

The resulting text value.

Example

`CURCredRef ("VEND-200002")` returns "**2200002**".

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

FA_BALANCE ER function

The `FA_BALANCE` function returns a *Container (record)* value that consists of data for the fixed asset balance for the specified fixed asset item, value model code, reporting year, and reporting date.

Syntax

VB

```
FA_BALANCE (fixed asset code, value model code, reporting year, reporting date)
```

Arguments

`fixed asset code`: *String*

A *String* value that represents the code of a fixed asset item that the balance is calculated for.

`value model code`: *String*

A *String* value that represents the code of a value model that the balance is calculated for.

`reporting year`: *Enumeration value*

An enumeration value of the **AssetYear** application enumeration that defines a period for the balance calculation.

`reporting date`: *Date*

A *Date* value that defines a date for the balance calculation.

Return values

Container (record)

The resulting record value.

Example

`FA_ BALANCE ("COMP-000001", "Current", AxEnumAssetYear.ThisYear, SESSIONTODAY ())` returns the data container of balances for fixed asset **COMP-000001** that has been prepared for the

Current value model on the current application session date.

Additional resources

Other (business domain–specific) functions

Last updated on 04/13/2026

FA_SUM ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `FA_SUM` function returns a *Container (record)* value that consists of data for the fixed asset amounts for the specified fixed asset item, value model code, and period of dates.

Syntax

VB

```
FA_SUM (fixed asset code, value model code, start date, end date)
```

Arguments

`fixed asset code`: *String*

A *String* value that represents the code of a fixed asset item that the balance is calculated for.

`value model code`: *String*

A *String* value that represents the code of a value model that the balance is calculated for.

`start date`: *Date*

A *Date* value that represents the start date of a period that the fixed asset amounts are calculated for.

`end date`: *Date*

A *Date* value that represents the end date of a period that the fixed asset amounts are calculated for.

Return values

Container (record)

The resulting record value.

Example

`FA_SUM ("COMP-000001", "Current", Date1, Date2)` returns the data container for fixed asset **COMP-000001** that has been prepared for the **Current** value model and for a period from **Date1** to **Date2**.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

GETCURRENTCOMPANY ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `GETCURRENTCOMPANY` function returns a *String* value that represents the code for the legal entity (company) that a user is currently signed in to.

Syntax

VB

```
GETCURRENTCOMPANY ()
```

Return values

String

The resulting text value.

Example

`GETCURRENTCOMPANY ()` returns **USMF** for a user who is signed in to the **Contoso Entertainment System USA** company.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

ISOCREDREF ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ISOCREDREF` function returns a *String* value that represents an International Organization for Standardization (ISO) creditor reference, based on the digits and alphabetic symbols of the specified invoice number.

Syntax

```
VB

ISOCREDREF (invoice number digits)
```

Arguments

`invoice number digits`: *String*

A text value that represents the digits of an invoice number.

Return values

String

The resulting text value.

Usage notes

> **Note**
>
> To eliminate symbols from alphabets that are't ISO-compliant, the `invoice number digits` argument must be translated before it's passed to this function.

Example

`ISOCredRef ("VEND-200002")` returns "**RF23VEND-200002**".

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

ISVALIDCHARACTERISO7064 ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ISVALIDCHARACTERISO7064` function returns a *Boolean* value of **TRUE** if the specified string represents a valid international bank account number (IBAN). Otherwise, it returns a *Boolean* value of **FALSE**.

Syntax

VB

```
ISVALIDCHARACTERISO7064 (text)
```

Arguments

`text`: *String*

A text value that represents an IBAN.

Return values

String

The resulting text value.

Example

`ISVALIDCHARACTERISO7064 ("AT61 1904 3002 3457 3201")` returns **TRUE**.

`ISVALIDCHARACTERISO7064 ("AT61")` returns **FALSE**.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

MOD_97 ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `MOD_97` function returns a *String* value that represents a creditor reference as a MOD97 expression, based on the digits of the specified invoice number.

Syntax

VB

```
MOD_97 (invoice number digits)
```

Arguments

`invoice number digits`: *String*

A text value that represents the digits of an invoice number.

Return values

String

The resulting text value.

Example

`MOD_97 ("VEND-200002")` returns **"20000285"**.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

NUMSEQVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NUMSEQVALUE` function returns a *String* value that represents the new generated value of a number sequence, based on the specified number sequence, scope, and scope ID. The scope ID equals the company code that is supplied by the context that the Electronic reporting (ER) format is run under.

Syntax 1

```
VB

NUMSEQVALUE (number sequence code)
```

Syntax 2

```
VB

NUMSEQVALUE (number sequence record ID)
```

Syntax 3

```
VB

NUMSEQVALUE (number sequence code, scope type, scope ID)
```

Arguments

`number sequence code`: *String*

A text value that represents the code of the number sequence that a new value is required in.

`number sequence record ID`: *Int64*

An *Int64* value that represents the record ID of a record in the NumberSequenceTable table that contains the definition of the number sequence that a new value is required in.

`scope type`: *Enum value*

An enumeration value of the **ERExpressionNumberSequenceScopeType** enumeration that defines the scope of the number sequence that a new value is required in. The available scope types are **Shared**, **Legal entity**, and **Company**.

`scope ID`: *String*

A *String* value that identifies the scope, based on the specified scope type.

Return values

String

The resulting text value.

Usage notes

For the **Shared** scope type, specify an empty string as the scope ID.

For the **Company** and **Legal entity** scope types, specify the company code as the scope ID. If you specify an empty string as the scope ID for these scope types, the current company code is used.

When syntax 1 is used, the number sequence is requested for the **Company** scope type, and the company code is supplied by the context that the ER format is run under.

Example 1

In your ER format, you define the **AskNumSeq** data source of the *User input parameter* type. This data source refers to the **Description** extended data type (EDT). Next, you define the **NumSeq** data source of the *Calculated field* type. This data source contains the expression `NUMSEQVALUE (AskNumSeq)`. When the **NumSeq** data source is called, it returns the new generated value of the number sequence that was specified at runtime by entering its code in the dialog box. The number sequence is requested for the **Company** scope type. The company code is supplied by the context that the ER format is run under.

Example 2

The following data sources are defined in your model mapping:

- The **LedgerParms** data source of the *Table* type. This data source refers to the LedgerParameters table.
- The **NumSeq** data source of the *Calculated field* type. This data source contains the expression `NUMSEQVALUE ( LedgerParameters.'numRefJournalNum()'.NumberSequenceId)`.

When the **NumSeq** data source is called, it returns the new generated value of the number sequence that has been configured in the General ledger parameters for the company that supplies the context that the ER format is run under. This number sequence uniquely identifies journals and acts as a batch number that links the transactions together.

Example 3

The following data sources are defined in your model mapping:

- The **enumScope** data source of the Microsoft Dynamics 365 Finance *enumeration* type. This data source refers to the **ERExpressionNumberSequenceScopeType** enumeration.
- The **NumSeq** data source of the *Calculated field* type. This data source contains the expression `NUMSEQVALUE ("Gene_1", enumScope.Company, "")`.

When the **NumSeq** data source is called, it returns the new generated value of the **Gene_1** number sequence that has been configured for the company that supplies the context that the ER format is run under.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

ROUNDAMOUNT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `ROUNDAMOUNT` function returns a *Real* value as the result of the rounding of the specified number to the nearest multiple of another number according to the specified rounding rule.

Syntax

VB

```
ROUNDAMOUNT (number, decimals, round rule)
```

Arguments

`number`: *Int* or *Real*

A numeric value that must be rounded.

`decimals`: *Int* or *Real*

The number that the value of the `number` parameter must be rounded to a multiple of.

`round rule`: *Enum value*

An enumeration value of the **RoundOffType** enumeration that defines the rounding rule. This enumeration offers the following values:

- Normal (Ordinary)
- Downward (RoundDown)
- Rounding-up (RoundUp)

Return values

Real

The resulting numeric value is a multiple of the value specified by the `decimals` parameter and is closest to the value specified by the `number` parameter.

Usage notes

When the `number` parameter is zero, this function always returns zero.

When the `decimals` parameter is zero, this function rounds to the default round-off value. When the `round rule` parameter is set to **RoundOffType.Ordinary**, the default round-off value is **0.01**. Otherwise, the default round-off value is **1.0**.

When the `round rule` parameter is set to **RoundOffType.Ordinary**, this function rounds to the nearest round-off amount.

When the `round rule` parameter is set to **RoundOffType.RoundDown**, this function rounds towards zero to the nearest round-off amount.

When the `round rule` parameter is set to **RoundOffType.RoundUp**, this function rounds away from zero to the nearest round-off amount.

When the `round rule` parameter is set to **RoundOffType.Ordinary**, this function behaves like the MROUND Excel function and the ROUND X++ function.

Remarks

To round a numeric value to a specified number of decimal places, use the ROUND function.

Example

If the **model.RoundOff** parameter is set to **RoundOffType.Ordinary**, `ROUNDAMOUNT (7.45, 1.05, model.RoundOff)` returns 7.35.

If the **model.RoundOff** parameter is set to **RoundOffType.RoundDown**, `ROUNDAMOUNT (7.45, 1.05, model.RoundOff)` returns 7.35.

If the **model.RoundOff** parameter is set to **RoundOffType.RoundUp**, `ROUNDAMOUNT (7.45, 1.05, model.RoundOff)` returns 8.4.

Additional resources

Other (business domain–specific) functions

Mathematical functions

Last updated on 04/08/2026

TABLENAME2ID ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `TABLENAME2ID` function returns a numeric representation of the table ID for the specified table name as an *Integer* value.

Syntax

```
VB

TABLENAME2ID (text)
```

Arguments

`text`: *String*

A text value that represents a valid table name.

Return values

Integer

The resulting numeric value.

Usage notes

Execution of this function can have different results in different instances of Microsoft Dynamics 365 Finance, even if the same company name is used.

Example

`TABLENAME2ID ("Intrastat")` returns **1510**.

Additional resources

Other (business domain–specific) functions

Last updated on 04/08/2026

TAXUNITCONVERSION ER function

The `TAXUNITCONVERSION` function returns a *Real* value that represents the quantity result of converting the specified unit from transaction to the specified unit in tax configuration. And it should be noticed that this function requires context that from an original sourcing system. The method `ConverUnit` in interface `IExternalAPIProvider` must be implemented in a sourcing Enterprise resource planning (ERP) environment.

Syntax

VB

```
TAXUNITCONVERSION(itemId, quantity, base unit, tax unit, invent dimension ID,
packaging)
```

Arguments

`itemId`: *String*

The Item ID defined in Finance and Operations Apps.

`quantity`: *Real*

The quantity of the original unit.

`base unit`: *String*

The unit of the original transaction.

`tax unit`: *String*

The unit is defined in the tax configuration.

`invent dimension Id`: *String*

The **invent dimension** is defined in the finance and operations apps transaction.

`packaging`: *Bool*

The packaging parameter is from the tax configuration.

Return values

Real

The converted quantity of the tax unit.

Example

`TAXUNITCONVERSION ("D001", 12, "ea", "dz", "", false)` returns the quantity converted from "**ea**" to "**dz**" which is **1**.

Additional resources

Other (business domain–specific) functions

Last updated on 04/13/2026

List of ER functions in the type conversion category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Electronic reporting (ER) type conversion functions can be used to convert values between types. This article provides a summary of these functions.

Type conversion functions

Expand table

Function	Description
Int64Value	This function returns an *Int64* value that represents the specified string.
IntValue	This function returns an *Int* value that represents the specified string.
NumberValue	This function returns a *Real* value that is converted from the specified *String* value. During the conversion, the specified decimal and digit grouping separators are considered.
Value	This function returns a *Real* value that is converted from the specified *String* value.

Type conversion functions in the container category

The following table describes the type conversion functions in the container category.

Expand table

Function	Description
Base64StringToContainer	This function converts the specified input of the *String* type to a data item of the *Container* type.

Type conversion functions in the date and time category

The following table describes the type conversion functions in the date and time category.

Expand table

Function	Description
DateTimeValue	This function returns a *DateTime* value that is converted from a given *String* value in the specified format and in an optionally specified culture to a date/time value.
DateToDateTime	This function returns a *DateTime* value that is converted from a given *Date* value to a date/time value in Coordinated Universal Time (Greenwich Mean Time [GMT]).
DateValue	This function returns a *Date* value that is converted from a given *String* value in the specified format and in an optionally specified culture to a date value.

Type conversion functions in the list category

The following table describes the type conversion functions in the list category.

Expand table

Function	Description
List	This function returns a *Record list* value as a new list that is created from specified arguments of the *Container (record)* type.
ListOfFields	This function returns a *Record list* value that is created based on the structure of a given argument of the *Enumeration* or *Container (record)* type.
Split	This function splits the specified *String* value into substrings and returns the result as a new *Record list* value.
StringJoin	This function returns a *String* value that consists of concatenated values of the specified field from the specified *Record list* value. The values can be separated by the specified delimiter.

Type conversion functions in the text category

The following table describes the type conversion functions in the text category.

Expand table

Function	Description
Char	This function returns a *String* value that represents a single character that is referenced by the specified Unicode number.
GuidValue	This function converts the specified input of the *String* type to a data item of the *GUID* type.
NumberFormat	This function returns a *String* value that represents the specified number in the specified format and in an optionally specified culture.
QrCode	This function returns a *Container* value that presents the Quick Response code (QR code) image for the specified string in binary format.
Text	This function returns a *String* value that represents the specified number after it has been converted to a text string that is formatted according to the server locale settings of the current application instance.

Additional resources

Electronic Reporting overview

Formula designer in Electronic reporting

Electronic reporting formula language

Last updated on 04/08/2026

INT64VALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `INT64VALUE` function returns an *Int64* value that represents the specified string.

Syntax 1

```
VB

INT64VALUE (text)
```

Syntax 2

```
VB

INT64VALUE (number)
```

Arguments

`text`: *String*

A text value that must be converted to an *Int64* number.

`number`: *Real* or *Integer*

A numeric *Real* or *Integer* value that must be converted to an *Int64* number.

Return values

Int64

The resulting numeric value.

Usage notes

Any decimal places are truncated.

Example 1

`INT64VALUE ("22565422744")` returns the *Int64* value **22565422744**.

Example 2

`INT64VALUE ( VALUE("22565422744.77"))` returns the *Int64* value **22565422744**.

Additional resources

Type conversion functions

Last updated on 04/08/2026

INTVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `INTVALUE` function returns an *Int* value that represents the specified string.

Syntax 1

```
VB

INTVALUE (text)
```

Syntax 2

```
VB

INTVALUE (number)
```

Arguments

`text`: *String*

A text value that must be converted to an *Int* number.

`number`: *Real* or *Integer*

A numeric *Real* or *Integer* value that must be converted to an *Int* number.

Return values

Int

The resulting numeric value.

Usage notes

Any decimal places are truncated.

Example 1

`INTVALUE ("100.77")` returns the *Int* value **100**.

Example 2

`INTVALUE (-100.77)` returns the *Int* value **-100**.

Additional resources

Type conversion functions

Last updated on 04/08/2026

NUMBERVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NUMBERVALUE` function returns a *Real* value that is converted from the specified *String* value. During the conversion, the specified decimal and digit grouping separators are considered.

Syntax

VB

```
NUMBERVALUE (text, decimal separator, digit grouping separator)
```

Arguments

`text`: *String*

A text value that must be converted to a *Real* number.

`decimal separator`: String

A decimal separator. It's used to separate the integer and fractional parts of a decimal number.

`digit grouping separator`: *String*

A digit grouping separator. It's used as the thousands separator.

Return values

Real

The resulting numeric value.

Example

`NUMBERVALUE( "1 234,56", ",", " ")` returns **1234.56**.

Additional resources

Type conversion functions

Last updated on 04/08/2026

VALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `VALUE` function returns a *Real* value that is converted from the specified string.

Syntax

VB

```
VALUE (text)
```

Arguments

`text`: *String*

A string value that must be converted to a numeric value.

Return values

Real

The resulting numeric value.

Usage notes

Commas and dot characters (.) are considered decimal separators, and a leading hyphen (-) is used as a negative sign. An exception is thrown at runtime if the specified string contains other non-numeric characters.

Example 1

`VALUE ("1 234,56")` throws an exception.

Example 2

`VALUE ("1234,56")` returns **1234.56**.

Additional resources

Type conversion functions

Last updated on 04/08/2026

Base64StringToContainer ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `BASE64STRINGTOCONTAINER` function converts the specified input of the *String* type to a data item of the *Container* type.

Syntax

VB

```
BASE64STRINGTOCONTAINER (input)
```

Arguments

`input`: *String*

The valid path of a data source of the *String* type.

Return values

Container

The resulting binary value in binary large object (BLOB) format.

Usage notes

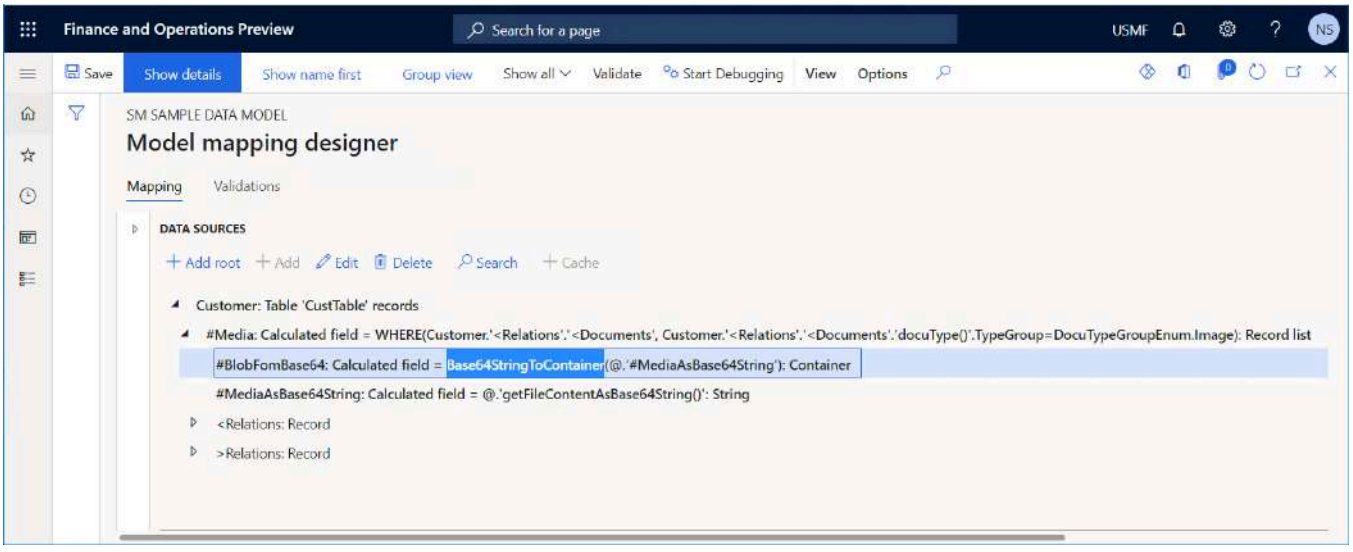
The "Parameter is not valid" exception is thrown if the input string doesn't provide a correct Base64 group of binary-to-text encoding schemes.

Example

You define the following data sources in your model mapping:

- The root **DocuTypeGroupEnum** data source of the *Dynamics 365 for Operations / Enumeration* type that refers to the **DocuTypeGroup** application enumeration
- The root **Customer** data source of the *Dynamics 365 for Operations / Table records* type that refers to the **CustTable** application table
- The **#Media** data source of the *Calculated field* type that is configured in the following way:
  - It's added as a child data source of the **Customer** data source.
  - It contains the expression `WHERE(@.'<Relations'.'<Documents', @.'<Relations'.'<Documents'.'docuType()'.TypeGroup = DocuTypeGroupEnum.Image)`.
- The **#MediaAsBase64String** data source of the *Calculated field* type that is configured in the following way:
  - It's added as a child data source of the **Customer.'#Media'** data source.
  - It contains the expression `Customer.'#Media'.'getFileContentAsBase64String()'`.
- The **#BlobFomBase64** data source of the *Calculated field* type that is configured in the following way:
  - It's added as a child data source of the **Customer.'#Media'** data source.
  - It contains the expression `Base64StringToContainer(Customer.'#Media'.'#MediaAsBase64String')'`.

In this example, the **#MediaAsBase64String** data source encodes the binary content of the current media attachment as text that represents a Base64 group of binary-to-text encoding schemes. The **#BlobFomBase64** data source decodes the Base64 string and returns a binary value in BLOB format.



Additional resources

Container functions

Last updated on 04/08/2026

DATETIMEVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `DATETIMEVALUE` function returns a *DateTime* value that is converted from a given text value in the specified format and in an optionally specified culture to a date/time value. For information about the supported formats, see standard and custom.

Syntax 1

```
VB

DATETIMEVALUE (text, format)
```

Syntax 2

```
VB

DATETIMEVALUE (text, format, culture)
```

Arguments

`text`: *String*

Text that represents the value to format.

`format`: *String*

The format of the given text. For information about the supported formats, see standard and custom.

`culture`: *String*

The culture that is used for formatting of the given text. For information about the supported cultures, see culture.

Return values

DateTime

The resulting date/time value.

Usage notes

When the culture isn't defined as an argument of the called function, the value of `culture` is defined by the calling context. For example, if the `DATETIMEVALUE` function is called by using syntax 1 in an Electronic reporting (ER) format for a **FILE** element that is configured to use the German culture, the conversion will be done by using the German culture. The default `culture` value is **EN-US**.

Example 1

`DATETIMEVALUE ("21-Dec-2016 02:55:00", "dd-MMM-yyyy hh:mm:ss")` returns **2:55:00 AM on December 21, 2016**, based on the specified custom format and the default application's **EN-US** culture.

Example 2

`DATETIMEVALUE ("21-Gen-2016 02:55:00", "dd-MMM-yyyy hh:mm:ss", "IT")` returns **2:55:00 AM on December 21, 2016**, based on the specified custom format and culture.

However, `DATETIMEVALUE ("21-Gen-2016 02:55:00", "dd-MMM-yyyy hh:mm:ss", "EN-US")` throws an exception to inform the user that the specified string isn't recognized as a valid date/time value for the specified culture.

Additional resources

Date and time functions

Last updated on 04/08/2026

DATETODATETIME ER function

The `DATETODATETIME` function returns a *DateTime* value that is converted from a given date value to a date/time value in Coordinated Universal Time (Greenwich Mean Time [GMT]).

Syntax

VB

```
DATETODATETIME (date)
```

Arguments

`date`: *Date*

A date value that represents the date to convert.

Return values

DateTime

The resulting date/time value.

Example 1

`DATETODATETIME (CompInfo. 'getCurrentDate()')` returns the date of the current Microsoft Dynamics 365 Finance session, December 24, 2025, as **12/24/2025 12:00:00 AM**. In this example, **CompInfo** is an Electronic reporting (ER) data source of the **finance and operations/Table** type, and it refers to the CompanyInfo table.

Example 2

`DATETODATETIME (DATEVALUE ("2024-11-12T16:00:00.0000000-07:00", "O"))` returns the date and time value **11/12/2024 12:00:00 AM**.

Additional resources

Date and time functions

Last updated on 08/04/2026

DATEVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `DATEVALUE` function returns a *Date* value that is converted from a given text value in the specified format and in an optionally specified culture to a date value. For information about the supported formats, see standard and custom.

Syntax 1

```
VB

DATEVALUE (text, format)
```

Syntax 2

```
VB

DATEVALUE (text, format, culture)
```

Arguments

`text`: *String*

Text that represents the value to format.

`format`: *String*

The format of the given text. For information about the supported formats, see standard and custom.

`culture`: *String*

The culture that is used for formatting of the given text. For information about the supported cultures, see culture.

Return values

Date

The resulting date value.

Usage notes

When the culture isn't defined as an argument of the called function, the value of `culture` is defined by the calling context. For example, if the `DATEVALUE` function is called by using syntax 1 in an Electronic reporting (ER) format for a **FILE** element that is configured to use the German culture, the conversion will be done by using the German culture. The default `culture` value is **EN-US**.

Example 1

`DATEVALUE ("21-Dec-2016", "dd-MMM-yyyy")` returns the date value **December 21, 2016**, based on the specified custom format and the default application's **EN-US** culture.

Example 2

`DATEVALUE ("21-Gen-2016", "dd-MMM-yyyy", "IT")` returns the date value **January 21, 2016**, based on the specified custom format and culture.

However, `DATEVALUE ("21-Gen-2016", "dd-MMM-yyyy", "EN-US")` throws an exception to inform the user that the specified string isn't recognized as a valid date for the specified culture.

Additional resources

Date and time functions

Last updated on 04/08/2026

LIST ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LIST` function returns a *Record list* value that consists of a new list of records that is created from the specified arguments.

Syntax

VB

```
LIST (record 1 [, record 2, …, record N])
```

Arguments

`record 1`: *Container (record)*

A reference to a data source of the *Record* data type. This argument is required.

`record N`: *Container (record)*

A reference to a data source of the *Record* data type. These additional arguments are optional.

Return values

Record list

The resulting list of records.

Usage notes

The structure of the list that is created contains only the fields that are presented in the structure of every record that is mentioned in the arguments.

Example

You enter data source **Record 1** of the *Container* type. This data source contains the following nested fields of the *Calculated field* type:

- **Code:** This field contains an expression that returns a value of the *String* type.
- **Amount:** This field contains an expression that returns a value of the *Real* type.

You then enter data source **Record 2** of the *Container* type. This data source contains the following nested fields of the *Calculated field* type:

- **Amount:** This field contains an expression that returns a value of the *Real* type.
- **IsValid:** This field contains an expression that returns a value of the *Boolean* type.

In this case, the expression `LIST('Record 1', 'Record 2')` returns a new list that contains two records. The structure of this list consists of a single **Amount** field of the *Real* type, because this field is the only field that is presented in every argument of the called function.

Additional resources

List functions

Last updated on 04/08/2026

LISTOFFIELDS ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `LISTOFFIELDS` function returns a *Record list* value that is created based on the structure of the specified argument of the *Enumeration* or *Container (record)* type.

Syntax 1

```
VB

LISTOFFIELDS (path)
```

Syntax 2

```
VB

LISTOFFIELDS (path, language)
```

Arguments

`path`: Data source reference

The valid reference path of a data source of one of the following data types:

- Model enumeration
- Format enumeration
- Application enumeration
- Container (record)

`language`: *String*

Text that represents a language code.

Return values

Record list

The resulting list of records.

Usage notes

The list that is created consists of records that have the following fields:

- **Name** (*String* data type)
- **Label** (*String* data type)
- **Description** (*String* data type)
- **IsTranslated** (*Boolean* data type)

If the `path` argument refers to a data source of the *Container (Record)* type, for every field of the referenced container record, a new record is added to the list that is created. For every record that is created, the **Name** field returns the name of the field of the referenced container record that the current record was created for.

If the `path` argument refers to a data source of one of the *Enumeration* types, for every enumeration value of the referenced enumeration, a new record is added to the list that is created. For every record that is created, the **Name** field returns the value of the referenced enumeration that the current record was created for, the **Description** field returns the description of that enumeration, and the **Label** field returns the label of that enumeration.

At runtime, when syntax 1 is used, the **Label** and **Description** fields must return values that are based on the language settings of the Electronic reporting (ER) format that is running:

- If the labels and descriptions for the requested language are available, the **Label** and **Description** fields return values that are based on that language, and the **IsTranslated** field returns **True**.
- If the labels and descriptions for the requested language aren't available, the **Label** and **Description** fields return values that are based on the default **EN-US** language, and the **IsTranslated** field returns **False**.

At runtime, when syntax 2 is used, the **Label** and **Description** fields must return values that are based on the language that is defined as the second argument of the called function:

- If the labels and descriptions for the requested language are available, the **Label** and **Description** fields return values that are based on that language, and the **IsTranslated**

field returns **True**.

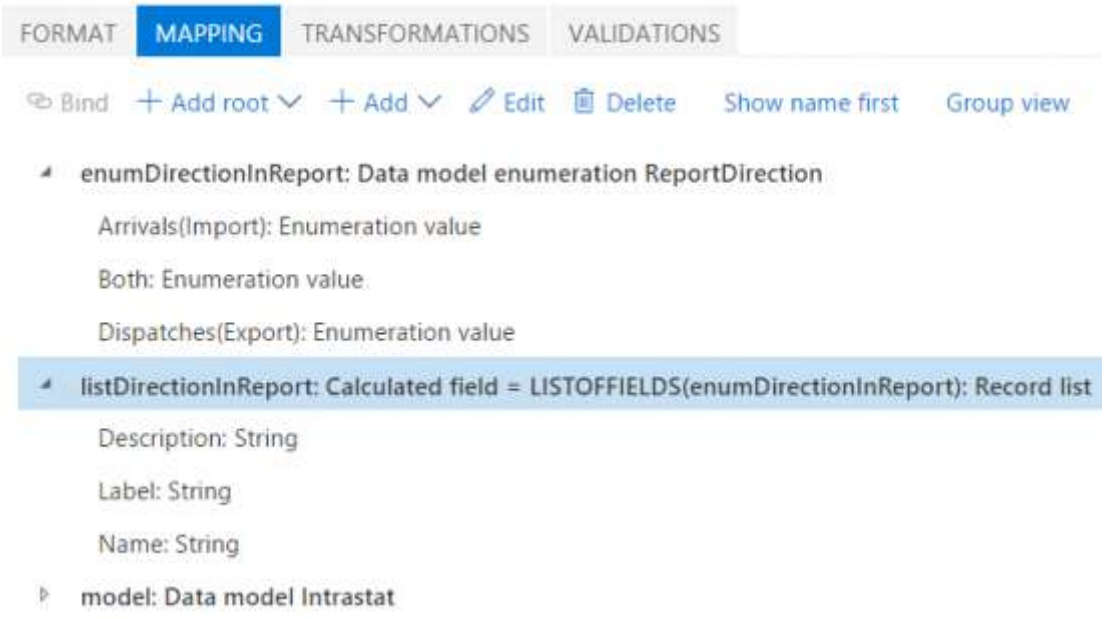
- If the labels and descriptions for the requested language aren't available, the **Label** and **Description** fields return values that are based on the **EN-US** language, and the **IsTranslated** field returns **False**.

Example 1

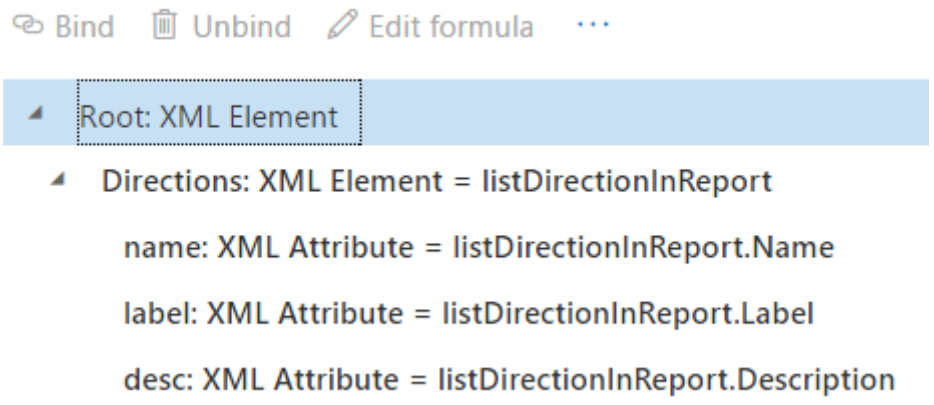
In the following illustration, an enumeration is introduced in an ER data model.

The following illustration shows these details:

- The model enumeration is inserted into a report as a data source.
- An ER expression uses the model enumeration as a parameter of the `LISTOFFIELDS` function.
- A data source of the *Record list* type is inserted into a report by using the ER expression that is created.



The following example shows the ER format elements that are bound to the data source of the *Record list* type that was created by using the `LISTOFFIELDS` function.



The following illustration shows the result when the designed format is run.

```
<?xml version="1.0" encoding="UTF-8"?>
- <Root>
    <Directions desc="" label="Both" name="Both"/>
    <Directions desc="" label="Dispatches" name="Export"/>
    <Directions desc="" label="Arrivals" name="Import"/>
  </Root>
```

> ⓘ Note
>
> Based on the language settings of the parent **FILE** and **FOLDER** format elements, translated text for labels and descriptions is entered in the output of the ER format.

Example 2

You use the *Calculated field* data source type to configure **enumType_de** and **enumType_deCH** data sources for the **enumType** data model enumeration:

- **enumType_de** = `LISTOFFIELDS (enumType, "de")`
- **enumType_deCH** = `LISTOFFIELDS (enumType, "de-CH")`

In this case, you can use the following expression to get the label of the enumeration value in Swiss German, if that translation is available. If the Swiss German translation isn't available, the label is in German.

VB

```
IF (NOT (enumType_deCH.IsTranslated), enumType_de.Label, enumType_deCH.Label)
```

Additional resources

List functions

Last updated on 04/08/2026

SPLIT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `SPLIT` function splits the specified input string into substrings and returns the result as a new *Record list* value.

Syntax 1

```
VB

SPLIT (input, length)
```

This syntax is used to split the specified input string into substrings, each of which has the specified length.

Syntax 2

```
VB

SPLIT (input, delimiter)
```

This syntax is used to split the specified input string into substrings, based on the specified delimiter.

Arguments

`input`: *String*

The text to split.

`length`: *Integer*

The maximum length of a single substring.

`delimiter`: *String*

A delimiter that is used to separate substrings.

Return values

Record list

The resulting list of records.

Usage notes

The record structure of the list that is returned consists of the **Value** field of the *String* type. Every record of the list that is returned contains generated substrings in this field.

If the `delimiter` argument is empty, the new list that is returned consists of one record that has the **Value** field of the *String* type. This field contains the input text.

If the `input` argument is empty, a new empty list is returned. If either the `input` or `delimiter` argument is unspecified (null), an application exception is thrown.

Example 1

`SPLIT ("abcd", 3)` returns a new list that consists of two records that have the **Value** field of the *String* type. The **Value** field in the first record contains the text "**abc**", and the **Value** field in the second record contains the text "**d**".

Example 2

`SPLIT ("XAb aBy", "aB")` returns a new list that consists of three records that have the **Value** field of the *String* type. The **Value** field in the first record contains the text "**X**", the **Value** field in the second record contains the text " ", and the **Value** field in the third record contains the text "**y**".

Example 3

Yo can use the INDEX function to access individual elements of the specified input string. If you enter the **MyList** data source of the **Calculated field** type and configure for it the `SPLIT("abc", 1)` expression, the expression `INDEX(MyList,2).Value` returns the text "**b**".

Example 4

The ENUMERATE function can also help you access individual elements of the specified input string. If you first enter the **MyList** data source of the **Calculated field** type and configure for it the `SPLIT("abc", 1)` expression, and then enter the **EnumeratedList** data source of the **Calculated field** type and configure for it the `ENUMERATE(MyList)` expression, the expression `FIRSTORNULL(WHERE(EnumeratedList, EnumeratedList.Number=2)).Value` returns the text "**b**".

Additional resources

List functions

Last updated on 04/08/2026

STRINGJOIN ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `STRINGJOIN` function returns a *String* value that consists of concatenated values of the specified field from the specified list. The values can be separated by the specified delimiter.

Syntax

```
VB

STRINGJOIN (list, field, delimiter)
```

Arguments

`list`: *Record list*

The valid path of a data source of the *Record list* data type.

`field`: *Field*

The valid path of a field of the *String* data type in the specified list.

`delimiter`: *String*

A delimiter that is used to separate substrings.

Return values

String

The resulting text value.

Example

If you enter `SPLIT("abc" , 1)` as data source **DS**, the expression `STRINGJOIN (DS, DS.Value, "-")` returns "**a-b-c**".

Additional resources

List functions

Last updated on 04/08/2026

CHAR ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `CHAR` function returns a *String* value that presents a single character that is referenced by the specified Unicode number.

Syntax

VB

```
CHAR (number)
```

Arguments

`number`: *Integer*

A number that corresponds to an expected single character.

Return values

String

The resulting text value.

Usage notes

The string that this function returns depends on the encoding that is selected in the parent **FILE** format element. For a list of the supported encodings, see Encoding class.

Example

`CHAR (255)` returns "ÿ".

Additional resources

Text functions

Last updated on 04/08/2026

GUIDVALUE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `GUIDVALUE` function converts the specified input of the *String* type to a data item of the *GUID* type.

Syntax

```
VB

GUIDVALUE (input)
```

Arguments

`input`: *String*

The valid path of a data source of the *String* type.

Return values

GUID

The resulting globally unique identifier (GUID) value.

Usage notes

To do a conversion in the opposite direction (that is, to convert specified input of the *GUID* data type to a data item of the *String* data type), you can use the TEXT function.

Example

You define the following data sources in your model mapping:

- A **myID** data source of the *Calculated field* type that contains the expression `GUIDVALUE ("AF5CCDAC-F728-4609-8C8B- A4B30B0C0AA0")`
- A **Users** data source of the *Table records* type that refers to the UserInfo table

You can then use an expression such as `FILTER (Users, Users.objectId = myID)` to filter the UserInfo table by the **objectId** field of the *GUID* data type.

Additional resources

Text functions

Last updated on 04/08/2026

NUMBERFORMAT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `NUMBERFORMAT` function returns a *String* value that presents the specified number in the specified format and in an optionally specified culture. For information about the supported formats, see standard and custom.

Syntax 1

```
VB

NUMBERFORMAT (number, format)
```

Syntax 2

```
VB

NUMBERFORMAT (number, format, culture)
```

Arguments

`number`: *Integer* or *Real*

A numeric value that specifies the number that must be formatted.

`format` : *String*

A *String* value that represents the format.

`culture`: *String*

A *String* value that represents the culture to use for formatting.

Return values

String

The resulting text value.

Usage notes

If the culture isn't defined as an argument of the called function, the context that this function is run in determines the culture that is used to format numbers.

Example 1

For the **EN-US** culture, `NUMBERFORMAT (0.45, "p")` returns "**45.00 %**", and `NUMBERFORMAT (10.45, "#")` returns "**10**".

Example 2

`NUMBERFORMAT (10/3, "F2", "de")` returns **3,33**, whereas `NUMBERFORMAT (10/3, "F2", "en-us")` returns **3.33**.

Additional resources

Text functions

Last updated on 04/08/2026

QRCODE ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `QRCODE` function returns a *Container* value that presents the Quick Response code (QR code) image for the specified string in binary format.

Syntax

VB

```
QRCODE (text)
```

Arguments

`text`: *String*

A *String* value that represents the original text.

Return values

Container

The resulting binary stream.

Example

You can configure an Electronic reporting (ER) format to generate an outbound document in Microsoft Office format (Excel workbooks or Word documents) by using a predefined template. This template might contain a **Picture** object (Excel workbook) or a **Picture Content Control** (Word document) as a placeholder for a QR code image. Add a **Cell** element to the configured ER format that fills this placeholder. To specify what information the QR code stores, define a

binding for this **Cell** element. For example, you can configure the binding to contain the following expression:

```
VB

QRCODE (model.ListOfShelfLabels.LabelText)`
```

When you run the configured ER format, the text value of the **LabelText** field of the **model.ListOfShelfLabels** data source generates a QR code image. This image replaces a QR code image placeholder in the document template that you use to generate an outbound document. When you scan this image of the generated document, it returns the text that was taken from the **LabelText** field of the **model.ListOfShelfLabels** data source. For more information, see Embed images and shapes in documents that you generate by using ER.

Additional resources

Text functions

Last updated on 04/13/2026

TEXT ER function

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The `TEXT` function returns the specified number as a *String* value after it converts the number to a text string that's formatted according to the server locale settings of the current application instance.

Syntax

```
VB

TEXT (number)
```

Arguments

`number`: *Integer* or *Real*

The number to convert to a text string.

Return values

String

The resulting text value.

Usage notes

For *Real* type values, the string conversion includes up to two decimal places.

Example

If the server locale of the Microsoft Dynamics 365 Finance instance is defined as **EN-US**, `TEXT (NOW ())` returns the current Finance session date, December 17, 2015, as the text string "**12/17/2015 07:59:23 AM**". `TEXT (1/3)` returns "**0.33**".

Additional resources

Text functions

Last updated on 04/13/2026

ER Design a configuration for generating reports in OPENXML format (November 2016)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user in the System Administrator or Electronic Reporting Developer role can create a new Electronic reporting (ER) configuration that contains a template for generating electronic documents in OPENXML format. Use this configuration for processing vendor payments.

In this example, you create a configuration for sample company, Litware, Inc. You can perform these steps in GBSI company.

To complete these steps, you must first complete the steps in the "Create a configuration provider and mark it as active" procedure. You must also have an Excel file that you import when creating the template. You can access this file from the Template of Payment Report.

Upload the Payments data model configuration

1. In the navigation pane, go to **Modules** > **Organization administration** > **Workspaces** > **Electronic reporting**.
2. In the list, select the configuration provider for sample company, Litware, Inc. If you don't see this configuration provider, you must first complete the steps in Create configuration providers and mark them as active.
3. Select **Set active**.
4. Select **Repositories**. Select a repository for the Operations Resources type, if available. If it's available, skip the following steps about creating a new repository.
5. Select **Add** to open the drop dialog.
6. In the **Configuration repository type** field, enter `Operations resourcesdd`.

7. Select **Create repository**.
8. Select **OK**.
9. Select **Open**.
10. In the tree, select **Payment model**.
11. Select **Import**. Import this data model. It serves as a data source in a new format configuration. Skip this step if this configuration is already imported.
12. Select **Yes**.
13. Close the pages until you return to the Electronic reporting page.

Create a new format configuration

1. Select **Reporting configurations**.
2. In the tree, select **Payment model**.
3. Select **Create configuration** to open the drop dialog.
4. In the **New** field, enter `Format based on data model PaymentModel`. Create a format that is based on the PaymentModel data model.
5. In the **Name** field, type `Sample worksheet report`. Sample worksheet report
6. In the **Description** field, type `Sample worksheet report for vendors' payments`. Sample worksheet report for vendors' payments.
7. In the **Data model definition** field, enter or select a value. Select the **CustomerCreditTransferInitiation** definition.
8. Select **Create configuration**.

Design a new document in OPENXML worksheet format

1. In the tree, select **Payment model\Sample worksheet report**.
2. Select **Designer**.
3. On the Action Pane, select **Import**.
4. Select **Import from Excel**.
5. Select **Attachments**. Attach the existing Excel document as a template.
6. Select **New**.
7. Select **File**. Point to the existing Excel file.
8. Close the page.
9. In the **Template** field, enter or select a value. Select the attached Excel file to be used as a template.

10. Select **OK**. Note that ER format components are created in the designing format based on the structure of the referring MS Excel document (named ranges).

Create a new data source to calculate totals by currency codes

1. Select the **Mapping** tab.
2. Select **Add root** to open the drop dialog.
3. In the tree, select **Functions\Group by**.
4. In the **Name** field, type `PaymentByCurrency`.
5. Select **Edit group by**.
6. In the tree, expand **model**, and then select **model\Payments**.
7. Select **Add field to**.
8. Select **What to group**.
9. In the tree, expand **model\Payments**, and then select **model\Payments\Currency**.
10. Select **Add field to**.
11. Select **Grouped fields**.
12. In the tree, select **model\Payments\Instructed Amount(InstructedAmount)**.
13. Select **Add field to**, and then select **Aggregation fields**.
14. In the **Method** field, select an option. Select the **SUM aggregation** function.
15. In the **Name** field, type `TotalInstructuredAmount`.
16. Select **Save**.
17. Close the page.
18. Select **OK**.

Map format components to data sources

1. In the tree, select **model\Payments\Initiating Party(InitiatingParty)\Name** and **Excel\ReportHeader\CompanyName**.
2. Select **Bind**.
3. In the tree, select **model\Payments\Creditor\Identification\Source ID(SourceID)** and **Excel\PaymLines\VendAccountName**.
4. Select **Bind**.
5. In the tree, select **model\Payments\Creditor\Name** and **Excel\PaymLines\VendName**.
6. Select **Bind**.
7. In the tree, select **model\Payments\Creditor Agent(CreditorAgent)\Name** and **Excel\PaymLines\Bank**.

8. Select **Bind**.
9. In the tree, select **model\Payments\Creditor Agent(CreditorAgent)\Routing Number(RoutingNumber)** and **Excel\PaymLines\RoutingNumber**.
10. Select **Bind**.
11. In the tree, select **model\Payments\Creditor Account(CreditorAccount)\Identification\Number** and **Excel\PaymLines\AccountNumber**.
12. Select **Bind**.
13. In the tree, select **model\Payments\Instructed Amount(InstructedAmount)** and **Excel\PaymLines\Debit**.
14. Select **Bind**.
15. In the tree, select **model\Payments\Currency** and **Excel\PaymLines\Currency**.
16. Select **Bind**.
17. In the tree, select **PaymentByCurrency\grouped\Currency** and **Excel\SummaryLines\SummaryCurrency**.
18. Select **Bind**.
19. In the tree, select **PaymentByCurrency\aggregated\TotalInstructuredAmount** and **Excel\SummaryLines\SummaryAmount**.
20. Select **Bind**.
21. In the tree, select **PaymentByCurrency** and **Excel\SummaryLines**.
22. Select **Bind**.
23. In the tree, select **model\Payments** and **Excel\PaymLines**.
24. Select **Bind**.
25. Select **Save**, and then close the page.

Use the created configuration for payments processing

1. Select **Change status**.
2. Select **Complete**.
3. Select **OK**.
4. In the navigation pane, go to **Modules** > **Accounts payable** > **Payment setup** > **Methods of payment**.
5. Use the Quick Filter to filter on the **Method of payment** field with a value of **Electronic**.
6. Select **Edit**.
7. Expand the **File formats** section.
8. Select **Yes** in the **Generic electronic reporting** field.

9. In the **Export format configuration** field, enter or select a value. Select the **Sample worksheet report** configuration.
10. Select **Save**.
11. Close the page.

Use the created configuration to test payment journal processing

1. In the navigation pane, go to **Modules** > **Accounts payable** > **Payments** > **Payment journal**.
2. Select **New** to create a new payment journal.
3. In the **Name** field, type **VendPay**.
4. Select **Lines**.
5. In the **Account** field, enter the value `GB_SI_000001`.
6. Set **Debit** to `1000`.
7. Select **New**.
8. In the **Account** field, enter the value `GB_SI_000005`.
9. Set **Debit** to `2000`.
10. In the **Currency** field, type `EUR`.
11. In the **Offset account** field, enter the value `GBSI OPER`.
12. In the **Method of payment** field, type `Electronic`.
13. Select **Save**.
14. Select **Generate payments**.
15. In the **Method of payment** field, type `Electronic`.
16. In the **File name** field, type `Payments`.
17. In the **Bank account** field, type `GBSI OPER`.
18. Select **OK**, and then select **OK** again. Review the created worksheet, including details of payment lines as well as totals for each currency code used in this payment message.

Last updated on 04/13/2026

Design a configuration for generating documents in Excel format

You can design an Electronic reporting (ER) format configuration that has an ER format component that you can configure to generate an outbound document in a Microsoft Excel workbook format. You must use specific ER format components for this purpose.

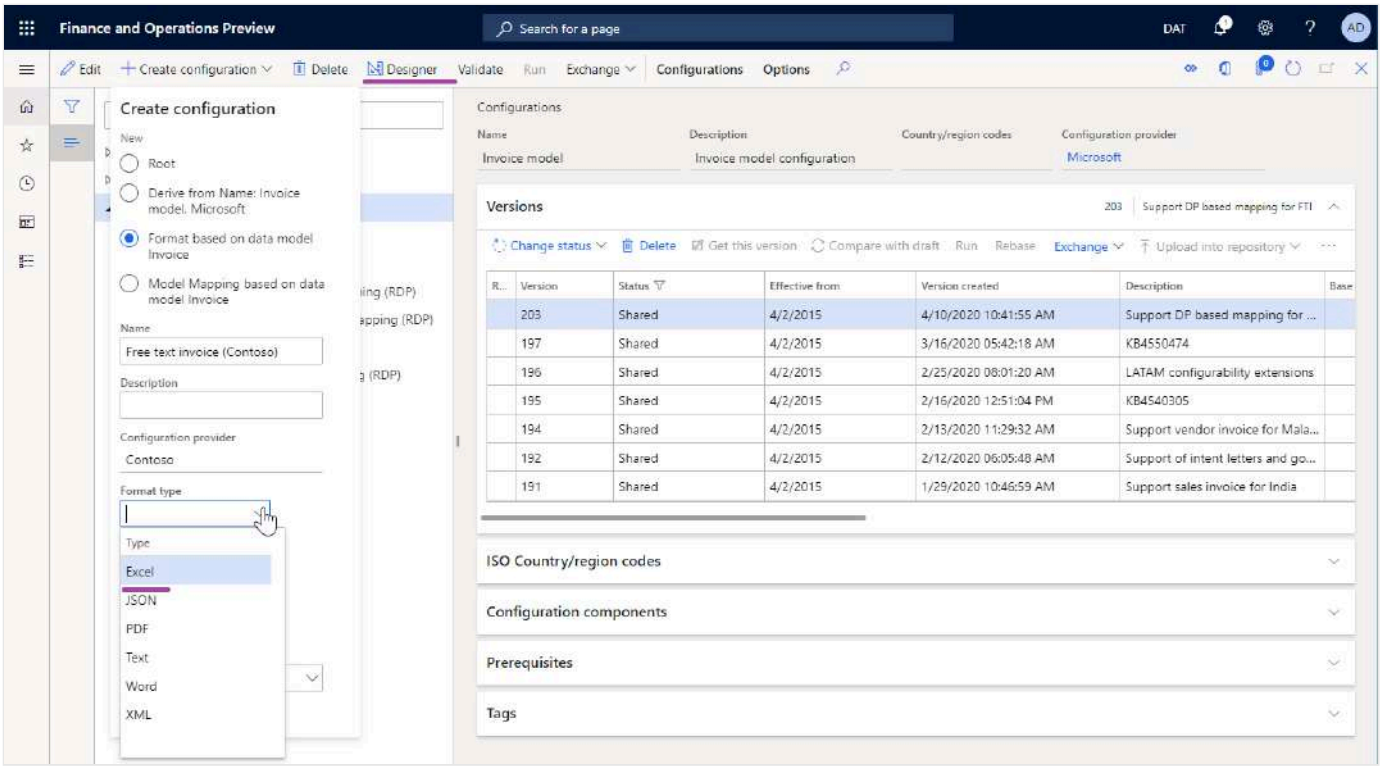
To learn more about this feature, see Design a configuration for generating reports in OPENXML format.

Add a new ER format

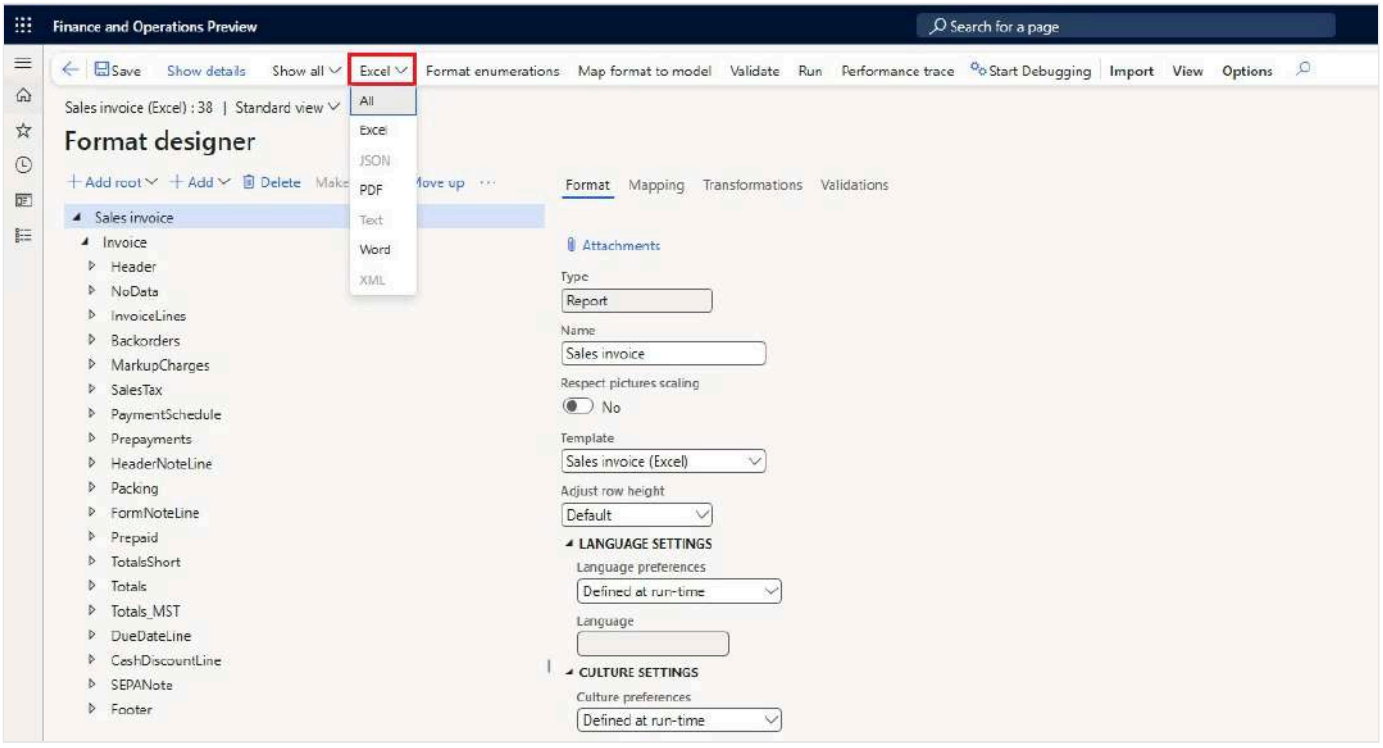
When you add a new ER format configuration to generate an outbound document in an Excel workbook format, select the **Excel** value for the **Format type** attribute of the format or leave the **Format type** attribute blank.

- If you select **Excel**, you can configure the format to generate an outbound document only in Excel format.
- If you leave the attribute blank, you can configure the format to generate an outbound document in any format that the ER framework supports.

To configure the ER format component of the configuration, select **Designer** on the Action Pane, and open the ER format component for editing in the ER Operation designer.



In Finance version **10.0.34**, you can view and change the **Format type** value for the selected format configuration version in the designer.

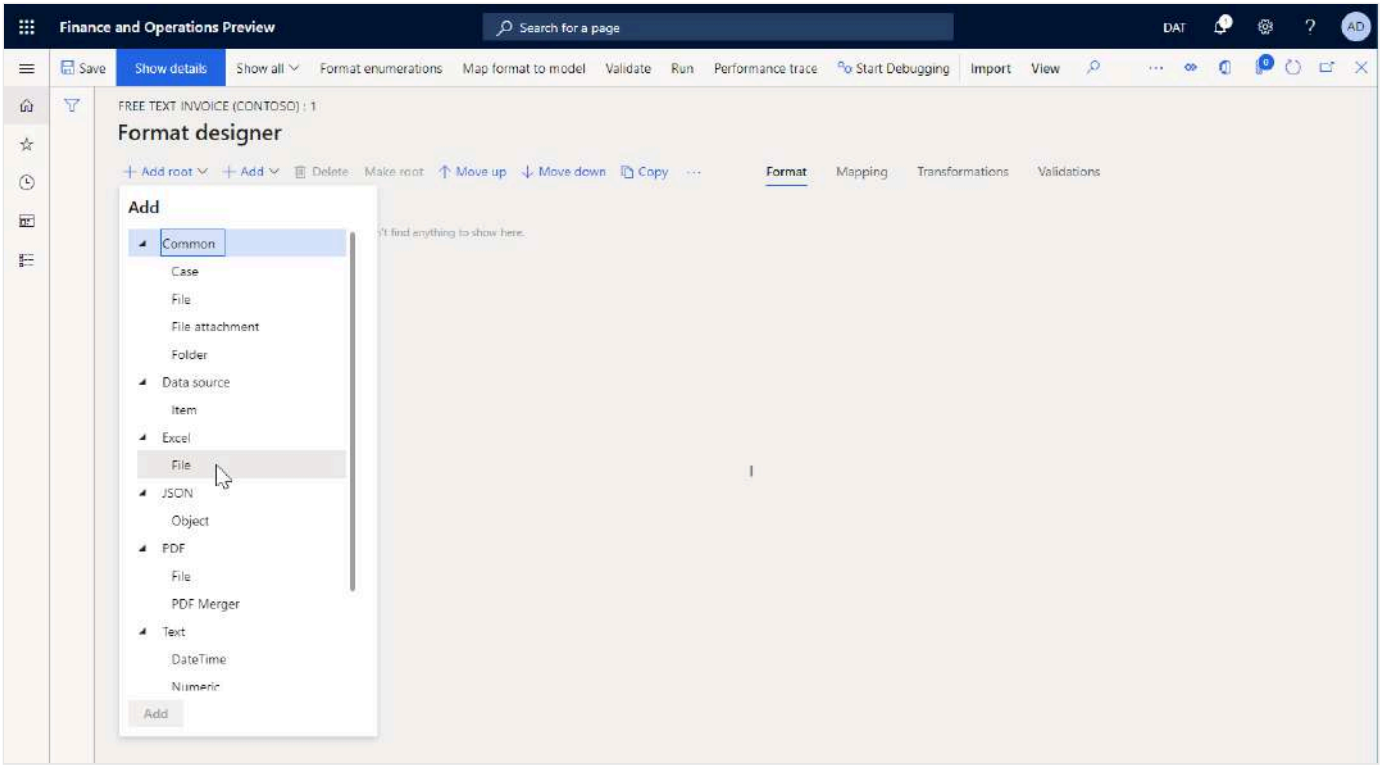


> **Note**
>
> The system suggests a selection of allowed format types, based on the current content of the format configuration.

Excel file component

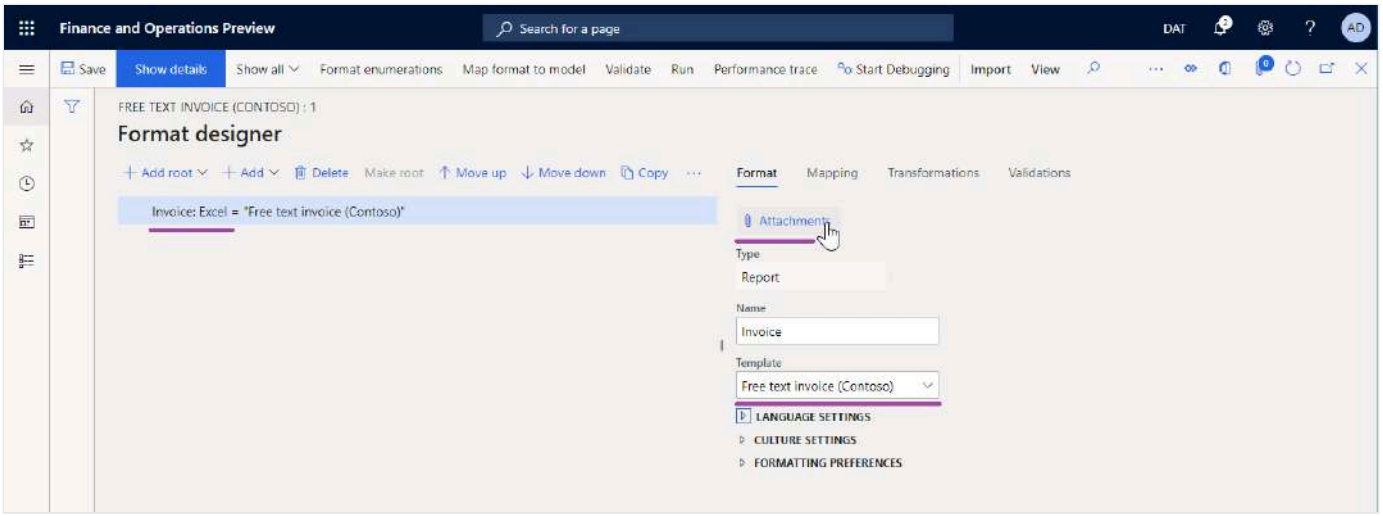
Manual entry

To generate an outbound document in Excel format, add an **Excel\File** component to the configured ER format.



To specify the layout of the outbound document, attach an Excel workbook that has the .xlsx extension to the **Excel\File** component as the template for outbound documents.

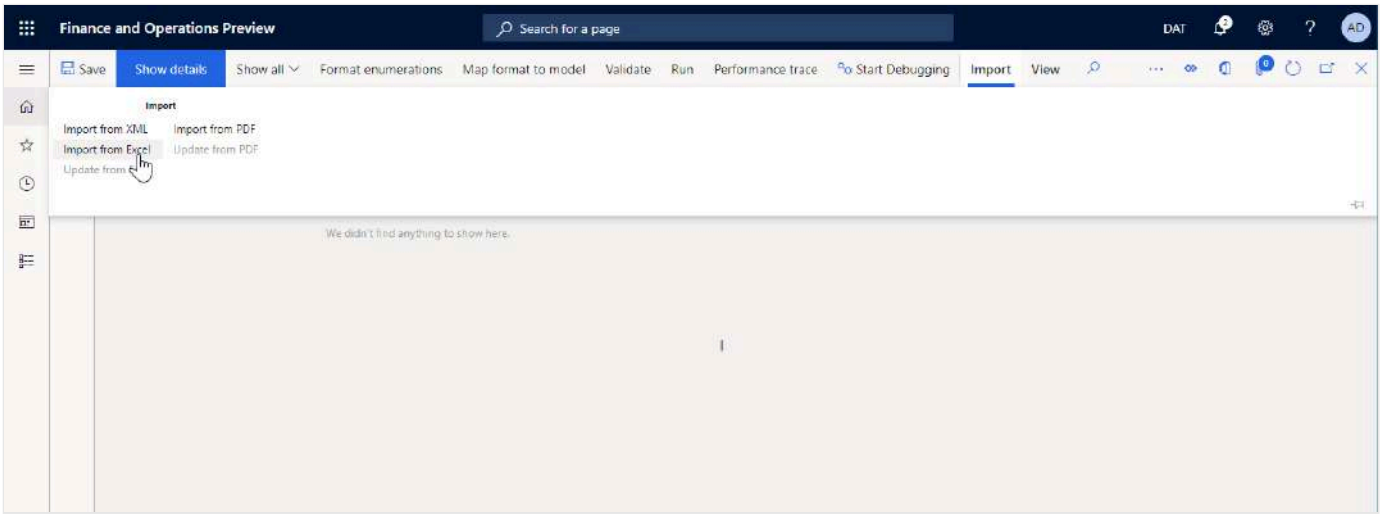
> **Note**
>
> When you manually attach a template, use a document type that's configured for this purpose in the ER parameters.



To specify how the attached template fills in when you run the configured ER format, add nested **Sheet**, **Range**, and **Cell** components to the **Excel\File** component. Associate each nested component with an Excel named item.

Template import

To import a new template into a blank ER format, select **Import from Excel** on the **Import** tab of the Action Pane. In this example, you automatically create an **Excel\File** component and attach the imported template to it. The process also automatically creates all required ER components, based on the list of Excel named items that it discovers.



> **Note**
>
> If you want to create the optional **Sheet** element in the editable ER format, set the **Create Excel Sheet format element** option to **Yes**.

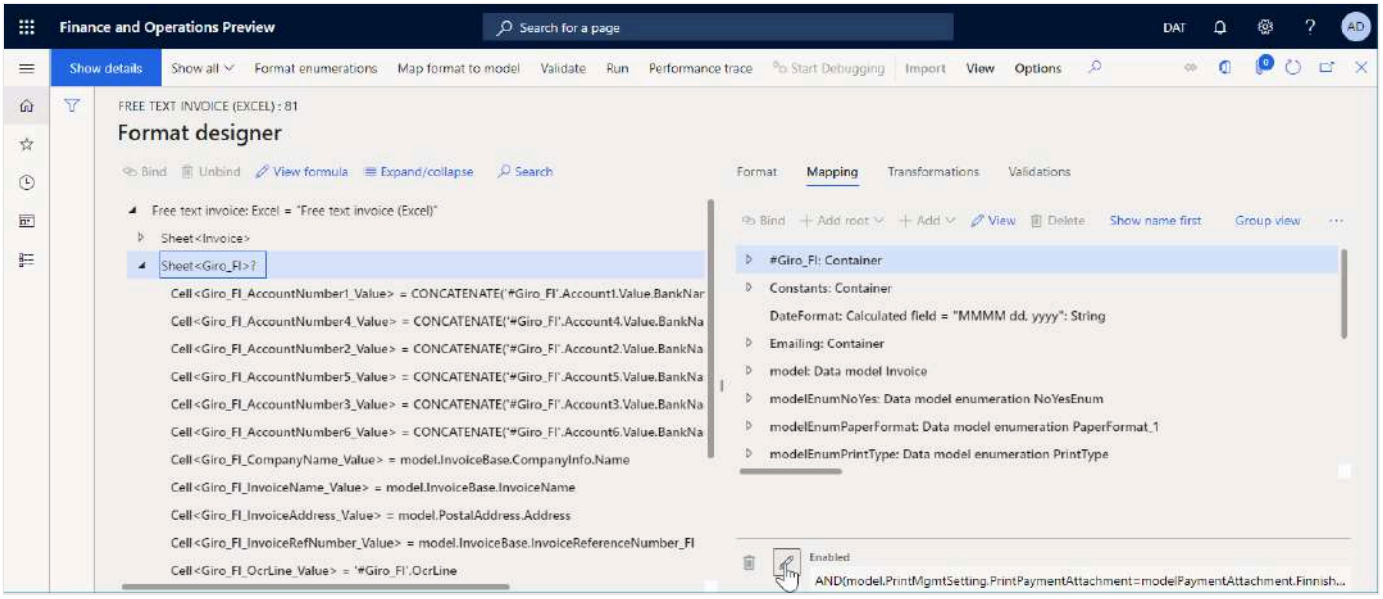
Sheet component

The **Sheet** component represents a worksheet in the attached Excel workbook that you need to fill in. You set the name of the worksheet in an Excel template through the **Sheet** property of this component.

> **Note**
>
> This component is optional for Excel workbooks that contain a single worksheet.

On the **Mapping** tab of the ER Operation designer, you can configure the **Enabled** property for a **Sheet** component to specify whether to include the component in a generated document:

- If you configure an expression for the **Enabled** property that returns **True** at runtime, or if you don't configure an expression, the appropriate worksheet is included in the generated document.
- If you configure an expression for the **Enabled** property that returns **False** at runtime, the generated document doesn't contain the worksheet.



Range component

Nested components

Data typing

The **Range** component can include other nested ER components that you use to enter values in the appropriate ranges.

- If you use any component from the **Text** group to enter values, you enter the value in an Excel range as a text value.

  > ⓘ **Note**
  >
  > Use this pattern to format entered values based on the locale that the application defines.

- If you use the **Cell** component from the **Excel** group to enter values, you enter the value in an Excel range as a value of the data type that the binding of that **Cell** component defines. For example, the data type might be **String**, **Real**, or **Integer**.

  > ⓘ **Note**
  >
  > Use this pattern to enable the Excel application to format entered values based on the locale of the local computer that opens the outbound document.

Row handling

You can configure the **Range** component as vertically replicated, so that it generates multiple rows in an Excel worksheet. The parent **Range** component or its nested **Range** components can generate the rows.

In version 10.0.26 and later, you can force a generated worksheet to keep the generated rows on the same page. In the ER format designer, set the **Keep rows together** option to **Yes** for the parent **Range** component in the editable ER format. ER then tries to keep all the content that it generates by that range on the same page. If the height of the content exceeds the remaining space on the current page, ER adds a page break, and the content starts at the top of the next new page.

> ⓘ **Note**
>
> Configure the **Keep rows together** option only for ranges that span the whole width of a generated document.
>
> The **Keep rows together** option applies only to **Excel** > **File** components that you configure to use an Excel workbook template.

You can use the **Keep rows together** option only when the **Enable usage of EPPlus library in Electronic reporting framework** feature is enabled.

This feature can be used for **Range** components that reside under the **Page** component. However, there's no guarantee that page footer totals are correctly calculated by using Data collection data sources.

To learn how to use this option, follow the example steps in Design an ER format to keep rows together on the same Excel page.

Replication

The **Replication direction** property specifies whether and how a range is repeated in a generated document:

- **No replication** – The appropriate Excel range isn't repeated in the generated document.
- **Vertical** – The appropriate Excel range is repeated vertically in the generated document. Each replicated range is placed below the original range in an Excel template. The number of repetitions is defined by the number of records in a data source of the **Record list** type that is bound to this ER component.
- **Horizontal** – The appropriate Excel range is repeated horizontally in the generated document. Each replicated range is placed to the right of the original range in an Excel template. The number of repetitions is defined by the number of records in a data source of the **Record list** type that is bound to this ER component.

  To learn more about horizontal replication, see Use horizontally expandable ranges to dynamically add columns in Excel reports.

Enabling

On the **Mapping** tab of the ER Operation designer, you can configure the **Enabled** property for a **Range** component to specify whether the component is included in a generated document:

- If you configure an expression for the **Enabled** property to return **True** at runtime, or if you don't configure an expression, the appropriate range is filled in the generated document.
- If you configure an expression for the **Enabled** property to return **False** at runtime, and if this range doesn't represent the entire rows or columns, the appropriate range isn't filled in the generated document.

- If you configure an expression for the **Enabled** property to return **False** at runtime, and this range represents the entire rows or columns, the generated document contains those rows and columns as hidden rows and columns.

Resizing

You can configure your Excel template to use cells to present textual data. To ensure that the whole text in a cell is visible in a generated document, you can configure that cell to automatically wrap the text inside it. You can also configure the row that contains that cell to automatically adjust its height if the wrapped text isn't fully visible. For more information, see the "Wrap text in a cell" section in Fix data that is cut off in cells.

> **Note**
>
> Because of a known Excel limitation, even if you configure cells to wrap text, and you configure the rows that contain those cells to automatically adjust their height to fit the wrapped text, you might not be able to use the **AutoFit** and **Wrap text** Excel features for merged cells and the rows that contain them.

As of Dynamics 365 Finance version 10.0.23, when you're working in a generated document, you can force ER to calculate the height of every row that was configured to automatically fit its height to the content of nested cells whenever that row contains at least one merged cell that was configured to wrap the text inside it. The calculated height is then used to resize the row to ensure that all cells in the row are visible in the generated document.

> **Note**
>
> Be aware that this functionality might not work as expected when a custom font is used to format a merged cell. Because Excel doesn't embed custom fonts, it doesn't provide information about custom font size. Therefore, the size of the merged cell can be incorrectly estimated.

To start to use this functionality when you run any ER formats that were configured to use Excel templates to generate outbound documents, follow these steps:

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Related links** section, select **Electronic reporting parameters**.

3. On the **Electronic reporting parameters** page, on the **Runtime** tab, set the **Autofit row height** option to **Yes**.

When you want to change this rule for a single ER format, update the draft version of that format by following these steps.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Configurations** section, select **Reporting configurations**.
3. On the **Configurations** page, in the configurations tree in the left pane, select an ER configuration that is designed to use an Excel template to generate outbound documents.
4. On the **Versions** FastTab, select the configuration version that has a status of **Draft**.
5. On the Action Pane, select **Designer**.
6. On the **Format designer** page, in the format tree in the left pane, select the Excel component that is linked to an Excel template.
7. On the **Format** tab, in the **Adjust row height** field, select a value to specify whether ER should be forced, at runtime, to change the height of rows in an outbound document that is generated by the edited ER format:
    - **Default** – Use the general setting that is configured in the **Autofit row height** field on the **Electronic reporting parameters** page.
    - **Yes** – Override the general setting, and change the row height at runtime.
    - **No** – Override the general setting, and don't change the row height at runtime.

Cell component

Use the **Cell** component to fill in Excel named cells, shapes, and pictures. To indicate an Excel named object that the **Cell** ER component must fill in, specify the name of that object in the **Excel range** property of the **Cell** component.

On the **Mapping** tab of the ER Operation designer, you can configure the **Enabled** property for a **Cell** component to specify whether the object is filled in in a generated document:

- If you configure an expression for the **Enabled** property that returns **True** at runtime, or if you don't configure an expression, the appropriate object is filled in in the generated

document. The binding of this **Cell** component specifies a value that goes in the appropriate object.

- If you configure an expression for the **Enabled** property that returns **False** at runtime, the appropriate object isn't filled in in the generated document.

When you configure a **Cell** component to enter a value in a cell, you can bind it to a data source that returns the value of a primitive data type (for example, **String**, **Real**, or **Integer**). In this case, the value is entered in the cell as a value of the same data type.

When you configure a **Cell** component to enter a value in an Excel shape, you can bind it to a data source that returns a value of a primitive data type (for example, **String**, **Real**, or **Integer**). In this case, the value is entered in the Excel shape as the text of that shape. For values of data types that aren't **String**, the conversion to text is done automatically.

> ⓘ **Note**
>
> You can configure a **Cell** component to fill in a shape only in cases where a shape text property is supported.

When you configure a **Cell** component to enter a value in an Excel picture, you can bind it to a data source that returns a value of the **Container** data type that represents an image in binary format. In this case, the value is entered in the Excel picture as an image.

> ⓘ **Note**
>
> Every Excel picture and shape is anchored by its upper-left corner to a specific Excel cell or range. To replicate an Excel picture or shape, you must configure the cell or range that it's anchored to as a replicated cell or range.

> 💡 **Tip**
>
> If you plan to use an image with custom scaling and aspect ratio at runtime, set the option **Respect pictures scaling** of the parent **Excel file component** to **Yes** to apply the scaling and aspect ratio as a placeholder of that image in the Excel template you're using.

To learn more about how to embed images and shapes, see Embed images and shapes in documents that you generate by using ER.

Page break component

The **PageBreak** component forces Excel to start a new page. This component isn't required when you want to use Excel's default paging, but use it when you want Excel to follow your ER format to structure paging.

Page component

Overview

Use the **Page** component when you want Excel to follow your ER format and structure pagination in a generated outbound document. When an ER format runs components that are under the **Page** component, it automatically adds the required page breaks. During this process, it considers the size of the generated content, the page setup of the Excel template, and the paper size that you select in the Excel template.

If you need to split a generated document into different sections, each with a different pagination, configure several **Page** components in every Sheet component.

Structure

If the first component under the **Page** component is a Range component where you set the **Replication direction** property to **No replication**, the system considers this range the page header for pagination that is based on the settings of the current **Page** component. The Excel range that is associated with this format component repeats at the top of every page that the system generates by using the settings of the current **Page** component.

> **Note**
>
> For correct pagination, if you configure the Rows to repeat at top range in your Excel template, the address of this Excel range must equal the address of the Excel range that is associated with the previously described **Range** component.

If the last component under the **Page** component is a **Range** component where you set the **Replication direction** property to **No replication**, the system considers this range the page footer for pagination that is based on the settings of the current **Page** component. The Excel range that is associated with this format component repeats at the bottom of every page that the system generates by using the settings of the current **Page** component.

> **Note**
>
> For correct pagination, don't resize the Excel ranges that are associated with the **Range** components at runtime. Don't format cells of this range by using the **Wrap text in a cell** and **Autofit row height** Excel options.

You can add multiple other **Range** components between the optional **Range** components to specify how a generated document is filled in.

If the set of nested **Range** components under the **Page** component doesn't comply with the previously described structure, a validation error occurs at design time in the ER format designer. The error message informs you that the issue can cause problems at runtime.

> **Note**
>
> To generate correct output, don't specify a binding for any **Range** component under the **Page** component if you set the **Replication direction** property for that **Range** component to **No replication**, and the range is configured to generate page headers or page footers.

If you want pagination-related summing and counting to compute running totals and totals per page, configure the required Data collection data sources. To learn how to use the **Page** component to paginate a generated Excel document, complete the procedures in Design an ER format to paginate a generated document in Excel format.

Limitations

When you use the **Page** component for Excel pagination, you don't know the final number of pages in a generated document until the pagination is complete. Therefore, you can't calculate the total number of pages by using ER formulas, and you can't print the correct number of pages of a generated document on any page before the last page.

> **Tip**
>
> To achieve this result in an Excel header or footer, use the special Excel formatting for headers and footers.

Dynamics 365 Finance version 10.0.22 doesn't consider configured **Page** components when you update an Excel template in the editable format. This functionality is considered for future

releases of Finance.

If you configure your Excel template to use conditional formatting, it might not work as expected in some cases.

Applicability

The **Page** component works for the Excel file format component only when you configure that component to use a template in Excel. If you replace the Excel template with a Word template and then run the editable ER format, the **Page** component is ignored.

The **Page** component works only when the **Enable usage of EPPlus library in Electronic reporting framework** feature is enabled. An exception is thrown at runtime if ER tries to process the **Page** component while this feature is disabled.

> ⓘ Note
>
> An exception is thrown at runtime if an ER format processes the **Page** component for an Excel template that contains at least one formula that refers to a cell that isn't valid. To help prevent runtime errors, fix the Excel template as described in How to correct a #REF! error.

Footer component

Use the **Footer** component to add footers at the bottom of a generated worksheet in an Excel workbook.

> ⓘ Note
>
> Add this component for every **Sheet** component to specify different footers for different worksheets in a generated Excel workbook.

When you configure an individual **Footer** component, use the **Header/footer appearance** property to specify the pages that the component applies to. The following values are available:

- **Any** – Run the configured **Footer** component for any page of the parent Excel worksheet.
- **First** – Run the configured **Footer** component for only the first page of the parent Excel worksheet.

- **Even** – Run the configured **Footer** component for only the even pages of the parent Excel worksheet.
- **Odd** – Run the configured **Footer** component for only the odd pages of the parent Excel worksheet.

For a single **Sheet** component, you can add several **Footer** components, each of which has a different value for the **Header/footer appearance** property. By using this approach, you can generate different footers for different types of pages in an Excel worksheet.

> ⓘ Note
>
> Make sure that each **Footer** component that you add to a single **Sheet** component has a different value for the **Header/footer appearance** property. Otherwise, a validation error occurs. The error message that you receive notifies you about the inconsistency.

Under the added **Footer** component, add the required nested components of the **Text\String**, **Text\DateTime**, or other type. Configure the bindings for those components to specify how your page footer is filled in.

You can also use special formatting codes to correctly format the content of a generated footer. To learn how to use this approach, follow the steps in Example 1, later in this article.

> ⓘ Note
>
> When you configure ER formats, be sure to consider the Excel limit and the maximum number of characters for a single header or footer.

Header component

Use the **Header** component to add headers at the top of a generated worksheet in an Excel workbook. Use it like the **Footer** component.

Edit an added ER format

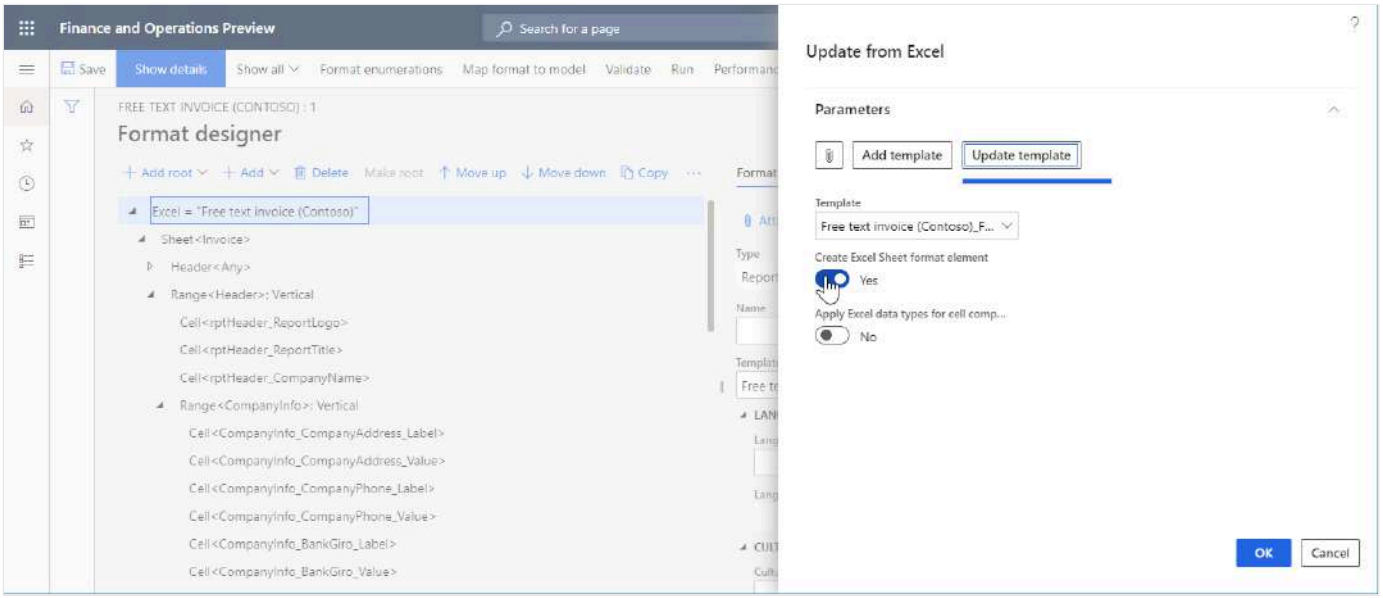
Update a template

Select **Update from Excel** on the **Import** tab of the Action Pane to import an updated template into an editable ER format. During this process, the template for the selected **Excel\File**

component is replaced by a new template. The content of the editable ER format syncs with the content of the updated ER template.

- If the process doesn't find the ER format component in the editable format, it automatically creates a new ER format component for every Excel name.
- If the process doesn't find the appropriate Excel name for an ER format component, it deletes that component from the editable ER format.

> **Note**
>
> Set the **Create Excel Sheet format element** option to **Yes** if you want to create the optional **Sheet** element in the editable ER format.
>
> If the editable ER format originally contained **Sheet** elements, set the **Create Excel Sheet format element** option to **Yes** when you import an updated template. Otherwise, the process creates all nested elements of the original **Sheet** element from scratch. Therefore, all bindings of the re-created format elements are lost in the updated ER format.



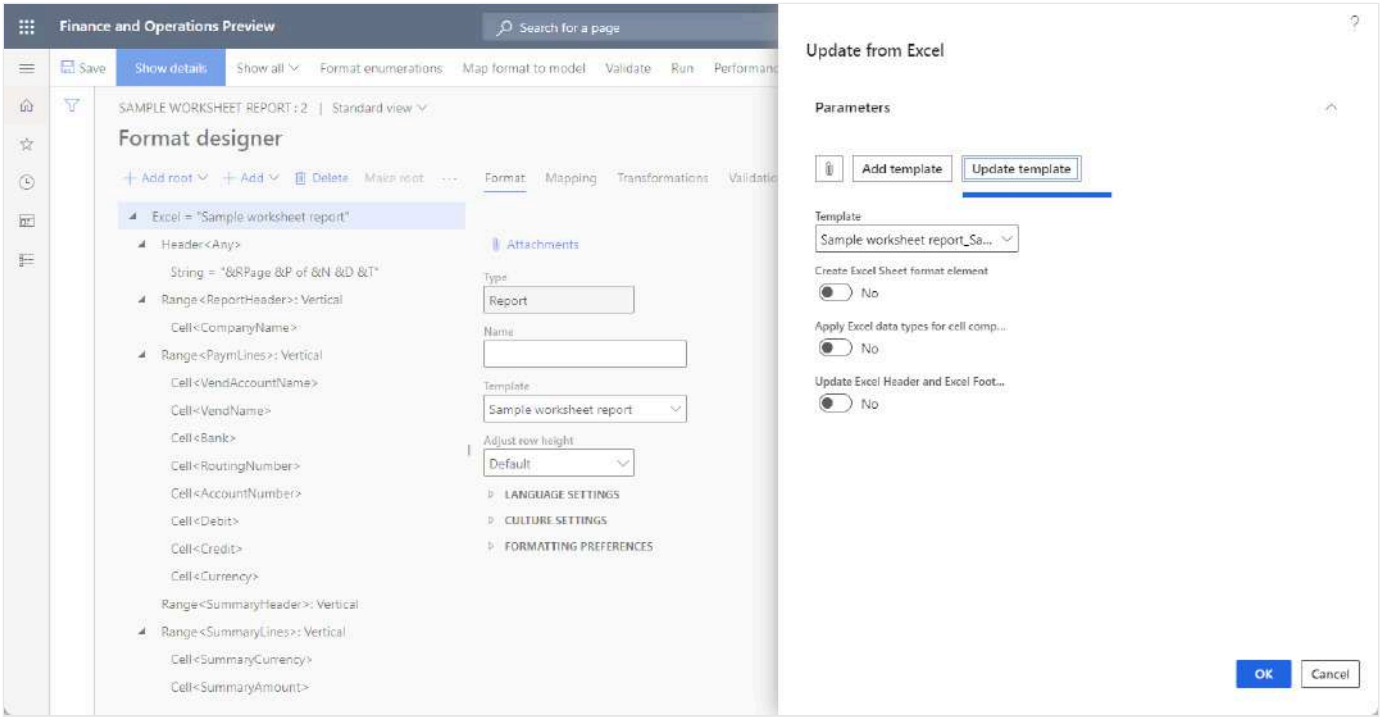
In version 10.0.28 and later, you can use the **Update Excel Header and Excel Footer format elements** option.

- When you set this option to **No**, the Excel Header and Excel Footer format elements remain unchanged, even if the corresponding headers or footers are updated in the worksheets of the imported template in the Excel workbook format.

- When you set this option to **Yes**, Excel Header and Excel Footer format elements change when the corresponding headers or footers are updated in worksheets of the imported

template in the Excel workbook format.

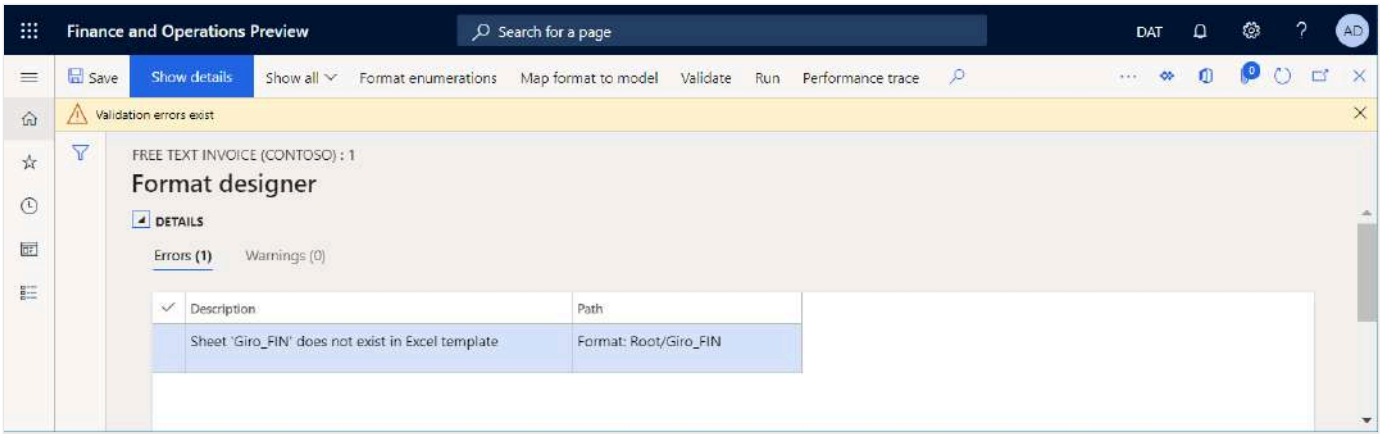
- If the structure of a worksheet header or footer isn't changed, or if it's only appended, the structure of the corresponding Excel Header or Excel Footer format element is updated. Bindings of format elements that are nested under this Excel Header or Excel Footer format element are preserved.
- If the structure of a worksheet header or footer is changed, the corresponding Excel Header or Excel Footer format element is re-created. Bindings of format elements that are nested under this Excel Header or Excel Footer format element are removed.



To learn more about this feature, see Modify Electronic reporting formats by reapplying Excel templates.

Excel-specific validation of ER format

When you validate an ER format that you can edit, the system checks for consistency to make sure that the Excel name exists in the Excel template that you're currently using. The system notifies you about any inconsistencies. For some inconsistencies, the option to automatically fix problems is available.



Control the calculation of Excel formulas

When you generate an outbound document in a Microsoft Excel workbook format, some cells in this document might contain Excel formulas. When you enable the **Enable usage of EPPlus library in Electronic reporting framework** feature, you can control when the formulas are calculated by changing the value of the **Calculation Options** parameter in the Excel template that you're using:

- Select **Automatic** to recalculate all dependent formulas every time new ranges, cells, and other elements are added to a generated document.

  > **Note**
  >
  > This choice might cause a performance problem for Excel templates that contain multiple related formulas.

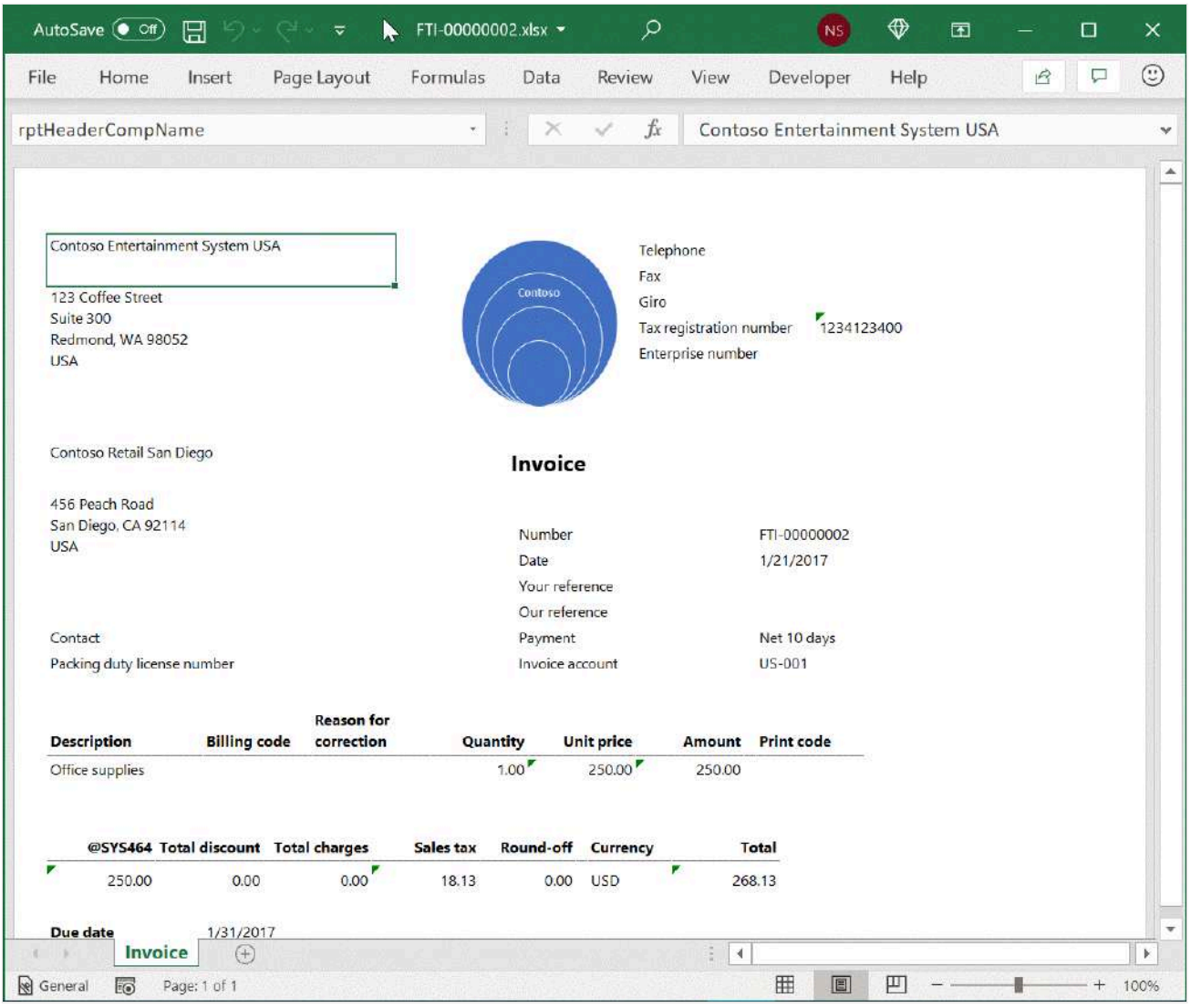
- Select **Manual** to avoid formula recalculation when a document is generated.

  > **Note**
  >
  > Formula recalculation is manually forced when a generated document is opened for preview by using Excel. Don't use this option if you configure an ER destination that assumes the usage of a generated document without its preview in Excel (PDF conversion, emailing, and so on) because the generated document might not contain values in cells that contain formulas.

Example 1: Format footer content

1. Use the provided ER configurations to generate a printable free text invoice (FTI) document.

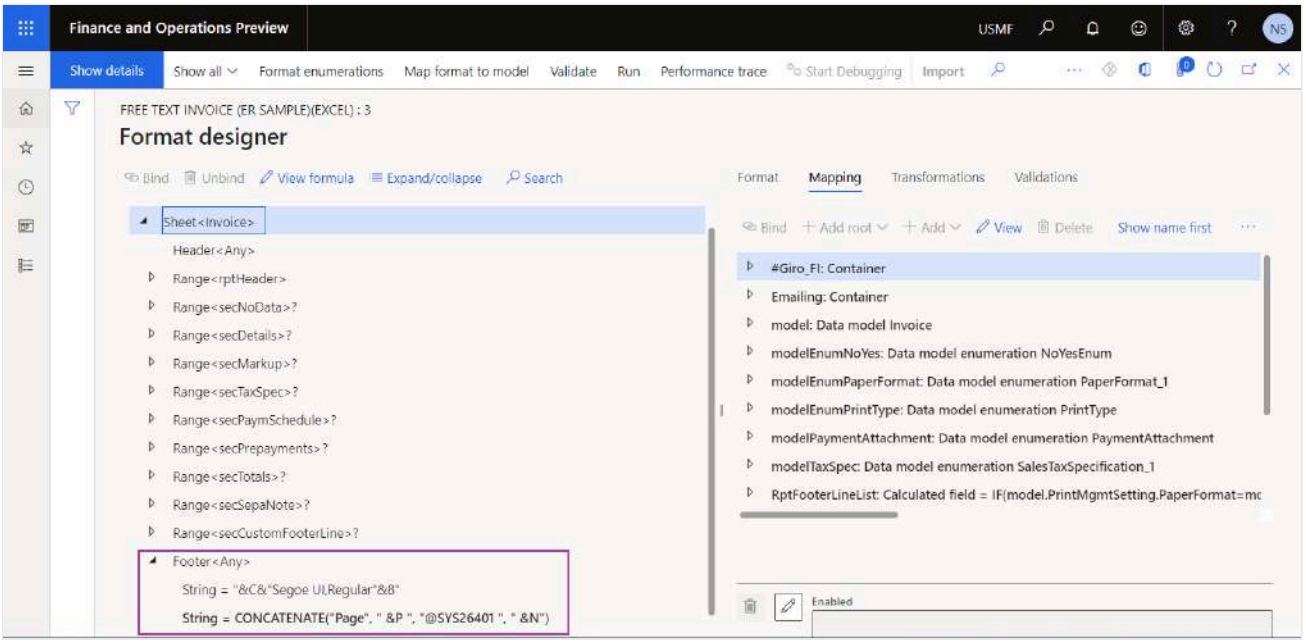
2. Review the footer of the generated document. Notice that it contains information about the current page number and the total number of pages in the document.



3. In the ER format designer, open the sample ER format for review.

   The footer of the **Invoice** worksheet is generated based on the settings of two **String** components that reside under the **Footer** component:

   - The first **String** component fills in the following special formatting codes to force Excel to apply specific formatting:
     - **&C** – Align the footer text in the center.
     - **&"Segoe UI,Regular"&8** – Present the footer text in the "Segoe UI Regular" font at a size of 8 points.
   - The second **String** component fills in the text that contains the current page number and the total number of pages in the current document.



4. Customize the sample ER format to modify the current page footer:

   a. Create a derived **Free text invoice (Excel) custom** ER format that is based on the sample ER format.

   b. Add the first new pair of **String** components for the **Footer** component of the **Invoice** worksheet:

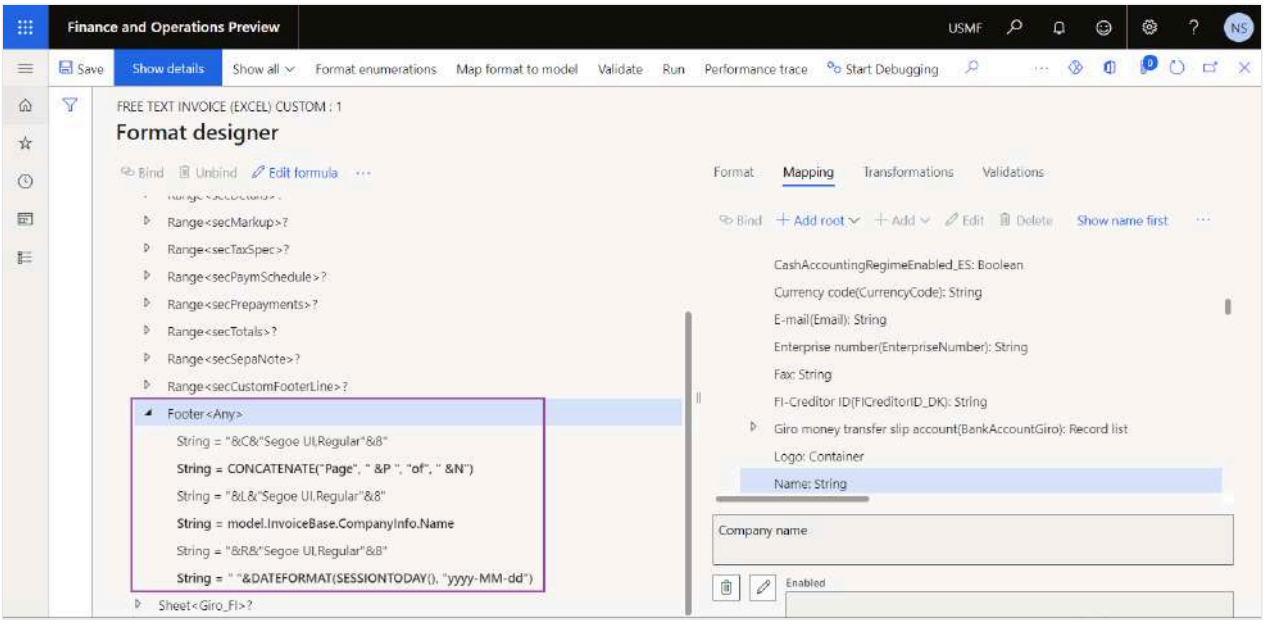
      i. Add a **String** component that aligns the company name on the left and presents it in 8-point "Segoe UI Regular" font (**"&L&"Segoe UI,Regular"&8"**).

      ii. Add a **String** component that fills in the company name (**model.InvoiceBase.CompanyInfo.Name**).

   c. Add the second new pair of **String** components for the **Footer** component of the **Invoice** worksheet:

      i. Add a **String** component that aligns the processing date on the right and presents it in 8-point "Segoe UI Regular" font (**"&R&"Segoe UI,Regular"&8"**).

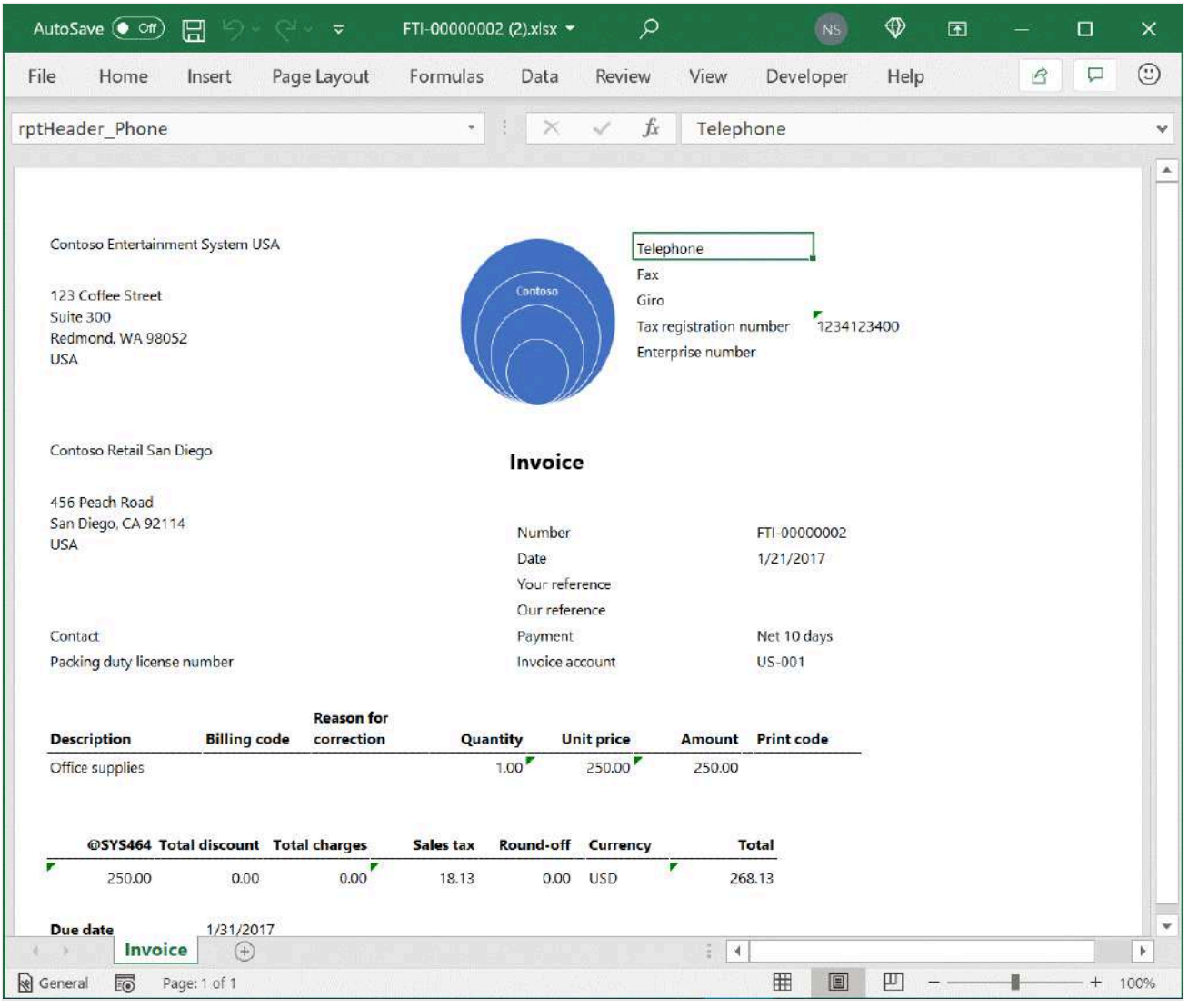
      ii. Add a **String** component that fills in the processing date in a custom format (**" "&DATEFORMAT(SESSIONTODAY(), "yyyy-MM-dd")**).



d. Complete the draft version of the derived **Free text invoice (Excel) custom** ER format.

5. Configure Print management to use the derived **Free text invoice (Excel) custom** ER format instead of the sample ER format.

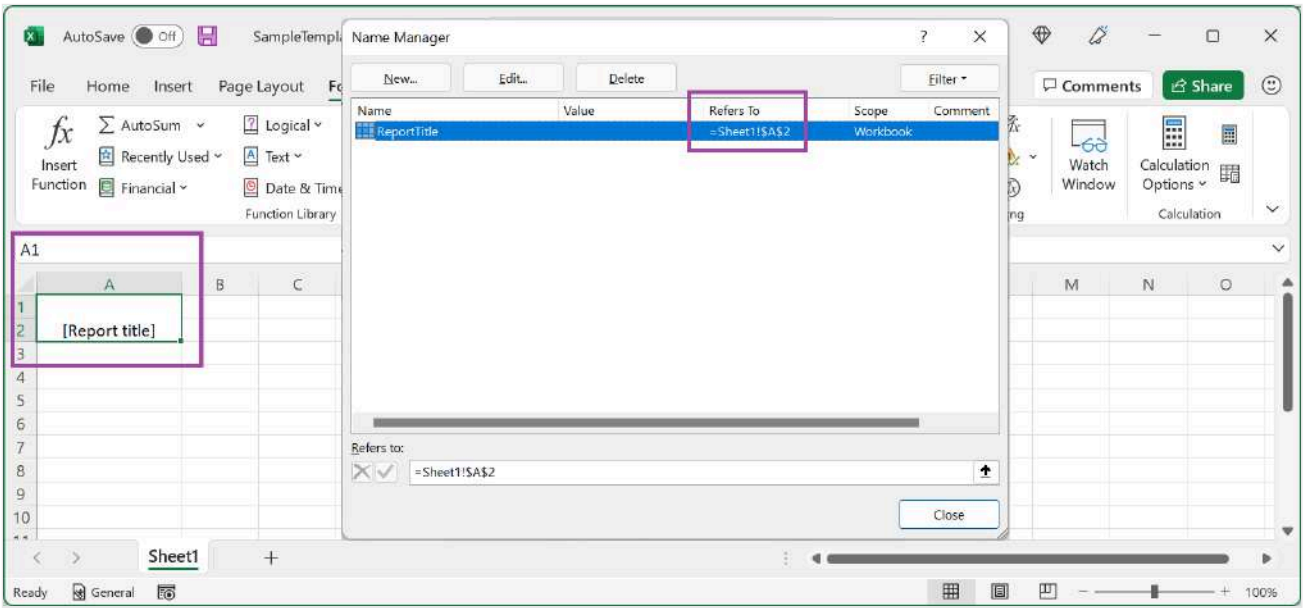
6. Generate a printable FTI document, and review the footer of the generated document.



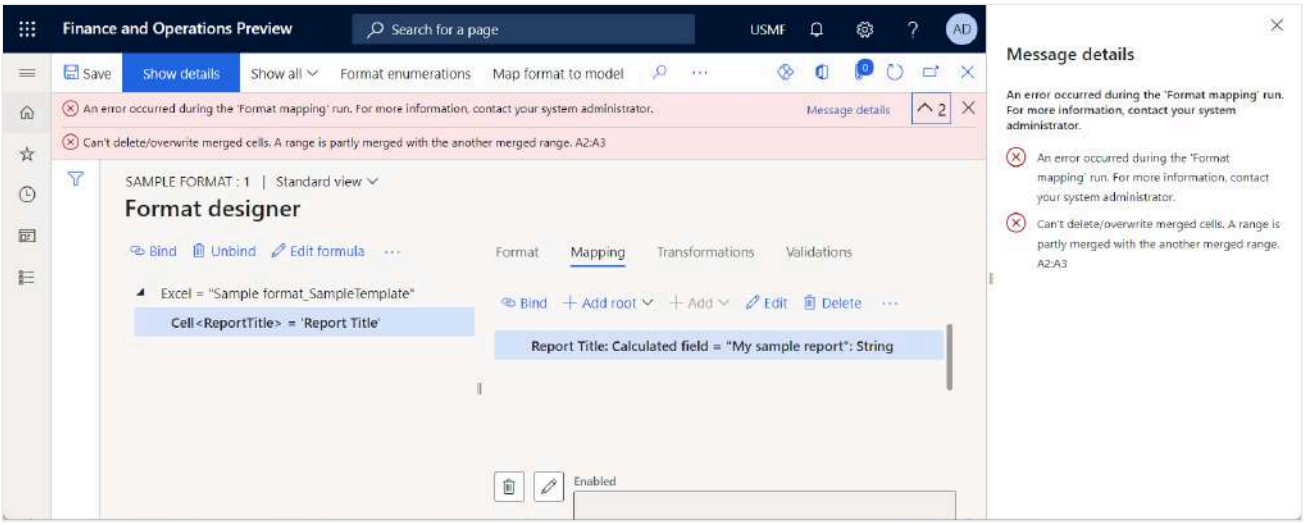
Example 2: Fixing the merged cells EPPlus issue

You can run an ER format to generate an outbound document in an Excel workbook format. When you enable the **Enable usage of EPPlus library in Electronic reporting framework** feature in the **Feature management** workspace, the EPPlus library is used to make Excel output. However, because of known Excel behavior and a limitation of the EPPlus library, you might encounter the following exception: "Can't delete/overwrite merged cells. A range is partly merged with the another merged range." To learn what kind of Excel templates might cause this exception and how you can fix the issue, complete the following example.

1. In the Excel desktop application, create a new Excel workbook.
2. On worksheet **Sheet1**, add the **ReportTitle** name for cell **A2**.
3. Merge cells **A1** and **A2**.



4. On the **Configurations** page, add a new ER format to generate an outbound document in an Excel workbook format.
5. On the **Format designer** page, import the designed Excel workbook into the added ER format as a new template for outbound documents.
6. On the **Mapping** tab, configure the binding for the **ReportTitle** component of the Cell type.
7. Run the configured ER format. Notice that the following exception is thrown: "Can't delete/overwrite merged cells. A range is partly merged with the another merged range."

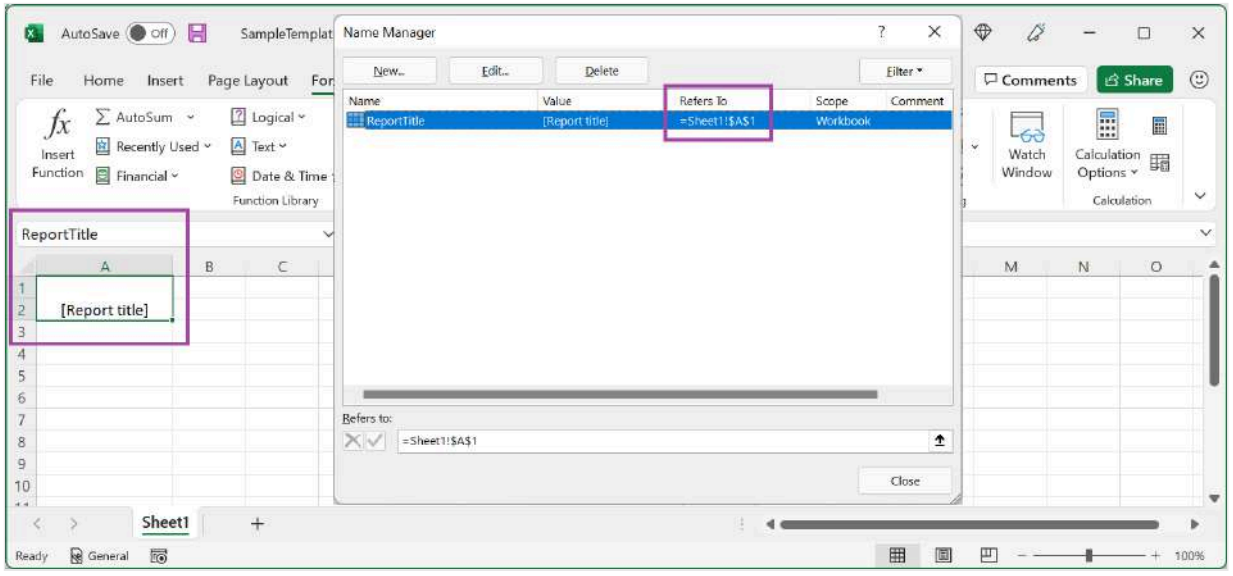


You can fix the issue in either of the following ways:

- **Easier but not recommended:** In the **Feature management** workspace, turn off the **Enable usage of EPPlus library in Electronic reporting framework** feature. Although this approach is easier, you might experience other issues if you use it, because some ER functionality is supported only when the **Enable usage of EPPlus library in Electronic reporting framework** feature is enabled.

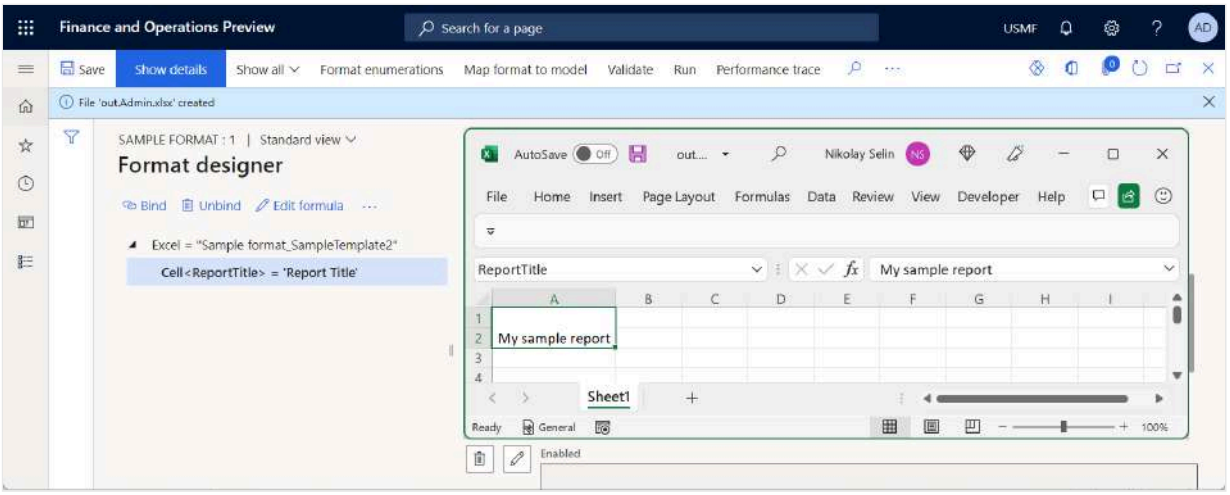
- **Recommended:** Follow these steps:

    1. In the Excel desktop application, modify the Excel workbook in one of the following ways:
        - On worksheet **Sheet1**, unmerge cells **A1** and **A2**.
        - Change the reference for the **ReportTitle** name from **=Sheet1!$A$2** to **=Sheet1!$A$1**.



    2. On the **Format designer** page, import the modified Excel workbook into the editable ER format to update the existing template.

3. Run the modified ER format.



Limitations

Known EPPlus library limitations

External data sources

If one of your templates contains a PivotTable that is based on a PowerPivot model that refers to an external data source, and you enable the **Enable usage of EPPlus library in Electronic reporting framework** feature, you receive the following error message when you run an ER format that uses that template to generate an outbound document in Excel format: "The cachesource is not a worksheet." To fix this issue, use the following options:

- **Recommended:** Redesign the Excel solution that you're using:

  1. Isolate the part that contains pivots in a separate Excel workbook (workbook A).
  2. Use ER to generate a second Excel workbook (workbook B) from Finance that has the required details.
  3. Refer to workbook B in workbook A as soon as workbook B is generated.

- Turn off the feature, **Enable usage of EPPlus library in Electronic reporting framework** to use an option other than EPPlus.

Additional resources

Electronic Reporting overview

Design a configuration for generating reports in OPENXML format

Modify Electronic reporting formats by reapplying Excel templates

Use horizontally expandable ranges to dynamically add columns in Excel reports

Embed images and shapes in documents that you generate by using ER

Configure Electronic reporting (ER) to pull data into Power BI

Last updated on 07/17/2026

Modify electronic reporting formats by reapplying Excel templates

Use the Electronic reporting (ER) tool to generate business documents in an electronic format. To generate a business document, create an ER format, and then use the ER designer to define the layout of the business document and specify the data that should be included in it. Then, run the ER format to generate the business document.

Use the ER tool to generate business documents as Microsoft Excel files. You can use an Excel document as a template for these documents. To define the document layout in the ER designer, import the contents of the Excel document that you want to use as a template into the defined ER format. For more details, and to practice this scenario, see the task guide **ER Design a configuration for generating reports in OPENXML format** (part of the 7.5.4.3 Acquire/Develop IT service/solution components (10677) business process). In the task guide step where you import an Excel template, use the initial template of the Payment Report Excel file, SampleVendPaymWsReport.

If you edit the Excel document that you use as a template for a business document, new ER functionality lets you reapply the updated template to the ER format. The ER format is then updated so that it adheres to the updated template. For more details about this functionality, see the task guide **ER Modify a format by reapplying an Excel template** (part of the 7.5.5.3 Acquire/Develop IT service/solution components (10683) business process). In the task guide step where you import an updated template, use the modified template of the Payment Report Excel file, SampleVendPaymWsReport2.

Additional resources

Update a template

Last updated on 06/26/2026

ER Use horizontally expandable ranges to dynamically add columns in Excel reports (Part 1 - Design format)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to generate reports as OPENXML worksheets (Excel) files in which the required columns can be created dynamically as horizontally expandable ranges. These steps can be performed in any company.

To complete these steps, you must first complete these three task guides:

"ER Create a configuration provider and mark it as active"

"ER Use financial dimensions as a data source (Part 1: Design data model)"

"ER Use financial dimensions as a data source (Part 2: Model mapping)"

You must also download and save a local copy of the template with a sample report found here, Sample Financial Dimensions Web Service Report.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Create a new report configuration

1. Go to Organization administration > Electronic reporting > Configurations.
2. In the tree, select `Financial dimensions sample model`.
3. Click Create configuration to open the drop dialog.
4. In the New field, enter `Format based on data model Financial dimensions sample model`.

    - Use the model created in advance as the data source for your new report.

5. In the Name field, type `Sample report with horizontally expandable ranges`.
    - Sample report with horizontally expandable ranges
6. In the Description field, type `To make Excel output with dynamically adding columns`.
    - To make Excel output with dynamically adding columns
7. In the Data model definition field, select Entry.
8. Click Create configuration.

Design the report format

1. Click Designer.
2. Turn on the `Show details` toggle button.
3. On the Action Pane, click Import.
4. Click Import from Excel.
5. Click Attachments.
    - Import the report`s template. Use Excel file that you downloaded for that.
6. Click New.
7. Click File.
8. Close the page.
9. In the Template field, enter or select a value.
    - Select the downloaded template.
10. Click OK.
    - Add a new range to dynamically create Excel output with as many columns as you selected (in the user dialog form) for financial dimensions. Each cell for every column will represent a single financial dimension`s name.
11. Click Add to open the drop dialog.
12. In the tree, select `Excel\Range`.
13. In the Excel range field, type `DimNames`.
    - DimNames
14. In the Replication direction field, select `Horizontal`.
15. Click OK.

16. In the tree, select `Excel = "SampleFinDimWsReport"\Range<DimNames>: Horizontal`.
17. Click Move up.
18. In the tree, select `Excel = "SampleFinDimWsReport"\Cell<DimNames>`.
19. Click Cut.
20. In the tree, select `Excel = "SampleFinDimWsReport"\Range<DimNames>: Horizontal`.
21. Click Paste.
22. In the tree, expand `Excel = "SampleFinDimWsReport"\Range<DimNames>: Horizontal`.
23. In the tree, expand `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical`.
24. In the tree, expand `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical`.
25. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical`.

    - Add a new range to dynamically create Excel output with as many columns as you selected (in the user dialog form) for financial dimensions. Each cell for every column will represent a single financial dimension`s value for each reporting transaction.

26. Click Add Range.
27. In the Excel range field, type `DimValues`.

    - DimValues

28. In the Replication direction field, select `Horizontal`.
29. Click OK.
30. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Cell<DimValues>`.
31. Click Cut.
32. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Range<DimValues>: Horizontal`.
33. Click Paste.
34. In the tree, expand `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Range<DimValues>: Horizontal`.

Map format elements to data sources

1. Click the Mapping tab.
2. In the tree, expand `model: Data model Financial dimensions sample model`.

3. In the tree, expand `model: Data model Financial dimensions sample model\Journal: Record list`.
4. In the tree, expand `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list`.
5. In the tree, expand `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list`.
6. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Range<DimValues>: Horizontal\Cell<DimValues>`.
7. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list\Code: String`.
8. Click Bind.
9. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Range<DimValues>: Horizontal`.
10. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list`.
11. Click Bind.
12. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Cell<Credit>`.
13. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Credit: Real`.
14. Click Bind.
15. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Cell<Debit>`.
16. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Debit: Real`.
17. Click Bind.
18. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Cell<Currency>`.
19. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Currency: String`.
20. Click Bind.
21. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Cell<TransDate>`.
22. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Date: Date`.
23. Click Bind.

24. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Cell<TransVoucher>`.
25. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Voucher: String`.
26. Click Bind.
27. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical\Cell<TransBatch>`.
28. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Batch: String`.
29. Click Bind.
30. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Range<TransactionLine>: Vertical`.
31. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list`.
32. Click Bind.
33. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical\Cell<Batch>`.
34. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list\Batch: String`.
35. Click Bind.
36. In the tree, select `Excel = "SampleFinDimWsReport"\Range<JournalLine>: Vertical`.
37. In the tree, select `model: Data model Financial dimensions sample model\Journal: Record list`.
38. Click Bind.
39. In the tree, expand `model: Data model Financial dimensions sample model\Dimensions setting: Record list`.
40. In the tree, select `model: Data model Financial dimensions sample model\Dimensions setting: Record list\Code: String`.
41. In the tree, select `Excel = "SampleFinDimWsReport"\Range<DimNames>: Horizontal\Cell<DimNames>`.
42. Click Bind.
43. In the tree, select `model: Data model Financial dimensions sample model\Dimensions setting: Record list`.
44. In the tree, select `Excel = "SampleFinDimWsReport"\Range<DimNames>: Horizontal`.
45. Click Bind.
46. In the tree, select `Excel = "SampleFinDimWsReport"\Cell<CompanyName>`.

47. In the tree, select `model: Data model Financial dimensions sample model\Company: String`.
48. Click Bind.
49. Click Save.
50. Close the page.

Last updated on 04/10/2026

ER Use horizontally expandable ranges to dynamically add columns in Excel reports (Part 2 - Run format)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to generate reports as OPENXML worksheets (Excel) files in which the required columns can be created dynamically as horizontally expandable ranges. You can perform these steps in the DEMF company.

To complete these steps, you must first complete the steps in the "ER Use horizontally expandable ranges to dynamically add columns in Excel reports (Part 1: Design format)" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Find created format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Financial dimensions sample model**.
3. In the tree, select **Financial dimensions sample model\Sample report with horizontally expandable ranges**.

Execute format to create Excel output

1. Select **Run**.
2. In the **Dimension name** field, type `BusinessUnit;CostCenter;Department`.

    - In the **Dimension name** field, enter or select a value. To select all dimensions for the current company, enter the following value:

`BusinessUnit;CostCenter;Department;ItemGroup;MainAccount;Project`

3. Expand the **Records to include** section.
4. Select **Filter**.
5. Select the row for the **Ledger journal** table and the **Journal batch number** field.
6. In the **Criteria** field, type `00057..00058`.

    - 00057..00058

7. Select **OK**.
8. Select **OK**.

    - Review the generated output. The newly created Excel file contains the same number of columns that you selected for financial dimensions. The report header in those columns represents financial dimensions' names. The transactions' lines in those columns represent financial dimensions. Run this report and select different dimensions to see that the report isn't dependent on the number of selected dimensions or the number of dimensions configured for this instance.

Last updated on 04/13/2026

Design an ER format to paginate generated documents in Excel

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user in the System Administrator or Electronic Reporting Functional Consultant role can configure an Electronic reporting (ER) format to generate outbound documents in Microsoft Excel and manage document pagination.

In this example, you will modify the Microsoft-provided ER format that is used to print the control report when the Intrastat declaration is generated. This report lets you observe reported Intrastat transactions. Your modifications will let you manage the pagination of control reports that are generated.

The procedures in this article can be completed in the **DEMF** company. No coding is required. Before you begin, download and save the following files.

Expand table

Description	File name
Report template 1	ERIntrastatReportDemo1.xlsx
Report template 2	ERIntrastatReportDemo2.xlsx

Configure the ER framework

Follow the steps in Configure the ER framework to set up the minimal set of ER parameters. You must complete this setup before you start to use the ER framework to design a custom version of a standard ER format.

Import the standard ER format configuration

Follow the steps in Import the standard ER format configuration to add the standard ER configurations to your current instance of Dynamics 365 Finance. Import version **1.9** of the **Intrastat report** format configuration. Base version 1 of the base **Intrastat model** configuration is automatically imported from the repository.

Customize the standard ER format

Create the custom ER format

In this scenario, you're the representative of Litware, Inc., which is currently selected as the active ER configuration provider. You must create a new ER format configuration by using the **Intrastat report** configuration as a base.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, in the configuration tree in the left pane, expand **Intrastat model**, and select **Intrastat report**. Litware, Inc. will use version 1.9 of this ER format configuration as the base for the custom version.
3. Select **Create configuration** to open the drop-down dialog box. You can use this dialog box to create a new configuration for a custom payment format.
4. In the **New** field group, select **Derive from Name: Intrastat report, Microsoft**.
5. In the **Name** field, enter **Intrastat report Litware**.
6. Select **Create configuration** to create the new format.

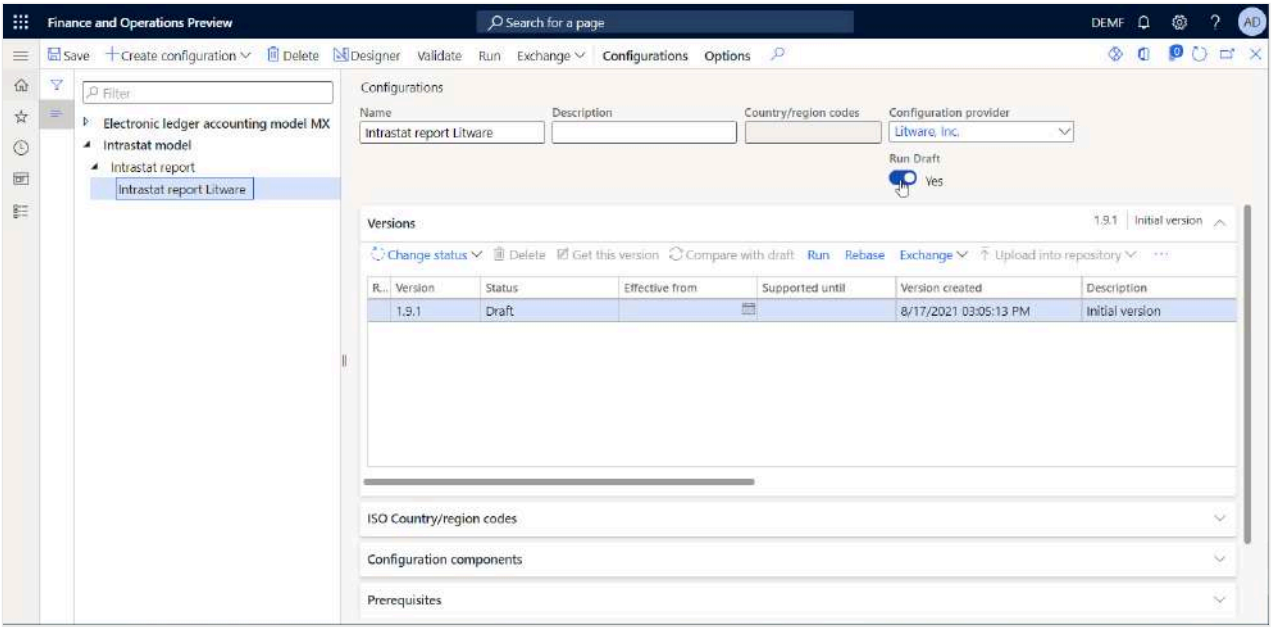
Version 1.9.1 of the **Intrastat report Litware** ER format configuration is created. This version has a status of **Draft** and can be edited. The current content of your custom ER format matches the content of the format that is provided by Microsoft.

Make the custom format runnable

Now that the first version of your custom format has been created and has a status of **Draft**, you can run the format for testing purposes. To run the report, process a vendor payment by using the payment method that refers to your custom ER format. By default, when you call an ER format from the application, only versions that have a status of **Completed** or **Shared** are considered. This behavior helps prevent ER formats that have unfinished designs from being used. However, for your test runs, you can force the application to use the version of your ER format that has a status of **Draft**. In this way, you can adjust the current format version if any modifications are required. For more information, see Applicability.

To use the draft version of an ER format, you must explicitly mark the ER format.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In the **User parameters** dialog box, set the **Run settings** option to **Yes**, and then select **OK**.
4. Select **Edit** to make the current page editable, as required.
5. In the configuration tree in the left pane, select **Intrastat report Litware**.
6. Set the **Run Draft** option to **Yes** and then select **Save**.



Set up Foreign trade parameters to use the custom ER format

Follow these steps to configure Foreign trade parameters so that you can use the custom format.

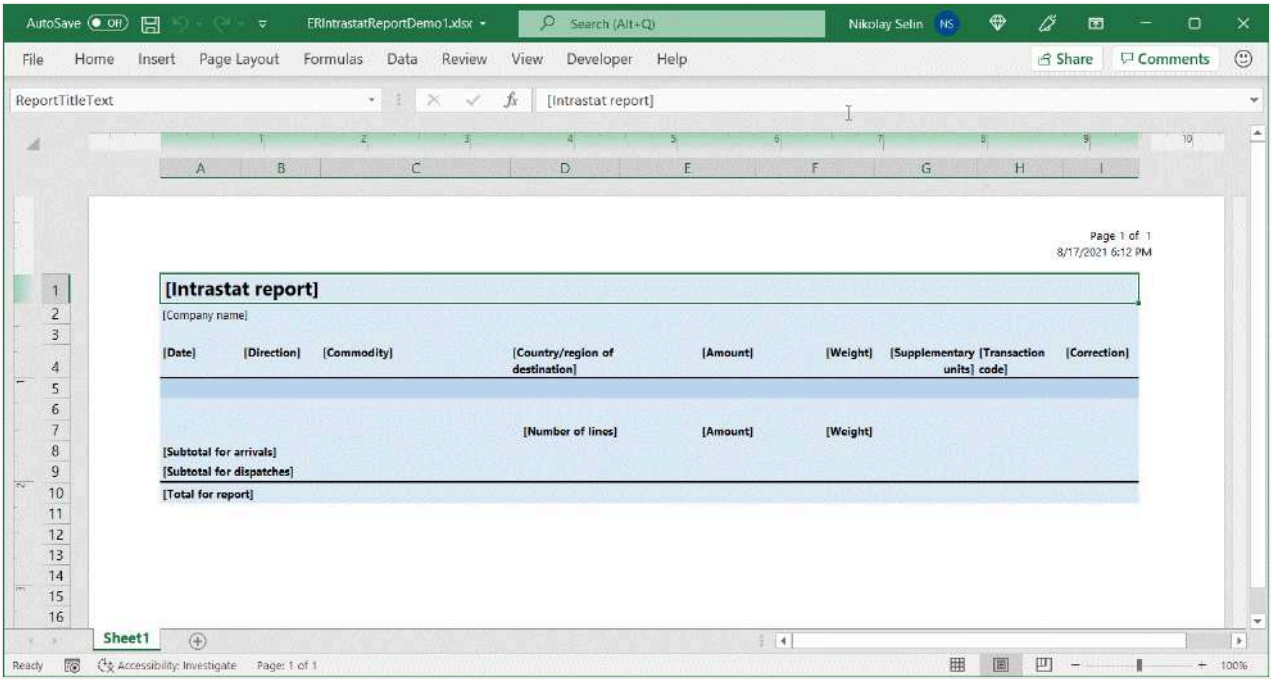
1. Go to **Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**.
2. On the **Foreign trade parameters** page, on the **Electronic reporting** FastTab, in the **File format mapping** field, select **Intrastat report Litware**.
3. In the **Report format mapping** field, select **Intrastat report Litware**.
4. Select **Save**.

Configure the custom format to use the downloaded report template

Review the first downloaded Excel template

1. In the Excel desktop application, open the **ERIntrastatReportDemo1.xlsx** template file that you downloaded earlier.

2. Verify that the template contains named ranges to create report header, report details, and report footer sections in generated documents.

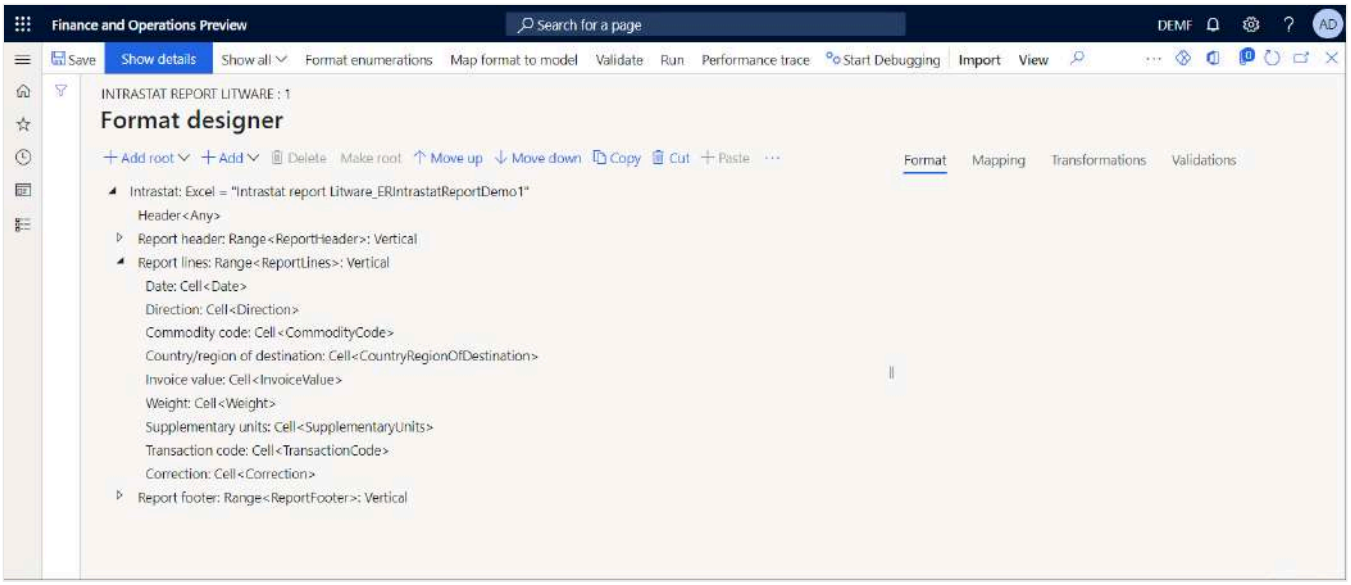


Replace the current Excel template in the custom ER format

You must add a new Excel template to the custom ER format.

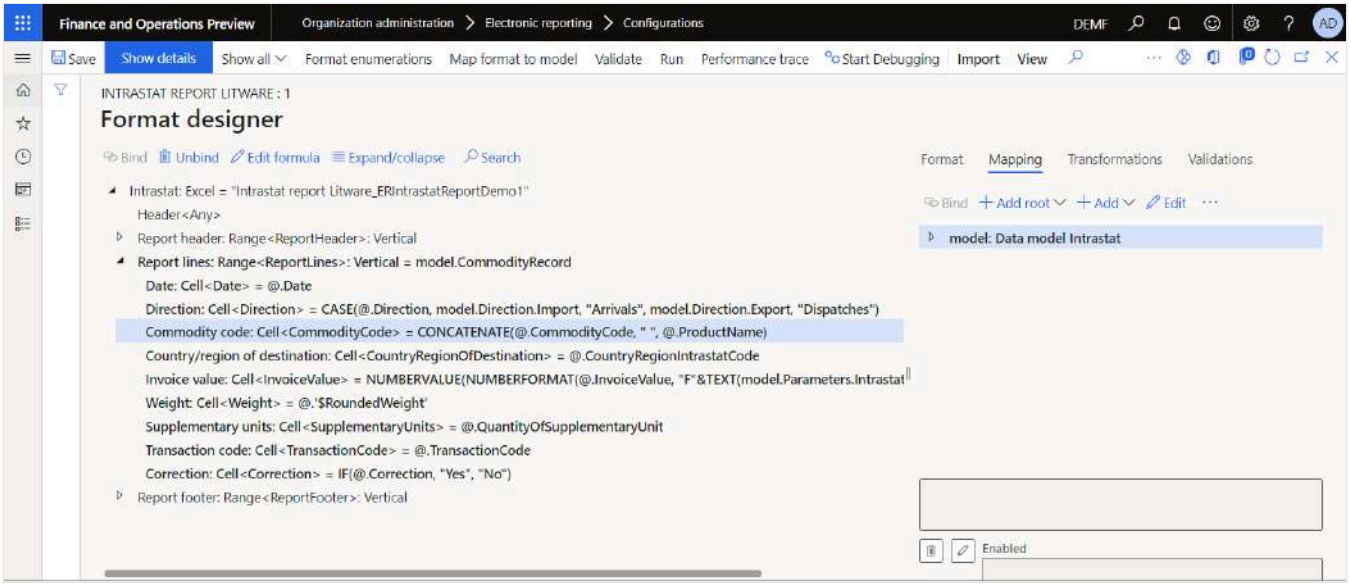
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, in the configuration tree in the left pane, expand **Intrastat model** > **Intrastat report**, and select the **Intrastat report Litware** configuration.
3. Select **Designer**.
4. On the **Format designer** page, on the Action Pane, select **Show details**.
5. Make sure that the **Intrastat: Excel** root format component is selected, and then, on the Action Pane, on the **Import** tab, in the **Import** group, select **Update from Excel**.
6. In the **Update from Excel** dialog box, select **Update template**.
7. In the **Open** dialog box, browse to and select the **ERIntrastatReportDemo1.xlsx** file that you downloaded earlier, and then select **Open**.

8. Select **OK**.
9. Select **Save**.



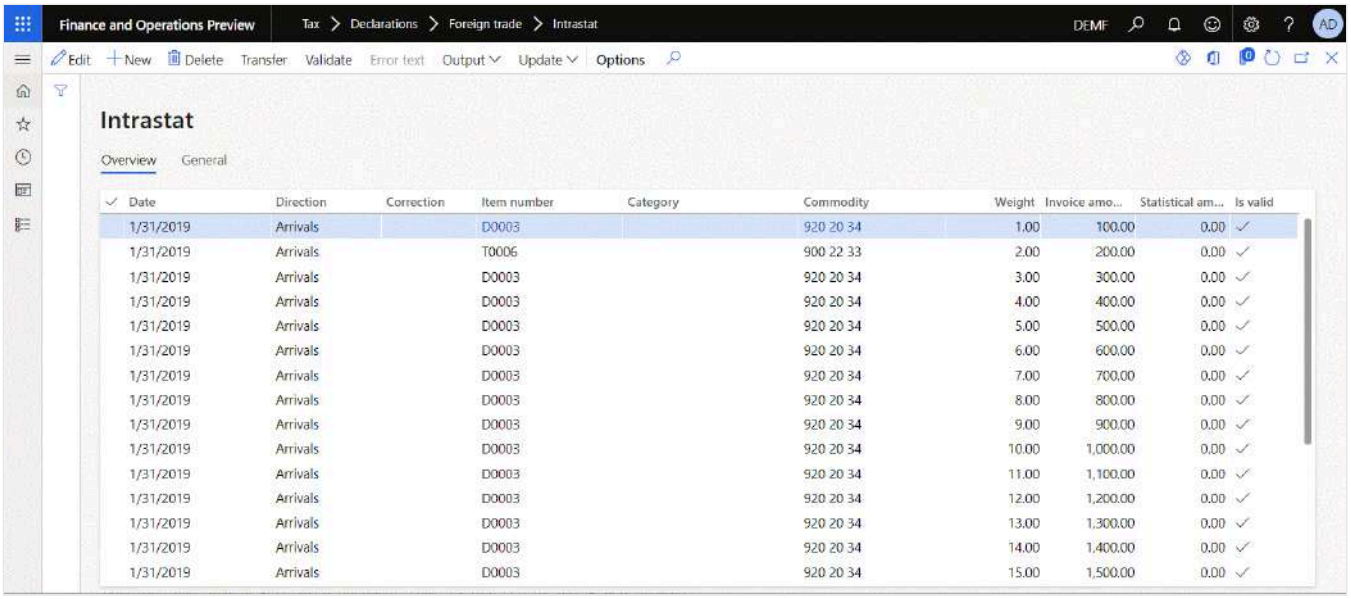
Change the data binding to show the item description on a generated report

1. On the **Format designer** page, select the **Mapping** tab.
2. Expand **Intrastat** > **Report lines**, and select the **Commodity code** component.
3. Select **Edit formula**.
4. Change the binding formula from `@.CommodityCode` to `CONCATENATE(@.CommodityCode, " ", @.ProductName)`.
5. Select **Save**.



Generate an Intrastat declaration control report

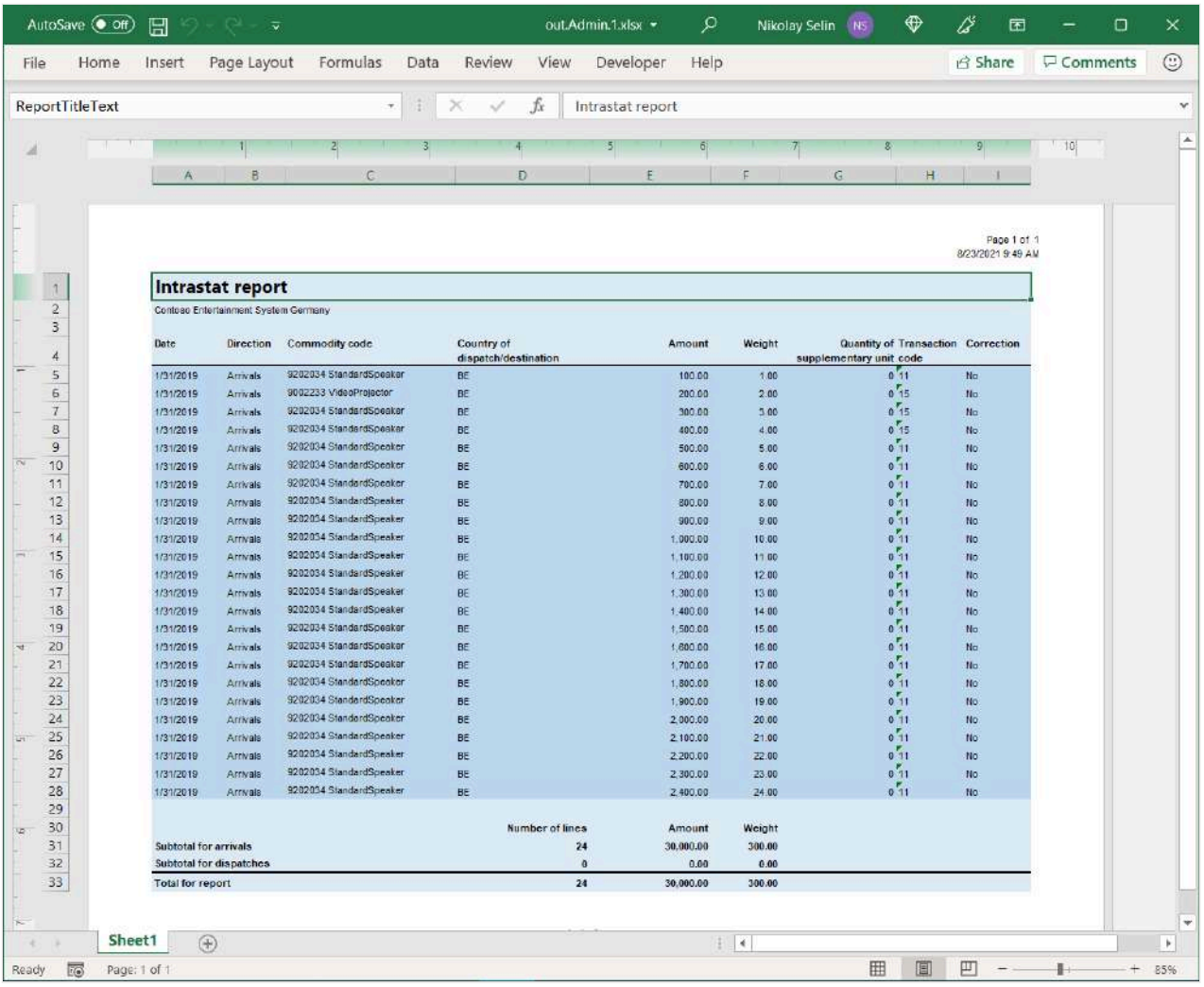
First, make sure that you have Intrastat transactions for reporting on the **Intrastat** page.



Date	Direction	Correction	Item number	Category	Commodity	Weight	Invoice amo...	Statistical am...	Is valid
1/31/2019	Arrivals		D0003		920 20 34	1.00	100.00	0.00	✓
1/31/2019	Arrivals		T0006		900 22 33	2.00	200.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	3.00	300.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	4.00	400.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	5.00	500.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	6.00	600.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	7.00	700.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	8.00	800.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	9.00	900.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	10.00	1,000.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	11.00	1,100.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	12.00	1,200.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	13.00	1,300.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	14.00	1,400.00	0.00	✓
1/31/2019	Arrivals		D0003		920 20 34	15.00	1,500.00	0.00	✓

Then use the custom ER format to generate the control report of the Intrastat declaration.

1. Go to **Tax** > **Declarations** > **Foreign trade** > **Intrastat**.
2. On the **Intrastat** page, on the Action Pane, select **Output** > **Report**.
3. In the **Intrastat Report** dialog box, follow these steps to run the report:
   a. Set the **From date** and **To date** fields to include specific Intrastat transactions to the report.
   b. Set the **Generate file** option to **No**.
   c. Set the **Generate report** option to **Yes**.
   d. Select **OK**.
4. Download and save the document that is generated.
5. Open the document in Excel, and review it.



Configure the custom format to paginate generated documents

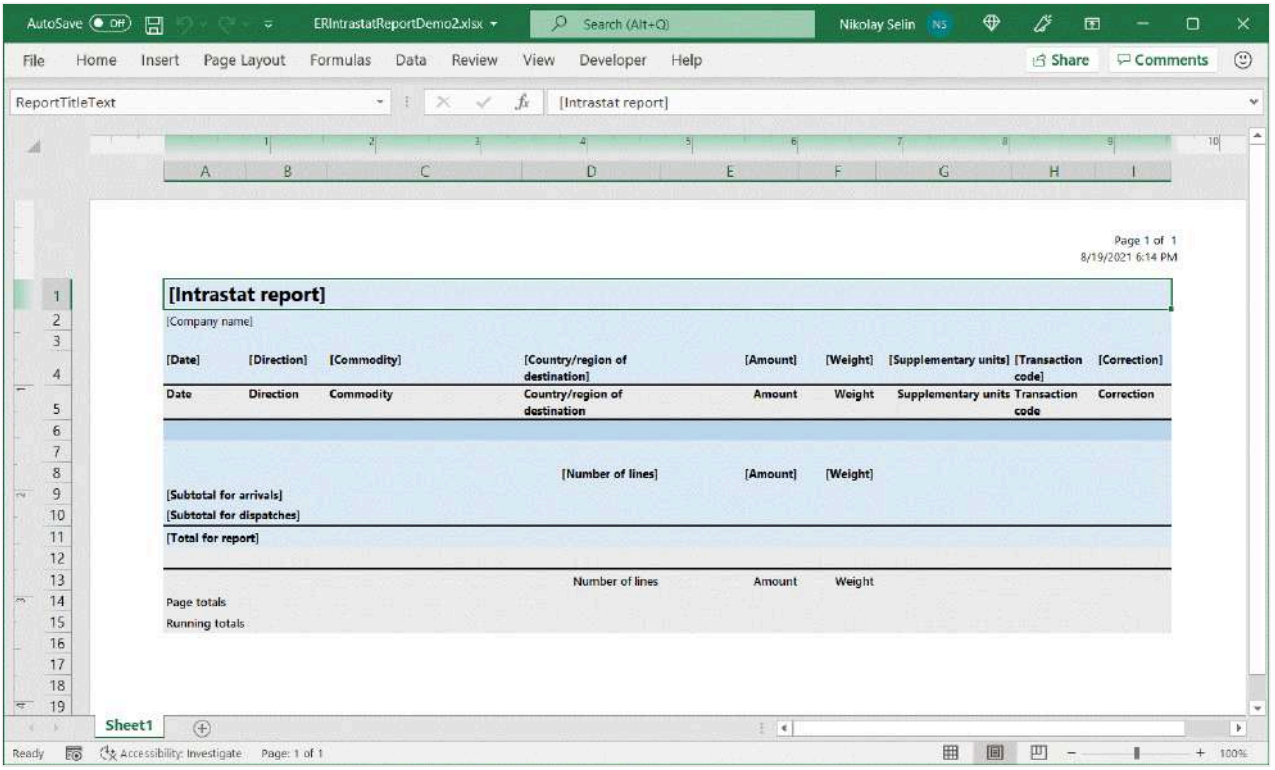
Review the second downloaded Excel template

1. In Excel, open the **ERIntrastatReportDemo2.xlsx** template file that you downloaded earlier.

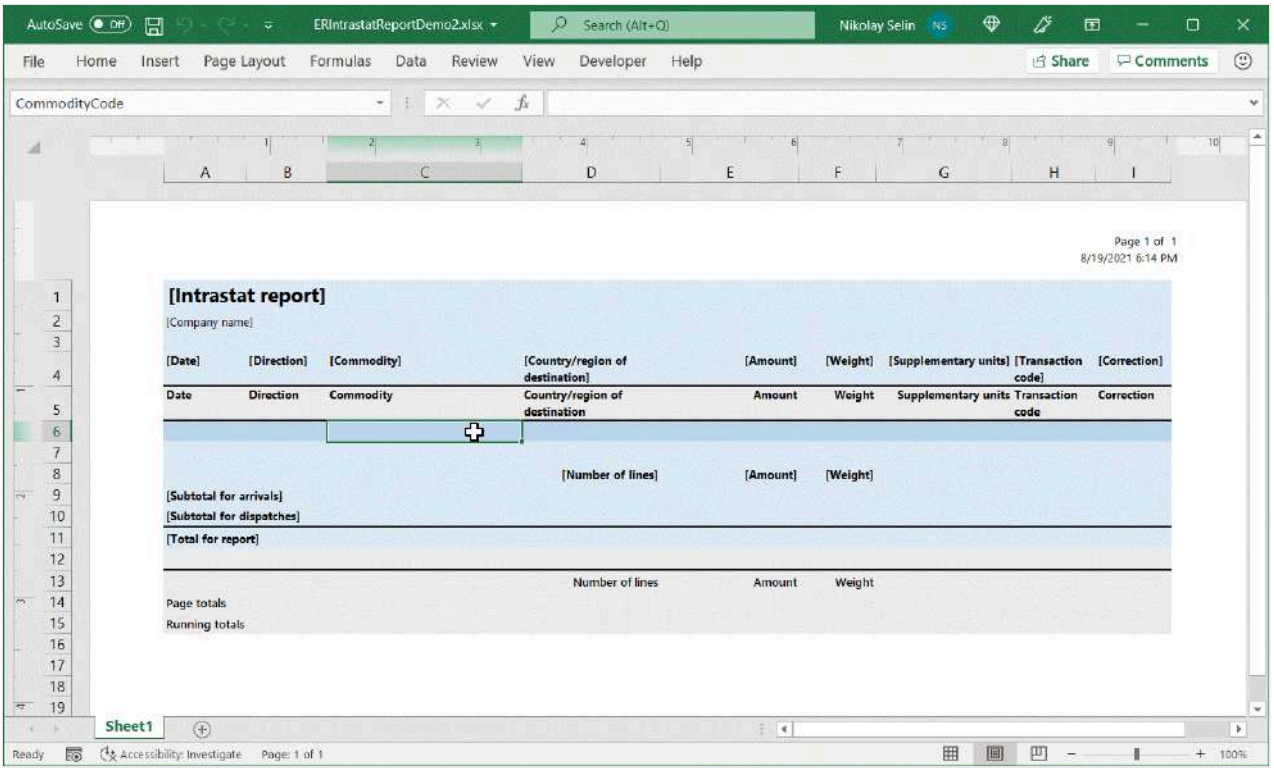
2. Compare this template with the **ERIntrastatReportDemo1.xlsx** template, and verify that it contains several new Excel names to create and fill in page-specific sections in generated documents:

    - The **ReportPageHeader** range is added to create page headers.
    - The **ReportPageFooter** range is added to create page footers.
    - The **ReportPageFooter_PageLines** cell is configured to show the number of transactions per page.
    - The **ReportPageFooter_PageAmount** cell is configured to show the total amount of transactions per page.

- The **ReportPageFooter_PageWeight** cell is configured to show the total weight of transactions per page.
- The **ReportPageFooter_RunningCounterLines** cell is configured to show the running counter of transactions from the beginning of the report through the current page.
- The **ReportPageFooter_RunningTotalAmount** cell is configured to show the amount running total for all transactions from the beginning of the report through the current page.
- The **ReportPageFooter_RunningTotalWeight** cell is configured to show the weight running total for transactions from the beginning of the report through the current page.



The **CommodityCode** cell of this template is configured to wrap cell text. Because the transaction details row is configured so that it will automatically fit the height of a row, the height of the whole row must automatically change when the text in the **CommodityCode** cell is wrapped.



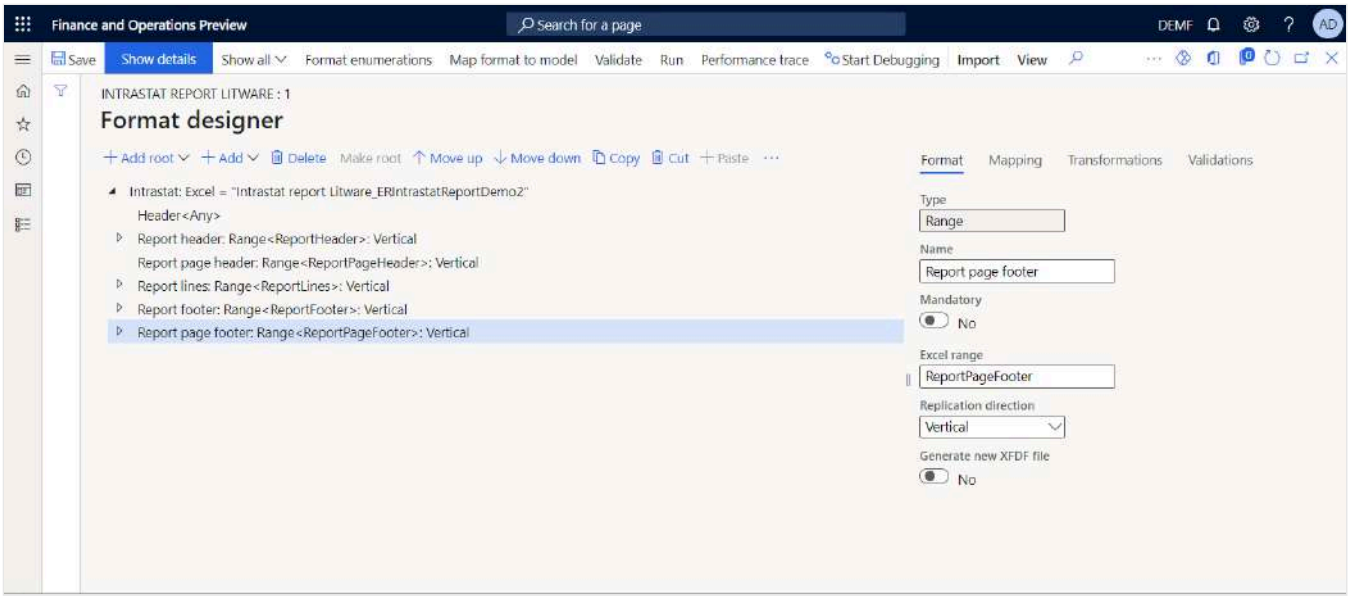
Repeat the replacement of the current Excel template in the custom ER format

1. Follow the steps in the Replace the current Excel template in the custom ER format section of this article. However, in step 7, select the **ERIntrastatReportDemo2.xlsx** file.

2. On the **Format designer** page, expand **Intrastat**.

3. Name the Range format components that have been added to the editable ER format to sync the structure with the structure of the applied Excel template:
   a. Select the **Range** component that is associated with the Excel name **ReportPageHeader**.
   b. On the **Format** tab, in the **Name** field, enter **Report page header**.
   c. Select the **Range** component that is associated with the Excel name **ReportPageFooter**.
   d. On the **Format** tab, in the **Name** field, enter **Report page footer**.

4. Select **Save**.



Change the format structure to implement document pagination

1. On the **Format designer** page, in the format tree in the left pane, select the **Intrastat** root component.

2. Select **Add**.

3. In the **Add** dialog box, select the Page component in the **Excel** group of components.

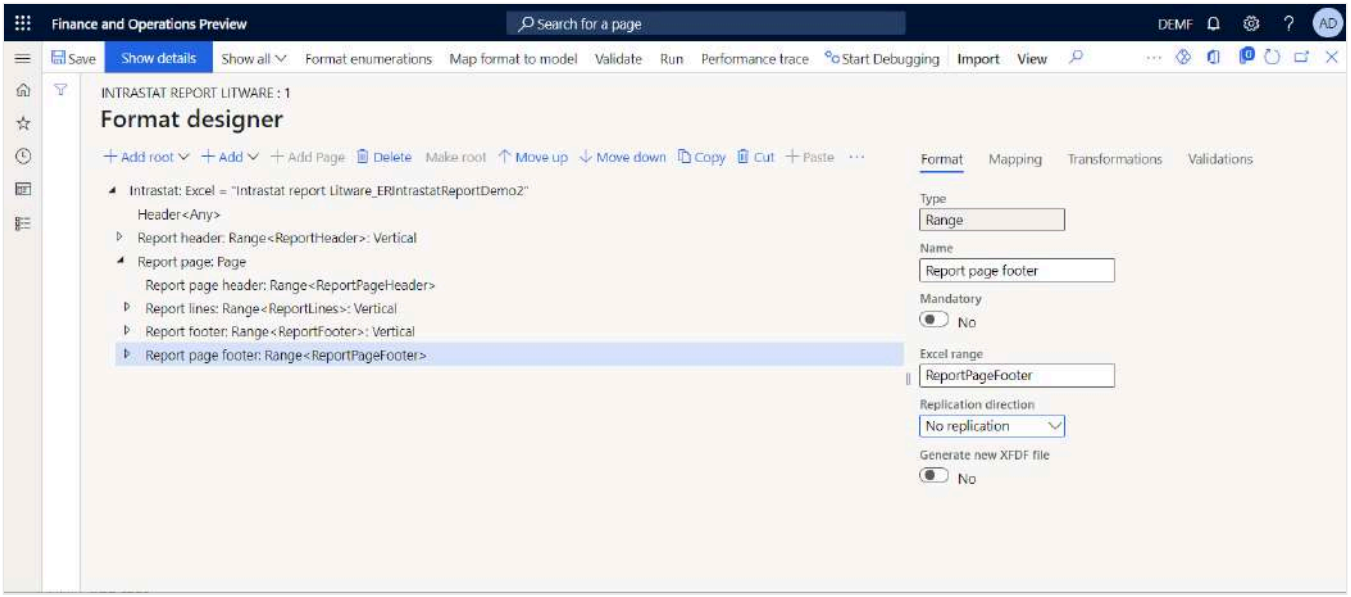
4. In the **Component properties** dialog box, in the **Name** field, enter **Report page**. Then select **OK**.

5. To use the **Report page header** component as a page header on every generated page, follow these steps:
   a. Select the **Report page header** component, and then select **Cut**.
   b. Select the **Report page** component, and then select **Paste**.
   c. Expand **Report page**.
   d. To force the **Page** component to consider this range a page header, select **Report page header**, and then, on the **Format** tab, in the **Replication direction** field, select **No replication**.

6. To paginate a generated document so that the content on report lines is considered, follow these steps:
   a. Select the **Report lines** component, and then select **Cut**.
   b. Select the **Report page** component, and then select **Paste**.

7. To include the report footer after report lines but before the final page footer, follow these steps:
   a. Select the **Report footer** component, and then select **Cut**.
   b. Select the **Report page** component, and then select **Paste**.

8. To use the **Report page footer** component as a page footer on every generated page, follow these steps:
   a. Select the **Report page footer** component, and then select **Cut**.
   b. Select the **Report page** component, and then select **Paste**.
   c. To force the **Page** component to consider this range a page footer, select **Report page footer**, and then, on the **Format** tab, in the **Replication direction** field, select **No replication**.



Add data sources to calculate page footer totals

You must configure new data sources to calculate the page total, the running counter, and the running total values, and to show them in the page footer section. We recommend that you use Data collection data sources for this purpose.

1. On the **Format designer** page, select the **Mapping** tab.

2. Select **Add root**, and then follow these steps:
   a. In the **Add data source** dialog box, in the **General** section, select **Empty container**.
   b. In the **'Empty container' data source properties** dialog box, in the **Name** field, enter **Total**.
   c. Select **OK**.

3. Select the **Total** data source, select **Add**, and then follow these steps:

a. In the **Add data source** dialog box, in the **General** section, select **Empty container**.
b. In the **'Empty container' data source properties** dialog box, in the **Name** field, enter **Page**.
c. Select **OK**.

4. Select **Add** again, and then follow these steps:
   a. In the **Add data source** dialog box, in the **General** section, select **Empty container**.
   b. In the **'Empty container' data source properties** dialog box, in the **Name** field, enter **Running**.
   c. Select **OK**.

5. Select the **Total.Page** data source, select **Add**, and then follow these steps:
   a. In the **Add data source** dialog box, in the **Functions** section, select **Data collection**.
   b. In the **'Data collection' data source properties** dialog box, in the **Name** field, enter **Amount**.
   c. In the **Item type** field, select **Real**.
   d. Set the **Collect all values** option to **Yes**.
   e. Select **OK**.

6. Select **Add** again, and then follow these steps:
   a. In the **Add data source** dialog box, in the **Functions** section, select **Data collection**.
   b. In the **'Data collection' data source properties** dialog box, in the **Name** field, enter **Weight**.
   c. In the **Item type** field, select **Real**.
   d. Set the **Collect all values** option to **Yes**.
   e. Select **OK**.

7. Select the **Total.Running** data source, select **Add**, and then follow these steps:
   a. In the **Add data source** dialog box, in the **Functions** section, select **Data collection**.
   b. In the **'Data collection' data source properties** dialog box, in the **Name** field, enter **Amount**.
   c. In the **Item type** field, select **Real**.
   d. Set the **Collect all values** field to **Yes**.
   e. Select **OK**.

8. Select **Add** again, and then follow these steps:
   a. In the **Add data source** dialog box, in the **Functions** section, select **Data collection**.
   b. In the **'Data collection' data source properties** dialog box, in the **Name** field, enter **Weight**.

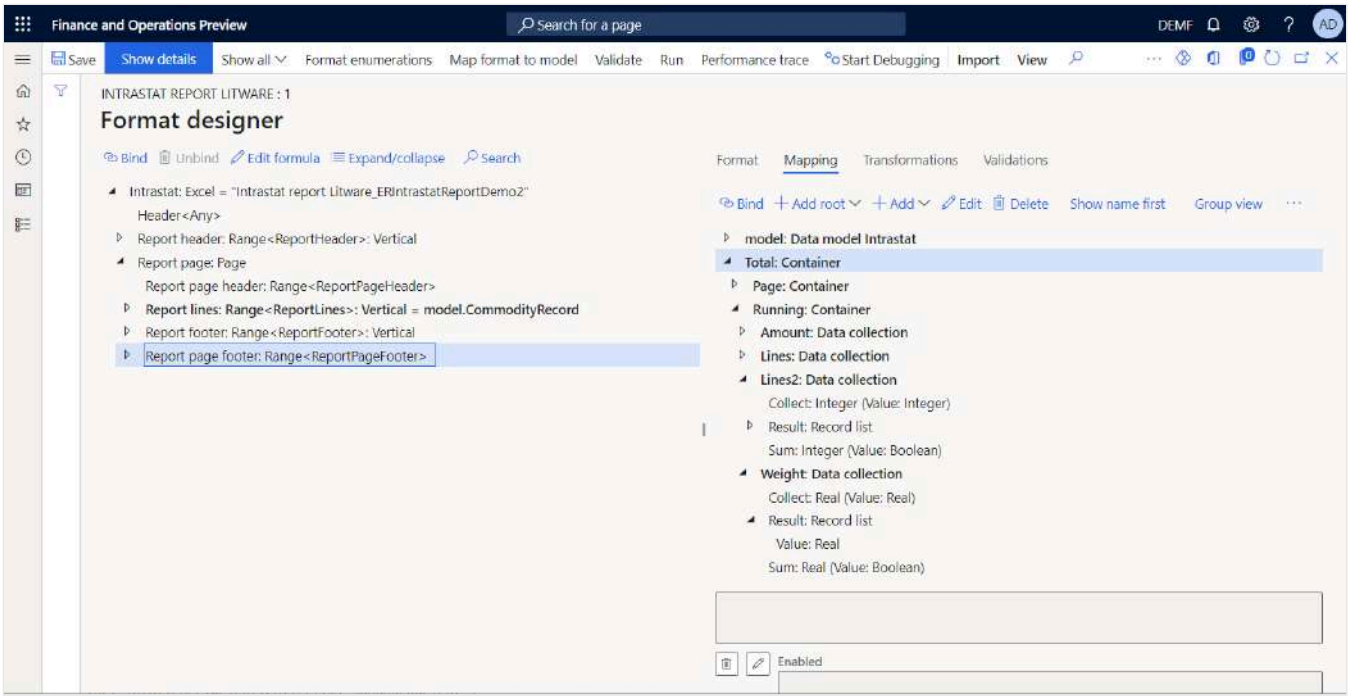
c. In the **Item type** field, select **Real**.
d. Set the **Collect all values** field to **Yes**.
e. Select **OK**.

9. Select **Add** again, and then follow these steps:
   a. In the **Add data source** dialog box, in the **Functions** section, select **Data collection**.
   b. In the **'Data collection' data source properties** dialog box, in the **Name** field, enter **Lines**.
   c. In the **Item type** field, select **Integer**.
   d. Set the **Collect all values** field to **Yes**.
   e. Select **OK**.
10. Select **Save**.

Add data sources to control page footer visibility

If you plan to control page footer visibility, and you don't plan to include the footer on the final page if it contains transactions, configure a new data source to calculate the required running counter.

1. On the **Format designer** page, select the **Mapping** tab.
2. Select the **Total.Running** data source, select **Add**.
3. In the **Add data source** dialog box, in the **Functions** section, select **Data collection**.
4. In the **'Data collection' data source properties** dialog box, in the **Name** field, enter **Lines2**.
5. In the **Item type** field, select **Integer**.
6. Set the **Collect all values** option to **Yes**.
7. Select **OK**.
8. Select **Save**.



Configure bindings to collect total values

1. On the **Format designer** page, in the format tree, expand the **Report lines** component, and select the nested **Invoice value** component.

2. Select **Edit formula**.

3. Change the binding formula from `NUMBERVALUE(NUMBERFORMAT(@.InvoiceValue, "F"&TEXT(model.Parameters.IntrastatAmountDecimals)), ".", "")` to `Total.Page.Amount.Collect(NUMBERVALUE(NUMBERFORMAT(@.InvoiceValue, "F"&TEXT(model.Parameters.IntrastatAmountDecimals)), ".", ""))`.

   > **Note**
   >
   > In addition to putting the amount value in an Excel cell for every iterated transaction, this binding collects the value in the data collection **Total.Page.Amount** data source.

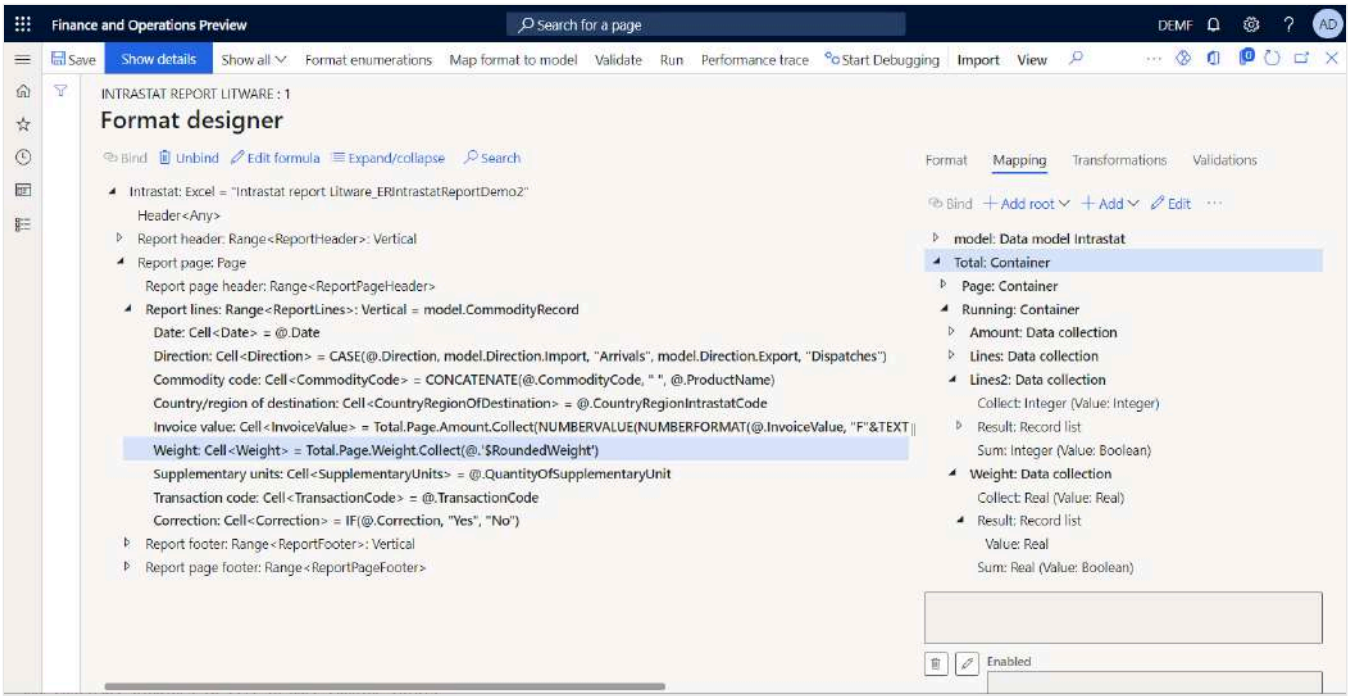
4. Select the nested **Weight** component.

5. Select **Edit formula**.

6. Change the binding formula from `@.'$RoundedWeight'` to `Total.Page.Weight.Collect(@.'$RoundedWeight')`.

   > **Note**

In addition to putting the weight value in an Excel cell for every iterated transaction, this binding collects the value in the **Total.Page.Weight** data source.



Configure bindings to fill in page footer totals

1. On the **Format designer** page, in the format tree, expand the **Report page footer** component, select the nested **Range** component that refers to the Excel **ReportPageFooter_PageAmount** cell, and then follow these steps:
   a. In the data sources tree in the right pane, select the **Total.Page.Amount.Sum()** item.
   b. Select **Bind**.
   c. Select **Edit formula**.
   d. Update the formula to `Total.Page.Amount.Sum(false)`.

> **Note**
>
> You must specify the argument of this function as **False** to keep collected data for the current page, because this data is required to calculate the amount running total, total number of lines per page, and running counter of lines.

2. In the format tree, select the nested **Range** component that refers to the Excel **ReportPageFooter_PageWeight** cell, and then follow these steps:
   a. In the data sources tree in the right pane, select the **Total.Page.Weight.Sum()** item.
   b. Select **Bind**.
   c. Select **Edit formula**.

d. Update the formula to `Total.Page.Weight.Sum(false)`.

Configure bindings to fill in page running totals

1. On the **Format designer** page, in the format tree, expand the **Report page footer** component, select the nested **Range** component that refers to the Excel **ReportPageFooter_RunningTotalAmount** cell, and then follow these steps:
   a. In the data sources tree in the right pane, select the **Total.Running.Amount.Collect()** item.
   b. Select **Bind**.
   c. Select **Edit formula**.
   d. Update the formula to `Total.Running.Amount.Sum(false)+Total.Running.Amount.Collect(Total.Page.Amount.Sum(true))`.

   > ⓘ **Note**
   >
   > The `Total.Running.Amount.Sum(false)` operand returns the previously collected amount running total. The `Total.Running.Amount.Collect(Total.Page.Amount.Sum(true))` operand returns the total amount of the current page. You must specify the argument of the nested function of the second operand as **True** to reset the `Total.Page.Amount` data collection as soon as this value is put in the `Total.Running.Amount` running total collection. The specified argument must start to collect the next page total from a 0 (zero) value.
   >
   > The `Total.Running.Amount.Sum(false)` function is called to enter the amount running total in the Excel **ReportPageFooter_RunningTotalAmount** cell on the current page.

2. In the format tree, select the nested **Range** component that refers to the Excel **ReportPageFooter_RunningTotalWeight** cell, and then follow these steps:
   a. In the data sources tree in the right pane, select **Total.Running.Weight.Collect()** item.
   b. Select **Bind**.
   c. Select **Edit formula**.
   d. Update the formula to `Total.Running.Weight.Sum(false)+Total.Running.Weight.Collect(Total.Page.Weight.Sum(true))`.

Configure bindings to fill in the page running counter

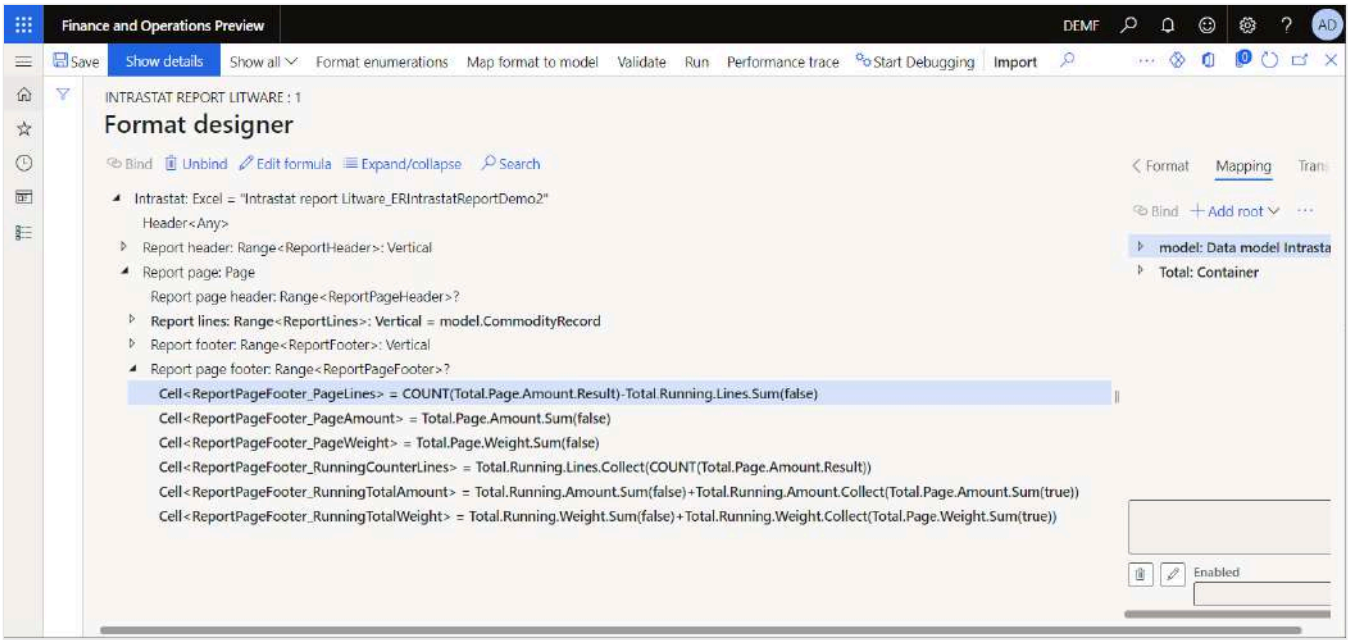
1. On the **Format designer** page, in the format tree, expand the **Report page footer** component, and select the nested **Range** component that refers to the Excel **ReportPageFooter_RunningCounterLines** cell.
2. Select **Edit formula**.
3. Add the formula `Total.Running.Lines.Collect(COUNT(Total.Page.Amount.Result))`.

> ⓘ Note
>
> This formula returns the number of collected amount values for the whole report. This number equals the number of transactions that have been iterated at the current moment. At the same time, the formula collects the returned value in the **Total.Running.Lines** collection.

Configure bindings to fill in the page footer counter

1. On the **Format designer** page, in the format tree, expand the **Report page footer** component, and select the nested **Range** component that refers to the Excel **ReportPageFooter_PageLines** cell.
2. Select **Edit formula**.
3. Add the formula `COUNT(Total.Page.Amount.Result)-Total.Running.Lines.Sum(false)`.

> ⓘ Note
>
> This formula calculates the number of transactions on the current page as the difference between the number of transactions that was collected in **Total.Page.Amount.Result** for the whole report and the number of transactions that has been stored at this stage in **Total.Running.Lines.Sum**. Because the number of transactions for the current page is stored in **Total.Running.Lines** in the binding of the **Range** component that refers to the Excel **ReportPageFooter_RunningCounterLines** cell, the number of transactions on the current page isn't yet included. Therefore, this difference equals the number of transactions on the current page.



Configure component visibility

You can change the visibility of the page header and footer on a specific page of a generated document to hide the following elements:

- The page header on the first page, because the report header already contains column titles
- The page header on any page that doesn't have transactions that can occur for the last page
- The page footer on any page that doesn't have transactions that can occur for the last page

To change the visibility, update the **Enabled** property of the **Report page header** and **Report page footer** components.

1. On the **Format designer** page, in the format tree, expand the **Report page** component, select the nested **Report page header** component, and then follow these steps:

   a. Select **Edit** for the **Enabled** field.

   b. On the **Formula designer** page, in the **Formula** field, enter the following expression:

   ```
   AND(
   COUNT(Total.Page.Amount.Result)<>0,
   COUNT(Total.Page.Amount.Result)<>COUNT(model.CommodityRecord)
   )
   ```

2. In the format tree, select the nested **Report page footer** component, and then follow these steps:

   a. Select **Edit** for the **Enabled** field.

   b. On the **Formula designer** page, in the **Formula** field, enter the following expression:

```
(
COUNT(Total.Page.Amount.Result)-Total.Running.Lines2.Sum(false)+
0*Total.Running.Lines2.Collect(COUNT(Total.Page.Amount.Result))
)<>0
```

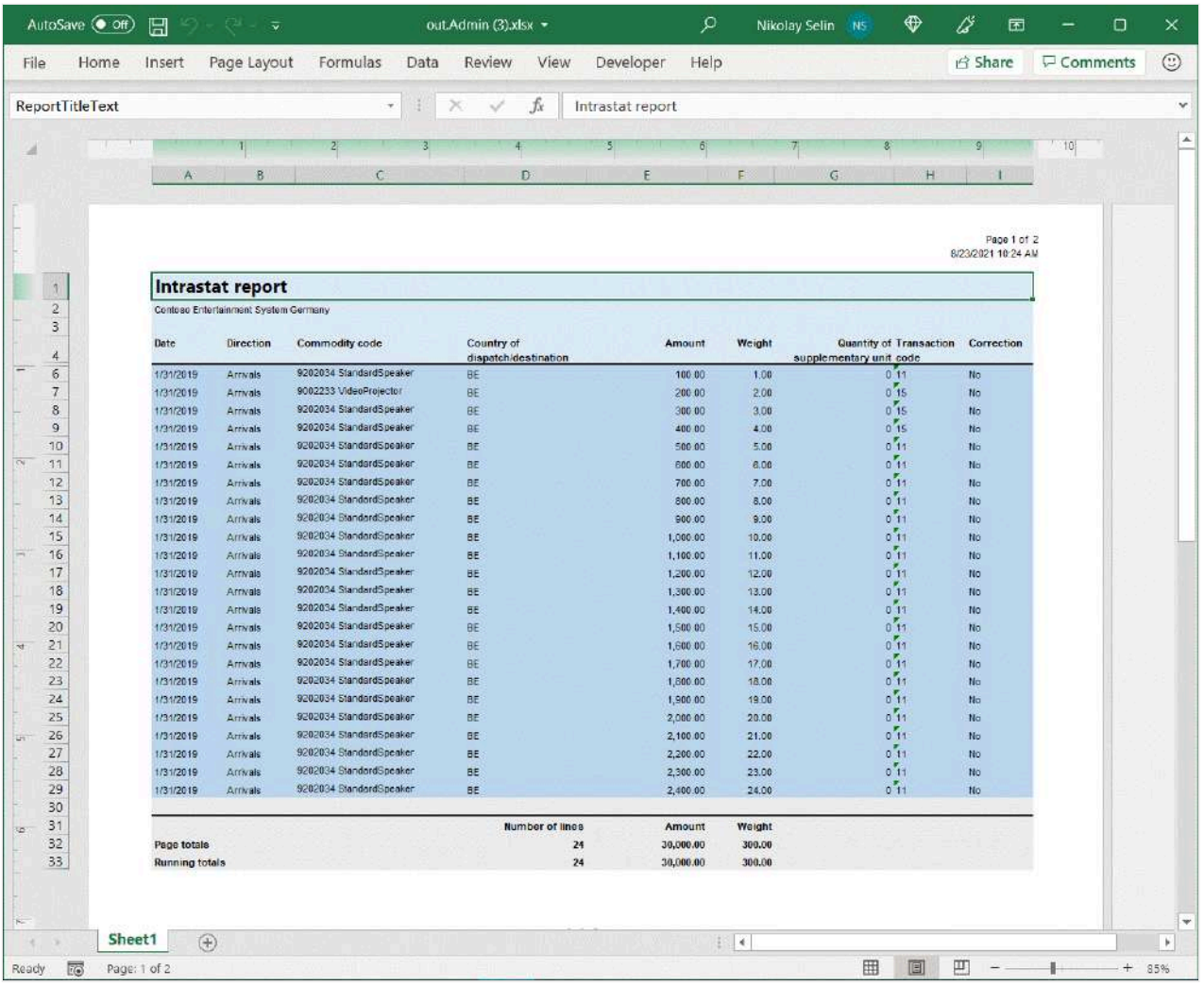
> **Note**
>
> The `COUNT(Total.Page.Amount.Result)-Total.Running.Lines2.Sum(false)` construction is used to calculate the number of transactions on the current page. The `0*Total.Running.Lines2.Collect(COUNT(Total.Page.Amount.Result)` construction is used to add the number of transactions on the current page to the collection, to correctly handle the visibility of the next page footer.
>
> The `Total.Running.Lines` collection can't be reused here, because the **Enabled** property of a base component is processed **after** the bindings of nested components are processed. When the **Enabled** property is processed, the `Total.Running.Lines` collection is already incremented by the number of transactions on the current page.

3. Select **Save**.

Generate an Intrastat declaration control report (updated)

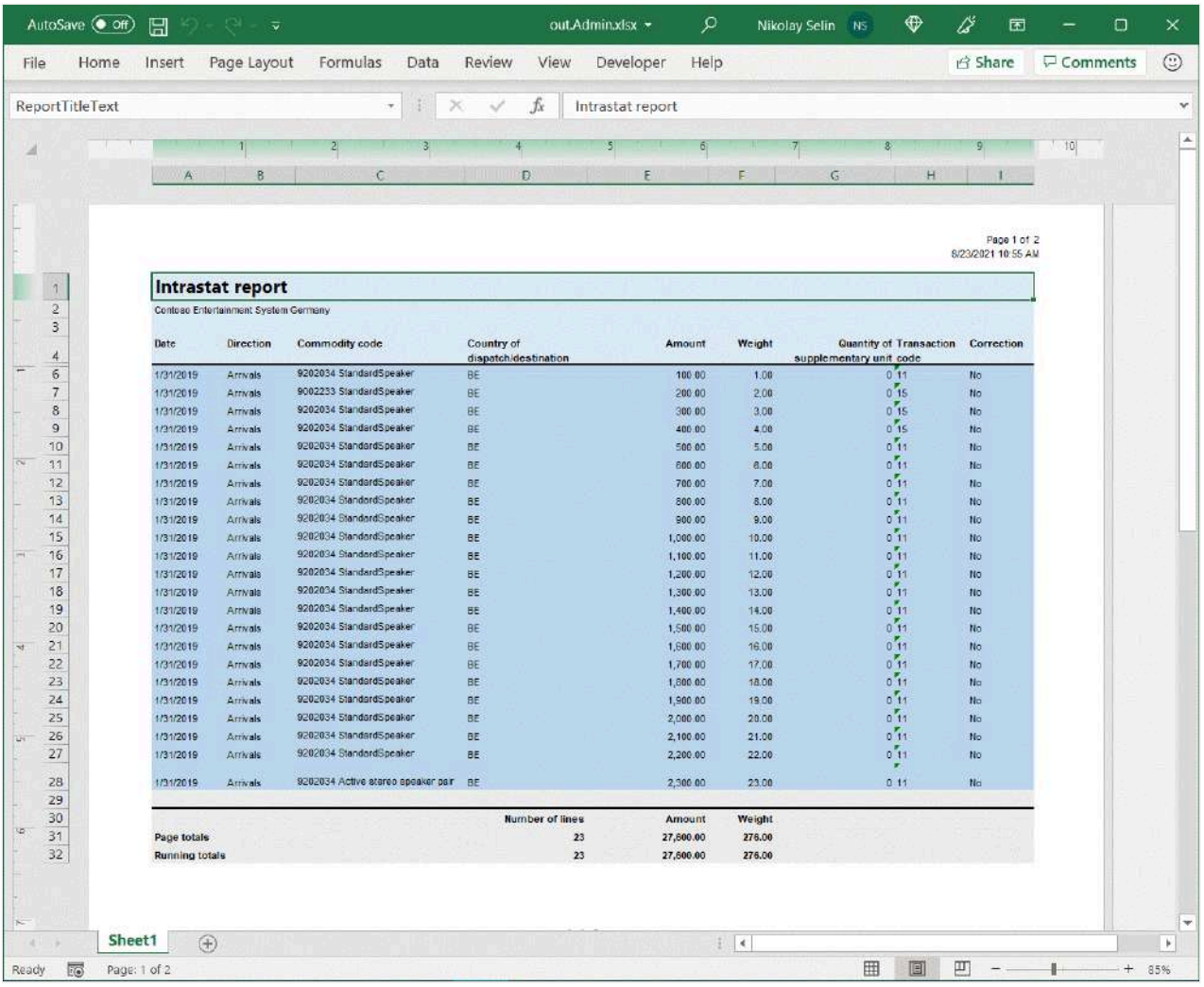
1. Make sure that you have 24 transactions on the **Intrastat** page. Repeat the steps in the Generate an Intrastat declaration control report section of this article to generate and review the control report.

   All transactions are presented on the first page. The page totals and counters equal the report totals and counters. The page header range is hidden on the first page because the report header already contains column titles. The page header and footer are hidden on the second page because that page contains no transactions.



2. Update two transactions on the **Intrastat** page by changing the **Item number** code from **D00006** to **L0010**. Notice that the product name of the new item, **Active stereo speaker pair**, is longer than the product name of the original item, **Standard speaker**. This situation forces text wrapping in the corresponding cell of the generated document. Document pagination, and page-related summing and counting, must now be updated. Repeat the steps in the Generate an Intrastat declaration control report section to generate and review the control report.

   Currently, transactions are presented on two pages, and page totals and counters are correctly calculated. The page header range is correctly hidden on the first page and visible on the second page. The page footer is visible on both pages because they contain transactions.



Frequently asked questions

Is there any way to recognize when the final page is processed by the Page format component?

The **Page** component doesn't expose information about the number of the processed page and the total number of pages in a generated document. Nevertheless, you can configure ER formulas to recognize the final page. Here is an example:

- Calculate the total number of transactions that have already been processed by using the **Report page** component. You can do this calculation by using the formula `COUNT(Total.Page.Amount.Result)`.
- Calculate the total number of transactions that must be processed based on the `model.CommodityRecord` binding that is configured for the **Report lines** component. You can do this calculation by using the formula `COUNT(model.CommodityRecord)`.
- Compare two numbers to recognize the final page. When both values are equal, the final page is generated.

> **Note**
>
> We recommend that you use this approach only when the **Enabled** property of the **Report lines** component contains no formula that might return False at runtime for some of the iterated records of the bound Record list.

Additional resources

- Design a configuration for generating documents in Excel format
- Use DATA COLLECTION data sources in Electronic reporting (ER) formats

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Last updated on 04/08/2026

Design an ER format to keep rows together on the same Excel page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user in the System Administrator or Electronic Reporting Functional Consultant role can configure an Electronic reporting (ER) format that generates outbound documents in Microsoft Excel and manage document pagination so that rows that are created stay on the same page.

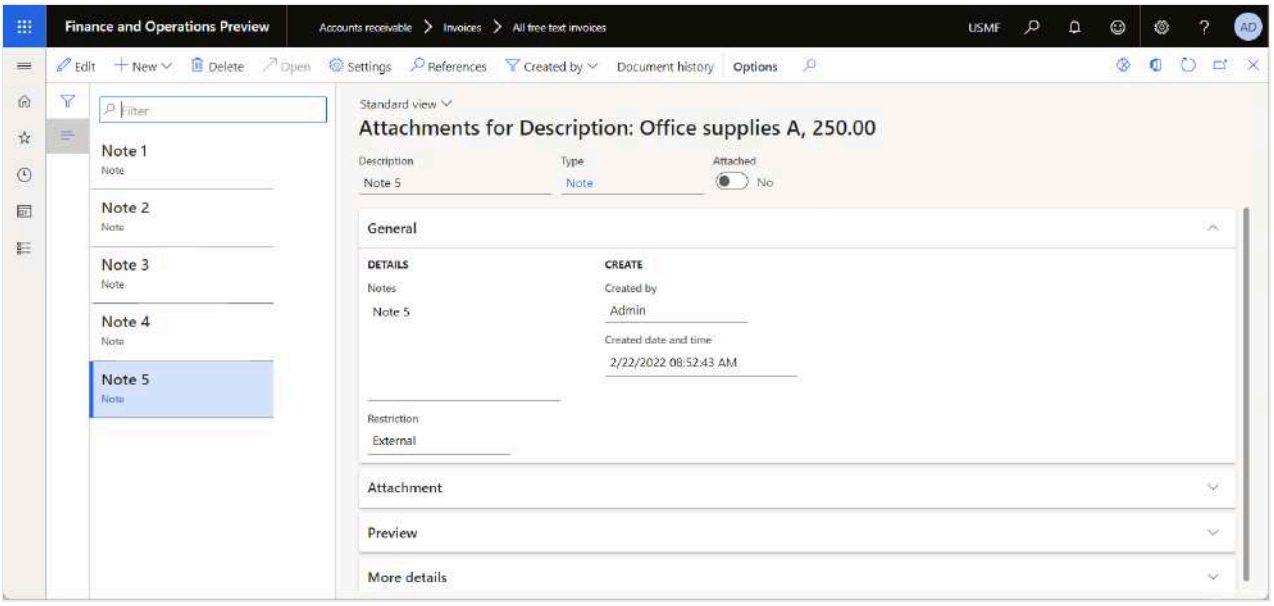
In this example, you modify the Microsoft-provided ER format that's used to print free text invoices in Excel. Your modifications let you manage the pagination of a generated free text invoice report so that all the rows of a single invoice line stay on the same page when possible.

You can complete the procedures in this article in the **USMF** company. No coding is required.

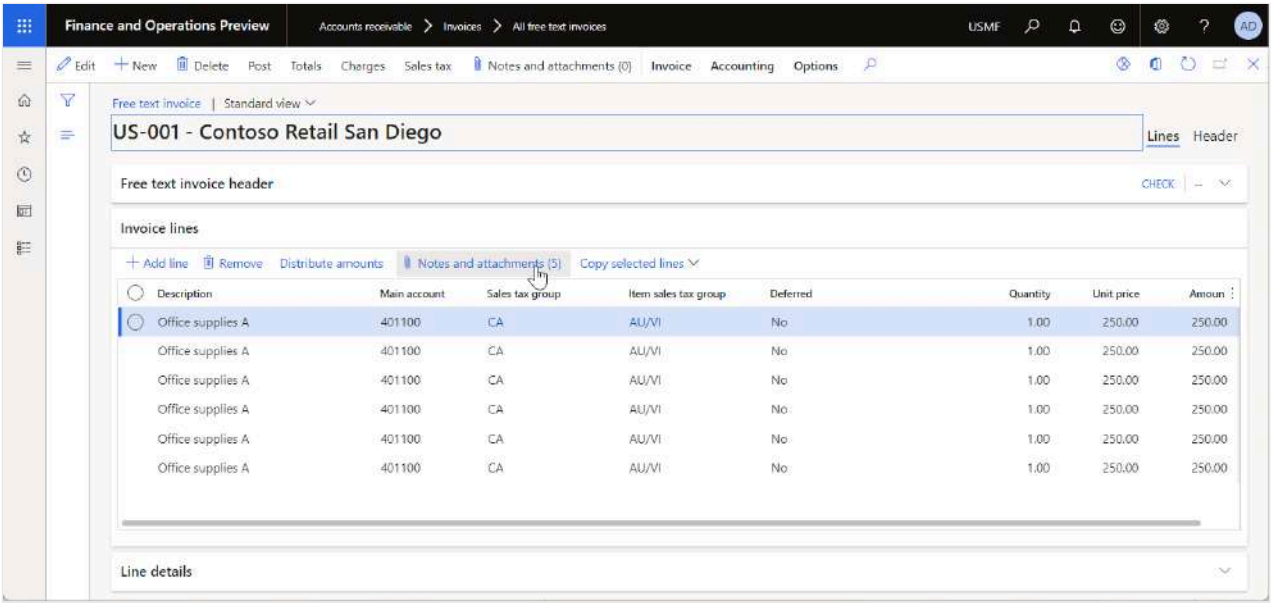
In this example, you create the required ER configurations for the **Litware, Inc.** sample company. Make sure that the configuration provider for the **Litware, Inc.** (`http://www.litware.com`) sample company is listed for the ER framework, and that it's marked as **Active**. If this configuration provider isn't listed, or if it isn't marked as **Active**, follow the steps in Create a configuration provider and mark it as active.

Enter a new free text invoice

1. Follow the steps in Create a free text invoice to add a free text invoice.
   a. Add a single line to the invoice.
   b. Add five notes for the invoice line.



2. Follow the steps in Copy lines to create five additional invoice lines that are copies of the invoice line that you added in the previous step.



Configure the ER framework

Follow the steps in Configure the ER framework to set up the minimal set of ER parameters. You must complete this setup before you start to use the ER framework to design a custom version of a standard ER format.

Import the standard ER format configuration

Follow the steps in Import the standard ER format configuration to add the standard ER configurations to your current instance of Dynamics 365 Finance. For example, import version **252.116** of the **Free text invoice (Excel)** format configuration. The base version **252** of the base

Invoice model configuration is automatically imported from the repository together with the required **Invoice model mapping** configuration.

Set up print management to use the standard ER format

Follow the steps in Set up print management to configure print management for the **Accounts receivable** module so that the standard ER format is used to print free text invoices.

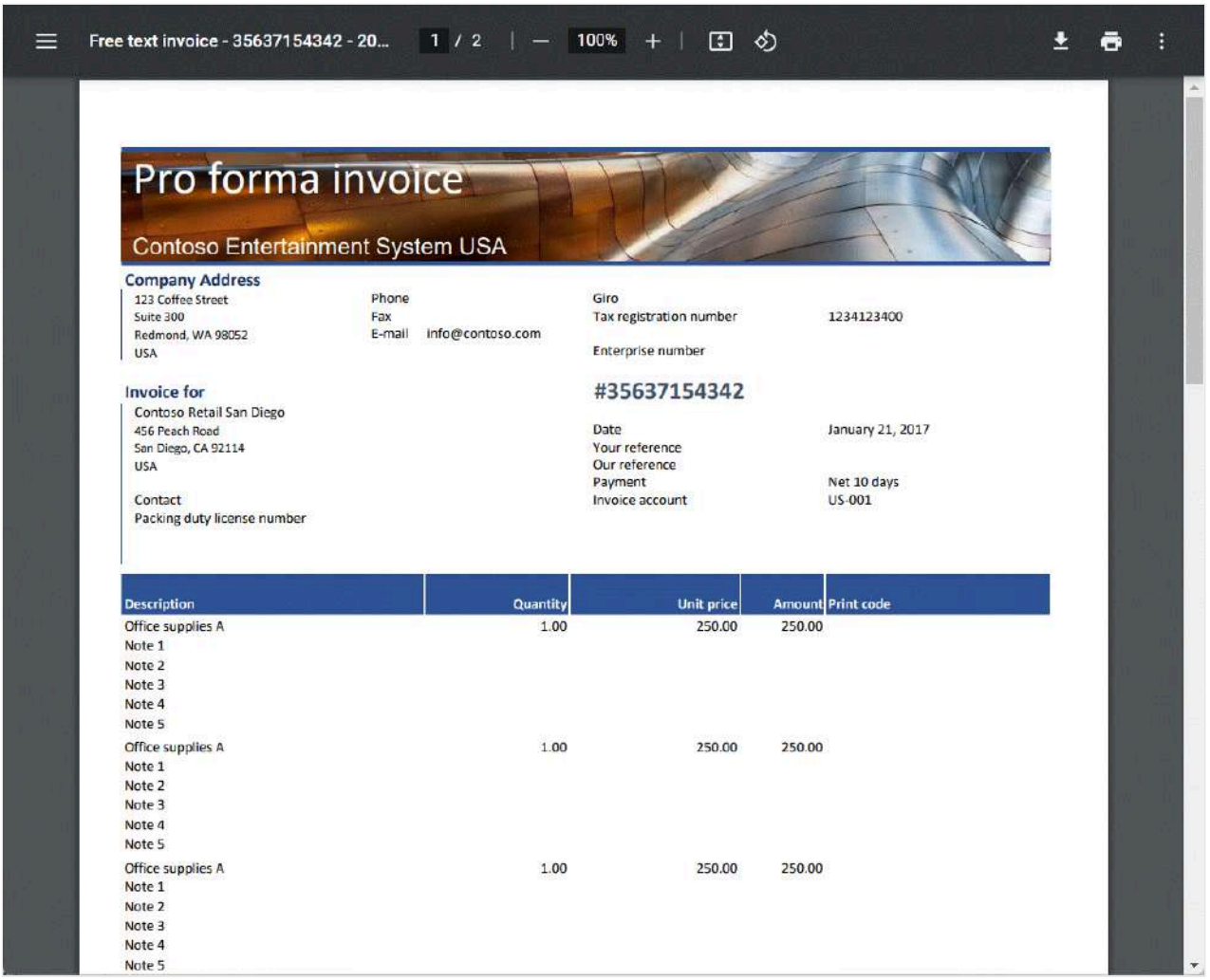
Configure a format destination for the standard ER format

Follow the steps in Configure a format destination for on-screen preview to configure the Screen ER destination of the standard ER format so that generated reports are converted to PDF format and previewed on a new browser tab.

Print a free text invoice by using the standard ER format

1. Follow the steps in Print a free text invoice to use the standard ER format to generate a free text invoice report in Excel format for the added invoice.
2. Download the generated Excel workbook, and review it in the Excel desktop application.

   Notice that the sixth line of the invoice starts on the first page of the report and continues on the second page. The last note appears on the second page, and it isn't obvious that it belongs to the sixth invoice line. Therefore, the page break in the middle of the content for the invoice line makes this document more difficult to read.



The remaining procedures in this article show how you can tune the standard ER format to improve the look and readability of the invoice report by keeping all the content for a single invoice line on the same page.

Create a custom format

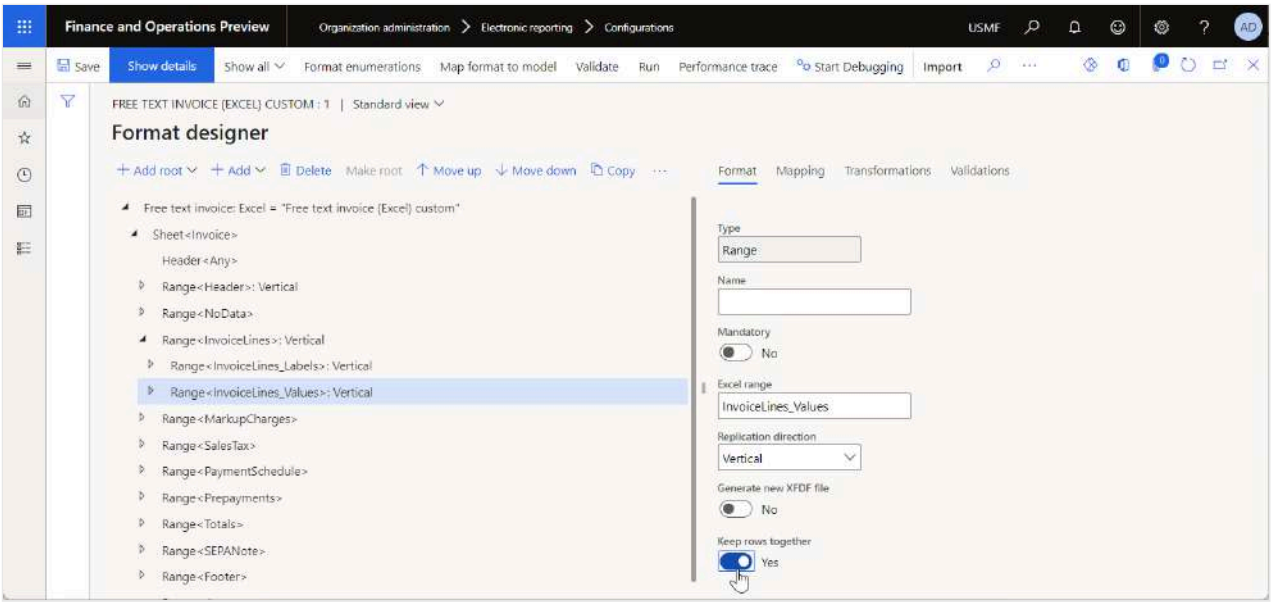
Follow the steps in Create a custom format to derive a format from the imported ER format and create a **Free text invoice (Excel) custom** ER configuration that you can edit and modify.

Edit the custom format

1. Follow the steps in Create a custom format to open the derived ER format for editing in the ER format designer.

2. On the **Format designer** page, in the format component tree in the left pane, expand **Free text invoice** > **Sheet** > **InvoiceLines**, and select the **InvoiceLines_Values** component.

3. On the **Format** tab, set the **Keep rows together** option to **Yes**.



4. Select **Save**, and close the page.

Mark the custom format as runnable

Follow the steps in Mark the custom format as runnable to make the modified version of the custom ER format runnable.

Set up print management to use the custom ER format

Follow the steps in Set up print management to configure print management for the **Accounts receivable** module so that the custom ER format is used to print free text invoices.

Configure a format destination for the custom ER format

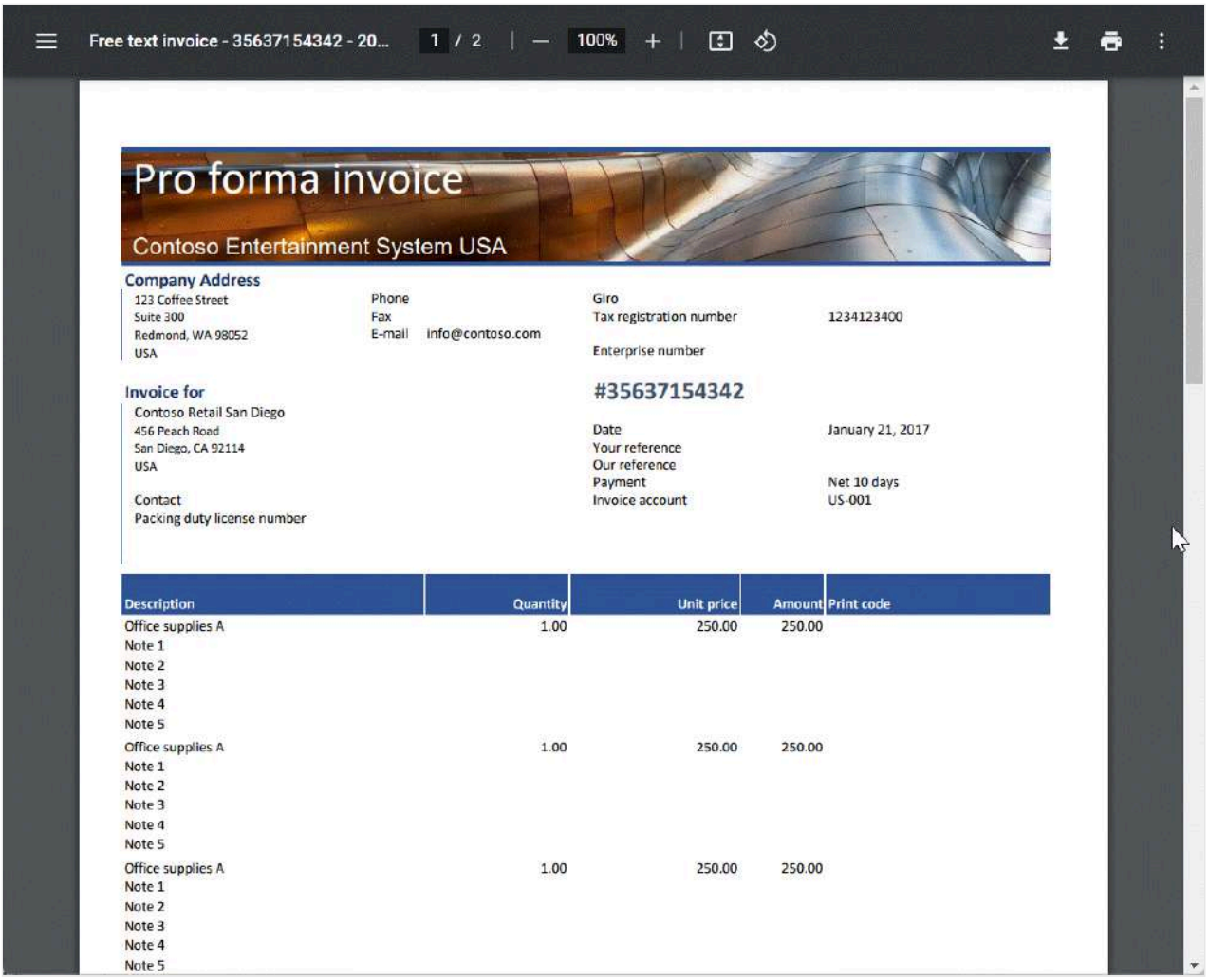
Follow the steps in Configure a format destination for on-screen preview to configure the **Screen** ER destination of the custom ER format so that generated reports are converted to PDF format and previewed on a new browser tab.

Print a free text invoice by using the custom ER format

1. Follow the steps in Print a free text invoice to use the custom ER format to generate a free text invoice report in Excel format for the added invoice.

2. Download the generated Excel workbook, and review it in the Excel desktop application.

Notice that the sixth line of the invoice starts on the second page, and all the Excel rows that represent this invoice line appear together on that page.



Additional resources

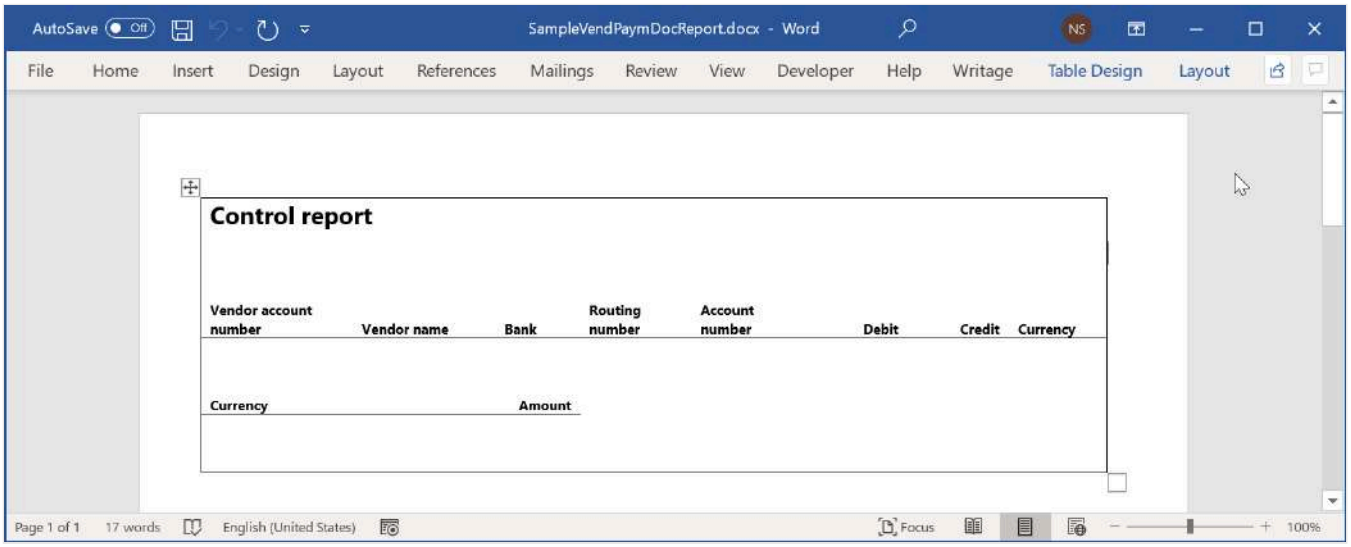
Design a configuration for generating documents in Excel format

Last updated on 04/08/2026

Design a new ER configuration to generate reports in Word format

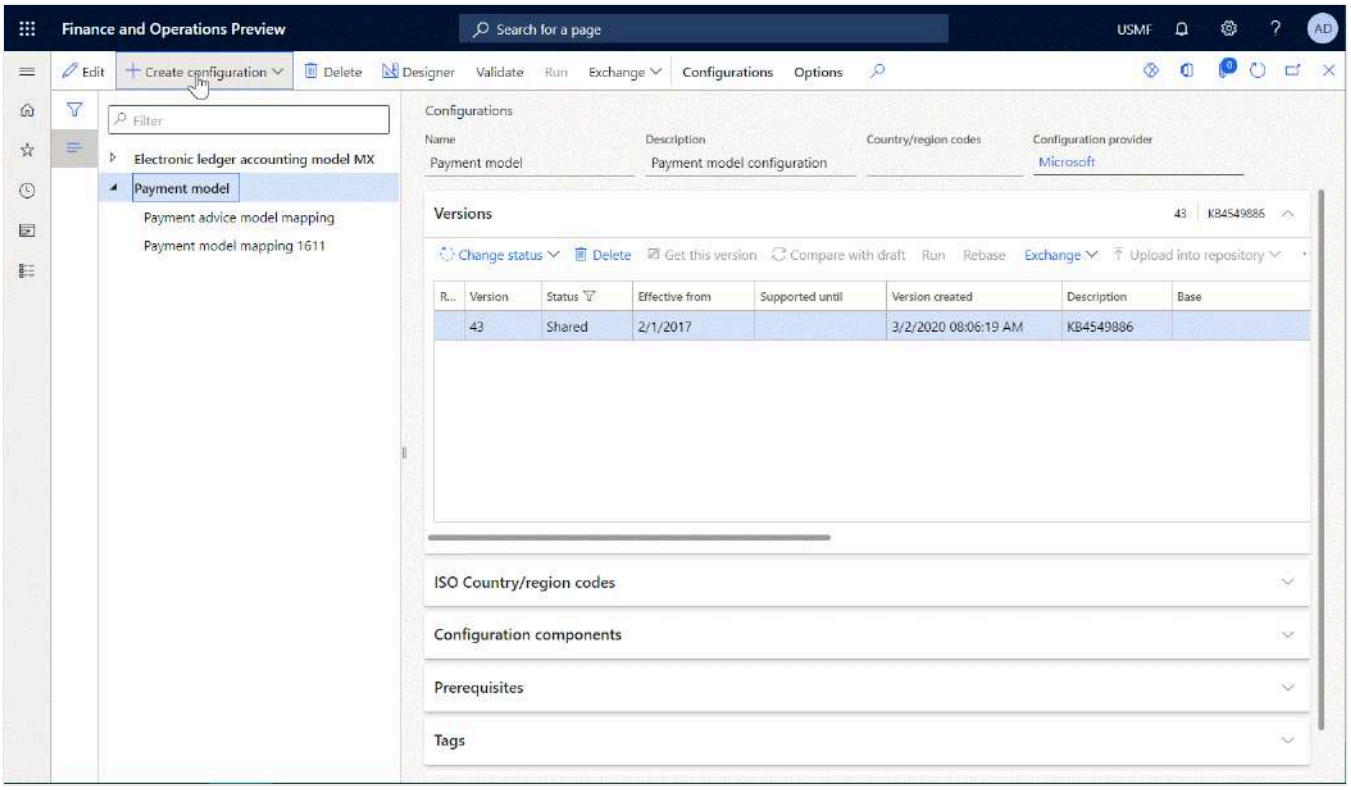
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To generate reports as Microsoft Word documents, design a template for the reports by using, for example, the Word desktop application. The following illustration shows the sample template for the control report that you can generate to show details of processed vendor payments.



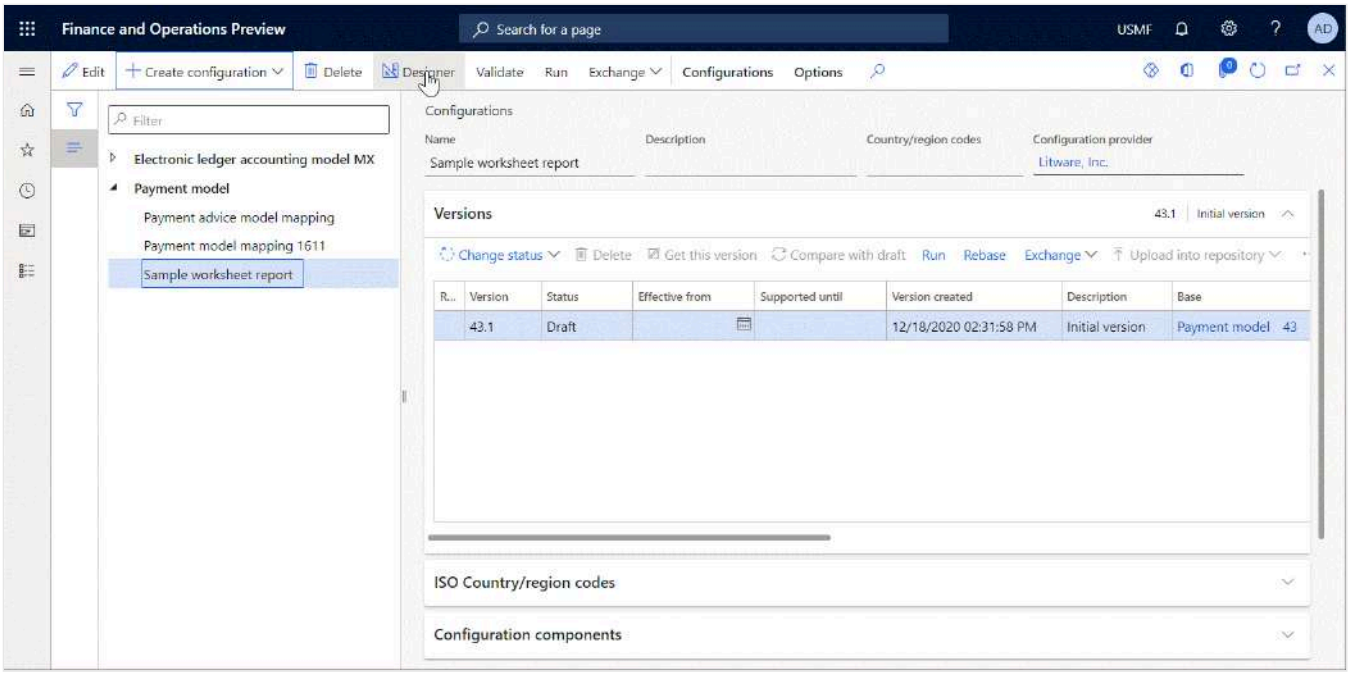
To use a Word document as a template for reports in Word format, configure a new Electronic reporting (ER) solution. This solution must include an ER configuration that contains an ER format component.

> **Note**
>
> When you create a new ER format configuration to generate reports in Word format, select **Word** as the format type in the **Create configuration** dropdown list or leave the **Format type** field blank.

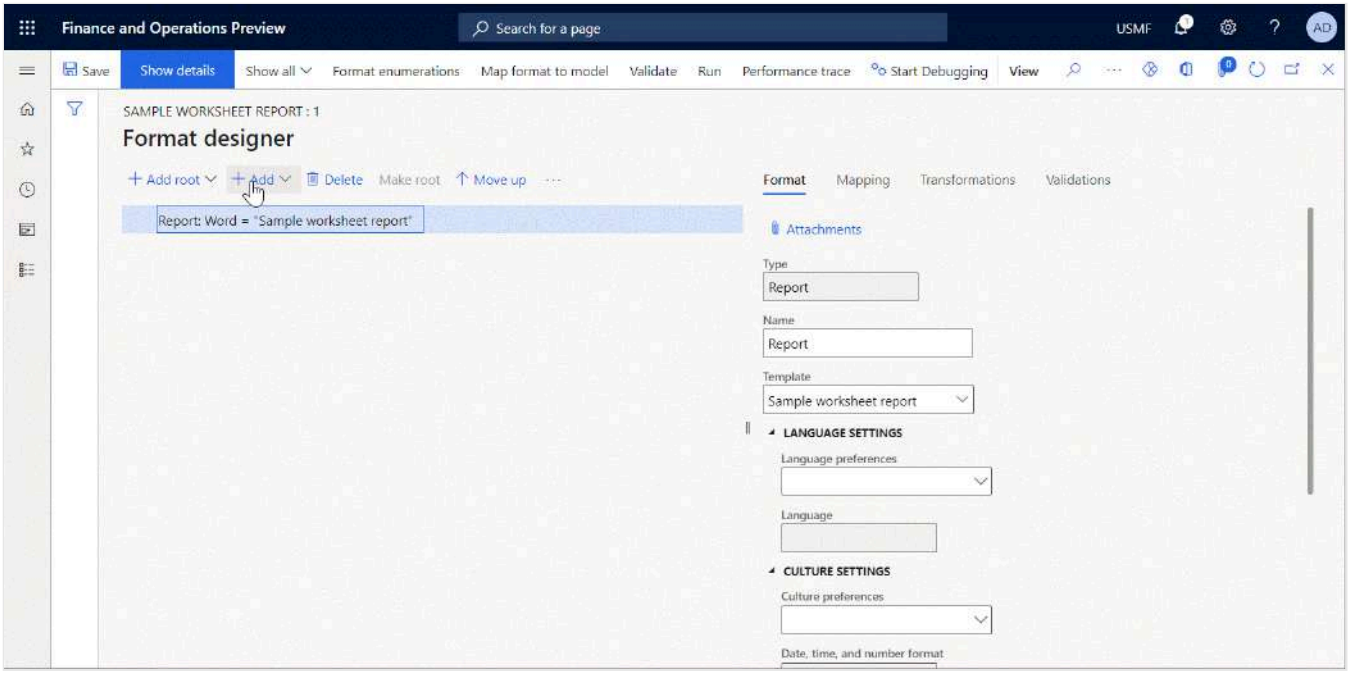


The ER format component of the solution must contain the **Excel\File** format element. Link that format element to the Word document that you use as the template for generated reports at runtime. To configure the ER format component, open the draft version of the created ER configuration in the ER format designer. Then add the **Excel\File** element, attach your Word template to the editable ER format, and link that template to the **Excel\File** element that you added.

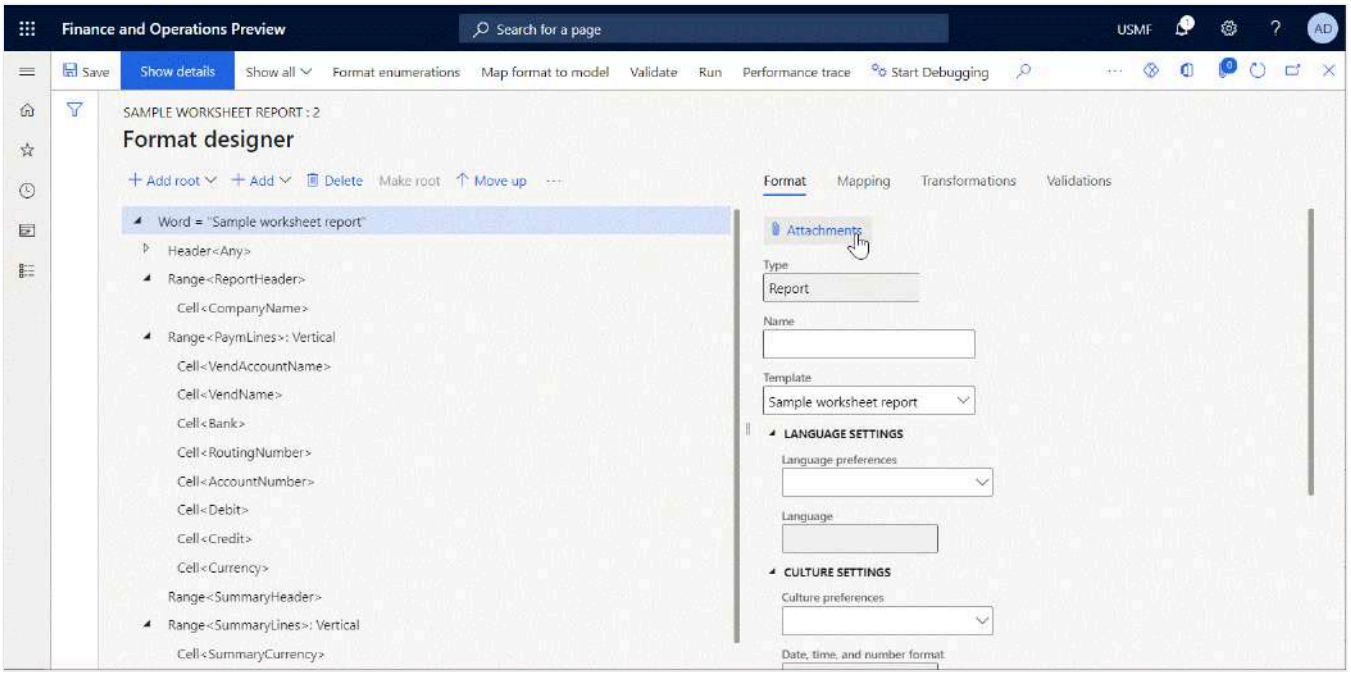
> **Note**
>
> When you attach a template, use a document type that is previously configured in the ER parameters to store templates of ER formats.



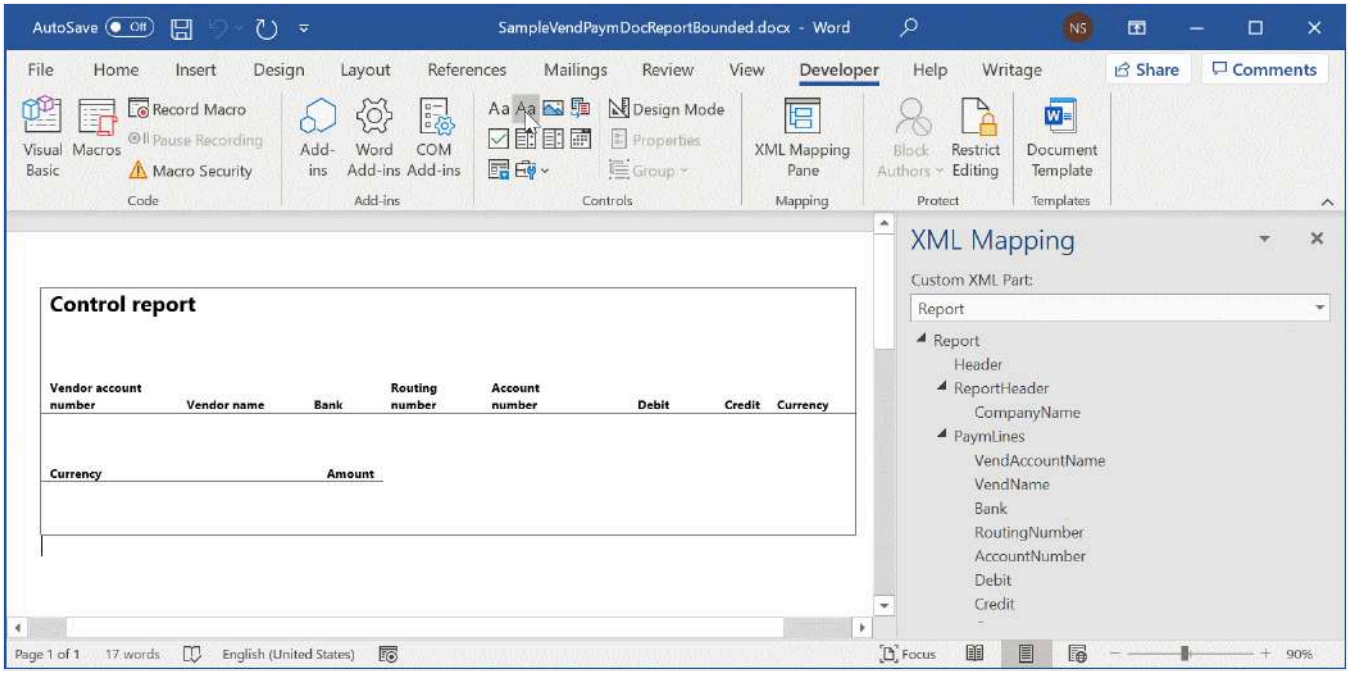
You can add **Excel\Range** and **Excel\Cell** nested elements for the **Excel\File** element to specify the structure of data to enter in generated reports at runtime. You must then bind those elements to data sources of the editable ER format to specify the actual data to enter in generated reports at runtime.



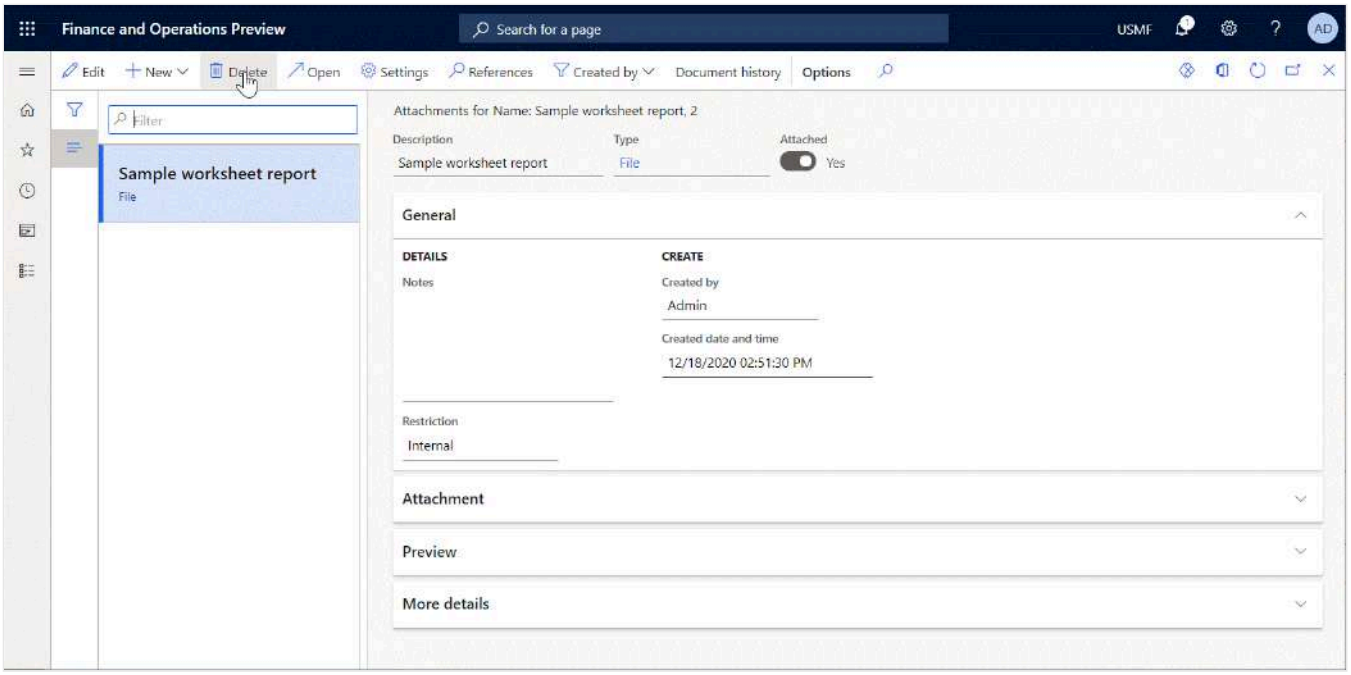
When you save your changes to the ER format at design time, the hierarchical format structure is stored in the attached Word template as a custom XML part that is named **Report**. You must access the modified template, download it from Finance, store it locally, and open it in the Word desktop application. The following illustration shows the locally stored sample template for the control report that contains the **Report** custom XML part.



When bindings of **Excel\Range** and **Excel\Cell** format elements run at runtime, the data that every binding delivers comes into the generated Word document as an individual field of the **Report** custom XML part. To enter the values from the fields of the custom XML part in a generated document, add the appropriate Word content controls to your Word template to serve as placeholders for data that's filled in at runtime. To specify how content controls are filled in, map every content control to the appropriate field of the **Report** custom XML part.



You must then replace the original Word template of the editable ER format with the modified template that now contains Word content controls that were mapped to the fields of the **Report** custom XML part.



When you run the configured ER format, the attached Word template is used to generate a new report. The actual data is stored in the Word report as a custom XML part that is named **Report**. When the generated report is opened, the Word content controls are filled in with data from the **Report** custom XML part.

Frequently asked questions

Question: I configured an ER format to print a Word document that contains a company address. In the Word template for this ER format, I inserted a rich text content control to present a company address. In Finance, I entered the company address as multiline text and selected **Enter** to add a carriage return for every line that I entered. Therefore, I expect the company address to appear as multiline text in generated documents. However, the address appears as a single line of text that contains special symbols instead of carriage returns. What is wrong with my settings?

Answer: Instead of using a rich text content control, use a plain text content control and select the **Allow carriage returns (multiple paragraphs)** check box in the control's properties.

Additional resources

- Reuse ER configurations with Excel templates to generate reports in Word format
- Embed images and shapes in documents that you generate by using ER

Last updated on 04/15/2026

Reuse ER configurations with Excel templates to generate reports in Word format

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To generate reports as Microsoft Word documents, you can configure a new Electronic reporting (ER) format. Alternatively, you can reuse an ER format that was originally designed to generate reports as Excel workbooks. In this case, you must replace the Excel template with a Word template.

The following procedures show how a user in either the System administrator role or the Electronic reporting developer role can configure an ER format to generate reports as Word files by reusing an ER format that was designed to generate reports as Excel files.

You can complete these procedures in the GBSI company.

Prerequisites

To complete these procedures, first follow the steps in the Design a configuration for generating reports in OPENXML format task guide.

You must also download and locally save the following templates for the sample report:

- Template of Payment Report (SampleVendPaymDocReport.docx)
- Bounded Template of Payment Report (SampleVendPaymDocReportBounded.docx)

These procedures are for a feature that was added in Dynamics 365 for Operations version 1611 (November 2016).

Select the existing ER report configuration

1. In Dynamics 365 Finance, go to **Organization administration** > **Workspaces** > **Electronic reporting**.

2. Make sure that the **Litware, Inc.** configuration provider is selected as **Active**. If it isn't, follow the steps in the Create configuration providers and mark them as active task guide.

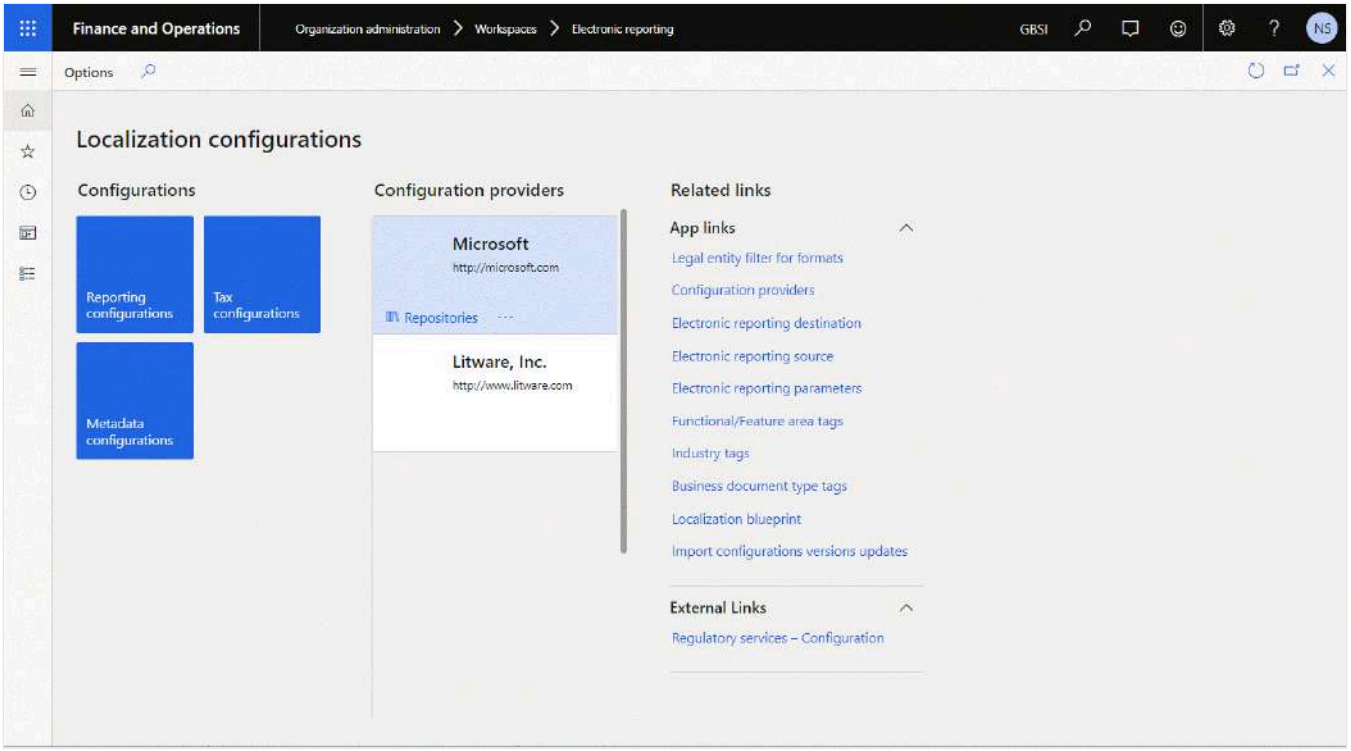
3. Select **Reporting configurations**. You reuse the existing ER configuration that was designed to generate the report output in OPENXML format.

4. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**, and select **Sample worksheet report**.

   > ⓘ **Note**
   >
   > You can edit the draft version of the selected ER format on the **Versions** FastTab.

5. Select **Designer**.

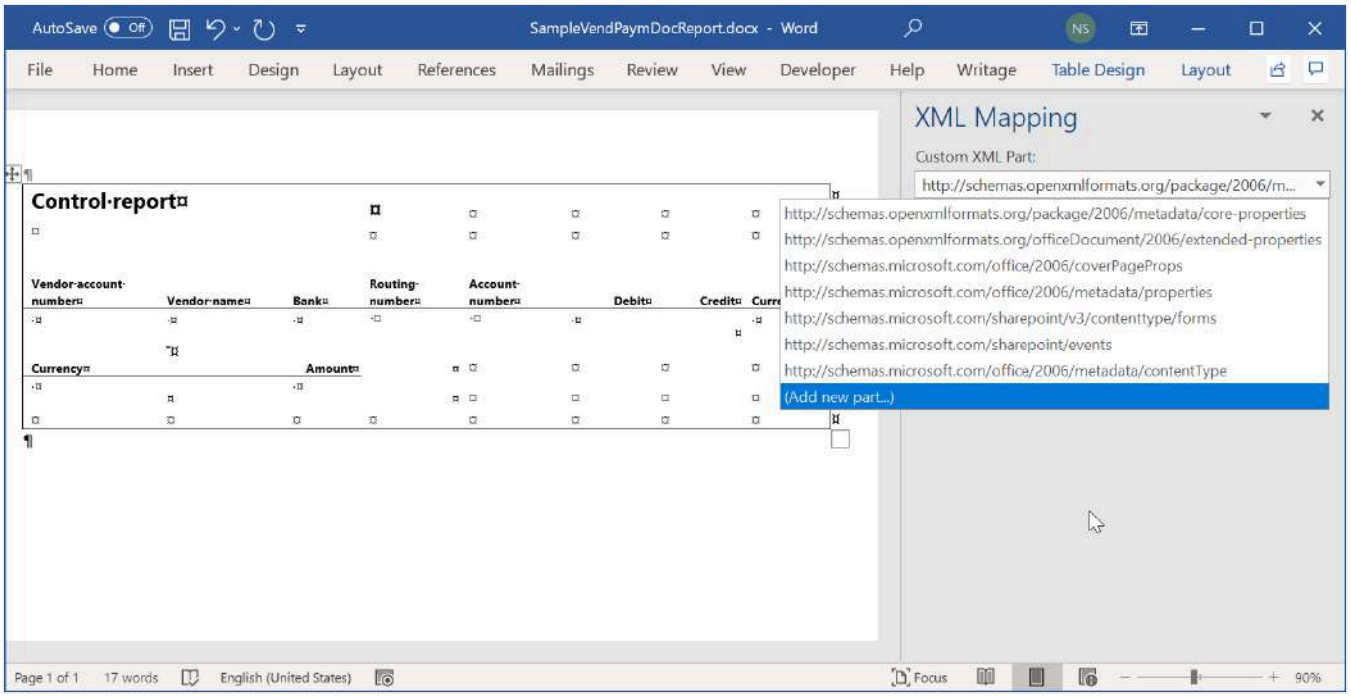
6. On **Format designer**, the title of the root format element indicates that an Excel template is currently used.



Review the downloaded Word template

1. In the Word desktop application, open the **SampleVendPaymDocReport.docx** template file that you downloaded earlier.

2. Verify that the template contains only the layout of the document that you want to generate as ER output.



Replace the Excel template with the Word template and add a custom XML part

Currently, the Excel document is used as a template to generate the output in OPENXML format. Replace this template with the **SampleVendPaymDocReport.docx** Word template file that you downloaded earlier. You can also extend the Word template by adding a custom XML part.

1. In Finance, on the **Format designer** page, on the **Format** tab, select **Attachments**.
2. On the **Attachments** page, select **Delete** to remove the existing Excel template. Select **Yes** to confirm the change.
3. Select **New** > **File**.

> **Note**
>
> To store templates of ER formats, select a document type that is configured in the ER parameters.

4. Select **Browse**, and then browse to and select the **SampleVendPaymDocReport.docx** file that you downloaded earlier.

5. Select **OK**.

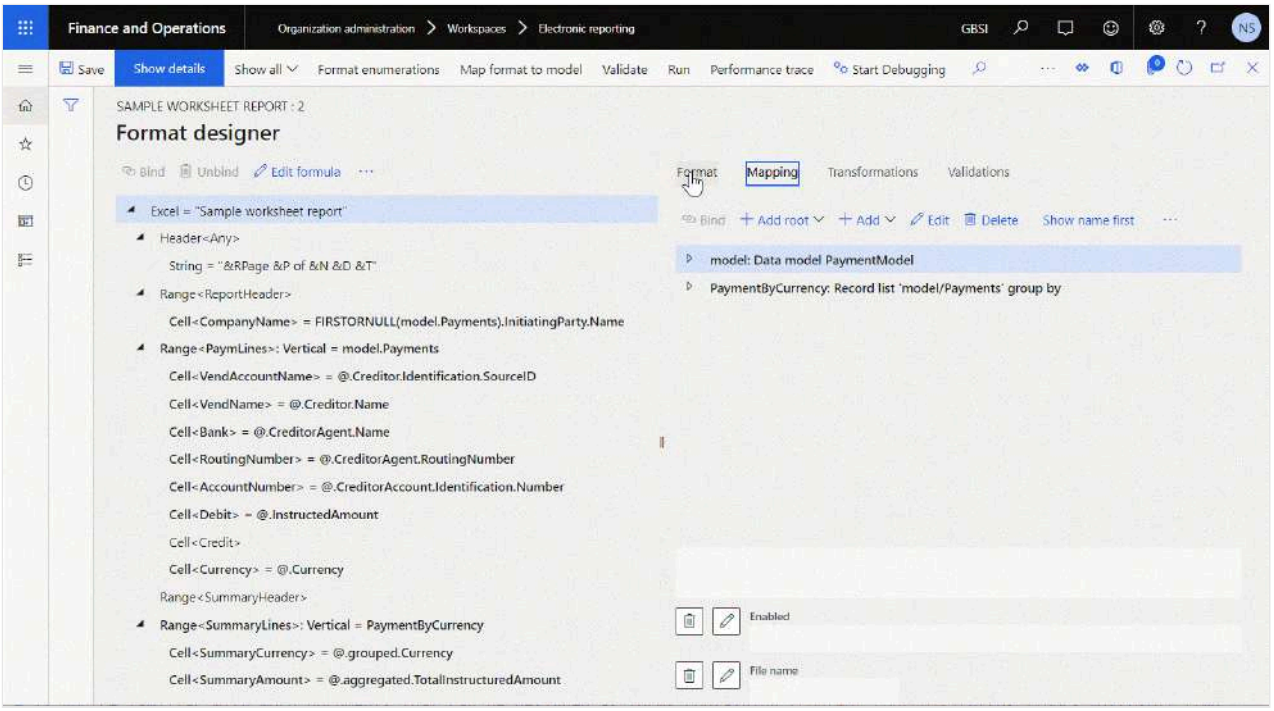
6. Close the **Attachments** page.

7. On the **Format designer** page, in the **Template** field, enter or select the **SampleVendPaymDocReport.docx** file to use that Word template instead of the Excel template that was previously used.

8. Select **Save**.

   In addition to storing configuration changes, the **Save** action updates the attached Word template. The hierarchical structure of the designed format is added to the attached Word document as a new custom XML part that is named **Report**. The attached Word template contains the layout of the document that is generated as ER output and the structure of data that ER enters in that template at runtime.

9. Notice that the title of the root format element indicates that a Word template is currently used.



10. On the **Format** tab, select **Attachments**.

You can now map the elements of the **Report** custom XML part to the content controls of the Word document.

If you're familiar with the process of designing Word documents as forms that contain content controls that are mapped to elements of custom XML parts, complete all steps in the next

procedure to create the document. For more information, see Create forms that users complete or print in Words. Otherwise, skip the next procedure.

Get a Word document that has a custom XML part and do data mapping

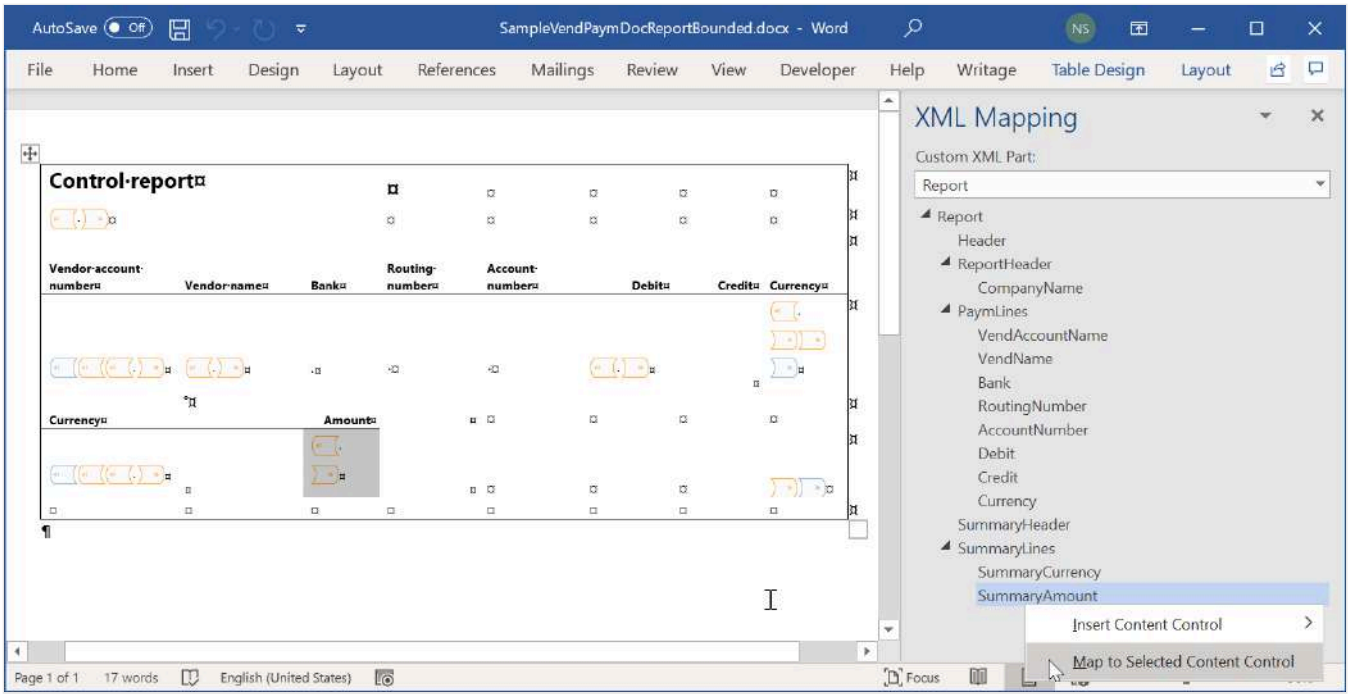
1. In Finance, on the **Attachments** page, select **Open** to download the selected template from Finance and store it locally as a Word document.
2. In the Word desktop application, open the document that you just downloaded.
3. On the **Developer** tab, select **XML Mapping Pane**.

   > **Note**
   >
   > If the **Developer** tab doesn't appear on the ribbon, customize the ribbon to add it.

4. In the **XML Mapping** pane, in the **Custom XML Part** field, select the **Report** custom XML part.
5. Map the elements of the selected **Report** custom XML part and the content controls of the Word document.
6. Save the updated Word document locally as **SampleVendPaymDocReportBounded.docx**.

Review the Word template where the custom XML part is mapped to content controls

1. In the Word desktop application, open the **SampleVendPaymDocReportBounded.docx** template file.
2. Verify that the template contains the layout of the document that you want to generate as ER output. The content controls that are used as placeholders for data that ER enters in this template at runtime are based on the mappings that you configure between elements of the **Report** custom XML part and the content controls of the Word document.



Upload the Word template where the custom XML part is mapped to content controls

1. In Finance, on the **Attachments** page, select **Delete** to remove the Word template that has no mappings between elements of the **Report** custom XML part and content controls. Select **Yes** to confirm the change.

2. Select **New** > **File** to add a new template file that contains mappings between elements of the **Report** custom XML part and content controls.

   > **Note**
   >
   > To store templates of ER formats, select a document type that is **configured** in the ER parameters.

3. Select **Browse**, and then browse to and select the **SampleVendPaymDocReportBounded.docx** file that you downloaded or prepared by completing the procedure in the Get a Word that has a custom XML part to do data mapping section.

4. Select **OK**.

5. Close the **Attachments** page.

6. On the **Format designer** page, in the **Template** field, select the document that you just downloaded.
7. Select **Save**.
8. Close the **Format designer** page.

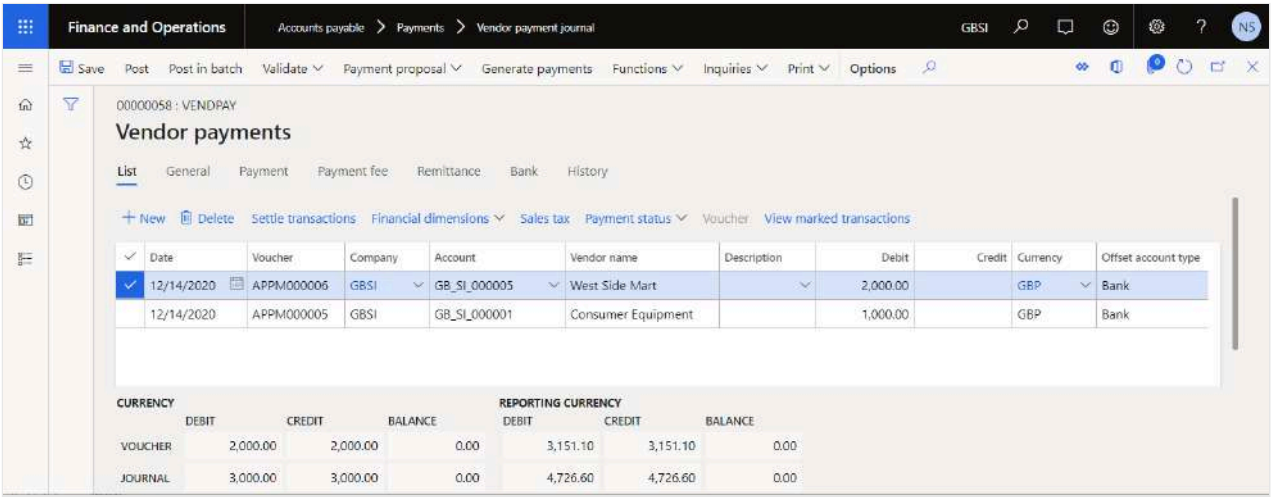
Mark the configured format as runnable

To run the draft version of the editable format, you must make it runnable.

1. In Finance, on the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
2. In the **User parameters** dialog box, set the **Run settings** option to **Yes**, and then select **OK**.
3. Select **Edit** to make the current page editable, as required.
4. For the currently selected **Sample worksheet report** configuration, set the **Run Draft** option to **Yes**.
5. Select **Save**.

Run the format to create output in Word format

1. In Finance, go to **Accounts payable** > **Payments** > **Payment journal**.
2. In a payment journal that you entered earlier, select **Lines**.
3. On the **Vendor payments** page, select all rows in the grid.
4. Select **Payment status** > **None**.

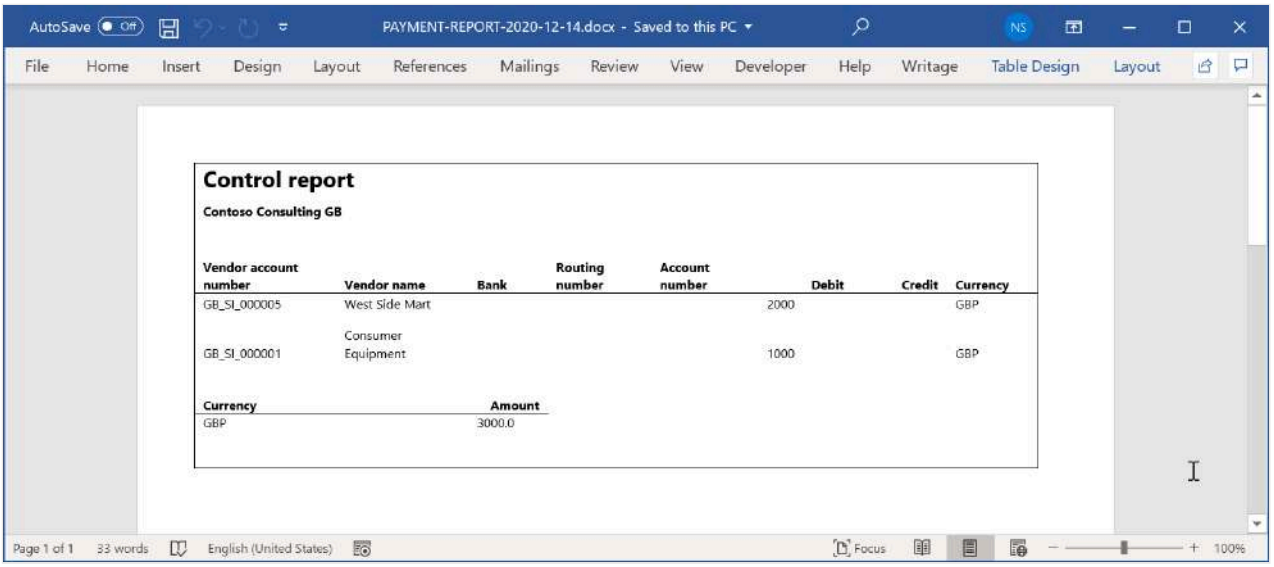


5. On the Action Pane, select **Generate payments**.

6. In the dialog box that appears, follow these steps:
    a. In the **Method of payment** field, select **Electronic**.
    b. In the **Bank account** field, select **GBSI OPER**.
    c. Select **OK**.

7. In the **Electronic report parameters** dialog box, select **OK**.

8. The generated output is presented in Word format and contains the details of the processed payments. Analyze the generated output.



Additional resources

- Design a new ER configuration to generate reports in Word format
- Embed images and shapes in documents that you generate by using ER

Last updated on 04/10/2026

Suppress Word content controls in generated reports

To generate reports as Microsoft Word documents, you must design a template for the reports as a Word document. This template must contain Word content controls as placeholders for data that's filled in at runtime. To use the created Word document as a template for your reports, configure a new Electronic reporting (ER) solution. The solution must include an ER configuration that contains an ER format component. You must configure this ER format to use the designed template for report generation.

In version 10.0.6 and later of Dynamics 365 Finance, you can configure formulas in your ER format to suppress some Word content controls in generated documents.

The following steps explain how a user assigned to the System administrator or Electronic reporting functional consultant role can configure an ER format that generates reports as Word files and suppresses some of the content controls in the generated reports that are configured by using a Word template.

You can complete these steps in the GBSI company.

Prerequisites

To complete these steps, first complete the steps in the following task guides:

- Design a configuration for generating reports in OPENXML format
- Re-use ER configurations with Excel templates to generate reports in Word format

When you complete the steps in these task guides, the following items are prepared:

- A **Sample worksheet report** ER format that is configured to generate a document in Word format
- A draft version of the **Sample worksheet report** ER format that is marked as **Runnable**
- An **Electronic** method of payments that is configured to use the **Sample worksheet report** ER format for vendor payment processing

You must also download and save the following template for the sample report:

- Bounded Template 2 of Payment Report (SampleVendPaymDocReportBounded2.docx)

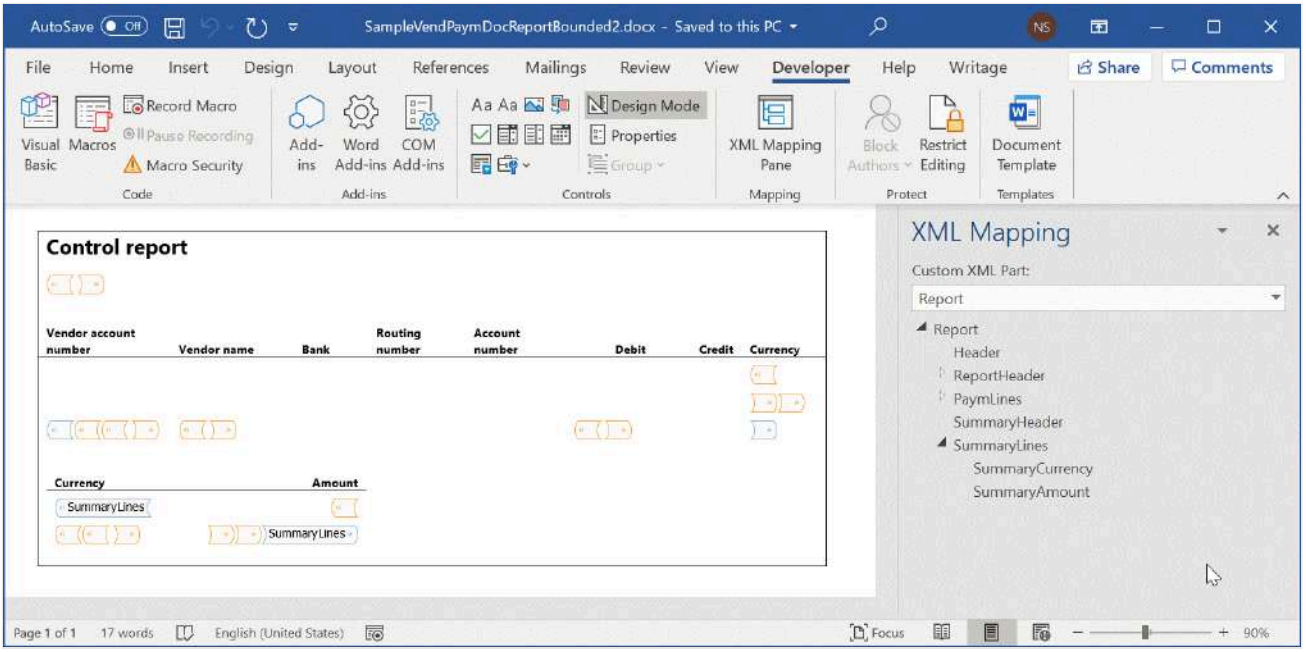
Review the downloaded Word template

1. In the Word desktop application, open the **SampleVendPaymDocReportBounded2.docx** template file that you downloaded earlier.

2. Verify that the template file contains a summary section that shows the total payment amounts for every currency code encountered in the processed payments.

    - The summary section resides in a separate table of the Word document.
    - The first row of this table holds the table columns headings as the section header.
    - The second row of this table holds the repeating content control as the section details.
    - This content control is mapped to the **SummaryLines** field of the **Report** custom XML part.
    - Based on this mapping, the content control is associated with the **SummaryLines** element of the editable ER format.

    > **Note**
    >
    > The repeating content control is tagged by the **SummaryLines** key that matches the field of the custom XML part that it maps to.



Select the existing ER report configuration

For the following steps, reuse the existing ER configuration that you configured when you completed the steps in the previously mentioned task guides.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. On **Configurations**, in the configuration tree, expand **Payment model**, and select **Sample worksheet report**.
4. Select **Designer** to edit the draft version of the selected ER format.

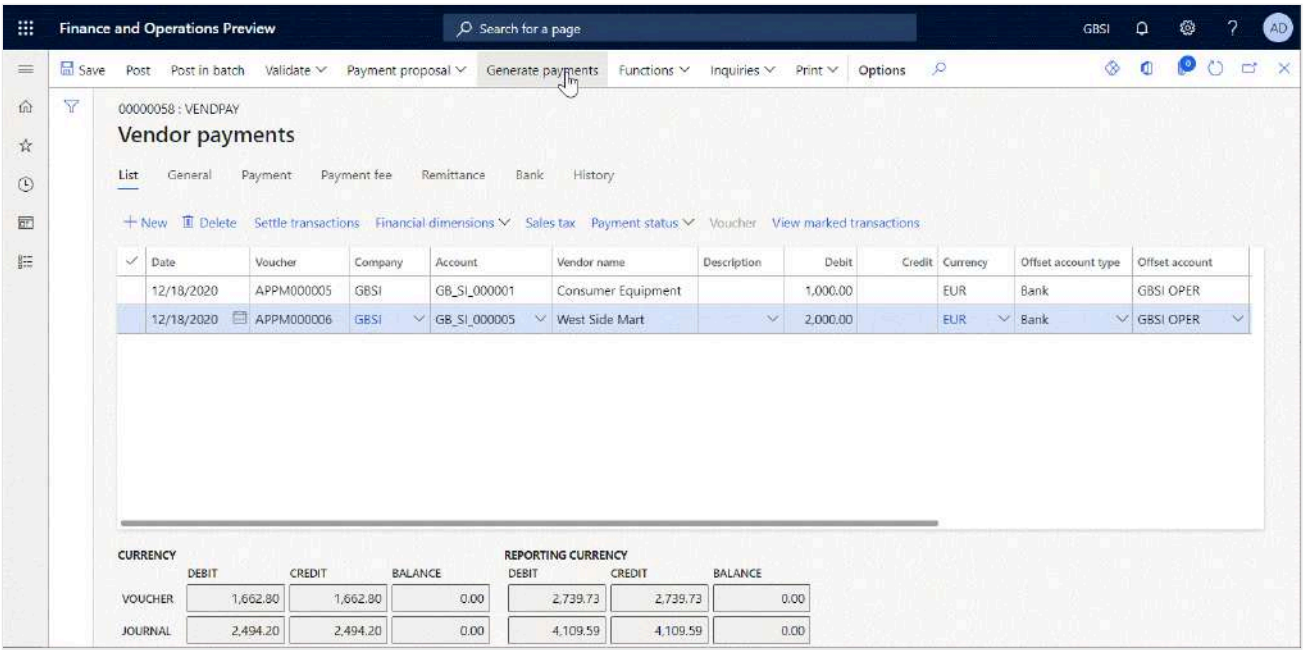
Replace the current template with the new template

Currently, the **SampleVendPaymDocReportBounded.docx** file is used as a template to generate the output in Word format. In the following steps, you replace this Word template with the new Word template, **SampleVendPaymDocReportBounded2.docx**, that you downloaded earlier.

1. On **Format designer**, select **Attachments**.
2. On **Attachments**, select **Delete** to remove the existing template.
3. Select **Yes** to confirm the deletion.
4. Select **New** > **File**.
5. Select **Browse**, and browse to and select the **SampleVendPaymDocReportBounded2.docx** file that you downloaded earlier.
6. Select **OK**.
7. Close **Attachments**.
8. On **Format designer**, in the **Template** field, enter or select the **SampleVendPaymDocReportBounded2.docx** file.

Run the format to create Word output

1. Go to **Accounts payable** > **Payments** > **Payment journal**.
2. On **Vendor payments**, on the **List** tab, select all the payments.
3. Select **Payment status** > **None**.
4. Select **Generate payments**.
5. In the **Method of payment** field, select **Electronic**.
6. In the **Bank account** field, select **GBSI OPER**.
7. Select **OK**.
8. In the **Electronic report parameters** dialog box, select **OK**, and analyze the generated output.



The output is presented in Word format and contains the summary section.

Configure the editable format to suppress the summary section

To suppress the summary section in a generated document, based on the request of a user who runs this ER format, modify the editable ER format.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**, and open the draft version of the ER format for editing.

2. Select **Reporting configurations**.

3. On **Configurations**, in the configuration tree, expand **Payment model** > **Sample worksheet report**.

4. Select **Designer**.

5. On **Format designer**, expand **Word**, and select **SummaryLines**.

6. On the **Mapping** tab, add a new data source to ask the user, at runtime, whether the summary section should be suppressed:
   a. Select **Add root**.
   b. In **Add data source**, select **General\User input parameter** to open **User input parameter data source properties**.
   c. In the **Name** field, enter `uipSuppress`.
   d. In the **Label** field, enter **Suppress summary section**.

e. In the **Operations data type name** field, select or enter **NoYes**.
f. Select **OK**.

7. Add a new data source of the **NoYes** application enumeration type:
   a. Select **Add root**.
   b. In **Add data source**, select **Dynamics 365 for Operations\Enumeration** to open **Enumeration data source properties**.
   c. In the **Name** field, enter `enumNoYes`.
   d. In the **Label** field, enter **Suppress options**.
   e. In the **Operations data type name** field, select or enter **NoYes**.
   f. Select **OK**.

8. For the selected **SummaryLines** format element, configure the formula to specify when the Word content control that is associated with the selected format element should be suppressed:

   a. On the **Mapping** tab, in the **Removed** section, select **Edit** to open the Formula designer.

   b. In the **Formula** field, enter the formula `uipSuppress = enumNoYes.Yes`.

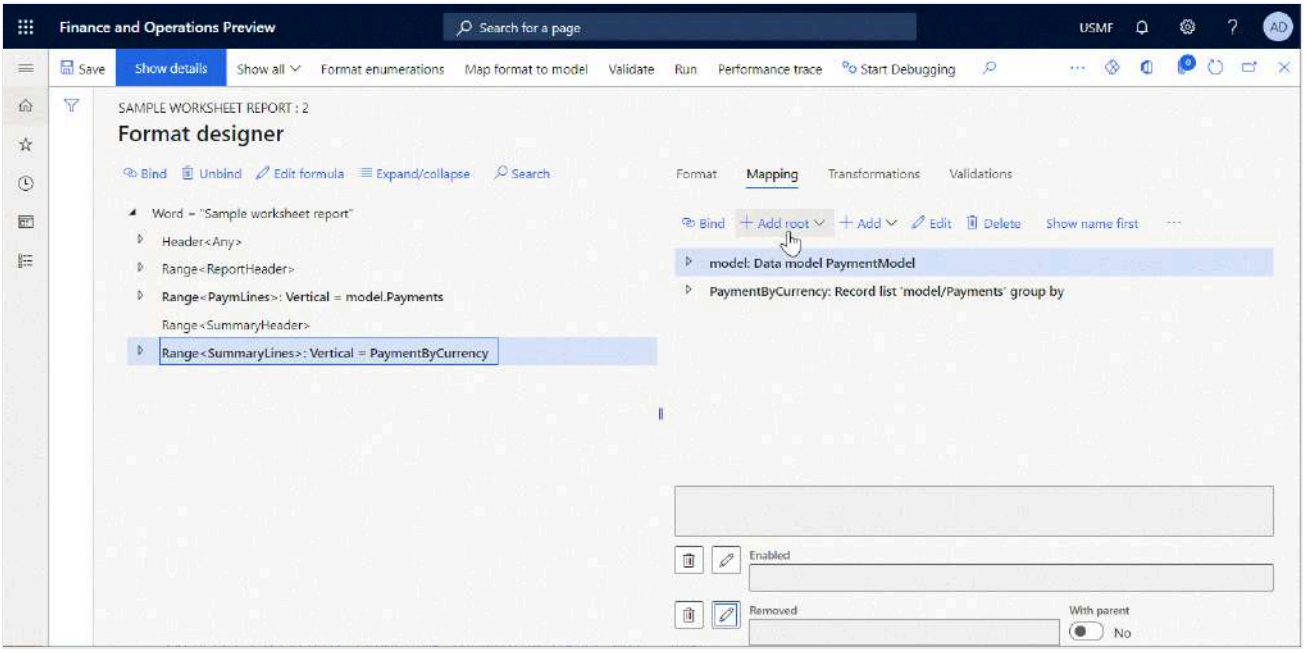
   c. Select **Save**, and close the **Formula designer**.

> ⓘ Note
>
> This formula is applied to a generated document **after all other format elements run**. To apply this formula, a Word content control that is tagged as a format element that the formula is configured for (**SummaryLines** in this case) is found in a generated document. The content control is then completely removed, together with the row in the Word table that holds it. The details row of the summary section is removed from the generated document.
>
> At design time, you might configure the **Removed** formula for a format element, even though no content control in the Word template that you're using has a tag that matches the name of a format element that the **Removed** property is configured for. When you validate the format at design time, you receive a warning about this inconsistency.
>
> At runtime, an exception is thrown if no content control in the Word template that you're using has a tag that matches the name of a format element that the **Removed** property is configured for.

d. On the **Mapping** tab, in the **Removed** section, set the **With parent** option to **Yes**.

> **Note**
>
> You must set this option to **Yes** to remove the whole Word table as the parent object of the row that holds the summary section details. If you set this option to **No**, the section header row remains in the generated document.

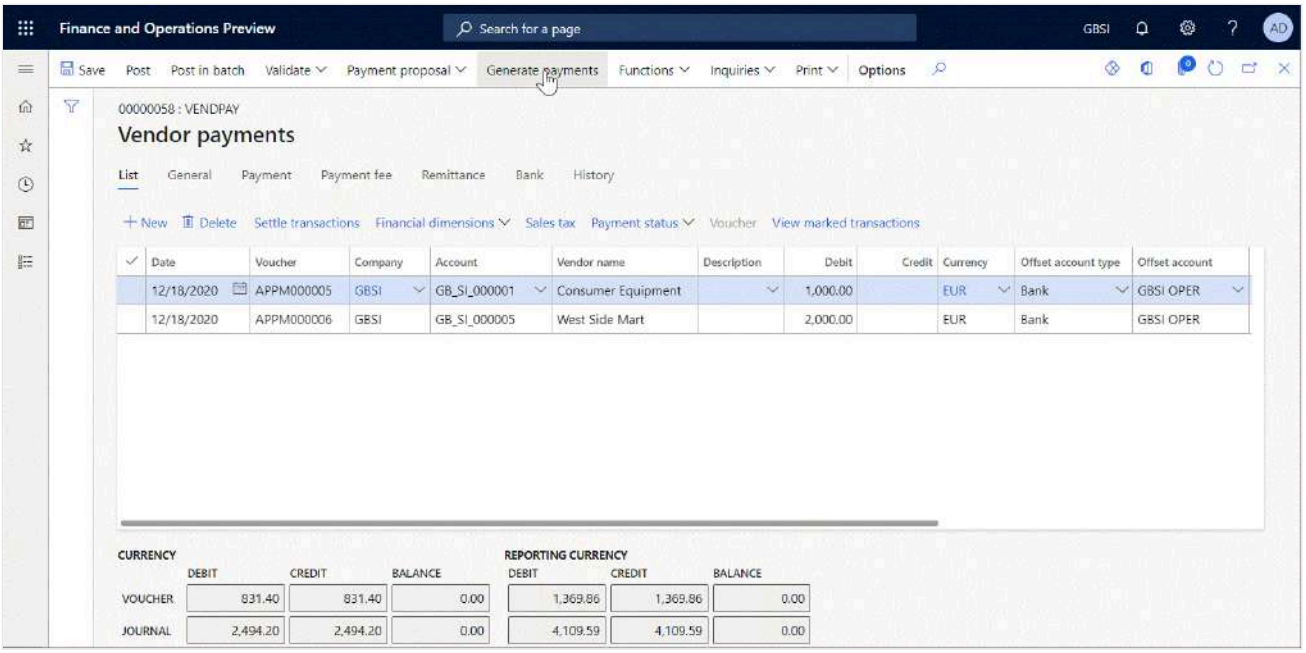
9. Select **Save** to save your changes to the editable format.



Run the modified format to create Word output

1. Go to **Accounts payable** > **Payments** > **Payment journal**.
2. Select the payment journal that you created, and then select **Lines**.
3. On the **Vendor payments** page, select all the rows, and then select **Payment status** > **None**.
4. Select **Generate payments**.
5. In the **Method of payment** field, select **Electronic**.
6. In the **Bank account** field, select **GBSI OPER**.
7. Select **OK**.
8. In the **Electronic report parameters** dialog box, in the **Suppress summary section** field, select **Yes**.

9. Select **OK**, and analyze the generated output.



The output doesn't contain the summary section, because it's suppressed.

Additional resources

- Design a configuration for generating reports in OPENXML format
- Design a new ER configuration to generate reports in Word format
- Re-use ER configurations with Excel templates to generate reports in Word format
- Inspect the configured ER component to prevent runtime issues

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Last updated on 07/17/2026

Design ER configurations to fill in PDF templates

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The procedures in this article provide examples that show how a user in either the **System administrator** role or the **Electronic reporting developer** role can configure an Electronic reporting (ER) format that generates reports as PDF files by using fillable PDF documents as report templates. You can perform these steps in any company of Dynamics 365 Finance or Regulatory Configuration Services (RCS).

Prerequisites

Before you begin, you must have one of the following types of access, depending on the service that you use to complete the procedures in this article:

- Access to Finance for one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- Access to RCS for one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator

You must also complete the Create configuration providers and mark them as active procedure.

Finally, download the following files.

Expand table

Content description	File name
Template for the first page of the report	IntrastatReportTemplate1.pdf
Template for other pages of the report	IntrastatReportTemplate2.pdf
Sample ER format - PDF	Intrastat report (PDF).version.1.1.xml
Sample ER format - Excel	Intrastat (import from Excel).version.1.1.xml
Sample dataset	Intrastat sample data.xlsx

Design the format configuration

Get access to the list of configurations provided by Microsoft

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Make sure that the **Litware, Inc.** provider is available and marked as active.
3. On the tile for the **Microsoft** provider, select **Repositories**.

> ⓘ **Note**
>
> If a repository of the **LCS** type already exists, skip the remaining steps of this procedure.

4. Select **Add**.
5. In the drop-down dialog box, in the **Configuration repository type** field group, select the **LCS** option.
6. Select **Create repository**.
7. Select **OK**.

Get the model configurations provided by Microsoft

1. On the left side of the **Configuration repositories** page, select the **Show filters** button (the funnel symbol).
2. Add a filter for a value of **LCS** in the **Type** field, and use the **begins with** operator.
3. Select **Apply**.

4. Select **Open**.
5. In the tree, select **Intrastat model**.
6. On the **Versions** FastTab, select version **1**.

> ⓘ Note
>
> If the **Import** button on the **Versions** FastTab is unavailable, skip the remaining steps of this procedure.

7. Select **Import**.
8. Select **Yes** to confirm the import of the selected version of the **Intrastat model** model configuration.

Create a new format configuration

1. In the **Electronic reporting** workspace, select the **Reporting configurations** tile.
2. In the tree, select **Intrastat model**.
3. Select **Create configuration**.

> ⓘ Note
>
> If the **Create configuration** button isn't available, you must turn on design mode on the **Electronic reporting parameters** page that can be opened from the **Electronic reporting** workspace.

4. In the drop-down dialog box, in the **New** field group, select the **Format based on data model Intrastat** option.
5. In the **Name** field, enter **Intrastat report (PDF)**.
6. In the **Description** field, enter **Intrastat report in PDF format**.

> ⓘ Note
>
> The active configuration provider is automatically entered. This provider will be able to maintain this configuration. Although other providers can use this configuration,

they won't be able to maintain it.

7. Optional: In the **Format type** field, you can select a specific format of electronic document. If you select **PDF**, at design time, the ER Operations designer will offer just the format elements that are applicable only to documents that are generated in PDF format (**PDF\File**, **PDF\PDF Merger**, etc.). If you leave this field blank, a format of electronic document will be specified at design time in the ER Operations designer when a first format element will be added. For example, if you add the **Excel\File** as the first format element, the ER Operations designer will offer you just the format elements that are applicable only to documents that are generated in Excel format (**Excel\Cell**, **Excel\Range**, etc.). format.

8. Select **Create configuration**.

A new ER format configuration is created. You can use the draft version of this configuration to store the ER format component that is designed to generate electronic documents in PDF format.

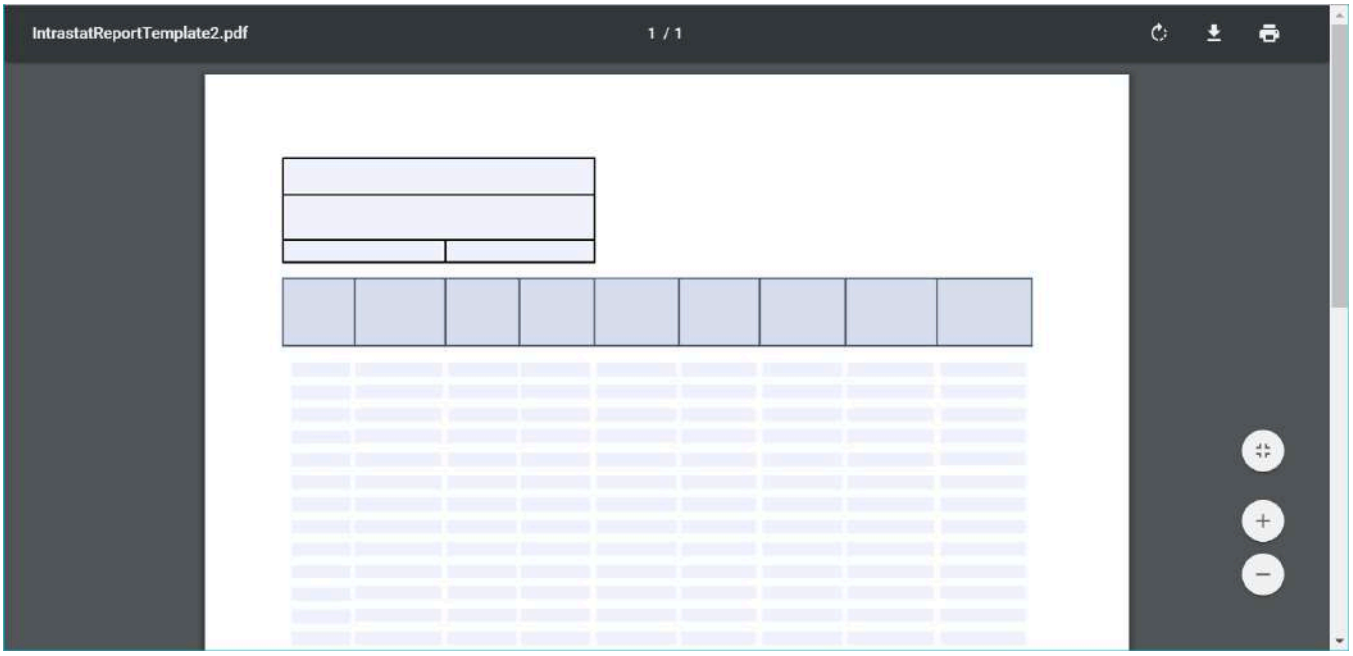
Design the format of an electronic document

Next, in the ER format configuration that you created, design the ER format that generates the **Intrastat control** report in PDF format. The first page of this report must show a summary of the report and details of the foreign trade transactions that are reported on. The other pages must show only details of the foreign trade transactions that are reported on. Because the report pages that are generated must have different layouts, the ER format uses two different templates in PDF format.

In any PDF viewer, open the PDF templates that you downloaded. Each template contains multiple fields that you fill in. In each template, details of foreign trade transactions are presented as 42 rows, each of which has nine fields. The row number appears at the end of each field's name (for example, **Date 1**...**Date 42** and **Commodity 1**...**Commodity 42**).

The following illustration shows the PDF template for the first page of the report.

The following illustration shows the PDF template for other pages of the report.



1. On the **Configurations** page, select **Designer**.
2. Select **Add root**.
3. In the tree, select **PDF** > **PDF Merger**.

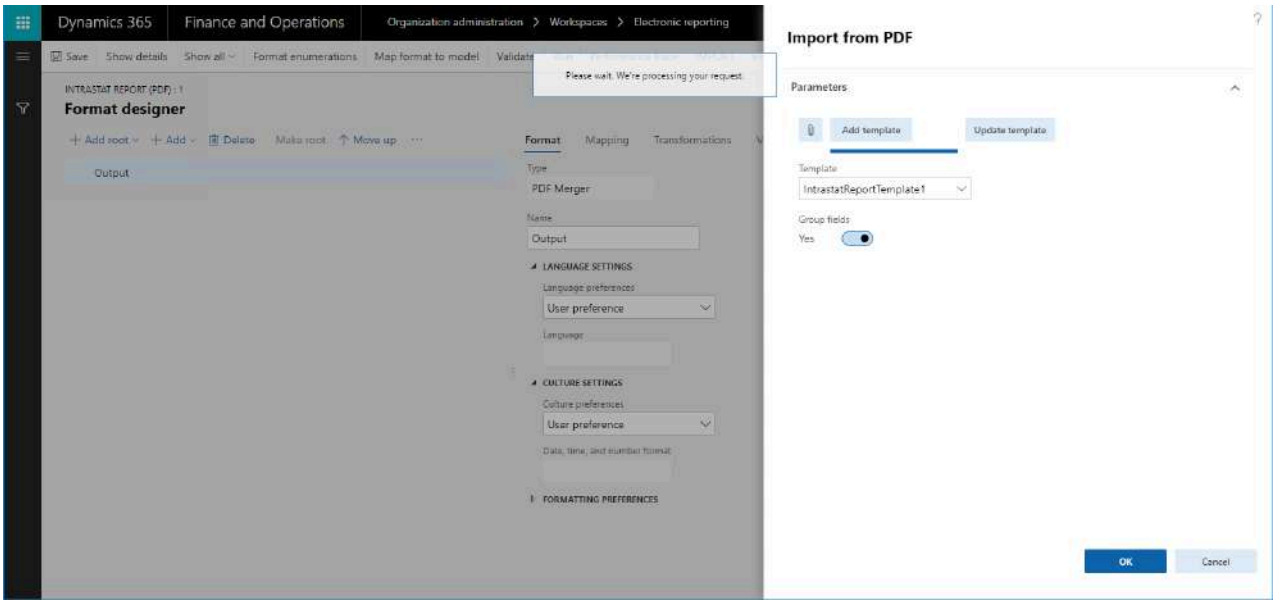
   When you select the **PDF Merger** element as the root element of the format, all PDF documents that are generated at run time are merged into a single final PDF document. If you need only one PDF template to generate all the required documents by using the ER format that you design, select **PDF file** as the root element.
4. In the **Name** field, enter **Output**.

5. In the **Language preferences** field, select **User preference**. The report is generated in the preferred language of the user who runs it.
6. In the **Culture preferences** field, select **User preference**. Values and dates that are presented on the pages of the report are formatted based on the preferred locale of the user who runs the report.
7. Select **OK**.
8. On the Action Pane, on the **Import** tab, select **Import from PDF**.

   When you import a fillable PDF document as a template for this ER format, the system automatically creates all the required ER format elements (**PDF file**, **Field group**, and **Field** elements) in the format that you design, based on the structure of the PDF document that you import.
9. Select **Browse**. Go to and select the **IntrastatReportTemplate1.pdf** file that you downloaded earlier as a prerequisite.
10. Select **OK**.

    The selected file is loaded, and the **Template** field in the **Import from PDF** dialog box is filled in.
11. Set the **Group fields** option to **Yes**. If the selected PDF document contains any field groups, the system uses them to group the ER format elements that it creates. The system creates a **Field group** format element for this purpose.

    If you set this option to **No**, the system creates the required ER format elements as a flat list of elements that are nested under the **PDF File** format element that it creates.
12. Select **OK**.



13. In the tree, expand **Output**.

    The **PDF File** component is automatically created to manage the creation of the first page of the report that is generated at run time.

14. In the tree, expand **Output** > **PDF File**.

    The structured list of format elements is automatically created in this ER format, based on the structure of the fillable PDF document that you imported earlier.

15. In the tree, select **Output** > **PDF File**.

16. In the **Name** field, enter **Page 1**.

    This format element generates the first page of the **Intrastat control** report. That page shows a summary of the report and details of foreign trade transactions.

    If you leave the **Language preferences** field blank, the **Language preferences** setting of the parent **PDF Merger** element determines the language of the report that is generated by using this format element. You can select another value to override the setting of the parent element.

    If you leave the **Culture preferences** field blank, the **Culture preferences** setting of the parent **PDF Merger** element determines the locale of the report that is generated by using this format element. The locale determines the format of values and dates on the pages of the report. You can select another value to override the setting of the parent element.

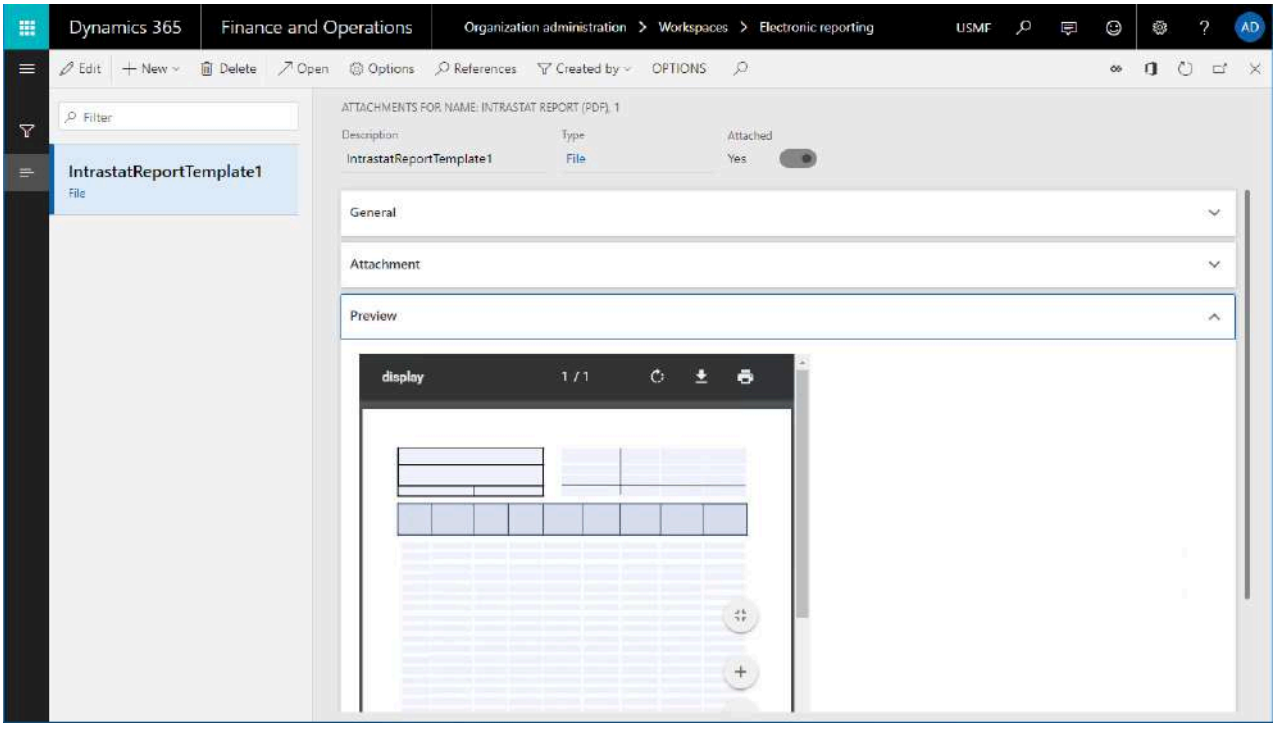
17. On the Action Pane, select the **Import** tab. Notice that the **Update from PDF** button is available for selected format element, **PDF File**.

    Use this button to import the updated PDF template to the edited format. When you import the updated PDF template, the list of format elements changes accordingly:

    - For any new fields in the updated PDF template, the system creates new format elements in the edited ER format.
    - If the updated PDF template no longer includes fields that correspond to any existing format elements in the edited ER format, the system deletes those format elements from the ER format.

18. On the **Format** tab, select **Attachments**.

    The imported PDF document is attached to the edited ER format.



19. Continue to design this format by importing the second PDF template, adding necessary bindings to data sources, and so on.

20. Select **Save**.

21. Close the page.

22. Select **Delete**.

23. Select **Yes**.

To learn how new **PDF Merger**, **PDF File**, **Field group**, and **Field** format elements can be used to generate documents in PDF format, import and analyze the sample ER format.

Import the format configuration

Next, import the sample ER format that you previously downloaded to generate the **Intrastat control** report in PDF format. The first page of the report shows a summary of the report and details of the foreign trade transactions that are reported on. The other pages show only details of the foreign trade transactions that are reported on.

1. On **Configurations**, select **Exchange** > **Load from XML file**.
2. Select **Browse**. Go to and select the **Intrastat report (PDF).version.1.1.xml** file that you downloaded earlier as a prerequisite.
3. Select **OK**.

Analyze the format configuration

Format layout

1. On **Configurations**, in the tree, select **Intrastat model** > **Intrastat report (PDF)**.
2. Select **Designer**.
3. Select **Show details**.
4. In the tree, expand **Output: PDF Merger**.

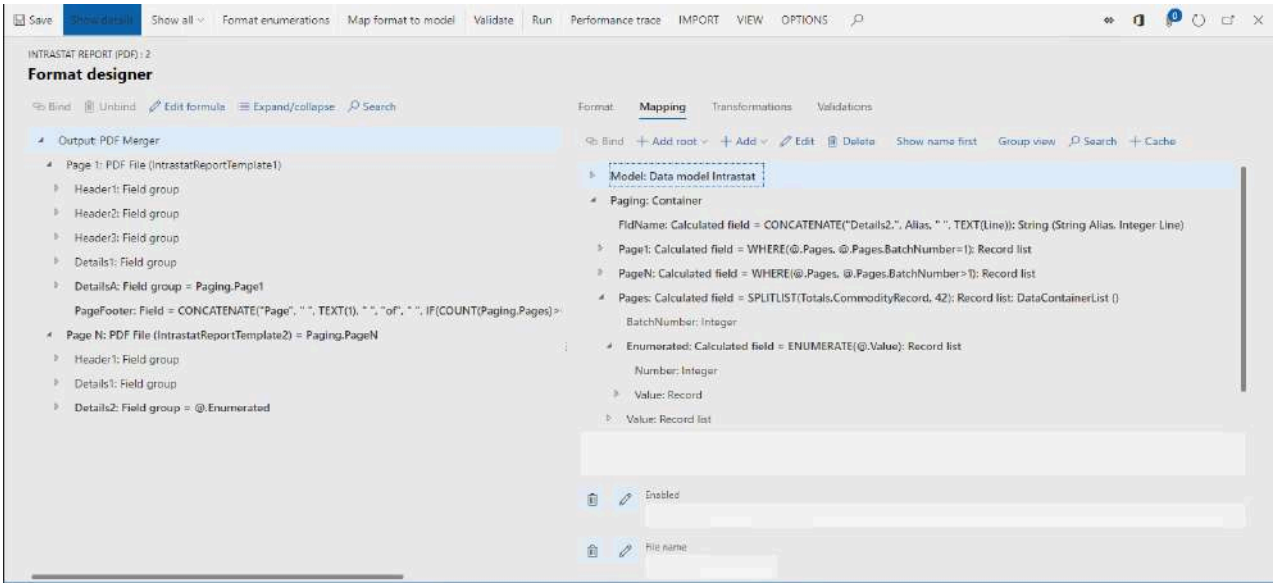
   This ER format contains two **PDF File** elements, each of which uses a different PDF template. One template generates the first page of the report in PDF format, and the other template generates the other pages.
5. In the tree, expand **Output: PDF Merger** > **Page 1: PDF File (IntrastatReportTemplate1)**.
6. In the tree, expand **Output: PDF Merger** > **Page N: PDF File (IntrastatReportTemplate2)**.

   Because the content of the two PDF templates differs, the structure of the nested format elements for the two **PDF File** elements also differs.

Format mapping

1. On **Format designer**, select the **Mapping** tab.

2. In the tree, expand **Paging** > **Pages**.



Note the following details:

- The **Output** > **Page 1** format element of the **PDF File** type isn't bound to any data source, and the **Enabled** expression of this format element is empty. Therefore, at run time, the **IntrastatReportTemplate1** PDF template is filled in only one time when an individual PDF document is generated.

- The **Output** > **Page N** format element of the **PDF File** type is bound to the **Paging.PageN** data source of the **Record list** type, and the **Enabled** expression of this format element is empty. Therefore, at run time, the **IntrastatReportTemplate2** PDF template is filled in for each record from the bound record list when an individual PDF document is generated.

- Because the **Page 1: PDF File** and **Page N: PDF File** format elements are children of the **Output: PDF Merger** format element, all PDF documents that are filled in are merged into a single final PDF document.

- The **Paging.Page1** and **Paging.PageN** data sources are configured as filters of records from the **Paging.Pages** data source. This data source is configured to split the whole set of foreign trade transactions into batches. Each batch contains up to 42 records. The following ER expression is used to split the transactions into batches:

  SPLITLIST(Totals.CommodityRecord,42)

- The **Paging.Pages** data source contains the **Paging.Pages.Enumerated** element that returns the details of each record that is included in a batch. These details include

the record's sequence in the current batch (the **Paging.Pages.Enumerated.Number** field). The **Paging.Pages.Enumerated.Number** field is used in the **Name** expression of **PDF Field** format elements to dynamically generate a field name that is based on the transaction number in a batch. The field name that is generated is then used to fill in the correct PDF field in the PDF template that is used.

- The **Output** > **Page N** > **Details 2** format element of the **PDF Group** type is bound to the **Paging.PageN.Enumerated** data source (or **@.Enumerated** if the **Relative path** view mode is used) of the **Record list** type. Therefore, at run time, the nested elements of this PDF group are filled in for each record from the bound record list. In this way, individual PDF lines are virtually generated when for each Nth of 42 records of the **Paging.PageN.Enumerated** list the following PDF fields are filled in: Date N, Direction N, Commodity N, and so on. Therefore, in this respect, the behavior of this **Field group** format element resembles the behavior of the **XML** > **Sequence** and **Text** > **Sequence** format elements.

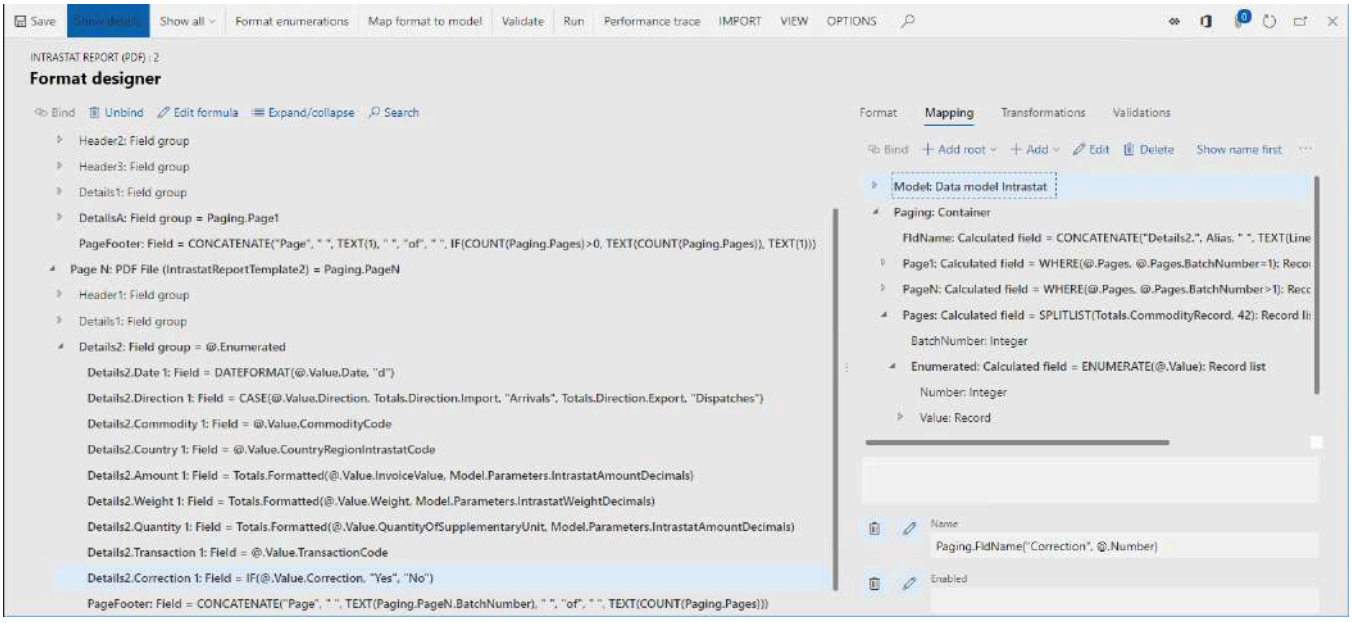
3. In the tree, expand **Output** > **Page N** > **Details2**.

4. In the tree, select **Output** > **Page N** > **Details2** > **PageFooter**.

   Notice that the **Name** attribute of this format element is defined as **PageFooter**. Also notice that the **Name** expression of the format element is empty.

5. In the tree, select **Output** > **Page N** > **Details2** > **Correction 1**.

   Notice that the **Name** attribute of this format element is defined as **Correction 1**. Also notice that the **Name** expression is defined as **Paging.FldName("Correction",@.Number)**.



Note that the **Field** format element is used to fill in an individual field of a fillable PDF document that is defined as a template of the parent **PDF File** format element. The binding of the **PDF File** format element or its nested elements, if it has any nested elements, specifies the value that is entered in corresponding PDF fields. Different properties of the **Field** format element can be used to specify which PDF field is filled in by an individual format element:

- On the **Format** tab, the **Name** attribute of the format element
- On the **Mapping** tab, the **Name** expression of the format element

Because both properties are optional for a **Field** format element, the following rules are applied to specify the target PDF field:

- If the **Name** attribute is blank, and the **Name** expression returns an empty string at run time, an exception is thrown at run time to notify the user that a PDF field can't be found in the PDF template that is being used to fill in the PDF document.
- If the **Name** attribute is defined, and the **Name** expression is blank, the PDF field that has the same name as the **Name** attribute of the format element is filled in.
- If the **Name** attribute is defined, and the **Name** expression is configured, the PDF field that has the same name as the value that is returned by the **Name** expression of the format element is filled in.

> ⓘ **Note**
>
> When a checkbox in the PDF template doesn't belong to a group of checkboxes, it's represented in the editable ER format as a **Field** element that is nested under the **PDF File** element. This type of PDF checkbox can be set as selected in the following ways:
>
> - The corresponding **Field** format element is bound to a data source field of the ***Boolean*** data type that has a value of **True**.
> - The corresponding **Field** format element contains a nested **String** format element that is bound to a data source field that has a text value of **1**, **True**, or **Yes**.
>
> Your template can contain a group of checkboxes where only one checkbox can be selected at a time. Those checkboxes are represented in a PDF template as multiple form fields of the *CHECKBOX* type. Each field has the same name but a different export value. When you import the template into the editable ER format, each checkbox is represented in the format hierarchical structure as a **Check box group item** element that is nested under the same **Check box group** element. The name of the **Check box group** element equals the name of the checkbox fields in the PDF template. The name of each **Check box**

group item element equals the export value of the corresponding checkbox field in the PDF template.

You can bind a **Check box group item** element to a data source field of the *Boolean* data type only.

Run the format configuration

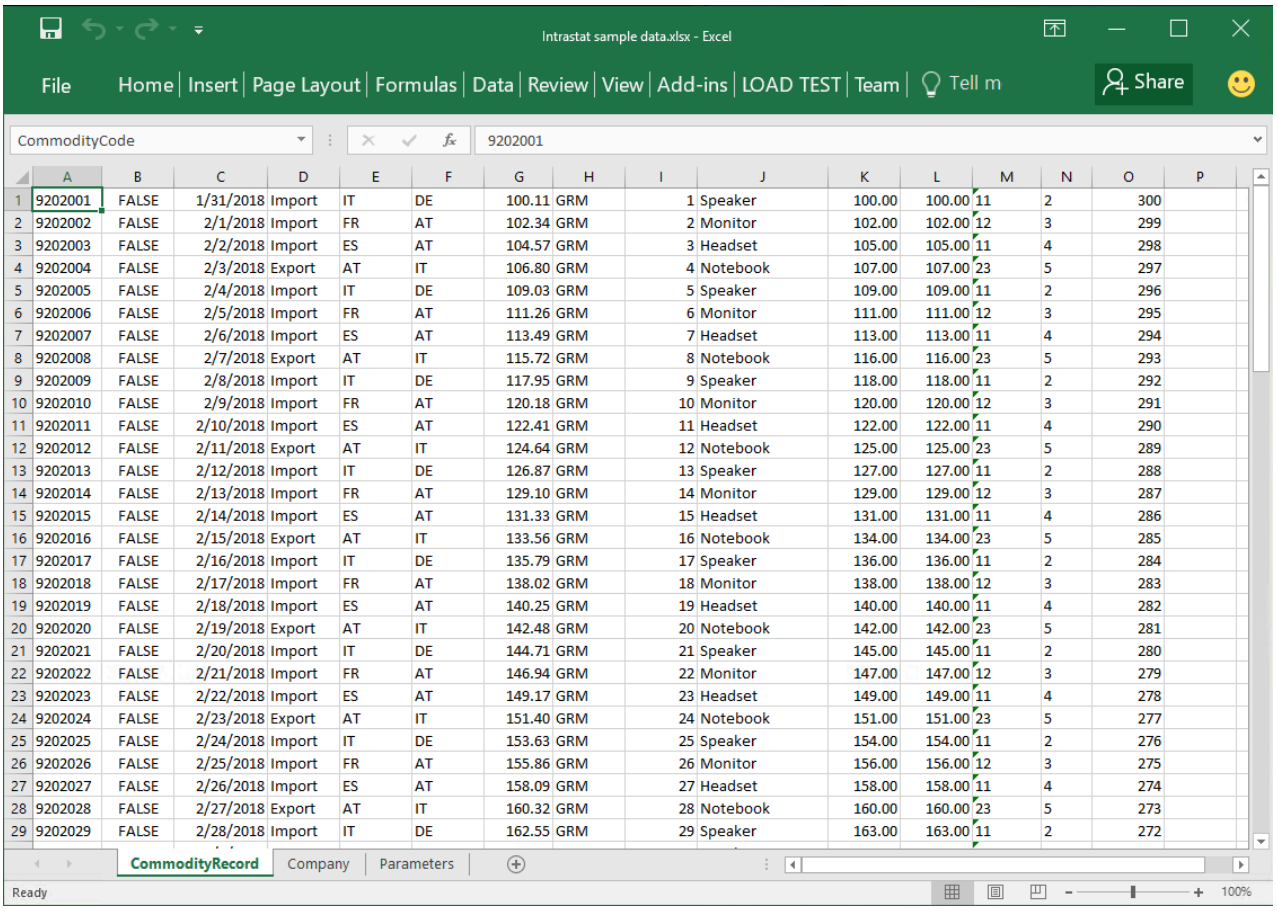
Import the format configuration

Next, load the **Intrastat (import from Excel)** sample ER format. This format is designed to parse a user-selected Microsoft Excel workbook that simulates foreign trade transactions.

1. On **Configurations**, select **Exchange** > **Load from XML file**.
2. Select **Browse**. Go to and select the **Intrastat (import from Excel).version.1.1.xml** file that you downloaded earlier as a prerequisite.
3. Select **OK**.
4. In the tree, select **Intrastat model** > **Intrastat (import from Excel)**.
5. Select **Edit**.
6. Set the **Default for model mapping** option to **Yes**.

   > ⓘ **Note**
   >
   > If you previously set the **Default for model mapping** option to **Yes** for the **Intrastat model** configuration or another configuration that is nested under the **Intrastat model** configuration, set this option to **No**.

   When you set the **Default for model mapping** option to **Yes**, the imported **Intrastat (import from Excel)** ER format becomes the default data source for the **Intrastat report (PDF)** format configuration. Then, when you run the **Intrastat report (PDF)** format configuration, the content of the Excel workbook that the **Intrastat (import from Excel)** ER format parses simulates foreign trade transactions that you must report. The following illustration shows an example of an Excel workbook.

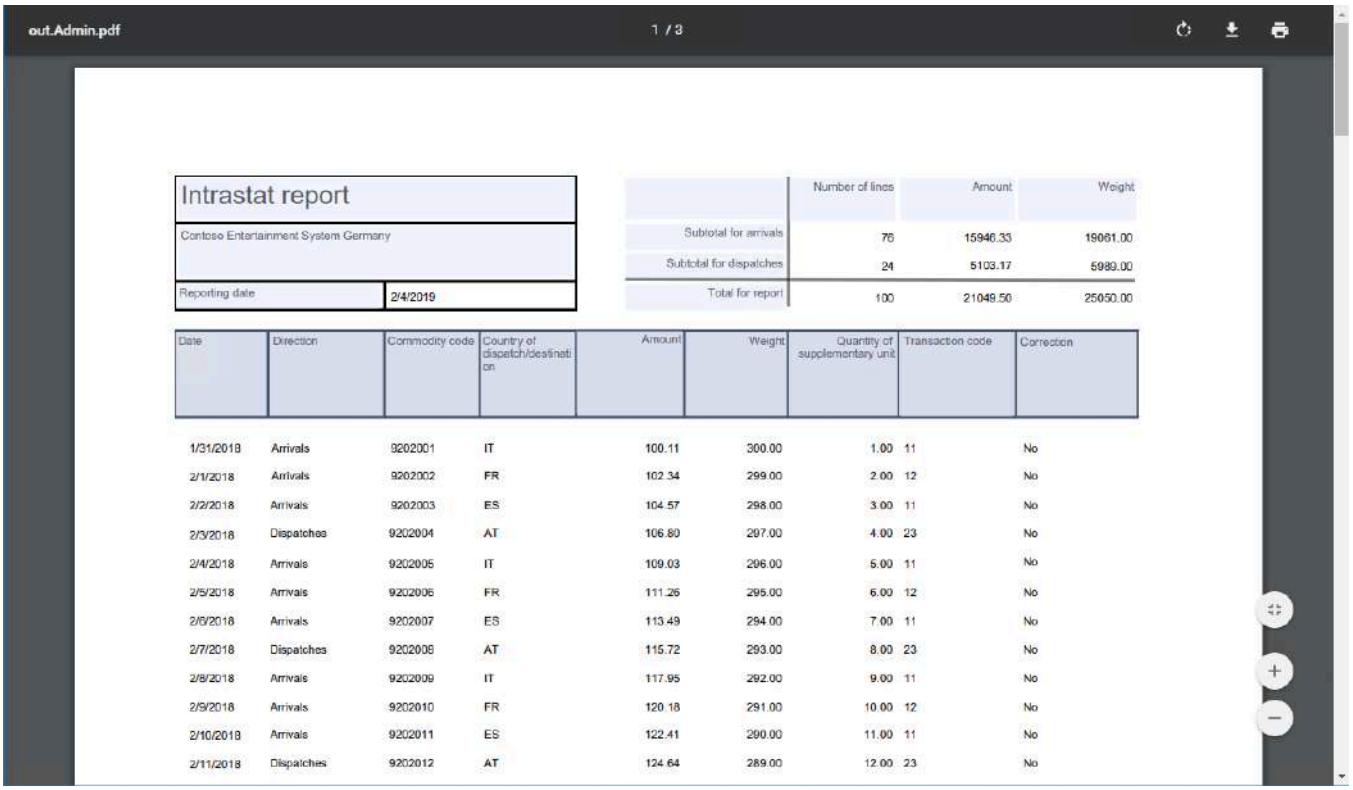


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	9202001	FALSE	1/31/2018	Import	IT	DE	100.11	GRM	1	Speaker	100.00	100.00	11	2	300	
2	9202002	FALSE	2/1/2018	Import	FR	AT	102.34	GRM	2	Monitor	102.00	102.00	12	3	299	
3	9202003	FALSE	2/2/2018	Import	ES	AT	104.57	GRM	3	Headset	105.00	105.00	11	4	298	
4	9202004	FALSE	2/3/2018	Export	AT	IT	106.80	GRM	4	Notebook	107.00	107.00	23	5	297	
5	9202005	FALSE	2/4/2018	Import	IT	DE	109.03	GRM	5	Speaker	109.00	109.00	11	2	296	
6	9202006	FALSE	2/5/2018	Import	FR	AT	111.26	GRM	6	Monitor	111.00	111.00	12	3	295	
7	9202007	FALSE	2/6/2018	Import	ES	AT	113.49	GRM	7	Headset	113.00	113.00	11	4	294	
8	9202008	FALSE	2/7/2018	Export	AT	IT	115.72	GRM	8	Notebook	116.00	116.00	23	5	293	
9	9202009	FALSE	2/8/2018	Import	IT	DE	117.95	GRM	9	Speaker	118.00	118.00	11	2	292	
10	9202010	FALSE	2/9/2018	Import	FR	AT	120.18	GRM	10	Monitor	120.00	120.00	12	3	291	
11	9202011	FALSE	2/10/2018	Import	ES	AT	122.41	GRM	11	Headset	122.00	122.00	11	4	290	
12	9202012	FALSE	2/11/2018	Export	AT	IT	124.64	GRM	12	Notebook	125.00	125.00	23	5	289	
13	9202013	FALSE	2/12/2018	Import	IT	DE	126.87	GRM	13	Speaker	127.00	127.00	11	2	288	
14	9202014	FALSE	2/13/2018	Import	FR	AT	129.10	GRM	14	Monitor	129.00	129.00	12	3	287	
15	9202015	FALSE	2/14/2018	Import	ES	AT	131.33	GRM	15	Headset	131.00	131.00	11	4	286	
16	9202016	FALSE	2/15/2018	Export	AT	IT	133.56	GRM	16	Notebook	134.00	134.00	23	5	285	
17	9202017	FALSE	2/16/2018	Import	IT	DE	135.79	GRM	17	Speaker	136.00	136.00	11	2	284	
18	9202018	FALSE	2/17/2018	Import	FR	AT	138.02	GRM	18	Monitor	138.00	138.00	12	3	283	
19	9202019	FALSE	2/18/2018	Import	ES	AT	140.25	GRM	19	Headset	140.00	140.00	11	4	282	
20	9202020	FALSE	2/19/2018	Export	AT	IT	142.48	GRM	20	Notebook	142.00	142.00	23	5	281	
21	9202021	FALSE	2/20/2018	Import	IT	DE	144.71	GRM	21	Speaker	145.00	145.00	11	2	280	
22	9202022	FALSE	2/21/2018	Import	FR	AT	146.94	GRM	22	Monitor	147.00	147.00	12	3	279	
23	9202023	FALSE	2/22/2018	Import	ES	AT	149.17	GRM	23	Headset	149.00	149.00	11	4	278	
24	9202024	FALSE	2/23/2018	Export	AT	IT	151.40	GRM	24	Notebook	151.00	151.00	23	5	277	
25	9202025	FALSE	2/24/2018	Import	IT	DE	153.63	GRM	25	Speaker	154.00	154.00	11	2	276	
26	9202026	FALSE	2/25/2018	Import	FR	AT	155.86	GRM	26	Monitor	156.00	156.00	12	3	275	
27	9202027	FALSE	2/26/2018	Import	ES	AT	158.09	GRM	27	Headset	158.00	158.00	11	4	274	
28	9202028	FALSE	2/27/2018	Export	AT	IT	160.32	GRM	28	Notebook	160.00	160.00	23	5	273	
29	9202029	FALSE	2/28/2018	Import	IT	DE	162.55	GRM	29	Speaker	163.00	163.00	11	2	272	

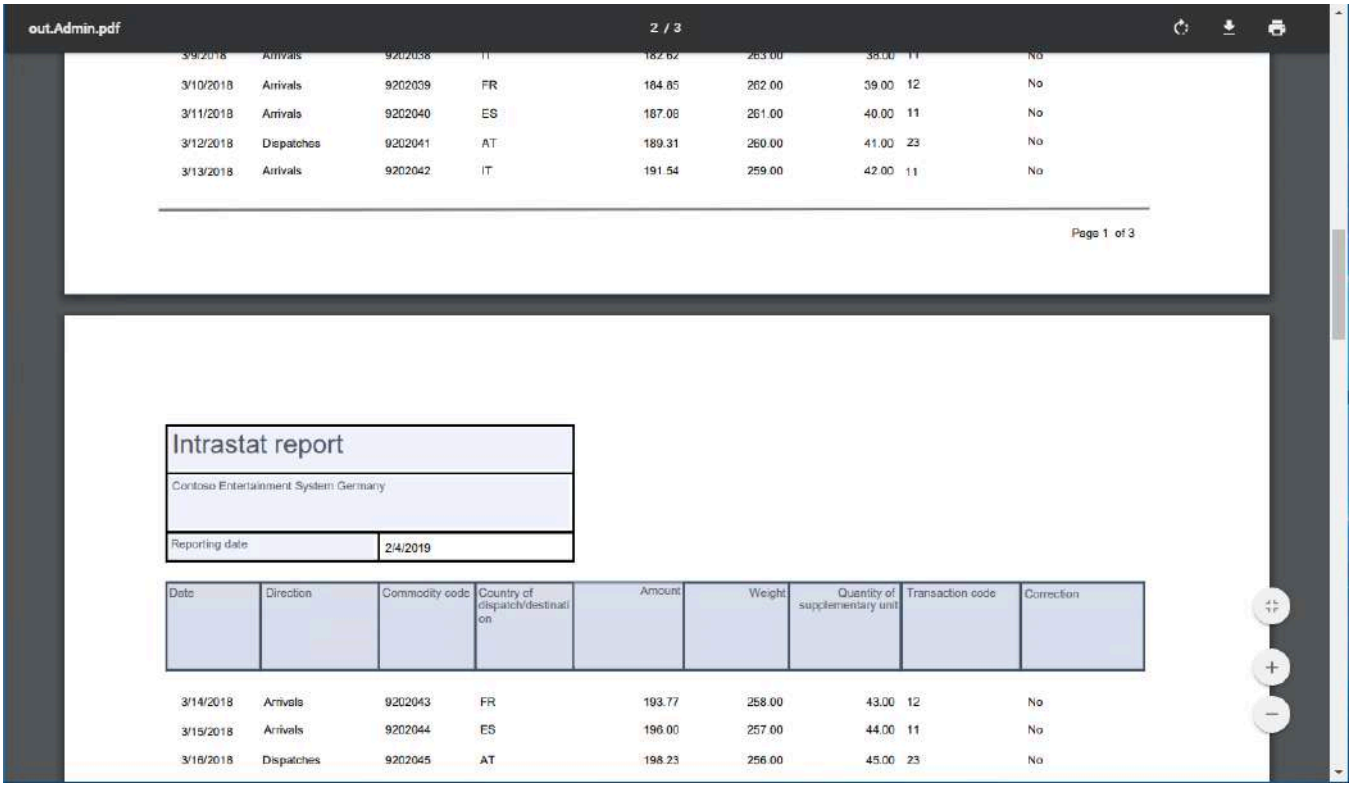
Run the format configuration

1. On **Configurations**, in the tree, select **Intrastat model** > **Intrastat report (PDF)**.
2. Select **Run**.
3. Select **Browse**. Go to the **Intrastat sample data.xlsx** file that you downloaded earlier as a prerequisite, and select it.
4. Select **OK**.
5. In the **Report direction** field, select **Both** to include all transactions from the imported Excel workbook in the generated PDF report.
6. Select **OK**.
7. Review the generated PDF document.

The following illustration shows an example of the first page of the generated report.



The following illustration shows an example of another page of the generated report.



Limitations

The names of fillable fields need to be unique in the PDF form that you plan to use as a report template. When you import a PDF form, the editable ER format creates an individual format element with the corresponding name for every such field. If a PDF form contains several fields

with the same name, the import process creates a single format element for those fields. This single format element doesn't allow you to fill in the fields individually at runtime.

Frequently asked questions

When I run the ER format to generate a report in PDF format, why do I get the following errors: Cannot handle iref streams. The current implementation of PDFSharp cannot handle this PDF feature introduced with Acrobat 6. and A PDF name must start with a slash (/).

The ER framework uses version 1.5 of the PDFSharp library to generate these PDF reports. Some features of PDF 1.5 (Adobe Reader 6.0) aren't yet implemented in this library. Therefore, PDFSharp can't yet open some files that are marked as **for PDF 1.5 or higher** and can result in the errors received. Use one of the following solutions to resolve the issue:

- When you use your own PDF template: Downgrade the template to an earlier Adobe version and start using a new template in your ER format.
- When you use an ER format template that another configuration provider shared with you as part of an ER solution: Contact the owner of this ER solution and provide a description of the issue.
- When you use the ISV solution that contains an earlier version of the PDFSharp library: Contact the owner of the solution and suggest an upgrade to the newer PDFSharp version.

Additional resources

- ER Design a configuration for generating reports in OPENXML format (November 2016)
- Design ER configurations to generate reports in Word format

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Last updated on 04/13/2026

Embed images and shapes in documents that you generate by using ER

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can use the Electronic reporting (ER) tool to design reports that you can run to generate required electronic documents. Use Microsoft Excel or Microsoft Word documents to specify the layout of a report. The ER Operations designer lets you attach the Excel or Word document as a template for the report. The named elements in the attached template are associated with the format elements of the ER report. Format elements of the report are bound to data sources. These elements specify the data that will be entered, at runtime, in the documents that are generated.

This new functionality goes beyond existing ER capabilities for creating documents in Microsoft Office formats. For more information, see the following task guides. You can find these task guides under the **7.5.4.3 Acquire/Develop IT service/solution components (10677)** business process.

- ER Design a configuration for generating reports in OPENXML format
- ER Design a configuration for generating reports in Microsoft WORD format

Embed an image in an Excel document

Embed an image in an Excel worksheet

First, add a placeholder for the image in an Excel document. Open an Excel workbook, and add a picture as a placeholder for the image that you add later. Then use the ER tool to add a new ER format configuration to include the report that you're designing. Attach the Excel workbook as a template for the format of the report, and then import the content of the workbook into the ER format. The format definition is created automatically. The image placeholder that you added is included in the ER format definition as a **CELL** element.

> **Note**

You can manually specify the format definition instead of importing it. When you save your changes, the format is validated.

Next, bind the **CELL** element of the ER format to the field from the format's data source that provides the picture's data in binary format at runtime. When an ER data model is used as a format's data source, the data type of the field must be Container. Currently, an ER data model field that has the Container data type can be bound to several types of data sources that return images in binary format. You can access a field in a data table and a file that is attached to the data table's record by using the Document management framework.

Important

- To fill the image placeholder in the document that you're creating by using the Excel template, the ER format must contain the **CELL** element that refers to the named picture element in the Excel template. Otherwise, no image placeholder appears in the report's output. If the binding of a **CELL** element returns no data at runtime, the generated document shows the image placeholder from the template. To hide an image in the document that you're generating, define a **CELL** element, and specify that the **Enabling** expression returns **FALSE**.
- In the Excel template, use a unique name for every element. These elements include pictures and cells. If you duplicate an element name, the content of the generated report is ambiguous and confusing.

Embed images in the header and footer of an Excel worksheet

Use the Excel workbook document as a template to specify the layout of a report. The workbook can contain multiple worksheets, each of which can be designed to have a header and a footer. Excel supports up to three images in the header and footer of every worksheet. The images can be aligned to the left, right, or center.

In Finance release 10.0.21, you can manage the header and footer images generated by an ER solution that has an Excel template.

If you want a default picture to appear in the header or footer of a generated document, add an image to the header or footer of a worksheet of an ER template. To access this image in your ER format, add the **Picture** component under the Header or Footer component of the format. Configure the alignment of the **Picture** component as appropriate.

You can also use the **Picture** component to put an image into the header or footer of a generated document at runtime, even if the template contains no default image. To specify the media content that you want to put into the header or footer of a generated document as either a new image or a replacement for a default image, bind the **Picture** component to a data source of the Container type that represents an image in binary format.

Every **Header** or **Footer** component can hold one **Picture** component for each supported alignment: **Left**, **Center**, and **Right**.

The **Scale the height** property of the **Picture** component lets you use the configured binding to control the size of an image:

- Select **Original** to keep the initial size of the image.
- Select **Relative** to scale the height of the image in proportion to the height of the header or footer that holds that image.
  - In this case, define the height of the image as a percentage of the parent header or footer.
  - If the scaling value exceeds 100 percent, the image might overlap the data area of the corresponding worksheet.
  - If the scaling changes the height of the image, the width also changes to maintain the image's original aspect ratio.
- Select **Absolute** to resize the image according to the height and width values (in pixels) provided at design time.
  - If the specified height exceeds the height of the parent header or footer, the image might overlap the data area of the corresponding worksheet.

You can also use the **Enabled** property of the **Picture** component to control the visibility of an image that you put into a generated document by using this component.

> ⓘ **Note**
>
> You must use the ER format designer to add a **Picture** component to the editable ER format. You can't add the component when you use the Business document management workspace to edit the template of a business document.

To learn more about this functionality, see Design an ER format to generate a report in Excel format with embedded images in page headers or footers.

Embed a shape in an Excel document

First, add a placeholder for the shape in an Excel document. Open an Excel workbook, and select **Shape**, **Text box**, or **WordArt** as a placeholder for the shape. Then use the ER tool to add a new ER format configuration to include the report that you're designing. Attach the Excel workbook as a template for the format of the report, and then import the content of the workbook into the ER format. The format definition is created automatically. The shape placeholder that you added is included in the ER format definition as a **CELL** element that refers to the named Excel shape element.

> ⓘ **Note**
>
> You can manually specify the format definition instead of importing it. When you save your changes, the format is validated.

Next, bind the **CELL** element in the ER format to the field from the format's data source that provides the data at runtime. You can convert this data to a text string. When the **CELL** element in the ER format refers to a shape element in the Excel template that supports text, the text provided through this binding at runtime appears in a shape in the generated document.

> ⓘ **Important**
>
> - To use the Excel template that includes the shape placeholder to generate a new document, the ER format must contain a **CELL** element that refers to the Excel shape element. Otherwise, no shape placeholder appears in the report's output. If the binding of a **CELL** element that refers to the named Excel shape element returns no data at runtime, the generated document shows the text of the shape placeholder from the Excel template. To hide a shape in the generated document, define a **CELL** element that refers to the named Excel shape element, and specify that the **Enabling** expression returns **FALSE**.
> - In the Excel template, use a unique name for every element. These elements include shapes and cells. If you duplicate an element name, the content of the generated report is ambiguous and confusing.

Embed an image in a Word document

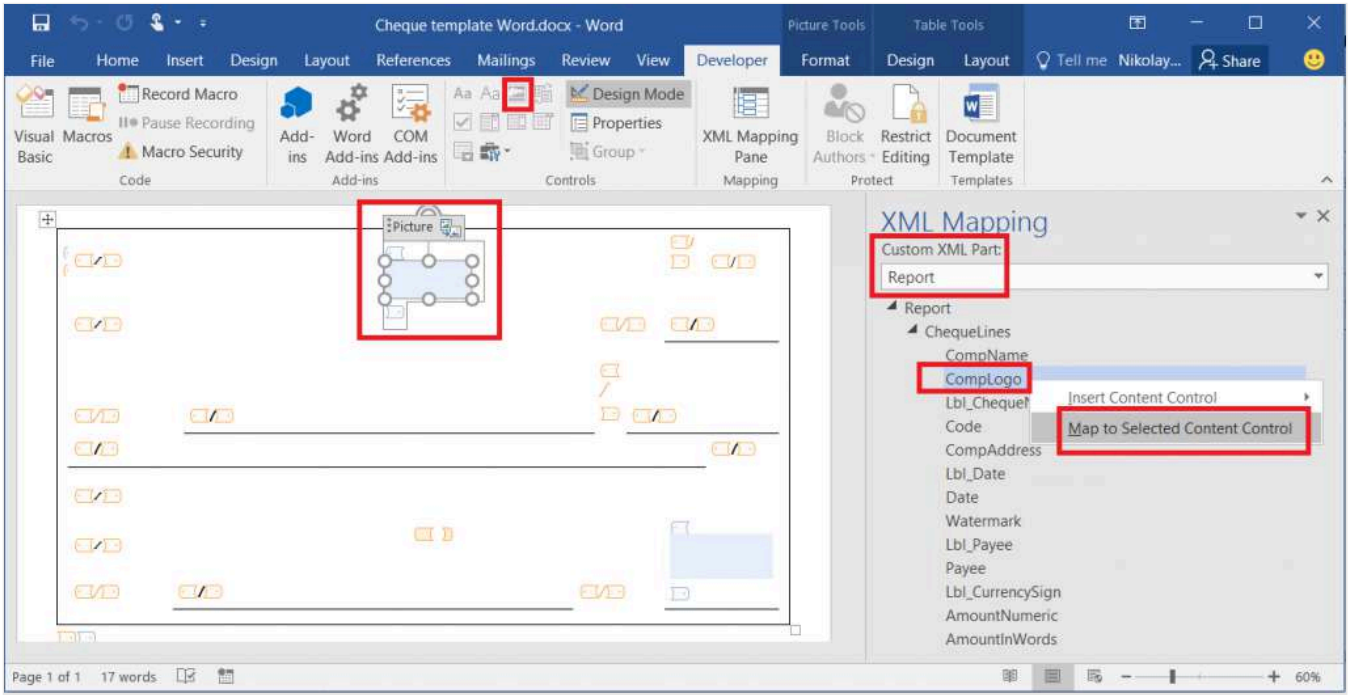
> ⓘ **Important**
>
> You can reuse the ER format that uses an Excel template to create documents that include embedded images. In the ER format, make sure that a name is specified for the **CELL**

element that refers to a named picture element in Excel, and that is bound to a data source that returns a picture at runtime.

First, configure the Word document's layout. Use the **Picture Content** control to create a placeholder for the embedded image. To access this control, you must first make the **Developer** tab visible on the Word Ribbon.

Next, delete the Excel template from the ER format, and attach the Word template document. Update the reference to the template, and save your changes. The structure of the current ER format is saved to the Word template as a new custom XML part that is named **Report**.

Next, save the Word template for the current ER format to your local computer. Open the template, and open the **XML Mapping** pane. Find the custom XML part that is named **Report**, and then point to the **CELL** element in the ER report that is bound to a data source that returns an image in binary format. Map this XML part's item to the selected **Picture Content** control, and save your changes.



Finally, delete the Word template from the ER format, and attach the Word document that includes the mapped custom XML parts. Update the format reference to the template, and save the changes that you made to this ER format.

More information

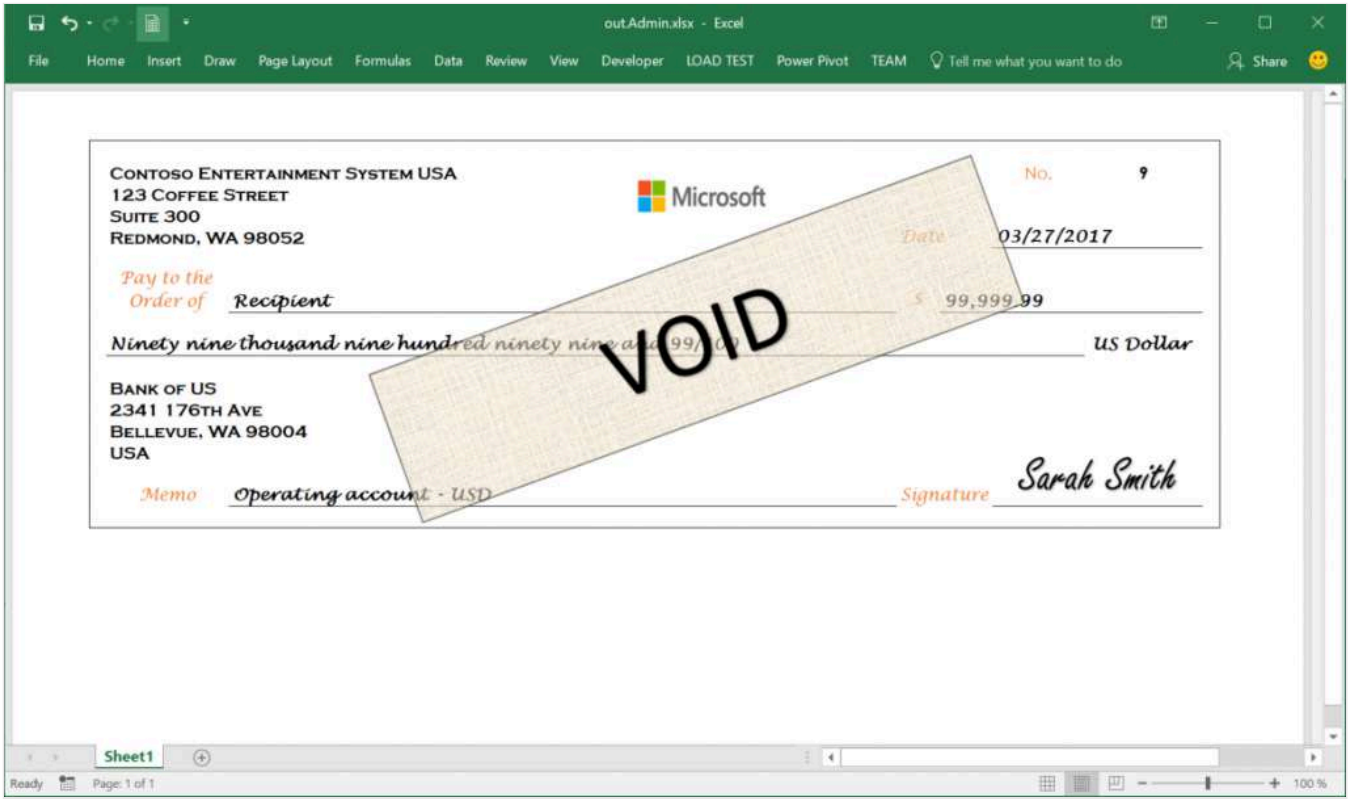
To learn more about this feature, play the set of task guides, **ER Make reports in MS Office formats with embedded images**. These task guides show how you can embed the images of a company logo and an authorized person's signature in the payment checks that the ER tool generates in Excel and Word documents.

The following table lists the files that you need to complete the **ER Make reports in MS Office formats with embedded images** task guides. Download and save the files to your local computer.

Expand table

Description	File name
ER data model configuration	Model for cheques.xml
ER format configuration	Cheques printing format.xml
Company logo image	Company logo.png
Signature image	Signature image.png
Alternative signature image	Signature image 2.png
Microsoft Word template for printing payment checks	Cheque template Word.docx
Microsoft Excel template for printing payment checks	Cheque template.xlsx

The following graphic provides an example of the test printout for a payment check that is generated from the Excel template.



Last updated on 03/17/2026

Design configurations to generate reports in Office format that have embedded images

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete the steps in this procedure, first complete the procedure, "ER Create a configuration provider and mark it as active." This procedure explains how to design Electronic reporting (ER) configurations to generate a Microsoft Excel or Word document that contains embedded images. In this procedure, you create the required ER configurations for the sample company, Litware, Inc. You can complete these steps by using the USMF dataset. This procedure is created for users with the assigned role of system administrator or electronic reporting developer. Before you begin, download and save the following files:

Expand table

Description	File name
ER data model configuration	Model for cheques.xml
ER format configuration	Cheques printing format.xml
Company logo image	Company logo.png
Signature image	Signature image.png
Alternative signature image	Signature image 2.png
Microsoft Word template for printing payment checks	Cheque template Word.docx

Verify prerequisites

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

2. Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the procedure, "Create a configuration provider and mark it as active."
3. Select **Reporting configurations**.

Add a new ER model configuration

1. Instead of creating a new model, load the ER model configuration file (`Model for cheques.xml`) that you saved earlier. This file contains the sample data model for payment cheques and the mapping of the data model to the data components of the Dynamics 365 for Operations application.
2. On the **Versions** FastTab, click **Exchange**.
3. Click **Load from XML file**.
4. Click **Browse**, and then select `Model for cheques.xml`.
5. Click **OK**.
6. Use the loaded model as a data source of information to generate documents that contain images in Excel and Word.

Add a new ER format configuration

1. Instead of creating a new format, load the ER format configuration file (`Cheques printing format.xml`) that you saved earlier. This file contains the sample layout of the format to print cheques using the pre-printed form and the mapping of this format to the **Model for cheques** data model.
2. Click **Exchange**.
3. Click **Load from XML file**.
4. Click **Browse** and select the `Cheques printing format.xml` file.
5. Click **OK**.
6. In the tree, expand **Model for cheques**.
7. In the tree, select **Model for cheques\Cheques printing format**.
8. Use the loaded format to generate documents that contain images in Excel and Word.

Configure ER user parameters

1. On the Action Pane, click **Configurations**.
2. Click **User parameters**.
3. Select **Yes** in the **Run settings** field.
   Turn on the **Run draft** flag to start the draft version of the selected format instead of the

completed one.
4. Click **OK**.

Configure Cash and bank management parameters

1. Go to **Cash and bank management** > **Bank accounts** > **Bank accounts**.
2. Use the Quick Filter to filter on the **Bank account** field with a value of `USMF OPER`.
3. On the Action Pane, select **Set up**.
4. Select **Check**.
5. Expand the **Setup** section.
6. Select **Edit**.
7. Select **Yes** in the **Company logo** field.
8. Select **Company logo**.
9. Select **Change**.
10. Select **Browse** and select the file that you downloaded earlier, **Company logo.png**.
11. Select **Save**.
12. Close the page.
13. Expand the **Signature** section.
14. Select **Yes** in the **Print first signature** field.
15. Select **Change**.
16. Select **Browse** and select the file that you downloaded earlier, **Signature image.png**.
17. Expand the **Copies** section.
18. In the **Watermark** field, select an option.
19. Select **Yes** in the **Generic electronic Export format** field.
20. Select **Cheques printing form** configuration.
21. Now the selected ER format is used for printing cheques.
22. Select **Attach**.
23. Select **New**.
24. Select **File**.
25. Select **Browse** and select the file that you downloaded earlier, **Signature image 2.png**.
26. Close the page.
27. Close the page.
28. Close the page.
29. Go to **Cash and bank management** > **Setup** > **Cash and bank management parameters**.
30. Select **Yes** in the **Allow prenote creation on inactive bank accounts:** field.
31. Select **Save**.
32. Close the page.

Last updated on 04/13/2026

Review configurations to generate reports in Office format that have embedded images

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete these steps, you must first complete the steps in the "ER Make reports in MS Office formats with embedded images (Part 1: Set up parameters)" task guide.

This procedure shows how to design Electronic reporting (ER) configurations to generate electronic documents that contain embedded images in Microsoft Excel and Microsoft Word. In this example, you review ER configurations for the sample company Litware, Inc.

This procedure is intended for users who have the System administrator or Electronic reporting developer role assigned to them. You can complete the steps by using the USMF data set.

Review the imported data model

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, select **Model for cheques**.
3. Select **Designer**. * This model represents payment cheques from the business standpoint and the mapping of this model to the application's data sources. Review this model by the ER Operations designer. The attributes of the model elements include structure, name, description, data type, and more.
4. In the tree, expand **root**.
5. In the tree, select **root\cheques**.
6. In the tree, expand **root\cheques**.
7. In the tree, expand **root\cheques\attributes**.
8. In the tree, expand **root\payer**.
9. In the tree, select **root\istestrun**.
10. In the tree, select **root\layout**.

    - The layout element of this model represents the details of the printing cheque form layout for the selected bank account. It also includes two nodes of the Container

data type to store images.
11. In the tree, expand **root\layout**.
12. In the tree, select **root\layout\company logo**.
13. In the tree, expand **root\layout\company logo**.
14. In the tree, select **root\layout\company logo\image**.
15. In the tree, select **root\layout\company logo\isprinted**.
16. In the tree, select **root\layout\signature**.
17. In the tree, expand **root\layout\signature**.
18. In the tree, select **root\layout\signature\image**.
19. In the tree, select **root\layout\signature\isprinted**. * Two image data model elements are bound to the fields of the tables that contain images of the company logo and the authorized person's signature in binary format.
20. In the tree, expand **root\layout\watermark**.
21. Select **Map model to datasource**.
22. Select **Designer**.
23. In the tree, expand **chequesselected**.
24. In the tree, expand **layout**.
25. In the tree, expand **layout\company logo**.
26. In the tree, expand **layout\signature**.
27. In the tree, expand **layout\watermark**.
28. Toggle **Show details** on. * The cheques data model element is bound to the TmpChequePrintout table that, at runtime, contains records for cheques that the user selects for printing.
29. Close the page.
30. Close the page.
31. Close the page.

Review the imported format

1. In the tree, expand **Model for cheques**.
2. In the tree, select **Model for cheques\Cheques printing format**.
3. Select **Designer**.
4. Select **Attachments**.
5. Select **Open**.

- Open the attached report's template in Excel.
  - Review the attached report's Excel template that you use to print cheques. The template contains two cheques per page and is designed to print cheques to the preprinted form. Two blank images are embedded. These blank images are for the company logo and the signature of the person who is authorizing a payment.

Verify that the images are named **CompLogo** and **SignatureImage**, respectively, in Excel.

6. Close the page.
7. In the tree, expand **Report**.
8. In the tree, expand **Report\ChequeLines**.
9. In the tree, select **Report\ChequeLines\CompLogo**.
10. Toggle **Show details** on.

    - The **CompLogo** format's cell element represents the Excel item that populates the company logo image in the report. This format element is bound to the image data model element that, at runtime, contains a company logo image in binary format.

11. Select the **Mapping** tab.
12. Select **Edit enabled**. * You can make the **CompLogo** format's cell element so that it's no longer enabled. In this case, the associated Excel image element hides a company logo in the generated report. If the enabled expression returns **TRUE** and the defined binding brings no image, the associated Excel image element shows an image saved in the Excel template.
13. Close the page.
14. In the tree, expand **labels: Container**. * Some labels presented in the preprinted cheque form are included in the report when you create it for testing purposes. However, the preprinted form already includes these labels, so the labels don't print during real printing.
15. Close the page.

Last updated on 03/17/2026

Generate reports in Office format that have embedded images

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user playing either 'System administrator' or 'Electronic reporting developer' role can design Electronic reporting (ER) configurations to generate electronic documents in MS office formats (Excel and Word) containing embedded images.

In this example, you use the created ER configurations for sample company, 'Litware, Inc.'. To complete these steps, you must first complete the steps in the "ER Make reports in MS Office formats with embedded images (Part 2: Review configurations)" task guide. You can perform these steps in 'USMF' company.

Run format with initial model mapping

1. Go to **Cash and bank management** > **Bank accounts** > **Bank accounts**.
2. Use the Quick Filter to filter on the Bank account field with a value of **USMF OPER**.
3. On the Action Pane, select **Set up**.
4. Select **Check**.
5. Select **Print test**.

    - Run the format for testing purposes.

6. Select **Yes** in the Negotiable check format field.
7. Select **OK**.

    - Review the created output. The company logo is presented in the report as well as the authorized person's signature. The signature image is taken from the field of the 'Container' data type of the cheque layout record that is associated with the selected bank account.

8. Expand the Copies section.
9. Select **Edit**.
10. In the Watermark field, enter **Print watermark as Void**.

- Change the watermark layout setting to show the watermark text in generating document in an Excel shape element.

11. Select **Print test**.
12. Select **OK**.

    - Review the created output. The watermark is shown in the created report in accordance to the selection option.

13. Close the page.
14. On the Action Pane, select **Manage payments**.
15. Select **Checks**.
16. Select **Show filters**.
17. Apply the following filters: Enter a filter value of "381","385","389" on the "Check number" field using the "is one of" filter operator.
18. In the list, mark all rows.
19. Select **Print check copy**.

    - Run the format to reprint the selected cheques.
    - Review the created output. The selected cheques are reprinted. The company logo and labels don't print out since they appear on the pre-printed form.

Modify the mapping of the imported data model

1. Close the page.
2. Close the page.
3. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
4. In the tree, select *Model for cheques*.
5. Select **Designer**.
6. Select **Map model to datasource**.
7. Select **Designer**. * Change the binding of the data model's signature item to get the signature image from the file that's attached to the cheque layout record associated with the selected bank account.
8. Turn off **Show details**.
9. In the tree, expand *layout*.
10. In the tree, expand *layout\signature*.
11. In the tree, select *layout\signature\image = chequesaccount.'<Relations'.BankChequeLayout.Signature1Bmp'*.
12. In the tree, expand *chequesaccount*.
13. In the tree, expand *chequesaccount.'<Relations'*.
14. In the tree, expand *chequesaccount.'<Relations'.BankChequeLayout'*.
15. In the tree, expand *chequesaccount.'<Relations'.BankChequeLayout.'<Relations'*.

16. In the tree, expand *chequesaccount.'<Relations'.BankChequeLayout.'<Relations'.<Documents'*.
17. In the tree, select *chequesaccount.'<Relations'.BankChequeLayout.'<Relations'.<Documents.getFileContentAsContainer()*.
18. Select **Bind**.
19. Select **Save**.
20. Close the page.
21. Close the page.
22. Close the page.
23. Close the page.

Run format using the adjusted model mapping

1. Go to **Cash and bank management** > **Bank accounts** > **Bank accounts**.
2. Use the Quick Filter to find records. For example, filter on the **Bank account** field with a value of `USMF OPER`.
3. On the Action Pane, select **Set up**.
4. Select **Check**.
5. Select **Print test**.
6. Select **OK**.

    - Review the created output. The image from the Document Management attachment is presented as the signature of an authorized person.

Use MS Word document as a template in the imported format

1. Close the page.
2. Close the page.
3. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
4. In the tree, expand *Model for cheques*.
5. In the tree, select *Model for cheques\Cheques printing format*.
6. Select **Designer**.
7. Select **Attachments**.
8. Select **Delete**.
9. Select **Yes**.
10. Select **New**.
11. Select **File**.

    - Select **Browse** and select the downloaded in advance *Cheque template Word.docx* file.
12. Close the page.
13. In the **Template** field, enter or select a value.
14. Select **Save**.
15. Close the page.
16. Select **Edit**.
17. Select **Yes** in the **Run Draft** field.
18. Close the page.
19. Go to **Cash and bank management** > **Bank accounts** > **Bank accounts**.
20. Use the **Quick Filter** to filter on the **Bank account** field with a value of *USMF OPER*.
21. Select **Check**.
22. Select **Print test**.
23. Select **OK**.

    - Review the created output. The output is generated as a Word document with embedded images presenting the company logo, the signature of an authorized person, and the selected text of the watermark.

Last updated on 03/17/2026

Design an ER format to generate a report in Excel format with embedded images in page headers or footers

This article explains how a user in the System Administrator or Electronic Reporting Functional Consultant role can perform these tasks:

- Configure parameters for the Electronic reporting (ER) framework.
- Import ER configurations that Microsoft provides and are used to generate free text invoices, based on a template in Microsoft Excel format.
- Create a custom (derived) version of a standard ER format configuration that Microsoft provides.
- Modify the custom ER format configuration so that it generates a free text invoice report that has a company logo image in the footer.

You can complete the procedures in this article in the **USMF** company. No coding is required. Before you begin, download and save the following file.

Expand table

Description	File name
Company logo image	Company logo.png

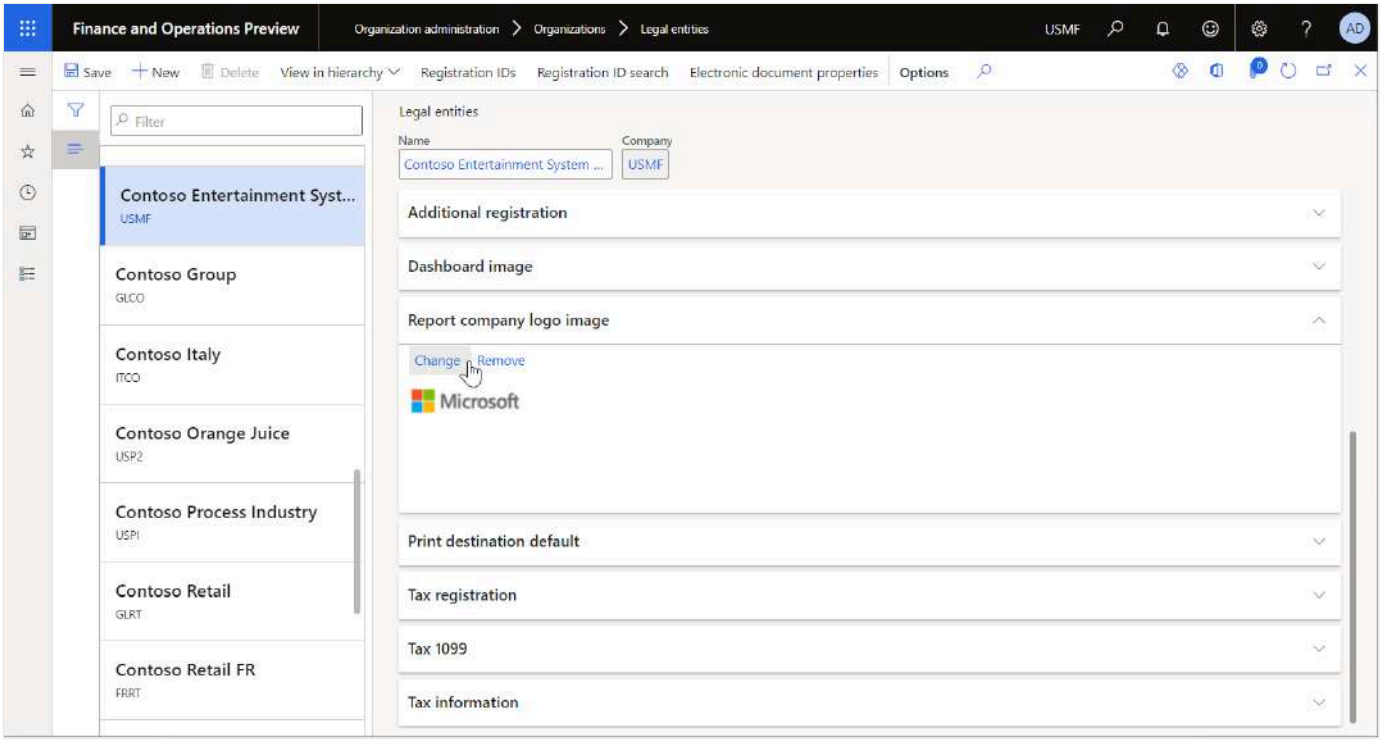
Content

- Configure the legal entity
- Configure the ER framework
  - Configure ER parameters
  - Activate an ER configuration provider
    - Review the list of ER configuration providers
    - Add a new ER configuration provider
    - Activate the new ER configuration provider
- Import the standard ER format configurations

    - Import the standard ER configurations
    - Review the imported ER configurations
- Print a free text invoice by using the standard ER format
    - Set up print management
    - Print a free text invoice
- Customize the standard ER format
    - Create a custom format
    - Edit the custom format
    - Mark the custom format as runnable
- Print a free text invoice by using the custom ER format
    - Set up print management
    - Print a free text invoice
- Additional resources

Configure the legal entity

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. On **Legal entities**, on the **Report company logo image** FastTab, select **Change**.
3. In **Select image file to upload**, select **Browse**, and select the **Company logo.png** file that you downloaded earlier.
4. Select **Save**, and then close **Legal entities**.



Configure the ER framework

As a user in the Electronic Reporting Functional Consultant role, you must configure the minimal set of ER parameters before you can start to use the ER framework to design a custom version of a standard ER format.

Configure ER parameters

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Electronic reporting parameters**.
3. On **Electronic reporting parameters**, on the **General** tab, set the **Enable design mode** option to **Yes**.
4. On the **Attachments** tab, set the following parameters:
    - In the **Configurations** field, select the **File** type for the **USMF** company.
    - In the **Job archive**, **Temporary**, **Baseline**, and **Others** fields, select the **File** type.

For more information about ER parameters, see Configure the ER framework.

Activate an ER configuration provider

Every ER configuration that you add is marked as owned by an ER configuration provider. Use the ER configuration provider activated in the **Electronic reporting** workspace for this purpose. Therefore, you must activate an ER configuration provider in the **Electronic reporting** workspace before you start to add or edit ER configurations.

> ⓘ **Note**
>
> You can only edit an ER configuration if you are the configuration owner. Before you can edit an ER configuration, you must activate the appropriate ER configuration provider in the **Electronic reporting** workspace.

Review the list of ER configuration providers

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Configuration providers**.

3. On **Configuration provider table**, each provider record has a unique name and URL. Review the contents of this page. If a record for **Litware, Inc.** (`https://www.litware.com`) already exists, skip the next procedure, Add a new ER configuration provider.

Add a new ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Configuration providers**.
3. On **Configuration providers**, select **New**.
4. In the **Name** field, enter **Litware, Inc.**
5. In the **Internet address** field, enter `https://www.litware.com`.
6. Select **Save**.

Activate the new ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, select the **Litware, Inc.** tile, and then select **Set active**.

For more information about ER configuration providers, see Create configuration providers and mark them as active.

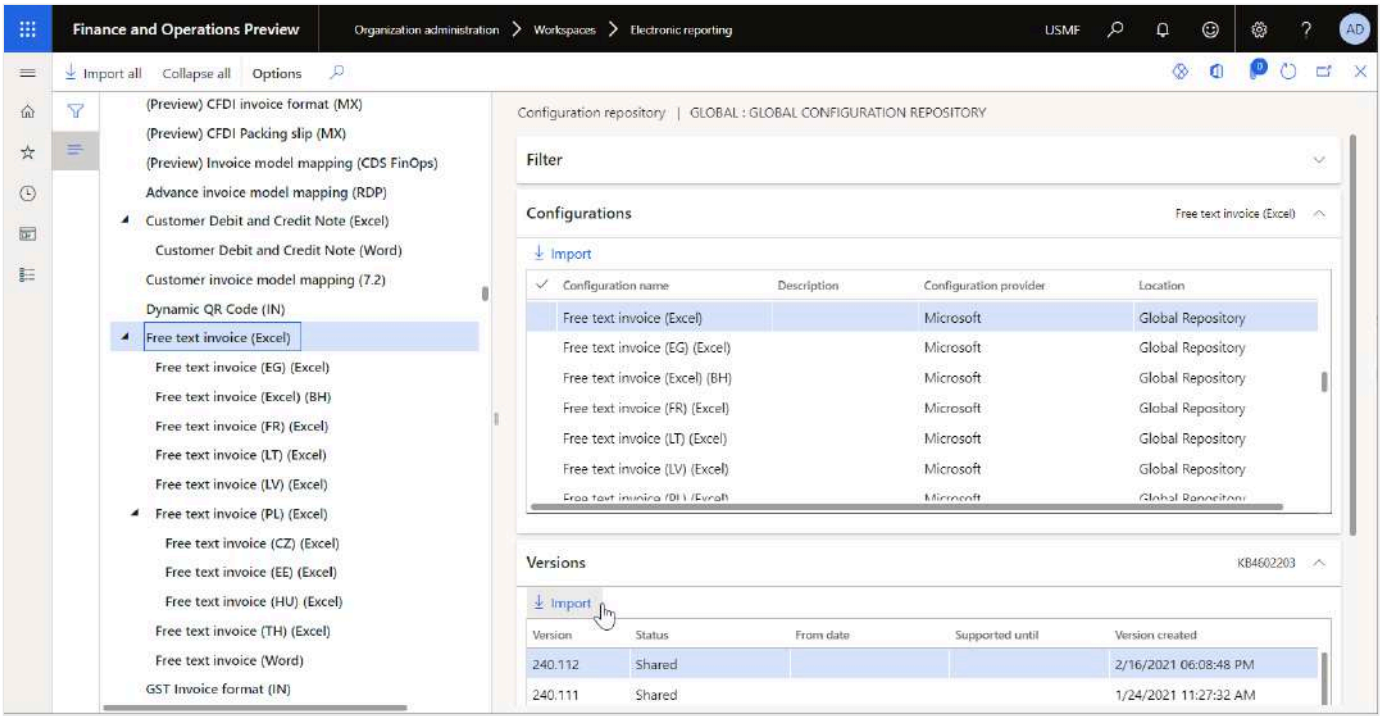
Import the standard ER format configurations

Import the standard ER configurations

To add the standard ER configurations to your current instance of Dynamics 365 Finance, you must import them from the ER repository that was configured for that instance.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, select the **Microsoft** tile, and then select **Repositories** to view the list of repositories for the **Microsoft** provider.
3. On **Configuration repositories**, select the repository of the **Global** type, and then select **Open**. If you're prompted for authorization to connect to Regulatory Configuration Service, follow the authorization instructions.
4. On **Configuration repository**, in the configuration tree in the left pane, select the **Free text invoice (Excel)** format configuration.
5. On the **Versions** FastTab, select the latest version (for example, **240.112**) of the selected ER format configuration.

6. Select **Import** to download the selected version from the Global repository to the current Finance instance.



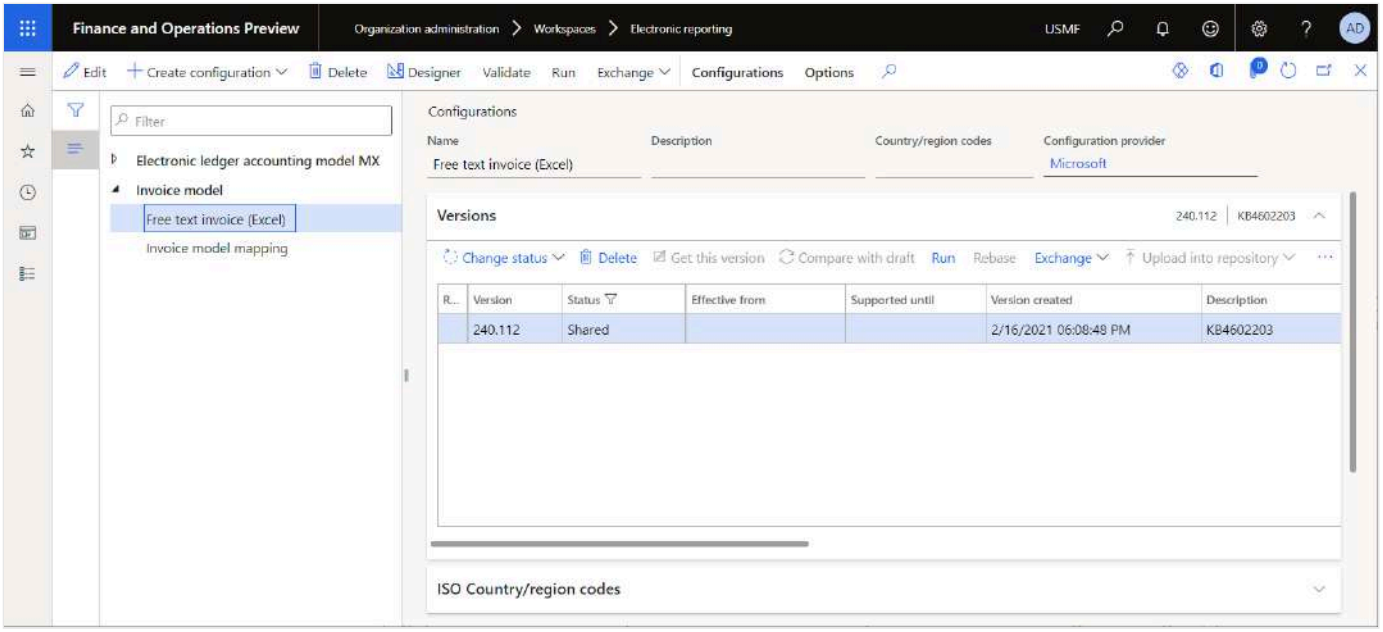
> **Tip**
>
> If you have trouble accessing the Global repository, you can download configurations from Microsoft Dynamics Lifecycle Services (LCS) instead.

Review the imported ER configurations

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configurations** section, select the **Reporting configurations** tile.
3. On **Configurations**, in the configuration tree in the left pane, expand **Invoice model**.
4. In addition to the selected **Free text invoice (Excel)** ER format, other required ER configurations were imported. Make sure that the following ER configurations are available in the configuration tree:
    - **Invoice model** – This configuration contains the data model ER component that represents the data structure of the invoicing business domain.
    - **Invoice model mapping** – This configuration contains the model mapping ER component that describes how the data model is filled in with application data at

runtime.

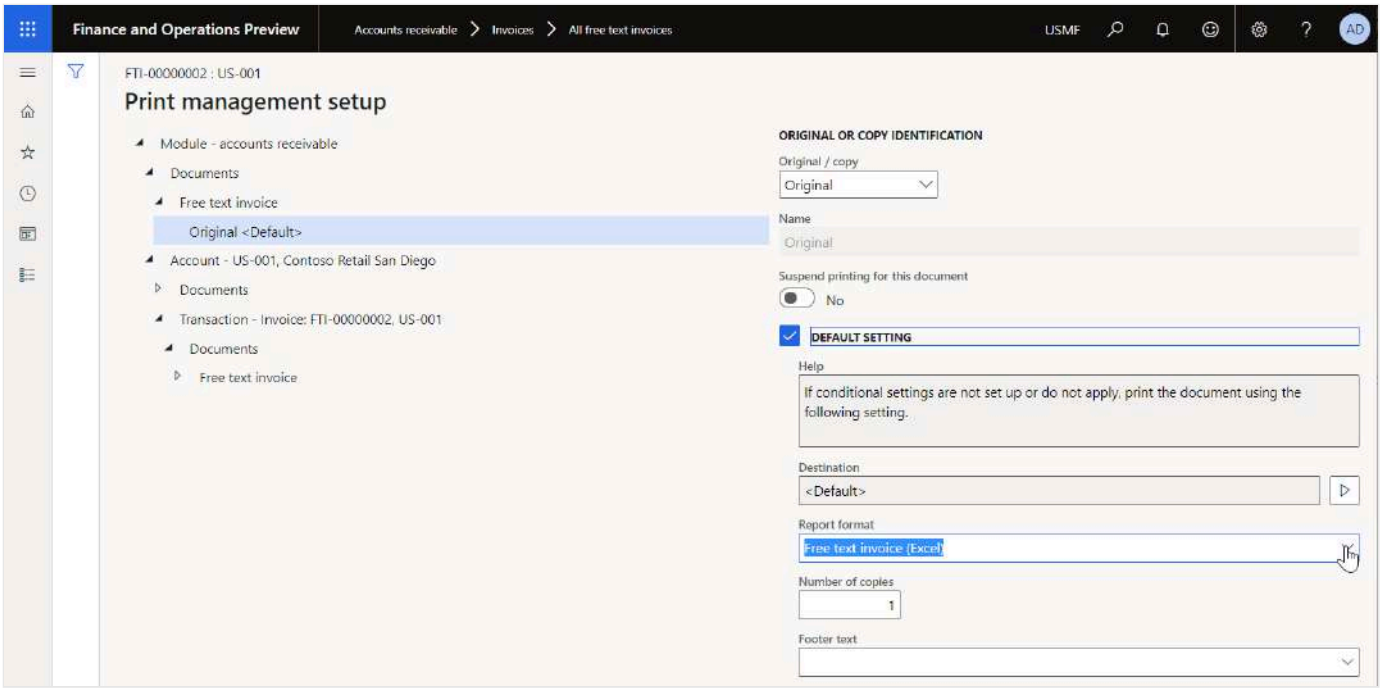
- **Free text invoice (Excel)** – This configuration contains the format and format mapping ER components. The format component specifies the report layout, based on a template in Excel format. The format mapping component contains the model data source and specifies how this data source is used to fill in the report layout at runtime.



Print a free text invoice by using the standard ER format

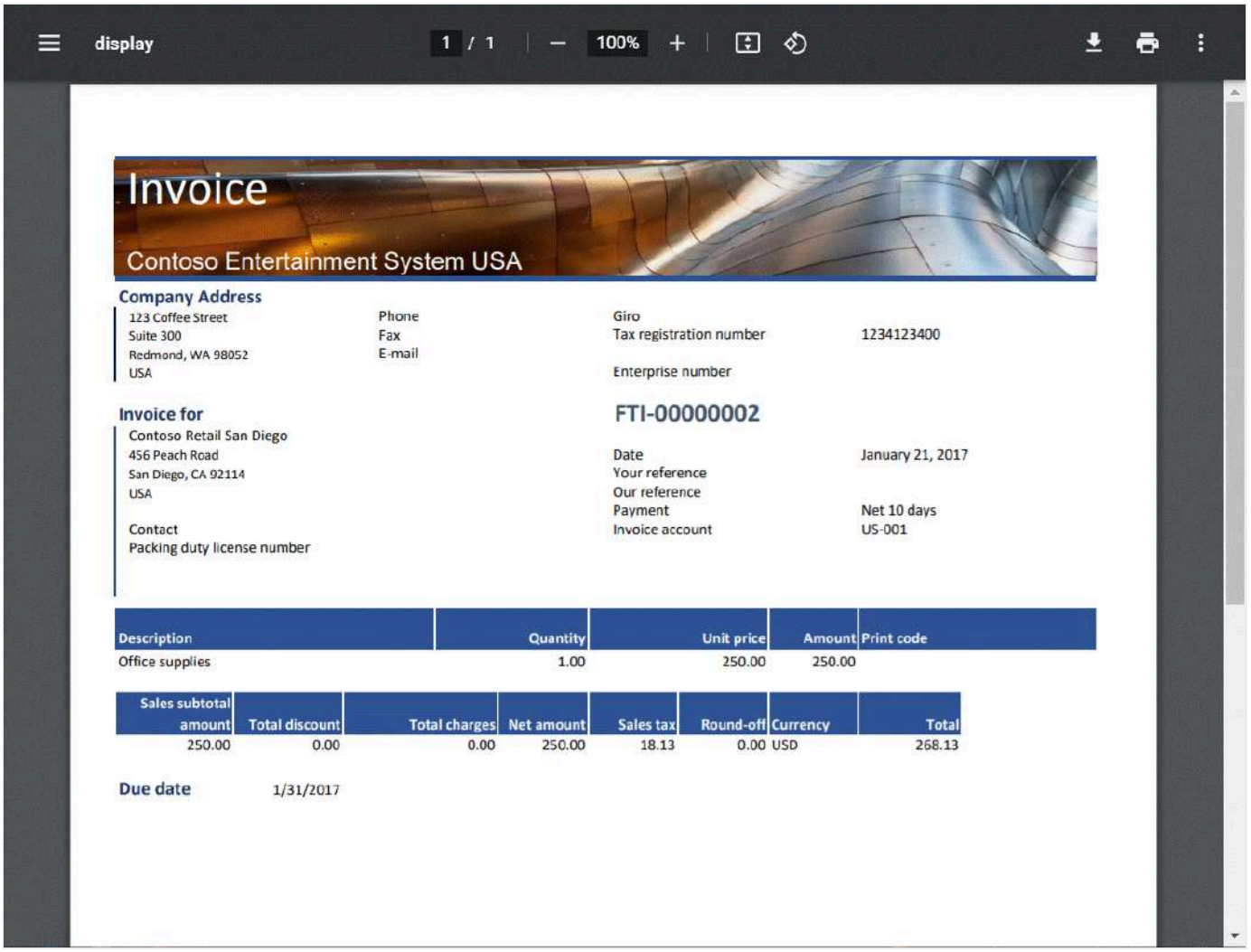
Set up print management

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. On **Free text invoice**, select the **FTI-00000002** invoice. Then, on the Action Pane, on the **Invoice** tab, in the **Print management** group, select **Print management**.
3. On **Print management setup**, in the tree on the left, expand **Module - accounts receivable** > **Documents** > **Free text invoice**, and then select the **Original *Default*** item.
4. In the **Report format** field, select **Free text invoice (Excel)**.
5. Select the **Esc** key to leave **Print management setup** and return to **Free text invoice**.



Print a free text invoice

1. On **Free text invoice**, make sure that the **FTI-00000002** invoice is still selected. Then, on the Action Pane, on the **Invoice** tab, in the **Document** group, select **Print** > **Selected**.
2. Download the generated invoice in Excel format, and open it for preview.
3. Notice that, in accordance with the structure of the Excel template for the provided ER format, the page footer of the generated invoice contains information about the current page number and the total number of pages in the report.



Customize the standard ER format

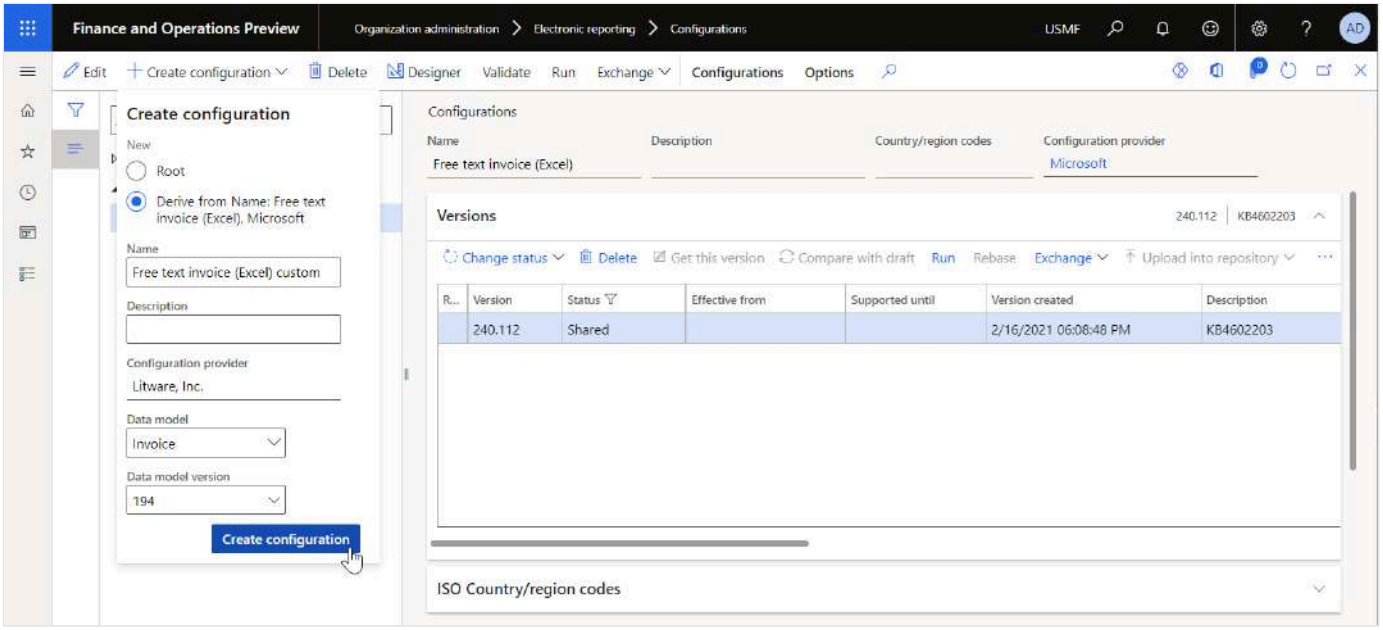
For the example in this section, use the ER configurations that Microsoft provides to generate a free text invoice in Excel format. However, you must add a customization to put a company logo image in the page footer of generated invoices.

In this case, as the representative of Litware, Inc., you must create (derive) a new ER format configuration based on the Microsoft-provided **Free text invoice (Excel)** configuration.

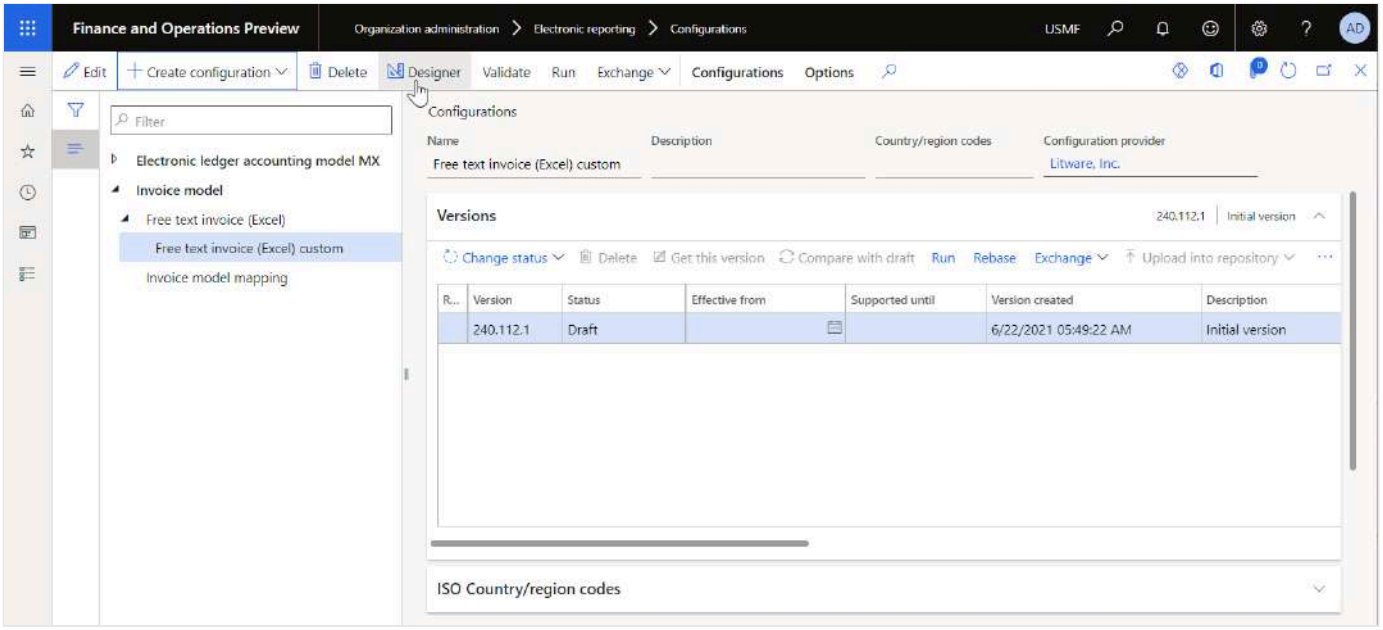
Create a custom format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Invoice model**, and then select **Free text invoice (Excel)**. Litware, Inc. uses the imported version (for example, **240.112**) of this ER format configuration as the base for the custom version.
3. Select **Create configuration** to open the drop-down dialog box. Use this dialog box to create a new configuration for a custom payment format.

4. In the **New** field group, select the **Derive from Name: Free text invoice (Excel), Microsoft** option.
5. In the **Name** field, enter **Free text invoice (Excel) custom**.
6. Select **Create configuration**.



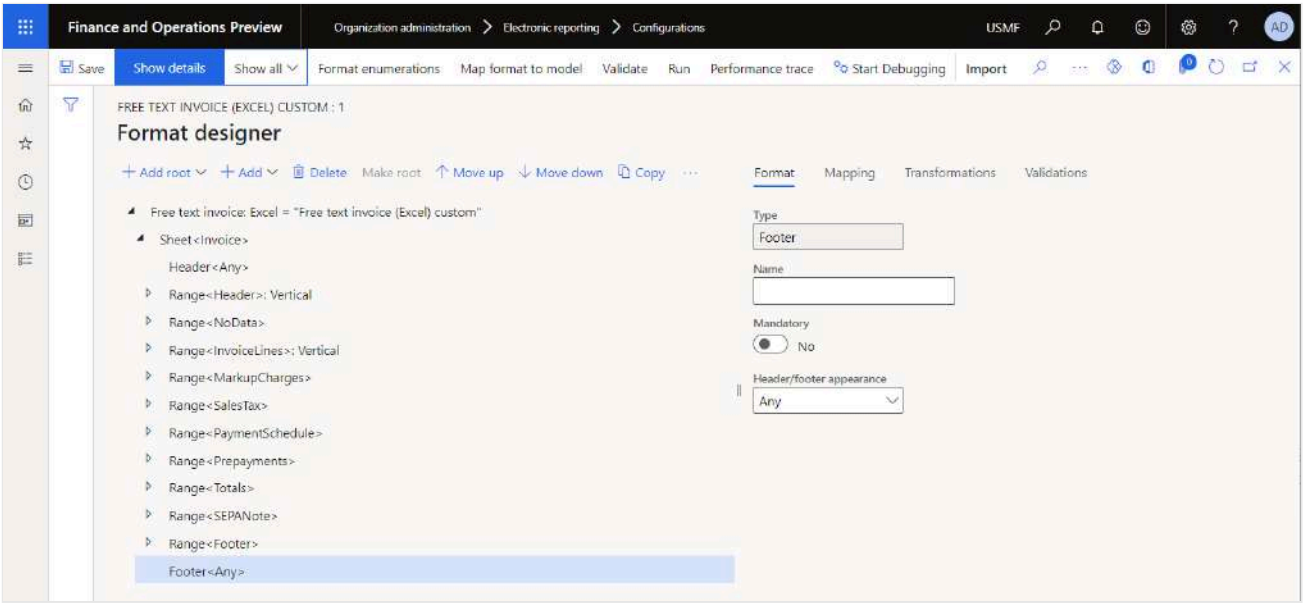
Version 240.112.1 of the **Free text invoice (Excel) custom** ER format configuration is created. This version has a status of **Draft** and can be edited. The current content of your custom ER format matches the content of the format that Microsoft provides.



Edit the custom format

Configure your custom format so that a company logo image appears in the footer on every page of the report.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**, and then select **Free text invoice (Excel) custom**.
3. On the **Versions** FastTab, select version **240.112.1** of the selected configuration.
4. Select **Designer**.
5. On **Format designer**, select **Show details** to view more information about the format elements.
6. Expand and review the following elements:
    - The **Free text invoice** element of the **Excel** type. This element generates an invoice in Excel workbook format.
    - The **Free text invoice \ Invoice** element of the **Sheet** type. This element generates a worksheet of the generated Excel workbook.
    - The **Free text invoice \ Invoice \ Footer** element of the **Footer** type. This element fills in an invoice footer.
7. Select the **Free text invoice \ Invoice \ Footer** element.



> **Note**
>
> Every page footer of a generated invoice contains information about the current page number and the total number of pages in the report. As you can see, the **Free text invoice \ Invoice \ Footer** element contains no child elements. Therefore, the Excel

template that you're using is configured to show paging details in the center of every report's footer.

8. Select **Add**, and then select the **Excel \ Picture** type of the format element that you're adding:

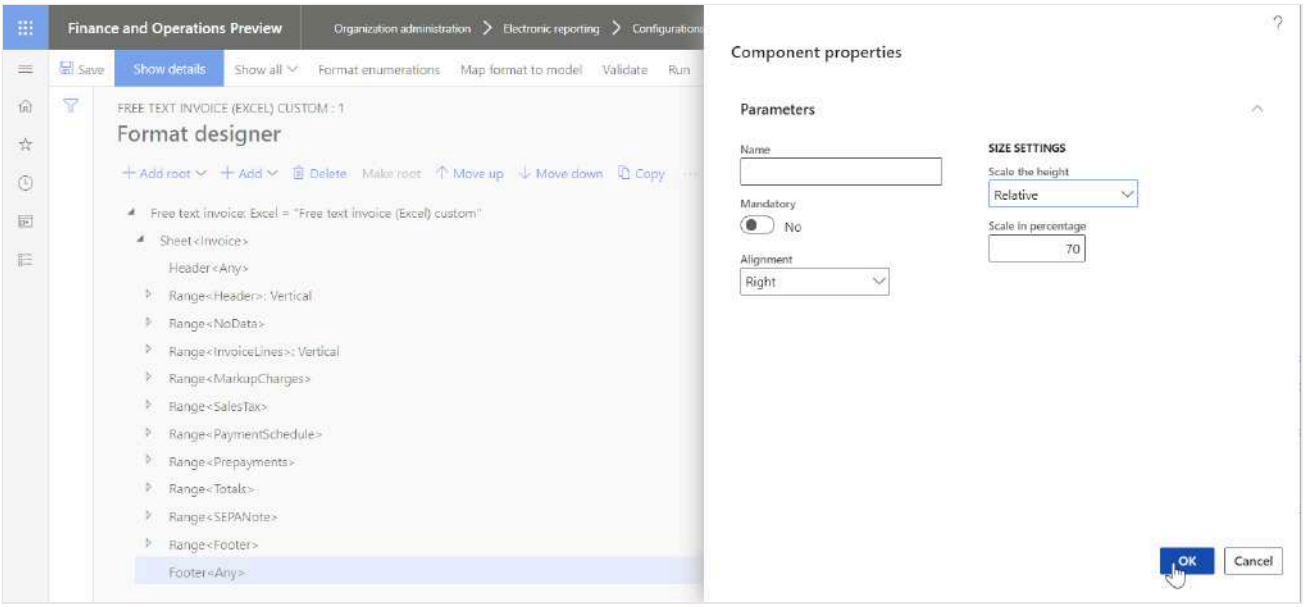
   a. In the **Alignment** field, select **Right**.

   b. In the **Scale the height** field, select **Relative**.

   c. In the **Scale in percentage** field, enter **70**.

   d. Select **OK**.

   > **Note**
   >
   > The **Excel \ Picture** element is used to add a company logo image and align it on right side of the page footer.

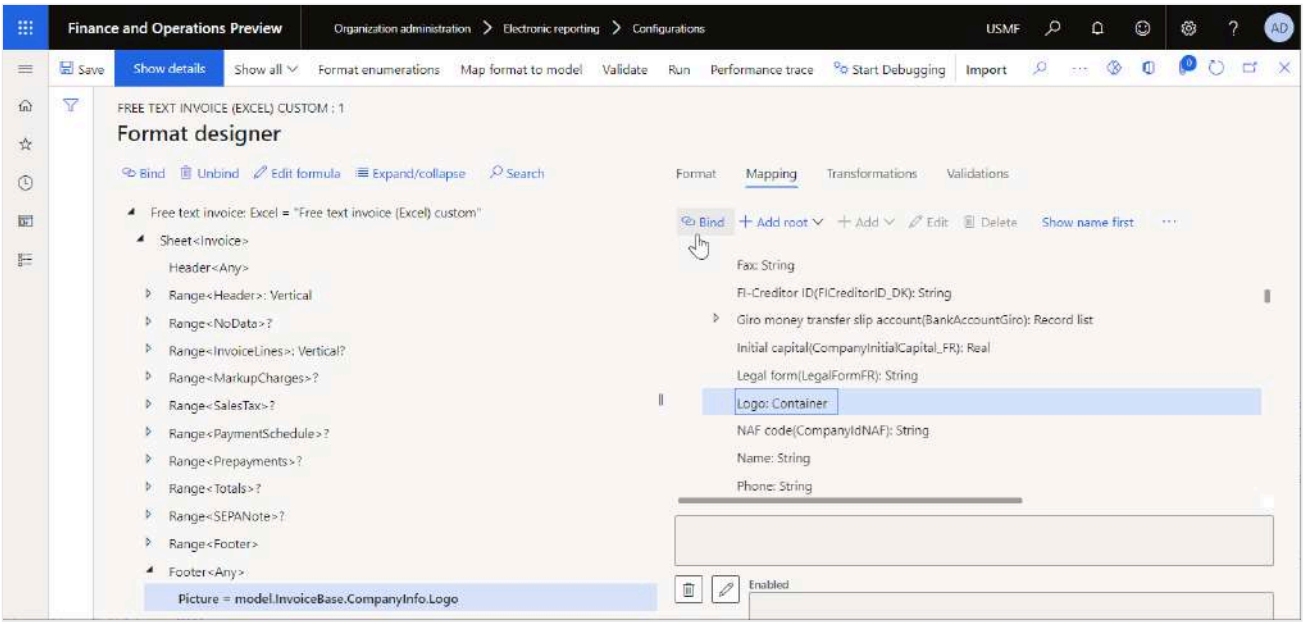


9. In the format structure tree on the left, select the **Picture** element that you just added. On the **Mapping** tab, expand the **model** data source.

10. Expand **model.Payment** > **model.InvoiceBase** > **model.InvoiceBase.CompanyInfo**, and then select the **model.InvoiceBase.CompanyInfo.Logo** data source field. The data source field of the Container type exposes the company logo image as media content.

11. Select **Bind**. The **Picture** format element is now bound with the **model.InvoiceBase.CompanyInfo.Logo** data source field. Therefore, at runtime, a company

logo image appears in the footer of generated invoices.



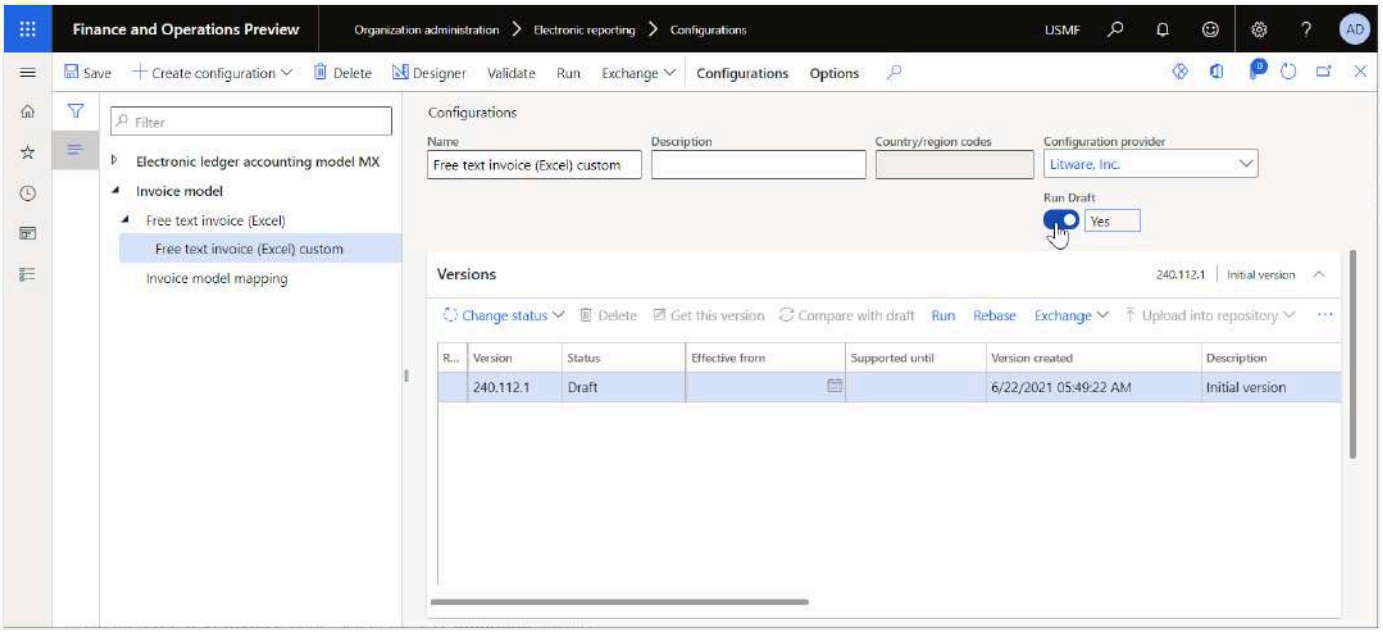
12. Select **Save**, and then close the **Designer** page.

Mark the custom format as runnable

After you create the first version of the custom format, it has a status of **Draft**. You can run the format to test it. To run the report, process a vendor payment by using the payment method that refers to your custom ER format. By default, when you call an ER format from the application, only versions that have a status of **Completed** or **Shared** are considered. This behavior helps prevent ER formats that have unfinished designs from being used. However, for your test runs, you can force the application to use the version of your ER format that has a status of **Draft**. In this way, you can adjust the current format version if any modifications are required. For more information, see Applicability.

To use the draft version of an ER format, you must explicitly mark the ER format.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In the **User parameters** dialog box, set the **Run settings** option to **Yes**, and then select **OK**.
4. Select **Edit** to make the current page editable. In the configuration tree in the left pane, select **Free text invoice (Excel) custom**.
5. Set the **Run Draft** option to **Yes**.



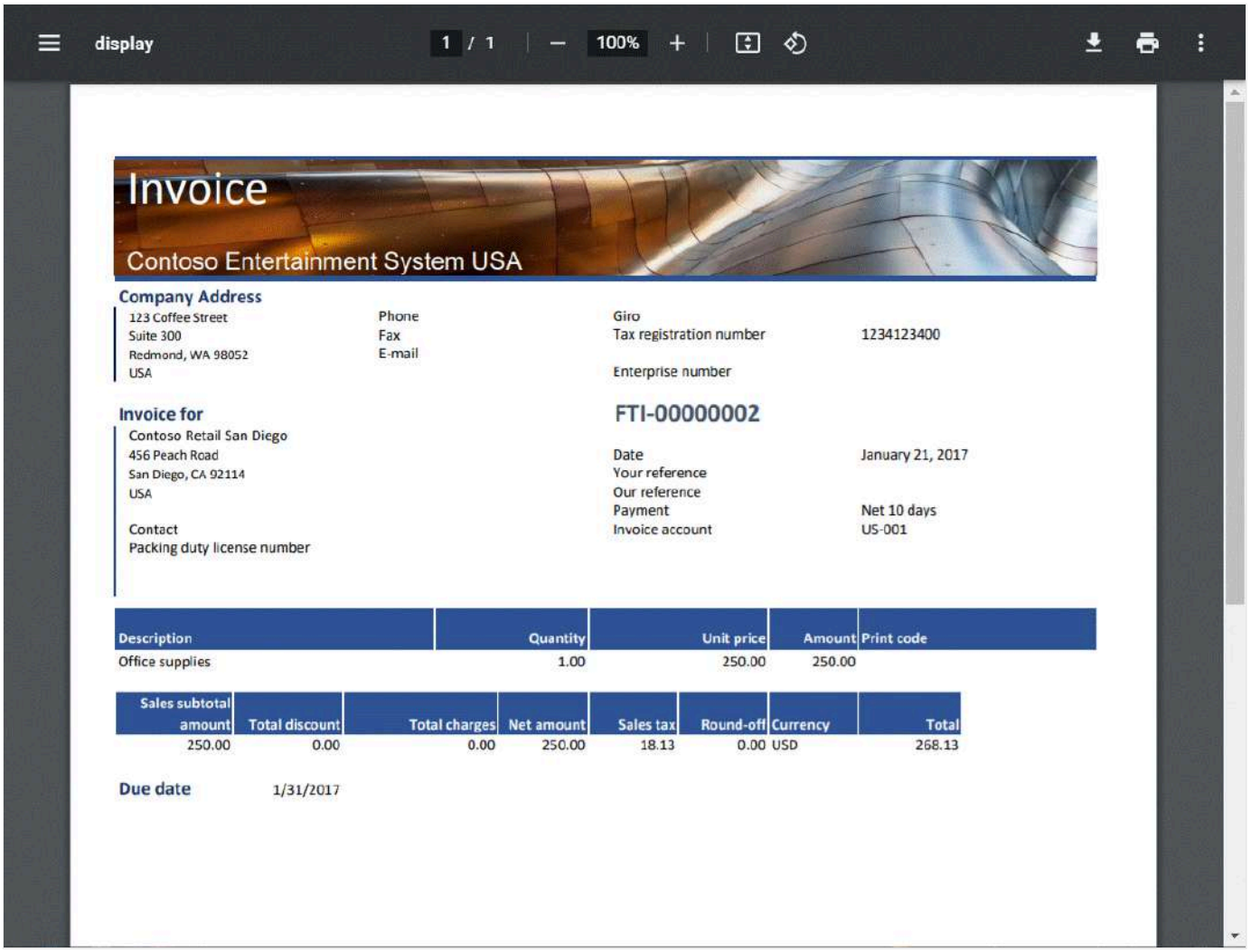
Print a free text invoice by using the custom ER format

Set up print management

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. On **Free text invoice**, select the **FTI-00000002** invoice. Then, on the Action Pane, on the **Invoice** tab, in the **Print management** group, select **Print management**.
3. On **Print management setup**, in the tree on the left, expand **Module - accounts receivable** > **Documents** > **Free text invoice**, and then select the **Original <Default>** item.
4. In the **Report format** field, select **Free text invoice (Excel) custom**.
5. Select **Esc** to leave **Print management setup** and return to **Free text invoice**.

Print a free text invoice

1. On **Free text invoice**, make sure that the **FTI-00000002** invoice is still selected. Then, on the Action Pane, on the **Invoice** tab, in the **Document** group, select **Print** > **Selected**.
2. Download the generated invoice in Excel format, and open it for preview.
3. In accordance with the structure of the custom ER format, the page footer of the generated invoice contains a company logo image in addition to information about the report's paging.



Additional resources

- Design a configuration for generating documents in Excel format
- Embed images and shapes in documents that you generate by using ER

Last updated on 04/13/2026

Split generated XML files based on file size and content quantity

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can design Electronic reporting (ER) formats to generate outgoing documents in XML format. Sometimes, recipients accept those documents only when they meet specific criteria, such as a maximum file size or a maximum number of some XML nodes. You can design ER formats to generate electronic documents that satisfy the requirements that the recipients of those documents specify.

- For the FILE format element, you can define a limit on the file size as an ER expression. If the defined limit is exceeded when an ER report is generated, ER finishes creating the current file and then moves on to create the next file.
- For any XML ELEMENT format, you can define a limit on the number of elements as an ER expression. If the number of XML nodes in the file that is generated exceeds the defined limit when an ER report is run, ER finishes creating the current file and then moves on to create the next file.
- For any XML SEQUENCE format element, you can define a limit on the number of child elements as an ER expression. If the number of nested XML nodes of the format element in the generated file exceeds the defined limit when an ER report is run, ER finishes creating the current file and then moves on to create the next file.
- You can mark any XML ELEMENT format element as non-breakable. In this way, you keep the nested items of XML nodes that are generated under the format element in a single generated file.

In addition to using the XML ELEMENT and XML SEQUENCE format elements to add XML nodes to the generated file, you can use the RAW XML format element. However, nodes that you add by using the RAW XML format element aren't considered when the number of nodes is calculated to evaluate the limits on the number of elements.

If you configure file destinations for a FILE format element that you configure to split the generated output whenever specific limits are exceeded, each piece of generated output is sent to the configured file destination as an individual file. To uniquely name the files that are created by splitting the output, you must configure an ER expression for the FILE format element. If you include an ER data source of the NUMBER SEQUENCE type, the number sequence increments for each piece of the split output.

To learn more about this feature, play the **ER Split XML files based on the file size or content item quantity** task guide, which is part of the **7.5.4.3 Acquire/Develop IT service/solution components (10677)** business process and can be downloaded from the Microsoft Download Center. This task guide walks you through the process of configuring an ER format to split generated files based on limits on the file size and content item quantity. To complete the task guide, you must download the following files:

- ER model configuration - XmlFilesSplittingModel.xml
- ER format configuration - XmlFilesSplittingFormat.xml

Additional resources

Electronic reporting (ER) destinations

Formula designer in Electronic reporting (ER)

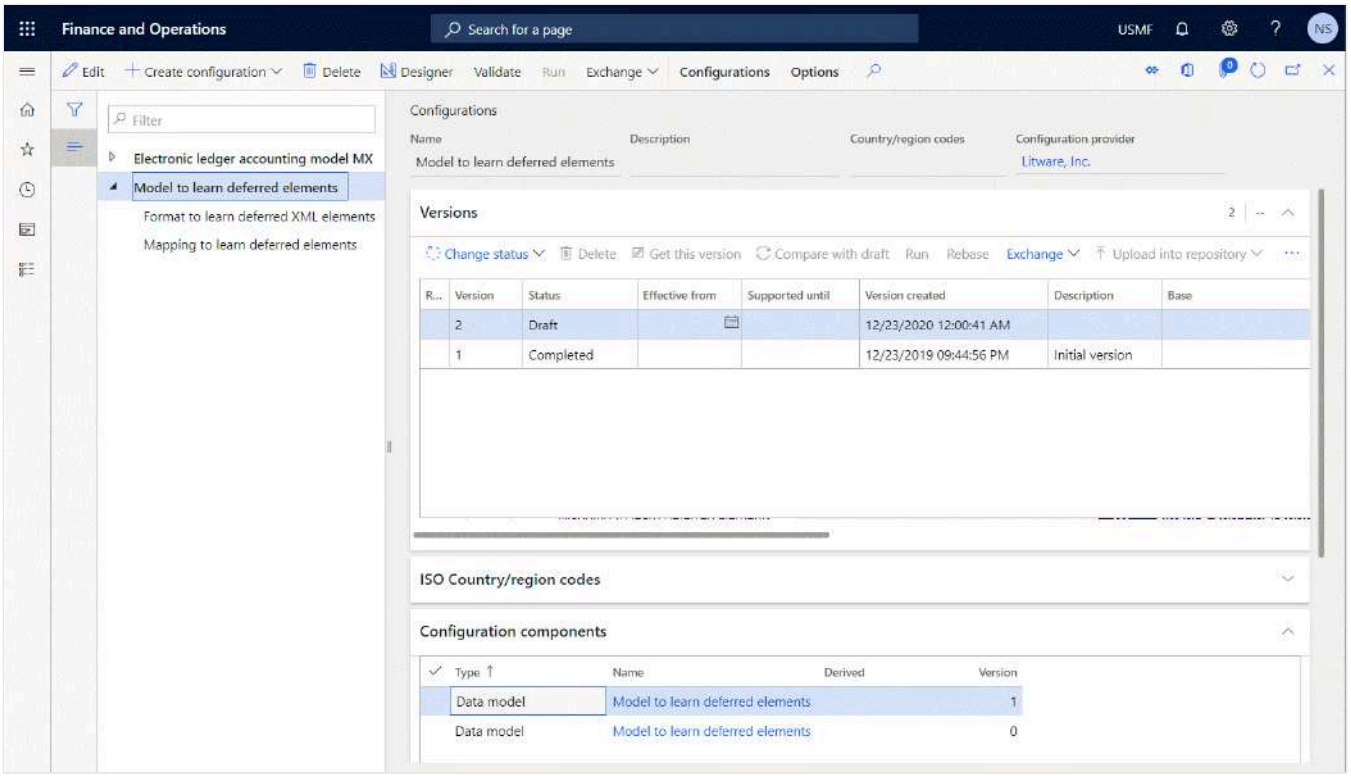
Last updated on 04/13/2026

Design ER configurations to suppress BOM characters in generated files

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

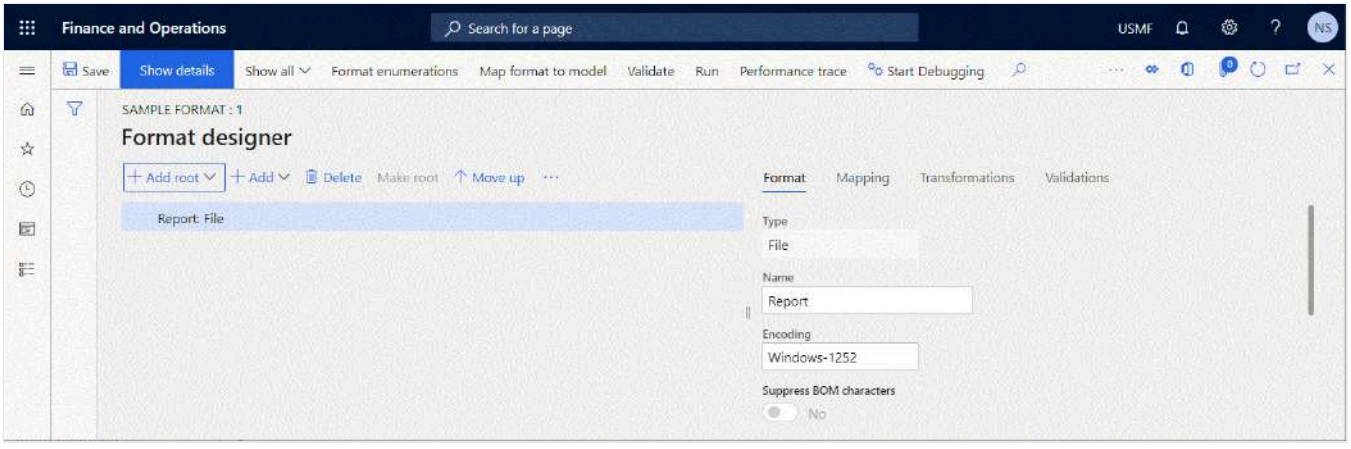
You can design an Electronic reporting (ER) solution to generate outgoing documents. To generate the documents as text or XML files, the solution must include an ER configuration that contains an ER format component. To specify the character encoding that represents the set of characters in generated files, the ER format must contain the **Common\File** format element. To configure the ER format component, open the draft version of the ER configuration in the ER format designer, and add the **Common\File** element. In the **Encoding** field, specify the encoding of outbound files that are generated at runtime by using this component.

> **Note**
>
> If the format contains an incorrect encoding name, an error is thrown when you save your changes to the format's settings.



If you specify **UTF-8**, **UTF-16**, or **UTF-32** as the encoding, the **Suppress BOM characters** option becomes available. Set this option to **Yes** to suppress byte order mark (BOM) characters in outbound files that are generated at runtime when the editable ER format is run.

> **Note**
>
> If you leave the **Encoding** field blank, the default **UTF-8** encoding is used.



To review the functionality at runtime, complete the appropriate procedure. For example, complete the steps in the Defer the execution of XML elements in ER formats article. After you complete the steps in the Modify the format so that the calculation is based on generated output section of that article, follow these additional steps.

1. Specify the UTF encoding:

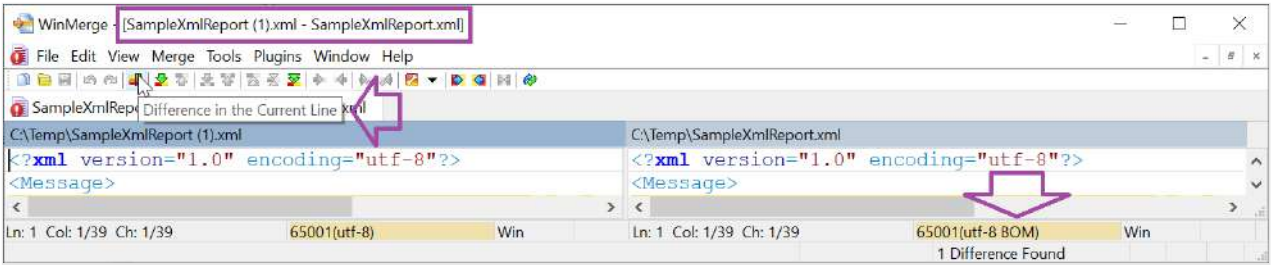
a. Select the **Report** element of the **Common\File** type.
b. In the **Encoding** field, specify the **UTF-8** encoding.

2. Generate an XML file that includes a BOM character:
   a. Set the **Suppress BOM characters** option to **No** to include BOM characters in generated XML files.
   b. Complete the steps in the Defer the execution of the summary XML element so that the calculated total is used section of the Defer the execution of XML elements in ER formats article, and save the generated file as **SampleXmlReport.xml**.

3. Generate an XML file that doesn't include a BOM character:
   a. Set the **Suppress BOM characters** option to **Yes** to suppress BOM characters in generated XML files.
   b. Complete the steps in the Defer the execution of the summary XML element so that the calculated total is used section of the Defer the execution of XML elements in ER formats article, and save the generated file as **SampleXmlReport (1).xml**.

4. In a file comparison utility, compare the generated files.

   The first difference that you notice is in the file header. The **SampleXmlReport.xml** file contains a BOM character, where the **SampleXmlReport (1).xml** file doesn't.



See also

- Defer the execution of XML elements in ER formats

Last updated on 04/08/2026

Generate electronic documents and update application data by using ER

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can design Electronic reporting (ER) formats that the application uses to generate outgoing electronic documents. You can also design ER formats that parse incoming electronic documents and use the content in those documents to update application data.

By using this functionality, you can use a single ER format to generate outgoing electronic documents and then update the application data. Use this feature in the following scenarios:

- To prevent repeated usage of application data in subsequent processes, mark an application's data immediately after you use it to generate electronic documents. For example, mark payment transactions as already processed immediately after you include them in a generated payment message.
- To store the processing details of electronic documents that you generate by using ER logic. For example, a unique payment message identification that is generated by using the ER expression. The expression is based on information entered in the ER dialog box when the ER format is run to generate documents.

To learn more about this feature, watch the set of ER Generate documents with application data update Task guides (part of the 7.5.4.3 Acquire/Develop IT service/solution components (10677) business process). The guides walk you through the details of Intrastat reporting and archiving. The following files are required to complete certain steps in these Task guides. Download and save these files to your local machine.

- ER data model configuration: Intrastat (model)
- ER model mapping configuration: Intrastat (mapping)
- ER format configuration: Intrastat (format)

Last updated on 04/13/2026

Import configurations to generate documents that have application data

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete the steps in this procedure, you must first complete the procedure, "ER Create a configuration provider and mark it as active".

The steps in this procedure explain how to design Electronic reporting (ER) configurations to generate an electronic document. In this procedure, you import the required ER configurations that are created for the sample company, Litware, Inc., and use them to generate electronic documents. This procedure is created for users with the assigned role of system administrator or electronic reporting developer. You can complete these steps by using the DEMF dataset. Before you begin, download and save the files listed in the Help article, "Generate electronic documents and update application data with ER tool" (generate-electronic-documents-update-application-data/). The files are Intrastat (model).xml, Intrastat (mapping).xml, and Intrastat (format).xml.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

    - Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the procedure, Create a configuration provider and mark it as active.
    - The steps in this procedure show how to use ER capabilities to complete an application data update and how to generate an Intrastat report. The details of the reporting process are archived in the application tables. Currently, when you activate the Intrastat reporting process from the Intrastat form, archiving is done based on the logic programmed in the existing source code. In this procedure, you configure a similar yet simplified logic of application data by using only the ER framework. You make no changes to the source code.

Import ER configurations

1. Select **Reporting configurations**.
2. Select **Exchange**.
3. Select **Load from XML file**.

    - Import the ER model configuration that contains the data model designed to be used as the data source for generating the Intrastat report. Later, you can extend this data model definition to use it for an application data update to archive details of the Intrastat reporting process.
    - Select **Browse** and select the Intrastat (model).xml file.

4. Select **OK**.
5. In the tree, select **Intrastat (model)**.
6. Select **Designer**.
7. In the tree, expand **For outgoing document**.
8. In the tree, expand **For outgoing document\Transactions**.

    - Review the structure of the imported data model. The root item **For outgoing document** specifies the data flow for getting data from the application and using it as the data source to generate the Intrastat report. The **Transactions (Record list)** represents the list of Intrastat transactions that you must report. Because you archive reported commodity codes, the unique identifier of a single commodity code **Commodity rec id (Int64)** is needed in this data flow.

9. Close the page.
10. Select **Exchange**.
11. Select **Load from XML file**.

    - Import the ER mapping configuration that specifies the data flow for getting data from the application and then using it to generate the Intrastat report. Later, you can extend this model mapping definition to get data from the Intrastat report and use it for the application data update to archive details of Intrastat reporting process.
    - Select **Browse** and select the Intrastat (mapping).xml file.

12. Select **OK**.
13. In the tree, expand **Intrastat (model)**.
14. In the tree, select **Intrastat (model)\Intrastat (mapping)**.
15. Select **Designer**.

- The current model mapping contains the value **To model** in the Direction field. This value means that this model mapping is designed for getting data from the application and storing it in the data model.

16. Select **Designer**.
17. In the tree, expand **List**.
18. In the tree, expand **Transactions= List**.

    - Review the structure of the model mapping that uses the data model filtered based on the root item, **For outgoing document**. The added data source, **List** provides access to the required application data, which is the list of records from the Intrastat table.

19. Close the page.
20. Close the page.
21. Select **Exchange**.
22. Select **Load from XML file**.

    - Import the ER format configuration that specifies the layout of the Intrastat report and the process of populating data to the report. Later, you can extend this format definition to put data from the Intrastat report into the data model and then use it to update application data to archive the details of Intrastat reporting process.
    - Click Browse and select the Intrastat (format).xml file.

23. Select **OK**.
24. In the tree, select 'Intrastat (model)\Intrastat (format)'.
25. Select **Designer**.
26. Click Expand/collapse.
27. In the tree, select 'File\Declaration'.
28. Click the Mapping tab.
29. In the tree, select 'File'.

    - Review the structure of the format used to generate the Intrastat report. Note that it is designed to generate an XML file by populating data from the data model, which is based on the root item 'For outgoing document'. Verify that the name for generated file is defined on the user dialog form ('fn' data source is used for that).

30. Close the page.

Last updated on 04/13/2026

Design configurations to generate documents that have application data

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete the steps in this procedure, you must first complete the procedure, Import configurations to generate documents that have application data.

The steps in this procedure explain how to design Electronic reporting (ER) configurations to generate an electronic document. In this procedure, you run the ER imported format configuration that you created for the sample company, Litware, Inc. to generate electronic documents.

This procedure is created for users with the assigned role of system administrator or electronic reporting developer. You can complete these steps by using the DEMF dataset.

Before you begin, change the country/region context for the DEMF company from DEU (Germany) to BEL (Belgium). Select **Organization administration** > **Organizations** > **Legal entities** to update the country/region code in the primary address of the legal entity DEMF. Restart your application.

Run imported ER format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Intrastat (model)**.
3. In the tree, select **Intrastat (model)\Intrastat (format)**.
4. Select **Run**.
    - Run the draft version of the ER format configuration to generate the Intrastat report.
5. In the **Enter file name** field, type `intrastat.xml`.
    - Specify the name of the file.
6. Select **OK**.
    - Review the generated XML file.

7. Close the page.
8. Go to **Tax** > **Declarations** > **Foreign trade** > **Intrastat**.
    - Open this form to view the Intrastat transactions that are included in the generated electronic document.
9. Select **Intrastat archive**.
    - Because the executed ER format doesn't contain any settings for application data update, the details of the completed Intrastat report aren't archived.
10. Close the page.
11. Close the page.

Last updated on 03/17/2026

Modify models and mappings to generate documents that have application data

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete the steps in this procedure, you must first complete the procedure, Design configurations to generate documents that have application data.

This procedure explains how to design Electronic reporting (ER) configurations to generate an electronic document and update application data. In this procedure, you modify the ER configurations to start using them to generate electronic documents and update application data. This procedure is for users assigned the role of system administrator or electronic reporting developer. You can complete these steps by using the DEMF dataset.

Modify data model

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, select **Intrastat (model)**. * You can extend how you use the data model. Besides using it as data source to generate the Intrastat report, the data model collects details about the Intrastat reporting process. These details help to update the application data.
3. Select **Designer**.
4. Select **New** to open the **Drop** dialog.
5. In the **New node as a field** box, enter **Model root**.
6. In the **Name** field, type **For application data update**.

    - For application data update

7. Select **Add**.
8. In the tree, select **For application data update**.

    - This new root item is added to specify the data flow for moving data from the Intrastat report (used as a data source) to the application tables (the update destination). Different root items must be used for getting data that is posted to the

outgoing document and for getting data from the document that is used to update application data.

9. Select **New** to open the **Drop** dialog.
10. In the **Name** field, type **Archive header**.

    - Archive header

11. In the **Item type** field, select **Record list**.
12. Select **Add**.

    - Create a new item because you're creating a record for each generated Intrastat report.

13. Select **New** to open the **Drop** dialog.
14. In the **Name** field, type **File name**.

    - File name

15. In the **Item type** field, select **String**.
16. Select **Add**.
17. Select **New** to open the **Drop** dialog.
18. In the **Name** field, type **Number of lines**.

    - Number of lines

19. In the **Item type** field, select **Integer**.
20. Select **Add**.

    - Add this item to represent the number of Intrastat transactions reported during the current reporting process.

21. Select **New** to open the **Drop** dialog.
22. In the **Name** field, type **Archive lines**.

    - Archive lines

23. In the **Item type** field, select **Record list**.
24. Select **Add**.

    - Add this item to represent the list of Intrastat transactions reported during the current reporting process.

25. Select **New** to open the **Drop** dialog.
26. In the **Name** field, type **Amount**.

    - Amount

27. In the **Item type** field, select **Real**.
28. Select **Add**.
29. Select **New** to open the **Drop** dialog.
30. In the **Name** field, type **Commodity rec id**.

    - Commodity rec id

31. In the **Item type** field, select **Int64**.
32. Select **Add**.
33. Select **New** to open the **Drop** dialog.
34. In the **Name** field, type **Item number**.

    - Item number

35. In the **Item type** field, select **String**.
36. Select **Add**.
37. Select **Save**.
38. Close the page.

Modify model mapping

1. In the tree, expand **Intrastat (model)**.
2. In the tree, select **Intrastat (model)\Intrastat (mapping)**.

    - Modify the existing model mapping to start using it for the application data update and to archive Intrastat reporting details.

3. Select **Designer**.
4. Select **New**.
5. In the **Name** field, type **Update archive**.

    - Update archive

6. In the **Direction** field, select **To destination**.
7. Select **Save**.

    - This new mapping specifies the data flow for moving data (Intrastat reporting details) from the data model to the application tables (the update destination). Use the root items of different models to get data from the application for the reporting process. Then, use the data from the data model for the application data update.

8. Select **Designer**.
9. In the tree, select **Data model\Data model**.

- Add the required data source. This source is the data model that contains details of the reported Intrastat transactions to archive.

10. Select **Add root**.
11. In the **Name** field, type **model**.

    - model

12. In the **Definition** field, enter or select the value **For application data update**.

    - For application data update

13. Select **OK**.
14. In the tree, expand **model**.
15. In the tree, select **Functions\Calculated field**.
16. In the tree, select **model\Archive header**.
17. Select **Add**.

    - Because you want to enumerate reported Intrastat transactions for archiving, ensure that you add the appropriate data source.

18. In the **Name** field, type **Enumerated lines**.

    - Enumerated lines

19. Select **Edit** formula.
20. In the tree, select **List\ENUMERATE**.
21. Select **Add** function.
22. In the tree, expand **model**.
23. In the tree, expand **model\Archive header**.
24. In the tree, select **model\Archive header\Archive lines**.
25. Select **Add data source**.
26. In the **Formula** field, enter **ENUMERATE(model.'Archive header'.'Archive lines')**.

    - ENUMERATE(model.'Archive header'.'Archive lines')

27. Select **Save**.
28. Close the page.
29. Select **OK**.
30. Select **Add destination**.

    - Add application tables as required destinations that require updates to archive details of reported Intrastat transactions.

31. In the **Name** field, type **Archive**.

- Archive

32. In the **Table name** field, type **IntrastatArchiveGeneral**.

    - IntrastatArchiveGeneral
    - Keep the record action **Insert** so you can add records during the detail archiving of each Intrastat reporting process.

33. In the **Record infolog** field, select **Yes**.

    - To get information about issues with the application data update, select **Yes**.

34. In the **Skip record action validation** field, select **Yes**.

    - To suppress validation errors about the empty **Intrastat archive ID** field, select **Yes**. These errors are suppressed after the addition of records, based on the sequence number settings configured for this table in the **Foreign trade parameters** form.

35. Select **OK**.

    - Bind elements of the added data source (the filtered model based on the selected root item) with elements from the added destination.

36. In the tree, expand **Archive**.
37. In the tree, expand **Archive<Relations**.
38. In the tree, expand **Archive<Relations\IntrastatArchiveDetail**.
39. In the tree, select **Archive<Relations\IntrastatArchiveDetail\Amount(AmountMST)**.
40. In the tree, expand **model\Archive header\Enumerated lines**.
41. In the tree, expand **model\Archive header\Enumerated lines\Value**.
42. In the tree, select **model\Archive header\Enumerated lines\Value\Amount**.
43. Select **Bind**.
44. In the tree, select **Archive<Relations\IntrastatArchiveDetail\Commodity(IntrastatCommodity)**.
45. In the tree, select **model\Archive header\Enumerated lines\Value\Commodity rec id**.
46. Select **Bind**.
47. In the tree, select **Archive<Relations\IntrastatArchiveDetail\Item number(ItemId)**.
48. In the tree, select **model\Archive header\Enumerated lines\Value\Item number**.
49. Select **Bind**.
50. In the tree, select **Archive<Relations\IntrastatArchiveDetail\Line number(LineNumber)**.
51. In the tree, select **model\Archive header\Enumerated lines\Number**.
52. Select **Bind**.
53. In the tree, select **Archive<Relations\IntrastatArchiveDetail**.
54. In the tree, select **model\Archive header\Enumerated lines**.
55. Select **Bind**.
56. In the tree, select **Archive\File name(FileName)**.

57. In the tree, select **model\Archive header\File name**.
58. Select **Bind**.
59. In the tree, select **Archive\Number of lines(NumberOfLines)**.
60. In the tree, select **model\Archive header\Number of lines**.
61. Select **Bind**.
62. In the tree, select **Archive**.
63. In the tree, select **model\Archive header**.
64. Select **Bind**.
65. Select **Save**.
66. Close the page.
67. Close the page.

Last updated on 03/17/2026

Modify formats to generate documents that have application data

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Before you start, complete the procedure, Modify models and mappings to generate documents that have application data.

This procedure shows how to design Electronic reporting (ER) configurations to generate an electronic document and update application data. Change the ER configurations so they generate electronic documents and update application data. This procedure is for users with the system admin or electronic reporting developer role. Use the DEMF dataset for these steps.

Modify format to collect details of reporting

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Intrastat (model)**.
3. In the tree, select **Intrastat (model)\Intrastat (format)**.
4. Select **Designer**.
5. In the tree, expand **File**.
6. In the tree, expand **File\Declaration**.
7. In the tree, select **File\Declaration\Data**.
8. In the **Multiplicity** field, select **One many**.

    - Configure this format element to archive details of the Intrastat reporting process. This item represents the archive's header record.

9. In the tree, expand **File\Declaration\Data**.
10. In the tree, select **File\Declaration\Data\Item**.
11. In the **Multiplicity** field, select **Zero many**.

    - Configure this format element to archive details of the Intrastat reporting process. This item represents the list of archived lines.

12. In the tree, expand **File\Declaration\Data\Item**.
13. In the tree, select **File\Declaration\Data\Item\Dim1**.

14. Select **Yes** in the **Excluded** field.

    - Don't archive this data, so you can exclude this format element from the data source of Intrastat reporting details.

15. In the tree, expand **File\Declaration\Data\Item\Dim1**.
16. In the tree, select **File\Declaration\Data\Item\Dim1\property**.
17. Select **Yes** in the **Excluded** field.
18. In the tree, select **File\Declaration\Data\Item\Dim1\date**.
19. Select **Yes** in the **Excluded** field.
20. In the tree, select **File\Declaration\Data\Item\Dim2**.
21. Select **Yes** in the **Excluded** field.
22. In the tree, expand **File\Declaration\Data\Item\Dim2**.
23. In the tree, select **File\Declaration\Data\Item\Dim2\property**.
24. Select **Yes** in the **Excluded** field.
25. In the tree, select **File\Declaration\Data\Item\Dim2\code**.
26. Select **Yes** in the **Excluded** field.
27. In the tree, select **File\Declaration\Data\Item\Dim3**.

    - Several format elements can have the same name. For example, Dim. You can't explicitly recognize them when you use this format as a data source for archiving Intrastat reporting details, so you need to define the alternative names for these format elements.

28. In the **Name** field, type **Amount**.

    - Amount

29. In the **Multiplicity** field, select **Exactly one**.
30. In the tree, select **File\Declaration\Data\Item\Dim4**.
31. In the **Name** field, type **Item**.

    - Item

32. In the **Multiplicity** field, select **Exactly one**.

    - In addition to the design format elements, archive the following Intrastat reporting details: unique record identification of each reported commodity item, and the name of the generated file. Because this data isn't in the Intrastat report, add the format related to these detail elements as data source items.

33. In the tree, select **File\Declaration\Data**.
34. Select **Add** to open the **Drop** dialog.
35. In the tree, select **Data source\Item**.

36. In the **Name** field, type **File name**.

    - File name

37. In the **Data type** field, select **String**.
38. Select **OK**.
39. In the tree, select **File\Declaration\Data\Item**.
40. Select **Add Item**.
41. In the **Name** field, type **Commodity rec ID**.

    - Commodity rec ID

42. In the **Data type** field, select **Int64**.
43. Select **OK**.
44. Select the **Mapping** tab.
45. In the tree, select **File\Declaration\Data\File name**.
46. Select **Bind**.
47. In the tree, expand **model**.
48. In the tree, expand **model\Transactions**.
49. In the tree, select **File\Declaration\Data\Item = model.Transactions\Commodity rec ID**.
50. In the tree, select **model\Transactions\Commodity rec ID**.
51. Select **Bind**.
52. Select **Save**.

Modify format to memorize details of reporting

1. Select **Map format to model**.
2. Select **New**.
3. In the **Definition** field, enter or select the **For application data update** root item.

    - For application data update.

4. In the **Name** field, type **Mapping to update data**.

    - Mapping to update data

5. Select **Save**.

    - This mapping defines how the details of the Intrastat report are collected in the data model, which uses the selected root item 'For application data update'. These details, the model mapping with the same root item 'For application data update', and the direction 'To destination' are used for the application data update. The application data update starts immediately after the outgoing Intrastat report is

generated. You can skip the application data update at run time, but the data model must be empty (containing an empty record list).

6. Select **Designer**.

    - The outgoing Intrastat report format is added by default as a data source for this model mapping.
    - Bind elements of the designed report (presented as data source) to elements of the data model, which is filtered based on the selected model's root item.

7. In the tree, expand **Archive header**.
8. In the tree, expand **Archive header\Archive lines**.
9. In the tree, expand `format`.
10. In the tree, expand `format\Declaration: XML Element(Declaration)`.
11. In the tree, expand `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)`.
12. In the tree, expand `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)`.
13. In the tree, expand `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)\Dim3: XML Element 1..1 (Amount)`.
14. In the tree, expand `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)\Dim4: XML Element 1..1 (Item)`.
15. In the tree, select **Archive header\Number of lines**.
16. Select **Edit**.
17. In the tree, select **List\COUNT**.
18. Select **Add function**.
19. In the tree, expand `format`.
20. In the tree, expand `format\Declaration: XML Element(Declaration)`.
21. In the tree, expand `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)`.
22. In the tree, select `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)`.
23. Select **Add data source**.
24. In the **Formula** field, enter `COUNT(format.Declaration.Data.Item)`.

    - COUNT(format.Declaration.Data.Item)

25. Select **Save**.
26. Close the page.
27. In the tree, select **Archive header\File name**.
28. In the tree, select `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\File name: Item String(File name)`.

29. Select **Bind**.
30. In the tree, select `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)\Dim4: XML Element 1..1 (Item)\number: String(number)`.
31. In the tree, select **Archive header\Archive lines\Item number**.
32. Select **Bind**.
33. In the tree, select `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)\Dim3: XML Element 1..1 (Amount)\value: Numeric Real(value)`.
34. In the tree, select **Archive header\Archive lines\Amount**.
35. Select **Bind**.
36. In the tree, select `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)\Commodity rec ID: Item Int64(Commodity rec ID)`.
37. In the tree, select **Archive header\Archive lines\Commodity rec ID**.
38. Select **Bind**.
39. In the tree, select **Archive header\Archive lines**.
40. In the tree, select `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)\Item: XML Element 0..* (Item)`.
41. Select **Bind**.
42. In the tree, select **Archive header**.
43. In the tree, select `format\Declaration: XML Element(Declaration)\Data: XML Element 1..* (Data)`.
44. Select **Bind**.
45. Select **Save**.
46. Close the page.
47. Close the page.
48. Close the page.

Last updated on 03/17/2026

Generate documents that have application data

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete the steps in this procedure, you must first complete the procedure, Modify formats to generate documents that have application data.

The steps in this procedure explain how to design Electronic reporting (ER) configurations to generate an electronic document and update application data. In this procedure, you execute the ER format configuration to generate the Intrastat report and update application data for archiving details of the reporting process.

This procedure is created for users with the assigned role of system administrator or electronic reporting developer. You can complete these steps by using the DEMF dataset. Before you begin, make sure that the country/region context for the DEMF company is BEL (Belgium).

Set up foreign trade parameters

1. Go to **Tax** > **Setup** > **Foreign trade** > **Foreign trade parameters**.
2. Select the **Number sequences** tab.

   To archive details of the Intrastat reporting process, you need to identify records of each archive. You must configure a special number sequence for this process.
3. Select the **Intrastat archive ID** reference.
4. In the **Number sequence code** field, type a value.

   In the **Number sequence code** field, enter or select the value **Fore_2**.
5. Resolve changes to the Number sequence code.
6. Select **Save**.
7. Close the page.

Run modified ER format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Intrastat (model)**.
3. In the tree, select **Intrastat (model)\Intrastat (format)**.
4. Select **Run**.
5. In the **Enter file name** field, type **intrastat2.xml**.
6. Select **OK**.

Review ER format execution results

Review the generated XML file.

1. Close the page.
2. Go to **Tax** > **Declarations** > **Foreign trade** > **Intrastat**.

   Open this form containing the Intrastat transactions included in the generated electronic document.
3. Select **Intrastat archive**.

   Since the executed ER format now contains settings for application data update, the details of the completed Intrastat reporting are archived. In this form, you can see the header record of the created archive.
4. Select **Details**.

   In this form, you can see the details for the created archive.
5. Close the page.
6. Close the page.
7. Close the page.

Last updated on 03/17/2026

Parse incoming documents

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article covers the following three tasks:

- Parse incoming documents to update application data
- Parse incoming documents in Excel format
- Parse incoming documents in CSV format

Parse incoming documents to update application data

You can design Electronic reporting (ER) formats and run them in the application to parse incoming electronic documents and then use their content to update application data.

The following new ER functionality improves the parsing of incoming electronic documents in XML format:

- Use the **CASE** format element as a root element of the ER format to parse incoming electronic documents in XML format. The **FILE** format element is supported as a nested element of the **CASE** element. Therefore, you can configure a single ER format to parse incoming electronic documents that might contain different root XML elements.

- A **Parsing order of nested elements** attribute is available for XML format elements in ER formats. Use this attribute to define a single XML element that you expect in the incoming file. There are two valid sequences of the nested elements:
  - **As in format** – The incoming file is valid when the sequence of nested elements in the file is the same as the order that is described in the ER format.
  - **Any** – The incoming file is valid when all nested elements in the ER format are present in the parsing file, regardless of their sequence in that file.

To become more familiar with the details of this feature, play the task guide, **ER - Parse incoming documents to update application data** (part of the 7.5.4.3 Acquire/Develop IT

service/solution components (10677) business process). This task guide shows how the responses from a web service can be parsed by using an ER format.

To complete some steps of the task guide, you must download the following files:

Expand table

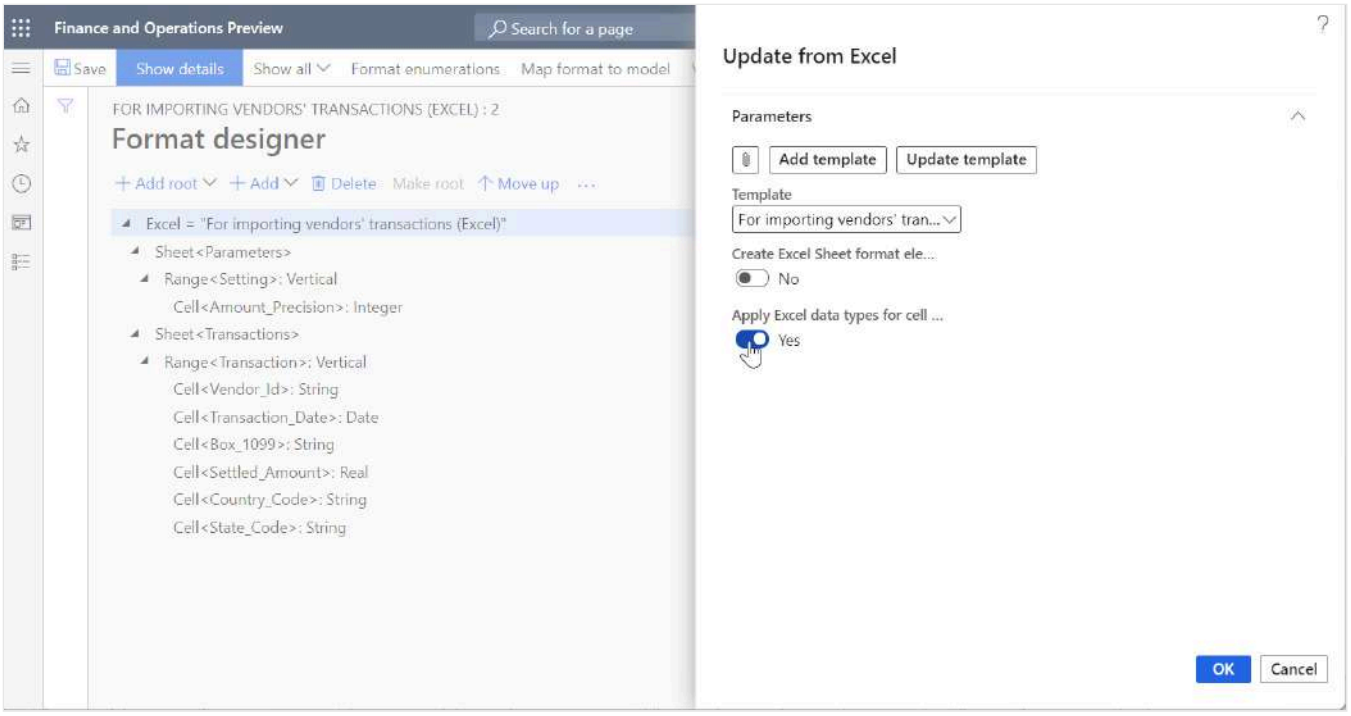
Content description	File
ER data model configuration	EFSTAmodel.xml
ER format configuration	EFSTAformat.xml
Web service response sample 1	Response1.xml
Web service response sample 2	Response2.xml
Web service response sample 3	Response3.xml
Web service response sample 4	Response4.xml

Parse incoming documents in Excel format

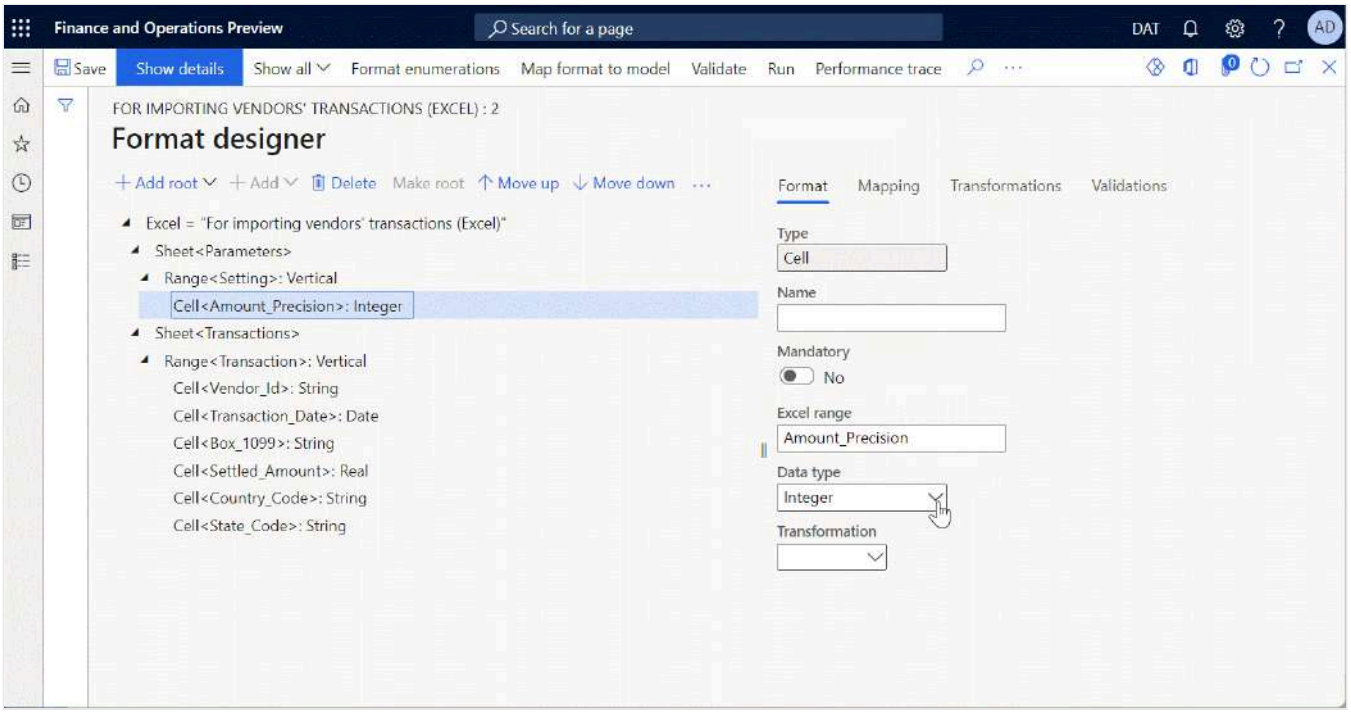
You can design Electronic reporting (ER) formats to parse incoming Microsoft Excel files that represent data in Microsoft Excel workbooks (files in XLSX format). You can then use the content from these files to update application data. This approach is useful if you:

- Design a new model and format and want to test them at run-time. In this case, Excel simulates the actual application data.
- Manage data beyond your application in Excel and want to import this data to submit a specific report.

When you design a new ER format to parse inbound documents, you can import an Excel workbook into it as a template for inbound files. To automatically detect the data type that is stored in Excel named cells, set the **Apply Excel data types for cell components** option to **Yes** in the **Update from Excel** dialog box.



You can manually change the cells later in the **Data type** field on the **Format** tab, if either no data type (void) or an incorrect data type is automatically detected for a single Cell component of the edited ER format. These data types are used in the format model mapping when you specify how data from an inbound file is used to fill in a data model.



By default, when you run this format to parse a real inbound file, ER tries to detect the data type of each cell, just as it did when you imported an Excel template. In some cases, an incorrect data type might be detected and cause incorrect data conversion or a runtime exception. In Finance version 10.0.25 and later, you can force the ER framework at runtime to use the data conversion data types that you specify for each cell component in the **Data type**

field on the **Format** tab of the edited ER format. To force this behavior, enable the **Forcing to use for data parsing only cell data types that are defined in an ER format** feature in the **Feature management** workspace.

To learn more about this feature, watch the task guides **ER Import data from a Microsoft Excel file (Part 1: Design format)** and **ER Import data from a Microsoft Excel file (Part 2: Import data)** (parts of the 7.5.4.3 Acquire/Develop IT service/solution components (10677) business process). These task guides walk through how the incoming Excel file can be parsed by using the ER format to import information from incoming documents and update application data. You can download the task guide files from the Microsoft Download Center.

Download the following files to complete the task guides mentioned earlier.

Expand table

Content description	File
Incoming file in .XLSX format - template	1099import-template.xlsx
Incoming file in .XLSX format - sample data	1099import-data.xlsx

If you didn't already, to learn how to create the required configurations to import data from an external file, see ER Create required configurations to import data from an external file in the current finance and operations application. To download the file, see the following table.

Expand table

Content description	File
ER model configuration	1099model.xml

Parse incoming documents in CSV format

You can design Electronic reporting (ER) formats to parse incoming electronic documents that represent tabular data in plain text (files in CSV format). Then, use the content from these documents to update application data. Use the following approach:

- Begin your format design by adding a new root sequence element to specify that each line in the parsing file is a separate record.
  - In the added sequence element, select the appropriate value, such as **New line - Windows (CR LF)**, in the **Special characters** field in the **Sequence element delimiter** field group.

- Continue your format design by adding a nested sequence element of the added root sequence element to specify that each line in the parsing file is a set of fields.
  - Specify the character in the **Custom delimiter** field that the parser recognizes in the parsing line as a fields separator.

    > ⓘ **Note**
    > - You can define different field separators for different sequence elements to parse specific file lines in which fields are separated by different characters.
    > - You can leave the **Custom delimiter** field blank for certain sequence elements. An empty field means that any file line that the parser uses this sequence to parse is like a .txt (fixed length text) file line.

  - In the **Quotation application** field, select the value when you expect that some fields of any line that the parser uses this sequence element to parse are enclosed by certain characters. The following options are available:
    - **All** – Include quotation characters in the parsing line for any field of any data type. If you select this option, define the desired characters in the **Quotation mark** field that the parser uses for fields quotation. For example, the double quotes character.
    - **Text only** – Include quotation characters in the parsing line for any field of the **String** data type. If you select this option, define the desired characters in the **Quotation mark** field that the parser uses for fields quotation.
    - **Derive from parent only** – Use the same **Quotation application** field settings that you define for the parent sequence element. The **Quotation mark** field setting comes from the settings of the parent sequence element as well.
    - **None** – Exclude quotation characters in the parsing line for any field of any data type.

- Complete your format design by adding nested elements for the added sequence element that represents the set of fields of the parsing line. Add the required elements of the **Text** group, such as **String**, **DateTime**, and **Numeric**, to describe the structure of the parsing line as a set of individual fields of different data types.
  - For each format element that represents an individual field of the parsing line, by default, nothing is selected in the **Multiplicity** field. This default means that the value of the field in the parsing line is required. In the **Multiplicity** field, select **Zero one** to make the value of this field in the parsing line optional.

> **Note**
>
> The data source item **isMatch** is available when you map this format to ER data model for each **String**, **DateTime**, or **Numeric** format element with the option **Zero one** selected in the **Multiplicity** field. When this field contains a value, **isMatch** returns **True**. If there's no value in the field, it returns **False**.

To learn more about this feature, watch the task guide, **ER Create a format configuration to import data from an external CSV file** (part of the 7.5.4.3 Acquire/Develop IT service/solution components (10677) business process). The guide walks through how the incoming CSV file can be parsed by using the ER format to update application data.

Download the following files to complete the task guide mentioned earlier.

Expand table

Title	File name
ER format configuration	1099formatcsv.xml
Sample of incoming file in .csv format	1099entriescsv.csv

Download the following file to complete the task guide mentioned earlier if you didn't watch the task guide, **ER Create required configurations to import data from an external file for electronic reporting**, in the current finance and operations application.

Expand table

Title	File name
ER model configuration	1099model.xml

Last updated on 04/13/2026

ER Create required configurations to import data from an external file

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System administrator or Electronic reporting developer role can design Electronic reporting (ER) configurations to import data into the application from an external file. In this example, you create the required ER configurations for the sample company, Litware, Inc. To complete these steps, you must first complete the steps in the task guide, "ER Create a configuration provider and mark it as active." You can complete these steps by using the USMF data set. You must also download and save the following files locally:

Expand table

Content description	File name
ER data model configuration - 1099	1099model.xml
ER format configuration - 1099	1099format.xml
Sample of the incoming document in XML format	1099entries.xml
Sample of the workbook to manage data of incoming document	1099entries.xlsx

ER offers business users the ability to configure the process of importing external data files to tables in either .XML or .TXT format. First, you must design an abstract data model and an ER data model configuration to represent the data that you're importing. Next, you need to define the structure of the file that you're importing and the method that you use to port the data from the file to the abstract data model. You must create the ER format configuration that maps to the designed data model for that abstract data model. Then, you must extend the data model configuration with a mapping that describes how the imported data is persisted as abstract data model data and how it is used to update tables. You must append the ER data

model configuration with a new model mapping that describes the binding of the data model to the application's destinations.

The following scenario shows the ER data import capabilities. This scenario includes vendor transactions that are tracked externally and then imported to be reported later in Vendor's settlement for 1099's.

Add a new ER model configuration

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

   Verify that the configuration provider for sample company **Litware, Inc.** is available and marked as active. If you don't see this configuration provider, first complete the steps in the procedure, "Create a configuration provider and mark it as active."

2. Select **Reporting configurations**.

   Instead of creating a new model to support data import, load the file, `1099model.xml`, that you previously downloaded. This file contains the custom data model of vendors' transactions. You map this data model to the data components that are in the AOT data entity.

3. Select **Exchange**.

4. Select **Load from XML file**.

   Select **Browse** and go to the `1099model.xml` file that you previously downloaded.

5. Select **OK**.

6. In the tree, select **1099 Payments model**.

Review data model settings

1. Select **Designer**.

   This model represents vendors' transactions from the business standpoint. It's separate from the implementation.

2. In the tree, expand **1099-MISC**.

3. In the tree, select **1099-MISC\Transactions**.

4. In the tree, expand **1099-MISC\Transactions**.

   The **Transactions** element of this model represents individual transactions. Use the child elements to specify required details, such as vendor account and transaction date, for each transaction.

5. Close the page.

Add a new ER format configuration that supports data import

The steps in this subtask show you how to create a new format configuration to manage data import from external files.

1. Select **Create configuration** to open the drop dialog.
2. In the **New** field, enter *Format based on data model 1099 Payments model*.
3. Select **Yes** in the **Supports data import** field.
4. Press **ESC** key to close this page.

   Instead of creating a new format to support data import, load the 1099format.xml file that you previously downloaded. This file contains the defined structure of the file you are importing and the mapping of the structure to the custom data model of vendors' transactions.

5. Select **Exchange**.
6. Select **Load from XML file**. Select **Browse** and go to the 1099format.xml file that you previously downloaded.
7. Select **OK**.
8. In the tree, expand *1099 Payments model*.
9. In the tree, select *1099 Payments model\Format for importing vendors' transactions*.

Review format settings

1. Select **Designer**.
2. Toggle **Show details** on.

3. Select **Expand/collapse**.
4. Select **Expand/collapse**.

   The designed format represents the expected structure of the external file. This file must be in XML format and have the settlement root element. Each vendor's transaction is represented by the transaction element that is defined as having zero-to-many multiplicity. This definition means that the incoming file can contain zero to multiple transactions. Nested elements of the *transaction* element represent a single transaction's attributes. All attributes, except country/region, are mandatory, so the importing file must include them.

Review the settings of the format mapping to the data model

1. Select **Map format to model**.

   The mapping **For importing vendors' transactions** contains the data transfer rules from the incoming XML file to the selected part of the custom data model, which you define by selecting the 1099-MISC definition.
2. Select **Designer**.
3. Toggle **Show details** on.
4. In the tree, expand **format: Record**.
5. In the tree, select **format: Record**.

   You see the designed format as a data source component.
6. In the tree, expand `format: Record*settlement: XML Element 1..1 (settlement): Record`.
7. In the tree, expand `format: Record*settlement: XML Element 1..1 (settlement): Record\transaction: XML Element 0..* (transaction): Record list`.
8. In the tree, expand `format: Record*settlement: XML Element 1..1 (settlement): Record\transaction: XML Element 0..* (transaction): Record list*vendor: XML Element 1..1 (vendor): Record`.
9. In the tree, expand `format: Record*settlement: XML Element 1..1 (settlement): Record\transaction: XML Element 0..1 (country): Record`.

10. In the tree, select `format: Record*settlement: XML Element 1..1 (settlement): Record\transaction: XML Element 0..* (transaction): Record list*vendor: XML Element 1..1 (vendor): Record`.

    The predefined **format** data source component presents mandatory and optional format elements differently.

11. In the tree, expand **Transactions: Record list= format.settlement.'$enumerated'**.

    The elements of the format that defines the structure of the imported file are bound to the elements of the custom data model. Based on these bindings, the content of the imported XML file is stored at run-time in the existing data model. Pay attention to the binding of the country element. For any transaction element in the incoming file that has no such element, the default country code **USA** is populated in the data model.

12. Select the **Validations** tab.

    This format mapping can contain user-defined logic to validate the accuracy of the imported data from a business standpoint. For example, based on the setting, for any transaction in the importing file without a defined country code, a warning message is generated in the Infolog that informs the user about the case and indicates the transaction's sequence number.

13. Close the page.

Run the format mapping to the data model

Run this format mapping for testing purposes. Use the `1099entries.xml` file that you previously downloaded. You can export this file from the `1099entries.xlsx` workbook that you use to manage vendor transactions. The generated output imports from the selected XML file and populates the custom data model at real import.

1. Select **Run**.

    Select **Browse** and go to the `1099entries.xml` file that you previously downloaded.

2. Select **OK**.

    Note the warning message about a missing country code for a transaction in the imported file.

Review the output in XML format, which represents the data that you imported from the selected file and ported to the data model.

3. Close the page.

4. Close the page.

Review the settings for the model mapping to the destinations

1. In the tree, select **1099 Payments model**.

2. Select **Designer**.

3. Select **Map model to datasource**.

   The mapping **For 1099 manual transactions import** is defined with the **To destination** direction type. This mapping supports data import and contains the setting of rules that define how the imported external file and persisted abstract data model data are used to update tables in the application.

4. Select **Designer**.

5. In the tree, expand **model: Data model 1099 Payments model**.

6. In the tree, expand **model: Data model 1099 Payments model\Transactions: Record list**.

   The designed model appears as a data source element. At runtime, it contains the data that you import from the external file. Add several tables as data source elements to ensure that the imported data is compliant with the data of the current application, including whether the importing transaction vendor account is available in the system, whether the combination of the importing country/region and state codes exists, and other checks.

7. In the tree, select `model: Data model 1099 Payments model\Transactions: Record list\$failed: Calculated field = IF(OR(ISEMPTY(model.Transactions.'$refs'.vendor), ISEMPTY(model.Transactions.'$refs'.vendor1099), ISEMPTY(model.Transactions.'$refs'.box1099), ISEMPTY(model.Transactions.'$refs'.country), ISEMPTY(model.Transactions.'$refs'.state), ISEMPTY(model.Transactions.'$refs'.location)), true, false): Boolean`.

8. Select **Edit**.
9. Select **Edit formula**.

   When at least one validation fails for a single imported transaction, the data source attribute `$failed` marks this transaction as failed.
10. Close the page.
11. Select **Cancel**.
12. In the tree, select `tax1099trans: Table 'VendSettlementTax1099' records= model.Validated`.
13. Select **Edit destination**.

    Add this ER destination to specify how the imported data updates the application tables. In this case, select the data table `VendSettlementTax1099`. Because the record action `Insert` is selected, the imported transactions are inserted in the table `VendSettlementTax1099`. A single model mapping can contain several destinations. This design means that the imported data can update multiple application's tables at once. You can use tables, views, and data entities as ER destinations.

    If you call the mapping from a point in the application (such as button or menu item) that you specifically designed for this action, mark the ER destination as the integration point. In this example, this point is `ERTableDestination#VendSettlementTax1099`.
14. Select **Cancel**.
15. Select **Show all**.
16. Select **Show mapped only**.
17. In the tree, expand `tax1099trans: Table 'VendSettlementTax1099' records= model.Validated`.

    The data source element that contains only validated transactions is bound to the created destination. You can filter the imported transactions to skip the ones that are incompatible with the applications' data.
18. In the tree, select `failed: Table 'VendSettlementTax1099Entity' records= model.Failed`.
19. Select the **Validations** tab.

This model mapping can contain user-defined logic to validate the correctness of the imported data from the existing application data. For example, based on the present setting, for any transaction in the imported file with a vendor account that isn't in the system, a warning message is generated that informs the user and indicates the incorrect vendor account code.

Use the **Post validation action** option for each validation to specify whether the import process must continue or stop, and if the already performed inserts or updates can be kept or rolled back.

20. Select **Show mapped only**.
21. Select **Show all**.
22. Close the page.

    Execute this model mapping to test the designed format and model mappings. Use the file `1099entries.xml`. The data from the selected file is imported into the system.

23. Select **Run**.

    The dialog box contains no additional questions about the format mapping that must be used to parse the imported file and then port the data to the data model. This condition exists because there's currently only one format that uses this model, which is marked as designed to support data import.

    Define the voucher ID to differentiate the imported transactions from other transactions that you might already entered manually or imported.

24. In the **Enter voucher id** field, type `IMPORT-001`.

    Browse to get the `1099entries.xml` file.

25. Select **OK**.

    The list of generated warnings provides information about incorrect vendor accounts, an incorrect tax 1099 box code, missing country codes, and other issues. Compare this list of warnings to the content that the execution XML file includes.

26. Close the page.
27. Close the page.

28. Close the page.
29. Close the page.
30. Go to **Accounts payable** > **Periodic tasks** > **Tax 1099** > **Vendor settlement for 1099s**.

    This form shows the cumulative transactions in the **Tax1099Summary** table that the system created based on imported transactions.
31. In the **From date** field, set the date to 2000-01-01.
32. Select **Manual 1099 transactions**.

    This form contains the list of transactions that you added manually and those that you just imported.
33. Open **Voucher** column filter.
34. Enter a filter value of `IMPORT-001` on the **Voucher** field by using the **begins with** filter operator.

Review the relationship between model and format mappings

1. Close the page.
2. Close the page.
3. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
4. Select **Reporting configurations**.
5. In the tree, select **1099 Payments model**.

    Assume that you want to support importing the same data but from a .TXT file format.
6. Select **Create configuration** to open the dialog box.
7. In the **New** field, enter *Format based on data model 1099 Payments model*.
8. In the **Name** field, type *Import data from TXT file*.
9. Select **Yes** in the **Supports data import** field.
10. Select **Create configuration**.

11. Select **Designer**.
12. Select **Map format to model**.
13. Select **New**.
14. In the **Definition** field, enter or select a value.

    Select **1099-MISC** option.
15. In the **Name** field, type *Import data from TXT file*.
16. In the **Description** field, type *Import data from TXT file*.
17. Select **Save**.
18. Close the page.
19. Close the page.
20. Select **Edit**.

    If you installed the hotfix "KB 4012871 Support of GER model mappings in separated configurations with an ability to specify different kinds of prerequisites for deploying them on different versions of Dynamics 365 Finance" (KB 4012871), execute the next step **Turn the flag 'Default for model mapping' on** for the entered format configuration. Skip the next step otherwise.
21. Select **Yes** in the **Default for model-mapping** field.
22. In the tree, select **1099 Payments model**.
23. Select **Designer**.
24. Select **Map model to datasource**.
25. Select **Run**.

    If you installed the hotfix, KB 4012871 Support of GER model mappings in separated configurations with an ability to specify different kinds of prerequisites for deploying them on different versions (KB 4012871), select the preferred model mapping in the lookup field. If you didn't install the hotfix, skip to the next step as the mapping is already selected by the definition of the default format configuration.

If you didn't install the hotfix, KB 4012871, the dialog box contains an additional model-mapping question that is used to parse the file that you're importing. The data is then ported from the dialog box to the data model. Currently, you can choose which format mapping to use depending on the type of file that you plan to import.

If you plan to call this model mapping from a point in the application that is specifically designed for the action, the ER destination and the format mapping must be marked as part of the integration.

26. Select **Cancel**.
27. Close the page.
28. Close the page.

Last updated on 04/13/2026

Import data from manually selected files in batch mode

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To use the Electronic reporting (ER) framework to import data from manually selected inbound files in batch mode, configure an ER format that supports the import. Then run a model mapping of the **To destination** type that uses that format as a data source. To import data, browse to the file that you want to import, and manually select it.

The new ER capability that supports data import in batch mode enables you to configure this process as unattended. You can use ER configurations to perform data import by scheduling a new batch job from the ER user interface (UI).

This article explains how to complete data import from a manually selected file in batch mode. The examples use vendor transactions as business data. You can complete the steps of these examples in the **USMF** company. No coding is required.

Prerequisites

To complete the examples in this article, you must have the following access:

- One of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- ER format and model configurations for 1099 payments

Create the required ER configurations

To create the required ER configurations and obtain other prerequisites, follow one of these steps:

- Play the **ER Import data from a Microsoft Excel file** task guides, which are part of the **7.5.4.3 Acquire/Develop IT service/solution components (10677)** business process. These task guides walk you through the process of designing and using ER configurations that interactively import vendor transactions from Microsoft Excel files. For more information, see Parse incoming documents in Excel format.
- Complete the examples in Configure data import from SharePoint. These examples walk you through the process of designing and using ER configurations that interactively import vendor transactions from Excel files that are stored in a SharePoint folder.

Download the required ER configurations

1. Download the following ER configurations, and save them locally.

Expand table

Content description	File
1099 Payments model ER data model configuration	1099model.xml
For importing vendors' transactions (Excel) ER format configuration	1099format-import-from-excel.xml

2. Use the Load from XML file ER option to import the downloaded ER configurations into your instance of Dynamics 365 Finance in the following order:
   a. ER data model configuration
   b. ER format configuration

Download the required Excel files

- Download the following sample data set, and save it locally.

Expand table

Content description	File
Inbound **1099import-data.xlsx** file that contains sample data for import	1099import-data.xlsx

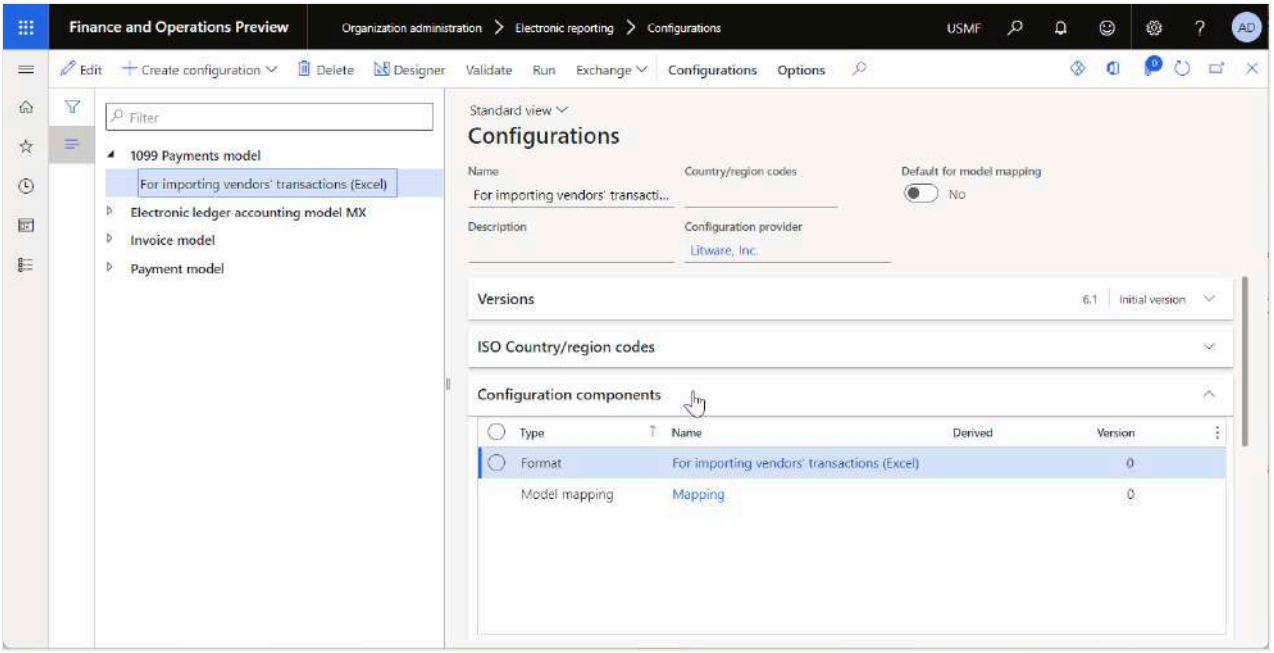
Review the prerequisites

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.

2. On **Configurations**, review the prepared ER solution for data import in batch mode.

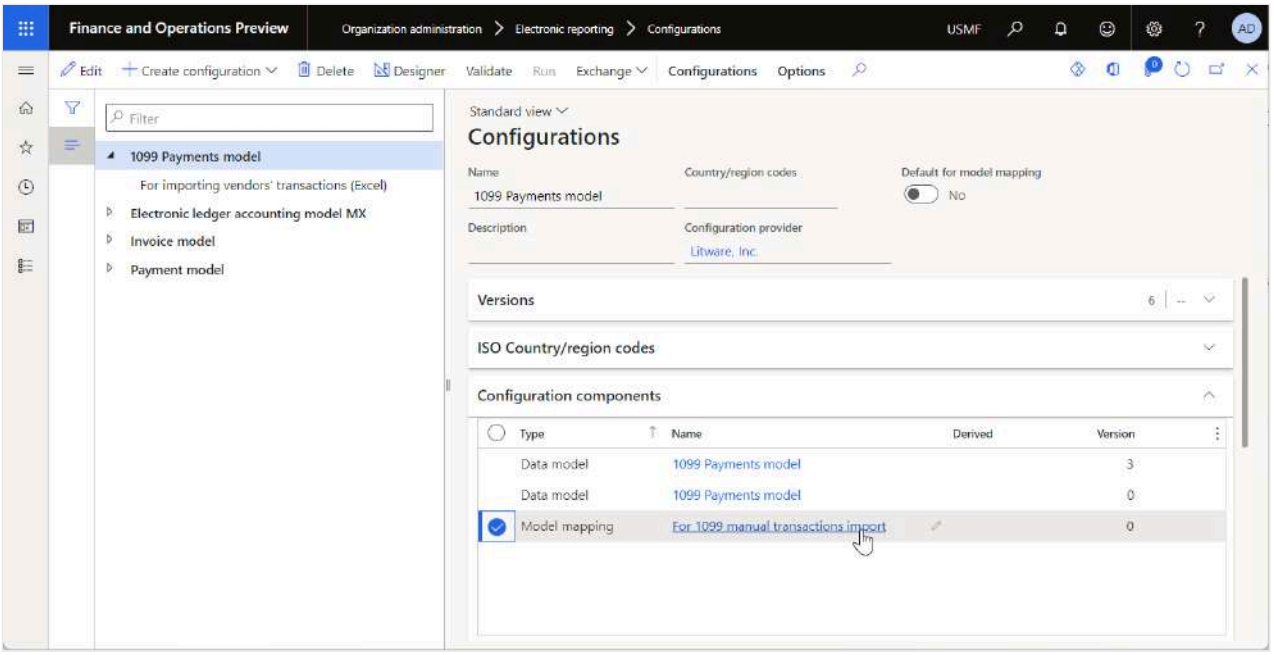
3. Review the **For importing vendors' transactions (Excel)** format configuration.

    - The format component of this configuration is designed to parse an inbound Excel file.
    - The model mapping component of this configuration is used to fill in the base data model by using data from the parsed Excel file.

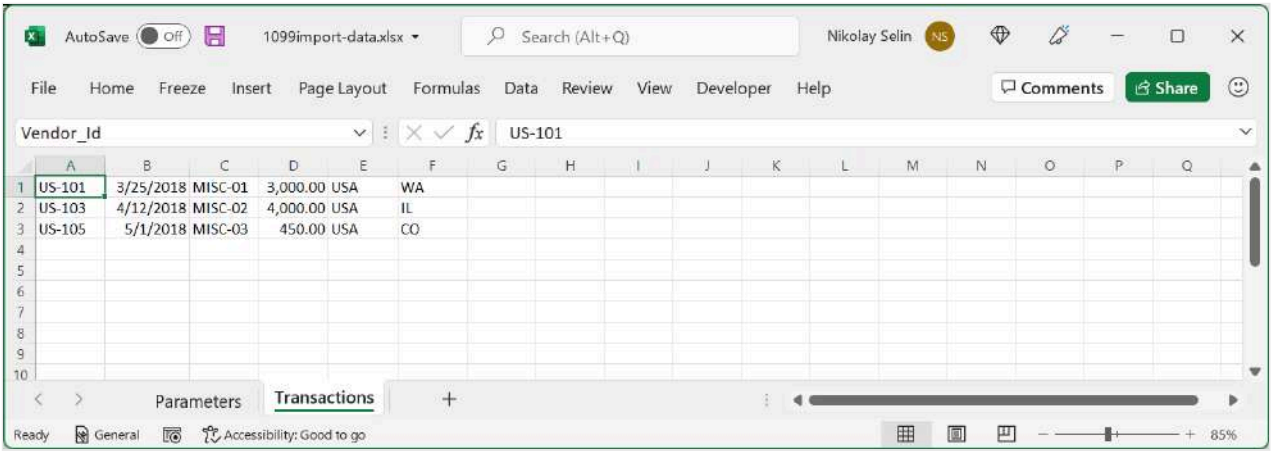


4. Review the **1099 Payments model** data model configuration.

    - The model component of this configuration represents the structure of the data model that is used to pass data between running ER components.
    - The model mapping component of this configuration is used to pull data from the executed format component and then update application tables.



5. Open the **1099import-data.xlsx** file in Excel.



Enable batch data import for ER from the UI

1. Go to **System administration** > **Workspaces** > **Feature management**.
2. In **Feature management**, select the **Run ER import of manually uploaded documents in batch** feature, and then select **Enable now**.

Import data from manually selected Excel files

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, select the **1099 Payments model** data model configuration.
3. On the **Configuration components** FastTab, select the **For 1099 manual transactions import** link.

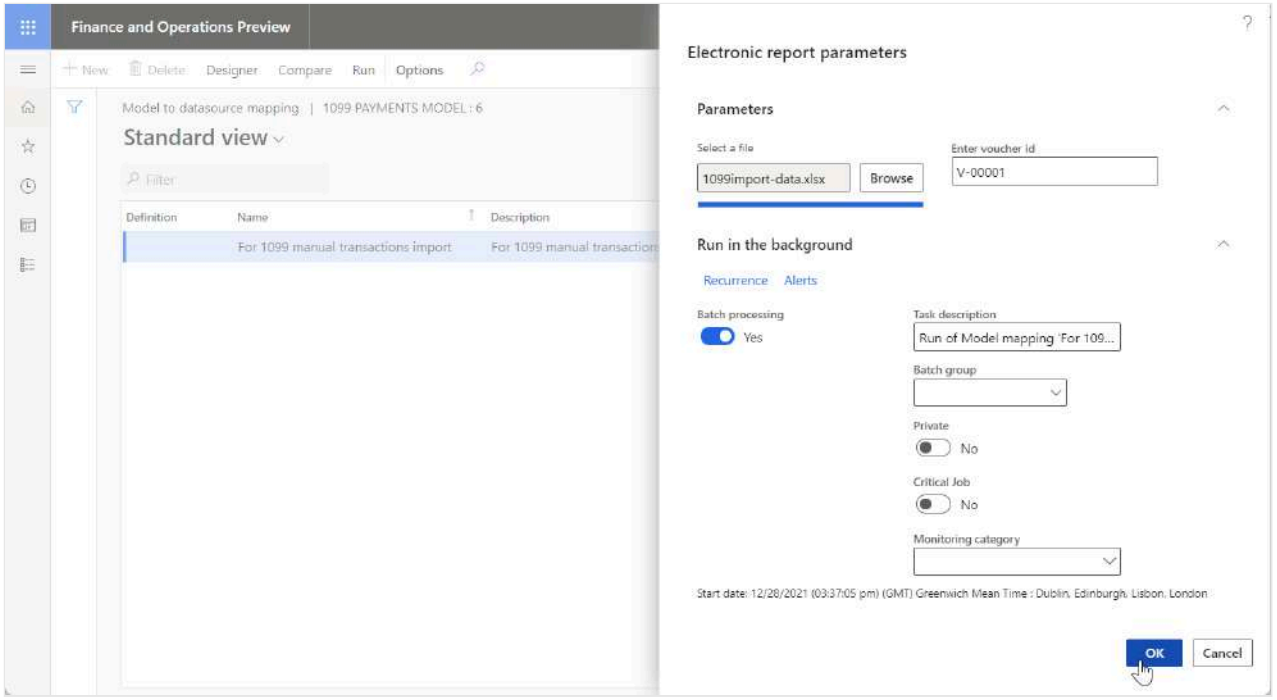
4. On **Model to datasource mapping**, the **For 1099 manual transactions import** model mapping is preselected. Select **Run**.

5. On the **Parameters** tab, select **Browse**. Find and select the **1099import-data.xlsx** file, and then select **OK**.

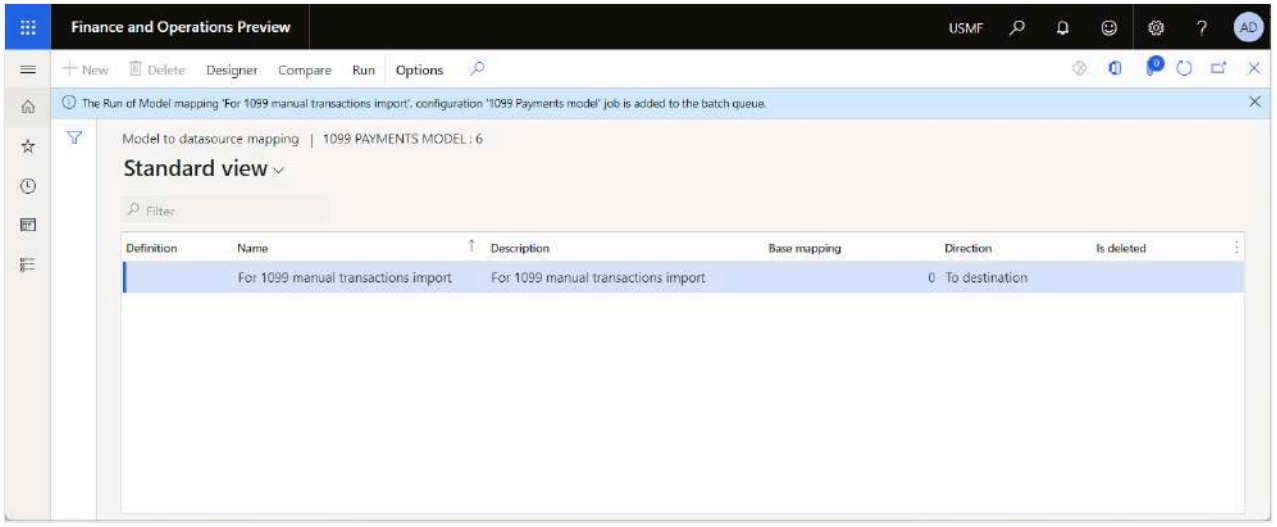
6. In the **Enter voucher id** field, enter **V-00001**.

7. On the **Run in the background** tab, set the **Batch processing** option to **Yes**.

   Notice that the **Task description** field is set to **Run of Model mapping 'For 1099 manual transactions import', configuration '1099 Payments model'**. This value indicates that the execution of the selected model mapping is scheduled as a new batch job.

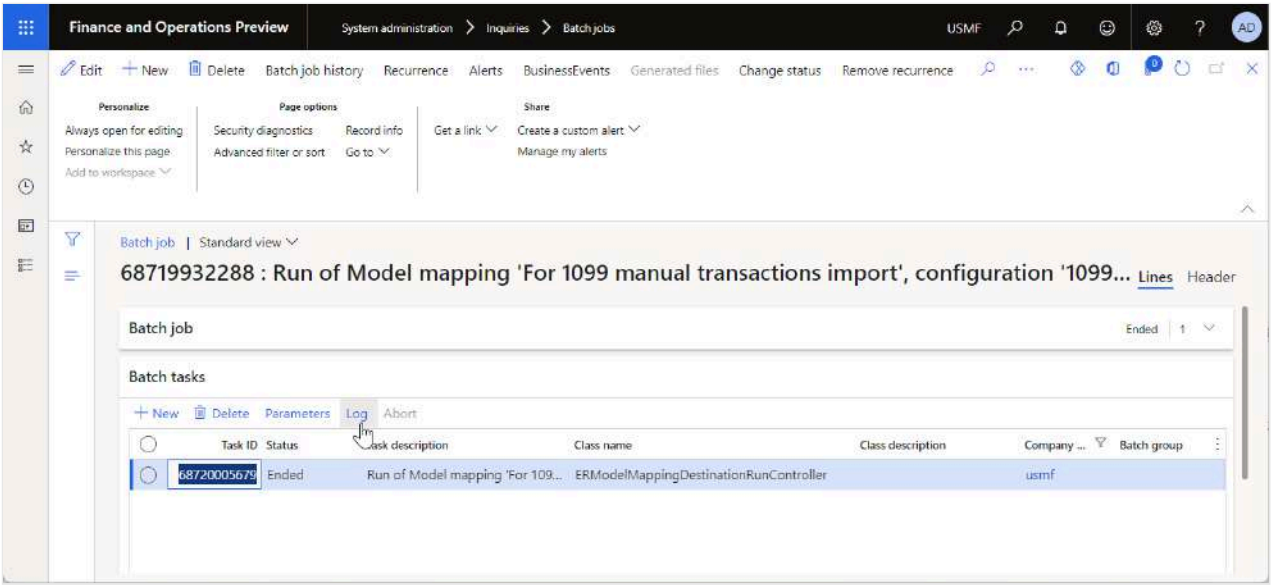


8. Select **OK**. A message notifies you that a new batch job is scheduled.

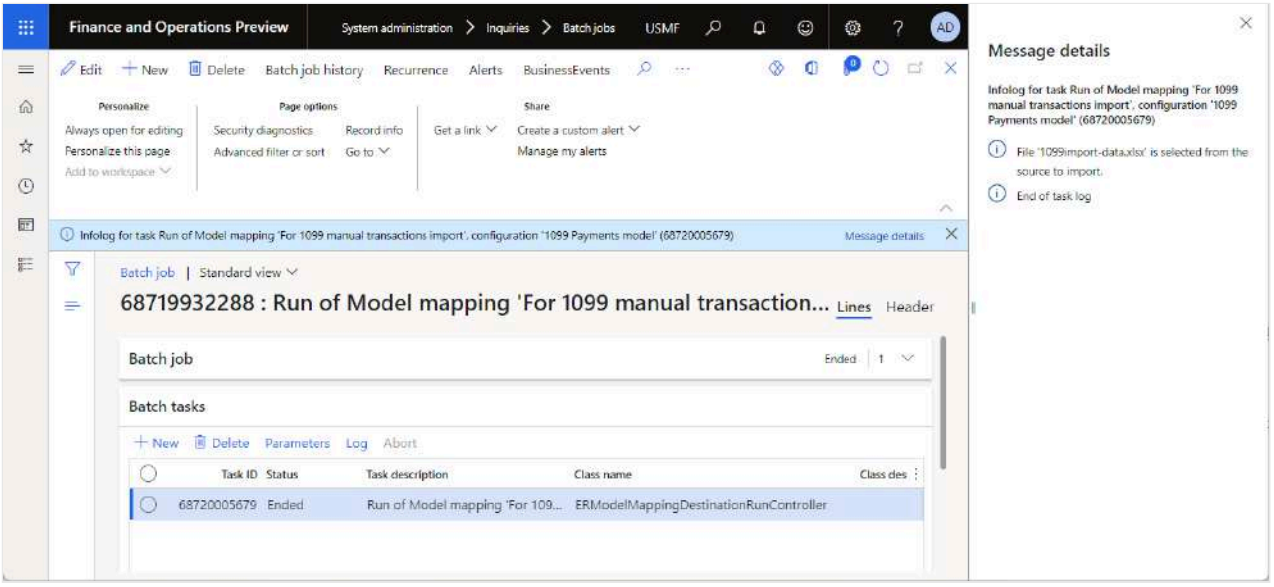


Review the data import results on the Batch job page

1. Go to **Common** > **Inquiries** > **Batch jobs** > **My batch jobs**.
2. On the **Batch job** page, filter the list of batch jobs to find the scheduled batch, and then select it.
3. Select the **Job ID** link to review job details.
4. On the **Batch tasks** FastTab, select **Log**.



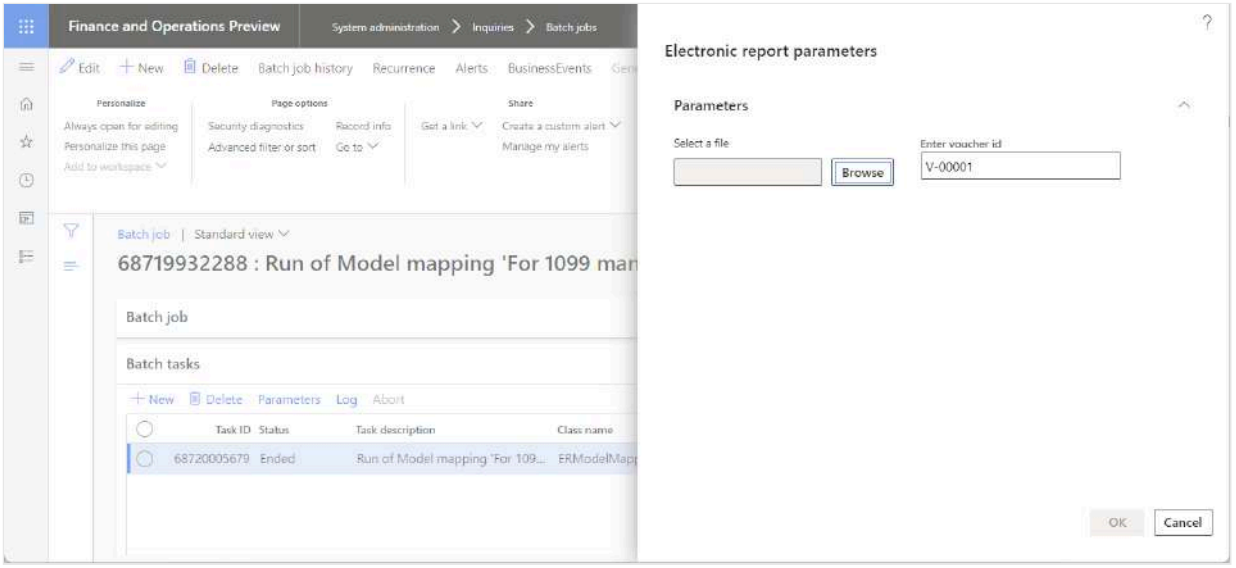
5. Review the execution details.



Change the data import parameters

After your batch is scheduled but before it runs, you can change the parameters of the scheduled data import.

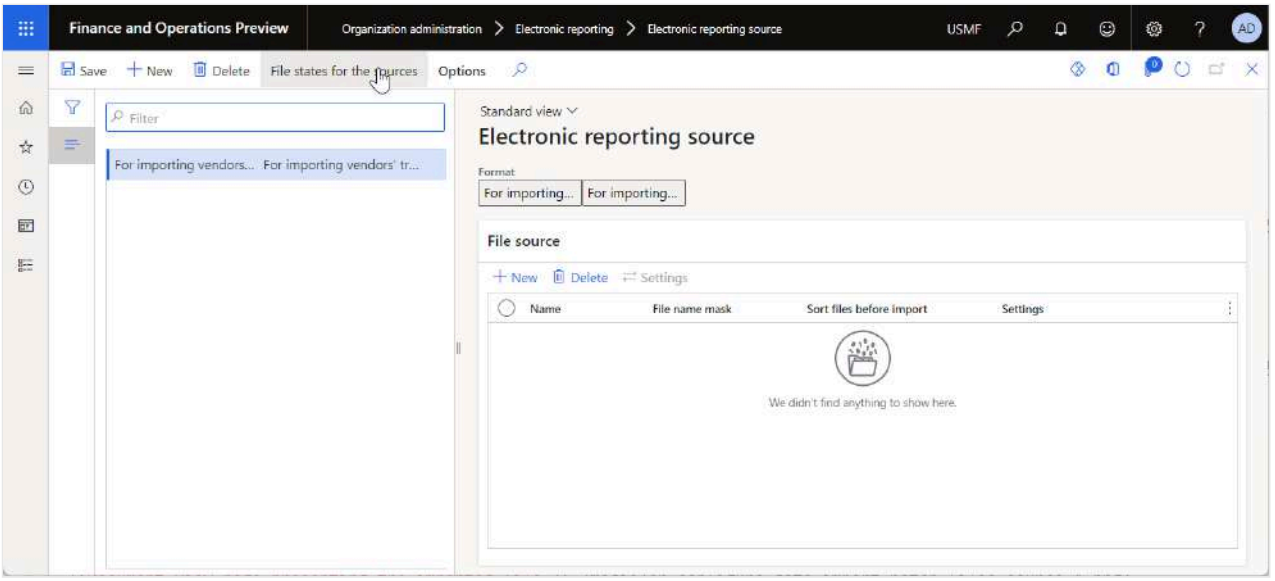
1. Go to **Common** > **Inquiries** > **Batch jobs** > **My batch jobs**.
2. On the **Batch job** page, filter the list of batch jobs to find the scheduled batch, and then select it.
3. Select **Change status**.
4. In the **Select new status** dialog box, select **Withhold**, and then select **OK**.
5. Select the **Job ID** link to access the job details.
6. On the **Batch tasks** FastTab, select **Parameters**.
7. In the **Electronic report parameters** dialog box, follow these steps:
    a. Select **Browse** to select the alternative file for data import.
    b. Select **Enter voucher id** to change the voucher number for importing vendor transactions.



    c. Select **OK**.
8. Make sure that the batch is still selected, and then select **Change status** again.
9. In the **Select new status** dialog box, select **Waiting**, and then select **OK**.

Review the data import results on the ER source page

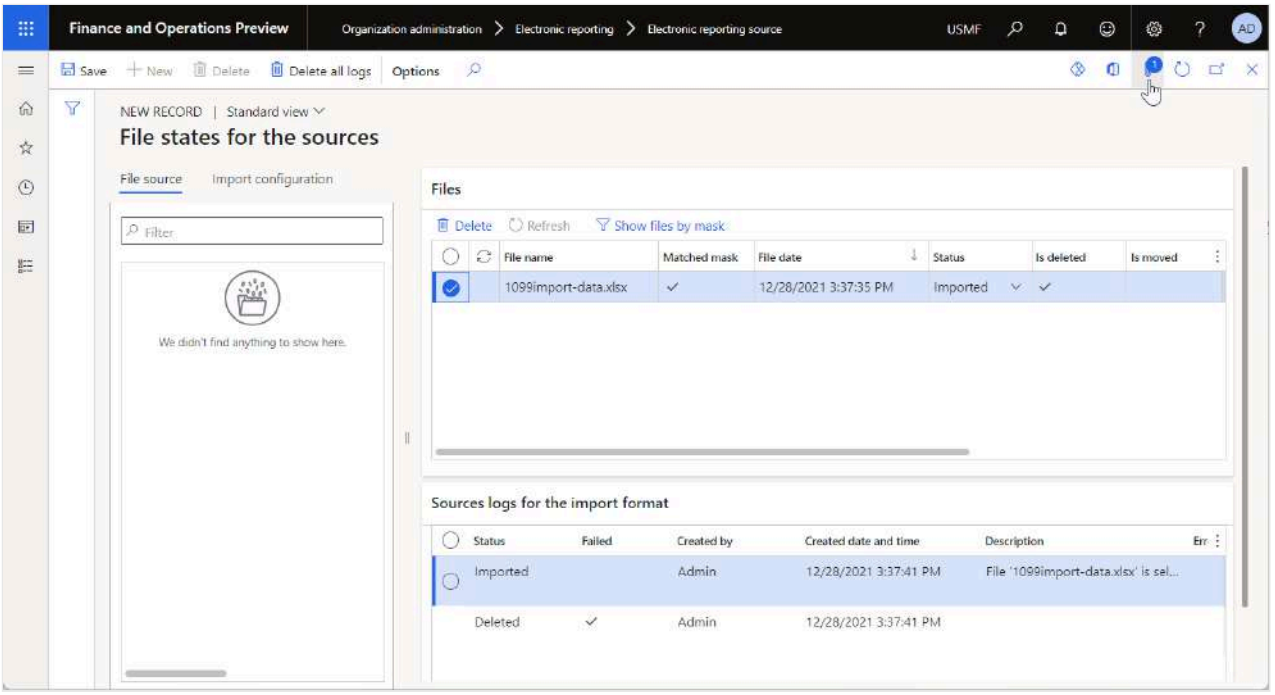
1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting source**.
2. Select the **For importing vendors' transactions (Excel)** record. This record is created automatically during the data import.



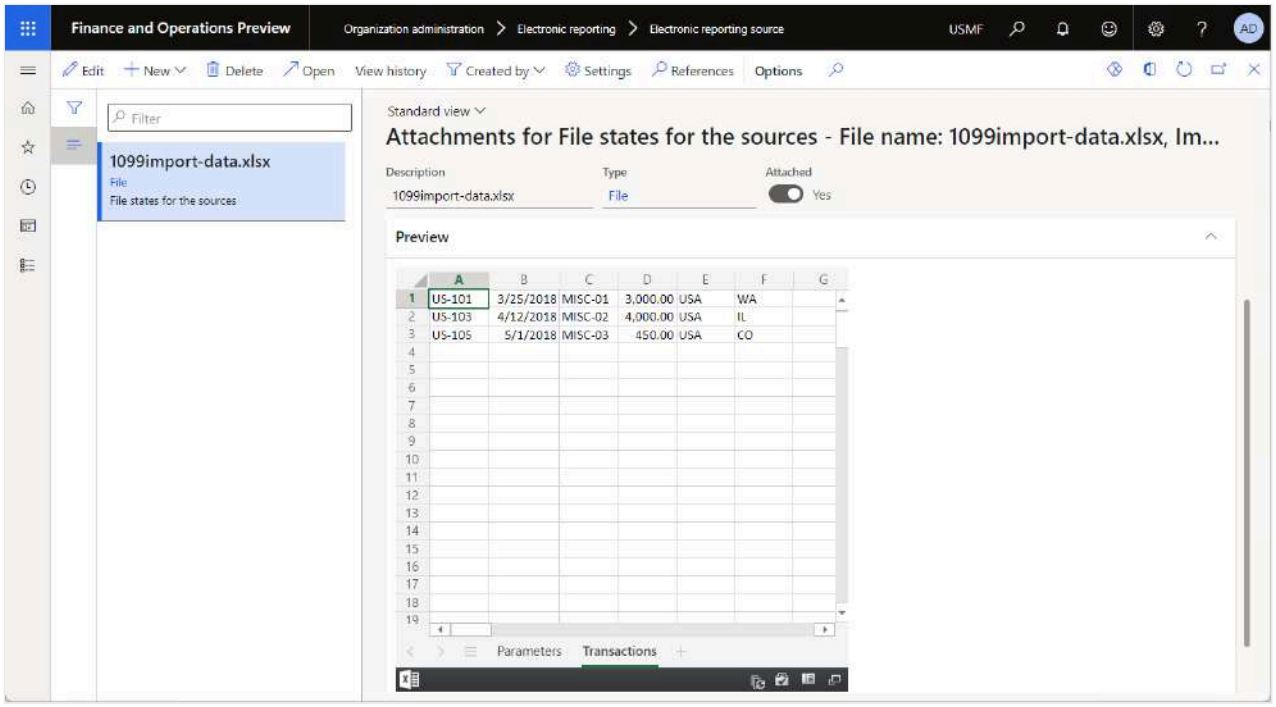
3. Select **File states for the sources**.

4. On the **Files** and **Source logs for the import format** FastTabs, review the import details.

5. On the **Files** FastTab, select the record. The imported file is attached to this record.



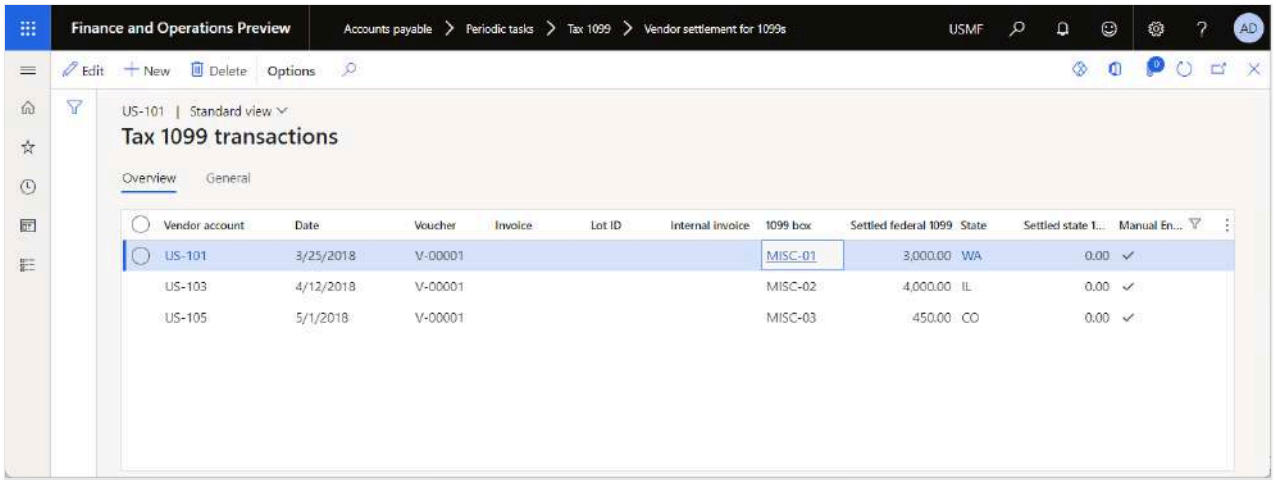
6. Select **Attachments** to review the imported file.



> **Tip**
>
> To keep these attachments, the ER framework uses a document type that is **set** for the current company in the **Others** field of the ER parameters.

Review the data import results on the Vendor settlement for 1099s page

1. Go to **Accounts payable** > **Periodic tasks** > **Tax 1099** > **Vendor settlement for 1099s**.
2. In the **From date** field, enter **12/31/2017** (December 31, 2017).
3. Select **Manual 1099 transactions**.



Additional resources

Electronic Reporting overview

Last updated on 04/15/2026

Configure data import from SharePoint

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To import data from an incoming file by using the Electronic reporting (ER) framework, you must configure an ER format that supports the import and then run a model mapping of the **To destination** type that uses that format as a data source. To import data, you must navigate to the file that you want to import. You can manually select the incoming file. By using the new ER capability to support importing data from Microsoft SharePoint, you can configure this process as unattended. You can use ER configurations to perform data import from files that are stored in Microsoft SharePoint folders. This article explains how to complete the import from SharePoint. The examples use vendor transactions as business data.

Prerequisites

To complete the examples in this article, you must have the following access:

- Access one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- Access to the instance of Microsoft SharePoint Server that is configured for use with the application.
- ER format and model configurations for 1099 payments.

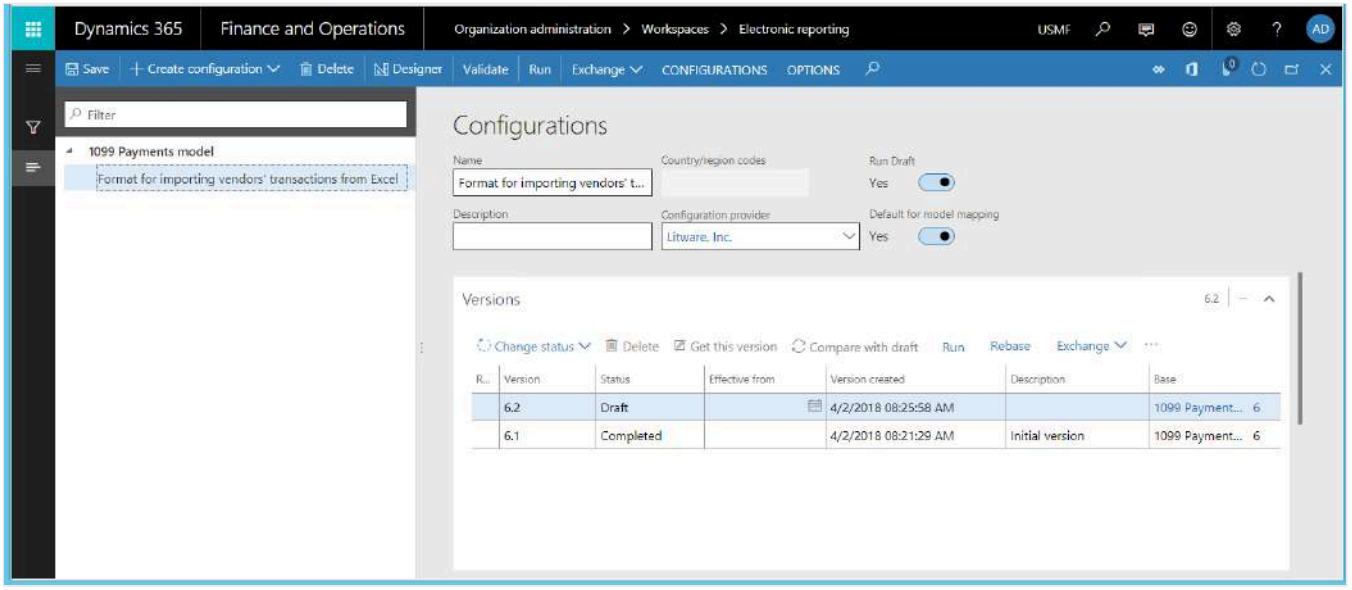
Create required ER configurations

The **ER Import data from a Microsoft Excel file** task guide is part of the **7.5.4.3 Acquire/Develop IT service/solution components (10677)** business process. This task guide walks you through the process of designing and using ER configurations to interactively import

vendor transactions from Microsoft Excel files. For more information, see Parse incoming documents in Excel format. When you complete the task guide, you have the following setup.

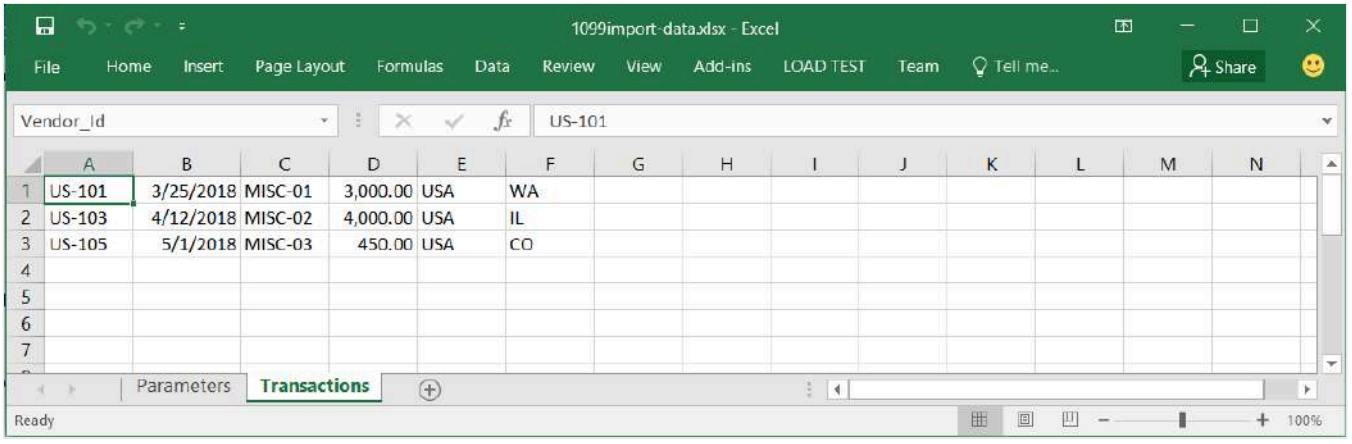
ER configurations

- ER model configuration, **1099 Payments model**
- ER format configuration, **Format for importing vendors' transactions from Excel**



Sample of the incoming file for data import

- Excel file **1099import-data.xlsx**, with vendor transactions that you want to import.

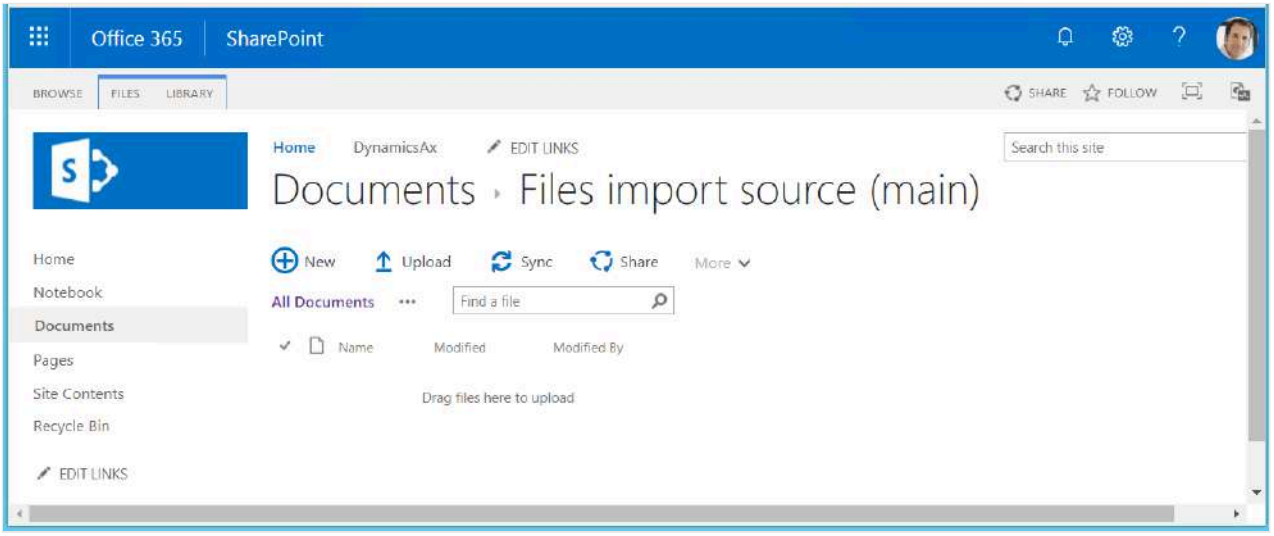


> **Note**
>
> The format for importing vendor transactions is the default model mapping. Therefore, if you run a model mapping of the **1099 Payments model**, and that model mapping is of the **To destination** type, the model mapping runs this format to import data from external files. It then uses that data to update application tables.

Configure access to SharePoint for file storage

To store electronic report files in a SharePoint location, you must configure access to the SharePoint Server instance used by the current company. In this example, the company is USMF. For instructions, see Configure SharePoint storage.

1. Complete the steps in Configure SharePoint storage.
2. Open the configured SharePoint site.
3. Create the following folders where you can store incoming electronic reporting files:
    - Files import source (main) (Example shown in screenshot below)
    - Files import source (alternative)

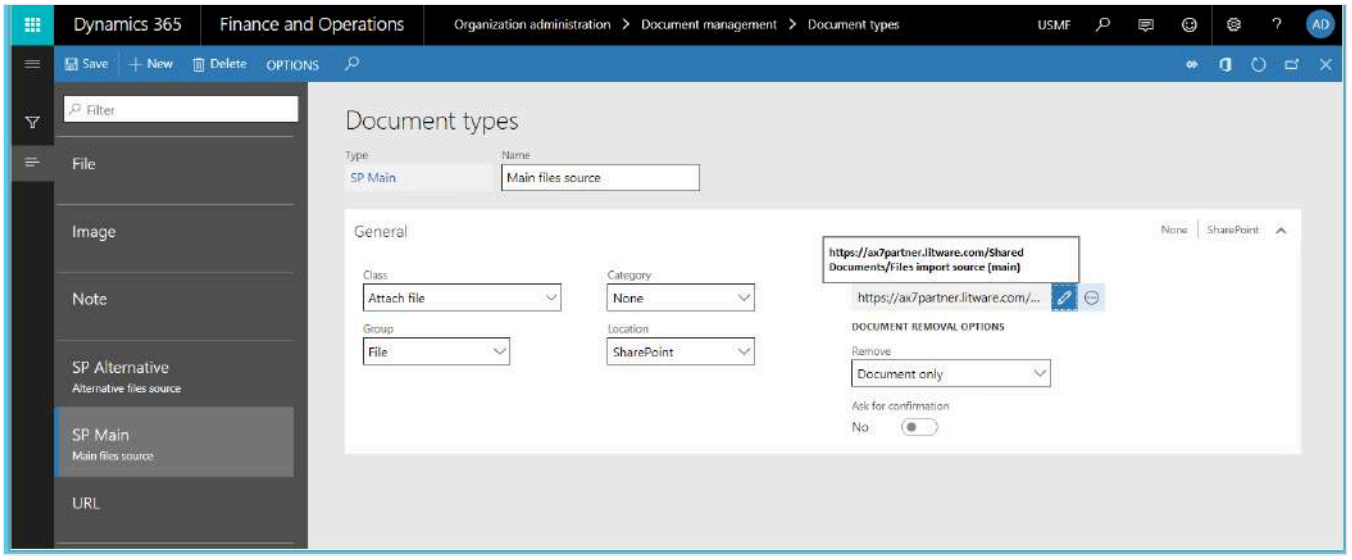


4. (Optional) Create the following folders where you can store files after import.
    - Files archive folder - This folder is for successfully imported files.
    - Files warning folder - This folder is for files that were imported with a warning.
    - Files error folder - This folder is for files that failed to import.
5. Go to **Organization administration** > **Document management** > **Document types**.
6. Create the following document types that help to access the SharePoint folders that you created. For instructions, see Configure document types.

Expand table

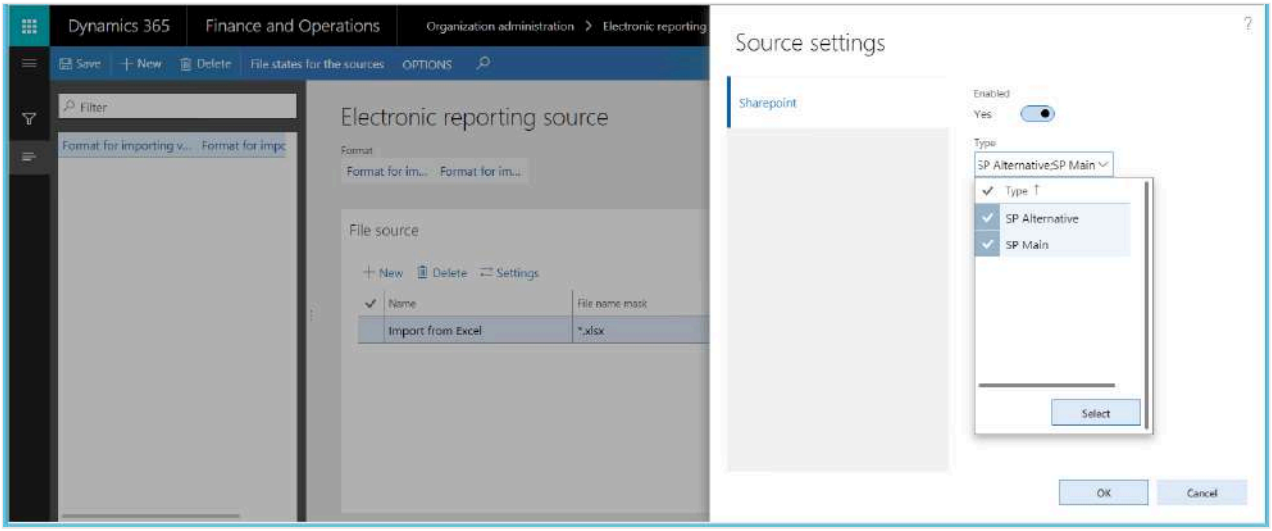
Document type	Group	Location	SharePoint folder
SP Main	File	SharePoint	Files import source (main)

Document type	Group	Location	SharePoint folder
SP Alternative	File	SharePoint	Files import source (alternative)
SP Archive	File	SharePoint	Files archive folder
SP Warning	File	SharePoint	Files warning folder
SP Error	File	SharePoint	Files error folder



Configure ER sources for the ER format

1. Select **Organization administration** > **Electronic reporting** > **Electronic reporting source**.
2. On **Electronic reporting source**, configure the source files for data import by using the configured ER format.
3. Define a file name mask so that only files with the `.xlsx` extension are imported. The file name mask is optional and works only it's defined. You can define only one mask for each ER format.
4. Change **Sort files before import** to **Do not sort** if there are several files for import and the import order isn't important.
5. Select all SharePoint folders that you created earlier.



> **Note**
>
> - The ER *source* is defined for each application company individually. By contrast, ER *configurations* are shared across companies.
> - When you delete an ER source setting for an ER format, all the connected file states are also deleted by confirmation. (see the following section).

Review the file states for the ER format

1. On **Electronic reporting source**, select **File states for the sources** to review the content of the configured file sources for the current ER format.

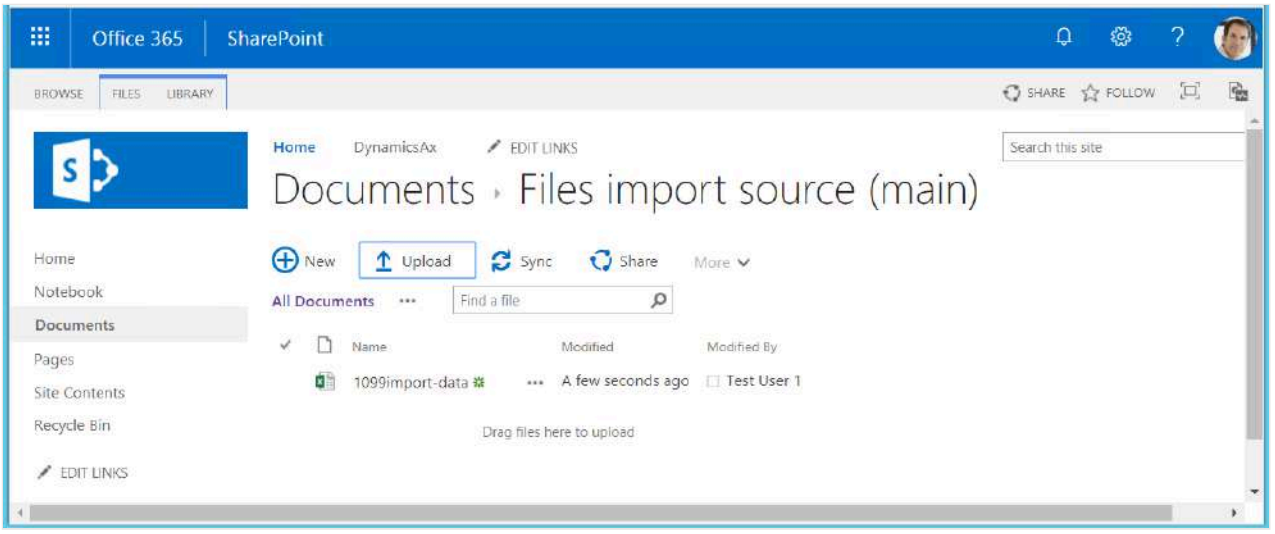
2. In the **Files** section, review the list of files. This list presents the following information:

    - Source files that are applicable, based on the file name mask (if you define one), and that are ready for data import. For these files, the **Sources log for the import format** section is blank.
    - Previously imported files. For each of these files, in the **Sources log for the import format** section, you can review the history of import for this file.

You can also open **File states for the sources** by selecting **Organization administration** > **Electronic reporting** > **File states for the sources**. In this case, the page provides information about file sources for all ER formats with configured file sources in the company that you're currently signed in to.

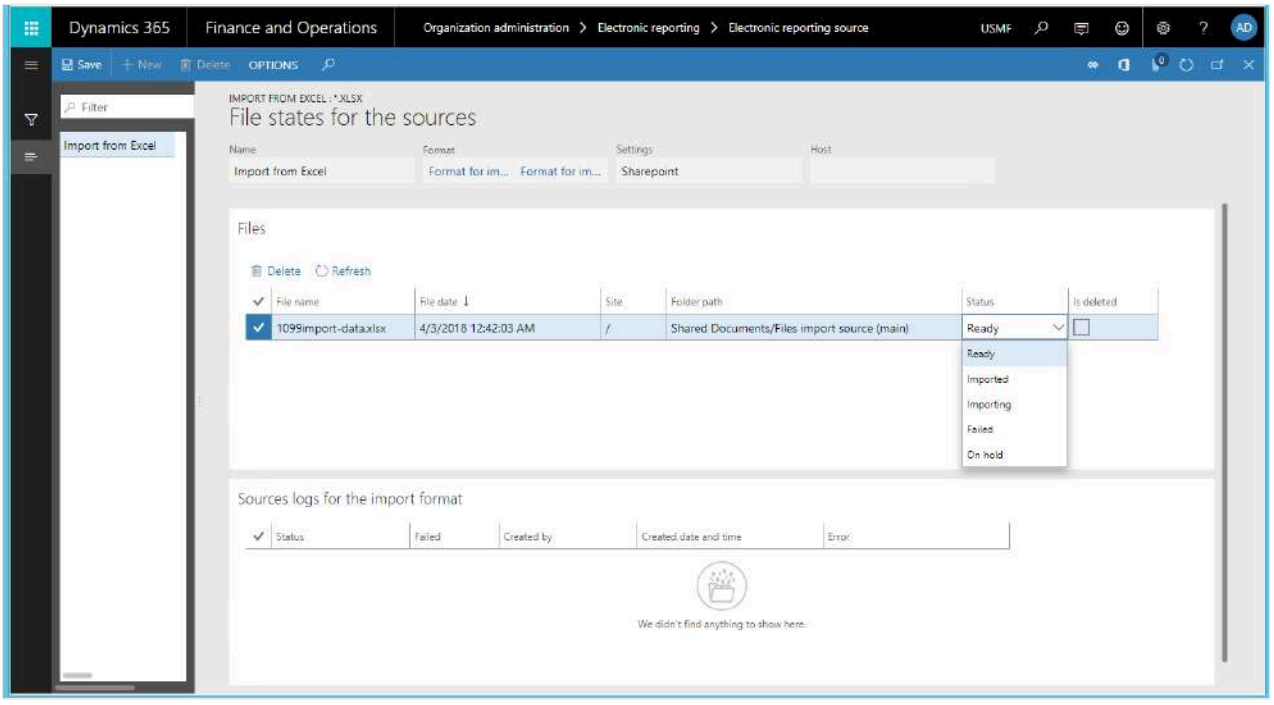
Import data from Excel files in a SharePoint folder

1. In SharePoint, upload the Microsoft Excel file **1099import-data.xlsx** that contains vendor transactions to the **Files import source (main)** SharePoint folder that you created earlier.



2. On **File states for the sources**, select **Refresh** to refresh the page. The Excel file that you uploaded to SharePoint appears on this page with the status **Ready**. The following statuses are currently supported:

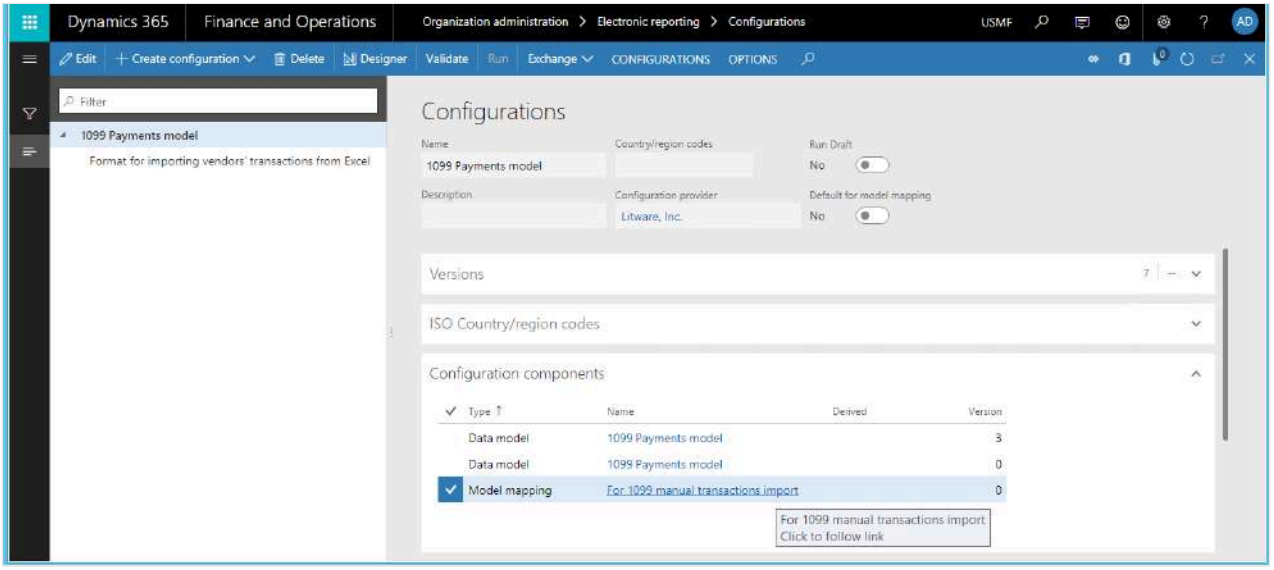
    - **Ready** – Assigned automatically for each new file in a SharePoint folder. This status means that the file is ready for import.
    - **Importing** – Assigned automatically by an ER report when the file is locked by the import process to prevent its usage by other processes (if many of them are running simultaneously).
    - **Imported** – Assigned automatically by an ER report when the file import is successfully completed. This status means that the imported file is deleted from the configured files source (SharePoint folder).
    - **Failed** – Assigned automatically by an ER report when the file import completes with errors or exceptions.
    - **On hold** – Assigned manually by user on this page. This status means that the file isn't imported for now. Use this status to postpone the import of some files.



Import data from SharePoint files

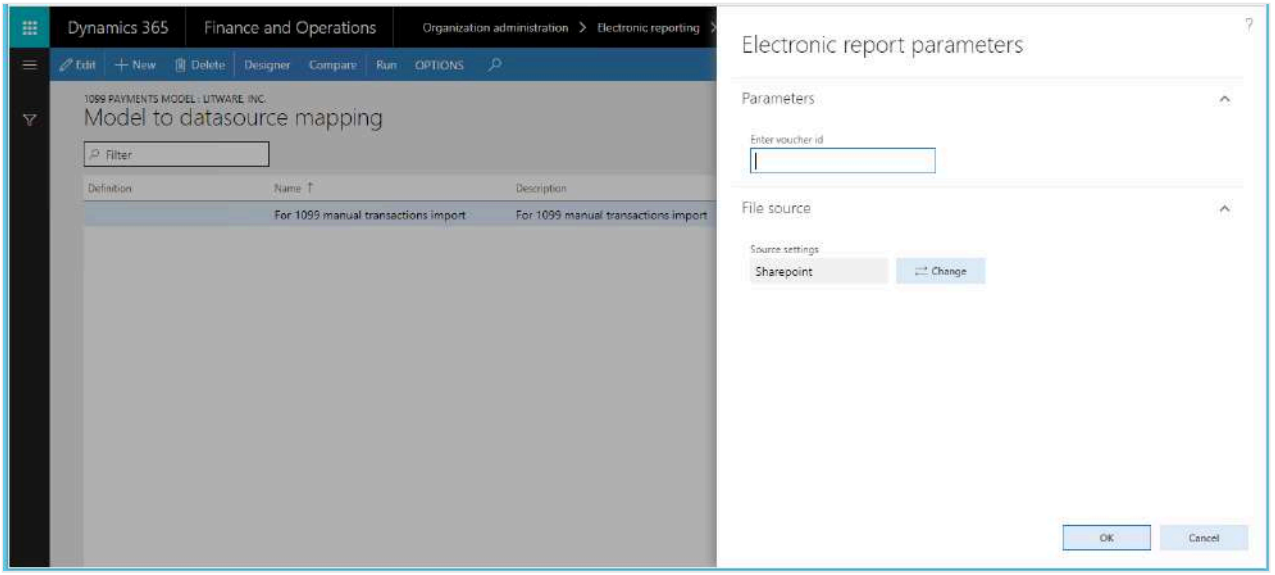
1. Open the ER configurations tree, select the **1099 Payment model**, and expand the list of ER model components.

2. Select the name of the model mapping to open the list of model mappings for the selected ER model configuration.



3. Select **Run** to run the selected model mapping. Because you configured file sources for the ER format, you can change the setting of the **File source** option, if needed. If you keep the setting of this option, the .xslx files are imported from the configured sources (the SharePoint folders, in this example).

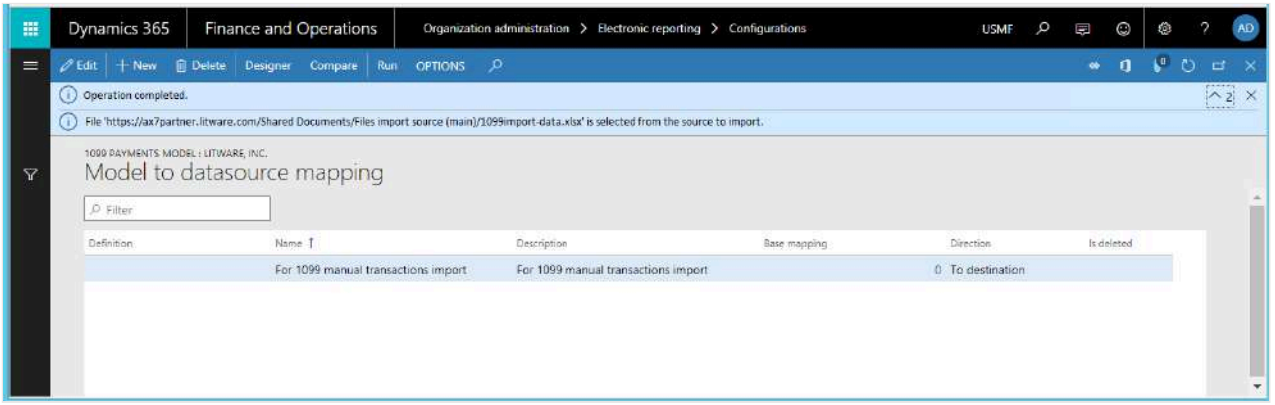
In this example, you're importing only one file. However, if there are multiple files, the files are selected for importing in the order in which you added them to the SharePoint folder. Every run of an ER format imports a single selected file.



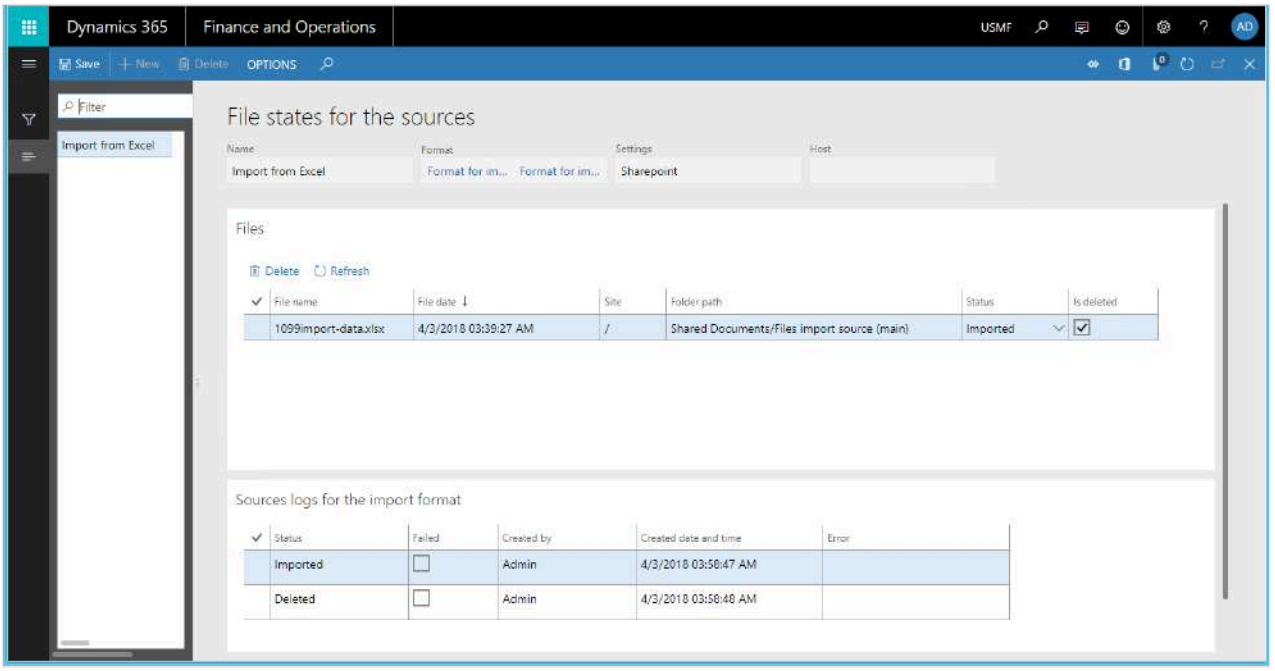
4. The model mapping can run unattended in batch mode. In this case, every time a batch runs this ER format, a single file is imported from the configured file sources.

   When a file is successfully imported from the SharePoint folder, it's deleted from that folder and moved to folder for successful imported files or to the folder to imported files with warnings. Otherwise, it's moved to folder for failed files or stays in this folder if the folder for failed files isn't set up.

5. Enter the voucher ID, such as **V-00001**, and then select **OK**.



6. On the **File states for the sources** page, select **Refresh** to refresh the page.



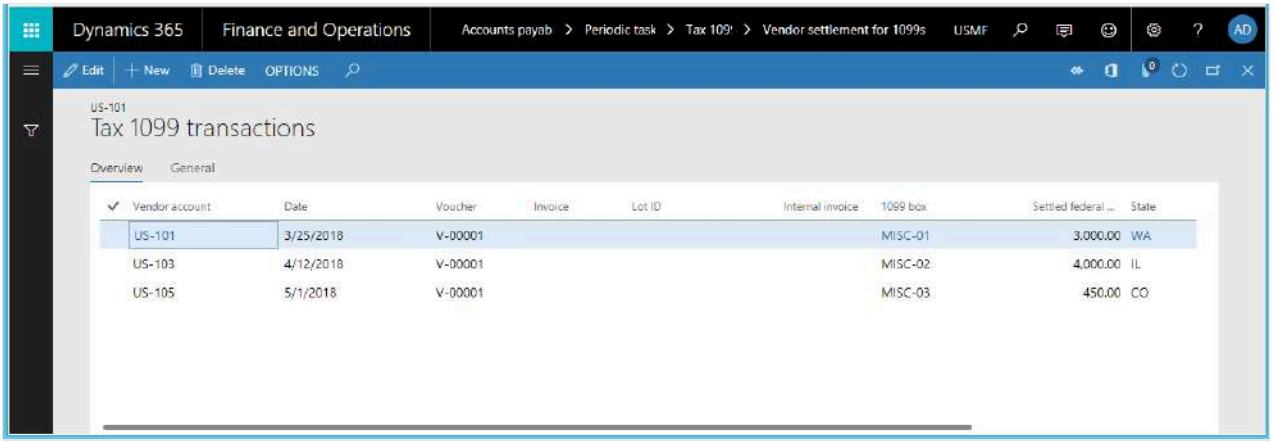
7. In the **Files** section, review the list of files. The **Sources log for the import format** section provides the history of the Excel file import. Because this file was successfully imported, it's marked as **Deleted** in the SharePoint folder.

8. Review the **Files import source (main)** SharePoint folder. The Excel files that were successfully imported are deleted from this folder.

9. Select **Accounts payable** > **Periodic tasks** > **Tax 1099** > **Vendor settlement for 1099s**.

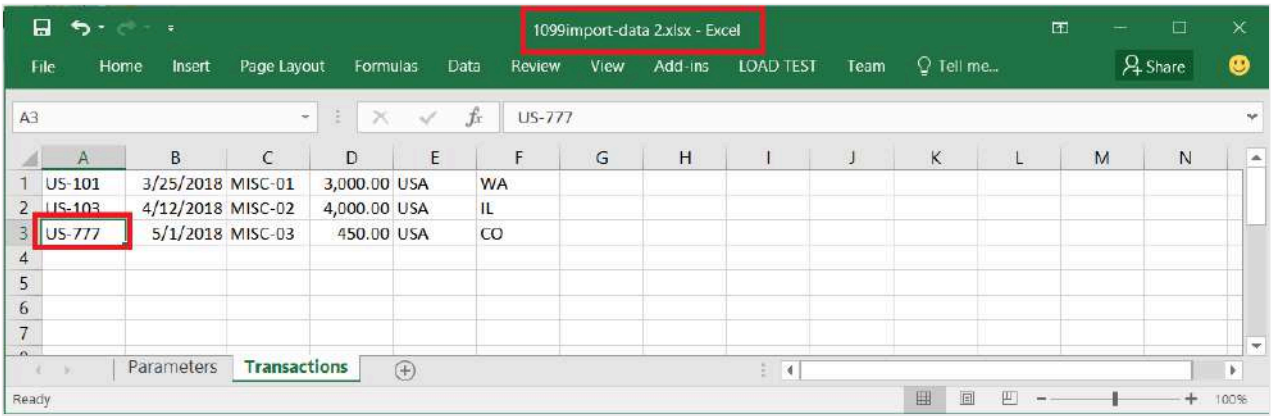
10. In the **From date** and **To date** fields, enter appropriate values. Then select **Manual 1099 transactions**.

    The vendor transactions imported from the Excel files on SharePoint for voucher **V-00001**, are presented on the page.



Prepare an Excel file for import

1. Open the Excel file that you previously used. In row 3, column 1, add a vendor code that doesn't exist in the application. Add additional false vendor information to the row.



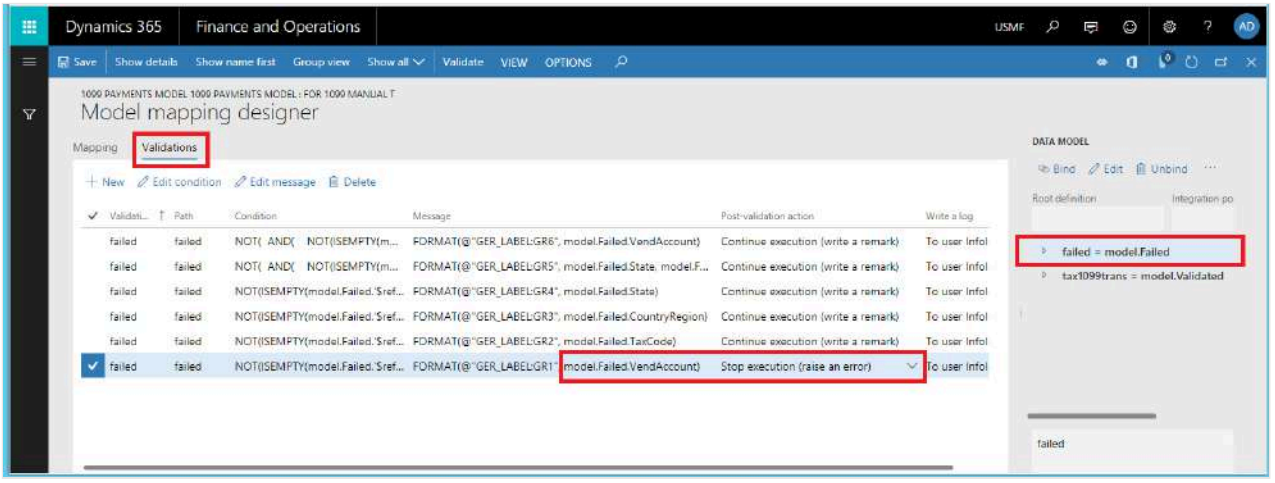
2. Upload the updated Excel file that contains vendor transactions to the **Files import source (main)** SharePoint folder.

3. Open the ER configurations tree, select the **1099 Payment model**, and expand the list of ER model components.

4. Select the name of the model mapping to update the model mapping so that the incorrect vendor code is considered an error during the data import process.

5. Select **Designer**.

6. On the **Validations** tab, change the post-validation action for the validation rule that checks whether the vendor account exists in the application. Update the value of the **Post-validation action** field to **Stop execution**, save your changes, and close the page.

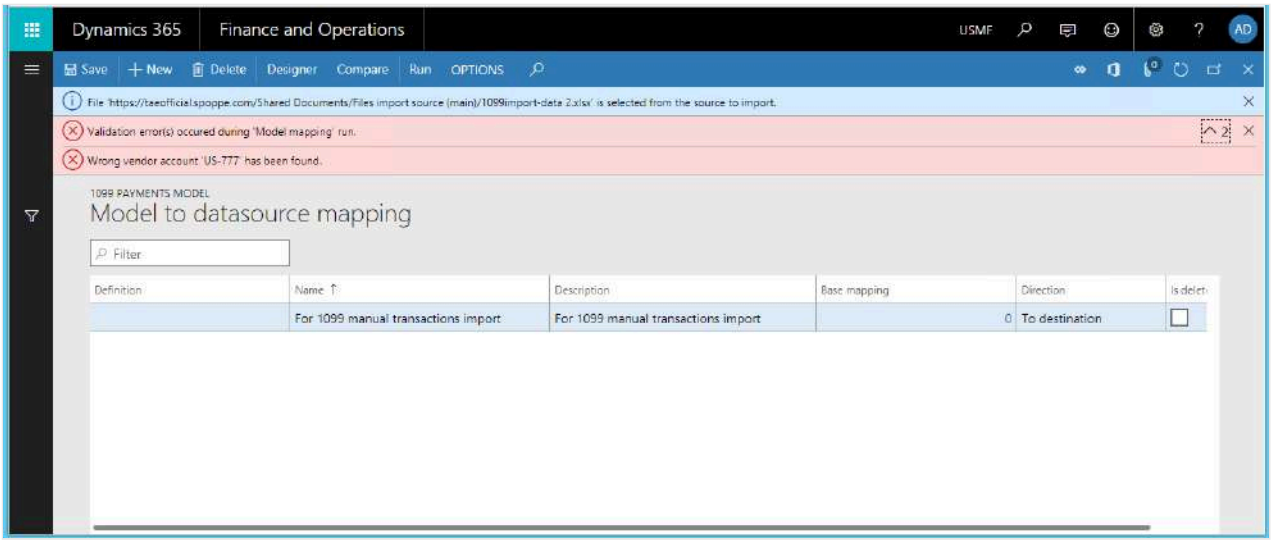


7. Save your changes, and close the ER model mapping designer.

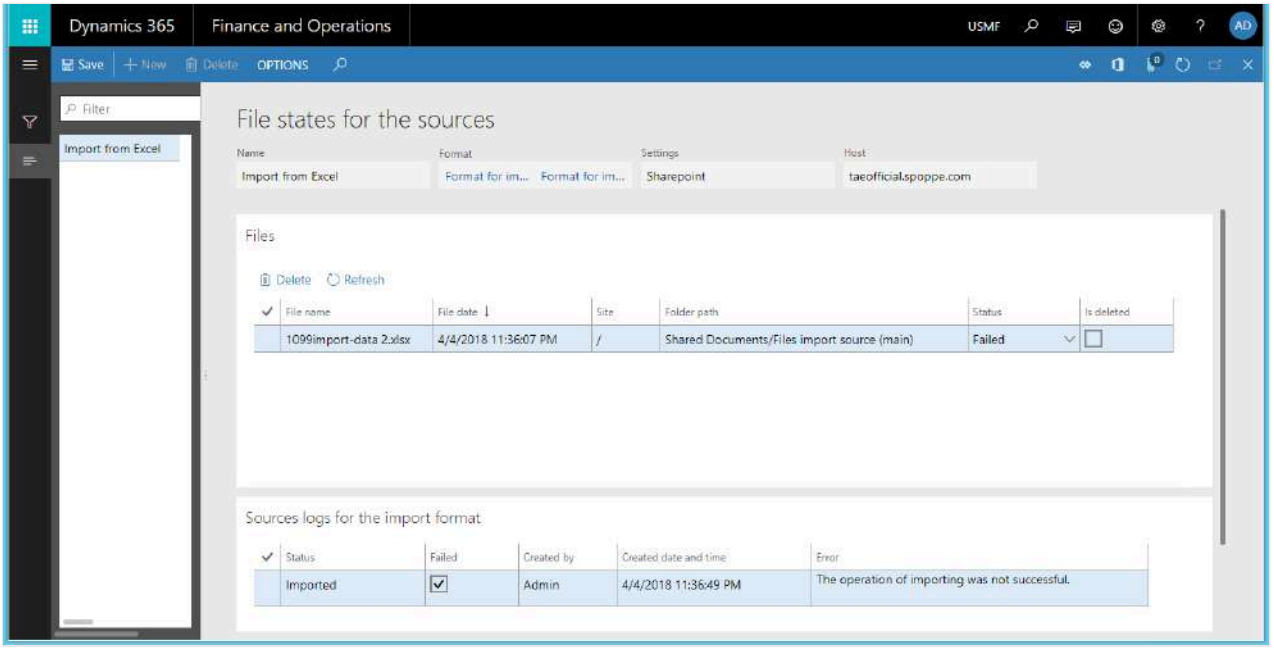
8. Select **Run** to run the modified ER model mapping.

9. Enter the voucher ID, such as **V-00002**, and then select **OK**.

The Infolog contains a notification that there's a file in the SharePoint folder that contains incorrect vendor account and can't be imported.



10. On the **File states for the sources** page, select **Refresh**, and then, in the **Files** section, review the list of files.



The **Sources log for the import format** section indicates that the import process failed and that the file is in the **Files error** SharePoint folder (the **Is deleted** check box isn't selected). If you fix this file on SharePoint by adding the proper vendor code and then move it to the **Files import source (main)** SharePoint folder, you can import the file again.

11. Select **Accounts payable** > **Periodic tasks** > **Tax 1099** > **Vendor settlement for 1099s**, enter appropriate values in the **From date** and **To date** fields, and then select **Manual 1099 transactions**.

Only transactions for voucher V-00001 are available. No transactions for voucher V-00002 are available even though the error for the last imported transaction is found in the Excel file.

Limitations

In versions of Dynamics 365 Finance before version 10.0.25, the user interface (UI) of the ER framework doesn't offer the capability to initiate a new batch job that runs a model mapping for data import in unattended mode. Instead, you must develop new logic so that the configured ER model mapping can be called from the application UI to import data from inbound files. To develop this logic, some engineering work is required.

For more information about the relevant ER API, see the Code to run a format mapping for data import section in ER framework API changes for Application update 7.3. Review the code in the `BankImport_RU` class of the `Application Suite` model to see how you can implement your custom logic. The `BankImport_RU` class extends the `RunBaseBatch` class. In particular, review the `runER()` method, where the `ERIModelMappingDestinationRun` object is created as the runner of an ER model mapping.

In Finance version 10.0.25 and later, the ER framework UI offers the capability to initiate a new batch job that runs a model mapping for data import in unattended mode. For more information about this process, see Import data in batch mode from manually selected files.

Additional resources

Electronic Reporting overview.

ER framework API changes for Application update 7.3.

ER framework API changes for Application update 10.0.23.

ER framework API changes for Application update 10.0.25.

Last updated on 04/13/2026

Import files in XML format with optional attributes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can design Electronic reporting (ER) formats to parse incoming electronic documents in XML format. You can specify certain attributes of XML elements as optional in the designed ER format. This option allows you to handle incoming files with and without these XML attributes properly. You can then use the content from these files to update application data.

To learn more about this feature, complete the steps in the article, (RCS) Import files in XML format with optional attributes, which is part of the 7.5.4.3 Acquire/Develop IT service/solution components (10677) business process. You can download this task guide and associated sample files from the Microsoft Download Center.

Expand table

Content description	File
Sample file in XML format	IncomingDocumentToLearnHowToHandleOptionalAttributes.xml
Task guide	RCS Import files in XML format with optional attributes.axtr

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can design ER format configuration to import files in XML format containing optional attributes. To complete these steps, you must first complete the steps in the procedure, Create configuration providers and mark them as active. Before you begin, download and save locally the IncomingDocumentToLearnHowToHandleOptionalAttributes.xml file from Microsoft Download Center (https://go.microsoft.com/fwlink/?linkid=874684).

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

2. Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the article, Create configuration providers and mark them as active.
3. Click **Reporting configurations**.

Create a new data model configuration

1. Select **Create configuration** to open the form.
2. In the **Name** field, enter *Model to import xml file*.
3. Select **Create configuration**.
4. Select **Designer**.
5. Select **New** to open the form.
6. In the **Name** field, enter *Root*.
7. Select **Add**.
8. Select **New** to open the form.
9. In the **Name** field, enter *List*.
10. In the **Item type** field, select **Record list**.
11. Select **Add**.
12. Select **New** to open the form.
13. In the **Name** field, enter *Code*.
14. In the **Item type** field, select **String**.
15. Select **Add**.
16. Select **Save**.
17. Close the page.
18. Select **Change status**.
19. Select **Complete**.
20. Select **OK**.

Create a format for data import

1. Select **Create configuration** to open the form.
2. In the **New** field, enter *Format based on data model Model to import xml file*.
3. In the **Name** field, type *Format to import xml file*.
4. Select **Yes** in the **Supports data import** field.
5. Select **Create configuration**.

Design a format to parse incoming file in XML format

1. Select **Designer**.
2. Select **Add root** to open the form.
3. In the tree, select **XML\Element**.
4. In the **Name** field, type *root*.
5. Select **OK**.
6. Select **Add** to open the form.
7. In the tree, select **XML\Element**.
8. In the **Name** field, type *document*.
9. In the **Multiplicity** field, select **One many**.
10. Select **OK**.
11. In the tree, select **root\document**.
12. Select **Add** to open the form.
13. In the tree, select **XML\Attribute**.
14. In the **Name** field, type *id*.
15. Select **OK**.
16. Select **Save**.

Design a format mapping to save parsed information to data model

1. Select **Map format to model**.
2. Select **New**.
3. In **Definition**, enter or select a value.
4. In **Name**, type `Mapping`.
5. Select **Save**.
6. Select **Designer**.
7. In the tree, expand **format**.
8. In the tree, expand **format\root: XML Element(root)**.
9. In the tree, select **format\root: XML Element(root)\document: XML Element 1.. (document)**.
10. Select **Bind**.
11. In the tree, expand **format\root: XML Element(root)\document: XML Element 1.. (document)**.
12. In the tree, select *format\root: XML Element(root)\document: XML Element 1.. (document)\id*.
13. In the tree, expand **List = format.root.document**.
14. In the tree, select **List = format.root.document\Code**.

15. Select **Bind**.
16. Select **Save**.
17. Close the page.

Run format mapping

1. Select **Run**.
2. Select **Browse**, and select the file **IncomingDocumentToLearnHowToHandleOptionalAttributes.xml**.
3. Select **OK**.

> **Note**
>
> The selected file isn't imported as the format design assumes the existence of `id` attribute for the `document` element, but the imported file contains no such attribute.

Modify format structure to handle XML attribute as optional

1. Close the page.
2. In the tree, expand **root\document**.
3. In the tree, select **root\document\id**.
4. In the **Empty string for missing attribute** field, select **Yes**.
5. Select **Save**.

Run format mapping to test changes

1. Select **Map format to model**.
2. Select **Run**.
3. Select **Browse**, and select the file **IncomingDocumentToLearnHowToHandleOptionalAttributes.xml**.
4. Select **OK**.
5. Review the generated file. The same file is imported as the format design now considers the `id` attribute for the `document` element as optional.

Last updated on 04/13/2026

(RCS) Import files in XML format with optional attributes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can design ER format configuration to import files in XML format containing optional attributes. To complete these steps, you must first complete the steps in the "Create a configuration provider and mark it as active" procedure. Before you begin, download and save locally the IncomingDocumentToLearnHowToHandleOptionalAttributes.xml file from Microsoft Download Center.

1. Go to **All workspaces** > **Electronic reporting**.
2. Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the procedure Create configuration providers and mark them as active.
3. Click **Reporting configurations**.

Create a new data model configuration

1. Select **Create configuration** to open the form.
2. In the **Name** field, enter *Model to import xml file*.
3. Select **Create configuration**.
4. Select **Designer**.
5. Select **New** to open the form.
6. In the **Name** field, enter *Root*.
7. Select **Add**.
8. Select **New** to open the form.
9. In the **Name** field, enter *List*.
10. In the **Item type** field, select **Record list**.

11. Select **Add**.
12. Select **New** to open the form.
13. In the **Name** field, enter *Code*.
14. In the **Item type** field, select **String**.
15. Select **Add**.
16. Select **Save**.
17. Close the page.
18. Select **Change status**.
19. Select **Complete**.
20. Select **OK**.

Create a format for data import

1. Select **Create configuration** to open the form.
2. In the **New** field, enter *Format based on data model Model to import xml file*.
3. In the **Name** field, type *Format to import xml file*.
4. Select **Yes** in the **Supports data import** field.
5. Select **Create configuration**.

Design a format to parse incoming file in XML format

1. Select **Designer**.
2. Select **Add root** to open the form.
3. In the tree, select **XML\Element**.
4. In the **Name** field, type *root*.
5. Select **OK**.
6. Select **Add** to open the form.
7. In the tree, select **XML\Element**.
8. In the **Name** field, type *document*.
9. In the **Multiplicity** field, select **One many**.
10. Select **OK**.
11. In the tree, select **root\document**.
12. Select **Add** to open the form.
13. In the tree, select **XML\Attribute**.
14. In the **Name** field, type *ID*.
15. Select **OK**.
16. Select **Save**.

Design a format mapping to save parsed information to data model

1. Select **Map format to model**.
2. Select **New**.
3. In the **Definition** field, enter or select a value.
4. In the list, select the link in the selected row.
5. In the **Name** field, type `Mapping`.
6. Select **Save**.
7. Select **Designer**.
8. In the tree, expand **format**.
9. In the tree, expand **format\root: XML Element(root)**.
10. In the tree, select **format\root: XML Element(root)\document: XML Element 1.. (document)**.
11. Select **Bind**.
12. In the tree, expand **format\root: XML Element(root)\document: XML Element 1.. (document)**.
13. In the tree, select *format\root: XML Element(root)\document: XML Element 1.. (document)\id*.
14. In the tree, expand **List = format.root.document**.
15. In the tree, select **List = format.root.document\Code**.
16. Select **Bind**.
17. Select **Save**.
18. Close the page.

Run format mapping

1. Select **Run**.
2. Select **Browse** and select **IncomingDocumentToLearnHowToHandleOptionalAttributes.xml**.
3. Select **OK**.

> **Note**
>
> The selected file isn't imported as the format design assumes the existence of `id` attribute for the `document` element, but the imported file contains no such attribute.

Modify format structure to handle xml attribute as optional

1. Close the page.
2. In the tree, expand **root\document**.
3. In the tree, select **root\document\id**.
4. Select **Yes** in the **Empty string for missing attribute** field.
5. Select **Save**.

Run format mapping to test changes

1. Select **Map format to model**.
2. Select **Run**.
3. Click **Browse** and select the **IncomingDocumentToLearnHowToHandleOptionalAttributes.xml** file.
4. Select **OK**.
5. Review the generated file.

> **Note**
>
> The same file is imported as the format design now considers the 'id' attribute for the 'document' element as optional.

Last updated on 04/13/2026

Manage the Electronic reporting (ER) configuration lifecycle

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to manage the lifecycle of Electronic reporting (ER) configurations for Dynamics 365 Finance.

Overview

Electronic reporting (ER) is an engine that supports statutory required and country/region-specific electronic documents. In general, ER assumes an ability to perform the following tasks for a single electronic document. For more details, see Electronic reporting (ER) overview.

- Design a template for an electronic document:
  - Identify the required sources of data that you can present in the document:
    - Underlying data, such as data tables, data entities, and classes.
    - Process-specific properties, such as execution date and time, and time zone.
    - User input parameters, specified by the end user at run time.
  - Define the required document elements and their topology to specify a final document format.
  - Configure the desired flow of data from selected data sources to defined document elements (via data source bindings to document format components), and specify process control logic.
- Make a template available so that you can use it in other instances:
  - Transform a document template that you created into an ER configuration, and export the configuration from the current application instance as an XML package that you can store either locally or in Microsoft Dynamics Lifecycle Services.

    - Transform an ER configuration into an application document template.
    - Import an XML package that you store either locally or in Lifecycle Services into the current instance.

- Customize the template of an electronic document:
    - Bring a template from Lifecycle Services into the current instance as an ER configuration.
    - Design a custom version of an ER configuration, and keep a reference to the base version.

- Integrate a template with a particular business process, so that it's available in the application:
    - Configure settings so that the application starts to use an ER configuration, by referring to that configuration in a process-related parameter. For example, refer to the ER configuration in a specific Accounts payable payment method to generate an electronic payment message for processing invoices.

- Use a template in a specific business process:
    - Run an ER configuration in a specific business process. For example, to generate an electronic payment message for processing invoices when a payment method that references the ER configuration is selected.

Concepts

The following roles and related activities are associated with the ER configuration lifecycle.

Expand table

Role	Activities	Description
Electronic reporting functional consultant	Create and manage ER components (models and formats).	A business person who designs ER domain–specific data models, designs the required templates for electronic documents, and binds them accordingly.
Electronic reporting developer	Create and manage data model mappings.	A specialist who selects the required Finance data sources and binds them to ER domain–specific data models.
Accounting supervisor	Configure process-related settings that reference ER artifacts.	For example, an **Accounting supervisor** role that allows the settings of an ER configuration to be used in a particular Accounts payable payment method to generate an electronic payment message for processing invoices.

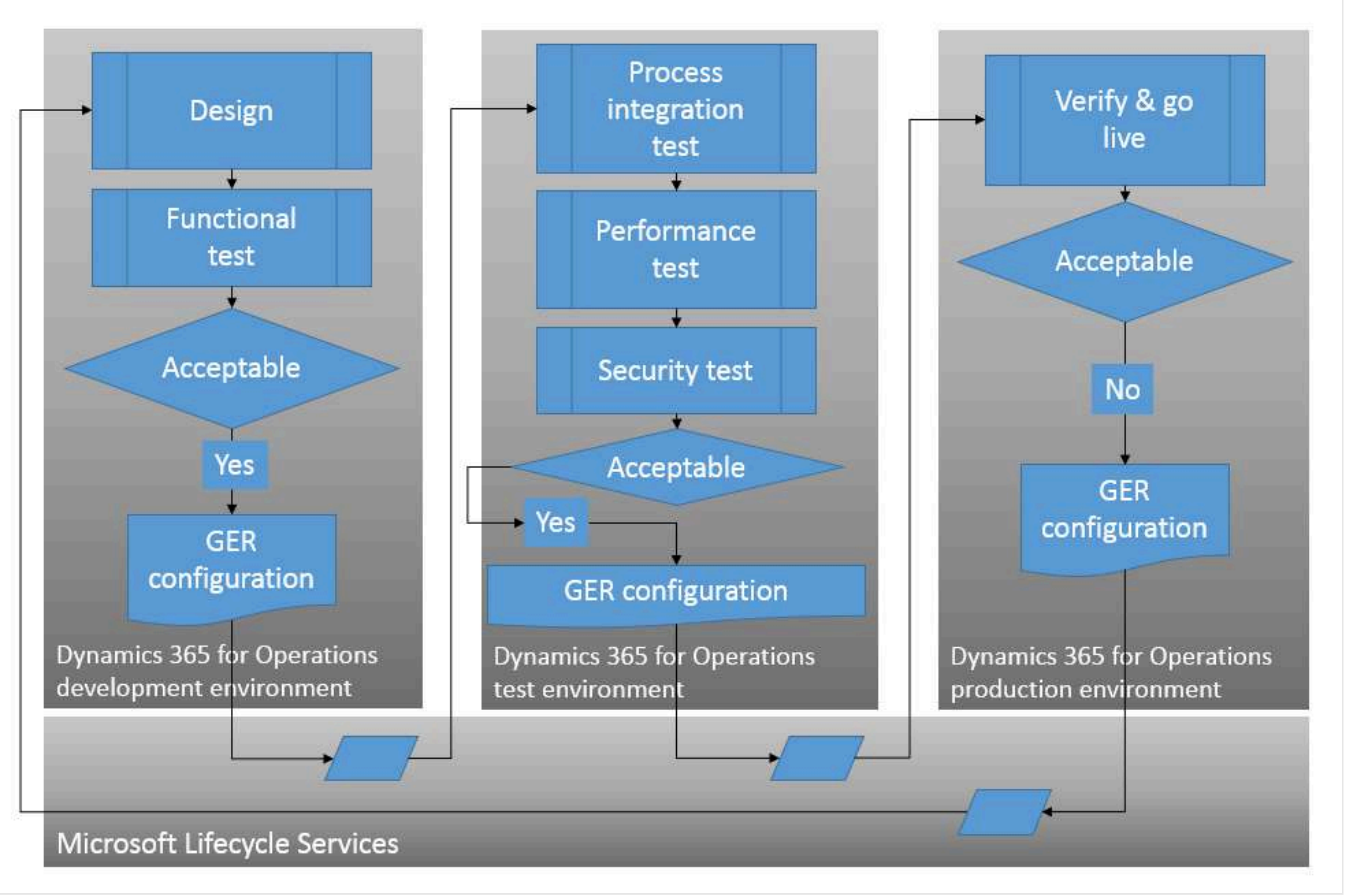
Role	Activities	Description
Accounts payable payments clerk	Use ER artifacts in a specific business process.	For example, an **Accounts payable payments clerk** role that allows electronic payment messages to be generated for processing invoices, based on the ER format that is configured for a specific payment method.

ER configuration development lifecycle

For the following ER-related reasons, design ER configurations in the development environment, as a separate instance of finance and operations:

- Users in either the **Electronic reporting developer** role or the **Electronic reporting functional consultant** role can edit configurations and run them for testing purposes. This scenario can cause calls to methods of classes and tables that might harm business data and the performance of the instance.
- Calls to methods of classes and tables as ER data sources of ER configurations aren't restricted by entry points and logged company content. Therefore, users in either the **Electronic reporting developer** role or the **Electronic reporting functional consultant** role can access business-sensitive data.

You can upload ER configurations that you design in the development environment to the test environment for configuration evaluation (proper process integration, correctness of results, and performance) and quality assurance, such as correctness of role-driven access rights and segregation of duties. Use the features that enable ER configuration interchange for this purpose. Proven ER configurations can be uploaded to Lifecycle Services to share them with service subscribers, or they can be imported to the production environment for internal use.



Data persistence consideration

You can individually import different versions of an ER configuration to your Finance instance. When you import a new version of an ER configuration, the system controls the content of the draft version of this configuration:

- If the imported version is lower than the highest version of this configuration in the current Finance instance, the content of the draft version of this configuration remains unchanged.
- If the imported version is higher than any other version of this configuration in the current Finance instance, the content of the imported version is copied to the draft version of this configuration so you can continue editing the last completed version.

If the configuration belongs to the configuration provider that you currently activate, you can view the draft version of this configuration on the **Versions** FastTab of the **Configurations** page (**Organization administration** > **Electronic reporting** > **Configurations**). You can select the draft version of the configuration and modify its content by using the relevant ER designer. When you edit the draft version of an ER configuration, its content no longer matches the content of the highest version of this configuration in the current Finance instance. To prevent the loss of your changes, the system displays an error that the import can't continue because the version of this configuration is higher than the highest version of this configuration in the

current Finance instance. When this error occurs, for example with the format configuration **X**, the **Format 'X' version is not completed** error is displayed.

To undo the changes that you introduced in the draft version, select the highest completed or shared version of your ER configuration in Finance on the **Versions** FastTab, and then select the **Get this version** option. The content of the selected version is copied to the draft version.

Applicability consideration

When you design a new version of an ER configuration, you can define its dependency on other software components. This step is a prerequisite for controlling the download of this configuration's version from an ER repository or an external XML file, and for any further use of the version. When you try to import a new version of an ER configuration, the system uses the configured prerequisites to control whether the version can be imported.

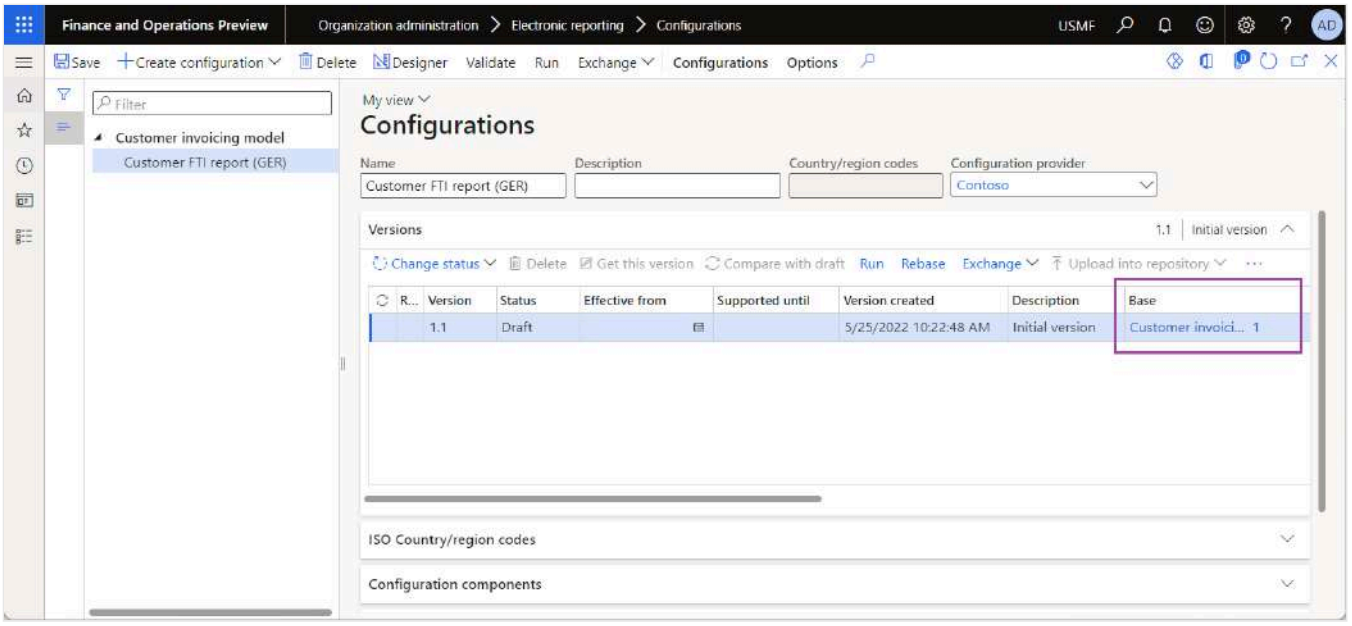
In some cases, you might require that the system ignore the configured prerequisites when you import new versions of ER configurations. To have the system ignore the prerequisites during import, follow these steps:

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. Set the **Skip product updates and version prerequisite check during import** option to **Yes**.

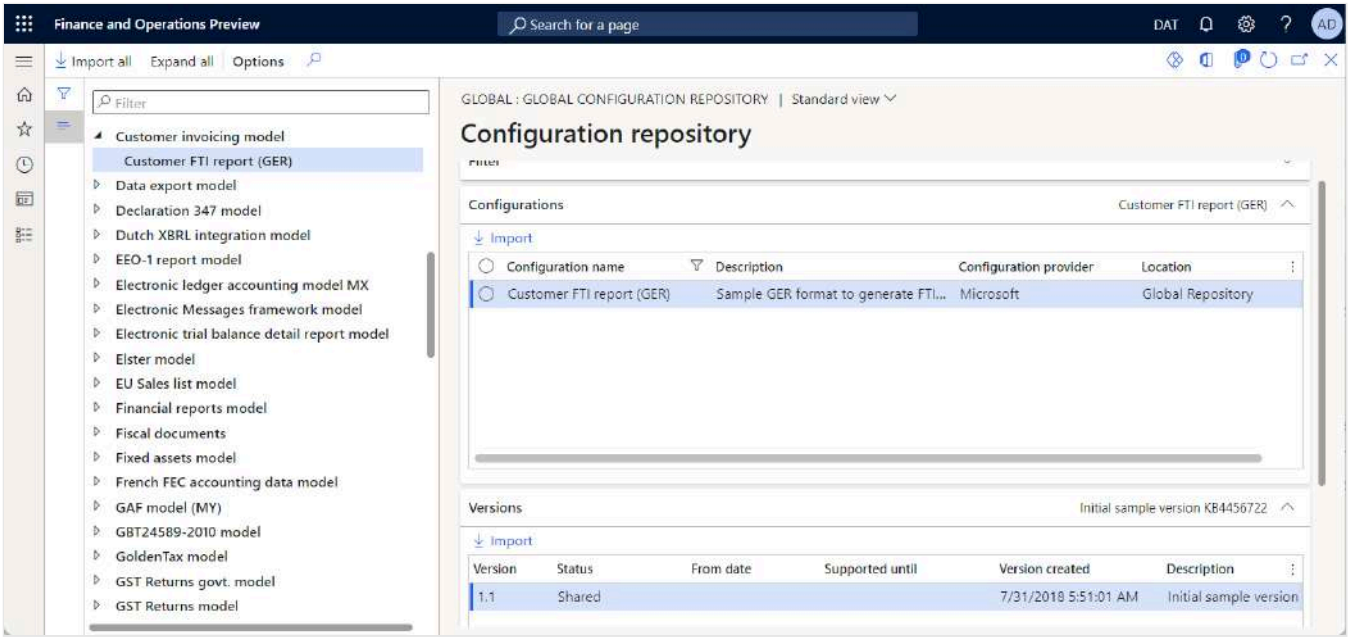
> **Note**
>
> This parameter is user-specific and company-specific.

Dependencies on other components

You can configure ER configurations as dependent on other configurations. For example, you can import an ER data model configuration from the Global repository into your Microsoft Regulatory Configuration Services (RCS) or Dynamics 365 Finance instance. Then, you can create a new ER format configuration that is derived from the imported ER data model configuration. The derived ER format configuration depends on the base ER data model configuration.

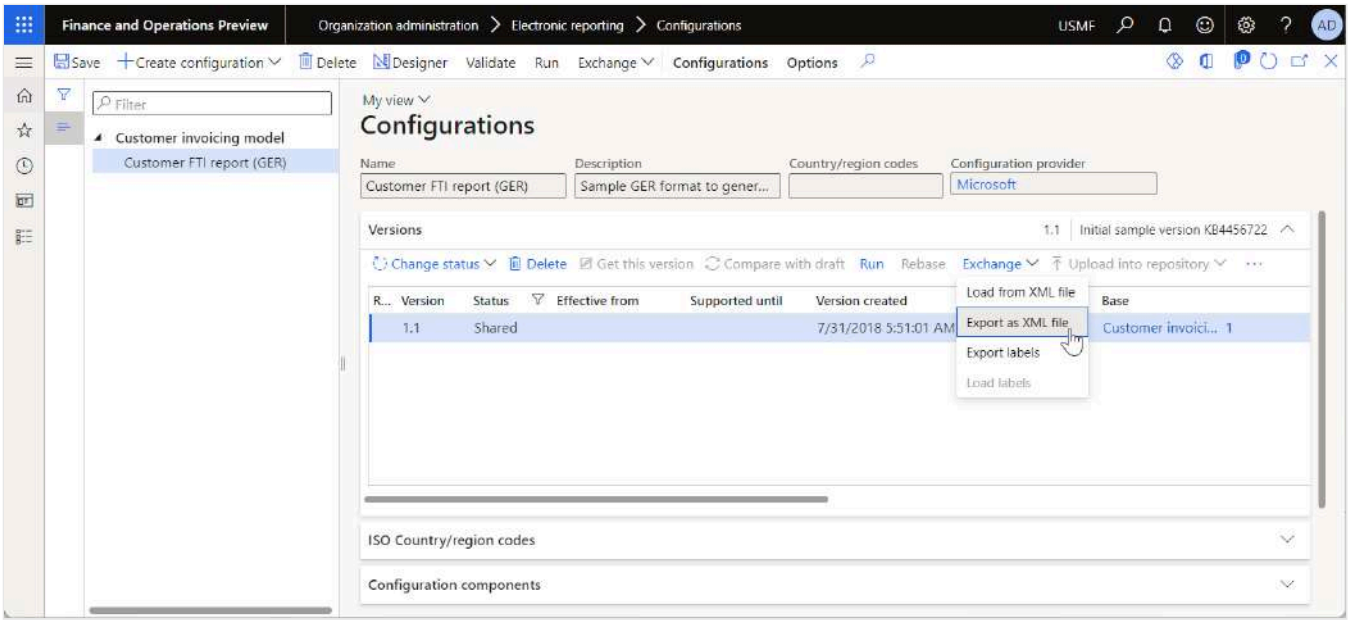


When you finish designing the format, change the status of your initial version of the ER format configuration from **Draft** to **Completed**. Then, share the completed version of the ER format configuration by publishing it to the Global repository. Next, you can access the Global repository from any RCS or Finance cloud instance. You can then import any ER configuration version that is applicable to the application from the Global repository into that application.



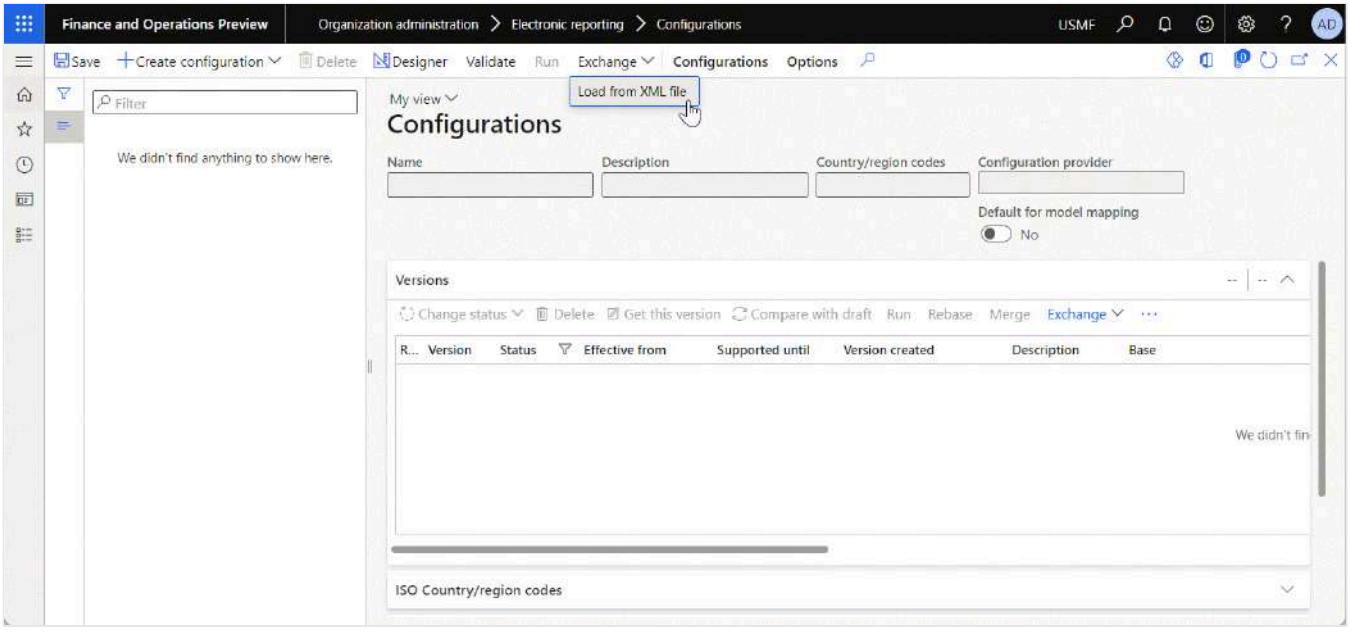
Based on the configuration dependency, when you select the ER format configuration in the Global repository to import it into a newly deployed RCS or Finance instance, the base ER data model configuration is automatically found in the Global repository and imported together with the selected ER format configuration as the base configuration.

You can also export your ER format configuration version out of your current RCS or Finance instance, and store it locally as an XML file.



In versions of Finance **before version 10.0.29**, when you try to import the ER format configuration version from that XML file or from any repository other than the Global repository into a newly deployed RCS or Finance instance that doesn't yet contain any ER configurations, the following exception is thrown to inform you that a base configuration can't be obtained:

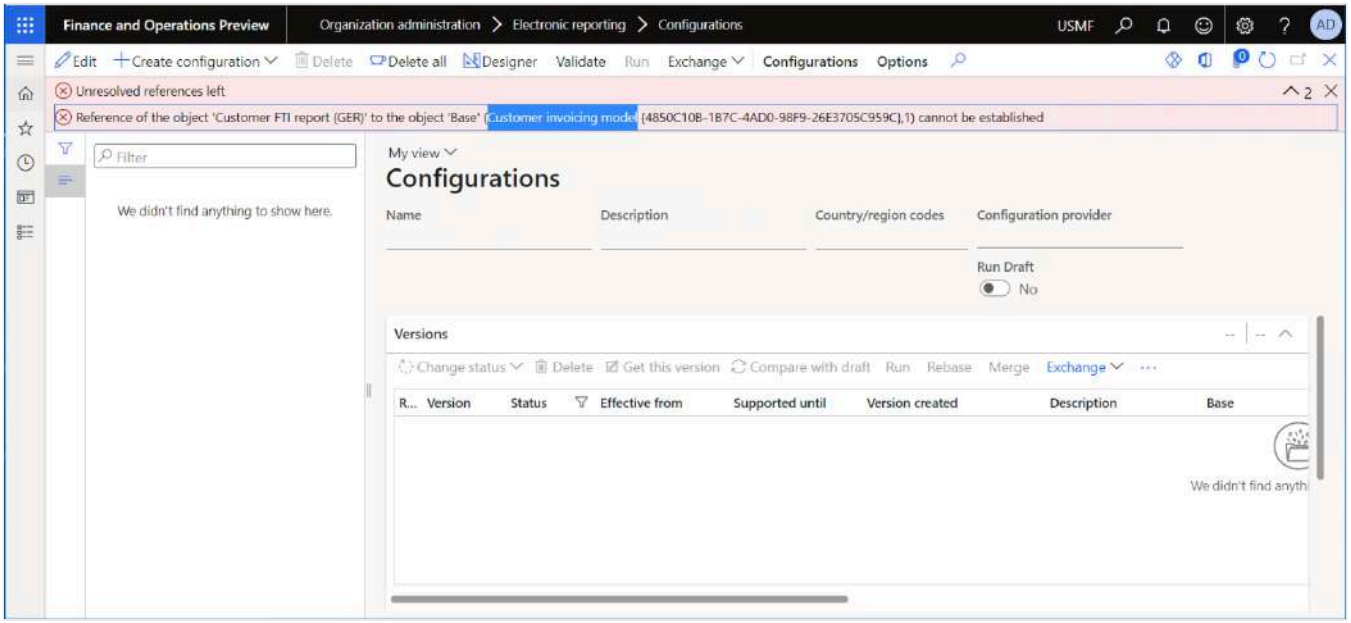
> Unresolved references left
> Reference of the object '<imported configuration name>' to the object 'Base' (<globally unique identifier of the missed base configuration>,<version of the missed base configuration>) can't be established



In version **10.0.29 and later**, when you try to do the same configuration import, if a base configuration can't be found in the current application instance or in the source repository that

you're currently using (if applicable), the ER framework automatically tries to find the name of the missing base configuration in the Global repository cache. It then presents the name and globally unique identifier (GUID) of the missing base configuration in the text of the exception that is thrown.

> Unresolved references left
> Reference of the object '<imported configuration name>' to the object 'Base' (<name of the missed base configuration> <globally unique identifier of the missed base configuration>,<version of the missed base configuration>) can't be established



You can use the provided name to find the base configuration and then manually import it.

> **Note**
>
> This new option works only when at least one user already signs in to the Global repository by using the **Configuration repositories** page or one of the Global repository **lookup** fields in the current Finance instance, and when the Global repository content is cached.

Additional resources

Electronic reporting (ER) overview

Define the dependency of ER configurations on other components

Last updated on 04/13/2026

Upload a configuration into Lifecycle Services

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user in the System administrator or Electronic reporting developer role can create a new Electronic reporting (ER) configuration and upload it into the project-level Asset library in Microsoft Dynamics Lifecycle Services.

> **Important**
>
> The use of Lifecycle Services as a storage repository for ER configurations is being deprecated. For more information, see Regulatory Configuration Service (RCS) – Lifecycle Services storage deprecation.

In this example, you create a configuration and upload it into Lifecycle Services for a sample company named Litware, Inc. You can complete these steps in any company, because ER configurations are shared among companies. To complete these steps, you must first complete the steps in Create configuration providers and mark them as active. You also need access to Lifecycle Services.

1. Sign in to the application by using one of the following roles:
    - Electronic reporting developer
    - System administrator
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. Select **Litware, Inc.**, and mark it as **Active**.
4. Select **Configurations**.

> **Note**
>
> Make sure that the current Dynamics 365 Finance user is a member of the Lifecycle Services project that contains the **Asset library** that is used to import ER configurations.
>
> You can't access an Lifecycle Services project from an ER repository that represents a different domain than the domain that is used in Finance. If you try, an empty list of Lifecycle Services projects is shown, and you can't import ER configurations from the project-level Asset library in Lifecycle Services. To access project-level Asset libraries from an ER repository that is used to import ER configurations, sign in to Finance by using the credentials of a user who belongs to the tenant (domain) that the current Finance instance is provisioned for.

Create a new data model configuration

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, select **Create configuration** to open the drop-down dialog box.

   In this example, you create a configuration that contains a sample data model for electronic documents. You upload this data model configuration into Lifecycle Services later.
3. In the **Name** field, enter **Sample model configuration**.
4. In the **Description** field, enter **Sample model configuration**.
5. Select **Create configuration**.
6. Select **Model designer**.
7. Select **New**.
8. In the **Name** field, enter **Entry point**.
9. Select **Add**.
10. Select **Save**.
11. Close the page.
12. Select **Change status**.

13. Select **Complete**.
14. Select **OK**.
15. Close the page.

Register a new repository

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. In the **Configuration providers** section, select the **Litware, Inc.** tile.
3. On the **Litware, Inc.** tile, select **Repositories**.

   You can now open the list of repositories for the Litware, Inc. configuration provider.
4. Select **Add** to open the drop-down dialog box.

   You can now add a new repository.
5. In the **Configuration repository enter** field, select **LCS**.
6. Select **Create repository**.
7. In the **Project** field, enter or select a value.

   For this example, select the desired Lifecycle Services project. You must have access to the project.
8. Select **OK**.

   Complete a new repository entry.
9. In the list, mark the selected row.

   For this example, select the **LCS** repository record.

   Note that the current provider marks a registered repository. In other words, only configurations that the provider owns can go in this repository and be uploaded into the selected Lifecycle Services project.
10. Select **Open**.

    You open the repository to view the list of ER configurations. If the selected project isn't yet used for ER configurations sharing, the list is empty.

11. Close the page.
12. Close the page.

Upload a configuration into Lifecycle Services

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configurations tree, select **Sample model configuration**.

   Select a configuration that you created and completed.
3. In the list, find and select the desired record.

   For this example, select the version of the selected configuration that has a status of **Completed**.
4. Select **Change status**.
5. Select **Share**.

   When you publish the configuration in Lifecycle Services, the status changes from **Completed** to **Shared**.
6. Select **OK**.
7. In the list, find and select the desired record.

   For this example, select the configuration version that has a status of **Shared**.

   The status of the selected version changes from **Completed** to **Shared**.
8. Close the page.
9. Select **Repositories**.

   You can now open the list of repositories for the Litware, Inc. configuration provider.
10. Select **Open**.

    For this example, select the **LCS** repository, and open it.

    The selected configuration appears as an asset of the selected Lifecycle Services project.
11. Open Lifecycle Services by going to https://lcs.dynamics.com.

12. Open a project that you used earlier for repository registration.
13. Open the Asset library of the project.
14. Select the **GER configuration** asset type.

    The ER configuration that you uploaded appears in the list.

    If providers have access to this Lifecycle Services project, they can import the uploaded Lifecycle Services configuration into another instance.

Last updated on 04/13/2026

Import a configuration from Lifecycle Services

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user in the System administrator or Electronic reporting developer role can import a new version of an Electronic reporting (ER) configuration from the project-level Asset library in Microsoft Dynamics Lifecycle Services.

> **Important**
>
> The use of Lifecycle Services as a storage repository for ER configurations is being deprecated. For more information, see Regulatory Configuration Service (RCS) – Lifecycle Services storage deprecation.

In this example, you select the desired version of the ER configuration and import it for a sample company named Litware, Inc. You can complete these steps in any company, because ER configurations are shared among companies. To complete these steps, you must first complete the steps in Upload a configuration into Lifecycle Services. You also need access to Lifecycle Services.

1. Sign in to the application by using one of the following roles:
    - Electronic reporting developer
    - System administrator
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. Select **Configurations**.

> **Note**

Make sure that the current Dynamics 365 Finance user is a member of the Lifecycle Services project that contains the Asset library that you want to access to import ER configurations.

You can't access an Lifecycle Services project from an ER repository that represents a different domain than the domain that is used in Finance. If you try, an empty list of Lifecycle Services projects is shown, and you can't import ER configurations from the project-level Asset library in Lifecycle Services. To access project-level Asset libraries from an ER repository that is used to import ER configurations, sign in to Finance by using the credentials of a user who belongs to the tenant (domain) that the current Finance instance is provisioned for.

Delete a shared version of a data model configuration

1. On **Configurations**, in the configurations tree, select **Sample model configuration**.

   You created the first version of a sample data model configuration and published it to Lifecycle Services when you completed the steps in Upload a configuration into Lifecycle Services. In this procedure, you delete that version of the ER configuration. You can import that version from Lifecycle Services later in this article.

2. In the list, find and select the desired record.

   For this example, select the version of the configuration that has a status of **Shared**. This status indicates that the configuration is published to Lifecycle Services.

3. Select **Change status**.

4. Select **Discontinue**.

   When you change the status of the selected version from **Shared** to **Discontinued**, you make the version available for deletion.

5. Select **OK**.

6. In the list, find and select the desired record.

   For this example, select the version of the configuration that has a status of **Discontinued**.

7. Select **Delete**.

8. Select **Yes**.

The only draft version 2 of the selected data model configuration is now available.

9. Close the page.

Import a shared version of a data model configuration from Lifecycle Services

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. In the **Configuration providers** section, select the **Litware, Inc.** tile.
3. On the **Litware, Inc.** tile, select **Repositories**.

   You can now open the list of repositories for the Litware, Inc. configuration provider.
4. Select **Open**.

   For this example, select the **Lifecycle Services** repository, and open it. You must have access to the Lifecycle Services project and to the Asset library that the selected ER repository accesses.
5. In the list, select the row.

   For this example, select the first version of **Sample model configuration** in the version list.
6. Select **Import**.
7. Select **Yes** to confirm the import of the selected version from Lifecycle Services.

   An informational message confirms that the selected version was successfully imported.
8. Close the page.
9. Close the page.
10. Select **Configurations**.
11. In the tree, select **Sample model configuration**.
12. In the list, find and select the desired record.

    For this example, select the version of the configuration that has a status of **Shared**.

Notice that shared version 1 of the selected data model configuration is also available now.

Last updated on 04/13/2026

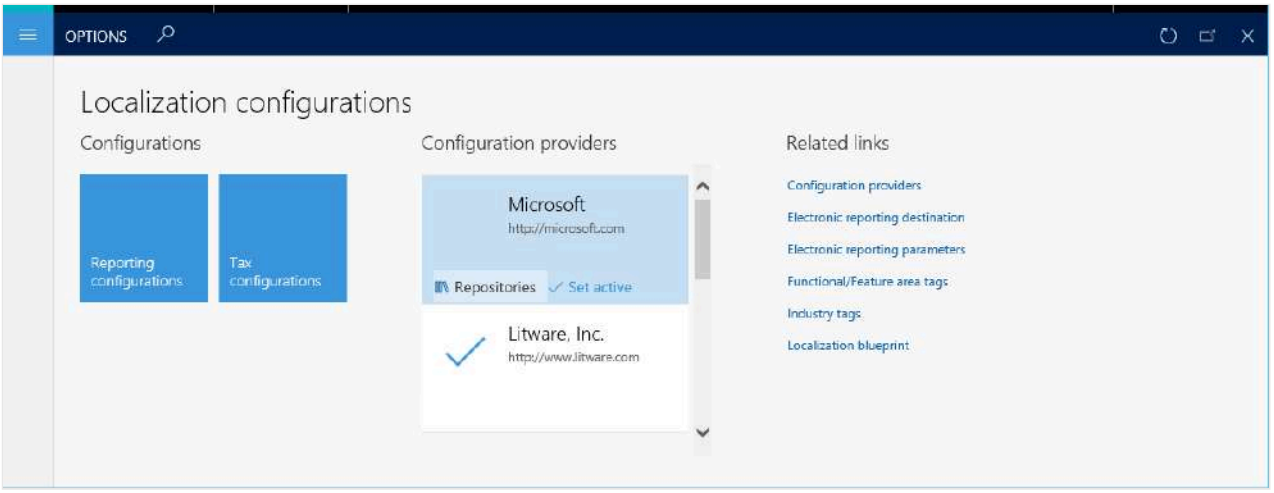
Download electronic reporting configurations from Lifecycle Services

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to download the newest version of Electronic reporting (ER) configurations from the Shared asset library in Microsoft Dynamics Lifecycle Services.

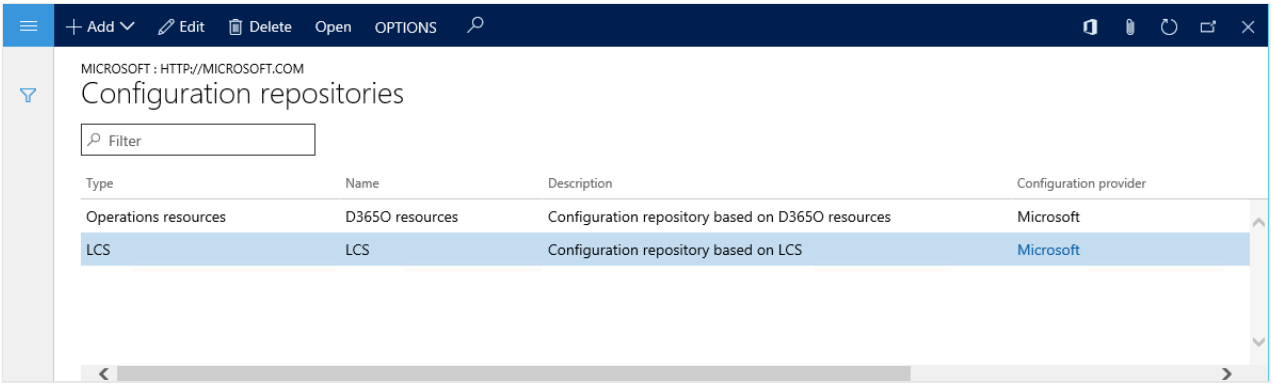
> **Important**
>
> The use of Lifecycle Services as a storage repository for ER configurations is being deprecated. For more information, see Regulatory Configuration Service (RCS) – Lifecycle Services storage deprecation.

1. Sign in to the application by using one of the following roles:
    - Electronic reporting developer
    - Electronic reporting functional consultant
    - System administrator
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Configuration providers** section, select the **Microsoft** tile.
4. On the **Microsoft** tile, select **Repositories**.



5. On the **Configuration repositories** page, in the grid, select the existing repository of the **LCS** type. If this repository doesn't appear in the grid, follow these steps:
   a. Select **Add** to add a repository.
   b. Select **LCS** as the repository type.
   c. Select **Create repository**.
   d. If you're prompted about authorization, follow the on-screen instructions.
   e. Enter a name and description for the repository.
   f. Select **OK** to confirm the new repository entry.
   g. In the grid, select the new repository of the **LCS** type.

6. Select **Open** to view the list of ER configurations for the selected repository.



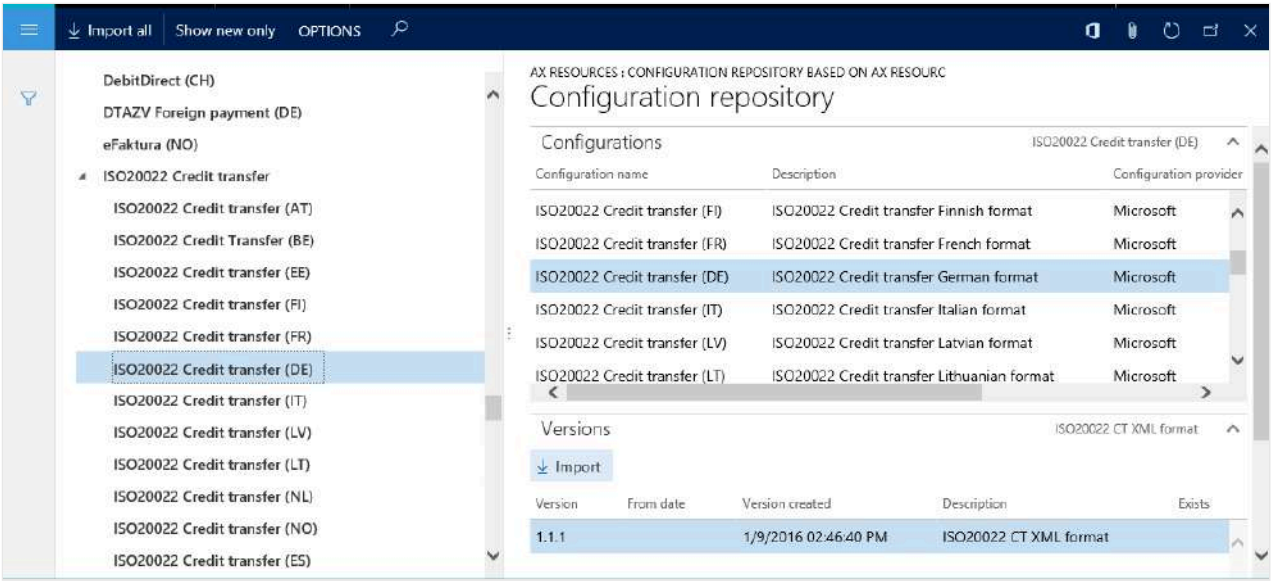
> **Tip**
>
> If you have trouble accessing the Lifecycle Services repository to download configurations from the Shared asset library in Lifecycle Services, you can download configurations from the Global repository instead.

7. In the configurations tree in the left pane, select the required ER configuration.

8. On the **Versions** FastTab, select the required version of the selected ER configuration.

9. Select **Import** to download the selected version from Lifecycle Services to the current instance.

> **Note**
>
> The **Import** button is unavailable for ER configuration versions that are already present in the current instance.



> **Note**
>
> Depending on the ER settings, configurations are validated after they are imported. You might be notified about any inconsistency issues that are discovered. You must resolve those issues before you can use the imported configuration version. For more information, see the list of related topics for this article.

Additional resources

- Electronic reporting (ER) overview
- Download ER configurations from the Global repository of Configuration service

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Last updated on 04/13/2026

Import Electronic reporting (ER) configurations

> ⓘ Note
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to download Electronic reporting (ER) configurations from Microsoft Dynamics Lifecycle Services to a local business data application or Tier-1 environment.

1. Sign in to your application by using one of the following roles:
    - Electronic reporting developer
    - Electronic reporting functional consultant
    - System administrator
2. Go to **Organization administration** > **Electronic reporting**.
3. In the **Configuration providers** section, select the card for the ER provider that is associated with your company.

    > ⓘ Note
    >
    > To learn how to register a new ER solution provider, see the **Create a configuration provider and mark it as active** task guide.

4. On the selected tile, select **Repositories**.
5. On the **Configuration repositories** page, in the grid, select the existing repository of the **File system** type. If the repository doesn't appear in the grid, follow these steps:
    a. Select **Add** to add a new repository.
    b. Select **FILE SYSTEM** as the repository type.
    c. Select **Create repository**.
    d. Enter a name and description for the repository.

e. Enter the path of the working directory for this repository. This path should point to a folder of the local file system where the ER configurations that belong to the repository are stored.
f. Select **OK** to confirm and save the new repository.
g. In the grid, select the new repository of the **File system** type.

6. In your browser, open another tab, and sign in to Lifecycle Services.
7. In the Shared asset library, select the **GER Configuration** asset type, and then download file **GER Configurations - All**.

> ⓘ **Note**
>
> All the ER configurations are in a zip file for download.

8. Open the file, select all the ER configurations, and then copy them to the working directory for the repository of the **File system** type.
9. On the **ER repositories** page, on the **Dynamics 365 Finance** tab, select **Open** to view the list of ER configurations for the selected repository.
10. In the **Configurations** tree in the left pane, select an ER configuration.
11. On the **Versions** FastTab, select the required version of the ER configuration.
12. Select **Import** to download the selected version from this repository to the current instance.

> ⓘ **Note**
>
> The **Import** button is unavailable for existing ER configuration versions.

> ⓘ **Note**
>
> Depending on the ER settings, configurations are validated after they are imported. You might see a notification about inconsistencies or issues that are discovered. You must resolve these inconsistencies or issues before you can use the imported configuration version.

Last updated on 04/13/2026

Adjust an ER format to generate a custom electronic document

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The procedures in this article explain how a user in the System Administrator or Electronic Reporting Functional Consultant role can perform these tasks:

- Configure parameters for the Electronic reporting (ER) framework.
- Import ER configurations that Microsoft provides and that are used to generate a payment file while a vendor payment is being processed.
- Create a custom version of a standard ER format configuration that Microsoft provides.
- Modify the custom ER format configuration so that it generates payment files that meet the requirements of a specific bank.
- Adopt changes that are made to the standard ER format configuration in the custom ER format configuration.

You can perform all the following procedures in the **GBSI** company. No coding is required.

- Configure the ER framework
  - Configure ER parameters
  - Activate an ER configuration provider
    - Review the list of ER configuration providers
    - Add a new ER configuration provider
    - Activate an ER configuration provider
- Import the standard ER format configurations
  - Import the standard ER configurations
  - Review the imported ER configurations

- Prepare a vendor payment for processing
  - Add bank information for a vendor account
  - Enter a vendor payment
- Process a vendor payment by using the standard ER format
  - Set up the electronic payment method
  - Process a vendor payment
- Customize the standard ER format
  - Create a custom format
  - Edit a custom format
  - Mark a custom format as runnable
- Process a vendor payment by using the custom ER format
  - Set up the electronic payment method
  - Process a vendor payment
- Import new versions of the standard ER format configurations
  - Import new versions of the standard ER configurations
  - Review the imported ER format configurations
- Adopt the changes in the new version of an imported format in a custom format
  - Complete the current draft version of a custom format
  - Rebase a custom format to a new base version
  - Process a vendor payment by using a rebased ER format
- Additional resources

Configure the ER framework

As a user in the Electronic Reporting Functional Consultant role, you must configure the minimal set of ER parameters before you can start to use the ER framework to design a custom version of a standard ER format.

Configure ER parameters

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Electronic reporting parameters**.

3. On **Electronic reporting parameters**, on the **General** tab, set the **Enable design mode** option to **Yes**.

4. On the **Attachments** tab, set the following parameters:

    - In the **Configurations** field, select the **File** type for the **USMF** company.
    - In the **Job archive**, **Temporary**, **Baseline**, and **Others** fields, select the **File** type.

For more information about ER parameters, see Configure the ER framework.

Activate an ER configuration provider

Each ER configuration that you add is owned by an ER configuration provider. The ER configuration provider that you activate in the **Electronic reporting** workspace serves this role. Therefore, you must activate an ER configuration provider in the **Electronic reporting** workspace before you start to add or edit ER configurations.

> ⓘ **Note**
>
> Only the owner of an ER configuration can edit it. Therefore, before you can edit an ER configuration, you must activate the appropriate ER configuration provider in the **Electronic reporting** workspace.

Review the list of ER configuration providers

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Configuration providers**.
3. On **Configuration provider table**, each provider record has a unique name and URL. Review the contents of this page. If a record for **Litware, Inc.** (`https://www.litware.com`) already exists, skip the next procedure, Add a new ER configuration provider.

Add a new ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Configuration providers**.
3. On **Configuration providers**, select **New**.
4. In the **Name** field, enter **Litware, Inc.**
5. In the **Internet address** field, enter `https://www.litware.com`.

6. Select **Save**.

Activate an ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, select the **Litware, Inc.** tile, and then select **Set active**.

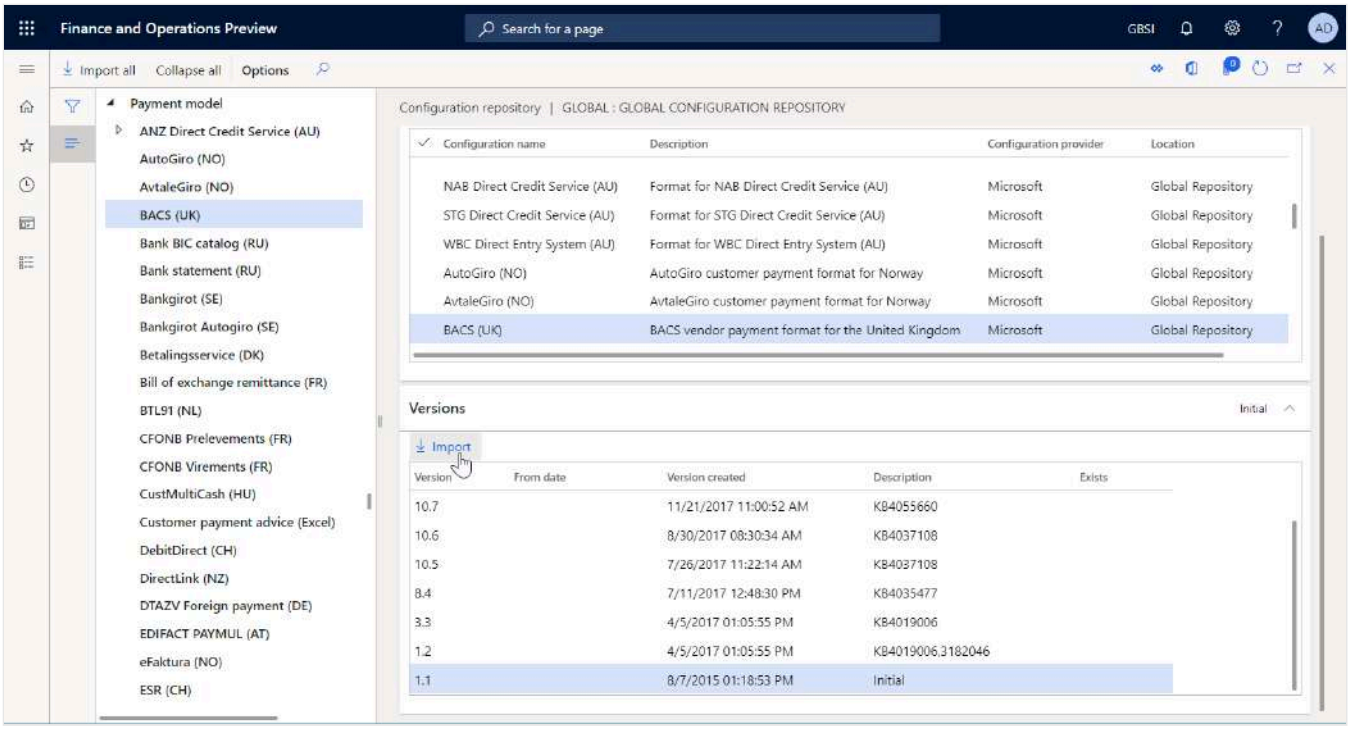
For more information about ER configuration providers, see Create configuration providers and mark them as active.

Import the standard ER format configurations

Import the standard ER configurations

To add the standard ER configurations to your current instance of Microsoft Dynamics 365 Finance, import them from the ER repository that you configured for that instance.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, select the **Microsoft** tile, and then select **Repositories** to view the list of repositories for the Microsoft provider.
3. On **Configuration repositories**, select the repository of the **Global** type, and then select **Open**. If you're prompted for authorization to connect to Regulatory Configuration Service, follow the authorization instructions.
4. On **Configuration repository**, in the configuration tree in the left pane, select the **BACS (UK)** format configuration.
5. On the **Versions** FastTab, select version **1.1** of the selected ER format configuration.
6. Select **Import** to download the selected version from the Global repository to the current Finance instance.



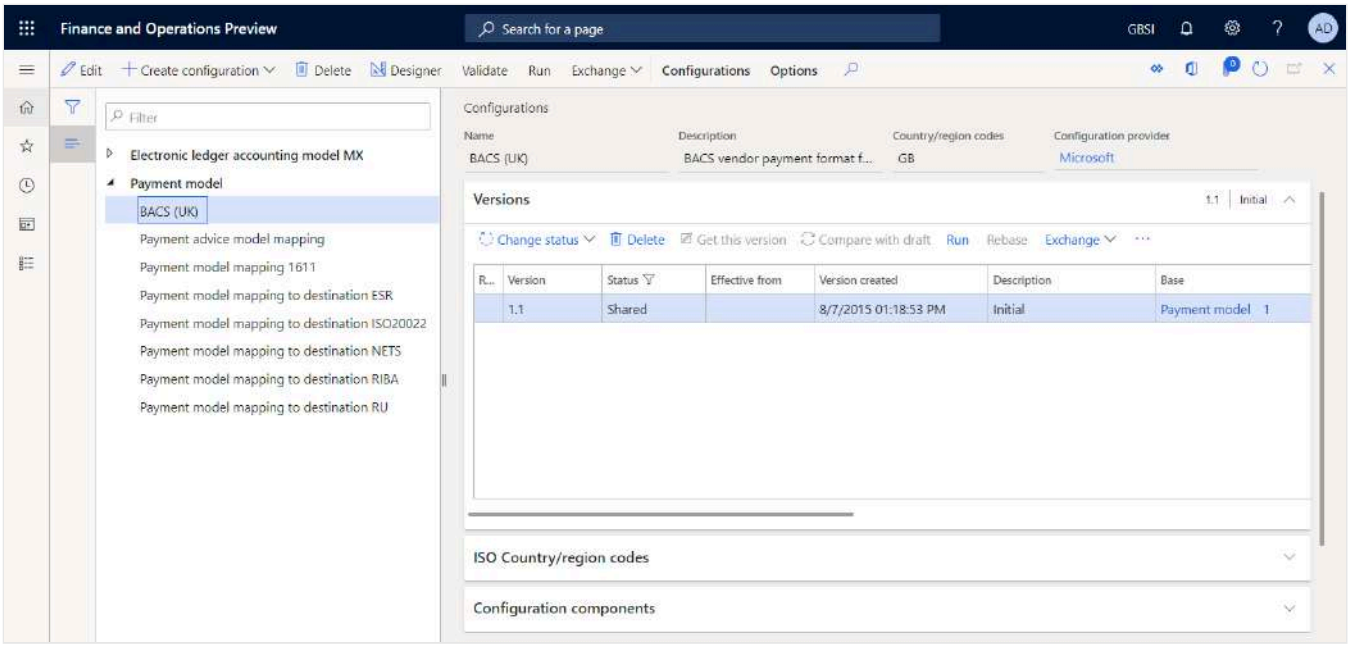
> **Tip**
>
> If you have trouble accessing the Global repository, you can download configurations from Microsoft Dynamics Lifecycle Services instead.

Review the imported ER configurations

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configurations** section, select the **Reporting configurations** tile.
3. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**.
4. In addition to the selected **BACS (UK)** ER format, the import process adds other required ER configurations. Make sure that the following ER configurations are available in the configuration tree:

    - **Payment model** – This configuration contains the data model ER component that represents the data structure of the payment business domain.
    - **Payment model mapping 1611** – This configuration contains the model mapping ER component that describes how the data model is filled in with application data at runtime.
    - **BACS (UK)** – This configuration contains the format and format mapping ER components. The format component specifies the report layout. The format

mapping component contains the model data source and specifies how the report layout is filled in by using this data source at runtime.

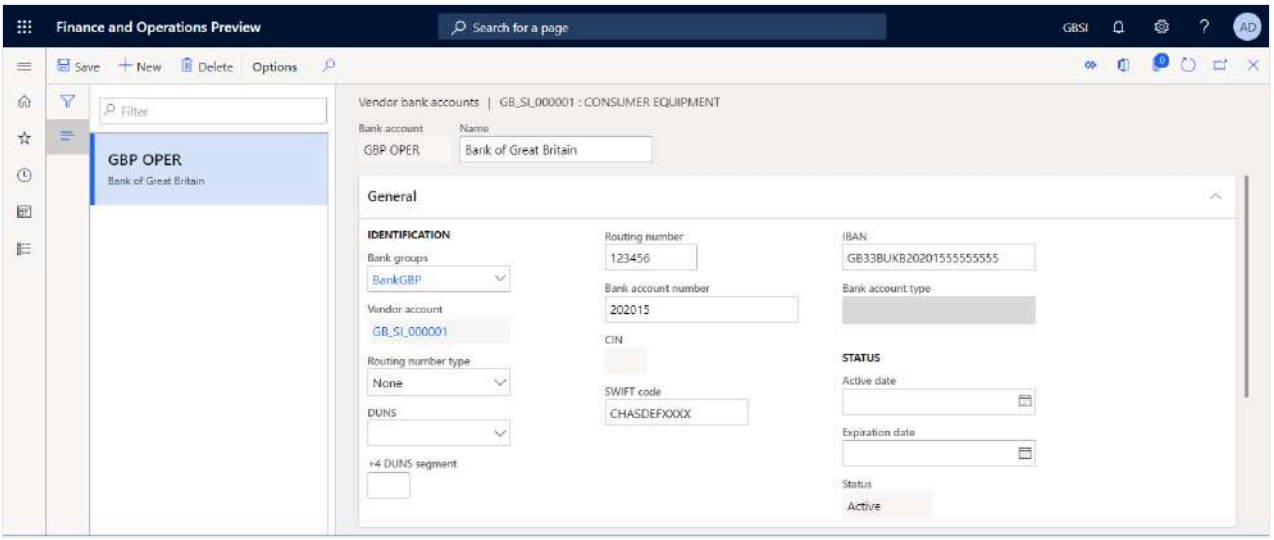


Prepare a vendor payment for processing

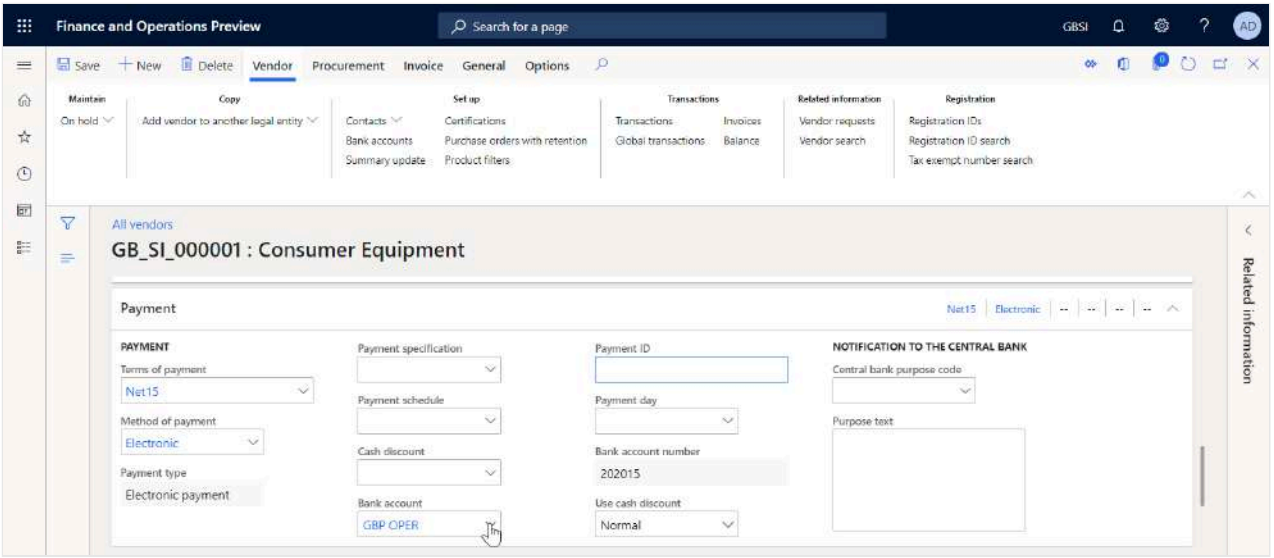
Add bank information for a vendor account

Add bank information for a vendor account that you refer to later in a registered payment.

1. Go to **Accounts payable** > **Vendors** > **All vendors**.
2. On **All vendors**, select the **GB_SI_000001** vendor account. On the Action Pane, on the **Vendor** tab, in the **Set up** group, select **Bank accounts**.
3. On **Vendor bank accounts**, select **New**, and then enter the following information:
   a. In the **Bank account** field, enter **GBP OPER**.
   b. In the **Bank groups** field, select **BankGBP**.
   c. In the **Bank account number** field, enter **202015**.
   d. In the **SWIFT code** field, enter **CHASDEFXXXX**.
   e. In the **IBAN** field, enter **GB33BUKB20201555555555**.
   f. In the **Routing number** field, keep the default value, **123456**.



4. Select **Save**.
5. Close the page.
6. On **All vendors**, open the **GB_SI_000001** vendor account.
7. On the vendor details page, select **Edit** to make the page editable, if required.
8. On the **Payment** FastTab, in the **Bank account** field, select **GBP OPER**.



9. Select **Save**.
10. Close the page.

Enter a vendor payment

Enter a new vendor payment by using a payment proposal.

1. Go to **Accounts payable** > **Payments** > **Vendor payment journal**.

2. On the **Vendor payment journal** page, select **New**.

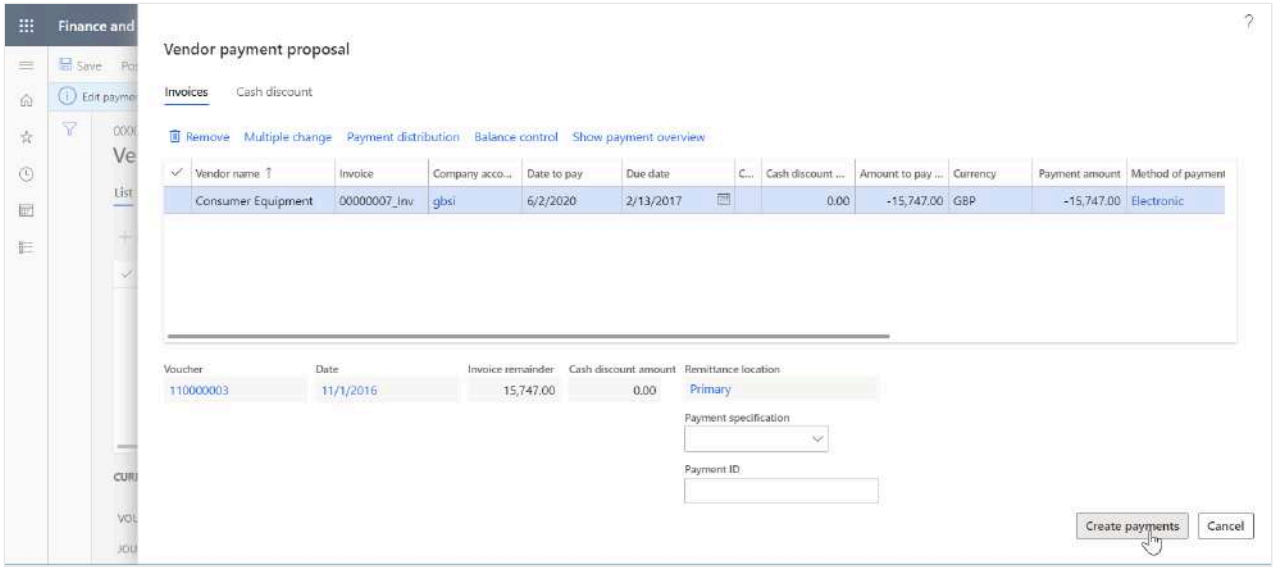
3. In the **Name** field, select **VendPay**.

4. Select **Lines**.

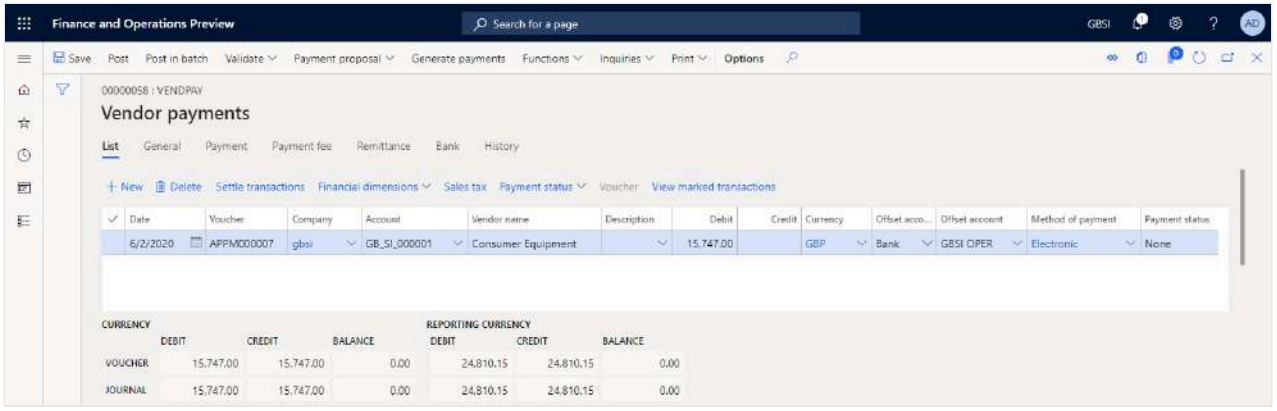
5. Select **Payment proposal** > **Create payment proposal**.

6. In the **Vendor payment proposal** dialog box, set conditions to filter for records for the **GB_SI_000001** vendor account only, and then select **OK**.

7. Select the line for the **00000007_Inv** invoice, and then select **Create payment**.



8. Verify that the payment that you entered is configured to use the **Electronic** method of payment.

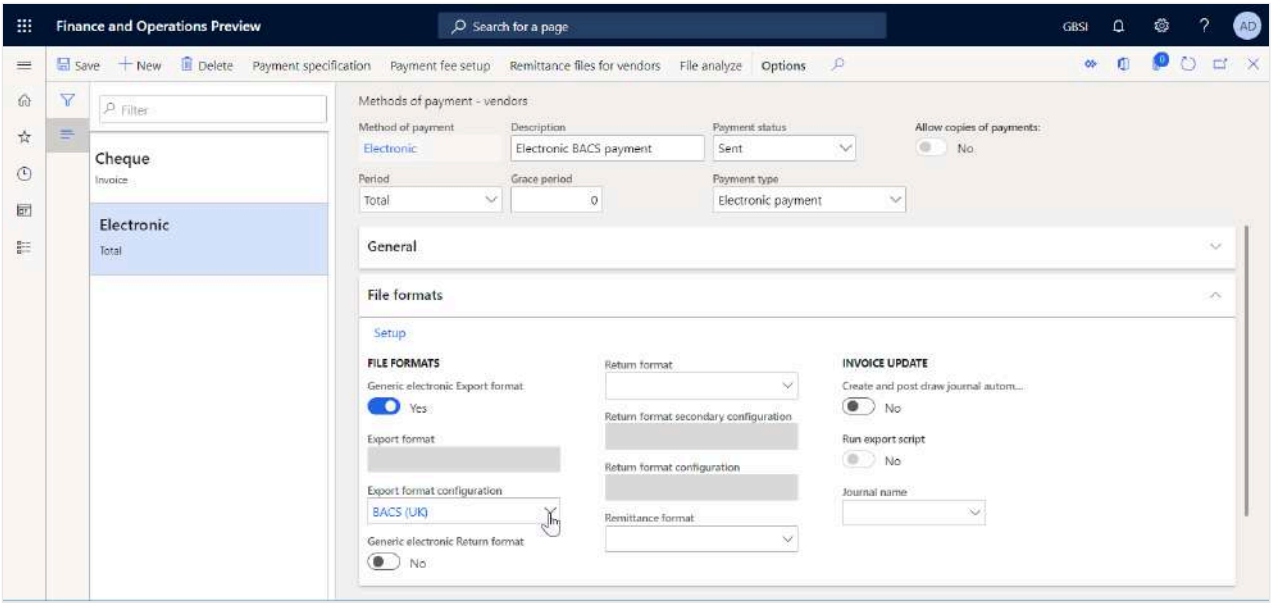


Process a vendor payment by using the standard ER format

Set up the electronic payment method

You must configure the electronic method of payment so that it uses the imported ER format configuration.

1. Go to **Accounts payable** > **Payment setup** > **Methods of payment**.
2. On **Methods of payment - vendors**, select the **Electronic** method of payment in the left pane.
3. Select **Edit**.
4. On the **File formats** FastTab, set the **General electronic Export format** option to **Yes**.
5. In the **Export format configuration** field, select the **BACS (UK)** format configuration.

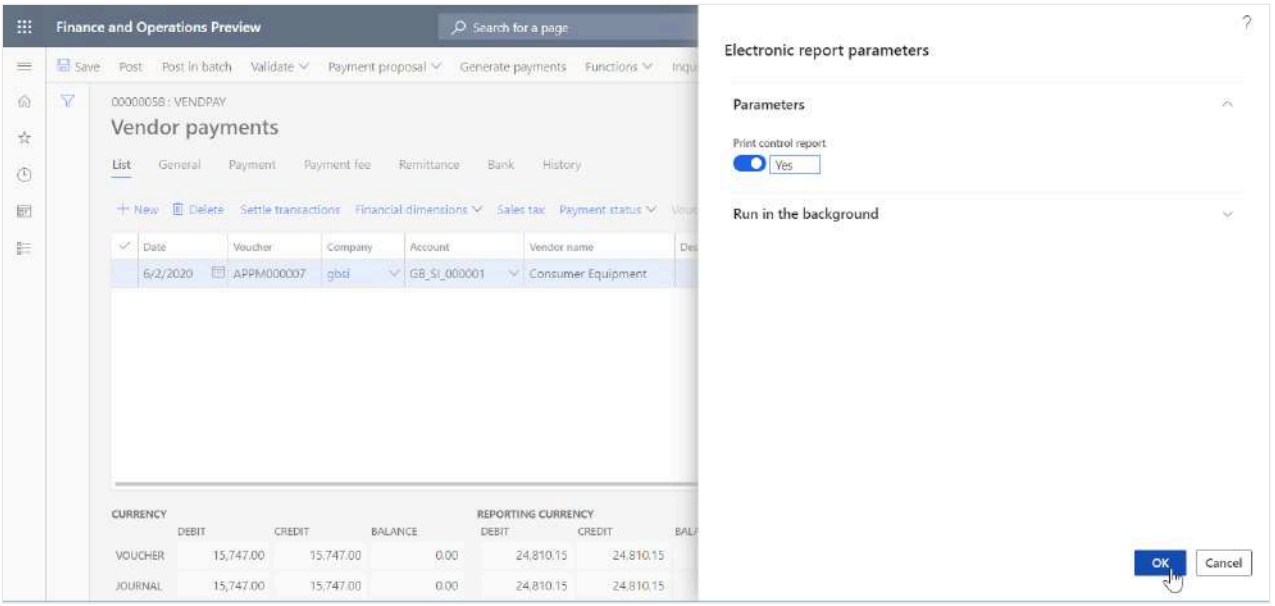


6. Select **Save**.

Process a vendor payment

1. Go to **Accounts payable** > **Payments** > **Vendor payment journal**.
2. On **Vendor payment journal**, select the payment journal that you added earlier, and then select **Lines**.
3. On **Vendor payments**, select **Generate payments**.
4. In **Generate payments**, enter the following information:
    - In **Method of payment**, select **Electronic**.
    - In **Bank account**, select **GBSI OPER**.
5. Select **OK**.

6. In **Electronic report parameters**, set the **Print control report** option to **Yes**, and then select **OK**.



> **Note**
>
> In addition to the payment file, you can now generate the control report.

7. Download the ZIP file, and then extract the following files from it:

   - The control report in Excel format
   - The payment file in TXT format

     Notice that, in accordance with the structure of the provided ER format, the payment line in the generated file starts with the routing number that you defined for the configured bank account.

Customize the standard ER format

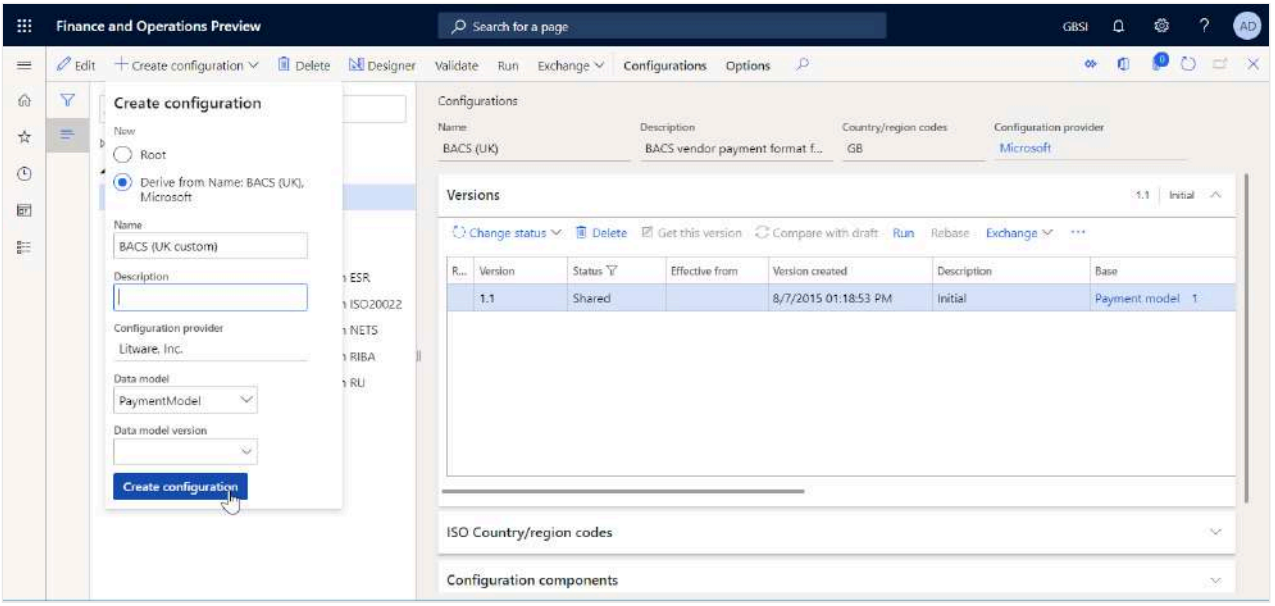
For the example in this section, you want to use the ER configurations that Microsoft provides to generate vendor payment files in BACS format, but you must add a customization to support the requirements of a specific bank. You also want to be able to upgrade your custom

format when new versions of ER configurations become available. However, you want to do the upgrade at the lowest cost.

In this case, as the representative of Litware, Inc., you must create (derive) a new ER format configuration by using the **BACS (UK)** Microsoft-provided configuration as a base.

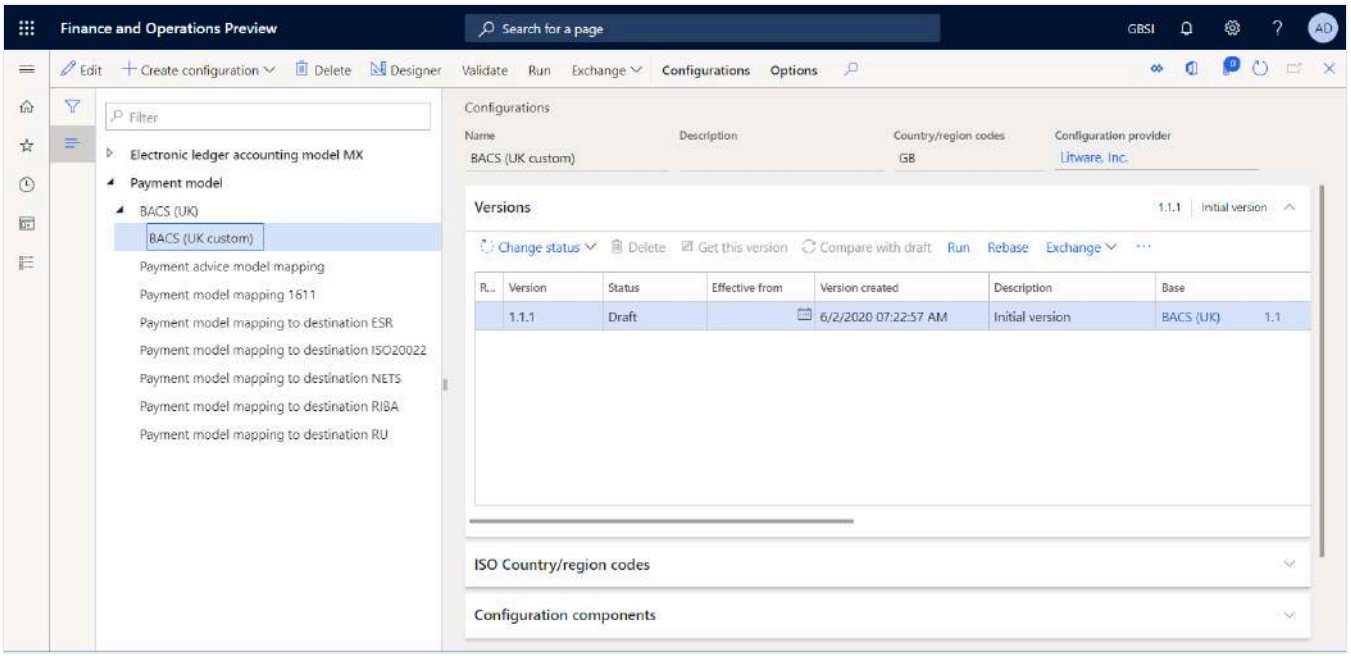
Create a custom format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**, and then select **BACS (UK)**. Litware, Inc. uses version 1.1 of this ER format configuration as the base for the custom version.
3. Select **Create configuration** to open the drop-down dialog box. Use this dialog box to create a new configuration for a custom payment format.
4. In the **New** field group, select the **Derive from Name: BACS (UK), Microsoft** option.
5. In the **Name** field, enter **BACS (UK custom)**.



6. Select **Create configuration**.

Version 1.1.1 of the **BACS (UK custom)** ER format configuration is created. This version has a status of **Draft** and can be edited. The current content of your custom ER format matches the content of the format that Microsoft provides.



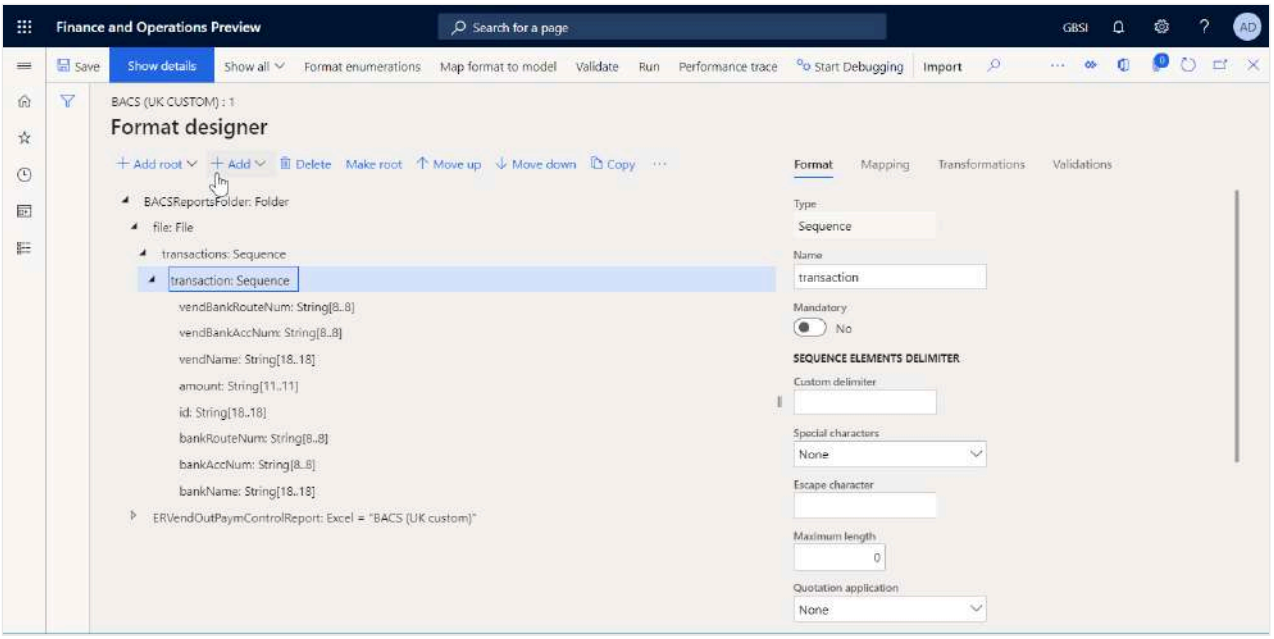
Edit a custom format

You must configure your custom format so that it meets bank-specific requirements. For example, a bank might require that generated payment files include the Society for Worldwide Interbank Financial Telecommunication (SWIFT) code of a bank that is assigned the agent role in the processed vendor payment. SWIFT codes are international bank codes that identify specific banks worldwide. They're also known as Bank Identifier Codes (BICs). The SWIFT code must be 11 characters long, and it must be entered at the beginning of every payment line in a generated payment file.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**, and then select **BACS (UK custom)**.
3. On the **Versions** FastTab, select version **1.1.1** of the selected configuration.
4. Select **Designer**.
5. On **Format designer**, select **Show details** to view more information about the format elements.
6. Expand and review the following elements:
    - The **BACSReportsFolder** element of the **Folder** type. This element generates output in ZIP format.
    - The **file** element of the **File** type. This element generates a payment file in TXT format.

- The **transactions** element of the **Sequence** type. This element generates a single payment line in a payment file.
- The **transaction** element of the **Sequence** type. This element generates individual fields of a single payment line.

7. Select the **transaction** element.



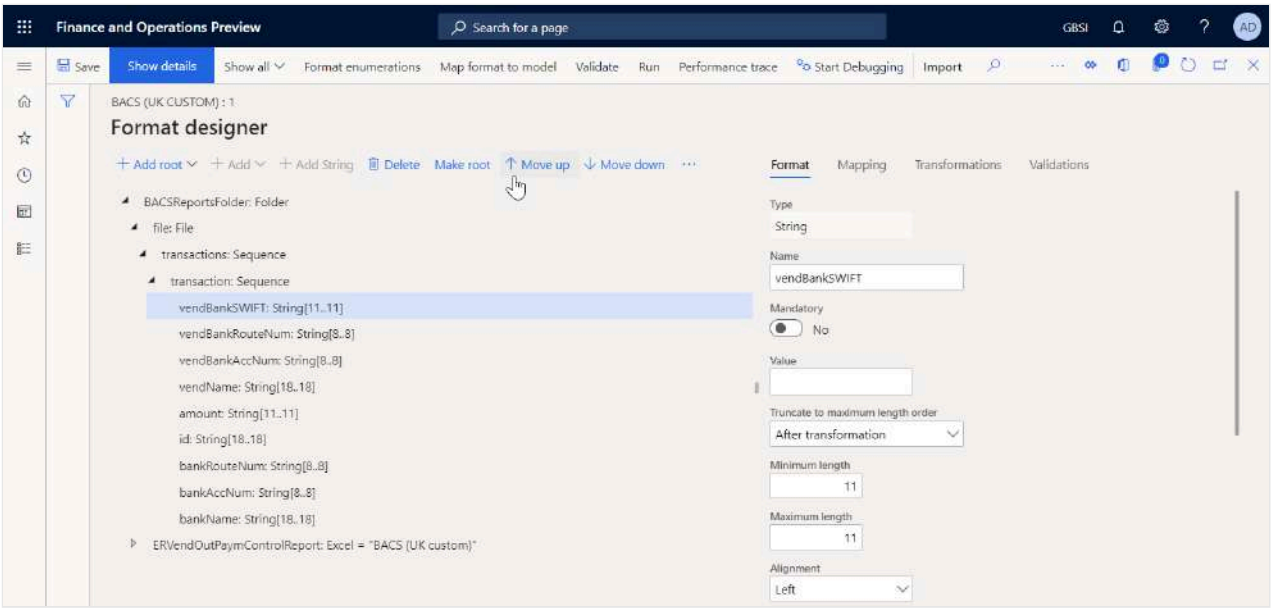
> ⓘ **Note**
>
> The provided report is configured so that every payment line starts with the bank routing number. The **vendBankRouteNum** format element is used for this purpose.

8. Select **Add**, and then select the **Text\String** type of the format element that you're adding:
   a. In the **Name** field, enter **vendBankSWIFT**.
   b. In the **Minimum length** field, enter **11**.
   c. In the **Maximum length** field, enter **11**.
   d. Select **OK**.

> ⓘ **Note**
>
> Use the **vendBankSWIFT** element to enter the SWIFT code of a vendor bank in generated files.

9. In the format structure tree, select **vendBankSWIFT**.

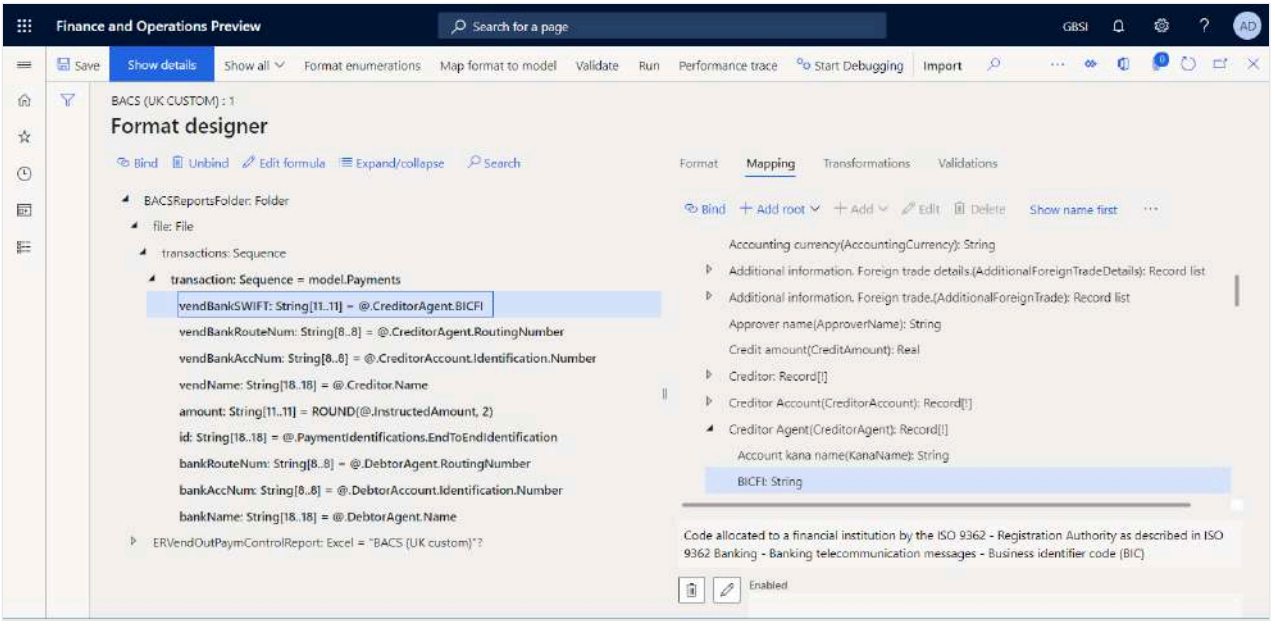
10. Select **Move up** to move the selected format element up one level. Repeat this step until the **vendBankSWIFT** element is the first element under the parent **transaction** element.



11. While the **vendBankSWIFT** is still selected in the format structure tree, select the **Mapping** tab, and then expand the **model** data source.

12. Expand **model.Payment** > **model.Payment.CreditorAgent**, and select the **model.Payment.CreditorAgent.BICFI** data source field. This data source field exposes the SWIFT code of a vendor bank that is assigned the agent role in the processed vendor payment.

13. Select **Bind**. The **vendBankSWIFT** format element is now bound with the **model.Payment.CreditorAgent.BICFI** data source field, so that SWIFT codes are entered in generated payment files.



14. Select **Save**.

15. Close the designer page.

Mark a custom format as runnable

After you create the first version of your custom format and set its status to **Draft**, you can run it for testing purposes. To run the report, you must process a vendor payment by using the payment method that refers to your custom ER format. By default, when you call an ER format from the application, only versions that have a status of **Completed** or **Shared** are considered. This behavior helps prevent ER formats that have unfinished designs from being used. However, for your test runs, you can force the application to use the version of your ER format that has a status of **Draft**. By using this approach, you can adjust the current format version if any modifications are required. For more information, see Applicability.

To use the draft version of an ER format, you must explicitly mark the ER format.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.

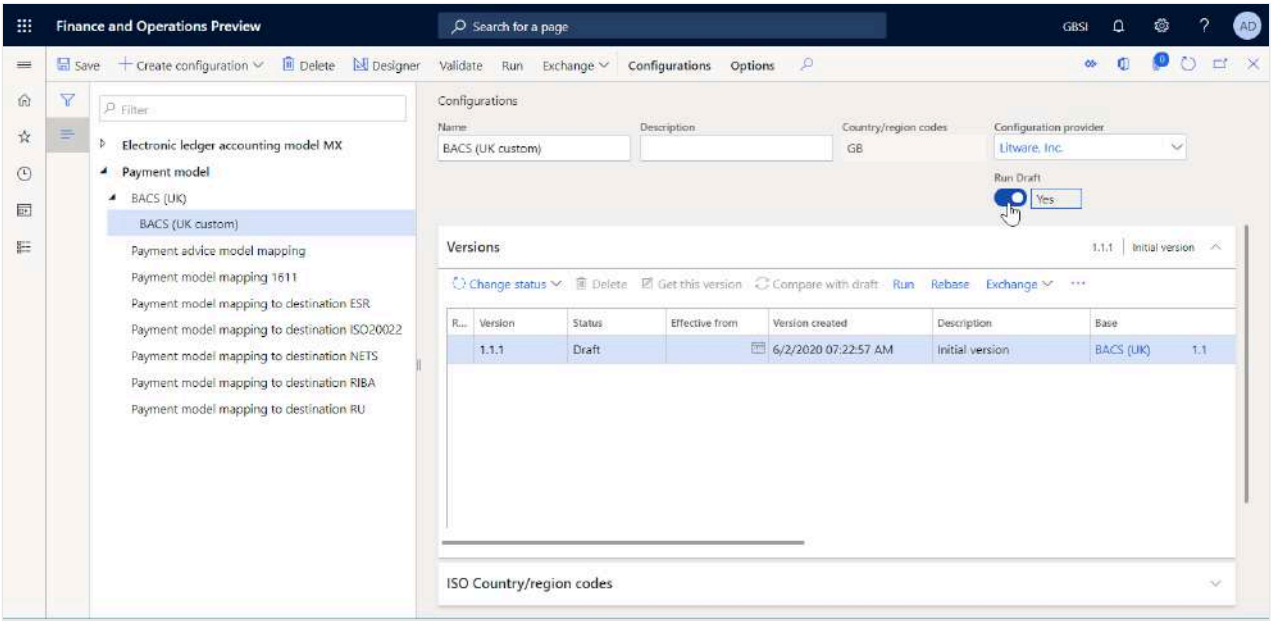
2. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.

3. In **User parameters**, set the **Run settings** option to **Yes**, and then select **OK**.

4. Select **Edit** to make the current page editable, as required.

5. In the configuration tree in the left pane, select **BACS (UK custom)**.

6. Set the **Run Draft** option to **Yes**.

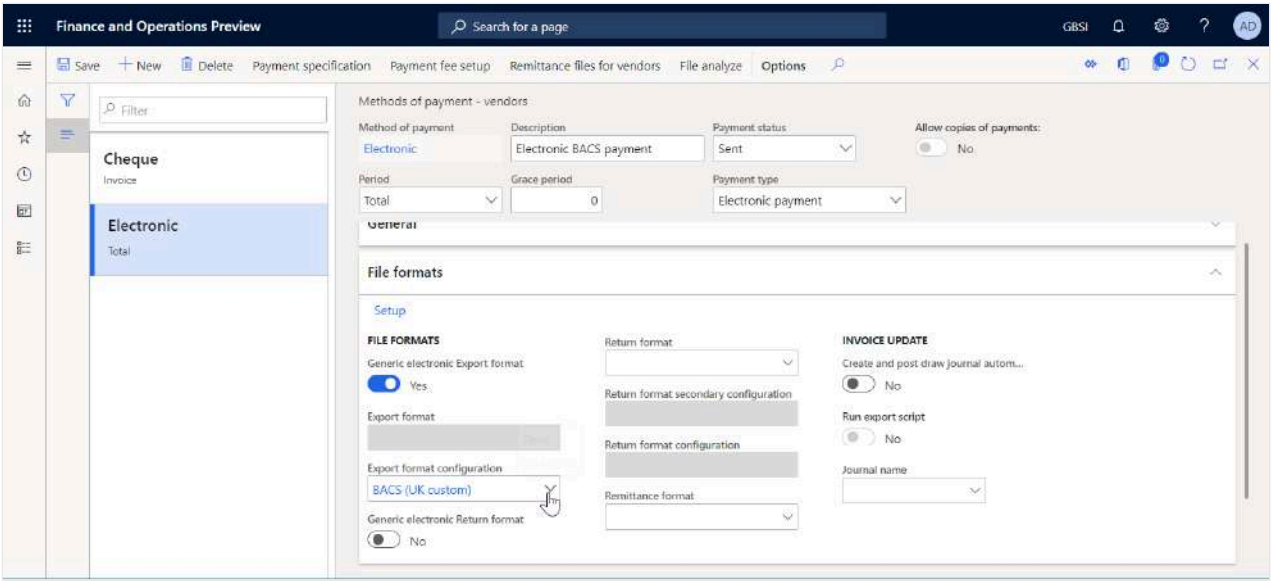


Process a vendor payment by using the custom ER format

Set up the electronic payment method

You must configure the electronic method of payment so that your custom ER format is used to process vendor payments.

1. Go to **Accounts payable** > **Payment setup** > **Methods of payment**.
2. On **Methods of payment - vendors**, select the **Electronic** method of payment in the left pane.
3. Select **Edit**.
4. On the **File format** FastTab, set the **General electronic export format** option to **Yes**.
5. In the **Export format configuration** field, select the **BACS (UK custom)** format configuration.



6. Select **Save**.

Process a vendor payment

1. Go to **Accounts payable** > **Payments** > **Vendor payment journal**.
2. On **Vendor payment journal**, select the payment journal that you created earlier.
3. Select **Lines**.

4. On **Vendor payments**, above the grid, select **Payment status** > **None**.

5. Select **Generate payment**.

6. In **Generate payments**, enter the following information:

    - In **Method of payment**, select **Electronic**.
    - In **Bank account**, select **GBSI OPER**.

7. Select **OK**.

8. In **Electronic report parameters**, set the **Print control report** option to **Yes**, and then select **OK**.

    > **Note**
    >
    > In addition to the payment file, you can generate only the control report.

9. Download the ZIP file, and then extract the following files from it:

    - The control report in Excel format
    - The payment file in TXT format

      Notice that, in accordance with the structure of your custom ER format, the payment line in the generated file now starts with the SWIFT code that you entered for the bank account of the vendor whose payment you processed.

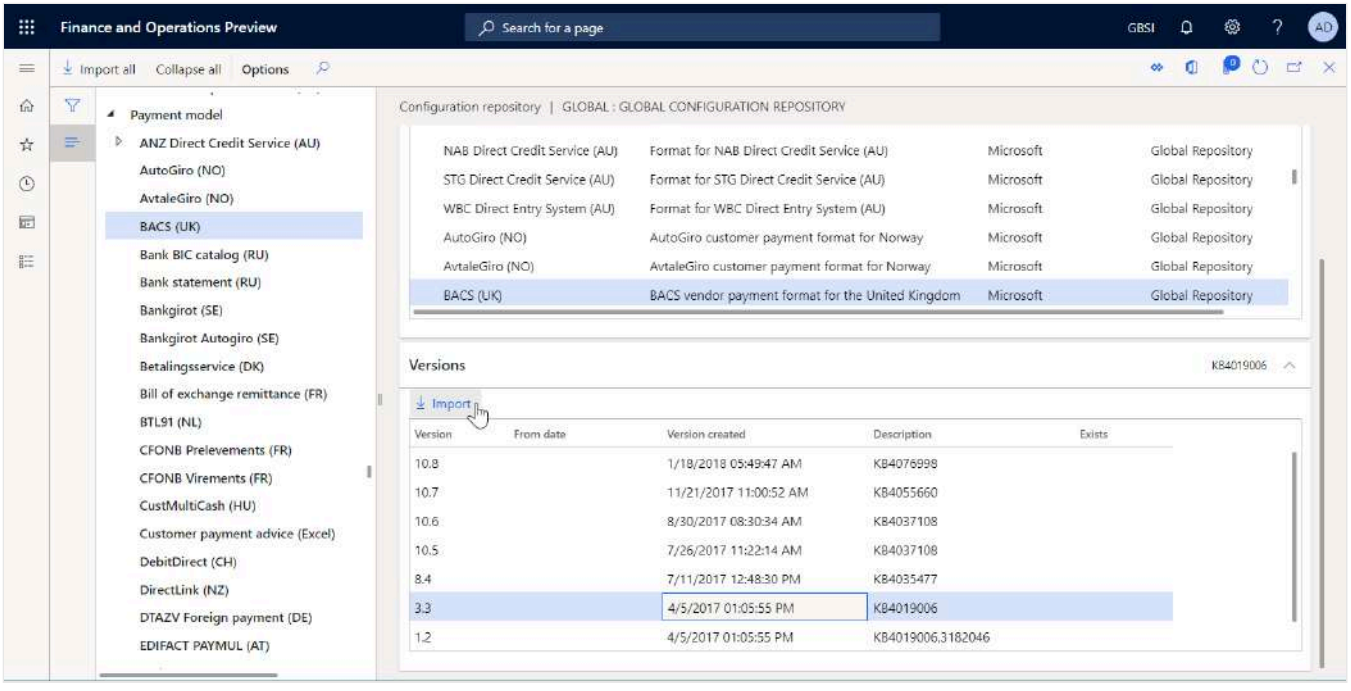
Import new versions of the standard ER format configurations

In the following example, you receive a notification about Knowledge Base article KB3763330. This notification informs you about the new version of the **BACS (UK)** ER format that Microsoft published. In addition to the control report, this new version lets users generate the payment advice report and the attending note report while a vendor payment is being processed. You want to start using that functionality.

Import new versions of the standard ER configurations

To add new versions of the ER configurations to the current Finance instance, import them from the ER repository that you configured.

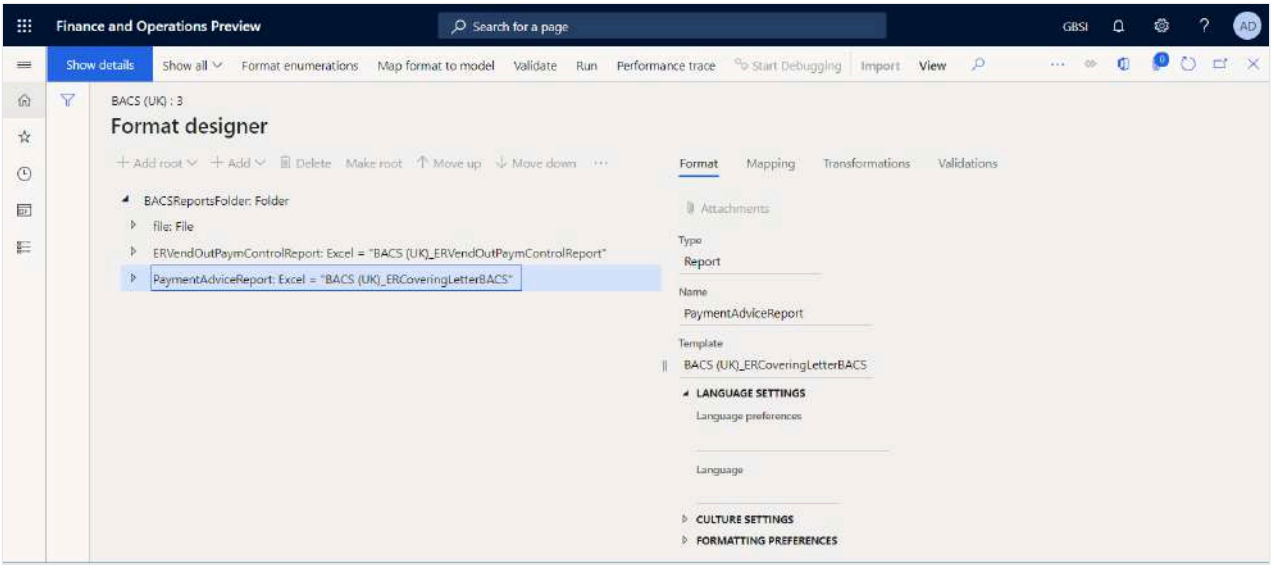
1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, select the **Microsoft** tile, and then select **Repositories** to view the list of repositories for the Microsoft provider.
3. On **Configuration repositories**, select the repository of the **Global** type, and then select **Open**. If you're prompted for authorization to connect to Regulatory Configuration Service, follow the authorization instructions.
4. On **Configuration repository**, in the configuration tree in the left pane, select the **BACS (UK)** format configuration.
5. On the **Versions** FastTab, select version **3.3** of the selected ER format configuration.
6. Select **Import** to download the selected version from the Global repository to the current Finance instance.



> **Tip**
>
> If you have trouble accessing the Global repository, you can download configurations from Lifecycle Services instead.

Review the imported ER format configurations

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configurations** section, select the **Reporting configurations** tile.
3. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**, and then select **BACS (UK)**.
4. On the **Versions** FastTab, select version **3.3**.
5. Select **Designer**.
6. On **Format designer**, expand the **BACSReportsFolder** format element.
7. Notice that version 3.3 contains the **PaymentAdviceReport** format element that you use to generate a payment advice report when a vendor payment is processed.



8. Close the designer page.

Adopt the changes in the new version of an imported format in a custom format

Complete the current draft version of a custom format

If you want to keep the current state of your custom format, complete the draft version 1.1.1 by changing its status from **Draft** to **Completed**.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configurations** section, select the **Reporting configurations** tile.

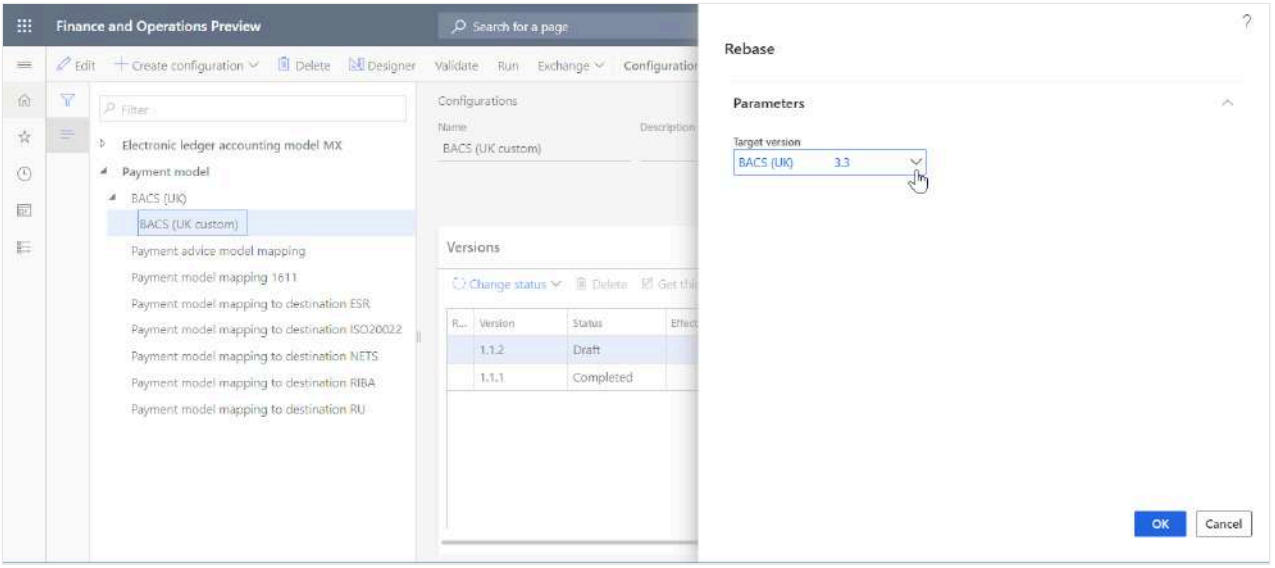
3. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**, expand **BACS (UK)**, and then select **BACS (UK custom)**.
4. On the **Versions** FastTab, select **Change status** > **Complete**, and then select **OK**.

The status of version 1.1.1 changes from **Draft** to **Completed**, and the version becomes read-only. A new editable version, 1.1.2, is added and has a status of **Draft**. Use this version to make further changes in your custom ER format.

Rebase a custom format to a new base version

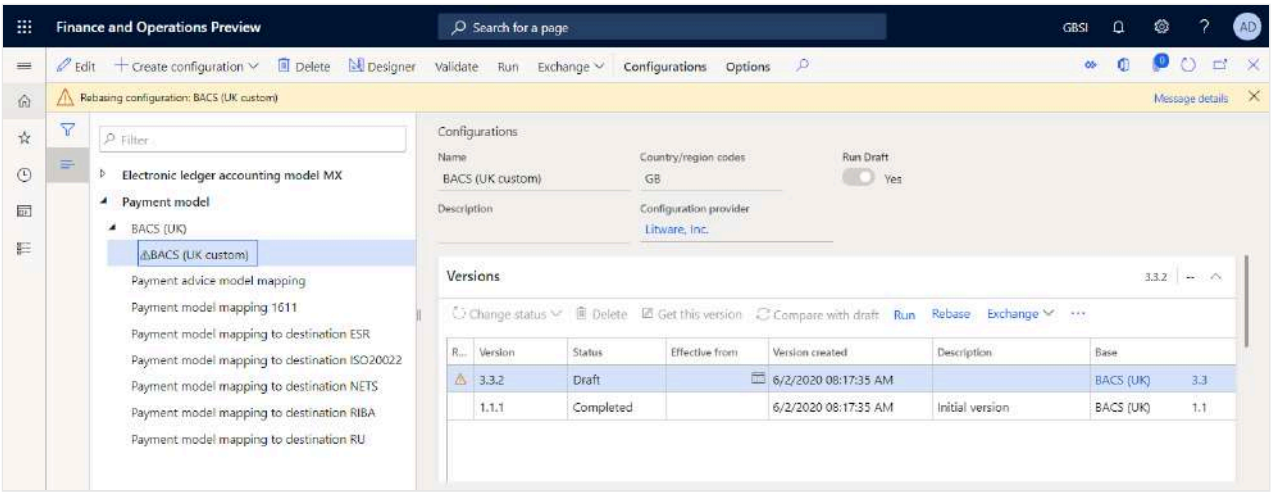
To use the new functionality of version 3.3 of the **BACS (UK)** format in your customization, change the base configuration version for the custom configuration, **BACS (UK custom)**. This process is known as rebasing. Instead of version 1.1 of **BACS (UK)**, use version 3.3.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Payment model**, and then select **BACS (UK custom)**.
3. On the **Versions** FastTab, select version **1.1.2**, and then select **Rebase**.
4. In **Rebase**, in the **Target version** field, select version **3.3** of the base configuration to apply it as the new base and use it to update the configuration.



5. Select **OK**.
6. The number of the draft version changes from **1.1.2** to **3.3.2** to reflect the change in the base version.

When the custom version and a new base version merge, some conflicts might be discovered because of format changes that can't be merged automatically.

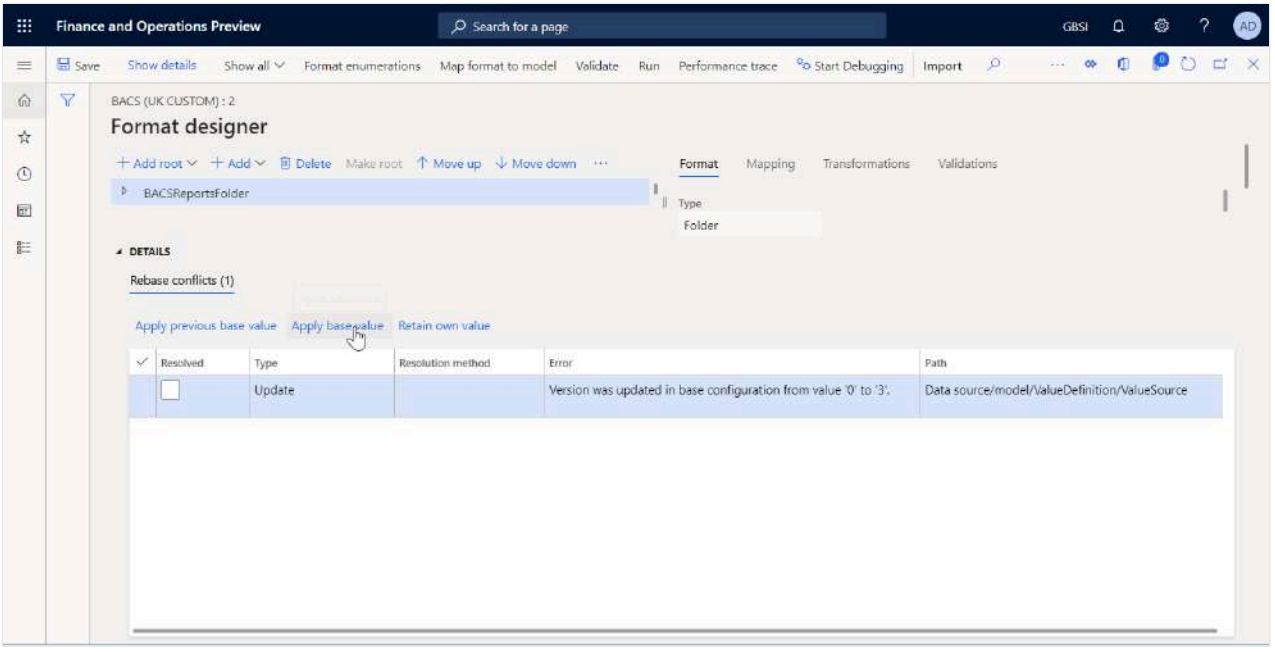


If conflicts are discovered, you must resolve them manually in the format designer.

7. On the **Versions** FastTab, select version **3.3.2**.

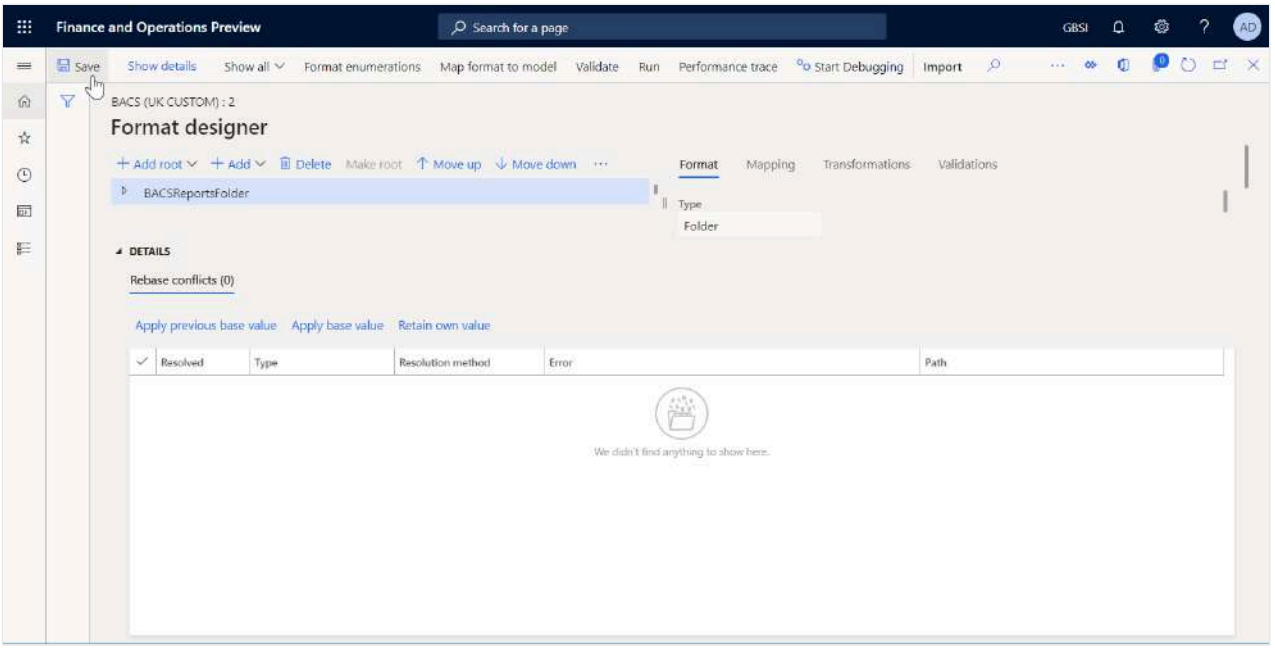
8. Select **Designer**.

9. On **Format designer**, on the **Details** FastTab, select a rebase conflict record, and then select **Apply base value**.



10. Select **Save**.

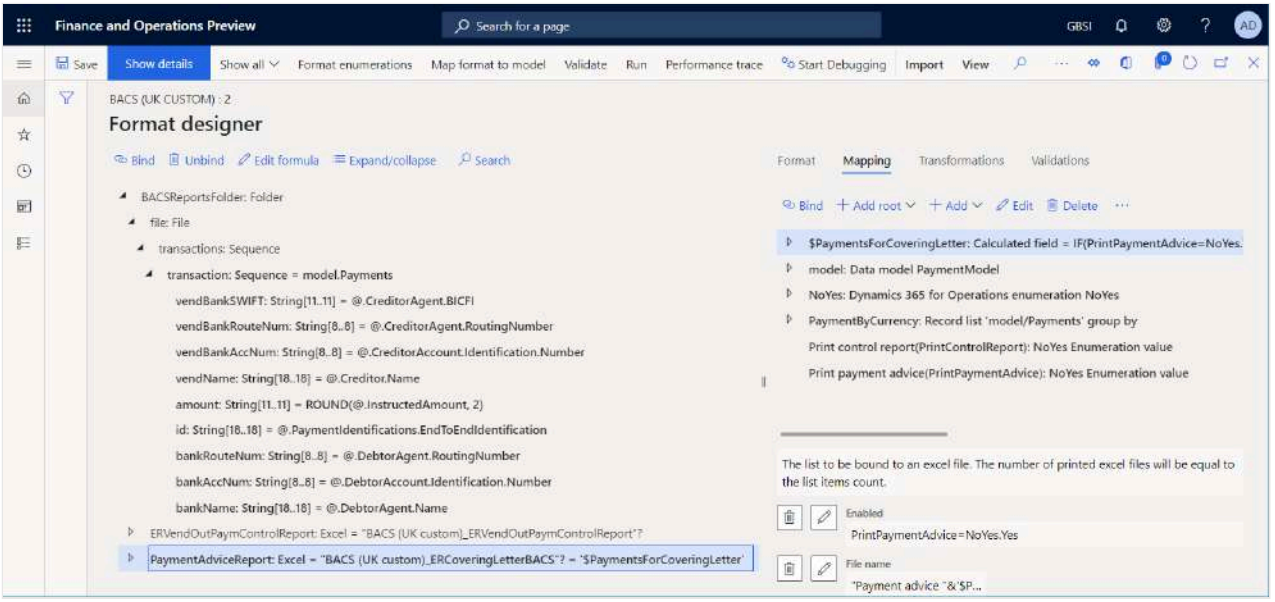
The rebase conflict record no longer appears on the **Details** FastTab.



> **Note**
>
> You resolved the conflict by confirming that version 3 of the base model must be used in this ER format.

11. Expand **BACSReportsFolder** > **file** > **transactions** > **transaction**.

12. On the **Mapping** tab, version 3.3.2 of your custom ER format contains both your customization (the **vendBankSWIFT** format element and its binding) and the new functionality of version 3.3 of the base ER format that Microsoft provides (the **PaymentAdviceReport** format element together with its nested elements and configured bindings). In just a few mouse clicks, you adopted the modifications of a new base version by merging them with your customization.

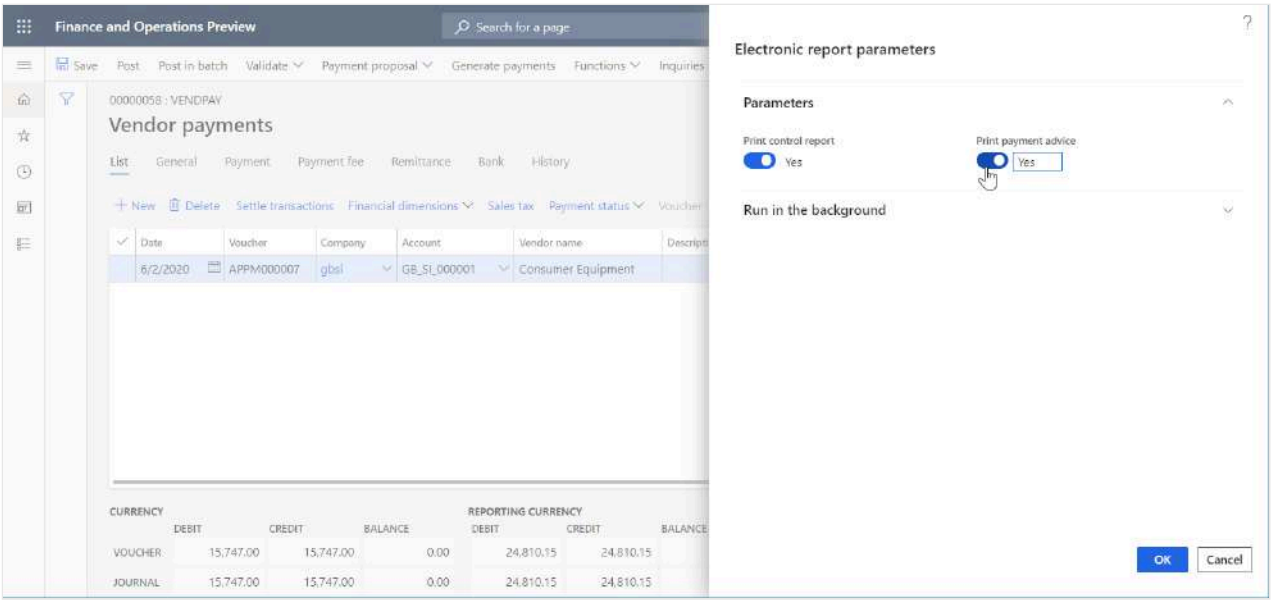


13. Close the designer page.

> ⓘ **Note**
>
> The rebase action is reversible. To cancel this rebase, select version **1.1.1** of the **BACS (UK custom)** format on the **Versions** FastTab, and then select **Get this version**. Version 3.3.2 is renumbered 1.1.2, and the content of draft version 1.1.2 matches the content of version 1.1.1.

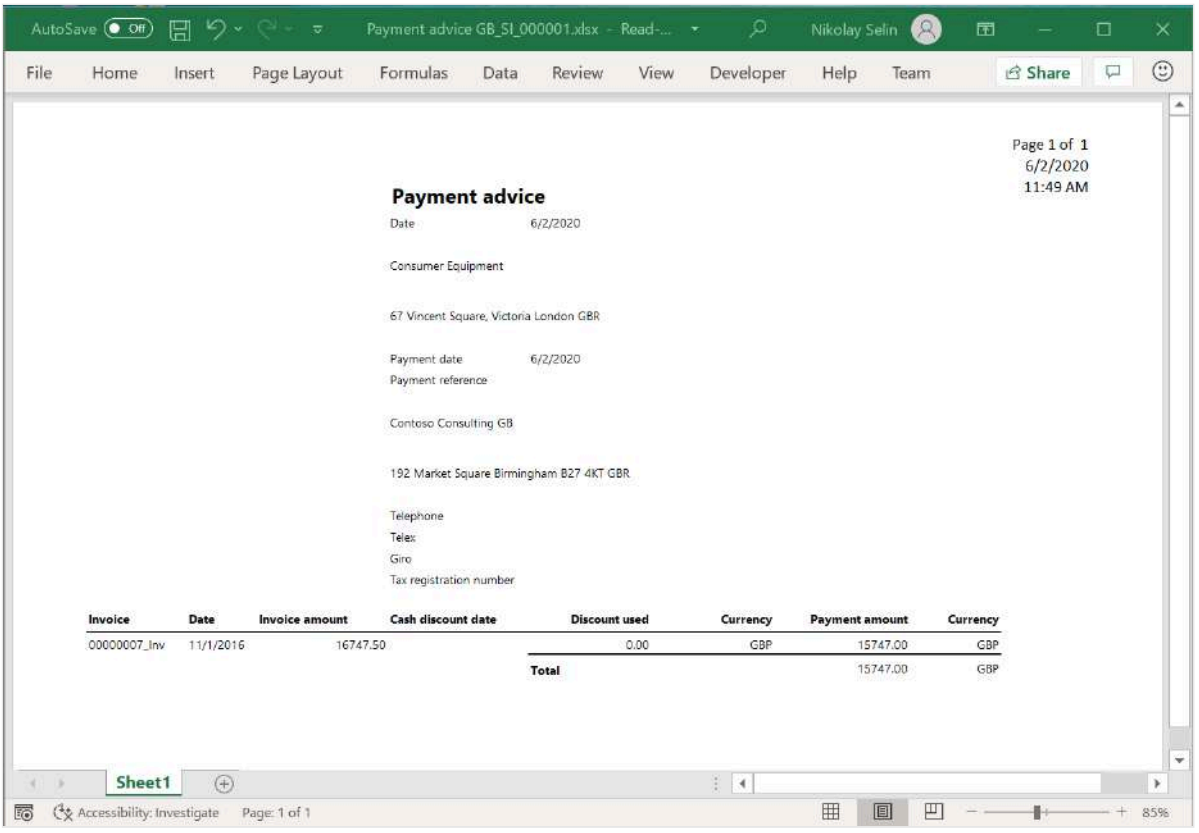
Process a vendor payment by using a rebased ER format

1. Go to **Accounts payable** > **Payments** > **Vendor payment journal**.
2. On **Vendor payment journal**, select the payment journal that you created earlier.
3. Select **Lines**.
4. On **Vendor payments**, above the grid, select **Payment status** > **None**.
5. Select **Generate payment**.
6. In **Generate payments**, enter the following information:
    - In **Method of payment**, select **Electronic**.
    - In **Bank account**, select **GBSI OPER**.
7. Select **OK**.
8. In **Electronic report parameters**, enter the following information:
    - Set the **Print control report** option to **Yes**.
    - Set the **Print payment advice** option to **Yes**.



> **Note**
>
> In addition to the payment file, you can now generate both the control report and the payment advice report.

9. Select **OK**.
10. Download the ZIP file, and then extract the following files from it:
    - The control report in Excel format
    - The payment advice report in Excel format



- The payment file in TXT format

  Notice that the payment line in the generated file starts with the SWIFT code that you entered for the bank account of a vendor whose payment you processed.

Additional resources

- Electronic Reporting overview
- Download ER configurations from Lifecycle Services
- Download ER configurations from Global repository of Configuration service

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Last updated on 04/13/2026

Design a new ER solution to print a custom report

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user in the System Administrator, Electronic Reporting Developer, or Electronic Reporting Functional Consultant role can configure parameters of the ER framework, design the required ER configurations of a new ER solution to access the data of a particular business domain, and generate a custom report in Microsoft Office format. You can complete these steps in the **USMF** company.

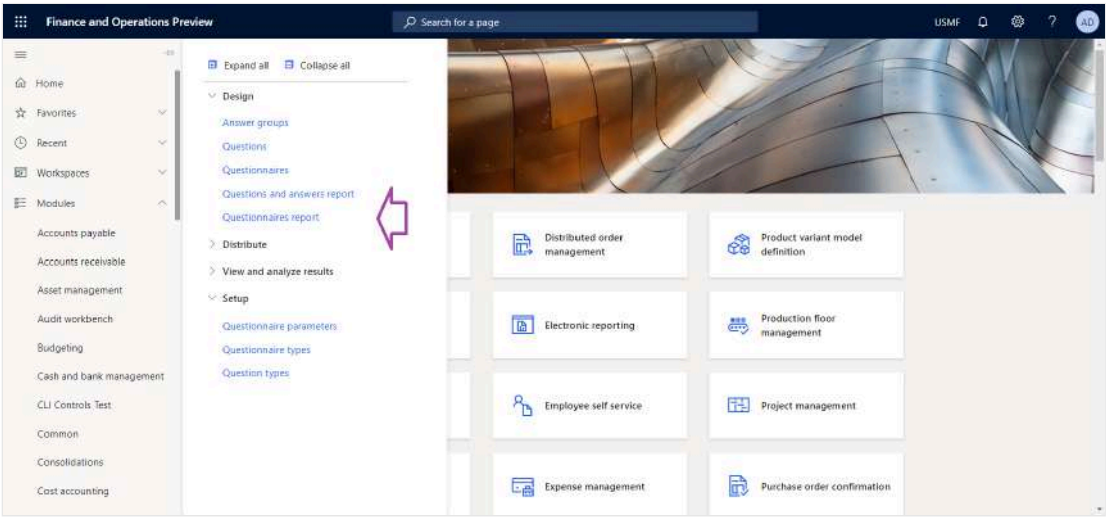
- Configure the ER framework
  - Configure ER parameters
  - Activate an ER configuration provider
    - Review the list of ER configuration providers
    - Add a new ER configuration provider
    - Activate an ER configuration provider
- Design a domain-specific data model
  - Import a new data model configuration
  - Create a new data model configuration
    - Name the data model
    - Add data model fields
    - Complete the design of the data model
- Design a model mapping for the configured data model
  - Import a new model mapping configuration
  - Create a new model mapping configuration
    - Design a new model mapping component
    - Add data sources to access application tables
    - Add data sources to access application enumerations
    - Add ER labels to generate a report in a specified language

        - Add a data source to transform the results of comparing enumeration values to a text value
        - Bind data sources to data model fields
        - Complete the design of the model mapping
- Design a template for a custom report
- Design a format
    - Import a designed format configuration
    - Create a new format configuration
        - Import a report template
        - Configure a format
        - Define the data binding for a report title
        - Review the model data source
        - Bind format elements to data source fields
    - Run a designed format from ER
- Tune a designed format
    - Modify a format to change the name of a generated document
    - Modify a format to change the order of questions
    - Run a modified format from ER
    - Complete the format design
- Develop application artefacts to call the designed report
    - Modify source code
        - Add a data contract class
        - Add a UI builder class
        - Add a data provider class
        - Add a labels file
        - Add a report service class
        - Add a report controller class
        - Add a menu item
        - Add a menu item to a menu
        - Build a Visual Studio project
    - Run a format from the application
- Tune a designed ER solution
    - Modify a model mapping
        - Add data sources to access a data contract object
        - Add a data source to access ER format mapping records
        - Add a data source to access a format mapping record of a running ER format

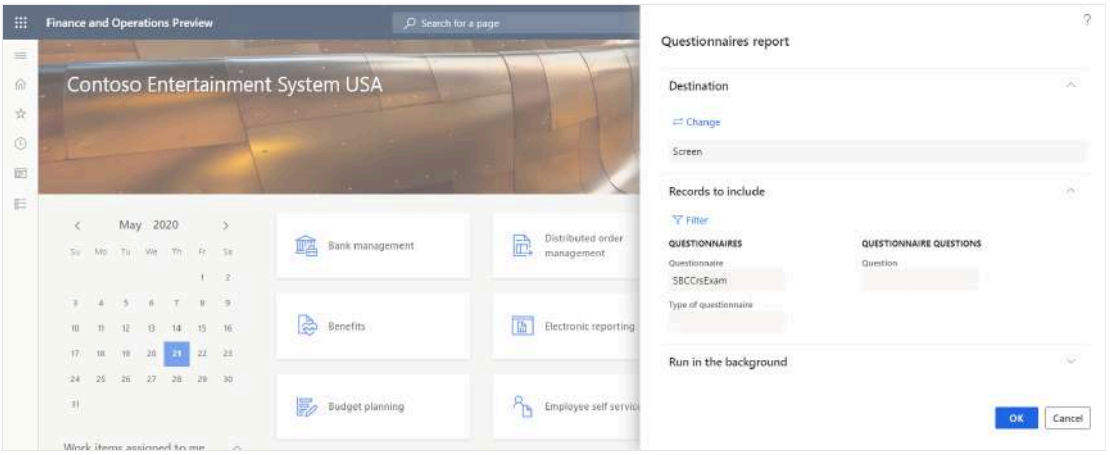
    - Enter the name of the running ER format in the data model
    - Complete the design of the model mapping
  - Modify a format
    - Add a new format element
    - Bind the added format element
    - Complete the format design
  - Run a format from the application
  - Run a format from ER
  - Configure a format destination for on-screen preview
  - Run a format from the application to preview it as a PDF document
- Additional resources

In this example, you create a new ER solution for the Questionnaire module. This new ER solution lets you design a report by using a Microsoft Excel worksheet as a template. You can then generate the **Questionnaire** report in Excel or PDF format, in addition to generating the existing SQL Server Reporting Services (SSRS) report. You can also modify the new report later, upon request. No coding is required.

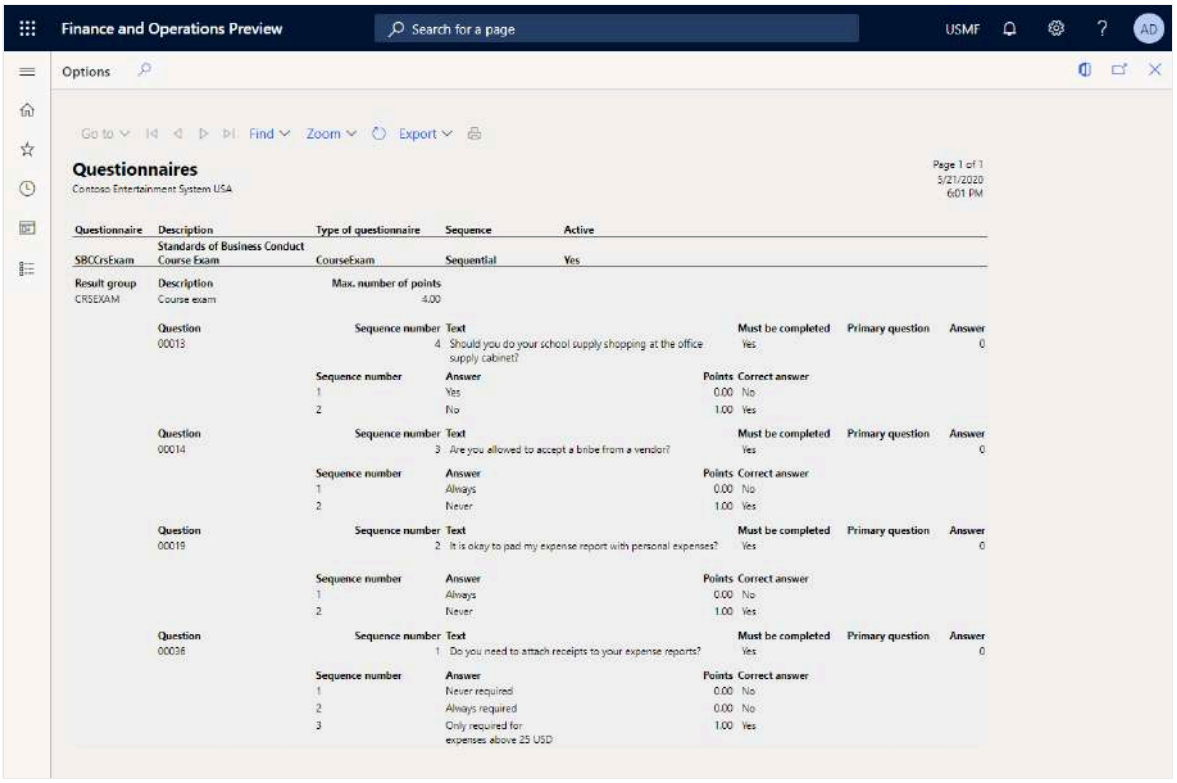
1. To run the existing report, go to **Questionnaire** > **Design** > **Questionnaires report**.



2. In the **Questionnaires report** dialog box, specify selection criteria. Apply a filter so that the report includes only the **SBCCrsExam** questionnaire.



The following illustration shows the generated version of the SSRS report for the **SBCCrsExam** questionnaire.



Configure the ER framework

As a user in the Electronic Reporting Developer role, you must configure the minimal set of ER parameters before you can start to use the ER framework to design your new ER solution.

Configure ER parameters

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. In the **Electronic reporting** workspace, select **Electronic reporting parameters**.
3. On the **Electronic reporting parameters** page, on the **General** tab, set the **Enable design mode** option to **Yes**.
4. On the **Attachments** tab, set the following parameters:

- Set the **Configurations** field to **File** for the **USMF** company.
- Set **Job archive**, **Temporary**, **Baseline**, and **Others** fields to **File**.

For more information about ER parameters, see Configure the ER framework.

Activate an ER configuration provider

Each ER configuration belongs to an ER configuration provider. Therefore, you must activate an ER configuration provider in the **Electronic reporting** workspace before you can add or edit any ER configurations.

> ⓘ **Note**
>
> Only the owner of an ER configuration can edit it. Therefore, before you can edit an ER configuration, activate the appropriate ER configuration provider in the **Electronic reporting** workspace.

Review the list of ER configuration providers

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. In the **Electronic reporting** workspace, in the **Related links** section, select **Configuration providers**.
3. On the **Configuration providers** page, each configuration provider record has a unique name and URL. Review the contents of this page. If a record for **Litware, Inc.** (`https://www.litware.com`) already exists, skip the next procedure, Add a new ER configuration provider.

Add a new ER configuration provider

1. On the **Configuration providers** page, select **New**.
2. In the **Name** field, enter **Litware, Inc.**
3. In the **Internet address** field, enter `https://www.litware.com`.
4. Select **Save**.

Activate an ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. In the **Electronic reporting** workspace, select the **Litware, Inc.** configuration provider.
3. Select **Set active**.

For more information about ER configuration providers, see Create configuration providers and mark them as active.

Design a domain-specific data model

You must create a new ER configuration that contains a data model component for the **Questionnaire** business domain. You use this data model as a data source when you design an ER format to generate the **Questionnaire** report.

By completing the steps in the Import a new data model configuration section, you can import the required data model from the provided XML file. Alternatively, you can complete the steps in the Create a new data model configuration section to design this data model from scratch.

Import a new data model configuration

1. Download the Questionnaires model.version.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Questionnaires model.version.1.xml** file.
6. Select **OK** to import the configuration.

To continue, skip the next procedure, Create a new data model configuration.

Create a new data model configuration

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. In the **Electronic reporting** workspace, select **Reporting configurations**.
3. Select **Create configuration**.
4. In the drop-down dialog box, in the **Name** field, enter **Questionnaire model**.
5. Select **Create configuration** to create the configuration.

Name the data model

1. On **Configurations**, in the configuration tree, select **Questionnaire model**.
2. Select **Designer**.
3. On **Data model designer**, on the **General** FastTab, in the **Name** field, enter **Questionnaires**.

Add new data model fields

1. On **Data model designer**, select **New**.

2. In the drop-down dialog box for adding a data model node, follow these steps:
   a. Select **Model root** as the type of the new node.
   b. In the **Name** field, enter **Root**.
   c. Select **Add** to add the new node.

   This root descriptor provides data for the **Questionnaire** report. A single data model can have multiple descriptors. Each descriptor can be specified for a single ER format, to

identify data that is required to generate the report.

3. Select **New** again. In the drop-down dialog box for adding a data model node, follow these steps:
   a. Select **Child of an active node** as the type of the new node.
   b. In the **Name** field, enter **CompanyName**.
   c. In the **Item type** field, select **String**.
   d. Select **Add** to add the new field.

   This field passes the name of the current company to an ER report that consumes this data model as a data source.

4. Select **New** again. In the drop-down dialog box for adding a data model node, follow these steps:
   a. Select **Child of an active node** as the type of the new node.
   b. In the **Name** field, enter **Questionnaire**.
   c. In the **Item type** field, select **Record list**.
   d. Select **Add** to add the new field.

   This field passes the list of questionnaires to an ER report that consumes this data model as a data source.

5. Select the **Questionnaire** node.

6. Continue to add the required fields of the editable data model in the same manner until you complete the following data model structure.

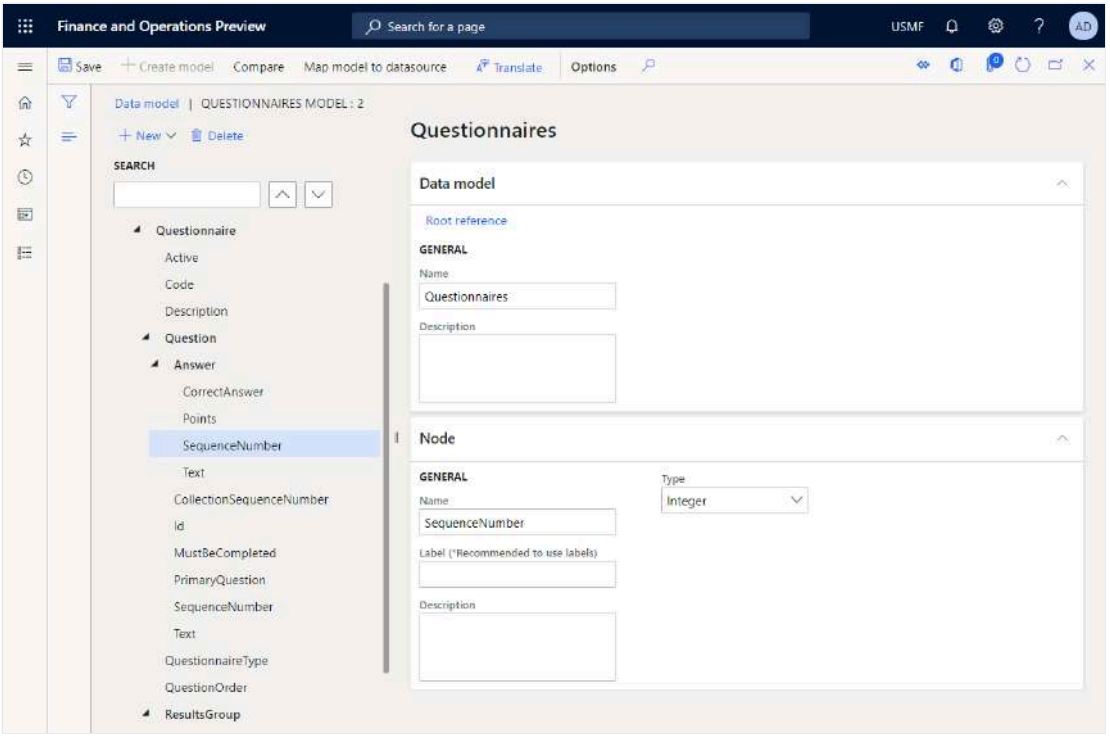
Expand table

Field path	Data type	Field designation/returned value
Root		The reference point to request questionnaire data.
Root\CompanyName	String	The name of the current company.
Root\ExecutionContext	Record	Format execution details.
Root\ExecutionContext\FormatName	String	The name of the ER format that is being run.
Root\Questionnaire	Record list	The list of questionnaires
Root\Questionnaire\Active	String	The status of the current questionnaire.
Root\Questionnaire\Code	String	The code of the current questionnaire.

Field path	Data type	Field designation/returned value
Root\Questionnaire\Description	String	The description of the current questionnaire.
Root\Questionnaire\QuestionnaireType	String	The type of the current questionnaire.
Root\Questionnaire\QuestionOrder	String	The numerical order of the current questionnaire.
Root\Questionnaire\ResultsGroup	Record	The result parameters of the current questionnaire.
Root\Questionnaire\ResultsGroup\Code	String	The identification code of the current result group.
Root\Questionnaire\ResultsGroup\Description	String	The description of the current result group.
Root\Questionnaire\ResultsGroup\MaxNumberOfPoints	Real	The maximum number of points that can be earned.
Root\Questionnaire\Question	Record list	The list of questions for the current questionnaire.
Root\Questionnaire\Question\CollectionSequenceNumber	Integer	The sequence number of the current answer collection.
Root\Questionnaire\Question\Id	String	The identification code of the current question.
Root\Questionnaire\Question\MustBeCompleted	String	A flag that indicates whether the current question must be answered.
Root\Questionnaire\Question\PrimaryQuestion	String	A flag that indicates whether the current question is primary.
Root\Questionnaire\Question\SequenceNumber	Integer	The sequence number of the current question.
Root\Questionnaire\Question\Text	String	The text of the current question.
Root\Questionnaire\Question\Answer	Record list	The list of answers for the current question.
Root\Questionnaire\Question\Answer\CorrectAnswer	String	A flag that indicates whether the current answer is correct.
Root\Questionnaire\Question\Answer\Points	Real	The points that are earned when the current answer is selected.

Field path	Data type	Field designation/returned value
Root\Questionnaire\Question\Answer\SequenceNumber	Integer	The sequence number of the current answer.
Root\Questionnaire\Question\Answer\Text	String	The text of the current answer.

The following illustration shows the completed editable data model on the **Data model designer** page.



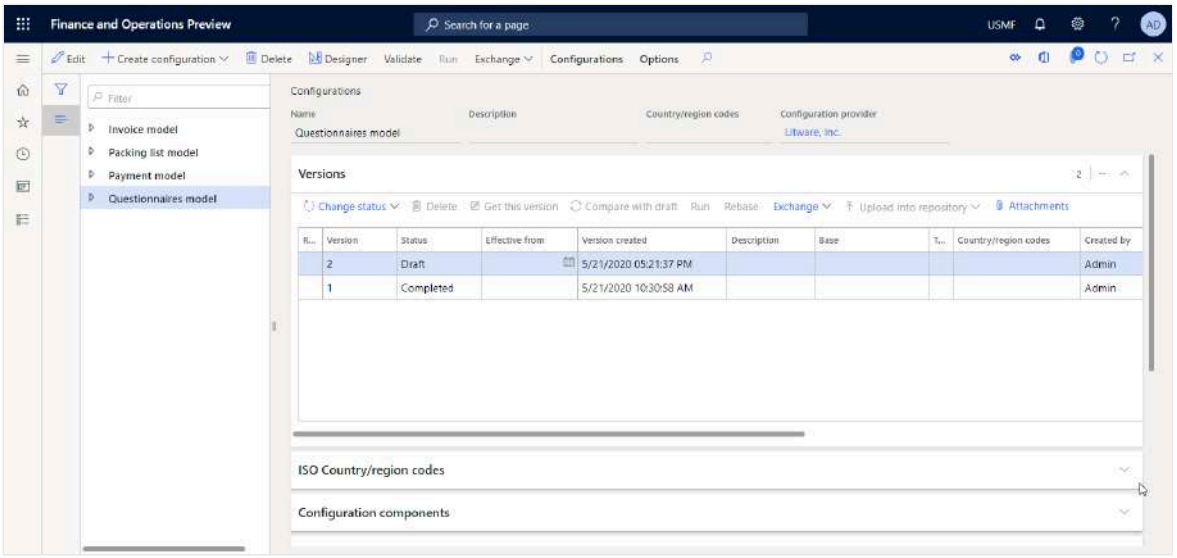
7. Save your changes.

8. Close the **Data model designer** page.

Complete the design of the data model

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, select **Questionnaire model**.
3. On the **Versions** FastTab, select the configuration version that has a status of **Draft**.
4. Select **Change status** > **Complete**.

The status of version 1 of this configuration changes from **Draft** to **Completed**. You can't change version 1 anymore. This version contains the configured data model and can be used as the basis for other ER configurations. Version 2 of this configuration is created and has a status of **Draft**. You can edit this version to adjust the **Questionnaire** data model.



For more information about versioning for ER configurations, see Electronic reporting (ER) overview.

> **Note**
>
> The configured data model is your abstract representation of the **Questionnaire** business domain and contains no relations to artifacts that are specific to Microsoft Dynamics 365 Finance.

Design a model mapping for the configured data model

As a user in the Electronic Reporting Developer role, you must create a new ER configuration that contains a model mapping component for the **Questionnaire** data model. Because this component implements the configured data model for Finance, it's Finance-specific. You must configure the model mapping component to specify the application objects that fill the configured data model with application data at runtime. To complete this task, you must be aware of the implementation details of the data structure of the **Questionnaire** business domain in Finance.

By completing the steps in the Import a new model mapping configuration section, you can import the required model mapping configuration from the provided XML file. Alternatively, you can complete the steps in the Create a new model mapping configuration section to design this model mapping from scratch.

Import a new model mapping configuration

1. Download the Questionnaires mapping.version.1.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Electronic reporting** workspace, select **Reporting configurations**.

4. On the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Questionnaires mapping.version.1.1.xml** file.
6. Select **OK** to import the configuration.

To continue, skip the next procedure, Create a new model mapping configuration.

Create a new model mapping configuration

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, select **Questionnaire model**.
3. Select **Create configuration**.
4. In the drop-down dialog box, follow these steps:
   a. In **New**, select **Model Mapping based on data model Questionnaires**.
   b. In **Name**, enter **Questionnaire mapping**.
   c. In **Data model definition**, select the **Root** definition.
   d. Select **Create configuration** to create the configuration.

Design a new model mapping component

1. On **Configurations**, in the configuration tree, select **Questionnaire mapping**.
2. Select **Designer** to open the list of mappings.
3. Select the **Questionnaires mapping** mapping that was automatically added for the **Root** definition.
4. Select **Designer** to start to configure the selected mapping.

A new mapping is automatically added for the **Root** definition. This mapping has the **To model** direction. Therefore, you can use this mapping to fill in a data model with required data.

Add data sources to access application tables

You must configure data sources to access the application tables that contain questionnaire details.

1. On **Model mapping designer**, in the **Data source types** pane, select **Dynamics 365 for Operations\Table records**.
2. Add a new data source that you use to access the KMCollection table, where every record represents a single questionnaire:
   a. In the **Data sources** pane, select **Add root**.
   b. In the dialog box, in **Name**, enter **Questionnaire**.
   c. In **Table**, enter **KMCollection**.
   d. Set the **Ask for query** option to **Yes**. You can then specify filtering options for this table in the system query dialog box at runtime.
   e. Select **OK** to add the new data source.

3. In the **Data source types** pane, select **Dynamics 365 for Operations\Table records**.

4. Add a new data source that you use to access the KMQuestion table, where every record represents a single question in a questionnaire:
   a. In the **Data sources** pane, select **Add root**.
   b. In the dialog box, in **Name**, enter **Question**.
   c. In **Table**, enter **KMQuestion**.
   d. Select **OK** to add the new data source.

5. In the **Data source types** pane, select **Dynamics 365 for Operations\Table records**.

6. Add a new data source that you use to access the KMAnswer table, where every record represents a single answer to a question in a questionnaire:
   a. In the **Data sources** pane, select **Add root**.
   b. In **Name**, enter **Answer**.
   c. In **Table**, enter **KMAnswer**.
   d. Select **OK** to add the new data source.

7. In the **Data source types** pane, select **Functions\Calculated field**.

8. Add a new calculated field that you use to access a record of the KMQuestionResultGroup table from every record of the parent KMCollection table:
   a. In the **Data sources** pane, select **Questionnaire**.
   b. Select **Add**.
   c. In the dialog box, in **Name**, enter **$ResultGroup**.
   d. Select **Edit formula**.
   e. In the ER formula editor, in **Formula**, enter **FIRSTORNULL(@.'<Relations'.KMQuestionResultGroup)** to use the path of the one-to-many relation between the KMCollection and KMQuestionResultGroup tables.
   f. Select **Save**, and close the formula editor.
   g. Select **OK** to add the new calculated field.

9. In the **Data source types** pane, select **Functions\Calculated field**.

10. Add a new calculated field that you use to access question records of the KMQuestion table from every record of the parent KMCollectionQuestion table:
    a. In the **Data sources** pane, select **Questionnaire**.
    b. Expand the **<Relations** node that contains one-to-many relations of the KMCollection table.
    c. Select the related **KMCollectionQuestion** table, and then select **Add**.
    d. In the dialog box, in the **Name** field, enter **$Question**.
    e. Select **Edit formula**.
    f. In the formula editor, in the **Formula** field, enter **FIRSTORNULL (FILTER(Question, Question.kmQuestionId = @.kmQuestionId))** to return the appropriate question records from the KMQuestion table.
    g. Select **Save**, and close the formula editor.

h. Select **OK** to add the new calculated field.

11. In the **Data source types** pane, select **Functions\Calculated field**.

12. Add a new calculated field that you use to access answer records of the KMAnswer table from every record of the parent KMQuestion table:
    a. In the **Data sources** pane, select **Questionnaire.<Relations.KMCollectionQuestion.$Question**, and then select **Add**.
    b. In the dialog box, in the **Name** field, enter **$Answer**.
    c. Select **Edit formula**.
    d. In the formula editor, in the **Formula** field, enter **FILTER (Answer, Answer.kmAnswerCollectionId = @.kmAnswerCollectionId)** to return the appropriate answer records from the KMAnswer table.
    e. Select **Save**, and close the formula editor.
    f. Select **OK** to add the new calculated field.

13. In the **Data source types** pane, select **Dynamics 365 for Operations\Table**.

14. Add a new data source that you use to access methods of the CompanyInfo table. Note that the **find()** method of this table returns a record that represents a company of the current Finance instance that this mapping is called in the context of.
    a. In the **Data sources** pane, select **Add root**.
    b. In the dialog box, in the **Name** field, enter **CompanyInfo**.
    c. In the **Table** field, enter **CompanyInfo**.
    d. Select **OK** to add the new data source.

Add data sources to access application enumerations

You must configure data sources to access application enumerations and compare their values with values of fields of the **Enumeration** type in the application tables. Use the result of the comparison to fill in appropriate fields of the data model.

1. On the **Model mapping designer** page, in the **Data source types** pane, select **Dynamics 365 for Operations\Enumeration**.

2. Add a new data source that you use to access values of the **EnumAppNoYes** enumeration:
    a. In the **Data sources** pane, select **Add root**.
    b. In the dialog box, in the **Name** field, enter **EnumAppNoYes**.
    c. In the **Enumeration** field, enter **NoYes**.
    d. Select **OK** to add the new data source.

3. In the **Data source types** pane, select **Dynamics 365 for Operations\Enumeration**.

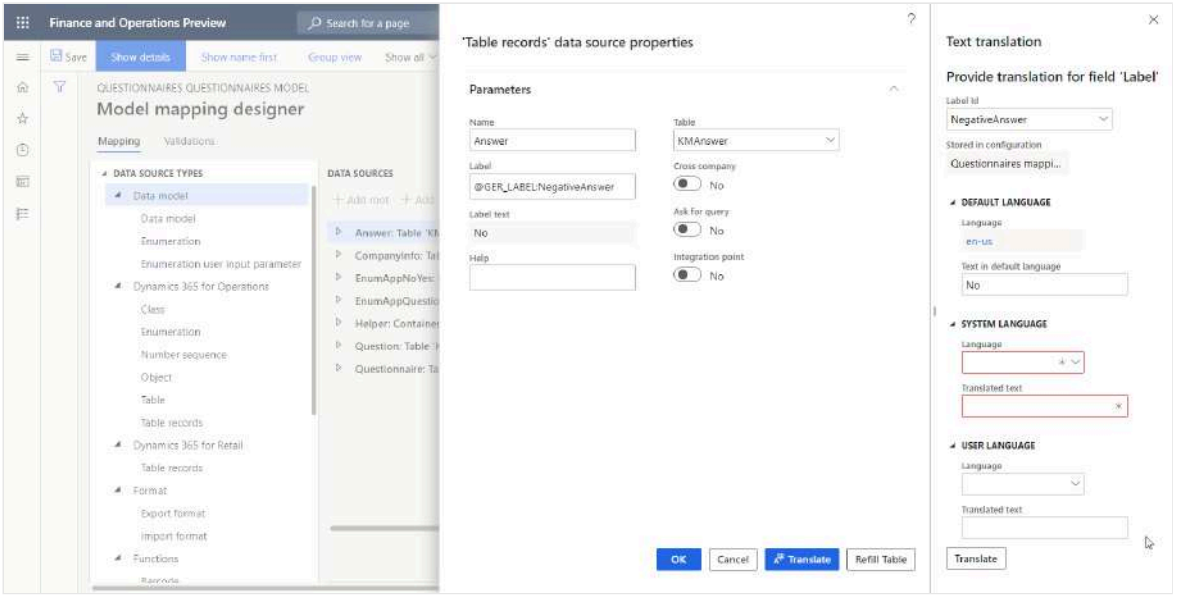
4. Add a new data source that you use to access the values of the **KMCollectionQuestionMode** enumeration:
    a. In the **Data sources** pane, select **Add root**.

b. In the dialog box, in the **Name** field, enter **EnumAppQuestionOrder**.
c. In the **Enumeration** field, enter **KMCollectionQuestionMode**.
d. Select **OK** to add the new data source.

Add ER labels to generate a report in a specified language

You can add ER labels to configure some of your data sources to return values that depend on the language that is defined in the context of the model mapping's call.

1. On the **Model mapping designer** page, in the **Data sources** pane, select **Answer**, and then select **Edit**.
2. Activate the **Label** field.
3. Select **Translate**.
4. In the **Text translation** dialog box, follow these steps:
   a. In the **Label Id** field, enter **PositiveAnswer**.
   b. In the **Text in default language** field, enter **Yes**.
   c. Select **Translate**.
   d. In the **Label Id** field, enter **NegativeAnswer**.
   e. In the **Text in default language** field, enter **No**.
   f. Select **Translate**.
5. Close the **Text translation** dialog box.
6. Select **Cancel**.

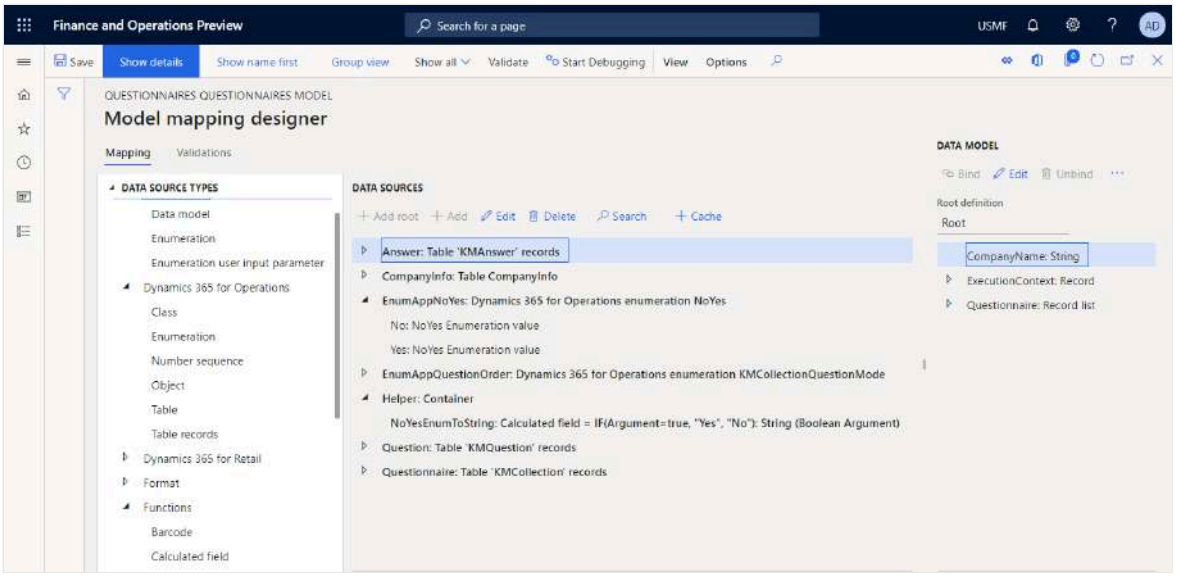


You entered ER labels only for the default language. For information about how to translate ER labels into other languages, see Design multilingual reports.

Add a data source to transform the results of comparing enumeration values to a text value

Because you must transform the results of the comparison between enumeration values and text values several times for different sources, it's a good idea to configure this logic as a single data source. However, to make this data source reusable, you must then configure it as the parameterized data source. For more information, see Support parameterized calls of ER data sources of the Calculated field type.

1. On the **Model mapping designer** page, in the **Data source types** pane, select **General\Empty container**.
2. Add a new container data source:
   a. In the **Data sources** pane, select **Add root**.
   b. In the dialog box, in the **Name** field, enter **Helper**.
   c. Select **OK** to add the new container data source.
3. In the **Data source types** pane, select **Functions\Calculated field**.
4. Add a new data source:
   a. In the **Data sources** pane, select **Helper**.
   b. Select **Add**.
   c. In the dialog box, in the **Name** field, enter **NoYesEnumToString**.
   d. Select **Edit formula**.
   e. In the formula editor, select **Parameters**.
   f. Follow these steps to specify parameters for the configured expression:
      i. Select **New**.
      ii. In the dialog box, in the **Name** field, enter **Argument**.
      iii. In the **Type** field, select the **Boolean** data type.
      iv. Select **OK**.
   g. In the **Formula** field, enter **IF (Argument = true, @"GER_LABEL:PositiveAnswer", @"GER_LABEL:NegativeAnswer")** to return the text of the appropriate ER label, depending on the language of the execution context and value of the specified parameter.
   h. Select **Save**, and close the formula editor.
   i. Select **OK** to add the new data source.



Bind data sources to data model fields

You must bind the configured data sources to the fields of the data model to specify how the data model is filled in with application data at runtime.

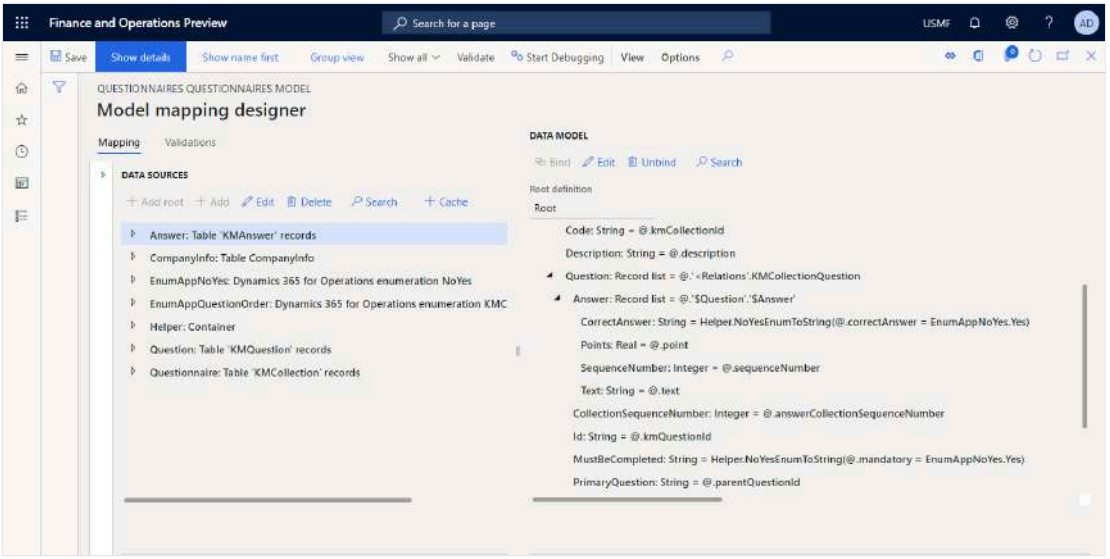
1. On the **Model mapping designer** page, in the **Data model** pane, select **CompanyName**.
2. In the **Data sources** pane, expand **CompanyInfo**, and then follow these steps:
   a. Expand the **CompanyInfo.find()** node that represents the **find()** method of the CompanyInfo table.
   b. Select **CompanyInfo.find().Name**.
   c. Select **Bind** to fill in the name of the company that the configured model mapping is called in the context of at runtime.
3. In the **Data model** pane, select **Questionnaire**.
4. In the **Data sources** pane, select **Questionnaire**, and then select **Bind** to fill in questionnaire records.
5. In the **Data model** pane, expand **Questionnaire**, and then follow these steps:
   a. In the **Data model** pane, select **Active**.
   b. In the **Data model** pane, select **Edit**.
   c. In the **Formula** field, enter **Helper.NoYesEnumToString (@.Active = EnumAppNoYes.Yes)** to fill the text-dependent and language-dependent result of the comparison between enumeration values.
6. Continue to bind data sources to data model fields in the same manner until you achieve the following result.

Expand table

Field path	Data type	Action	Binding expression
CompanyName	String	Bind	CompanyInfo.'find()'.Name
Questionnaire	Record list	Bind	Questionnaire
Questionnaire\Active	String	Edit	Helper.NoYesEnumToString(@.active = EnumAppNoYes.Yes)
Questionnaire\Code	String	Bind	@.kmCollectionId
Questionnaire\Description	String	Bind	@.Description
Questionnaire\QuestionnaireType	String	Bind	@.'>Relations'.kmCollectionTypeId.Desc
Questionnaire\QuestionOrder	String	Edit	CASE (@.questionMode, EnumAppQuestionOrder.Conditional, "Conditional", EnumAppQuestionOrder.Random, "Ran (percentage in questionnaire)", EnumAppQuestionOrder.RandomGroup "Random (percentage in result groups)" EnumAppQuestionOrder.Sequence, "Sequential", "")
Questionnaire\ResultsGroup	Record		
Questionnaire\ResultsGroup\Code	String	Bind	@.'$ResultGroup'.kmQuestionResultGrc
Questionnaire\ResultsGroup\Description	String	Bind	@.'$ResultGroup'.description
Questionnaire\ResultsGroup\MaxNumberOfPoints	Real	Bind	@.'$ResultGroup'.maxPoint
Questionnaire\Question	Record list	Bind	@.'<Relations'.KMCollectionQuestion
Questionnaire\Question\CollectionSequenceNumber	Integer	Bind	@.answerCollectionSequenceNumber
Questionnaire\Question\Id	String	Bind	@.kmQuestionId
Questionnaire\Question\MustBeCompleted	String	Edit	Helper.NoYesEnumToString(@.mandatc EnumAppNoYes.Yes)
Questionnaire\Question\PrimaryQuestion	String	Bind	@.parentQuestionId
Questionnaire\Question\SequenceNumber	Integer	Bind	@.SequenceNumber
Questionnaire\Question\Text	String	Bind	@.'$Question'.text
Questionnaire\Question\Answer	Record list	Bind	@.'$Question'.'$Answer'
Questionnaire\Question\Answer\CorrectAnswer	String	Edit	Helper.NoYesEnumToString(@.correctA = EnumAppNoYes.Yes)
Questionnaire\Question\Answer\Points	Real	Bind	@.point

Field path	Data type	Action	Binding expression
Questionnaire\Question\Answer\SequenceNumber	Integer	Bind	@.sequenceNumber
Questionnaire\Question\Answer\Text	String	Bind	@.text

The following illustration shows the final state of the configured model mapping on the **Model mapping designer** page.

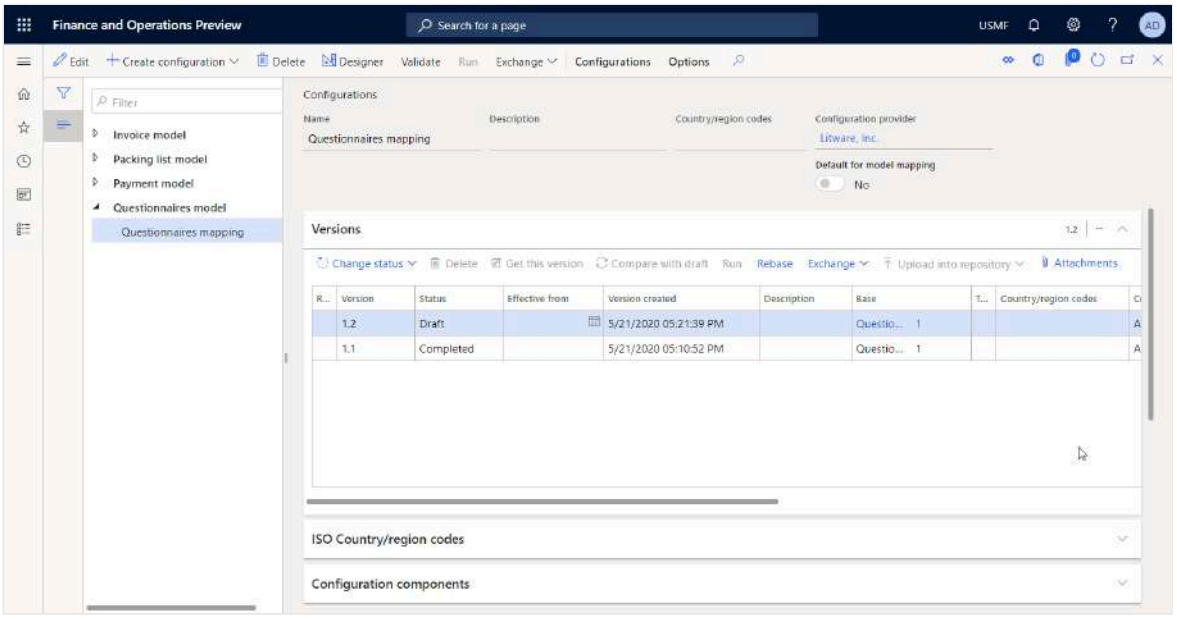


7. Save your changes.
8. Close the **Model mapping designer** page.

Complete the design of the model mapping

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, select **Questionnaire mapping**.
3. On the **Versions** FastTab, select the configuration version that has a status of **Draft**.
4. Select **Change status** > **Complete**.

The status of version 1.1 of this configuration changes from **Draft** to **Completed**. You can no longer change version 1.1. This version contains the configured model mapping and can be used as the basis for other ER configurations. Version 1.2 of this configuration is created and has a status of **Draft**. You can edit this version to adjust the **Questionnaire mapping** configuration.



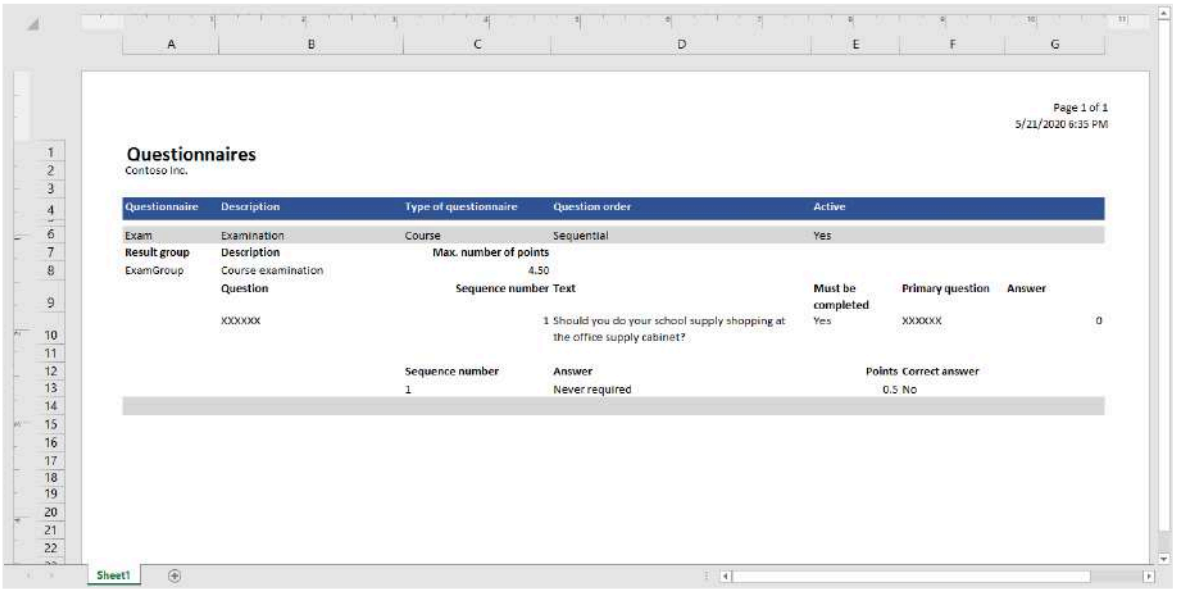
> **Note**
>
> The configured model mapping is your Finance-specific implementation of the abstract data model that represents the **Questionnaire** business domain.

Design a template for a custom report

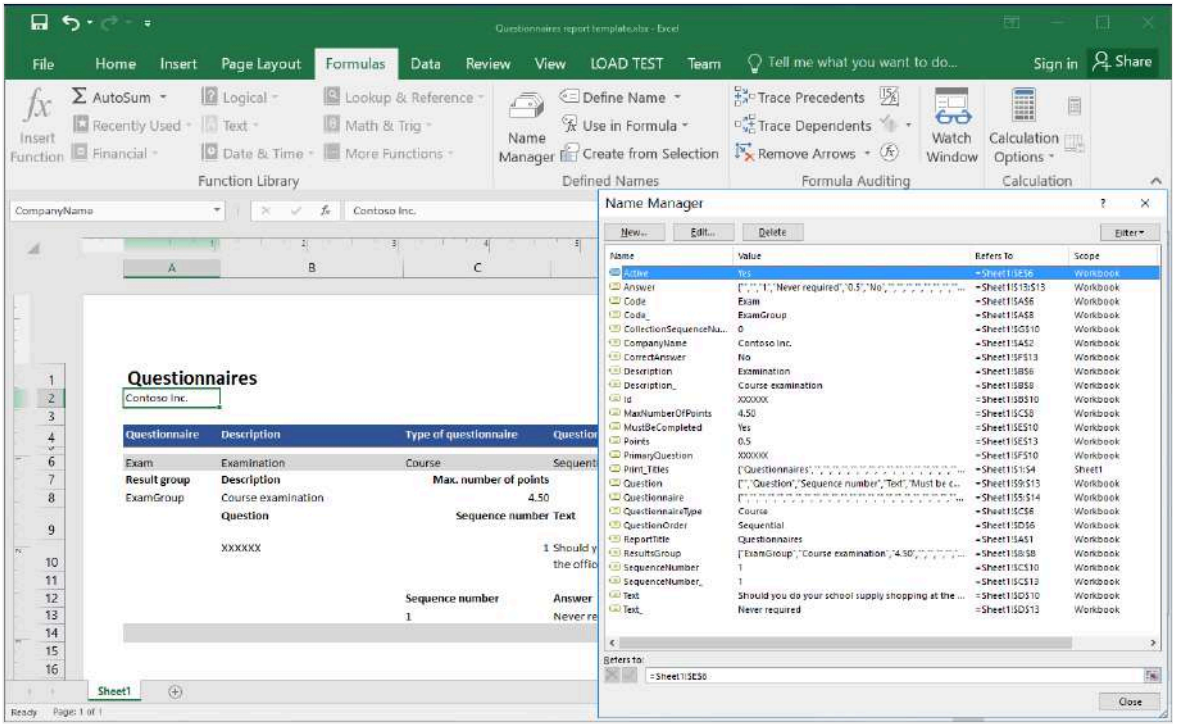
The ER framework uses predefined templates to generate reports in Microsoft Office formats (Excel workbooks or Word documents). While generating the required report, the framework fills the template with the required data according to the configured dataflow. Therefore, you must first design a template for your custom report. Design this template as an Excel workbook, the structure of which represents the layout of a custom report. Name every Excel item that you plan to fill with required data.

1. Download the Questionnaires report template.xlsx file, and save it to your local computer.
2. Open the file in Excel, and review the structure of the workbook.

As the following illustration shows, the downloaded template is designed to print specified questionnaires that present a questionnaire's questions together with appropriate answers.

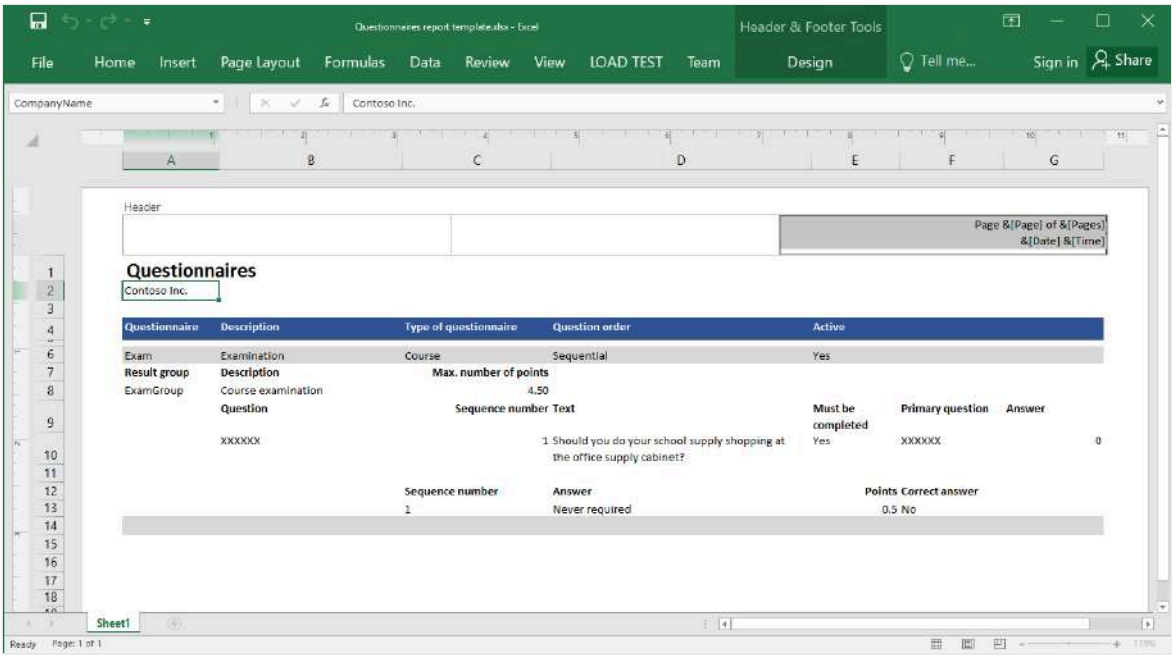


Excel names are added to this template to fill in the questionnaire details. Use Name Manager to review the Excel names.



Report labels are added as fixed text in the English language. You can replace the report labels with new Excel names that fill in the labels with language-dependent text by using the ER format labels, as you did for language-dependent expressions in the configured model mapping. In this case, ER labels must be added in the editable ER format.

As the following illustration shows, the custom report header is specified to enable Excel to do paging.



Design a format

As a user in the Electronic Reporting Functional Consultant role, you must create a new ER configuration that contains a format component. You must configure the format component to specify how a report template is filled with required data at runtime.

By completing the steps in the Import a designed format configuration section, you can import the required format from the provided XML file. Alternatively, you can complete the steps in the Create a new format configuration section to design this format from scratch.

Import a designed format configuration

1. Download the Questionnaires format.version.1.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the Action pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Questionnaires format.version.1.1.xml** file.
6. Select **OK** to import the configuration.

To continue, skip the next procedure, Create a new format configuration.

Create a new format configuration

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, select **Questionnaire model**.
3. Select **Create configuration**.
4. In the drop-down dialog box, follow these steps:

a. In **New**, select **Format based on data model Questionnaires**.

b. In **Name**, enter **Questionnaire report**.

c. In **Data model version**, select **1**.

> **Note**
>
> - If you select a specific version of the base data model, the structure of the corresponding version of the data model appears as the structure of the **Model** data source in the format that you create.
> - You can leave this field blank. In that case, the structure of the **Draft** version of the data model appears as the structure of the **Model** data source in the format that you create. You can then adjust your model and immediately see those adjustments in your format. This approach might improve the efficiency of ER solution design when you configure your data model, model mapping, and format simultaneously.
> - If you select a specific version of the base data model, you can switch to using the **Draft** version later, when you start to edit a format.

d. In **Data model definition**, select the **Root** definition.

5. Select **Create configuration** to create the configuration.

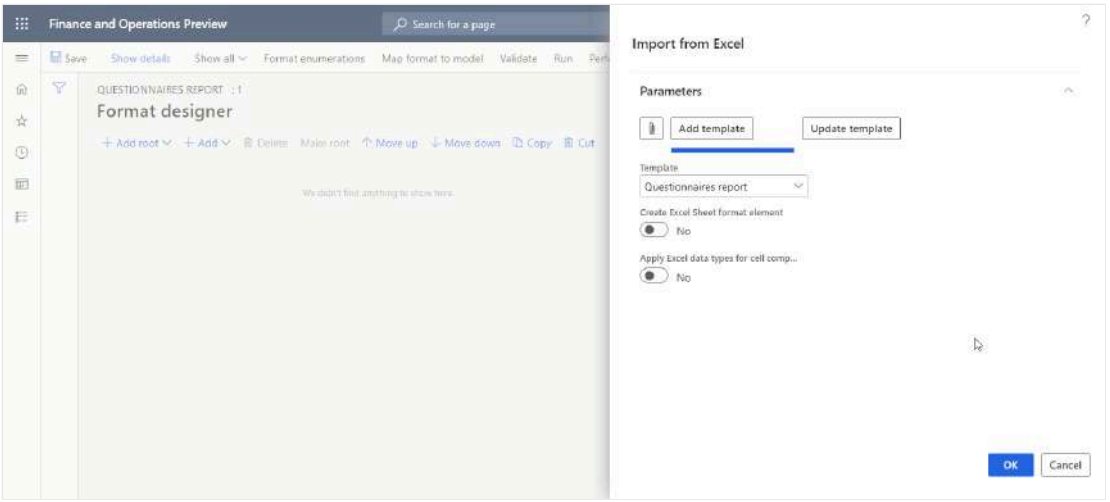
Import a report template

1. On **Configurations**, in the configuration tree, select **Questionnaire report**.

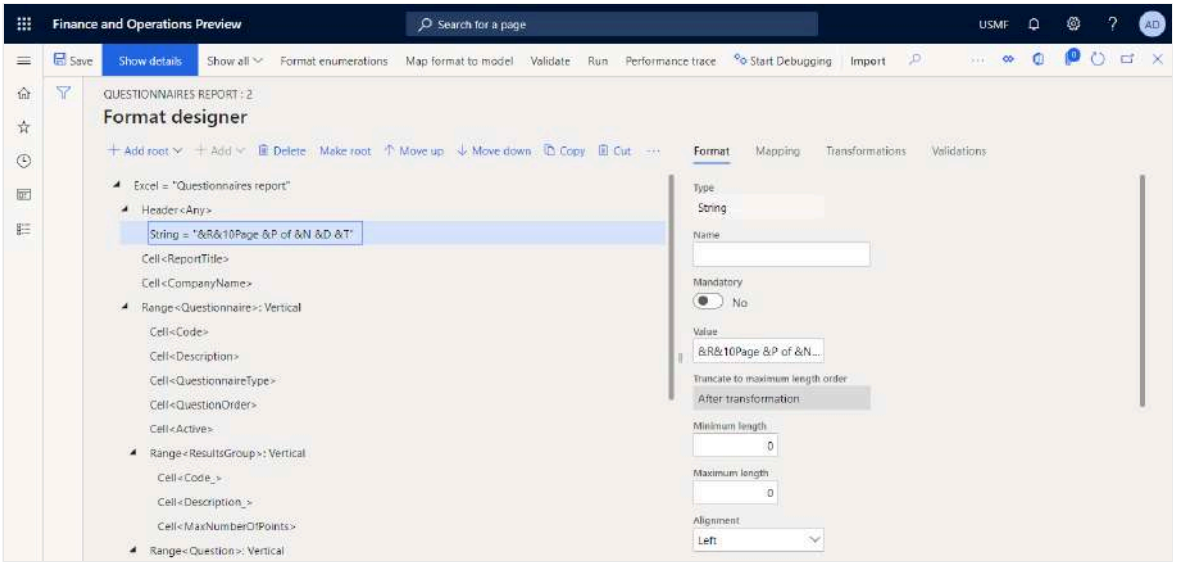
2. Select **Designer** to start to configure a custom format.

3. On **Format designer**, on the Action Pane, select **Import** > **Import from Excel**.

4. In the dialog box, follow these steps:
   a. Select **Add template**.
   b. Find and select the locally saved **Questionnaires report template.xslx** file, and then select **Open**.
   c. Select **OK** to import the template.



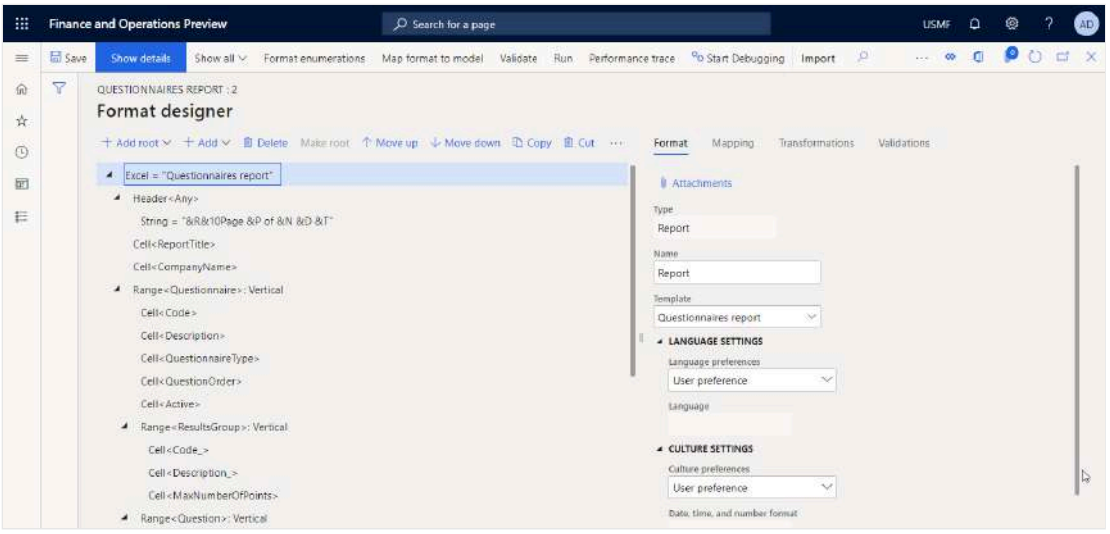
The **Excel\File** format element is automatically added to the editable format as a root element. Additionally, either the **Excel\Range** format element or the **Excel\Cell** format element is automatically added for every recognized Excel name of the imported template. The **Excel\Header** format that has the nested **String** element is automatically added to reflect the header settings of the imported template.



Configure a format

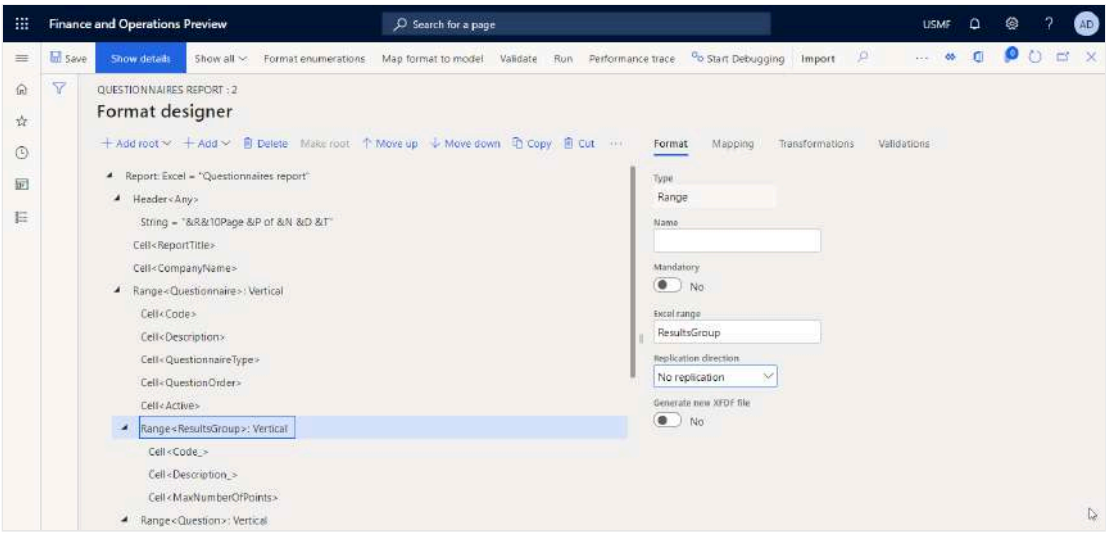
1. On **Format designer**, in the format tree, select the **Excel** root element.
2. On the **Format** tab, in the **Name** field, enter **Report**.
3. In the **Language preference** field, select **User preference** to run the report in the user's preferred language.
4. In the **Culture preference** field, select **User preference** to run the report in the user's preferred culture.

   For information about how to specify the language and culture contexts for an ER process, see Design multilingual reports.



5. In the format tree, expand the root node, and then select **ResultsGroup**.

6. On the **Format** tab, in the **Replication direction** field, select **No replication**, because you don't expect to have multiple result groups for a single questionnaire.



7. Select **Save**.

Define the data binding for a report title

You must specify a data binding for a format element that fills in the title of a generated report.

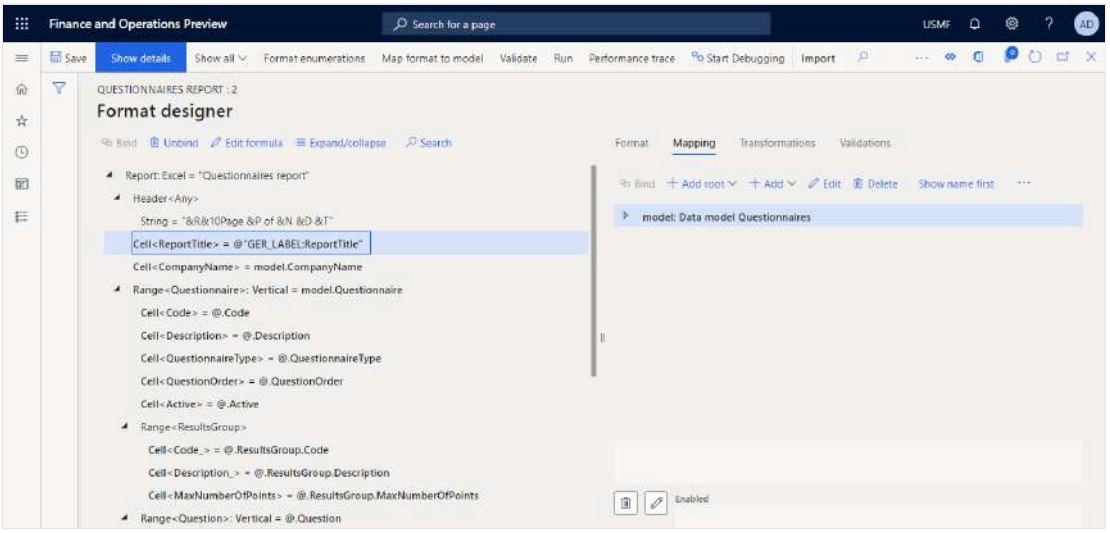
1. On **Format designer**, on the **Mapping** tab, select the **Report\ReportTitle** element.

2. Select **Edit formula**.

3. In the formula editor, select **Translate**.

4. In the **Text translation** dialog box, follow these steps:
   a. In the **Label ID** field, enter **ReportTitle**.
   b. In the **Text in default language** field, enter **Questionnaires report**.
   c. Select **Translate**, and then select **Save**.
   d. Select **Translate** to close the **Text translation** dialog box.

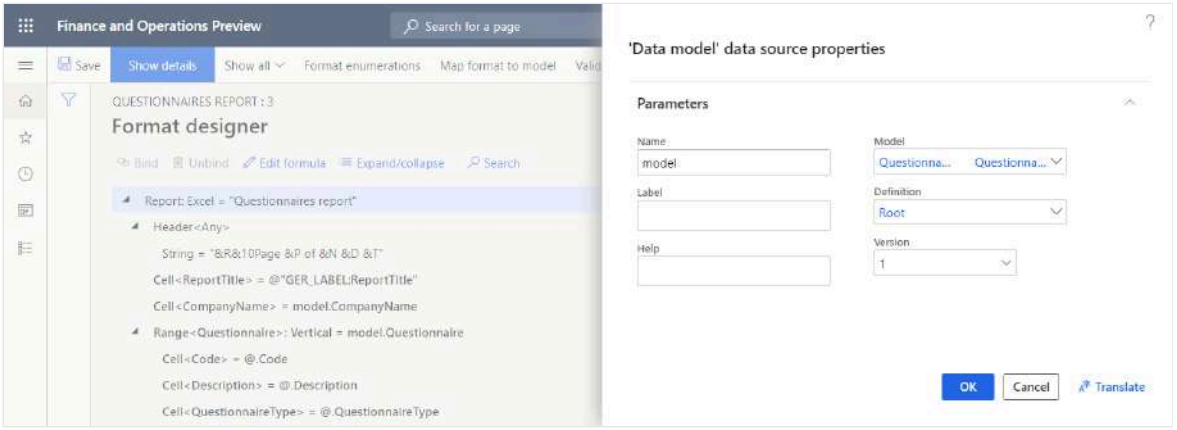
5. Close the formula editor.



Use this technique to make all other labels of the current template language-dependent. For information about how to translate the added labels of a single ER configuration into all supported languages, see Design multilingual reports.

Review model data source

1. On the **Format designer** page, on the **Mapping** tab, select the **model** data source that represents the base data model of this ER format.
2. Select **Edit**.
3. Review the information in the **Data source properties** dialog box. This data source represents version 1 of the **Questionnaires** data model component that resides in the **Questionnaires model** ER configuration.



Bind format elements to data source fields

To specify how a template is filled in at runtime, bind every format element that is associated with an appropriate Excel name to a single field of this format's data source.

1. On the **Format designer** page, in the format tree, select the **Report\CompanyName** format element.

2. On the **Mapping** tab, select the **model.CompanyName** data source field of the **String** type.

3. Select **Bind** to enter a company name in a template.

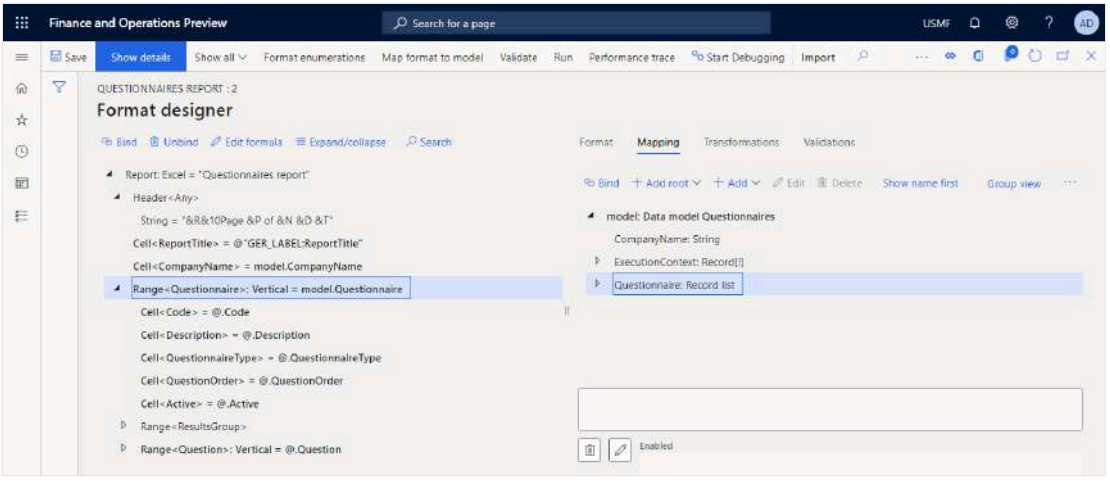
4. In the format tree, select the **Report\Questionnaire** element.

5. On the **Mapping** tab, select the **model.Questionnaire** data source field of the **Record list** type.

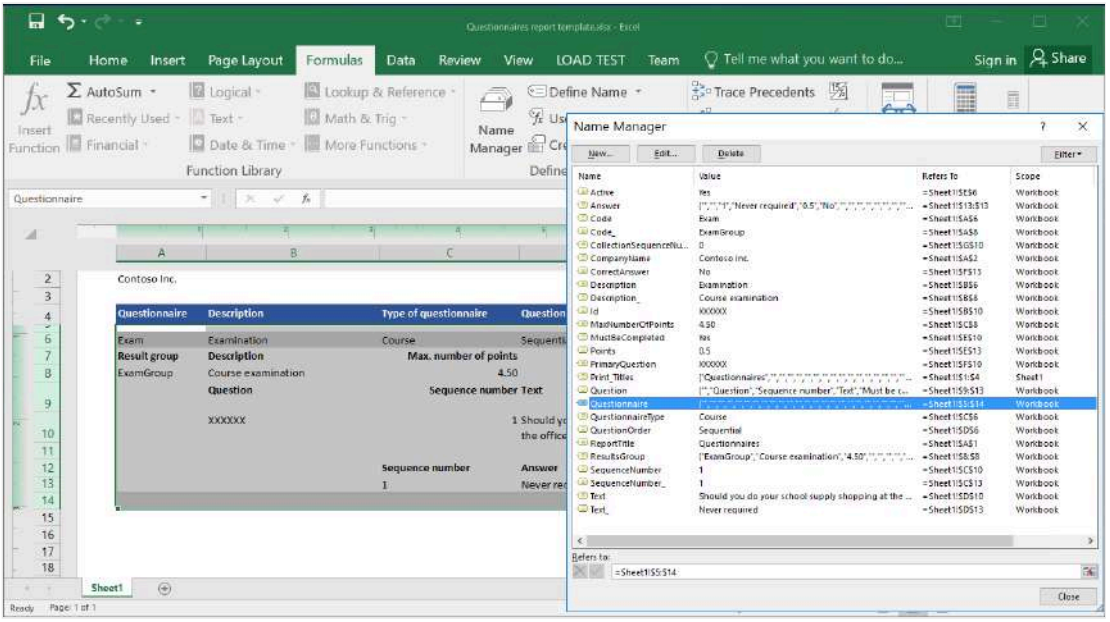
6. Select **Bind**.

7. Select **Show details** to see more details for format elements.

   The **Questionnaire** range format element is configured as vertically replicated. When it's bound to a data source of the **Record list** type, the appropriate **Questionnaire** range of the Excel template is repeated for every record of the bound data source.



Because the **Questionnaire** range of the Excel template is defined between rows 5 through 14, these rows are repeated for every reported questionnaire.



8. Configure similar bindings for the remaining format elements, as described in the following table.

> **Note**
>
> In this table, the information in the "Data source path" column assumes that the relative path ER feature is turned on.

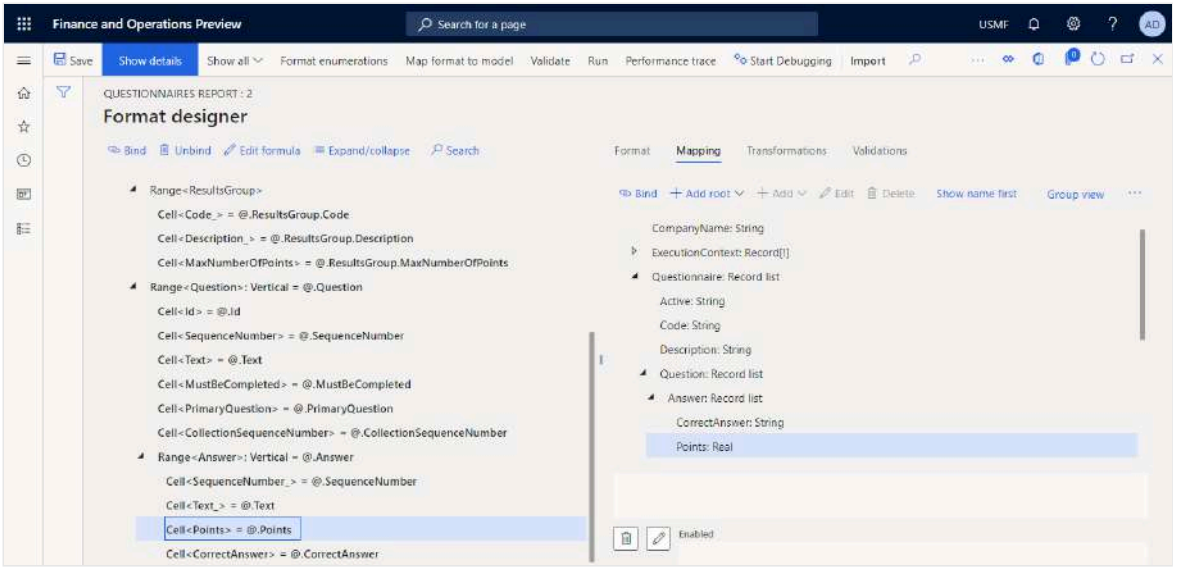
Expand table

Format element path	Data source path
Excel\ReportTitle	**@"GER_LABEL:ReportTitle"**
Excel\CompanyName	**model.CompanyName**
Excel\Questionnaire	**model.Questionnaire**
Excel\Questionnaire\Active	**@.Active**, where @ is **model.Questionnaire**
Excel\Questionnaire\Code	**@.Code**
Excel\Questionnaire\Description	**@.Description**
Excel\Questionnaire\QuestionnaireType	**@.QuestionnaireType**
Excel\Questionnaire\QuestionOrder	**@.QuestionOrder**
Excel\Questionnaire\ResultsGroup\Code_	**@.ResultsGroup.Code**
Excel\Questionnaire\ResultsGroup\Description_	**@.ResultsGroup.Description**
Excel\Questionnaire\ResultsGroup\MaxNumberOfPoints	**@.ResultsGroup.MaxNumberOfPoint**
Excel\Questionnaire\Question	**@.Question**
Excel\Questionnaire\Question\CollectionSequenceNumber	**@.CollectionSequenceNumber**, where @ is **model.Questionnaire.Question**
Excel\Questionnaire\Question\Id	**@.Id**
Excel\Questionnaire\Question\MustBeCompleted	**@.MustBeCompleted**
Excel\Questionnaire\Question\PrimaryQuestion	**@.PrimaryQuestion**
Excel\Questionnaire\Question\SequenceNumber	**@.SequenceNumber**
Excel\Questionnaire\Question\Text	**@.Text**
Excel\Questionnaire\Question\Answer	**@.Answer**
Excel\Questionnaire\Question\Answer\CorrectAnswer	**@.CorrectAnswer**, where @ is **model.Questionnaire.Answer**
Excel\Questionnaire\Question\Answer\Points	**@.Points**

Format element path	Data source path
Excel\Questionnaire\Question\Answer\Text	**@.Text**

9. When you finish, select **Save**.

The following illustration shows the final state of the configured data bindings on the **Format designer** page.



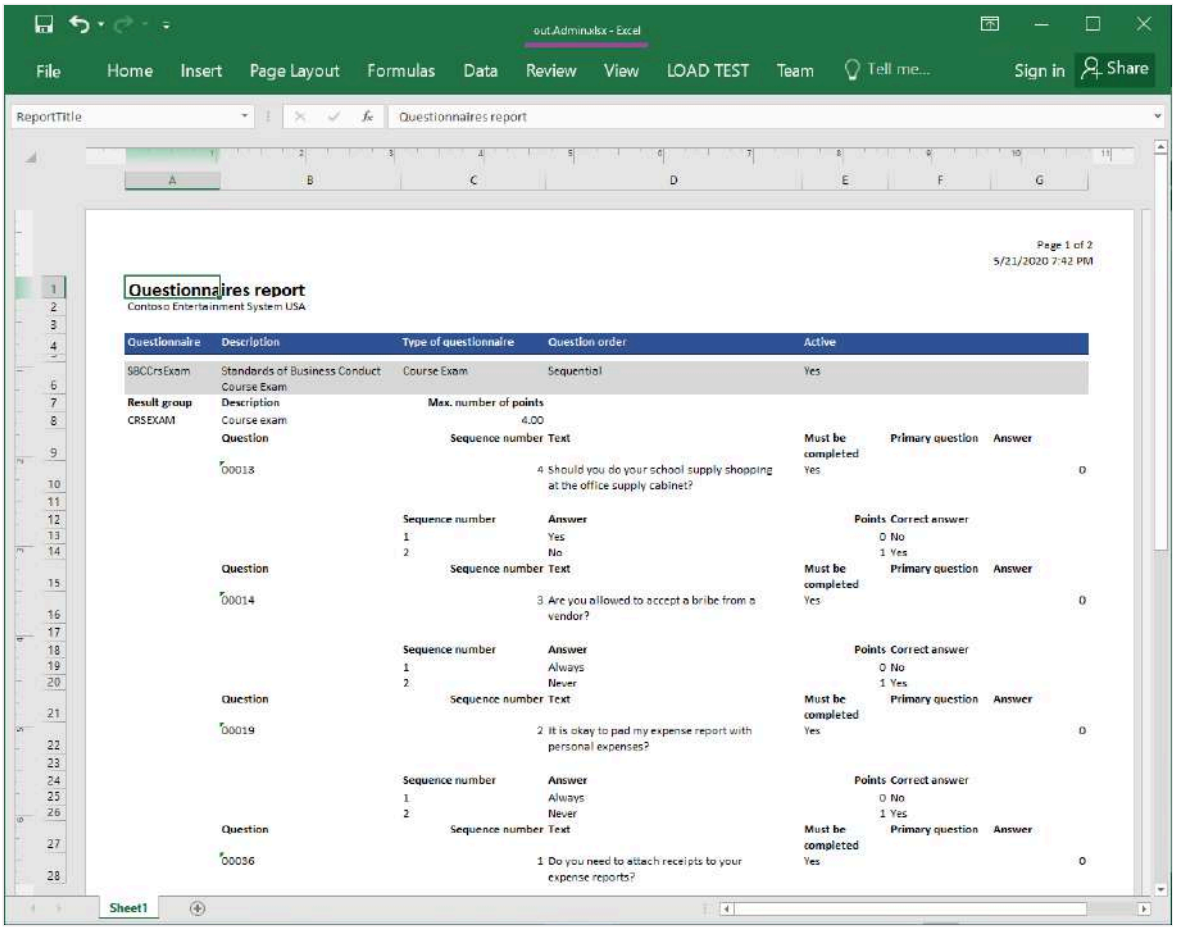
> **Important**
>
> The whole collection of specified data sources and bindings represents a format mapping component of the configured format. This format mapping is called when you run the configured format for report generation.

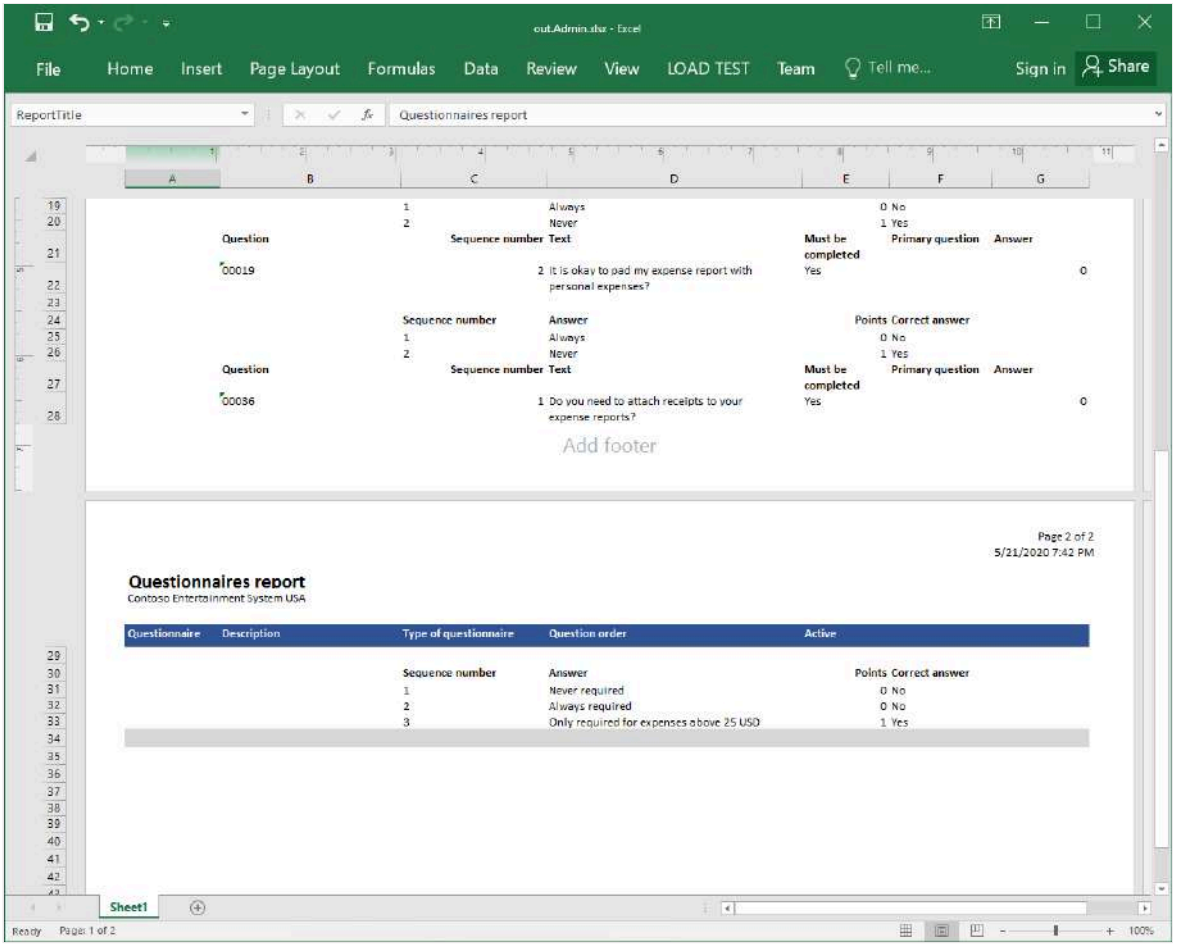
Run a designed format from ER

You can now run a designed format for testing purposes from the **Configurations** page.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configuration** page, in the configuration tree, expand **Questionnaire model**, and then select **Questionnaire report**.
3. Select **Designer** for the format version that has a status of **Draft**.
4. On the **Format designer** page, select **Run**.
5. In the **ER parameters** dialog box, on the **Records to include** FastTab, configure the filtering option so that only the **SBCCrsExam** questionnaire is included.
6. Select **OK** to confirm the filter.
7. Select **OK** to run the report.
8. Review the generated report.

By default, a generated report is delivered as an Excel file that you can download. The following illustrations show two pages of the generated report in Excel format.





Tune a designed format

Modify a format to change the name of a generated document

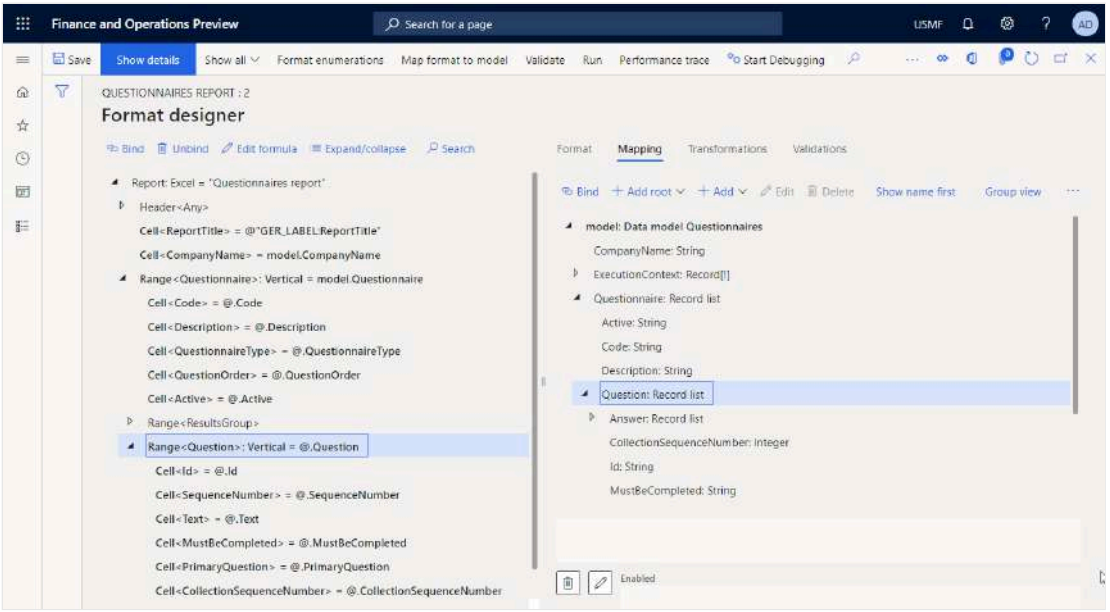
By default, the alias of the current user names a generated document. When you modify the format, you can change this behavior so that a generated document is named based on your custom logic. For example, the name of a generated document can be based on the current session date and time, and on the report's title.

1. On **Format designer**, select the **Report** root item.
2. On the **Mapping** tab, select **Edit file name**.
3. In the **Formula** field, enter `CONCATENATE (\@"GER_LABEL:ReportTitle", " - ", DATETIMEFORMAT(SESSIONNOW(), "yyyy-MM-dd hh-mm-ss"))`.
4. Select **Save**, and close the formula editor.
5. Select **Save**.

Modify a format to change the order of questions

The questions don't appear in the correct order in a generated report. You can change the order by modifying the format.

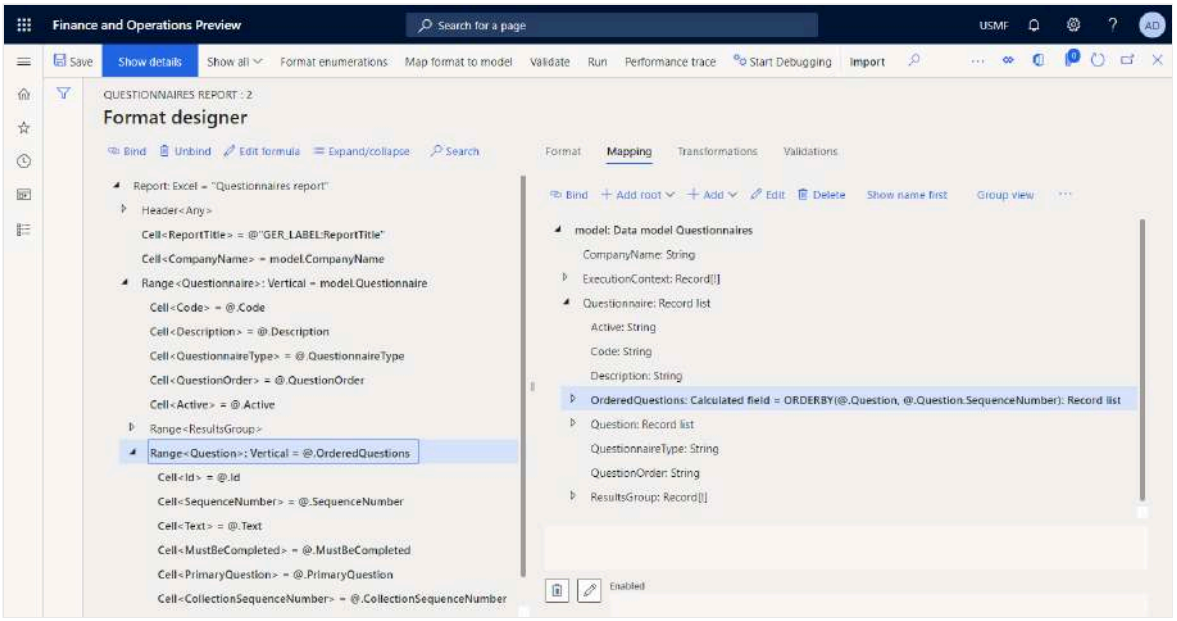
1. On **Format designer**, select the **Report** root item.
2. On the **Mapping** tab, in the format tree, expand **Report\Questionnaire\Question**.



3. On the **Mapping** tab, select **model.Questionnaire**.
4. Select **Add** > **Functions\Calculated field**, and then, in the **Name** field, enter **OrderedQuestions**.
5. Select **Edit formula**.
6. In the formula editor, in the **Formula** field, enter **ORDERBY (model.Questionnaire.Question, model.Questionnaire.Question.SequenceNumber)** to

order the list of questions of the current questionnaire by the sequence order number.

7. Select **Save**, and close the formula editor.
8. Select **OK** to complete the entry of a new calculated field.
9. On the **Mapping** tab, select **model.Questionnaire.OrderedQuestions**.
10. In the format tree, select **Excel\Questionnaire\Question**.
11. Select **Bind**, and then confirm that the current **model.Questionnaire.Questions** path is replaced by the new **model.Questionnaire.OrderedQuestions** path in all bindings of nested elements.
12. Select **Save**.

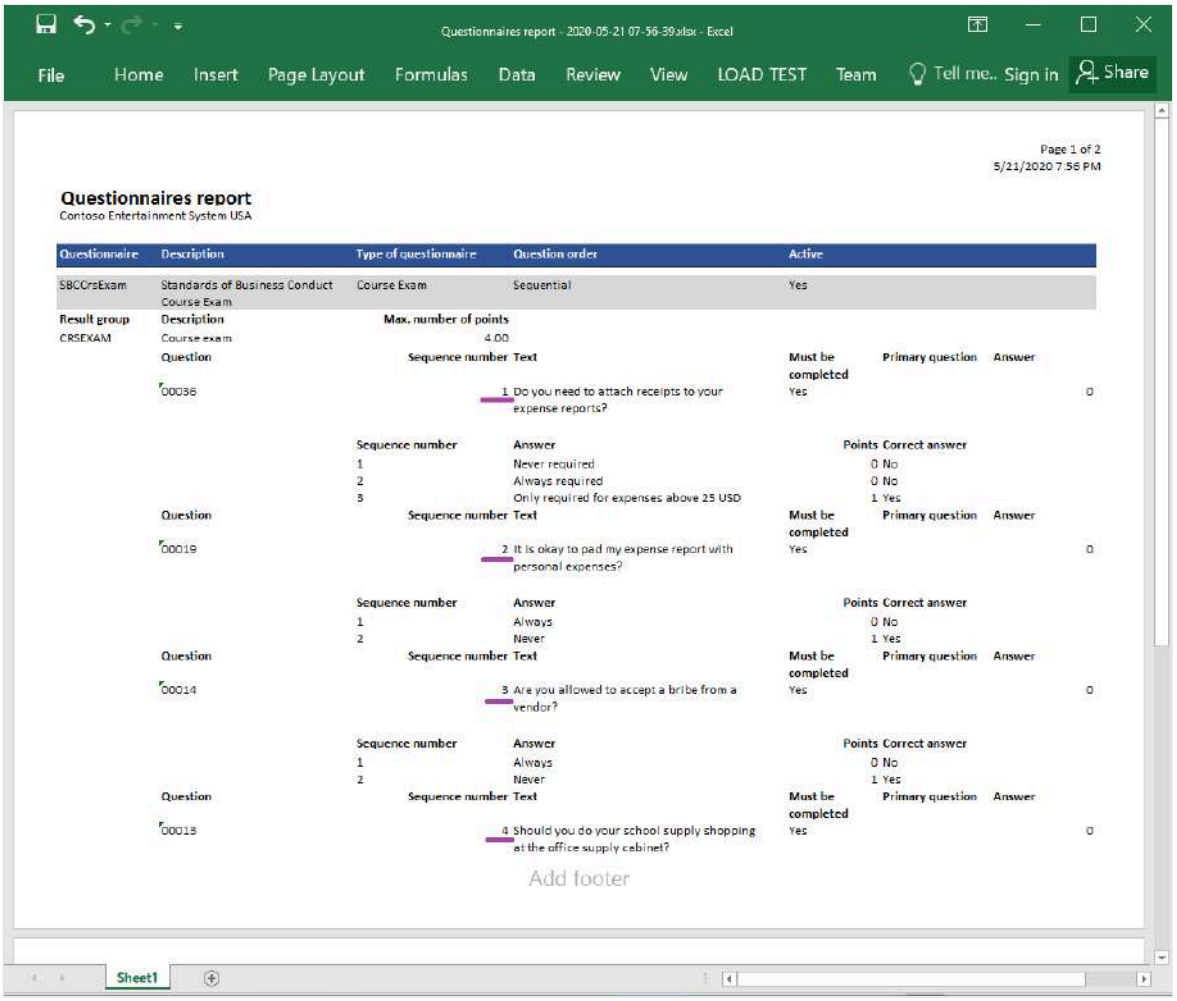


Run a modified format from ER

You can now run a modified format for testing purposes from the ER framework.

1. On the **Format designer** page, select **Run**.
2. In the **ER parameters** dialog box, on the **Records to include** FastTab, configure the filtering option so that only the **SBCCrsExam** questionnaire is included.
3. Select **OK** to confirm the filter.
4. Select **OK** to run the report.
5. Review the generated report.

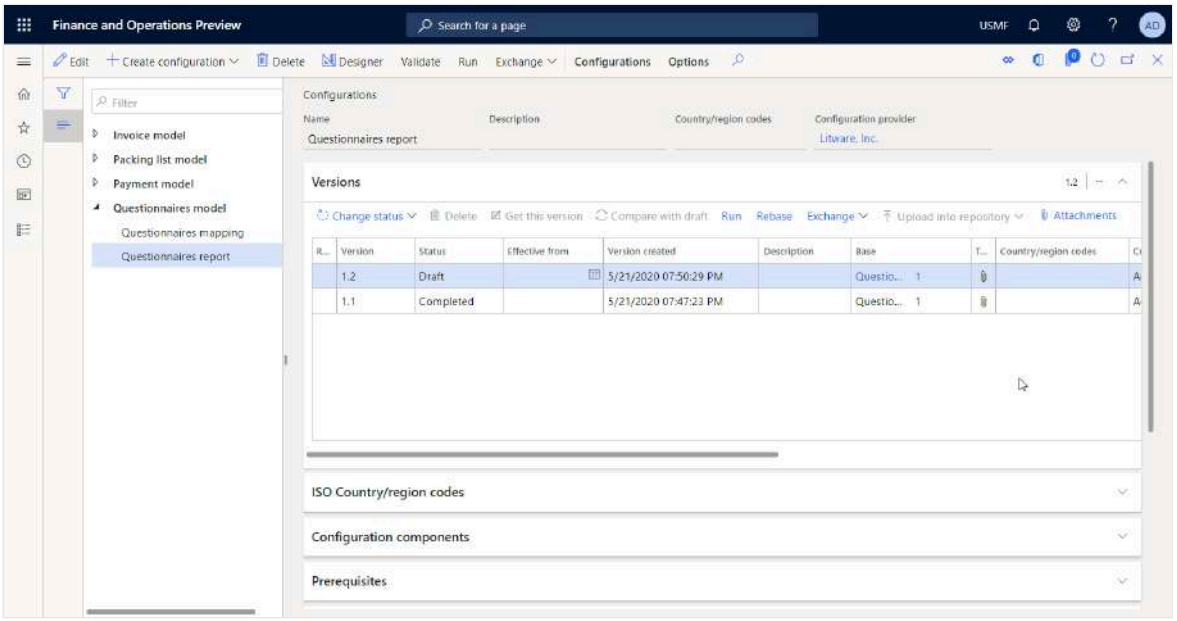
The following illustration shows a generated report in Excel format where the questions are correctly ordered.



Complete the format design

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, expand **Questionnaire model**, and then select **Questionnaire report**.
3. On the **Versions** FastTab, select the configuration version that has a status of **Draft**.
4. Select **Change status** > **Complete**.

The status of version 1.1 of this configuration changes from **Draft** to **Completed**. You can no longer change version 1.1. This version contains the configured format and can be used to print your custom report. Version 1.2 of this configuration is created and has a status of **Draft**. You can edit this version to adjust the format of your **Questionnaire** report.



> **Note**
>
> The configured format is your design of the **Questionnaire** report and contains no relations to the Finance-specific artefacts.

Develop application artifacts to call the designed report

As a user in the System Administrator role, you must develop new logic so that the configured ER format can be called from the application user interface (UI) to generate your custom report. Currently, ER doesn't offer any capability for configuring this type of logic. Therefore, some engineering work is required.

To develop the new logic, you must deploy a topology that supports continuous build. For more information, see Deploy topologies that support continuous build and test automation. You must also have access to the development environment for this topology. For more information about the available ER API, see ER framework API.

Modify source code

Add a data contract class

Add the new **QuestionnairesErReportContract** class to your Visual Studio project. Write code that specifies the data contract to use for running the configured ER format.

X++

```
/// <summary>
///     This class is the data contract class for the
<c>QuestionnairesErReportDP</c> class.
/// </summary>
/// <remarks>
```

```
///    This is the data contract class for the Questionnaires ER report.
/// </remarks>
[
    DataContractAttribute,

SysOperationContractProcessingAttribute(classStr(QuestionnairesErReportUIBuilder))
    ]
    public class QuestionnairesErReportContract extends
ERFormatMappingRunBaseContract implements SysOperationValidatable
{
    ERFormatMappingId formatMapping;

    /// <summary>
    ///    Validates the report parameters.
    /// </summary>
    /// <returns>
    ///    true if no errors; otherwise, false.
    /// </returns>
    public boolean validate()
    {
        boolean ret = true;
        if (!formatMapping)
        {
            ret = checkFailed(strFmt("@SYS26332", new
SysDictType(extendedTypeNum(ERFormatMappingId)).label()));
        }
        return ret;
    }
    [
        DataMemberAttribute('FormatMapping'),

SysOperationLabelAttribute(literalstr("@ElectronicReporting:FormatMapping")),

SysOperationHelpTextAttribute(literalstr("@ElectronicReporting:FormatMapping"))
    ]
    public ERFormatMappingId parmFormatMapping(ERFormatMappingId _formatMapping =
formatMapping)
    {
        formatMapping = _formatMapping;
        return formatMapping;
    }
}
```

Add a UI builder class

Add the new **QuestionnairesErReportUIBuilder** class to your Visual Studio project. Write code to generate a runtime dialog box that looks up the format mapping ID of the ER format to run. The provided code looks up only ER formats that contain a data source of the **Data model** type that refers to the **Questionnaires** data model by using the **Root** definition.

> ⓘ Note
>
> Alternatively, you can use ER integration points to filter ER formats. For more information, see API to show a format mapping lookup.

```
X++
```

```
/// <summary>
/// The UIBuilder class for Questionnaires ER report
/// </summary>
class QuestionnairesErReportUIBuilder extends SysOperationAutomaticUIBuilder
{
    public const str ERQuestionnairesModel = 'Questionnaires';
    public const str ERQuestionnairesDataContainer = 'Root';

    /// <summary>
    /// Action after build of the dialog UI.
    /// </summary>
    public void postBuild()
    {
        DialogField formatMapping;
        super();
        formatMapping = this.bindInfo().getDialogField(this.dataContractObject(),
            methodStr(QuestionnairesErReportContract, parmFormatMapping));
        formatMapping.registerOverrideMethod(
            methodStr(FormReferenceControl, lookupReference),
            methodStr(QuestionnairesErReportUIBuilder, formatMappingLookup),
            this);
    }

    /// <summary>
    /// Performs the lookup form for format mapping.
    /// </summary>
    /// <param name="_referenceGroupControl">
    /// The control to perform lookup form.
    /// </param>
    public void formatMappingLookup(FormReferenceControl _referenceGroupControl)
    {
        ERObjectsFactory::createFormatMappingTableLookupForControlAndModel(
            _referenceGroupControl,
            ERQuestionnairesModel,
            ERQuestionnairesDataContainer).performFormLookup();
    }
}
```

Add a data provider class

Add the new **QuestionnairesErReportDP** class to your Visual Studio project. Write code that introduces the data provider to use for running the configured ER format. The provided code includes only the data contract for this data provider.

X++

```
/// <summary>
/// Data provider class for Questionnaires ER report.
/// </summary>
public class QuestionnairesErReportDP
{
    QuestionnairesErReportContract contract;
    public static QuestionnairesErReportDP construct()
    {
        QuestionnairesErReportDP  dataProvider;
        dataProvider = new QuestionnairesErReportDP();
        return dataProvider;
```

```
    }
}
```

Add a labels file

Add the new **QuestionnairesErReportLabels_en-US** labels file to your Visual Studio project. Specify the following labels for new UI resources:

- The **@QuestionnairesReport** label for a new menu item that contains the following text in US English (en-US): **Questionnaires report (powered by ER)**
- The **@QuestionnairesReportBatchJobDescription** label for a batch job title if a selected ER format is scheduled for execution as a batch job

Add a report service class

Add the new **QuestionnairesErReportService** class to your Visual Studio project. Write code that calls an ER format, identifies it by a format mapping ID, and provides a data contract as a parameter.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
/// <summary>
/// The electronic reporting service class for Questionnaires ER report
/// </summary>
class QuestionnairesErReportService extends SysOperationServiceBase
{
    public const str ERModelDataSourceName = 'model';
    public const str DefaultExportedFileName = 'Questionnaires report';
    public const str ParametersDataSourceName = 'RunTimeParameters';

    /// <summary>
    /// Generates report by using Electronic reporting framework
    /// </summary>
    /// <param name = "_contract">The Questionnaires report contract</param>
    public void generateReportByGER(QuestionnairesErReportContract _contract)
    {
        ERFormatMappingId formatMappingId;
        QuestionnairesErReportDP  dataProvider;
        dataProvider = QuestionnairesErReportDP::construct();
        formatMappingId = _contract.parmFormatMapping();
        if (formatMappingId)
        {
            try
            {
                ERIModelDefinitionParamsAction parameters = new
ERModelDefinitionParamsUIActionComposite()
                    .add(new
ERModelDefinitionObjectParameterAction(ERModelDataSourceName,
ParametersDataSourceName, _contract, true));

                // Call ER to generate the report.
                ERIFormatMappingRun formatMappingRun =
ERObjectsFactory::createFormatMappingRunByFormatMappingId(formatMappingId,
DefaultExportedFileName);
                if (formatMappingRun.parmShowPromptDialog(true))
```

```
                {
                    formatMappingRun.withParameter(parameters);

formatMappingRun.withFileDestination(_contract.getFileDestination());
                    formatMappingRun.run();
                }
            }
            catch
            {
                // An error occurred while exporting data.
                error("@SYP4861341");
            }
        }
        else
        {
            // There is no data available.
            info("@SYS300117");
        }
    }
}
```

When you use an ER format that runs application data, you must configure a data source of the **Data model** type in the format mapping. This data source refers to a specific part of the specified data model by using a single root definition. When the ER format runs, it calls this data source to access the appropriate ER model mapping that is configured for a given model and root definition.

You can pass all the information you prepare in the source code and store as part of the data contract to the running ER format by using an ER model mapping of this type. In the ER model mapping, you must configure a data source of the **Object** type that refers to the QuestionnairesErReportContract class. To identify a model mapping, you must specify a data source that calls this model mapping. In the provided code, this data source is specified by the **ERModelDataSourceName** constant that has the model value. To identify which data source is used to expose the data contract in the model mapping, you must specify a data source name. In the provided code, this name is specified by the **ParametersDataSourceName** constant that has **RunTimeParameters** value.

> ⓘ Note
>
> In a new environment, you might have to refresh the ER metadata so that this type of class is available in the ER model mapping designer. For more information, see Configure the ER framework.

Add a report controller class

Add the new **QuestionnairesErReportController** class to your Visual Studio project. Write code that runs an ER format in either synchronous mode or batch mode, as you prefer, in the dialog box that is built based on the logic of the provided **QuestionnairesErReportUIBuilder** class.

X++

```
/// <summary>
/// The controller for Questionnaires ER report
/// </summary>
class QuestionnairesErReportController extends ERFormatMappingRunBaseController
{
    /// <summary>
    /// The main entrance of the controller
    /// </summary>
    /// <param name = "args">The arguments</param>
    public static void main(Args args)
    {
        QuestionnairesErReportController operation;
        operation = new QuestionnairesErReportController(
            classStr(QuestionnairesErReportService),
            methodStr(QuestionnairesErReportService, generateReportByGER),
            SysOperationExecutionMode::Synchronous);
        operation.startOperation();
    }

    /// <summary>
    /// Gets caption of the dialog.
    /// </summary>
    /// <returns>Caption of the dialog</returns>
    public ClassDescription defaultCaption()
    {
        ClassDescription batchDescription;
        batchDescription = "Questionnaires report (powered by ER)";
        return batchDescription;
    }
}
```

Add a menu item

Add the new **QuestionnairesErReport** menu item to your Visual Studio project. In the **Object** property, this menu item refers to the **QuestionnairesErReportController** class. Use it to specify a user permission to select and run an ER format. In the **Label** property, this menu item refers to the **@QuestionnairesReport** label that you created earlier, so that correct text is presented in the application UI.

Add a menu item to a menu

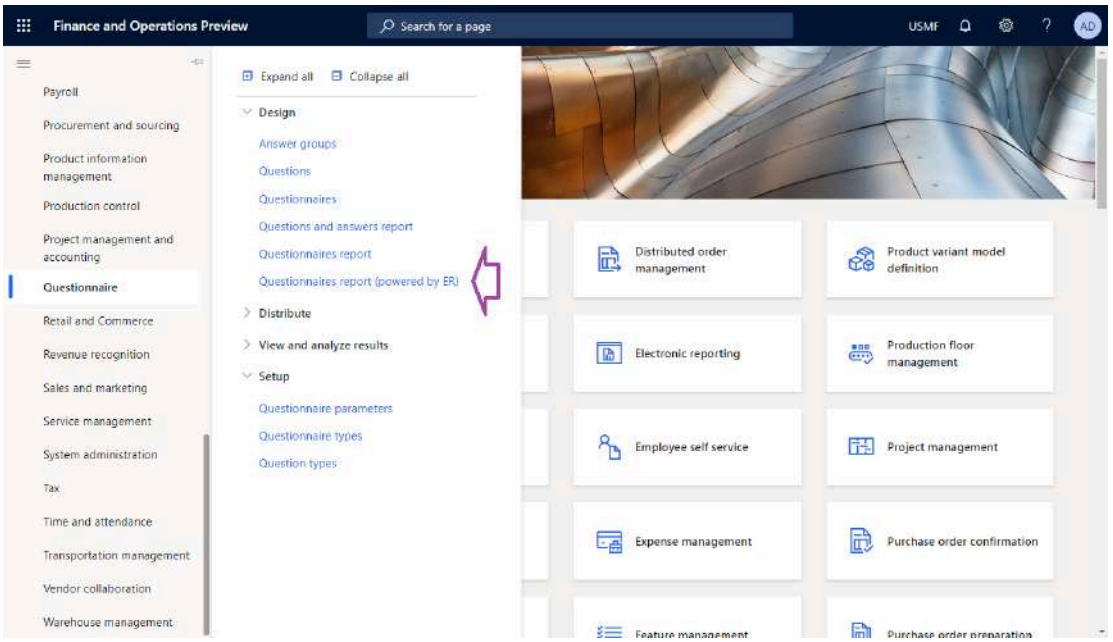
Add the existing **KM** menu to your Visual Studio project. Add a new **QuestionnairesErReport** item of the **Output** type to this menu. This item must refer to the **QuestionnairesErReport** menu item that is described in the previous section.

Build a Visual Studio project

Build your project to make the new menu item available to users.

Run a format from the application

1. Go to **Questionnaire** > **Design** > **Questionnaires report (powered by ER)**.



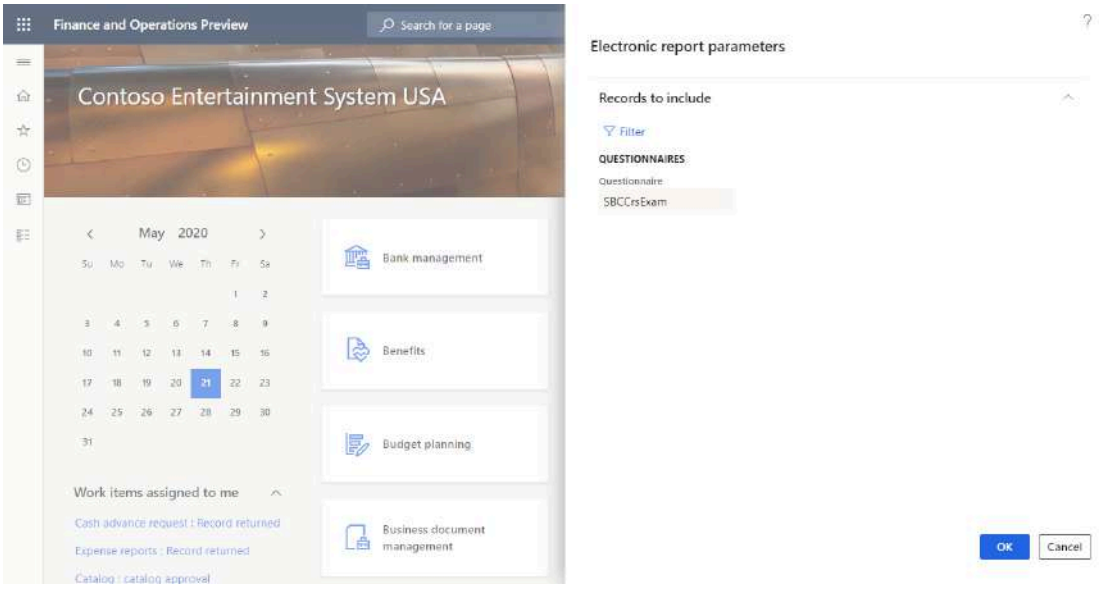
2. In the dialog box, in the **Format mapping** field, select **Questionnaires report**.

3. Select **OK**.

4. In the **Electronic report parameters** dialog box, on the **Records to include** FastTab, set the filter so that it includes only the **SBCCrsExam** questionnaire.

5. Select **OK** to confirm the filter.

6. Select **OK** to run the report.



7. Review the generated report.

Tune a designed ER solution

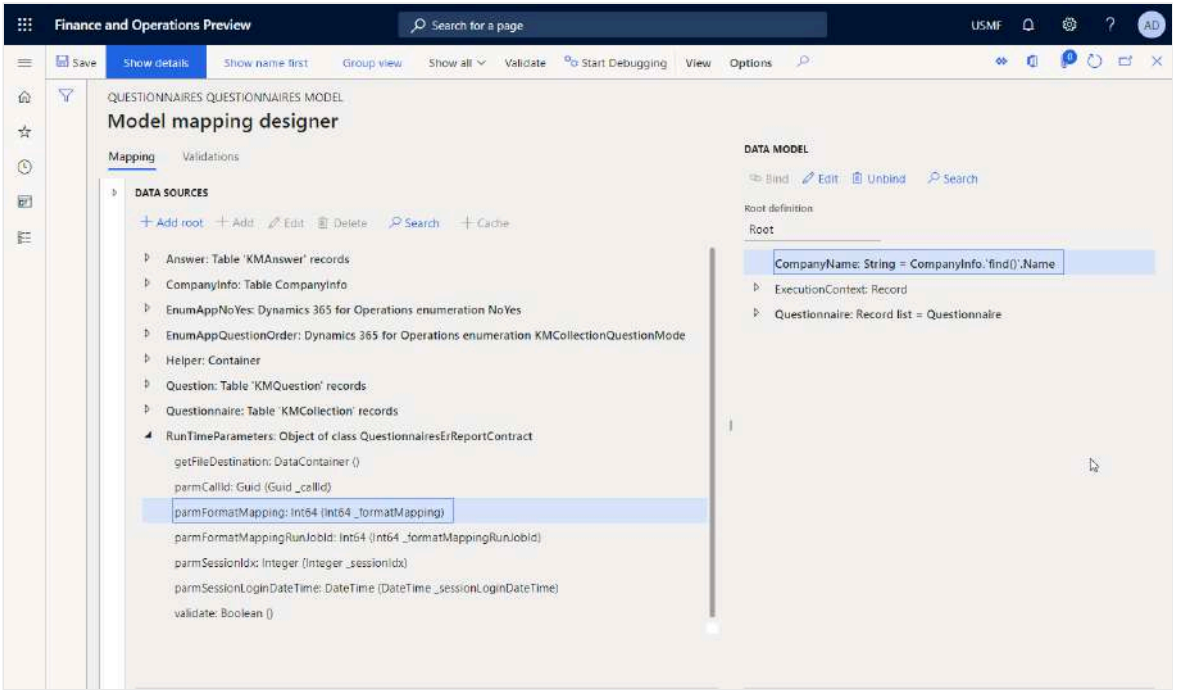
Modify the configured ER solution so that it uses the data provider class you developed to access details of the running ER format. Also, make it enter the name of this ER format in a generated report.

Modify a model mapping

Add data sources to access a data contract object

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, expand **Questionnaire model**, and then select **Questionnaire mapping**.
3. Select **Designer** to open **Model to datasource mapping**.
4. Select **Designer** to open the selected mapping in the model mapping designer.
5. On **Model mapping designer**, in the **Data source types** pane, select **Dynamics 365 for Operations\Object**.
6. In the **Data sources** pane, select **Add root**.
7. In the dialog box, in the **Name** field, enter RunTimeParameters, as defined in the source code of the **QuestionnairesErReportService** class.
8. In the **Class** field, enter QuestionnairesErReportContract, which you coded earlier.
9. Select **OK**.
10. Expand **RunTimeParameters**.

The added data source provides information about the record ID of the running ER format mapping.



Add a data source to access ER format mapping records

Continue to edit the selected model mapping by adding a data source to access ER format mapping records.

1. On **Model mapping designer**, in the **Data source types** pane, select **Dynamics 365 for Operations\Table records**.
2. In the **Data sources** pane, select **Add root**.

3. In the dialog box, in the **Name** field, enter **ER1**.
4. In the **Table** field, enter **ERFormatMappingTable**.
5. Select **OK**.

Add a data source to access a format mapping record of a running ER format

Continue to edit the selected model mapping by adding a data source to access the format mapping record of the running ER format.

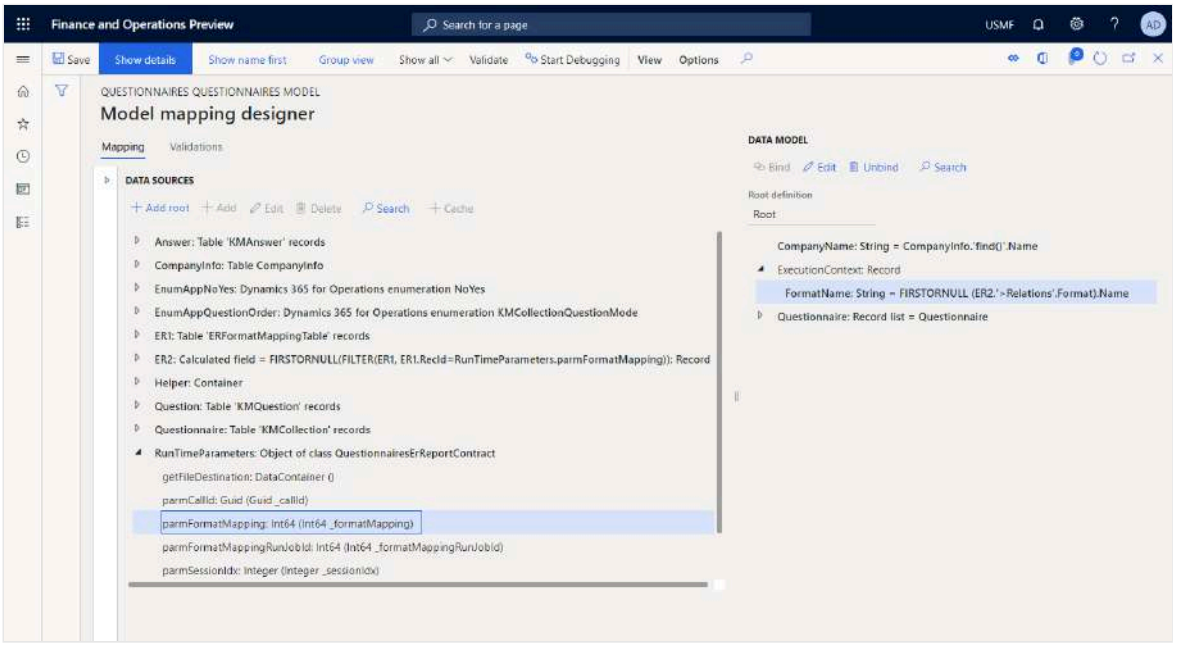
1. On **Model mapping designer**, in the **Data source types** pane, select **Functions\Calculated field**.
2. In the **Data sources** pane, select **Add root**.
3. In the dialog box, in the **Name** field, enter **ER2**.
4. Select **Edit formula**.
5. In the formula editor, in the **Formula** field, enter **FIRSTORNULL (FILTER(ER1, ER1.RecId = RunTimeParameters.parmFormatMapping))**.
6. Select **Save**, and close the formula editor.
7. Select **OK**.

Enter the name of the running ER format in the data model

Continue to edit the selected model mapping so that the name of the running ER format is entered in the data model.

1. On **Model mapping designer**, in the **Data model** pane, expand **ExecutionContext**, and then select **ExecutionContext\FormatName**.
2. In the **Data model** pane, select **Edit** to configure a data binding for the selected data model's field.
3. In the formula editor, in the **Formula** field, enter **FIRSTORNULL (ER2.'>Relations'.Format).Name**.
4. Select **Save**, and close the formula editor.

Because you used the **FormatName** field, the configured model mapping now exposes the name of an ER format that calls this model mapping during execution.



Complete the design of the model mapping

1. On **Model mapping designer**, select **Save**.
2. Close the page.
3. Close the model mappings page.
4. On **Configurations**, in the configuration tree, make sure that the **Questionnaire mapping** configuration is still selected. Then, on the **Versions** FastTab, select the configuration version that has a status of **Draft**.
5. Select **Change status** > **Complete**.

The status of version 1.2 of this configuration changes from **Draft** to **Completed**. You can't change version 1.2 anymore. This version contains the configured model mapping and can be used as the basis for other ER configurations. Version 1.3 of this configuration is created and has a status of **Draft**. You can edit this version to adjust the **Questionnaire** model mapping.

Modify a format

You can modify the configured ER format so that its name appears in the footer of a report that is generated when the ER format runs.

Add a new format element

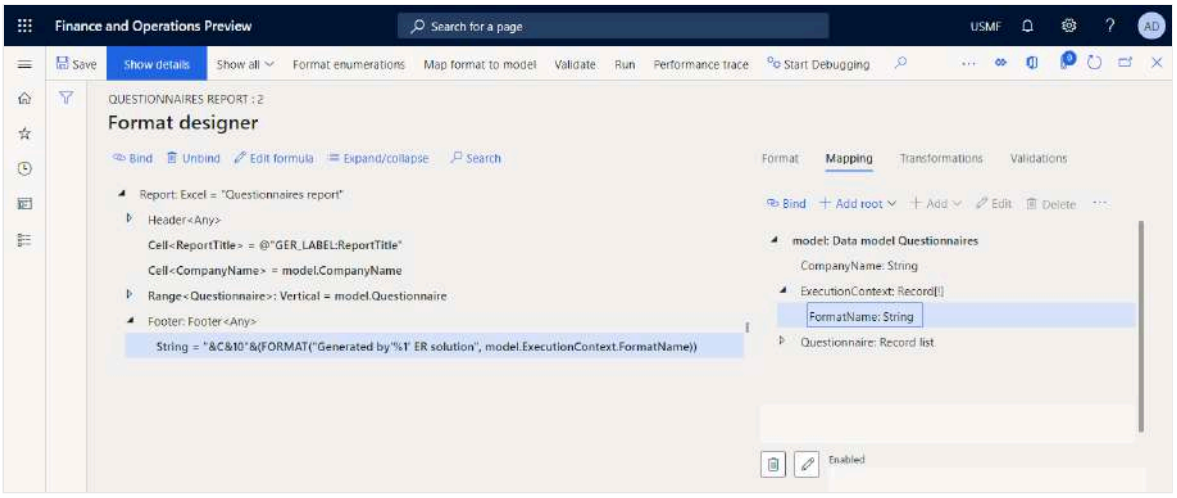
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, expand **Questionnaire model**, and then select **Questionnaire report**.
3. Select **Designer**.
4. On **Format designer**, select the **Report** root item.
5. Select **Add** to add a new nested format element for the selected **Report** root item.
6. Select **Excel\Footer**.
7. In the **Name** field, enter **Footer**.

8. Select **Report\Footer**, and then select **Add**.
9. Select **Text\String**.

Bind the added format element

1. On **Format designer**, on the **Mapping** tab, in the format tree, for the active **Footer\String** element, select **Edit formula**.
2. In the formula editor, in the **Formula** field, enter `CONCATENATE ("\&C\&10", FORMAT("Generated by'\%1' ER solution", model.ExecutionContext.FormatName))`.
3. Select **Save**, and close the formula editor.
4. Select **Save**.

You modified the configured format so that its name is entered in the footer of a generated report by using the **Footer\String** element.



Complete the format design

1. Close **Format designer**.
2. On **Configurations**, in the configuration tree, make sure that the **Questionnaire report** configuration is still selected. Then, on the **Versions** FastTab, select the configuration version that has a status of **Draft**.
3. Select **Change status** > **Complete**.

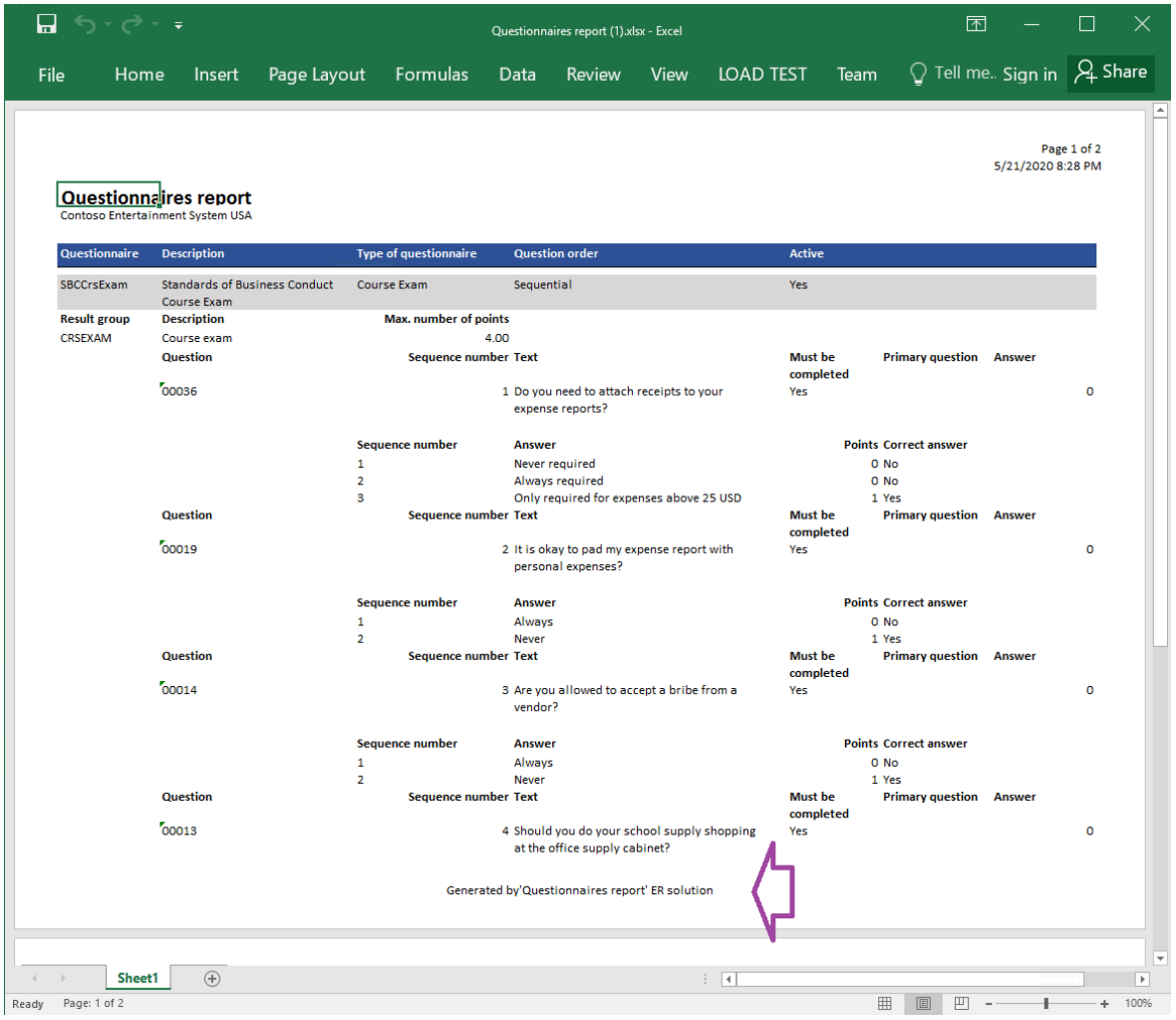
The status of version 1.2 of this configuration changes from **Draft** to **Completed**. You can no longer change version 1.2. This version contains the configured format and can be used as the basis for other ER configurations. Version 1.3 of this configuration is created and has a status of **Draft**. You can edit this version to adjust the **Questionnaire** report.

Run a format from the application

1. Go to **Questionnaire** > **Design** > **Questionnaires report (powered by ER)**.
2. In the dialog box, in the **Format mapping** field, select **Questionnaires report**.
3. Select **OK**.

4. In the **ER parameters** dialog box, on the **Records to include** FastTab, set the filter so that it includes only the **SBCCrsExam** questionnaire.
5. Select **OK** to confirm the filter.
6. Select **OK** to run the report.
7. Review the generated report in Excel format.

The footer of the generated report contains the name of the ER format that was used to generate it.



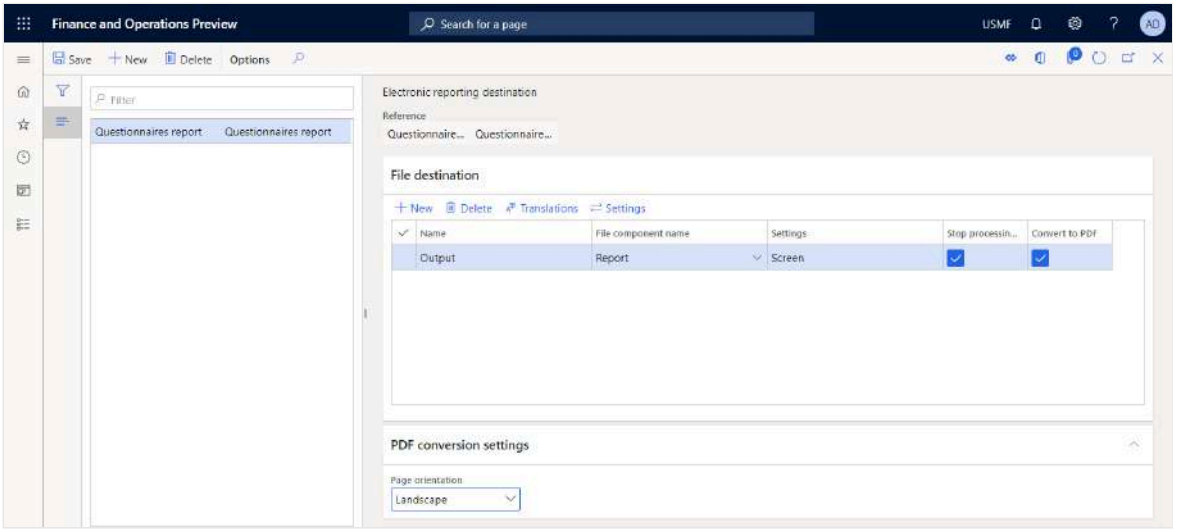
Run a format from ER

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, expand **Questionnaire model**, and then select **Questionnaire report**.
3. On the Action Pane, select **Run**.
4. In the **Electronic report parameters** dialog box, on the **Records to include** FastTab, set the filter so that it includes only the **SBCCrsExam** questionnaire.
5. Select **OK** to confirm the filter.
6. Select **OK** to run the report.
7. Review the generated report in Excel format.

The footer of the generated report doesn't contain the name of ER format that was used to generate it, because the running model mapping wasn't passed the data contract object when the ER format called it.

Configure a format destination for on-screen preview

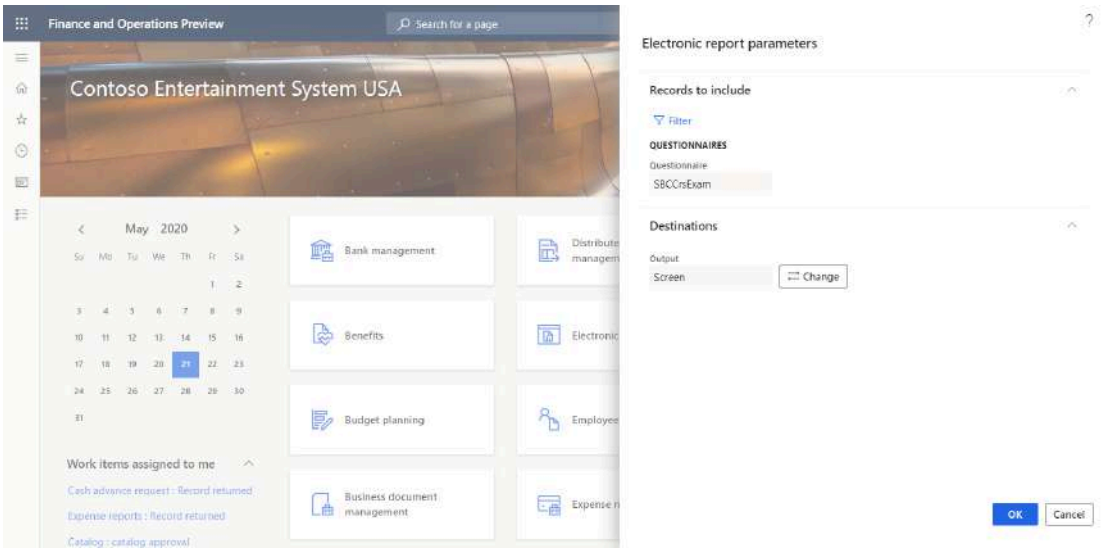
1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting destination**.
2. On **Electronic reporting destination**, add a destination record for the configured **Questionnaire report** ER format.
3. On the **File destination** FastTab, set up the **Screen** destination for the **Report** format component that you added as the root element of the configured **Questionnaire report** ER format.
4. On the **PDF conversion settings** FastTab, configure the destination to convert a report to PDF format that uses the **Landscape** page orientation.



Run a format from the application to preview it as a PDF document

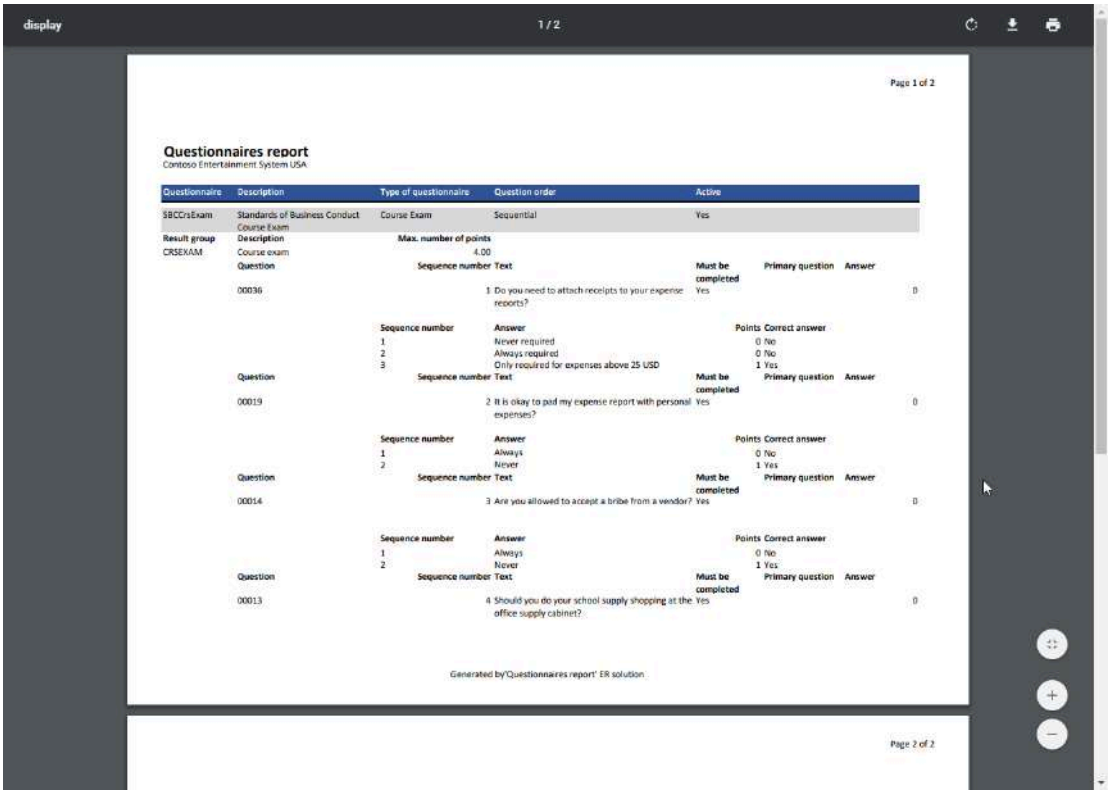
1. Go to **Questionnaire** > **Design** > **Questionnaires report (powered by ER)**.
2. In the dialog box, in the **Format mapping** field, select **Questionnaires report**.
3. Select **OK**.
4. In the **Electronic report parameters** dialog box, on the **Records to include** FastTab, set the filter so that it includes only the **SBCCrsExam** questionnaire.
5. Select **OK** to confirm the filter.

   On the **Destinations** FastTab, notice that the **Output** field is set to **Screen**. If you want to change the configured destination, select **Change**.



6. Select **OK** to run the report.

7. Review the generated report in PDF format.



Additional resources

- Electronic Reporting overview
- Electronic reporting formula language
- Design multilingual reports
- ER framework API
- CASE function
- CONCATENATE function
- DATETIMEFORMAT function
- FILTER function

- FIRSTORNULL function
- FORMAT function
- IF function
- ORDERBY function
- SESSIONNOW function

Last updated on 04/13/2026

Customize Electronic reporting configurations to generate an electronic document

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Electronic reporting (ER) framework lets you upload the ER configurations that Microsoft provides into your Microsoft Dynamics 365 Finance instance. By using these Microsoft-provided configurations, you can generate electronic customer invoices (e-invoices). Use this ER solution to configure your custom ER solution to access your custom database fields and generate e-invoices that comply with your specific requirements, without editing the source code.

Overview

For the example in this article, you must specify a federal tax identification code as a new custom attribute for every customer that you electronically invoice. Therefore, you must customize the structure of the invoice that is currently used by adding a new item that you fill with the tax code in every e-invoice that you generate.

The procedures in this article explain how a user in the System Administrator, Electronic Reporting Developer, or Electronic Reporting Functional Consultant role can perform the following tasks in your Finance instance:

- Configure the minimal set of ER parameters that is required to start to use the ER framework.
- Import the initial versions of the standard ER configurations that are provided to generate e-invoices.
- Configure the Accounts receivable parameters to start to use the standard ER configurations.

- Configure the legal entity parameters to invoice customers.
- Prepare a customer record for electronic invoicing.
- Add, post, and send a customer invoice by using the standard ER configurations.
- Add a custom database field to manage a federal tax identification code for customers.
- Refresh the ER metadata to enable database changes for the ER model mapping designer.
- Design the custom ER configurations to generate e-invoices that contain the new tax code.
- Configure the Accounts receivable parameters to start to use the custom ER configurations.
- Update a customer record for electronic invoicing.
- Add, post, and send a customer invoice by using the custom ER configurations.
- Import the new versions of the standard ER configurations that are provided to generate e-invoices.
- Adopt the changes to the new versions of the standard ER configurations in your custom ER configurations.
- Add, post, and send a customer invoice by using the new versions of the custom ER configurations.

You can complete all these procedures in the **DEMF** company.

Configure the ER framework

As a user in the Electronic Reporting Functional Consultant or Electronic Reporting Developer role, you must configure the minimal set of ER parameters. You can then start to use the ER framework to design custom versions of the standard configurations of the ER solution that is used to generate e-invoices.

Configure ER parameters

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization blueprint** page, in the **Related links** section, select **Electronic reporting parameters**.
3. On the **Electronic reporting parameters** page, on the **General** tab, set the **Enable design mode** option to **Yes**.
4. On the **Attachments** tab, in the **Configurations** field, select **File**.
5. In the **Job archive**, **Temporary**, **Baseline**, and **Others** fields, select the **File** type.

For more information about ER parameters, see Configure the ER framework.

Activate an ER configuration provider

Each ER configuration you add is owned by an ER configuration provider. The ER configuration provider that you activate in the **Electronic reporting** workspace is used for this purpose. Therefore, you must activate an ER configuration provider in the **Electronic reporting** workspace before you start to add or edit ER configurations.

> ⓘ **Note**
>
> Only the owner of an ER configuration can edit it. Therefore, before you can edit an ER configuration, you must activate the appropriate ER configuration provider in the **Electronic reporting** workspace.

Review the list of ER configuration providers

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization blueprint** page, in the **Related links** section, select **Configuration providers**.
3. On the **Configuration provider table** page, each provider record has a unique name and URL. Review the contents of this page. If a record for **Litware, Inc.** (`https://www.litware.com`) already exists, skip the next procedure, Add a new ER configuration provider.

Add a new ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization blueprint** page, in the **Related links** section, select **Configuration providers**.
3. On the **Configuration providers** page, select **New**.
4. In the **Name** field, enter **Litware, Inc.**
5. In the **Internet address** field, enter `https://www.litware.com`.
6. Select **Save**.

Activate an ER configuration provider

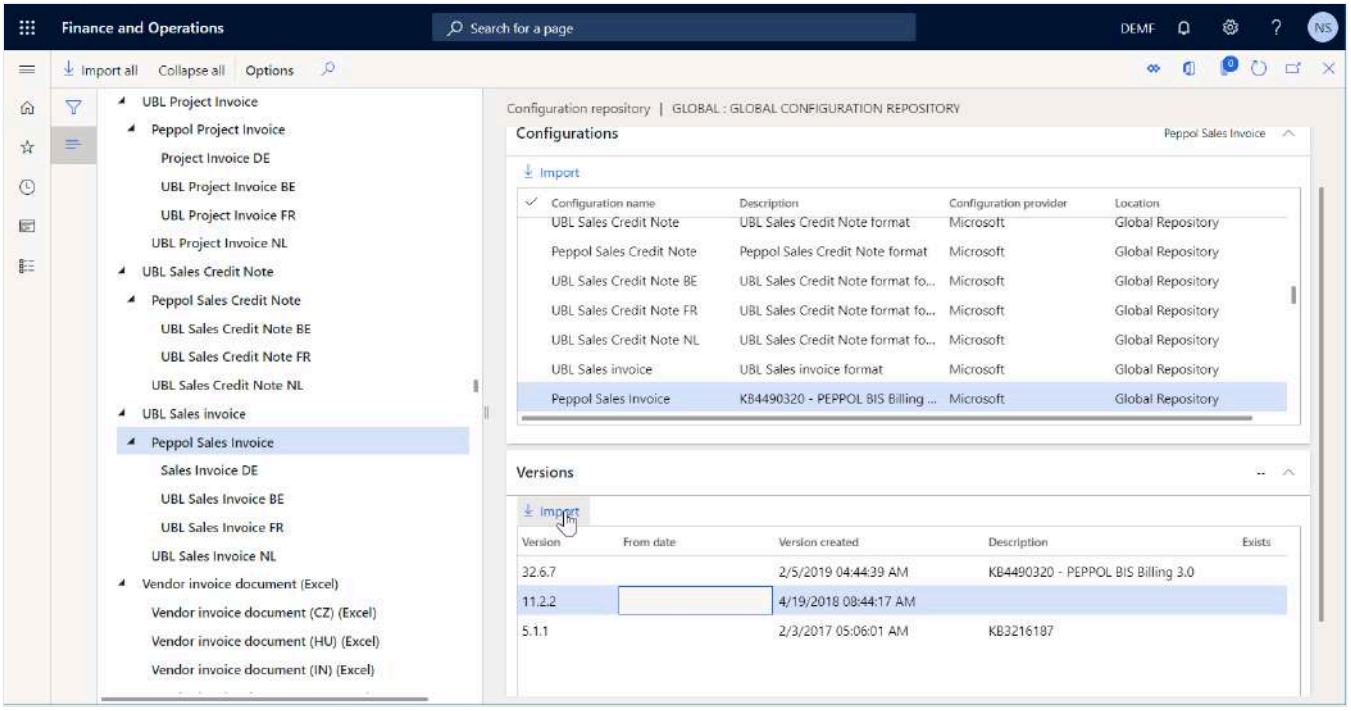
1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization blueprint** page, in the **Configuration providers** section, select the **Litware, Inc.** tile, and then select **Set active**.

For more information about ER configuration providers, see Create configuration providers and mark them as active.

Import the initial versions of standard ER configurations

To add the standard ER configurations to your current Finance instance, import them from the ER repository that you configured for that instance.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization blueprint** page, in the **Configuration providers** section, select the **Microsoft** tile, and then select **Repositories** to view the list of repositories for the Microsoft provider.
3. On **Configuration repositories**, select the repository of the **Global** type, and then select **Open**. If you're prompted for authorization to connect to Regulatory Configuration Service, follow the authorization instructions.
4. On **Configuration repository**, in the configuration tree in the left pane, select the **Peppol Sales Invoice** format configuration.
5. On the **Versions** FastTab, select version **11.2.2**.
6. Select **Import** to download the selected version from the Global repository.



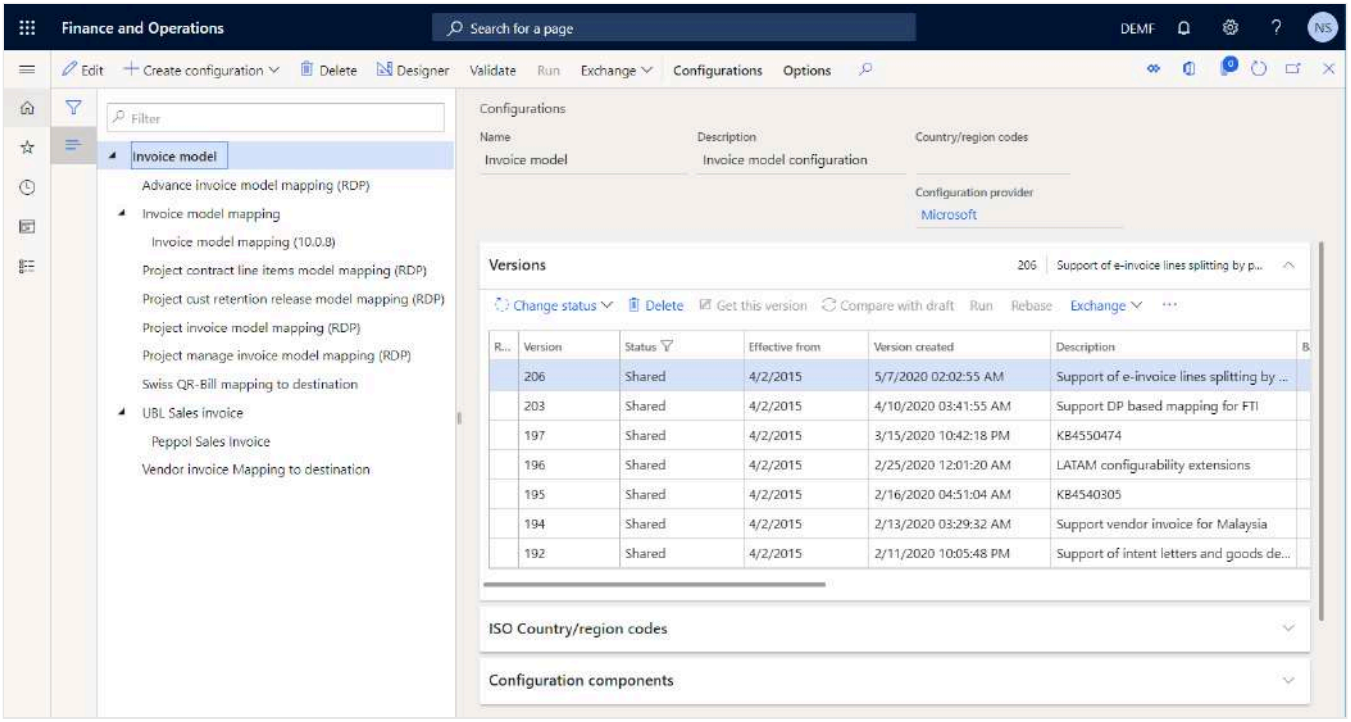
> **Tip**

If you have trouble accessing the Global repository, you can download configurations from Microsoft Dynamics Lifecycle Services instead.

Review the imported ER configurations

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization blueprint** page, in the **Configurations** section, select the **Reporting configurations** tile.
3. On the **Configurations** page, expand the **Configuration components** FastTab.
4. In the configuration tree in the left pane, expand **Invoice model**, and then expand **UBL Sales invoice**.

You see that, in addition to the selected **Peppol Sales Invoice** ER format, the import process brings in other required ER configurations. Because new versions of ER configurations are constantly published to the Global repository and Lifecycle Services to keep the corresponding solutions compliant with new requirements, the latest versions of the required data model configuration and its model mapping configurations are imported.



To simulate the state that ER configurations in the current Finance instance would be in if you imported version **11.2.2** of the **Peppol Sales Invoice** ER format in the past (for example, on August 7, 2019), follow these steps:

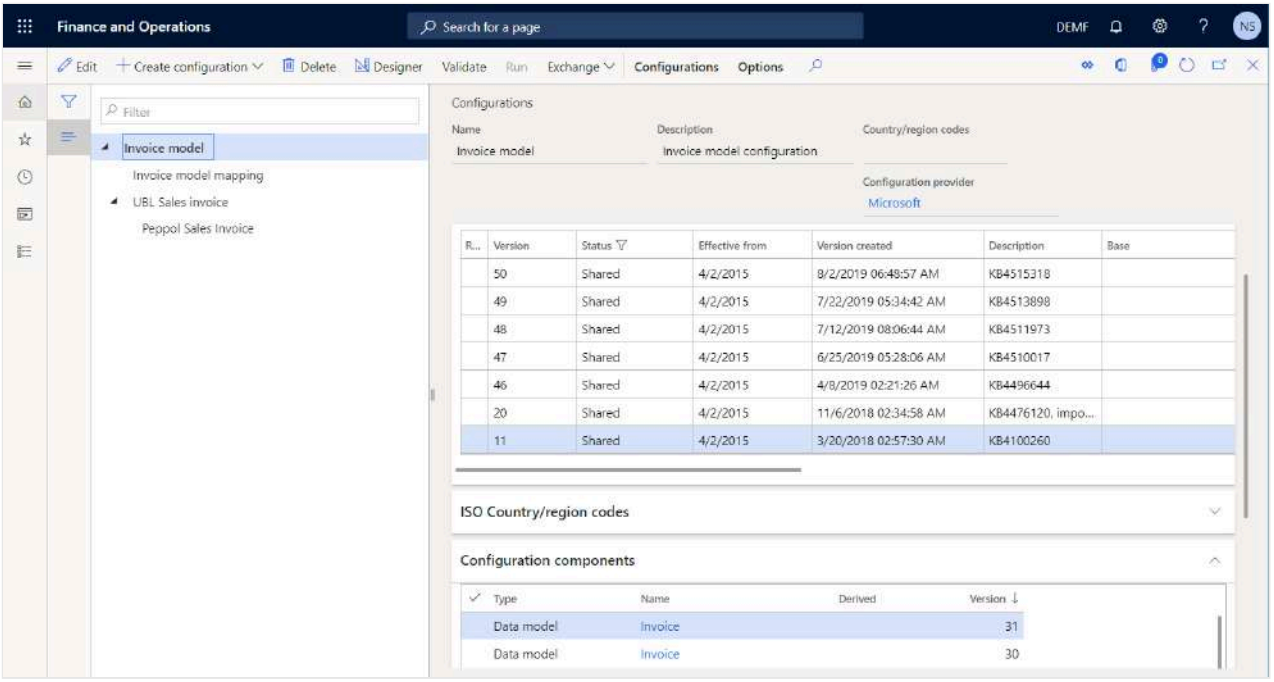
- On the Action Pane, select **Delete** to delete all ER configurations that were published after August 7, 2019. The only **Invoice model**, **Invoice model mapping** (initially named

Customer invoice model mapping), **UBL Sales invoice**, and **Peppol Sales Invoice** configurations remain.

- For the remaining ER configurations, on the **Versions** FastTab, select **Delete** to delete all versions of ER configurations that were published after August 7, 2019.

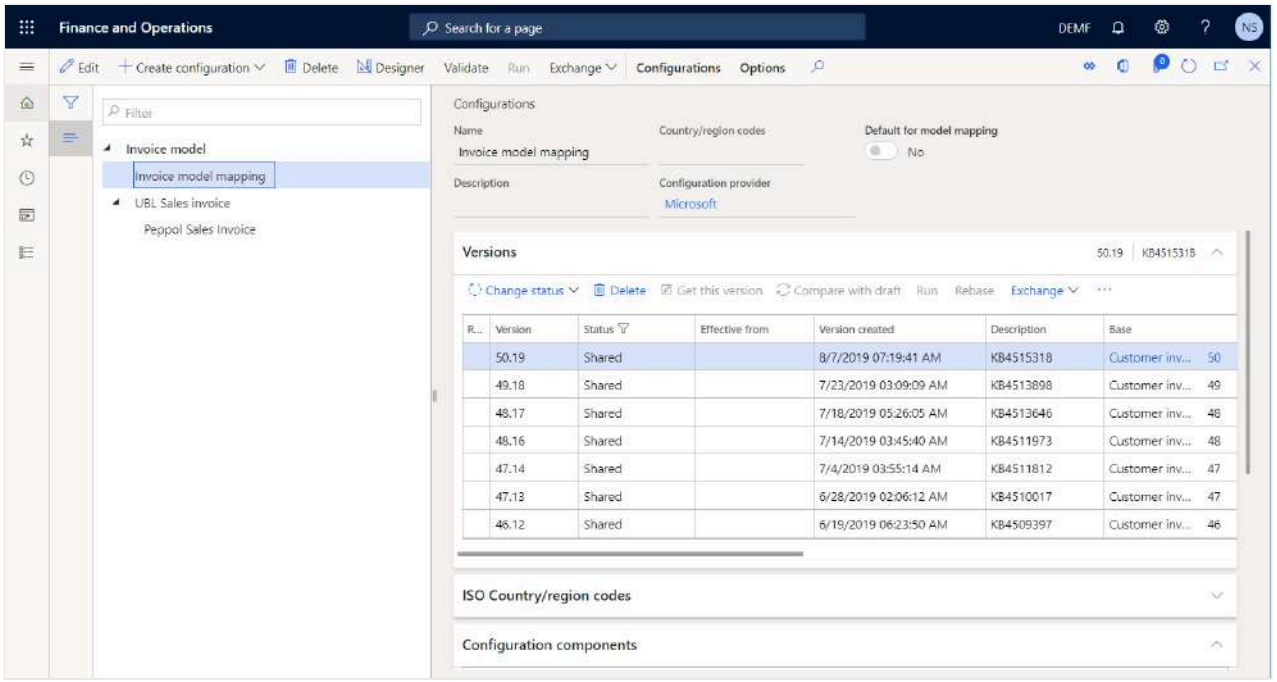
Then verify that the following configurations are available in the configuration tree:

- **Invoice model** ER data model configuration (initially named **Customer invoice model**):
  - Version 11 contains version 10 of the data model ER component that represents the data structure of the invoicing business domain. This ER configuration is imported as an ancestor of the **Peppol Sales Invoice** ER format that you selected for import.
  - Version 50 contains version 31 of the data model ER component. This ER configuration is imported as an ancestor of the August 7, 2019, version of the **Invoice model mapping** ER model mapping configuration.



> **Tip**
>
> If you don't see version 50 of this data model, open the Global repository, and import version 50.19 of the **Invoice model mapping** ER configuration.

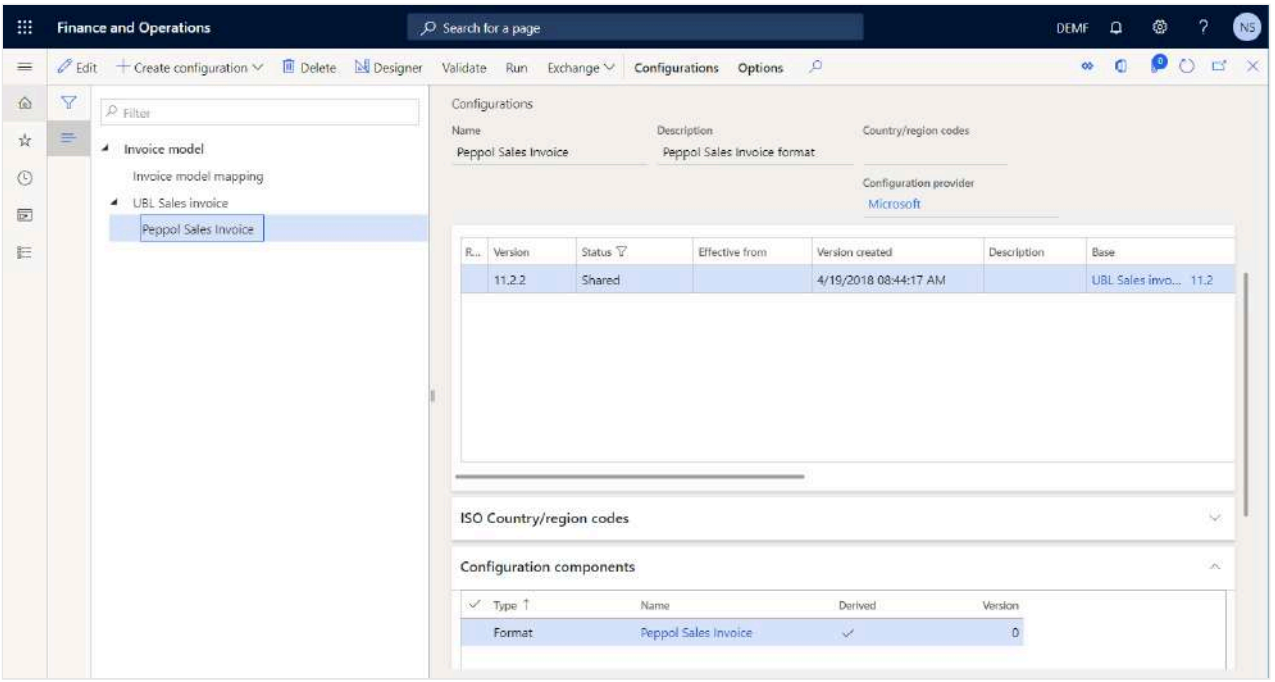
- **Invoice model mapping** ER model mapping configuration (initially named **Customer invoice model mapping**):
  - Version 50.19 is imported as the latest implementation of version 50 of the **Invoice model** ER data model configuration. It contains two model mapping ER components that describe how the data model is filled in with application data at runtime.



> **Tip**
>
> If you don't see version 50.19 of this model mapping, open the Global repository, and import version 50.19 of the **Invoice model mapping** ER configuration.

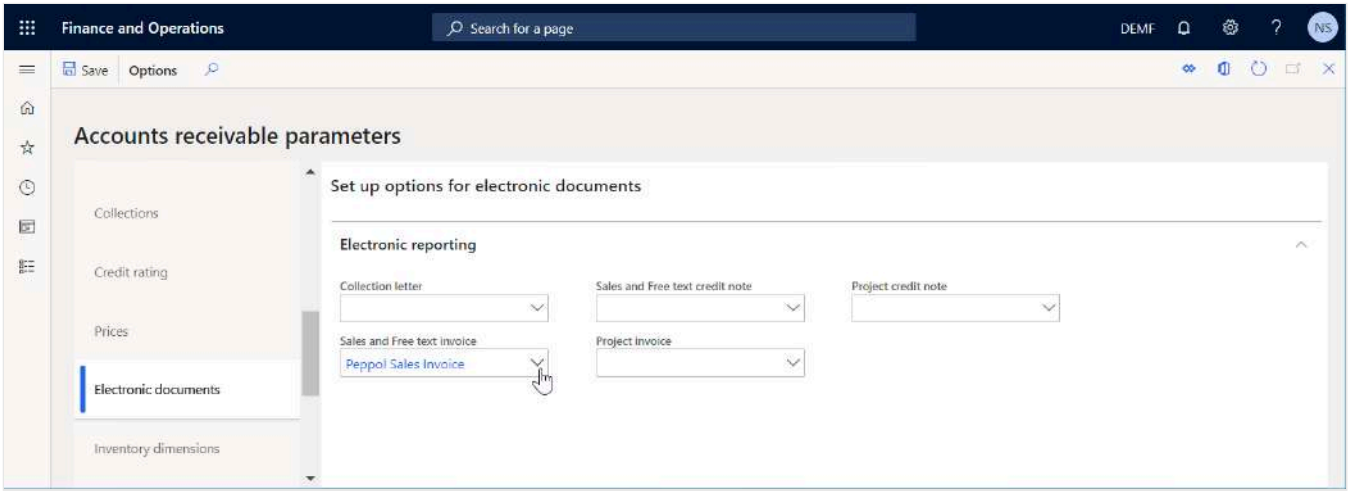
- **UBL Sales invoice** ER format configuration:
  - Version 11.2 contains the format and format mapping ER components. The format component specifies the report layout. The format mapping component contains the model data source and specifies how this data source is used to fill in the report layout at runtime. This ER format is configured to generate e-invoices in Universal Business Language (UBL) format. It's imported as a parent of the **Peppol Sales Invoice** ER format that you selected for import.

- **Peppol Sales Invoice** ER format configuration:
  - Version 11.2.2 contains the format and format mapping ER components that are configured to generate e-invoices in Pan-European Public Procurement OnLine (PEPPOL) format.



Configure the Accounts receivable parameters

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Electronic documents** tab, in the **Electronic reporting** section, select **Peppol Sales Invoice** in the **Sales and Free text invoice** field.
3. Select **Save**.



Configure the legal entity parameters

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. For the selected **DEMF** company, in the **Bank account information** section, enter **1234** in the **Routing number** field.
3. Select **Save**.
4. Close the **Legal entities** page.

Prepare a customer record

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. On the **All customers** page, select the **DE-014** customer account link.

Add a customer contact

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. On the Action Pane, on the **Customer** tab, in the **Accounts** group, select **Contacts**.
3. Select **Add contacts**.
4. On the **Contacts** page, in the **First name** field, open the lookup, select **Adam Carter**, and then select **Select** to close the lookup.
5. Select **Save**.
6. Close the **Contacts** page.

Define a primary contact

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. On the **Sales demographics** FastTab, in the **Primary contact** field, select **Adam Carter**.

Set the e-invoicing option

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. On the **Invoice and delivery** FastTab, set the **eInvoice** option to **Yes**.
3. Select **Save**.
4. Close the **All customers** page.

Process a customer invoice by using the standard ER configurations

Use the standard ER configurations that you imported to electronically send a free text invoice to a customer.

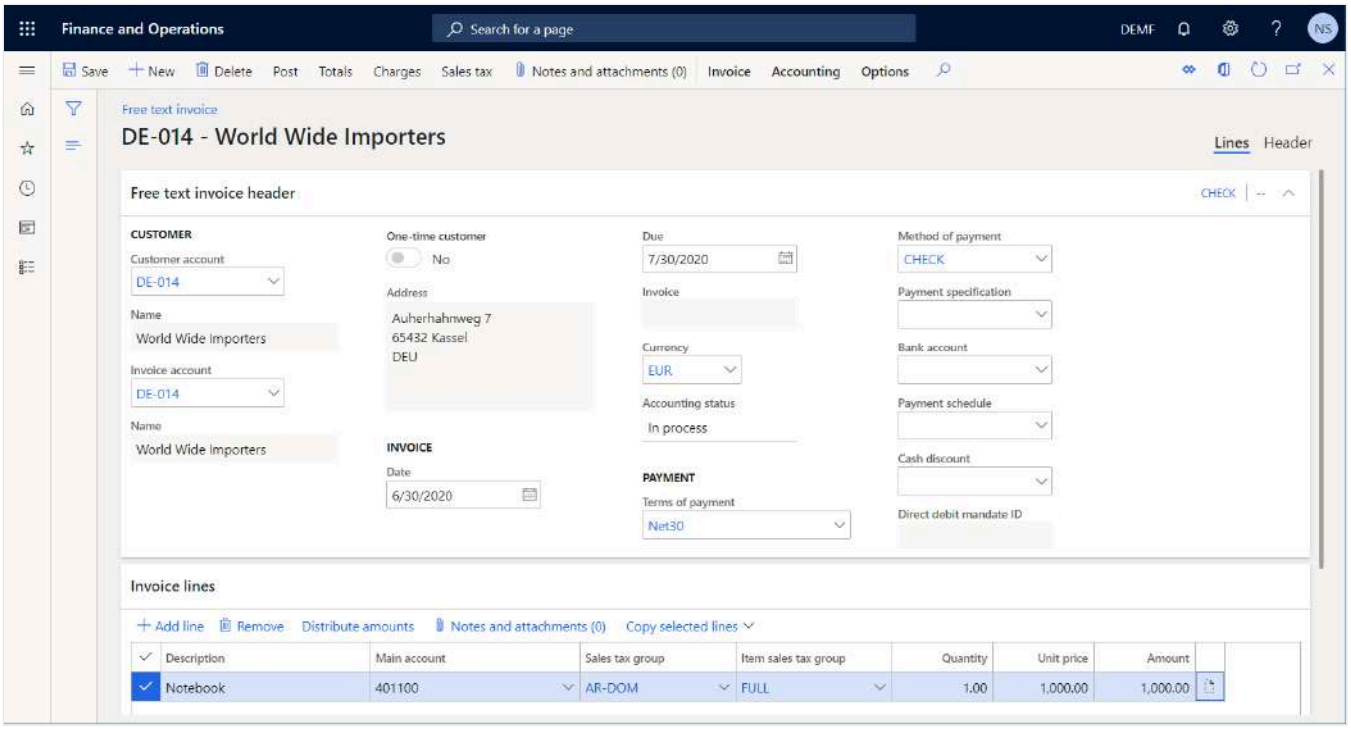
Add a new invoice

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. On **Free text invoice**, select **New**.
3. On the **Free text invoice header** FastTab, in the **Customer account** field, select **DE-014**.

4. On the **Invoice lines** FastTab, an invoice line is automatically added. On this line, set the following fields:

    - In the **Description** field, enter **Notebook**.
    - In the **Main account** field, select **401100**.
    - In the **Unit price** field, enter **1000**.

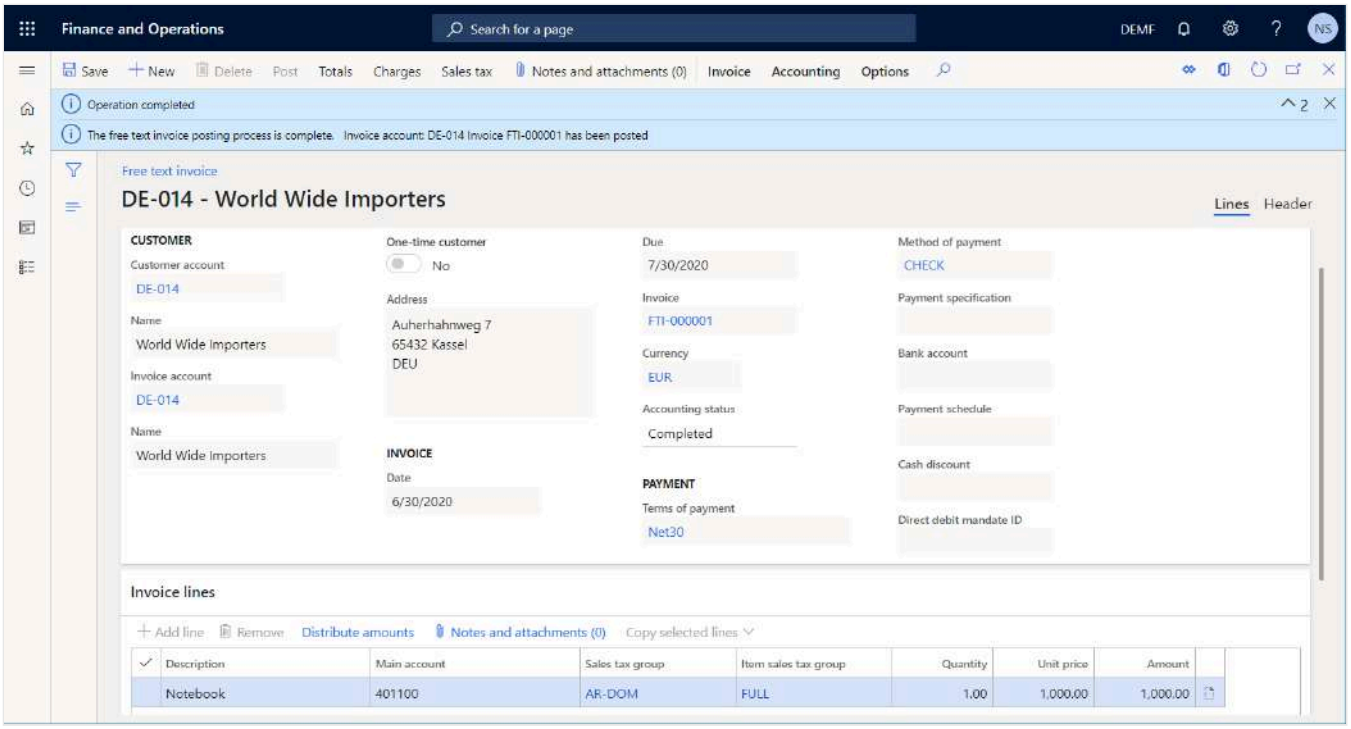
5. Select **Save**.



For more information, see Create a free text invoice.

Post a new invoice

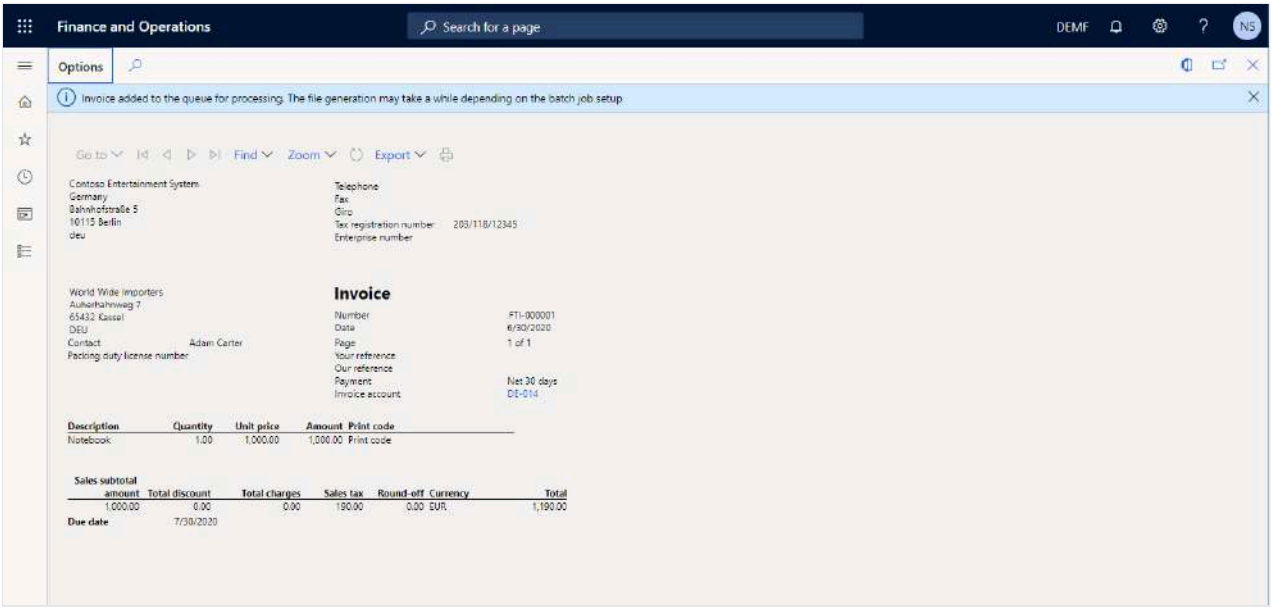
1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. On **Free text invoice**, on the Action Pane, select **Post**.
3. In the **Post free text invoice** dialog box, select **OK**.



Send a posted invoice

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.

2. On **Free text invoice**, on the Action Pane, in the **Document** group, select **Send** > **Original**.



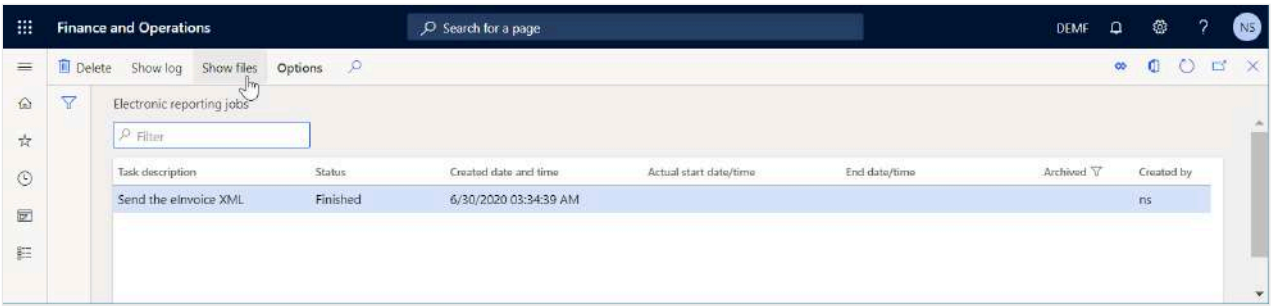
3. Close **Free text invoice**.

Analyze a generated e-invoice

1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting jobs**.

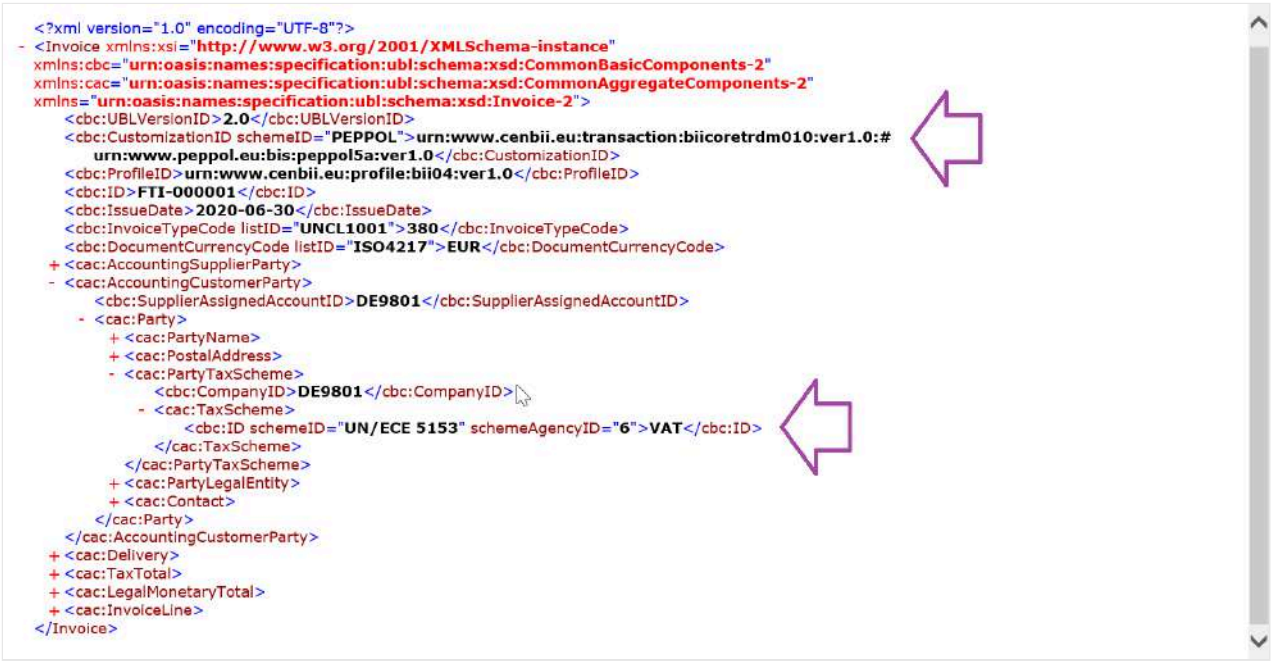
2. On **Electronic reporting jobs**, select the initial record that has the task description **Send the eInvoice XML**.

3. Select **Show files** to access the list of generated files.



4. Select **Open** to download the e-invoice XML file.

5. Analyze the e-invoice XML file. The customer tax schema is currently represented by the **schemeID** and **schemeAgencyID** XML attributes. The **cbc:CustomizationID** XML element currently contains the following text: `urn:www.cenbii.eu:transaction:biicoretrdm010:ver1.0:#` `urn:www.peppol.eu:bis:peppol5a:ver1.0`.



Add a custom database field

Use the Custom field feature to customize the current Finance instance:

- Customize the database structure by adding a new custom database field that stores a federal tax identification code for every customer.

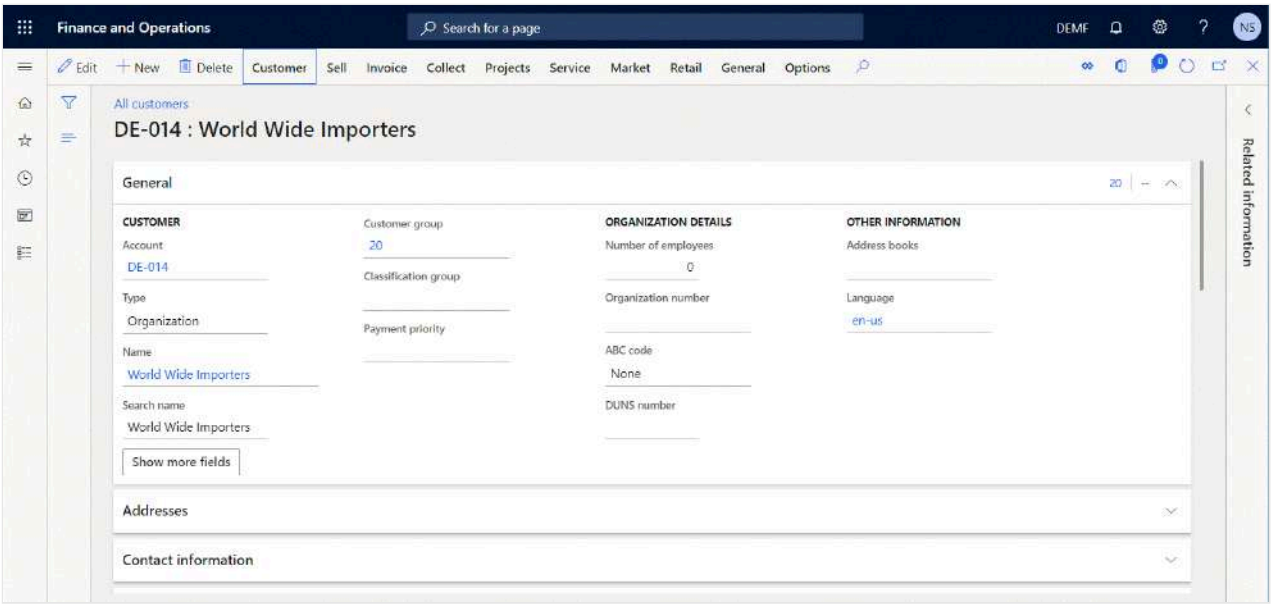
- Customize the **Customer** page by adding a new data entry field that you can use to enter a tax code value in the new custom database field.

Follow these steps to do the customization.

1. Go to **Accounts receivables** > **Customers** > **All customers**.
2. On the **All customers** page, select the **DE-014** customer account link.
3. On the **General** FastTab, right-click any blank area under the **Language** field, and then select **Personalize: UpperGroup**.

   The contents of the **General** FastTab are highlighted, and a **Personalize** menu appears.
4. On the **Personalize** menu, select **Add a field**.
5. In the **Add columns** dialog box, select **Create new field**.
6. In the **Create new field** dialog box, in the **Table name** field, select **Customers**.
7. In the **Name prefix** field, enter **FederalTaxID**.

   > ⓘ **Note**
   >
   > The whole field name is automatically defined as **FederalTaxID_Custom**.
8. In the **Label** field, enter **Federal Tax ID**.
9. In the **Type** field, select **Text**.
10. In the **Length** field, enter **20**.
11. Select **Save**.
12. In the message box that appears, select **Yes** to confirm that you want to create a new **FederalTaxID** field entry for the **Customers** table.
13. Select **Insert** to add the **FederalTaxID_Custom** field to the current page.



14. Close the **All customers** page.

Refresh the ER metadata

To make the custom field you added visible in the ER model mapping designer, refresh the ER metadata.

1. Go to **Organization administration** > **Electronic reporting** > **Rebuild table references**.
2. In **Update data model**, select **OK**.

Design the custom ER configurations

Use the Microsoft-provided ER configurations as a starting point to design your custom ER configurations. This way, you can generate e-invoices that contain the new tax code.

Customize the data model configuration

As a user in the Electronic Reporting Functional Consultant role, you can design your custom ER data model.

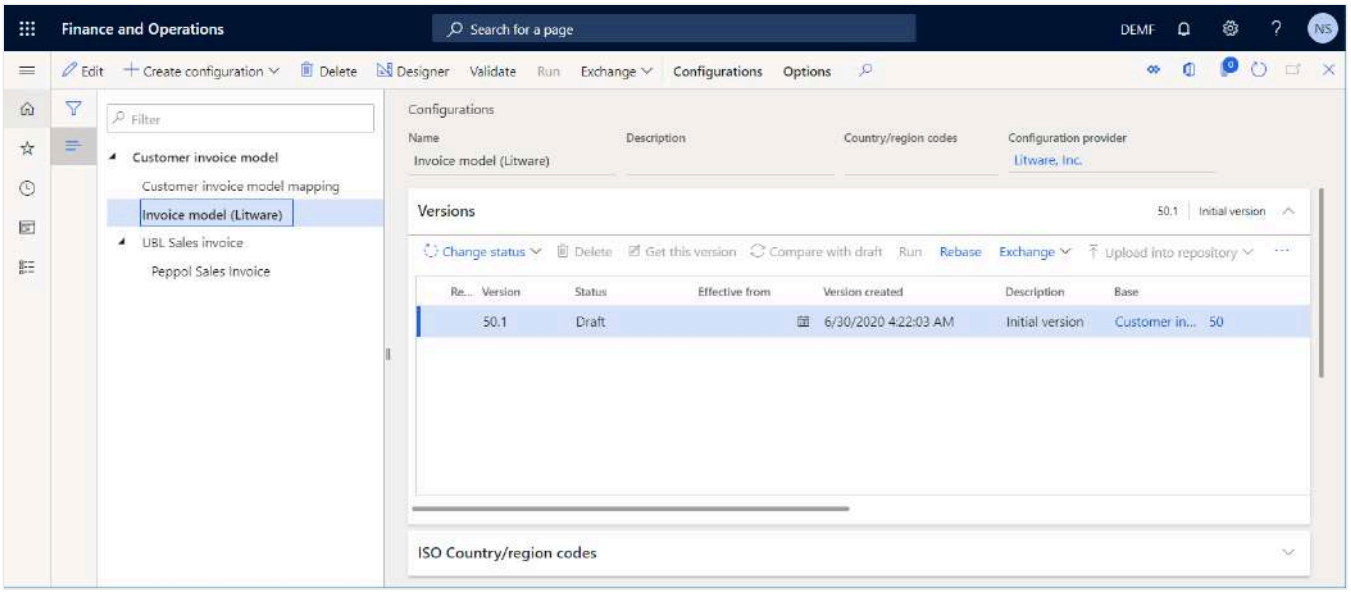
Add a custom data model configuration

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, select **Customer invoice model**.
3. On the Action Pane, select **Create configuration**.
4. In the drop-down dialog box, in the **New** field, select **Derive from Name: Customer invoice model, Microsoft** to indicate that your new custom ER data model configuration

should be based on the ER data model configuration.

5. In the **Name** field, enter **Invoice model (Litware)**.
6. Select **Create configuration** to add the new ER configuration.

You can now use the ER data model designer to edit version 50.1 of the **Invoice model (Litware)** ER configuration in **Draft** status.



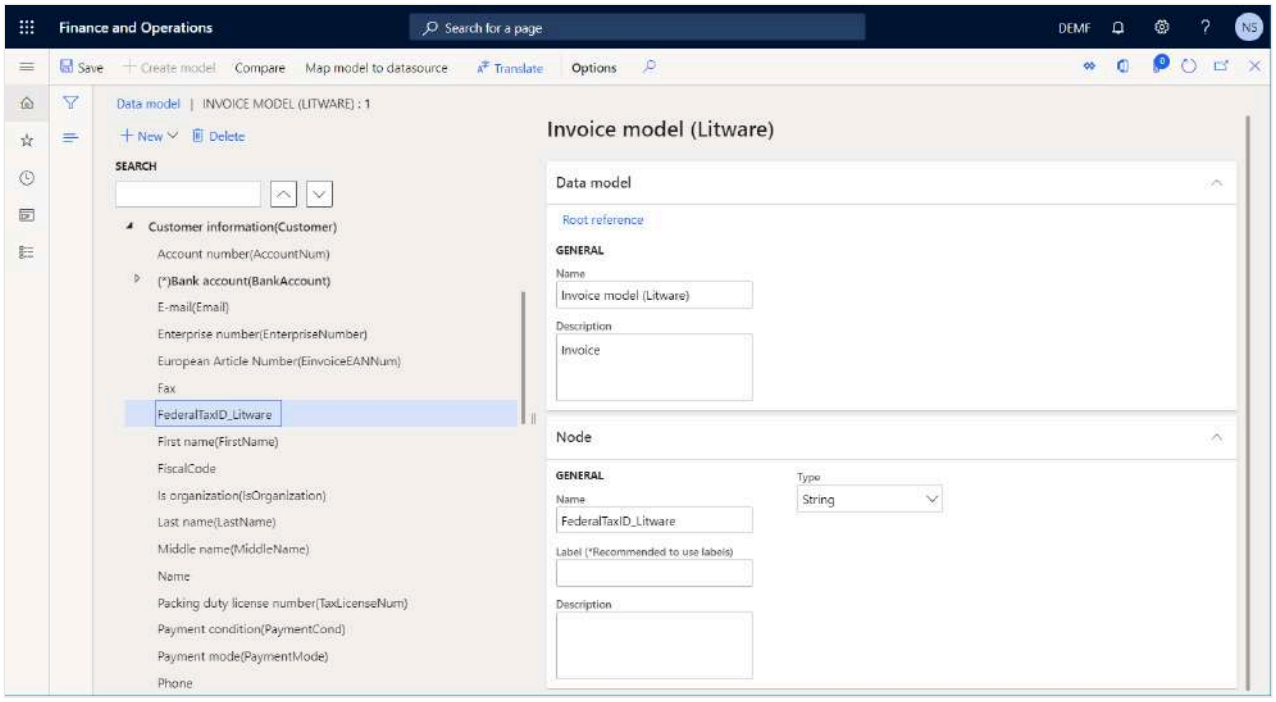
Configure a custom data model

Modify your custom data model by adding a new field to provide the value of a federal tax identification code. This code is part of the customer data for every ER format that uses this data model as a data source.

1. On **Configurations**, in the configuration tree in the left pane, select **Invoice model (Litware)**.
2. On the **Versions** FastTab, select version **50.1** of the selected ER data model configuration in **Draft** status.
3. On the Action Pane, select **Designer** for the selected configuration version.
4. On the **Data model designer** page, in the data model tree, select **Customer information (Customer)**.
5. Select **New**.
6. In the drop-down dialog box, in the **New node as a** field, accept the default value, **Child of an active node**.
7. In the **Name** field, enter **FederalTaxID_Litware**.

8. In the **Item type** field, accept the default value, **String**.

9. Select **Add**, and then select **Save**.



> **Note**
>
> The **Label** and **Description** fields describe the purpose of the new field. You can fill in these fields in multiple languages. For more information, see Design multilingual reports in Electronic reporting.

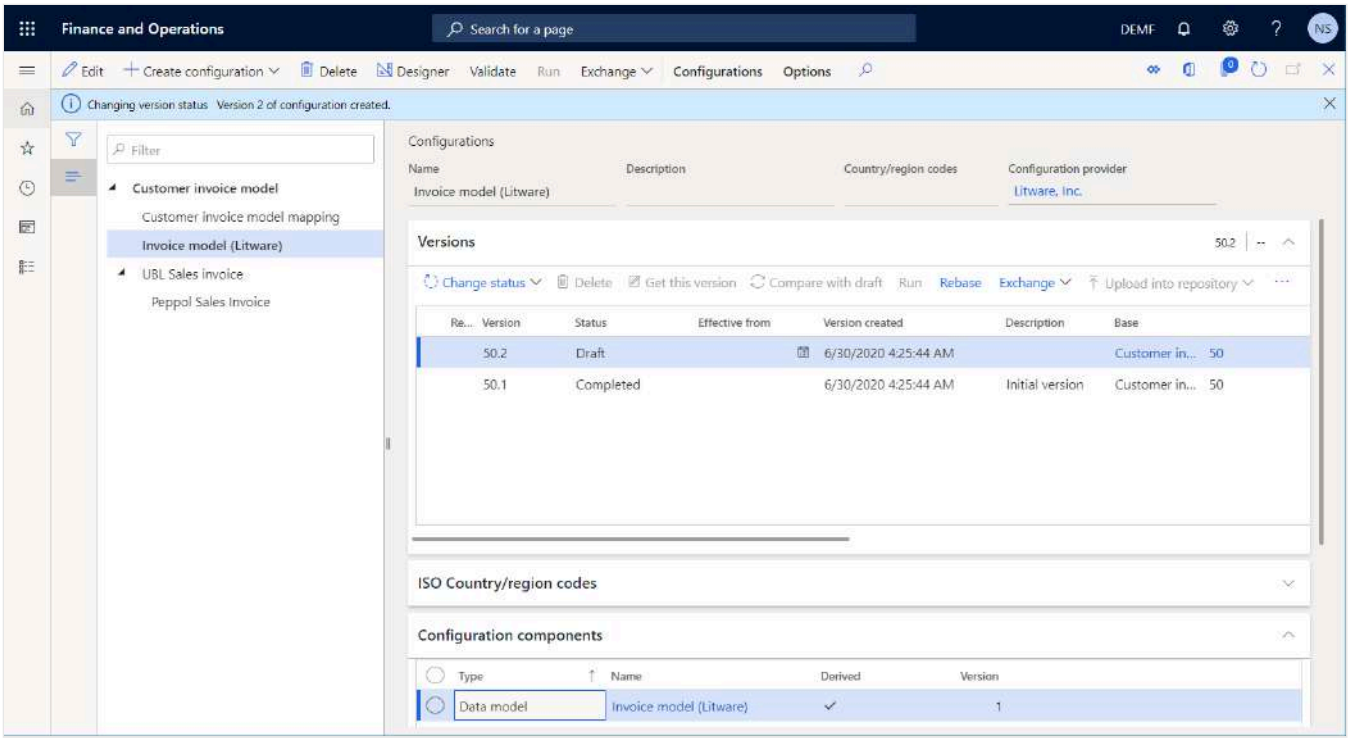
10. Close the **Data model designer** page.

Complete a custom data model configuration

Complete your work with version 50.1 of your custom ER data model configuration to make it available so that other custom ER configurations can be added.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Invoice model**, and select **Invoice model (Litware)**.
3. On the **Versions** FastTab, select **Change status** > **Complete**, and then select **OK**.

The status of version 50.1 changes from **Draft** to **Completed**, and the version becomes read-only. A new editable version, 50.2, is added and has a status of **Draft**. You can use this version to make further changes in your custom ER data model configuration.



Customize the model mapping configuration

As a user in the Electronic Reporting Developer role, you can design your custom ER model mapping.

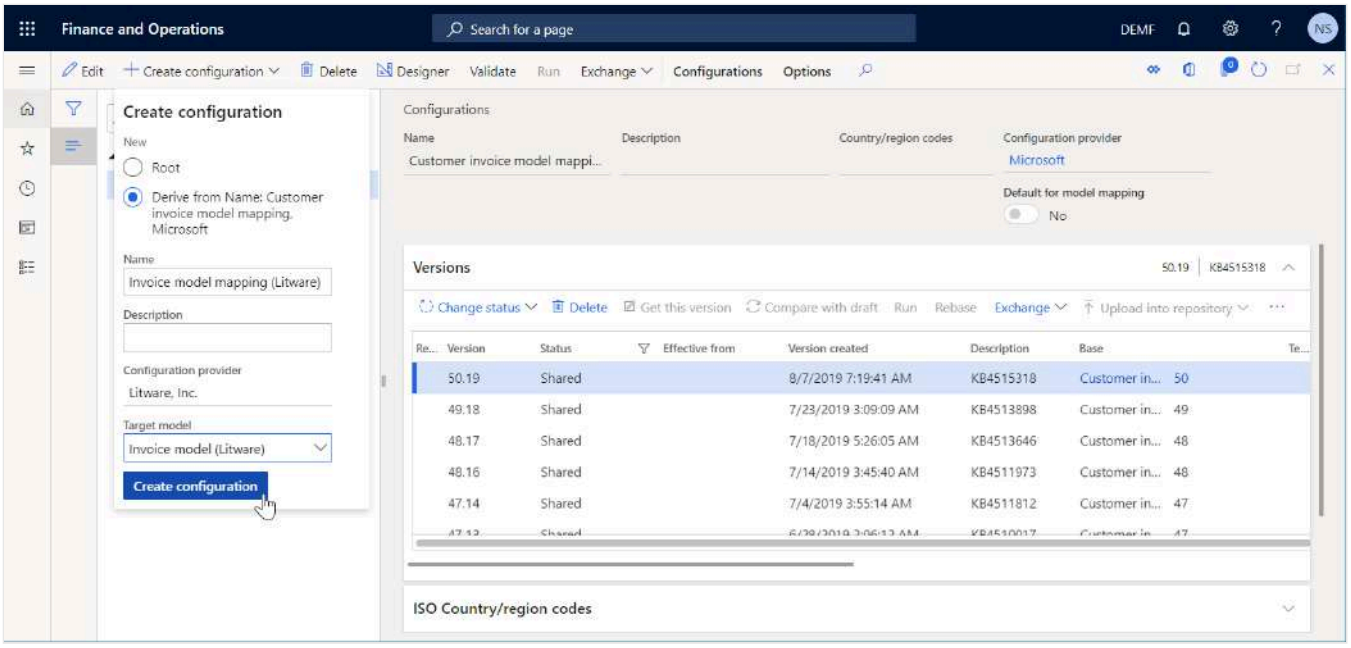
Add a custom model mapping configuration

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Customer invoice model**, and select **Customer invoice model mapping**.
3. On the Action Pane, select **Create configuration**.
4. In the drop-down dialog box, in the **New** field, select **Derive from Name: Customer invoice model mapping, Microsoft** to indicate that your new custom ER model mapping configuration should be based on the ER model mapping configuration.
5. In the **Name** field, enter **Invoice model mapping (Litware)**.
6. In the **Target model** field, select **Invoice model (Litware)**.

> **Important**

> This step is very important. If you omit it, you can't use your custom data model in the configured model mapping.

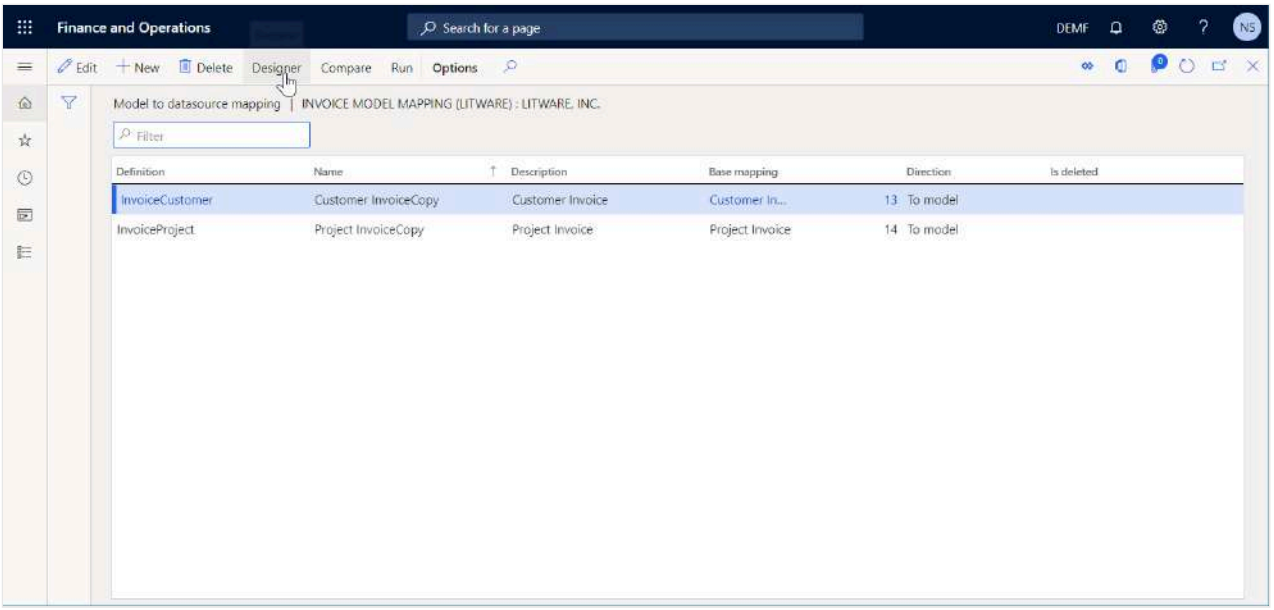
7. Select **Create configuration** to add the new ER configuration.



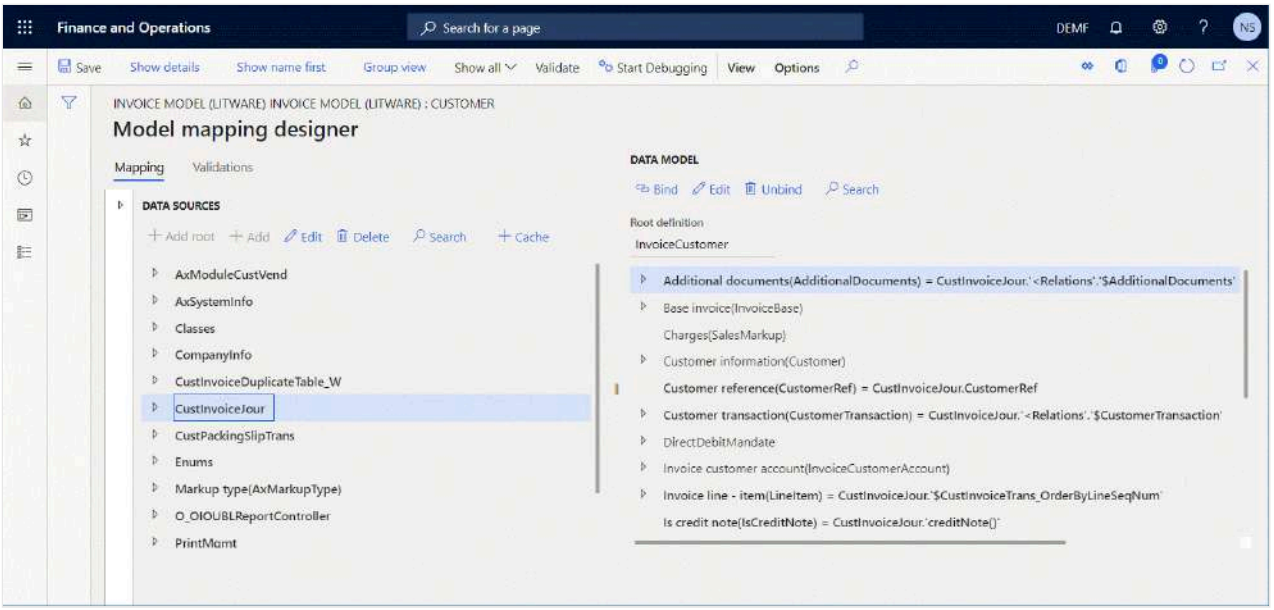
Configure a custom model mapping

You must modify your custom model mapping and specify how the custom **FederalTaxID_Litware** field of the custom data model should be filled in with application data at runtime.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Customer invoice model** > **Customer invoice model mapping**, and select **Invoice model mapping (Litware)**.
3. On the Action Pane, select **Designer**.
4. On the **Model to datasource mapping** page, select the **Customer invoice** mapping.



5. Select **Designer**.
6. On the **Model mapping designer** page, in the **Data sources** pane, expand the **CustInvoiceJour** data source that represents the **CustInvoiceJour** application table.
7. Under **CustInvoiceJour**, expand **Relations** to review the list of relations of the many-to-one (N:1) type for the **CustInvoiceJour** table.
8. Under **CustInvoiceJour** > **Relations**, expand **Customers (CustTable)** to access the fields and relations of the **Customers (CustTable)** table.
9. Select the **FederalTaxID_Custom** data source field that you implemented earlier.
10. In the **Data model** pane, expand **Customer information (Customer)**, and select the **FederalTaxID_Litware** data model field.
11. Select **Bind**.



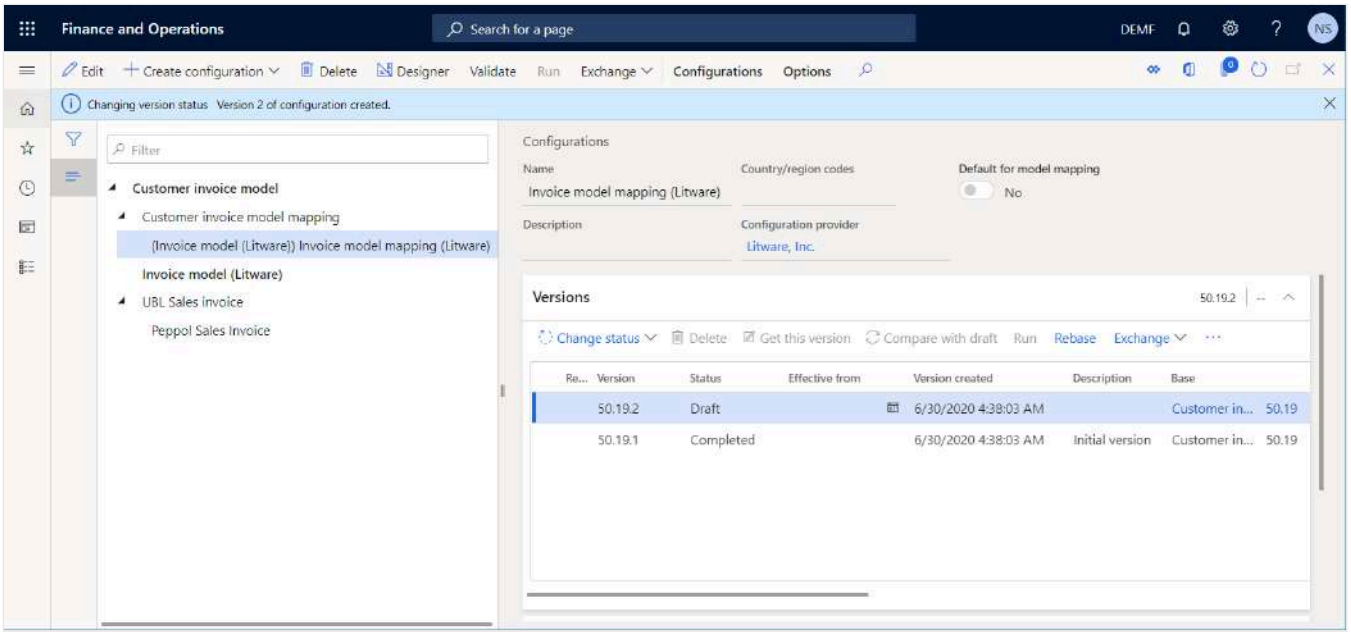
12. Select **Save**.
13. Close the **Model mapping designer** page.
14. Close the **Model to datasource mapping** page.

Complete a custom model mapping configuration

You must complete your work with version 50.19.1 of your custom ER model mapping configuration to make it available for use.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Customer invoice model** > **Customer invoice model mapping**, and select **Invoice model mapping (Litware)**.
3. On the **Versions** FastTab, select **Change status** > **Complete**, and then select **OK**.

The status of version 50.19.1 changes from **Draft** to **Completed**, and the version becomes read-only. A new editable version, 50.19.2, is added and has a status of **Draft**. You can use this version to make further changes in your custom ER model mapping configuration.



> **Note**
>
> The supported configuration lifecycle doesn't cover the lifecycle of database changes. If you export version 50.19.1 of the **Invoice model mapping (Litware)** configuration from the current Finance instance and try to import it into another instance that doesn't contain the custom **FederalTaxID_Custom** field in the **CustTable** table, an exception occurs. The exception states that the imported ER configuration isn't compatible with the metadata of the target Finance instance.

Customize the format configuration

As a user in the Electronic Reporting Functional Consultant role, you can design your custom ER format.

Add a custom format configuration

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Customer invoice model** > **UBL Sales invoice**, and select **Peppol Sales Invoice**.
3. On the Action Pane, select **Create configuration**.
4. In the drop-down dialog box, in the **New** field, select **Derive from Name: Peppol Sales Invoice, Microsoft** to indicate that your custom ER format configuration should be based on the ER format configuration.

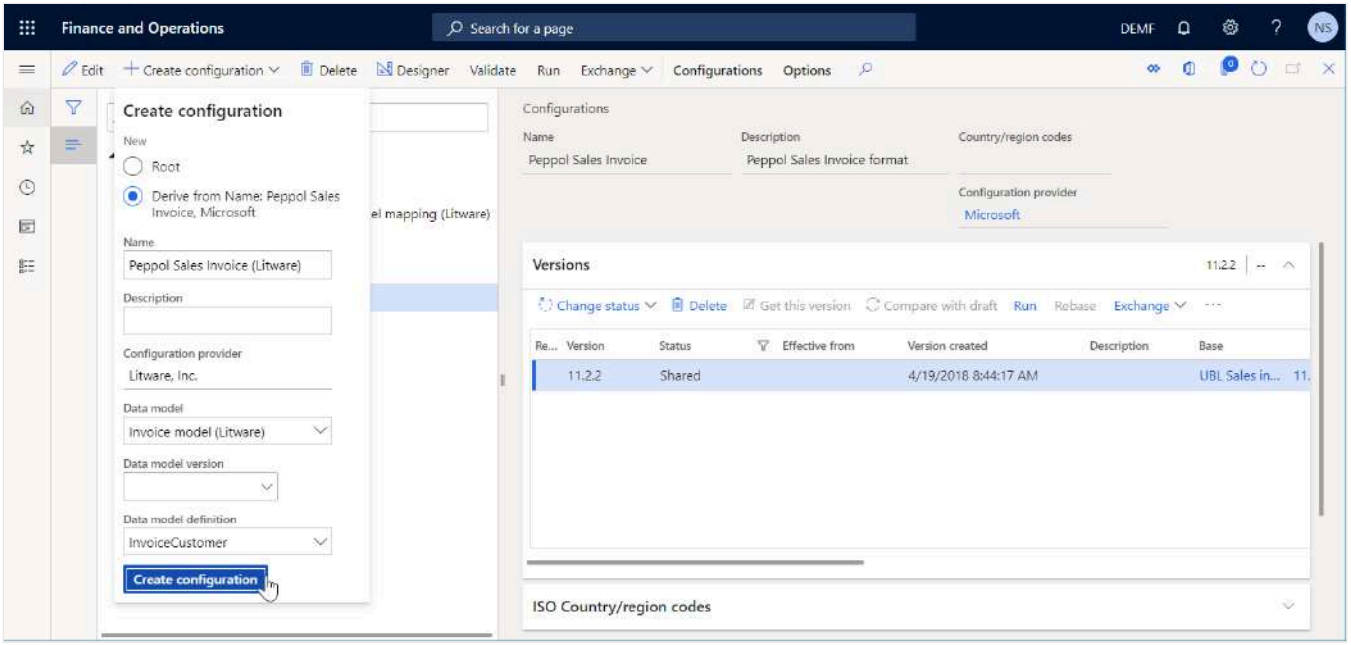
5. In the **Name** field, enter **Peppol Sales Invoice (Litware)**.

6. In the **Data model** field, select **Invoice model (Litware)** to use your custom data model and model mapping components.

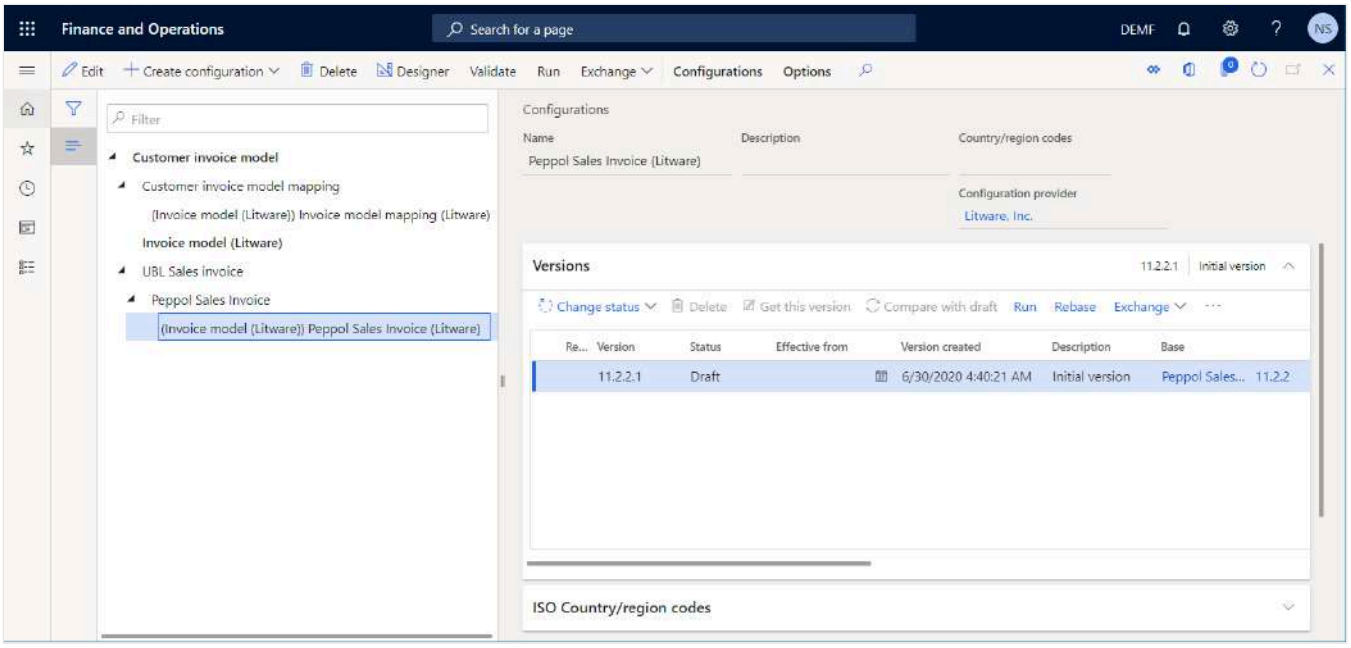
> **Note**
>
> This step is very important. If you omit it, you can't use your custom data model in the configured format.

7. In the **Data model** field, select the **InvoiceCustomer** root definition.

8. Select **Create configuration** to add the new ER configuration.



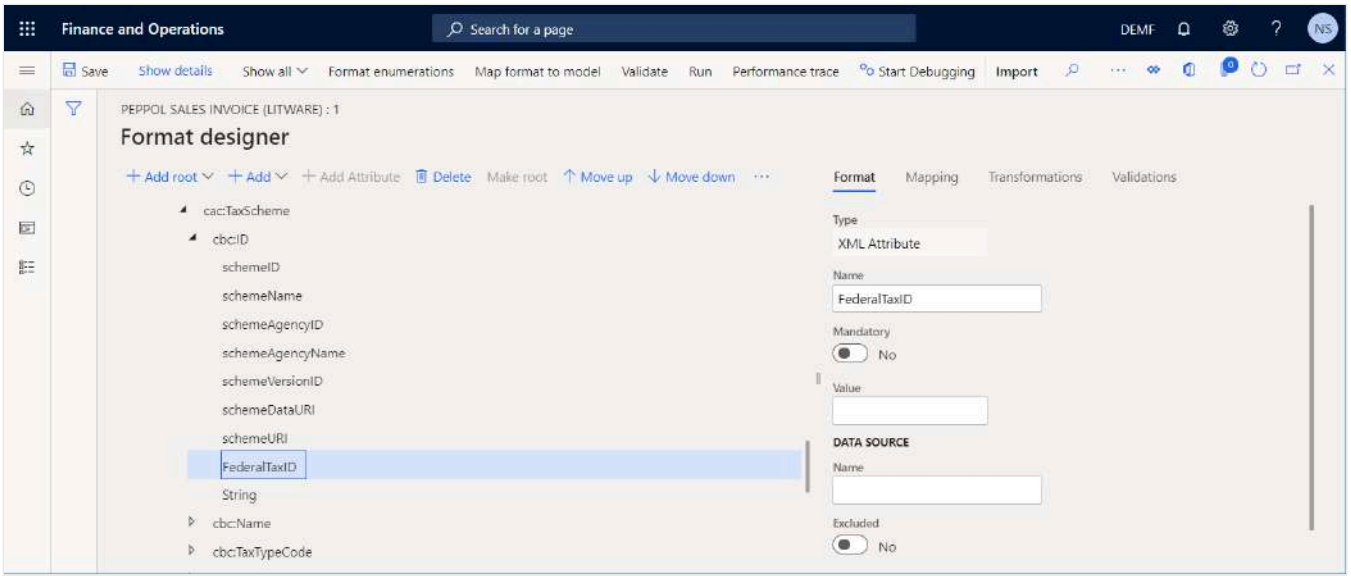
You can now use the ER Operations designer to edit version 11.2.2.1 of the **Peppol Sales Invoice (Litware)** ER configuration in **Draft** status.



Configure a custom format

Modify your custom format by adding a new format element to fill in the value of an invoiced customer's federal tax identification code.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Customer invoice model** > **UBL Sales invoice** > **Peppol Sales Invoice**, and select **Peppol Sales Invoice (Litware)**.
3. On the Action Pane, select **Designer**.
4. In the format tree, expand **XMLHeader** > **Invoice** > **cac:AccountingCustomerParty** > **cac:Party** > **cac:PartyTaxScheme** > **cac:TaxScheme**, and select **cbc:ID**.
5. Select **Add**, and then select **XML** > **Attribute**.
6. In the **Component property** dialog box, in the **Name** field, enter **FederalTaxID**.
7. Select **OK** to add a new format element to create a new **FederalTaxID** XML attribute in a generated XML file.
8. In the format tree, under **XMLHeader** > **Invoice** > **cac:AccountingCustomerParty** > **cac:Party** > **cac:PartyTaxScheme** > **cac:TaxScheme** > **cbc:ID**, select **FederalTaxID**.
9. Select **Move up**.

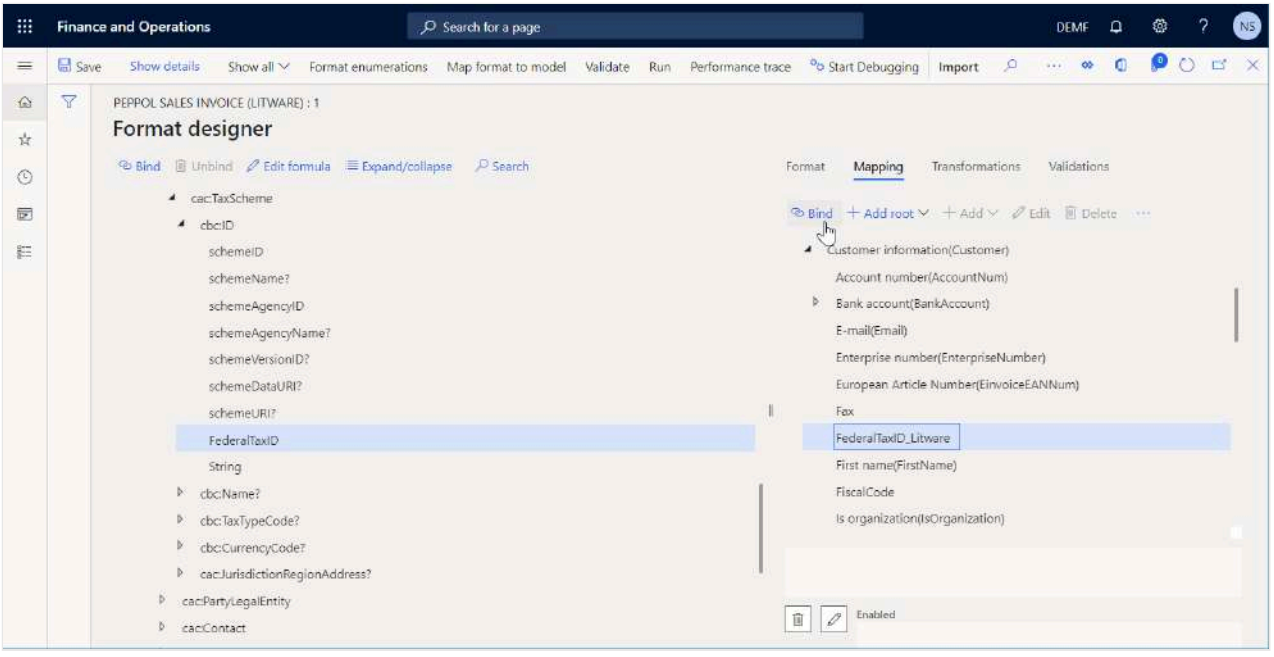


Configure a custom format mapping

1. On the **Format designer** page, on the **Mapping** tab, expand the **Invoice** data source of the **Model** type.

2. Under **Invoice**, expand **Customer information (Customer)**, and select **FederalTaxID_Litware**.

3. Select **Bind**.



4. Select the **Invoice** data source of the **Model** type, and then select **Edit**.

5. In the **Version** field, select version **1** of your custom data model, and then select **OK**.

6. Select **Save**.

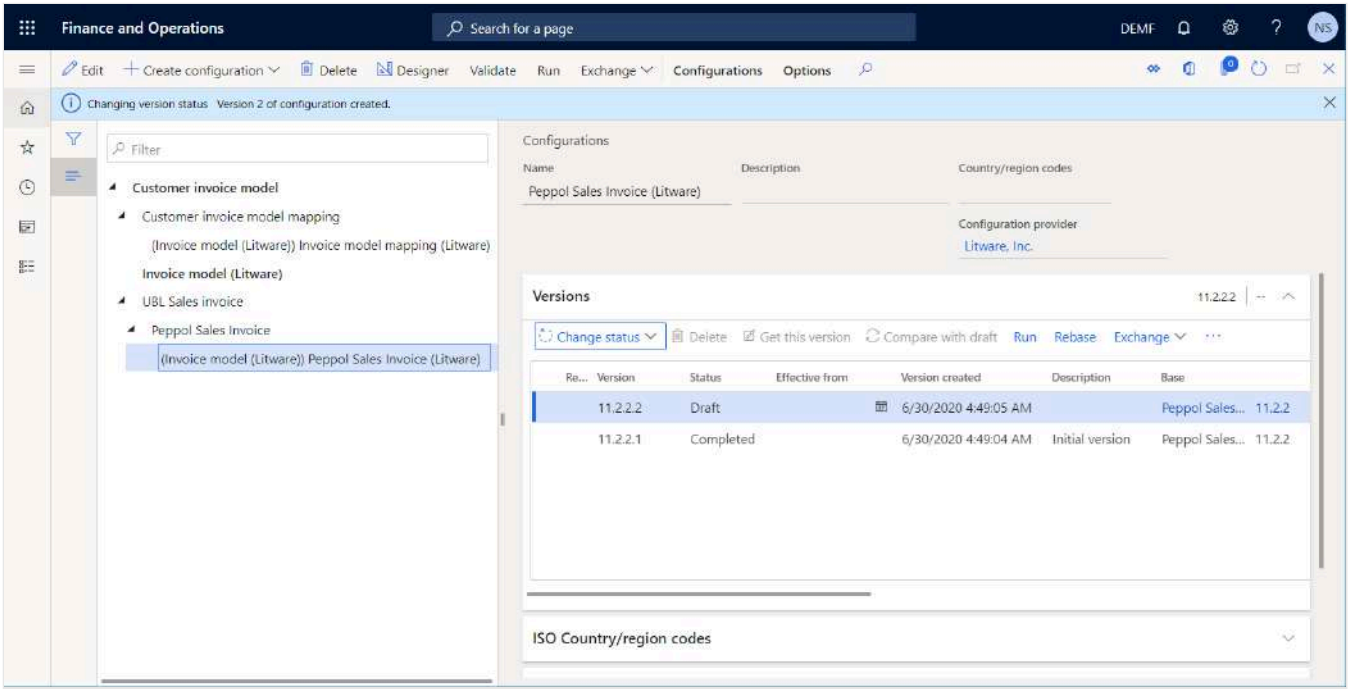
7. Close the **Format designer** page.

Complete a custom format configuration

Complete your work with version 11.2.2.1 of your custom ER format configuration to make it available for use.

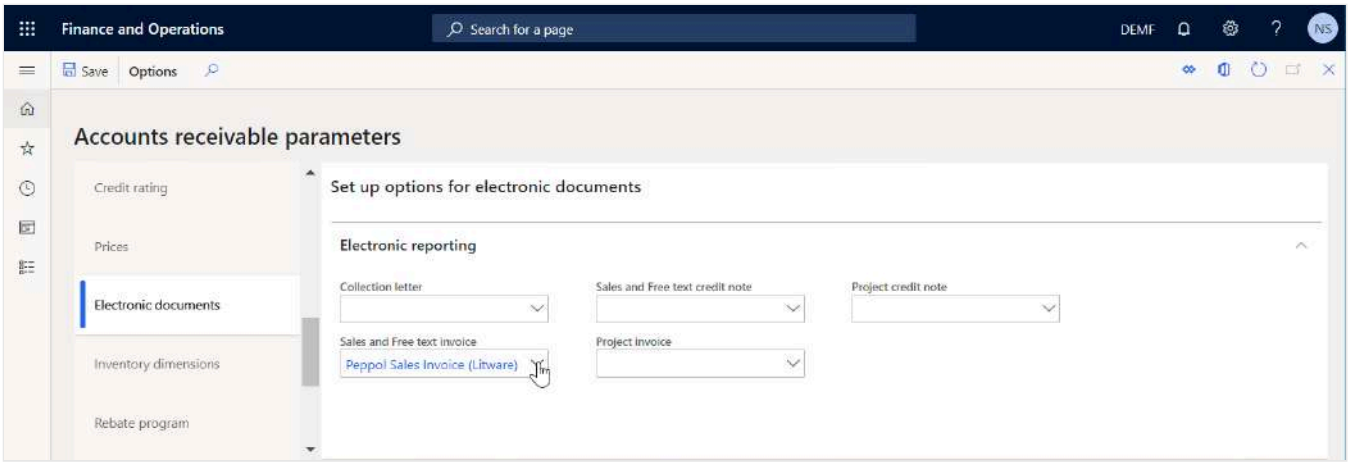
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Customer invoice model** > **UBL Sales invoice** > **Peppol Sales Invoice**, and select **Peppol Sales Invoice (Litware)**.
3. On the **Versions** FastTab, select **Change status** > **Complete**, and then select **OK**.

The status of version 11.2.2.1 changes from **Draft** to **Completed**, and the version becomes read-only. The system adds a new editable version, 11.2.2.2, with a status of **Draft**. Use this version to make further changes in your custom ER format configuration.



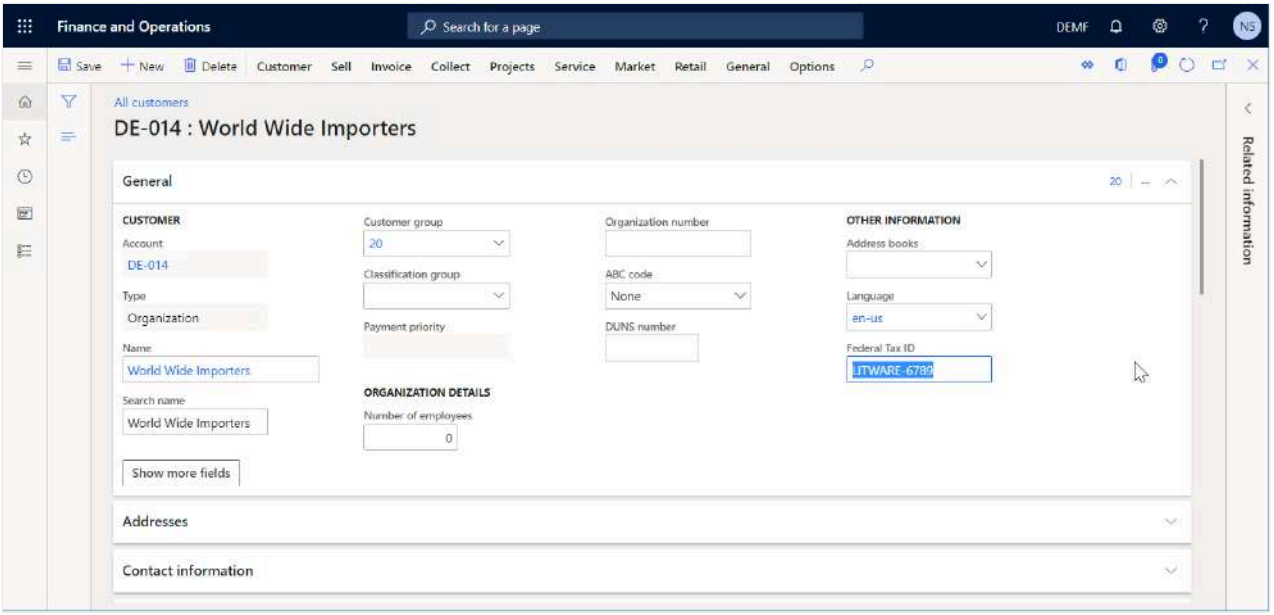
Configure the Accounts receivable parameters to start using custom ER configurations

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Electronic documents** tab, in the **Electronic reporting** section, select **Peppol Sales Invoice (Litware)** in the **Sales and Free text invoice** field.
3. Select **Save**.



Update a customer record by adding a federal tax identification code

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. On the **All customers** page, select the **DE-014** customer account link.
3. On the **General** section, enter **LITWARE-6789** in the **Federal Tax ID** field.
4. Select **Save**.



5. Close the **All customers** page.

Process a customer invoice by using custom ER configurations

Select and send a posted invoice

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. On **Free text invoice**, select the invoice that you previously added and posted.
3. On the Action Pane, in the **Document** group, select **Send** > **Original**.
4. Close **Free text invoice**.

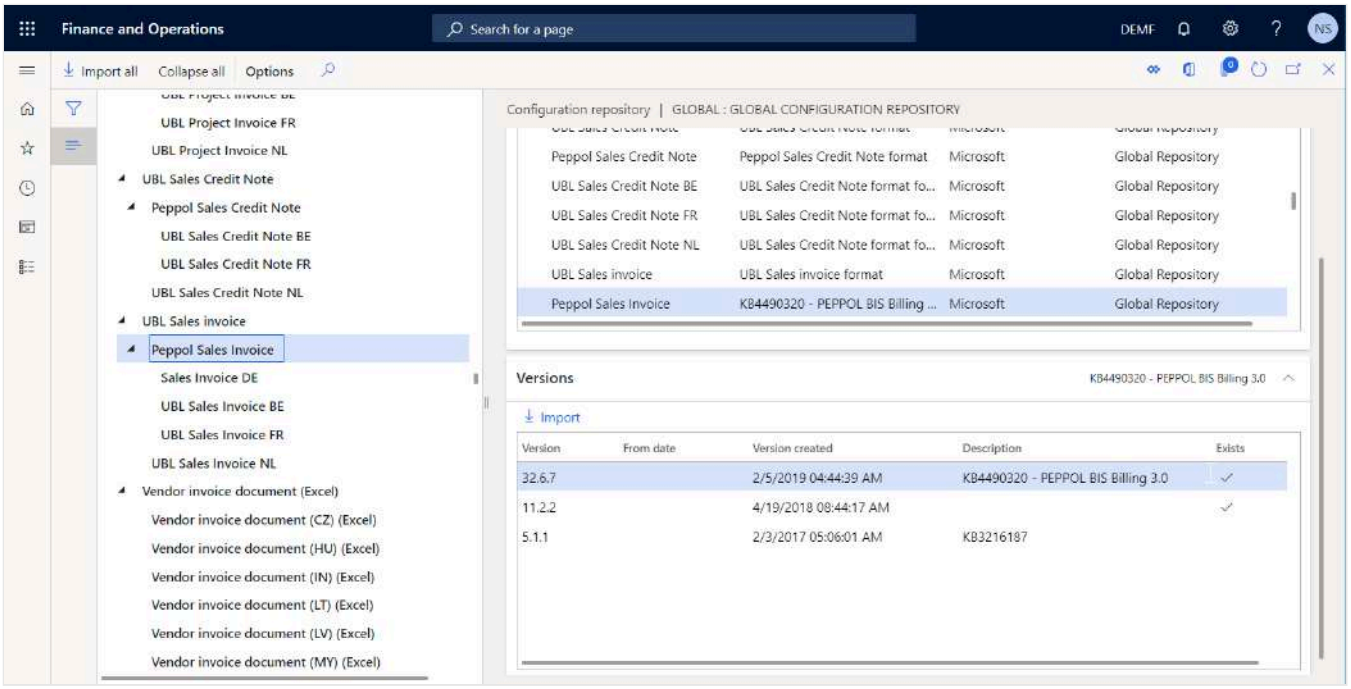
Analyze a generated e-invoice

1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting jobs**.
2. On **Electronic reporting jobs**, select the latest record that has the task description **Send the eInvoice XML**.
3. Select **Show files** to access the list of generated files.
4. Select **Open** to download the e-invoice XML file.
5. Analyze the e-invoice XML file. Notice that, in accordance with your customization, the customer tax schema includes the custom **FederalTaxID** XML attribute in addition to the **schemeID** and **schemeAgencyID** XML attributes. The value of this new XML attribute is specified by the **LITWARE-6789** federal tax ID that you entered for an invoiced customer.

Import the latest versions of standard ER configurations

To keep the set of standard ER configurations in your Finance instance up to date, import new versions whenever they become available in the ER repository that you configured for the instance.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, select the **Microsoft** tile, and then select **Repositories** to view the list of repositories for the Microsoft provider.
3. On **Configuration repositories**, select the repository of the **Global** type, and then select **Open**. If you're prompted for authorization to connect to Regulatory Configuration Service, follow the authorization instructions.
4. On **Configuration repository**, in the configuration tree in the left pane, select the **Peppol Sales Invoice** format configuration.
5. On the **Versions** FastTab, select version **32.6.7** of the selected ER format configuration that supports customer electronic invoices in PEPPOL BIS 3 format. For more information, see KB4490320.
6. Select **Import** to download the selected version from the Global repository to the current Finance instance.



For information about how to automate this process, see Import updated versions of ER configurations.

Review the imported ER configurations

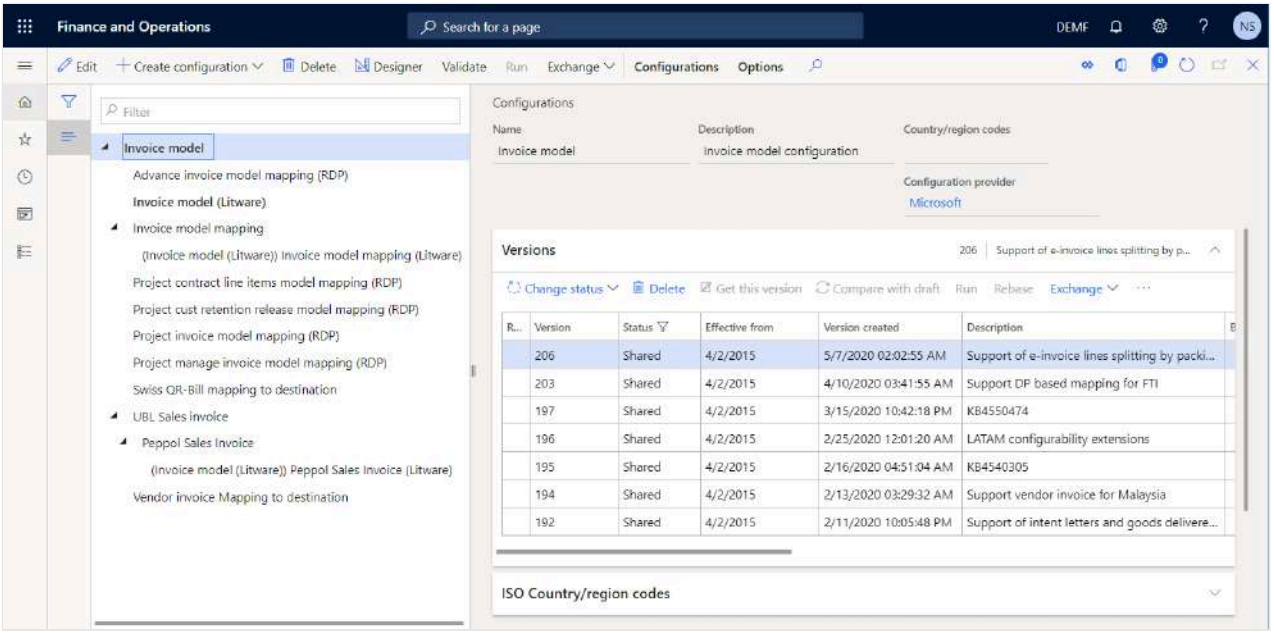
1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configurations** section, select the **Reporting configurations** tile.
3. Expand the **Configuration components** FastTab.

4. In the configuration tree in the left pane, expand **Invoice model**. The name of **Customer invoice model** changed to **Invoice model** in one of the imported ER data model configurations.
5. In the configuration tree in the left pane, expand **Invoice model mapping**. The name of **Customer invoice model mapping** changed to **Invoice model mapping** in one of the imported ER model mapping configurations.
6. Expand **UBL Sales invoice** > **Peppol Sales Invoice**.

In addition to the selected **Peppol Sales Invoice** ER format, the latest versions of other required ER configurations are imported. Because new versions of ER configurations are constantly published to the Global repository and Lifecycle Services to keep the corresponding ER solutions compliant with new requirements, the latest versions of the required data model configuration and its model mapping configurations are imported.

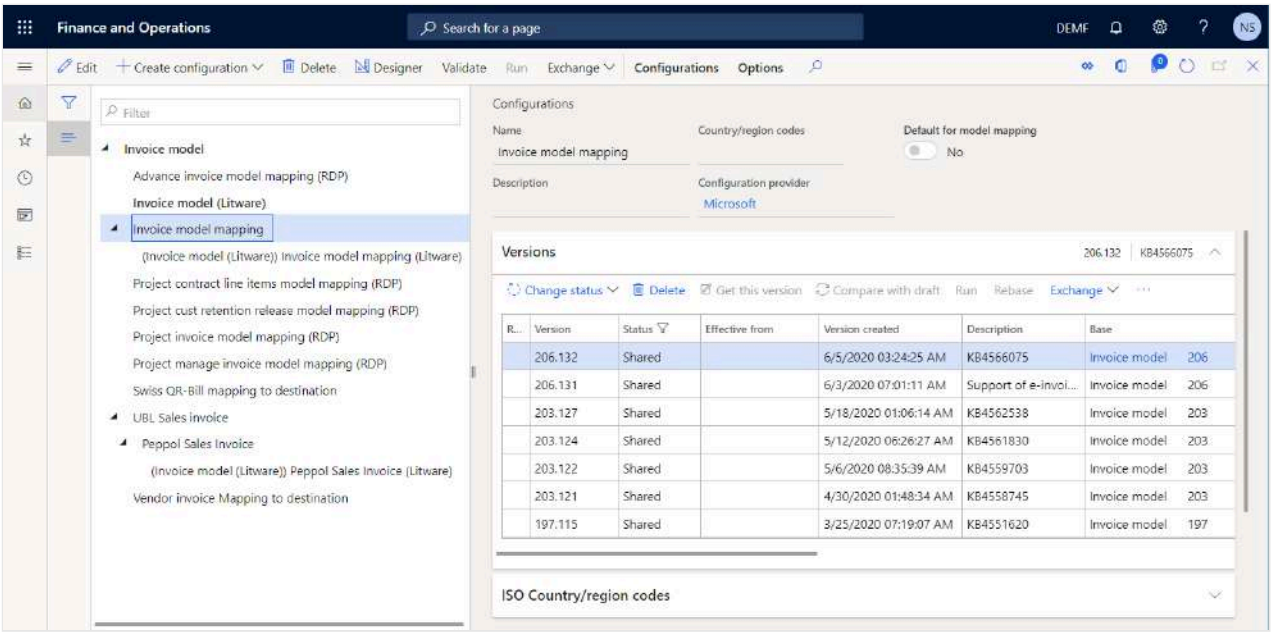
Make sure that the following ER configurations are eventually available in the configuration tree:

- **Invoice model** ER data model configuration:
  - Version 206 (or later) contains version 24 (or later) of the data model ER component that represents the data structure of the invoicing business domain. This ER configuration is imported as an ancestor of the latest available **Invoice model mapping** ER model mapping configuration.



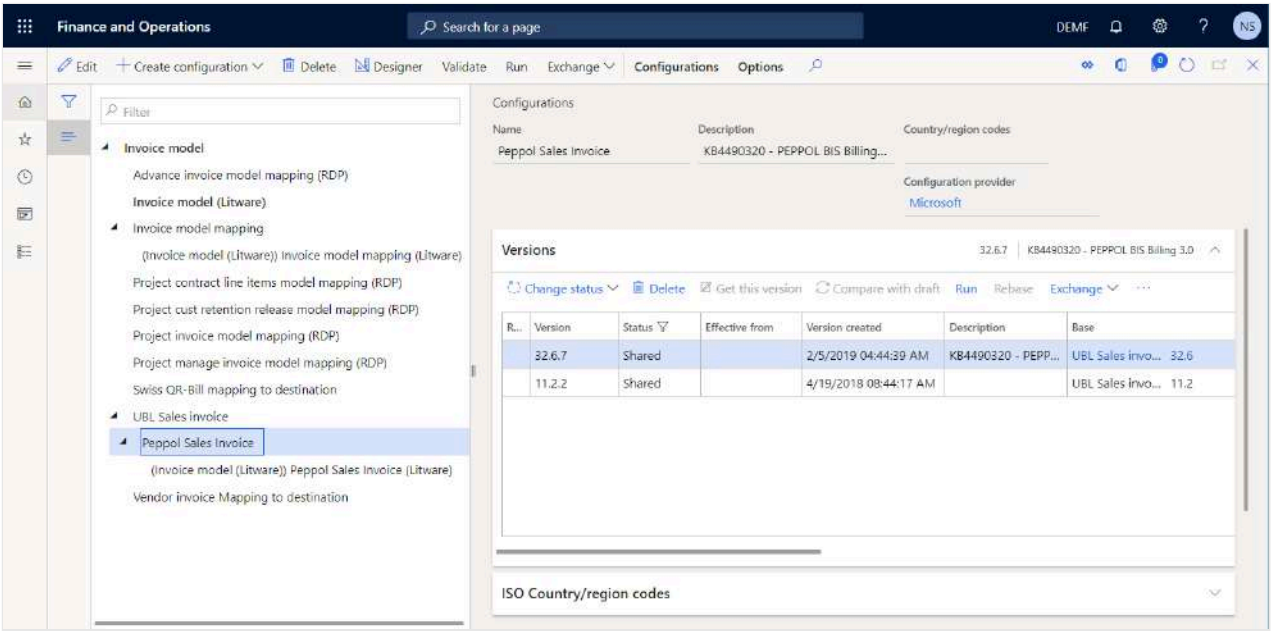
- **Invoice model mapping** ER model mapping configuration:
  - Version 206.132 (or later) is imported as the latest implementation of version 206 of the **Invoice model** ER data model configuration. It contains several model mapping ER

components that describe how the data model is filled in with application data at runtime.



- **UBL Sales invoice** ER format configuration:
  - Version 32.6 contains the format and format mapping ER components. The format component specifies the report layout. The format mapping component contains the model data source and specifies how this data source is used to fill in the report layout at runtime. This ER format is configured to generate e-invoices in UBL format. It's imported as a parent of the **Peppol Sales Invoice** ER format that you selected for import.

- **Peppol Sales Invoice** ER format configuration:
  - Version 32.6.7 contains the format and format mapping ER components that are configured to generate e-invoices in PEPPOL format.



Adopt the changes to the new standard ER configurations in your custom ER configurations

Adopt your custom ER data model

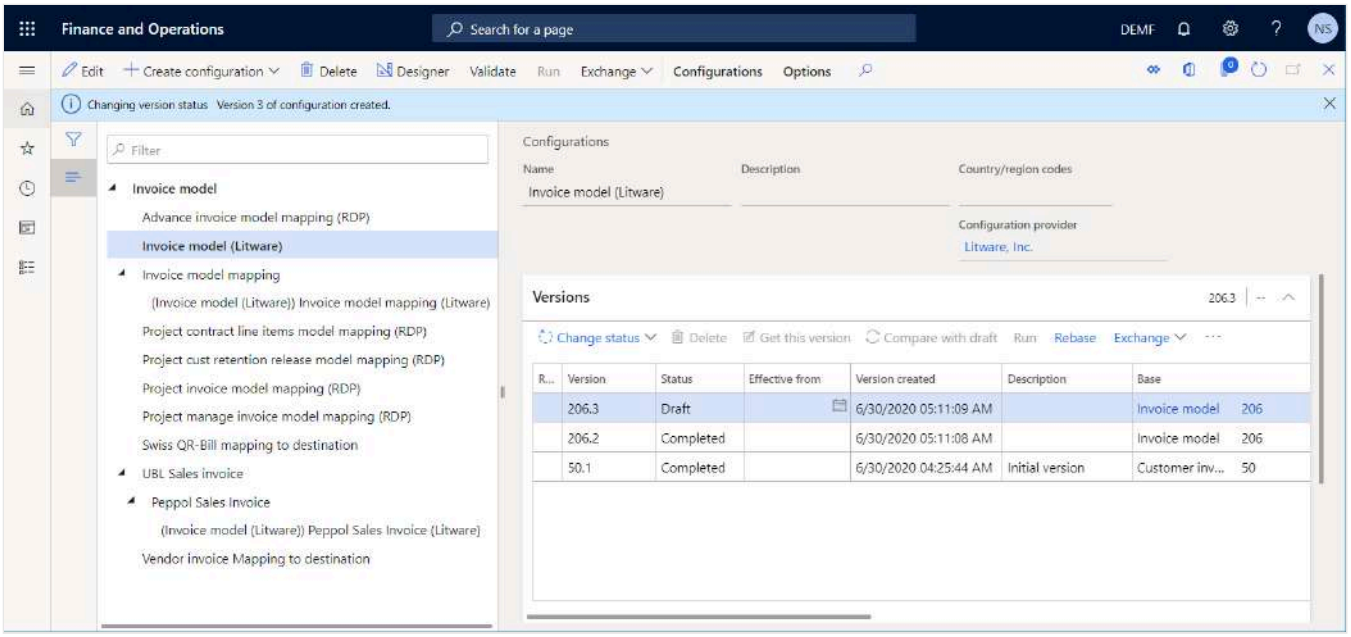
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Invoice model**, and select **Invoice model (Litware)**.
3. On the **Versions** FastTab, for draft version **50.2** of the selected data model configuration, select **Rebase**.
4. In the **Target version** field, confirm the selection of version **206** of the base ER data model configuration, and then select **OK**.

   The system renumbers the draft version of your custom ER data model configuration from **50.2** to **206.2** to indicate that it now contains your customization that was merged with the changes in the latest version (206) of the base ER data model configuration.

   > ⓘ **Note**
   >
   > The rebase action is reversible. To cancel this rebase, select version **50.1** of the **Invoice model (Litware)** model on the **Versions** FastTab, and then select **Get this version**. Version 206.2 is renumbered back to **50.2**, and the content of draft version 50.2 matches the content of version 50.1.

5. On the **Versions** FastTab, select **Change status** > **Complete**, and then select **OK**.

The status of version 206.2 changes from **Draft** to **Completed**, and the version becomes read-only. A new editable version, 206.3, is added and has a status of **Draft**. You can use this version to make further changes in your custom ER data model configuration.



Adopt your custom ER model mapping

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.

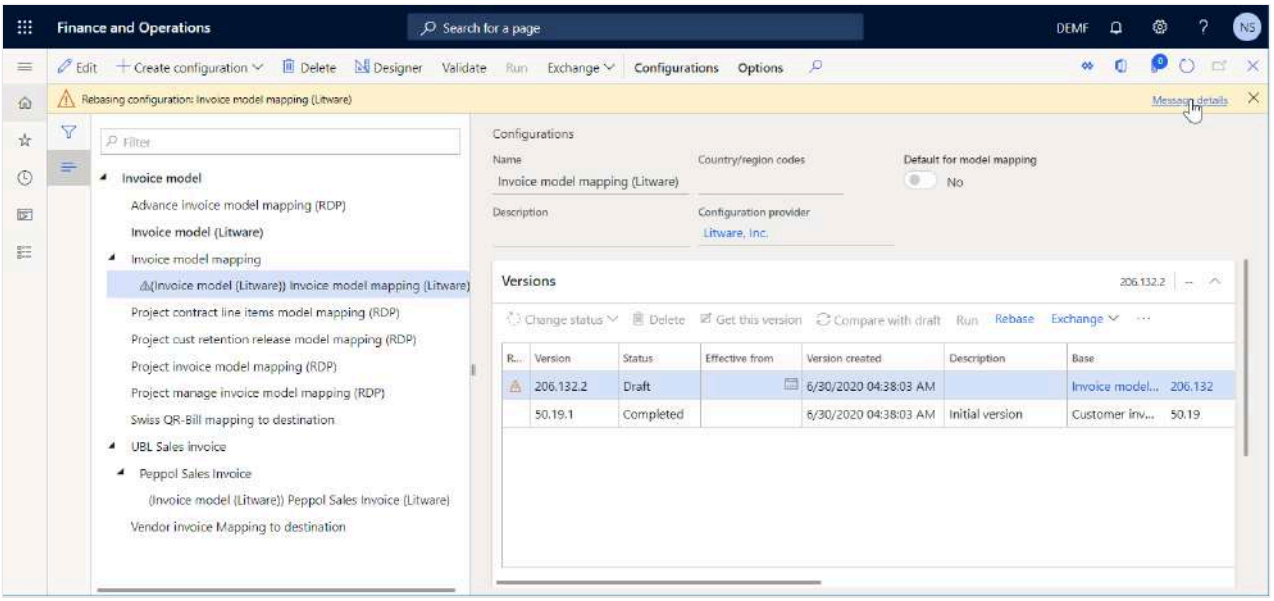
2. On **Configurations**, in the configuration tree in the left pane, expand **Invoice model** > **Invoice model mapping**, and select **Invoice model mapping (Litware)**.

3. On the **Versions** FastTab, for draft version **50.19.2** of the selected model mapping configuration, select **Rebase**.

4. In the **Target version** field, confirm the selection of version **206.132** of the base ER model mapping configuration, and then select **OK**.

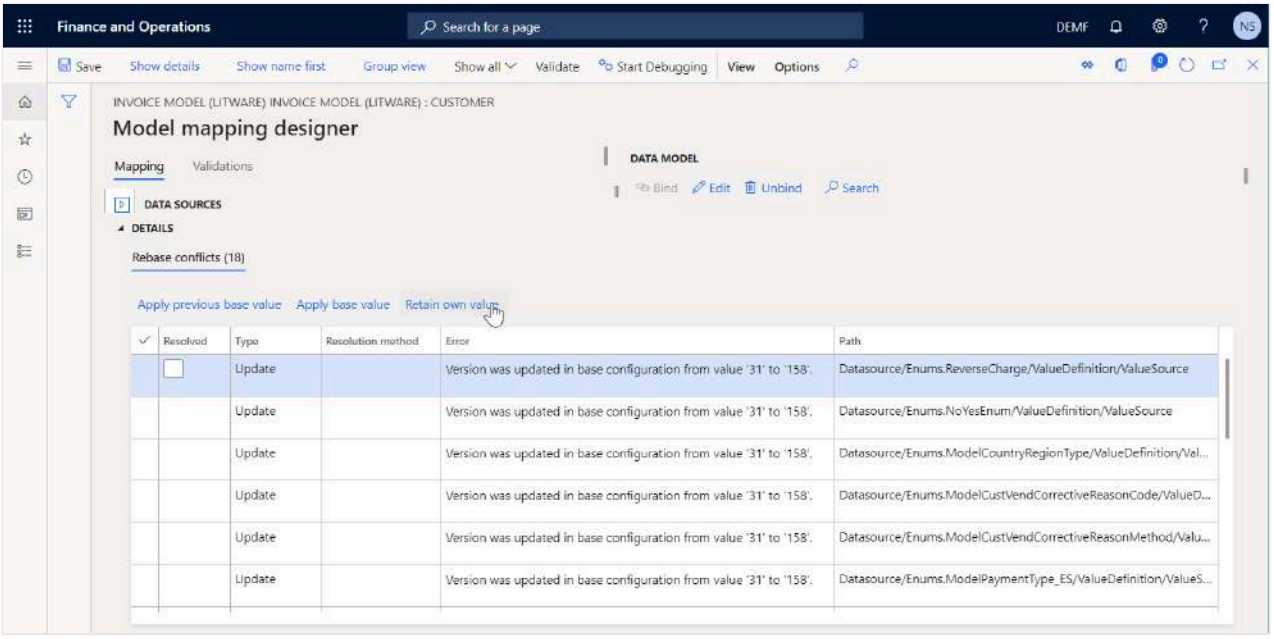
   The system renumbers the draft version of your custom ER model mapping configuration from **50.19.2** to **206.132.2** to indicate that it now contains your customization that was merged with the changes in the latest version (206.132) of the base ER model mapping configuration.

   The system discovers some rebase conflicts. You must now manually resolve those conflicts.



5. On the Action Pane, select **Designer**, and then, in the list of mappings, select **Customer Invoice**.

6. For each rebase conflict, select **Retain own value**, because you must keep the version number of your custom data model for every component that the system mentions.



7. Select **Save**, and then close the **Model mapping designer** page.

8. In the list of mappings, select **Project Invoice**.

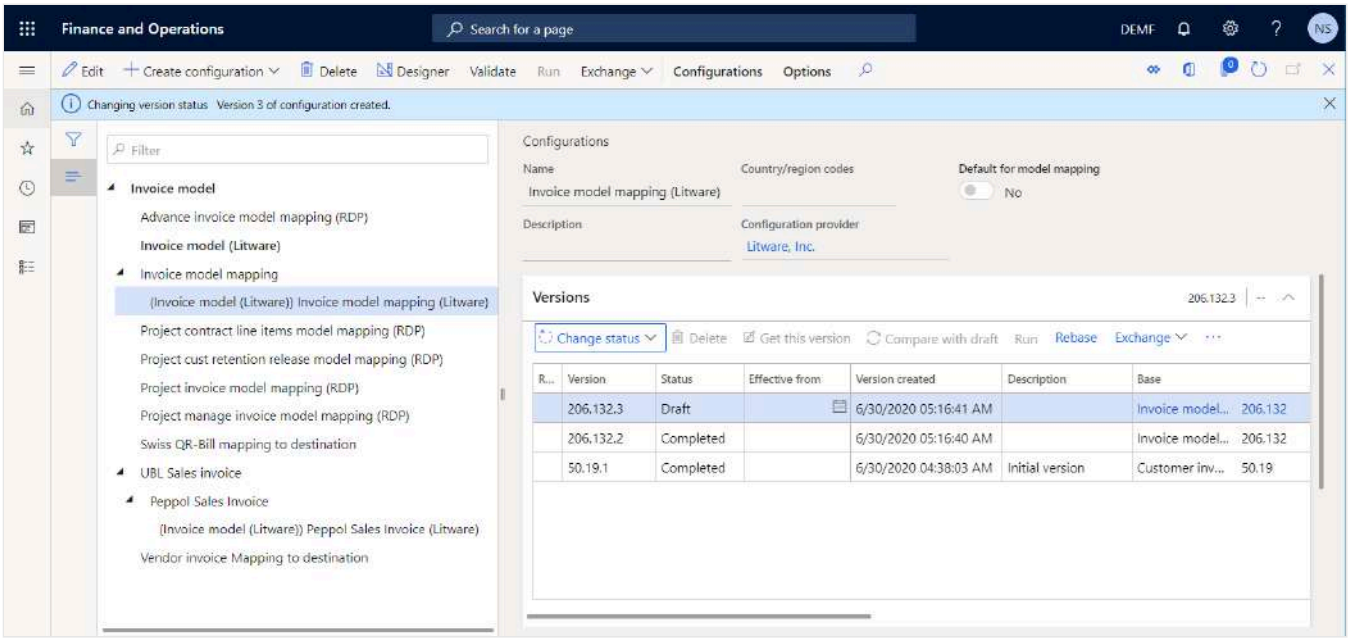
9. For each rebase conflict, select **Retain own value**, because you must keep the version number of your custom data model for every component that the system mentions.

10. Select **Save**, and then close the **Model mappings** page.

> **Note**
>
> The rebase action is reversible. To cancel this rebase, select version **50.19.1** of the **Invoice model mapping (Litware)** model mapping on the **Versions** FastTab, and then select **Get this version**. The system renumbers version 206.132.2 back to **50.19.2**, and the content of draft version 50.19.2 matches the content of version 50.19.1.

11. On the **Versions** FastTab, select **Change status** > **Complete**, and then select **OK**.

The system changes the status of version 206.132.2 from **Draft** to **Completed**, and makes the version read-only. The system adds a new editable version, 206.132.3, and sets its status to **Draft**. You can use this version to make further changes in your custom ER model mapping configuration.



Adopt your custom ER format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree in the left pane, expand **Invoice model** > **UBL Sales invoice** > **Peppol Sales Invoice**, and select **Peppol Sales Invoice (Litware)**.
3. On the **Versions** FastTab, for draft version **11.2.2.2** of the selected format configuration, select **Rebase**.
4. In the **Target** version field, confirm the selection of version **32.6.7** of the base ER format configuration, and then select **OK**.

The system renumbers the draft version of your custom ER format configuration from **11.2.2.2** to **32.6.7.2** to indicate that it now contains your customization that was merged with the changes in the latest version (32.6.7) of the base ER format configuration.

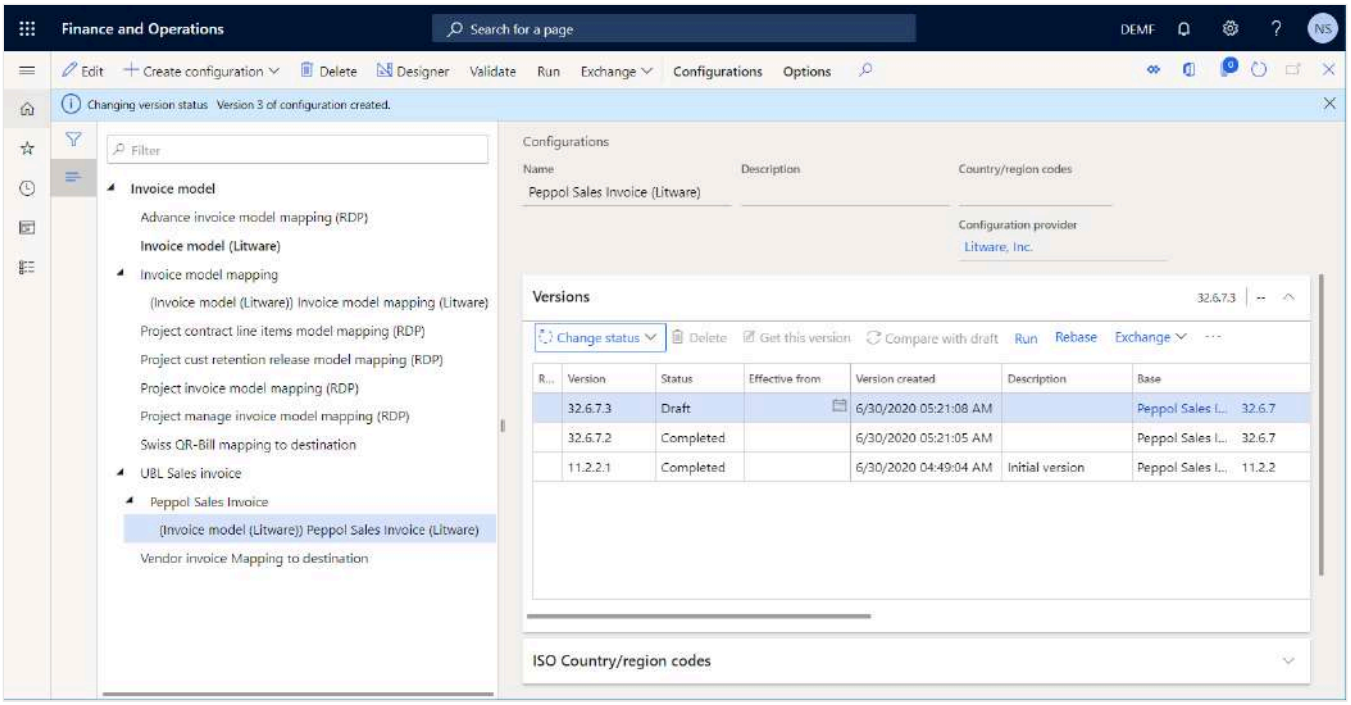
The system discovers some rebase conflicts. You must now manually resolve those conflicts.

5. On the Action Pane, select **Designer**.
6. For each rebase conflict, select **Retain own value**, because you must keep the version number of your custom data model for every component that the system mentions.
7. Select **Save**.
8. On the **Mapping** tab, select the **Invoice** data source of the **Model** type, and then select **Edit**.
9. In the **Version** field, select version **2** of your custom data model, and then select **OK**.
10. Select **Save**.

> ⓘ **Note**
>
> The rebase action is reversible. To cancel this rebase, select version **11.2.2.1** of the **Peppol Sales Invoice (Litware)** format on the **Versions** FastTab, and then select **Get this version**. Version 32.6.7.2 is renumbered back to **11.2.2.2**, and the content of draft version 11.2.2.2 matches the content of version 11.2.2.1.

11. Close the **Format designer** page.
12. On the **Versions** FastTab, select **Change status** > **Complete**, and then select **OK**.

The status of version 32.6.7.2 changes from **Draft** to **Completed**, and the version becomes read-only. A new editable version, 32.6.7.3, is added and has a status of **Draft**. You can use this version to make further changes in your custom ER format configuration.



Process a customer invoice by using new versions of the custom ER configurations

Select and send a posted invoice

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. On **Free text invoice**, select the invoice that you previously added and posted.
3. On the Action Pane, in the **Document** group, select **Send** > **Original**.

> **Note**
>
> Because you now have two versions of the **Peppol Sales Invoice (Litware)** ER format configuration, and neither version has an effective date value, the latest version is used to generate an e-invoice.

4. Close **Free text invoice**.

Analyze a generated e-invoice

1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting jobs**.
2. On **Electronic reporting jobs**, select the most recent record that has the task description **Send the eInvoice XML**.

3. Select **Show files** to access the list of generated files.

4. Select **Open** to download the e-invoice XML file.

5. Analyze the e-invoice XML file. Notice that, in accordance with your customization, the customer tax schema still contains the custom **FederalTaxID** XML attribute in addition to the **schemeID** and **schemeAgencyID** XML attributes. Additionally, because the changes in the new version of the base **UBL Sales invoice** format were merged with your customization, the text of the **cbc:CustomizationID** XML element changed from `urn:www.cenbii.eu:transaction:biicoretrdm010:ver1.0:#` `urn:www.peppol.eu:bis:peppol5a:ver1.0` to `urn:cen.eu:en16931:2017#compliant#urn:fdc:peppol.eu:2017:poacc:billing:3.0`.

Additional resources

- Electronic Reporting overview
- Download ER configurations from Lifecycle Services
- Download ER configurations from Global repository of Configuration service
- Create a free text invoice
- Create and work with custom fields

Last updated on 04/13/2026

Use JOIN data sources to get data from multiple application tables in Electronic reporting (ER) model mappings

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

While configuring Electronic reporting (ER) model mappings or formats, you can add required data sources of the **Join** type. At design time, you configure a **Join** data source as a set of several data sources, each of which returns a list of records. For every data source except the first one, you need to define the necessary conditions to join records of the current and previous data sources. At runtime, a configured data source of **Join** type returns a single joined list of records containing fields from the records of nested data sources.

The following types of joins are currently supported:

- Outer (left) join:
  - Join all records of the first (left-most) data source and then any matching records of the second (right-most) data source according to the configured conditions.
- Inner (right) join:
  - Join only records of the first (left-most) data source and only records of the second (right-most) data source that match each other according to the configured conditions.

In the configured **Join** data source, when all data sources are the **Table records** type, execution of the Join data source can be performed at the database level using a single SQL statement. This statement reduces the number of database calls, which improves model-mapping performance. Otherwise, execution of **Join data** source is performed in memory.

> **Note**

Using the **VALUEIN** function in ER expressions that specify conditions for joining records in data sources of Join type isn't supported yet. For more information about this function, see Formula designer in Electronic reporting.

To learn more about this feature, complete the example in this article.

Example: Use JOIN data sources in ER model mappings

The following steps explain how the system administrator or Electronic reporting developer can configure an Electronic reporting (ER) model mapping to get data from multiple application tables at once by using data sources of the **Join** type to improve data access performance. You can perform these steps for any company of Dynamics 365 Finance or Regulatory Configuration Services (RCS).

Prerequisites

To complete the examples in this article, you must have access to one of the following roles, depending on the service you use to complete these steps:

Access to Finance for one of the following roles:

- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

Access to RCS for one of the following roles:

- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

You must also complete the steps in the Create a configuration provider and mark it as active procedure.

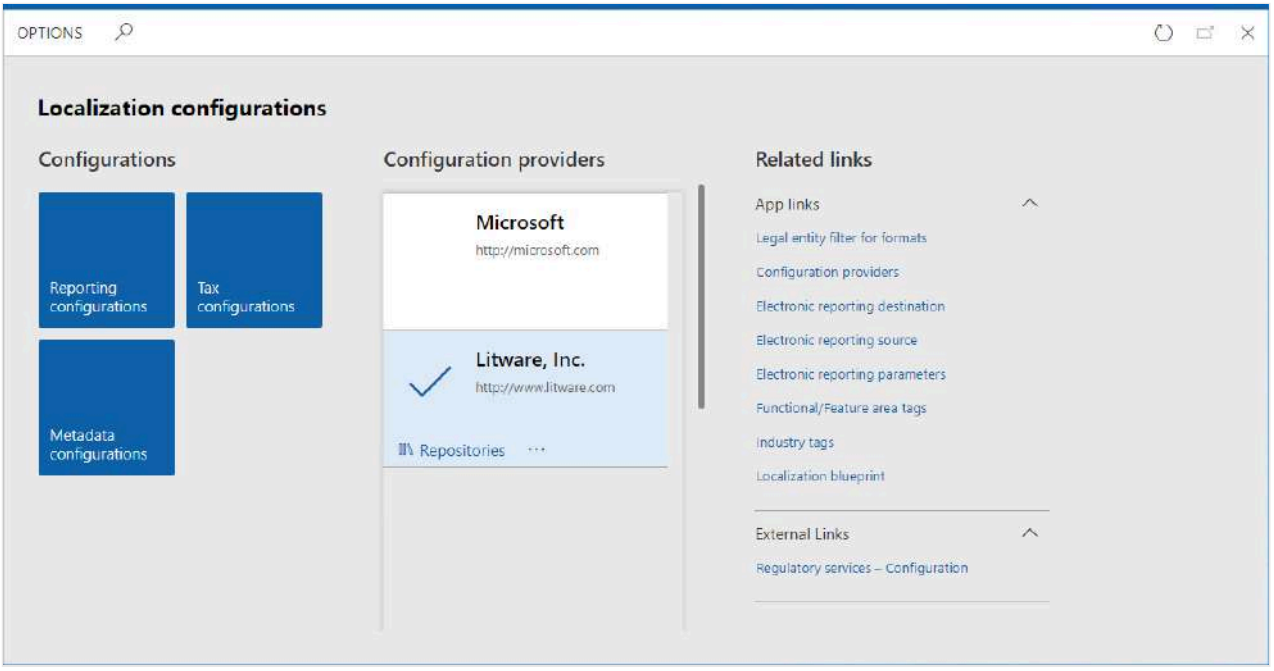
In advance, download and save the following sample ER configuration files:

Expand table

Content description	File name
Sample **ER data model** configuration file, which is used as the data source for the examples.	Model to learn JOIN data sources.version.1.1.xml
Sample **ER model mapping** configuration file, which implements the ER data model for the examples.	Mapping to learn JOIN data sources.version.1.1.xml
Sample **ER format** configuration file. This file describes the data to populate the ER format component for the examples.	Format to learn JOIN data sources.version.1.1.xml

Activate a configurations provider

1. Access either Finance or RCS in the first session of your web browser.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. On the **Localization configurations** page, in the **Configuration providers** section, make sure that the configuration provider for the Litware, Inc. sample company is listed, and that it's marked as **Active**. If you don't see this configuration provider, follow the steps in Create a configuration provider and mark it as active procedure.



Import sample ER configuration files

1. Select **Reporting configurations**.
2. Import the ER data model configuration file.
   a. Select **Exchange**.
   b. Select **Load from XML file**.

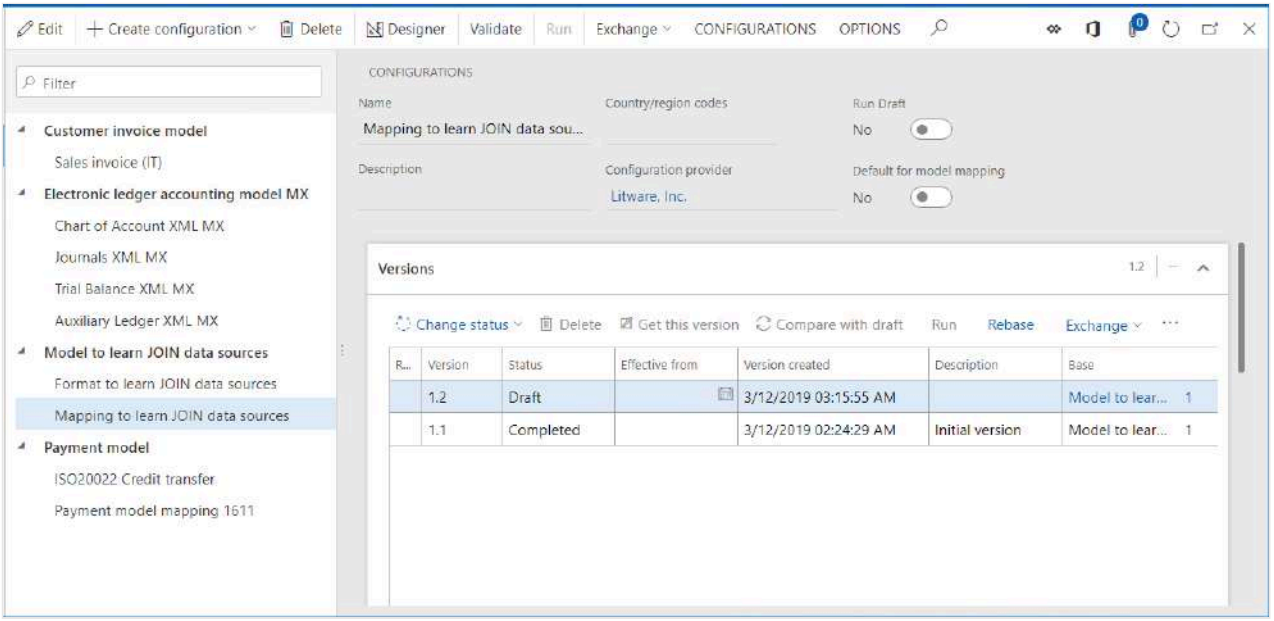
c. Select **Browse** to find the **Model to learn JOIN data sources.version.1.1.xml** file.
d. Select **OK**.

3. Import the ER model-mapping configuration file.
   a. Select **Exchange**.
   b. Select **Load from XML file**.
   c. Select **Browse** to find the **Mapping to learn JOIN data sources.version.1.1.xml** file.
   d. Select **OK**.

4. Import the ER format configuration file.
   a. Select **Exchange**.
   b. Select **Load from XML file**.
   c. Select **Browse** to find the **Format to learn JOIN data sources.version.1.1.xml** file.
   d. Select **OK**.

5. In the configurations tree, expand the **Model to learn JOIN data sources** item as well as other model items (when available).

6. Observe the list of ER configurations in the tree as well as version details on the **Versions** FastTab – they are the source of data for your sample report.

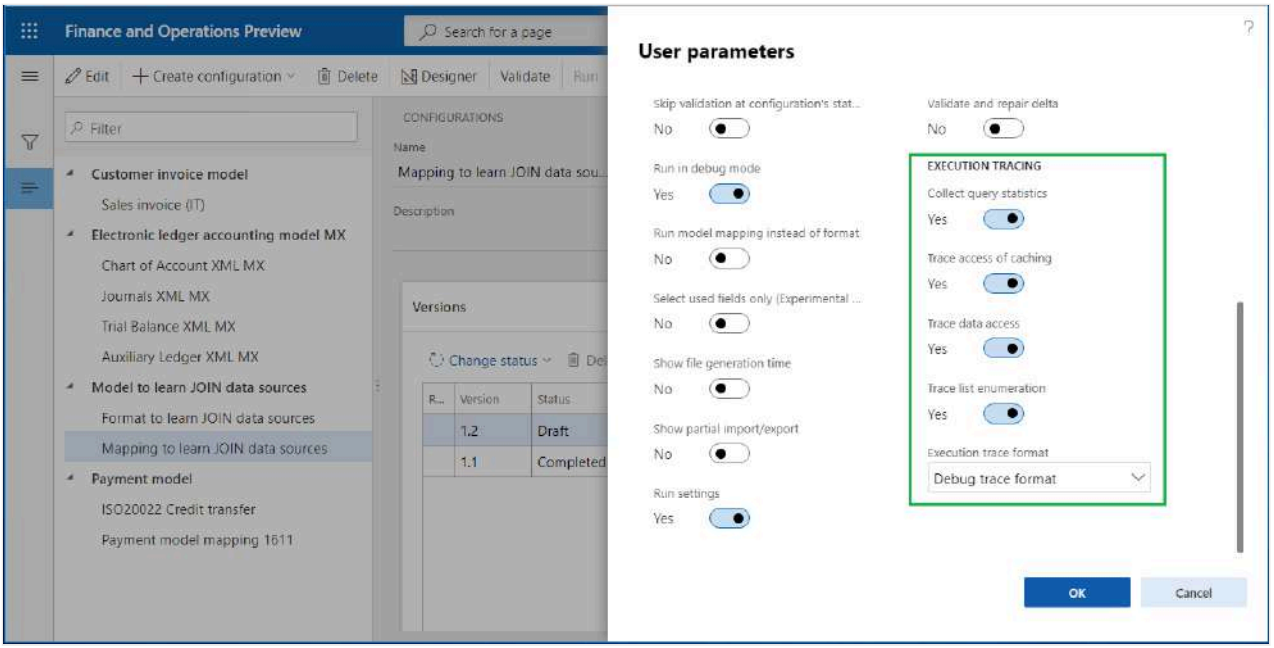


Turn on execution trace options

1. Select **CONFIGURATIONS**.

2. Select **User parameters**.

3. Set execution trace parameters as shown in the following screenshot.



When you turn on these parameters, the system generates an execution trace for every execution of the imported ER format file. By using the details in the generated execution trace, you can analyze the execution of ER format and ER model-mapping components. For more information about the ER execution trace feature, see Trace execution of ER format to troubleshoot performance issues.

Review ER model mapping (part 1)

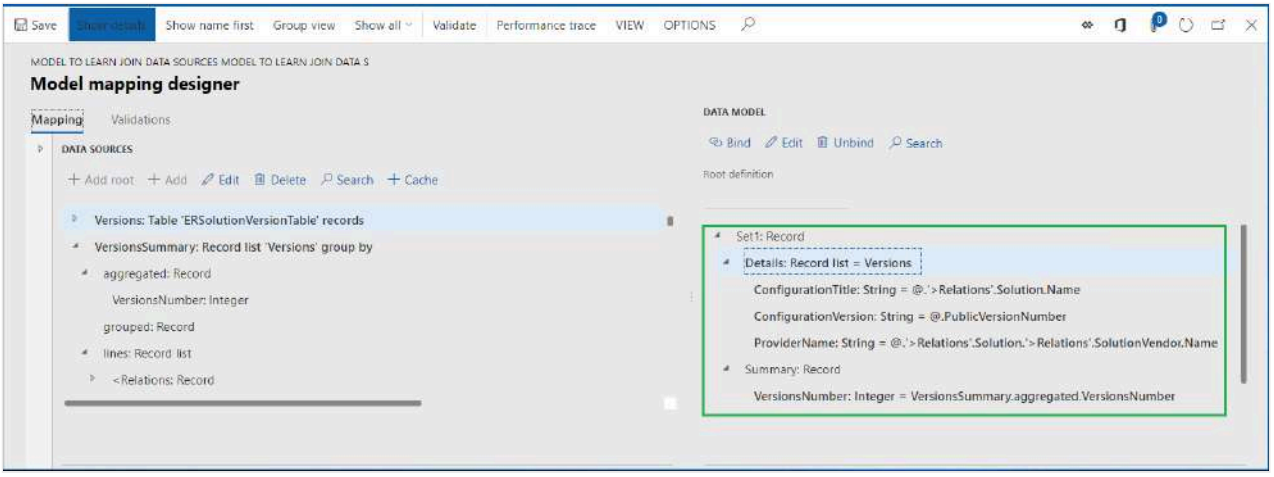
Review the settings of the ER model-mapping component. The component is configured to access information about versions of ER configurations, details of configurations, and configuration providers without using data sources of the **Join** type.

1. Select **Mapping to learn JOIN data sources** configuration.
2. Select **Designer** to open the list of mappings.
3. Select **Designer** to review the mapping details.
4. Select **Show details**.
5. In the configurations tree, expand the **Set1** and **Set1.Details** data model items:
   a. Binding **Details: Record list = Versions** indicates that the **Set1.Details** item is bound to the **Versions** data source returning records of the **ERSolutionVersionTable** table. Each record of this table represents a single version of an ER configuration. The content of this table is presented in the **Versions** FastTab on the **Configurations** page.
   b. Binding **ConfigurationVersion: String = @.PublicVersionNumber** means that the value of the public version of each ER configuration's version comes from the

PublicVersionNumber field of the **ERSolutionVersionTable** table and goes to the **ConfigurationVersion** item.

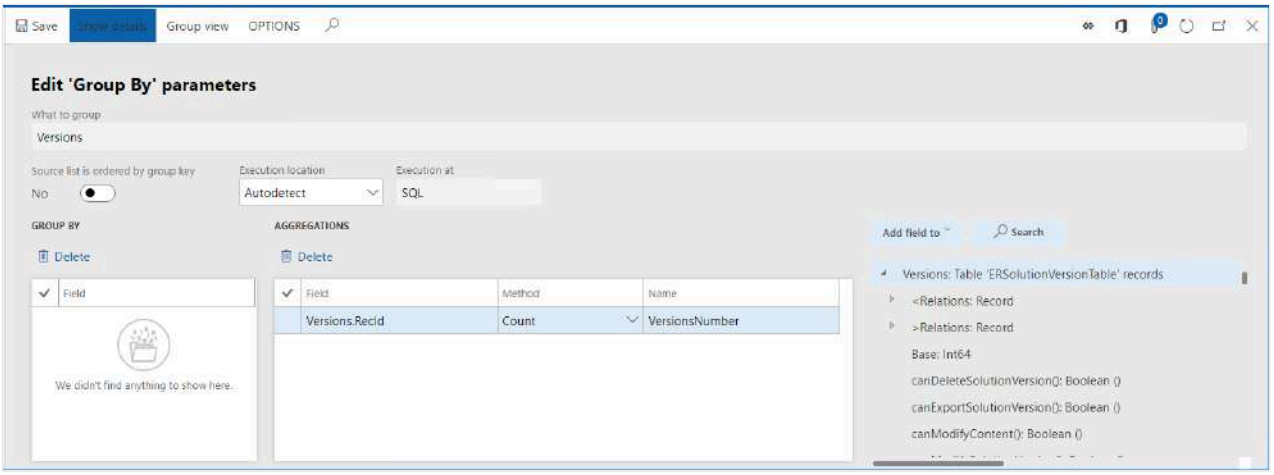
c. Binding **ConfigurationTitle: String = @.'>Relations'.Solution.Name** indicates that the name of an ER configuration comes from the **Name** field of the **ERSolutionTable** table, accessed by using the many-to-one relation ('>**Relations**') between the **ERSolutionVersionTable** and **ERSolutionTable** tables. Names of ER configurations of the current application instance are presented in the configurations tree on the **Configurations** page.

d. Binding **@.'>Relations'.Solution.'>Relations'.SolutionVendor.Name** means that the name of the configuration provider that owns the current configuration comes from the **Name** field of the **ERVendorTable** table, accessed by using the many-to-one relation between **ERSolutionTable** and **ERVendorTable** tables. Names of ER configuration providers are presented in the configurations tree on the **Configurations** page on the page header for each configuration. The entire list of ER configuration providers can be found on the **Organization administration** > **Electronic reporting** > **Configuration provider** table page.



6. In the configurations tree, expand the **Set1.Summary** data model item:

a. Binding **VersionsNumber: Integer = VersionsSummary.aggregated.VersionsNumber** indicates that the **Set1.Summary.VersionsNumber** item is bound to the **VersionsNumber** aggregation field of the **VersionsSummary** data source of the **GroupBy** type that you configured to return the number of records of the **ERSolutionVersionTable** table via the **Versions** data source.

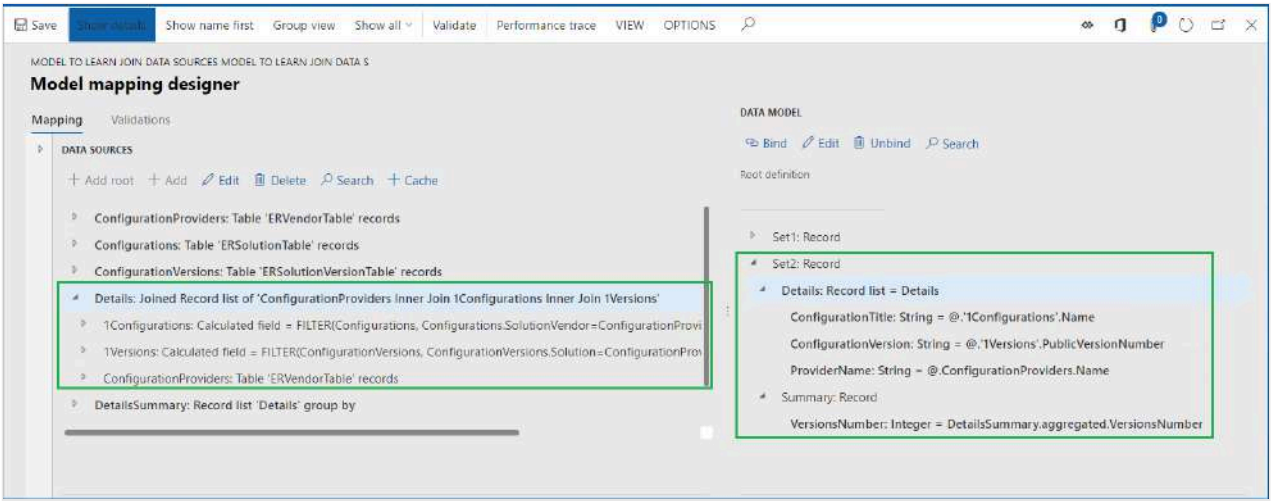


7. Close the page.

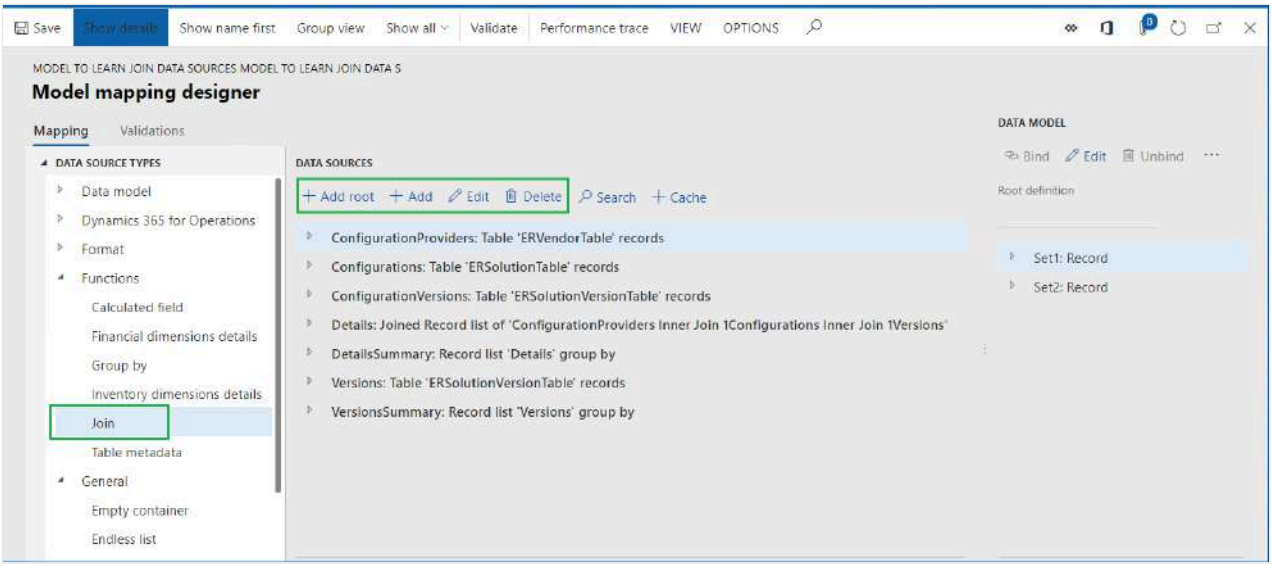
Review ER model mapping (part 2)

Review the settings of the ER model-mapping component. The component is configured to access information about versions of ER configurations, details of configurations, and configuration providers by using a data source of the **Join** type.

1. In the configurations tree, expand the **Set2** and **Set2.Details** data model items. The binding **Details: Record list = Details** indicates that the **Set2.Details** item is bound to the **Details** data source configured as the data source of the **Join** type.



Add the **Join** data source by selecting the **Functions\Join** data source:

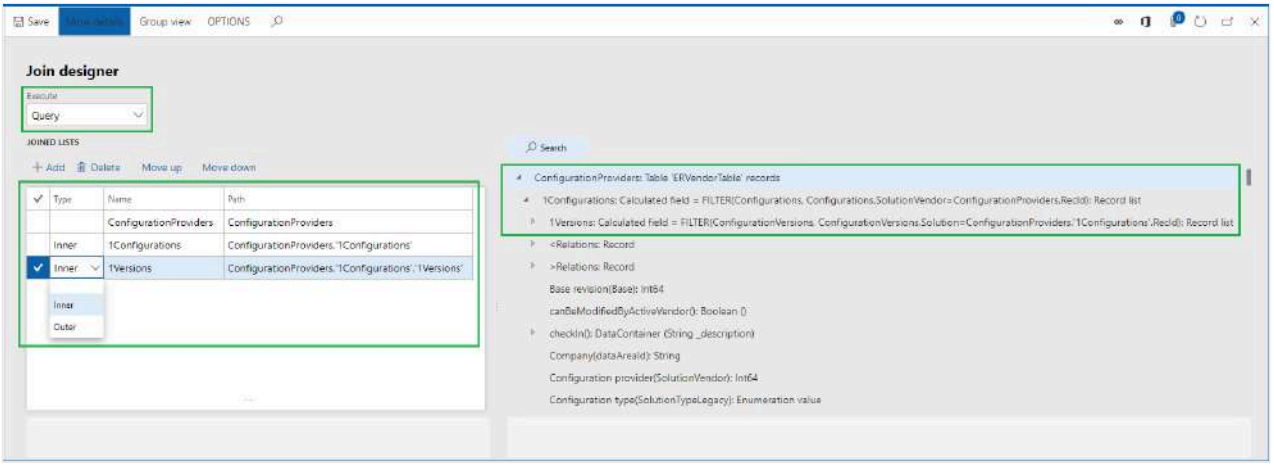


2. Select **Details** data source.

3. Select **Edit** in the **Data sources** pane.

4. Select **Edit join**.

5. Select **Show details**.



Use this page to design the required data source of the **Join type**. At runtime, this data source creates a single joined list of records from the data sources in the **Joined list** grid. The join of records starts from the **ConfigurationProviders** data source that is in the grid as the first one (the **Type** column is blank for it). Records of every other data source are joined consequently to records of the parent data source based on its order in this grid. You must configure every joining data source as a data source nested under a target data source (`1Versions` data source is nested under `1Configurations` one; `1Configurations` data source is nested under **ConfigurationProviders** one). Each configured data source must contain the conditions for the join. In the data source for this particular **Join**, the following joins are defined:

- Each record of the **ConfigurationProviders** data source (referred to the **ERVendorTable** table) is joined with only records of the **1Configurations** one (referred to in the **ERSolutionTable** table) that have the same value in the **SolutionVendor** and **RecId** fields. The **Inner join** type is used for this join as well as the following conditions for matching records:

FILTER (Configurations, Configurations.SolutionVendor = ConfigurationProviders.RecId)

- Each record of the **1Configurations** data source (referred to the **ERSolutionTable** table) is joined with the only records of the **1Versions** one (referred to the **ERSolutionVersionTable** table) that have the same value in the **Solution** and **RecId** fields. **Inner join** type is used for this join as well as the following conditions for matching records:

FILTER (ConfigurationVersions, ConfigurationVersions.Solution = ConfigurationProviders.'1Configurations'.RecId)

- The **Execute** option is configured as **Query** meaning that this join data source is executed at runtime on database level as a direct SQL call.

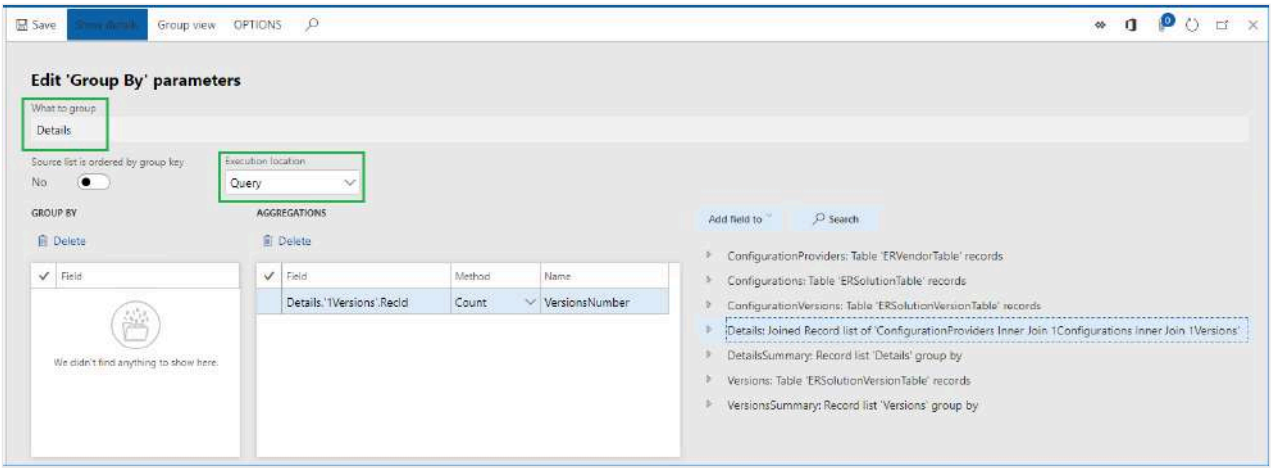
To join records of data sources that represent application tables, specify join conditions by using pairs of fields other than ones that describe existing in AOT relations between these tables. You can configure this type of join to execute at the database level as well.

6. Close the page.

7. Select **Cancel**.

8. In the configurations tree, expand the **Set2.Summary** data model item:

- Binding **VersionsNumber: Integer = DetailsSummary.aggregated.VersionsNumber** indicates that the **Set2.Summary.VersionsNumber** item is bound to the **VersionsNumber** aggregation field of the **DetailsSummary** data source of the **GroupBy** type that you configured to return the number of joined records of the **Details** data source of the **Join** type.
- The **Execution** location option is configured as **Query** meaning that this **GroupBy** data source runs at runtime as a direct SQL call at the database level. This behavior is possible because the base data source **Details** of the **Join** type is configured as executed at the database level.

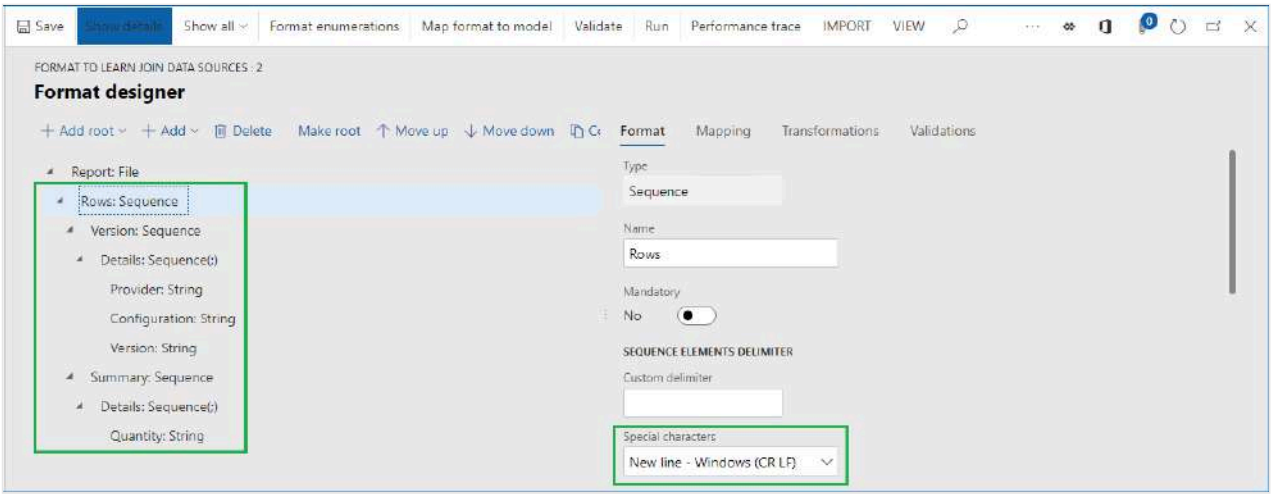


9. Close the page.
10. Select **Cancel**.

Execute ER format

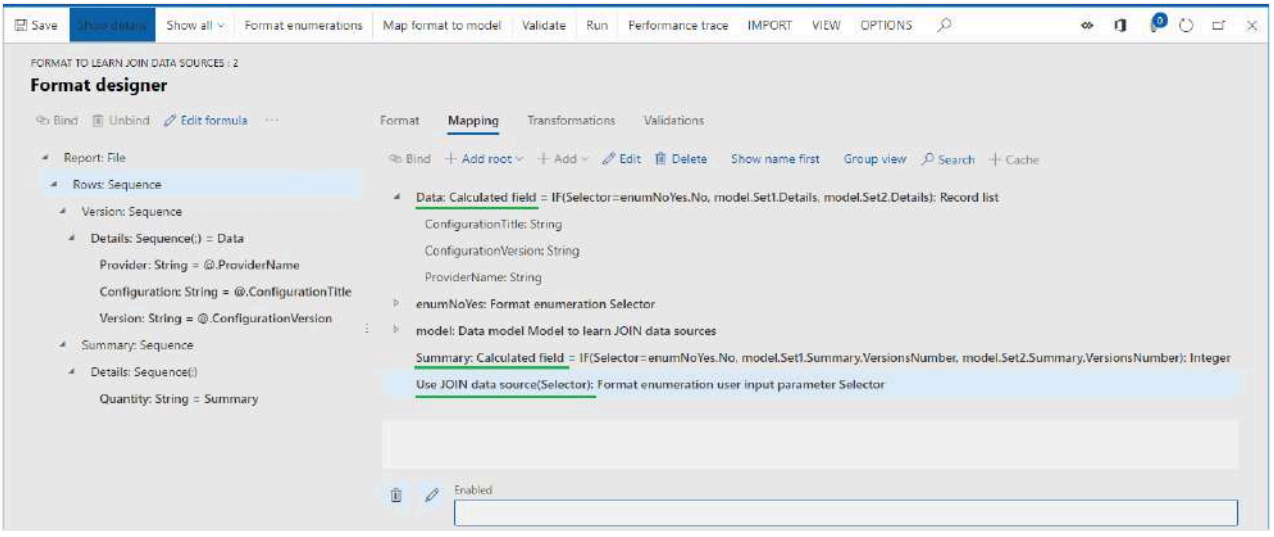
1. Access Finance or RCS in the second session of your web browser by using the same credentials and company as in the first session.
2. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
3. Expand **Model to learn JOIN data sources** configuration.
4. Select **Format to learn JOIN data sources** configuration.
5. Select **Designer**.
6. Select **Show details**.
7. Select **Mapping**.
8. Select **Expand/Collapse**.

   This format populates a generated text file with a new line for every version of an ER configuration (**Version** sequence). Each generated line contains the name of a configuration provider that owns the current configuration, the configuration name, and the configuration version separated by semicolon mark. The final line of generated file contains the number of discovered versions of ER configurations (**Summary** sequence).

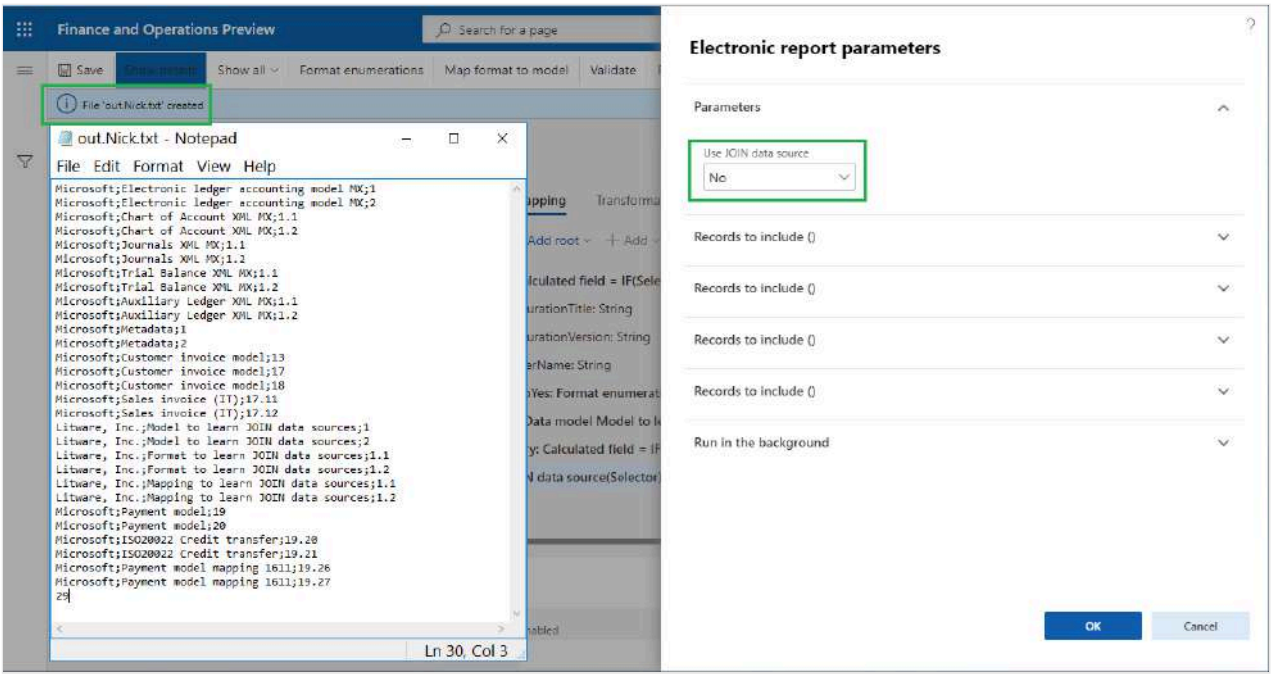


The **Data** and **Summary** data sources populate configuration version details to the generated file:

- The **Set1** data model provides information when you choose **No** for the **Selector** data source at runtime on the user dialog page when running ER format.
- The **Set2** data model provides information when you choose **Yes** for the **Selector** data source at runtime on the user dialog page.



9. Select **Run**.
10. On the dialog page, select **No** in the **Use JOIN data source** field.
11. Select **OK**.
12. Review generated file.

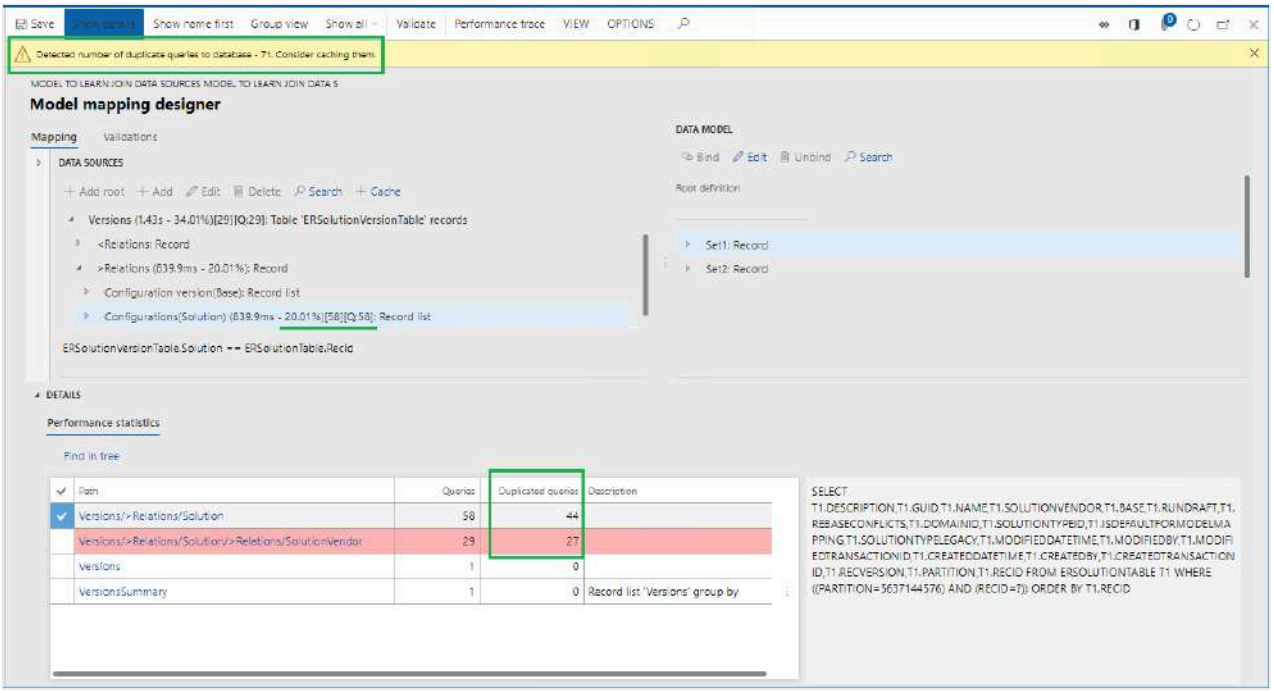


Analyze ER format execution trace

1. In the first session of Finance or RCS, select **Designer**.
2. Select **Performance trace**.
3. In the **Performance trace** grid, select the top-most record of the latest execution trace of an ER format that used the current model mapping component.
4. Select **OK**.

   Execution statistics inform you about duplicated calls to application tables:

   - The process calls **ERSolutionTable** as many times as you have configuration version records in the **ERSolutionVersionTable** table, while the number of such calls could be reduced for performance improvement.
   - The process calls **ERVendorTable** twice for every configuration version record that the process discovers in the **ERSolutionVersionTable** table, while the number of such calls could be reduced.



5. Close the page.

Execute ER format

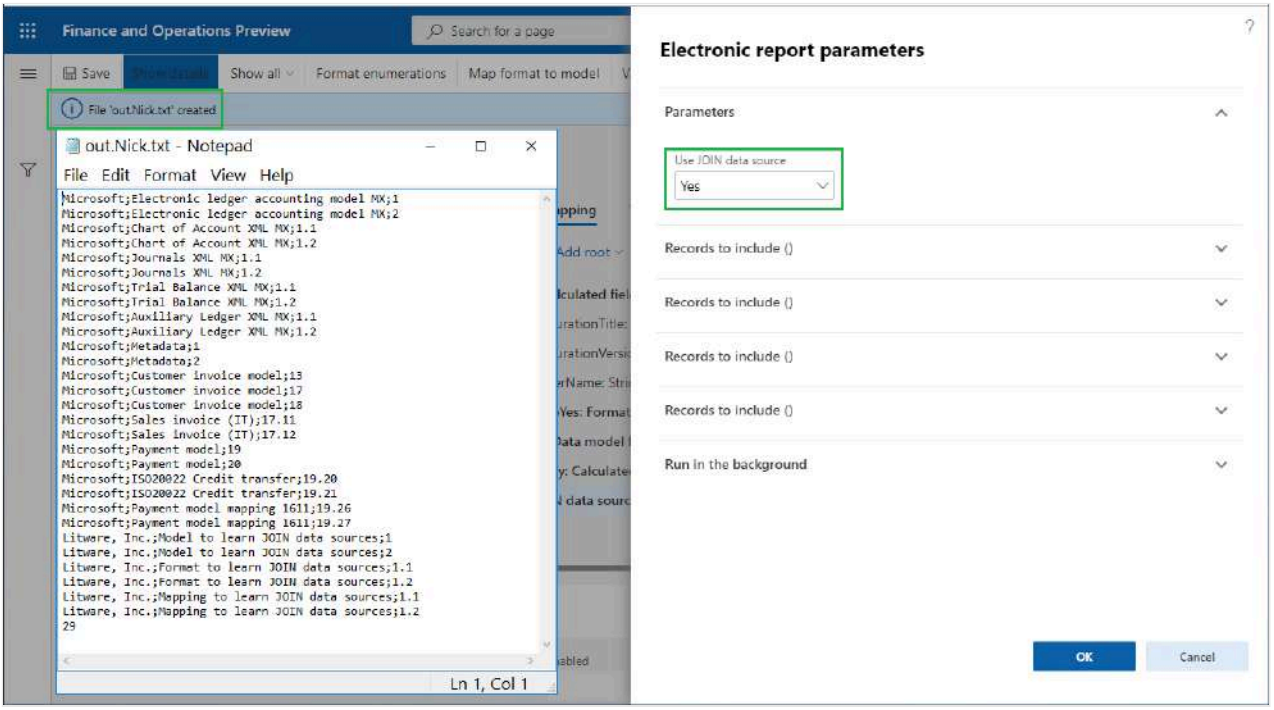
1. Switch to your web browser tab with the second session of Finance or RCS.

2. Select **Run**.

3. On the dialog page, select **Yes** in the **Use JOIN data source** field.

4. Select **OK**.

5. Review generated file.

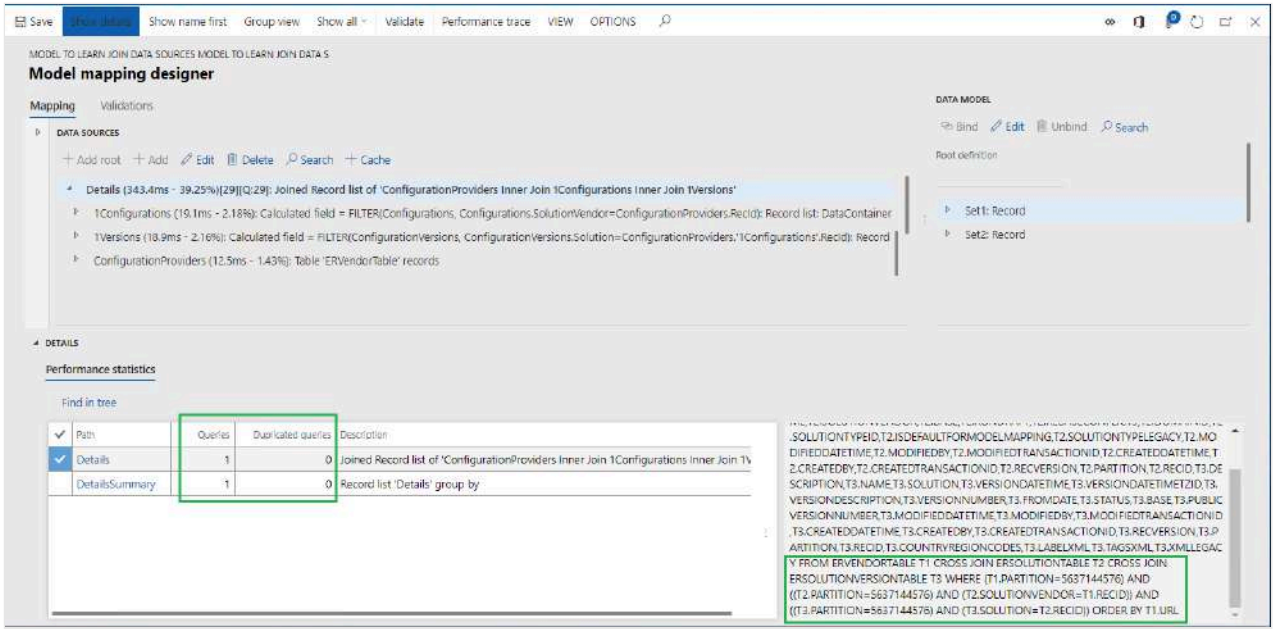


Analyze ER format execution trace

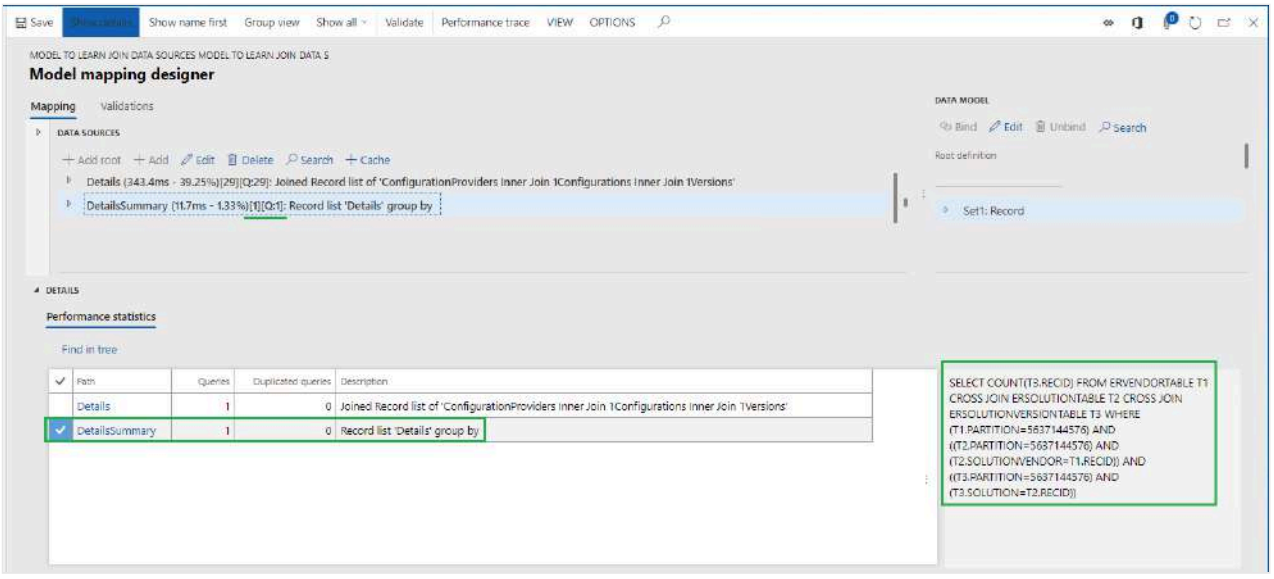
1. In the first session of Finance or RCS, select **Designer**.
2. Select **Performance trace**.
3. In the **Performance trace** grid, select the top-most record representing the latest execution trace of an ER format that used the current model mapping component.
4. Select **OK**.

   The statistics inform you about the following information:

   - The application database is called once to get records from **ERVendorTable**, **ERSolutionTable**, and **ERSolutionVersionTable** tables to access required fields.



   - The application database is called once to calculate the number of configuration versions by using joins that are configured in the **Details** data source.



Limitations

As you can see from the example in this article, you can build the **JOIN** data source from several data sources that describe the individual datasets of the records that you eventually join. Use the built-in ER FILTER function to configure those data sources. When you configure the data source so that it calls beyond the **JOIN** data source, you can use company ranges as part of the condition for data selection. The initial implementation of the **JOIN** data source doesn't support data sources of this type. For example, if you call a FILTER-based data source within the scope of execution of a **JOIN** data source and the called data source contains company ranges as part of the condition for data selection, an exception occurs.

In Microsoft Dynamics 365 Finance version 10.0.12 (August 2020), you can use company ranges as part of the condition for data selection in FILTER-based data sources that you call within the scope of execution of a **JOIN** data source. Because of the limitations of the application query builder, the company ranges are supported only for the first data source of a **JOIN** data source.

Example

For example, you must make a single call to the application database to get the list of foreign trade transactions of multiple companies and the details of the inventory item that is referred to in those transactions.

In this case, you configure the following artifacts in your ER model mapping:

- **Intrastat** root data source that represents the **Intrastat** table.
- **Items** root data source that represents the **InventTable** table.
- **Companies** root data source that returns the list of companies (**DEMF** and **GBSI** in this example) where transactions must be accessed. The company code is available from the

Companies.Code field.

- **X1** root data source that has the expression `FILTER (Intrastat, VALUEIN(Intrastat.dataAreaId, Companies, Companies.Code))`. As part of the condition for data selection, this expression contains the definition of company ranges `VALUEIN(Intrastat.dataAreaId, Companies, Companies.Code)`.
- **X2** data source as a nested item of the **X1** data source. It includes the expression `FILTER (Items, Items.ItemId = X1.ItemId)`.

Finally, you can configure a **JOIN** data source where **X1** is the first data source and **X2** is the second data source. You can specify **Query** as the **Execute** option to force ER to run this data source on the database level as a direct SQL call.

When you run the configured data source while the ER execution is traced, the following statement is shown in the ER model mapping designer as part of the ER performance trace.

```
SELECT ... FROM INTRASTAT T1 CROSS JOIN INVENTTABLE T2 WHERE ((T1.PARTITION=?) AND (T1.DATAAREAID IN (N'DEMF',N'GBSI') )) AND ((T2.PARTITION=?) AND (T2.ITEMID=T1.ITEMID AND (T2.DATAAREAID = T1.DATAAREAID) AND (T2.PARTITION = T1.PARTITION))) ORDER BY T1.DISPATCHID,T1.SEQNUM
```

> **Note**
>
> An error occurs if you run a **JOIN** data source that you configured so that it contains data selection conditions that have company ranges for additional data sources of the executed **JOIN** data source.

Additional resources

Formula designer in Electronic reporting

Trace execution of ER format to troubleshoot performance issues

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Last updated on 04/08/2026

Generate reports by adding content as raw XML

> ⓘ Note
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the new **RAW XML** format element to design Electronic reporting (ER) formats that generate outgoing documents in XML format. In some cases, you might prefer to add raw XML data to these reports for one or more of the following reasons:

- It's more convenient to use raw XML for the original design and ongoing maintenance of a report because the XML structure can be automatically generated by executing a runtime expression. Therefore, you don't need to determine multiple bindings for multiple format elements at design time. This approach is possible when the data sources you use contain information that can be used to make XML elements while the report is generated.
- No other method can fill the report with XML content that the system previously received and stored. For example, the XML response that is generated might have to contain the content of an XML request that was sent earlier.
- No other method can insert characters into the generated document based on their numeric codes. For some languages and characters, codes of this type don't exist. Examples include the Greek letter rho (ρ) and HTML entity codes such as é for an *e* that has an acute accent (é).

> ⓘ Note
>
> The framework doesn't validate whether the XML content that you add to the generated document by using the **RAW XML** format element is correct.

To learn more about this feature, watch the **ER Use raw XML data to generate XML reports (Part 1: Design data model)** and **ER Use raw XML data to generate XML reports (Part 2:**

Design and run report) task guides. These task guides are part of the **7.5.4.3 Acquire/Develop IT service/solution components (10677)** business process, and you can download them from the Microsoft Download Center. The task guides walk you through the process of configuring an ER format to insert raw XML data into generated files.

Last updated on 04/13/2026

Support parameterized calls of ER data sources of the Calculated field type

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how you can design an Electronic reporting (ER) data source by using the **Calculated field** type. This data source can contain an ER expression that, when executed, can be controlled by the values of the parameter arguments configured in a binding that calls this data source. By configuring parameterized calls of such a data source, you can reuse a single data source in many bindings. This approach reduces the total number of data sources that you must configure in ER model mappings or ER formats. It also simplifies the configured ER component, which reduces maintenance costs and the cost of use by other consumers.

Prerequisites

To complete the examples in this article, you must have the following access:

- Access to one of these roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- Access to Regulatory Configuration Services (RCS) that are provisioned for the same tenant as finance and operations for one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator

You must also download and locally store the following files.

Expand table

Content	File name
Sample ER data model configuration	Model to learn parameterized calls.version.1.xml
Sample ER metadata configuration	Metadata to learn parameterized calls.version.1.xml
Sample ER model mapping configuration	Mapping to learn parameterized calls.version.1.1.xml
Sample ER format configuration	Format to learn parameterized calls.version.1.1.xml

Sign in to your RCS instance

In this example, you create a configuration for the sample company, Litware, Inc. First, in RCS, complete the steps in the Create configuration providers and mark them as active procedure:

1. On the default dashboard, select **Electronic reporting**.
2. Select **Reporting configurations**.
3. Import the downloaded configurations to RCS in the following sequence: data model, metadata, model mapping, format. Complete the following steps for each ER configuration:
   a. Select **Exchange**.
   b. Select **Load from XML file**.
   c. Select **Browse**, and then select the required ER configuration in XML format.
   d. Select **OK**.

Review the provided ER solution

Review model mapping

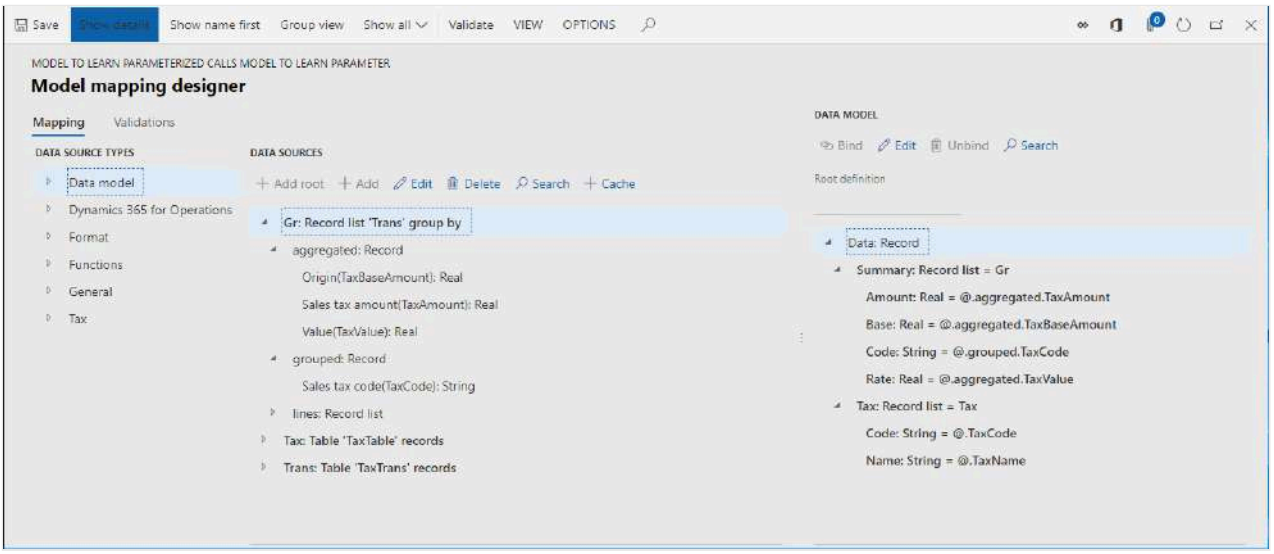
1. In the configuration tree, expand the content of the **Model to learn parameterized calls** item.
2. Select **Mapping to learn parameterized calls**.
3. Select **Designer**.
4. Select **Designer**.

   This ER model mapping is designed to do the following tasks:

   - Fetch the list of tax codes (**Tax** data source) residing in the **TaxTable** table.

- Fetch the list of tax transactions (**Trans** data source) residing in the **TaxTrans** table.
  - Group the list of fetched transactions (**Gr** data source) by tax code.
    - Calculate the following aggregated values per tax code, for grouped transactions:
    - Sum of tax base values.
    - Sum of tax values.
    - Minimum value of applied tax rate.

The model mapping in this configuration implements the base data model for any of the ER formats created for this model and executed in finance and operations. As a result, the content of the **Tax** and **Gr** data sources is exposed for ER formats such as abstract data sources.

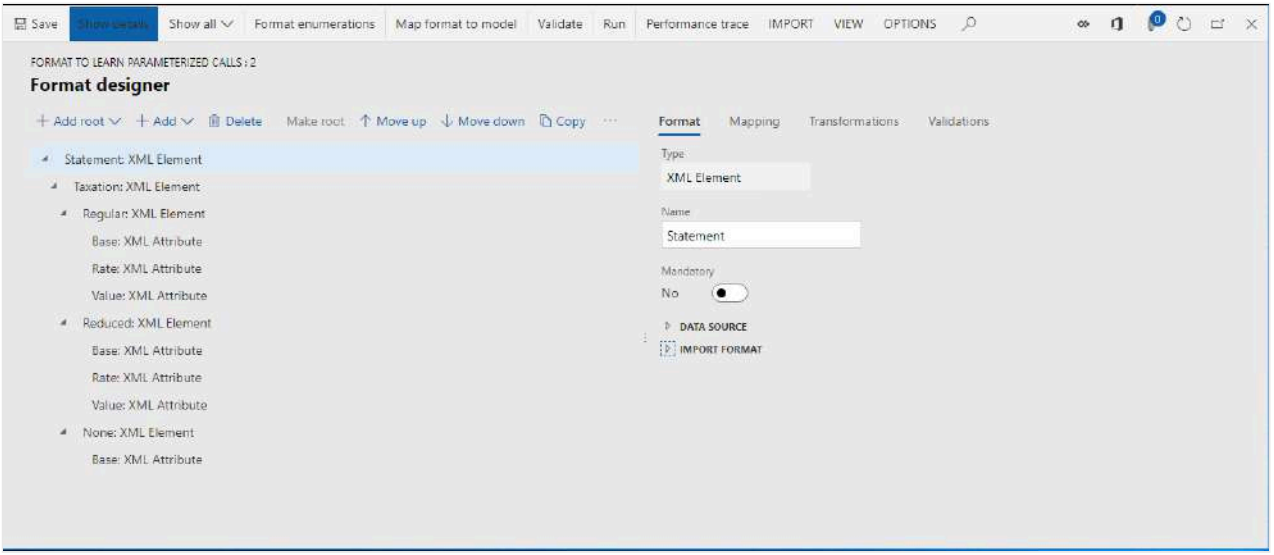


5. Close the **Model mapping designer** page.
6. Close the **Model mapping** page.

Review format

1. In the configuration tree, expand the content of the **Model to learn parameterized calls** item.
2. Select **Format to learn parameterized calls**.
3. Select **Designer**. This ER format is designed to do the following:
   - Generate a tax statement in XML format.

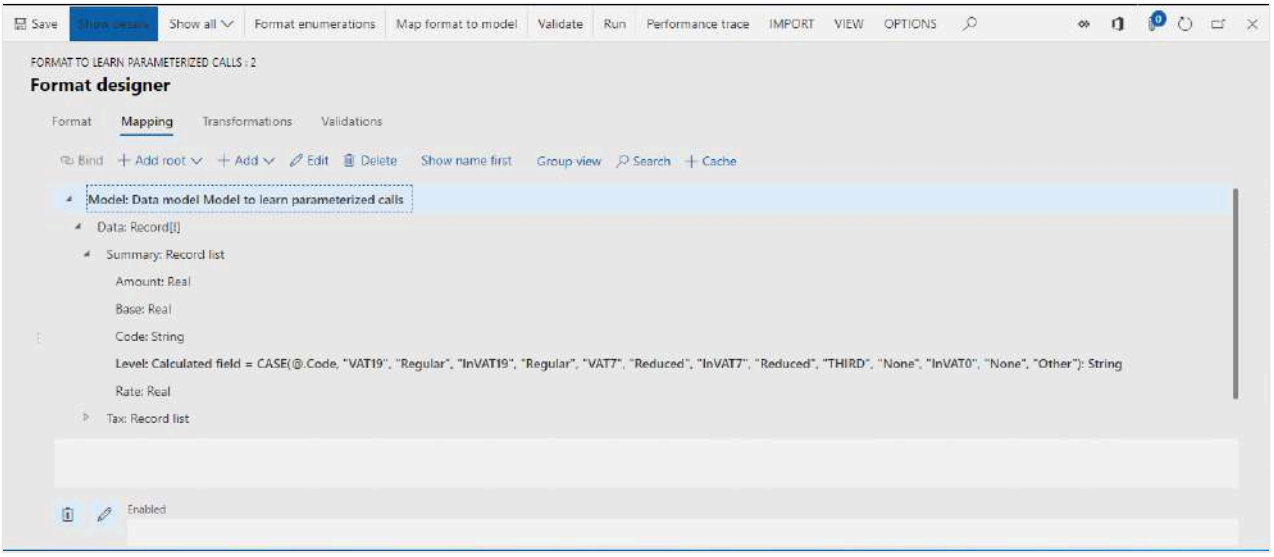
- Present the following levels of taxation in the tax statement: regular, reduced, and none.
- Present multiple details at each taxation level, having a different number of details in each level.



4. Select **Mapping**.

5. Expand the **Model**, **Data,** and **Summary** items.

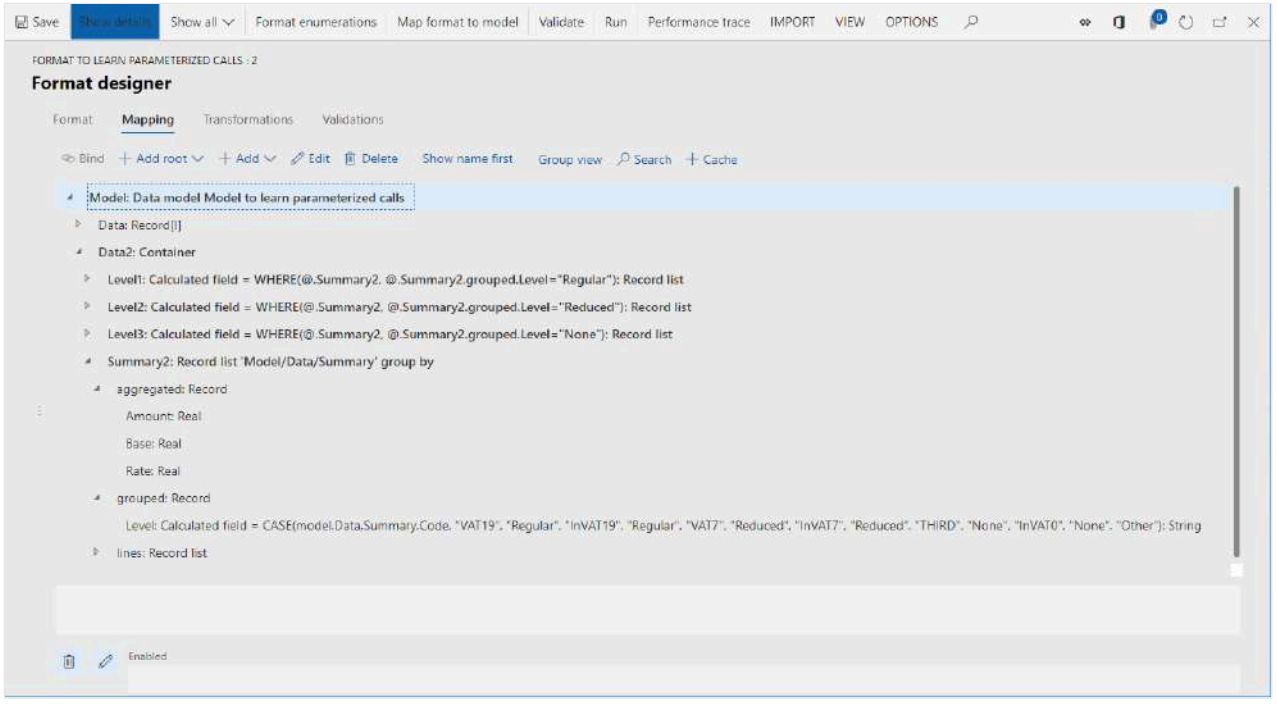
   The calculated field **Model.Data.Summary.Level** contains the expression that returns the code of the taxation level (**Regular**, **Reduced**, **None**, or **Other**) as a text value for any tax code that can be retrieved from the **Model.Data.Summary** data source at run time.



6. Expand the **Model.Data2** item.

7. Expand the **Model.Data2.Summary2** item.

The **Model.Data2.Summary2** data source is configured to group the **Model.Data.Summary** data source transaction details by taxation level (returned by the **Model.Data.Summary.Level** calculated field) and compute the aggregations.



8. Review the calculated fields **Model.Data2.Level1**, **Model.Data2.Level2**, and **Model.Data2.Level3**. Use these calculated fields to filter the **Model.Data2.Summary2** records list and return only records that represent a particular taxation level.

9. Close the **Format designer** page.

Create a derived format

You can improve the provided format by adding one calculated field to filter the required taxation level instead of using the existing three fields: **Model.Data2.Level1**, **Model.Data2.Level2**, and **Model.Data2.Level3**. The required taxation level can be specified in the location where this new calculated field is called.

1. In the configuration tree, expand the content of the **Model to learn parameterized calls** item.
2. Select **Format to learn parameterized calls**.
3. Select **Create configuration**.
4. Select **Derive from Name: Format to learn parameterized calls, Microsoft**.
5. In the **Name** field, enter **Format to learn parameterized calls (custom)**.
6. Select **Create configuration**.

Configure a parameterized calculated field that returns a list of records

Start adding a new calculated field

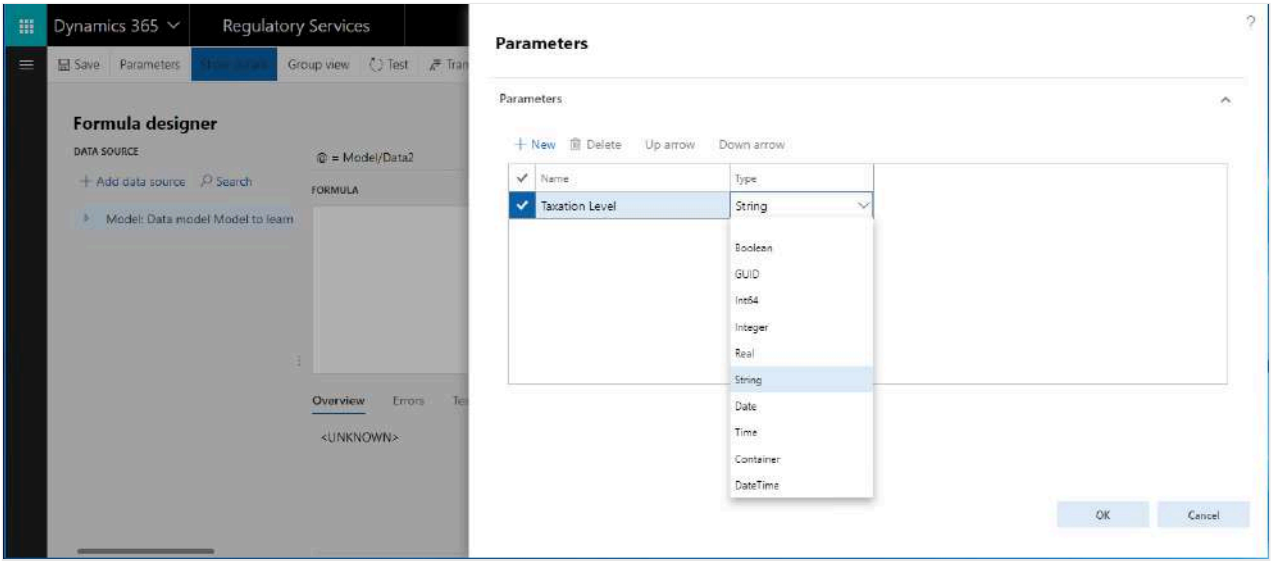
1. Select **Designer**.
2. Select **Expand/collapse** to expand all format items.
3. Select **Mapping**.
4. Expand the **Model** item.
5. Select the **Model.Data2** item.
6. Select **Add**.
7. Select **Functions\Calculated field**.
8. In the **Name** field, enter **Levels**.
9. Select **Edit formula**.

Define a parameter for adding a calculated field

1. Select **Parameters**.
2. Select **New**.
3. In the **Name** field, enter **Taxation Level**.
4. In the **Type** field, select **String**.

   You can only use primitive data types to specify the type of the parameter's argument. Therefore, you can't use **Record list**, **Record**, or **Enum** types for this purpose.

   You can specify up to eight parameters for a single calculated field.

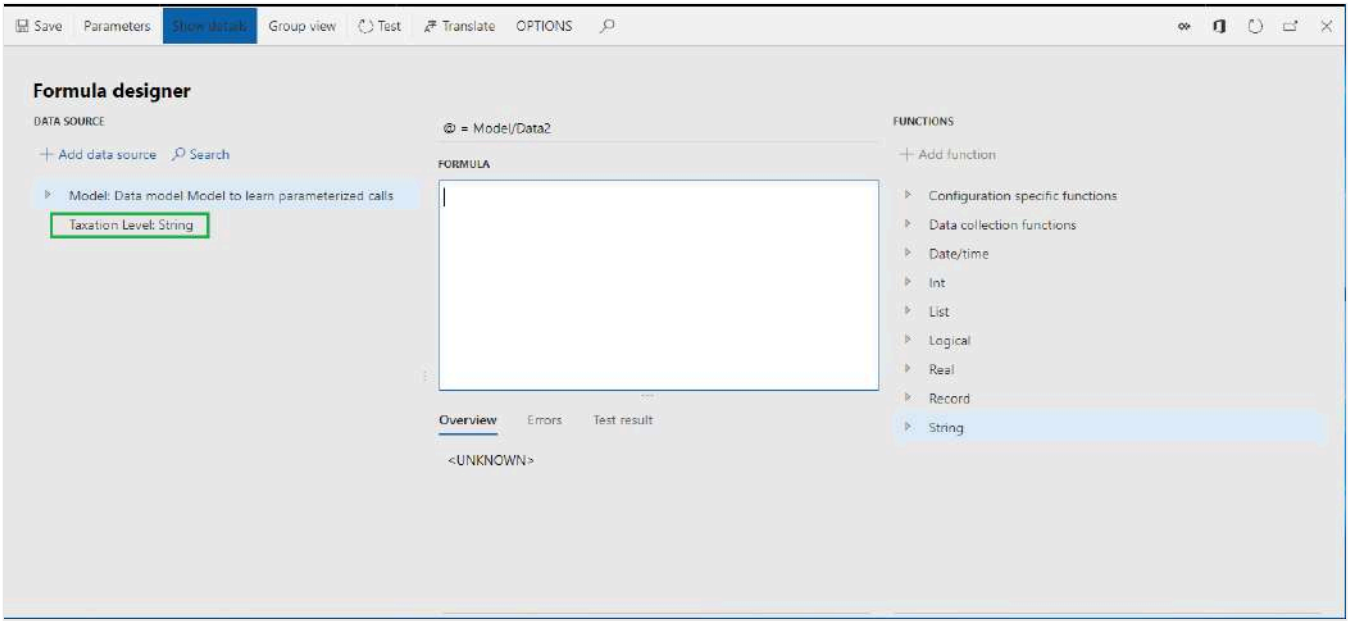


5. Select **OK**.

By adding this parameter, you specify the condition that must be in place to call this calculated field. When you call this calculated field, you need to specify the argument of the **Taxation Level** parameter as a value with **String** format.

Make sure that you define parameters only for those calculated fields that reside in a container (either **Record list**, **Record**, or **Container**).

The configured parameter is available in the list of data sources for this calculated field. You can add the parameter to the configured expression by selecting **Add data source**.



Define an expression for adding a calculated field

1. In the **Formula** field, enter:

   ```
   WHERE(\@.Summary2, \@.Summary2.grouped.Level =
   ```

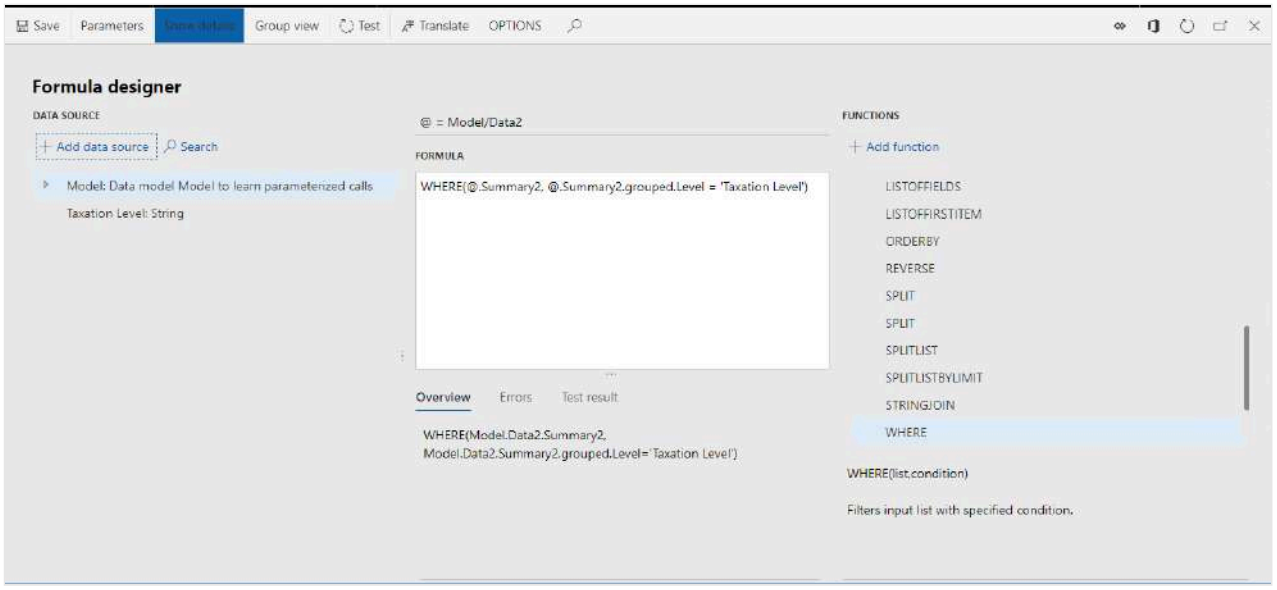
2. Select the **Taxation Level** parameter in the list of data sources.

3. Select **Add data source**.

4. In the **Formula** field, finalize the expression as:

   ```
   WHERE(\@.Summary2, \@.Summary2.grouped.Level = 'Taxation Level')
   ```

5. Select **Save**.

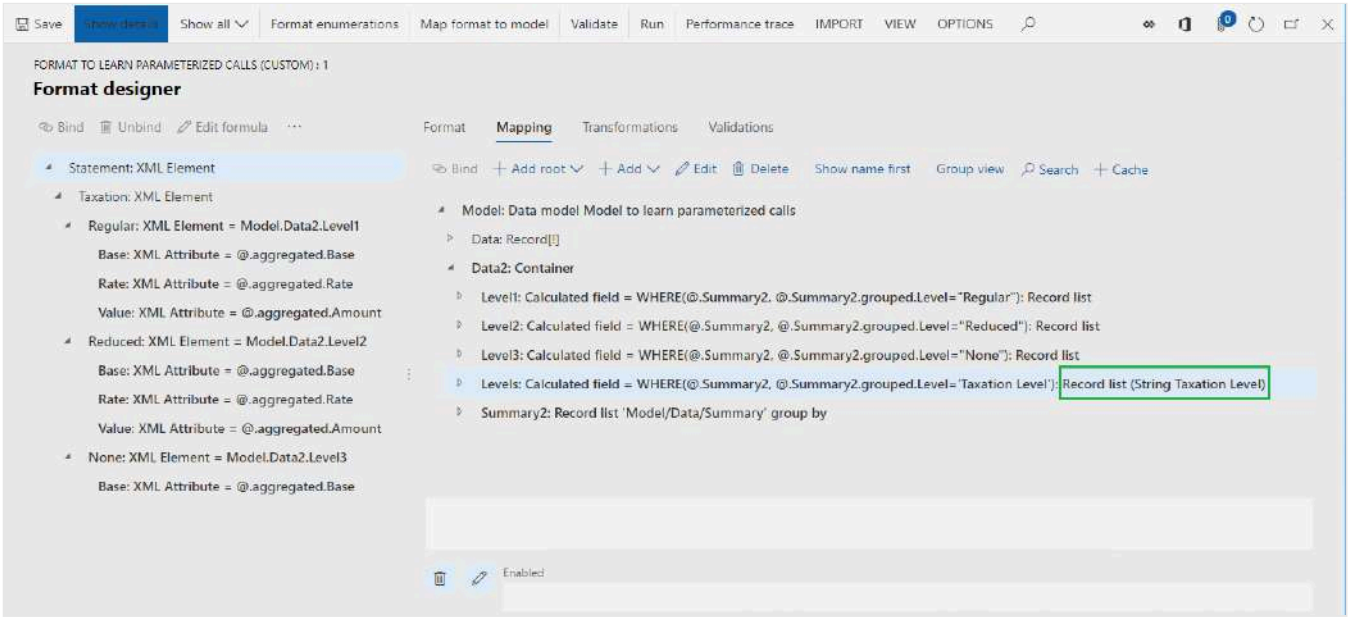


6. Close the **Formula designer** page.

Finish adding a new calculated field

- Select **OK**.

On the **Format designer** page, the configured parameterized calculated field **Levels** requires a **String** argument.



Use the configured calculated field for binding format elements

1. Select **Model.Data2.Levels** to select the configured calculated field.

2. Select the **Statement.Taxation.Regular** format element.

3. Select **Bind**.
4. Select **Yes** to confirm the replacement of the currently used data source, **Level1**, by the new data source, **Levels**, in all nested format elements of the selected format element.

   Applied binding is a call of the parameterized calculated field. By default, the name of the bound format element is used as an argument for the parameterized calculated field under the following conditions:

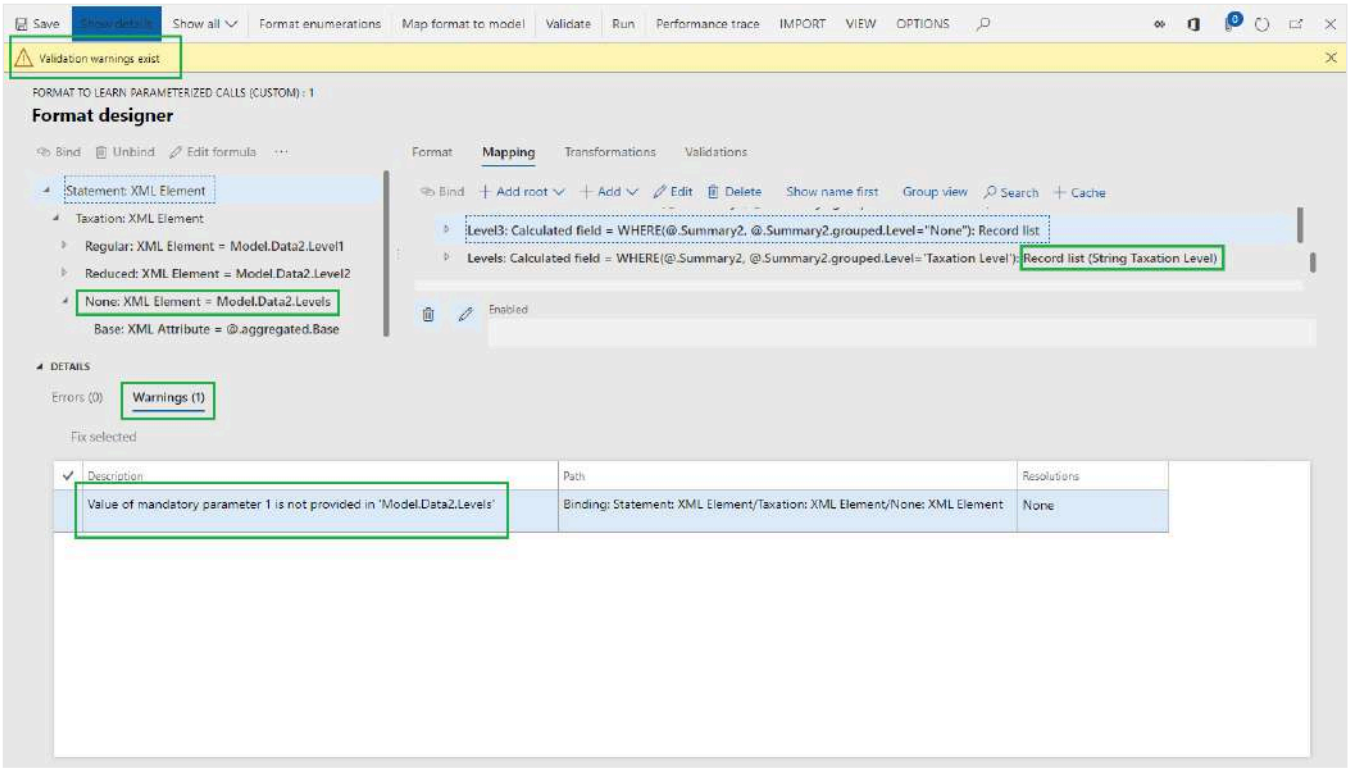
   - The calculated field is configured to use a single parameter.
   - The data type of this parameter is defined as **String**.

   When the name of the bound format element is blank, the data source name of this element is used in applied binding.
5. Select the **Statement.Taxation.Reduced** format element.
6. Select **Bind**.
7. Select **Yes** to confirm the replacement of the currently used data source, **Level2**, by the new data source, **Levels**, in all nested format elements under the selected format element.
8. Select the **Statement.Taxation.None** format element.
9. Select **Bind**.
10. Select **Yes** to confirm the replacement of the currently used data source, **Level3**, by the new data source, **Levels**, in all nested format elements under the selected format element.

    When you specify the argument of the parameterized calculated field for the XML element representing taxation level (for example, **Model.Data2.Levels("Reduced")** as a text value), you don't need to specify the argument for nested XML attributes—their bindings automatically inherit the value of the argument defined on the parent level (**Model.Data2.Levels.aggregated.Base**, not **Model.Data2.Levels("Reduced").aggregated.Base**).

Recurrent calls of any parameterized calculated field aren't supported.

You can select **Edit formula** and change the applied-by-default argument of the parameterized calculated field in the selected binding. If this argument is missing, it can cause errors at run time. Users are informed about such a situation when the current format is validated.



Configure a parameterized calculated field to return a record

When a parameterized calculated field returns a record, you need to support binding of individual fields of this record to format elements. In such cases, there's no parent binding that contains the value of an argument to call a parameterized calculated field — this value must be defined in the binding of a single record's field.

Start adding a new calculated field

5. Select the **Model.Data2** item.
6. Select **Add**.
7. Select **Functions\Calculated field**.
8. In the **Name** field, enter **LevelRecord**.
9. Select **Edit formula**.

Define a parameter for adding a calculated field

1. Select **Parameters**.
2. Select **New**.
3. In the **Name** field, enter **Taxation Level**.
4. In the **Type** field, select **String**.
5. Select **OK**.

Define an expression for adding a calculated field

1. In the **Formula** field, enter the following formula:

   `FIRSTORNULL(@.Levels(`

2. Select the **Taxation Level** parameter.

3. Select **Add data source**.

4. In the **Formula** field, append `'Taxation Level'))` to the formula you entered in Step 1 to finalize the expression:

   `FIRSTORNULL(@.Levels('Taxation Level'))`

5. Select **Save**.

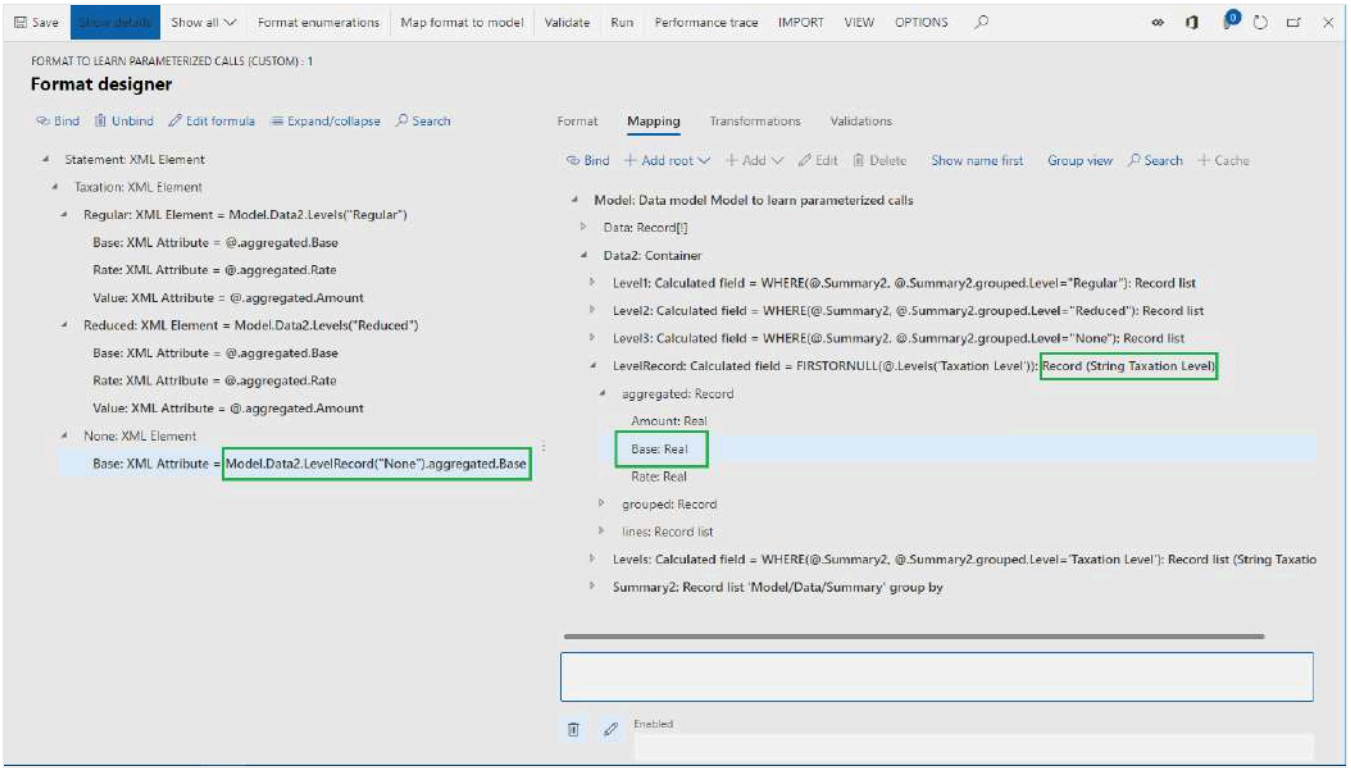
6. Close **Formula designer**.

Finish adding a new calculated field

1. Select **OK**.

Use the configured calculated field to bind format elements

1. Expand **Model.Data2.LevelRecord** to select the configured calculated field.
2. Expand the **Model.Data2.LevelRecord.aggregated** container of the configured calculated field.
3. Select the **Model.Data2.LevelRecord.aggregated.Base** field.
4. Select the **Statement.Taxation.None** format element.
5. Select **Unbind**.
6. Select the **Statement.Taxation.None.Base** format element.
7. Select **Bind**.
8. Select **Edit formula**.
9. Change the expression to **Model.Data2.LevelRecord("None").aggregated.Base**.



Remove calculated fields that are not used

1. Select **Model.Data2.Level1**.
2. Select **Delete**.
3. Select **Model.Data2.Level2**.
4. Select **Delete**.
5. Select **Model.Data2.Level3**.
6. Select **Delete**.
7. Select **Save**.

> **Note**
>
> You reused the same calculated field **Model.Data2.Levels** several times in format bindings. It's much easier to use and maintain a single calculated field instead of doing this for multiple similar fields.

1. Close the **Format designer** page.

Complete adjusted version of a derived format

1. In the **Versions** tab, select **Change status**.
2. Select **Complete**.

Export completed version of a derived format

1. Select **Format to learn parameterized calls (custom)** format in the configurations tree.
2. In the **Versions** tab, select the completed version 1.1.1.
3. Select **Exchange**.
4. Select **Export as XML file**.
5. Store the downloaded configuration locally, in XML format.

Test ER formats

To make sure that configured parameterized calculated fields work properly, run the initial and improved ER formats.

Import ER configurations

You can import reviewed configurations from RCS by using the ER repository of the **RCS** type. If you already went through the steps in the article, Import Electronic reporting (ER) configurations from Regulatory Configuration Services (RCS), use the configured ER repository to import configurations discussed earlier in this article to your environment. Otherwise, follow these steps:

1. Select the **DEMF** company and on the default dashboard, select **Electronic reporting**.

2. Select **Reporting configurations**.

3. Import the configurations from Microsoft Download Center in the following sequence: data model, model mapping, format. Complete the following steps for each ER configuration:
   a. Select **Exchange**.
   b. Select **Load from XML file**.
   c. Select **Browse** to select the required ER configuration in XML format.
   d. Select **OK**.

4. Import the exported from RCS completed version 1.1.1 of the **Format to learn parameterized calls (custom)** format:
   a. Select **Exchange**.
   b. Select **Load from XML file**.
   c. Select **Browse** to select the locally stored **Format to learn parameterized calls (custom)** file in XML format.
   d. Select **OK**.

Run ER formats

1. In the configuration tree, expand the content of the **Model to learn parameterized calls** item.
2. Select **Format to learn parameterized calls**.
3. Select **Run** on the top-most ribbon.
4. Save the locally generated output.
5. Select the **Format to learn parameterized calls (custom)** item.
6. Select **Run** on the top-most ribbon.
7. Save the generated output locally.
8. Compare the contents of the generated outputs.

Additional resources

- Formula designer in Electronic reporting (ER)
- Improve performance of ER solutions by adding parameterized CALCULATED FIELD data sources

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Last updated on 04/09/2026

Improve the performance of ER solutions by adding parameterized CALCULATED FIELD data sources

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how you can take performance traces of Electronic reporting (ER) formats that are run, and then use the information from those traces to help improve performance by configuring a parameterized **Calculated field** data source.

As part of the process of designing ER configurations to generate business documents, you define the method to retrieve data from the application and enter it in the generated output. By designing a parameterized ER data source of the **Calculated field** type, you can reduce the number of database calls and significantly reduce the time and cost that are involved in collecting the details of ER format execution.

Prerequisites

- To complete the examples in this article, you must have access to one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- The company must be set to **DEMF**.
- Follow the steps in Appendix 1 of this article to download the components of the sample Microsoft ER solution that you need to complete the examples in this article.
- Follow the steps in Appendix 2 of this article to configure the minimal set of ER parameters that you need to use the ER framework to help improve the performance of

the sample Microsoft ER solution.

Import the sample ER solution

Imagine that you must design an ER solution to generate a new report that shows vendor transactions. Currently, you can find the transactions for a selected vendor on the **Vendor transactions** page (go to **Account payable** > **Vendors** > **All vendors**, select a vendor, and then, on the Action Pane, on the **Vendor** tab, in the **Transactions** group, select **Transactions**). However, you want to have all vendor transactions together in one electronic document in XML format. This solution consists of several ER configurations that contain the required data model, model mapping, and format components.

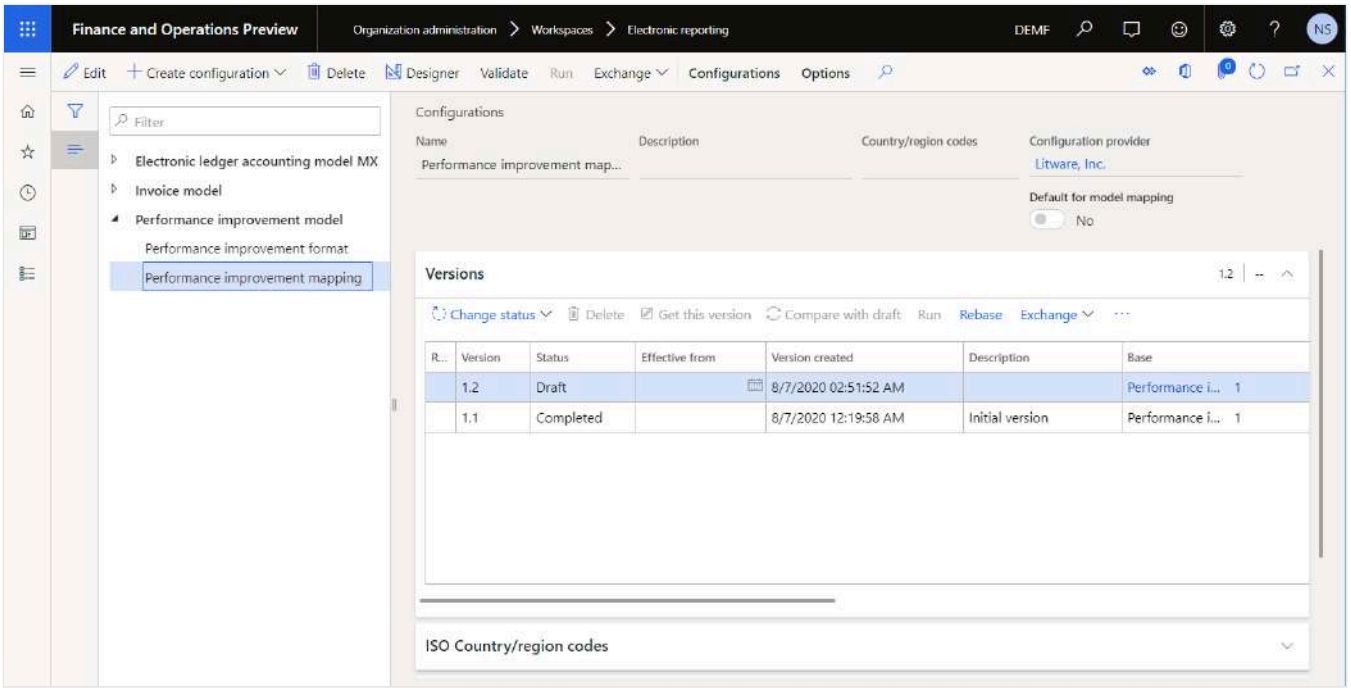
The first step is to import the sample ER solution to generate a vendor transactions report.

1. Sign in to the instance of Microsoft Dynamics 365 Finance that is provisioned for your company.

2. In this article, you create and modify configurations for the **Litware, Inc.** sample company. Make sure that this configuration provider is added to your Finance instance and is marked as active. For more information, see Create configuration providers and mark them as active.

3. In the **Electronic reporting** workspace, select the **Reporting configurations** tile.

4. On the **Configurations** page, import the ER configurations that you downloaded as a prerequisite into Finance, in the following order: data model, model mapping, format. For each configuration, follow these steps:
   a. On the Action Pane, select **Exchange** > **Load from XML file**.
   b. Select **Browse**, and select the appropriate file for the ER configuration in XML format.
   c. Select **OK**.



Review the sample ER solution

Review the model mapping

1. On **Configurations**, in the configuration tree, expand **Performance improvement model**, and select **Performance improvement mapping**.

2. On the Action Pane, select **Designer**.

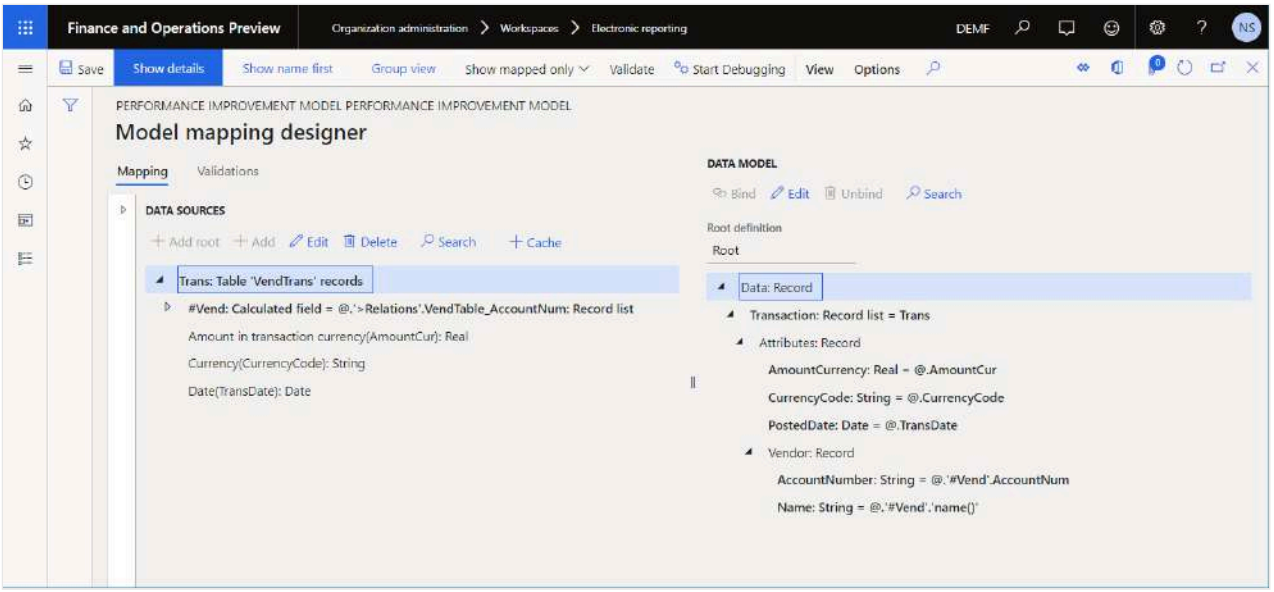
3. On **Model to datasource mapping**, on the Action Pane, select **Designer**.

   This ER model mapping is designed to perform the following actions:

   - Fetch the list of vendor transactions that are stored in the VendTrans table (**Trans** data source).
     - For every transaction, fetch, from the VendTable table, the record of a vendor that the transaction is posted for (**#Vend** data source).

   > ⓘ **Note**
   >
   > The **#Vend** data source is configured to fetch the corresponding vendor record by using the existing many-to-one relation **@.'>Relations'.VendTable_AccountNum**.

   The model mapping in this configuration implements the base data model for any ER formats that are created for this model and run in Finance. Therefore, the content of the **Trans** data source is exposed for ER formats such as abstract **model** data sources.



4. Close **Model mapping designer**.

5. Close **Model to datasource mapping**.

Review format

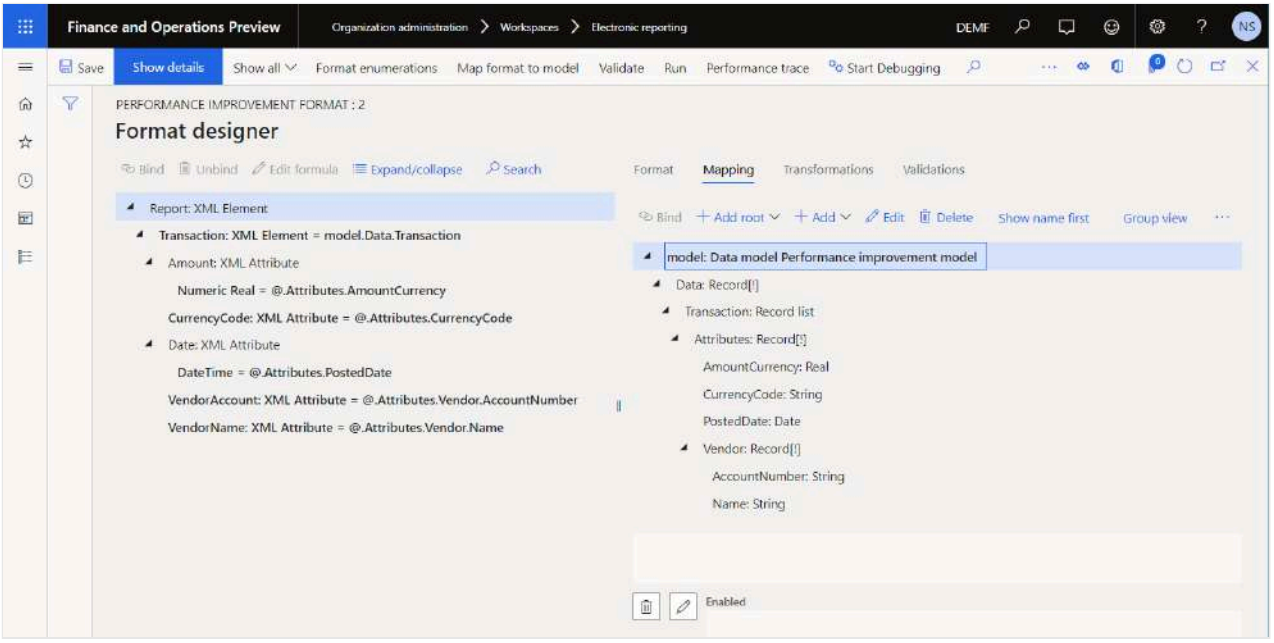
1. On **Configurations**, in the configuration tree, expand **Performance improvement model**, and select **Performance improvement format**.

2. On the Action Pane, select **Designer**.

3. On **Format designer**, on the **Mapping** tab, select **Expand/Collapse**.

4. Expand the **Model**, **Data**, and **Transaction** items.

   This ER format generates a vendor transactions report in XML format.



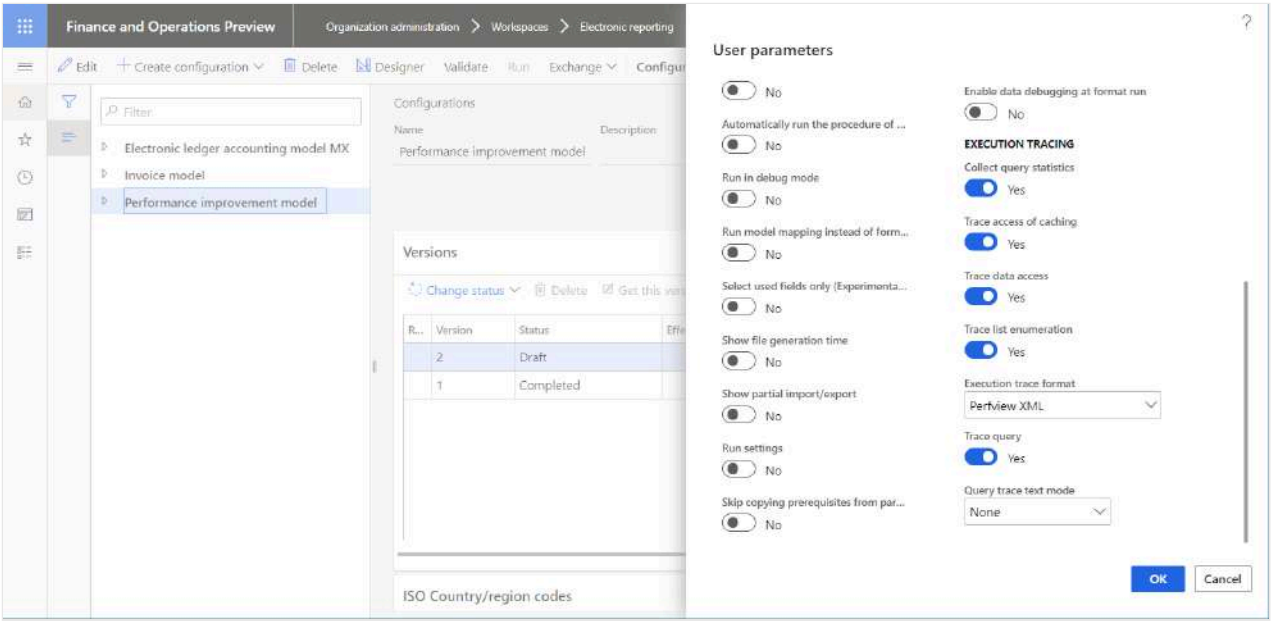
5. Close **Format designer**.

Run the sample ER solution to trace execution

Imagine that you finish designing the first version of the ER solution. You want to test the solution in your Finance instance and analyze the execution performance.

Turn on the ER performance trace

1. Select the **DEMF** company.
2. Follow the steps in Turn on the ER performance trace to generate a performance trace while an ER format runs.



Run the ER format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, select **Performance improvement format**.
3. On the Action Pane, select **Run**.

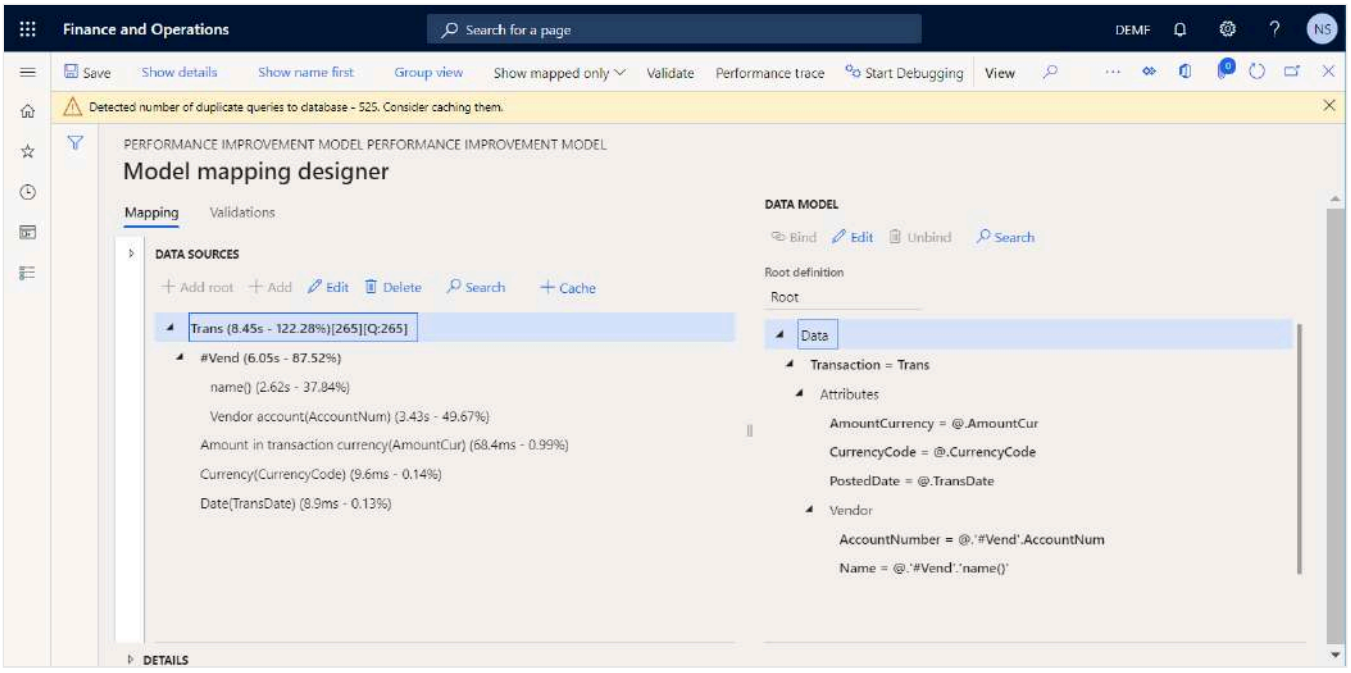
Use the performance trace to analyze model mapping performance

1. On **Configurations**, in the configuration tree, select **Performance improvement mapping**.
2. On the Action Pane, select **Designer**.
3. On **Model mappings**, on the Action pane, select **Designer**.
4. On the **Model mapping designer** page, on the Action Pane, select **Performance trace**.

5. Select the most recently generated trace, and then select **OK**.

New information is now available for some data source items of the current model mapping:

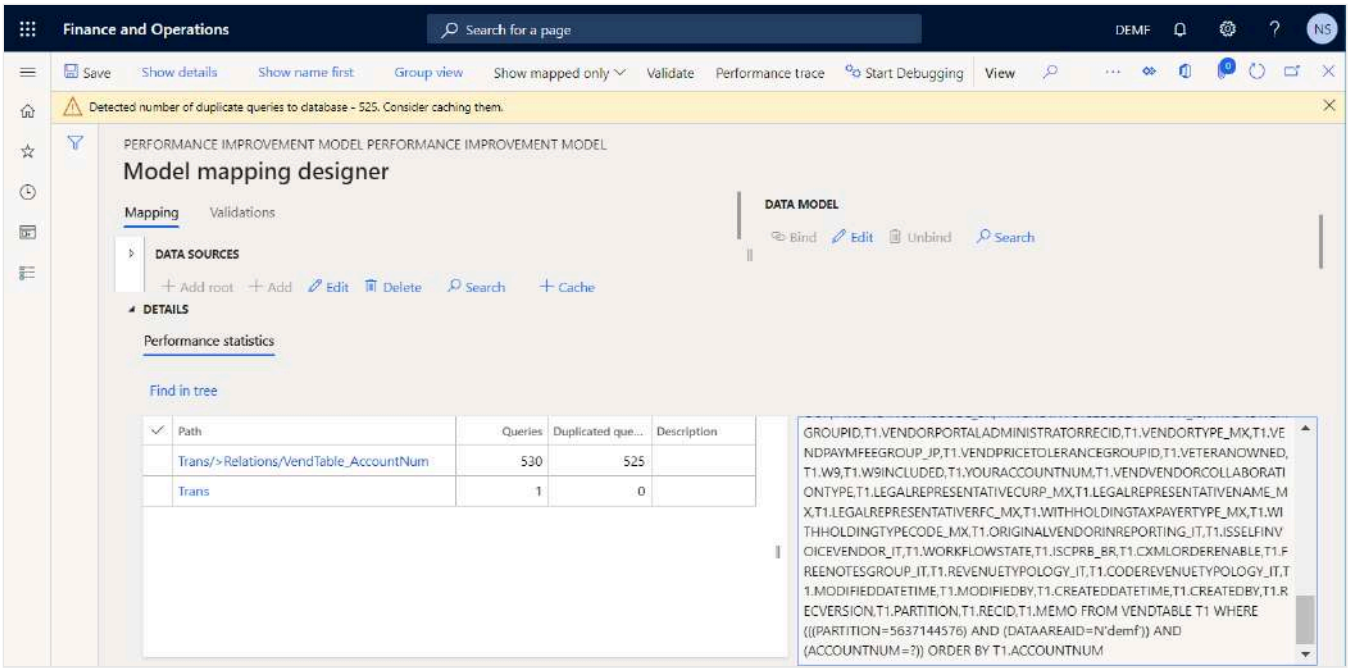
- The actual time spent getting data by using the data source
- The same time expressed as a percentage of the total time spent running the whole model mapping



The **Performance statistics** grid shows that the **Trans** data source calls the VendTrans table one time. The value **[265][Q:265]** of the **Trans** data source indicates that 265 vendor transactions are fetched from the application table and returned to the data model.

The **Performance statistics** grid also shows that the current model mapping duplicates database requests while the **#Vend** data source runs. This duplication occurs for the following reasons:

- The vendor table is called two times for each of the 265 iterated vendor transactions, for a total of 530 calls:
  - One call is made to enter the vendor account number.
  - One call is made to enter the vendor name.
- The vendor table is called for each iterated vendor transaction, even though the fetched transactions are posted for only five vendors. Of the 530 calls, 525 are duplicates. The following illustration shows the message that you receive about duplicate calls (database requests).



Of the total model mapping execution time (approximately eight seconds), more than 80 percent (approximately six seconds) is spent retrieving values from the VendTable application table. That percentage is too large for two attributes of five vendors, compared with the volume of information from the VendTrans application table.

To reduce the number of calls that get the vendor details for every transaction and to improve the performance of the model mapping, you can use caching for the **#Vend** data source.

> **Note**
>
> At runtime, the **Trans\#Vend** data source is cached in the scope of the current transaction of the **Trans** data source.

By caching the **#Vend** data source, you reduce the number of duplicated calls from 525 to 260, but you don't completely eliminate the duplication. To completely eliminate duplication, you can configure a new parameterized data source of the **Calculated field** type.

Improve the model mapping based on information from the execution trace

Change the logic of the model mapping

Follow these steps to use caching and a data source of the **Calculated field** type, to help prevent duplicate calls to the database.

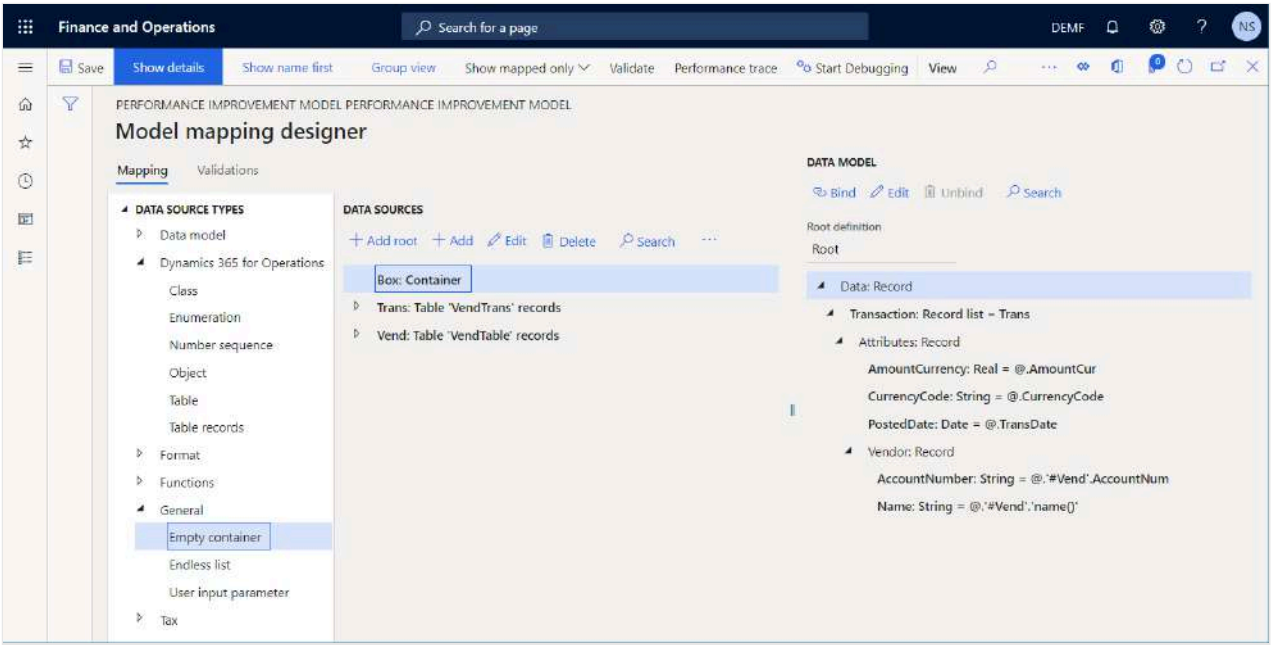
1. On **Configurations**, in the configuration tree, select **Performance improvement mapping**.

2. On the Action Pane, select **Designer**.

3. On **Model mappings**, on the Action pane, select **Designer**.

4. On **Model mapping designer**, add a data source of the **Table records** type to access records in the VendTable application table:
   a. In the **Data source types** pane, expand **Dynamics 365 for Operations**, and select **Table records**.
   b. Select **Add root**.
   c. In the dialog box, enter **Vend** in the **Name** field.
   d. Enter **VendTable** in the **Table** field.
   e. Select **OK**.

5. You can parameterize calls to data sources of the **Calculated field** type only if those data sources reside in a container. Therefore, add a data source of the **Empty container** type to hold a new parameterized data source of the **Calculated field** type:
   a. In the **Data source types** pane, expand **General**, and select **Empty container**.
   b. Select **Add root**.
   c. In the dialog box, enter **Box** in the **Name** field.
   d. Select **OK**.



6. Add a parameterized data source of the **Calculated field** type:
   a. In the **Data sources** pane, select **Box**.
   b. In the **Data source types** pane, expand **Functions**, and select **Calculated field**.
   c. Select **Add**.

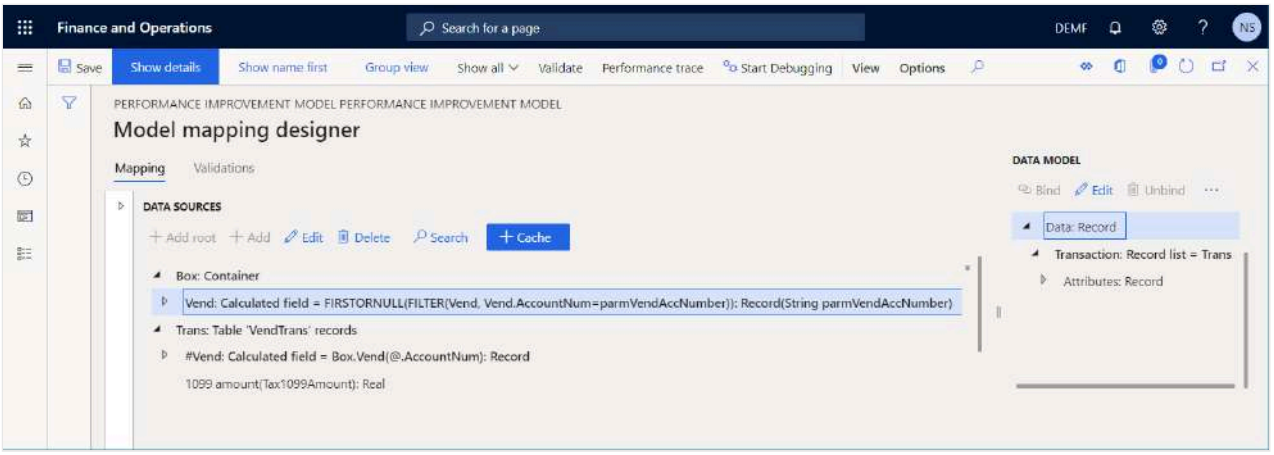
d. In the dialog box, enter **Vend** in the **Name** field.
e. Select **Edit formula**.
f. On **Formula designer**, select **Parameters** to specify the parameters that must be provided when this data source is called.
g. In **Parameters**, select **New**.
h. Enter **parmVendAccNumber** in the **Name** field.
i. Select **String** in the **Type** field.
j. Select **OK**.
k. In the **Formula** field, enter `FIRSTORNULL(FILTER(Vend, Vend.AccountNum=parmVendAccNumber))`.
l. Select **Save**, and close **Formula designer**.
m. Select **OK** to add the new data source.

7. Mark the added data source as cached during the execution:
   a. In the **Data sources** pane, select **Box\Vend**.
   b. Select **Cache**.

> ⓘ **Note**
>
> The **Box\Vend** data source is cached in the scope of all vendor transactions of the **Trans** data source at runtime.

8. Update the nested **Trans\#Vend** data source so that it uses the **Box\Vend** data source:
   a. In the **Data sources** pane, expand **Trans**.
   b. Select the **Trans\#Vend** data source, and then select **Edit** > **Edit formula**.
   c. On **Formula designer**, enter `Box.Vend(@.AccountNum)` in the **Formula** field.
   d. Select **Save**, and then close **Formula designer**.
   e. Select **OK** to complete your changes to the selected data source.

9. Select **Save**.



10. Close **Model mapping designer**.

11. Close the **Model mappings** page.

Complete the modified version of the ER model mapping

1. On **Configurations**, on the **Versions** FastTab, select version **1.2** of the **Performance improvement mapping** configuration.
2. Select **Change status** > **Complete**, and then select **OK**.

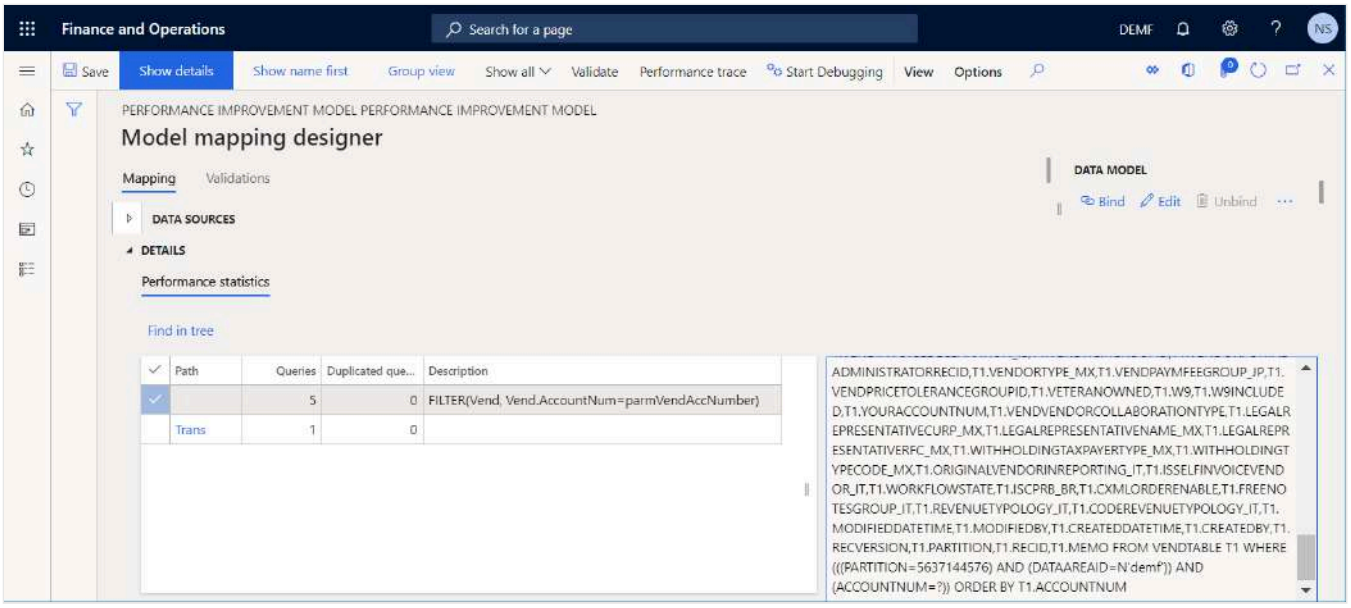
Run the modified ER solution to trace execution

Repeat the steps in the Run the ER format section earlier in this article to generate a new performance trace.

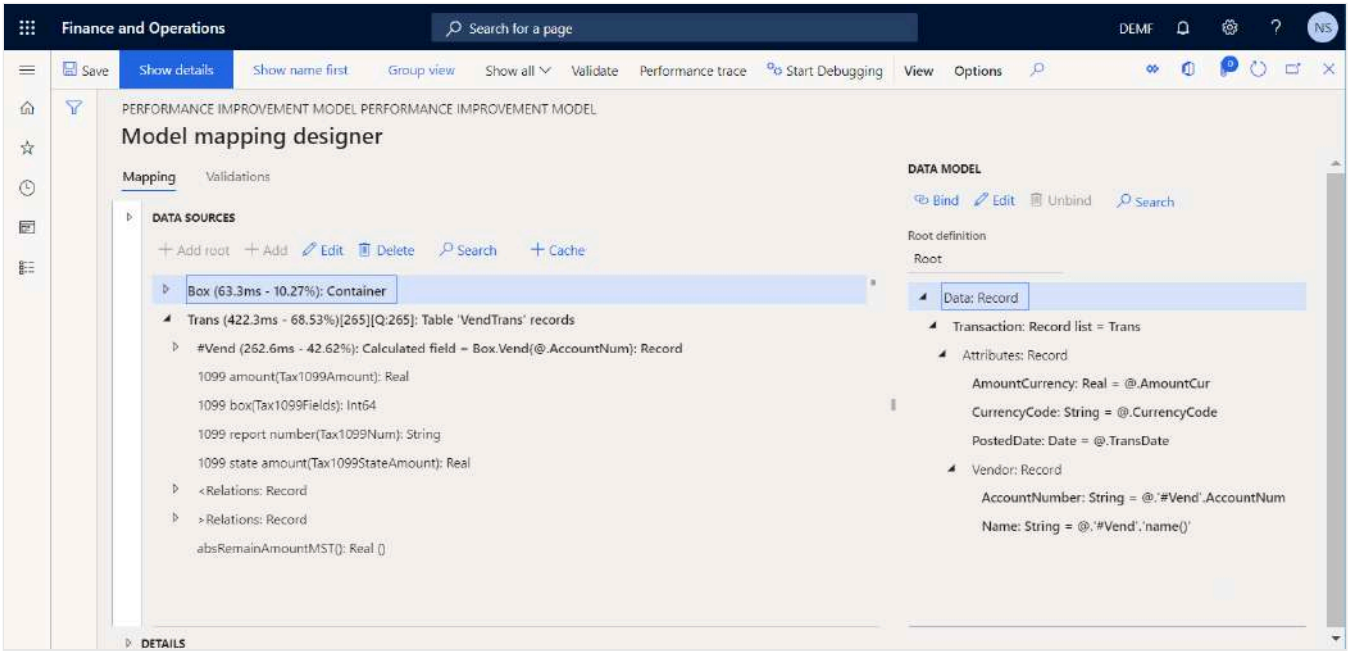
Use the performance trace to analyze adjustments to the model mapping

1. On **Configurations**, in the configuration tree, select **Performance improvement mapping**.
2. On the Action Pane, select **Designer**.
3. On **Model mappings**, on the Action Pane, select **Designer**.
4. On **Model mapping designer**, on the Action Pane, select **Performance trace**.
5. Select the most recent trace that was generated, and then select **OK**.

Notice that the adjustments you made to the model mapping eliminated duplicate queries to the database. The number of calls to database tables and data sources for this model mapping is also reduced.



The total execution time is reduced by about 20 times (from about 8 seconds to about 400 milliseconds). Therefore, the performance of the whole ER solution is improved.



Appendix 1: Download the components of the sample Microsoft ER solution

Download the following files and store them locally.

Expand table

File	Content
Performance improvement model.version.1	Sample ER data model configuration
Performance improvement mapping.version.1.1	Sample ER model mapping configuration

File	Content
Performance improvement format.version.1.1	Sample ER format configuration

Appendix 2: Configure the ER framework

Before you can start to use the ER framework to improve the performance of the sample Microsoft ER solution, you must configure the minimal set of ER parameters.

Configure ER parameters

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Related links** section, select **Electronic reporting parameters**.
3. On the **Electronic reporting parameters** page, on the **General** tab, set the **Enable design mode** option to **Yes**.
4. On the **Attachments** tab, set the following parameters:
    - In the **Configurations** field, select the **File** type for the **DEMF** company.
    - In the **Job archive**, **Temporary**, **Baseline**, and **Others** fields, select the **File** type.

For more information about ER parameters, see Configure the ER framework.

Activate an ER configuration provider

Every ER configuration that you add is marked as owned by an ER configuration provider. Use the ER configuration provider activated in the **Electronic reporting** workspace for this purpose. Therefore, you must activate an ER configuration provider in the **Electronic reporting** workspace before you start to add or edit ER configurations.

> **Note**
>
> Only the owner of an ER configuration can edit the configuration. Therefore, before editing an ER configuration, you must activate the appropriate ER configuration provider in the **Electronic reporting** workspace.

Review the list of ER configuration providers

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Configuration providers**.
3. On **Configuration provider table**, each provider record has a unique name and URL. Review the contents of this page. If a record for **Litware, Inc.** already exists, skip the next procedure, Add a new ER configuration provider.

Add a new ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Related links** section, select **Configuration providers**.
3. On **Configuration providers**, select **New**.
4. In the **Name** field, enter **Litware, Inc.**.
5. In the **Internet address** field, enter `https://www.litware.com`.
6. Select **Save**.

Activate an ER configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, select the **Litware, Inc.** tile, and then select **Set active**.

For more information about ER configuration providers, see Create configuration providers and mark them as active.

Additional resources

- Electronic Reporting overview
- Trace the execution of ER formats to troubleshoot performance issues
- Support parameterized calls of ER data sources of the Calculated field type

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Last updated on 04/13/2026

Cross-company data sources in Electronic reporting (ER)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can design Electronic reporting (ER) formats to generate outgoing documents in various formats. When you generate a document, an ER format calls data sources that you configured in a corresponding ER model mapping. To configure access to application tables for record retrieval, use ER data sources of the **Table records** type. When the accessing table is a shared table (that is, a table where data is saved without a company identifier), this data source returns all records. When the accessing table is a company-dependent table (that is, a table where data is saved per company), this data source returns only the records saved for the current company (that is, the company context that the ER format is running under).

You can now mark every data source of the **Table records** type in a model mapping as a cross-company data source. Therefore, you can use data sources of the **Table records** type to access cross-company data in application tables.

If you mark a data source as cross-company, the following behavior occurs:

- For a data source that refers to any shared table except CompanyInfo, the data source returns all records that exist in the referenced table.
- For a data source that refers to the CompanyInfo table, even though CompanyInfo is a shared table, the data source returns the records that contain the identifier of a company from the defined scope.
- For any company-dependent table, the data source returns the records of the referenced table that contain the identifier of a company from the defined scope.

In the system query dialog box, when the **Ask for query** option is turned on for any data source marked as cross-company, you can manually select one or more companies to include on the **Company range** tab.

> **ⓘ Important**
>
> Like other filters, the company filter is persisted as a last-used value for queries when you run an ER format. The filter isn't automatically changed if you change the cross-company value for a data source. To use a different cross-company value for another data source, delete the corresponding user-specific selection.

For every data source that is marked as cross-company, you can select the records that you want by using the **FILTER** and **WHERE** functions in ER expressions. The **dataAreaID** field can also be used as a company identifier. Currently, the **dataAreaID** field is limited to the following types of conditions when the **FILTER** function is used:

- Only conditions that have a single **dataAreaID** field comparison are supported.
- Only comparisons that have expressions that don't depend on records list items are allowed.

Therefore, the following expression is valid.

ER

```
FILTER (MyTable, MyTable.dataAreaID = $StringUserInputParameter)
While shown below expressions will not pass the validation:
FILTER (MyTable, MyTable.dataAreaID = MyTable2RecordsList.MyField)
FILTER (MyTable,
    OR(
        MyTable.dataAreaID = $StringUserInputParameter1,
        MyTable.dataAreaID = $StringUserInputParameter2
    )
)
```

By default, the scope includes all companies of the current application. However, you can restrict it. To restrict the scope of cross-company data access for a single ER format, assign a specific organization hierarchy to the format. When you define a hierarchy for an ER format, only records for legal entities that are presented in the assigned hierarchy are returned, even though the format calls cross-company data sources. When a reference to a hierarchy that no longer exists is defined for an ER format, the default scope is applied, and the format calls cross-company data sources. In this situation, records for all application companies are returned.

When you turn on the **Use draft** option for the assigned to a single ER format organization hierarchy, the legal entities from the draft version of this hierarchy are used to identify the

scope for cross-company data sources. If the draft version doesn't exist, the legal entities from the last published version of this organization hierarchy are used.

When you turn off the **Use draft** option for the assigned to a single ER format organization hierarchy, the legal entities from the last published version of this organization hierarchy are used to identify the scope for cross-company data sources. Date effectiveness of organization hierarchies isn't supported yet in the ER framework.

Assign the hierarchy to a format in a specific page that you can access from the ER workspace or by using the **Organization administration** > **Electronic reporting** > **Legal entity filter for formats** menu item. To access the page, grant the **Maintain legal entity filters for format** privilege (ERMaintainFormatMappingLegalEntityFilters) to a user. The scope restriction of hierarchy-based legal entities for the format is applied in addition to the restriction that the user can manually specify in the system query dialog box. The intersection of these restrictions is used when the format is run.

To learn more about this feature, play the task guide, **ER Access records of company dependent tables in cross-company mode**, which is part of the 7.5.4.3 Acquire/Develop IT service/solution components (10677) business process, and can be downloaded from the Microsoft Download Center. This task guide walks you through the process of configuring an ER model mapping and ER format to access application tables in cross-company mode.

Download the following files to complete the task guide:

- ER model configuration - CrossCompanyDataAccessModel.xml
- ER format configuration - CrossCompanyDataAccessFormat.xml

Last updated on 04/15/2026

ER Use financial dimensions as a data source (Part 1 - Design data model)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a system administrator or electronic reporting developer can configure an Electronic reporting (ER) model to use financial dimensions as a data source for ER reports. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the procedure, "Create a configuration provider and mark it as active".

Create a new data model

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
    - Make sure that the **Litware, Inc.** provider is available and marked as active.
2. Select **Reporting configurations**.
3. Select **Create configuration** to open the form.
4. In the **Name** field, type *Financial dimensions sample model*.
5. Select **Create configuration**.
6. Select **Designer**.
7. Select **New** to open the form.
8. In the **Name** field, type *Entry*.
9. Select **Add**.
10. Select **New** to open the form.
11. In the **Name** field, type *Company*.
12. Select **Add**.
    - Add a new record list to your model. This list exposes the settings of selected financial dimensions for any ER reports that use this model as a data source. Each

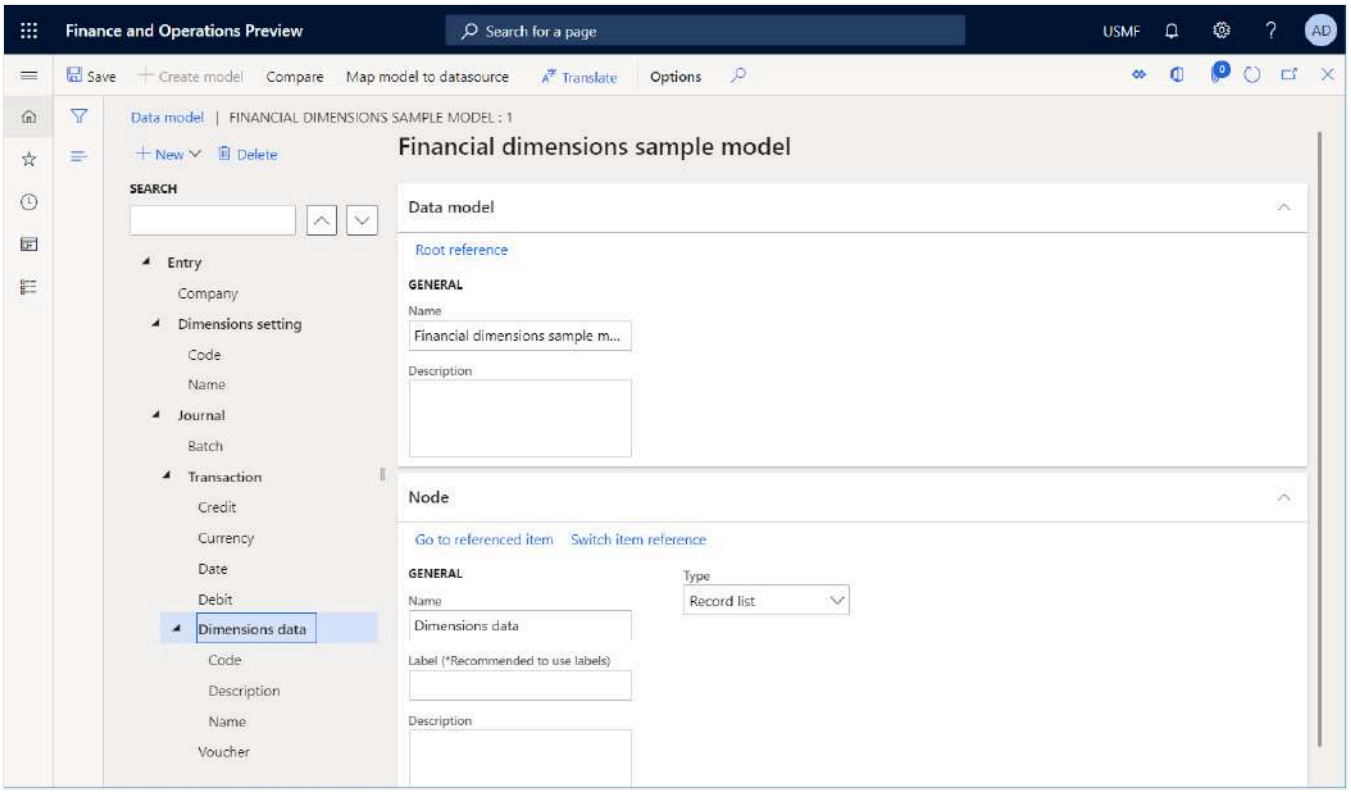
financial dimension appears in this list as a record with fields that represent the dimension's settings.

13. Select **New** to open the form.
14. In the **Name** field, type *Dimensions setting*.
15. In the **Item type** field, select **Record list**.
16. Select **Add**.
17. Select **New** to open the form.
18. In the **Name** field, type *Code*.
19. In the **Item type** field, select **String**.
20. Select **Add**.
21. Select **New** to open the form.
22. In the **Name** field, type *Name*.
23. Select **Add**.
24. In the tree, select **Entry**.
25. Select **New** to open the form.
26. In the **Name** field, type *Journal*.
27. In the **Item type** field, select **Record list**.
28. Select **Add**.
29. Select **New** to open the form.
30. In the **Name** field, type *Batch*.
31. In the **Item type** field, select **String**.
32. Select **Add**.
33. Select **New** to open the form.
34. In the **Name** field, type *Transaction*.
35. In the **Item type** field, select **Record list**.
36. Select **Add**.
37. Select **New** to open the form.
38. In the **Name** field, type *Date*.
39. In the **Item type** field, select **Date**.
40. Select **Add**.
41. Select **New** to open the form.
42. In the **Name** field, type *Debit*.
43. In the **Item type** field, select **Real**.
44. Select **Add**.
45. Select **New** to open the form.
46. In the **Name** field, type *Credit*.
47. Select **Add**.

48. Select **New** to open the form.
49. In the **Name** field, type *Currency*.
50. In the **Item type** field, select **String**.
51. Select **Add**.
52. Select **New** to open the form.
53. In the **Name** field, type *Voucher*.
54. Select **Add**.
55. Select **New** to open the form.
56. In the **Name** field, type *Dimensions data*.
57. In the **Item type** field, select **Record list**.
58. Select **Add**.

    - You add a new record list to your model. This list exposes the values of selected financial dimensions for any ER reports that use this model as a data source. Each financial dimension appears in this list as a record with fields that represent the dimension's values. The dimension name also appears in this record as a field that you can use for selection purposes.

59. Select **New** to open the form.
60. In the **Name** field, type *Code*.
61. In the **Item type** field, select **String**.
62. Select **Add**.
63. Select **New** to open the form.
64. In the **Name** field, type **Description**.
65. Select **Add**.
66. Select **New** to open the form.
67. In the **Name** field, type *Name*.
68. Select **Add**.
69. Select **Save**.
70. Close the page.



Last updated on 04/13/2026

ER Use financial dimensions as a data source (Part 2 - Model mapping)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) model to use financial dimensions as a data source for ER reports. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the "ER Use financial dimensions as a data source (Part 1: Design data model" procedure.

Add required data sources to model mapping

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, select **Financial dimensions sample model**.
3. Select **Designer**.
4. Select **Map model to datasource**.
5. Select **New**.
6. In the **Definition** field, select **Entry**.
7. In the **Name** field, type **Dimensions data mapping**.
8. In the **Description** field, type **Dimensions data mapping**.
9. Select **Save**.
10. Select **Designer**.
11. In the tree, select **Dynamics 365 for Operations\Table**.

12. Select **Add root**.

13. In the **Name** field, type **Company**.

14. In the **Table** field, type **CompanyInfo**.

15. Select **OK**.

16. In the tree, select **Functions\Financial dimensions details**.

17. Select **Add root**.

    - This data source specifies how the scope of financial dimensions is defined for any report that uses this model as a data source.

18. In the **Name** field, type a value.

19. Select **Yes** in the **Ask for dimensions** field.

    - Select **Yes** to allow the user to select dimensions at run-time on the User dialog form. If set to **No**, all financial dimensions of the current instance are used by default.

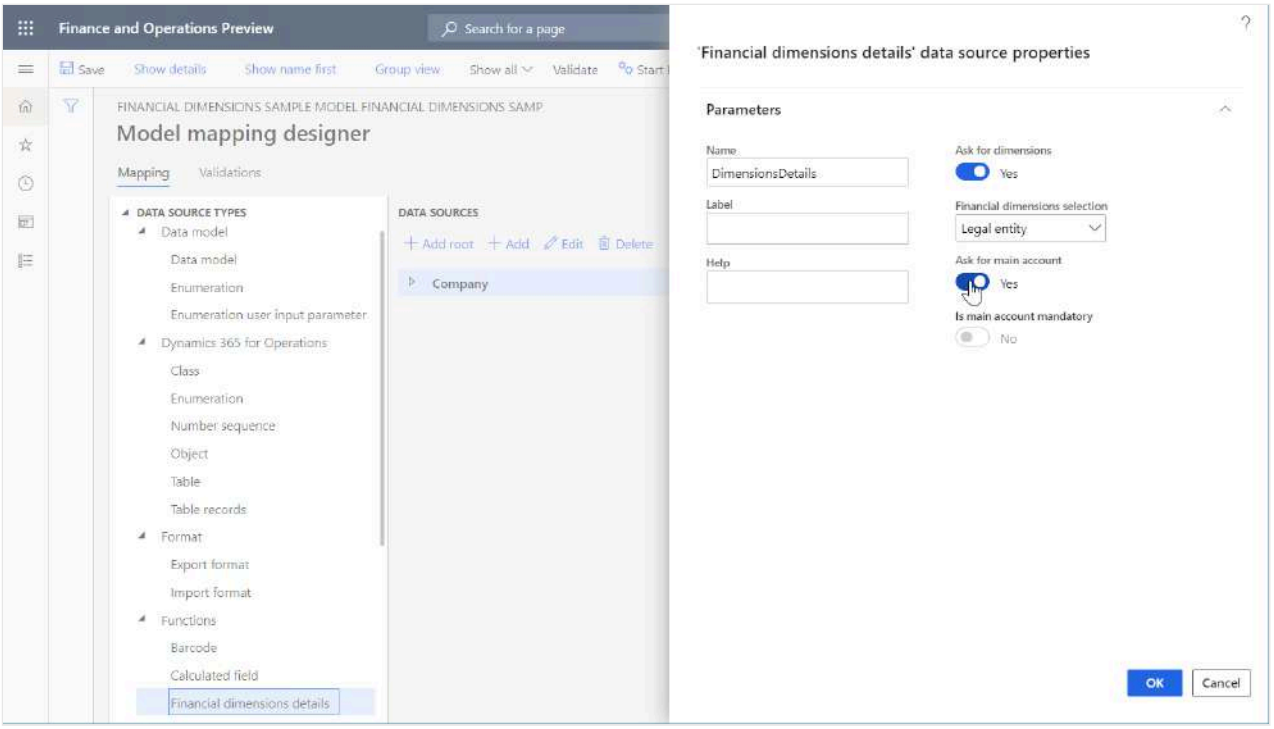
20. In the **Financial dimensions selection** field, select **Legal entity**.

    - Select **All** to allow the user to select desired dimensions for the current instance in the **Lookup** field. Select **Legal entity** to allow the user to select dimensions for the company in the **Lookup** field. Select **Dimension** to allow the user to select dimensions using a single dimension set.

21. Select **Yes** in the **Ask for main account** field.

    - Set **Ask for main account** to **Yes** to allow users to select the main account as part of the list of dimensions. If set to **No**, the main account isn't included in the list of dimensions and the **Is main account mandatory** option is enabled. If **Is main account mandatory** is set to **Yes**, include the main account in the list of dimensions regardless of the user's selection.

22. Select **OK**.



23. In the tree, select **Dynamics 365 for Operations\Table records**.

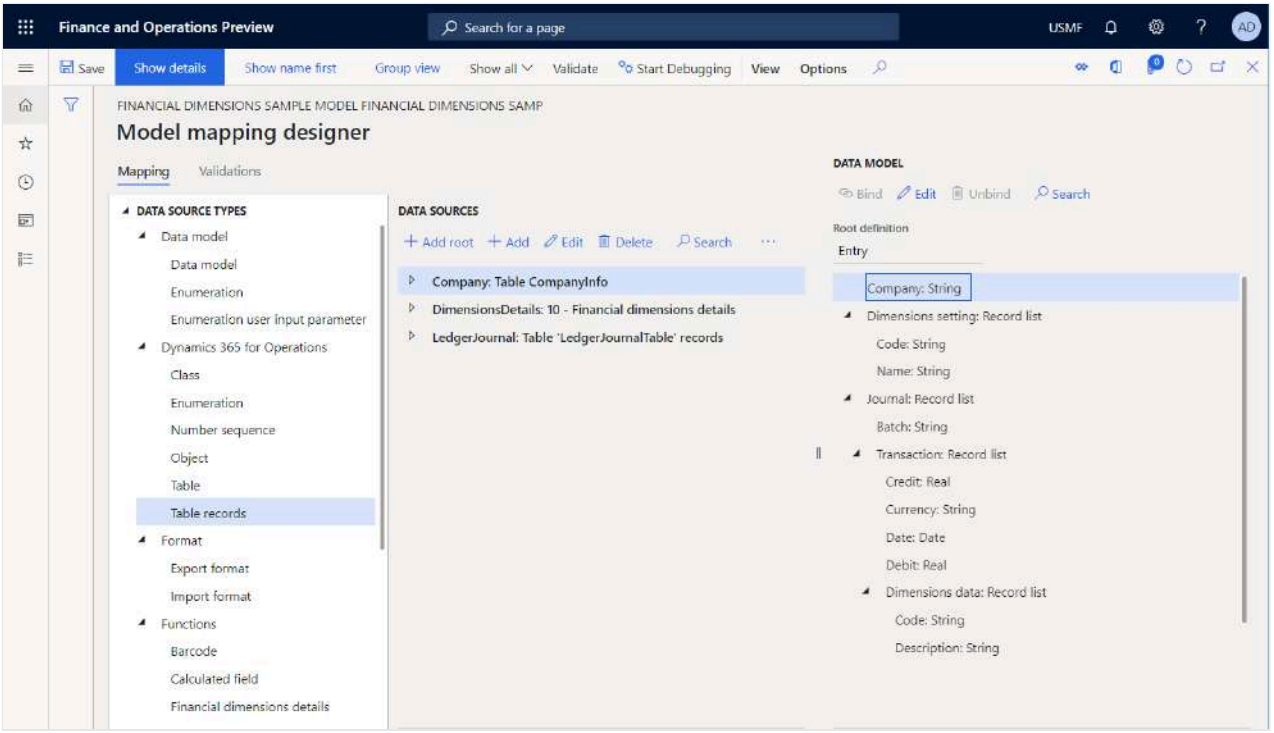
24. Select **Add root**.

25. In the **Name** field, type **LedgerJournal**.

26. Select **Yes** in the **Ask for query** field.

27. In the **Table** field, type **LedgerJournalTable**.

28. Select **OK**.



Map data model elements to added data sources

1. In the tree, expand **Journal**.
2. In the tree, expand **Journal\Transaction**.
3. In the tree, expand **Journal\Transaction\Dimensions data**.
4. In the tree, expand **Dimensions setting**.
5. In the tree, expand **LedgerJournal**.
6. In the tree, expand **LedgerJournal<Relations**.
7. In the tree, expand **LedgerJournal<Relations\LedgerJournalTrans**.
8. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Voucher**.
9. In the tree, select **Journal\Transaction\Voucher**.
10. Select **Bind**.
11. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)**.
    - For any reference to financial dimensions that you set to, for example, `LedgerDimension`, a corresponding data source item is available (`LedgerDimension.Dimension`). This data source item offers the financial dimensions of that dimensions set as the record's list.
12. In the tree, expand **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)**.
13. In the tree, expand **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)\Main account and dimensions**.
14. In the tree, expand **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)\Main account and dimensions\Value**.

15. In the tree, expand **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)\Main account and dimensions\Definition**.

16. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)\Main account and dimensions\Definition\Name**.

17. In the tree, select **Journal\Transaction\Dimensions data\Name**.

18. Select **Bind**.

19. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)\Main account and dimensions\Value\Description**.

20. In the tree, select **Journal\Transaction\Dimensions data\Description**.

21. Select **Bind**.

22. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)\Main account and dimensions\Value\Code**.

23. In the tree, select **Journal\Transaction\Dimensions data\Code**.

24. Select **Bind**.

25. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Account.Dimension(LedgerDimension.Dimension)\Main account and dimensions**.

26. In the tree, select **Journal\Transaction\Dimensions data**.

27. Select **Bind**.

28. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Debit(AmountCurDebit)**.

29. In the tree, select **Journal\Transaction\Debit**.

30. Select **Bind**.

31. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Date(TransDate)**.

32. In the tree, select **Journal\Transaction\Date**.

33. Select **Bind**.

34. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Currency(CurrencyCode)**.

35. In the tree, select **Journal\Transaction\Currency**.

36. Select **Bind**.

37. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans\Credit(AmountCurCredit)**.

38. In the tree, select **Journal\Transaction\Credit**.

39. Select **Bind**.

40. In the tree, select **LedgerJournal<Relations\LedgerJournalTrans**.

41. In the tree, select **Journal\Transaction**.

42. Select **Bind**.

43. In the tree, select **LedgerJournal\Journal batch number(JournalNum)**.

44. In the tree, select **Journal\Batch**.

45. Select **Bind**.

46. In the tree, select **LedgerJournal**.

47. In the tree, select **Journal**.

48. Select **Bind**.

49. In the tree, expand **Dimensions**.

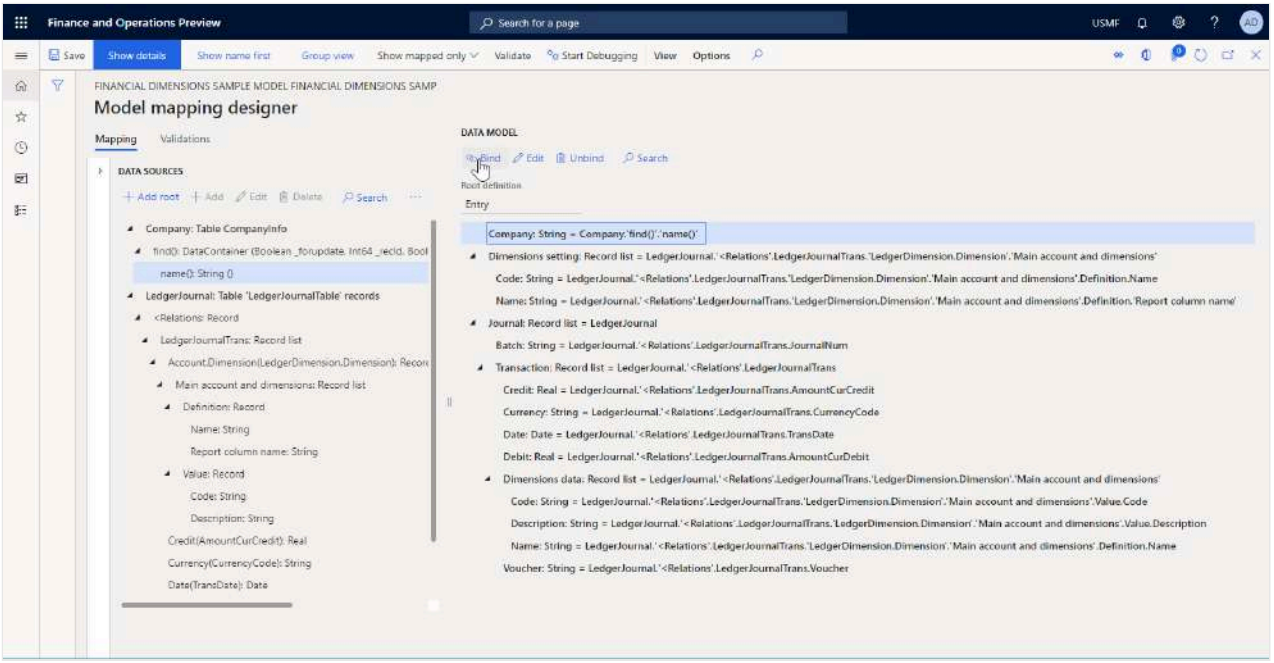
50. In the tree, expand **Dimensions\Main account and dimensions**.

51. In the tree, expand **Dimensions\Main account and dimensions\Definition**.

52. In the tree, select **Dimensions\Main account and dimensions\Definition\Name**.

53. In the tree, select **Dimensions setting\Code**.

54. Select **Bind**.
55. In the tree, select **Dimensions\Main account and dimensions\Definition\Report column name**.
56. In the tree, select **Dimensions setting\Name**.
57. Select **Bind**.
58. In the tree, select **Dimensions\Main account and dimensions**.
59. In the tree, select **Dimensions setting**.
60. Select **Bind**.
61. In the tree, select **Company**.
62. Click **Edit**.
63. In the **expressionAsStringText** field, enter `Company.'find()'.'name()'`.
    - `Company.'find()'.'name()'`
64. Select **Save**.



65. Close the page.
66. Select **Save**.
67. Close the page.

Complete this draft model's version

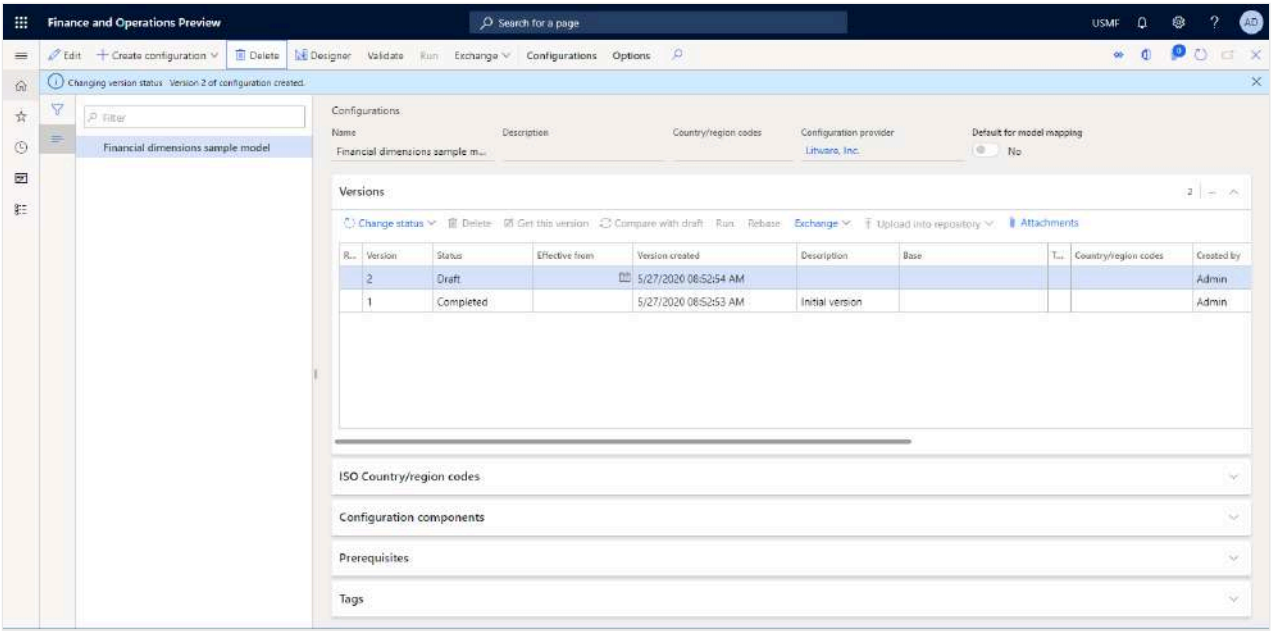
1. Close the page.

2. Close the page.

3. Select **Change status**.

4. Select **Complete**.

5. Select **OK**.



Last updated on 04/13/2026

ER Use financial dimensions as a data source (Part 3 - Design the report)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) model to use financial dimensions as a data source for ER reports. You can perform these steps in any company.

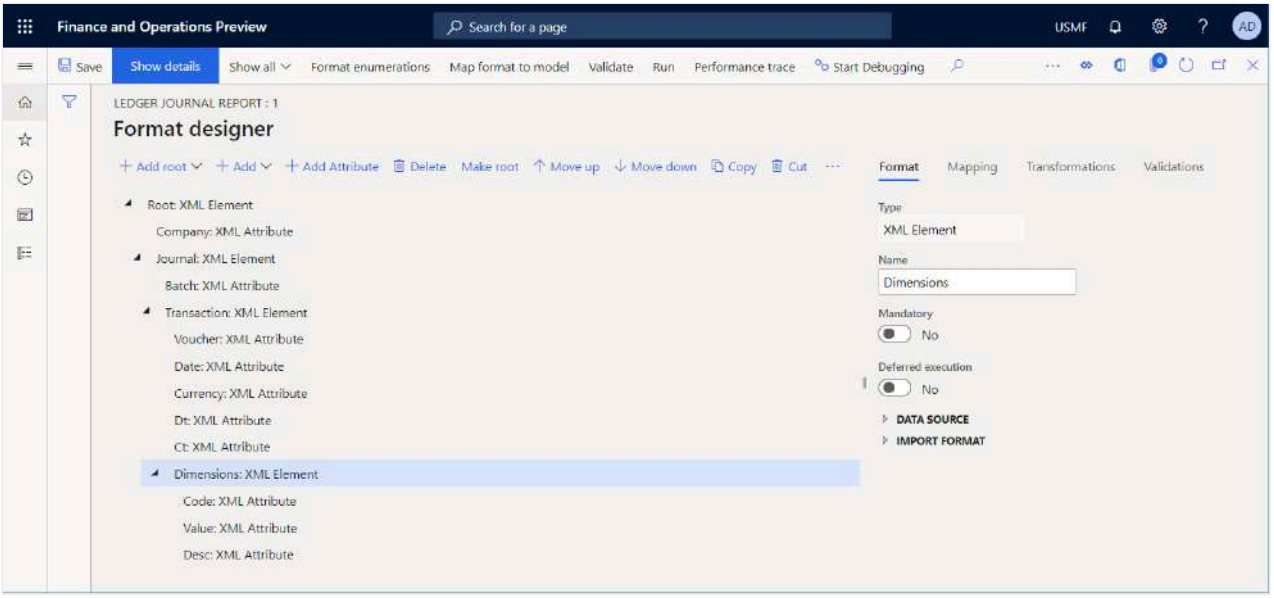
To complete these steps, you must first complete the steps in the "ER Use financial dimensions as a data source (Part 2: Model mapping)" procedure.

Design a report to present financial dimensions

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, select **Financial dimensions sample model**.
3. Select **Create configuration** to open the form.
4. In **New**, enter *Format based on data model Financial dimensions sample model*.
    - Use the model that you created in advance as the data source for your new report.
5. In **Name**, type *Ledger journal report*.
6. In **Data model definition**, select *Entry*.
7. Select **Create configuration**.
8. Select **Designer**.
9. Select **Add root** to open the form.
10. In the tree, select **XML\Element**.
11. In **Name**, type *Root*.
12. Select **OK**.
13. Select **Add** to open the form.
14. In the tree, select **XML\Attribute**.

15. In **Name**, type *Company*.
16. Select **OK**.
17. Select **Add** to open the form.
18. In the tree, select **XML\Element**.
19. In **Name**, type *Journal*.
20. Select **OK**.
21. In the tree, select **Root: XML Element\Journal: XML Element**.
22. Select **Add** to open the form.
23. In the tree, select **XML\Attribute**.
24. In **Name**, type *Batch*.
25. Select **OK**.
26. Select **Add** to open the form.
27. In the tree, select **XML\Element**.
28. In **Name**, type *Transaction*.
29. Select **OK**.
30. In the tree, select **Root: XML Element\Journal: XML Element\Transaction: XML Element**.
31. Select **Add** to open the form.
32. In the tree, select **XML\Attribute**.
33. In **Name**, type *Voucher*.
34. Select **OK**.
35. Select **Add Attribute**.
36. In **Name**, type *Date*.
37. Select **OK**.
38. Select **Add Attribute**.
39. In **Name**, type *Currency*.
40. Select **OK**.
41. Select **Add Attribute**.
42. In **Name**, type *Dt*.
43. Select **OK**.
44. Select **Add Attribute**.
45. In **Name**, type *Ct*.
46. Select **OK**.
47. Select **Add** to open the form.
48. In the tree, select **XML\Element**.
49. In **Name**, type *Dimensions*.
50. Select **OK**.

51. In the tree, select **Root: XML Element\Journal: XML Element\Transaction: XML Element\Dimensions: XML Element**.
52. Select **Add** to open the form.
53. In the tree, select **XML\Attribute**.
54. In **Name**, type *Code*.
55. Select **OK**.
56. Select **Add Attribute**.
57. In **Name**, type *Value*.
58. Select **OK**.
59. Select **Add Attribute**.
60. In the **Name** field, enter `Desc`.
61. Select **OK**.



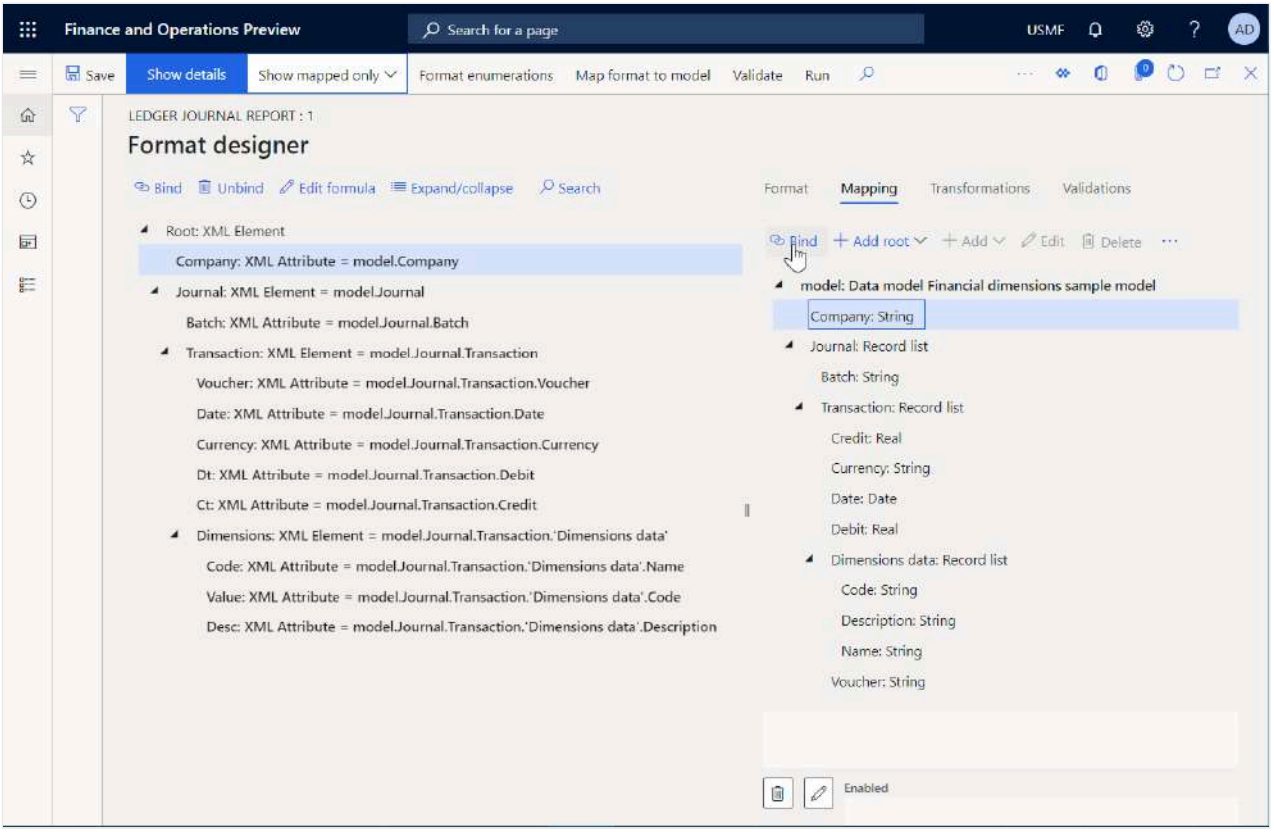
Map report elements to data sources

1. Select the **Mapping** tab.
2. In the tree view, expand `model: Data model Financial dimensions sample model`.
3. In the tree view, expand `model: Data model Financial dimensions sample model\Journal: Record list`.
4. In the tree view, expand `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list`.
5. In the tree view, expand `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list`.
6. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Dimensions: XML Element\Desc: XML Attribute`.

7. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list\Description: String`.
8. Select **Bind**.
9. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Dimensions: XML Element\Value: XML Attribute`.
10. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list\Code: String`.
11. Select **Bind**.
12. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Dimensions: XML Element\Code: XML Attribute`.
13. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list\Name: String`.
14. Select **Bind**.
15. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Dimensions data: Record list`.
16. In the tree, select **Root: XML Element\Journal: XML Element\Transaction: XML Element\Dimensions: XML Element**.
17. Select **Bind**.
18. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Ct: XML Attribute`.
19. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Credit: Real`.
20. Select **Bind**.
21. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Dt: XML Attribute`.
22. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Debit: Real`.
23. Select **Bind**.
24. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Currency: XML Attribute`.
25. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Currency: String`.
26. Select **Bind**.
27. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Date: XML Attribute`.

28. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Date: Date`.
29. Select **Bind**.
30. In the tree view, select `Root: XML Element\Journal: XML Element\Transaction: XML Element\Voucher: XML Attribute`.
31. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list\Voucher: String`.
32. Select **Bind**.
33. In the tree, select **Root: XML Element\Journal: XML Element\Transaction: XML Element**.
34. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Transaction: Record list`.
35. Select **Bind**.
36. In the tree view, select `Root: XML Element\Journal: XML Element\Batch: XML Attribute`.
37. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list\Batch: String`.
38. Select **Bind**.
39. In the tree, select **Root: XML Element\Journal: XML Element**.
40. In the tree view, select `model: Data model Financial dimensions sample model\Journal: Record list`.
41. Select **Bind**.
42. In the tree view, select `Root: XML Element\Company: XML Attribute`.
43. In the tree view, select `model: Data model Financial dimensions sample model\Company: String`.
44. Select **Bind**.
45. Select **Save**.

46. Close the page.



Last updated on 04/13/2026

ER Use financial dimensions as a data source (Part 4 - Run the report)

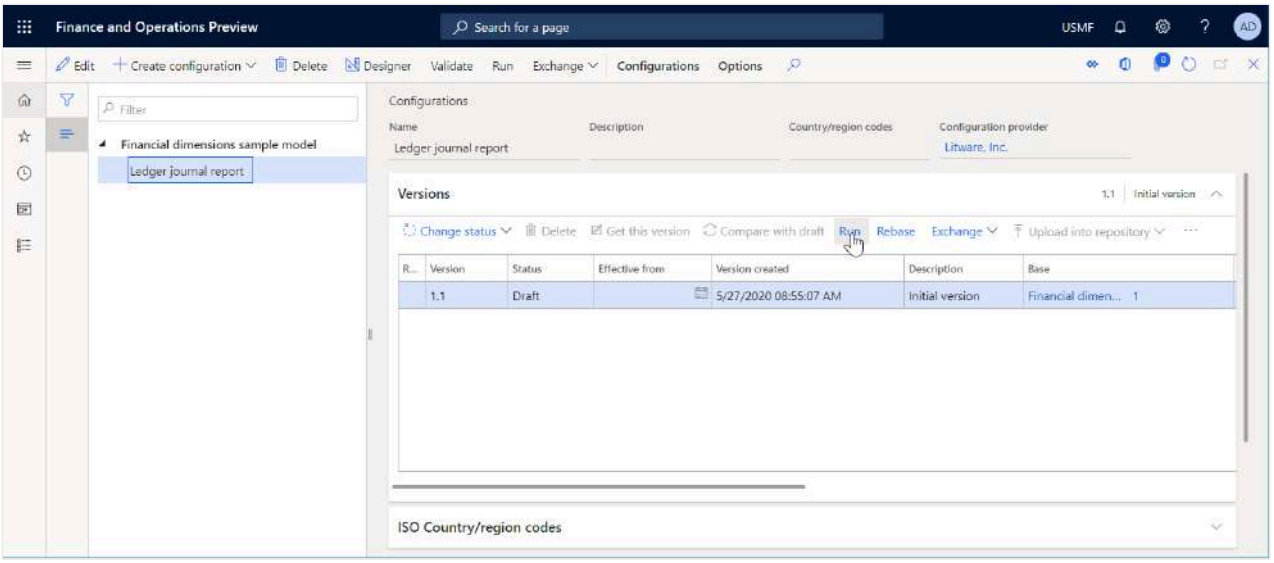
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) model to use financial dimensions as a data source for ER reports. You can perform these steps in the DEMF company.

To complete these steps, you must first complete the steps in the "ER Use financial dimensions as a data source (Part 3: Design the report)" procedure. You must also configure default document types on the Electronic reporting parameters page. When you download and import any ER configuration, you also set default document types.

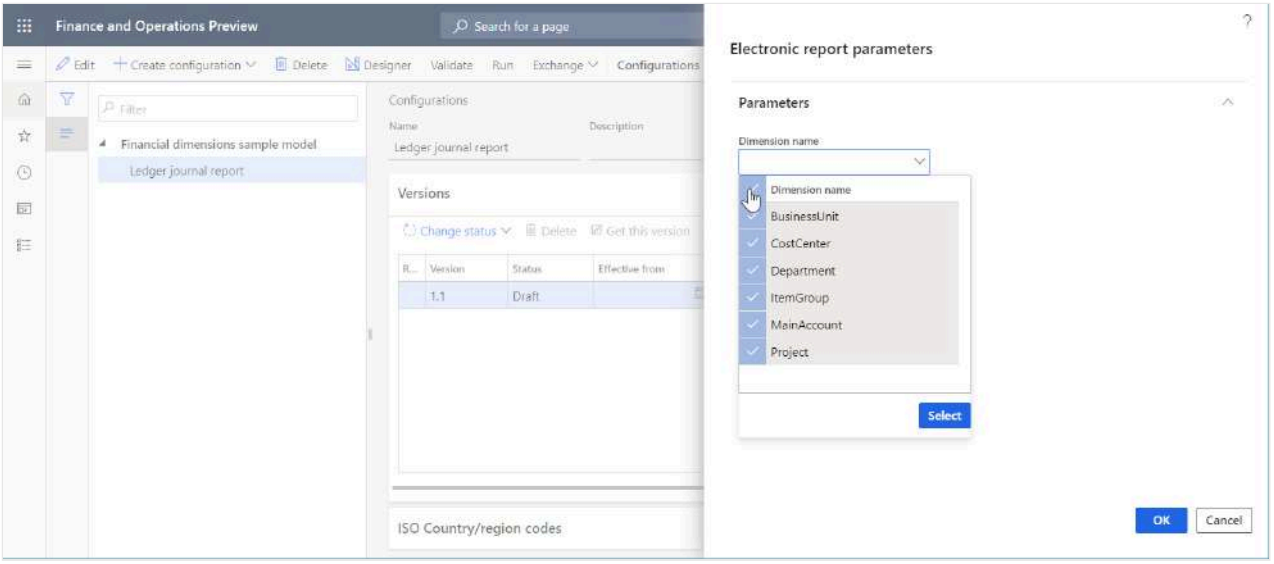
Run report

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Financial dimensions sample model**.
3. In the tree, select **Financial dimensions sample model\Ledger journal report**.
4. Select **Run**.



5. In the **Dimension name** field, enter or select a value.

    - To select all dimensions in the current company, enter the following information: BusinessUnit;CostCenter;Department;ItemGroup;MainAccount;Project



6. Expand the **Records to include** section.

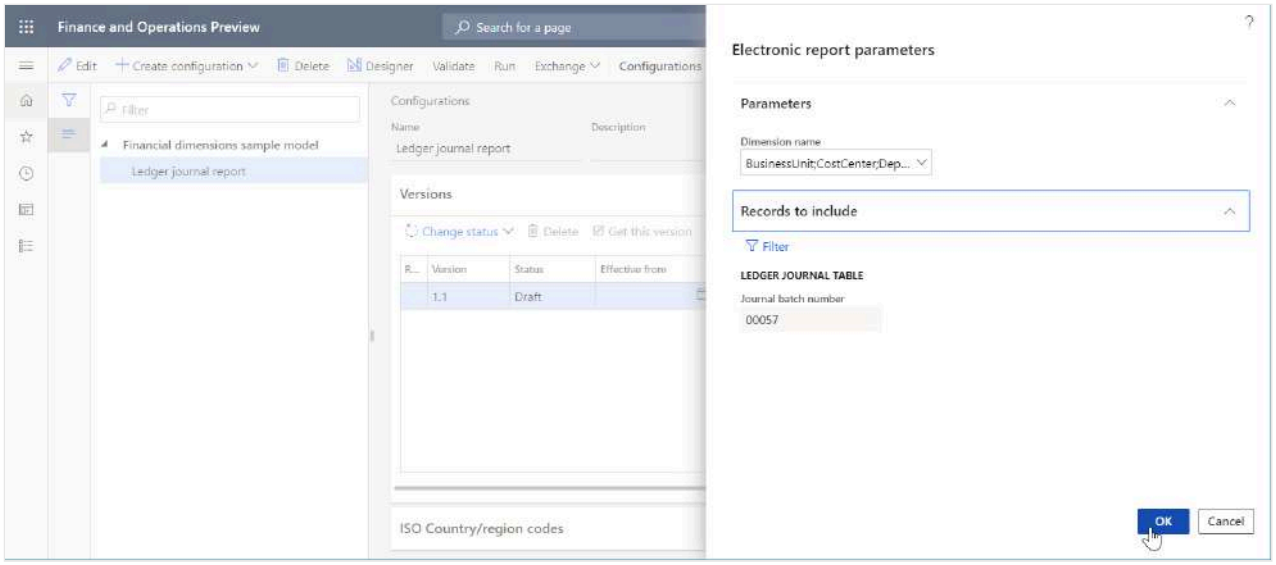
7. Select **Filter**.

8. Select the row for the **Ledger journal** table and the **Journal batch number** field.

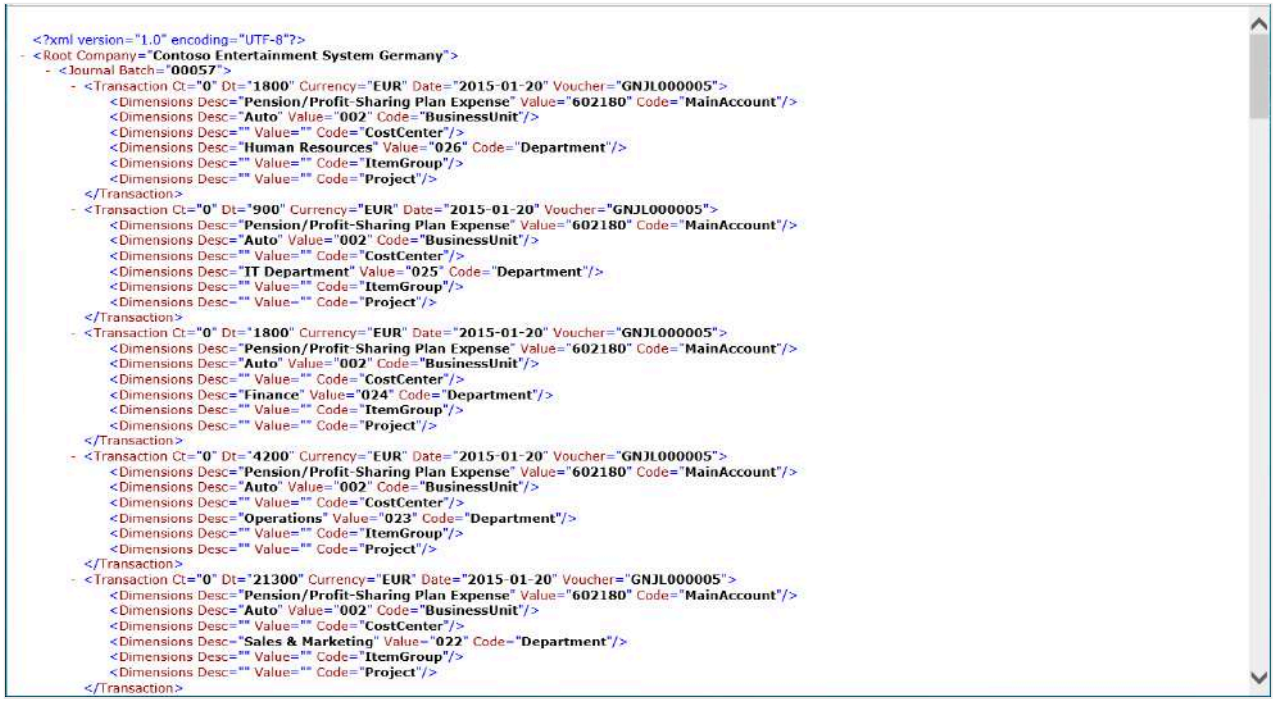
9. In the **Criteria** field, type `00057`.

10. Select **OK**.

11. Select **OK**.



- Review the generated output. For each transaction of the selected batch, the financial dimensions from the corresponding dimensions set are presented. Run this report and select different dimensions to see that the report isn't dependent on the number of selected dimensions or the number of dimensions configured for this instance.



```
<?xml version="1.0" encoding="UTF-8"?>
- <Root Company="Contoso Entertainment System Germany">
  - <Journal Batch="00057">
    - <Transaction Ct="0" Dt="1800" Currency="EUR" Date="2015-01-20" Voucher="GNJL000005">
        <Dimensions Desc="Pension/Profit-Sharing Plan Expense" Value="602180" Code="MainAccount"/>
        <Dimensions Desc="Auto" Value="002" Code="BusinessUnit"/>
        <Dimensions Desc="" Value="" Code="CostCenter"/>
        <Dimensions Desc="Human Resources" Value="026" Code="Department"/>
        <Dimensions Desc="" Value="" Code="ItemGroup"/>
        <Dimensions Desc="" Value="" Code="Project"/>
      </Transaction>
    - <Transaction Ct="0" Dt="900" Currency="EUR" Date="2015-01-20" Voucher="GNJL000005">
        <Dimensions Desc="Pension/Profit-Sharing Plan Expense" Value="602180" Code="MainAccount"/>
        <Dimensions Desc="Auto" Value="002" Code="BusinessUnit"/>
        <Dimensions Desc="" Value="" Code="CostCenter"/>
        <Dimensions Desc="IT Department" Value="025" Code="Department"/>
        <Dimensions Desc="" Value="" Code="ItemGroup"/>
        <Dimensions Desc="" Value="" Code="Project"/>
      </Transaction>
    - <Transaction Ct="0" Dt="1800" Currency="EUR" Date="2015-01-20" Voucher="GNJL000005">
        <Dimensions Desc="Pension/Profit-Sharing Plan Expense" Value="602180" Code="MainAccount"/>
        <Dimensions Desc="Auto" Value="002" Code="BusinessUnit"/>
        <Dimensions Desc="" Value="" Code="CostCenter"/>
        <Dimensions Desc="Finance" Value="024" Code="Department"/>
        <Dimensions Desc="" Value="" Code="ItemGroup"/>
        <Dimensions Desc="" Value="" Code="Project"/>
      </Transaction>
    - <Transaction Ct="0" Dt="4200" Currency="EUR" Date="2015-01-20" Voucher="GNJL000005">
        <Dimensions Desc="Pension/Profit-Sharing Plan Expense" Value="602180" Code="MainAccount"/>
        <Dimensions Desc="Auto" Value="002" Code="BusinessUnit"/>
        <Dimensions Desc="" Value="" Code="CostCenter"/>
        <Dimensions Desc="Operations" Value="023" Code="Department"/>
        <Dimensions Desc="" Value="" Code="ItemGroup"/>
        <Dimensions Desc="" Value="" Code="Project"/>
      </Transaction>
    - <Transaction Ct="0" Dt="21300" Currency="EUR" Date="2015-01-20" Voucher="GNJL000005">
        <Dimensions Desc="Pension/Profit-Sharing Plan Expense" Value="602180" Code="MainAccount"/>
        <Dimensions Desc="Auto" Value="002" Code="BusinessUnit"/>
        <Dimensions Desc="" Value="" Code="CostCenter"/>
        <Dimensions Desc="Sales & Marketing" Value="022" Code="Department"/>
        <Dimensions Desc="" Value="" Code="ItemGroup"/>
        <Dimensions Desc="" Value="" Code="Project"/>
      </Transaction>
```

Last updated on 04/13/2026

Group records and aggregate calculations by using GROUPBY data sources

When you configure Electronic reporting (ER) model mappings or formats, you can add required data sources of the **GroupBy** type.

At design time, you configure a **GroupBy** data source to identify the following elements:

- A base data source that contains records grouped at runtime.
- Grouping fields of the base data source that are used for record grouping at runtime.
- Aggregate functions that specify the aggregate calculations that are done for each discovered group at runtime.

At runtime, a configured **GroupBy** data source groups records that have the same values in the grouping fields, and then returns a list of records. Each record represents a single group. For each group, the data source exposes the field values that the initial records were grouped by, the values of the calculated aggregate functions, and the list of records of the base data source that belongs to the group.

Aggregate functions

At runtime, the system performs every aggregate calculation for each group of records. This calculation uses the value of a single field or an expression in the records of a data source that you selected for grouping in the editable data source of the **GroupBy** type. The following aggregate functions are currently supported:

- **Avg** – This function returns the average of the values in a group. You can use it only with numeric fields.
- **Count** – This function returns the number of items that it found in a group.
- **Min** – This function returns the minimum value among the values in a group.
- **Max** – This function returns the maximum value among the values in a group.
- **Sum** – This function returns the sum of all the values in a group. You can use it only with numeric fields.
- **Any** - This function returns any value in a group. You can use it for any type of fields, including Enum. This function is available as of 10.0.46 version of Finance.

Execution location

When you edit a **GroupBy** data source and specify the base data source that contains the records to group, the system automatically detects the most efficient location for executing that **GroupBy** data source. If the base data source is queryable (that is, if it can run at the database level), the application database is also the execution location of the editable **GroupBy** data source. Otherwise, the application server memory is the execution location.

You can manually change the automatically detected execution location by selecting the location that is applicable to the configured data source. If the execution location that is selected isn't applicable, a validation error is thrown at design time.

> **Tip**
>
> Use the database location to group data sources that expose a large number of records.

Memory consumption

By default, if you run a **GroupBy** data source in memory, the application server memory stores records of the base data source that belong to each discovered group as records of a single group. To help reduce memory consumption, you can suppress record storage for **GroupBy** data sources if they're configured to compute only aggregated functions and their group's records aren't used at runtime. To reduce memory consumption in this way, enable the **Reduce memory usage in ER when records grouping is only used to compute aggregations** feature in the **Feature management** workspace.

Alternatives

You can calculate similar aggregations by using different types of data sources or ER built-in functions.

To learn more about this feature, complete the example that follows.

Example: Use a GROUPBY data source for aggregate calculations and record grouping

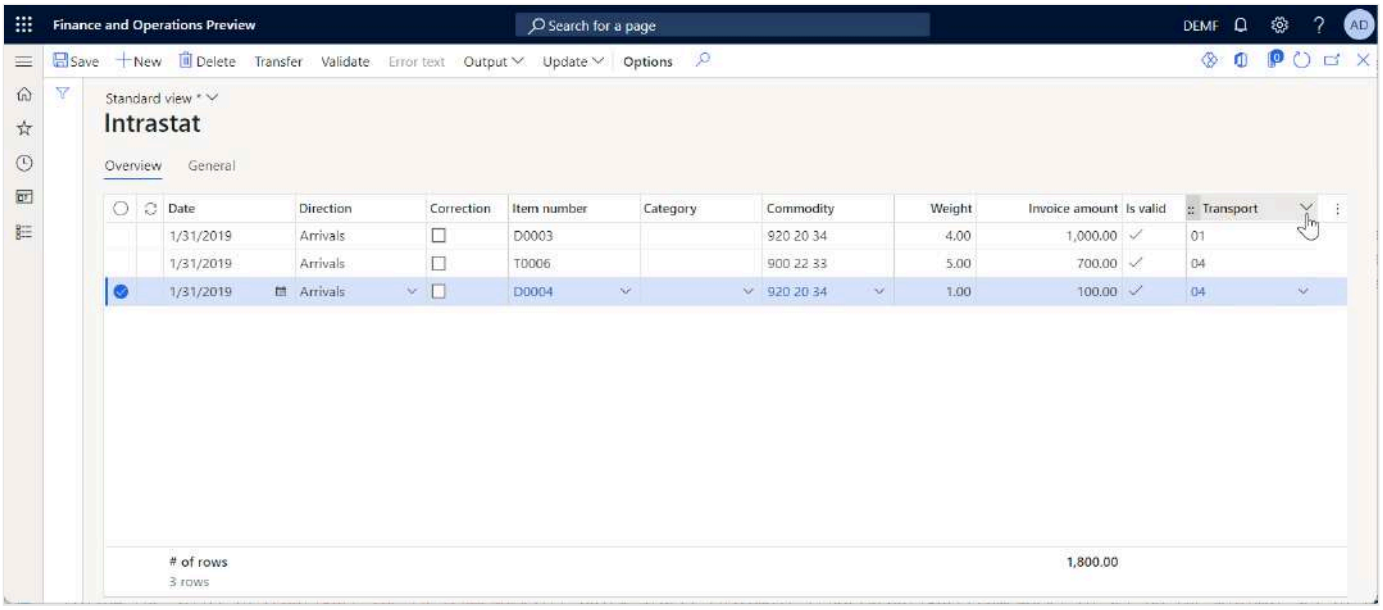
This example shows how a user in the System administrator or Electronic reporting functional consultant role can configure an ER model mapping that has a **GROUPBY** data source that calculates aggregate functions and groups records. This model mapping prints the control report

when the Intrastat declaration is generated. That report lets you review reported Intrastat transactions.

You can complete the procedures in this example in the **DEMF** company in Microsoft Dynamics 365 Finance.

Prepare sample data

Make sure that you have Intrastat transactions for reporting on the **Intrastat** page. You must have transactions for different transport codes, because you group transactions by the **Transport** field in this example.



Configure the ER framework

Follow the steps in Configure the ER framework to set up the minimal set of ER parameters. You must complete this setup before you start to use the ER framework to design an ER model mapping.

Import the standard ER format configuration

Follow the steps in Import the standard ER format configuration to add the standard ER configurations to your current instance of Dynamics 365 Finance. Import version 1 of the **Intrastat model** configuration from the repository.

Create a custom data model configuration

Follow the steps in Add a custom data model configuration to manually add a new **Intrastat model (Litware)** ER data model configuration that you derive from the imported **Intrastat model**

configuration.

Configure a custom data model component

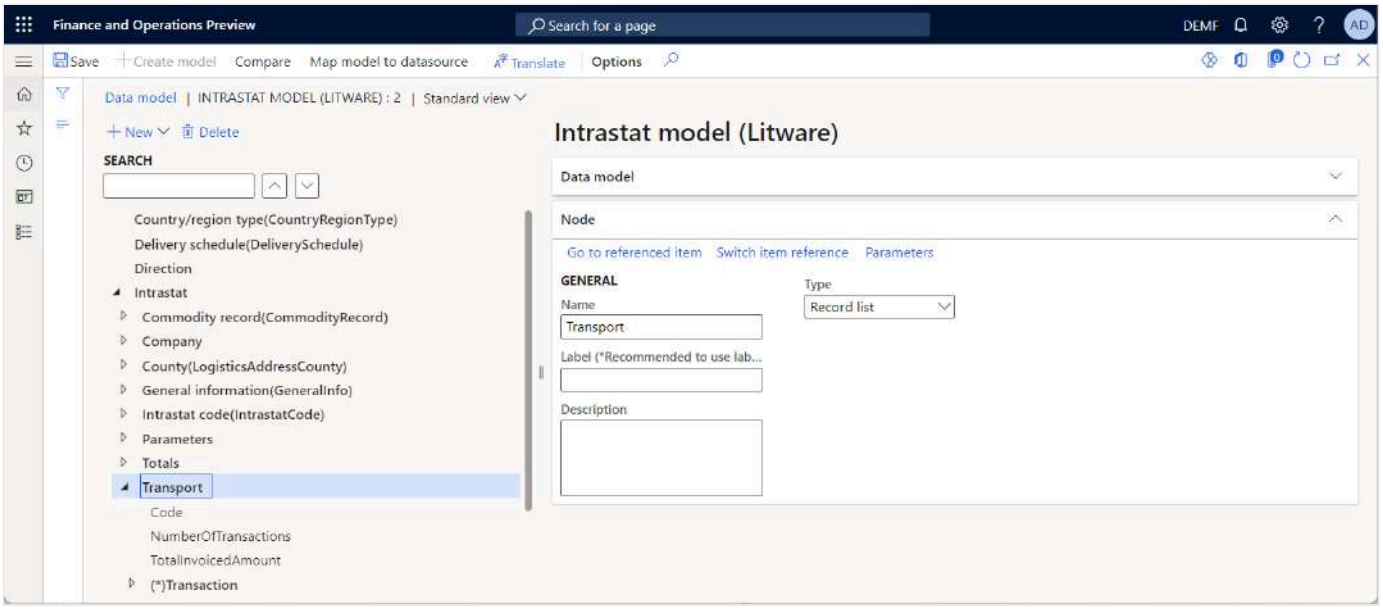
Follow these steps to make the required changes to the derived **Intrastat model (Litware)** data model, so that you can use it to expose transport codes that have the required details.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, select **Intrastat model (Litware)**.
3. Select **Designer**.
4. On **Data model designer**, in the model tree, select **Intrastat**.
5. Select **New** to add a new nested node for the selected **Intrastat** node. In the drop-down dialog box for adding a data model node, follow these steps:
   a. In the **Name** field, enter **Transport**.
   b. In the **Item type** field, select **Record list**.
   c. Select **Add** to add the new node.
6. Select **New** to add a new nested node for the **Transport** node that you just added. In the drop-down dialog box for adding a data model node, follow these steps:
   a. In the **Name** field, enter **Code**.
   b. In the **Item type** field, select **String**.
   c. Select **Add** to add the new node.
7. Select **New** to add another new nested node for the **Transport** node. In the drop-down dialog box for adding a data model node, follow these steps:
   a. In the **Name** field, enter **TotalInvoicedAmount**.
   b. In the **Item type** field, select **Real**.
   c. Select **Add** to add the new node.
8. Select **New** to add another new nested node for the **Transport** node. In the drop-down dialog box for adding a data model node, follow these steps:
   a. In the **Name** field, enter **NumberOfTransactions**.
   b. In the **Item type** field, select **Integer**.
   c. Select **Add** to add the new node.
9. Select **New** to add another new nested node for the **Transport** node. In the drop-down dialog box for adding a data model node, follow these steps:

a. In the **Name** field, enter **Transaction**.
b. In the **Item type** field, select **Record list**.
c. Select **Add** to add the new node.

10. For the **Transaction** node that you just added, on the **Node** FastTab, select **Switch item reference**.

11. In the **Switch item reference** dialog box, in the data model tree, select **CommodityRecord**. Then select **OK**.



Complete the design of a custom data model

Follow the steps in Complete the design of the data model to complete the design of the derived **Intrastat model (Litware)** data model.

Create a new model mapping configuration

Follow the steps in Create a new model mapping configuration to manually add a new **Intrastat sample mapping** ER model mapping configuration for the derived **Intrastat model (Litware)** configuration.

Add a new model mapping component

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, expand the **Intrastat model** configuration.
3. Select the **Intrastat sample mapping** configuration.
4. Select **Designer** to open the list of mappings.
5. Select **Delete** to remove the existing mapping component.

6. Select **New** to add a new mapping component.
7. In the **Definition** field, select **Intrastat**.
8. In the **Name** field, enter **Intrastat mapping**.
9. Select **Designer** to configure the new mapping.

Design the added model mapping component

Add a data source to access an application table

Configure a data source to access the application tables that contain the details of Intrastat transactions.

1. On the **Model mapping designer** page, in the **Data source types** pane, select **Dynamics 365 for Operations\Table records**.
2. In the **Data sources** pane, select **Add root** to add a new data source that accesses the **Intrastat** table. Each record in the **Intrastat** table represents a single Intrastat transaction.
3. In the **Data source properties** dialog box, in the **Name** field, enter **Transaction**.
4. In the **Table** field, enter **Intrastat**.
5. Select **OK** to add the new data source.

Add a data source to group Intrastat transactions

Configure a **GroupBy** data source to group Intrastat transactions and compute aggregate functions.

1. On the **Model mapping designer** page, in the **Data source types** pane, select **Functions\Group by**.
2. In the **Data sources** pane, select **Add root** to add a new data source that groups Intrastat transactions and computes aggregate functions.
3. In the **Data source properties** dialog box, in the **Name** field, enter **TransportRecord**.
4. Select **Edit group by** to configure grouping conditions.
5. On the **Edit 'Group By' parameters** page, in the data sources list in the right pane, select the **Transaction** data source, and expand it.
6. Select **Add field to** > **What to group** to indicate that the **Transaction** data source is selected as the base data source for the configured **GroupBy** data source. The records of the

Transaction data source are grouped, and the field values of this data source are used for calculations in aggregate functions.

7. Select the **Transaction\Transport** field, and then select **Add field to** > **Grouped field** to indicate that the **Transport** field of the base data source is selected as the grouping criterion for the configured **GroupBy** data source. In other words, the records of the **Transaction** data source are grouped based on the value of the **Transport** field. Every record of the configured **GroupBy** data source represents a single transport code found in records of the base data source.

8. Select the **Transaction\AmountMST** field, and then follow these steps:
   a. Select **Add field to** > **Aggregate fields** to indicate that an aggregate function is calculated for this field.
   b. In the **Aggregations** pane, in the record that you added for the selected **Transaction\AmountMST** field, in the **Method** field, select the **Sum** function.
   c. In the **Name** optional field, enter **TotalInvoicedAmount**.

   These settings specify that, for every transport group, the total amount of the **Transaction\AmountMST** field is calculated.

9. Select the **Transaction\RecId** field, and then follow these steps:
   a. Select **Add field to** > **Aggregate fields** to indicate that an aggregate function is calculated for this field.
   b. In the **Aggregations** pane, in the record that you added for the selected **Transaction\RecId** field, in the **Method** field, select the **Count** function.
   c. In the **Name** optional field, enter **NumberOfTransactions**.

   These settings specify that, for every transport group, the number of transactions in the group is calculated.

10. Select **Save**.

11. Review the execution parameters of the editable data source. Notice that **Autodetect** is automatically selected in the **Execution location** field, and the **Execution at** field contains the value **SQL**. These settings specify that the selected **Transaction** base data source is currently queryable, and you can run the editable **GroupBy** data source at database level.

12. Open the lookup for the **Execution location** field to review the list of available values. Notice that you can select **Query** or **In memory** to force this **GroupBy** data source to run on the database level or in the application server memory.

13. Select **Save**, and close the **Edit 'Group By' parameters** page.

14. Select **OK** to complete the settings of the **GroupBy** data source.

Bind the GroupBy data source to data model fields

Bind the configured data source to the fields of the data model to specify how the data model is filled in with application data at runtime.

1. On the **Model mapping designer** page, in the **Data model** pane, expand the **Transport** node.

2. In the **Data sources** pane, expand the **TransportRecord** data source.

3. Add a binding to expose the list of discovered transport groups:
   a. In the **Data model** pane, select the **Transport** item.
   b. In the **Data sources** pane, select the **TransportRecord** data source.
   c. Select **Bind**.

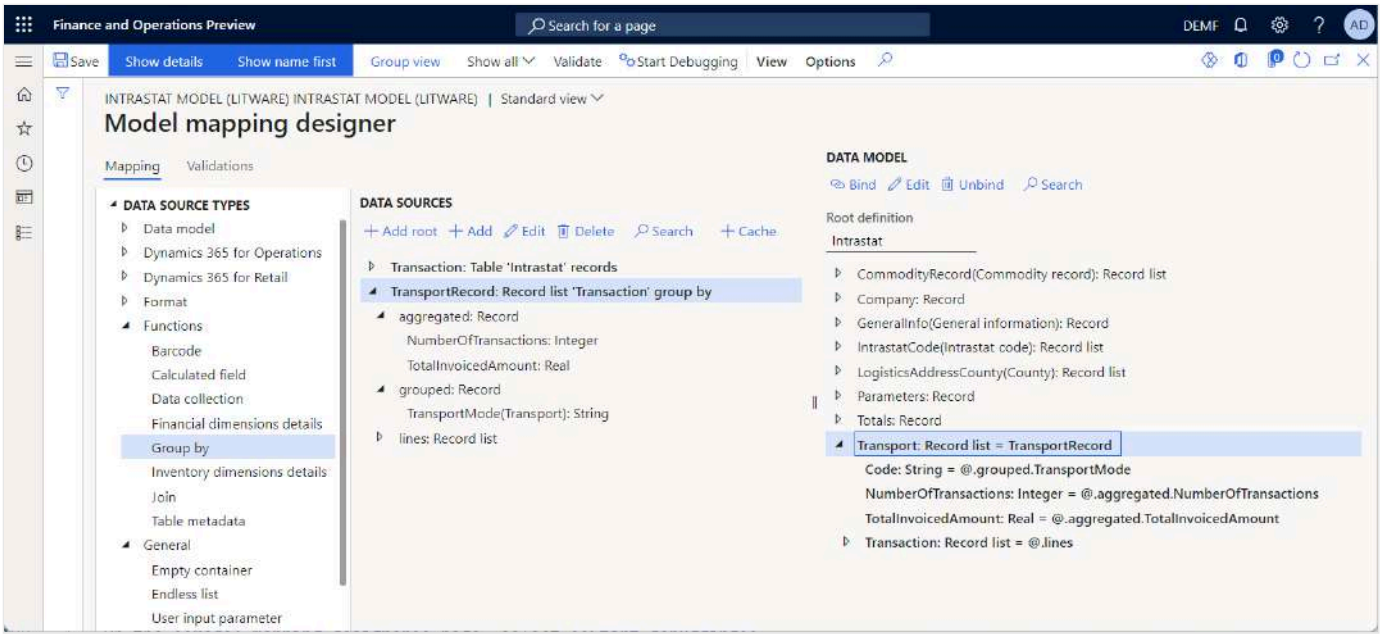
4. Add a binding to expose the transport code of each discovered transport group:
   a. Select the **Transport.Code** data model item.
   b. Select the **TransportRecord.grouped.TransportMode** grouped field.
   c. Select **Bind**.

5. Add a binding to expose the values of calculated aggregate functions for each discovered transport group:
   a. Select the **Transport.NumberOfTransactions** data model item.
   b. Select the **TransportRecord.aggregated.NumberOfTransactions** aggregated field.
   c. Select **Bind**.
   d. Select the **Transport.TotalInvoicedAmount** data model item.
   e. Select the **TransportRecord.aggregated.TotalInvoicedAmount** aggregated field.
   f. Select **Bind**.

6. Add a binding to expose transaction records that belong to each discovered transport group:
   a. Select the **Transport.Transaction** data model item.
   b. Select the **TransportRecord.lines** field.
   c. Select **Bind**.

   You can continue to configure bindings for the nested items of the **Transport.Transaction** data model item and the **TransportRecord.lines** data source field to expose, at runtime,

details of the Intrastat transactions that belong to each discovered transport group.



Debug the added model mapping component

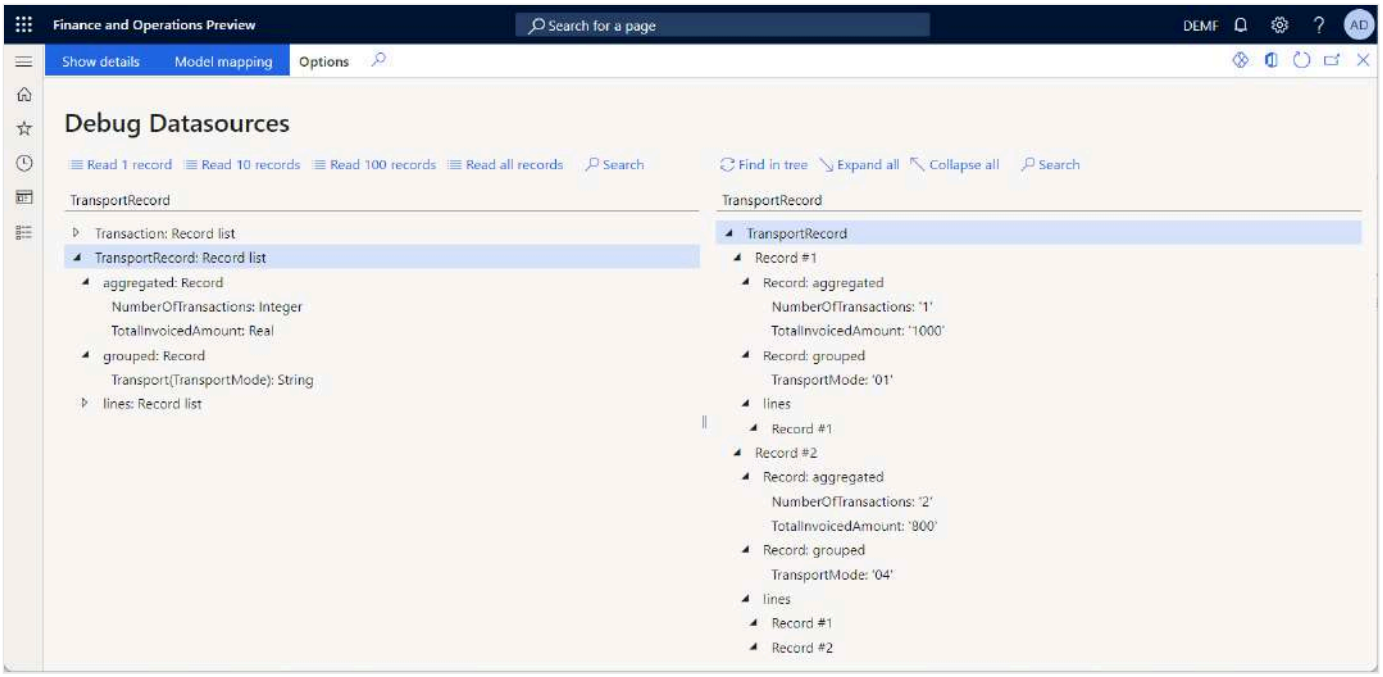
Use the ER data source debugger to test the configured model mapping.

1. On the **Model mapping designer** page, select **Start debugging**.
2. On the **Debug datasources** page, in the left pane, select the **TransportRecord** data source, and then select **Read all records**.
3. Expand the **TransportRecord** data source, and then follow these steps:
   a. Select the **TransportRecord.grouped.TransportMode** data source.
   b. Select **Get value**.
   c. Select the **TransportRecord.grouped.NumberOfTransactions** data source.
   d. Select **Get value**.
   e. Select the **TransportRecord.grouped.TotalInvoicedAmount** data source.
   f. Select **Get value**.
4. In the right pane, select **Expand all**.

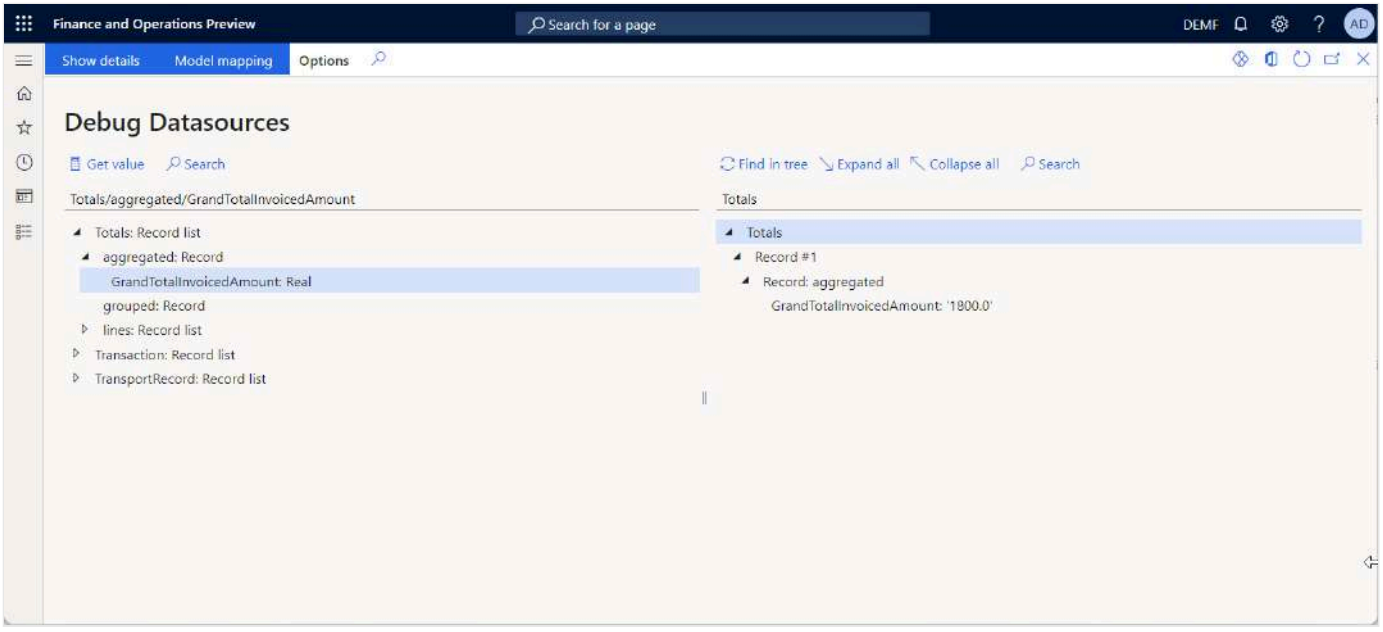
The **TransportRecord** data source exposes two records and presents two transport codes. For each transport code, the number of transactions and the total invoiced amount are calculated.

> **Note**
>
> The "lazy reading" approach is used when a **GroupBy** data source is called to optimize database calls. Therefore, some of the field values in a **GroupBy** data source are calculated in

the ER data source debugger only when they're bound to data model fields.



The following illustration shows the results of the **Totals** data source debugging.



Best practices for working with GroupBy data sources

Calling table extension methods in a GroupBy data source

When you design expressions for an in-memory **GroupBy** data source in an ER model mapping, avoid calling application (X++) table extension methods that require parameters on grouped table records. In the current implementation, the runtime might not resolve these method calls when applied to grouped records. As a result, the expression can return an error or an empty value. This behavior occurs only when all the following conditions are met:

- The table extension method declares one or more parameters, in addition to the table variable (`this`) that it extends.
- You call the method as an instance method on the table record (for example, `record.myExtensionMethod(parameter)`).
- The in-memory **GroupBy** data source wraps the table record.
- You define both the **GroupBy** data source and the underlying table records data source at the model mapping level.

> ⓘ **Note**
>
> This limitation doesn't affect:
>
> - Table extension methods that don't require parameters.
> - Methods called on table records that aren't wrapped by a **GroupBy** data source.

Recommendation

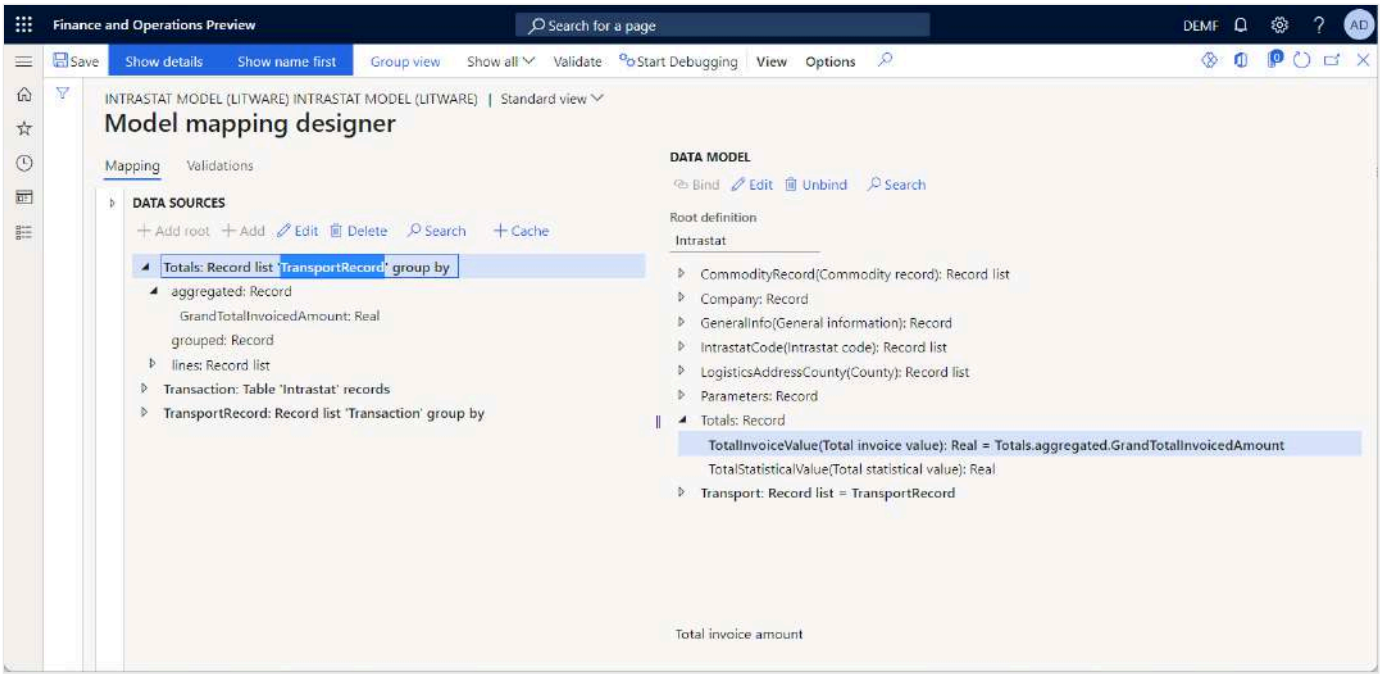
To achieve the expected behavior, consider one of the following approaches:

- Refactor the logic into a parameterless method when possible.
- Evaluate the method on the base data source (before applying **GroupBy**) and expose the result as a calculated field.
- Move the required calculation logic into ER expressions or computed fields that don't depend on parameterized method calls.

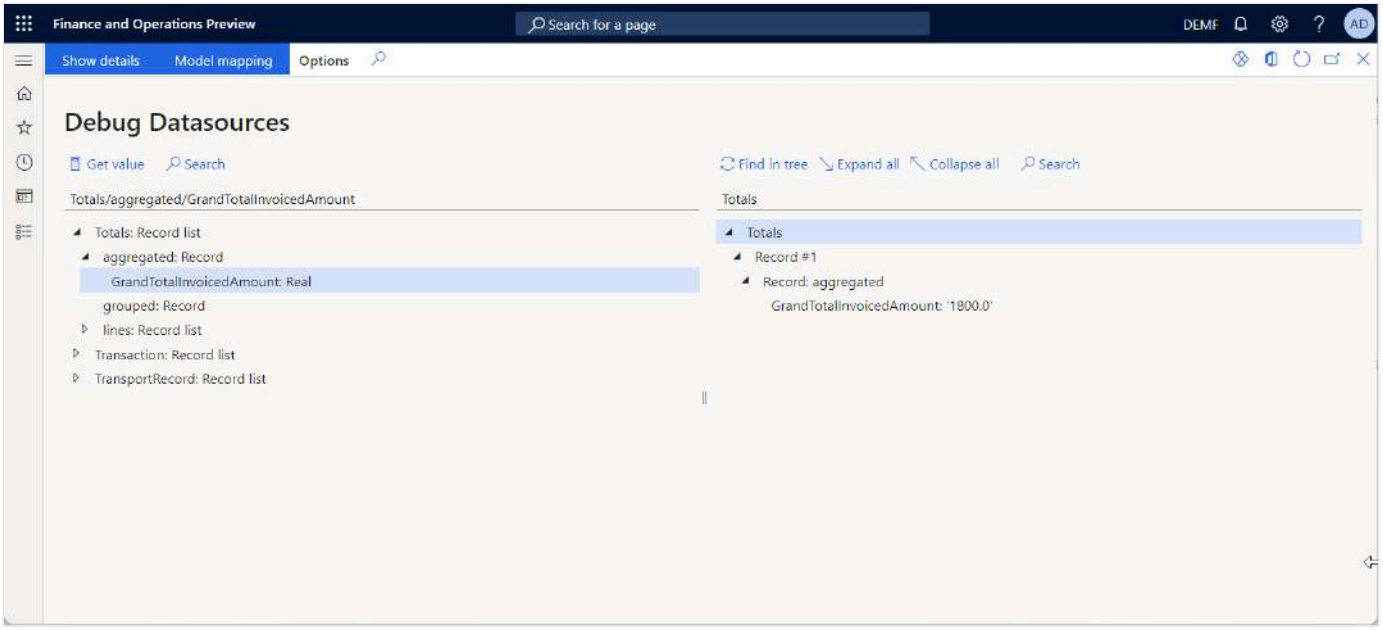
Frequently asked questions

Is there a way to calculate grand totals when the group totals are calculated?

Yes. To calculate grand totals, configure another **GroupBy** data source where the previously configured **GroupBy** data source is the base data source. The following illustration shows the **Totals** data source of the **GroupBy** type that calculates the aggregate **SUM** function, based on the **SUM** aggregation of the **TransportRecord** data source of the **GroupBy** type.



The following illustration shows the results of the **Totals** data source debugging.



Additional resources

- Electronic Reporting overview
- Debug data sources of an executed ER format to analyze data flow and transformation
- Use DATA COLLECTION data sources in Electronic reporting formats

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Last updated on 07/17/2026

ER Configure format to do counting and summing (Part 1 - Create format)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to do counting and summing based on data of the already generated text output. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the "Create a configuration provider and mark it as active" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Get access to the list of configurations provided by Microsoft

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
    - Make sure that the **Litware, Inc.** provider is available and marked as active.
2. Select the **Litware, Inc.** provider.
3. Click **Repositories**.
    - If a repository of the **Operations resources** type already exists, skip the remaining steps of the current sub-task.
4. Click **Add** to open the drop dialog.
5. In the **Configuration repository type** field, enter **Operations resources**.
6. Click **Create repository**.
7. Click **OK**.

Get the Intrastat configurations provided by Microsoft

1. Select **Open**.
2. In the tree, select **Intrastat model\Intrastat (DE)**.
3. Select **Import**.
    - Select **Import** for version 1.1 of the selected configuration.
4. Select **Yes**.
5. Close the page.
6. Close the page.
7. Select **Reporting configurations**.
8. In the tree, expand **Intrastat model**.
9. In the tree, select **Intrastat model\Intrastat (DE)**.

Last updated on 04/13/2026

ER Configure format to do counting and summing (Part 2 - Configure computations)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to do counting and summing based on data of the already generated text output. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the "ER Configure format to do counting and summing (Part 1: Create format)" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Create a format configuration to add counting and summing details

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the tree, expand **Intrastat model**.
4. In the tree, select **Intrastat model\Intrastat (DE)**.
    - To customize the format that Microsoft provides, add lines with summary details at the end of the Intrastat report. Derive your own instance of the Intrastat configuration from the Microsoft instance to make modifications.
5. Select **Create configuration** to open the dialog.
6. In the **New** field, enter `Intrastat (DE), Microsoft`.
7. In the **Name** field, type `Intrastat (DE) with counting & summing`.

8. Select **Create configuration**.

Configure this report to count and sum based on output details

1. Select **Designer**.
2. Select **Yes** in the **Collect output details** field.

    - This flag activates at run-time the process of collecting output details for generating the Intrastat file.
    - You need to count for different Intrastat directions, so add a dedicated model enumeration to the data sources' list of this format configuration.

3. Select the **Mapping** tab.
4. Select **Add root** to open the drop dialog.
5. In the tree, select `Data model\Enumeration`.
6. In the **Name** field, type `Direction`.
7. In the **Model enumeration** field, enter or select a value.

    - Select the value `Direction`.

8. Select **OK**.
9. Select **Add root** to open the drop dialog.
10. In the tree, select `Functions\Calculated field`.
11. In the **Name** field, type `$BlockName`.
12. Select **Edit formula**.
13. In the **Formula** field, enter `"block"`.
14. Select **Save**.
15. Close the page.
16. Select **OK**.
17. Select **Add root** to open the drop dialog.
18. In the tree, select `Functions\Calculated field`.
19. In the **Name** field, type `$RecName`.
20. Select **Edit formula**.
21. In the **Formula** field, enter `"record"`.
22. Select **Save**.
23. Close the page.
24. Select **OK**.
25. Select **Add root** to open the drop dialog.

26. In the tree, select `Functions\Calculated field`.
27. In the **Name** field, type `$InvName`.
28. Select **Edit formula**.
29. In the **Formula** field, enter `"InvoicedAmountEUR"`.
30. Select **Save**.
31. Close the page.
32. Select **OK**.
33. In the tree, select `Intrastat\Data`.
34. Select **Edit** for the **Collected data key name** field.
35. Select **Add data source**.

    - Use `$BlockName`.

36. Select **Save**.
37. Close the page.
38. Select **Edit** for the **Collected data key value** field.
39. In the **Formula** field, enter `IF(Intrastat.CommodityRecord.Direction=Direction.Import, "Import", "Export")`.

    - Use `IF(Intrastat.CommodityRecord.Direction=Direction.Import, "Import", "Export")`.

40. Select **Save**.
41. Close the page.

    - Count the lines of this sequence. The results are used with the name `block` separately for different directions. Value `Import` is used for any arrivals Intrastat transactions. The value `Export` is used for any Intrastat dispatches transactions. Consider this to be a virtual Excel spreadsheet. For each transaction a row where the first column `block` is filled with the values `Import` and `Export` accordingly.

42. In the tree, expand `Intrastat\Data: Sequence`.
43. In the tree, select `Intrastat\Data: Sequence\Arrivals?`.
44. Select **Edit** for the **Collected data key name** field.

    - Count the lines of this sequence. The results are stored using the name `record`.

45. In the tree, select `$RecName`.
46. Select **Add data source**.
47. Select **Save**.

48. Close the page.
49. Select **Edit** for the **Collected data key value** field.
50. In the **Formula** field, enter `Intrastat.CommodityRecord.CommodityCode`.
51. Select **Save**.
52. Close the page.

    - Count the lines in this sequence. Use the results with the name "record" for different commodity codes. Consider this count as a virtual Excel spreadsheet. For each transaction, create a row where the first column "block" contains the values "Import" and "Export" accordingly, and the second column "record" contains the commodity code value.

53. In the tree, select `Intrastat\Data: Sequence\Dispatches?`.
54. Select **Edit** for the **Collected data key name** field.
55. In the tree, select `$RecName`.
56. Select **Add data source**.
57. Select **Save**.
58. Close the page.
59. Select **Edit** for the **Collected data key value** field.
60. In the **Formula** field, enter `Intrastat.CommodityRecord.CommodityCode`.
61. Select **Save**.
62. Close the page.
63. In the tree, expand `Intrastat\Data: Sequence\Dispatches: Sequence?`.
64. In the tree, expand `Intrastat\Data: Sequence\Dispatches: Sequence?\Record = Intrastat.CommodityRecord`.
65. Select the **Format** tab.
66. In the tree, select `Intrastat\Data\Dispatches\Record\Invoice amount EUR`.
67. Select the **Mapping** tab.
68. Select **Edit** for the **Collected data key name** field.
69. In the tree, select `$InvName`.
70. Select **Add data source**.
71. Select **Save**.
72. Close the page.

    - Summarize the invoiced amount values for lines in this sequence. Use the results with the name "InvoicedAmountEUR" for different Intrastat directions and commodity codes. Consider this summary as a virtual creation in an Excel spreadsheet. For each transaction, create a row where the first column "block"

contains the values "Import" and "Export" accordingly. The second column "record" contains the commodity code value, and the third column "InvoicedAmountEUR" contains the invoice amount value.

73. In the tree, expand `Intrastat\Data\Arrivals?`.
74. In the tree, expand `Intrastat\Data\Arrivals?\Record = Intrastat.CommodityRecord`.
75. Select the **Format** tab.
76. In the tree, select `Intrastat\Data\Arrivals\Record\Invoice amount EUR`.
77. Select the **Mapping** tab.
78. Select **Edit** for the **Collected data key name** field.
79. In the tree, select `$InvName`.
80. Select **Add data source**.
81. Select **Save**.
82. Close the page.
83. Select **Save**.
84. Close the page.

Last updated on 04/13/2026

ER Configure format to do counting and summing (Part 3 - Use computations to make the output)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to do counting and summing based on data of the already generated text output. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the "ER Configure format to do counting and summing (Part 2: Configure computations)" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Configure this report to use counting and summing info

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the tree, expand **Intrastat model**.
4. In the tree, expand **Intrastat model\Intrastat (DE)**.
5. In the tree, select **Intrastat model\Intrastat (DE)\Intrastat (DE) with counting & summing**.
6. Select **Designer**.
7. Select the **Mapping** tab.
8. Select **Add root** to open the drop dialog.

    - Add a new data source to get the list of memorized blocks.

9. In the tree, select **Functions\Calculated field**.
10. In the **Name** field, type `$BlocksList`.

    - `$BlocksList`

11. Select **Edit formula**.
12. In the tree, select **Data collection functions\COLLECTEDLIST**.
13. Select **Add function**.
14. Select **Add data source**.
15. In the **Formula** field, enter `COLLECTEDLIST('$BlockName', '`.

    - `COLLECTEDLIST('$BlockName',`

16. In the **Formula** field, enter `COLLECTEDLIST('$BlockName', "*")`.

    - `COLLECTEDLIST('$BlockName', "*")`

17. Select **Save**.

    - The pattern `*` means that all blocks are included in the list for this record.

18. Close the page.
19. Click OK.
20. Select the **Format** tab.
21. In the tree, select 'Intrastat\Data'.
22. Click Add to open the drop dialog.
23. In the tree, select 'Text\Sequence'.
24. In the Name field, type 'Totals by blocks'.

    - Totals by blocks

25. In the Special characters field, select 'New line - Windows (CR LF)'.
26. Click OK.
27. In the tree, select 'Intrastat\Data\Totals by blocks'.
28. Click Add to open the drop dialog.
29. In the tree, select 'Text\String'.
30. In the Name field, type 'Block code'.

    - Block code

31. Click OK.
32. Click Add String.

33. In the Name field, type 'Lines counting'.

    - Lines counting

34. Click OK.
35. Click Add String.
36. In the Name field, type 'Total amount'.

    - Total amount

37. Click OK.
38. Select the **Mapping** tab.
39. In the tree, select '$BlocksList'.
40. Click Bind.

    - Create a summary line for each memorized block.

41. Select the **Format** tab.
42. In the tree, select 'Intrastat\Data\Totals by blocks\Block code'.
43. Select the **Mapping** tab.
44. Select **Edit formula**.
45. In the Formula field, enter '"Block id: " & '.

    - "Block id: " &

46. In the tree, expand `$BlocksList`.
47. In the tree, select `$BlocksList\Value`.
48. Select **Add data source**.
49. Select **Save**.
50. Close the page.
51. Select the **Format** tab.
52. In the tree, select `Intrastat\Data\Totals by blocks\Lines counting`.
53. Select the **Mapping** tab.
54. Select **Edit formula**.

    - Create output for the number of lines for each block presented in this report.

55. In the **Formula** field, enter `"Number of lines in this block: " &` .

    - `"Number of lines in this block: " &`

56. In the **Formula** field, enter `"Number of lines in this block: " & TEXT(` .

- `"Number of lines in this block: " & TEXT(`

57. In the tree, select `Data collection functions\COUNTIFS`.
58. Select **Add function**.
59. Select **Add data source**.
60. In the **Formula** field, enter `"Number of lines in this block: " & TEXT(COUNTIFS('$BlockName', `.

    - `"Number of lines in this block: " & TEXT(COUNTIFS('$BlockName',`

61. In the tree, expand `$BlocksList`.
62. In the tree, select `$BlocksList\Value`.
63. Select **Add data source**.
64. In the **Formula** field, enter `"Number of lines in this block: " & TEXT(COUNTIFS('$BlockName', '$BlocksList'.Value, `.

    - `"Number of lines in this block: " & TEXT(COUNTIFS('$BlockName', '$BlocksList'.Value,`

65. In the tree, select `$RecName`.
66. Select **Add data source**.
67. In the **Formula** field, enter `"Number of lines in this block: " & TEXT(COUNTIFS('$BlockName', '$BlocksList'.Value, '$RecName', "*"))`.

    - `"Number of lines in this block: " & TEXT(COUNTIFS('$BlockName', '$BlocksList'.Value, '$RecName', "*"))`

68. Select **Save**.
69. Close the page.
70. Select the **Format** tab.
71. In the tree, select `Intrastat\Data\Totals by blocks\Total amount`.
72. Select the **Mapping** tab.
73. Select **Edit formula**.

    - Create output that is the total of the invoiced amount for each block presented in this report.

74. In the **Formula** field, enter `"Sum of invoiced amount: " & TEXT(SUMIFS('$InvName', '$BlockName', '$BlocksList'.Value, '$RecName', "*"))`.

    - `"Sum of invoiced amount: " & TEXT(SUMIFS('$InvName', '$BlockName', '$BlocksList'.Value, '$RecName', "*"))`

75. Select **Save**.
76. Close the page.
77. Select **Save**.
78. Close the page.

Last updated on 04/13/2026

ER Configure format to do counting and summing (Part 4 - Run format)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to do counting and summing based on data from the already generated text output. You can perform these steps in the DEMF company.

To complete these steps, you must first complete the steps in the "ER Configure format to do counting and summing (Part 3: Use computations to make the output)" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Test this configuration for generation of the Intrastat reports

1. Go to Organization administration > Workspaces > Electronic reporting.
2. Click Reporting configurations.
3. In the tree, expand **Intrastat model**.
4. In the tree, expand **Intrastat model\Intrastat (DE)**.
5. In the tree, select **Intrastat model\Intrastat (DE)\Intrastat (DE) with counting & summing**.
6. On the Action Pane, select **Configurations**.
7. Select **User parameters**.
8. Select Yes in the Run settings field.
9. Select **OK**.
10. Click Edit.
11. Select Yes in the Run Draft field.

12. Click Save.
13. Go to Tax > Setup > Foreign trade > Foreign trade parameters.
14. Expand the Electronic reporting section.
15. Select the "Intrastat (DE) with counting & summing" configuration.
16. Select the "Intrastat (DE) with counting & summing" configuration.
17. Click Save.
18. Close the page.
19. Go to **Tax** > **Declarations** > **Foreign trade** > **Intrastat**.
20. Select **Output**.
21. Select **Report**.

    - Run the Intrastat report generation process.

22. In the From date field, set the date to '2000-01-01'.

    - Define start and end dates for the reporting period that include the existing on the form transactions.

23. In the To date field, set the date to '2022-12-31'.

    - Define start and end dates for the reporting period that include the existing on the form transactions.

24. In the Direction field, select 'Arrivals'.
25. Select Yes in the Generate file field.
26. Select **OK**.

    - Review the created output with the summary lines in the end.

27. Click New.
28. In the list, mark the selected row.
29. In the Direction field, select 'Dispatches'.
30. In the Item number field, enter or select a value.
31. In the Commodity field, enter or select a value.
32. Set Weight to '10'.
33. Set Invoice amount to '10000'.
34. Set Statistical amount to '10000'.
35. Select **Output**.
36. Select **Report**.
37. In the Direction field, select 'Dispatches'.

38. Select **OK**.

    - Review the created output with the summary lines in the end. Note that it has been changed in comparison to the first run.

Run this configuration in debug mode to review the collected counting and summing data

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Intrastat model**.
3. In the tree, expand **Intrastat model\Intrastat (DE)**.
4. In the tree, select **Intrastat model\Intrastat (DE)\Intrastat (DE) with counting & summing**.
5. On the Action Pane, select **Configurations**.
6. Select **User parameters**.
7. Select **Yes** in the **Run in debug mode** field.
8. Select **OK**.
9. Close the page.
10. Go to **Tax** > **Declarations** > **Foreign trade** > **Intrastat**.
11. Select **Output**.
12. Select **Report**.
13. Select **OK**.
14. Close the page.
15. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
16. In the tree, expand **Intrastat model**.
17. In the tree, expand **Intrastat model\Intrastat (DE)**.
18. In the tree, select **Intrastat model\Intrastat (DE)\Intrastat (DE) with counting & summing**.
19. Select **Debug logs**.

    - A debug log record is created for the execution process of the selected configuration.

20. Select **Attach**.
21. Select **Open**.

    - Review the created XML file that contains counting and summing details that the system collects during the execution of the selected configuration.

Last updated on 04/13/2026

ER use Document Management files in format outputs (Part 1 - Prepare data model)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to use Document Management files (attachments) in ER output. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the "Create a configuration provider and mark it as active" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Get access to the list of configurations provided by Microsoft

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

   Make sure that the **Litware, Inc.** provider is available and marked as active.

2. Select the **Litware, Inc.** provider.

3. Select **Repositories**.

   If a repository of the **Operations resources** type already exists, skip the remaining steps of the current sub-task.

4. Select **Add** to open the drop dialog.

5. In the **Configuration repository type** field, enter **Operations resources**.

6. Select **Create repository**.
7. Select **OK**.

Get the customer invoice model configurations provided by Microsoft

1. Select **Show filters**.
2. Apply the following filters: Enter a filter value of **Operations resources** on the **Name** field by using the **begins with** filter operator. Enter a filter value of "" on the **Description** field by using the **begins with** filter operator.
3. Select **Show filters**.
4. Select **Open**.
5. In the tree, select **Customer invoice model**.

   Select the model configuration **Customer invoice model** to import it.
6. Select **Import**.

   Select **Import** for version 1 of the selected configuration.
7. Select **Yes**.
8. Close the page.
9. Close the page.
10. Select **Reporting configurations**.
11. In the tree, select **Customer invoice model**.

Create the derived model to support access to the Document Management files

Create your own configuration of the Customer invoice model by deriving it from the configuration provided by Microsoft. Use this configuration to implement access to the Document Management files and make them available for electronic documents that you create based on this model.

1. Select **Create configuration** to open the drop dialog.
2. In the **New** field, enter `Derive from Name: Customer invoice model, Microsoft`.
3. In the **Name** field, type `Customer invoice model (custom)`.
4. Select **Create configuration**.

Last updated on 04/13/2026

ER Use Document Management files in format outputs (Part 2 - Extend data model)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the System Administrator or Electronic Reporting Developer role can configure an Electronic reporting (ER) format to use Document Management files (attachments) in ER output. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the "ER Use Document Management files in format outputs (Part 1: Prepare data model)" task guide.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Extend data model to present the Document Management files in it

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the tree, expand **Customer invoice model**.
4. In the tree, select **Customer invoice model\Customer invoice model (custom)**.
5. Select **Designer**.
6. In the tree, select **Customer invoice(InvoiceCustomer)**.
    - Extend this data model to expose any files that you attached to a sales order that is related to an electronically processing invoice.
7. Select **New** to open the drop dialog.
8. In the **Name** field, type **Invoice attachments**.

- Invoice attachments

9. In the **Item type** field, select **Record list**.
10. Select **Add**.
11. Select **New** to open the drop dialog.
12. In the **Name** field, type **File content**.

    - File content

13. In the **Item type** field, select **Container**.
14. Select **Add**.
15. Select **New** to open the drop dialog.
16. In the **Name** field, type **File name**.

    - File name

17. In the **Item type** field, select **String**.
18. Select **Add**.

Map new data model elements to data sources

1. Select **Map model to datasource**.
2. Use the Quick Filter to filter on the Definition field with a value of `InvoiceCustomer`.

    - InvoiceCustomer
    - Map new model elements to appropriate data sources.

3. Select **Designer**.
4. In the tree, select **Invoice attachments**.
5. In the tree, expand **Invoice attachments**.
6. In the tree, select **Invoice attachments\File name**.
7. Select **Edit**.
8. In the Formula field, enter `CustInvoiceJour.'>Relations'.SalesTable.'<Relations'.'<Documents'.'originalFileName()'`.

    - `CustInvoiceJour.'>Relations'.SalesTable.'<Relations'.'<Documents'.'originalFileName()'`

9. Select **Save**.
10. Close the page.
11. In the tree, select **Invoice attachments\File content**.

12. Select **Edit**.
13. In the Formula field, enter `CustInvoiceJour.'>Relations'.SalesTable.'<Relations'.'<Documents'.'getFileContentAsContainer()'`.

    - `CustInvoiceJour.'>Relations'.SalesTable.'<Relations'.'<Documents'.'getFileContentAsContainer()'`

14. Select **Save**.
15. Close the page.
16. In the tree, select **Invoice attachments**.
17. Select **Edit**.
18. In the Formula field, enter `CustInvoiceJour.'>Relations'.SalesTable.'<Relations'.'<Documents'`.

    - `CustInvoiceJour.'>Relations'.SalesTable.'<Relations'.'<Documents'`

19. Select **Save**.
20. Close the page.
21. Select **Save**.
22. Close the page.
23. Close the page.
24. Close the page.
25. Select **Change status**.
26. Select **Complete**.
27. Select **OK**.

Last updated on 04/13/2026

ER use Document Management files in format outputs (Part 3 - Create format)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to use Document Management files (attachments) in ER output. You can perform these steps in any company.

To complete these steps, you must first complete the steps in the "ER Use Document Management files in format outputs (Part 2: Extend data model" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Create a format to process invoices

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the tree, expand **Customer invoice model**.
4. In the tree, select **Customer invoice model\Customer invoice model (custom)**.

    - You create a format to generate electronic messages with information about any files that you attach to a sales order that relates to an electronically processing invoice.

5. Select **Create configuration** to open the drop dialog.
6. In the **New** field, enter *Format based on data model Customer invoice model (custom)*.
7. In the **Name** field, type *Electronic invoice sample message*.

    - Electronic invoice sample message

8. In the **Data model definition** field, enter or select a value.

- InvoiceCustomer

9. Select **Create configuration**.

Design a format to populate attachments into generating a message in MIME format

1. Click **Designer**.
2. Click **Add root** to open the drop dialog.
3. In the tree, select **XML\Element**.
4. In the **Name** field, type **Invoice**.

    - Invoice

5. Click **OK**.
6. Click **Add** to open the drop dialog.
7. In the tree, select **XML\Attribute**.
8. In the **Name** field, type **SalesOrder**.

    - SalesOrder

9. Click **OK**.
10. Click **Add Attribute**.
11. In the **Name** field, type **InvoiceNumber**.

    - InvoiceNumber

12. Click **OK**.
13. Click **Add Attribute**.
14. In the **Name** field, type **InvoiceAmount**.

    - InvoiceAmount

15. Click **OK**.
16. Click **Add** to open the drop dialog.
17. In the tree, select **XML\Element**.
18. In the **Name** field, type **EnclosedDocs**.

    - EnclosedDocs

19. Click **OK**.
20. In the tree, select **Invoice\EnclosedDocs**.

21. Click **Add Element**.
22. In the **Name** field, type **Document**.

    - Document

23. Click **OK**.
24. In the tree, select **Invoice\EnclosedDocs\Document**.
25. Click **Add** to open the drop dialog.
26. In the tree, select **XML\Attribute**.
27. In the **Name** field, type **FileName**.

    - FileName

28. Click **OK**.
29. Click **Add** to open the drop dialog.
30. In the tree, select **XML\Element**.
31. In the **Name** field, type **FileContent**.

    - FileContent

32. Click **OK**.
33. In the tree, select **Invoice\EnclosedDocs\Document\FileContent**.
34. Click **Add** to open the drop dialog.
35. In the tree, select **Text\Base64**.
36. Click **OK**.

Map format elements to data model as data source

1. In the tree, select `Invoice\SalesOrder`.
2. Select the **Mapping** tab.
3. In the tree, expand `model`.
4. In the tree, select `model\Sales order number(SalesId)`.
5. Select **Bind**.
6. In the tree, select `Invoice\InvoiceNumber`.
7. In the tree, expand `model\Base invoice(InvoiceBase)`.
8. In the tree, select `model\Base invoice(InvoiceBase)\Invoice number(Id)`.
9. Select **Bind**.
10. In the tree, select `Invoice\InvoiceAmount`.
11. In the tree, select `model\Base invoice(InvoiceBase)\Invoice amount(Amount)`.
12. Select **Bind**.

13. In the tree, expand `model\Invoice attachments`.
14. In the tree, select `model\Invoice attachments\File content`.
15. In the tree, select `Invoice\EnclosedDocs\Document\FileContent\Base64`.
16. Select **Bind**.
17. In the tree, select `model\Invoice attachments\File name`.
18. In the tree, select `Invoice\EnclosedDocs\Document\FileName`.
19. Select **Bind**.
20. In the tree, select `model\Invoice attachments`.
21. In the tree, select **Invoice\EnclosedDocs\Document**.
22. Select **Bind**.
23. Select **Save**.
24. Close the page.

Last updated on 04/13/2026

ER Use Document Management files in format outputs (Part 4 - Run format)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to use Document Management files (attachments) in ER output. You can perform these steps in the DEMF company.

To complete these steps, you must first complete the steps in the "ER Use Document Management files in format outputs (Part 3: Create format)" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Add necessary attachments for sales order of a single invoice

1. Go to **Accounts receivable** > **Invoices** > **Open customer invoices**.
2. Use the Quick Filter to find records. For example, filter on the **Invoice** field with a value of `CIV-000148`.
    - CIV-000148
3. Select the invoice link.
    - CIV-000148
4. Select the link in the **Sales order** field.
    - 000148
5. In the **Lines or header** field, select the option of **Header**.

- Select **Header** to indicate that this sales order is the target for adding attachments.

6. Select **Attach**.

    - Add a few files as attachments for this sales order. Use the files of the document types that are supported by the Document Management system, such as DOCX, PDF, XML, JPG, and more. Browse and select files to attach and further process with the related invoice in the ER electronic message.

7. Select **New**.
8. Select **File**.
9. Select **New**.
10. Select **File**.
11. Close the page.
12. Close the page.
13. Close the page.
14. Close the page.

Run the designed report for the selected invoice

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Customer invoice model**.
3. In the tree, expand **Customer invoice model\Customer invoice model (custom)**.
4. In the tree, select **Customer invoice model\Customer invoice model (custom)\Electronic invoice sample message**.
5. Select **Run**.
6. Expand the **Records to include** section.
7. Select **Filter**.
8. Select the row of the **Customer invoice journal** and the **Sales order** field.
9. In the **Criteria** field, type `000148`.

    - In the criteria **Sales order** field, type the order number `000148`.

10. Select **OK**.
11. Select **OK**.

    - Review the generated output. For each attachment, the report creates a single XML node. The attachment's content is added to the XML output in MIME (base64) text format.

Last updated on 04/13/2026

ER Use Document Management files in format outputs (Part 5 - Modify and run format)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the system administrator or electronic reporting developer role can configure an Electronic reporting (ER) format to use Document Management files (attachments) in ER output. You can perform these steps in the DEMF company.

To complete these steps, you must first complete the steps in the "ER Use Document Management files in format outputs (Part 4: Run format)" procedure.

This procedure is for a feature that was added in Dynamics 365 for Operations version 1611.

Modify the format to populate attachments into generating messages in binary format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Customer invoice model**.
3. In the tree, expand **Customer invoice model\Customer invoice model (custom)**.
4. In the tree, select **Customer invoice model\Customer invoice model (custom)\Electronic invoice sample message**.
5. Select **Designer**.

    - You populate the invoice message in the generating output as an XML file using UNICODE encoding.

6. Select **Add root** to open the drop dialog.
7. In the tree, select **Common\File**.

8. In the **Name** field, type **Xml message**.
    - Xml message
9. In the **Encoding** field, type **UTF-8**.
    - UTF-8
10. Select **OK**.
    - Configure the generating output as a zipped file.
11. Select **Add root** to open the drop dialog.
12. In the tree, select **Common\Folder**.
13. In the **Name** field, type **Zip output**.
    - Zip output
14. Select **OK**.
15. In the tree, select **Zip output**.
    - Add attachments to the generating zipped file as files with original names and extensions.
16. Select **Add** to open the drop dialog.
17. In the tree, select **Common\File**.
18. In the **Name** field, type **Attached file**.
    - Attached file
19. Select **OK**.
20. In the tree, select **Zip output\Attached file**.
21. Select **Add** to open the drop dialog.
22. In the tree, select **Text\Base64**.
23. Select **OK**.

Map new format elements to data model

1. Select the **Mapping** tab.
2. In the tree, expand `model`.
3. In the tree, expand `model\Invoice attachments`.
4. In the tree, select `Zip output\Attached file\Base64`.

5. In the tree, select `model\Invoice attachments\File content`.
6. Select **Bind**.
7. In the tree, select **Zip output\Attached file**.
8. Select **Edit filename**.
9. In the tree, expand `model`.
10. In the tree, expand `model\Invoice attachments`.
11. In the tree, select `model\Invoice attachments\File name`.
12. Select **Add data source**.
13. Select **Save**.
14. Close the page.
15. In the tree, select `model\Invoice attachments`.
16. Select **Bind**.
17. Select **Save**.
18. Close the page.

Run the designed report for the selected invoice

1. Select **Run**.
2. Expand the **Records to include ()** section.
3. Select **Filter**.
4. Select the row of the **Customer invoice journal** and the **Sales order** field.
5. In the **Criteria** field, enter the order number `000148`.

    - 000148

6. Select **OK**.
7. Select **OK**.

    - Review the generated output. In addition to the invoice message in XML format, the process creates a single file for each attachment. The attachment files are populated with the zipped output in binary format.

Last updated on 04/13/2026

Use Barcode data sources to generate bar code images

10/02/2025

You can use the Electronic reporting (ER) framework to design ER format components that you can run to generate electronic and printable outbound documents that you require. To generate an outbound document in Microsoft Office format, you must specify the layout of the report by using either a Microsoft Excel document or a Microsoft Word document as a report template. The ER Operations designer lets you attach an Excel or Word document as a template for an ER format. The following named elements in the attached template are associated with the elements of the configured format component:

- Content controls in Word
- Named sheets, ranges, cells, shapes, and images in Excel

These named elements are used as placeholders for data that is entered in a generated document when an ER format is run. ER format elements are bound to data sources. These data sources specify the data that will be entered in the generated documents at runtime. For more information, see Embed images and shapes in documents that you generate by using ER.

ER now supports the **Barcode** data source type. Therefore, you can now generate an image that represents the bar code for specified text. When you configure an ER format, you can specify data sources of the **Barcode** type to generate bar code images. You can then add those images to generated business documents, such as orders, invoices, packing slips, and receipts. You can also add them to various kind of labels, such as product and shelf labels, and packaging and shipping labels.

The following placeholders can be used in report templates to enter bar code images:

- Picture content control for Word
- Picture object in Excel

By using a data source of the **Barcode** type, you can generate bar codes in the following formats:

- One-dimensional bar codes:
  - Codabar
  - Code 39
  - Code 93
  - Code 128
  - EAN-8
  - EAN-13

    - ITF-14
    - Intelligent Mail
    - MSI
    - Plessey
    - PDF417
    - UPC-A
    - UPC-E

- Two-dimensional bar codes:
    - Aztec
    - Data Matrix
    - QR Code

When you configure a **Barcode** data source, you can define specific rendering parameters that are used to generate an image:

- **Width** – Specify the bar code's width in pixels. A value of **0** (zero) indicates that the default width is used. The meaning can vary for different formats.
- **Height** – Specify the bar code's height in pixels. A value of **0** (zero) indicates that the default height is used. The meaning can vary for different formats.
- **Margin** – Specify the size of the bar code's margin in pixels. The margin is the area on each side of a bar code that must be kept clear (quiet zone). A value of **0** (zero) indicates that the default margin is used. The meaning can vary for different formats.
- **Output content** – Set the value to **Yes** to generate a bar code image that contains the encoded information as text. The default value is **No**.
- **Encoding** – Specify the type of characters that are encoded in the generated bar code image. By default, the **UTF-8** encoding is used.

> ⓘ **Important**
>
> When you add a new **Barcode** data source, you must place it under another item (container) as a nested element.
>
> When you bind a **Barcode** data source to a cell element in a format, and the cell element represents either a Word content control or an Excel picture, the data source is presented in that binding as a function that has a single parameter of the **String** type. You must use this parameter to specify the text that should be transformed into a bar code image and read when a generated bar code is scanned.

For more information about this feature, complete the examples in this article.

Example: Generate a payment check that contains a bar code that encodes the payable amount

This example shows how a user in the **System administrator** or **Electronic reporting functional consultant** role can configure an ER format that contains a template that is used to generate an outbound document in Excel format that contains a bar code. Here is an overview of the steps that are involved.

1. Complete the prerequisites.
2. Activate a configuration provider.
3. Import the provided ER solution.
4. Generate a payment check.
5. Review the generated payment check.
6. Modify the format of the provided ER solution.
    a. Apply a new check template.
    b. Add a new Barcode data source.
    c. Bind a new format element.
    d. Make the modified version available for test runs.
        i. Complete the modified format version.
        ii. Make the draft version available for use.
7. Generate a payment check.
8. Convert the generated check to a PDF.

In this example, you will use the provided ER solution that has been configured to generate payment checks. This solution generates payment checks where the payable amount is written both as a number and as text. You will modify this ER solution so that the check also includes a generated bar code where the payable amount is encoded and can be read by using a bar code scanner.

The steps can be completed in the **USMF** company in Microsoft Dynamics 365 Finance.

Complete the prerequisites

To complete this example, you must have access to the USMF company in Finance for one of the following roles:

- Electronic reporting functional consultant
- System administrator

If you haven't yet completed the example in the Embed images and shapes in documents that you generate by using ER article, download the following configurations of the sample ER solution.

Expand table

Content description	File name
ER data model configuration	Model for cheques.xml
ER format configuration	Cheques printing format.xml

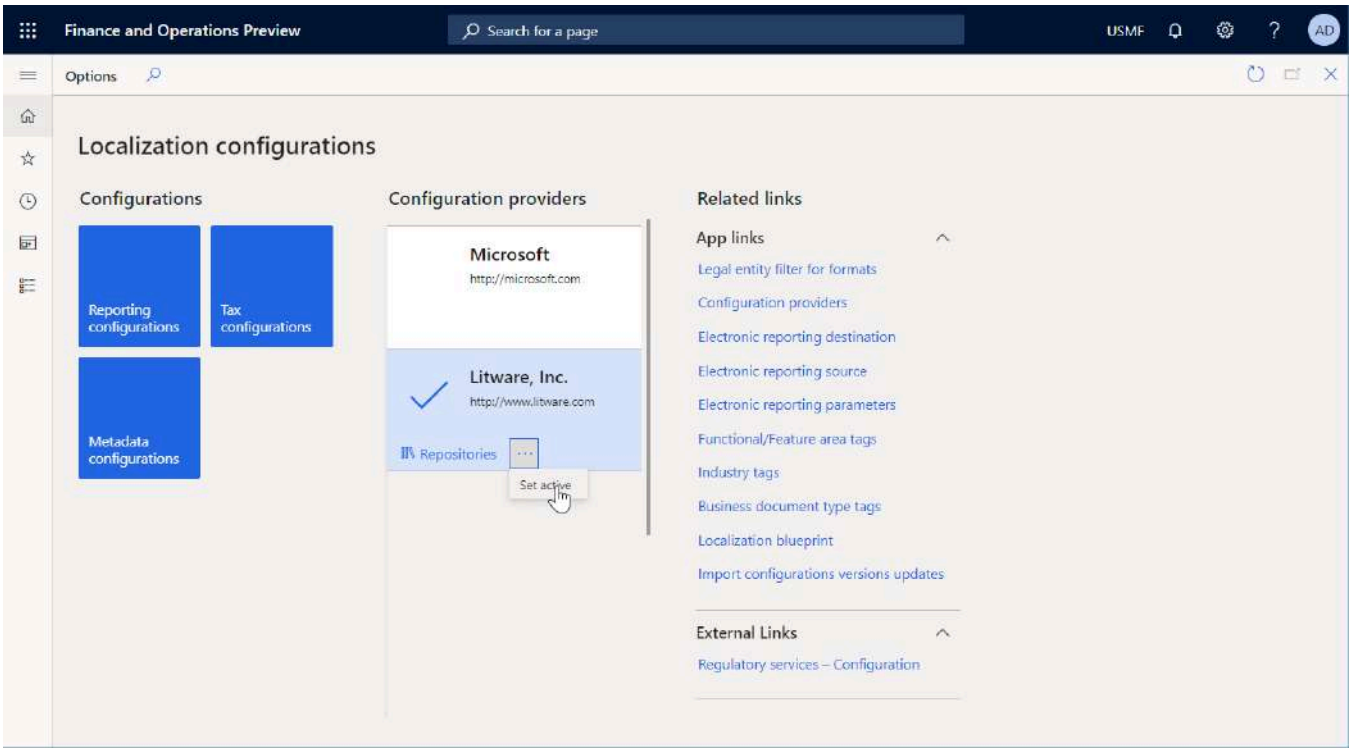
Additionally, download the following Excel file that contains the modified template for the provided ER solution.

Expand table

Content description	File name
Report template	Check template Excel.xlsx

Activate a configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Configuration providers** section, make sure that the configuration provider for the **Litware, Inc.** sample company is listed, and that it's marked as active. If it isn't listed, or if it isn't marked as active, follow the steps in the Create a configuration provider and mark it as active article.



Import the provided ER solution

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Configurations** section, select the **Reporting configurations** tile.
3. On the **Configurations** page, if the **Model for cheques** configuration isn't available in the configuration tree, follow these steps to import the ER data model configuration:
   a. On the Action Pane, select **Exchange** > **Load from XML file**.
   b. In the dialog box, select **Browse**, find and select the **Model for cheques.xml** file, and then select **OK**.
4. If the **Cheques printing format** configuration isn't available in the configuration tree, follow these steps to import the ER format configuration:
   a. On the Action Pane, select **Exchange** > **Load from XML file**.
   b. In the dialog box, select **Browse**, find and select the **Cheques printing format.xml** file, and then select **OK**.
5. In the configuration tree, expand **Model for cheques**.
6. Review the list of imported ER configurations in the configuration tree.

Generate a payment check

1. Go to **Cash and bank management** > **Bank accounts** > **Bank accounts**.

2. On the **Bank accounts** page, select the **USMF OPER** account.

3. On the bank account details page, on the Action Pane, on the **Set up** tab, in the **Layout** group, select **Check**.

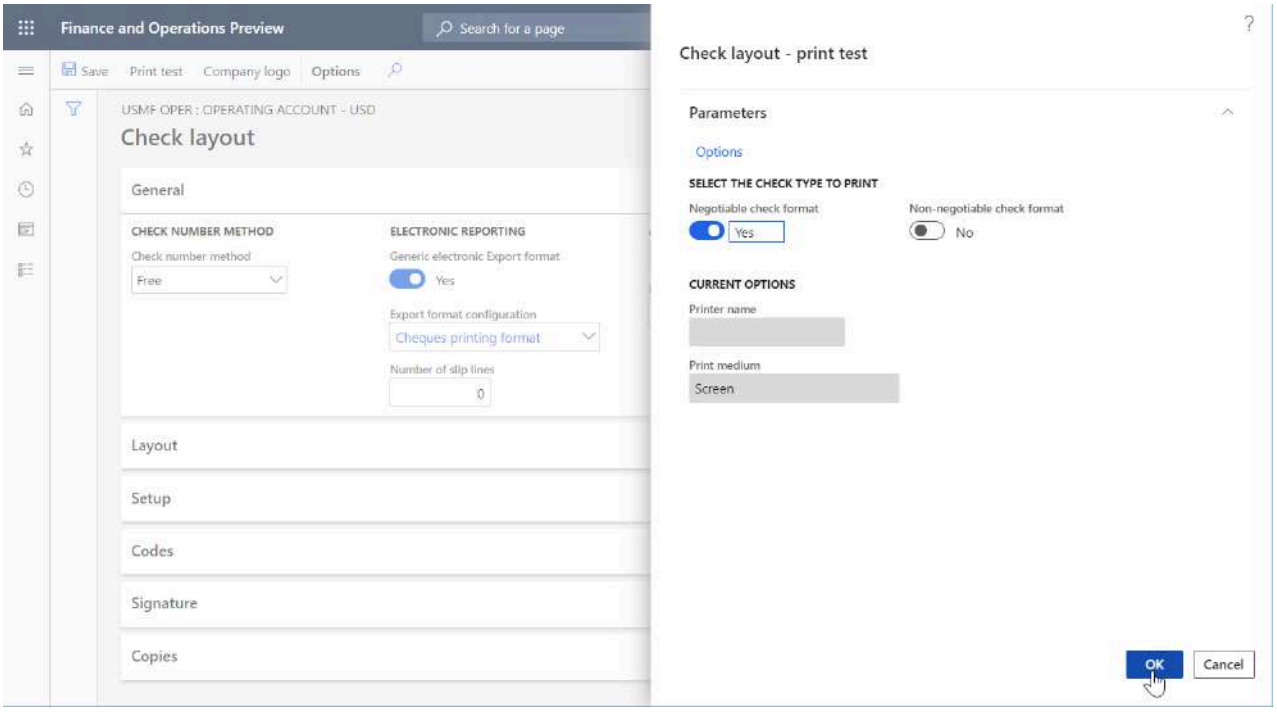
4. On the **Check layout** page, select **Edit**.

5. On the **General** FastTab, set the **Generic electronic Export format** option to **Yes**.

6. In the **Export format configuration** field, select the **Cheques printing format** ER format that you imported earlier.

7. On the Action Pane, select **Print test**.

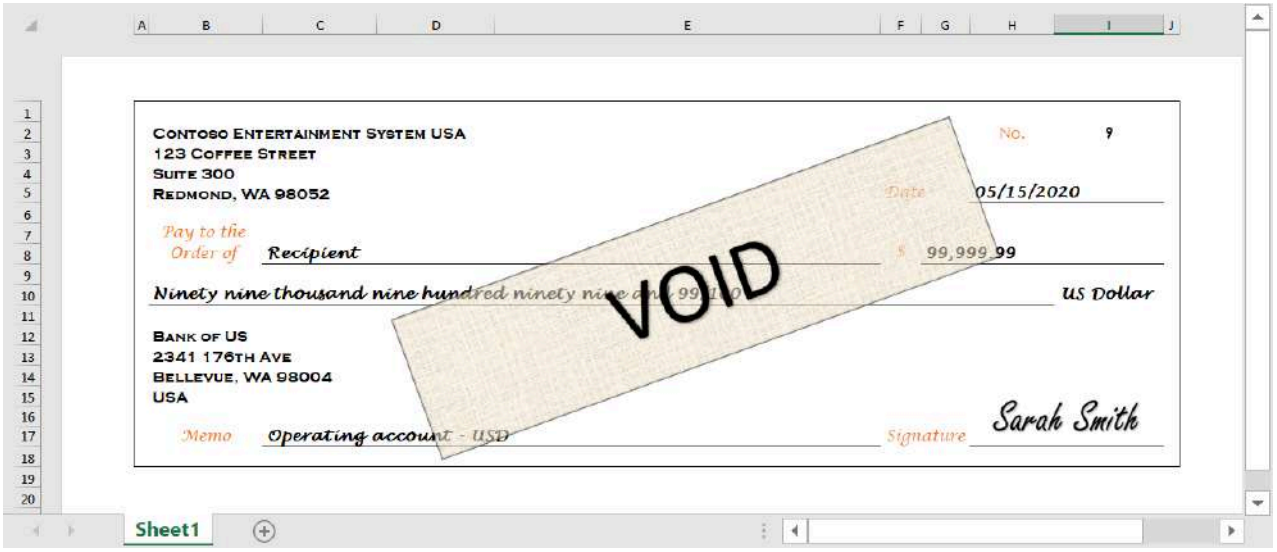
8. In the dialog box, set the **Negotiable check format** option to **Yes**, and then select **OK**.



Review the generated payment check

- Open the generated check in Excel.

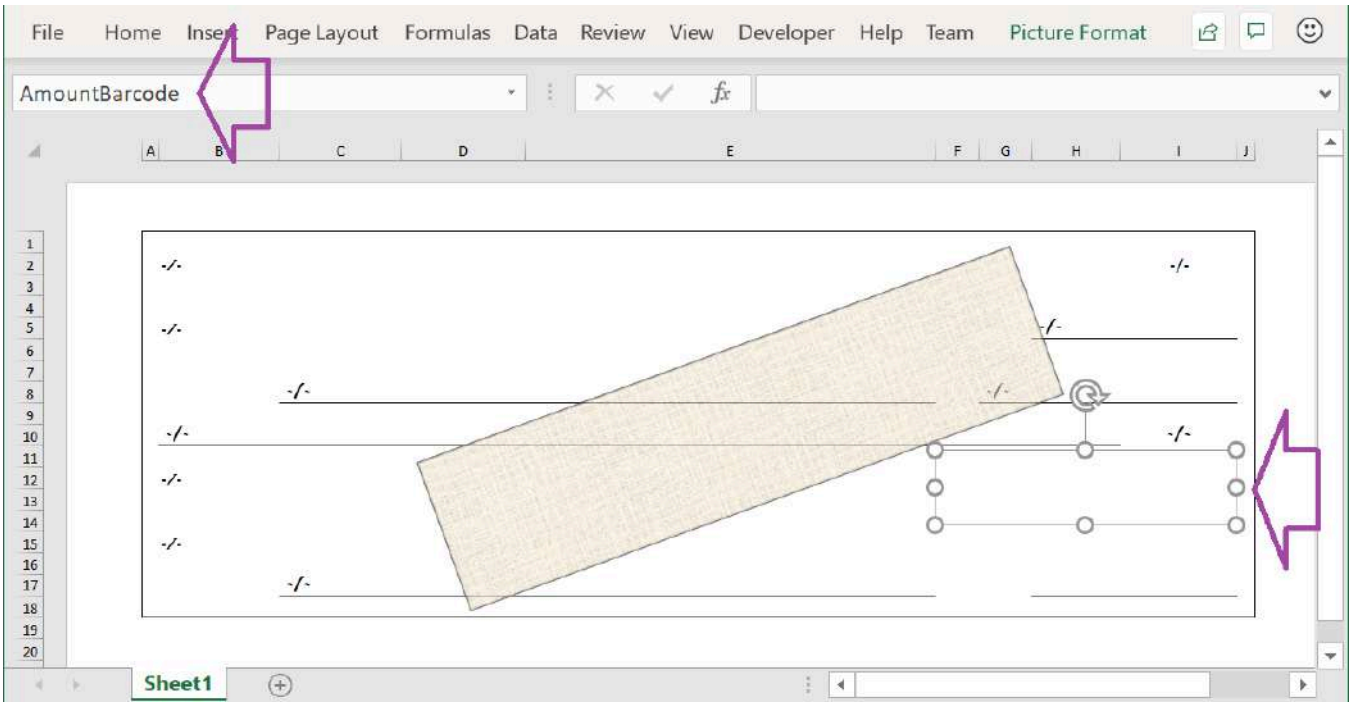
2. Review the generated check.



Modify the format of the provided ER solution

Apply a new check template

You can use the Excel desktop application to open the **Cheque template Excel.xlsx** file that you imported earlier. Notice that this template differs from the template that you used to generate a payment check in the provided ER solution. In addition, it includes an **AmountBarcode** element for the bar code image.



You must now modify the ER solution and then reapply the modified template.

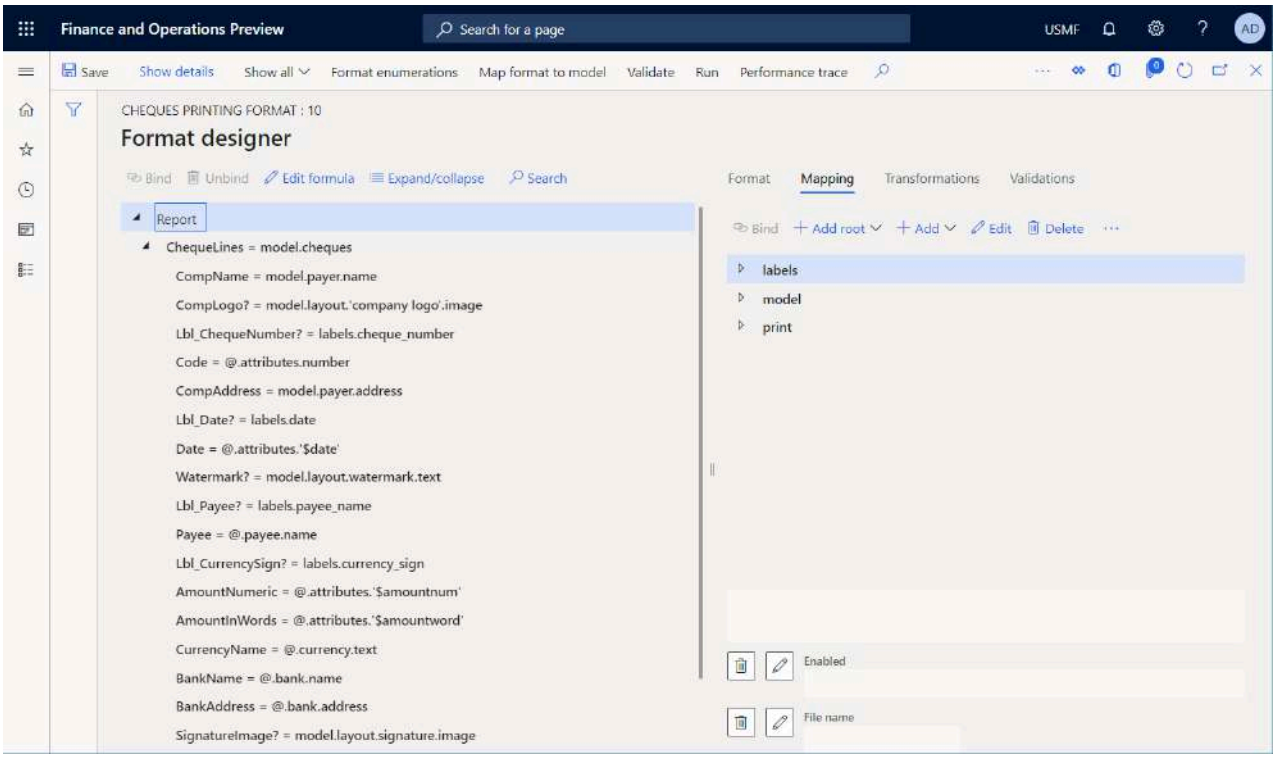
1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Configurations** section, select **Reporting configurations**.

3. On the **Configurations** page, in the configuration tree, expand **Model for cheques**, and select **Cheques printing format**.

4. On the Action Pane, select **Designer**.

5. In the ER Operations designer, select the **Mapping** tab on the right side of the page, and then, in the format tree pane on the left, select **Expand/collapse**.

6. Notice that all cell format elements are bound to the appropriate data sources.



7. Select the **Format** tab on the right side of the page.

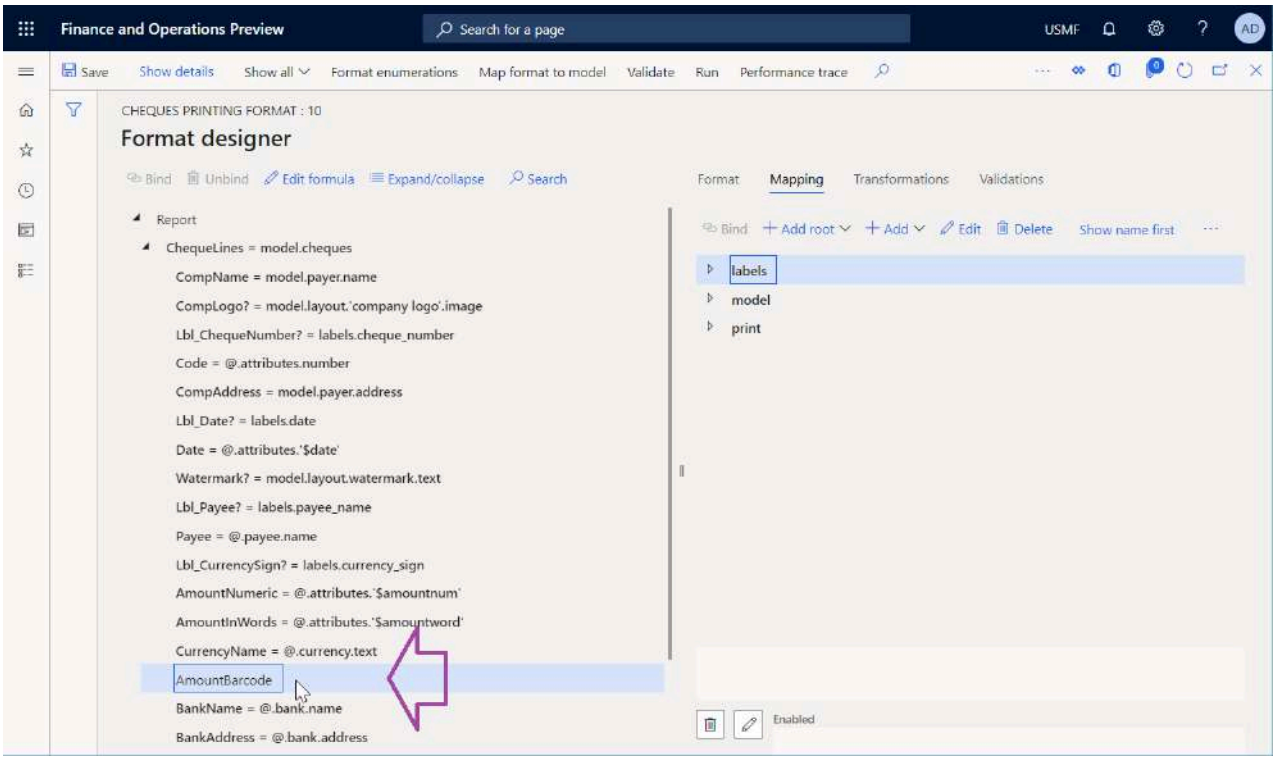
8. On the Action Pane, select the ellipsis (**...**), and then select **Import**.

9. In the **Import** group, select **Update from Excel**, and then select **Update template**.

10. In the dialog box, browse to the **Cheque template Excel.xlsx** file that is saved on your computer, select it, and then select **OK** to confirm that the selected template should be applied.

11. Select the **Mapping** tab on the right side of the page, and then, in the format tree pane on the left, select **Expand/collapse**.

12. Notice that the **AmountBarcode** cell element has been added to the format. This element is associated with the **AmountBarcode** element that has been added to the modified Excel template as a placeholder for a bar code image.

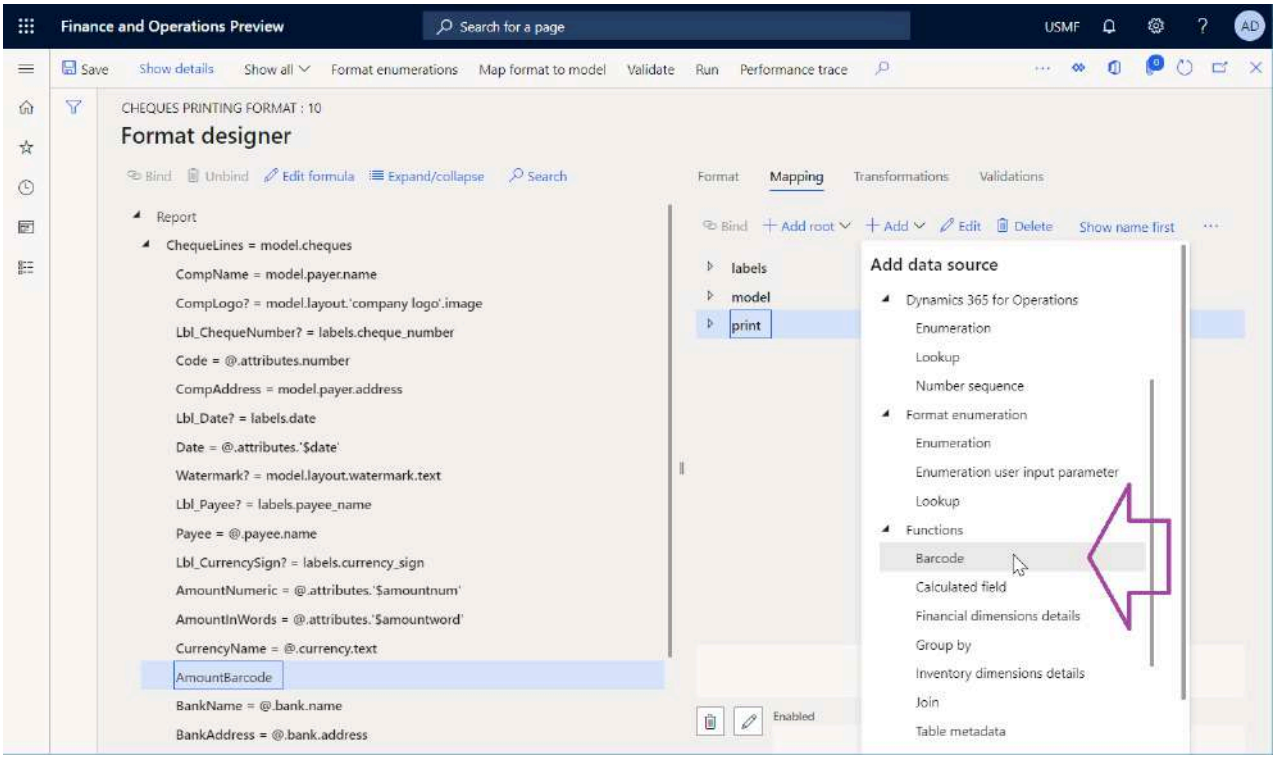


Add a new Barcode data source

Next, you must add a new data source of the **Barcode** type.

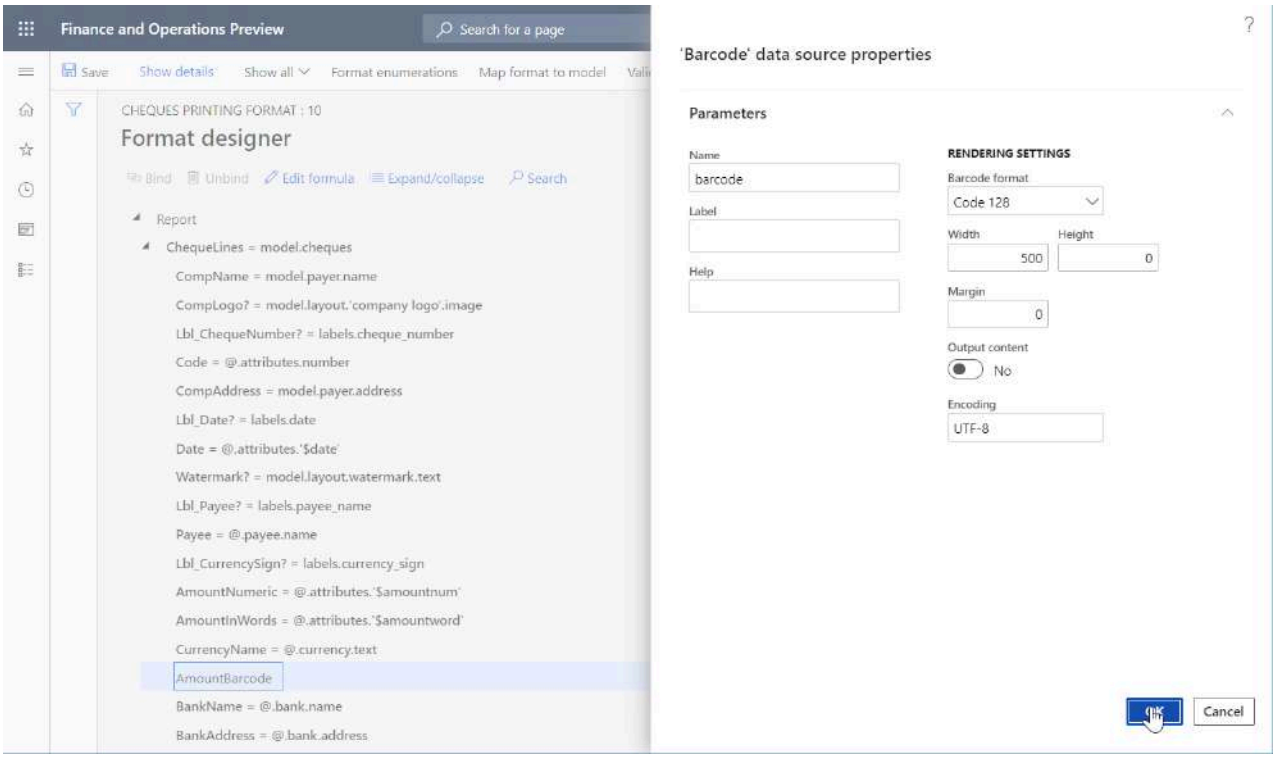
1. In the ER Operations designer, on the **Mapping** tab on the right side of the page, select the **print** data source.

2. Select **Add**, and then, in the **Functions** group, select the **Barcode** data source type.



3. In the dialog box, in the **Name** field, enter **barcode**.

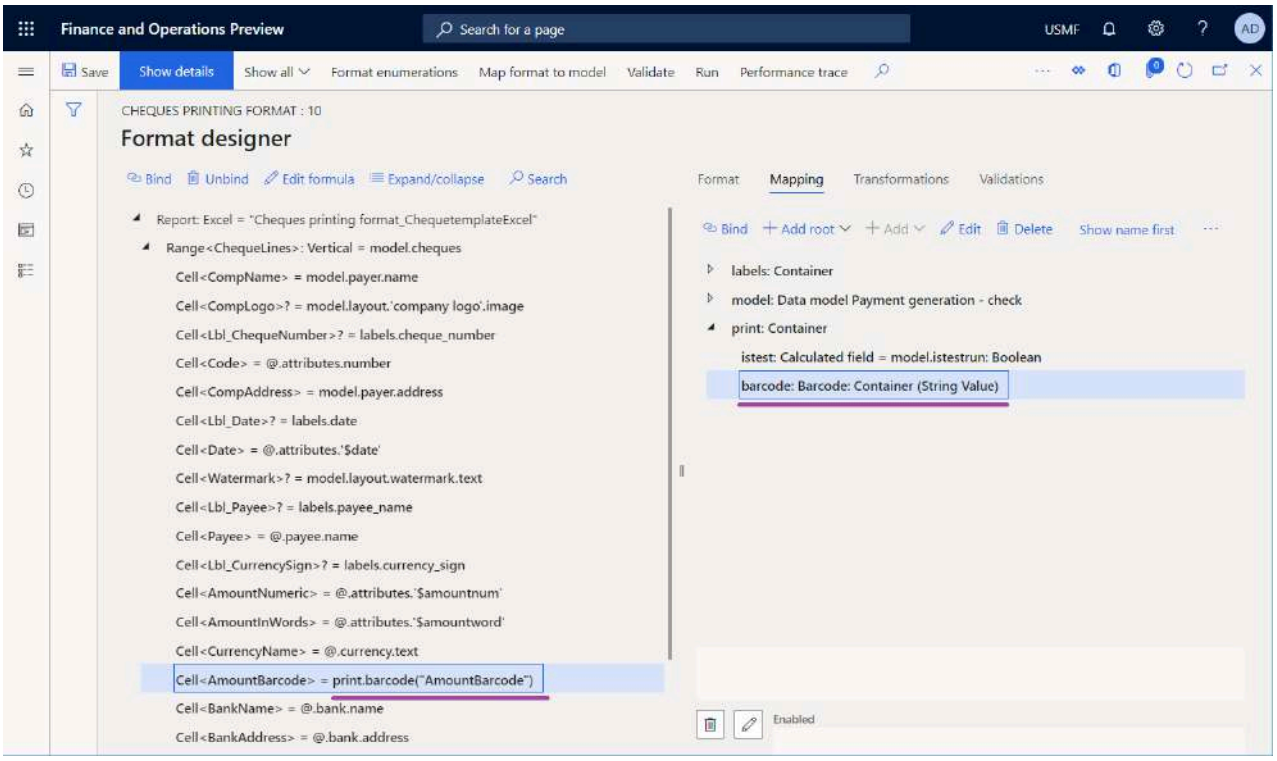
4. In the **Barcode format**, select **Code 128**.
5. In the **Width** field, enter **500**.
6. Select **OK**.



Bind a new format element

Next, you must bind the new format element to the data source that you just added.

1. In the ER Operations designer, on the **Mapping** tab on the right side of the page, select the **print\barcode** data source.
2. In the format tree pane on the left, select the **AmountBarcode** cell element, and then select **Bind**.
3. On the Action Pane, select **Show details**.
4. Notice that, because the **Barcode** data source is represented in the binding as a function that contains a single parameter, the name of the bound format element has been automatically taken as the argument of that parameter.



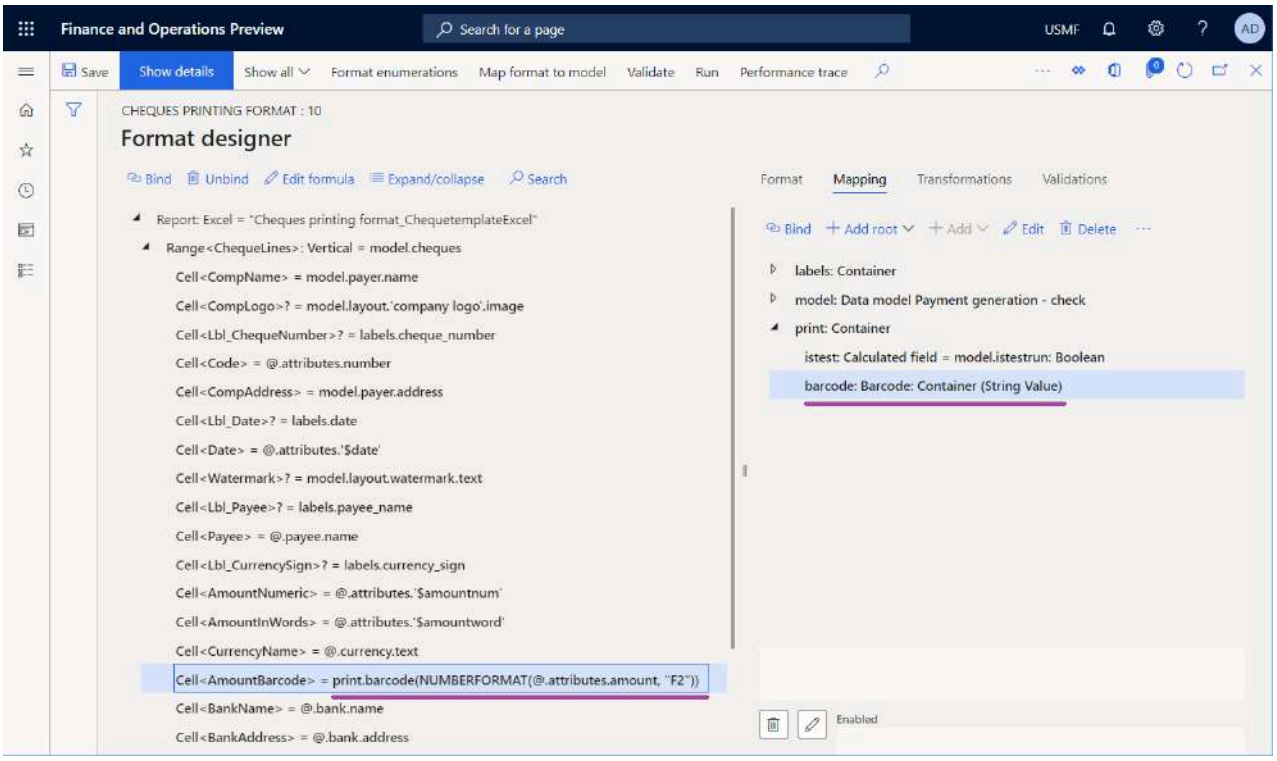
5. Select **Edit formula** to adjust the binding.

   You don't want the name of the cell element to be returned. Therefore, you must configure an expression that returns text that contains the payable amount of the current check. Because the parent **ChequeLines** range is bound to the **model.cheques** data source, the payable amount of the current check is available in the **model.cheques.attributes.amount** field of the **Real** data type.

6. In the **Formula** field, enter **print.barcode(NUMBERFORMAT(@.attributes.amount, "F2"))**.

7. Select **Save**, and then close the ER Formula designer.

8. Notice that the binding has been adjusted.



9. Select **Save**, and then close the ER Operations designer.

Make the modified version available for test runs

By default, the only versions that have a status of **Completed** and **Shared** are used when you run an ER format.

If you've finalized your changes, you can complete your work with the current draft version and make your changes available for use. For instructions, see the Complete the modified format version section that follows.

If you want to continue to work with the current draft version, but you have to use it to generate checks, you must explicitly specify that you want to use the draft version of the format for execution. For instructions, see the Make the draft version available for use section.

Complete the modified format version

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Configurations** section, select **Reporting configurations**.
3. On the **Configurations** page, in the configuration tree, expand **Model for cheques**, and select **Cheques printing format**.
4. On the **Versions** FastTab, select the record that has a status of **Draft**.
5. Select **Change status**, and then select **Complete**.
6. In the dialog box, select **OK**.

The status of the current version is changed from **Draft** to **Completed**, and a new version that has a status of **Draft** is created. You can use this new draft version to apply additional changes.

Make the draft version available for use

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, in the **Configurations** section, select **Reporting configurations**.
3. On the **Configurations** page, on the Action Pane, in the **Configurations** tab, in the **Advance settings** group, select **User parameters**.
4. In the dialog box, set the **Run setting** options to **Yes**, and then select **OK**.
5. In the configuration tree, expand **Model for cheques**, and select **Cheques printing format**.
6. Set the **Run draft** option to **Yes**.
7. Select **Save**.

The draft version of the selected format is marked as available for use when the selected format is run.

Generate a payment check

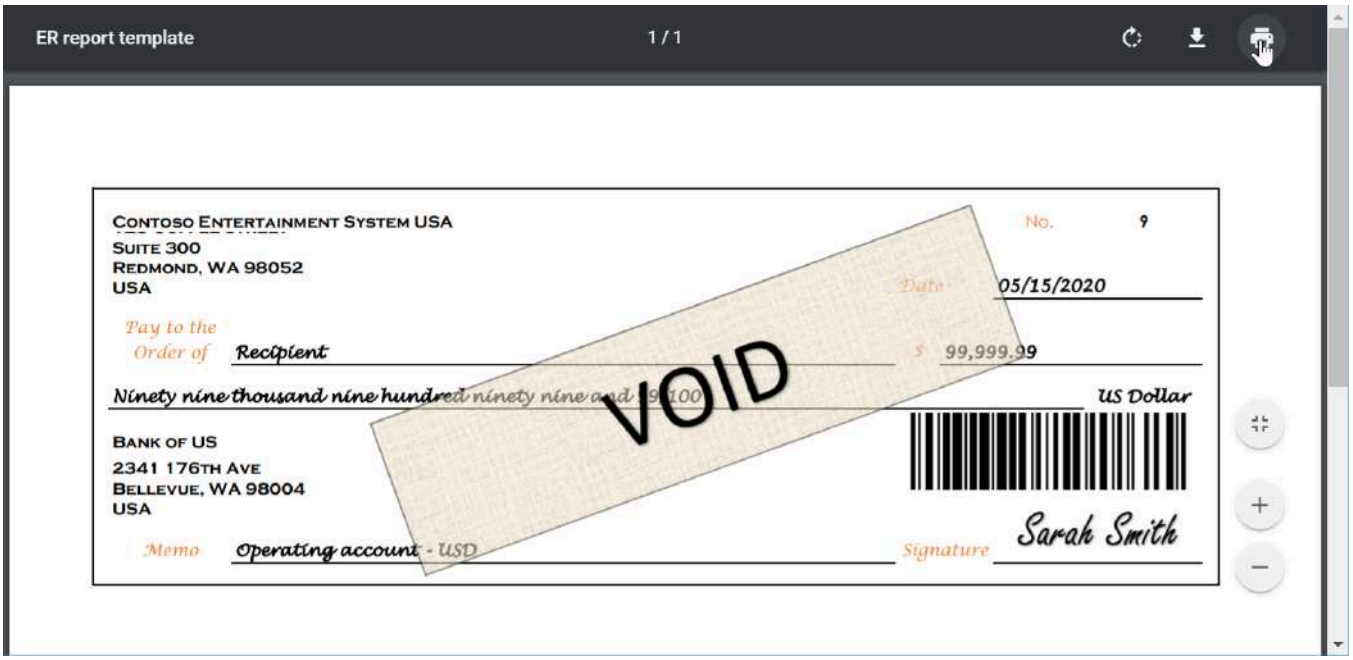
1. Go to **Cash and bank management** > **Bank accounts** > **Bank accounts**.
2. On the **Bank accounts** page, select the **USMF OPER** account.
3. On the bank account details page, on the Action Pane, on the **Set up** tab, in the **Layout** group, select **Check**.
4. On the **Check layout** page, on the Action Pane, select **Print test**.
5. In the dialog box, set the **Negotiable check format** option to **Yes**.
6. Select **OK**.
7. Review the generated check. Notice that a bar code has been generated to encode the payable amount of the check.

> **Important**
>
> An exception is thrown if the argument of a **Barcode** data source doesn't comply with the appropriate requirements that are specific to the bar code format. For example, when the **Barcode** data source is called to generate an EAN-8 bar code for the provided text, an exception is thrown if the length of the text exceeds seven characters.

Convert the generated check to a PDF

As described in the Generate printable FTI forms article, you can use a special font to produce bar codes in a generated document. In this case, additional transformations of the generated document might depend on the availability of that font in the transformation environment. For example, if you try to convert a document to PDF format or preview it in an environment where the font is missing, bar codes won't be rendered correctly.

However, when you use the **Barcode** data source to produce bar codes, the rendering of those bar codes doesn't depend on any font. Therefore, you can easily convert documents that contain the bar codes to PDF format. The following illustration shows the preview of a generated payment check that has been converted to a PDF, based on the setting of the configured ER destination.



Limitations

> **Note**
>
> Some types of bar codes that are generated have a fixed aspect ratio. This behavior makes sense if you've turned on the **Enable usage of EPPlus library in Electronic reporting framework** feature to work with Excel documents in ER. In that case, an image is entered into a placeholder that has a locked aspect ratio. Therefore, when the dimensions of a placeholder in a template correspond to the ratio of an image that is entered, a real picture in a generated document might be resized to maintain the required aspect ratio. To prevent picture resizing, use a placeholder that has an expected aspect ratio.

Additional resources

- Electronic Reporting overview
- Electronic Reporting destinations
- Electronic reporting formula language
- NUMBERFORMAT function

Configure Lookup data sources to use ER application-specific parameters

The Electronic reporting (ER) application-specific parameters feature lets you configure data filtering in an ER format so that it's based on a set of abstract rules. You can configure this set of rules to use the data source of the **Lookup** type that is available in an ER format. Use the user interface (UI) that is automatically generated based on the settings of the **Lookup** data source of the corresponding ER format and the current legal entity data to specify real rules beyond the ER components designers. Eventually, the ER format's **Lookup** data source accesses the specified rules when this ER format is executed.

> **Note**
>
> Use the configured data sources of the editable ER format to specify what application data is used to configure real rules.

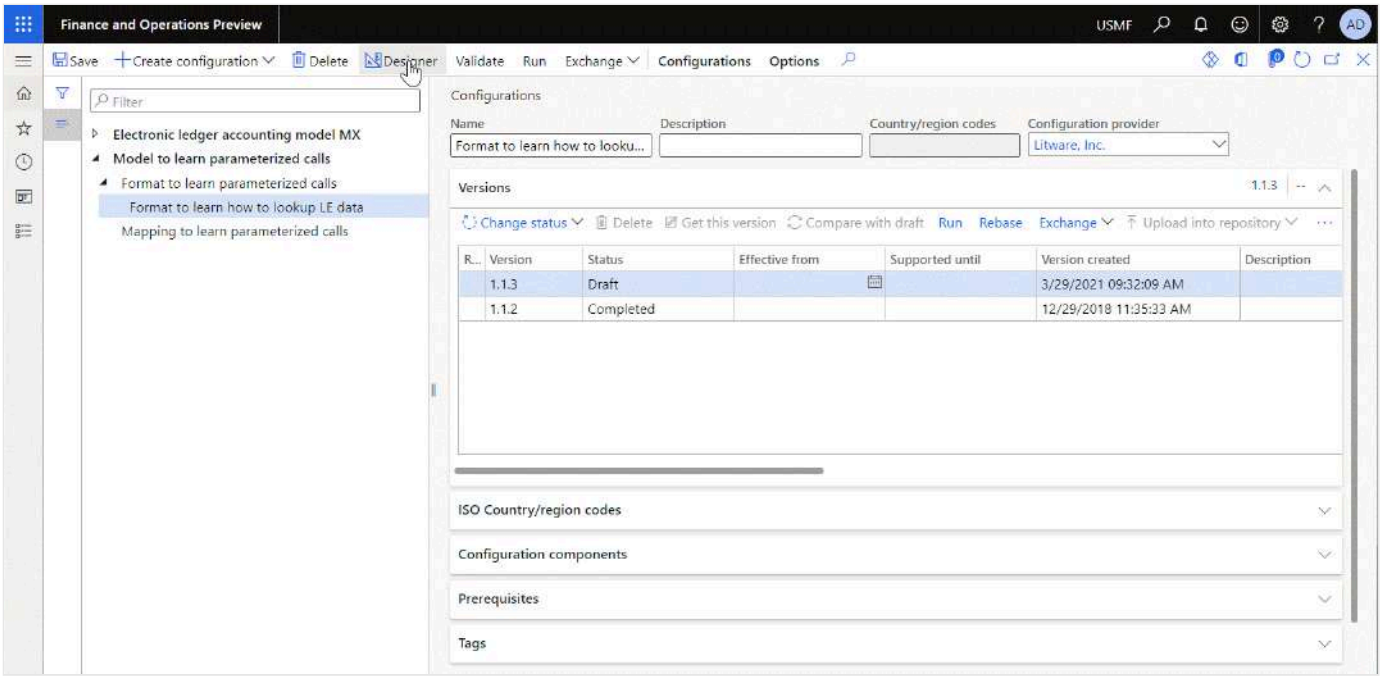
Use the ER Operations designer to bring a data source of the **Lookup** type into your ER format. You must configure the data source to describe how your abstract rules look, including the following elements:

- The parameter set of a certain data type whose value specifies a single rule.
- The value type of a certain data type that a single rule returns when this rule is considered the most appropriate among others.

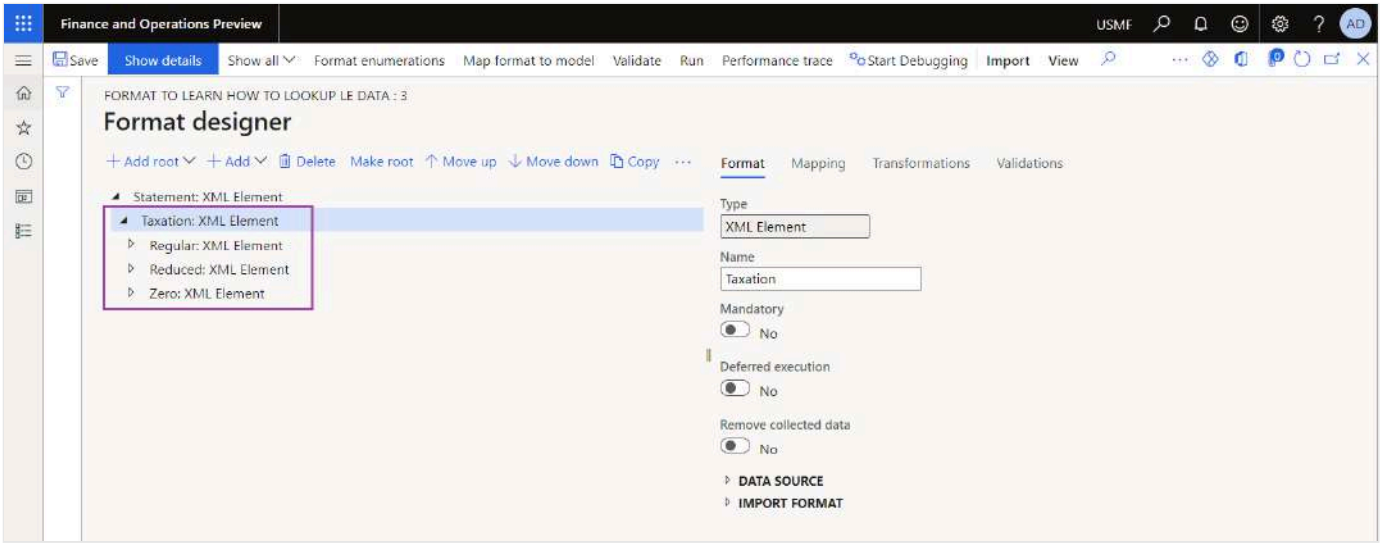
Depending on the type of value that any configured rule returns, you can configure the following types of **Lookup** data sources:

- Use the **Data model\Lookup** type when you must return a model enumeration value.
- Use the **Dynamics 365 Operations\Lookup** type when you must return an application enumeration value or an application extended data type value.
- Use the **Format enumeration\Lookup** type when you must return a format enumeration value.

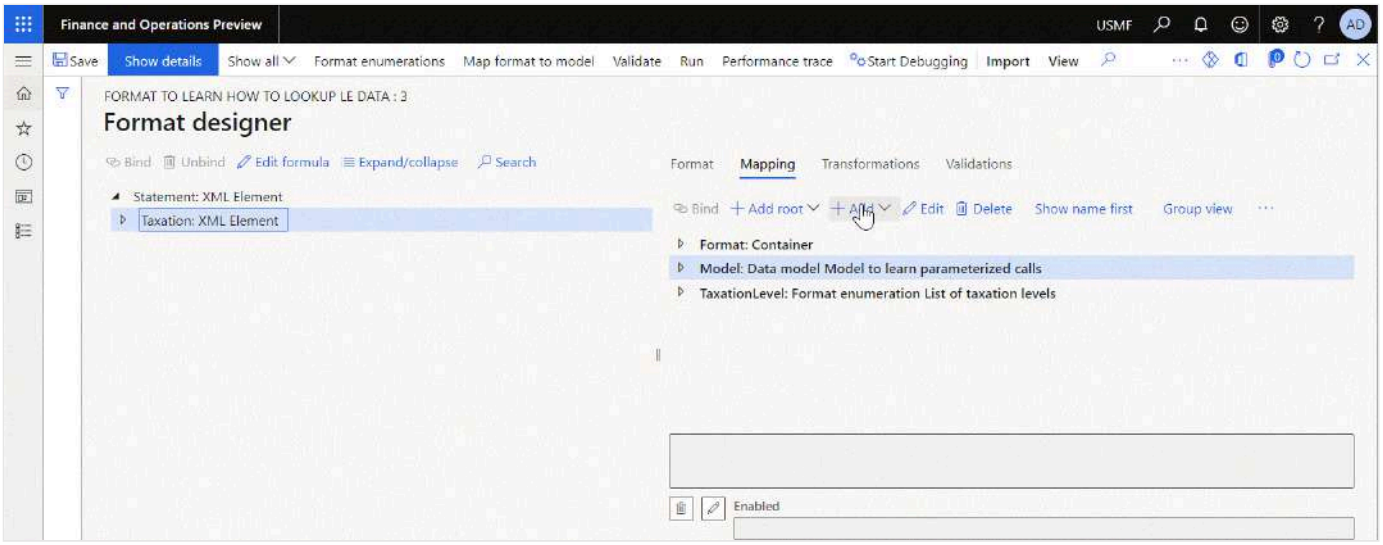
The following illustration shows how you can configure a format enumeration in the sample ER format.



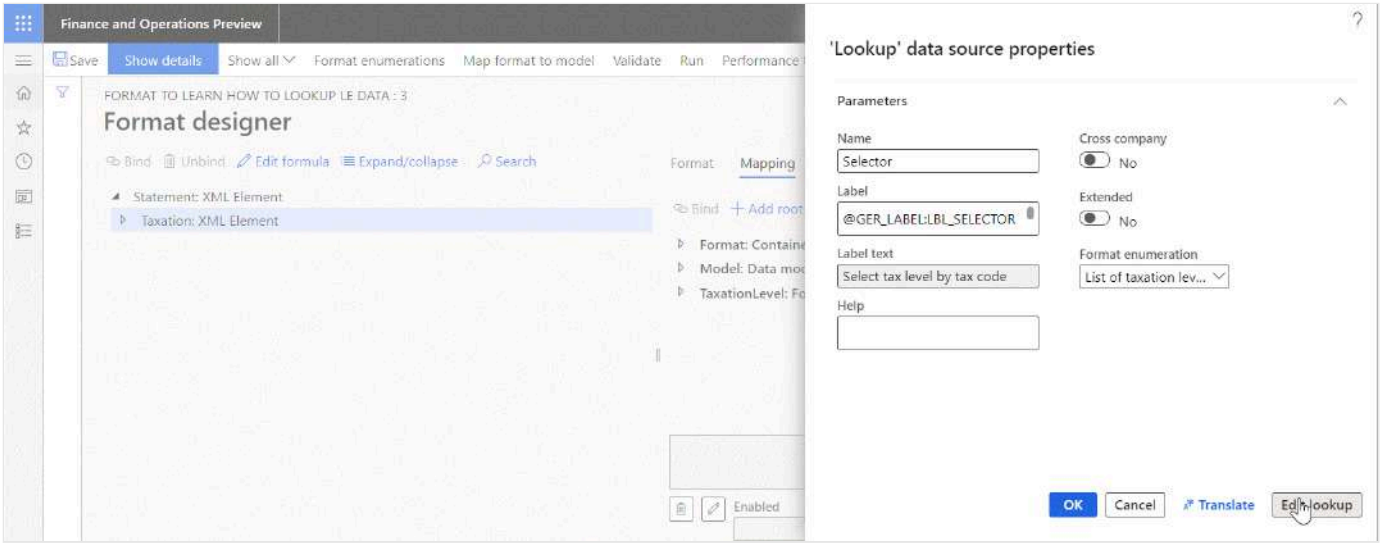
The following illustration shows the format components that you configured to report different types of taxes in different sections of a generated report.



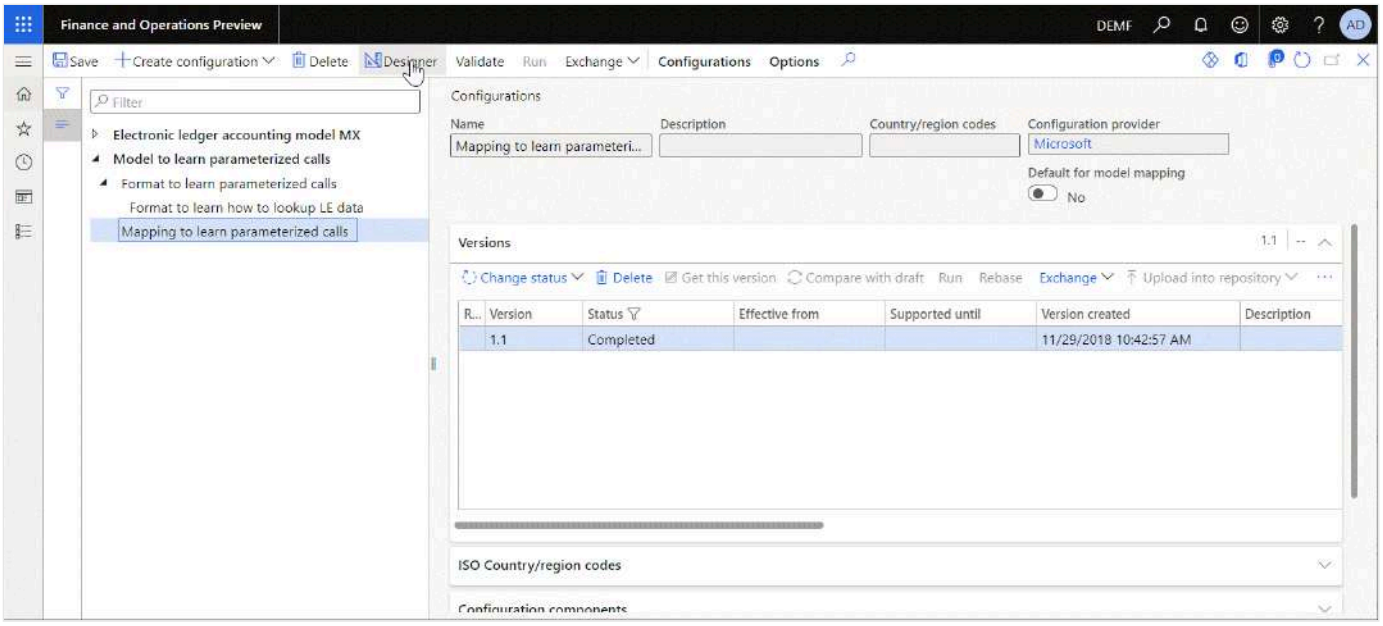
The following illustration shows how the ER Operations designer allows you to add a data source of the **Format enumeration\Lookup** type. The added data source is configured as returning a value of the `List of taxation levels` format enumeration.



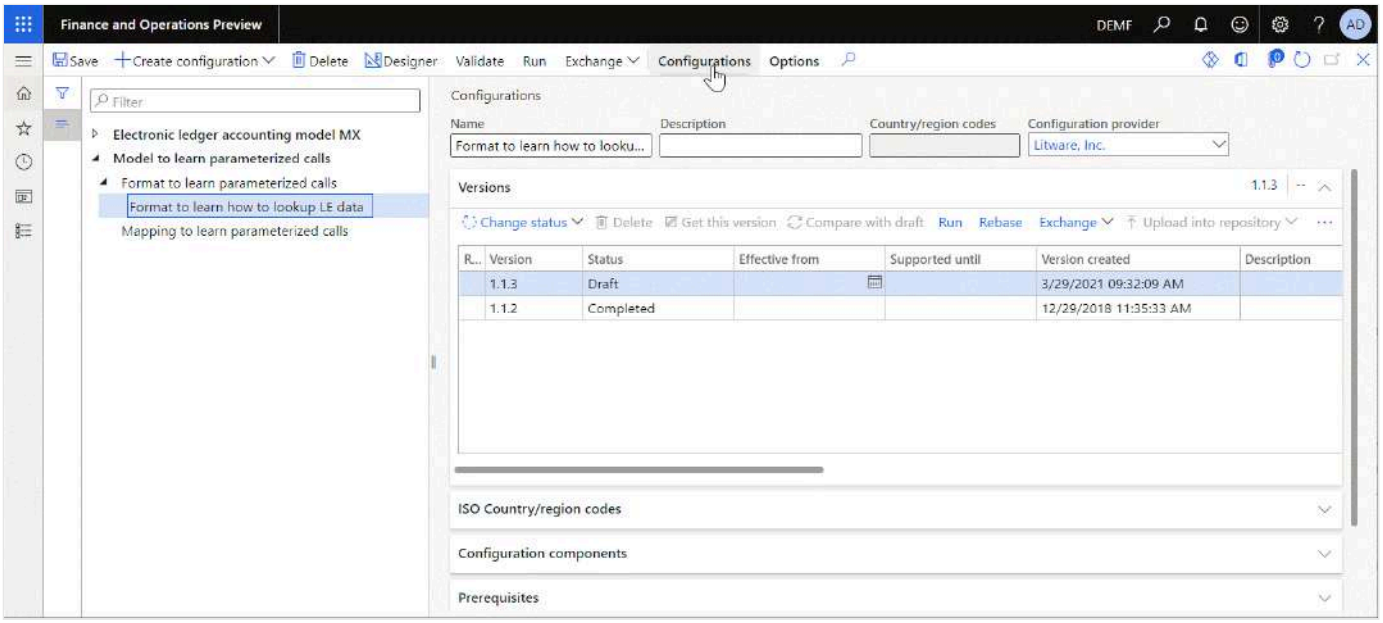
The following illustration shows how you configure the added data source to use the **Code** field of the **Model.Data.Tax** record list of the **Model** data source as a parameter that you must specify for every configured rule.



You configure the added `Model.Data.Tax` data source to specify a tax code for every configured rule by accessing records of the **TaxTable** application table.

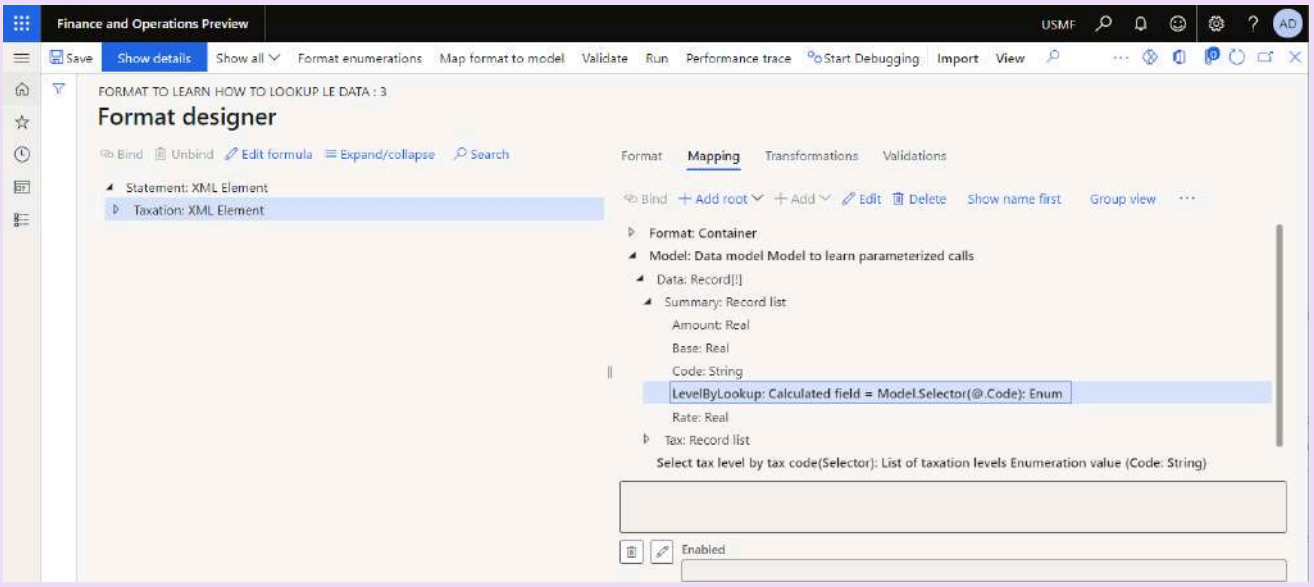


Set up the lookup rules for the selected ER format by using the UI that automatically aligns with the structure of the configured data source. Currently, this UI requires that for each rule, you specify the returned value as the `List of taxation levels` format enumeration value as well as the tax code as a parameter.

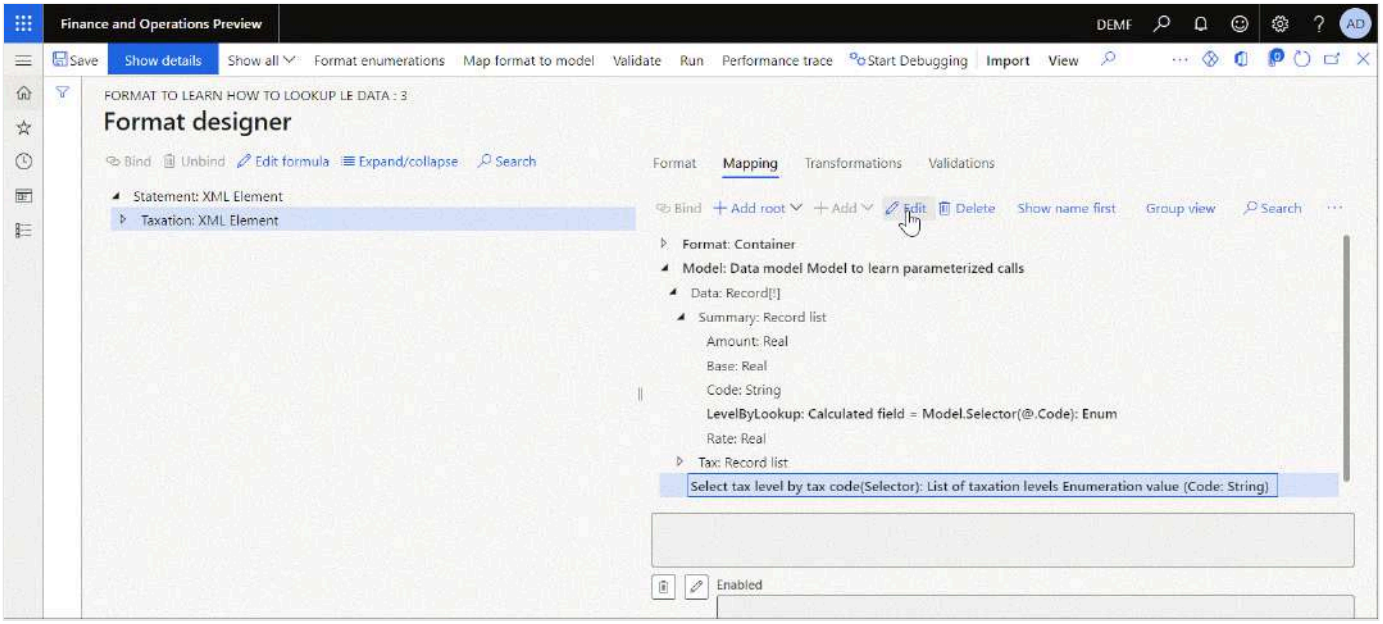


The following illustration shows how you can configure the `Model.Data.Summary.LevelByLookup` data source of the **Calculated field** type to call the configured **Lookup** data source by providing the required parameters. To process this call at runtime, ER goes through the list of configured rules in the defined sequence to locate the first rule that satisfies the provided conditions. In this example, it's the rule that contains the tax code that matches the provided one. As the result, ER finds the most appropriate rule and returns the enumeration value that you configured for the found rule.

> **Note**
>
> An exception is thrown when no applicable rule is found. To prevent these exceptions, configure additional rules at the end of the rules list to handle cases when a non-configured value or no value is provided. Use the ***Not blank*** and ***Blank*** options accordingly.

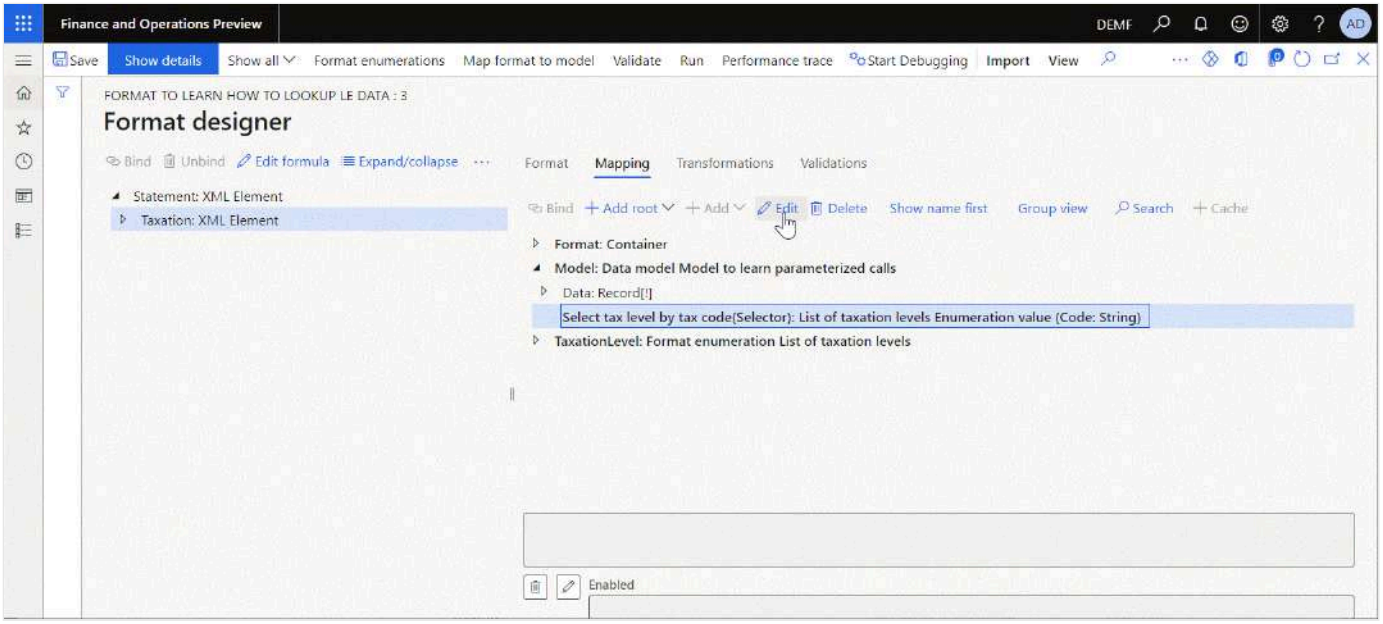


When you set the **Cross-company** option to **Yes** for the editable lookup data source, you add a new required **Company** parameter to the set of parameters for this data source. You must specify the value of the **Company** parameter at runtime when the lookup data source is called. When you specify the company code at runtime, the rules configured for this company are used to find the most appropriate rule, and the corresponding value is returned. The following illustration shows how you can do this and how the set of parameters of the editable data source is changed.



> **Note**
>
> Select every company separately to configure the set of rules for this lookup data source of the editable ER format. An exception is thrown at runtime when the cross-company lookup is called with the code of the company for which the lookup setting wasn't completed.
>
> Make sure that you grant permissions for a user who runs the ER format with the cross-company **Lookup** data source to access the data of every company that is in scope of this data source. Otherwise, an exception is thrown at runtime.

Starting in version 10.0.19, the **Lookup** data sources include extended capabilities. When you set the **Extended** option to **Yes** for the editable lookup data source, the configured lookup data source transforms into a structured data source that offers extra capabilities to analyze the configured set of rules. The following illustration shows this transformation.



- The **Lookup** subitem works as a function that finds the most appropriate rule from the set of configurable rules based on the provided set of parameters.
- The **IsLookupResultSet** subitem works as a function that accepts the provided value of the base enumeration data source and returns the *Boolean* value of **True** when the set of rules contains at least one rule for which the provided enumeration value is configured as a returned value. This function returns the *Boolean* value of **False** when there are no rules configured to return the provided enumeration value.
- The **Setting** subitem works as a function that returns the set of configured rules as records of a record list. The returned values and the set of parameters of the configured rules are

presented in every returned record as fields of the relevant data types:

- The returned value appears as the **Lookup result** field.
- The configured parameters appear as fields with names of parameters (**Code** field in this example).

For more information about how to configure the **Lookup** data source, see Configure ER formats to use parameters that are specified per legal entity. To learn how to set the Lookup rules, see Set up the parameters of an ER format per legal entity.

Additional resources

Configure ER formats to use parameters that are specified per legal entity

Set up the parameters of an ER format per legal entity

Last updated on 06/26/2026

Use DATA COLLECTION data sources in Electronic reporting formats

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the Operations designer of the Electronic reporting (ER) framework to configure the format component of an ER solution that generates outbound documents in different formats. The hierarchical structure of the configured format component consists of various types of format elements. These format elements fill generated documents with the required information at runtime. By default, when you run an ER format, the format elements run in the same order as they appear in the format hierarchy: one by one, from top to bottom.

When ER runs a format element that contains a binding, the formula of that binding is run. The format element adds the value to a generated document. For example, the binding can pass the value of a data model field to a format element. You can configure a DATA COLLECTION data source to collect values of data model fields at runtime, sum values, and fill a generated document with the collected values. To use this approach, change the initial binding so that the configured DATA COLLECTION data source passes the value of a data model field to a format element. By passing values through the DATA COLLECTION data source, you can collect required details for further use.

When you configure a DATA COLLECTION data source, specify a value type that will be managed in the data source. The following data types are currently supported for collecting values:

- Boolean
- Date
- DateTime
- GUID
- Int64
- Integer

- Real
- String
- Time

Use the `Collect(Value)` method of a DATA COLLECTION data source to pass a value to a data source for collection. In this method, the `Value` argument is either a constant or the valid path of a data source field of the relevant data type.

Use the `Result` property of a DATA COLLECTION data source to access the list of collected values. This property returns a record list. The records of the record list contain the `Value` field that you can use to access collected values.

By default, a DATA COLLECTION data source collects only unique values.

To collect all values, set the **Collect all values** field of the configured DATA COLLECTION data source to **Yes**. When the **Collect all values** field is set to **Yes**, the `Sum(Flag)` parameterized property becomes available. You can use this property to get the total amount of all currently collected values. In this property, the `Flag` argument is a Boolean value that is used to indicate whether the total value must be reset.

- When the value **False** is provided, summing continues from the previously collected amount.
- When the value **True** is provided, a new summing starts.

The following data types are currently supported for summing:

- Int64
- Integer
- Real

To learn more about this feature, complete the example that follows.

Example: Configure an ER format to do counting and summing by using a DATA COLLECTION data source

This example shows how a user in the System administrator or Electronic reporting functional consultant role can configure an ER format that has a DATA COLLECTION data source that calculates running totals and collects summed values.

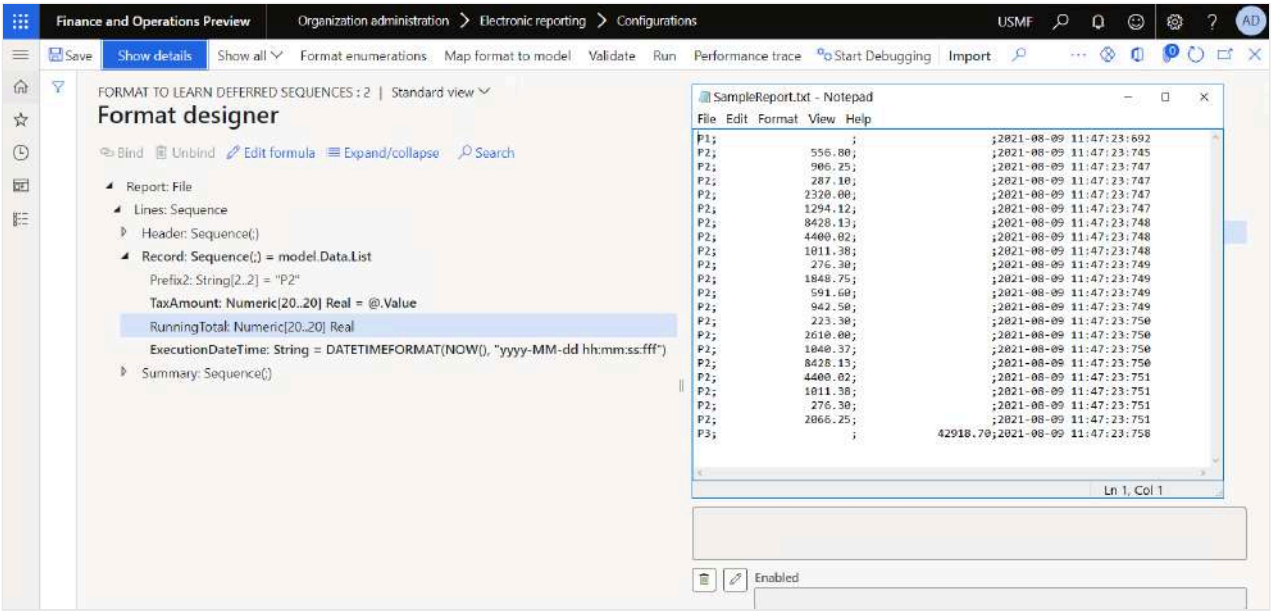
You can complete the procedures in this example in the USMF company in Microsoft Dynamics 365 Finance.

Upload and use the provided ER solution

1. Import the sample ER configurations.
2. Activate a configuration provider.
3. Review the imported model mapping.
4. Review the imported format.
5. Run the imported format.

Run the format of the provided ER solution

1. On the **Format designer** page, select **Run**.
2. In the **Electronic report parameters** dialog box, select **OK**.
3. Download and review the file that the web browser offers.



Modify the format of the ER solution to calculate the running tax total

If the volume of transactions is much larger than the volume in the current example, the time that summing requires might increase and cause performance issues. By changing the settings of the format, you can help prevent these performance issues. Because you access tax values to include them in the generated report, you can reuse that information to sum tax values.

1. On the **Format designer** page, on the **Mapping** tab, select **Add root**.
2. In the **Add data source** dialog box, select **Functions** > **Data collection**.

3. In the **'Data collection' data source properties** dialog box, follow these steps:
   a. In the **Name** field, enter **CollectedTaxValues**.
   b. In the **Item type** field, select **Real**.
   c. In the **Collect all values** field, select **Yes**.
   d. Select **OK**.

4. Select the **Report\Lines\Record\TaxAmount** numeric format element.

> ⓘ **Note**
>
> Currently, the `@.Value` binding is configured for this element. Therefore, a generated document is filled with tax values from the `model.Data.List.Value` field.

5. Select **Edit formula**.

6. On **Formula designer**, follow these steps:
   a. In the **Formula** field, replace `@.Value` with `CollectedTaxValues.Collect(@.Value)`.
   b. Save your changes, and close the page.

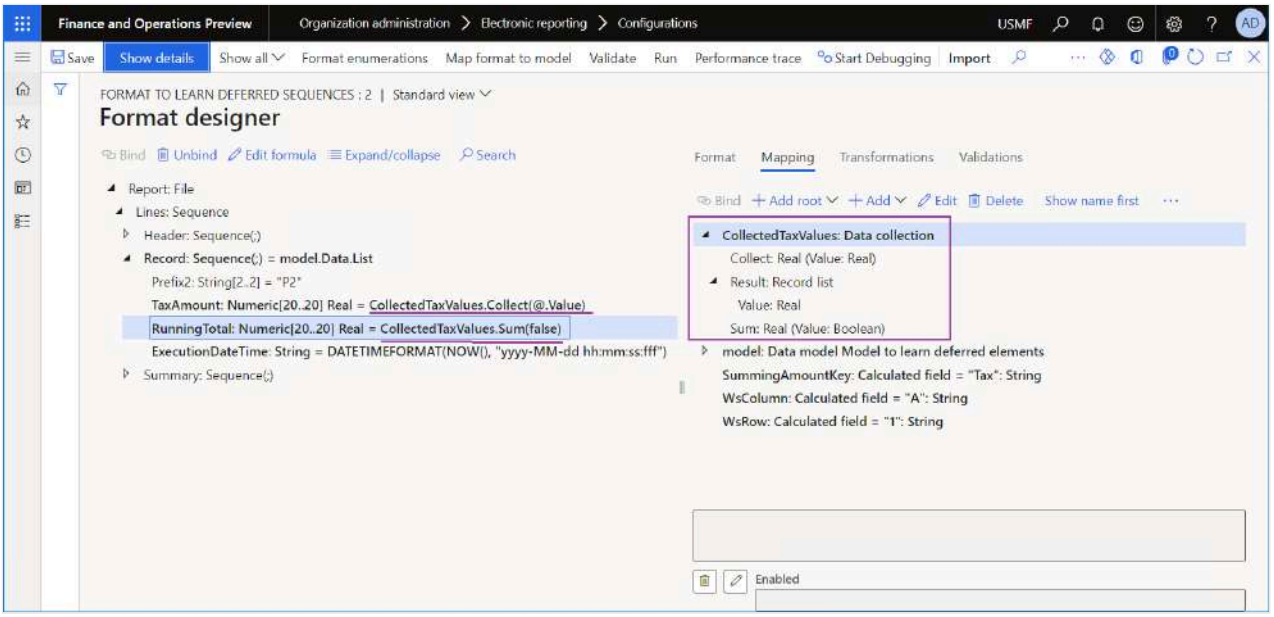
> ⓘ **Note**
>
> The new binding passes the same tax values to a generated document. However, those values are also collected in the **CollectedTaxValues** data source.

7. Select the **Report\Lines\Record\RunningTotal** numeric format element.

8. Select **Edit formula**.

9. On **Formula designer**, follow these steps:
   a. In the **Formula** field, enter `CollectedTaxValues.Sum(false)`.
   b. Save your changes, and close the page.

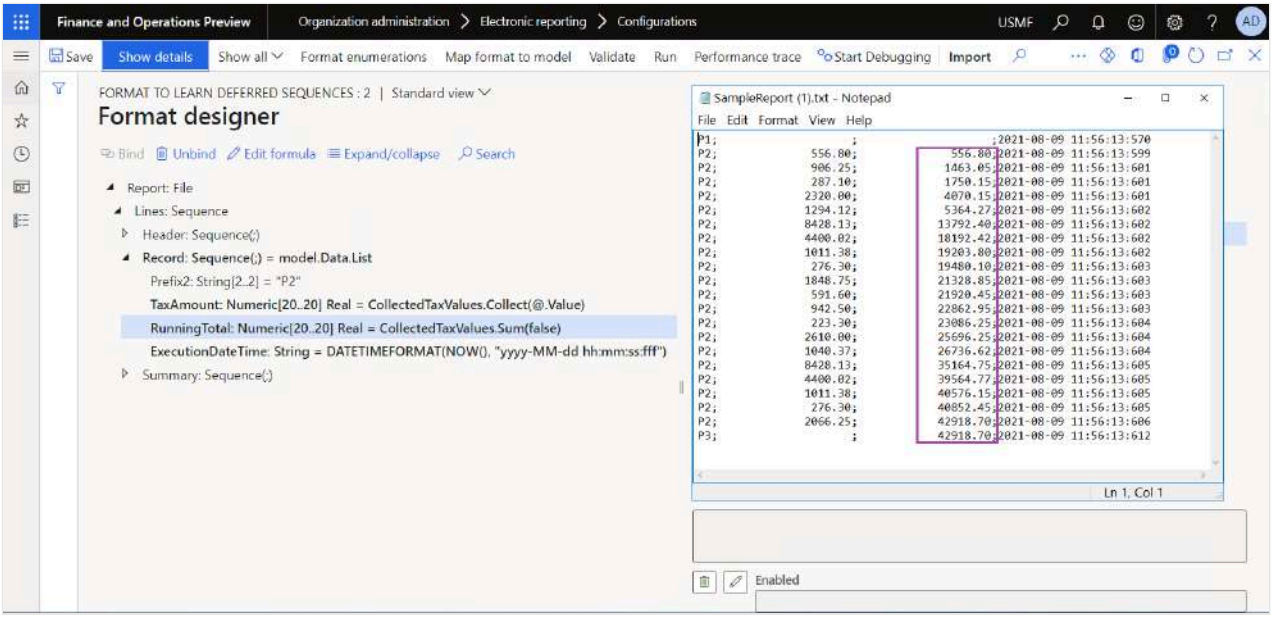
> ⓘ **Note**
>
> The new binding passes, to a generated document, the total amount of tax values that are already entered.



10. Select **Save**, and then select **Run**.

11. In the **Electronic report parameters** dialog box, select **OK**.

12. Download and review the file that the web browser offers.



Modify the format to evaluate the list of collected tax values

1. On **Format designer**, on the **Format** tab, select the **Report\Lines\Record\RunningTotal** numeric format element, and then follow these steps:
   a. In the **Numeric type** field, change the value from **Real** to **Integer**.
   b. In the **Numeric format** field, change the value from **F2** to **F0**.

2. On the **Mapping** tab, select **Edit formula**.

3. On **Formula designer**, follow these steps:

a. In the **Formula** field, enter `COUNT(CollectedTaxValues.Result)`.

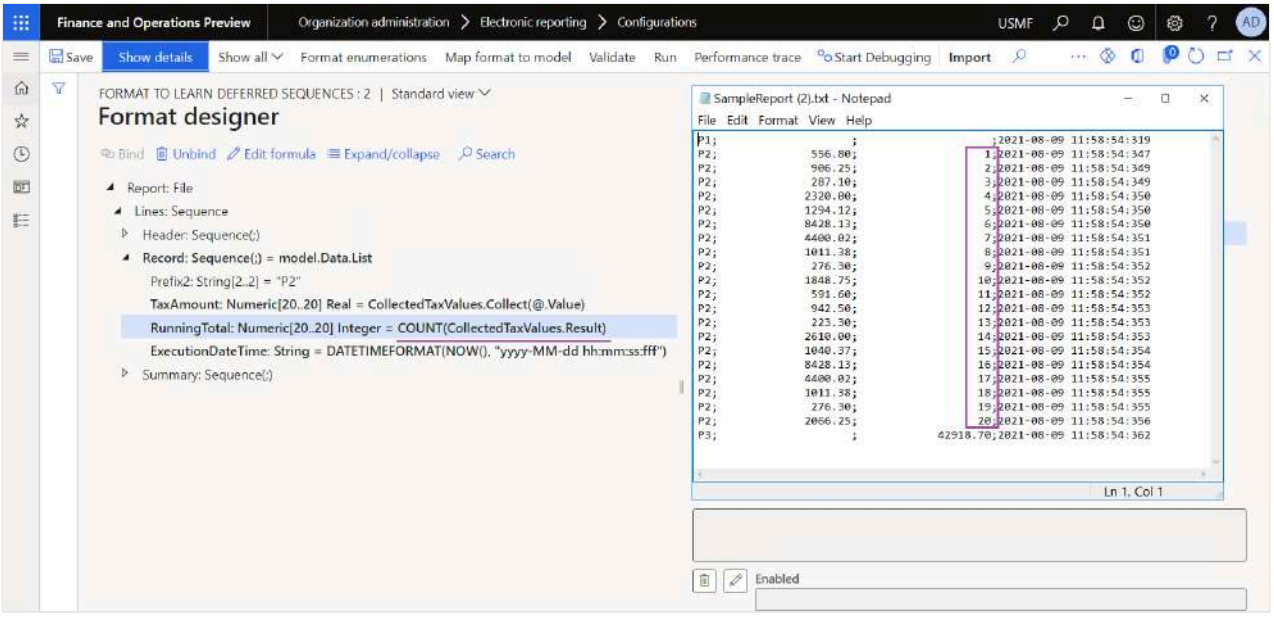
b. Save your changes, and close the page.

> **Note**
>
> The new binding passes the number of records in the list where the tax values are collected to a generated document.

4. Select **Save**, and then select **Run**.

5. In the **Electronic report parameters** dialog box, select **OK**.

6. Download and review the file that the web browser offers.



Frequently asked questions

If I need to calculate running totals and collect data, what's the difference between using a DATA COLLECTION data source and using the built-in DATA COLLECTION functions?

You can use both a DATA COLLECTION data source and the built-in DATA COLLECTION functions for data collection, summing, and counting, based on information that is passed to a generated outbound document. However, consider the following points when deciding which technique to use.

Expand table

Data source	Built-in functions
Only values are collected.	Named values are collected. Therefore, totals can be computed for separate groups of values. Additionally, groups can be extracted as a list. Text values can also be collected.
Unique values are automatically collected.	Additional settings are required to extract a list of unique values from the collected values.
Performance depends on the volume of collected values.	In practice, performance doesn't depend on the volume of collected values.
This technique works for all types of outbound documents.	This technique works only for text and XML documents.

Additional resources

- Configure format to do counting and summing
- Defer the execution of sequence elements in ER formats

Last updated on 04/13/2026

Use USER INPUT PARAMETER data sources to specify parameters for a report

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you design Electronic reporting (ER) model mapping and ER format components, use data sources of the *USER INPUT PARAMETER* type to get the values you need. You can specify these values in data entry fields in the dialog box at runtime, before execution of an ER format begins. This article describes the *USER INPUT PARAMETER* data sources that are currently supported.

Mandatory properties

Specify the following properties for data sources of every *USER INPUT PARAMETER* type:

- In the **Name** field, enter the internal name of the data source. Use this name in other expressions and bindings of the configured model mapping or format component.

Optional properties

Optionally, specify the following properties for data sources of a *USER INPUT PARAMETER* type:

- In the **Label** field, enter the label for the related data entry field in the dialog box at runtime. To add different label text for different language codes, activate the **Label** field and then select **Translate**.
- In the **Help** field, enter the help text that appears at design time at the bottom of the **Format designer** page or the **Model mapping designer** page when you select an editable data source of a *USER INPUT PARAMETER* type. This text might provide extra details about the data source to help users when they configure the editable format or model mapping component. To add different help text for different language codes, select **Translate**.

> **Note**
>
> The **Translate** button that you use to add language specific labels and text becomes available only after you add the data source, save your changes, and then reopen the data source for editing.

- In the **Read only** field, configure an expression that returns a *Boolean* value.
  - If the configured expression returns **True** at runtime, the related data entry field appears dimmed in the dialog box, and you can't change its value.
  - If the configured expression returns **False** at runtime, or if you don't configure an expression, the related data entry field is available in the dialog box, and you can change its value.

- In the **Default value** field, configure an expression that returns the value of the referenced parameter type. This value fills in a default value for the related data entry field in the dialog box at runtime.

  When you run an ER format, the value that you enter in the related data entry field in the dialog box at runtime is saved in memory as the previously used value. The system saves previously used values individually for each field, running ER format, current user, and current organization (company).

  - Set the **Always reset to default value** option to **Yes** if the value that the **Default value** expression returns should always be used as the default value, regardless of the previously used value.

  - Set the **Always reset to default value** option to **No** if the value that the **Default value** expression returns should be used as the default value only when the previously used value is missing.

    > **Note**
    >
    > If you set the **Always reset to default value** option to **Yes**, you must configure an expression in the **Default value** field.

- If you set the **Allow multiple selection** option to **Yes**, you can select multiple values for the configured parameter at runtime. If you set it to **No**, you can select only a single value.

> **Note**
>
> This option isn't applicable to all *USER INPUT PARAMETER* types. At design time, an exception is thrown to inform the user that the configured user input parameter doesn't support multiple selection, and that only a single value can be selected or entered.
>
> If you set the **Allow multiple selection** option to **Yes**, and you specify an expression in the **Default value** field, that expression can set only a single default value.

- Select the **Edit visibility** option to specify whether the configured parameter should be shown in the dialog box at runtime.

  > **Note**
  >
  > The default visibility of data sources of a *USER INPUT PARAMETER* type depends on the ER component that holds them.
  > - If you configure a data source in the format component, it's visible by default.
  > - If you configure a data source in the model mapping component, it's visible only if the value of the data source affects the outcome when an ER component is run. For example, you added a data source but didn't use it in expressions and bindings of the current model mapping component. In this case, by default, the relevant data entry field isn't shown in the dialog box at runtime.

  On the **Formula designer** page, in the **Formula** field, configure an expression that returns a *Boolean* value.
  - If the configured expression returns **True** at runtime, or if you don't configure an expression, the related data entry field is visible in the dialog box at runtime.
  - If the configured expression returns **False**, the related data entry field is hidden in the dialog box at runtime. When other expressions call it at runtime, it returns the default value, the previously used value, or the default for the current data type value, depending on other settings.

Type-specific properties

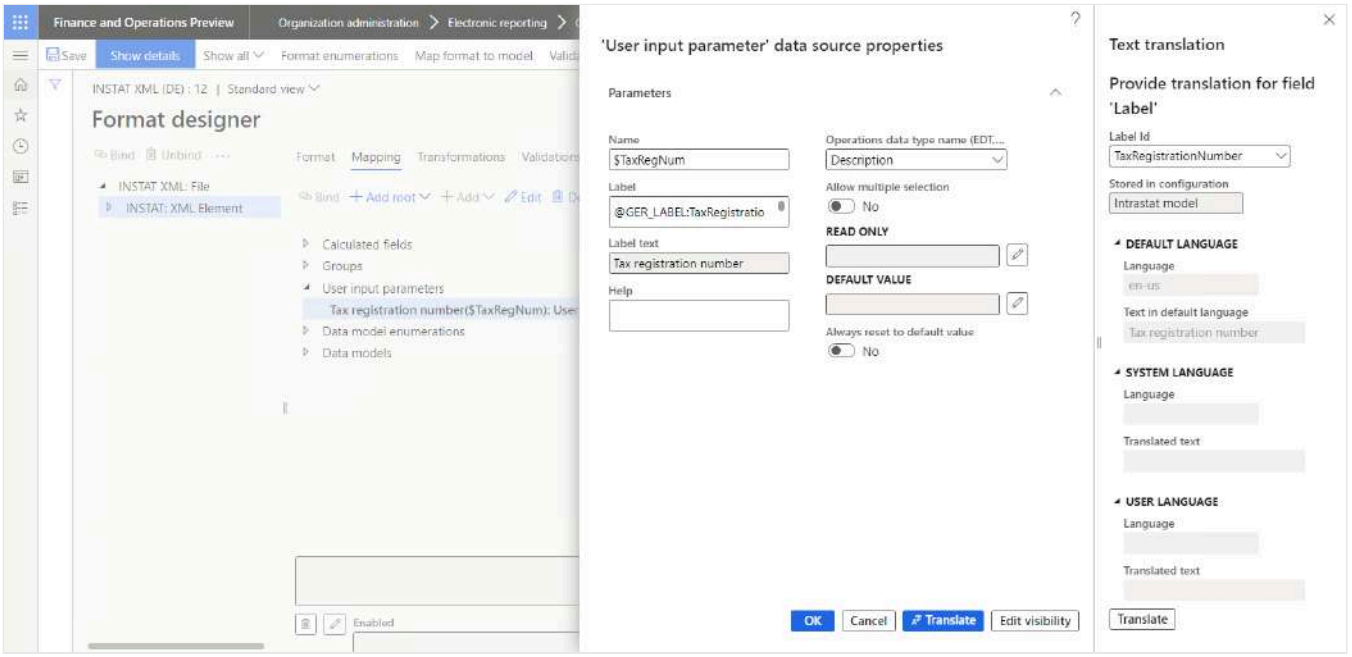
Application-dependent user input parameter

Use a data source of the **General** > **User input parameter** type to get the required value or values of a data type that you specify for the current instance of the Microsoft Dynamics 365 Finance application. When you add a data source of this type to an ER component, specify the following properties:

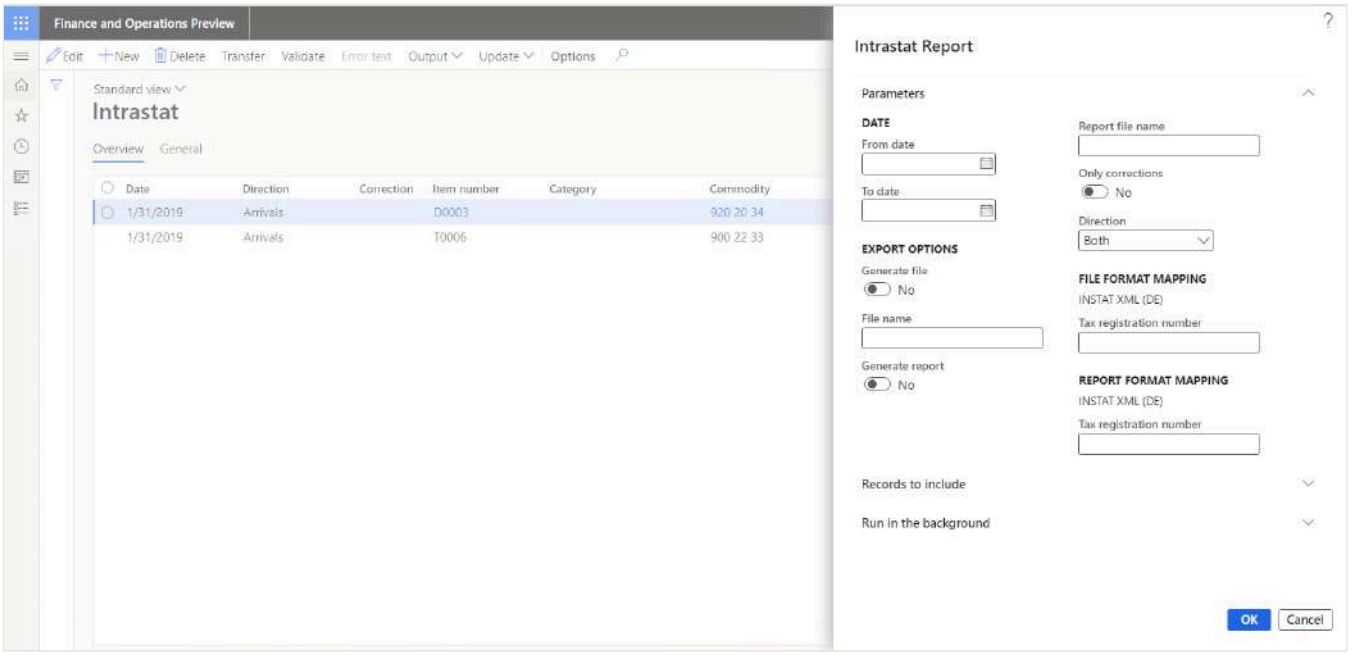
- In the **Operations data type name (EDT, enum)** field, select an application extended data type (EDT) or an application enumeration.

> **Note**
>
> Review the expressions that are configured in the **Read only** and **Default value** fields when you change the **Operations data type name (EDT, enum)** reference in an editable data source of this *USER INPUT PARAMETER* type.

The following illustration shows properties of the `$TaxRegNum` data source that you configure in the **Instat XML (DE)** ER format configuration. This data source uses the *Description* EDT to offer the **Tax registration number** data entry field in the dialog box at runtime.



The following illustration shows the dialog box that appears at runtime when you run the **Instat XML (DE)** ER format configuration to generate the Intrastat declaration. The configured **Tax registration number** field is available for data entry.



Data model enumeration user input parameter

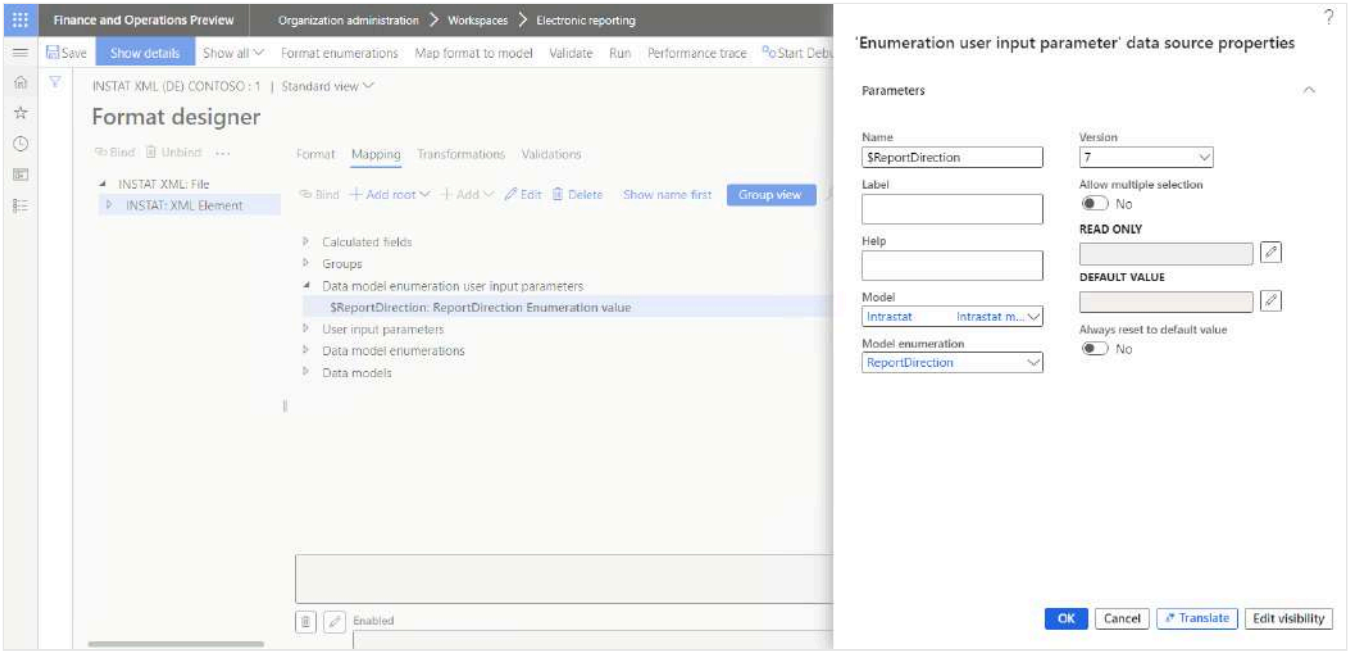
Use a data source of the **Data model** > **Enumeration user input parameter** type to get the required value or values for a single data model enumeration. When you add a data source of this type to an ER component, specify the following properties:

- In the **Model** field, specify a reference to the base data model.
- In the **Model enumeration** field, specify a reference to an enumeration of the referenced data model.
- In the **Version** field, select the revision number of the ER data model component that contains the referenced model enumeration.

> **Tip**
>
> At design time, you can leave the **Version** field blank to access the list of enumerations for the referenced data model component that resides in the draft version of the corresponding ER data model configuration. By using this approach, you can simultaneously edit the draft version of your model mapping or format component and the draft version of the base data model component.
>
> However, you can leave the **Version** field blank only in the draft version of a model mapping or format component. When you change the status of an ER model mapping or format configuration from **Draft** to **Completed**, the highest model revision number that's available in the current Finance instance automatically fills in this field. If you introduce a new enumeration or a new enumeration value in the

draft version of your base data model and refer to it in the editable model mapping or format component, complete that draft version of the base data model configuration before the draft version of your ER model mapping or format configuration is completed. Otherwise, a "Path not found" exception is thrown when you change the status of the model mapping or format configuration from **Draft** to **Completed**. The message informs you that the referenced enumeration or enumeration value is missing in the base data model.

The following illustration shows properties of the `$ReportDirection` data source that you configured in the **Instat XML (DE) Contoso** ER format configuration. The **Instat XML (DE) Contoso** configuration is derived from the **Instat XML (DE)** configuration. This data source is configured to use the *ReportDirection* model enumeration to offer the appropriate lookup field in the dialog box at runtime.



Format enumeration user input parameter

Use a data source of the **Format enumeration** > **Enumeration user input parameter** type to get the required value or values for a single format enumeration. When you add a data source of this type to an ER component, specify the following property:

- In the **Format enumeration** field, specify an enumeration of the editable format.

> **Note**

You can configure data sources of this type only in the scope of the editable format component.

Additional resources

Formula designer in Electronic reporting

Initiate data source values of the USER INPUT PARAMETER type from source code

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Last updated on 04/13/2026

Debug data sources of an executed ER format to analyze data flow and transformation

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you configure an Electronic reporting (ER) solution to generate outbound documents, you define the methods that get data out of the application and enter it in the generated output. To make the life cycle support of the ER solution more efficient, your solution should consist of an ER data model and its mapping components, and also an ER format and its mapping components. The model mapping is application-specific, whereas other components remain application-agnostic. Therefore, several ER components might affect the process of entering data in the generated output.

Sometimes, the data in the generated output looks different from the same data in the application database. In these cases, you want to determine which ER component is responsible for the data transformation. The ER data source debugger feature significantly reduces the time and cost that are involved in this investigation. You can interrupt the execution of an ER format and open the data source debugger interface. There, you can browse the available data sources and select an individual data source for execution. This manual execution simulates the execution of the data source during the real run of an ER format. The result is presented on a page where you can analyze the data that is received.

To turn on the data source debugging feature, set the **Enable data debugging at format run** option to **Yes** in the ER user parameters. You can then start data source debugging while you run an ER format to generate outbound documents. You can also use the **Start debugging** option to initiate data source debugging for an ER format that is configured in the ER Operation designer.

This article provides guidelines for initiating data source debugging for executed ER formats. It explains how the information can help you understand the data flow and data transformations. The examples in this article use the business process for vendor payments processing.

Limitations

You can use the data source debugger to access the data of data sources that are used in ER formats that run to generate outbound documents. You can't use it to debug data sources of ER formats that are designed to parse inbound documents.

The following settings of ER formats aren't currently accessible for data source debugging:

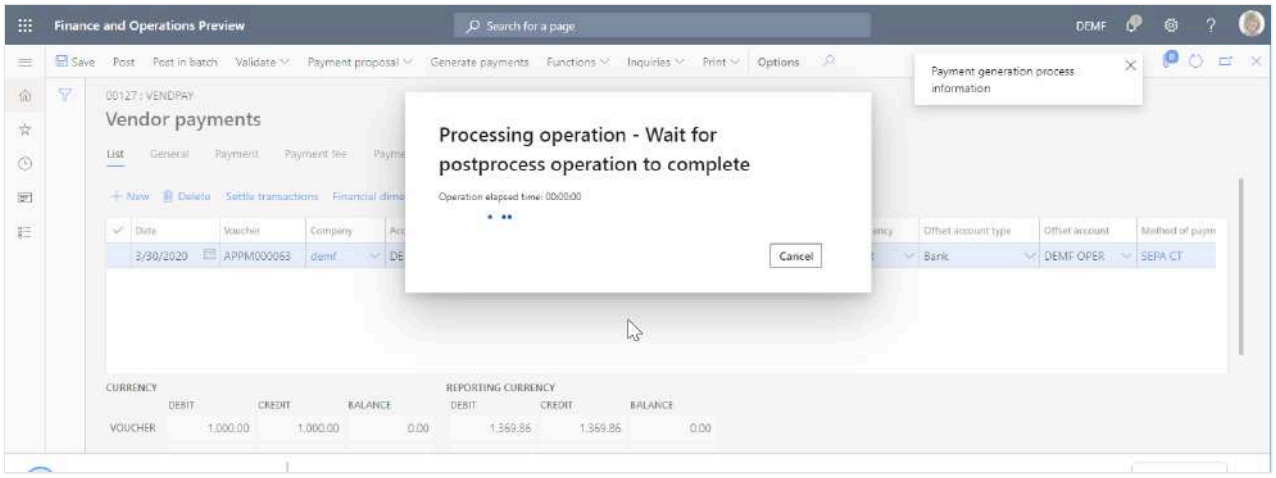
- Format transformations
- Format and mapping validation rules
- Enabling expressions
- Details of in-memory data collection

Prerequisites

- To complete the examples in this article, you must have access to one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- Set the company to **DEMF**.
- Follow the steps in Appendix 1 of this article to download the components of the Microsoft ER solution that are required to process vendor payments.
- Follow the steps in Appendix 2 of this article to prepare Accounts payable for vendor payment processing by using the ER solution that you download.

Process a vendor payment to get a payment file

1. Follow the steps in Appendix 3 of this article to process vendor payments.



2. Download and save the ZIP file to your local computer.

3. Extract the **ISO20022 Credit transfer.xml** payment file from the ZIP file.

4. Open the extracted payment file by using the XML file viewer.

   In the payment file, the International Bank Account Number (IBAN) code of the vendor bank account contains no spaces. Therefore, it differs from the value that was entered on the **Bank accounts** page.

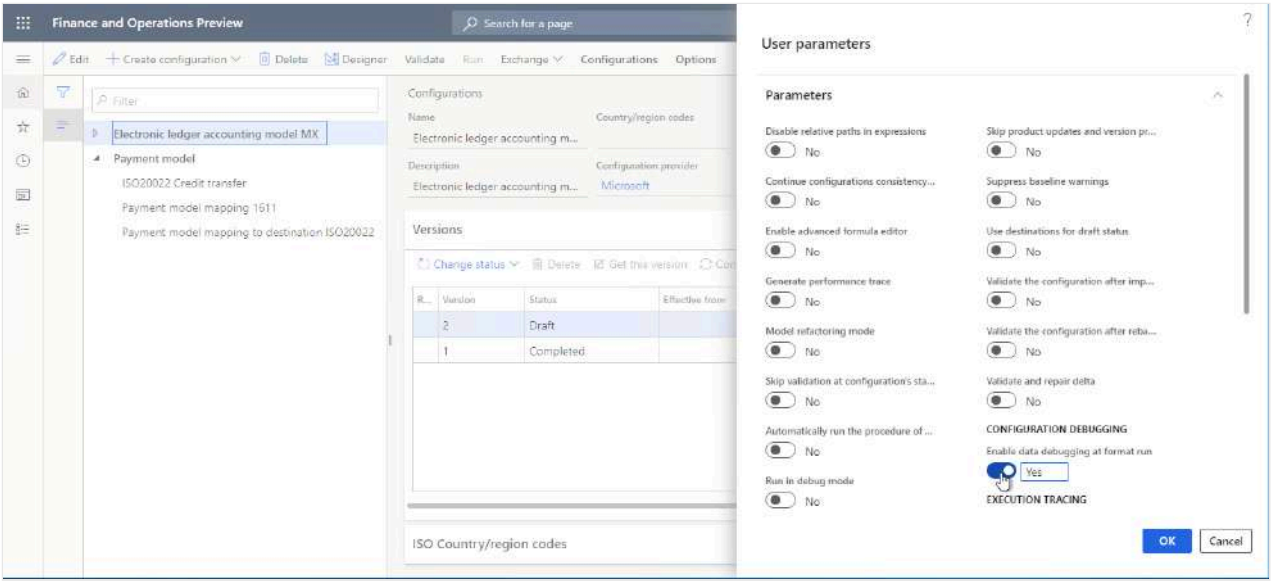
```
<?xml version="1.0" encoding="UTF-8"?>
- <Document xsi:schemaLocation="urn:iso:std:iso:20022:tech:xsd:pain.001.001.03 pain.001.001.03.xsd"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="urn:iso:std:iso:20022:tech:xsd:pain.001.001.03">
  - <CstmrCdtTrfInitn>
    - <GrpHdr>
        <MsgId>00127-20200331082108</MsgId>
        <CreDtTm>2020-03-31T08:21:08</CreDtTm>
        <NbOfTxs>1</NbOfTxs>
        <CtrlSum>1000</CtrlSum>
      + <InitgPty>
      </GrpHdr>
    - <PmtInf>
        <PmtInfId>000127020200330-05</PmtInfId>
        <PmtMtd>TRF</PmtMtd>
        <BtchBookg>false</BtchBookg>
        <NbOfTxs>1</NbOfTxs>
        <CtrlSum>1000</CtrlSum>
        <ReqdExctnDt>2020-03-30</ReqdExctnDt>
      + <Dbtr>
      + <DbtrAcct>
      + <DbtrAgt>
      - <CdtTrfTxInf>
        + <PmtId>
        - <Amt>
            <InstdAmt Ccy="EUR">1000</InstdAmt>
          </Amt>
        - <CdtrAgt>
          - <FinInstnId>
            - <ClrSysMmbId>
                <MmbId>1458</MmbId>
              </ClrSysMmbId>
              <Nm>EURBANK Munich</Nm>
            </FinInstnId>
          </CdtrAgt>
        - <Cdtr>
            <Nm>City-wide Advertising</Nm>
          - <PstlAdr>
              <Ctry>DE</Ctry>
              <AdrLine>Rebhuhnweg 561 79539 Berlin DEU</AdrLine>
            </PstlAdr>
          </Cdtr>
        - <CdtrAcct>
          - <Id>
              <IBAN>GB33BUKB20201555555555</IBAN>
            </Id>
          </CdtrAcct>
        </CdtTrfTxInf>
      </PmtInf>
  </CstmrCdtTrfInitn>
</Document>
```

Use the ER data source debugger to learn which component of the ER solution truncates spaces in the IBAN code.

Turn on data source debugging

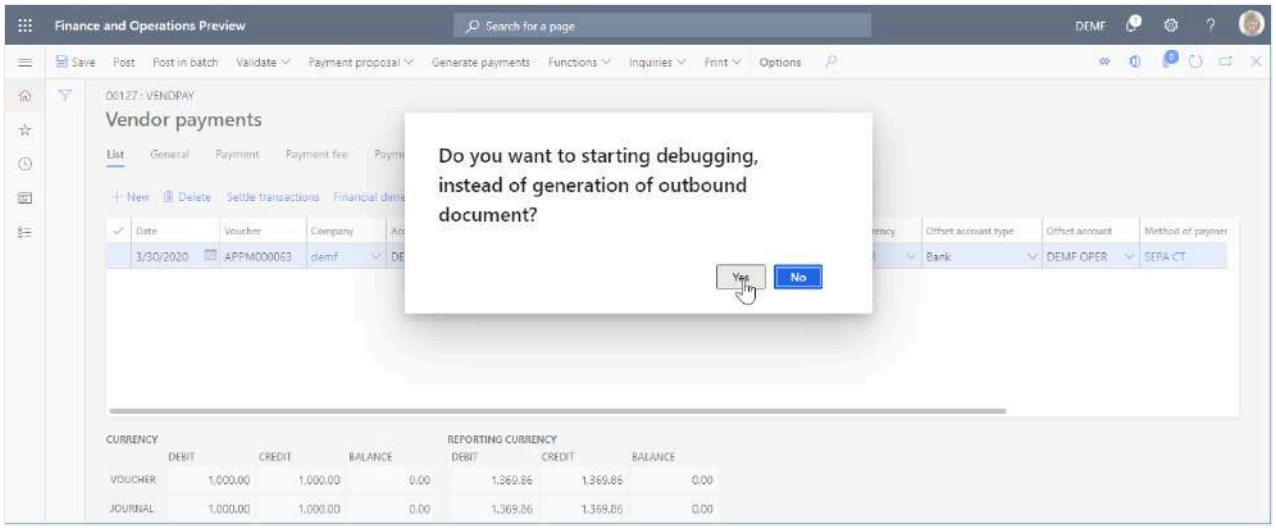
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. Set the **Enable data debugging at format run** option to **Yes**.

> **Note**
>
> This parameter is user-specific and company-specific.



Process a vendor payment for debugging

1. Follow the steps in Appendix 3 of this article to process vendor payments.
2. In the message box, select **Yes** to confirm that you want to interrupt vendor payment processing and instead start data source debugging on the **Debug Datasources** page.



Debug data sources that are used in payment processing

Debug the model mapping

1. On **Debug Datasources**, on the Action Pane, select **Model mapping** to start debugging from this ER component.

2. In the data source pane on the left, select the **$notSentTransactions** data source, and then select **Read all records**.

   To force the appropriate number of records to be read from the selected data source, select **Read 1 record**, **Read 10 records**, **Read 100 records**, or **Read all records**. By using this approach, you can simulate access to the data source from the running ER format.

3. In the data pane on the right, select **Expand all**.

   You see that the selected data source of the **Record list** type contains a single record.

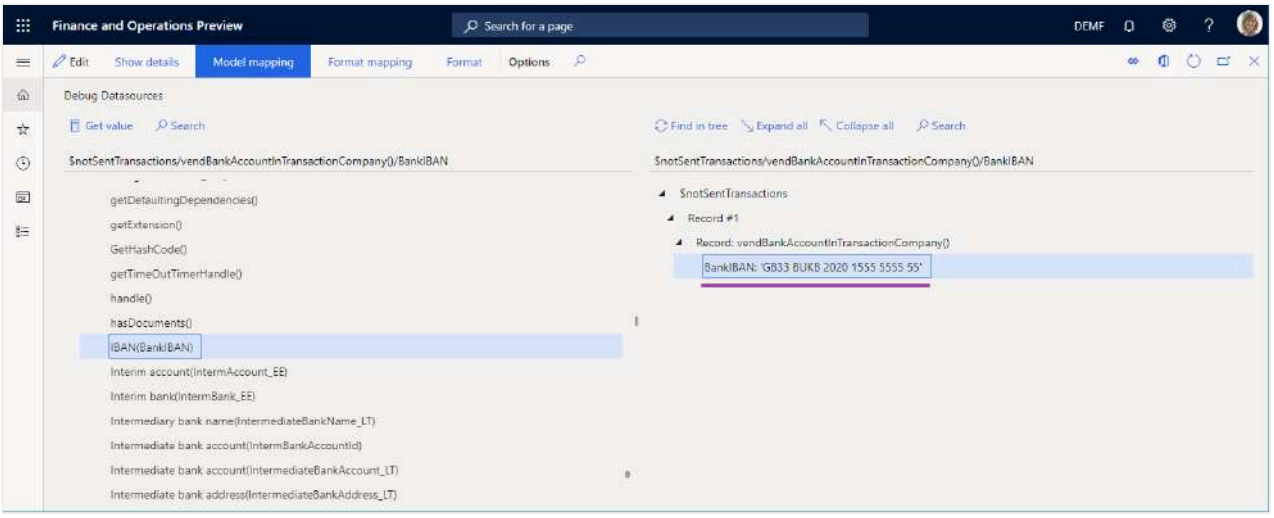
4. Expand the **$notSentTransactions** data source, and select the nested **vendBankAccountInTransactionCompany()** method.

5. Expand the **vendBankAccountInTransactionCompany()** method, and select the nested **IBAN** field.

6. Select **Get value**.

   Select **Get value** to force the value of a selected field of the selected data source to be read. By using this approach, you can simulate access to this field from the running ER format.

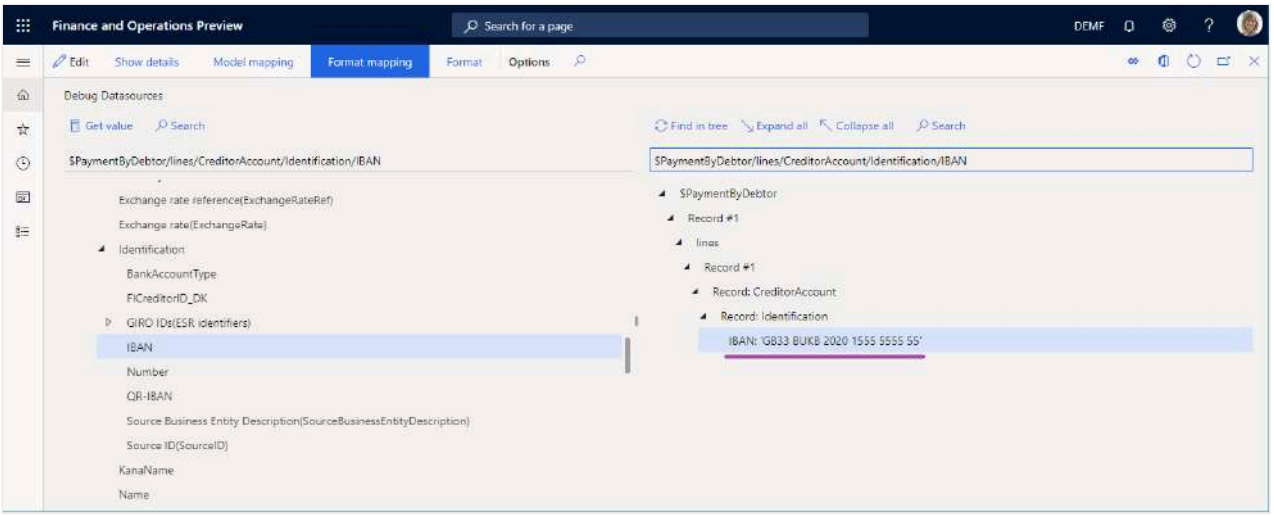
7. Select **Expand all**.



As you can see, the model mapping isn't responsible for the truncated spaces, because the IBAN code that it returns for the vendor bank account includes spaces. Therefore, you must continue data source debugging.

Debug the format mapping

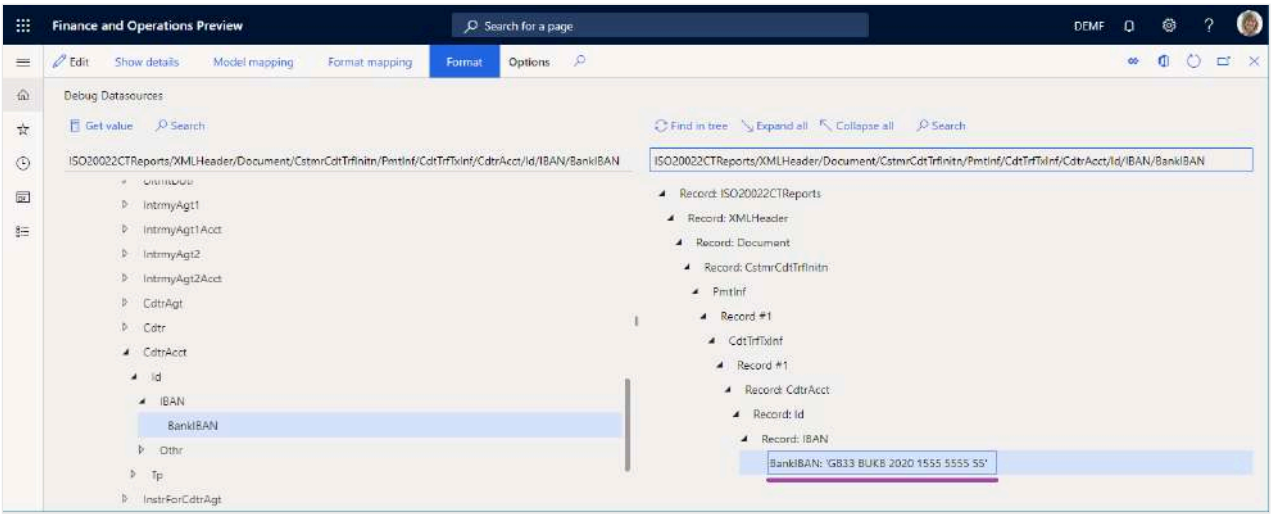
1. On **Debug Datasources**, on the action pane, select **Format mapping** to continue debugging from this ER component.
2. Select the **$PaymentByDebtor** data source, and then select **Read all records**.
3. Expand **$PaymentByDebtor**.
4. Expand **$PaymentByDebtor.Lines**, and then select **Read all records**.
5. Expand **$PaymentByDebtor.Lines.CreditorAccount**.
6. Expand **$PaymentByDebtor.Lines.CreditorAccount.Identification**, and then select **$PaymentByDebtor.Lines.CreditorAccount.Identification.IBAN**.
7. Select **Get value**.
8. Select **Expand all**.



As you can see, the data sources of the format mapping aren't responsible for the truncated spaces, because the IBAN code that they return for the vendor bank account includes spaces. Therefore, you must continue data source debugging.

Debug the format

1. On **Debug Datasources**, on the action pane, select **Format** to continue debugging from this ER component.
2. Expand the format elements to select **ISO20022CTReports** > **XMLHeader** > **Document** > **CstmrCdtTrfInitn** > **PmtInf**, and then select **Read all records**.
3. Expand the format elements to select **ISO20022CTReports** > **XMLHeader** > **Document** > **CstmrCdtTrfInitn** > **PmtInf** > **CdtTrfTxInf**, and then select **Read all records**.
4. Expand the format elements to select **ISO20022CTReports** > **XMLHeader** > **Document** > **CstmrCdtTrfInitn** > **PmtInf** > **CdtTrfTxInf** > **CdtrAcct** > **Id** > **IBAN** > **BankIBAN**, and then select **Get value**.
5. Select **Expand all**.

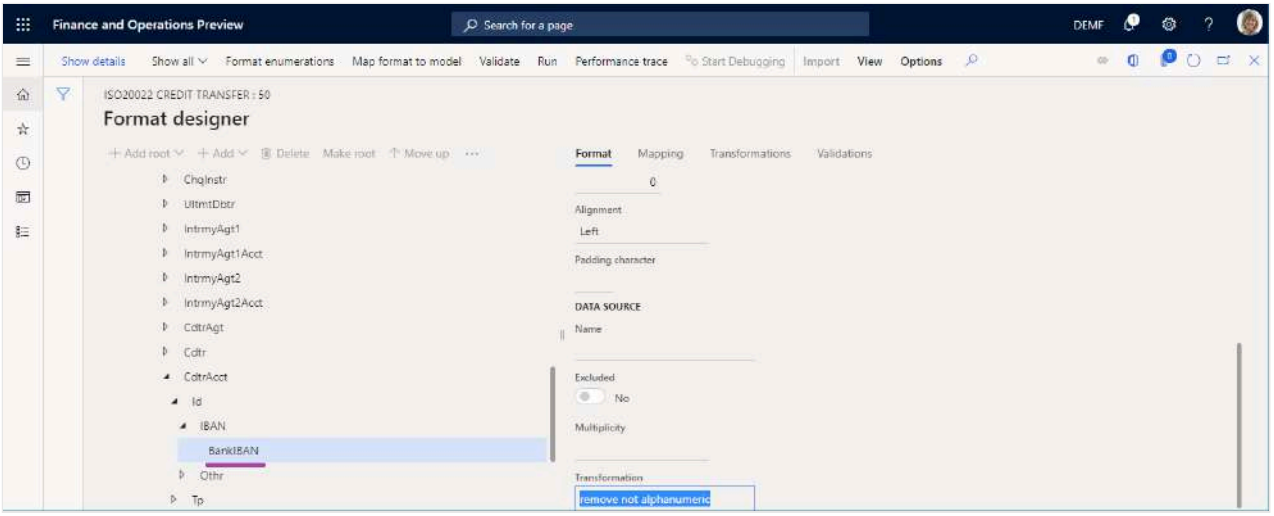


As you can see, the format binding isn't responsible for the truncated spaces because the IBAN code that it returns for the vendor bank account includes spaces. Therefore, the **BankIBAN** element is configured to use a format transformation that truncates spaces.

6. Close the data source debugger.

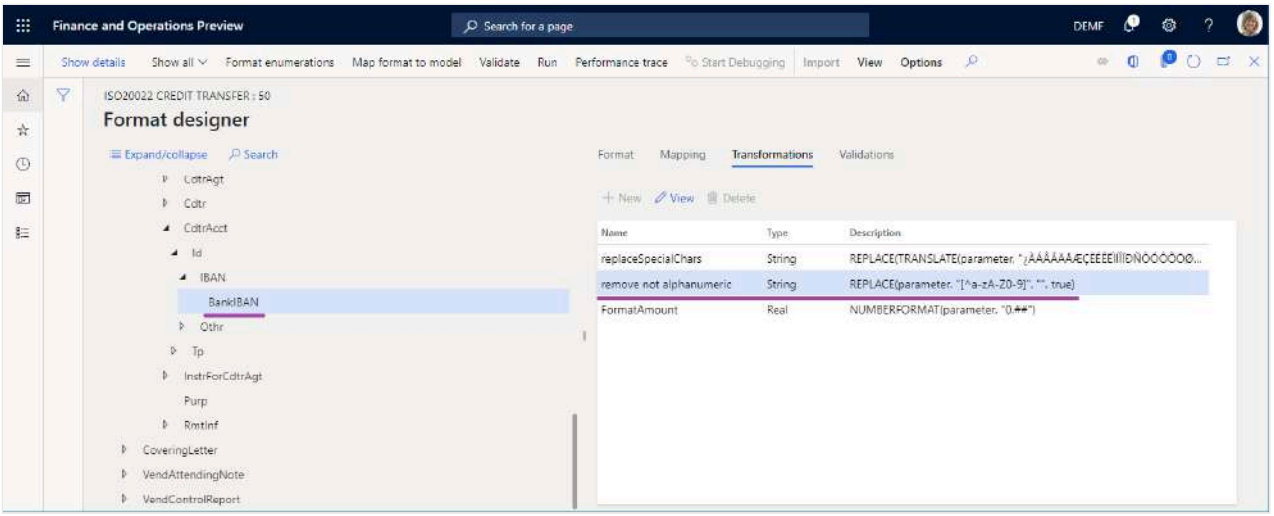
Review the format transformations

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, select **Payment model** > **ISO20022 Credit transfer**.
3. Select **Designer**, and then expand the elements to select **Document** > **CstmrCdtTrfInitn** > **PmtInf** > **CdtTrfTxInf** > **CdtrAcct** > **Id** > **IBAN** > **BankIBAN**.



As you can see, the **BankIBAN** element is configured to use the **remove not alphanumeric** transformation.

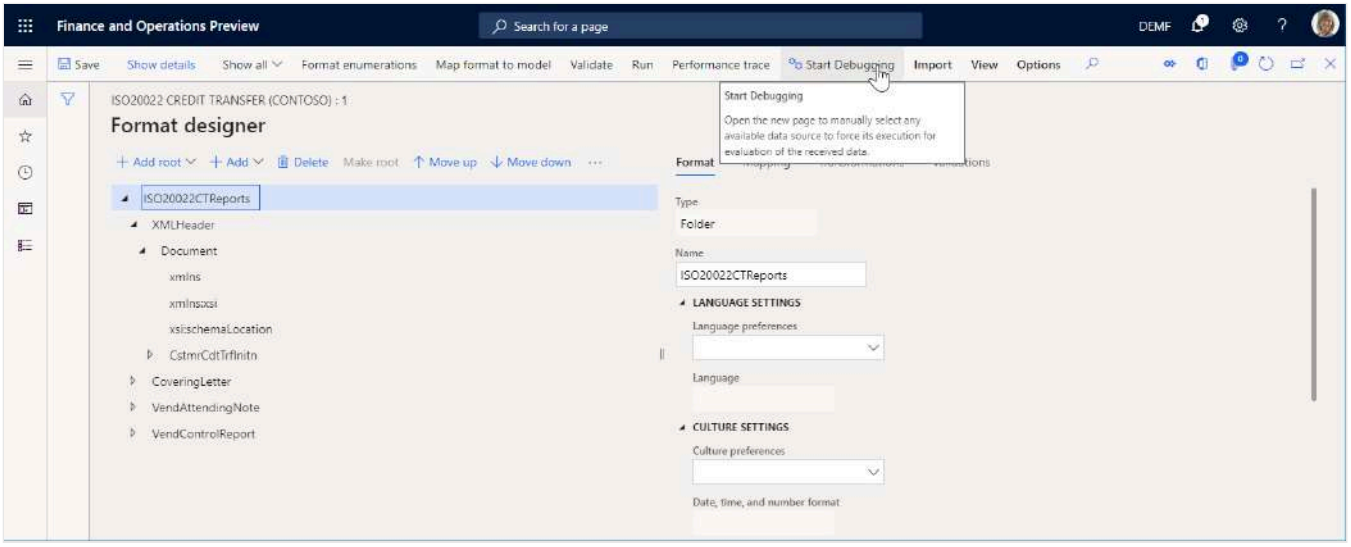
4. Select the **Transformations** tab.



As you can see, the **remove not alphanumeric** transformation uses an expression that truncates spaces from the provided text string.

Start to debug in the Operation designer

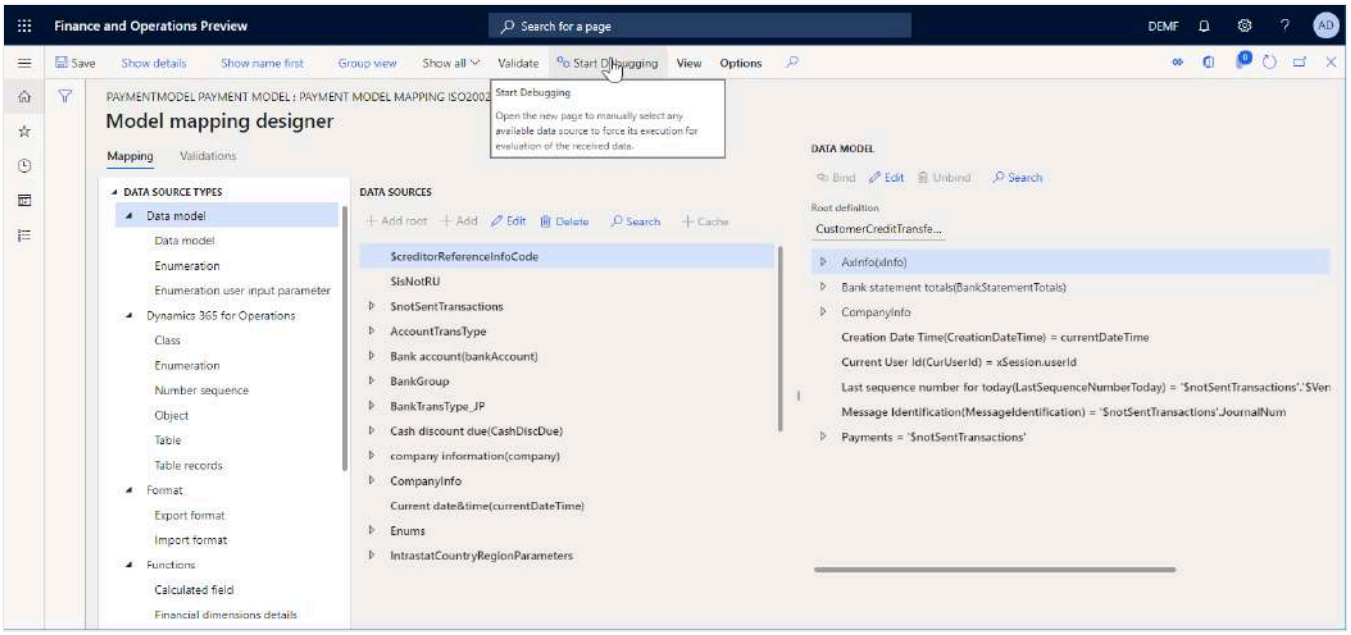
When you configure a draft version of the ER format that you can run directly from the Operation designer, you can access the data source debugger by selecting **Start Debugging** on the Action Pane.



You can debug the format mapping and format components of the ER format that you're editing.

Start debugging in the Model mapping designer

When you configure an ER model mapping that you can run from the **Model mapping** page, select **Start Debugging** on the Action Pane to access the data source debugger.

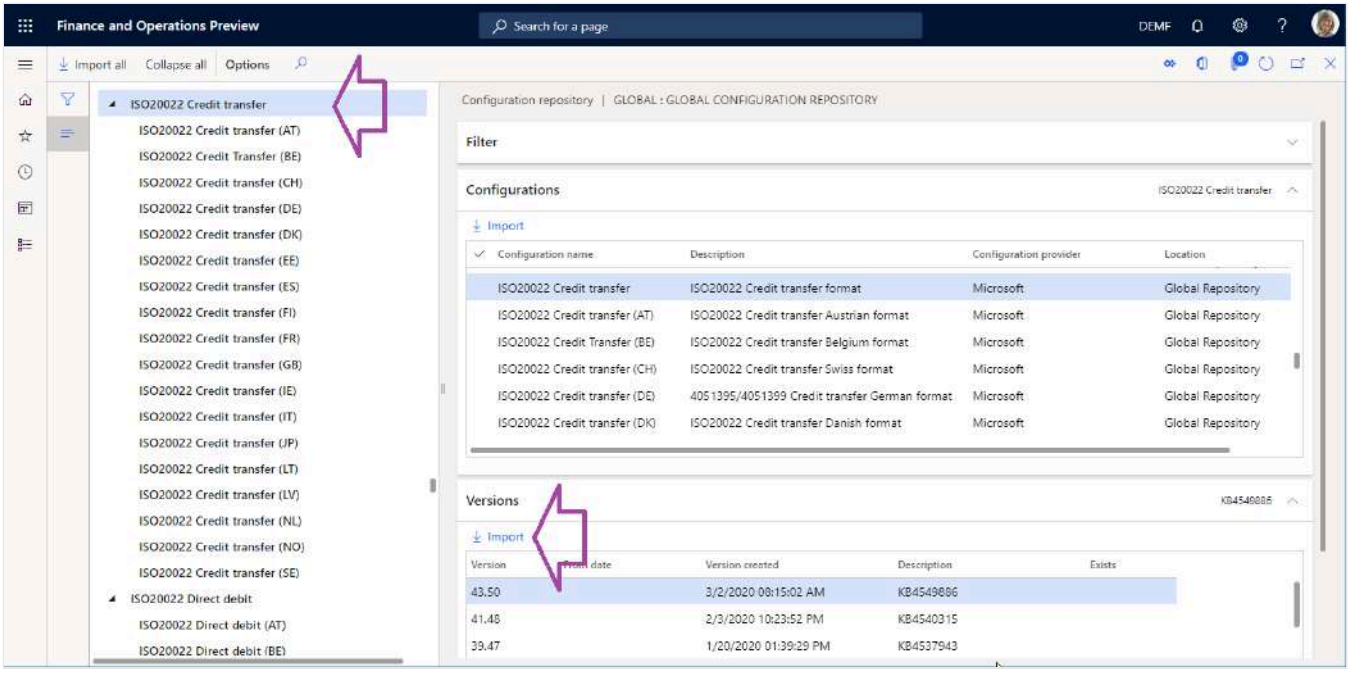


You can debug the model mapping component of the ER mapping that you're editing.

Appendix 1: Get an ER solution

Download an ER solution

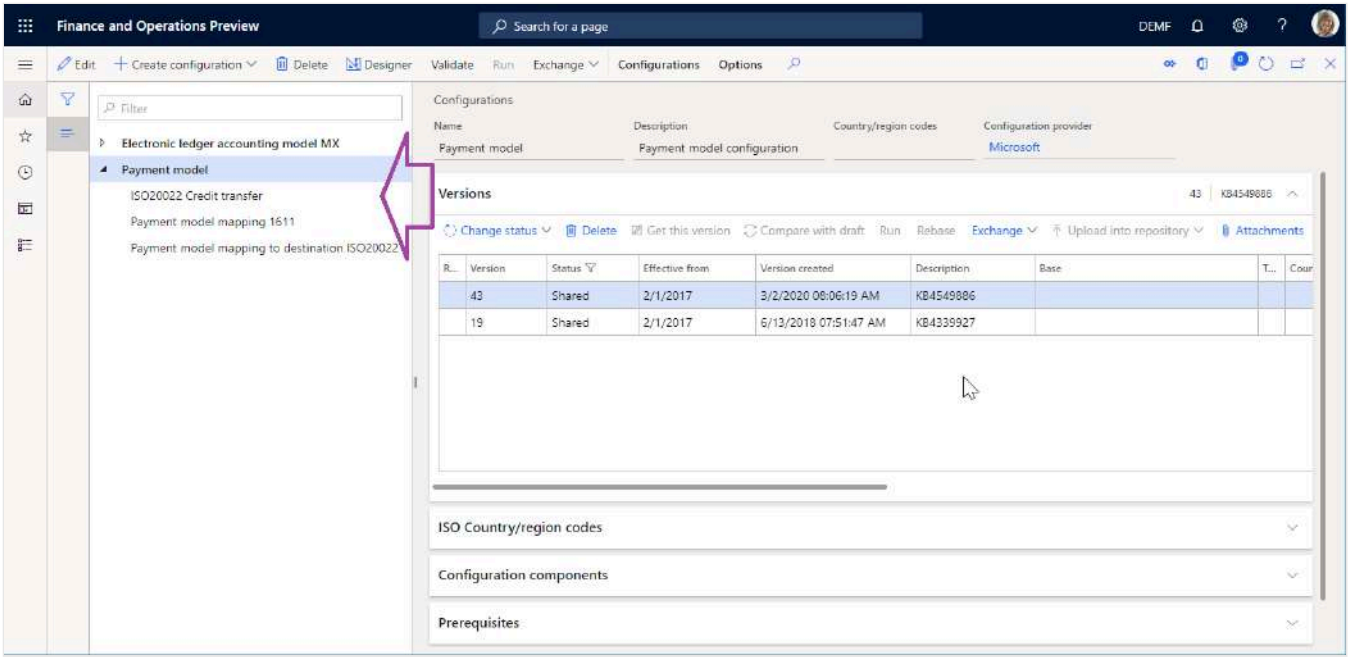
To use an ER solution to generate an electronic payment file for a processed vendor payment, download the **ISO20022 Credit transfer** ER payment format from the Shared asset library in Microsoft Dynamics Lifecycle Services (LCS) or from the Global repository.



In addition to the selected ER format, automatically import the following configurations into your Microsoft Dynamics 365 Finance instance as part of the **ISO20022 Credit transfer** ER solution:

- **Payment model** ER data model configuration
- **ISO20022 Credit transfer** ER format configuration
- **Payment model mapping 1611** ER model mapping configuration
- **Payment model mapping to destination ISO20022** ER model mapping configuration

You can find these configurations on the **Configurations** page of the ER framework (**Organization administration** > **Electronic reporting** > **Configurations**).



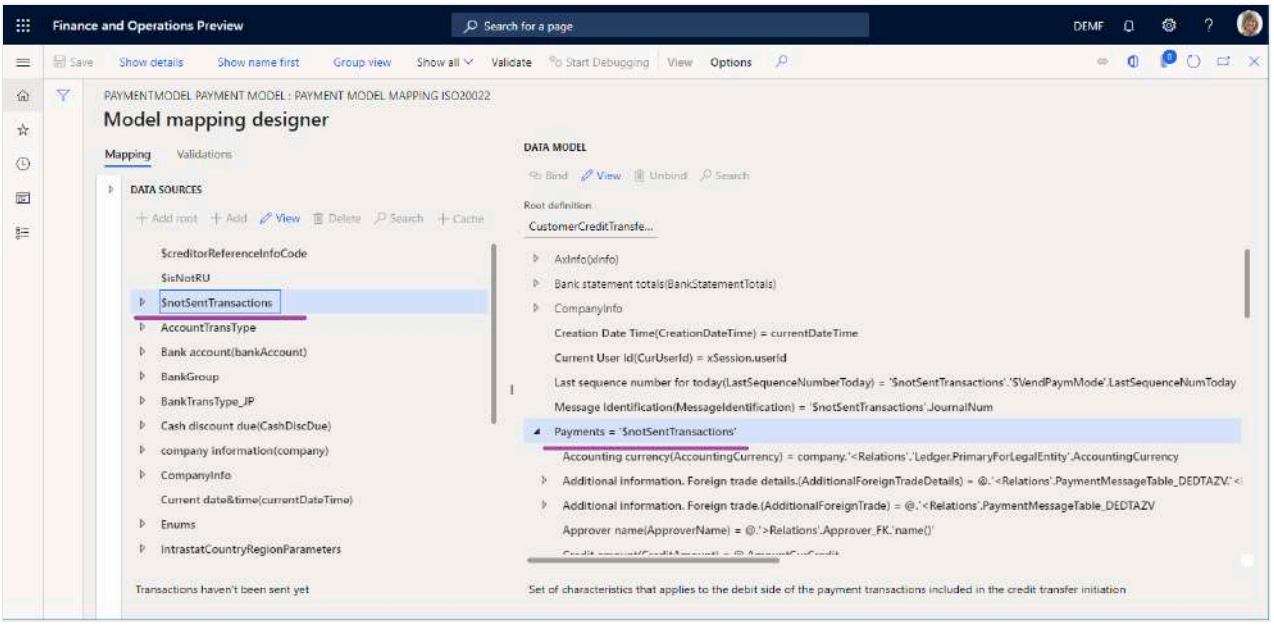
If any of the previously listed configurations are missing in the configuration tree, manually download them from the LCS Shared asset library in the same way that you downloaded the **ISO20022 Credit transfer** ER payment format.

Analyze the downloaded ER solution

Review the model mapping

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. Select **Payment model** > **Payment model mapping 1611**.
3. Select **Designer**.
4. Select the **Payment model mapping ISO20022 CT** mapping record.
5. Select **Designer**, and then review the model mapping that opens.

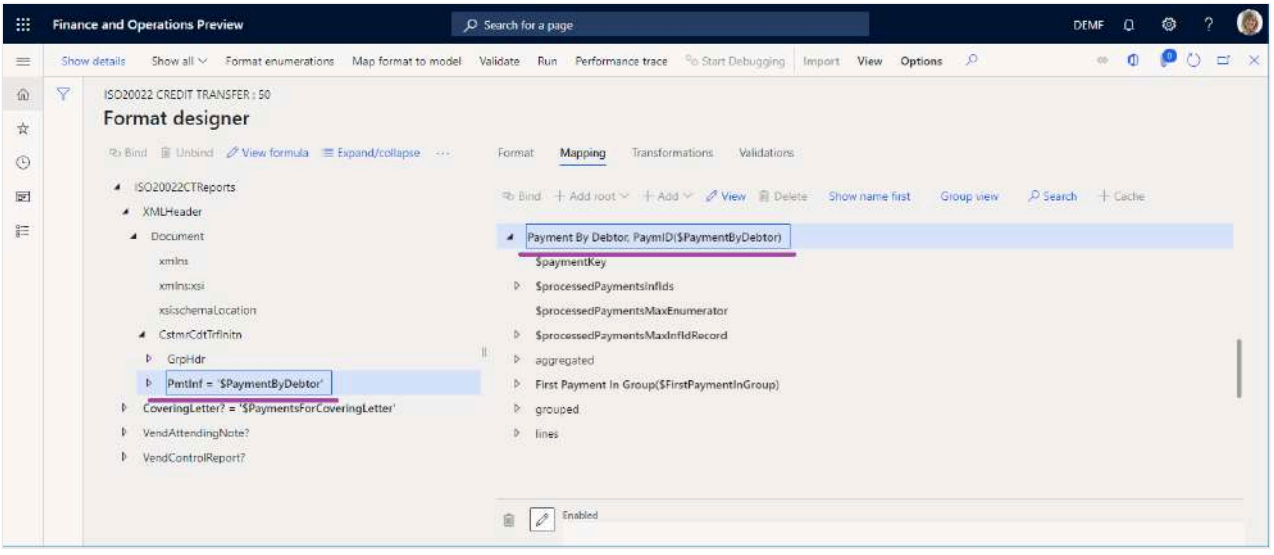
   Notice that the **Payments** field of the data model is bound to the **$notSentTransactions** data source that returns the list of vendor payment lines that are being processed.



Review the format mapping

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. Select **Payment model** > **ISO20022 Credit transfer**.
3. Select **Designer**.
4. On the **Mapping** tab, review the format mapping that opens.

   Notice that the **Document** > **CstmrCdtTrfInitn** > **PmtInf** element of the **ISO20022CTReports** > **XMLHeader** file is bound to the **$PaymentByDebtor** data source that is configured to group records of the data model's **Payments** field.



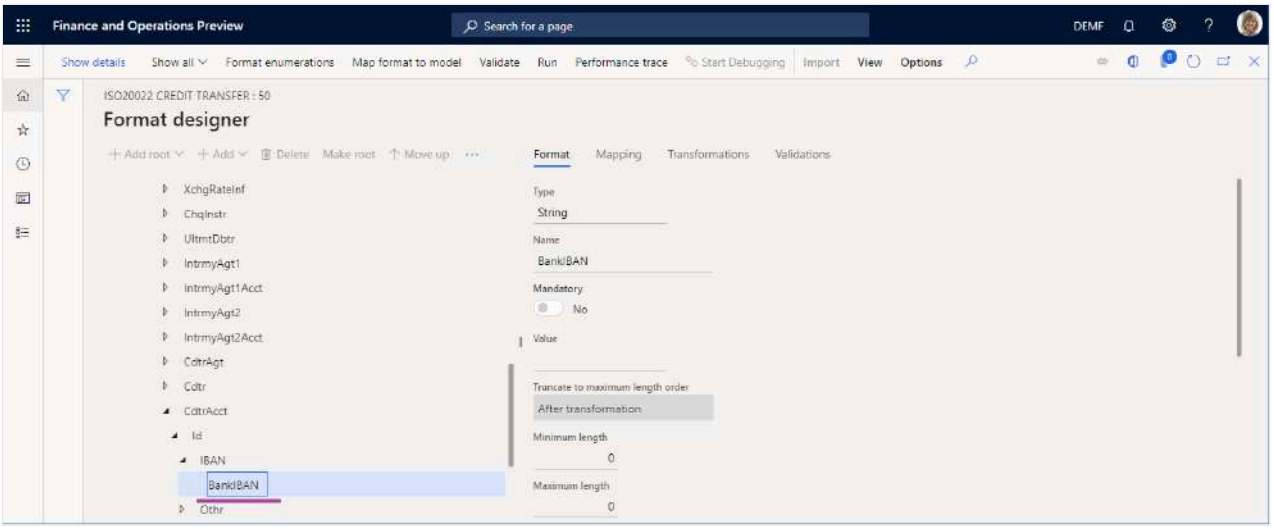
Review the format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.

2. Select **Payment model** > **ISO20022 Credit transfer**.

3. Select **Designer**, and then review the format that opens.

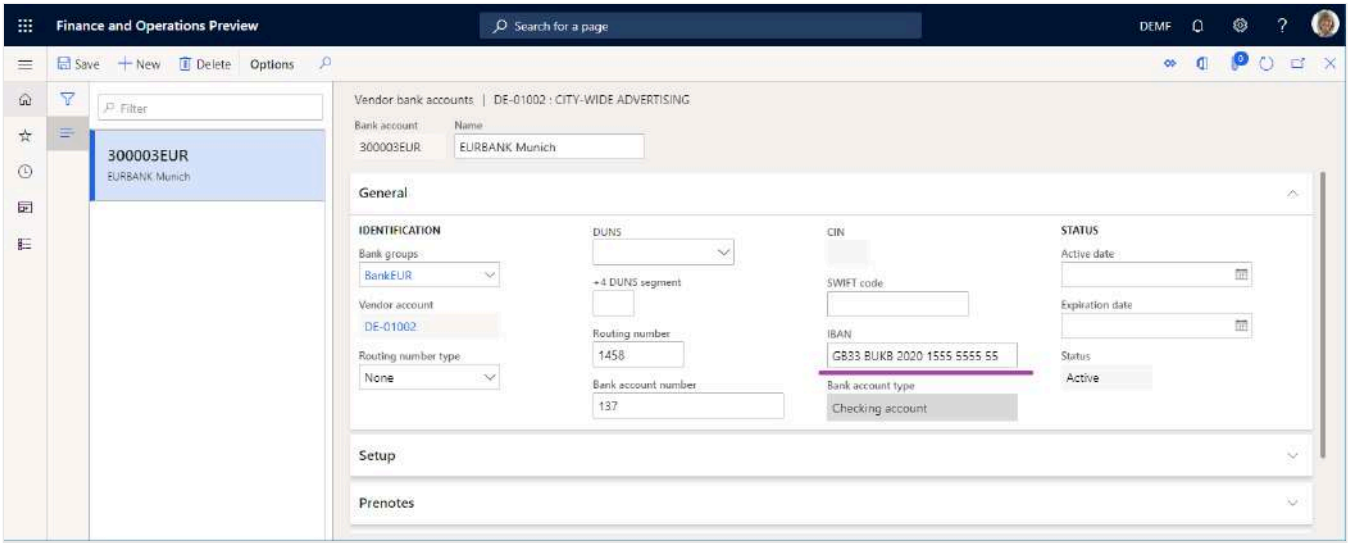
   Notice that the format element under **Document** > **CstmrCdtTrfInitn** > **PmtInf** > **CdtTrfTxInf** > **CdtrAcct** > **Id** > **IBAN** > **BankIBAN** is configured to enter the IBAN code of the vendor account in the payment file.



Appendix 2: Configure Accounts payable

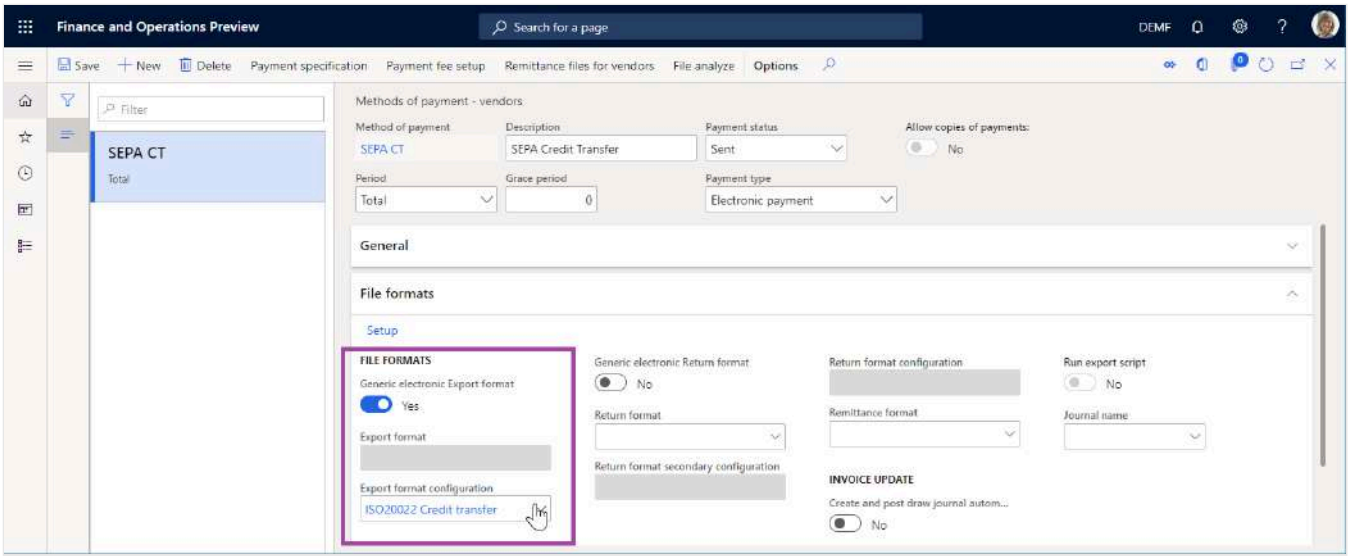
Modify a vendor property

1. Go to **Accounts payable** > **Vendors** > **All vendors**.
2. Select vendor **DE-01002**. On the Action Pane, on the **Vendor** tab, in the **Set up** group, select **Bank accounts**.
3. On the **Identification** FastTab, in the **IBAN** field, enter **GB33 BUKB 2020 1555 5555 55**.
4. Select **Save**.



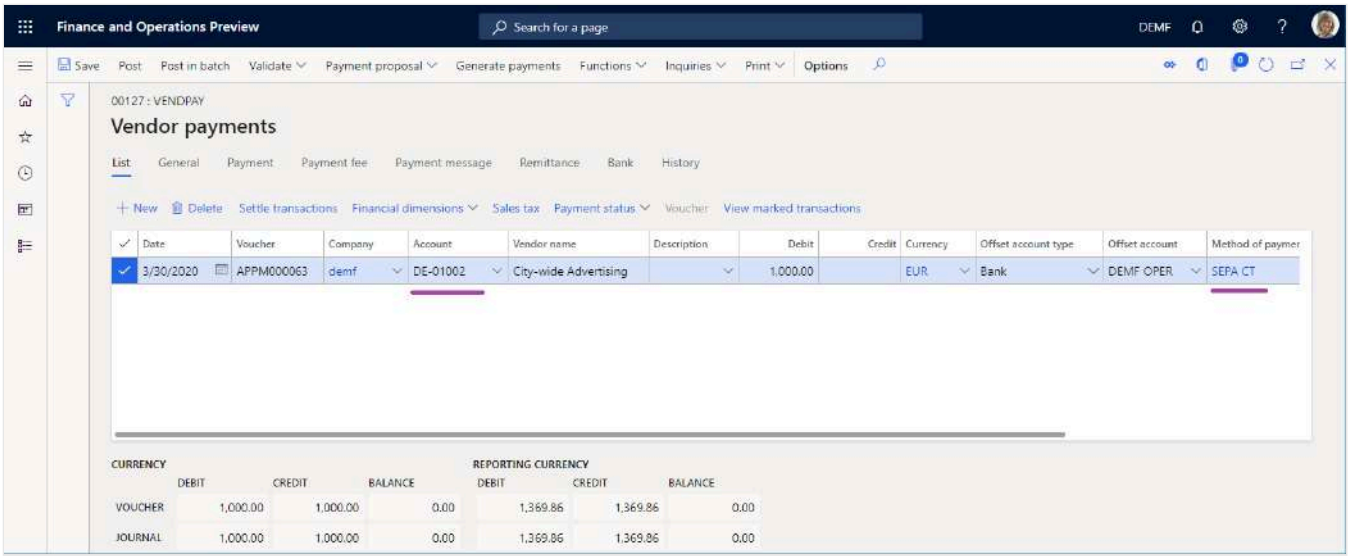
Set up a method of payment

1. Go to **Accounts payable** > **Payment setup** > **Methods of payment**.
2. Select the **SEPA CT** payment method.
3. On the **File formats** FastTab, in the **File formats** section, set the **Generic electronic Export format** option to **Yes**.
4. In the **Export format configuration** field, select the **ISO20022 Credit transfer** ER format.
5. Select **Save**.



Add a vendor payment

1. Go to **Accounts payable** > **Payments** > **Vendor payment journal**.
2. Add a new payment journal.
3. Select **Lines**, and add a new payment line.
4. In the **Account** field, select vendor **DE-01002**.
5. In the **Debit** field, enter a value.
6. In the **Method of payment** field, select **SEPA CT**.
7. Select **Save**.



Appendix 3: Process a vendor payment

1. Go to **Accounts payable** > **Payments** > **Vendor payment journal**.
2. On the **Vendor payment journal** page, select the payment journal that you previously created, and then select **Lines**.
3. Select the payment line, and then select **Payment status** > **None**.
4. Select **Generate payments**.
5. In the **Method of payment** field, select **SEPA CT**.
6. In the **Bank account** field, select **DEMF OPER**.
7. In the **Generate payments** dialog box, select **OK**.
8. In the **Electronic report parameters** dialog box, select **OK**.

Last updated on 04/13/2026

Support parameterized calls of ER data models

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To generate business documents, configure an Electronic reporting (ER) solution that contains the following ER components:

- **Format** – This component specifies the structure of a business document.
- **Format mapping** – This component controls how a business document is filled in at runtime.
- **Model mapping** – This component specifies what data pulls from the application into a format mapping.
- **Data model** – This component passes information between individual components.

When you run an ER format, every format element runs, starting from the root format element. Whenever a format element that runs contains a binding to a data source, the data source runs to deliver the expected data and uses it to fill in the format element. When a data source of the *Model* type is called, the appropriate model mapping is called to pull data out of the application, based on the model mapping settings.

Previously, you couldn't parameterize these model mapping calls to make them dependent on the logic of the format that runs. Only the following data flow was supported.

Expand table

Format	***Binding***	Format mapping	***Data model***	Model mapping	***Binding***	Table
Format element	> request >	Data source	> request >	Data source	> request >	Record
	< value <		< value <		< value <	Field

However, in version 10.0.15 and later, you can configure data model fields that can be called only when the values of the configured parameters are provided. When such a field is configured and called from a format component, the required parameters must be provided in a format binding as arguments of the call. In these cases, the values of the arguments can be specified based on format-specific values. Therefore, you can use **dynamic runtime parametrization** for data model calls to support the following data flow.

Expand table

Format	***Binding***	Format mapping	***Data model***	Model mapping	***Binding***	Tal
Format element 1	> request 1 >	Data source 1	> field1 >	Data source 1	> request 1 >	Re
	< value 1 <		< value 1 <		< value 1 <	Fie
Format element 2	> request 2 >	**value2=Func(value1)**	**> field2(value2) >**	Data source 2	> request 2 >	**Tal**
	< value 3 <		< value 3 <		< value 3 <	Re
						Fie

The new functionality lets you parametrize the call to any data model field of the *Record* or *Record list* type. The following data types are supported for the parameters of a data model field:

- Boolean
- Container
- Date
- DateTime
- GUID
- Int64
- Integer
- Real
- String

You can specify every parameter of a data model field for which the argument can be provided as a single value of the defined data type and the *list* of such values.

> **Note**
>
> The default value for the parameter of a data model field isn't supported. If you add a parameter to a field in a data model, and the version of that data model is already released and published, you must rebase all corresponding model mappings and formats to the new version of this model, because this data model change isn't backward compatible.

You can configure parametrized data model fields to make model mapping calls format-specific. This approach can help you reduce the number of model mappings that must be configured for many formats of a single data model. You can also use this approach to improve the execution performance of your formats and reduce the time that is required to generate business documents. To learn more about this feature, complete the example in this article.

Example: Use parameterized calls of ER data models

The following steps explain how a user in the System Administrator or Electronic Reporting Developer role can design an ER solution that contains a data model, a model mapping, and a format ER component. These can be configured to call a model mapping by providing an argument for the call. The value of the call is calculated at runtime by using the formula of the running format.

You can complete these steps in the **DEMF** company. No code modifications are required.

In this example, you create the required ER configurations for the **Litware, Inc.** sample company. Make sure that the configuration provider for the **Litware, Inc.** (`http://www.litware.com`) sample company is listed for the ER framework, and that it's marked as **Active**. If this configuration provider isn't listed, or if it isn't marked as **Active**, follow the steps in Create a configuration provider and mark it as active.

Business scenario

You have an ER solution that includes a format that you can run to generate an electronic document for audit purposes. This format contains tax transactions that are related to sales orders and purchase orders, and that have the required details, such as the transaction date and tax value. As of the new fiscal year, the structure of this document has changed. You must now submit an extended document that includes additional details (names) of all parties (customers and vendors) whose transactions are presented on generated reports. Therefore, you must modify your ER solution so that it generates documents that comply with this new requirement.

Configure the ER framework

Follow the steps in Configure the ER framework to set up the minimal set of ER parameters. You must complete this setup before you start to use the ER framework to design a new ER solution.

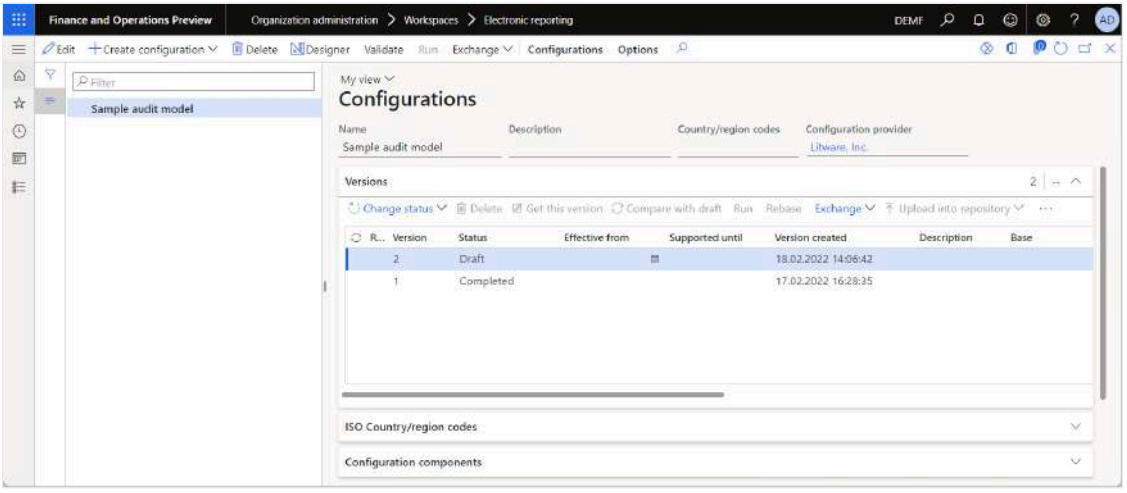
Design a domain-specific data model

Create a new ER configuration that contains the required data model component. Later, use this data model as a data source when you design an ER format to generate an audit report.

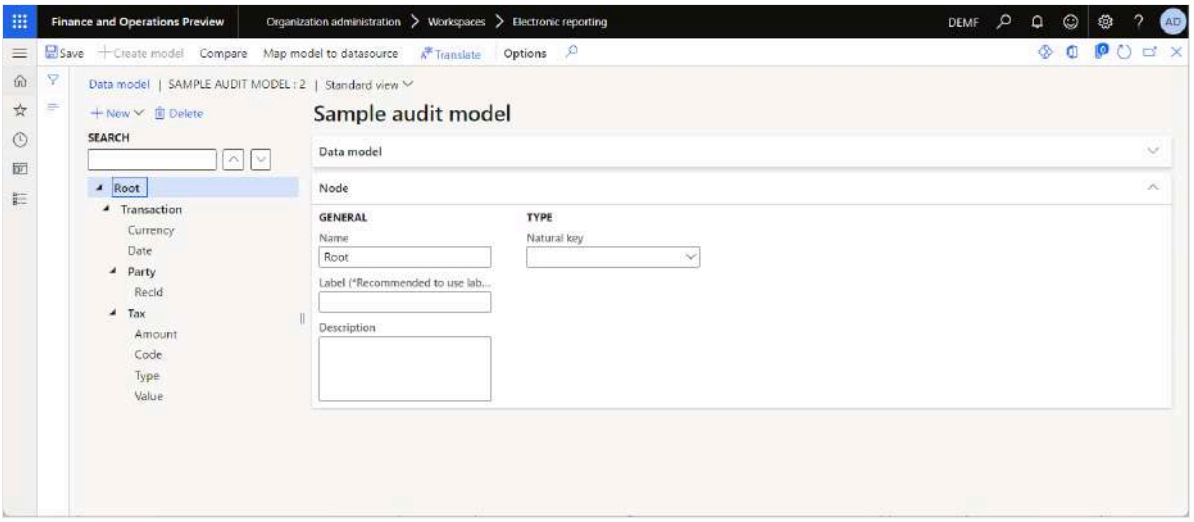
Follow these steps to import the required data model from an XML file that Microsoft provides.

1. Download the Sample audit model.version.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the **Configurations** page, on the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Sample audit model.version.1.xml** file.

6. Select **OK** to import the configuration.



The following illustration shows the editable version of the configured data model on the **Data model designer** page.



Currently, the model is designed to expose only tax transactions that have the required details.

Design a model mapping for the configured data model

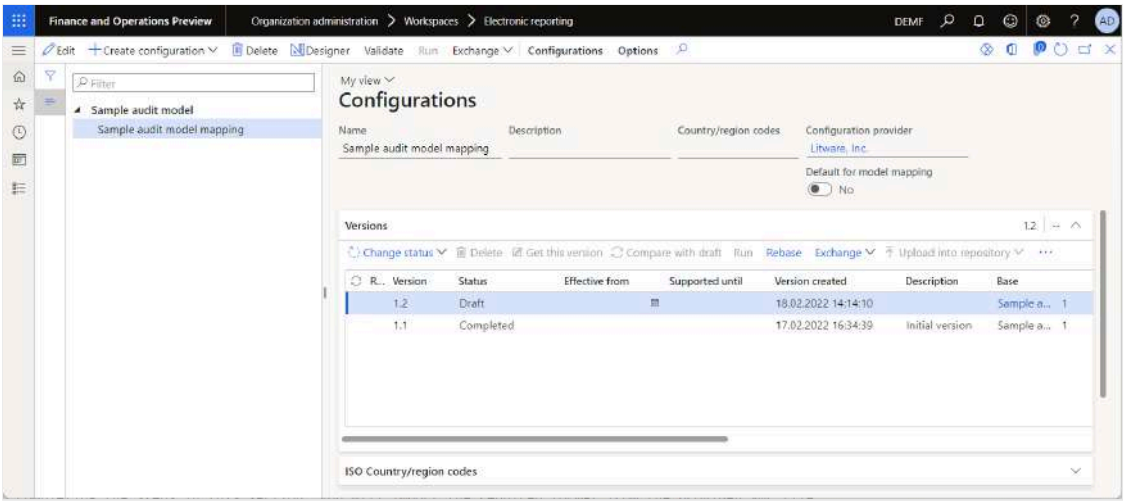
As a user in the Electronic Reporting Developer role, you must create a new ER configuration that contains a model mapping component for the Sample audit data model. This component implements the configured data model for Microsoft Dynamics 365 Finance and is specific to that app. You must configure the model mapping component to specify the application objects that fill the configured data model with application data at runtime. To complete this task, you must understand how the data structure of the tax business domain is implemented in Finance.

Follow these steps to import the required model mapping from an XML file that Microsoft provides.

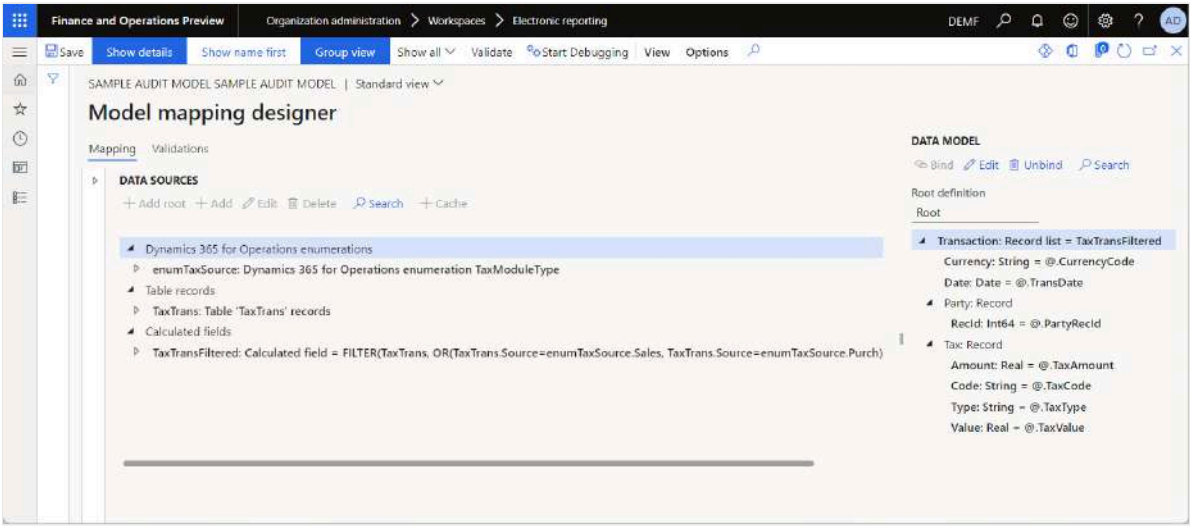
1. Download the Sample audit model mapping.version.1.1.xml file, and save it to your local computer.

2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the **Configurations** page, on the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Sample audit model mapping.version.1.1.xml** file.
6. Select **OK** to import the configuration.



The following illustration shows the editable version of the configured model mapping on the **Model mapping designer** page.



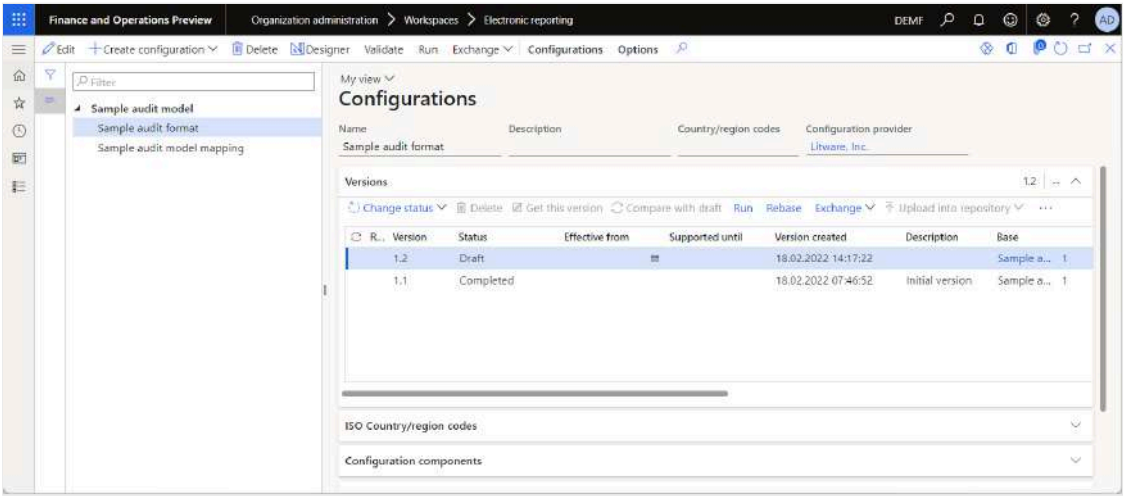
Currently, the model mapping exposes tax transactions that have the required details. It fetches this information from the `TaxTrans` application table by using the configured `TaxTrans` and `TaxTransFiltered` ER data sources.

Design a new format

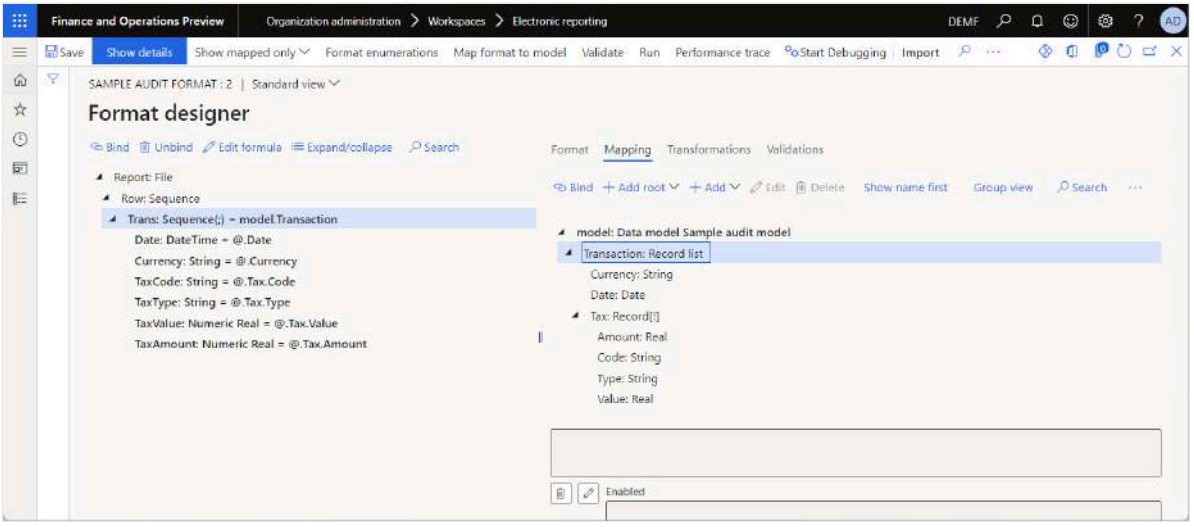
As a user in the Electronic Reporting Functional Consultant role, you must create a new ER configuration that contains a format component. You must configure the format component to fill in generated reports with tax transactions that have all the required details.

Follow these steps to import the required format from an XML file that Microsoft provides.

1. Download the Sample audit format.version.1.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the **Configurations** page, on the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Sample audit format.version.1.1.xml** file.
6. Select **OK** to import the configuration.



The following illustration shows the editable version of the configured format on the **Format designer** page.

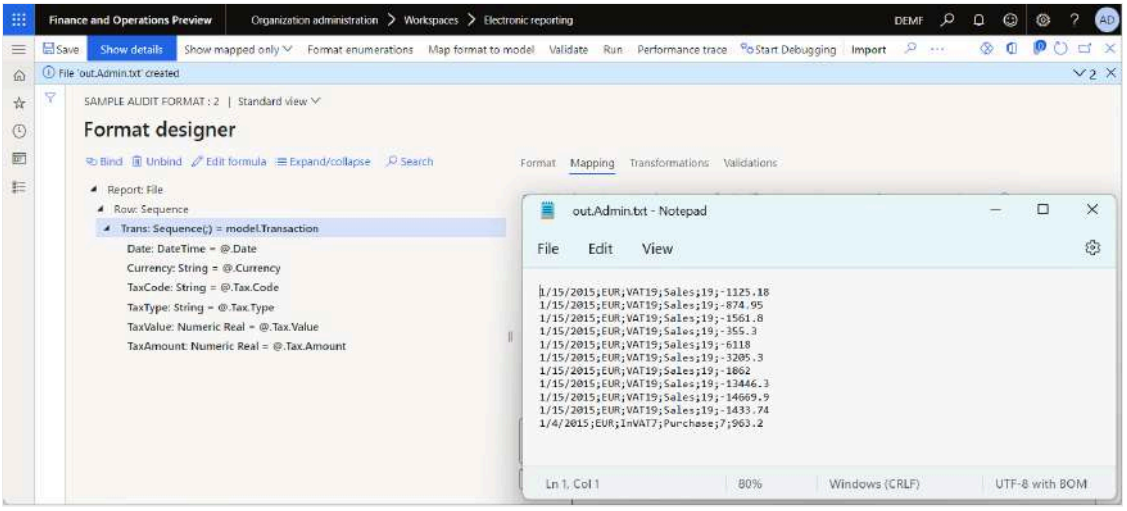


The ER format is configured to generate a report as a text file in comma-separated values (CSV) format.

Run the imported format

1. On **Configurations**, select the **Sample audit format** configuration. On the Action Pane, select **Run**.

2. In **Electronic report parameters**, on the **Records to include** tab, select **Filter**.
3. Specify the conditions to select the tax transactions for the **PIV-110000004** and **INV-10000001** vouchers.
4. Select **OK**.
5. Select **OK**.
6. Review the generated document that contains tax transactions for the selected vouchers.



Adjust the imported ER solution

According to the new requirement, the document that you must submit must contain the names of customers and vendors whose transactions are included. Therefore, you must modify the imported ER solution so that it generates a document that complies with this new requirement.

Decide how you want to implement the required modifications of the imported ER components.

The obvious approach is to implement the following modifications:

- In your data model, add the new `Transaction.Party.Name` data model field of the *String* type.
- In your model mapping, configure the binding for the new data model field by using available table relations to access the relevant record of the `DirPartyTable` application table and fetch the value of the `Name` field from it.

Although this approach works, it might cause performance issues in the SQL database, because `TaxTrans` is the transaction table and can therefore contain a large volume of records. In this case, the number of calls to `DirPartyTable` must equal the number of records in the `TaxTrans` table, which can cause performance issues.

Alternatively, you might implement the following modifications:

- In your data model, add the new `Party` root and the new `Party.Name` field.
- In your model mapping, add a new data source that joins all the records of tables that are used in table relations to access the relevant record of the `DirPartyTable` application table, starting from the `TaxTrans` table.

Although this approach works, it might cause some memory consumption issues. Even when a new JOIN data source runs as a single SQL request to the application database to prevent database performance problems, the result must be fetched to the memory of the application server. Because the number of records and the number of fields in those records are quite large, this approach might cause very heavy memory consumption. An out-of-memory runtime exception might even be thrown.

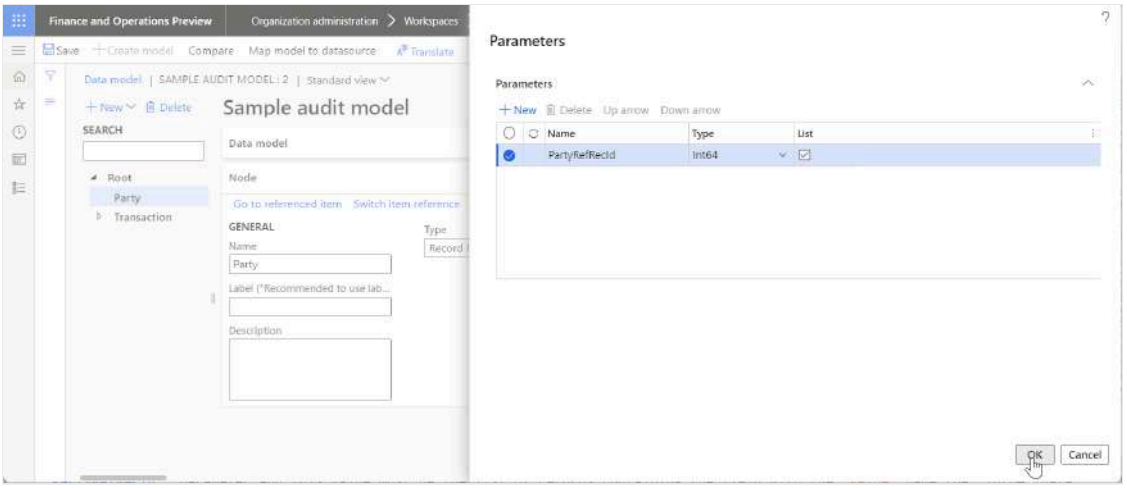
You can implement the modifications when a running format collects, in memory, the unique identification codes of customers and vendors for all tax transactions presented on a generated report. Because only unique codes are kept, the final set of codes isn't large enough to affect memory consumption. The set of codes is then passed to the model mapping as the argument of another call of the data source of the *Model* type. Based on that call, the model mapping runs a new ER data source that makes a single SQL request to the application database to fetch, from the `DirPartyTable` table, records only for those parties whose codes are presented in the provided set of codes.

Adjust the imported data model

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configurations tree in the left pane, select **Sample audit model**.
3. On the **Versions** FastTab, select version **2** that has a status of **Draft**.
4. Select the **Configuration components** FastTab.
5. Select **Designer** to open the data model for editing.
6. On **Data model**, make sure that the `Root` field is selected, and then select **New**.
7. In the drop-down dialog box, follow these steps:
   a. In the **New node as** field group, select the **Child of an active node** option.
   b. In the **Name** field, enter **Party**.
   c. In the **Item type** field, select **Record list**.
   d. Select **Add** to finish adding the new field.
8. Make sure that the `Root.Party` field is selected, and then, on the **Node** tab, select **Parameters**.
9. In the **Parameters** dialog box, follow these steps:
   a. On the **Parameters** tab, select **New**.
   b. In the **Name** field, enter **PartyRefRecId**.

c. In the **Type** field, select **Int64**.
d. Select the **List** checkbox.
e. Select **OK** to finish entering the parameters.

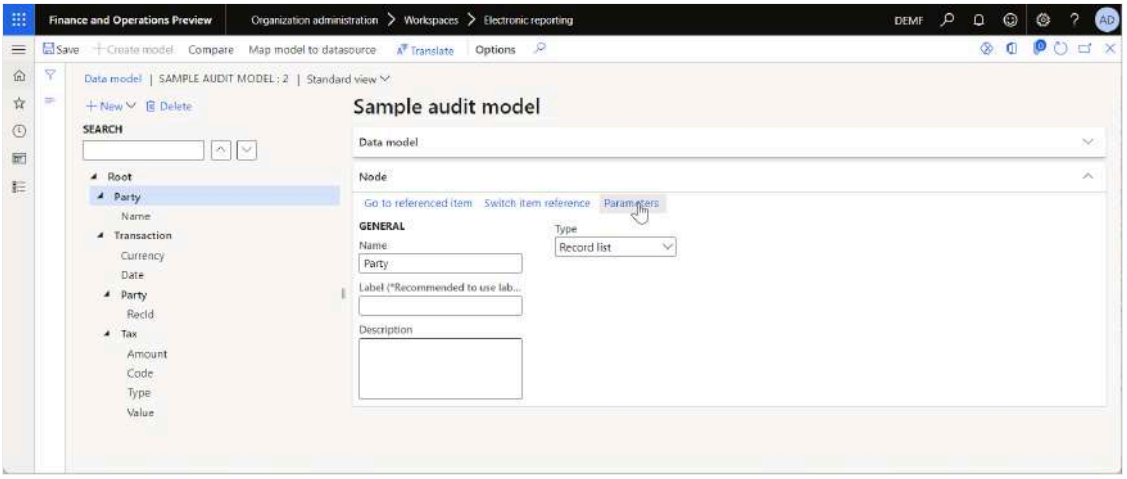
> **Note**
>
> You just configured the `Root.Party` data model field as a field that is called when a value is provided in the **PartyRefRecId** parameter. This value must be present in the list of records that contain a `Value` field of the *Int64* data type.



10. Make sure that the `Root.Party` field is still selected, and then select **New**.

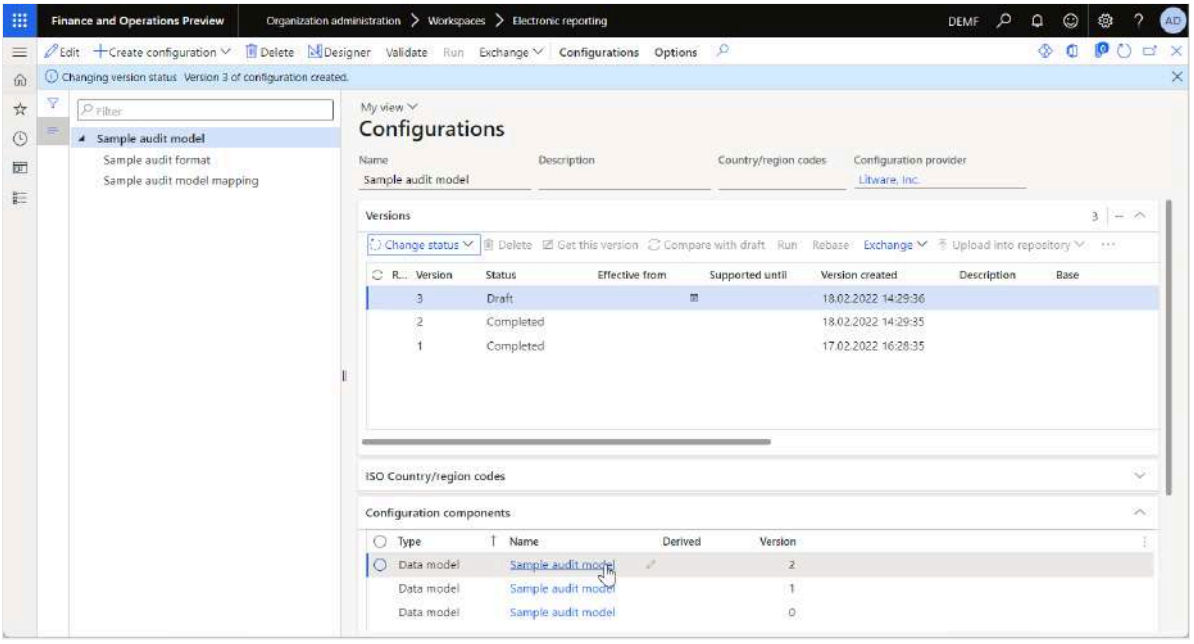
11. In the drop-down dialog box, follow these steps:
    a. In the **New node as** field group, select the **Child of an active node** option.
    b. In the **Name** field, enter **Name**.
    c. In the **Item type** field, select **String**.
    d. Select **Add** to finish adding the new field.

12. Select **Save**, and close the **Data model** page.



13. On the **Versions** FastTab, for version **2**, select **Change status** > **Complete**. Then select **OK**.

> **Note**
>
> Your data model changes are stored in the second revision of the **Sample audit model** data model component that is located in the second version of the **Sample audit model** ER configuration.



Adjust the imported model mapping

1. On **Configurations**, in the configurations tree in the left pane, expand **Sample audit model**.

2. Select **Sample audit model mapping**. On the **Versions** FastTab, select the second mapping version that is based on the first data model version (version **1.2**) and has a status of **Draft**.

3. Select **Rebase**.

4. In the **Target version** field, leave version **2** of the **Sample audit model** base model.

5. Select **OK** to rebase and align your model mapping with recent data model changes.

   The version number of your editable model mapping changes from **1.2** to **2.2** to indicate that the second model version is currently used as a base.

6. Select **Designer**.

7. On **Model to datasource mapping**, select **Designer** for the current model mapping.

8. Follow these steps to add a new data source to access the `DirPartyTable` application table:
   a. In the **Data source type** pane, select **Dynamics 365 for Operations** > **Table records**.

b. In the **Data sources** pane, select **Add root**.
c. In the **Name** field, enter **PartyTable**.
d. In the **Table** field, enter **DirPartyTable**.
e. Select **OK** to finish adding the new data source.

9. Follow these steps to add a new data source to request `DirPartyTable` records, based on the provided list of record identification codes:

a. In the **Data source type** pane, select **General** > **Empty container**.

b. In the **Data sources** pane, select **Add root**.

c. In the **Name** field, enter **Data**.

d. Select **OK** to finish adding the new data source.

e. In the **Data sources** pane, select the **Data** data source.

f. In the **Data source type** pane, select **Functions** > **Calculated field**.

g. In the **Data sources** pane, select **Add**.

h. In the **Name** field, enter **DirParty**.

i. Select **Edit formula**.

j. On **Formula designer**, select **Parameters**.

k. In the **Parameters** dialog box, follow these steps:
   i. On the **Parameters** tab, select **New**.
   ii. In the **Name** field, enter **DirPartyId**.
   iii. In the **Type** field, select **Int64**.
   iv. Select the **List** checkbox.
   v. Select **OK**.

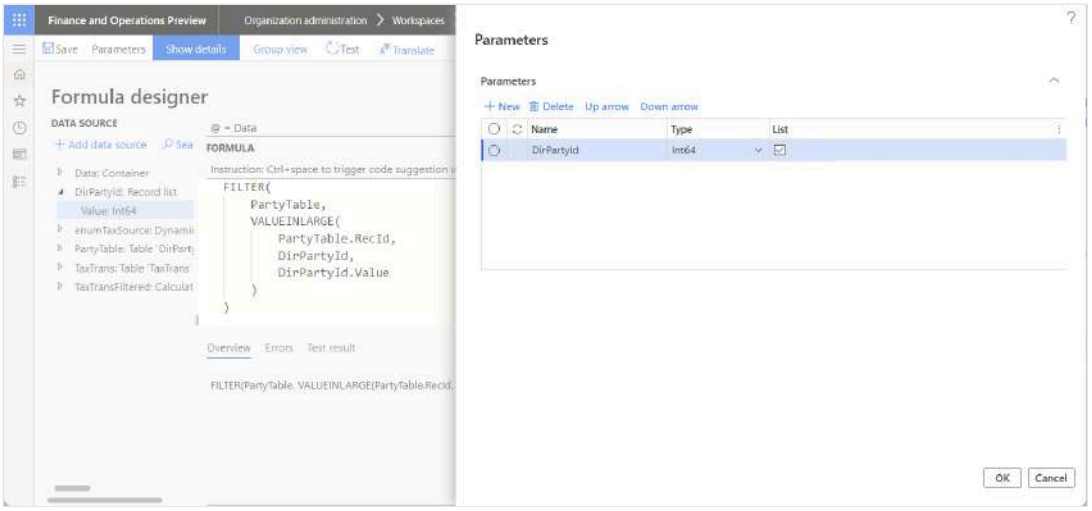
> ⓘ **Note**
>
> You just configured this calculated field so that it accepts, at runtime, the argument of a single parameter that is configured as a record list that has a single `Value` field of the *Int64* type.

l. In the **Formula** field, enter the following expression:

```
FILTER(PartyTable, VALUEINLARGE(PartyTable.RecId, DirPartyId, DirPartyId.Value))
```

> ⓘ **Note**

> You just configured the `DirParty` calculated field to fetch only `DirPartyTable` records where the record identification codes are included in the `DirPartyId` list that is provided as an argument when the `DirParty` field is called.



m. Select **Save**, and close **Formula designer**.

n. Select **Save**, and then select **OK** to finish adding the new data source.

10. Follow these steps to bind the new data source to the new data model field, so that the data model exposes party names:

    a. In the **Data model** pane, select the `Root.Party` data model field.

    b. In the **Data model** pane, select **Edit**.

    c. On **Formula designer**, in the **Formula** field, enter the expression `Data.DirParty(PartyRefRecId)`.

    d. Select **Save**, and close **Formula designer**.

    > ⓘ **Note**
    >
    > You just configured the binding to call the configured `Data.DirParty` data source and provide the list of record identification codes specified in the format when the `Root.Party` data model field is called.

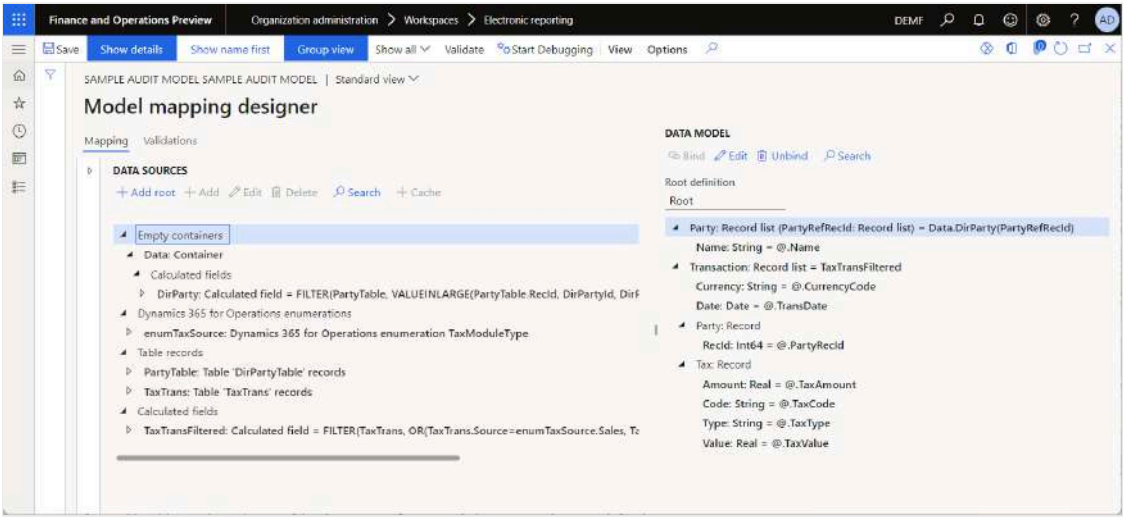
    e. In the **Data model** pane, select the `Root.Party.Name` data model field.

    f. In the **Data model** pane, select **Edit**.

    g. On the **Formula designer** page, in the **Data source** pane, expand **Data** > **DirParty**, and select **Name**.

    h. Select **Add data source**.

i. Select **Save**, and close **Formula designer**.



11. Select **Save**, and close the **Model mapping designer** page.
12. Close the **Model to datasource mapping** page.
13. On the **Versions** FastTab, for version **2.2**, select **Change status** > **Complete**. Then select **OK**.

Adjust the imported format

1. On **Configurations**, select **Sample audit format**.
2. On the **Versions** FastTab, select version **1.2** that has a status of **Draft**.
3. Select **Rebase**.
4. In the **Target version** field, leave version **2** of the **Sample audit model** base model.
5. Select **OK** to rebase and align your format with the recent data model changes.
6. Select **Designer**.
7. On the **Format designer** page, in the format structure tree in the left pane, select **Expand/Collapse**.
8. Follow these steps to add a new format element to collect record identification codes of parties whose transactions are presented on generated reports.
   a. In the format structure tree, select the **Report.Row.Trans** sequence element.
   b. Select **Add**.
   c. In **Add**, select **Data source** > **Item**.
   d. In **Component properties**, in the **Name** field, enter **Id**.
   e. In the **Data type** field, select **Int64**.
   f. Select **OK**.

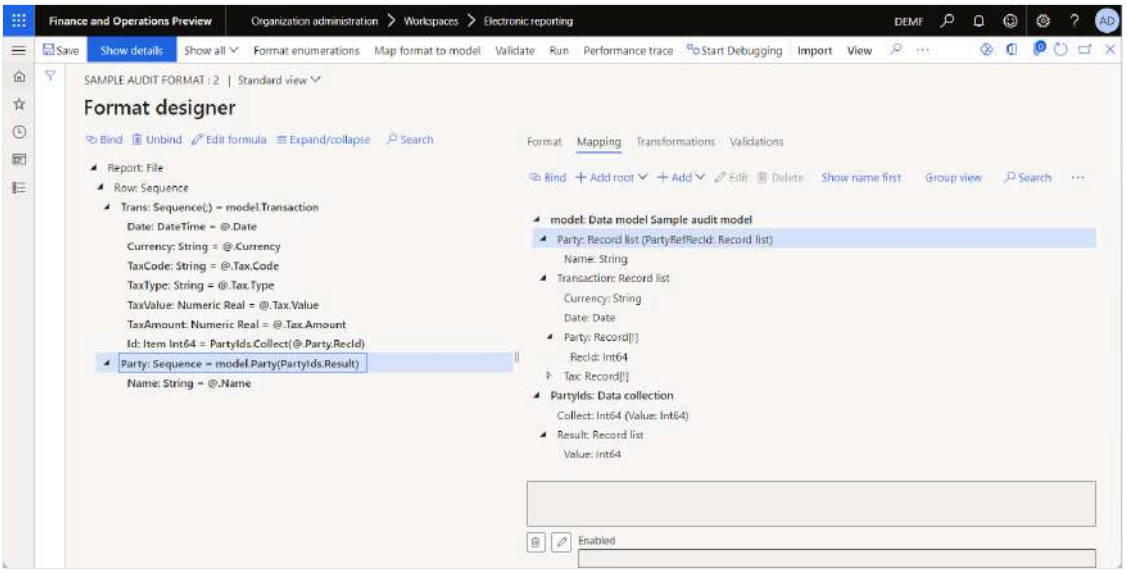
> ⓘ **Note**
>
> Use the **Data source** > **Item** element to perform internal calculations and data transformation only in the scope of the running format. Therefore, by adding this format element, you don't change the content of a generated document.

9. Follow these steps to add new format elements to enter party names on generated reports:
   a. Select the **Report.Row** sequence element.
   b. Select **Add**.
   c. In **Add**, select **Text** > **Sequence**.
   d. In **Component properties**, in the **Name** field, enter **Party**.
   e. Select **OK**.
   f. Select the **Report.Row.Party** sequence element.
   g. Select **Add**.
   h. In **Add**, select **Text** > **String**.
   i. In **Component properties**, in the **Name** field, enter **Name**.
   j. Select **OK**.

10. Follow these steps to add a new data source to collect record identification codes of parties whose transactions are presented on generated reports:
    a. On the **Mapping** tab, select **Add root**.
    b. In **Add data source**, select **Functions** > **Data collection**.
    c. In **'Data collection' data source properties**, in the **Name** field, enter **PartyIds**.
    d. In the **Item type** field, select **Int64**.
    e. Select **OK**.

11. Follow these steps to add a new binding to collect record identification codes of parties whose transactions are presented on generated reports:
    a. In the format structure tree, select the **Report.Row.Trans.Id** data item element.
    b. Select **Edit formula**.
    c. On the **Formula designer** page, enter the expression `PartyIds.Collect(model.Transaction.Party.RecId)`.
    d. Select **Save**, and close **Formula designer**.

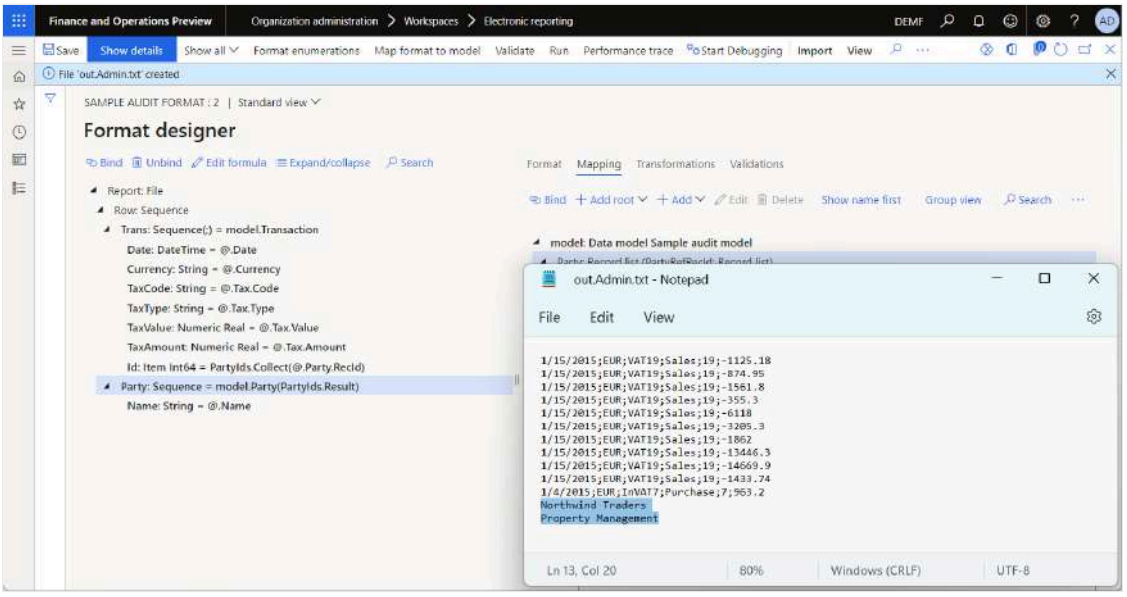
12. Follow these steps to add new bindings to enter party names on generated reports:
    a. In the format structure tree, select the **Report.Party** sequence element.
    b. Select **Edit formula**.
    c. On the **Formula designer** page, enter the expression `model.Party(PartyIds.Result)`.
    d. Select **Save**, and close **Formula designer**.
    e. In the format structure tree, select the **Report.Party.Name** sequence element.
    f. On the **Mapping** tab, select the `model.Party.Name` data model field.
    g. Select **Bind**.



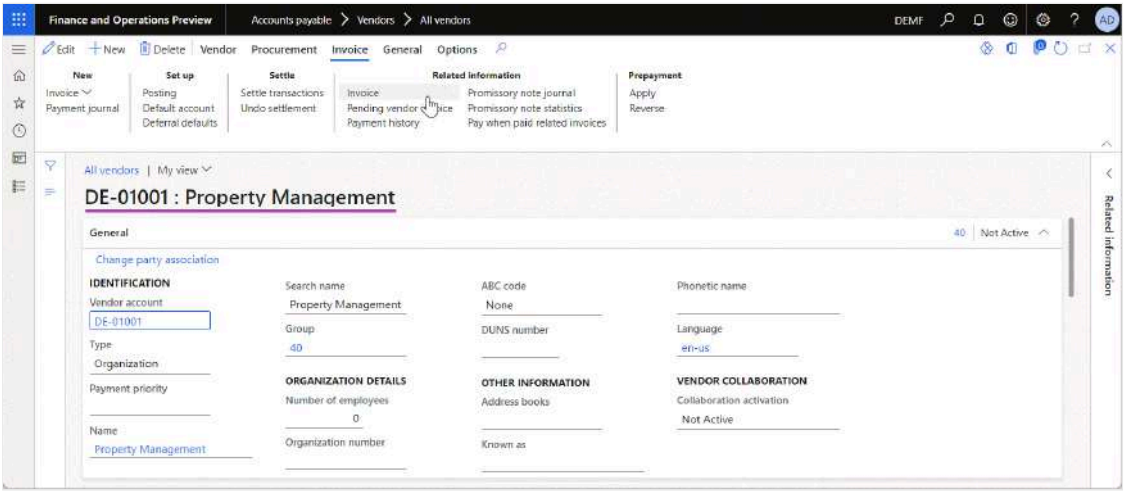
13. Select **Save**, and close the **Format designer** page.
14. Close the **Model to datasource mapping** page.
15. On the **Versions** FastTab, for version **2.2**, select **Change status** > **Complete**. Then select **OK**.

Run the adjusted format

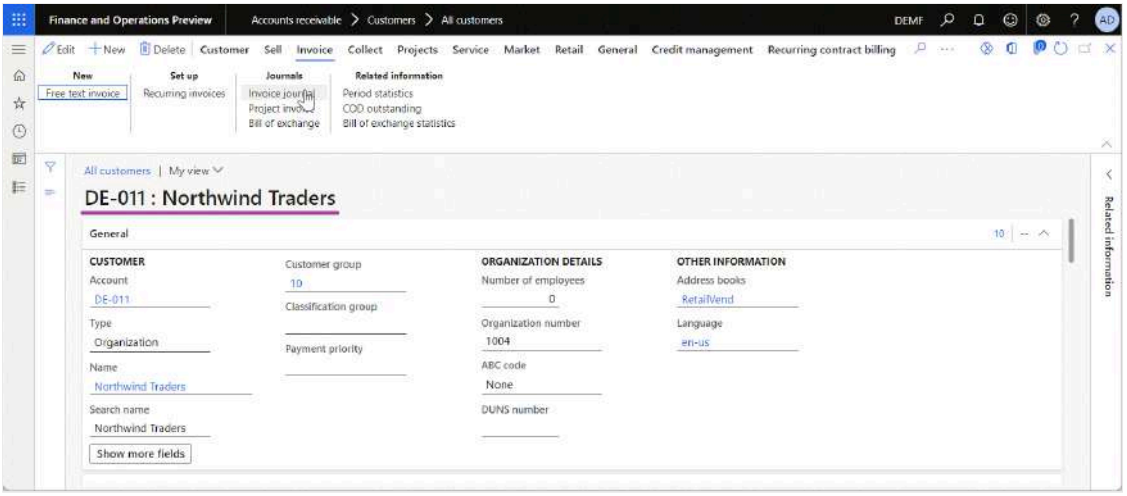
1. On **Configurations**, select **Sample audit format**. On the action pane, select **Run**.
2. In **Electronic report parameters**, on the **Records to include** tab, select **Filter**.
3. Specify the conditions to select tax transactions for the **PIV-110000004** and **INV-10000001** vouchers.
4. Select **OK**.
5. Select **OK**.
6. Review the generated document that contains tax transactions for the selected vouchers and the names of the corresponding customer and vendor.



7. Go to **Accounts payable** > **Vendors** > **All vendors**, and review the details of the selected **PIV-110000004** voucher, including the vendor name.



8. Go to **Accounts receivable** > **Customers** > **All customers**, and review the details of the selected **INV-10000001** voucher, including the customer name.



For more details about this execution of the ER solution, use the ER built-in performance trace parser.

Additional resources

- Support parameterized calls of ER data sources of the Calculated field type
- Trace the execution of ER formats to troubleshoot performance issues
- Use DATA COLLECTION data sources in Electronic reporting formats
- ValueInLarge ER function

Last updated on 04/15/2026

Configure country/region context dependent ER model mappings

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can configure Electronic reporting (ER) model mappings so that they implement a generic ER data model but are specific to Dynamics 365 Finance. This article explains how to design multiple ER model mappings for an ER data model to control how corresponding ER formats use them when you run them from companies that have different country/region contexts.

Prerequisites

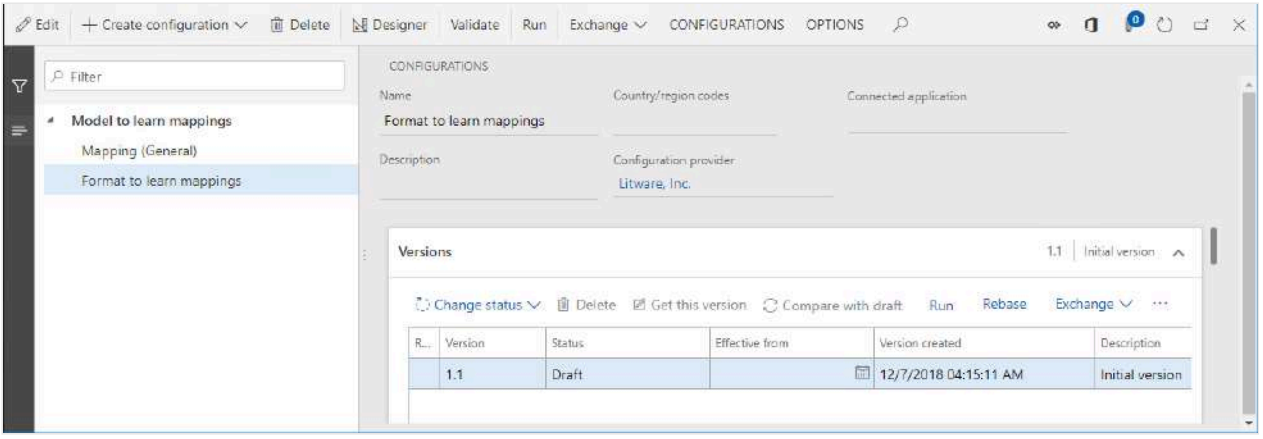
To complete the examples in this article, you must have the following access:

- Access to Finance for one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- Access to the instance of Regulatory Configuration Services (RCS) that is provisioned for the same tenant as Finance for one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator

Some steps in this article require execution of an ER format. In some cases, the country/region context of the company that you're currently signed in to affects execution of an ER format. You can run an ER format in the current RCS instance if the company that has the required country/region context is available in RCS. Otherwise, you must upload a completed version of the ER model mapping and ER format configurations that use the ER data model to your Finance instance, and then run the ER format in that Finance instance. For information about how to import configurations that reside in RCS into a Finance instance, see Import configurations from RCS.

Single model mapping case

Follow the steps in Appendix 1 of this article to design the required ER components. You now have the **Mapping (General)** model mapping configuration that contains the model mapping for the **Entry point 1** definition.



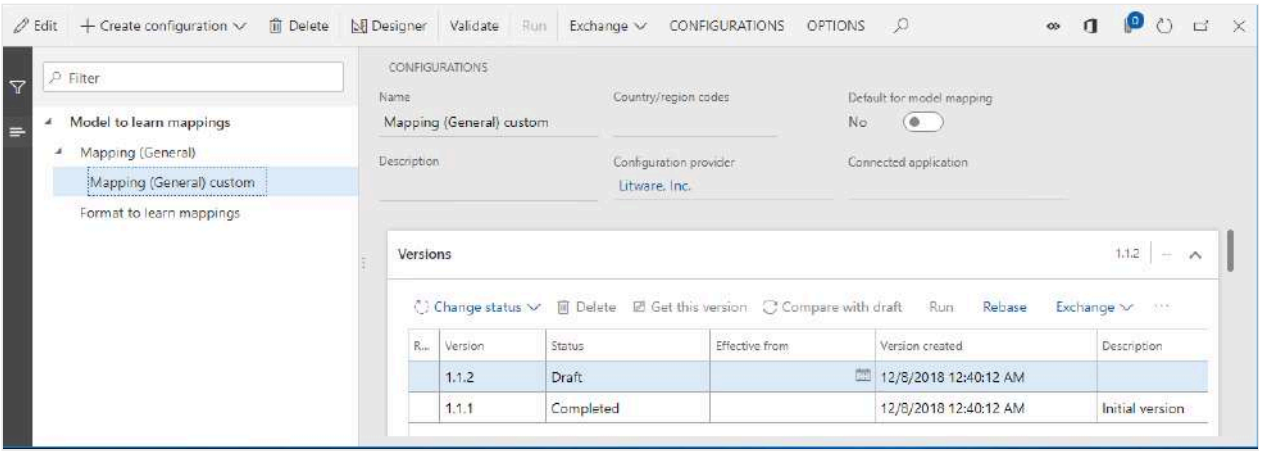
Run the configured format

1. On the **Configurations page**, on the **Versions** FastTab, select **Run**.
2. Select **OK**.

The web browser offers to download the text file that the executed ER format generated. Because you configured this format to use the **Entry point 1** definition, and only a single model mapping is currently available for the base model that contains a mapping for this definition, the executed ER format uses the **Mapping (General)** model mapping of the **Mapping (General)** configuration as a data source. Therefore, the downloaded file contains the **Generic functionality 1** text.

Multiple shared model mappings case

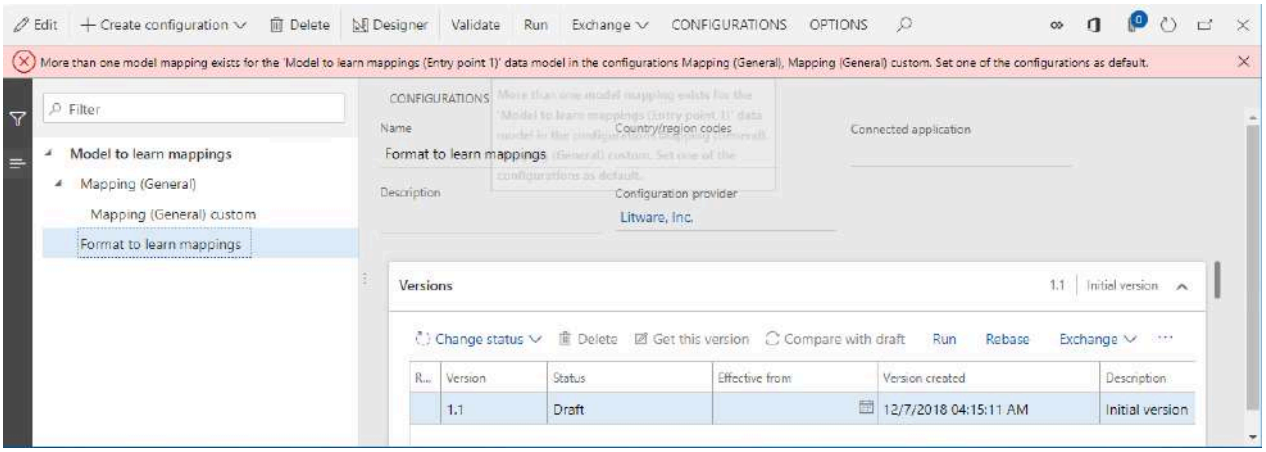
Follow the steps in Appendix 2 of this article to design the required ER components. You now have **Mapping (General)** and **Mapping (General) custom** model mapping configurations, each of which contains the model mapping for the **Entry point 1** definition.



Run the configured format

1. On **Configurations**, in the configurations tree, select **Format to learn mappings**.
2. On the **Versions** FastTab, select **Run**.
3. Select **OK**.

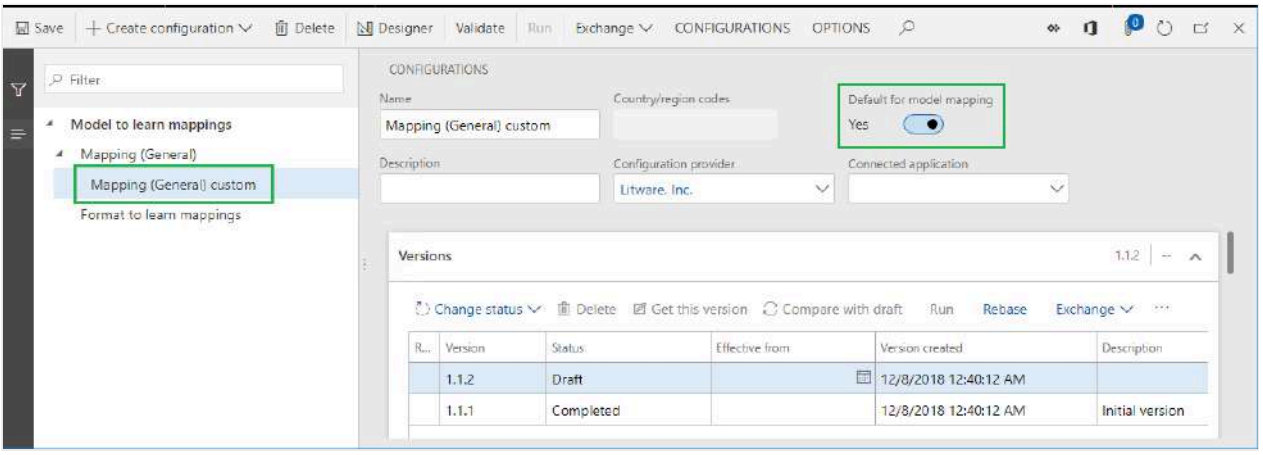
Execution of the selected ER format isn't successful. An error message informs you that more than one model mapping exists for the **Model to learn mappings** model and the **Entry point 1** definition in the **Mapping (General)** and **Mapping (General) custom** model mapping configurations. The message also recommends that you select one of those configurations as the default configuration.



Define a default mapping configuration

Follow these steps to define the **Mapping (General) custom** model mapping configuration as the default configuration, so that you can use its mappings as data sources for the **Format to learn mappings** ER format.

1. On **Configurations**, in the configurations tree, select **Mapping (General) custom**.
2. Select **Edit** to make the current page ready for editing.
3. Set the **Default for model mapping** option to **Yes**.
4. Select **Save**.



Run the configured format

1. On **Configurations**, in the configurations tree, select **Format to learn mappings**.
2. On the **Versions** FastTab, select **Run**.
3. Select **OK**.

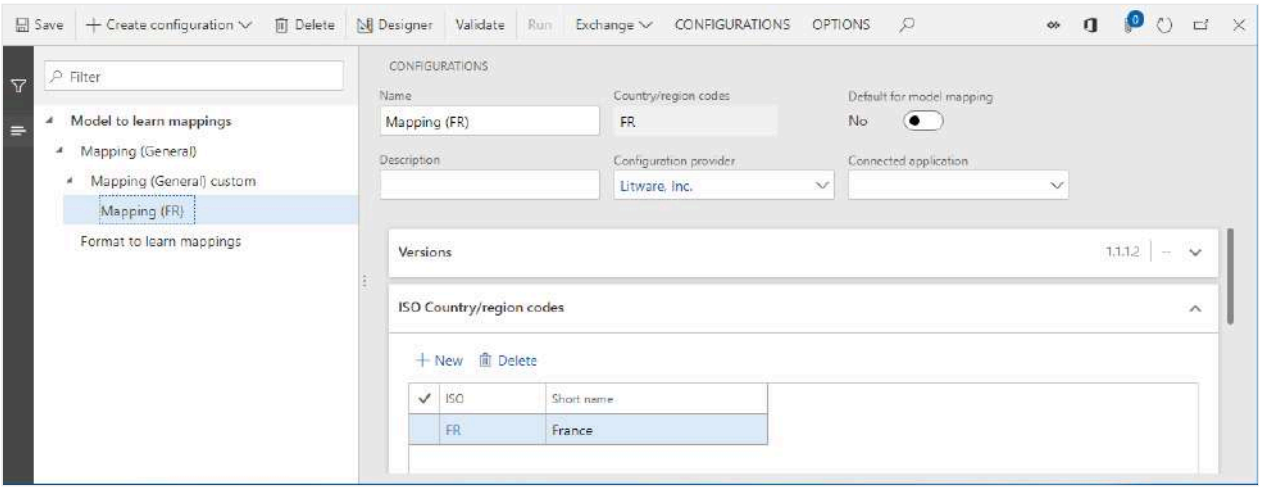
The execution of the selected ER format succeeds. The web browser offers to download the text file that the executed ER format generated. Because you configured this format to use the **Entry point 1** definition, and you selected the **Mapping (General) custom** model mapping configuration as the default configuration, the executed ER format uses the **Mapping (General) copy** model mapping of the **Mapping (General) custom** configuration as a data source. Therefore, the downloaded file contains the **Generic functionality 1 custom** text.

> **Note**
>
> If you change the company that you're currently signed in to and run this ER format again, you get the same content in the generated file, because the default ER model mapping configuration doesn't contain any company-dependent restrictions.

Multiple mixed model mappings case

Follow the steps in Appendix 3 of this article to design the required ER components. You now have **Mapping (General)**, **Mapping (General) custom**, and **Mapping (FR) model mapping** configurations that contain the model mapping for the **Entry point 1** definition.

Version 1 of the **Mapping (FR)** model mapping configuration is configured so that it applies only to ER formats of the **Model to learn mappings** model that run in Finance companies that have French country/region context.



Run the configured format

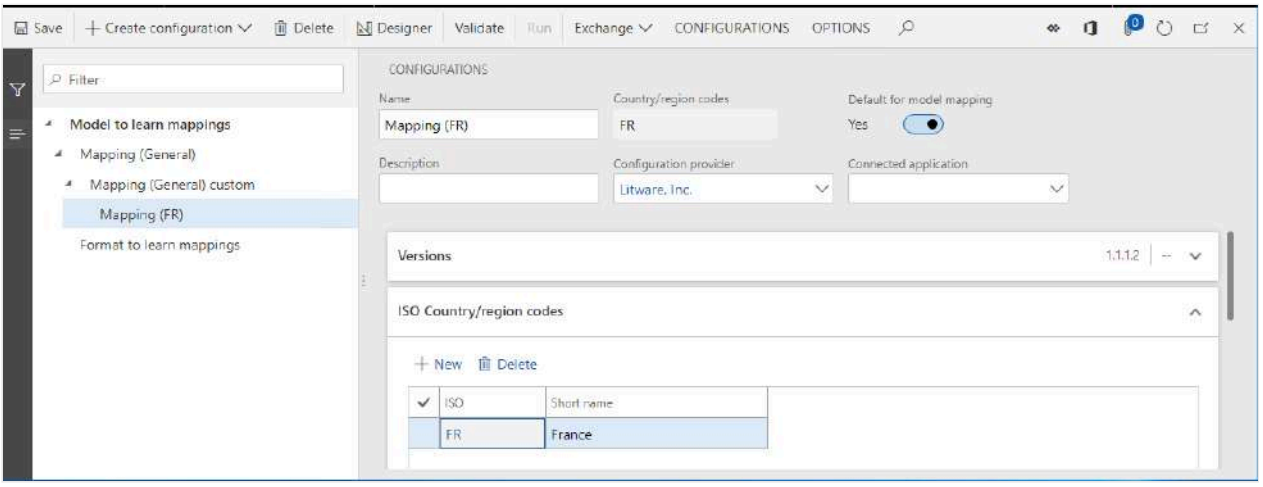
1. Change the company to **FRSI**.
2. On **Configurations**, in the configurations tree, select **Format to learn mappings**.
3. On the **Versions** FastTab, select **Run**.
4. Select **OK**.

The execution of the selected ER format succeeds. The web browser offers to download the text file that the executed ER format generated. Because this format uses the **Entry point 1** definition and the **Mapping (General) custom** model mapping configuration is selected as the default configuration, the executed ER format uses the **Mapping (General) copy** model mapping of the **Mapping (General) custom** configuration as a data source. Therefore, the downloaded file contains the **Generic functionality 1 custom** text.

Define the France-specific mapping configuration as the default configuration

Follow these steps to define the custom **Mapping (FR)** model mapping configuration as the default configuration. Because this mapping is specific to France, it's the default mapping between all model mapping configurations that have the **FR** country code specified in the **ISO country/region codes** field.

1. On **Configurations**, in the configurations tree, select **Mapping (FR)**.
2. Select **Edit** to make the current page ready for editing.
3. Set the **Default for model mapping** option to **Yes**.
4. Select **Save**.



Run the configured format

1. On **Configurations**, in the configurations tree, select **Format to learn mappings**.
2. On the **Versions** FastTab, select **Run**.
3. Select **OK**.

The execution of the selected ER format succeeds. The web browser offers to download the text file that the executed ER format generated. Because this format uses the **Entry point 1** definition and the **Mapping (FR)** model mapping configuration is selected as the default configuration, the executed ER format uses the **Mapping (FR)** model mapping of the **Mapping (FR)** configuration as a data source. Therefore, the downloaded file contains the **FR functionality 1** text.

> **Note**
>
> If you change the company that you're currently signed in to and run this ER format again, the output depends on the country/region context of the selected company.

Other model mapping cases

As you saw, the selection of a model mapping for the execution of an ER format works in the following way:

1. You specify the model mapping definition that an ER format uses (**Entry point 1** in the examples in this article).
2. You can potentially use all mapping configurations that contain a mapping that has the specified definition and satisfy any country/region context restrictions that you configure to run the ER format (**Mapping (General)**, **Mapping (General) custom**, and **Mapping (FR)** in the examples in this article).

3. Any default model mapping that has country/region context restrictions has the highest priority for selection (**Mapping (FR)** in the examples in this article).
4. Any default model mapping that doesn't have country/region context restrictions has the next higher priority for selection (**Mapping (General) custom** in the examples in this article).
5. Any model mapping that has country/region context restrictions has higher priority for selection than a model mapping that doesn't have country/region context restrictions.

The following table provides information about the results of model mapping selection for all possible cases for model mapping settings:

- Column 1 indicates whether the first model mapping that doesn't have country/region context restrictions (for example, the shared **Mapping (General)** mapping) is presented and, if it is, whether the **Default for model mapping** option is set to **Yes** for it.
- Column 2 indicates whether the second model mapping that doesn't have country/region context restrictions (for example, the shared **Mapping (General) custom** mapping) is presented and, if it is, whether the **Default for model mapping** option is set to **Yes** for it.
- Column 3 indicates whether the first model mapping that has country/region A context restrictions (for example, the France-specific **Mapping (FR)** mapping) is presented and, if it is, whether the **Default for model mapping** option is set to **Yes** for it.
- Column 4 indicates whether the second model mapping that has country/region A context restrictions is presented and, if it is, whether the **Default for model mapping** option is set to **Yes** for it.
- Column 5 presents the result of a model mapping selection for execution of an ER format under the control of a company that has country/region A context.
- Column 6 presents the result of a model mapping selection for execution of an ER format under the control of a company that has country/region B context.

In the table, a plus sign (+) indicates the presence of a model mapping configuration in the current instance of the Microsoft Azure service that is used to run an ER format (either Finance or RCS).

Expand table

Case	Model mapping 1 without country/region context (MM1)	Model mapping 2 without country/region context (MM2)	Model mapping 1 with country/region A context (MM1A)	Model mapping 2 with country/region A context (MM2A)	Run under control of a company that has country/region A context	Run under th control of a company tha has country/regi B context
	1	2	3	4	5	6

Case	Model mapping 1 without country/region context (MM1)	Model mapping 2 without country/region context (MM2)	Model mapping 1 with country/region A context (MM1A)	Model mapping 2 with country/region A context (MM2A)	Run under control of a company that has country/region A context	Run under th control of a company tha has country/regi B context
1					Error (missing mapping)	Error (missing mapping)
2	+				MM1	MM1
3	+	+			Error (multiple mappings)	Error (multiple mappings)
4	+		+		MM1A	MM1
5	+		+	+	Error (multiple mappings)	MM1
6	+	default	+	+	MM2	MM2
7	+		default		MM1A	MM1
8	+		default	+	MM1A	MM1
9	+		default	default	Error (multiple mappings)	MM1
10	default				MM1	MM1
11	default	+			MM1	MM1
12	default		+		MM1	MM1
13	default	default			Error (multiple mappings)	Error (multiple mappings)
14	default		default		MM1A	MM1
15	default		default	default	MM1A	MM2A
16			+	+	MM1A	MM2A
17			default	default	MM1A	MM2A

Learn what mapping was used in the execution of an ER format

Configure ER user parameters

1. On **Configurations**, on the Action Pane, on the **CONFIGURATIONS** tab, select **User parameters**.

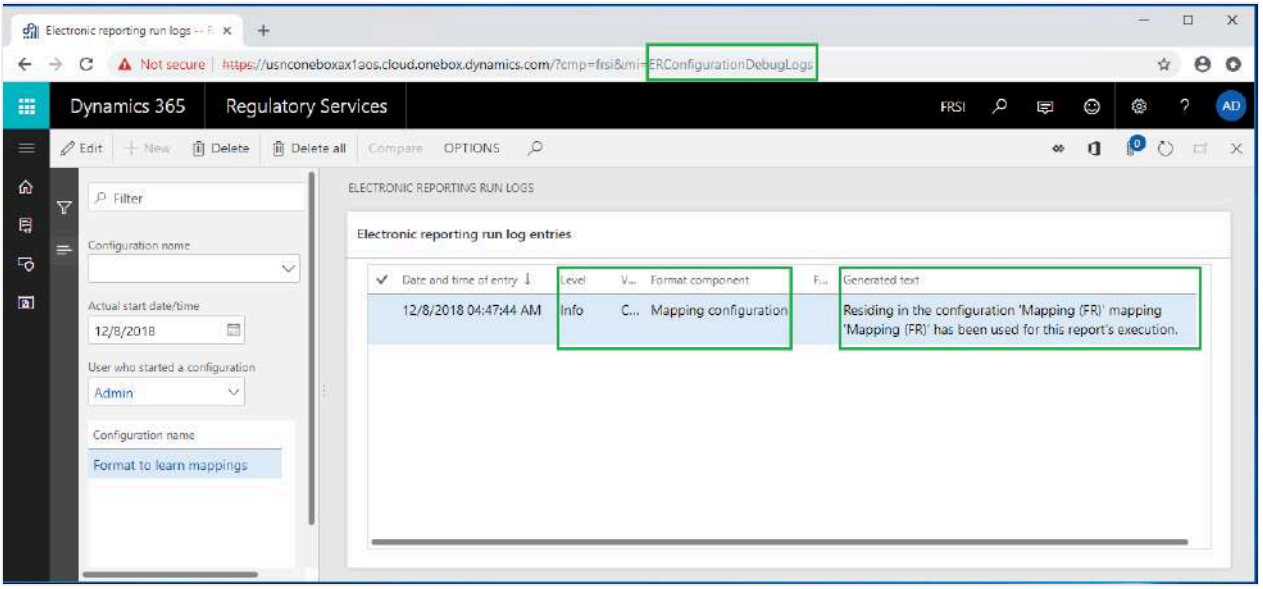
2. Set the **Run in debug mode** option to **Yes**.
3. Select **Ok**.

Run the configured format

1. On **Configurations**, in the configurations tree, select **Format to learn mappings**.
2. On the **Versions** FastTab, select **Run**.
3. Select **Ok**.

Review the ER debug log

1. In the navigation pane, go to **Modules** > **Organization administration** > **Electronic reporting** > **Configuration debug log**.
2. Select the **Reload this page** button.



A new record is added to the ER debug log for the executed ER format. Because the **Level** field of this record is set to **Info**, the record is informational. Because the **Format component** field is set to **Mapping configuration**, the record informs you about a model mapping that was used during execution of the **Format to learn mappings** ER format (selected in the **Configuration name** field). The content of the **Generated text** field informs you that the **Mapping (FR)** mapping component that resides in the **Mapping (FR)** configuration is used to run this report.

Appendix 1

Configure a sample data model

Sign in to your RCS instance.

In this example, you create a configuration for sample company, Litware, Inc. To complete these steps, you must first complete, in RCS, the steps in the Create a configuration provider and

mark it as active procedure.

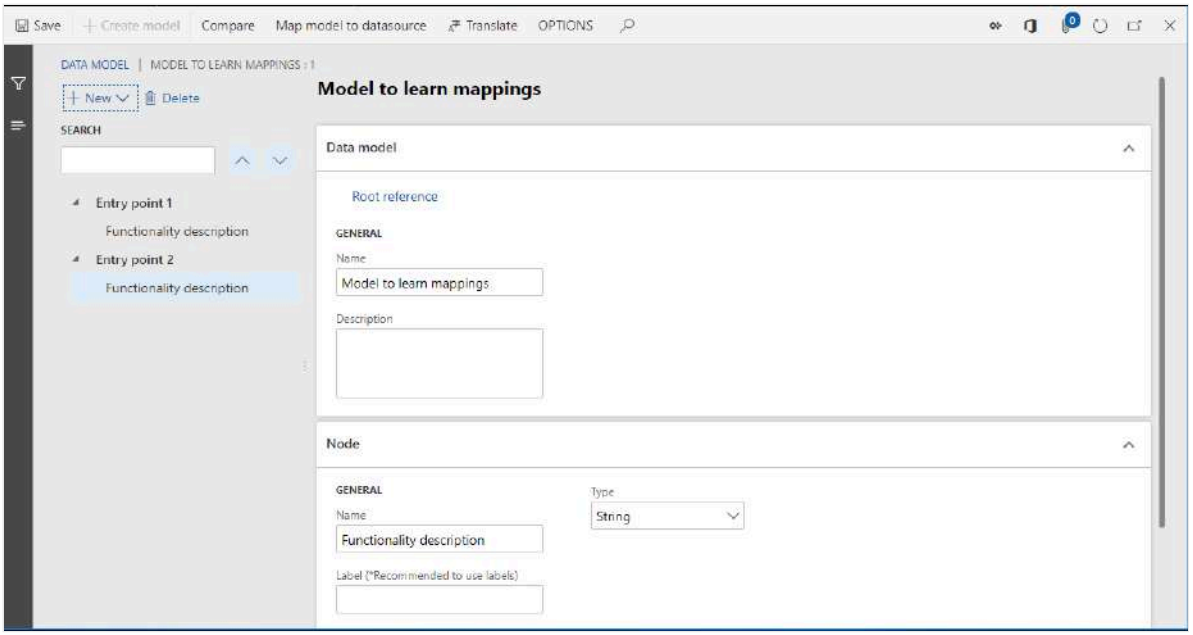
Create an ER data model configuration

1. On the default dashboard, select **Electronic reporting**.
2. Select the **Reporting configurations** tile.
3. On **Configurations**, select **Create configuration**.
4. In the drop-down dialog box, in the **Name** field, enter **Model to learn mappings**.
5. Select **Create configuration**.
6. Select the **Configuration components** FastTab.

Draft version 1 of this ER configuration is ready for editing. This version contains the data model component.

Design a sample data model

1. On the **Configurations page**, select **Designer**.
2. Select **New**.
3. In the drop-down dialog box, in the **Name** field, enter **Entry point 1**.
4. Select **Add**.
5. Select **New**.
6. In the drop-down dialog box, in the **Name** field, enter **Functionality description**.
7. Select **Add**.
8. Select **New**.
9. In the drop-down dialog box, in the **New node** field group, select **Model root**.
10. In the **Name** field, enter **Entry point 2**.
11. Select **Entry point 2**.
12. Select **Add**.
13. Select **New**.
14. In the drop-down dialog box, in the **Name** field, enter **Functionality description**.
15. Select **Add**.



16. Select **Save**.

17. Close the page.

Complete the modified version of the model configuration

1. On **Configurations**, on the **Versions** FastTab, select **Change status**.

   > Change the status of designed model configuration from **Draft** to **Completed**, so that you can use it to design the required model mappings and formats.

2. Select **Complete**.

3. Select **OK**.

The configuration that you created is saved as completed version 1.

Configure a sample model mapping

Create an ER model mapping configuration

1. On **Configurations**, select **Create configuration**.
2. In the drop-down dialog box, in the **New** field group, select **Model mapping based on data model Model to learn mappings**.
3. In the **Name** field, enter **Mapping (General)**.
4. In the **Data model definition** field, select **Entry point 1**.
5. Select **Create configuration**.

Draft version 1 of this ER configuration is ready for editing. This version contains the model mapping component.

Design a sample model mapping

1. On **Configurations**, select **Designer**.

   The model mapping of the **To model** direction type is automatically added to this component for the **Entry point 1** definition.

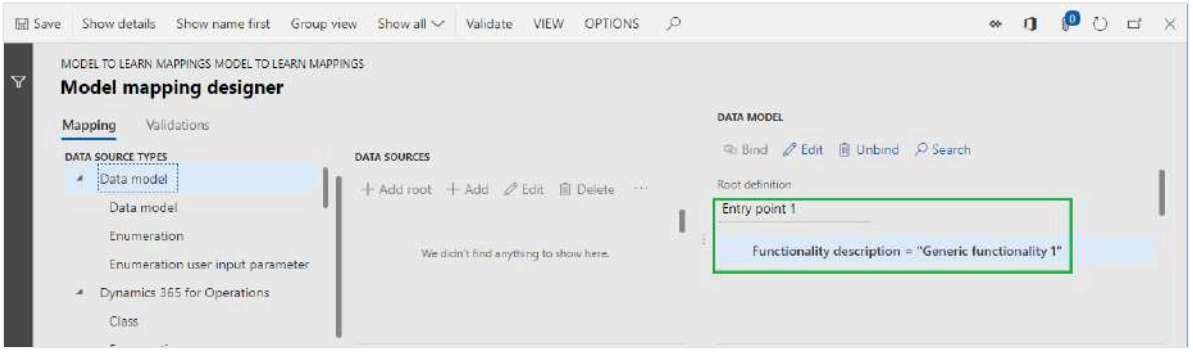
2. Select **Designer** to start editing the added model mapping.

3. In the **Data model** section, select **Edit**.

4. In the **Formula** field, enter **"Generic functionality 1"**.

5. Select **Save**.

6. Close the **Formula designer**.



7. Select **Save**.

8. Close the **Model mapping designer**.

9. Select **New**.

10. In the **Definition** field, select **Entry point 2**.

11. In the **Name** field, enter **Mapping (General) 2**.

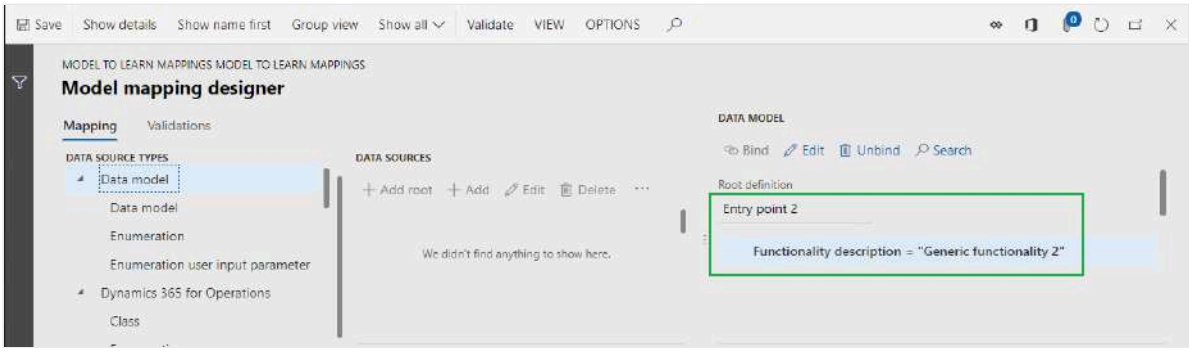
12. Select **Designer**.

13. In the **Data model** section, select **Edit**.

14. In the **Formula** field, enter **"Generic functionality 2"**.

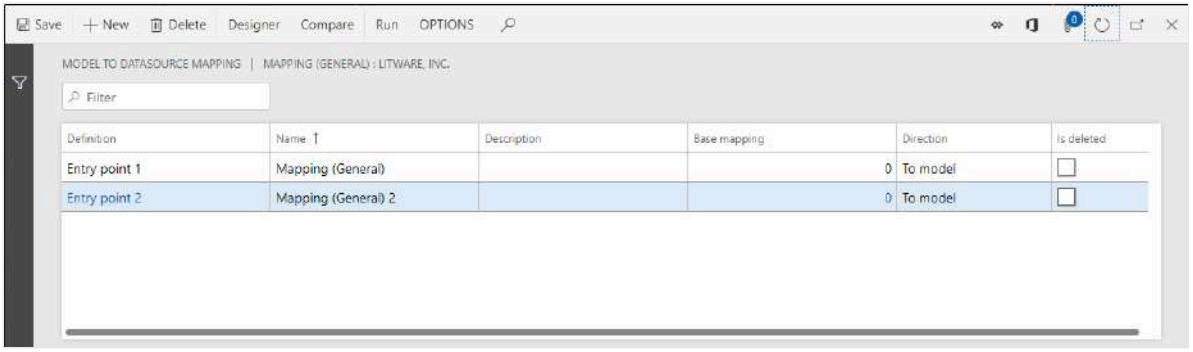
15. Select **Save**.

16. Close the **Formula designer**.



17. Select **Save**.

18. Close the **Model mapping designer**.



19. Close the **Model mappings**.

Complete the modified version of the model mapping configuration

1. On **Configurations**, on the **Versions** FastTab, select **Change status**.

   > Change the status of designed model mapping configuration from **Draft** to **Completed**, so that ER formats can use it.

2. Select **Complete**.

3. Select **OK**.

The configuration is saved as completed version 1.

Configure a sample format

Create an ER format configuration

1. On **Configurations**, in the configurations tree, select **Model to learn mappings**.
2. Select **Create configuration**.
3. In the drop-down dialog box, in the **New** field group, select **Format based on data model Model to learn mappings**.
4. In the **Name** field, enter **Format to learn mappings**.

5. In the **Data model definition** field, select **Entry point 1**.
6. Select **Create configuration**.

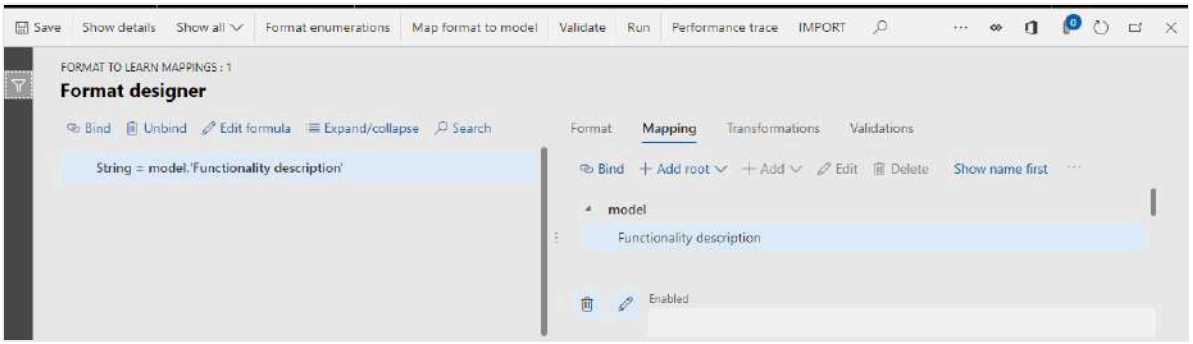
Draft version 1 of this ER configuration is ready for editing. This version contains the format component.

Design a sample format

1. On **Configurations**, select **Designer**.
2. Select **Add root**.
3. In the **Text** group, select the **String** item.
4. Select **OK**.

Bind format elements to a data source

1. On **Format designer**, on the **Mapping** tab, expand the model data source.
2. Select the **Functionality description** field.
3. Select **Bind**.



4. Select **Save**.
5. Close the page.

Appendix 2

Configure a sample model mapping for general customization

You might want to customize a model mapping that a configuration provider (partner) provided to you, and then use the customized version as a data source for your ER formats. In this case, you must create a custom ER model mapping configuration to make the required changes in existing model mappings. The procedures in this appendix use the **Mapping (General)** model mapping as an example.

Create an ER model mapping configuration

1. On the **Configurations** page, in the configurations tree, select **Mapping (General)**.
2. Select **Create configuration**.
3. In the drop-down dialog box, in the **New** field group, select **Derive from Name: Mapping (General), Litware, Inc.**.
4. In the **Name** field, enter **Mapping (General) custom**.
5. Select **Create configuration**.

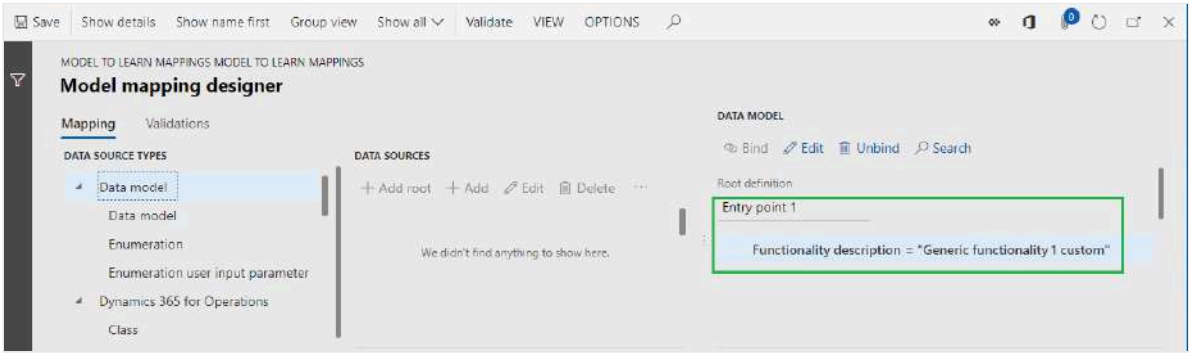
Draft version 1 of this ER configuration is ready for editing.

Design a sample model mapping

1. On **Configurations**, select **Designer**.

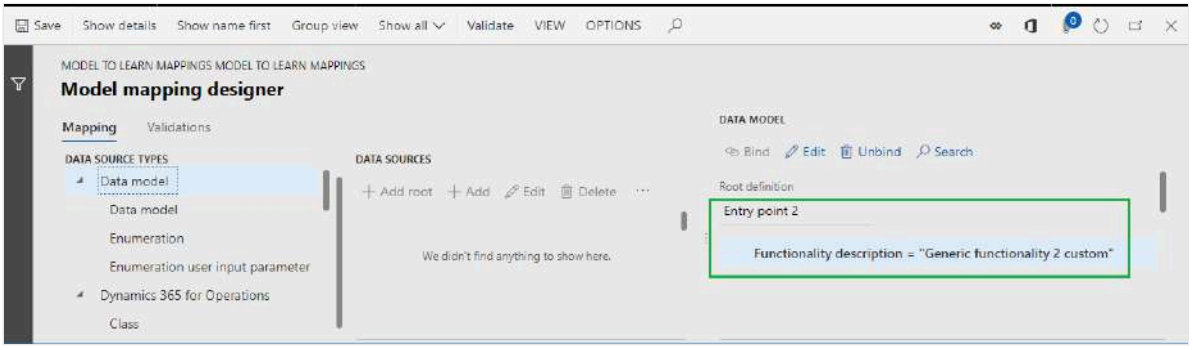
   > The model mappings of the base configuration are automatically copied to this configuration.

2. Select the **Mapping (General) Copy** mapping.
3. Select **Designer**.
4. In the **Data model** section, select **Edit**.
5. In the **Formula** field, enter **"Generic functionality 1 custom"**.
6. Select **Save**.
7. Close the page.

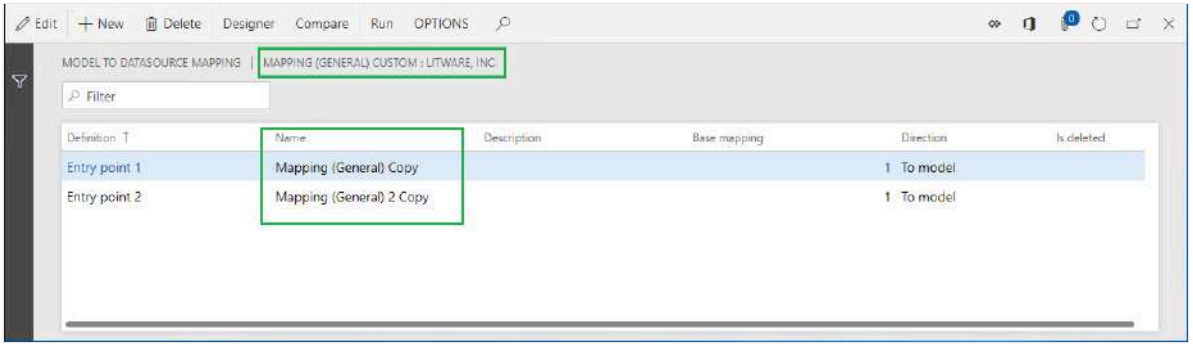


8. Select **Save**.
9. Close the page.
10. Select the **Mapping (General) 2 Copy** mapping.
11. Select **Designer**.

12. In the **Data model** section, select **Edit**.
13. In the **Formula** field, enter **"Generic functionality 2 custom"**.
14. Select **Save**.
15. Close the page.



16. Select **Save**.
17. Close the page.



18. Close the page.

Complete the modified version of the model mapping configuration

1. On **Configurations**, on the **Versions** FastTab, select **Change status**.

   > Change the status of designed model mapping configuration from **Draft** to **Completed**, so that ER formats can use it.

2. Select **Complete**.
3. Select **OK**.

The configuration is saved as completed version 1.

Appendix 3

Configure a sample model mapping for country/region-specific customization

For some ER formats, country/region-specific requirements exist for data preparation. In this case, manage a separate ER model mapping configuration to isolate the implementation of these country/region-specific requirements from the general implementation. The procedures in this appendix use the **Format to learn mappings** ER format and French-specific requirements as an example.

Create an ER model mapping configuration

First, create a new ER model mapping configuration to implement the country/region-specific requirements. Use your custom ER model mapping configuration as a base.

1. On **Configurations**, in the configurations tree, select **Mapping (General) custom**.
2. Select **Create configuration**.
3. In the drop-down dialog box, in the **New** field group, select **Derive from Name: Mapping (General) custom, Litware, Inc**.
4. In the **Name** field, enter **Mapping (FR)**.
5. Select **Create configuration**.

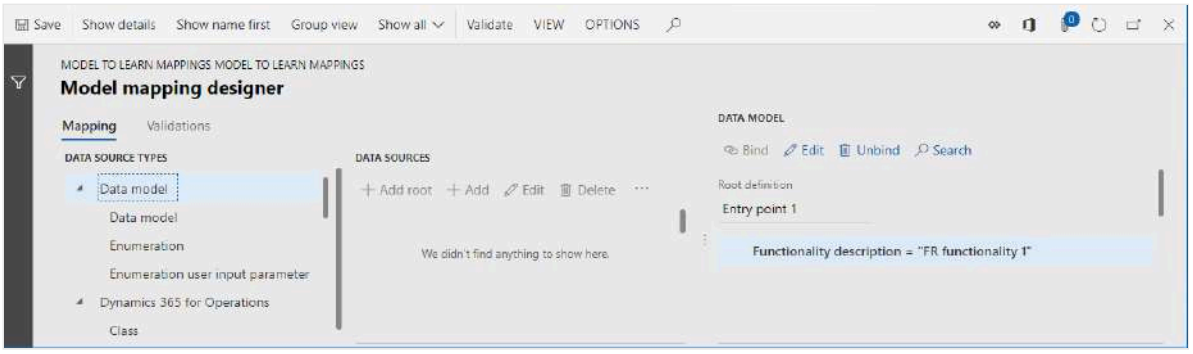
Draft version 1 of this ER configuration is ready for editing.

Design a sample model mapping

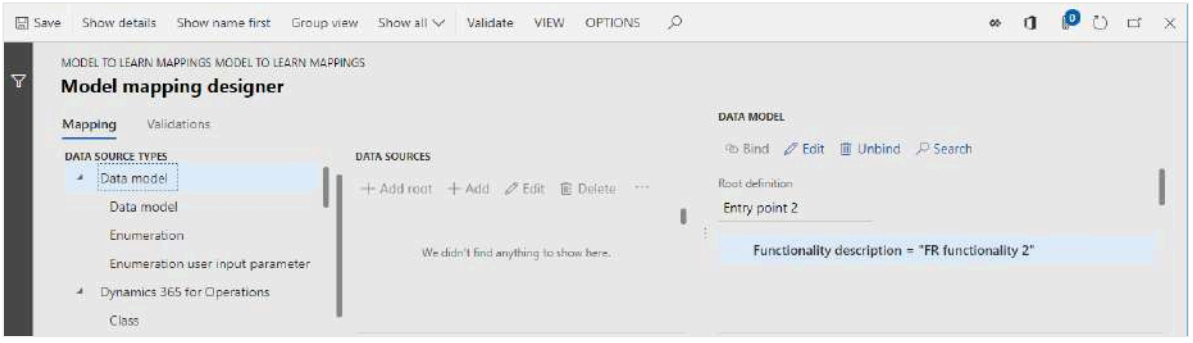
1. On **Configurations**, select **Designer**.

   > Model mappings of the base configuration are automatically copied to this configuration.

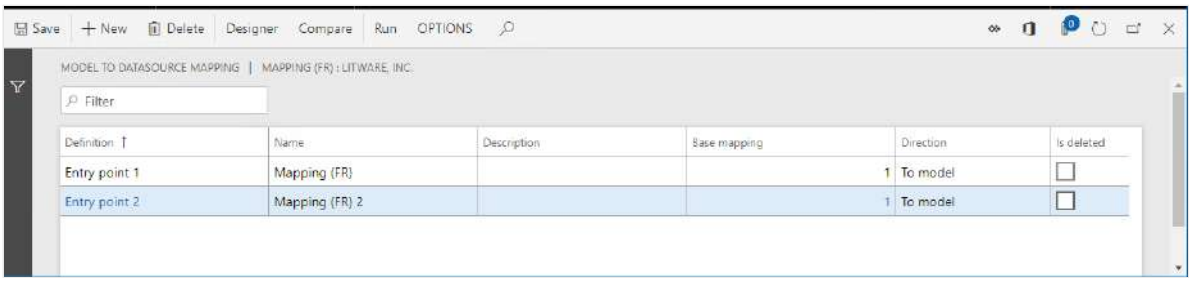
2. Select the **Mapping (General) Copy Copy** mapping.
3. Rename it **Mapping (FR)**.
4. Select **Designer**.
5. In the **Data model** section, select **Edit**.
6. In the **Formula** field, enter **"FR functionality 1"**.
7. Select **Save**.
8. Close the page.



9. Select **Save**.
10. Close the page.
11. Select the **Mapping (General) 2 Copy Copy** mapping.
12. Rename it **Mapping (FR) 2**.
13. Select **Designer**.
14. In the **Data model** section, select **Edit**.
15. In the **Formula** field, enter **"FR functionality 2"**.
16. Select **Save**.
17. Close the page.



18. Select **Save**.
19. Close the page.



20. Close the page.

Specify country/region context restrictions for use

1. On **Configurations**, on the **ISO Country/region codes** FastTab, select **New**.
2. In the **ISO** field, select **FR**.
3. Select **Save**.

You must sign in to a specific company in Finance to run an ER format. Therefore, this company controls both ER format execution and selection of the correct ER model mapping of the base ER data model. By adding the **FR** country code, you specify that this model mapping is available for selection by an ER format of the base data model only when that format runs under the control of a company that has French country/region context.

You can add multiple country/region codes for a single version of an ER model mapping configuration. In this way, model mappings that reside in that model mapping configuration can be used for an ER format that runs under the control of companies that have a different country/region context.

The list of country/region codes can vary between versions. Each version of an ER model mapping configuration specifies its own list.

Complete the modified version of the model mapping configuration

1. On **Configurations**, on the **Versions** FastTab, select **Change status**.

   > Change the status of designed model mapping configuration from **Draft** to **Completed**, so that ER formats can use it.

2. Select **Complete**.
3. Select **OK**.

The configuration is saved as completed version 1.

Additional resources

Electronic reporting (ER) overview

Manage ER model mapping in separate ER configurations

Apply country/region context

Frequently asked questions

I configured two shared ER model mapping configurations in RCS and marked one of them as the default model mapping configuration. I successfully ran an ER format that was created for the same base ER data model configuration, to test model mappings. I then imported the whole ER solution (ER data model, two ER model mapping configurations, and ER format configuration) into Finance. Why do I receive an error message when I try to run the same ER format in Finance?

The default model mapping setting is environment-specific. You configure it in RCS but it isn't exported to Finance. To successfully run this ER format, you must mark one of ER model mapping configurations as the default model mapping configuration in Finance too.

I configured one model mapping as a shared model mapping and completed the draft version of it. I then added a new model mapping configuration for same data model and configured it as French-specific. Why is the shared model mapping selected when I run an ER format, even though this ER format uses the correct root definition and execution is done under the control of the company that has French country/region context?

Make sure that you don't mark the shared model mapping configuration as the default model mapping configuration. Otherwise, it has higher priority during mapping selection. Also make sure that the French-specific model mapping configuration is considered when a mapping is selected during ER format execution. An ER model mapping configuration is available for selection only if at least one of the following conditions is met:

- At least one version of the ER model mapping configuration has either **Completed** or **Shared** status. In this case, the version that has the highest version number is used for ER format execution.
- The **Run draft** option for the ER model mapping configuration is turned on. In this case, the version that has **Draft** status is used for ER format execution.

> The **Run draft** option becomes available on the **Configurations** page for each ER model mapping configuration when the **Run setting** ER user parameter is turned on.

Last updated on 04/13/2026

Manage ER model mapping in separate ER configurations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following steps explain how a user assigned to the System administrator or Electronic reporting developer role can manage Electronic reporting (ER) model mappings in separate ER configurations. In this task guide, you create required ER configurations for the sample company, Litware, Inc. To complete this task guide, you must first complete the steps in the Create configuration providers and mark them as active article.

Because ER configurations are shared among companies, you can complete this task guide using the company data set of your choice. The functionality for this task guide is available if you install one of the following hotfixes: https://fix.lcs.dynamics.com/Issue/Resolved?kb=4012872 for the Dynamics AX 7.0 version or https://fix.lcs.dynamics.com/Issue/Resolved?kb=4012871 for the Dynamics 365 for Operations version.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

    - Verify that the configuration provider for the sample company Litware, Inc. is available and marked as active. If you don't see this configuration provider, you must first complete the steps in the task guide, Create a configuration provider, and mark it as active.

Add a new ER model configuration

1. Select **Reporting configurations**.

    - Add a new model configuration. The name must be unique in the configurations tree.

2. Select **Create configuration** to open the form.
3. In the **Name** field, type *Sample data model*.

    - Sample data model

4. Select **Create configuration**.

5. Select **Designer**.
6. Select **New** to open the form.
7. In the **Name** field, type *Root*.

    - Root

8. Select **Add**.
9. Select **New** to open the form.
10. In the **Name** field, type *Company*.

    - Company

11. Select **Add**.
12. In the **Description** field, enter the text, *Description of the legal entity or company in which a user logged at run-time*.

    - Description of the legal entity or company in which a user logged at run-time.

13. Select **Root reference**.
14. Select **OK**.
15. Select **Save**.
16. Close the page.
17. Select **Change status**.
18. Select **Complete**.
19. Select **OK**.

Add a new ER model-mapping configuration

1. Select **Create configuration** to open the form.
2. In the **New** field, enter *Model Mapping based on data model Sample data model*.
3. In the **Name** field, type *Sample mapping*.

    - Sample mapping

4. Select **Create configuration**.
5. Expand the **Prerequisites** section.

    - The **Implementations prerequisites** group is added automatically. The group contains the prerequisite component that refers to the parent data model configuration and is marked as **Implementation**. This status means that the **Sample mapping** model-mapping configuration is considered the implementation of the data model, **Sample data model**. Therefore, this component forces ER to download the model-mapping configuration, **Sample mapping** from an ER repository when the model configuration, **Sample data model**, is downloaded.

6. Select **Designer**.

    - The created model-mapping configuration contains a new blank mapping with the same name as the created configuration. When a selected parent model configuration contains model mappings, they are copied to a new model-mapping configuration.

7. Select **Designer**.
8. In the tree, select *Dynamics 365 for Operations\Table*.
9. Select **Add root**.
10. In the **Name** field, type *Company*.

    - Company

11. In the **Table** field, type *CompanyInfo*.

    - CompanyInfo

12. Select **OK**.
13. In the tree, expand *Company*.
14. In the tree, expand *Company\find()*.
15. In the tree, select *Company\find()\Name*.
16. Select **Bind**.
17. Select **Save**.
18. Close the page.
19. Close the page.
20. On the Action Pane, select **Configurations**.
21. Select **User parameters**.
22. Select **Yes** in the **Run settings** field.
23. Select **OK**.
24. Select **Edit**.
25. Select **Yes** in the **Run Draft** field.

Add a new ER format configuration

1. In the tree, select **Sample data model**.
2. Select **Create configuration** to open the form.
3. In the **New** field, enter `Format based on data model Sample data model`.
4. In the **Name** field, type `Sample format`.

    - Sample format

5. Select **Create configuration**.

6. Select **Designer**.
7. Select **Add root** to open the **Drop** dialog.
8. In the tree, select `Text\String`.
9. Select **OK**.
10. Select the **Mapping** tab.
11. In the tree, expand `model`.
12. In the tree, select `model\Company`.
13. Select **Bind**.
14. Select **Save**.
15. Close the page.

    - Run the draft version of the created format for testing purposes.

16. Select **Run**.

    - On the **Versions** FastTab, select **Run**.

17. Select **OK**.

    - Review the output that contains the name of the company in which the user who is running this format configuration is logged into. The created model-mapping configuration is used by this format configuration because there's only one configuration available that contains required model mappings.

Add alternative ER model-mapping configuration

1. In the tree, select **Sample data model**.
2. Select **Create configuration** to open the form.
3. In the **New** field, enter *Model Mapping based on data model Sample data model*.
4. In the **Name** field, type *Sample mapping (alternative)*.

    - Sample mapping (alternative)

5. Select **Create configuration**.
6. Select **Designer**.
7. Select **Designer**.
8. In the tree, select *Dynamics 365 for Operations\Table*.
9. Select **Add root**.
10. In the **Name** field, type *Company*.

    - Company

11. In the **Table** field, type *CompanyInfo*.

- CompanyInfo

12. Select **OK**.
13. Select **Edit**.
14. In the tree, select **String\CONCATENATE**.
15. Select **Add function**.
16. In the tree, expand *Company*.
17. In the tree, expand *Company\find()*.
18. In the tree, select *Company\find()\Name*.
19. Select **Add data source**.
20. In the **Formula** field, type a value.

    - `CONCATENATE(Company.'find()'.Name, ";",`

21. In the tree, select **Company\find()\Company(DataArea)**.
22. Select **Add data source**.
23. In the **Formula** field, type a value.

    - `CONCATENATE(Company.'find()'.Name, ";", Company.'find()'.DataArea)`

24. Select **Save**.
25. Close the page.
26. Select **Save**.
27. Close the page.
28. Close the page.
29. Select **Yes** in the **Run Draft** field.

Use an existing ER model-mapping configuration

1. In the tree, select `Sample data model\Sample format`.
2. Select **Run**.

    - You can't run the selected draft version of the ER format configuration because more than one model-mapping configuration exists for the undefined data model that is selected as the data source of the running ER format.
    - Next, you need to define the alternative model-mapping configuration as the one from which model mappings are used as data sources for running ER format.

3. In the tree, select `Sample data model\Sample mapping (alternative)`.
4. Select **Yes** in the **Default for model-mapping** field.
5. In the tree, select `Sample data model\Sample format`.
6. Select **Run**.
7. Select **OK**.

- This format configuration uses the default model-mapping configuration to generate the electronic document. The created output contains the company code.

Last updated on 03/17/2026

Manage several derived mappings for a single model root

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use an Electronic reporting (ER) data model component as the data source in every configured ER format component to generate outbound documents. To describe a single business domain, configure a data model component that has many root definitions.

Each root definition lets you represent data of that domain in the way that best suits specific reporting purposes. For every root definition, you can configure an ER model mapping component as the Microsoft Dynamics 365 Finance–specific implementation of your data model. In this way, you describe how your data model is filled in at runtime.

ER model mapping components can reside in ER data model configurations and ER model mapping configurations. A single ER configuration can contain many mapping components, each of which is configured for a single root definition. Alternatively, a single ER configuration can contain just one mapping component that is configured for a single root definition.

Many configuration providers might offer ER model mapping configurations for the same ER data model. Those model mapping configurations might contain mapping components for different root definitions. You might use a model mapping for one root definition that one provider offers and use a model mapping for another root definition that another provider offers.

The procedures in this article explain how to manage multiple ER model mapping configurations of an ER data model when they contain different model mapping components configured for the same root definition.

To complete the procedures in this article, you must be assigned to the System administrator or Electronic reporting developer role.

You can perform all the following procedures in the USMF company. No coding is required.

Configure the ER framework

As a user in the Electronic reporting developer role, configure the minimal set of ER parameters before you use the ER framework to generate business documents.

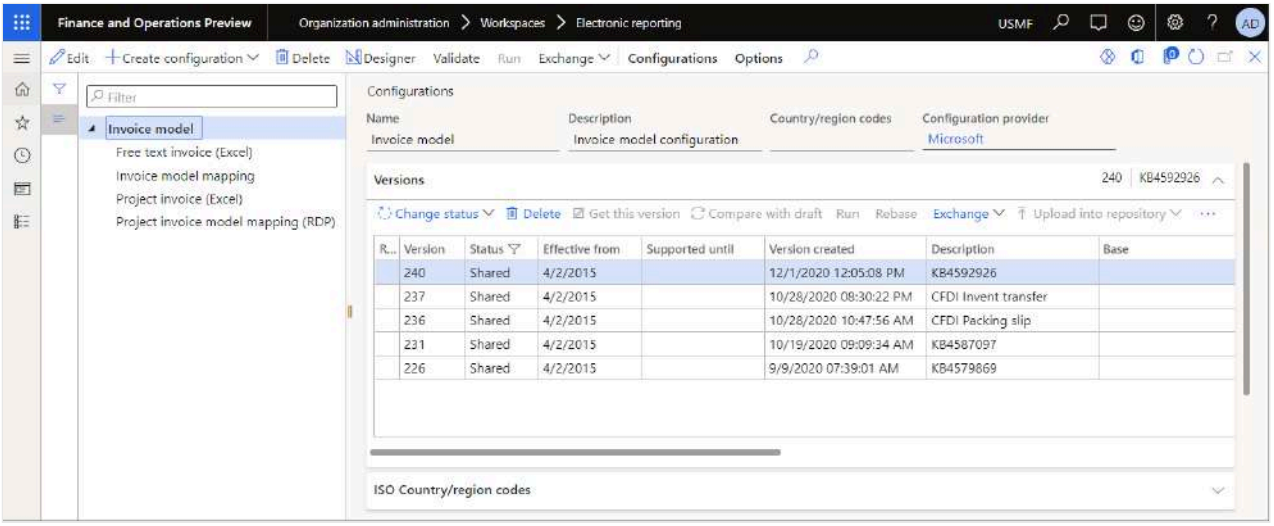
Import the standard ER format configurations

To add the standard ER configurations to your current instance of Finance, import them from the ER repository that you configured for that instance. Follow the steps in Download ER configurations from the Global repository of Configuration service to import the following ER format configurations:

- **Free text invoice (Excel)**, version 220.106
- **Project invoice (Excel)**, version 226.27

Review the imported ER configurations

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configurations** section, select the **Reporting configurations** tile.
3. On **Configurations**, in the configuration tree in the left pane, expand **Invoice model**.



4. Review the **Free text invoice (Excel)** format:

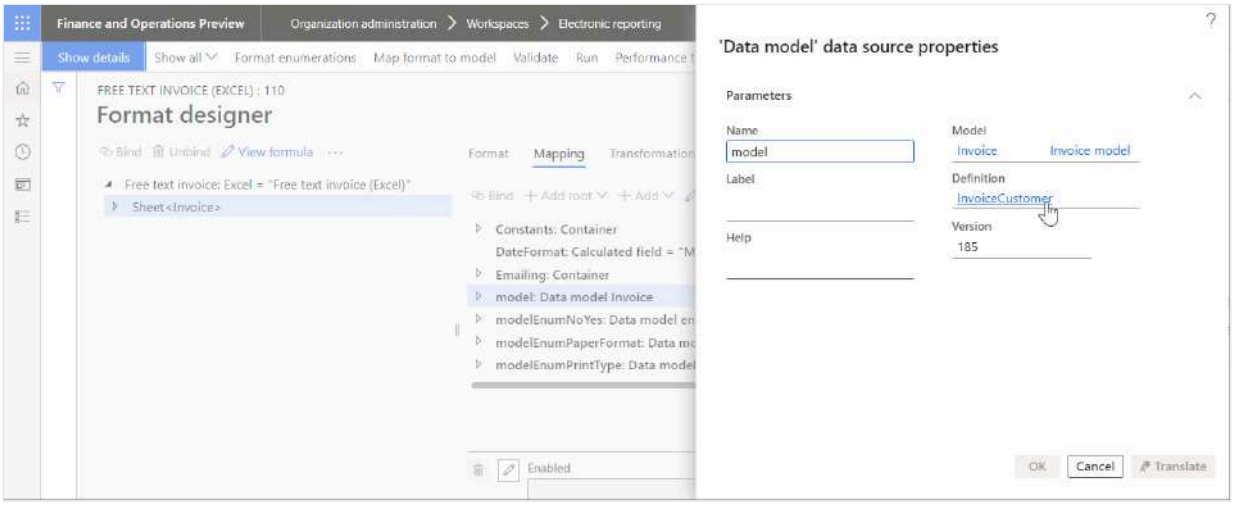
   a. In the configuration tree in the left pane, select **Free text invoice (Excel)**.

b. On the Action Pane, select **Designer**.

c. On **Format designer**, on the **Mapping** tab, in the data sources list, select **Model**.

d. Select **View**.

The current ER format is configured to use the **InvoiceCustomer** root definition of **Invoice model**. When you run this format and call the **Model** data source, the model mapping that you configured for the **InvoiceCustomer** root definition is used to access application data and fill in the data model.



e. Close **Format designer**.

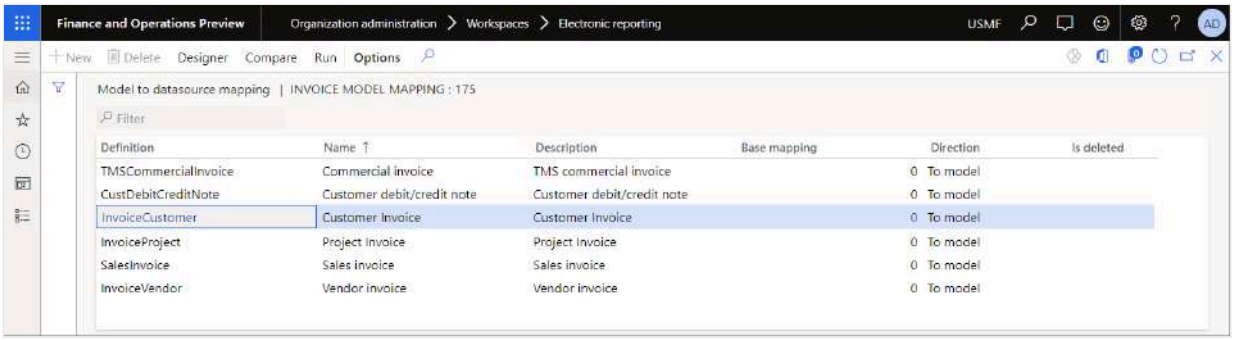
5. Review the content of the **Invoice model mapping** configuration:

a. In the configuration tree in the left pane, select **Invoice model mapping**.

b. On the Action Pane, select **Designer**.

c. On **Model to datasource mapping**, notice that the current ER model mapping configuration contains several model mapping components:

- The **Customer Invoice** model mapping is configured for the **InvoiceCustomer** root definition of **Invoice model**. Therefore, when you run the **Free text invoice (Excel)** ER format, you can choose the **Customer Invoice** model mapping of this ER configuration to access application data and fill in the data model.
- The **Project Invoice** model mapping is configured for the **InvoiceProject** root definition of **Invoice model**. Therefore, when you run the **Project invoice (Excel)** ER format, you can choose the **Project Invoice** model mapping of this ER configuration to access application data and fill in the data model.



d. Close **Model to datasource mapping**.

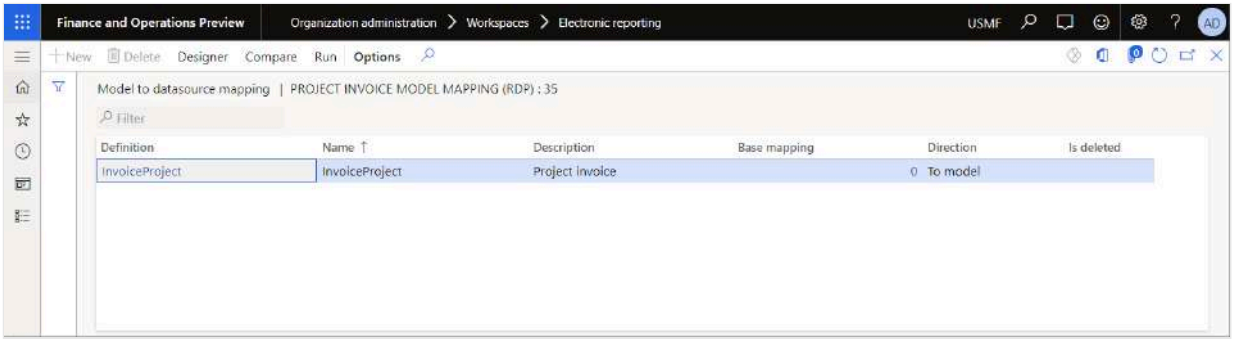
e. On the **Versions** FastTab, select **Delete** to delete all versions of this ER configuration that are later than version 240.175.

6. Review the content of the **Project invoice model mapping (RDP)** configuration:

a. In the configuration tree in the left pane, select **Project invoice model mapping (RDP)**.

b. On the Action Pane, select **Designer**.

c. On **Model to datasource mapping**, notice that the current ER model mapping configuration contains the **InvoiceProject** model mapping, and that this model mapping is configured for the **InvoiceProject** root definition of **Invoice model**. When you run the **Project invoice (Excel)** ER format, select the **InvoiceProject** model mapping of this ER configuration to access application data and fill in the data model.



d. Close **Model to datasource mapping**.

e. On the **Versions** FastTab, select **Delete** to delete all versions of this ER configuration that are later than version 226.35.

Customize the imported ER configurations

This section explains how to customize the model mappings that Microsoft provides. For example, you might need to customize the model mappings to implement your custom logic or add missing bindings.

Customize the Invoice model mapping configuration

1. On **Configurations**, in the configuration tree in the left pane, select **Invoice model mapping**.

2. On the action pane, select **Create configuration**.

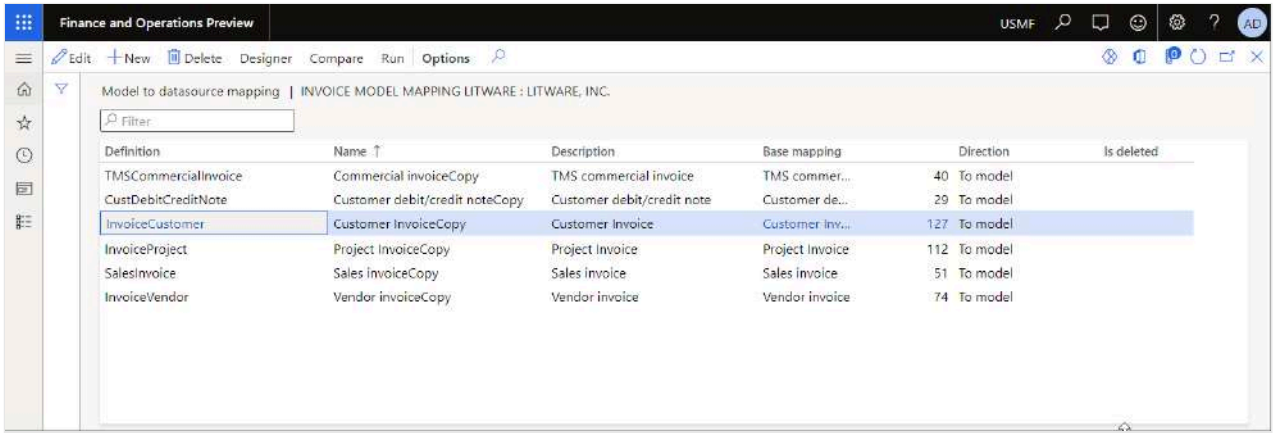
3. In **Create configuration**, select **Derive from Name: Invoice model mapping, Microsoft** in the **New** field.

4. Enter **Invoice model mapping Litware** in the **Name** field.

5. Select **Create configuration**.

6. Mark the draft version of the derived mapping as available for use at runtime:
   a. On the action pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
   b. In **User parameters**, set the **Run settings** option to **Yes**, and then select **OK**.
   c. Select **Edit** to make the page editable, as required.
   d. For the **Invoice model mapping Litware** configuration that you selected in the configuration tree, set the **Run Draft** option to **Yes**.

7. On the action pane, select **Designer** to review the model mappings of this configuration.

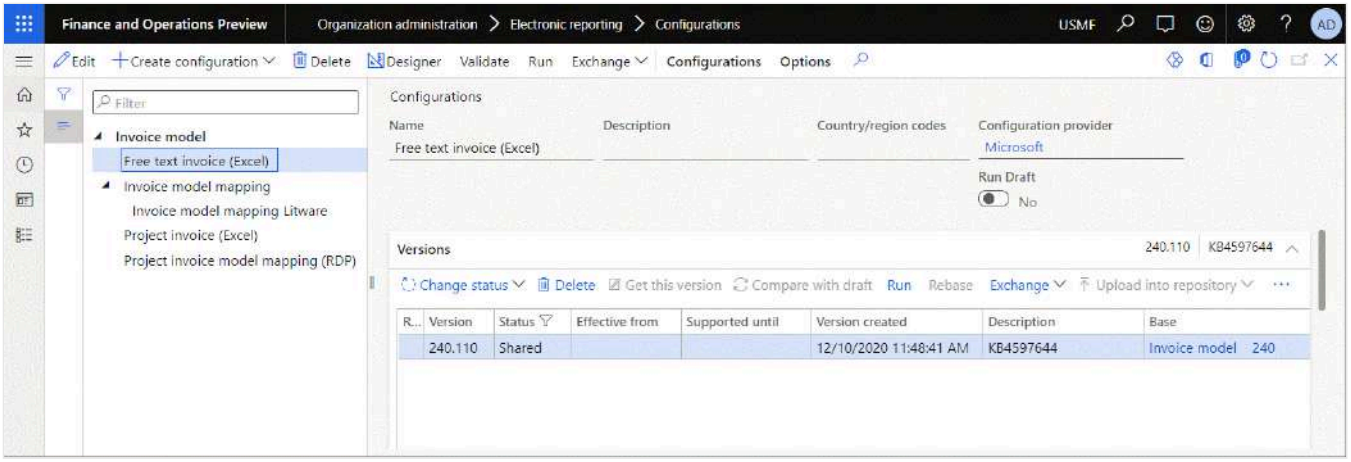


> **Tip**
>
> You can now open any of the ER model mapping components of this ER configuration in the designer to configure your custom logic. For more information, see Customize the model mapping configuration.

8. Close **Model to datasource mapping**.

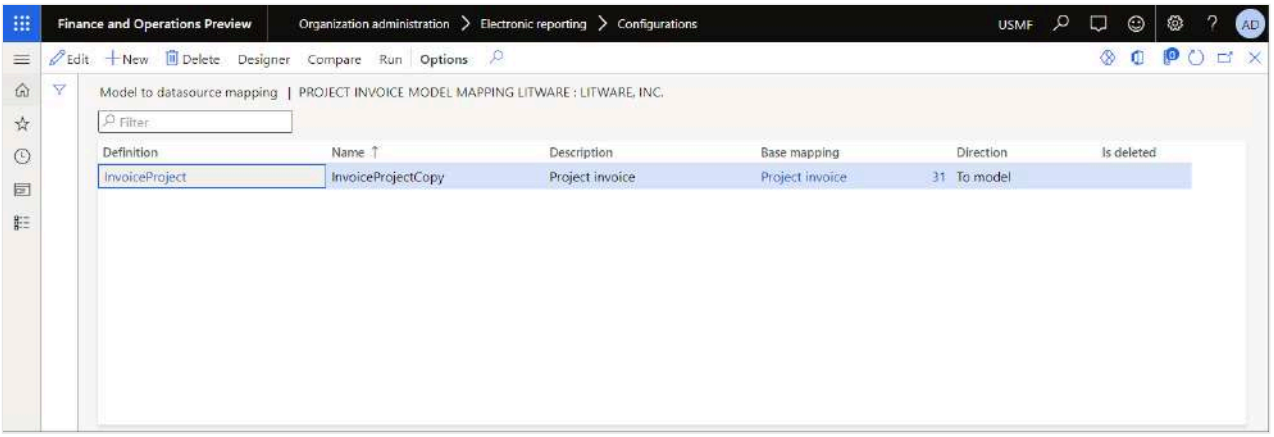
You now have **Invoice model mapping** and **Invoice model mapping Litware** configurations, each of which has a model mapping that is configured for the **InvoiceCustomer** root definition. Explicitly assign one of the model mappings as the default model mapping that any of the ER formats uses, such as the **Free text invoice (Excel)** format that contains a model data source that has the **InvoiceCustomer** root definition. Otherwise, when you run, edit, or validate one of the ER formats, the following exception is thrown to notify you that no default model mapping is explicitly assigned:

> More than one model mapping exists for the '<model name> (<root descriptor>)' data model in the configurations <configuration names separated by commas>. Set one of the configurations as default.



Customize the Project invoice model mapping (RDP) configuration

1. On **Configurations**, in the configuration tree in the left pane, select **Project invoice model mapping (RDP)**.
2. On the action pane, select **Create configuration**.
3. In **Create configuration**, in the **New** field, select **Derive from Name: Project invoice model mapping (RDP), Microsoft**.
4. In the **Name** field, enter **Project invoice model mapping Litware**.
5. Select **Create configuration**.
6. For the **Project invoice model mapping Litware** configuration that you selected in the configuration tree, set the **Run Draft** option to **Yes**.
7. On the action pane, select **Designer** to review the model mappings of this configuration.



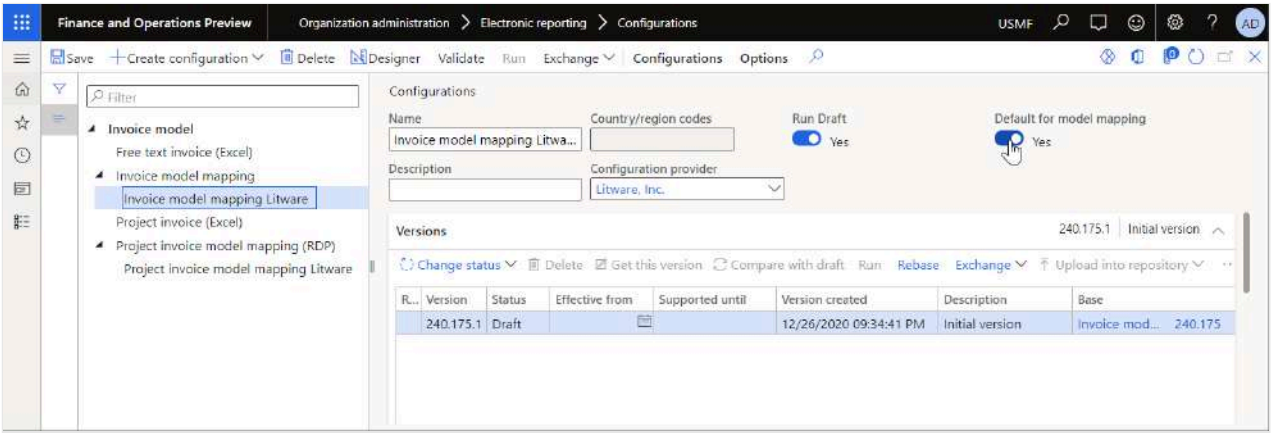
8. Close **Model to datasource mapping**.

You now have **Invoice model mapping**, **Project invoice model mapping (RDP)**, and **Project invoice model mapping Litware** configurations. Each of these configurations has a model mapping configured for the **InvoiceProject** root definition. You must explicitly assign one of the model mappings as the default model mapping that any of the ER formats uses. For example, use the **Project invoice (Excel)** format that contains a model data source that has the **InvoiceProject** root definition. Otherwise, when you run or edit one of the ER formats, an exception is thrown to notify you that no default model mapping is explicitly assigned.

Select the derived Invoice model mapping Litware configuration as the configuration that contains default model mappings

1. On **Configurations**, in the configuration tree in the left pane, select **Invoice model mapping Litware**.

2. Set the **Default for model mapping** option to **Yes**.



Because of this setting, the **Customer Invoice Copy** model mapping is used when you run the **Free text invoice (Excel)**, or when you edit or validate it. The **Customer invoice** model

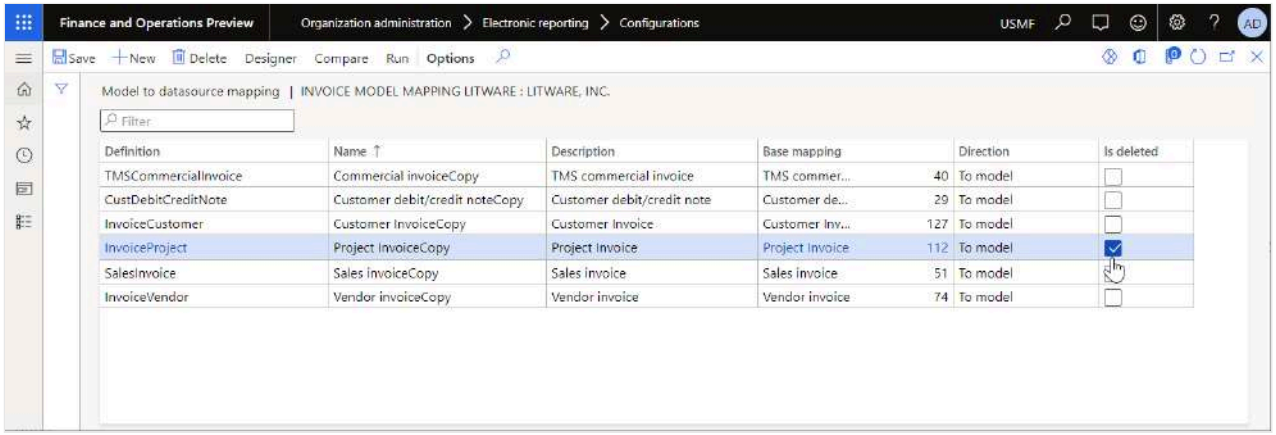
mapping from the **Invoice model mapping** configuration is ignored.

You can now open the **Free text invoice (Excel)** format for review in the format designer.

Select the derived Project invoice model mapping Litware configuration as the configuration that contains default model mappings

1. On **Configurations**, in the configuration tree in the left pane, select **Project invoice model mapping Litware**.
2. Set the **Default for model mapping** option to **Yes**.

   In this case, unlike the case that is described for the **Invoice model mapping Litware** configuration in the previous section, you can't start to use the **InvoiceProject Copy** model mapping from the **Project invoice model mapping Litware** configuration. Two configurations that contain a model mapping for the **InvoiceProject** root definition are currently marked as the default configuration. Therefore, they have equal priority for use. To resolve this issue, complete the remaining steps of this procedure.
3. In the configuration tree in the left pane, select **Invoice model mapping Litware**.
4. On the Action Pane, select **Designer**.
5. On the **Model to datasource mapping** page, select **Edit** to make the page editable, as required.
6. Select the **Project Invoice Copy** model mapping, and then select the **Is deleted** check box for it.



Because of this setting, the **Invoice model mapping Litware** configuration is treated as though it has no model mapping for the **InvoiceProject** root definition. The

InvoiceProject Copy model mapping is the default. The configuration, **Project invoice model mapping Litware**, which contains this model mapping, is marked as the default configuration. Because it is marked as default, it has a higher priority than the **InvoiceProject** model mapping from the **Project invoice model mapping (RDP)** configuration.

Other considerations

The **InvoiceProject Copy** model mapping of the **Project invoice model mapping Litware** configuration is designed to use the **ReportDataProvider** data source. The data source is part of the *Object* type that refers to the **PsaProjInvoiceDP** application class. This class is implemented as the data provider for the project invoice SQL Server Reporting Services (SSRS) report of the Print management framework. Select this data source as the ER integration point. The current ER implementation for Print management reports takes this setting into account. For more details, review the source code of the **ERPrintMgmtDataProviderReport** application class. At runtime, the assignment of the **ReportDataProvider** data source as the model mapping integration point forces Finance to treat this mapping component with a higher priority than the **InvoiceProject** mapping component from the **Project invoice model mapping (RDP)** configuration.

See also

- Manage ER model mapping in separate ER configurations
- Configure country/region context dependent ER model mappings
- Electronic reporting framework API changes

Last updated on 04/13/2026

Select data model definitions when you create formats

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To complete the steps in this procedure, you must first complete the procedure, Create configuration providers and mark them as active.

This procedure shows how to select a model's root item as a data model definition for inserting an Electronic reporting (ER) format configuration that's designed to generate electronic documents. In this procedure, you add a new ER format configuration for the sample company Litware, Inc.

This procedure is intended for users who have the System administrator or Electronic reporting developer role assigned to them. You can complete the steps by using any dataset.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
    - Make sure that the configuration provider for the sample company, Litware, Inc., is available and marked as **Active**. If you don't see this configuration provider, complete the steps in the procedure, Create configuration providers and mark them as active.
2. Select **Reporting configurations**.

Add a new ER data model configuration

1. Select **Create configuration** to open the **Drop** dialog.
    - Add a new ER model configuration containing a data model that is designed for use as a data source in generating ER reports.
2. In the **Name** field, enter `Payment model (fictitious)`.
    - Payment model (fictitious)
3. Select **Create configuration**.

4. Select **Designer**.

    - Open the ER designer to specify the structure of the data model for this configuration.
    - Design the data model for the payments business domain to support two payment methods – credit transfer and direct debit.

5. Select **New** to open the form.
6. In the **Name** field, enter `Payments – credit transfer`.

    - Payments – credit transfer

7. Select **Add**.
8. Select **New** to open the form.
9. In the **New node as a field**, enter `Model root`.
10. In the **Name** field, enter `Payments – direct debit`.

    - Payments – direct debit

11. Select **Add**.
12. Select **Save**.
13. Close the page.
14. Select **Change status**.

    - Complete the draft version of the model to make it available in new model mappings and formats.

15. Select **Complete**.
16. Select **OK**.

Start to enter a new ER format configuration

1. Select **Create configuration** to open the **Drop** dialog.
2. In the **New** field, enter *Format based on data model Payment model (fictitious)*.
3. In the **Data model definition** field, enter or select a value.

    - You can select any root item from the chosen data model as the data model definition. You can use any required root item from the data model to design your format. If a model mapping is missing for the selected root item, you can still continue.

4. Close the page.

Add a new ER model mapping configuration

1. Select **Create configuration** to open the **Drop** dialog.
2. In the **New** field, enter *Model Mapping based on data model Payment model (fictitious)*.
3. In the **Name** field, enter *Payment model mappings (fictitious)*.

    - Payment model mappings (fictitious)

4. In the **Data model definition** field, enter or select a value.
5. Select **Create configuration**.

Design ER model mappings

1. Select **Designer**.

    - Use the ER designer to specify the model mappings for the required root items.

2. Select **Designer**.

    - Simulate setting of selected model mapping for the selected model's root item.

3. In the tree, select `Dynamics 365 for Operations\Table records`.
4. Select **Add root**.
5. In the **Name** field, type `Ledger`.
6. In the **Table** field, type `LedgerJournalTrans`.

    - LedgerJournalTrans

7. Select **OK**.
8. Select **Save**.
9. Close the page.
10. Close the page.

Start to enter another new ER format configuration

1. In the tree, select `Payment model (fictitious)`.
2. Select **Create configuration** to open the **Drop** dialog.
3. In the **New** field, enter *Format based on data model Payment model (fictitious)*.
4. In the **Data model definition** field, enter or select a value.

    - Only one root item is available to map to the application data sources. When you add at least one model mapping, you can select only the model's root items that are

mapped to application data sources as a model definition while the ER format is added.

5. Close the page.

Last updated on 03/17/2026

Configure ER formats to use parameters that are specified per legal entity

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Overview

In many of the Electronic reporting (ER) formats that you design, you must filter data by using a set of values that are specific to each legal entity of your instance (for example, a set of tax codes to filter tax transactions). Currently, when you configure this type of filtering in an ER format, you use values that depend on the legal entity (for example, tax codes) in expressions of the ER format to specify data filtering rules. Therefore, the ER format is legal entity–specific. To generate the required reports, you must create derived copies of the original ER format for each legal entity where you want to run the ER format. You must edit each derived ER format to add legal entity–specific values. You must rebase each derived ER format whenever the original (base) version is updated. You must export each derived ER format from a test environment and import it into a production environment when it must be deployed for production use. Maintenance of this type of configured ER solution is complex and time-consuming for several reasons:

- The more legal entities there are, the more ER format configurations you need to maintain.
- Maintenance of ER configurations requires that business users have ER knowledge.

By using the ER application-specific parameters feature, power users can configure data filtering in an ER format so that it's based on a set of abstract rules. They can configure this set of rules can to use the data sources available in an ER format. Business users can then specify real rules beyond the ER framework by using the user interface (UI) that is automatically generated based on the settings of the corresponding ER format and the current legal entity data that the ER format's dat asources can access. The set of rules specified for an ER format can be exported from the current legal entity of the Dynamics 365 Finance (Finance) instance. It can then be imported into another legal entity of either the same Finance instance or a different instance as a set of rules for the same ER format.

Prerequisites

To complete the examples in this article, you must have access to the instance of Regulatory Configuration Services (RCS) provisioned for the same tenant as Finance for one of the following roles:

- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

You need to complete the steps in the Support parameterized calls of ER data sources of CALCULATED FIELD type article. If you completed those steps, you can skip the steps in the **Import ER configurations into RCS** section that follows.

Import ER configurations into RCS

Download and save the following ER configurations to your local device.

Expand table

Content description	File name
Sample **ER data model** configuration file	Model to learn parameterized calls.version.1.xml
Sample **ER metadata** configuration file	Metadata to learn parameterized calls.version.1.xml
Sample **ER model mapping** configuration file	Mapping to learn parameterized calls.version.1.1.xml
Sample **ER format** configuration	Format to learn parameterized calls.version.1.1.xml

Next, sign in to your RCS instance.

In this example, you create a configuration for the Litware, Inc sample company. Before you can complete this procedure, you must complete the steps in the Create a configuration provider and mark it as active article in RCS.

1. On the default dashboard, select **Electronic reporting**.
2. Select **Reporting configurations**.
3. Import the ER configurations that you downloaded earlier into RCS in the following order: data model, metadata, model mapping, and format. For each ER configuration, follow these steps:
   a. Select **Exchange**.
   b. Select **Load from XML file**.

c. Select **Browse** to select the file for the required ER configuration in XML format.
d. Select **OK**.

Review the ER solution provided.

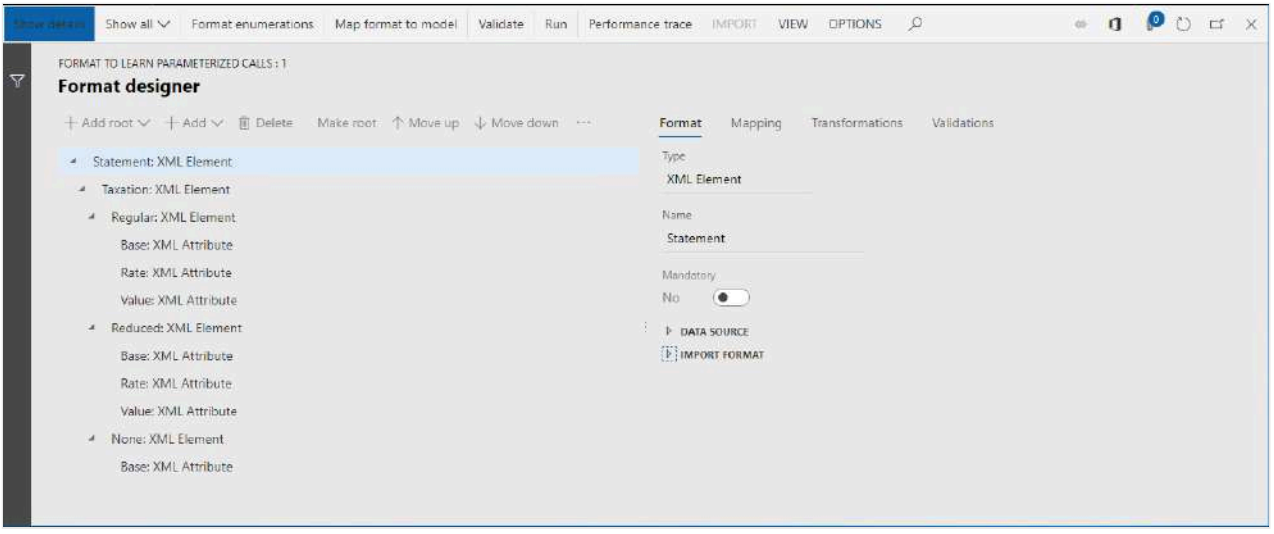
1. In the configuration tree, expand the contents of the **Model to learn parameterized calls** item.

2. Select the **Format to learn parameterized calls** item.

3. Select **Designer**.

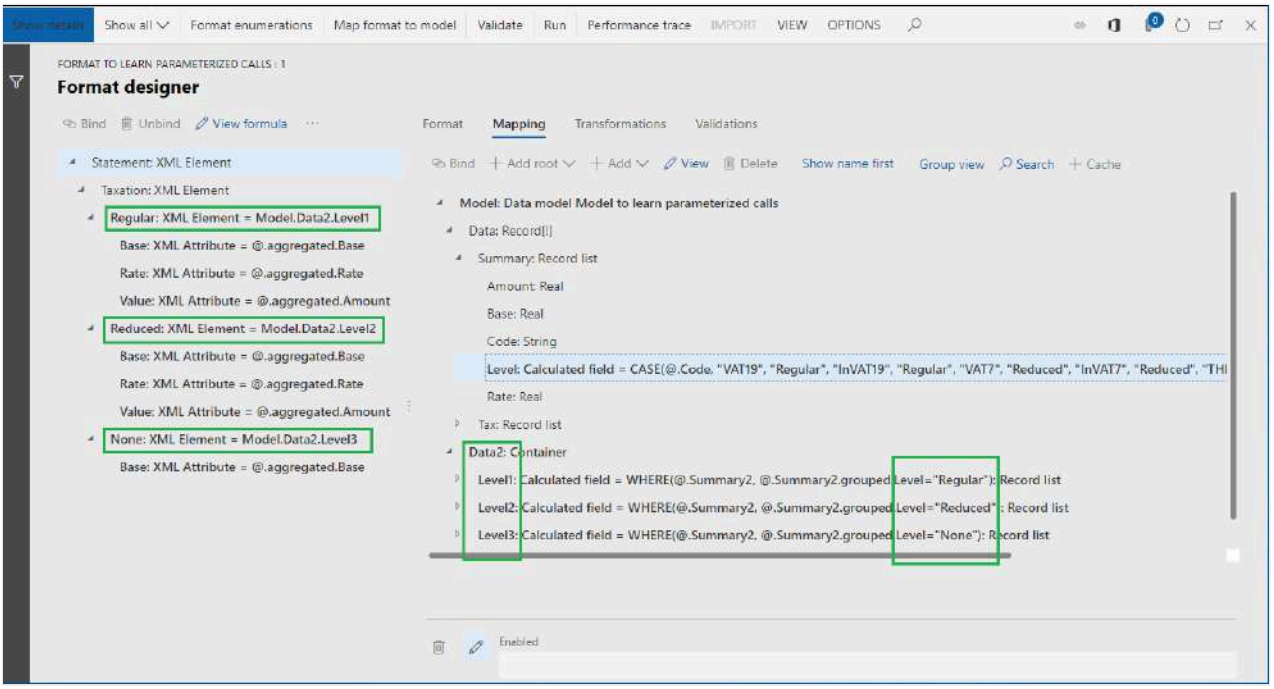
4. Select **Expand/Collapse**.

   The **Format to learn parameterized calls** ER format is designed to generate a tax statement in XML format that presents several levels of taxation (regular, reduced, and none). Each level has a different number of details.

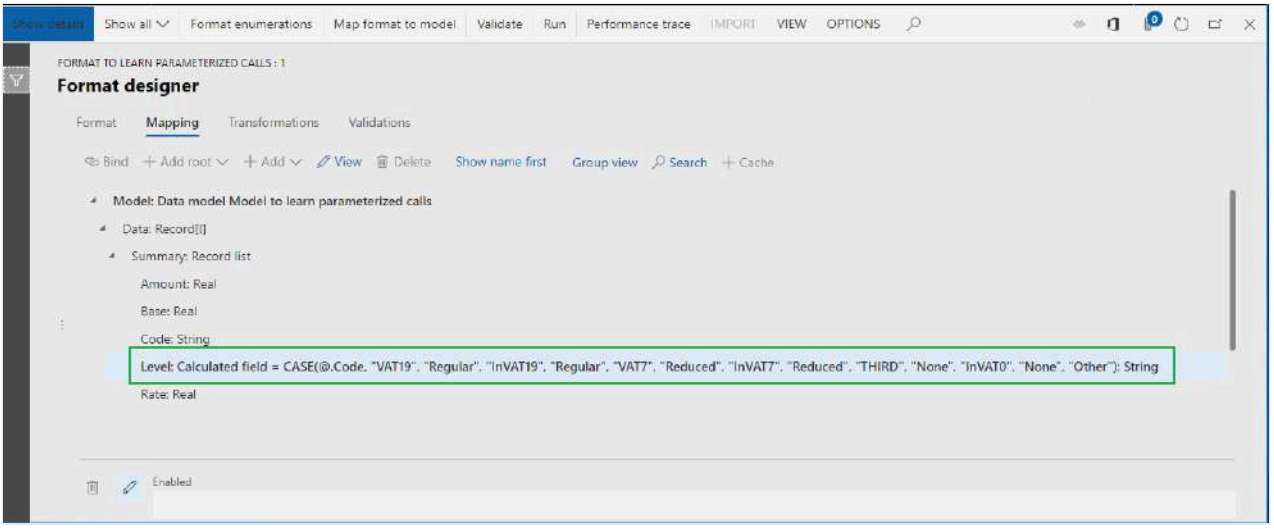


5. On the **Mapping** tab, expand the **Model**, **Data**, and **Summary** items.

   The **Model.Data.Summary** data source returns the list of tax transactions. These transactions are summarized by tax code. For this data source, the **Model.Data.Summary.Level** calculated field is configured to return the code for the taxation level of each summarized record. For any tax code retrieved from the **Model.Data.Summary** data source at runtime, the calculated field returns the taxation level code (**Regular**, **Reduced**, **None**, or **Other**) as a text value. The **Model.Data.Summary.Level** calculated field filters records of the **Model.Data.Summary** data source and enters the filtered data in each XML element that represents a taxation level by using the **Model.Data2.Level1**, **Model.Data2.Level2**, and **Model.Data2.Level3** fields.



The **Model.Data.Summary.Level** calculated field contains an ER expression. Tax codes (**VAT19**, **InVAT19**, **VAT7**, **InVAT7**, **THIRD**, and **InVAT0**) are hardcoded into this configuration. Therefore, this ER format depends on the legal entity where the tax codes were configured.



To support a different set of tax codes for each legal entity, you must follow these steps:

- Create a derived version of the ER format for each legal entity.
- Update the tax codes in the **Model.Data.Summary.Level** calculated field, based on the legal entity setting.

6. Close the **Format designer** page.

Create a derived format

Next, you will use the ER application-specific parameters feature to support a different set of tax codes for each legal entity in a single ER format.

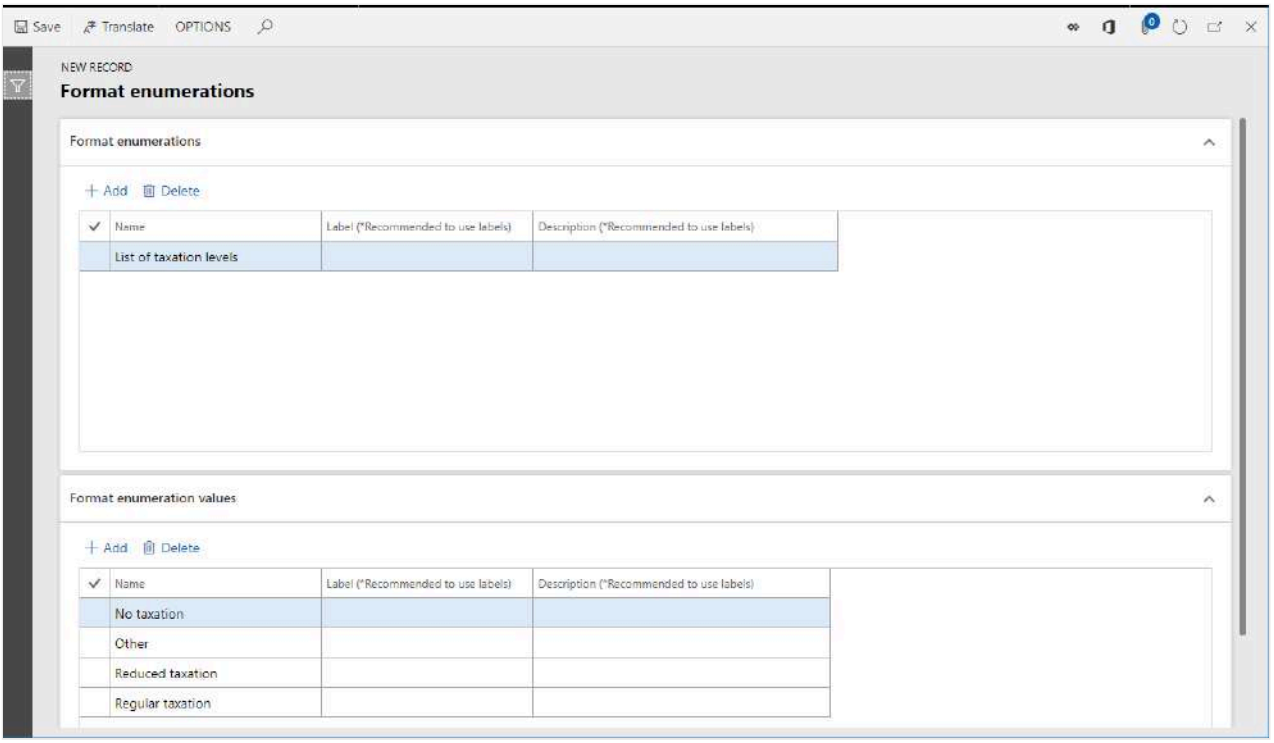
1. In the configuration tree, expand the contents of the **Model to learn parameterized calls** item.
2. Select the **Format to learn parameterized calls** item.
3. Select **Create configuration**.
4. Select the **Derive from Name: Format to learn parameterized calls, Microsoft** option.
5. In the **Name** field, enter **Format to learn how to lookup LE data**.
6. Select **Create configuration**.

Configure a derived format

Add a format enumeration

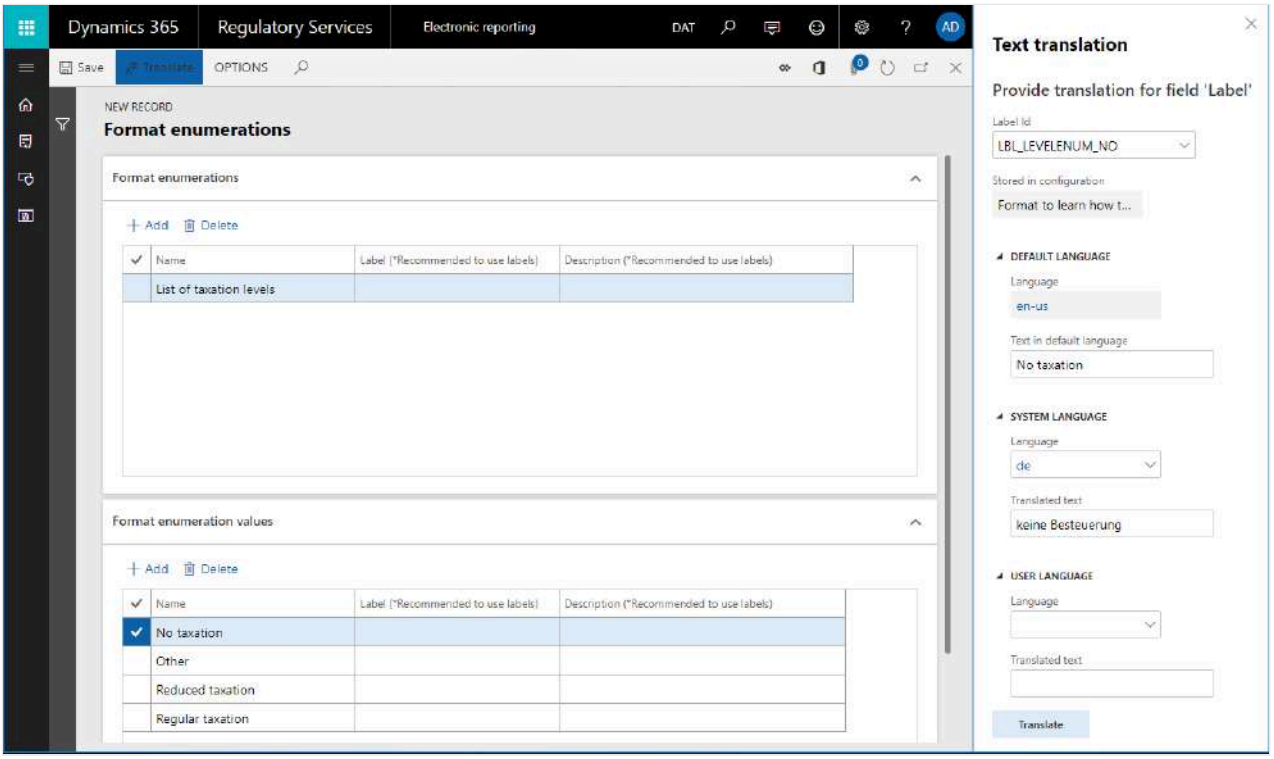
Next, add a new ER format enumeration. The values of this format enumeration are presented to business users so they can specify legal entity–dependent sets of tax codes for the various taxation levels that are used in the ER format.

1. Select **Designer**.
2. Select **Format enumerations**.
3. Select **Add**.
4. In the **Name** field, enter **List of taxation levels**.
5. Select **Save**.
6. On the **Format enumeration values** tab, select **Add**.
7. In the **Name** field, enter **Regular taxation**.
8. Select **Add** again.
9. In the **Name** field, enter **Reduced taxation**.
10. Select **Add** again.
11. In the **Name** field, enter **No taxation**.
12. Select **Add** again.
13. In the **Name** field, enter **Other**.



Because the business users might use different languages to specify legal entity–dependent sets of tax codes, we recommend that you translate the values of this enumeration into the languages that are configured as the preferred languages for those users in Finance.

14. Select the **No taxation** record.
15. Click in the **Label** field.
16. Select **Translate**.
17. In the **Text translation** pane, in the **Label ID** field, enter **LBL_LEVELENUM_NO**.
18. In the **Text in default language** field, enter **No taxation**.
19. Select **DE** in the **Language** field.
20. In the **Translated text** field, enter **keine Besteuerung**.
21. Select **Translate**.



22. Select **Save**.

23. Close the **Format enumerations** page.

Add a new lookup data source

Next, add a new data source to specify how business users can create legal entity–dependent rules to recognize the correct taxation level for each summarized transaction record.

1. On the **Mapping** tab, select **Add**.

2. Select **Format enumeration\Lookup**.

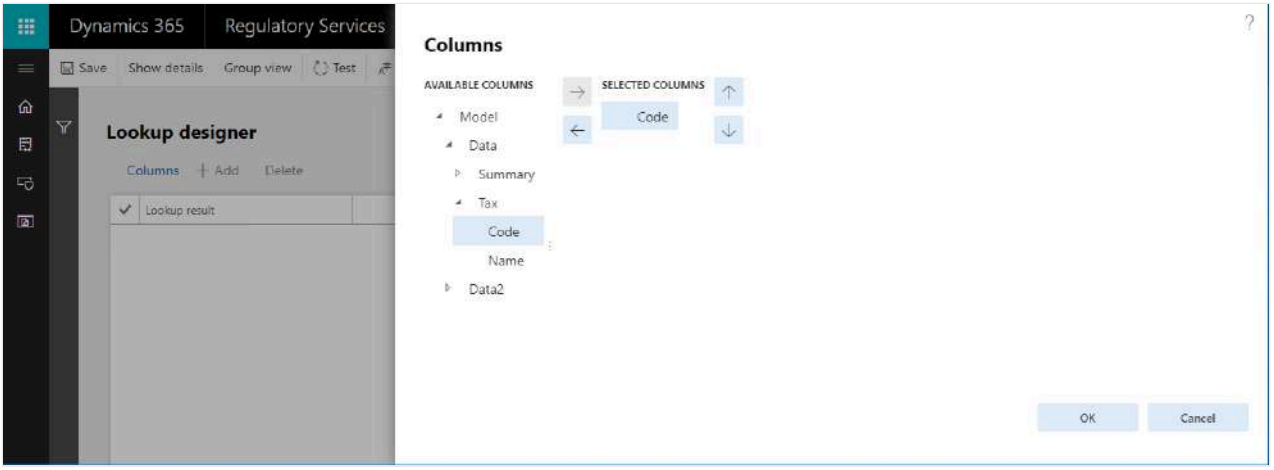
   You just identified that each rule business users specify for taxation level recognition returns a value of an ER format enumeration. Notice that the **Lookup** data source type can be accessed under the **Data model** and **Dynamics 365 for Operations** blocks in addition to the **Format enumeration** block. Therefore, you can use ER data model enumerations and application enumerations to specify the type of values returned for the data sources of that type. To learn more about the **Lookup** data sources, see Configure Lookup data sources to use the ER application-specific parameters feature.

3. In the **Name** field, enter **Selector**.

4. In the **Format enumeration** field, select **List of taxation levels**.

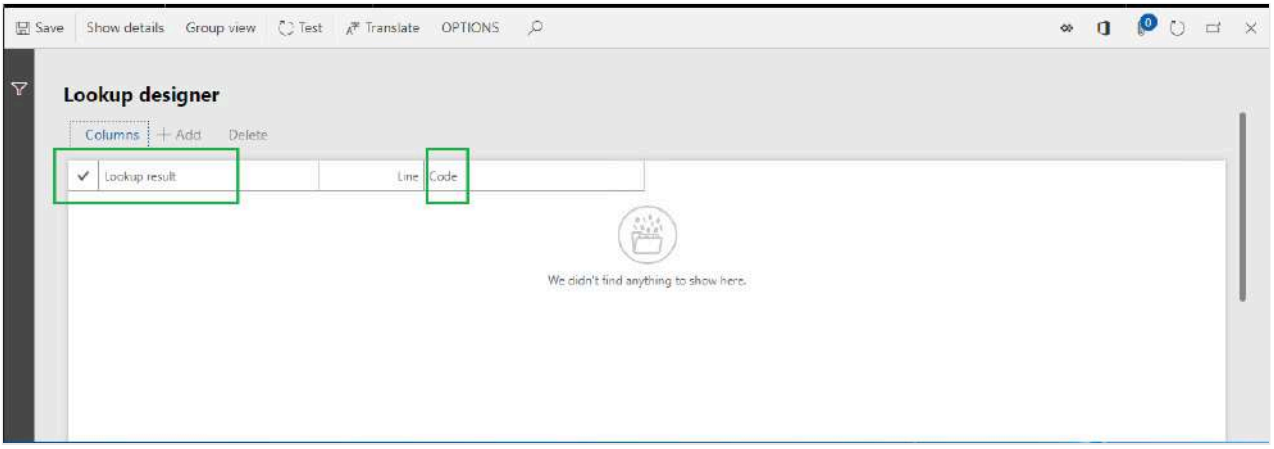
   You specified that for each rule in this data source, a business user must select one of the values of the **List of taxation levels** format enumeration as a returned value.

5. Select **Edit lookup**.
6. Select **Columns**.
7. Expand the **Model** item.
8. Expand the **Data** item.
9. Expand the **Tax** item.
10. Select the **Model.Data.Tax.Code** item.
11. Select the **Add** button (the right arrow).



You just specified that for each rule in this data source for taxation level recognition, a business user must select one of the tax codes as a condition. The **Model.Data.Tax** data source returns the list of tax codes that the business user can select. Because this data source contains the **Name** field, the name of the tax code appears for each tax code value in the lookup that is presented to the business user.

12. Select **OK**.



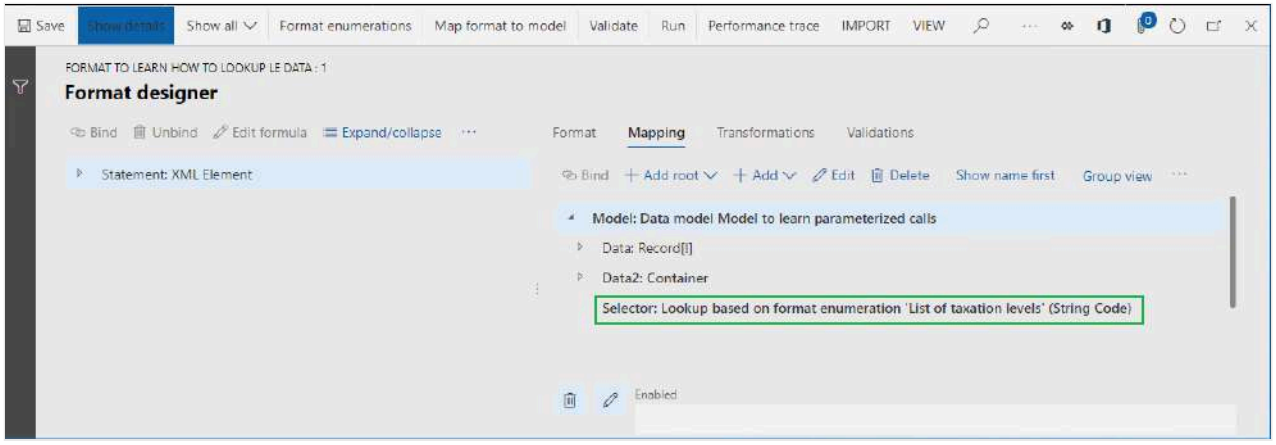
Business users can add multiple rules as records of this data source. Each record is numbered by a line code. Rules are evaluated in order of increasing line number.

Because you selected the **Tax code** field as a condition for rules in this lookup data source, and because **Tax code** is set up as a field of the **String** data type, each rule is evaluated at runtime by comparing the tax code that is passed to the data source with the tax code that is defined in this record of the data source.

When a rule that satisfies the configured condition is found, this data source returns the lookup value of the rule that is defined in the **Lookup result** field. If no rule is found, an exception is thrown to notify the user that the current data source can't return a correct value.

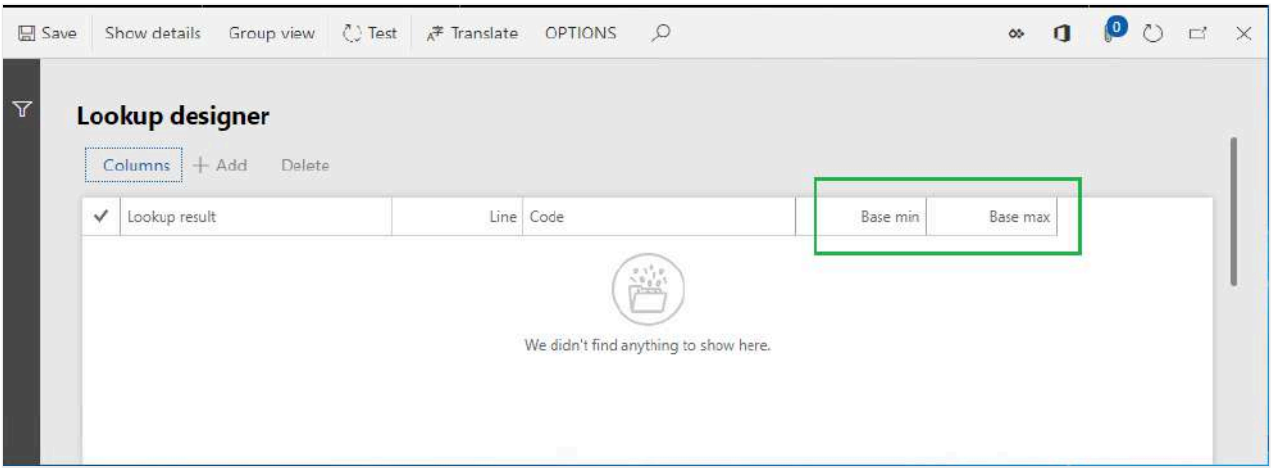
13. Select **Save**.
14. Close the **Lookup designer** page.
15. Select **OK**.

You added a new data source that returns the taxation level as the value of the **List of taxation levels** format enumeration for any tax code that is passed to the data source as the argument of the **Code** parameter of the **String** data type.



The evaluation of configured rules depends on the data type of the fields that you select to define conditions of those rules. When you select a field that is configured as a field of either the **Numeric** or **Date** data type, the criteria differ from the criteria described earlier for the **String** data type. For **Numeric** and **Date** fields, the rule must be specified as a range of values. The condition of the rule is considered satisfied when a value that is passed to the data source is in the configured range.

The following illustration shows an example of this type of setup. In addition to the **Model.Data.Tax.Code** field of the **String** data type, the **Model.Tax.Summary.Base** field of the **Real** data type is used to specify conditions for a lookup data source.



Because the **Model.Data.Tax.Code** and **Model.Tax.Summary.Base** fields are selected for this lookup data source, each rule of this data source is configured in the following way:

- In the list that is presented, select the value of the **List of taxation levels** format enumeration as a returned value.
- Enter the tax code as a condition of this rule. Only tax codes that the **Model.Data.Tax** data source provides are applicable.
- Enter minimum and maximum values of the tax base amount as conditions of this rule.

Here is how each rule of this data source will be evaluated at runtime:

- Does the code of the **String** data type that was passed to this data source equal the tax code of a rule?
- Does the value of the **Real** data type that was passed to this data source fall between specific minimum and maximum values?

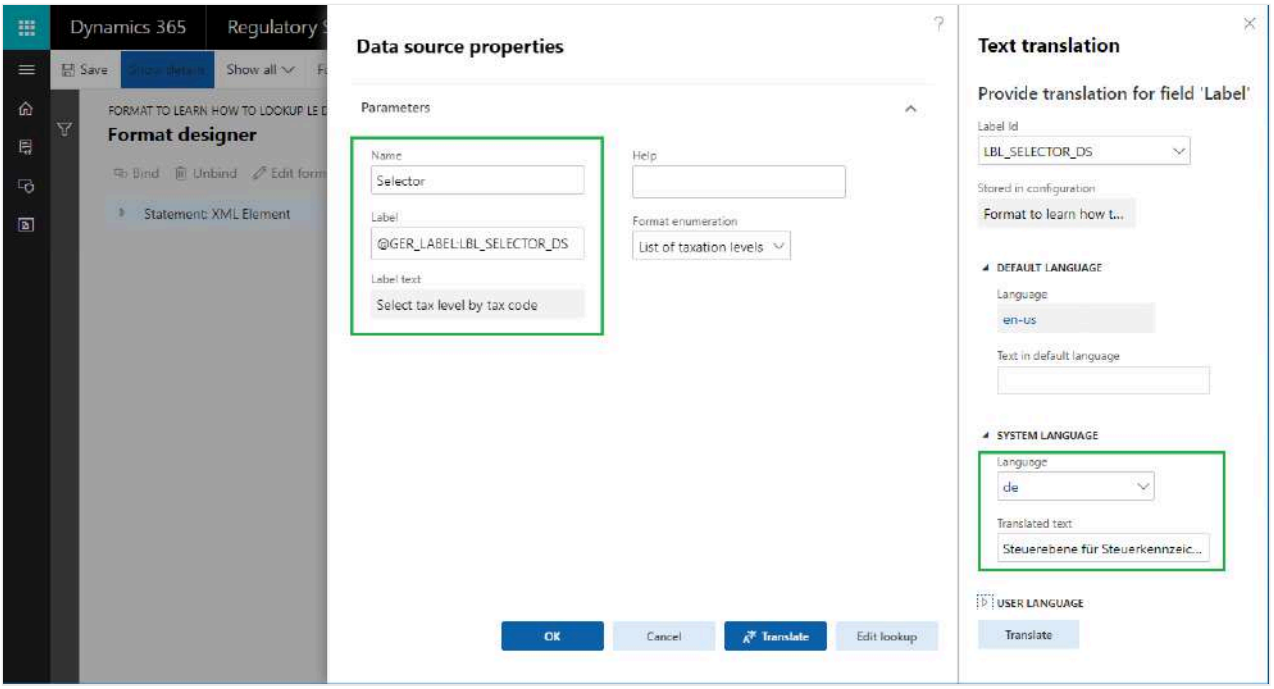
A rule is applicable when both conditions are satisfied.

Translate the label of the lookup data source that was added

Because business users might use different languages to specify legal entity–dependent sets of tax codes, translate the label of any lookup data source that you add. This way, it's presented in each user's preferred language on the corresponding page.

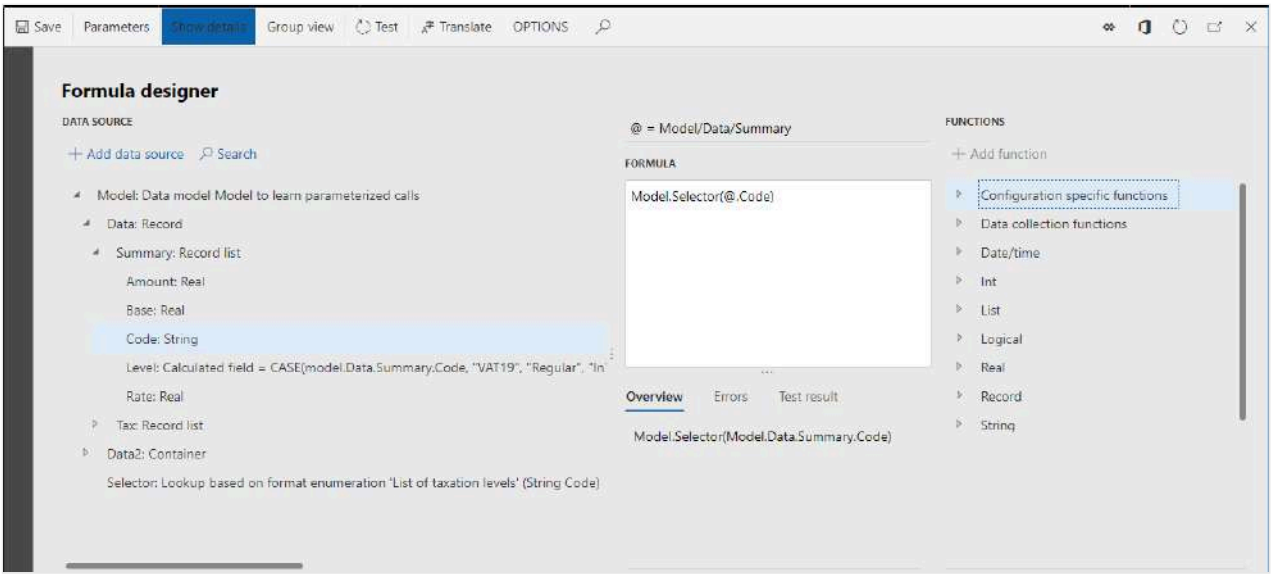
1. Select the **Model.Data.Selector** data source.
2. Select **Edit**.
3. Click in the **Label** field.
4. Select **Translate**.
5. In the **Text translation** pane, enter **LBL_SELECTOR_DS** in the **Label ID** field.

6. Enter **Select tax level by tax code** in the **Text in default language** field.
7. Select **DE** in the **Language** field.
8. Enter **Steuerebene für Steuerkennzeichen auswählen** in the **Translated text** field.
9. Select **Translate**.
10. Select **OK**.

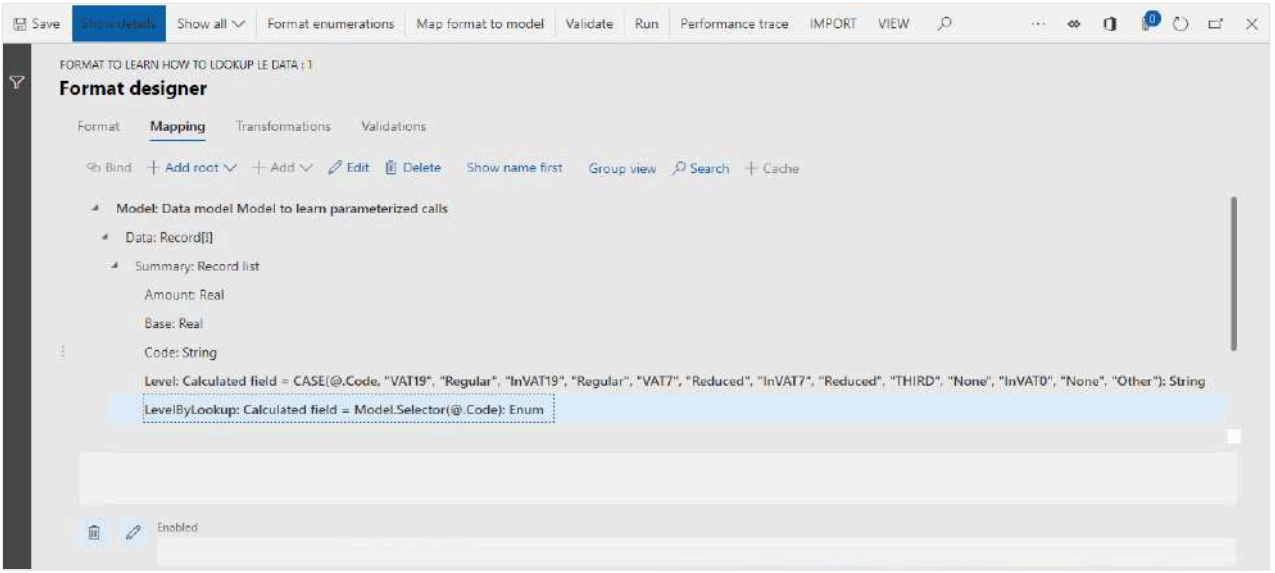


Add a new field to consume the configured lookup

1. Expand the **Model.Data** item.
2. Select the **Model.Data.Summary** item.
3. Select **Add**.
4. Select **Functions/Calculated field**.
5. In the **Name** field, enter **LevelByLookup**.
6. Select **Edit formula**.
7. In the **Formula field**, enter **Model.Selector(Model.Data.Summary.Code)**.
8. Select **Save**.



9. Close the **Formula editor** page.
10. Select **OK**.



The **LevelByLookup** calculated field that you added returns the taxation level as the value of the **List of taxation levels** format enumeration for each summarized tax transactions record. The tax code of the record is passed to the **Model.Selector** lookup data source, and the set of rules for this data source selects the correct taxation level.

Add a new format enumeration-based data source

Next, add a new data source that refers to the format enumeration that you added earlier. Values of this data source are used in an ER format expression later.

1. Select **Add root**.
2. Select **Format enumerations\Enumeration**.
3. In the **Name** field, enter **TaxationLevel**.
4. In the **Format enumeration** field, select **List of taxation levels**.

5. Select **Save**.

Modify an existing field to start using the lookup

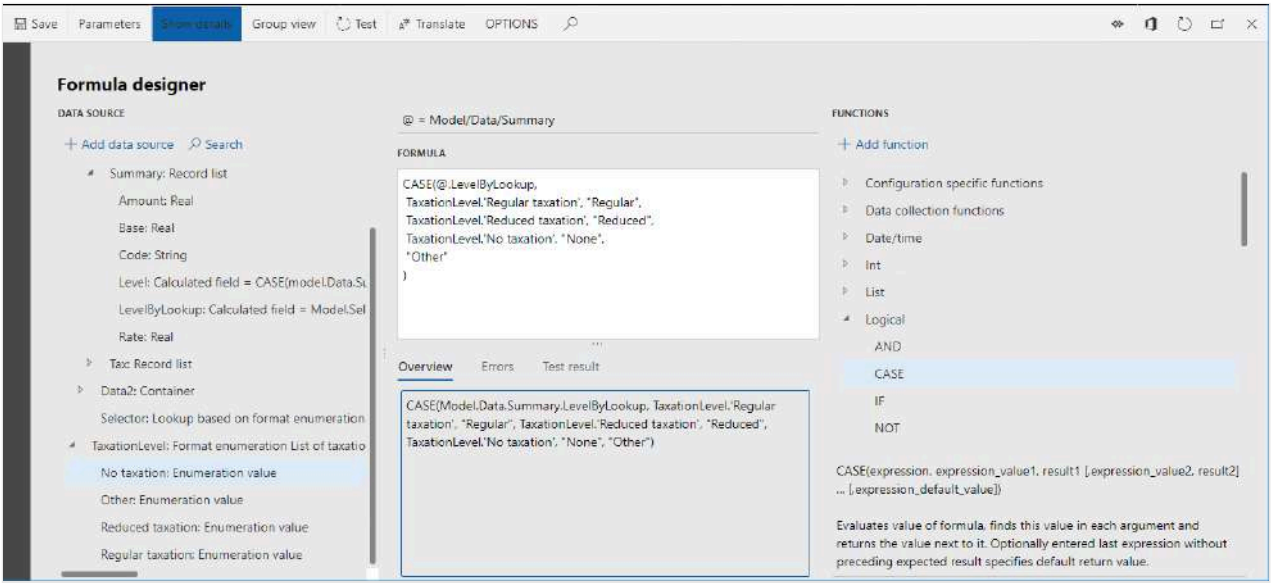
Next, you modify the existing calculated field so that it uses the configured lookup data source to return the correct taxation level value, depending on the tax code.

1. Select the **Model.Data.Summary.Level** item.
2. Select **Edit**.
3. Select **Edit formula**.

   Notice that the current expression of the **Model.Data.Summary.Level** field includes the following hard-coded tax codes:

   ```
   CASE (@.Code, "VAT19", "Regular", "InVAT19", "Regular", "VAT7", "Reduced", "InVAT7", "Reduced", "THIRD", "None", "InVAT0", "None", "Other")
   ```

4. In the **Formula** field, enter `CASE(@.LevelByLookup, TaxationLevel.'Regular taxation', "Regular", TaxationLevel.'Reduced taxation', "Reduced", TaxationLevel.'No taxation', "None", "Other")`.

   The expression of the **Model.Data.Summary.Level** field now returns the taxation level, based on the tax code of the current record and the set of rules that a business user configures in the **Model.Data.Selector** lookup data source.

5. Select **Save**.
6. Close **Formula designer**.
7. Select **OK**.

8. Select **Save**.
9. Close **Format designer**.

Complete the draft version of a derived format

1. On the **Versions** FastTab, select **Change status**.
2. Select **Complete**.
3. Select **OK**.

Export completed version of modified format

1. In the configuration tree, select the **Format to learn how to look up LE data** item.
2. On the **Versions** FastTab, select the record that has a status of **Completed**.
3. Select **Exchange**.
4. Select **Export as XML file**.
5. Select **OK**.
6. The web browser downloads a **Format to learn how to look up LE data.xml** file. Store this file locally.

Repeat the preceding steps for parent items of the **Format to learn how to look up LE data** format, and store the following files locally:

- Format to learn parameterized calls.xml
- Mapping to learn parameterized calls.xml
- Model to learn parameterized calls.xml

To learn how to use the configured **Format to learn how to look up LE data** ER format to set up legal entity–dependent sets of tax codes to filter tax transactions by different taxation levels, complete the steps in the Set up the parameters of an ER format per legal entity article.

Additional resources

Formula designer in Electronic reporting

Set up the parameters of an ER format per legal entity

Configure Lookup data sources to use the ER application-specific parameters feature

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Last updated on 03/17/2026

Set up the parameters of an ER format per legal entity

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Prerequisites

To complete these steps, first complete the steps in Configure ER formats to use parameters that are specified per legal entity.

To complete the examples in this article, you must have access to Microsoft Dynamics 365 Finance for one of the following roles:

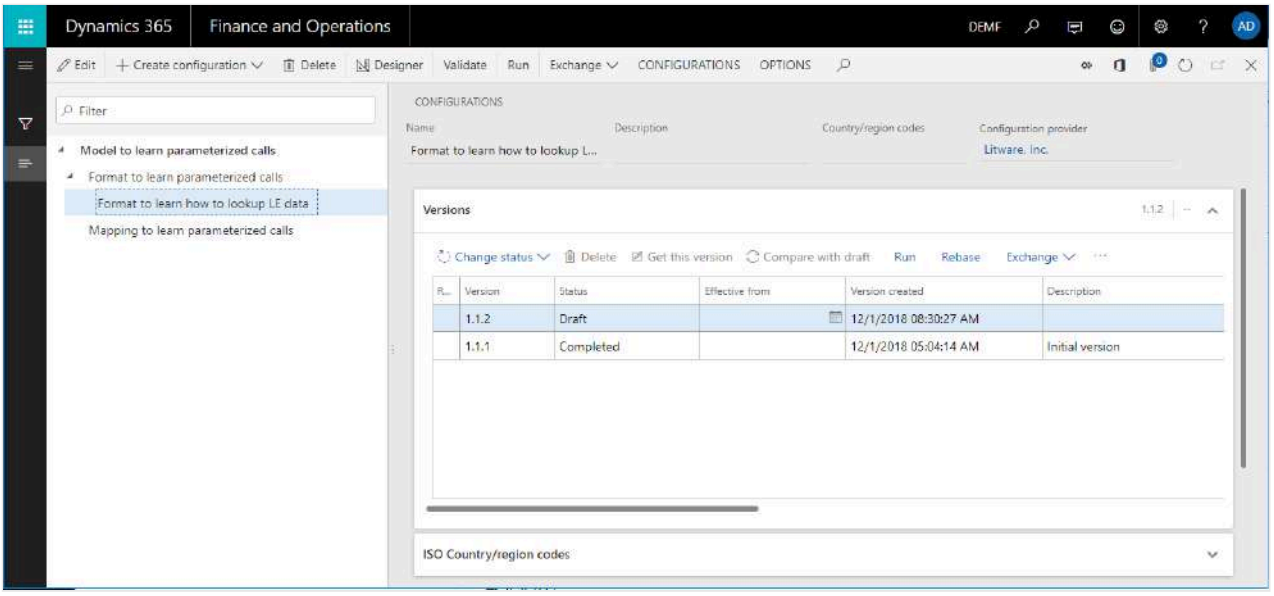
- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

Import ER configurations

To import ER configurations, follow these steps:

1. Sign in to your environment.
2. On the default dashboard, select **Electronic reporting**.
3. Select **Reporting configurations**.
4. In the current instance of Finance, import the configurations that you exported from Regulatory Configuration Services (RCS) while you were completing the steps in Configure ER formats to use parameters that are specified per legal entity. Follow these steps for each Electronic reporting (ER) configuration in the following order: data model, model mapping, and formats.
   a. Select **Exchange** > **Load from XML file**.
   b. Select **Browse** to select the file for the required ER configuration in XML format.
   c. Select **OK**.

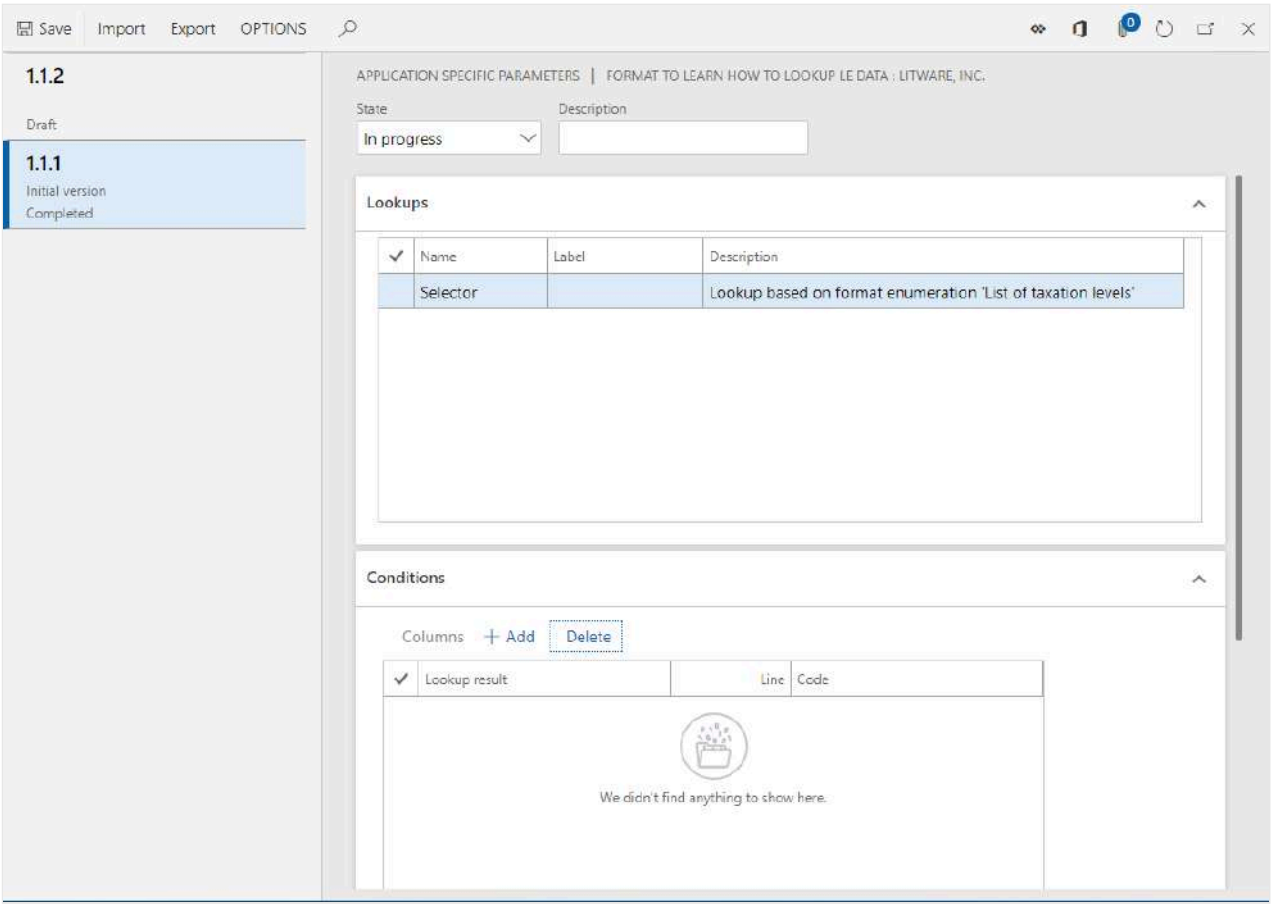
The following illustration shows the configurations that you must have when you finish.



Set up parameters for the DEMF company

You can use the ER framework to set up application-specific parameters for an ER format.

1. Select the **DEMF** legal entity.
2. In the configurations tree, select the **Format to learn how to lookup LE data** format.
3. On the Action Pane, on the **Configurations** tab, in the **Application specific parameters** group, select **Setup**.



On the **Application specific parameters** page, you can configure the rules for the **Selector** data source of the **Format to learn how to lookup LE data** format.

If the base ER format contains several data sources of the **Lookup** type, select the desired data source on the **Lookups** FastTab before you can start to configure the set of rules for the data source.

For each data source, you can configure separate rules for each version of the selected ER format.

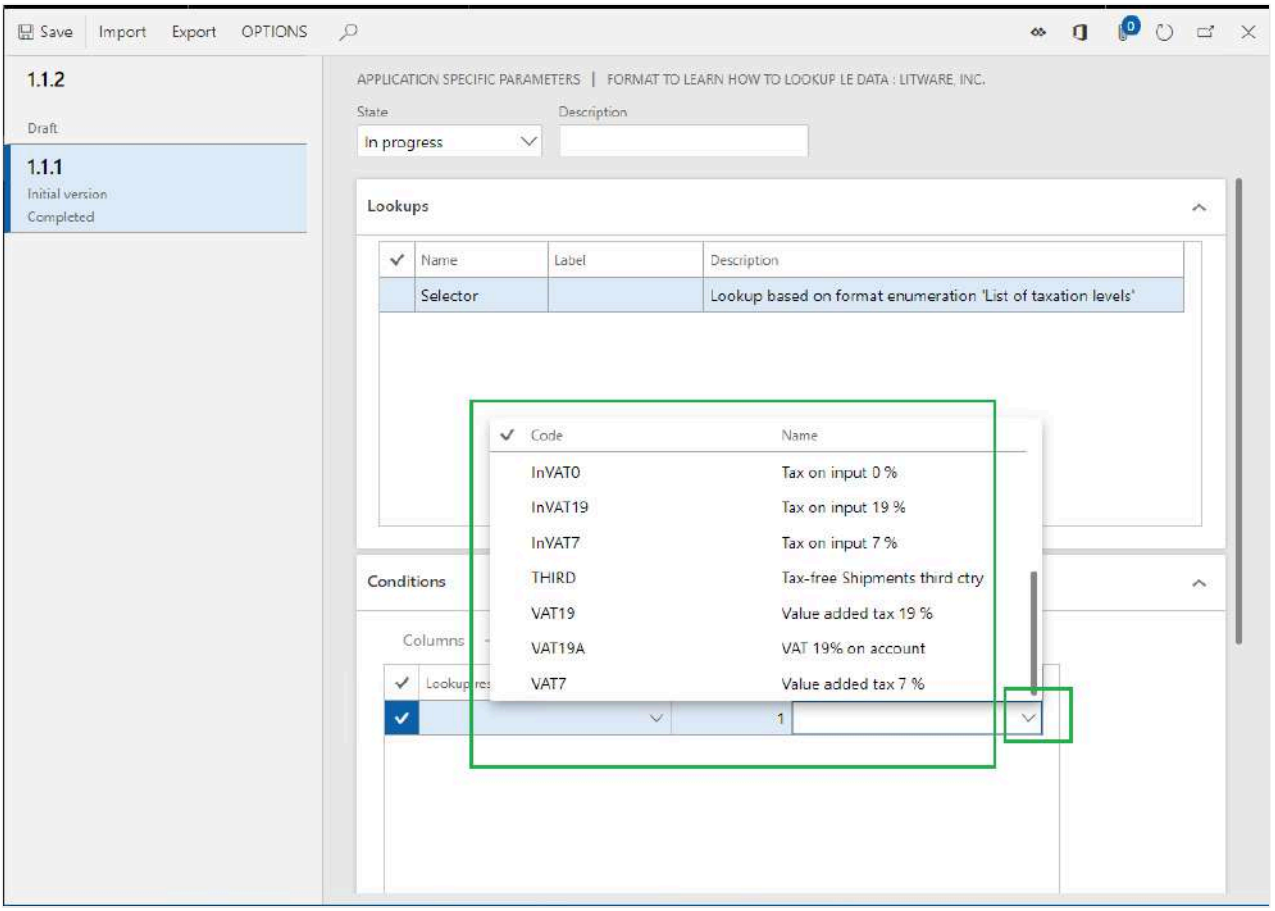
The whole set of rules for all lookup data sources that are available in the selected version of the base ER format makes up the application-specific parameters for the ER format.

4. Select version **1.1.1** of the ER format.

5. On the **Conditions** FastTab, select **Add**.

6. In the **Code** field of the new record, select the drop-down arrow to open the lookup.

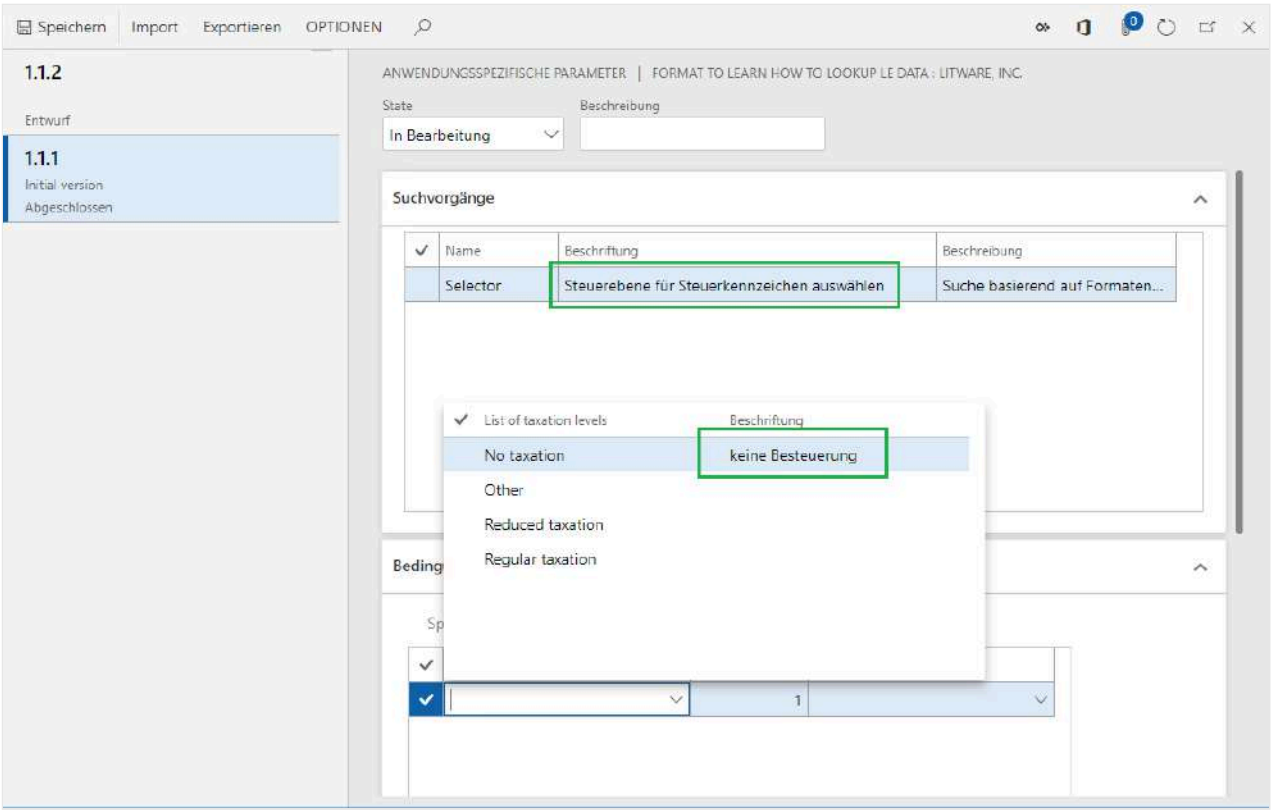
   The lookup presents the list of tax codes for selection. This list is returned by the **Model.Data.Tax** data source that's configured in the base ER format. Because this data source contains the **Name** field, the name of each tax code appears in the lookup.



7. Select the **VAT19** tax code.

8. In the **Lookup result** field of the new record, select the drop-down arrow to open the lookup. The lookup presents the list of values for the TaxationLevel format enumeration for selection.

   If you selected German as the preferred language for the user that you're signed in as, the labels of the values in the lookup appear in German, provided that they are translated in the base ER format. Additionally, if the label of a lookup data source is translated, that label appears in the user's preferred language on the **Lookups** tab.



9. Select the **Regular taxation** value.

   By adding this record, you define the following rule: When the **Selector** lookup data source is requested, and the **VAT19** tax code is passed as an argument, **Regular taxation** is returned as the requested taxation level.

10. Select **Add**, and then follow these steps:
    a. In the **Code** field, select the **InVAT19** tax code.
    b. In the **Lookup result** field, select the **Regular taxation** value.

11. Select **Add**, and then follow these steps:
    a. In the **Code** field, select the **VAT7** tax code.
    b. In the **Lookup result** field, select the **Reduced taxation** value.

12. Select **Add** , and then follow these steps:
    a. In the **Code** field, select the **InVAT7** tax code.
    b. In the **Lookup result** field, select the **Reduced taxation** value.

13. Select **Add**, and then follow these steps:
    a. In the **Code** field, select the **THIRD** tax code.
    b. In the **Lookup result** field, select the **No taxation** value.

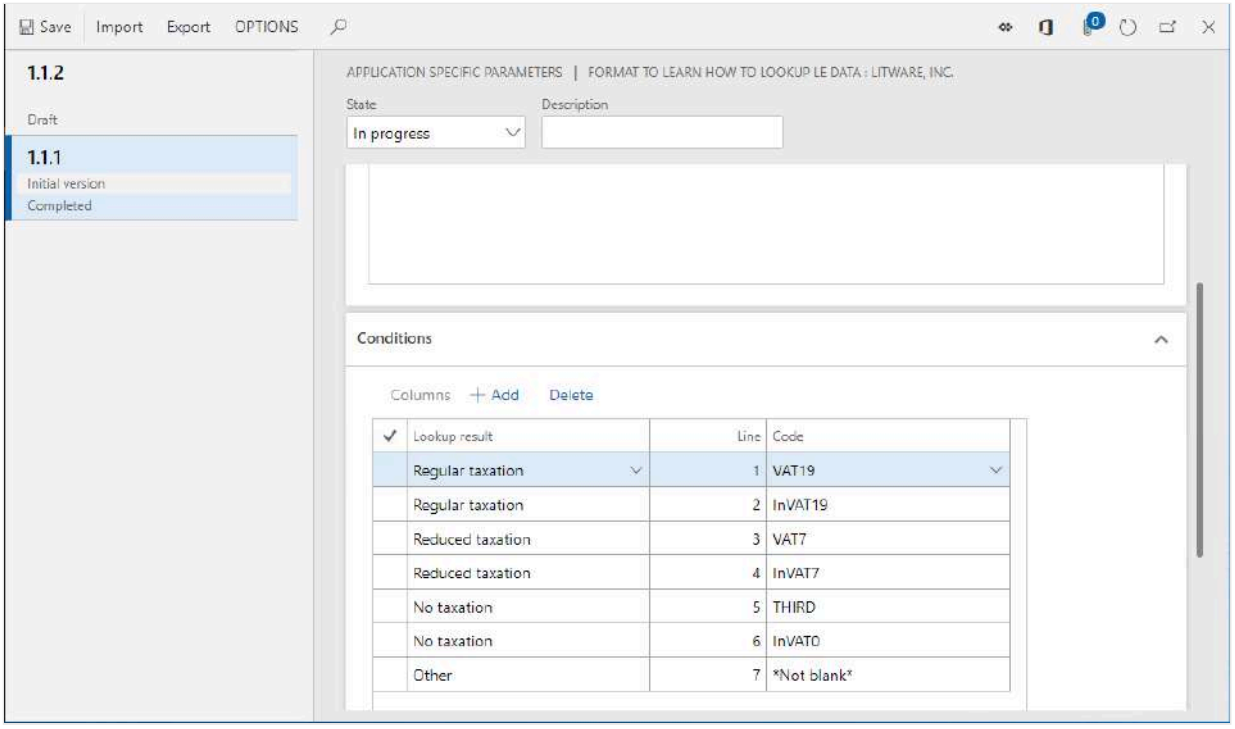
14. Select **Add**, and then follow these steps:
    a. In the **Code** field, select the **InVAT0** tax code.
    b. In the **Lookup result** field, select the **No taxation** value.

15. Select **Add**, and then follow these steps:

a. In the **Code** field, select the ***Not blank*** option.

b. In the **Lookup result** field, select the **Other** value.

By adding this last record, you define the following rule: When the tax code passed as an argument doesn't satisfy any of the previous rules, the lookup data source returns **Other** as the requested taxation level.



16. In the **State** field, select **Completed**.

    When you run an ER format version that has a status of either **Completed** or **Shared**, this set of rules must be in the **Completed** state. Otherwise, execution of the base ER format is interrupted when the format tries to load data from this set of rules while the **Selector** lookup data source is running.

    When you run an ER format version that has a status of **Draft**, the base ER format can access this set of rules, regardless of its state.

17. Select **Save**.

18. Close the **Application specific parameters** page.

Run the ER format in the DEMF company

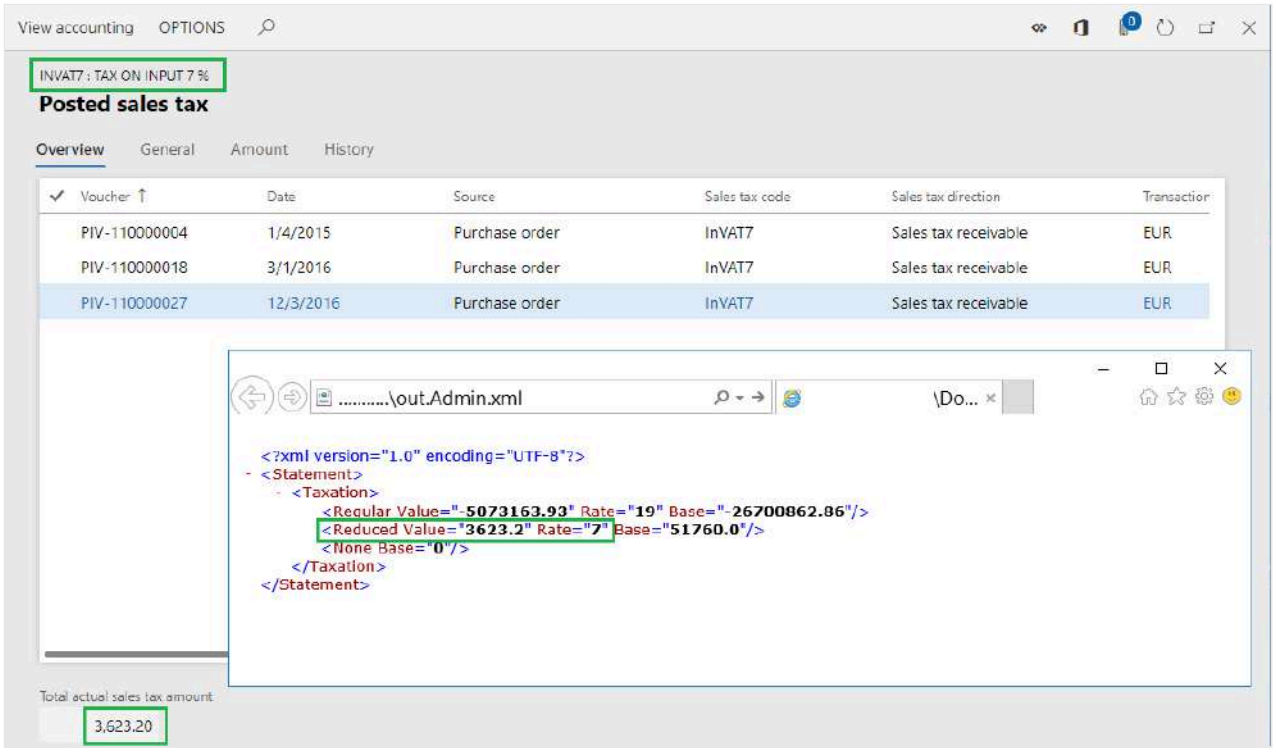
To run the ER format in the DEMF company, follow these steps:

1. In the configurations tree, select the **Format to learn how to lookup LE data** format.

2. On the action pane, select **Run**.

3. In the dialog box that appears, select **OK**.
4. Download the generated statement and store it locally.

   In the generated statement, the summary of the **InVAT7** tax code is on the **Reduced** level, and the summaries of the **VAT19** and **InVA19** tax codes are on the **Regular** level. The configuration in the legal entity–dependent set of rules determines this behavior.
5. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax codes**.
6. Select the **InVAT7** tax code.
7. On the Action Pane, on the **Sales tax code** tab, in the **Inquiries** group, select **Posted sales tax** to view information about the tax value and applied tax rate per tax code.



8. Close the **Posted sales tax** page.

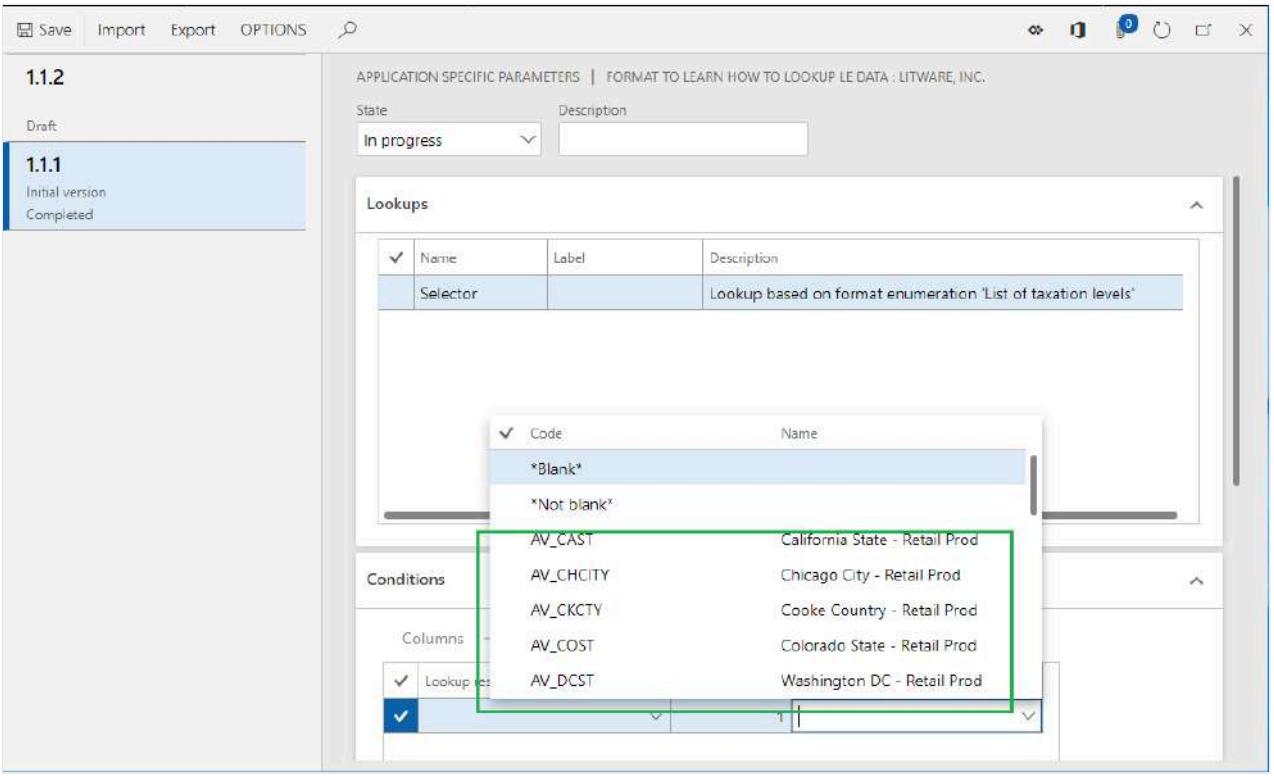
Set up parameters for the USMF company

To set up parameters for the USMF company, complete the following steps:

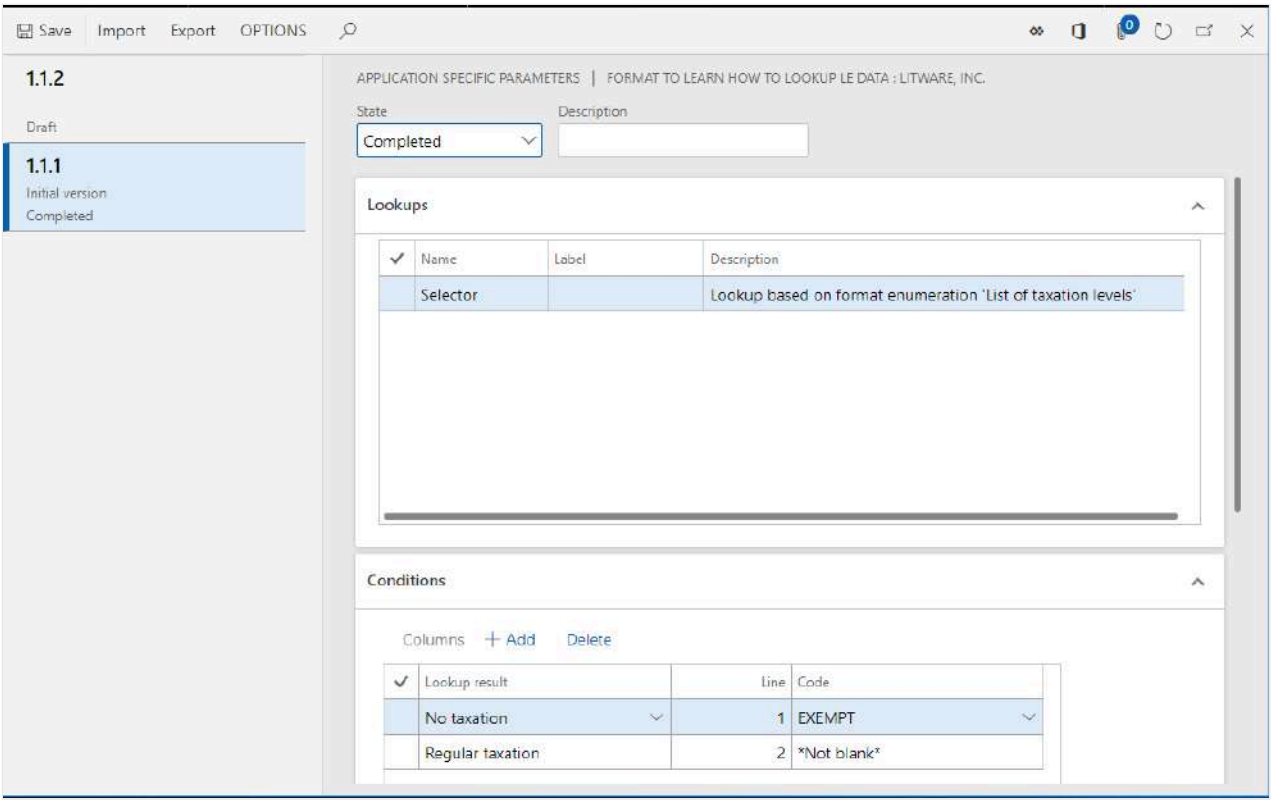
1. Select the **USMF** legal entity.
2. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
3. In the configurations tree, expand the **Model to learn parameterized calls** item, expand the **Format to learn parameterized calls** item, and select the **Format to learn how to lookup LE data** format.

4. On the **Configurations** tab, in the **Application specific parameters** group, select **Setup**.
5. Select version **1.1.1** of the selected ER format.
6. On the **Conditions** FastTab, select **Add**.
7. In the **Code** field of the new record, select the drop-down arrow to open the lookup.

   The lookup now presents the list of tax codes for the **USMF** company tax for selection.



8. Select the **EXEMPT** tax code.
9. In the **Lookup result** field of the new record, select the **No taxation** value.
10. Select **Add**.
11. In the **Code** field of the new record, select the ***Not blank*** option.
12. In the **Lookup result** field of the new record, select the **Regular taxation** value.
13. In the **State** field, select **Completed**.
14. Select **Save**.



15. Close the **Application specific parameters** page.

Run the ER format in the USMF company

To run the ER format in the USMF company, complete the following steps:

1. In the configurations tree, select the **Format to learn how to lookup LE data** format.
2. On the action pane, select **Run**.
3. In the dialog box that appears, select **OK**.
4. Download the generated statement and store it locally.

   In the generated statement, observe that you reused the same ER format for a different legal entity, but you didn't make any adjustments to the ER format.

Reuse legal entity–dependent parameters

Duplicate existing parameters

Export parameters

To export parameters, complete the following steps:

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the configurations tree, select the **Format to learn how to lookup LE data** format.
4. On the **Configurations** tab, in the **Application specific parameters** group, select **Setup**.
5. Select version **1.1.1** of the ER format.
6. On the Action Pane, select **Export**.
7. Download the generated file and store it locally.

   The configured set of application-specific parameters is exported as an XML file.

Import parameters

To import parameters, complete the following steps:

1. Select version **1.1.2** of the ER format.
2. On the Action Pane, select **Import**.
3. Select **Yes** to confirm that you want to override the existing application-specific parameters for this format version.
4. Select **Browse** to find the file that contains the exported application-specific parameters for version **1.1.1**.
5. Select **OK**.

   Version **1.1.2** of the ER format now has the same application-specific parameters that you originally configured for version **1.1.1**.

Applicability considerations

The application-specific parameters of an ER format are legal entity–dependent. To reuse the application-specific parameters that are configured for one legal entity in a different legal entity, export them while you're signed in to the first legal entity. Then import them after you sign in to the other legal entity.

You can also use this export-import approach to transfer ER format–related application-specific parameters originally configured in one instance of Finance to another instance of Finance.

If you configure application-specific parameters for one version of an ER format and then import a later version of the same format into the current Finance instance, the existing application-specific parameters aren't applied to the imported version unless you use the **Use application specific parameters from previous versions of ER formats** feature. For more information, see the Reuse existing parameters section later in this article.

When you select a file for import, the structure of the application-specific parameters in that file is compared with the structure of the corresponding data sources of the **Lookup** type in the ER format selected for import. By default, the import completes only if the structure of each application-specific parameter matches the structure of the corresponding data source in the ER format selected for import. If the structures don't match, a warning message informs you that the import can't be completed. If you force the import, the existing application-specific parameters for the selected ER format are cleaned up, and you must set them up from the beginning.

As of Finance version 10.0.24, you can change the default behavior and avoid receiving a warning message by enabling the **Align ER application specific parameters while importing** feature in the **Feature management** workspace. When this feature is enabled, if the structure of application-specific parameters that you're importing differs from the structure of the corresponding data sources in the target ER format selected for import, the import succeeds in the following cases:

- The structure of the target ER format is changed by adding new condition columns to any existing data sources of the **Lookup** type. When the import completes, the application-specific parameters are updated. In all the imported records of application-specific parameters, the values in every added condition column are initialized with the default value for the data type of that column.
- The structure of the target ER format is changed by removing some condition columns from any existing data sources of the **Lookup** type. When the import completes, the application-specific parameters are updated. In all the imported records of application-specific parameters, the values in every removed condition column are deleted.
- The structure of the target ER format is changed by adding new data sources of the **Lookup** type. When the import completes, the added lookups are appended to the application-specific parameters.
- The structure of the target ER format is changed by removing some of the existing data sources of the **Lookup** type. When the import completes, all artifacts that are related to the data sources of the **Lookup** type that were removed from the target ER format are deleted from the imported application-specific parameters.

When the import completes, in addition to the changes just described, the state of the imported application-specific parameters is changed to **In progress**. A warning message

informs you that you must manually edit the automatically adjusted application-specific parameters.

Replicate parameters

As of Finance version 10.0.27, you can copy the parameters that you configured in one company to other companies at the same time.

To copy parameters, complete the following steps.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. In the configurations tree, select the **Format to learn how to lookup LE data** format.
4. On the **Configurations** tab, in the **Application specific parameters** group, select **Setup**.
5. Select version **1.1.1** of the ER format.
6. On the Action Pane, select **Replicate**.
7. In the **Replicate** dialog box, on the **Companies** tab, select the companies that you want to copy parameters to.

   > ⓘ **Note**
   >
   > You see the list of target companies only if you're assigned a security role that grants access to all organizations.

8. Select **OK**.

   > ⓘ **Note**
   >
   > The confirmation dialog box informs you if some target companies contain previously configured parameters for the selected version of an ER format. Select **Yes** to override the parameters by copying them from the current company.

   The configured set of application-specific parameters is now copied to the selected companies.

Reuse existing parameters

Starting with Finance version 10.0.23, you can reuse application-specific parameters configured for one version of an ER format when you run a higher version of the same format. To reuse existing parameters, enable the **Use application specific parameters from previous versions of ER formats** feature in the **Feature management** workspace. When you enable this feature and run one version of an ER format that tries to read application-specific parameters, ER tries to find application-specific parameters that are configured for the running version of the format. If they aren't available, ER tries to find them for the nearest lower version of the format.

> ⓘ **Note**
>
> You can reuse application-specific parameters only in the scope of the current legal entity.
>
> An error is displayed at runtime when you run a higher version of an ER format that tries to reuse application-specific parameters configured for a lower version of the same format and the structure of at least one data source of the **Lookup** type in the higher format version changed.

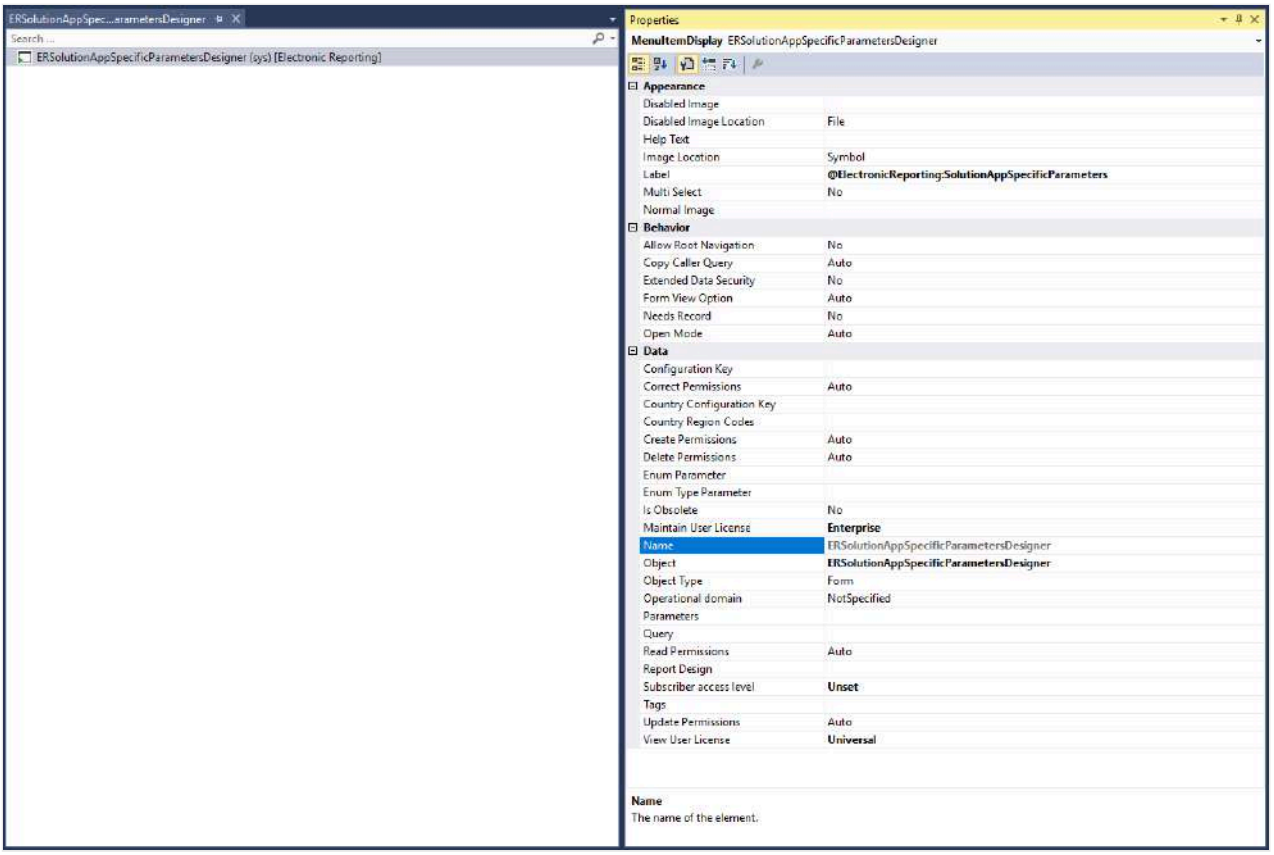
Relationship between application-specific parameters and an ER format

The relationship between an ER format and its application-specific parameters is established by the ER format's instance-independent unique identification code. Therefore, when you remove an ER format from Finance, the application-specific parameters configured for the ER format are kept in the current instance of Finance. You can access them on reimporting the base ER format into this instance of Finance.

Access application-specific parameters by using the ER framework

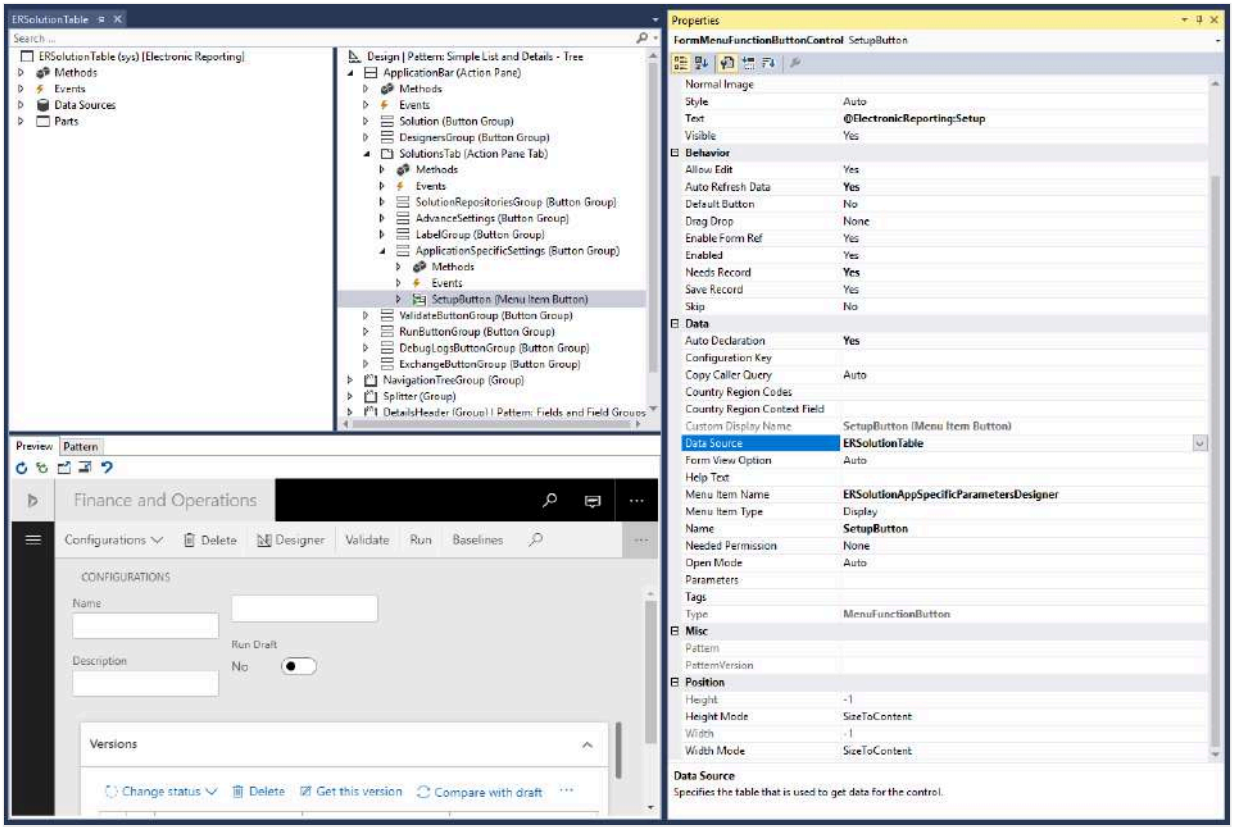
In the preceding example, you accessed application-specific parameters of an ER format by using the ER framework. This approach doesn't let you restrict access to the application-specific parameters of a specific ER format. If you need to apply such restrictions, follow these steps:

1. Either reuse an existing **ERSolutionAppSpecificParametersDesigner** menu item, or implement your own **ERSolutionAppSpecificParametersDesigner** menu item.



2. Follow one of these steps:

a. Create a new menu item button, and link it to the corresponding record from the **ERSolutionTable** table by setting its **Data Source** property to **ERSolutionTable**.



b. Create a simple button, and override the **Clicked** method as shown in the following example.

By using this approach, you can specify a unique solution ID (defined through the **GUID** value) to allow access to the application-specific parameters of only a specific ER format and descendant copies that are derived from it.

X++

```
public void clicked()
    {
        super();

        ERSolutionTable solutionTableRecord =
ERSolutionTable::findByGUID(str2Guid('ADACCB2F-EFD1-4C90-877D-
7E1E5D1AEE92'));

        Args args = new Args();
        args.record(solutionTableRecord);
        args.caller(this);

        new
MenuFunction(menuItemDisplayStr(ERSolutionAppSpecificParametersDesigner),
MenuItemType::Display)
            .run(args);
    }
```

Additional resources

Formula designer in Electronic reporting

Configure ER formats to use parameters that are specified per legal entity

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Last updated on 03/17/2026

Defer the execution of sequence elements in ER formats

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Overview

Use the Operations designer of the Electronic reporting (ER) framework to configure the format component of an ER solution that generates outbound documents in a text format. The hierarchical structure of the configured format component consists of format elements of various types. These format elements fill generated documents with the required information at runtime. By default, when you run an ER format, the format elements run in the same order as they appear in the format hierarchy: one by one, from top to bottom. However, at design time, you can change the execution order for any sequence elements of the configured format component.

When you turn on the **Deferred execution** option for a sequence format element in the configured format, you defer (postpone) the execution of that element. The element isn't run until all other elements of its parent are run.

To learn more about this feature, complete the example in this article.

Limitations

The **Deferred execution** option supports only sequence elements configured for an ER format that generates **outbound** documents in text format.

The **Deferred execution** option isn't applicable to sequences configured as trimmed sequences where the maximum length is limited.

Example: Defer the execution of a sequence element in an ER format

The following steps explain how a user in the System administrator or Electronic reporting functional consultant role can configure an ER format that contains a sequence element where order of execution differs from the order in the format hierarchy.

You can perform these steps in the **USMF** company in Microsoft Dynamics 365 Finance.

Prerequisites

To complete this example, you must have access to the **USMF** company in Finance for one of the following roles:

- Electronic reporting functional consultant
- System administrator

If you didn't yet complete the example in the Defer the execution of XML elements in ER formats article, download the following configurations of the sample ER solution.

Expand table

Content description	File name
ER data model configuration	Model to learn deferred elements.version.1.xml
ER model mapping configuration	Mapping to learn deferred elements.version.1.1.xml

Before you begin, download and save the following configuration of the sample ER solution.

Expand table

Content description	File name
ER format configuration	Format to learn deferred sequences.version.1.1.xml

Import the sample ER configurations

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.

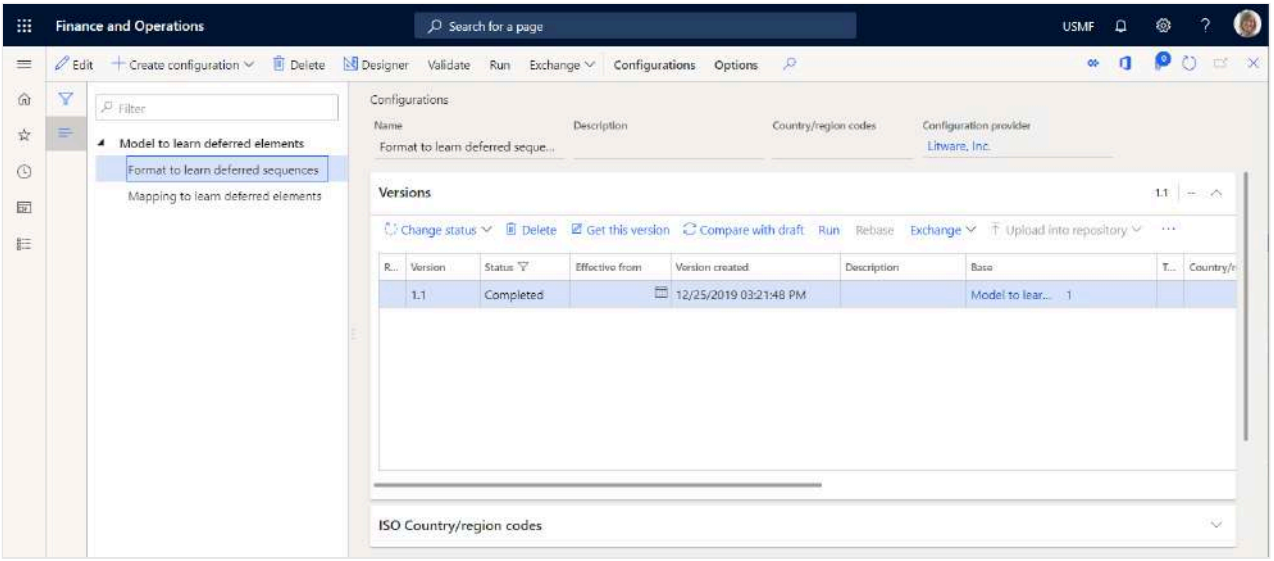
3. On **Configurations**, if the **Model to learn deferred elements** configuration isn't available in the configuration tree, import the ER data model configuration:
   a. Select **Exchange**, and then select **Load from XML file**.
   b. Select **Browse**, find and select the **Model to learn deferred elements.1.xml** file, and then select **OK**.

4. If the **Mapping to learn deferred elements** configuration isn't available in the configuration tree, import the ER model mapping configuration:
   a. Select **Exchange**, and then select **Load from XML file**.
   b. Select **Browse**, find and select the **Mapping to learn deferred elements.1.1.xml** file, and then select **OK**.

5. Import the ER format configuration:
   a. Select **Exchange**, and then select **Load from XML file**.
   b. Select **Browse**, find and select the **Format to learn deferred sequences.1.1.xml** file, and then select **OK**.

6. In the configuration tree, expand **Model to learn deferred elements**.

7. Review the list of imported ER configurations in the configuration tree.

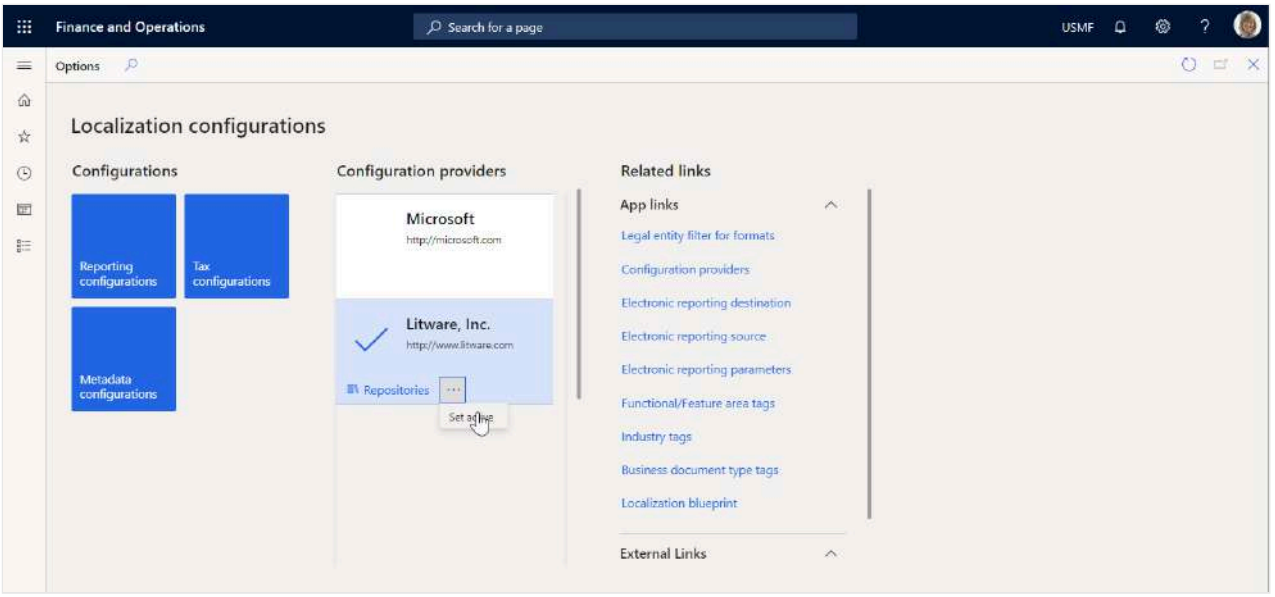


Activate a configurations provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

2. On **Localization configurations**, in the **Configuration providers** section, make sure that the configuration provider for the Litware, Inc. (`http://www.litware.com`) sample company is listed and marked as active. If this configuration provider isn't listed, or if it isn't marked

as active, follow the steps in the Create a configuration provider and mark it as active article.

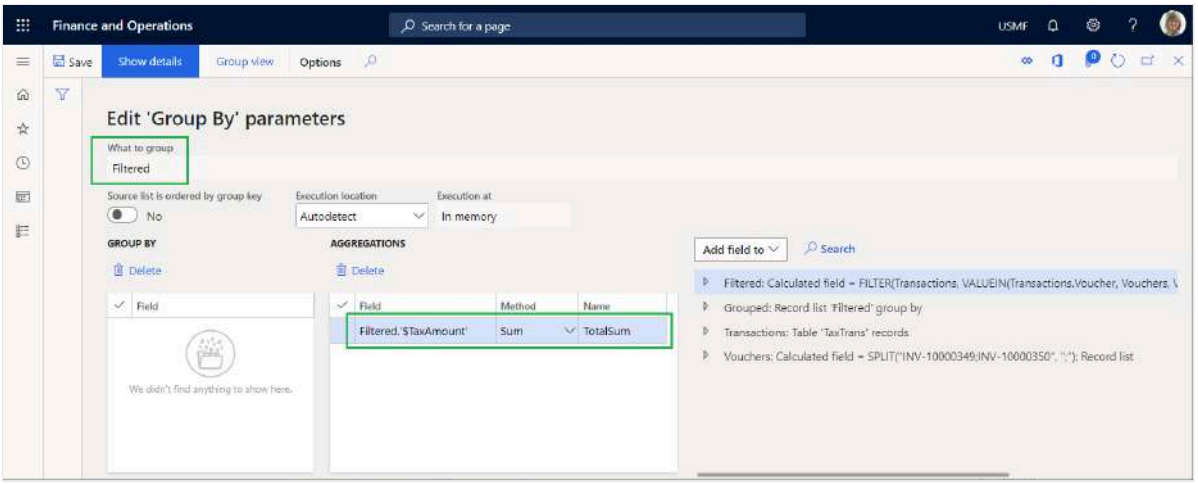


Review the imported model mapping

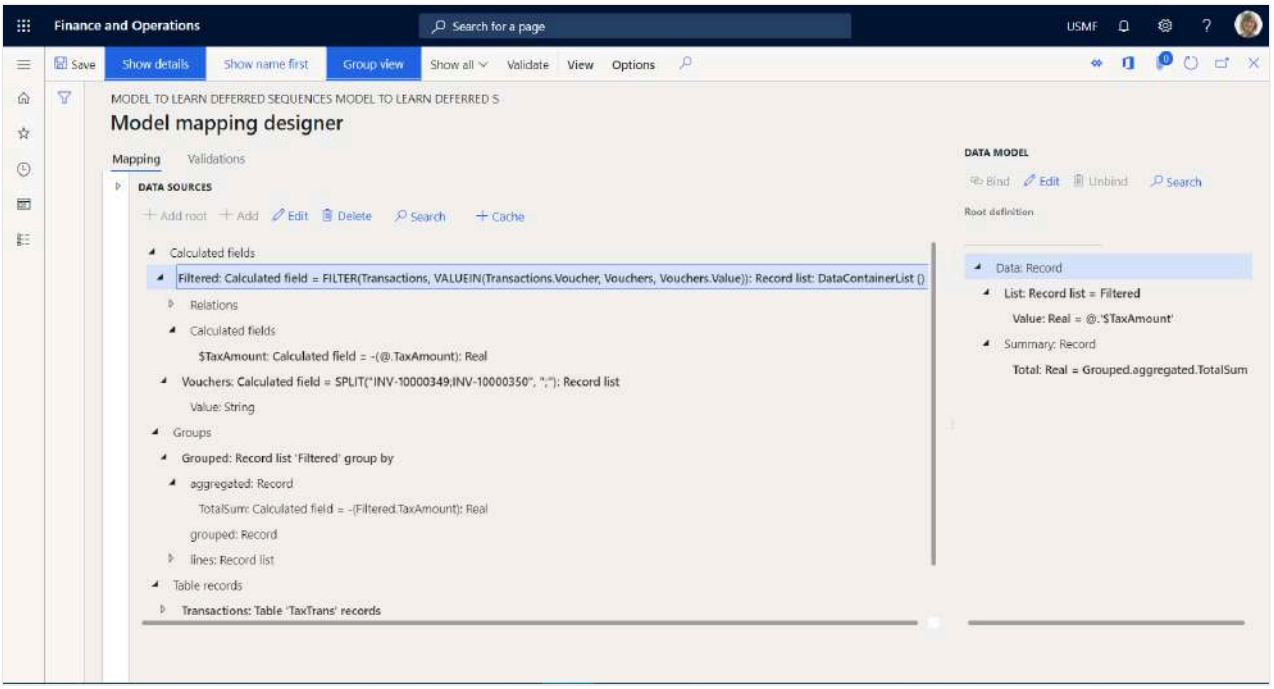
Review the settings of the ER model mapping component configured to access tax transactions and expose accessed data on request.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. On **Configurations**, in the configuration tree, expand **Model to learn deferred elements**.
4. Select the **Mapping to learn deferred elements** configuration.
5. Select **Designer** to open the list of mappings.
6. Select **Designer** to review the mapping details.
7. Select **Show details**.
8. Review the data sources configured to access tax transactions:
    - The **Transactions** data source of the *Table record* type is configured to access records of the **TaxTrans** application table.
    - The **Vouchers** data source of the *Calculated field* type is configured to return the required voucher codes (**INV-10000349** and **INV-10000350**) as a list of records.

- The **Filtered** data source of the *Calculated field* type is configured to select, from the **Transactions** data source, only tax transactions of the required vouchers.
- The **$TaxAmount** field of the *Calculated field* type is added for the **Filtered** data source to expose the tax value that has the opposite sign.
- The **Grouped** data source of the *Group By* type is configured to group filtered tax transactions of the **Filtered** data source.
- The **TotalSum** aggregation field of the **Grouped** data source is configured to summarize values of the **$TaxAmount** field of the **Filtered** data source for all filtered tax transactions of that data source.



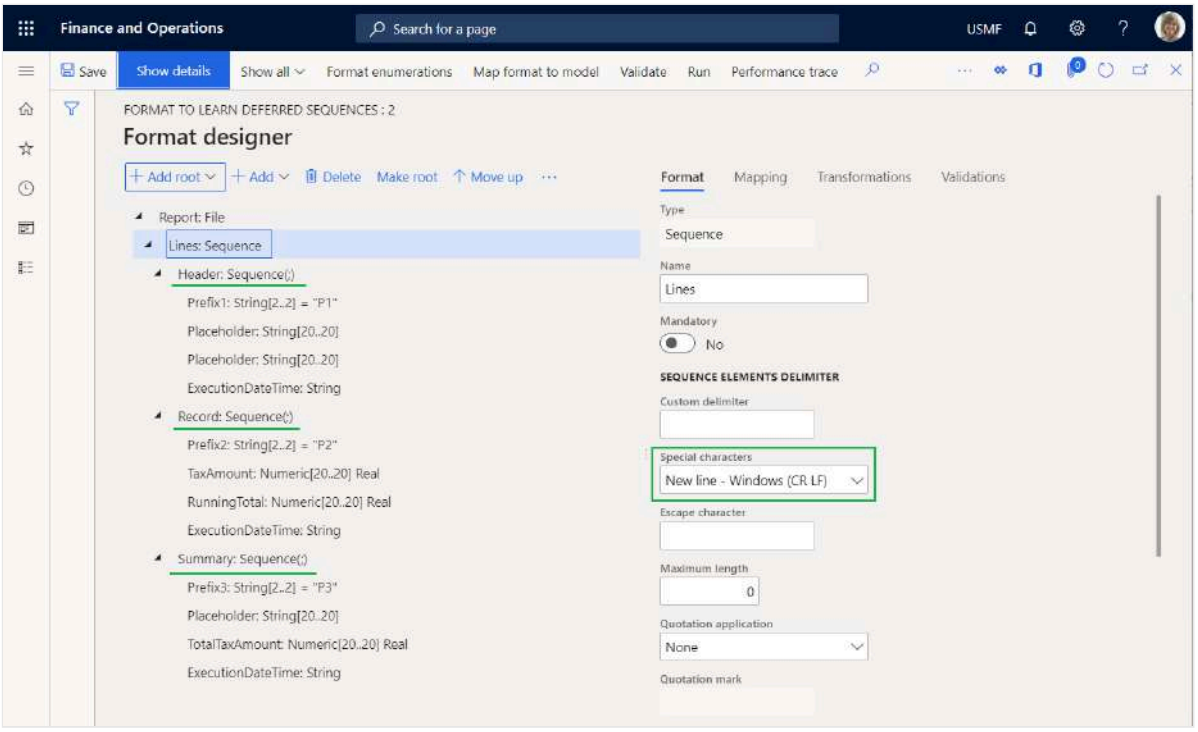
9. Review how the configured data sources are bound to the data model, and how they expose accessed data to make it available in an ER format:
   - The **Filtered** data source is bound to the **Data.List** field of the data model.
   - The **$TaxAmount** field of the **Filtered** data source is bound to the **Data.List.Value** field of the data model.
   - The **TotalSum** field of the **Grouped** data source is bound to the **Data.Summary.Total** field of the data model.



10. Close the **Model mapping designer** and **Model mappings** pages.

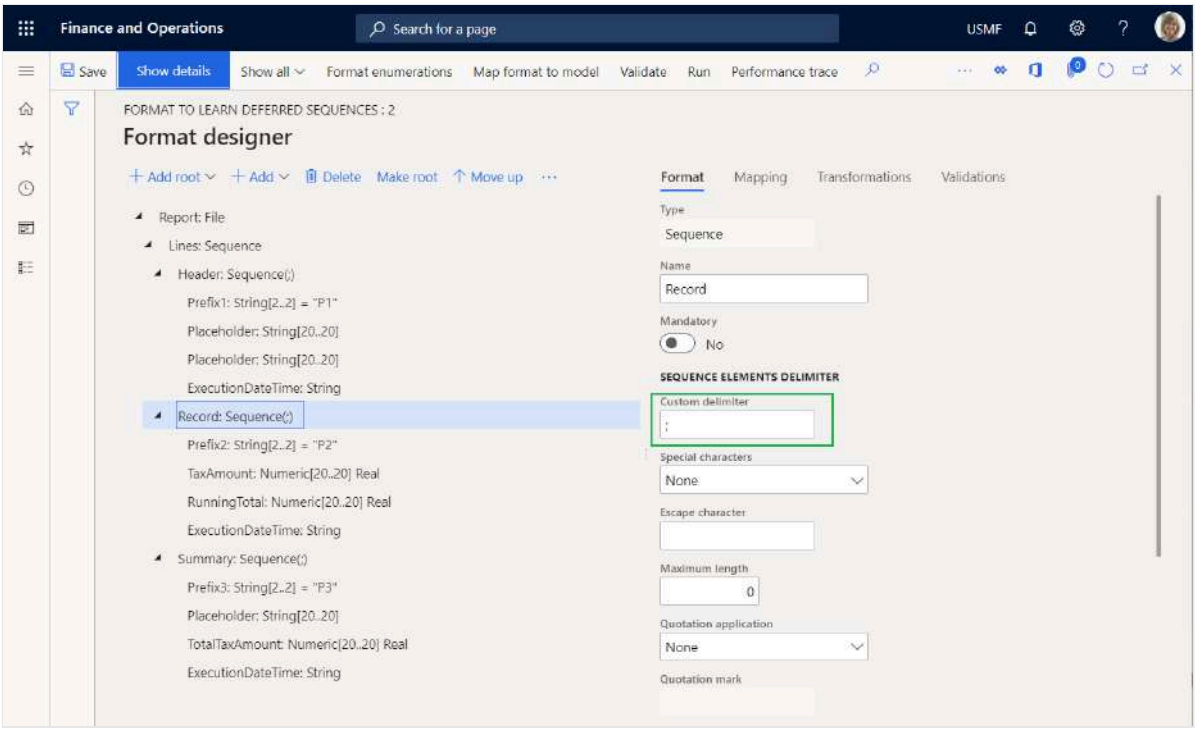
Review the imported format

1. On **Configurations**, in the configuration tree, select the **Format to learn deferred sequences** configuration.
2. Select **Designer** to review the format details.
3. Select **Show details**.
4. Review the settings of the ER format components that are configured to generate an outbound document in text format that includes details of the tax transactions:
    - The **Report\Lines** sequence format element is configured to fill the outbound document with a single line that is generated from the nested sequence elements (**Header**, **Record**, and **Summary**).



- The **Report\Lines\Header** sequence format element is configured to fill the outbound document with a single header line that shows the date and time when the processing starts.

- The **Report \Lines\Record** sequence format element is configured to fill the outbound document with a single line that shows the details of individual tax transactions. These tax transactions are separated by a semicolon.

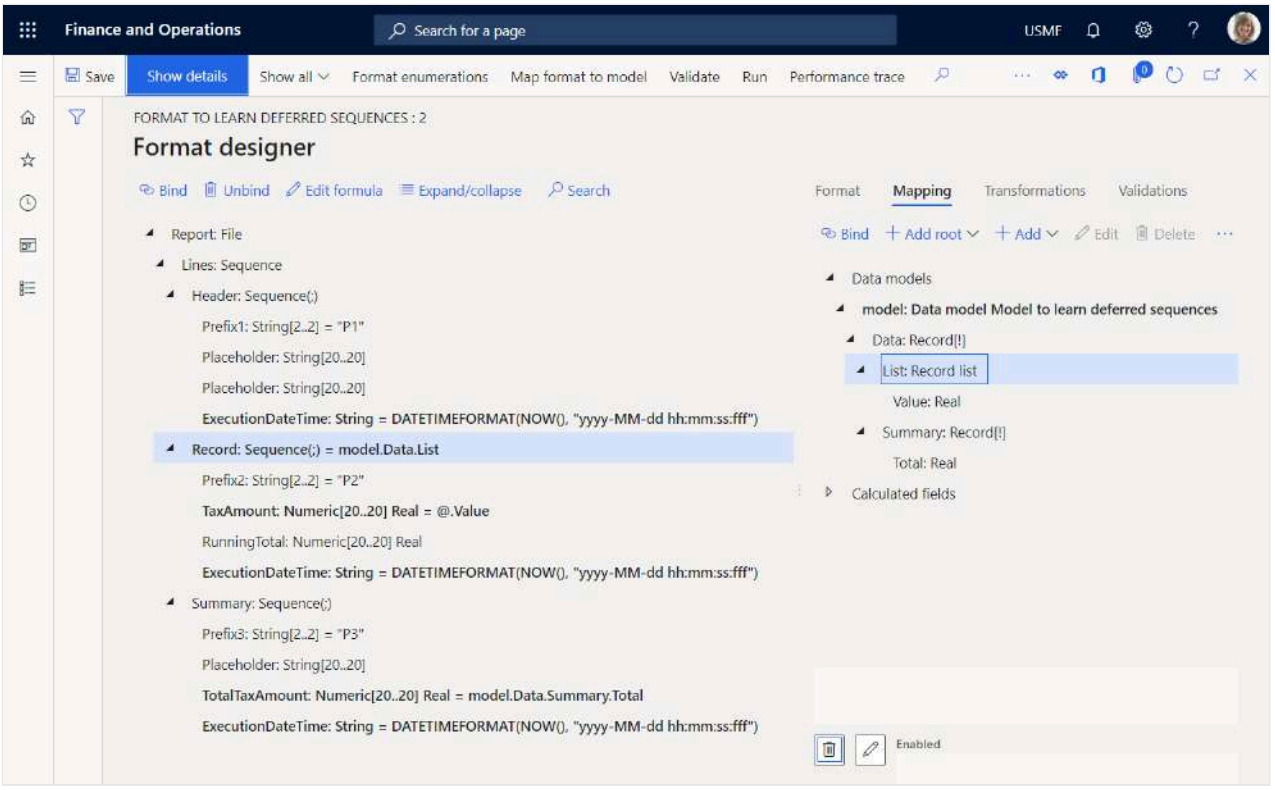


- The **Report\Lines\Summary** sequence format element is configured to fill the outbound document with a single summary line that includes the sum of the tax

values from the processed tax transactions.

5. On the **Mapping** tab, review the following details:

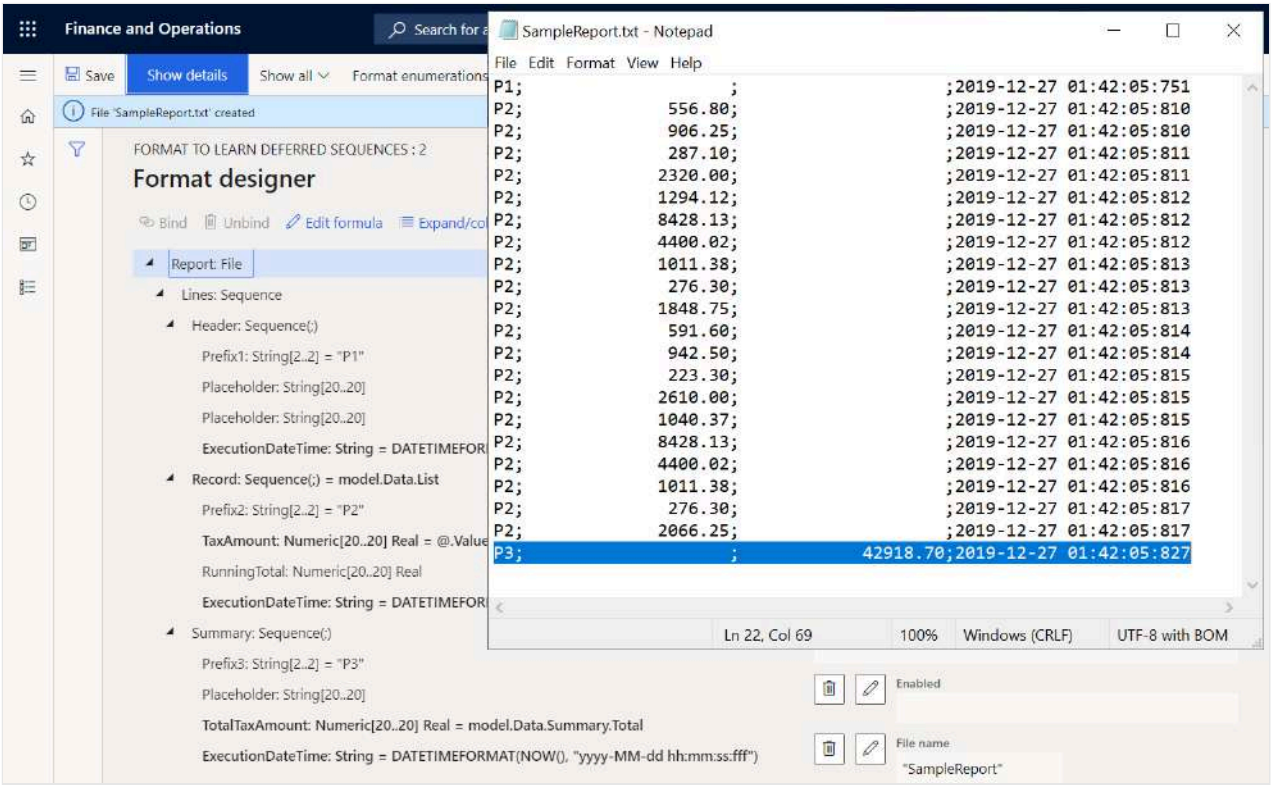
- The **Report\Lines\Header** element doesn't need to be bound to a data source to generate a single line in an outbound document.
- The **Prefix1** element generates **P1** symbols to indicate that the line that is added is the report header line.
- The **ExecutionDateTime** element generates the date and time (including milliseconds) when the header line is added.
- The **Report\Lines\Record** element is bound to the **model.Data.List** list to generate a single line for every record from the bound list.
- The **Prefix2** element generates **P2** symbols to indicate that the line that is added is for the tax transaction details.
- The **TaxAmount** element is bound to **model.Data.List.Value** (which is shown as **@.Value** in the relative path view) to generate the tax value of the current tax transaction.
- The **RunningTotal** element is a placeholder for the running total of the tax values. Currently, this element has no output, because neither a binding nor a default value is configured for it.
- The **ExecutionDateTime** element generates the date and time (including milliseconds) when the current transaction is processed in this report.
- The **Report\Lines\Summary** element doesn't need to be bound to a data source to generate a single line in an outbound document.
- The **Prefix3** element generates **P3** symbols to indicate that the line that is added contains the total tax value.
- The **TotalTaxAmount** element is bound to **model.Data.Summary.Total** to generate the sum of the tax values of the processed tax transactions.
- The **ExecutionDateTime** element generates the date and time (including milliseconds) when the summary line is added.



Run the imported format

1. On **Format designer**, select **Run**.

2. Download the file that the web browser offers, and open it for review.



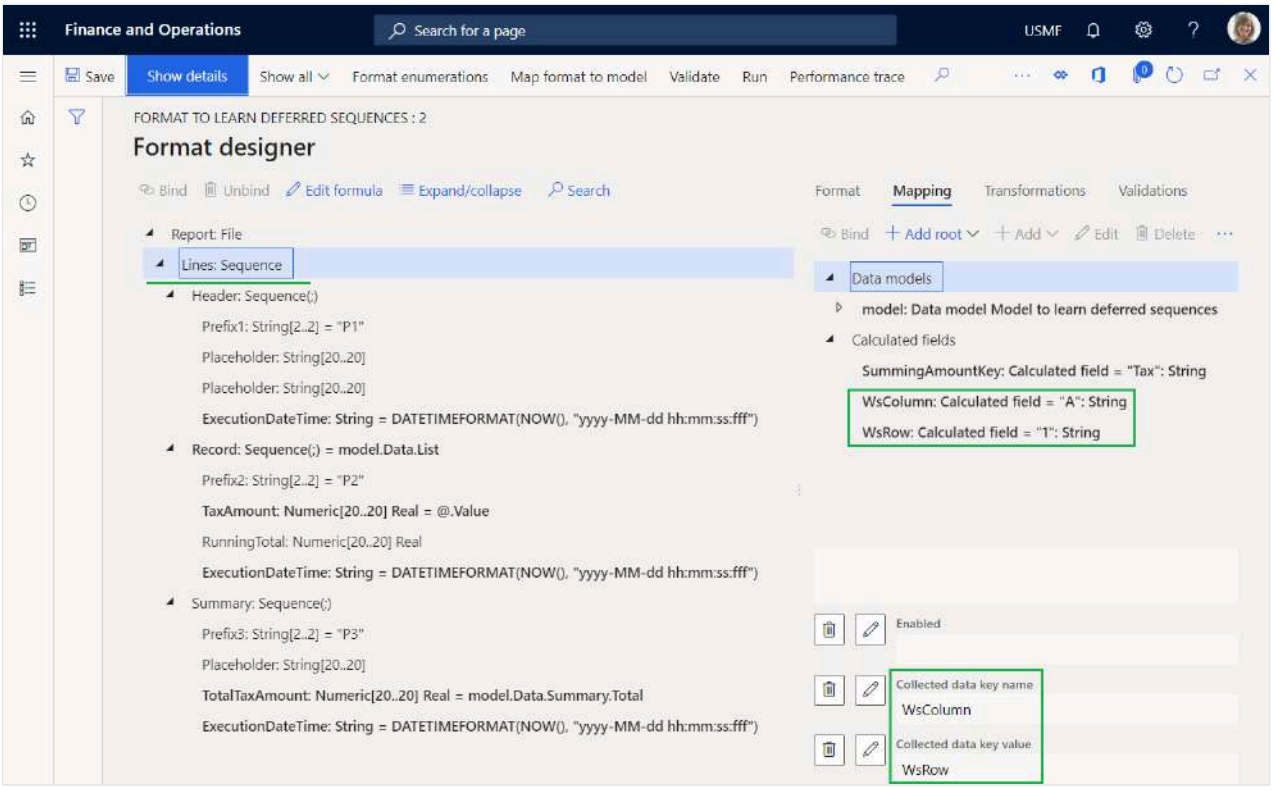
Summary line 22 presents the sum of the tax values for the processed transactions. Because the format uses the **model.Data.Summary.Total** binding to return this sum, the sum is calculated

by calling the **TotalSum** aggregation of the **Grouped** data source of the *GroupBy* type that uses the model mapping. To calculate this aggregation, model mapping iterates over all transactions selected in the **Filtered** data source. By comparing the execution times of lines 21 and 22, you can determine that calculation of the sum took 10 milliseconds (ms). By comparing the execution times of lines 2 and 21, you can determine that generation of all transactional lines took 7 ms. Therefore, a total of 17 ms was required.

Modify the format so that the summing is based on generated output

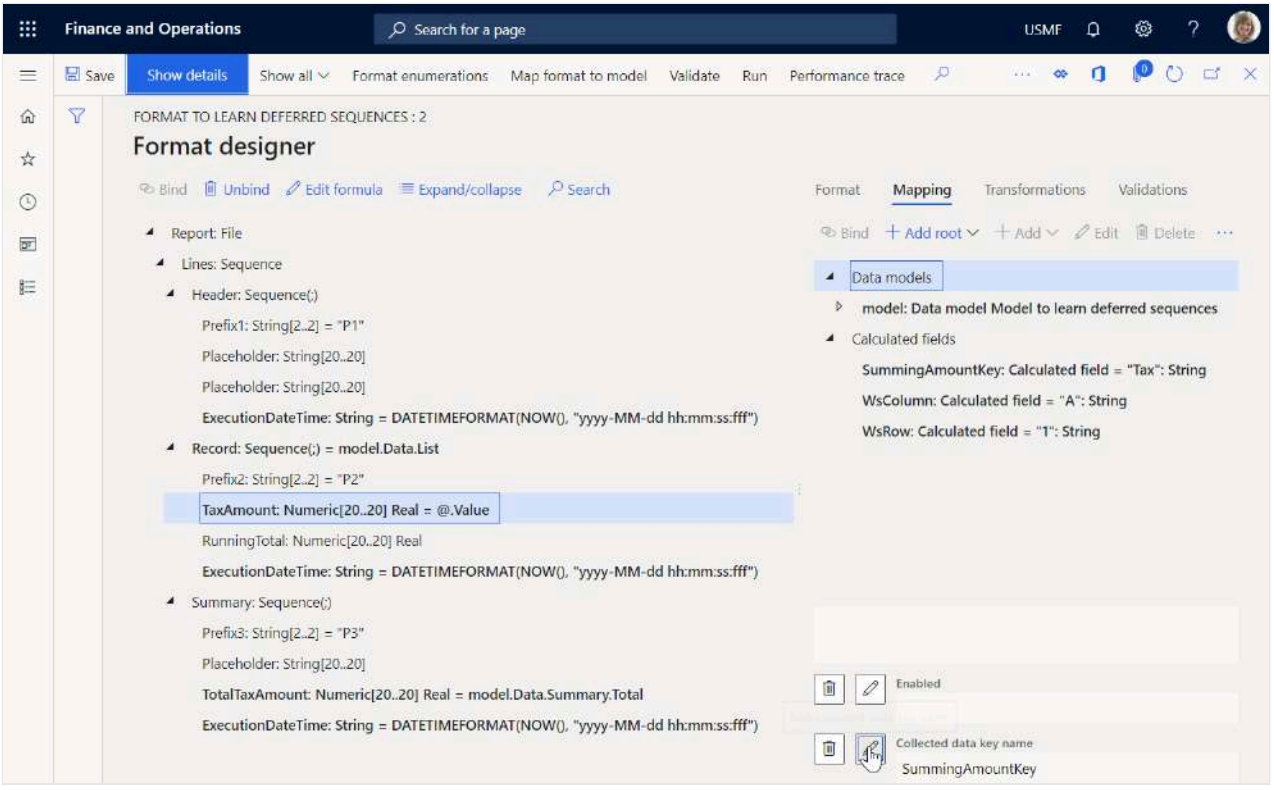
If the volume of transactions is much larger than the volume in the current example, the summing time might increase and cause performance issues. By changing the setting of the format, you can help prevent these performance issues. Because you access tax values to include them in the generated report, you can reuse this information to sum tax values. For more information, see Configure format to do counting and summing.

1. On the **Format designer** page, on the **Format** tab, select the **Report** file element in the format tree.
2. Set the **Collect output details** option to **Yes**. You can now configure this format by using the content of an existing report as a data source that can be accessed by using the built-in ER functions in the Data collection category.
3. On the **Mapping** tab, select the **Report\Lines** sequence element.
4. Configure the **Collected data key name** expression as `WsColumn`.
5. Configure the **Collected data key value** expression as `WsRow`.



6. Select the **Report\Lines\Record\TaxAmount** numeric element.

7. Configure the **Collected data key name** expression as `SummingAmountKey`.

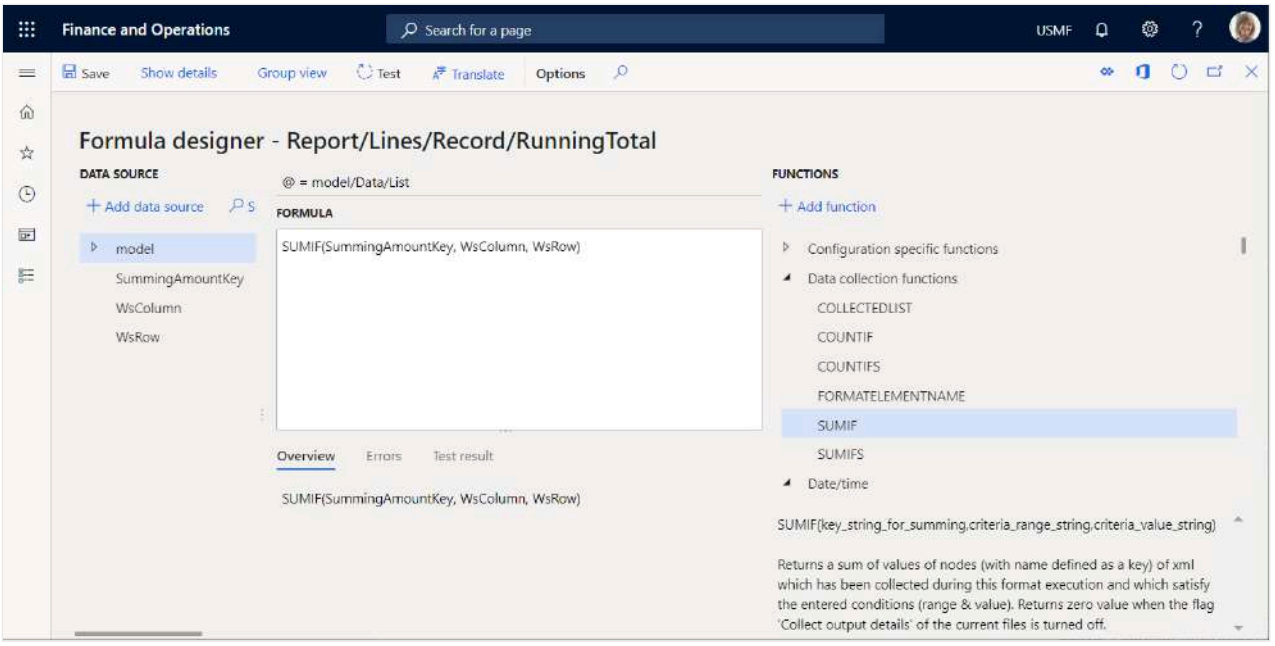


You can consider this setting the fulfillment of a virtual worksheet, where the value of cell A1 is appended with the value of the tax amount from every processed tax transaction.

8. Select the **Report\Lines\Record\RunningTotal** numeric element, and then select **Edit formula**.

9. Configure the `SUMIF(SummingAmountKey, WsColumn, WsRow)` expression by using the built-in SUMIF ER function.

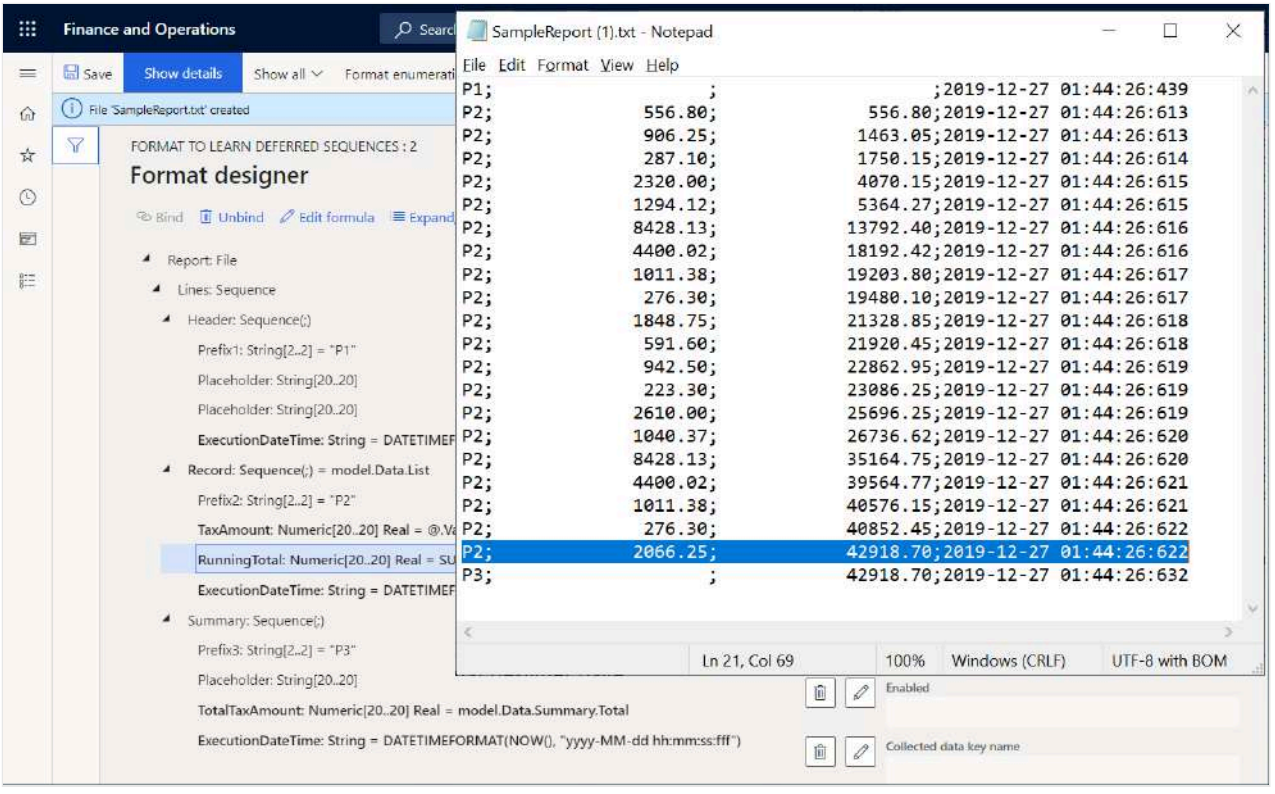
10. Select **Save**.



11. Close the **Formula designer** page.

12. Select **Save**, and then select **Run**.

13. Download and review the file that the web browser offers.



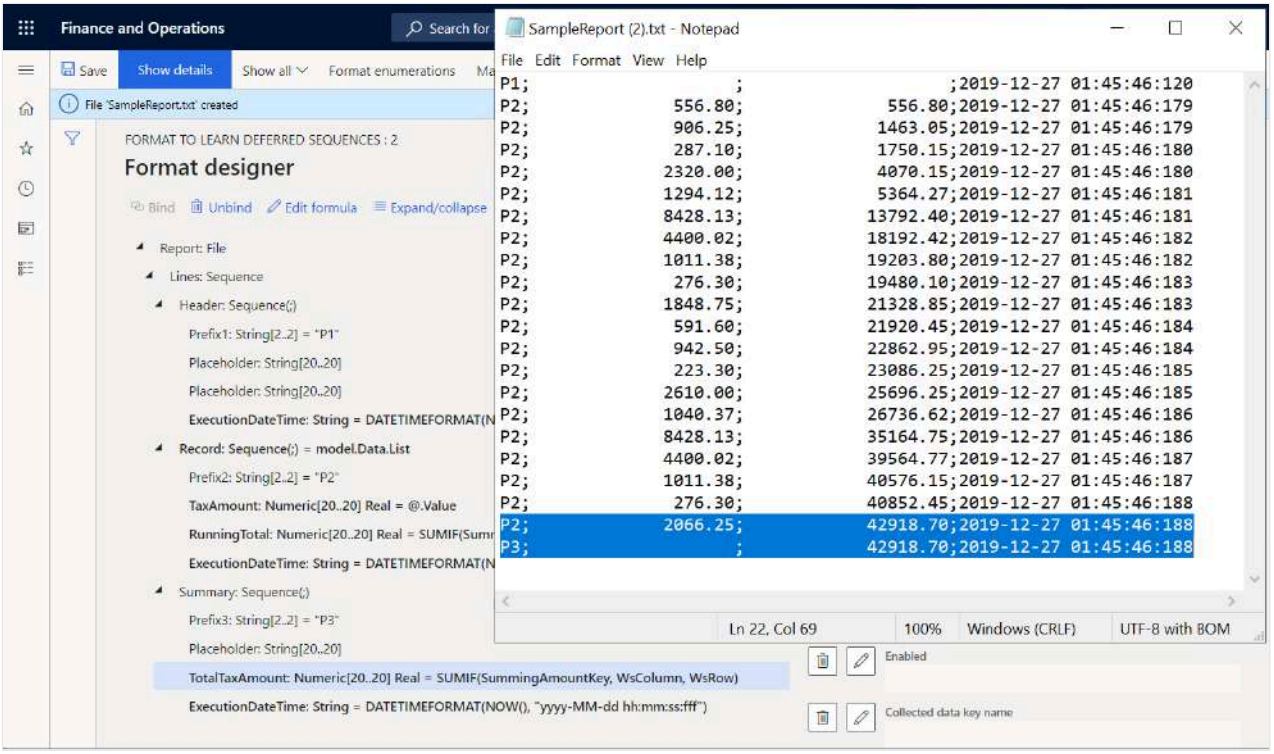
Line 21 contains the running total of tax values that is calculated for all processed transactions by using the generated output as a data source. This data source starts from the beginning of the report and continues through the last tax transaction. Line 22 contains the sum of the tax values for all processed transactions that are calculated in the model mapping by using the data source of the *GroupBy* type. Notice that these values are equal. Therefore, you can use output-based summing instead of **GroupBy**. By comparing the execution times of lines 2 and 21, you can determine that generation of all the transactional lines and summing took 9 ms. Therefore, as far as the generation of detailed lines and the summing of tax values are concerned, the modified format is approximately two times faster than the original format.

14. Select the **Report\Lines\Summary\TotalTaxAmount** numeric element, and then select **Edit formula**.

15. Enter the `SUMIF(SummingAmountKey, WsColumn, WsRow)` expression instead of the existing expression.

16. Select **Save**, and then select **Run**.

17. Download and review the file that the web browser offers.

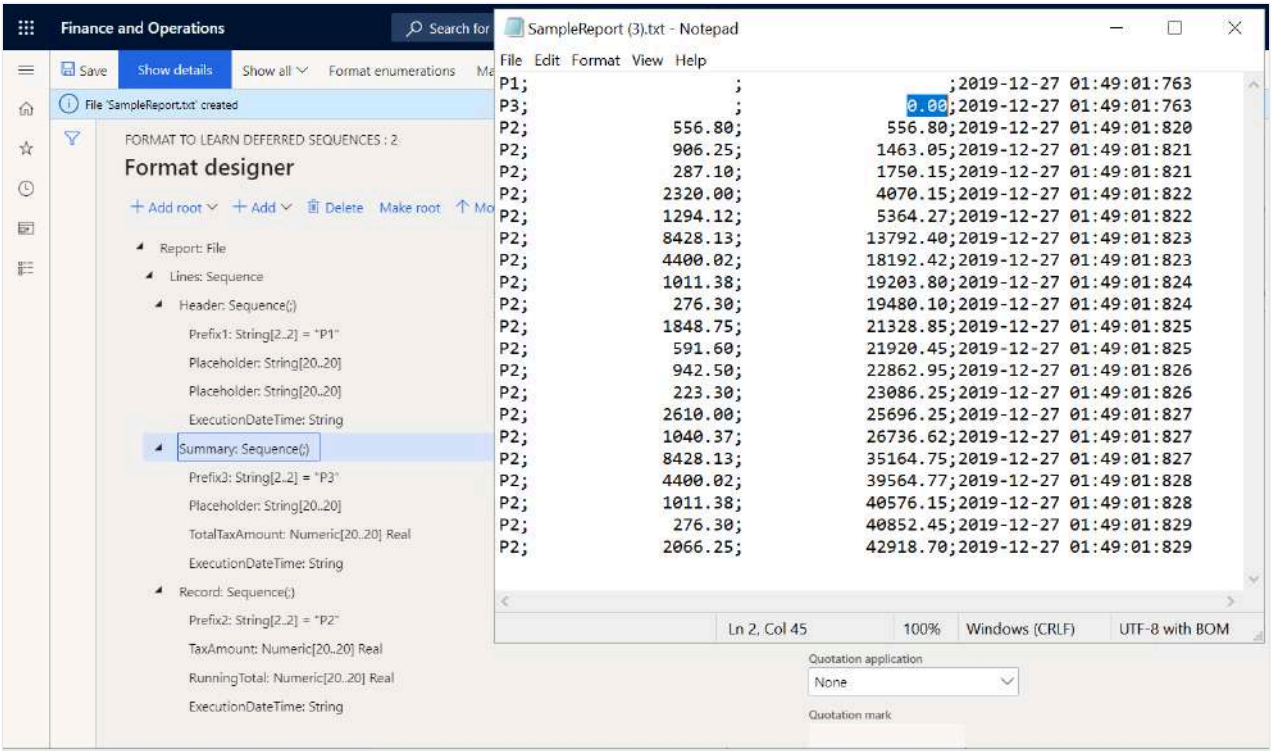


Notice that the running total of tax values on the last transaction details line now equals the sum on the summary line.

Put values of output-based summing in the report header

If, for example, you need to present the sum of tax values in the header of your report, you can modify your format.

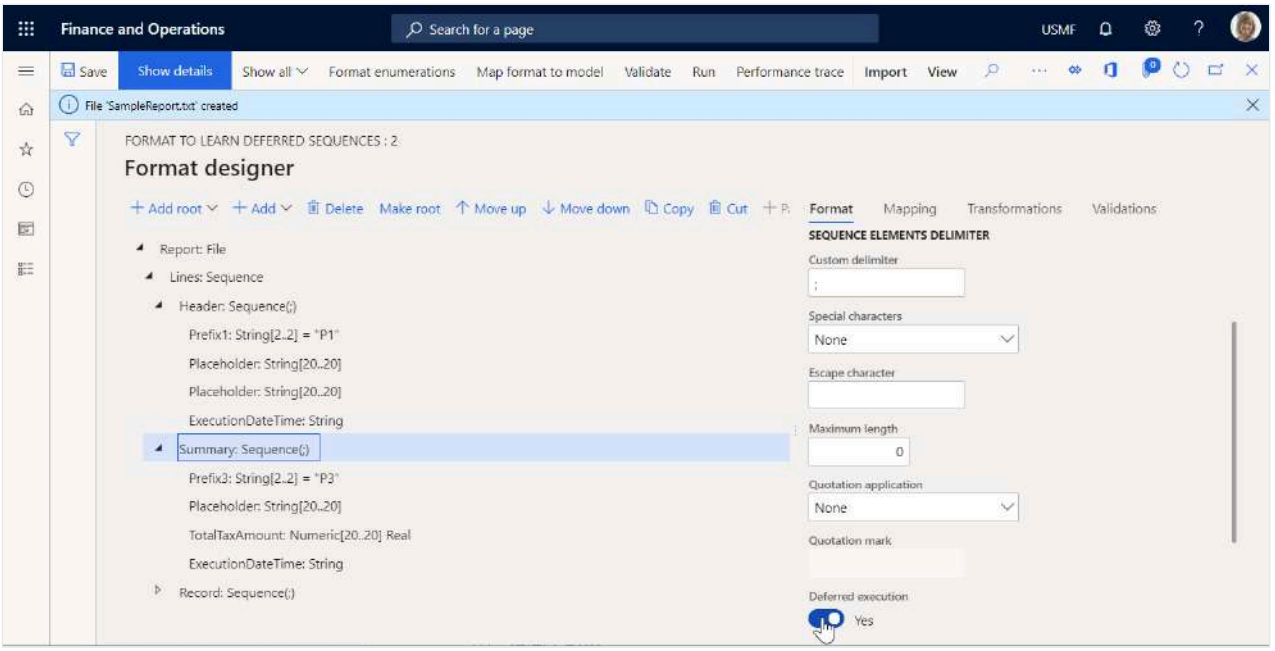
1. On the **Format designer** page, on the **Format** tab, select the **Report\Lines\Summary** sequence element.
2. Select **Move up**.
3. Select **Save**, and then select **Run**.
4. Download and review the file that the web browser offers.



The sum of tax values on summary line 2 now equals 0 (zero), because this sum is now calculated based on the generated output. When line 2 is generated, the generated output doesn't yet contain lines that have transaction details. You can configure this format to defer the execution of the **Report\Lines\Summary** sequence element until the **Report\Lines\Record** sequence element runs for all tax transactions.

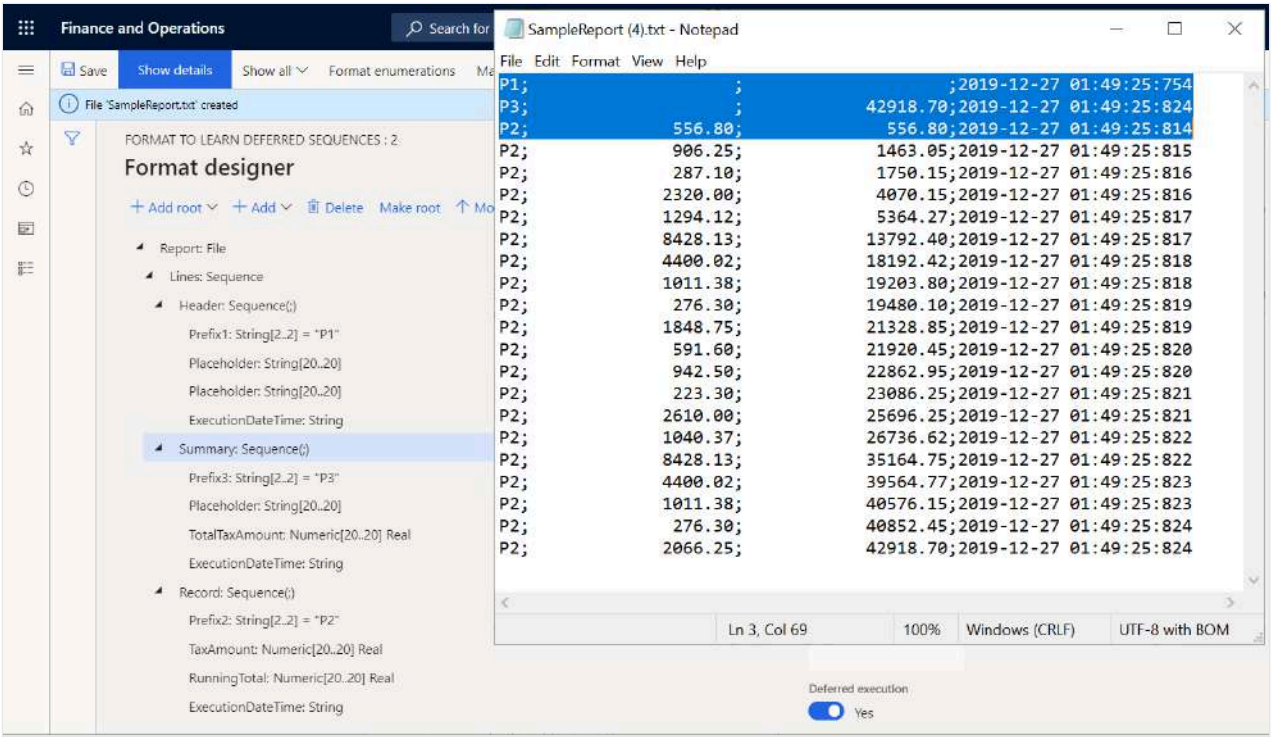
Defer the execution of the summary sequence so that the calculated total is used

1. On the **Format designer** page, on the **Format** tab, select the **Report\Lines\Summary** sequence element.
2. Set the **Deferred execution** option to **Yes**.



3. Select **Save**, and then select **Run**.

4. Download and review the file that the web browser offers.



The **Report\Lines\Summary** sequence element now runs only after all other items that are nested under its parent element, **Report\Lines**, run. Therefore, it runs after the **Report\Lines\Record** sequence element runs for all tax transactions of the **model.Data.List** data source. The execution times of lines 1, 2, and 3, and of the last line, 22, reveal this fact.

Additional resources

- Configure format to do counting and summing
- Trace execution of ER format to troubleshoot performance issues
- Defer the execution of XML elements in ER formats

Last updated on 04/09/2026

Defer the execution of XML elements in ER formats

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Overview

Use the Operations designer of the Electronic reporting (ER) framework to configure the format component of an ER solution that generates outbound documents in XML format. The hierarchical structure of the configured format component consists of format elements of various types. These format elements fill generated documents with the required information at runtime. By default, when you run an ER format, the format elements run in the same order as they appear in the format hierarchy: one by one, from top to bottom. However, at design time, you can change the execution order for any XML elements of the configured format component.

When you turn on the **Deferred execution** option for an XML element in the configured format, you defer (postpone) the execution of that element. In this case, the element isn't run until all other elements of its parent run.

To learn more about this feature, complete the example in this article.

Limitations

The **Deferred execution** option supports only XML elements that are configured for an ER format that generates **outbound** documents in XML format.

The **Deferred execution** option supports only XML elements that reside in one other XML element. Therefore, it doesn't apply to XML elements that reside in other types of format elements (for example, in an **XML sequence** element).

The **Deferred execution** option isn't supported for XML elements that reside in the **Common\File** format element when the **Split file** option is set to **Yes**. For more information about how to split XML files, see Split generated XML files based on file size and content quantity.

Example: Defer the execution of an XML element in an ER format

The following steps explain how a user in the System administrator or Electronic reporting functional consultant role can configure an ER format that contains an XML element where the order of execution differs from the order in the format hierarchy.

You can perform these steps in the **USMF** company in Microsoft Dynamics 365 Finance.

Prerequisites

To complete this example, you must have access to the **USMF** company in Finance for one of the following roles:

- Electronic reporting functional consultant
- System administrator

If you didn't yet complete the example in the Defer the execution of sequence elements in ER formats article, download the following configurations of the sample ER solution.

Expand table

Content description	File name
ER data model configuration	Model to learn deferred elements.version.1.xml
ER model mapping configuration	Mapping to learn deferred elements.version.1.1.xml

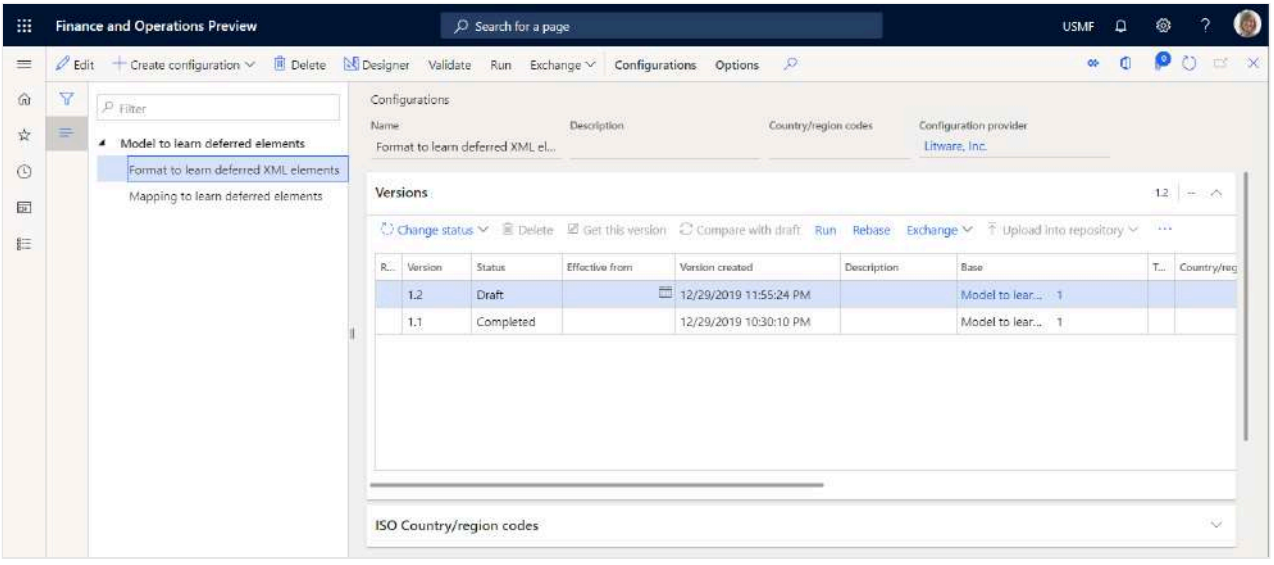
Before you begin, also download and save the following configuration of the sample ER solution to your local computer.

Expand table

Content description	File name
ER format configuration	Format to learn deferred XML elements.version.1.1.xml

Import the sample ER configurations

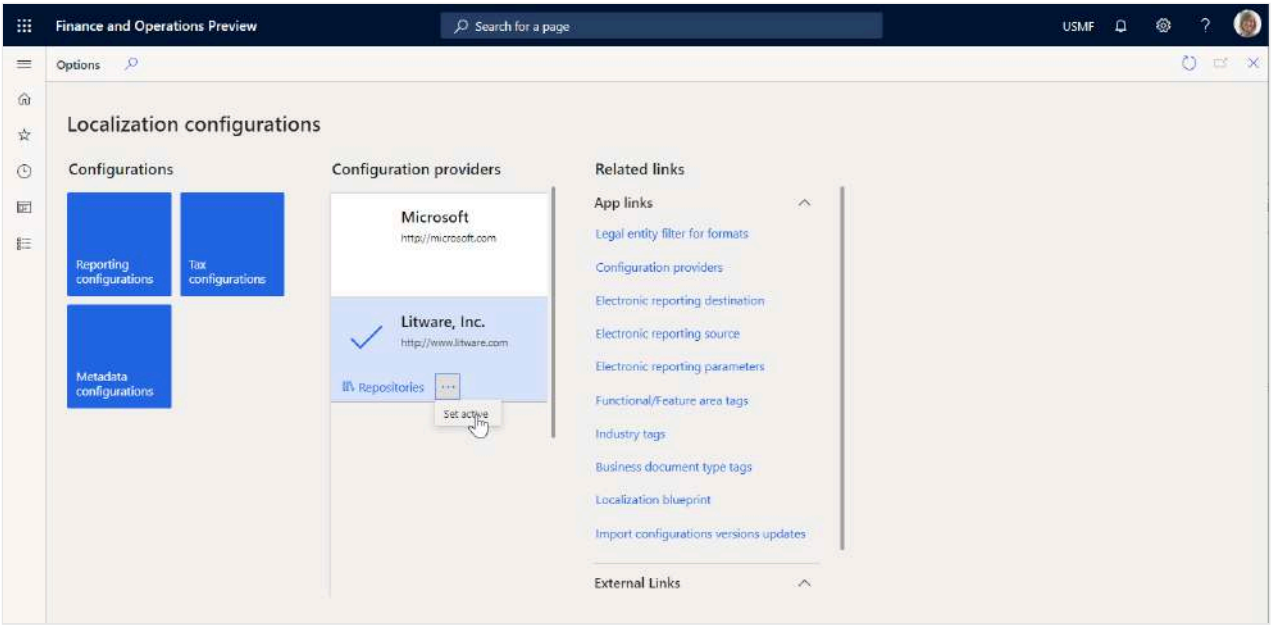
1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. On **Configurations**, if the **Model to learn deferred elements** configuration isn't available in the configuration tree, import the ER data model configuration:
   a. Select **Exchange**, and then select **Load from XML file**.
   b. Select **Browse**, find and select the **Model to learn deferred elements.1.xml** file, and then select **OK**.
4. If the **Mapping to learn deferred elements** configuration isn't available in the configuration tree, import the ER model mapping configuration:
   a. Select **Exchange**, and then select **Load from XML file**.
   b. Select **Browse**, find and select the **Mapping to learn deferred elements.1.1.xml** file, and then select **OK**.
5. Import the ER format configuration:
   a. Select **Exchange**, and then select **Load from XML file**.
   b. Select **Browse**, find and select the **Format to learn deferred XML elements.1.1.xml** file, and then select **OK**.
6. In the configuration tree, expand **Model to learn deferred elements**.
7. Review the list of imported ER configurations in the configuration tree.



Activate a configuration provider

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

2. On **Localization configurations**, in the **Configuration providers** section, make sure that the configuration provider for the Litware, Inc. (`http://www.litware.com`) sample company is listed and marked as active. If this configuration provider isn't listed, or if it isn't marked as active, follow the steps in the Create a configuration provider and mark it as active article.



Review the imported model mapping

Review the settings of the ER model mapping component that you configured to access tax transactions and expose accessed data on request.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

2. Select **Reporting configurations**.

3. On **Configurations**, in the configuration tree, expand **Model to learn deferred elements**.

4. Select the **Mapping to learn deferred elements** configuration.

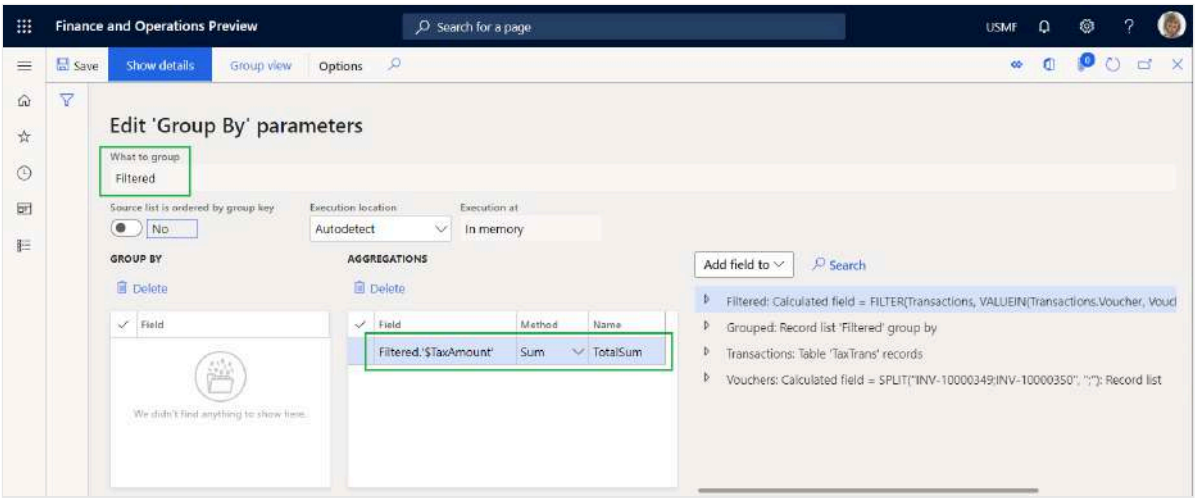
5. Select **Designer** to open the list of mappings.

6. Select **Designer** to review the mapping details.

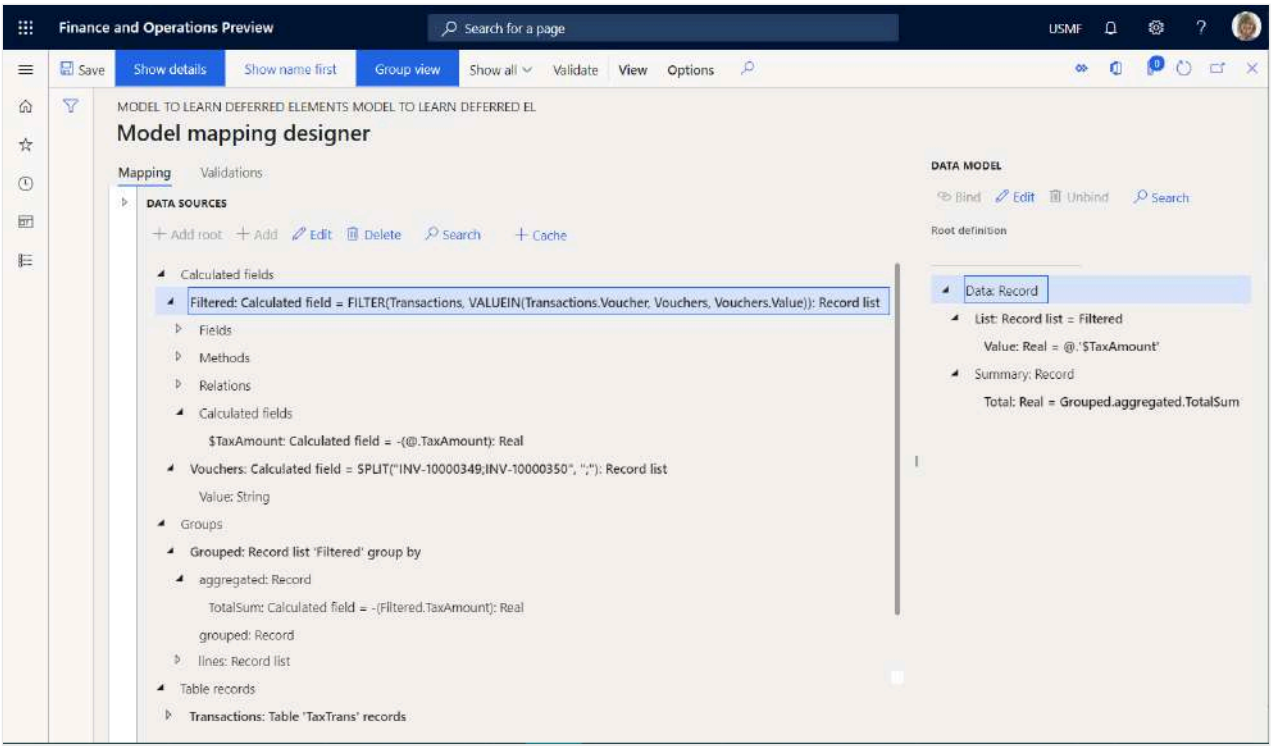
7. Select **Show details**.

8. Review the data sources that are configured to access tax transactions:

- The **Transactions** data source of the *Table record* type is configured to access records of the **TaxTrans** application table.
- The **Vouchers** data source of the *Calculated field* type is configured to return the required voucher codes (**INV-10000349** and **INV-10000350**) as a list of records.
- The **Filtered** data source of the *Calculated field* type is configured to select, from the **Transactions** data source, only tax transactions of the required vouchers.
- The **$TaxAmount** field of the *Calculated field* type is added for the **Filtered** data source to expose the tax value that has the opposite sign.
- The **Grouped** data source of the *Group By* type is configured to group filtered tax transactions of the **Filtered** data source.
- The **TotalSum** aggregation field of the **Grouped** data source is configured to summarize values of the **$TaxAmount** field of the **Filtered** data source for all filtered tax transactions of that data source.



9. Review how the configured data sources are bound to the data model, and how they expose accessed data to make it available in an ER format:
   - The **Filtered** data source is bound to the **Data.List** field of the data model.
   - The **$TaxAmount** field of the **Filtered** data source is bound to the **Data.List.Value** field of the data model.
   - The **TotalSum** field of the **Grouped** data source is bound to the **Data.Summary.Total** field of the data model.



10. Close the **Model mapping designer** and **Model mappings** pages.

Review the imported format

1. On **Configurations**, in the configuration tree, select the **Format to learn deferred XML elements** configuration.

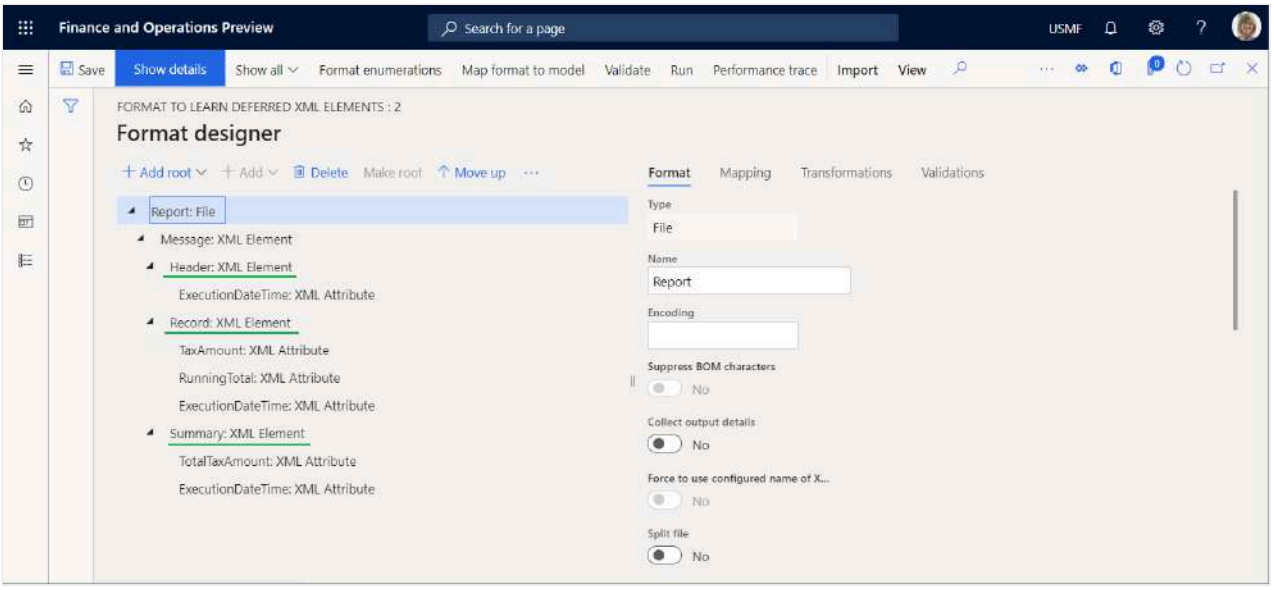
2. Select **Designer** to review the format details.

3. Select **Show details**.

4. Review the settings of the ER format components that are configured to generate an outbound document in XML format that includes details of the tax transactions:

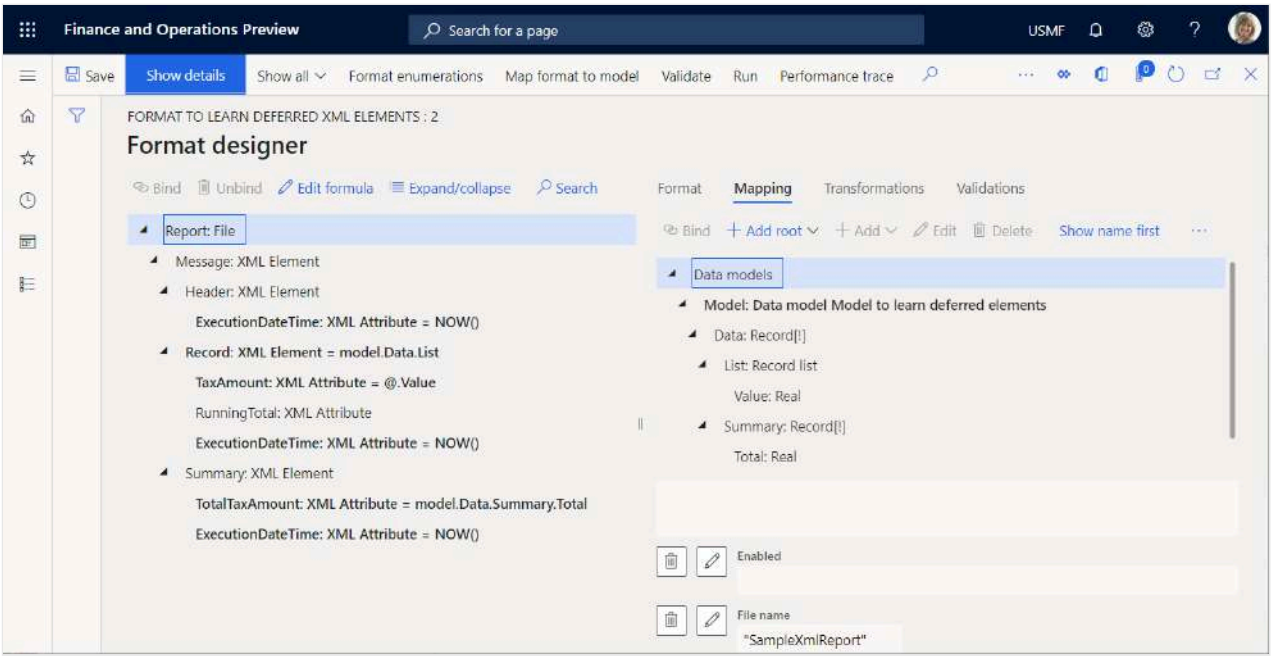
    - The **Report\Message** XML element is configured to fill the outbound document with a single node that includes the nested XML elements (**Header**, **Record**, and **Summary**).
    - The **Report\Message\Header** XML element is configured to fill the outbound document with a single header node that shows the date and time when the processing starts.
    - The **Report \Message\Record** XML element is configured to fill the outbound document with a single record node that shows the details of a single tax transaction.
    - The **Report\Message\Summary** XML element is configured to fill the outbound document with a single summary node that includes the sum of the tax values from

the processed tax transactions.



5. On the **Mapping** tab, review the following details:

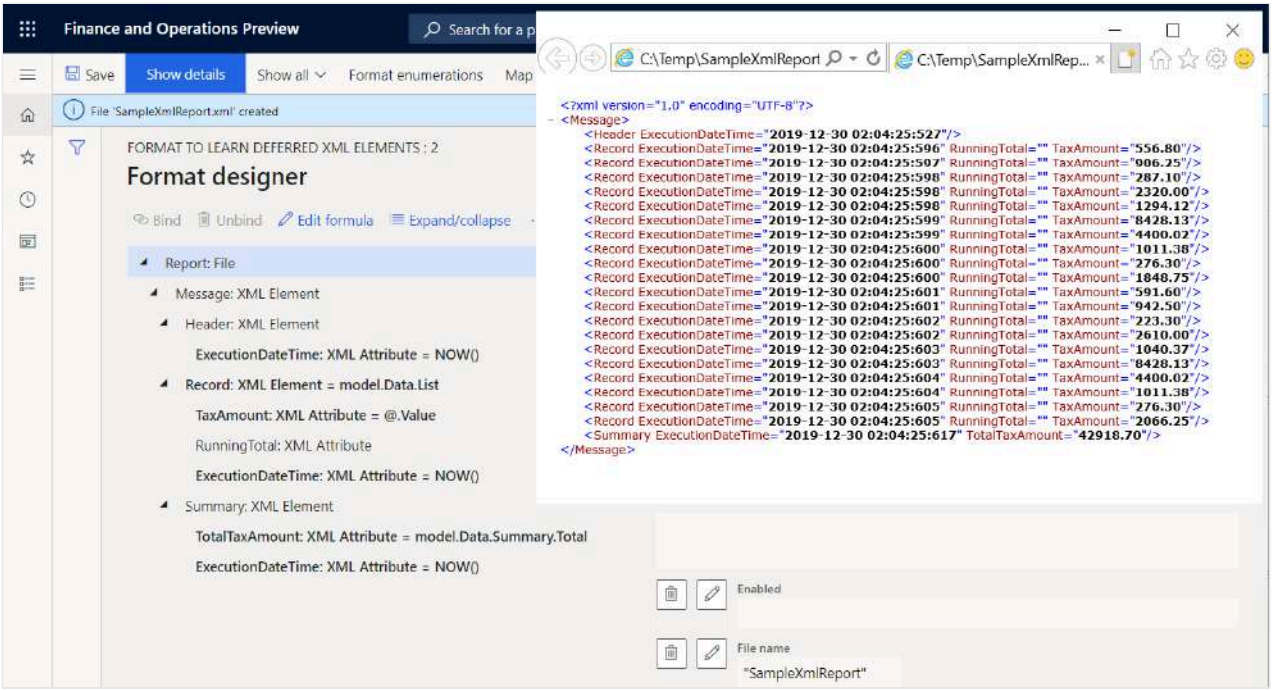
- The **Report\Message\Header** element doesn't need to be bound to a source to generate a single node in an outbound document.
- The **ExecutionDateTime** attribute generates the date and time (including milliseconds) when the header node is added.
- The **Report\Message\Record** element is bound to the **model.Data.List** list to generate a single record node for every record from the bound list.
- The **TaxAmount** attribute is bound to **model.Data.List.Value** (which is shown as **@.Value** in the relative path view) to generate the tax value of the current tax transaction.
- The **RunningTotal** attribute is a placeholder for the running total of the tax values. Currently, this attribute has no output, because neither a binding nor a default value is configured for it.
- The **ExecutionDateTime** attribute generates the date and time (including milliseconds) when the current transaction is processed in this report.
- The **Report\Message\Summary** element doesn't need to be bound to a data source to generate a single node in an outbound document.
- The **TotalTaxAmount** attribute is bound to **model.Data.Summary.Total** to generate the sum of the tax values of the processed tax transactions.
- The **ExecutionDateTime** attribute generates the date and time (including milliseconds) when the summary node is added.



Run the imported format

1. On **Format designer**, select **Run**.

2. Download the file that the web browser offers, and open it for review.



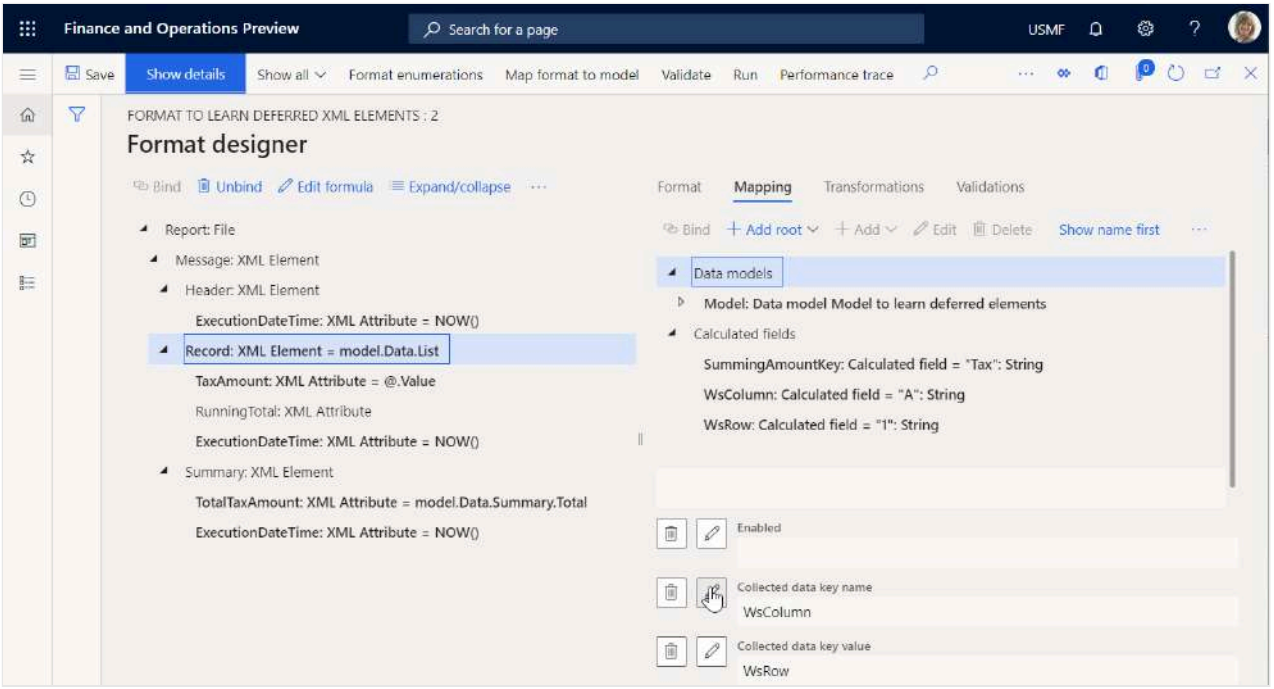
The summary node presents the sum of the tax values for the processed transactions. Because the format uses the **model.Data.Summary.Total** binding to return this sum, the sum is calculated by calling the **TotalSum** aggregation of the **Grouped** data source of the *GroupBy* type in the model mapping. To calculate this aggregation, the model mapping iterates over all transactions that you select in the **Filtered** data source. By comparing the execution times of the summary node and the last record node, you can determine that calculation of the sum

took 12 milliseconds (ms). By comparing the execution times of the first and last record nodes, you can determine that generation of all record nodes took 9 ms. Therefore, a total of 21 ms was required.

Modify the format so that the calculation is based on generated output

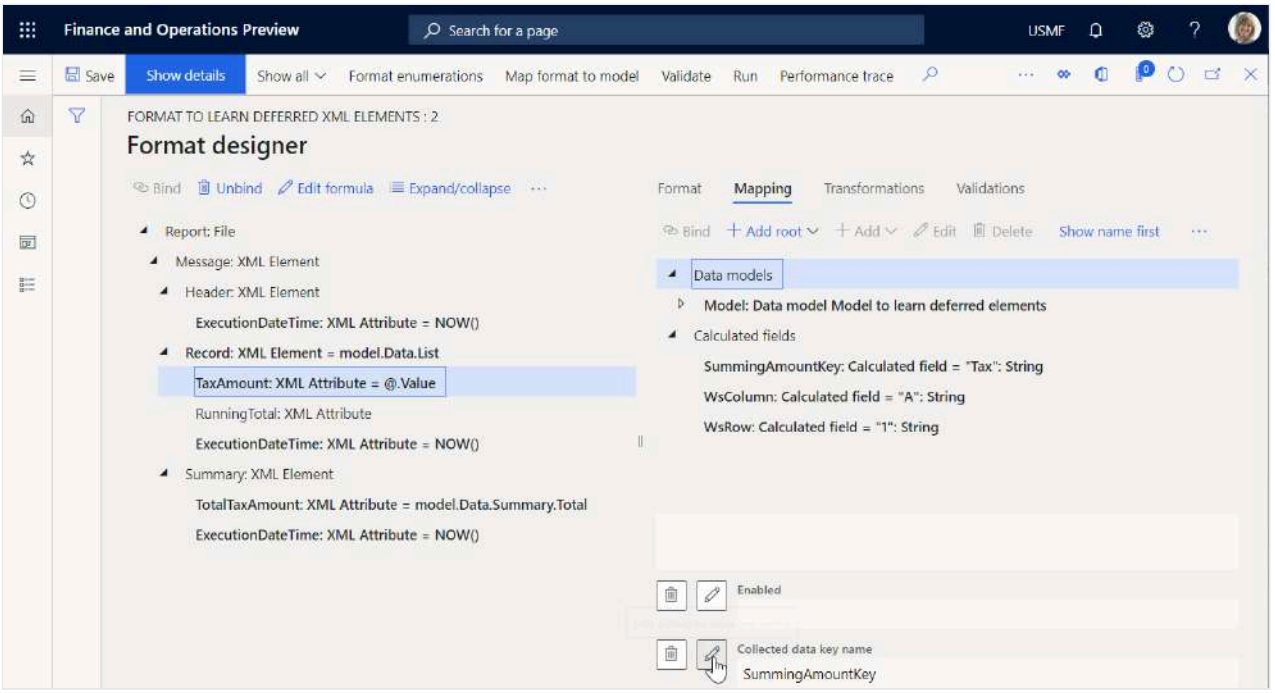
If the transaction volume is much larger than the volume in the current example, the calculation time might increase and cause performance problems. By changing the setting of the format, you can help prevent these performance problems. Because you access tax values to include them in the generated report, you can reuse this information to calculate tax values. For more information, see Configure format to do counting and summing.

1. On the **Format designer** page, on the **Format** tab, select the **Report** file element in the format tree.
2. Set the **Collect output details** option to **Yes**. You can now configure this format by using the content of a generated report as a data source that the built-in ER functions in the Data collection category can access.
3. On the **Mapping** tab, select the **Report\Message\Record** XML element.
4. Configure the **Collected data key name** expression as `WsColumn`.
5. Configure the **Collected data key value** expression as `WsRow`.



6. Select the **Report\Message\Record\TaxAmount** attribute.

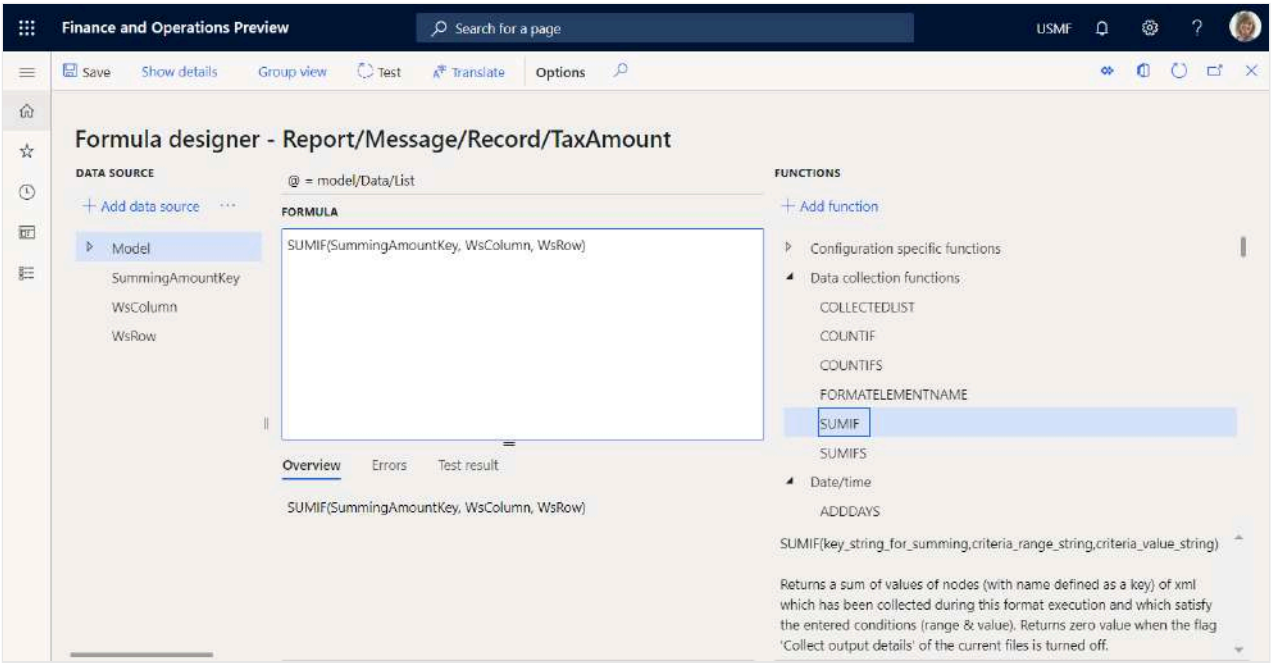
7. Configure the **Collected data key name** expression as `SummingAmountKey`.



You can consider this setting the fulfillment of a virtual worksheet, where the value of cell A1 is appended with the value of the tax amount from every processed tax transaction.

8. Select the **Report\Message\Record\RunningTotal** attribute, and then select **Edit formula**.

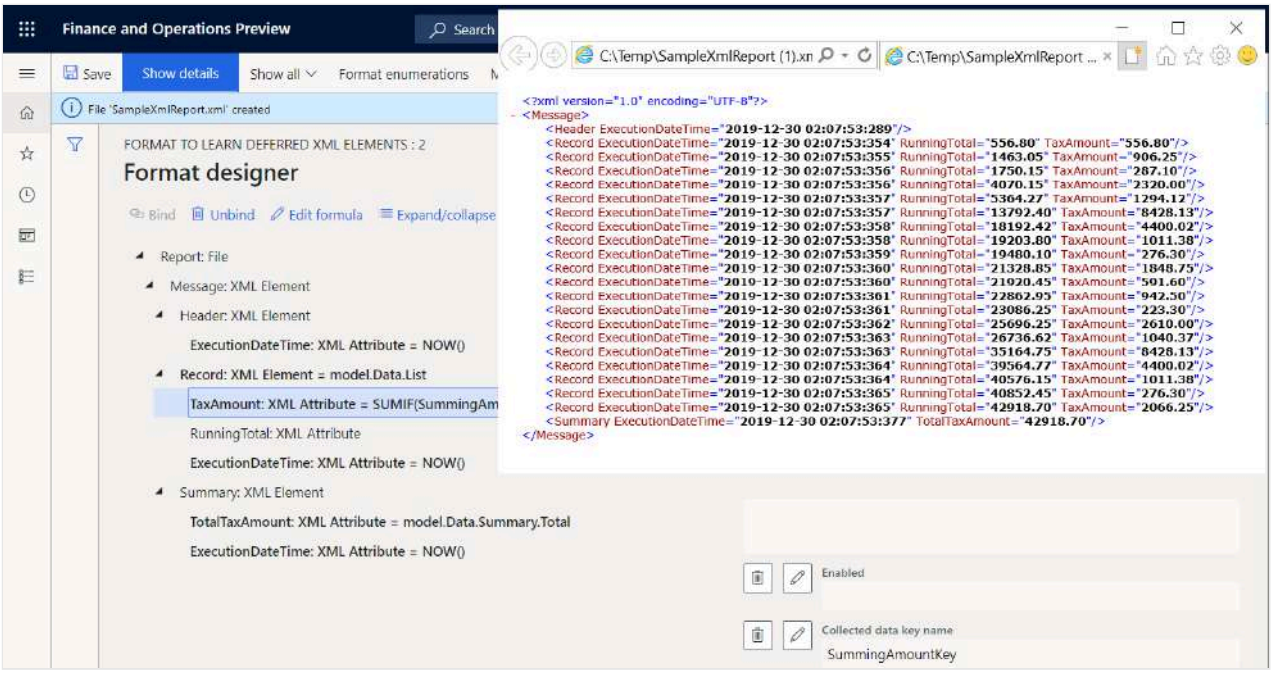
9. Configure the `SUMIF(SummingAmountKey, WsColumn, WsRow)` expression by using the built-in SUMIF ER function, and then select **Save**.



10. Close the **Formula designer** page.

11. Select **Save**, and then select **Run**.

12. Download and review the file that the web browser offers.



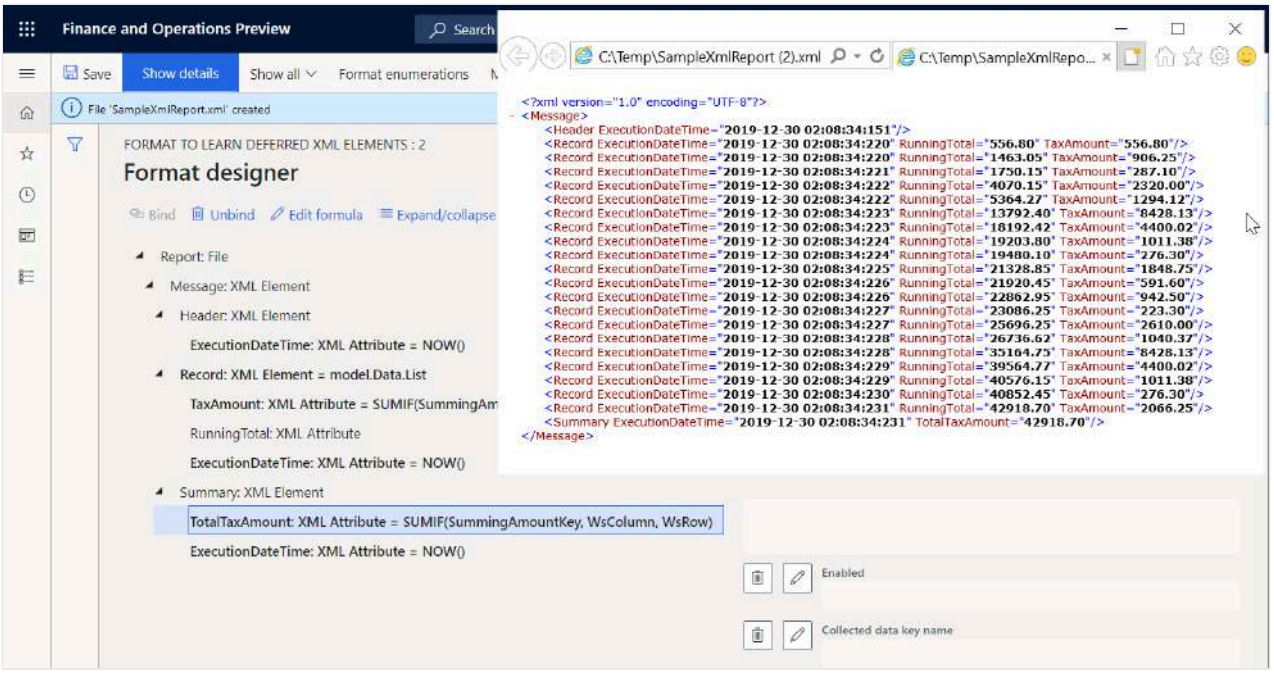
The last record node contains the running total of tax values that is calculated for all processed transactions by using the generated output as a data source. This data source starts from the beginning of the report and continues through the last tax transaction. The summary node contains the sum of the tax values for all processed transactions that are calculated in the model mapping by using the data source of the *GroupBy* type. Notice that these values are equal. Therefore, you can use output-based summing instead of **GroupBy**. By comparing the execution times of the first record node and the summary node, you can determine that generation of all the record nodes and summing took 11 ms. Therefore, as far as the generation of record nodes and the summing of tax values are concerned, the modified format is approximately two times faster than the original format.

13. Select the **Report\Message\Summary\TotalTaxAmount** attribute, and then select **Edit formula**.

14. Enter the `SUMIF(SummingAmountKey, WsColumn, WsRow)` expression instead of the existing expression.

15. Select **Save**, and then select **Run**.

16. Download and review the file that the web browser offers.

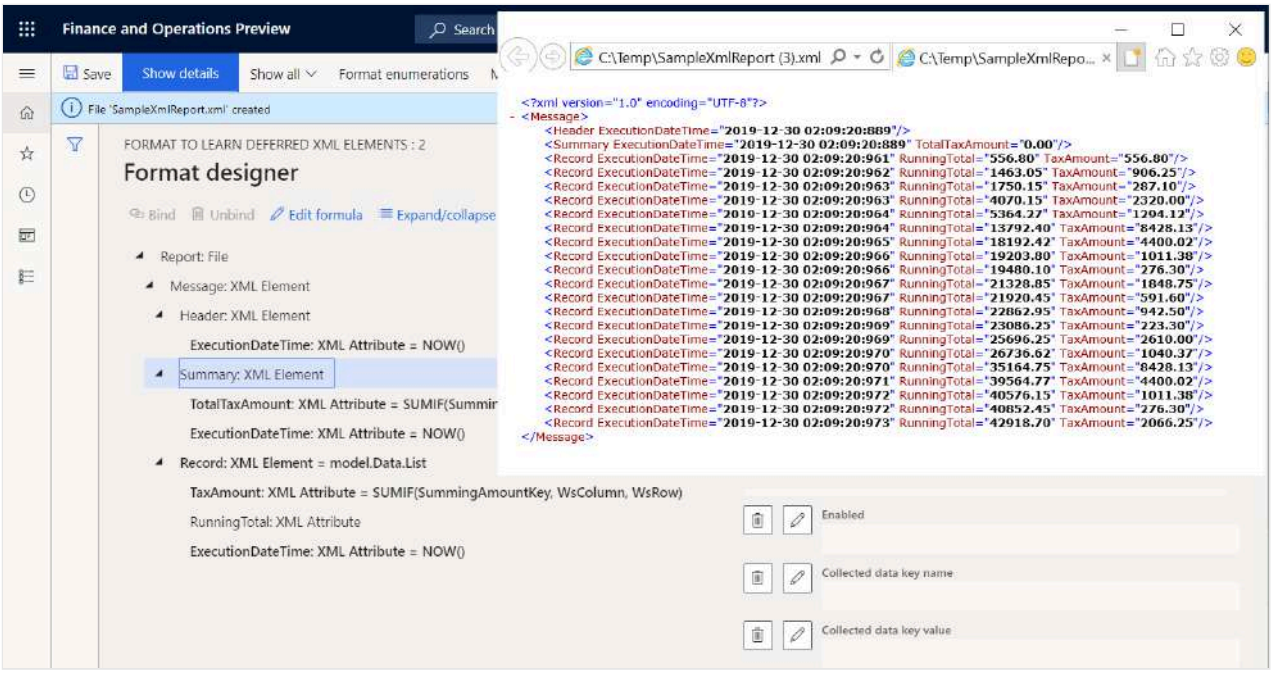


Notice that the running total of tax values in the last record node now equals the sum in the summary node.

Put values of output-based summing in the report header

If you need to present the sum of tax values in the header of your report, you can modify your format.

1. On the **Format designer** page, on the **Format** tab, select the **Report\Message\Summary** XML element.
2. Select **Move up**.
3. Select **Save**, and then select **Run**.
4. Download and review the file that the web browser offers.

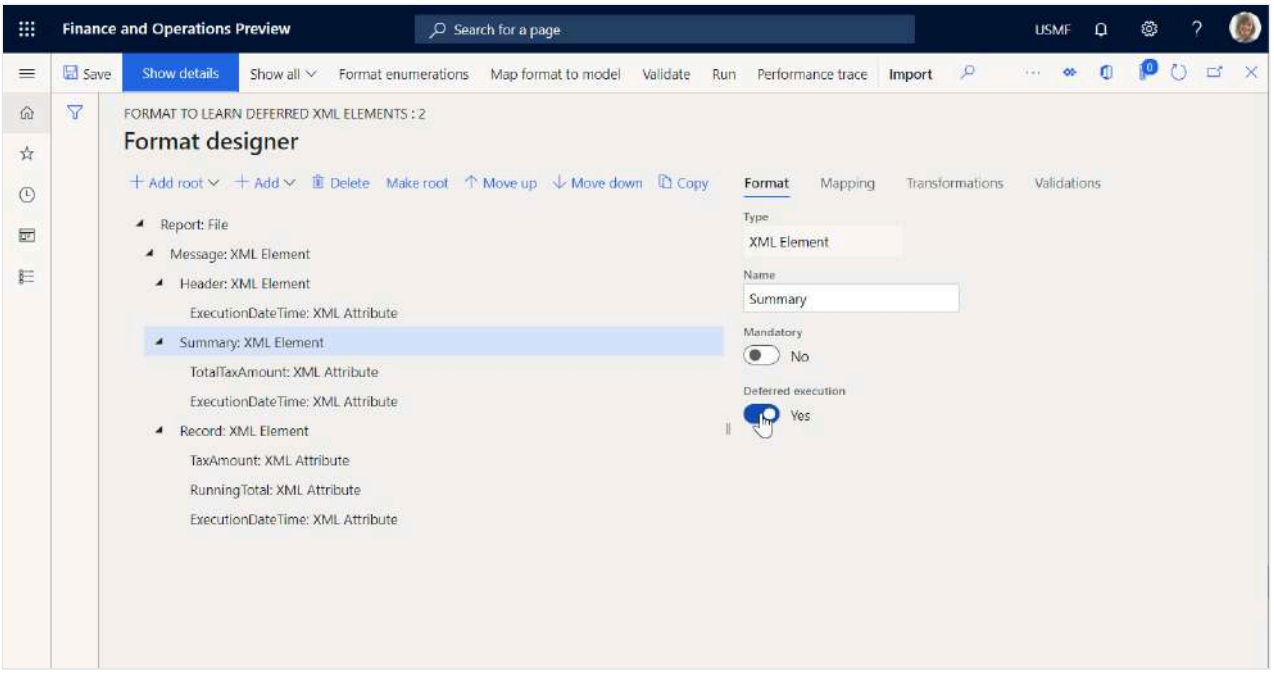


The sum of tax values in the summary node now equals 0 (zero), because this sum is now calculated based on the generated output. When the first record node is generated, the generated output doesn't yet contain record nodes that have transaction details. You can configure this format to defer the execution of the **Report\Message\Summary** element until the **Report\Message\Record** element runs for all tax transactions.

Defer the execution of the summary XML element so that the calculated total is used

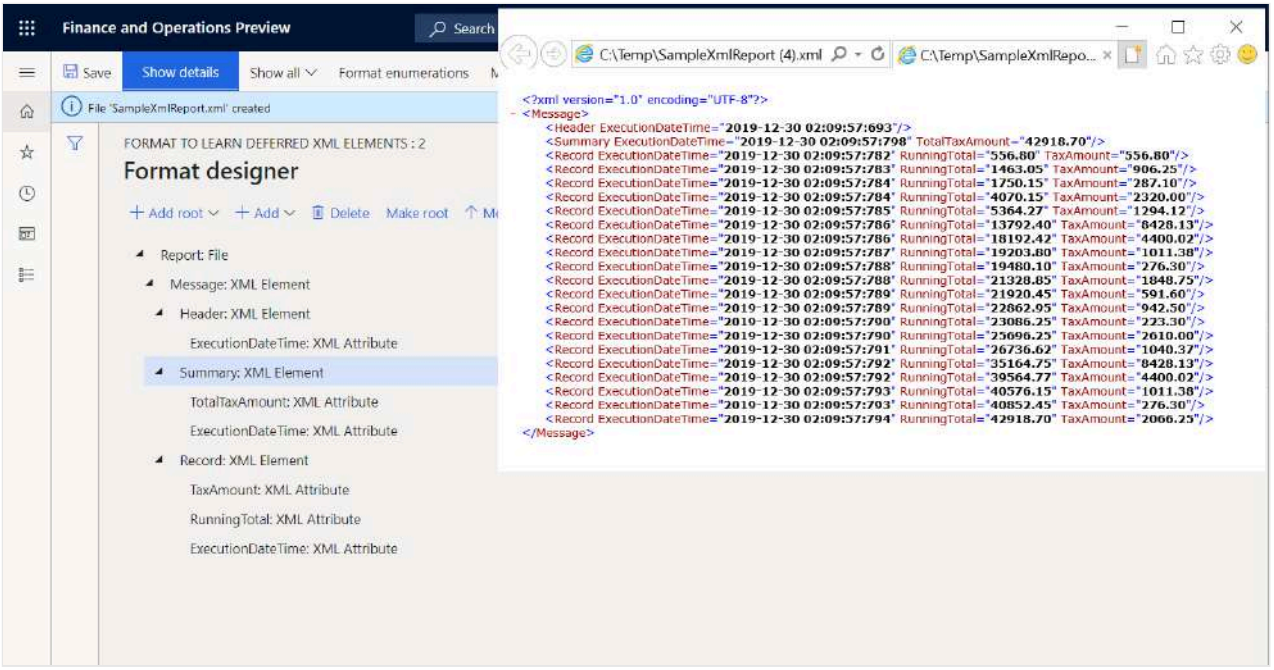
1. On the **Format designer** page, on the **Format** tab, select the **Report\Message\Summary** XML element.

2. Set the **Deferred execution** option to **Yes**.



3. Select **Save**, and then select **Run**.

4. Download and review the file that the web browser offers.



The **Report\Message\Summary** element now runs only after all other items that are nested under its parent element, **Report\Message**, run. Therefore, it runs after the **Report\Message\Record** element runs for all tax transactions of the **model.Data.List** data source. The execution times of the first and last record nodes, and of the header and summary nodes, reveal this fact.

Additional resources

- Configure format to do counting and summing

- Trace execution of ER format to troubleshoot performance issues
- Defer the execution of sequence elements in ER formats

Last updated on 04/08/2026

Design multilingual reports in Electronic reporting

10/02/2025

Overview

As a business user, you can use the Electronic reporting (ER) framework to configure formats for outbound documents that must be generated in accordance with the legal requirements of various countries or regions. When these requirements demand that outbound documents be generated in different languages for different countries or regions, you can configure a single ER format that contains language-dependent resources. In that way, you can reuse the format to generate outbound documents for various countries or regions. You might also want to use a single ER format to generate an outbound document in different languages for corresponding customers, vendors, subsidiaries, or any other parties.

You can configure ER data models and model mappings as the data sources of configured ER formats to define the data flow that specifies what application data is put into generated documents. As an ER configuration provider, you can publish configured data models, model mappings, and formats as components of an ER solution to generate specific outbound documents. You can also allow customers to upload the published ER solution so that it can be used and customized. If you expect that customers might speak other languages, you can configure the ER components so that they contain language-dependent resources. In that way, the content of an editable ER component can be presented in a customer's user-preferred language at design time.

You can configure language-dependent resources as ER labels. You can then use those labels to configure ER components for the following purposes:

- At design time:
  - Present the content of configured ER components in the user-preferred language.
- At runtime:
  - Generate language-dependent content for outbound documents.
  - Provide warning and error messages in the user-preferred language.
  - Prompt for required fields in the user-preferred language.

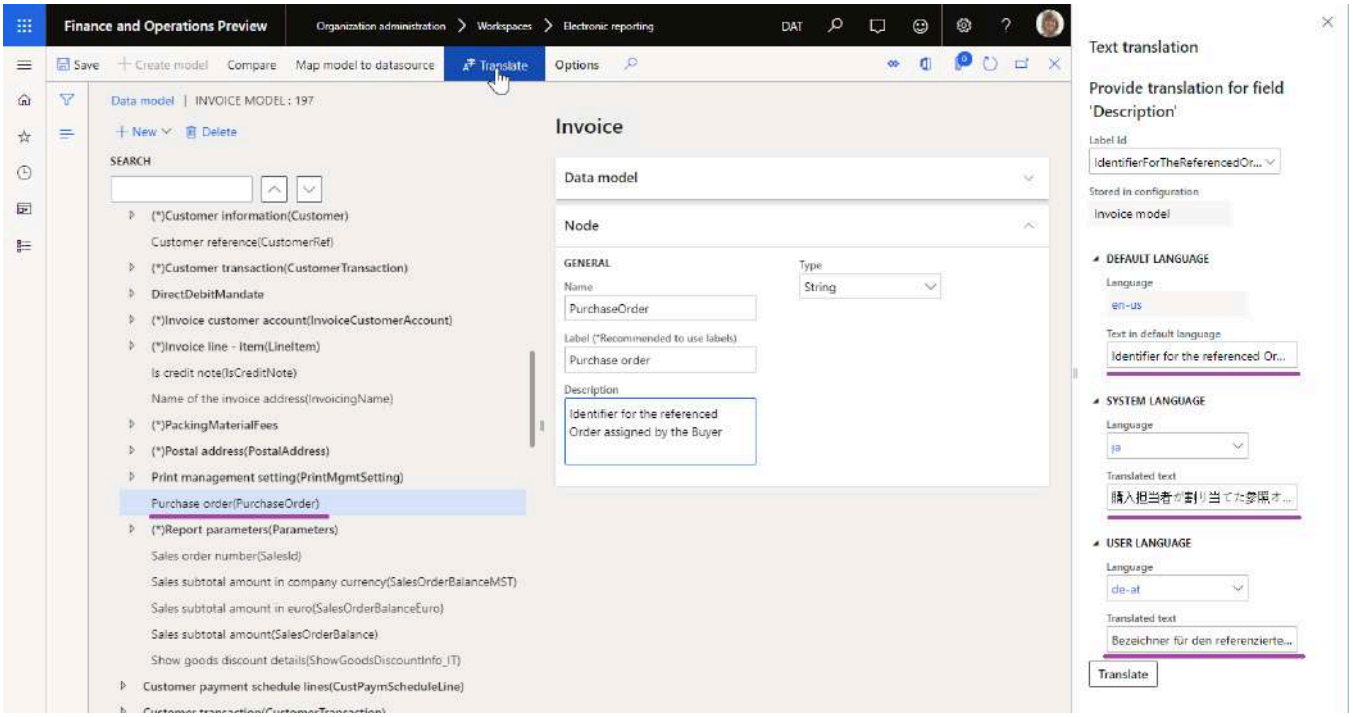
ER labels can be configured in every ER configuration that contains different components. The labels can be maintained independently of the configured logic of ER data models, ER model mappings, and ER format components.

Every ER label is identified by an ID that is unique in the scope of the ER configuration that holds that label. Every label can contain label text for every language that is supported in the current instance of Microsoft Dynamics 365 Finance. These supported languages include the languages of deployed customizations.

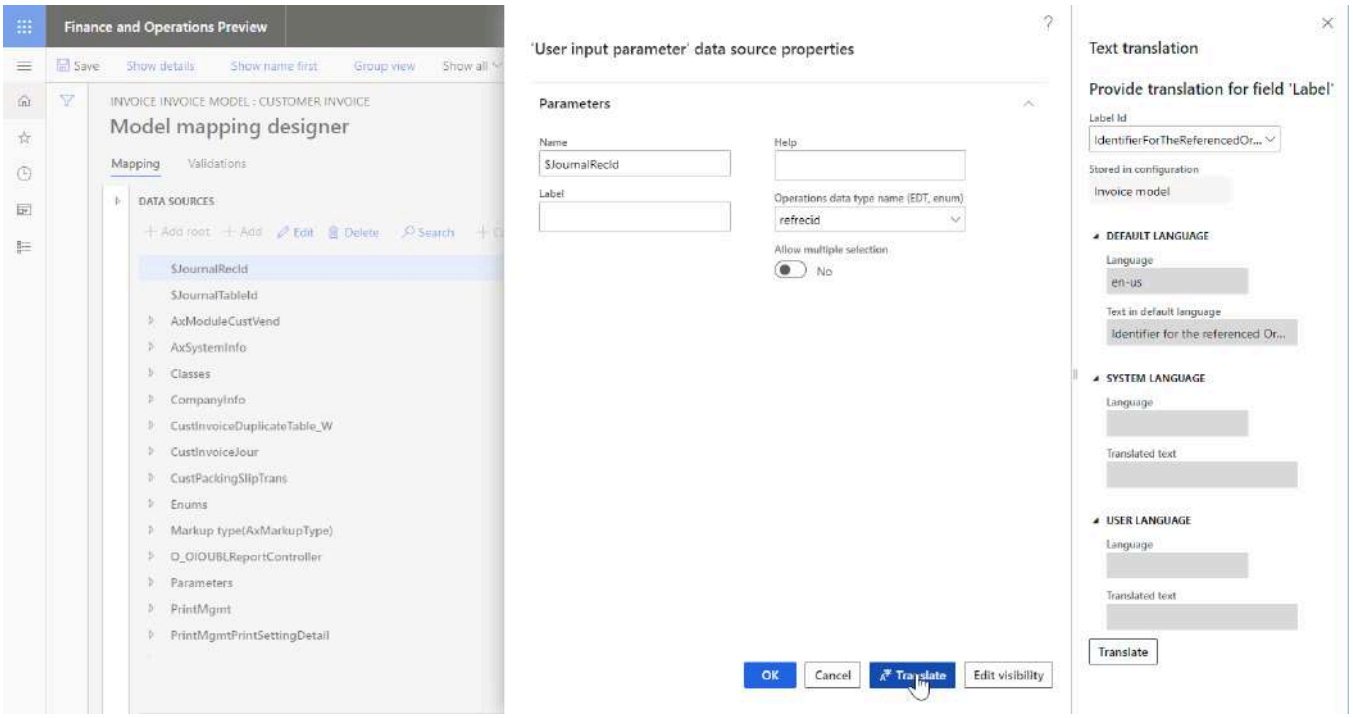
Entry

When you design an ER data model, an ER model mapping, or an ER format, the **Translate** option is shown whenever you select a field that might contain the translatable context. When you select this option, you can link the selected field to an ER label on the **Text translation** pane. You can select an existing ER label, or you can add a new ER label if it isn't available yet. When you select or add an ER label, you can add related text for every language that is supported in the current Finance instance.

The following illustration shows how this translation is done in an editable ER data model. In this example, the **Description** attribute of the **PurchaseOrder** field for the editable **Invoice model** is translated into the Austrian German (DE-AT) and Japanese (JA) languages.



Only label text for labels that reside in an editable ER component can be translated. For example, if you select **Translate** for the label attribute of an ER model mapping data source, and you then select an ER label that resides in the parent ER data model, you will see the content of the label, but you can't change it. In these cases, the **Translated text** field is unavailable, as shown in the following illustration.



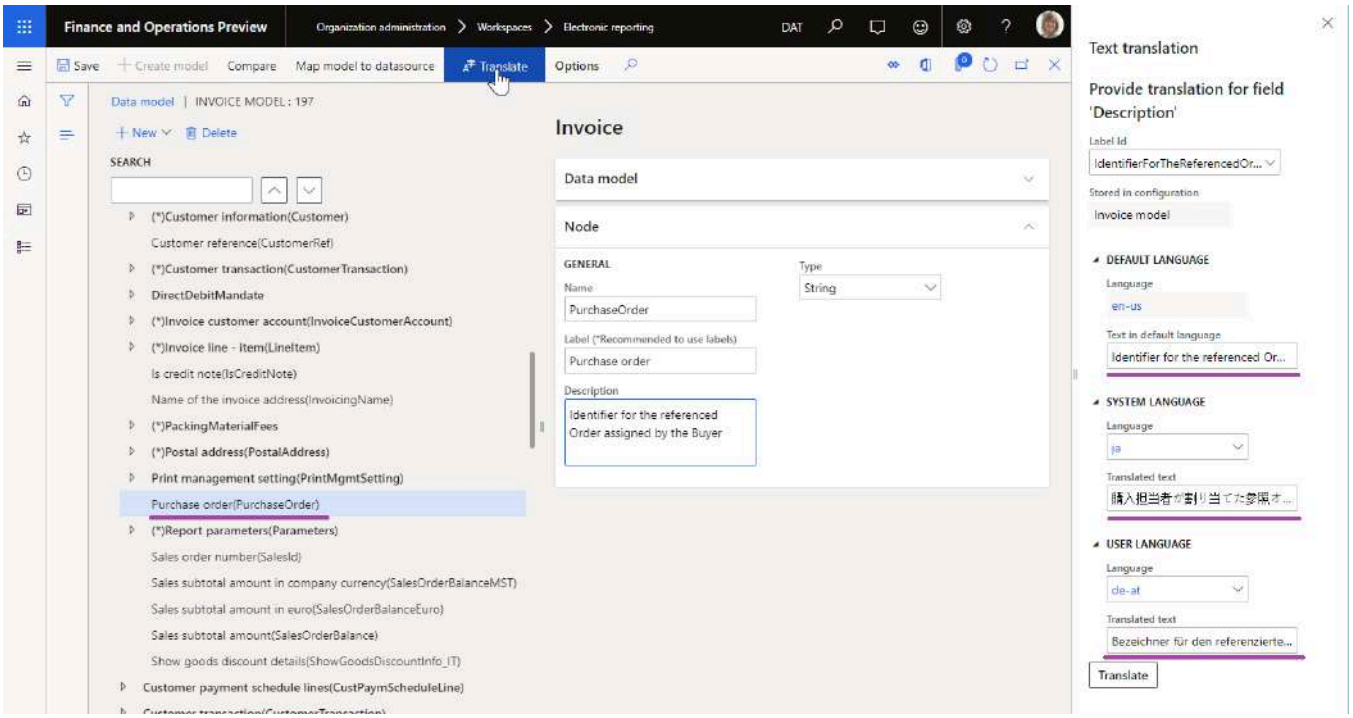
> **Note**
>
> You can't use the designers to delete label that has been entered in an editable ER component.

Scope

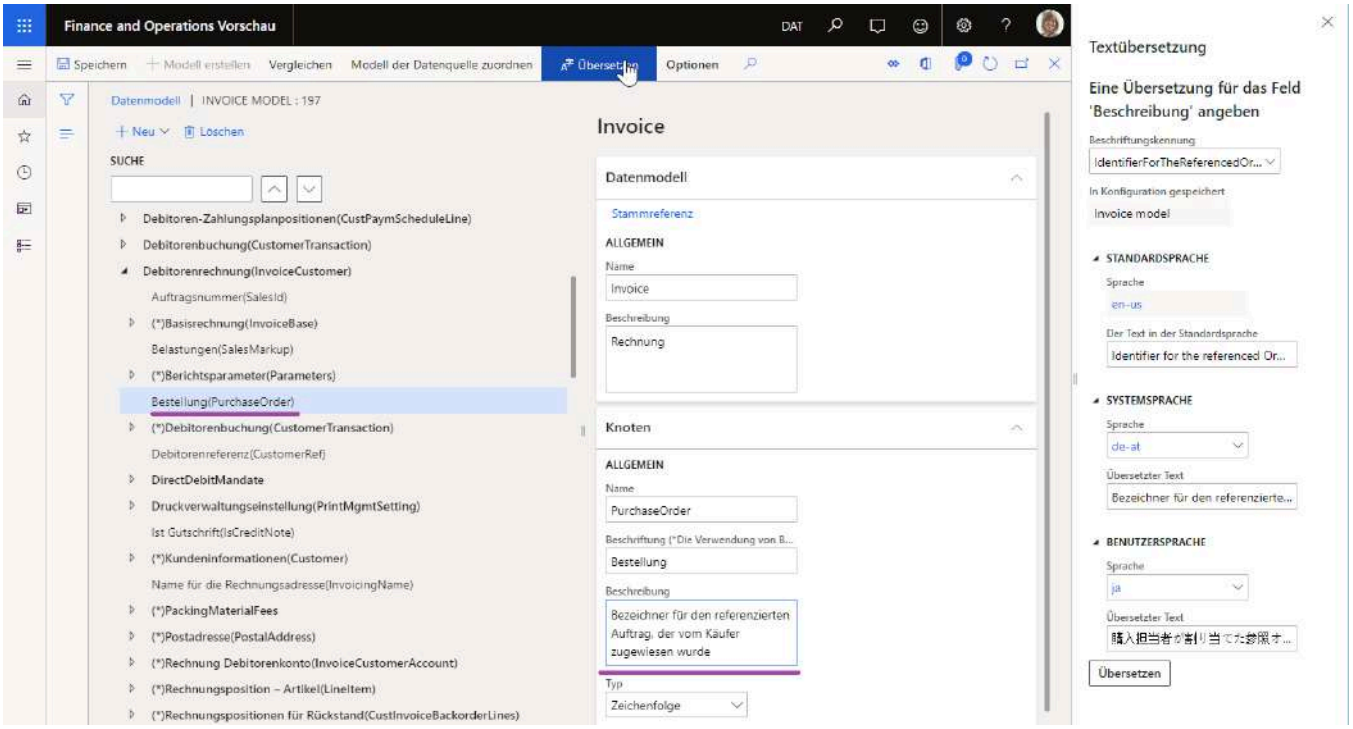
ER labels can be referred to in several translatable attributes of ER components.

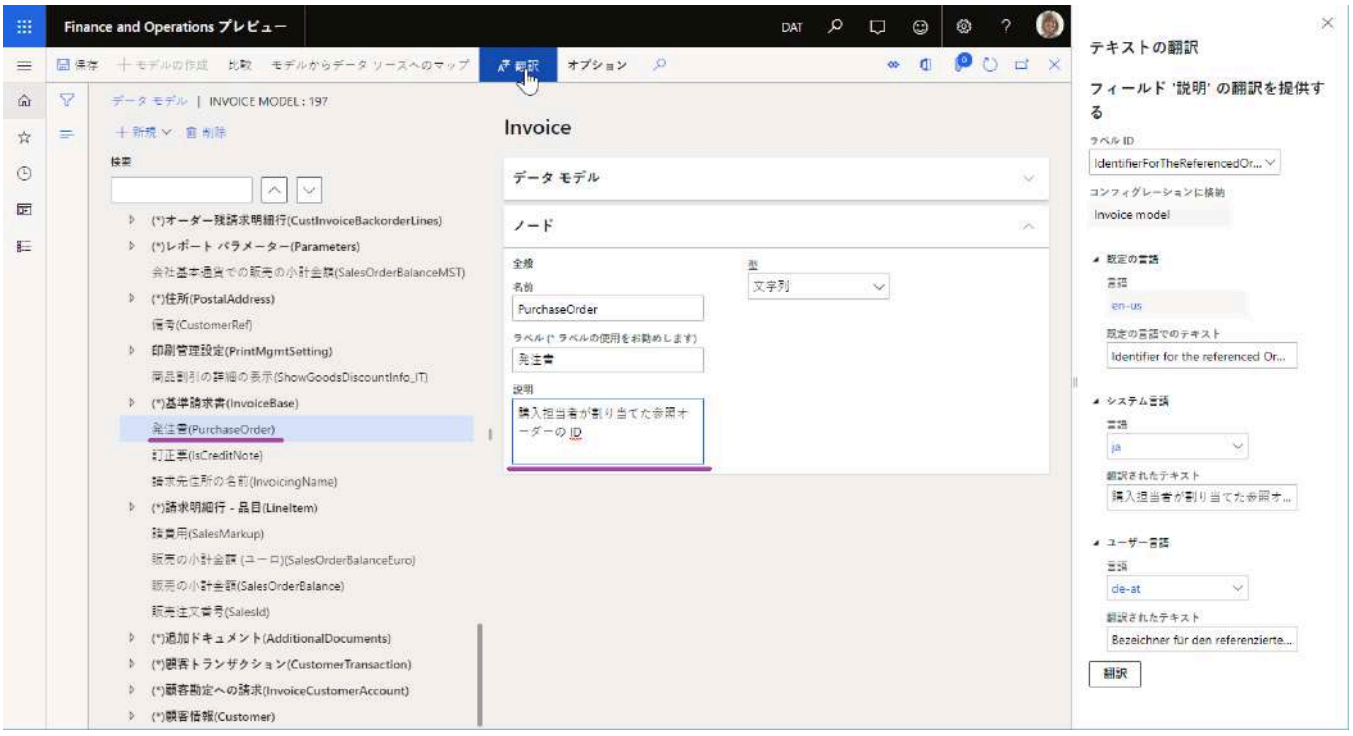
Data model component

When you configure an ER data model, you can add ER labels for it. **Label** and **Description** attributes of the model item, every model's field, and every model enumeration value can be linked to an ER label that is added to the ER data model.



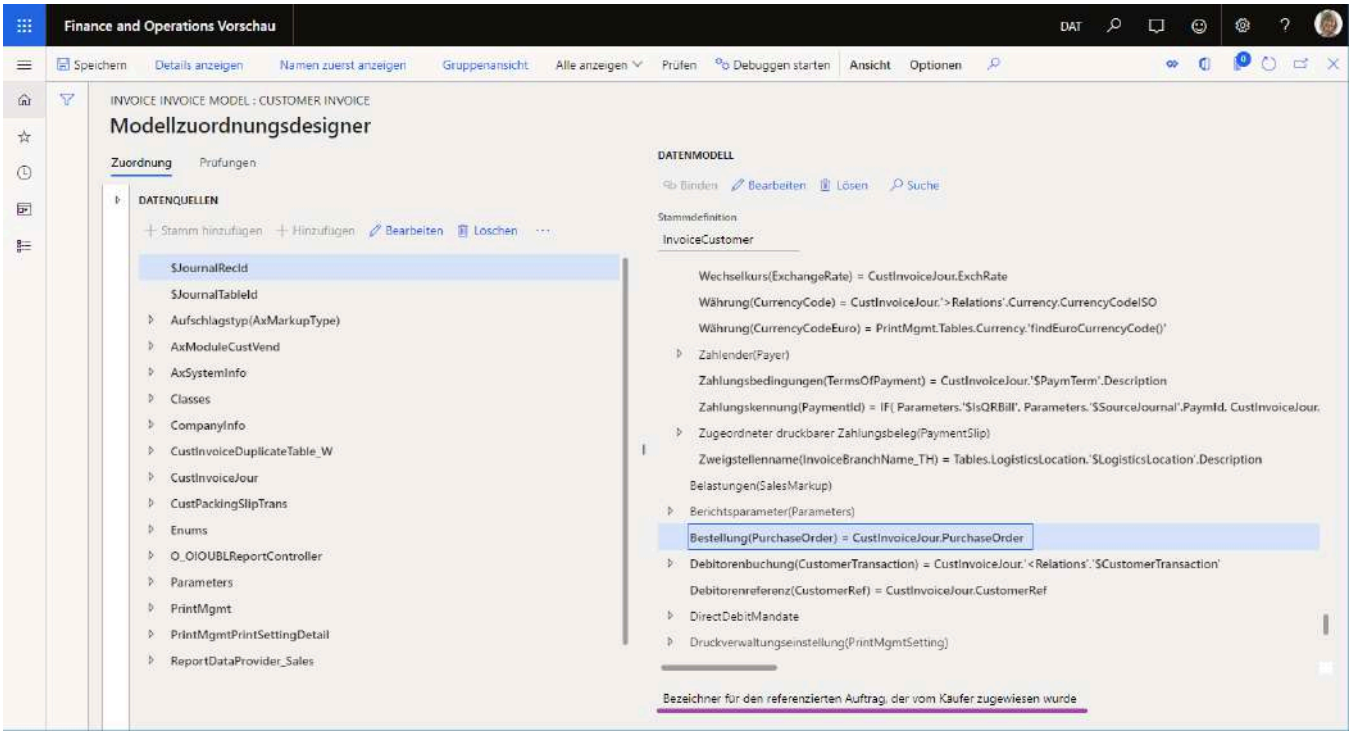
When an ER data model is configured in this way, its content will be presented to users of the ER data model designer in each user's preferred language. Therefore, model maintenance is simplified. The following illustrations show how this functionality works for users who have DE-AT and JA set as their preferred language.





Model mapping component

Because the ER model mapping is based on an ER data model, the labels of the data model elements that are referred to appear in the user's preferred language in the model mapping designer. The following illustration shows how the meaning of the **PurchaseOrder** field is explained in the editable model mapping by using the label of the **Description** attribute that has been added to the configured data model. Notice that this label is presented in the user's preferred language (DE-AT in this example).



When the **Label** attribute of the **User input parameter** data source is configured as linked to an ER label, the parameter field that corresponds to this data source is presented in the user dialog box at runtime to users in their preferred language.

Format component

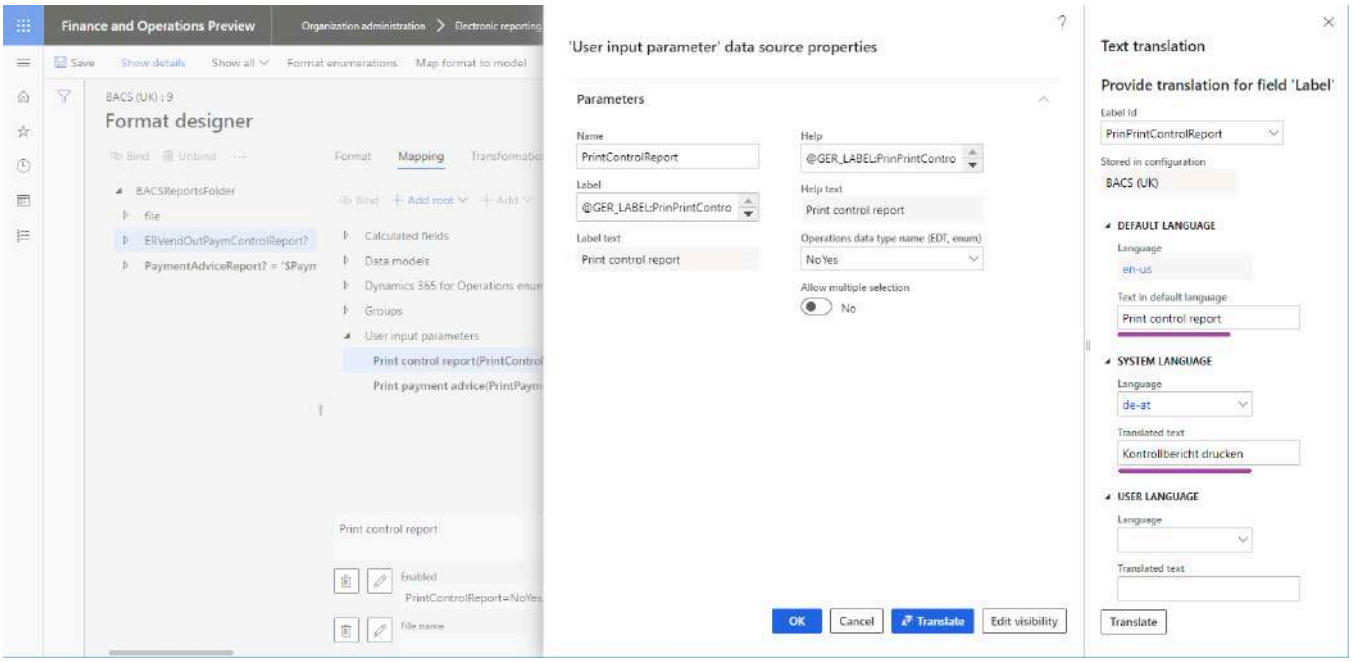
When you configure an ER format, you can add ER labels for it. The **Label** and **Help text** attributes of every configured data source can be linked to an ER label that is added to the ER format. The **Label** and **Description** attributes of every format enumeration value can also be linked to an ER label that is accessible from the editable ER format.

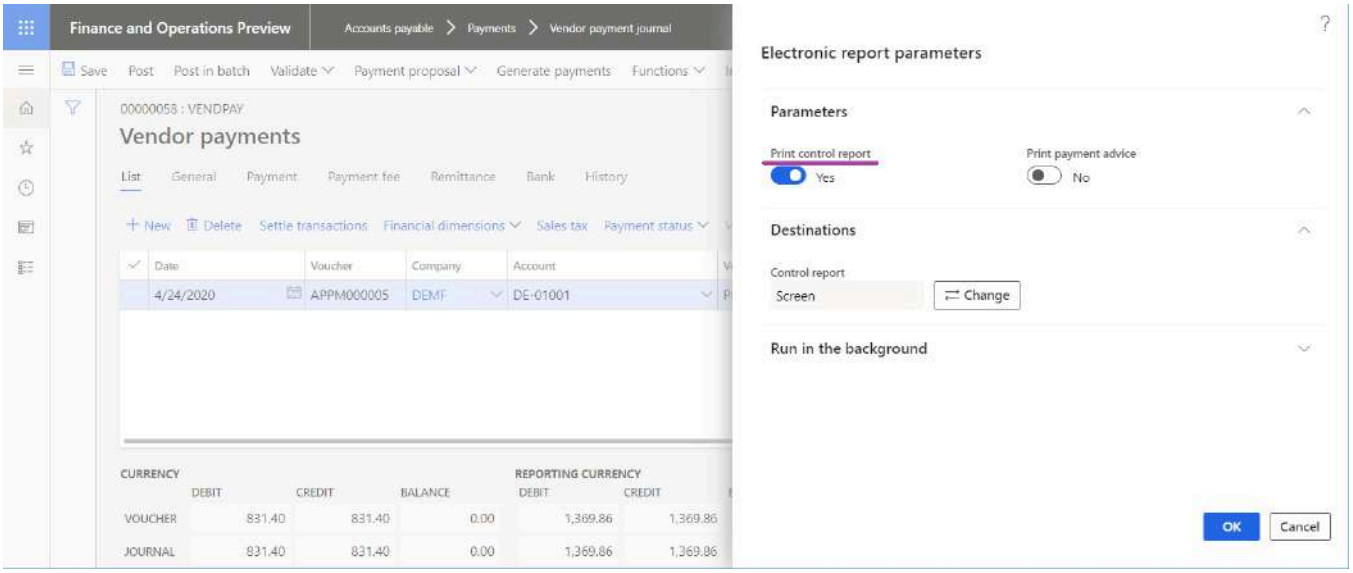
> **Note**
>
> You can also link these attributes to an ER label of the parent ER data model that reuses the model's labels in every ER format that is configured for this ER data model.

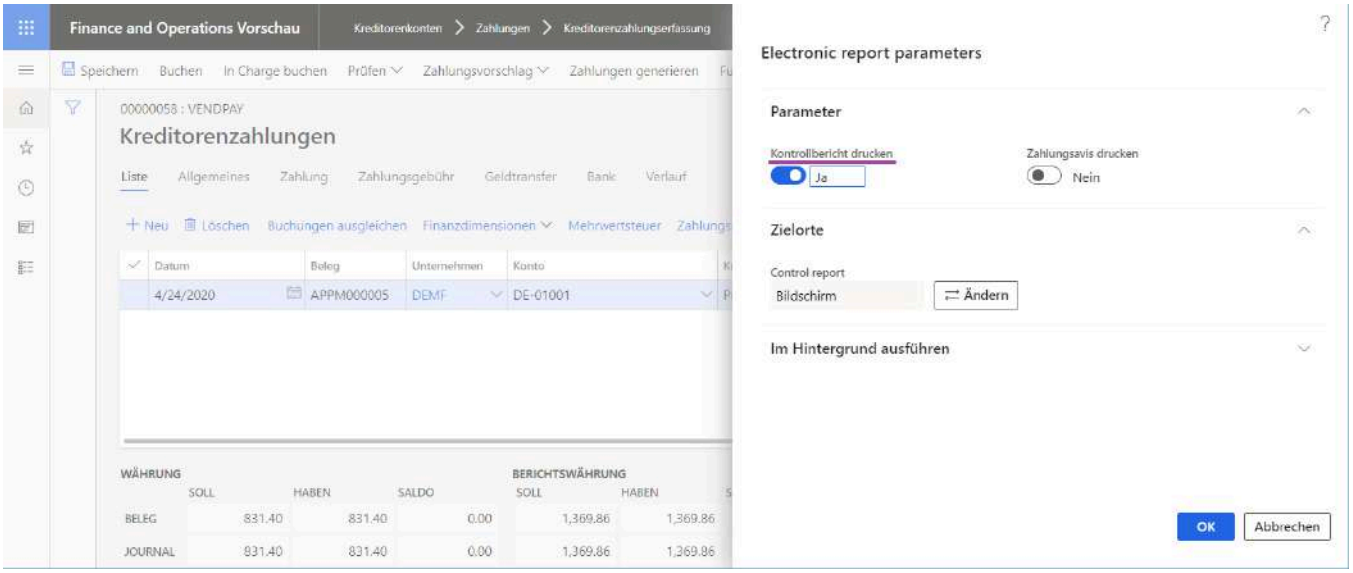
When an ER format is configured in this way, the content of the format will be presented to users of the ER Operation designer in each user's preferred language. Therefore, format maintenance and analysis of the configured logic are simplified.

Because an ER format is based on an ER data model, the labels that are referred to in the data model elements are presented in the ER format designer in the user-preferred language.

When the **Label** attribute of the **User input parameter** data source is linked to an ER label, the field that corresponds to the parameter in the user dialog box at runtime is presented to the user as a prompt. The following illustrations show how you can link the **Label** attribute of the **User input parameter** data source at design time to an ER label, so that users are prompted for the parameter in different user-preferred languages (shown for English United States (EN-US) and DE-AT languages) at runtime.

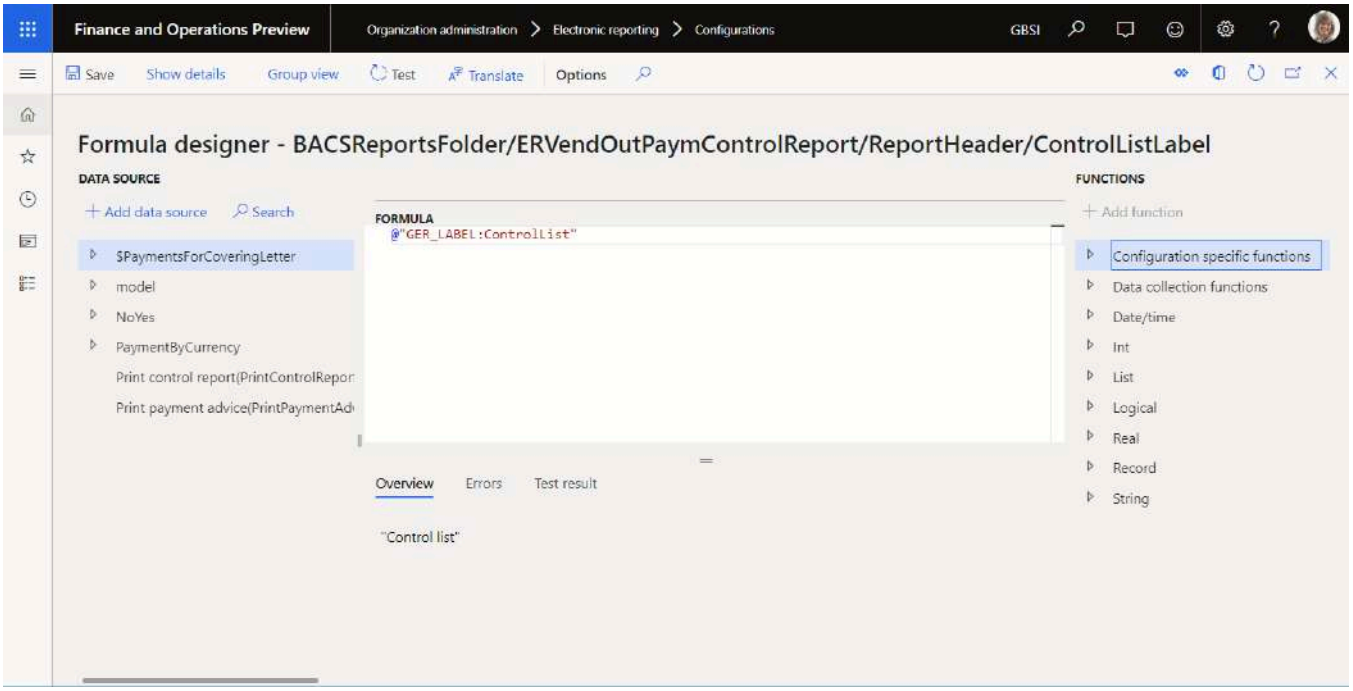




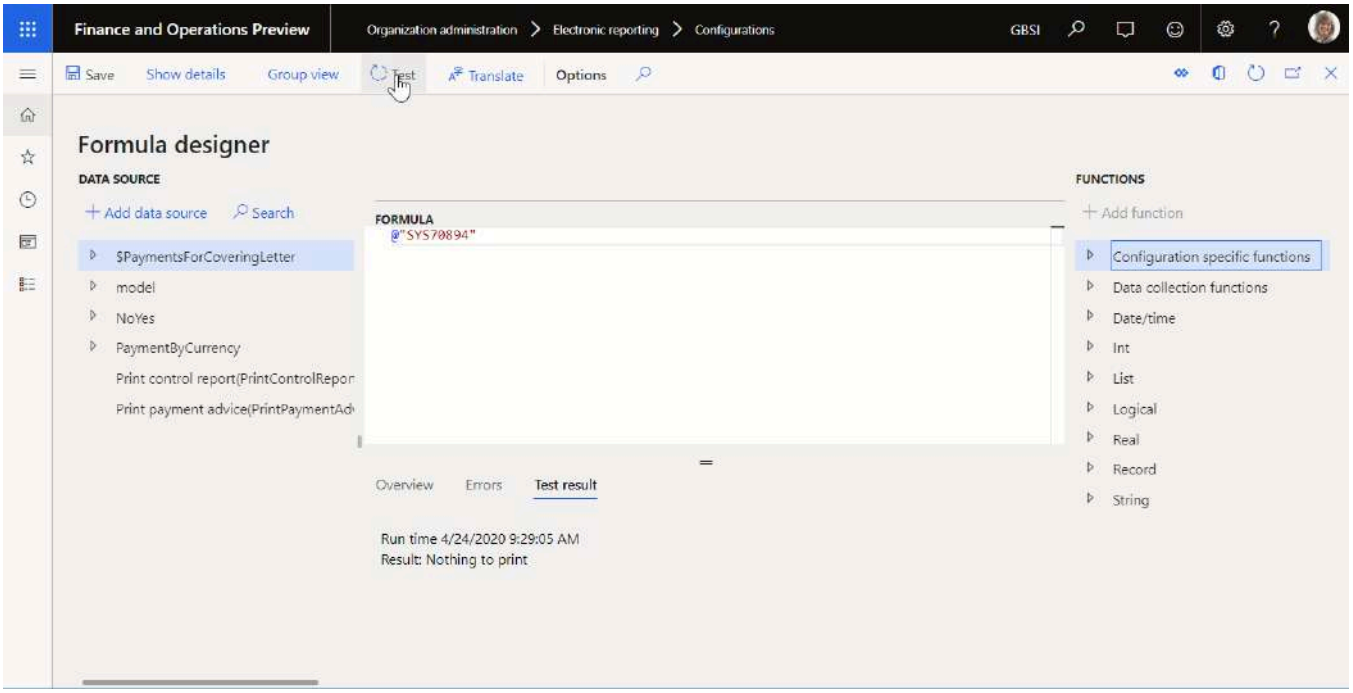


Expressions

To use a label in an ER expression, you must use the syntax @"**GER_LABEL:X**", where the prefix @ indicates that the operand refers to a label, **GER_LABEL** indicates that an ER label is involved, and **X** is the ER label ID.



To refer to a system (application) label, use the syntax @"**X**", where the prefix @ indicates that the operand refers to a label, and **X** is the system label ID.



Model mapping

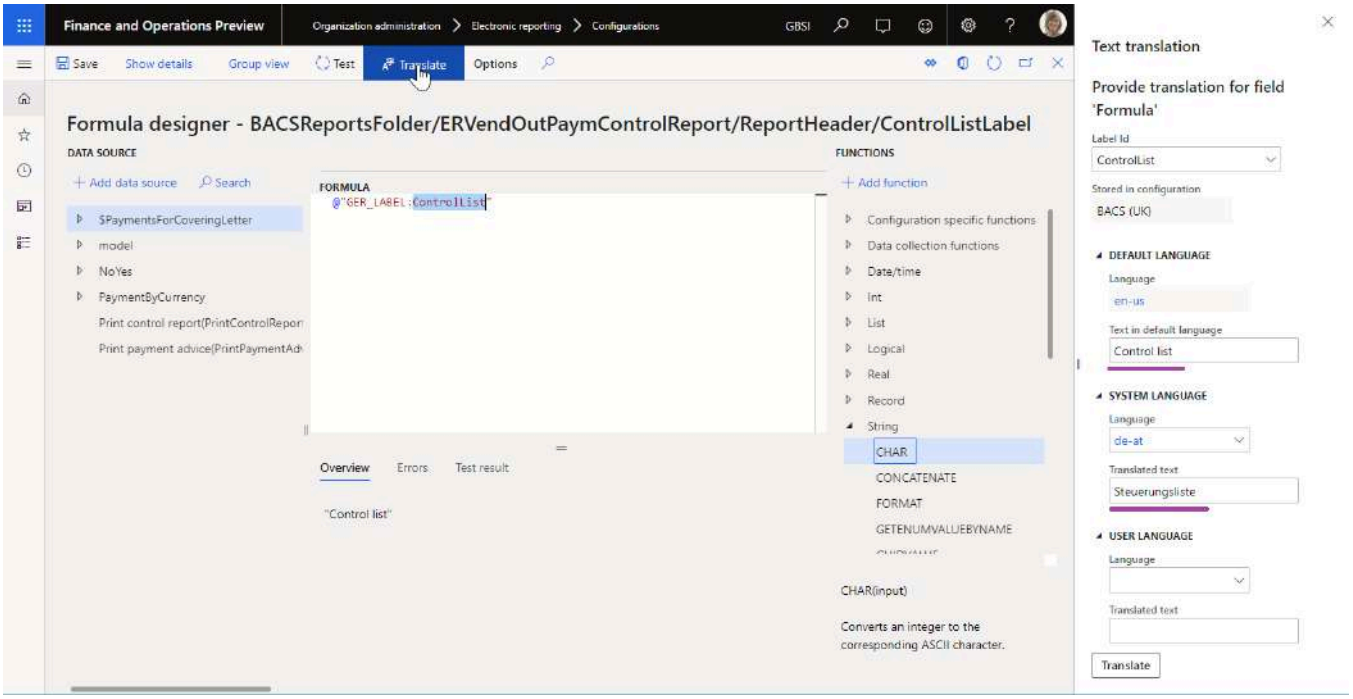
An expression of an ER model mapping can be configured by using a label. When this mapping is called by an ER format that is run to generate an outbound document, the context of the

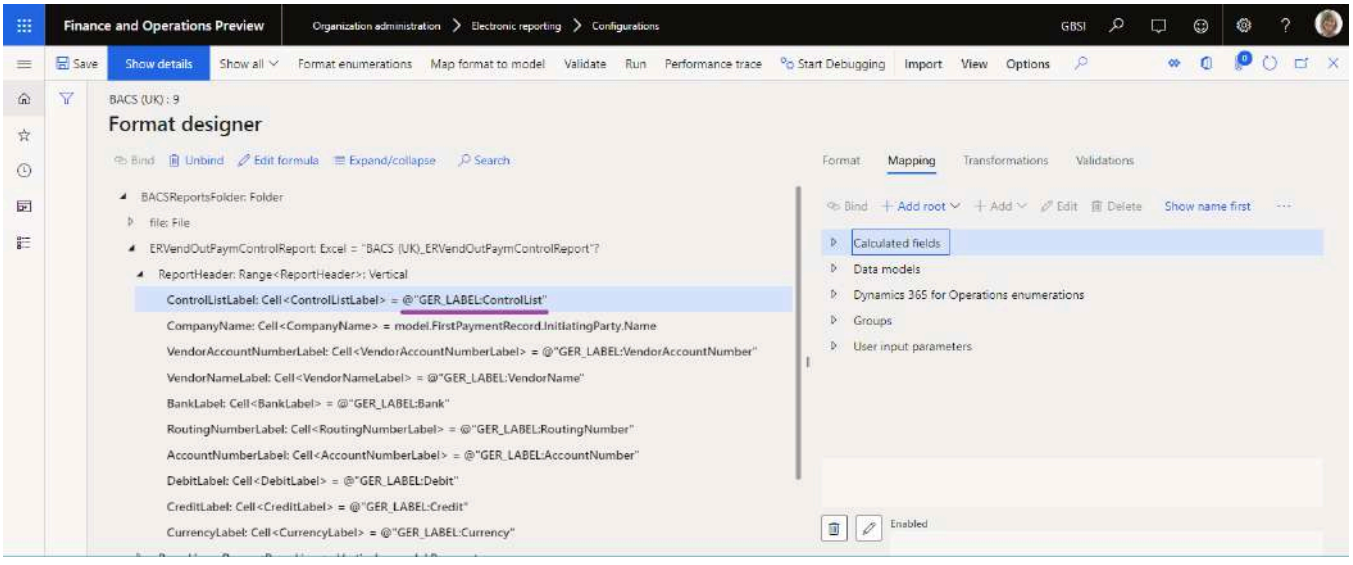
execution includes a language code. A configured expression label will be filled in with the label text that has been configured for the language of that context.

If a referenced label has no translation for the language of the format execution context that calls the model mapping, the label text in the EN-US language is used instead.

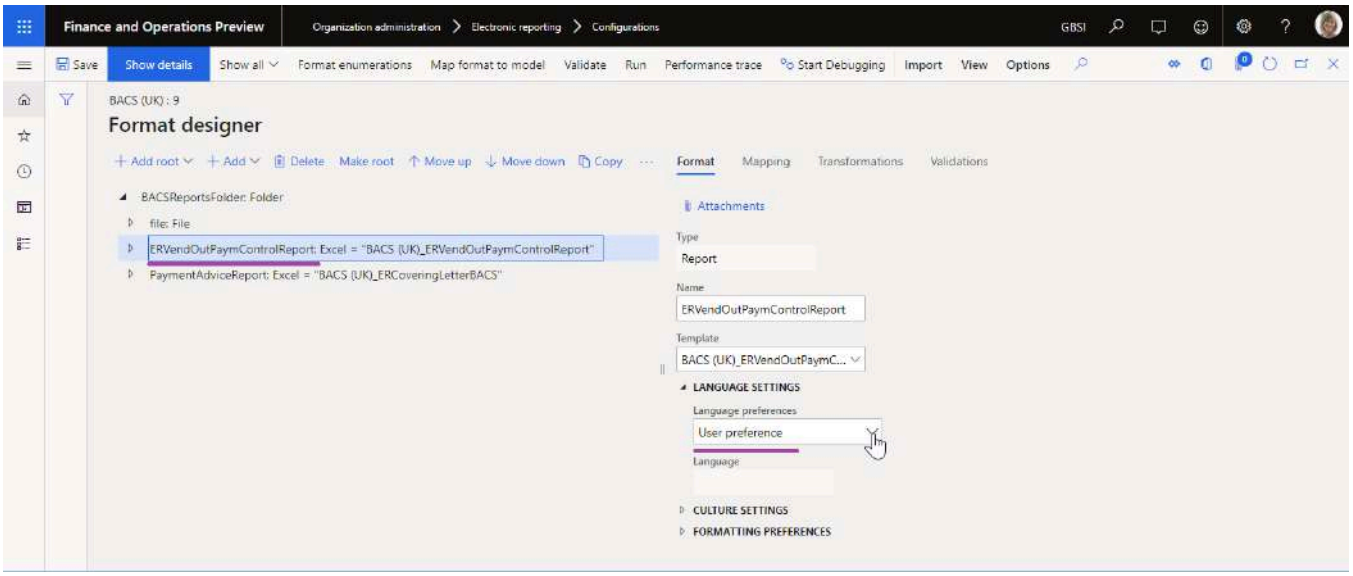
Format

An ER expression of an ER format can be configured by using labels. When this format is run to generate an outbound document, the context of the execution includes a language code. A configured expression label will be filled in with the label text that has been configured for the language of that context.

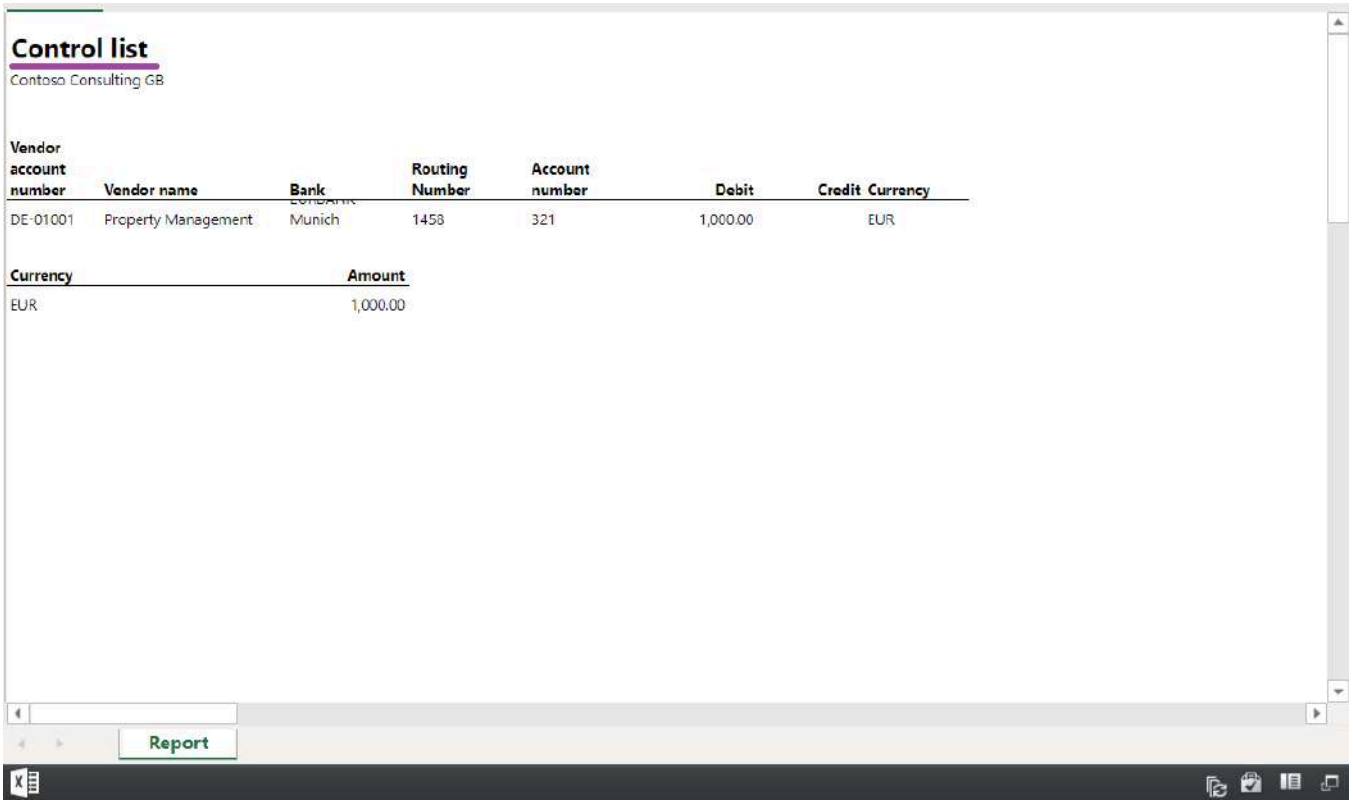


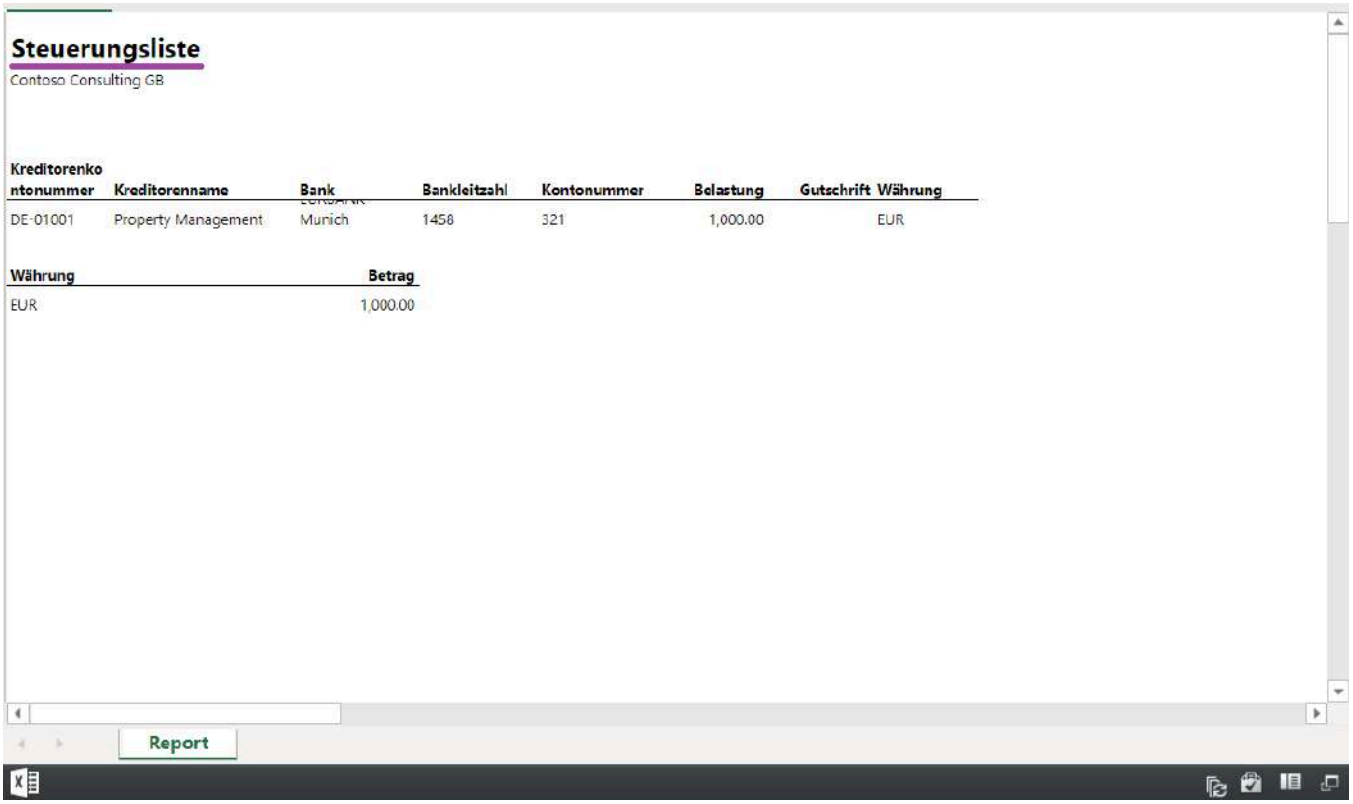


You can configure the **FILE** component of an ER format to generate the report in the user's preferred language.



If you configure an ER format in this way, the report is generated by using the corresponding text of the ER labels. The following illustrations show examples of reports for the EN-US and DE-AT user languages.





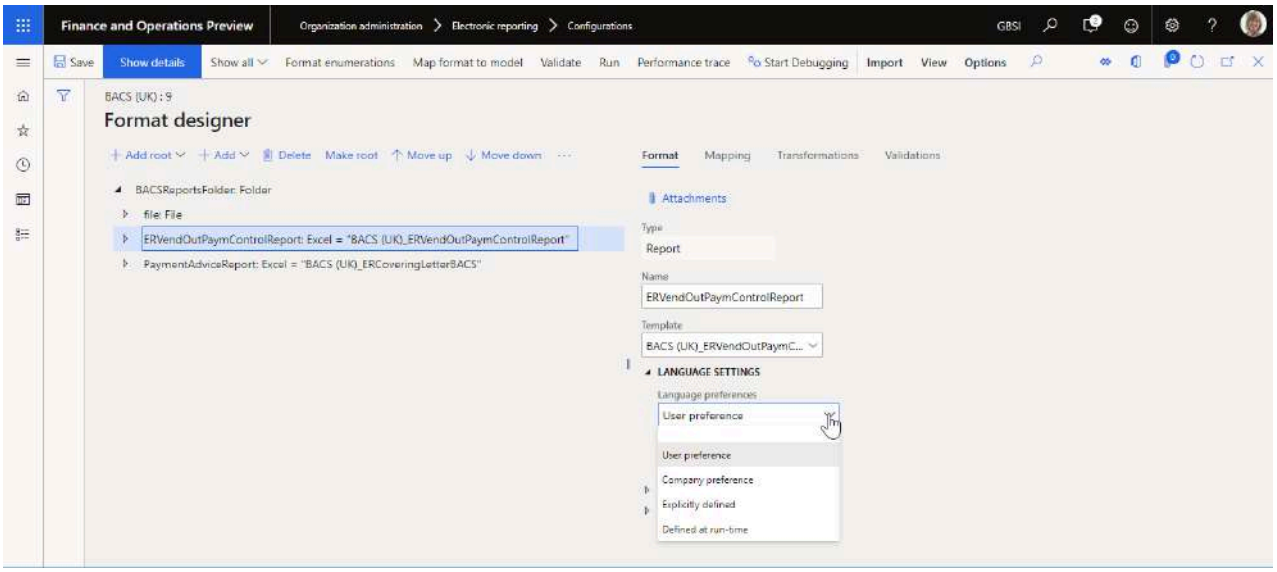
If a referenced label has no translation for the language of the format execution context, the label text in the EN-US language is used instead.

> **Tip**
>
> You can use the **FOLDER** and distinct types of **FILE** components in the editable ER format to specify how an outbound file is generated. To name a generated file, configure the ER expression for the **File name** parameter of the component. You can use labels in the configured expression. Because the **File name** parameter is language agnostic by default, the text of all labels that you refer to in this expression are exposed in the default EN-US language at runtime. However, in version 10.0.28 and later, you can enable the **Apply the 'Language preference' parameter to the 'File name' expression** feature. The **File name** expression then takes the **Language preferences** parameter into account when it's computed.

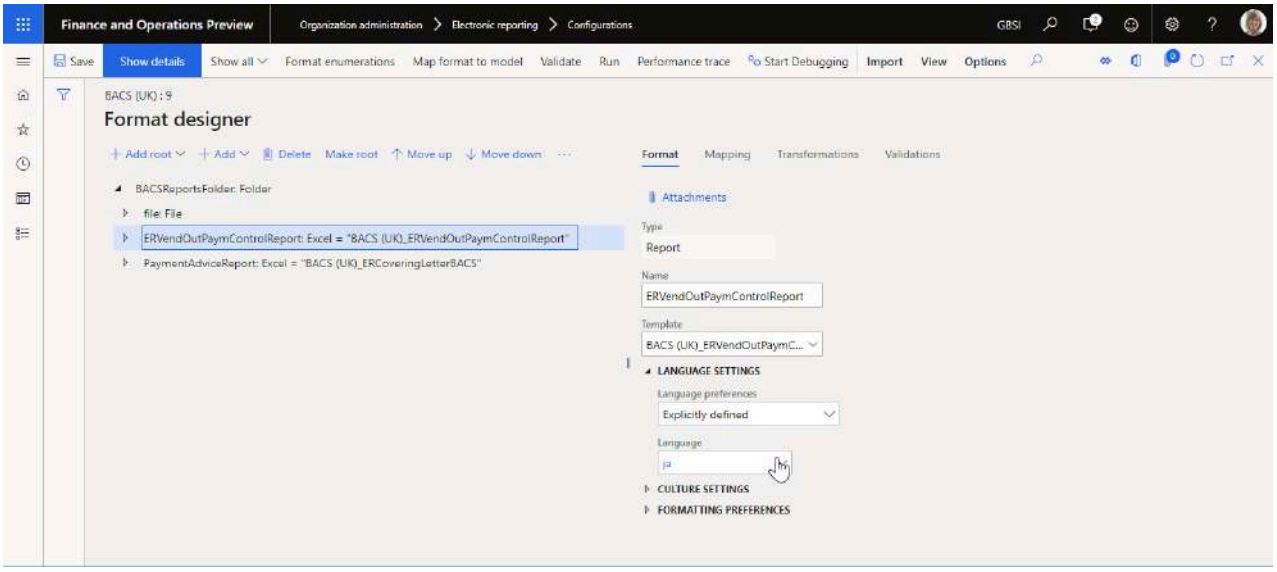
Language

ER supports different ways to specify a language for a generated report. In the **Language preferences** field on the **Format** tab, you can select the following values:

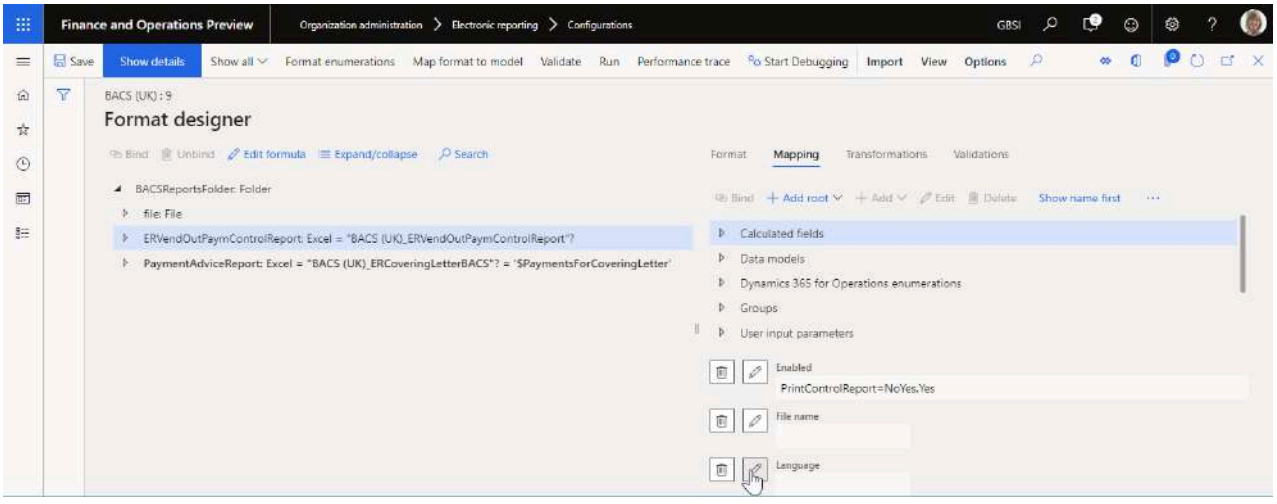
- **Company preference** – Generate a report in a company-specified language.



- **User preference** – Generate a report in the user's preferred language.
- **Explicitly defined** – Generate a report in a language that is specified at design time.



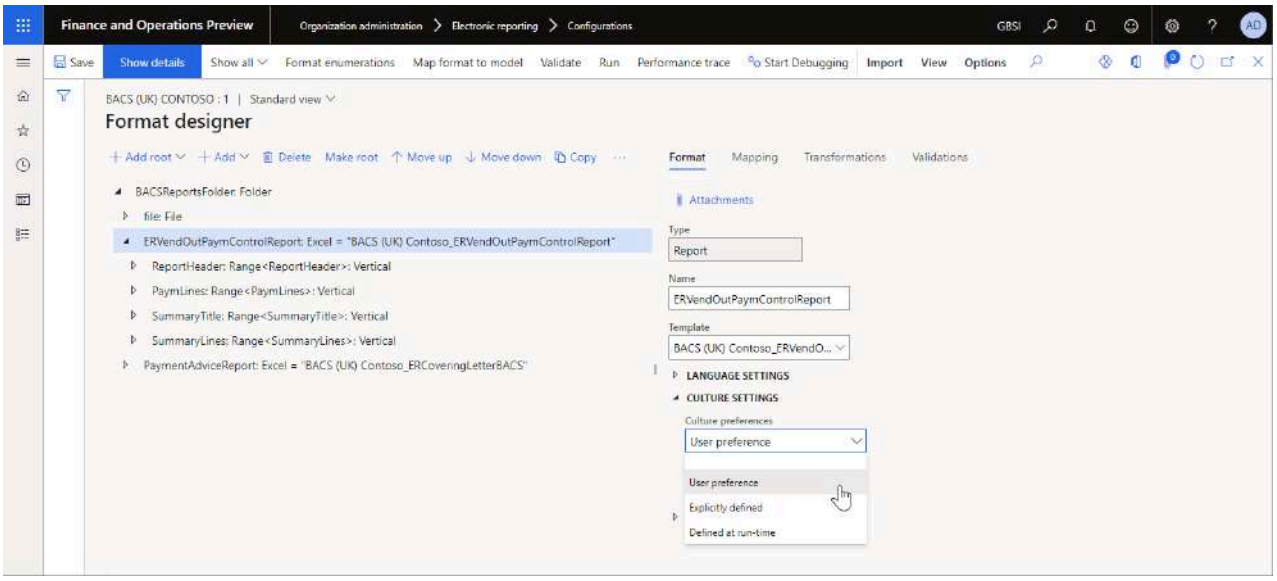
- **Defined at run-time** – Generate a report in a language that is specified at runtime. If you select this value, in the **Language** field, configure an ER expression that returns the language code for the language, such as the language of the corresponding customer.



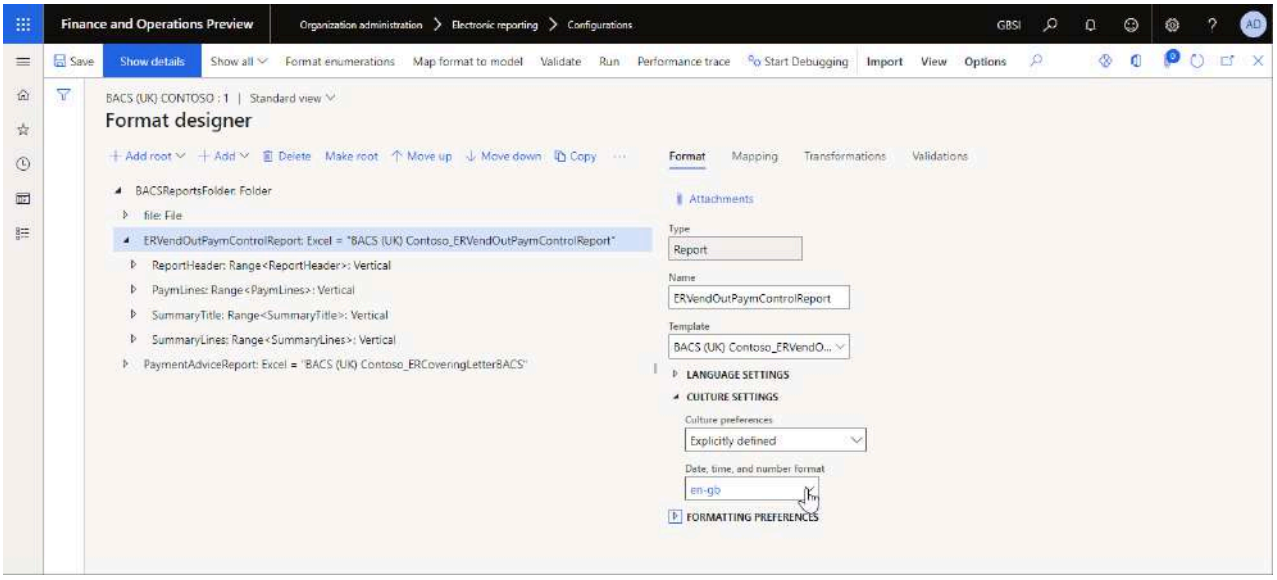
Culture-specific formatting

ER supports different ways to specify the culture for a generated report. Therefore, the correct culture-specific formatting can be used for date, time, and numeric values. When you design an ER format, on the **Format** tab, in the **Culture preferences** field, you can select one of the following values for every format component of the **Common\File**, **Excel\File**, **PDF\File**, or **PDF\Merger** type:

- **User preference** – Format the values according to the user's preferred culture. That culture is defined in the **Date, time, and number format** field on the **Preferences** tab of the **User options** page.

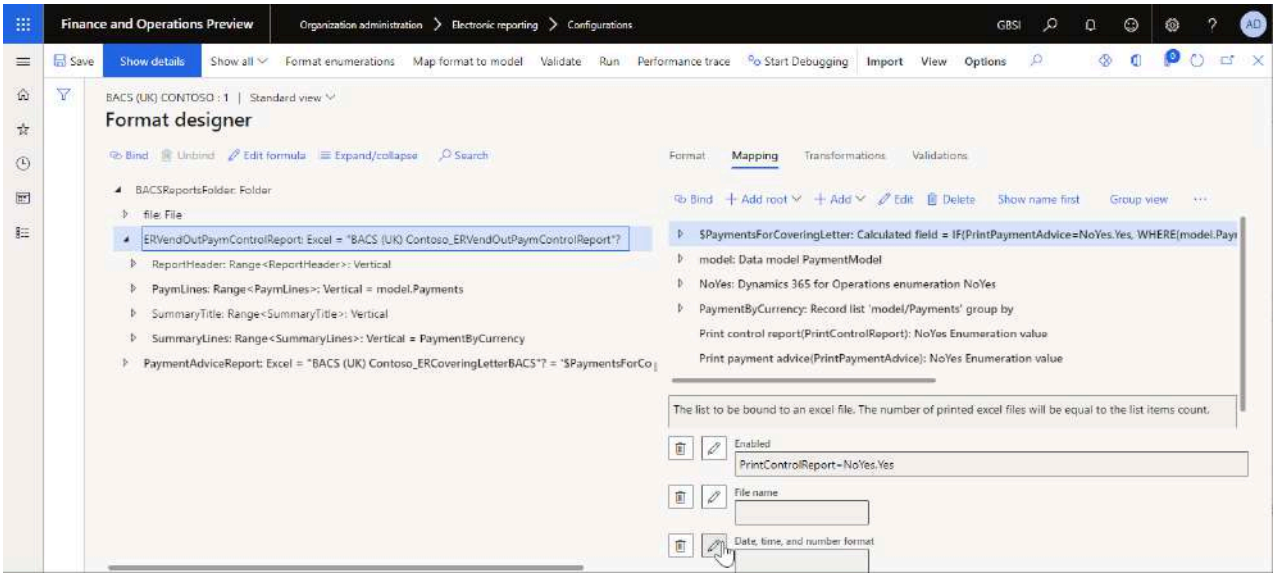


- **Explicitly defined** – Format the values according to the culture that is specified at design time.



- **Defined at run-time** – Format the values according to the culture that is specified at runtime. If you select this value, on the **Mapping** tab, in the **Date, time, and number**

format field, configure an ER expression that returns the culture code for the culture, such as the culture of the corresponding customer.



> **Note**
>
> An ER component that you define a specific culture for might contain child ER components that were configured to fill in a text value. By default, the culture of the parent component is used to format the values of those components. You can use the following built-in ER functions to configure bindings for those components and apply an alternative culture for value formatting:
>
> - DATEFORMAT
> - DATETIMEFORMAT
> - NUMBERFORMAT
>
> In version 10.0.20 and later, the locale of format components of the **Common\File** and **Excel\File** types is used to format values during PDF conversion of a generated document.

Translation

You can add required ER labels to an editable ER component. When an ER label is added, it can be translated in two ways: manually and automatically.

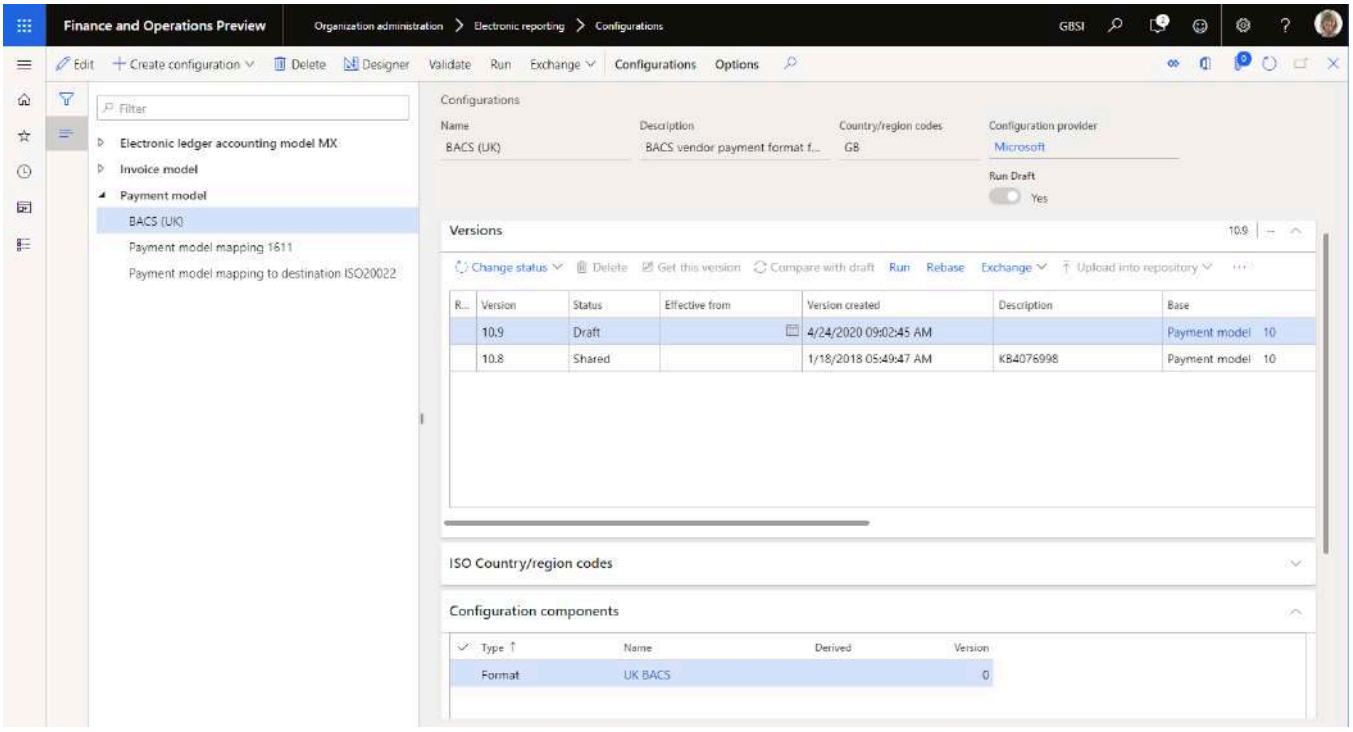
Manual translation

When you add an ER label on the **Text translation** pane, you can manually translate it into all languages that are supported in the current Finance instance. You can select the preferred language in the **Language** field in the **System language** or **User language** section, enter the

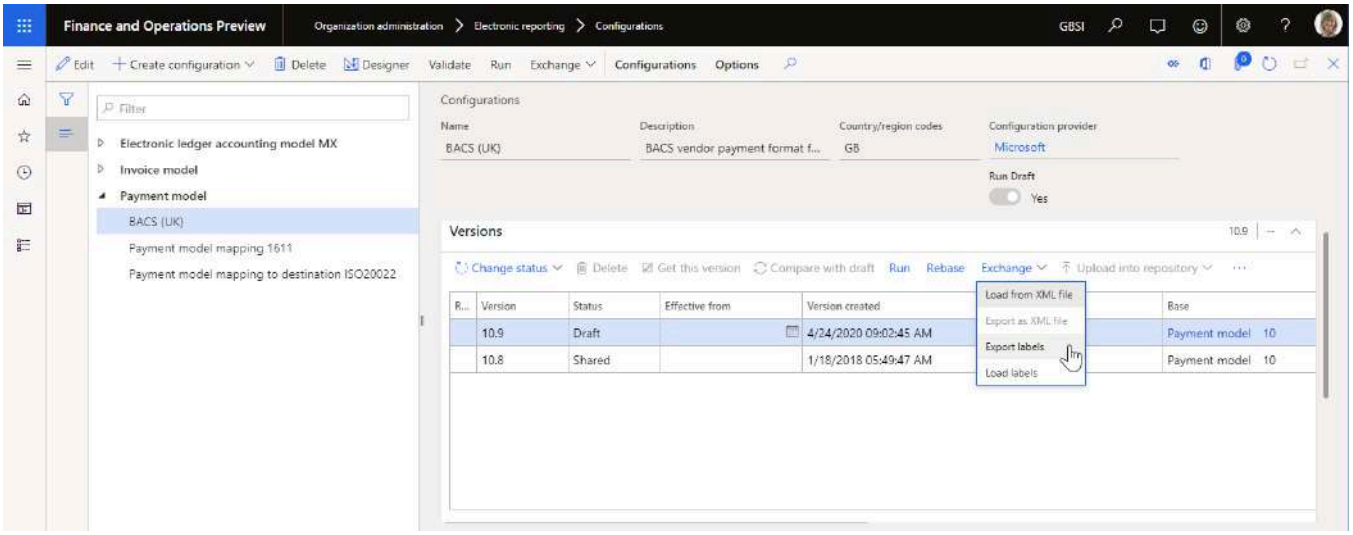
appropriate text in the corresponding **Translated text** field, and then select **Translate**. This process must be repeated for every required language and every label that you add.

Automatic translation

Configuration of an ER component is done in the draft version of the ER configuration that the editable ER component resides in.



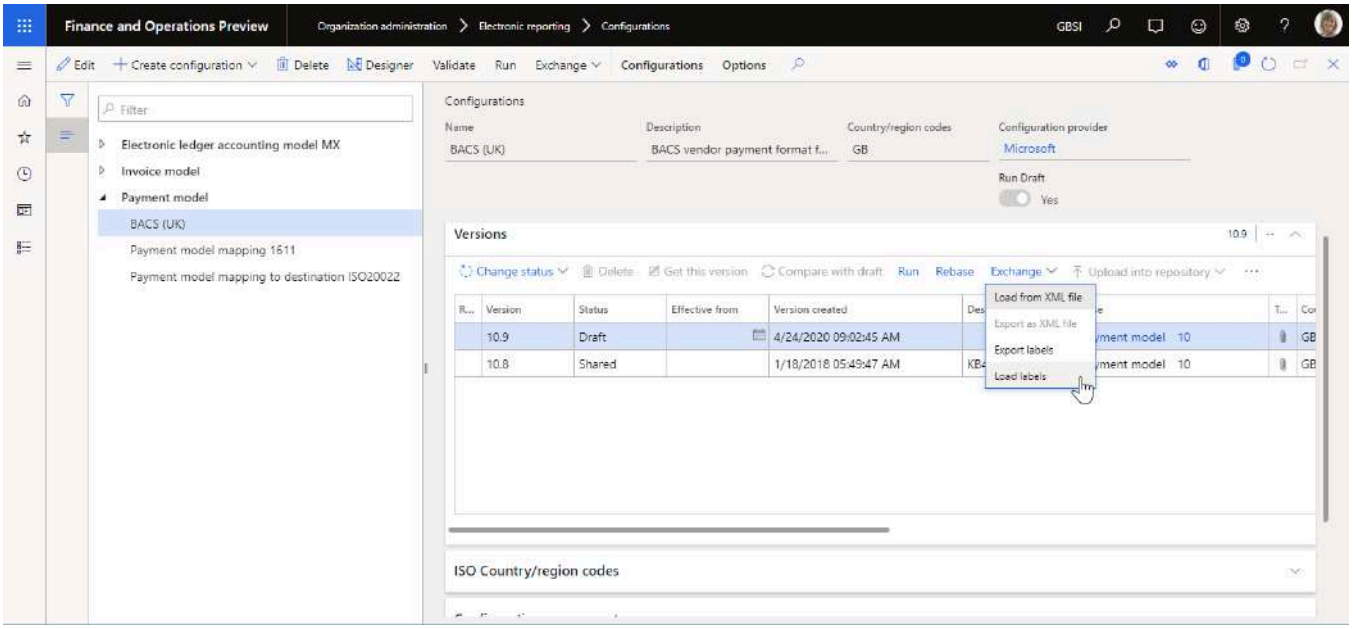
As described earlier in this article, you can add required ER labels to an editable ER component. In this way, you can specify the text of the ER labels in the EN-US language. You can then export the labels of the ER component by using the built-in ER function. Select the draft version of an ER configuration that contains the editable ER component, and then select **Exchange** > **Export labels**.



You can export either all labels or the labels for a single language that you specify at the beginning of export. Labels are exported as a zip file that contains XML files. Every XML file contains labels for a single language.

```
<?xml version="1.0" encoding="UTF-8"?>
- <StringResources>
  - <String _locID="guid{423CC719-54A0-4B2B-AD78-E5AC9F86DCEE}">
        <!--_locComment_text="Label file for configuration BACS (UK)"-->
      de-AT
    </String>
  - <String _locID="AccountNumber">
        <!--_locComment_text=""-->
      Kontonummer
    </String>
  - <String _locID="Bank">
        <!--_locComment_text=""-->
      Bank
    </String>
  - <String _locID="ControlList">
        <!--_locComment_text=""-->
      Steuerungsliste
    </String>
  - <String _locID="DiscountUsed">
        <!--_locComment_text=""-->
      Gewährter Rabatt
    </String>
  - <String _locID="FirstPaymentRecord">
        <!--_locComment_text=""-->
      Erster Zahlungsdatensatz
    </String>
  - <String _locID="Giro">
        <!--_locComment_text=""-->
      Girokonto
    </String>
  - <String _locID="InvoiceAmount">
        <!--_locComment_text=""-->
      Rechnungsbetrag
    </String>
```

This format is used for automatic translation of labels by external translation services such as Dynamics 365 Translation Service. When you receive the translated labels, you can import them back into the draft version of an ER configuration that contains the ER components that own those labels. Select the draft version of an ER configuration that contains the editable ER component, and select **Exchange** > **Load labels**.



Translated labels will be imported into the selected ER configuration. Translated labels that exist in this ER configuration are replaced. If any translated label is missing in the ER configuration, it's appended.

Lifecycle

Labels of an ER component that can be edited are kept, together with other content for the component, in the appropriate version of an ER configuration.

Labels of a base ER component can be referred to in a derived version of the ER component that you create to introduce your modifications.

> **Tip**
>
> When you design a ER solution, you can derive your own ER **data model** component from the one that is provided. In this derived data model, you can introduce your own ER labels and use them in all ER formats that will use the data model as the data source. You can then derive your own ER **format** component from the one that is provided by selecting your derived ER data model instead of the provided one. In version 10.0.28 and later, you can enable the **Enhanced access to labels of the ascendent ER data model** feature to access labels of an ascendent ER data model in derived ER format components, even when the ER data model that you selected for the derived ER component differs from the one that was used in the base ER component.
>
> When the same label name is used in your derived component and its ascendent components, your translation of that label is used as the most relevant one.

ER versioning controls label assignment to any attribute in an ER component. Changes to the label assignment are recorded in the list of changes (delta) of an editable ER component that has been created as a derived version of the provided ER component. These changes will be validated when a derived version is rebased to a new base version.

Functions

The built-in LISTOFFIELDS ER function can access ER labels that have been configured for some items of ER components.

As described earlier in this article, the **Label** and **Description** attributes of every model or format ER enumeration's value can be linked to an ER label that is accessible in the appropriate ER component. You can configure an ER expression where you call the **LISTOFFIELDS** function by using the ER enumeration as an argument. This expression returns a list that contains a record for every value of an ER enumeration that has been defined as an argument of this function. Every record contains the value of an ER label that is linked to an ER enumeration value:

- The value of an ER label that is linked to the **Label** attributes is stored in the **Label** field of the returned record.
- The value of an ER label that is linked to the **Description** attributes is stored in the **Description** field of the returned record.

Performance

When you configure an ER format component to generate a report in your preferred language, or to import an inbound document where the content is parsed by your preferred language, we recommend that you enable the **Cache the preferred language of the current user for ER runs** feature in the Feature management workspace. This feature helps improve performance, especially for ER format components that contain multiple references to labels in ER formulas and bindings and many validation rules to generate user messages in your preferred language.

When you change the status of an ER configuration version from **Draft** to **Completed**, if the configuration version contains ER labels, those labels are stored in the application database. The storage schema depends on the state of the **Accelerate the ER labels storage** feature:

- If the feature isn't enabled, all labels are stored in the **LABELXML** field of the **ERSOLUTIONVERSIONTABLE** table as a single XML snippet.
- If the feature is enabled, a separate record is created for each language in the **ERSOLUTIONVERSIONLABELSTABLE** table. The **CONTENTS** field of this table stores labels per language as a compressed XML snippet.

We recommend that you enable the **Accelerate the ER labels storage** feature in the **Feature management** workspace. This feature helps improve network bandwidth utilization and overall system performance because, in most cases, ER labels of a single language are used when you work with a single ER configuration.

To apply the selected storage schema for keeping labels of all ER configurations in the current Finance instance, complete the following steps.

1. Go to **Organization administration** > **Periodic** > **Apply the selected labels storing schema for all ER configurations**.
2. Select **OK**.

Additional resources

- Electronic Reporting overview
- Electronic Reporting functions

Compress large documents that are generated in Electronic reporting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can use the Electronic reporting (ER) framework) to configure a solution that fetches transactional data to generate an outbound document. This generated document might be quite large. When this type of document is generated, the Application Object Server (AOS) memory is used to hold it. At some point, you must download the document from your Microsoft Dynamics 365 Finance application. Currently, the maximum size of a single document that generated in ER is limited to 2 gigabytes (GB). Additionally, Finance currently limits the size of a downloaded file to 1 GB. Therefore, you must configure an ER solution that reduces the likelihood that these limitations might be exceeded and that you receive a **Stream was too long** or **Overflow or underflow in the arithmetic operation** exception.

When you configure a solution, you can adjust your ER format in the Operations designer by adding a root element of the **Folder** type to compress the content that any of its nested elements generates. Compression works "just in time," so it reduces peak memory usage and the size of the file that you download.

> **Note**
>
> File compression takes an additional percentage of CPU usage.

To learn more about this approach, complete the example in this article.

Example: Compress an outbound document

This example shows how a user assigned to the **System administrator** or **Electronic reporting functional consultant** role can configure an ER format to compress a generated document.

Prerequisites

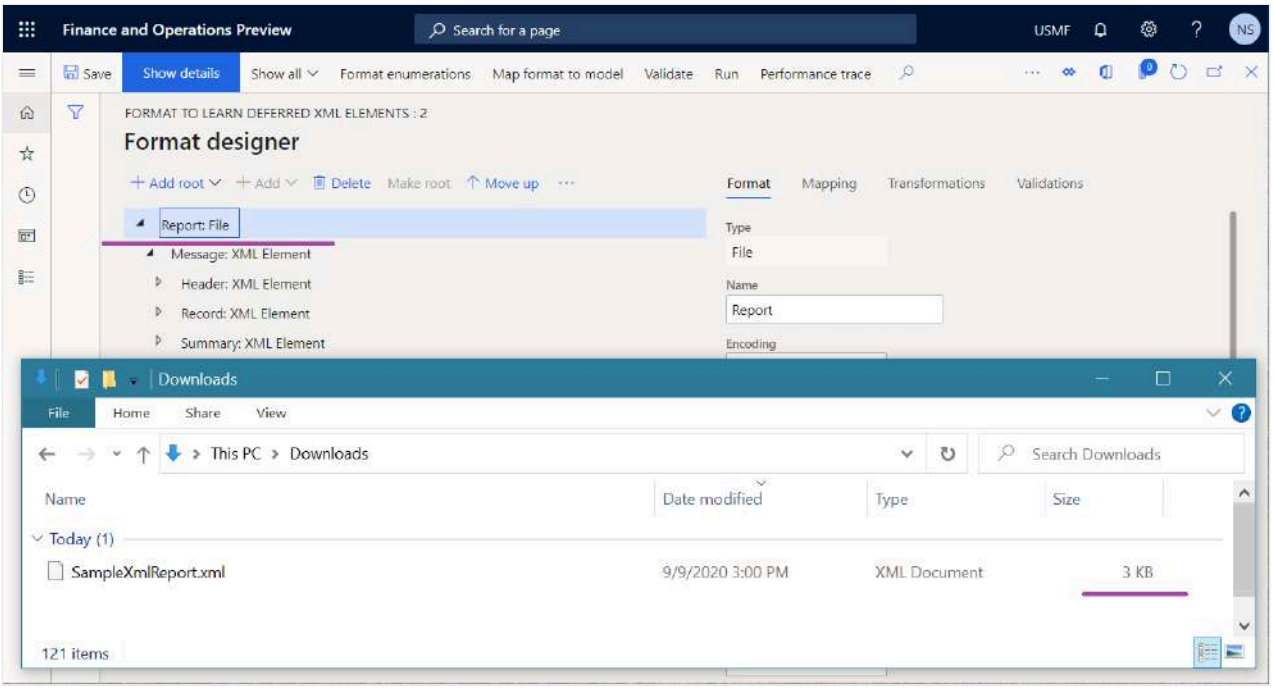
Before you complete the procedures in this article, complete the following steps.

1. Activate a configuration provider.
2. Import the sample ER configurations.
3. Review the imported format.

> **Note**
>
> Currently, the format structure starts from the **Report** element of the **File** type and contains XML elements. Therefore, an outbound document is generated in XML format, and it doesn't use compression.

Generate an ER format to get an uncompressed document

1. Run the imported format.
2. Notice that the size of the generated document in XML format is 3 kilobytes (KB).



Modify the format to compress the generated output

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, in the configuration tree, expand **Model to learn deferred elements**.
3. Select the **Format to learn deferred XML elements** configuration.

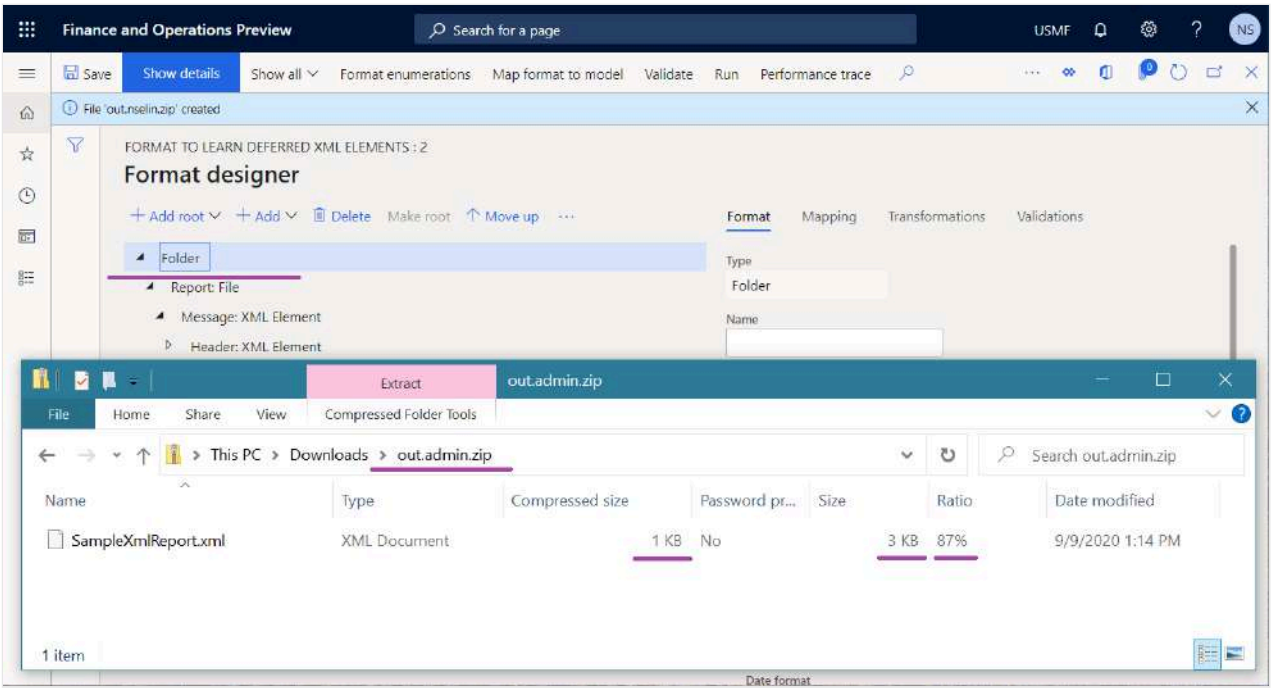
4. Select **Designer** to modify the format structure.
5. On the **Format designer** page, on the **Format** tab, select **Add root** to add a root format element.
6. In the **Add** dialog box, select **Common\Folder**.
7. Select **OK** to confirm the addition of the new root element.
8. Select **Save**.

> ⓘ **Note**
>
> The format structure starts from the element of the **Folder** type. This element generates output as a compressed (zip) file. When a document that the **Report** element generates is placed in an outbound zip file, the content is compressed to reduce the size of the outbound file.

Generate an ER format to get a compressed document

1. On the **Format designer** page, select **Run**.
2. Download the ZIP file that the web browser offers, and open it for review.
3. Notice that the size of the generated document in ZIP format is 1 KB.

   > ⓘ **Note**
   >
   > The compression ratio of the XML file that this ZIP file holds is 87 percent. The compression ratio depends on the data that is being compressed.



> **Note**
>
> If the ER destination is configured for the format element that generates output (the **Report** element in this example), compression of the output is bypassed.

Additional resources

Electronic reporting (ER) overview

Electronic reporting (ER) destinations

Defer the execution of XML elements in ER formats

Last updated on 04/15/2026

Validate ER format and model mapping to prevent runtime issues

You can validate each configured Electronic reporting (ER) format and model mapping at design time. During this validation, a consistency check runs to help prevent runtime problems, such as execution errors and performance degradation. For every problem that the check finds, it provides the path of a problematic element. For some problems, an automatic fix is available.

By default, the validation automatically runs in the following cases for an ER configuration that contains the previously mentioned ER components:

- You import a new version of an ER configuration into your instance of Microsoft Dynamics 365 Finance.
- You change the status of the editable ER configuration from **Draft** to **Completed**.
- You rebase an editable ER configuration by applying a new base version.

Starting in version 10.0.49, enhanced validation management lets you validate multiple completed configuration versions in a single operation, run that validation in the background as a batch job, and review the results later through a persistent validation history.

> **Note**
>
> Enhanced validation management is a feature that you turn on in **Feature management**.
>
> 1. Go to **Workspaces** > **Feature management**.
> 2. In the list, find and select **Enhanced validation management for Electronic Reporting configurations**.
> 3. Select **Enable now**.

Validate a single configuration version

To validate one version of a configuration, follow these steps:

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the configurations tree in the left pane, select the desired ER configuration that contains the ER format or ER model mapping component.
3. On the **Versions** FastTab, select the version of the selected ER configuration.

4. On the **Versions** FastTab or on Action Pane (if **Enhanced validation management for Electronic Reporting configurations** feature is disabled), select **Validate**.

The system validates the selected version and reports any errors or warnings.

> ⓘ **Note**
>
> When the **Enhanced validation management for Electronic Reporting configurations** feature is disabled, the **Validate** button is available on Action Pane and validates the selected version of the selected configuration.
>
> When the **Enhanced validation management for Electronic Reporting configurations** feature is enabled, the **Validate** button is available on the **Versions** FastTab and validates the selected version of the selected configuration.

Validate multiple configurations

> ⓘ **Note**
>
> This function is available starting in version 10.0.49 when the **Enhanced validation management for Electronic Reporting configurations** feature is enabled.

When you turn on the feature, the Action Pane on the **Configuration repository** page includes a **Validation** menu that you can use to validate several configurations at once.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the Action Pane, select **Validation** > **Validate configurations**.
3. In the **Configuration validation** dialog box, select the completed configuration versions that you want to validate.
4. Choose how to run the validation:
    - To run validation immediately in interactive mode, start the validation and confirm when prompted. Interactive validation can take some time, depending on how many configurations you select.
    - To run validation in the background, schedule it as a batch job by using the **Batch** button. When the job is queued, a notification confirms that the ER configuration validation job is added to the batch queue.

> **Tip**
>
> Run validation as a batch job when you select many configurations or large configurations, so that the work doesn't block your interactive session.

Only completed configuration versions are available for validation.

Validate ER format in the Designer

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the configurations tree in the left pane, select the desired ER configuration that contains the ER format component.
3. On the **Versions** FastTab, select the version of the selected ER configuration.
4. On the Action Pane, select **Designer**.
5. On the **Format designer** page, on the Action Pane, select **Validate**.

Validate ER model mapping in the Designer

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the configurations tree in the left pane, select the ER configuration that contains the ER model mapping component.
3. On the **Versions** FastTab, select the version of the selected ER configuration.
4. On the Action Pane, select **Designer**.
5. On the **Model to datasource mapping** page, on the Action Pane, select **Designer**.
6. On the **Model mapping designer** page, on the Action Pane, select **Validate**.

Skip the validation when importing the configuration

To skip the validation when importing the configuration, follow these steps:

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. Set the **Validate the configuration after importing** option to **No**.

Skip the validation when you change or rebase the version's status

To skip the validation when you change or rebase the version's status, follow these steps:

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. Set the **Skip validation at configuration's status change and rebase** option to **Yes**.

ER uses the following categories to group consistency check inspections:

- **Executability** – Inspections that detect critical issues that might occur at runtime. These issues are mostly at an **Error** level.
- **Performance** – Inspections that detect issues that might cause inefficient execution of configured ER components. These issues are mostly at a **Warning** level.
- **Data integrity** – Inspections that detect issues that might cause data loss or runtime issues. These issues are mostly at a **Warning** level.

Additional resources

ALLITEMS ER function

ALLITEMSQUERY ER function

INT64VALUE ER function

INTVALUE ER function

FILTER ER function

WHERE ER function

Use JOIN data sources to get data from multiple application tables in ER model mappings

Trace the execution of ER formats to troubleshoot performance issues

Business document management overview

- Suppress Word content controls in generated reports
- Manage several derived mappings for a single model root

Last updated on 07/22/2026

List of validations

The following table provides an overview of the validations that ER provides. For more information about these validations, use the links in the first column to go to the relevant sections of this article. Those sections explain the types of components that ER provides inspections for and how you can reconfigure ER components to help prevent issues.

Expand table

Name	Category	Level	Message
Type conversion	Executability	Error	Can't convert expression of type <type> to field of type <type>. **Runtime error:** Exception for type
Type compatibility	Executability	Error	The configured expression can't be used as the binding of the current format element to a data source as this expression returns value of the data type <type> that is beyond the scope of data types that are supported by the current format element of type <type>. **Runtime error:** Exception of type
Missing configuration element	Executability	Error	Path not found <path>. **Runtime error:** Element of the configuration <path> not found
Executability of an expression with FILTER function	Executability	Error	The list expression of FILTER function isn't queryable. **Runtime error:** Filtering isn't supported. Validate the configuration to get more details about this.
Executability of a GROUPBY data source	Executability	Error	Path <path> doesn't support querying.
Executability of a GROUPBY data source	Executability	Error	Group by function can't be executed with query. **Runtime error:** Group by function can't be executed with query.
Executability of a JOIN data source	Executability	Error	Can't join a list <path> that isn't a filter in query. **Runtime error:** Function Joined datasource should be a

Name	Category	Level	Message
			filter expression calculated field has been incorrectly called.
Preferability of FILTER vs WHERE function	Performance	Warning	Using FILTER function for the expression is preferable than WHERE from the performance perspective. Select Fix to replace it automatically.
Preferability of ALLITEMSQUERY vs ALLITEMS function	Performance	Warning	Using ALLITEMSQUERY function for the expression is preferable than ALLITEMS from the performance perspective. Select Fix to replace it automatically.
Consideration of empty list cases	Executability	Warning	List <path> doesn't have any check for empty list case, it can result an error at run time. Add a check for empty list case. **Runtime error:** List is empty at <path> **Potential issue:** Line is getting populated once while a data source it's populated from contains multiple records
Executability of an expression with FILTER function (caching)	Executability	Error	FILTER function can't be applied to the selected type of data source. A data source of the Table records type is applicable only when it isn't cached and has no manually added nested data sources. **Runtime error:** Filtering isn't supported. Validate the configuration to get more details about this.
Missing binding	Executability	Warning	Path <path> has no binding to any datasource in using model's mapping. **Runtime error:** Path <path> isn't bound
Not linked template	Data integrity	Warning	File <name> isn't linked to any file components and will be removed after changing status of configuration version.
Not synced format	Data integrity	Warning	Defined name <component name> doesn't exist in Excel sheet <sheet name>
Not synced format	Data integrity	Warning	<Tagged Word content control> tag doesn't exist in Word template file **Runtime error:** <Tagged Word content control> tag doesn't exist in Word template file.
No default mapping	Data integrity	Error	More than one model mapping exists for the <model name (root descriptor)> data model in the

Name	Category	Level	Message
			configurations <configuration names separated by comma>. Set one of the configurations as default **Runtime error:** More than one model mapping exists for the <model name (root descriptor)> data model in the configurations <configuration names separated by comma>. Set one of the configurations as default.
Inconsistent setting of Header or Footer components	Data integrity	Error	Headers/footers (<component type: Header or Footer>) are inconsistent **Runtime:** The last configured component is used at runtime if the draft version of the configured ER format is executed.
Inconsistent setting of Page component	Data integrity	Error	There are more than two range components without replication. Remove unnecessary components.
Executability of an expression with ORDERBY function	Executability	Error	The list expression of ORDERBY function isn't queryable. **Runtime error:** Sorting isn't supported. Validate the configuration to get more details about this.
Obsolete application artifact	Data integrity	Warning	The element <path> is marked as obsolete. or The element <path> is marked as obsolete with message <message text>. **Runtime error sample:** Class '<path>' not found.

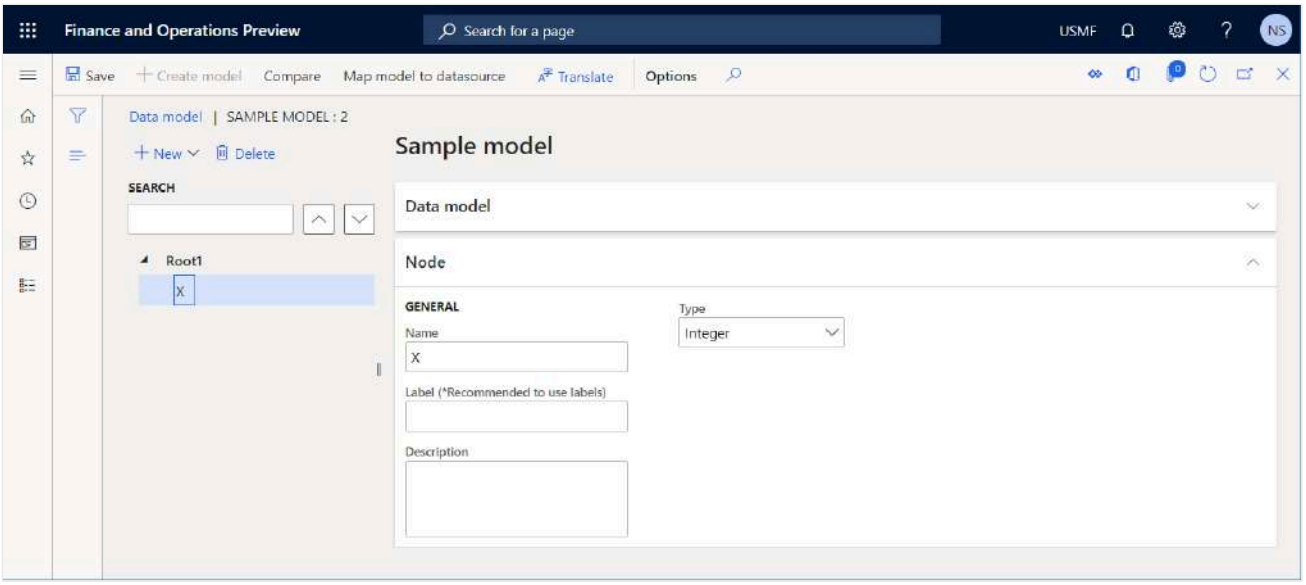
Type conversion

ER checks whether the data type of a data model field is compatible with the data type of an expression that you configure as the binding for that field. If the data types are incompatible, the ER model mapping designer shows a validation error. The error message states that ER can't convert an expression of type A to a field of type B.

The following steps show how this issue might occur.

1. Start to configure the ER data model and the ER model mapping components simultaneously.

2. In the data model tree, add a field that you name **X**, and select **Integer** as the data type.



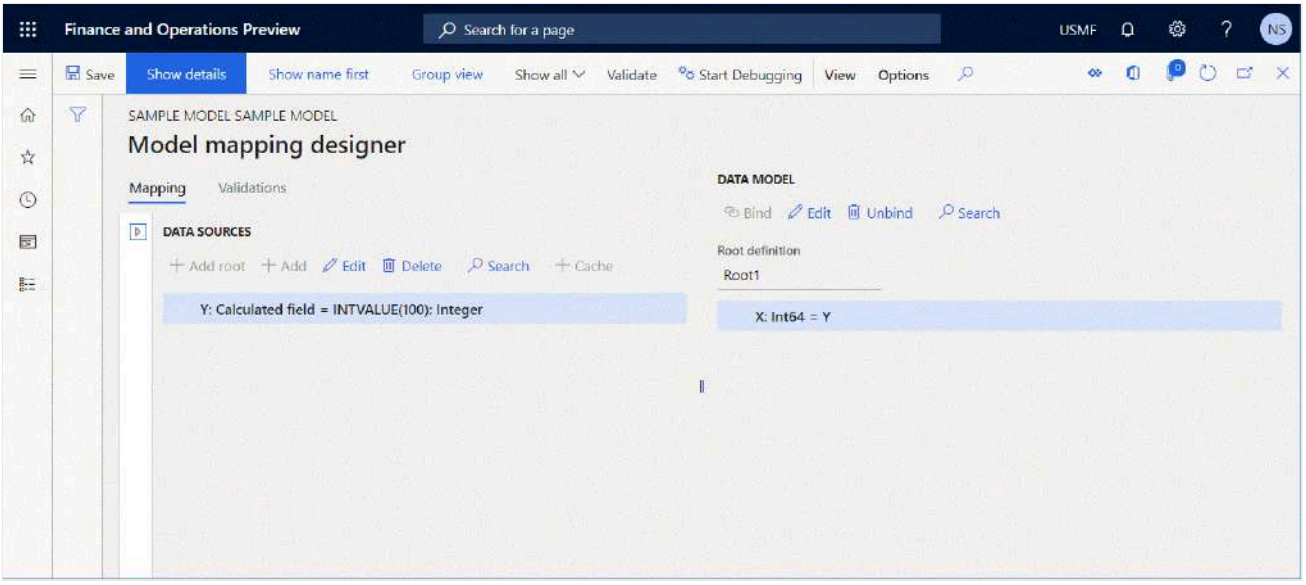
3. In the model mapping designer, in the **Data sources** pane, add a data source of the **Calculated field** type.

4. Name the new data source **Y**, and configure it so that it contains the expression `INTVALUE(100)`.

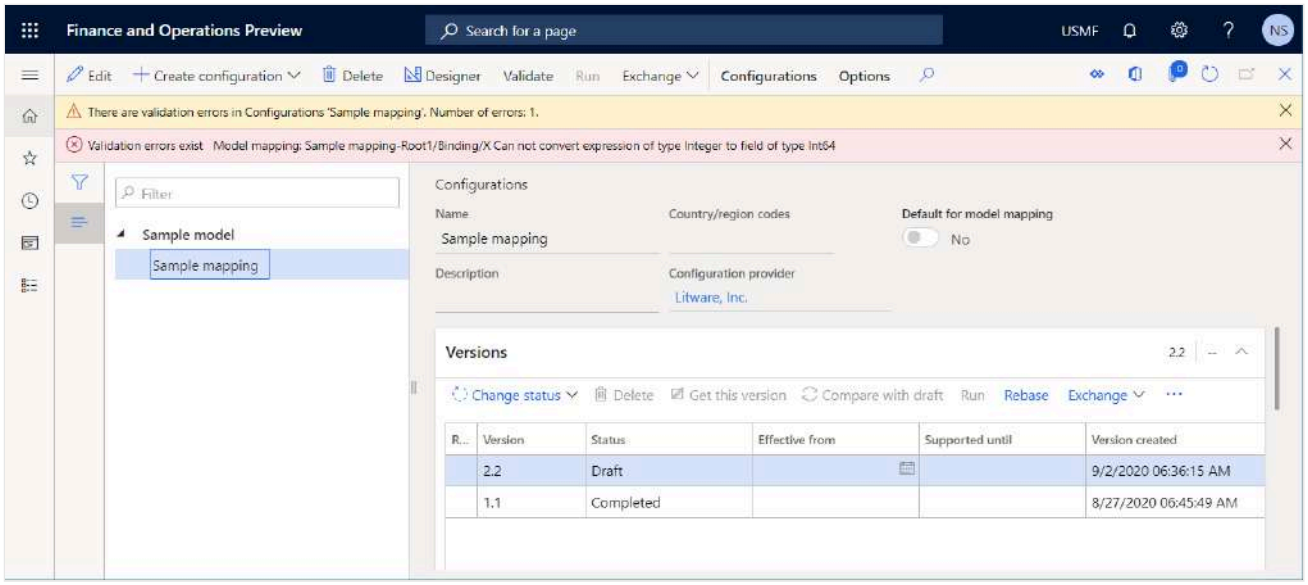
5. Bind **X** to **Y**.

6. In the data model designer, change the data type of the **X** field from **Integer** to **Int64**.

7. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page.

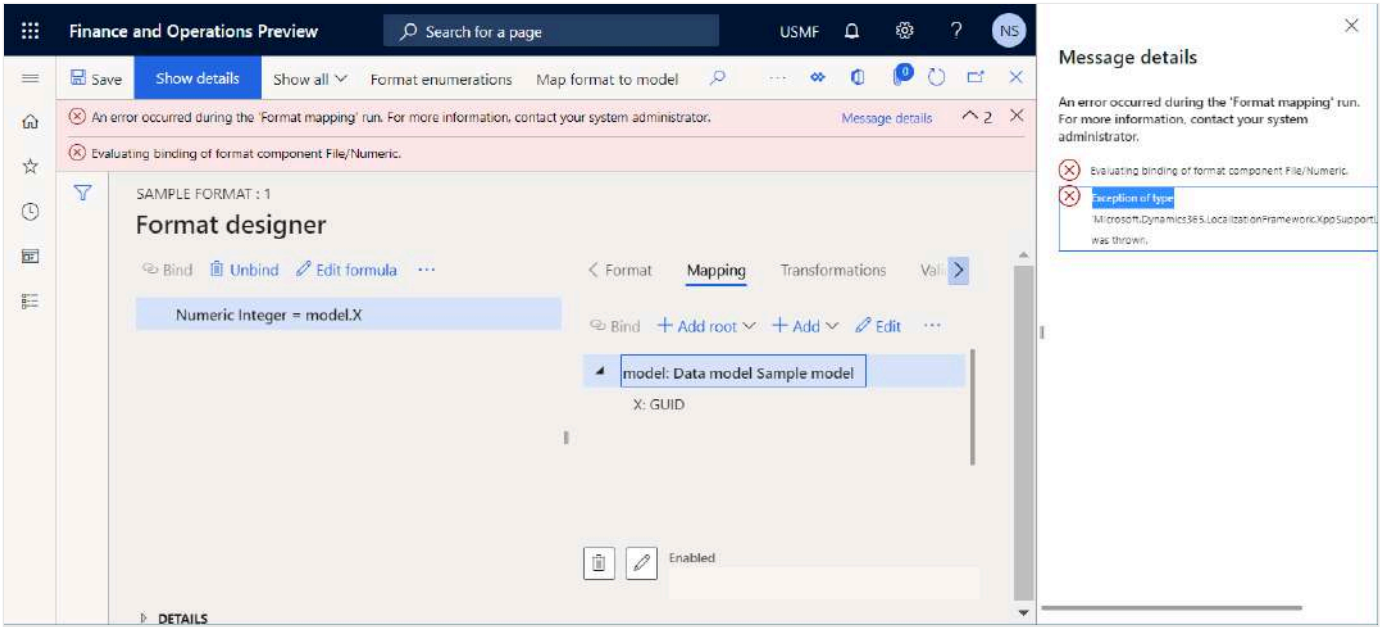


8. Select **Validate** to inspect the model mapping component of the selected ER configuration on the **Configurations** page.



9. Notice that a validation error occurs. The message states that the value of the **Integer** type that the `INTVALUE(100)` expression of the **Y** data source returns can't be stored in the **X** data model field of the **Int64** type.

The following illustration shows the runtime error that occurs if you ignore the warning and select **Run** to run a format that is configured to use the model mapping.



Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Update the data model structure by changing the data type of the data model field so that it matches the data type of the expression that you configure for the binding of that field. For the preceding example, change the data type of the **X** field back to **Integer**.

Option 2

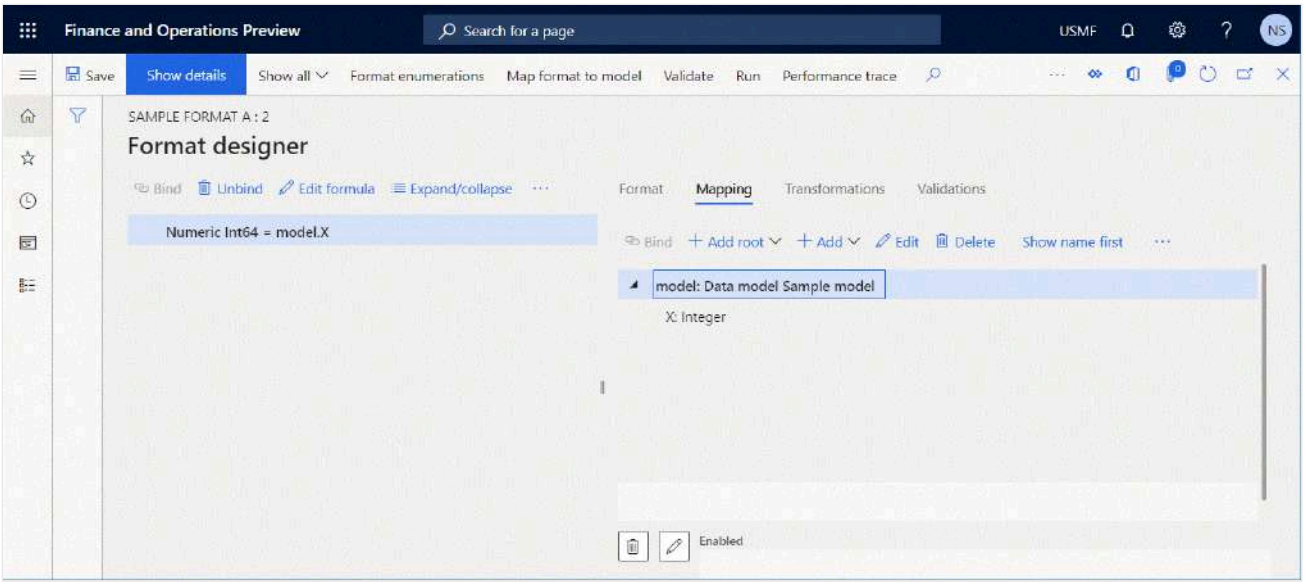
Update the model mapping by changing the expression of the data source that you bind to the data model field. For the preceding example, change the expression of the **Y** data source to `INT64VALUE(100)`.

Type compatibility

ER checks whether the data type of a format element is compatible with the data type of an expression that you configure as the binding of that format element. If the data types are incompatible, a validation error occurs in the ER Operations designer. The message that you receive states that the configured expression can't be used as the binding of the current format element to a data source, because the expression returns a value of data type A that is beyond the scope of the data types that the current format element of type B supports.

The following steps show how this issue might occur.

1. Start to configure the ER data model and the ER format components simultaneously.
2. In the data model tree, add a field that you name **X**, and select **Integer** as the data type.
3. In the format structure tree, add a format element of the **Numeric** type.
4. Name the new format element **Y**. In the **Numeric type** field, select **Integer** as the data type.
5. Bind **X** to **Y**.
6. In the format structure tree, change the data type of the **Y** format element from **Integer** to **Int64**.
7. Select **Validate** to inspect the editable format component on the **Format designer** page.



8. Notice that a validation error occurs. The message states that the configured expression can accept only **Int64** values. Therefore, the value of the **X** data model field of the **Integer** type can't be entered in the **Y** format element.

Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Update the format structure by changing the data type of the **Numeric** format element so that it matches the data type of the expression that you configured for the binding of that element. In the preceding example, change the **Numeric type** value of the **X** format element back to **Integer**.

Option 2

Update the format mapping of the **X** format element by changing the expression from `model.X` to `INT64VALUE(model.X)`.

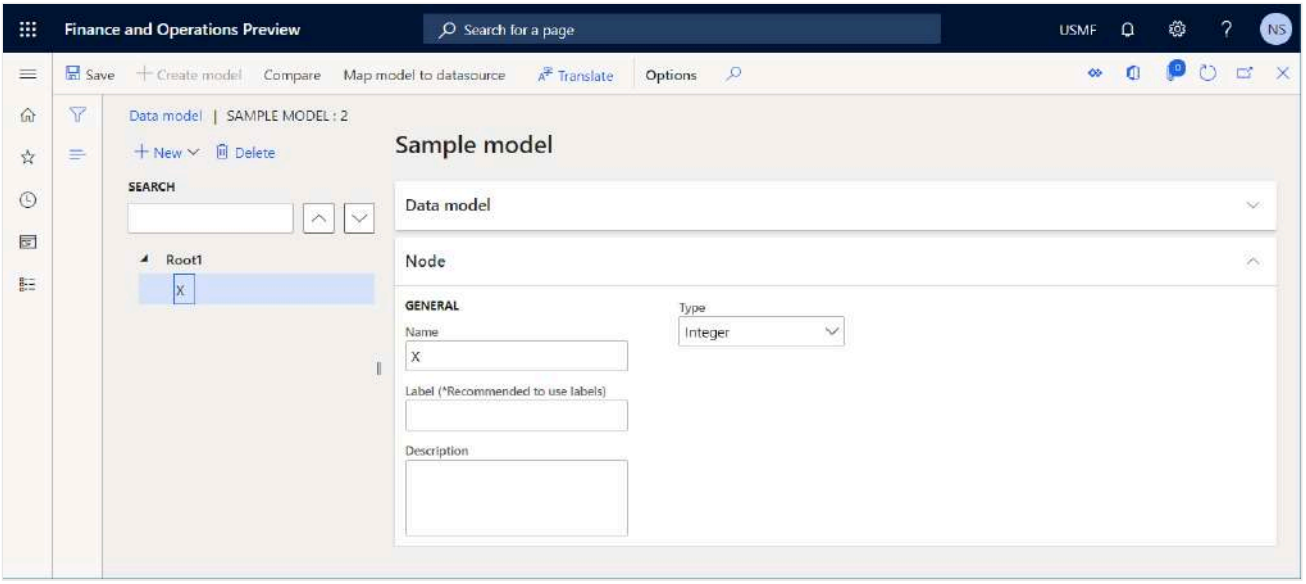
Missing configuration element

ER checks whether the binding expressions contain only data sources that the editable ER component configures. For every binding that contains a data source that the editable ER component is missing, a validation error occurs in the ER Operations designer or the ER model mapping designer.

The following steps show how this issue might occur.

1. Start to configure the ER data model and the ER model mapping components simultaneously.

2. In the data model tree, add a field that you name **X**, and select **Integer** as the data type.



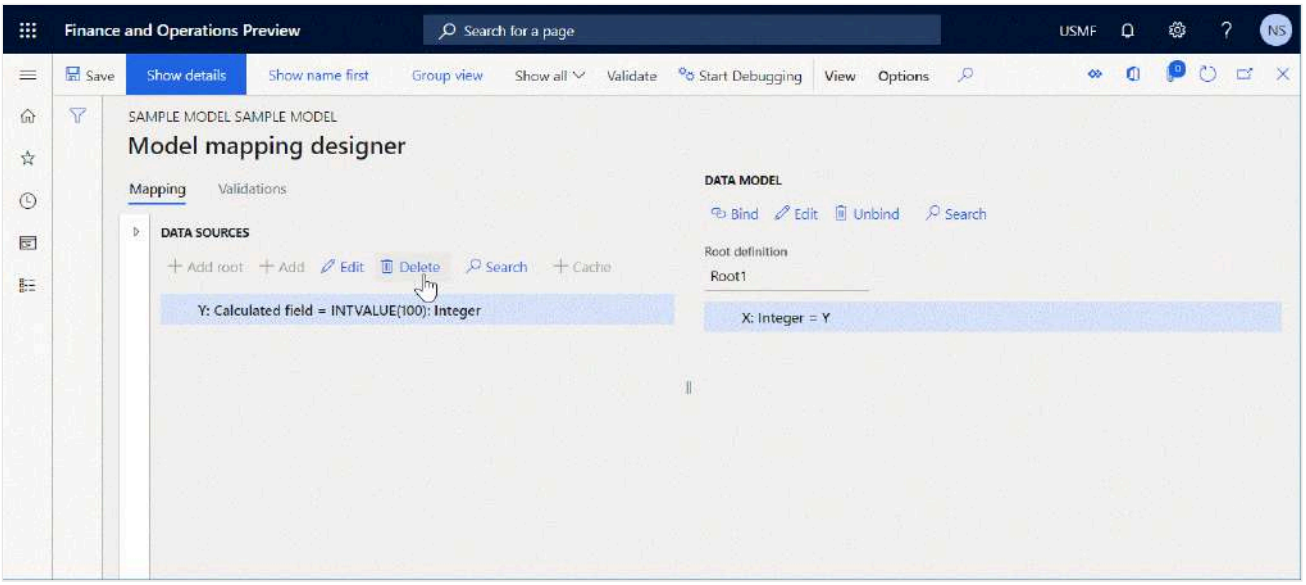
3. In the model mapping designer, in the **Data sources** pane, add a data source of the **Calculated field** type.

4. Name the new data source **Y**, and configure it so that it contains the expression `INTVALUE(100)`.

5. Bind **X** to **Y**.

6. In the model mapping designer, in the **Data sources** pane, delete the **Y** data source.

7. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page.



8. Notice that a validation error occurs. The message states that the binding of the **X** data model field contains the path that refers to the **Y** data source, but this data source isn't found.

Automatic resolution

Select **Unbind** to automatically fix this issue by removing the missing data source binding.

Manual resolution

Option 1

Unbind the **X** data model field to stop referring to the nonexistent **Y** data source.

Option 2

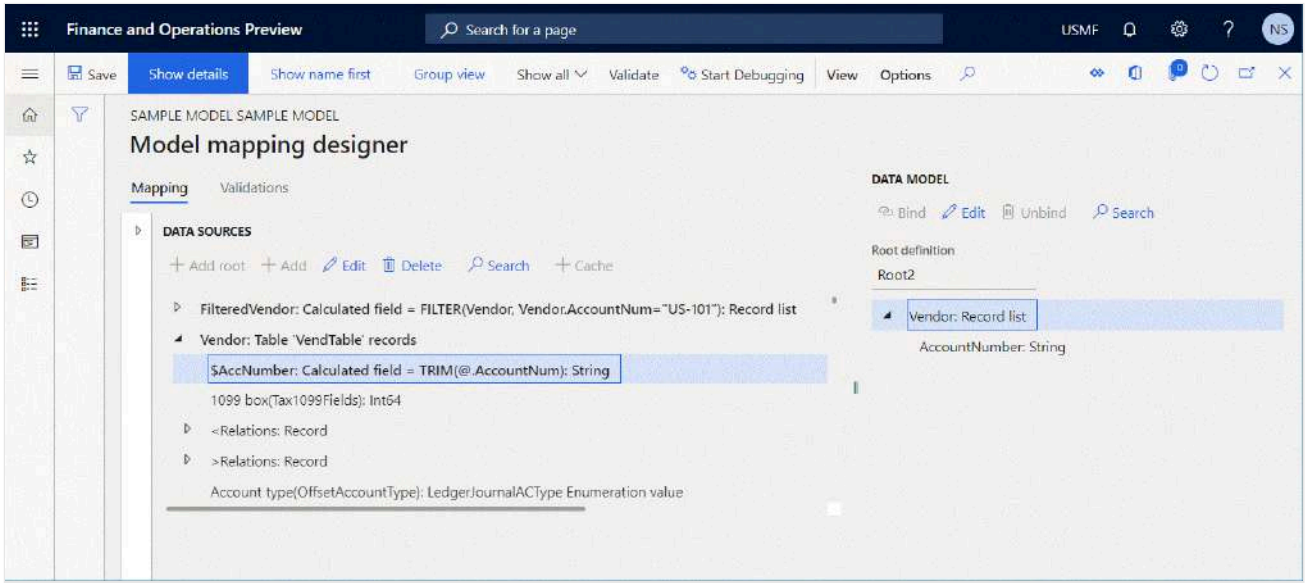
In the model mapping designer, in the **Data sources** pane, add the **Y** data source again.

Executability of an expression with FILTER function

Use the built-in FILTER ER function to access application tables, views, or data entities. The function places a single SQL call to get the required data as a list of records. Use a data source of the **Record list** type as an argument of this function to specify the application source for the call. ER checks whether a direct SQL query can be established to a data source that you refer to in the `FILTER` function. If a direct query can't be established, the ER model mapping designer shows a validation error. The error message states that the ER expression that includes the `FILTER` function can't be run at runtime.

The following steps show how this issue might occur.

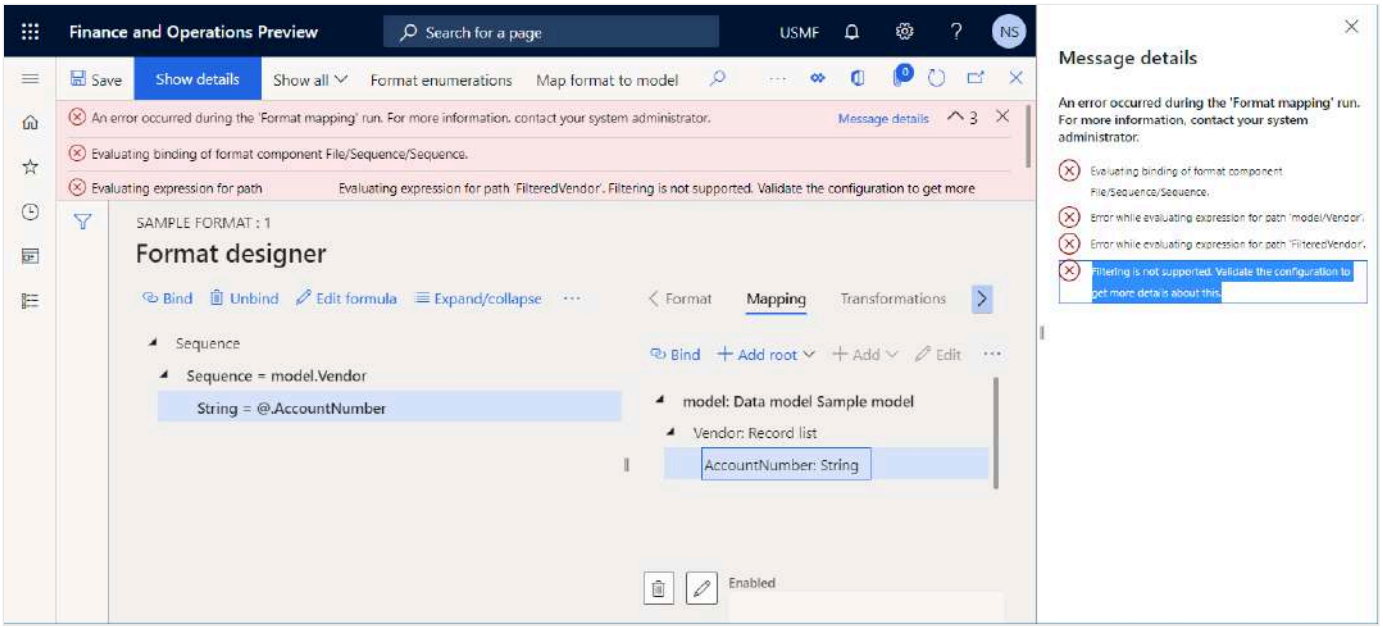
1. Start to configure the ER model mapping component.
2. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
3. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the VendTable table.
4. Add a data source of the **Calculated field** type.
5. Name the new data source **FilteredVendor**, and configure it so that it contains the expression `FILTER(Vendor, Vendor.AccountNum="US-101")`.
6. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page and verify that the `FILTER(Vendor, Vendor.AccountNum="US-101")` expression in the **Vendor** data source can be queried.
7. Modify the **Vendor** data source by adding a nested field of the **Calculated field** type to get the trimmed vendor account number.
8. Name the new nested field **$AccNumber**, and configure it so that it contains the expression `TRIM(Vendor.AccountNum)`.
9. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page and verify that the `FILTER(Vendor, Vendor.AccountNum="US-101")` expression in the **Vendor** data source can be queried.



10. Notice that a validation error occurs, because the **Vendor** data source contains a nested field of the **Calculated field** type that doesn't allow the expression of the **FilteredVendor**

data source to be translated to the direct SQL statement.

The following illustration shows the runtime error that occurs if you ignore the warning and select **Run** to run a format that is configured to use the model mapping.



Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Instead of adding a nested field of the **Calculated field** type to the **Vendor** data source, add the **$AccNumber** nested field to the **FilteredVendor** data source, and configure it so that it contains the expression `TRIM(FilteredVendor.AccountNum)`. By using this approach, you can run the `FILTER(Vendor, Vendor.AccountNum="US-101")` expression at the SQL level and calculate the **$AccNumber** nested field afterward.

Option 2

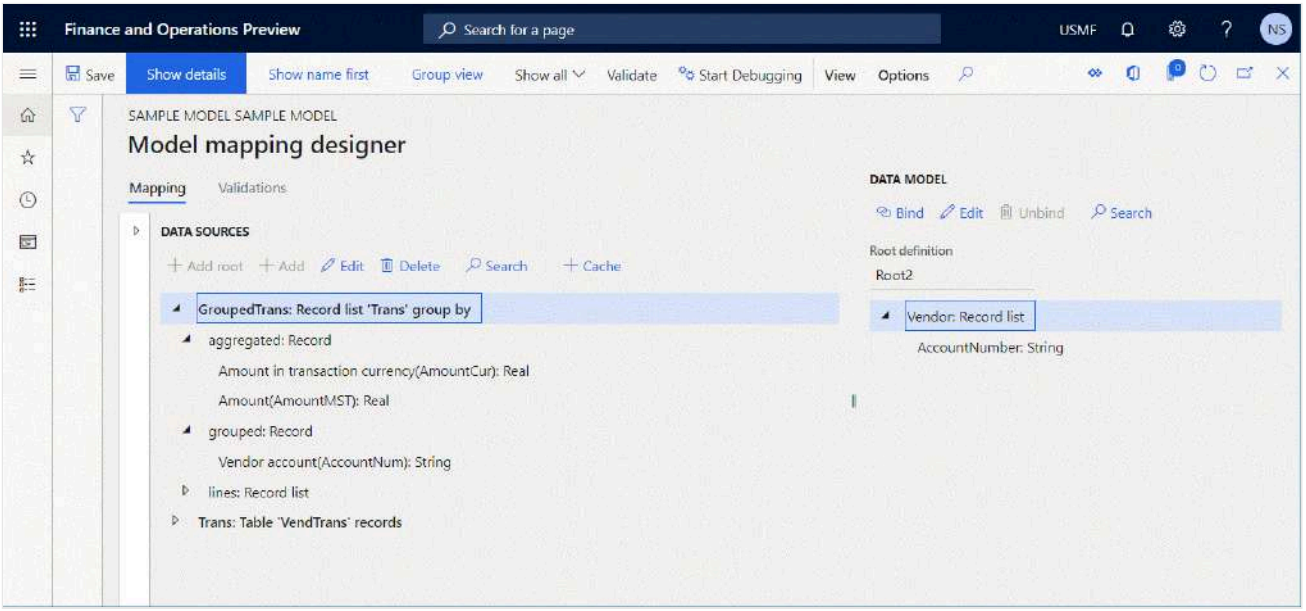
Change the expression of the **FilteredVendor** data source from `FILTER(Vendor, Vendor.AccountNum="US-101")` to `WHERE(Vendor, Vendor.AccountNum="US-101")`. Don't change the expression for a table that has a large volume of data (transactional table), because this change fetches all records and selects the required records in memory. Therefore, this approach can cause poor performance. For more information, see WHERE ER function.

Executability of a GROUPBY data source

The **GROUPBY** data source divides the query result into groups of records, usually to perform one or more aggregations on each group. You can configure every **GROUPBY** data source to run either at the database level or in memory. When you configure a **GROUPBY** data source to run at the database level, ER checks whether it can create a direct SQL query to a data source that the **GROUPBY** data source refers to. If ER can't create a direct query, the ER model mapping designer shows a validation error. The error message states that the configured **GROUPBY** data source can't run at runtime.

The following steps show how this issue might occur.

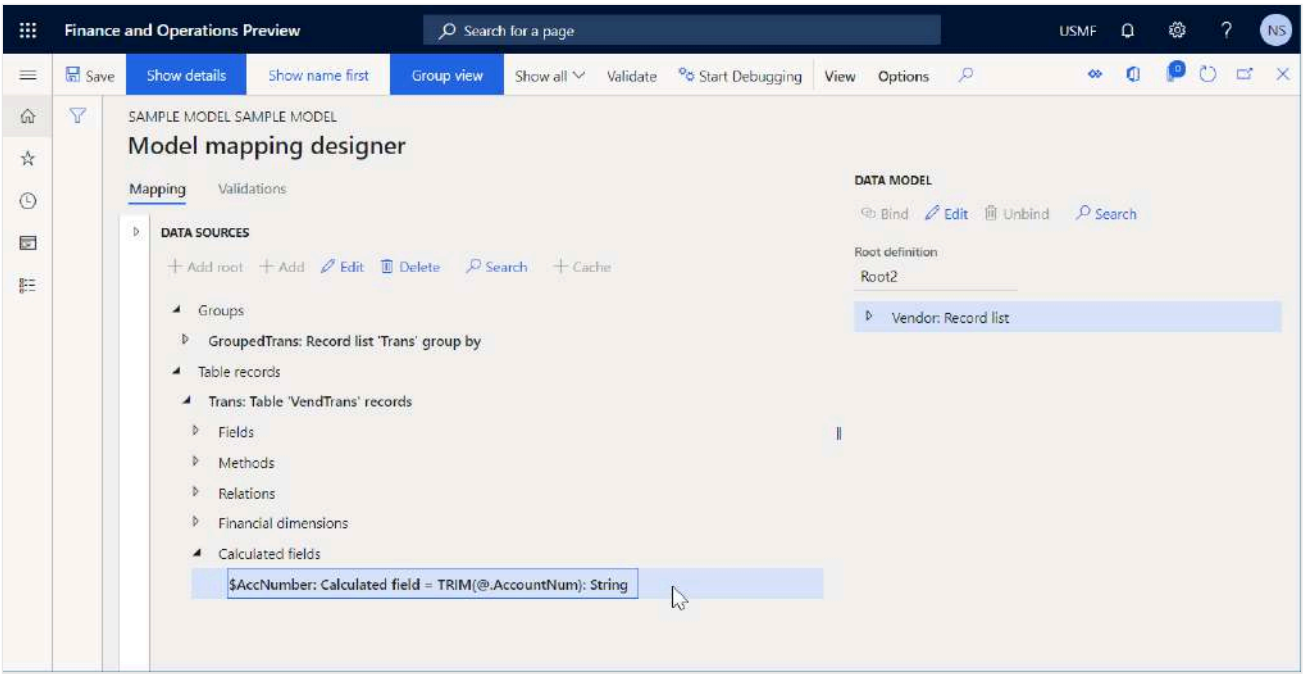
1. Start to configure the ER model mapping component.
2. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
3. Name the new data source **Trans**. In the **Table** field, select **VendTrans** to specify that this data source requests the VendTrans table.
4. Add a data source of the **Group by** type.
5. Name the new data source **GroupedTrans**, and configure it in the following way:
    - Select the **Trans** data source as the source of records that should be grouped.
    - In the **Execution location** field, select **Query** to specify that you want to run this data source at the database level.



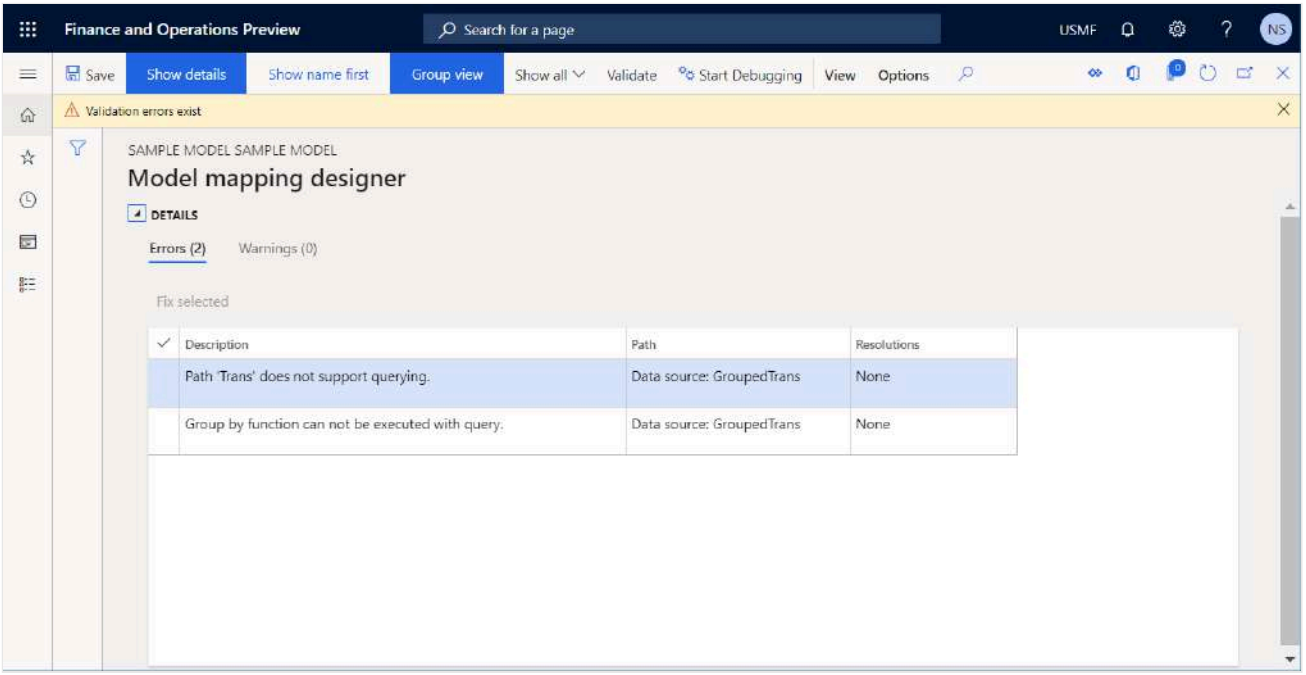
6. Select **Validate** to check the editable model mapping component on the **Model mapping designer** page and verify that the configured **GroupedTrans** data source can be queried.

7. Modify the **Trans** data source by adding a nested field of the **Calculated field** type to get the trimmed vendor account number.

8. Name the new data source **$AccNumber**, and configure it so that it contains the expression `TRIM(Trans.AccountNum)`.

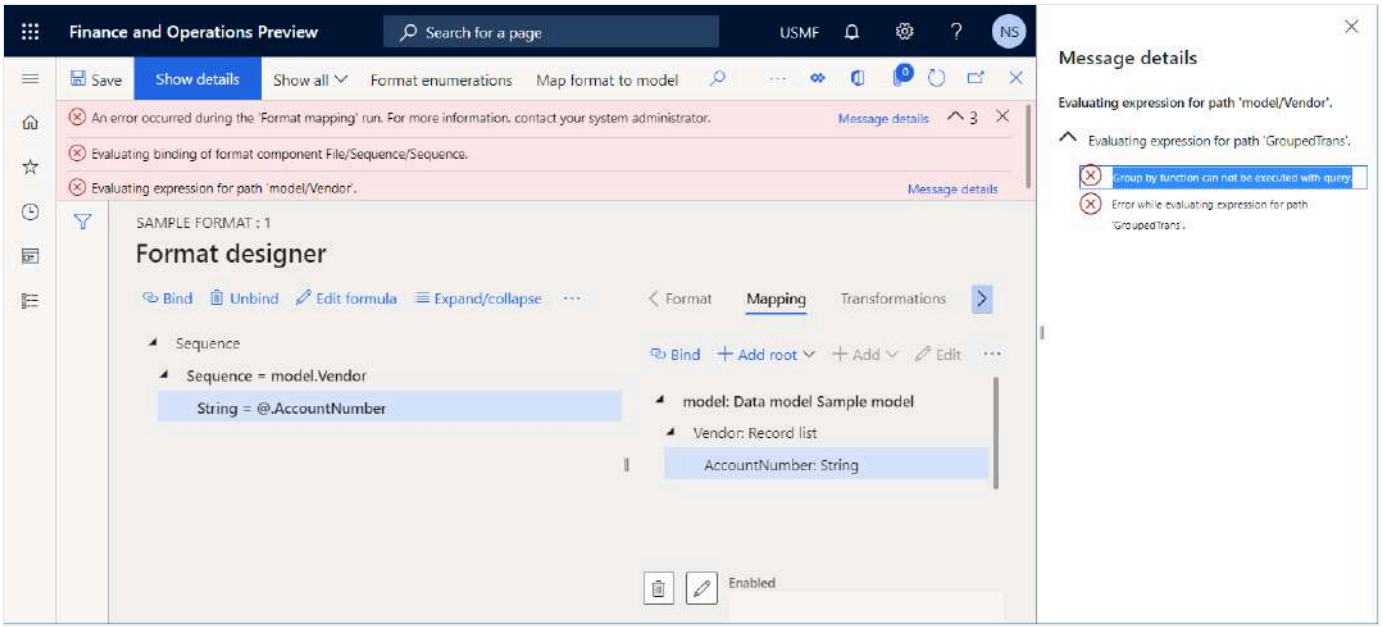


9. Select **Validate** to check the editable model mapping component on the **Model mapping designer** page and verify that the configured **GroupedTrans** data source can be queried.



10. Notice that a validation error occurs, because the **Trans** data source contains a nested field of the **Calculated field** type that doesn't allow the call for the **GroupedTrans** data source to be translated to the direct SQL statement.

The following illustration shows the runtime error that occurs if you ignore the warning and select **Run** to run a format that is configured to use the model mapping.



Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Instead of adding a nested field of the **Calculated field** type to the **Trans** data source, add the **$AccNumber** nested field for the **GroupedTrans.lines** item of the **GroupedTrans** data source, and configure it so that it contains the expression `TRIM(GroupedTrans.lines.AccountNum)`. In this way, you can run the **GroupedTrans** data source at the SQL level and calculate the **$AccNumber** nested field afterward.

Option 2

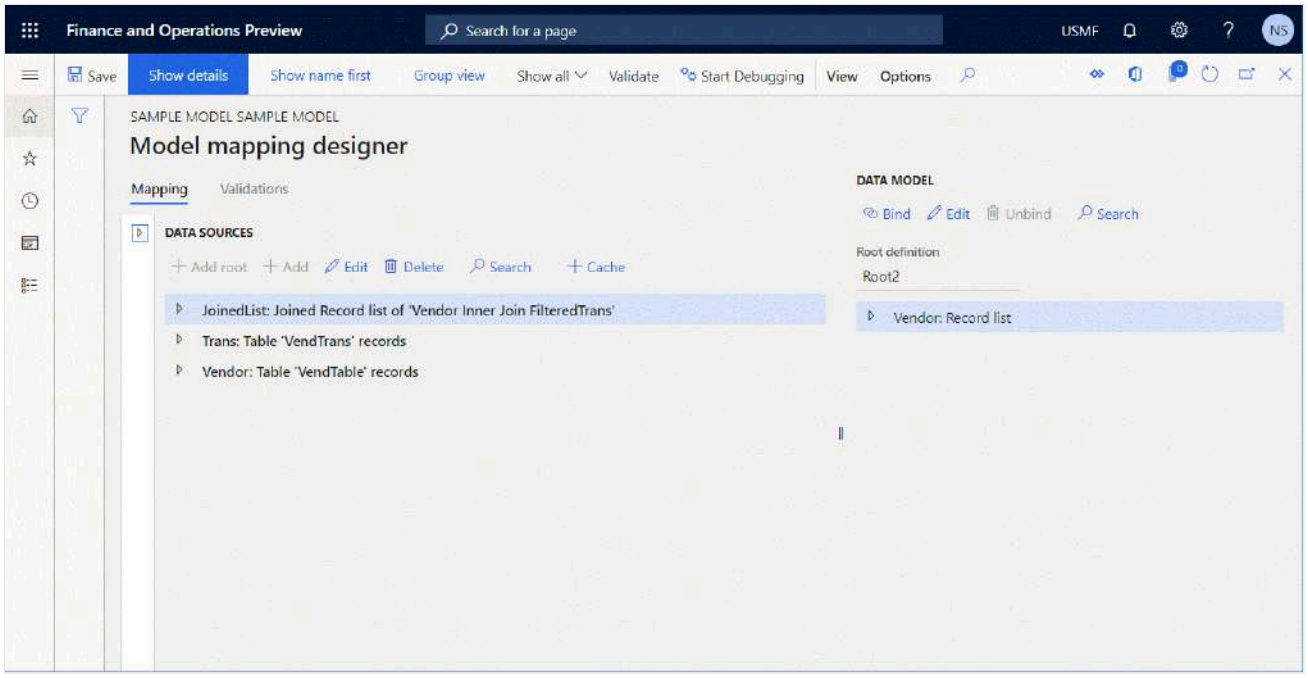
Change the value of the **Execution location** field for the **GroupedTrans** data source from **Query** to **In memory**. Don't change the value for a table that has a large volume of data (transactional table), because this change fetches all records and does grouping and aggregation in memory. Therefore, this approach can cause poor performance.

Executability of a JOIN data source

The JOIN data source combines records from two or more database tables, based on related fields. You can configure every **JOIN** data source to run either at the database level or in memory. When you configure a **JOIN** data source to run at the database level, ER checks whether it can create a direct SQL query to data sources that the data source refers to. If ER can't create a direct SQL query to at least one referenced data source, the ER model mapping designer shows a validation error. The error message states that the configured **JOIN** data source can't run at runtime.

The following steps show how this issue might occur.

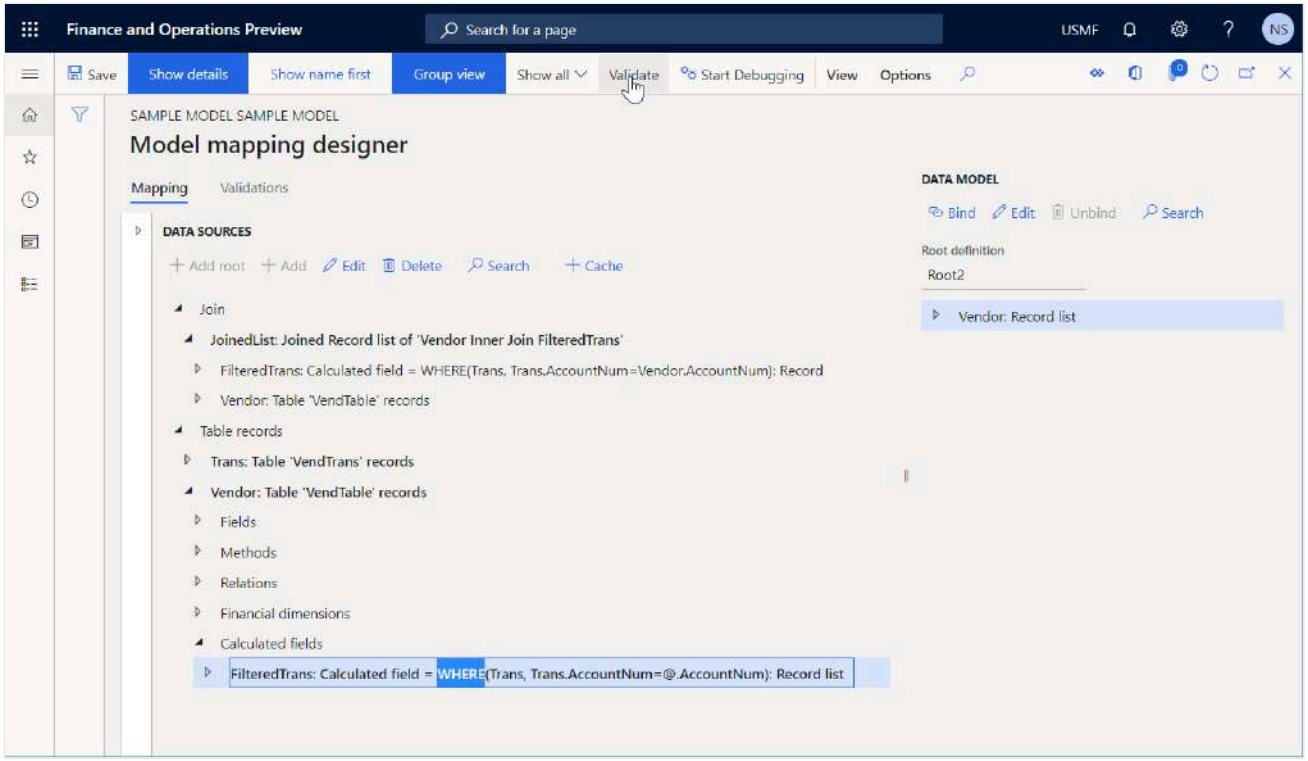
1. Start to configure the ER model mapping component.
2. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
3. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the VendTable table.
4. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
5. Name the new data source **Trans**. In the **Table** field, select **VendTrans** to specify that this data source requests the VendTrans table.
6. Add a data source of the **Calculated field** type as the nested field of the **Vendor** data source.
7. Name the new data source **FilteredTrans**, and configure it so that it contains the expression `FILTER(Trans, Trans.AccountNum=Vendor.AccountNum)`.
8. Add a data source of the **Join** type.
9. Name the new data source **JoinedList**, and configure it in the following way:
   a. Add the **Vendor** data source as the first set of records to join.
   b. Add the **Vendor.FilteredTrans** data source as the second set of records to join. Select **INNER** as the type.
   c. In the **Execute** field, select **Query** to specify that you want to run this data source at the database level.



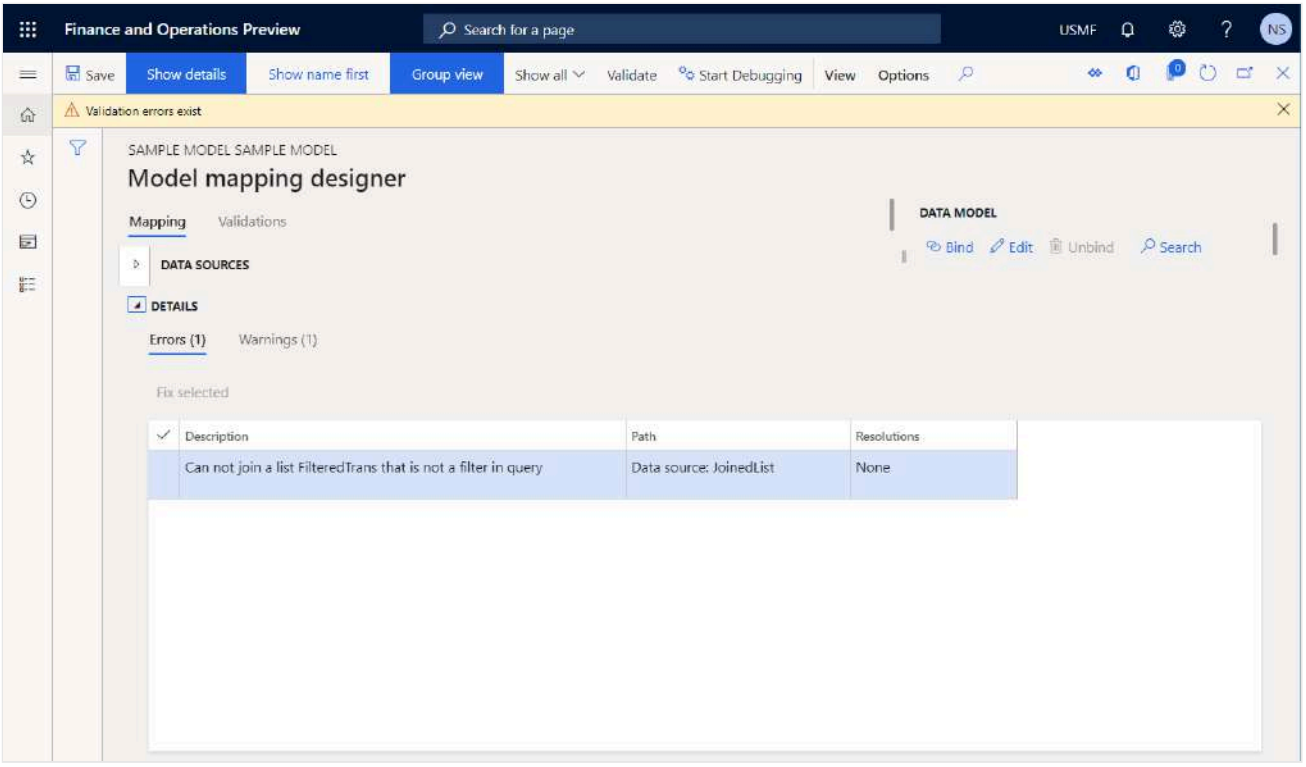
10. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page and verify that the configured **JoinedList** data source can be queried.

11. Change the expression of the **Vendor.FilteredTrans** data source from `FILTER(Trans, Trans.AccountNum=Vendor.AccountNum)` to `WHERE(Trans, Trans.AccountNum=Vendor.AccountNum)`.

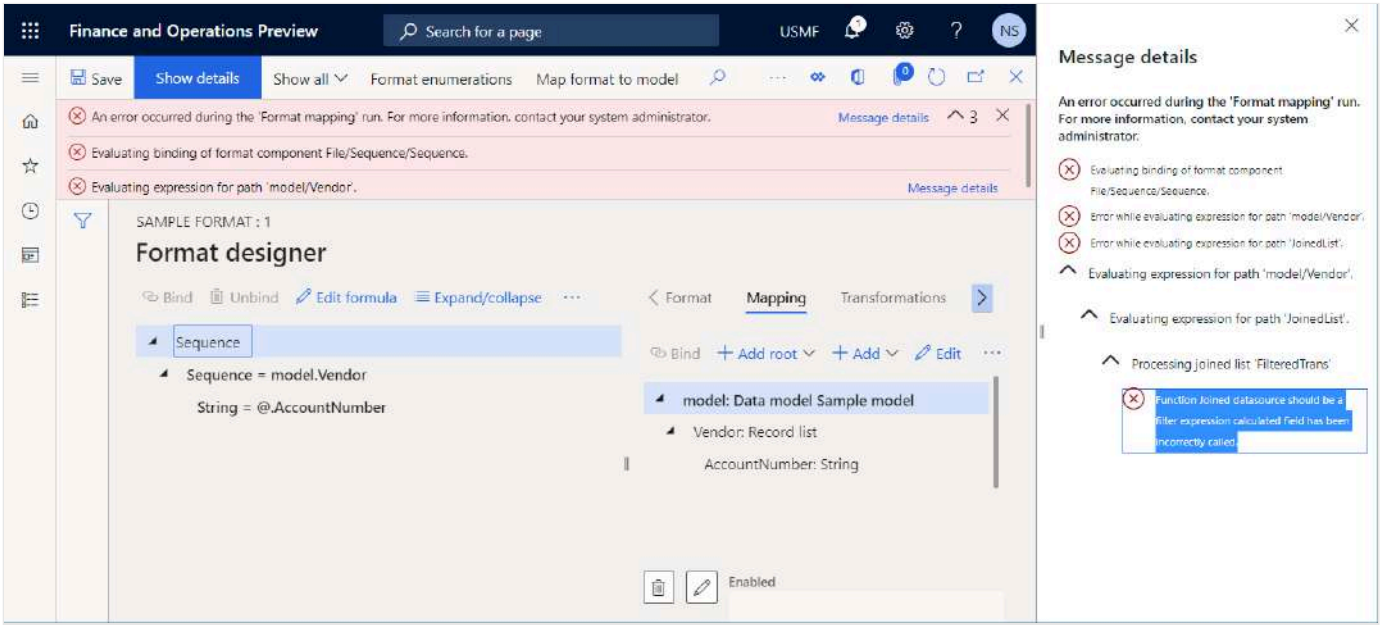
12. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page and verify that the configured **JoinedList** data source can be queried.



13. Notice that a validation error occurs, because the expression of the **Vendor.FilteredTrans** data source can't be translated to the direct SQL call. Additionally, the direct SQL call doesn't allow the call for the **JoinedList** data source to be translated to the direct SQL statement.



The following illustration shows the runtime error that occurs if you ignore the warning and select **Run** to run a format that is configured to use the model mapping.



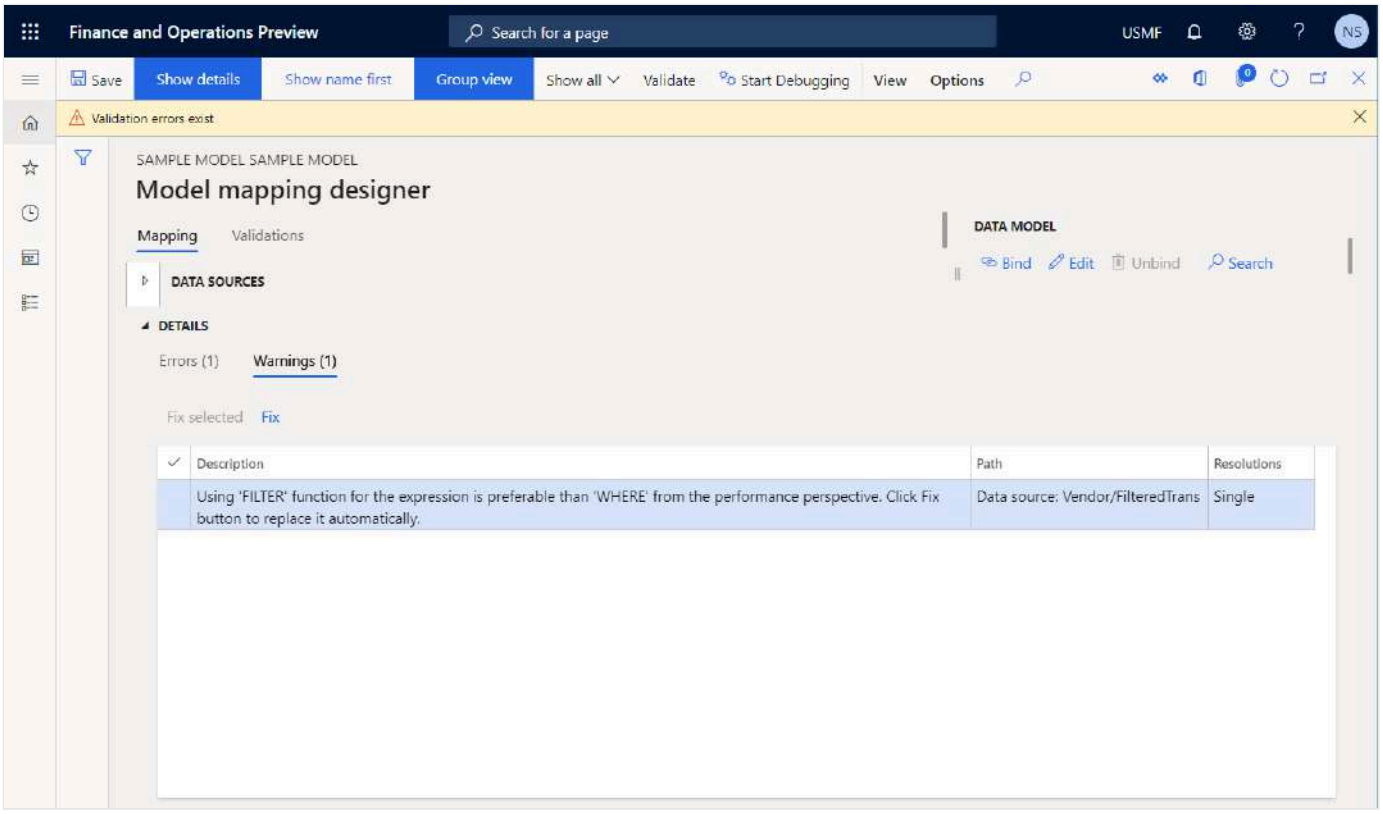
Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Change the expression of the **Vendor.FilteredTrans** data source from `WHERE(Trans, Trans.AccountNum=Vendor.AccountNum)` back to `FILTER(Trans, Trans.AccountNum=Vendor.AccountNum)`, as the warning advises.



Option 2

Change the value of the **Execute** field for the **JoinedList** data source from **Query** to **In memory**. Don't change the value for a table that has a large volume of data (transactional table), because this change fetches all records, and the join happens in memory. Therefore, this approach can cause poor performance. A validation warning informs you about this risk.

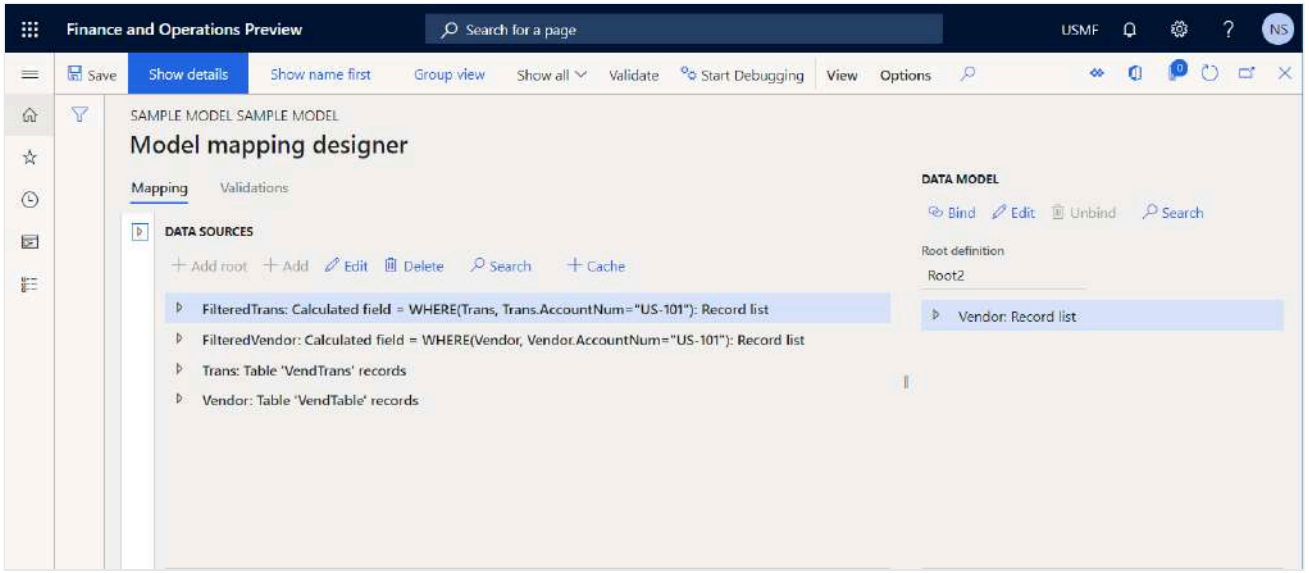
Preferability of FILTER vs WHERE function

Use the built-in FILTER ER function to access application tables, views, or data entities. It places a single SQL call to get the required data as a list of records. The WHERE function fetches all records from the given source and does record selection in memory. Both functions use a data source of the **Record list** type as an argument and specify a source for getting records. ER checks whether a direct SQL call can be established to a data source that you refer to in the **WHERE** function. If a direct call can be established, a validation warning occurs in the ER model mapping

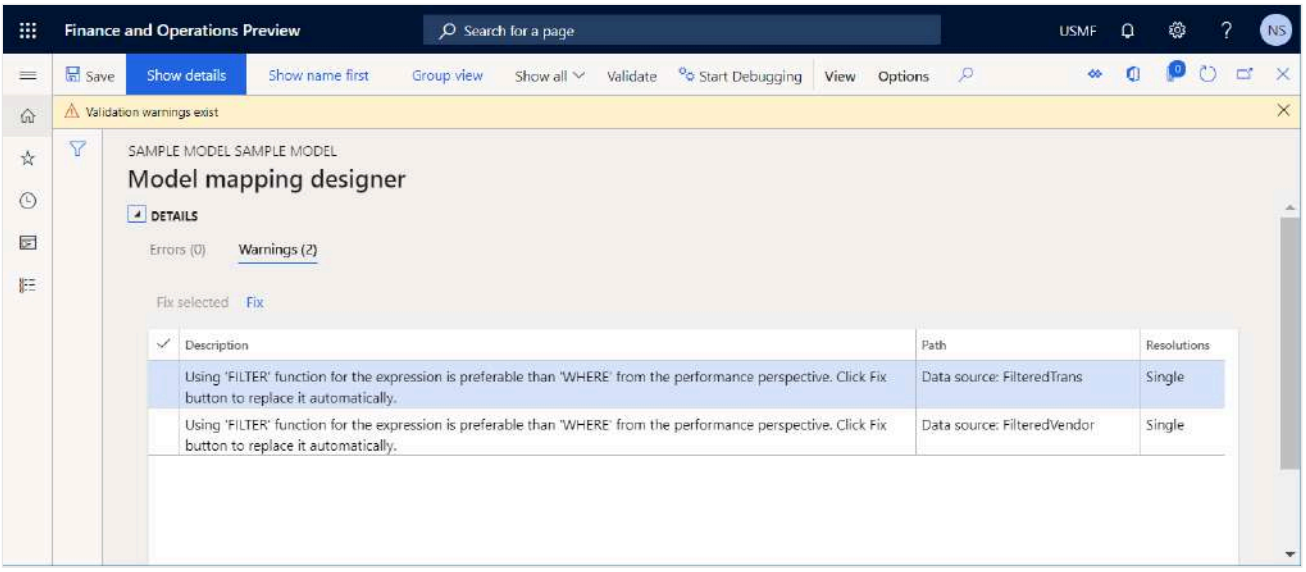
designer. The message that you receive recommends that you use the **FILTER** function instead of the **WHERE** function to help improve efficiency.

The following steps show how this issue might occur.

1. Start to configure the ER model mapping component.
2. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
3. Name the new data source **Trans**. In the **Table** field, select **VendTrans** to specify that this data source requests the VendTrans table.
4. Add a data source of the **Calculated field** type as the nested field of the **Vendor** data source.
5. Name the new data source **FilteredTrans**, and configure it so that it contains the expression `WHERE(Trans, Trans.AccountNum="US-101")`.
6. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
7. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the VendTable table.
8. Add a data source of the **Calculated field** type.
9. Name the new data source **FilteredVendor**, and configure it so that it contains the expression `WHERE(Vendor, Vendor.AccountNum="US-101")`.
10. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page.



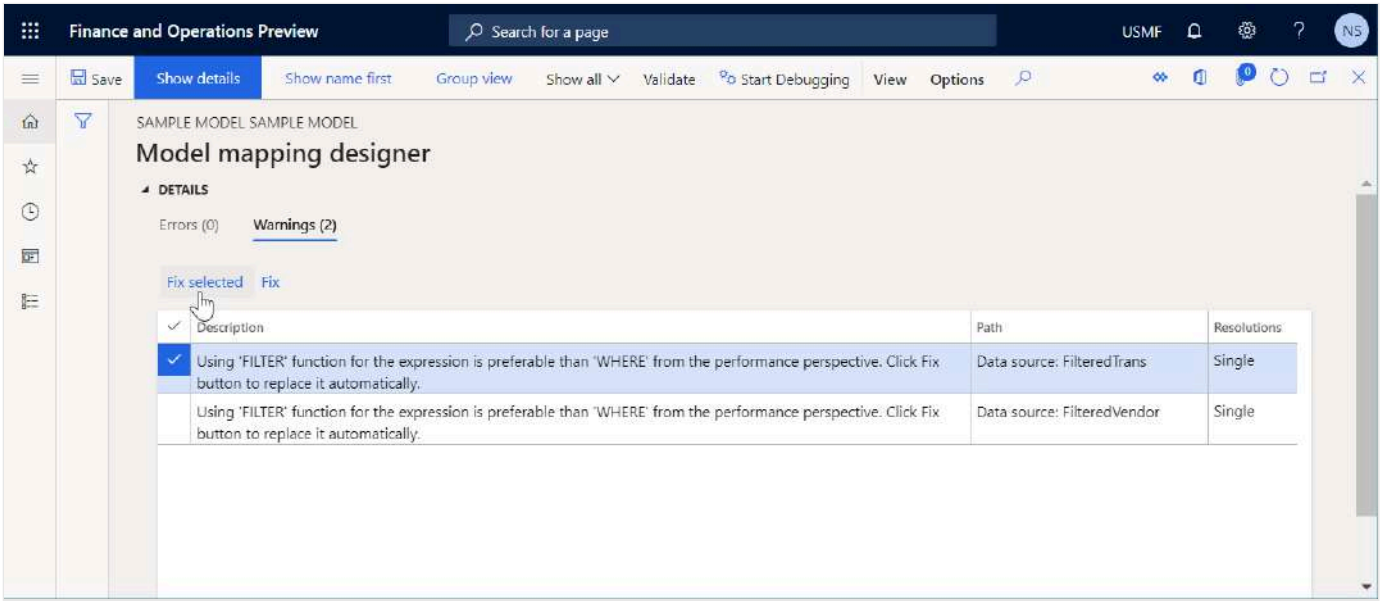
11. Notice that validation warnings recommend that you use the **FILTER** function instead of the **WHERE** function for the **FilteredVendor** and **FilteredTrans** data sources.



Automatic resolution

Select **Fix** to automatically replace the **WHERE** function with the **FILTER** function in the expression of all data sources that appear in the grid on the **Warnings** tab for this type of inspection.

Alternatively, select the row for a single warning in the grid and then select **Fix selected**. In this case, the expression is automatically changed only in the data source that the selected warning mentions.



Manual resolution

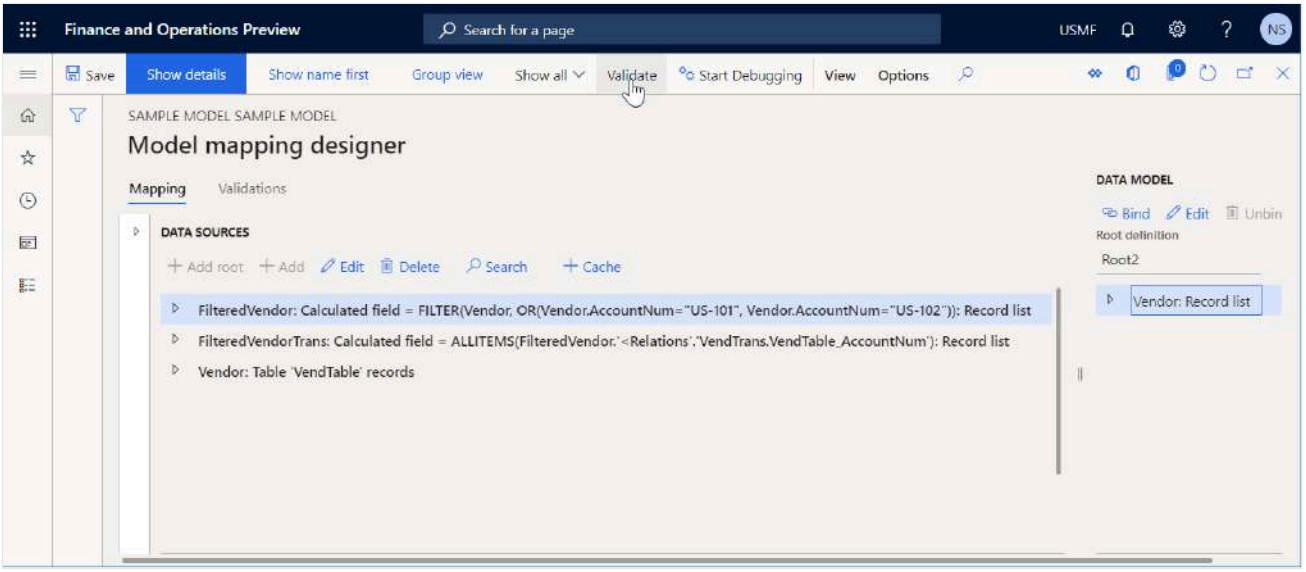
Manually adjust the expressions of all the data sources in the validation grid by replacing the **WHERE** function with the **FILTER** function.

Preferability of ALLITEMSQUERY vs ALLITEMS function

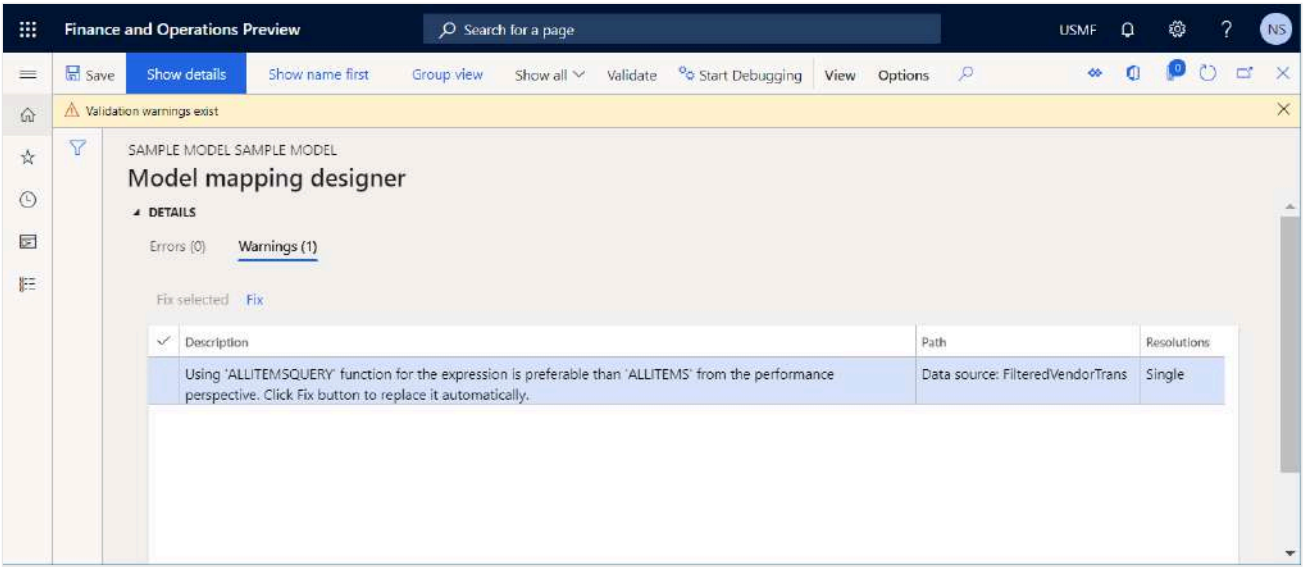
The built-in ALLITEMS and ALLITEMSQUERY ER functions return a flattened **Record list** value that consists of a list of records that represent all items that match the specified path. ER checks whether it can establish a direct SQL call to a data source that the **ALLITEMS** function refers to. If ER can establish a direct call, a validation warning occurs in the ER model mapping designer. The message that you receive recommends that you use the **ALLITEMSQUERY** function instead of the **ALLITEMS** function to help improve efficiency.

The following steps show how this issue might occur.

1. Start to configure the ER model mapping component.
2. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
3. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the VendTable table.
4. Add a data source of the **Calculated field** type to get records for several vendors.
5. Name the new data source **FilteredVendor**, and configure it so that it contains the expression `FILTER(Vendor, OR(Vendor.AccountNum="US-101",Vendor.AccountNum="US-102"))`.
6. Add a data source of the **Calculated field** type to get the transactions of all filtered vendors.
7. Name the new data source **FilteredVendorTrans**, and configure it so that it contains the expression `ALLITEMS(FilteredVendor.'<Relations'.'VendTrans.VendTable_AccountNum')`.
8. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page.



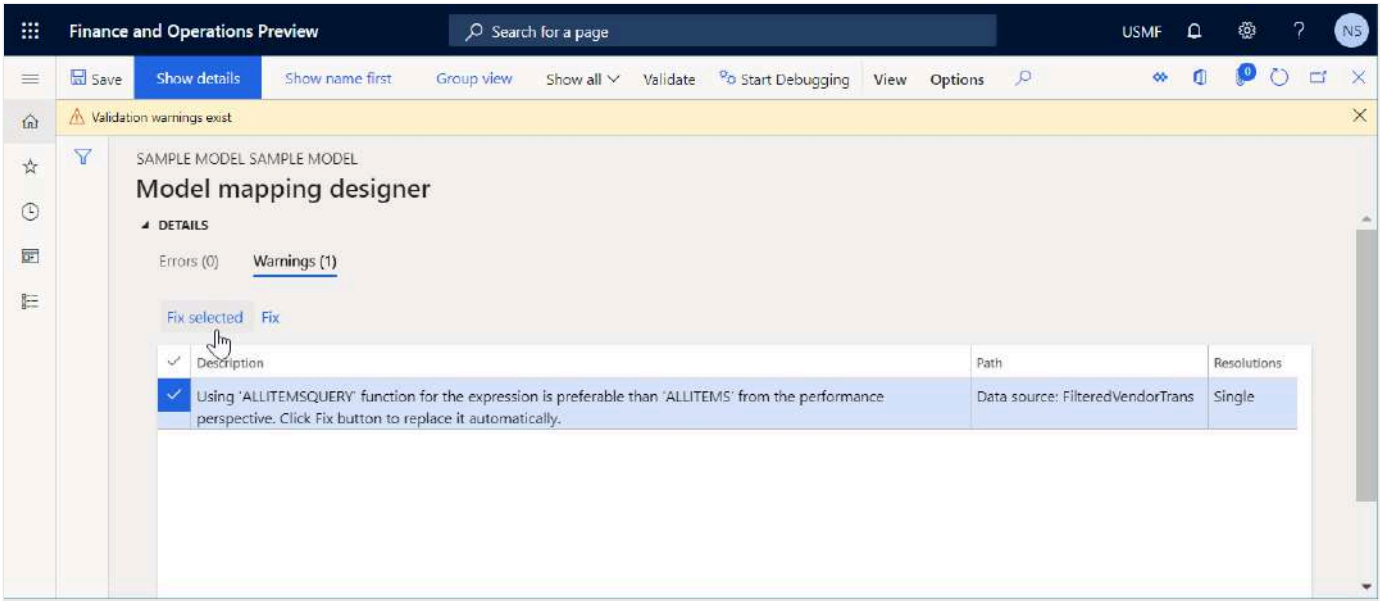
9. Notice that a validation warning occurs. The message recommends that you use the **ALLITEMSQUERY** function instead of the **ALLITEMS** function for the **FilteredVendorTrans** data source.



Automatic resolution

Select **Fix** to automatically replace the **ALLITEMS** function with the **ALLITEMSQUERY** function in the expression of all data sources that appear in the grid on the **Warnings** tab for this type of inspection.

Alternatively, select the row for a single warning in the grid and then select **Fix selected**. In this case, the expression is automatically changed only in the data source that the selected warning mentions.



Manual resolution

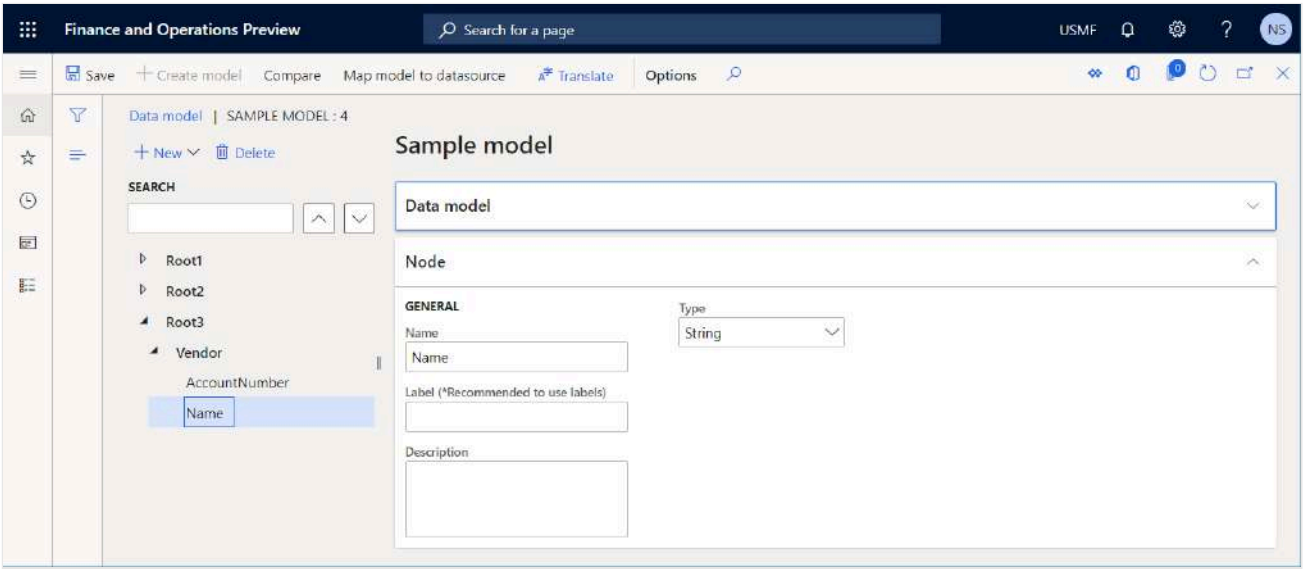
To resolve the issue, manually adjust the expressions for all data sources mentioned in the validation grid. Replace the **ALLITEMS** function with the **ALLITEMSQUERY** function.

Consider empty list cases

You can configure your ER format or model mapping component to get the field value of a data source of the **Record list** type. ER checks whether your design considers the case where a data source that you call contains no records (that is, it's empty). This check prevents runtime errors when a value is fetched from a field of a nonexistent record.

The following steps show how this issue might occur.

1. Start to configure the ER data model, the ER model mapping, and the ER format components simultaneously.
2. In the data model tree, add a root item that you name **Root3**.
3. Modify the **Root3** item by adding a nested item of the **Record list** type.
4. Name the new nested item **Vendor**.
5. Modify the **Vendor** item in the following way:
    - Add a nested field of the **String** type, and name it **Name**.
    - Add a nested field of the **String** type, and name it **AccountNumber**.



6. In the model mapping designer, in the **Data sources** pane, add a data source of the **Dynamics 365 for Operations \ Table records** type.

7. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the VendTable table.

8. Add a data source of the **General \ User input parameter** type to search for a vendor account in the runtime dialog box.

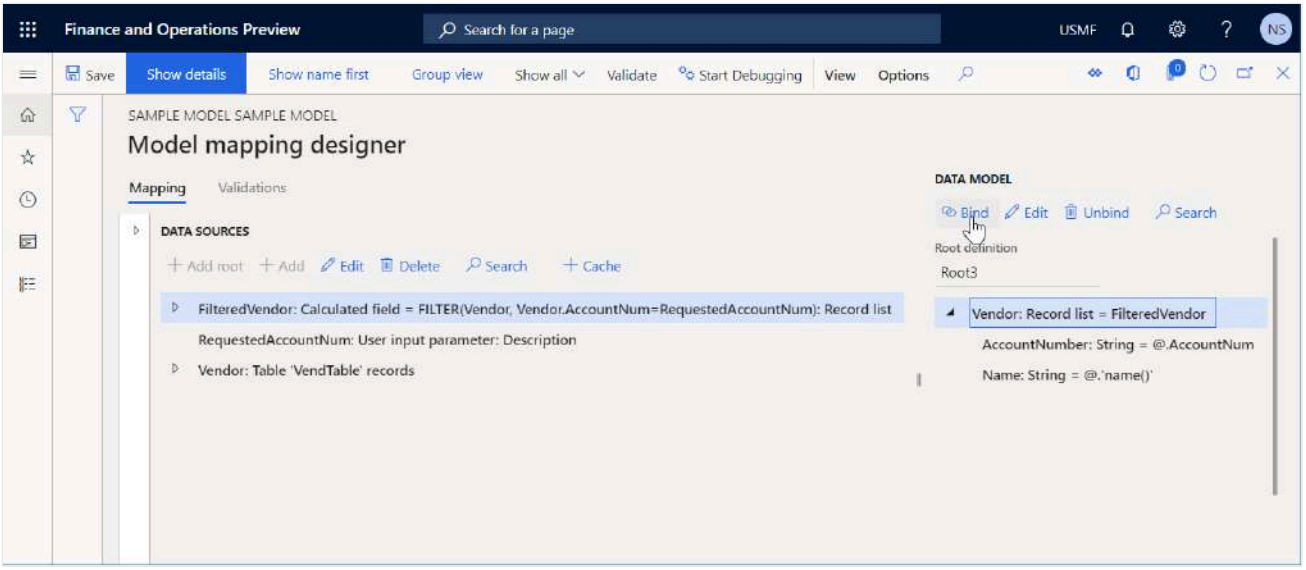
9. Name the new data source **RequestedAccountNum**. In the **Label** field, enter **Vendor account number**. In the **Operations data type name** field, keep the default value, **Description**.

10. Add a data source of the **Calculated field** type to filter a vendor that the user inquires about.

11. Name the new data source **FilteredVendor**, and configure it so that it contains the expression `FILTER(Vendor, Vendor.AccountNum=RequestedAccountNum)`.

12. Bind the data model items to configured data sources in the following way:

    - Bind **FilteredVendor** to **Vendor**.
    - Bind **FilteredVendor.AccountNum** to **Vendor.AccountNumber**.
    - Bind **FilteredVendor.'name()'** to **Vendor.Name**.



13. In the format structure tree, add the following items to generate an outbound document in XML format that contains the vendor details:

    a. Add the **Statement** root XML element.

    b. For the **Statement** XML element, add the nested **Party** XML element.

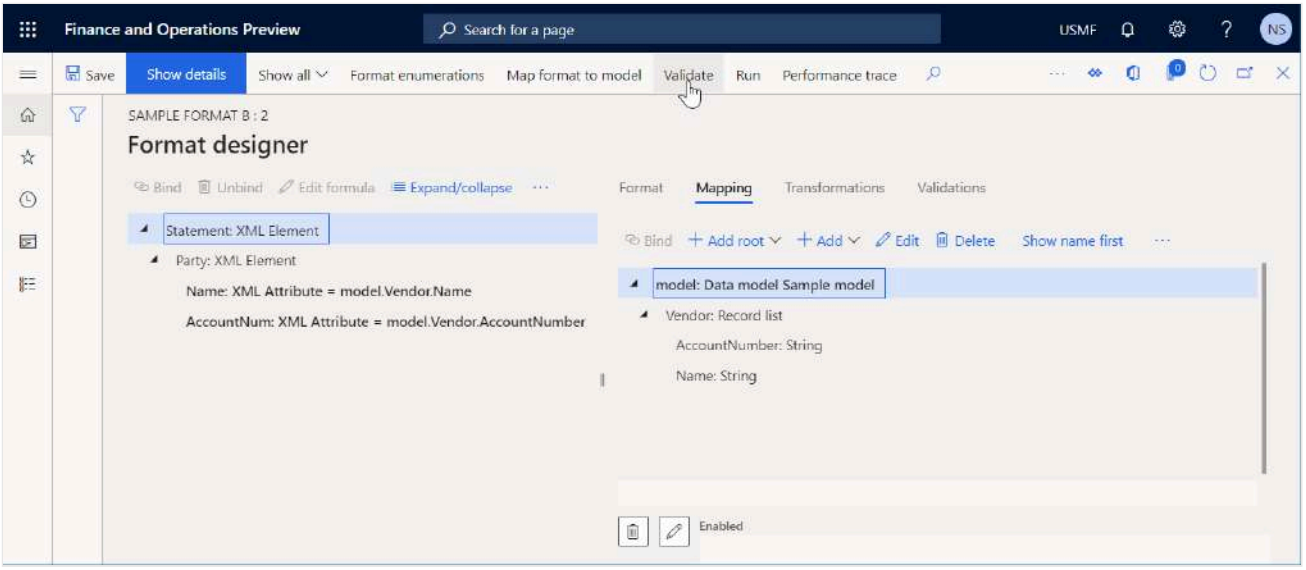
    c. For the **Party** XML element, add the following nested XML attributes:

- Name
- AccountNum

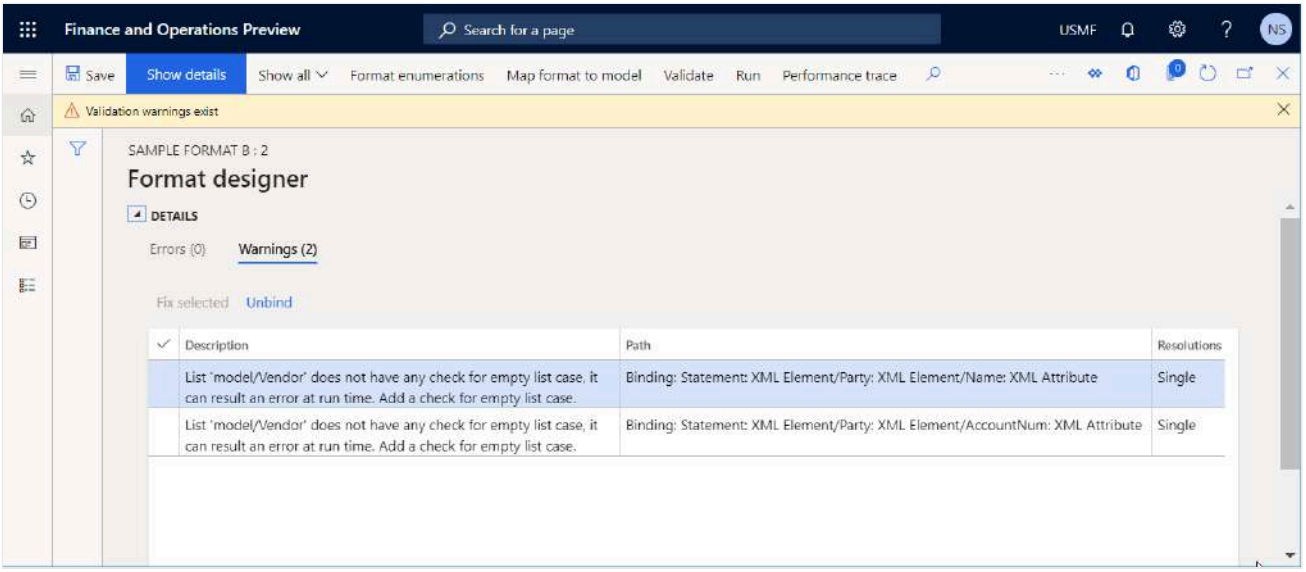
14. Bind format elements to provided data sources in the following way:

    - Bind the **Statement\Party\Name** format element to the **model.Vendor.Name** data source field.
    - Bind the **Statement\Party\AccountNum** format element to the **model.Vendor.AccountNumber** data source field.

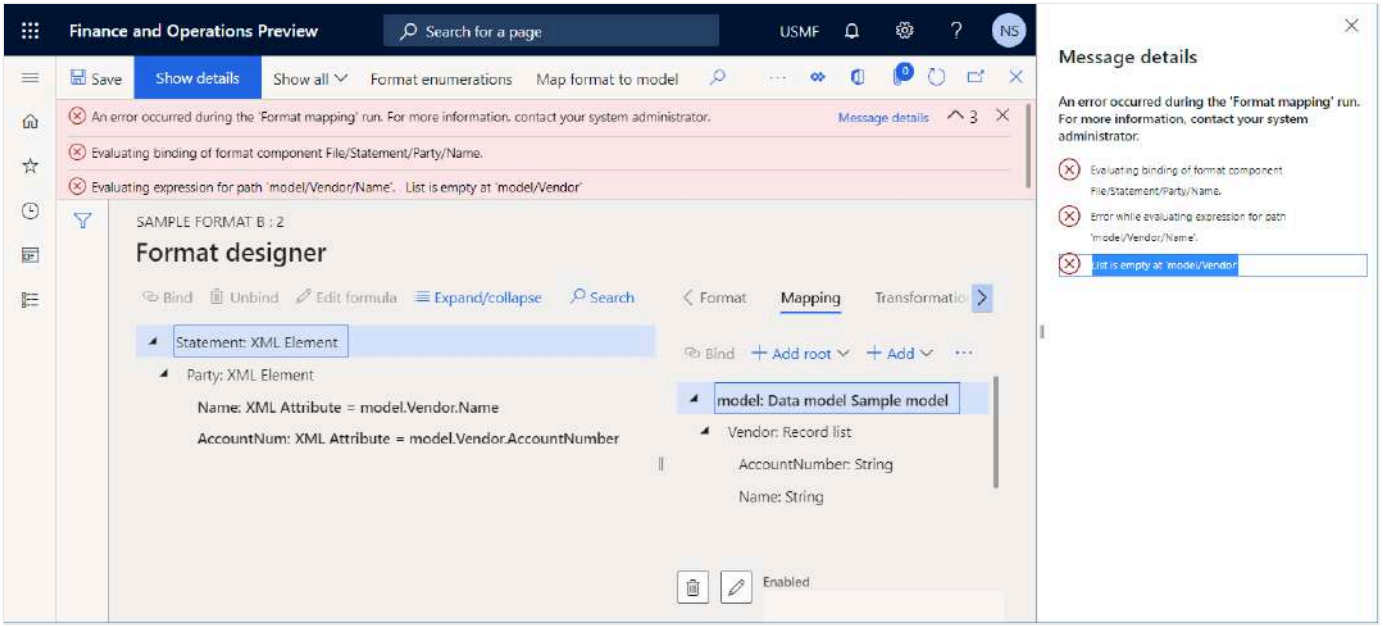
15. Select **Validate** to inspect the editable format component on the **Format designer** page.



16. Notice that a validation error occurs. The message states that an error might be thrown for the configured **Statement\Party\Name** and **Statement\Party\AccountNum** format components at runtime if the `model.Vendor` list is empty.



The following illustration shows the runtime error that occurs if you ignore the warning, select **Run** to run the format, and select the account number of a nonexistent vendor. Because the requested vendor doesn't exist, the `model.Vendor` list is empty (that is, it contains no records).



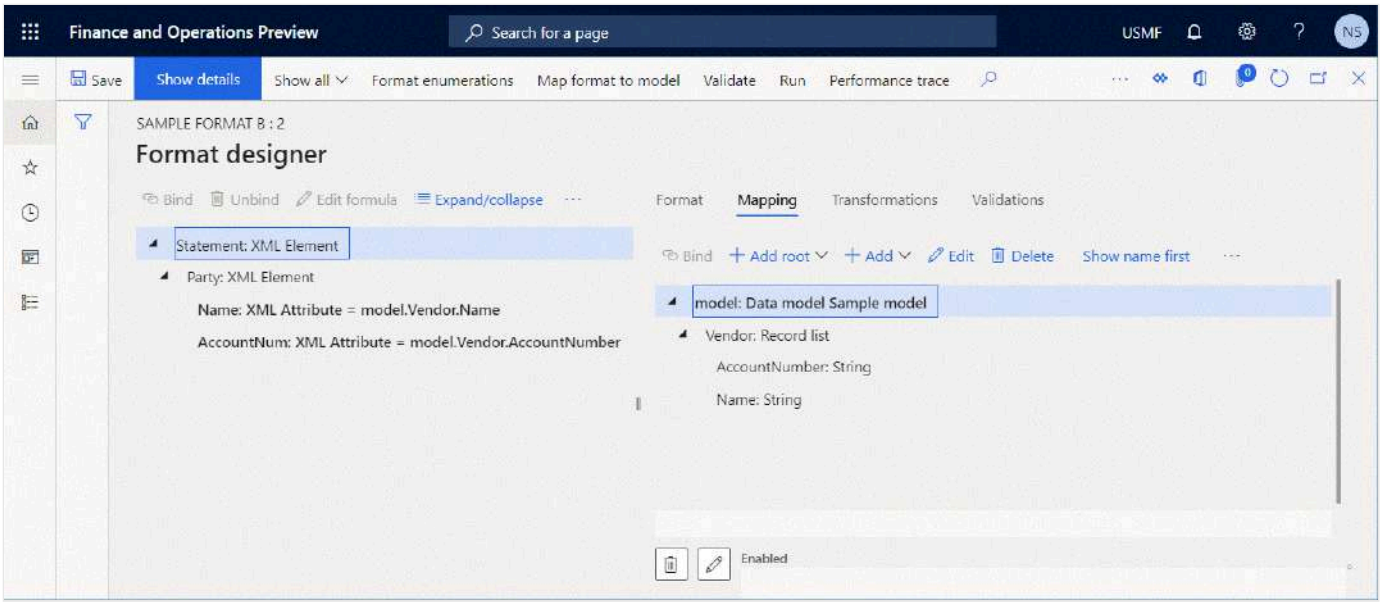
Automatic resolution

For the selected row in the grid on the **Warnings** tab, select **Unbind**. This action automatically removes the binding from the format elements that the **Path** column points to.

Manual resolution

Option 1

Bind the **Statement\Party\Name** format element to the `model.Vendor` data source item. At runtime, this binding calls the `model.Vendor` data source first. When `model.Vendor` returns an empty record list, the nested format elements don't run. Therefore, no validation warnings occur for this format configuration.

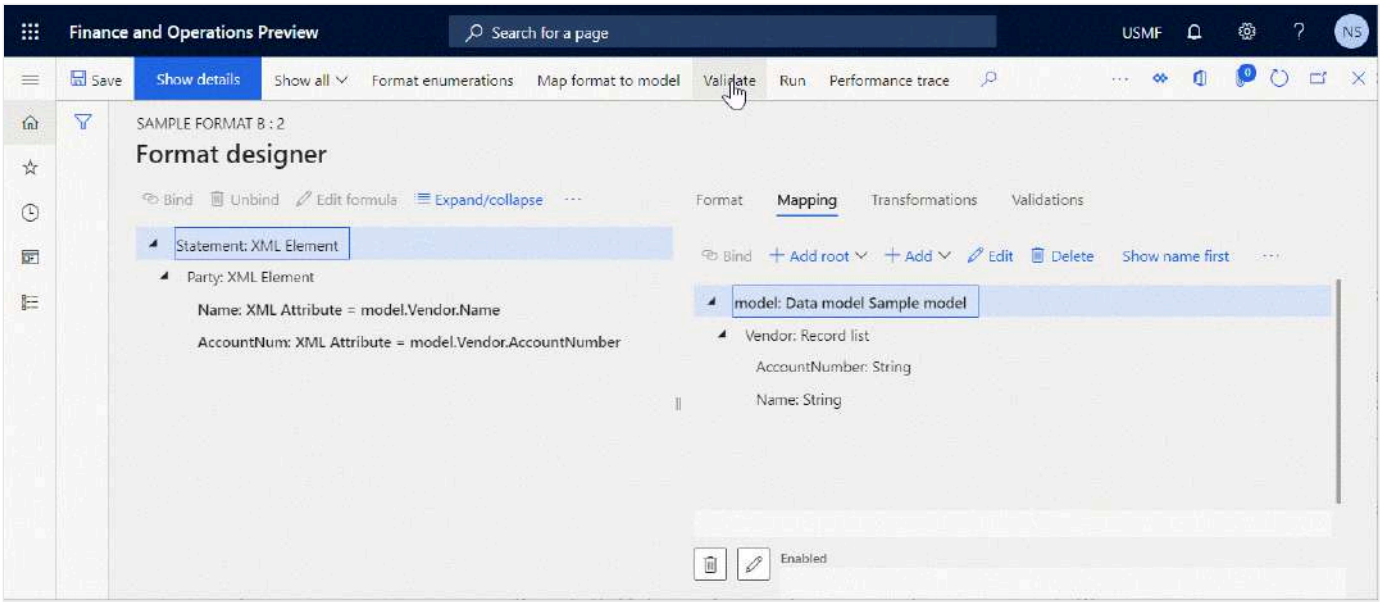


Option 2

Change the binding of the **Statement\Party\Name** format element from `model.Vendor.Name` to `FIRSTORNULL(model.Vendor).Name`. The updated binding conditionally converts the first record of the `model.Vendor` data source of the **Record list** type to a new data source of the **Record** type. This new data source contains the same set of fields.

- If at least one record is available in the `model.Vendor` data source, the fields of that record are filled with the values of the fields of the first record of the `model.Vendor` data source. In this case, the updated binding returns the vendor name.
- Otherwise, every field of the record that is created is filled with the default value for the data type of that field. In this case, the blank string is returned as the default value of the **String** data type.

Therefore, no validation warnings occur for the **Statement\Party\Name** format element when it's bound to the `FIRSTORNULL(model.Vendor).Name` expression.



Option 3

If you want to explicitly specify the data that's entered in a generated document when the `model.Vendor` data source of the **Record list** type returns no records (the text **Not available** in this example), change the binding of the **Statement\Party\Name** format element from `model.Vendor.Name` to `IF(NOT(ISEMPTY(model.Vendor)), model.Vendor.Name, "Not available")`. You can also use the expression `IF(COUNT(model.Vendor)=0, model.Vendor.Name, "Not available")`.

Additional consideration

The inspection also warns you about another potential problem. By default, when you bind the **Statement\Party\Name** and **Statement\Party\AccountNum** format elements to the appropriate fields of the `model.Vendor` data source of the **Record list** type, you run those bindings and take the values from the appropriate fields of the first record of the `model.Vendor` data source, if that list isn't empty.

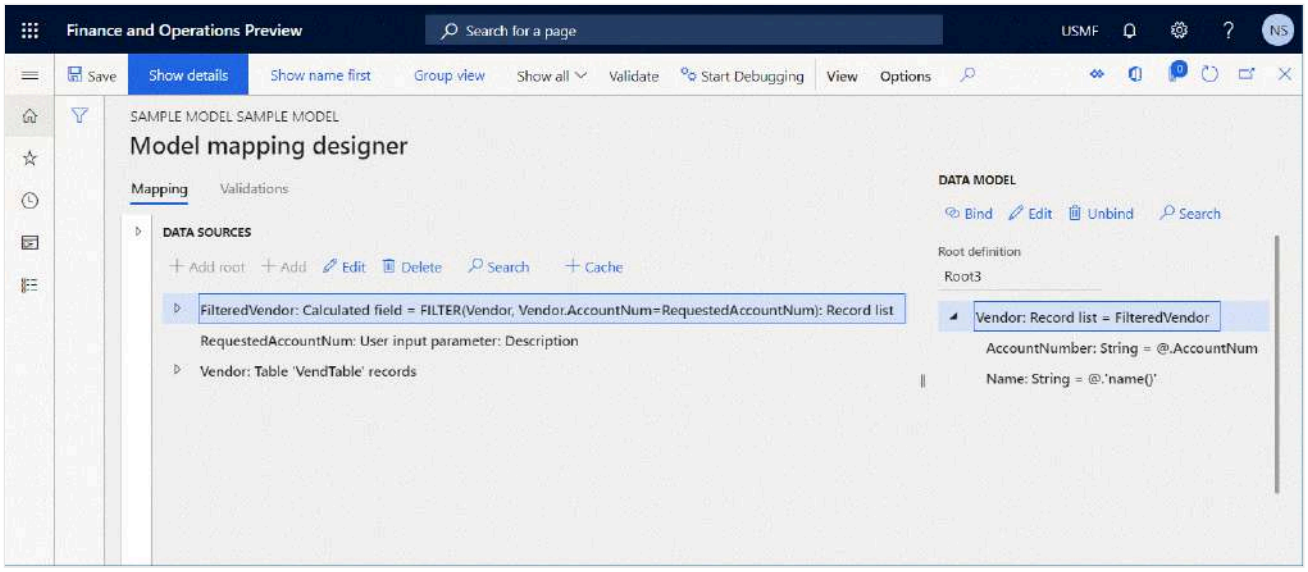
If you don't bind the **Statement\Party** format element with the `model.Vendor` data source, the **Statement\Party** element doesn't iterate for every record of the `model.Vendor` data source during format execution. Instead, the generated document fills with information from only the first record of the record list, if that list contains multiple records. To fix this problem, bind the **Statement\Party** element with the `model.Vendor` data source.

Executability of an expression with FILTER function (caching)

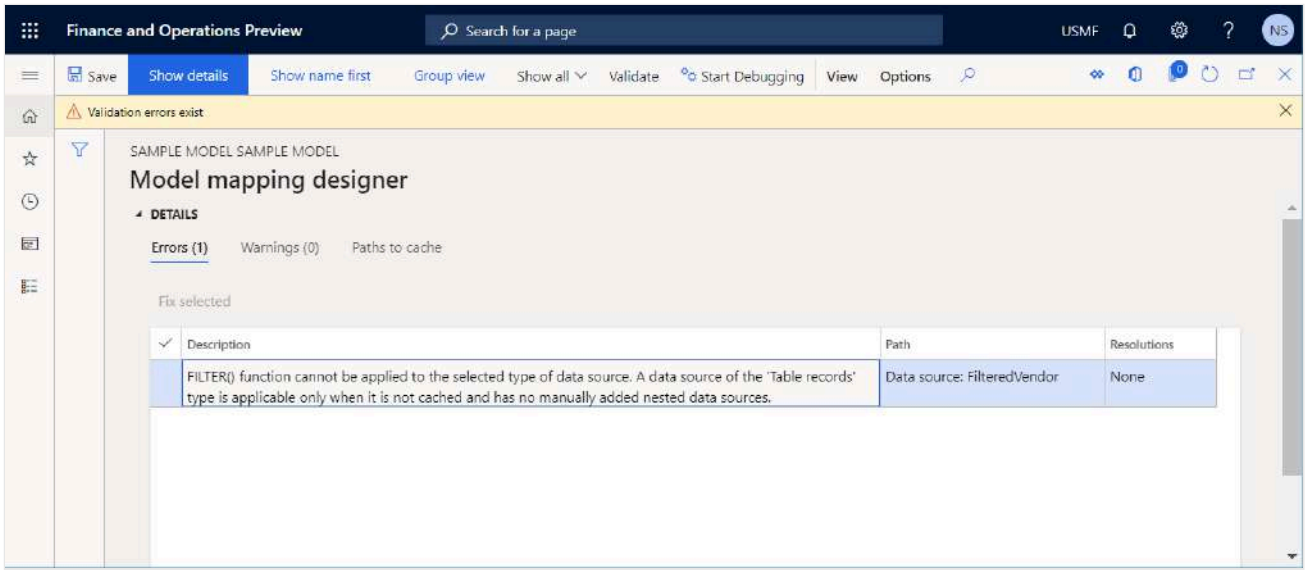
To access application tables, views, or data entities, several built-in ER functions, including FILTER and ALLITEMSQUERY, make a single SQL call to get the required data as a list of records. You use a data source of the **Record list** type as an argument for each of these functions to specify an application source for the call. ER checks whether it can establish a direct SQL call to a data source that you refer to in one of these functions. If ER can't establish a direct call because you marked the data source as cached, the ER model mapping designer shows a validation error. The message states that the ER expression that contains one of these functions can't run at runtime.

The following steps show how this issue might occur.

1. Start to configure the ER model mapping component.
2. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
3. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the VendTable table.
4. Add a data source of the **General \ User input parameter** type to search for a vendor account in the runtime dialog box.
5. Name the new data source **RequestedAccountNum**. In the **Label** field, enter **Vendor account number**. In the **Operations data type name** field, keep the default value, **Description**.
6. Add a data source of the **Calculated field** type to filter a vendor that the user inquires about.
7. Name the new data source **FilteredVendor**, and configure it so that it contains the expression `FILTER(Vendor, Vendor.AccountNum=RequestedAccountNum)`.
8. Mark the configured **Vendor** data source as cached.

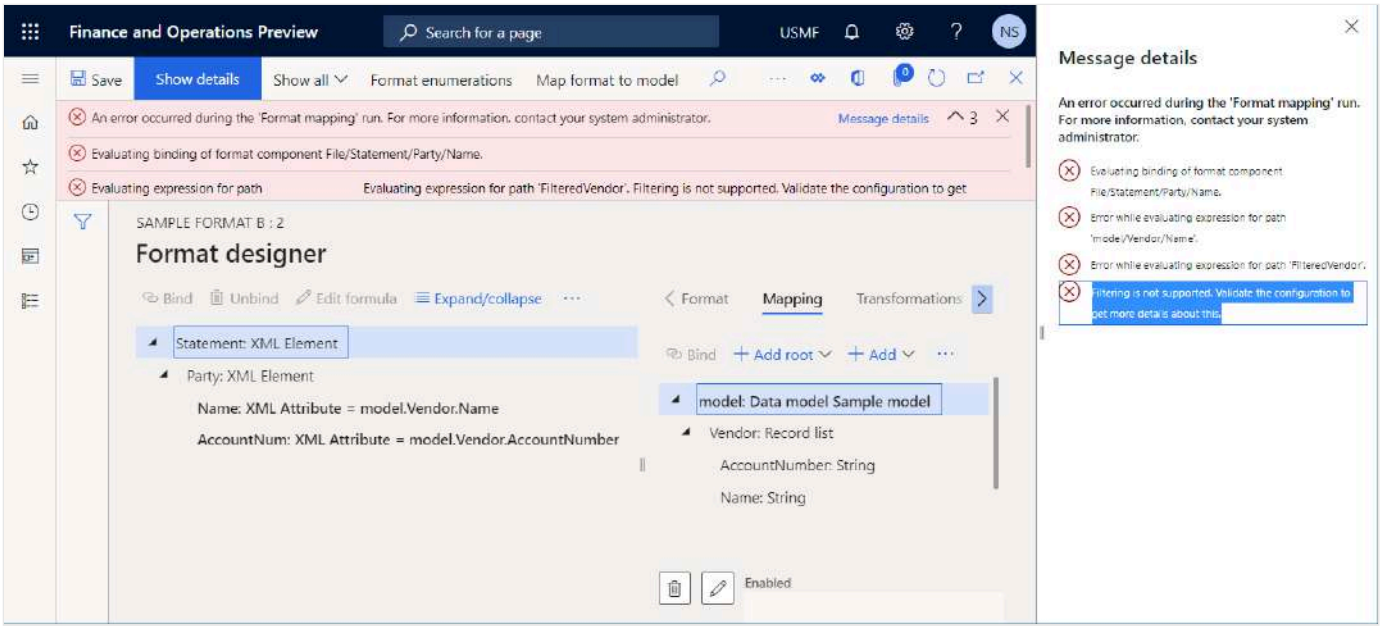


9. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page.



10. Notice that a validation error occurs. The message states that the **FILTER** function can't be applied to the cached **Vendor** data source.

The following illustration shows the runtime error that occurs if you ignore the warning and select **Run** to run the format.



Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Remove the **Cache** flag from the **Vendor** data source. The **FilteredVendor** data source then becomes executable, but the **Vendor** data source that the VendTable table refers to is accessed every time the **FilteredVendor** data source is called.

Option 2

Change the expression of the **FilteredVendor** data source from `FILTER(Vendor, Vendor.AccountNum="US-101")` to `WHERE(Vendor, Vendor.AccountNum="US-101")`. In this case, the **Vendor** data source that the VendTable table refers to is accessed only during the first call of the **Vendor** data source. However, the selection of records is done in memory. Therefore, this approach can cause poor performance.

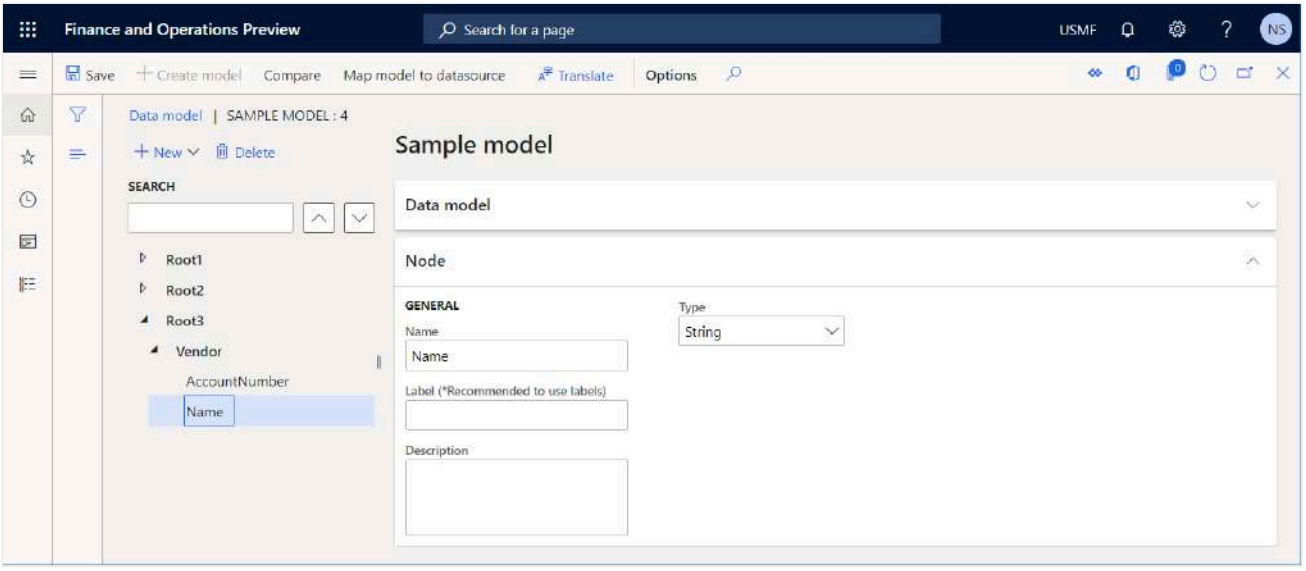
Missing binding

When you configure an ER format component, the base ER data model is the default data source for the ER format. When you run the configured ER format, the default model mapping for the base model fills the data model with application data. The ER format designer shows a warning if you bind a format element to a data model item that isn't bound to any data source in the model mapping currently selected as the default model mapping for the editable format. You can't run

this type of binding at runtime, because the format that runs can't fill a bound element with application data. Therefore, an error occurs at runtime.

The following steps show how this issue might occur.

1. Start to configure the ER data model, the ER model mapping, and the ER format components simultaneously.
2. In the data model tree, add a root item that you name **Root3**.
3. Modify the **Root3** item by adding a new nested item of the **Record list** type.
4. Name the new nested item **Vendor**.
5. Modify the **Vendor** item in the following way:
    - Add a nested field of the **String** type, and name it **Name**.
    - Add a nested field of the **String** type, and name it **AccountNumber**.



6. In the model mapping designer, in the **Data sources** pane, add a data source of the **Dynamics 365 for Operations \ Table records** type.
7. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the VendTable table.
8. Add a data source of the **General \ User input parameter** type to inquire about a vendor account in the runtime dialog box.
9. Name the new data source **RequestedAccountNum**. In the **Label** field, enter **Vendor account number**. In the **Operations data type name** field, keep the default value, **Description**.

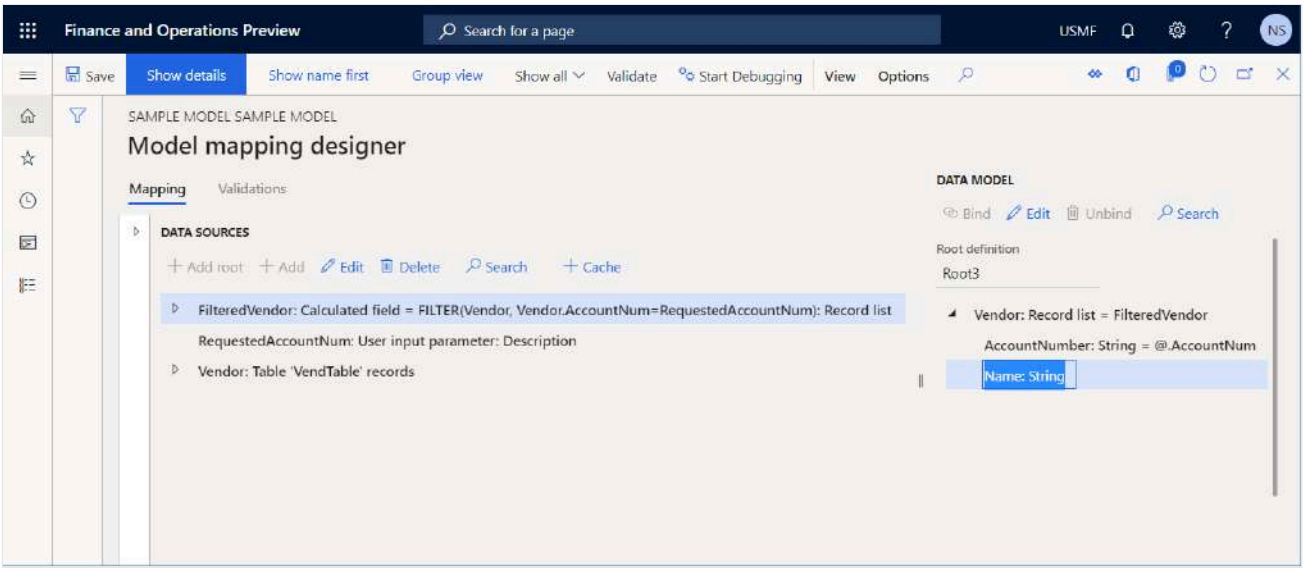
10. Add a data source of the **Calculated field** type to filter a vendor that the user inquires about.

11. Name the new data source **FilteredVendor**, and configure it so that it contains the expression `FILTER(Vendor, Vendor.AccountNum=RequestedAccountNum)`.

12. Bind the data model items to configured data sources in the following way:

    - Bind **FilteredVendor** to **Vendor**.
    - Bind **FilteredVendor.AccountNum** to **Vendor.AccountNumber**.

    > **Note**
    >
    > The **Vendor.Name** data model field remains unbound.



13. In the format structure tree, add the following items to generate an outbound document in XML format that contains the details of the vendors that the user inquires about:

    a. Add the **Statement** root XML element.

    b. For the **Statement** XML element, add the nested **Party** XML element.

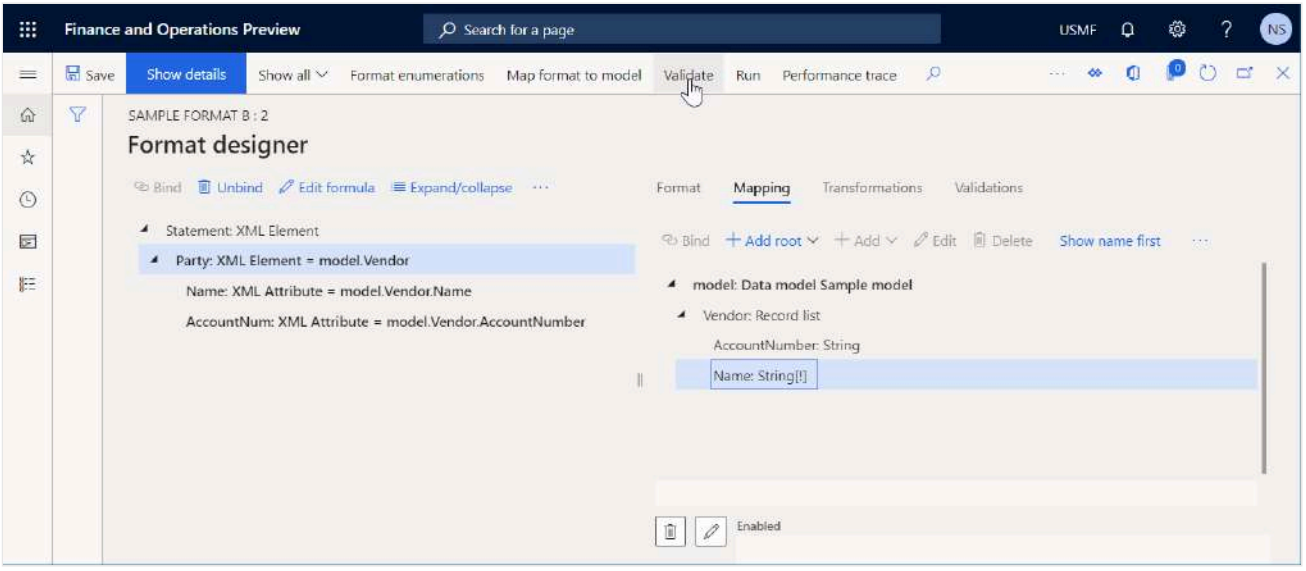
    c. For the **Party** XML element, add the following nested XML attributes:

    - Name
    - AccountNum

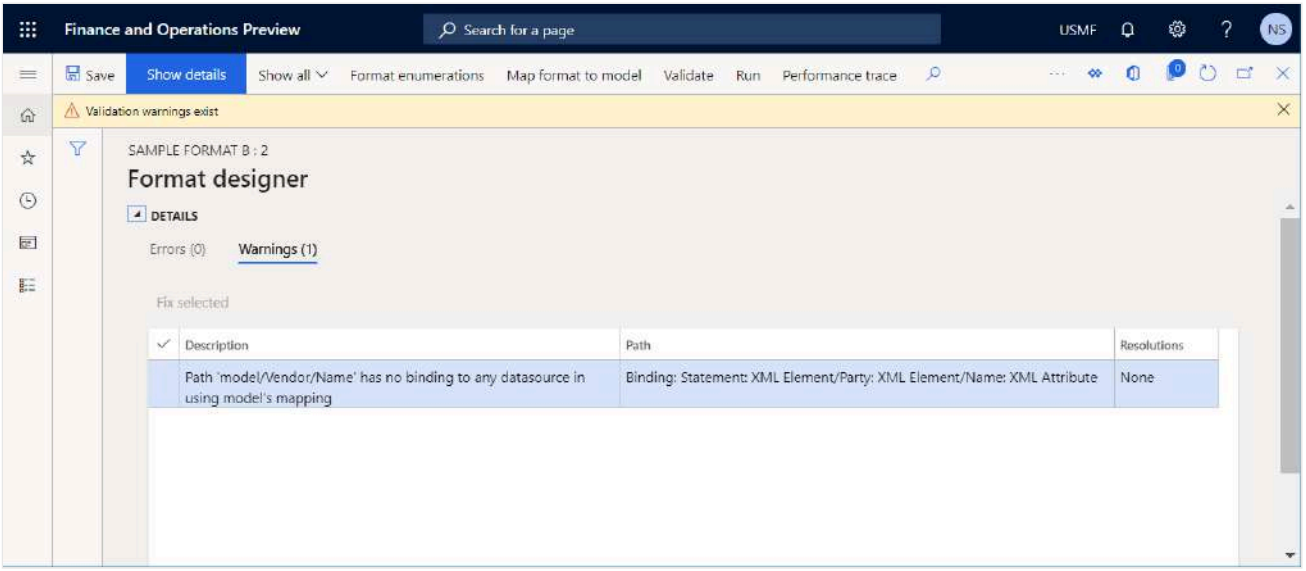
14. Bind the format elements to provided data sources in the following way:

- Bind the **Statement\Party** format element to the `model.Vendor` data source item.
- Bind the **Statement\Party\Name** format element to the **model.Vendor.Name** data source field.
- Bind the **Statement\Party\AccountNum** format element to the **model.Vendor.AccountNumber** data source field.

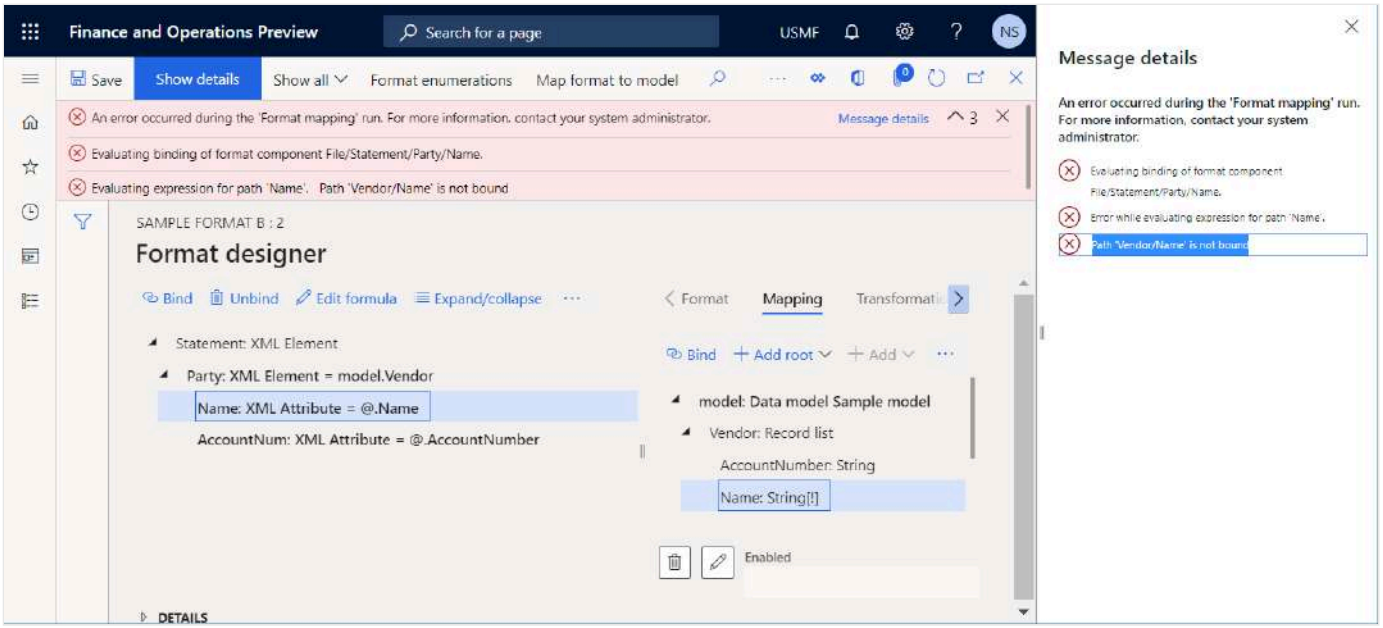
15. Select **Validate** to inspect the editable format component on the **Format designer** page.



16. Notice that a validation warning occurs. The message states that the **model.Vendor.Name** data source field isn't bound to any data source in the model mapping that you configured to be used by the format. Therefore, the **Statement\Party\Name** format element might not be filled at runtime, and a runtime exception might occur.



The following illustration shows the runtime error that occurs if you ignore the warning and select **Run** to run the format.



Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Modify the configured model mapping by adding a binding for the **model.Vendor.Name** data source field.

Option 2

Modify the configured format by removing the binding for the **Statement\Party\Name** format element.

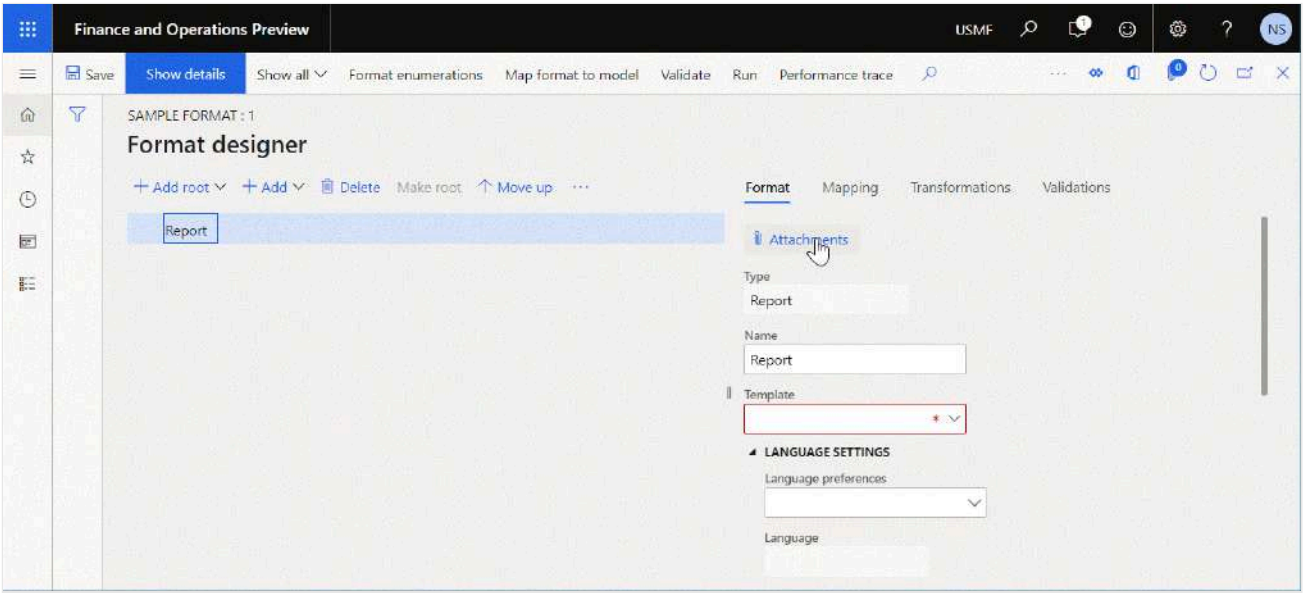
Not linked template

When you manually configure an ER format component to use a template to generate an outbound document, you must manually add the **Excel\File** element, add the required template as an attachment of the editable component, and select that attachment in the added **Excel\File** element. By using this process, you indicate that the added element fills the selected template at runtime. When you configure a format component version in **Draft** status, you can add several templates to the editable component and then select each template in the **Excel\File** element to run the ER format. By using this process, you can see how different templates are filled at runtime. If you have templates that aren't selected in any **Excel\File** elements, the ER format

designer warns you that those templates are deleted from the editable ER format component version when its status changes from **Draft** to **Completed**.

The following steps show how this issue might occur.

1. Start to configure the ER format component.
2. In the format structure tree, add the **Excel\File** element.
3. For the **Excel\File** element that you just added, add an Excel workbook file, **A.xlsx**, as an attachment. Use the document type that is configured in the ER parameters to specify the storage of ER format templates.
4. For the **Excel\File** element, add another Excel workbook file, **B.xlsx**, as an attachment. Use the same document type that you used for workbook file A.
5. In the **Excel\File** element, select workbook file A.
6. Select **Validate** to inspect the editable format component on the **Format designer** page.



7. Notice that a validation warning occurs. The message states that workbook file B.xlsx isn't linked to any components, and that it's removed after the status of the configuration version changes.

Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Modify the configured format by removing all templates that aren't linked to any **Excel\File** element.

Not synced format

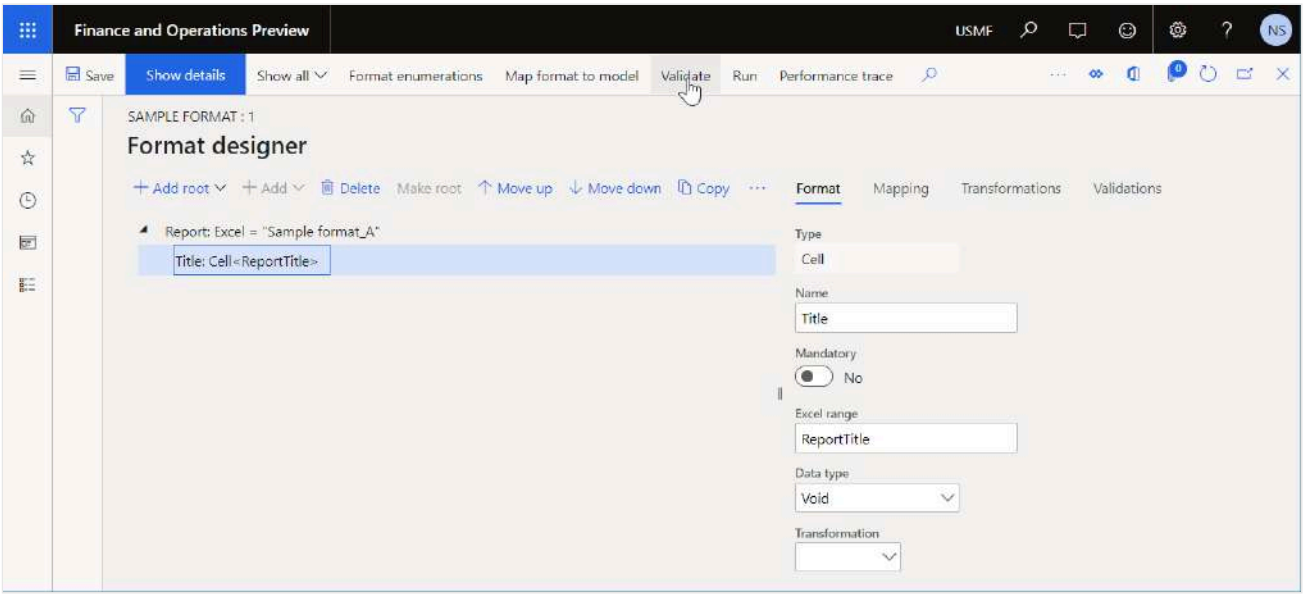
When you configure an ER format component to use an Excel template to generate an outbound document, you can manually add the **Excel\File** element, add the required template as an attachment of the editable component, and select that attachment in the added **Excel\File** element. In this way, you indicate that the added element fills the selected template at runtime. Because you designed the added Excel template externally, the editable ER format might contain Excel names that are missing from the added template. The ER format designer warns you about any inconsistencies between the properties of the ER format elements and the names that aren't included in the added Excel template.

The following steps show how this issue might occur.

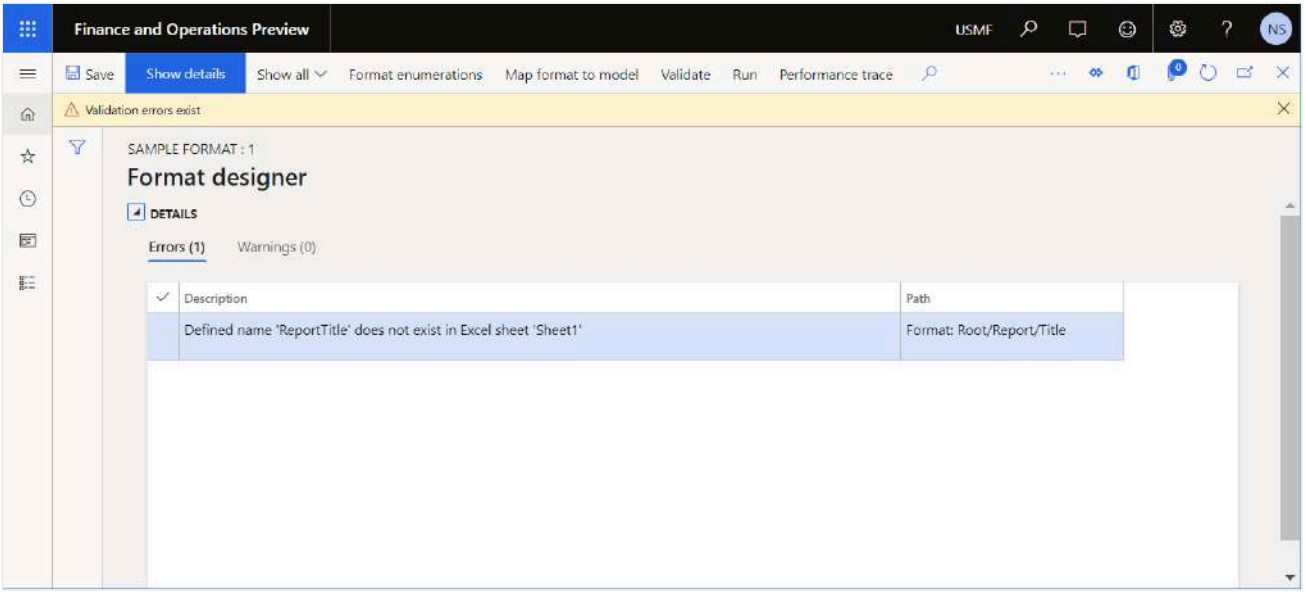
1. Start to configure the ER format component.
2. In the format structure tree, add the **Excel\File** element **Report**.
3. For the **Excel\File** element that you just added, add an Excel workbook file, **A.xlsx**, as an attachment. Use the document type that is configured in the ER parameters to specify the storage of ER format templates.

   > ⓘ **Important**
   >
   > Make sure that the added Excel workbook doesn't contain the name **ReportTitle**.

4. Add the **Excel\Cell** element **Title** as a nested element of the **Report** element. In the **Excel range** field, enter **ReportTitle**.
5. Select **Validate** to inspect the editable format component on the **Format designer** page.



6. Notice that a validation warning occurs. The message states that the name **ReportTitle** doesn't exist on sheet **Sheet1** of the Excel template that you're using.



Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Modify the configured format by removing all elements that refer to Excel names that are missing from the template.

Option 2

Update the editable ER format by importing an Excel template. The structure of the editable ER format is synced with the structure of the imported Excel template.

Additional consideration

To learn how the format structure can be synced with an ER template in the template editor of Business document management, see Update the structure of a business document template.

Not synced with a Word template format

When you configure an ER format component to use a Word template to generate an outbound document, you can manually add the **Excel\File** element, add the required Word template as an attachment of the editable component, and select that attachment in the added **Excel\File** element.

> **Note**
>
> When the Word document is attached, the ER format designer presents the editable element as **Word\File**.

In this way, you indicate that the added element fills the selected template at runtime. Because you externally design the added Word template, the editable ER format might contain references to Word content controls that are missing from the added template. The ER format designer warns you about any inconsistencies between the properties of the ER format elements and the content controls that aren't included in the added Word template.

For an example that shows how this issue might occur, see Configure the editable format to suppress the summary section.

Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Modify the configured format by deleting the **Removed** formula from the format element that the validation warning mentions.

Option 2

Modify the Word template by adding the required tag to the relevant Word content control.

No default mapping

When the Missing binding inspection runs, it evaluates the inspected format bindings against the bindings of the relevant model mapping component. You can import several ER model mapping configurations to your Finance instance, and each configuration might contain the applicable model mapping component. You must select one configuration as the default configuration. Otherwise, an exception occurs when you try to run, edit, or validate the inspected ER format. You receive the following message: "More than one model mapping exists for the <model name (root descriptor)> data model in the configurations <configuration names separated by comma>. Set one of the configurations as default."

For an example that shows how this issue might occur and how it can be fixed, see Manage several derived mappings for a single model root.

Inconsistent setting of Header or Footer components

When you configure an ER format component to use an Excel template to generate an outbound document, you can add the **Excel\Header** component to fill in headers at the top of a worksheet in an Excel workbook. You can also add the **Excel\Footer** component to fill in footers at the bottom of a worksheet. For every **Excel\Header** or **Excel\Footer** component that you add, you must set the **Header/footer appearance** property to specify the pages that the component runs for. Because you can configure several **Excel\Header** or **Excel\Footer** components for a single **Sheet** component, and you can generate different headers or footers for different type of pages in an Excel worksheet, you must configure a single **Excel\Header** or **Excel\Footer** component for a specific value of the **Header/footer appearance** property. If you configure more than one **Excel\Header** or **Excel\Footer** component for a specific value of the **Header/footer appearance** property, a validation error occurs, and you receive the following error message: "Headers/footers (<component type: Header or Footer>) are inconsistent."

Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Modify the configured format by deleting one of the inconsistent **Excel\Header** or **Excel\Footer** components.

Option 2

Modify the value of the **Header/footer appearance** property for one of the inconsistent **Excel\Header** or **Excel\Footer** components.

Inconsistent setting of Page component

When you configure an ER format component to use an Excel template to generate an outbound document, you can add the **Excel\Page** component to paginate a generated document by using ER formulas. For every **Excel\Page** component that you add, you can add many nested Range components and still remain compliant with the following structure:

- The first nested **Range** component can have the **Replication direction** property set to **No replication**. Use this range to make page headers in generated documents.
- You can add many other nested **Range** components where the **Replication direction** property is set to **Vertical**. Use these ranges to fill in generated documents.
- The last nested **Range** component can have the **Replication direction** property set to **No replication**. Use this range to make page footers in generated documents and to add the required page breaks.

If you don't follow this structure for an ER format in the ER format designer at design time, a validation error occurs, and you receive the following error message: "There are more than two range components without replication. Remove unnecessary components."

Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Modify the configured format by changing the **Replication direction** property for all inconsistent **Excel\Range** components.

Executability of an expression with ORDERBY function

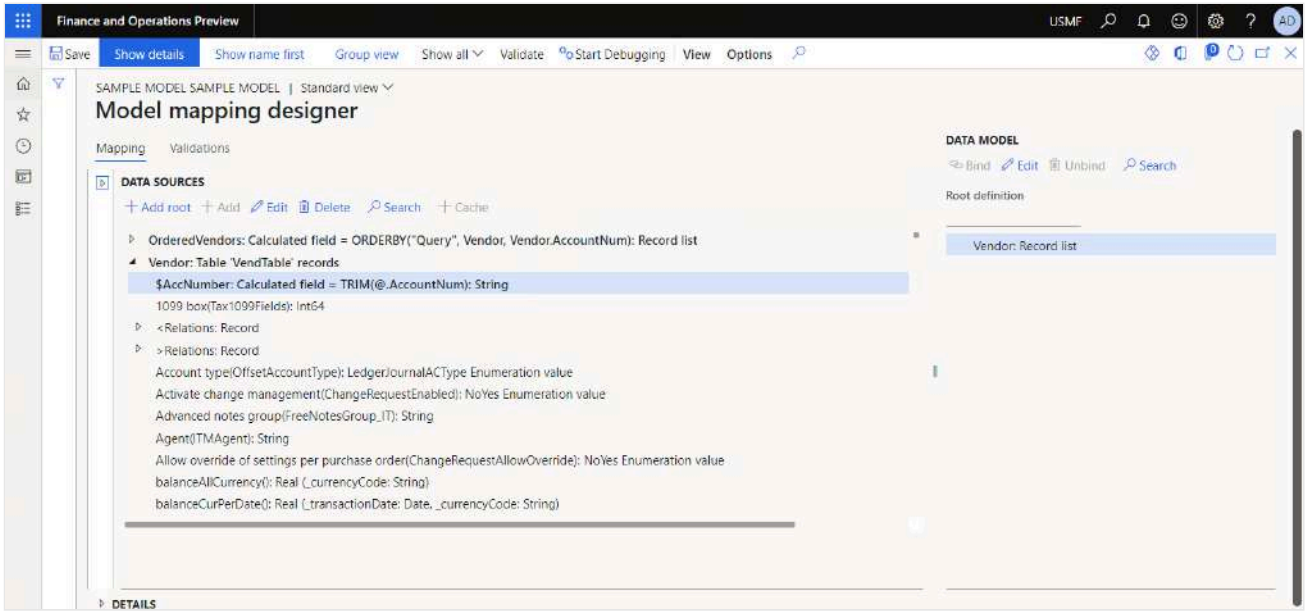
Use the built-in ORDERBY ER function to sort the records of an ER data source of the **Record list** type. You specify the data source as an argument of the function.

Specify arguments of the `ORDERBY` function to sort records of application tables, views, or data entities. By using this function, you make a single database call to get the sorted data as a list of records. Use a data source of the **Record list** type as an argument of the function to specify the application source for the call.

ER checks whether it can establish a direct database query to a data source that you refer to in the `ORDERBY` function. If ER can't establish a direct query, a validation error occurs in the ER model mapping designer. The message that you receive states that the ER expression that includes the `ORDERBY` function can't be run at runtime.

The following steps show how this issue might occur.

1. Start to configure the ER model mapping component.
2. Add a data source of the **Dynamics 365 for Operations \ Table records** type.
3. Name the new data source **Vendor**. In the **Table** field, select **VendTable** to specify that this data source requests the **VendTable** table.
4. Add a data source of the **Calculated field** type.
5. Name the new data source **OrderedVendors**, and configure it so that it contains the expression `ORDERBY("Query", Vendor, Vendor.AccountNum)`.

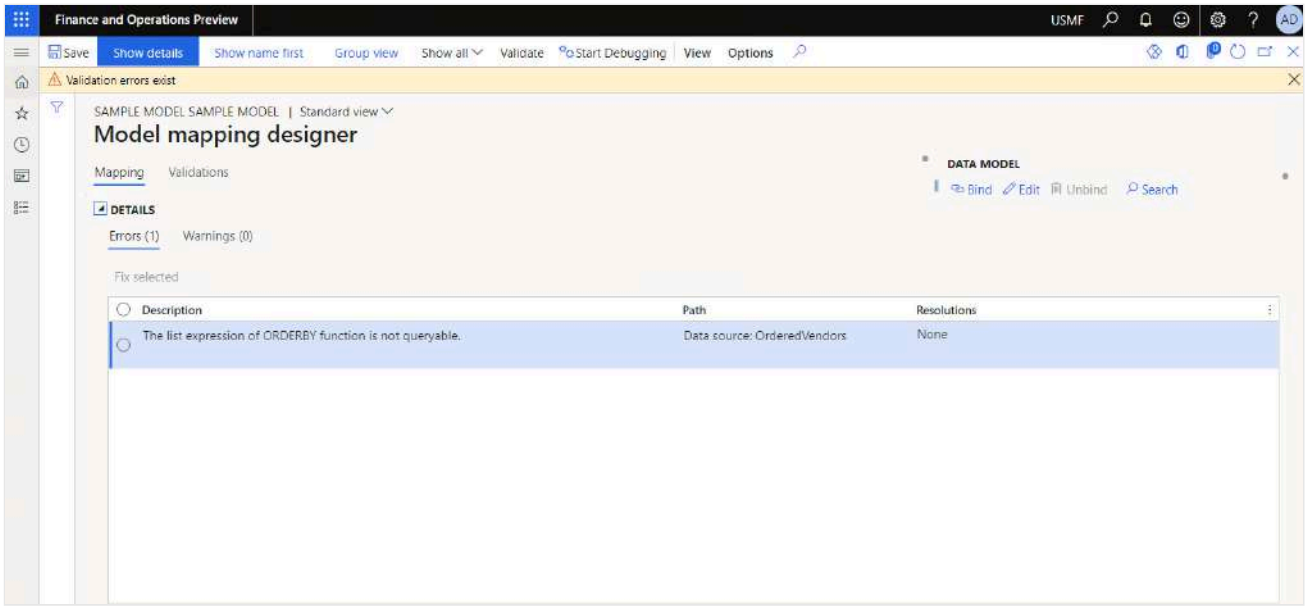


6. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page and verify that the expression in the **OrderedVendors** data source can be queried.

7. Modify the **Vendor** data source by adding a nested field of the **Calculated field** type to get the trimmed vendor account number.

8. Name the new nested field **$AccNumber**, and configure it so that it contains the expression `TRIM(Vendor.AccountNum)`.

9. Select **Validate** to inspect the editable model mapping component on the **Model mapping designer** page and verify that the expression in the **Vendor** data source can be queried.



10. Notice that a validation error occurs, because the **Vendor** data source contains a nested field of the **Calculated field** type. This nested field doesn't allow the expression of the **OrderedVendors** data source to be translated to the direct database statement. The same error occurs at runtime if you ignore the validation error and select **Run** to run this model mapping.

Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Option 1

Instead of adding a nested field of the **Calculated field** type to the **Vendor** data source, add the **$AccNumber** nested field to the **FilteredVendors** data source, and configure the field so that it contains the expression `TRIM(FilteredVendor.AccountNum)`. By using this approach, you can run the `ORDERBY("Query", Vendor, Vendor.AccountNum)` expression at the database level, and you can calculate the **$AccNumber** nested field after.

Option 2

Change the expression of the **FilteredVendors** data source from `ORDERBY("Query", Vendor, Vendor.AccountNum)` to `ORDERBY("InMemory", Vendor, Vendor.AccountNum)`. Don't change the expression for a table that has a large volume of data (transactional table), because this change fetches all records and orders the required records in memory. Therefore, this approach can cause poor performance.

Obsolete application artifact

When you design an ER model mapping component or an ER format component, you can configure an ER expression to call an application artifact in ER, such as a database table, a method of a class, and more. In Finance version 10.0.30 and later, you can configure ER to warn you when the referred application artifact is marked in source code as obsolete. This warning is useful because usually, obsolete artifacts are eventually removed from source code. Being informed about an artifact's status can stop you from using the obsolete artifact in the editable ER component before its removal from the source code. This action helps prevent errors of calling non-existing application artifacts from an ER component at runtime.

To start evaluating the obsolete attribute of application artifacts during the inspection of an editable ER component, enable the **Validate obsolete elements of Electronic Reporting data sources** feature in the **Feature management** workspace. The obsolete attribute is currently evaluated for the following types of application artifacts:

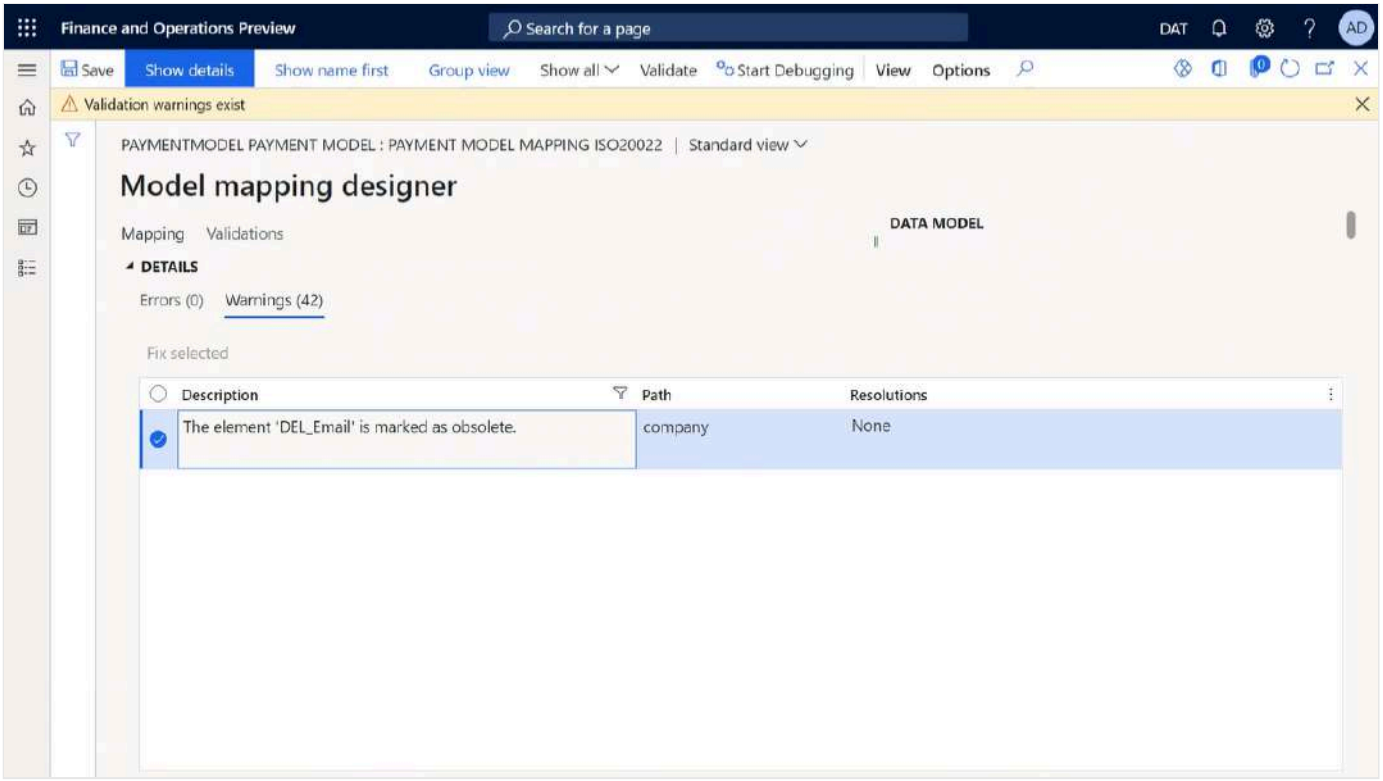
- Database table
  - Field of a table
  - Method of a table
- Application class
  - Method of a class

> ⓘ **Note**

A warning occurs during the inspection of the editable ER component for a data source that refers to an obsolete artifact only when this data source is used in at least one binding of this ER component.

> **Tip**
>
> When the SysObsoleteAttribute class is used to notify the compiler to issue warning messages instead of errors, the inspection warning presents the specified in source code warning at design time in the **Details** FastTab on the **Model mapping designer** or **Format designer** page.

The following illustration shows the validation warning that occurs when the obsolete `DEL_Email` field of the `CompanyInfo` application table is bound to a data model field by using the configured `company` data source.



Automatic resolution

No option to automatically fix this issue is available.

Manual resolution

Modify the configured model mapping or format by removing all bindings to a data source that refers to an obsolete application artifact.

Last updated on 07/22/2026

Review Electronic reporting (ER) validation history

This article describes how to review the validation history for Electronic reporting (ER) configurations. Validation history lets you return to previous validation runs to confirm which configuration versions were validated, when they were validated, and what errors or warnings were reported.

Overview

When you validate ER configurations, the system retains the results in a persistent validation history. Instead of re-running validation to confirm a configuration's state, you can open the history at any time and review the outcome of earlier runs.

The history is organized by configuration and by version, so you can review the individual errors and warnings that were reported for that run. Because the history is persistent and shared, you and other users in the environment see the same results.

Validation history is part of the enhanced validation management feature that's introduced in version 10.0.49. For information about turning on the feature and running validation, see Validate ER format and model mapping to prevent runtime issues.

Prerequisites

- Turn on the **Enhanced validation management for Electronic Reporting configurations** feature in **Feature management**.
- Complete at least one validation run so that there are results to review.

Open the validation history

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the Action Pane, select **Validation** > **Validation history**.

The **Validation history** opens and lists the validation runs that are completed in the current environment.

Review validation runs

The **History** tab shows every validation run that completed in the current environment. Each row provides the following information.

Expand table

Column	Description
Completed	The date and time when the validation run finished.
Created by	The user who started the validation run.
Mode	How the run was performed - **Interactive** or **Batch**.
Batch job	The ID of the batch job, when the run was performed in batch mode.
Configurations checked	The number of configurations that the run validated.
Passed	The number of configurations that passed validation.
Warnings	The number of configurations that reported warnings.
Failed	The number of configurations that failed validation.

To review a run, select its row in the **History** grid. The configurations that the run checked appear in the lower **Configurations** grid.

Review the configurations that were checked

When you select a run on the **History** tab, the **Configurations** grid lists the configurations that the run validated. Each row provides the following information.

Expand table

Column	Description
Configuration	The name of the validated ER configuration.
Version	The configuration version that the run validated.
Status	The status of the version, such as **Draft** or **Completed**.
Configuration type	The component type - for example, **Format**, **Model mapping**, or **Data model**.
Result	The validation result for the version, such as **Valid**.
Messages	The number of validation messages that the run reported for the version.

Show message details

To review the individual errors and warnings that a validation run reported, select **Show message details**. You can use this action in both the **History** grid and the **Configurations** grid. You can review messages for an entire run or for a single configuration version.

Delete validation runs

To remove a validation run from the history, select the run in the **History** grid, and then select **Delete**.

Because the system tracks validation for each configuration version, both users and system processes can identify which versions are already validated in the current environment. This information helps you avoid redundant validation runs and focus on the versions that still need attention.

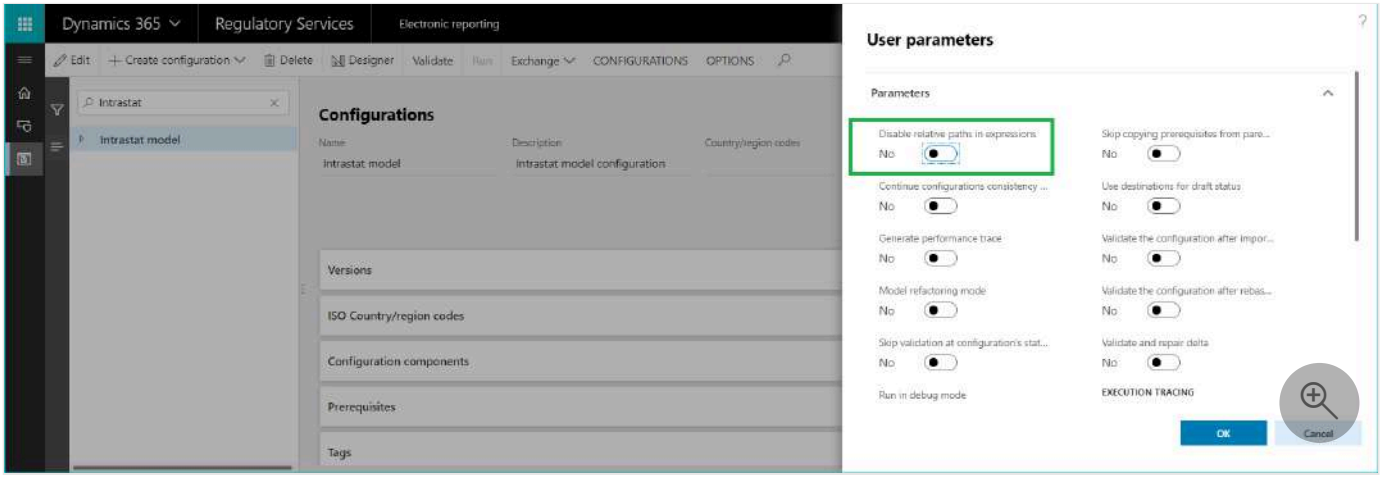
Last updated on 07/22/2026

Use a relative path in data bindings of ER models and formats

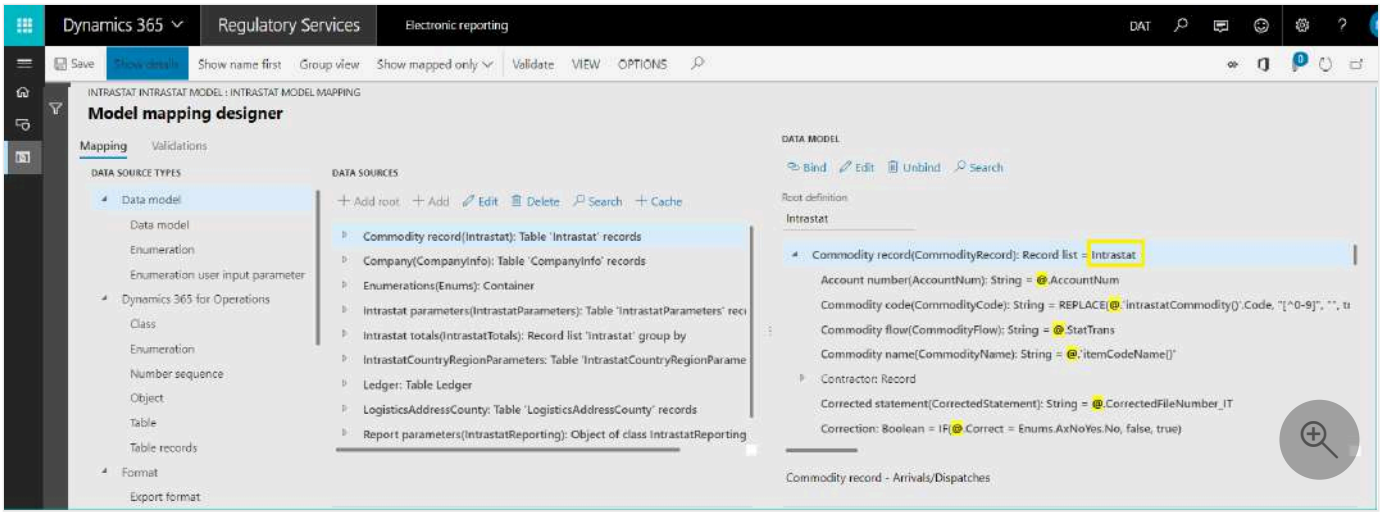
The Electronic reporting (ER) tool lets you define electronic format structures and then describe how to fill those structures by using data and algorithms that exist in the application. For more information, see Create Electronic reporting (ER) configurations. To specify the data flow for retrieving finance and operations data and using it to generate an electronic document, complete the following steps:

- Bind configured data sources to elements of the designed domain-specific data model. The model structure and selected data sources might be part of a complex hierarchical structure. Because of this complexity, final bindings can be quite large and contain many elements of different types, such as relations, tables, and methods. The bindings can become less readable and quite complex to review and understand, especially for non-owners.
- Bind data model elements with format components to define what data populates from the data model to the generated format's output.

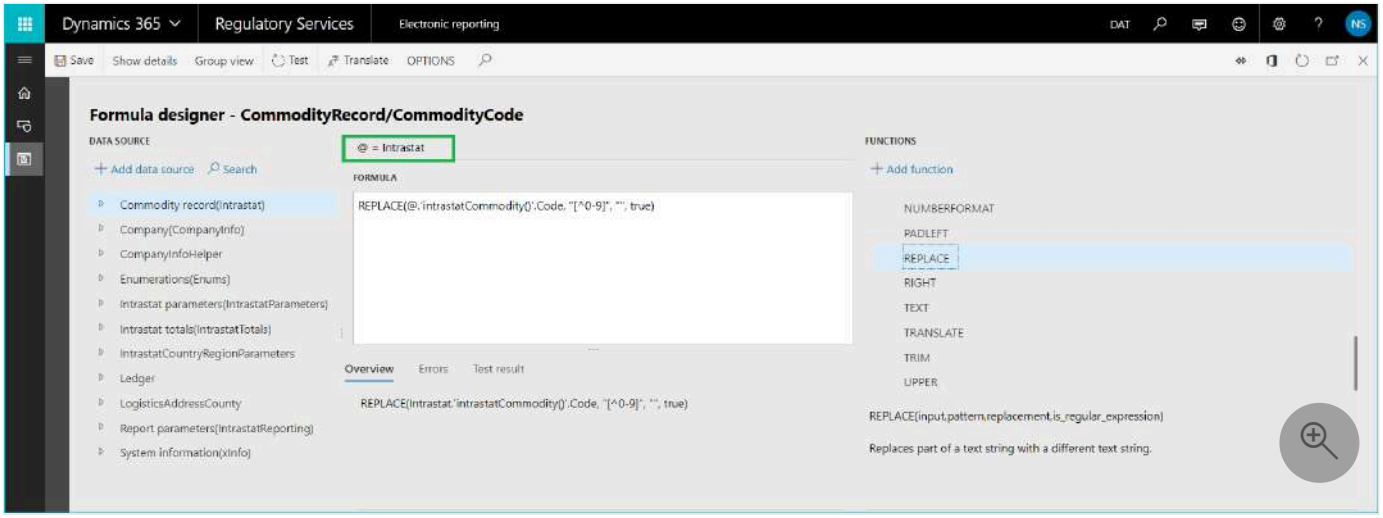
To improve the usability of ER mapping designers, the relative path feature was released. By default, the relative path representation option is turned on for any new instance of the application where ER design experience is enabled (Microsoft Dynamics 365 Finance, Microsoft Regulatory Configuration Service). The relative path parameter enables users to keep using the full path when working with this presentation of ER bindings.



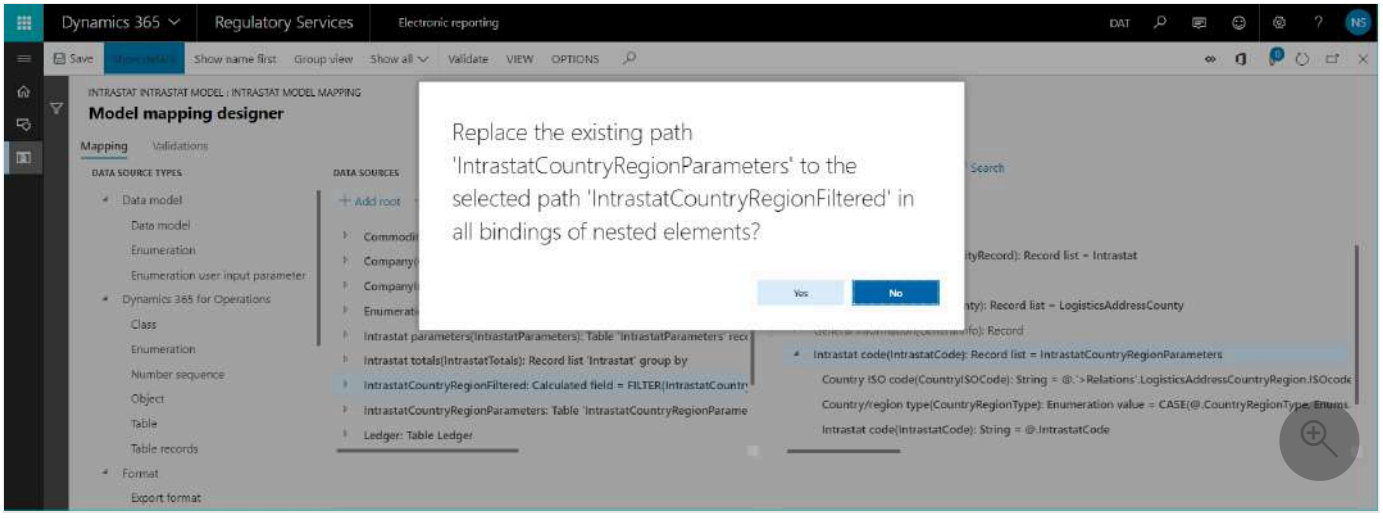
When you turn on the relative path usage parameter, a single `@` character replaces the path to the parent item in the binding of the current model element. The entire binding path becomes shorter, which makes the entire mapping more obvious and easier to understand. In most cases, no additional scrolling is required in the ER designer to view all the bindings of the data model.



When you start designing a new ER expression, you need to enter only one character to define a binding to a field of the parent item.



When you decide to change the data source of the parent model item, with absolute path usage, you have to manually rebind this model item, as well as all nested items, to a new data source. When you turn on relative path usage and select a new data source to bind to a parent item, you are offered an option to automatically rebind all nested elements of this parent item with one click.



If you confirm rebinding of nested items, the new parent item is placed to the path of each nested item containing the existing parent item. This feature doesn't break the backward compatibility of the ER framework. All previously designed ER configurations work with this new feature, and no upgrades or conversions are required.

> **Note**
>
> All changes that are introduced by mass modification of bindings of nested elements in model mappings are correctly saved in a configuration delta (trace of changes). This process allows customers to rebase their derived version of model mappings to any new base version of it that is modified by using this new feature.

Additional resources

ER formula language

Last updated on 06/26/2026

Design ER expressions to call application class methods

This article describes how to reuse existing application logic in Electronic reporting (ER) configurations by calling required methods of application classes in ER expressions. You can dynamically define values of arguments for calling classes at runtime. For example, you can base values on information in the parsing document to ensure its correctness.

For the example in this article, you design a process that parses incoming bank statements for an application data update. You receive the incoming bank statements as text (.txt) files that contain International Bank Account Number (IBAN) codes. As part of the process of importing the bank statements, you must validate the correctness of the IBAN code by using the logic that's already available.

Prerequisites

The procedures in this article are intended for users who are assigned the **System administrator** or **Electronic reporting developer** role.

You can complete the procedures by using any data set.

To complete the procedures, you must download and save the following file: SampleIncomingMessage.txt .

In this article, you will create the required ER configurations for the Litware, Inc. sample company. Therefore, before you complete the procedures in this article, you must follow these steps:

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On the **Localization configurations** page, verify that the configuration provider for the **Litware, Inc.** sample company is available and marked as active. If you don't see this configuration provider, you must first complete the steps in Create configuration providers and mark them as active.

Import a new ER model configuration

1. On **Localization configurations**, in the **Configuration providers** section, select the tile for the **Microsoft** configuration provider.

2. Select **Repositories**.
3. On **Localization repositories**, select **Show filters**.
4. To select the Global repository record, add a **Name** filter field.
5. In the **Name** field, enter **Global**. Then select the **contains** filter operator.
6. Select **Apply**.
7. Select **Open** to review the list of ER configurations in the selected repository.
8. On **Configuration repository**, in the configuration tree, select **Payment model**.
9. On the **Versions** FastTab, if the **Import** button is available, select it, and then select **Yes**.

   If the **Import** button isn't available, you already imported the selected version of the **Payment model** ER configuration.
10. Close **Configuration repository**, and then close **Localization repositories**.

Add a new ER format configuration

Add a new ER format to parse incoming bank statements in TXT format.

1. On **Localization configurations**, select the **Reporting configurations** tile.
2. On **Configurations**, in the configuration tree in the left pane, select **Payment model**.
3. Select **Create configuration**.
4. In the drop-down dialog box, follow these steps:
   a. In the **New** field, enter **Format based on data model PaymentModel**.
   b. In the **Name** field, enter **Bank statement import format (sample)**.
   c. In the **Supports data import** field, select **Yes**.
   d. Select **Create configuration** to finish creating the configuration.

Design the ER format configuration – Format

Design an ER format that represents the expected structure of the external file in TXT format.

1. For the **Bank statement import format (sample)** format configuration that you added, select **Designer**.

2. On the **Format designer** page, in the format structure tree in the left pane, select **Add root**.

3. In the dialog box that appears, follow these steps:
   a. In the tree, select **Text\Sequence** to add a **Sequence** format component.
   b. In the **Name** field, enter **Root**.
   c. In the **Special characters** field, select **New line - Windows (CR LF)**. Based on this setting, each line in the parsing file is a separate record.
   d. Select **OK**.

4. Select **Add**.

5. In the dialog box that appears, follow these steps:
   a. In the tree, select **Text\Sequence**.
   b. In the **Name** field, enter **Rows**.
   c. In the **Multiplicity** field, select **One many**. Based on this setting, the parsing file contains at least one line.
   d. Select **OK**.

6. In the tree, select **Root\Rows**, and then select **Add Sequence**.

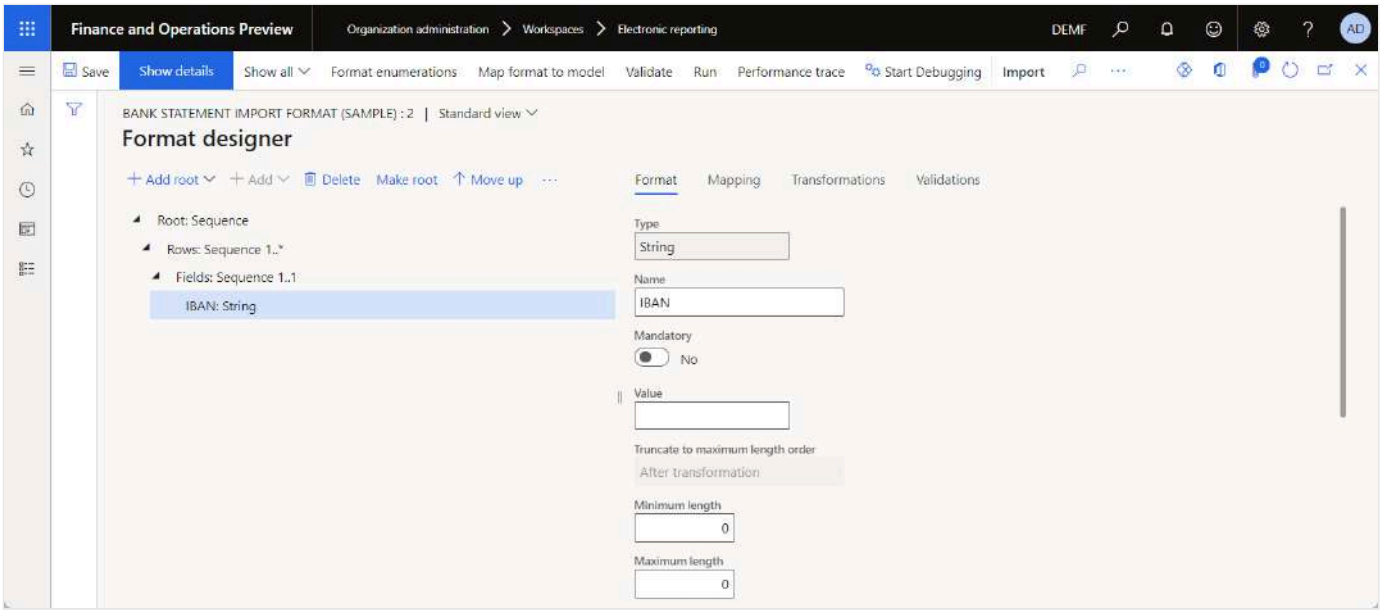
7. In the dialog box that appears, follow these steps:
   a. In the **Name** field, enter **Fields**.
   b. In the **Multiplicity** field, select **Exactly one**.
   c. Select **OK**.

8. In the tree, select **Root\Rows\Fields**, and then select **Add**.

9. In the dialog box that appears, follow these steps:
   a. In the tree, select **Text\String**.
   b. In the **Name** field, enter **IBAN**.
   c. Select **OK**.

10. Select **Save**.

The configuration is now set up so that each line in the parsing file contains only the IBAN code.



Design the ER format configuration – Mapping to a data model

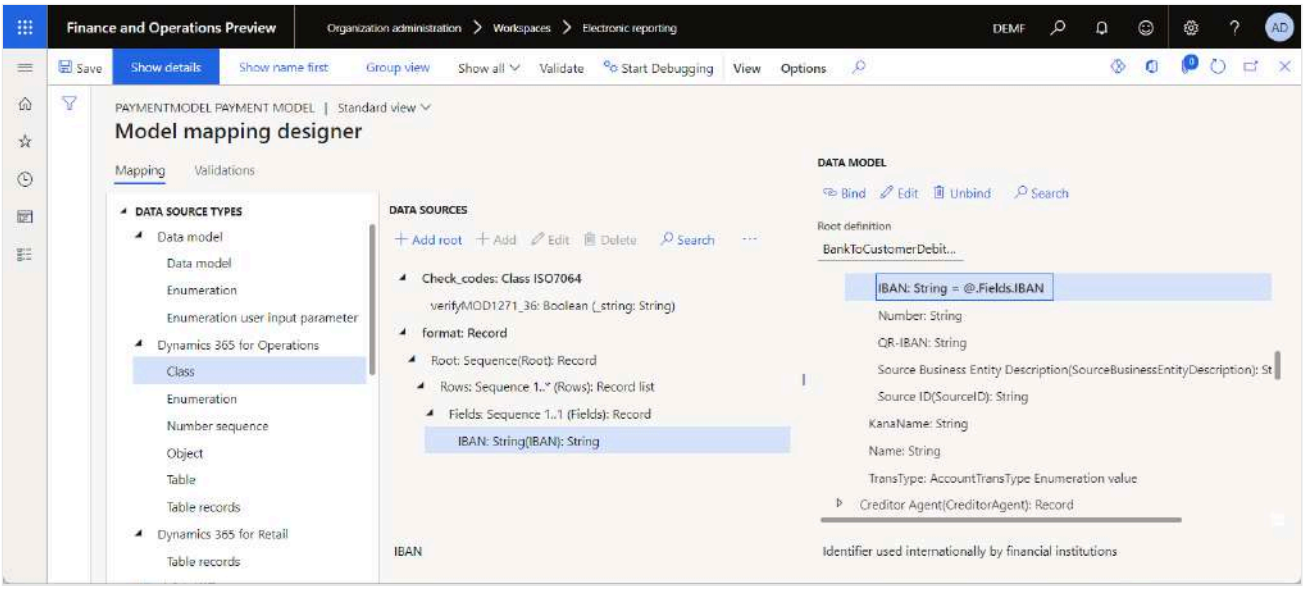
Design an ER format mapping that uses information from the parsing file to fill in a data model.

1. On the **Format designer** page, on the Action Pane, select **Map format to model**.
2. On the **Model to datasource mapping** page, on the Action Pane, select **New**.
3. In the **Definition** field, select **BankToCustomerDebitCreditNotificationInitiation**.
4. In the **Name** field, enter **Mapping to data model**.
5. Select **Save**.
6. Select **Designer**.
7. On the **Model mapping designer** page, in the **Data source types** tree, select **Dynamics 365 for Operations\Class**.
8. In the **Data sources** section, select **Add root** to add a data source that calls the existing application logic for IBAN codes validation.
9. In the dialog box that appears, follow these steps:
   a. In the **Name** field, enter **Check_codes**.
   b. In the **Class** field, enter or select **ISO7064**.
   c. Select **OK**.
10. In the **Data source types** tree, follow these steps:

a. Expand the **format** data source.
b. Expand **format\Root: Sequence(Root)**.
c. Expand **format\Root: Sequence(Root)\Rows: Sequence 1..* (Rows)**.
d. Expand **format\Root: Sequence(Root)\Rows: Sequence 1..* (Rows)\Fields: Sequence 1..1 (Fields)**.

11. In the **Data model** tree, follow these steps:
    a. Expand the **Payments** field of the data model.
    b. Expand **Payments\Creditor Account(CreditorAccount)**.
    c. Expand **Payments\Creditor Account(CreditorAccount)\Identification**.
    d. Expand **Payments\Creditor Account(CreditorAccount)\Identification\IBAN**.

12. Follow these steps to bind components of the configured format to data model fields:
    a. Select **format\Root: Sequence(Root)\Rows: Sequence 1..* (Rows)**.
    b. Select **Payments**.
    c. Select **Bind**. Based on this setting, each line in the parsing file is a single payment.
    d. Select **format\Root: Sequence(Root)\Rows: Sequence 1..* (Rows)\Fields: Sequence 1..1 (Fields)\IBAN: String(IBAN)**.
    e. Select **Payments\Creditor Account(CreditorAccount)\Identification\IBAN**.
    f. Select **Bind**. Based on this setting, the **IBAN** field of the data model is filled with the value from the parsing file.



13. On the **Validations** tab, follow these steps to add a validation rule that shows an error message for any line in the parsing file that contains an invalid IBAN code:
    a. Select **New**, and then select **Edit condition**.
    b. On the **Formula designer** page, in the **Data source** tree, expand the **Check_codes** data source that represents the **ISO7064** application class to view the available methods of

this class.

c. Select **Check_codes\verifyMOD1271_36**.

d. Select **Add data source**.

e. In the **Formula** field, enter the following expression: **Check_codes.verifyMOD1271_36(format.Root.Rows.Fields.IBAN)**.

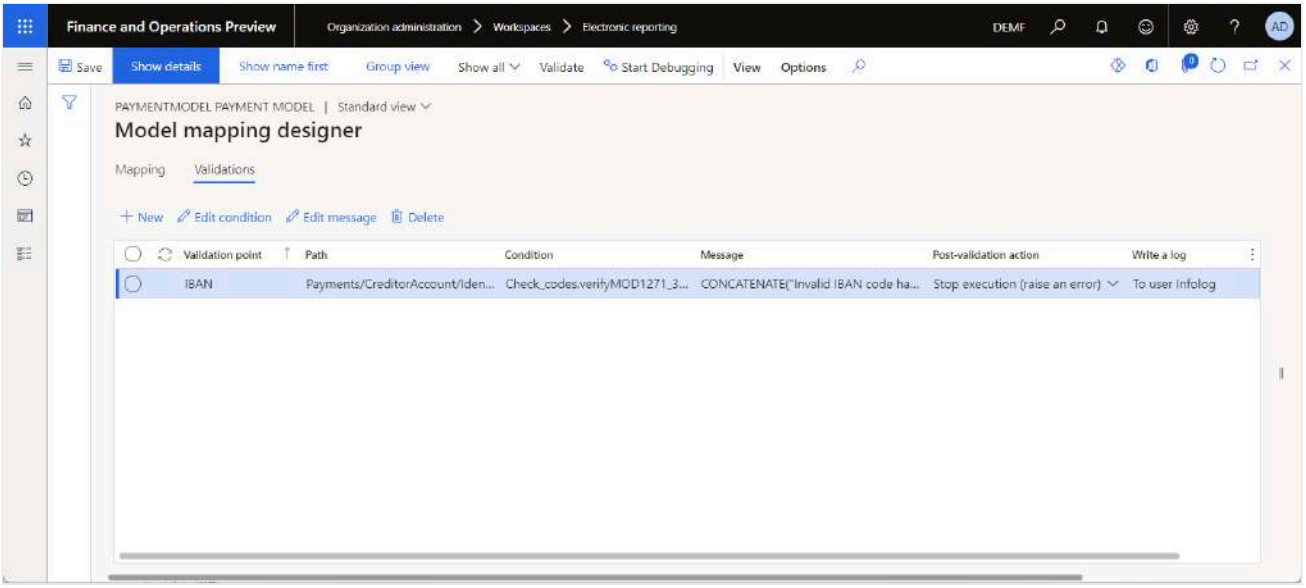
f. Select **Save**, and then close the page.

g. Select **Edit message**.

h. On the **Formula designer** page, in the **Formula** field, enter **CONCATENATE("Invalid IBAN code has been found: ", format.Root.Rows.Fields.IBAN)**.

i. Select **Save**, and then close the page.

Based on these settings, the validation condition returns *FALSE* for any invalid IBAN code by calling the existing **verifyMOD1271_36** method of the **ISO7064** application class. The value of the IBAN code is dynamically defined at runtime as the argument of the calling method, based on the content of the parsing text file.



14. Select **Save**.

15. Close the **Model mapping designer** page, and then close the **Model to datasource mapping** page.

Run the format mapping

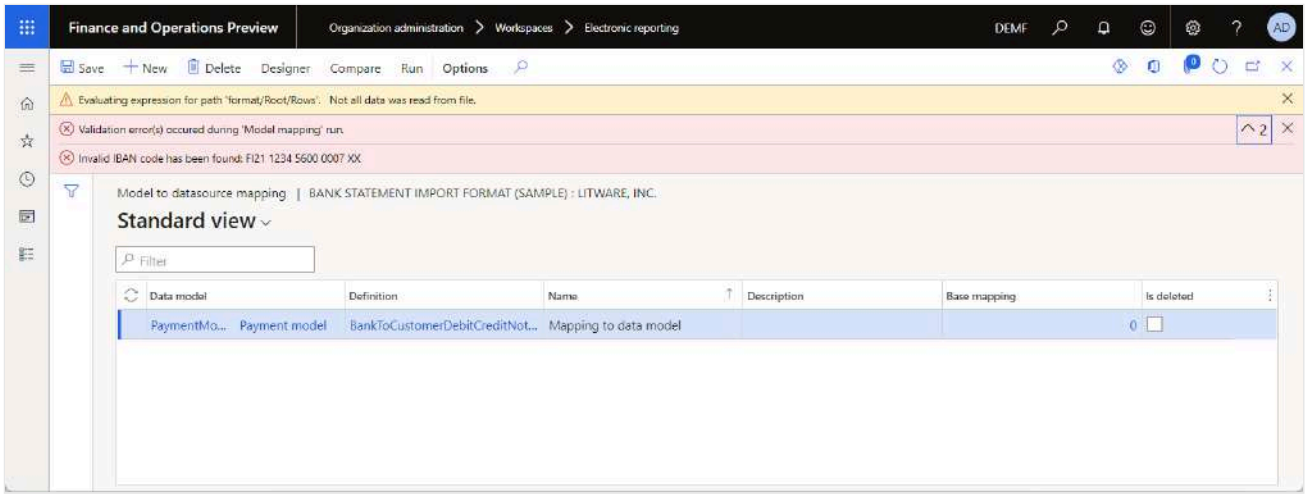
For testing purposes, run the format mapping by using the SampleIncomingMessage.txt file that you downloaded earlier. The generated output includes data that the process imports from the selected text file and ports to the custom data model during the real import.

1. On **Model to datasource mapping**, select **Run**.

2. On **Electronic report parameters**, select **Browse**, browse to the **SampleIncomingMessage.txt** file that you downloaded, and select it.

3. Select **OK**.

4. The **Model to datasource mapping** page shows an error message about an invalid IBAN code.



5. Review the output in XML format that represents the data imported from the selected file and ported to the data model. Notice that the process handles only three lines of the imported text file without errors. The IBAN code on line 4 isn't valid and is skipped.

Last updated on 04/13/2026

Configure the Electronic reporting (ER) framework

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up the basic functionality for Electronic reporting (ER). It also describes the steps that you must complete before you can set up ER.

Prerequisites for ER setup

Before you can set up ER, you must set up the required document types in Document management:

- A document type for Microsoft Office documents that are used as templates for ER reports.
- A document type that is used to store the output of ER reports in the jobs archive.
- A document type that is used to store the output of ER reports so that they can be viewed in other programs.
- A document type that is used to keep baselines of outputs of ER configurations.
- A document type that is used to handle files in the ER framework for all other purposes.

For each document type, the following attribute values can be selected.

Expand table

Attribute name	Attribute value
Class	**Attach file**
Group	**File**
Location	**Azure storage** or **SharePoint**

Set up ER

Use the following procedure to set up the basic functionality of ER for all legal entities.

1. Open the **Electronic reporting** workspace page.
2. Click **Electronic reporting parameters**.

Main parameters

On the **General** tab of the **Electronic reporting parameters** page, set the following ER parameters:

- **Enable design mode**
  - Set this option to **Yes** to enable embedded ER designers, so that users can create their own ER configurations.
  - Set this option to **No** to require that users access the functionality of ER designers by signing up for Configuration service.

- **Disable tracing of ER performance in data handling**
  - Set this option to **No** to allow Microsoft Telemetry to collect information about the average time that is required to process a single incoming or outgoing record as an ER configuration. This information is tracked as a specific health metric of the environment, and it will help Microsoft quickly identify and address issues that affect customers who use the ER framework.
  - Set this option to **Yes** to stop collecting telemetry information.

Parameters to manage documents

On the **Attachments** tab of the **Electronic reporting parameters** page, set the following ER parameters:

- **Configurations** – Select a document type to specify the storage of templates of ER formats. This document type is selected within the scope of a specific company. This document type will be used, regardless of the company that a user is signed in to while an ER format is being used.

- **Job archive** – Select a document type to specify the storage of generated documents that are attached to the records of the ER jobs archive. This document type is selected as a company-specific document type. You must make sure that the selected document type is

configured for every company that you plan to run ER formats for, and then store the results in the ER jobs archive.

- **Temporary** – Select a document type to specify the storage of generated documents that are used for other purposes. For example, documents might be generated for preview by other services. This document type is selected as a company-specific document type. You must make sure that the selected document type is configured for every company that you plan to run ER formats for, and then store the results in the ER jobs archive.
- **Baseline** – Select a document type to specify the storage of documents that are used as baselines during the automated testing of ER configurations. This document type is selected as a company-specific document type. You must make sure that the selected document type is configured for every company that you plan to run ER formats for, and then store the results in the ER jobs archive.
- **Others** – Select a document type to specify the storage of generated documents that are used for all other purposes. This document type is selected as a company-specific document type. You must make sure that the selected document type is configured for every company that you plan to run ER formats for, and then store the results in the ER jobs archive.
- **Stop making backup copies of templates**
    - Set this option to **No** to automatically create a backup copy of any ER format configuration template and store the copy in the database storage.
    - Set this option to **Yes** to stop making backup copies of ER formation configuration templates.

      For more information, see Backup storage of ER templates.

Runtime parameters

On the **Runtime** tab of the **Electronic reporting parameters** page, set the following ER parameters:

- **Autofit row height** - When selected, this setting ensures that the height of a row configured in an Excel template as an *AutoFit Row Height* and containing at least one merged cell with wrap text alignment, is automatically adjusted at runtime. This adjustment occurs when the merged cell is filled with large text, allowing all wrapped content to be fully visible without manual resizing.

- **Query timeout (seconds)** - As of version 10.0.46 of Finance, the ER engine supports manual configuration of query timeouts during runtime. The default value of 0 retains the system's standard timeout, but users can now define custom duration for queries in ER runtime, improving flexibility and execution control.

Active ER configurations provider

On the **Configuration provider table** page, create ER provider records. Each provider can be marked as **Active**. The active provider's name and internet address are stored in an ER configuration, as attributes of the configuration owner.

Optional setup for ER

In addition to the basic functionality, ER has other functionality that you can set up.

- On the **Electronic reporting destination** page, define the ER output destinations for each file output of each ER format configuration. Use the document types of the Document management framework that you set up earlier. You can also use this page to set up the optional ER functionality for each legal entity. For more information, see Electronic reporting (ER) destinations.
- When you add new Application Object Tree (AOT) artifacts or update existing AOT artifacts that are used as data sources (tables, views, or data entities) in ER, use the **Rebuild table references** menu item (**Organization administration** > **Electronic reporting** > **Rebuild table references**) to bring your AOT changes into the ER metadata.

Frequently asked questions

Question: I have added a custom table to the AOT. I created a new ER model mapping configuration for my ER data model. During the design of the model mapping, I tried to add a new data source type, **Table records**, that refers to my table. I could manually add my table name to the **Table** lookup, and the ER model mapping accepted it without errors or warnings. However, my table's name isn't included in the list of available choices that the **Table** lookup of this data source offers. How do I include the name of my table?

Answer: To include the name of your custom table in the **Table** lookup, use the **Rebuild table references** menu item as described in the "Optional setup for ER" section earlier in this article.

Question: Why can't I mark the Microsoft provider as **Active** in the **Electronic reporting** workspace in my production environment?

Answer: The Microsoft provider is used to mark ER configurations that have been designed and maintained by Microsoft. We expect that Microsoft will release new versions of the configurations in the future. We recommend that you not mark the Microsoft provider as **Active**. Otherwise, you can update the configurations. (For example, you can change the content and register new versions.) These updates will cause issues in the future, when Microsoft provides new versions of the configurations, and those new versions must be imported and adopted. Instead, register a new ER provider for your company, and use it for your ER configurations maintenance. To reuse a Microsoft configuration, select it as the base for your derived copy. To incorporate changes that are provided by Microsoft, rebase your configuration to a new version of the Microsoft configuration when it becomes available.

Question: I successfully ran an ER format in one company. However, when I ran the same ER format in another company and used the same settings, I received the following error message: "The provided document type is not a File type." Why?

Answer: Most likely, the second company doesn't contain document types that have been selected in the **Job archive**, **Temporary**, **Baseline**, and **Others** ER parameters. To fix this issue, configure these document types in the second company.

Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) destinations
- Configuration service of Regulatory Services

Last updated on 04/09/2026

Backup storage of ER templates

By using the Electronic reporting (ER) overview, business users can configure formats for outbound documents that meet the legal requirements of various countries and regions. Configured ER formats can use predefined templates to generate outbound documents in various formats, such as Microsoft Excel workbooks, Microsoft Word documents, or PDF documents. The templates are filled with data that the configured dataflow for generated documents requires.

You can publish each configured format as part of an ER solution. You can export each ER solution from one instance of finance and operations and import it into another instance.

The ER framework uses the Configure document management to keep the required templates for the current finance and operations instance. Depending on the settings of the ER framework, Microsoft Azure Blob storage or a Microsoft SharePoint folder can be selected as the physical primary storage location for templates. (For more information, see Configure the Electronic reporting (ER) framework.) The DocuValue table holds an individual record for each template. In each record, the **AccessInformation** field stores the path of a template file that is located in the configured storage location.

When you manage your finance and operations instances, you might decide to migrate the current instance to another location. For example, you might migrate your production instance to a new sandbox environment. If you configure the ER framework to store templates in Blob storage, the DocuValue table in the new sandbox environment refers to the instance of Blob storage in the production environment. However, you can't access this instance from the sandbox environment, because the migration process doesn't support the migration of artifacts in Blob storage. Therefore, if you try to run an ER format that uses a template to generate business documents, an exception occurs, and you're notified about the missing template. You're also guided to use the ER cleanup tool to delete and then re-import the ER format configuration that contains the template. Because you might have several ER format configurations, this process can be time consuming.

The Backup storage of ER templates feature can help you make your templates always available to generate business documents.

> ⓘ **Note**

You can use this feature only when Blob storage is selected as the physical storage location for ER templates.

Automated recovery and notification

For this feature, every template of a new ER format configuration in the current environment is automatically saved to the backup storage location for templates (the ERDocuDatabaseStorage database table) when the following events occur:

- You import a new ER format configuration that contains a template.
- You complete the draft version of an ER format configuration that contains a template.

Backup copies of templates are migrated to a new instance of finance and operations as part of the application database.

If an ER format template is required to generate outbound documents, such as to process vendor payments including generation of payment advice and control reports, but the required template isn't found in the primary storage location, the following events occur:

- If the template is available in the backup storage location, it is automatically taken from the backup storage location, restored to the primary storage location, and used for the current execution.

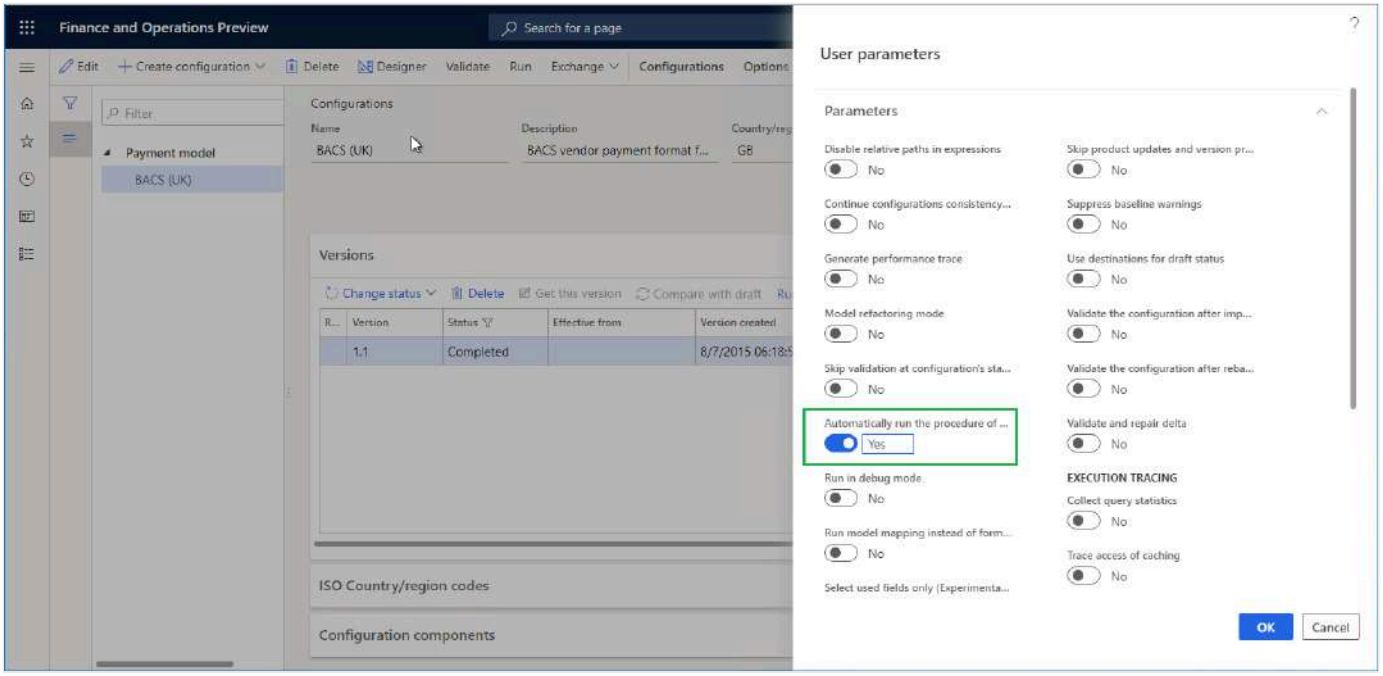
- Every user who is assigned to the **Electronic reporting developer** or **System administrator** role is notified about the missing template issue through the Action center. The message that appears depends on the value of the **Automatically run the procedure of restoring the broken templates in batch** parameter:
  - If this parameter is set to **Off**, the message recommends that you start the batch process to automatically fix similar issues for other ER format configuration templates. The message includes a link that you can use to start the batch process.
  - If this parameter is set to **On**, the message notifies you that a missing templates issue has been discovered, and that a new batch process, **Restore broken templates from internal database backup**, has been automatically scheduled. This batch process automatically fixes similar issues for other templates.

To set up the **Automatically run the procedure of restoring the broken templates in batch** parameter, complete the following steps:

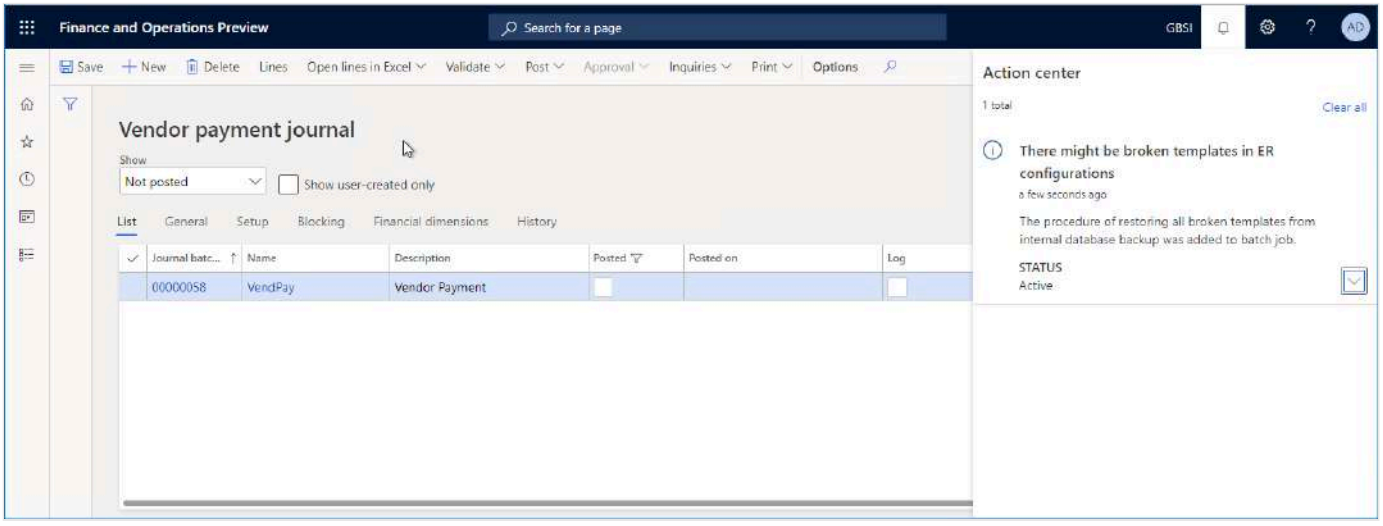
1. In Finance and Operations, open the **Organization administration** > **Electronic reporting** > **Configurations** page.

2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In the **User parameters** dialog box, set the required value for the **Automatically run the procedure of restoring the broken templates in batch** parameter.

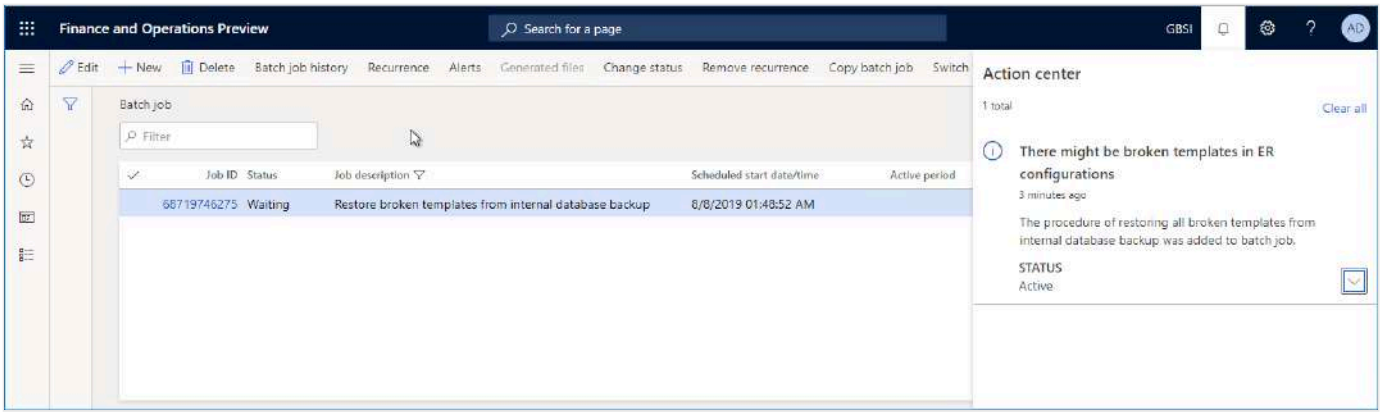
> **Note**
>
> This parameter is defined as application user and logged company specific.



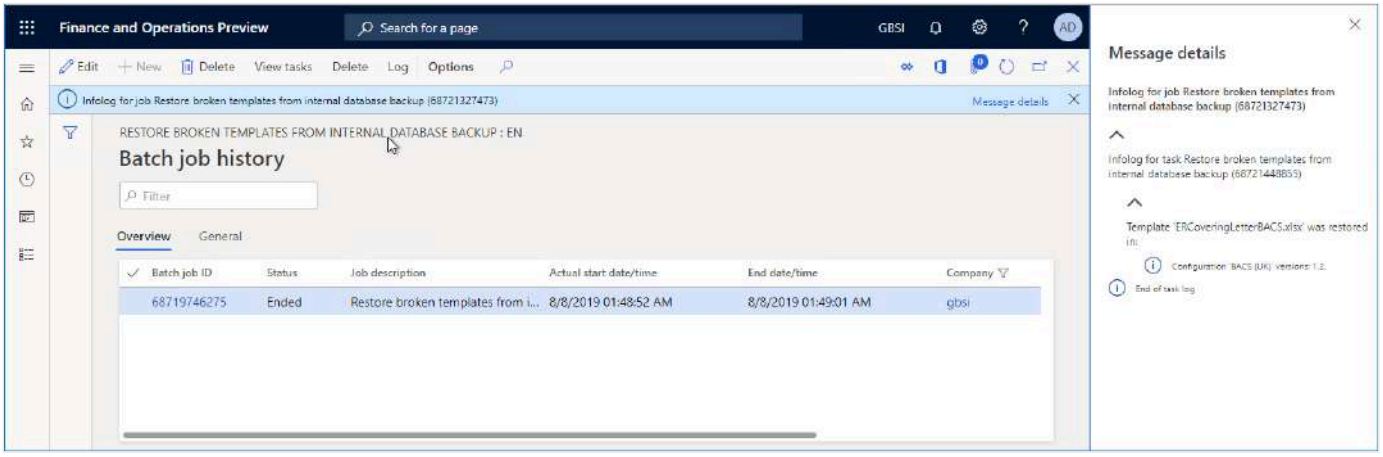
The following illustration shows an example of the message that appears when the **Automatically run the procedure of restoring the broken templates in batch** parameter is set to **On**.



The following illustration shows the **Restore broken templates from internal database backup** batch process on the **Batch job** page.



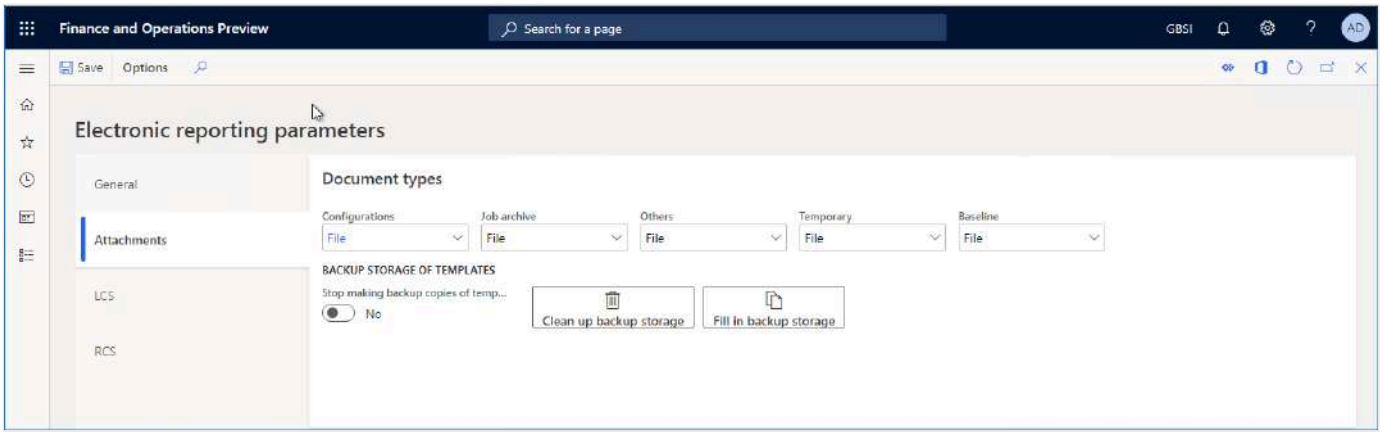
The execution log of the completed **Restore broken templates from internal database backup** batch process includes information about the templates that the process restores from the backup storage location to the primary storage location.



By default, the process of automatically creating backup copies of templates that reside in ER format configurations is turned on. To stop making backup copies of templates, set the **Stop making backup copies of template** option to **Yes** on the **Attachments** tab of the **Electronic reporting parameters** page. You can open this page from the **Electronic reporting** workspace.

If you set the **Stop making backup copies of templates** option to **Yes** and don't want to keep the backup copies that were previously made of templates, select **Clean up backup storage** on the **Electronic reporting parameters** page.

If you upgraded your environment to Finance and Operations version 10.0.5 (October 2019) and want to migrate to a new environment that includes ER format configurations that can be run, select **Fill in backup storage** on the **Electronic reporting parameters** page before the migration occurs. This button starts the process of making backup copies of all available templates, so that they can be stored in the ER backup storage location for templates.



Manual recovery

Go to **Organization administration** > **Electronic reporting** > **Restore broken templates** to manually restore ER templates from the backup storage location to the primary storage location. Before you start this process, use the **Restore broken templates** page to specify whether to perform the restoration interactively or schedule it as a batch process.

Supported deployments

In finance and operations version 10.0.5, the **Backup storage of ER templates** feature is available only in cloud deployments.

Additional resources

- Electronic reporting (ER) overview
- Configure the Electronic reporting (ER) framework

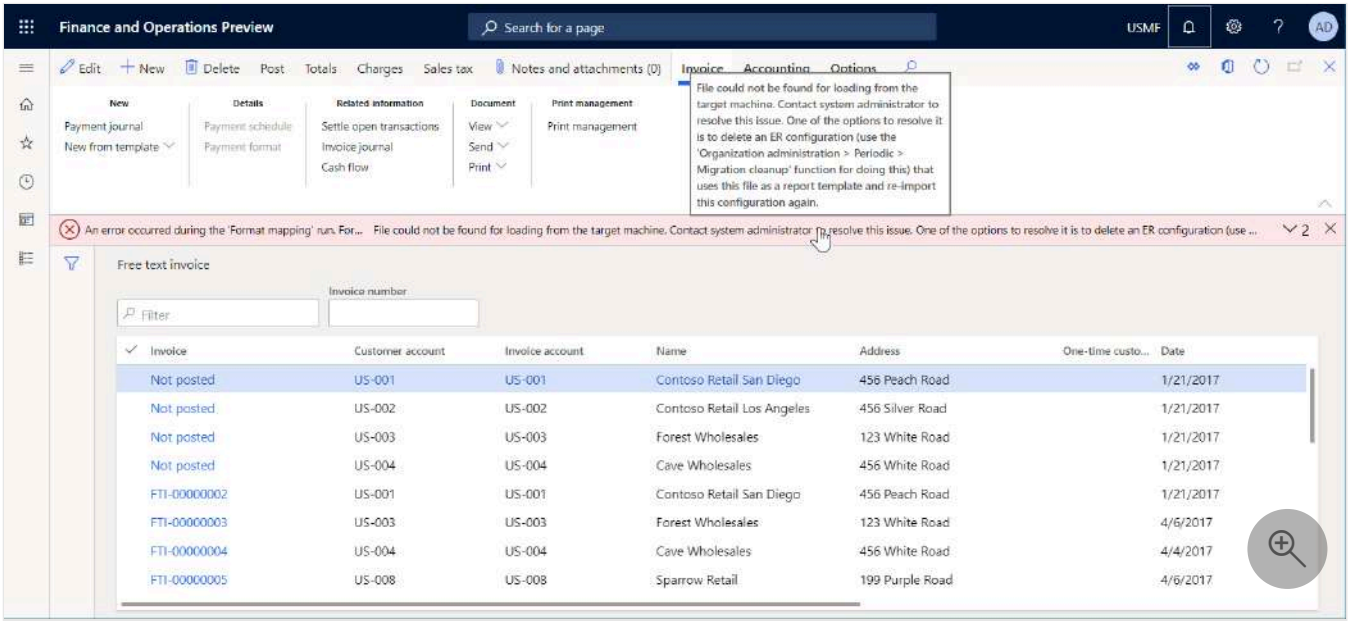
Last updated on 06/26/2026

ER migration cleanup

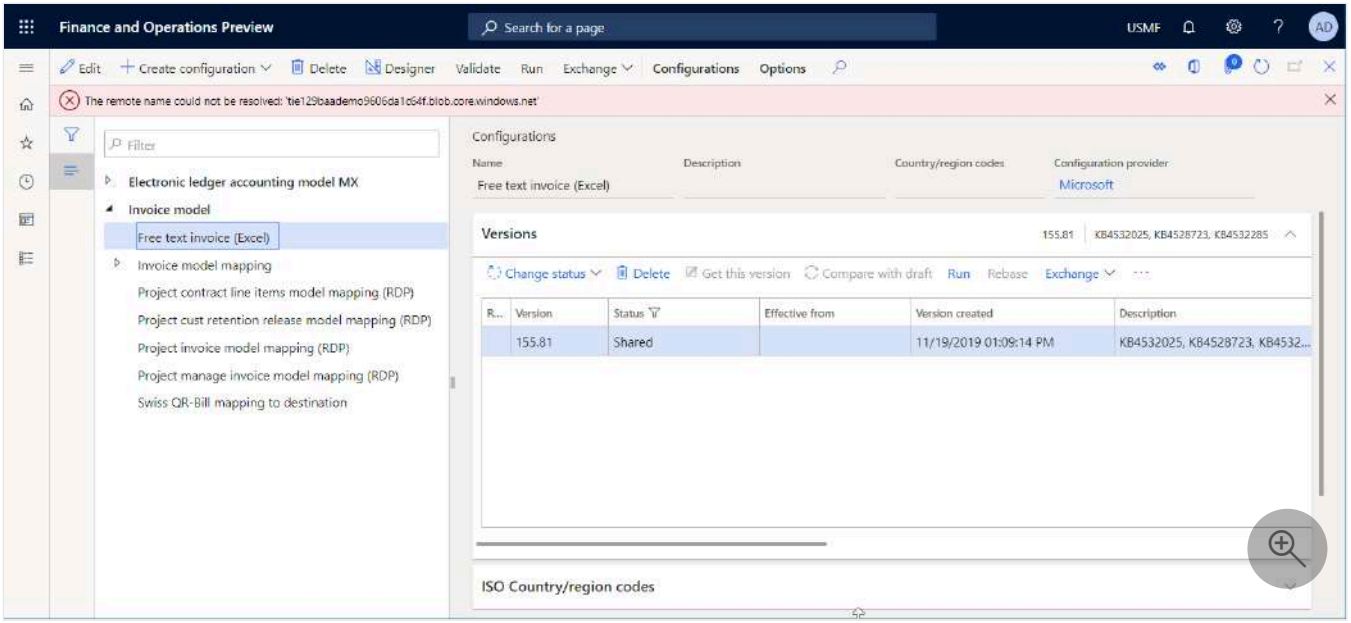
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you manage your Finance instances, you might decide to migrate your current instance to another location. For example, you might migrate your production instance to a new sandbox environment. If you configured the Electronic reporting (ER) framework to store templates in Microsoft Azure Blob storage, the **DocuValue** table in the new sandbox environment refers to the instance of Blob storage in the production environment. However, you can't access this instance from the sandbox environment because the migration process doesn't support the migration of artifacts in Blob storage. Instead, in the new sandbox environment, you refer to the instance of Blob storage in the sandbox environment that doesn't yet have the ER templates.

If you try to run an ER format that uses a template to generate business documents, an exception occurs, and you're notified about the missing template. You're also guided to use the ER migration cleanup option to delete and then re-import the ER format configuration that contains the template.



You receive a similar error if you go to the **Configurations** page (**Organization administration** > **Electronic reporting** > **Configurations**) and in the configurations tree, try to delete an ER format configuration that uses a template.



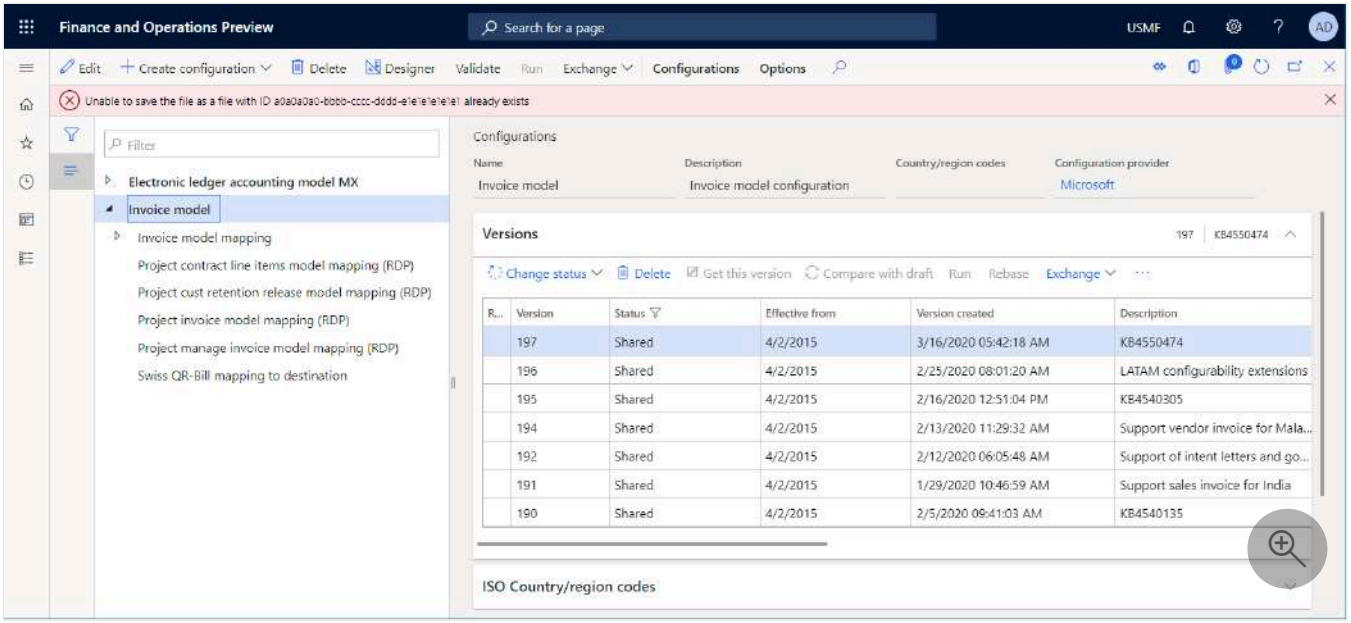
Complete the following steps to resolve issues with ER templates that you can't access.

1. Go to **Organization administration** > **Periodic** > **Migration cleanup**.
2. Select an ER format configuration that you can't execute or delete.
3. Select **Delete**.
4. Confirm the deletion of the selected ER format configuration.
5. Import the deleted ER format configuration to the current Finance instance.

Applicability

> **Important**
>
> The **Migration cleanup** option is only for ER format configurations that contain non-accessible ER templates. When you use the **Migration cleanup** option to delete an ER format configuration, ER deletes the templates that are related to the configuration artifacts in the only application database. ER doesn't validate the existence of the appropriate physical files in Blob storage. Instead, ER assumes that there are none. Therefore, don't use the **Migration cleanup** option as an alternative to the ER configuration deletion option on the **Configurations** page. Use the **Migration cleanup** option only when the ER configuration deletion option on the **Configurations** page failed.
>
> If you use the **Migration cleanup** option to delete an ER format configuration when the referred template is available in the Blob storage, you only delete related configuration

artifacts in the application database. The physical file of the template in the Blob storage remains. File overwriting in Blob storage is no longer allowed. For more information, see KB4557217. Additionally, you can't re-import the configurations deleted by using the Migration cleanup in this environment. To resolve this issue, you need to find the corresponding file in Blob storage and manually delete it.



A similar problem can occur if you migrate your application instance to another location that you used as a migration target more than once and for which the Blob storage already contains ER template files.

Because you can have several ER format configurations, this process can be time consuming. Therefore, using the Backup storage of ER templates feature to automatically recover templates with broken references is preferred.

Last updated on 04/13/2026

Create configuration providers and mark them as active

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user assigned to the System Administrator or Electronic Reporting Developer role can create a configuration provider for Electronic reporting (ER). Each ER configuration refers to the provider as the author of the configuration. In this example, you create a configuration provider for sample company, Litware, Inc. You can perform these steps in any company because ER configuration providers are shared among all companies.

Create a provider

1. Go to the **navigation pane** in the upper left corner and select **Organization administration**.
2. Go to **Workspaces** > **Electronic reporting**.
3. Go to **Related links** > **Configuration providers**.
4. Select **New**.

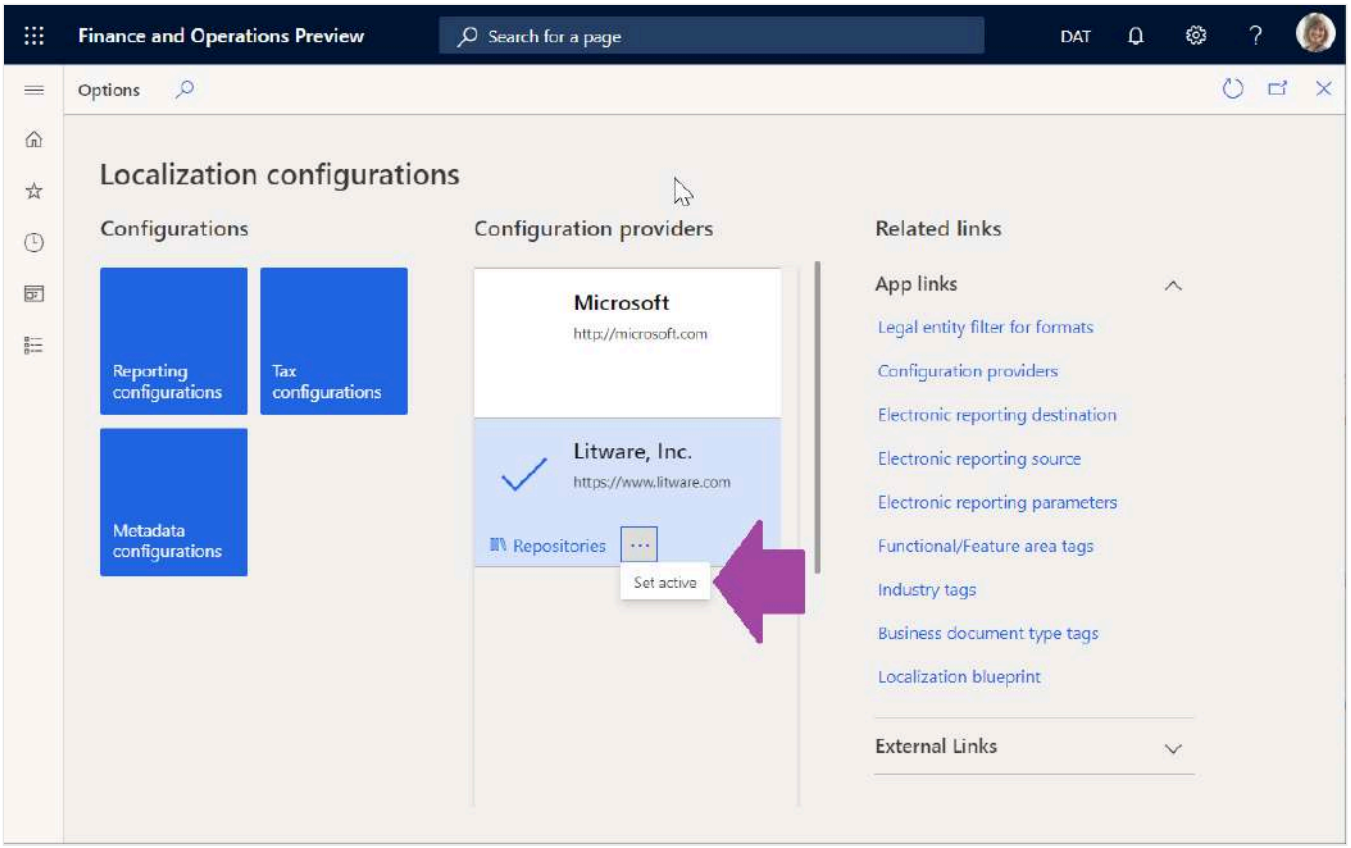
    - A provider record has a unique name and URL. Review the content of this page and skip this procedure if a record for Litware, Inc. (https://www.litware.com) already exists.

5. In the Name field, type `Litware, Inc.`.
6. In the Internet address field, type `https://www.litware.com`.
7. Select **Save**.
8. Close the page.

Select as an active provider

1. Select the Litware, Inc. provider.

2. Select **Set active**.



Last updated on 04/10/2026

Electronic reporting (ER) destinations

10/02/2025

You configure a destination for each Electronic reporting (ER) format configuration and its output component (a folder or a file). Users who have appropriate access rights can also modify destination settings at runtime. This article explains ER destination management, the types of destinations that are supported, and security considerations.

ER format configurations usually contain at least one output component: a file. Typically, configurations contain multiple file output components of different types (for example, XML, TXT, XLSX, DOCX, or PDF) that are grouped into either a single folder or multiple folders. ER destination management lets you preconfigure what occurs when each component runs. By default, when you run a configuration, a dialog box appears that lets you save or open the file. The same behavior also occurs when you import an ER configuration and don't configure any specific destinations for it. After you create a destination for a main output component, that destination overrides the default behavior, and the folder or file is sent according to the destination's settings.

Overview

You can set up destinations only for ER configurations that you import into the current Finance instance and for the formats that are available on the **Electronic reporting configurations** page. You can access ER destination management at **Organization administration** > **Electronic reporting** > **Electronic reporting destination**.

Default behavior

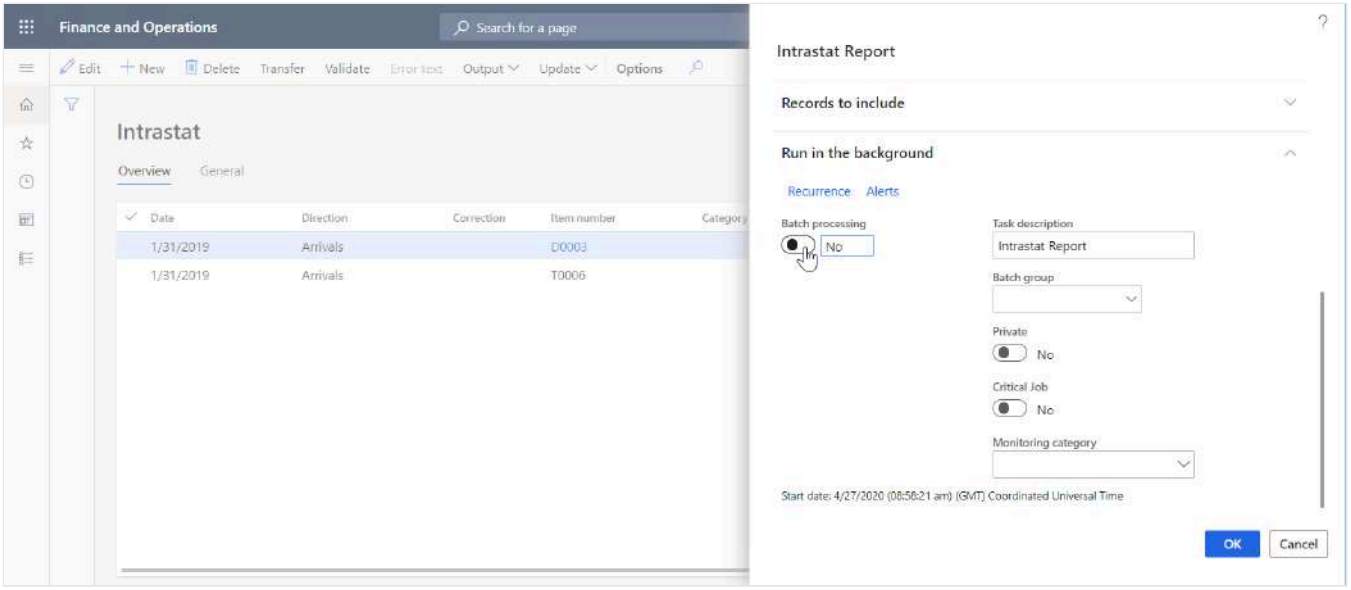
The default behavior for an ER format configuration depends on the execution type that you specify when an ER format starts.

For example, in the **Intrastat Report** dialog box, on the **Run in the background** FastTab, if you set the **Batch processing** option to **No**, an ER format runs immediately in interactive mode. When this execution completes successfully, a generated outbound document is available for download.

If you set the **Batch processing** option to **Yes**, an ER format runs in batch mode. The appropriate batch job is created based on the parameters that you specify on the **Run in the background** tab of the **ER parameters** dialog box.

> ⓘ **Note**

The job description informs you about the run of an ER format mapping. It also contains the name of the ER component that runs.

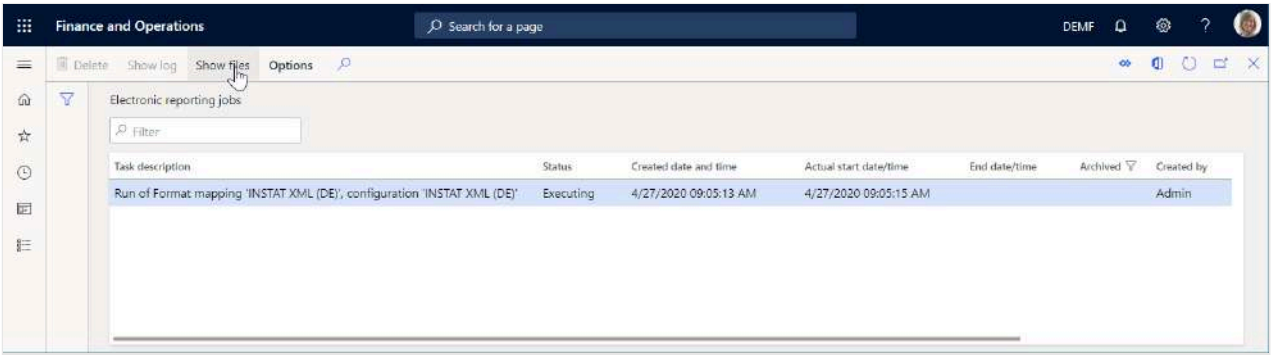


You can find information about this job in several places:

- Go to **Common** > **Inquiries** > **Batch jobs** > **My batch jobs** to check the status of the scheduled job.
- Go to **Organization administration** > **Electronic reporting** > **Electronic reporting jobs** to check the status of the scheduled job and the execution results of the completed job. When job execution completes successfully, select **Show files** on the **Electronic reporting jobs** page to get a generated outbound document.

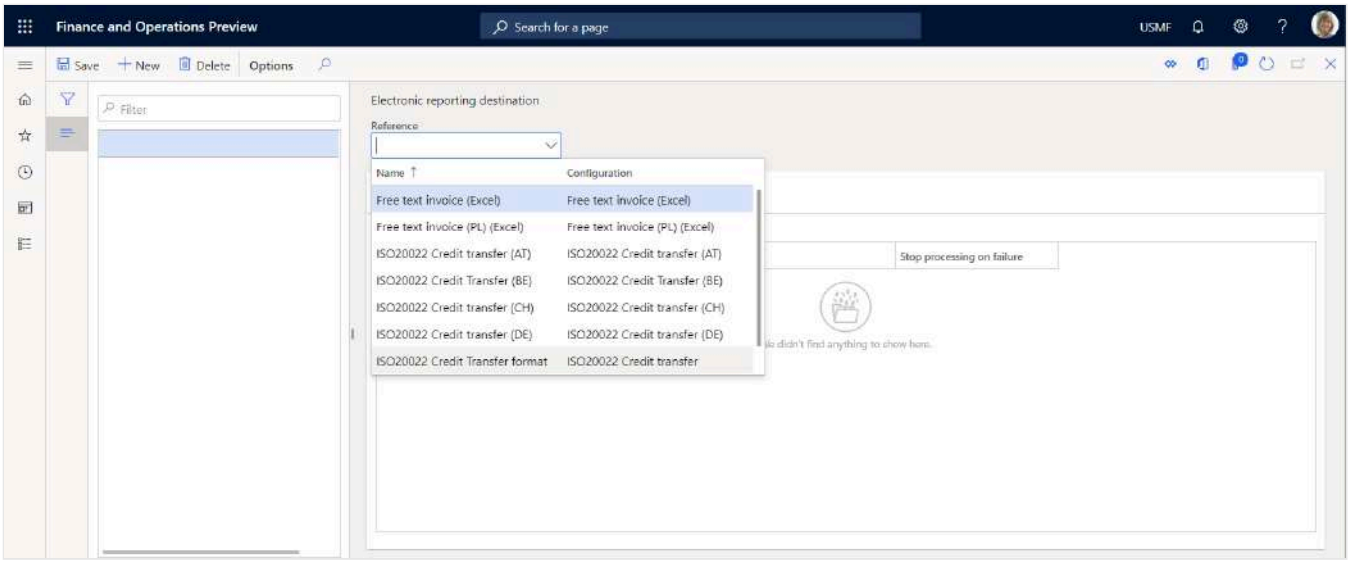
  > **Note**
  >
  > This document is stored as an attachment of the current job record and is controlled by the Document management framework. The document type that is used to store ER artifacts of this type is configured in the ER parameters.

- On the **Electronic reporting jobs** page, select **Show files** to view the list of any errors and warnings that were generated during job execution.

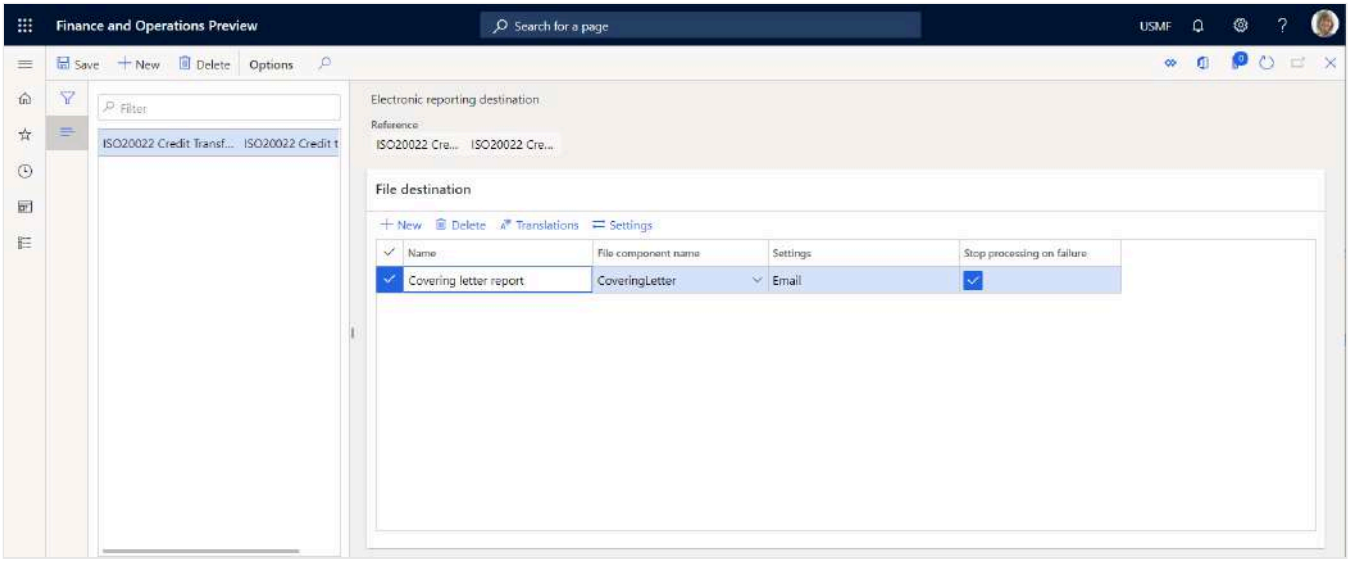


User-configured behavior

On the **Electronic reporting destination** page, you can override the default behavior for a configuration. Imported configurations don't appear on this page until you select **New** and then, in the **Reference** field, select a configuration to create destination settings for.



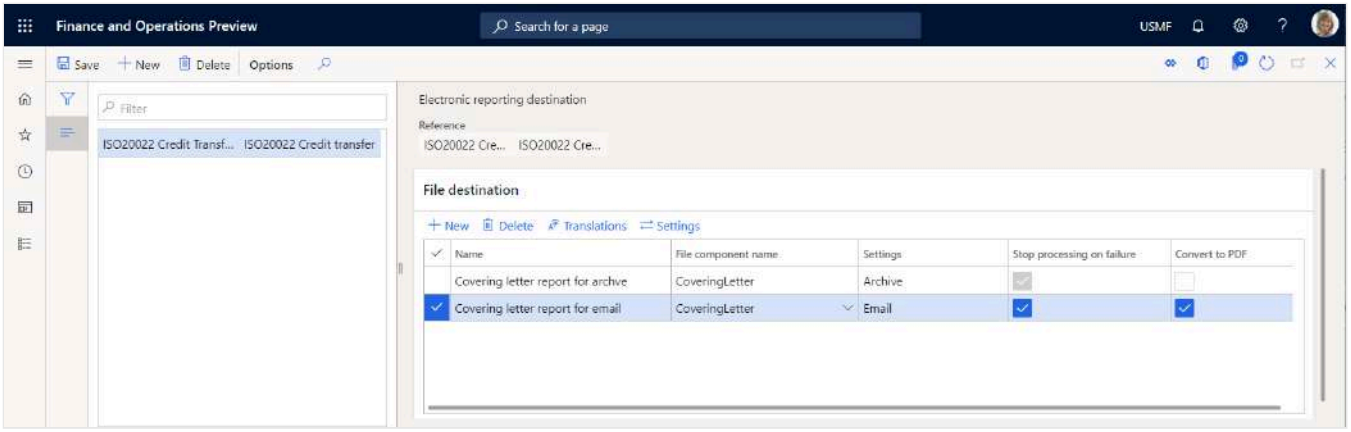
After you create a reference, you can create a file destination for each **Folder** or **File** output component of the referenced ER format.



Next, in the **Destination settings** dialog box, you can enable and disable individual destinations for the file destination. Use the **Settings** button to control all the destinations for a selected file destination. In the **Destination settings** dialog box, you can control each destination separately by setting the **Enabled** option for it.

You can create **multiple file destinations** for each output component of the same format. For example, you can use this capability to configure file destinations for a file component that's used to generate an outbound document in Excel format. One destination can be configured to store the original Excel file in the ER jobs archive, and another destination can be configured

to simultaneously convert the Excel file to PDF format and send the PDF file by email. Learn more in Archive, Email, and Convert.



When you run an ER format, the system always runs all destinations that you configured for components of the format. In addition, the ER destinations functionality lets you configure different sets of destinations for a single ER format. This configuration marks each set as configured for a particular user action. The ER API was extended so that an action can be provided that the user performs by running an ER format. The action code that is provided is passed to ER destinations. You can run different destinations of an ER format, depending on the action code that is provided. Learn more in Configure action-dependent ER destinations.

Destination types

The following destinations are currently supported for ER formats. You can disable or enable all types at the same time. In this way, you can either do nothing or send the component to all configured destinations.

- Email
- Archive
- File
- Screen
- Power BI
- Print

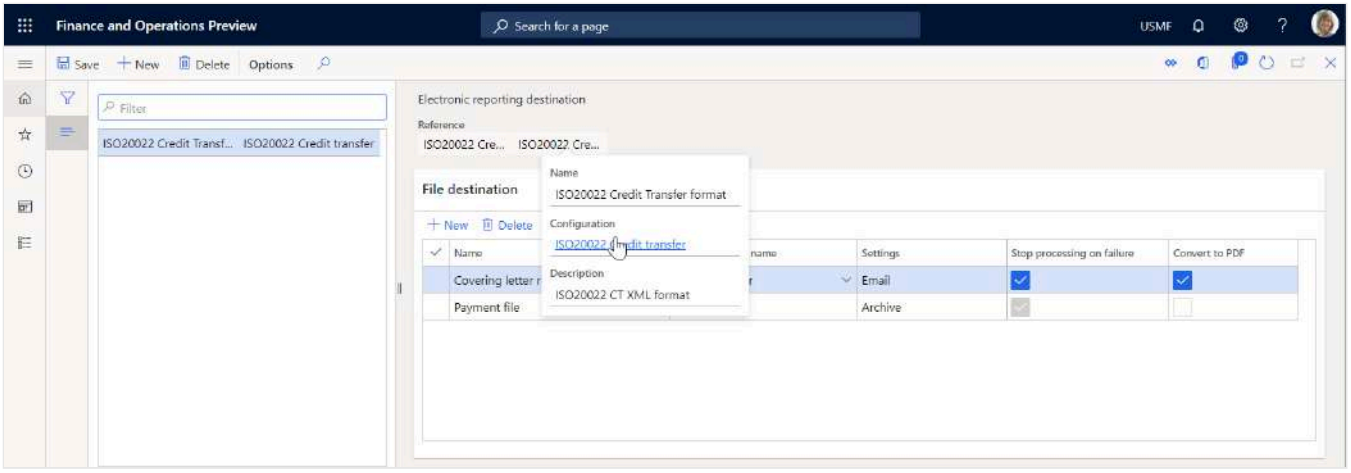
Applicability

You can set up destinations only for ER configurations that you import and for the formats that are available on the **Electronic reporting configurations** page.

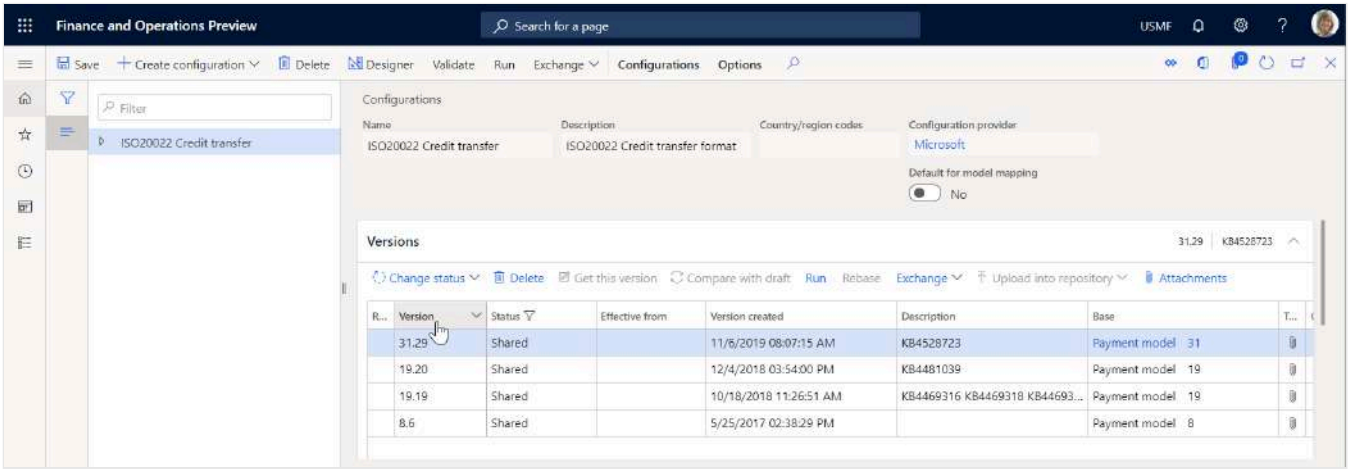
> ⓘ **Note**

Configured destinations are company-specific. If you plan to use an ER format in different companies of the current Finance instance, you must configure destinations for that ER format for each of those companies.

When you configure file destinations for a selected format, you configure them for the whole format.



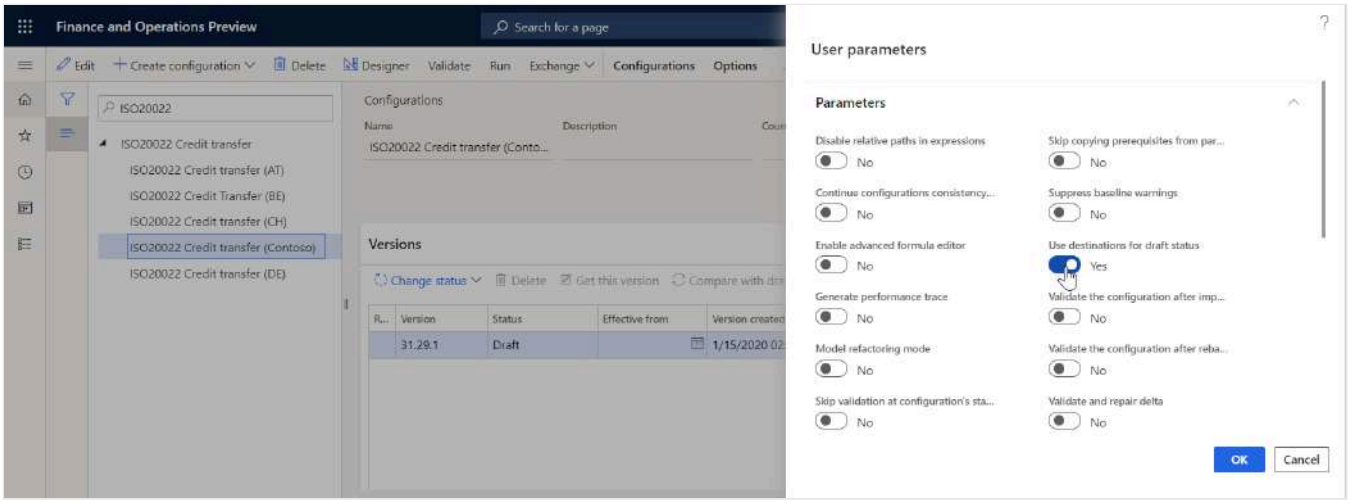
At the same time, you might have multiple versions of the format that you import into the current Finance instance. You can view them if you select the **Configuration** link that is offered when you select the **Reference** field.



By default, configured destinations are applied only when you run an ER format version that has a status of either **Completed** or **Shared**. However, you sometimes must use configured destinations when you run the draft version of an ER format. For example, you modify a draft version of your format, and you want to use configured destinations to test how the system delivers generated . Follow these steps to apply destinations for an ER format when the draft version is run.

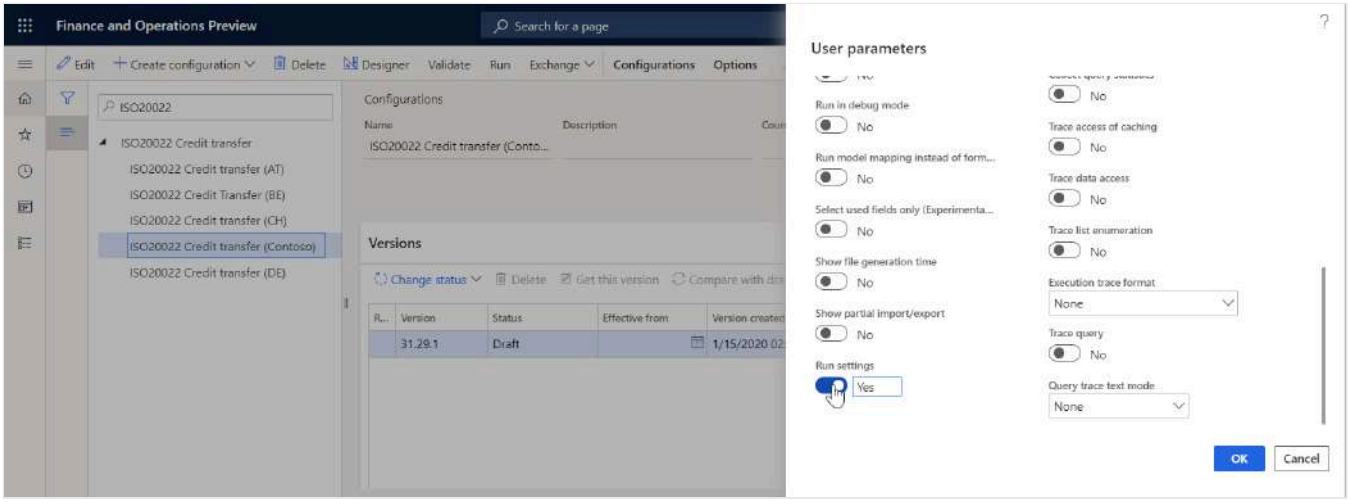
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.

3. Set the **Use destinations for draft status** option to **Yes**.

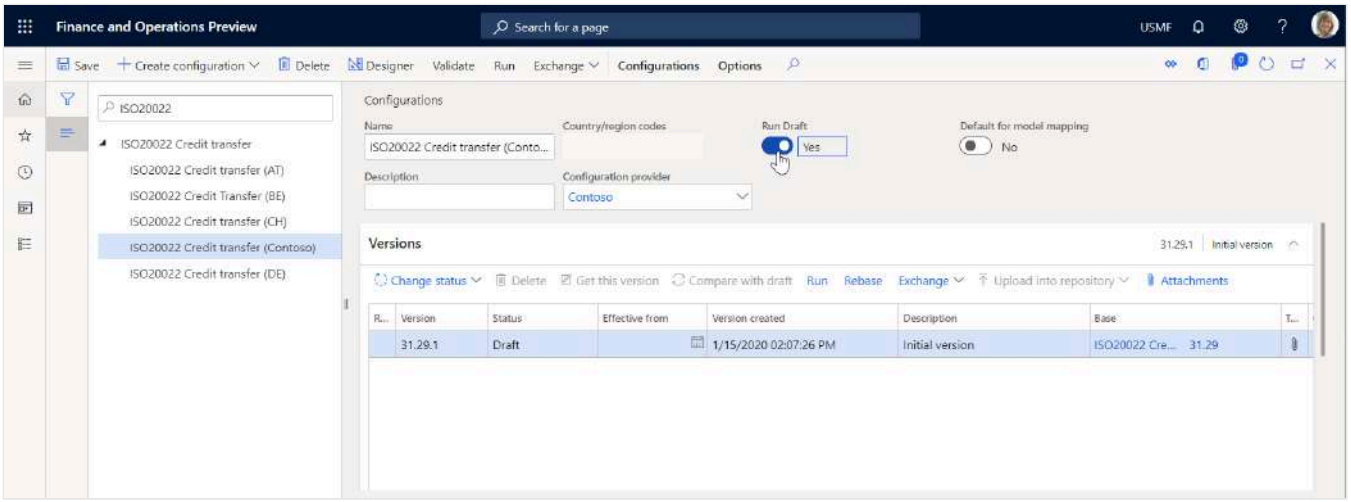


To use the draft version of an ER format, you must mark the ER format accordingly.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. Set the **Run setting** option to **Yes**.



After you complete this setup, the **Run draft** option is available for ER formats that you modify. Set this option to **Yes** to start to use the draft version of the format when the format is run.



Security considerations

Two types of privileges and duties are used for ER destinations. One type controls a user's overall ability to maintain the destinations configured for a legal entity (that is, it controls access to the **Electronic reporting destinations** page). The other type controls an application user's ability to override the destination settings that an ER developer or ER functional consultant configures at runtime.

Expand table

Role (AOT name)	Role name	Duty (AOT name)	Duty name
ERDeveloper	Electronic reporting developer	ERFormatDestinationConfigure	Configure electronic reporting format destination
ERFunctionalConsultant	Electronic reporting functional consultant	ERFormatDestinationConfigure	Configure electronic reporting format destination
PaymAccountsPayablePaymentsClerk	Accounts payable payments clerk	ERFormatDestinationRuntimeConfigure	Configure electronic reporting format destination during runtime
PaymAccountsReceivablePaymentsClerk	Accounts receivable	ERFormatDestinationRuntimeConfigure	Configure electronic

Role (AOT name)	Role name	Duty (AOT name)	Duty name
	payments clerk		reporting format destination during runtime

> **Note**
>
> The preceding duties use two privileges. These privileges have the same names as the corresponding duties: **ERFormatDestinationConfigure** and **ERFormatDestinationRuntimeConfigure**.

Frequently asked questions

I imported electronic configurations, and I see them on the Electronic reporting configurations page. But why don't I see them on the Electronic reporting destinations page?

Make sure that you select **New** and then select a configuration in the **Reference** field. The **Electronic reporting destinations** page shows only configurations that you configure destinations for.

Is there any way to define which Microsoft Azure Storage account and Azure Blob storage are used?

No. The default Microsoft Azure Blob storage that is defined and used for the document management system is used.

What is the purpose of the File destination in the destination settings? What does that setting do?

The **File** destination controls a dialog box of your web browser when you run an ER format in interactive mode. If you enable this destination, or if you don't define a destination for a configuration, an open or save dialog box appears in your web browser after an file is created.

Can you give me an example of the formula that refers to a vendor account that I can send email to?

The formula is specific to the ER configuration. For example, if you use the ISO 20022 Credit Transfer configuration, you can use **'$PaymentsForCoveringLetter'.Creditor.Identification.SourceID** or **model.Payments.Creditor.Identification.SourceID** to get an associated vendor account.

One of my format configurations contains multiple files that are grouped into one folder (for example, Folder1 contains File1, File2, and File3). How do I set up destinations so that Folder1.zip isn't created at all, File1 is sent by email, File2 is sent to SharePoint, and I can open File3 immediately after the configuration runs?

Your format must first be available in the ER configurations. If this prerequisite is met, open the **Electronic reporting destination** page and create a new reference to the configuration. You must then have four file destinations, one for each component. Create the first file destination, give it a name such as **Folder**, and select a file name that represents a folder in your configuration. Then select **Settings**, and make sure that all the destinations are disabled. For this file destination, the folder isn't created. By default, because of hierarchical dependencies between files and parent folders, the files behave in the same way. In other words, they aren't sent anywhere. To override that default behavior, you must create three more file destinations, one for each file. In the destination settings for each, you must enable the destination that the file should be sent to.

More information

Electronic reporting (ER) overview

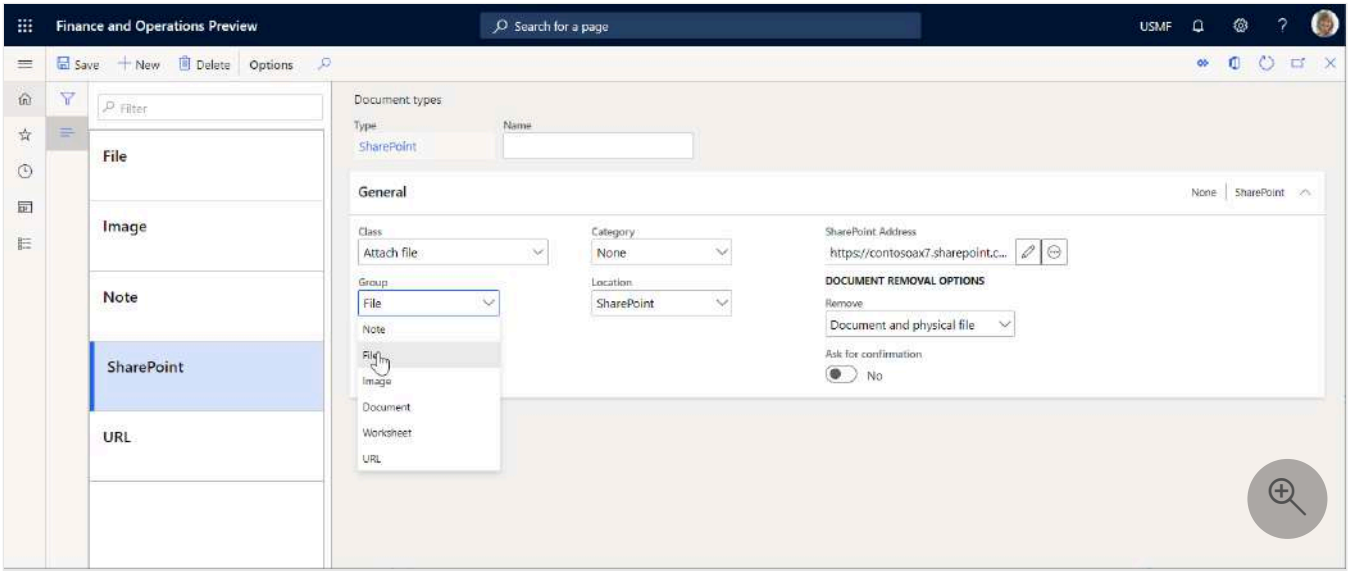
Configure action-dependent ER destinations

Archive ER destination type

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can configure an archive destination for each **Folder** or **File** component of an Electronic reporting (ER) format configured to generate outbound documents. Based on the destination setting, the system stores a generated document as an attachment of a record in the ER jobs list. To view the results, go to **Organization administration** > **Electronic reporting** > **Electronic reporting jobs**.

Use this option to send the generated document to a Microsoft SharePoint folder or Microsoft Azure Storage. Set **Enabled** to **Yes** to send output to a destination that the selected document type defines. You can only select document types where the group is set to **File**. You define document types at **Organization administration** > **Document management** > **Document types**. The configuration for ER destinations is the same as the configuration for the document management system.

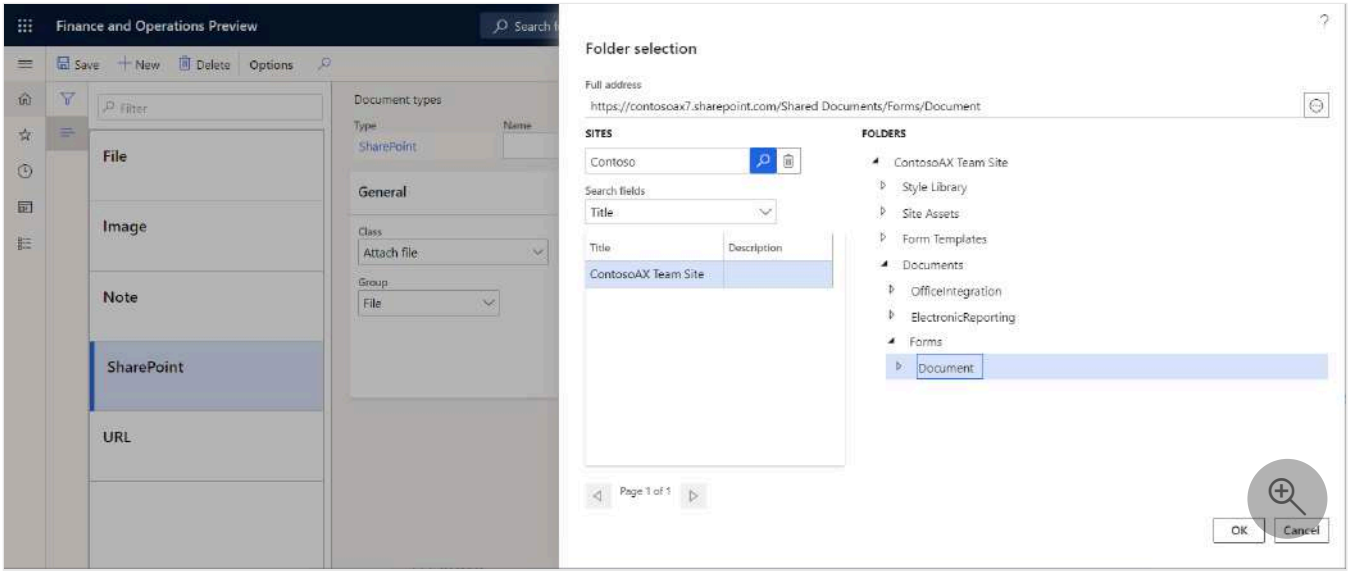


The location determines where the file is saved. After you enable the **Archive** destination, the results can be saved in the Job archive. You can view the results at **Organization administration** > **Electronic reporting** > **Electronic reporting archived jobs**.

> **Note**
>
> Select a document type for the Job archive by navigqting to **Organization administration** > **Workspaces** > **Electronic reporting** > **Electronic reporting parameters**. For more information, see Configure the Electronic reporting (ER) framework.

SharePoint

You can save a file in a designated SharePoint folder. To define the default SharePoint server, go to **Organization administration** > **Document management** > **Document management parameters**. On the **SharePoint** tab, configure the SharePoint folder. Then, you can select it as the folder where the ER output is saved. You must select the **SharePoint** location in this document type.



Azure Storage

When the document type location is set to **Azure storage**, you can save a file to Azure Storage.

> **Note**
>
> Unlike the Data management framework, which applies a seven-day retention policy for documents that must be processed, the ER framework permanently stores files in Azure Blob storage. For more information, see API for getting message status and Status check API. The ER-related files are stored in Azure Blob storage as attachments of application

table records for as long as necessary. A single file is deleted from Azure Blob storage along with the application table record that this file was attached to.

Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) destinations
- Configure document management

Last updated on 04/15/2026

Email ER destination type

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you run an Electronic reporting (ER) format, you can generate one or more outbound documents. Use **Folder** or **File** format components in ER formats to specify the structure of outbound documents. You can configure an email destination for these types of components to send outbound documents as email attachments.

You can configure an email destination for each **Folder** or **File** component of an ER format. In this case, **each outbound document is emailed individually**. Based on this destination setting, a generated document is delivered as an attachment of an email.

> **Note**
>
> If no document is generated because the **Enabled** expression for the relevant **File** component returns a **False** Boolean value, no email is sent, even if an email destination an email destination is configured and enabled for the component.

You can also group several **Folder** or **File** components together, and then configure an email destination for all the components in the group. In this case, all outbound documents that components in the group generate **are sent as multiple attachments of a single email**. Based on this destination setting, each generated document is delivered as an attachment of a single email.

> **Note**
>
> If at least one document is generated by a **File** component in a group of components, an email is sent. If no document is generated by grouped components because the **Enabled** expression for each **File** component returns a **False** Boolean value, no email is sent, even if an email destination is configured and enabled for that group of components.

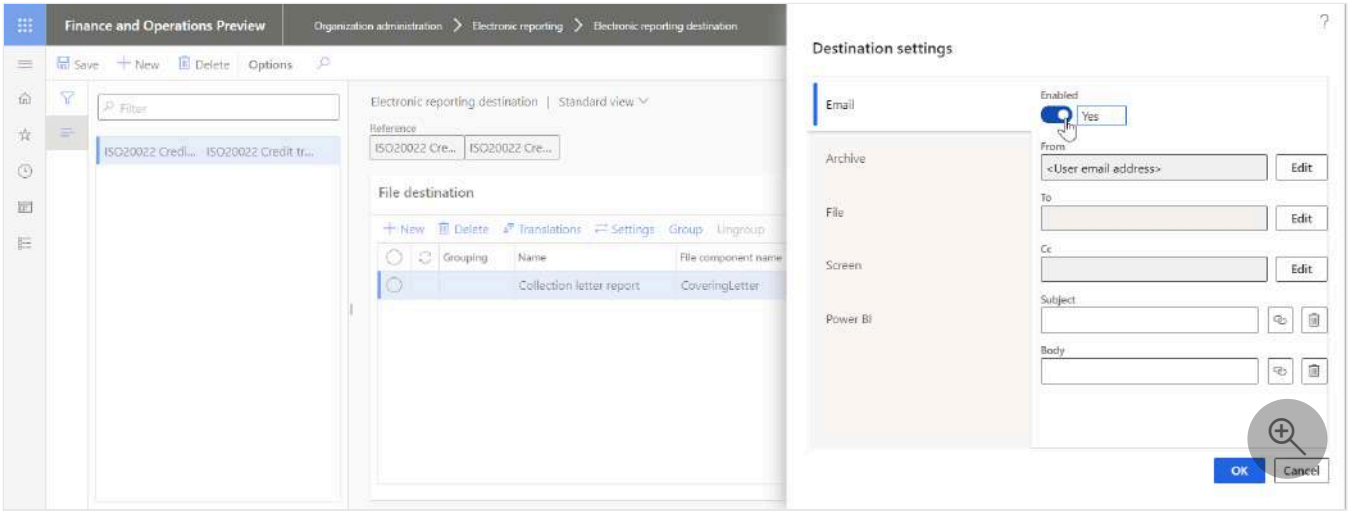
Email is the only destination that you can configure for a group of components. To deliver a document that is emailed based on the email destination setting for a group, add one more destination record, select the component that you want, and then configure another destination for this record.

You can configure multiple groups of components for a single ER format configuration. In this way, you can configure an email destination for every group of components and an email destination for every component.

Enable an email destination

To send one or more output files by email, follow these steps:

1. On the **Electronic reporting destination** page, on the **File destination** FastTab, select a component or group of components in the grid.
2. Select **Settings**, and then, in the **Destination settings** dialog box, on the **Email** tab, set the **Enabled** option to **Yes**.



Configure an email destination

Email content

You can edit the subject and body of the email message.

In the **Subject** field, enter the text of the email subject that should appear in the subject field of an electronic message that is generated at runtime. In the **Body** field, enter the text of the email body that should appear in the body field of an electronic message. You can set up constant text for the email subject and body, or you can use ER formulas to dynamically create email text at runtime. The configured formula must return a value of the String type.

The body of your email is composed in TEXT or HTML format, depending on the email client. You can use any layout, styling, and branding that HTML and inline Cascading Style Sheets (CSS) allow for.

> ⓘ **Note**
>
> Email clients impose layout and style limitations that might require adjustments to the HTML and CSS that you use for the message body. Familiarize yourself with the best practices for creating HTML that the most popular email clients support.
>
> Use the correct encoding to implement a carriage return, depending on the body formatting. For more information, see the definition of the **String** data type.

Email addresses

You can specify the email sender and email recipients. By default, an email is sent on behalf of the current user. To specify a different email sender, configure the **From** field.

> ⓘ **Note**
>
> When you configure an email destination, only users with the `ERFormatDestinationSenderEmailConfigure` security privilege, **Configure the sender email address for ER format destinations**, can see the **From** field.
>
> When an email destination is offered for modification at **runtime**, the **From** field is visible only to users who have the `ERFormatDestinationSenderEmailMaintain` security privilege, **Maintain the sender email address for ER format destination**.
>
> When you configure the **From** field to use an email address other than the current user's, you must correctly **set** either the **Send as** or **Send on behalf** permission in advance. Otherwise, the following exception is thrown at runtime: "Unable to send email as <from email account> from the <current user account> account, please check the 'Send as' permissions on the <from email account>."

You can configure the **From** field to return more than one email address. In this case, the first address in the list is used as the email sender address.

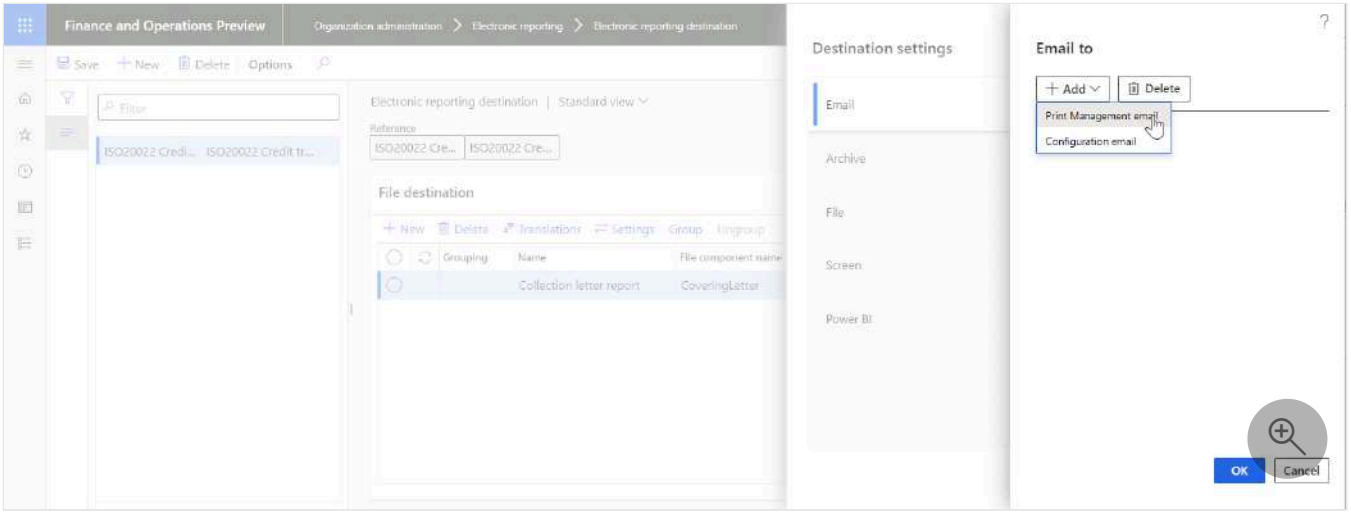
To specify email recipients, configure the **To** and **Cc** (optional) fields.

As of version 10.0.48, you can also configure **BCC** (optional) field.

You can configure email addresses for ER in two ways. You can complete the configuration in the same way as the Print Management feature, or you can resolve an email address by using a direct reference to the ER configuration through a formula.

Email address types

If you select **Edit** next to the **From**, **To**, **Cc**, or **BCC** field in the **Destination settings** dialog box, the appropriate **Email from**, **Email to**, **Email cc**, or **Email BCC** dialog box appears. You can configure the email sender and email recipients. Select **Add**, and then select the type of email address to use. Two types are currently supported: **Print Management email** and **Configuration email**.



Print Management email

If you select **Print Management email** as the email address type, enter fixed email addresses in the **Email from**, **Email to**, **Email cc**, or **Email BCC** dialog box by setting the following fields:

- In the **Email source** field, select **None**.
- In the **Additional email addresses, separated by ";"** field, enter the fixed email addresses.

Alternatively, you can get email addresses from the contact details of the party that you generate an outbound document for. To use email addresses that aren't fixed, in the **Email source** field, select the role of the party for a file destination. The following roles are supported:

- Customer
- Vendor
- Prospect
- Contact

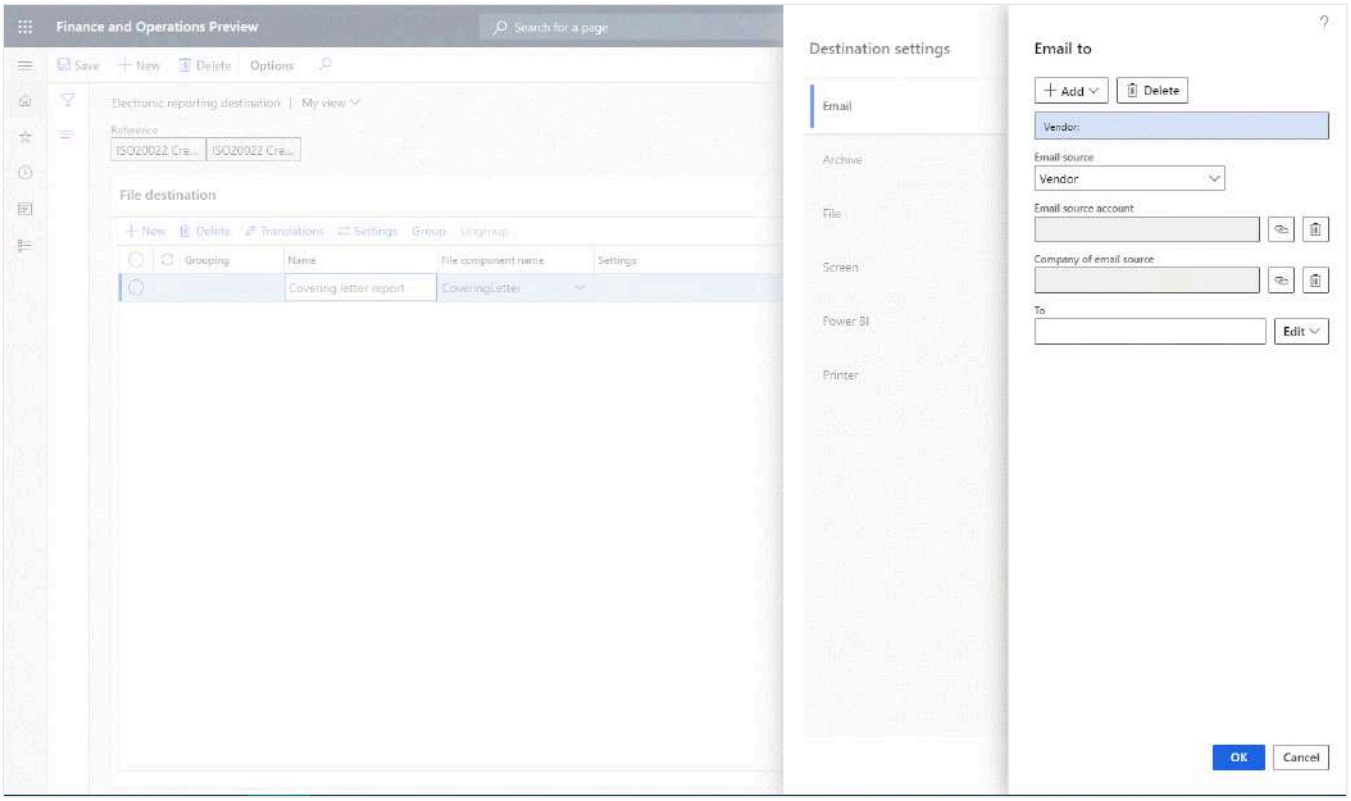
- Competitor
- Worker
- Applicant
- Prospective vendor
- Disallowed vendor
- Legal entity

For example, to configure an email destination for an ER format that is used to process vendor payments, select the **Vendor** role.

After you select the desired role, select the **Bind** button (chain symbol) next to the **Email source account** field to open the Formula designer page. You can then use this page to configure a formula that returns, at runtime, the account number of the party that is assigned to the configured role from the processed document to the email destination.

> **Note**
>
> Formulas are specific to the ER configuration.

On the **Formula designer** page, in the **Formula** field, enter a document-specific reference to a supported role. Instead of typing the reference, in the **Data source** pane, find and select the data source node that represents an account of the configured role, and then select **Add data source** to update the formula. For example, if you configure the email destination for the **ISO 20022 Credit Transfer** configuration that is used to process vendor payments, the node that represents a vendor account is `'$PaymentsForCoveringLetter'.Creditor.Identification.SourceID`.



If the account numbers of the configured role are unique for the whole instance of Microsoft Dynamics 365 Finance, the **Company of email source** field in the **Email to** dialog box can remain blank.

Alternatively, you might have a situation where different parties in the Global address book are registered in different companies (legal entities) in such a way that they all use the same account number to fill the configured role. In this case, account numbers for the configured role aren't unique for the whole Finance instance. Therefore, to explicitly select a party, you can't specify only an account number. You must also specify the company with the scope that the party is registered in to fill the configured role. Select the **Bind** button (chain symbol) next to the **Company of email source** field in the **Email to** dialog box to open the Formula designer page. You can then use this page to configure a formula that returns, at runtime, the code of the company that the desired source must be found in the scope of.

> **Tip**
>
> If you must use the company code to run an ER format, but the ER format doesn't provide any data source that the company code can be obtained from, configure the `GetCurrentCompany()` formula by using the built-in GETCURRENTCOMPANY ER function.

> **Note**

Formulas are specific to the ER configuration.

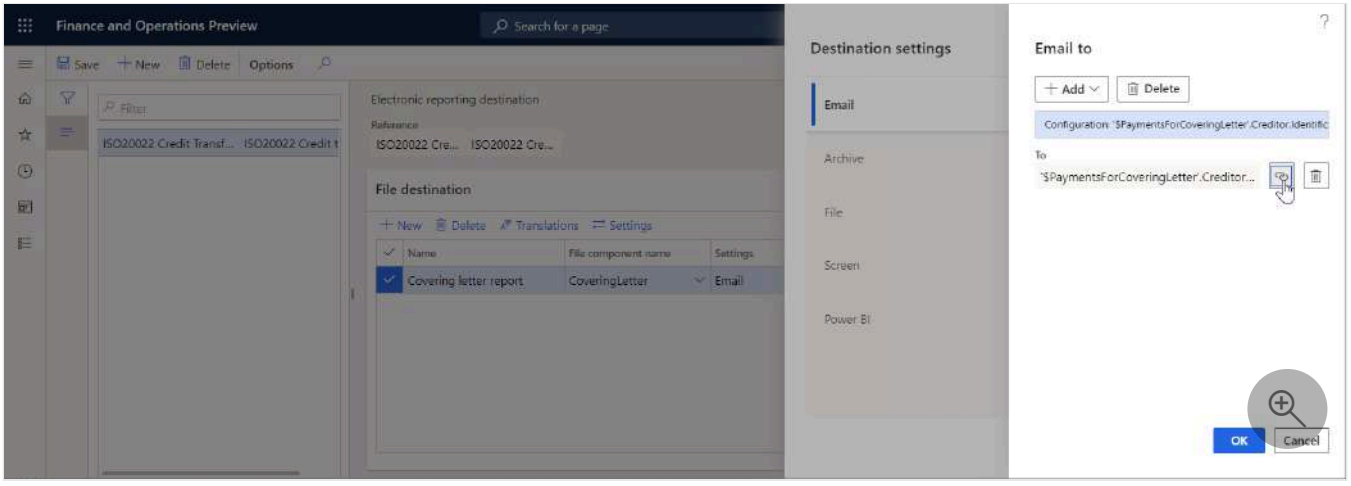
To specify the type of email addresses that must be used at runtime, in the **Email to** dialog box, select **Edit** next to the **To** field to open the **Assign email address** drop-down dialog box. Then set the following fields:

- In the **Purpose** field, select the desired purposes. Only email addresses of the selected purposes from contacts of the discovered party are used.
- Set the **Primary contact** option to **Yes** to use an email address that is configured for the discovered party as the primary email address.

> **Note**
>
> If you select purposes in the **Purpose** field and set the **Primary contact** option to **Yes** at the same time, every email that satisfies at least one configured criterion is used at runtime.

Configuration email

Select **Configuration email** as the email address type if the configuration that you use has a node in the data sources that returns either a single email address or multiple email addresses that are separated by semicolons (;). You can use data sources and functions in the formula designer to get a correctly formatted email address or correctly formatted email addresses that are separated by semicolons. For example, if you use the **ISO 20022 Credit Transfer** configuration, the node that represents the primary email address of a vendor from the vendor contact details that the covering letter should be sent to is `'$PaymentsForCoveringLetter'.Creditor.ContactDetails.Email`.



Group format components

To group format components, on the **Electronic reporting destination** page, on the **File destination** FastTab, select the components in the grid, and then select **Group**.

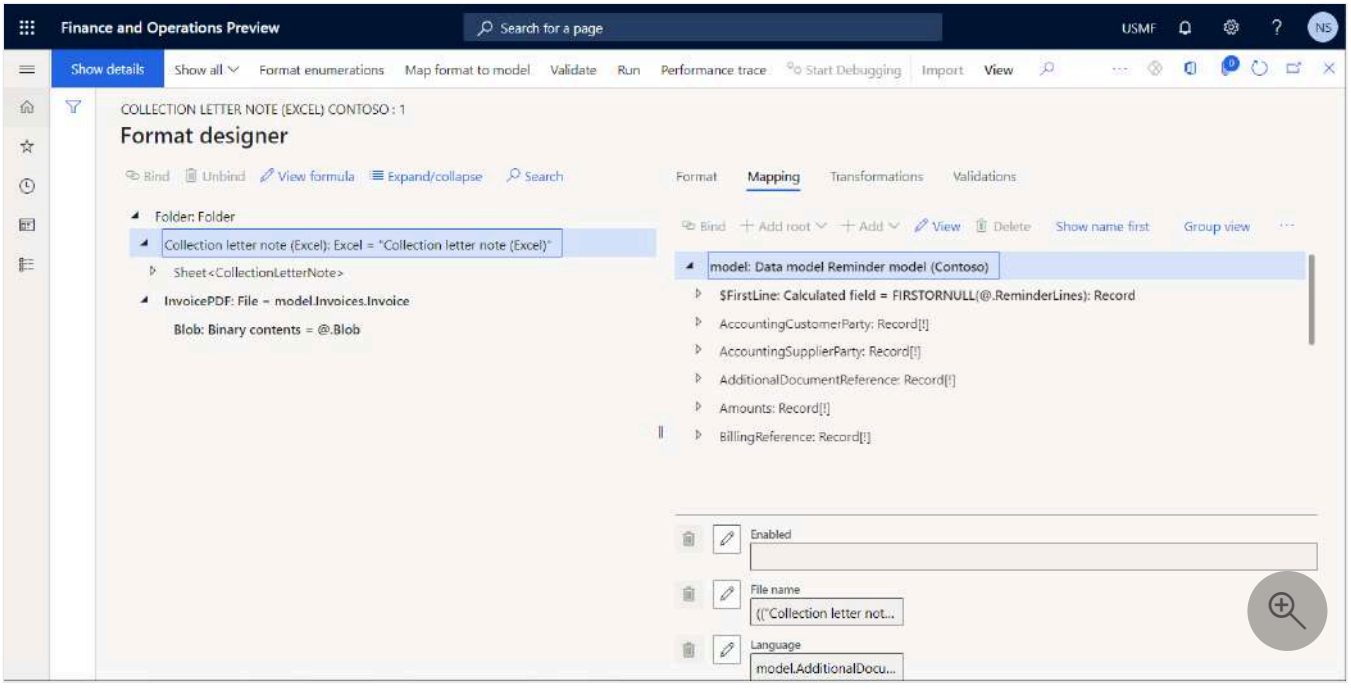
Email is the only previously configured destination that is still available for the selected components. No other previously configured destinations are available, because they're considered unsupported for a group of components. You're notified about these changes as appropriate.

The record that you previously added is considered the header of the created group. This header record holds the email destination settings for the group. Other records are group members that use the email destination settings of the group's header record.

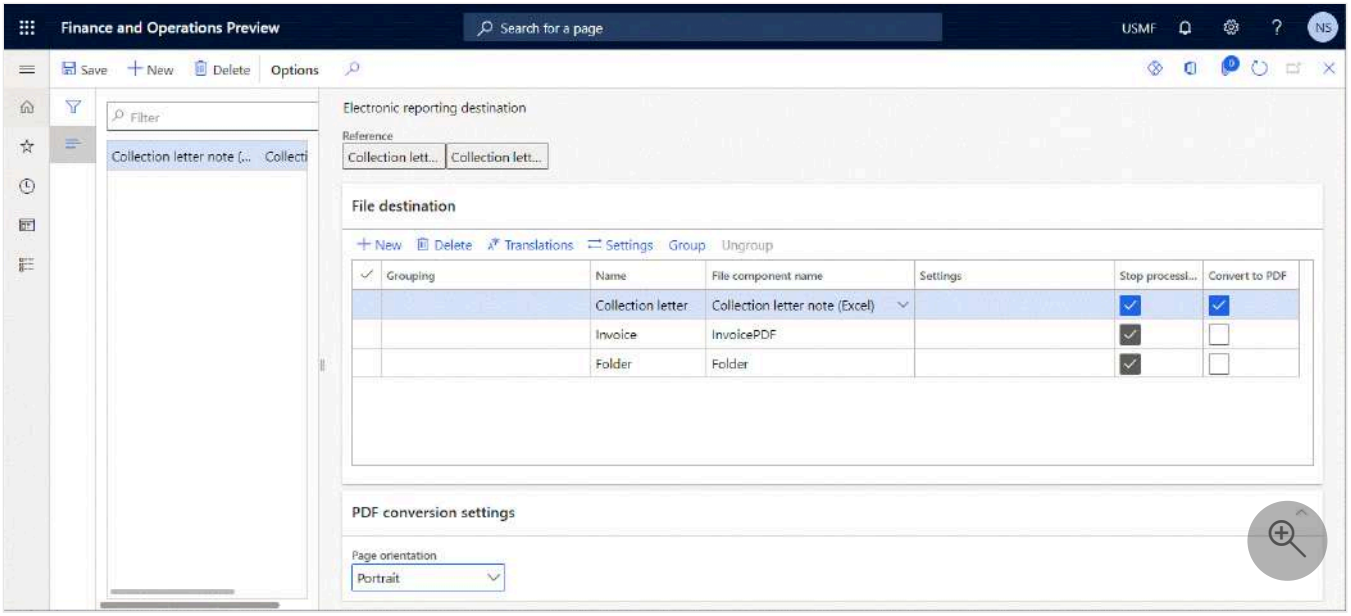
To ungroup format components, on the **File destination** FastTab, select a record that belongs to the group, and then select **Ungroup**.

- If you select a header record, the whole group is ungrouped.
- If you select a member record, and it's the last member record in a group, the whole group is ungrouped.
- If you select a member record that isn't the last member record in a group, that record is excluded from the current group.

The following illustration shows the structure of an ER format configured to produce a zipped outbound file that contains a collection letter note and appropriate customer invoices in PDF format.



The following illustration shows the process, as described in this article, of grouping individual components and enabling the **Email** destination for the new group, so that a collection letter note is sent together with appropriate customer invoices as email attachments.



Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) destinations
- Formula designer in Electronic reporting (ER)

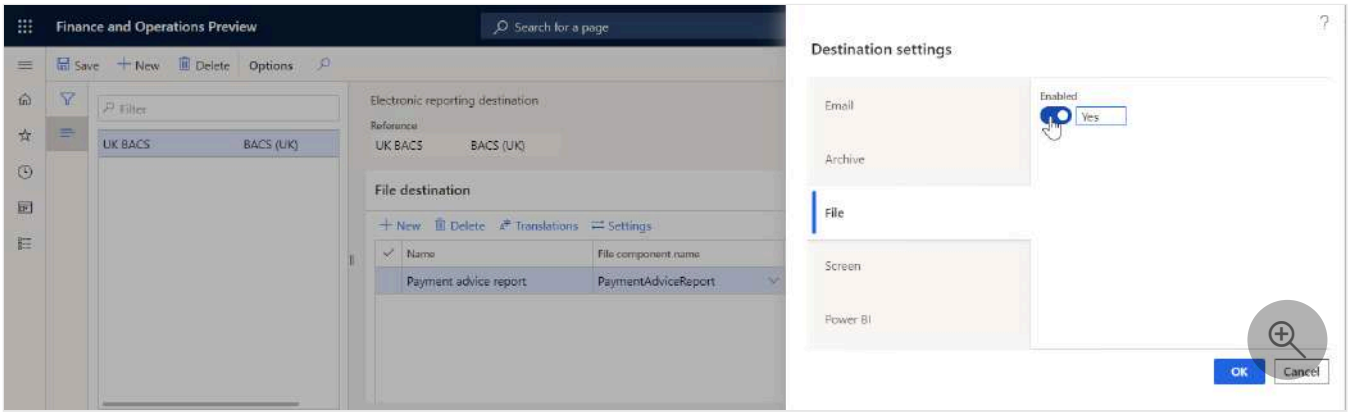
Last updated on 04/15/2026

File destination

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can configure a file destination for each FOLDER or FILE component of an Electronic reporting (ER) format configured to generate outbound documents. Based on the setting of the destination, the generated document is available for download from the web browser.

On the **Destination settings** page, if you set **Enabled** to **Yes**, an open or save dialog box opens when the configuration finishes running.



Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) destinations

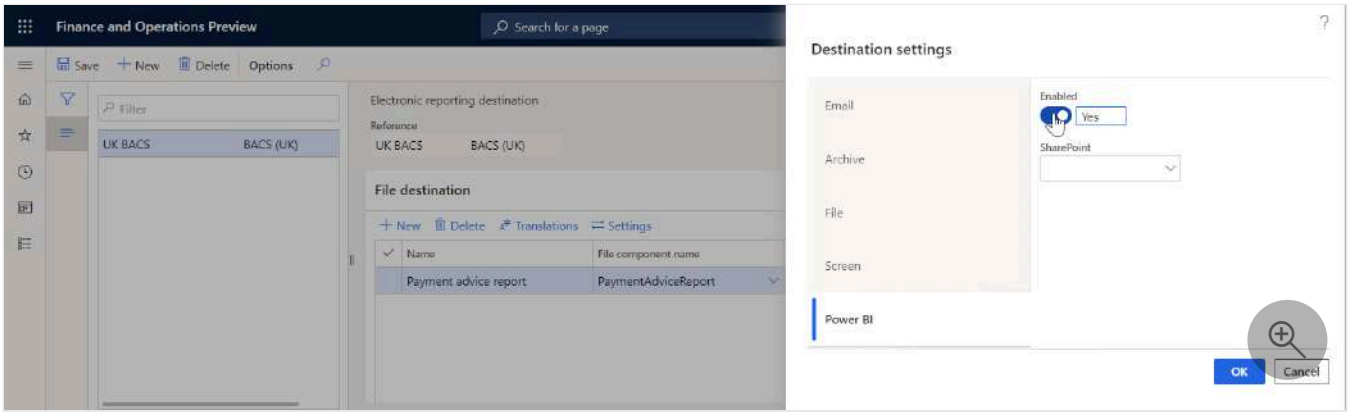
Last updated on 04/15/2026

Power BI destination

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can configure a Microsoft Power BI destination for each folder or file component of an Electronic reporting (ER) format configured to generate outbound documents. Based on the setting of the destination, a generated document is stored in a previously configured SharePoint folder.

Set **Enabled** to **Yes** to use your ER configuration to arrange the transfer of data from your Dynamics 365 Finance instance to Microsoft Power BI services. The transferred files are stored on a Microsoft SharePoint Server instance that you must configure for that purpose. For more information, see Configure Electronic reporting (ER) to pull data into Power BI.



Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) destinations

Last updated on 04/15/2026

Printer destination

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can send a generated document directly to a network printer for direct printing.

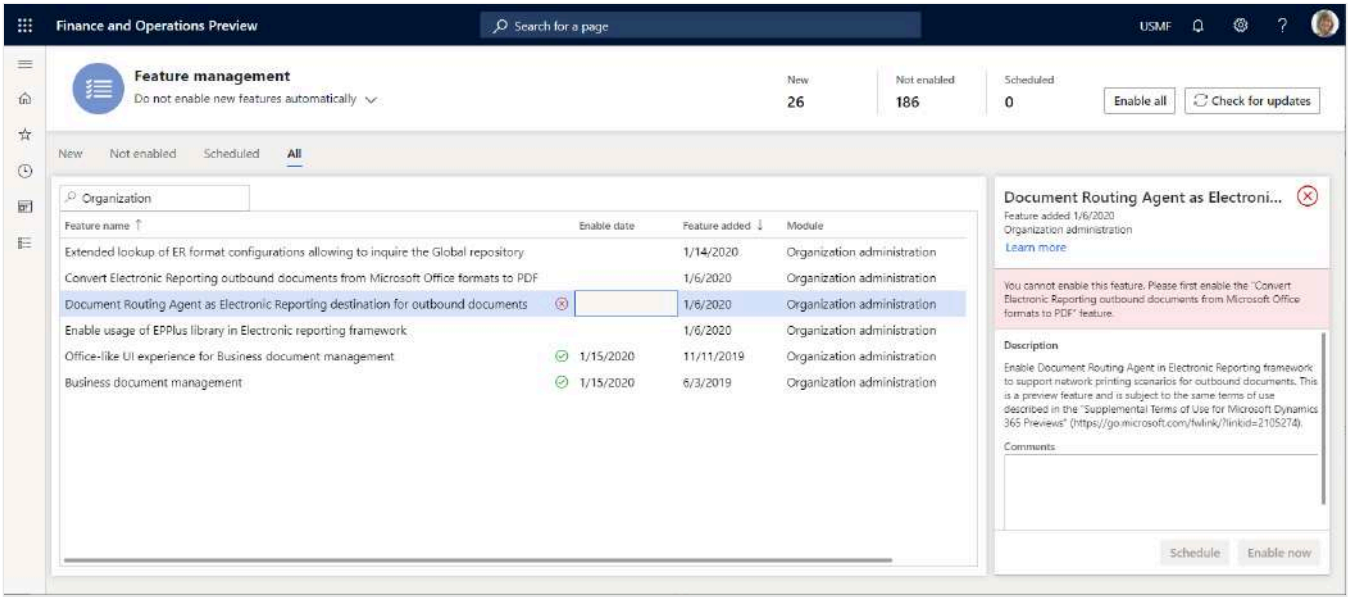
Prerequisites

Before you begin, install and configure the Document Routing Agent, and then register the network printers. For more information, see Install the Document Routing Agent to enable network printing.

Make the Printer destination available

To make the **Printer** destination available in the current instance of Microsoft Dynamics 365 Finance, go to the **Feature management** workspace, and turn on the following features, in this order:

1. Convert Electronic Reporting outbound documents from Microsoft Office formats to PDF
2. Document Routing Agent as Electronic Reporting destination for outbound documents

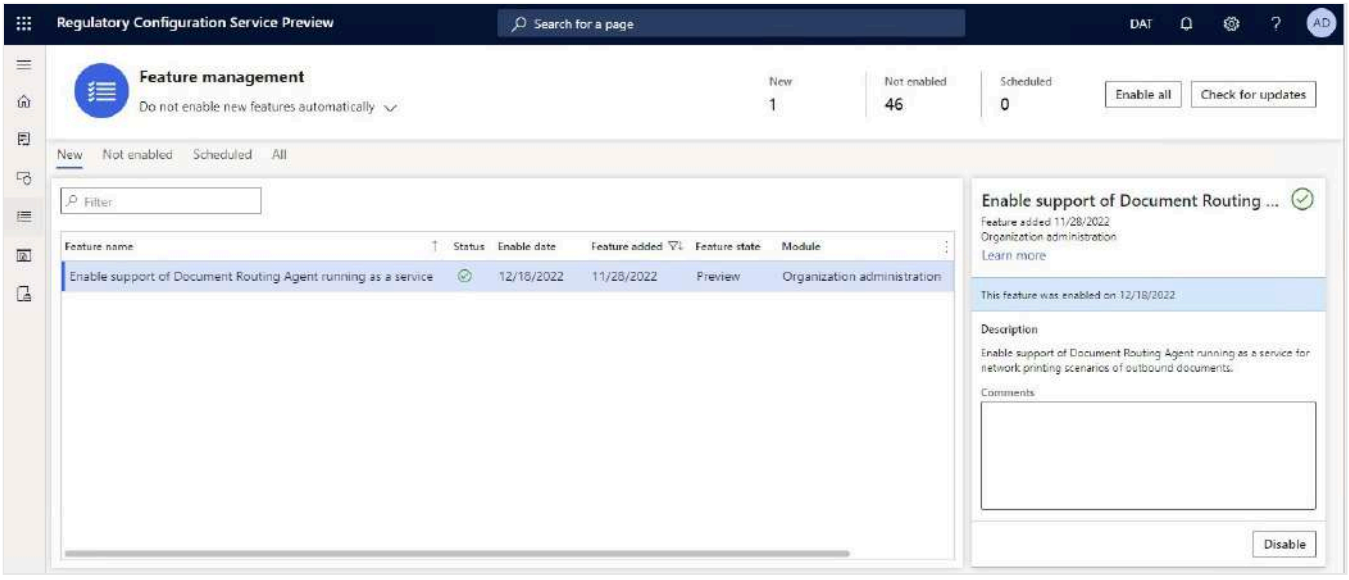


> **Note**
>
> As of Finance version 10.0.32, the **Convert Electronic Reporting outbound documents from Microsoft Office formats to PDF** and **Document Routing Agent as Electronic Reporting destination for outbound documents** features are removed. The functionality that these features controlled is now enabled by default.

Run the Document Routing Agent as a service

The Document Routing Agent lets you select the mode of execution. The process can run either as a desktop application or as a Windows service. For more information, see Run the Document Routing Agent as a Windows service.

As of Finance version 10.0.32, the capability to support running the Document Routing Agent as a Windows service is implemented. To activate this capability in the system, turn on the **Enable support of Document Routing agent running as a service** feature in the **Feature management** workspace.



Applicability

PDF printing

> **Note**
>
> You can print PDF documents directly to a network printer only if the Document Routing Agent is running as a desktop application. If the Document Routing Agent is running as a service, PDF documents aren't sent directly to a printer. Instead, they're stored in a target

folder. You must configure this folder in advance in the Document Routing Agent's parameters. For more information, see **Update the execution mode for the Document Routing Agent**.

To continue sending printable documents directly to a printer when the Document Routing Agent is running as a service, use **EMF printing**.

In versions of Finance before version 10.0.18, you can configure the **Printer** destination only for file components that generate output in printable PDF format (**PDF Merger** or **PDF file** format elements) or Microsoft Office Excel and Word format (**Excel file** format element). When you generate the output in PDF format, it's sent to a printer. When you generate the output in Office format by using the **Excel file** format element, it's automatically converted to PDF format and then sent to a printer.

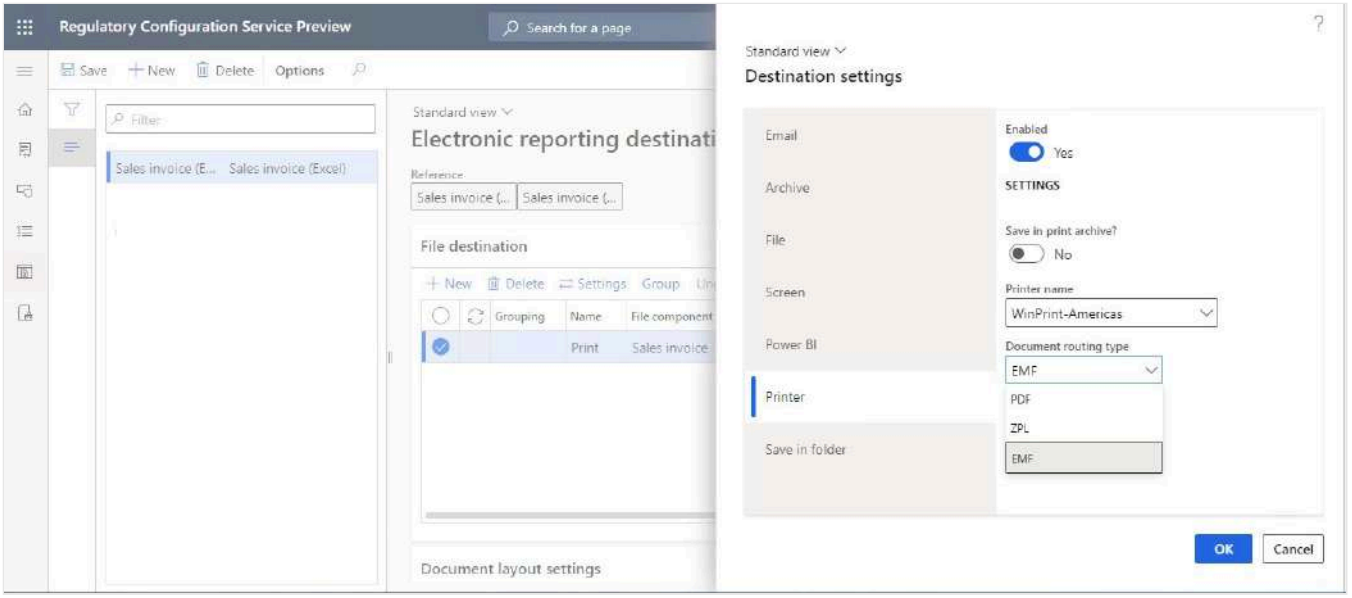
However, as of version 10.0.18, you can configure the **Printer** destination for the **Common file** format element. This format element mostly generates output in either TXT or XML format. You can configure an ER format that contains the **Common file** format element as the root format element and the **Binary contents** format element as the only nested element under it. In this case, the **Common file** format element produces output in the format that you specify by the binding that you configure for the **Binary contents** format element. For example, you can configure this binding to fill in this element with the content of a Document management attachment in PDF or Office (Excel or Word) format. You can print out the output by using the configured **Printer** destination.

> ⓘ **Note**
>
> When you select the **Common\File** format element to configure the **Printer** destination, there's no way to guarantee, at design time, that the selected element produces output in PDF format or output that can be converted to PDF format. Therefore, you receive the following warning message: "Please, make sure that the output that is generated by selected format component can be converted to PDF. Otherwise, uncheck the 'Convert to PDF' option." You must take steps to help prevent runtime issues when non-PDF or non-PDF-convertible output is provided for printing at runtime. If you expect to receive output in Office (Excel or Word) format, the **Convert to PDF** option must be selected.
>
> In version 10.0.26 and later, to use the **Convert to PDF** option, you must select **PDF** for the **Document routing type** parameter of the configured **Printer** destination.

EMF printing

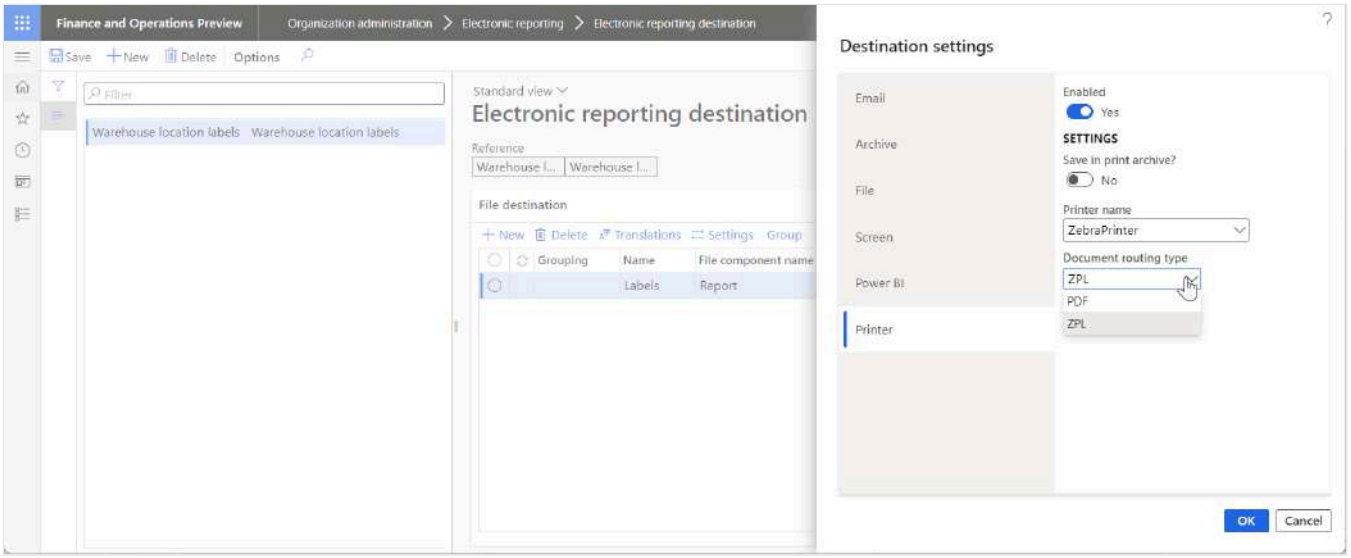
As of Finance version 10.0.32, the capability to generate outbound documents in Enhanced Metafile (EMF) format is implemented. To activate this capability, follow these configuration steps.

1. Make sure that the Enable support of Document Routing agent running as a service feature is turned on in Feature management.
2. Configure the **Printer** destination for the required format component by selecting **EMF** for the **Document routing type** parameter.



ZPL printing

In version 10.0.26 and later, you can configure the **Printer** destination for the **Common\File** format element by selecting **ZPL** for the **Document routing type** parameter. In this case, the **Convert to PDF** option is ignored at runtime, and the TXT or XML output is sent directly to a selected printer by using the Zebra Programming Language (ZPL) contract of the Document routing agent (DRA). Use this feature for an ER format that represents a ZPL II label layout to print various labels.



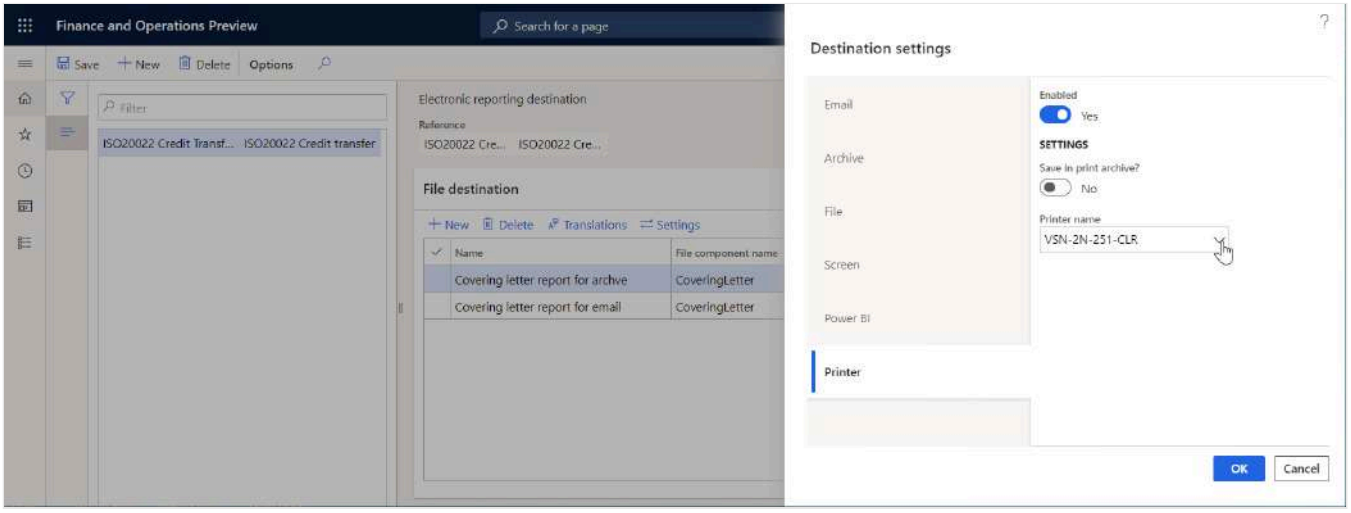
For more information about this feature, see Design a new ER solution to print ZPL labels.

Limitations

The **Printer** destination works only for cloud deployments.

Use the Printer destination

1. Set the **Enabled** option to **Yes** to send a generated document to a printer.
2. In the **Printer name** field, select the required network printer.
3. Set the **Save in print archive?** option to **Yes** to store the generated output in the print archive, so that it's available for further printing. To access archived output later, go to **Organization administration** > **Inquiries and reports** > **Report archive**.



> **Note**
>
> You don't need to turn on the **Convert to PDF** option when you configure the **Printer** destination. The PDF conversion for printing purposes happens even if the option is

turned off.

To use a specific page orientation when you print an outbound document in Excel format, you must turn on the **Convert to PDF** option. When you set the **Convert to PDF** option to **Yes**, the **Page orientation** field becomes available. In the **Page orientation** field, you can select a page orientation.

Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) destinations

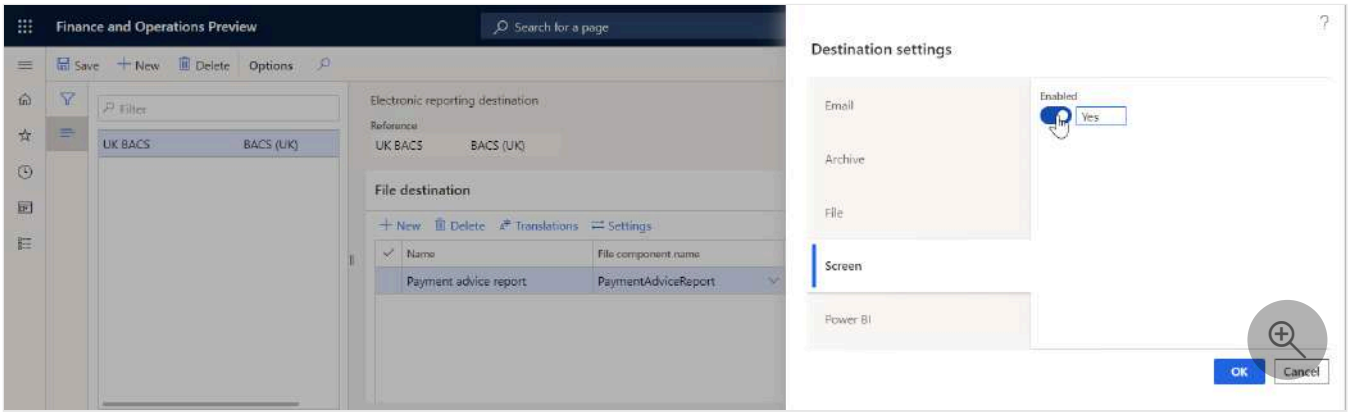
Last updated on 04/13/2026

Screen destination

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can configure a screen destination for each folder or file component of an Electronic reporting (ER) format configured to generate outbound documents. Based on the setting of the destination, a generated document opens for preview in a separate browser tab.

If you set **Enabled** to **Yes**, a preview of the output is created. You can view some file types, such as XML, TXT, or PDF, directly in a browser window. For other file types, such as Microsoft Excel or Word, the Microsoft Office Online service is used.



Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) destinations

Last updated on 04/15/2026

ER Configure destinations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to set up and use different destinations for Electronic reporting (ER) output components, such as a folder or a file. The demo data company used to create this procedure is DEMF. Germany is the country\region of the legal entity's primary address, but you can use any legal entity for this procedure.

The format used in this example is ISO20022 Credit transfer, but you can use any format that you already imported. This procedure is an example of a single file and a single destination setup. For more information about Electronic reporting destination management, see Dynamics 365 Finance Help.

1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting destination**.
2. Select **New** to create a new set of destinations for a format.
3. In the **Reference** field, select a format for which you want to configure destinations.

    - If you don't see a value to select, it means that you didn't import any Electronic reporting format configurations. You must import a format configuration before setting up destinations.

4. Select **New** to create a new file destination.

    - You can create one file destination for each output component of the same format, such as a folder or a file. You can enable and disable destinations separately in the settings.

5. In the **Name** field, enter the user-friendly name of output component.

    - Use meaningful names, such as "Payment file" or "Control report". Users see these names at configuration runtime along with the destination settings.

6. In the **File name**, select a file or folder that is specific to the format.
7. Select **Settings**.
8. Select **Yes** in the **Enabled** field.

    - The **Enabled** check box on each tab enables and disables each destination separately. In this example, you enable sending an output file to a mail recipient when the file is generated.

9. Select **Edit**, to set up email recipients.
10. Select **Add**.
11. Select **Print Management email**.
12. In the **Email source** field, select an option.

    - You can select different email source types, such as a customer or a vendor type. This option defines the type of argument that the **Email source account** formula returns. The **Email source account** formula, described in a following step, is the place where you bind an email source. Select **Vendor** if your formula returns a vendor account. Use **Vendor** if you're using the ISO 20022 Credit Transfer configuration example.

13. Select **Email source** bind button.
14. In the **Formula**, enter a document-specific reference to a party type that you selected earlier.

    - Instead of typing, you can find a data source node that represents the party account, and select the **Add data source** button to update the formula. For example, if you use the ISO 20022 Credit Transfer configuration, the node representing a vendor account is `$PaymentsForCoveringLetter.Creditor.Identification.SourceID`. Otherwise, enter any string value, such as "DE-001", to save a formula.

15. Select **Save**.
16. Close the page.
17. Select **Edit** to configure contact details for the party.
18. Select **Yes** in the **Primary contact** field.

    - You can use different options to indicate what contact type of the party should be used as an email address for this destination. This example uses primary contact.

19. Select **OK**.
20. Select **OK**.

21. In the **Subject** field, type a value.
22. Select **OK**.

Last updated on 04/13/2026

Configure action-dependent ER destinations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can configure destinations for each output component (folder or file) of an Electronic reporting (ER) format configuration that helps generate an outbound document. Users who run an ER format of this type and have appropriate access rights can also change the configured destination settings at runtime.

In Microsoft Dynamics 365 Finance **version 10.0.17 and later**, an ER format can be run by provisioning an action code that the user performs by running that ER format. For example, in the **Accounts receivable** module, in the Print management settings, you can select an ER format that generates a specific business document, such as a free text invoice. You can then select **View** to preview the invoice or **Print** to send it to a printer. If a user action is passed for the running ER format at runtime, you can configure different ER destinations for different user actions. This article explains how to configure ER destinations for this type of ER format.

Make action-dependent ER destinations available

To configure action-dependent ER destinations in the current Finance instance and enable the new ER API, open the Feature management workspace, and turn on the **Configure specific ER destinations to be used for different PM actions** feature. To use configured ER destinations for reports at runtime, enable the **Route output of PM reports based on ER destinations that are user action specific (wave 1)** feature.

Configure action-dependent ER destinations

When you turn on the **Configure specific ER destinations to be used for different PM actions** feature, you can configure action-dependent ER destinations on the **File destination** FastTab of the **Electronic reporting destination** page. For each component, you can add a record and enable specific ER destinations. For each record, you must specify the type of document that the configured ER destinations should apply to. In the **Document type** field, select one of the following values:

- Select **Electronic** to apply the configured destinations if no user action code is provided at runtime.
- Select **Print management** to apply the configured destinations if a user action code is provided at runtime.
- Select **Any** to always apply the configured destinations, regardless of whether a user action is provided at runtime.

If you select the **Print management** document type, you must specify the user actions that the configured ER destinations should apply to. In the **Print management action** field, select one of the following values:

- Select **View** to apply the configured destinations if the **View** user action is provided at runtime.
- Select **Print** to apply the configured destinations if the **Print** user action is provided at runtime.
- Select **Send** to apply the configured destinations if the **Send** user action is provided at runtime.

> **Note**
>
> Multiple actions can be selected for a single destination record.

If you select the **Any** document type, **Autodetect** is automatically selected in the **Print management action** field as a user action, and the following behavior occurs:

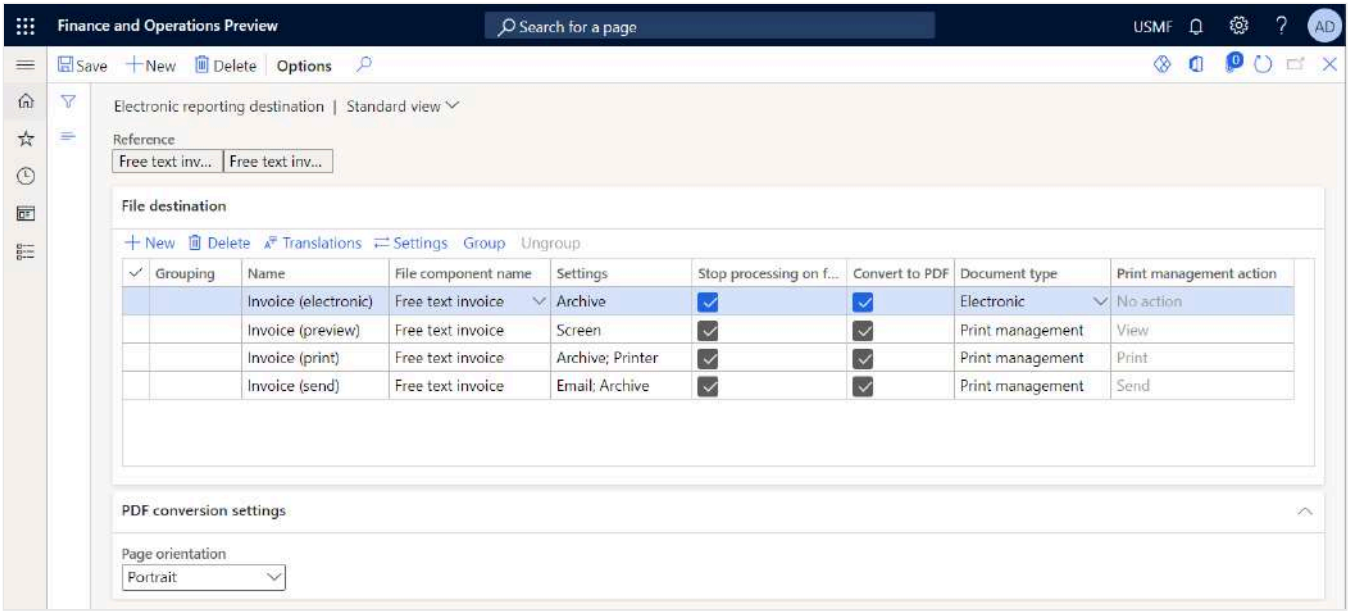
- If no user action code is provided at runtime, all configured ER destinations are applied.
- If a user action code is provided at runtime, an ER destination that is predefined for a specific action is applied, **regardless of whether it's enabled**:
  - When the **View** action is provided at runtime, the **Screen** ER destination is applied.
  - When the **Send** action is provided at runtime, the **Email** ER destination is applied.
  - When the **Print** action is provided at runtime, the **Printer** ER destination is applied.

For example, you can use the **Free text invoice (Excel)** ER format to print a free text invoice when you post it. To route a generated document, you must configure ER destinations for this ER format. For example, you might need to configure these ER destinations to perform the following actions on a generated document:

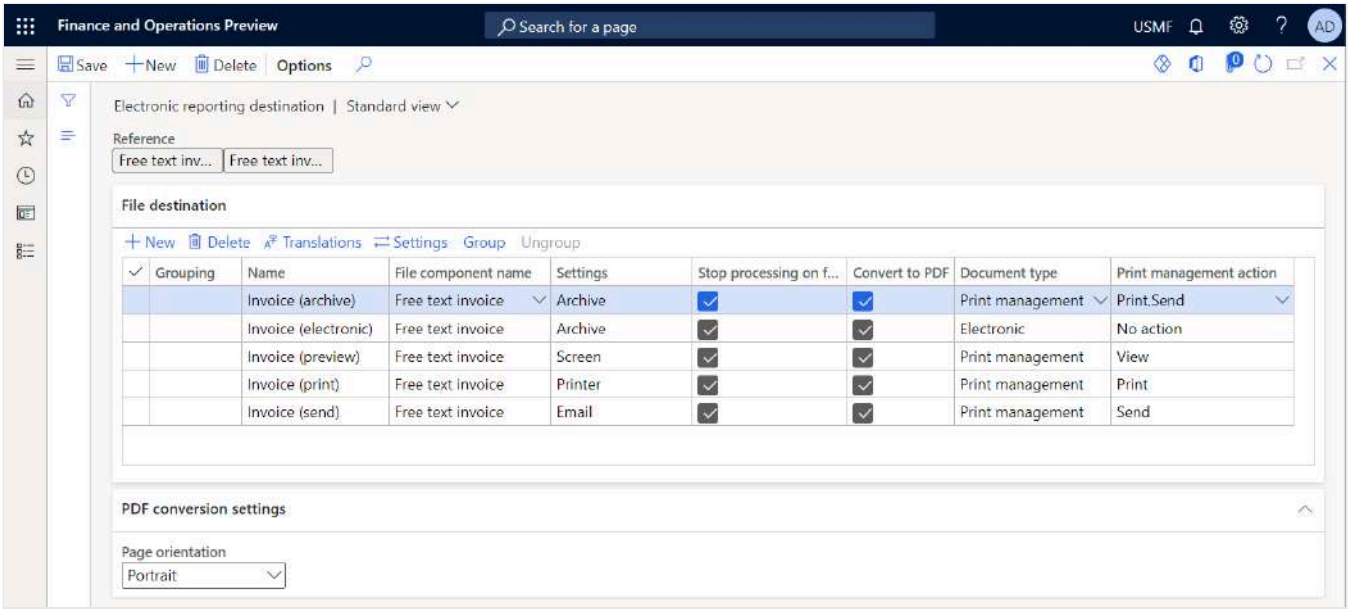
- Archive the document if the ER format is run but no action code is provided (for example, when the document is sent electronically).
- Preview the document in a web browser when a user performs the **View** action.
- Archive and print the document when a user performs the **Print** action.

- Archive the document and email it as the attachment of an outbound email message when a user performs the **Send** action.

The following illustration shows how you can achieve this by configuring ER destinations as the set of individual destination records when every record is configured for an individual user action:



The following illustration shows how you can achieve the same result by configuring ER destinations as the set of individual destination records when every record is configured for an individual destination:

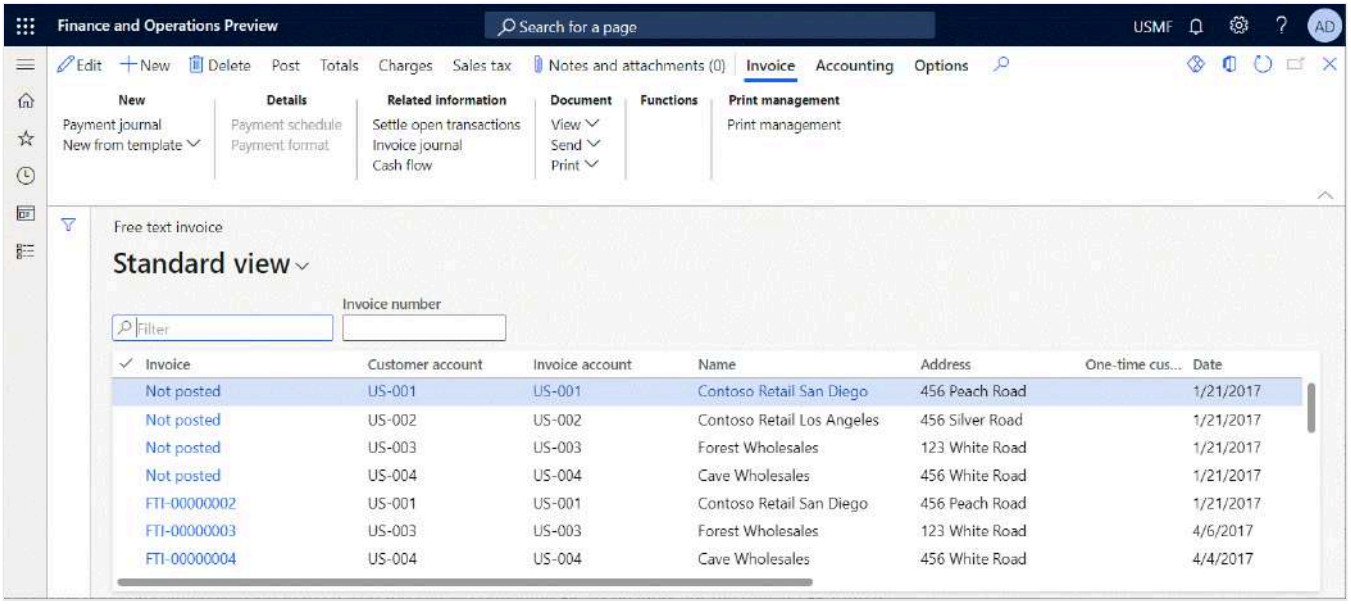


> **Note**
>
> If you provide an action code for the running ER format but don't configure destinations for that action code, the default destination behavior is applied.

Change action-dependent ER destinations at runtime

When an ER format runs, if you provision user actions and have the appropriate permissions to change the configured destination settings at run time, a dialog box appears providing you with the option to change the configured destination settings. This dialog box is optional, and its appearance depends on how you implement the call that the ER framework makes to run an ER format. If this dialog box appears, the corresponding ER destinations are enabled according to the provided user actions.

The following illustration shows an example of the **Electronic reporting format destinations** dialog box that appears when a free text invoice is posted and the **Free text invoice (Excel)** ER format is run to generate this document, if the **Printer** action was provisioned and ER destinations were configured for this format as shown earlier in this article.



> **Note**
>
> If you configured ER destinations for several components of the running ER format, an option will be offered separately for every configured component of the ER format.

If several ER formats are applicable as report templates for the selected document, all ER destinations for all applicable ER report templates are shown in the dialog box and available for manual adjustment at runtime.

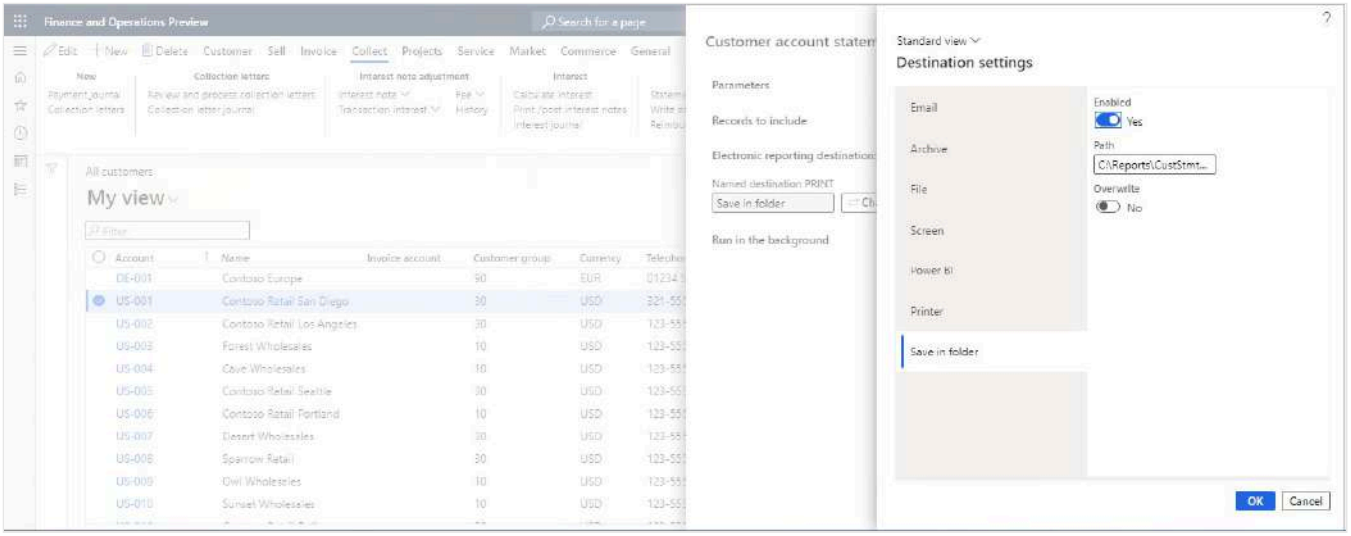
If no SQL Server Reporting Services (SSRS) report templates are applicable to the selected document, the standard selection of Print management destinations is dynamically hidden.

As of Finance version **10.0.31**, you can manually change the assigned ER destinations at runtime for the following business documents:

- Customer account statement
- Interest note
- Collection letter note
- Customer Payment advice
- Vendor Payment advice

To activate the capability to change ER destinations at runtime, enable the **Allow ER destinations adjustment at runtime** feature in the Feature management workspace.

> **Important**
>
> For the **Customer Payment advice** and **Vendor Payment advice** reports, the capability to manually change ER destinations is available only if the **ForcePrintJobSettings** flight is enabled.



> **Note**
>
> When the **Use print management destination** option is set to **Yes**, the system uses the default ER destinations that are configured for specific ER reports. All manual changes that are made in the dialog box are ignored. Set the **Use print management destination** option to **No** to process documents to the ER destinations that are defined in the dialog box immediately before you run the reports.

The following business documents don't assume explicit user selection of an action when they're run:

- Customer account statement

- Interest note
- Collection letter note
- Customer Payment advice
- Vendor Payment advice

The following logic is used to determine which action is used while the preceding reports are processed:

- If the **ForcePrintJobSettings** flight is enabled:
  - If the **Use print management destination** option is set to **Yes**, the **Print** action is used.
  - If the **Use print management destination** option is set to **No**, the **View** action is used.

- If the **ForcePrintJobSettings** flight isn't enabled:
  - If the **Use print management destination** option is set to **Yes**, the **Print** action is used for the **Customer Payment advice** and **Vendor Payment advice** reports.
  - If the **Use print management destination** option is set to **No**, the default SSRS report template is always used for the **Customer Payment advice** and **Vendor Payment advice** reports, regardless of any ER settings that are configured.
  - The **Print** action is always used for the **Customer account statement**, **Interest note**, and **Collection letter note** reports.

For the preceding logic, the **Print** or **View** actions can be used to configure action-dependent ER report destinations. At runtime, only ER destinations that are configured for a specific action are filtered in the dialog box.

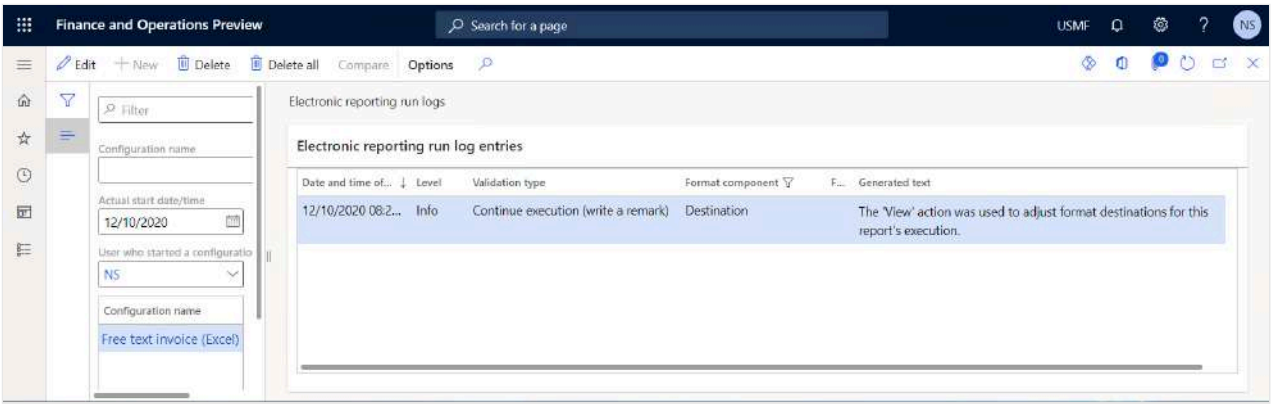
Verify the provided user action

You can verify what user action, if any, is provided for the running ER format when you perform a specific user action. This verification is important when you must configure action-dependent ER destinations, but you aren't sure which user action code, if any, is provided. For example, when you start to post a free text invoice and set the **Print invoice** option to **Yes** in the **Post free text invoice** dialog box, you can set the **Use print management destination** option to **Yes** or **No**.

Follow these steps to verify the user action code that is provided.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In the **User parameters** dialog box, set the **Run in debug mode** option to **Yes**.

4. Perform a user action by running an ER format. Remember that ER user parameters are company-specific and user-specific.
5. Go to **Organization administration** > **Electronic reporting** > **Configuration debug logs**.
6. On the **Configuration debug logs** page, filter the ER run logs to find the log for your ER format run.
7. Review the log entries that must contain the record that presents the provided user action code, if any action has been provided for the ER format run.



Additional resources

Electronic reporting (ER) overview

Electronic reporting (ER) destinations

Electronic reporting framework API changes for Application update 10.0.17

Last updated on 03/11/2026

Configure print management record-specific ER destinations

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how a user in the System Administrator or Accounts Receivable Clerk role can perform the following tasks:

- Configure named Electronic reporting (ER) destinations for an ER solution that generates free text invoices.
- Assign an ER destination to a single print management record.

Complete the procedures in the USMF company. No coding is required.

Introduction

You can configure destinations for each folder in the file output component of an ER format configuration that generates an outbound document. When you run an ER format of this type, if you have the appropriate access rights, you can also change the configured destination settings at runtime.

In Microsoft Dynamics 365 Finance **version 10.0.17 and later**, you can set up an action code for an ER format to specify the action that users perform by running that ER format. For example, in the **Accounts receivable** module, in the print management settings, you can select an ER format that generates a specific business document, such as a free text invoice. You can then select **View** to preview the invoice or **Print** to send it to a printer. If you pass an action for the running ER format at runtime, you can configure different ER destinations for different user actions.

In Finance **version 10.0.21 and later**, you can set up a named destination for an ER format and assign it to the print management record that the ER format processes when it runs. For

example, in the **Accounts receivable** module, in the print management settings, you want to configure the **Original** record to perform the following actions:

- Run an ER format.
- Email the generated invoice to a customer.
- Configure the **Copy** record to run the same format.
- Print a copy of the invoice to a network printer.

In this case, you must configure different ER destinations for the ER format that runs, and you must select the destinations for different print management records.

Prerequisites

Before you begin, the **Allow to set up ER destinations per print management item** feature must be turned on in the Feature management workspace. The source code must also be changed to allow for the configuration of named ER destinations in the current Finance instance and to enable the new ER API.

Additionally, the **Free text invoice (Excel)** ER configuration must be imported into your Finance instance.

Configure a new ER destination

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.

2. Select an invoice for customer account **US-001**.

3. On the **Free text invoice** page, on the **Invoice** tab, in the **Print management** group, select **Print management**.

4. On the **Print management setup** page, expand **Module - accounts receivable** > **Documents** > **Free text invoice**, select the **Original** record, and then follow these steps:
   a. Observe the current settings of the selected record:

      - The default SSRS report **FreeTextInvoice.Report** is selected in the **Report format** field.
      - The **<Default>** name is shown in the **Destination** field informing that there is no custom destination selected for the assigned SSRS report.

   b. In the **Report format** field, select the **Free text invoice (Excel)** ER format configuration.

- The name of the **Destination** field is changed to **Destination name**.
- The **<No named destination>** name is shown in the **Destination name** field informing that there is no named destination selected for the assigned ER format.

c. Select the arrow button to the right of the **Destination name** field.

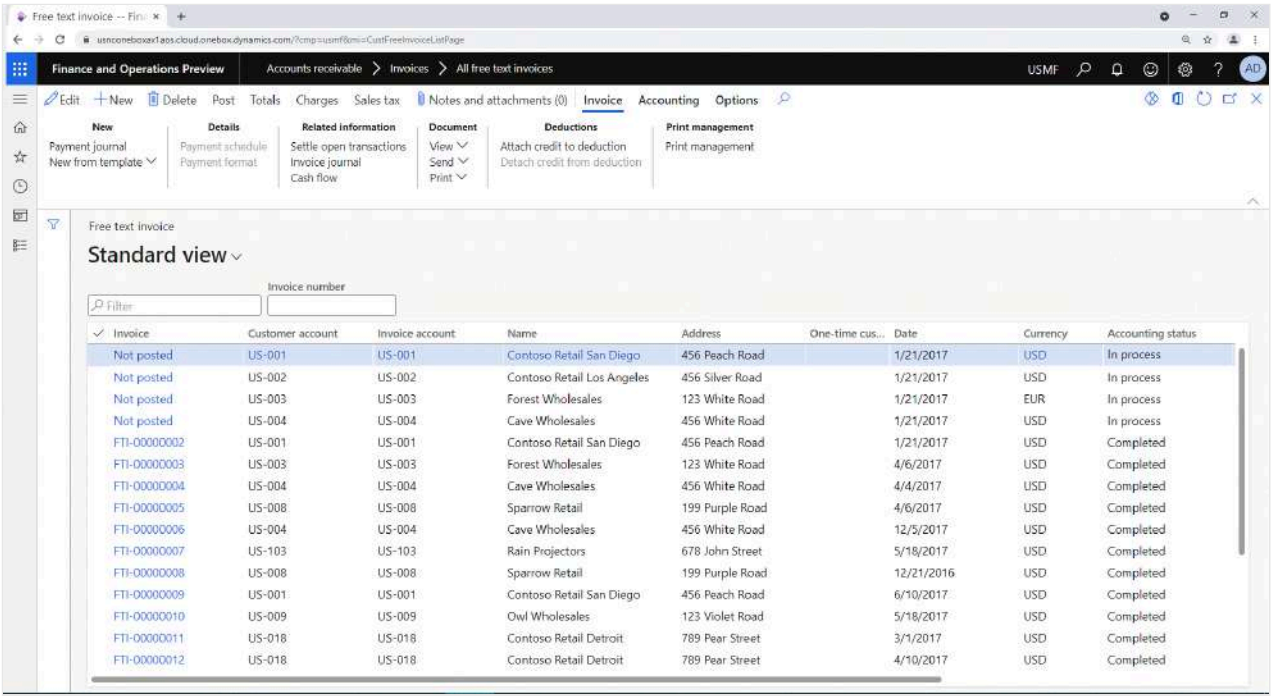
d. On the **Electronic reporting named destination** page, in the **Name** field, enter **Main destination**.

e. Select **Save**.

f. On the **File destination** FastTab, configure the **Email** destination for the **Report** component.

g. Close the **Electronic reporting named destination** page.

h. On the **Print management setup** page, in the **Destination name** field, select the **Main destination** named destination.



You've now configured the named ER destination **Main destination** for the **Free text invoice (Excel)** format and assigned it to the **Original** print management record under **Module - accounts receivable** > **Documents** > **Free text invoice**.

5. Expand **Module - accounts receivable** > **Account - US-001** > **Free text invoice**, select the **Original** record, and then follow these steps:

a. Select and hold (or right-click) the record, and then select **Override**.

b. Select the arrow button to the right of the **Destination name** field.

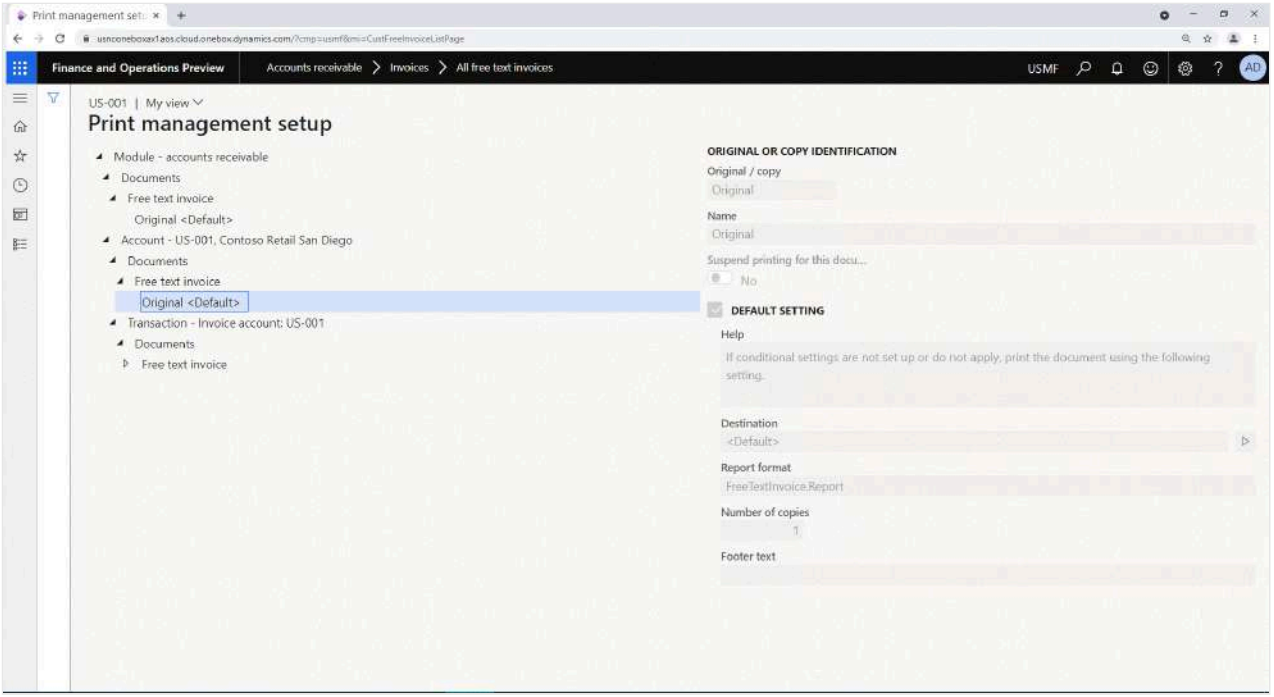
c. On the **Electronic reporting named destination** page, on the Action Pane, select **New**.

d. In the **Name** field, enter **US-001 destination**.

e. On the **File destination** FastTab, configure the **Printer** destination for the **Report** component.

f. Close the **Electronic reporting named destination** page.

g. On the **Print management setup** page, in the **Destination name** field, select the **US-001 destination** named destination.



You've now configured the named ER destination **US-001 destination** for the **Free text invoice (Excel)** format and assigned it to the **Original** print management record under **Module - accounts receivable** > **Account - US-001** > **Free text invoice**.

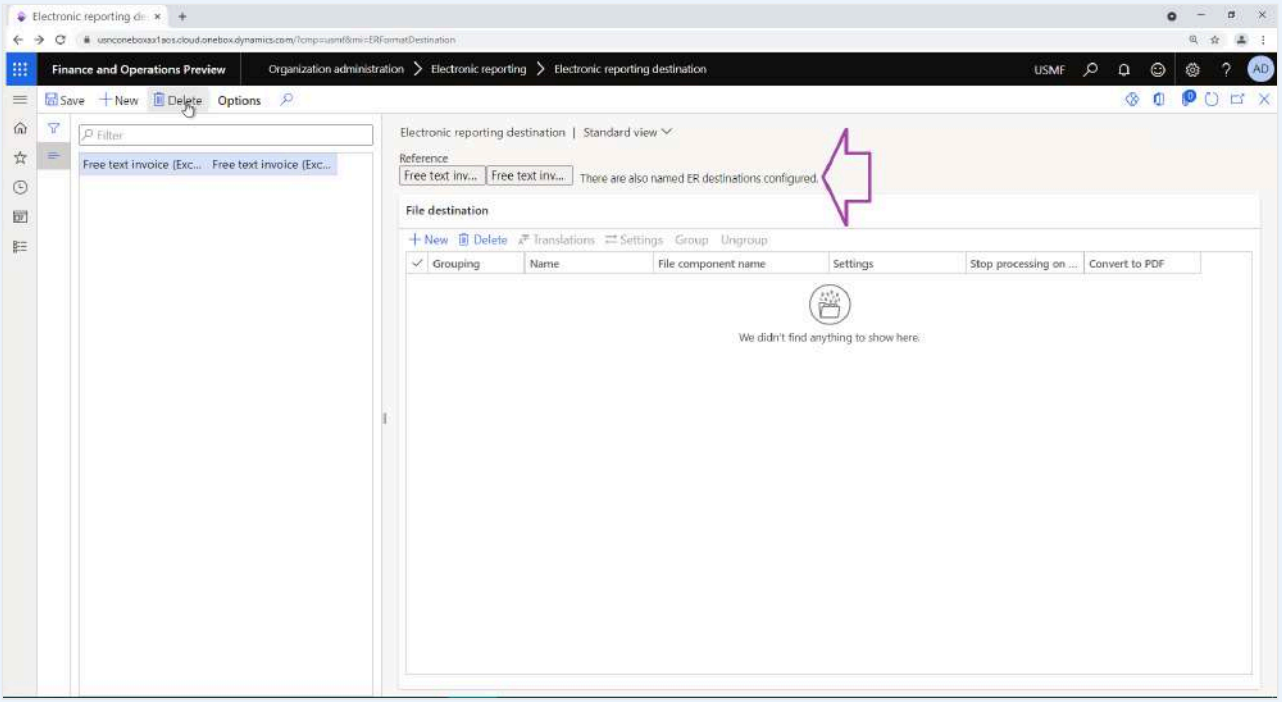
Now, when a free text invoice is processed, the **Free text invoice (Excel)** ER format is run, and one of the following events occurs:

- If the free text invoice is for a customer other than US-001, the generated document is sent by email.
- If the free text invoice is for customer US-001, the generated document is printed.

> ⓘ **Note**
>
> When a named destination hasn't been defined for a print management record that is processed at runtime, the default ER destinations are used if they are available for the ER format that is running.

> ⓘ **Important**
>
> On the **Electronic reporting destination** page (**Organization administration** > **Electronic reporting** > **Electronic reporting destination**), if you select an ER format that at least one

named destination was configured for, you're notified about the presence of named destinations.



If you delete the default destination, all named destinations are also deleted. If those named destinations were selected for print management records, the selection of those named destinations is canceled.

Copy an existing ER destination as a new named destination

When the default named ER destination is available for an ER format, it can be used as a template for new ER destinations. To create a new ER destination that is based on the default destination, select **Copy from default** on the **Electronic reporting named destination** page. The new destination is named **Default**. You can repeat this step as many times as required.

You can select any existing named destination as a template for a new named destination. To create a new named ER destination that is based on a selected named destination, select **Copy** on the **Electronic reporting named destination** page. The new destination has the same name as the selected named destination, but the suffix "Copy" is added. You can repeat this step as many times as required.

Security considerations

You can set up named ER destinations on the **Print management setup** page only if you have permission to perform this task. Permission can be granted to you if the **Configure electronic reporting format destination** duty is assigned to one of your security roles. For more information, see Security considerations.

Additional resources

Electronic reporting (ER) overview

Electronic reporting (ER) destinations

Electronic reporting framework API changes for Application update 10.0.17

Electronic reporting framework API changes for Application update 10.0.21

Change code to enable users to configure and use named ER destinations

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Last updated on 04/08/2026

Output unfolding for Folder component of ER format

When you configure a destination for the **Folder** component of your ER format, you specify how the output of that component is delivered to the configured destination.

Applicability

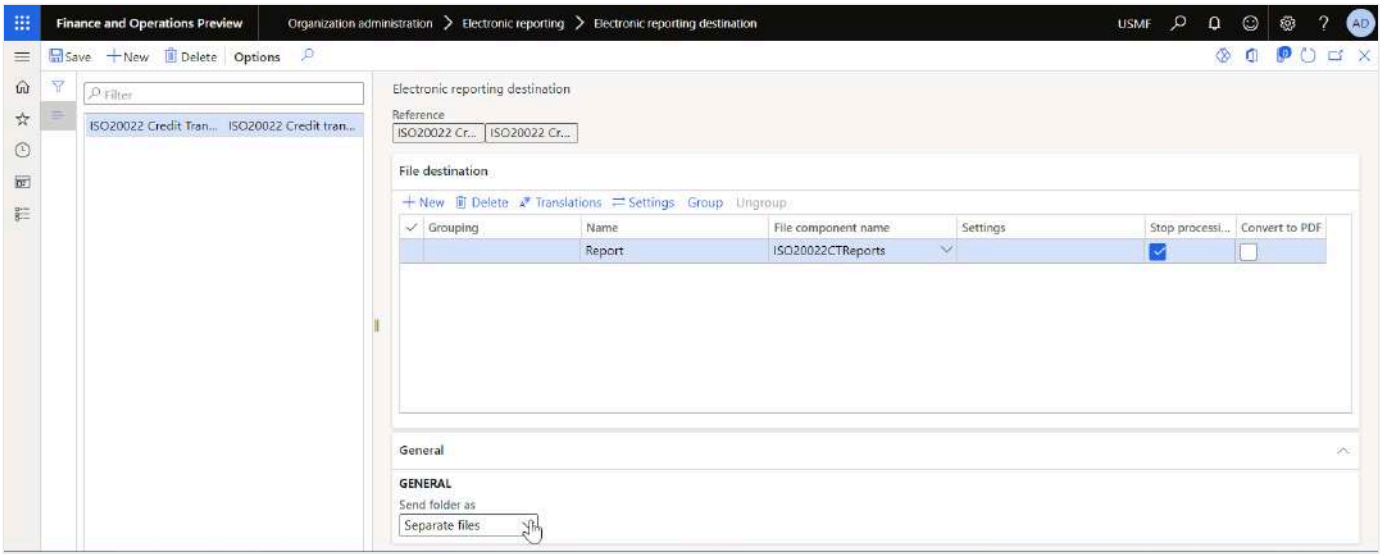
You can configure the output unfolding option only for the format components of the **Folder** type. When you start to configure a **Folder** component, the **General** FastTab is available on the **Electronic reporting destination** page.

Use the output unfolding option

On the **General** FastTab, in the **Send folder as** field, select one of the following values:

- **ZIP archive** – Deliver a generated file as a zip file.
- **Separate files** – Deliver every file of a generated zip file as an individual file.

> ⓘ **Note**
>
> When you select **Separate files**, the generated output is collected in memory in a zipped state. Therefore, the maximum file size limit is applied for zipped output when the real file size can exceed this limit. Select this value when you expect the size of the generated output to be large.



Limitations

If you set the **Send folder as** field to **Separate files** for a **Folder** component that contains other nested **Folder** components, the setting doesn't apply recursively to the nested **Folder** components.

Last updated on 10/02/2025

Change page layout properties of a template in ER destinations

You can configure an ER destination for an ER format component that's designed to use a template in a Microsoft Office (Excel or Word) format for report generation. If you aren't the owner of this format and you need to change page layout properties of the format's template, you can change the page layout properties of the template at runtime to avoid creating and maintaining the derived format configuration. To do this, set up the desired properties as part of the settings of the configured ER destination. When you run an ER format and execute an ER destination that's configured to use certain page layout properties, the values of page layout properties of the executed destination are applied to the template you use, replacing the original template's properties. You can configure different destinations for the same format's component configuring different page layout properties for the template in use.

The following properties can be configured in an ER destination for a format component that's designed to use a template in Excel or Word format:

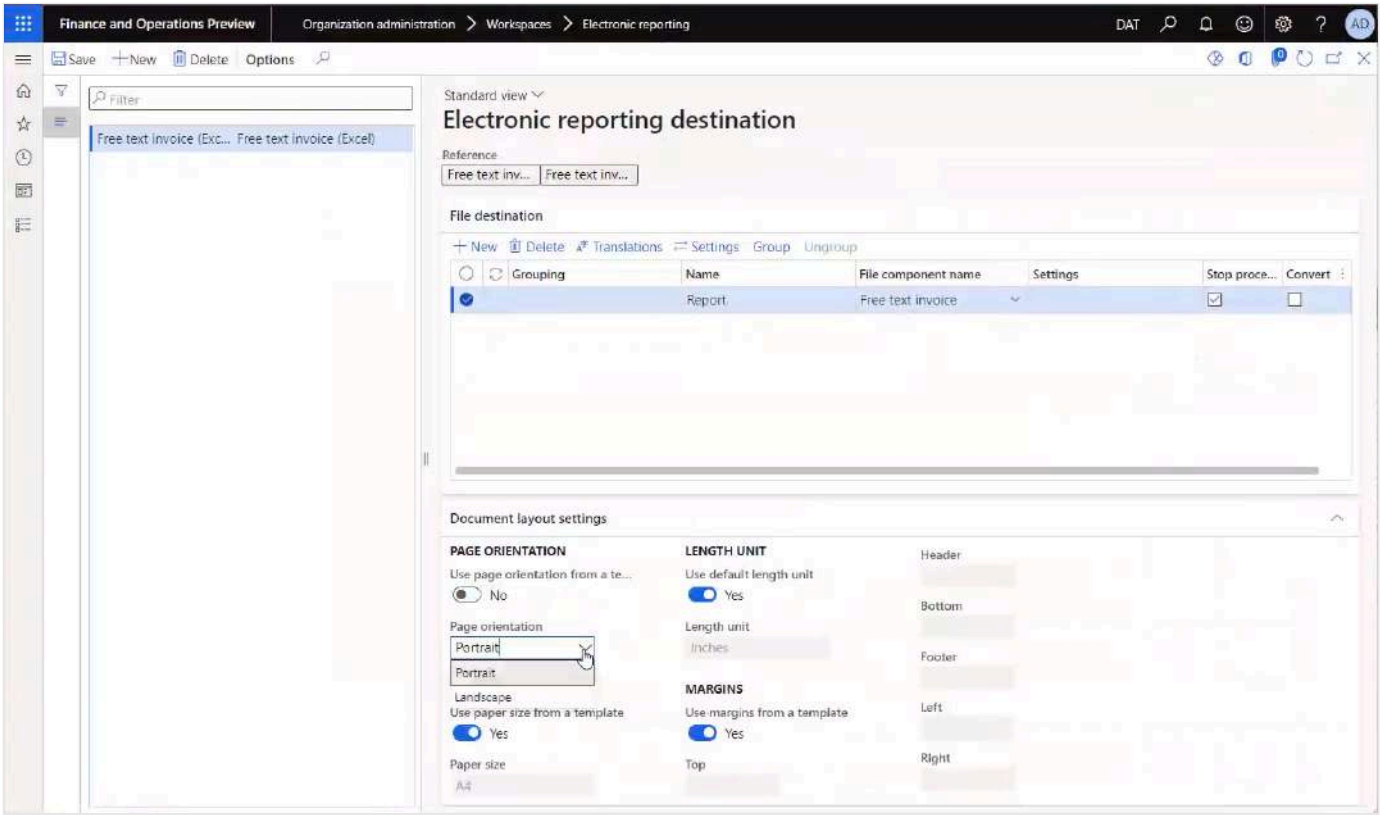
- Page orientation
  - Portrait
  - Landscape
- Paper size
  - A3
  - A4
  - A5
  - B4
  - B5
  - Executive
  - Legal
  - Letter
  - Statement
  - Tabloid
- Page margins
  - Top
    - Header
  - Bottom
    - Footer

- Left
- Right

> **Note**
>
> The page orientation of the template that's configured in this way must be aligned with the page orientation for PDF conversion if the PDF conversion is configured.

You must select the length unit for setting page margins:

- Inches
- Centimeters
- Millimeters



> **Tip**
>
> When you specify a margin value in centimeters with multiple decimals, the value is rounded at runtime to the nearest value with one decimal point.
>
> When you specify a margin value in millimeters with decimals for Excel, the value is rounded at runtime to the nearest integer value with no decimal point.

When you specify a margin value in millimeters with multiple decimals for Word, the value is rounded at runtime to the nearest value with one decimal point.

Last updated on 04/13/2026

Output conversion to PDF

Use the PDF conversion option to convert output in Microsoft Office (Excel or Word) format to PDF format.

Turn on the PDF conversion option for **Excel\File** and **Common\File** components that generate output in Office (Excel or Word) format. When you turn on this option, the output that you generate in Office format is automatically converted to PDF format.

> ⓘ **Note**
>
> Pay attention to the warning message that you receive when you turn on the PDF conversion option for an ER component of the **Common\File** type. This message informs you that there's no way to guarantee, at design time, that the selected file component exposes the content in PDF format or the PDF-convertible content at runtime. Therefore, turn on the option only if you're sure that the selected file component was configured to expose the content in PDF format or the PDF-convertible content at runtime.
>
> If you turn on the PDF conversion option for a format component, and if that component exposes content in a format other than PDF that can't be converted to PDF format, an exception occurs at runtime. The message that you receive informs you that the generated content can't be converted to PDF format.

Limitations

In versions earlier than Dynamics 365 Finance version 10.0.43, you can only convert PDFs outside the current Finance instance. A generated file is sent out of Finance to the conversion service, and that service returns the converted document. This solution has the following limitations:

- You can use the PDF conversion option for cloud deployments and on-premises deployment that has Internet connectivity enabled.
- The PDF document that you produce has a maximum length of 300 pages.

In Dynamics 365 Finance version **10.0.43** and later, you can enable the **In-App PDF conversion for Configurable Business Documents (CBD)** feature in **Feature management** to facilitate the seamless conversion of configurable business documents from Word or Excel formats to PDF.

> **Note**
>
> The **In-App PDF conversion for Configurable Business Documents (CBD) (preview)** feature is available in Finance version 10.0.43 and the 10.0.2095.**124** build and later of the 10.0.42 version.

This feature uses Application Object Server (AOS) resources to remove the need for external conversion services. By using in-app capabilities, you get efficient and secure document processing and reduce dependency on tools outside of Finance while maintaining high performance and reliability. This enhancement supports a wide range of business scenarios to provide users with the flexibility to generate and distribute professional-grade PDF documents directly within the application.

The following advantages come with in-app PDF conversion when you enable the **In-App PDF conversion for Configurable Business Documents (CBD)** feature:

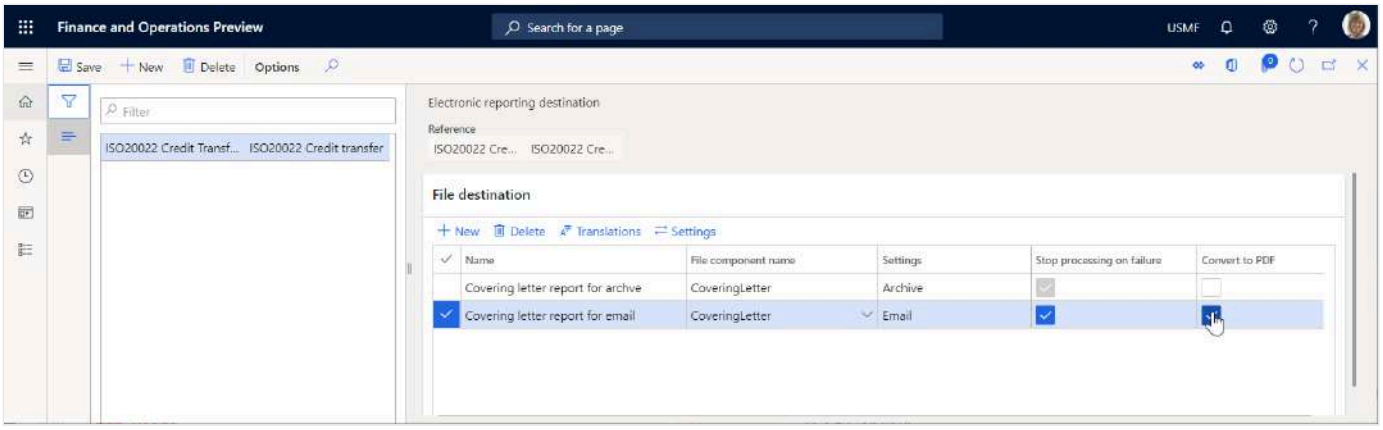
- The PDF document that you produce doesn't have a maximum of 300 pages.
- The Word document that you convert can have a large number of content controls.
- On-premises deployments don't require Internet connectivity.

> **Note**
>
> Only the common system fonts of the Windows operating system are used to convert output that contains no embedded fonts.

> **Tip**
>
> When using the **In-App PDF conversion for Configurable Business Documents (CBD)**, you might notice the font in the generated PDFs has a lighter weight compared to the one in Excel templates. This difference can result in small visual discrepancies between your original template and the final PDF output. Configuration engineers should preview the generated PDF before completing and sharing their configurations.

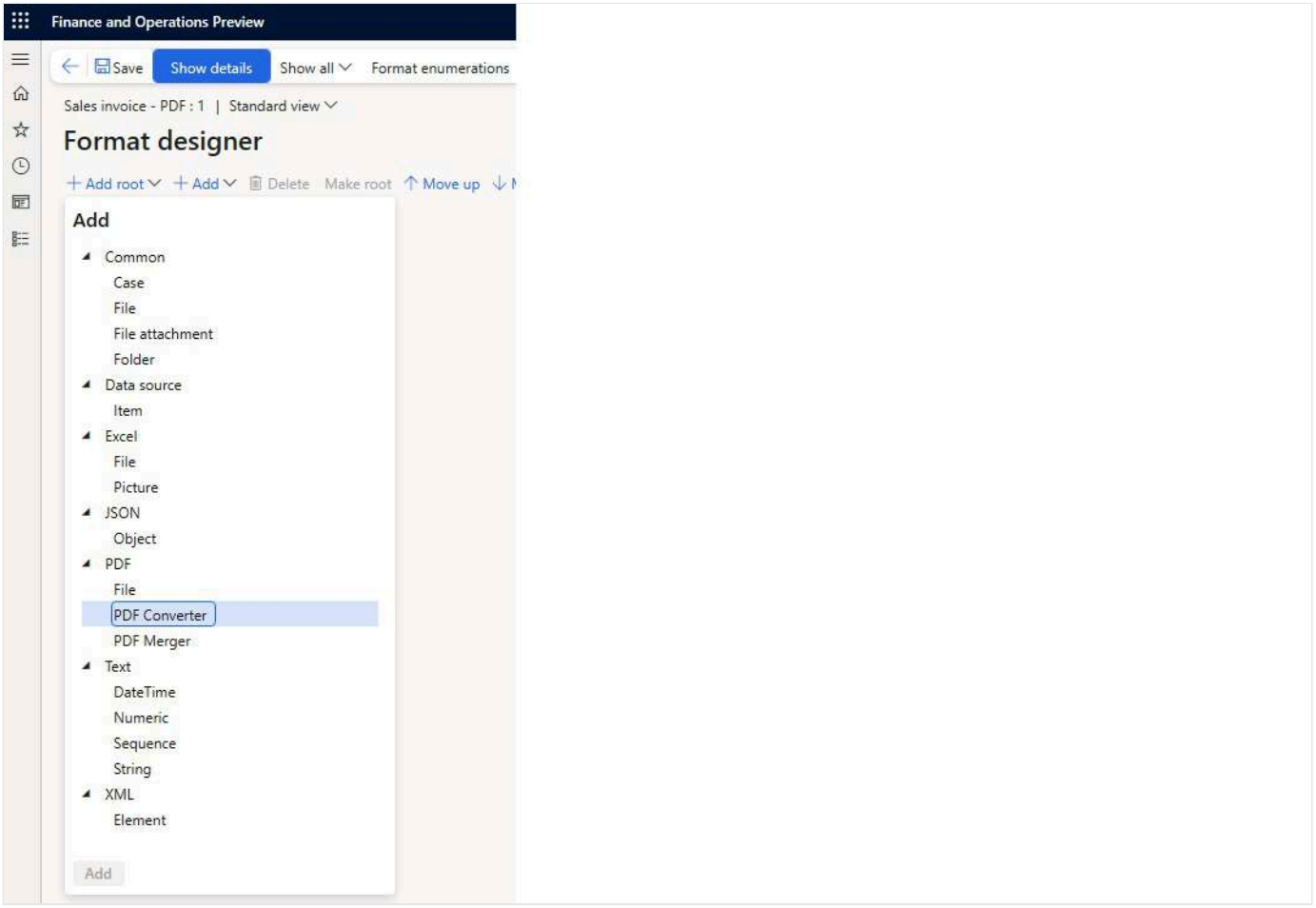
Use the PDF conversion option

To turn on PDF conversion for a file destination, select the **Convert to PDF** check box.



Use the PDF conversion component in ER format configurations

You can use PDF conversion directly in format configurations. In this case, you don't have to use the **Convert to PDF** checkbox on the **Electronic reporting destination** page. A format component, **PDF Converter**, is available.

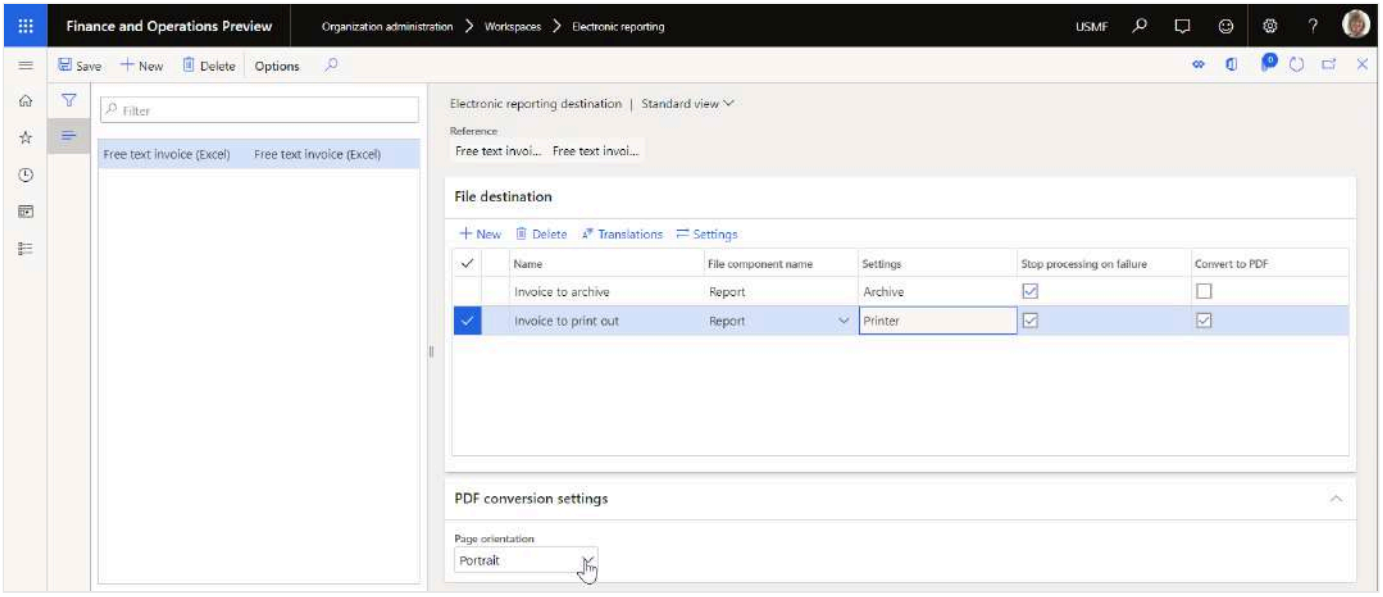


> **Note**
>
> You can add the **PDF Converter** component only in the format configurations of a **PDF** format type, or when the format type isn't explicitly defined but is left blank. The **PDF**

Converter component can contain only one **Excel\File** subcomponent.

Select a page orientation for PDF conversion

If you generate an ER configuration in Excel format and want to convert it to PDF format, you can specify the page orientation of the PDF document. When you select the **Convert to PDF** checkbox to turn on PDF conversion for a file destination that produces an output file in Excel format, the **Page orientation** field becomes available on the **PDF conversion settings** FastTab. In the **Page orientation** field, select your preferred orientation.



The following page orientation options are supported:

- Portrait
- Landscape
- Worksheet specific

The selected page orientation applies to all pages of an outbound document that you generate in Excel format and convert to PDF format. When you select the **Worksheet specific** option, every worksheet of a generated Excel workbook is converted to PDF using the page orientation that you configured for this worksheet in the Excel template. You might have a final PDF document containing portrait and landscape pages.

> ⓘ **Note**
>
> For ER configuration in Word format converted to PDF format, the page orientation of the PDF document always comes from the Word document.

Last updated on 10/02/2025

Export, import, and replicate ER destinations

As of version **10.0.46** of Microsoft Dynamics 365 Finance, you can use these capabilities to streamline the process of configuring destinations across multiple legal entities or environments.

- **Export and import** destination configurations by using the Data Management Framework (DMF).
- **Replicate** destinations across multiple legal entities within the same environment.

These capabilities reduce manual effort, improve consistency, and make ER destination setup more scalable.

Export and import ER destinations

You can now use a dedicated **Data entity** to export and import ER destination configurations between environments.

Expand table

Data entity name	Target data entity	Tables included
File destination	**ERFormatFileDestinationEntity**	ERFormatDestinationTable ERFormatFileDestinationTable ERFormatDestinationNamedTable

For more information about data entities in Finance, see Data entities overview.

Replicate ER destinations across legal entities

1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting destination**.
2. In the list on the left, select one or more ER formats.
3. On the action pane, select the **Replicate** button.
4. Select one or more legal entities to copy destinations into.
5. (Optional) To also replicate **named destinations** that are used in **Print management** for configurable business documents, select the **Include named destinations** checkbox. For more information about named destinations used in **Print management**, see Configure print management record-specific ER destinations.
6. Select **OK** to replicate.

The process copies all destinations for the selected formats into the chosen companies, including those destinations that use container fields.

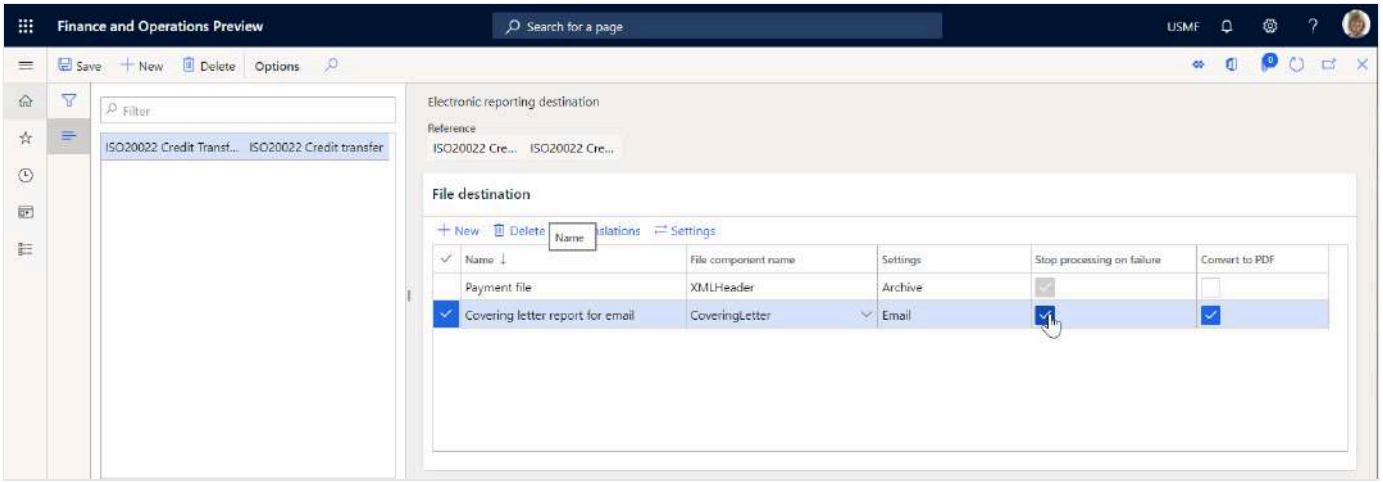
> **Note**
>
> You can apply replication to multiple formats and multiple legal entities in a single operation, which saves significant setup time.

Last updated on 10/02/2025

Electronic reporting destination failure handling

Usually, you run an ER format within the scope of a specific business process. However, sometimes you need to consider the delivery of an outbound document that the ER format generates during execution as part of that business process. In this case, if delivery of a generated outbound document to a configured destination isn't successful, you must cancel execution of the business process. To configure the appropriate ER destination, select the **Stop processing on failure** option.

For example, you configure vendor payment processing so that the **ISO20022 Credit Transfer** ER format runs to generate the payment file and supplementary documents (such as the covering letter and control report). If a payment should be considered successfully processed only if the covering letter is successfully delivered by email, you must select the **Stop processing on failure** checkbox for the **CoveringLetter** component in the appropriate file destination, as shown in the following illustration. In this case, the status of the payment that you select for processing changes from **None** to **Sent** only when the covering letter that's generated is successfully accepted for delivery by an email provider that's configured in the Finance instance.



If you clear the **Stop processing on failure** checkbox for the **CoveringLetter** component in the destination, a payment is considered successfully processed even if the covering letter isn't successfully delivered by email. The status of the payment changes from **None** to **Sent** even if the covering letter can't be sent because, for example, the email address of the recipient or sender is missing or incorrect.

Last updated on 10/02/2025

Design a new ER solution to print ZPL labels

This article explains how a user in the System Administrator, Electronic Reporting Developer, or Electronic Reporting Functional Consultant role can configure parameters of the Electronic reporting (ER) framework, design the required ER configurations of a new ER solution to access the data of the Warehouse management system, and generate custom warehouse location labels in Zebra Programming Language (ZPL) II format. You can complete these steps in the **USRT** company.

Business scenario

You represent a company that implemented warehouse management in Microsoft Dynamics 365 Finance. Every warehouse location must be labeled with a self-adhesive label that includes a bar code. Warehouse workers use handheld bar code readers to scan the bar codes.

All warehouse locations are labeled in the scope of pre-go-live activities. However, you must also be able to print warehouse location labels on demand if existing labels become damaged or warehouse shelving is reconfigured. By using recently released ER functionality, you can configure a new ER solution that lets a warehouse supervisor print labels directly to a thermal label printer.

Configure the ER framework

Follow the steps in Configure the ER framework to set up the minimal set of ER parameters. You must complete this setup before you start to use the ER framework to design a new ER solution.

Design a domain-specific data model

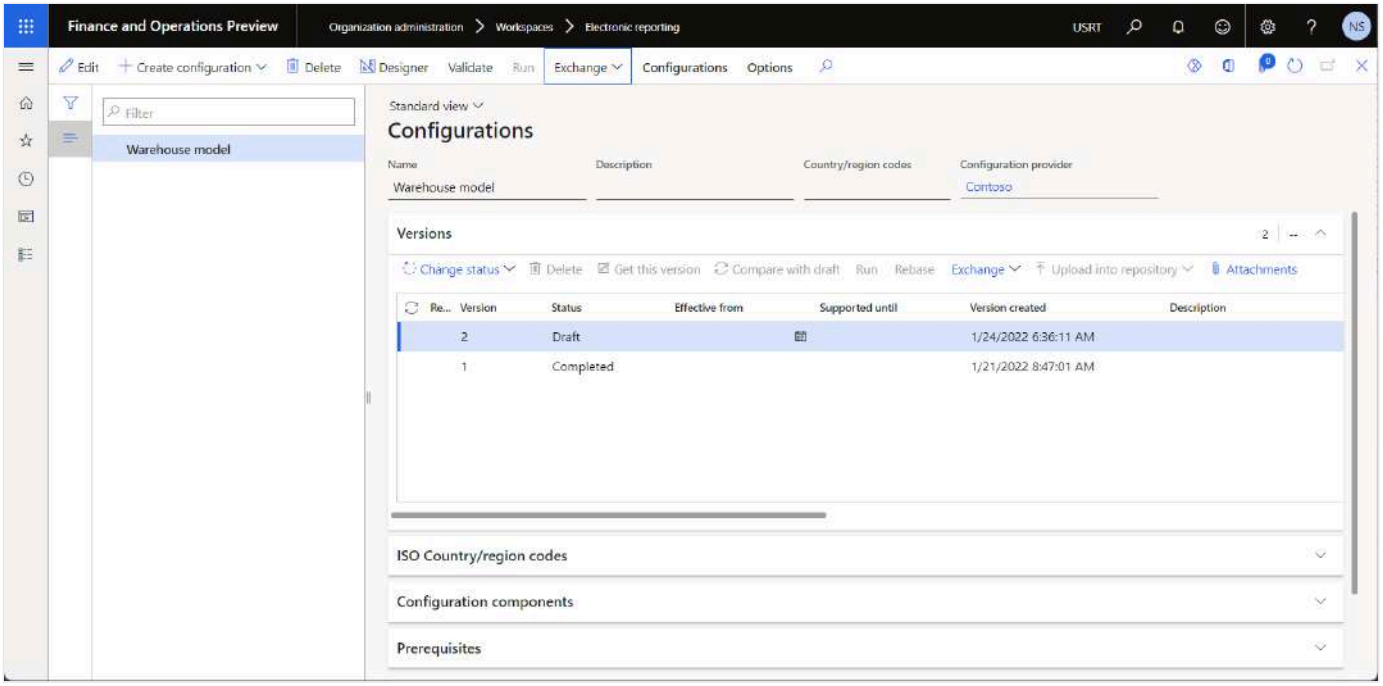
Create a new ER configuration that contains a data model component for the Warehouse management domain. You use this data model as a data source later, when you design an ER format to generate warehouse location labels.

Import a data model configuration

Follow these steps to import the required data model from an XML file that Microsoft provides. Alternatively, you can create your own data model as described in the next section.

1. Download the Warehouse model.version.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.

3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the **Configurations** page, on the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Warehouse model.version.1.xml** file.
6. Select **OK** to import the configuration.

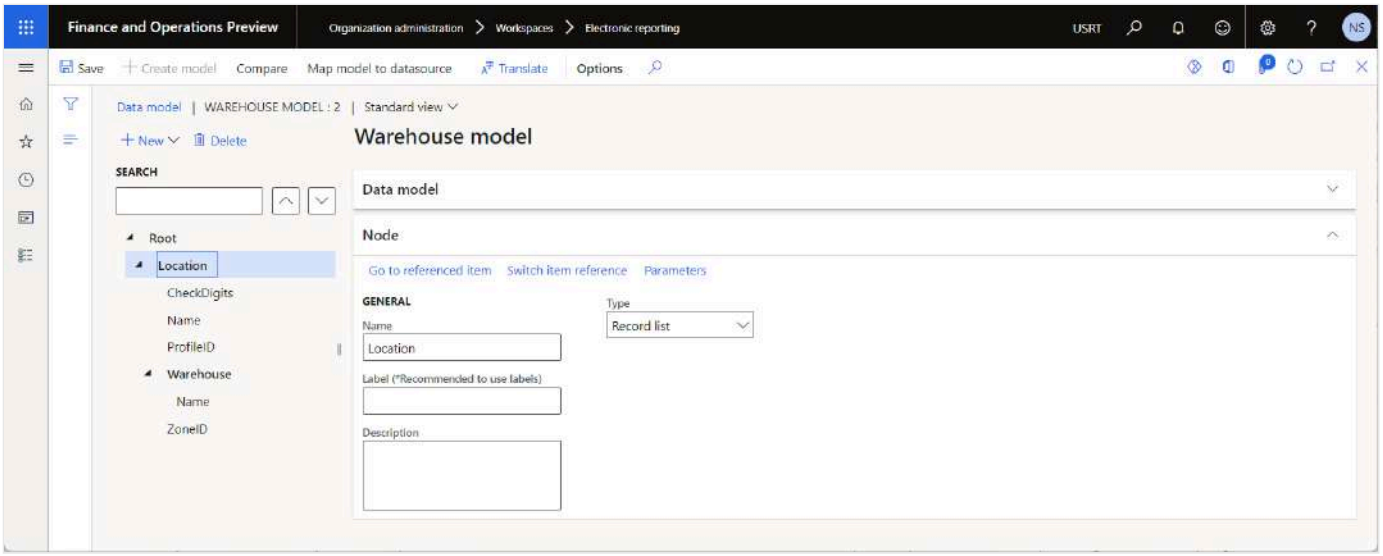


Create a data model configuration

Instead of importing the Microsoft-provided data model file, you can create a data model from scratch. For an example that shows how to complete this task, see Create a new data model configuration.

Review the data model

You can view an editable version of the configured data model on the **Data model designer** page.



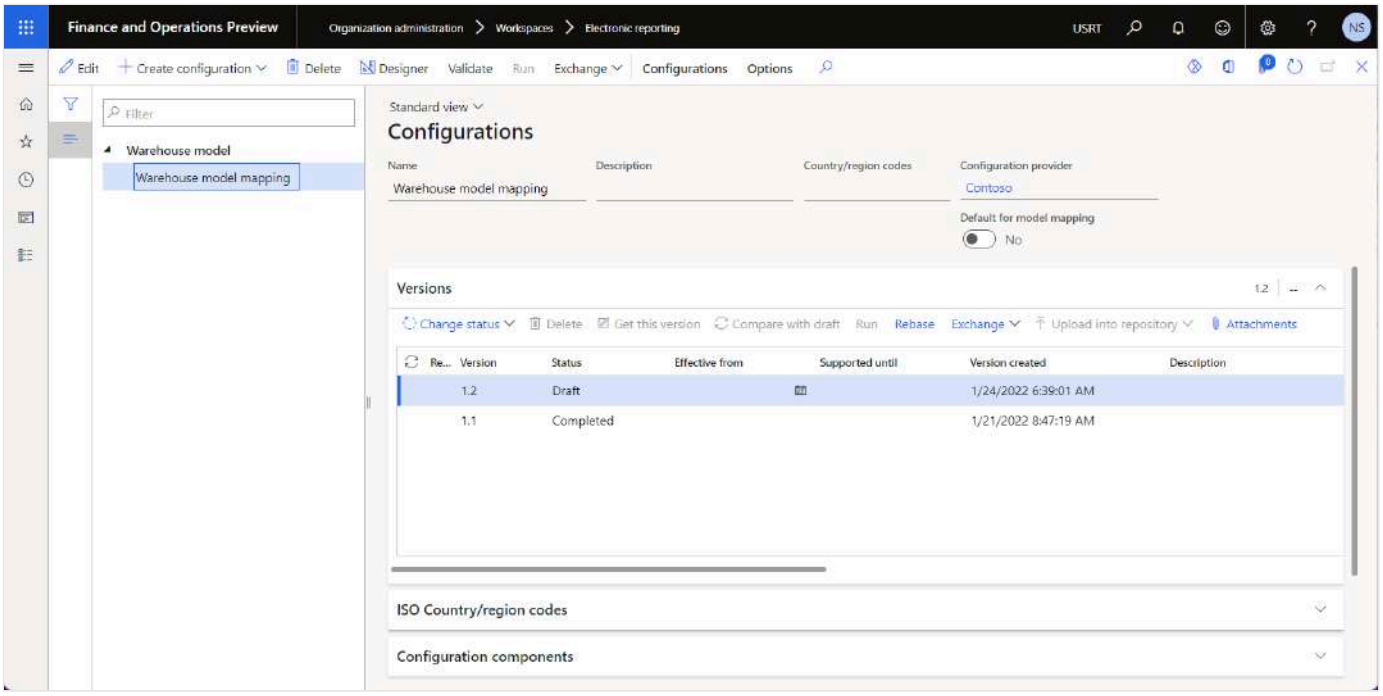
Design a model mapping for the configured data model

As a user in the Electronic Reporting Developer role, you must create a new ER configuration that contains a model mapping component for the Warehouse data model. This component implements the configured data model for Dynamics 365 Finance and is specific to that app. You must configure it to specify the application objects that fill the configured data model with application data at runtime. To complete this task, you must understand how the data structure of the Warehouse management business domain is implemented in Finance.

Import a model mapping configuration

Follow these steps to import the required model mapping from an XML file that Microsoft provides. Alternatively, you can create your own model mapping as described in the next section.

1. Download the Warehouse model mapping.version.1.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the **Configurations** page, on the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Warehouse model mapping.version.1.1.xml** file.
6. Select **OK** to import the configuration.

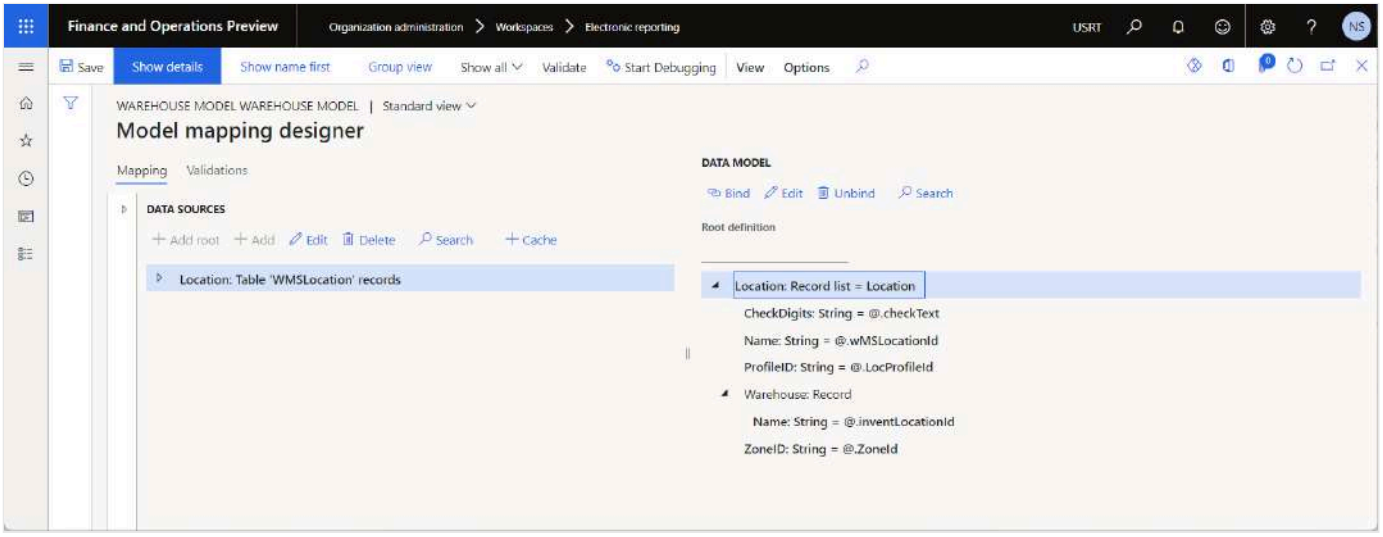


Create a model mapping configuration

Instead of importing the Microsoft-provided model mapping file, you can create a model mapping from scratch. For an example that shows how to complete this task, see Create a new model mapping configuration.

Review the model mapping

You can view an editable version of the configured model mapping on the **Model mapping designer** page.



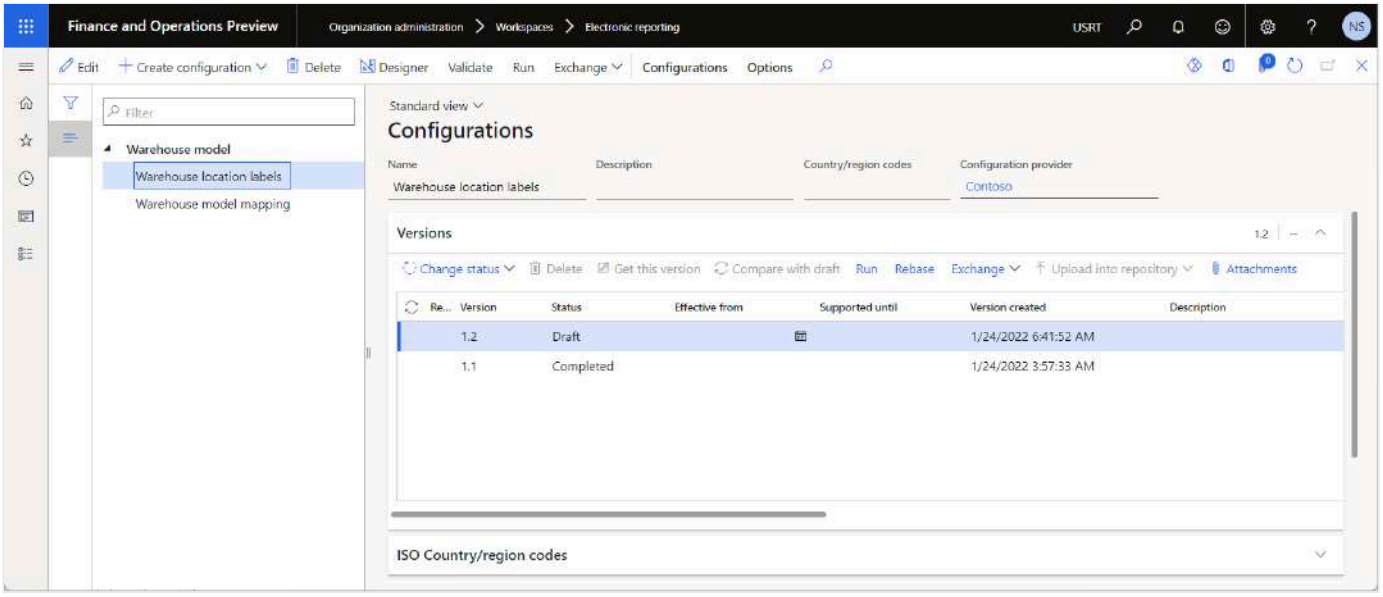
Design a format

As a user in the Electronic Reporting Functional Consultant role, you must create a new ER configuration that contains a format component. To configure this component, use ZPL II code to specify the layout of your warehouse location label.

Import a format configuration

Follow these steps to import the required format from an XML file that Microsoft provides. Alternatively, you can create your own format as described in the next section.

1. Download the Warehouse location labels.version.1.1.xml file, and save it to your local computer.
2. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
3. In the **Electronic reporting** workspace, select **Reporting configurations**.
4. On the **Configurations** page, on the Action Pane, select **Exchange** > **Load from XML file**.
5. Select **Browse**, and then find and select the **Warehouse location labels.version.1.1.xml** file.
6. Select **OK** to import the configuration.

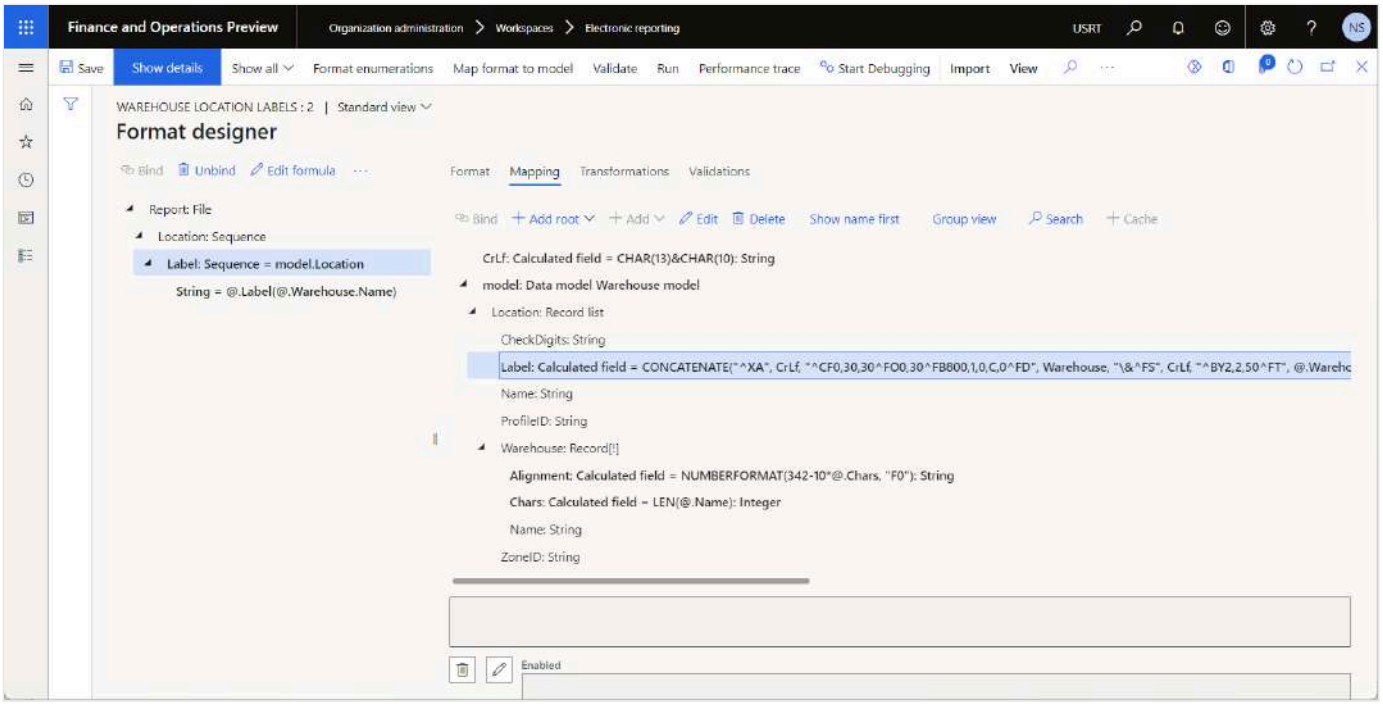


Create a format configuration

Instead of importing the Microsoft-provided format file, you can create a format from scratch. For an example that shows how to complete this task, see Create a new format configuration.

Review the format

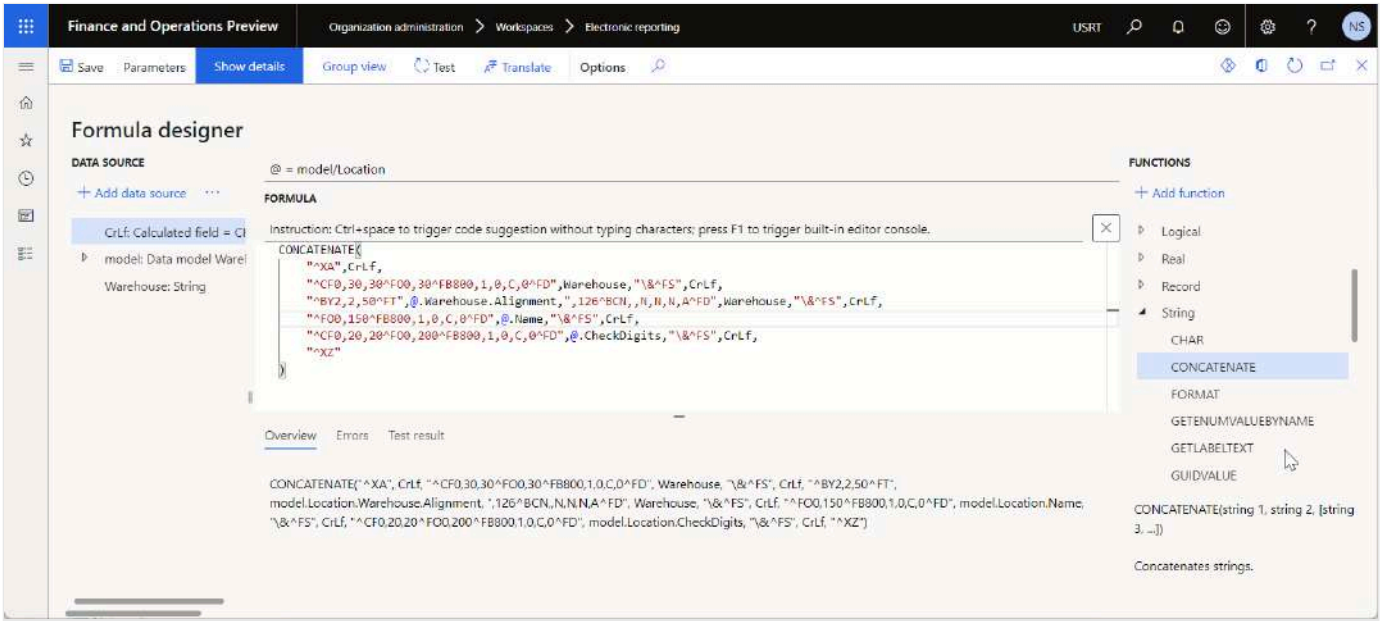
You can view an editable version of the configured format on the **Format designer** page.



The `model.Location.Label` data source of this format is configured to generate labels that contain the following information:

- The warehouse title as text
- The warehouse title as a bar code
- The location title
- Check digits

On the **Formula designer** page for the data source, the ER formula that generates labels includes a `CONCATENATE` function that combines the information in the desired layout.



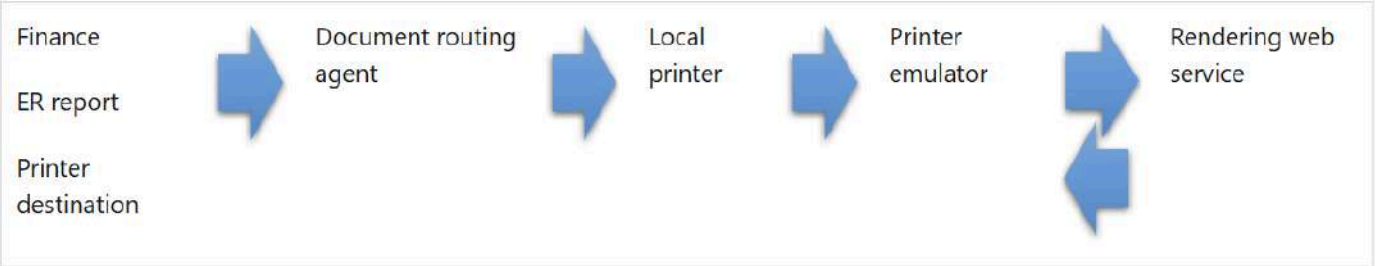
> **Tip**

The label layout is designed so that the location title and the check digits are aligned in the center of the label. However, ZPL II doesn't support center alignment for bar codes. Therefore, the formula of the `model.Location.Warehouse.Alignment` data source aligns the bar code in the center of the label. This formula calculates the left offset of the bar code, based on the number of characters in the warehouse title.

Prepare your environment for previewing generated labels

The following example uses a printer emulator application for ZPL labels to show a preview of generated labels on the screen. Follow these steps to enable this option.

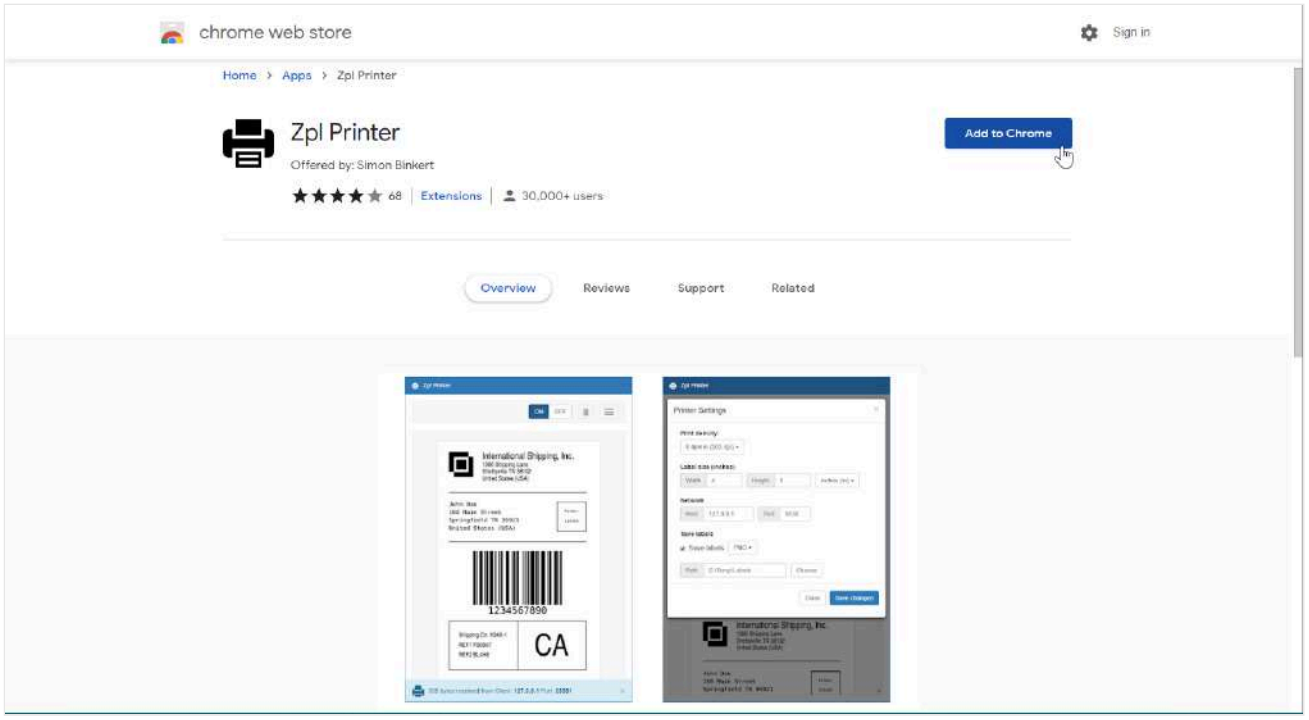
1. Add the Printer ER destination for the **Warehouse location label** ER format, and configure it to send generated labels from Finance to the Document routing agent (DRA).
2. Install and configure the DRA to route generated labels from Finance to a local printer that you can access from the current workstation.
3. Add a local printer for the current workstation, and configure it to pass generated labels from the DRA to a printer emulator application.
4. Install a printer emulator application as an extension of the Chrome web browser, and configure it to pass generated labels from a local printer to a web service that renders generated labels and returns them to the printer emulator for preview.



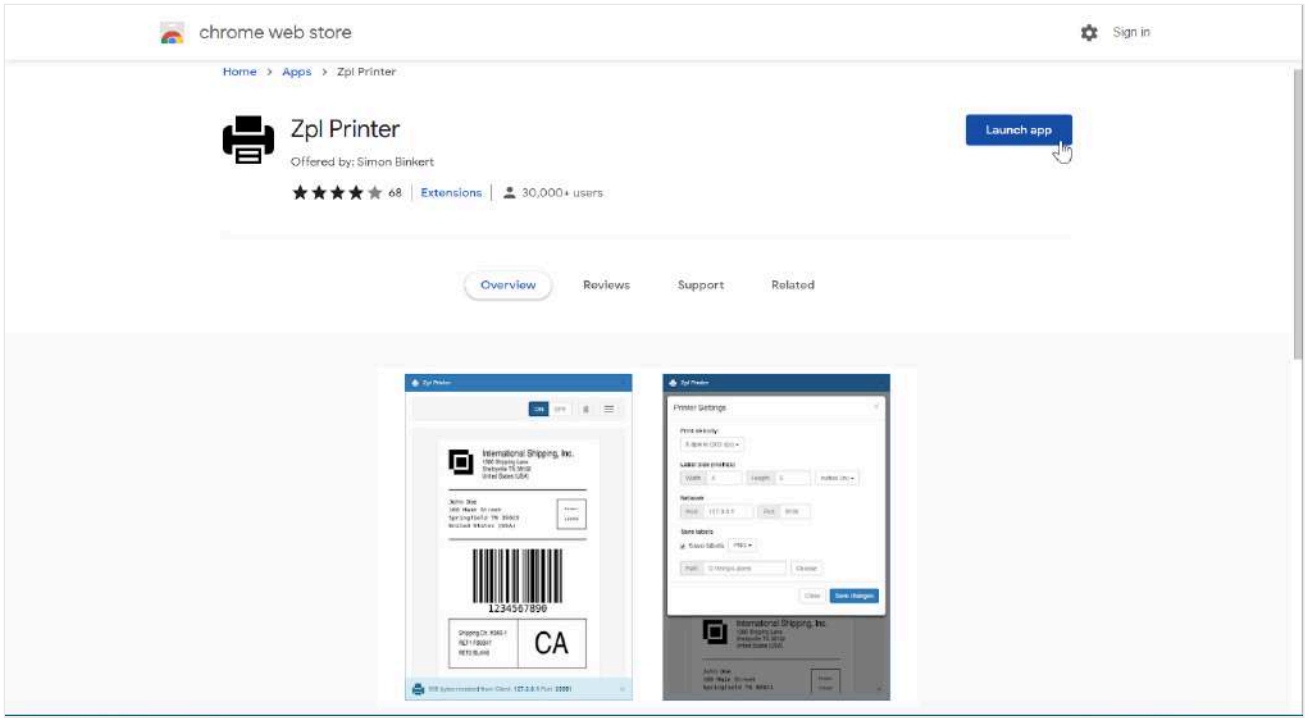
Install and configure a printer emulator application

Add a printer emulator application for the ZPL rendering engine to your Chrome web browser. This example uses the Zpl Printer emulator that is based on the Labelary ZPL web service. The printer emulator application passes generated labels in ZPL format from a local printer to the web service and then returns labels as PDF or PNG files for preview.

1. In the Chrome web store, find and select the printer emulator application that you want to use. Then select **Add to Chrome** to add it to your Chrome web browser.



2. Select **Launch app** to run the printer emulator application from the Chrome web browser.

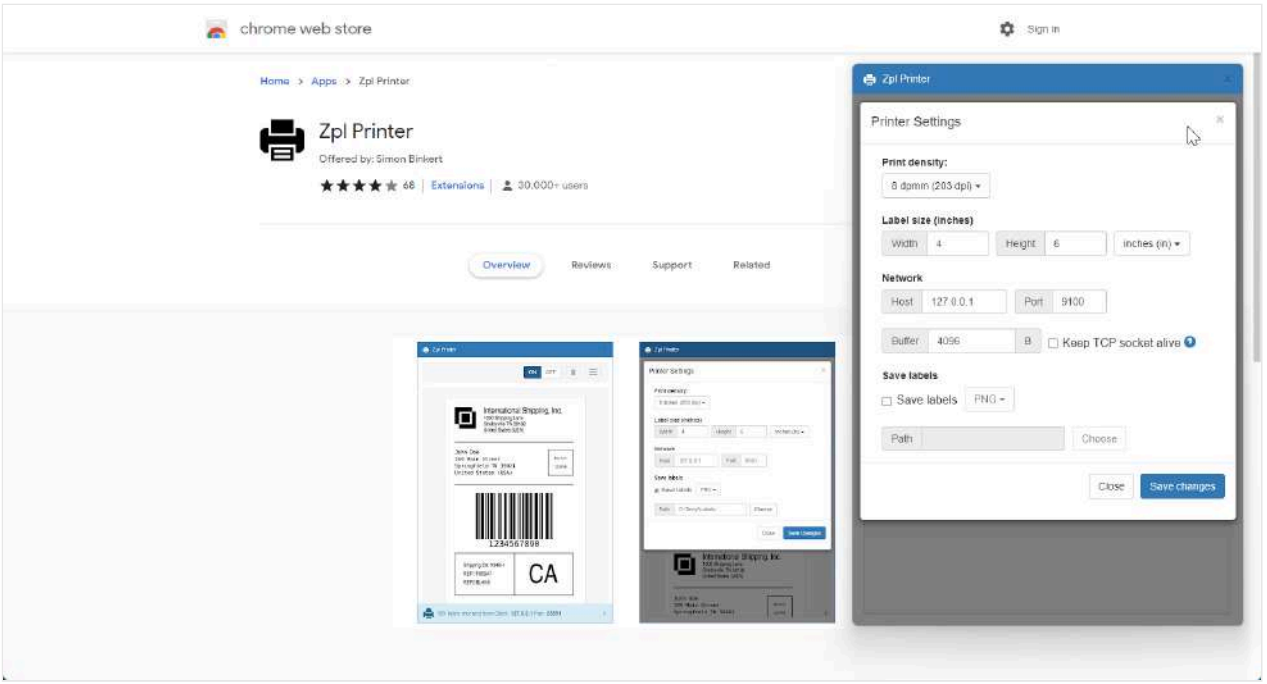


3. Configure the running application:

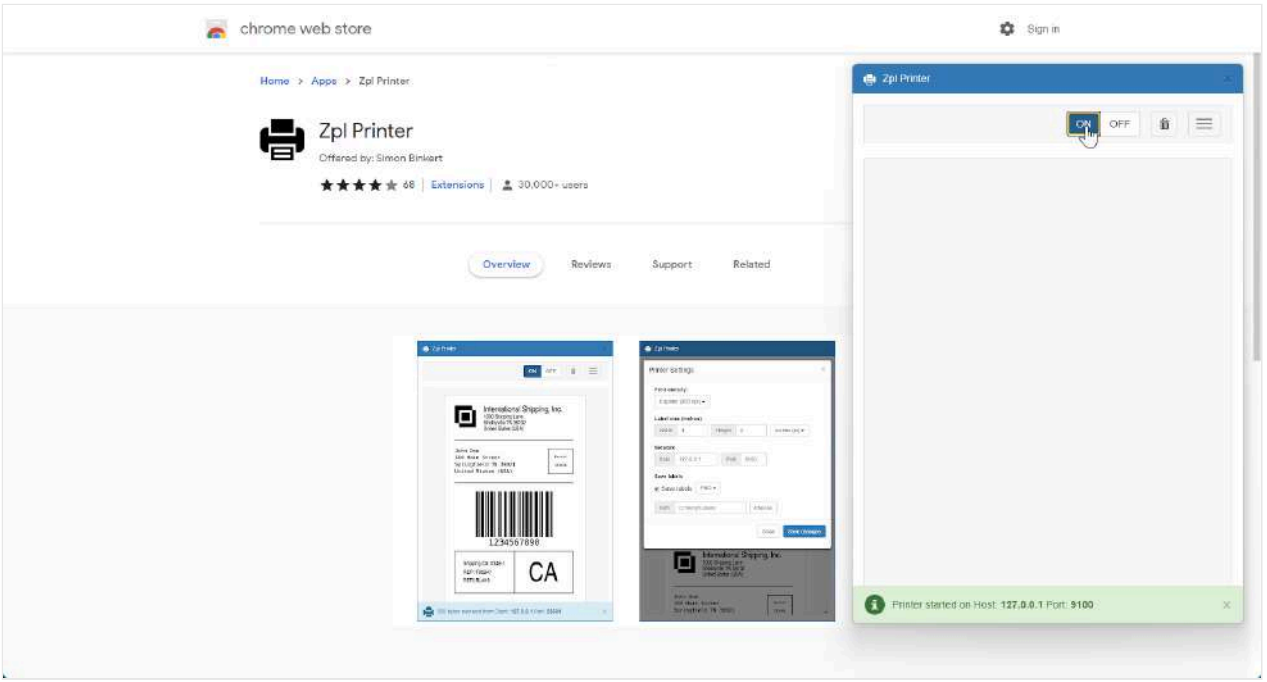
   a. Turn the application off.

   b. In the printer settings, set the host to **127.0.0.1**.

   c. Set the port to **9100**.



d. Turn the application back on. You receive a message that states that the printer was started on the specified host and port.

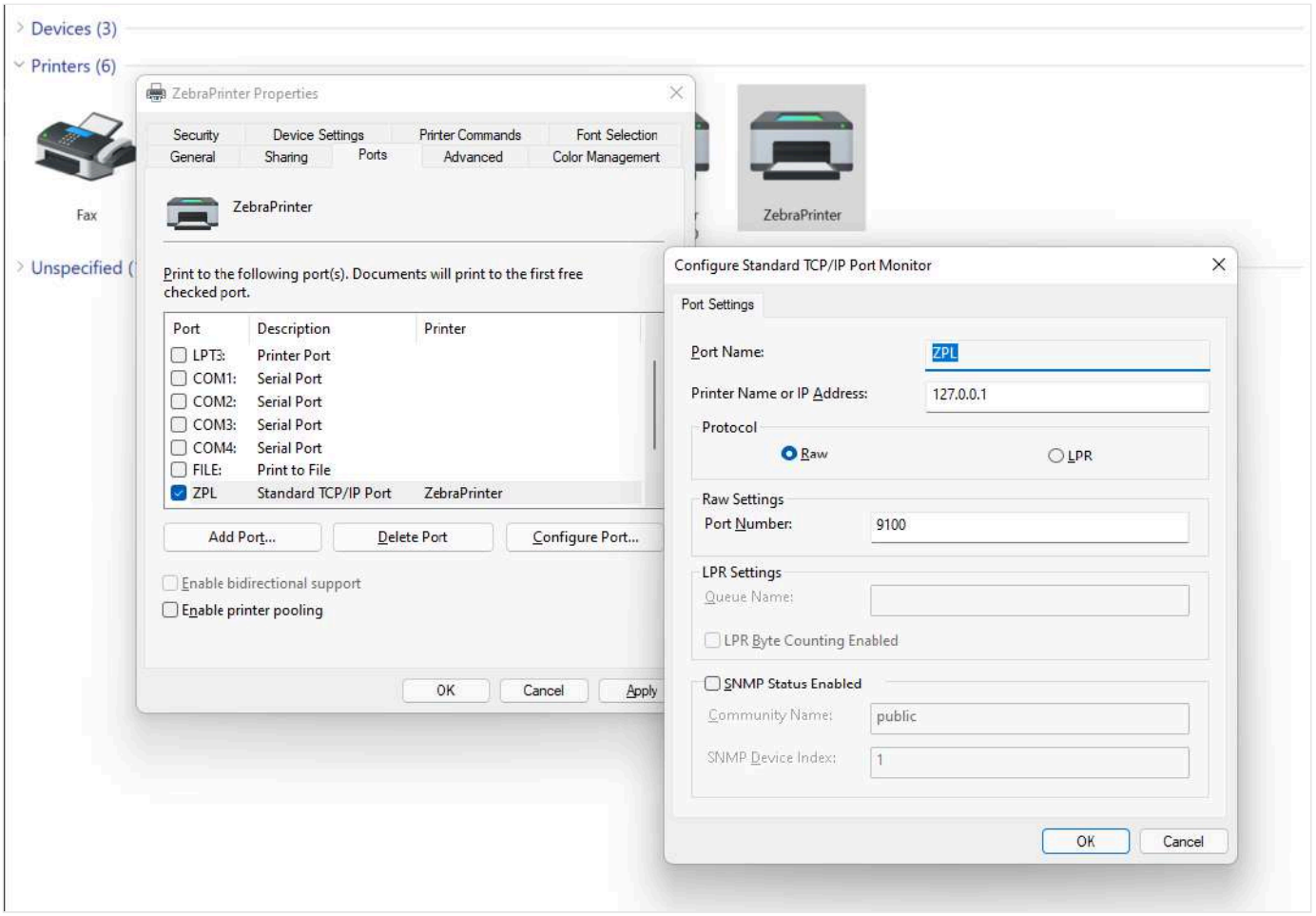


> **Note**
>
> Because the printer emulator application that this example uses relies on a web service to render labels, make sure that your security settings allow you to communicate with the service. Otherwise, the application doesn't receive the rendered labels, and no preview of those labels is available.

Add and configure a local printer

Add a new local printer that the current device can use to pass generated labels from the DRA to the printer emulator application.

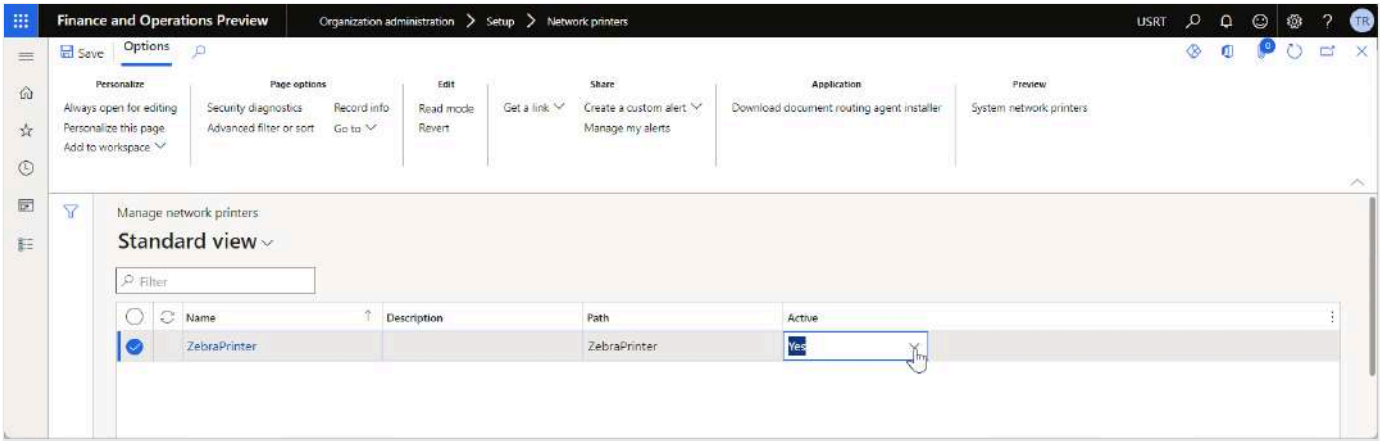
1. In Windows, select **Start** > **Settings** > **Devices** > **Printers & scanners**.
2. Select **Printers & scanners settings**.
3. For **Add a printer or scanner**, select **Add device**.
4. For **The printer that I want isn't listed**, select **Add manually**.
5. In the **Find a printer by other options** field, select **Add a local printer or network printer with manual settings**.
6. In the **Choose a printer port** field, select **Create a new port**, and then follow these steps:
   a. In the **Type of port** field, select **Standard TCP/IP Port**.
   b. In the **Hostname or IP address** field, enter **127.0.0.1**.
   c. In the **Port name** field, enter **ZPL**.
   d. Wait until the **Detecting TCP/IP port** operation completes.
   e. In the **Device type** field, select **Custom**, and then select **Settings**.
   f. Make sure that the following port settings are specified:
      - **Port name:** ZPL
      - **Printer name or IP address:** 127.0.0.1
      - **Protocol:** Raw
      - **Port number:** 9100
7. In the **Install the printer driver** field, select **Generic / Text only**.
8. In the **Printer name** field, enter **ZebraPrinter**.



Install and configure the DRA

Prepare the DRA to pass generated labels from Finance to the configured local printer.

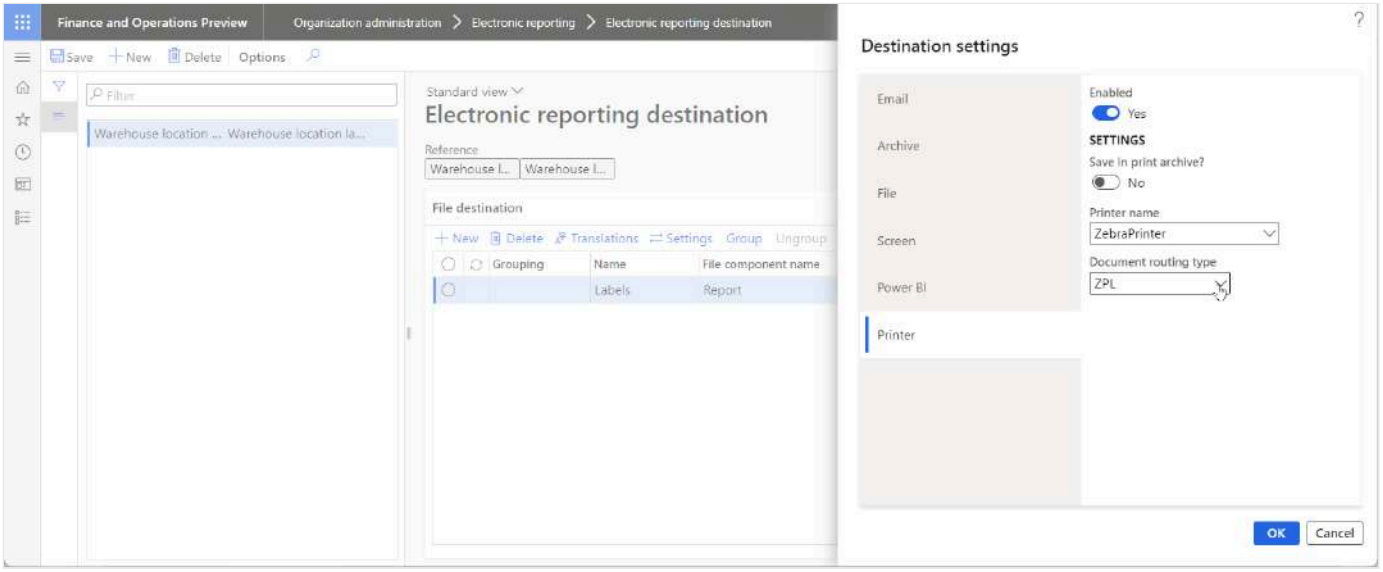
1. Install the DRA.
2. Configure the DRA.
3. Register the local printer in the DRA.
4. Activate the local printer in your Finance environment.



Configure the ER destination

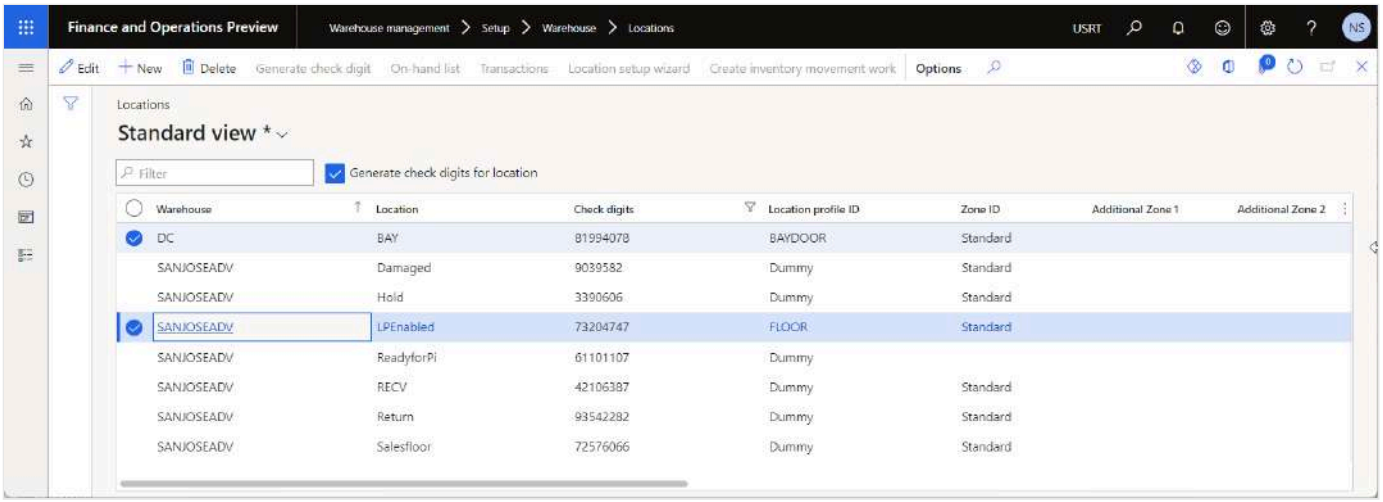
Prepare the ER destination to pass generated labels from Finance to the DRA.

1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting destination**.
2. On the **Electronic reporting destination** page, on the Action Pane, select **New**.
3. In the **Reference** field, select **Warehouse location labels**.
4. On the **File destination** FastTab, select **New**.
5. In the **Name** field, enter **Labels**.
6. In the **File component name** field, select **Report**.
7. Select **Settings**.
8. In the **Destination settings** dialog box, on the **Printer** tab, set the **Enabled** option to **Yes**.
9. In the **Printer name** field, select **ZebraPrinter**.
10. In the **Document routing type** field, select **ZPL**.
11. Select **OK**.



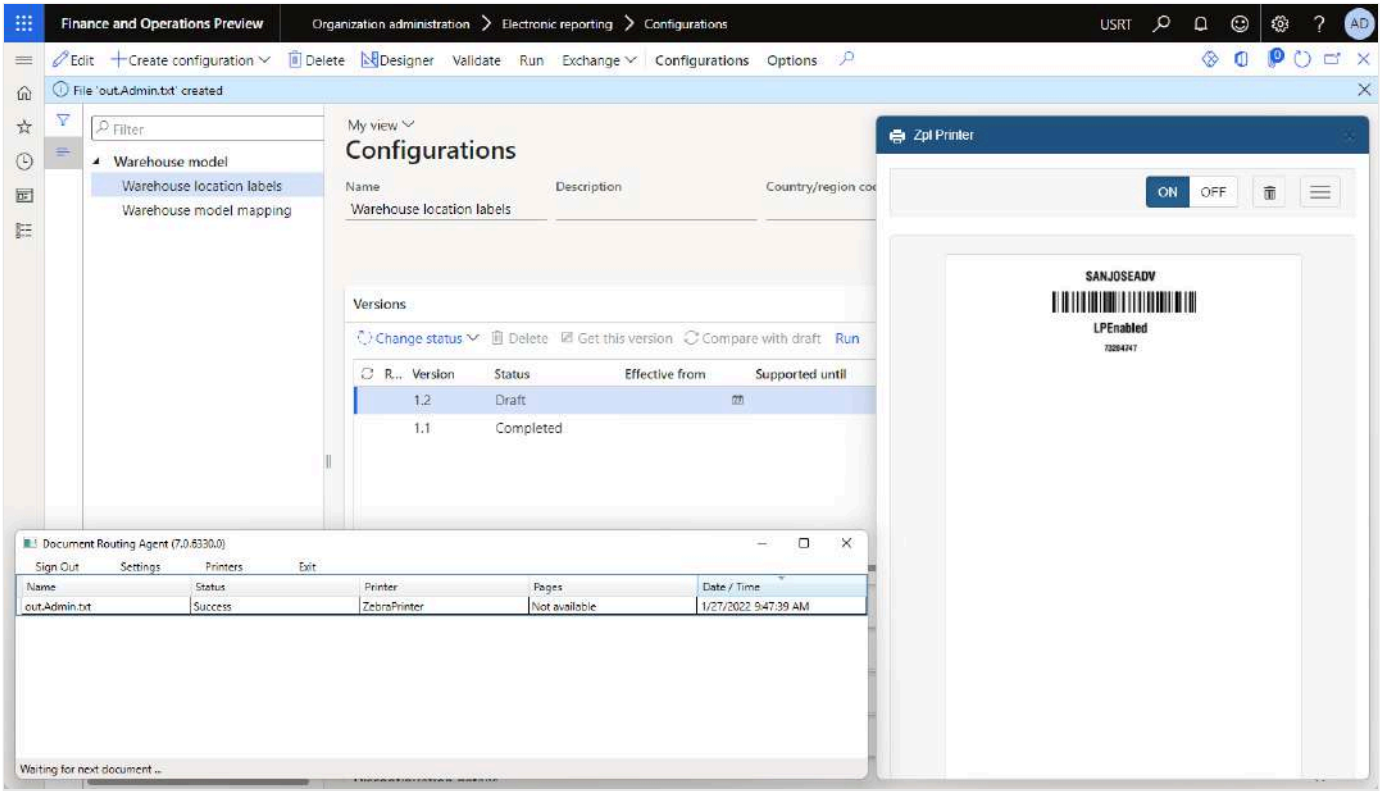
Review warehouse locations

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.
2. On **Locations**, filter to view only locations that have a value in the **Check digits** field.



Print warehouse location labels

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, in the configuration tree, expand **Warehouse model**, and select **Warehouse location labels**.
3. On the Action Pane, select **Run**.
4. In the **Electronic report parameters** dialog box, on the **Records to include** tab, select **Filter**.
5. On the **Range** tab, find the row where the **Table** field is set to **Locations** and the **Field** field is set to **Location**. In the **Criteria** field, enter **LPEnabled**.
6. Select **OK**.
7. Select **OK**. A label is generated and shown on the preview page in the printer emulator application.



Modify the layout of a label

You can change the current layout of your warehouse location labels. The following example shows how to change the layout so that generated labels include a location profile ID.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. Set the **Use destinations for draft status** ER user parameter to **Yes**.
3. On **Configurations**, in the configuration tree, expand **Warehouse model**, and select **Warehouse location labels**.
4. Select **Designer**.
5. On **Format designer**, on the **Mapping** tab, select the `model.Location.Label` data source.
6. In the **Data source properties** dialog box, select **Edit** > **Edit formula**.
7. On **Formula designer**, in the **Formula** field, review the ER formula that generates labels.

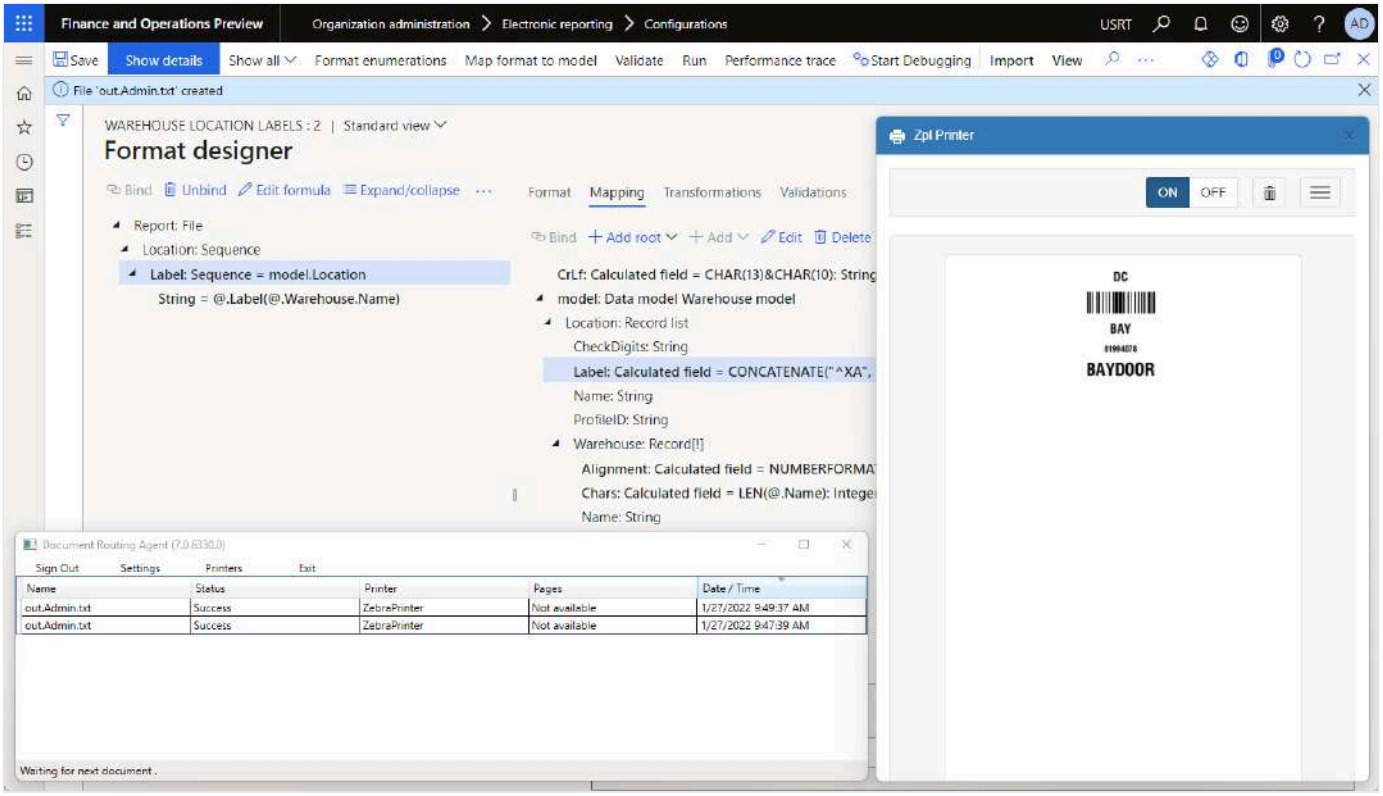
```VB
CONCATENATE(
"^XA",CrLf,
"^CF0,30,30^FO0,30^FB800,1,0,C,0^FD",Warehouse,"\&^FS",CrLf,
"^BY2,2,50^FT",@.Warehouse.Alignment,",126^BCN,,N,N,N,A^FD",Warehouse,"\&^FS",CrL
f,
"^FO0,150^FB800,1,0,C,0^FD",@.Name,"\&^FS",CrLf,
"^CF0,20,20^FO0,200^FB800,1,0,C,0^FD",@.CheckDigits,"\&^FS",CrLf,
"^XZ")
```

8. Update the formula to add a location profile ID to generated labels.

```VB
CONCATENATE(
"^XA",CrLf,
"^CF0,30,30^FO0,30^FB800,1,0,C,0^FD",Warehouse,"\&^FS",CrLf,
"^BY2,2,50^FT",@.Warehouse.Alignment,",126^BCN,,N,N,N,A^FD",Warehouse,"\&^FS",CrL
f,
"^FO0,150^FB800,1,0,C,0^FD",@.Name,"\&^FS",CrLf,
"^CF0,20,20^FO0,200^FB800,1,0,C,0^FD",@.CheckDigits,"\&^FS",CrLf,
"^CF0,40,40^FO0,240^FB800,1,0,C,0^FD",@.ProfileID,"\&^FS",CrLf,
"^XZ")
```

9. Select **Save**.

10. Select **OK**.
11. On the Action Pane, select **Run**.
12. In the **Electronic report parameters** dialog box, on the **Records to include** tab, select **Filter**.
13. On the **Range** tab, find the row where the **Table** field is set to **Locations** and the **Field** field is set to **Location**. In the **Criteria** field, enter **Bay**.
14. Select **OK**.
15. Select **OK**. A label is generated and shown on the preview page in the printer emulator application.



Encoding

> **Note**
>
> You must synchronize the encoding setting of the **Common\File** component of the editable ER format and the appropriate setting of the designed label. The value of the Encoding field of the **Common\File** component shouldn't contradict a ZPL command that controls the label's encoding (for example, the `^CI` command). ER doesn't validate that these settings are synchronized.

Additional resources

Printer destination

Last updated on 04/08/2026

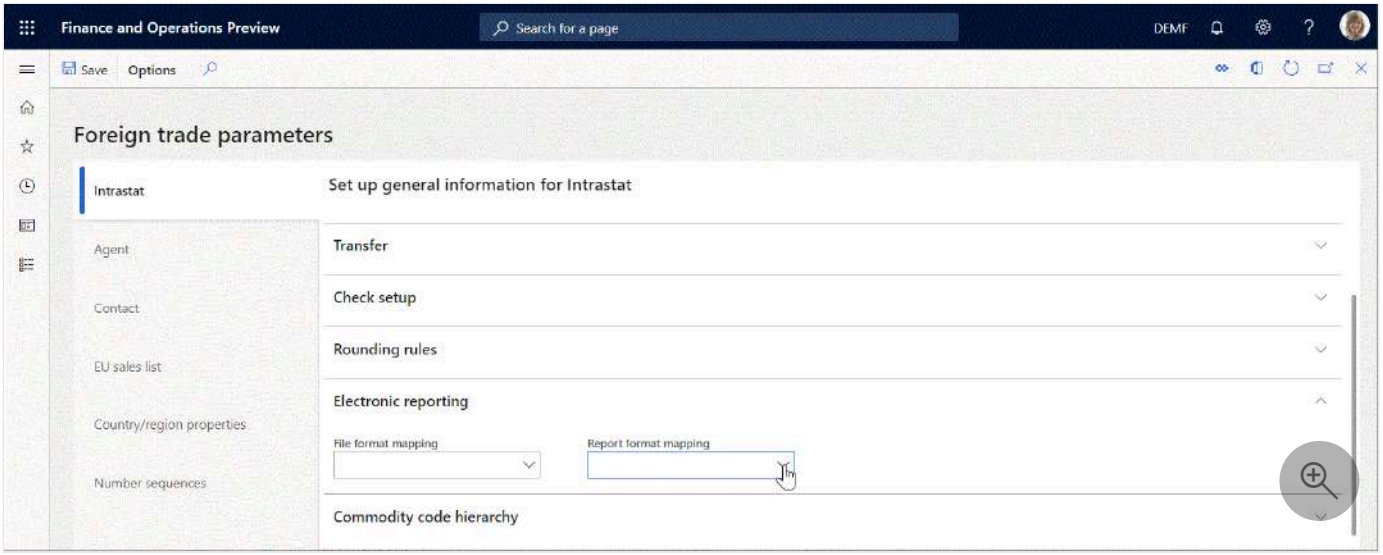
Allow users to set up an ER format reference by inquiring a format from the Global repository

Use the Electronic reporting (ER) framework to configure formats for outbound documents that comply with the legal requirements of various countries/regions. You can also use the ER framework to configure formats for parsing inbound documents and use the information from those documents to append or update application data. You can use each of these formats in your Dynamics 365 Finance instance for handling inbound or outbound business documents as part of a certain business process.

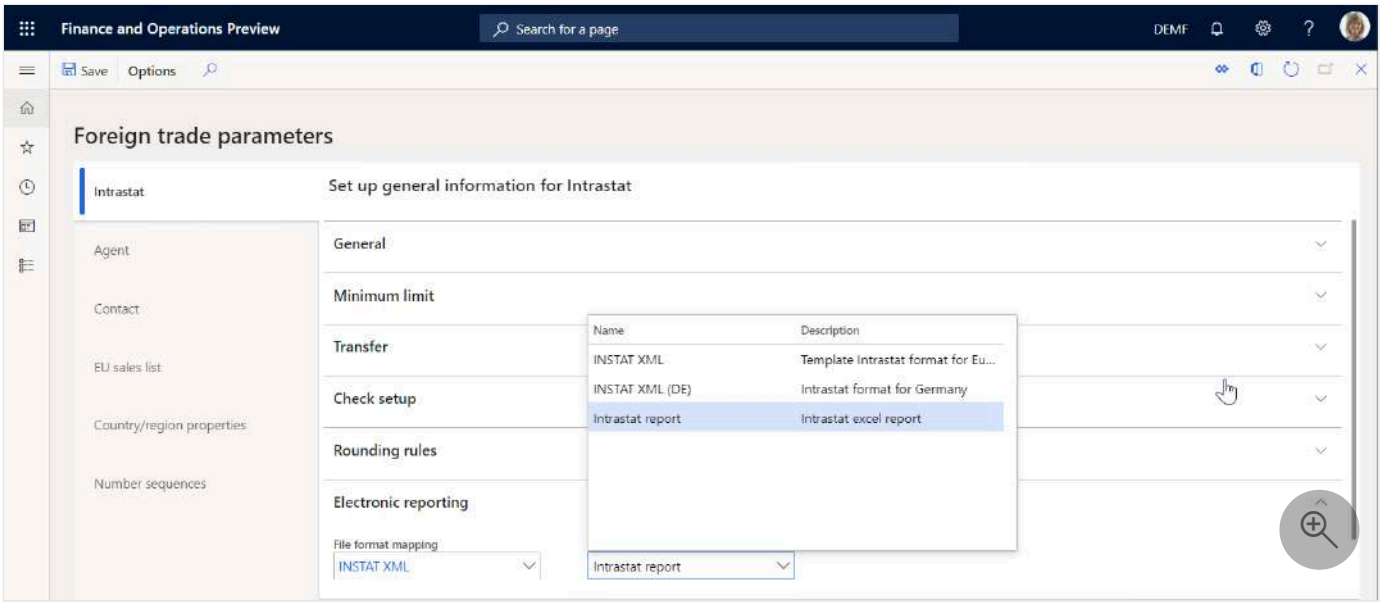
Usually, you must specify what ER format to use in a certain business process. To do that, select a single ER format in a lookup field configured as part of business process-specific parameters. Use the appropriate API of the ER framework to implement these lookup fields. For more information, see ER framework API - code to display a format mapping lookup.

For example, when you configure foreign trade parameters, you need to set up the references to individual ER formats that generate the Intrastat declaration and the Intrastat declaration control report. The following screenshots show how the ER formats lookup field looks in the **Foreign trade parameters** page.

If the current Finance instance contains no Intrastat business process-related ER formats, this lookup field is empty.

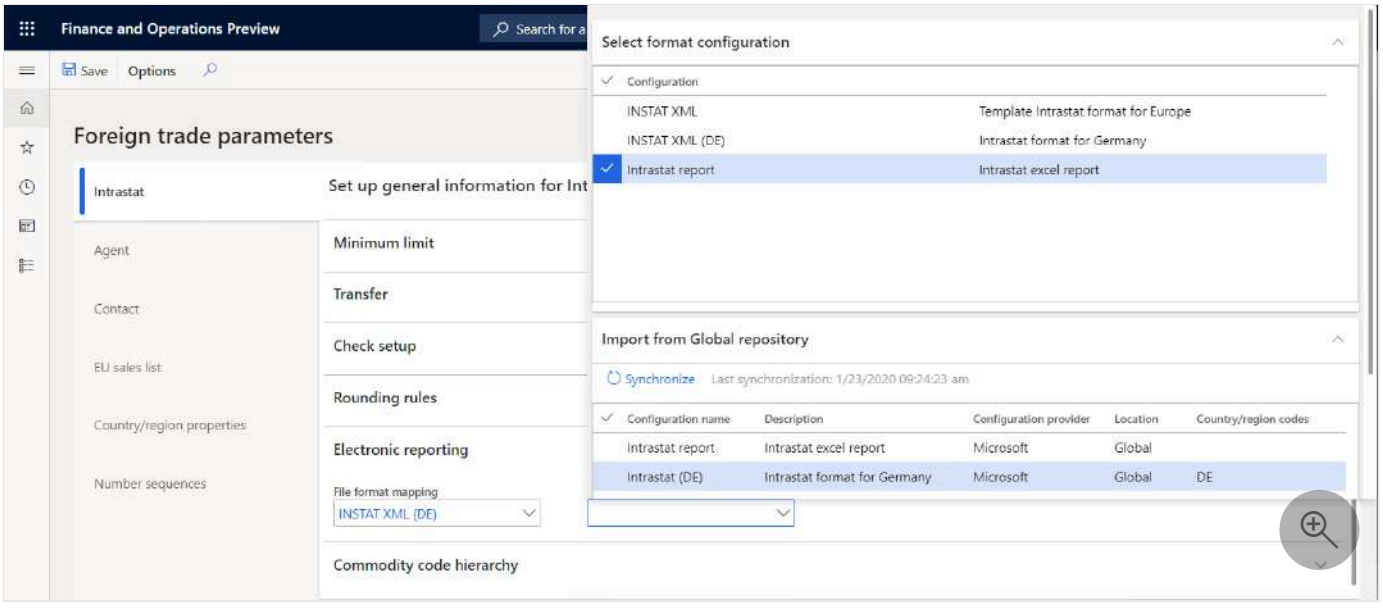


If the current Finance instance contains Intrastat business process related ER formats, this lookup field offers the ER formats.



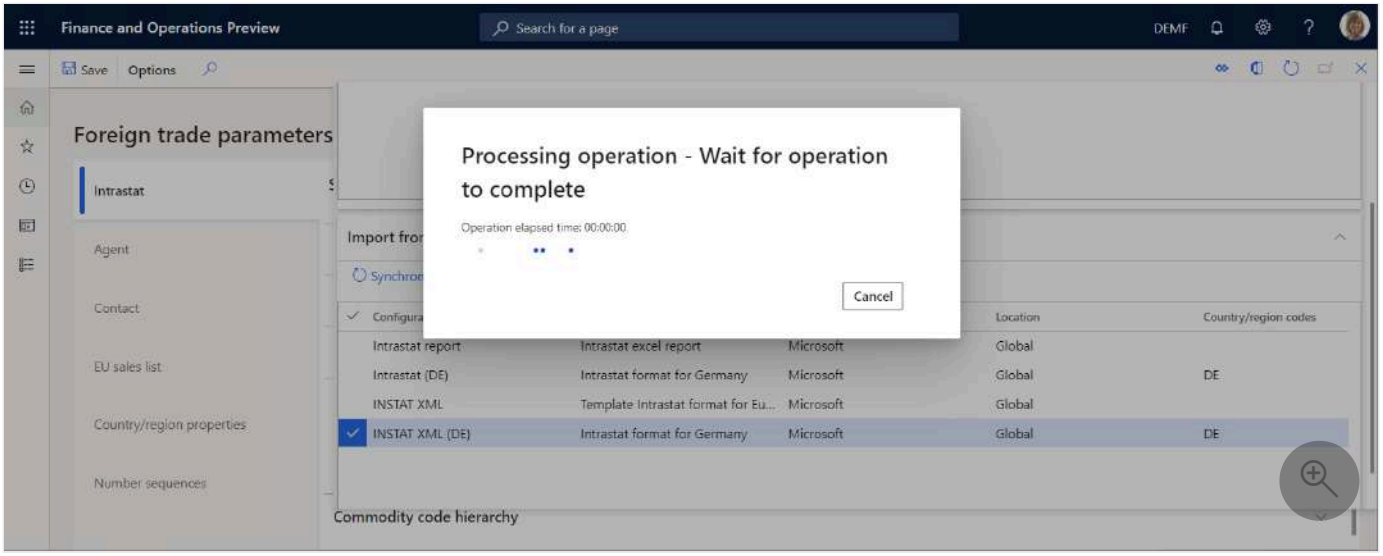
This lookup offers only the ER formats that are already imported to the current Finance instance. To import ER solutions to the current Finance instance, you need permissions to run the appropriate function of the ER framework that supports the lifecycle of ER solutions that contain ER formats.

Starting in Finance version 10.0.9 (April 2020 release), the user interface of the ER format lookup that the ER framework API implements is extended. You can still select the existing ER formats, which are on the **Select format configuration** FastTab. In addition, the extended lookup offers the new option to search the Global repository (GR) to locate specific ER formats. All ER formats of the GR are offered on the **Import from Global repository** FastTab.

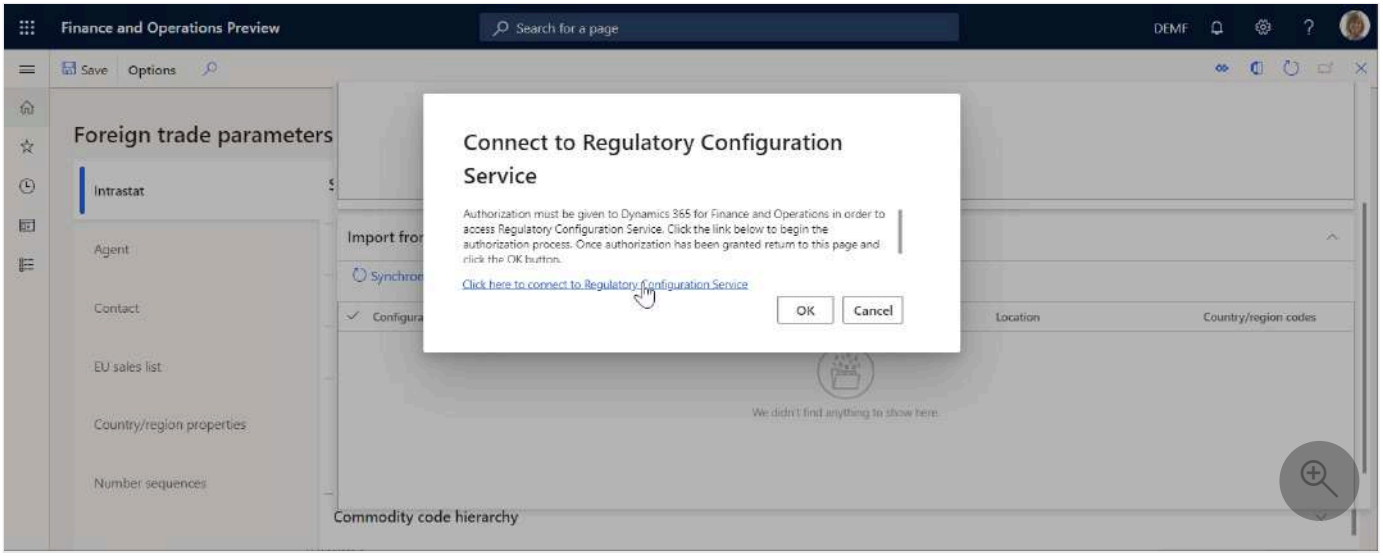


Similar to the **Select format configuration** FastTab, the **Import from Global repository** FastTab shows only the ER formats that are applicable to the business process for which an ER format is selected in this lookup field. In this example, the generation of Intrastat declaration. The ER format is applicable for the company to which the user is currently signed in, depending on the company country/region context.

When you select an ER format on the **Import from Global repository** FastTab, the selected ER format configuration is imported from the GR to the current Finance instance.



If the import completes successfully, the reference to the imported ER format is stored in this lookup field. When you access the GR for the first time, you need to follow the link provided to sign up for the Globalization studio workspace that is used to manage access to the GR storage.

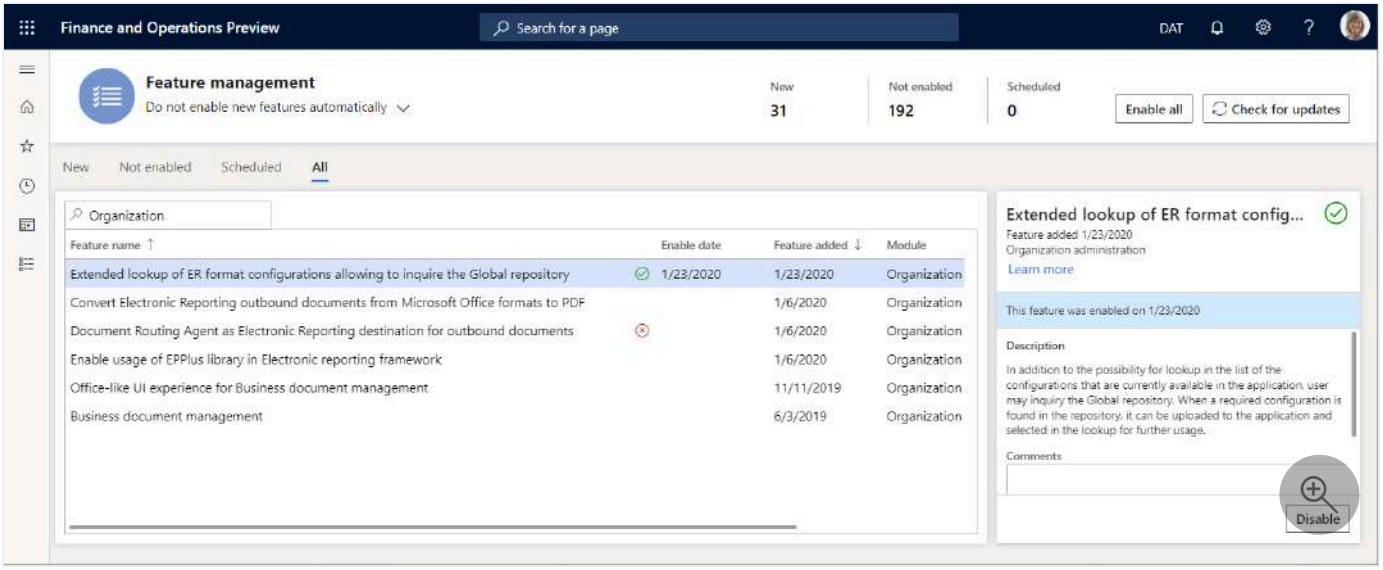


By default, the **Import from Global repository** FastTab presents the list of ER formats from the temporary storage that is automatically created based on the GR content for performance improvements. This happens when the **Import from Global repository** FastTab is opened the first time, which might take several seconds.

If you don't see the required ER format in the **Import from Global repository** FastTab, but you're sure that this ER format is stored in the GR, select the **Synchronize** option. This option updates the temporary storage and synchronizes it with the current content of the GR.

Feature activation

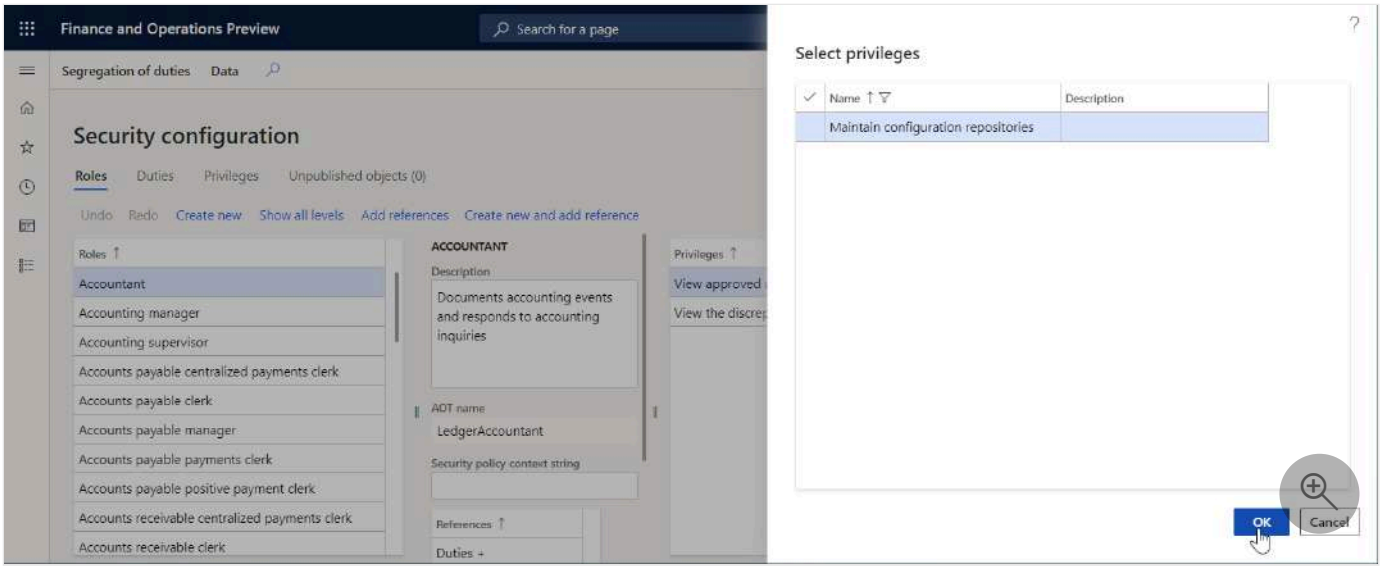
The availability of this functionality is controlled by the feature **Extended lookup of ER format configurations allowing to inquire the Global repository** in **Feature management**. This feature is enabled by default.



Security considerations

The **Maintain configuration repositories** (**ERMaintainSolutionRepositories**) privilege controls access to the GR for a user opening the ER format lookup with the enabled **Import from Global repository** FastTab. To allow users to access the GR content from the ER format lookups, change the security settings by granting the **ERMaintainSolutionRepositories** privilege to users either directly or by using already assigned roles and duties.

The following screenshot shows how to grant this privilege to users who are assigned to the **Accountant** role. This role allows users to configure foreign trade parameters and set up references to the ER formats in the **File format mapping** and **Report format mapping** fields on the **Foreign trade parameters** page.



Limitations

Access to the GR in the ER format lookup is currently only supported for the selection of ER formats that are used to generate outbound documents.

Frequently asked questions

Why can't I access the Global repository from the ER format lookup?

If you enable the **Extended lookup of ER format configurations allowing to inquire the Global repository** feature on the **Feature management** page, but users can't see ER formats on the **Import from Global repository** FastTab and the **Synchronize** option is visible but disabled, make sure that the **Maintain configuration repositories** (**ERMaintainSolutionRepositories**) privilege is granted to the user. Contact your system administrator to receive this privilege.

Additional resources

- Electronic reporting (ER) overview
- Electronic reporting (ER) framework API
- Manage Electronic reporting (ER) configurations lifecycle

Last updated on 06/26/2026

Trace generated report results and compare them with baseline values

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can trace the results of Electronic reporting (ER) formats that generate outgoing electronic documents. When you turn on trace generation by using the **Run in debug mode** ER user parameter, the system creates a new trace record in the ER format execution log every time it runs an ER report. Each trace record stores the following details:

- All warnings that validation rules generate
- All errors that validation rules generate
- All generated files that are stored as attachments of the trace record

You can store individual baseline application files for any ER format. Files are baseline files when they describe the expected results of reports that are run. If a baseline file exists for an ER format that you run while trace generation is turned on, the trace stores, in addition to the earlier mentioned details, the result of the comparison of the generated electronic document with the baseline file. In one click, you can also get the generated electronic document and its baseline file in a single zip file. You can then do detailed comparison by using an external tool such as WinDiff.

You can evaluate the trace to analyze whether the electronic documents that are generated include the expected content. You can do this evaluation in a user acceptance testing (UAT) environment when the code base changes (for example, when you migrated to a new instance of the application, installed hotfix packages, or deployed code modifications). In this way, you make sure that the evaluation doesn't affect the execution of ER reports that are used. For many ER reports, the evaluation can be done in unattended mode.

To learn more about this feature, watch the **ER Generate reports and compare results (Part 1)** and **ER Generate reports and compare results (Part 2)** task guides. These task guides are part

of the **7.5.4.3 Test IT services/solutions (10679)** business process and can be downloaded from the Microsoft Download Center. These task guides walk you through the process of configuring the ER framework to use baseline files to evaluate generated electronic documents.

Example: Trace generated report results and compare them with baseline values

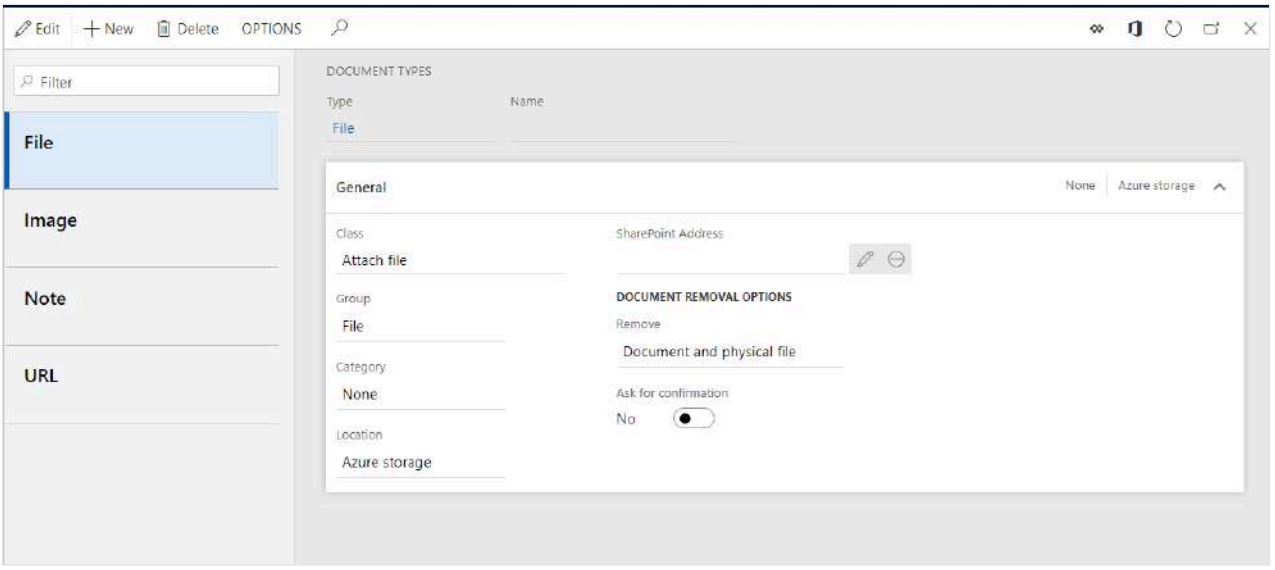
This procedure explains how to configure the ER framework to collect information about ER format executions and then evaluate the results of those executions. As part of that evaluation, generated documents are compared with their baseline files. In this example, you create the required ER configurations for the Litware, Inc. sample company. This procedure is intended for users who have the System administrator or Electronic reporting developer role assigned to them. You can complete these steps by using any data set.

To complete the steps in this example, you must first complete the steps in Create configuration providers and mark them as active.

1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. On **Localization configurations**, in the **Configuration providers** section, verify that the configuration provider for the Litware, Inc. sample company is listed and marked as **Active**. If you don't see this configuration provider, follow the steps in Create configuration providers and mark them as active.

Configure document management parameters

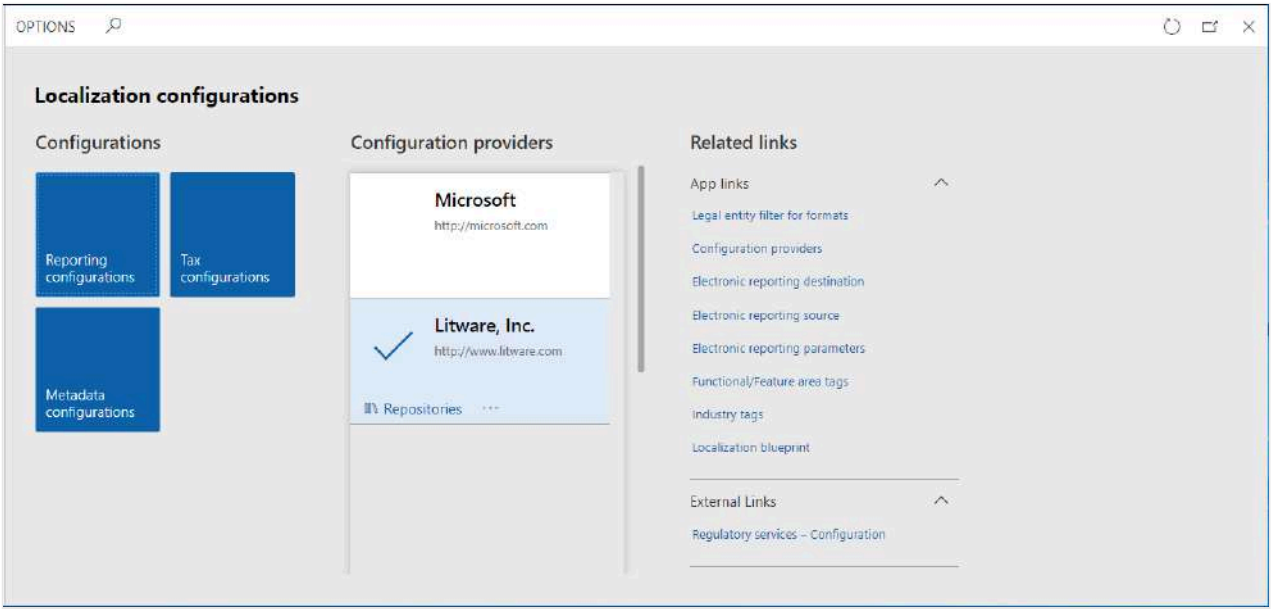
1. Go to **Organization administration** > **Document management** > **Document types**, and create a new document type to store baseline files.
2. In the **Class** field, enter **Attach file**.
3. In the **Group** field, enter **File**.



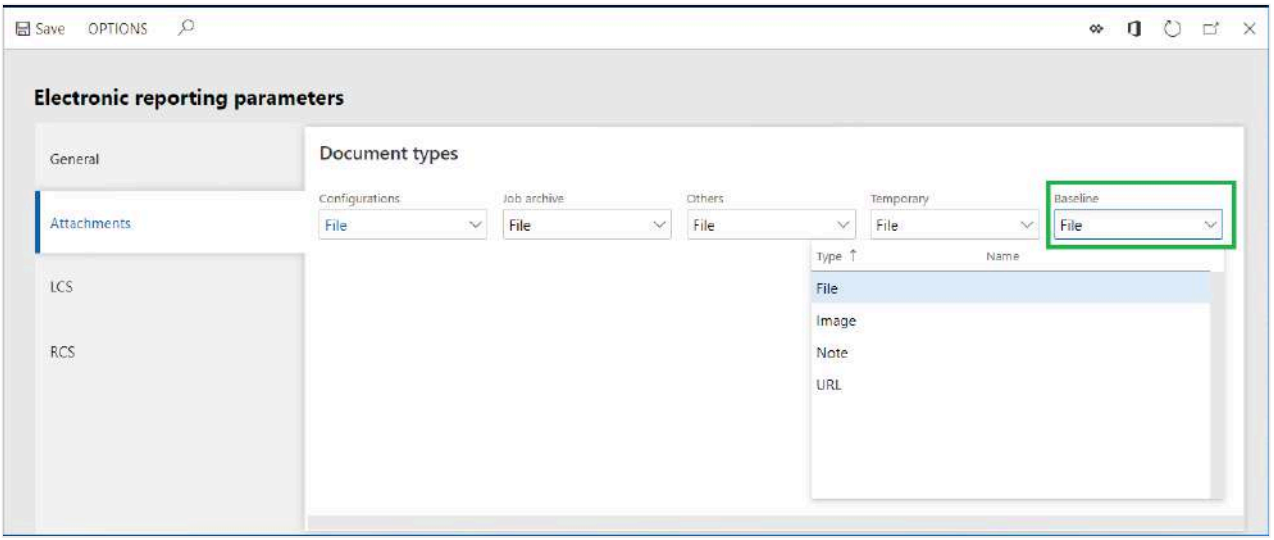
> **Note**
>
> You must configure a new document type with the same name for each data set where you plan to use the ER baseline feature.

Configure ER parameters to start using the baseline feature

1. In the **Electronic reporting** workspace, in the **Related links** section, select **Electronic reporting parameters**.



2. On the **Attachments** tab, in the **Baseline** field, enter or select the document type that you just created.



3. Select **Save**, and then close the **Electronic reporting parameters** page.

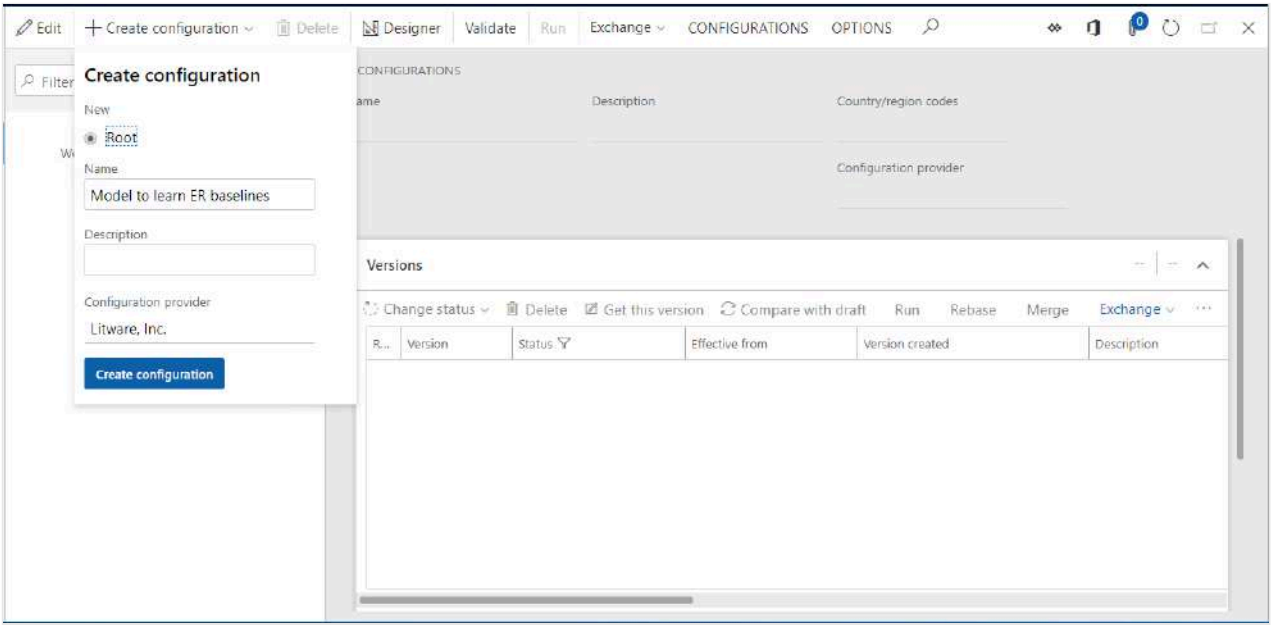
Add a new ER model configuration

1. In the **Electronic reporting** workspace, in the **Configurations** section, select the **Reporting configurations** tile.

2. On the Action Pane, select **Create configuration**.

3. In the drop-down dialog box, in the **Name** field, enter **Model to learn ER baselines**.

4. Select **Create configuration** to confirm the creation of a new ER data model entry.



Design a data model

1. On **Configurations**, on the Action Pane, select **Designer**.

2. Select **New**.

3. In the drop-down dialog box, in the **Name** field, enter **Root**.

4. Select **Add**.

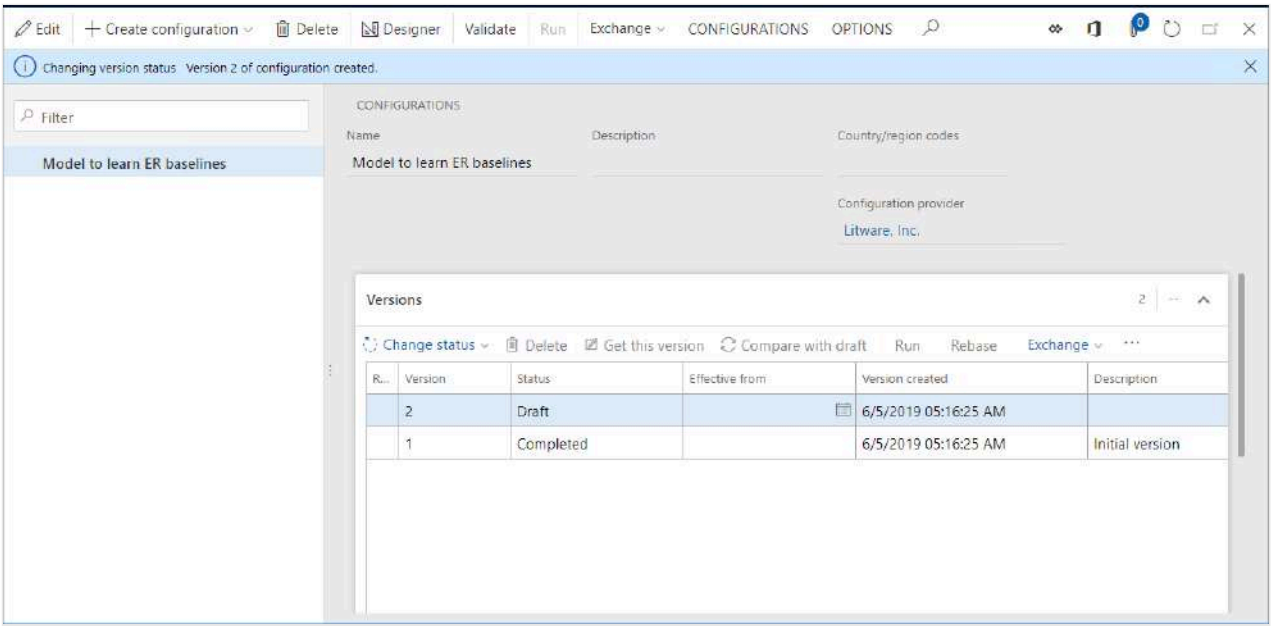
5. Select **Root reference**.

6. Select **OK**, and then select **Save**.

7. Close the **Model designer** page.

8. Select **Change status**.

9. Select **Complete**, and then select **OK**.



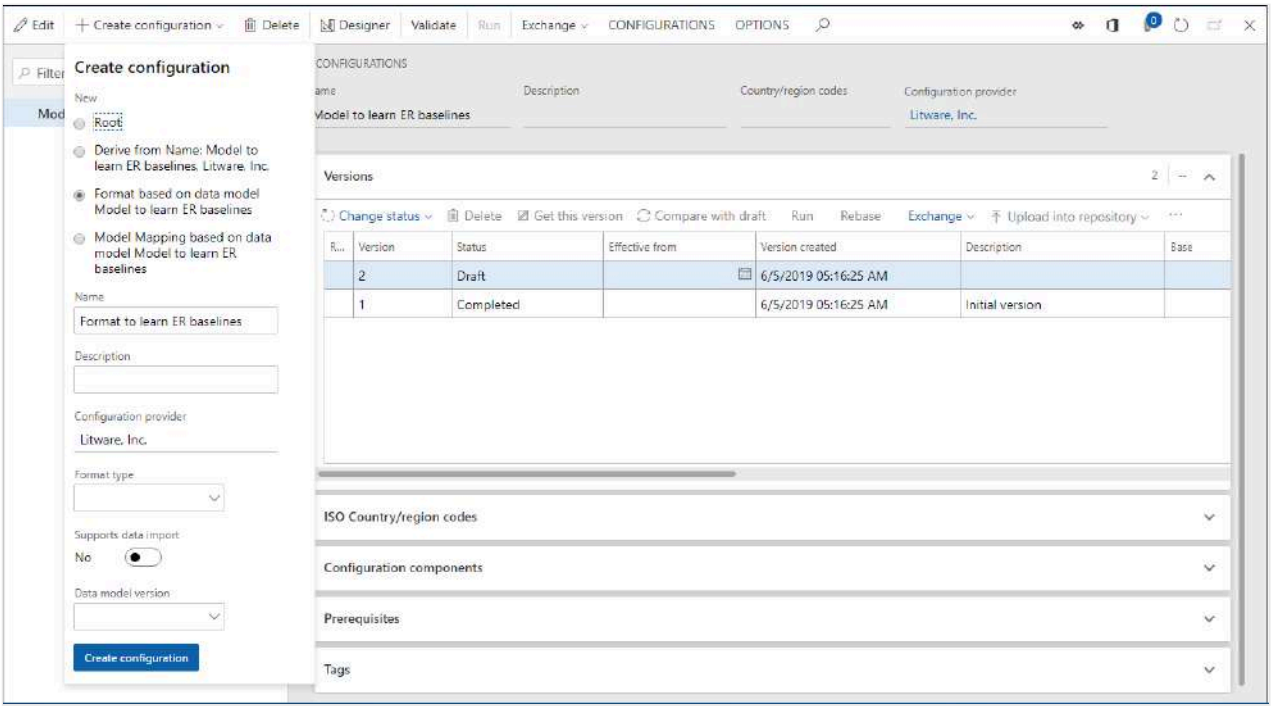
Add a new ER format configuration

1. On **Configurations**, on the action pane, select **Create configuration**.

2. In the drop-down dialog box, in the **New** field group, select **Format based on data model Model to learn ER baselines**.

3. In the **Name** field, enter **Format to learn ER baselines**.

4. Select **Create configuration** to confirm the creation of a new ER format entry.



Design a format

For this example, create a simple ER format to generate XML documents.

1. On **Configurations**, on the Action Pane, select **Designer**.

2. Select **Add root**.

3. In the drop-down dialog box, follow these steps:
   a. In the tree, select **Common\File**.
   b. In the **Name** field, enter **Output**.
   c. Select **OK**.

4. Select **Add**.

5. In the drop-down dialog box, follow these steps:
   a. In the tree, select **XML\Element**.
   b. In the **Name** field, enter **Document**.
   c. Select **OK**.

6. In the tree, select **Output\Document**.

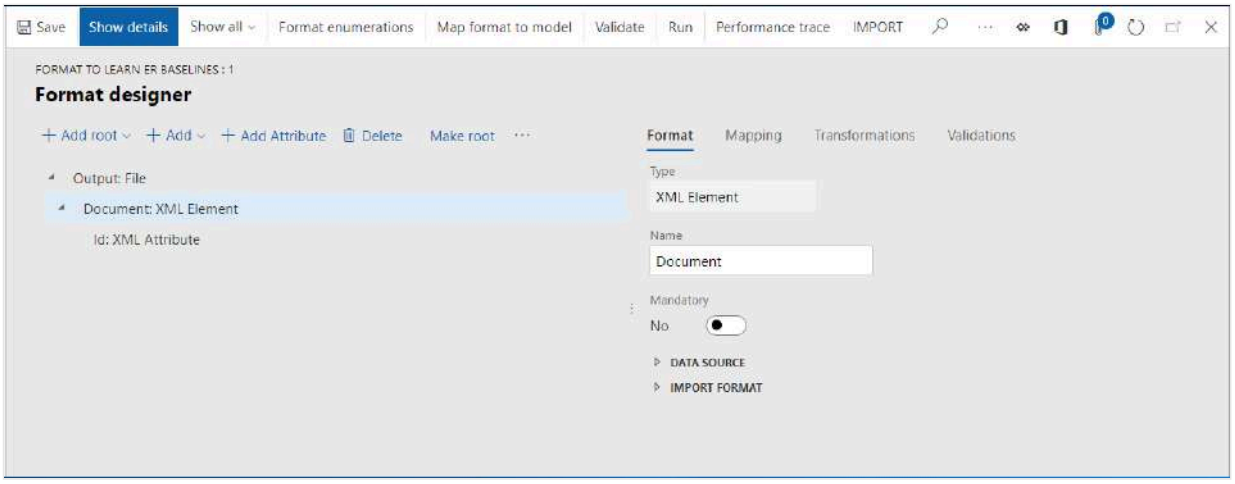
7. Select **Add**.

8. In the drop-down dialog box, follow these steps:

   a. In the tree, select **XML\Attribute**.

b. In the **Name** field, enter **ID**.

c. Select **OK**.



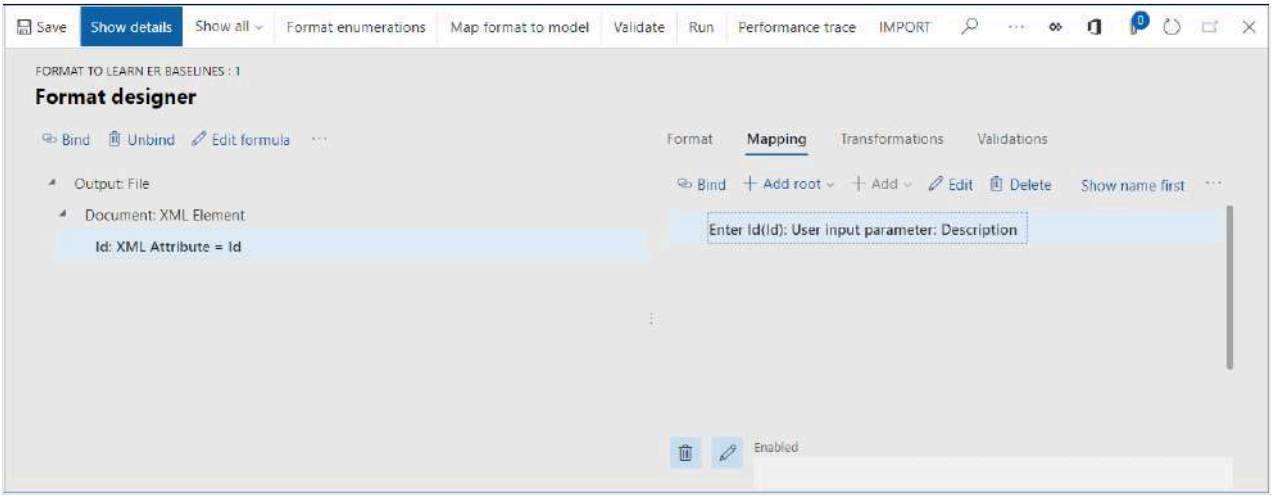
9. On the **Mapping** tab, select **Delete**.

10. Select **Add root**.

11. In the drop-down dialog box, in the tree, select **General\User input parameter**, and then follow these steps:
    a. In the **Name** field, enter **ID**.
    b. In the **Label** field, enter **Enter ID**.
    c. Select **OK**.

12. In the tree, select **Output\Document\Id**.

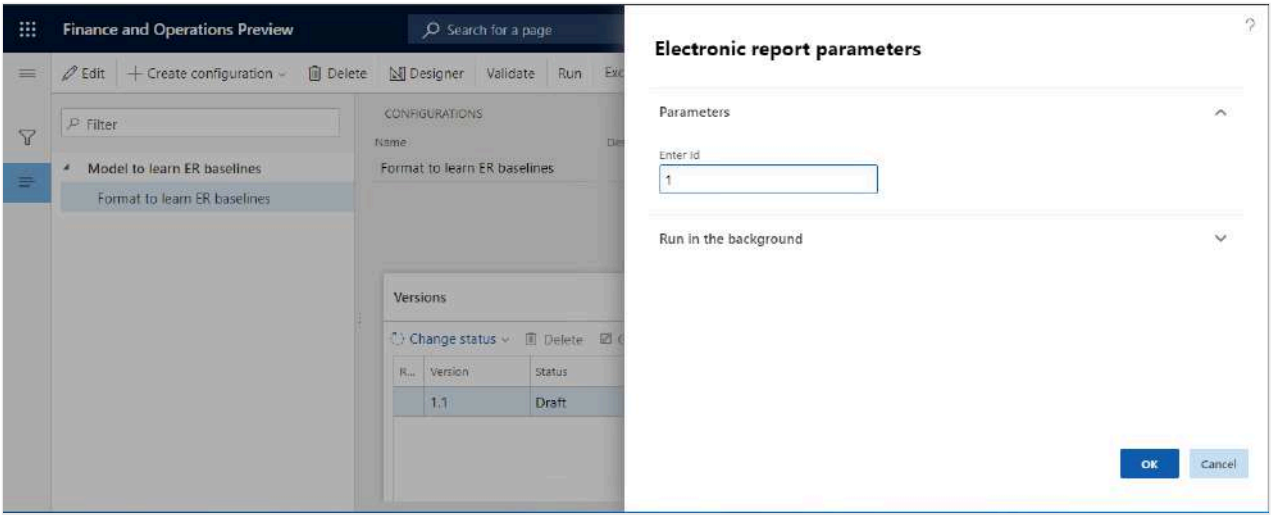
13. Select **Bind**, and then select **Save**.



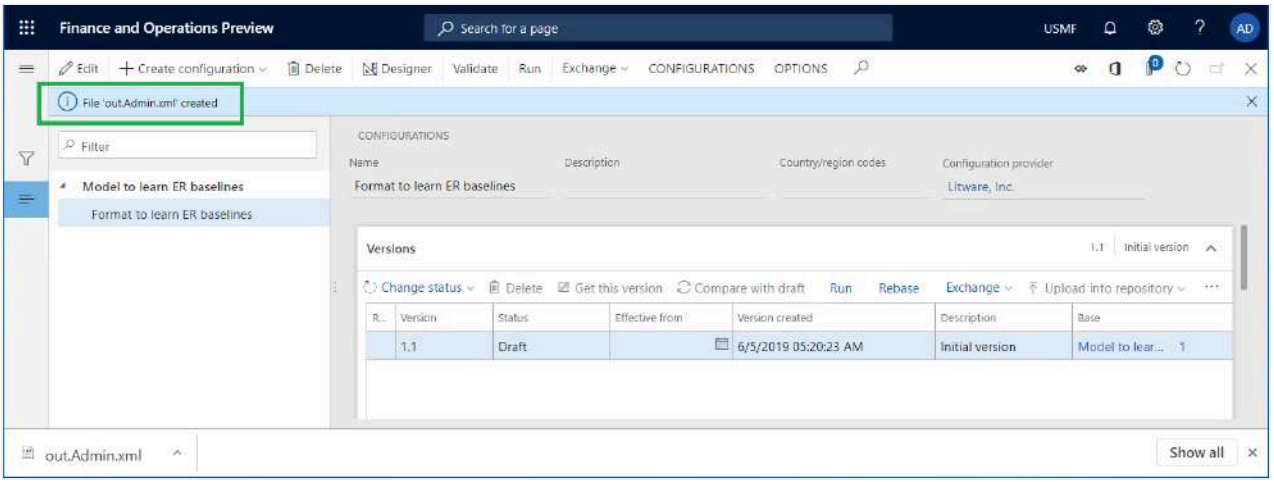
Based on the designed structure, the configured format generates an XML file. This XML contains the **Root** element that has the **ID** attribute set to the value that the user enters in the ER runtime dialog box.

Generate a new baseline file for a designed ER format

1. On **Configurations**, on the **Versions** FastTab, select **Run**.
2. In the **Enter ID** field, enter **1**.
3. Select **OK**.

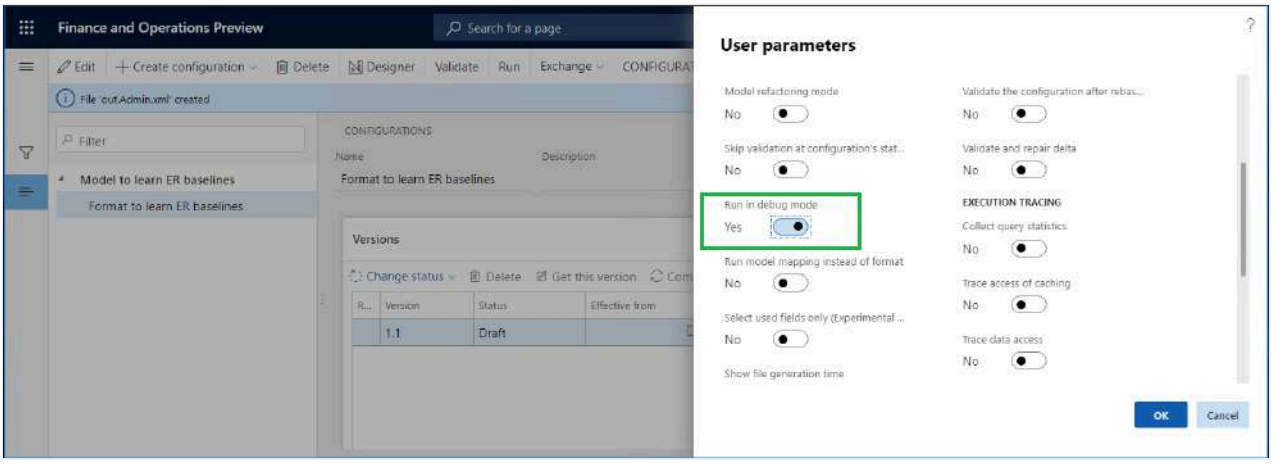


4. Save a local copy of the **out.Admin.xml** file that is generated, so you can use it later as a baseline for this ER format.



Configure ER parameters to use the baseline feature

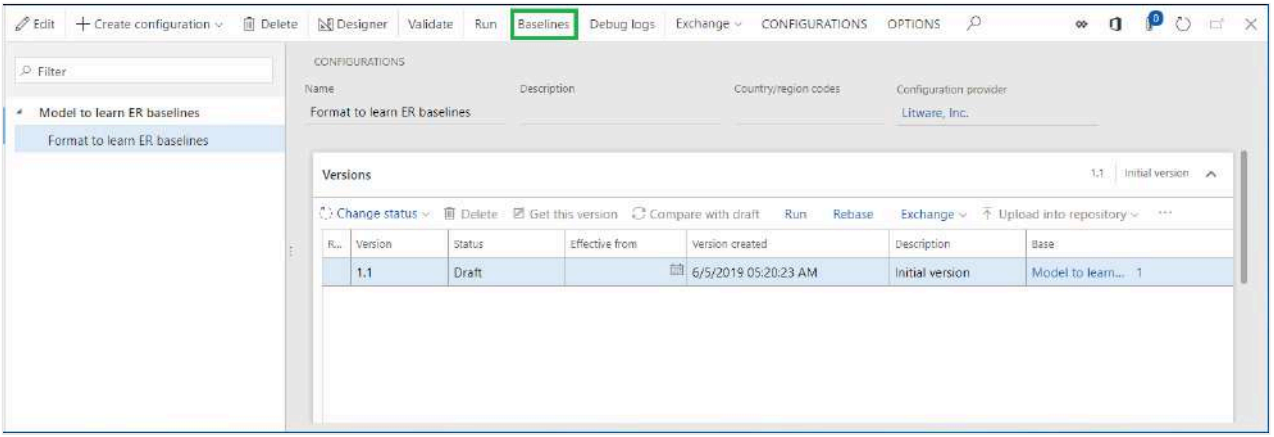
1. On **Configurations**, on the Action Pane, on the **Configurations** tab, select **User parameters**.
2. Set the **Run in debug mode** option to **Yes**.
3. Select **OK**.



Add a new baseline for designed ER format

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.

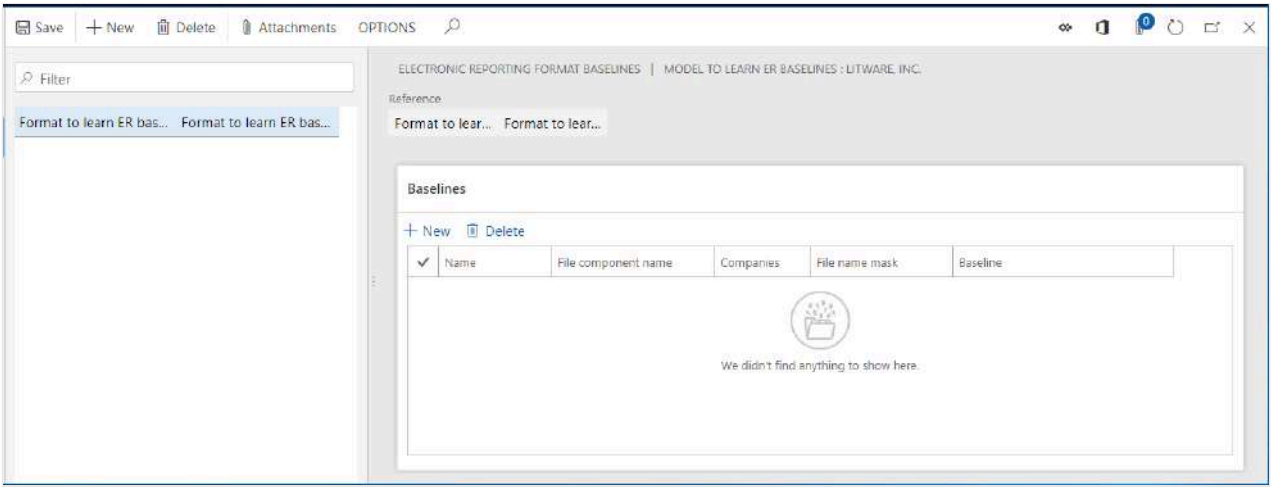
2. On the Action Pane, select **Baselines**.



3. On the Action Pane, select **New**.

4. Select the **Format to learn ER baselines** ER format that you designed earlier.

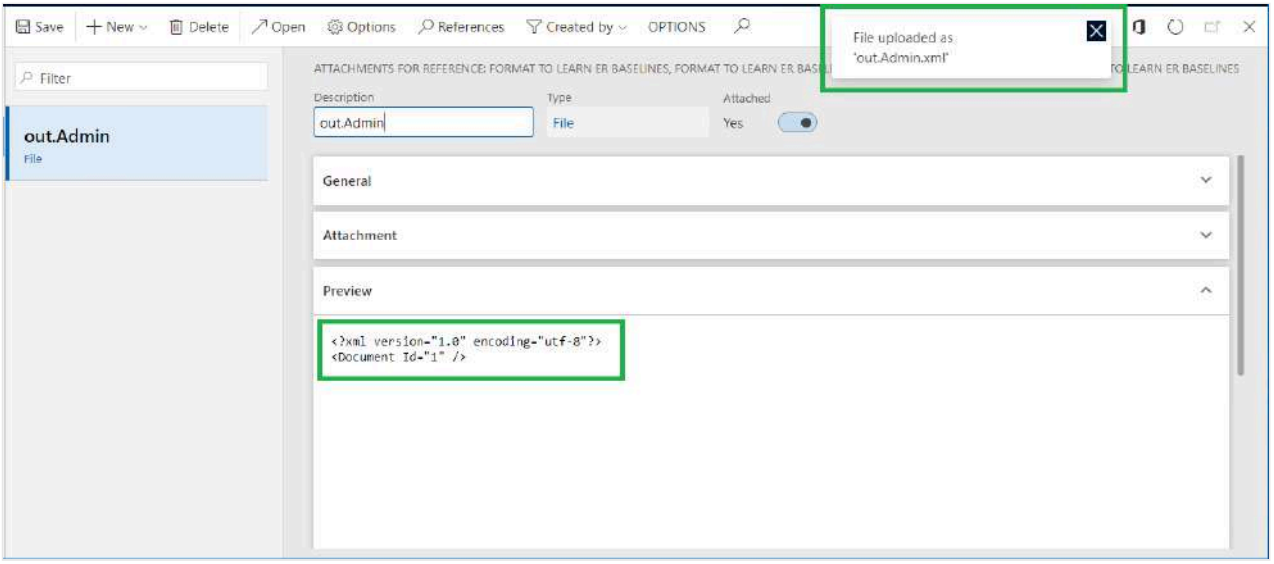
5. Select **Save**.



The baseline is added for the **Format to learn ER baselines** format.

Configure a baseline rule for the added baseline

1. On **Electronic reporting format baselines**, on the Action Pane, select the **Attachments** button (the paper clip symbol).
2. On the Action Pane, select **New** > **File**. In the ER parameters, select the **File** document type as the document type that stores baseline files.
3. Select **Browse**, and select the **out.Admin.xml** file that was generated when you ran the configured ER format earlier.



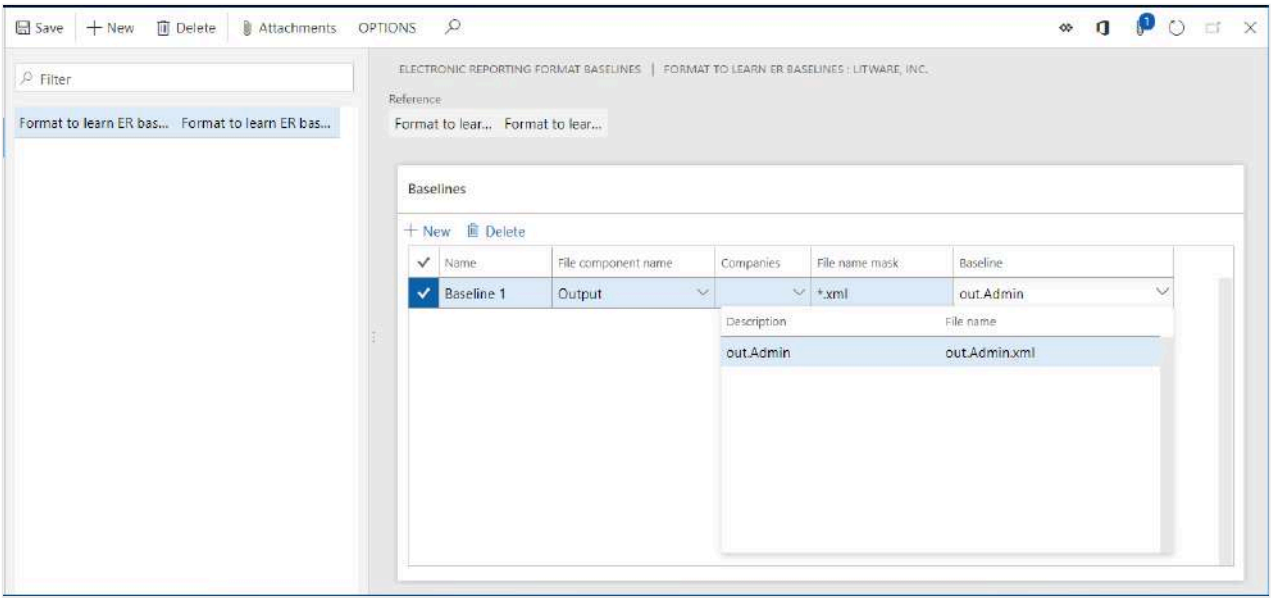
4. Close the **Attachments** page.
5. On the **Baselines** FastTab, select **New**.
6. In the **Name** field, enter **Baseline 1**.
7. In the **File component name** field, enter or select **Output**. This value indicates that the configured baseline is compared with a file that uses the **Output** format element.
8. In the **File name mask** field, enter ***.xml**.

> **Note**
>
> You can define the file name mask. When you define the file name mask, the baseline record evaluates the generated output only when the name of the output file satisfies that mask.

9. If users must run the **Format to learn ER baselines** ER format and sign in to specific companies to use the configured baseline, select those companies in the **Companies** field.

10. In the **Baseline** field, enter or select the **out.Admin** attachment.

11. Select **Save**.



Run the designed ER format and review the log to analyze the results

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.

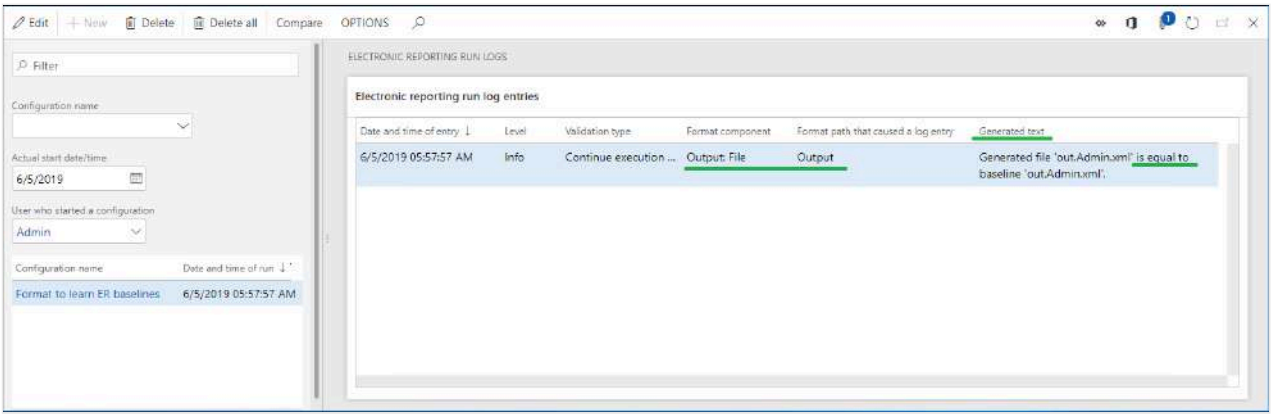
2. In the tree, expand **Model to learn ER baselines**, and then select **Model to learn ER baselines\Format to learn ER baselines**.

3. On the **Versions** FastTab, select **Run**.

4. In the **Enter ID** field, enter **1**.

5. Select **OK**.

6. Go to **Organization administration** > **Electronic reporting** > **Configuration debug logs**.

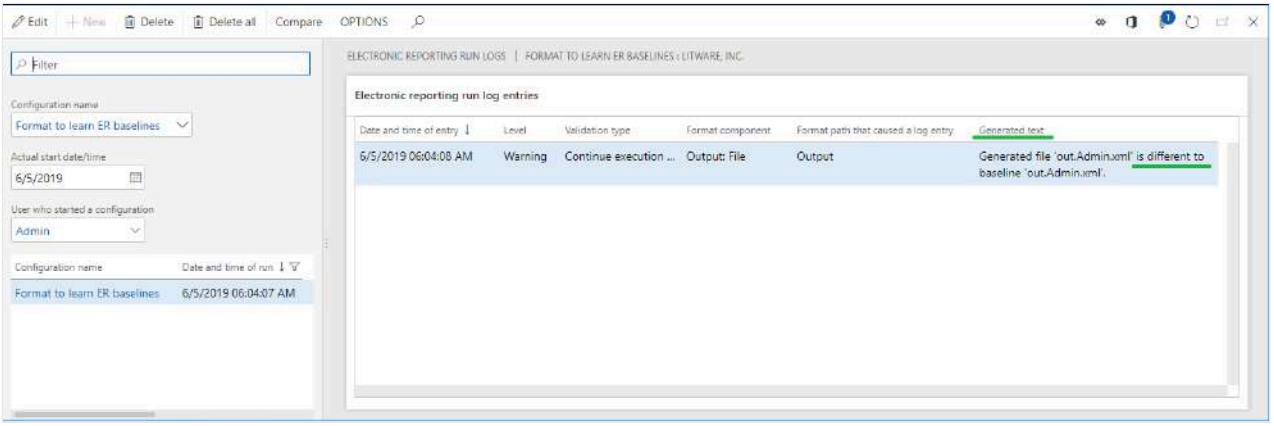


> **Note**
>
> The execution log contains information about the results of the comparison of the generated file with the configured baseline. In this example, the log indicates that the generated file and the baseline are equal.

7. Select **Delete all**.

Run the designed ER format and review the log to analyze the results

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Model to learn ER baselines**, and then select **Model to learn ER baselines\Format to learn ER baselines**.
3. On the **Versions** FastTab, select **Run**.
4. In the **Enter ID** field, enter **2**.
5. Select **OK**.
6. Go to **Organization administration** > **Electronic reporting** > **Configuration debug logs**.



> **Note**
>
> The execution log contains information about the results of the comparison of the generated file with the configured baseline. In this example, the log indicates that the generated file and the baseline differ.

7. Select **Compare**.

> **Note**
>
> The generated file and the baseline file are offered as a zip file. You can use external comparison tools such as WinDiff to compare the files and review the differences.

Additional resources

- Configure the Electronic reporting (ER) framework

Last updated on 04/13/2026

Improve tracing the results of generated ER reports to compare with baseline values

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the first set of improvements to the baseline feature of the Electronic reporting (ER) framework. These improvements are available in Microsoft Dynamics 365 Finance version 10.0.3 (June 2019) and later.

Automate the setting of baseline rules

The Trace generated report results and compare them with baseline values article explains how to configure the ER framework to collect information about ER format executions and evaluate the results of those executions. The example in this article shows the steps that you must complete.

Here are some of the steps:

- Run an ER format to generate an outbound file, and store the file locally.
- Add the locally stored file as an attachment of the baseline that you added for an ER format.
- Configure the baseline rule for the added baseline. This configuration includes the following steps:
  - Specify an ER format element that you use to generate an outbound file.
  - Select the attachment that refers to the generated outbound file.

> **Note**

You must complete these steps manually, even though the new ER capabilities enable you to automate them. To learn more about this feature, complete the following example.

Example: Automate the setting of baseline rules

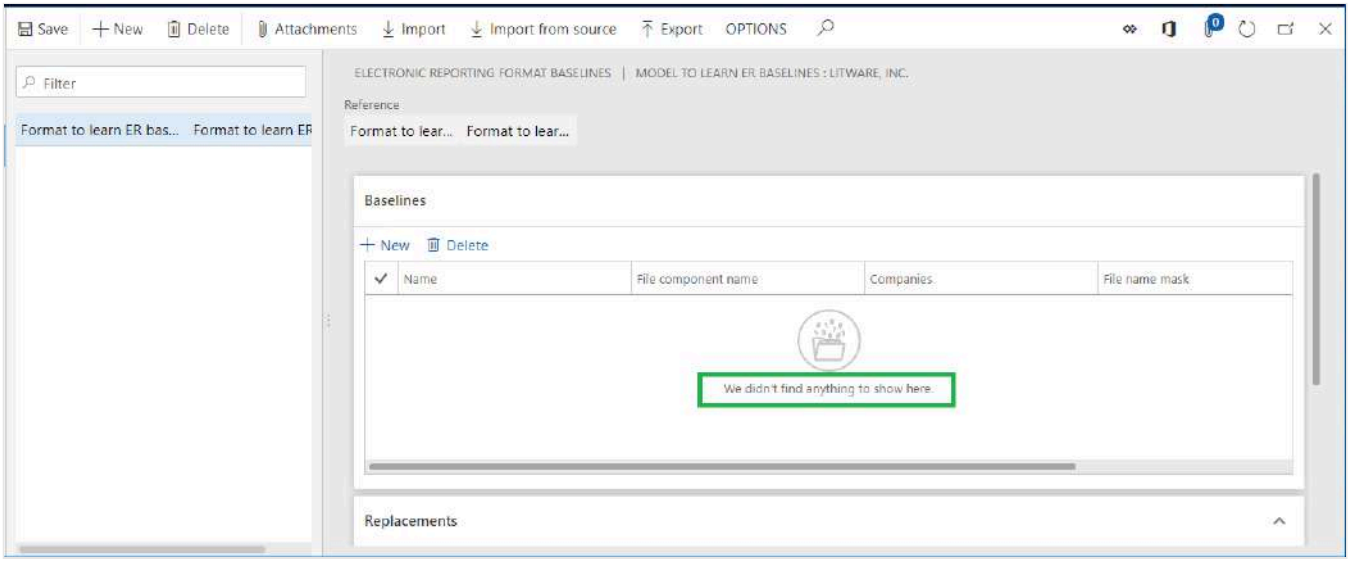
To complete the steps in this example, you must first complete the steps in the example in the Trace generated report results and compare them with baseline values article, up through the "Add a new baseline for a designed ER format" section.

Review added baseline

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. Select **Baselines**.

> **Note**
>
> The **Baselines** button on the Action Pane is available only when the **Run in debug mode** ER user parameter is turned on for the current company.

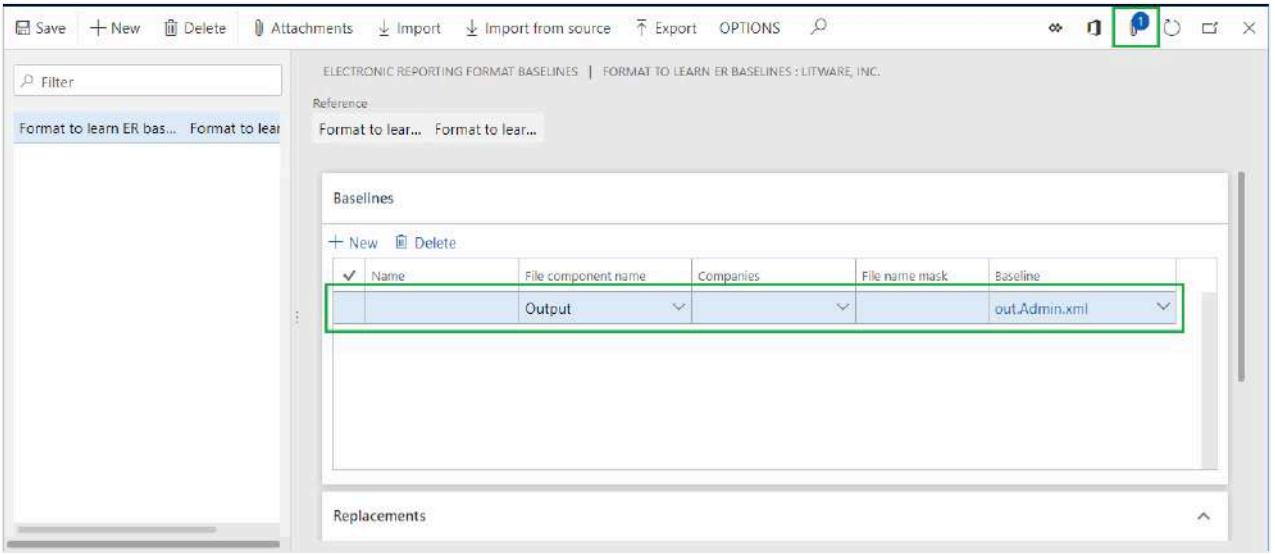
The baseline is added for the selected **Format to learn ER baselines** format, but you didn't add baseline rules for this baseline yet.



Create a new baseline rule

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Model to learn ER baselines**.

3. In the tree, select **Model to learn ER baselines\Format to learn ER baselines**.
4. On the **Versions** FastTab, select **Run**.
5. In the **Enter Id** field, enter **1**.
6. Set the **Make baseline files** option to **Yes**.
7. Select **OK**.
8. Select **Baselines**.



The generated outbound file is automatically attached to the baseline of the executed ER format. The baseline rule is automatically added to this baseline and also contains the reference to the attached file.

9. In the **Name** field, enter **Baseline 1**.
10. In the **File name mask** field, enter **.xml**.
11. Select **Save**.

Run the format

You're now ready to complete the remaining steps in the example in the Trace generated report results and compare them with baseline values article, starting from the "Run the designed ER format and review the log to analyze the results" section.

> **Note**

When you delete the automatically added baseline rule on the **Baselines** FastTab, the referenced attachment isn't automatically deleted.

Configure the baseline to ignore constantly changing parts of the ER output

When you design an ER format to contain information that changes when the format runs, you must use the ER baseline feature to compare the generated results with baseline values. For example, the information might be the processing date and time or the unique identifier of a generated document in different formats (globally unique identifier [GUID], and so on). The new ER capabilities let you configure the baseline rule to ignore changeable elements of an ER format when the format runs with the purpose of comparing baseline values with the results of the format's execution. To learn more about this feature, complete the following example.

Example: Configure the baseline to ignore constantly changing parts of the ER output

To complete the steps in this example, you must first complete the steps in the example in the Trace generated report results and compare them with baseline values article.

Modify a configured ER format

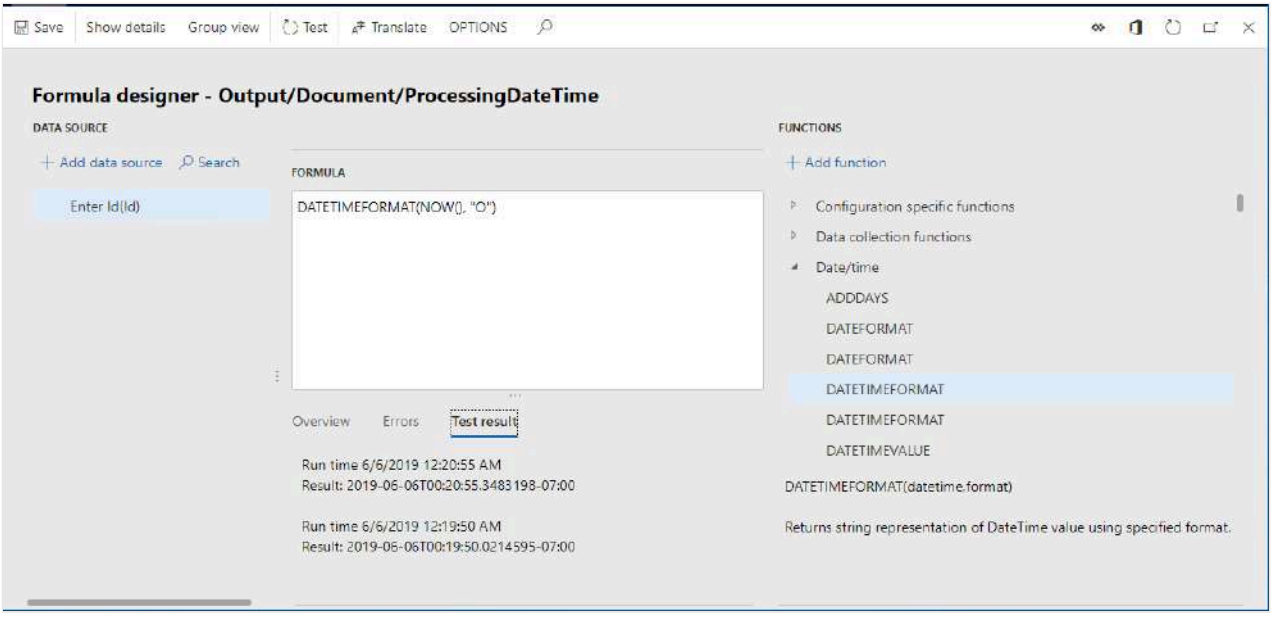
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Model to learn ER baselines**.
3. In the tree, select **Model to learn ER baselines\Format to learn ER baselines**.
4. Select **Designer**.
5. In the tree, select **Output\Document**.
6. Select **Add**.
7. In the drop-down dialog box, in the tree, select **XML\Attribute**.
8. In the **Name** field, enter **ProcessingDateTime**.
9. Select **OK**.
10. On the **Mapping** tab, in the tree, select **Output\Document\ProcessingDateTime**.

11. Select **Edit formula**.

12. In the **Formula** field, enter the following expression: **DATETIMEFORMAT(NOW(), "O")**

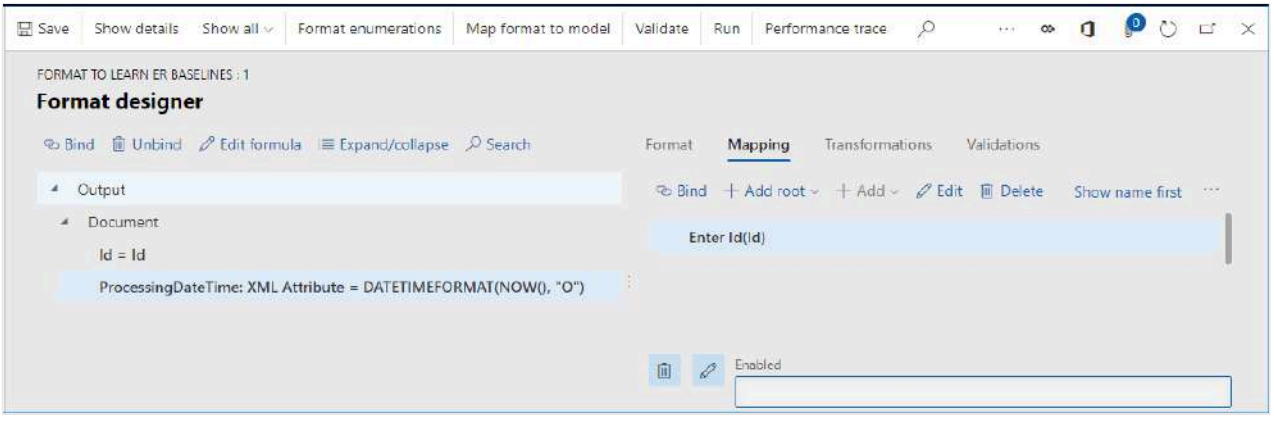
13. Select **Save**, and then select **Test**.

14. Select **Test** again to retest the configured expression.



> ⓘ Note
>
> The **Test result** tab shows that the configured expression returns a different date and time value whenever it's called.

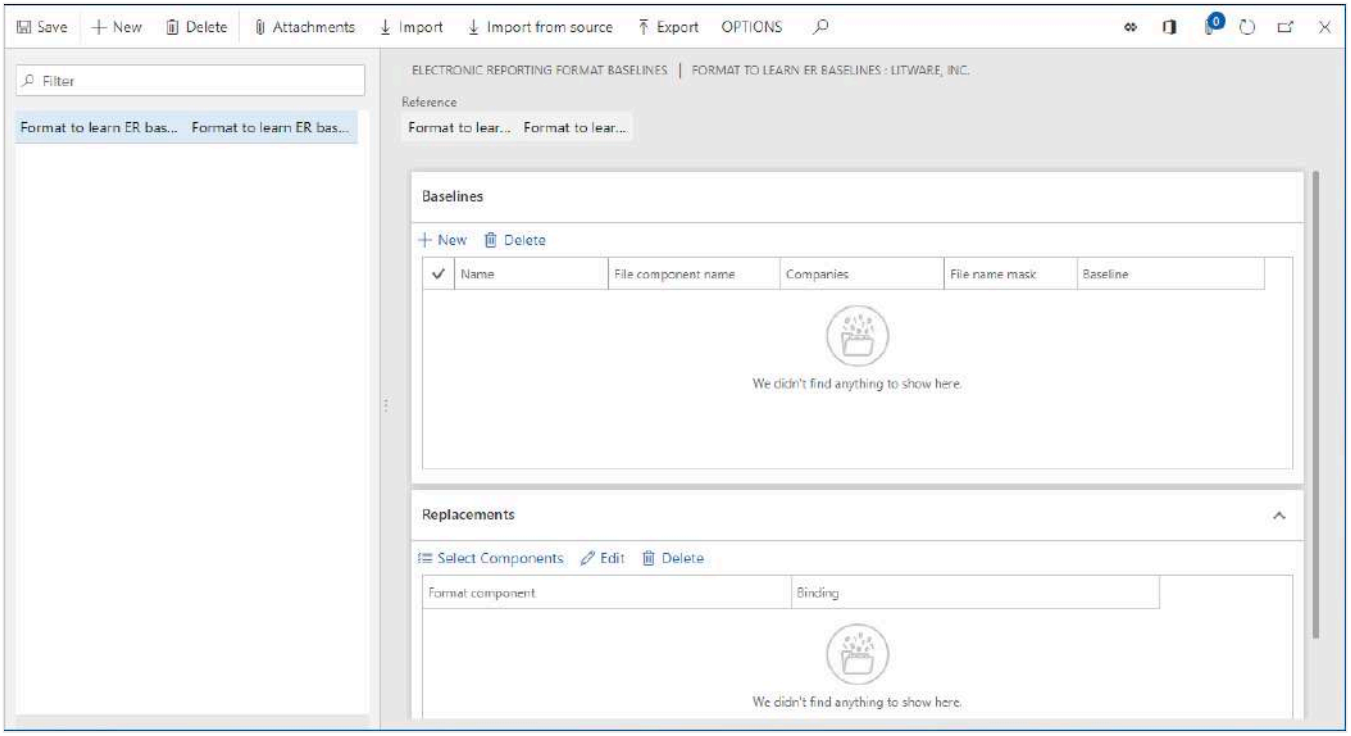
15. Close the **Formula designer** page, and then select **Save**.



16. Close the **Format designer** page.

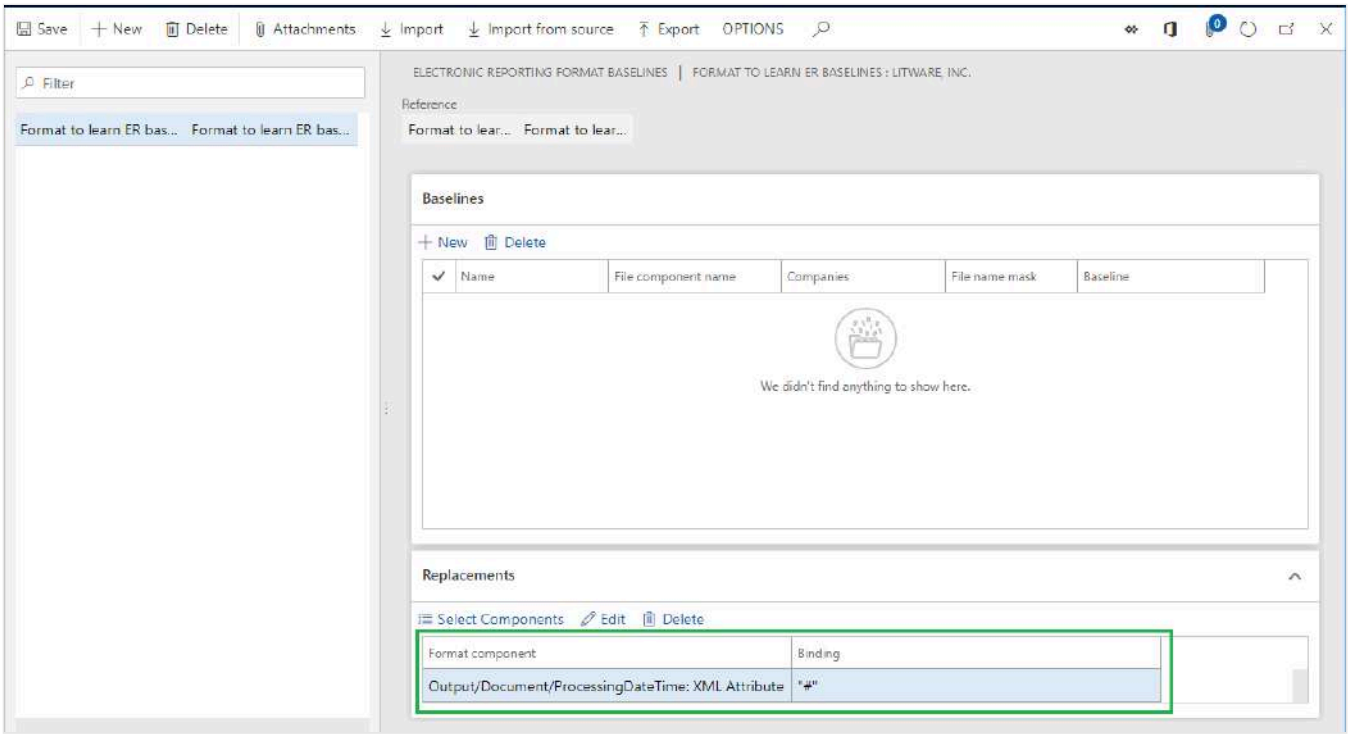
Remove an existing baseline rule

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. Select **Baselines**.
3. In the list of baselines, select the baseline that you configured for the **Format to learn ER baselines** format.
4. On the **Baselines** FastTab, select **Delete** to remove the baseline rule that you configured earlier.



Define replacements for bindings of designed ER format

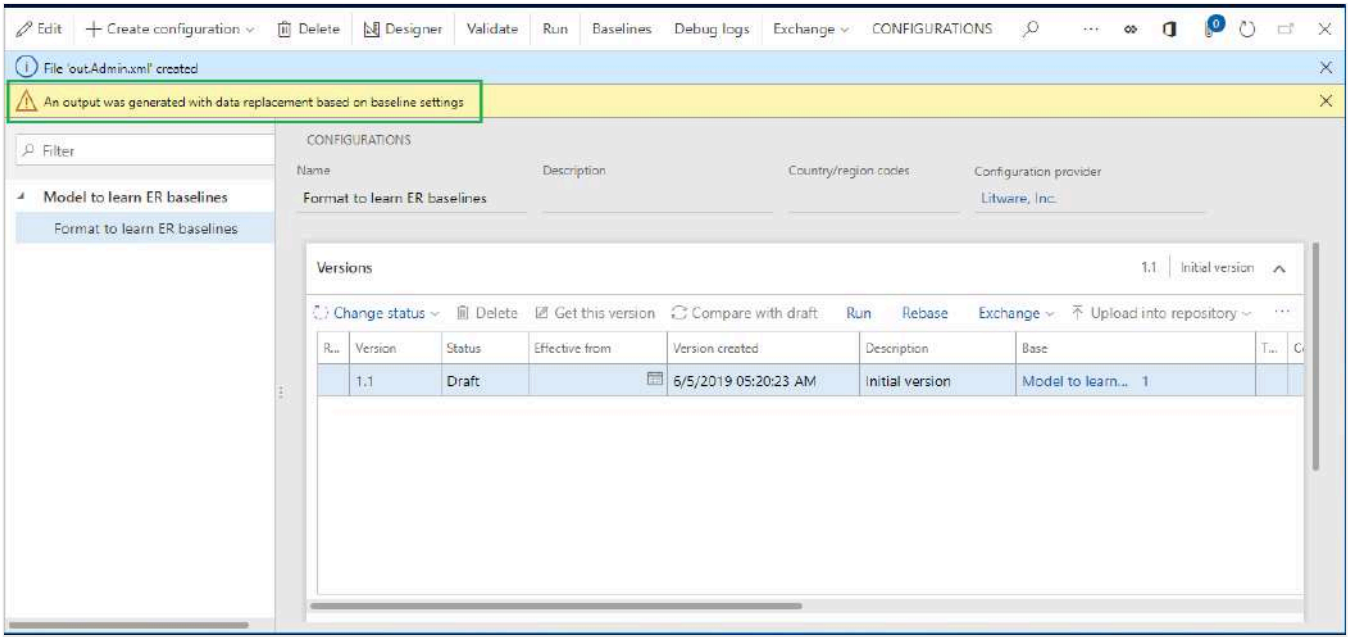
1. On the **Configurations** page, on the **Replacements** FastTab, select **Select components**.
2. In the format components tree, expand **Output**, expand **Output\Document**, and then select the check box for **Output\Document\ProcessingDateTime**.
3. Select **OK**.



The selected ER format component is added to the list of components on the **Replacements** FastTab. When you run the base ER format in debug mode, the format's binding for each component is replaced by the binding that appears in the **Binding** column. To change the default binding for a component that appears on the **Replacements** FastTab, select **Edit**.

Create a new baseline rule

Follow the steps in the "Example: Automate the setting of baseline rules" section earlier in this article. A notification warns you that the outbound file is generated by using baseline settings, and that a forced replacement of the format bindings occurs.



Suppress warnings about the replacement of format bindings

Set specific ER parameters to suppress notifications about the replacement of format bindings. This suppression is useful when the Regression Suite Automation Tool replaces format bindings in unattended mode. In this case, the warning can cause the test case to fail.

1. On **Configurations**, on the Action Pane, on the **Configurations** tab, select **User parameters**.
2. Set the **Suppress baseline warnings** option to **Yes**, and then select **OK**.

Review the generated baseline file

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. Select **Baselines**.
3. Select **Attachments**.

   > ⓘ **Note**
   >
   > The generated file contains the processing date and time text ("#") from the binding that you configured in the added baseline rule, not from the format's binding.

4. Close **Attachments**.

Run the designed ER format and review the log to analyze the results

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. In the tree, expand **Model to learn ER baselines**.
3. In the tree, select **Model to learn ER baselines\Format to learn ER baselines**.
4. On the **Versions** FastTab, select **Run**.
5. In the **Enter Id** field, type **1**.
6. Select **OK**.
7. Go to **Organization administration** > **Electronic reporting** > **Configuration debug logs**.

The execution log contains information about the results of the comparison of the generated file with the configured baseline. The log indicates that the generated file and the baseline are equal, even though the executed format contains the binding to enter a constantly changing date and time value in the outbound file.

> **Note**
>
> Although the outbound file is generated by using baseline settings that force the replacement of the format's bindings, you don't receive any warnings about the replacement.

Exchange baseline settings between environments

Export baseline settings

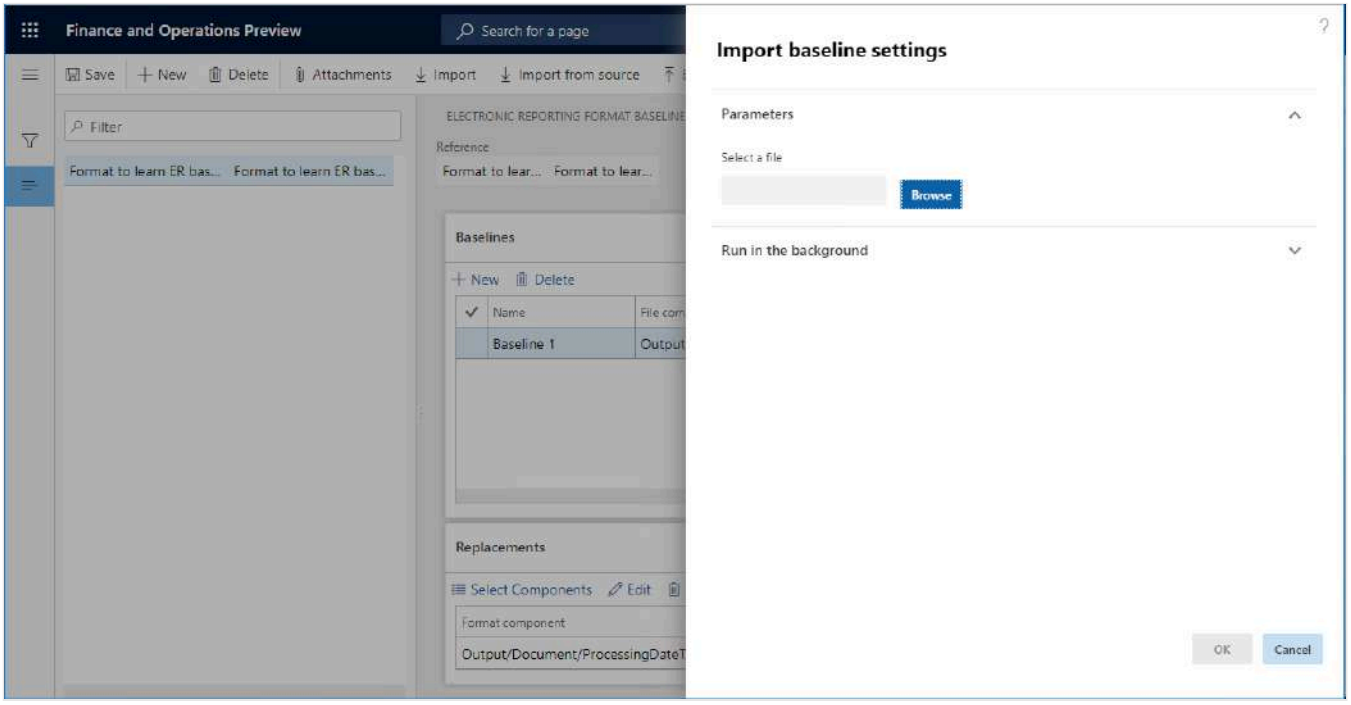
The new ER capabilities let you export baseline settings for the selected ER format from the current environment and store them as XML files.

To export baseline settings, on the **Electronic reporting format baselines** page, select **Export**.

Import baseline settings

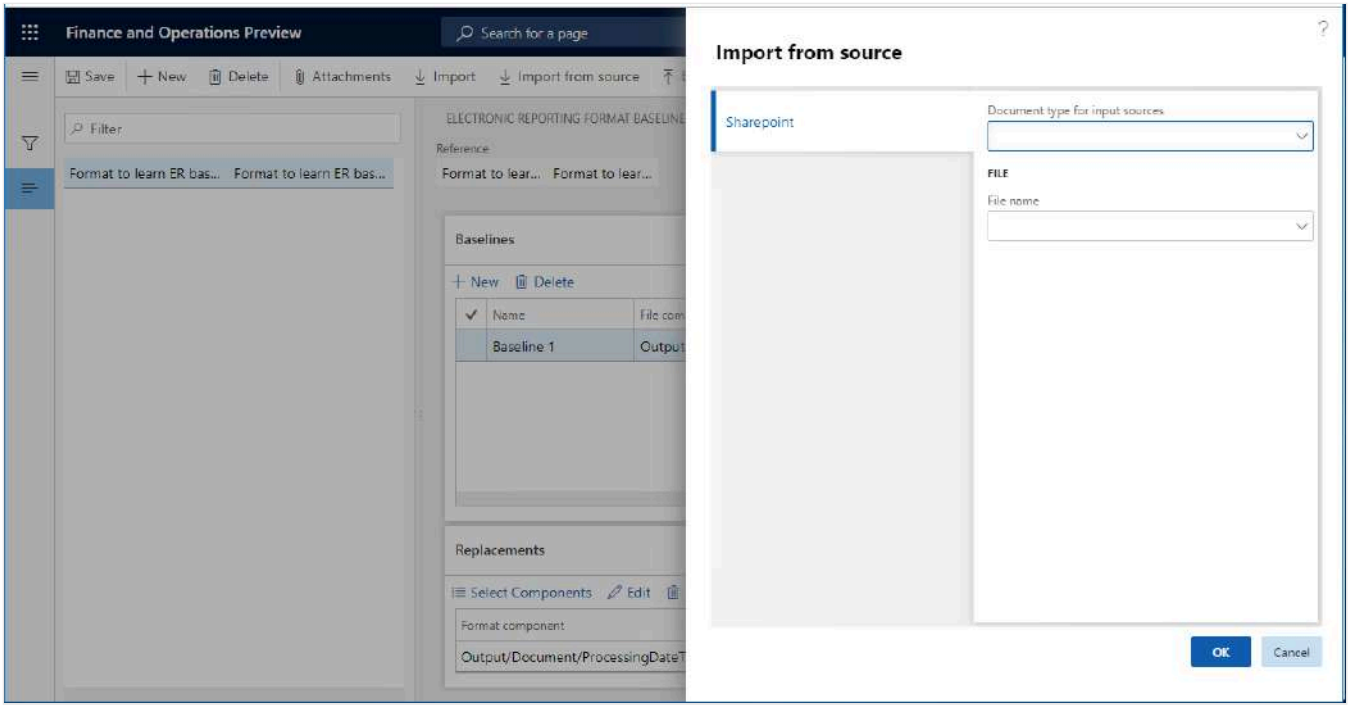
You can import exported baseline settings into a different environment. First, import the environment as an ER format. Then, import the baseline settings.

To import baseline settings from a locally stored XML file, on the **Electronic reporting format baselines** page, select **Import**, and then select **Browse** to select the XML file.



To import baseline settings from an XML file that is stored on the Microsoft SharePoint Server, based on the current Document management settings and the selected document type, on the

Electronic reporting format baselines page, select **Import from source**. Then select the document type and the XML file. You must configure the required document type to access the SharePoint folder.



> **Note**
>
> Use Task recorder to record the steps for selecting the required document type and the file name in the **Import from source** dialog box. By using this method, you can keep required baseline settings on SharePoint Server and then automatically import them by playing a task recording when you run automated tests by using the Regression Suite Automation Tool.

Additional resources

- Trace generated report results and compare them with baseline values
- Task recorder resources

Last updated on 04/13/2026

Automate testing by using Electronic reporting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how you can use the Electronic reporting (ER) framework to automate testing of some functionality. The example in this article shows how to automate the testing of vendor payment processing.

The application uses the ER framework to generate payment files and corresponding documents during vendor payment processing. The ER framework consists of a data model, model mappings, and format components that support payment processing for different payment types and the generation of documents in different formats. These components can be downloaded from Microsoft Dynamics Lifecycle Services (LCS) and imported into the instance.

You can also customize each Microsoft component and use it as the basis of your own custom component. By creating a custom version, you can make changes that support specific requirements. For example, you can adjust the ER data model and ER model mapping to access customer-specific application data, or you can change an ER format to modify the layout of a generated document.

You can use customized ER formats to process payment files that generate vendor payments and also to process control reports. ER components support versioning. Therefore, Microsoft can provide updated versions of ER solutions for vendor payment processing, and you can automatically merge the updated version with your customized component by rebasing it. However, you must test the rebased version to make sure that it works as you expect.

ER data models and ER model mappings are common for many ER formats that are used to process payments of different types and to generate country/region-specific payment documents. Therefore, it's highly desirable to automate user acceptance and integration testing so that it automatically happens in multiple companies but considers the country/region context of each target company, uses different datasets, and so on.

For more information about how to create a custom version of a format that's based on the format that you received from a configuration provider, see ER Upgrade your format by

adopting a new, base version of that format.

Key concepts

Functional power users can create user acceptance and integration testing without writing source code.

- Use the ER baseline feature to compare generated documents to master copies. For more information, see Trace generated report results and compare them with baseline values.
- Use Task recorder to record test cases, and include baseline assessment. For more information, see Task recorder resources.
- Group test cases for required test scenarios. For more information, see Create and automate user acceptance tests.
  - Use Business process modeler (BPM) in LCS to make libraries for user acceptance tests.
  - Use BPM test libraries to create a test plan and test suites in Microsoft Azure DevOps Services (Azure DevOps).

Functional power users can run user acceptance and integration tests.

- Use Regression suite automation tool (RSAT) to run test cases of the desired test suite.
- Report the results of the testing to Azure DevOps, and use this service to investigate those results.

Prerequisites

Before you can complete the tasks in this article, complete the following prerequisites:

- Deploy a topology that supports test automation. You must have access to the instance of this topology for the **System administrator** role. This topology must contain the demo data used in this example. For more information, see Deploy and use an environment that supports continuous build and test automation.
- To run user acceptance and integration tests automatically, install RSAT in the topology that you're using and configure it in the appropriate manner. For information about how to install and configure RSAT and configure it to work with finance and operations apps and Azure DevOps, see Regression Suite Automation Tool. Pay attention to the prerequisites for using the tool.
- To organize test cases in suites to help guarantee the correct execution sequence, so that you can collect logs of test executions for further reporting and investigation, you must have access to Azure DevOps from the deployed topology.

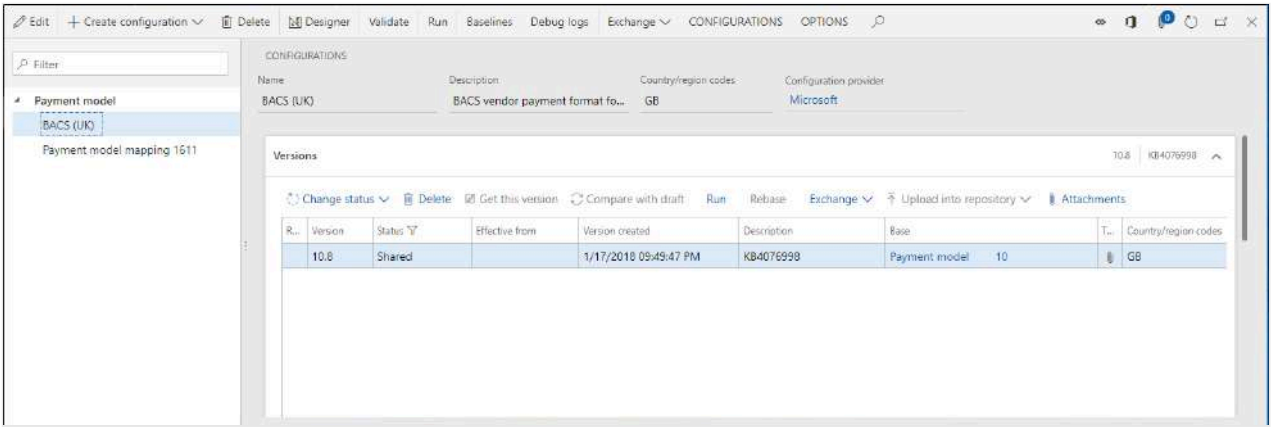
- To complete the example in this article, download ER usage for RSAT tests. This zip file contains the following task guides:

Expand table

Content	File name and location
Sample task recording to prepare data for testing	Prepare\Recording.xml
Sample task recording to process vendor payment	Process\Recording.xml

Prepare the Accounts payable module to process vendor payments

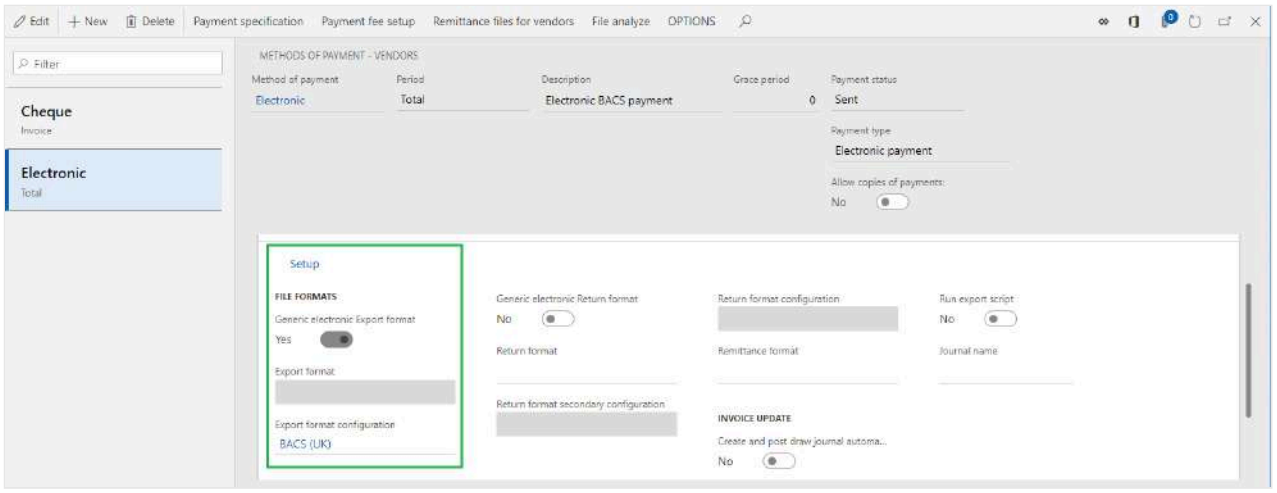
1. Sign in to your instance.
2. Download the following ER configurations from LCS. For instructions, see ER Import a configuration from Lifecycle Services.
    - **Payment model** ER model configuration
    - **Payment model mapping 1611** ER model mapping configuration
    - **BACS (UK)** ER format configuration



3. Select the **GBSI** demo data company, which has a country/region context in the United Kingdom.
4. Configure Accounts payable parameters:
    a. Go to **Accounts payable** > **Payment setup** > **Methods of payment**.
    b. Select the **Electronic** method of payment.
    c. Configure the selected method of payment so that it uses the **BACS (UK)** ER format that you downloaded earlier for vendor payment processing:

i. On the **File formats** FastTab, set the **Generic electronic Export format** option to **Yes**.

ii. In the **Export format configuration** field, select **BACS (UK)**.

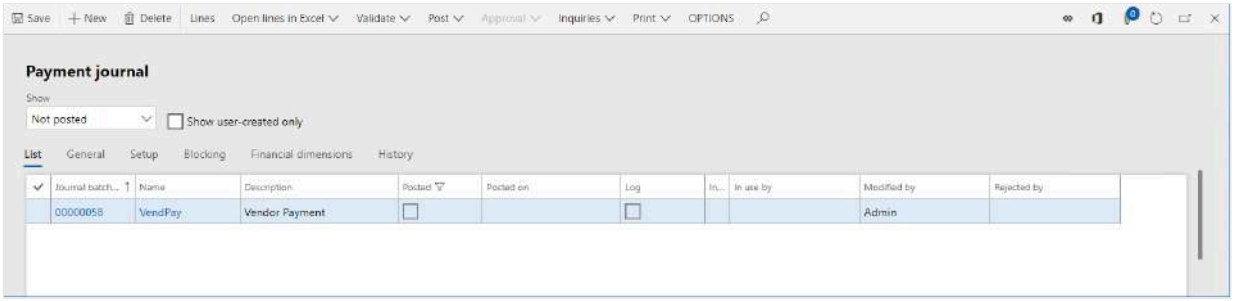


> **Note**
>
> If you have the derived version of this ER format that was created to support customizations, you can select this configuration in the **Electronic** method of payment.

5. Create an example vendor payment:

a. Go to **Accounts payable** > **Payments** > **Payment journal**.

b. Make sure that you didn't post the payment journal.

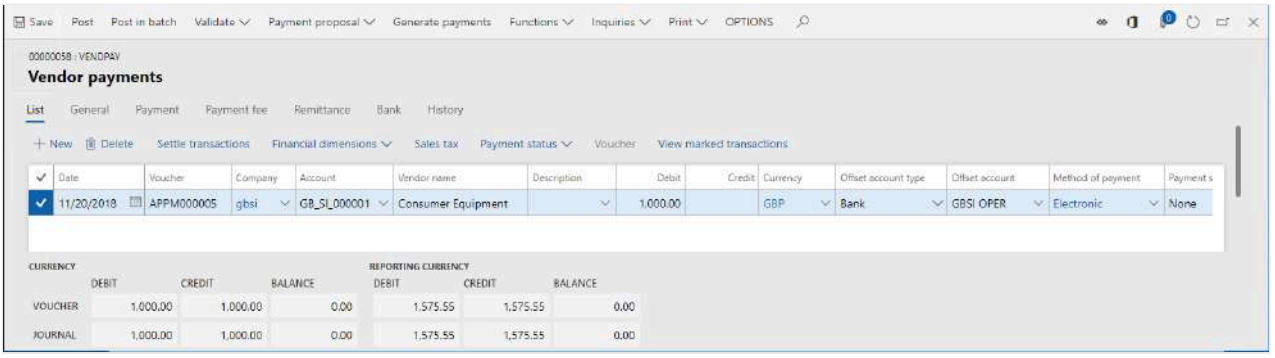


c. Select **Lines**, and enter a line that has the following information.

Expand table

Field	Example value
Vendor name	GB_SI_000001
Debit	1,000.00
Currency	GBP

Field	Example value
Offset account type	Bank
Offset account	GBSI OPER
Method of payment	Electronic

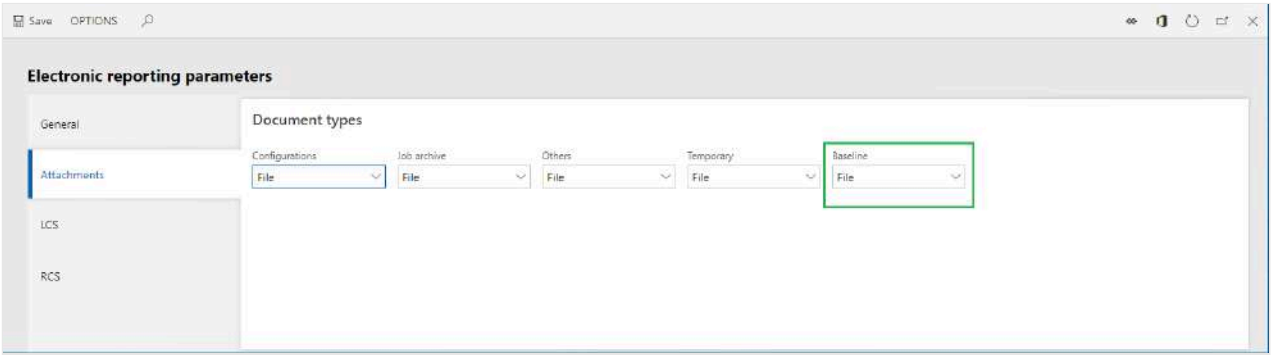


Prepare the ER framework to test vendor payment processing

Configure ER parameters

1. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting parameters**.

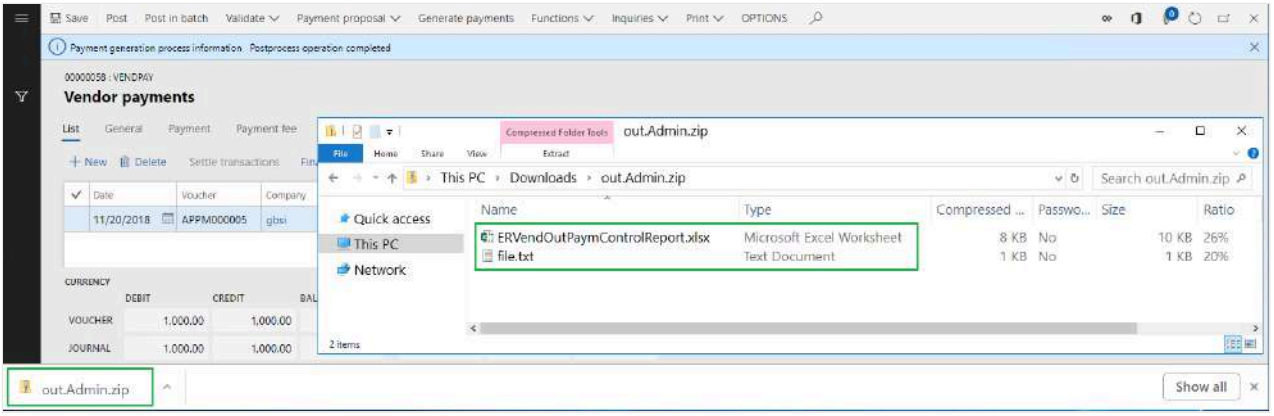
2. On the **Attachments** tab, in the **Baseline** field, select **File** as the document type. The Document management (DM) framework uses this type to keep documents that are related to the baseline feature as DM attachments.



Generate baseline copies of vendor payment–related documents

1. Go to **Accounts payable** > **Payments** > **Payment journal**.

2. Select **Lines**.
3. Select **Generate payments**.
4. Select the **Electronic** method of payment.
5. Select the **GBSI OPER** bank account.
6. Set the **Print control report** option to **Yes**.
7. Download the generated output as a ZIP file.
8. Open the downloaded file.
9. Extract the following files from the downloaded file:
    - **File** payment file in text format
    - **ERVendOutPaymControlReport** control report file in XLSX format



Turn on the ER baseline feature

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** tab, select **User parameters**.
3. Set the **Run in debug mode** option to **Yes**.

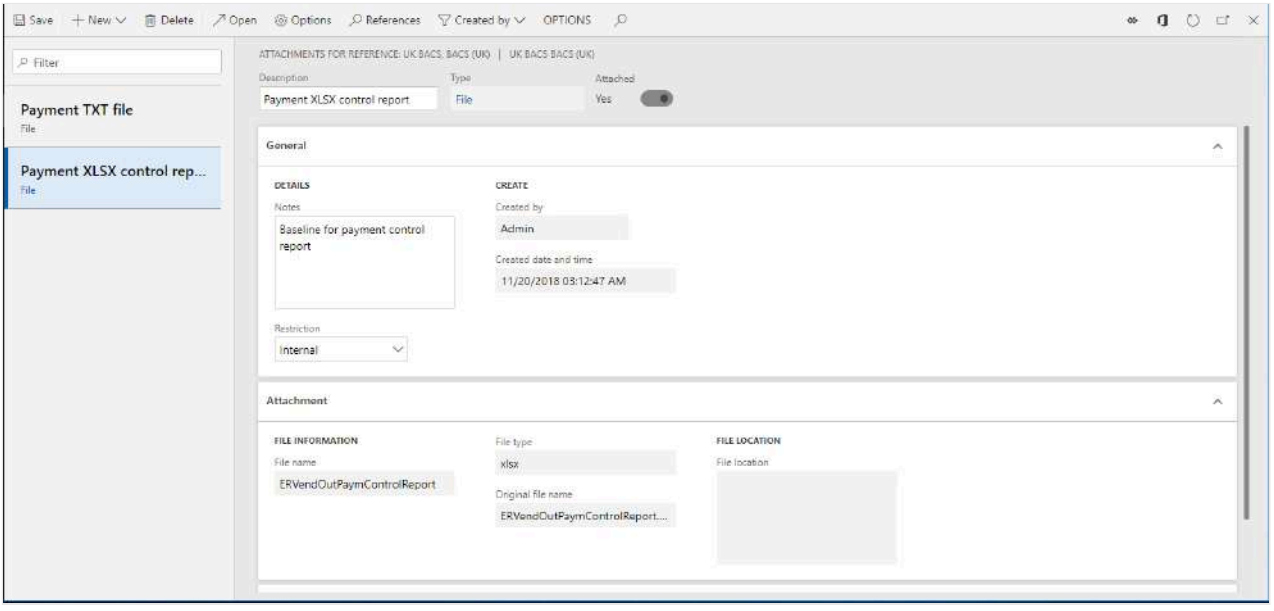
When you turn on the **Run in debug mode** parameter, the ER framework takes these actions after it runs any ER format that generates outgoing documents:

1. It checks whether a baseline is configured for any components of the ER format that ran.
2. It checks whether each configured baseline is applicable in the current conditions, like the company code of the signed-in company, the file name, and the file name extension of the generated output.
3. For each applicable baseline, it takes these steps:
    a. It compares the output that the ER format generates with the corresponding baseline.

b. It stores the results of the comparison in the ER configurations debug log.

Configure ER baselines for vendor payment processing

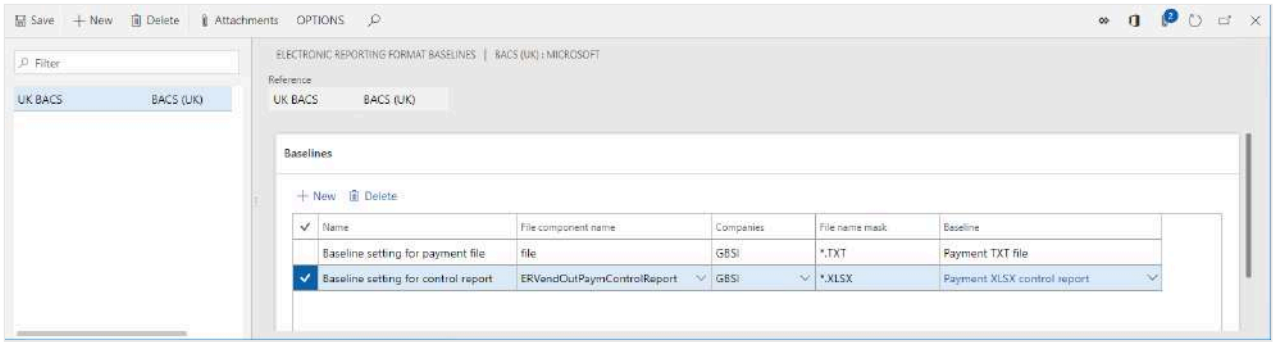
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. Select **Baselines**.
3. Select **New**.
4. In the **Reference** field, select the **BACS (UK)** format.
5. Select **Attachments**.
6. Add a new baseline for the vendor payment file:
   a. Select **New**.
   b. In the **Type** field, select the **File** DM document type that you configured in the ER parameters to store baseline artifacts.
   c. Browse to select the locally saved **File** payment file in text format.
   d. In the **Description** field, enter **Payment TXT file**.
7. Add a new baseline for the control report for the vendor payment:
   a. Select **New**.
   b. In the **Type** field, select the **File** DM document type that you configured in the ER parameters to store baseline artifacts.
   c. Browse to select the locally saved **ERVendOutPaymControlReport** control report file in XLSX format.
   d. In the **Description** field, enter **Payment XLSX control report**.



8. Close the page.

9. On the **Baselines** tab, select **New** to configure a baseline for the payment file:
    a. Name the line **Baseline setting for payment file**.
    b. In the **File component name** field, select **file** to apply this baseline to the ER format output that generates the payment file in BACS (UK) text format.
    c. In the **Companies** field, select **GBSI** to apply this baseline when the **BACS (UK)** ER format is run in the GBSI company.
    d. In **File name mask** field, enter ***.TXT** to apply this baseline only to outputs of the **file** format component that have the **.txt** file name extension.
    e. In the **Baseline** field, select **Payment TXT file** so that this baseline is used for comparison with the generated output.

10. Select **New** to configure a baseline for the control report:
    a. Name the line **Baseline setting for control report**.
    b. In the **File component name** field, select **ERVendOutPaymControlReport** to apply this baseline to the ER format output that generates the control report.
    c. In the **Companies** field, select **GBSI** to apply this baseline when the **BACS (UK)** ER format is run in the GBSI company.
    d. In **File name mask** field, enter ***.XLSX** to apply this baseline only to outputs of the **ERVendOutPaymControlReport** format component that have the **.xslx** file name extension.
    e. In the **Baseline** field, select **Payment XLSX control report** so that this baseline is used for comparison with the generated output.



Record tests to validate vendor payment processing

As a functional power user, you can record your own steps to test vendor payment processing. Play (and edit, as required) the **Prepare\Recording.xml** task recording that you downloaded earlier. Use this recording to set all testing data to the correct state. This step is required because you can run the testing many times, and every test must use data that's in the same state.

Reset user settings

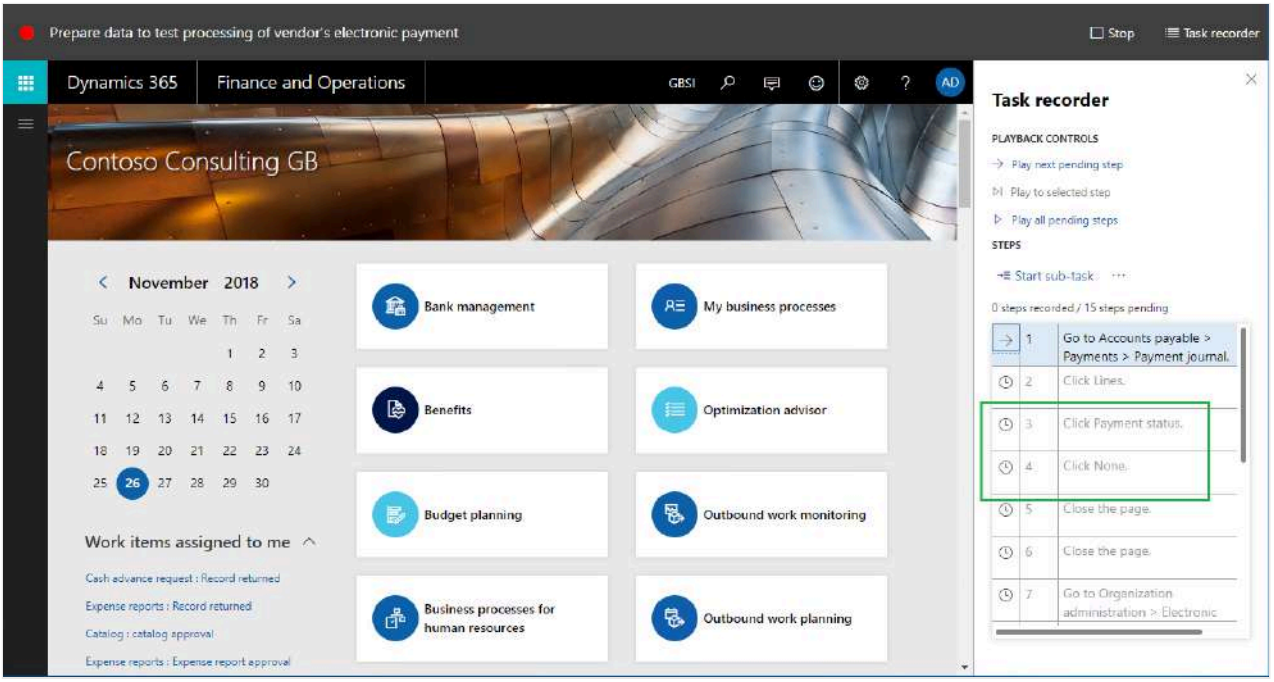
1. Open the default dashboard.
2. Select the **Settings** button (the gear symbol).
3. Select **User options**.
4. Select **Usage data**.
5. Select **Reset**.
6. Select **Yes** to confirm that you want to reset usage data.
7. Close the page.

Record the steps to prepare data for testing

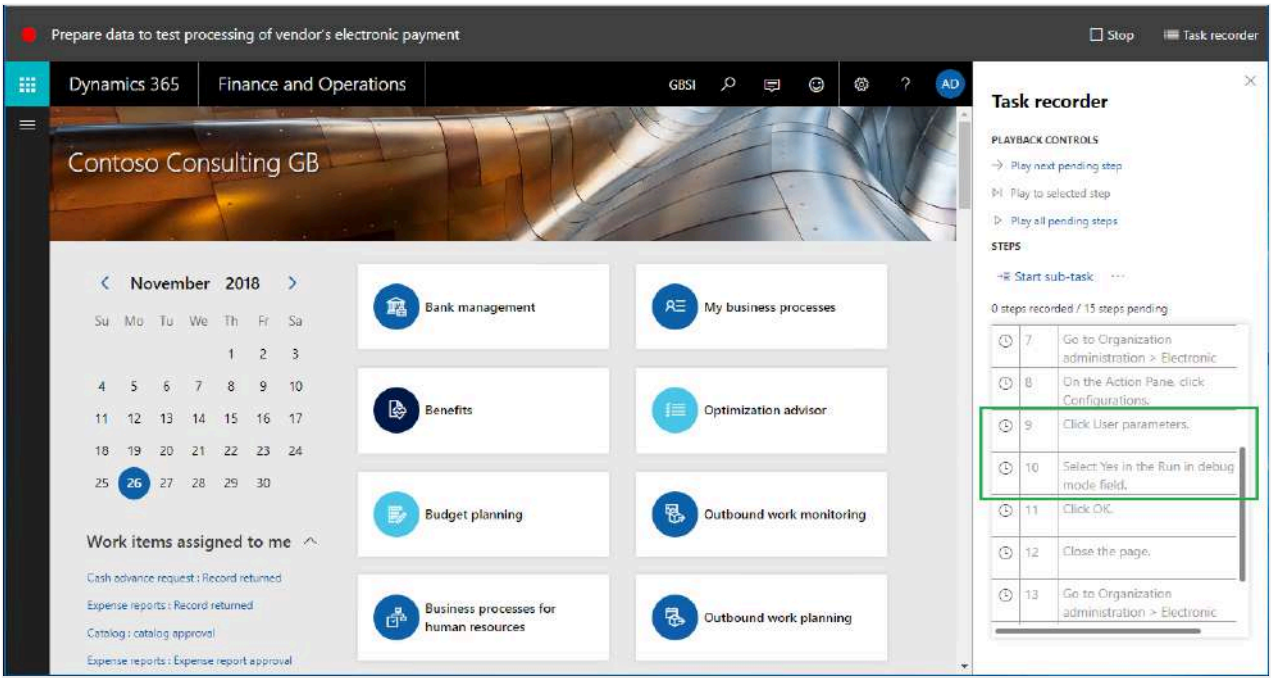
1. Select the **Settings** button (the gear symbol).
2. Select **Task recorder**.
3. Select **Playback recording**.
4. Select **Open from this PC**.
5. Select **Browse**, and select the locally saved **Prepare\Recording.xml** file.
6. Select **Start**.
7. Keep selecting **Play next pending step** until you play all the steps in the recording.

This task recording performs the following actions:

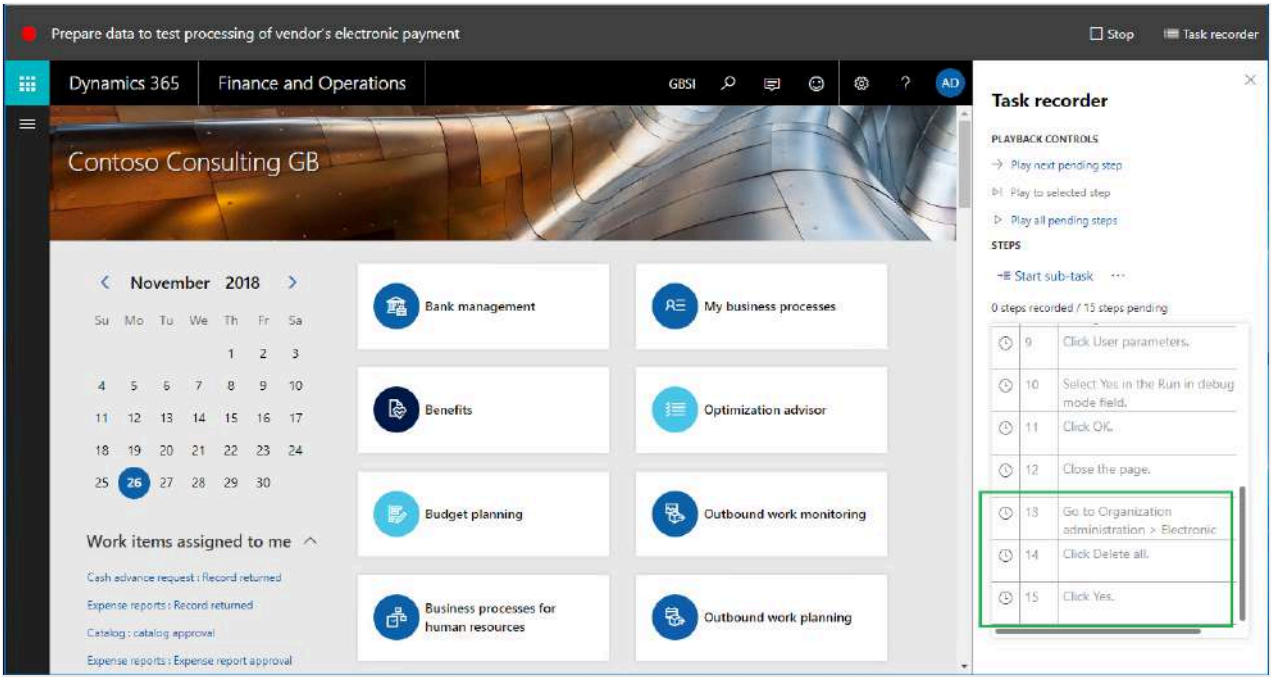
1. Set the status of the processed payment line to **None**.



2. Turn on the **Run in debug mode** ER user parameter.



3. Clean up the ER debug log that contains the results of the comparison of generated files to baselines.



Record the steps to test vendor payment processing

Play (and edit, if needed) the **Process\Recording.xml** task recording that you downloaded earlier. This recording processes vendor payments and validates the results of the comparison of generated documents to corresponding baselines.

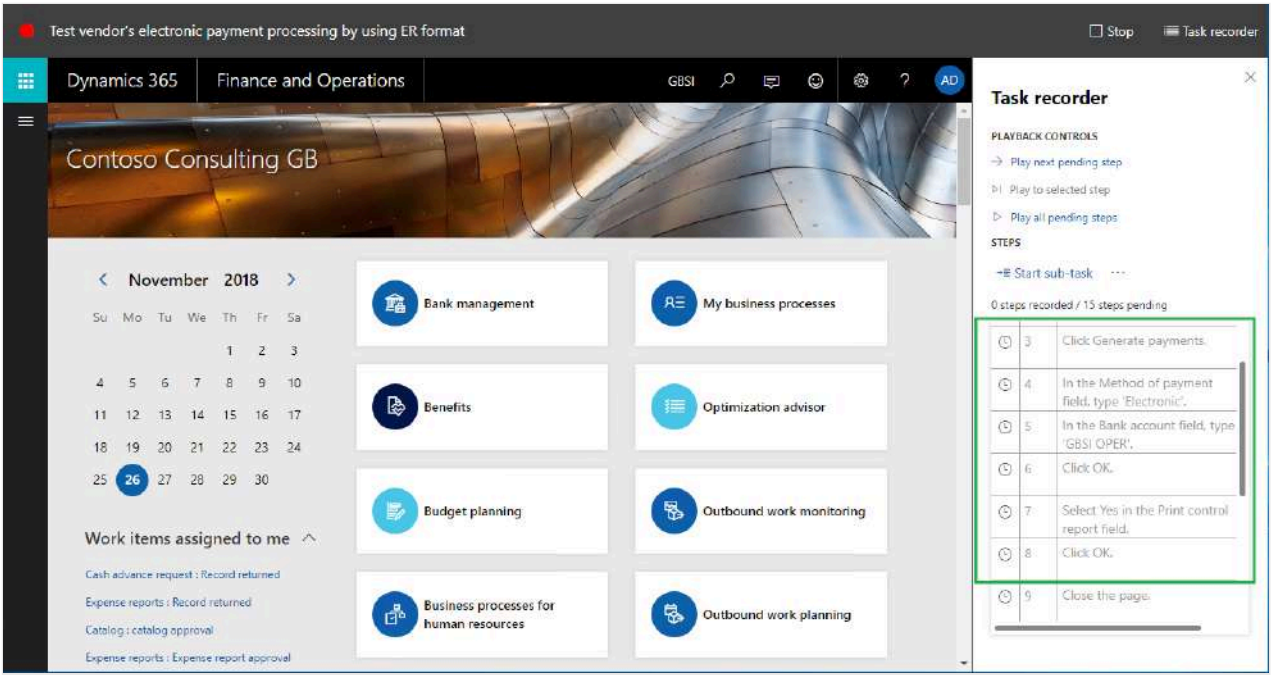
1. Select the **Settings** button (the gear symbol).
2. Select **Task recorder**.
3. Select **Playback recording**.
4. Select **Open from this PC**.

5. Select **Browse**, and select the locally saved **Process\Recording.xml** file.
6. Select **Start**.
7. Keep selecting **Play next pending step** until you play all the steps in the recording.

This task recording performs the following actions:

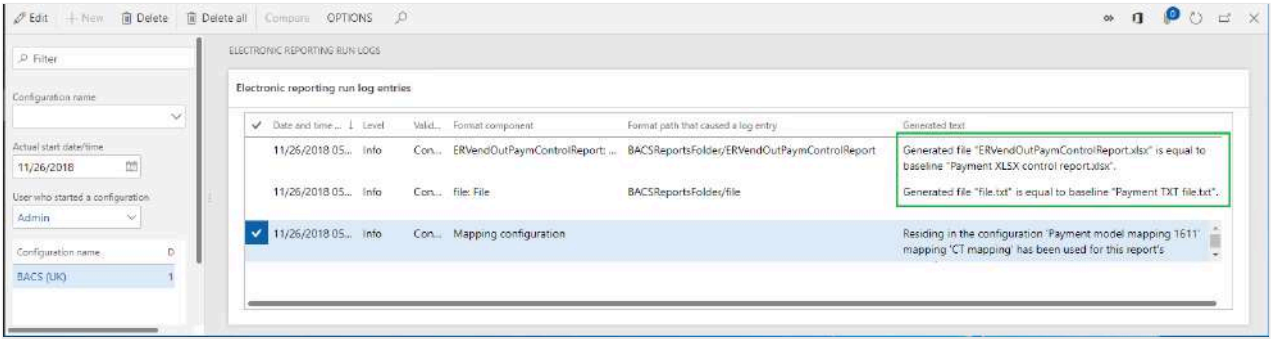
1. Starts vendor payment processing.

2. Selects the correct runtime parameters, and turns on generation of a control report.

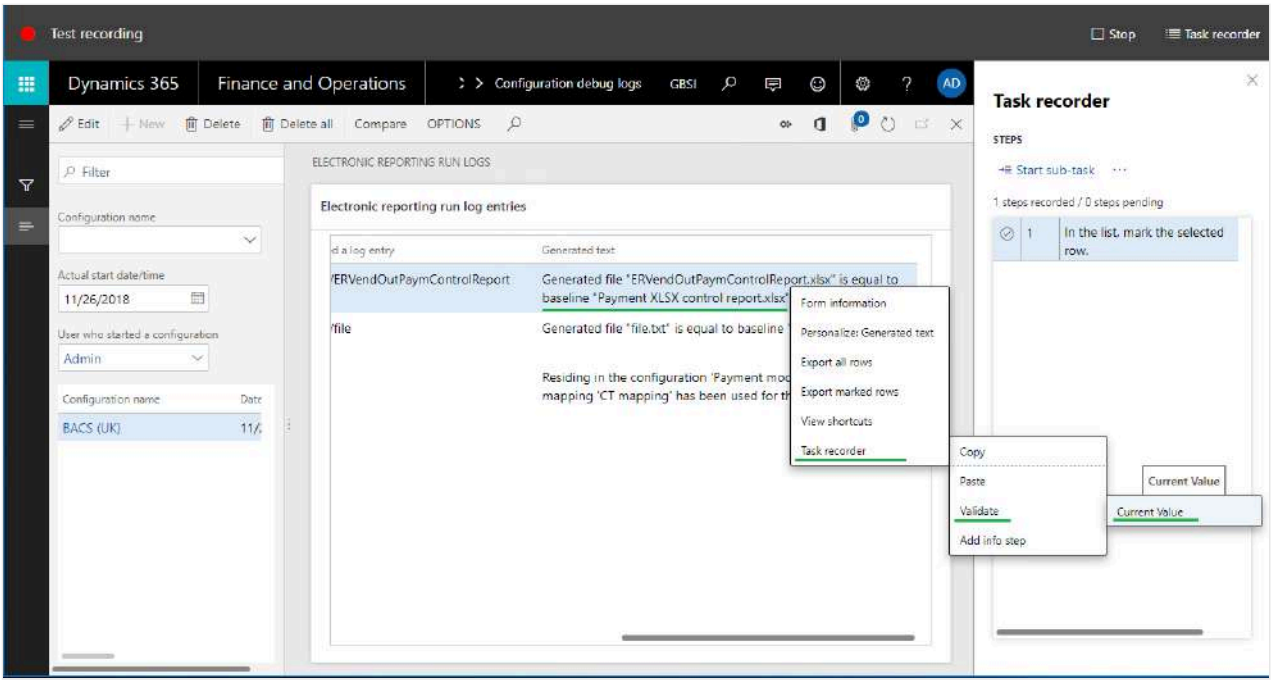


3. Accesses the ER debug log to record the results of the comparison of generated outputs to corresponding baselines.

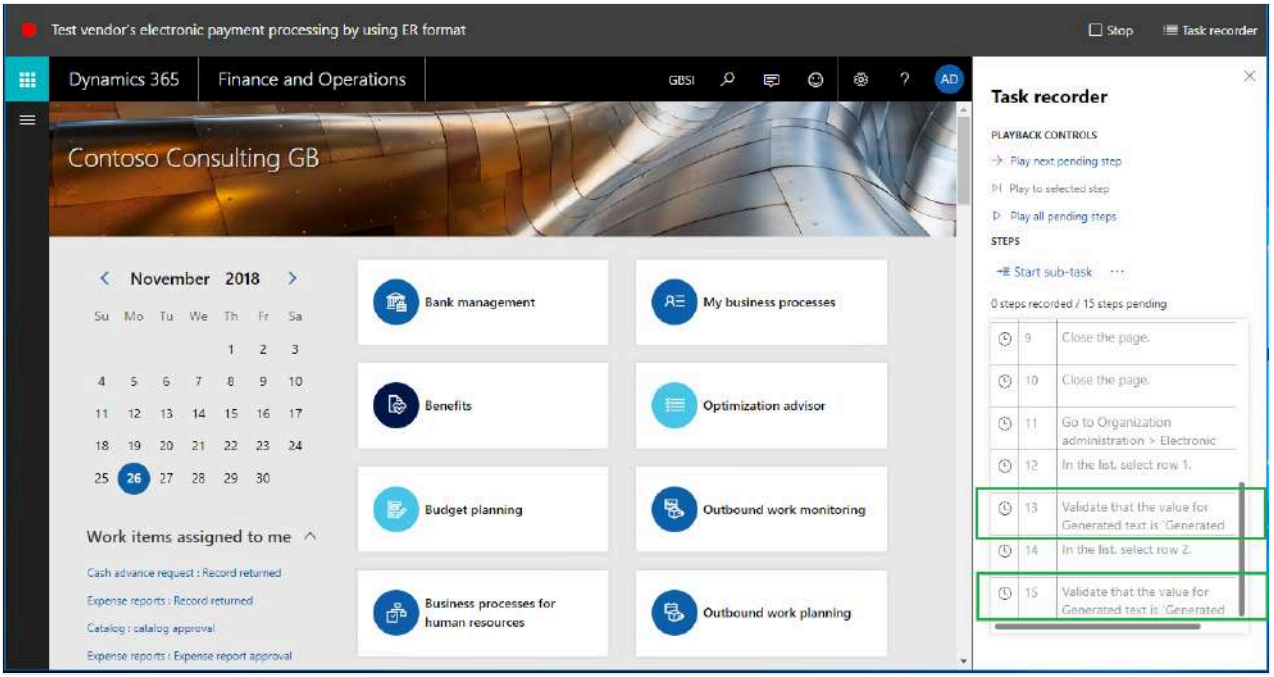
   In the ER debug log, the results of the comparison appear in the **Generated text** field. The **Format component** and **Format path that caused a log entry** fields refer to the file component for which the generated output is compared to the baseline.



4. Records the comparison of the current output to the baseline by using the **Validate** Task recorder option and selecting **Current Value**.



The following illustration shows what the recorded validation steps look like in the task recording.

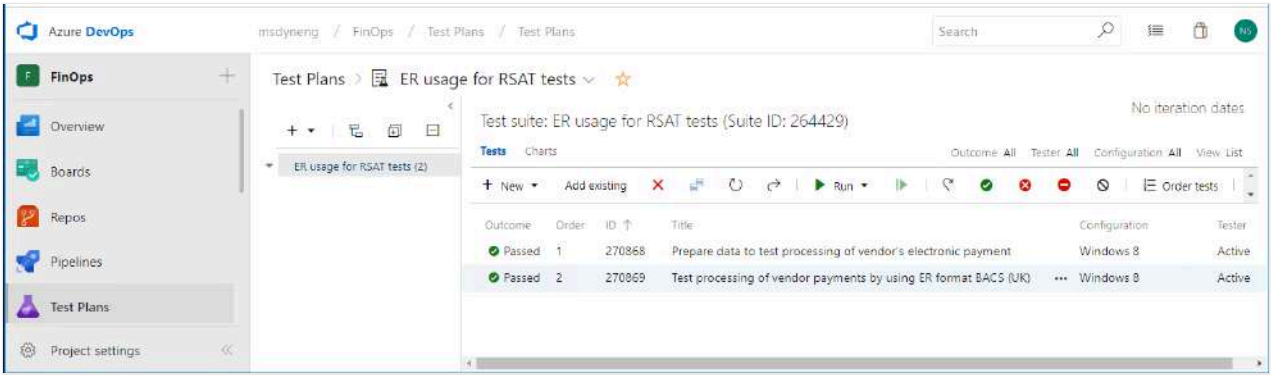


Add the recorded tests to Azure DevOps

1. Open the Azure DevOps environment.
2. Select the project that you defined in the RSAT parameters when you configured the tool.
3. Select the test plan that you defined in the RSAT parameters when you configured the tool.
4. Create a new test case for the selected test plan:

a. Name the test case **Prepare data to test processing of vendor's electronic payment**.
b. Attach the **Recording.xml** file from the **Prepare** folder that you downloaded earlier.

5. Create a new test case for the selected test plan:
   a. Name the test case **Test processing of vendor payments by using ER format BACS (UK)**.
   b. Attach the **Recording.xml** file from the **Process** folder that you downloaded earlier.



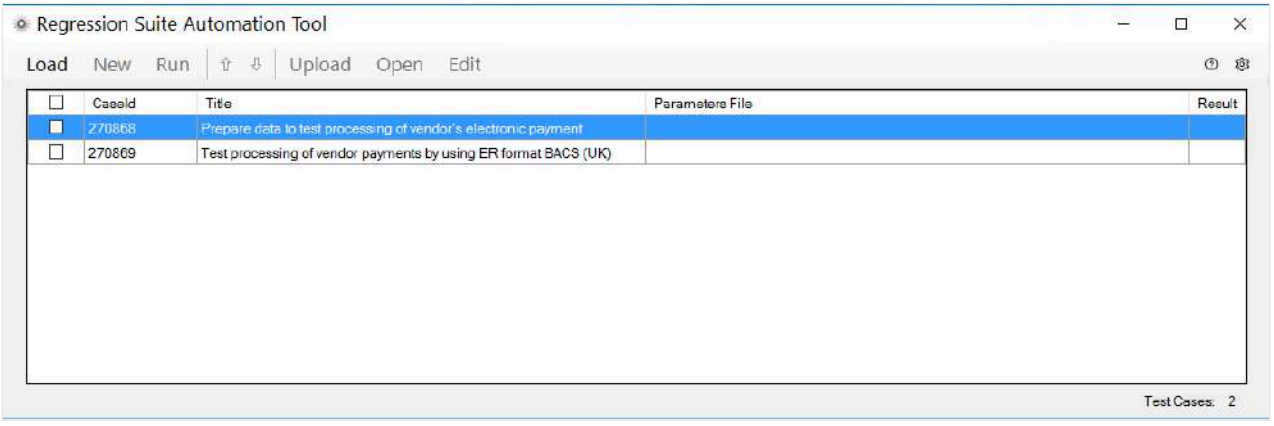
> ⓘ **Note**
>
> Pay attention to the correct execution order of the tests that are added.

Prepare RSAT to run the recorded tests

Load the tests from Azure DevOps to RSAT

1. Open the local RSAT application in the current topology.

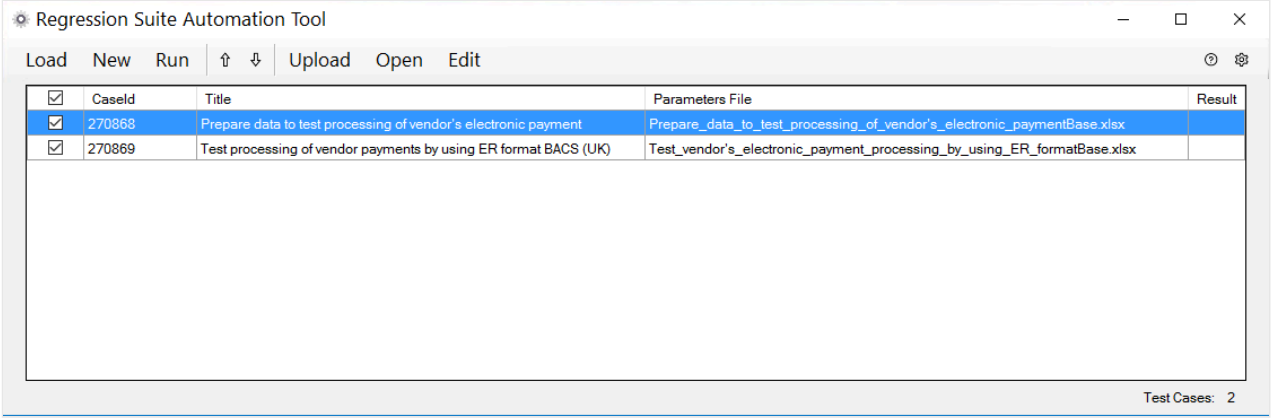
2. Select **Load** to load the tests that currently reside in Azure DevOps into RSAT.



Create automation and parameters files

1. In RSAT, select the tests that you loaded from Azure DevOps.

2. Select **New** to create RSAT automation and parameters files.



Modify the parameters files

1. In RSAT, select the **Prepare data to test processing of vendor's electronic payment** test case.

2. Select **Edit**.

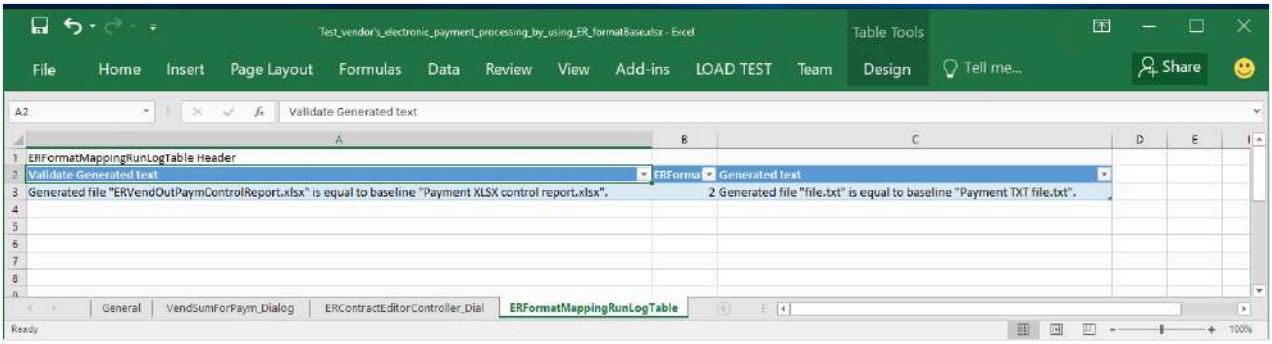
3. In the Microsoft Excel workbook that is opened, on the **General** worksheet, change the company code to **GBSI**, because this company will be used for test execution.

4. In RSAT, select the **Test processing of vendor payments by using ER format BACS (UK)** test case.

5. Select **Edit**.

6. In the Excel workbook that is opened, on the **General** worksheet, change the company code to **GBSI**.

7. On the **ERFormatMappingRunLogTable** worksheet, notice that cells A:3 and C:3 contain the text of the fields in the ER debug log table that are used to validate the results of the comparison of the output to the baseline. These texts will be used to evaluate ER debug log records that are created during test execution.



Run the tests and analyze the results

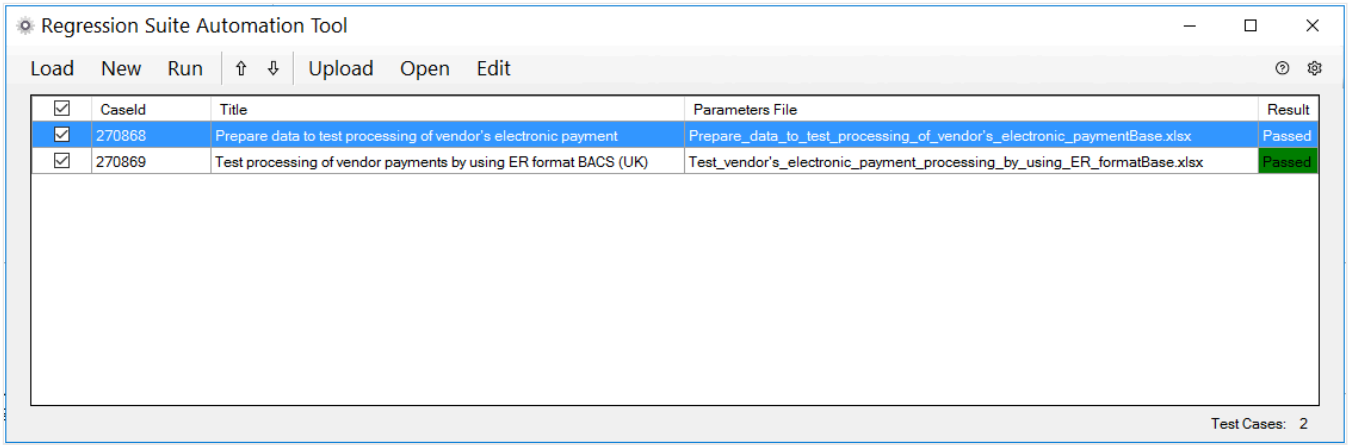
Run the tests in RSAT

1. In RSAT, select the loaded tests.
2. Select **Run**.

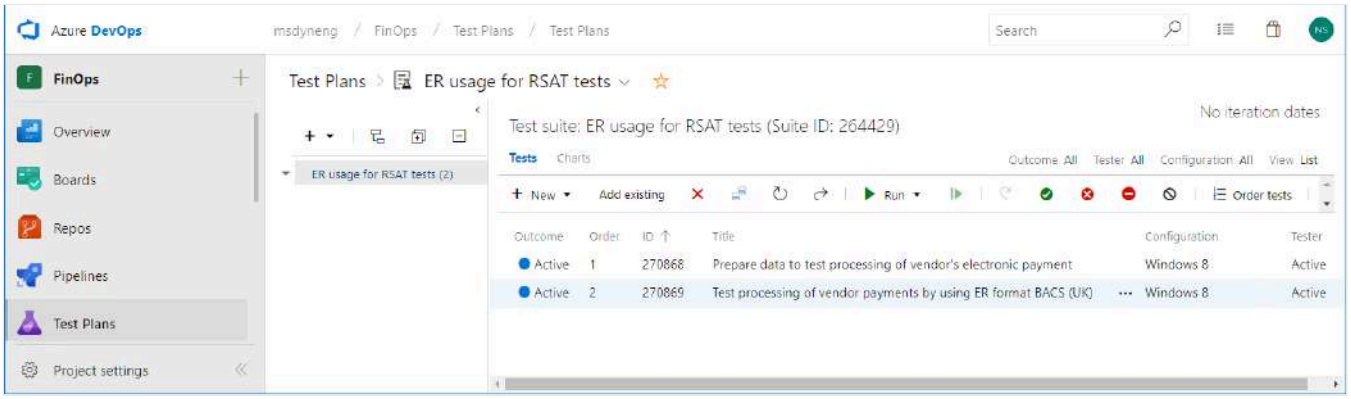
Test cases automatically run in the application by using a web browser.

Analyze the results of test execution

The results of the test execution are stored in RSAT. Notice that both tests were passed.



The test execution results are also sent to Azure DevOps so that you can do further analysis.



Simulate a situation where tests fail

This test suite fails when at least one of the generated outputs doesn't match the corresponding baseline. To create this condition, use your derived version of the **BACS (UK)** format that generates a payment file with different content than the corresponding baseline. To simulate this condition, use the same **BACS (UK)** format but change the payment amount on the processed payment line.

1. Open the application and go to **Accounts payable** > **Payments** > **Payment journal**.
2. Select **Lines**.
3. Select the payment line, and then select **Payment status** > **None**.
4. In the **Debit** field, change the value from **1,000.00** to **2,000.00**.
5. Select **Save** to save your changes.

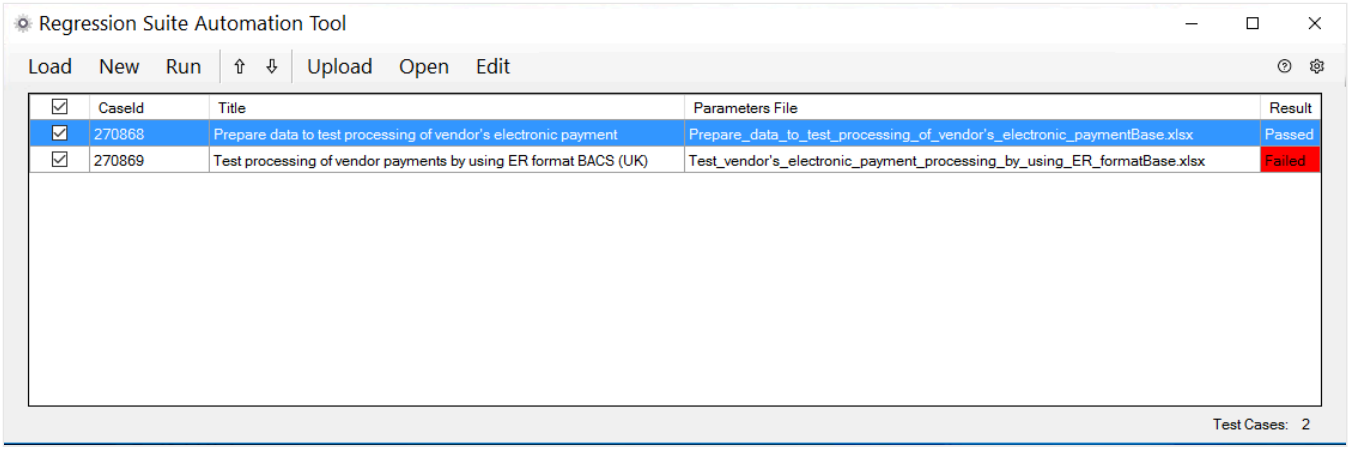
Run the tests in RSAT

1. In RSAT, select the loaded tests.
2. Select **Run**.

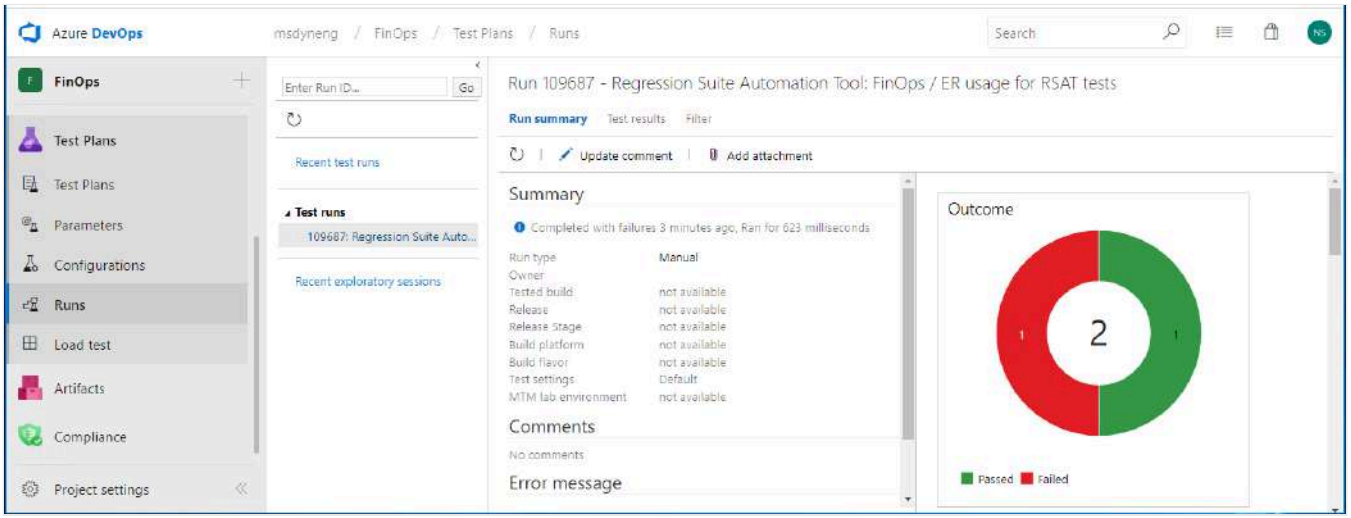
Test cases automatically run in the application by using a web browser.

Analyze the results of test execution

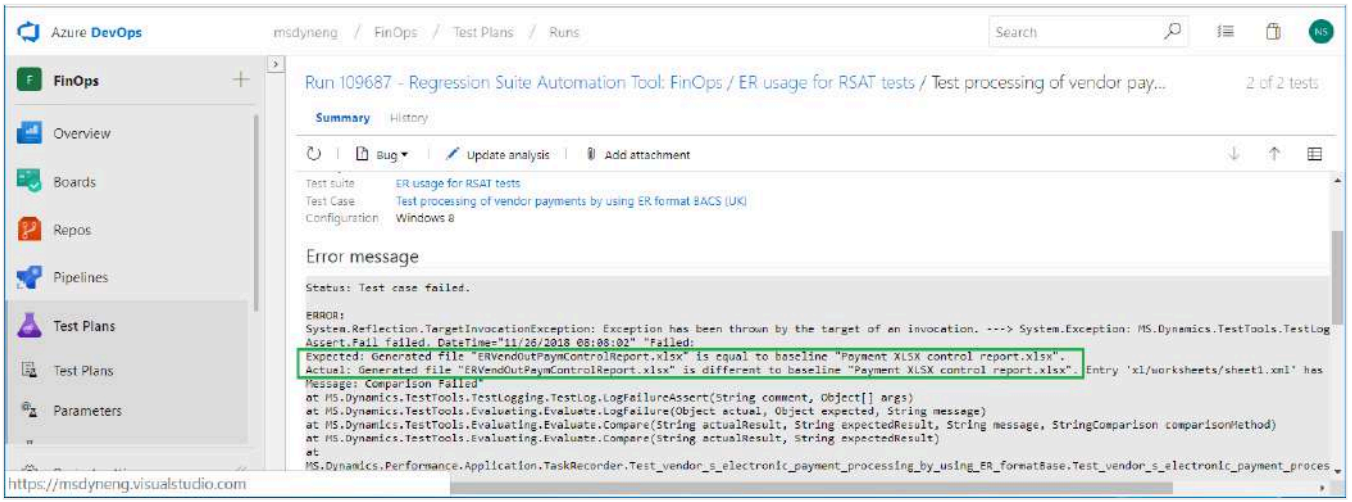
RSAT stores the results of the test execution. Notice that the second test failed during the second execution.



The test execution results are also sent to Azure DevOps so that you can do further analysis.



You can access the status of each test. You can also access the execution log so that you can analyze the reasons for any failure. In the following illustration, the execution log shows that the failure occurred because of the difference in content between the generated payment file and its baseline.



Therefore, as you saw, you can automatically evaluate the functioning of any ER format by using RSAT as the testing platform and by using Task recorder-based test cases that use the ER baseline feature.

Additional resources

- Task recorder resources
- Regression suite automation tool
- Create and automate user acceptance tests
- Deploy and use an environment that supports continuous build and test automation
- Trace generated report results and compare them with baseline values
- ER Upgrade your format by adopting a new, base version of that format
- ER Import a configuration from Lifecycle Services

Last updated on 03/17/2026

Configure Electronic reporting (ER) to pull data into Power BI

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to use your Electronic reporting (ER) configuration to arrange the transfer of data from your instance to Power BI services. As an example, this article uses Intrastat transactions as business data that you must transfer. The Power BI map visualization uses this Intrastat transaction data to present a view for analysis of company import and export activities on the Power BI report.

Overview

Microsoft Power BI is a collection of software services, apps, and connectors that work together to turn external sources of data into coherent, visually immersive, and interactive insights. Electronic reporting (ER) lets users easily configure data sources and arrange the transfer of data from the application to Power BI. The data is transferred as files in the OpenXML worksheet (Microsoft Excel workbook file) format. You store the transferred files on a Microsoft SharePoint Server that you configure for this purpose. Use the stored files in Power BI to make reports that include visualizations (tables, charts, maps, and so on). Share Power BI reports with Power BI users, and access them in Power BI dashboards and on the application pages. This article explains the following tasks:

- Configure Microsoft Dynamics 365 Finance.
- Prepare your ER format configuration to get data from the Finance application.
- Configure the ER environment to transfer data to Power BI.
- Use transferred data to create a Power BI report.
- Make the Power BI report accessible in Finance.

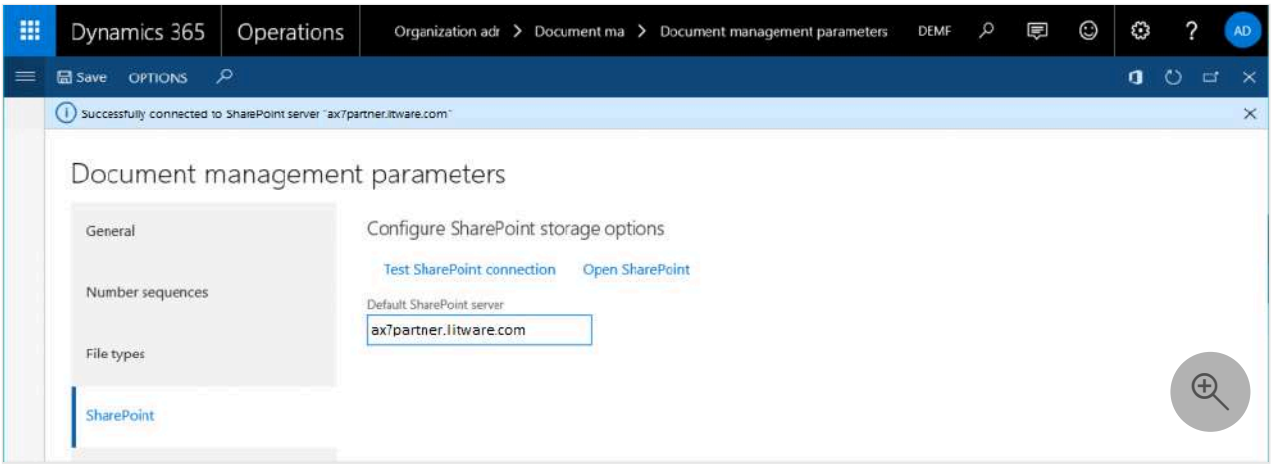
Prerequisites

To complete the example in this article, you must have the following access:

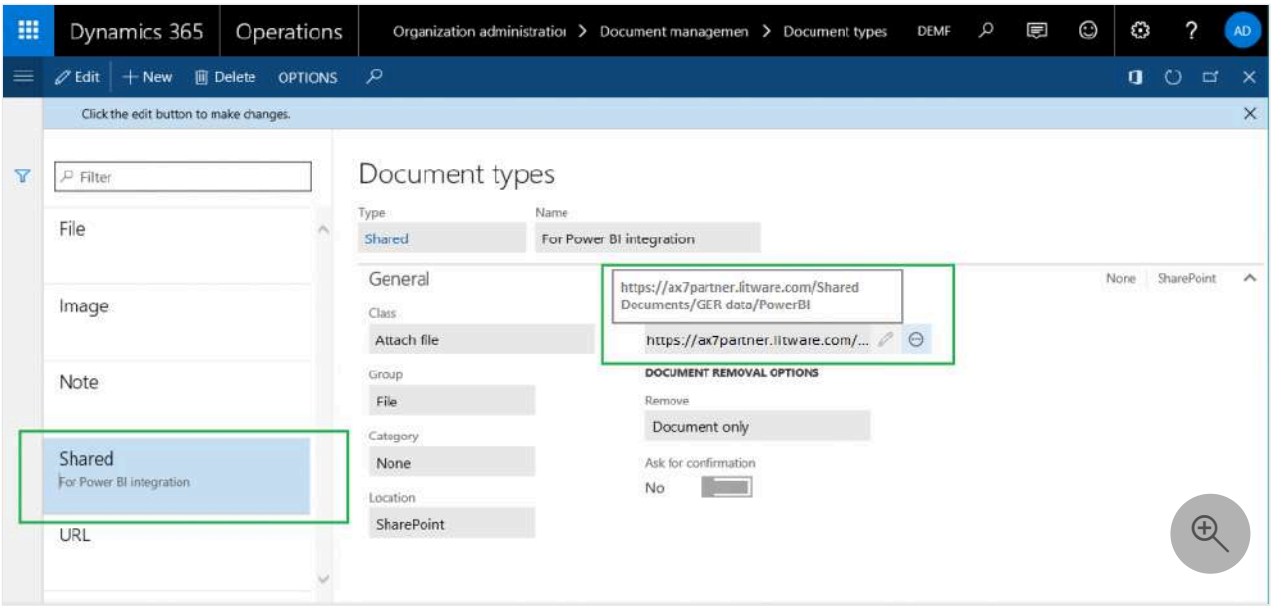
- Access for one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- Access to the SharePoint Server that you configure for use with the application
- Access to the Power BI framework

Configure document management parameters

1. On **Document management parameters**, configure access to the SharePoint Server that the company uses and that you sign in to (the DEMF company in this example).
2. Test the connection to the SharePoint Server to make sure that you have access.



3. Open the configured SharePoint site. Create a new folder where ER stores Excel files that have the business data that the Power BI reports require as a source of Power BI datasets.
4. On **Document types**, create a new document type that you use to access the SharePoint folder that you just created. Enter **File** in the **Group** field and **SharePoint** in the **Location** field, and then enter the address of the SharePoint folder.

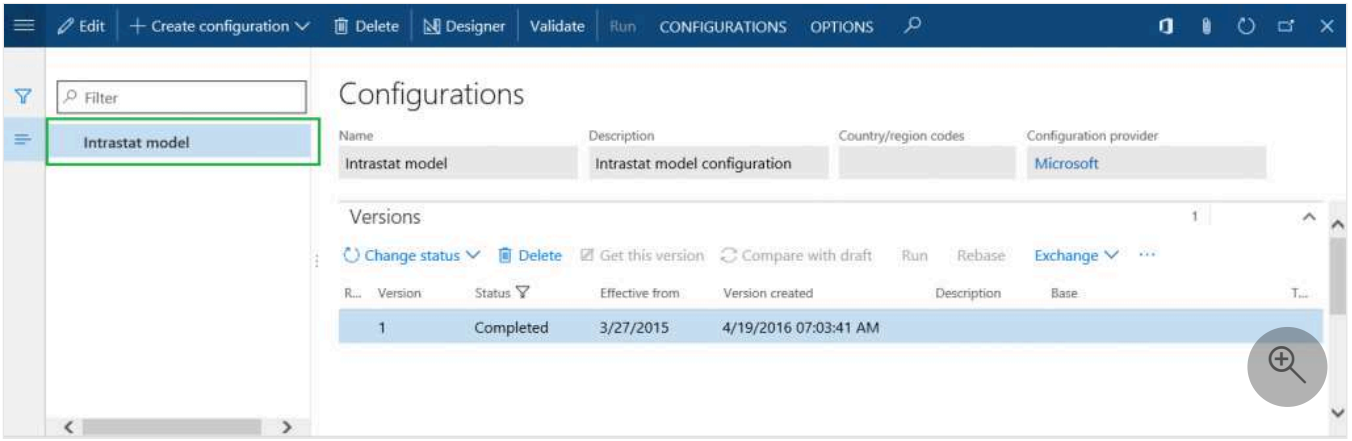


Configure ER parameters

1. In the **Electronic reporting** workspace, select the **Electronic reporting parameters** link.
2. On the **Attachments** tab, select the **File** document type for all the fields.
3. In the **Electronic reporting** workspace, make the required provider active by selecting **Set active**. For more information, see the **ER Select service provider** task guide.

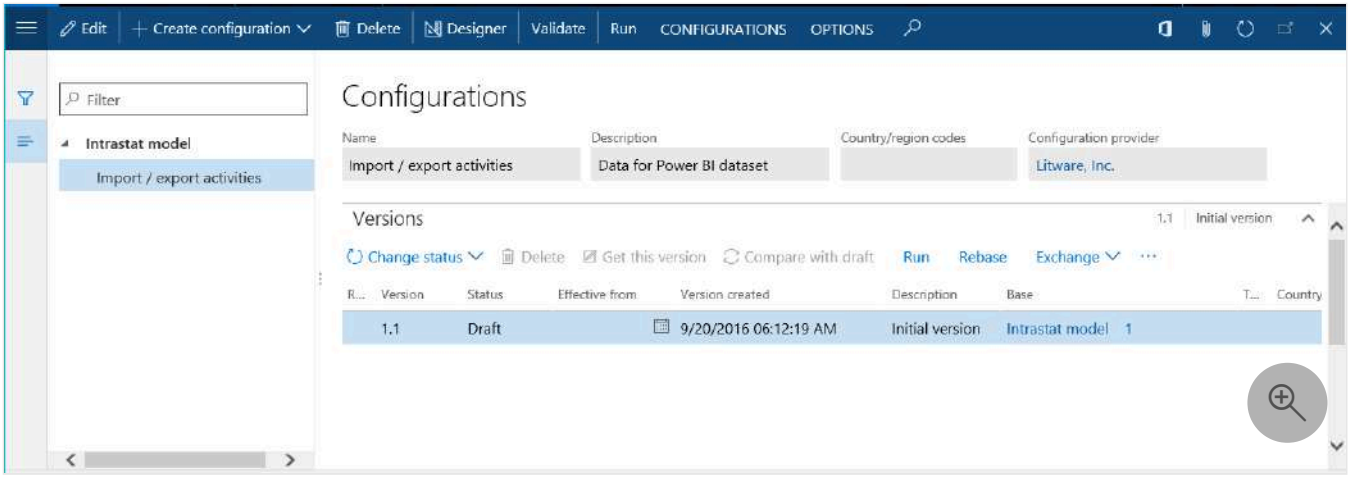
Use an ER data model as the source of data

You must have an ER data model as the source of business data that Power BI reports use. You upload this data model from the ER configurations repository. For more information, see Download Electronic reporting configurations from Lifecycle Services or see the **ER Import a configuration from Lifecycle Services** task guide. Select **Intrastat** as the data model that you upload from the selected ER configurations repository. (In this example, version 1 of the model is used.) You can then access the **Intrastat** ER model configuration on the **Configurations** page.



Design an ER format configuration

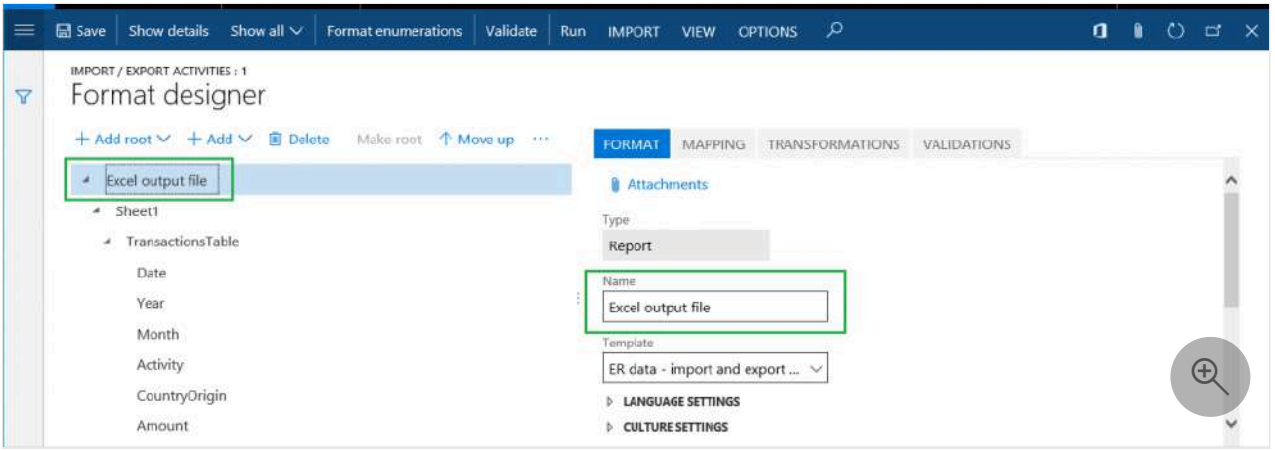
Create a new ER format configuration that uses the **Intrastat** data model as the source of business data. This format configuration generates output results as electronic documents in OpenXML (Excel file) format. For more information, see the **ER Create a configuration for reports in OPENXML format** task guide. Name the new configuration **Import / export activities**, as shown in the following illustration. Use the ER data - import and export details Excel file as a template when you design the ER format. For information about how to import a format template, see the task guide.



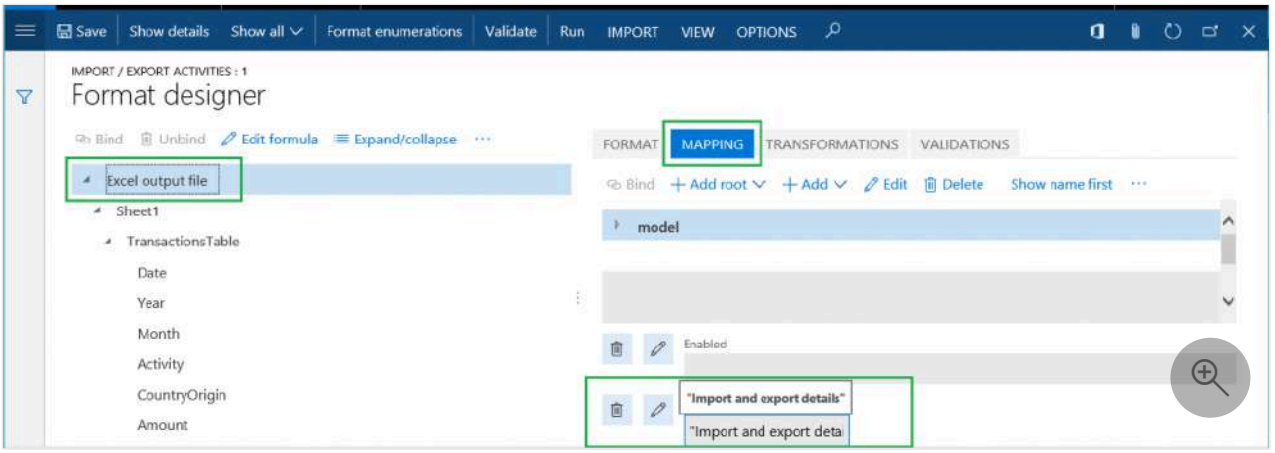
To modify the **Import / export activities** format configuration, follow these steps:

1. Select **Designer**.

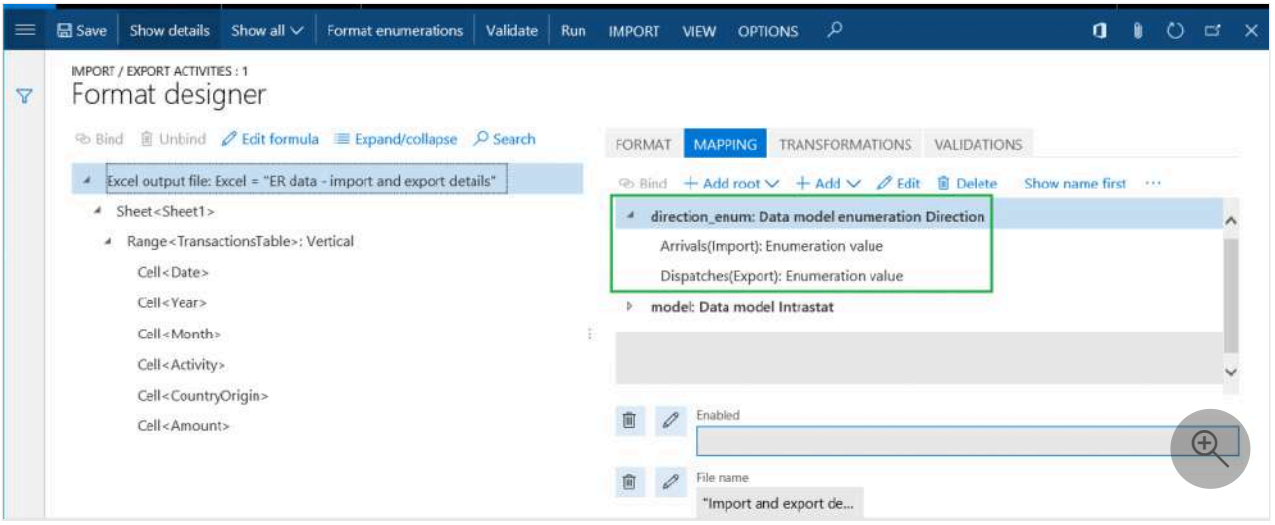
2. On the **Format** tab, name the file element for this format **Excel output file**.



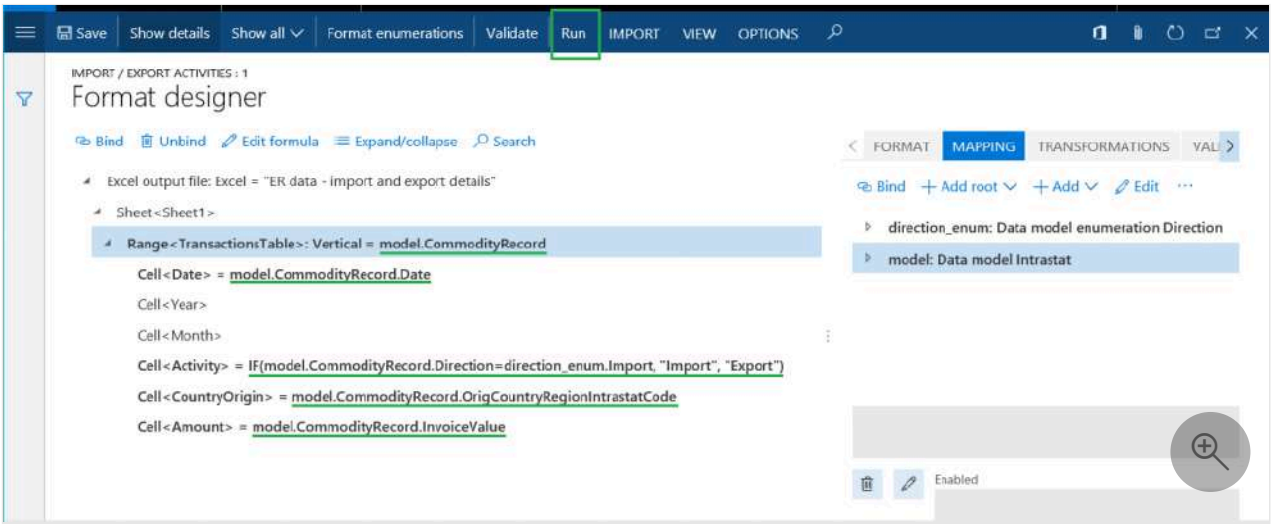
3. On the **Mapping** tab, specify the name of the Excel file that this format generates whenever it runs. Configure the related expression to return the value **Import and export details**. The .xlsx file name extension is added automatically.



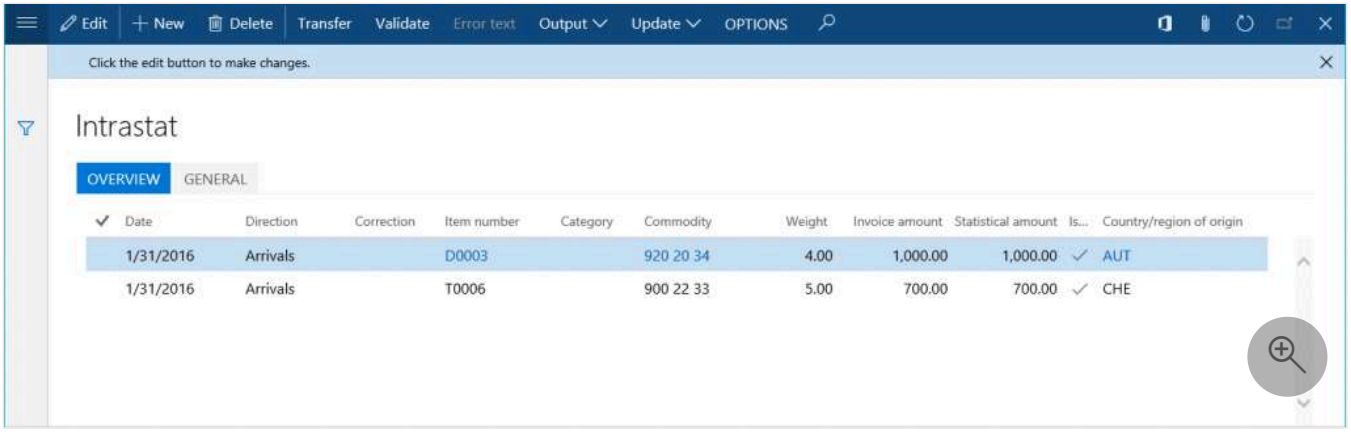
4. Add a new data source item for this format. This enumeration is required for further data binding.
   a. Name the data source **direction_enum**.
   b. Select **Data model enumeration** as the data source type.
   c. Refer to the **Direction** data model enumeration.



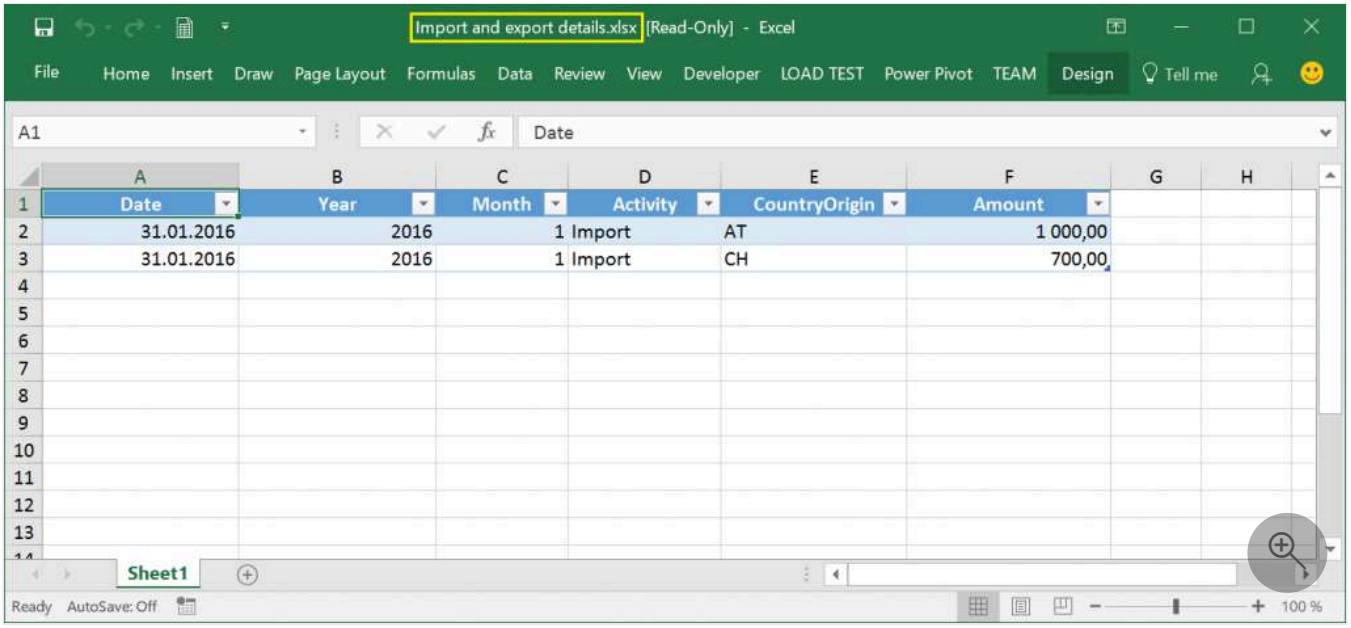
5. Bind elements of the **Intrastat** data model to elements of the designed format, as shown in the following illustration.



After you run it, the ER format generates the output result in Excel format. It sends the details of the Intrastat transactions to the output result, and separates them as transactions that describe either import activities or export activities. Select **Run** to test the new ER format for the list of Intrastat transactions on the **Intrastat** page (**Tax** > **Declarations** > **Foreign trade** > **Intrastat**).



The following output result is generated. The file is named **Import and export details.xlsx**, as you specified in the format settings.



Configure the ER destination

You must configure the ER framework to send the output result of the new ER format configuration in a special way.

- The output result must go to the folder of the selected SharePoint Server.
- Each execution of the format configuration must create a new version of the same Excel file.

On **Electronic reporting** (**Organization administration** > **Electronic reporting**), select **Electronic reporting destination**, and add a new destination. In the **Reference** field, select the **Import / export activities** format configuration that you created earlier. Follow these steps to add a new file destination record for the reference.

1. In the **Name** field, enter the title of the file destination.
2. In the **File name** field, select the name **Excel output file** for the Excel file format component.

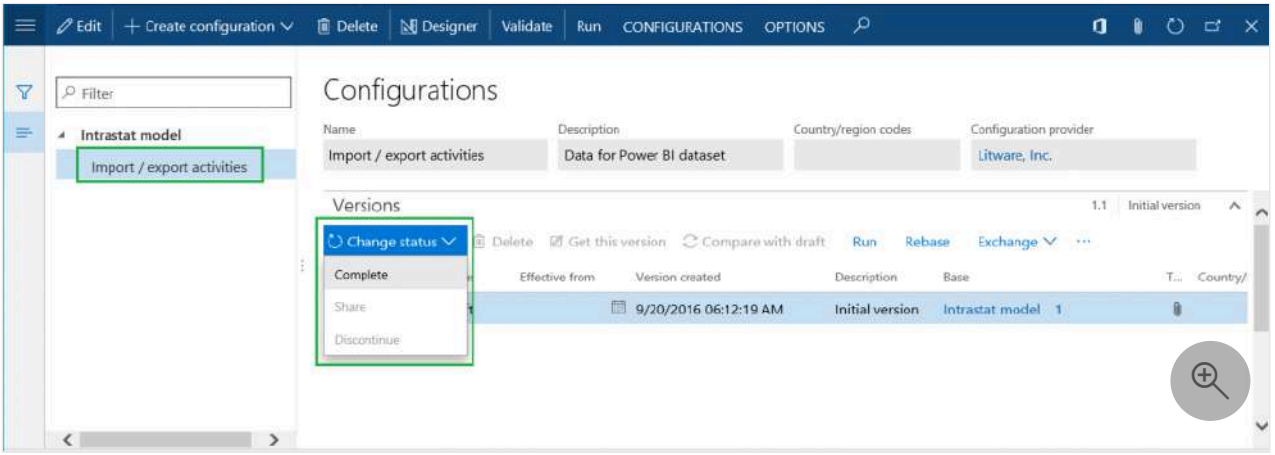
Select the **Settings** button for the new destination record. Then, in the **Destination settings** dialog box, follow these steps:

1. On the **Power BI** tab, set the **Enabled** option to **Yes**.
2. In the **SharePoint** field, select the **Shared** document type that you created earlier.

Schedule execution of the configured ER format

1. On **Configurations** (**Organization administration** > **Electronic reporting** > **Configurations**), in the configurations tree, select the **Import / export activities** configuration that you created earlier.

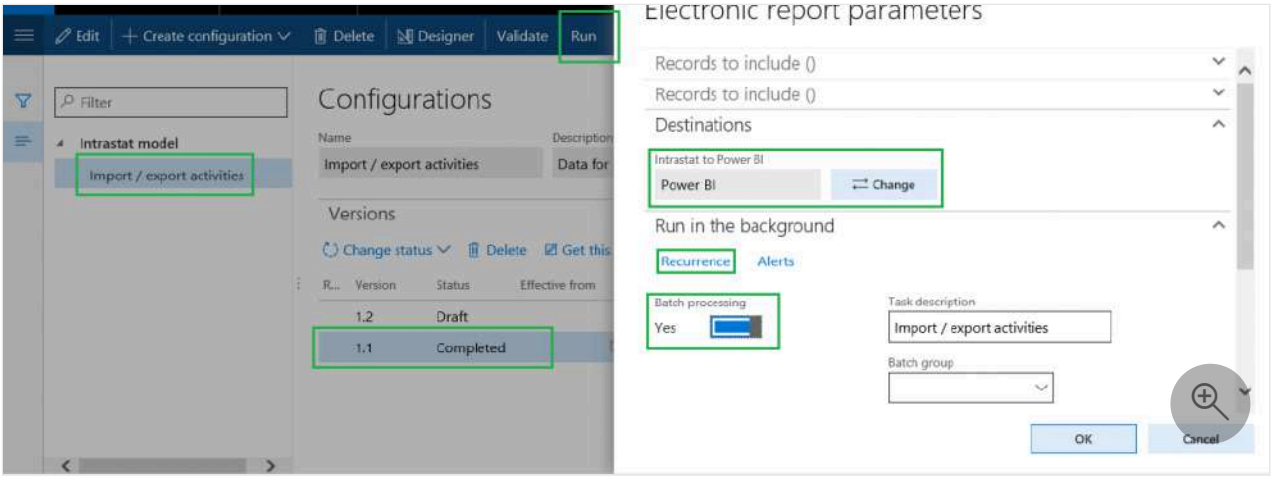
2. Change the status of version 1.1 from **Draft** to **Complete** to make this format available for use.



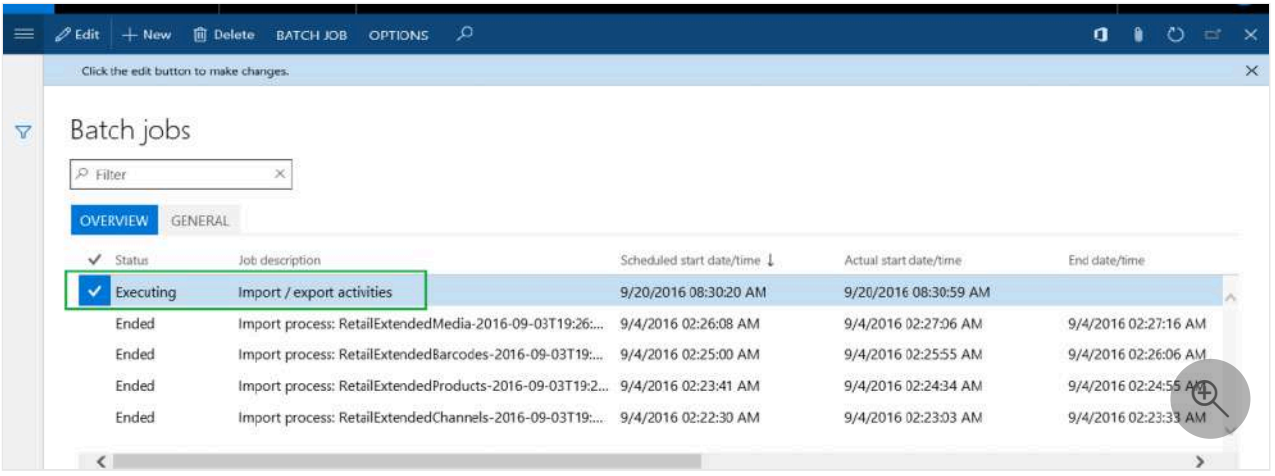
3. Select the completed version of the **Import / export activities** configuration, and then select **Run**. The configured destination is applied to the output result that is generated in Excel format.

4. Set the **Batch processing** option to **Yes** to run this report in unattended mode.

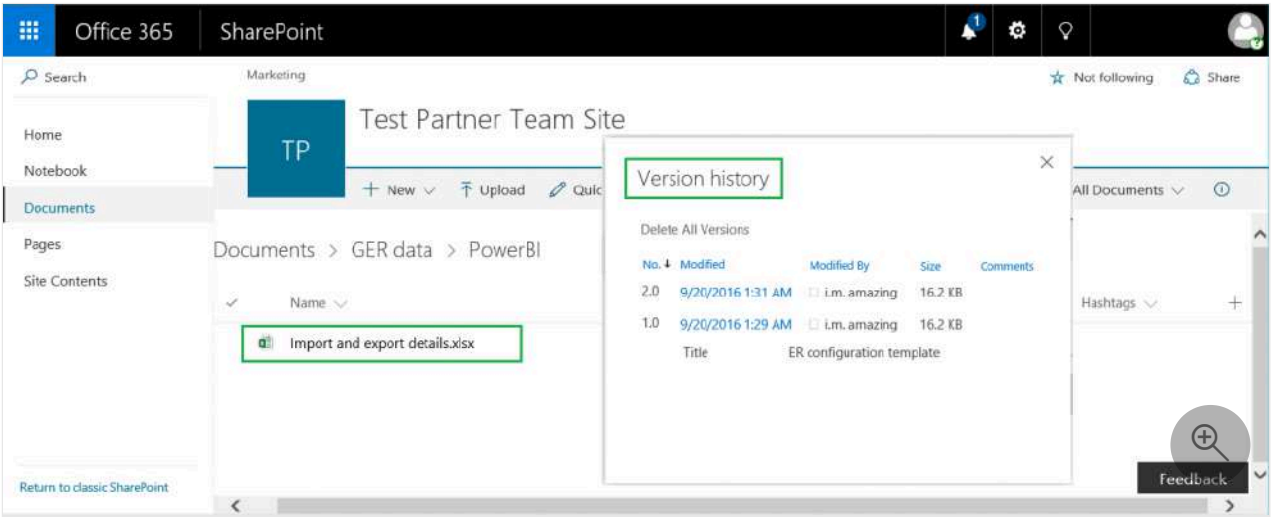
5. Select **Recurrence** to schedule the required recurrence of this batch execution. The recurrence defines how often the updated data is transferred to Power BI.



6. After you configure it, you can find the ER report execution job on the **Batch jobs** page (**System administration** > **Inquiries** > **Batch jobs**).

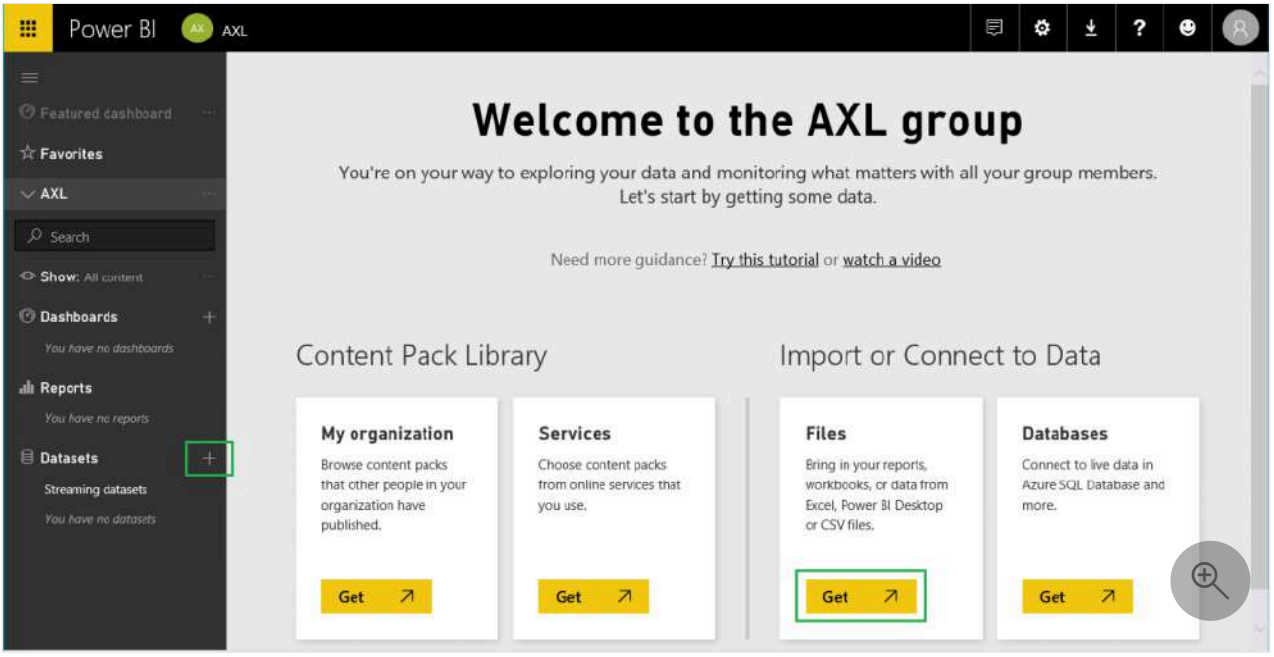


7. When you run this job for the first time, the destination creates a new Excel file that has the configured name in the selected SharePoint folder. Every subsequent time that you run the job, the destination creates a new version of this Excel file.



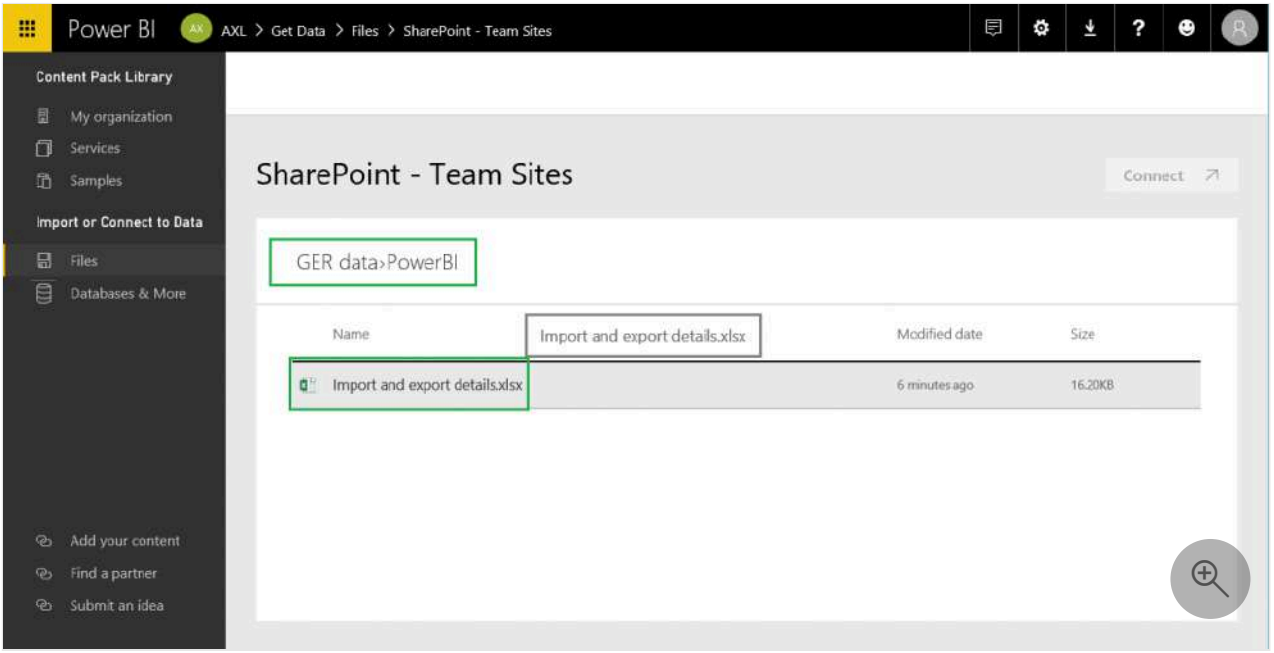
Create a Power BI dataset by using the output result of the ER format

1. Sign in to Power BI, and either open an existing Power BI group (workspace) or create a new group. Either click **Add** under **Files** in the **Import or Connect to Data** section, or click the plus sign (+) next to **Datasets** in the left pane.

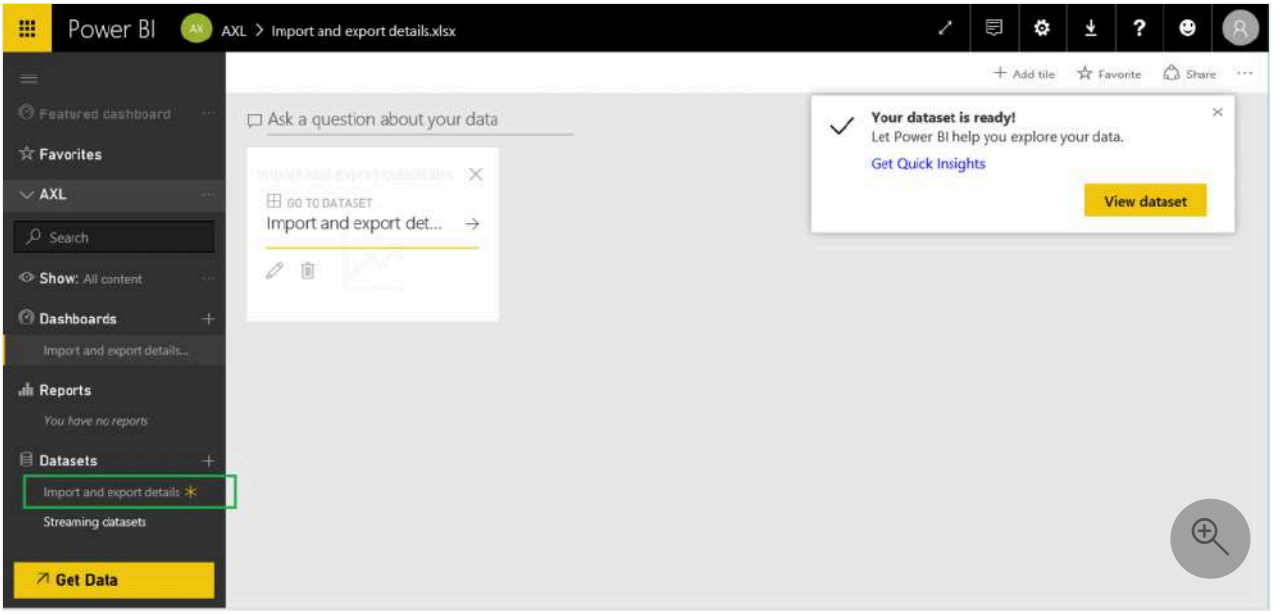


2. Select the **SharePoint – Team sites** option, and then enter the path of SharePoint Server that you're using (`https://ax7partner.litware.com` in our example).

3. Browse to the **/Shared Documents/GER data/PowerBI** folder, and select the Excel file that you created as the source of data for the new Power BI dataset.



4. Click **Connect**, and then click **Import**. A new dataset is created that is based on the selected Excel file. The dataset can also be added automatically to the newly created dashboard.



5. Configure the refresh schedule for this dataset to force a periodic update. Periodic updates enable the consumption of new business data that comes via periodic execution of the ER report through new versions of the Excel file that are created on the SharePoint Server.

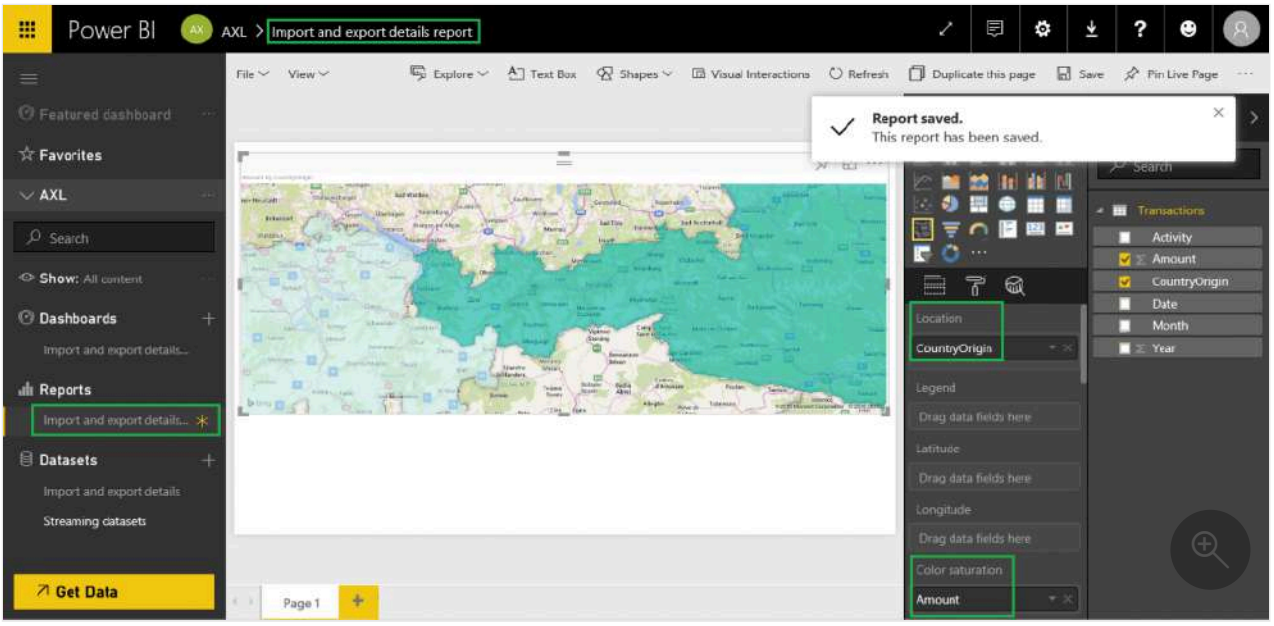
Create a Power BI report by using the new dataset

1. Select the **Import and export details** Power BI dataset that you created.

2. Configure the visualization. For example, select the **Filled map** visualization, and configure it as follows:

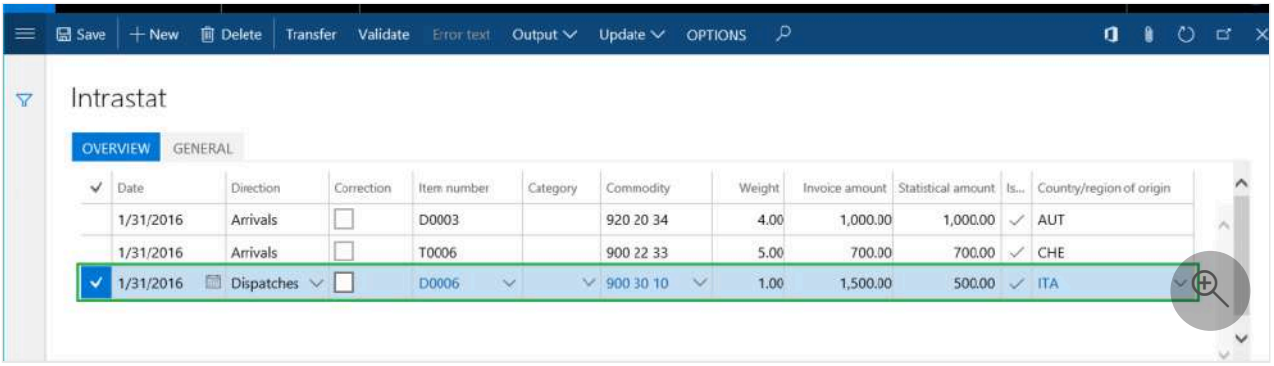
    - Assign the **CountryOrigin** dataset field to the **Location** field of the map visualization.
    - Assign the **Amount** dataset field to the **Color saturation** field of the map visualization.
    - Add the **Activity** and **Year** dataset fields to the **Filters** fields collection of the map visualization.

3. Save the Power BI report as **Import and export details report**.

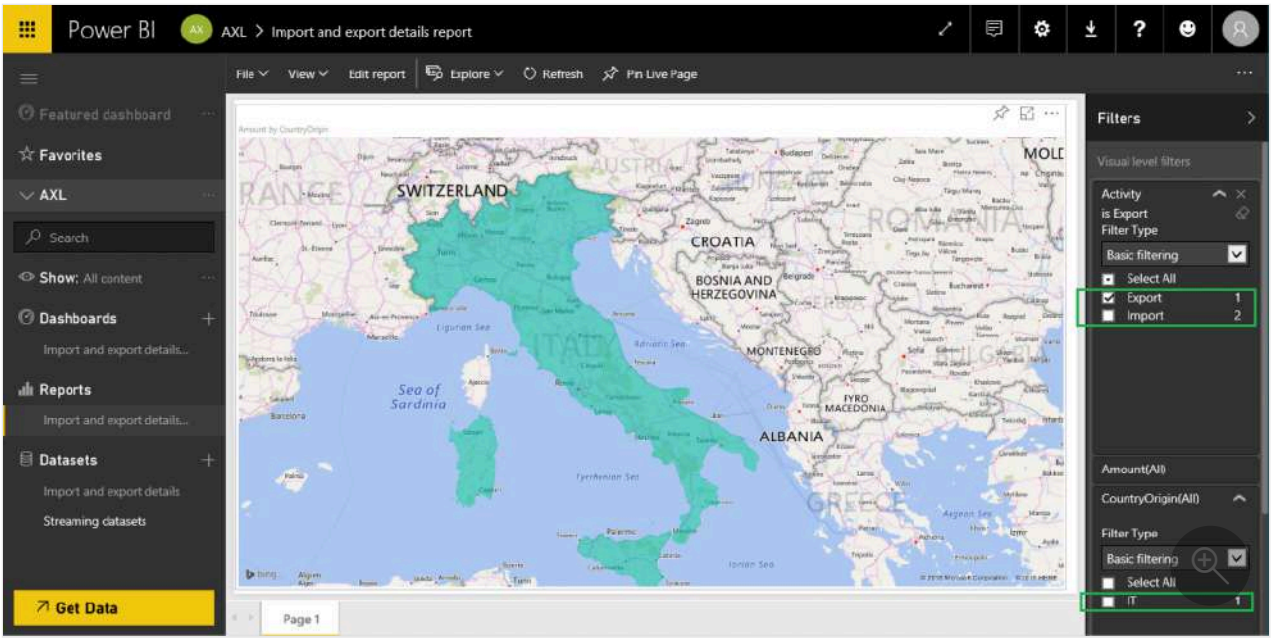


The map shows the countries and regions that are mentioned in the Excel file (Austria and Switzerland in this example). These countries and regions are colored to show the proportion of invoiced amounts for each.

4. Update the list of Intrastat transactions. The export transaction that originated from Italy is added.



5. Wait for the next scheduled execution of the ER report and the next scheduled update of the Power BI dataset. Then review the Power BI report (select to show import transactions only). The updated map now shows Italy.



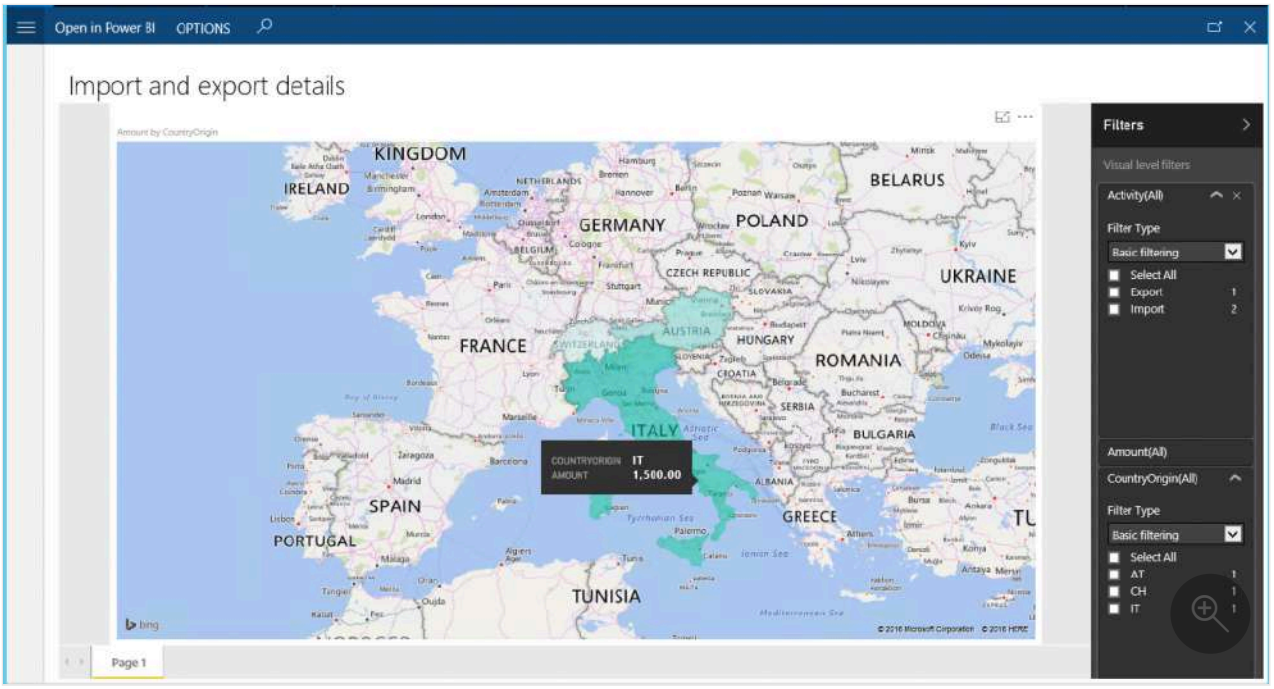
Access Power BI report in Finance

Set up the integration with Power BI. For more information, see Configure Power BI integration for workspaces.

1. On the **Electronic reporting** workspace page that supports Power BI integration (**Organization administration** > **Workspaces** > **Electronic reporting workspace**), select **Options** > **Open report catalog**.

2. Select the **Import and export details** Power BI report that you created, to show that report as an action item on the selected page.

3. Select the action item to open the page that shows the report that you designed in Power BI.



Additional resources

Electronic reporting (ER) destinations

Electronic reporting (ER) overview

Last updated on 04/13/2026

Generate printable FTI forms

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Electronic reporting (ER) framework lets you generate printable free text invoice (FTI) forms as Microsoft Office documents. This article provides information about how to build your own configurations as well as details of available configuration templates.

Overview

In addition to the existing capability of generating printable FTI forms by using Microsoft SQL Server Reporting Services (SSRS), you can now use the ER framework. You can manage printable FTI forms in Microsoft Office Excel and Word. You can also modify the layout, data flow, and formatting to meet specific requirements without making code changes.

> **Note**
>
> If you want to start with an overview of existing ER configurations for this sample of the printable FTI forms solution, you can go directly to section **Download sample ER configurations to generate printable FTI forms** later in this article.

Create customized configurations for FTI printable forms

As part of your customized solution for printable FTI forms, you must create a set of ER configurations.

Configure the ER data model

Your application must include the ER data model configuration that contains a data model that describes the customer invoicing business domain. As a requirement, the name of the data

model must be **CustomersInvoicing**. For information about how to design ER data models, see ER Design domain specific data model.

Configure the ER model mapping

Your application must include the ER model mapping for the CustomersInvoicing data model. You can add the model mapping to either the ER data model configuration or the ER model mapping configuration. However, the name of the root descriptor of the model mapping must be **FreeTextInvoice**.

The mapping must contain the following data sources:

- Data source type: **Table records**
  - This data source must be named **CustInvoiceJour**.
  - It must refer to the CustInvoiceJour application table.
  - At runtime, it passes from the application to the ER model mapping the list of invoices that you select for printing.
- Data source type: **Object**
  - This data source must be named **PrintMgmtPrintSettingDetail**.
  - It must refer to the **PrintMgmtPrintSettingDetail** application class.
  - At runtime, it passes from the application to the ER model mapping details of the print management settings for the ER format that is running.

You can find the details of the application integration with the ER framework in the **ERPrintMgmtReportFormatSubscriber** class (ER Application Suite integration model) in the source code of the application.

For more information about the design of ER model mappings, see Define ER model mappings and select data sources for them.

Configure the ER format

In your application instance, you must have the ER format configuration that you use to generate FTI forms.

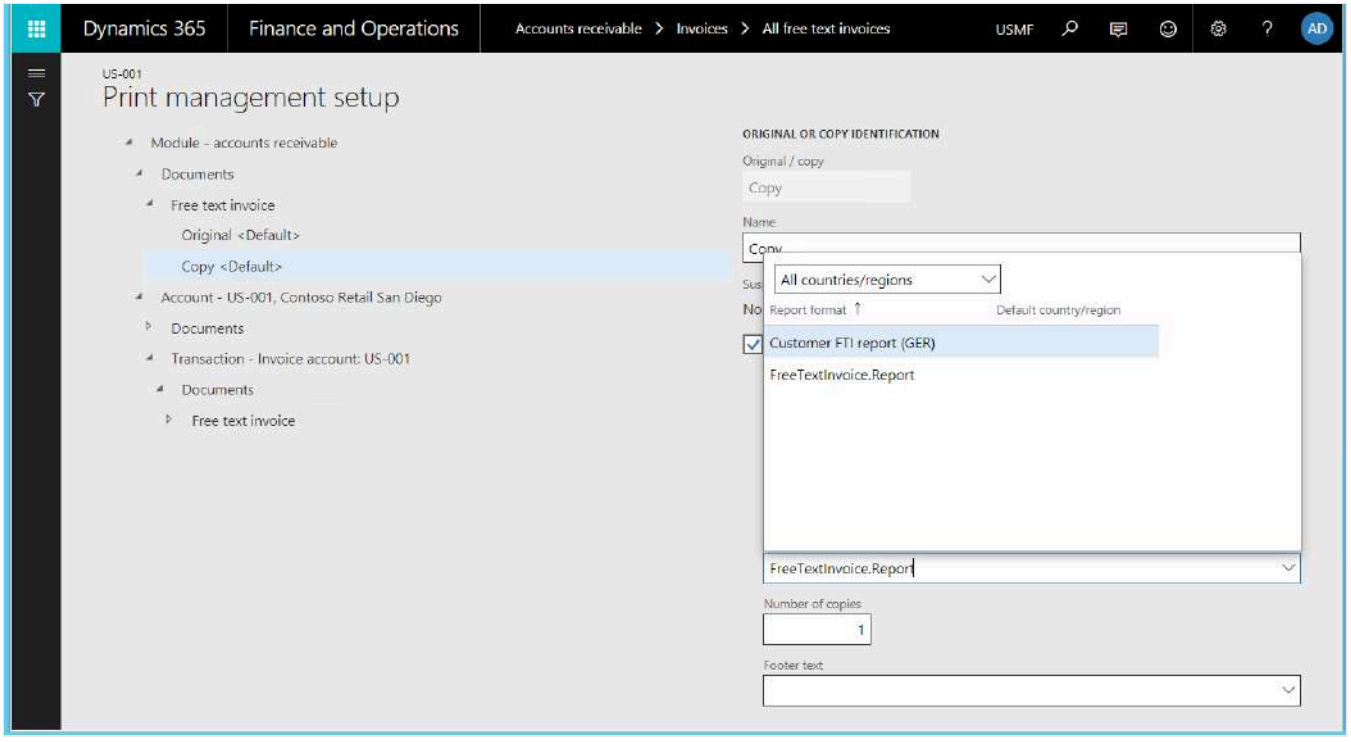
> **Note**
>
> Create this format configuration for the **CustomersInvoicing** data model. Use the model mapping that has the **FreeTextInvoice** root descriptor.

For information about how to configure ER formats, see ER Create a format configuration (November 2016). For information about how to design ER formats to generate reports in OpenXML format, see ER Design a configuration for generating reports in OPENXML format (November 2016).

Configure print management

To generate FTI forms by using the ER framework, assign ER formats in the same way that you assign SSRS reports. To associate the ER format with all Accounts receivable FTIs, go to **Accounts receivable** > **Setup** > **Forms** > **Form setup** > **General** > **Print management** > **Free text invoice** > **Original**. To associate the ER format with a specific customer or invoice, follow these steps:

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. Select the FTI to associate the ER format with, and open the **Print management setup** page.
3. Select the document level to specify the scope of invoices for processing.
4. Select the ER format for the specified document level.

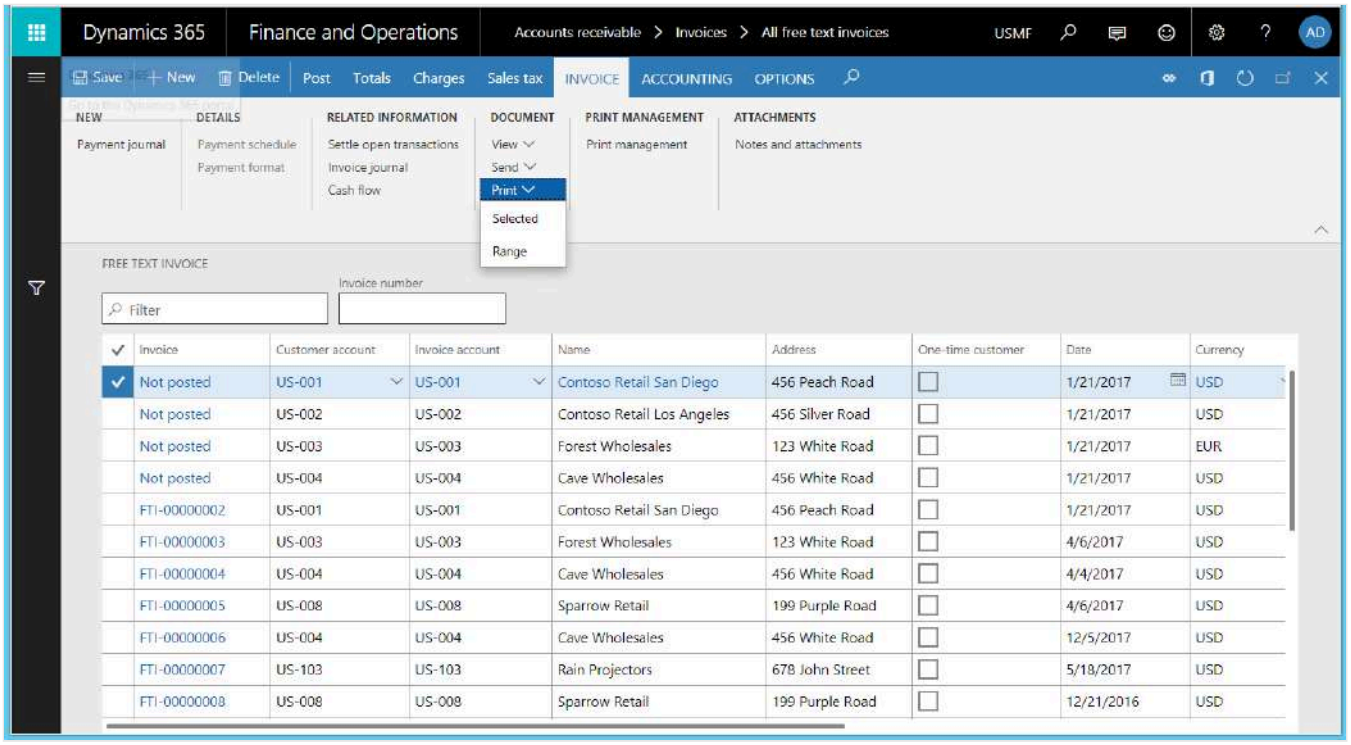


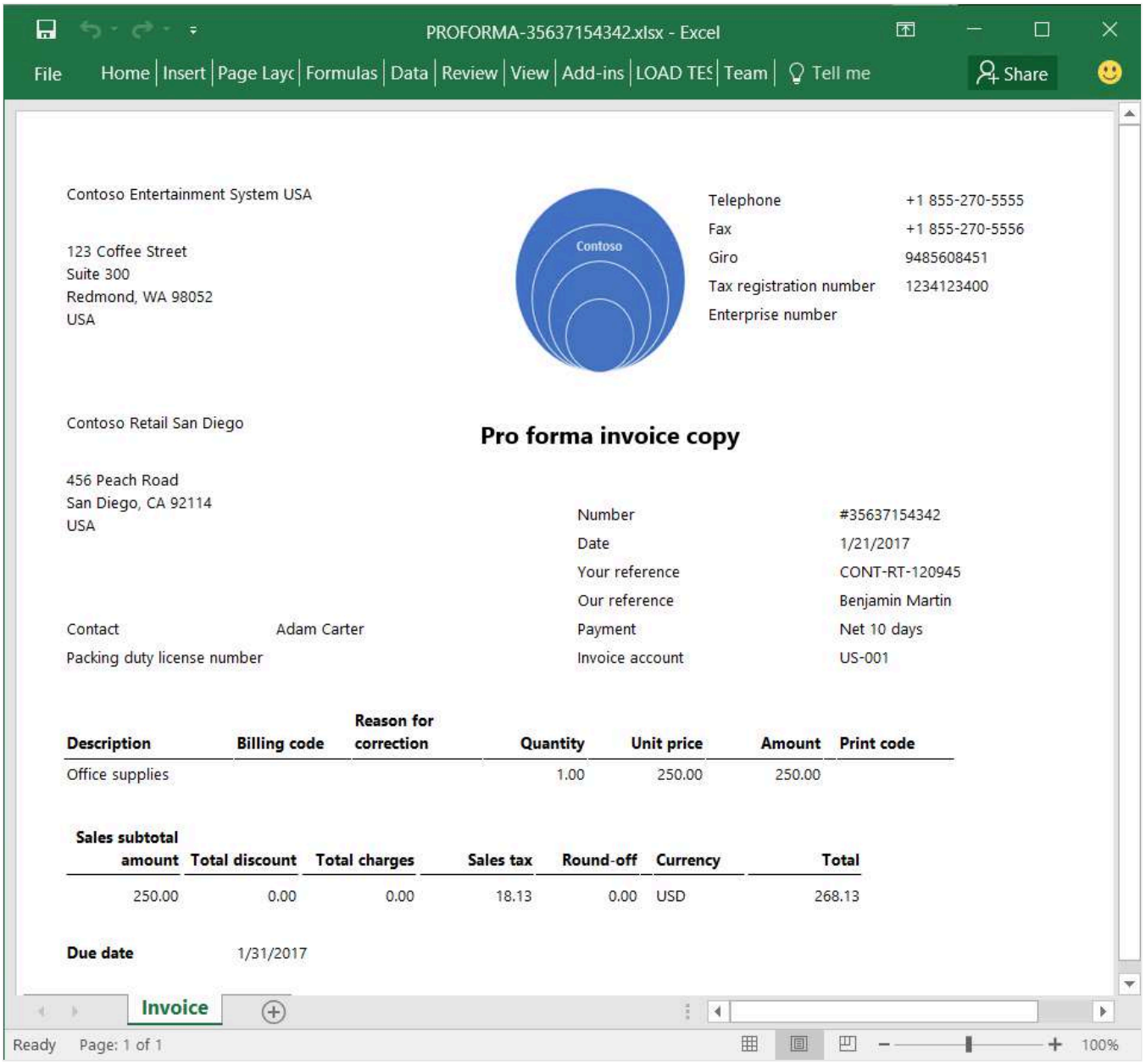
> **Note**
>
> Only ER formats that use the **FreeTextInvoice** root descriptor of the **CustomersInvoicing** data model appear in the **Report format lookup** field for the selected format.

Generate FTI forms

The ER framework generates FTI forms the same way it generates SSRS reports.

To generate FTI forms, select invoices by range or by selection.



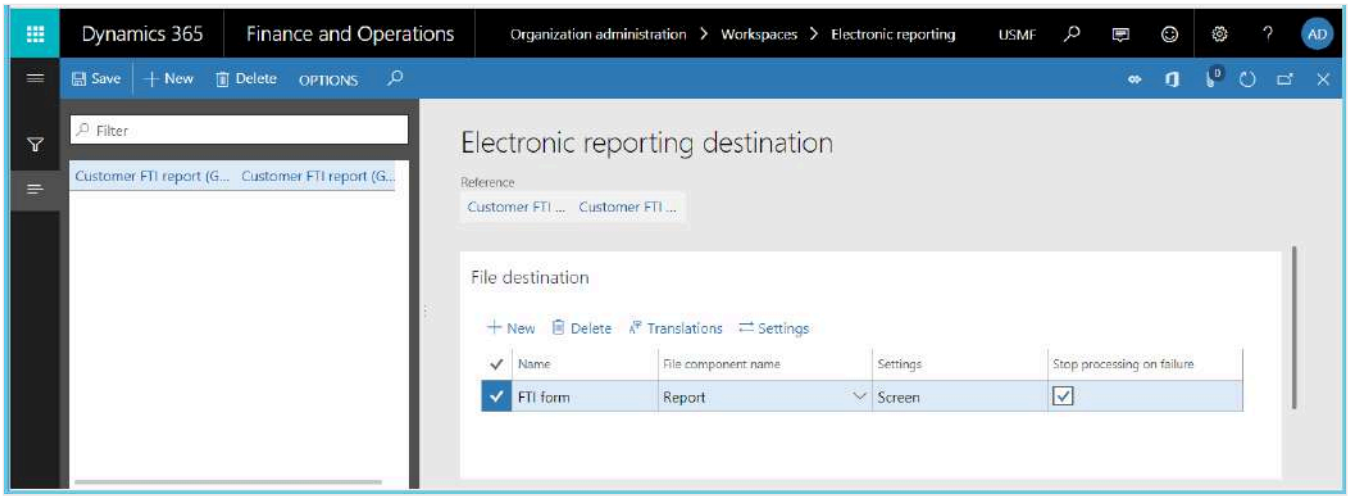


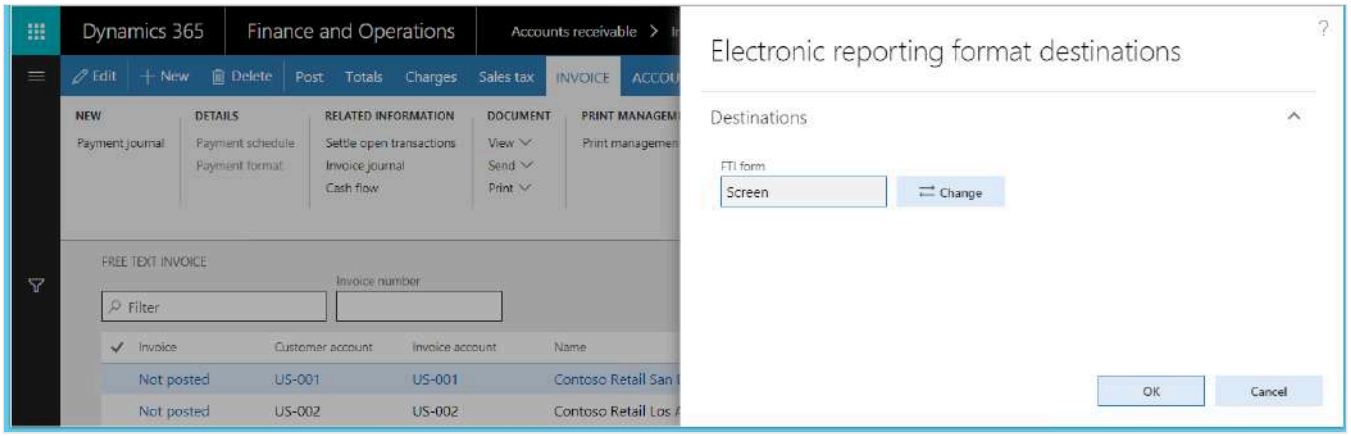
When you use ER formats to print FTI forms, the default ER file destinations are used. You can't change the destination. For more information about how to configure the ER destinations for ER formats, see Electronic reporting (ER) destinations.

You can also generate FTI forms when you post an FTI, by turning **Print invoice** on and turning **Use print management destinations** off.

> ⓘ Note
>
> When you use ER formats to print FTI forms, the default ER file destinations are used. You can change the default destination at runtime if the destination is already configured. To change the destination, you must have the following security privilege:
>
> - **Name:** ERFormatDestinationRuntimeMaintain

- **Label:** Maintain electronic reporting format destination during runtime





The ER framework currently supports the following destinations for generated documents:

- **Downloaded file** – You can save generated forms by downloading them through the browser.
- **Screen** – Microsoft 365 Excel previews generated FTI forms in Excel format.
- **SharePoint folder** – The Document management framework settings store generated forms.
- **Application archive** – Microsoft Azure Storage stores generated forms as attachments of execution log records.
- **Email** – Generated forms are sent as email attachments.

> ⓘ **Note**
>
> You can't send the generated FTI forms directly to the printer, because direct printing that uses the Dynamics Printer Routing Agent isn't currently supported.

Download sample ER configurations to generate printable FTI forms

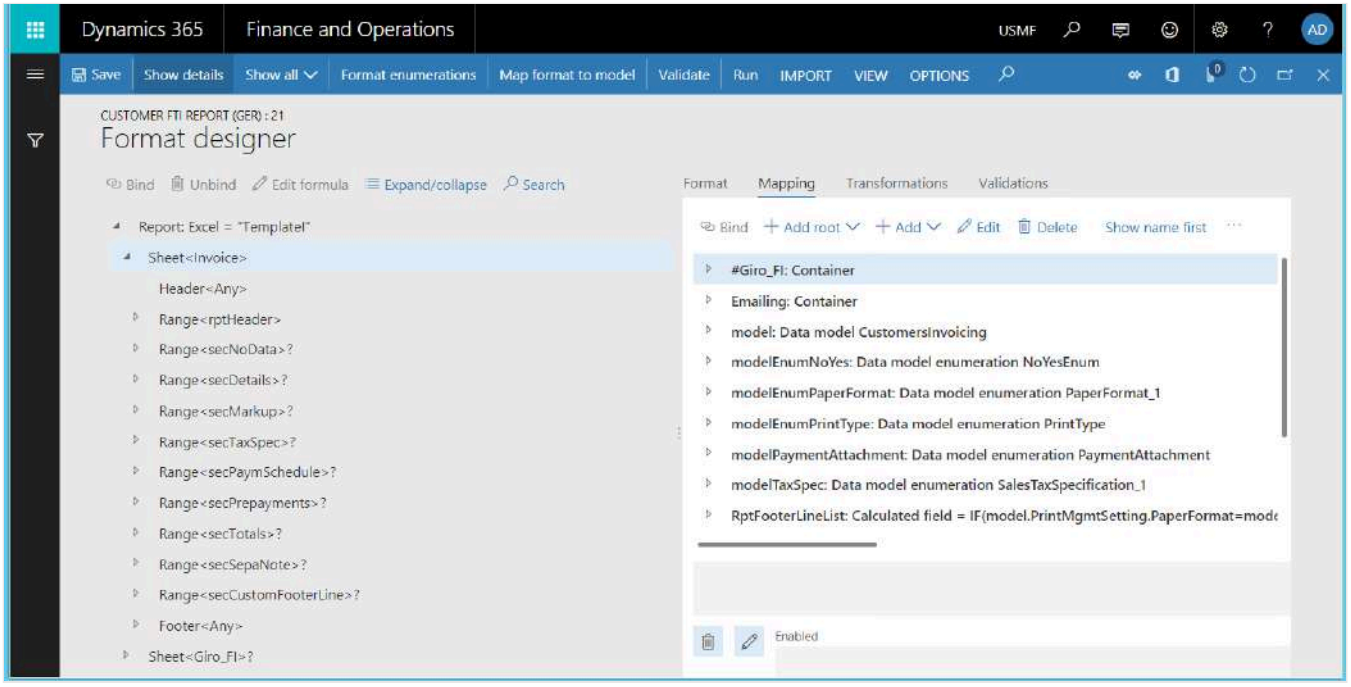
You can download sample ER configurations to use as a template for your FTI solution. Microsoft Dynamics Lifecycle Services stores these configurations in the Shared asset library. The configurations include:

- The **Customer invoicing model** configuration contains the required data model and model mapping.
- The **Customer FTI report (GER)** configuration contains the sample format.

> **Note**
>
> These configurations are samples to help clarify possible scenarios. The future of these configurations depends on the results of this evaluation and any feedback that you provide.

Features implemented in the sample ER format

In the sample ER format configuration, an Excel file is used as a template to generate FTI forms.



Currently, this sample ER format supports the following features to generate FTI forms:

- The system generates FTI forms for both original invoices that you posted and original invoices that you didn't post yet. Corrected invoices and credit notes aren't supported.

- The system generates FTI forms in the invoice language. The format of values and dates in the generated forms is based on the settings of the user's client locale.
- Generated invoices show data unavailability notifications if there are no lines in the invoices that are processed.
- Generated invoice headers are based on the paper format that you select for the FTI form on the **Accounts receivable parameters** page. Company details appear in the header of the generated invoice form only if the paper format is blank.
- Generated invoice forms show company and customer tax exempt numbers when you select the appropriate option for the FTI form on the **Accounts receivable parameters** page.
- The generated invoice lines and invoice totals sections show the default invoice's monetary details in the invoice registration currency.
- The generated invoice totals section can show monetary details in the euro currency and the invoice registration currency when you enable the **Print amount in currency representing the euro** option on the **Accounts receivable parameters** page.
- Generated invoice forms show any process invoice notes that are available, based on settings on the **Accounts receivable parameters** page. Notes are included for both the whole invoice and each invoice line.
- Generated invoice forms include notes for the customer FTI form and the processing invoice language when you configure them in the AR form notes list.
- Depending on the Print management settings, generated invoices include custom footer text when you configure it for the invoice language, the ER format, and the FTI document scope.
- The totals section of generated invoice forms includes any cash discount information that is available.
- The payment schedule section of generated invoice forms includes any payment schedule details that are available.
- The markup section of generated invoice forms includes any charges transactions that are available.
- Generated invoice forms include sales tax details, based on the **Sales tax specification** setting on the **Accounts receivable parameters** page. This section can show tax details either in the invoice registration currency only, or in the invoice registration currency and the company accounting currency at the same time.
- Generated invoice forms show direct debit notification details. For example, they show when the method of payment that has the mandatory direct debit mandate ID was selected for the invoice, when the processing invoice was registered in the euro currency, and when the direct debit mandate ID was defined for the invoice.

- Generated invoices show any prepayment details that are available for posted invoices.
- Generated invoice forms can be sent to an invoice customer as an email attachment. You should configure the appropriate ER file destination for the ER format that you're using.

Country/region-specific features

The sample ER format includes the following country/region-specific features to show how you can handle specific requirements in ER configurations.

Norway

The generated invoice form header includes the Enterprise register term when you process the invoice for a legal entity that you configure as follows:

- Use the country/region context for Norway.
- Activate the **Print Foretaksregisteret** parameter on sales documents.

Spain

The generated invoice form header includes the **Special regime for cash accounting method** term when you process the invoice for a legal entity that you configure as follows:

- Use the country/region context for Spain.
- Enable the special regime for the cash accounting method on the invoice processing date.

When cash discount details, such as the cash discount amount and invoice line net amount, are available, present them in the invoice totals section of the generated invoice form when you process the invoice for a legal entity that you configure as follows:

- Use the country/region context for Spain.
- Turn on **Cash discount is applied in the invoice** in the invoice option (**General ledger parameters** > **Sales tax section**).

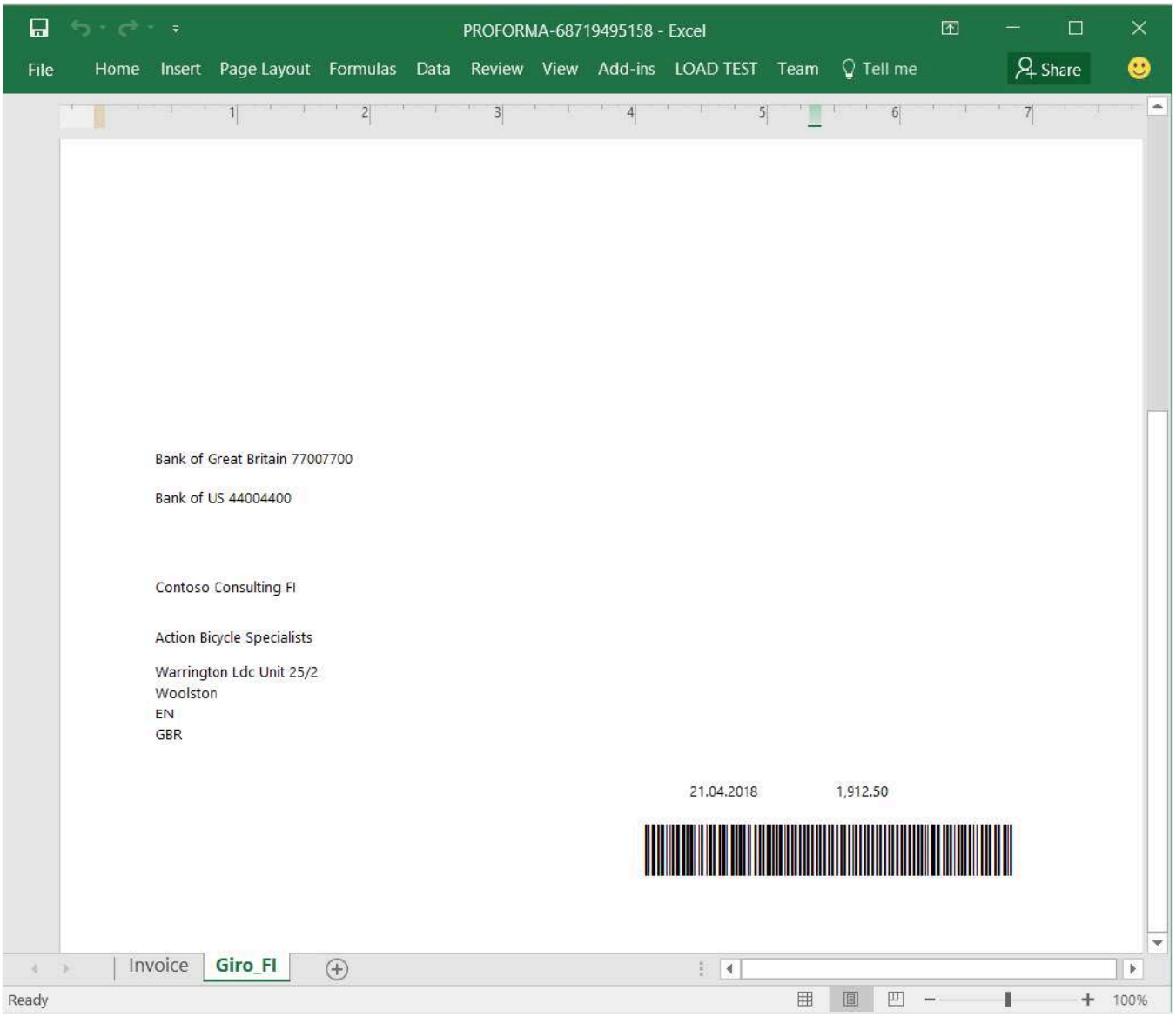
Italy

Include the goods discount mark on the invoice lines of the generated invoice when you process the invoice for a legal entity that you configure by using the country/region context for Italy.

Finland

In addition to the generated invoice form, you can generate Giro money transfer slips as follows:

- For the legal entity that uses the country/region context for Finland, and that has at least one bank account that is marked as **Giro account** and **Bank bar code**.
- For an invoice that is marked as required for the **Finnish** associated payment attachment.



> ⓘ **Note**
>
> The sample ER format is configured to optionally generate the Giro money transfer slips in the separate worksheet.

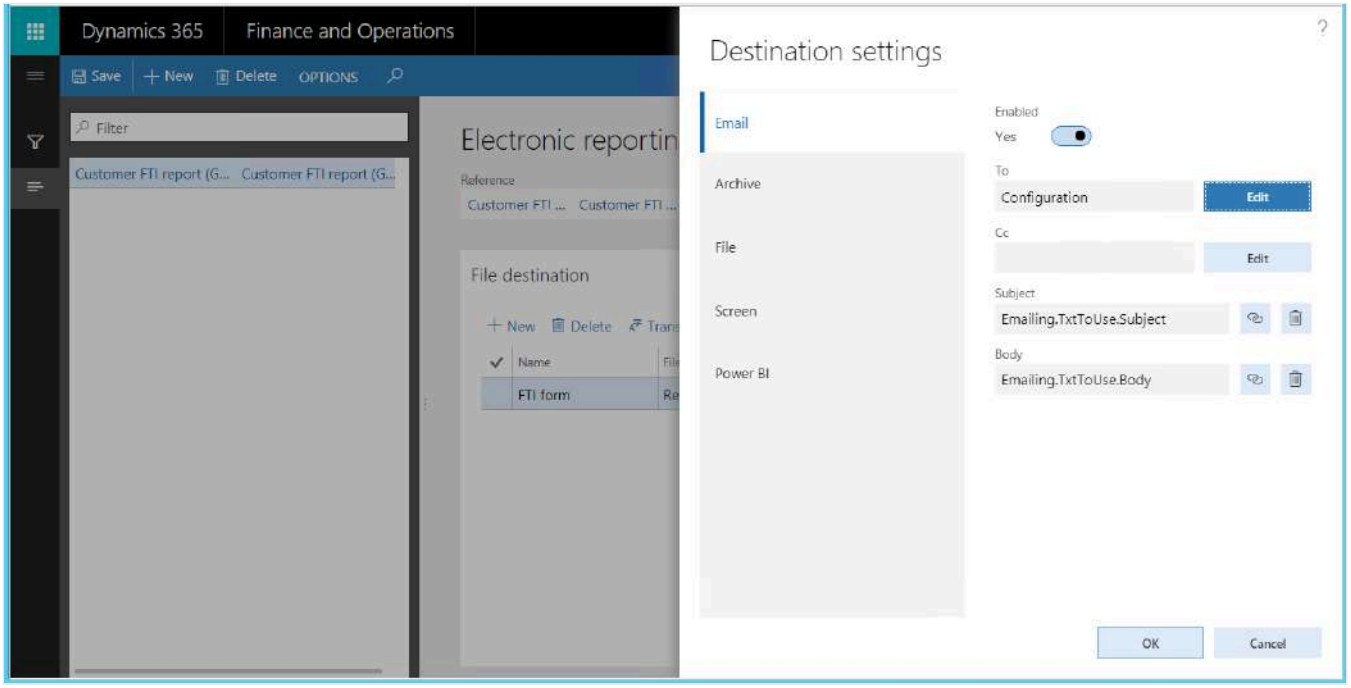
> ⓘ **Note**

You must first install the font that is used to generate the bar code on the local machine where you preview the generated invoice form in Excel format.

Use the sample ER format to configure email destinations

Use the following elements of the sample ER format to configure email destinations:

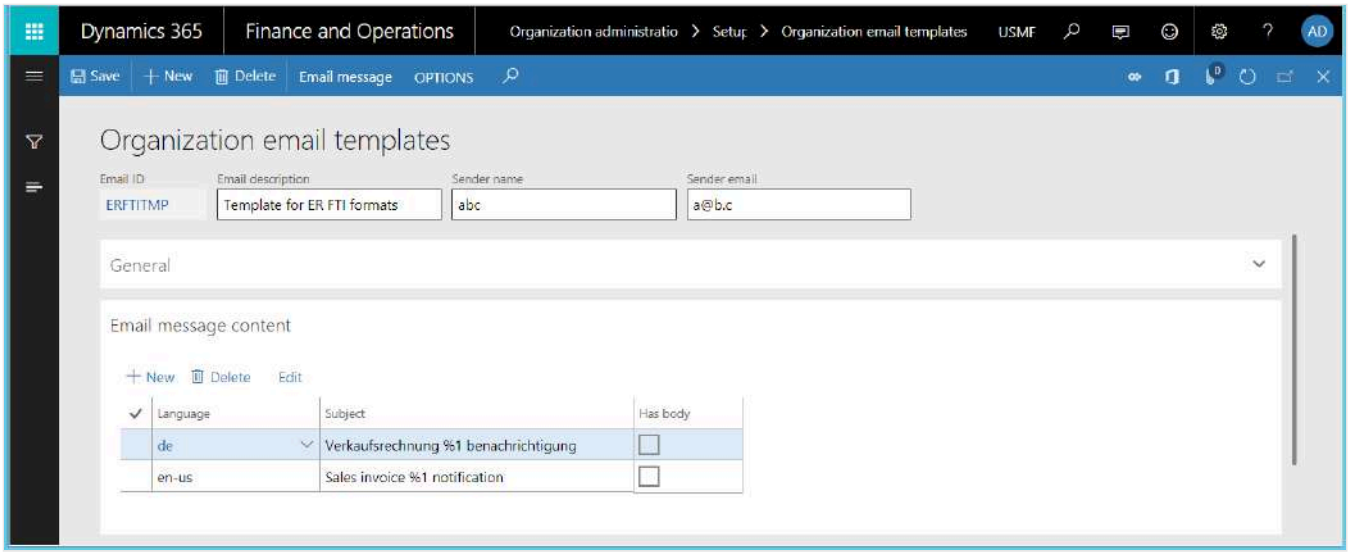
- Access the email address of a customer contact through the following ER expression: **model.InvoiceBase.Contact.ElectronicMail**.
- Access the email subject text through the following ER expression: **Emailing.TxtToUse.Subject**.
- Access the email body text through the following ER expression: **Emailing.TxtToUse.Body**.

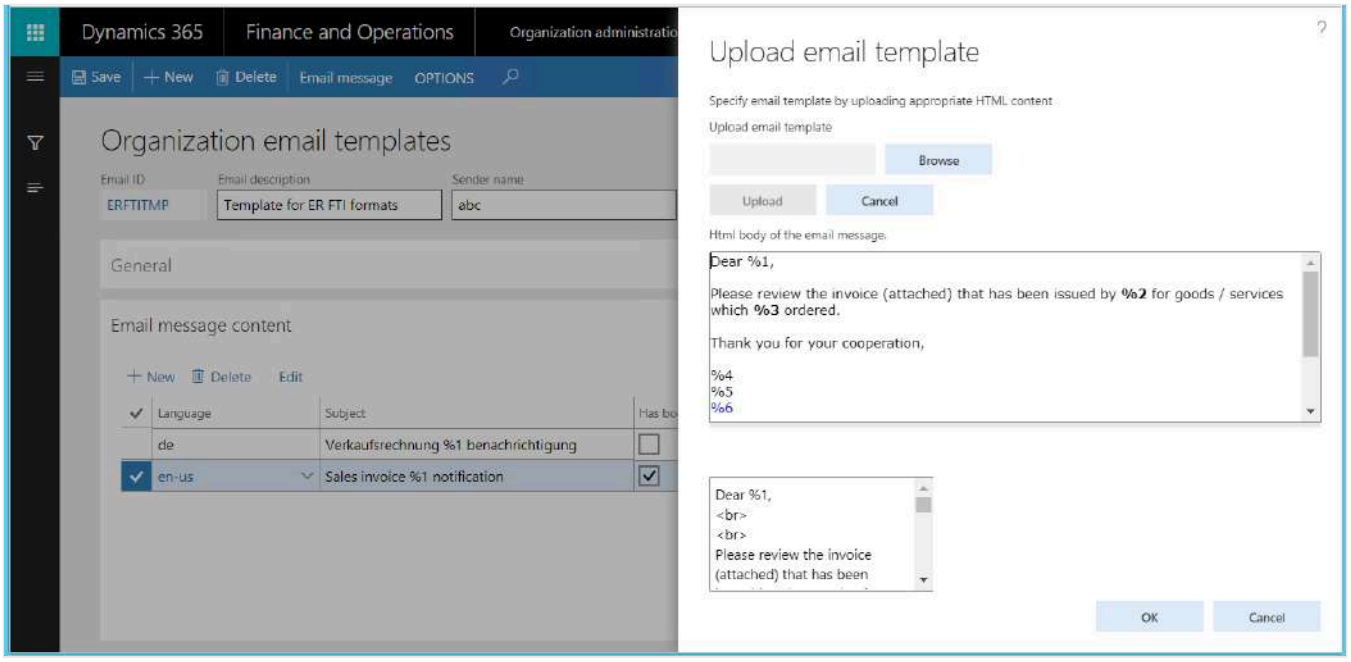


The sample ER format defines the default text of the email's subject and body. The language depends on the format's labels. If you don't add a custom organization email template that uses the predefined **ERFTITMP** ID, this default text is used for emails.

> **Note**
>
> The sample ER format defines the **ERFTITMP** email template ID. You can change this ID as needed in a new ER format that you create from this sample format.

If you add the organization email template that uses the predefined **ERFTITMP** ID for the legal entity that you're processing the invoice for, the template for the email subject and body text is used to generate the email.

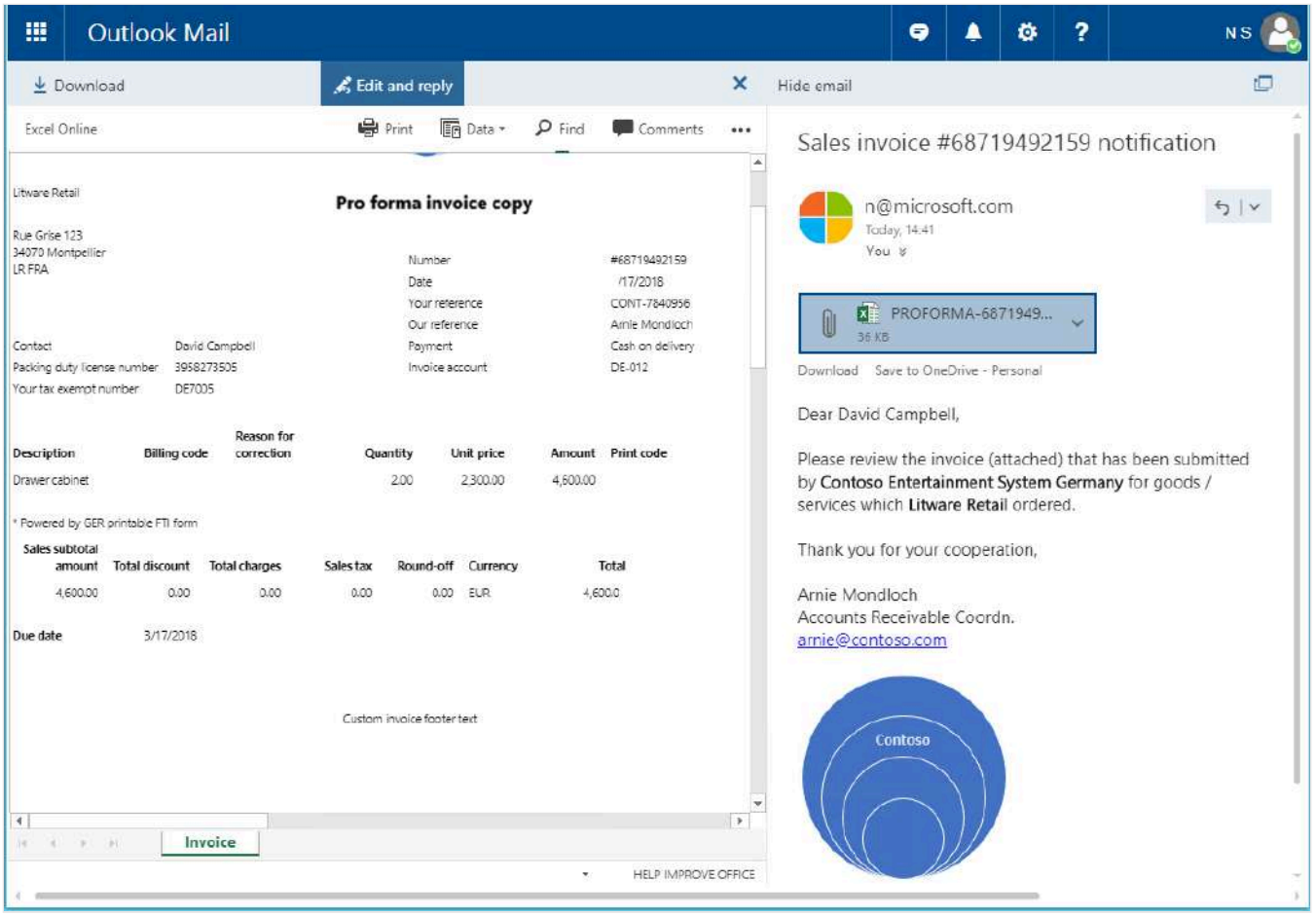




The **Emailing.TxtToUse.Subject** ER expression of the sample ER format is configured to replace any occurrences of the placeholder %1 with the processing invoice ID.

The **Emailing.TxtToUse.Body** expression of the sample format is configured for the following substitutions for placeholders:

- "%1" is replaced with the name of the customer's contact person.
- "%2" is replaced with the company name.
- "%3" is replaced with the customer name.
- "%4" is replaced with the name of the company's contact person.
- "%5" is replaced with the job title of the company's contact person.
- "%6" is replaced with the email address of the company's contact person.



Additional resources

Electronic reporting (ER) overview

Last updated on 04/13/2026

Trace the execution of ER formats to troubleshoot performance issues

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

As part of the process of designing Electronic reporting (ER) configurations to generate electronic documents, you define the method that gets data out of the application and puts it in the generated output. The ER performance trace feature helps you significantly reduce the time and cost involved in collecting the details of ER format execution and using them to troubleshoot performance problems. This tutorial provides guidelines about how to take performance traces for executed ER formats, and how to use the information from these traces to help improve performance.

Prerequisites

To complete the examples in this tutorial, you need the following access:

- Access to one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator
- Access to the instance of Regulatory Configuration Services (RCS) that's provisioned for the same tenant as the application, for one of the following roles:
  - Electronic reporting developer
  - Electronic reporting functional consultant
  - System administrator

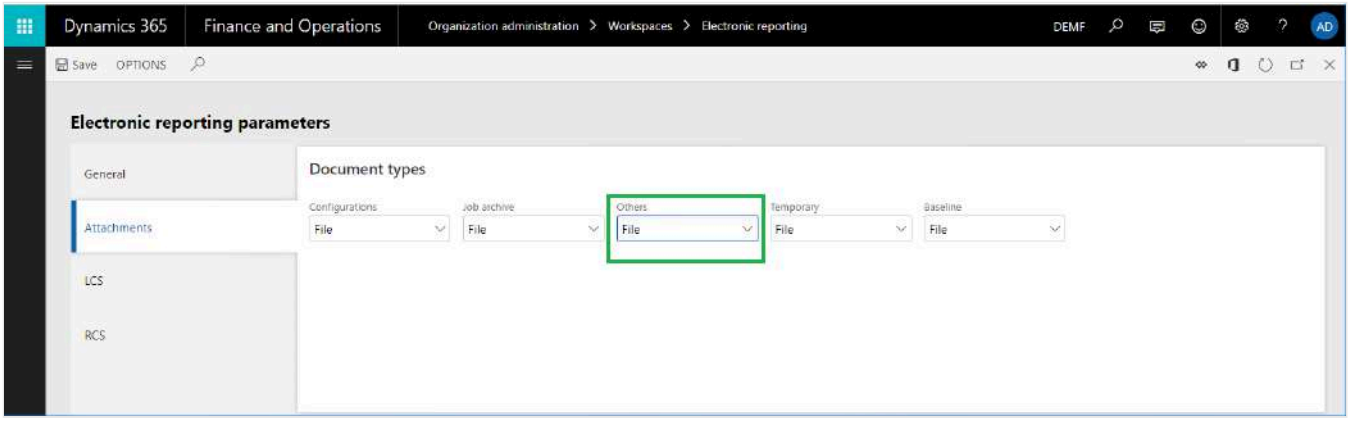
Also, download and store the following files locally.

Expand table

File	Content
Performance trace model.version.1	Sample ER data model configuration
Performance trace metadata.version.1	Sample ER metadata configuration
Performance trace mapping.version.1.1	Sample ER model mapping configuration
Performance trace format.version.1.1	Sample ER format configuration

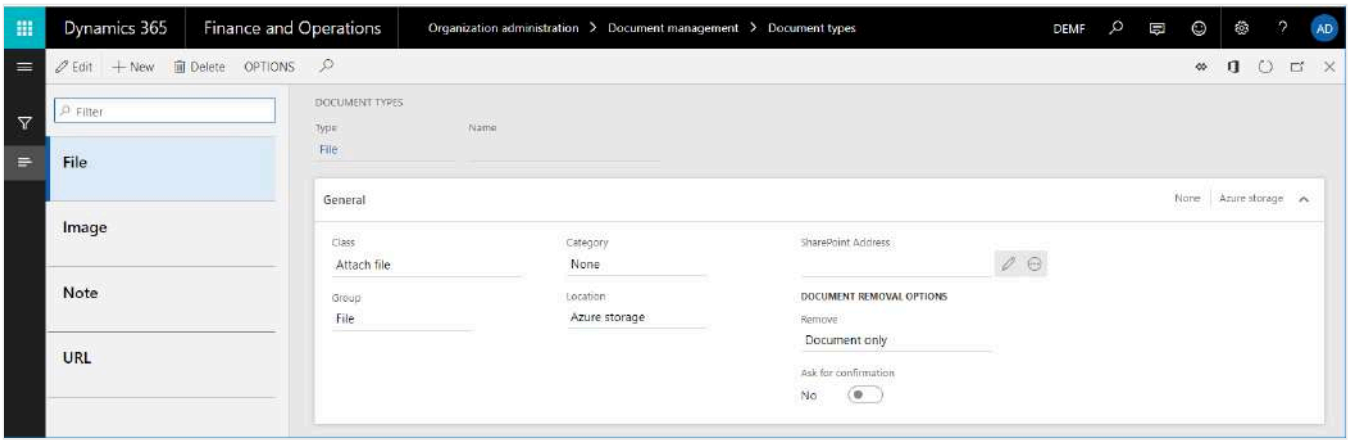
Configure ER parameters

Each ER performance trace that the application generates is stored as an attachment of the execution log record. The Document management (DM) framework manages these attachments. You must configure ER parameters in advance to specify the DM document type that should be used to attach performance traces. In the **Electronic reporting** workspace, select **Electronic reporting parameters**. Then, on the **Electronic reporting parameters** page, on the **Attachments** tab, in the **Others** field, select the DM document type to use for performance traces.



To be available in the **Others** lookup field, a DM document type must be configured in the following manner on the **Document types** page (**Organization administration** > **Document management** > **Document types**):

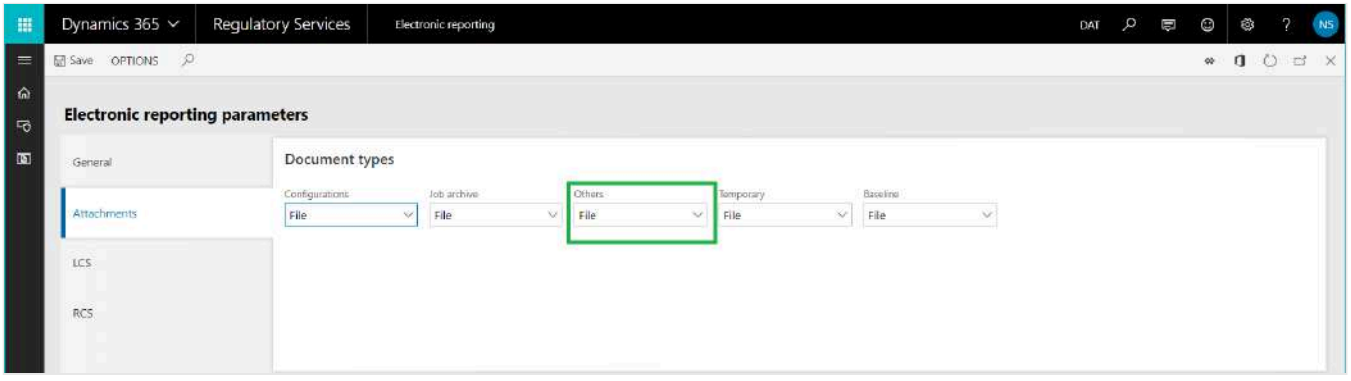
- **Class:** Attach file
- **Group:** File



> **Note**
>
> The selected document type must be available in every company of the current instance, because DM attachments are company-specific.

Configure RCS parameters

You import ER performance traces into RCS for analysis by using the ER format designer and the ER mapping designer. Because ER performance traces are stored as attachments of the execution log record that relates to the ER format, you must configure RCS parameters in advance to specify the DM document type that should be used to attach performance traces. In the instance of RCS that you provisioned for your company, in the **Electronic reporting** workspace, select **Electronic reporting parameters**. Then, on the **Electronic reporting parameters** page, on the **Attachments** tab, in the **Others** field, select the DM document type to use for performance traces.



To be available in the **Others** lookup field, a DM document type must be configured in the following manner on the **Document types** page (**Organization administration** > **Document management** > **Document types**):

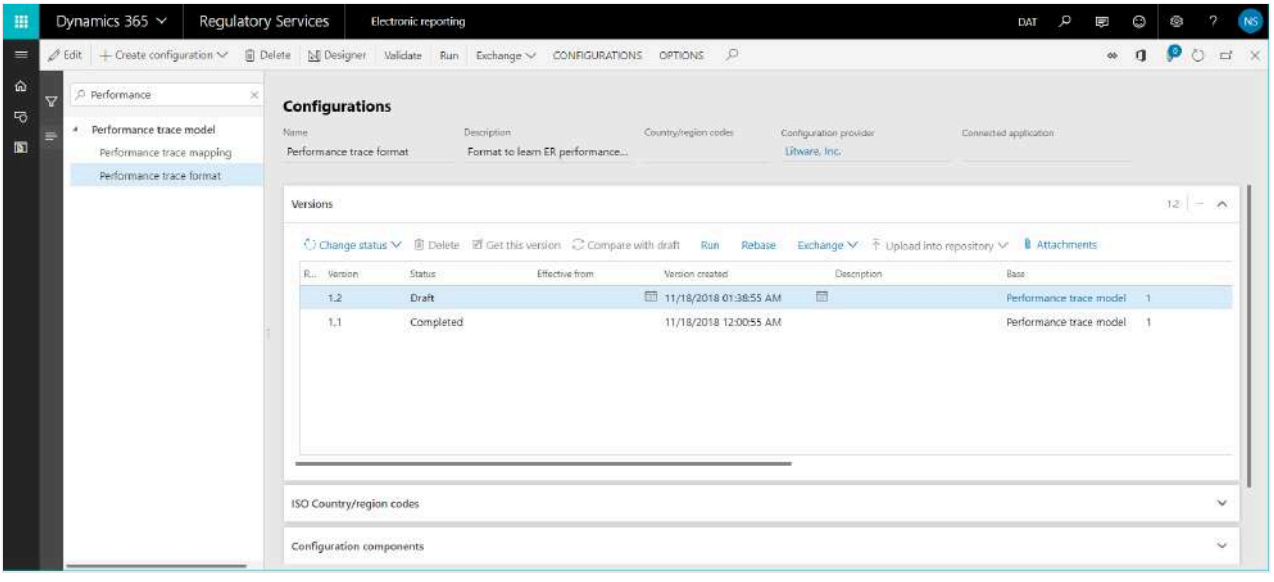
- **Class:** Attach file

- **Group:** File

Design an ER solution

Assume that you start designing a new ER solution to generate a new report that presents vendor transactions. Currently, you can find the transactions for a selected vendor on the **Vendor transactions** page (go to **Account payable** > **Vendors** > **All vendors**, select a vendor, and then, on the Action Pane, on the **Vendor** tab, in the **Transactions** group, select **Transactions**). However, you want to have all vendor transactions at the same time in one electronic document in XML format. This solution consists of several ER configurations that contain the required data model, metadata, model mapping, and format components.

1. Sign in to the instance of RCS that is provisioned for your company.
2. In this tutorial, you create and modify configurations for the **Litware, Inc.** sample company. Therefore, make sure that you add this configuration provider to RCS and select it as active. For instructions, see the Create configuration providers and mark them as active procedure.
3. In the **Electronic reporting** workspace, select the **Reporting configurations** tile.
4. On the **Configurations** page, import the ER configurations that you downloaded as a prerequisite into RCS, in the following order: data model, metadata, model mapping, format. For each configuration, follow these steps:
   a. On the Action Pane, select **Exchange** > **Load from XML file**.
   b. Select **Browse** to select the appropriate file for the required ER configuration in XML format.
   c. Select **OK**.



Run the ER solution to trace execution

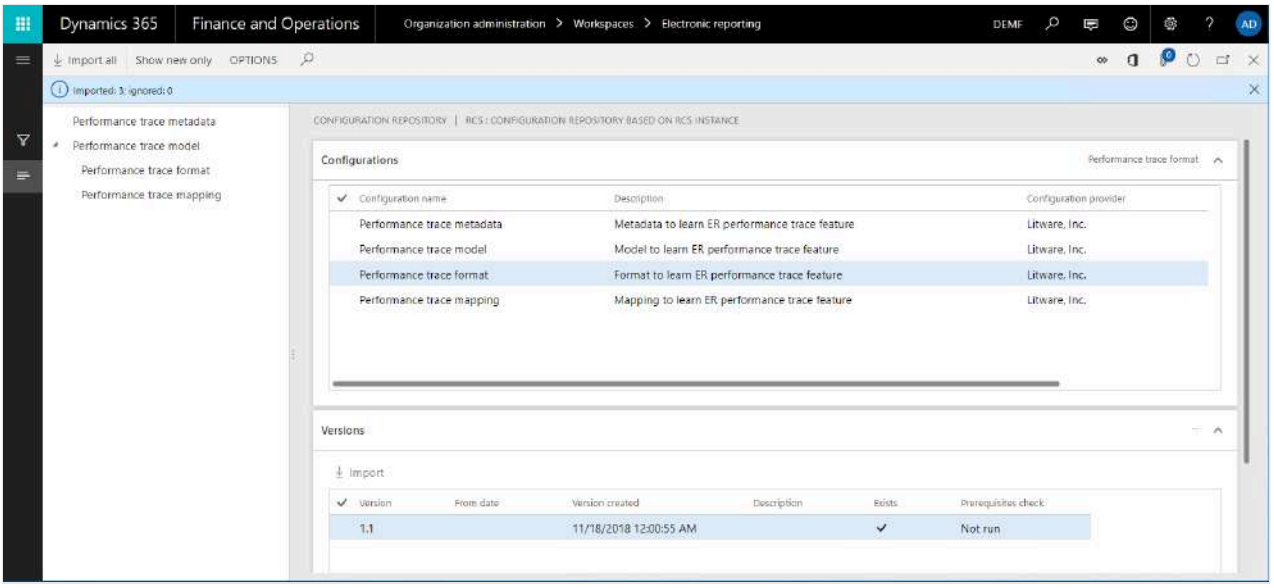
Assume that you finish designing the first version of the ER solution. You now want to test it in your instance and analyze execution performance.

Import an ER configuration from RCS into finance and operations

1. Sign in to your application instance.

2. For this tutorial, import configurations from your RCS instance (where you design your ER components) into your instance (where you test and finally use them). Therefore, make sure that you prepare all the required artifacts. For instructions, see the Import Electronic reporting (ER) configurations from Regulatory Configuration Services (RCS) procedure.

3. Follow these steps to import the configurations from RCS into the application:
   a. In the **Electronic reporting** workspace, on the tile for the **Litware, Inc.** configuration provider, select **Repositories**.
   b. On the **Configuration repository** page, select the repository of the **RCS** type, and then select **Open**.
   c. On the **Configurations** FastTab, select the **Performance trace format** configuration.
   d. On the **Versions** FastTab, select version **1.1** of the selected configuration, and then select **Import**.



The corresponding versions of the data model and model mapping configurations are automatically imported as prerequisites for the imported ER format configuration.

Turn on the ER performance trace

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.

2. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.

3. In **User parameters**, in the **Execution tracing** section, follow these steps:

    a. In the **Execution trace format** field, specify the format of the generated performance trace that stores execution details for ER format and mapping elements:

    - **Debug trace format** – Select this value if you plan to interactively run an ER format that has a short execution time. The collection of details about ER format execution starts. When you select this value, the performance trace collects information about the time spent on the following actions:
        - Running each data source in the model mapping that the process calls to get data
        - Processing each format item to enter data in the output that is generated

      If you select the **Debug trace format** value, you can analyze the content of the trace in the ER Operation designer. There, you can view the ER format or mapping elements that are mentioned in the trace.

    - **Aggregated trace format** – Select this value if you plan to run an ER format that has a long execution time in batch mode. The collection of the aggregated details about ER format execution starts. When you select this value, the performance trace collects information about the time spent on the following actions:
        - Running each data source in the model mapping that the process calls to get data
        - Running each data source in the format mapping that the process calls to get data
        - Processing each format item to enter data in the output that is generated

      The **Aggregated trace format** value is available in Microsoft Dynamics 365 Finance version 10.0.20 and later.

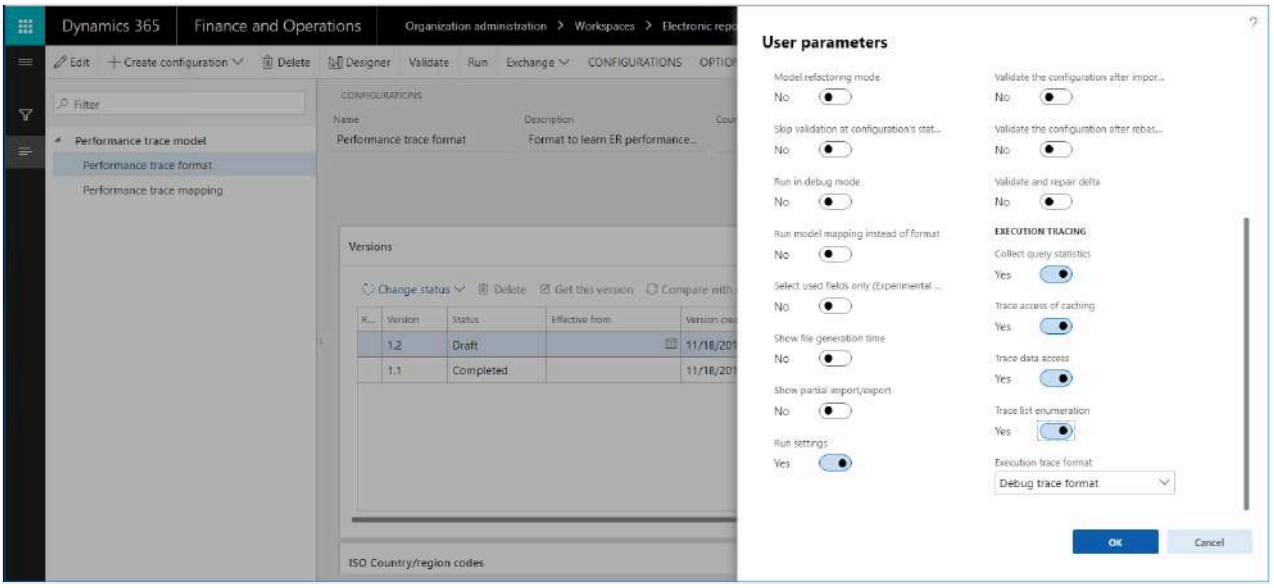
      In the ER format designer and ER model mapping designer, you can view the total execution time for a single component. Additionally, the trace contains details about the execution, such as the number of executions, and the minimum and maximum time of a single execution.

> **Note**
>
> This trace is collected based on the traced components path. Therefore, the statistics might be incorrect when a single parent component contains several unnamed child components, or when several child components have the same name.

b. Set the following options to **Yes** to collect specific details of the execution of the ER model mapping and ER format components:

- **Collect query statistics** – When you turn on this option, the performance trace collects the following information:
  - The number of database calls that data sources make
  - The number of duplicate calls to the database
  - Details of the SQL statements that were used to make database calls
- **Trace access of caching** – When you turn on this option, the performance trace collects information about the ER model mapping's cache usage.
- **Trace data access** – When you turn on this option, the performance trace collects information about the number of calls to the database for executed data sources of the record list type.
- **Trace list enumeration** – When you turn on this option, the performance trace collects information about the number of records that are requested from data sources of the record list type.

> **Note**
>
> The parameters in the **User parameters** dialog box are specific to the user and the current company.



Run the ER format

1. Select the **DEMF** company.
2. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
3. On **Configurations**, in the configuration tree, select the **Performance trace format** item.
4. On the Action Pane, select **Run**.

The generated file presents information about 265 transactions for six vendors.

Review the execution trace

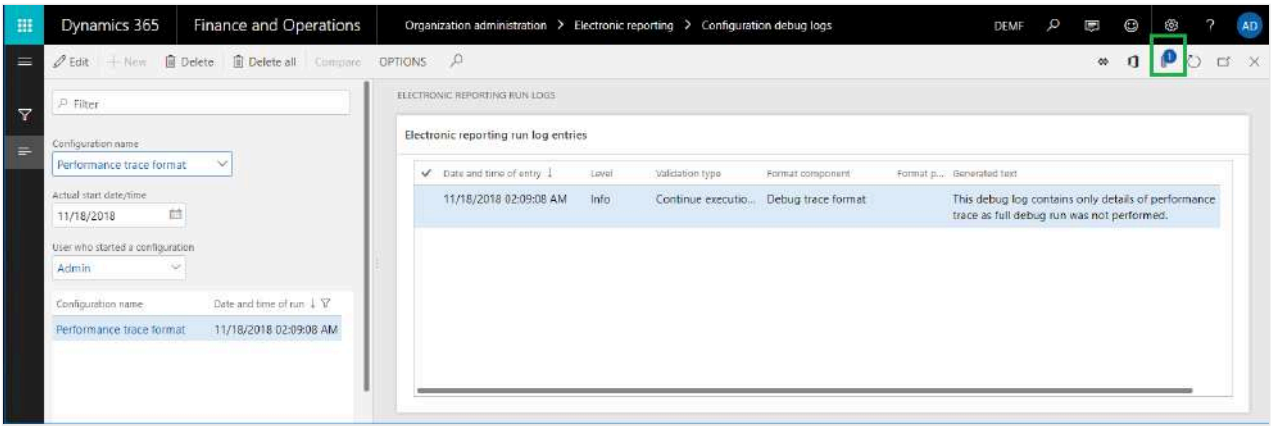
Export the generated trace from the application

Performance traces are decoupled from the source ER format and can be serialized to an external zip file.

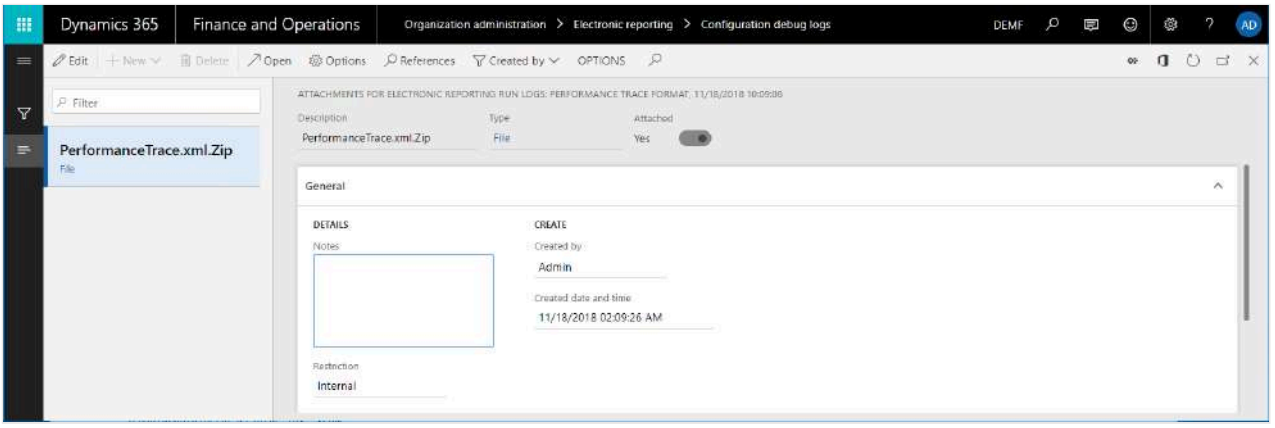
1. Go to **Organization administration** > **Electronic reporting** > **Configuration debug logs**.

2. On **Electronic reporting run logs**, in the left pane, in the **Configuration name** field, select **Performance trace format** to find the log records that the execution of the **Performance trace format** configuration generates.

3. Select the **Attachments** button (the paper clip symbol) in the upper-right corner of the page, or press **Ctrl+Shift+A**.



4. On **Attachments for Electronic reporting run logs**, on the Action Pane, select **Open** to get the performance trace as a zip file and store it locally.



> ⓘ **Note**
>
> The generated trace has a reference to the source ER report via a unique report identifier in **GUID** format only. The version numbering of the format isn't considered.

The association between the performance trace that the application generates for the executed ER format and the ER model mapping is based on the root descriptor that the application uses and the common data model. The version numbering of the format and model mapping isn't considered. The setting of the **Default for model mapping** flag for the model mapping also isn't considered.

Import the generated trace into RCS

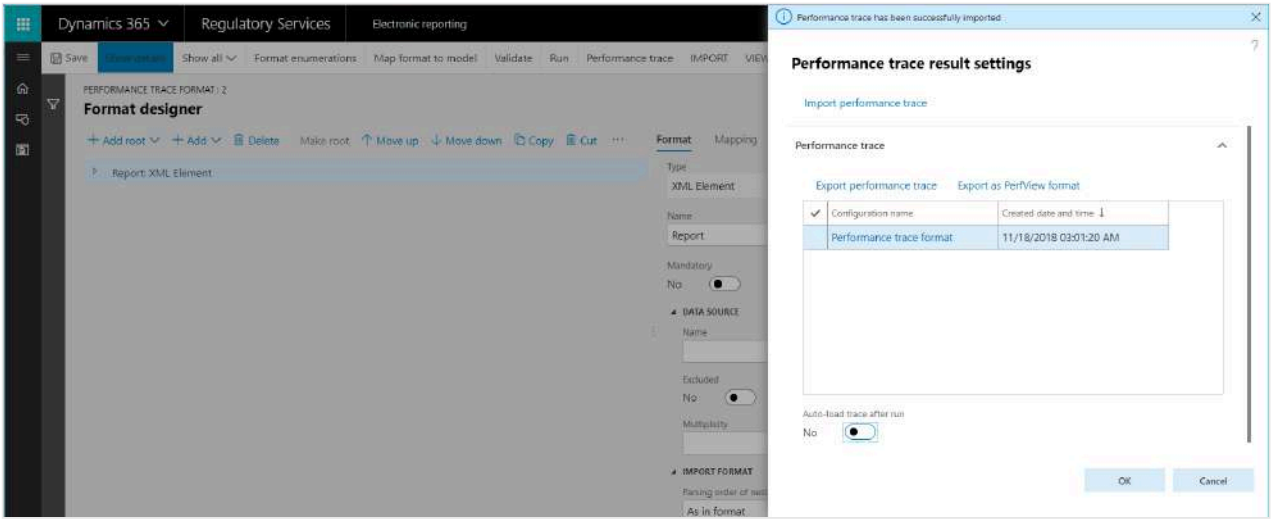
1. In RCS, in the **Electronic reporting** workspace, select the **Reporting configurations** tile.
2. On the **Configurations** page, in the configuration tree, expand the **Performance trace model** item, and select the **Performance trace format** item.
3. On the Action Pane, select **Designer**.

4. On the **Format designer** page, on the Action Pane, select **Performance trace**.

5. In the **Performance trace result settings** dialog box, select **Import performance trace**.

6. Select **Browse** to select the zip file that you exported earlier.

7. Select **OK**.

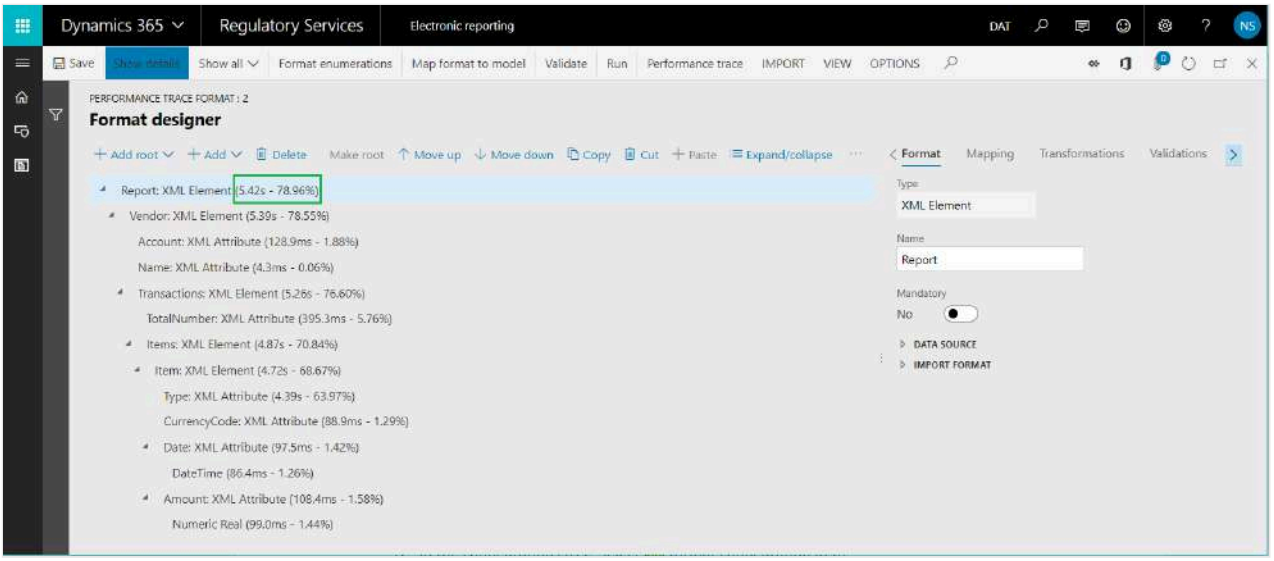


Use the performance trace for analysis in RCS – Format execution

1. In RCS, on the **Format designer** page, select **Expand/collapse** to expand the content of all format items.

   Notice that the trace shows extra information for some items of the current format:

   - The actual time that was spent entering data in the generated output by using the format item
   - The same time expressed as a percentage of the total time that was spent generating the whole output



2. Close **Format designer** page.

Use the performance trace for analysis in RCS – Model mapping

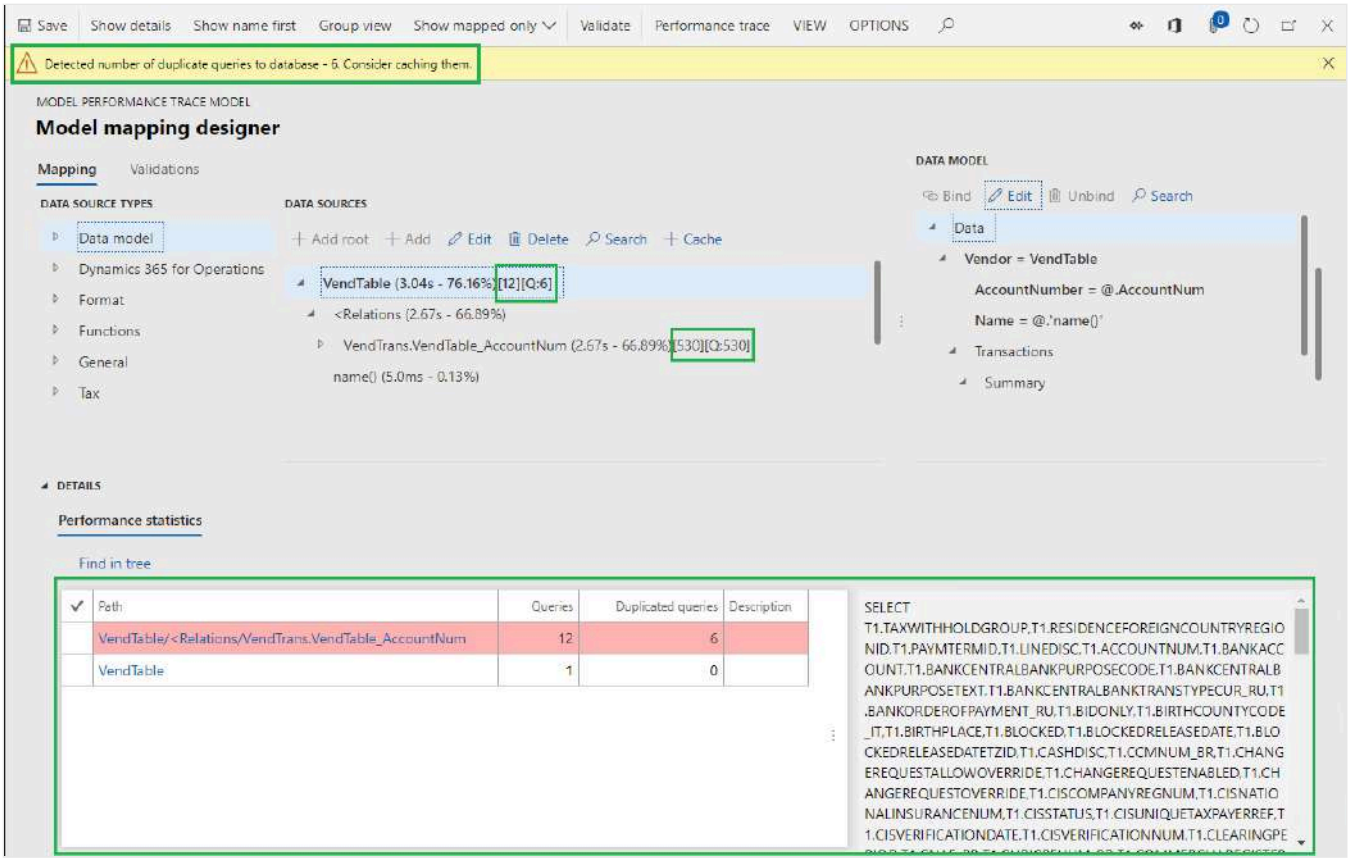
1. In RCS, on the **Configurations** page, in the configuration tree, select the **Performance trace mapping** item.
2. On the Action Pane, select **Designer**.
3. Select **Designer**.
4. On the **Model mapping designer** page, on the Action Pane, select **Performance trace**.
5. Select the trace that you imported earlier.
6. Select **OK**.

You see new information for some data source items of the current model mapping:

- The actual time that the data source spends getting data
- The same time expressed as a percentage of the total time that was spent running the whole model mapping

ER informs you that the current model mapping duplicates database requests while the VendTable/<Relations/VendTrans.VendTable_AccountNum data source runs. This duplication occurs because the list of vendor transactions is called two times for each iterated vendor record:

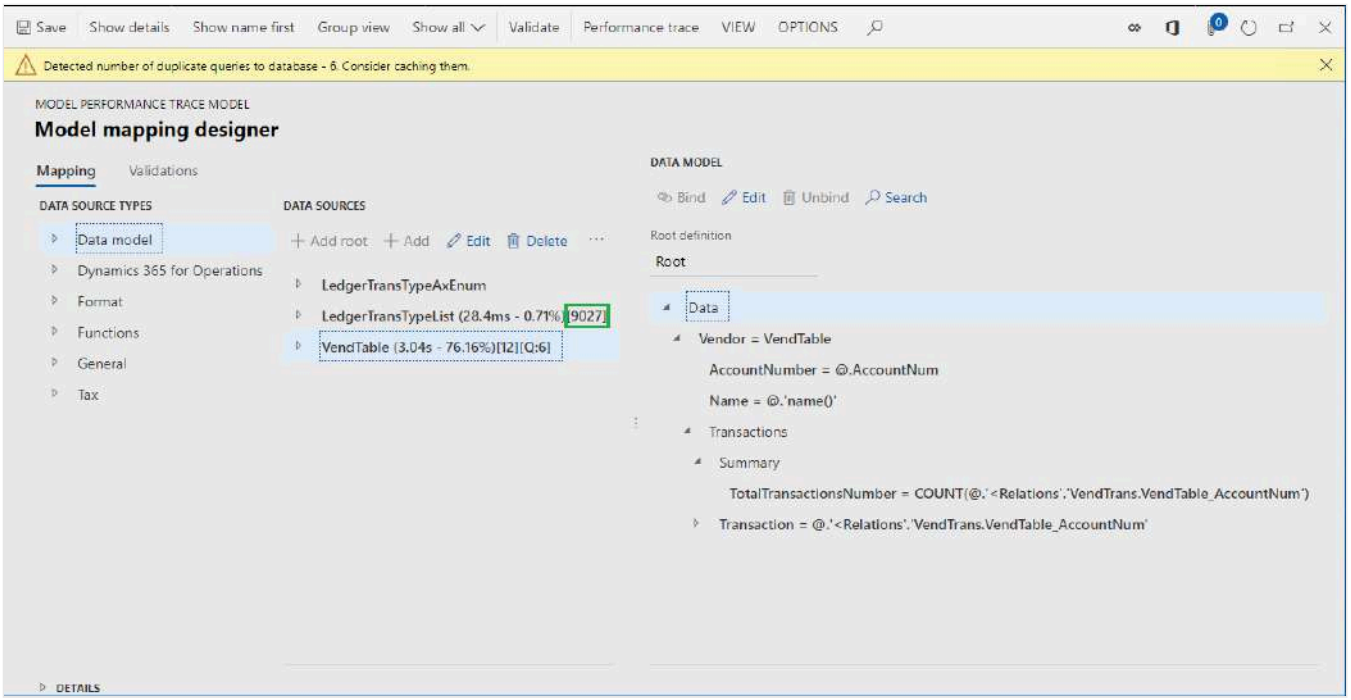
- One call enters details of each transaction in the data model, based on configured bindings.
- One call enters the calculated number of transactions per vendor in the data model.



The value **[Q:530]** indicates that the VendTrans table was called 530 times to return a record from that table to the VendTable/<Relations/VendTrans.VendTable_AccountNum data source. The value **[530]** indicates that the VendTable/<Relations/VendTrans.VendTable_AccountNum data source was called 530 times to return a record from that data source and enter the details from it in the data model.

Use caching for the VendTable/<Relations/VendTrans.VendTable_AccountNum data source to reduce the number of calls that are made to get the details for 265 transactions and help improve the performance of the model mapping.

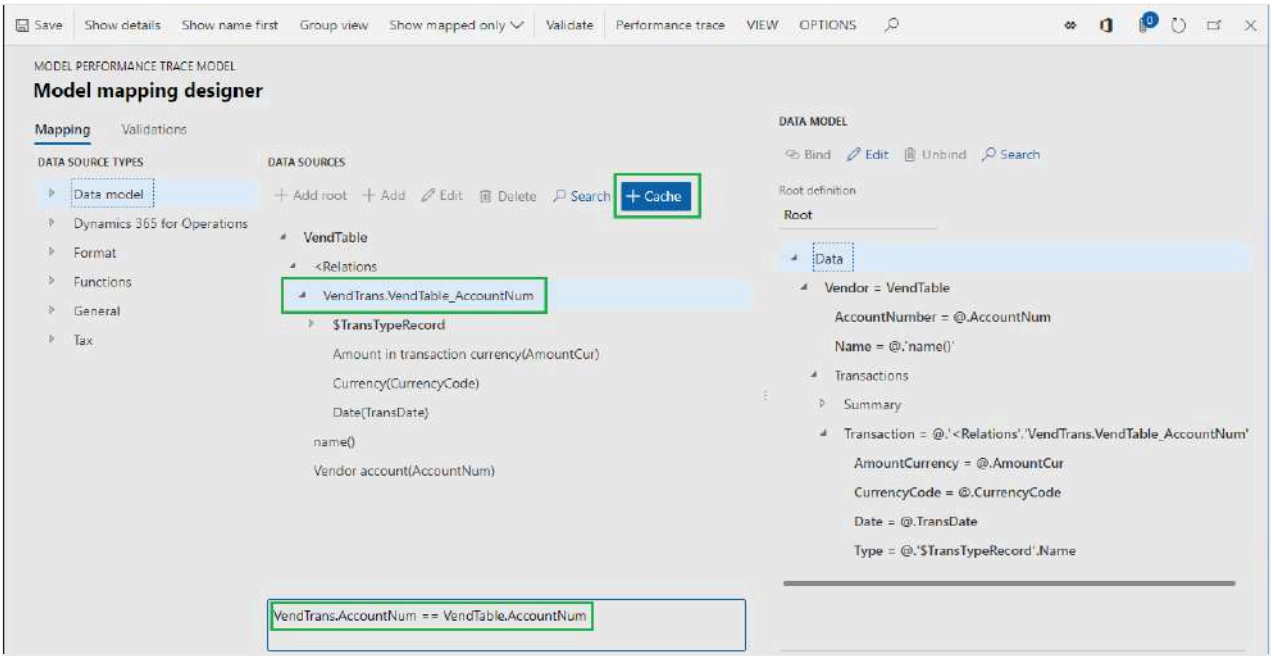
It can also be useful to reduce the number of calls that are made to the LedgerTransTypeList data source. This data source associates each value of the **LedgerTransType** enumeration with its label. By using this data source, you can find an appropriate label and enter it in the data model for each vendor transaction. The current number of calls to this data source (9,027) is quite high for 265 transactions.



Improve the model mapping based on information from the execution trace

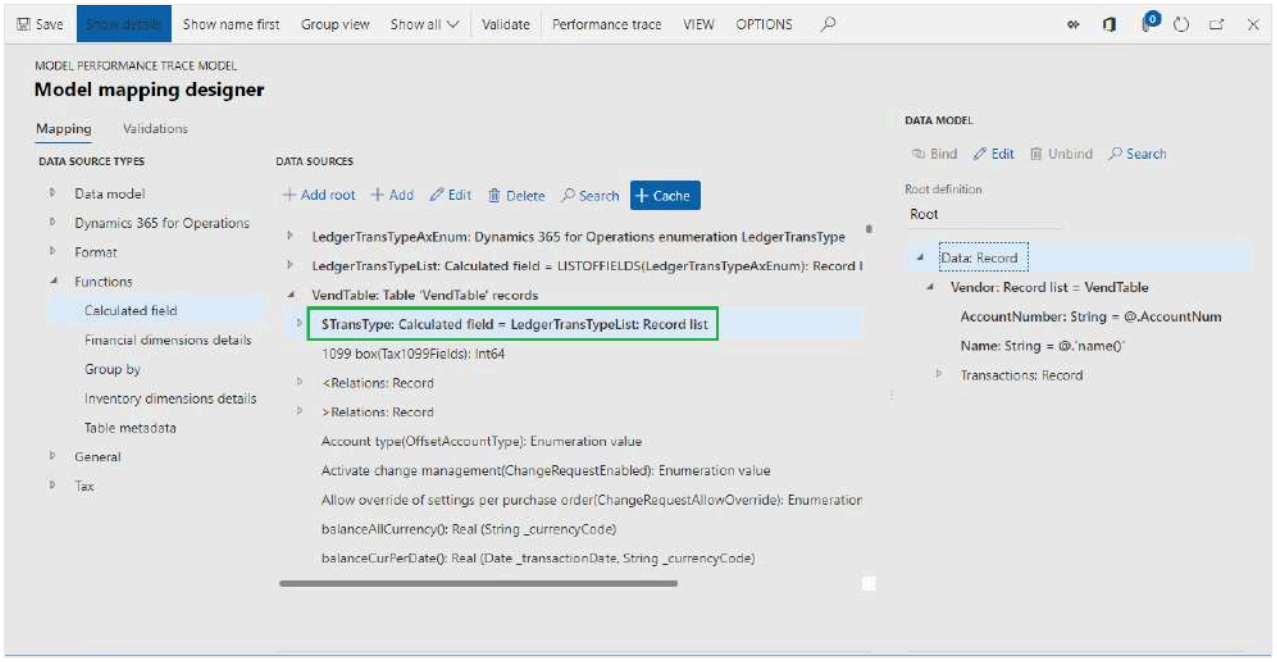
Modify the logic of the model mapping

1. To use caching and help prevent duplicate calls to the database, follow these steps:
   a. In RCS, on the **Model mapping designer** page, in the **Data sources** pane, select the **VendTable** item.
   b. Select **Cache**.
   c. Expand the **VendTable** item, expand the list of one-to-many relations for the VendTable data source (the **<Relations** item), and select the **VendTrans.VendTable_AccountNum** item.
   d. Select **Cache**.



2. To bring the LedgerTransTypeList data source into the scope of the VendTable data source, follow these steps:
   a. In the **Data source types** pane, expand the **Functions** item, and select the **Calculated field** item.
   b. In the **Data sources** pane, select the **VendTable** item.
   c. Select **Add**.
   d. In the **Name** field, enter **$TransType**.
   e. Select **Edit formula**.
   f. In the **Formula** field, enter **LedgerTransTypeList**.
   g. Select **Save**.
   h. Close the **Formula editor** page.
   i. Select **OK**.

3. To cache the **$TransType** field, follow these steps:
   a. Select the **LedgerTransTypeList** item.
   b. Select **Cache**.
   c. Select the **VendTable.$TransType** item.
   d. Select **Cache**.



4. To change the **$TransTypeRecord** field so that it starts to use the cached **$TransType** field, follow these steps:

   a. In the **Data sources** pane, expand the **VendTable** item, expand the **<Relations** item, expand the **VendTrans.VendTable_AccountNum** item, and select the **VendTable.VendTrans.VendTable_AccountNum.$TransTypeRecord** item.

   b. Select **Edit**.

   c. Select **Edit formula**.

   d. In the **Formula** field, find the following expression:

      FIRSTORNULL (WHERE (LedgerTransTypeList, LedgerTransTypeList.Enum = @.TransType))

   e. In the **Formula** field, enter the following expression:

      FIRSTORNULL (WHERE (VendTable.'$TransType', VendTable.'$TransType'.Enum = @.TransType)).

   f. Select **Save**.

   g. Close the **Formula editor** page.

   h. Select **OK**.

5. Select **Save**.

6. Close the **Model mapping designer** page.

7. Close the **Model mappings** page.

Complete the modified version of the ER model mapping

1. In RCS, on the **Configurations** page, on the **Versions** FastTab, select version **1.2** of the **Performance trace mapping** configuration.
2. Select **Change status**.
3. Select **Complete**.

Import the modified ER model mapping configuration from RCS into the application

To import version 1.2 of the **Performance trace mapping** configuration, see Import an ER configuration from RCS into finance and operations.

Run the modified ER solution to trace execution

Run the ER format

To generate a new performance trace, see Run the ER format.

Work with the execution trace

Export the generated trace from the application

To save a new performance trace locally, see Export the generated trace from the application.

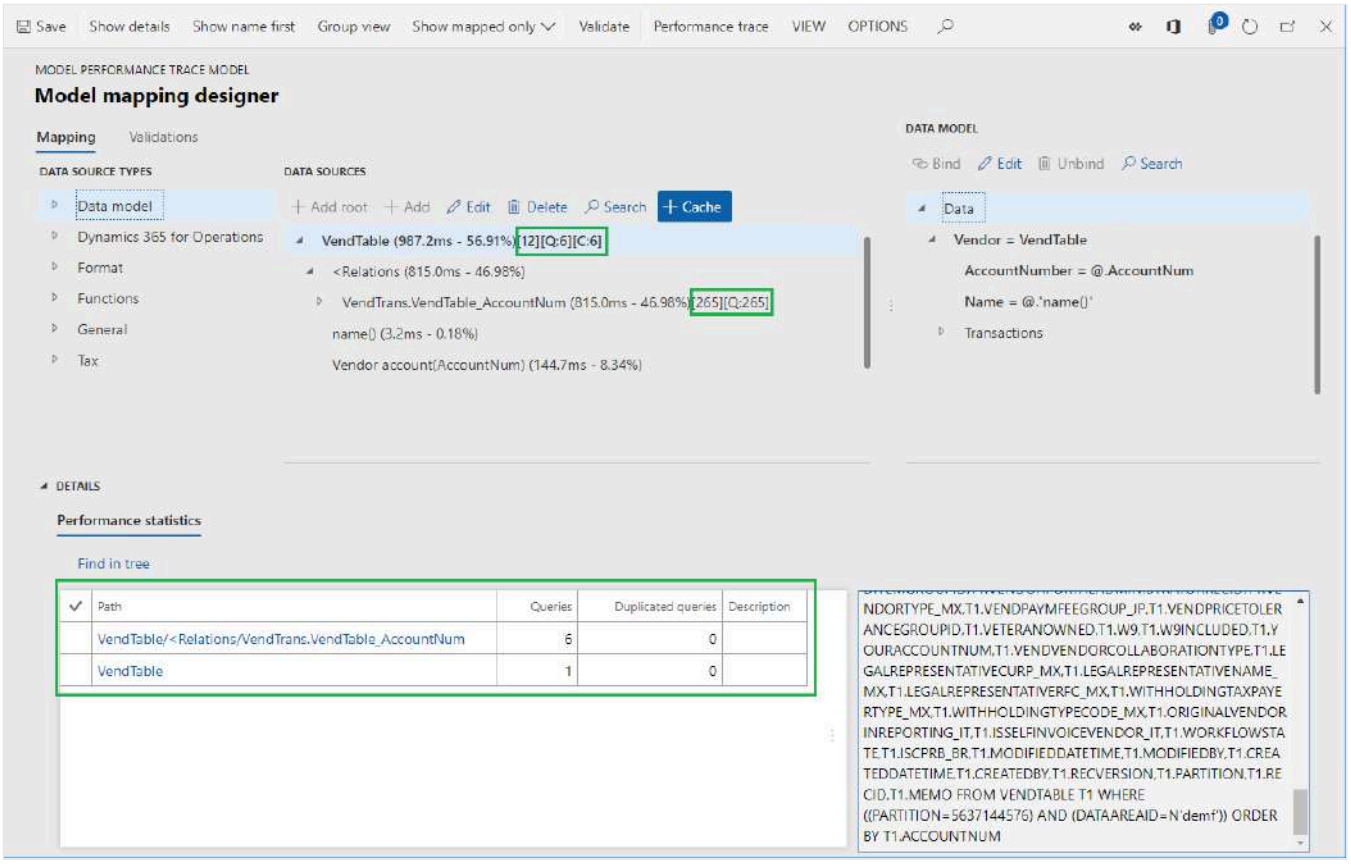
Import the generated trace into RCS

To import the new performance trace into RCS, see Import the generated trace into RCS.

Use the performance trace for analysis in RCS – Model mapping

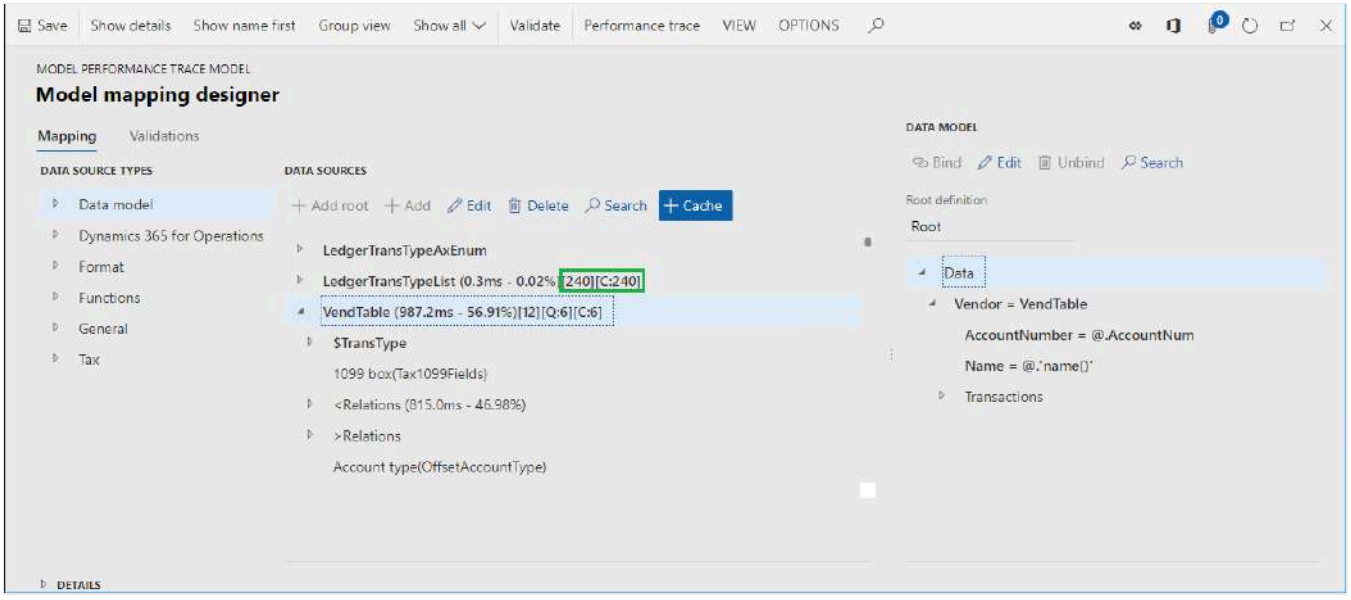
To analyze the latest performance trace, repeat the steps in the Use the performance trace for analysis in RCS – Model mapping section earlier in this article.

The adjustments you made to the model mapping eliminate duplicate queries to the database. The number of calls to database tables and data sources for this model mapping is also reduced. Therefore, the performance of the whole ER solution improves.



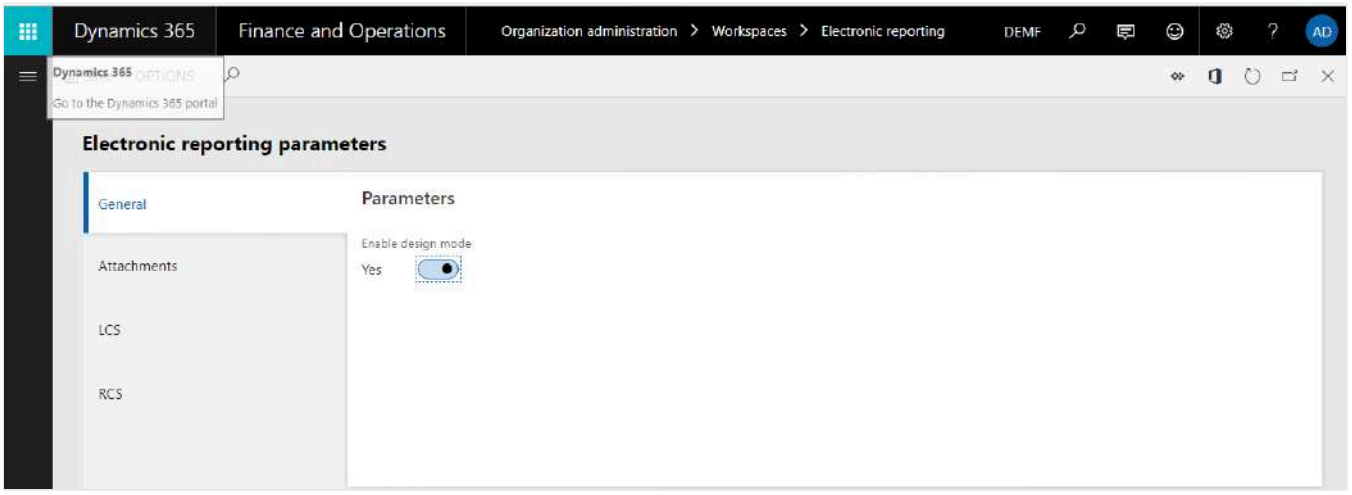
In the trace information, the value **[12]** for the VendTable data source indicates that this data source was called 12 times. The value **[Q:6]** indicates that six calls were translated to database calls to the VendTable table. The value **[C:6]** indicates that the records fetched from the database were cached, and six other calls were processed by using the cache.

The number of calls to the LedgerTransTypeList data source is reduced from 9,027 to 240.



Review the execution trace in the application

In addition to RCS, some versions might offer capabilities for an ER framework designer experience. These versions have an **Enable design mode** option that you can turn on. You can find this option on the **General** tab of the **Electronic reporting parameters** page, which you can open from the **Electronic reporting** workspace.



If you use one of these versions, you can analyze the details of generated performance traces directly in the application. You don't have to export them from the application and import them into RCS.

Review the execution trace by using external tools

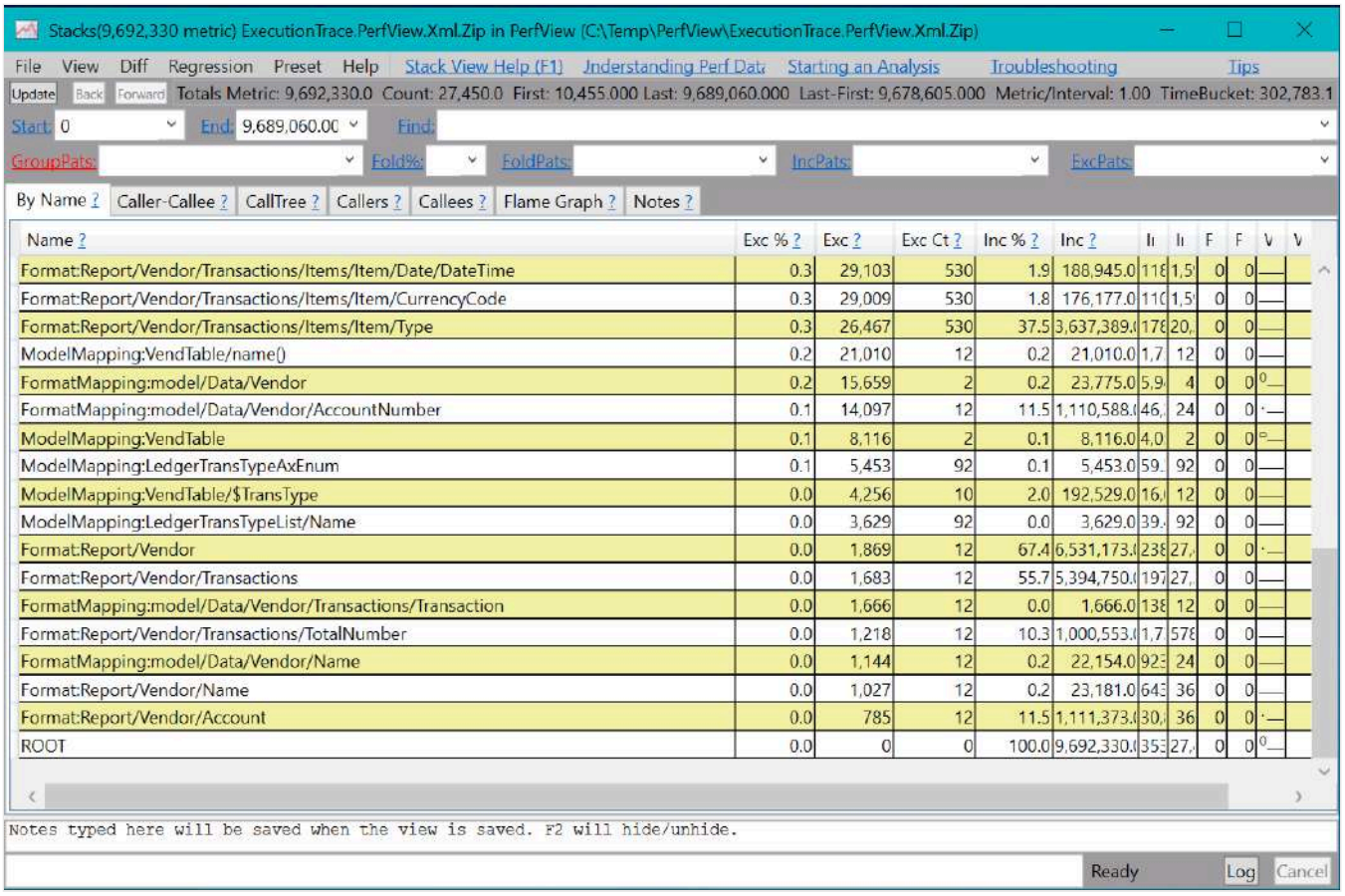
Configure user parameters

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In the **User parameters** dialog box, in the **Execution tracing** section, in the **Execution trace format** field, select **PerfView XML**.

Run the ER format

To generate a new performance trace, see Run the ER format.

The web browser offers a zip file for download. This file contains the performance trace in PerfView format. You can use the PerfView performance analysis tool to analyze the details of ER format execution.

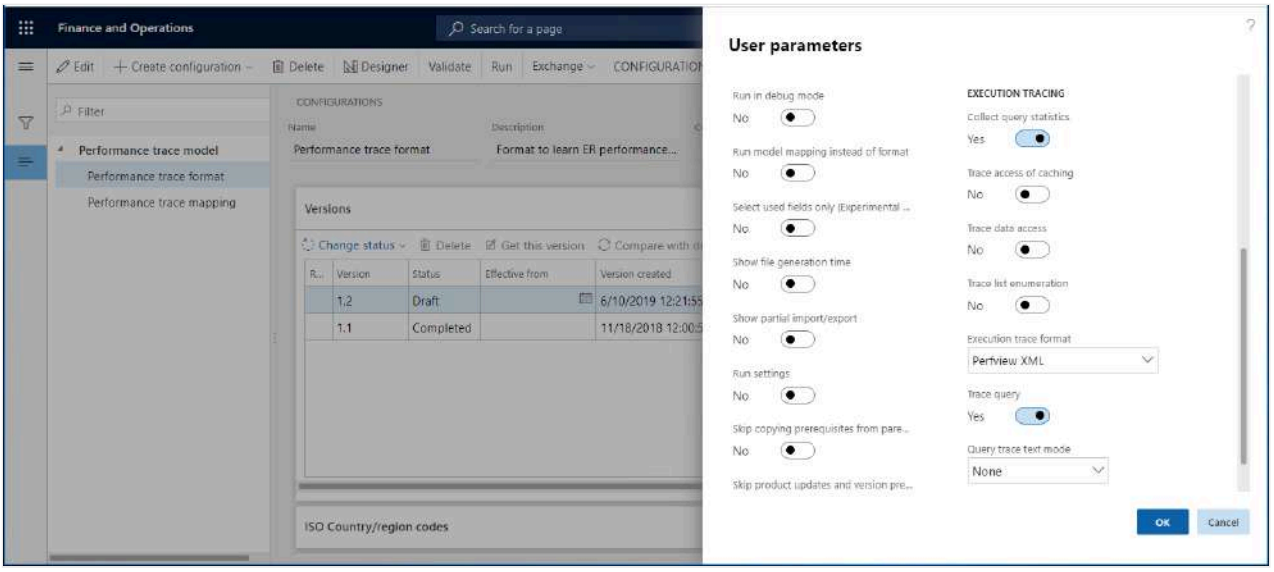


Use external tools to review an execution trace that includes database queries

Improvements to the ER framework mean that the performance trace generated in PerfView format now offers more details about ER format execution. In Microsoft Dynamics 365 Finance version 10.0.4 (July 2019), this trace can also include details of executed SQL queries to the application database.

Configure user parameters

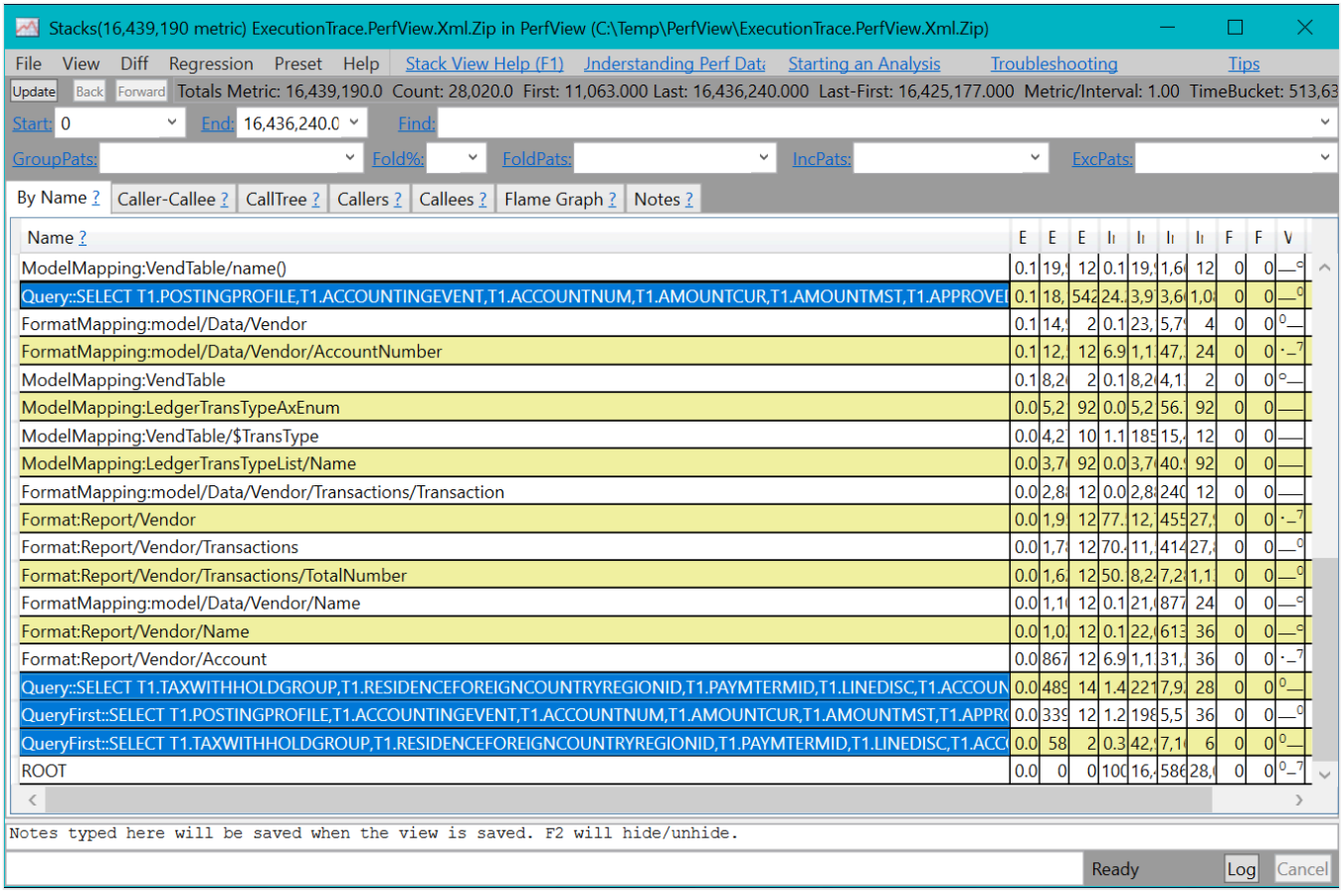
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On **Configurations**, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In the **User parameters** dialog box, in the **Execution tracing** section, set the following parameters:
    - In the **Execution trace format** field, select **PerfView XML**.
    - Set the **Collect query statistics** option to **Yes**.
    - Set the **Trace query** option to **Yes**.



Run the ER format

To generate a new performance trace, see Run the ER format.

When you finish, the web browser offers a ZIP file for download. This file contains the performance trace in PerfView format. You can use the PerfView performance analysis tool to analyze the details of ER format execution. This trace now includes the details of SQL database access during the execution of the ER format.



Additional resources

- Electronic Reporting overview
- Improve performance of ER solutions by adding parameterized CALCULATED FIELD data sources

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Last updated on 04/13/2026

Performance and memory considerations for ER

This article describes how the Electronic reporting (ER) engine manages memory during report execution, and how that management affects reports that process large volumes of data.

Overview

When you run an ER format, the ER engine loads data from the configured data sources, transforms it according to the format mapping, and produces the output document. For reports that process large volumes of data such as statutory declarations or high-volume transaction extracts, the amount of memory used during execution can become a limiting factor. High peak memory consumption can slow down execution and, in extreme cases, cause a report run to fail.

Optimized memory consumption during ER execution

Starting in version 10.0.49, the ER reporting engine includes targeted optimizations that improve memory efficiency during execution. These optimizations reduce peak memory consumption by:

- **Minimizing unnecessary object creation** during data processing and document generation.
- **Avoiding redundant duplication of data in memory** so that the same data isn't held in multiple places at the same time.
- **Reusing existing data structures more effectively** instead of allocating new ones.

These improvements are broadly applicable across ER scenarios and are beneficial for reports that process large volumes of data.

> ⓘ **Note**
>
> These optimizations are built into the ER engine. You automatically benefit from them after you update to the applicable version. No configuration changes, parameter settings, or report redesign are required.

What to expect

- **Reduced peak memory usage** during execution, which lowers the risk of out-of-memory failures for high-volume runs.
- **More consistent completion** of large reports, because the engine holds less data in memory at any one time.
- **No change to report output.** The optimizations affect only how the engine uses memory internally. The generated documents are identical to those produced by earlier versions.

Recommendations for large reports

To get the most benefit when you run reports that process large volumes of data, consider the following practices:

- **Run high-volume reports in batch mode.** Batch execution offloads processing from interactive sessions. For more information, see Electronic reporting (ER) destinations.
- **Filter the source data** to the records that the report actually requires, so that the engine processes only the data that you need.
- **Keep your ER configurations up to date** so that you receive the latest engine optimizations and fixes.

Last updated on 07/22/2026

Troubleshooting performance issues in ER configurations

This article explains how to find and solve performance issues in Electronic reporting (ER) configurations.

Typically, performance investigation consists of several steps.

1. Collect data.
2. Analyze the collected data.
3. Based on the results of the analysis, use ER configurations to make changes, or decide to collect more data.

Troubleshooting

Analyze execution time

Execution time can depend on unpredictable factors, such as other tasks that run in the same environment and caching that uses data when it's called for the first time. Therefore, you should repeat the execution and measurement several times.

Sometimes, performance issues aren't caused by an ER format configuration that you use for reporting. Instead, X++ code that you use to open that ER format configuration causes the issues.

1. In either the Action center or the event log, look at the execution time of the report.
2. Compare the execution time of the report with the total execution time in the scenario.
3. If the execution time of the report is much less than the total execution time, consider the amount of data that the report processes:

    - If the report processes a small amount of data, the issue might involve the loading time of the configuration.
    - If the report processes a large amount of data, the issue might involve preprocessing X++. Use Trace parser to analyze this case.

    For other cases, see the next sections.
4. Run multiple tests that involve different amounts of data to determine how the execution time depends on the amount of data.

Analyze Trace parser traces

Prepare a small example, or collect several traces during random parts of the report generation.

Then, in Trace parser, perform a standard bottom-up analysis and answer the following questions:

- What methods consume the most time?
- What part of the overall time do those methods use?

Answer the same questions for queries.

If you see that methods are prefixed with "ER," move on to the next section.

If you see that methods or queries originated in the application suite, consider generic optimizations (for example, create indexes).

Analyze the number of calls. If the number is higher than expected, consider caching the corresponding nodes of the configuration.

Analyze database calls

Prepare an example that has a small amount of data, so that you can collect an ER trace.

Then open the trace in the ER model mapping designer, and look at the bottom of the page. Answer the following questions:

- Is there any query duplication? If there is, consider one of the following fixes:
  - Use caching if you think that there are several calls inside one parent record.
  - Use a cached, parameterized calculated field if you think that there are calls to the same record inside different records.
  - Use a **JOIN** data source if you have to read to a substantial number of different records from a database.

- Does the number of queries and fetched records correspond to the overall amount of data? For example, if a document has 10 lines, do the statistics show that the report extracts 10 lines or 1,000 lines? If you have a substantial number of fetched records, consider one of the following fixes:
  - Use the **FILTER** function instead of the **WHERE** function to process data on the Microsoft SQL Server side.
  - Use caching to avoid fetching the same data.
  - Use collected data functions to avoid fetching the same data for summarization.

Analyze PerfView traces

PerfView is a tool for experienced developers. For more detailed information about the following steps, see Wall Clock Time Investigation Basics.

1. Collect a trace by using thread time.
2. Include only stacks that use **runUnattended**, to filter only the thread that has configuration execution. (Add **runUnattended** to the **IncPats** input box.)
3. Fold all CPU, network, and blocked time.
4. Load ER presets for PerfView.
5. Select **ER** > **Other preset**.
6. Look at the names:
    - You probably see the platform code that consumes the most time.
    - You can double-tap (or double-click) and go up through **callees**.

      If you find classes that have the prefix "ERExpression," and if they're functions related to formulas, you can guess the function name based on the class name, and you can look at the ER repo to view the attributes.

Fixes

- If queries consume most of the CPU time, try to reduce the number of queries:
    - Review the ER trace for duplicated queries.
    - Check how many records are fetched, and evaluate how much data should theoretically be fetched.
- If functions consume most of the CPU time, try to find the place in the configuration that consumes the most resources.
- If data collection functions consume most of the CPU time, consider replacing them with *SQL group by* on the model mapping side.

Collecting data

Depending on your environment, you can collect available data in several ways:

- Get the total running time:
    - From the Action center

    - From the event log

- Profile the execution:
    - By using ER tools
    - By using Trace parser
    - By using PerfView

Collecting data in a production environment

Sometimes, you can reproduce issues only in a production environment. Collect data in the following ways:

- By using Trace parser traces
- By using ER execution traces
- By using the total execution time

Collecting data in a development environment

In addition to the tools that you can use in a production environment, use several tools in a development environment:

- Event log (Microsoft-Dynamics-ElectronicReporting). This log can give you the total execution time.
- Common .NET tools, such as PerfView.

Additionally, a development environment gives you more flexibility to experiment. For example, you can turn off parts of reports to see how the execution time is affected.

Tools

Execution time in the Action center

ER can show the execution time of the configuration in the Action center. This option works only for a specific user and a specific company, and only for interactive sessions. To make this feature available, follow these steps.

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, on the Action Pane, on the **Configurations** tab, in the **Advanced settings** group, select **User parameters**.
3. In the **User parameters** dialog box, set the **Show file generation time** option to **Yes**.

Execution time in the event log

1. Open Windows Event Viewer.
2. Under **Applications and Services logs**, open **Microsoft-Dynamics-ElectronicReporting/Operational**.
3. Look for **FormatMapingRun** events where **EventID=2**, because these events contain the information about elapsed time.

Trace parser traces

Because ER is implemented in X++, you can use common X++ tools to analyze performance. For more information, see Take traces by using Trace parser.

There are a few limitations to this approach. Because part of ER is implemented in C#, not all the details are available. However, you can view the data usage details. Additionally, long report runs can exceed trace storage limitations.

ER traces

ER can collect its own traces, and it has visualization and analysis tools for those traces. For more information, see Trace the execution of ER formats to troubleshoot performance issues.

PerfView

Because both X++ and ER are implemented on top of the .NET platform, you can use common .NET tools. For example, you can use the free PerfView tool.

You can also collect data from the command line. For example, the following Windows PowerShell script collects the execution time until any format execution is stopped on the machine.

PowerShell

```
c:\programs\PerfView collect "e:\traces\$(date -format "ddMMyyyy_hhmm").etl" `
    -CircularMB:20000 -ThreadTime `
    -NoNGenRundown `
    -StopOnEtwEvent:Microsoft-Dynamics-ElectronicReporting/FormatMappingRun/Stop
```

There are a few limitations to this approach. You must have administrative access to the machine. Additionally, you must be an experienced developer, because there's a steep learning curve.

Making changes

Use caching

Although caching reduces the amount of time that it takes to fetch data again, it costs memory. Use caching in cases where the amount of fetched data isn't large. For more information and an example that shows how to use caching, see Improve the model mapping based on information from the execution trace.

Reduce volume of data fetched

You can reduce memory consumption for caching by limiting the number of fields in the records of an application table that you fetch at runtime. In this case, you fetch only those field values of an application table that you need in your ER model mapping. You don't fetch other fields in that table. Therefore, the volume of memory that is required to cache fetched records is reduced. For more information, see Improve performance of ER solutions by reducing the number of table fields that are fetched at runtime.

Use a cached, parameterized calculated field

Sometimes, you need to look up values repeatedly. Examples include account names and account numbers. To help save time, you can create a calculated field that has parameters on the top level, and then add the field to the cache.

Use this approach only when the size of the cached data is small. For more information, see Improve the performance of ER solutions by adding parameterized CALCULATED FIELD data sources.

Use a JOIN data source

A **JOIN** data source enables one query to fetch several connected records. You don't need a separate query to fetch each connected record. For example, if you have 1,000 lines, and you fetch item data for each line by relation, you have 1,001 queries (= 1,000 + 1). If you use a **JOIN** data source, you use only one query to fetch the same data. For more information, see Use JOIN data sources in ER model mappings to get data from multiple application tables.

Use the FILTER function instead of the WHERE function

The **FILTER** function runs conditions on SQL Server, whereas the **WHERE** function fetches all data from the list, one record at a time, and applies the condition for each record. For example,

you want to select one record out of 1,000 records. If you use **WHERE**, you fetch all 1,000 records. However, if you use **FILTER**, you fetch exactly one record. **FILTER** can also use indexes on the database side.

Use collected data functions or an accumulated data source

If your configuration has a *group by* component that summarizes previously fetched data by report, the component fetches all the data again. By using collected data functions, you enable ER to accumulate data during the first fetch. For more information, see ER Configure format to do counting and summing.

Rewrite parts of the configuration in X++

ER supports calling methods of tables and classes, including extensions. Consider rewriting parts of the model mapping in X++ to help improve performance.

ER can consume data from the following sources:

- Classes (**object** and **class** data sources)
- Tables (**table** and **table records** data sources)

The ER application programming interface (API) also provides a way to send precalculated data from the calling code. The application suite contains numerous examples of this approach.

Last updated on 12/02/2025

Improve performance of ER solutions by reducing the number of table fields that are fetched at runtime

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can design Electronic reporting (ER) formats that generate outgoing documents in various formats. When a document is generated, an ER format calls data sources that were configured in a corresponding ER model mapping. To configure access to application tables, queries, or entities for record retrieval, you can use ER data sources of the *Table records* type. By default, a data source of the *Table records* type retrieves the values of all fields in the requested records. However, you can configure this type of data source so that it fetches only the field values that are required for the running ER format. This configuration helps reduce memory consumption of the application server that performs data retrieval and further record caching.

To learn more about how to limit the list of fetched fields of data sources of the *Table records* type, complete the example in this article.

Example: Reduce the number of table fields that are fetched at runtime

The following procedures show how a user in the System administrator or Electronic reporting developer role can configure an ER model mapping so that it fetches only the fields that are required to run the ER format, to help reduce the consumption of application server memory.

These procedures can be completed in the **USMF** company in Microsoft Dynamics 365 Finance. No coding is required.

To complete this example, you must have access to the **USMF** company for one of the following roles:

- Electronic reporting functional consultant
- System administrator

In this example, you will use the ER configurations that are provided for the **Litware, Inc.** sample company. Make sure that the configuration provider for the **Litware, Inc.** (`http://www.litware.com`) sample company is listed for the ER framework, and that it's marked as **Active**. If this configuration provider isn't listed, or if it isn't marked as **Active**, follow the steps in Create a configuration provider and mark it as active.

Configure the ER framework

Follow the steps in Configure the ER framework to set up the minimal set of ER parameters. You must complete this setup before you start to use the ER framework to modify data sources of the provided ER solution.

Import the sample ER configurations

If you haven't yet completed the example in the Design a new ER solution to print a custom report article, download and locally store the XML files for the following configurations of the provided ER solution.

Expand table

Content description	File name
ER data model configuration	Questionnaires model.version.1.xml
ER model mapping configuration	Questionnaires mapping.version.1.1.xml
ER format configuration	Questionnaires format.version.1.1.xml

Then follow these steps to upload the configurations of the provided ER solution to your Finance instance.

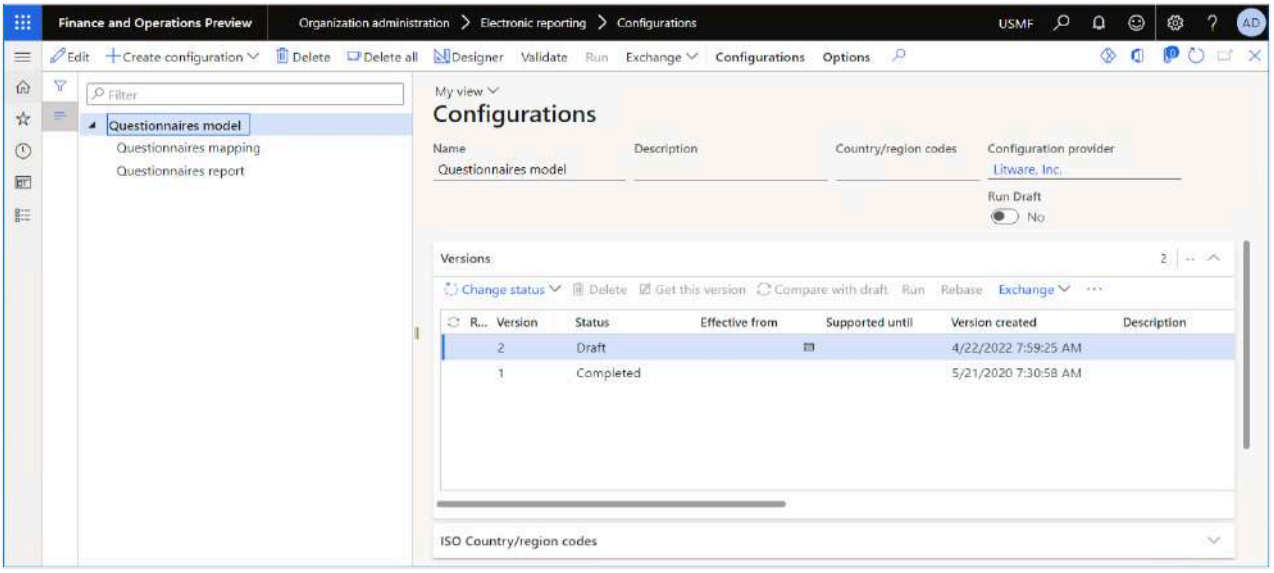
1. Go to **Organization administration** > **Workspaces** > **Electronic reporting**.
2. Select **Reporting configurations**.
3. On the **Configurations** page, import the ER data model configuration.
   a. Select **Exchange**, and then select **Load from XML file**.
   b. Select **Browse**, find and select the **Questionnaires model.version.1.xml** file, and then select **OK**.

4. Import the ER model mapping configuration.
    a. Select **Exchange**, and then select **Load from XML file**.
    b. Select **Browse**, find and select the **Questionnaires mapping.1.1.xml** file, and then select **OK**.

5. Import the ER format configuration.
    a. Select **Exchange**, and then select **Load from XML file**.
    b. Select **Browse**, find and select the **Questionnaires format.1.1.xml** file, and then select **OK**.

6. In the configuration tree, expand **Questionnaires model**.

7. Review the list of imported ER configurations in the configuration tree.



Review the provided ER model mapping

1. On the **Configurations** page, select **Questionnaires mapping**.

2. On the Action Pane, select **Designer**.

3. On the **Model to datasource mapping** page, select **Designer**.

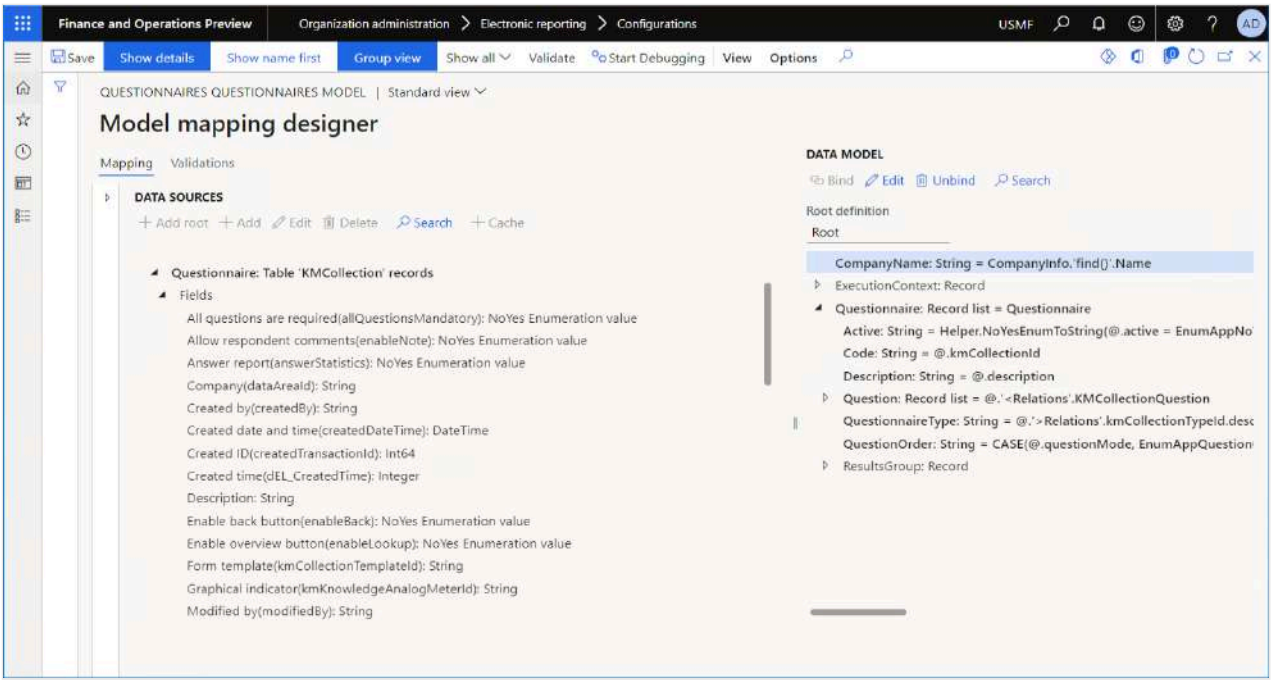
4. On the **Model mapping designer** page, on the Action Pane, select **Group view** to turn on **Group** view.

5. In the **Data model** pane, expand **Questionnaire**.

    Notice that the **Questionnaire** data source has been configured to access the `KMCollection` application table.

6. In the **Data sources** pane, expand **Table records** > **Questionnaire** > **Fields**.

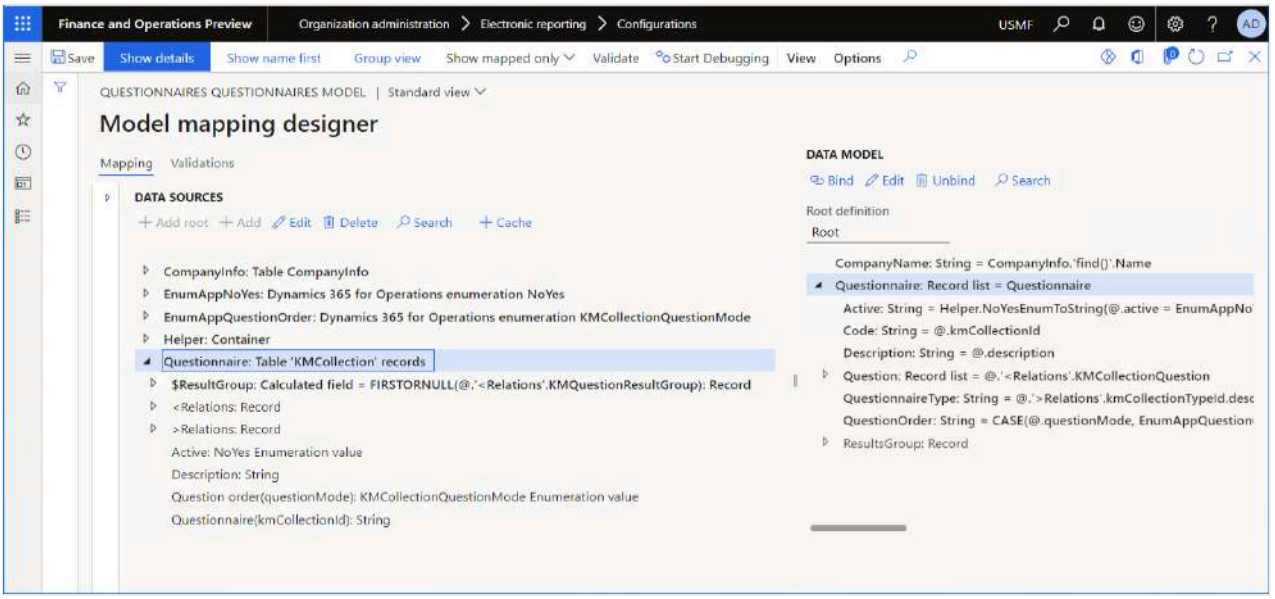
   Notice how many fields from the `KMCollection` application table are exposed by the **Questionnaire** data source of the *Table records* type.



7. On the Action Pane, select **Group view** again to turn off **Group** view, and then select **Show all** > **Show mapped only**.

   Notice that a few fields of the `KMCollection` application table are used to fill in the **Questionnaire** record list in the ER data model:

   - `Active`
   - `Description`
   - `questionMode`
   - `kmCollectionId`



Turn on the ER performance trace

Follow the steps in Turn on the ER performance trace to set the ER user parameters that enable the execution of ER components to be traced.

Run the provided ER format by using the provided model mapping

Follow the steps in Run a designed format from ER to run the provided ER format for a single questionnaire from the **Configurations** page.

Review the execution trace of the first run

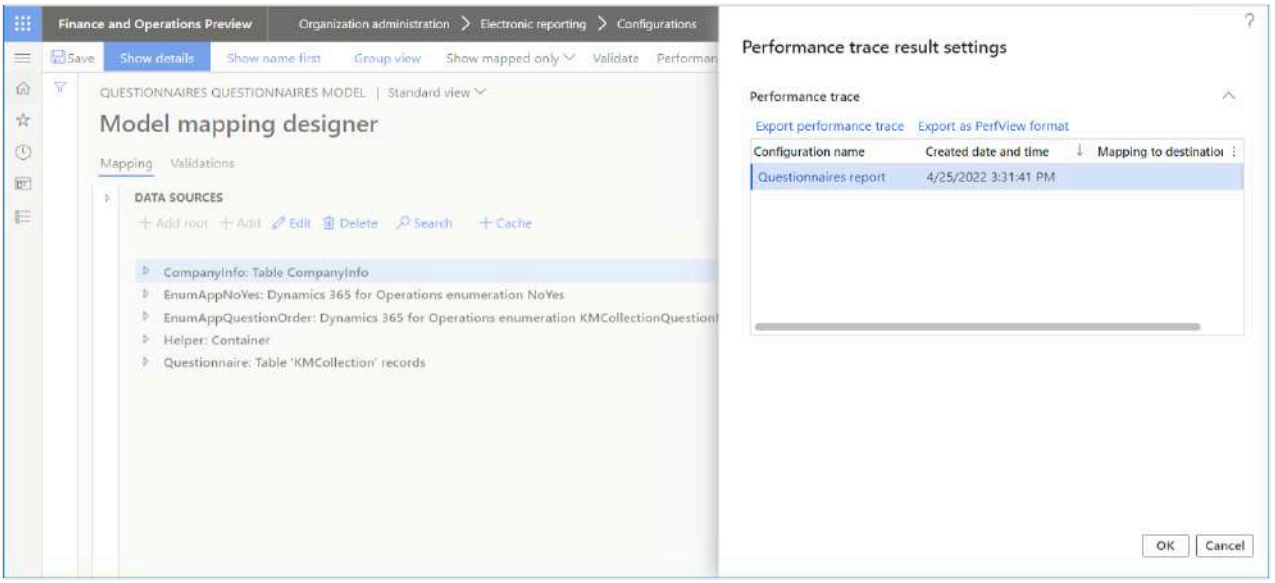
1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, expand **Questionnaires model**, and select **Questionnaires mapping**.

   > **Note**
   >
   > The details on the **Versions** FastTab indicate that you selected the draft version of the **Questionnaires mapping** configuration. Therefore, you can modify the content of this model mapping.

3. On the Action Pane, select **Designer**.
4. On the **Model to datasource mapping** page, select **Designer**.

5. On the **Model mapping designer** page, on the Action Pane, select **Performance trace**.

6. In the **Performance trace result settings** dialog box, select the trace that was generated during the last format run.

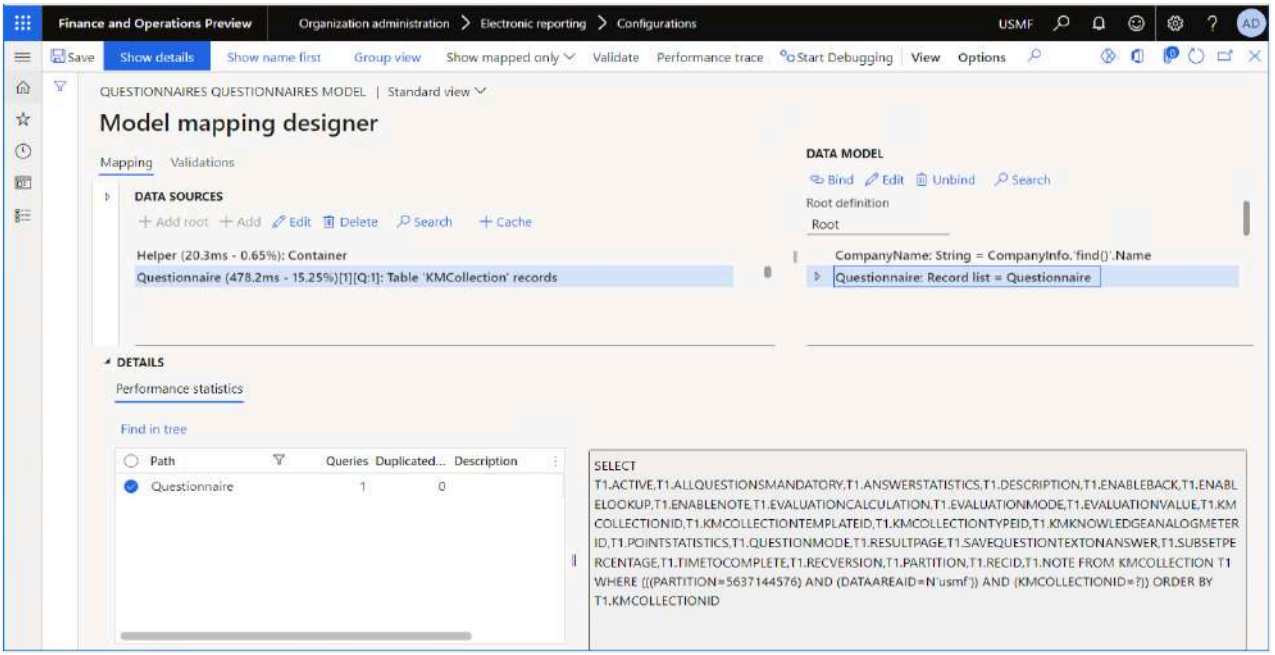


7. Select **OK**.

8. On the **Details** FastTab, filter the **Questionnaire** path that points to the **Questionnaire** data source.

9. Review the details of the database query that was generated when the **Questionnaire** data source was called.

   Notice that all fields of the `KMCollection` application table were fetched at runtime when the **Questionnaire** data source was called.

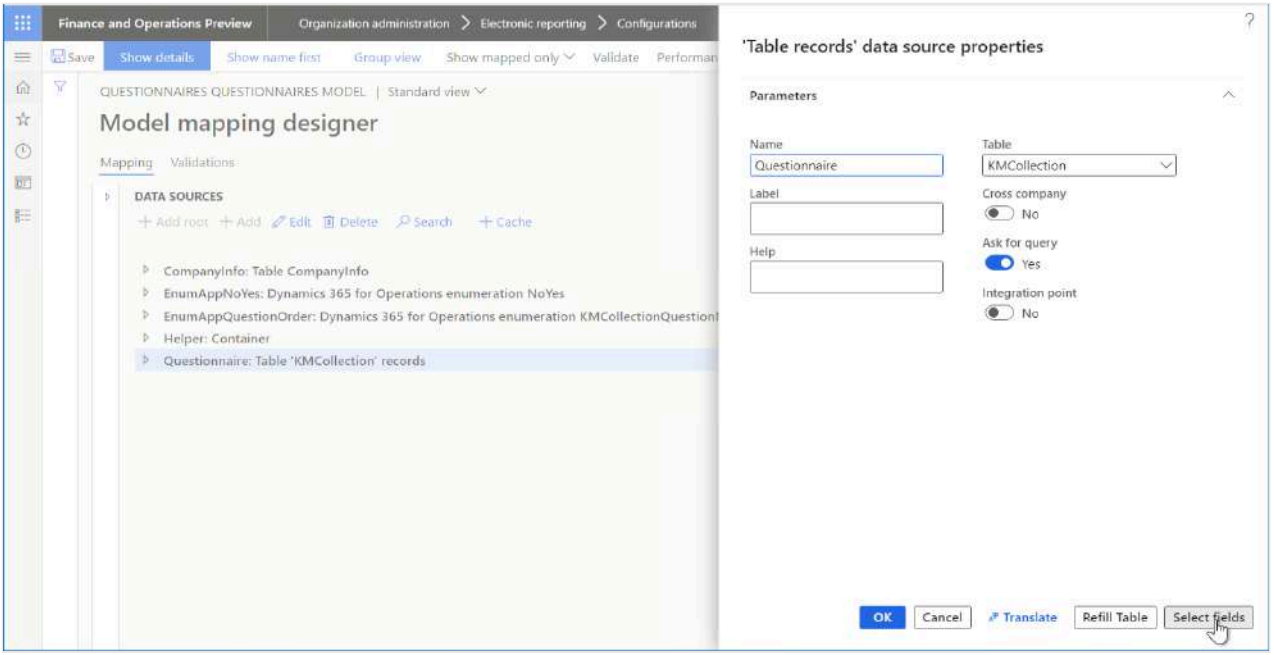


Modify the provided ER model mapping

1. On the **Model mapping designer** page, in the **Data sources** pane, select the **Questionnaire** data source.

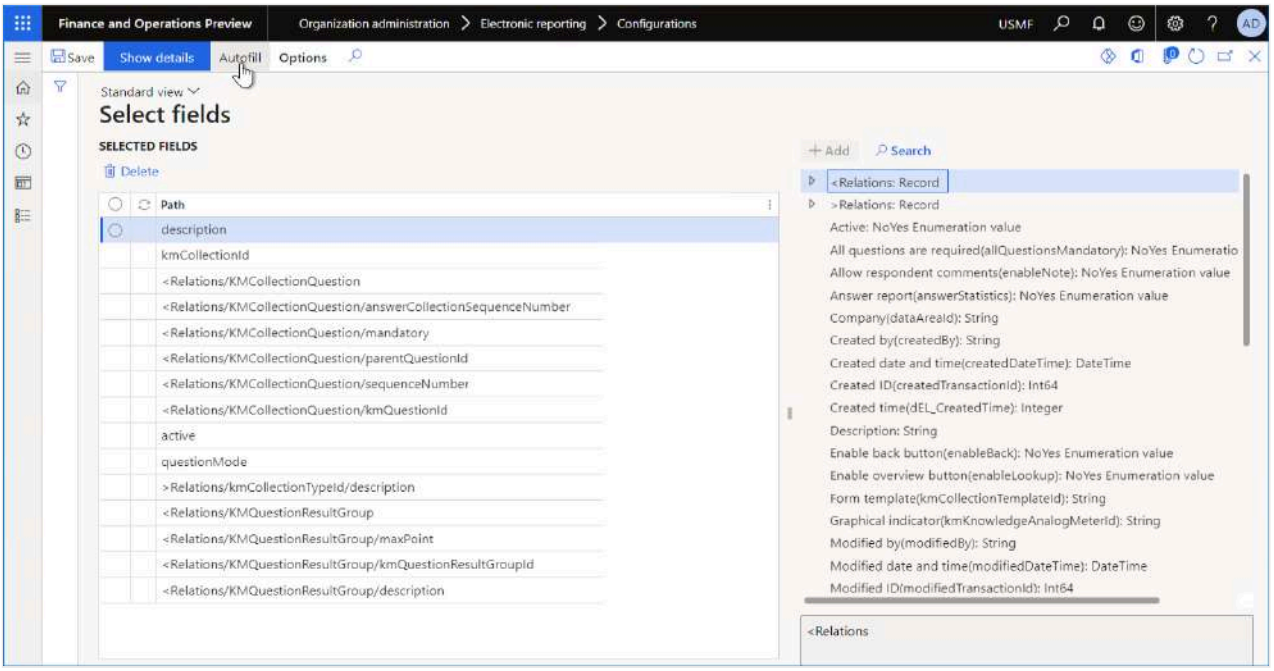
2. In the **Data sources** pane, select **Edit**.

3. In the **Data source properties** dialog box, select **Select fields** to specify the list of fields of the referenced `KMCollection` application table that will be fetched at runtime when the editable **Questionnaire** data source is called.



4. On the **Select fields** page, select **Autofill**.

   The **Selected fields** list is automatically filled in, based on preconfigured artefacts of the model mapping. All fields and relations of the referenced table that are mentioned in any binding, formula, or data source of the model mapping are added to the list.

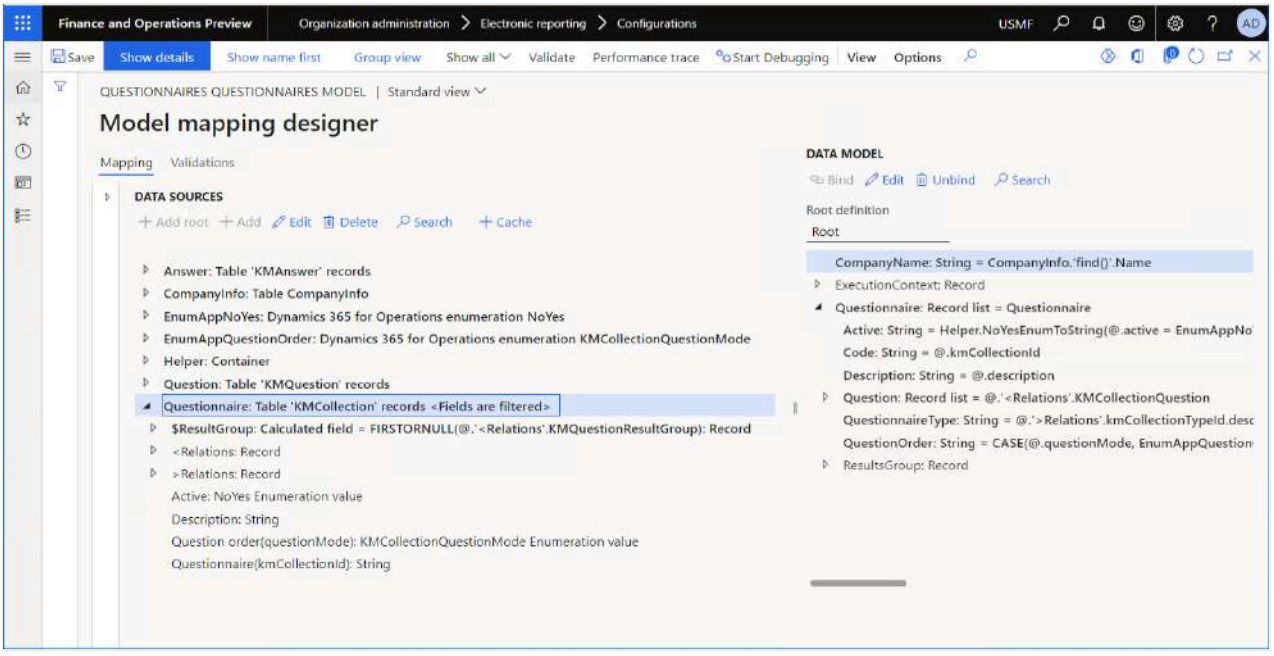


5. Select **Save**, and close the **Select fields** page.

6. Select **OK** to store the changes that you've made to the data source settings.

7. On the Action Pane, select **Show all**.

   Notice that the **Questionnaire** data source now shows the text **<Fields are filtered>**. This text indicates that the data source has been configured to fetch a limited number of fields from the referenced application table.



8. Select **Save** to store the changes that you've made to the editable model mapping.

> **Note**

At runtime, ER analyzes the added relations and adds all fields that are used in them to the database query, even if those fields weren't explicitly added to the list of fetched fields at design time.

Run the provided ER format by using the updated model mapping

Follow the steps in Run a designed format from ER to run the provided ER format for a single questionnaire from the **Configurations** page.

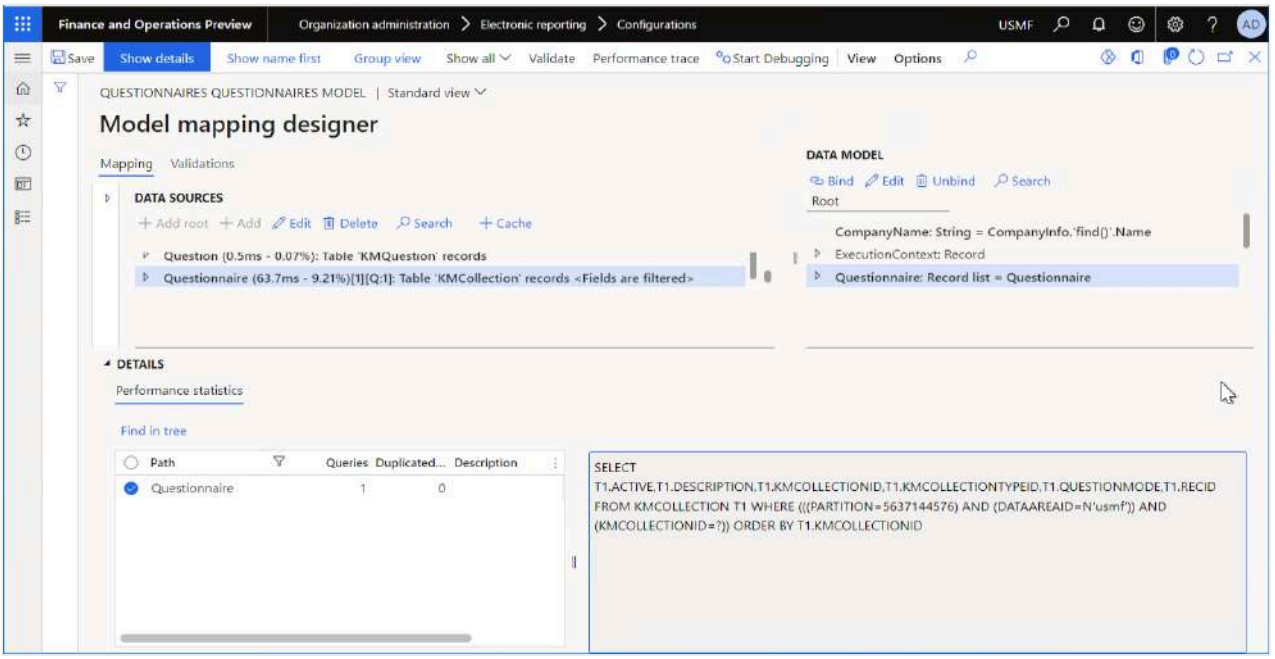
Review the execution trace of the second run

1. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
2. On the **Configurations** page, expand **Questionnaires model**, and select **Questionnaires mapping**.
3. On the Action Pane, select **Designer**.
4. On the **Model to datasource mapping** page, select **Designer**.
5. On the **Model mapping designer** page, on the Action Pane, select **Performance trace**.
6. In the **Performance trace result settings** dialog box, select the trace that was generated during the last format run.
7. Select **OK**.
8. On the **Details** FastTab, filter the **Questionnaire** path that points to the **Questionnaire** data source.
9. Review the details of the database query that was generated when the **Questionnaire** data source was called.

   Notice that only the fields that are required to fill in the data source were fetched at runtime from the `KMCollection` application table when the **Questionnaire** data source was called.

   Note

Some fields, such as the fields for the partition ID, data area ID, and record ID, are automatically added by the Data Management framework of the Finance app.



You can use this technique to reduce the number of fetched records when you must reduce memory consumption by the running ER model mapping and ER format.

Limitations

When you limit the number of fetched fields for a data source of the *Table records* type, you can't use the methods of an application table that the data source refers to, because the application metadata doesn't provide information about table fields that are required to call those methods.

Usage notes

Although the **Autofill** command automatically adds fields, it doesn't automatically delete previously added fields, even if they are no longer used in bindings, formulas, and data sources of the editable model mapping.

When you select **Autofill**, ER analyzes bindings, formulas, and the data sources that the editable model mapping had when you opened it for editing. If you change bindings, formulas, and data sources of the editable model mapping, and you want to use the **Autofill** command, close the model mapping designer, and then re-open it to edit your model mapping.

Additional resources

- Trace the execution of ER formats to troubleshoot performance issues
- Troubleshooting performance issues in ER configurations

Last updated on 04/08/2026

Business document management overview

Business users use the **Business document management** (BDM) workspace to configure formats for outbound documents in accordance with the legal requirements of various countries or regions or company-specific needs. **Business document management** is built on top of the Electronic reporting (ER) framework and enables business users to edit business document templates by using Microsoft 365 Apps for business or an appropriate Microsoft 365 desktop applications. Edits to the documents might include changing business document designs and adding placeholders for extra data without source code changes and new deployments. Updating business document templates doesn't require any prior knowledge of the ER framework.

The ER framework generates outbound documents in Excel and Word formats by using predefined templates. The templates are populated with required data according to the configured dataflow while required documents are generated. Each configured format can be published as part of an ER solution to generate specific outbound documents. This process is represented by an ER format configuration that can contain templates you can use to generate different outbound documents.

> **Note**
>
> **Business document management** enables you to modify templates that are used to produce business documents such as orders, invoices, and more. When you modify a template and publish a new version, the system uses this version to generate required business documents. You can't use Business document management to modify already generated business documents.

Supported deployments

Currently, the **Business document management** workspace is implemented only for cloud deployments. If this feature is critical to your on-premises deployment, provide feedback on the Ideas site.

Supported Microsoft 365 and Office applications

To use **Business document management** for editing templates in Excel or Word formats by using Microsoft Office desktop applications, you must have Microsoft Office 2010 or later installed and

a configured **SharePoint** integration to host the files during editing. This support is available in cloud and on-premises deployments.

To use **Business document management** for editing templates in Excel or Word formats to edit templates online in Microsoft 365 (Word or Excel for the web), you must have a Microsoft 365 Office for the web subscription and a configured **SharePoint** integration to host the files during editing.

Additional resources

Electronic reporting (ER) overview

ER Design a configuration for generating reports in OPENXML format (November 2016)

Design ER configurations to generate reports in Word format

Embed images and shapes in documents that you generate by using ER

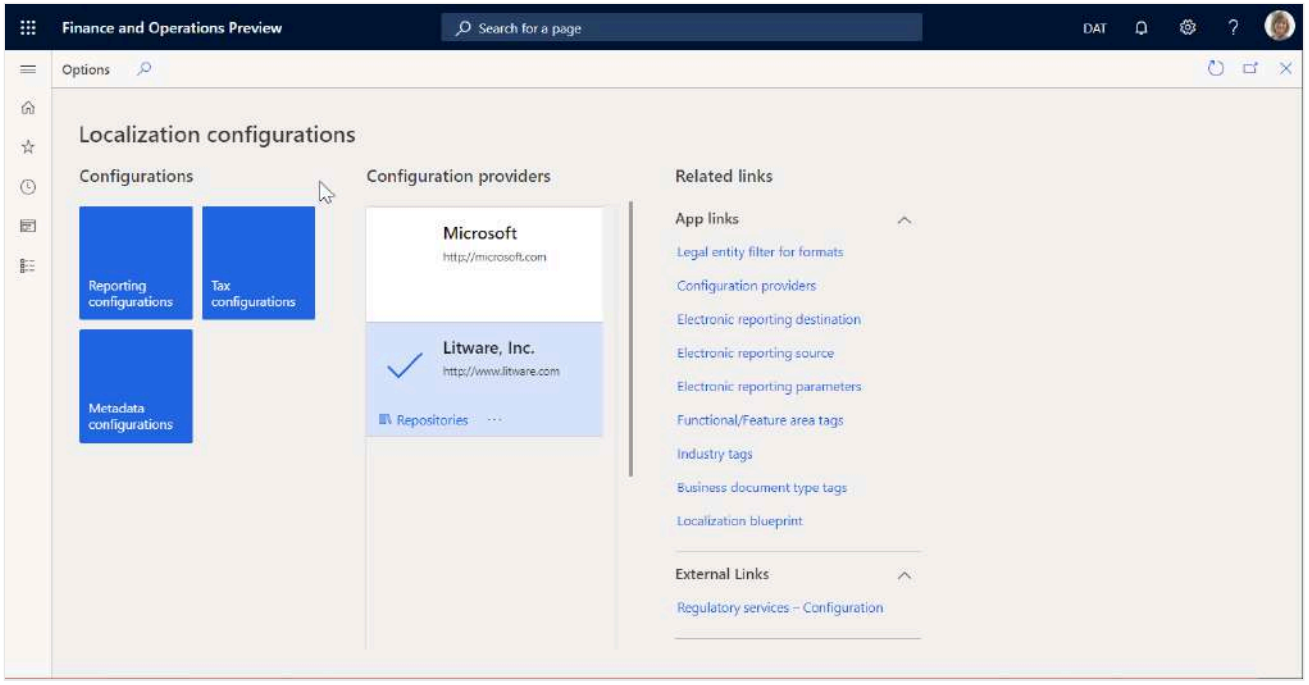
Configure Electronic reporting (ER) to pull data into Power BI

Last updated on 08/14/2026

Configure Electronic reporting parameters for business documents

Business document management is built on top of the ER framework. Complete configuring the ER parameters to start working with **Business document management**.

- Set up the ER parameters as described in Configure the Electronic reporting (ER) framework.
- Add a new configuration provider as described in Create configuration providers and mark them as active.



Import ER solutions

This procedure uses sample ER configurations. You must import into your current instance of Dynamics 365 Finance the ER configurations that contain the business document templates you can edit by using Business document management. To complete this procedure, download and locally store the following files.

Sample ER customer invoicing solution

Expand table

File	Content
Customer invoicing model.version.2.xml	ER data model configuration
Customer FTI report (GER).version.2.3.xml	Free text invoice ER format configuration

Sample ER payment checks solution

Expand table

File	Content
Model for cheques.version.10.xml	ER data model configuration
Cheques printing format.version.10.9.xml	Payment cheque ER format configuration

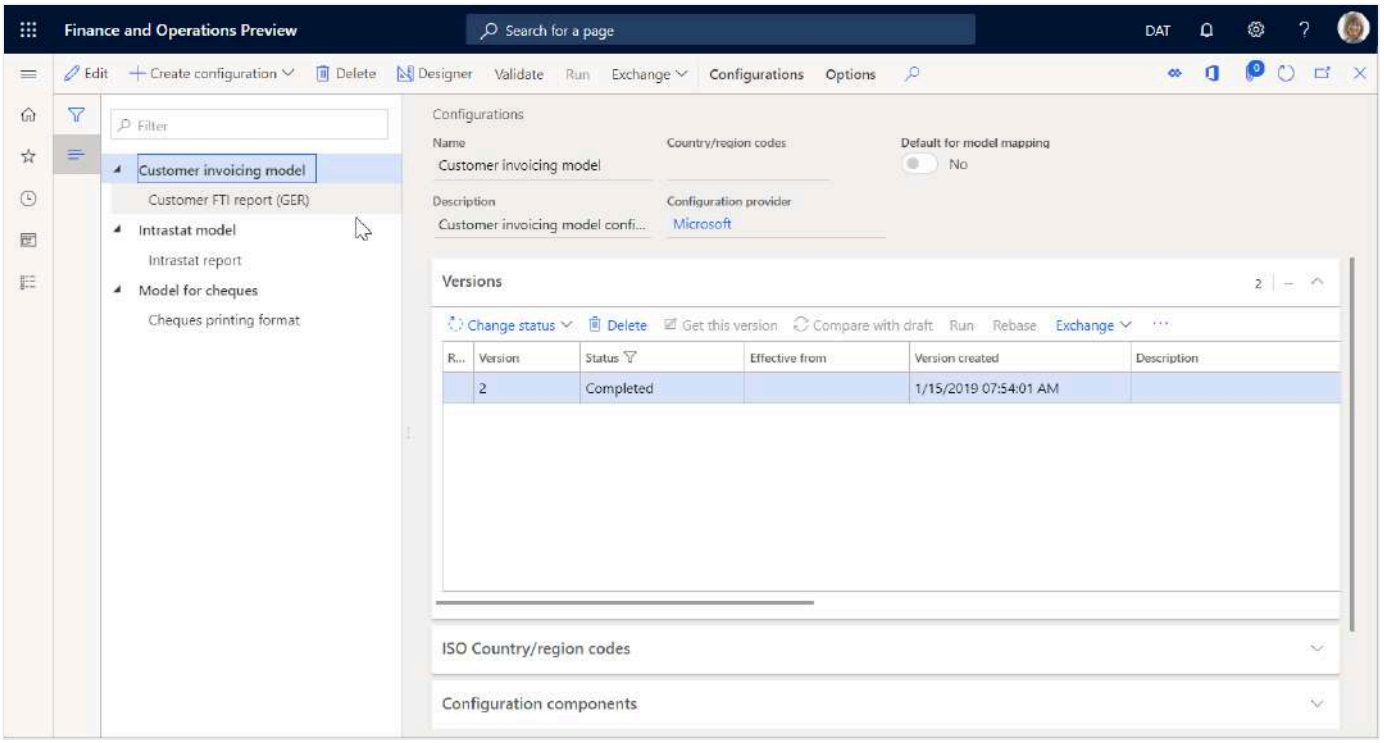
Sample ER foreign trade solution

Expand table

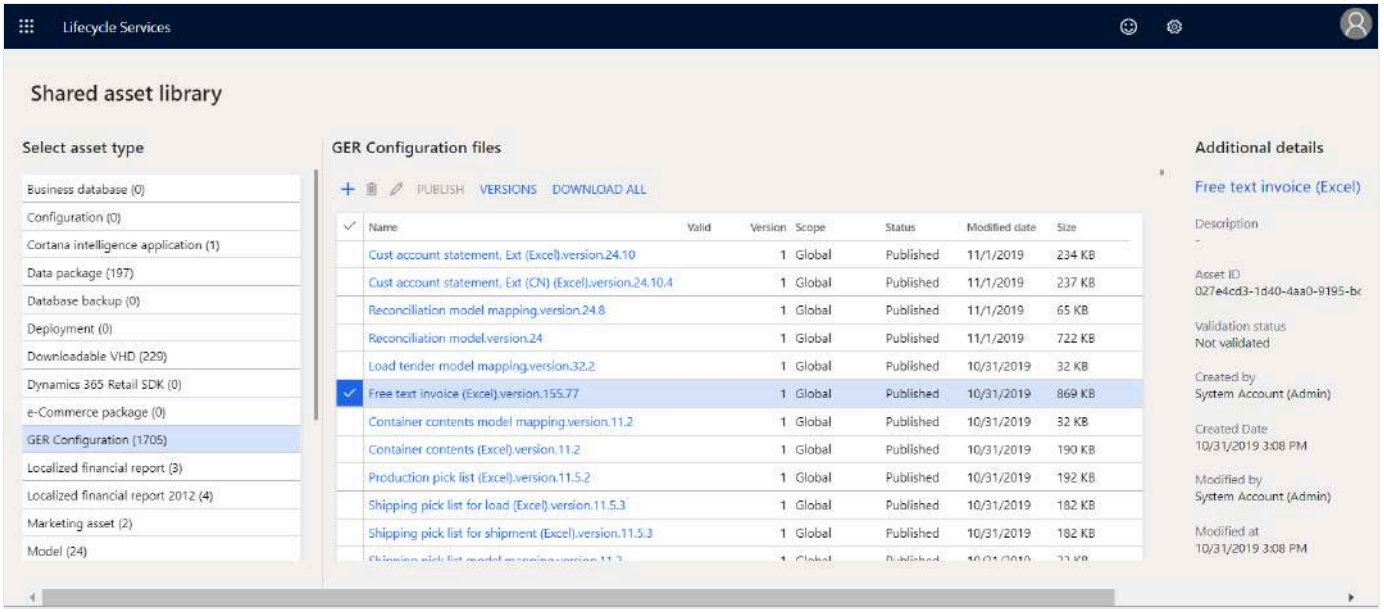
File	Content
Intrastat model.version.1.xml	ER data model configuration
Intrastat report.version.1.9.xml	Intrastat control report ER format configuration

Use the following procedure to import each file. Import the ER *data model* configuration of each ER solution in the preceding tables before you import the corresponding ER *format* configuration.

1. Open **Organization administration** > **Electronic reporting** > **Configurations**.
2. At the top of the page, select **Exchange**.
3. Select **Load from XML file**.
4. Select **Browse** to load the required XML file.
5. Select **OK** to confirm the configuration import.



Alternatively, you can import the officially published ER format configurations from Microsoft Dynamics Lifecycle Services. For example, to complete this procedure, you can import the latest version of the **Free text invoice (Excel)** ER format configuration. The corresponding ER data model and ER model mapping configurations are imported automatically.



For more information about importing ER configurations, see Manage the ER configuration lifecycle.

Configure parameters

Use the information in the following sections to set up the basic parameters for **Business document management**.

Prerequisites for parameter setup

Before you set up **Business document management**, configure **SharePoint** storage in the **Document management** framework. This configuration is required so you can host templates in SharePoint for online editing in Word or Excel for the web (and for desktop editing). To edit templates online in Microsoft 365 (Word or Excel for the web), you must have a Microsoft 365 Office for the web subscription and a configured **SharePoint** integration to host the files during editing.

Configure SharePoint storage

On the **Document management parameters** page, on the **SharePoint** tab, enter the Microsoft SharePoint address in the **SharePoint server** field and select **Test interactive SharePoint connection** to verify valid connection with SharePoint.

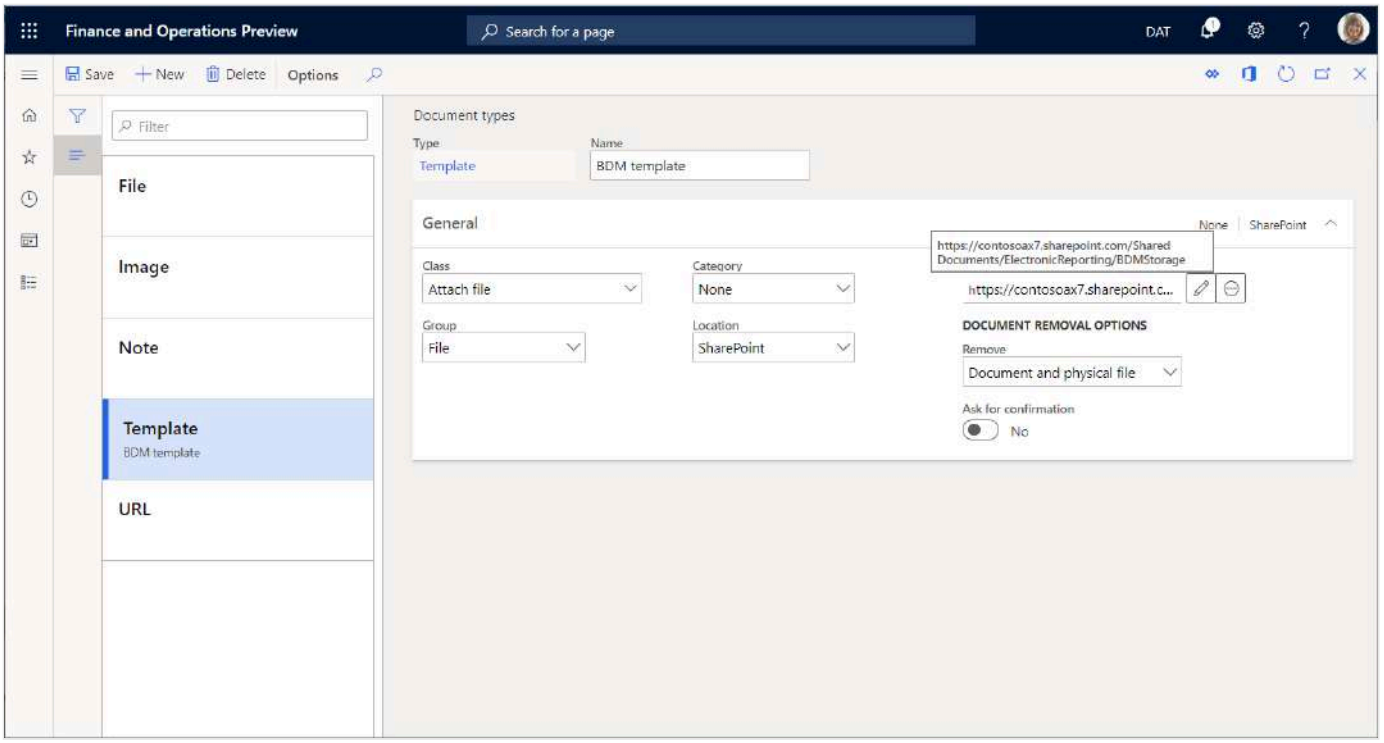
Set up document type in the Document management framework

Create a document type that **Business document management** can use for template editing. This document type specifies a temporary storage of documents in Office formats (Excel and Word) that are used as templates for ER reports. For this document type, go to the **Document types** page, select the following attribute values.

Expand table

Attribute name	Attribute value
Class	Attach file
Group	File
Location	SharePoint

For information about how to set up the required document management parameters and document types, see Configure document management.



Set up parameters

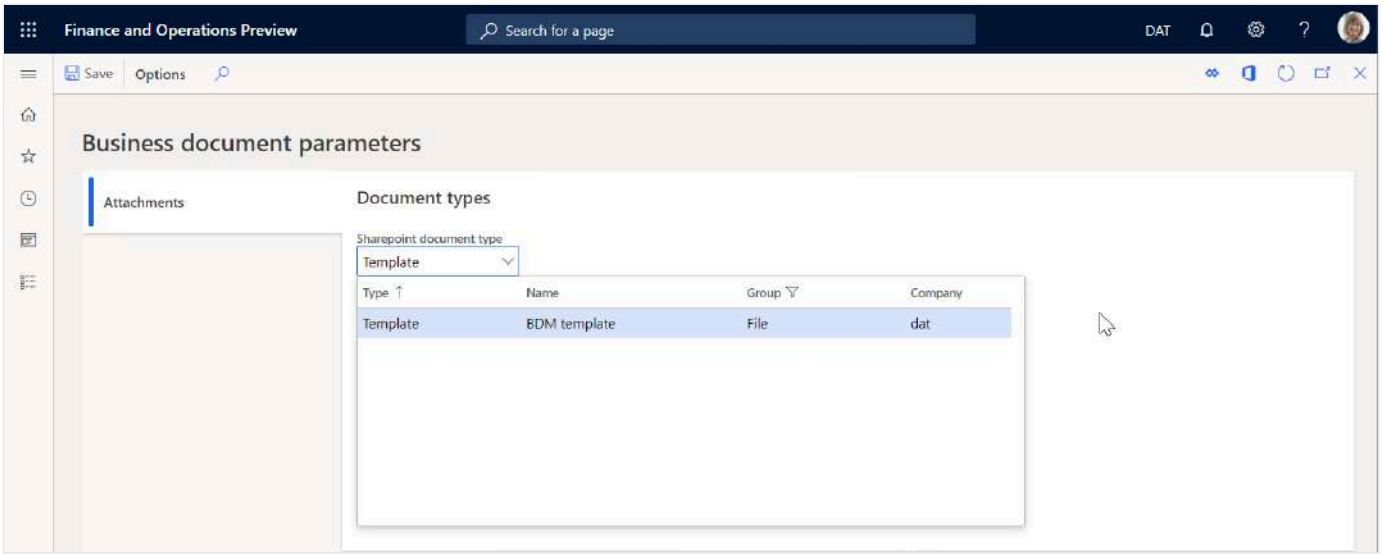
Set up basic **Business document management** parameters on the **Business document parameters** page. Only specific users can access the page. This access includes:

- Users assigned to the **System administrator** role.
- Users assigned to any role that is configured to perform the duty, **Maintain Business document management parameters** (AOT name **ERBDMaintainParameters**).

Use the following procedure to set up the basic parameters for all legal entities.

1. Sign in as a user with access to the **Business document parameters** page.
2. Go to **Organization administration** > **Electronic reporting** > **Business document management** > **Business document parameters**.
3. On the **Business document parameters** page, on the **Attachments** tab, in the **SharePoint document type** field, define the document type that you want to use to temporarily store templates in Office formats while you edit them online in Microsoft 365 (Word/Excel for the web) or in the Office desktop applications.

> ⓘ **Note**
>
> Only document types that are configured by using a SharePoint location are available for this parameter.



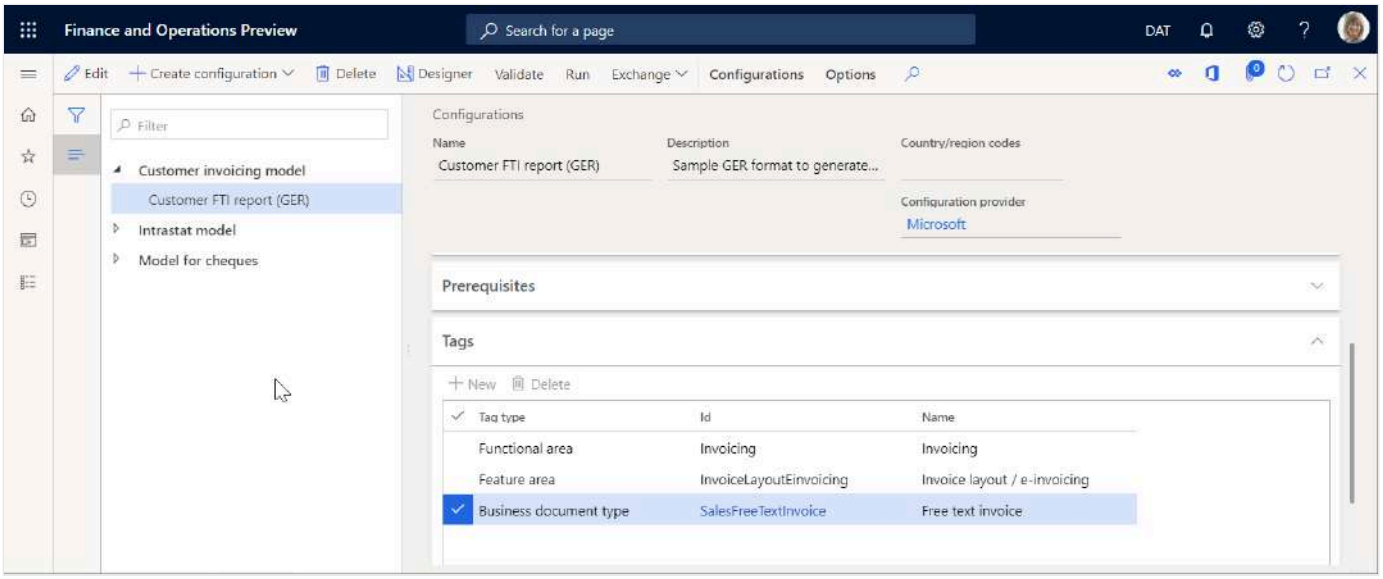
The selected document type is company-specific and is used when the user works with **Business document management** in the company for which the selected document type is configured. When the user works with **Business document management** in another company, the same selected document type is used if one isn't configured for this company. When you configure a document type, it replaces the one selected in the **SharePoint document type** field.

> **Note**
>
> The **SharePoint document type** parameter defines a SharePoint folder as temporary storage for templates that are editable by using either Microsoft Excel or Word. Set up this parameter if you plan to use these Office desktop applications for editing templates. For more information, see Edit a Business document template.

Last updated on 08/07/2026

Manage access permissions to edit templates for configurable business documents

By default, when you don't enable access to **Business document management** permissions, every user with access to the **Business document management** workspace sees all of the ER solution templates that are available. The **Business document management** workspace shows only those templates that reside in ER format configurations and that are marked by a **Business document type** tag.



You can restrict the list of templates available in the **Business document management** workspace by configuring access permissions. This restriction is important when different templates produce business documents for different business domains (functional areas), and you want to allow specific users access to different templates for editing in the **Business document management** workspace.

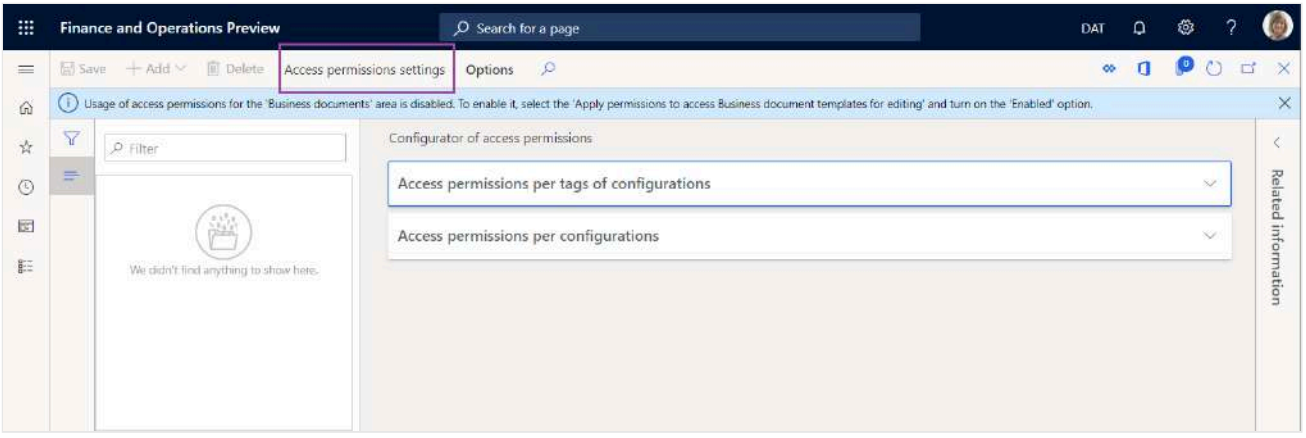
Set **Business document management** access permissions on the **Configurator of access permissions**. Only the following users can access the page:

- Users assigned to the **System administrator** role.
- Users assigned to any other role that is configured to perform the duty, **Configure permissions to access Business document templates for editing** (AOT name **ERBDTemplatesSecurity**).

Use the following procedure to set up the access **Business document management** permissions for all legal entities.

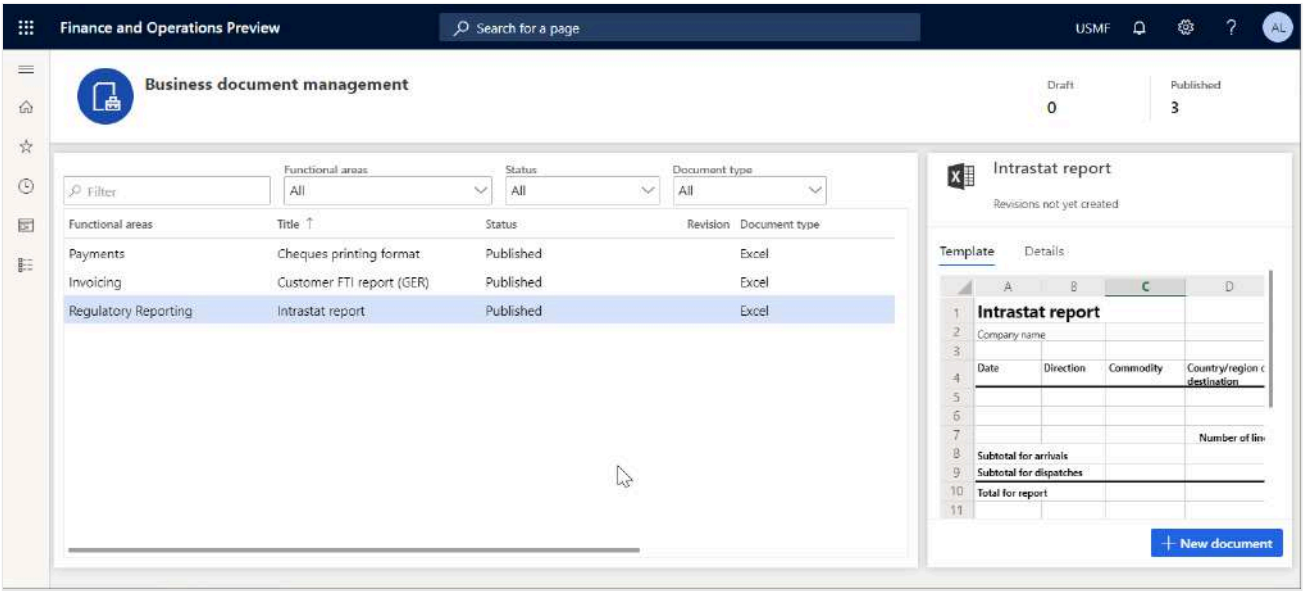
1. Sign in as a user with access to the **Configurator of access permissions** page.
2. Go to **Organization administration** > **Electronic reporting** > **Business document management** > **Manage access permissions**.

   Pay attention to the notification that informs you that the usage of access permissions for **Business document management** isn't currently enabled.

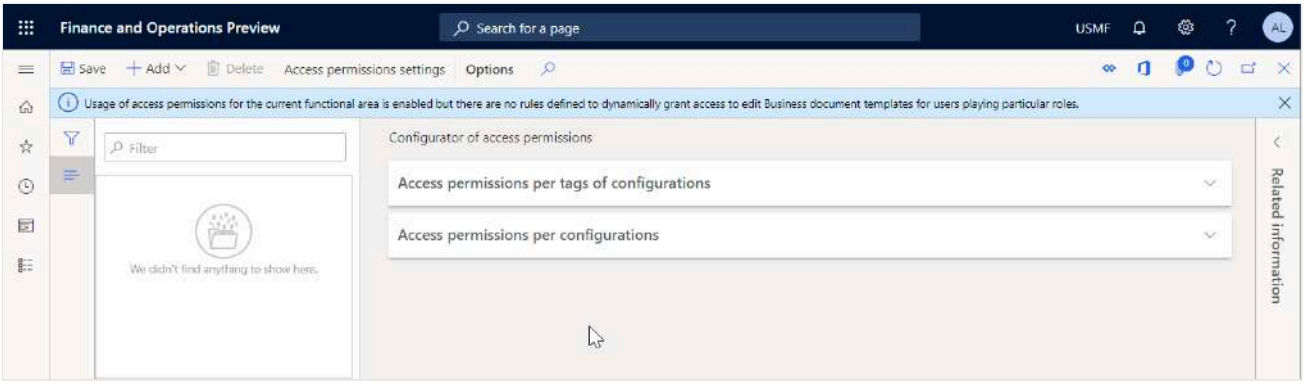


   With this setting, every user assigned to any security role that is configured to perform the **Manage Business document templates** (AOT name **ERBDManageTemplates**) duty can open the Business document management workspace and can edit any template that is available.

   The following graphic shows what is available in the **Business document management** workspace for users assigned to the **Accounts receivable clerk** role. With the current access permissions setting, the user can edit business document templates from different functional areas including invoicing, regulatory reporting, and payments.

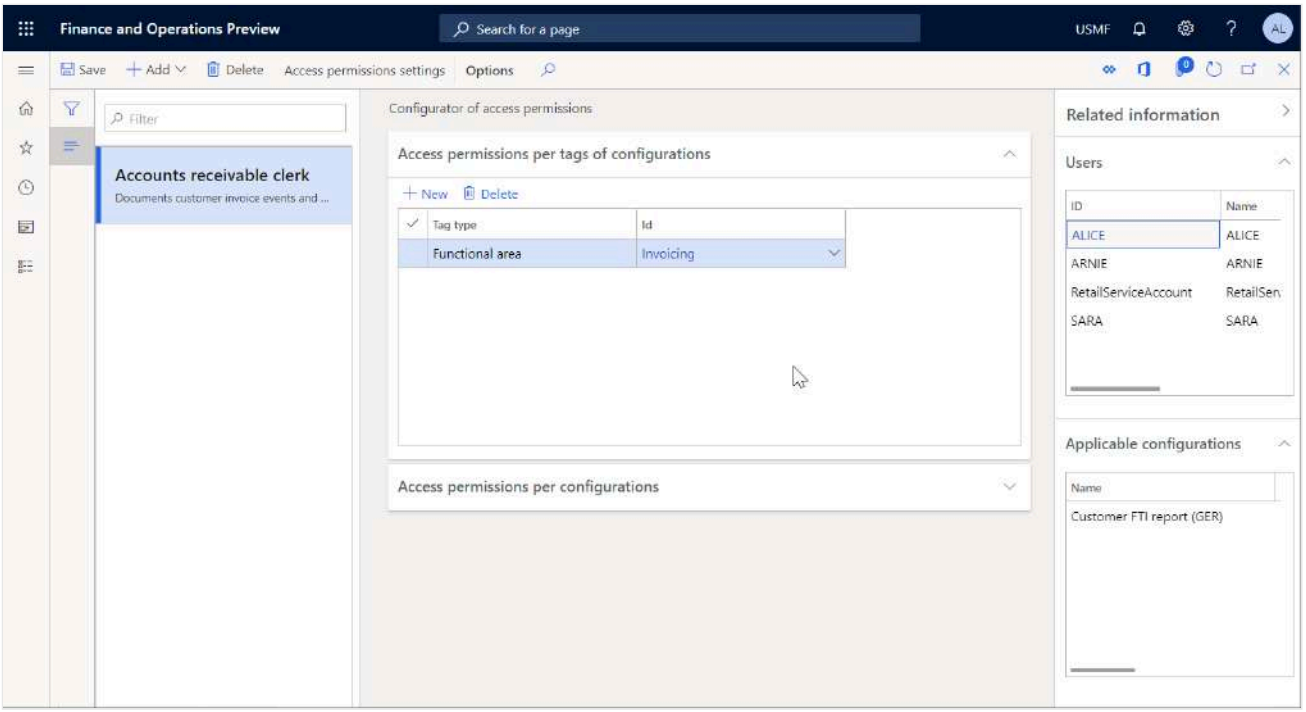


3. On the **Configurator of access permissions** page, select **Access permissions setting**.
4. In the **Settings of access permissions to edit templates** dialog box, enable the **Apply configured access permissions** option.
5. Select **OK** to confirm that Business document management access permissions are enabled.



6. Select **Add** to enter a new business role for which permissions to access Business document management templates must be configured.
7. In the **Security roles** dialog box, select the **Accounts receivable clerk** role and then select **OK** to confirm the role selection.
8. On the **Access permissions per tags of configurations** tab, select **New**.
9. In the **Tag type** field, select **Functional area**, and in the **ID** field, select **Invoicing**.
10. Select **Save** to store configured access permissions for the selected role.

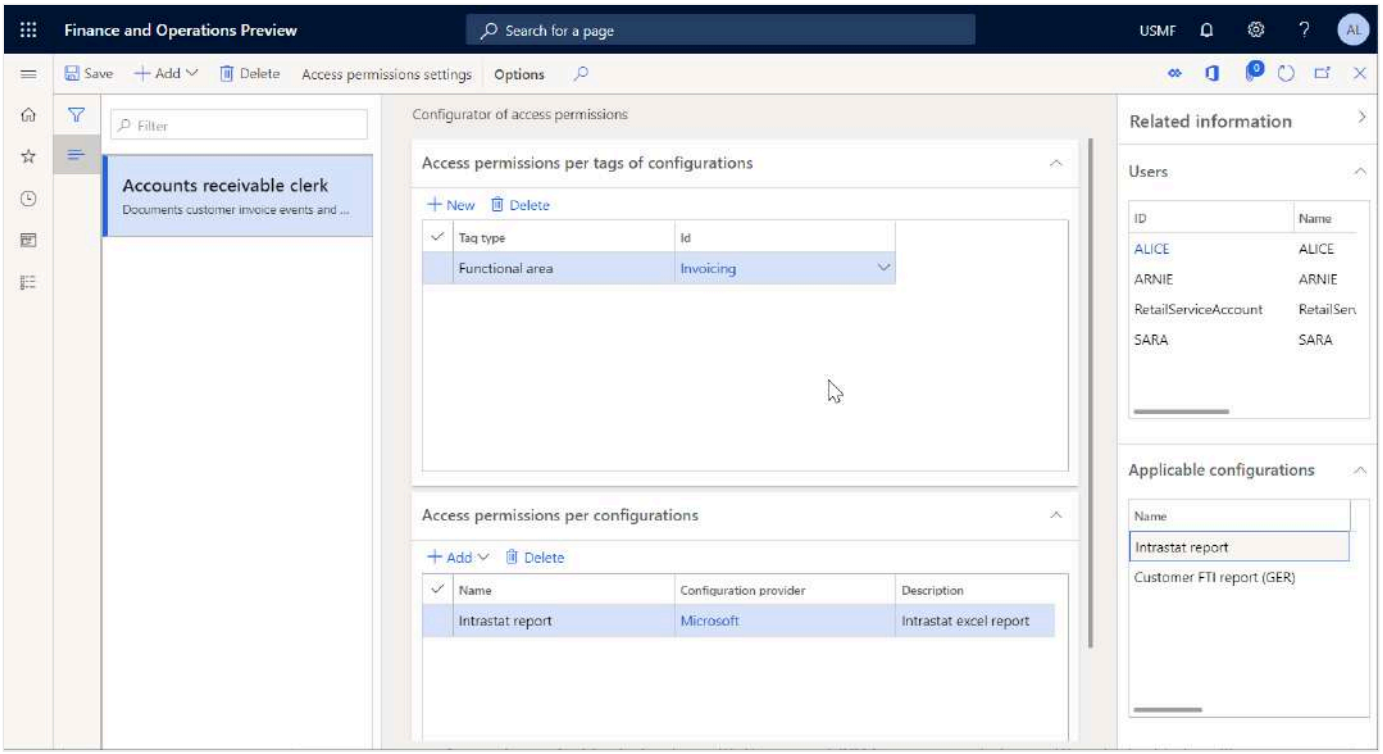
    The current setting means that for any user who is assigned to the **Accounts receivable clerk** role and performing the duty, **Manage Business document templates** (AOT name **ERBDManageTemplates**), ER format configuration templates that have the **Invoicing** value for the **Functional area** tag are available to edit in the Business document management workspace.
11. Switch the **Related information** pane from the right side of the current page. The **Related information** pane shows how the configured access permissions are applied, including what ER configuration templates are available for users that are assigned to the **Accounts receivable clerk** role.



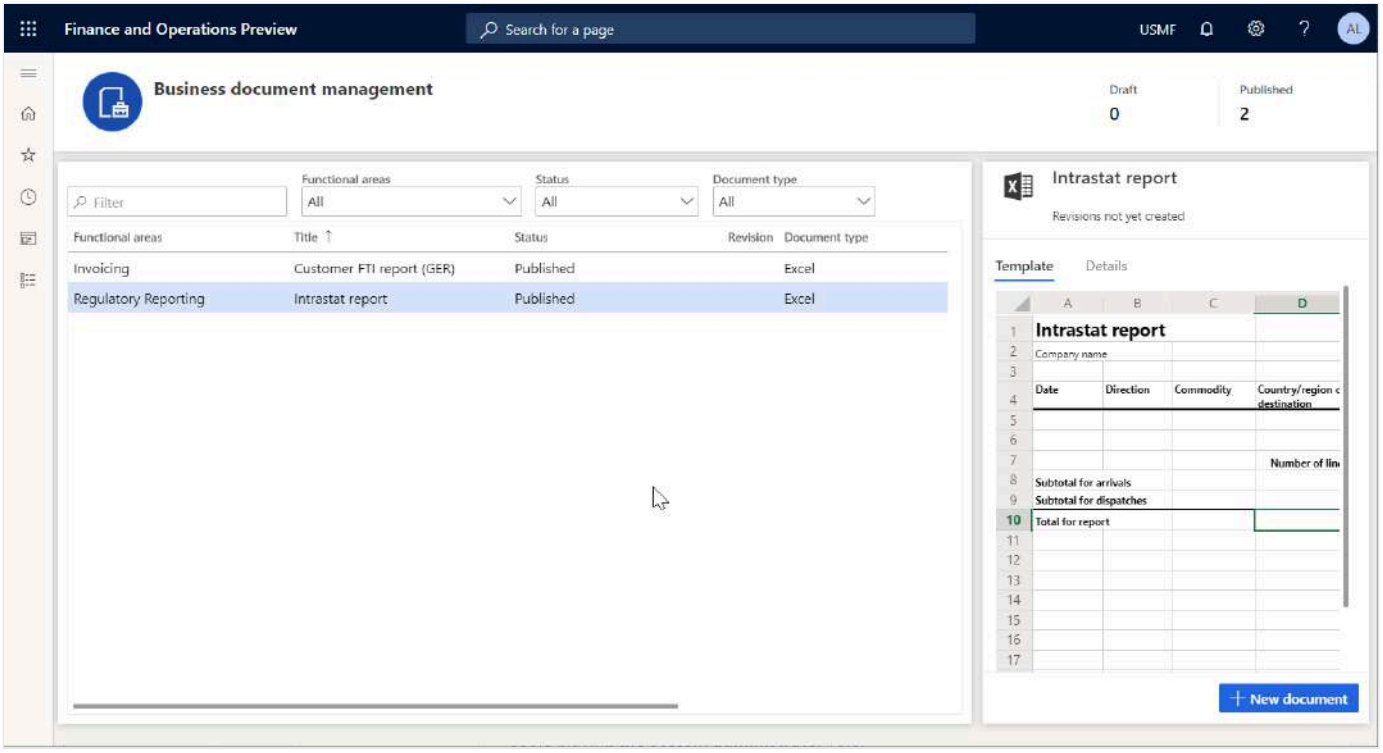
12. On the **Access permissions per configurations** tab, select the **Add** option.
13. In the **Select configuration** dialog box, mark the **Intrastat report** ER format configuration.
14. Select **OK** to confirm the entry of the selected configurations and then select **Save** to store the configured access permissions for the selected role.

The current setting means that for any user assigned to the **Accounts receivable clerk** role and performing the duty, **Manage Business document templates** (AOT name **ERBDManageTemplates**), the following templates are available to edit in the Business document management workspace:

- Templates that have the value, **Invoicing** for the **Functional area** tag.
- Templates from ER format configurations that are listed on the **Access permissions per configurations** tab (templates from the **Intrastat report** format configuration of the **Statutory reporting** domain in this example).



The following graphic shows what the **Business document management** workspace provides to a user assigned to the **Accounts receivable clerk** role. With the current Business document management access permissions setting, the user can edit business document templates from the **Invoicing** domain and the **Intrastat report** ER format configuration. Templates from the **Payments** domain aren't accessible for the **Accounts receivable clerk** role.



> **Note**

The **Access permissions per configurations** rules use the unique identification ID of an ER format configuration. These rules stay even if you delete an ER configuration that refers to them. When you import deleted configurations back to this instance, these rules refer to them again. You don't need to set up the rules again after you import the deleted configurations.

Last updated on 08/07/2026

Use Business document management to edit templates

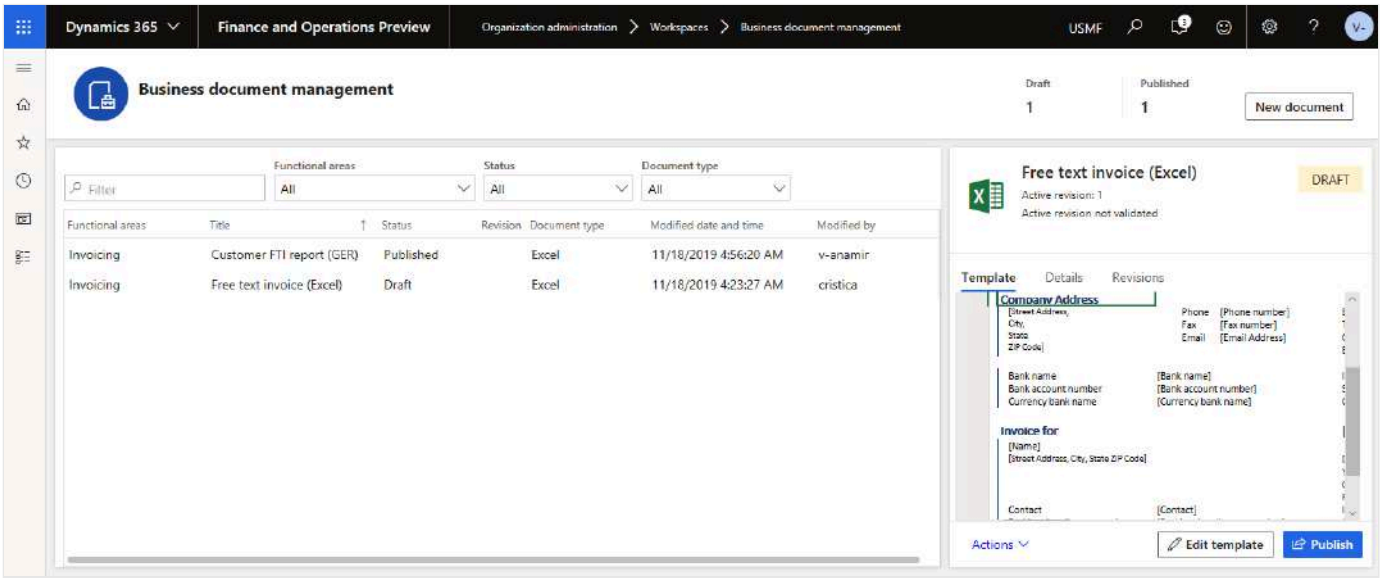
Business users can access business document templates for editing in the **Business document management** workspace. Only the following users can access the **Business document management** workspace:

- Users assigned to the role, **System administrator**.
- Users assigned to any role that is configured to perform the duty, **Manage Business document templates** (AOT name **ERBDManageTemplates**).

Use the following procedure to edit free text invoice templates in the **Business document management** workspace. Before you complete this procedure, you must complete all of the preceding procedures described in Configure Electronic reporting parameters for business documents.

1. Sign in as a user with access to the **Business document management** workspace.
2. Open the **Organization administration** > **Business document management** workspace.

The **Business document management** workspace shows only the templates that the current active provider owns and that are located in the current instance of Dynamics 365 Finance.



On the **Template** tab, you can preview the content of the selected template. Select the **Details** tab to review details of the selected template as well as details of an ER format configuration this template resides in.

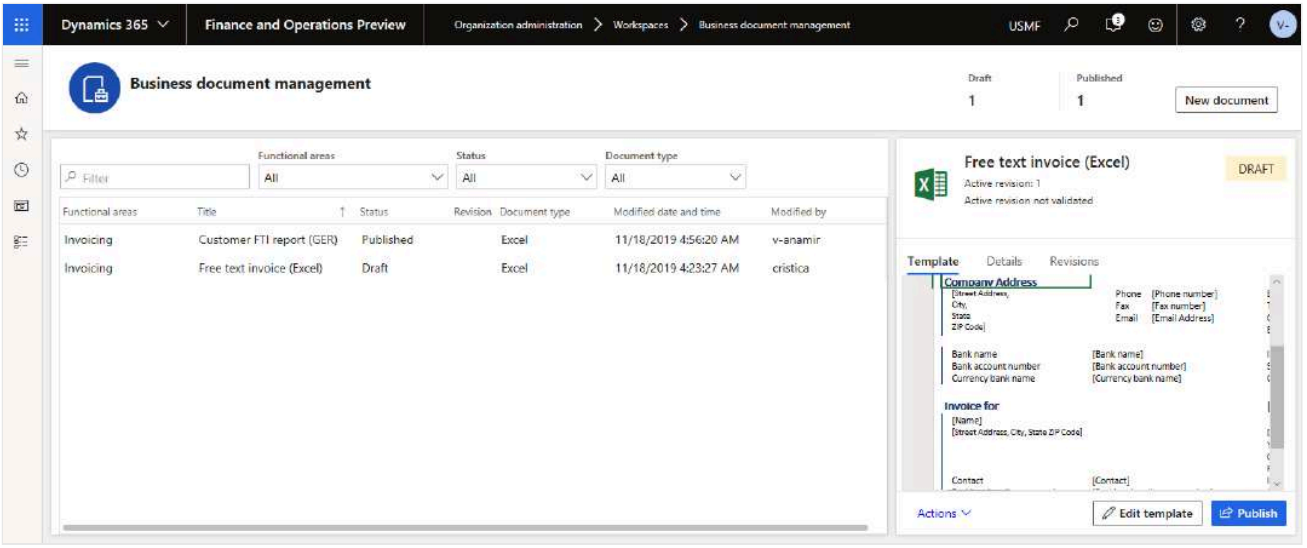
You can use the **New document** button in the **Business document management** workspace to create and edit a template in an Electronic reporting (ER) format configuration that another provider provides and that is located in the current Finance instance. You can also use the **New document** button to upload a new template from an Excel workbook. Additionally, you can use the **New document** button to create and edit a template in an ER format configuration that is stored in the Global repository.

Edit a template that another provider owns

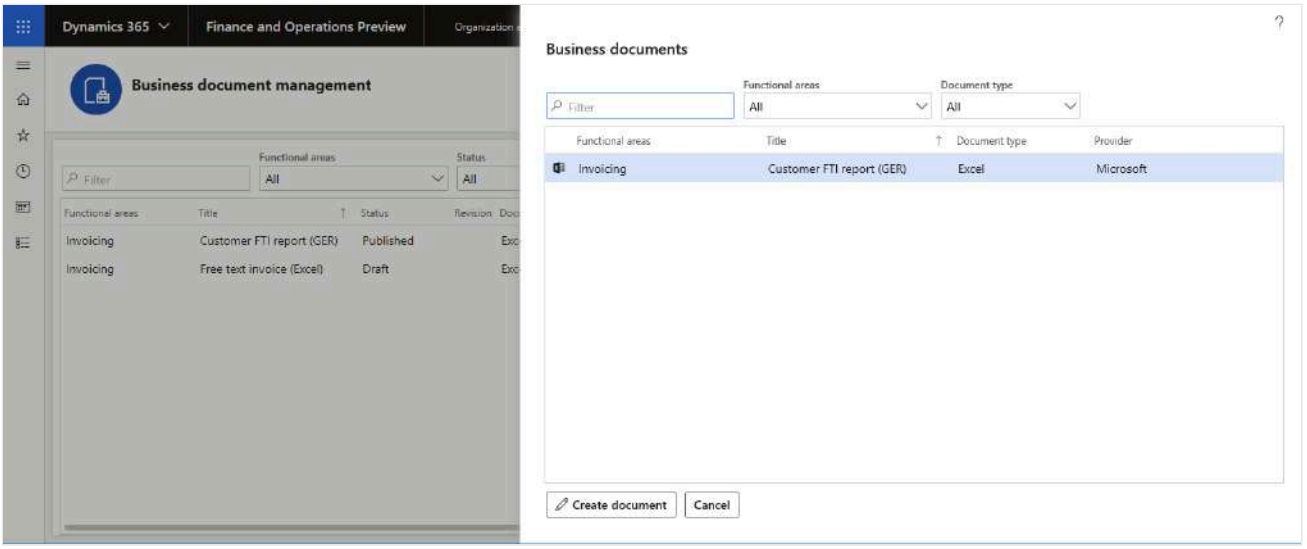
This example shows how to use the **New document** button in the **Business document management** workspace to create and edit a template in an ER format configuration that another provider supplies and that is located in the current Finance instance. In this example, the active provider is Contoso, which uses the ER format configuration that Microsoft provides. After you select **New document**, the **Select** tab on the **Create a new template** page shows all the templates of the current Finance instance that the current provider and other providers own. Select a template to open it. You can then create a new editable copy of the template. The edited template is stored in a new ER format configuration that is automatically generated.

To edit a template that another provider owns, follow these steps:

1. In the **Business document management** workspace, select **New document**.



2. On the **Create a new template** page, on the **Select** tab, select the document to use as a template, and then select **Create document**.

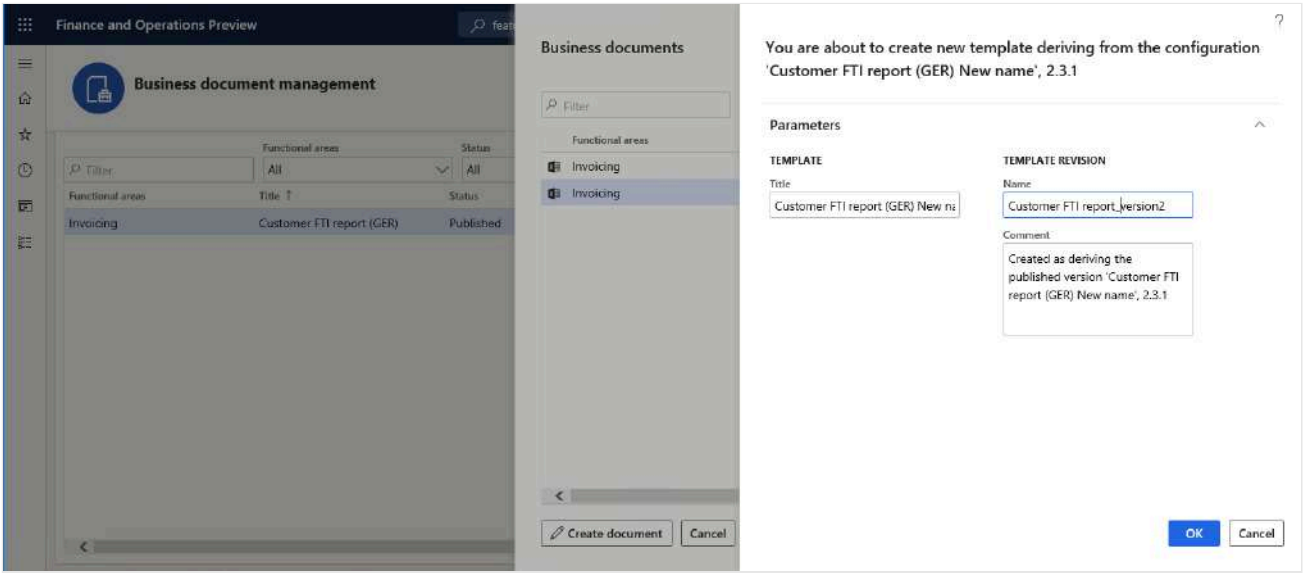


3. In the new dialog box, in the **Title** field, change the title as you require. The title text names the new ER format configuration that is automatically created. The draft version of this configuration (**Customer FTI report (GER) Copy**) contains the edited template and is used to run this ER format for the current user. The original template from the base ER format configuration is used to run this ER format for every other user.

4. In the **Name** field, change the name of the first revision of the editable template that is automatically created.

5. In the **Comment** field, update the remarks for the revision of the editable template that is automatically created.

6. Select **OK** to confirm the start of the editing process.



Upload a template that uses an existing Excel workbook

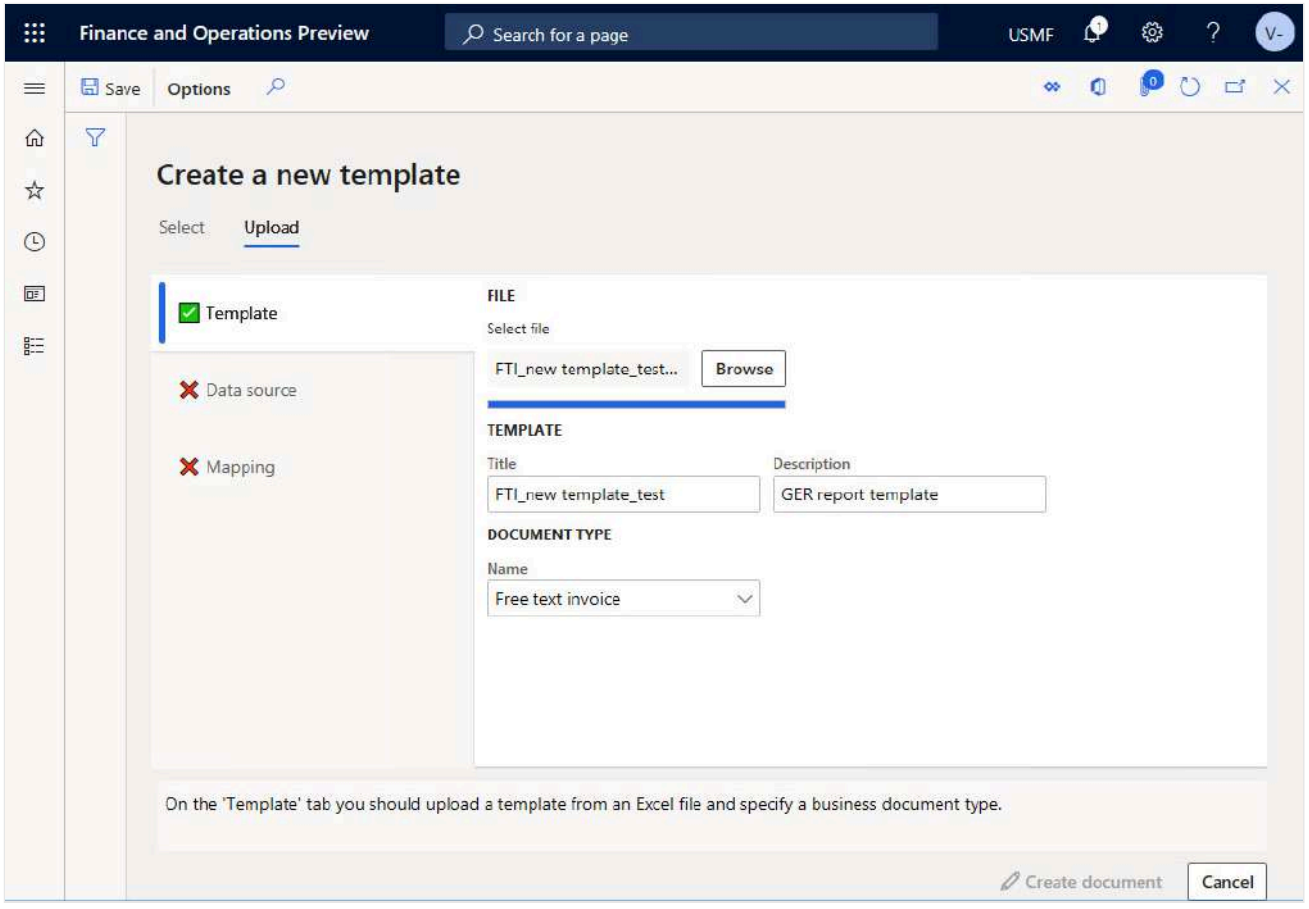
This example shows how to use the **New document** button in the **Business document management** workspace to create and edit a template in an ER format configuration, based on the available Excel workbook. In this example, the active provider is Contoso, and you use the ER data model and ER model mapping configurations that Microsoft provides. After you select **New document**, select the **Upload** tab on the **Create a new template** page. You can specify the details of an Excel workbook upload. When you upload the Excel workbook, the system transforms it into a business document template and opens it for editing. The edited template is stored in a new ER format configuration that the system automatically generates.

To provide the required information before you upload a template, follow these steps:

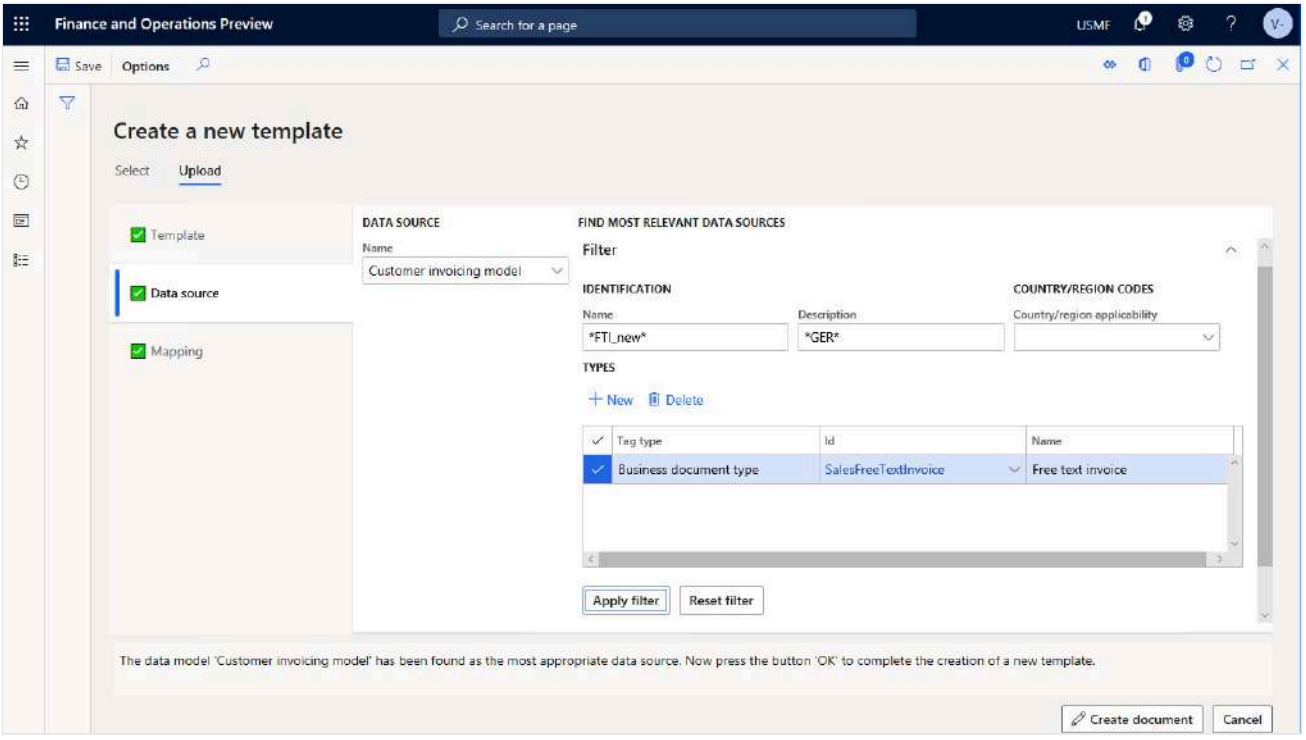
1. In the **Business document management** workspace, select **New document**.

2. On the **Create a new template** page, on the **Upload** tab, on the **Template** tab, select **Browse** to find and select the Excel file that you want to use as a template. In the **Template** section, the **Title** and **Description** fields are automatically filled in. They specify the name and description of the new ER format configuration that the system automatically creates. You can edit these fields as needed.

3. In the **Document Type** section, in the **Name** field, specify the type of business document. This value is used to search the correct data source (that is, the ER model configuration).

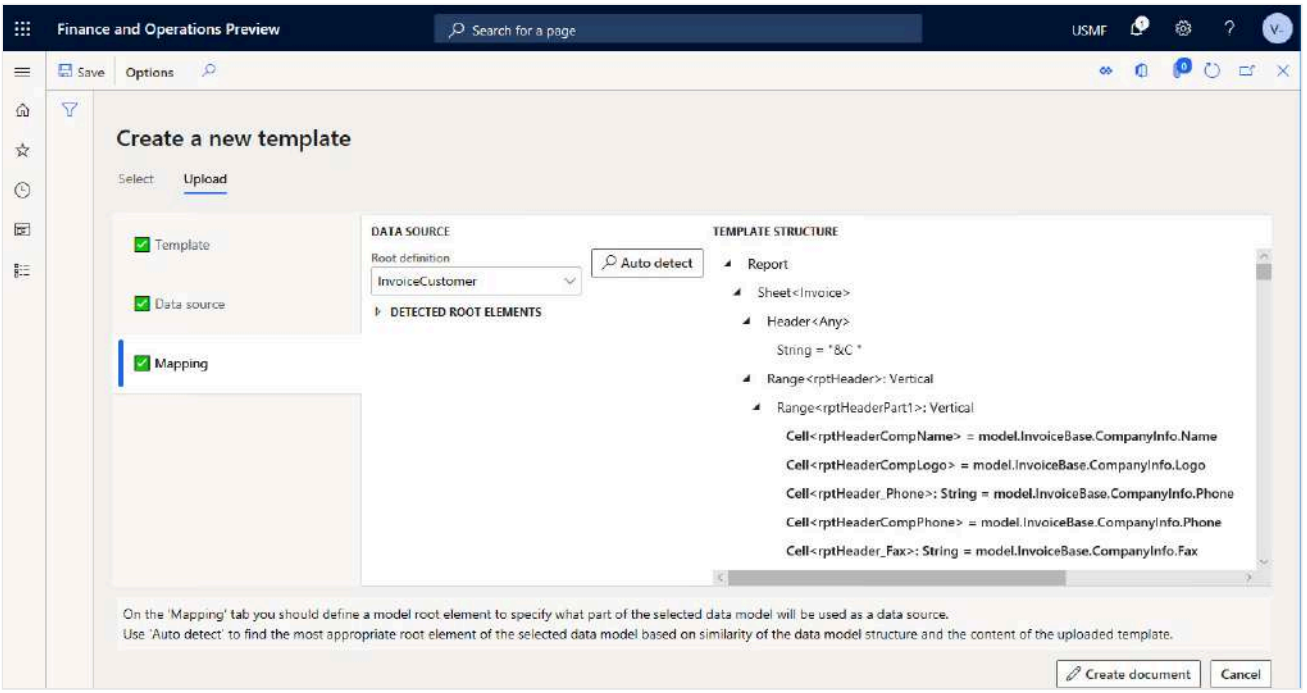


4. On the **Data source** tab, on the **Filter** FastTab, select **Apply filter**. In the **Data source** section, the **Name** field is automatically filled in, or you can manually select a value. You can use the filter to search for the appropriate data source name by name, description, country/region code, and business document type.



> **Note**
>
> Use the **Filter** FastTab to search the correct data source (that is, the ER model configuration). You can edit all filter fields to find the most appropriate data source for the document that you're uploading.
>
> The conditions on the **Filter** FastTab are used as **OR** conditions.

5. On the **Mapping** tab, select **Auto detect**. The **Root definition** field is automatically filled in, or you can manually select a value. This tab shows the end mapping for the elements from the template and the model.



> **Note**
>
> The mapping in the **Template structure** section uses the full match of the labels or descriptions in the data source in the user's language, and in the cell name in the template.

6. Select **Create document** to confirm that you want to create a template and start the editing process.

Learn more in Business document management overview.

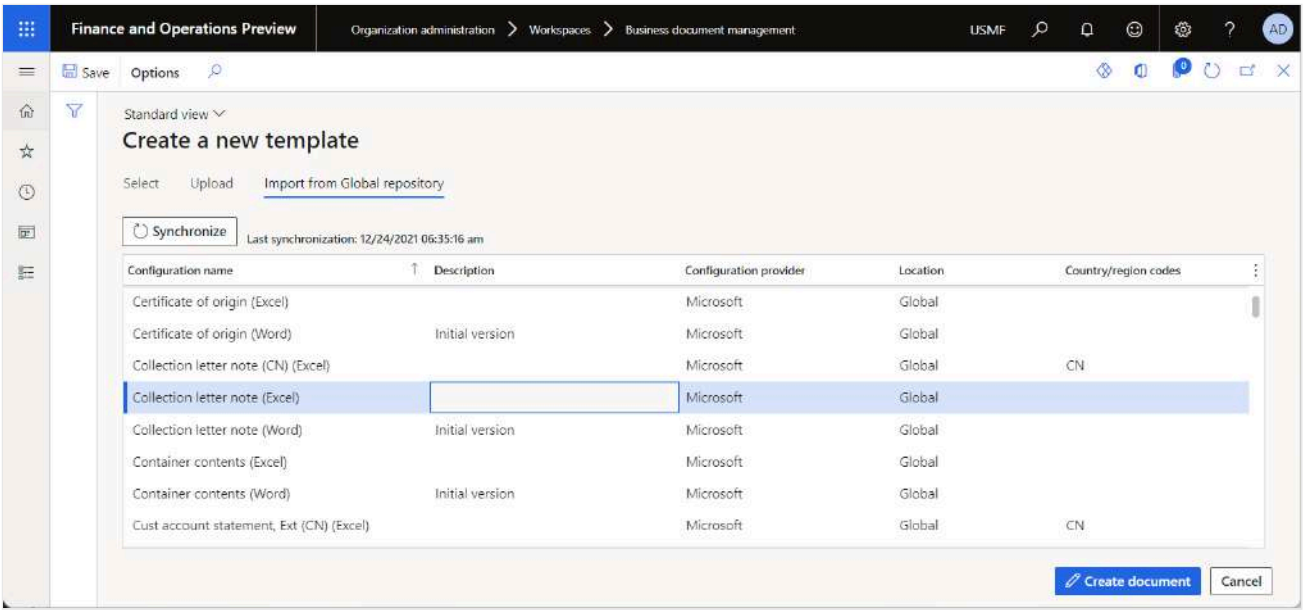
Upload a template from the Dataverse repository

This example shows how to use the **New document** button in the **Business document management** workspace to create and edit a template in an ER format configuration that Microsoft provides and stores in the Microsoft Dataverse repository. In this example, the active provider is Contoso, which uses the ER format configuration that Microsoft provides. After you select **New document**, the **Import from Dataverse repository** tab on the **Create a new template** page shows all the business document templates that are stored in the Dataverse repository but missing in the current Finance instance. After you select a template, you import it from the Dataverse repository into the current Finance instance to create a new editable copy. The edited template is stored in a new ER format configuration that is automatically generated.

To upload a template from the Dataverse repository, follow these steps:

1. In the **Business document management** workspace, select **New document**.

2. On the **Create a new template** page, on the **Import from Dataverse repository** tab, select the document to use as a template, and then select **Create document**.



3. In the message box, select **Yes** to confirm that you want to import the selected document from the Dataverse repository into the current Finance instance. If you're prompted for authorization, follow the on-screen instructions.

4. In the new dialog box, in the **Title** field, change the title as you require. The title text is used to name the new ER format configuration that is automatically created. The draft version of this configuration (**Collection letter note (Excel) Copy**) contains the edited template and is used to run this ER format for the current user. The original template from the base ER format configuration is used to run this ER format for every other user.

5. In the **Name** field, change the name of the first revision of the editable template that is automatically created.

6. In the **Comment** field, update the remarks for the revision of the editable template that is automatically created.

Initiate editing templates owned by your configuration provider

To initiate editing templates owned by your configuration provider, follow these steps:

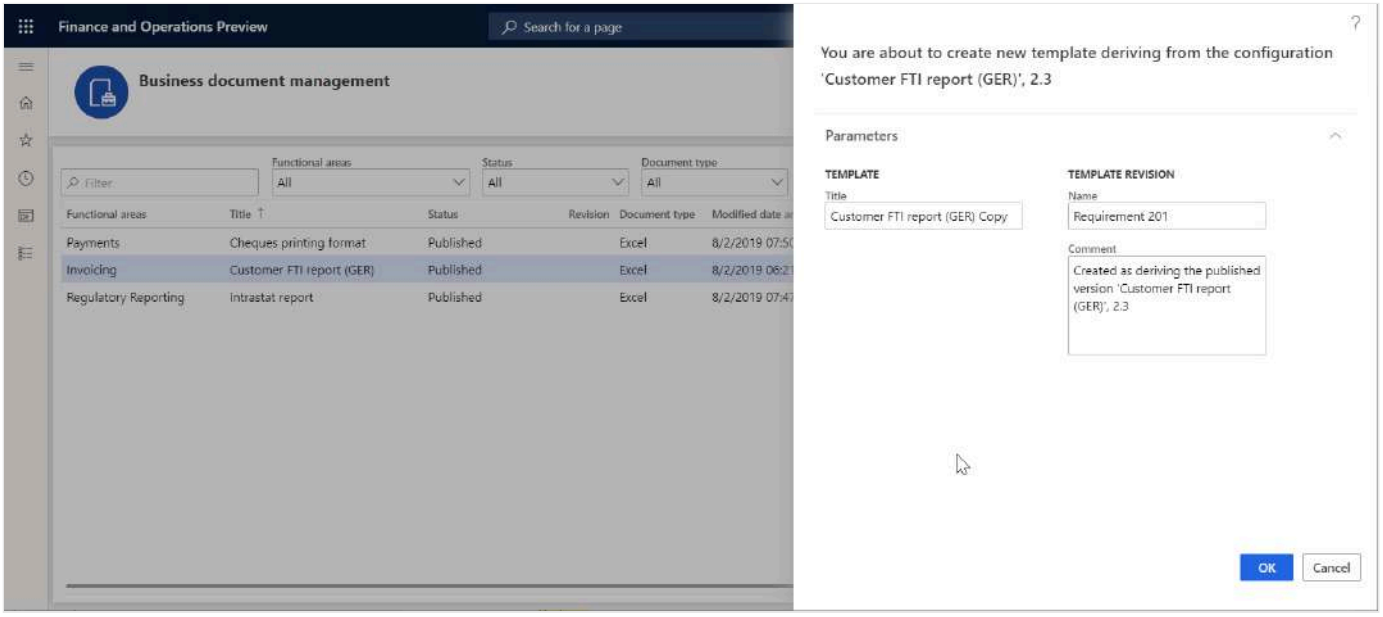
1. In the **Business document management** workspace, select the **Cheques printing format** template in the list.

2. Select the **Details** tab. The **Edit** option is available for the selected template. This option is always available for a template in an ER format configuration that the active ER configuration provider owns. When you select **Edit** for the template, the existing template from the draft version of the underlying ER format configuration is available to edit.

Initiate editing templates owned by other providers

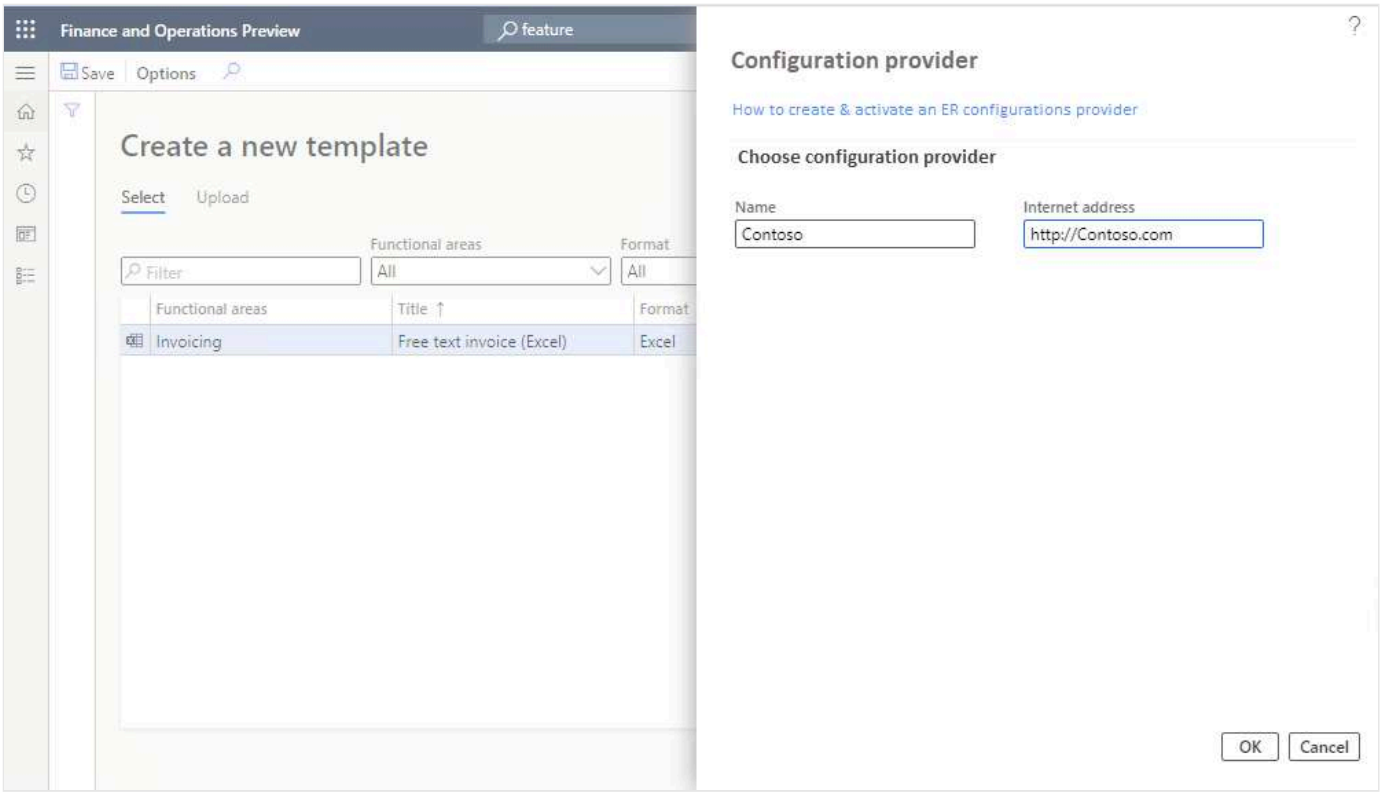
To initiate editing templates owned by other providers, follow these steps:

1. In the **Business document management** workspace, select the document that you want to use as a template.
2. Select **New document**, and in the **Title** field, change the title of the editable template if needed. The text names the ER format configuration that is automatically created. The draft version of this configuration (**Customer FTI report (GER) Copy**) automatically runs this ER format for the current user. At the same time, the non-modified original template from the base ER format configuration runs this ER format for any other user.
3. In the **Name** field, change the name of the first revision of the editable template that you create automatically.
4. In the **Comment** field, change the comment for the automatically created revision of the editable template.
5. Select **OK** to confirm the start of the editing process.

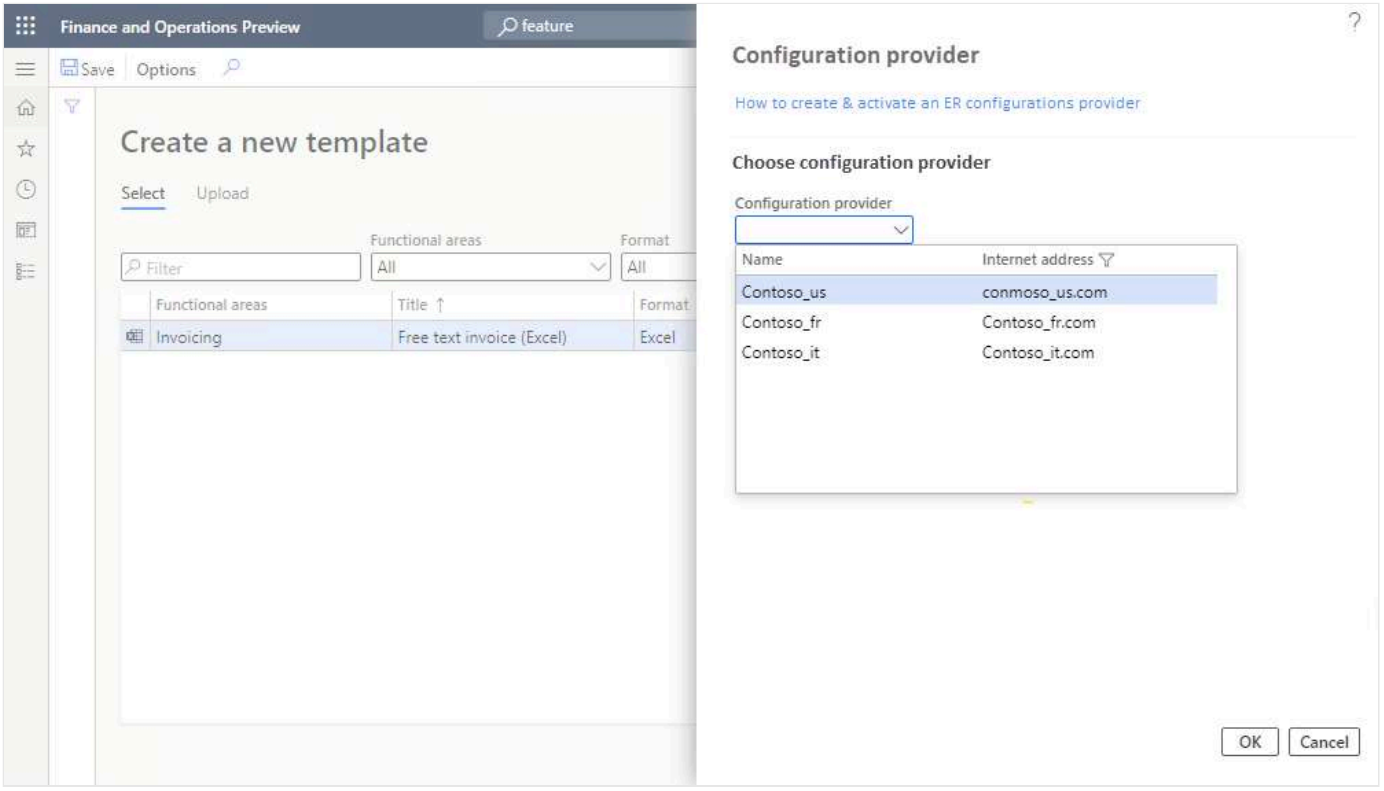


If there's no provider, the system offers to create one. If there's no active provider, the system offers to choose one for activation.

To create a provider, change the name of the provider in the **Name** field, update the internet address of the new provider in the **Internet address** field, and select **OK** to confirm.



To activate an existing provider, choose the name of the provider in the **Configuration provider** field, and select **OK** to set the provider as active.



> **Note**
>
> Each BDM template refers to the provider as the author of the configuration. This is why an active provider is required for the template.

The **New document** option is always available for a template in an ER format configuration provided by current and another provider (Microsoft in this example) that doesn't have any revision. The edited template is stored in a new ER format configuration that is automatically generated.

Start editing a template

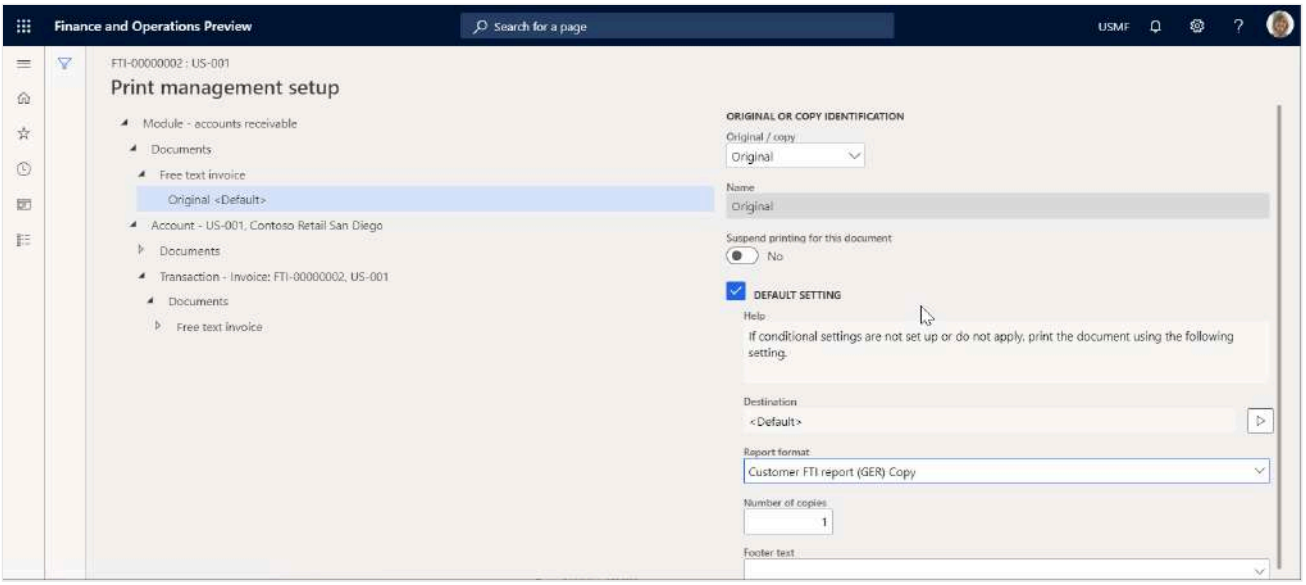
To start editing a template, follow these steps:

1. Select **Edit** from the selected template.
2. In the **Name** field, change the name of the first revision of the editable template that you create automatically.
3. In the **Comment** field, enter a remark for the automatically created revision of the editable template.
4. Select **OK** to confirm the start of the editing process. The selected template is available for online editing by using Microsoft 365. You can modify the template, for example, change the font of the field prompts in the template header from **Regular** to **Bold**.

Test the modified template

To test the modified template, follow these steps:

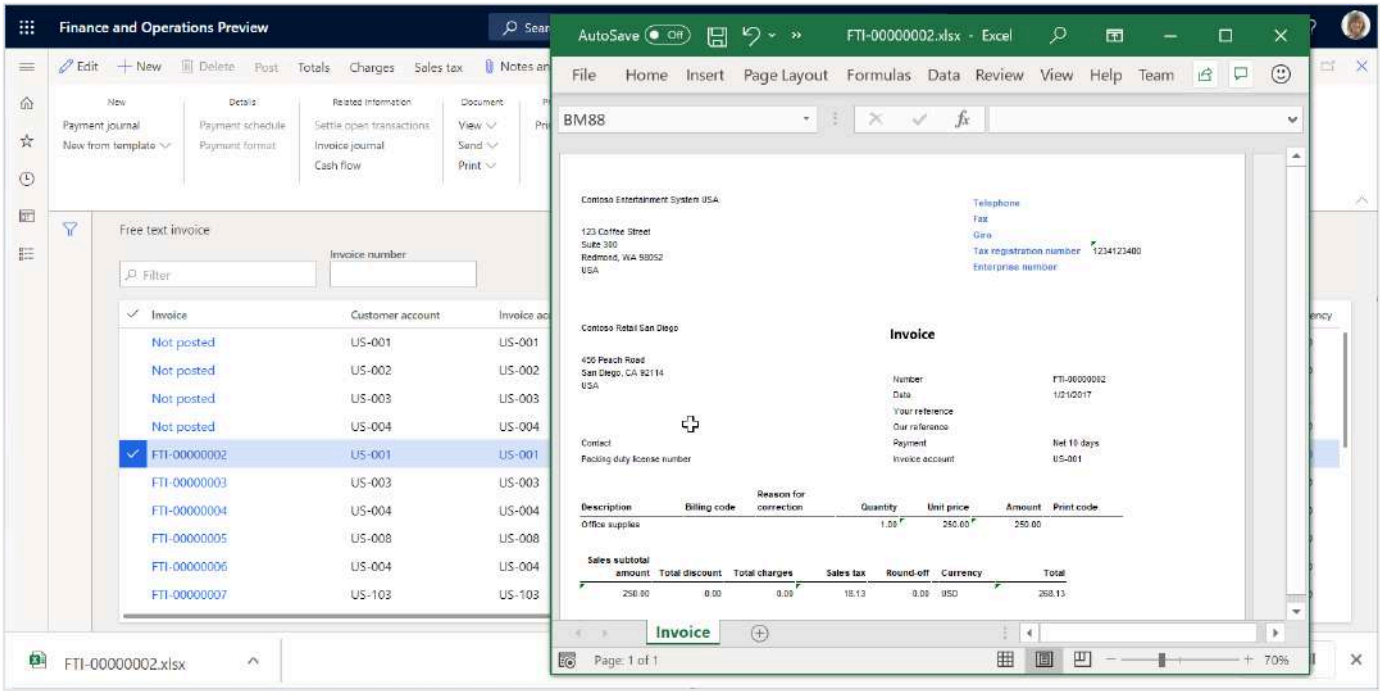
1. In the application, change to the company, **USMF**.
2. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
3. Select the **FTI-00000002** invoice, and then select **Print management**.
4. Select the **Module - accounts receivable** > **Documents** > **Free text invoice** > **Original document** level to specify the scope of invoices for processing.
5. In the **Report format** field, select the **Customer FTI report (GER) Copy** ER format for the specified document level.



6. Press **Escape** to close the current page.

7. Select **Print**, and then select **Selected**.

8. Download the document and open it by using the Excel desktop application.



The modified template is used to generate the free text invoice report for the selected item. To analyze how this report is affected by the changes that you introduced to the template, you can run this report in one application session right after you modified the template in another application session.

Create an alternative template revision

To create an alternative template revision, follow these steps:

1. Open the **Business document management** workspace and select the **Customer FTI report (GER) Copy** template.
2. On the **Revisions** tab, select **New**.
3. If needed, change the name of the second revision in the **Name** field. Base it on the currently active first revision.
4. If needed, change the remark in the **Comment** field for the automatically created revision of the editable template. You created a new revision of your template that is stored in the permanent template's storage. Now you can continue editing the template of the second revision that is currently selected as active.
5. Select the first revision and then select **Set active**. You can select another revision as active if you want to return to that revision of the template.
6. Select the second revision, and then select **Delete**.
7. Select **OK** to confirm that you want to delete the selected revision. You can delete any of the non-active revisions when they're no longer needed.

Delete a modified template

To delete a modified template, follow these steps:

1. In the **Business document management** workspace, select the **Template** tab.
2. Select **Delete**.
3. If you select **OK** to confirm deletion, the **Customer FTI report (GER) Copy** ER format with the modified template is deleted. Select **Cancel** to explore other options.

Revoke changes of template

To revoke changes of template, follow these steps:

When you edit the template from an ER format that the current active provider owns, the option to revoke changes introduced for the template appears.

1. In the **Business document management** workspace, select the **Template** tab.
2. Select **Undo**.
3. If you select **OK** to revoke the changes introduced for the template, the modified template is replaced by the original template and all changes are removed. When you revoke changes to the template, you can delete the template. Select **Cancel** to explore other options.

Publish a modified template

To publish a modified template, follow these steps:

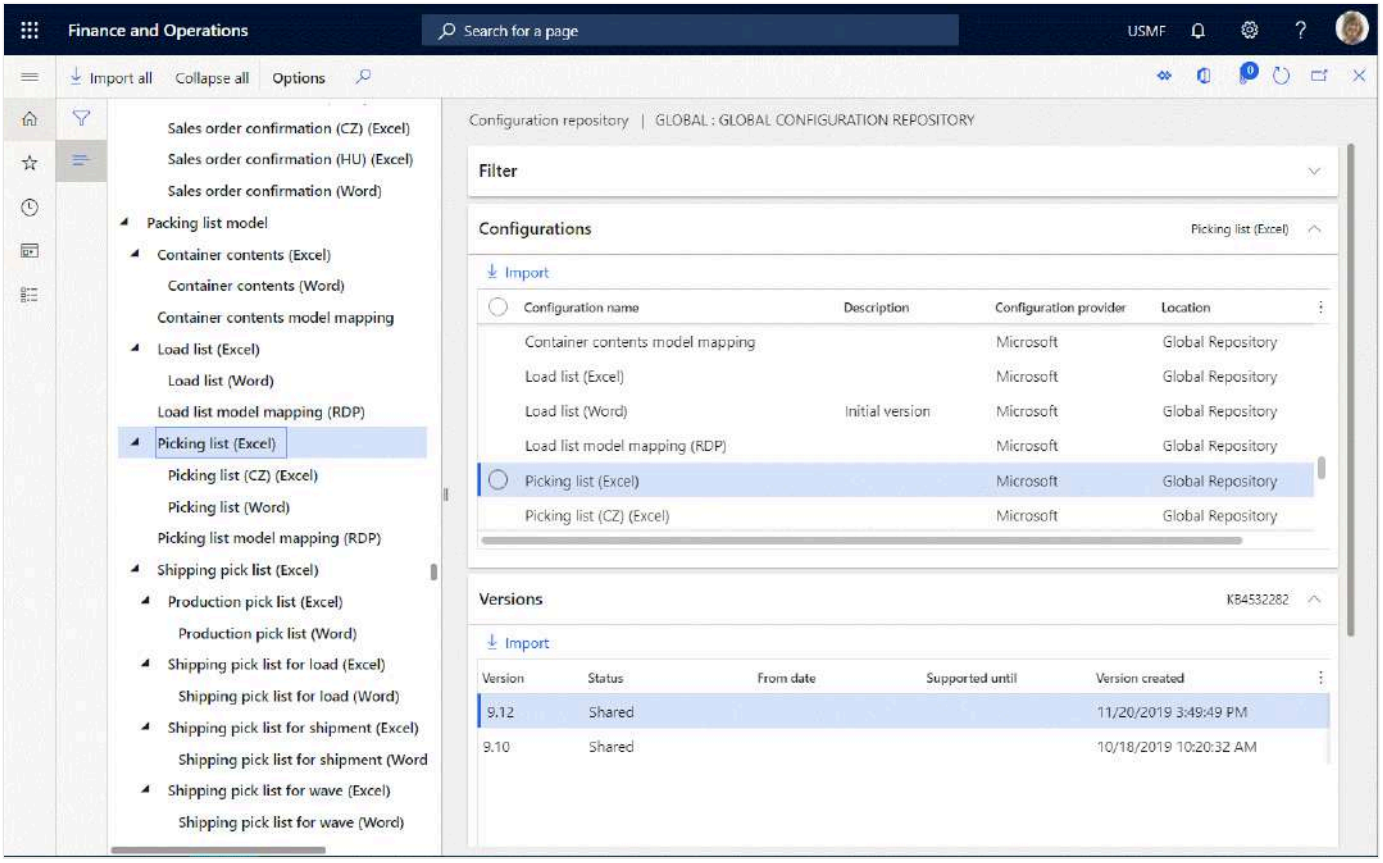
1. In the **Business document management** workspace, on the **Template** tab, select **Publish**.
2. Select **OK** to confirm publishing. The draft version of the derived **Customer FTI report (GER) Copy** ER format that contains the modified template is marked as completed. The modified template becomes available for other users. The completed versions of this ER format keep only the last active revision of your template. Other revisions are deleted. Select **Cancel** to explore other options.

Last updated on 08/07/2026

List of configurable business documents

This article provides the list of Electronic reporting (ER) configurations that are released in Dynamics 365 Finance to support configurable business documents.

The list of ER configurations for Finance is updated regularly. Open the configuration repository of Microsoft configuration provider to review the list of ER configurations that are currently supported. You can filter the repository to review the list of ER configurations that support configurable business documents.



The following table shows the list of ER configurations that support configurable business documents and that are released in Finance up until December 2020.

Expand table

Data model configuration	Format configurations
Bill of lading model	Bill of lading (Excel)
	Bill of lading (Word)
Certificate of origin model	Certificate of origin (Excel)

Data model configuration	Format configurations
	Certificate of origin (Word)
Invoice model	Customer Debit and Credit Note (Excel)
	Customer Debit and Credit Note (Word)
	Free text invoice (Excel)
	Free text invoice (Excel) (BH)
	Free text invoice (FR) (Excel)
	Free text invoice (LT) (Excel)
	Free text invoice (LV) (Excel)
	Free text invoice (PL) (Excel)
	Free text invoice (CZ) (Excel)
	Free text invoice (EE) (Excel)
	Free text invoice (HU) (Excel)
	Free text invoice (TH) (Excel)
	Free text invoice (Word)
	Project contract line items (Excel)
	Project contract line items (CZ) (Excel)
	Project contract line items (Excel) (BH)
	Project contract line items (HU) (Excel)
	Project contract line items (LT) (Excel)
	Project contract line items (PL) (Excel)
	Project contract line items (Word)
	Project customer retention release (Excel)
	Project customer retention release (CZ) (Excel)
	Project customer retention release (HU) (Excel)
	Project customer retention release (LT) (Excel)
	Project customer retention release (PL) (Excel)

Data model configuration	Format configurations
	Project customer retention release (TH) (Excel)
	Project customer retention release (Word)
	Project invoice (Excel)
	Project Invoice (Word)
	Project invoice (AE) (Excel)
	Project invoice (CZ) (Excel)
	Project invoice (Excel) (BH)
	Project invoice (HU) (Excel)
	Project invoice (JP) (Excel)
	Project invoice (LT) (Excel)
	Project invoice (PL) (Excel)
	Project invoice (TH) (Excel)
	Project invoice full (MY) (Excel)
	Project invoice simple (MY) (Excel)
	Project manage invoice (Excel)
	Project manage invoice (CZ) (Excel)
	Project manage invoice (Excel) (BH)
	Project manage invoice (HU) (Excel)
	Project manage invoice (JP) (Excel)
	Project manage invoice (LT) (Excel)
	Project manage invoice (PL) (Excel)
	Project manage invoice (Word)
	Purchase advance invoice (Excel)
	Purchase advance invoice (Word)
	Sales advance invoice (Excel)
	Sales advance invoice (Word)

Data model configuration	Format configurations
	Sales advance invoice (PL) (Excel)
	Sales invoice (Excel)
	Sales invoice (Excel) (BH)
	Sales invoice (Excel) (CZ)
	Sales invoice (Excel) (EE)
	Sales invoice (Excel) (FR)
	Sales invoice (Excel) (HU)
	Sales invoice (Excel) (IN)
	Sales invoice (Excel) (LT)
	Sales invoice (Excel) (LV)
	Sales invoice (Excel) (PL)
	Sales invoice (Excel) (TH)
	Sales invoice (Word)
	TMS Commercial Invoice (Excel)
	TMS Commercial Invoice (Word)
	Vendor invoice document (Excel)
	Vendor invoice document (CZ) (Excel)
	Vendor invoice document (HU) (Excel)
	Vendor invoice document (IN) (Excel)
	Vendor invoice document (LT) (Excel)
	Vendor invoice document (LV) (Excel)
	Vendor invoice document (MY) (Excel)
	Vendor invoice document (Word)
Order model	Agreement confirmation (Excel)
	Agreement confirmation (Word)
	Purchase agreement confirmation (Excel)

Data model configuration	Format configurations
	Purchase agreement confirmation (Word)
	Purchase order (Excel)
	Purchase order (CZ) (Excel)
	Purchase order inquiry (CZ) (Excel)
	Purchase order (HU) (Excel)
	Purchase order inquiry (HU) (Excel)
	Purchase order (Word)
	Purchase order inquiry (Excel)
	Purchase order inquiry (Word)
	Sales order confirmation (Excel)
	Sales order confirmation (CZ) (Excel)
	Sales order confirmation (HU) (Excel)
	Sales order confirmation (Word)
Packing list model	Container contents (Excel)
	Container contents (Word)
	Load list (Excel)
	Load list (Word)
	Picking list (Excel)
	Picking list (CZ) (Excel)
	Picking list (Word)
	Production pick list (Excel)
	Production pick list (Word)
	Shipping pick list for load (Excel)
	Shipping pick list for load (Word)
	Shipping pick list for shipment (Excel)
	Shipping pick list for shipment (Word)

Data model configuration	Format configurations
	Shipping pick list for wave (Excel)
	Shipping pick list for wave (Word)
Payment model	Customer payment advice (Excel)
	Customer payment advice (Word)
	Vendor payment advice (Excel)
	Vendor payment advice (Word)
Quotation model	Project quotation (Excel)
	Project quotation (Word)
	Request for quotation (Excel)
	Request for quotation (Accept) (Excel)
	Request for quotation (Accept) (Word)
	Request for quotation (Reject) (Excel)
	Request for quotation (Reject) (Word)
	Request for quotation (Return) (Excel)
	Request for quotation (Return) (Word)
	Request for quotation (Word)
	Sales quotation (Excel)
	Sales quotation (CZ) (Excel)
	Sales quotation (HU) (Excel)
	Sales quotation (Word)
	Sales quotation confirmation (Excel)
	Sales quotation confirmation (Word)
Reconciliation model	Cust account statement, Ext (Excel)
	Cust account statement, Ext (CN) (Excel)
	Cust account statement, Ext (Word)
	Cust account statement, France (Excel)

Data model configuration	Format configurations
Reminder model	Collection letter note (Excel)
	Collection letter note (CN) (Excel)
	Collection letter note (Word)
	Customer interest note (Excel)
	Customer interest note (Word)
Waybill model	Load tender (Excel)
	Load tender (Word)
	Purchase order packing slip (Excel)
	Purchase order packing slip (CZ) (Excel)
	Purchase order packing slip (Word)
	Route (Excel)
	Route (Word)
	Sales order packing slip (Excel)
	Sales order packing slip (CZ) (Excel)
	Sales order packing slip (LT) (Excel)
	Sales order packing slip (PL) (Excel)
	Sales order packing slip (Word)

Last updated on 08/07/2026

Extend the list of Electronic reporting (ER) functions

Electronic reporting (ER) expressions support various types of functions for data transformation, including text, date and time, mathematical, logical, information, data type conversion, and other business domain–specific functions. In addition to built-in functions, you can extend the list of available functions in ER. This article provides an overview of key tasks that you must complete to introduce a new function.

> **Important**
>
> In Microsoft Dynamics 365 Finance, Enterprise edition 7.3, extensions of the **ERExpression** class are deprecated. Instead of using extensions of the **ERExpression** class to add custom built-in ER functions, implement your custom logic as public methods of your custom classes. Then, call those methods from ER formats and ER model mappings that configure the required ER data sources of either the ***Class*** type or the ***Object*** type. For more information, see Design ER expressions to call application class methods.

All ER functions in application code are represented as classes that extend the **ERExpression** class. Two types of functions exist:

- **Fixed number of arguments** – These functions are represented by classes that include methods with the prefix **parm** (see **parmInput**, **parmStartNum** in the sample code that follows). The **SysOperationDisplayOrderAttribute** attribute sets the order of arguments.
- **Variable number of arguments** – These functions (see the **ERExpressionGenericCase** class) are represented by classes that implement the **ERIObjectContainer** interface. An additional **Add** method declares the types that a function accepts.

To introduce a new function for ER expressions, follow these steps:

- Select a base class for your function, based on the return value type (see **ERExpressionString** in the sample code that follows).
    - Create a new class that extends the selected class (see **ERExpressionStringMid** in the sample code that follows).
        - Add the required attributes:

        - **SysOperationLabelAttribute** – Defines the function's name.
        - **SysOperationHelpTextAttribute** – Defines the function's Help text.
        - **ERComponentGroupAttribute** – Defines the group that the function belongs to. For more information, see Formula designer in Electronic reporting (ER).

    - Add arguments:
        - For a fixed number of arguments function, add methods with the prefix **parm**, and use the **SysOperationDisplayOrderAttribute** attribute to set the order of the arguments.
        - For a variable number of arguments function, implement the **ERIObjectContainer** interface.

    - Add an evaluation method.

Here's an example.

X++

```
/// <summary>
/// Returns the characters from the middle of a text string, given a starting position
and length.
/// </summary>
[
    SysOperationLabelAttribute ('MID'),
    SysOperationHelpTextAttribute
("@ElectronicReporting:ExpressionStringMidHelpText"),
    ERComponentGroupAttribute ("@ElectronicReporting:String")
]
class ERExpressionStringMid extends ERExpressionString
{
    ERExpressionString input;
    ERExpressionInt startNum;
    ERExpressionInt numChars;
    public str evaluateString(ERIDataContext _dataContext)
    {
        return subStr(
            this.parmInput().evaluateString(_dataContext),
            this.parmStartNum().evaluateInt(_dataContext),
            this.parmNumChars().evaluateInt(_dataContext));
    }
    [DataMemberAttribute, SysOperationLabelAttribute ("@ElectronicReporting:Input"),
SysOperationDisplayOrderAttribute ("1")]
    public ERExpressionString parmInput(ERExpressionString _input = input)
    {
        input = _input;
        return input;
    }
    [DataMemberAttribute, SysOperationLabelAttribute
("@ElectronicReporting:NumChars"), SysOperationDisplayOrderAttribute ("3")]
```

```
    public ERExpressionInt parmNumChars(ERExpressionInt _numChars = numChars)
    {
        numChars = _numChars;
        return numChars;
    }
    [DataMemberAttribute, SysOperationLabelAttribute
("@ElectronicReporting:StartNum"), SysOperationDisplayOrderAttribute ("2")]
    public ERExpressionInt parmStartNum(ERExpressionInt _startNum = startNum)
    {
        startNum = _startNum;
        return startNum;
    }
    public str toString()
    {
        return ERExpressionStringPresenter::namedFunctionToStr(this);
    }
}
```

Suggested guidance

The following guidance helps you design your custom ER functions:

- Reuse the names of Microsoft Excel functions whenever you can, so that ER formulas remain Excel-like. In this way, you keep ER formulas intelligible for end users.
- ER doesn't support list types for primitive data types. Therefore, use a data container list that has a single **Value** item in it.
- Release a new function's list extension as a new application hotfix. ER designers refer to the hotfix number in ER configurations that use that new custom function. Whenever you import a configuration of this type into a new instance, ER evaluates whether the required hotfix is installed, to maintain compliance between the ER configuration and the version that configuration is imported into.

Additional resources

Electronic reporting (ER) overview

Formula designer in Electronic reporting (ER)

Last updated on 06/26/2026

ER framework API changes for Application update 7.3

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how the API of the Electronic reporting (ER) framework is changed in Dynamics 365 Finance, Enterprise edition Application update 7.3.

Two types of changes affect the ER APIs:

- Several X++ classes moved from X++ to an external assembly.
- The rest of X++ classes are internal.

How to access classes that moved from X++ to an external assembly

To access external classes, add the **using** directive to the beginning of your file.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
```

You can then access an external class without any additional changes.

X++

```
var destination = new ERFileDestinationMemory();
```

You can also create an alias for your namespace.

X++

```
using LF = Microsoft.Dynamics365.LocalizationFramework;
```

You can then refer to an external class by using the namespace alias that you created.

X++

```
var destination = new LF.ERFileDestinationMemory();
```

How to access internal X++ objects by using ERObjectsFactory

In Application update 7.3 and later updates, the calling code must access the ER objects by using the methods of the **ERObjectsFactory** class. Several examples of these changes are shown.

Code to display a format mapping lookup

Before application update 7.3

X++

```
// pattern
ERFormatMappingTableLookup::lookupFormatMapping(<form control>, <model name>[,
<data container name>]);
// sample code
ERFormatMappingTableLookup::lookupFormatMapping(_referenceGroupControl,
bankLCMiscChargeReportERModelName);
```

Application update 7.3 and later

X++

```
// pattern
ERObjectsFactory::createFormatMappingTableLookupForControlAndModel(<form control>,
<model name>[, <data container name>]).performFormLookup();
// sample code
ERObjectsFactory::createFormatMappingTableLookupForControlAndModel(_referenceGroupC
ontrol, bankLCMiscChargeReportERModelName).performFormLookup();
```

Code to run a format mapping for data export

Before application update 7.3

X++

```
// pattern
ERFormatMappingRun::constructByFormatMappingId(<format mapping id>, <file name>,
<show prompt dialog>).run();
```

```
// sample code
ERFormatMappingRun::constructByFormatMappingId(erBinding, '', true).run();
```

Application update 7.3 and later

X++

```
// pattern
ERObjectsFactory::createFormatMappingRunByFormatMappingId(<format mapping id>,
<file name>, <show prompt dialog>).run();
// sample code
ERObjectsFactory::createFormatMappingRunByFormatMappingId(erBinding, '',
true).run();
```

Code to run a format mapping for data import

Before application update 7.3

X++

```
// pattern
ERModelMappingDestinationRun::constructByImportFormatMappingId(<mapping id>,
<integration point>).run();
// sample code
ERModelMappingDestinationRun::constructByImportFormatMappingId(custPaymModeTable.ER
ModelMappingTable, CustVendOutPaymConstants::IntegrationPoint).run();
```

Application update 7.3 and later

X++

```
// pattern
ERObjectsFactory::createMappingDestinationRunByImportFormatMappingId(<mapping id>,
<integration point>).run();
// sample code
ERObjectsFactory::createMappingDestinationRunByImportFormatMappingId(custPaymModeTa
ble.ERModelMappingTable, CustVendOutPaymConstants::IntegrationPoint).run();
```

Code to create a browser file destination

Before application update 7.3

X++

```
// sample code
new ERFileDestinationBrowser();
```

Application update 7.3 and later

```
X++
```

```
// sample code
ERObjectsFactory::createFileDestinationBrowser();
```

Code to create an attachment file destination

Before application update 7.3

```
X++
```

```
// pattern
ERFileDestinationAttachment::construct(<record>,
ERDocuManagement::instance().otherDocuType());
// sample code
ERFileDestinationAttachment::construct(_cashRegisterFiscalTrans_W,
ERDocuManagement::instance().otherDocuType());
```

Application update 7.3 and later

```
X++
```

```
// pattern
ERObjectsFactory::createFileDestinationAttachmentWithOtherDocuType(<record>);
// sample code
ERObjectsFactory::createFileDestinationAttachmentWithOtherDocuType(_cashRegisterFis
calTrans_W);
```

Last updated on 03/17/2026

Electronic reporting framework API changes for application update 10.0.11

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how the application programming interfaces (APIs) of the Electronic reporting (ER) framework are changed in Microsoft Dynamics 365 Finance version 10.0.11.

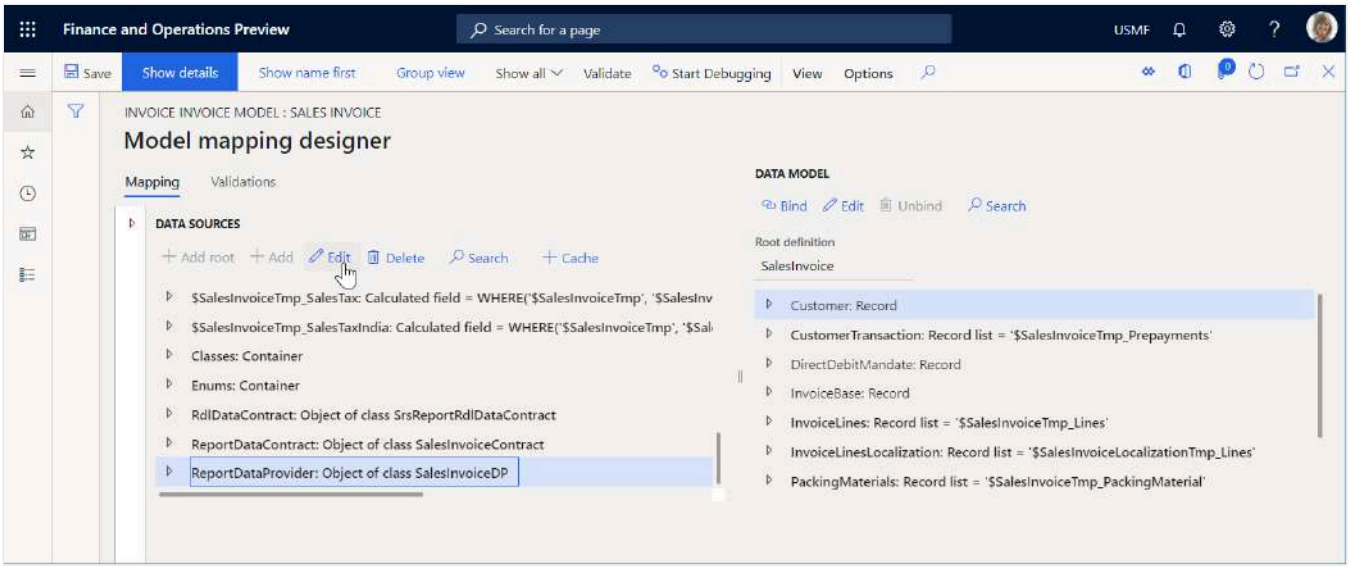
API to run a format mapping for the generation of outbound documents

To generate an outbound document, run an ER format mapping. Most ER format mappings contain a data source of the **Data model** type. At runtime, a specific model mapping must be identified as the implementation of the data model for which the data source is configured.

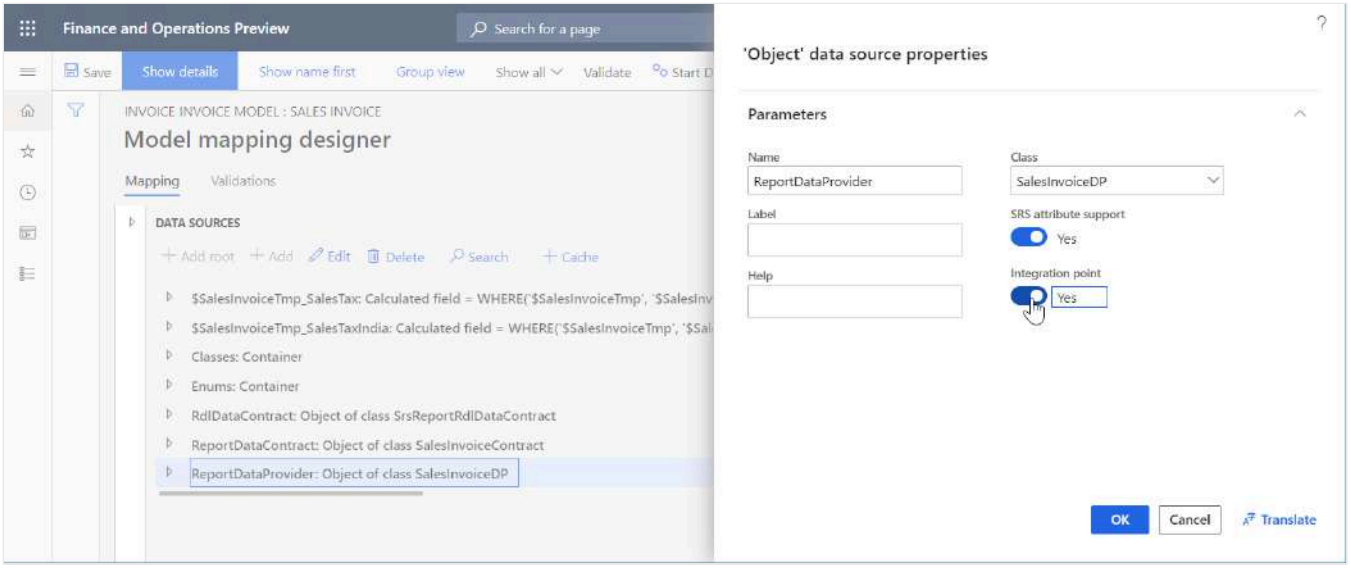
When you use the initial API of the ER framework to call an ER format mapping from an execution point in the source code, the API finds a model mapping by using the settings (the default model mapping flag and country/region codes) of the corresponding ER configurations that contain a model mapping component. For more information, see Configure country/region context dependent ER model mappings.

In some cases, when an ER format mapping is called from a specific place in the X++ code, you must specify additional criteria to find the most appropriate model mapping. For example, you configure an ER format mapping by using a data model that has two model mappings. The first model mapping is marked as a default model mapping, because it ensures the smooth execution of this ER format mapping whenever it's called, from every execution point that's currently available. In X++, you implement a new execution point from which to call this ER format mapping. By using this new execution point, you force the data model to use the second model mapping, because that model mapping performs better given the other conditions of this call (for example, the arguments that are provided and the available resources of the current application instance).

To complete this setup, configure your model mapping with an integration point. In the **Data sources** pane, select a single data source of your model mapping as the component of this integration point. Then select **Edit** to change the properties of this data source.



In the **Data source properties** dialog box, set the **Integration point** option to **Yes** to mark the data source as the component of the integration point.



You can then use the new API of the ER framework to call an ER format mapping and force it to use a model mapping that has a specific integration point. The following example shows how to use this new API.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using Microsoft.Dynamics365.LocalizationFramework.XppSupportLayer;

class ERIntegrationPointCodeSamples extends RunBaseBatch
{
    private ERFormatMappingId formatMappingId;

    public void run()
    {
        var runner =
ERObjectsFactory::createFormatMappingRunByFormatMappingId(this.formatMappingId,
'OutboundFileName', true);
```

```
        var integration_point = new
ERIntegrationPointFactory().WithObjectIntegrationPoint('SalesInvoiceDP').ToIntegrat
ionPoint();
        /// new
ERIntegrationPointFactory().WithTableRecordsIntegrationPoint(tableName)
        /// new ERIntegrationPointFactory().WithTableIntegrationPoint(tableName)
        /// new ERIntegrationPointFactory().WithObjectIntegrationPoint(className)
        /// new ERIntegrationPointFactory().WithClassIntegrationPoint(className)
        runner.withIntegrationPoint(integration_point).run();
    }
}
```

> ⓘ **Note**
>
> - Only a data source of the **Table record**, **Table**, **Class**, or **Object** type can be the component of the model mapping integration point.
> - Currently, only a single data source can be the component of the model mapping integration point.
> - In addition to other existing criteria, the integration point is considered during the selection of the most appropriate model mapping.
> - At runtime, an error occurs if more than one model mapping that has the requested integration point is found among the model mappings that are applicable to the ER format mapping that is running.

API to show a format mapping lookup

Use the initial API of the ER framework to look up the ER format mappings. It uses the data model name and container name as text constants. This lookup offers only the ER format mappings of the one data model that meets both these conditions:

- It has the specified names.
- Its criteria are met first in the ER configurations list.

The new API of the ER framework lets you use the configured integration point to implement the same lookup. This lookup offers all the ER format mappings that contain a data source of the **Data model** type that a model mapping that has the specified integration point is available for. The ER format mappings that are offered in this lookup can use different model mappings as data sources.

The following example shows how you can use this new API.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using Microsoft.Dynamics365.LocalizationFramework.XppSupportLayer;

class ERIntegrationPointCodeSamples extends RunBaseBatch
{
    private ERFormatMappingId formatMappingId;

    public void formatMappingLookup(FormReferenceControl _referenceGroupControl)
    {
        var integration_point = new
ERIntegrationPointFactory().WithIntegrationPointKey('SalesInvoiceDP').ToIntegration
Point();
        ERObjectsFactory::createFormatMappingTableLookupForIntegrationPoint(
            integration_point,
            _referenceGroupControl
            ).performFormLookup();
    }
}
```

Last updated on 03/17/2026

Electronic reporting framework API changes for Application update 10.0.17

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how the application programming interfaces (APIs) of the Electronic reporting (ER) framework changed in Microsoft Dynamics 365 Finance version 10.0.17.

API to run a format mapping that provides a user action code to run action-dependent destinations

To generate an outbound document, you must run an ER format mapping. When you use the initial API of the ER framework to call an ER format mapping, all destinations configured for components of the format are always run. To review the sample code for a call of this type, see Add a report service class.

X++

```
// Call ER to generate the report.
ERIFormatMappingRun formatMappingRun =
ERObjectsFactory::createFormatMappingRunByFormatMappingId(formatMappingId,
DefaultExportedFileName);
if(formatMappingRun.parmShowPromptDialog(true))
{
    formatMappingRun.withParameter(parameters);
    formatMappingRun.withFileDestination(_contract.getFileDestination());
    formatMappingRun.run();
}
```

In some cases, when an ER format mapping is called from a specific place in the X++ code, you need to specify an action that the user performs by running an ER format so that only action-dependent destinations are run instead of all the ER destinations configured for that format.

For example, you have an ER format that is based on Print management settings. When you call this ER format to preview a generated document, you expect that only the Screen destination is run. However, when you call the same ER format to send a generated document

as the attachment of an outbound email message, you expect that only the Email destination is run.

To achieve these results, you must configure action-dependent ER destinations for the ER format. For more information, see Configure action-dependent ER destinations.

After you complete the configuration, you can use the new API of the ER framework to call an ER format mapping that provides a user action code, which runs only destinations configured for the provided action. The following example shows how you can change the previously mentioned sample code to use this new API to run an ER format that provides the **View** action.

X++

```
// Call ER to generate the report.
ERIFormatMappingRun formatMappingRun =
ERObjectsFactory::createFormatMappingRunByFormatMappingId(formatMappingId,
DefaultExportedFileName);
if(formatMappingRun.parmShowPromptDialog(true))
{
    formatMappingRun.withParameter(parameters);
    formatMappingRun.withFileDestination(_contract.getFileDestination());

    var formatMappingRunWithAction = formatMappingRun as
ERIFormatMappingRunWithDestinationAction;
    formatMappingRunWithAction.withDestinationAction(ERDestinationAction::View);
    formatMappingRun.run();
}
```

> **ⓘ Important**
>
> To set up action-dependent destinations and force the ER framework to use the provided action code, you must first turn on the **Configure specific ER destinations to be used for different PM actions** feature in the Feature management workspace.

Additional resources

Electronic reporting (ER) overview

Electronic reporting (ER) destinations

Configure action-dependent ER destinations

Design a new ER solution to print a custom report

Last updated on 03/17/2026

Electronic reporting framework API changes for Application update 10.0.19

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how the application programming interfaces (APIs) of the Electronic reporting (ER) framework is changed in Microsoft Dynamics 365 Finance version 10.0.19.

API to extend the list of ER destinations

To implement a custom ER destination, use the `ERIFormatFileDestinationSettings` public interface.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using Microsoft.Dynamics365.LocalizationFramework.Format.FileGeneration;

using SL = Microsoft.Dynamics365.LocalizationFramework.XppSupportLayer;

/// <summary>
/// Destination settings are required to store settings of a FILE destination type
for the format file component.
/// </summary>
/// <remarks>
/// The implementation object should have static create(container) method for a
proper packing.
/// </remarks>
public interface ERIFormatFileDestinationSettings extends SysPackable
{
    /// <summary>
    /// Determines if current settings are enabled.
    /// </summary>
    /// <returns>True if enabled; otherwise - false.</returns>
    boolean isEnabled()
    {
    }

    /// <summary>
    /// Sets settings enabling state.
    /// </summary>
```

```
    /// <param name = "_value">An enabling state.</param>
    void setEnabled(boolean _value)
    {
    }

    /// <summary>
    /// Gets a name of the destination.
    /// </summary>
    /// <returns>The name.</returns>
    str getName()
    {
    }

    /// <summary>
    /// Creates a file destination for current settings.
    /// </summary>
    /// <param name = "_destinationContext">A destination context.</param>
    /// <param name = "_isGrouped">Determines whether the current file destination
is grouped with others.</param>
    /// <returns>A new instance of a file destination.</returns>
    ERIFileDestination createDestination(ERDestinationExecutionContext
_destinationContext, boolean _isGrouped)
    {
    }

    /// <summary>
    /// Gets a settings key. This key should start with or be an implementation
class name. It should also be persistent and should not be changed over time.
    /// This key will be used to retrieve this type of settings from a settings
storage.
    /// </summary>
    /// <remarks>Examples of good keys: MyCustomDestinationSettings,
MyCustomDestinationSettings#UniqueKey.</remarks>
    /// <returns>The settings key.</returns>
    str getKey()
    {
    }

    /// <summary>
    /// Validates the settings.
    /// </summary>
    void validate()
    {
    }
}
```

This interface lets you offer parameters of a custom destination in the ER destinations dialog box, so that they can be set at design time. The configured destination can then be used at runtime to store generated outbound documents.

To learn more about this interface, see the example in Implement a custom destination for generated documents.

Additional resources

Electronic reporting (ER) overview

Electronic reporting (ER) destinations

Last updated on 03/17/2026

Electronic reporting framework API changes for Application update 10.0.21

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how the application programming interfaces (APIs) of the Electronic reporting (ER) framework changed in Microsoft Dynamics 365 Finance version 10.0.21.

API to enable users to configure ER destinations that are print management record specific

This new API lets you configure ER destinations for a specific type of business document and assign a configured named destination to a print management record. Therefore, you can make ER destinations print management record specific.

X++

```
/// <summary>Class for enabling NamedDestinationFeature for current document type.
</summary>
/// <remarks>To enable the document type you should wrap a
<c>isNamedDestinationEnabledByDocumentType</c> method.</remarks>
public class ERNamedDestinationReportEnabler
{
    /// <summary>Checks that feature enabled for current document type. Make
extension for this method.</summary>
    /// <param name = "_typeId">Print mgmt document type</param>
    /// <returns>True if feature enabled.</returns>
    [Wrappable(true), SuppressBPWarning('BPParameterNotUsed', 'Required by
signature')]
    public static boolean
isNamedDestinationEnabledByDocumentType(PrintMgmtDocumentType _typeId)
    {
        return false;
    }
}
```

API to run a format mapping where the destination that is provided is print management record specific

Use this new API to force ER to deliver generated documents by using a named ER destination.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;

/// <summary>
/// Format mapping run interface to provide an ability to specify destination
preset.
/// </summary>
public interface ERIFormatMappingRunWithNamedDestination
{
    /// <summary>
    /// Sets the named destination to the format mapping run object.
    /// </summary>
    /// <param name = "_destinationNamedId">A rec id of named destination.</param>
    /// <returns>The format mapping run object.</returns>
    ERIFormatMappingRun withDestinationNamed(RefRecId _destinationNamedId)
    {
    }
}
```

Applicability

To set up named destinations and force the ER framework to use the named destination, turn on the **Allow to set up ER destinations per print management item** feature in the Feature management workspace.

Additional resources

Electronic reporting (ER) overview

Electronic reporting (ER) destinations

Change code to enable users to configure and use named ER destinations

Configure action-dependent ER destinations

Configure print management record-specific ER destinations

Last updated on 03/17/2026

Electronic reporting framework API changes for Application update 10.0.23

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how the application programming interfaces (APIs) of the Electronic reporting (ER) framework are changed in Microsoft Dynamics 365 Finance version 10.0.23.

API to extend the list of ER sources of inbound documents

To implement a custom ER source of inbound documents, use the following public interfaces:

- `ERIImportFile`
- `ERIImportFileSourceSettings`
- `ERIImportFileSourceSettingsStorage`

cs

```cs
using Microsoft.Dynamics365.LocalizationFramework.XppSupportLayer;

namespace Microsoft.Dynamics365.LocalizationFramework
{
    /// <summary>
    /// The import file object.
    /// </summary>
    public interface ERIImportFile : ERIFile
    {
        /// <summary>
        /// Gets or sets a file source settings key.
        /// </summary>
        string SourceSettingsKey { get; set; }

        /// <summary>
        /// Sets a file id. File ID should be unique string with max length 300.
        /// </summary>
        string FileId { set; }

        /// <summary>
```

```
        /// Gets or sets a file name. File name with max length 259.
        /// </summary>
        string Name { get; set; }

        /// <summary>
        /// Gets or sets a file created date time.
        /// </summary>
        utcdatetime CreatedDateTime { get; set; }

        /// <summary>
        /// Sets a file modified date time.
        /// </summary>
        utcdatetime ModifiedDateTime { set; }
    }
}
```

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using System.IO;

/// <summary>
/// Format source settings required to store settings for a specific sources for
the import format.
/// </summary>
/// <remarks>
/// The implementation object should have static create(container) method for a
proper packing.
/// </remarks>
public interface ERIImportFileSourceSettings extends SysPackable
{
    /// <summary>
    /// Gets a name of the source.
    /// </summary>
    /// <returns>The name.</returns>
    str getName()
    {
    }

    /// <summary>
    /// Validates the settings.
    /// </summary>
    /// <returns>True, if validated successful; otherwise - false.</returns>
    boolean validate()
    {
    }

    /// <summary>
    /// Determines if current settings are enabled.
    /// </summary>
    /// <returns>True if enabled; otherwise - false.</returns>
    boolean isEnabled()
    {
    }
```

```
    /// <summary>
    /// Sets settings enabling state.
    /// </summary>
    /// <param name = "_value">An enabling state.</param>
    void setEnabled(boolean _value)
    {
    }

    /// <summary>
    /// Gets a settings key, this key should start with or be an implementation
class name. This key should also be persistent and shouldn't be changed over time.
    /// This key will be used to retrieve this type of settings from a settings
storage.
    /// </summary>
    /// <remarks>Examples of good keys: MyCustomFileSourceSettings,
MyCustomFileSourceSettings#UniqueKey.</remarks>
    /// <returns>The settings key.</returns>
    str getKey()
    {
    }

    /// <summary>
    /// Creates a file source for current settings.
    /// </summary>
    /// <param name = "_fileMask">A file mask.</param>
    /// <returns>A new instance of a file source.</returns>
    ERIFileSource createFileSource(str _fileMask)
    {
    }

    /// <summary>
    /// Post processing for resulted file. An action after file importing.
    /// </summary>
    /// <param name = "_args">Arguments for file post import process.</param>
    void fileImported(ERFileImportedArgs _args)
    {
    }

}
```

X++

```
/// <summary>
/// Provides access to the source settings storage.
/// </summary>
public interface ERIImportFileSourceSettingsStorage
{
    /// <summary>
    /// Adds settings to the format source settings storage.
    /// </summary>
    /// <param name = "_settings">A given settings to add.</param>
    public void addSettings(ERIImportFileSourceSettings _settings)
    {
```

```
    }

    /// <summary>
    /// Gets source settings by the key.
    /// </summary>
    /// <param name = "_key">A given key.</param>
    /// <returns>The source settings.</returns>
    public ERIImportFileSourceSettings getSettingsByKey(str _key)
    {
    }

}
```

These interfaces let you offer custom source parameters in the **ER source settings** dialog box, so that they can be set at design time. The configured source can be used to access, in a configured custom source, inbound documents that must be imported at runtime.

To learn more about this interface, see the example in Implement a custom source of inbound documents.

Additional resources

Electronic reporting (ER) overview

Parse incoming documents

Configure access to SharePoint for file storage

Last updated on 03/17/2026

Electronic reporting framework API changes for Application update 10.0.25

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how the application programming interfaces (APIs) of the Electronic reporting (ER) framework are changed in Microsoft Dynamics 365 Finance version 10.0.25.

API to enable a model mapping to run in batch mode

The initial API of the ER framework lets you run an ER model mapping for data import in interactive mode. To specify an ER format for inbound file parsing and an ER model mapping for application data update, this API uses the format name and the integration point.

The new `ERIModelMappingDestinationWithVirtualSourceRun` public interface of the ER framework lets you run an ER model mapping in batch mode for data import of an inbound file. To provide the file, you can either manually select it in the ER user interface (UI) or provide it programmatically.

The code of the `ERModelMappingDestinationRun` class shows how you can use the new API to include the **Run in the background** tab in the dialog box so that data import can run in batch mode.

X++

```
private ERModelMappingDestinationRunController getControllerInternal()
    {
        if (!controller)
        {
            controller = ERModelMappingDestinationRunController::construct();
            controller.parmShowDialog(showPromptDialog);
            controller.showBatchTab(showPromptDialog && showBatchTab &&
this.isBatchEnabledForFileSource());
            controller.parmDialogCaption(this.getBatchJobCaption());
            controller.parmModelMappingRun(this);
            controller.parmModelDefinitionParameters(parameters);
```

```
            controller.parmHiddenParameters(hiddenParameters);

controller.parmIsImportLoadingFilesFromSource(this.getImportFormatFileSourceContrac
t() != null);
        }
        return controller;
    }

    private boolean isBatchEnabledForFileSource()
    {
        var fileSource = this.getImportFormatFileSourceContract();
        if (ERCast::asAny(fileSource) is
ERImportFormatDatasourceByVirtualFileSourceContract)
        {
            return useVirtualImportFormatSource;
        }
        else
        {
            return ERCast::asAny(fileSource) is
ERImportFormatDatasourceByFileSourceContract;
        }
    }
```

To learn more about this interface, see the examples in Import data from manually selected files in batch mode.

Last updated on 03/17/2026

Specify a custom storage location for generated documents

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The application programming interface (API) of the Electronic reporting (ER) framework lets you extend the list of storage locations for documents that ER formats generate. This article provides an overview of the main tasks you must complete to add a custom storage location.

Prerequisites

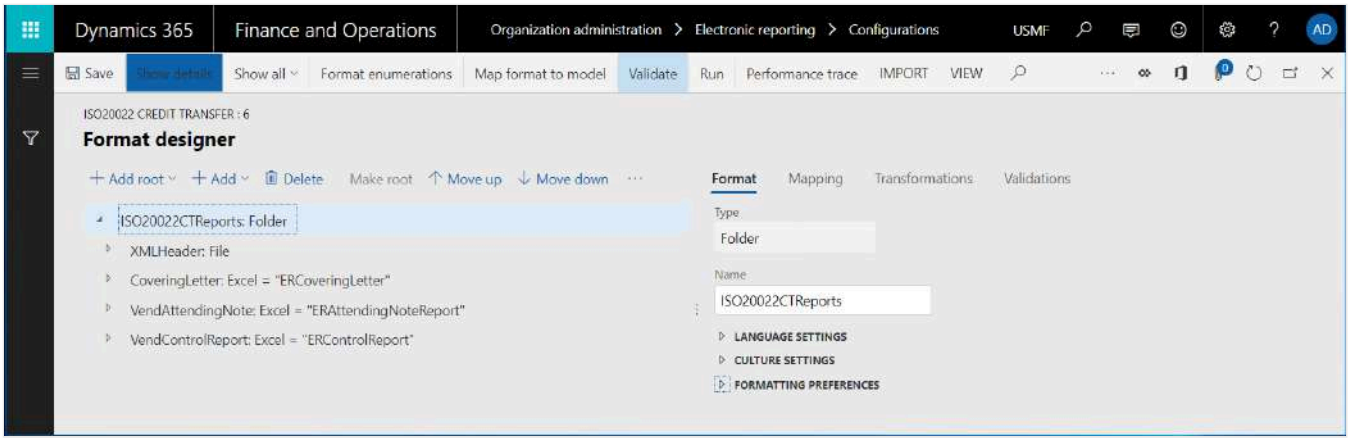
You must deploy a topology that supports continuous build. For more information, see Deploy topologies that support continuous build and test automation. You must have access to this topology for one of the following roles:

- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

You must also have access to the development environment for this topology.

Create or import an ER format configuration

In the current topology, create a new ER format to generate documents that you plan to add a custom storage location for. Alternatively, import an existing ER format into this topology.

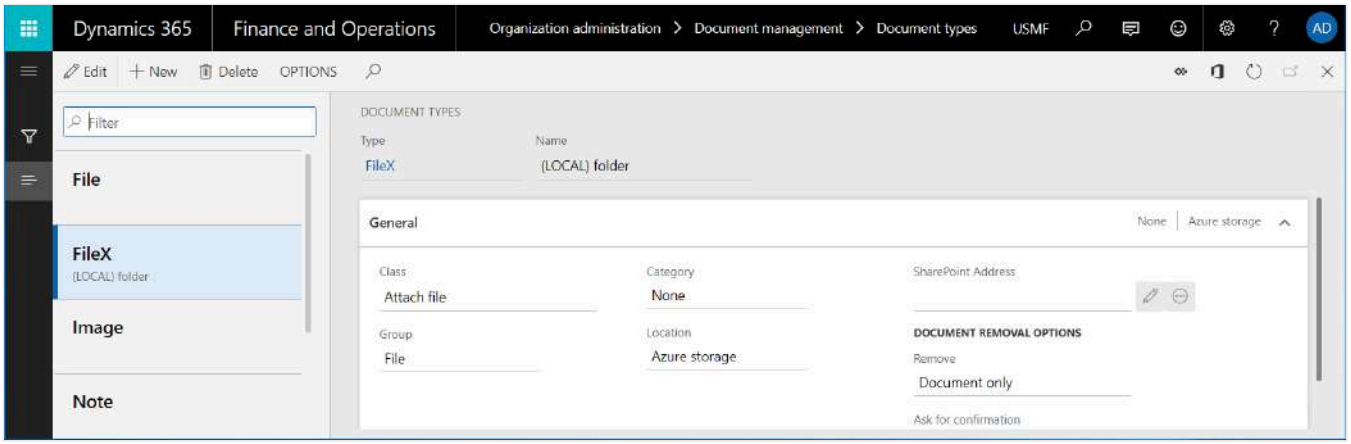


> **Important**
>
> The ER format that you create or import must contain at least one of the following format elements:
>
> - File
> - Folder
> - Merger
> - Attachment

Create a new document type

To specify how to route documents that an ER format generates, configure Electronic reporting (ER) destinations. In each ER destination that you configure to store generated documents as files, specify a document type of the Document management framework. Use different document types to route documents that different ER formats generate.

1. Add a new document type for the ER format that you created or imported earlier. In the following illustration, the document type is **FileX**.
2. To differentiate this document type from other document types, include a specific keyword in its name. For example, in the following illustration, the name is **(LOCAL) folder**.
3. In the **Class** field, specify **Attach file**.
4. In the **Group** field, specify **File**.



> **Note**
>
> Document types are company-specific. To use an ER format with a configured destination in multiple companies, you must configure a separate document type in each company.

Review source code

Review the code of the **insertFile()** method of the **ERDocuManagement** class. Notice that the **AttachingFile()** event is raised while the generated file is attached to a record.

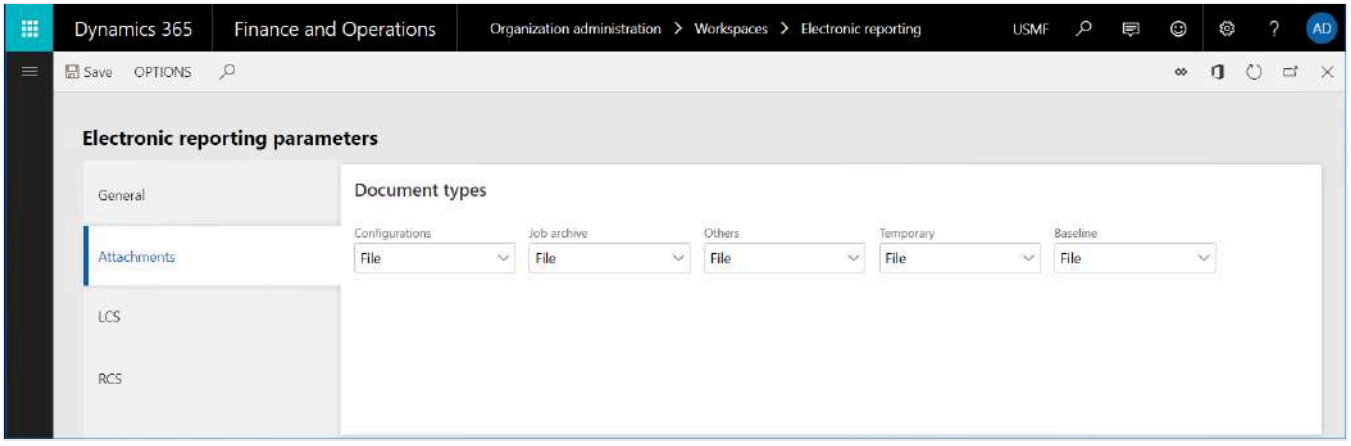
X++

```
/// <summary>
/// Inserts file as attachment in Document Management.
/// </summary>
/// <param name = "_owner">A record as the attachment owner.</param>
/// <param name = "_stream">The file stream.</param>
/// <param name = "_filePath">The file path with name.</param>
/// <param name = "_attachmentName">The name of file attachment.</param>
/// <returns>The reference to inserted file.</returns>
[Hookable(false)]
public DocuRef insertFile(
    Common _owner,
    System.IO.Stream _stream,
    str _filePath,
    str _attachmentName,
    DocuTypeId _docuTypeId)
{
    DocuRef docuRef;
    if (_stream)
    {
        DocuType::createDefaults();
        if (!this.isDocuTypeValid(_docuTypeId))
        {
            throw error(strFmt("@ElectronicReporting:DocuTypeIsNotValid",
_docuTypeId));
```

```
        }
        var args = ERDocuManagementAttachingFileEventArgs::construct(_owner,
_stream, _filePath, _attachmentName, _docuTypeId);
        ERDocuManagementEvents::onAttachingFile(args);
        if (args.isHandled())
        {
            docuRef = args.getDocuRef();
        }
        else
        {
            docuRef = this.attachFile(_owner, _stream, _filePath, _attachmentName,
_docuTypeId);
        }
    }
    return docuRef;
}
```

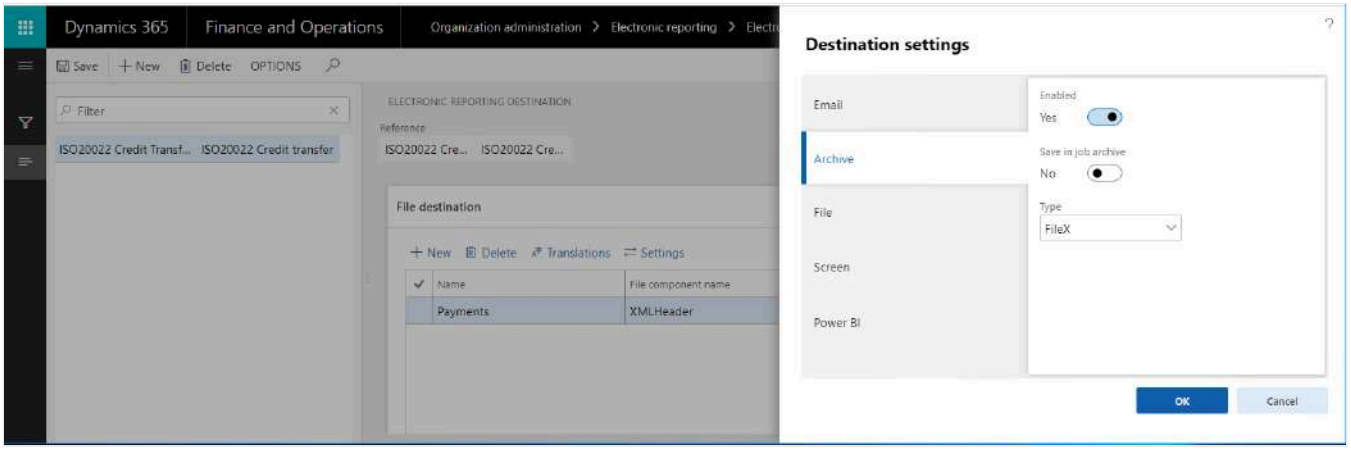
The **AttachingFile()** event is raised when the following ER destinations are processed:

- **Archive** – When you use this destination, a new record for the ER format that runs is created in the ERFormatMappingRunJobTable table. The **Archived** field in this record is set to **False**. If the ER format successfully runs, the generated document is attached to this record, and the **AttachingFile()** event is raised. The document type that is selected in this ER destination determines the storage location for the attached file (Microsoft Azure Storage or a Microsoft SharePoint folder).
- **Job archive** – When you use this destination, a new record for the ER form that runs is created in the ERFormatMappingRunJobTable table. The **Archived** field in this record is set to **True**. If the ER format successfully runs, the generated document is attached to this record, and the **AttachingFile()** event is raised. The document type that is configured in the ER parameters determines the storage location for the attached file (Azure Storage or a SharePoint folder).



Configure an ER destination

1. Configure the archived destination for one of the previously mentioned elements (file, folder, merger, or attachment) of the ER format that you created or imported. For guidance, see ER Configure destinations.
2. Use the document type that you added earlier for the configured destination. (For the example in this article, the document type is **FileX**.)



Modify source code

1. Add a new class to your Microsoft Visual Studio project, and write code to subscribe to the **AttachingFile()** event that was mentioned earlier. (For more information about the extensibility pattern that is used, see Respond by using EventHandlerResult.) For example, in the new class, write code that performs the following actions:
   a. Store generated files in a folder of the local file system of the server that runs the Application Object Server (AOS) service.
   b. Store these generated files only when the new document type (for example, the **FileX** type that has the "(LOCAL)" keyword in its name) is used while a file is attached to the record in the ER execution job log.

X++

```
class ERDocuSubscriptionSample
{
    void new()
    {
    }
    [SubscribesTo(classStr(ERDocuManagementEvents),
    staticDelegateStr(ERDocuManagementEvents,
    attachingFile))]
    public static void
ERDocuManagementEvents_attachingFile(ERDocuManagementAttachingFileEventArgs
_args)
    {
        if (!_args.isHandled())
        {
```

```
            DocuType docuType = DocuType::find(_args.getDocuTypeId());
            if (strContains(docuType.Name, '(LOCAL)'))
            {
                _args.markAsHandled();
                var stream = _args.getStream();
                if (stream.CanSeek)
                {
                    stream.Seek(0, System.IO.SeekOrigin::Begin);
                }
                using (var localStream = System.IO.File::OpenWrite(@'c:\0\' +
_args.getAttachmentName()))
                {
                    stream.CopyTo(localStream);
                }
            }
        }
    }
}
```

2. Rebuild your project.

Run the ER format that you created or imported

1. Execute the ER format that you created or imported.
2. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting jobs**. Find the record that was created for this execution job, and that has the generated file attached to it.
3. Explore the local **C:\0** folder to find same generated file.

Additional resources

- Electronic reporting (ER) destinations
- Extensibility home page

Last updated on 04/13/2026

Specify custom storage locations for generated documents

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The application programming interface (API) of the Electronic reporting (ER) framework lets you extend the list of storage locations for documents that ER formats generate. This article explains how to add a custom storage location for generated documents by delegating the task of creating ER destinations to the default destination factory and then implementing a custom class that has its own destination logic.

Prerequisites

Deploy a topology that supports continuous build. For more information, see Deploy topologies that support continuous build and test automation. You must have access to this topology for one of the following roles:

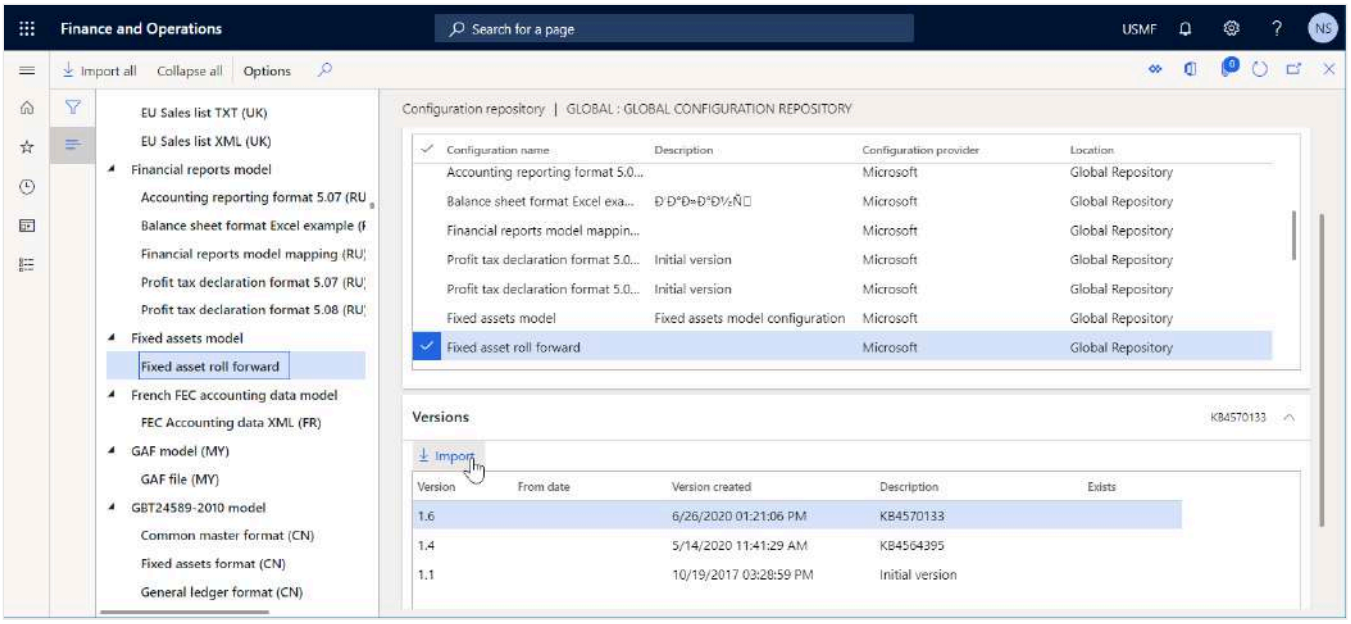
- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

You must also have access to the development environment for this topology.

You can complete all the tasks in this article in the **USMF** company.

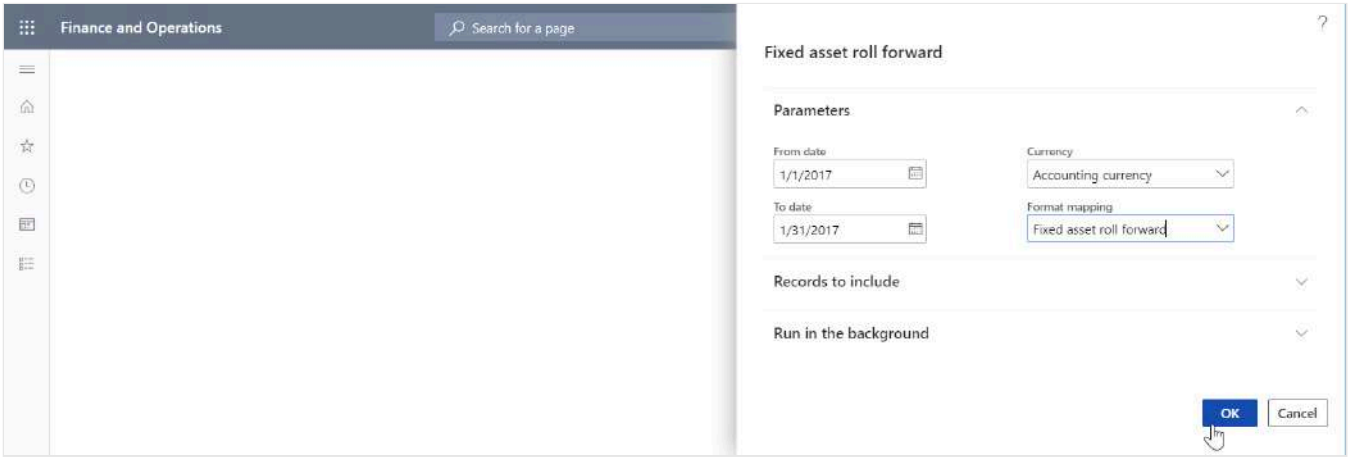
Import the Fixed asset roll forward ER format

To generate the documents that you plan to add a custom storage location for, import the **Fixed asset roll forward** ER format configuration into the current topology.



Run the Fixed asset roll forward report

1. Go to **Fixed assets** > **Inquiries and reports** > **Transaction reports** > **Fixed asset roll forward**.
2. In the **From date** field, enter **1/1/2017** (January 1, 2017).
3. In the **To date** field, enter **1/31/2017** (January 31, 2017).
4. In the **Currency** field, select **Accounting currency**.
5. In the **Format mapping** field, select **Fixed asset roll forward**.
6. Select **OK**.



In Microsoft Excel, review the outbound document that is generated and available for download. This behavior is the default behavior for an ER format that no destinations are configured for, and that is running in interactive mode.

Review the source code

Review the code of the `generateReportByGER()` method of the `AssetRollForwardService` class. Notice that the `Run()` method is used to call the ER framework and generate the **Fixed asset roll forward** report.

X++

```
class AssetRollForwardService extends SysOperationServiceBase
{
    public const str ERFormatModelName = 'Fixed assets';
    public const str ERModelDataSourceName = 'model';
    public const str DefaultExportedFileName = 'AssetRollForward';

    /// <summary>
    /// Generates report by general electronic reporting
    /// </summary>
    /// <param name = "_contract">The Asset Period Statement contract</param>
    public void generateReportByGER(AssetRollForwardContract _contract)
    {
        ERFormatMappingId   formatMappingId;
        AssetRollForwardDP  dataProvider;
        AssetRollForwardTmp assetRollForwardTmp;
        Query               query;

        query = new Query(SysOperationHelper::base64Decode(_contract.parmQuery()));
        dataProvider = AssetRollForwardDP::construct();
        formatMappingId = _contract.parmFormatMapping();
        assetRollForwardTmp = dataProvider.getAssetRollForwardTmp(_contract,
query);

        if (assetRollForwardTmp)
        {
            try
            {
                ERIModelDefinitionParamsAction parameters = new
ERModelDefinitionParamsUIActionComposite()
                    .add(new
ERModelDefinitionDatabaseContext().addTemporaryTable(assetRollForwardTmp))
                    .add(new
ERModelDefinitionObjectParameterAction(ERModelDataSourceName, 'MyParameters',
_contract, true));

                // Call ER to generate the report.

EREObjectsFactory::createFormatMappingRunByFormatMappingId(formatMappingId,
DefaultExportedFileName)
                    .withParameter(parameters)
                    .withFileDestination(_contract.getFileDestination())
                    .run();
            }
            catch
            {
                // An error occurred while exporting data.
                error("@SYP4861341");
```

```
            }
        }
        else
        {
            // There is no data available.
            info("@SYS300117");
        }
    }
}
```

Modify the source code

1. In your Visual Studio project, add a new class (`AssetRollForwardDestination` in this example), and write code to implement your custom destination for **Fixed asset roll forward** reports that are generated.

    - The `new()` method gets the original ER destination object and the application logic–driven parameter that specifies the custom location where generated reports should be stored. In this example, the custom location is the name of a folder on the local file system of the server that runs the Application Object Server (AOS) service.
    - The `saveFile()` method saves a generated document to a folder on the local file system of the server that runs the AOS service.

    X++

```
using Microsoft.Dynamics365.LocalizationFramework;

/// <summary>
/// Destination class for <c>AssetRollForwardDestinationFactory</c> that
stores a generated report.
/// </summary>
public class AssetRollForwardDestination implements ERIFileDestination
{
    private ERIFileDestination originDestination;
    private str TargetFolder;

    /// <summary>
    /// Creates a stream for new file.
    /// </summary>
    /// <param name = "_fileName">Name of a new file.</param>
    /// <returns>Stream for new file.</returns>
    public System.IO.Stream newFileStream(System.String _fileName)
    {
        return originDestination.newFileStream(_fileName);
    }

    /// <summary>
    /// Saves file in destination.
```

```
    /// </summary>
    /// <param name = "_stream">A stream to save.</param>
    /// <param name = "_fileName">A file name.</param>
    /// <returns>Saved stream.</returns>
    public System.IO.Stream saveFile(System.IO.Stream _stream, System.String
_fileName)
    {
        _stream.Seek(0, System.IO.SeekOrigin::Begin);
        using (var localStream = System.IO.File::OpenWrite(TargetFolder +
_fileName))
        {
            _stream.CopyTo(localStream);
        }
        return _stream;
    }

    /// <summary>
    /// Constructs destination for fixed asset roll forward report.
    /// </summary>
    /// <param name = "_originDestination">The original destination.</param>
    /// <param name = "_TargetFoder">The folder to store a report that's being
created.</param>
    /// <returns>The fixed asset roll forward destination.</returns>
    public static AssetRollForwardDestination construct(ERIFileDestination
_originDestination, str _TargetFoder)
    {
        return new AssetRollForwardDestination(_originDestination,
_TargetFoder);
    }

    protected void new(ERIFileDestination _originDestination, str
_TargetFoder)
    {
        originDestination = _originDestination;
        TargetFolder = _TargetFoder;
    }
}
```

2. In your Visual Studio project, add a new class (`AssetRollForwardDestinationFactory` in this example), and write code to set up a custom destination factory that delegates the creation of a destination to the default destination factory, and to wrap a file destination with your own destination.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using Microsoft.Dynamics365.LocalizationFramework.Format.FileGeneration;

/// <summary>
/// Destination factory for using <c>AssetRollForwardDestinationFactory</c>.
/// </summary>
public class AssetRollForwardDestinationFactory implements
```

```
ERIFileDestinationFactory, ERIFileDestinationFactoryPostProcessor
{
    private ERIFileDestinationFactory originDestinationFactory;
    private str TargetFolder;

    /// <summary>
    /// Creates file destination for print management.
    /// </summary>
    /// <param name = "_fileDestination">A default file destination.</param>
    /// <param name = "_identification">An identification strategy.</param>
    /// <param name = "_rootContext">A root context.</param>
    /// <param name = "_root">A root element.</param>
    /// <returns>A file destination.</returns>
    public ERIDataDrivenFileDestination createPrintMgmtDestination(
        ERIFileDestination _fileDestination,
        ERIObjectIdentificationStrategy _identification,
        ERTextFormatDataContext _rootContext,
        ERTextFormatIFileComponent _root)
    {
        ERIDataDrivenFileDestination dataDrivenDestination =
originDestinationFactory.createPrintMgmtDestination(
            _fileDestination,
            _identification,
            _rootContext,
            _root);

        IFileDestinationHost fileDestinationHost = dataDrivenDestination as
IFileDestinationHost;
        if (fileDestinationHost)
        {
            fileDestinationHost.FileDestination =
AssetRollForwardDestination::construct(
                fileDestinationHost.get_FileDestination(),
                TargetFolder);
        }

        return dataDrivenDestination;
    }

    /// <summary>
    /// Constructs the fixed asset roll forward destination factory.
    /// </summary>
    /// <param name = "_originDestinationFactory">The original destination.
</param>
    /// <param name = "_TargetFolder">The string containing a folder name that
corresponds to the report, that's being created.</param>
    /// <returns>The destination factory for the fixed asset roll forward
report.</returns>
    public static AssetRollForwardDestinationFactory
construct(ERIFileDestinationFactory _originDestinationFactory, str
_TargetFolder)
    {
        AssetRollForwardDestinationFactory destinationFactory = new
AssetRollForwardDestinationFactory(_originDestinationFactory, _TargetFolder);
```

```
        ERIFileDestinationFactoryPostProcessorsHost factoryPostProcessing =
ERCast::asObject(destinationFactory.originDestinationFactory) as
ERIFileDestinationFactoryPostProcessorsHost;

        if (factoryPostProcessing)
        {

factoryPostProcessing.addDestinationPostProcessor(destinationFactory);
        }
        return destinationFactory;
    }

    public ERIFileDestination
processDestinationAfterCreation(ERIFileDestination _sourceDestination)
    {
        return AssetRollForwardDestination::construct(_sourceDestination,
TargetFolder);
    }

    protected void new(ERIFileDestinationFactory _originDestinationFactory,
str _TargetFolder)
    {
        originDestinationFactory = _originDestinationFactory;
        TargetFolder = _TargetFolder;
    }
}
```

3. Modify the existing `AssetRollForwardService` class, and write code to set up a custom destination factory for the report runner. When a custom destination factory is constructed, the application-driven parameter that specifies a target folder is passed. In this way, the target folder stores generated files.

> **Note**
>
> Make sure that the specified folder (**c:\0** in this example) is present in the local file system of the server that runs the AOS service. Otherwise, a DirectoryNotFoundException exception is thrown at runtime.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
/// <summary>
/// The electronic reporting service class for fixed asset roll forward report
/// </summary>
class AssetRollForwardService extends SysOperationServiceBase
{
    public const str ERFormatModelName = 'Fixed assets';
    public const str ERModelDataSourceName = 'model';
```

```
    public const str DefaultExportedFileName = 'AssetRollForward';

    /// <summary>
    /// Generates report by general electronic reporting
    /// </summary>
    /// <param name = "_contract">The Asset Period Statement contract</param>
    public void generateReportByGER(AssetRollForwardContract _contract)
    {
        ERFormatMappingId   formatMappingId;
        AssetRollForwardDP  dataProvider;
        AssetRollForwardTmp assetRollForwardTmp;
        Query               query;

        query = new
Query(SysOperationHelper::base64Decode(_contract.parmQuery()));
        dataProvider = AssetRollForwardDP::construct();
        formatMappingId = _contract.parmFormatMapping();
        assetRollForwardTmp = dataProvider.getAssetRollForwardTmp(_contract,
query);

        if (assetRollForwardTmp)
        {
            try
            {
                ERIModelDefinitionParamsAction parameters = new
ERModelDefinitionParamsUIActionComposite()
                    .add(new
ERModelDefinitionDatabaseContext().addTemporaryTable(assetRollForwardTmp))
                    .add(new
ERModelDefinitionObjectParameterAction(ERModelDataSourceName, 'MyParameters',
_contract, true));

                // Call ER to generate the report.
                ERIFormatMappingRun formatMappingRun =
ERObjectsFactory::createFormatMappingRunByFormatMappingId(formatMappingId,
DefaultExportedFileName);
                formatMappingRun.withParameter(parameters);

formatMappingRun.withFileDestination(_contract.getFileDestination());

                ERIFileDestinationFactoryHost factoryHost =
ERCast::asObject(formatMappingRun) as ERIFileDestinationFactoryHost;
                if (factoryHost)
                {
                    ERIFileDestinationFactory fileDestinationFactory =
factoryHost.getFileDestinationFactory();

factoryHost.setFileDestinationFactory(AssetRollForwardDestinationFactory::cons
truct(fileDestinationFactory, @'c:\0\'));
                    formatMappingRun.run();
                }
            }
            catch
            {
                // An error occurred while exporting data.
```

```
                error("@SYP4861341");
            }
        }
        else
        {
            // There is no data available.
            info("@SYS300117");
        }
    }
}
```

4. Rebuild your project.

Re-run the Fixed asset roll forward report

1. Go to **Fixed assets** > **Inquiries and reports** > **Transaction reports** > **Fixed asset roll forward**.
2. In the **From date** field, enter **1/1/2017**.
3. In the **To date** field, enter **1/31/2017**.
4. In the **Currency** field, select **Accounting currency**.
5. In the **Format mapping** field, select **Fixed asset roll forward**.
6. Select **OK**.
7. Browse the local **C:\0** folder to find the generated file.

> **Note**
>
> Because the `originDestination` object isn't used in the `AssetRollForwardDestination` object in this example, the configurations for the ER format destinations is ignored at runtime.

Additional resources

- Electronic reporting (ER) destinations
- Extensibility home page

Last updated on 04/13/2026

Implement a custom destination for generated documents

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The application programming interface (API) of the Electronic reporting (ER) framework lets you extend the list of ER destinations that you can use to store documents that ER formats generate. This article provides an overview of the main tasks that you must complete to implement a custom ER destination.

Prerequisites

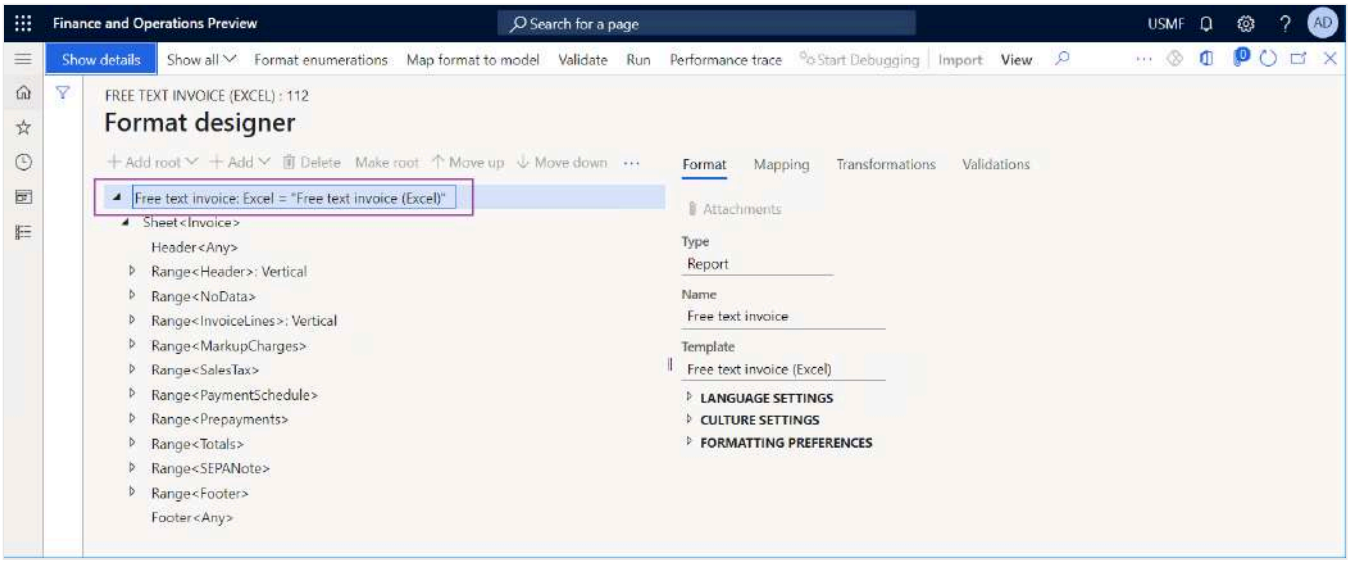
You must deploy a topology that supports continuous build. For more information, see Deploy topologies that support continuous build and test automation. You must have access to this topology for one of the following roles:

- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

You must also have access to the development environment for this topology.

Create or import an ER format configuration

In the current topology, create a new ER format to generate the documents that you plan to store by using a custom ER destination. Alternatively, you can import an existing ER format into this topology.



> **ⓘ Important**
>
> The ER format that you create or import must contain at least one of the following format elements:
>
> - Common\File
> - Common\File attachment
> - Common\Folder
> - Excel\File
> - PDF\Merger
> - PDF\File

Extend the source code

1. Append your Microsoft Visual Studio project with a new class (`ERFileDestinationFolder` in this example), and write code that uses the `ERIFileDestination` public interface to implement your custom destination. In the following example, the class is designed to store generated documents as files in a folder of the local file system.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;

public class ERFileDestinationFolder implements ERIFileDestination
{
    private str folderPath;
    private boolean overwrite;

    public str parmFolderPath(str _value = folderPath)
```

```
    {
        folderPath = _value;
        return folderPath;
    }

    public boolean parmOverwrite(boolean _value = overwrite)
    {
        overwrite = _value;
        return overwrite;
    }

    [Hookable(false)]
    public System.IO.Stream saveFile(System.IO.Stream _stream, str _filePath)
    {
        this.sendFileToFolder(_stream,
System.IO.Path::GetFileName(_filePath));
        return _stream;
    }

    [Hookable(false)]
    public System.IO.Stream newFileStream(str _filePath)
    {
        return new System.IO.MemoryStream();
    }

    private str getFileName(str _fileName)
    {
        if (overwrite)
        {
            return _fileName;
        }
        System.DateTime now = DateTimeUtil::utcNow();
        return strFmt('%1_%2%3',
System.IO.Path::GetFileNameWithoutExtension(_fileName), now.ToString('yyyy-M-
d_HH_mm_ss'), System.IO.Path::GetExtension(_fileName));
    }

    private void sendFileToFolder(System.IO.Stream _stream, str _fileName)
    {
        var resultPath = System.IO.Path::Combine(this.folderPath,
this.getFileName(_fileName));
        using(var fileStream = System.IO.File::Create(resultPath))
        {
            if( _stream.CanSeek)
            {
                _stream.Seek(0, System.IO.SeekOrigin::Begin);
            }
            _stream.CopyTo(fileStream);
        }
    }

    /// <summary>
    /// Finalizes files processing.
    /// </summary>
    [Hookable(false)]
```

```
    public void finalize()
    {
    }
}
```

2. Add a new class to your Visual Studio project (`ERFormatDestinationSaveInFolderSettings` in this example) to use the `ERIFormatFileDestinationSettings` public interface to write code that specifies how a custom destination is created, how its parameters are packed for storage, and how its parameters are unpacked for presentation in the user interface (UI).

X++

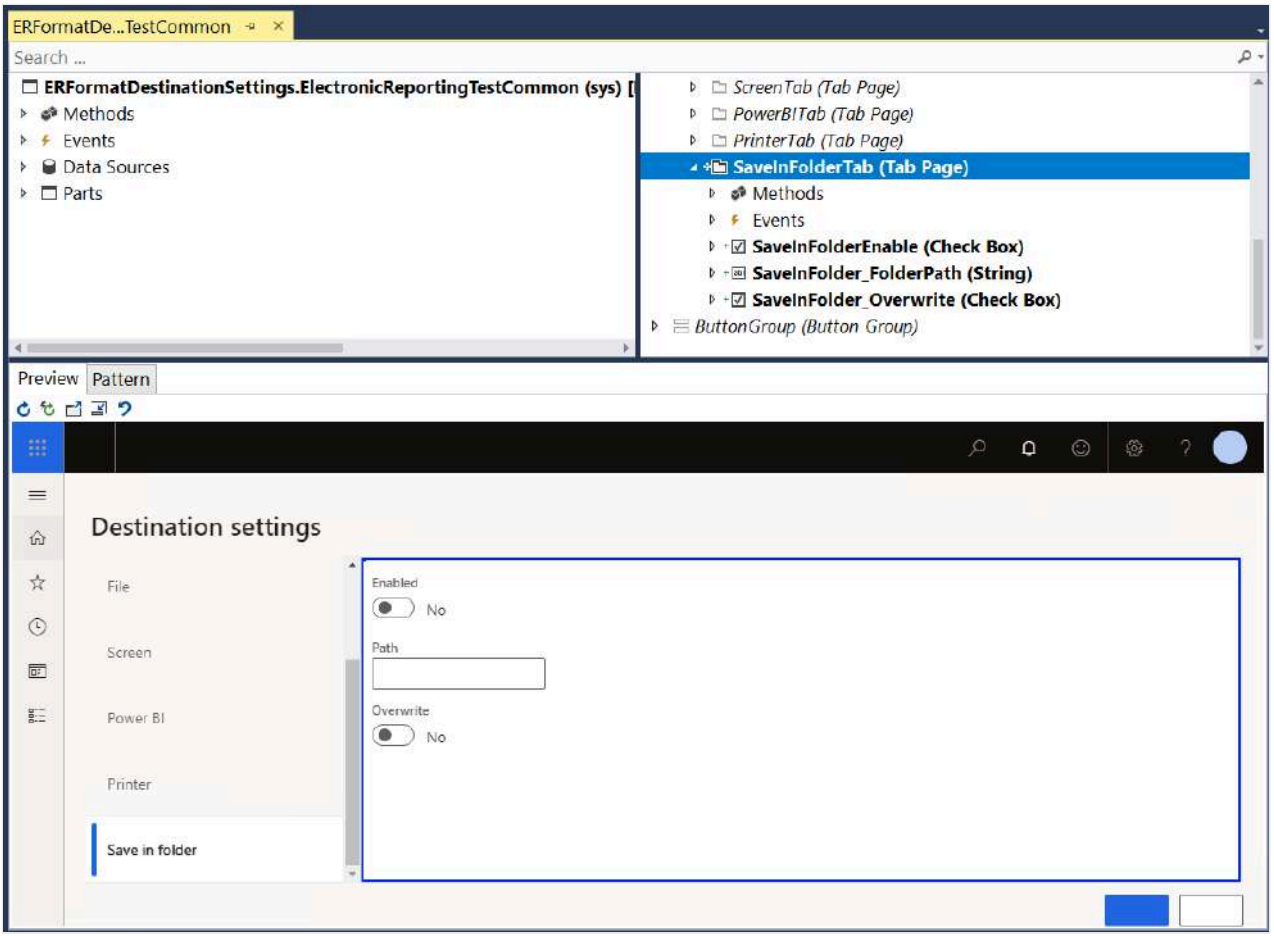
```
using Microsoft.Dynamics365.LocalizationFramework;

public class ERFormatDestinationSaveInFolderSettings implements
ERIFormatFileDestinationSettings
{
    internal const str SettingsKey =
classStr(ERFormatDestinationSaveInFolderSettings);
    public const int CurrentVersion = 1;

    private boolean isEnabled;
    private boolean overwrite;
    private str folderPath;

    [Hookable(false)]
    public str parmFolderPath(str _value = folderPath)
    {
        folderPath = _value;
        return folderPath;
    }

    [Hookable(false)]
    public boolean parmIsEnabled(boolean _value = isEnabled)
    {
        isEnabled = _value;
        return isEnabled;
    }

    [Hookable(false)]
    public boolean parmOverwrite(boolean _value = overwrite)
    {
        overwrite = _value;
        return overwrite;
    }

    [Hookable(false)]
    public void setEnabled(boolean _value)
    {
        isEnabled = _value;
```

```
    }

    [Hookable(false)]
    public boolean isEnabled()
    {
        return isEnabled;
    }

    [Hookable(false)]
    public str getName()
    {
        return 'Save in folder';
    }

    [Hookable(false)]
    public void validate()
    {
    }

    [Hookable(false)]
    public ERIFileDestination createDestination(ERDestinationExecutionContext
_destinationContext, boolean _isGrouped)
    {
        ERFileDestinationFolder ret = new ERFileDestinationFolder();
        ret.parmFolderPath(this.parmFolderPath());
        ret.parmOverwrite(this.parmOverwrite());
        return ret;
    }

    [Hookable(false)]
    public str getKey()
    {
        return SettingsKey;
    }

    [Hookable(false)]
    public container pack()
    {
        return [CurrentVersion, isEnabled, folderPath];
    }

    [Hookable(false)]
    public boolean unpack(container packedClass)
    {
        int version = RunBase::getVersion(packedClass);
        switch (version)
        {
            case CurrentVersion:
                [version, isEnabled, folderPath] = packedClass;
                return true;
            default:
                return false;
        }
    }
```

```
    /// <summary>
    /// Creates a new instance of an objected from a packed class container.
    /// </summary>
    /// <param name = "_packedClass">A packed class container.</param>
    /// <returns>The new instance of an object.</returns>
    public static ERFormatDestinationSaveInFolderSettings create(container
_packedClass)
    {
        var ret = new ERFormatDestinationSaveInFolderSettings();
        ret.unpack(_packedClass);
        return ret;
    }
}
```

> **Note**
>
> In the preceding code, the `ERFormatDestinationSaveInFolderSettings` class is introduced as the `static` class. The implementation object should have a `static create(container)` method that creates an object and unpacks the container that the object requires for a correct pack-unpack process. For more information, see Pack-Unpack Design Pattern.

3. In your Visual Studio project, add a new extension for the `ERFormatDestinationSettings` form, and write code that implements a custom UI for your custom destination. The following illustration shows what this UI looks like in the Visual Studio designer.



4. Add another new class (`ERFormatDestinationSettingsEventHandlers` in this example) to your Visual Studio project, and write the event handler code for an extended destination form. This step requires that the public `ERIFormatFileDestinationSettingsStorage` interface be implemented.

X++

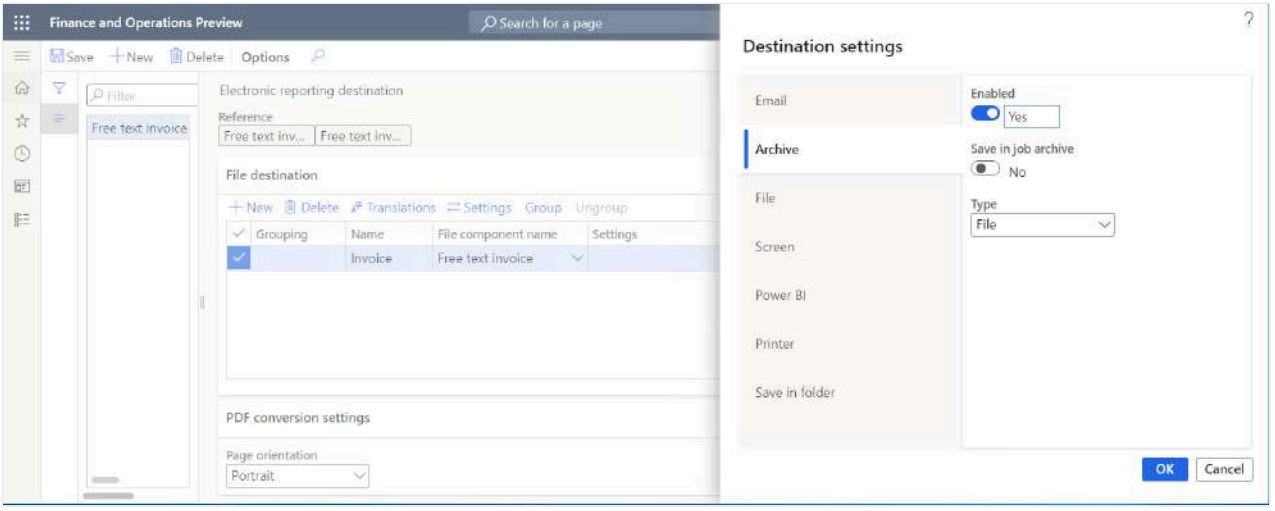
```
public class ERFormatDestinationSettingsEventHandlers
{
    [PostHandlerFor(formStr(ERFormatDestinationSettings),
formMethodStr(ERFormatDestinationSettings, init))]
    public static void ERFormatDestinationSettings_Post_init(XppPrePostArgs
args)
    {
        FormRun formRun = args.getThis();
        ERIFormatFileDestinationSettingsStorage settingsStorage =
args.getThis();
        var settings =
ERFormatDestinationSettingsEventHandlers::getSaveInFolderSettings(settingsStor
age);

        FormStringControl folderPathControl =
formRun.design().controlName(formControlStr(ERFormatDestinationSettings,
SaveInFolder_FolderPath));
        folderPathControl.text(settings.parmFolderPath());

        FormCheckBoxControl enabledControl =
```

```
formRun.design().controlName(formControlStr(ERFormatDestinationSettings,
SaveInFolderEnable));
        enabledControl.checked(settings.isEnabled());

        FormCheckBoxControl overwriteControl =
formRun.design().controlName(formControlStr(ERFormatDestinationSettings,
SaveInFolder_Overwrite));
        overwriteControl.checked(settings.parmOverwrite());
    }

    [PreHandlerFor(formStr(ERFormatDestinationSettings),
formMethodStr(ERFormatDestinationSettings, closeOk))]
    public static void ERFormatDestinationSettings_Pre_closeOk(XppPrePostArgs
args)
    {
        FormRun formRun = args.getThis();
        ERIFormatFileDestinationSettingsStorage settingsStorage =
args.getThis();
        var settings =
ERFormatDestinationSettingsEventHandlers::getSaveInFolderSettings(settingsStor
age);

        FormStringControl folderPathControl =
formRun.design().controlName(formControlStr(ERFormatDestinationSettings,
SaveInFolder_FolderPath));
        settings.parmFolderPath(folderPathControl.text());

        FormCheckBoxControl enabledControl =
formRun.design().controlName(formControlStr(ERFormatDestinationSettings,
SaveInFolderEnable));
        settings.setEnabled(enabledControl.checked());

        FormCheckBoxControl overwriteControl =
formRun.design().controlName(formControlStr(ERFormatDestinationSettings,
SaveInFolder_Overwrite));
        settings.parmOverwrite(overwriteControl.checked());
    }

    private static ERFormatDestinationSaveInFolderSettings
getSaveInFolderSettings(ERIFormatFileDestinationSettingsStorage
_settingsStorage)
    {
        ERFormatDestinationSaveInFolderSettings ret =
_settingsStorage.getSettingsByKey(ERFormatDestinationSaveInFolderSettings::Set
tingsKey);
        if (ret == null)
        {
            ret = new ERFormatDestinationSaveInFolderSettings();
            _settingsStorage.addSettings(ret);
        }
        return ret;
    }
}
```

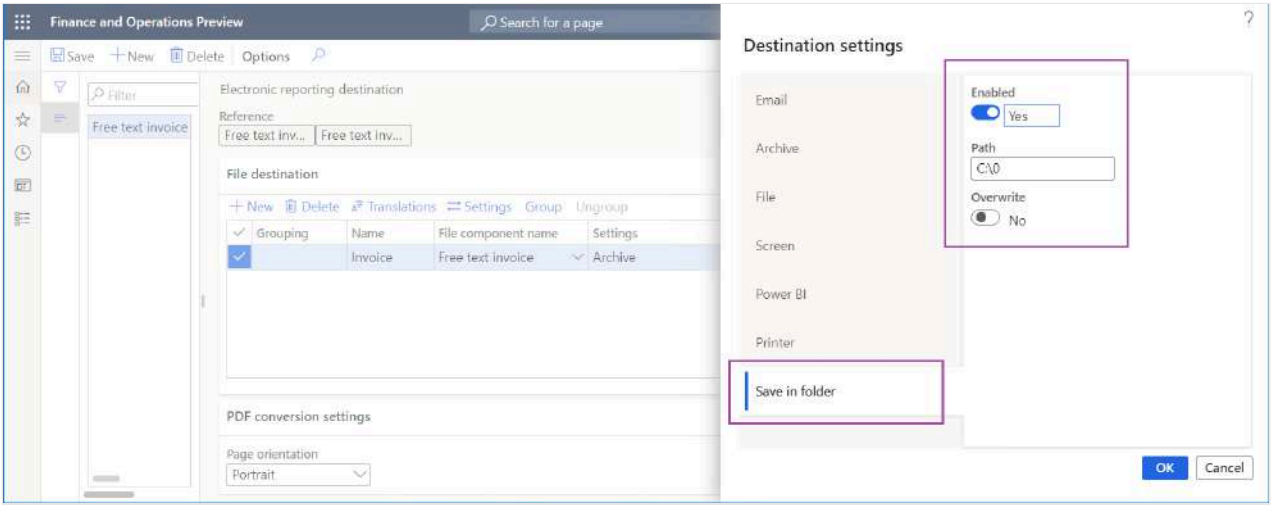
5. Rebuild your Visual Studio project.

Configure ER destinations for the ER format that you created or imported

1. Configure the Archive destination for one of the previously mentioned components (file, folder, merger, or attachment) of the ER format that you created or imported. For more information, see ER Configure destinations.



2. For the same component of the selected ER format, enable and configure the custom **Save in folder** destination.



> **Note**
>
> Make sure that the specified custom destination folder (**c:\0** in this example) is present in the local file system of the server that runs the AOS service. Otherwise, a DirectoryNotFoundException exception is thrown at runtime.

Run the ER format that you created or imported

1. Run the ER format that you created or imported.
2. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting jobs**, and find the record that was created for this execution job and that has the generated file attached to it.
3. Browse to the local **C:\0** folder to find the generated file.

Additional resources

Electronic reporting (ER) destinations

Electronic reporting framework API changes for Application update 10.0.19

Last updated on 04/09/2026

Initialize data source values of the USER INPUT PARAMETER type from source code

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you design ER model mapping and ER format components, use the data sources of the *User input parameter* type to get the necessary values at runtime before the execution of an ER format begins. You can programmatically turn off this dialog box when another page provides the required parameters or when an ER format runs in unattended (batch) mode. When the ER dialog box is turned off, you must initialize the data source values of the *User input parameter* type from source code.

Format components for outgoing electronic documents

You can configure an ER format to generate an outbound document. When you configure the format, select an ER data model as the data source to fill the outbound document. Configure an ER model mapping to specify how the selected data model is filled by application data at runtime. When you design your model mapping, specify data sources to collect the desired values from different sources and populate them to your data model. Among other data sources, you might use data sources of the *User input parameter* type to fill your data model with values that users enter on the ER user dialog box that appears when you first run an ER format.

For an ER format that you configure to generate an outbound document, you can programmatically turn off the ER dialog box by using the `_showPromptDialog` parameter of the `createFormatMappingRunByFormatMappingId()` method of the `ERObjectsFactory` class. When you turn off the ER dialog box, you need to initialize values of data sources of the *User input parameter* type from the source code. To do this, use the following pattern.

X++

```
ERModelDefinitionInputParametersAction Obj = new
ERModelDefinitionInputParametersAction();
Obj.addParameter(<path to an ER data source>, <data source value>);
ERObjectsFactory::createFormatMappingRunByFormatMappingId(formatMappingID,
fileName, false)
            .withParameter(Obj)
            .run();
```

For ER formats that you configure to generate an outbound document, the path to an ER data source is constructed from the prefix and the suffix that are separated by a slash (**/**) character. The prefix represents the name of a data source of the *Model* type that resides in the running ER format. The suffix represents the path to a data source of the *User input parameter* type that resides in a model mapping that the running ER format uses as the implementation of the relevant data model.

> **Tip**
>
> The nodes of a data source path are separated by a slash (**/**) character. Use the `ERPath::Combine()` method to construct a path.

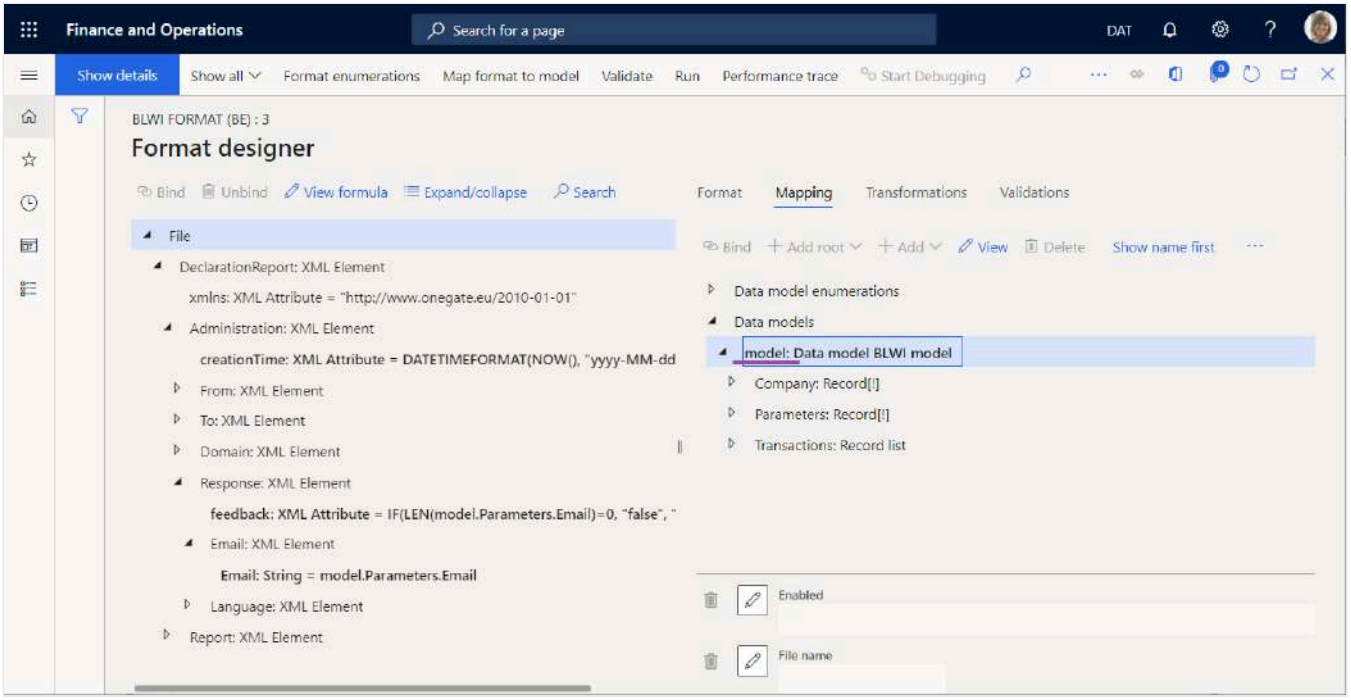
For more information, review the source code of the `BankPaymBalanceXML` application class that you can use as an example. From the Global repository, import the **BLWI format (BE)** ER format configuration that this application class calls to your Finance instance. The base **BLWI model** configuration is imported automatically with the imported ER format configuration.

x++

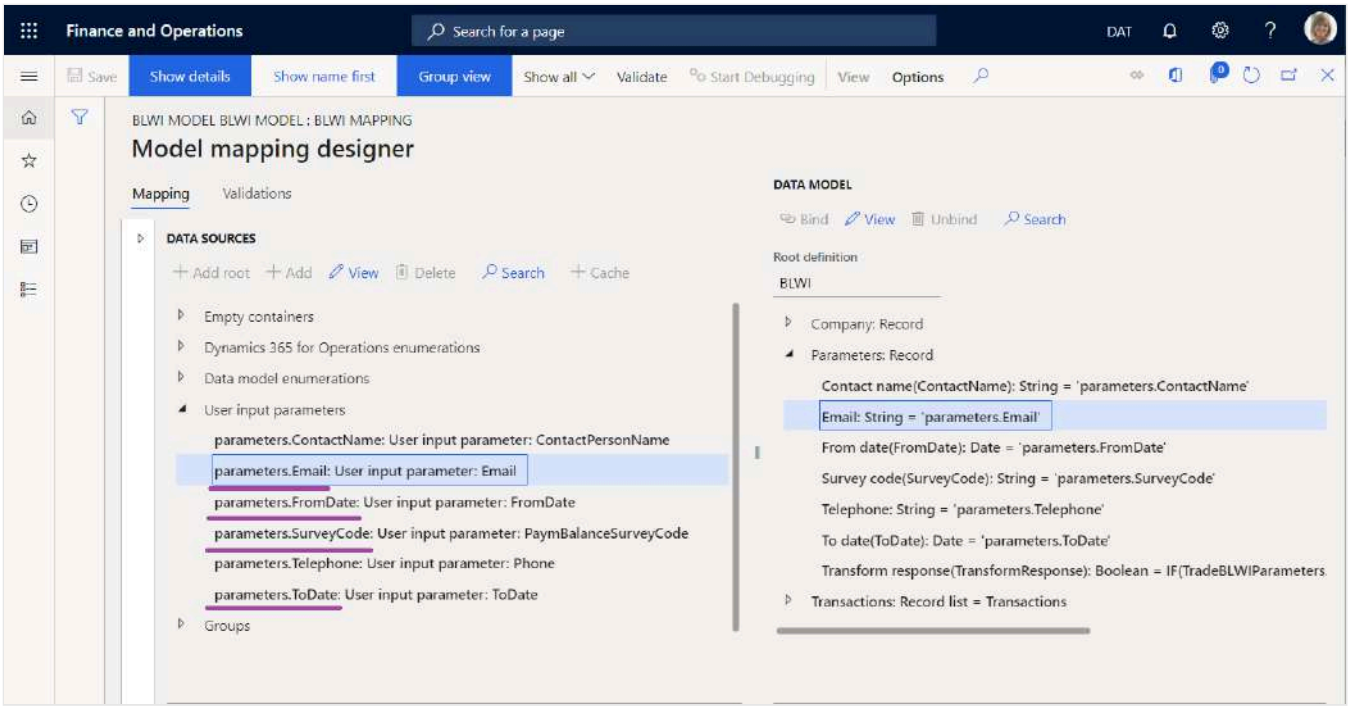
```
    /// <summary>
    /// Runs export via ER solution.
    /// </summary>
    [Microsoft.Dynamics.BusinessPlatform.SharedTypes.InternalUseOnlyAttribute]
    protected void runER()
    {
        const str erParmEmail = 'model/parameters.Email';
        const str erParmFromDate = 'model/parameters.FromDate';
        const str erParmToDate = 'model/parameters.ToDate';
        const str erParmSurveyCode = 'model/parameters.SurveyCode';

        ERModelDefinitionInputParametersAction modelDefinitionInputParametersAction
= new ERModelDefinitionInputParametersAction();

        modelDefinitionInputParametersAction.addParameter(erParmEmail, email)
            .addParameter(erParmFromDate, fromDate)
            .addParameter(erParmToDate, toDate)
```

```
            .addParameter(erParmSurveyCode,
BankPaymBalanceSurvey::find(surveyCodeRecId).SurveyCode);


EROjectsFactory::createFormatMappingRunByFormatMappingId(tradeBLWIParameters.ERFor
matMappingID, fileName)
            .withFileDestination(fileDestination)
            .withParameter(modelDefinitionInputParametersAction)
            .run();
    }
```

The `email`, `fromDate`, and `toDate` variables store values that users enter on other pages before an ER format is called. The `runER()` method of this class runs an ER format mapping that uses these variables to initialize values of several data sources. The prefix of every path of such a data source (such as `erParmEmail`, `erParmFromDate`, or `erParmToDate` constants in the sample code) is defined as the name of a format data source of the *Model* type (**model** value).



The suffix of every path of such a data source is defined as the path to the relevant data source of a model mapping that is used at runtime. In the sample code, this path can include `parameters.Email`, `parameters.FromDate`, or `parameters.ToDate`.



Format components for inbound electronic documents

You can also configure an ER format that can be run to parse an inbound document and update application data. This format must contain a format mapping that refers to a model mapping of the *To destination* type that is used to update application data based on the content of an inbound document. In this type of model mapping, you can also use data sources of the *User input parameter* type to get values at runtime from the ER user dialog box and then use them for application data update.

When an ER format is executed to parse an inbound document, you can turn off the ER dialog box programmatically by using the `_showPromptDialog` parameter of the `createMappingDestinationRunByImportFormatMappingId()` method of the `ERObjectsFactory` class.

For ER formats that are configured to parse an inbound document, the path to an ER data source is constructed as the path to a data source of the *User input parameter* type that resides in a model mapping of the *To destination* type that the running ER format calls to update application data.

For more information, review the source code of the `BankStatementImportBatch` application class that you can use as an example.

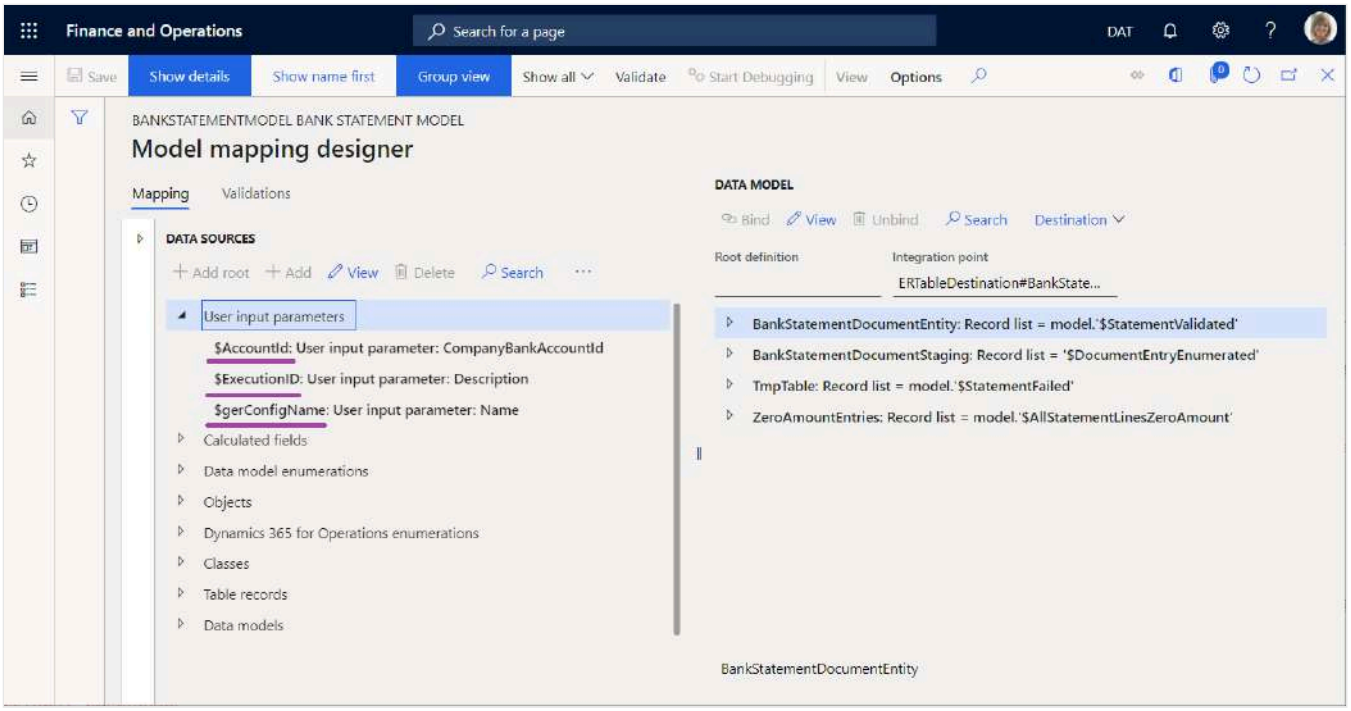
X++

```
        var integrationPoint = classStr(ERTableDestination) + '#' +
tableStr(BankStatementDocumentEntity);
```

```
        ERmodelDefinitionInputParametersAction inputParameters = new
ERmodelDefinitionInputParametersAction();
        inputParameters.addParameter('$ExecutionID', _executionID)
            .addParameter('$gerConfigName', _gerConfigName)
            .addParameter('$AccountId', bankAccount);

        var runner =
ERObjectsFactory::createMappingDestinationRunByImportFormatMappingId(_erModelMappin
gId, integrationPoint);
        runner.withParameter(inputParameters);
        runner.init();

        if (runner.promptsContractedModelMapping())
        {
            var parameters = runner.getParameters();
            var traverser = new ERModelDefinitionParametersTraverser(parameters);
            while (traverser.moveNext())
            {
                ERIImportFormatDataSourceContract current =
ERCast::asObject(traverser.current()) as ERIImportFormatDataSourceContract;
                if (current)
                {

current.parmInputDataStream(File::UseFileFromURL(DMFStagingWriter::getDownloadURLFr
omFileId(_uploadedStatement)));
                }
            }
        }

        runner.runUnattended();
```

Pay attention to the `_executionID`, `_gerConfigName`, and `bankAccount` variables that are used to initiate the values of the `$ExecutionID`, `$gerConfigName`, and `$AccountId` data sources.

From the Global repository, import the **Camt.053 Format** ER format configuration that this application class calls to your Finance instance. The **Bank statement model** configuration is imported automatically with the imported ER format configuration. Additionally, import the **Bank statement mapping to destination** ER configuration that contains a model mapping used to update application data.



Limitations

You can only initiate data source values of the *User input parameter* type that you configure in an ER model mapping that is used at runtime. You can't initiate data source values of the *User input parameter* type that you configure in an ER format from the source code.

Additional resources

Electronic reporting (ER) overview

Parse incoming documents

Last updated on 04/13/2026

Change code so users can configure and use named ER destinations

To generate an outbound document, run an ER format mapping. When you use the initial API of the ER framework to call an ER format mapping, it runs all destinations configured for components of the format. To review the sample code for a call of this type, see Add a report service class.

You can also configure action-dependent ER destinations for the ER format. You can then use the extended API of the ER framework to call an ER format mapping that provides a user action code. The action code runs only the destinations configured for the action provided.

In some cases, you must configure print management and select the same ER format for different print management records, so that you can apply different ER destinations for generated documents when you run that format. For example, you might configure print management so that it runs the same ER format to email original documents and to print document copies to a network printer. Use the new API of the ER framework in Microsoft Dynamics 365 Finance version 10.0.21 for this purpose.

API to enable users to configure ER destinations that are print management record specific

Currently, you can open the **Print management setup** page for every type of business document, to specify how the business document is generated and delivered. For every available record of the print management tree, you can perform the following actions:

- Use the **Report format** field to select either a SQL Server Reporting Services (SSRS) report or an ER format that runs.
- Use the **Destination** button to open the **Print destination settings** dialog box, so that you can change the destination of the generated report.
- Review the **Destination** field, which shows the defined destination.

> **Important**
>
> If you select an ER format in the **Report format** field, the configured SSRS destination isn't applied for a generated report.

You can change this behavior and configure ER destinations that are specific to a print management record. Unlike general ER destinations, these destinations are named. You can select a single named destination for several print management records. For more information, see Configure print management record-specific ER destinations.

First, turn on the **Allow to set up ER destinations per print management item** feature in the Feature management workspace.

Next, enable users to configure print management record–specific ER destinations for a specific type of business document, if this functionality isn't enabled by default. In Finance version 10.0.21, the functionality isn't enabled for every type of business document, because the `isNamedDestinationEnabledByDocumentType` method of the `ERNamedDestinationReportEnabler` class returns *false* for any `PrintMgmtDocumentType` document type that is provided.

X++

```
/// <summary>Class for enabling NamedDestinationFeature for current document type.
</summary>
/// <remarks>To enable the document type you should wrap a
<c>isNamedDestinationEnabledByDocumentType</c> method.</remarks>
public class ERNamedDestinationReportEnabler
{
    /// <summary>Checks that feature enabled for current document type. Make extension
for this method.</summary>
    /// <param name = "_typeId">Print mgmt document type</param>
    /// <returns>True if feature enabled.</returns>
    [Wrappable(true), SuppressBPWarning('BPParameterNotUsed', 'Required by
signature')]
    public static boolean
isNamedDestinationEnabledByDocumentType(PrintMgmtDocumentType _typeId)
    {
        return false;
    }
}
```

You must wrap the `isNamedDestinationEnabledByDocumentType` method for a specific document type to enable users to configure print management record–specific ER destinations for documents of that type. The following example shows how to enable the functionality for free text invoices.

X++

```
[ExtensionOf(classStr(ERNamedDestinationReportEnabler))]
final class ERNamedDestinationReportEnablerContoso_Extension
{
    public static boolean
```

```
isNamedDestinationEnabledByDocumentType(PrintMgmtDocumentType _typeId)
    {
        var ret = next isNamedDestinationEnabledByDocumentType(_typeid);
        if(_typeId == PrintMgmtDocumentType::SalesFreeTextInvoice)
        {
            return true;
        }
        else
        {
            return ret;
        }
    }
}
```

API to run a format mapping where the destination that is provided is print management record specific

When you enable users to configure print management record–specific destinations, they expect a named destination configured for a print management record. When you use the print management record to run an ER format and generate an outbound document, you must pass that named destination to the ER framework to force ER to run only that destination, not any other destinations that might be configured for the ER format that is running.

The following code for the `ERPrintMgmtReportFormatSubscriber` class shows how the ER framework currently gets called to generate a free text invoice.

X++

```
/// <summary>
/// Runs a format mapping based on incoming arguments.
/// </summary>
/// <param name = "_args">Event arguments object.</param>
public static void
runFreeTextInvoiceReport(NonSSRSPrintMgmtAdapterReportExecutedEventArgs _args)
{
    ERIFormatMappingRun formatMappingRun =
ERPrintMgmtReportFormatSubscriber::buildFreeTextInvoiceReportMappingRun(_args);

    if (_args.parmArgs() && _args.parmArgs().parmEnumType() ==
enumNum(PrintCopyOriginal))
    {
        Args args = _args.parmArgs();
        boolean doUsePrintManagement = true;

        if (args.caller() is SalesFormLetter)
        {
            SalesFormLetter caller = args.caller();
            doUsePrintManagement = caller.usePrintManagement();
        }
```

```
        ERDestinationAction action;
        if (doUsePrintManagement ||
!isFlightEnabled(LocalizationFlights::ForcePrintJobSettings))
        {
            action =
ReportDestinationContract::getPrintDestinationAction(args.parmEnum());
        }
        else
        {
            action = ERDestinationAction::View;
        }

        ERIFormatMappingRunWithDestinationAction mappingRunWithAction =
formatMappingRun as ERIFormatMappingRunWithDestinationAction;

        if (mappingRunWithAction && action != ERDestinationAction::NoAction)
        {
            mappingRunWithAction.withDestinationAction(action);
        }
    }

    formatMappingRun.withCreatingObjectParameter(
        CustomersInvoicingModelName,
        classStr(PrintMgmtPrintSettingDetail),
        _args.parmSettingDetail());
    formatMappingRun.run();
}
```

The following example shows how you can change the sample code to use the new API to run an ER format and provide an ER named destination that is print management record specific.

X++

```
/// <summary>
/// Runs a format mapping based on incoming arguments.
/// </summary>
/// <param name = "_args">Event arguments object.</param>
public static void
runFreeTextInvoiceReport(NonSSRSPrintMgmtAdapterReportExecutedEventArgs _args)
{
    ERIFormatMappingRun formatMappingRun =
ERPrintMgmtReportFormatSubscriber::buildFreeTextInvoiceReportMappingRun(_args);

    if (_args.parmArgs() && _args.parmArgs().parmEnumType() ==
enumNum(PrintCopyOriginal))
    {
        Args args = _args.parmArgs();
        boolean doUsePrintManagement = true;

        if (args.caller() is SalesFormLetter)
        {
            SalesFormLetter caller = args.caller();
```

```
            doUsePrintManagement = caller.usePrintManagement();
        }

        ERDestinationAction action;
        if (doUsePrintManagement ||
!isFlightEnabled(LocalizationFlights::ForcePrintJobSettings))
        {
            action =
ReportDestinationContract::getPrintDestinationAction(args.parmEnum());
        }
        else
        {
            action = ERDestinationAction::View;
        }

        ERIFormatMappingRunWithDestinationAction mappingRunWithAction =
formatMappingRun as ERIFormatMappingRunWithDestinationAction;

        if (mappingRunWithAction && action != ERDestinationAction::NoAction)
        {
            mappingRunWithAction.withDestinationAction(action);
        }

        // Check whether a named destination is provided.
        // If so, pass it to ER to specify a single destination for a generated
document

        RefRecId NamedDestination;
        if (doUsePrintManagement ||
!isFlightEnabled(LocalizationFlights::ForcePrintJobSettings))
        {
            NamedDestination =
ReportDestinationContract::getNamedDestination(args.parmEnum());
        }
        else
        {
            NamedDestination = 0;
        }

        ERIFormatMappingRunWithNamedDestination RunWithNamedDestination =
formatMappingRun as ERIFormatMappingRunWithNamedDestination;

        if (RunWithNamedDestination && NamedDestination)
        {
            RunWithNamedDestination.withDestinationNamed(NamedDestination);
        }
    }

    formatMappingRun.withCreatingObjectParameter(
        CustomersInvoicingModelName,
        classStr(PrintMgmtPrintSettingDetail),
         _args.parmSettingDetail());
    formatMappingRun.run();
}
```

The `NamedDestination` field of the `PrintMgmtSettings` table includes the reference to a named destination. You must pass this reference to the `runFreeTextInvoiceReport` method from the processed print management record that extends `ReportDestinationContract`.

Applicability

To set up named destinations and force the ER framework to use the named destination, first turn on the **Allow to set up ER destinations per print management item** feature in the Feature management workspace.

The feature enables named ER destinations for all types of business documents that can be generated in Finance by using the ER framework. For more information, see the **Configurable business documents – specific destinations via printer management settings in the reports** feature in the "Globalization" section for the "Dynamics 365 Finance" application in the Dynamics 365 and industry clouds release plan for the 2021 release wave 2.

Additional resources

Electronic reporting (ER) overview

Electronic reporting (ER) destinations

Configure action-dependent ER destinations

Configure print management record-specific ER destinations

Electronic reporting framework API changes for Application update 10.0.21

Last updated on 03/17/2026

Implement a custom ER source of inbound documents

To import data from an inbound document by using the Electronic reporting (ER) framework, configure an ER format that supports the import, and then run a model mapping of the **To destination** type that uses that format as a data source. To import data, go to the document that you want to import, and use a SharePoint folder as the standard ER source of inbound documents that you can import in unattended mode. For more information about this process, see Configure data import from SharePoint.

The application programming interface (API) of the ER framework now lets you extend the list of ER sources that you can use to access inbound documents that ER formats parse for data import. Therefore, you can use ER configurations to perform data import from documents that are stored in your custom source.

This article provides an overview of the main tasks that you must complete to implement and start to use a custom ER source of inbound documents. To complete these tasks, use the same example for importing vendor transactions that is described in Configure data import from SharePoint. You can complete the steps of these tasks in the **USMF** company in Microsoft Dynamics 365 Finance.

Prerequisites

Deploy a topology

You must deploy a topology that supports continuous build. For more information, see Deploy topologies that support continuous build and test automation. You must have access to this topology for one of the following roles:

- Electronic reporting developer
- Electronic reporting functional consultant
- System administrator

You must also have access to the development environment for this topology.

Configure the ER framework

Follow the steps in Configure the ER framework to set up the minimal set of ER parameters. Complete this setup before you start to use the ER framework to import inbound documents from a custom source.

Import an ER format configuration

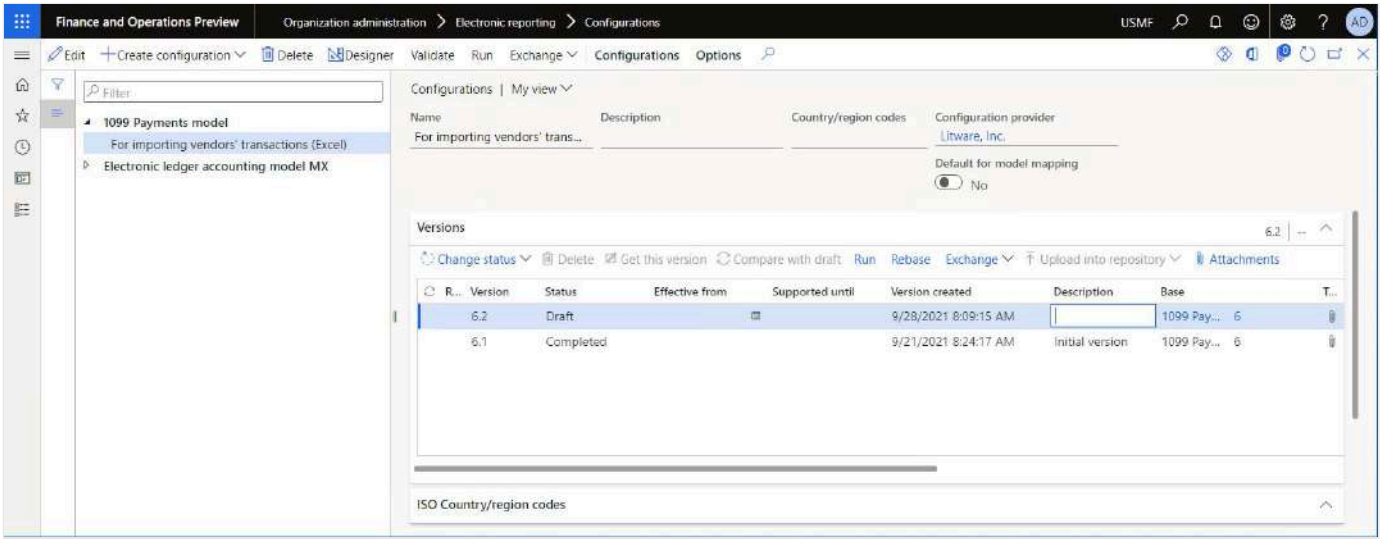
Additional tasks use the ER configurations that are mentioned in Configure data import from SharePoint. If you didn't follow the task guide in that article, download the following files.

Expand table

Content description	File
ER data model configuration	1099model.xml
ER format configuration	1099format-import-from-excel.xml
Inbound file in .XLSX format containing sample data for import	1099import-data.xlsx

Use the Load from XML file ER option to import the downloaded ER configurations into your Finance instance in the following order:

1. ER data model configuration
2. ER format configuration



> **Important**
>
> The ER format that you create or import must contain at least one of the following format elements:
>
> - Common\File

- Excel\File

Extend the source code

1. Append a new class to your Visual Studio project. In this example, use `ContosoInboundFile`.

2. Write code that uses the `ERIImportFile` public interface to describe a custom inbound file. To comply with the ER framework's requirements for handling inbound files, specify the following information:

   - A key to uniquely identify a custom source at runtime (the `sourceSettingsKey` variable in this example), and the methods to get and set this key. The key is important because there might be many sources.
   - *DateTime* values to specify the date and time of inbound file creation and modification (the `createdDateTime` and `modifiedDateTime` variables in this example), and the methods to get and set these dates. The values are used at runtime to sort and filter inbound documents that are recognized in a custom source.
   - A method to load an inbound file from a custom source (the `Load()` method in this example).
   - A method to delete an imported file in custom storage (the `Delete()` method in this example).
   - A method to expose additional information about an inbound file (the `parmAdditionalInfo()` method in this example), so that this information can be stored and then extracted upon request.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using SL = Microsoft.Dynamics365.LocalizationFramework.XppSupportLayer;
using System.IO;

/// <summary>
/// Import file
/// </summary>
public class ContosoInboundFile implements ERIImportFile
{
    private str folderPath;
    private System.IO.Stream fileContent;
    private str additionalInfo;
    private str fileId;
    private str name;
    private SL.utcdatetime createdDateTime;
    private SL.utcdatetime modifiedDateTime;
    private str sourceSettingsKey;
```

```
    protected void new(
        str _fileId,
        str _name,
        utcdatetime _dateTime,
        str _folderPath,
        Stream _fileContent
        )
    {
        fileId = _fileId;
        name = _name;
        createdDateTime = ERLFConvert::fromUtcdatetime(_dateTime);
        modifiedDateTime =  ERLFConvert::fromUtcdatetime(_dateTime);
        folderPath = _folderPath;
        fileContent = _fileContent;

this.set_SourceSettingsKey(ContosoInboundFileSourceSettings::SettingsKey);

    }

    [Hookable(false)]
    public str get_SourceSettingsKey()
    {
        return sourceSettingsKey;
    }

    [Hookable(false)]
    public void set_SourceSettingsKey(str _sourceSettingsKey = sourceSettingsKey)
    {
        sourceSettingsKey = _sourceSettingsKey;
    }

    [Hookable(false)]
    public str parmAdditionalInfo(str _additionalInfo = additionalInfo)
    {
        additionalInfo = _additionalInfo;
        return additionalInfo;
    }

    /// <summary>
    /// Construct new object.
    /// </summary>
    /// <param name = "_fileId">File id.</param>
    /// <param name = "_name">Name.</param>
    /// <param name = "_dateTime">Date time.</param>
    /// <param name = "_folderPath">Folder path.</param>
    /// <param name = "_fileContent">File content as stream.</param>
    /// <returns>Import file.</returns>
    public static ContosoInboundFile construct(str _fileId,
        str _name,
        utcdatetime _dateTime,
        str _folderPath,
        Stream _fileContent
        )
    {
```

```
        return new ContosoInboundFile(_fileId, _name, _dateTime, _folderPath,
_fileContent);
    }

    [Hookable(false)]
    public void set_FileId(str _fileId = fileId)
    {
        fileId = _fileId;
    }

    [Hookable(false)]
    public void set_Name(str _name = name)
    {
        name = _name;
    }

    [Hookable(false)]
    public str get_Name()
    {
        return name;
    }

    [Hookable(false)]
    public SL.utcdatetime get_CreatedDateTime()
    {
        return createdDateTime;
    }

    [Hookable(false)]
    public void set_CreatedDateTime(SL.utcdatetime _createdDateTime =
createdDateTime)
    {
        createdDateTime = _createdDateTime;
    }

    [Hookable(false)]
    public void set_ModifiedDateTime(SL.utcdatetime _modifiedDateTime =
modifiedDateTime)
    {
        modifiedDateTime = _modifiedDateTime;
    }

    /// <summary>
    /// Loads the file from the source.
    /// </summary>
    /// <returns>A file operation result.</returns>
    public ERFileLoadResult Load()
    {
        return ERFileLoadResult::Success(fileContent);
    }

    public ERFileDeleteResult Delete()
    {
        if (System.IO.File::Exists(Path::Combine(folderPath, name)))
        {
```

```
            System.IO.File::Delete(Path::Combine(folderPath, name));
            return ERFileDeleteResult::Success();
        }
        return ERFileDeleteResult::Fail();
    }

    [Hookable(false)]
    public str GetFileId()
    {
        return fileId;
    }

    [Hookable(false)]
    public SL.utcdatetime GetModifiedDateTime()
    {
        return modifiedDateTime;
    }

}
```

3. Append a new class to your Visual Studio project. In this example, use `ContosoInboundFileSource`.

4. Write code that uses the `ERIFileSource` public interface to describe a custom source of inbound files. To comply with the ER framework's requirements for handling inbound files, specify the following information:

    - A file mask (the `fileMask` variable in this example), so that users can specify the file name pattern to filter only specific files from a custom source, based on a file name and extension.
    - A method to load inbound files from a custom source (the `GetFiles()` method in this example). This method refers to the `ContosoInboundFile()` method for specifying an inbound file.

    X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using System.IO;

/// <summary>
/// File source as a folder
/// </summary>
public class ContosoInboundFileSource implements ERIFileSource
{
    private str folderPath;
    private str fileMask;

    /// <summary>
    /// Creates a new instance of <C>ERFileSourceFolder</C>.
```

```
    /// </summary>
    /// <param name = "_folderPath">A folder path.</param>
    /// <param name = "_fileMask">A file mask to filter; optional.</param>
    protected void new(str _folderPath, str _fileMask)
    {
        folderPath = _folderPath;
        fileMask = _fileMask;
    }

    /// <summary>
    /// Construct new instance of <c>ERFileSourceFolder</c>.
    /// </summary>
    /// <param name = "_folderPath">A folder path.</param>
    /// <param name = "_fileMask">A file mask to filter; optional.</param>
    /// <returns>New instance of class.</returns>
    internal static ContosoInboundFileSource construct(str _folderPath, str
_fileMask)
    {
        return new ContosoInboundFileSource(_folderPath, _fileMask);
    }

    public str parmFolderPath(str _value = folderPath)
    {
        folderPath = _value;
        return folderPath;
    }

    [Hookable(false)]
    public ERIFiles GetFiles()
    {
        if (fileMask == '')
        {
            fileMask = '*.*';
        }
        var ret = new ERFiles();
        var directoryInfo = new System.IO.DirectoryInfo(folderPath);
        var files = directoryInfo.GetFiles(fileMask);
        var i = files.GetEnumerator();

        while (i.MoveNext())
        {
            System.IO.FileInfo fileInfo = i.get_Current();
            if (fileInfo)
            {
                using (StreamReader sr = fileInfo.OpenText())
                {
                    var localStream = new System.IO.MemoryStream();

                    System.IO.Stream base = sr.BaseStream;
                    base.CopyTo(localStream);
                    if (localStream.CanSeek)
                    {
                        localStream.Seek(0, System.IO.SeekOrigin::Begin);
                    }
                    ERIFile file = ContosoInboundFile::construct(fileInfo.Name,
```

```
fileInfo.Name, DateTimeUtil::getSystemDateTime(), folderPath, localStream);
                    ret.Add(file);

                }
            }
        }
        return ret;
    }

}
```

> **Note**
>
> The file name is the unique file identifier in the configured custom source when the file object is constructed while the `GetFiles()` method is called. If you use something other than a folder of the local file system source to store inbound files, you can implement another way to uniquely identify inbound files.

5. Append a new class to your Visual Studio project. In this example, use `ContosoERFileDestinationFolder`.

6. Write code that uses the `ERIFileDestination` public interface to describe a custom source of inbound files as a folder. You can reuse the code sample from step 1 of the "Extend the source code" procedure in Implement a custom destination for generated documents. In the example, the class is designed to look for inbound documents as files in a folder of the local file system.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;

/// <summary>
/// File destination as a folder
/// </summary>
public class ContosoERFileDestinationFolder implements ERIFileDestination
{
    private str folderPath;
    private boolean overwrite;

    public str parmFolderPath(str _value = folderPath)
    {
        folderPath = _value;
        return folderPath;
    }

    public boolean parmOverwrite(boolean _value = overwrite)
    {
```

```
        overwrite = _value;
        return overwrite;
    }

    [Hookable(false)]
    public System.IO.Stream saveFile(System.IO.Stream _stream, str _filePath)
    {
        this.sendFileToFolder(_stream, _filePath);
        return _stream;
    }

    [Hookable(false)]
    public System.IO.Stream newFileStream(str _filePath)
    {
        return new System.IO.MemoryStream();
    }

    private str getFileName(str _fileName)
    {
        if (overwrite)
        {
            return _fileName;
        }
        System.DateTime now = DateTimeUtil::utcNow();
        return strFmt('%1_%2%3',
System.IO.Path::GetFileNameWithoutExtension(_fileName), now.ToString('yyyy-M-
d_HH_mm_ss'), System.IO.Path::GetExtension(_fileName));
    }

    private void sendFileToFolder(System.IO.Stream _stream, str _filePath)
    {
        using(var fileStream = System.IO.File::Create(_filePath))
        {
            if( _stream.CanSeek)
            {
                _stream.Seek(0, System.IO.SeekOrigin::Begin);
            }
            _stream.CopyTo(fileStream);
        }
    }

    /// <summary>
    /// Finalizes files processing.
    /// </summary>
    [Hookable(false)]
    public void finalize()
    {
    }

}
```

7. Append a new class to your Visual Studio project. For this example, use `ContosoInboundFileSourceSettings`.

8. Write code that uses the `ERIImportFileSourceSettings` public interface to specify how a custom source is created, how its parameters are packed for storage, and how its parameters are unpacked for presentation in the user interface (UI). To comply with the ER framework's requirements for handling inbound files, specify the following information:

    - A key (the `SettingsKey` text constant in this example) to uniquely identify settings of a custom source at runtime. Use a class name to guarantee the uniqueness of keys of this type.
    - The name of a custom source (the `Folder` text constant in this example) to present a custom source in the appropriate UI dialog box at design time.
    - A parameter (the `isEnabled` variable in this example) to indicate whether a custom source is enabled at design time for use at runtime.
    - Because all these parameters must be stored as parts of a single binary object in the application database, they must be packed for this purpose at design time and then unpacked when they are requested at runtime. The appropriate methods must be implemented to support this process.
    - Pay attention to the `fileImported()` method that is used to implement the logic of any required post-processing that must be applied to the imported file. The argument of this method represents an imported file itself and information about the import that is performed. A state manager is included. This state manager includes a logger that collects information about whether an imported file was deleted or moved to another location. In this example, the code moves successfully imported files to another folder (**C:\InboundFiles\Archive**) of the local file system and keeps the original file names.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;
using System.IO;

/// <summary>
/// Folder file source settings
/// </summary>
public class ContosoInboundFileSourceSettings implements SysPackable,
ERIImportFileSourceSettings
{
    internal const str SettingsKey = classStr(ContosoInboundFileSourceSettings);
    public const int CurrentVersion = 1;

    private boolean isEnabled;
    private str folderPath;

    [Hookable(false)]
    public str parmFolderPath(str _value = folderPath)
    {
```

```
        folderPath = _value;
        return folderPath;
    }

    [Hookable(false)]
    public boolean parmIsEnabled(boolean _value = isEnabled)
    {
        isEnabled = _value;
        return isEnabled;
    }

    [Hookable(false)]
    public void setEnabled(boolean _value)
    {
        isEnabled = _value;
    }

    [Hookable(false)]
    public boolean isEnabled()
    {
        return isEnabled;
    }

    [Hookable(false)]
    public str getName()
    {
        /// Name of settings of a custom source.
        /// This name will be presented in the dialog box that is offered at
runtime for user to adjust configured settings.
        return 'Folder';
    }

    [Hookable(false)]
    public boolean validate()
    {
        return true;
    }

    [Hookable(false)]
    public str getKey()
    {
        return SettingsKey;
    }

    [Hookable(false)]
    public container pack()
    {
        return [CurrentVersion, isEnabled, folderPath];
    }

    [Hookable(false)]
    public boolean unpack(container packedClass)
    {
        int version = RunBase::getVersion(packedClass);
        switch (version)
```

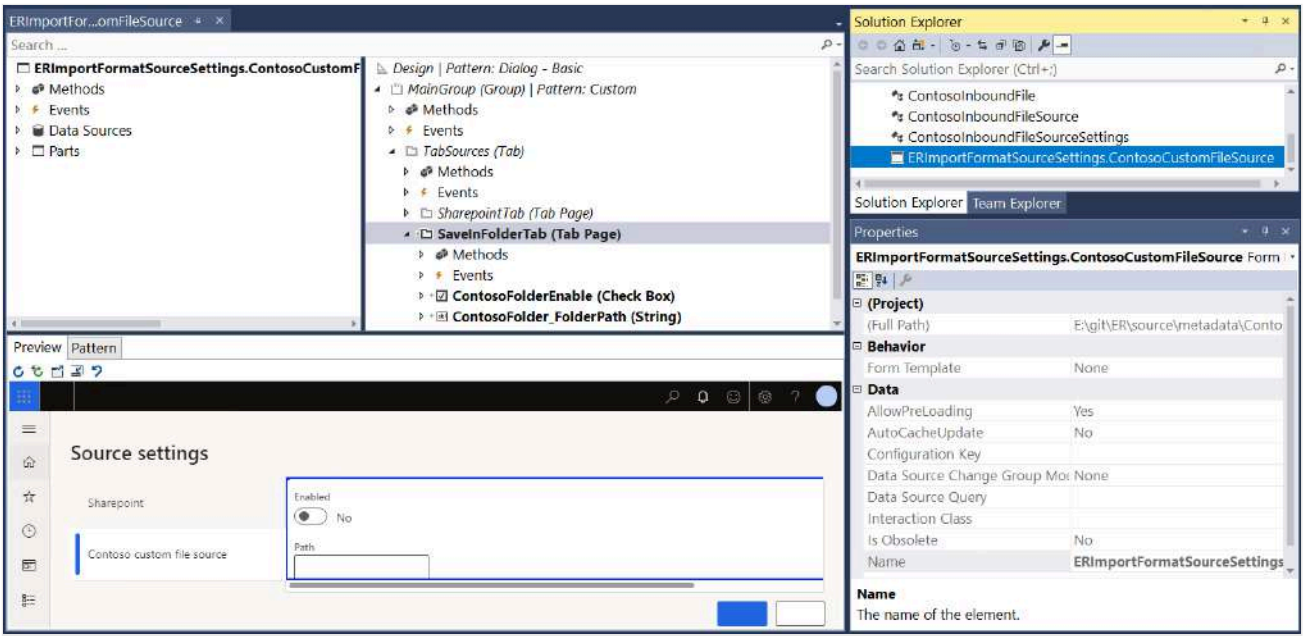
```
        {
            case CurrentVersion:
                [version, isEnabled, folderPath] = packedClass;
                return true;
            default:
                return false;
        }
    }

    /// <summary>
    /// Creates a file source for current settings.
    /// </summary>
    /// <param name = "_fileMask">File mask.</param>
    /// <returns>A new instance of a file source.</returns>
    public ERIFileSource createFileSource(str _fileMask)
    {
        return ContosoInboundFileSource::construct(folderPath, _fileMask);
    }

    /// <summary>
    /// Creates a file destination for current settings.
    /// </summary>
    /// <param name = "_destination">A destination folder to move the file.
</param>
    /// <returns>A new instance of a file destination.</returns>
    private ERIFileDestination
createPostProcessingDestination(ERFormatFileDestinationAfterImport _destination)
    {
        ContosoERFileDestinationFolder ret = new
ContosoERFileDestinationFolder();
        var path = folderPath;
        switch (_destination)
        {
            case ERFormatFileDestinationAfterImport::None :
                break;
            default:
                path = System.IO.Path::Combine(path, enum2Str(_destination));
                break;
        }
        ret.parmFolderPath(path);
        return ret;
    }

    public void fileImported(ERFileImportedArgs _args)
    {
        if (_args.IsImportOperationSuccessful())
        {
            /// copy an outbound file to the alternative location
            var destination = this.createPostProcessingDestination(
_args.IsImportOperationSuccessful() ? ERFormatFileDestinationAfterImport::Archive
: ERFormatFileDestinationAfterImport::None);
            ERIImportFile file = _args.getFile();
            destination.saveFile(_args.getFileStream(),
@'C:\InboundFiles\Archive\'+file.Name);
```

```
            /// do logging
            ttsbegin;
            _args.getState().markAsMoved();

_args.getState().getLogger().logMoved(!_args.IsImportOperationSuccessful(),
'Custom error message', 'Custom description');
            ttscommit;

            /// delete file
            ttsbegin;
            _args.getFile().Delete();
            _args.getState().markAsDeleted();
            ttscommit;
        }
    }

    /// <summary>
    /// Create new instance of <c>ERImportFormatSourceFolderSettings</c> from
packed presentation.
    /// </summary>
    /// <param name = "_packedClass">A container with packed presentation.
</param>
    /// <returns>New instance.</returns>
    internal static ContosoInboundFileSourceSettings create(container
_packedClass)
    {
        var settings = new ContosoInboundFileSourceSettings();
        settings.unpack(_packedClass);
        return settings;
    }

    /// <summary>
    /// Event.
    /// </summary>
    /// <param name="_record">Record.</param>
    /// <param name="_file">File.</param>
    ///
    /// This code and code below presents the sample how you can memorize
additional info about an imported file
    /// in the extended field of the ERImportFormatFileSourceStateTable table.
    ///
    [SubscribesTo(tableStr(ERImportFormatFileSourceStateTable),
delegateStr(ERImportFormatFileSourceStateTable, objectInitialized))]
    public static void
ERImportFormatFileSourceStateTable_objectInitialized(ERImportFormatFileSourceStat
eTable _record, ERIImportFile _file)
    {
        anytype file =_file;
        var folderFile = file as ContosoInboundFile;
        if (folderFile != null)
        {
///             folderFile.parmAdditionalInfo(_record.additionalInfo);
        }
    }
```

```
    /// <summary>
    /// Event.
    /// </summary>
    /// <param name="_record">Record.</param>
    /// <param name="_file">File</param>
    [SubscribesTo(tableStr(ERImportFormatFileSourceStateTable),
delegateStr(ERImportFormatFileSourceStateTable, tableRecordInitialized))]
    public static void
ERImportFormatFileSourceStateTable_tableRecordInitialized(ERImportFormatFileSourc
eStateTable _record, ERIImportFile _file)
    {
        anytype file =_file;
        var folderFile = file as ContosoInboundFile;
        if (folderFile != null)
        {
///             _record.AdditionalInfo = folderFile.parmAdditionalInfo();
        }
    }

}
```

9. In your Visual Studio project, add a new extension for the `ERImportFormatSourceSettings` form. In this example, use `ERImportFormatSourceSettings.ContosoCustomFileSource`.

10. Write code that implements a custom UI for your custom source of inbound files. The following illustration shows what this UI looks like in the Visual Studio designer.



11. Append a new class to your Visual Studio project. For this example, use `ContosoImportFormatSourceSettingsEventHandlers`.

12. Write the event handler code for an extended form of settings of a custom source. This step requires implementation of the public `ERIImportFileSourceSettingsStorage` interface.

X++

```
using Microsoft.Dynamics365.LocalizationFramework;

/// <summary>
/// Handler for <c>ERImportFormatSourceSettings</c> form
/// </summary>
public class ContosoImportFormatSourceSettingsEventHandlers
{
    /// <summary>
    /// Post init method event handler.
    /// </summary>
    /// <param name = "args">Args.</param>
    [PostHandlerFor(formStr(ERImportFormatSourceSettings),
formMethodStr(ERImportFormatSourceSettings, init))]
    public static void ERImportFormatSourceSettings_Post_init(XppPrePostArgs
args)
    {
        FormRun formRun = args.getThis();
        ERIImportFileSourceSettingsStorage settingsStorage = args.getThis();
        var settings =
ContosoImportFormatSourceSettingsEventHandlers::getSaveInFolderSettings(settingsS
torage);

        FormStringControl folderPathControl =
formRun.design().controlName(formControlStr(ERImportFormatSourceSettings,
ContosoFolder_FolderPath));
        folderPathControl.text(settings.parmFolderPath());

        FormCheckBoxControl enabledControl =
formRun.design().controlName(formControlStr(ERImportFormatSourceSettings,
ContosoFolderEnable));
        enabledControl.checked(settings.isEnabled());

    }

    /// <summary>
    /// CloseOk method event handler.
    /// </summary>
    /// <param name = "args">Args.</param>
    [PreHandlerFor(formStr(ERImportFormatSourceSettings),
formMethodStr(ERImportFormatSourceSettings, closeOk))]
    public static void ERFormatDestinationSettings_Pre_closeOk(XppPrePostArgs
args)
    {
        FormRun formRun = args.getThis();
        ERIImportFileSourceSettingsStorage settingsStorage = args.getThis();
        var settings =
ContosoImportFormatSourceSettingsEventHandlers::getSaveInFolderSettings(settingsS
torage);

        FormStringControl folderPathControl =
formRun.design().controlName(formControlStr(ERImportFormatSourceSettings,
ContosoFolder_FolderPath));
        settings.parmFolderPath(folderPathControl.text());
```

```
        FormCheckBoxControl enabledControl =
formRun.design().controlName(formControlStr(ERImportFormatSourceSettings,
ContosoFolderEnable));
        settings.setEnabled(enabledControl.checked());

    }

    private static ContosoInboundFileSourceSettings
getSaveInFolderSettings(ERIImportFileSourceSettingsStorage _settingsStorage)
    {
        ContosoInboundFileSourceSettings ret =
_settingsStorage.getSettingsByKey(ContosoInboundFileSourceSettings::SettingsKey);
        if (ret == null)
        {
            ret = new ContosoInboundFileSourceSettings();
            _settingsStorage.addSettings(ret);
        }
        return ret;
    }

}
```

13. Rebuild your Visual Studio project.

Configure ER sources for the imported ER format

1. Configure the custom source of inbound files for one of the previously mentioned components (Common\File or Excel\File) of the ER format that you imported.

    a. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting source**.

    b. On **Electronic reporting source**, select **New**.

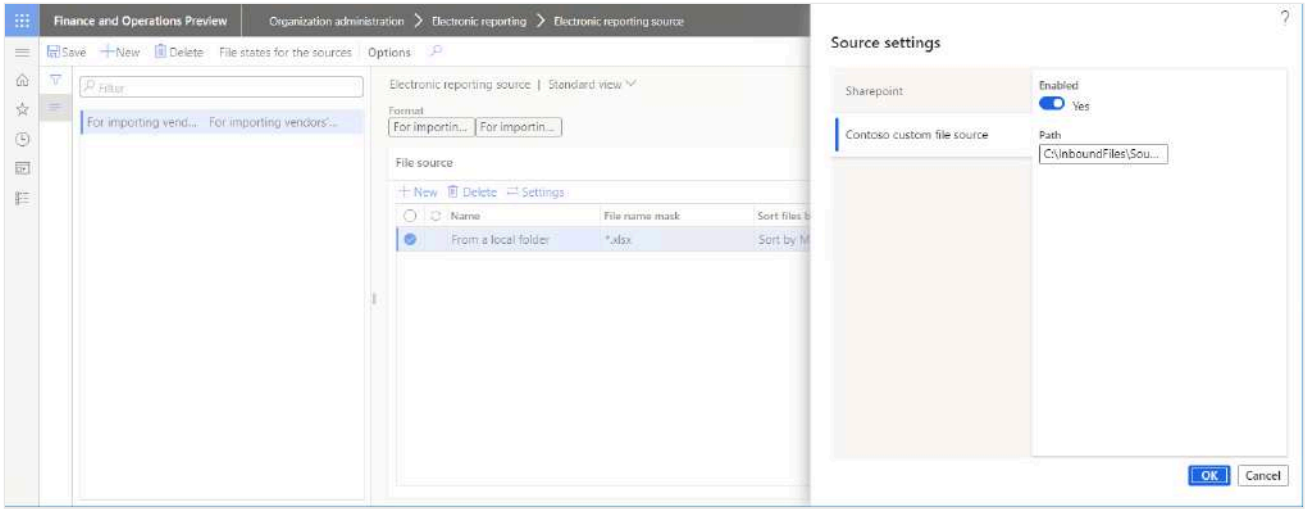
    c. In the **Format** field, select the imported ER format.

    d. On the **File source** FastTab, select **New**, and then follow these steps:
        i. In the **Name** field, enter **From a local folder** as the name of a custom source.
        ii. In the **File name mask** field, enter the ***.xlsx** mask to filter only files that have the .xlsx extension in a custom source.
        iii. In the **Sort files before import** field, select **Sort by Modified date/time**.

    e. For the source record that you entered, select the **Settings** option, and then follow these steps:
        i. In the **Source settings** dialog box, on the **Contoso custom file source** tab, set the **Enabled** option to **Yes**.

ii. In the **Path** field, enter **C:\InboundFiles\Source** to specify the local folder that contains inbound files.

iii. Select **OK**, and then select **Save**.



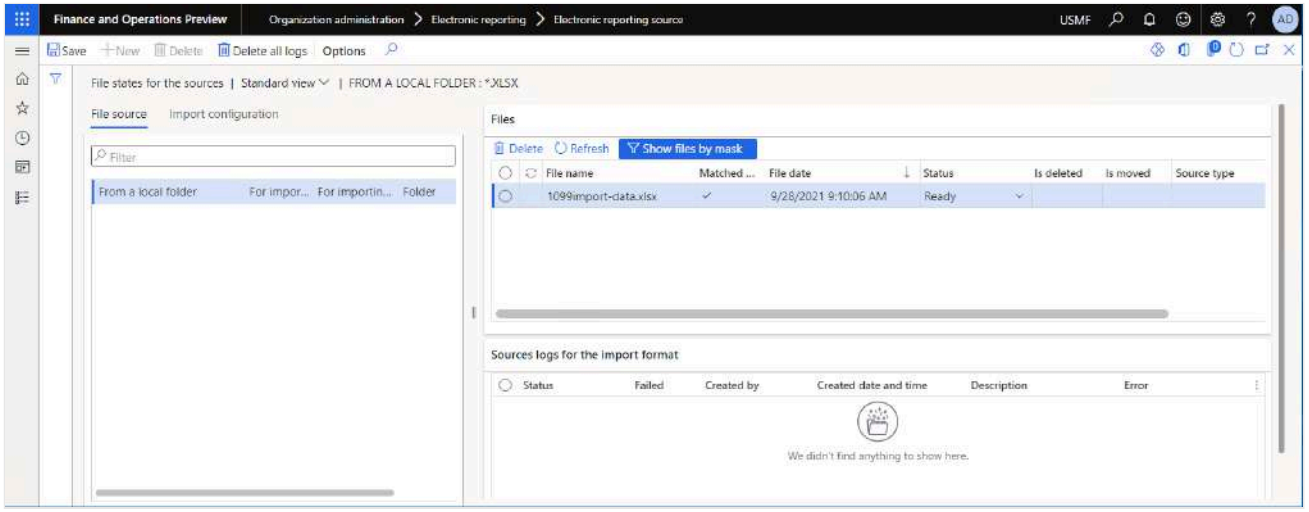
2. Copy the imported file, **1099import-data.xlsx**, to the **C:\InboundFiles\Source** folder of the file system of the server that runs the first instance of Application Object Server (AOS).

3. Make sure that the **C:\InboundFiles\Archive** folder exists in the file system of the server that runs the first AOS instance.

4. On **Electronic reporting source**, select **File states for the sources**, and then follow these steps to review the content of the configured custom file source for the current ER format:

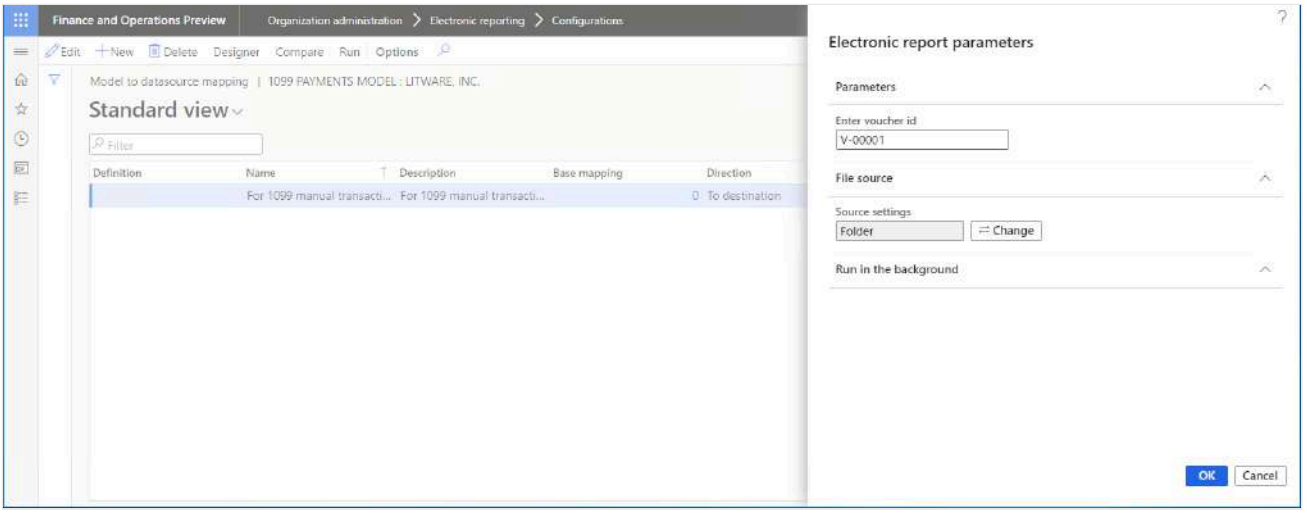
   a. On **File states for the sources**, in the **File source** list, select the **From a local folder** source.

   b. On the **Files** tab, select **Refresh** to see the **1099import-data.xlsx** file that is ready for import from a custom source.



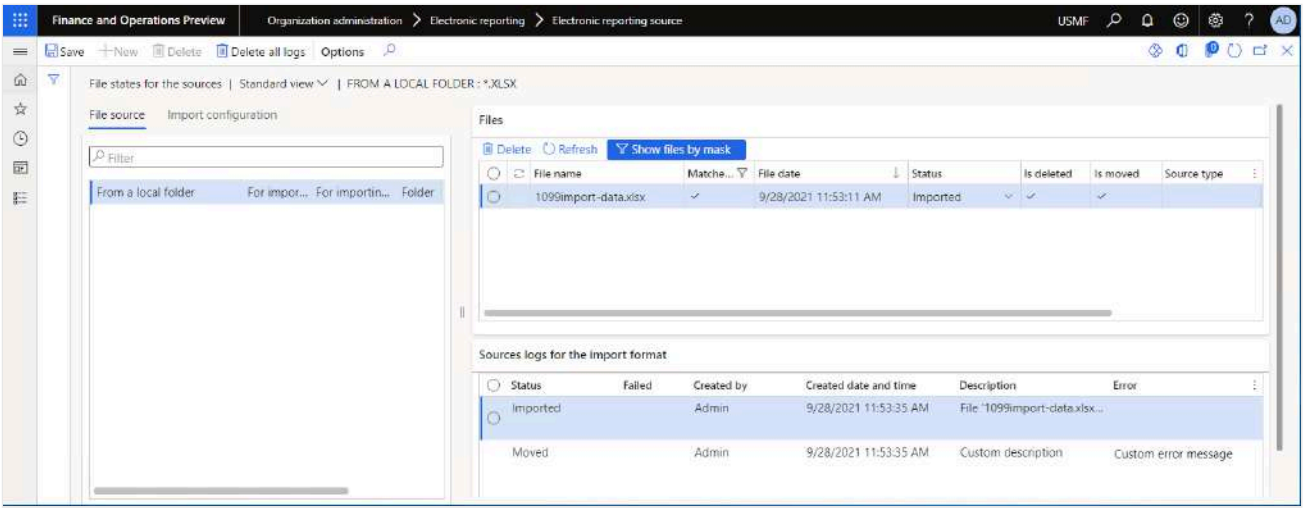
Run the ER mapping that you imported

1. To import vendor transactions from an inbound file that is located in a custom source, follow these steps to run the ER model mapping that you imported:
    a. Go to **Organization administration** > **Electronic reporting** > **Configurations**.
    b. In the **Configurations** tree, select **1099 Payments model**.
    c. On the **Configuration components** FastTab, select the name of the **Model mapping** component to open the list of model mappings for the selected ER model configuration.
    d. Select **Run** to run the selected model mapping. Because you configured file sources for the ER format, you can change the setting of the **File source** option as you require. If you keep the setting of this option, the .xslx files are imported from the configured custom sources (the **1099import-data.xlsx** file from the **C:\InboundFiles\Source** local folder in this example).
    e. Enter the voucher ID, such as **V-00001**, and then select **OK**.



2. Browse to the local **C:\InboundFiles\Archive** folder to find the imported **1099import-data.xlsx** file that is moved from the **C:\InboundFiles\Source** local folder, based on the post-processing logic that is defined in the source code.

3. Follow these steps to review the current status of the configured custom source of inbound files.
    a. Go to **Organization administration** > **Electronic reporting** > **Electronic reporting source**.
    b. On the **Electronic reporting source** page, select **File states for the sources** to review the content of the configured custom file source for the current ER format.
    c. On **File states for the sources**, in the **File source** list, select the **From a local folder** source.
        i. On the **Files** tab, select **Refresh** to see the previously imported **1099import-data.xlsx** file that is moved from the **C:\InboundFiles\Source** folder to the **C:\InboundFiles\Archive** folder.



Additional resources

[Configure data import from SharePoint]

[Electronic reporting framework API changes for Application update 10.0.23]

Last updated on 04/15/2026

Inventory management overview

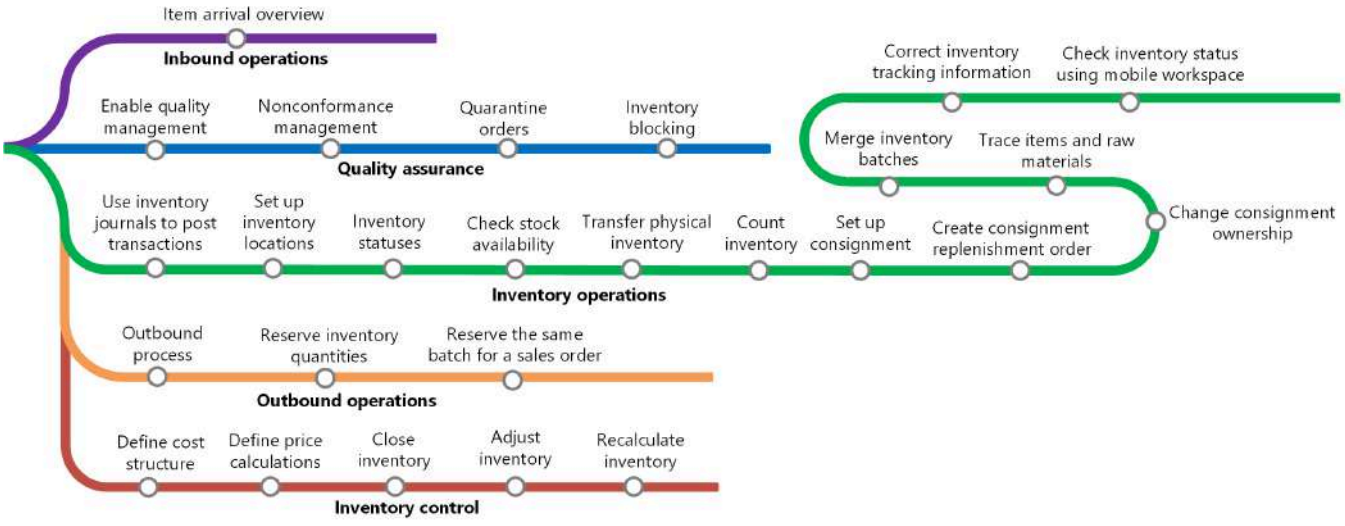
08/29/2025

With Inventory management, you can perform the following tasks:

- Inbound operations
- Quality assurance
- Inventory activities
- Outbound operations
- Inventory control

Learning map

The following learning map shows the major concepts and tasks that make up the framework of the Inventory management module.



Related information

What's new and in development

Go to the Dynamics 365 Release Planner to see what new features are released and what new features are in development.

Inventory accounting

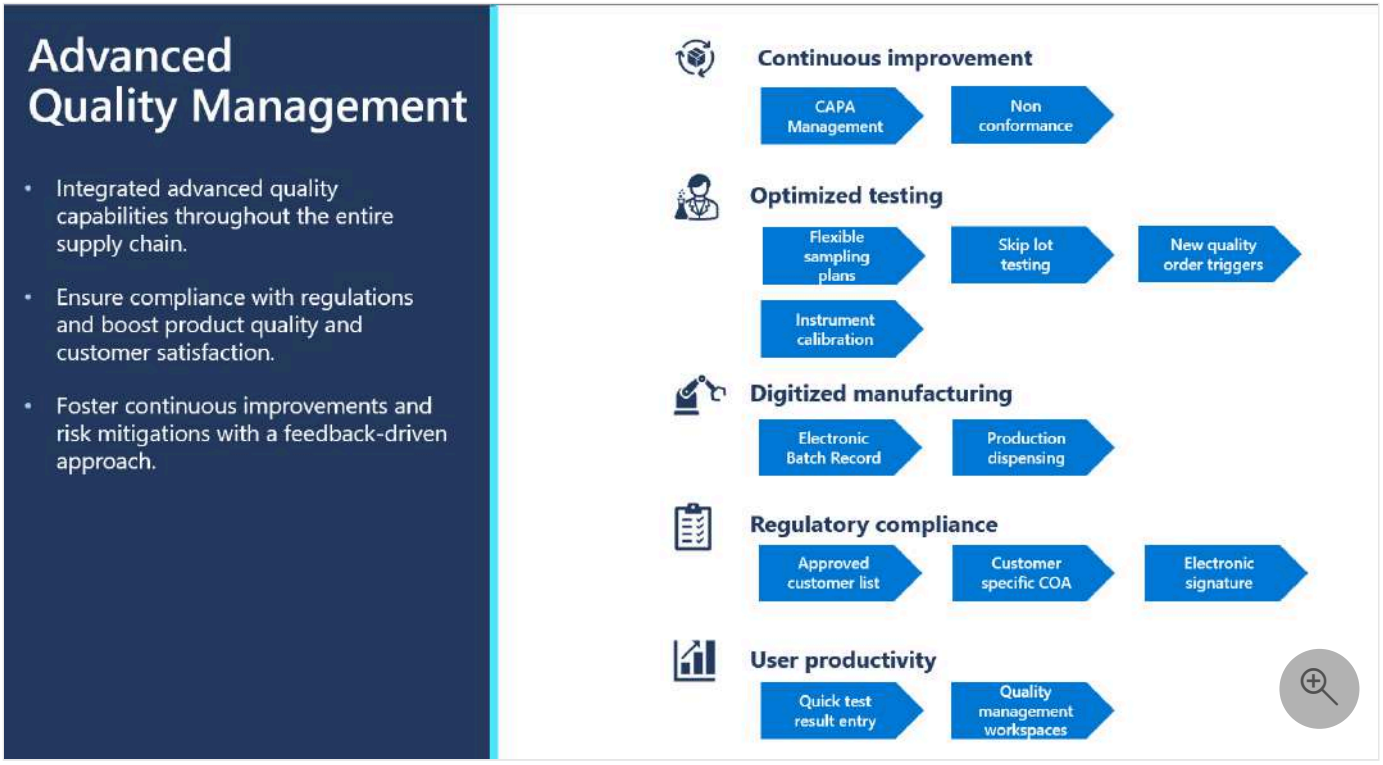
Learn more in Inventory close.

Manufacturing and Supply Chain Management blogs

You can find opinions, news, and other information about Inventory management and other solutions on the Dynamics AX Manufacturing R&D Team Blog and Supply Chain Management in Dynamics AX R&D Team Blog.

Advanced quality management overview

Advanced quality management is a suite of quality management features introduced in Microsoft Dynamics 365 Supply Chain Management version 10.0.44. It enhances the existing quality management features by adding many new tools. These tools address core issues that manufacturers face as they try to meet the United States Food and Drug Administration's (FDA's) Quality System Regulation (QSR), part 11 of Title 21 of the Code of Federal Regulations (21 CFR Part 11), current Good Manufacturing Practice (cGMP), and other regulatory and international standards, such as the International Organization for Standardization (ISO) and Six Sigma.



Prerequisites

Before you can use the features that this article describes, ensure that your system meets the following requirements:

- You're running Supply Chain Management version 10.0.44 or later.
- The following features are turned on in Feature management. As of Supply Chain Management version 10.0.47, all of these features are turned on by default.
  - *Approved customer list* – This feature lets administrators set up rules that limit the sale of specific products to specific customers. The feature applies to sales orders, sales quotations, and sales agreements.

- *Dispense management* – This feature lets users separate dispensing activities from picking activities. In this way, it ensures effective management of controlled and regulated products. The feature supports over-picking where residual products are returned to inventory, identifies items that require dispensing control, and requires authorized personnel for dispensing. Additionally, it captures electronic signatures for dispensing postings, establishes dispensing tolerance parameters, generates pick lists and dispensing tickets at production release, and enables the quantity that is issued to exceed the proposed quantity. As of Supply Chain Management version 10.0.49, this feature is mandatory and can't be turned off.
- *Electronic signature improvements* – This feature introduces new electronic signature requirements for various management tasks, including dispense management, corrective and preventive actions (CAPA) management, instrument calibration, and validation of quality orders. The feature also defines passkey expiration and passphrase constraints for electronic signatures.
- *Advanced quality management* – This feature integrates advanced quality capabilities throughout the Supply Chain Management solution. Key capabilities include optimized testing strategies (flexible sampling plans, skip lot testing, new quality order triggers, and instrument calibration), regulatory compliance (customer-specific certificates of analysis [COAs] and enhancements to electronic signatures), digitized manufacturing for precision and compliance (electronic batch records), continuous improvement, risk mitigation (CAPA management and enhancements to nonconformance), and an enhanced user experience (simplified entry of quality test results and new workspaces for quality management). As of Supply Chain Management version 10.0.49, this feature is mandatory and can't be turned off.

Key features of advanced quality management

This section describes some of the key industry-specific capabilities of the solution.

- **Corrective and preventive actions (CAPA)** – Users can automate processes for managing, tracking, and correcting root causes of issues. They can also implement action plans to prevent the same issues and similar issues from occurring in the future. For more information, see Corrective and preventive action (CAPA) management overview.
- **Electronic signature requirement for CAPA case closure** – Users can establish CAPA case closure and cancellation as a trigger for electronic signature sign-off. For more information, see CAPA management administration.
- **Flexible sampling plans** – Companies can inspect a fraction of a lot, or skip lots completely, to reduce quality and material handling costs, accelerate material throughput, and make

testing less redundant and destructive. For more information, see Flexible sampling plans.

- **Approved customer lists (ACLs)** – This feature provides more control over what products specific customers can buy and when they can buy those products. This capability is especially important for a product line that involves controlled substances and branded products. For more information, see Approved customer lists.
- **Electronic signature** – This feature enhancement improves the security of digital signatures to meet the latest government requirements and regulations. For more information, see CAPA management administration and Electronic batch records (EBRs).
- **Electronic batch records (EBRs)** – Users can view all the versions, methods, and characteristics for consumed bill of materials (BOM) and formula items. This view is referred to as a *master manufacturing record* (MMR). It maintains all the methods that can be used to produce an item. Users can also view and print reports of the actual versions, test values, attribute values, and quality order results of all the items that are produced and the ingredients that are consumed. This view is referred to as a *batch production record* (BPR). It captures the actual version and methods that are used to produce an item. For more information, see Electronic batch records (EBRs).
- **Instrument calibration** – This new feature adds a capability for tracking individual test instrument tags. It also supports an ongoing calibration process for instrument tags and tracks the use of tags against quality order tests. For more information, see Test instrument calibration.
- **Quality associations for return and transfer orders** – This feature enhances the capability of quality associations by enabling automatic creation of quality orders that are triggered from a sales return or a transfer order. For more information, see Quality associations.
- **Customer-specific certificates of analysis (COAs)** – This feature supports the setup and application of customer-specific COA requirements. Each customer drives details such as whether a specific test should be included, whether minimum and maximum values should be suppressed, whether customer-specific batch attribute ranges should be used instead of the standard range, and whether pass and fail results should be replaced with standard text. In addition, customer-specific COAs can be automatically generated from the sales order packing slip posting process. For more information, see Customer specific certificate of analysis (COA).
- **Production dispensing** – Users can segregate the dispensing area and activities from the picking activities. For more information, see Production dispensing.
- **Quick test results entry** – Users can quickly and easily enter test results on a quality order from a consolidated view that provides visibility into minimum, maximum, and target values for a test. For more information, see Enter quality test results quickly.

Quality order usability improvements

Advanced quality management adds several usability improvements to the quality order process. Quality order tests can now be skipped. A user who decides not to perform a test on a given quality order can make a note of that decision on the quality order without losing visibility into the original test. When a test is skipped, all validations on the test are bypassed, and the test is excluded from the results. Acceptable quality level (AQL) calculations are then performed on the remaining tests.

The priority level of the quality order and the tester that the quality order is assigned to are new data elements on the quality order. Use them to sort and filter the quality work that must be completed. Additionally, you can now record result notes for a given test result line. Finally, a parameter now controls the validation that determines whether a selected test instrument tag is already used on an open quality order. Previously, this validation always ran. However, it can now be skipped if it isn't important to a business.

Skip tests as required

A common business process is to create a test group that has several tests, but intentionally skip some tests when the test group runs within a quality order. Previously, you had to delete tests that you didn't want to enter results for. However, some businesses prefer to mark the test as *Skipped* instead of deleting it, so that the decision to skip it can be logged in an audit trail.

When you skip a test, all result data is cleared and can no longer be edited. In addition, no validation is done on associated test result data, such as tag numbers and test result values. The skipped test is also excluded from AQL calculations.

You can reverse the decision to skip a test. In this case, the test returns to its original state, and you can enter test results as usual.

Skip tests from a quality order test line

To mark a test as *Skipped*, select the desired test lines, and then, on the Action Pane, select **Skip**. The test result symbol, order line result, and batch attribute values are cleared. The **Update inventory batch attribute** and **Include results** checkboxes are cleared, and the **Skip test** checkbox is selected. You can't edit the test while it is in the *Skipped* state.

If test results were already entered, you receive a warning message when you try to skip the test.

If you no longer want to skip a test that was previously skipped, select the line, and then select **Skip** again. The line returns to its original state.

Skip tests during quick results entry

When you use the **Skip** option from quick results entry, you can't select multiple test lines at a time. If there are multiple rows of result data for the same test, the **Skip** option collapses them to their original quantities before it updates the data and makes the line uneditable.

Quality order priority and tester assignment

Quality orders now include **Quality order priority** and **Worker** fields on the **General** tab. Both fields are optional.

- **Quality order priority** – This free-form field can be user-defined based on business requirements. For example, a business might express priority as a number between 0 (zero) and 100, where 100 represents the highest priority.
- **Worker** – This field must be set to a valid worker.

Additionally, quality orders now include fields for entering a test outcome description and result notes for each test line.

Skip test instrument tags validation

Previously, the system always ran a validation to check whether the test instrument tags that were selected for use on a new quality order are already used on an existing open quality order. If the system found a selected test instrument tag on an open quality order, it showed a warning message.

This validation is now optional, and a parameter lets you control whether it's run or skipped. You can configure this parameter for each tag individually. You can also set a default value that applies to all new tags.

To set the validation option for individual tags, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality Control** > **Test instrument tags**.
2. For each tag, set the **Skip check for test instrument on open quality order** option to one of the following values:
    - *Yes* – Skip the validation that determines whether the tag is already assigned to an open quality order. If the tag is already assigned to an open quality order, no warning

message is shown.

- *No* – The system runs the validation and shows a warning message if the tag is already assigned to an open quality order.

To set the default validation setting for all new tags, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory and Warehouse Management parameters**.
2. On the **Quality management** tab, in the **Skip check for test instrument on open quality order** field, specify whether the validation should be skipped for all new test instrument tags by default. You can change the setting for individual tags.

Learn more in Test instrument calibration.

Last updated on 07/27/2026

Arrival overview

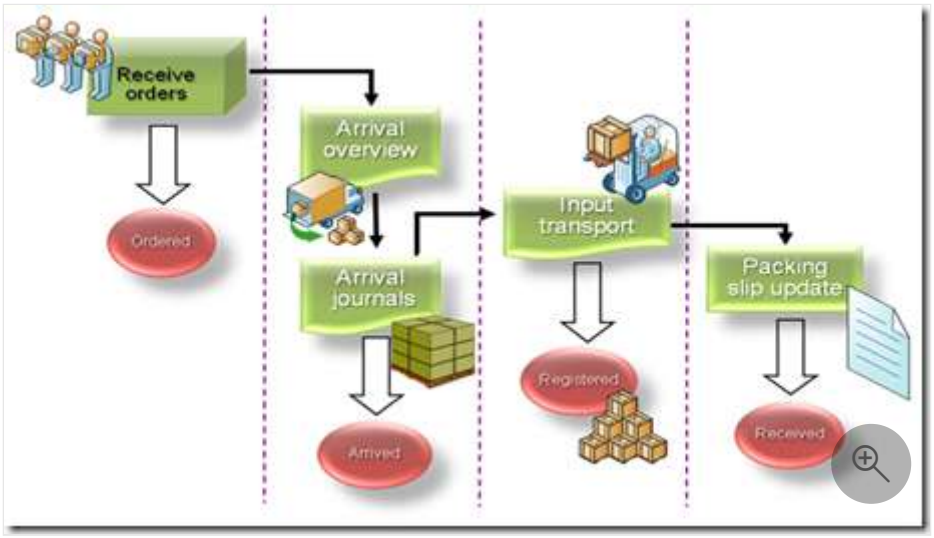
08/29/2025

This article provides information about the Arrival overview feature. The Arrival overview page is part of this feature and provides an overview of all items that are expected to arrive as incoming items.

The **Arrival overview** page provides an overview of all expected incoming items. It also shows arrivals that you can initialize based on the overview. This article focuses on the receiving process.

Business scenario

Consider the following scenario in the inbound processes.



Sammy, a receiving clerk, wants to know what is expected to be received on the current day. On the **Arrival overview** page, Sammy can get an overview of the current tasks, and a rough estimate of quantities, volume, weight, different order types, and so on. Later, a delivery arrives at one of the inbound docks, and Sammy receives a list of the delivery. On the **Arrival overview** page, Sammy can perform the following tasks:

- Identify the matching receipt order, and register the receipt as *In progress*. The lines that are required for a registration are automatically generated, and the receipt can be monitored, even though the transactions aren't yet posted as *Registered*.
- Access the appropriate arrival journal reference (that is, the **Item arrival** journal or the **Production input** journal), and identify journals that are ready for a product receipt update.

Arrival overview page

To open the **Arrival overview** page, select **Inventory management** > **Inbound orders** > **Arrival overview**. You can view a list of orders that are expected to be received. The overview is divided into a header and lines. The header information is grouped by the order type, expected receipt date, and delivery destination. When you select a header line for arrival, all the detail lines that are related to the receipt reference are selected for arrival in the line details part of the page. When you post all related journal lines, this information isn't shown.

Arrival overview profiles

The **Arrival overview** page provides an overview of items that are expected to arrive and the date when they're expected to arrive. As part of the arrival process, you can use multiple sets of personal settings. Individual users can define their personal settings on the **Arrival overview profiles** page.

Set up item arrival

In this example, Sammy sets up a new computer at a location that receives finished goods from production at Site 1. On the **Arrival overview profiles** page, Sammy creates a new setup named **Site 1 production** and specifies the following settings.

1. On the **Arrival options** FastTab, in the **Range** field group, enter information about a day interval and the warehouses to include in the overview.
2. On the **Arrival options** FastTab, in the **Journal** field group, enter a receiving warehouse, a location, and a journal name (item arrival/production input).
3. On the **Arrival query details** FastTab, in the **Site** field group, in the **Restrict to site** field, enter a site to limit the view in the overview area.
4. On the **Arrival query details** FastTab, in the **Transaction types shown** field group, set the **Production orders** option to *Yes*.
5. On the **Arrival query details** FastTab, in the **Miscellaneous** group, set the **Update on startup** option to *Yes* if the view should be automatically updated at startup. Set the **Update on range change** option to **Yes** if the view should be automatically updated when range values are changed.

For this example, the **Arrival overview profile name** field on the **Arrival options** FastTab of the **Arrival overview** page is set to *Site 1 production*.

Prerequisites for arrival journals

To automatically create arrival journals from the **Arrival overview** page, define appropriate information in the **Journal** field group on the **Arrival options** FastTab.

- You must specify a journal name to create a journal.
- If you specify values in the **Warehouse** and **Location** fields, those values apply on the journal lines. If you don't specify values, the system uses the values from the dimension that is specified on the inventory transactions.

Items that you receive from one expected receipt order

On the **Receipts** FastTab, Sammy selects a line and then selects **Start arrival**. All related lines that are in the specified range and that have a quantity to register are automatically selected. An item arrival journal is generated that matches the expected receipt order and the journal. The quantity is automatically initialized on all lines that are created.

Items that you receive from more than one expected receipt order

On the **Receipts** FastTab, Sammy selects multiple lines and then selects **Start arrival**. An item arrival journal is generated that matches all the expected receipt orders and the journal. All lines are created on one item arrival journal header, and the quantity is automatically initialized.

Receive items from one or more expected receipt orders

View information

To get an overview of expected receipts in a date interval, Sammy enters the following information in the fields on the **Arrival options** FastTab on the **Arrival overview** page and then selects **Update** to update the view:

- Arrival overview profile name: *Inquiry*
- Show lines: *All*
- Days back: (Blank)
- Days forward: *0*
- Warehouses: *GW, MW*

Sammy can view the following information:

- All related receipt orders for the system date and an infinite number of days back from it (the **InventTrans.StatusDate** interval), and receipts to warehouses GW and MW, regardless of status.
- Detailed line information for more than one order. Sammy can select multiple header lines in the overview and then select **Show all selected** to view all the corresponding line detail information for all selected orders.

- Information about a specific purchase order. To show only information that is related to a specific reference number in the overview, Sammy can enter an account number in the **Account number** field and an order number in the **Reference number** field.
- An overview of the registration tasks that are due for all the order lines that an item arrival journal has been created for but hasn't yet been posted. To view this information, Sammy can select **In progress** in the **Show lines** field.

Update journals

To register one or more order lines that are due to be processed, Sammy can select the lines in the overview grid or in the line grid, and then select **Journals** > **Show arrivals from receipts**. The item arrival headers that match the lines are shown. To update the purchase order product receipt for the registered items, Sammy can access the item arrival journal headers that are ready for update. To access these item arrival journal headers, Sammy selects **Journals** > **Product receipt ready journals**. All the header lines that are ready for product receipt update in the specified warehouse range are shown. (The header lines that are shown aren't related to the day interval).

Start an arrival registration

By selecting multiple lines on the **Arrival overview** page, Sammy can start an arrival of more than one receipt reference. When Sammy selects a line from the receipts overview, the corresponding line details are selected. If a quantity for registration exists, the **Start arrival** button is available. Sammy can use two methods to start the arrival registration:

- To filter the page so that it shows only records that meet specific criteria, in the **Vendor reference** field, scan a reference number from a vendor, such as the bar code for a delivery note.
- In the overview part or the details part of the **Arrival overview** page, manually select or cancel the selection of records for arrival registration. Then, when Sammy selects **Start arrival**, the selected records are automatically created in an item arrival journal. The records include line information, and all unique field information is assigned.

Update arrival information and process a product receipt

When all goods have been registered, the warehouse manager or purchasing manager can update the received items with a product receipt to add the physical cost. To update arrival information and post a product receipt, follow these steps.

1. Select **Inventory management** > **Inbound orders** > **Arrival overview**.
2. On the **Arrival overview** page, select **Journals** > **Product receipt ready journals** to show a list of the journals that are ready for product receipt update.
3. Select the journals that you want to update, then select **Functions** > **Product receipt**.
4. On the **Posting** page, enter the product receipt number, if it isn't already available on the journal, then select **OK** to process the product receipt.

Summary

The **Arrival overview** page helps the warehouse manager and warehouse workers get an overview of expected work that must be done as part of an inbound process. You can also use the page to start the item arrival process, to help guarantee that items are tracked at the first entry into the warehouse.

Set up an item arrival overview profile

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article focuses on the setup of an arrival overview profile. The arrival overview profile is a collection of rules that will be applied when the Arrival overview page is opened by a user. You can use this procedure in demo data company USMF. This procedure would typically be carried out by a receiving clerk.

1. Go to **Inventory management** > **Setup** > **Distribution** > **Arrival overview profiles**.
2. Select **New**. Because you'll almost always work in the same warehouse offloading full truck loads, you should create an arrival overview profile to simplify the process of registering received items.
3. In the **Arrival overview profile name** field, type a value.
4. In the **Show lines** field, select an option. Select which lines to show for the receipts:
    - *All* – Show all lines, regardless of status.
    - *In progress* – Show lines for receipts in which the progress is *Complete* or *Partly*. This means that for each line, either the full quantity or part of the quantity has been registered in an arrival journal.
    - *Not complete* – Show lines for receipts in which the progress is *None* or *Partly*. This means that for each line, nothing or only part of the quantity has been registered in an arrival journal.
5. Expand or collapse the **Arrival options** section.
6. In the **Days forward** field, type a value. This sets a filter to show the receipt lines expected to be received within the next few days (depending on the number you set).
7. In the **Days back** field, type a value. This sets a filter to show the receipt lines expected to be received within the specified number of days before today.
8. In the **Warehouses** field, type a value. Filter on one or more warehouses.
9. In the **Mode of delivery** field, select a value. This sets a filter to show only the receipt lines with this Mode of delivery.

10. In the **Name** field, select *WHS*.
11. In the **Warehouse** field, select warehouse *24*. This is the default warehouse used for registered receipt lines that use this profile.
12. In the **Location** field, select *Baydoor*. This is the default location used for registered receipt lines that use this profile.
13. Expand or collapse the **Arrival query details** section.
14. In the **Restrict to site** field, select site *2*. This sets a filter to show only the receipt lines with this site.
15. Set the **Purchase orders** option to *Yes*. Select receipt lines from purchase orders.
16. Set the **Transfer** orders option to *Yes*. Select receipt lines from transfer orders.
17. Select **Save**.
18. Close the page.

Last updated on 04/10/2025

Quality and nonconformance management overview

08/13/2025

This article introduces the quality and nonconformance management features in Microsoft Dynamics 365 Supply Chain Management and explains how they can help improve product quality in your supply chain.

Quality assurance involves product testing and the management of nonconforming material. Quality management processes help guarantee a high level of product quality in your supply chain. These processes also help optimize supply chain processes and increase customer satisfaction. Quality management can help you manage turnaround times when you're dealing with nonconforming products, regardless of point of origin of those products. You can link diagnostic results to correction tasks. The system can schedule tasks to correct problems and therefore help prevent recurrences of those problems in the future. Quality management also lets you track issues (including internal problems) by problem type, and lets you identify solutions as either short-term or long-term. Statistics about key performance indicators (KPIs) provide insight into nonconformance problems that have previously occurred and the solutions that were applied to correct them. You can use historical data to help review the effectiveness of quality measures that have previously been taken and to determine appropriate measures in the future.

Quality management can help you manage turnaround times when you handle nonconforming products, regardless of their point of origin. Because diagnostic types are linked to correction reporting, Supply Chain Management can schedule tasks to correct problems and prevent them from recurring.

In addition to functionality for managing nonconformance, quality management includes functionality for tracking issues by problem type (even when the issues are internal problems), and for identifying solutions as short-term or long-term. Statistics about key performance indicators (KPIs) provide insight into the history of previous nonconformance issues and the solutions that were used to correct them. You can use historical data to review the effectiveness of previous quality measures and determine appropriate measures to use in the future.

When you set up a quality association, Supply Chain Management can generate quality orders for various business processes, events, and conditions. The quality association can cover a specific item, a specific group of items, or all items.

Examples of the use of quality management

Quality management is flexible and can be implemented in various ways to meet the requirements of specific levels of supply chain operations. The following examples illustrate possible uses of these features:

- Automatically start a quality control process, based on predefined criteria (for example, when warehouse registration of a purchase order from a specific vendor occurs).
- Block inventory during inspection to prevent non-approved inventory from being used (full blocking of purchase order quantities).
- Use item sampling as part of a quality association to define the amount of current physical inventory that must be inspected. Sampling can be based on fixed quantities, a percentage, or a full license plate.
- Create quality orders for partial receipts. To create a quality order that is based on the quantity that is physically received with an order, you must select the **Per updated quantity** check box on the **Item sampling** page.
- Create test types that include minimum, maximum, and target test values, and perform qualitative-versus-quantitative testing that has predefined validation results.
- Specify an acceptable quality level (AQL) to control quality measure tolerances.
- Specify the resources that an inspection operation requires, such as a test area and test instruments.

> ⓘ **Note**
>
> The *Quality management for warehouse processes* feature extends the capabilities of quality management. If you're using this feature, see **Quality management for warehouse processes** for examples that show how quality management works when it's enabled.

Controlling the quality management process

Here are some of the ways that you can control the quality management process:

- Create quality orders that are based on trigger points, such as "at product receipt" for inbound operations or "at product pick-up" for outbound operations.
- Document test results, and determine whether the results meet the established test criteria and AQLs.
- Use document management for detailed product specifications and user-specific notes as part of reporting during the inspection process.
- Maintain nonconforming products, and correlate those products with additional nonconformance information to track down the original cause of a problem.
- Document the cost of managing a nonconformance. This cost can include the items (such as spare parts), miscellaneous charges, and the timesheet hours that are required to

correct the nonconformance.

- Schedule error correction processes by using correction handling that is linked to quality orders.

Product testing and quality orders

Product testing is typically referred to as quality control and uses quality orders. By using the quality control functionality, you can do the following:

- Define the tests that must be performed for material. These tests include the quality specifications, the applicable test instrument, documents that describe the test, a sampling plan, and the AQLs.
- Create a quality order that identifies the tests that must be performed for a specific order, such as a purchase or production order, or for a specific inventory quantity. You can manually create a quality order, or a quality order can be automatically generated based on quality guidelines.
- Define the quality guidelines that are related to purchase, quarantine, production, or sales orders in each business process, so that a quality order that identifies the testing requirements for incoming or outgoing material can be automatically generated.

- Record the test results in a quality order, validate the test results against the AQL, and print a certificate of analysis that shows the test results.

Nonconformance

A nonconformance describes an item that has a quality problem. The nonconformance process lets you create a nonconformance order that describes a quantity of nonconforming material, the problem source, the problem type, and explanatory notes. You can define a classification of problem types to make analysis of nonconforming material easier. You can also print a nonconformance tag and a nonconformance report to guide the disposition of nonconforming material. For example, the tag and report might indicate a condition of **Unusable** or **Restricted usage**.

The following table lists the six default nonconformance types and describes the information that must be recorded for each type.

Expand table

Nonconformance type	Source information
Customer	The customer account number, the sales order number, or a lot number of a sales order transaction. For example, the nonconformance might be related to a specific sales order shipment or to customer feedback about product quality.
Service request	The customer account number, the sales order number, or a lot number of a sales order transaction. For example, the nonconformance might be related to a specific sales order shipment or to a customer's complaint about item quality.
Vendor	The vendor account number, the purchase order number, or a lot number of a purchase order transaction. For example, the nonconformance might be related to a purchase order receipt or to a vendor's concern about a part that it supplies.
Production	The production order number or a lot number of a production order transaction. For example, the nonconformance might be related to a specific batch that was produced.
Internal	The quality order number or a lot number of a quality order transaction. For example, the nonconformance might be related to the tests that are performed as part of a quality order or to an employee's concern about product quality.
Co-product production	A co-product production order nonconformance that is related to batch production orders.

Nonconformances are associated with a problem type. Problem types are defined on the **Problem types** page, where you specify which problem types can be associated with each

nonconformance type. For example, the problem types for nonconformances of the **Service request** type might reflect a classification of customer complaints, whereas the problem types for nonconformances of the **Internal** type might represent a classification of defect codes.

When you create a new nonconformance, you select the nonconformance type and the problem type. The initial approval status is **New**, which represents a request for action. The next step is to change the approval status to **Approved** or **Refused**, to indicate that you will or won't take action on the nonconformance. You can also close a nonconformance (by selecting a separate check box) to indicate that you're finished with it, or you can reopen a nonconformance to indicate that additional consideration is required.

You can enter comments for a nonconformance by attaching a document. It's a good idea to define a unique document type for nonconformances by using the **Document type** page. You can then use the **Report setup** page to define whether comments for this document type should be printed on the nonconformance report and nonconformance tag. The conformance report and nonconformance tag can help with material disposition. You can selectively generate reports and tags, based on selection criteria that are associated with a nonconformance. These criteria include the nonconformance number, item, customer, vendor, and status.

The nonconformance report displays the nonconformance number, item, and problem type. Depending on your report setup policy, the report might also display related notes about the nonconformance. The nonconformance tag displays similar information, and also includes the quarantine zone and type (such as **Restricted usage** or **Unusable**) that you assigned to the nonconformance to help guide disposition of the defective material.

Approved nonconformance

You can optionally define one or more related operations for an approved nonconformance. A related operation describes the work that should be performed, and contains a list of the quality operations that you've defined and descriptive text about the reason for the work. After you define an operation, you can optionally define the miscellaneous charges, items, and timesheet labor hours that are required in order to perform the work. The calculated costs are shown for the related operation, and the total calculated costs are shown for the nonconformance. The calculated costs and the underlying details (about items, labor hours, and miscellaneous charges) represent reference information, and they're used only in the quality management function.

You can optionally create a quality order from a nonconformance by first performing an inquiry for quality orders and then creating the new quality order. For example, a quality order might identify the need to test (or retest) the defective material. The newly created quality order displays the link to the originating nonconformance.

You can optionally link one nonconformance to another and create a new nonconformance from an existing one. For example, the link can reflect the interconnection between quality problems.

Correction handling

The **Corrections** page lets you create a list of nonconformances that must be corrected. Each correction item is associated with the diagnostic type that caused the problem to be discovered. The **Corrections** page also contains information about who must perform a corrective action, and when. You can describe the details of the problem and the corrective action that is required by attaching a document to the correction. After the nonconformance has been addressed or corrected, you "close" the correction item by selecting the **Completed** option. You can also indicate that the solution was a short-term solution.

It's a good idea to define a unique document type for corrections by using the **Document type** page. You can then use the **Report setup** page to define whether comments for this document type are printed on the correction report. A printed correction report displays information about the nonconformance and the related nonconformance notes. The report also includes correction information, such as the diagnostic type, and the related correction notes.

Next steps

- Enable quality and nonconformance management
- Inventory blocking
- Quarantine orders
- Inspect the quality of goods

Enable quality and nonconformance management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an overview of the process for setting up and configuring quality and nonconformance management features in Microsoft Dynamics 365 Supply Chain Management.

Enable the feature

Follow these steps to enable quality management on your system.

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. On the **Quality management** tab, set the **Use quality management** option to *Yes*.
3. In the **Hourly rate** field, enter an hourly labor rate in the local currency. The hourly rate is used to calculate costs for operations that are related to a nonconformance. The hourly rate and calculated costs provide reference information for a nonconformance. They don't interact with other functionality.
4. Select **Report setup**.
5. Add new lines for the various report types, and select the type of document to use for each report.
6. Close the page.
7. Close the page.

Configure quality management

Before you can use the quality management features and generate quality orders, you must configure the system and prerequisites. Here's a list of the steps required to configure quality management.

1. Enable quality and nonconformance management.
2. Optional: Configure test instruments.
3. Configure tests.
4. Configure test variables and outcomes.
5. Configure test groups.
6. Optional: Configure quality groups, and link to products.
7. Optional: Configure quality associations.

After you complete the configuration, you can start to create and process quality orders. For more information about how to create and work with quality orders, see Quality orders.

Configure nonconformance management

Before you can use the nonconformance management features and generate nonconformances, you must configure the system and prerequisites. Here's a list of the steps required to configure nonconformance management.

1. Enable quality and nonconformance management.
2. Configure workers who are responsible for approving nonconformances.
3. Configure problem types.
4. Configure quarantine zones.
5. Configure diagnostic types.
6. Configure operations.
7. Optional: Configure quality charges.

After you complete the configuration, you can start to create and process nonconformances. For more information about how to create and work with nonconformances, see Create and process nonconformances.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Quality management for warehouse processes

Last updated on 05/27/2026

Quality management test instruments

This article describes how to create test instruments or instrument types that you can use for tests on quality orders in Microsoft Dynamics 365 Supply Chain Management.

Use the **Test instruments** page to define and view details about the instruments or instrument types that you use to perform tests on quality orders. Test instruments are optional. However, they can help quality workers determine which device or tool they should use to perform a specific test.

Test instrument example

You're performing various tests on electrical components. Some tests are for the voltage output of the components, one test is for their temperature, and one test is for their weight. Different tools, devices, or equipment are used to perform each test. For example, a voltage meter is used to measure voltage, a thermometer is used to measure temperature, and a scale is used to measure weight. You can configure each of these device types as a test instrument and indicate the unit of measure that the test results should be recorded in. For example, you record results from a voltage meter in volts, results from a thermometer in degrees Fahrenheit or degrees Celsius, and results from a scale in pounds or kilograms.

Prerequisites for calibration-related settings

Most of the features that are described in this article are available as a standard part of all current versions of Supply Chain Management. However, the calibration-related settings (the **Tag number required**, **Used for calibration**, and **Calibration label layout** fields) add the following requirements:

- You must be running Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Learn more in Test instrument calibration.

Create a test instrument

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test instruments**.

2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:

    - **Test instrument** – Enter a unique ID or name for the test instrument.
    - **Description** – Enter a detailed description of the test instrument.
    - **Unit** – Select the unit that the instrument measures results in. The **Precision** field is automatically set based on the unit that you select.
    - **Tag number required** – Select this checkbox if a test instrument tag number (in addition to an instrument type) must be specified for the test instrument on quantity orders. The test instrument tag identifies the specific physical instrument that is used in the test. Selection of this checkbox triggers other functionality that is related to tracking and calibration of test instruments. To view, create, and edit the available test instrument tags, select **Test instrument tags** on the Action Pane.
    - **Used for calibration** – Select this checkbox for test instrument types that are used to calibrate other instruments and equipment.
    - **Calibration label layout** – For test instrument types where the **Tag number required** checkbox is selected, select the layout that is used to print calibration labels.

3. If you're using Asset Management to manage test instruments, then make the following additional settings. Skip this step for test instrument types that you aren't managing with Asset Management. These settings are only available when the **Tag number required** checkbox is selected.

    - **Asset type** – If you want to manage test instruments of this type using Asset Management, then select the related asset type here.
    - **Test quantity counter type** – Select the counter to increment each time a quality order for a test instrument of this type is validated. The system increments the counter by the quantity that was tested on the quality order. If you select a counter type in this field, you can't select a value in the **Fixed increment counter type** field. Learn more about counters in Counters.
    - **Fixed increment counter type** – Select the counter to increment each time a quality order for a test instrument of this type is validated. The system increments the counter by the value specified in the **Usage increment** field, regardless of the quantity that was tested on the quality order. If you select a counter type in this field, you can't select a value in the **Test quantity counter type** field. Learn more about counters in Counters.
    - **Usage increment** – If you chose a fixed-increment counter, then enter the value by which to increment the asset counter when a quality order is validated.

- **Calibration state** – Select the asset lifecycle state that indicates that the asset is being calibrated. When an asset of the type specified in the **Asset type** column changes status to the value set in this field, the system automatically updates the related test instrument tag to *Calibration*. Learn more about asset stages in Asset lifecycle states.
- **Out of service state** – Select the asset lifecycle state that indicates that the asset is out of service. When an asset of the type specified in the **Asset type** column changes status to the value set in this field, the system automatically updates the related test instrument tag to *Out of service*. Learn more about asset stages in Asset lifecycle states.
- **Auto-create test instrument tag from asset** – Select this checkbox to automatically create a test instrument tag when you create a new asset of the linked **Asset type** (existing assets aren't affected).

4. Close the page.

Related information

- Quality management tests
- Quality management test groups
- Test instrument calibration
- Manage test instrument calibration with Asset Management (preview)

Last updated on 07/23/2026

Quality management tests

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to create tests that you can use for quality orders in Microsoft Dynamics 365 Supply Chain Management.

You use the **Tests** page to define and view the individual tests that determine whether your products meet quality specifications. You can assign one or more individual tests to a test group. In this case, you also specify test-specific information, such as the acceptable measurement values. Measurement values are used for quantitative tests. For qualitative tests, test variables are used.

- For quantitative tests, the **Type** field is set to *Integer* or *Fraction*. A unit of measure is also specified. Quality specifications and test results are expressed as numbers.
- For qualitative tests, the **Type** field is set to *Option*. Qualitative tests require additional information about the test variable that is being measured and its enumerated options. Quality specifications and test results are expressed according to an outcome.

You can optionally assign a test instrument to a test. However, test instruments aren't required to create and use tests with quality orders. Learn more in Quality management test instruments.

You can also optionally specify a unit for a test, to indicate the unit of measure that test results are recorded in. For example, a test for temperature might include a unit of either degrees Fahrenheit or degrees Celsius.

Example of a test

A manufacturing company performs two tests on purchased material: a quantitative test for material quality and a qualitative test for packaging damage. The company specifies additional information about the qualitative test to define the test variable (for example, damaged packaging) and its outcomes. The company uses the **Test groups** page to assign the two tests to

a test group and to specify the test-specific information. The test group is assigned to a quality order so that the company can report test results for the two tests.

Create a test

Follow these steps to create a test.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Tests**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Test** – Enter a unique ID or name for the test.
    - **Description** – Enter a detailed description of the test.
    - **Type** – Select the type of results that the test produces. For quantitative tests, select *Fraction* or *Integer*. For qualitative tests, select *Option*.
    - **Test instrument** – Select a test instrument that should be used for the test.
    - **Unit** – If you selected *Fraction* or *Integer* in the **Type** field (that is, if the test is a quantitative test), select the unit of measure that test results should be recorded in.
    - **Defect type** – If you're using acceptance sampling, then choose a defect type that applies for this test. Learn more about defect types, how to set them up, and how to use them in Acceptance sampling.
3. Close the page.

> **Note**
>
> After you save a test, you can no longer edit the **Type** field in the grid. If you must change the type of test results that will be recorded for a test, select **Change quality test type** on the Action Pane. In the **Change quality test type** dialog box, set the **New type** field to the desired type, and then select **OK** to close the dialog box.

Related information

- Quality management test instruments
- Quality management test groups
- Quality management test variables
- Quality associations
- Acceptance sampling

Last updated on 05/27/2026

Quality management test variables

Article • 06/03/2022

This article describes how to create test variables that can be used for qualitative tests on quality orders in Microsoft Dynamics 365 Supply Chain Management.

For every qualitative test that is defined on the **Tests** page, you must define a test variable and its possible outcomes (results). (For qualitative tests, the **Type** field on the **Tests** page is set to *Option*.)

You use the **Test variables** page to set up, edit, and view the possible outcomes for a test variable that is associated with a qualitative test. For each outcome, you assign an outcome status of either *Pass* or *Fail* to indicate whether the test is passed or failed when that outcome is selected as a test result. You use the **Test groups** page to assign a test variable and a default outcome for it to an individual qualitative test.

For every test variable, we recommend that you define at least two outcomes: one that has an outcome status of *Pass* and one that has an outcome status of *Fail*. There is no limit on the total number of variables or outcomes that can be defined. Additionally, multiple tests can use the same test variables to record results.

Examples of test variables

Example 1

A manufacturing company performs two tests on manufactured materials. In one test, the pH level is associated with a color strip. Acceptable pH levels are in lighter colors, and unacceptable pH levels are in darker colors. In another test, multiple visual inspections are performed, and quality workers use their judgement to determine whether the item passes or fails the visual inspection.

Example 2

A manufacturing company that produces cookies uses an inspection test for the finished product. This inspection test has several variables. One variable is taste, and the possible outcomes for it are good and bad. A second variable is color, and the possible outcomes for it are too dark, too light, and correct.

Create a test variable

Follow these steps to create a test variable.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test variables**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Variable** – Enter a unique ID or name for the variable.
    - **Description** – Enter a detailed description of the variable.
3. While the new row is still selected, select **Outcomes** on the Action Pane.
4. On the **Test variable outcomes** page, on the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Outcome** – Enter a unique ID or name for the outcome.
    - **Description** – Enter a detailed description of the outcome.
    - **Outcome status** – Select either *Pass* or *Fail* to indicate whether the test is passed or failed when the outcome is selected as a test result.
5. Repeat the previous step for each additional outcome. Make sure that at least one outcome has an outcome status of *Pass* and at least one has an outcome status of *Fail*.
6. Close the pages.

Related information

- Quality management test instruments
- Quality management tests
- Quality management test groups

Feedback

Was this page helpful? Yes No

Provide product feedback

Quality management test groups

09/19/2025

This article describes how to create test groups so you can use multiple tests with quality orders in Microsoft Dynamics 365 Supply Chain Management.

Use the **Test groups** page to set up, edit, and view test groups and the individual tests that you assign to them. The upper part of the page shows the test groups, and the lower part shows the tests that are assigned to a selected test group.

Assign several policies to a test group, such as a sampling plan, an acceptable quality level (AQL), and the requirement for destructive testing. When you assign an individual test to a test group, define additional information, such as the sequence, documents, and validity dates. For a quantitative test, the information that you define also includes the acceptable measurement values. For a qualitative test, the information includes the test variable and default outcome.

The test group that you assign to a quality order defines the default set of tests that you must perform on the specified item. However, you can add, delete, or change tests for the quality order. The system reports test results for each test on a quality order.

When you define a test group, you can optionally specify an item sampling. Use item samplings to define the amount of the product that must be tested. Learn more in Quality management item sampling. You can also indicate whether the tests in a test group are destructive. In a destructive test, the item sampling is destroyed, and the quantity is removed from the on-hand inventory.

Example of a test group

A manufacturing company defines a test group for each variation of its quality guidelines. The various test groups reflect differences in the sampling plans, the sets of tests that must be performed together, the AQL, and other factors. For quantitative tests, there are also differences in the acceptable measurement values. To enforce its quality guidelines, the company assigns a test group to each rule that is used to automatically generate quality orders on the **Quality associations** page. It also assigns a test group to quality orders that are manually created.

Create a test group

To create a test group, follow these steps.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**.

2. On the Action Pane, select **New** to add a row to the grid in the upper part of the page. Then set the following fields for the new row:

    - **Test group** – Enter a unique ID or name for the test group.
    - **Description** – Enter a detailed description of the test group.
    - **Acceptable quality level** – Enter the total percentage of test results that must pass for the group of tests to be considered passed.
    - **Item sampling** – Select an item sampling. Learn more in Quality management item sampling.
    - **Destructive** – Select this check box to indicate that the test group destroys the items that are tested.

3. While the new row is still selected, select the **General** tab in the upper part of the page. Some of the settings for the test group that you select on the **Overview** tab are repeated here. In addition, you can set the following fields for the group:

    - **Update inventory batch attribute** – When you set this option to *Yes* here, it automatically sets to *Yes* for every new test that you add to the test group in the lower part of the page.
    - **Update batch disposition** – When you set this option to *Yes*, if the item that you're testing is batch controlled, the batch disposition automatically updates with the value selected in the **Failed quality order batch disposition** or **Passed quality order batch disposition** field.
    - **Failed quality order batch disposition** – Select the batch disposition code that should be assigned when a quality order that includes this test group fails because it doesn't meet the AQL.
    - **Passed quality order batch disposition** – Select the batch disposition code that should be assigned when a quality order that includes this test group passes because it meets the AQL.
    - **Update inventory status** – When you set this option to *Yes*, if **Inventory status** is enabled on the storage dimension group for the item that you're testing, the status automatically updates with the status selected in the **Failed quality order status** or **Passed quality order status** field.
    - **Failed quality order status** – Select the inventory status that should be assigned to the item when a quality order that includes this test group fails because it doesn't meet the AQL.
    - **Passed quality order status** – Select the inventory status that should be assigned to the item when a quality order that includes this test group passes because it meets the AQL.

Add a qualitative test to a test group

To add a qualitative test to a test group, follow these steps.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**.
2. On the **Overview** tab in the upper part of the page, select the test group that you want to add a test to.
3. In the lower part of the page, on the **Overview** tab, select **Add** on the toolbar to add a row to the grid. Then set the following fields for the new row:
    - **Sequence number** – Enter an integer to specify the order that the tests should be performed in. Tests that have smaller sequence numbers run before tests that have larger sequence numbers.

      > ⓘ **Note**
      >
      > Leave a gap between sequence numbers. For example, set the **Sequence number** field to *10* for your first test, and then increment the value by 10 for each additional test. In this way, you can add new tests later without having to delete and re-create the lines to put them in the desired order.

    - **Test** – Select the test that you want to perform. For a qualitative test, select a test where the **Type** field is set to *Option*.
    - **Effective** – Select the first date when the test is valid. If you leave this field blank, there's no limit.
    - **Expiration** – Select the last date when the test is valid. If you leave this field blank or set it to *Never*, there's no limit.
    - **Test value determination** – Select how an AQL should be determined when multiple results are recorded for the same test. For a qualitative test, only *Manual* can be selected. If you select one of the other values (*Average*, *Minimum*, or *Maximum*), you receive an error message when you save the test group.
    - **Attribute** – If you want to record test results on a batch attribute, select the attribute here, and select the **Update inventory batch attribute** check box.
    - **Update inventory batch attribute** – Select this check box to record test results for the batch attribute that is selected in the **Attribute** field.
4. In the lower part of the page, select the **General** tab. Some of the settings for the test selected on the **Overview** tab are repeated here. In addition, you can set the following fields for the test:

- **Certificate of analysis report** – Set this option to *Yes* to indicate that the test results should be printed on the certificate of analysis (CoA). This report can be generated from the quality order.
- **Action on failure** – Select either *Accept* or *Fail* to indicate whether the test should pass or fail if the AQL for it isn't met.
- **Acceptable quality level** – Enter the total percentage of test results that must pass for the test to be considered passed.

5. In the lower part of the page, on the **Test** tab, set the following fields:

    - **Variable** – Select the test variable to record the qualitative test results for.
    - **Default outcome** – Select the default outcome that should be recorded for the test results.
    - **Test instrument** – Select the device that should be used for the test. If a test instrument is defined, it's automatically entered for the test in the test group.

6. Close the page.

Add a quantitative test to a test group

To add a quantitative test to a test group, follow these steps.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**.

2. On the **Overview** tab in the upper part of the page, select the test group that you want to add a test to.

3. In the lower part of the page, on the **Overview** tab, select **Add** on the toolbar to add a row to the grid. Then set the following fields for the new row:

    - **Sequence number** – Enter an integer to specify the order that the tests should be performed in. Tests that have smaller sequence numbers run before tests that have larger sequence numbers.

        > **Tip**
        >
        > You should leave a gap between sequence numbers. For example, set the **Sequence number** field to *10* for your first test, and then increment the value by 10 for each additional test. In this way, you can add new tests later without having to delete and re-create the lines to put them in the desired order.

    - **Test** – Select the test that you want to perform. For a quantitative test, select a test where the **Type** field is set to *Fraction* or *Integer*.

- **Effective** – Select the first date when the test is valid. If you leave this field blank, there's no limit.
- **Expiration** – Select the last date when the test is valid. If you leave this field blank or set it to *Never*, there's no limit.
- **Test value determination** – Select how an AQL should be determined when multiple results are recorded for the same test. The available options are *Average*, *Minimum*, *Maximum*, and *Manual*.
- **Attribute** – If you want to record test results on a batch attribute, select the attribute here, and select the **Update inventory batch attribute** check box.
- **Update inventory batch attribute** – Select this check box to record test results for the batch attribute that is selected in the **Attribute** field.

4. In the lower part of the page, select the **General** tab. Some of the settings for the test selected on the **Overview** tab are repeated here. In addition, you can set the following fields for the test:
    - **Certificate of analysis report** – Set this option to *Yes* to indicate that the test results should be printed on the CoA. This report can be generated from the quality order.
    - **Action on failure** – Select either *Accept* or *Fail* to indicate whether the test should pass or fail if the AQL for it isn't met.
    - **Acceptable quality level** – Enter the total percentage of test results that must pass for the test to be considered passed.
5. In the lower part of the page, on the **Test** tab, set the following fields:
    - **Standard** – Enter the amount (integer or fraction) that is expected for the test results. The value that you enter is entered by default in the test results. Additionally, it automatically enters as a default value in the **Min** and **Max** fields.
    - **Min** – Enter the minimum allowed value for the test results. The value that you enter must be less than the amount that is set in the **Standard** field. When you update the **Min** field, the **Min tolerance (%)** field is automatically updated. Likewise, when you update the **Min tolerance (%)** field, the **Min** field is automatically updated.
    - **Max** – Enter the maximum allowed value for the test results. The value that you enter must be more than the amount that is set in the **Standard** field. When you update the **Max** field, the **Max tolerance (%)** field is automatically updated. Likewise, when you update the **Max tolerance (%)** field, the **Max** field is automatically updated.
    - **Test instrument** – Select the device that should be used for the test. If a test instrument is defined, it's automatically entered for the test in the test group.

6. Close the page.

Related information

- Quality management test instruments
- Quality management tests
- Quality management test variables
- Quality associations
- Batch attributes
- Acceptance sampling

Item quality groups

A quality group represents common testing requirements for items. This article describes how to use and create item quality groups to logically group products so that they can be assigned to quality associations for the automatic generation of quality orders.

To set up, edit, and view the items that are assigned to a quality group, or the quality groups that are assigned to an item, go to **Inventory management** > **Setup** > **Quality groups**. After you define the test requirements on the **Test groups** page, you can define the rules for automatically generating quality orders. To simplify the process, you don't define rules for individual items. Instead, you can define the rules for a quality group on the **Quality associations** page.

Example of an item quality group

A manufacturing company purchases various raw materials that have the same testing requirements for incoming inspection. Therefore, the company defines a quality group and then assigns the item numbers that are associated with the raw materials to that group. Later, the company purchases a new type of raw material that has the same testing requirements. Instead of setting up new testing requirements for the new material, the company adds the item number for the new material to the existing quality group.

The same company also produces items that have the same production testing requirements and ships items that have the same requirement for pre-shipment testing. Therefore, the company defines two additional quality groups and then assigns the relevant item numbers to each group.

Create a quality group

To create a quality group, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Quality groups**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Quality group** – Enter a unique ID or name for the quality group.
    - **Description** – Enter a detailed description of the quality group.
3. Close the page.

Manually add items to a quality group

To manually add items to a quality group, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Quality groups**.
2. Select the quality group that you want to add items to.
3. On the Action Pane, select **Items**.
4. On the **Items in quality groups** page, on the Action Pane, select **New** to add a row to the grid. Then, for the new row, set the **Item number** field to the item number that you want to add.
5. Repeat the previous step for each additional item that must be added.
6. Close the pages.

Add multiple items to a quality group

To add multiple items to a quality group, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Quality groups**.
2. Create or select the quality group that you want to add items to.
3. On the Action Pane, select **Add items**.
4. In the **Inquiry** dialog box, enter the filter criteria for the items that you want to add. For example, you might filter for all items in a specific item group.
5. Select **OK**.
6. In the **Add items** dialog box, a grid shows all the items that match your query. Review the results.

   If any changes are required, select **Select** to return to the **Inquiry** dialog box, and adjust your query.
7. When the results include all the items that you want to add, select **OK**.
8. Close the page.

Manually associate an item with a quality group

To manually associate an item with a quality group, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Item quality groups**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Item number** – Select the item number to add.
    - **Quality group** – Select the quality group to assign to the item.
3. Repeat the previous step for each additional combination of an item and a quality group that must be added.
4. Close the pages.

Manually add a quality group from the Released products page

To manually add a quality group from the **Released products** page, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product that you want to assign a quality group to.
3. On the Action Pane, on the **Manage inventory** tab, in the **Quality** group, select **Item quality groups**.
4. On the **Items in quality groups** page, on the Action Pane, select **New** to add a row to the grid. Then, for the new row, set the **Quality group** field to the quality group that you want to assign to the product.
5. Repeat the previous step for each additional quality group that you want to assign to the product.
6. Close the pages.

Related information

- Quality management test instruments
- Quality management tests
- Quality management test groups
- Quality management test variables
- Quality associations

Last updated on 12/30/2025

Quality management item sampling

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Item sampling is used as part of a quality association. It defines the amount of current physical inventory that must be inspected. Sampling can be based on fixed quantities, a percentage, or a full license plate.

Set up item sampling

Follow these steps to set up item sampling.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Item sampling**.
2. Select **New**.
3. Make the following settings in the header of your new record:

    - **Item sampling** – Enter a name for the record.
    - **Description** – Enter a short description of the record.
    - **Sampling scope** – Select the scope to use when evaluating if quality work should be created. When set to *Shipment* or *Load*, those entities are used if available. If not, the *Order* is used.
    - **Use acceptance sampling charts** – Choose whether to use acceptance sampling with this record. Learn more in Acceptance sampling.

4. If you're using acceptance sampling, then fill out the settings provided on the **Acceptance sampling** FastTab. Learn more about how to use these settings in Acceptance sampling.
5. If you're using sample management, then fill out the settings provided on the **Sample management** FastTab to set up inline or continuous sampling. Learn more about how to use these settings in Enable and configure sample management.
6. On the **Sampling quantity** FastTab, in the **Value** field, enter a number. This value is related to the quantity specification value that is selected in the adjacent field.
7. On the **Process** FastTab, make the following settings:

- **Full blocking** – Set to *Yes* if the entire lot or order line quantity should be blocked when a test fails. Set this option to *No* if only the items in the quality order should be blocked when a test fails.
- For each dimension (such as **License plate** or **Batch number**) that should be automatically populated on the **Inventory dimensions** tab of a quality order, set it's option to *Yes*.

8. Select **Save**.
9. Close the page.

> ⓘ **Note**
>
> The *Quality management for warehouse processes* feature provides additional item sampling capabilities. It adds the concept of *item sampling scope* and lets you define a full license plate as the quantity specification. If you've turned on this feature, go to **Quality management for warehouse processes**.

Last updated on 03/11/2026

Quality associations

This article describes how you can use quality associations in Microsoft Dynamics 365 Supply Chain Management to automatically generate quality orders that are related to your sales, purchase, inbound shipment order, and production processes.

A quality association defines all the following information for a quality order that is generated:

- The transaction event
- The set of tests that must be performed on the items
- The acceptable quality level (AQL)
- The sampling plan

You must define a quality association for each variation in a business process that requires automatic generation of quality orders. For example, a quality order can be generated in the business processes for purchase orders, quarantine orders, sales orders, and production orders.

Prerequisites for sales returns and transfer orders (preview)

[This section is prerelease documentation and is subject to change.]

Most of the features that are described in this article are available as a standard part of all current versions of Supply Chain Management. However, if you want to set up quality associations for sales returns and transfer orders, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Important

- This is a preview feature.
- Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are

available before an official release so that customers can get early access and provide feedback.

Working with quality associations

The business process that uses a quality association can be related to various source documents, such as purchase orders, sales orders, or production orders.

Each quality association record defines the set of tests, the AQL, and the sampling plan that applies to the quality orders that are generated. You must define a quality association record for each variation in a business process. For example, you can set up a quality association that generates a quality order when a purchase order product receipt is updated. Depending on the setup of the execution plan, the triggering process itself can be blocked while there's an open quality order. Alternatively, subsequent processes, such as purchase order invoicing, can be blocked.

> **Note**
>
> While there are open quality orders, inventory quantities are automatically blocked from being issued. Depending on the setting of the **Full blocking** field on the **Item samplings** page, the quantity is either the quantity on the quality order or the quantity on the source document line. Learn more in Quality management item sampling.

For a given business process, the quality association record identifies the event and conditions that a quality order is generated for. The conditions can be specific to either a site or a legal entity. A quality order that involves destructive tests can be generated only when on-hand inventory exists for the event.

To work with quality associations, go to **Inventory management** > **Setup** > **Quality control** > **Quality associations**. The following examples show how a quality association record is defined for the variations in each business process. For each example, the following table summarizes the events and conditions that are defined by a quality association record.

> **Note**
>
> The *Quality management for warehouse processes* feature adds capabilities for quality order processing for production where the **Event type** field is set to *Report as finished* and the

Execution field is set to *After*, and for purchases where the **Event type** field is set to *Registration*. Learn more in Quality management for warehouse processes.

The following table provides more information about how quality orders can be generated for specific types of processes.

Expand table

Type of process	When quality orders can be automatically generated	When quality orders can be generated if destructive testing is required	Condition information	Manual generation information
Purchase order	Before or after a receipts list or product receipt is posted for the material that is received	After the product receipt is posted for the material that is received, because the material must be available for destructive testing	The requirement for a quality order can reflect a specific site, item, or vendor, or a combination of these conditions.	A manually generated quality order that refers to a purchase order can use information in a quality association record, such as the test sampling plan.
Inbound shipment order	As part of the registration process (warehouse receiving)	As part of the registration process (warehouse receiving)	The requirement for a quality order can reflect a specific site, item, or consigner, or a combination of these conditions.	A manually generated quality order that refers to an inbound shipment order can use information in a quality association record, such as the test sampling plan.
Quarantine order	Before or after the quarantine order is reported as finished or ended	Quality orders that require destructive tests can't be generated. The assumption is that the quarantine order functionality	The requirement for a quality order can reflect a specific site, item, or vendor, or a combination of these conditions.	A manually generated quality order that refers to a quarantine order can use information in a quality association

Type of process	When quality orders can be automatically generated	When quality orders can be generated if destructive testing is required	Condition information	Manual generation information
		handles the disposition of the material that is destroyed.		record, such as the test sampling plan.
Sales order	Before a scheduled picking process or packing slip update for the items that are being shipped	At any step	The requirement for a quality order can reflect a specific site, item, or customer, or a combination of these conditions.	A manually generated quality order that refers to a sales order can use information in a quality association record, such as the test sampling plan.
Production order	Before or after the finished quantity is reported for the production order	After the finished quantity is reported for the production order	The requirement for a quality order can reflect a specific site or item, or a combination of these conditions.	A manually generated quality order that refers to a production order can use information in a quality association record, such as the test sampling plan.
Production order that has a route operation	Before or after the report is finished for an operation	After the reporting production is finished for the last operation	The requirement for a quality order can reflect a specific site, item, or operations resource, or a combination of these conditions.	A manually generated quality order that refers to a route operation can use information in a quality association record, such as the test sampling plan.
Inventory	A quality order can't be			A quality order must be manually

Type of process	When quality orders can be automatically generated	When quality orders can be generated if destructive testing is required	Condition information	Manual generation information
	automatically generated for a transaction in an inventory journal or for transfer order transactions.			created for an item's inventory quantity. Physical on-hand inventory is required.
Sales return	A quality order is generated when the packing slip on the sales return order is posted.	After the product receipt is posted for the material that is received, because the material must be available for destructive testing.	Before the packing slip is posted on the return order, the quantity on the return line must be registered. To register the quantity, on the **Update line** menu, select **Registration**. Then, on the **Assign disposition code** dropdown menu, select one of the applicable disposition codes. The selected disposition code must be set up to search for an applicable quality association. To complete this setup, select the **Check for quality association** checkbox in the definition of the disposition code.	
Transfer order	A quality order can be generated when you do the picking, when you ship the product, or when you receive it.	After registration of the receipt of the material, because the material must be available for destructive testing.		

Note

When you filter quality associations for the *Inbound shipment order* reference type, and the **Account code** value is *Table* or *Group*, you must create consigners (for *Table*) or consigner groups (for *Group*) beforehand.

Set up a quality association

Follow these steps to create a quality association:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Quality associations**.
2. On the Action Pane, select **New**.
3. On the header, make the following settings:
    - **Reference type** – Select the business process for which quality orders are generated by the system, such as *Purchase* or *Production*.
    - **Item code** – Select which products should be included in the rule. Choose one of the following values:
        - *Table* – A specific product.
        - *Group* – A group of products.
        - *All* – All products.
    - **Item** – Depending on what you selected for **Item code**, choose the specific product or product group for which this rule should apply. Leave blank if **Item code** is set to *All*.
4. On the **Conditions** FastTab, make the following settings:
    - **Site** – Select the site where the rule applies.
    - **Applicable warehouse type** – Select the type of warehouse for which the quality association is used for. Choose one of the following values:
        - *All* – Includes all warehouses. Quality order functionality for warehouse processes isn't available for the quality association when **Applicable warehouse type** is set to *All*.
        - *Quality order for warehouse processes only* – Includes only warehouses where **Enable quality order for warehouse processes** is set to *Yes*.
    - **Account code** – Select the vendor or customer account to whom this rule applies. Available options depend on the selected **Reference type**.
    - **Account selection** – Depending on which **Account code** is selected, choose a specific account or account group. Leave this blank if **Account code** is set to *All*.

- **Resource code** – Select the scope of resources to which the rule applies. This option only applies if the **Reference type** is *Route operation*. Choose one of the following values:
  - *All* – Applies to all resources.
  - *Group* – Applies to a group of resources.
  - *Table* – Applies to a single, selected resource.
- **Resource** – Depending on what you selected for **Resource code**, choose the specific resource or resource group to which this rule should apply. Leave this blank if **Resource code** is set to *All*.

5. On the **Process** FastTab, make the following settings:

- **Event type** – Depending on the **Reference type**, select the business event that should generate the quality order. For example, if the **Reference type** is *Production* you can choose between *Report as finished* and *Registration*.
- **Execution** – Choose one of the following values:
  - *Before* – Automatically create the quality order before the business event occurs.
  - *After* – Automatically create the quality order after the business event occurs.
- **Show info** – Select whether an info-log should be shown after the quality order is automatically created based on the business event.

6. On the **Quality order process** FastTab, make the following settings:

- **Event blocking** – Choose whether the document should be set on hold until the associated quality order is validated and closed. Choose one of the following values:
  - *End* – Blocks processing of the document until the associated quality order is validated and closed.
  - *None* – Don't block the document from being processed.
- **Quarantine upon validation failure** – Set to *Yes* to automatically create a quarantine order if the associated quality order fails validation.
- **Use for certificate of analysis** – Choose one of the following values:
  - *No* – Don't make it possible to generate a certificate of analysis for this quality association.
  - *Yes* – Make it possible to generate a certificate of analysis for this quality association.
- **Quality processing policy** – Choose one of the following values:
  - *Create quality order* – Uses the full quality process and create quality orders.
  - *Create item sampling work only* – Uses a minimal quality process that creates quality item sampling work but doesn't create quality orders or inventory blocking. This

option only applies when the **Applicable warehouse type** is set to *Quality order for warehouse processes only*.

7. On the **Specifications** FastTab, make the following settings:

- **Test group** – Select a test group with tests defined for acceptance sampling or sample management. Learn more in Acceptance sampling and Enable and configure sample management.
- **Item sampling** – Select an item sampling defined for acceptance sampling or sample management. Learn more in Acceptance sampling and Enable and configure sample management.
- **Acceptable quality level** – Choose the acceptable quality level (AQL) to be used for the quality association. The AQL determines whether the quality order passes or fails validation.
- **Flexible sampling** – Choose whether to use the flexible sampling plans as part of this quality association.
- **Flexible sampling plan code** – If **Flexible sampling** is set to *Yes*, then select a flexible sampling plan code.
- **Effective** – Select the date on which this quality association becomes effective.
- **Expiration** – Select the date on which the quality association expires.
- **Worker** – Select the default worker to be assigned to quality orders generated by this quality association.
- **Quality order priority** – Select the default priority that should be assigned by default to the quality order generated for this quality association.
- **Description** – Displays the description of the selected quality order priority.

Quality associations and flexible sampling plans

Instead of selecting a fixed test group and item sampling, you can set up a *flexible sampling plan*. A flexible sampling plan is a test strategy where you can change both the test and sample sizes over time, based on successful test validation. To use flexible sampling, on the **Specification** FastTab, select **Flexible sampling**. Then, in the **Flexible sampling plan code** field, select the code for a flexible sampling plan. Learn more in Flexible sampling plans.

Examples of automatic generation of quality orders

Purchasing

In purchasing, if you set the **Event type** field to *Product receipt* and the **Execution** field to *After* on the **Quality associations** page, you get the following results:

- If the **Per updated quantity** option is set to *Yes*, a quality order is generated for every receipt against the purchase order, based on the received quantity and settings in the item sampling. Every time that a quantity is received against the purchase order, new quality orders are generated based on the newly received quantity.
- If the **Per updated quantity** option is set to *No*, a quality order is generated for the first receipt against the purchase order, based on the received quantity. Additionally, one or more quality orders are created based on the remaining quantity, depending on the tracking dimensions. Quality orders aren't generated for subsequent receipts against the purchase order.

Production

In production, if you set the **Event type** field to *Report as finished* and the **Execution** field to *After* on the **Quality associations** page, you get the following results:

- If the **Per updated quantity** option is set to *Yes*, a quality order is generated based on every finished quantity and settings in the item sampling. Every time that a quantity is reported as finished against the production order, new quality orders are generated based on the newly finished quantity. This generation logic is consistent with purchasing.
- If the **Per updated quantity** option is set to *No*, a quality order is generated the first time that a quantity is reported as finished, based on the finished quantity. Additionally, one or more quality orders are created based on the remaining quantity, depending on the tracking dimensions of the item sampling. Quality orders aren't generated for subsequent finished quantities.

Expand table

Quality specification	Per updated quantity	Per tracking dimension	Result
Percentage: 10%	Yes	Batch number: No Serial number: No	Order quantity: 100 1. Report as finished for 30 • Quality order 1 for 3 (10% of 30) 2. Report as finished for 70

Quality specification	Per updated quantity	Per tracking dimension	Result
			• Quality order 2 for 7 (10% of the remaining order quantity, which is 70 in this case)
Fixed quantity: 1	No	Batch number: No Serial number: No	Order quantity: 100 1. Report as finished for 30 • Quality order 1 for 1 (for the first reported-as-finished quantity, which has a fixed value of 1) • Quality order 2 for 1 (for the remaining quantity, which still has a fixed value of 1) 2. Report as finished for 10 • No quality orders are created. 3. Report as finished for 60 • No quality orders are created.
Fixed quantity: 1	Yes	Batch number: Yes Serial number: Yes	Order quantity: 10 1. Report as finished for 3: 1 for batch number b1, serial number s1; 1 for batch number b2, serial number s2; and 1 for batch number b3, serial number s3 • Quality order 1 for 1 of batch number b1, serial number s1 • Quality order 2 for 1 of batch number b2, serial number s2 • Quality order 3 for 1 of batch number b3, serial number s3 2. Report as finished for 2: 1 for batch number b4, serial number s4; and 1 for batch number b5, serial number s5 • Quality order 4 for 1 of batch number b4, serial number s4 • Quality order 5 for 1 of batch number b5, serial number s5 **Note:** The batch can be reused.
Fixed quantity: 2	No	Batch number: Yes	Order quantity: 10 1. Report as finished for 4: 1 for batch number b1, serial number s1; 1 for batch number b2, serial number s2; 1

Quality specification	Per updated quantity	Per tracking dimension	Result
		Serial number: Yes	for batch number b3, serial number s3; and 1 for batch number b4, serial number s4 • Quality order 1 for 1 of batch number b1, serial number s1 • Quality order 2 for 1 of batch number b2, serial number s2 • Quality order 3 for 1 of batch number b3, serial number s3 • Quality order 4 for 1 of batch number b4, serial number s4 • Quality order 5 for 2, without a reference to a batch number and a serial number 2. Report as finished for 6: 1 for batch number b5, serial number s5; 1 for batch number b6, serial number s6; 1 for batch number b7, serial number s7; 1 for batch number b8, serial number s8; 1 for batch number b9, serial number s9; and 1 for batch number b10, serial number s10 • No quality orders are created.

Related information

- Quality management processes
- Enable quality and nonconformance management
- Quality management for warehouse processes

Last updated on 07/27/2026

Workers responsible for approving nonconformances

Article • 06/03/2022

This article describes how to configure workers that are responsible for approving nonconformances.

Nonconformances must be approved before users can start to enter details such as corrections or operations. Before users can approve or reject nonconformances, their user ID (user record) must be linked to a worker record. You can optionally configure workers that are responsible for quality and then allow one worker to approve work on behalf of another worker.

Enable a user for nonconformance processing

Before a user can approve or reject nonconformances, you must link their user record to a worker record. The approval fields can then be automatically set, and the current user can be automatically filled in for the timesheet. Before the user can use the document notes, you must activate document handling in their user options.

1. Go to **System administration** > **Users** > **Users**.
2. Find and open the user who should be able to approve or reject nonconformances.
3. Set the **Person** field to the name of the worker that is related to the current user record.
4. On the Action Pane, select **User options**.
5. On the **Options** page for the current user record, on the **Preferences** tab, set the **Enable document handling** option to *Yes*.
6. Close the pages.

Define workers that are responsible for quality

1. Go to **Inventory management** > **Setup** > **Quality control** > **Workers responsible for quality**.
2. On the Action Pane, select **New**.
3. In the **Worker** field, select the worker that enters quality data.
4. In the **Worker responsible** field, select the worker that the selected worker enters work on behalf of. When nonconformances are created and updated, this worker will be entered by default in **Worker** fields.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Quality charges
- Quarantine zones for nonconformances
- Diagnostic types for nonconformances
- Quality charges for nonconformances
- Operations for nonconformances
- Quality management for warehouse processes

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Problem types for nonconformances

Article • 06/03/2022

This article describes how to create and use problem types for nonconformances.

You use the **Problem types** page to define a classification of the quality problems that might occur for the various nonconformance types. For each problem type that you create, you must specify the types of nonconformances that the problem type can be used with. You can set up the following nonconformance types:

- **Internal** – Nonconformances of this type are related to on-hand inventory (for example, quality orders or quarantine orders).
- **Customer** – Nonconformances of this type are related to sales orders.
- **Vendor** – Nonconformances of this type are related to purchase orders.
- **Service request** – Nonconformances of this type are related to service orders.
- **Production** – Nonconformances of this type are related to batch orders or production orders.
- **Co-product production** – Nonconformances of this type are related to batch orders for co-products.

When you create a new problem type, select **Non conformance types** on the Action Pane to create a list of one or more nonconformance types that are authorized for the problem type. For example, a problem type that is related to a defect code might apply to all nonconformance types. However, a problem type that is related to customer complaints might apply only to the **Customer** and **Service request** nonconformance types.

Examples of problem types

Here are some examples of scenarios for problem types that can be used with the different nonconformance types:

- **Customer:** A customer filed a complaint, a customer made a return, or quality specifications weren't met.
- **Vendor:** The product was damaged, quality specifications weren't met, or the wrong product was received.
- **Service request:** Quality specifications weren't met, a customer filed a complaint, or machine maintenance is required.
- **Production:** Quality specifications weren't met, machine maintenance is required, or the product was damaged.

- **Co-product production:** Quality specifications weren't met, machine maintenance is required, or the product was damaged.

Create a problem type and assign it to nonconformance types

1. Go to **Inventory management** > **Setup** > **Quality management** > **Problem types**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Problem type** – Enter a unique ID or name for the problem type.
    - **Description** – Enter a detailed description of the problem type.
3. While the new row is still selected, select **Non conformance types** on the Action Pane.
4. On the Action Pane, select **New** to add a row to the grid. Then, for the new row, set the **Non conformance type** field to the nonconformance type that you want to allow for the problem type.
5. Repeat the previous step for each additional nonconformance type that you want to authorize for the problem type.
6. Close the pages.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Workers responsible for approving nonconformances
- Quarantine zones for nonconformances
- Diagnostic types for nonconformances
- Quality charges for nonconformances
- Operations for nonconformances
- Quality management for warehouse processes

Feedback

Was this page helpful?

Yes No

Provide product feedback

Quarantine zones for nonconformances

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to create and use quarantine zones for nonconformances.

You use the **Quarantine zones** page to define zones that can be assigned to nonconformances. When you create a nonconformance, you can set the **Quarantine zone** and **Quarantine type** fields on the **General** tab of the **Non conformances** page. The **Quarantine zone** field typically indicates the area or location where the item is located. The **Quarantine type** field defines the item as either *Restricted usage* or *Unusable*.

When you print a nonconformance or corrections report for a nonconformance, you can view the quarantine zone where the item is located.

You can also print a nonconformance tag for a nonconformance. This report shows the assigned quarantine zone and information about usage to provide guidance about how defective material should be handled. The quarantine zones might correspond to inventory locations or operations resources.

Examples of quarantine zones

Example 1

You work at an electronics manufacturing company that produces and distributes televisions, speakers, and media players. In this case, you can configure a quarantine zone to represent each type of product.

Example 2

Three bins and two racks are used to store items that are nonconforming. In this case, you can configure five quarantine zones, one for each bin and each rack.

Create a quarantine zone

1. Go to **Inventory management** > **Setup** > **Quality management** > **Quarantine zones**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Quarantine zone** – Enter a unique ID or name for the quarantine zone.
    - **Description** – Enter a detailed description of the quarantine zone.
3. Close the page.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Workers responsible for approving nonconformances
- Problem types for nonconformances
- Diagnostic types for nonconformances
- Quality charges for nonconformances
- Operations for nonconformances
- Quality management for warehouse processes

Last updated on 05/07/2025

Diagnostic types for nonconformances

This article describes how to use and create diagnostic types that can be used with nonconformances.

Diagnostic types define classifications of diagnostic actions. When you create a *correction* for a nonconformance, you select a diagnostic. A correction specifies what type of diagnostic action should be taken for an approved nonconformance and who should take that action. It also specifies the requested completion date and the planned completion date.

Examples of diagnostic types

You work for a manufacturing company, and several items have failed quality tests. Those items are moved into a quarantine area and marked as nonconforming products. A nonconformance record is created in Microsoft Dynamics 365 Supply Chain Management to track the details through issue resolution.

In this case, the quality issues are occurring because bearings in the machine that packages the items have gone bad and are overheating. The correction for this problem is to replace the parts in the machine.

When you configure the diagnostic types, you can create multiple records, each of which represents a different type of problem that the machine might have. Alternatively, you can create one generic diagnostic type that represents machine repair.

Create a diagnostic type

1. Go to **Inventory management** > **Setup** > **Quality management** > **Diagnostic types**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Diagnostic** – Enter a unique ID or name for the diagnostic type.
    - **Description** – Enter a detailed description of the diagnostic type.
3. Close the page.

Diagnostic groups

Use *diagnostic groups* to collect related diagnostic types so that you can quickly apply them all at once as corrections to a nonconformance. For example, you might create a group of diagnostic types that together describe how to diagnose an issue with raw material quality (such as repair and retest).

Prerequisites

Before you can use diagnostic groups, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Manage diagnostic groups

1. Go to **Inventory management** > **Setup** > **Quality management** > **Diagnostic groups**.
2. Use the buttons on the Action Pane to create a new diagnostic group in the list pane or edit an existing one. (You can also delete existing groups.)
3. On the heading of the new or selected group, enter a name and description for the group.
4. On the **Diagnostic** FastTab, use the buttons on the toolbar to add and remove diagnostics that belong to the current group. For each row, select the diagnostic that should be performed and the worker who should perform it.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Workers responsible for approving nonconformances
- Quarantine zones for nonconformances
- Problem types for nonconformances
- Quality charges for nonconformances
- Operations for nonconformances
- Quality management for warehouse processes

Last updated on 07/27/2026

Nonconformance root cause codes

Use *nonconformance root cause codes* to establish standard codes for root causes that are associated with corrections for nonconformance issues that are often encountered at your company. For example, you might create codes for wind damage, water damage, training issues, or other common issues that you encounter. Root cause code give you a consistent way to identify causes that you can apply to the relevant corrective actions.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Manage root cause codes

1. Go to **Inventory management** > **Setup** > **Quality management** > **Correction root cause codes**.

2. Use the buttons on the Action Pane to add new root cause codes or edit existing ones. (You can also delete existing codes.)

3. Set the following fields for each row:

    - **Root cause code** – Enter a unique ID or name for the root cause code.
    - **Description** – Enter a short description of the root cause code.

Assign a root cause code to a nonconformance correction

You can assign a root cause code to each correction that you create for a nonconformance. Learn more in Create and process nonconformances.

Last updated on 07/23/2026

Nonconformance work groups

Use *nonconformance work groups* to define group names that you can assign to nonconformance records to make your nonconformances easier to classify. For example, you might create a group for inspecting receipts and another one for inspecting production output.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Manage nonconformance work groups

1. Go to **Inventory management** > **Setup** > **Quality management** > **Non conformance work groups**.

2. Use the buttons on the Action Pane to create new nonconformance work groups in the grid or edit existing ones. (You can also delete existing work groups.)

3. Set the following fields for each row:

    - **Work group** – Enter a unique ID or name for the work group.
    - **Description** – Enter a short description of the work group.

Assign a work group to a nonconformance

To assign a work group to a nonconformance, edit the **Work group** field when you create or edit a nonconformance.

Last updated on 07/23/2026

Operations for nonconformances

This article describes how to create and use operations for nonconformances.

Operations are classifications of work that can be performed for an approved nonconformance. When you assign operations to a nonconformance, you can provide details such as the associated material, labor hours, and charges that are required to do the operation. The system uses this information to calculate an estimated cost for the operation. The detailed information and estimated costs are for reference. The related operations for quality differ from the operations that can be defined for a production route.

> **Note**
>
> Although you can track costs, time, and the items that are used in an operation that is related to a nonconformance, the data that you enter is only informational. It isn't automatically integrated with the general ledger, inventory subledger, or the **Time and attendance** module.

Examples of operations

You work for a manufacturing company. A nonconformance was created and approved for items that failed a quality test. A correction was created to indicate that the problem was related to a bad bearing on a machine. Several steps are required to replace the bearing, and the responsibility for each operation is tracked. For example, the following steps might be required:

1. The production line is stopped, and the clean-out routine is performed.
2. Maintenance personnel replace the bearing.
3. Quality assurance personnel inspect the machine before it's turned back on.

For this example, the following operations can be created to represent the work that must be done:

- Stop the production line.
- Clean out the production line.
- Perform machine maintenance.
- Inspect the machine.

Create an operation

1. Go to **Inventory management** > **Setup** > **Quality management** > **Operations**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Operation** – Enter a unique ID or name for the operation.
    - **Description** – Enter a detailed description of the operation.
    - **Type** – If the operation can be used only with nonconformances that are related to a specific type of order, select the order type (*Purchase order* or *Sales order*).
3. Close the page.

Nonconformance operation groups

Use *nonconformance operation groups* to collect related operations that you can apply quickly to a nonconformance all at once. For example, you might create a group of operations that together describe the steps that your company takes to fix issues with purchased raw materials (research, repair, and retest).

Prerequisites

Before you can use nonconformance operation groups, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Manage nonconformance operation groups

1. Go to **Inventory management** > **Setup** > *Quality management* > *Non conformance operation groups*.
2. Use the buttons on the Action Pane to create new nonconformance operation groups in the list pane or edit existing ones. (You can also delete existing groups.)
3. On the heading of a new or selected group, enter a name and description for the group.

4. On the **Operations** FastTab, use the buttons on the toolbar to add and remove operations that belong to the current group.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Create and process nonconformances
- Workers responsible for approving nonconformances
- Quarantine zones for nonconformances
- Diagnostic types for nonconformances
- Quality charges for nonconformances
- Problem types for nonconformances
- Quality management for warehouse processes

Last updated on 07/27/2026

Charges for nonconformance operations

Article • 06/03/2022

This article describes how to create quality charges that can be used with operations for a nonconformance.

You use the **Quality charges** page to define the types of charges that can be added to operations for a nonconformance. Charges let you track details about fees or charges that you owe to a customer for nonconforming products, or that a vendor owes to you for nonconforming products that you received. You might also use charges to track costs that are required for external vendors or services to perform an operation.

Examples of quality charges

You work for a manufacturing company. Your company has contracts with several vendors. Those contracts outline standards for on-time shipment, damages, and product quality for items. Likewise, several of your customers have agreements with your company about returns, damages, and product quality.

A fee structure is defined for each circumstance and specified in the contract. You can configure quality charges to outline the various types of charges, such as *Damages*, *Late shipment*, and *Quality*. Then, when you create a nonconformance, you add one or more operations. For each operation, you select **Charges** to define the details about the fees.

Create a quality charge

1. Go to **Inventory management** > **Setup** > **Quality management** > **Quality charges**.
2. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:
    - **Charges code** – Enter a unique ID or name for the quality charge.
    - **Description** – Enter a detailed description of the quality charge.
3. Close the page.

Add a quality charge to an operation for a nonconformance

For details about how to add operations to a nonconformance and apply charges to them, see Operations for nonconformances.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Workers responsible for approving nonconformances
- Quarantine zones for nonconformances
- Diagnostic types for nonconformances
- Quality charges for nonconformances
- Operations for nonconformances
- Quality management for warehouse processes

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Sample management overview

Summarize this article for me

Sample management enables manufacturers to systematically handle, track, and test product samples throughout their lifecycle. This functionality supports industries such as pharmaceuticals, food and beverage, chemicals, and life sciences, where quality assurance and regulatory compliance are critical.

Sample management provides tools to:

- Define and register samples.
- Execute sampling and testing plans.
- Track sample lifecycle states.
- Analyze test results.
- Control inventory disposition based on quality outcomes.

The following video provides a high-level overview of sample management capabilities and a demo of how to work with them.

https://learn-video.azurefd.net/vod/player?id=36124ac5-86c0-44f6-a57e-3beb10c753ca&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Finventory%2Fquality-sample-management-overview

Why sample management matters

In regulated and quality-sensitive industries, manufacturers must ensure that products meet strict specifications before release. Sampling is a key part of this process. It allows organizations to test small quantities of product at various stages of production to detect issues early and ensure consistency.

Sample management helps organizations:

- Maintain traceability and auditability of samples.
- Automate sample registration and testing.
- Integrate quality control with production workflows.
- Comply with standards like GMP, GLP, ISO, FDA, and EMA.

Sample inspection methods

Sample management supports two primary sample inspection methods:

- **Inline process** – Manually collect samples during production, before the item is reported as finished. This method detects quality issues early and ensures stable quality before the finished product. Here are the main characteristics for inline sampling:
  - *When used* – During production, before inventory is reported as finished.
  - *Purpose* – Detect and correct quality issues early (such as pH imbalance or viscosity).
  - *How it works* – Manually register samples on a production or batch order. Each sample generates a quality order. Track quality test results over time to monitor trends.
  - *Inventory impact* – No inventory transactions are associated with inline samples.

- **Continuous process** – Automatically take samples at regular intervals throughout batch production to monitor quality trends and maintain consistency in product quality during a production run. Here are the main characteristics for continuous sampling:
  - *When used* – During batch production, triggered automatically when items are reported as finished.
  - *Purpose* – Monitor quality trends and control inventory release.
  - *How it works* – Register samples at defined intervals, such as every two license plates. Test a subset of samples. Based on the test results, release or block the inventory by using batch disposition codes.
  - *Inventory impact* – Inventory transactions are associated with continuous samples.

Each sample type is configured with its own lifecycle states, label layouts, and procedures. These configurations ensure that you handle samples consistently and according to business rules.

Related quality management tools

Sample management both leverages and enhances existing quality management entities and capabilities within Supply Chain Management, while also introducing new tools and features to support advanced quality processes.

Existing entities and capabilities

Sample management builds on the following existing quality management entities and capabilities:

- **Quality orders** – Automatically or manually created for each sample to record test results.
- **Item sampling** – Enhanced to support sample registration and testing plans.
- **Test groups** – Define which tests are performed during quality orders and control how inventory is updated for samples, based on whether tests pass or fail.
- **Quality associations** – Link sampling logic to the report-as-finished production event.

- **Batch disposition codes** – Used to block or release produced inventory batches based on test outcomes.
- **Inventory status** – Used to block or release produced license plates based on test outcomes.

New entities and capabilities

Sample management introduces the following new entities and capabilities:

- **Sample label layouts** – Enable flexible design of sample label content to accommodate specific business requirements.
- **Sample procedures** – Define the specific steps or actions required for handling samples.
- **Sample life cycle states** – Represent the various stages a sample goes through during its lifecycle.
- **Sample types** – Define the classification and processing methods for samples.
- **Sample associations** – Define how and when quality samples are taken for a specific item.
- **Sample procedures** – Define the specific steps or actions required for handling samples.
- **Sample procedure types** – Define categories and groups for organizing sample procedures.
- **Sample management workbench** – A centralized interface for managing samples, viewing audit trails, and performing lifecycle actions.
- **Audit trail** – Tracks all sample-related events including registration, testing, and inventory updates.

Last updated on 03/11/2026

Enable and configure sample management

This article provides step-by-step instructions for enabling and configuring sample management. It covers key aspects such as designing sample label layouts, setting up parameters and number sequences, defining lifecycle states, and configuring quality associations. By following these instructions, manufacturing managers and system administrators can ensure that they manage samples efficiently, accurately, and in compliance with industry standards.

Prerequisites

To use sample management, your system must meet the following requirements:

- To run the preview version of this feature, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- To run the general availability version of this feature, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.47 or later.
- The following features must be turned on in feature management:
  - *Advanced quality management* (As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.)
  - *Sample management* (As of Supply Chain Management version 10.0.49, this feature is turned on by default.)

Set up the sample ID number sequence

The sample ID number sequence ensures that each sample has a unique identifier for traceability and compliance. As with other number sequences in Supply Chain Management, you can define the format and scope of the sample ID number sequence to meet your organization's requirements.

Follow these steps to create and assign a number sequence for sample IDs:

1. Go to **Organization administration** > **Number sequences** > **Number sequences**.
2. Create and configure a number sequence for generating sample IDs, as described in Set up number sequences on an individual basis. The number sequence must be defined to use a non-shared scope.
3. Note the **Number sequence code** that you assign to this number sequence.

4. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
5. Open the **Number sequences** tab.
6. Find the row with a **Reference** of *Sample ID* and set the **Number sequence code** field for that row to the number sequence that you created previously in this procedure.
7. On the Action Pane, select **Save**.

Set up sample labels

Sample labels play an important role in the sample management process. They make sure that each sample is properly identified and traceable throughout its lifecycle. A well-designed label layout includes essential information such as the sample ID, type, and other relevant details. This information helps with efficient handling and reduces the risk of errors.

Sample labels are designed and printed in Zebra Programming Language (ZPL) format, which is a standard format for printing labels on Zebra printers. Sample labels can include information such as the sample ID, sample type, lot ID, and more.

Design sample label layouts

To design and configure sample label layouts, follow these steps:

1. Go to **Inventory management** > **Setup** > **Sample management** > **Sample label layouts**.

2. Use the buttons on the Action Pane to create a new calibration label layout or edit an existing one as required. You can also delete existing layouts.

3. On the header of the new or selected layout, enter details such as the name, description, and dimensions.

4. On the **Label layout** FastTab, use the large field to design your label layout. For help, consult a ZPL reference.

5. The page helps you find the codes that you need to insert field values. To add a field value to your layout, follow these steps:
   a. Set the **Use fields from** field to the type of database record that you want to use a field from (such as *Samples*).
   b. In the **Labels** list, select a field name.
   c. Select **Insert at end of text** to add the reference code to the end of the label layout text.

d. Use a cut-and-paste operation to move the reference code to the desired location in the label layout text.

Assign which labels to use when printing

Make one or both of the following assignments to specify which label layout to use for printing sample labels:

- Assign the default label layout, which applies to all samples that don't have a more specific layout assigned to them.

  To do this, go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**. Open the **Quality management** tab, and then set the **Default sample label layout** field to the desired label layout.

- Assign a custom label layout to each specific sample type as needed (overrides the default).

  To do this, go to **Inventory management** > **Setup** > **Sample management** > **Sample types**. Select a sample type, and then set the **Label layout** field to the desired label layout.

Configure lifecycle states

Lifecycle states represent the various stages a sample goes through during its lifecycle. Lifecycle states ensure that you track samples accurately and update their status consistently as they progress through the management process.

Follow these steps to manage your lifecycle states:

1. Go to **Inventory management** > **Setup** > **Sample management** > **Sample lifecycle states**.
2. Use the buttons on the Action Pane to create a new lifecycle state or edit an existing one as required. You can also delete existing lifecycle states.
3. Enter the following details for your new or selected lifecycle state:
   - **Sample lifecycle state** – Enter a unique name for the lifecycle state, such as *Registered*, *Tested*, *Archived*, or *Scrapped*.
   - **Sample lifecycle state description** – Add a brief description of the lifecycle state.
   - **Quality order created** – Select this checkbox for the lifecycle state that a sample should be updated to when you create a quality order for it.

- **Quality order validated** – Select this checkbox for the lifecycle state that a sample should be updated to when you validate a quality order against it.
- **Sample scrapped** – Select this checkbox for the lifecycle state that, when you assign it manually, triggers an audit trail logging for *Scrapped by* and *Scrapped date/time*.

  Only samples with this property selected can be deleted from the **Sample management workbench**.

4. On the Action Pane, select **Save**.

Configure sample procedure types

Sample procedure types define categories and groups for organizing procedures. These configurations ensure that you process samples consistently and in compliance with organizational and regulatory requirements.

Follow these steps to manage your procedure types:

1. Go to **Inventory management** > **Setup** > **Sample management** > **Sample procedure types**.
2. Use the buttons on the Action Pane to create a new procedure type or edit an existing one as required. You can also delete existing procedure types.
3. Enter the following details for your new or selected procedure type:
   - **Sample procedure type** – Provide a unique name for the procedure type.
   - **Description** – Add a brief description of the procedure type.
4. On the Action Pane, select **Save**.

Define sample procedures

Sample procedures define the specific steps or actions required for handling samples. These configurations ensure that you process samples consistently and in compliance with organizational and regulatory requirements.

Follow these steps to manage your sample procedures:

1. Go to **Inventory management** > **Setup** > **Sample management** > **Sample procedures**.
2. Use the buttons on the Action Pane to create a new procedure or edit an existing one as required. You can also delete existing procedures.

3. Enter the following details in the header of your new or selected procedure:

    - **Sample procedure ID** – Provide a unique name for the procedure.
    - **Description** – Add a brief description of the procedure.
    - **Sample procedure type** – Select the relevant procedure type for the procedure.
    - **Sample inspections method** – Choose which inspection methods the procedure is intended for (*Inline process*, *Continuous process*, or *Both*).

4. On the **Procedure** FastTab, enter or paste detailed instructions for the procedure.

5. On the Action Pane, select **Save**.

Define sample types

Sample types define the classification and processing methods for samples. When you configure sample types, you ensure that samples are managed according to their specific requirements and that appropriate workflows are applied.

To manage your sample types, follow these steps:

1. Go to **Inventory management** > **Setup** > **Sample management** > **Sample types**.

2. Use the buttons on the Action Pane to create a new sample type or edit an existing one as required. You can also delete existing sample types.

3. Enter the following details in the header of your new or selected sample type:

    - **Sample type** – Enter a unique name for the sample type, such as *Inline* or *Continuous*.
    - **Description** – Add a brief description of the sample type.
    - **Sample inspections method** – Choose which inspection methods the sample type applies to (*Inline process*, *Continuous process*, or *Both*).
    - **Label layout** – Select the label layout to use for this sample type.
    - **Description** – Shows the description configured for your selected label layout (read only).

4. On the Action Pane, select **Save**.

5. Use the **Sample lifecycle states** FastTab to select the lifecycle states that are relevant to this sample type. Use the buttons between the columns to move the relevant states between the **Available** and **Selected** lists as needed. Selecting lifecycle states ensures that the sample type follows a defined progression through its lifecycle, from creation to completion or disposal. These states trigger specific system behaviors, such as updating the sample when

a quality order is created or validated, or logging audit trail details when a sample is scrapped. By configuring the relevant states, you enable consistent handling, tracking, and automation of sample-related processes.

6. On the Action Pane, select **Save**.

Set up sample associations

A sample association defines how and when quality samples are taken for a specific item. It supports both inline and continuous sampling methods, using designated sample types with configurable retention periods and automated expiration tracking. Linking sampling procedures to items and test groups ensures consistent quality control, supports regulatory compliance, and enables efficient inspection workflows tailored to operational needs.

Follow these steps to define and configure sample associations:

1. Go to **Inventory management** > **Setup** > **Sample management** > **Sample associations**.

2. Use the buttons on the Action Pane to create a new sample association or edit an existing one as required. You can also delete existing sample associations.

3. Enter the following details in the top section to set up your new or selected sample association:

    - **Item code** – Select the scope of products to include in the rule. Choose one of the following values:
        - *Table* – A specific product.
        - *Group* – A group of products.
        - *All* – All products.
    - **Item** – Depending on what you selected for **Item code**, choose the specific product or product group for which this rule should apply. Leave blank if **Item code** is set to *All*.
    - **Inline sample type** – Select the sample type that defines how inline samples for this item should be inspected, including their applicable lifecycle states. This sample type governs the inspection method used for inline sampling and ensures that the correct procedures and validations are applied throughout the sample's lifecycle.
    - **Inline sample scrap in days** – Specify how many days inline samples should be retained before being automatically scrapped. This setting helps manage sample lifecycle and storage by defining the retention period for inline samples associated with the item.

- **Default item sampling for inline** – Specifies the default item sampling policy applied to inline samples during production. Item sampling policies define the criteria for selecting samples. When initiating inline sampling from the production orders list or details page, this value is prefilled in the create dialog but can be adjusted as needed.
- **Default inline test group** – Specify the default test group used for inline samples from production. When initiating inline sampling from the production orders list or details page, this value is prefilled in the create dialog but can be adjusted as needed.
- **Continuous sample type** – Select the sample type that defines how continuous samples for this item should be inspected, including their applicable lifecycle states. This sample type governs the inspection method used for continuous sampling and ensures that the correct procedures and validations are applied throughout the sample's lifecycle.
- **Continuous sample scrap in days** – Select the number of days that continuous samples for this item should be saved before being scrapped.

4. In the bottom section, with the relevant row still selected in the top section, add each of the sample procedures that workers should follow when handling samples for this association. Use the toolbar buttons to add or remove procedures as needed.

5. On the Action Pane, select **Save**.

Configure item sampling policies

Item sampling policies define the criteria for selecting samples. Configuring these settings ensures that you select samples consistently and in compliance with quality standards.

Follow these steps to define an item sampling policy:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Item sampling**.

2. Use the buttons on the Action Pane to create a new item sampling policy or edit an existing one as required. You can also delete existing policies.

3. Set up the item sampling policy as described in Quality management item sampling.

4. On the **Sample management** FastTab, set up the settings that are specific for sample management. Make the following settings here.

    - To set up inline sampling, make the following settings on the **Sample management** FastTab:
        - **Sample inspections method** – Select *Inline process*.

    - **Sample size** – Enter the desired sample size.
    - **Unit** – Select the unit of measurement for the sample size (such as *Pieces* or *Milliliters*).

- To set up continuous sampling, make the following settings on the **Sample management** FastTab:
    - **Sample inspections method** – Select *Continuous process*.
    - **Sample size** – Enter the desired sample size.
    - **Unit** – Select the unit of measurement for the sample size (such as *Pieces* or *Milliliters*).
    - **One sample every** – Specify how often samples should be taken and then choose whether to count batches or license plates (for example, enter *10* and select *Batch* to take a sample from every tenth batch that's reported as finished).
    - **Number of samples per quality order generated** – Specify how often samples should be tested (for example, enter *4* to test every fourth sample).

5. On the Action Pane, select **Save**.

Configure related quality management records

The following subsections describe other important settings for sample management. These records and settings are shared by other kinds of quality management processes supported by Supply Chain Management, but they are also essential for sample management.

Quality associations

Configure quality associations to trigger sample management when production orders are reported as finished for specific items. This setup references test groups and item sampling policies to ensure consistency and automation in quality control. For inline samples, you don't need a quality association because you collect these samples on demand directly from the production order. Learn more in Quality associations. The following settings for quality associations are particularly relevant for sample management:

- **Reference type** – Select *Production*, which is the only supported type for sample management.
- **Event type** – Select *Report as finished*.
- **Execution** – Select *After*.
- **Test group** – Select the test group that should be used for quality orders generated for samples.

- **Item sampling** – Select the item sampling configuration used for sample management. This configuration determines whether the *Inline* or *Continuous* sampling method is applied, along with related details for how each method is used.

Configure default item status

The default item status establishes the initial inventory status assigned to new items in inventory. This is relevant when samples are configured to be generated for each license plate produced. Typically, you should define the default inventory status as *On-hold*, which indicates that the license plate is awaiting inspection. You can configure the default item status to apply for an individual item or for all transactions at a specific warehouse.

To configure the default item status for a specific item, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Inventory** > **Default item status**.
2. On the Action Pane, select **New** to create a new record.
3. Enter the following details for your new record:

    - **Item number** – Select the product you want to configure for sample management.
    - **Module** – Select *Inventory*, which is the only applicable value for sample management.
    - **Default inventory status ID** – From the list, select the inventory status that should be used as default when a sample with an associated license plate is created in sample management.

To configure the default item status that is applicable for all item transactions at a specific warehouse, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Inventory breakdown** > **Warehouses**.
2. Find and select the warehouse where you want to configure the default item status.
3. On the **Warehouse** FastTab, set the default inventory status in the **Default inventory status ID** field.

Test groups

Test groups define the set of tests applied while processing quality orders. On each test group, you also define the status the system should apply to license plates or product batches based on the outcome of the quality order test results.

If you configured sample management to generate samples per license plate, then the **Inventory status** is used to control the status of the license plate. To control the initial status of the license

plates generated during production report as finished, set the following fields on the relevant test group:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**.
2. Select an existing test group or create a new one.
3. Select the **General** tab and make the following settings:

    - **Update inventory status** – Set to *Yes* to enable inventory status updates based on test results.
    - **Failed quality order status** – Specify the inventory status to apply to the license plate when a quality order fails. This setting only applies when **Update inventory status** is set to *Yes*.
    - **Passed quality order status** – Specify the inventory status to apply to the license plate when a quality order passes. This setting only applies when **Update inventory status** is set to *Yes*.

If you configured sample management to generate samples per product batch number, then the batch disposition master is used to control the status of the individual batch number. To control the initial status on batch numbers generated during report as finished, set the following fields on each test group:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**.
2. Select an existing test group or create a new one.
3. Select the **General** tab and make the following settings:

    - **Update batch disposition** – Set to *Yes* to enable update of the tested batch number based on whether the quality order is failing or passing.
    - **Failed quality order batch disposition** – Specify the batch disposition master to apply to the tested batch number when a quality order fails. This setting only applies when **Update batch disposition** is set to *Yes*.
    - **Passed quality order batch disposition** – Specify the batch disposition master to apply to the tested batch number when a quality order passes. This setting only applies when **Update batch disposition** is set to *Yes*.

Learn more in Batch disposition codes.

Last updated on 07/27/2026

Set up continuous sampling (preview)

10/24/2025

[This article is prerelease documentation and is subject to change.]

Prerequisites

To prepare your environment for inline sampling, set up the following elements as described in Enable and configure sample management (preview):

- A number sequence for generating sample IDs.
- Sample life cycle states.
- Sample type used for continuous sampling.
- Sample procedures and sample procedure types.
- An item sampling policy set up for continuous sampling. It must have the following settings on the **Sample management** FastTab:
  - **Sample inspection method** – Select *Continuous process*.
  - **Sample size** – Set to the amount of material you want to sample.
  - **Unit** – Set the unit that applies to the selected sample size.
  - **One sample every** – Specify how often samples should be taken and then choose whether to count batches or license plates.
  - **Number of samples per quality order generated** – Specify how often samples should be tested.
- Sample associations configured to link the continuous sample type, continuous item sampling, and continuous sample procedures to the relevant products. Each sample association also defines how long samples can be kept before being scrapped.

Continuous sampling example scenario

This section illustrates how to configure and use continuous sampling. The example assumes the following setup for generating samples and quality orders: - A sample is created for every second license plate that is reported as finished. - A quality order is created for every second sample that is generated.

To work through a scenario that shows how to set up and use continuous sampling, follow these steps:

1. Create a batch-controlled product with the following characteristics:

   - **Enabled for advanced warehouse processes** – Associate the product with a **Storage dimension group** that has **Use warehouse management processes** set to *Yes*.

- **Batch enabled** – Associate the product with a **Tracking dimension group** that has the **Batch number** dimension enabled.
- **Automatic batch numbering** – Associate the product with a **Batch number group** that automatically generates a batch number from a number sequence when quantities are reported as finished from a production or batch order.

2. Set up a sample association for a batch-controlled product as described in Enable and configure sample management.

3. Create a production or batch order for the product with the following information:

    - Choose a warehouse that is enabled for the advanced warehouse processes. Make sure that **Use warehouse management processes** is set to *Yes* for the selected warehouse.
    - Set the quantity for the order to *2,000 pieces*.

4. Go to **Inventory management** > **Setup** > **Quality control** > **Item sampling**. Set up an item sampling policy as described in Quality management item sampling. On the **Sample management** FastTab, make the following settings to set up continuous sampling:

    - **Sample inspection method** – Set to *Continuous process*.
    - **Sample size** – Set to *1*.
    - **Unit** – Set to *Sp* (spoonful).
    - **One sample per every** – Set to *2* and *License plate*.
    - **Number of samples per quality order generated** – Set to *2*.

5. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**. Set up a test group as described in Quality management test groups. Make sure at least one test is assigned in the bottom section. On the **General** tab in the top section, make the following settings to set up the test group for continuous sampling:

    - **Update inventory status** – Set to *Yes* to enable inventory status updates based on test results.
    - **Failed quality order status** – Specify the inventory status to apply to the license plate when a quality order fails.
    - **Passed quality order status** – Specify the inventory status to apply to the license plate when a quality order passes.

6. Go to **Inventory management** > **Setup** > **Quality control** > **Quality associations**. Set up a quality association as described in Quality associations. Make the following settings:

    - **Reference type** – Select **Production*.
    - **Item code** – Select *Table*.

- **Item** – Select the item from the production or batch order you created for this scenario.
- **Event type** – Select *Report as finished*
- **Execution** – Select *After*.
- **Item sampling** – Select the item sampling policy that you created for this exercise.
- **Test group** – Select the test group that you created for this exercise.

7. Return to the production or batch order that you created for this scenario.

8. Bring the order into status *Started*.

9. Report four license plates as finish by following these steps:
   a. On the Action Pane, open the **Production order** tab and, from the **Process** group, select **Report as finished**.
   b. The **Report as finished** dialog opens. Check that the **License plate** dimension is visible in the grid. If it's not visible, select **Inventory** > **Display dimensions** from the toolbar and then select the **License plate** check box in the **Display dimensions** dialog.
   c. Create four new license plates by following these steps:

      - Right-click on the right side of the **License plate** field. Then select **View details** from the context menu to open the **License plates** page.
      - On the Action Pane, select **New** to create a new license plate.
      - In the **License plate** field, enter a unique and recognizable ID for the license plate.
      - Repeat the last two steps until you have created four license plates.
      - On the Action Pane, select the **Back** button to return to the **Report as finished** dialog.

   d. Complete the report-as-finished operation by providing the following information in the **Report as finished** dialog:

      - In the **Good quantity** field, set the value to *20*.
      - In the **License plate** field, select the first license plate you created in the previous step.
      - Make sure that the **End job** checkbox isn't selected.
      - Select **OK** to confirm report as finished of the first license plate.

   e. Complete three more report-as-finished operations for this production order—one for each of the remaining three license plates. Make sure to set the **Good quantity** field to *20* for each license plate.
   f. Verify that two samples and one quality order were created by following these steps:

      - On the Action Pane, open the **View** tab and, from the **Manage quality** group, select **Sample management workbench**.

- Verify that two samples were created for separate license plates, and that these license plates are blocked according their default inventory status. Learn more about configuring the default item status in Enable and configure sample management.
- Select the last sample and open the **References** tab in the bottom section to confirm that the sample has an associated **Quality order**.

Initiate an inline sample

Summarize this article for me

Prerequisites

To prepare your environment for inline sampling, be sure to set up the following elements as described in Enable and configure sample management

- A number sequence for generating sample IDs.
- Sample lifecycle states.
- Sample type used for inline sampling.
- Sample procedures and sample procedure types.
- An item sampling policy set up for inline sampling. It must have the following settings on the **Sample management** FastTab:
  - **Sample inspection method** – Select *Inline process*.
  - **Sample size** – Set to the amount of material you want to sample.
  - **Unit** – Set the unit that applies to the selected sample size.
- Sample associations configured to link the inline sample type, inline item sampling, and inline sample procedures to the relevant products.

Initiate the sample

To initiate an inline sample, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Open or create a production or batch order for a batch-controlled product that is configured for inline sampling.
3. Bring the production or batch order into status *Started*.
4. On the Action Pane, open the **View** tab and, from the **Manage quality** group, select **Initiate inline sample management**.
5. The **Initiate inline sample management** dialog opens. Review and adjust the following settings:

   - **Item number** – The product for which you want to take a sample. This field is defaulted based on the production or batch order that you had open.
   - **Product name** – The name of the product for which you want to take a sample. This field is read-only and comes from the selected **Item number**.

- **Sample type** – Set to *Inline*.
- **Item sampling** – Item samplings define the criteria for selecting samples. This field is defaulted based on the product that you selected, but you choose a new one if necessary.
- **Test group** – Test groups define the set of tests applied while processing quality orders. Select the test group that defines the tests that need to be performed on this sample. This field is defaulted based on the selected item sampling.
- **Sample size** – The amount of material that you want to sample. This field is read-only and comes from the selected item sampling.
- **Unit** – The unit that applies to the selected sample size. This field is read-only and comes from the selected item sampling.

6. Select **OK**.

Work with samples in the sample management workbench

To manage the inline samples related to a production or batch order, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Open or create a production or batch order for a batch-controlled product that is configured for inline sampling.
3. Initiate an inline sample as described in the previous section.
4. On the production order details page, go to the Action Pane, open the **View** tab and, from the **Manage quality** group, select **Sample management workbench**.
5. On the sample management workbench, go to the Action Pane and open the **Sample** tab. Select the various actions here to update the sample and view related information. Learn more in Manage and process samples.

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Manage and process samples (preview)

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[This article is prerelease documentation and is subject to change.]

Sample management is a component of quality control in manufacturing. It ensures that products meet regulatory and quality standards. The processes described in this article serve as an example of how you can implement sample management. However, the specific details might vary depending on your company's branch, industry, and practices. By tailoring the sample management process to your organization's needs, you can maintain compliance, improve operational efficiency, and ensure product integrity. Use this guide as a framework to understand the key steps involved, and adapt them to align with your unique requirements and workflows. Proper implementation of sample management can significantly enhance traceability, accountability, and overall quality assurance.

The following video provides a high-level overview of sample management capabilities and a demo of how to work with them.
https://learn-video.azurefd.net/vod/player?id=36124ac5-86c0-44f6-a57e-3beb10c753ca&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Finventory%2Fquality-sample-management-use

Understanding the lifecycle of a sample in sample management

In modern manufacturing and quality control environments, the ability to systematically manage product samples is essential for ensuring compliance, traceability, and consistent product quality. Sample management provides a structured framework for handling samples from their creation through to disposal. It integrates seamlessly with production workflows and quality assurance protocols. The lifecycle of a sample begins with its registration. You can trigger registration either manually or automatically based on predefined sampling plans. Tailor these plans to the production context, whether *inline sampling*, which occurs during active production before inventory is finalized, or *continuous sampling*, which monitors quality at regular intervals throughout batch production.

Once registered, the system assigns each sample a unique identifier and categorizes it by sample type. The sample type determines its handling procedure, retention policy, and applicable lifecycle states. These lifecycle states, such as *Quality order created*, *Quality order validated*, and *Sample scrapped*, are system-defined and visually tracked to ensure clarity and accountability throughout the sample's journey.

Samples undergo testing through *quality orders*. You can automatically generate or manually initiate these quality orders. The outcome of these tests directly influences inventory control decisions. Passing results might release blocked inventory, while failed tests can trigger automatic blocking of related batches or license plates. This dynamic interaction between sample status and inventory disposition ensures that only quality-approved goods proceed through the supply chain.

Throughout the lifecycle, you manage samples by using tools like the *Sample management workbench*. It offers capabilities such as label printing, audit trail review, lifecycle state changes, and manual overrides. These features empower quality teams to maintain control and visibility over sample data, enabling informed decision-making and regulatory compliance.

In essence, sample management transforms quality control from a reactive process into a proactive, data-driven discipline. By embedding sample lifecycle tracking into production and inventory systems, organizations can uphold high standards of product integrity while streamlining operations across departments.

The remaining sections of this article provide an overview of the activities connected to what could be typical lifecycle states of a sample.

Sample registration

Sample registration is the first step in the sample management process. It serves as the foundation for ensuring quality control in manufacturing. This step involves creating a database record that uniquely identifies each sample, which you can use for accurate tracking and management throughout its lifecycle. By registering samples, manufacturing managers can ensure that the system records all quality-related data and makes it easily accessible. You can register samples in multiple ways, but usually production personnel register them manually as inline samples, or periodically based on a continuous sampling setup.

Register new samples automatically with continuous sampling

Use the continuous sampling method to register new samples automatically based on predefined sampling plans.

Register new samples manually with inline sampling

The inline sampling method lets you register new samples directly during the production or inspection process.

Register new samples manually from the sample management workbench

You can also use the sample management workbench to manually register new samples, providing flexibility for ad-hoc or planned sampling activities. You can only create manual samples from the workbench for existing license plates or batch numbers for products that are set up for continuous sampling.

Follow these steps to use the sample management workbench to manually register a new sample for a license plate or batch number for a product set up for the continuous sampling process:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample Management Workbench**.
2. On the Action Pane, open the **Sample** tab, and select **Create sample**.
3. The **Create manual sample** page opens, showing a list of license plates or batch numbers that are available for sampling. Select the license plate or batch number from which you want to register a new sample.
4. From the Action Pane, select **Create sample**.
5. You return to the sample management workbench, which should now list the newly created sample.
6. Usually, you should also create a quality order for the newly created sample. To do so, go to the Action Pane, open the **Sample** tab, and select **Create manual quality order**. The **Create manual quality order** dialog opens. Review the information, edit it as needed, and select **OK** to create the quality order.

Collect, label, and test a sample

Accurate sample collection and labeling are foundational to reliable testing and traceability. Proper collection ensures that the sample truly represents the intended material, while clear and consistent labeling safeguards identification throughout analysis and storage, preventing costly mix-ups. Learn more in Enable and configure sample management.

Collect samples

Collecting samples is the physical process of extracting a small fraction of the produced material to represent the entire batch or license plate. This step is essential for quality control and traceability, ensuring that subsequent testing reflects the actual production output.

Workers typically use the sample management workbench to find sample registrations that require collection and labeling. It provides visibility and filtering options for finding sample

registrations generated through either inline or continuous sampling methods. The sample registration includes details such as the sample type, size, license plate, associated production order, and sampling procedures, guiding workers on how to collect the sample correctly.

Print labels

Printed labels ensure proper identification and traceability of a sample throughout the testing process. The label includes the sample ID and other information, such as batch or license plate ID, preventing mix-ups and supporting compliance. After you register a sample in the system, you can generate and print a label for it directly from the sample management workbench. First collect the sample, and then attach the label before storage or analysis. Use the following procedure to print labels for one or more samples.

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample Management Workbench**.
2. Find the sample registrations that you want to print a label for by using the **Filter** field or column filters.
3. Select one or more samples you want to print labels for.
4. Usually, use the default label layout, but if you want to choose an alternative layout, open the **Samples** tab on the Action Pane, and select **Change label layout**. Choose the desired layout in the drop-down dialog and select **OK**.
5. On the Action Pane, open the **Samples** tab and select **Print sample label**.
6. The **Print sample label** dialog opens. Make the following settings:

    - **Print destination** – Select *File* to save the label or *Print* to print it now.
    - **Name** – If you chose *Print* as the destination, select the printer you want to use.

7. Select **OK** to print (or save) the labels.

Test your samples

The purpose of collecting samples is to test them to ensure your production run meets its defined specifications. Sample management uses your recorded test results to influence which produced items can be used or shipped by updating the status of associated license plates or batch numbers based on the outcome.

Test inline samples

When you create an inline sample from a production order, the system automatically links it to a new quality order. This link ensures that every inline sample has a defined testing process from the start.

Workers can identify inline samples that are ready for testing by using the sample management workbench and applying filters to find samples according to their inspection method or lifecycle state. For example, you might have a sample lifecycle state named *Quality order generated*, which indicates a sample that has an untested quality order, and your system could be configured so that any inline sample automatically moves to this state when a quality order is created for it. Learn more about configuring lifecycle states in Configure lifecycle states.

Here's an example of how to test an inline sample:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample Management Workbench**.
2. Use the **Filter** field or column filters to find and select the sample registration that you want to test.
3. On the Action Pane, open the **Sample** tab and select **Quality orders**.
4. On the Action Pane, select **Quick result entry**.
5. In the **Result value** field, enter the result of your test. The test passes or fails based on whether your measure value is within the acceptable limits defined by the test.
6. On the Action Pane, select the **Back** button to return to the quality order. Then select **Validate** from the Action Pane.
7. Select **OK** to confirm the dialog and verify that the quality order shows the expected **Status** (either *Pass* or *Fail*).

Test continuous samples

As with inline samples, workers can identify continuous samples that are ready for testing by using the sample management workbench and applying filters to find samples according to their inspection method or lifecycle state.

When you use continuous sampling, the system automatically generates samples according to a predefined sampling plan set up on the item sampling configuration. Learn more about item sampling in Configure item sampling policies. For example, if the item sampling plan specifies that a sample should be created for every second license plate produced and a quality order should be produced for every second sample, then for every four license plates manufactured, the system creates two sample registrations and one quality order. The one quality order covers all four license plates. This means that the validation of that single quality order determines the inventory status for all four plates. If the quality order fails, the system updates the status of all four plates to *Blocked*, ensuring that none of them can be used or shipped. If a worker wants to perform more detailed testing, they can manually create a quality order from the sample management workbench for a sample that doesn't yet have one associated. When this manual test passes, the system automatically updates the inventory status of the related plates, such as the first two, to indicate availability. This method reduces the number of quality orders required for routine checks while still allowing targeted testing when necessary, and it

ensures that inventory status always reflects the outcome of quality validations across the linked items.

Here's an example of how to test a continuous sample:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample Management Workbench**.
2. Select a row that shows the **Item number** you want to test.
3. On the Action Pane, open the **Sample** tab and select **Sample relationships**. The **Sample relationships** page shows information about samples and license plates related to the selected sample. You can find the following information here:

    - There is record for each license plate that has been reported as finished and is waiting for quality testing.
    - Some license plates have samples registered for them (for example, every second license plate), as configured in the item sampling setup.
    - Some of the sampled license plates also have quality orders created for them (for example, every second sample), as configured in the item sampling setup.

4. Go back to the sample management workbench and select the specific sample that you want to test. The sample should be for the item you selected that has a quality order associated with it.
5. On the Action Pane, open the **Sample** tab and select **Quality orders**.
6. On the Action Pane, select **Quick result entry**.
7. In the **Result value** field, enter the result of your test. The test passes or fails based on whether your measure value is within the acceptable limits defined by the test.
8. On the Action Pane, select the **Back** button to return to the quality order. Then select **Validate** from the Action Pane.
9. The **Validate the quality order** dialog opens. The **Sample management** section of the dialog shows the following information:

    - If the test passes, then the **Update inventory status to** field is set to *Available*. If the test fails, then the **Update inventory status to** field is set to *Blocked*. The statuses for each type of result are configured on the relevant test group.
    - All the license plates listed in the grid will be assigned the **Update inventory status to** status, which is based on the test result.
    - A failed status typically prevents the license plate from being used in any warehouse or inventory transactions.

10. Select **OK** to confirm the dialog and assign the inventory status to each of the listed license plates.
11. Go back to the sample management workbench and confirm that the license plates for each sample has the updated inventory status that you expect to see. If you don't see the

inventory status in the grid, select **Dimensions display** on the Action Pane to add it.

Sample archiving

After you collect, label, and successfully test a sample, you can archive it for a defined period until it reaches its expiration date. The system sets the expiration date when you create the sample and calculates it based on the rules specified in the sample association. The system uses the sample lifecycle state to show when a sample is archived, marking its transition from active testing to storage. For details on how expiration periods are determined through sample associations, go to Set up sample associations.

To support archiving, make sure you define a lifecycle state that represents the archived state and select this state on the sample type configured for the product. Learn more in Configure lifecycle states and Define sample types.

Use the following procedure to change the lifecycle state:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample Management Workbench**.
2. Use the **Filter** field or column filters to find and select the sample registration that you want to archive.
3. On the Action Pane, open the **Sample** tab and select **Change lifecycle state**. Choose the desired lifecycle state in the drop-down dialog and select **OK**.

Sample disposal

Disposal involves safely and compliantly disposing of samples that reached the end of their retention period or are no longer needed for testing or quality assurance. Proper disposal minimizes environmental impact, prevents contamination, and ensures compliance with regulatory requirements.

The system uses the sample lifecycle state to show when a sample is disposed, marking its transition from active testing to disposal. To identify samples that have expired in the sample management workbench, you can filter for those whose expiration date has passed.

To support sample disposal, make sure you define a lifecycle state that represents the disposed state and select this state on the sample type configured for the product. Learn more in Configure lifecycle states and Define sample types.

To dispose of a sample, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample Management Workbench**.
2. Use the **Filter** field or column filters to find and select the sample registration that you want to archive.
3. On the Action Pane, open the **Sample** tab and select **Change lifecycle state**. Choose the desired lifecycle state in the drop-down dialog and select **OK**.

The *Disposed* state is typically the final stage in a sample's lifecycle. You can only delete samples from the sample management workbench after they reach a state that is configured with the property **Sample scrapped**. Learn more about how to define the lifecycle state in Configure lifecycle states.

Sample management reports and auditing (preview)

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[This article is prerelease documentation and is subject to change.]

This article describes tools that support quality management through reporting, auditing, and procedural guidance. Reports provide insights into trends in test results for specific tests, helping you monitor quality performance over time. Audit trails track changes and actions for compliance and traceability. Sample procedures are step-by-step instructions that workers can access while processing samples through different life cycle states, ensuring consistency in activities such as collection and testing.

Analyze sample test results for a production order

Sample reports provide visibility into how the results of a specific test evolve over time across multiple quality orders and samples for a given production or batch order. By analyzing these trends, you can identify patterns, monitor process stability, and detect potential quality issues early. You can perform this type of analysis for both inline sampling and continuous sampling, giving you flexibility to evaluate quality performance across different sampling strategies. These reports typically aggregate historical data for the selected test, allowing you to compare performance across different periods and make informed decisions for continuous improvement.

To analyze sample test results for a production order, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Find and select the order you want to analyze.
3. From the Action Pane, open the **View** tab and, from the **Manage quality** group, select **Analyze inline sample results** or **Analyze continuous sample results**, depending on which type of sample you want to analyze.
4. In the dialog, set the following fields

    - **Test** – Select the test you want to analyze.
    - **Item number** – For production orders, this value is read-only (the order you selected). For batch orders, select a formula item or co-product.

5. Select **OK** to see the report.
6. The report provides a graph that shows the upper and lower limits required to pass the test, along with the actual test results for each sample taken over time. You can hover over each data point to see more details about the specific sample and test result.

Sample procedures

Supply Chain Management stores detailed procedures for working with each type of sample. Each procedure provides step-by-step guidance for different stages of sample handling, such as collecting samples and performing tests. By following these instructions, workers ensure that sampling and testing activities are consistent and comply with defined quality standards.

To see the procedures defined for any sample, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample management workbench**.
2. Find and select the sample for which you want to view the procedures.
3. From the Action Pane, open the **Sample** tab and select **Sample procedures**.

Audit trail

Supply Chain Management maintains a full audit trail of all changes made to samples and sample testing records. It provides a detailed log of all actions performed on a selected sample, including the date and time of each change. For updates, the audit trail shows both the previous value and the new value, so you can track the full history of modifications. Each record in the audit trail corresponds to an insert or update event, providing a complete picture of the sample's lifecycle changes.

To view the audit trail for a sample, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Sample management workbench**.
2. Find and select the sample for which you want to view the audit trail.
3. From the Action Pane, open the **Sample** tab and select **Audit trail**.

Test instrument calibration

Calibration is the process of evaluating and adjusting the precision and accuracy of measurement equipment. It's usually defined as a performance comparison against a standard of known accuracy. Proper calibration of test instruments helps ensure a safe working environment. It also helps ensure that valid data is produced for future reference. Test instruments that aren't regularly calibrated can lead to false-positive and/or false-negative quality control tests.

The instrument calibration feature in Microsoft Dynamics 365 Supply Chain Management supports an ongoing calibration process. You can use it to track the calibration schedule and history of individual test instruments. You can also track the use of each test instrument based on the completion of quality order tests.

Instrument calibration is managed primarily through *test instrument tags*. Each test instrument tag is a digital record that represents a specific physical test instrument. This record holds settings for the test instrument and information about it.

Each test instrument tag includes a calibration history for the test instrument, indicates whether the instrument requires calibration. Each tag also provides a calibration procedure (which might include attached documents) and manages other calibration details.

Test instrument tags are assigned to a *calibration group*, which defines the calibration schedule. There are two ways to define the calibration schedule for the test instrument tags in a calibration group:

- **Based on usage** – The system tracks the numbers of times that the test instrument tag is used (that is, assigned to a quality order). It automatically creates calibration records when the usage count reaches a specified threshold. For example, calibration records might be created after every 1,000 uses.
- **Based on a periodic schedule** – The system tracks the date of the last calibration. It automatically creates a calibration record when the next scheduled calibration date is reached. For example, calibration records might be created monthly.

The feature also lets you generate and print calibration reports, such as calibration certificates and calibration schedule reports. In addition, you can design calibration label layouts that you print and attach to individual test instruments as required.

Prerequisites

Before you can use test instrument calibration features in Supply Chain Management, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Set up instrument calibration parameters

The settings on the **Inventory and warehouse management parameters** page affect the way that instrument calibration works across the system. Follow these steps to set up the instrument calibration parameters.

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.

2. On the **Quality management** tab, in the **Test instrument calibration** section, set the following fields:

    - **Automatically assign test instrument tag number on quality order line** – Set this option to *Yes* if the system should try to automatically assign the best available test instrument tag to new quality order test lines. If you set this option to *No*, test instrument tags must be manually assigned when they are needed.

    - **Check if test instrument is being calibrated** – Select one of the following values to specify what action the system takes if the test instrument tag that is assigned to a quality order line has a usage status of *Calibration*. (This status indicates that the instrument is being calibrated and therefore can't be used.) The value that you select is used as the default setting for new test instrument tags.
        - *No check* – Skip this check.
        - *Warning only* – If the instrument is being calibrated, show a warning message, but allow the instrument to be used.
        - *Not allowed* – If the instrument is being calibrated, show an error message that states that the instrument can't be used.

    - **Check for maximum usage before calibration when assigning tag numbers** – Select one of the following values to specify what action the system takes if the maximum usage count for a test instrument tag is exceeded. (When the maximum usage count is exceeded, the instrument is due for calibration.)

    - *No check* – Skip this check.
    - *Warning only* – If the instrument is due for calibration, show a warning message, but allow the instrument to be used.
    - *Not allowed* – If the instrument is due for calibration, show an error message that states that the instrument can't be used.

- **Check for lifetime usage maximum when assigning tag numbers** – Select one of the following values to specify what action the system takes if the lifetime usage maximum for a test instrument tag is exceeded.
    - *No check* – Skip this check.
    - *Warning only* – If the instrument exceeds its lifetime usage maximum, show a warning message, but allow the instrument to be used.
    - *Not allowed* – If the instrument exceeds its lifetime usage maximum, show an error message that states that the instrument can't be used.

- **Skip check for test instrument on open quality order** – Set this option to *No* to prevent test instrument tags that are already assigned to open quality orders from being assigned to new quality orders. Set it to *Yes* to skip this check. In this case, a test instrument tag might be assigned to more than one open quality order at a time.

- **Label layout** – Select the default calibration label layout to assign to new test instrument type records. Learn more in the Design and print calibration labels section.

- **Print destination** – Select one of the following values to specify the default destination for calibration labels that are printed:
    - *Printer* – Print physical labels to a printer.
    - *File* – Download and save labels as a ZPL file.

- **Printer name** – If you set the **Print destination** field to *Printer*, select the name of the default printer to use to print calibration labels.

Set up test instrument types

Use the **Test instruments** page to define and view details about the types of devices that are used to perform tests on quality orders and calibrate other test instruments. *Test instrument types* help quality workers determine which type of device or tool they should use to run tests for a quality order and calibrate test instruments. For each test instrument type, you can specify the following details:

- Whether the instrument is used for calibration

- Whether a test instrument tag is required to identify the specific physical instrument that must be used
- What layout should be used when calibration labels are printed

Learn more in Quality management test instruments.

Set up test locations

Use the **Test locations** page to establish a list of locations where test instruments can be kept. In this way, workers who perform tests on quality orders can find the test instruments that they need. Later, you can assign locations to each test instrument tag as required.

To set up test locations, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test locations**.
2. Use the buttons on the Action Pane to create new test locations or edit existing ones as required. (You can also delete existing test locations.)
3. For each test location, set the following fields:
    - **Test location** – Enter a unique name or number that identifies the test location.
    - **Description** – Enter a short description of the test location.

Set up test departments

Use the **Test departments** page to establish a list of departments that use test instruments. In this way, users can find a test instrument and/or the people who work with it. Later, you can assign departments to each test instrument tag as required.

To set up test departments, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test departments**.
2. Use the buttons on the Action Pane to create new test departments or edit existing ones as required. (You can also delete existing test departments.)
3. For each test department, set the following fields:
    - **Test departments** – Enter a unique name or number that identifies the test department.
    - **Description** – Enter a short description of the test department.

Set up calibration procedures

For each type of instrument calibration that you perform, you can define a *calibration procedure* in Supply Chain Management. Calibration procedures help you track the steps that are required to calibrate a specific type of instrument. You can also use them to track the external vendors that you use to calibrate instruments. Later, when you set up a test instrument tag, you can assign a calibration procedure to it.

To set up a calibration procedure, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Calibration procedures**.
2. Create a new calibration procedure, or select an existing one that you want to edit.
3. On the header of the new or selected procedure, set the following fields:
    - **Calibration procedure name** – Enter a name for the calibration procedure.
    - **Description** – Enter a short description of the calibration procedure.
    - **External calibrating vendor** – Enter or select the account number of the external vendor that you use to calibrate instruments. Leave this field blank for calibration procedures that you perform in-house. After you select a value in this field, the read-only **External vendor name** field shows the name of the selected external vendor.
4. On the **Procedure** FastTab, enter details about the steps that are required to calibrate instruments where this procedure must be used.
5. Optional: If you want to attach a document that provides even more details about the calibration procedure, select the **Attachments** button (paperclip symbol) on the right side of the Action Pane. The **Attachments for Calibration procedures** page shows all the documents that are attached to the calibration procedure. Use the **New** menu on the Action Pane to upload a file as a new attachment. Attachments can be viewed from instrument and calibration records that use this procedure.

Set up calibration groups

Use calibration groups to group test instrument tags and assign calibration schedules to them. All test instrument tags that are assigned to a given calibration group share the same calibration method, calibration schedule, and other settings.

To set up a calibration group, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Calibration groups**.

2. Create a new calibration group, or select an existing one that you want to edit.

3. On the header of the new or selected group, set the following fields:

    - **Calibration group** – Enter the name of the group.
    - **Description** – Enter a short description of the group.

4. On the **General** FastTab, set the following fields:

    - **Calibration method** – Select one of the following values to specify the method that determines when the instrument is calibrated:
        - *Manual* – Manually calibrate the instrument as it's required.
        - *Usage* – Calibrate the instrument based on the number of times that it's used.
        - *Periodic* – Calibrate the instrument on a regular schedule, based on calendar days, months, or years.

    - **Calibration period** – This field is available only when the **Calibration method** field is set to *Periodic*. Select one of the following values to specify the unit for the amount of time between calibrations:
        - *Day* – Specify the period in days.
        - *Month* – Specify the period in months.
        - *Year* – Specify the period in years.

    - **Calibration period frequency** – This field is available only when the **Calibration method** field is set to *Periodic*. Enter the amount of time between calibrations, in the unit that is selected in the **Calibration period** field. For example, if test instrument tags for this group are assigned to instruments that should be calibrated every three months, set the **Calibration period** field to *Month* and the **Calibration period frequency** field to *3*.

    - **Calibration usage update method** – This field is available only when the **Calibration method** field is set to *Usage*. Select one of the following values to specify the method that is used to update the current usage of the test instrument:
        - *Fixed increment* – Each time a quality order that has a test instrument tag in this group is validated, update the test instrument tag's usage count by the value that is specified in the **Usage increment** field.
        - *Test quantity* – Each time a quality order that has a test instrument tag in this group is validated, update the test instrument tag's usage count by the test quantity that

is specified on the quality order.

- **Usage increment** – This field is available only when the **Calibration method** field is set to *Usage* and the **Calibration usage update method** field is set to *Fixed increment*. Specify the increment by which to increase the usage count of a test instrument tag each time an instrument is used. For example, if the **Usage increment** field is set to *2*, the usage count for a test instrument tag in this group is increased by two for every validated quality order that includes the test instrument tag, regardless of the number of units that are tested.

- **Usage count trigger for calibration** – This field is available only when the **Calibration method** field is set to *Usage*. Enter the usage value that triggers a calibration for test instrument tags in this group. The instrument can still be used until the usage count that is specified in the **Maximum usage before calibration** field is reached. For example, if the **Usage count trigger for calibration** field is set to *100*, and a test instrument tag in this group reaches 100 uses, the periodic job that creates calibration records creates a record for that test instrument tag the next time the job runs.

- **Maximum usage before calibration** – This field is available only when the **Calibration method** field is set to *Usage*. Enter the maximum number of times that a test instrument tag in this group can be used before it must be calibrated. For example, the **Usage count trigger for calibration** field might be set to *100*, whereas the **Maximum usage before calibration** field might be set to *125*. In this case, calibration is triggered after 100 uses, but the instrument can continue to be used until it reaches 125 uses.

- **Lifetime usage maximum** – This field is available only when the **Calibration method** field is set to *Usage*. Enter the maximum number of times that an instrument can be used before it's taken out of service. For example, if the **Lifetime usage maximum** field is set to *10,000*, test instrument tags in this group that reach 10,000 uses are marked as out of service and can no longer be calibrated.

> ⓘ **Note**
>
> Use the **Inventory and warehouse management parameters** page to configure how the system reacts when the **Maximum usage before calibration** and **Lifetime usage maximum** values are reached. You can choose to have the system do nothing, show a warning but still allow the instrument to be used, or show an error and block the instrument from being used. Learn more in the Set up instrument calibration parameters section.

Set up test instrument tags

Each physical test instrument is represented in Supply Chain Management by a test instrument tag record. This record specifies data such as the manufacturer, warranty number, acquisition date, and specifications. The test instrument tag is used to track the calibration schedule and history of the test instrument, and it can be assigned to quality order tests.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test instrument tags**.
2. Create a new test instrument tag, or select an existing one that you want to edit.
3. On the header of the new or selected test instrument tag, review or set the following fields:
    - **Tag number** – Enter a unique name, number, or code for the instrument.
    - **Test instrument type** – Select the type or category of test instrument that the test instrument tag is associated with. Examples include *Scale*, *Thermometer*, and *Oximeter*. The test instrument type defines options such as whether the instrument is used for calibration, whether a test instrument tag is required to identify the specific physical instrument that must be used, and what layout should be used when calibration labels are printed. Learn more in the Set up test instrument types section.
    - **Instrument usage status** – The current usage status of the test instrument. Although the field is read-only, you can manually change the status to *Out of service* or *Available* by using the buttons on the **Update usage status** tab of the Action Pane. The following status values are used:
        - *Available* – The test instrument is ready to be assigned to a new quality order.
        - *Calibration* – The test instrument is currently being calibrated. This status is automatically set when an open (in-progress) calibration record exists for this test instrument tag.
        - *Out of service* – The test instrument is out of service and can't be used for quality order tests. This status is automatically set when the lifetime usage maximum is reached.
4. On the **General** FastTab, review or set the following fields. These fields provide general information about the test instrument tag of the test instrument.
    - **Description** – Enter a brief description of the test instrument.
    - **Instrument description** – A description of the test instrument type that is selected in the **Test instrument type** field on the header.

- **Used for calibration** – A value that indicates whether the test instrument is used to calibrate other instruments. The value is inherited from the selected test instrument type.
- **Skip check for test instrument on open quality order** – Set this option to *No* to prevent test instrument tags that are already assigned to open quality orders from being assigned to new quality orders. Set it to *Yes* to skip this check. In this case, a test instrument tag might be assigned to more than one open quality order at a time.
- **Restricted use** – Set this option to *Yes* if this test instrument is restricted. This option is for information only.
- **Check if test instrument is being calibrated** – Select the action that is taken on a quality order when the **Instrument usage status** value of the instrument is *Calibration*. The default value for new tags is defined in Inventory and warehouse management parameters, but you can change it for individual test instrument tags. The following values are available:
  - *Not allowed* – If the **Instrument usage status** value is *Calibration*, don't allow the instrument to be used for new or existing quality orders.
  - *Warning only* – If the **Instrument usage status** value is *Calibration*, show a warning, but still allow the instrument to be used for new or existing quality orders.
  - *No check* – Always allow the instrument to be used for new or existing quality orders, regardless of its calibration status.
- **Serial number** – Enter the serial number of the test instrument.
- **Owner** – Enter the owner of the test instrument.
- **Acquisition date** – Enter the date when the test instrument was acquired.
- **Fixed asset number** – If the test instrument is linked to a fixed asset number, select that number. This field is for information only.
- **Customer account** – Select the customer account that the test instrument is related to. For example, the test instrument might belong to and/or be assigned to a specific customer.
- **Vendor account** – Select the vendor account that the test instrument is related to. For example, the test instrument might belong to and/or be assigned to a specific vendor.

- **Test area** – Select the test area that the test instrument is associated with. This field is a standard Supply Chain Management field. It's reused on this page based on standard logic.

- **Test department** – Select the department that is responsible for the test instrument. Learn more in the Set up test departments section.

- **Test location** – Select the location where the test instrument can be found. Learn more in the Set up test locations section.

- **Unit** – The unit that the test instrument measures in (such as grams or milliliters). The value is inherited from the selected test instrument type.

- **Precision** – The precision of measurement that can be used when results from the test instrument are recorded. Precision refers to the number of digits that are used to express a value. The value of this field is inherited from the selected test instrument type.

5. On the **Calibration details** FastTab, review or set the following fields. These fields provide information that is specific to the calibration of the test instrument.

    - **Calibration required** – Specify whether calibration is required for the test instrument. The other fields on this FastTab can be edited only when this option is set to *Yes*.

    - **Calibration procedure name** – Select the name of the procedure that is used to calibrate the test instrument. The procedure provides instructions for calibrating the instrument. To read the procedure, select the link. If the selected procedure includes attached documents, select the **Attachments** button (paperclip symbol) on the right side of the Action Pane to view them. Learn more in the Set up calibration procedures section.

    - **External calibrating vendor** – If an external vendor is used to calibrate the test instrument, the account number of the vendor. The value of this read-only field is inherited from the calibration procedure that is selected in the **Calibration procedure name** field.

    - **External vendor name** – If an external vendor is used to calibrate the test instrument, the name of the vendor. The value of this read-only field is inherited from the selected calibration procedure.

- **Calibration start date** – The date of the first calibration. Other calibration dates are calculated relative to this date.
- **Usage count since last calibration** – The number of times that the test instrument has been used since its last calibration. This field applies only when the **Calibration method** field is set to *Usage*. It's used to determine when the next calibration is due. Each time the instrument is calibrated and approved, the value of this field is reset to *0* (zero). Although this field is usually read-only, you can make it editable by selecting **Override usage data** on the **Functions** tab of the Action Pane.
- **Lifetime usage count** – The number of times that the test instrument has been used during its lifetime. Although this field is usually read-only, you can make it editable by selecting **Override usage data** on the **Functions** tab of the Action Pane.
- **Calibration group** – Select the calibration group that the test instrument belongs to. The calibration group establishes the calibration method (manual, periodic, or usage) and related settings that define the calibration schedule. Settings from the selected group are shown nearby in read-only form. They depend on the calibration method that the group uses. Learn more in the Set up calibration groups section.
- **Completion assignee** – Select the user who is usually assigned to complete the calibration of the test instrument.
- **Approval assignee** – Select the user who is usually assigned to approve the calibration of the test instrument.
- **Started date** – The date when the most recent calibration of the test instrument was started.
- **Completion date** – The date when the most recent calibration of the test instrument was completed.
- **Calibration result** – The result of the most recent calibration of the test instrument.
- **Approval date** – The date when the most recent calibration of the test instrument was approved.
- **Next calibration date** – The date when the next calibration of the test instrument is scheduled. This field applies only when the **Calibration method** field is set to *Periodic* or *Manual*. If you're using the *Manual* calibration method, this field is editable. If you're using the *Periodic* calibration method, the value is automatically calculated by using

the approval date of the most recent calibration and the settings of the calibration group schedule. If you must reschedule the date that is automatically calculated for the next calibration, select **Reschedule calculations** on the **Functions** tab of the Action Pane.

Here are some examples that show how dates are calculated:

- The **Approval date** value of the last closed calibration is *January 1, 2025*, the **Calibration period** value is *Day*, and the **Calibration period frequency** value is *10*. In this case, the **Next calibration date** value is calculated as *January 11, 2025*.
- The **Approval date** value of the last closed calibration is *January 1, 2025*, the **Calibration period** value is *Month*, and the **Calibration period frequency** value is *3*. In this case, the **Next calibration date** value is calculated as *April 1, 2025*.
- The **Approval date** value of the last closed calibration is *January 1, 2025*, the **Calibration period** value is *Year*, and the **Calibration period frequency** value is *1*. In this case, the **Next calibration date** value is calculated as *January 1, 2026*.

- **Date calibration certificate printed** – The date when the instrument calibration certificate was last printed.

6. Enter information on the following FastTabs as required. The fields on these FastTabs are for information only. They aren't used in any calculations or processes.

    - **Manufacturer data** – Enter information that is specific to the test instrument and its manufacturer.
    - **Test procedures** – Add more test procedures to support the calibration procedures.
    - **Specifications** – Add more specifications.
    - **Notes** – Add notes and other comments.

Create calibration records

When a test instrument requires calibration, a *calibration record* must be created. The calibration record is used to track the details of the calibration process, including the date and time when the calibration was started, completed, and approved, the person who performed each step, and the results of the calibration. To review the full calibration history, open the calibration record list. You can filter the list so that it shows the calibration history of a specific physical instrument, for example.

While a calibration record is open, its related test instrument tag shows an **Instrument usage status** value of *Calibration*. This status usually indicates that the instrument isn't available for use.

Manually create a calibration record

You can manually create a calibration record for a test instrument tag at any time by following these steps.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test instrument tags**.
2. Select the test instrument tag that you want to create a calibration record for.
3. On the Action Pane, on the **Functions** tab, select **Calibrate instrument**.
4. On the **Test instrument calibration record detail** page, most of the fields are preset so that the values match the settings of the selected test instrument tag. However, you can adjust the values as you require.
5. On the Action Pane, select **Save**.

Schedule a batch job to automatically create calibration records as required

The *Create calibration records* batch job automatically creates calibration records for test instrument tags that are due for calibration. The batch job can be scheduled to run regularly (for example, daily or weekly), or it can be manually run as required. Each time the batch job runs, it checks every test instrument tag and uses the following method to determine whether it should create an open calibration record:

- If the **Calibration method** value is *Usage*, the job checks the **Usage count since last calibration** and **Maximum usage before calibration** fields to determine whether a calibration record should be created.
- If the **Calibration method** value is *Periodic*, the job checks the **Next calibration date** field to determine whether a calibration record should be created.

To run or schedule the batch job, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Create calibration records**.
2. On the **Records to include** FastTab, set up filters and constraints to define which test instrument tags are checked. Select **Filter** to open a standard query editor dialog, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Supply Chain Management.
3. On the **Run in the background** FastTab, specify how, when, and how often the job runs. The fields work just as they do for other types of background jobs in Supply Chain Management.
4. Select **OK** to apply your settings and close the dialog.

Calibrate an instrument

Calibration work is created and traced by using calibration records. Each calibration record is associated with a specific test instrument tag. It contains information about the calibration process, including the date and time when the calibration was started, completed, and approved, the person who performed each step, and the results of the calibration.

Calibration records are listed on the **Calibrate test instruments** page. There, you can view, start, take notes about, complete, and approve the calibration of a test instrument.

The three possible calibration results are *Pass*, *Limited pass*, and *Fail*. For failed calibrations, you can mark the instrument as out of service or have it continue to show up as being calibrated. Calibrations that pass must also be approved. Electronic signatures can optionally be required for approval of a completed calibration record.

A completed or approved calibration can be reopened at any time. You can view a history of all reopened calibrations.

To perform a calibration, follow these steps:

1. Go to **Inventory management** > **Periodic** > **Quality management** > **Calibrate test instruments**.

2. All the open records that are listed on the page represent calibrations that are due. Either select one of these records or select **New** on the Action Pane to create a new one.

3. Review and set the following fields as required:

    - **Test instrument type** – The test instrument that is being calibrated.
    - **Description** – A description of the test instrument.
    - **Tag number** – The test instrument tag number of the test instrument that is being calibrated.
    - **Description** – A description of the test instrument tag.
    - **Instrument usage status** – The current status of the test instrument tag (*Available*, *Calibration*, or *Out of Service*).
    - **Calibration procedure name** – The calibration procedure to use. By default, this field is set to the procedure that is selected for test instrument tag. However, you can select a different procedure. Learn more in the Set up calibration procedures section.
    - **Description** – A description of the calibration procedure.

- **External calibrating vendor** – If an external vendor is used to calibrate the test instrument, the account number of the vendor. This value of this read-only field is inherited from the calibration procedure that is selected in the **Calibration procedure name** field.
- **External vendor name** – If an external vendor is used to calibrate the test instrument, the name of the vendor. This value of this read-only field is inherited from the selected calibration procedure.
- **Calibration procedure** – The procedures as they are specified on the **Calibration procedure** page. The value of this read-only field is inherited from the selected calibration procedure.
- **Started by** – The user who started the calibration. This field is updated by the **Start calibration** action.
- **Started date** – The date when the calibration was started. This field is updated by the **Start calibration** action.
- **Completion assignee** – The user who is responsible for assigning the user who will perform the calibration. The default value comes from the test instrument tag, but you can modify it.
- **Completion by** – The user who completed the calibration. This field is updated by the **Complete calibration** action.
- **Completion date** – The date when the calibration was completed. This field is updated by the **Complete calibration** action.
- **Calibration result** – The result of the calibration of the test instrument. This field is updated by the **Complete calibration** action.
- **Approval assignee** – The user who is responsible for assigning the user who will approve the calibration. The default value comes from the test instrument tag, but you can modify it.
- **Approval by** – The user who approved the calibration. This field is updated by the **Approve calibration** action.
- **Approval date** – The date when the calibration was approved. This field is updated by the **Approve calibration** action.
- **Test procedures** – More information about how to calibrate the test instrument that must be calibrated. The default information comes from the test instrument tag, but it can be updated during calibration.
- **Specifications** – Further specifications for the test instrument that must be calibrated. The default information comes from the test instrument tag, but it can be updated during calibration.

- **Calibration notes** – Further notes about the calibration. The default information comes from the test instrument tag, but it can be updated during calibration.

> **Tip**
>
> If more calibration instructions are available as attached documents, the **Attachments** button (paperclip symbol) on the right side of the Action Pane includes a badge that shows the number of attachments. Select the **Attachments** button to open the **Attachments for Calibration procedures** page, where you can view, add, and describe documents that are attached to the calibration record.

4. If you want to record details about the specific instruments that are required or used to calibrate the test instrument, on the Action Pane, on the **Calibrate instruments** tab, select **Assign calibration tools**. Then use the **Calibration tools used** page to record each instrument. For each instrument that you add, set the following fields:

   - **Calibration tool** – Select the test instrument type for the instrument that is used. This field lists only test instrument types where the **Use for calibration** option is set to *Yes*. Learn more in Quality management test instruments.
   - **Calibration tool tag number** – To identify a specific physical test instrument, select its test instrument tag number. This field is required if the **Requires test instrument tag** option is set to *Yes* for test instrument type that is selected in the **Calibration tool** field.

5. To start the calibration, on the Action Pane, on the **Calibrate instruments** tab, select **Start calibration**. Then select **OK** in the dialog. The **Started by** and **Started date** fields on the calibration record are updated.

6. Do your calibration tests as required. When you're finished, on the Action Pane, on the **Calibrate instruments** tab, select **Complete calibration**. Then, in the **Calibration action** dialog, set the following fields:

   - **Calibration result** – Select one of the following values to specify the result of the calibration:
     - *Pass* – The instrument passed the calibration.
     - *Limited pass* – The instrument can still be used, but within a limited scope.
     - *Fail* – The instrument failed the calibration.

- **Update associated test instrument tag status to 'Out of service'** – Set this option to *Yes* if the instrument failed the calibration, and you want to mark it as out of service. This option is available only when the **Calibration result** field is set to *Fail*.

7. The calibration must now be approved, either by you or by somebody else (such as a supervisor), depending on the setup of your system. The approver must open the calibration record, and then, on the Action Pane, on the **Calibrate instruments** tab, select **Approve calibration**. The instrument is now approved, and the **Approved by** and **Approved date** fields are updated. The related test instrument tag is updated with the test result, approval date, and next calibration date.

8. If you must print a calibration label and/or calibration certificate, on the Action Pane, on the **Calibration instruments** tab, in the **Print** group, select the appropriate button.

Reopen an approved calibration record

If you must edit an approved calibration record, follow these steps:

1. Go to **Inventory management** > **Periodic** > **Quality management** > **Calibrate test instruments**.
2. Find and select the relevant calibration record. The record must be in the *Fail* or *Approved* state.
3. On the Action Pane, on the **Calibrate instruments** tab, select **Reopen calibration**. The completed and approved fields are cleared, a *reopen calibration history* record is created, and the calibration record returns to the *Open* state. You can now edit and complete the calibration record in the usual way.

To view a history of reopened calibration records, follow these steps:

- Go to **Inventory management** > **Inquiries and reports** > **Quality management** > **Calibration reopen history log**.
- Go to **Inventory management** > **Periodic** > **Quality management** > **Calibrate test instruments**. On the Action Pane, on the **Calibrate instruments** tab, select **Calibration reopen history log**.

Apply test instrument types and tags to a quality order

You set up and manage quality tests by using quality orders. A quality order defines what must be tested and how. A test group is assigned to each quality order. The test group defines which tests must be done, and which test instrument type should be used for each test.

When you assign a test group to a quality order, each line from the selected test group becomes a line on the new quality order. The quality order also inherits the test instrument type from the test group. When you create a quality order, you can (and in some cases must) assign a specific test instrument tag to each line. In this way, you specify the physical instrument that must be used for the test.

To view and assign test instrument types for a test group, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**.
2. Create a new test group, or select an existing one.
3. At the bottom of the page, on the **Overview** tab, create or select a test group line.
4. On the **Test** tab, in the **Test instrument** field, select the test instrument type.

To view and assign test instrument tags to a quality order, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.
2. Create a new quality order, or select an existing one.
3. At the bottom of the page, the **Overview** tab has a line for each test that the quality order requires. It includes lines that are inherited from the test group that is assigned to the quality order. You can manually add more lines as you require.
4. On the **Overview** tab, select a quality order line.
5. On the **Test** tab, in the **Tag number** field, review or set the test instrument tag for the selected quality order line.

Here are some guidelines for using test instrument types and tags on quality orders:

- If the **Automatically assign test instrument tag number on quality order line** option is set to *Yes* in Inventory and warehouse management parameters, when a quality order is created, the system sorts through the test instrument tags for each selected instrument type and selects the best available test instrument tag for the quality test. If the option is set to *No*, you must manually assign test instrument tags as required.
- If a test instrument type that is selected for use on a quality order line is set up to require a test instrument tag, the quality order can't be validated until a test instrument tag is selected on the quality order line.
- Test instrument tags that have an **Instrument usage status** value of *Out of service* aren't available for selection on the quality test.
- Test instrument tags that have an **Instrument usage status** value of *Calibration* might be available for selection on the quality test, depending on the setup of the test instrument tag.

- Depending on how you set up the system to validate test instrument tags, you might receive a warning or error if you select a test instrument tag that is already assigned to another open quality order, or if the selected tag has exceeded its maximum usage and/or its lifetime usage.
- When a quality order is validated, if a selected test instrument tag is calibrated based on usage, the usage counts on the tag are updated based on the quality order. The current usage count and the lifetime usage count are incremented by either the test quantity or the usage increment, depending on the details of the calibration group on the test instrument tag.

Design and print calibration labels

Use calibration labels to attach calibration information directly to physical test instruments.

Set up calibration label layouts

Calibration label layouts are designed and printed in Zebra Programming Language (ZPL) format, a standard format for printing labels on Zebra printers. Calibration labels can include information such as the test instrument tag number, the test instrument type, the calibration procedure name, and the next calibration date.

To get started quickly, you can generate a predesigned ZPL layout for a calibration label. This layout is designed to be used with Zebra printers and can be modified as required. To add the predesigned ZPL layout to your system, go to **Inventory management** > **Setup** > **Quality control** > **Create sample calibration labels**. The system then generates a label layout that is named *SampleLabel_2X1*.

To customize the standard layout or create new ones, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Calibration label layout**.
2. Use the buttons on the Action Pane to create a new calibration label layout or edit an existing one as required. (You can also delete existing layouts.)
3. On the header of the new or selected layout, enter details such as the name, description, and dimensions.
4. On the **Label layout** FastTab, use the large field to design your label layout. For help, use a ZPL reference.

5. The page helps you find the codes that you need to insert field values. To add a field value to your layout, follow these steps:
    a. Set the **Use fields from** field to the type of database record that you want to use a field from (*Test instrument tags* or *Test instrument calibration records*).
    b. In the **Labels** list, select a field name.
    c. Select **Insert at end of text** to add the reference code to the end of the label layout text.
    d. Use a copy-and-paste operation to move the reference code to the desired location in the label layout text.

Assign layouts to test instrument types

When you print a calibration label, the system uses the layout that is assigned to the test instrument type of the test instrument tag. Learn how to assign a layout to a test instrument type in Quality management test instruments.

Print calibration labels

To print calibration labels for a physical test instrument, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test instrument tags**.
2. Select the test instrument tag that you want to print a calibration label for.
3. On the Action Pane, on the **Print** tab, select **Calibration label**.
4. In the **Print calibration label** dialog, set the following fields:
    - **Print destination** – Select *Printer* to print the label directly to a printer. Select *File* to download and save the label locally as a ZPL file.
    - **Name** – If you selected *Printer* in the **Print destination** field, select the printer to use.

Print calibration certificates

Calibration certificates can be printed for each calibration record. They document the calibration process and results for a test instrument, and are used to verify that the test instrument was calibrated according to the specified procedures and standards. Calibration certificates include information such as the test instrument tag number, the calibration procedure name, the calibration result, the name of the person who performed the calibration, and the next calibration date.

To print a certificate for the last closed calibration for a test instrument tag, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test instrument tags**.
2. Select the test instrument tag that you want to print a calibration label for.
3. On the Action Pane, on the **Print** tab, select **Calibration certificate**.
4. The certificate is generated and shown on your screen. You can review, print, and/or export it.

To print a certificate for any selected calibration record, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Calibrate test instruments**.
2. Select the test calibration record that you want to print a calibration label for.
3. On the Action Pane, on the **Calibrate instruments** tab, in the **Print** group, select **Calibration certificate**.
4. The certificate is generated and shown on your screen. You can review, print, and/or export it.

View or print the calibration schedule report

The calibration schedule report provides a list of test instruments that must be calibrated. However, you can customize the selection criteria to support many other uses. For example, you can filter by owner or calibration method. You can use the **Show all due today** option to create a list of test instrument tags where the due date of the next calibration is before the current date.

To print a calibration schedule report, follow these steps:

1. Go to **Inventory management** > **Inquiries and reports** > **Quality management** > **Test instrument calibration schedule**.

2. In the **Test instrument calibration schedule** dialog, on the **Parameters** FastTab, set the **Show all due today** option to one of the following values:

    - *Yes* – Show only test instrument tags where the next calibration date is before the current date.
    - *No* – Show all test instruments tags, regardless of when they are due for calibration.

3. The **Destination** FastTab shows the current destination for the report. The possible destinations are *Screen*, *Printer*, *File*, and *Email*. (The default destination is *Screen*.) To change the destination, select **Change**, select a new destination, and then configure settings that are related to that destination.

4. On the **Records to include** FastTab, set up filters and constraints to define which test instrument tags are included. Select **Filter** to open a standard query editor dialog, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Supply Chain Management.

5. On the **Run in the background** FastTab, specify how, when, and how often to run the job. The fields work just as they do for other types of background jobs in Supply Chain Management.

6. Select **OK** to apply your settings and close the dialog.

Last updated on 07/27/2026

Flexible sampling plans

A *flexible sampling plan* defines the criteria for inspecting randomly selected units of items from a lot. The results from the inspected sample determine whether the whole lot passes or fails testing. Typically, the number of units that are selected for inspection is proportional to the total size of the lot. The quality association that is defined for a plan identifies the activity for the inspection. For example, the activity might be a quality order to inspect a product before it's released to a customer on a sales order. Alternatively, it might be a quality order to inspect a manufactured product after production is completed.

Flexible sampling plans offer other features that you can use to modify your criteria over time, based on previous testing results.

- **Sampling size variations** – Flexible sampling lets you vary your item sampling over time, based on test findings. For example, for the first five rounds of testing, you inspect 50% of the receipt. If no quality failures are found in those rounds, you might decide to reduce the inspection sampling size for subsequent rounds to 10% of the receipt. Rounds identify the number of times that a test is performed at a specific level before testing moves to the next level in the testing plan. For example, if a level requires five rounds of testing, the assigned item sampling should be used for all five rounds before the testing moves to the next level. The next level might use a different item sampling.
- **Test group flexibility** – By using a flexible sampling plan, you can change a plan's current test group based on inspection results. The test group identifies the individual tests that are performed. For example, if previous testing results show minimal issues with the quality of receipts, you might change the test group to one that has a less stringent set of tests. You can also specify different test groups for different levels in a flexible sampling plan.
- **Skip lot inspections** – Skip lot inspections are an alternative to lot-by-lot inspections. They let you skip a set number of goods during testing. For example, a long-time supplier has demonstrated that the quality of their receipts is very good. Therefore, a quality manager might set up a flexible sampling plan to inspect a quantity of 10 units from this supplier and then skip the next 10 units. Receipts that don't require inspection are put away, and no quality order is generated.

Because flexible sampling plans give you the flexibility to perform lot-by-lot inspections instead of unit-by-unit inspections, and to use features such as sampling size variations, test group flexibility, and skip lot inspections, they help your company minimize time, costs, handling damage, and inspection errors.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Set up flexible sampling plans

To set up flexible sampling plans, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Flexible sampling plans**.

2. On the Action Pane, use the following buttons as required:

    - **Add** – Add a new sampling plan.
    - **Delete** – Remove an existing plan. (You can't delete a plan if quality orders use it or activities are tracked against it.)
    - **Copy plan** – Copy content from another plan to the selected plan.
    - **View activities** – Open the **Flexible sampling plan activities** page, where you can view summaries and detailed information about the plan activities.

3. On the header of the new or selected sampling plan, set the following fields:

    - **Flexible sampling plan code** – Enter a name for the sampling plan.
    - **Name** – Enter a short description of the sampling plan.
    - **Alert role** – Select the security role that should be notified if a specified number of failures for a level is reached.
    - **Last level** – Select the level number for the last activity that is performed in this flexible sampling plan process.
    - **Approved** – Set the plan to the *Approved* state.

4. On the **Details** FastTab, use the grid to set up the plan. Use the buttons on the toolbar to add and remove lines in the grid. For new or selected lines, set the following fields:

    - **Level** – Select or review the level number that is assigned to the activity. The plan is processed in descending order of levels.

- **Item sampling** – Select or review the item sampling. The item sampling determines the sampling size of the quality order for the activity. Examples include a 10% sampling or a 50-unit quantity sampling.

- **Test group** – Select or review the test group. The test group determines the set of tests for the quality order for the activity.

- **Rounds** – Set or review the number of testing rounds for the specific flexible sampling plan level that must pass quality order validation before testing can move to the next level. After you select a level in the **Last level** field, any **Rounds** value that you entered for that level is removed from the plan. In this way, the last level can continue indefinitely until several failures cause the plan to revert to a previous level.

  > ⓘ **Note**
  >
  > Unless rounds are removed from the last level in the testing plan, all future tests remain at that level indefinitely, until the enough failures occurred so that the plan can revert to a previous level.

  For example, a quality association is set up to test purchase order registrations for a specific vendor. The vendor's flexible sampling plan requires that 10% of every registration is tested for five rounds. If you receive 100 units, a quality order is generated to test 10 of them. The quality order for the first registration passes. The second quality order then fails, but the next three quality orders pass. If round six passes, the first level in the plan is accomplished, because the number of passed rounds met the requirement.

- **Time span (days)** – Enter or review the number of days within which the specified number of rounds must be completed. This field is optional. It works together with the **Rounds** field. If this field is set, testing can move to the next level only if the specified number of rounds pass inspection within the defined time span. For example, if you set the time span to 60 days and specify 10 rounds, testing can move to the next level in the plan only if 10 quality orders for the quality association pass within 60 days.

- **Skip** – Select this checkbox to enable skip lot sampling, where only a fraction of the submitted lots is inspected. If you select this checkbox, you must enter the skip range in the **Test frequency** and **Out of frequency** fields.

- **Test frequency** – If the **Skip** checkbox is selected, enter the first number in the skip range.
- **Out of frequency** – If the **Skip** checkbox is selected, enter the last number in the skip range.

  For example, the quality association is set for production order registrations. The flexible sampling plan that is associated with this reference type uses skip lot sampling. The **Test frequency** field is set to *1*, and the **Out of frequency** field is set to *5*. In this case, a quality order is generated to test one out of every five registrations. The system randomly selects one of the five registrations for testing, and the other four registrations are skipped.
- **Number of fails** – Enter or review the number times that tests can fail before the flexible sampling plan reverts to a previous testing level. That level might be more or less restrictive than the current test, based on previous testing results. If the **Consecutive** checkbox is selected, the plan reverts to a previous level only if the failures are consecutive. For example, a vendor might initially define a plan to test 10 out of 40 receipts, to minimize the costs that are associated with testing. However, if recent testing results show a notable increase in failures, the current level might be replaced by a level that tests 20 out of 40 receipts. On the other hand, if the vendor's testing results show consistent quality and no failures, the current level might be replaced by a level that requires fewer failures.
- **Consecutive** – Select this checkbox if the number of failures that is specified in the **Number of fails** field must be consecutive. If this checkbox is selected, and the number of consecutive failures is reached, the appropriate action is initiated, based on the specified fail action level and fail action code. For example, if the **Number of fails** field is set to *2*, and this checkbox is selected, two consecutive failures generate a predefined fail action, such as issuing an alert notification to the quality manager.
- **Fail action level** – Enter or review the level of action that is initiated when the number that is specified in the **Number of fails** field is reached. This action indicates that more testing is required. If the **Consecutive** checkbox is also selected, the action is initiated only if the failures are consecutive. For example, if two consecutive quality orders fail at level 3, the defined action might require that testing starts over at level 1. You can assign any level that is currently defined in the flexible sampling plan, provided that it's equal to or lower than the current level.

- **Fail action code** – Select the action code that defines the action that should be taken when the failure criteria are met or exceeded for the flexible sampling plan. All the actions return the test sampling process to the fail action level.

Set up the quality association for the flexible sampling plan

Quality orders that use flexible sampling plans are set up so that they are automatically generated by using a quality association.

Learn more about how to manage quality associations in Quality associations.

View flexible sampling plan activities

To open the **Flexible sampling plan activities** page, select the **View activities** button on the Action Pane of the **Flexible sampling plan** page. The **Flexible sampling plan activities** page provides an overview of the quality associations where flexible sampling plans are used.

The following buttons are available on the **Flexible sampling plan activities** page:

- **Flexible sampling plan activity summary** – View a summary of activity for the selected flexible sampling plan.
- **Flexible sampling plan activity details** – View the specific details of the flexible sampling plan that you select. On this page, you can view how the activities in the plan are progressing through the levels of the plan. For each activity, the page indicates the quality order that is used. Color-coded related text describes the pass and fail criteria for each level. For example, a red color indicates that a quality order failed. If a quality order fails, the page also indicates whether the test reverted to a previous level that requires more testing. From this page, you can also open pages that include summaries and details about activities, and view charts that show statistics about the performance of quality order testing.
- **Flexible sampling plans** – View the specific activity details for the flexible sampling plan that you select. You can also create a new flexible sampling plan from this page.

Last updated on 07/27/2026

Acceptance sampling

Acceptance sampling is a quality control method that uses statistical analysis to evaluate a batch or lot of products. Instead of checking every item, you examine a subset of items to determine whether the entire lot meets predefined quality standards. This approach is widely used in manufacturing and supply chain operations to balance inspection effort with quality assurance.

Acceptance sampling overview

Defect types and categories

In many practical applications, you inspect sampled items for different *types* of defects. Each defect type typically falls into one of the following *categories*:

- **Critical defects** – Might cause harm or render a product unsafe or unusable. Even a single critical defect can lead to rejection of the entire lot.
- **Major defects** – Significantly reduce the usability or performance of a product but don't pose safety risks. A limited number of major defects might be acceptable depending on the sampling plan.
- **Minor defects** – Don't affect a product's function or safety but might impact appearance or user satisfaction. A higher tolerance is usually allowed for minor defects.

You define acceptance criteria in advance, often using standards like ANSI/ASQ Z1.4 or ISO 2859-1. These standards specify permissible defects count per category before rejecting a lot. This structured, statistically supported approach ensures consistent product quality while optimizing inspection resources.

Single and double sampling plans

Single and double sampling plans use two different strategies to decide if a lot meets quality standards.

- **Single sampling plan** – You inspect one sample from the lot. Based on the number of defects found, you either accept or reject the lot. Use this method when simplicity and speed are important.
- **Double sampling plan** – If the first sample gives an inconclusive result, you take a second sample. You base the final decision on the combined results of both samples. This strategy can reduce inspection effort over time while maintaining decision accuracy.

The current version of Supply Chain Management supports only the single sampling strategy, where each lot is evaluated by using one sample and a fixed acceptance threshold.

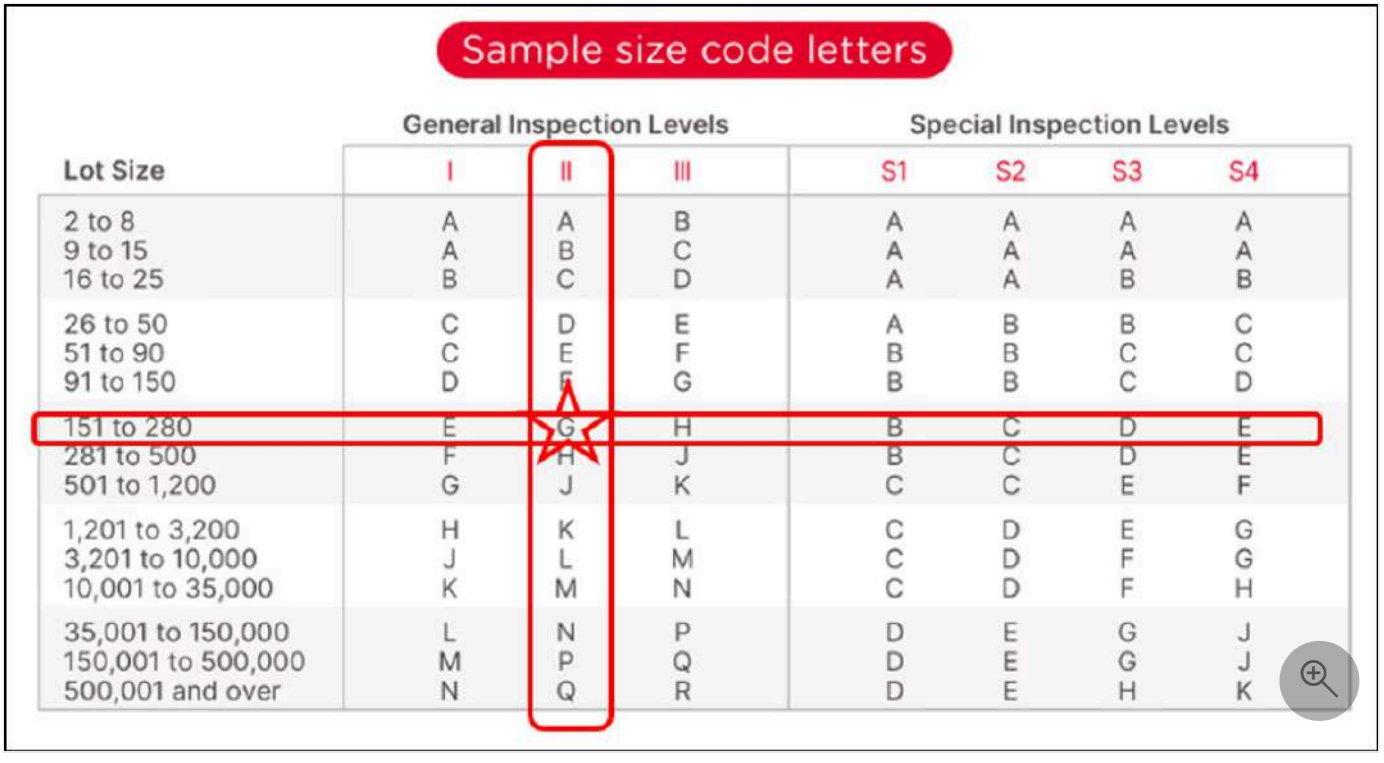
Acceptance sampling charts

Acceptance sampling uses a chart with the following two components to validate whether a lot passes or fails an inspection:

- **Sampling code letter chart** – Defines code letters for each combination of lot size and inspection level.
- **Acceptable quality limit (AQL) chart** – Uses the code letter and AQL index to determine sample size and defect thresholds.

An *inspection level* is a code or category defined by a business user. It indicates the extent of inspection based on the importance of the product, the production history, and customer requirements. There are two categories of inspection levels: - *General inspection level* – Used for routine, thorough inspections - *Special inspection level* – Apply to less critical checks with smaller sample sizes.

The following image shows an example of a code letter chart. In this example, if you have a lot size between 151 and 280 and use a general inspection level II, then use code letter G as input in the acceptable quality limit chart.



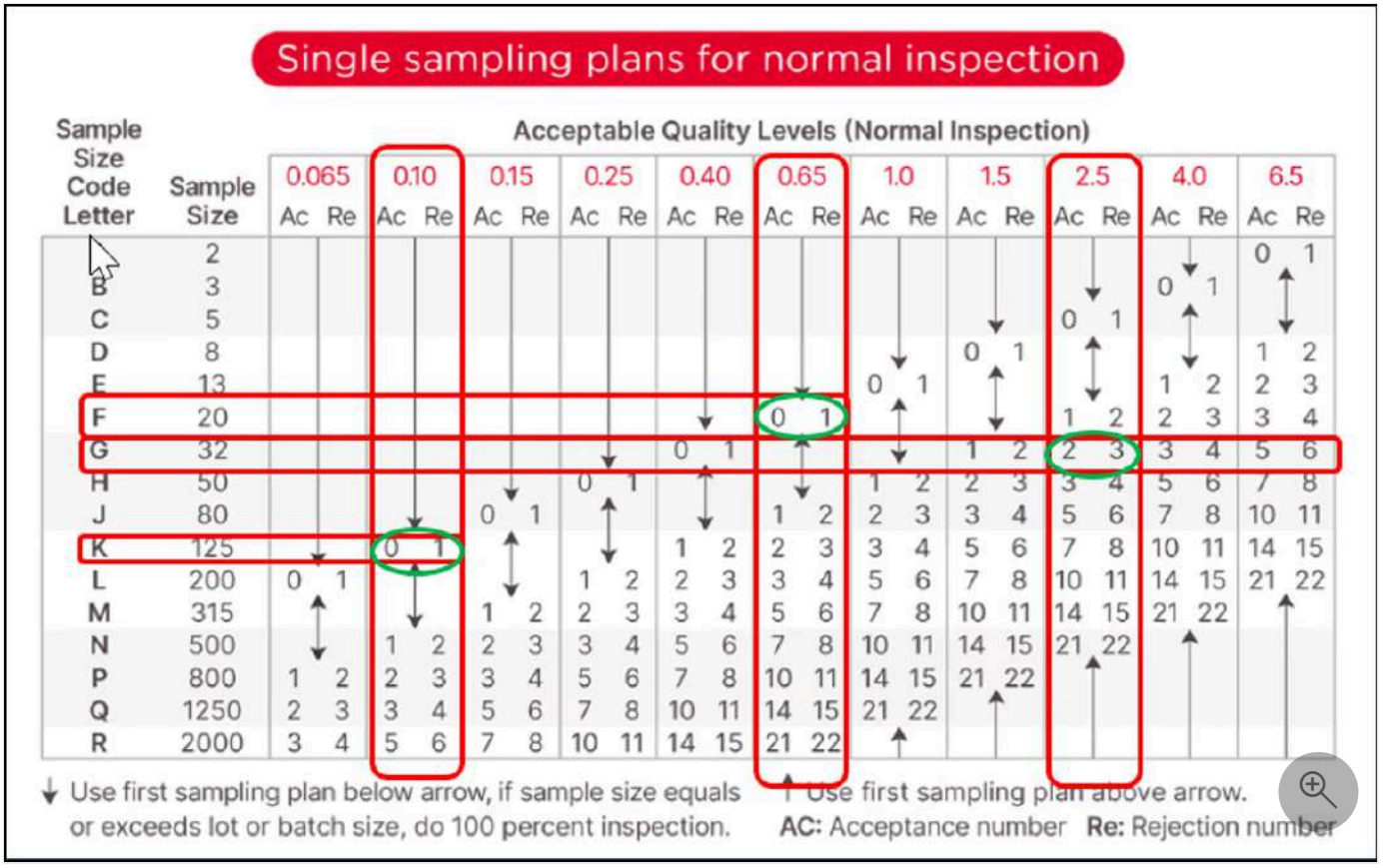
Sample size code letters

	General Inspection Levels			Special Inspection Levels			
Lot Size	I	II	III	S1	S2	S3	S4
2 to 8	A	A	B	A	A	A	A
9 to 15	A	B	C	A	A	A	A
16 to 25	B	C	D	A	A	B	B
26 to 50	C	D	E	A	B	B	C
51 to 90	C	E	F	B	B	C	C
91 to 150	D	F	G	B	B	C	D
151 to 280	E	G	H	B	C	D	E
281 to 500	F	H	J	B	C	D	E
501 to 1,200	G	J	K	C	C	E	F
1,201 to 3,200	H	K	L	C	D	E	G
3,201 to 10,000	J	L	M	C	D	F	G
10,001 to 35,000	K	M	N	C	D	F	H
35,001 to 150,000	L	N	P	D	E	G	J
150,001 to 500,000	M	P	Q	D	E	G	J
500,001 and over	N	Q	R	D	E	H	K

The AQL chart defines the acceptable quality level for each combination of sampling code letter and AQL index. Business users determine the AQL indexes based on how critical the product is,

industry standards, customer requirements, and the acceptable level of risk. Lower indexes are used for high-risk or critical items, while higher AQLs are acceptable for less critical products where some defects are tolerable. For example, businesses can use index 0.1% for testing for critical defects, 0.65% for major defects, and 2.5% for minor defects. The selected code letter and AQL index resolve to find the sample size and acceptable quality level for the test.

The following image shows an example of an acceptable quality limit chart. In this example, if you use code letter G with an AQL index of 2.5%, then the sample size should be 32 units and up to two defects are accepted for the lot to pass, but three or more defects cause the lot to fail inspection. If you use code letter G for AQL index 0.65%, then the arrow in the chart indicates that an AQL level for a different code letter should be used. In this case, the code letter with the nearest AQL values is F, which has a different sample size than code letter G (so 20 units should be tested instead of 32), and zero defects are accepted for the lot to pass.



In Supply Chain Management, you can load all the data for generating the sampling code letter chart and the acceptable quality limit chart from a template. The template generates inspection levels, AQL indexes, and lot size ranges. After you load these values from the template, you can customize the charts as needed.

You can learn more about how to use the charts in the following video:

https://learn-video.azurefd.net/vod/player?id=bb27c2c0-ffda-4768-9d06-2ca67c05d508&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Finventory%2Fquality-acceptance-sampling

Quality orders, item sampling methods, and quality associations

Supply Chain Management uses *quality orders* to schedule and manage testing based on acceptance sampling. To enable this functionality, you must define an appropriate item sampling method that supports acceptance sampling.

Item sampling methods define how much of a product you should inspect during a quality order. They specify the sample quantity or percentage to test from a lot, and are essential for enabling acceptance sampling in quality control processes. Learn more in Quality management item sampling.

Item sampling methods take effect when you assign them to *quality associations*. Supply Chain Management uses quality associations to establish and apply rules that automatically trigger a quality order based on specific events, such as product receipt, production output, or inventory movement. Each quality association links a product, process, or condition to a quality inspection requirement, ensuring consistent and automated quality control. Learn more in Quality associations.

When a quality order is triggered based on an event defined in a quality association, the system automatically creates a quality order that you can use for acceptance sampling. The quality association links the triggering event (such as a product receipt or production output) to a specific test process. If the associated item sampling record is configured for acceptance sampling, the generated quality order uses the defined sampling criteria to determine how much of the lot you should inspect.

On the quality order for acceptance sampling, the various categories of tests (critical, major, and minor), their sample sizes, and acceptable quality levels are clearly indicated. If a test falls outside a quality level, the entire quality order fails during validation.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.45 or later.

- The feature named *Acceptance sampling* must be turned on in feature management. As of Supply Chain Management version 10.0.49, this feature is turned on by default.

Set up acceptance sampling charts

To set up your acceptance sampling charts, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Acceptance sampling** > **Acceptance sampling chart**.

2. Use the buttons on the Action Pane to add a new chart or edit an existing one.

3. On the header of the new or selected chart, set the following fields:

    - **Acceptance sampling chart name** – Enter a name for the sampling chart.
    - **Description** – Enter a short description of the sampling chart.

4. On the Action Pane, select **Save**.

5. To get started quickly, you can either copy an existing chart or load a predefined template. If you choose to do so, select one of the following options in the Action Pane. Note that all the settings for the current chart (except name and description) are replaced with the template or copied chart.

    - **Load AQL chart template** – Load all required data from a predefined JSON file. The content of the file aligns with the ANSI/ASQ Z1.4 or ISO 2859-1 standards.
    - **Copy** – Copy data from another chart. The system asks you to choose an existing chart.

6. On the **Inspection levels to include** FastTab, use the buttons in the toolbar to add and remove inspection levels for the current chart. The inspection levels are used as a criteria to find the applicable code letter in the code letter chart. Learn more about inspection levels in Acceptance sampling charts.

7. On the **AQL indexes to include** FastTab, use the buttons in the toolbar to add and remove AQL indexes. The AQL index defines the maximum acceptable defect rate in a sample that determines whether a lot passes or fails inspection. You can learn about AQL indexes in Acceptance sampling charts.

8. On the **Lot/batch size ranges** FastTab, use the buttons in the toolbar to add and remove lot size ranges for the current chart. Use the **Unlimited** button in the toolbar to set a range to have no upper limit (this sets the **To** field to *Unlimited* for the selected row). The lot size

ranges group total units in a lot and help find the applicable code letter. Learn more about lot size ranges in Acceptance sampling charts.

9. When you're done setting up the chart, select **Validate** in the Action Pane to check data integrity. If the chart is valid, it sets the **Validated** field to *Yes*. Only validated charts can be used for acceptance sampling.

   > **Tip**
   >
   > Validated charts are locked and can't be edited. If you need to edit a validated chart, select **Invalidate** from the Action Pane, make the required changes, and then validate the chart again.

Set up inspection levels and acceptance sampling indexes

If you don't use the option to load your charts from a template, you must create the inspection levels and sampling indexes manually.

Set up inspection levels

Follow these steps to create inspection levels manually:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Acceptance sampling** > **Inspection levels**.

2. Select **New** to create a new inspection level and set the following fields:

   - **Inspection level** – The identification of the inspection level.
   - **Inspection level type** – Choose between the two types *General* or *Special*, which are used as a criteria to find the code letter in the code letter chart. Learn more about level designations in Acceptance sampling charts.
   - **Description** – User defined description of the inspection level.

3. On the Action Pane, select **Save**.

Set up acceptance sampling indexes

Business users determine AQL indexes based on how critical the product is, industry standards, customer requirements, and the acceptable level of risk. If you don't want to use the option to

generate the AQL indexes by loading the AQL chart template, then you can use the following steps to create them manually:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Acceptance sampling** > **Acceptance sampling indexes**.
2. Select **New** to create a new inspection level and set the following fields:
    - **Acceptable quality limit index** – The identification of the acceptance sampling index. The AQL index defines the maximum acceptable defect rate in a sample that determines whether a lot passes or fails inspection.
    - **Description** – User defined description of the acceptance sampling index.
3. On the Action Pane, select **Save**.

View the sampling size code letter chart

After you create and validate an acceptance sampling chart, you can view the sampling size code letter chart:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Acceptance sampling** > **Acceptance sampling chart**.
2. Select the chart you want to view.
3. On the Action Pane, select **Sampling size code letter chart**. You can open the corresponding acceptable quality limit chart by selecting **Acceptable quality limit chart**.

View or edit the acceptable quality limit chart

To view or edit the acceptable quality limit chart, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Acceptance sampling** > **Acceptance sampling chart**.
2. Select the chart you want to edit.
3. On the Action Pane, select *Acceptable quality limit chart* to open the chart.
4. Use the following options to make edits:
    - **Up** – Adds an upward arrow in the selected cell, indicating the direction to find the AQL level and sample size. Learn more in Acceptance sampling charts.
    - **Down** – Adds a downward arrow in the selected cell, indicating the direction to find the AQL level and sample size. Learn more in Acceptance sampling charts.

- **Accept** – Sets the acceptance level in the selected cell, defining the maximum number of defects allowed for a given sample size and AQL index. Learn more in Acceptance sampling charts.

Set up defect types

Defect types identify the types of defects that users test for, such as *paint chipped*, *screen cracked*, *packaging damaged*, and so on. Each defect type is categorized as *Critical*, *Major*, or *Minor*.

To set up defect types, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Acceptance sampling** > **Defect types**.
2. Select **New** and set the following fields:

    - **Defect type** – Identify the type of defect to test for.
    - **Description** – Enter a detailed description of the defect type.
    - **Defect category** – Categorize the defect type with one of the three following fixed categories:
        - *Critical* – Could cause harm or render a product unsafe or unusable. Even a single critical defect might lead to rejection of the entire lot.
        - *Major* – Significantly reduces the usability or performance of a product but don't pose safety risks. A limited number of major defects might be acceptable depending on the sampling plan.
        - *Minor* – Doesn't affect function or safety but might impact appearance or user satisfaction. A higher tolerance is usually allowed for minor defects

Set up a test for acceptance sampling

You must define each of the tests that you want to use on quality orders, and each test must be associated with one of the defect types that you set up in the previous section. Learn more about tests for quality orders in Quality management tests.

Set up a test group for acceptance sampling

Each of the tests that you use for acceptance sampling must be associated with a *test group*. A test group is a collection of tests needed on a quality order. Each quality association has an assigned test group, which it uses to automatically generate quality orders. Learn more about test groups in Quality management test groups

Set up item sampling for acceptance testing

Use *item sampling* records to define how much of a product to inspect during a quality order. Each record specifies the sample quantity or percentage to test from a lot. These records are essential for enabling acceptance sampling in quality control processes. Learn more about item sampling in Quality orders, item sampling methods, and quality associations and Quality management item sampling.

To set up item sampling for acceptance sampling, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Item sampling**.

2. Select **New** to create a new item sampling and set the following fields:

    - **Item sampling** – Enter a name for the item sampling record.
    - **Description** – Enter a short description of the item sampling record.
    - **Sampling scope** – Select the scope to use when evaluating if quality work should be created. When set to *Shipment* or *Load*, those entities are used if available. If not, the *Order* is used. When you use acceptance sampling charts, the only supported sampling scope is *Order*.
    - **Use acceptance sampling charts** – Choose one of the following options:
        - *Single* – Use acceptance sampling. This value indicates that you'll use a single sampling plan. Learn more in Single and double sampling plans.
        - *None* – Don't use acceptance sampling for quality inspection.

3. On the **Acceptance sampling** FastTab, set the following required fields:

    - **Acceptance sampling chart name** – Select the sampling chart you want to use for this item sampling.
    - **Description** – Displays the name of the selected acceptance sampling chart.
    - **Inspection level** – Select the inspection level you want to use for this item sampling. Learn more about the use of inspection levels in Acceptance sampling charts.
    - **Description** – Displays the description of the selected AQL index.
    - **Minor AQL%** – Enter the AQL index for minor defects for the item sampling. Learn more about how to use AQL indexes in Acceptance sampling charts.
    - **Description** – Displays the description of the selected AQL index.
    - **Major AQL%** – Enter the AQL index for major defects for the item sampling. Learn more about AQL indexes in Acceptance sampling charts.
    - **Description** – Displays the description of the selected AQL index.

- **Critical AQL%** – Enter the AQL index for critical defects for the item sampling. Learn more about AQL indexes in Acceptance sampling charts.
- **Description** – Displays the description of the selected AQL index.

Set up a quality association for acceptance sampling

Supply Chain Management uses quality associations to define and apply rules that automatically trigger a quality order based on specific events, such as product receipt, production output, or inventory movement. Each quality association links a product, process, or condition to a quality inspection requirement, ensuring consistent and automated quality control. Learn more in Quality associations.

When an event defined in a quality association triggers a quality order, the system automatically creates a quality order that you can use for acceptance sampling. The quality association links the triggering event (such as a product receipt or production output) to a specific test process. If the associated item sampling is configured for acceptance sampling, the generated quality order uses the defined sampling criteria to determine how much of the lot you should inspect.

To set up a quality association for acceptance sampling, follow the instructions in Quality associations. When you set up your quality associations on the **Quality associations** page, pay special attention to the following settings, which are specific to acceptance sampling:

- Use the **Item sampling** field on the **Specifications** FastTab to select an item sampling record. Choose an item sampling record that's configured to use acceptance sampling charts. This setting is required to enable acceptance sampling for the quality association.
- Use the **Test group** field on the **Specifications** FastTab to select a test group that contains the tests you want to use for acceptance sampling. The test group must contain tests that are associated with defect types that you set up in Set up defect types. The test group can include at most one of each defect type.

Using acceptance sampling on quality orders

Quality orders created for acceptance sampling include an **Acceptance sampling** tab on the **Quality orders** page. This tab contains a grid with line items for each test that you need to conduct. The grid provides the following information:

- **Defect category** – Indicates one of the three fixed categories of the test: *Critical*, *Major*, or *Minor*.

- **Defect type** – Indicates the type of the test defined in the configuration. Learn more in Set up defect types.
- **Acceptable quality level index** – Indicates the AQL index for the specific test as specified in the configuration of the item sampling for the quality order.
- **Sample size** – The number of items to test.
- **Target Ac** – Maximum number of defects accepted for the test to pass.
- **Target Re** – Minimum number of defects for the test to fail.
- **Test result** – Graphical indication of whether the test passed or failed.

To record test results for a selected quality order, either enter them directly into the test lines or select **Quick results entry** from the Action Pane to use the quick-entry form.

Last updated on 07/27/2026

Quality orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to manually or automatically create quality orders, and how to work with them to perform inspections and record test results in Microsoft Dynamics 365 Supply Chain Management.

Automatically created quality orders

Configure the system to automatically create quality orders based on item sampling rules. Learn more in Quality management item sampling.

Manually create quality orders

To manually create a quality order, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.
2. Select **New**.
3. In the **Quality orders** dialog box, in the **Reference type** field, select the inventory reference that your quality order is related to. For a description of the reference types that are available for selection, see the Quality order reference types section later in this article.

   > **Note**
   >
   > Inventory that is related to the selected reference must be available. If no inventory is available for the combination of the reference type, quantity, and inventory dimensions that you select, you'll receive an error message.

4. Follow one of these steps, depending on the value that you selected in the **Reference type** field:

    - If you selected **Inventory**, no additional reference information is required.
    - If you selected **Sales** or **Purchase**, specify the following information:
        - **Account selection** – Reference the customer or vendor number.
        - **Reference number** – Reference the sales order or purchase order number.
        - **Reference lot** – Reference the specific sales order line or purchase order line.
    - If you selected **Production**, **Quarantine**, or **Co-product production**, in the **Reference number** field, reference the production order, batch order, or quarantine order number.
    - If you selected **Route operation**, specify the following information:
        - **Reference number** – Reference the production order or batch order number.
        - **Oper. no.** – Reference the specific route operation number.
        - **Operation** – Reference the specific route operation.
        - **Resource** – Reference the specific resource that must be used with the route operation.

5. In the **Test group** field, select the group of tests that must be performed.
6. In the **Quantity** field, enter the quantity of items that must be tested.
7. In the **Inventory dimensions** section, enter any additional inventory dimensions that are required for the selected item.
8. Select **OK**.

After a quality order is created, you can begin to inspect the inventory and record the test results. For more information about how to record and validate test results, see Inspect the quality of goods.

Quality order reference types

Quality orders are used to track the details about inspections and test results for specific inventory in your warehouse. A quality order must include a reference that represents the source of the inventory that is being tested. Quality orders can be related to the following reference types:

- **Inventory** – This reference type indicates that you're inspecting inventory that is on hand. Inspections of this type are typically random or unplanned, and are done when a warehouse worker identifies a problem.
- **Sales** – This reference type indicates that you're inspecting inventory that is related to a specific sales order. Inspections of this type are typically related to customer specifications or the generation of a certificate of analysis (CoA) that must be provided to a customer as part of a shipment. (A CoA is sometimes also referred to as a certificate of compliance [CoC].)
- **Purchase** – This reference type indicates that you're inspecting inventory that is related to a purchase order. Inspections of this type are often used to inspect incoming goods before they're put into inventory or consumed in production processes.
- **Production** – This reference type indicates that you're inspecting inventory that is related to a production order. Inspections of this type are often used to inspect finished goods before they're put into inventory.
- **Quarantine** – This reference type indicates that you're inspecting inventory that is related to a quarantine order. Quarantine orders are a special type of order that tracks inventory in a segregated warehouse or a segregated area in your warehouse. Inspections of this type are often used to inspect goods that a customer has returned or that have been put into quarantine for further analysis. Quarantine orders can be generated from quality orders. Alternatively, they can be generated from other sources, and then quality orders can be related to the quarantine orders.
- **Route operation** – This reference type indicates that you're inspecting inventory that is related to a specific step of the route for a production order. Inspections of this type are typically used to analyze the work in process (WIP) of a product before it moves to the next step in the production process.
- **Co-product production** – This reference type indicates that you're inspecting inventory that is related to a co-product of a production order. Inspections of this type are typically used to inspect the co-product of a batch order before the co-product is added to inventory.
- **Goods in transit order** – This reference type indicates that you're inspecting inventory that is related to a goods-in-transit order from Landed Cost. Inspections of this type are often used to inspect incoming goods before they're put into inventory.

View and create quality orders from various parts of the system

Quality orders can be created manually. Alternatively, they can be automatically created based on the quality associations that you define. For more information about how to create and manage

the automatic creation of quality orders, see Quality associations.

Use the quality order page to manually create a new quality order or view the status of a quality order related to another record. There are several ways to access the quality order page.

From the Quality orders page

To manually create quality orders and view all existing quality orders, go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**. The remaining sections of this article provide more information about how to work with the **Quality orders** page.

From sales orders

To work with quality orders that are related to your sales orders, go to **Sales and marketing** > **Sales orders** > **All sales orders**, and then follow any of these steps:

- Open a sales order, or select it in the grid. Then, on the Action Pane, on the **Pick and pack** tab, in the **Quality management** group, select **Quality orders** to open the **Quality orders** page. There, you can view, create, or update quality orders that are related to the sales order.
- Open a sales order, and then, on the **Header** tab, select the **General** FastTab. The **Quality order status** field shows the overall status of all quality orders that are related to the sales order.
- Open a sales order, and then, on the **Lines** tab, select the **Sales order lines** FastTab. The **Quality order status** column shows the status of each line of the sales order.

From purchase orders

To work with quality orders that are related to your purchase orders, go to **Procurement and sources** > **Purchase orders** > **All purchase orders**, and then follow any of these steps:

- Open a purchase order, or select it in the grid. Then, on the Action Pane, on the **Receive** tab, in the **Quality management** group, select **Quality orders** to open the **Quality orders** page. There, you can view, create, or update quality orders that are related to the purchase order.
- Open a purchase order, and then, on the **Header** tab, select the **General** FastTab. The **Quality order status** field shows the overall status of all quality orders that are related to the purchase order.
- Open a purchase order, and then, on the **Lines** tab, select the **Purchase order lines** FastTab. The **Quality order status** column shows the status of each line of the purchase order.

From production orders

To work with quality orders that are related to your production orders, go to **Production control** > **Production orders** > **All production orders**, and then follow any of these steps:

- Open a production order, or select it in the grid. Then, on the Action Pane, on the **View** tab, in the **Manage quality** group, select **Quality orders** to open the **Quality orders** page. There, you can view, create, or update quality orders that are related to the production order.
- Open a production order, or select it in the grid. Then, on the Action Pane, on the **Production order** tab, in the **Production details** group, select **Route** to open the **Production route** page. To view the quality orders that are related to a route operation, follow one of these steps:
  - Select the target route operation in the grid. Then, on the Action Pane, select **Inquiries** > **Quality orders**.
  - Select the value in the **Oper. No.** field for the target route operation to open its **Production route** details page. On the **General** FastTab, the **Quality order status** field shows the status for the quality orders that are related to the route operation.
- Open a production order, and select the **General** FastTab. The **Quality order status** field shows the status of the quality orders that are related to the production order.

From quarantine orders

To work with quality orders that are related to your quarantine orders, go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quarantine orders**, and then follow any of these steps:

- Review the values in the **Quality order status** column. This way, you learn the overall status of all quality orders that are related to each quarantine order in the grid.
- Select a quarantine order in the grid, and then, on the Action Pane, select **Quality orders** to view, create, or update quality orders that are related to the quarantine order.

From goods-in-transit orders

If you use the **Landed cost** module, you can set up the system to automatically create quality orders when goods-in-transit orders are received. The quality control process can then be initiated for the goods-in-transit orders. This process ensures that products undergo stringent quality checks throughout transit and that high-quality goods are received at the warehouse. During quality control, inventory from goods-in-transit orders can be blocked to prevent

shipment of defective products. This function aligns with regulatory compliance and helps mitigate legal risks.

To automatically create quality orders based on goods-in-transit orders, enable the *Enable Quality Control for Goods In-Transit Order* feature. For more information, see Turn on the Landed cost module and related features for your system. In addition, configure quality associations.

To create quality orders for the received quantity only, set the **Per updated quantity** option to *Yes* on the **Item sampling** page (**Inventory management** > **Setup** > **Quality control** > **Item sampling**).

Users can view, manually create, or update quality orders directly from the **Goods-in-transit order** page. To work with quality orders that are related to your goods-in-transit orders, follow one of these steps:

- Go to **Landed cost** > **Periodic tasks** > **Goods in transit orders** and select a goods-in-transit order. Then, on the Action Pane, select **Quality orders** to open the **Quality orders** page. There, you can view, create, or update quality orders that are related to the goods-in-transit order.
- Go to **Landed cost** > **Voyages** > **All voyages** and open or select a voyage or select a voyage. On the Action pane, on the **General** tab, in the **Inquiries** group, select **Goods in transit orders** to open the **Goods in transit orders** page. Select a goods-in-transit order. Then, on the Action Pane, select **Quality orders** to open the **Quality orders** page, where you can view, create, or update quality orders that are related to the selected goods-in-transit order.

Advanced actions for quality orders

You can take several actions on quality orders to manage the status, generate documents, and inquire about additional details.

Reopen a quality order

By default, you can no longer edit or update a quality order after it has been validated and the tests have been passed. However, in some cases, you might have to reopen a quality order after it has been completed.

To reopen a quality order, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.

2. Open the validated quality order, or select it in the grid.
3. On the Action Pane, select **Reopen quality order**.

Create a certificate of analysis for a quality order

A certificate of analysis (CoA) is a report that an organization's quality assurance team generates. It validates that a product meets specific regulations or requirements. Your customers or regulatory establishments in your geopolitical location might require CoA reports. They might also be required based on your industry and the type of products that you handle, purchase, produce, or sell.

Supply Chain Management lets you generate a CoA from a quality order. The report includes the results of any tests on the quality order where the **Certificate of analysis report** option is set to *Yes*. You can set this option by default based on the test that you define on the **Tests** page. However, you can override the setting on specific tests for a specific quality order.

To generate a CoA for a quality order, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.
2. Select the quality order that you want to create a CoA for.
3. On the Action Pane, select **Inquiries** > **Certificate of analysis**.
4. On the **Certificate of analysis** page, on the Action Pane, select **New**.
5. Optional: In the **Contact** field, select the contact person that the certificate should be addressed to.
6. On the Action Pane, select **Print**.
7. In the **Certificate of analysis** dialog box, define how the report should be printed. Then select **OK** to print the report to a printer, to the screen, to a file, or to email.
8. Close the pages.

Block or unblock inventory for a quality order

When a quality order is automatically generated from a quality association, the item sampling assigned to the quality association can block the full inventory quantity of the reference that is being tested. For more information about item samplings, see Quality management item sampling.

If you aren't using full blocking, or if you're manually creating a quality order, the system automatically creates an inventory blocking record for the quantity of the item that is being tested on the quality order. In the record that is created on the **Inventory blocking** page, the **Inventory blocking type** field is set to *Quality order*.

To view and edit the inventory blocking for a quality order that is selected on the **Inventory blocking** page, select **Inquiries** > **Inventory blocking** on the Action Pane. Learn more in Inventory blocking.

Inquire about the details of a quality order

Use the following buttons on the Action Pane of the **Quality orders** page to view more information about or related to a quality order:

- **Inquiries** > **Work details** – Open a page where you can view warehouse work that is related to the quality order.
- **Inquiries** > **Non conformances** – Open a page where you can view any nonconformances that are related to the quality order.
- **Inventory** – The commands on this menu are common across all inventory transactions. Use them to view or update details such as transactions, on-hand inventory, reservations, and marking.

Create cases related to quality orders

You can create cases that are related to quality orders. This way, you can track details about issues and work toward a resolution. You can use the workflow features of case management to route a case through a predefined business process to obtain additional approvals or get more information about a specific issue. You can also use the knowledge articles feature to create a knowledge base of resolutions for common issues.

Case management examples for quality management

Example 1

You work for a manufacturing company that must follow strict regulations that are related to the production of regulated products such as food. Quality orders are used to record and track details about the quality of items throughout the production process. If a quality order fails specific tests, there might be a problem with the equipment. Cases are used to follow a business process and escalate the issue to the correct engineers so that they can determine the root cause. To make business processes easier to follow, the company keeps a knowledge base of common issues that are related to quality orders and equipment issues.

Example 2

You work for a distribution company that ships products that can be customized for various countries/regions. Some customers have strict specifications that you must follow. Otherwise, you might incur fees, returns, or chargebacks. You use quality orders to track the details about each test and results that match customer requirements. Cases are used to review and approve the details for the CoA before the document is generated and attached with other shipping paperwork.

Related information

- Quality management processes
- Quality test
- Quality test groups
- Quality associations
- Quality management processes
- Enable quality and nonconformance management
- Quality management for warehouse processes

Last updated on 06/22/2026

Customer-specific certificate of analysis (COA)

Microsoft Dynamics 365 Supply Chain Management lets you create a basic certificate of analysis (COA) either from the quality order or from the menu directly after you select a quality order. Learn more about how to use the basic COA in Quality orders. This article explains how to customize the content of a COA to meet specific customer requirements and automatically print the COA when a sales order packing slip is generated.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Set up a COA customer group

To manage COAs for different customers, you can group customers into COA customer groups. To create a COA customer group and add customers to it, follow these steps:

1. Go to **Inventory management** > **Setup** > **Certificate of analysis** > **Certificate of analysis customer groups**.

2. Use the buttons on the Action Pane to add a new COA customer group or edit an existing one. (You can also delete existing groups.)

3. For the new or selected COA customer group, set the following fields:

    - **COA customer group** – Enter the identifier of the COA customer group.
    - **Description** – Enter a description for the COA customer group.

Set up and maintain customer-specific COA requirements on test groups

To set up customer-specific COA requirements on test groups, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Test groups**.
2. In the lower section of the page, select a test.
3. On the toolbar, select **Customer COA requirements** to open the **Customer COA requirements** page.
4. Use the buttons on the Action Pane to add a new customer COA requirement or edit an existing one. (You can also delete existing requirements.)

   If you add a new customer COA requirement, the **Test** and **Test group** fields are automatically set. If the test is associated with a batch attribute, the **Attribute** field is also set.
5. For the new or selected customer COA requirement, set the following fields in the grid:

   - **Customer code** – Select *All* to make the requirement applicable to all customers, *Group* to make it applicable to a group of customers, or *Table* to make it applicable to a specific customer.
   - **Customer relation** – If you selected *Group* in the **Customer code** field, select a COA customer group. If you selected *Table*, select a specific customer account. (If you selected *All*, this field is unavailable.)
   - **Exclude** – Select whether the test should appear on the customer-specific COA. The assumption is that the customer-specific COA includes all tests except tests are explicitly marked as excluded.
   - **Use customer specific ranges** – Specify whether the customer-specific batch attribute range should be used for the customer-specific COA. If no customer -specific batch attribute range if found, the standard range is printed.
   - **Suppress Min/Max values** – Specify whether the minimum and maximum values of the test should be suppressed on the customer-specific COA. For example, the minimum value for a test is *1*, the maximum value is *10*, and the result is *1*. In this case, you might not want the range to be shown at all for some customers, to avoid drawing attention to the fact that the result barely passed the quality test.
   - **Replace pass results** – Enter text that should replace the test results on the customer-specific COA if the test is passed. In this situation, some businesses prefer to show standard text, such as "Within specification," instead of the actual test results.
   - **Replace fail results** – Enter text that should replace the test results on the customer-specific COA if the test is failed. In this situation, some businesses prefer to show

standard text, such as "Outside acceptable range," instead of the actual test results.

Set up and maintain customer-specific COA requirements from a quality order

To set up and maintain customer-specific COA requirements from a quality order, follow these steps:

1. Go to **Inventory management** > **Periodic** > **Quality management** > **Quality orders**.
2. Select a quality order.
3. Select a line on the selected quality order.
4. On the Action Pane, select **Customer COA requirements**.
5. Follow steps 4 and 5 of the previous procedure to add, edit, or delete customer-specific COA requirements.

Include product-specific and customer-specific batch attributes on a COA

You can mark product-specific and customer-specific batch attributes for inclusion on a COA, even if they aren't included in testing through a quality order.

1. Go to **Product information management** > **Products** > **Released products**.
2. On the **All released products** list page, find and select or open the batch-enabled product that you want to set up.
3. On the Action Pane, on the **Manage inventory** tab, in the **Batch attributes** group, select either **Product specific** or **Customer specific**.
4. Select an existing attribute in the list pane, or create a new one.
5. To include the new or selected attribute on a COA, set the **Include in COA independent of quality order** option to *Yes*.
6. Repeat the previous steps for every other attribute that you want to include on the COA.

After the customer-specific COA requirements are set for a test group, they automatically apply to all quality orders that use that test group. However, they can be adjusted on individual quality orders as required.

Access customer-specific COAs

The following table describes the different ways to access customer-specific COAs.

Expand table

Access method	Navigation path	Details
Directly from the menu	Go to **Inventory management** > **Inquiries** > **Quality management** > **Certificate of analysis**.	You can print either the standard COA or a customer-specific COA. (You trigger printing of a customer-specific COA by selecting a specific customer account.) In both cases, you must select a quality order.
From a quality order	Go to **Inventory management** > **Periodic** > **Quality management** > **Quality order**, and then, on the Action Pane, select **Inquiries** > **Certificate of analysis**.	You can print either the standard COA or a customer-specific COA. (You trigger printing of a customer-specific COA by selecting a specific customer account.) Because access is from a quality order, that quality order is assumed in both cases.
From an inventory batch	Go to **Inventory management** > **Inquiries** > **Dimensions** > **Batches**, and then, on the Action Pane, select **Inquiries** > **Certificate of analysis**.	You can print either the standard COA or a customer-specific COA. (You trigger printing of a customer-specific COA by selecting a specific customer account.) Because access is from an inventory batch, the COA quality order that is specified on the **General** tab of the inventory batch is used to generate the COA in both cases. If the COA quality order must be updated, you can use the **Reset COA Quality order** button on the Action Pane to change the quality order that drives the details of the COA.
As part of processing the sales order packing slip	Go to **Sales and marketing** > **Common** > **Sales orders** > **All sales orders**, and then, on the Action Pane, on the **Pick and pack** tab, select **Generate** > **Packing slip**. Alternatively, go to **Sales and marketing** > **Periodic** > **Sales update** > **Packing slip**.	Only customer-specific COAs are printed from this process. Depending on the customer who placed the order, a customer-specific COA is printed for every item/batch combination on the order that has a COA quality order specified for it in the batch. A COA is printed only for items that are batch controlled and items where the batch specifies a quality order.

> **Note**
>
> The setting of the option to print customer-specific COAs initially comes from the **Updates** tab of the **Accounts receivable parameters** page. By default, this setting is used for new customers. If you process the packing slip for a single order, the setting of the checkbox on the packing slip processing page comes from the customer. If you process packing slips for multiple orders, you must select the checkbox to print customer-specific COAs.

Last updated on 07/27/2026

Inspect the quality of goods

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to process quality orders. Quality inspections are typically done by a quality clerk.

If the standard demo data is installed, you can use it to complete the procedures in this article. To use the demo data, select the *USMF* legal entity before you begin. You must then confirm purchase order *000016* and post a product receipt. A quality order is automatically generated.

Step 1: Select a quality order

To select a quality order, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.
2. Select the quality order that was generated before you started this procedure.

Step 2: Record test results

To record test results, follow these steps:

1. Select **Results**.
2. Select **Edit**.
3. In the **Result quantity** field, enter a number.
4. In the **Outcome** field, select the desired record. In this example, the result is based on a predefined outcome. Usually, you will record a more specific test result, such as a size or other dimension.
5. Select **Save**.
6. Close the page.

Step 3: Validate the quality order

To validate the quality order, follow these steps:

1. Select **Validate**.
2. In the **Validated by** field, select the user who is doing the inspection.
3. Select **Select**.
4. Select **OK**.
5. Close the page.

Last updated on 12/30/2025

Enter quality test results quickly

The **Quick results entry** page makes the process of entering quality order test results faster and more flexible than the **Quality order line results** page. The **Quick results entry** page consolidates data from all tests into a single view to streamline the result entry process. Quantities are automatically populated based on the quality order quantities. Other details are also populated, such as minimum, maximum, and target values.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

Enter quality test results

To enter quality test results by using the **Quick results entry** page, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.
2. Select a quality order, and then, on the Action Pane, select **Quick results entry**.
3. The **Quick results entry** page shows the tests that are associated with the selected quality order. Use the following buttons on the Action Pane to prepare the page for result entry:
    - **New** and **Delete** – Use these buttons to separate the test results that were entered for different quantities. For example, if the quality order quantity is *5*, you can use the **New** button to enter a total of five lines for the test. In this way, a different test result can be entered for each quantity. The sum of the quantities for each test must equal the total quantity of the quality order.
    - **Split** – Split an existing line into multiple lines. This button opens a dialog where you can set the following fields:

- **Split quantity** – Split the selected line into two lines, one of which has the quantity that you enter here, and the other of which has the remaining quantity. For example, the quantity on the selected line is *8*, and you enter *3* in this field. In this case, the quantity on one of the resulting lines is *3*, and the quantity on the other line is *5*.

- **Result quantity per line** – Split the selected line into multiple lines, each of which has the specified test quantity. For example, the quantity on the selected is *8*, and you enter *2* in this field. In this case, the line is split into four lines, each of which has a quantity of *2*.

  If the quantity can't be divided evenly, the last line has the remainder that is required to ensure that the total quantity equals the original quantity. For example, the quantity on the selected line is *8*, and you enter *3* in this field. In this case, the line is split into three lines. Each of the first two lines has a quantity of *3*, and the last line has a quantity of *2*, to produce a total quantity of *8*.

  If the item is a catch weight item, you must enter the catch weight quantity instead of the standard quantity.

- **Skip** – Skip the selected test so that it doesn't affect whether the quality order passes of fails.

- **Validate** – Validate the selected line, and evaluate whether it passes or fails. This action isn't required. Validation for all rows is automatically done when the page is closed.

- **Reset test result** – Eliminate the test result for the selected line.

4. Fill in test results for each line.

5. Select the **Back** button to return to the **Quality orders** page.

6. On the Action Pane, select **Validate** to finalize the quality order with the appropriate status, based on the line results (*Pass* or *Fail*).

Last updated on 07/23/2026

Create and process nonconformances

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to perform nonconformance management based on an existing quality order. Typically, a quality clerk manages nonconformances. As a prerequisite, you must have a quality order available. For information about how to create a quality order, see Inspect the quality of goods.

Before a user can process the approval of a nonconformance, a worker must be assigned to them in the **Person** field on the **Users** page. Additionally, before the user can use the document notes, the **Enable document handling** option must be set to *Yes* in their user options.

Create a nonconformance

To create a nonconformance, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.
2. On the Action Pane, select **New**.
3. In the **Create non conformance** dialog, in the **Problem type** field, select the type of problem that you found during the inspection process.
4. Select **OK**.

Approve or reject a nonconformance

To approve or reject a nonconformance, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.

2. In the list, select the nonconformance that you want to update.

> ⓘ Note
>
> You can add a correction only to nonconformances that are approved.

3. On the Action Pane, select **Functions** > **Approve non conformance** to approve the nonconformance or **Functions** > **Refuse non conformance** to reject it. You can associate approved nonconformances with related operations. By using this association, you can record work that is done as part of the nonconformance handling and the processing of correction handling.

4. You're prompted to confirm your selection. Select **Yes** to continue.

Add operations and other details to nonconformances

After you approve a nonconformance, you can add related operations and specify additional information about them. You can only add related operations to nonconformances that are approved.

Besides the basic information, you can add the following details to an operation:

- **Items** – You can create a list of items that are consumed to perform the correction. For example, the items might be products that are required to repair equipment or ingredients that are required to rework a finished product.
- **Quality charges** – You can create a list of charges that are incurred or billed out to external sources.
- **Timesheet** – You can create a log of the time that each worker spends on the operation.

The remaining settings are optional. The cost for each item, quality charges, and the timesheet are summed and shown on the operation. You can't directly edit the **Cost** field on the operation.

Create an operation for a nonconformance

To create an operation for a nonconformance, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.

2. In the list, select the nonconformance that you want to update.

> ⓘ Note

> You can add or update operations only for nonconformances that are approved.

3. On the Action Pane, select **Related operations**.
4. On the **Related operations** page, add rows to the grid in one or both of the following ways:
    - To add a single row to the grid, select **New** on the Action Pane.
    - To add all the operations from a predefined operations group, select **Add operations from a group** on the Action Pane.
5. For each new row, set the following fields:
    - **Operation** – Select the code of the operation that will be performed for the nonconformance.
    - **Reason** – Enter a detailed description that explains why the operation is required.
    - **Sales order** – If the selected operation code is related to the sales order type, select the sales order that you're linking the operation to.
    - **Purchase order** – If the selected operation code is related to the purchase order type, select the purchase order that you're linking the operation to.
6. Close the pages.

Add items to an operation

To add items to an operation, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.
2. In the list, select the nonconformance that you want to update.

> ⓘ **Note**
>
> You can add or update operations only for nonconformances that are approved.

3. On the Action Pane, select **Related operations**.
4. On the **Related operations** page, select the operation that you want to add items to.
5. On the Action Pane, select **Items**.

6. On the **Related operations** page, on the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:

    - **Item number** – Select the product that the operation consumes.
    - **Quantity** – Enter the number of items that the operation consumes.

    > ⓘ Note
    >
    > You can view the cost for the item in the **Cost** field on the **General** tab. The cost that is shown comes from the cost that you set up on the **Released product** page. You can't update the cost directly on the **Related operation item** page. The cost of all items that you add on the **Related operation items** page automatically adds to the **Cost** field on the **Related operations** page.

7. Repeat the previous step for each additional item that you must add.

8. Close the pages.

Add quality charges to an operation

To add quality charges to an operation, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.

2. In the list, select the nonconformance that you want to update.

    > ⓘ Note
    >
    > You can add or update operations only for nonconformances that are approved.

3. On the Action Pane, select **Related operations**.

4. On the **Related operations** page, select the operation that you want to add quality charges to.

5. On the Action Pane, select **Quality charges**.

6. On the **Related operation charges** page, on the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:

- **Charges code** – Select the quality charge that you want to add.
- **Description** – Enter a detailed description of the charge.
- **Charges value** – Enter the amount of the charge.

7. Repeat the previous step for each additional charge that you must add.

8. Close the pages.

> ⓘ **Note**
>
> The system automatically sums the amount in the **Charges value** field for all quality charges and adds it to any other amounts in the **Cost** field on the **Related operations** page.

Create a timesheet on an operation

To add a timesheet to an operation, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.

2. In the list, select the nonconformance that you want to update.

   > ⓘ **Note**
   >
   > You can add or update operations only for nonconformances that are approved.

3. On the Action Pane, select **Related operations**.

4. On the **Related operations** page, select the operation that you want to add a timesheet to.

5. On the Action Pane, select **Timesheet**.

6. On the **Related operation time sheets** page, on the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:

   - **Date** – Specify the date when work was done. By default, this field is set to the current date.
   - **Worker** – Select the worker who did the work. By default, this field is set to the worker that is assigned to the current user.
   - **Operation hours** – Enter the number of hours that were worked on the selected operation.

- **Invoiced** – Select this checkbox if the time has been charged to a customer or vendor on an invoice.

> ⓘ Note
>
> You can view the cost in the **Cost** field on the **General** tab. The system calculates the cost by using the rate that you define on the **Inventory management parameters** page.

7. Repeat the previous step for each additional timesheet line that you must add.
8. Close the pages.

> ⓘ Note
>
> The amount in the **Cost** field is summed for all timesheet lines and added to any other amounts in the **Cost** field on the **Related operations** page.

Add a correction to a nonconformance

To add a correction to a nonconformance, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.
2. In the list, select the nonconformance that you want to update.

   > ⓘ Note
   >
   > You can add a correction only to nonconformances that are approved.

3. On the Action Pane, select **Corrections**.
4. On the **Corrections** page, add rows to the grid in one or both of the following ways:

   - To add a single row to the grid, select **New** on the Action Pane.
   - To add a correction for each diagnostic from a predefined diagnostic group, select **Apply diagnostic group** on the Action Pane.

5. For each new row, set the following fields:

- **Diagnostic** – Select the diagnostic type that identifies the type of correction that you're creating.
- **Worker** – Select the person who is responsible for the correction.
- **Correction priority** – Select an option to indicate how the correction should be prioritized (*Low*, *Normal*, or *High*).
- **Requested date** – Enter the date when the corrective action was requested.
- **Planned date** – Enter the date when the correction is expected to be completed.
- **Short term solution** – Select this checkbox if the corrective action corrects the nonconformance only for a short time.
- **Root cause code** – Select the root cause code that identifies the root cause of the issue that required the correction.

6. Close the pages.

Mark a correction as completed

To mark a nonconformance correction as completed, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.

2. In the list, select the nonconformance that you want to update.

   > ⓘ **Note**
   >
   > You can add a correction only to nonconformances that are approved.

3. On the Action Pane, select **Corrections**.

4. On the **Corrections** page, in the grid, select the correction that you want to mark as completed.

5. On the Action Pane, select **Mark complete**.

6. You're prompted to confirm your selection. Select **OK** to continue. The **Completion date and time** field is set to the current date and time, and the **Completed** checkbox is selected.

7. Close the page.

Reopen a completed correction

To reopen a completed correction, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Non conformances**.
2. In the list, select the nonconformance that you want to update.
3. On the Action Pane, select **Corrections**.
4. On the **Corrections** page, in the grid, select the correction that you want to reopen.
5. On the Action Pane, select **Reopen**.
6. You're prompted to confirm your selection. Select **OK** to continue. The value is cleared from the **Completion date and time** field, and the **Completed** checkbox is cleared.
7. Close the page.

You can now make additional edits or updates to the correction. When you finish, you can mark the correction as completed again.

Close a nonconformance

To close a nonconformance, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.
2. Select a quality order in the grid.
3. On the Action Pane, select **Inquiries** > **Non conformances**.
4. On the **Non conformances** page, select the target nonconformance in the grid.
5. On the Action Pane, select **Functions** > **Close non conformance**.
6. You're prompted to confirm your selection. Select **Yes** to continue.
7. Close the pages.

Related information

- Quality management overview
- Enable quality and nonconformance management
- Workers responsible for approving nonconformances
- Quarantine zones for nonconformances
- Diagnostic types for nonconformances
- Quality charges for nonconformances
- Operations for nonconformances
- Quality management for warehouse processes

Last updated on 05/27/2026

Quarantine orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to use quarantine orders to block inventory.

Quarantine orders block inventory. For example, you might want to quarantine items for quality control reasons. When you quarantine inventory, it gets transferred to a quarantine warehouse.

> **Note**
>
> If you're using warehouse management processes (WMS), quarantine order processing only applies to return sales orders.

Quarantine on-hand inventory items

When you quarantine items, you can either manually create the quarantine orders or set up the system to automatically create them during inbound processing.

To set up the system to automatically generate quarantine orders, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory** > **Item model groups**.
2. Select a relevant model group in the list pane, or create a new model group.
3. On the **Inventory policies** FastTab, select the **Quarantine management** check box.
4. Close the page.
5. In the **Quarantine warehouse** field for the receiving warehouses, specify a default quarantine warehouse.

When an item that is registered as received at the warehouse belongs to a model group where the **Quarantine management** check box is selected, the system generates a quarantine order for it. The quarantine order instructs workers to move the item to the quarantine warehouse.

When you manually create quarantine orders on the **Quarantine orders** page, the item doesn't need to be set up for quarantine management in the associated item model group. For this process, you must specify the on-hand inventory that you want to quarantine and the quarantine warehouse that you want to use. You can use the quarantine order statuses to help plan the process.

Quarantine order statuses

Quarantine orders can have the following statuses:

- Created
- Started
- Reported as finished
- Ended

Created

When you manually create a quarantine order but don't yet place the item in the quarantine warehouse, the quarantine order has a status of *Created*. This status generates two inventory transactions. One transaction is an issue transaction that can have a status of *On order*, *Reserved physical*, or *Picked*. The other transaction is a receipt transaction that can have a status of *Ordered* or *Registered* at the quarantine warehouse. You can reserve, pick, and register updates to the inventory by using the usual processes.

Started

When a quarantine order has a status of *Started*, the inventory is transferred from the regular warehouse to the quarantine warehouse, and two inventory transactions are generated. One transaction has a status of *Deducted*, and the other transaction has a status of *Received*. At the same time, two inventory transactions are created to handle the return transfer. These transactions aren't dated. One transaction has a status of *Reserved physical*, and the other transaction has a status of *Ordered*.

Reported as finished

To report a started quarantine order as finished, open the order, and select **Report as finished** on the Action Pane. The item is released from quarantine, but it isn't yet moved back to the regular warehouse. You can process the movement back to the regular warehouse through an item arrival journal that you can initialize during the report as finished process.

Ended

When you end a quarantine order, move the item from the quarantine warehouse back to the regular warehouse. The status of the item transaction is set to *Sold* at the quarantine warehouse and *Purchased* at the regular warehouse.

Quarantine order scrap

As part of the quarantine order process, you can scrap inventory. When you process scrap, the status of the inventory is set to *Sold* by an issue transaction from the quarantine warehouse.

Related information

- Inventory blocking

Last updated on 05/27/2026

Inventory blocking

08/14/2025

This article describes how to work with inventory blocking, which is part of the quality inspection process in Supply Chain Management. You can use inventory blocking to prevent items from being processed or consumed.

You can block inventory items in the following ways:

- Manually
- By creating a quality order
- By using a process that generates a quality order
- By using inventory status blocking

Blocking items manually

You can block a quantity of an item by creating a transaction on the **Inventory blocking** page. Only items that are available as on-hand inventory can be blocked manually. For manually blocked quantities, you must decide whether planning activities include expected receipts as an expected on-hand quantity. Expected receipts are blocked items that you expect to be available as on-hand inventory after inspection is completed. You can maintain the expected date. By default, the **Expected receipts** option is selected for items that are blocked through a quality order. You can cancel a manual block on a quantity by deleting the transaction on the **Inventory blocking** page.

Blocking items by creating a quality order

You can specify items that must be inspected by creating a quality order on the **Quality orders** page. When you create a quality order, the quantity that you specify for an item is blocked. The sampling plan that is associated with a quality order controls only the quantity of items that must be inspected, not the quantity that is blocked. The quantity that is entered on the quality order is the quantity that is blocked, regardless of the quantity that the sampling plan specifies should be sent for inspection.

> **Note**
>
> Master planning doesn't support using both the batch expiry date and blocking inventory status features. This could result in double exclusion of on-hand inventory, which can occur during master planning. We recommend that you rely on batch disposition codes, instead of inventory status, for blocking expired batches.

Blocking items by using a process that generates a quality order

If a quality process specifies that an item must be inspected, a quantity of the item is blocked automatically. Therefore, when a quality order is generated automatically, the item sampling plan that is associated with the quality order controls both the quantity of items that is blocked and the quantity that must be inspected. If the **Full blocking** option on the **Item sampling** page is selected, the full quantity of, for example, a purchase order line is blocked during inspection, regardless of the item sampling quantity.

Example

In the following example, a quality order is generated when a purchase order packing slip is posted. On the **Quality associations** page, you specified that posting of a purchase order packing slip is the process that activates a quality order.

Expand table

Setup	User action	Result
A quality association specifies that a quality order must be generated when a purchase order packing slip is posted. The item sampling setup of the quality order specifies that 10 percent of the quantity on the purchase order line must be inspected. Furthermore, because the **Full blocking** option selected in the item sampling setup, the full quantity of the purchase order line must be blocked during inspection, regardless of the quantity that is sent for inspection.	The packing slip is posted.	A quality order is generated. Ten percent of the purchase order quantity for the item is sent to inspection. The full quantity of the purchase order line is blocked.

Blocking items by using inventory status blocking

You can specify which inventory statuses are blocking statuses by using the **Inventory blocking** parameter on the **Inventory statuses** page. You can't use inventory statuses as blocking statuses for production orders, sales orders, transfer orders, outbound transactions, or project integrations. For outbound work, use items that have an available inventory status. If items have a status of **Broken**, and master planning is run on those items, the items are considered missing, and inventory is automatically replenished.

Take care when blocking items that use both inventory status blocking and quality order blocking

You can create a quality order associated with inventory having an inventory status that has its **Inventory blocking** parameter enabled. In this case, the quality order generates an extra inventory blocking record in addition to the one created by the inventory status. Because the quality order inventory blocking will have the parameter **Expected receipts** enabled, this generates an extra *Ordered inventory* transaction that is also blocked by its inventory status. This combination can lead to difficulties in understanding the meaning of generated inventory transactions, because it will appear as though the total blocked quantity is in excess of the total quantity on hand. Let us examine the transactions on an example of a receipt of 10 pieces of item A0001 with a quality order generated to sample 1 piece. The behavior also depends on

whether the option **Reserve ordered items** is enabled on the **Inventory and warehouse management parameters** page.

> **Note**
>
> If you're using inventory status blocking and quality orders together, we strongly recommend having the **Reserve ordered items** option enabled.

Example with "Reserve ordered items" enabled

When **Reserve ordered items** is enabled, all inventory blocking transactions has a status of either *Reserved physical* or *Reserved ordered*.

Expand table

Inventory transaction reference	Receipt	Issue	Quantity	Site	Warehouse	Inventory status	Location	License plate	Comment
Purchase order	Purchased		10 Pcs	2	24	Blocking	RECV	receiptLp1	
Inventory blocking		Reserved physical	-9 Pcs	2	24	Blocking			Transaction generated by the inventory status blocking
Inventory blocking		Reserved physical	-1 Pcs	2	24	Blocking	RECV	receiptLp1	Transaction generated by the quality order sampling blocking
Inventory blocking	Ordered		1 Pcs	2	24	Blocking	RECV	receiptLp1	Expected receipts from the quality order sampling blocking
Inventory blocking		Reserved ordered	1 Pcs	2	24	Blocking			Transaction generated by the inventory status blocking

Example with "Reserve ordered items" disabled

When **Reserve ordered items** is disabled, the expected receipts can't be reserved because they are in status *Ordered*, so some blocking are left in status *On order*.

Expand table

Inventory transaction reference	Receipt	Issue	Quantity	Site	Warehouse	Inventory Status	Location	License Plate	Comment
Purchase order	Purchased		10 Pcs	2	24	Blocking	RECV	receiptLp1	
Inventory blocking		Reserved physical	-9 Pcs	2	24	Blocking			Transaction generated by the inventory status blocking
Inventory blocking		Reserved physical	-1 Pcs	2	24	Blocking	RECV	receiptLp1	Transaction generated by the quality order sampling blocking
Inventory blocking	Ordered		1 Pcs	2	24	Blocking	RECV	receiptLp1	Expected receipts from the quality order sampling blocking
Inventory blocking		On order	1 Pcs	2	24	Blocking	RECV	receiptLp1	Transaction generated by the inventory status blocking

Note the difference in transaction status and dimensions between the two cases. For this reason, we recommend enabling the **Reserve ordered items** option.

Expected receipts from quality orders that sample blocked inventory are disabled

To simplify the inventory transactions when quality orders that sample inventory are blocked as a consequence of inventory status, the system disables expected receipts from such quality orders. Because inventory status blocking immediately blocks the expected receipt, there's no reduction of on-hand inventory because of this.

Related information

- Create and maintain an inventory blocking
- Quality management processes
- Inspect the quality of goods

Create and maintain an inventory blocking

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to use an inventory blocking to prevent physical on-hand inventory from being reserved by other outbound source documents. Before you start the procedures in this article, you must have an item that physical on-hand inventory is available for.

Block inventory

To create an inventory blocking record so that inventory is blocked, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Inventory blocking**.
2. On the Action Pane, select **New**.
3. On the header of the new blocking record, set the **Item number** field to the item that you want to block, and enter a description.
4. On the **General** FastTab, in the **Quantity** field, enter the number of items to block.
5. On the **Inventory dimensions** FastTab, specify the site and warehouse where the items that you want to block are currently located.
6. On the Action Pane, select **Save**.

Update the conditions of the inventory blocking

To update an inventory blocking record, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Inventory blocking**.
2. In the list pane, select the relevant blocking record.
3. Edit the record as required. For example, you might change the value of the **Expected date** field to indicate when the blocked inventory is expected to become available for reservation. If the **Expected receipts** option is selected, the date will appear on the expected transaction. (The **Expected receipts** option is selected by default when you manually create a blocking record.)
4. On the Action Pane, select **Save**.

Unblock inventory

To remove an inventory blocking record so that inventory is unblocked, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Inventory blocking**.
2. In the list pane, select the relevant blocking record.
3. On the Action Pane, select **Delete**.
4. You're prompted to confirm the operation. Select **Yes** to continue.
5. Close the page.

Last updated on 12/30/2025

Corrective and preventive action (CAPA) management overview

Corrective and preventive action (CAPA) management lets you create and maintain records of the actions taken to manage and resolve nonconformities or defects. By planning, tracking, and analyzing CAPA cases, you can develop efficient processes that can also help resolve similar issues. CAPA management provides a formal and controlled process for managing various types of issues that arise from manufacturing, engineering, quality, or data quality sources.

CAPA management provides the following benefits:

- It implements a case management infrastructure for identifying, correcting, and preventing future issues.
- It leads to corrective and/or preventive actions.
- It provides a tool that drives continuous improvement.
- It provides a closed-loop system that meets current Good Manufacturing Practice (cGMP) requirements and United States Food and Drug Administration (FDA) expectations. Both cGMP and FDA mandate repeatable, systematic failure investigation and root cause analysis.

CAPA management in Microsoft Dynamics 365 Supply Chain Management lets you capture issues and investigate, evaluate, and determine the root cause or causes of each issue. In this way, you can find a solution for the issue and decide how best to implement it. You can also capture and track corrective and preventive actions, and then monitor and review their effectiveness.

CAPA management provides features that automate the process to help drive continuous improvement. This approach supports the core disciplines for various quality standards, such as Six Sigma and ISO-9000. The solution is role based, supports electronic records and electronic signatures, and is fully compliant with part 11 of Title 21 of the Code of Federal Regulations (21 CFR Part 11).

CAPA management features in Supply Chain Management provide the following functionality:

- **Unlimited and flexible CAPA processes** – A CAPA process is a user-defined list of activities that must be performed to bring a CAPA case to a satisfactory conclusion.
- **A workflow-driven process** – The system automatically assigns CAPA work, based on the definition of your CAPA processes and CAPA worker groups.

- **Flexible assignment of activities** – Associate specific users or groups with each part of a CAPA process.
- **Integration with Outlook** – When work is assigned, the system can automatically send emails to notify the appropriate user that work needs their attention.
- **Create CAPA cases at the point of need** – The person who first recognizes an issue needs attention can create a CAPA case at the point of need in the system. The system automatically links each CAPA case to the record that it was created from (for example, the specific customer, quality order, or production order that was open when the case was created).
- **Trending analysis** – The CAPA workspace provides analytical tools to help you analyze CAPA trends.

The following table describes tasks that employees can perform when they work with CAPA management.

Expand table

Task	Description
Create a CAPA case	Create a new CAPA case record for a customer, vendor, product, or employee.
Work with a CAPA case	Add detailed information, such as activities, to a new or existing CAPA case.
Update the status of a CAPA case	Update the status of a CAPA case (for example, from *Opened* to *In process*).
Update the stage of a CAPA case	Update the stage in the process of working a CAPA case (for example, from *Evaluation* to *Investigation*).
Close a CAPA case	Change the status of an *Opened* case to *Closed* to indicate that the issue was resolved.
Rank a knowledge article	Rank a knowledge article to indicate whether it helped close a case.

Last updated on 06/22/2026

CAPA management administration

This topic explains how to enable and set up corrective and preventive action (CAPA) management features in Microsoft Dynamics 365 Supply Chain Management.

Prerequisites

Before you can use CAPA management features in Supply Chain Management, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.44 or later.
- The feature named *Advanced quality management* must be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.
- If you want to require electronic signatures when CAPA cases are closed, the feature named *Electronic signature improvements* must also be turned on in Feature management. As of Supply Chain Management version 10.0.47, this feature is turned on by default.

Set up CAPA parameters

Before users can set up and use CAPA features, you must configure a few administrator settings.

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.

2. On the **Quality management** tab, set the following fields:

    - **CAPA administrator** – Select the worker who should administer the CAPA program. For each CAPA worker group, the CAPA administrator that you specify here can be selected as an automatic group member and/or a default worker.
    - **Use case management workflow** – Select whether the case management workflow can work with CAPA cases.
    - **Require CAPA type** – Set this option to *Yes* if users must specify a CAPA type when they create a CAPA case. If you set this option to *No*, a CAPA type is optional.
    - **Empty case category** – Select the category to apply to new, empty cases. You must select a value if you want to use CAPA.
    - **Require root cause** – Set this option to *Yes* if users must specify a root cause when they create a CAPA case. If you set this option to *No*, a root cause is optional.

- **Copy documents** – Select whether the system should copy document handling details from CAPA processes to new CAPA cases.

Load standard worker groups and processes from a template

To get started quickly, you can load a set of standard CAPA processes and worker groups from a template that's provided with Supply Chain Management. You can use the standard records as-is or modify them to meet your specific needs. Learn more about how to work with CAPA processes in Set up CAPA case components.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **Create standard CAPA process templates**.

2. In the **Create standard CAPA process templates** dialog, set the following fields:

   - **Load** – Set this option to *Yes*.
   - **Name** – Leave this field set to *CAPAProcessTemplate* (the provided default template) unless you have another template that you want to use instead.

3. Select **OK**.

Require an electronic signature to close, cancel, or reopen a CAPA case

You can set up the CAPA features to require an electronic signature from any user who closes, cancels, and/or reopens a CAPA case.

An electronic signature confirms the identity of the person who is about to start or approve a computing process. In some industries, an electronic signature is as legally binding as a handwritten one.

Set up the electronic signature feature

CAPA cases use the standard electronic signature feature that's provided in finance and operations apps. Before you can use this feature, you must configure it and issue certificates to each relevant user. Learn more in Electronic signatures overview and Set up electronic signatures.

Turn the electronic signature requirement for CAPA cases on or off

To set up the electronic signature requirements for CAPA cases, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature requirements**.
2. In the left pane, select the row where the **Name** field is set to *Close/Cancel CAPA case*.
3. In the right pane, set the **Signature required** option to *Yes* to turn on the feature. Set it to *No* to turn off the feature.

Status changes that require a signature

The following table shows which status changes for CAPA cases require a signature when the electronic signature requirement is turned on.

Expand table

Initial status	New status	Signature required
Opened	In Process	No
Opened	Canceled	Yes
In Process	Opened	No
In Process	Canceled	Yes
In Process	Closed	Yes
Canceled	Opened	Yes
Canceled	In Process	Yes
Closed	Opened	Yes
Closed	In Process	Yes

Last updated on 07/27/2026

Set up CAPA case components (preview)

Article • 05/09/2025

[This article is prerelease documentation and is subject to change.]

Before you can start to create and work with corrective and preventive action (CAPA) cases, you must set up the components that are needed to categorize and process them.

Set up CAPA worker groups

CAPA worker groups let you group users together for CAPA purposes. You can assign a CAPA worker group to each activity and stage in your CAPA process. The system uses worker groups to validate worker assignments.

Each CAPA worker group can include a default worker. This worker is used if no specific worker is assigned.

When a CAPA case, CAPA process stage, or CAPA process activity that is assigned to a worker or worker group becomes active, the system sends the worker or worker group an email notification.

> **Tip**
>
> To get started quickly, you can load a set of standard CAPA worker groups from a template that is provided with Microsoft Dynamics 365 Supply Chain Management. You can use the standard groups as-is or modify them to meet your specific needs, as described in the rest of this section. Learn more in CAPA administration (preview).

To set up CAPA worker groups, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA worker groups**.
2. Use the buttons on the Action Pane to add new worker groups or edit existing ones as required. (You can also delete existing groups, but only if they aren't assigned to CAPA case activities.)
3. On the header of the new or selected worker group, set the following fields:
    - **CAPA worker group** – Enter a name for the worker group. Typical examples include *Investigator*, *Implementer*, *Approver*, and *Verifier*.
    - **Name** – Enter a short description of the worker group.

4. On the **Group settings** FastTab, set the following fields to set up details of the group. These details include whether all members of the group can change the stage of a CAPA case, and whether the CAPA administrator should be an implied member and/or default worker for the group.

    - **Allow any member to change stage** – Select this checkbox if all workers in the CAPA worker group are allowed to change the stage of CAPA cases that are assigned to the group. Clear this checkbox if you want to limit which workers in the group can move CAPA cases to the next stage.
    - **Implied member of this worker group** – Select this checkbox if the CAPA administrator who is assigned on the **Inventory management** parameters page should be an implied member of the CAPA worker group. Alternatively, the CAPA administrator can be assigned directly to the CAPA worker group. However, if the CAPA administrator is changed, extra maintenance is required.
    - **Default worker** – Select this checkbox if the CAPA administrator should be the default worker for this CAPA worker group.

5. On the **Worker assignments** FastTab, add or remove workers in the group.

    - To add a worker, use the **New** button on the **Overview** tab. Alternatively, on the **Assign** tab, use the right arrow button to move workers into the **Selected workers** list.
    - To remove a worker, use the **Delete** button on the **Overview** tab. Alternatively, on the **Assign** tab, use the left arrow button to move workers out of the **Selected workers** list.

6. On the **Overview** tab of the **Worker assignments** FastTab, set the following fields for each worker in the group:

    - **Allowed** – If the **Allow any member to change stage** checkbox is cleared on the **Group settings** FastTab, select this checkbox for each worker that should be allowed to move CAPA cases to the next stage. If the **Allow any member to change stage** checkbox is selected, this checkbox is unavailable.
    - **Default worker** – If the **Default worker** checkbox is cleared on the **Group settings** FastTab, select this checkbox for the worker who should be the default worker for the CAPA worker group. The default worker is used when CAPA cases are created and moved to the next stage. Each CAPA worker group can have a maximum of one default worker. Therefore, if you select this checkbox for a worker, it's automatically cleared for all other workers, including the administrator who is assigned on the **Group settings** FastTab.

Set up CAPA categories

CAPA categories let you group similar case types together. By organizing CAPA cases into categories, you help employees identify known solutions, such as knowledge articles, when similar issues frequently occur. This information is also used by the *Trending analysis by category* feature. Therefore, managers can review whether the frequency of specific issues is declining over time.

To set up CAPA categories, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA categories**.
2. Use the buttons on the Action Pane to add new CAPA categories as required. (You can also delete existing categories, but only if they aren't assigned to active CAPA cases.)
3. For each new CAPA category, set the following fields:
    - **CAPA category** – Enter the name of the CAPA category. The name must be unique.
    - **Description** – Enter a description of the CAPA category.
    - **CAPA subcategory type** – Select a subcategory for the CAPA category. The list includes several predefined subcategories (*None*, *Customer*, *Vendor*, and *Product*). It also includes the *User defined* subcategory type. If you select this subcategory type, users can select from a set of custom subcategories when they create a CAPA case that uses this CAPA category. The next procedure explains how to set up custom CAPA subcategories.

To set up custom CAPA subcategories for a category that uses the *User defined* subcategory type, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA categories**.
2. Select the category that you want to define subcategories for. The selected category must have the **CAPA subcategory type** field set to *User defined*.
3. On the Action Pane, select **User defined CAPA subcategory**.
4. Use the buttons on the Action Pane to add new subcategories as required. (You can also delete existing subcategories.)
5. For each subcategory, set the following fields:
    - **CAPA subcategory** – Enter the name of the subcategory.
    - **Description** – Enter a short description of the subcategory.

Set up CAPA processes

CAPA processes are hierarchical processes that let you manage and resolve CAPA cases. Each CAPA process consists of a series of sequential stages. Each stage can include one or more automatically generated activities. When you create a CAPA case, you assign the CAPA process that is required to guide it to completion.

> **Tip**
>
> To get started quickly, you can load a set of standard CAPA processes from a template that is provided with Supply Chain Management. You can use the standard processes as-is or modify them to meet your specific needs, as described in the rest of this section. Learn more in CAPA administration (preview).

Manage and approve CAPA processes

To name, describe, and approve CAPA processes, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA processes**.

   The **CAPA processes** page shows a list of existing CAPA processes. For each selected process, the following information is shown:

   - **Name** – The name of the CAPA process. For existing processes, this field is read-only.
   - **Description** – A short description of the CAPA process.
   - **Approved** – A value that indicates whether the CAPA process is approved for use in CAPA case management. Only approved CAPA processes are available to users when they create and process a CAPA case. You can leave this option set to *No* while you develop a new process, and then set it to *Yes* when you're ready to start to use it.

2. Use the buttons on the Action Pane to add new CAPA processes or edit existing ones. (You can also delete existing processes.)

   - To add a new process, select **New**.
   - To edit the name, description, or approval status of a selected process, select **Edit**.
   - To delete a selected process, select **Delete**.

Configure CAPA process details

To define the stages and activities that make up a CAPA process, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA processes**.

2. Select or create the CAPA process that you want to work with.

3. On the Action Pane, on the **Process** tab, select **Details**.

   You can use the page that appears to create or edit the structure of the new or selected process. The page has two panes.

   - The left pane shows a hierarchical list of stages and activities that make up the process. The top level is for the process itself. The second level is for the stages. One or more activities can then be nested below each stage. Usually, one level of stages is enough. However, you can create deeper levels for substages as you require.
   - The right pane shows details about the selected stage or activity. Workers must complete the stages in order, from top to bottom. However, the activities that belong to a stage can be done in any order or simultaneously. Activities can be set as mandatory. In this case, they must be completed before the process can move to the next stage.

   > **Tip**
   >
   > If you want to create a new process that is similar to an existing process, you can get started quickly by copying the existing process. On the Action Pane, on the **Process** tab, select **Copy from**. Then select the process that you want to copy.

4. Set up the top-level stages. To add a new stage at the bottom of the current stage list, select the top-level heading (the process name), and then, on the Action Pane, select **New** > **Create level**. (To create a substage for an existing stage, select the parent stage, and then select **New** > **Create level**.)

5. For each new stage, review or enter the following information in the right pane:

   - **Name** – This field is read-only. For a new stage, its value comes from the value that you enter in the **Purpose** field. It can't be changed later.
   - **Activity number** – This field is read-only. The value is automatically generated.
   - **CAPA worker group** – Select or view the CAPA worker group that represents the group of CAPA workers who are responsible for the activity. By default, this group is also assigned to the activities that belong to the stage. However, you can override the group for each stage as required.
   - **Responsible** – Select the worker from the selected CAPA worker group who is responsible for the activity. If you don't select a worker, the designated default worker for the worker group is used when a CAPA case is processed.
   - **Notify group** – Specify who should be notified by email when this stage becomes active. Select *Yes* to notify the whole worker group. Select *No* to notify only the

responsible worker. Select *Inherit* to inherit the setting from the parent stage in the hierarchy.
    - **Email status updates** – Specify whether email should be sent to the responsible worker or worker group when this stage is closed or reopened. The available values are *Yes*, *No*, and *Inherit*.
    - **Print on report** – Select whether information about this stage should be included on the printed CAPA case report. The available values are *Yes*, *No*, and *Inherit*.
    - **Check for required activities** – Select this checkbox if activities that belong to this stage and that are marked as mandatory must be completed before the stage can be closed.

6. Add activities to each stage. To add a new activity at the bottom of the current activity list for a stage, select the stage, and then, on the Action Pane, select **New** > **Create action**, **New** > **Create appointment**, **New** > **Create event**, or **New** > **Create task**, depending on the type of activity that you want. All the activity types work in nearly the same way, but slightly different scheduling settings are available for each.

7. For each new activity, enter the following information in the right pane:

    - The **Name**, **Purpose**, and **Activity number** fields work the same way that they work for stages.
    - Use the **Required** field to mark an activity as mandatory. If the **Check for required activities** field is set to *Yes* for the parent stage, mandatory activities must be completed before the parent stage can be closed.
    - Use the scheduling fields (such as **Start time** and **End time**) to set the start and end dates and times for the activity. These fields are used to calculate when the activity should be performed relative to the day when the activity is created. The availability of scheduling fields varies by activity type.
    - Set the notification options as required. All these settings work the same way that they work for stages. By default, they are set to inherit their setting from the parent stage.

8. Continue to work until you finish setting up all the elements that you need for your process. Then, on the Action Pane, select **Save**.

View all cases that use a selected CAPA process

To view all CAPA cases that use a selected CAPA process, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA processes**.
2. Select or open the CAPA process that you want to view cases for.
3. On the Action Pane, on the **Process** tab, select **All cases** or **All CAPA cases**.

Set up CAPA resolution types

CAPA resolution types let users select a resolution reason when they resolve a CAPA case. You can use CAPA resolution types to identify and categorize typical reasons why CAPA cases can be resolved.

To set up CAPA resolution types, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA resolution types**.
2. Use the buttons on the Action Pane to create new CAPA resolution types as required. (You can also delete existing resolution types.)
3. For each new CAPA resolution type, enter a name and a description.

Set up CAPA sources

CAPA sources let users select a source for each CAPA case. You can use CAPA sources to track the source of each case, such as *Internal audit*, *External audit*, or *Ad-hoc testing*.

To set up CAPA sources, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA sources**.
2. Use the buttons on the Action Pane to create new CAPA sources as required. (You can also delete existing sources.)
3. For each new CAPA source, enter a name and a description.

Set up CAPA types

CAPA types let users select a type to categorize each CAPA case.

To set up CAPA types, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA types**.
2. Use the buttons on the Action Pane to create new CAPA types as required. (You can also delete existing types.)
3. For each new CAPA type, enter a name and a description.

Set up major/minor classification codes

CAPA types let users select a *major/minor classification code* for each CAPA case.

To set up major/minor classification codes, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **Major/minor classification codes**.
2. Use the buttons on the Action Pane to create new major/minor classification codes as required. (You can also delete existing classification codes.)
3. For each new CAPA classification code, enter a name and a description.

Set up CAPA root causes

CAPA root cause codes let users select the root cause of an issue when they resolve a CAPA case.

To set up CAPA root cause codes, follow these steps.

1. Go to **Inventory management** > **Setup** > **CAPA management** > **CAPA root causes**.
2. Use the buttons on the Action Pane to create new CAPA root causes as required. (You can also delete existing root causes).
3. For each new CAPA root cause, enter a name and a description.

Manage CAPA cases (preview)

Article • 05/09/2025

[This article is prerelease documentation and is subject to change.]

Use corrective and preventive action (CAPA) cases to record, update, track, follow up on, and close issues that your customers, vendors, or employees raise.

Use the CAPA case workspace to view, open, manage, and analyze CAPA cases

The CAPA case workspace provides a central location where you can create, view, and manage all CAPA cases in the system.

The workspace provides several tabs where you can view various lists of cases and activities. Here are some examples:

- Cases and activities that are assigned to you
- Open cases and activities
- Cases and activities that are due during the current week

You can also search for CAPA cases that are related to a specific customer, vendor, sales order, purchase order, return order, or released product.

From the workspace, you can quickly view details, set the status of a listed case or activity, move a case to the next process stage, and more.

To open the workspace, go to **Inventory management** > **Workspaces** > **CAPA management**.

View CAPA cases on list pages

To view all CAPA cases that are related to a selected customer, vendor, sales order, purchase order, return order, or released product, open or select the relevant record, and then follow one of these steps.

- **For customers or vendors:** On the Action Pane, on the **General** tab, in the **Accounts** group, select **CAPA case**.
- **For sales orders, purchase orders, or return orders:** On the Action Pane, on the **General** tab, in the **CAPA** group, select **View CAPA cases**.
- **For released products:** On the Action Pane, on the **General** tab, in the **Maintain CAPA cases** group, select **View CAPA cases**.

To view and work with all CAPA cases in the system, follow one of these steps.

- Go to **Inventory management** > **Periodic tasks** > **CAPA management** > **All CAPA cases**.
- Go to **Common** > **Common** > **CAPA management** > **All CAPA cases**.

To view and work with only the CAPA cases that you're responsible for, follow one of these steps.

- Go to **Inventory management** > **Periodic tasks** > **CAPA management** > **My CAPA cases**.
- Go to **Common** > **Common** > **CAPA management** > **My CAPA cases**.

FastTabs on the CAPA case details page

When you open a CAPA case record from the workspace or any list page, the details page provides the following FastTabs. Use the fields and settings on these FastTabs to view or edit information about the CAPA case.

- **General** – View or edit general information about the selected CAPA case. Many of the values on this FastTab are set when the case is created.
- **Email notifications** – View, add, and edit people who should be emailed when selected events occur for the current CAPA case (for example, when the status of the case or one of its activities is changed to *Closed*, *Canceled*, or *Reopened*).
- **Case log** – View, add, or edit information about the CAPA case. The system automatically adds log entries when specific events occur (for example, when the case is created, its status is changed, or activities are completed). Additionally, you can manually add log entries.
- **Associations** – View or edit information about records that are associated with the CAPA case. Examples of these records include sales order, item, vendor, and batch order records. Often, one or more of these associations are automatically set, based on where the user was when they created the case. However, you can add more associations as you work with the case.
- **Knowledge article** – View or edit information about knowledge articles that are associated with the CAPA case.
- **Administration** – View administrative details about the CAPA case. Examples include the date when the case was opened and the user who opened it.
- **Resolution** – View or edit information about how the CAPA case was resolved.

Create a new CAPA case

You can create CAPA cases from many different pages in the system, including pages where you work with records that CAPA cases might originate from. Examples of those records

include customer, vendor, sales order, purchase order, return order, and released product records. You can also create CAPA cases from the **All CAPA cases** and **My CAPA cases** pages. When you create a CAPA case from a related or originating record, the case is automatically linked to that record.

1. Go to the customer, vendor, sales order, purchase order, return order, or released product record that the CAPA case originates from.

2. Depending on the page that you're working on, follow one of these steps.

    - **For customers or vendors:** On the Action, on the **General** tab, in the **New** group, select **CAPA case**.

    - **For sales orders, purchase orders, or return orders:** On the Action Pane, on the **General** tab, in the **CAPA** group, select **Create CAPA case**.

    - **For released products:** On the Action Pane, on the **General** tab, in the **Maintain CAPA cases** group, select **Create CAPA case**.

    - **For nonconformance records:** On the Action Pane, select **Functions** > **Create CAPA case**.

    - **From the All CAPA cases or My CAPA cases page:**
        - To create a CAPA case that isn't linked to an existing case, on the Action Pane, select **New** > **CAPA case**.
        - To create a CAPA case that is linked to an existing case, select the parent case in the grid, and then, on the Action Pane, select **New** > **Dependent CAPA case**.

    - **From the CAPA management workspace:** On the Action Pane, select **Create new** > **CAPA case**. Alternatively, on the **CAPA case summary** FastTab, select **New CAPA case**.

3. In the **New CAPA case** dialog, on the **General** tab, set the following fields:

    - **Name** – Enter or select the name of a person from the global address book to associate that person with the CAPA case. If you started from a customer, vendor, sales order, purchase order, or return order, this field is automatically set based on the record that was selected when you created the CAPA case.
    - **Case ID** – The system-generated identification number of the CAPA case.
    - **Status** – Select the status of the CAPA case. The default status is *Opened*. This status indicates that the case isn't being worked on yet. When work on the case begins, you or the assigned worker should change the status to *In process*.
    - **Description** – Enter a short description of the CAPA case. The description is used to identify the case in lists and on reports. It's also shown as the heading of the case

when you open it.

- **Priority** – Select the priority of the CAPA case.
- **Parent case** – If you want to associate the CAPA case with another case, select the parent case. If you selected a parent case when you created this case, the association is automatically set.
- **CAPA category** – Select a category for the CAPA case. The category is used to analyze trends for reporting purposes.
- **CAPA subcategory** – If subcategories are defined for the category that you selected, select one. The subcategory is used to analyze trends for reporting purposes.
- **Description** – Enter a brief description of the CAPA case.
- **CAPA process** – Select the CAPA process that should be followed to resolve this case. The CAPA process guides workers who are assigned to the case through the steps that are required to resolve it. It defines a series of stages. These stages must be completed in order. Within each stage, the CAPA process defines activities that should be completed before the process is moved to the next stage.

4. If you want to add a log entry to the case, on the **Case log** tab, enter the details of the log entry. You can view and add log entries from the details page for a CAPA case.

5. Select **OK** to create the case and open its details.

Work through a CAPA process

A CAPA process guides workers who are assigned to a case through the steps that are required to resolve it. It defines a series of stages that must be completed in order. Each stage can include one or more activities. The activities can be completed in any order or simultaneously. Some activities might be mandatory. All mandatory activities must be completed before the process can be moved to the next stage.

Based on the configuration of the CAPA process, a worker or worker group is automatically assigned to each stage and activity. The system uses email to notify each responsible worker or worker group when a case, stage, or activity is assigned to them. The email includes links so that the case, stage, or activity can be opened directly from the message. Workers can always track which cases and activities are assigned to them by opening the **CAPA management** dashboard.

CAPA processes are defined by an administrator or manager. The **CAPA process** field of the CAPA record is used to assign CAPA processes to cases as they are created. Your company can define as many CAPA processes as it needs. Learn more in Set up CAPA case components (preview).

The workflow for CAPA processes consists of the following steps.

1. Create a CAPA case, and assign a CAPA process to it. The case initially has a status of *Opened*.

2. Select or open the case (for example, from the **CAPA management** workspace). Then select **Change status** > **In process**. (If you're in the **CAPA management** workspace, the button is on the toolbar. If you're on the details page or another page, it's on the **CAPA case** tab of the Action Pane.) This status change indicates that the case is being actively worked on and enables it to move through the various stages of the process.

3. The process starts with the first stage (for example, *Definition*). The system generates the activities that are configured for the stage. (For example, there might be a single activity that is named *Identify the problem*.)

4. The system emails the **Employee responsible** contacts for the case, the active stage, and each active activity that belongs to that stage.

5. Workers can now work on each activity that they are assigned to, enter notes on the details page for the activity, and fill in information as required. Employees can access the details page for an activity in several ways:

    - Select a link in the email that was sent to them when the activity was generated.
    - Open the **CAPA management** workspace, and browse the activities that are listed on the various tabs (such as **My open activities**).
    - Open the **CAPA management** workspace, and select any tab that lists CAPA cases (such as **My open cases**). Select a case, and then, on the toolbar, select **View activities**. A list shows all activities for the selected CAPA case that belong to stages that have been started. Select an activity to open its details page.
    - Open the details of a CAPA case. On the Action Pane, on the **CAPA case** tab, select **Case process**. In the process tree, expand a stage, and select an activity. (The tree shows activities only for stages that have been started). Select the **Activity number** link to view the details for the selected activity.

6. As workers finish working on each activity, they close it (that is, mark it as completed). There are several ways to close an activity:

    - In the **CAPA management** workspace, select the activity on any tab where it appears. Then, on the toolbar, select **Set as complete**.
    - On the activity details page, on the Action Pane, on the **Activity** tab, select **Set as complete**.
    - On the activity details page, on the **Status** FastTab, set the **Closed** option to *Yes*.

7. After all mandatory activities for the current stage are closed, an employee who has the required permissions can move the case to the next stage. You can skip a stage as

required. However, after you move to a stage, you can't return to a previous one. There are several ways to move a CAPA case to the next stage in its process:

- In the **CAPA management** workspace, select the CAPA case on any tab where it appears. On the toolbar, select **Change stage**, and then select the stage that you want to move the case to.
- On the CAPA case details page, on the **CAPA case** FastTab, select **Change stage**, and then select the stage that you want to move the case to.

8. The selected new stage begins. The system generates the activities that are configured for the stage, sends out emails to the responsible employees, and updates the **CAPA management** workspace, just as it did for the first stage in the process.

9. The process continues until all the required activities are completed and you're ready to close the case. Often, you begin by opening the CAPA case details and filling in the fields on the **Resolution** FastTab to document the outcome of the case. You then mark the case as closed. There are several ways to close a CAPA case:

    - In the **CAPA management** workspace, select the CAPA case on any tab where it appears, and then, on the toolbar, select **Set as complete**.
    - On a CAPA case list or details page, on the Action Pane, on the **CAPA Case** tab, select **Change status** > **Closed**.

> **Tip**
>
> To get a graphical overview of the process hierarchy for the current CAPA case, follow these steps.
>
> 1. Open the details page for the relevant CAPA case.
> 2. In the rightmost pane, select the **Related information** button ⓘ to open the **Related information** FactBox pane.
> 3. On the **Process tree** FastTab, review the list of stages in the current process. You can expand any stage that has been started to view the activities that belong to it. Select and hold (or right-click) an activity to open its details.

Analyze CAPA case statistics

Microsoft Dynamics 365 Supply Chain Management provides graphical displays that you can use to analyze CAPA case generation and progress. Use the statistics to identify trends, frequent issues, bottlenecks, and areas for improvement in your CAPA processes.

Statistical charts are provided in the **CAPA management** workspace and on the **Trending analysis** page. Although the charts in both places are similar, there are a few minor differences in the data that they show.

Analyze statistics using the CAPA management workspace

To view statistics in the **CAPA management** workspace, follow these steps.

1. Go to **Inventory management** > **Workspaces** > **CAPA management**.
2. Select the **Statistics** FastTab. Two charts are shown.
3. Use the **Number of cases by** dropdown list to select the type of statistics that you want to view in the left chart.
4. Use the **Trending of cases by** dropdown list to select the period that you want to view in the right chart.
5. Use the buttons at the top of each chart to select its graphical style (for example, bar chart, line chart, or pie chart).

Analyze statistics using the Trending analysis page

To view statistics on the **Trending analysis** page, follow these steps.

1. Go to **Inventory management** > **Inquiries and reports** > **CAPA management** > **Trending analysis**.
2. On the **Data** tab, select the date range, and turn on each type of chart that you want to view.
3. Select the **Analytics by category** tab to view the selected charts.
4. Use the buttons at the top of each chart to select its graphical style (for example, bar chart, line chart, or pie chart).

Review signatures for a CAPA case

The CAPA features can optionally be set up to require an electronic signature from any user who closes, cancels, and/or reopens a CAPA case. To review the signatures that are applied to a CAPA case, follow these steps.

1. Open or select the CAPA case that you want to review (for example, on the **All CAPA cases** page).
2. On the Action Pane, on the **CAPA case** tab, select **Signature review**.

Inventory journals

08/29/2025

This article describes how you can use inventory journals to post various types of physical inventory transactions.

In Supply Chain Management, use inventory journals to post physical inventory transactions of various types, such as the posting of issues and receipts, inventory movements, the creation of bills of materials (BOMs), and the reconciliation of physical inventory. All these inventory journals work in a similar way, but they're divided into different types.

Types of inventory journals

The following types of inventory journals are available:

- Movement
- Inventory adjustment
- Transfer
- BOM
- Item arrival
- Production input
- Counting
- Tag counting

Movement

When you use an inventory movement journal, you can add cost to an item when you add inventory, but you must manually allocate the extra cost to a particular general ledger account by specifying a general ledger offset account when you create the journal. This inventory journal type is useful if you want to overwrite the default posting accounts.

Inventory adjustment

When you use an inventory adjustment journal, you can add cost to an item when you add inventory. The extra cost is automatically posted to a specific general ledger account, based on the setup of the item group posting profile. Use this inventory journal type to update gains and losses to inventory quantities when the item should keep its default general ledger offset account. When you post an inventory adjustment journal, you post an inventory receipt or issue, change the inventory values, and create ledger transactions.

Transfer

Use transfer journals to transfer items between stocking locations, batches, or product variants without associating any cost implications. For example, you can transfer items from one warehouse to another warehouse within the same company. When you use a transfer journal, you must specify both the "from" and "to" inventory dimensions (for example, for Site and Warehouse). The on-hand inventory for the defined inventory dimensions changes accordingly. Inventory transfers reflect the immediate movement of material. In-transit inventory isn't tracked. If you need to track in-transit inventory, use a transfer order instead. When you post a transfer journal, two inventory transactions are created for each journal line:

- An inventory issue at the "from" location.
- An inventory receipt at the "to" location.

BOM

When you report a BOM as finished, you can create a BOM journal. By using a BOM journal, you can post the BOM directly. This posting generates an inventory receipt of the product, together with an associated BOM and an inventory issue of the products that are included in the BOM. This inventory journal type is useful in simple or high-volume production scenarios where routes aren't required.

Item arrival

Use the item arrival journal to register the receipt of items (for example, from purchase orders). You can create an item arrival journal as part of arrival management from the **Arrival overview** page, or you can manually create a journal entry from the **Item arrival** page. If you enable the item arrival journal name to check for picking locations, Supply Chain Management looks for a location for received items and, if there's room, generates location destinations for the incoming items.

Production input

Production input journals work like the item arrival journals but are used for production orders.

Counting

Counting journals let you correct the current on-hand inventory that is registered for items or groups of items, and then post the actual physical count, so that you can make the adjustments that are required to reconcile the differences. You can associate counting policies with counting groups to help group items that have various characteristics, so that those items

can be included in a counting journal. For example, you can set up counting groups to count items that have a specific frequency, or to count items when stock falls to a particular level. For information about how to define counting groups, see Define inventory counting processes.

Tag counting

Use tag counting journals to assign a numbered tag to a count lot. The tag contains a tag number, item number, and item quantity. To ensure that a tag is used only one time, and that all tags are used, every item number has a unique set of tags with its own number sequence. You can set three status values for each tag:

- **Used** – The item number is counted for this tag.
- **Voided** – The item number is voided for this tag.
- **Missing** – The item number is missing for this tag.

When you post a tag counting journal, you create a new counting journal based on the tag counting journal lines. For more information about tag counting, see Inventory tag counting.

Working with journals

Only one user at a time can access a journal. If several users need to access journals at the same time to create journal lines, they must select journals that aren't currently being used. This approach prevents information from being overwritten. In situations where multiple departments use the same journal type, it's helpful to create multiple journal names (for example, one per department). It can also be helpful to divide journals so that each posting routine is entered in its own unique inventory journal. For posting routines that are associated with inventory transactions, create one journal for periodic inventory adjustments and another for inventory counting.

Posting journal lines

You can post the journal lines that you create at any time until you lock an item from additional transactions. The data that you enter in a journal remains in that journal, even if you close the journal without posting the lines.

Data entity support for inventory journals

Data entities support the following types of integration scenarios:

- Synchronous service (OData)
- Asynchronous integration

Learn more in Data entities.

> **Note**
>
> Not all inventory journals are OData-enabled, so you can't use the Excel data connector to get data published, updated, and imported back to Supply Chain Management.

Another difference between the journal data entities is the ability to use composite entities that include both the header and line data. Currently, you can use the composite entities for:

- Inventory adjustment journal
- Inventory movement journal

These two inventory journals only support the *Initialize stock* scenario as part of a data management import project:

- When you don't specify a journal header number but specify a number sequence for the journal type, the import job automatically creates journal headers per 1,000 lines. For example, importing 2,020 lines results in the following three journal headers:
  - Header 1: contains 1,000 lines
  - Header 2: contains 1,000 lines
  - Header 3: contains 20 lines
- The import process assumes that unique line information exists per inventory dimension, which can be a product, storage, and tracking dimension. Therefore, you can't import journal lines where only the date field differs on the lines within the same import project.

Related information

- Data entities

Inventory journal approval workflows

This article describes how to set up and use inventory journal approval workflows for various types of physical inventory transactions, such as issues and receipts, inventory movements, bills of materials (BOMs), and the reconciliation of physical inventory. Inventory journal workflows help ensure that only approved inventory journals can be posted to transactions.

> **Note**
>
> Inventory journal approval workflows apply only to transactions recorded using the Inventory Management module. They don't work with inventory journals triggered from the Warehouse Management module.

Create your inventory journal approval workflows

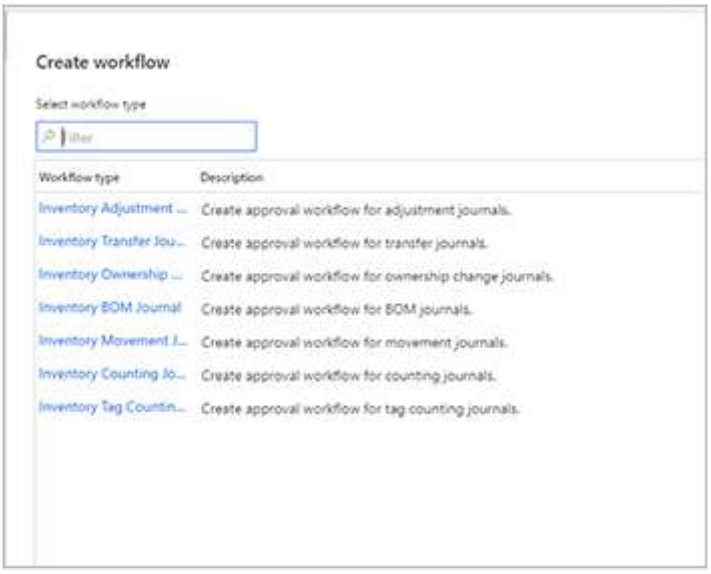
To set up this feature, create a workflow for each inventory journal type you want to control. Because different inventory journal types can have different approval hierarchies and workflow steps, you can configure individual workflows for each inventory journal type.

Workflows support version control, and each has a workflow ID and an active version. You can choose to activate each new workflow version immediately upon creation or keep it inactive. If you need different workflows for the same journal type, create multiple workflows for that journal type, and then assign each of these workflows to a different journal name that uses that type.

To create your inventory journal approval workflows:

1. Go to **Inventory Management** > **Setup** > **Inventory management workflows**.
2. Select **New** on the Action Pane.
3. Choose the inventory journal type for which you want to set up a workflow:
    - *Inventory tag counting journal*
    - *Inventory ownership change journal*
    - *Inventory movement journal*
    - *Inventory transfer journal*
    - *Inventory counting journal*
    - *Inventory BOM journal*

- *Inventory adjustment journal*



4. The workflow editor app launches on your machine. (You might be asked to approve this action.) Use it to design your workflow as needed. For details about how to use the workflow editor, see Workflow system overview.

5. After saving and closing the workflow editor app, choose whether to activate this workflow version or keep it inactive.

> **Note**
>
> Workflows provide version control, which means that you can view a list of versions you created and choose which one is active. To view the list of available versions and choose which to activate, select a workflow listed on the **Inventory management workflows** page. On the Action Pane, open the **Workflow** tab, and select **Versions**. Only one version can be active at a time for each workflow ID.

Assign approval workflows to inventory journal names

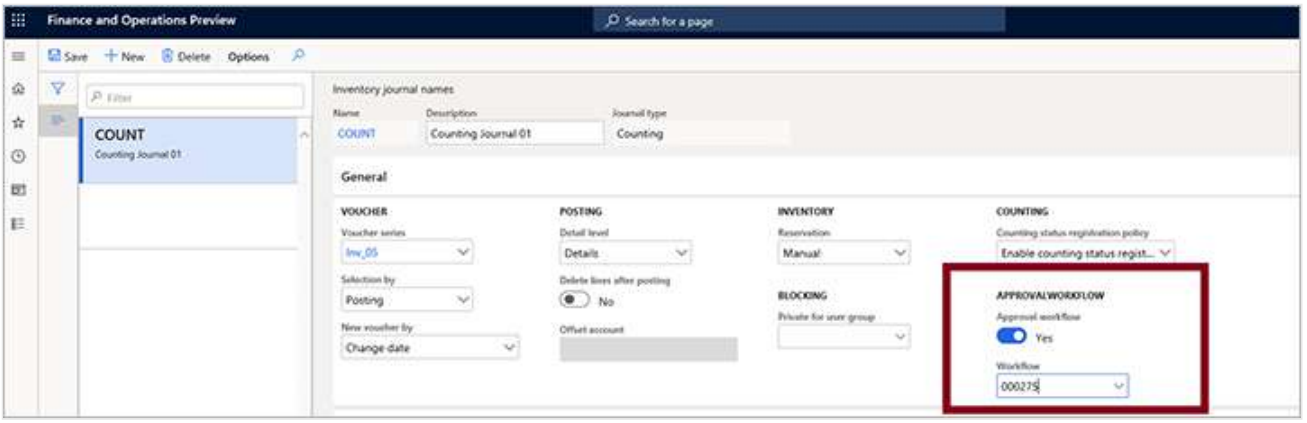
Next, assign an inventory journal workflow to each inventory journal name. For each inventory journal type, you can set up multiple inventory journal names.

To associate an inventory journal workflow with an inventory journal name:

1. Go to **Inventory management** > **Setup** > **Journal names** > **Inventory**.

2. Select a journal name from the list column to open its settings page.

3. On the **General** FastTab, set **Approval workflow** to **Yes**. If you're prompted to approve the action, select **Yes**.



4. Open the **Workflow** drop-down list and select the appropriate workflow. The list shows each active workflow that you created by using the workflow editor app.

Create an inventory journal and send it for approval

After you associate an inventory journal name with its matching inventory journal approval workflow, you can create new inventory journals that use that name and then send these journals for approval by using that workflow. You can't post the inventory journal until the approvers configured in the workflow approve it.

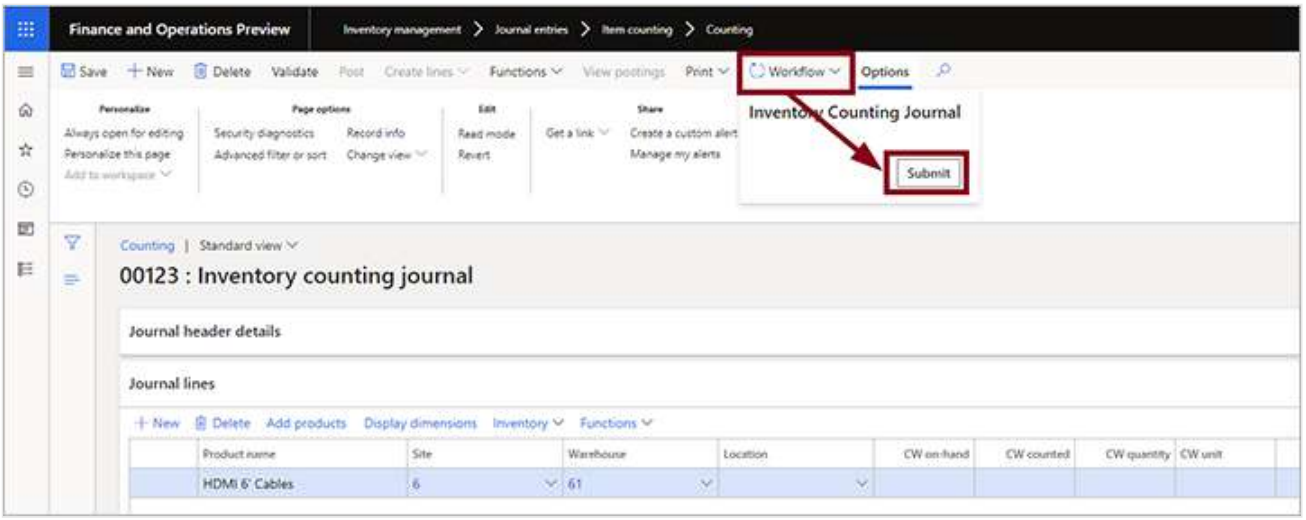
1. On the navigation pane, expand **Inventory management** > **Journal entries** > **Items** and then select an inventory journal type.

2. Select **New** to create a new journal of your selected type.

3. The **Create inventory journal** dialog box opens. Fill out the form as needed and then select **OK** to save the journal.

4. Complete the journal as required.

5. When you create or open an inventory journal with an approval workflow associated with it, the **Workflow** button is active in the Action Pane. When you're ready to submit the journal for approval, select the **Workflow** button to open a drop-down dialog box and then select **Submit**. The approval request routes to the relevant approver, who is alerted by using the notification method configured for the workflow.



To recall an approval request, open the relevant journal, select the **Workflow** button, and then select **Recall**. This action resets the workflow.

When your journal is approved, you can post it. To post the journal, select **Post** from the Action Pane. If the **Post** button isn't active, the journal isn't approved yet.

Respond to an inventory journal approval request

If you're an approver, you receive a message each time your approval is needed (as configured in the relevant workflow). Then, you can approve or reject a journal approval request by following these steps:

1. On the navigation pane, expand **Inventory management** > **Journal entries** > **Items** and then select an inventory journal type.
2. Open the relevant journal and review it.
3. Select the **Workflow** button on the Action Pane to open a drop-down dialog box. Select one of the following options:

    - **Approve** - To approve the request.
    - **Reject** - To reject the request.
    - **More** > **Request change** - To send a message to the requester asking them to change something specific and then resubmit.
    - **More** > **Delegate** - To delegate the approval to another user.
    - **More** > **Recall** - To recall the approval request (resets the workflow).
    - **More** > **Workflow history** - To view the history of this approval workflow so far.

Review the approval history

As with other types of workflows, you can use the **Workflow history** page to view the approval workflow history for any journal.

To review the workflow history for a journal:

1. On the navigation pane, expand **Inventory management** > **Journal entries** > **Items** and then select an inventory journal type.
2. Open the relevant journal.
3. Select the **Workflow** button on the Action Pane to open a drop-down dialog box. Select **Workflow history**. For more information, see View workflow history.

Last updated on 07/01/2026

Inventory locations

Inventory locations determine where you store and pick items in warehouses that don't use warehouse management processes (WMS).

This article applies to features in the Inventory management module. It doesn't apply to features in the Warehouse management module.

The term location refers to the place that items are stored and drawn from.

For each location, you can also specify the place the item is inserted. By default, they're the same. Items are usually inserted and drawn from the same side of a location, but not always. For example, items that are stored in live storage racks are inserted from one aisle and drawn from another. The main input is given by a location name, which is usually determined by its coordinates: warehouse, aisle, rack, shelf, and bin. You can enter this name or ID manually or generate it from the location coordinates. For example, `01-02-03-4` represents aisle 1, rack 2, shelf 3, and bin 4 in the Inventory locations page.

Location properties

A location has the following characteristics:

- Size (height, width, depth, and thereby volume)
- Warehouse, aisle, rack, shelf, and bin position
- Location type (bulk location, picking location, inbound dock, outbound dock, production input location, inspection location, or kanban supermarket)

Check text can be used in online systems to verify that the operator has selected the correct location for a specific item. This check text can be created manually or by default.

Sort codes

Use sort codes to optimize the handling of picking lines, which describe the information that is required for picking items from inventory, including the picking order. Sort codes can be specified by the aisle and other coordinates, or assigned manually for the location.

Blocked locations

Occasionally, you might want to indicate that a location is blocked for a period of time, for example, to allow for repairs. At other times, you might want to indicate blocking of only the input or only output.

Tree structure

In the Inventory locations page, you can view the warehouse layout in a tree structure based on the coordinates of inventory locations, in a defined display format.

Maintain inventory locations via the warehouse form

You can copy locations from one warehouse to another and create locations through a wizard. Before running the wizard, make sure that you define the default location names on the **Warehouse** page.

Related information

Create a new warehouse layout

Last updated on 07/01/2026

Inventory statuses

This article describes how to use inventory statuses to categorize and keep track of inventory.

Set up and use inventory statuses

You can use inventory statuses to categorize inventory. You can then initiate appropriate actions, such as replenishment or putaway work.

Here are some examples of ways that you can use inventory statuses:

- Create inventory statuses for on-hand inventory, inbound transactions, and outbound transactions.
- Specify a default inventory status for warehouse transactions.
- Change an inventory status for items before arrival, during arrival, or when the items are put away during inventory movement.
- Use an inventory status to price items that are returned and to plan item coverage during master planning.

An inventory status is one of the dimensions in the storage dimension group. Inventory statuses can be categorized as available or unavailable, and you can use the **Inventory blocking** parameter to block items that have an unavailable inventory status. Items that have a blocked status are considered physical inventory, and they can't be used on a production order, sales order, transfer order, or outbound transaction.

You can use warehouse items that have either available or unavailable inventory statuses for inbound work. For example, you create an available status that is named *Ready*, an unavailable status that is named *Damaged*, and a blocked status that is named *Blocked*. When you create a purchase order for received or returned items, if any items are damaged or broken, you can change the inventory status of those items to *Damaged* on the purchase order line. After these items are received, the status is automatically set to *Blocked*. If you scan the damaged items by using a mobile device, Supply Chain Management can use location directives and work templates to show information about an appropriate location or range of locations where you can put away those items. For returned items, an issue type of *Reservation* is created on the **Inventory transactions** page.

You can specify which inventory statuses are blocking statuses by using the **Inventory blocking** check boxes on the **Inventory statuses** page. You can't use inventory statuses as blocking

statuses for sales orders, transfer orders, or project integrations.

For outbound work, you can use different nonblocking inventory statuses to control which inventory to reserve against. If you have items that have a status of *Blocking*, and master planning is run on these items, the items are considered missing, and inventory is automatically replenished. Furthermore, for quality orders associated with outbound work, it isn't possible to update the **Inventory status** as part of the quality order validation.

> **Note**
>
> You can't change the status of inventory at locations where open work exists. For example, if you did a purchase receive for an item, but didn't do the putaway step, then open work would exist for the receiving location and you would get an error if you tried to change the status of inventory at that location. Completing or cancelling the related work would allow you to change the status.
>
> Usually, the status of on-hand inventory related to open warehouse work is only changed by workers using the Warehouse Management mobile app, for example while executing a movement process.

After you set up inventory statuses, you can set the default inventory status for a site, item, and warehouse. You can also set a default status for sales, transfer, and purchase orders. The default status for sales orders and outbound transfer order can't have the **Inventory blocking** option set to *Yes*. The inventory status that is inherited from the default settings on a site, warehouse, item, purchase order, transfer order, or sales order can be changed by using the mobile device, or on the purchase order, sales order, or transfer order line.

To plan coverage for items that have an available inventory status, select the **Coverage plan by dimension** option for a storage dimension on the **Storage dimension groups** page. When you open the **Item Coverage** wizard, items that have an available status appear on the **Status** page. To create coverage settings for these items, select the inventory status ID for the available inventory statuses. Based on the coverage settings, you can calculate the item requirements and forecast the supply and demand of available items during master planning. You can't create an item coverage setup that has a blocked inventory status. Alternatively, use the **Item coverage** page to create or modify the item coverage parameters.

Change inventory statuses

You can change inventory statuses either by using the **On-hand by location** page or by using a batch job.

To change inventory status of individually selected records, follow these steps:

1. Go to **Warehouse management** > **Inquiries and reports** > **On-hand by location**.
2. Use the **Site**, **Warehouse**, and **Location** fields to find the relevant location.
3. On the **Locations** tab, select the location.
4. On the **On-hand** tab, select the relevant records, and then select the **Inventory status change** button.

To set up a batch job task that selects and updates records, follow these steps:

1. Go to **Warehouse management** > **Periodic tasks** > **Warehouse management inventory status change**.
2. On the **Parameters** FastTab, set **To inventory status** to the status value you want to update the selected records to.
3. On the **Records to include** FastTab, select the **Filter** button and set up a query that selects which records to change.
4. On the **Run in background** FastTab, set **Batch processing** to *Yes*. If you want to run the job at a regular interval, select **Recurrence** and assign the interval.

Last updated on 07/01/2026

Use batch disposition codes to mark batches as available or unavailable

This article describes how to set up and use *batch disposition codes*. Each batch disposition code has a status of either *Available* or *Unavailable*. You assign batch disposition codes to inventory batches to indicate whether each batch is available for master planning, reservation, picking, and shipping.

To use batch disposition codes, you must set up the codes and assign them to the batches that you want to manage.

> ⓘ **Note**
>
> For items that are enabled for warehouse management processes (WMS) and that use a reservation hierarchy where the *Batch number* dimension is below the *Location* dimension, the system enforces batch disposition codes during reservation only when the line's warehouse is enabled for WMS, or when the **Batch reservation policy for non advanced warehouses** parameter is set to *Advanced*. Learn more in Batch reservation policy for non-WMS warehouses.

Set up batch disposition codes

Set up each batch disposition code that you want to use in your system. You can create as many codes as you want. For example, you can create codes to identify the different reasons why a batch might be available or unavailable. However, you often need just two codes: one for *available* and one for *unavailable*. You can also create custom codes that enable a batch to be used for some operations but not others.

Follow these steps to set up batch disposition codes.

1. Go to **Inventory management** > **Setup** > **Batch** > **Batch disposition master**.
2. The **Batch disposition master** page lists your current batch disposition codes, and lets you create, delete, and edit them. Follow one of these steps:
    - To edit an existing code, select it in the list pane.
    - To create a new code, select **New** on the Action Pane.

3. Set the following fields on the header of the new or selected code:

    - **Batch disposition code** – Enter the display name for the code.
    - **Description** – Describe how the code should be used.
    - **Batch disposition status** – Select the status that applies to batches that the code is assigned to:
        - *Unavailable* – The batches can't be used for master planning, reservation, picking, or shipping. When you select this value, all the **Block** options on the **Setup** FastTab are set to *Yes*, and all the **Nettable** options are set to *No*. However, you can change some of these settings to add exceptions.
        - *Available* – The batches can be used for master planning, reservation, picking, and/or shipping. When you select this value, all the **Block** options on the **Setup** FastTab are set to *No*, and all the **Nettable** options are set to *Yes*. These settings are read-only while the **Batch disposition status** field is set to *Available*.

4. If you set the **Batch disposition status** field to *Unavailable*, you can customize the block status of each operation (reserve, pick, and ship) for each type of order (sales, transfer, and production). For production orders, you can choose to block or unblock the production picking journal. You can also choose to block or unblock master planning. Use the options on the **Setup** FastTab to block or unblock each operation as you require. Set the **Nettable** option to *Yes* to enable master planning, or set it to *No* to block master planning.

Assign batch disposition codes to batches

After you define the batch disposition codes that you require, follow these steps to assign them to batches.

1. Go to **Warehouse management** > **Setup** > **Inventory** > **Batches**.
2. Select one or more batches to assign a batch disposition code to.
3. On the Action Pane, on the **Reset** tab, select **Reset batch disposition code**.
4. In the **Change the restrictions on the inventory batch** dialog box, set the **New batch disposition code** field to the name of the code that you want to assign.
5. Select **OK** to apply the new setting and save your change.

    On the **Batches** page, the values in the **Batch disposition code** and **Batch disposition status** columns are updated so that they reflect the new settings for the selected batches.

Master planning example

This example shows how batch disposition codes can affect master planning.

For this example, batch disposition codes are set up in the following way:

- *P-Available:*
  - **Batch disposition status:** *Available*
  - **Nettable:** *Yes*
- *P-Unavailable:*
  - **Batch disposition status:** *Unavailable*
  - **Nettable:** *No*

There's a product (*Product-1*) that has two batches: *Batch-A* and *Batch-B*. These batches are set up in the following way:

- *Batch-A:*
  - **Batch disposition code:** *P-Available*
  - **On-hand quantity:** 1
- *Batch-B:*
  - **Batch disposition code:** *P-Unavailable*
  - **On-hand quantity:** 1

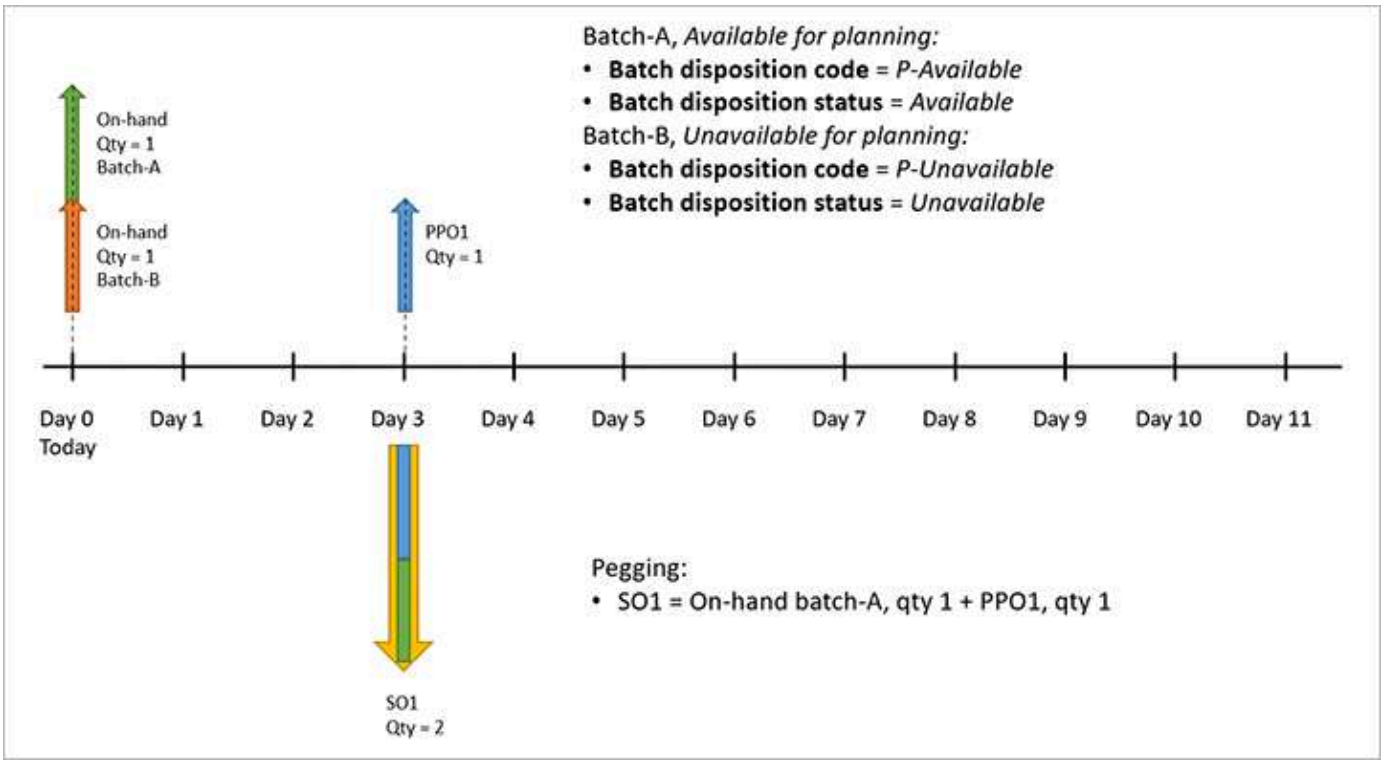
There's a sales order (*SO1*) for a quantity of 2 of product *Product-1*. The delivery date is three days from today.

You run master planning and set the following values that are relevant to this example:

- **Planned order:** *Purchase order*
- **Replenishment strategy:** *Requirement*
- **Lead time:** *0*

As a result of the planning run, the system uses the available batch (*Batch-A*) to cover a quantity of 1 of product *Product-1* for sales order *SO1*. However, it can't use batch *Batch-B* because that batch is marked as unavailable for planning. Therefore, to cover the remaining quantity, the system creates a planned purchase order (*PPO1*) for a new batch of product *Product-1*.

The following illustration shows a timeline for the planning result.



Last updated on 06/02/2026

Create a new warehouse layout

07/02/2025

This article describes how to set up information about the locations in a warehouse. These instructions only apply for the Inventory management module (when you aren't using warehouse management processes (WMS)), not when you're using the WMS-enabled warehouse functionality that's available in the Warehouse management module. You can use this procedure in demo data company USMF, or on your own data.

Set the default location capacity

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. Select the **Locations** tab.
3. In the **Standard width** field, enter a number.
4. In the **Standard depth** field, enter a number.
5. In the **Standard height** field, enter a number.
6. Select **Save**.
7. Close the page.

Define the location name format

1. Go to **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**.
2. Select **New**.
3. In the **Warehouse** field, type a value.
4. In the **Name** field, type a value.
5. In the **Site** field, select the desired record in the lookup.
6. Toggle the expansion of the **Location names** section. The options in this section define the default format for location names. In our example, we'll include the aisle number, rack number, and shelf number.
7. Set the **Include aisle** option to *Yes*.
8. Set the **Include rack** option to *Yes*.
9. In the **Format** field, for the rack, type a value.
10. Set the **Include shelf** option to *Yes*.
11. In the **Format** field, for the shelf, type a value.

Define warehouse locations

1. On the Action Pane, select **Warehouse**.

2. Select **Location Wizard**.
3. Select **Next**.
4. Deselect the **Outbound docks** option
5. Deselect the **Bulk locations** option
6. Select **Next** until you come to the option to select **Finish**.
7. Close the page.
8. Refresh the page.

Warehouse blocking limitations

The Inventory management module doesn't support blocking or closing an entire warehouse for future transactions. As a workaround, you can rename the warehouse to indicate that it shouldn't be used for future transactions. Alternatively, you can block individual locations within the warehouse as needed. For details about blocking specific locations, see Inventory locations.

Maintain bar code types

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows you how to set up a new bar code definition which can then be used as part of the picking list report. You can walk through this procedure in demo data company USMF, or using your own data. If you're using USMF you can use the example values that are shown. These tasks would typically be carried out by a warehouse manager.

1. Go to **Bar codes**.
2. Select **New**.
3. In the **Bar code setup** field, type a value.
4. In the **Description** field, type a value.
5. In the **Bar code type** field, select an option. If you're using USMF, you can select *Code 39*.
6. In the **Mask ID** field, specify the bar code mask ID. Bar code masks are used to create bar codes and to quickly identify bar codes that are scanned into a point of sale (POS) system. Learn more in Set up bar code masks.
7. In the **Size** field, enter a number.
8. In the **Maximum length** field, enter a number.
9. Select **Save**.
10. Close the page.
11. Go to **Inventory and warehouse management parameters**.
12. In the **Bar code setup** field, enter or select a value. Select the bar code setup that you created before, but be aware that the bar code format must match the format of the unique identifier for the record type used in the process. For example, for picking routes, the bar code format should match the format of the picking route reference, which is typically a number sequence.
13. Select **Save**.
14. Close the page.

Last updated on 04/10/2025

Check the availability of stock

08/29/2025

This procedure shows you how to check on-hand and physical on-hand inventory for a specific item number. It also shows you how to get supply information related to an item. Physical on-hand inventory is the on-hand inventory that's available (purchased, received, and registered). On-hand inventory includes the available on-hand inventory, but also the inventory that's been ordered and is expected, but not yet received or registered.

If you're using USMF demo data you can use the example values that are shown in this article.

Check on-hand inventory for an item

1. Go to **Inventory management** > **Inquiries and reports** > **On-hand list**.
2. On the **Filters** pane, enter search criteria to find the products and/or locations that you want to look up. For example, if you're using the USMF demo data company, you could set **Item number** to *M9201*.
3. On the **Filters** pane, select **Apply**.
4. On-hand inventory is displayed for your selected criteria.

    - If you want to change the set of columns shown in the list, select **Dimensions** on the Action Pane.
    - To see more information about a listed item, select the item and then choose the type of information you're looking for on the Action Pane.

Check physical on-hand inventory

1. Go to **Warehouse management** > **Inquiries and reports** > **Physical on-hand inventory**.
2. In the **Item number** field, specify the item you're looking for. You can also use the **Site** and **Warehouse** fields to filter the list of items.
3. Select **Refresh** at the right side of the Action Pane.
4. Physical on-hand inventory is displayed. If you want to change the set of columns shown in the list, select **Display Dimensions** on the Action Pane.

Check on-hand inventory by location

1. Go to **Warehouse management** > **Inquiries and reports** > **On-hand by location**.
2. Use the fields at the top of the page to define the site, warehouse, and location you want to look up. For example, if you're using the USMF demo data company, you could set **Warehouse** to *51*.

3. Select **Refresh** at the right side of the Action Pane.
4. Select a row in the **Locations** section to see the on-hand inventory for that location in the **On hand** section. To learn more about an item listed in the **On hand** section, select the item and then choose the type of information you're looking for from the **On hand** section toolbar.

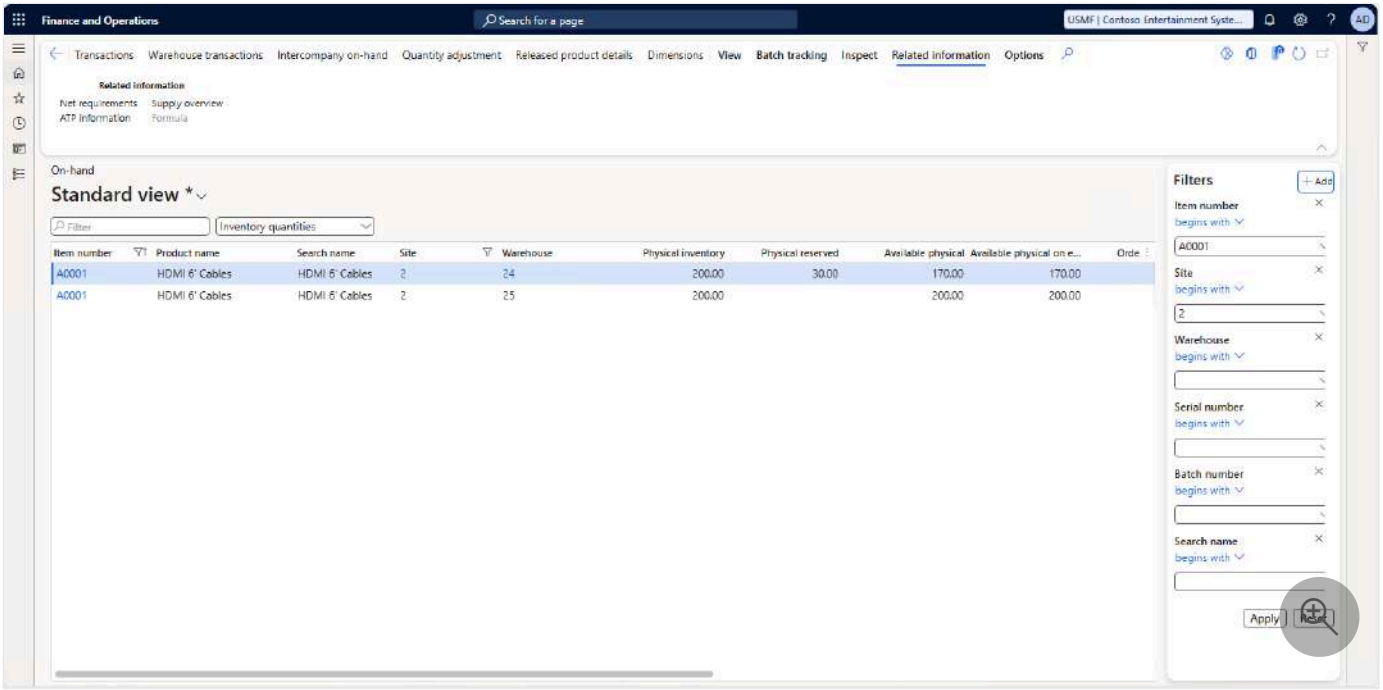
Inventory on-hand list

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to use the **On-hand list** page to inspect on-hand inventory details. Use this page to understand where the goods are, what's available, what's expected, and what's reserved for use.

The on-hand inventory list provides a summarized view of all inventory and warehouse transactions in your system. On the **On-hand list** page, you can filter by dimension values (product, storage, and/or tracking dimensions) and break down output values by dimension values. The values that are shown on the page change based on the filter and display dimensions that you set. This article shows a few of the ways that the various filtering and output options work together, and how they affect the results that are presented.

In addition, you can inquire about the inventory and warehouse transactions to get an explanation of the values in the list, change the on-hand values, and find related information.



Query your on-hand inventory

The **On-hand list** page is automatically updated when transactions are made in inventory. Those transactions might be forecasted, physical, or financial transactions.

To check the availability of inventory, follow these steps:

1. Go to **Inventory management** > **Inquiries and reports** > **On-hand list**.

2. In the **Filters** pane, specify the product and locations that you want to view on-hand information for. Add filters as needed by using the **Add** button. To remove filters, use the **Remove** (**X**) button. Select **Apply** to view the results. The more filters you specify, the more specific the on-hand results are. Therefore, you see fewer rows and/or lower on-hand values.

3. On the Action Pane, select **Dimensions** to open the **Dimensions display** dialog box, where you can specify the dimensions that are used to present your on-hand inventory. The more dimensions you specify, the more granular the results are. For example, if you add the **Location** storage dimension, a **Location** column is added to the grid, and the results are expanded. Instead of showing one row per warehouse, they now show one row per warehouse location.

   > **Tip**
   >
   > - You can save the set of dimensions that you use most often. The **On-hand list** page then automatically shows those dimensions each time that you open it. To save a dimension set, set the **Save setup** option to *Yes* in the **Dimensions display** dialog box before you close it.
   > - The on-hand information is easiest to understand when the dimensions that are shown align with the filters that you're using. Use the **examples** later in this article to understand how different combinations of filters and dimensions lead to different results.
   > - You can further filter the results by using the **Filter** field at the top of the grid and the dropdown lists in the column headers. Unlike the **Filters** pane, these filters apply only to the presentation of results. They don't change the result values.

The **On-hand** grid provides the following columns of inventory information.

Expand table

Column	Description
Physical inventory	The physical quantity that's available in inventory.
Physical reserved	The total quantity that was physically reserved.
Available physical	The available (not reserved) quantity that's available in physical inventory. **Available physical** is a calculated field. The value equals the **Physical inventory** value minus the **Physical reserved** value.
Available physical on exact dimensions	The available physical quantity for all the dimensions that are shown in the grid.
Ordered in total	The total quantity that's included on inbound orders or that has a positive quantity in various inventory journals.
On order	The total quantity that's included on outbound orders or that has a negative quantity in various inventory journals.
Ordered reserved	The total quantity that's reserved on ordered receipts. The value in this field represents the total quantity of items in outbound transactions that have a status of *Ordered reserved*. Items that are reserved as ordered aren't physically available in inventory. Therefore, they can't be directly picked and delivered.
Available for reservation	The total quantity of on-hand inventory that can be reserved. **Note:** If the **Reserve ordered items** checkbox is selected on the **Inventory and warehouse management parameters** page, the value in this field includes expected receipts. If the checkbox is cleared, the value excludes expected receipts.
Total available	The total available quantity. **Total available** is a calculated field. The value equals the **Available physical** value plus the **Ordered in total** value minus the **On order** value.

Apply filters to scope on-hand calculations to items and dimensions you're looking for

Use the **Filters** pane to filter the on-hand inventory list so that it includes only records where the field values match the filtering criteria. To define a filter, follow these steps:

1. In the **Filters** pane, find the field that you want to filter on.
2. In the field below the name of the target field, select a logical operator (for example, *starts with*, *equal to*, or *greater than*).
3. Enter or select the value to look for.

> **Important**
>
> The **On-hand list** page is assembled from a detailed on-hand inventory table that includes all available dimensions. However, the list on this page is a summary. Therefore, it might combine rows from the source table by aggregating values according to the dimensions that are shown.
>
> The filters that you define in the **Filters** pane apply to the source table, not to the aggregated list. This behavior can sometimes produce unexpected results. Use the **examples** later in this article to understand how this behavior can affect your results.
>
> However, the **filters that are provided in the grid** *do* apply to the aggregated list. These filters include both the QuickFilter at the top of the grid and the filter for each column heading.

You can modify the set of filters that is available in the **Filters** pane by following these steps.

- To remove a filter from the pane, select its **Close** button (**X**).
- To add a filter, select **Add** at the top of the **Filters** pane. The **Add filter fields** dialog box that appears shows a list of the available fields. It also shows information about the data type and table for each field. Use the column headings to filter and sort the list as you require, and then select the checkbox for each field that you want to add to the **Filter** pane. When you finish, select **Insert** to apply your changes.

Select the dimensions you want on-hand results for

Dimensions provide more information about each item in the on-hand inventory list. They also give you more ways to sort and filter the list. The dimensions that you select to show affect how rows are aggregated on the **On-hand list** page. This aggregation can affect how rows from the source tables are combined in the results that are shown. Use the examples later in this article to understand how this behavior can affect your results.

To customize the selection of inventory dimensions that is shown, follow these steps:

1. On the Action Pane, select **Dimensions**.

   The **Dimension display** dialog box that appears shows every dimension.

2. Select the checkbox for each dimension that you want to include in the grid.

3. If you want your selection to be the default the next time you open the **On-hand list** page, set the **Save setup** option to *Yes*. If you set this option to *No*, your selection is used only during the current session. Therefore, the next time you open the page, the current default selection is used.

4. Select **OK** to close the dialog box and apply your changes.

Investigate and act on the on-hand information

The on-hand information on the **On-hand list** page is aggregated based on the filters and dimensions that you specify. In SQL terms, filters control the `WHERE` clause of the on-hand calculation, and dimensions control the `GROUP BY` clause. These clauses control how many records and columns are shown in the on-hand list. Nevertheless, you might still need to further inspect a specific value in the on-hand information.

To learn more about the transactions that participated in the on-hand calculation, follow these steps:

1. On the Action Pane, select **Transactions**. The Inventory transactions page appears. The same filters and dimensions that you used in the on-hand list are applied to this page.
2. On the Action Pane, select **Warehouse transactions**. The Warehouse transactions page appears. The same filters and dimensions that you used in the on-hand list are applied to this page.

Inspect the Action Pane to find useful actions that are related to your on-hand inventory. For example, you can perform the following actions:

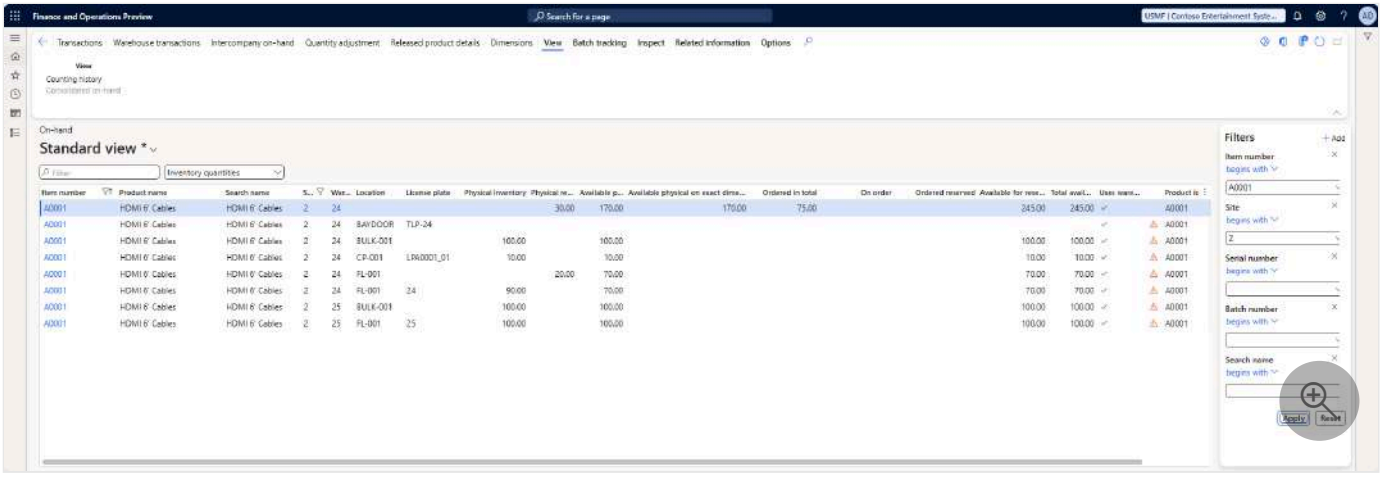
- Update your on-hand quantity by selecting **Quantity adjustment**.
- Investigate on-hand inventory in other companies by selecting **Intercompany on-hand**.

Understand on-hand information for warehouse-enabled items

On the **On-hand list** page, items that are enabled for warehouse management processes (WMS) differ from non-WMS items in the following ways:

- A check mark in the **Uses the warehouse management process** column indicates WMS-enabled items. They also show helpful warnings. You can learn more about each warning by hovering over it.

- For both WMS-enabled items and non-WMS items, values in the physical inventory columns follow the same aggregation principles of filtering and requested dimensions. The value is placed on the most specific storage dimension. In the example that follows this list, **Physical inventory** is shown per license plate, except in the case of location *BULK-001* in Warehouse *24* (in row 7), because that location isn't license plate–enabled.
- Values that are shown for **Physical reserved** work differently for WMS-enabled items and non-WMS items. Because WMS-enabled items use reservation hierarchies, **Physical reserved** values are presented at the hierarchy level that is associated with the item. Therefore, in the example that follows this list, additional rows 1 and 5 are added next to rows that represent **Physical inventory** values. Different processes reserve at different levels. In the example, row 1 is related to sales process reservations that are made at the warehouse level, whereas row 5 is related to a counting operation.
- The list might include empty rows (such as row 2 in the example that follows this list). These rows indicate that there's currently no on-hand inventory, related reservation, or related order on the dimension, but there was in the past.



Filter on the output of the inventory on-hand list

Several columns in the **On-hand** grid let you sort or filter by values in that column by selecting the column heading. A QuickFilter at the top of the grid provides additional filtering options. These filters apply to the results, not to the source tables. Use the examples later in this article to understand how this behavior can affect your results.

> **Note**
>
> You can't filter and sort by all columns. Most of the quantity columns don't include sorting and filtering controls, because they're calculated fields. The **On order** column is an exception.

Examples

Your system includes a detailed (non-aggregated) inventory table that shows the following on-hand inventory.

Expand table

Item Number	Site	Warehouse	Physical Inventory	Available Physical
IA0001	1	1	1	1
IA0001	1	2	2	2
IA0001	1	3	1	1

Scenario 1

The **On-hand list** page shows the following final dimensions:

- Item number
- Site
- Warehouse

In the **Filters** pane, set up the following filtering criteria:

- **Item Number** | *is exactly* | *IA0001*
- **Available Physical** | *less than or equal* | *1*

Here's the resulting output.

Expand table

Item Number	Site	Warehouse	Physical Inventory	Available Physical
IA0001	1	1	1	1
IA0001	1	3	1	1

Scenario 2

The **On-hand list** page shows the following final dimensions:

- Item number
- Site

In the **Filters** pane, set up the following filtering criteria:

- **Item Number** | *is exactly* | *IA0001*
- **Available Physical** | *less than or equal* | *1*

Here's the resulting output.

Expand table

Item Number	Site	Physical Inventory	Available Physical
IA0001	1	2	2

The settings in the **Filters** pane apply to the detailed (non-aggregated) inventory table that is shown at the beginning of this section. Therefore, the criterion **Available Physical** | *less than or equal* | *1* finds two rows from that table (the first and third rows, each of which shows an **Available Physical** value of *1*). However, in this scenario, the **On-hand list** page isn't set up to show the **Warehouse** dimension. Therefore, it aggregates the two original rows into a single resulting row, because both rows have identical values in all the dimensions that are shown. This row appears to violate the filtering criterion, because the **Available Physical** value is shown as *2*. However, the result is correct, because the settings in the **Filters** pane apply to the source table, not to the aggregated table that is shown on the **On-hand list** page.

Last updated on 06/15/2026

Inventory transaction details

Use the **Transactions details** page to view details of any selected inventory transaction.

> **Tip**
>
> Many transactions and updates are categorized as being either *physical* or *financial*. You can learn more about these categories in Physical and financial updates.

Inventory transactions page performance improvement

Supply Chain Management version 10.0.39 introduced the *Inventory transaction details page's performance improvement* feature, which improves the performance of the **Inventory transactions** page. As of version 10.0.43, this feature is turned on by default. As of Supply Chain Management version 10.0.45, it's mandatory and can't be turned off.

To allow the **Inventory transactions** list page to load more quickly, it now provides a **Filters** pane that lets you limit the sets of transactions listed. The next time you open the page, it will remember your last-used filter settings. The page won't list any transactions until you select **Apply** on the **Filters** pane. In addition, the page now shows only inventory-related fields.

> **Important**
>
> The *Inventory transaction details page's performance improvement* feature changes the form used to create the **Inventory transactions** page from `InventTrans` to `InventTransNew`. The feature also changes the navigation so that the new form opens when you go to **Inventory management** > **Inquiries and reports** > **Transactions** (which now opens a URL that resembles `https://contoso.operations.dynamics.com/?cmp=usmf&mi=InventTransNew`). This same form also creates the **Transaction details** page. After upgrading to a version of Supply Chain Management that has this feature turned on, you must reproduce any customizations that you made to the old form on the new form.

Open the Transaction details page

To open the **Transactions details** page, follow these steps.

1. Go to **Inventory management** > **Inquiries and reports** > **Transactions**.

2. The **Inventory transactions** page opens. Find and select the transaction that you want to inspect.
3. On the **Filters** pane, set filters to narrow down the list of inventory transactions and then select **Apply**.
4. On the Action Pane, select **Transaction details**.

FastTabs overview

The **Transaction details** page is split into several FastTabs. The following table describes the purpose of each FastTab.

Expand table

FastTab	Description
General	This FastTab shows basic information about the selected inventory transaction. In addition to the fields that appear on the **Inventory transactions** page, it includes the additional fields that are described later in this article.
Updates	This FastTab shows information that is related to the physical, financial, and settlement aspects of the selected inventory transaction. Depending on the current status of the transaction, some fields might be blank. However, those fields will automatically be set when transactions are posted. In addition to the fields that appear on the **Inventory transactions** page, this FastTab includes the additional fields that are described later in this article.
Ledger postings	This FastTab shows the posting type and ledger account that are used for the physical update, financial update, physical revenue, physical charges, financial revenue, and financial charges that are related to the selected inventory transaction.
References	This FastTab shows additional information about the source transaction that created the selected inventory transaction. For example, this information might include details from the purchase order or sales order.
Related information	This FastTab shows additional information about the selected inventory transaction. These fields might not be set if you aren't using some features, such as procurement categories or projects.
Financial dimensions – physical	This FastTab shows the financial dimension values that were used when the physical update was posted. If the physical update hasn't been posted, the fields will be blank.
Financial dimensions – financial	This FastTab shows the financial dimension values that were used when the financial update was posted. If the financial update hasn't been posted, the fields will be blank.
Inventory dimensions	This FastTab shows the inventory dimension values that are assigned to the selected inventory transaction. These values include the site, warehouse, size, color,

FastTab	Description
	configuration, location, batch number, serial number, inventory status, license plate, and owner.

Fields on the General FastTab

The following table describes the fields on the **General** FastTab that don't also appear on the **Inventory transactions** page.

Expand table

Field	Description
Party	The name of the relevant party for the selected inventory transaction. For example, a purchase order transaction shows the name of the vendor on the purchase order, and a sales order transaction shows the name of the customer. This field isn't available for all types of inventory transactions.
Inventory reference	If another inventory transaction is related to the selected inventory transaction, the type of that other transaction. For example, if a purchase order is marked against a sales order, the **Inventory reference** field of the purchase order transaction will be set to *Sales order*. Marking automatically occurs for direct delivery orders and intercompany orders. When you use marking with costing methods other than *standard cost* and *moving average*, the system will create settlements and adjustments to the exact transactions that are marked. In this way, you can achieve "actual" costing that overrides the logic for first in, first out (FIFO); last in, first out (LIFO); or weighted average.
Inventory number	The specific transaction that is related to the selected inventory transaction. For example, if a purchase order is marked against a sales order, the **Inventory number** field of the purchase order transaction will be set to the sales order number.
Project ID	If the selected inventory transaction is related to a project, this field is set to the project number.
Lot ID	The unique lot ID that the system has assigned to the selected inventory transaction.
Reference lot	If another inventory transaction is related to the selected inventory transaction, the lot ID of that other transaction. For example, if a purchase order is marked against a sales order, the **Reference lot** field of the purchase order transaction will be the **Lot ID** value of the sales order line's inventory transaction.
Dimension number	A unique ID that represents the combination of all the inventory dimension values that are assigned to the selected inventory transaction.
Value open	A value that indicates whether the selected inventory transaction has been settled by the inventory closing process. If this field is set to *Yes*, the transaction hasn't been settled.

Field	Description
Expected date	For receipt transactions, the date when the inventory receipt is expected to occur. For example, this field might show the expected receipt date on an inventory blocking order or the delivery date on a purchase order line.
Inventory date	This field is set when a registration transaction was done on the receipt order before a physical receipt was posted.
Non-financial transfer	A value that indicates whether the selected inventory transaction is a non-financial transfer. A non-financial transfer occurs when a change in inventory dimensions isn't financially tracked on the **Item model group** page.
Transfer lot ID	If the selected inventory transaction is a non-financial transfer, this field is set to the **Lot ID** value of the other inventory transaction that the selected transaction is settled against.

Fields on the Updates FastTab

The following table describes the fields on the **Updates** FastTab that don't also appear on the **Inventory transactions** page.

Expand table

Field	Description
Physical voucher	The voucher number from the general ledger where the physical posting for the selected inventory transaction was generated.
Route	The route that is related to the selected inventory transaction. This field is set only for production picking list transactions where the bill of materials (BOM) or formula line is related to a specific item.
Packing slip	The packing slip number that was entered for a purchase order product receipt transaction.
Physical cost amount	The amount that was posted to the general ledger for the physical update. For a standard cost product, this amount represents the standard cost. For other costing methods, it represents the weighted average for the product at the time of posting.
Physical revenue	The amount of deferred revenue that was posted to the general ledger for the physical update.
Load ID	If the current transaction is related to a load in Transportation management, this field shows the unique number of the load that included the item.
Physically posted	A value that indicates whether the selected inventory transaction has been physically posted. If the current transaction is posted to the general ledger, there will also be a physical voucher.

Field	Description
Financially posted	A value that indicates whether the selected inventory transaction has been financially posted. If the current transaction is posted to the general ledger, there will also be a financial voucher.
Physical charge posted	A value that indicates whether the physical charges that are related to the selected inventory transaction have been posted.
Financial charge posted	A value that indicates whether the financial charges that are related to the selected inventory transaction have been posted.
Financial voucher	The voucher number in the general ledger where the financial posting was generated.
Invoice	The invoice number that was generated, for example, for a purchase order or sales order.
Adjustment	The amount that was adjusted on the selected inventory transaction from the inventory close and adjustment process.
Profit and loss, posted amount	The amount of the selected inventory transaction that was posted to the profit and loss statement.
Unposted invoice	The invoice number of a related purchase order that appears on the **Pending vendor invoice** page.
Financially closed	The date when the transaction was fully settled.
Covered CW quantity	The catch weight quantity that is covered by the settlement.
Settled quantity	The quantity that has been settled for the selected inventory transaction.
Amount settled	The value that has been settled for the selected inventory transaction.

Last updated on 11/11/2025

Initialize stock levels in the warehouse

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows you how to manually update the on-hand inventory by using an Inventory movement journal. (You can also update on-hand inventory by importing transactions in data entities.) You can run this guide in demo data company *USMF* where all the prerequisites like journal name, item setup, posting profiles, and accounts are available. The guide suggests specific values for the item and dimensions that are used. If you choose a different item, you might need to enter values for different dimensions.

1. Go to **Inventory management** > **Journal entries** > **Items** > **Movement**.
2. Select **New**.
3. In the **Name** field, select *IMov*. Use different journal name templates for different business purposes.
4. In the list, select the link in the selected row.
5. In the **Offset account** field, enter the value *140200*. This offset account is the default account on the journal lines. You can override the default to assign different offset accounts per line.
6. Select **OK**.
7. Select **New**.
8. In the **Item number** field, select *A0001*.
9. In the list, select the link in the selected row.
10. Select the **Inventory dimensions** tab.
11. In the **Site** field, select site *1*.
12. In the **Warehouse** field, select warehouse *13*.
13. In the list, select the link in the selected row.
14. In the **Location** field, select location *13*.
15. In the **Quantity** field, enter a number.
16. Select **Save**.
17. Select **Post**.
18. Check or uncheck the **Transfer all posting errors to a new journal** check box. If you enable this option, any lines that fail to post are copied to a new journal. You can use the information in the log to correct the issues and then re-post the lines.
19. Select **OK**.

20. Close the page.
21. Close the page.

Last updated on 11/20/2025

Link transfer order lines with sales order lines

Microsoft Dynamics 365 Supply Chain Management lets you create transfer order lines directly from sales order lines. This functionality is useful when the goods that you must supply are located in a different warehouse than the warehouse that you're shipping from.

The feature described in this article expands on that functionality by linking and marking the related transfer and sales order lines together. This link makes it easier to track the relationship between the sales and transfer orders, and helps you manage your inventory more effectively. The feature adds the following capabilities and benefits:

- View all transfer orders that were generated for a specific sales order line.
- View all sales orders that are related to a specific transfer order line.
- When transfer lines are created from a sales order, you can choose to add them either to an existing transfer order or to a newly created transfer order.
- You receive a warning if you try to create multiple transfer orders for the same sales order line.
- Automatically mark new transfer order lines with their related sales order line. Marking matches specific inventory receipts (transfer order line inventory transactions) with inventory issues (sales order line inventory transactions) for the purpose of inventory costing.
- Automatically reserve transfer order quantities for the sales order that they were created from. In this way, you prevent them from being used to fulfill another sales order by mistake.

Prerequisites

Before you can use the feature described in this article, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management version 10.0.41 or later.
- The feature named *Pair transfer order lines with sales order lines* must be turned on in Feature management.

Add and link lines to a new or existing transfer order from a sales order

To create transfer order lines directly from sales order lines, link the lines together, and (optionally) reserve the transferred inventory, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Open the sales order for which you want to add lines to a transfer order.
3. On the **Sales order lines** FastTab, select the lines that you want to add to the transfer order.
4. On the toolbar of the **Sales order lines** FastTab, select **Product and supply** > **New** > **Transfer order**.
5. In the **Transfer order** dialog box, set the following fields:
    - **From warehouse** – Select the warehouse that the items should be transferred from (source warehouse).
    - **To warehouse** – Select the warehouse that the items should be transferred to (destination warehouse). This warehouse is the one that is associated with the sales order line. It should be selected by default.
    - **Add to an existing transfer order** – If you want to add the items to an existing transfer order, select the transfer order in the list. The list shows only open transfer orders that use the source and destination warehouses that are selected in this dialog box. If you want to create a new transfer order, leave this field blank.
    - **Ship date** – The date when the items should be shipped.
    - **Reserve items automatically** – If you want the items in the source warehouse to be reserved automatically for the transfer order, set this option to *Yes*.
6. Select **OK**.

The new or selected transfer order opens. You can now view the transfer order lines and other details.

View transfer order lines related to a sales order

When you review sales orders, you can confirm whether the required transfer order lines exist for each sales order line.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. Open the sales order that you want to review.

3. On the **Sales order lines** FastTab, on the toolbar, select **Product and supply** > **View** > **Related transfer orders**.

4. On the **Paired transfer order and sales order lines** page, you can review the following information about all the transfer order lines that are linked to the selected sales order:

    - **Item number** – The item number of product on the sales order line and the transfer order line. Select the link to view the item details.
    - **Sales order** – The identifier of the sales order. Select the link to open the sales order.
    - **Line number** – The line number of the sales order line.
    - **Sales quantity** – The sales quantity of the sales order line.
    - **Transfer number** – The identifier of the transfer order. Select the link to open the transfer order.
    - **From warehouse** – The source warehouse of the transfer order.
    - **To warehouse** – The destination warehouse of the transfer order. This warehouse is also the warehouse for the related sales order line.
    - **Line number** – The line number of the transfer order line.
    - **Transfer quantity** – The transfer quantity of the transfer order line.
    - **Ship date** – The ship date of the transfer order line.
    - **Receive date** – The receipt date of the transfer order line.
    - **Remaining** – The remaining status of the transfer order line. Possible values include *Nothing*, *Receive updates*, and *Shipping updates*.

5. To view more information about a selected line, select one of the following buttons on the Action Pane:

    - **Sales order line transactions** – View all inventory transactions that are related to the sales order line.
    - **Transfer order line transactions** – View all inventory transactions that are related to the transfer order line.

View sales orders related to a transfer order line

When you review transfer orders, you can check whether they contain lines that are related to a specific sales order. This process helps you understand the context of the transfer order and why it was created.

1. Follow one of these steps:

- Go to **Inventory management** > **Inbound orders** > **Transfer order**.
- Go to **Inventory management** > **Outbound orders** > **Transfer order**.

2. Open the transfer order that you want to review.
3. On the **Transfer order lines** FastTab, select the line that you want to view the related sales order for.
4. On the toolbar of the **Transfer order lines** FastTab, select **Inventory** > **Related sales order**.
5. On the **Paired transfer order and sales order lines** page, you can review information about the sales order line that is related to the selected transfer order line. The grid columns and Action Pane buttons are the same as the ones that are described in the previous section.

Automatic marking between sales and transfer order lines

Marking matches specific inventory receipts (transfer order line inventory transactions) with inventory issues (sales order line inventory transactions) for the purpose of inventory costing. When you create a transfer order line from a sales order line, the system automatically marks the two lines together, but only if they're for *nonsettled items*.

> **Important**
>
> Nonsettled items are items from transactions that haven't been matched or processed according to the inventory model that you're using. Examples of inventory models include standard cost, moving average, and non-valuated. Automatic marking isn't supported for settled items. It also isn't supported for catch-weight items. For settled and catch-weight items, you must do the marking manually.

To view and edit markings from a sale order line, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Open the sales order that you want to review.
3. On the **Sales order lines** FastTab, select the line that you want to view the marking for.
4. On the toolbar of the **Sales order lines** FastTab, select **Inventory** > **Maintain** > **Marking**.
5. The **Marking** dialog box shows the marked quantities. You can edit the markings as required.

To view and edit markings for a specific inventory transaction, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Open the sales order that you want to review.
3. On the **Sales order lines** FastTab, select the line that you want to view transactions for.
4. On the toolbar of the **Sales order lines** FastTab, select **Inventory** > **View** > **Transactions**.
5. On the **Inventory transactions** page, select the transaction that you want to inspect. You can inspect both shipment transactions and receipt transactions.
6. On the Action Pane, on the **Inventory** tab, in the **View** group, select **Marking**.
7. The **Marking** dialog box shows the marked quantities. You can edit the markings as required.

To view and edit markings from a transfer order line, follow these steps:

1. Follow one of these steps:

    - Go to **Inventory management** > **Inbound orders** > **Transfer order**.
    - Go to **Inventory management** > **Outbound orders** > **Transfer order**.

2. Open the transfer order that you want to review.

3. On the **Transfer order lines** FastTab, select the line that you want to view the related sales order for.

4. On the toolbar of the **Transfer order lines** FastTab, select **Inventory** > **Transactions**.

5. On the **Inventory transactions** page, select the transaction that you want to inspect.

6. On the Action Pane, on the **Inventory** tab, in the **View** group, select **Marking**.

7. The **Marking** dialog box shows the marked quantities. You can edit the markings as required.

Automatically reserve transferred items for their matching sales orders

You can automatically reserve transfer order quantities for the sales order that they were created from. This reservation prevents these quantities from being used to fulfill another sales order by mistake. For this functionality to work, the following conditions must be met:

- The **Reserve items automatically** option must be set to *Yes* when the transfer order line is created from the sales order line.

- The inventory transactions for the related sales order line and transfer order line must be marked together automatically. (Automatic marking is supported only for nonsettled items.)

If the preceding conditions are met, the transferred stock is automatically reserved against the sales order line when it's received. In this case, the inventory transaction record that is created for the sales order line has an issue status of *Reserved ordered*, and the reservation is triggered.

If the conditions aren't met, the transferred stock isn't automatically reserved against the sales order line. In this case, the inventory transaction record that is created for the sales order line has an issue status of *On order*.

Last updated on 12/30/2025

Transfer physical inventory within the warehouse

This procedure shows how to create and post an inventory transfer journal to register movement of an item from one location in a warehouse to another. Before you start, set up an inventory journal name for inventory transfers. You can follow this procedure in the demo data company USMF using the example values that are shown, or you can use your own data if you have products and locations set up. A warehouse employee normally carries out these tasks.

Create an inventory transfer journal

1. Go to **Inventory management** > **Journal entries** > **Items** > **Transfer**.
2. On the Action Pane, select **New**.
3. In the **Name** field, enter or select a value.
4. Select **OK**.

Create journal lines

You can specify *From* and *To* dimensions for each journal line. These dimensions are essential for this journal type. You can transfer items to locations by using different rules. In this example, you transfer an item within the same warehouse, from a license plate controlled location to a location that isn't license plate controlled.

1. On the **Journal lines** FastTab, select **New**.
2. In the **Item number** field, enter or select a value. If you're using USMF, select *A0001*.
3. In the **From site** field, enter or select a value. If you're using USMF, select *2*.
4. In the **To site** field, enter or select a value. If you're using USMF, select *2*.
5. In the **From warehouse** field, enter or select a value. If you're using USMF, select *24*.
6. In the **To warehouse** field, enter or select a value. If you're using USMF, select *24*.
7. In the **From location** field, enter or select a value. If you're using USMF, select *FL-001*.
8. In the **To location** field, enter or select a value. If you're using USMF, select *BULK-001*.
9. In the **Quantity** field, enter a number.
10. On the **Line details** FastTab, select the **Inventory dimensions** tab.
11. In **From inventory dimensions**, in the **License plate** field, enter or select a value. If you're using USMF, select *24*.
12. Select **Save**.

Post the inventory transfer journal

1. On the Action Pane, select **Post**.
2. Select **OK**.

View inventory transactions

To see the transactions that were created when you posted your journal, follow these steps:

1. On the **Journal lines** FastTab, select the line you want to inspect.
2. On the **Journal lines** FastTab toolbar, select **Inventory** > **Transactions**.

Last updated on 07/01/2026

Adjust stock levels in the warehouse (basic warehousing)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure walks you through the process of creating and posting an inventory adjustment journal in order to adjust stock levels of products in the warehouse. You need to have an inventory journal name set up for inventory adjustments before you start this. You can walk through this procedure in demo data company USMF, or using your own data. These tasks would normally be carried out by a warehouse employee.

Create an inventory adjustment journal

1. Go to **Inventory management** > **Journal entries** > **Items** > **Inventory adjustment**.
2. Select **New**.
3. In the **Name** field, select the drop-down button to open the lookup.
4. In the list, select on the inventory adjustment journal name you want to use. Some other fields are populated based on the setup of the inventory adjustment journal name you select.
5. Select **OK**.

Create journal lines

1. Select **New**.
2. In the list, mark the item number field.
3. In the **Item number** field, select an item. If you're using demo data company USMF, type *D0001*.
4. In the **Site** field, select the drop-down button to open the lookup.
5. In the list, select a site.
6. In the **Warehouse** field, select the drop-down button to open the lookup.
7. In the list, select a warehouse. If you selected an item with location as a mandatory dimension, you must specify the location here.
8. In the **Quantity** field, enter a number. The cost price field specifies the cost per unit for inventory receipts. If the cost isn't specified for the item number or if you wanted to

change it manually, you would do this here.

Validate and post the inventory adjustment journal

1. Select **Validate**.
2. Select **OK**.
3. Select **Post**. When you post this kind of journal, an inventory receipt or issue is posted, the inventory level and value are changed, and ledger transactions are generated.
4. Select **OK**.
5. Close the form.
6. Close the page.

Last updated on 04/10/2025

Inventory tag counting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about tag counting, which you use to compare the actual contents of a warehouse with the on-hand inventory.

By creating lines on the **Tag counting** page, you place a tag number on each inventory item, such as a number from 1 to 500. During the count, you enter the item number and the quantity on a corresponding tag. This tag can then be used as the basis for input in the tag counting journal.

After you post the tag counting journal, a new counting journal is created on the **Counting** page. The new journal is based on the tag counting journal lines that you created.

To tag-count items by a specific inventory dimension, select the dimension on the **Display dimension** page that is displayed when you create the tag counting journal. For example, to count items in a specific warehouse, select the **Warehouse** check box. If **Lock items during count** on the **General** tab of the **Inventory and warehouse management parameters** page is set to *Yes*, items can't be physically updated during counting. However, items in tag counting journals aren't locked during counting. Inventory transactions aren't created until the tag counting lines are posted and transferred to a counting journal. If tags are entered randomly, and you want to identify missing tags, select the **Tag** column header to sort the lines by tag.

Related information

Cycle counting

Last updated on 04/10/2025

Count inventory in a warehouse

This article describes the process of creating and posting an inventory counting journal to count a specific item at a location in the warehouse. The procedure applies to the Inventory management module when you're not using warehouse management processes (WMS). The procedure doesn't apply to WMS-enabled warehouse functionality that's available in the Warehouse management module. You can walk through this procedure in the demo data company USMF, or use your own data. If you use your own data, make sure that you set up products and locations, and create an inventory journal name for counting journals. A warehouse employee normally performs inventory counting.

Create an inventory counting journal

1. Go to **Inventory management** > **Journal entries** > **Item counting** > **Counting**.
2. Select **New**.
3. In the **Name** field, select the inventory counting journal name you want to use from the drop-down list. Some other fields are populated based on the setup of the inventory counting journal name that you select.
4. In the **Worker** field, select the drop-down button to open the lookup.
5. In the list, **Select** the worker you want to use.
6. Select **OK**.

Create journal lines

1. Select **New**.
2. In the **Item number** field, select the desired record from the drop-down list. If you're using demo data company USMF, select *A0001*.
3. In the **Site** field, select the desired record from the drop-down list. If you're using demo data company USMF, select site *2*.
4. In the **Warehouse** field, select the desired record from the drop-down list. If you're using demo data company USMF, select warehouse *24*.
5. In the **Location** field, select the desired record from the drop-down list. If you're using demo data company USMF, select location *BULK-001*.
6. In the **Counted** field, enter a number. If you enter a counted number that's different from the number shown in the **On-hand** field, the **Quantity** field is updated to show the discrepancy.

7. Select **Save**.

Post the inventory counting journal

1. Select **Post**. When you post an inventory counting journal, if the counted amount differs from the amount that's reported in the **On-hand** field, an inventory receipt or issue is posted. The inventory level and value change, and ledger transactions are generated.
2. Select **OK**.

View inventory transactions

1. Select **Inventory**.
2. Select **Transactions**. You can see related transactions that are created when you post your inventory counting journal.

Last updated on 07/01/2026

Define inventory counting processes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to configure basic inventory counting processes by creating a counting group and a counting journal. It also shows you how to enable counting policies at the warehouse and item levels.

For more information about advanced warehouse scenarios, see Cycle counting.

Create a counting group

1. Go to **Inventory management** > **Setup** > **Inventory** > **Counting groups**.
2. Select **New**.
3. In the **Counting group** field, enter a value.
4. In the **Name** field, type a value.
5. In the **Counting code** field, choose one of the following values:
    - *Manual* – Add lines to the counting journal every time you run the job. In other words, you decide the counting interval for the counting group.
    - *Period* – Add lines to the counting journal according to the period interval set in the **Counting period** field. When you choose this option, the job for creating new lines in the counting journal adds them only at the specified interval, regardless of how often you run the job. For example, suppose **Counting period** is set to *7* and counting journal lines were last generated on January 1. If another job is started on January 5, then no lines are generated in the counting journal because seven days haven't passed. However, if you start the job again on January 8, then lines are generated because seven days have passed.

- *Zero in stock* – If on-hand inventory reaches zero (0), generate lines in the counting journal when the job is run. If the on-hand inventory reaches 0 after a count, lines are generated the next time that you start the count.
- *Minimum* – Insert lines in the counting journal if the on-hand inventory is equal to or less than the minimum that is specified.

6. If you set **Counting code** to *Period*, enter the period interval (in days) in the **Counting period** field.

7. Select **Save**.

Create a counting journal name

1. Go to **Inventory management** > **Setup** > **Journal names** > **Inventory**.

2. Select **New**.

3. In the **Name** field, type a value.

4. In the **Description** field, enter a value.

5. In the **Journal type** field, select *Counting*.

6. In the **Voucher series** field, select the number sequence to use for generating voucher IDs for new counting journals.

7. In the **Detail level** field, select the level of detail to apply when posting the journal.

8. In the **Reservation** field, choose the method used to reserve items during counting. Choose one of the following values:

   - *Manual* – The items are reserved manually on the **Reservation** page.
   - *Automatic* – The order quantity is reserved from the available on-hand inventory for the item.
   - *Explosion* – The reservation is part of the master planning of the transaction.

9. Select **Save**.

Set standard counting journal name

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.

2. Open the **Journals** tab.
3. In the **Counting** field, select the journal you created for this purpose. This journal name is the default for inventory journals of the *Counting* type.
4. Open the **General** tab. If you want to lock items during the counting process to prevent updates for packing slips, picking lists, or picking list registrations, set **Lock items during count** to *Yes*. Otherwise, set it to *No*.

Set the counting policy for an item

1. Go to **Product information management** > **Products** > **Released products**.
2. Open the item that you want to set counting policies on. You must select an item that is inventory tracked. Non-stocked products can't be counted.
3. Select **Edit**.
4. Expand the **Manage inventory** FastTab.
5. In the **Counting group** field, assign a counting group. This product is now included when inventory counting journal lines are created by using this counting group.
6. Select **Save**.

Implement advanced counting scenarios

To prevent double-counting mistakes, the system normally only allows items to be part of a single active counting journal at a time. However, if your company processes require it, you can relax this constraint to count items independently per warehouse or turn this integrity check off completely. The following subsections explain how.

Count items independently per warehouse

To set the counting policy to allow items to be counted independently in each warehouse, and ensure that items with identical dimension values can only be included within the same counting journal, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Open the item that you want to be counted. You must select an item that is inventory tracked. Non-stocked products can't be counted.
3. On the Action Pane, open the **Manage inventory** tab and, from the **Warehouse** group, select **Warehouse items**.
4. On the Action Pane, select **New**.
5. On the **General** FastTab, set the **Warehouse** field to the warehouse you want to set up specific counting policies for.

6. On the Action Pane, select **Save**.
7. Repeat from step 4 for each warehouse where the item should be counted.
8. Repeat from step 2 for each item that should be counted independently in each warehouse.
9. Go to **Inventory management** > **Journal entries** > **Items counting** > **Counting**.
10. On the Action Pane, select **New**.
11. On the **Overview** FastTab, select the **Site** and **Warehouse**.
12. On the **Counting by** FastTab, set **Warehouse** to *Yes*.
13. Select **OK** to create the journal.
14. In the newly created counting journal, add items as usual.

Disable counting status registration

To disable counting status registration, which allows duplicate item and dimension combinations to be placed in multiple active counting journals, follow these steps:

1. Go to **Inventory management** > **Setup** > **Journal names** > **Inventory**.
2. Select or create a journal name with a **Journal type** of *Counting*.
3. Set **Counting status registration policy** to *Disable counting status registration*.

Last updated on 05/06/2026

Set up consignment

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to use the inbound consignment inventory processes.

Consignment inventory is inventory that a vendor owns but stores at your site. When you're ready to consume or use the inventory, you take over the ownership of the inventory. This article includes information about how to physically receive vendor-owned inventory on hand without creating general ledger transactions, how to start a production process where the vendor-owned inventory can be physically reserved, and how to change the ownership of the raw material to process the consumption as part of the production order processing. It also includes information about how vendors can monitor consumption of their inventory by using the vendor collaboration interface.

Overview of the consignment process

In this example scenario, company USMF has a consignment agreement with vendor *US-104* for the raw material *M9211CI*.

1. A USMF employee manually creates a consignment replenishment order based on the expected demand. The order is created for vendor *US-104* and a line is added for item *MS9211CI*.
2. The vendor is informed about the expected delivery. This notification can happen in one of three ways:

    - Someone working at USMF sends the order information to the vendor.
    - The vendor monitors the expected inventory on-hand by using the vendor collaboration interface.
    - Someone working at USMF filters the data on the **On-hand inventory** page to show only the records for vendor *US-104*, where the receipt status is *Ordered*, and then sends this information to the vendor.

3. The inventory is delivered from *US-104* to USMF.
4. When the material arrives at USMF, the consignment replenishment order is updated with a product receipt. Only the physical quantities of the vendor-owned inventory are recorded. No general ledger transactions are created, because the inventory is still owned by the vendor.
5. The vendor monitors updates to the physical on-hand inventory by using the **On-hand consignment inventory** page.
6. Now that the physical inventory is on-hand, the production process reserves the vendor-owned inventory and starts the production order for the finished goods that are going to consume the raw material M9211CI.
7. The owner of the reserved raw materials that are going to be consumed in today's production is changed from *US-104* to USMF. This change is made by using an Inventory ownership change journal. This process creates purchase orders where the **Origin** field is set to *Consignment*.
8. The vendor monitors the consumption (change of ownership) on the **Products received from consignment inventory** page and issues an invoice based on the agreements between the two companies.
9. The production process consumes the raw material via a production picking list. The physical reservation is automatically updated to reflect that the on-hand inventory is owned by USMF.
10. The invoice from *US-104* is processed against the purchase orders that were autogenerated when the inventory ownership change journal was processed. Payment is made to vendor *US-104* for the inventory that was consumed.

USMF carries out additional periodic processes:

- A transfer journal is used to process the physical movement of the vendor-owned inventory between different warehouses.
- An **Item counting** journal is used to update the physical inventory on-hand. The vendor can also use counting to update the on-hand inventory, if they have permission to do this.

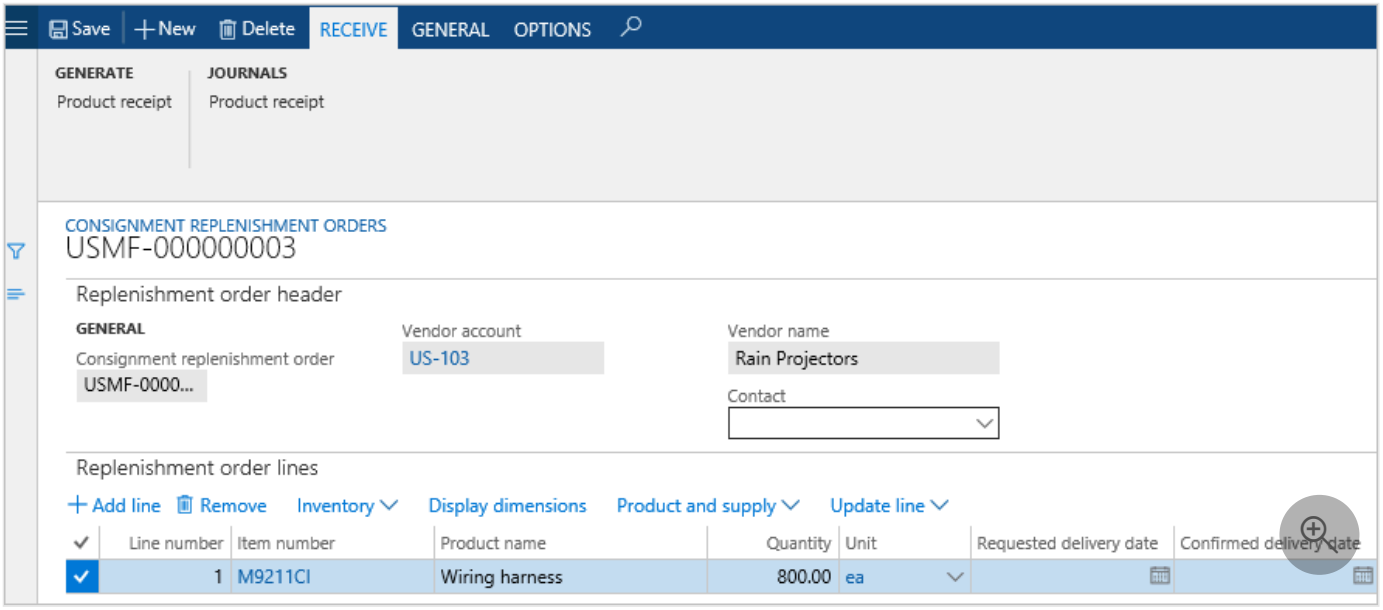
The vendor, *US-104*, can monitor the updates by using the **On-hand consignment inventory** page.

Consignment replenishment orders

A consignment replenishment order is a document that is used to request and keep track of inventory quantities of products that a vendor intends to deliver within a certain date interval by creating ordered inventory transactions. Typically, the order is based on the forecast and actual

demand of the specific products. The inventory to receive against the consignment replenishment order remains the vendor's ownership. You only record the possession of the products related to the physical receipt update, so no general ledger transaction updates occur.

The **Owner** dimension is used to separate information about which inventory is owned by the vendor and which is owned by the receiving legal entity. Consignment replenishment order lines have an *Open order* status as long as the full quantity of the lines isn't received or canceled. When the full quantity is received or canceled, the status is changed to *Completed*. The physical on-hand inventory that's related to a consignment replenishment order can be recorded using either a registration process or a product receipt update process. Registration can be done as part of the item arrival process or by manually updating the order lines. When the Product receipt update process is used, a record is made in the product receipt journal, which can be used to acknowledge the receipt of goods to the vendors.



Inventory ownership change journal

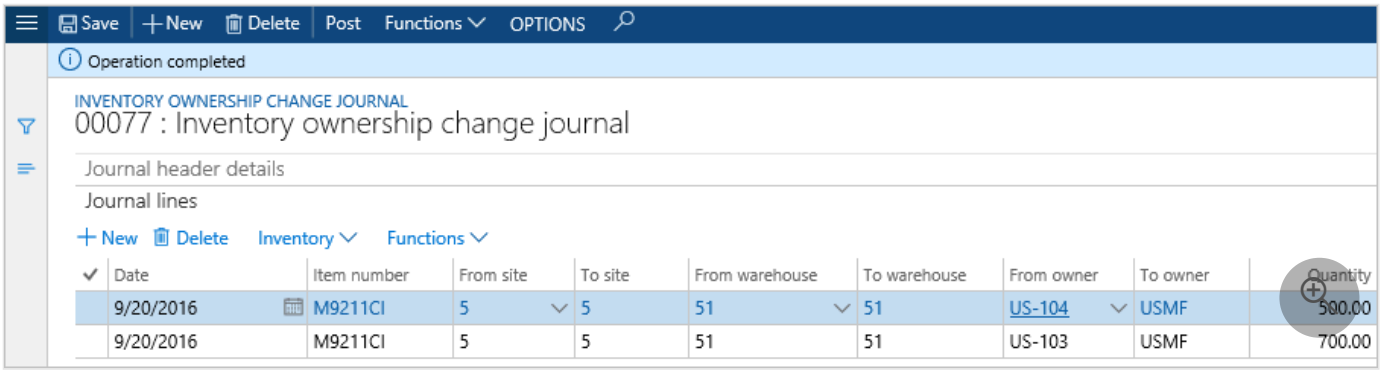
The **Inventory ownership change** journal is used to record the transfer of ownership of consignment inventory from the vendor to the legal entity that consumes it. The journal doesn't create expected inventory transactions. Like any inventory journal, it must be identified with an Inventory journal name. These names are created on the **Inventory journal names** page, and the **Journal type** must be set to *Ownership change*.

The only inventory transactions created are those that relate to a posted journal. When the journal is posted:

- The vendor-owned inventory is issued using an *Ownership change* reference with a *Sold* status.

- The legal entity that consumes the inventory receives on-hand inventory by using a product receipt updated inventory transaction on the purchase order. This action sets the status of the order to *Received*. Purchase orders used for consignment have the **Origin** field set to *Consignment*.

You can't update the quantity on consignment purchase order lines after the order is created.



Vendor collaboration in consignment processes

If your vendors use the vendor collaboration interface, they can use it to monitor the consumption of inventory at your site. The vendor collaboration interface has three pages related to the inbound consignment process:

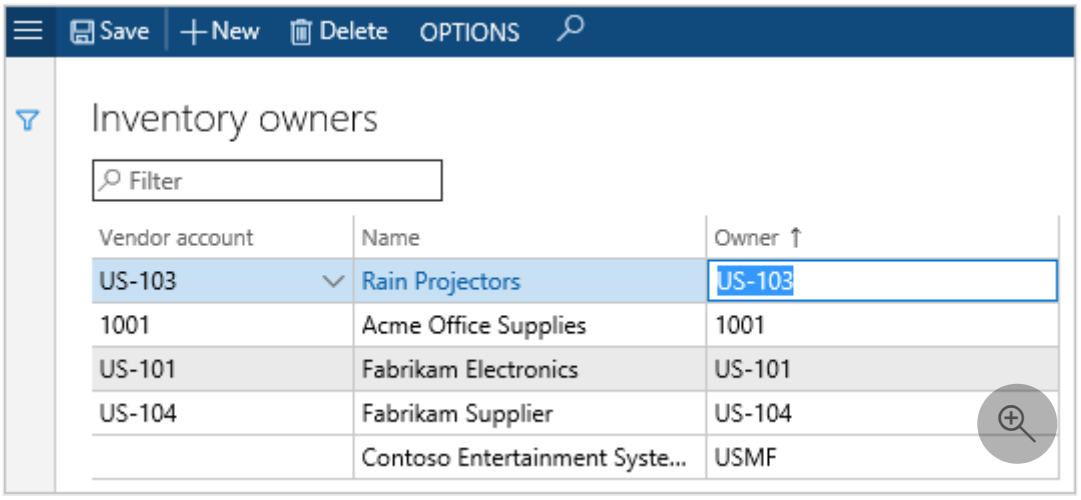
- **Purchase orders consuming consignment inventory** – Shows detailed purchase order information related to the ownership change from the consignment process.
- **Products received from consignment inventory** – Shows information about the items and quantities that have product receipts updated during the ownership change process.
- **On-hand consignment inventory** – Shows information about the consignment items that they're expected to deliver, and the items that are already physically available at the customer site.

For more information about setting up vendors to use vendor collaboration, see Manage vendor collaboration users.

Inventory owners

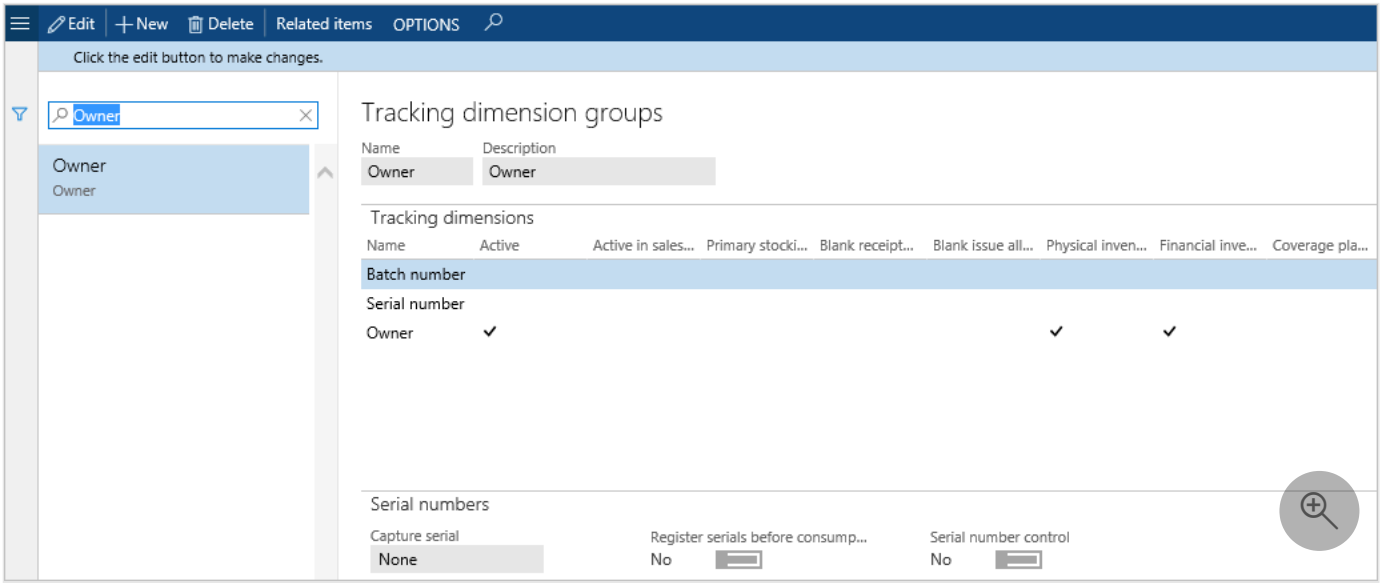
To record physical inbound consignment inventory, you need to define a vendor owner. You define the vendor owner on the **Inventory owner** page. When you select a **Vendor account**, it generates default values for the **Name** and **Owner** fields. The value in the **Owner** field is visible to the vendor, so you might want to change it if your vendor account names aren't easy for external people to recognize. You can edit the **Owner** field, but only up to the point when you save the

Inventory owner record. The **Name** field is populated with the name of the party that the vendor account is associated with, and you can't change it.



Tracking dimension group

Items for use in consignment processes must be associated with a **Tracking dimension group** where the **Owner** dimension is set to **Active**. The Owner dimension always has the **Physical inventory** and **Financial inventory** options selected. The **Coverage plan by dimension** is never selected.



Last updated on 05/06/2026

Change the ownership of consignment inventory based on production demand

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to change the owner of consignment inventory from the vendor to your legal entity when there's demand for the inventory in production. This change of ownership is done by creating and posting an inventory ownership change journal. The ownership change journal lines can be created manually or, as shown in this recording, based on existing production demand. Typically, a shop floor supervisor performs this task. You can use this procedure in the USMF demo data company or on your own data. If you're using your own data, make sure that you have the following prerequisites: an inventory journal name that has been set up for inventory ownership change, physically recorded vendor-owned on-hand items, and one or more production order lines for the material. This procedure is for a feature that was added in Dynamics 365 for Operations, version 1611.

> **Note**
>
> Outbound consignment processes aren't supported out-of-the-box and automatic ownership journal processing isn't supported.

Create an inventory ownership journal

1. Go to **Inventory management** > **Journal entries** > **Items** > **Inventory ownership change**.
2. Select **New**. The inventory ownership change journal is used to change the owner of consignment inventory from the vendor to the current legal entity. This change of ownership is done by releasing the on-hand inventory that is owned by the vendor and then receiving that inventory in the current legal entity.
3. In the **Name** field, enter or select a value. You can select only inventory journal names that have a journal type of *Ownership change*.
4. Select **OK**.
5. Select **Functions**.

6. Select **Create journal lines from production orders**. You can start the change of ownership process by querying all the production lines that have demand for consumption of raw material.
7. In the **Inventory issue statuses to include** field, select an option. This option lets you filter by the issue status of the inventory transactions. For example, you can create journal lines for inventory that has the *Picked* and *Reserved physical* statuses.
8. Expand the **Records to include** section. This section lets you apply additional filtering. For example, you can select a specific raw material date.
9. Select **OK**.

Post the inventory ownership change journal

1. Select **Post**. When the journal is posted, the vendor-owned inventory is released by using an *Ownership change* reference. The inventory is then received as on-hand by using an inventory transaction that is updated with a purchase order product receipt. Only transactions that are related to the posted journal are created. No expected inventory transactions are created.
2. Select **OK**.
3. Close the page.

Last updated on 04/10/2025

Create a consignment replenishment order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to create a consignment replenishment order where you can track the expected delivery from a vendor into your consignment inventory. It also shows how to record a receipt of products so that the consignment inventory is registered as on-hand inventory owned by the vendor. This procedure would typically be done by a procurement professional. You can use this guide in demo data company USMF. This procedure is for a feature that was added in Dynamics 365 for Operations, version 1611.

Create a consignment replenishment order

1. Go to **Procurement and sourcing** > **Consignment** > **Consignment replenishment orders**.
2. Select **New**.
3. In the **Vendor account** field, select vendor **US-104** (you must select a vendor that's registered as an owner on the **Inventory owners** page).
4. Select **OK**.
5. Select **Add line**.
6. In the **Item number** field, type *M9211CI* (you must select an item that is set up for consignment inventory).
7. In the **Quantity** field, enter a number.
8. In the **Requested delivery date** field, enter a date. The requested and confirmed dates are used by the MRP engine for the expected arrival of the goods.
9. In the **Confirmed delivery date** field, enter a date.
10. Expand the **Line details** section.
11. Select the **Inventory dimensions** tab.
12. To show the owner in the **Inventory dimensions owner** field, refresh the page. Vendor US-104 is now listed as the owner.

Check the inventory transaction status

1. Select **Inventory**.
2. Select **Transactions**.

3. In the desired row, notice that the **Receipt** field is set to *Ordered*.
4. Close the page.

Receive items

1. Select **Product receipt**.
2. In the **External product receipt** field, type a value.
3. In the **Quantity** field, enter a number that's lower than the number that's shown there.
4. Select **OK**.

Check the on-hand inventory

1. Select **Inventory**.
2. Select **On-hand**.
3. Select **Overview**. The items that have been received as consignment inventory owned by the vendor are available on-hand. The remaining quantity on the consignment replenishment order is shown in the **Ordered in total** field.
4. Close the page.

Last updated on 04/10/2025

Monitor consignment inventory using vendor collaboration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to use vendor collaboration to see information about the stock level of product that you placed in consignment with a customer. You can also monitor the consumption of the stock when the customer takes ownership of the inventory.

View consumed inventory

To view the inventory that the customer has consumed from the consignment stock, follow these steps:

1. Go to **Vendor collaboration** > **Consignment inventory** > **Products received from consignment inventory**. The list shows the product receipt lines that were generated when ownership of the consignment inventory changed from the vendor to the customer. You might have to scroll to the right to see quantities and other information. You can use the information in this list to generate invoices for your customer. You can also export the data to Microsoft Excel.
2. Select **View purchase order**.
3. Expand the **Line details** section. The line is marked as *Consignment*, and the header section shows that the purchase order has a status of *Received*.
4. Close the page.

View on-hand inventory

To view on-hand inventory, go to **Vendor collaboration** > **Consignment inventory** > **On-hand consignment inventory**. The **On-hand consignment inventory** page shows the stock that you own at the customer's warehouse. You can show additional dimensions, such as the site and warehouse, by selecting the Display dimensions tab.

Last updated on 11/20/2025

Item and raw material tracing in inventory, production, and sales

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how you can use item tracing to identify where items or raw materials have been used, are being used, or will be used in production and sales processes.

Item tracing functionality is available on the **Item tracing** page. The following sections describe how you can use item tracing, and what the options and limitations are.

What is item tracing?

Item tracing is a business intelligence (BI) tool that provides visibility into the source and destination of items and raw materials in the supply chain. Manufacturers can trace items, raw materials, or ingredients back to the vendor, and forward through the production and sale of the finished product. Item tracing helps manufacturers comply with regulatory requirements, and also helps quality officers and production managers analyze and take action to address variances in the quality of products and materials. Here are some examples of the ways that manufacturers can use item tracing:

- Identify the amount of an item or raw material that is currently in inventory, and where it's stored.
- Determine how much of the item or raw material is shipped, and which customers it was shipped to.
- Identify any planned shipments that include the item or raw material.
- Locate production orders that use the item or raw material, and that are planned, in progress, or reported as finished.
- Find out where the item or raw material was purchased.
- Investigate where an item or raw material was consumed in the production of another item.

What can I trace, and are there any limitations?

You can trace historical inventory transactions for items and raw materials, based on an item number and a tracking dimension, such as a serial number, batch number, or vendor batch number. You can trace an item or raw material only if you assign a tracking dimension to it. Because tracing is based on inventory transactions, some limitations exist when you trace items. For example, some limitations relate to transactions for projects, fixed assets, and commerce. Additionally, co-products appear in the trace details, but by-products aren't included. The trace includes all warehouse transactions from one location to another. Therefore, users might find the amount of information overwhelming. The trace displays for one legal entity at a time. The trace doesn't include cross-company capabilities in an intercompany context. You must start a new trace for each company where an item is received or issued.

What criteria can I specify for an item trace?

To perform an item trace, you need to provide the item number, a tracking dimension (such as a batch number or serial number), and the direction. The following table describes the criteria that you can use in an item trace.

Expand table

Field group	Description
Item number	Enter the identifier for the item or raw material that you're tracing.
Product dimensions	Optional: Trace specific aspects of the product definition, such as a configuration, size, color, or style.
Tracking dimensions	Enter a batch number, vendor batch number, or serial number tracking dimension. When you use the batch number as a criterion, the vendor batch number is displayed if you captured that information.
Storage dimensions	Optional: Trace items that are stored in a specific location.
Period	Optional: Enter dates to limit the trace to a specific period. The trace details show only documents and transactions that were created between these dates.
Forward or backward	Select the direction for the trace. You can trace forward or backward: • **Backward** – Trace downstream to identify the source, the quantity that remains on hand, and any production orders that are at least partially reported as finished.

Field group	Description
	• **Forward** – Trace upstream to identify the destination. You can find the sales orders, and the customers that the traced item or raw material is at least partially shipped to.

What information is included in the trace details?

The results of a trace appear in chronological order in the tree on the **Details** FastTab of the **Item tracing** page. The order varies, depending on the trace direction. The details include all transactions from the purchase of the item from the vendor to the sale of the item to the customer. Trace results also include interim products that are related to the item or the tracking dimension specified in the trace criteria. The top node shows the quantity of the item, in the inventory unit, that remains on hand, based on the storage dimensions specified in the trace criteria.

> **Note**
>
> The trace details provide a snapshot of the information that was available when the trace was done. The trace details aren't updated if the information changes after the trace is done.

Why don't some nodes contain any details?

To reduce the amount of information in the trace details, only the node for the first instance of the item or raw material includes details. If a selected node doesn't contain details, you can view the node that does contain details by selecting **Go to traced line**. You can then return to the node that you left by selecting **Go back**.

Can I view only a particular type of document? For example, can I view only production orders, customers, or vendors?

Yes, from the trace details, you can open list pages that include only a particular type of document or transaction. You can access these pages from the **Tracing** menu item on the Action Pane, in the **Item**, **Sales**, **Purchase**, **Production**, and **Quality** groups. For example, to view a list of the vendors in the trace details, select **Tracing** > **Purchase** > **Vendors**. The list pages summarize the documents or transactions from the trace details. The **Pending transactions**, **Pending orders**, and **Not shipped sales orders** list pages also show other information that isn't included in the

trace details. Additionally, they always show results as of the current date, even if a date range was specified. The following table describes the additional details that these pages can include.

Expand table

Lists	Description
Not shipped sales orders	View sales order lines that aren't picked, and that therefore aren't shown in the trace details.
Pending orders	View all pending production orders for the traced item, regardless of the tracking dimensions that the trace criteria used. You can also view pending production orders where the traced item is an ingredient, and where no registrations exist and no transactions are reported as finished for the order.
Pending transactions	View pending inventory transactions for the traced item that include the specified tracking dimensions or a blank tracking dimension. You can also view pending inventory transactions for items and tracking dimension combinations, or a blank value, in the trace details.

How many levels can I trace up or down in a BOM or formula?

You can trace one level up or down in a bill of materials (BOM) or formula. For example, if you run a trace on raw materials, you can see the finished product and any co-products. However, if you trace a co-product, the trace details don't include the finished product.

How can I view more details about a document or transaction in the trace details?

On the **Details** FastTab, there are two ways to view more information about a selected document or transaction in the trace details:

- When you select a node in the trace details on the **Details** FastTab, the other FastTabs on the page display information about the document or transaction in the node.
- Open the details page for the document in a selected node by selecting **View details**. For example, if you select a node that describes a production order, and you select **View details**, the **Production orders details** page appears.

Some FastTabs give you access to additional information about the selected node. For example, on the **Non conformances** FastTab, you can select **Inquiries** to investigate whether there's a history of non-conformance. On the **Batch** FastTab, you can select **On-hand list** to view the

amount of physical inventory that is currently on hand and any inventory transactions that involve the batch.

Can I run more than one trace and then compare the details?

After you run the trace, use the following options on the **Trace from node** button to run a new trace on the transaction in the selected node:

- **Backward** or **Forward** – Start a new trace for the selected node, and overwrite the details of the current trace.
- **New backward** or **New forward** – Start a new trace in a new window, and keep the details of the current trace.

To use the **New backward** or **New forward** option, use the **Open in a new window** functionality to have a new trace appear in a new window.

Can I save the trace details?

You can save the information on the **Details** tab as an XML file by selecting **Export** under the **Tracing** action on the Action Pane. In addition to the trace details, the XML file includes the trace criteria, parent node, and on-hand quantity. The capability to save the details of a trace is useful if, for example, you want to attach the information to a quality order or other compliance documentation. You can specify where the file is saved. To view the file immediately, select the **Show document** option.

> ⓘ **Note**
>
> The file is always saved, even if you only want to view it. By default, the XML file opens in a browser window. However, you can right-click the file, select **Open with**, and then select the program to use to display the contents.

Can I calculate a balance for a particular item or ingredient?

You can export the information from the summary pages to Microsoft Excel. Open the relevant page, select the **Open in Microsoft Office** icon, and then select **Export to Microsoft Excel**. This functionality is especially useful when you want to calculate a mass balance for an item or

ingredient from the **Transactions summary** page. On the **Transactions summary** page, you can filter on the item or ingredient, and optionally on the batch, and then export the information to Excel. In Excel, you can isolate, for example, the on-hand quantity, the quantity that was sold, and the amount that was used in production.

Can I investigate whether there's a history of issues with items or raw materials?

The trace details include information about quality orders and nonconformances that involve the item or raw material. To view a summary of quality orders and nonconformances, select **Quality orders** or **Non conformances** on the Action Pane.

> **Note**
>
> Destructive quality orders can appear more than one time in the trace details. When a destructive quality order is created for a document, such as a purchase order, it appears for each transaction for that document.

Are there any reporting capabilities that are related to item tracing?

You can generate the **Shipped to customers** report to identify the amount of the item or raw material that's shipped and the customers that it was shipped to. For an inquiry that's related to compliance, you can generate the report for all customers. For an inquiry that's related to customer service, you can generate the report for a selected customer. If the product was a raw material used in the production of a finished item, the finished item is also included.

> **Note**
>
> If you're using the features for deleting or archiving sales orders, the report results also include any sales orders that are deleted or archived.

Can I trace co-products and by-products?

You can trace co-products, but you can't trace a by-product, because tracking dimensions aren't typically assigned to by-products. When you trace an item, the trace details include any related co-products. A node that contains a co-product includes the word "co-product" in the details.

You can also view details about a co-product by selecting the node in the trace details and then selecting the **Production** FastTab.

Last updated on 05/22/2026

Trace an item or raw material

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure demonstrates how to use item tracing to identify where items or raw materials have been used or are being used. With this procedure, you can identify an item, trace it back to the source, and then trace forward through the production and sale of the finished product. The process can be used to investigate the customers impacted, the sales orders affected, and more. This procedure uses demo data company USP2.

Trace an item backwards using a known batch number

1. Go to **Inventory management** > **Inquiries and reports** > **Tracking dimensions** > **Item tracing**.
2. In the **Item number** field, select *P9100*.
3. In the list, select the link in the selected row.
4. In the **Forward or backward** field, select *Backward*.
5. In the **Batch number** field, select *as-12-344-01*.
6. In the list, select the link in the selected row.
7. Select **OK**.

Identify an item, trace it forward, and make an analysis

The top node of the tree represents the on hand quantity of the selected item and batch. You need to expand the nodes of the tree to find the item that the forward trace should be executed on.

1. In the tree, expand the following nodes and then select the last node.

   Expand: *P9100 / 1 / 10 / as-12-344-01 ● 2 keg ● 7.00 gal \P9100 ● Picked ● Sales order 000072 ● 12/22/2015 ● -1 keg ● -4.00 gal ● Site=1, Warehouse=10, Batch number=as-12-344-01 \P9100 ● Production B-000050 ● 12/9/2015● 7 keg ● 27.00 gal ●*

Site=1,Warehouse=10,Batch number=as-12-344-01 ● Co-products: P9101 and then select that node.

2. In the tree, expand the following node and then select that node.

   Starting from the node that you just selected, expand *M9103 ● Production line B-000050 ● 12/9/2015 ● -160.00 lb ● Size=70, Color=OK, Site=1, Warehouse=10, Batch number=App01* and then select that node.

3. Select **Trace from node**.

4. Select **Forward**.

5. On the Action Pane, select **Tracing**.

   There are several tracing options which provide information about the customers impacted by the item that you're tracing, and the sales orders related to the item which have and haven't been shipped.

6. Select **Customers**.

7. Close the page.

8. On the Action Pane, select **Tracing**.

9. Select **Shipped sales orders**.

10. Close the page.

Last updated on 04/10/2025

Merge inventory batches

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about how to consolidate two or more inventory batches into a merged batch.

When you merge batches, calculations can help optimize the characteristics and batch attributes of the merged batch. After you select the source batches, you can review and change the merged batch before you post it. You can also transfer the batch merge to an inventory journal for approval. Inventory can then be reserved or posted directly from that inventory journal. When you post a merged batch, the inventory is adjusted for the source batches and the merged batch.

Are there any prerequisites?

Yes, there are some things that you must set up before you can use the merge batch tools. The following table describes the prerequisites.

Expand table

Page	Description
Journal names, inventory	You must create a journal name of the type BOM that is used by default when you post batch merges in inventory journals. Optional but recommended: You can specify that reservations should be made automatically when the batch merge is transferred to the inventory journal. Otherwise, there's a risk that the on-hand inventory might be changed after the batch merge details are set up and the journal is posted. To enable automatic reservations for the journal name, select **Automatic** in the **Reservation** field.
Inventory and warehouse management parameters	You must specify the default journal name for the inventory journal.
Released products	Here are the recommended settings for the item:

Page	Description

	• To automatically generate batch numbers for merged batches, you must assign the released product to a batch number group. You can also enter a batch number manually when you create a merged batch, or select an existing batch number. If you select an existing batch number, make sure that the selected batch hasn't been included in any inventory transactions.
	• If you're using shelf life or best-before dates for the released product, the dates for a merged batch are calculated based on the selection in the **Batch merge date calculation** field. The following options are available:
	  ◦ **Earliest** – The calculation is based on the earliest date that is specified for a source batch that is selected for the batch merge.
	  ◦ **Latest** – The calculation is based on the latest date that is specified for a source batch that is selected for the batch merge.
	  ◦ **Manual** – No calculation is done. If a date is the same on all source batches, a date is suggested. You can change that date. If a date isn't the same on the source batches, you can manually enter the date.

Batch number group	Optional: To generate a batch number when you create a merged batch, you must assign a batch number group to the released product. Otherwise, you can enter a batch number manually.

When might I want to merge batches of inventory?

Here are some examples of scenarios where it might be useful to merge batches:

- Walking through the warehouse, Sammy notices that several batches of the same item have low quantities. Sammy is expecting to receive several new shipments, and realizes that some floor space can be freed by merging the odd quantities into a new batch.
- Sammy is receiving inventory, and wants to combine the new batch with one that has already been received, to improve the batch attribute value of the existing batch. By doing so, you create a new batch.

Can I merge batches across sites and legal entities?

No, you can merge only batches that have the same site and warehouse storage dimensions in one legal entity. However, you can specify a different location and pallet ID for the merged batch.

Can I merge partial quantities?

No, you can merge only the full quantity of batches. The batch merge functionality is intended as an inventory feature, not a production feature.

What if the batches have different batch attribute values?

When you select the source batches to combine in the merged batch, Supply Chain Management verifies whether all the batches have the characteristics or attribute values. When an attribute value is the same, a value is suggested for the merged batch. You can change that value. Attribute values that aren't the same are left blank for the merged batch, and you can enter those values manually. If the batch attribute type for the attribute value is an integer or a fraction, and the values aren't the same for all the source batches, the value is calculated by using a weighted average calculation. The calculated value is rounded up or down to the nearest increment. If the value is blank for a source batch, the batch and its quantity aren't included in the calculation. **Example** The following example shows a weighted average calculation for a merged batch. Two of the source batches have a blank value for a batch attribute type that is an integer. The following attribute is assigned to the source batches.

Expand table

Attribute	Minimum	Increment	Maximum
Grade	3	3	30

The source batches have the following attribute values for the **Grade batch** attribute.

Expand table

Batch	Quantity	Attribute	Attribute value
B1	10	Grade	Blank
B2	15	Grade	15

Batch	Quantity	Attribute	Attribute value
B3	20	Grade	20
B4	25	Grade	Blank
B5	30	Grade	25

When you add these batches as source batches, the following values are assigned to the merged batch.

Expand table

Batch	Quantity	Attribute	Attribute value
B6	100	Grade	21

The values and quantities for batches B1 and B4 aren't included in the weighted average calculation. Therefore, here's how the value for the merged batch is calculated.

Expand table

Value	Quantity (weight)	Relative weight	Relative weight × Value
15	15	0.230769231	3.461538462
20	20	0.307692308	6.153846154
25	30	0.461538462	11.53846154
	Total: 65, which is the sum of the weights		**Total:** 21.5384615, which is rounded to 21 (the nearest increment)

What if the batches have different batch dates?

If the batches have different batch dates, some of the dates are calculated based on the values in the **Batch dates** group on the **Merged batch** FastTab of the **Batch merge** page. The system calculates values for the fields in the **Batch dates** group. These values include the manufacturing date, expiration date, shelf advice date, and best-before date. The dates are calculated based on settings for the item in the **Item data** field group of the **Released product details** page. You can change the values or enter them manually. For all other dates, no calculation is done. The same principle is used for batch attribute values. If a date is the same for all the source batches, that date is suggested for the merged batch. If the date isn't the

same for all source batches, the date is blank on the merged batch, and you can enter it manually.

What if the dimensions are different on the batches that I want to merge?

Here's how product dimensions, tracking dimensions, and storage dimensions are handled:

- **Product dimensions** – All product dimensions for the selected item must be the same. You can't merge batches across product dimensions.
- **Tracking dimensions** – A new batch number is automatically generated if a batch number group is specified for the item. If a batch number group isn't assigned to an item, you can select an existing batch or enter the number manually. Serial numbers are transferred from the source batch to the inventory journal lines for the merged batch.
- **Storage dimensions** – The site and warehouse storage dimensions must be the same for all the source batches and the merged batch. However, you can specify a new location and pallet ID for the merged batch.

How does posting work?

Posting works in two ways, depending on whether you use an approval process for journals. You can use the **Transfer to journal** and **Post the batch merge** actions to transfer the batch merge to a journal where it can be verified and posted, or you can post the batch merge directly. The main difference between the two actions is that a transfer to a journal doesn't post the batch merge. Both actions create a new batch if an existing batch isn't selected, update all batch details and attribute values, and create an inventory journal.

- **Transfer to journal** – Transfer the batch merge details to a new inventory journal. If you've set up automatic reservations, the quantities in the source batches are reserved. The details of the batch merge can't be changed. To modify the batch merge, you must delete the journal. The journal can be used as a task that another employee must perform later. The reservation of the batch quantity to the journal line is secured. This allocation lets a quality planner or a warehouse manager create tasks for their employees.
- **Post the batch merge** – Post the batch merge directly. This action can be performed after the physical merge has occurred.

You can approve the inventory journal for the batch merge from the **All batch merges** list page. Select **Journal** > **Post**. After a journal is posted, you can't change the details in the merged batch. After you transfer a batch merge to an inventory journal, you can change the details only if the journal is deleted.

After I merged a catchweight item, why can’t I see the catchweight information in the inventory journal?

You can merge batches of catch-weight items just like all other items. However, the catch-weight information doesn't appear in the inventory journal. We recommend that you verify the catch-weight information before you transfer the batch merge to the inventory journal.

Last updated on 04/10/2025

Correct inventory tracking information

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure walks you through the process of creating and posting an inventory transfer journal to correct inventory tracking information. In this example, you update the information of a batch-controlled item by changing an incorrectly registered batch to another batch. You can walk through this procedure in demo data company *USPI*, or use your own data. If you use your own data, you need to have an item that's batch-enabled, and it must not be location-controlled. You also need to have an inventory journal name set up for inventory transfers. These tasks are normally carried out by a warehouse employee.

Create an inventory transfer journal

To create an inventory transfer journal, follow these steps:

1. Go to **Transfer**.
2. Select **New**.
3. In the **Name** field, enter or select a value.
4. Select **OK**.

Create journal lines

To create journal lines, follow these steps:

1. Select **New**.
2. In the **Item number** field, enter or select a value. If you're using *USPI* demo data, select item *M5003*.
3. In the **Quantity** field, enter a number.
4. Select the **Inventory dimensions** tab.
5. In the **Batch number** field, enter or select a value.
6. In the **Site** field, enter or select a value.
7. In the **Warehouse** field, enter or select a value.
8. In the **Batch number** field, enter or select a value.

Post the journal

To post the journal, follow these steps:

1. Select **Post**.
2. Select **OK**.

Check tracing information

To check tracing information, follow these steps:

1. Select **Inventory**.
2. Select **Trace**.
3. Select **OK**. With this tracing information, you can trace back to the batch you corrected inventory from. You can also use the Item tracing page to see this information.
4. Close the page.

Check inventory transactions

To check inventory transactions, follow these steps:

1. Select **Inventory**.
2. Select **Transactions**. Here you can see the transactions that were created when you posted your journal.

Last updated on 11/20/2025

Working with serialized items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to register serial numbers on packing slips or invoices during the sales process. This functionality is useful if a company wants to capture serial numbers for service and warranty purposes, but doesn't need to maintain serial numbers in inventory from receipt to issue.

Many companies just want to capture serial numbers for service and warranty purposes, and don't need to maintain serial numbers in inventory from receipt to issue. In these scenarios, you can register the serial numbers on the packing slips or invoices when products are sold. If products are later returned, you can trace each product to an invoice to determine whether you sold the product, and whether the service or warranty obligations are valid.

You must enable serial numbers for the sales process by selecting the **Active in sales process** option on the **Tracking dimension groups** page. The following events then occur in Supply Chain Management:

- On the **Serial numbers** FastTab, the **Serial number control** option is set to *Yes*. When you select this option, you must register one serial number for each item on the packing slip or invoice.
- All selections on the tracking dimension group for serial numbers are cleared, except the **Blank issue allowed** option. You can select the **Blank issue allowed** option to override the serial number control, and allow products to be packed and invoiced without registering serial numbers.

> **Important**
>
> When the **Physical inventory** option for the **Serial number** dimension is cleared, including the automatic clearing that occurs when you select **Active in sales process**, on-hand

inventory isn't tracked per serial number. This change can cause discrepancies between received and issued serial numbers for the same item.

When do I register serial numbers during the sales process?

You can register serial numbers either on the packing slip for a sales order or on the invoice. When you prepare an invoice for a serialized item that was shipped together with a packing slip, you can select which serial numbers on the packing slip to invoice. The number of registered serial numbers must not exceed the quantity of items that were shipped. If you're creating a partial invoice, you can select fewer serialized items than were registered on the packing slip. When you print a packing slip or an invoice, the serial numbers that were registered are included.

Can I enter serial numbers by scanning them, or do I have to type them?

You can either scan or type serial numbers. When you use a scanner, the scan mode determines whether the serial numbers are added to or removed from the list of serial numbers on the invoice or packing slip. If you want to scan serial numbers by using, for example, a hand-held bar code scanner, configure the scanner to send an Enter or TAB command after the serial number. This command indicates the end of the data stream. Otherwise, you must press Enter or TAB on the keyboard after you scan each serial number.

If I enable serial numbers for the sales process, do I have to register all serial numbers for all items?

The setup of the tracking dimension group that is assigned to the product determines whether serial numbers must be registered for all items on a packing slip or invoice. When you enable serial numbers for the sales process, the **Serial number control** option is automatically selected. You must then register one serial number, or register a blank registration for an unreadable number, for each item on the packing slip or invoice. If you don't want to require a serial number for each item, select the **Blank issue allowed** option on the tracking dimension group that is assigned to the item. You can then register fewer serial numbers than the quantity of items that are being shipped. If you register more serial numbers than the quantity of items that are being shipped, you won't be able to post the packing slip or invoice.

Can I register serial numbers for partial invoices and partial shipments?

You can create partial invoices and packing slips for sales orders, and register serial numbers only for the items that those invoices and packing slips include. If you want to create a partial invoice and you have more than one packing slip for the sales order, you can include serial numbers from more than one packing slip. However, there can be only one packing slip that doesn't include all serial numbers. For example, if you have three packing slips, and each packing slip includes two serialized items, you can't create a partial invoice for one item from each packing slip.

What do I do when a serial number isn't readable?

If a serial number can't be read or scanned, you can create a blank line for the item by selecting **Not readable** on the **Serial numbers** page. If the serial number becomes available later, you can update the invoice or packing slip. For more information, see the next section, "Can I correct or change the serial numbers that I registered for a sales order?"

Can I correct or change the serial numbers that I registered for a sales order?

Yes, you can correct serial numbers if the following conditions are met:

- **Invoices** – You can change the serial numbers for items that you didn't yet invoice. The packing slip is then also updated. However, if you corrected a sales order line by registering a negative quantity, you can't change serial numbers for the sales order line.
- **Packing slips** – You can't partially correct a packing slip line that contains serialized items. You must reverse the full quantity for the line. If you canceled or corrected a packing slip, you don't have to register the reversed serial numbers again when you create a new packing slip for the same serialized items. The numbers that you registered are used.

Can I view the serial numbers that shipped together with a specific packing slip, or that were included on an invoice?

Yes, you can run an inquiry on the packing slip journal line or invoice journal line to view a list of all serial numbers that the document includes.

Can I view the serialized items that I have on hand?

No, you can't view the serialized items that you have on hand, because serial numbers aren't registered for items until the items are sold.

Can I register serial numbers for catchweight items?

No, you can't register serial numbers for catch-weight items during the sales process. Additionally, if a product is set up as a catch-weight item, you can't assign the product to a tracking dimension group that is set up to use serial numbers only during the sales process.

Can I register serial numbers at the retail POS?

Yes, the retail point of sale (POS) prompts the user to enter a serial number when the user sells an item that is assigned a tracking dimension group that is set up to use serial numbers only during the sales process.

What security roles are required to register serial numbers during the sales process?

All roles that can maintain sales packing slips and sales invoices can use this functionality. The following duties let workers correct serial numbers and register blank entries for serial numbers that can't be read or scanned:

- Maintain serial number corrections
- Maintain registration of non-readable serial numbers

Last updated on 05/27/2026

Outbound process overview

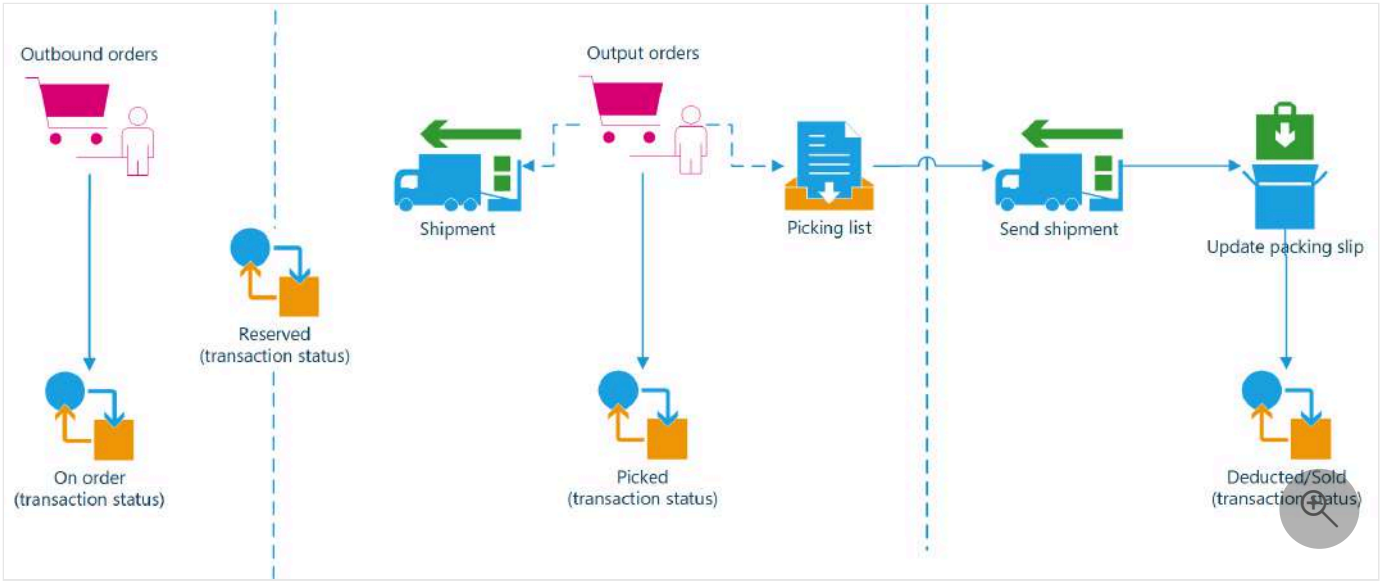
This article provides an overview of the outbound process in Inventory management.

Output orders

Use output orders to link sales order lines and transfer order lines with the outbound picking processes that use picking lists.

When picking lists are generated from either sales orders or transfer orders, output orders and shipments are automatically created. A picking list has a one-to-one relationship with a shipment. The transfer order shipment or the sales order packing slip can be processed from the shipment.

The following diagram shows an overview of the process for outbound orders.



Set up outbound rules to define how the program should handle the outbound process. Use these rules to control the shipment process. In particular, use the rules to control which stage in the process a shipment can be sent during. The following settings define how the outbound processes are handled.

Picking route status for sales and transfer orders

Go to **Account receivable** > **Setup** > **Account receivable parameters**, and then, on the **Updates** tab, select a value in the **Picking route status** field.

If the **Picking route status** field is set to **Completed**, the picking process occurs automatically as part of the process of generating picking lists. If the field is set to *Activated*, the picking list lines

must be manually updated.

The same setup applies to transfer orders. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**, and then, on the **Transport** tab, select a value in the **Picking route status** field.

End output inventory orders

Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**. On the **General** tab, set the **End output inventory order** option.

When the warehouse worker reduces the picking list quantities, the corresponding inventory order quantities are removed from the shipment. When the picking list is updated at a point in time, the remaining quantities are reported back to the order if the **End output inventory order** option is set to *Yes*. If the **End output inventory order** option is set to *No*, the remaining quantities stay as an open output order quantity. They must be added to a new picking list as part of the **Open output orders** functionality.

Reduce quantity

Use the **Reduce quantity** parameter as part of the process of generating picking lists. The setting of this parameter works together with the **Reservation** setting that triggers a reservation process as part of the release to the warehouse.

Example of an outbound process for a sales order

For this example, there's a sales order for two items. During picking list generation, you select the **Reduce quantity** parameter. Therefore, you release and create picking lines only for available on-hand inventory. You must report the picking via a registration process for picking lists (**Picking route status** = *Activated*).

The inventory, which isn't already reserved, is reserved during picking list generation. The unavailable inventory can be either removed from the sales order or released to the warehouse for outbound processing later, when inventory is available for picking.

As soon as you pick all the picking lines on the **Picking list registration** page, the associated shipment is completed. The process for sales order packing slips can then be initialized based on the picked inventory.

Last updated on 07/01/2026

Reserve inventory quantities

08/29/2025

This article describes the different options that are available for reserving inventory.

You can automatically reserve inventory quantities for a specific sales order. This reservation means that you can't withdraw reserved inventory from the warehouse for other orders unless you cancel the inventory reservation or part of the inventory reservation.

There are several reasons for reserving inventory:

- First ordered, first delivered, which means that customers get available items in the same order in which they place their orders.
- Shortage of items due to a long or unknown delivery time from the vendor. You might want to make sure that certain customers or orders get delivery of the first-available items.
- Certain customers and certain types of orders have first priority for delivery.
- Items with serial or batch numbers. You can mark certain items that are or will be delivered to specific orders.
- Specially ordered items that are reserved for certain orders.
- Production orders. For example, you can mark items that are produced for and adjusted to specific orders.

Inventory can be reserved automatically when you create a new order line, or you can reserve inventory manually on the individual orders. You can also reserve inventory at different stages in a production process. Only stocked products can be reserved. You can't reserve services because there's no on-hand inventory. You can reserve both physical on-hand inventory and ordered but not yet received inventory. If you reserve a larger quantity than what the on-hand inventory contains, a message displays that you can't reserve such a large quantity. You can then either reserve the quantity anyway or change the ordered quantity. The quantity can be either reserved or changed. If you reserve more items than are available, the shortage is covered the next time that items are available for delivery.

Inventory reservation policies

Set inventory reservation policies on the **Item model groups** page, the **Inventory and warehouse management parameters** page, or the **Production parameters** page.

Policies on the Item model groups page

The **Inventory policies** section contains the following reservation policies.

Expand table

Reservation policy	Description
FIFO date-controlled	If you select the **FIFO date-controlled** option, the inventory reservation is controlled by a sorting date according to the FIFO principle. Batches are reserved based on the earliest date of receipt of items, according to the principle of first in, first out (FIFO).
Backward from ship date	This option becomes available if you select the **FIFO date-controlled** option. If you select **Backward from ship date**, and if you're reserving available physical inventory, then the inventory reservation is controlled by the last update physical date according to the FIFO rule. If you're reserving available ordered inventory, the inventory is reserved backward from the desired ship date according to the principle of last in, first out (LIFO). If no receipts are available before the ship date, the reservation follows the LIFO rule using the transaction ship date.
Item sales reservation	Determines whether item reservation is manual or automatic. If a reservation is automatic, inventory is reserved when order lines are created. You can make reservations at the item number level for BOMs (**Automatic** option), or for the individual elements of a BOM (**Explosion** option). The default value for **Item sales reservation** might be inherited from **Accounts receivable parameters.** On that page, set the value in the Reservation field in the **Sales default values section** on the **General** tab.
Same batch selection	Same batch reservation lets you reserve inventory for a sales order line against a single batch of inventory. To use this option, you must also set the **Consolidate requirement** option to **Yes**. Additional settings are required for the tracking dimension group and storage dimension group. Learn more in Reserve the same batch for a sales order.
Consolidate requirement	This option is similar to the **Same batch selection** option, and it consolidates inventory that's reserved for sales order lines into a single requirement.
FEFO date-controlled	This option allows you to reserve batches that are close to their expiration date or best before date. You also need to set the **Pick criteria** field to select either **Expiration date** or **Best before date**.

Example for FIFO date-controlled and Backward from ship date

In this example, on-hand inventory for item number A exists for three different batch numbers.

Expand table

Item number	Batch number	Quantity	Date
A	1000	5	February 2, 2016
A	1001	3	January 1, 2016
A	1002	7	March 3, 2016

A sales order that should be automatically reserved and delivered on April 4, 2016, reserves the following batch:

- If both the **FIFO date-controlled** and **Backward from ship date** check boxes are cleared, batch 1000 is reserved because it's the batch with the lowest number.
- If the **FIFO date-controlled** check box is selected and the **Backward from ship date** check box is cleared, batch 1001 is reserved because it's the batch with the first date of receipt (FIFO).
- If both the **FIFO date-controlled** and **Backward from ship date** check boxes are selected, batch 1002 is reserved because it's the batch receipt closest to the sales order ship date.

Policies on the Inventory and warehouse management parameters page

The **Inventory and warehouse management parameters** page has two options related to reservations:

- The **Reserve ordered items** option on the **General** tab lets you reserve item receipts that you ordered against item issues in Accounts receivable, Project management and accounting, and Production control. If you clear this option, you can only reserve items that you physically received. If you set up a particular item to accept negative inventory, this field isn't relevant.
- The **Reserve items automatically** option on the **Transfer** tab determines the default setting for automatically reserving items for transfer orders. You can override the default setting on individual transfer orders.

Inventory reservation policies on the Production parameters page

The value of the **Reservation** field on the **General** tab on the **Production parameters** page determines the default point in the production process at which inventory should be reserved. For example, you can reserve inventory when work is scheduled or when work is started.

Reserve the same batch for a sales order

Article • 09/29/2021

This article explains how to set up a product to allow reservation of inventory against a single batch of inventory.

Same batch reservation lets you reserve inventory for a sales order line against a single batch of inventory. For example, a customer who orders wallpaper can request that the whole order be filled from the same batch or lot, to avoid inconsistencies among the rolls. To set up a product to use same batch reservation, the following settings must be active in the item model group, tracking dimension group, and storage dimension group that you assign to the product:

- **Item model groups** – The item model group must have the **Same batch selection** and **Consolidate requirement** fields selected in the **Reservation** field group for inventory policies.
- **Tracking dimensions groups** – The tracking dimension group must have the **Coverage plan by dimension** field selected for the batch number.
- **Storage dimensions groups** – The storage dimension group must have the **Coverage plan by dimension** field selected for **Site** and **Warehouse**.

When you reserve inventory for a product on a sales order line that is set up for same batch selection, the system tries to reserve the ordered quantity from a single inventory batch. Any specific batch attribute requirements are also considered. If the quantity can't be filled from a single batch, the **Same batch reservation conflict** page appears. This page describes the issues and also the actions that you can take to continue with the reservation. The following conditions might prevent the batch from being reserved:

- The batch disposition code has **Block reservation** for sales flagged as **Blocked**.
- The batch has expired, based on the expiration date and any applicable customer sellable days. The item can still be considered for reservation if the item model group for the item is First Expiry First Out (FEFO) date–controlled, and if the best-before date is selected as the pick criterion.
- The batch doesn't have enough shelf-life days remaining, based on the expiration date and best-before date, plus any customer sellable days.

For items associated with a storage dimension group that has **Use warehouse management processes** enabled, you can reserve specific batch numbers by using a reservation hierarchy where the batch number inventory dimension is defined above the location dimension. This type of reservation hierarchy is also known as a *Batch-above [location]* reservation hierarchy. The **Batch reservation** page for sales and transfer order

lines also lets you select and reserve multiple lines based on the available batch numbers. For more information about what to do if you are using a reservation hierarchy that has the batch number dimension below the location (*Batch-below [location]*), see Flexible warehouse-level dimension reservation policy.

Feedback

Was this page helpful? Yes No

Provide product feedback

Inventory close

As part of the process to settle issue transactions with receipt transactions, you can also choose to have the general ledger updated to reflect the adjustments that have been made.

The inventory close process settles issue transactions to receipt transactions, based on the inventory valuation method that is selected in the item's item model group. As part of the settlement process, you can specify that the general ledger should be updated, so that it reflects the adjustments that have been made. However, until inventory close or recalculation has been run, issue transactions are posted at the calculated running average cost price.

After inventory close, you can no longer post in periods that are before the inventory closing date that you set, unless you reverse a completed inventory close process. For example, if inventory close is run for the period that ends on January 31, you can't post transactions that have a date that is earlier than January 31.

Items in inventory are assigned to one of two inventory types: item or service. Inventory close performs the same functions for both types. However, for service items, inventory close still settles issues to receipts.

How often the inventory close process is run varies by company. However, transaction volume should help determine how often you decide to run inventory close. In general, most companies run inventory close as part of their month-end close and reconciliation procedures.

Inventory recalculation and the general ledger

If adjustments to inventory and the general ledger are required during a month or other inventory period, you can run inventory recalculation instead of inventory close. Inventory recalculation makes adjustments but doesn't make settlements to inventory transactions.

During inventory recalculation, on-hand inventory is adjusted, inventory transactions are adjusted, and inventory recalculations and inventory closes are run. These tasks affect any ledger account that is linked to the original inventory transaction.

Example

When you create a purchase order from a sales order, the general ledger accounts that are used for the original sales order are updated. Even if the ledger accounts for the item group that is

assigned to the item were changed after the sales order was posted, and an inventory recalculation created an adjustment amount, the adjustment amount is posted to the original ledger accounts. The adjusted amount isn't posted to the new ledger accounts that are assigned to the item.

After the update is completed, you can review the ledger voucher that is posted because of one of these tasks.

1. On the **Closing and adjustment** page, on the **Overview** tab, select the update to review.
2. Select **Details**, and then select **Voucher**.

Effects of the inventory close process on the general ledger

Several of the tasks that you can perform on the **Closing and adjustment** page cause an update to general ledger. For example, the general ledger is updated when you make inventory on-hand adjustments, make inventory transaction adjustments, run inventory recalculation, and run inventory close.

The ledger accounts that are updated because of these tasks are linked to the original inventory transaction. For example, if a sales order is settled to a purchase order, the general ledger accounts that were used for the original sales order are adjusted. This behavior occurs even if the ledger accounts for the item group that is assigned to the item have changed since the sales order was posted. After inventory close creates a settlement amount, the settlement amount is still posted to the original ledger accounts, not to the new ledger accounts that are assigned to the item. The general ledger might also be updated if you reverse an inventory close.

> ⓘ **Note**
>
> - Inventory close is a required step in the month-end closing procedure for all inventory models except moving average. You'll be warned if you try to close a financial period without first performing the inventory close as of the period end date.
> - Before you run the closing procedure, you can view a list of items that can't be settled during the update.
> - We recommend that you run inventory close during off-peak hours, to distribute computing resources more evenly.

The inventory close log

After the inventory close process has been completed, a message in the message center might inform you that a unit cost price might be incorrect because a transaction couldn't be fully settled.

Before this message is shown, the system reports the item number and the affected transaction. The message informs you that the cost amount that is used for this transaction wasn't updated because of the inventory close. This message appears when a transaction of the issue type can't be settled.

During the inventory close process, the system checks each financial dimension to see whether there are more issues than receipts up to the specified closing date. This type of imbalance can occur when an inventory transaction from a purchase order isn't fully posted financially, either because the vendor invoice hasn't been received, or because bill of materials (BOM) components that are included in a production on a higher level aren't financially posted. (The subproduction isn't cost-calculated.) In this case, the subsequent close won't adjust all issues to the correct cost price, because not enough receipt information is available.

For each run of the closing procedure, the system indicates whether a log that contains the warnings is stored and can be viewed.

If you receive many warnings in the message, we recommend that you perform the following actions:

- Update receipts financially.
- Advance the closing date.
- Reevaluate the business procedures.

In some circumstances, you might not be able to do anything about the warnings. For example, if marking is used, and the marked purchase order has a financial date that is after the closing date, the closing date can't be changed.

Reversing a completed inventory close

Occasionally, you might have to reverse a completed inventory close to return settlements to the state that they had before adjustments were made. When you reverse a completed inventory close, inventory is reopened to enable posting in the period that the inventory close covers. Related changes might also be made in the general ledger. After you've finished making adjustments, you can run inventory close again for the period that you're working with.

> ⓘ **Note**

Only the last inventory period that was closed can be reopened. To reverse an earlier inventory close, you must reverse each subsequent inventory close one at a time, starting with the most recent close.

Last updated on 11/26/2025

Running average cost price

The inventory close process settles issue transactions to receipt transactions, based on the inventory valuation method that is selected in the item's item model group. However, before inventory close is run, the system calculates a running average cost price that is typically used when issue transactions are posted.

The system estimates this running average cost price for an item by using the following formula:

Estimated price = (Physical amount + Financial amount) ÷ (Physical quantity + Financial quantity)

Default item cost

The default item cost for a released product can be configured in one of two ways on the **Released product details** page:

- Create a standard cost by selecting **Item price** in the **Set up** group on the **Manage costs** tab of the Action Pane. If you use this method, you must use a standard cost costing version, and the cost must be activated.
- Define a default item cost price for a released product by entering a value in the **Price** field on the **Manage costs** FastTab.

In addition to entering or creating a price, you can select the **Use latest cost price** checkbox on the **Manage costs** FastTab of the **Released product details** page. In this case, the system automatically updates the **Price** field when you post a financial update. For example, if you post a purchase order invoice, the field is set to the purchase price from that invoice.

If you have a cost price in an active costing version, and you enter a price on the **Manage costs** FastTab, the system uses the price from the active costing version before it uses the price that is defined on the **Manage costs** FastTab.

Using the running average cost price

The following table shows when the system posts inventory transactions by using the running average cost price, and when it uses the cost price that is defined on the item master record instead.

Expand table

Condition	The system uses the estimated running average cost price	The system uses the default item cost price
Both the numerator* and denominator** are positive.	Yes	No
The numerator*, denominator**, or both are negative.	No	Yes
The denominator** is 0 (zero).	No	Yes

* Numerator = (Physical amount + Financial amount)

** Denominator = (Physical quantity + Financial quantity)

> **Note**
>
> If the **Include physical value** option isn't selected for an item, the system uses 0 (zero) for both the physical amount and the physical quantity. For information about this option, see Options for including physical value in cost calculations.

Avoiding pricing amplification

On rare occasions, the system prices several issues before it has enough receipts to base the price on. This scenario can cause estimates of the running average cost price to be overly inflated. However, there are steps that you can take to avoid pricing amplification, or to reduce its impact when it does occur.

Scenario: The following transactions occur for an item where the **Include physical value** option is selected:

1. You financially receive a quantity of 100 at USD 100.00.
2. You financially issue a quantity of 200.
3. You physically receive a quantity of 101 at USD 202.00.

When you examine the estimated running average cost price for the item, you expect a cost price of USD 1.51. Instead, you find an estimated running average of USD 102.00, which is based on the following formula:

Estimated price = [202 + (-100)] ÷ [101 + (-100)] = 102 ÷ 1 = 102

This pricing amplification occurs because, when 200 items are financially issued in step 2, the system must price 100 of the items before it has any corresponding receipts. This situation causes

negative inventory. The system then estimates a unit price of USD 1.00, as you might expect. However, when the corresponding 100 receipts arrive, they are at a unit price of USD 2.00 each.

> **Note**
>
> Although the issues create negative inventory, inventory is positive when the issue price is calculated. Therefore, the running average cost price is used instead of the price on the item master. At this point, the system has an inventory value offset of USD 100.00. Although that offset was built up over 100 pieces, where there was a unit offset of USD 1.00 each, you now have only one piece in inventory. Therefore, the offset of USD 100.00 is allocated to this one piece. The result is the overly inflated estimated cost price.
>
> For comparison, notice that if steps 2 and 3 are reversed in the scenario, 200 items will be issued at a unit price of USD 1.51, and one piece will remain at a unit price of USD 1.51. Because this pricing amplification scenario can occur when negative inventory is involved, it's difficult to avoid in the following cases:
>
> - You must estimate issue prices on the on-hand value and quantity.
> - You must adjust the on-hand value and quantity on issues and receipts.
> - Your business model allows you to send out, or price, more pieces than you have.
> - You must accept any receipt value and quantity that are submitted to you.

However, if your business model allows for the following practices, they can help you avoid the negative quantities that make the pricing amplification scenario possible:

- If you select the **Include physical value** option for an item, clear the **Physical negative inventory** checkbox on the **Item model groups** page.
- If you do **not** select the **Include physical value** option for an item, clear the **Financial negative inventory** option on the **Item model groups** page.

Additionally, consider that the maximum offset in your physical inventory value is limited by the number of physical transactions, and the difference between physical and financial prices. If all physical transactions are eventually updated financially, the physical value can't rise to extreme levels. The amplification effect decreases significantly when the accumulated offset is spread out over several on-hand pieces instead of just one.

Avoid a zero cost price on issues

When the **Include physical value** option isn't selected on the **Item model group** page, an issue from inventory can have a zero cost price if there are no financially updated receipts in inventory. To help avoid this scenario, consider the following options:

- Select the **Include physical value** option on the **Item model group** page. This option prevents a zero cost price on an issue, provided that the receipt is physically updated. If you don't allow negative physical inventory, issues calculate the running average from the physically updated transactions.
- Create a default item cost price by activating a costing version that has a standard cost, or enter the price on the **Manage costs** FastTab of the **Released product details** page. This option is best when the **Include physical value** option isn't selected on the **Item model group** page, because the system will always have a fallback price to use.
- Select the **Use latest cost price** option on the **Manage costs** FastTab of the **Released product details** page. This option keeps the **Price** field up to date each time that you financially update a receipt. If you select this option, but you don't enter a default price or activate a cost in a standard costing version, you can still have a zero cost on an issue.

If you have issue transactions that have zero cost, the *Inventory close and adjustment* process will correct the cost price by creating an adjustment after the receipts are financially updated. Keep in mind that the financial period when this update occurs might differ from the financial period when the items were physically received or issued.

Last updated on 11/26/2025

Adjust on-hand inventory cost values

Use the **Adjustment of on-hand inventory** page to adjust the cost value of the on-hand inventory quantities after running an inventory close process. To open the **Adjustment of on-hand inventory** page, start on the **Closing and adjustment** page, select the record of a completed inventory close process, and then select **Adjustment** > **On-hand**.

For example, suppose you have the following transactions in February:

- February 1: An inventory financial receipt for a quantity of 2 at a cost of USD 10.00
- February 5: An inventory financial receipt for a quantity of 1 at a cost of USD 13.00
- February 19: An inventory financial issue for a quantity of 1 at a running average cost of USD 11.00

This item was set up with the first in, first out (FIFO) inventory model, and inventory close was performed as of February 28. The financial issue transaction of USD 11.00 is settled against the financial receipt that is dated February 1, and an adjustment of USD 1.00 is made. The following inventory receipts then contain open inventory quantities:

- February 1: A quantity of 1 at a cost of USD 10.00
- February 5: A quantity of 1 at a cost of USD 13.00

To set the cost of these two items to USD 15.00, use the on-hand adjustment option to adjust the open on-hand quantities as of the last inventory close period.

> **Note**
>
> The posting date of the on-hand adjustment transaction is the date of the last inventory close. You can't modify this date.

Last updated on 07/01/2026

Track running average cost per inventory dimension

An inventory dimension group is attached to every inventory item. Therefore, the running average cost price of an item is calculated based on the selected inventory dimensions that are being tracked financially.

There are three types of inventory dimensions: product, storage, and tracking. Product dimensions include configuration, size, and color. Product dimensions are always tracked financially. Storage and tracking dimensions include site, warehouse, location, inventory status, license plate, batch number, and serial number. You can decide which storage and tracking dimensions are tracked financially.

Example 1

If the storage dimension group that is attached to the item is financially tracked by warehouse, the running average cost price is calculated per warehouse. The following purchase orders have been invoiced:

- A purchase order for a quantity of 2 at a cost price of USD 10.00 has been invoiced for warehouse GW.
- A purchase order for a quantity of 3 at a cost price of USD 12.00 has been invoiced for warehouse GW.
- A purchase order for a quantity of 5 at a cost price of USD 15.00 has been invoiced for warehouse MW.

The running average cost price for warehouse GW is USD 11.20, and the running average cost price for warehouse MW is USD 15.00. A sales order invoice is posted for warehouse GW. The value of the inventory and the cost of goods sold (before inventory close is run and without marking) is USD 11.20. Another sales order is posted for warehouse MW. The value of the inventory and the cost of goods sold (before inventory close is run and without marking) is USD 15.00.

Example 2

If the storage dimension group that is attached to the item is financially tracked by warehouse, and the tracking dimension group is financially tracked by batch number, the running average

cost price is calculated for each batch. The following purchase orders have been invoiced:

- A purchase order for a quantity of 2 at a cost price of USD 10.00 has been invoiced for warehouse GW and batch AAA.
- A purchase order for a quantity of 3 at a cost price of USD 12.00 has been invoiced for warehouse GW and batch AAA.
- A purchase order for a quantity of 2 at a cost price of USD 15.00 has been invoiced for warehouse GW and batch BBB.

The running average cost price for warehouse GW and batch AAA is USD 11.20, and the running average cost price for warehouse GW and batch BBB is USD 15.00.

> **Note**
>
> We recommend that you always view the cost price for all financial dimensions that are being tracked.

Last updated on 11/26/2025

Configure and manage financial dimension links to sites

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To enable the system to calculate profit and loss figures for each site for your legal entity, link a financial dimension to the site inventory dimension and assign a financial dimension value to each site.

Set up a financial dimension link to a site

When you set up a financial dimension link to a site, you:

- Associate a financial dimension with the site inventory dimension.
- Associate a financial dimension value with each site.

To set up a financial dimension link to a site, follow these steps:

1. Go to **Inventory management** > **Setup** > **Posting** > **Dimension link**.
2. On the Action Pane, select **Edit**.
3. In the **Select the dimension to be linked to the storage dimension, Site** field, select the financial dimension to link to the site inventory dimension.
4. On the Action Pane, select **Sites** to open the **Sites** page.
5. On the list pane, select the site you want to link the financial dimension to.
6. Expand the **Financial dimensions** FastTab. The financial dimension that you selected in the previous page appears under the **Default financial dimensions** heading.
7. From the drop-down list under the **Default financial dimensions** heading, select the financial dimension value that you want to associate with the selected site.
8. Repeat steps 5 to 7 for each site.
9. When you're done, select **Save**.

Activate a financial dimension link

Activating a financial dimension link has the following effects:

- When you create new transactions that use both inventory dimensions and financial dimensions, the system uses the dimension value that you specify for the site as the dimension value for the linked financial dimension.
- You can post transactions that use both inventory dimensions and financial dimensions, even if the dimension value for the linked financial dimension doesn't match the value that you specified for the site.
- You can change the dimension value for the linked financial dimension on open transactions that use both inventory dimensions and financial dimensions.

> ⓘ **Note**
>
> You must assign a financial dimension value to each site before you can complete this procedure.

To activate a financial dimension link, follow these steps:

1. Go to **Inventory management** > **Setup** > **Posting** > **Dimension link**.
2. On the Action Pane, select **Activate link**.
3. The **Linked dimension update** dialog opens. Select **OK** to activate the link.
4. The **Current status** field updates to show that the *Dimension link is active* and the **Select the dimension to be linked to the storage dimension, Site** field becomes inactive.

Lock a financial dimension link

When you lock a financial dimension link, the following effects take place:

- New inventory transactions that use both inventory dimensions and financial dimensions get a financial dimension value based on the value specified for the site.
- You can post transactions that use both inventory dimensions and financial dimensions only when the dimension value of the linked financial dimension matches the value specified for the site.
- You can't change the dimension value of the linked financial dimension on transactions that use both inventory dimensions and financial dimensions.
- You can't modify the financial dimension value that's associated with a site.

You must set up a financial dimension link before you can lock it.

To lock a financial dimension link, follow these steps:

1. Go to **Inventory management** > **Setup** > **Posting** > **Dimension link**.
2. On the Action Pane, select **Lock link**.

3. The **Linked dimension update** dialog opens. Select **OK** to lock the link. The system updates all open transactions accordingly.
4. The **Current status** field updates to show that the *Dimension link is locked*.

Unlock a financial dimension link

You can unlock locked financial dimension links. When you unlock a link, you can complete these tasks:

- Post transactions that use both inventory dimensions and financial dimensions, even if the dimension value of the linked financial dimension doesn't match the value that is specified for the site.
- Modify the dimension value of the linked financial dimension on open transactions that use both financial dimensions and inventory dimensions.
- Modify the financial dimension values on the **Sites** page.

1. Go to **Inventory management** > **Setup** > **Posting** > **Dimension link**.
2. On the Action Pane, select **Unlock link**.
3. The **Current status** field updates to show that the *Dimension link is active*.

Deactivate a financial dimension link

When you deactivate a financial dimension link, the following changes occur:

- The link between the site inventory dimension and the financial dimension is removed.
- The linked financial dimension isn't assigned the dimension value specified for each site.
- You can choose a different financial dimension to link to the site inventory dimension.

To deactivate a financial dimension link, follow these steps:

1. Go to **Inventory management** > **Setup** > **Posting** > **Dimension link**.
2. On the Action Pane, select **Deactivate link**.
3. The **Current status** field updates to show that the *Dimension link is inactive*.

Last updated on 03/30/2026

Run an on-hand consistency check

The on-hand consistency check is a corruption-recovery tool that rebuilds the on-hand inventory tables (`InventSum` and, for warehouse management items, `WHSInventReserve`) from the inventory transactions table (`InventTrans`). Use it when you have confirmed on-hand data corruption—not as a periodic maintenance task.

This article explains when to run the check, how to run it safely, the performance characteristics you should plan for, and how the **Skip on-hand aggregation and clean up in on-hand consistency check** parameter interacts with the standard on-hand database cleanup jobs.

> **ⓘ Important**
>
> If Inventory Visibility is enabled on your system, complete the additional preparation steps described in Run on-hand consistency checks while Inventory Visibility is enabled before you run the on-hand consistency check as described in this article.

When to run an on-hand consistency check

Run an on-hand consistency check when you observe symptoms such as the following for one or more items:

- The **On-hand list** shows a quantity that doesn't match the sum of the related inventory transactions for the same dimensions.
- Inventory close or recalculation fails with an error referencing negative on-hand for an item whose item model group doesn't allow negative inventory (for example, *Inventory closing can't proceed because available physical on-hand inventory on item <Item> is currently negative*).
- Negative on-hand quantities exist that can't be explained by posted transactions.
- Physical inventory or available quantity doesn't match the sum of posted inventory transactions.

These symptoms typically indicate that the `InventSum` or `WHSInventReserve` tables are out of sync with `InventTrans`. The most common cause is a data inconsistency introduced by a script, external process, or user action that updated `InventTrans` directly without recalculating the summary tables. Running the consistency check in *Fix error* mode rebuilds the summary tables from the transactions, which resolves the discrepancy.

If none of these conditions apply, don't run the check. The dedicated on-hand entries cleanup jobs are the correct tool for routine maintenance.

When not to run the check

Here are some rules that illustrate when not to run the on-hand consistency check:

- Don't schedule the on-hand consistency check as a recurring batch job. It isn't a maintenance routine.
- Don't run the check across all items during business hours. The check replays the entire `InventTrans` history per item and takes locks on `InventSum` and `WHSInventReserve`.
- Don't run the check with **Check/Fix** set to *Fix error* as your first action. Always run it in *Check* mode first and review the log.
- Don't run the check as a precaution before or after upgrades, cleanup jobs, or routine close. None of those operations corrupt on-hand data.

How to run the check safely

To run the on-hand consistency check, follow these steps:

1. Go to **System administration** > **Periodic tasks** > **Database** > **Consistency check**.
2. Set **Module** to *Inventory management*.
3. Expand the **Item** tree, and select the **On-hand** check box. If you suspect transaction-level corruption, select the **Inventory transactions** checkbox too.
4. Click on the text in the **Item** row (not the check box) so that the entire row has a blue background.
5. Open the **More** menu (**...**) and select **Dialog**.
6. Use the dialog box to filter the run to the specific item or item range that's affected. Running on *all items* is rarely the right choice.

   > **Tip**
   >
   > Scope the check to a single item, or to as few items as possible, whenever you can. Running an on-hand consistency check for all items processes the entire `InventTrans`

table and can take a long time on large datasets. Limiting the scope through the **Dialog** filter makes the check finish much faster and reduces the risk of timeouts.

7. Select **OK** to close the filter dialog.
8. In the **Consistency check** dialog, set **Check/Fix** to *Check* and then select **OK** to run the check. Review the log when it completes.
9. Select **Show messages** (bell icon at the top-right of the page) to open the **Action center** and review the messages for any errors or warnings.
10. If errors are confirmed, repeat this procedure with **Check/Fix** set to *Fix error*, ideally as a batch job outside business hours.

If the *Fix error* run doesn't resolve the discrepancy, contact Microsoft Support or your partner.

Performance considerations

The run time of the on-hand consistency check scales with the volume of `InventTrans` history for the items in scope, not with the current on-hand quantity. An item with a long posting history takes longer to rebuild than an item with few transactions, even if both currently have similar on-hand values.

To keep the run predictable:

- **Filter by item** – Use the dialog filter to scope the run to the affected items. Running on *all items* should be reserved for documented disaster recovery scenarios.
- **Run in batch and outside business hours** – The fix run takes exclusive locks on `InventSum` and `WHSInventReserve` rows for each item it touches. Concurrent posting against the same item can deadlock or be blocked.
- **Be aware of the cleanup side effect** – With the default parameter setting, the fix run also runs the equivalent of an on-hand cleanup job for each item in scope, which adds time to the run. Set the **Skip on-hand aggregation and clean up in on-hand consistency check** parameter to *Yes* if you want the fix run to focus only on rebuilding the on-hand tables. Learn more about this option in Choose whether to apply on-hand aggregation and clean up.

Choose whether to apply on-hand aggregation and clean up

You can choose to either run or skip on-hand cleanup when the on-hand consistency check runs in *Fix error* mode.

For very large datasets, you can shorten the run time of the consistency check by skipping the aggregation and clean-up steps that the check performs alongside the core recalculation. The recalculation that recreates `InventSum` records still runs; only the non-essential clean up is skipped.

To set this option, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. Open the **General** tab.
3. In the **On-hand consistency check clean up** field group, set **Skip on-hand aggregation and clean up in on-hand consistency check** to one of the following values:

    - *Yes* – The consistency check skips the on-hand cleanup side effect and focuses solely on rebuilding the on-hand tables. This choice results in a faster run, but zero-quantity rows aren't cleaned up.
    - *No* – The consistency check includes the on-hand cleanup side effect. It also aggregates and deletes zero-quantity rows for the items in scope. This choice results in a longer run time, but it tidies up the on-hand tables. This value is the default setting.

What the check does in fix mode

When you run the on-hand consistency check with **Check/Fix** set to *Fix error*, the system performs the following steps for each item in scope:

1. **Always** – The system rebuilds `InventSum` (and `WHSInventReserve` for warehouse management items) from `InventTrans`. This is the corruption fix itself.
2. **Only when the parameter is *No* (default)** – The system also performs the following on-hand cleanup work for that item:

    - **On-hand entries aggregation by financial dimensions** – Aggregates zero-quantity `InventSum` rows up to the financial inventory dimension level. This operation uses the same logic as the standalone *On-hand entries aggregation by financial dimensions* cleanup job, scoped to the item.
    - **On-hand cleanup** – For warehouse management items, runs the same logic as the *Warehouse management on-hand entries cleanup* job to delete zero-quantity rows in

`InventSum` and `WHSInventReserve` for the item. For non-warehouse-management items, the system runs the same logic as the *On-hand entries cleanup* job for the item's tracking-dimension rows.

If the aggregation or cleanup step fails for a specific item, the system logs a warning (*On-hand clean up is failed for item <Item> in consistency check*) and continues with the next item. The recalculation work for that item is still completed—only the cleanup operation is skipped.

When to skip on-hand aggregation and clean up

Set **Skip on-hand aggregation and clean up in on-hand consistency check** to *Yes* in any of the following situations:

- You already scheduled the on-hand cleanup jobs and want the consistency check to focus only on rebuilding the on-hand tables.
- You want the fix run to finish faster, because the per-item cleanup work is skipped.
- You need to preserve zero-quantity rows that downstream historical reporting depends on, such as *Physical inventory by inventory dimension* with **As of date** set, or *On-hand list* with **Closed transactions** selected.
- You're running the fix on a tight maintenance window and can't accept the additional runtime of the inline cleanup.

When not to skip on-hand aggregation and clean up

Set **Skip on-hand aggregation and clean up in on-hand consistency check** to *No* in either of the following situations:

- The on-hand cleanup jobs aren't yet scheduled in the environment and you want the fix run to also tidy up zero-quantity rows for the affected items.
- The scope of the fix run is small (a few items) and the additional cleanup time isn't a concern.

Relationship to the standalone cleanup jobs

The standalone on-hand cleanup jobs are the correct tool for ongoing database maintenance. The consistency check, even with the cleanup side effect enabled, isn't a substitute for them.

Expand table

Job	Path	Purpose
On-hand entries cleanup	**Inventory management** > **Periodic tasks** > **Clean up** > **On-hand entries cleanup**	Deletes closed and unused on-hand entries for items with tracking dimensions.
Warehouse management on-hand entries cleanup	**Inventory management** > **Periodic tasks** > **Clean up** > **Warehouse management on-hand entries cleanup**	Deletes zero-quantity rows in `InventSum` and `WHSInventReserve` for warehouse management items.
On-hand entries aggregation by financial dimensions	**Inventory management** > **Periodic tasks** > **Clean up** > **On-hand entries aggregation by financial dimensions**	Aggregates zero-quantity `InventSum` rows up to the financial inventory dimension level.

Learn more in Cleanup routines in Dynamics 365 Finance and Dynamics 365 Supply Chain Management and Warehouse management on-hand entries cleanup job.

Reports that can be affected by cleanup

The cleanup process removes zero-quantity rows from the on-hand tables. This removal happens both when you run the standalone cleanup jobs and when you run the on-hand consistency check in *Fix error* mode with the **Skip on-hand aggregation and clean up in on-hand consistency check** parameter set to *No*. These rows don't contribute to current on-hand values, but several historical and "as of date" reports read them directly. After cleanup, those reports might return incomplete or missing data for periods where on-hand rows are removed.

Reports that depend on zero-quantity on-hand rows include those listed in the following table.

Expand table

Report	Impact when rows are deleted
Inventory aging report	"As of date" snapshots and aging buckets might be missing rows for inventory dimensions that have since dropped to zero quantity.
Historical inventory financial calculation (Russia)	Historical inventory value calculations rely on the full on-hand row history. Deleted zero-quantity rows can cause incorrect financial snapshots for past dates.
Physical quantity summary engine (China)	Physical quantity summaries that aggregate by inventory dimension might be missing closed or zero-quantity rows after cleanup.
Physical inventory by inventory dimension	When the **As of date** parameter is set, or when **Closed transactions** is selected, deleted rows no longer appear in

Report	Impact when rows are deleted
	the report output.
On-hand list (with the **Quantity <> 0** filter cleared or **Closed transactions** selected in **Dimensions display**)	Closed or zero-quantity dimensions no longer appear at the level (for example, license plate) where they were deleted.

How to mitigate the impact on reports

To mitigate the impact of cleanup on historical and "as of date" reports, use the following approaches:

- **Don't run the consistency check in *Fix error* mode with the default parameter setting if you rely on these reports for the items in scope** – Set **Skip on-hand aggregation and clean up in on-hand consistency check** to *Yes* for that run, so the recalculation doesn't also delete zero-quantity rows.
- **Run the historical reports before you schedule the cleanup jobs in environments where the reports are business-critical** – Reports that are already generated aren't affected retroactively.
- **Weigh the trade-off per environment** – The cleanup jobs and the inline cleanup in the consistency check exist for performance reasons. The performance gains are real, and the lost dimension-level visibility is usually acceptable. However, it isn't acceptable in environments where the reports listed earlier are part of statutory or business-critical reporting.

Recommended workflow when corruption is suspected

If you suspect on-hand data corruption, use the following workflow to investigate and resolve it:

1. **Identify the affected item** – Use the original error message, the **On-hand list**, and **Inventory transactions** to confirm which item has corrupted on-hand data.
2. **Back up if you're in a position to do so** – The fix is set-based and not easily reversible.
3. **Run the check in *Check* mode for that item only** – Use the dialog filter to scope the run. Review the log.
4. **Decide on the parameter setting** – If the environment has scheduled cleanup jobs that already manage on-hand rows, set **Skip on-hand aggregation and clean up in on-hand consistency check** to *Yes* for this run. Otherwise, leave it at *No* so the fix also tidies up zero-quantity rows.
5. **Rerun in *Fix error* mode as a batch job** outside business hours.

6. **Verify** – After the fix completes, re-check the **On-hand list** and the original error condition.

Related articles

- Warehouse management on-hand entries cleanup job
- Cleanup routines in Dynamics 365 Finance and Dynamics 365 Supply Chain Management
- Run on-hand consistency checks while Inventory Visibility is enabled

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Last updated on 06/04/2026

Inventory On-hand mobile app

08/28/2025

> **Important**
>
> The Inventory On-hand mobile app (preview) is no longer available on the Power Platform Admin Center and is instead being provided as a sample app that you can download from the scmsamples-InventoryOnHand repository on GitHub. The sample app is identical to the previously available first-party app, but is now distributed through GitHub rather than through the Power Platform Admin Center. You can fork the GitHub repository so you can customize it as needed for your scenarios. Previously installed instances of the first-party mobile app will continue to work. Learn more in Removed or deprecated features in Dynamics 365 Supply Chain Management.

The Inventory On-hand mobile app lets you query and view product inventory in Microsoft Dynamics 365 Supply Chain Management by using a mobile device. It was built by using Power Apps.

The app lets you perform the following tasks:

- Search product inventory by entering a product ID or name.
- Filter products and inventory by site, warehouse, location, batch, and default product dimensions.
- Filter products and inventory by inventory status and inventory quantity range.
- Specify your regular sites and locations to prioritize the search range.
- Switch among legal entities to check cross-organization inventory.

This article describes how administrators can prepare your Supply Chain Management and Dataverse environments to support the Inventory On-hand mobile app. It also describes how to install the app on your mobile devices.

Prerequisites

Before you can start to onboard the Inventory On-hand mobile app, you must meet the requirements that are outlined in this section.

System requirements

To run the latest Inventory On-hand mobile app, you must be using Supply Chain Management version 10.0.40 or later.

Set up Dataverse for your Supply Chain Management environment

The mobile app uses Dataverse virtual tables that are connected to Supply Chain Management. Therefore, a Dataverse environment must be integrated with your Supply Chain Management environment. If Dataverse isn't already set up for your environment, follow the instructions in Microsoft Power Platform integration with finance and operations apps.

When you create the Dataverse environment where you want to install the Inventory On-hand mobile app, be sure to enable Dynamics 365 apps.

The Power Apps component framework feature must be enabled for your environment.

Install the mobile app in Dataverse

For complete details about how to download, customize, install, and set up the app, go to the scmsamples-InventoryOnHand repository on GitHub.

Share the mobile app and assign security roles

After the mobile app solution is installed in your Dataverse environment, you must share it with your users. The Inventory On-hand mobile app is a canvas app. To share it, follow the instructions in Share an app.

The security roles that can access and query on-hand inventory through the mobile app are inherited from your Supply Chain Management setup. Therefore, any user (such as a sales manager, warehouse manager, or production manager) that has the security roles required to access the on-hand inventory list in Supply Chain Management can also query and view that data in the Inventory On-hand mobile app. For more information about how to set up roles and security in Supply Chain Management, see Security roles.

Open the Inventory On-hand mobile app

Follow these steps to use the Inventory On-hand mobile app on a mobile device.

1. Open the Power Apps mobile app, and sign in by using the same corporate account that you use to sign in to Supply Chain Management.
2. Use the **Search** field to search for *Inventory on-hand mobile app*. Because the app is a canvas app, you can add it to your favorites list in Power Apps by swiping left on it after you find it.
3. Open the Inventory On-hand mobile app, and start to use it.

4. The first time that you open the app, you can go to the settings and select a legal entity. You can also set the default site and warehouse where you usually work. These settings are especially useful for store managers.
5. When you search for products and inventory, the app first searches in the default site and warehouse. On the result page, you can easily switch between a full search and a search in the default site and warehouse.

Clear the Power Apps cache on a mobile device after an update of the back-end setup

If the back-end setup is updated in Supply Chain Management after you install the Power Apps mobile app on a mobile device (for example, if the legal entity setup is changed), you might have to clear the cache in Power Apps to update the local setup data. Power Apps then reloads the settings that otherwise might not take effect until the cache is cleared.

Follow these steps to clear the Power Apps cache on a mobile device.

1. Sign in to the Power Apps mobile app.
2. In the upper left of the page, select the image for your user account to open the settings menu.
3. Select **Clear cache** to open the **Clear cache** dialog box.
4. Select **Confirm**.

Frequently asked questions

What platforms and mobile devices are supported?

The Inventory On-hand mobile app runs within the Power Apps mobile app. All platforms supported by the Power Apps mobile app can also run the Inventory On-hand mobile app. Learn more in System requirements, limits, and configuration values for Power Apps.

Why don't I see anything when I sign in to the app?

If you don't see any features in the app, you probably lack the permissions required to access them. Access is controlled by the security roles assigned to your user account in Supply Chain Management. The roles you need are the same as those required to access the on-hand inventory list in Supply Chain Management. Learn more in Security roles.

Can I customize and extend the app?

Yes. You can customize and extend the app by working with the source code provided on GitHub.

Does the app support offline mode?

No, the Inventory On-hand mobile app doesn't currently support offline mode.

Is there multi-language support?

Yes, the Inventory On-hand mobile app is available in all the same languages as Supply Chain Management. The language is set according to your mobile device settings.

Consolidate inventory transactions

> **Important**
>
> Some or all of the functionality noted in this article is available as part of a preview release. The content and the functionality are subject to change. For more information about preview releases, see One version service updates FAQ.

Over time, the inventory transaction table (`InventTrans`) grows and consumes more database space. This article describes how to consolidate data about inventory transactions to help improve system performance and reduce storage consumption.

> **Note**
>
> Only financially updated inventory transactions can be consolidated in a selected closed ledger period. To be consolidated, financially updated outbound inventory transactions must have an issue status of *Sold*, and inbound inventory transactions must have a receipt status of *Purchased*.

When you consolidate inventory transactions, the system moves all related transactions to the `InventTransArchive` table. It then creates summary transactions in the `InventTrans` table, grouped by `ItemId` and `InventDimId`. If the combined quantity and cost fields for a group equal zero, the system doesn't create a new summary transaction.

If an `itemId` and `inventDimId` combination contains only one receipt or issue transaction, the transaction isn't consolidated.

> **Note**
>
> After consolidating your inventory transactions, you can further optimize storage and system performance by using the *Archive with Dataverse long term retention* feature to move `InventTransArchive` records to a Microsoft Azure data lake. Learn more in Archive inventory transaction data in Dynamics 365 Supply Chain Management.

Considerations before consolidating inventory transactions

Before you consolidate inventory transactions, consider the following impacts on your business processes:

- **Transaction auditing** – When you audit inventory transactions from related documents (such as purchase order lines), you can view the consolidated transactions under **Archived transactions** until they're moved to long-term storage by using the *Archive with Dataverse Long term retention* feature. To review all consolidated transactions, go to **Inventory management** > **Periodic tasks** > **Clean up** > **Inventory transaction consolidation**.
- **Inventory closing** – You can't cancel inventory closing for periods that are consolidated.
- **Cost conversion** – You can't perform standard cost conversion for periods that are consolidated.
- **Inventory reports** – Reports that rely on detailed inventory transaction data show consolidated values instead of individual transactions. Affected reports include the inventory aging report and inventory value reports.
- **Inventory forecasts** – Forecasts that run during consolidated periods might produce different results because they use summary data rather than detailed transaction history.
- **Transaction-dependent operations** – You can't perform business processes that require the original inventory transactions, such as creating return orders and adjustments, after a period is consolidated.

Prerequisites

Version requirements

To consolidate inventory transactions, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.

Feature management requirements

To consolidate inventory transactions, the feature named *(Preview) Optimized inventory transaction consolidation* must be turned on in feature management.

Configure a number sequence for the consolidation process

Before you start the first inventory transaction consolidation job, configure a number sequence for the consolidation process. Configure this number sequence under **Inventory management** >

Inventory and warehouse management parameters > **Number sequences** and assign a number sequence to the **Inventory transaction consolidation voucher** reference.

Conditions for consolidating inventory transactions

You can consolidate inventory transactions only for periods that meet all of the following conditions:

- The ledger period is on hold or permanently closed.
- The period is at least one year before the current system date.
- Inventory closing ran on or after the end date of the consolidation period.
- The period length is less than 365 days from the last consolidation or from the earliest transaction date.
- No inventory recalculations are currently running or pending.
- No license plate cleanup jobs, cost conversion jobs, or inventory recalculations are currently running.
- All inventory transactions and related data pass the validation checks described in the next section.

Validate inventory transactions

The system performs the following validations during the consolidation process to ensure data integrity and prevent inconsistencies:

- **Open issue transactions** – Settle all issue transactions through inventory closing before consolidation.
- **Physically updated transactions that aren't financially posted** – Financially post all transactions before consolidation to prevent downstream process issues.
- **Inconsistent inventory settlements** – Inventory settlements must sum to zero in the selected period.

> ⓘ **Note**
>
> The validation phase duration depends on the number of inventory transactions in your system. In environments with a high volume of inventory transactions, expect the validation to take considerable time.

Allow inventory transaction consolidation without inventory closing

To allow inventory transaction consolidation without inventory closing, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. Open the **General** tab.
3. Set **Allow inventory transaction consolidation without inventory closing** to one of the following values:
    - *No* – The system can only consolidate inventory transactions for closed inventory.
    - *Yes* – The system can also consolidate inventory transactions on unclosed inventory for items that use the *Standard cost* or *Moving average* inventory model.

    By default, this setting is permanently set to *No*. If you want to change it to *Yes*, contact Microsoft Technical Support for assistance.

> **Important**
>
> After you consolidate an inventory transaction, you can no longer close that inventory period.

Consolidate your inventory transactions

To consolidate inventory transactions, follow these steps:

1. Go to **Inventory management** > **Periodic tasks** > **Clean up** > **Inventory transaction consolidation**.

    The **Inventory transaction consolidation** page appears and shows a list of consolidated process records.
2. On the Action Pane, select **Inventory transaction consolidation** to create an inventory transaction consolidation.
3. In the **Inventory transaction consolidation** dialog box, on the **Parameters** FastTab, set the following fields:

- **To date in closed ledger period** – Select the end date of the consolidation period. The system consolidates all eligible transactions up to and including this date. Previously unconsolidated transactions before this date are included if they now meet the consolidation requirements.

> ⓘ **Note**
>
> Only periods that meet the conditions for consolidating inventory transactions are available for selection.

4. On the **Run in the background** FastTab, set up batch processing details as you require. Follow the usual steps for batch jobs in Microsoft Dynamics 365 Supply Chain Management.

5. Select **OK**.

6. You receive a message that prompts you to confirm that you want to continue. Read the message carefully, and then select **Yes** if you want to continue.

   You receive a message that states that your inventory transaction consolidation job is added to the batch queue. The job starts to consolidate inventory transactions from the selected period. You can find the batch job under **System administration** > **Inquiries** > **Batch jobs**, where you can also review any logs generated by the batch tasks.

View consolidated inventory transactions

The **Inventory transaction consolidation** page shows your full consolidating history. Each row in the grid shows information such as the date when the consolidation was created, the user who created it, and its status.

Filter the list of consolidations

In the drop-down list at the top of the page, select one of the following values to filter the consolidations that appear in the grid:

- *Active* – Show only active consolidations.
- *All* – Show all consolidations.

You can also filter and sort the grid by using the column headers.

Information shown in the grid

For each consolidation in the grid, the following information is provided:

- **Active** – A check mark indicates that the consolidation is active.
- **To date** – The date of the newest transaction that can be included in the consolidation.
- **Voucher** - The voucher for the consolidation session, generated by the configured number sequence.
- **Scheduled by** – The user account that created the consolidation.
- **Executed** – The date when the consolidation was created.
- **Stop current update** – A check mark indicates that the consolidation is in progress, but it has been paused.
- **State** – The processing status of the consolidation. The possible values are *Waiting*, *In progress*, and *Finished*.

The grid toolbar

The toolbar above the grid provides the following buttons that you can use to work with a selected consolidation:

- **Consolidated transactions** – View the full details of the selected consolidation. The **Consolidated transactions** page shows all the transactions in the consolidation. To view more information about a specific transaction on the **Consolidated transactions** page, select it in the grid. On the Action Pane, select **Consolidated transaction details**. The **Consolidated transaction details** page shows information such as the ledger posting, related subledger references, and financial dimensions.
- **Pause** – Pause a selected consolidation that is currently being processed. The pause takes effect only after the archiving task is generated. Therefore, there might be a short delay before the pause takes effect. If a consolidation is paused, a check mark appears in its **Stop current update** field.
- **Resume** – Resume processing for a selected consolidation that is currently paused.
- **Progress details** – Open a log that shows the progress of your inventory consolidation jobs.

Extend your code to support custom fields

If your `InventTrans` table contains one or more custom fields, you might need to extend the code to support them, depending on how they are named.

- If the custom field has the same name in both `InventTrans` and `InventTransArchive`, add it to the `InventoryArchiveFields` field group on `InventTrans`. No code is required—the engine automatically resolves the matching field on `InventTransArchive`.

- If the field names differ between the two tables, use the field mapper extension described in this section.

> **Important**
>
> The pattern for mapping custom field names changed in Supply Chain Management version 10.0.47. In version 10.0.46 and earlier, you mapped custom fields by extending the `InventTransArchiveProcessTask` and `InventTransArchiveSqlStatementHelper` classes. In version 10.0.47 and later, you map custom fields by extending the `InventTransArchiveFieldMapper` class. If your customizations still use the legacy extension pattern, you must migrate them to use the new pattern. You can't run inventory transaction consolidation until you migrate your customizations to use the new extension pattern.

In Supply Chain Management version 10.0.47 and later, custom field mapping is done through a dedicated `InventTransArchiveFieldMapper` class. Create a Chain of Command (CoC) extension and override the two list methods. Each method returns a `List` of field IDs that the consolidation engine adds to the source query (`InventTrans`) and the destination insert (`InventTransArchive`).

The following sample code includes the `CreatedBy` field from `InventTrans` and writes it to a custom `InventTransCreatedBy` field on `InventTransArchive`.

X++

```
[ExtensionOf(classStr(InventTransArchiveFieldMapper))]
final class InventTransArchiveFieldMapper_Foundation_Extension
{
    public List getInventTransFields()
    {
        List inventTransFields = next getInventTransFields();

        inventTransFields.addEnd(fieldNum(InventTrans, CreatedBy));

        return inventTransFields;
    }

    public List getInventTransArchiveFields()
    {
        List inventTransArchiveFields = next getInventTransArchiveFields();

        inventTransArchiveFields.addEnd(fieldNum(InventTransArchive,
InventTransCreatedBy));

        return inventTransArchiveFields;
    }
}
```

Keep the following points in mind when you write the extension:

- **The engine pairs the two lists positionally.** The first field returned by `getInventTransFields()` is paired with the first field returned by `getInventTransArchiveFields()`, and so on. The lists must contain the same number of entries in the same order.
- **Always call `next` and append to the returned list.** This step preserves the contributions of any other extensions and keeps your code forward-compatible.

Related information

- Consolidate inventory transactions FAQ

Last updated on 08/20/2026

Consolidate inventory transactions FAQ (preview)

[This article is prerelease documentation and is subject to change.]

This article provides answers to frequently asked questions about inventory transaction consolidation in Microsoft Dynamics 365 Supply Chain Management.

Why are there more rows in the InventTransOrigin table after consolidation?

This result is expected. The *Inventory transactions consolidation* job creates new inventory transaction (`InventTrans`) and inventory transactions originator (`InventTransOrigin`) records.

For transactions that aren't warehouse management process (WMS) transactions, the job creates a new record with reference category `InventTransArchive` in the `InventTransOrigin` table. This new record links to the new consolidating inventory transaction.

For WMS transactions, the job creates a new record with reference category `WHSInventTransArchiveOnlyAffectsLocationAndBelow` in the `InventTransOrigin` table. This type of record affects only location and below in the inventory reservation hierarchy. In other words, it's used only for transactions that affect specific locations in the warehouse. Here are some examples:

- `WHSWork` – Transactions related to warehouse work orders.
- `WHSContainer` – Transactions that involve containers in the warehouse.
- `WHSOrderCommittedReservation` – Transactions for orders that have committed reservations.
- `WHSWarehouseInventoryBlocking` – Transactions that block inventory in the warehouse.

Reduce the consolidation frequency to help minimize the increase in the `InventTransOrigin` table.

Why aren't some transactions moved to the InventTransArchive table during consolidation?

This result is expected. In the following scenarios, the system keeps transactions in the `InventTrans` table after inventory transaction consolidation:

- When the *Inventory transactions consolidation* job creates new inventory transactions that consolidate the archived inventory transactions, the job keeps these new transactions in the `InventTrans` table. Learn more in the previous section.
- If an `ItemId` and `InventDimId` combination contains only one receipt or issue transaction in the archive period, the job doesn't consolidate that transaction. Instead, it keeps the transaction in the `InventTrans` table.
- Inventory transactions are only included in the consolidation if all transactions for a given document can be consolidated. If, for example, not all inventory transactions for a sales order can be consolidated, no transactions for the sales order are consolidated. These transactions are automatically included in future consolidations when the prerequisites are met.

The following examples illustrate these aggregation rules.

Aggregation rule example 1

Before consolidation

Expand table

ID	Item number	Inventory dimension ID	Financial date	Reference	Receipt	Issue	Quantity
A1	WMS item A	Dim_1	January 10, 2020	Purchase order	Purchased		10
A2	WMS item A	Dim_1	January 15, 2020	Counting		Sold	-3
A3	WMS item A	Dim_1	January 20, 2020	Work		Sold	-7
A4	WMS item A	Dim_2	January 20, 2020	Work	Purchased		7
A5	WMS item A	Dim_2	January 20, 2020	Work		Sold	-7
A6	WMS item A	Dim_3	January 20, 2020	Work	Purchased		7
A7	WMS item A	Dim_3	January 25, 2020	Sales order		Sold	-7

After consolidation until January 31, 2020

Expand table

ID	Item number	Inventory dimension ID	Financial date	Reference	Receipt	Issue	Quantity
B1	WMS item A	Dim_1	January 31, 2020	Archived inventory transaction	Purchased		7
A3	WMS item A	Dim_1	January 20, 2020	Work		Sold	-7
B3	WMS item A	Dim_2	January 31, 2020	Archived warehouse transaction type that affects only location and below		Sold	0
A6	WMS item A	Dim_3	January 20, 2020	Work	Purchased		7
A7	WMS item A	Dim_3	January 25, 2020	Sales order		Sold	-7

Result summary

Note the following results of the consolidation:

- Line B1 consolidates line A1 and line A2, because A1 and A2 both have the same inventory dimension (Dim_1), and they aren't a warehouse transaction type that affects only location and below.
- Line B3 consolidates line A4 and line A5, because A4 and A5 both have the same inventory dimension (Dim_2), and they are a warehouse transaction type that affects only location and below. Typically, if the cost amount and quantity are 0, line B3 isn't inserted into the inventory transactions table.
- Line A3 isn't consolidated with line B1, because it's a warehouse transaction type that affects only location and below.

Last updated on 08/24/2026

Inventory Visibility Add-in overview

08/14/2025

The Inventory Visibility Add-in (also referred to as the *Inventory Visibility service*) provides an independent and highly scalable microservice that enables real-time on-hand inventory change postings and visibility tracking across all your data sources and channels. It provides a platform that lets you manage your global inventory by using functionality that includes (but isn't limited to) the following list:

- Centrally track the latest inventory status (such as on-hand, ordered, purchased, in-transit, returned, and quarantined) across all your data sources, warehouses, and locations by connecting your Supply Chain Management or third-party logistics data sources (such as order management systems, third-party enterprise resource planning [ERP] systems, point of sale [POS] systems, and warehouse management systems) to the Inventory Visibility service.
- Query on-hand stock availability and shortages, and obtain immediate responses by calling the Inventory Visibility service directly.
- Avoid overselling, especially when your demand comes from different channels, by making real-time soft reservations in the Inventory Visibility service.
- Better manage promised orders and customer expectations by providing accurate current or next-available dates, so that the omnichannel available-to-promise (ATP) feature can calculate expected order fulfillment dates.

Extensibility

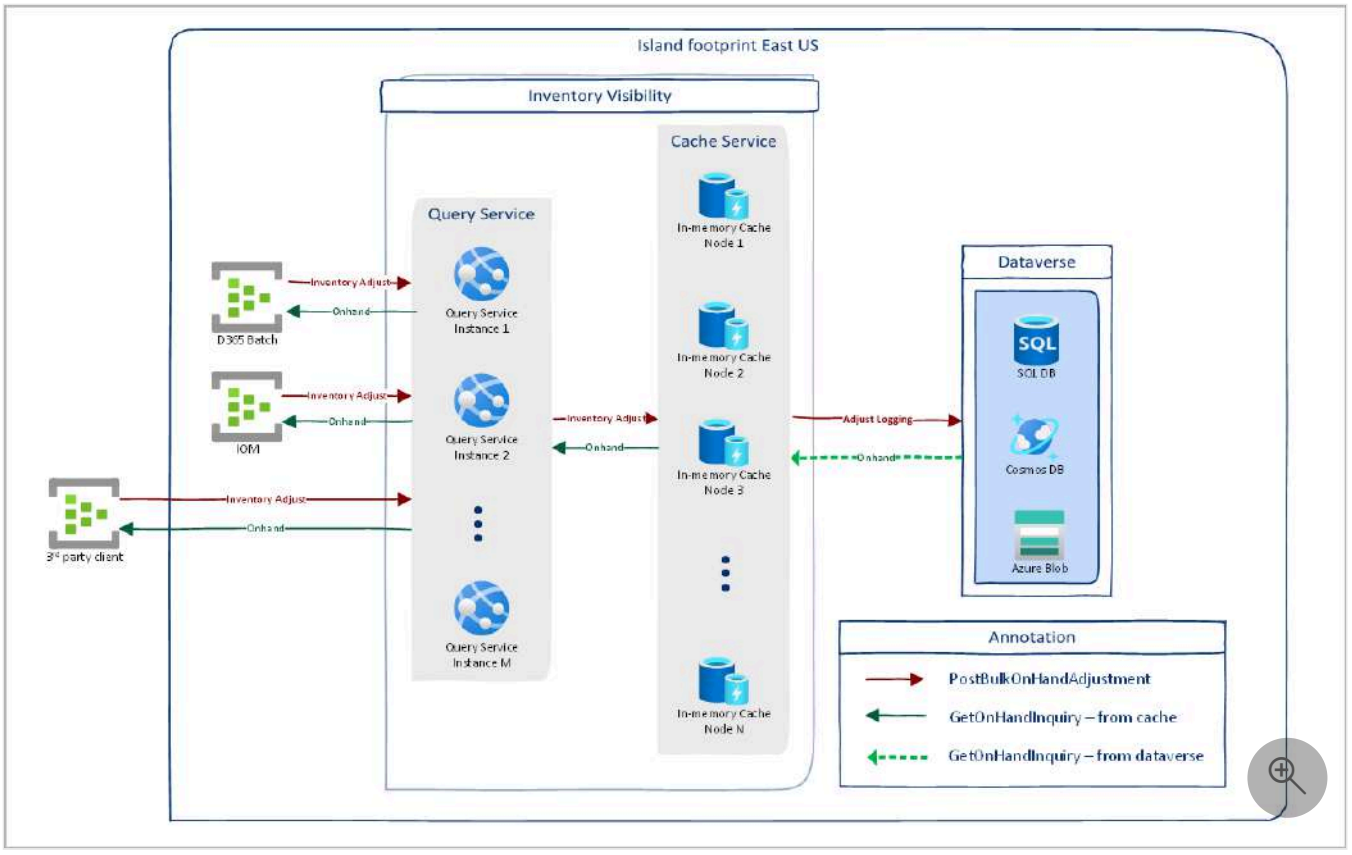
The Inventory Visibility service is highly extensible because data input and output aren't restricted to Microsoft applications. External systems can access the service through RESTful application programming interfaces (APIs). In addition to using the out-of-box mappings that are provided for the Supply Chain Management data source and dimensions, you can integrate Inventory Visibility with multiple third-party systems by setting up additional data sources, inventory statuses measures (referred to as *physical measures* in Inventory Visibility service), and inventory dimensions via the configuration app. In this way, you can flexibly query and change your multiple data sources and predefined inventory dimensions.

Additionally, because Inventory Visibility is built on Microsoft Dataverse, its data can be used to build and integrate with Power Apps. You can also use Power BI to create customized dashboards that meet your business requirements.

Scalability

The Inventory Visibility service can be scaled up or down, depending on your data volume. The scalability experience is mostly seamless and is conducted by the Microsoft platform team, based on automatic detection and assessment of your transaction data volume.

The following illustration shows the Inventory Visibility architecture.



Feature highlights

Get a global view of real-time inventory

Inventory Visibility ensures that you have access to the most up-to-date inventory quantities at all times, across all your channels, locations, and warehouses. You'll benefit most from using it to support your daily operational business whenever you must obtain inventory records. Physical on-hand inventory, quantities sold, and quantities purchased are all available out of the box. You can also configure other physical inventory measures (such as returned, quarantined, and posted data) as you require, to obtain those details in real time. Inventory Visibility can efficiently process millions of inventory change posts. This data can be aggregated and reflected in the latest inventory quantities in the service immediately, per second, or per minute, depending on the interval that data is posted at. Learn more in Inventory Visibility public APIs.

Central inventory adjustment

Inventory Visibility allows external systems to call its API to post inventory changes. The changes will immediately take effect in Inventory Visibility. Therefore, on-hand inventory is instantly deducted.

Soft reservation to avoid overselling across all order channels

A *soft reservation* lets you assign or flag specific quantities to fulfill an order or demand. A soft reservation doesn't affect physical inventory, but it does deduct from the *available for reservation* inventory quantity and provides an updated quantity for future order fulfillment. This feature will be useful if sales requests or orders come into your business from one or more channels or data sources that are outside your system-of-record enterprise resource planning (ERP) system.

If you don't use soft reservations in the Inventory Visibility service, you must wait until the order is synced to, and processed by, your ERP system to get a physical inventory quantity update. This process typically has huge latency. However, soft reservations take immediate effect each time that a sales request or order is generated in your sales channels. Therefore, they help prevent oversell situations by ensuring that your omnichannel orders won't interfere with each other when they eventually reach the ERP system. Soft reservations also ensure that you can fulfill all the orders that you've promised. Therefore, they help you meet customer expectations and maintain customer loyalty. Learn more in Inventory Visibility reservations.

Immediate response of ATP quantity and dates

Visibility into your near-future projected inventory (including supply, demand, and projected on-hand details) is important, because it helps your company achieve the following goals:

- Minimize inventory levels to reduce inventory management costs.
- Facilitate internal order processing, so that salespeople can calculate shipment and delivery dates, based on the availability of the products that are ordered.
- Provide transparency about when customers can expect an out-of-stock item to become available, by providing the next-available date.

The ATP feature is easy to adopt into your daily order fulfillment process. Most importantly, like other Inventory Visibility offerings, the ATP feature is *global and real-time*. Therefore, you can set up multiple ATP calculation formulas to have full inventory availability queries that cover all your business channels and data sources. Learn more in Inventory Visibility on-hand change schedules and available to promise.

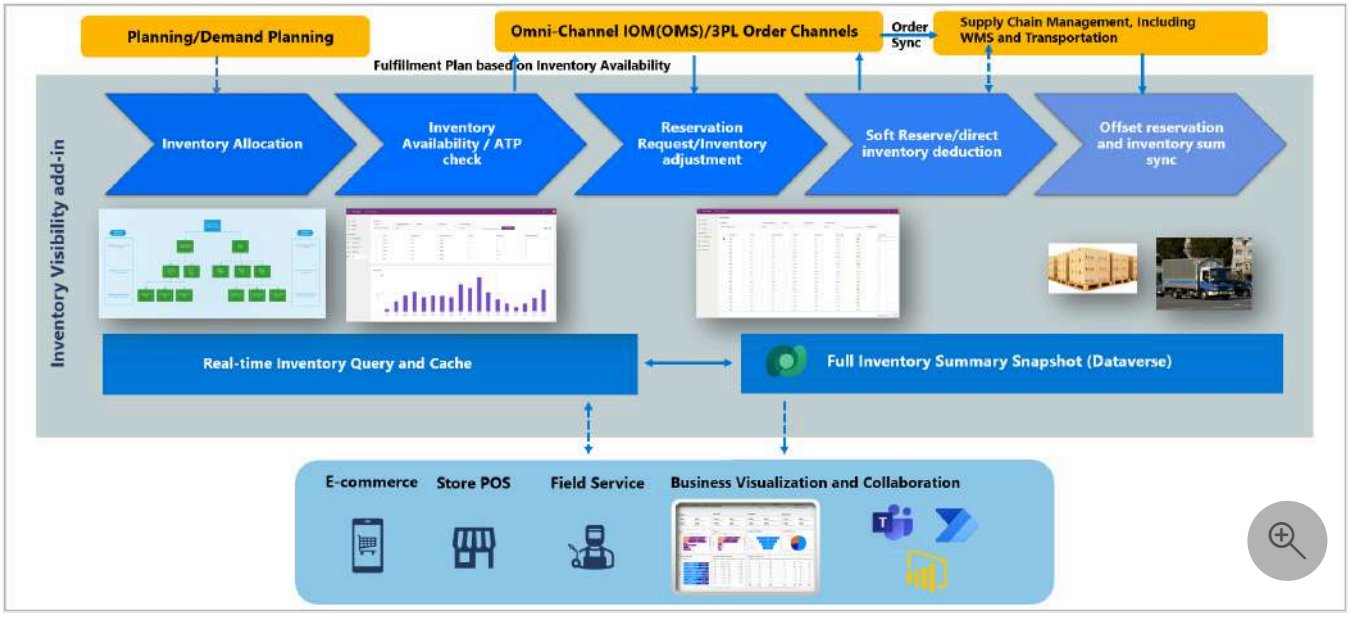
Preallocate your stock to important channels or customers with Inventory Allocation

The Inventory Visibility allocation feature lets you protect and ring-fence your valuable on-hand stock for important channels, customer groups, or locations. After stock is allocated, inventory consumption is restricted to the allocated pool, and the quantities that are left in the pool will be deducted in near-real time to reflect the quantity that is still available for consumption. Learn more in Inventory Visibility inventory allocation.

Compatibility with WMS items

Microsoft aims to provide out-of-box integration with warehouse management processes (WMS), so that WMS customers can also enjoy the benefits of the Inventory Visibility service. Per the 2022 Wave 1 release (public preview in March), inventory service supports WMS item on-hand queries and ATP. The soft reservation and allocation feature will be supported for WMS customers in the next wave. Learn more in Inventory Visibility support for WMS items.

The following illustration shows a high-level summary of existing features and how they can be positioned in the data flow.



Licensing, API Usage, and Storage for Inventory Visibility Service

The Inventory Visibility Service (IVS) is a first-party microservice available in two versions, which are described in the following subsections.

Inventory Visibility Add-in for Microsoft Dynamics 365 Supply Chain Management

The *Inventory Visibility Add-in for Microsoft Dynamics 365 Supply Chain Management* version is for companies that have a valid Supply Chain Management license (which always includes Inventory Visibility).

There's no throttling on how frequently you can call IVS APIs. Most on-hand inventory queries are served from an in-memory cache and therefore don't trigger Dataverse API calls. However, operations such as *inventory adjustments (post on-hand change)*, *soft reservations, or *ATP (scheduled on-hand change) queries* via IVS *do* result in Dataverse API calls. These Dataverse calls are subject to Dataverse service protection API limits, which are enforced *per web server*. These limits accumulate across web servers, and Microsoft automatically scales web server capacity based on your usage and performance needs. While the exact number of web servers isn't fixed or user-configurable, the platform scales to meet high-volume enterprise workloads as needed.

In terms of storage, IVS usually stores aggregated inventory data in Dataverse. However, there are two exceptions:

- **Transaction logging** – When enabled, this feature logs successful inventory API posts and stores them in Dataverse for audit and traceability.
- **Sync external inventory adjustments to Supply Chain Management** – This feature stores detailed omnichannel inventory transactions in Dataverse before syncing them to Supply Chain Management.

Even for these two exceptions, you can archive the transactions if you don't need them anymore.

For most use cases, therefore, the data footprint is typically small and is usually covered by the Dataverse storage included with your Supply Chain Management licenses. Each full-access user license for a Microsoft finance and operations app (such as Supply Chain Management) contributes additional Dataverse storage capacity, which accumulates across all licensed users (typically at 250 MB of database storage and 2 GB of file storage per user).

Your actual storage usage might vary depending on your configuration and transaction volume. Learn more in Dataverse capacity-based storage details. If you need to estimate your usage, you can enable Inventory Visibility on a sandbox environment, load a small subset of your data, and check the Dataverse usage. If your estimated API and storage usage exceeds the standard allocation, you can contact your Microsoft sales representative.

Inventory Visibility Service component for IOM

The *Inventory Visibility Service component for Intelligent Order Management (IOM)* version is for companies that use IOM or companies that aren't using Supply Chain Management as their

ERP system. The license is included in the Intelligent Order Management bundle. Learn more in Intelligent Order Management overview.

Inventory Visibility terminology

It's important that you understand the following concepts and terms when you're working with the Inventory Visibility Add-in:

- **Data source** – A data source represents the system that your data is from.
- **Dimensions** – Dimensions identify product characteristics. They can be storage dimensions (such as site or warehouse) or product dimensions (such as color, size, or style).
- **Physical measures** – Physical measures are quantities that measure different inventory statuses, such as on-hand, purchased, on order, or sold.
- **Calculated measures** – Calculated measures are quantitative measures that are calculated from a set of physical measures. For example, the *Total available* calculated measure is calculated as *On-hand* + *Purchased* – *On order* – *Sold*.
- **Partition** – A partition defines a hierarchy for how inventory visibility will distribute received data. Currently, the default partition is site and location.
- **Index hierarchy** – An index hierarchy further defines how you want to query inventory and obtain results that have more granularity.

For more information about these terms and concepts, see Configure Inventory Visibility.

Install and set up Inventory Visibility

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes how to install the Inventory Visibility Add-in for Microsoft Dynamics 365 Supply Chain Management.

> **Tip**
>
> If you're a feature consultant or solution consultant, join the D365 SCM Inventory Management and Inventory Visibility community on Microsoft Viva Engage. Read about the latest developments, exchange tips with other consultants and developers, and discuss features.
>
> For useful code samples and troubleshooting guides, visit the Inventory Visibility GitHub repo.
>
> If you're experiencing technical issues or running into exceptions, get help by contacting Microsoft Support through Microsoft Dynamics Lifecycle Services or the Power Platform admin center. Be sure to include your Supply Chain Management environment ID when you contact them.

Inventory Visibility prerequisites

Before you install Inventory Visibility, complete the following tasks:

- Get a Supply Chain Management environment that's linked to a Dataverse environment.
- Make sure that you complete the prerequisites for setting up add-ins. For information about these prerequisites, see Add-ins overview. Inventory Visibility doesn't require dual-write linking.

The following table lists the countries/regions where Inventory Visibility is currently supported.

Expand table

Azure region	Region short name
Asia East	eas
Asia Southeast	seas
Australia East	eau
Australia Southeast	seau
Brazil South	sbr
Canada Central	cca
Canada East	eca
China East 2	cne2
China East 3	cne3
China North 2	cnn2
China North 3	cnn3
Europe North	neu
Europe West	weu
France Central	cfr
France South	sfr
India Central	cin
India South	sin
Italy North	nit

Azure region	Region short name
Japan East	ejp
Japan West	wjp
Norway East	eno
Norway West	wno
Poland Central	cpl
South Africa West	wza
South Africa North	nza
Sweden Central	cse
Switzerland North	nch
Switzerland West	wch
UAE Central	cae
UAE North	nae
UK South	suk
UK West	wuk
US Central	cus
US East	eus
US East 2	eus2
US West	wus

Install the Inventory Visibility Add-in

Before you install the add-in, register an application and add a client secret to Microsoft Entra under your Azure subscription. For instructions, see Register an application and Add a client secret. Make sure to note the **Application (client) ID**, **Client secret**, and **Tenant ID** values, because you'll need them later.

> **Important**
>
> If you have more than one environment, create a different Microsoft Entra application for each environment. If you use the same application ID and tenant ID to install the Inventory

Visibility Add-in for different environments, a token issue occurs for older environments. As a result, only the latest installation is valid.

You can install the Inventory Visibility Add-in from either of two places:

- **Microsoft Dynamics Lifecycle Services** – Lifecycle Services is a collaboration portal that provides a unified collaborative environment together with a set of regularly updated services that help you manage the application lifecycle of your implementations.
- **Power Platform admin center** – The Power Platform admin center provides a unified portal where administrators can manage environments and settings for Power Apps, Power Automate, Power Pages, and Microsoft Copilot Studio.

Install the Inventory Visibility Add-in from Lifecycle Services

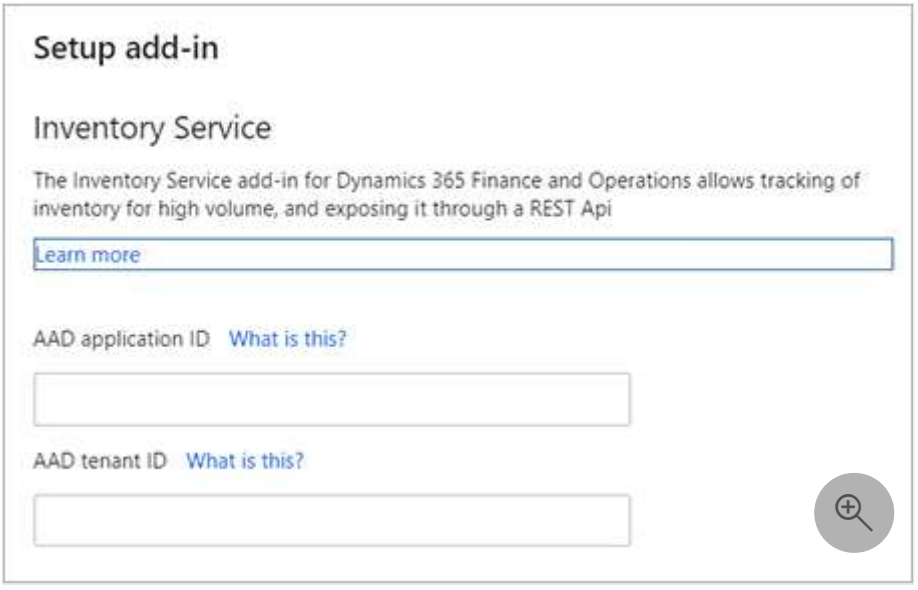
After you register an application and add a client secret to Microsoft Entra ID, follow these steps to install the Inventory Visibility Add-in.

1. Sign in to Lifecycle Services.
2. On the home page, select the project where your environment is deployed.
3. On the project page, select the environment where you want to install the add-in.
4. On the environment page, scroll down until you find the **Environment add-ins** section in the **Power Platform integration** section. There, you can find the Dataverse environment name. Confirm that the Dataverse environment name is the one that you want to use for Inventory Visibility.

   > **Important**
   >
   > Before you start the installation, you should check for linking mismatch warnings in Lifecycle Services. To do so, open the details page for your environment in Lifecycle Services and look for a warning that resembles the following example: "Microsoft has detected that your environment is linked via dual-write to a different destination than specified in Power Platform Integration, which isn't recommended." If you see this warning, it's possible that your dual-write environment is linked to a Dataverse instance, but Lifecycle Services isn't set up for Power Platform integration. This linking mismatch can cause unexpected behavior. For information about how to fix this issue,

see Linking mismatch. After the linking mismatch is fixed, you can install Inventory Visibility.

5. In the **Environment add-ins** section, select **Install a new add-in**.
6. Select the **Install a new add-in** link. A list of available add-ins appears.
7. In the list, select **Inventory Visibility**.
8. Set the following fields for your environment:
    - **Microsoft Entra application (client) ID** – Enter the Microsoft Entra application ID that you created and noted earlier.
    - **Microsoft Entra tenant ID** – Enter the tenant ID that you noted earlier.



9. Agree to the terms and conditions by selecting the **Terms and conditions** checkbox.
10. Select **Install**. The status of the add-in is shown as **Installing**. When the installation is completed, refresh the page. The status should change to **Installed**.
11. In Dataverse, select the **Apps** section in the left navigation, and verify that the **Inventory Visibility** Power Apps is installed successfully.

> ⓘ **Note**
>
> If the system warns you that you don't have permission to install Inventory Visibility on Lifecycle Services, contact the administrator to modify your permission.

If it takes more than an hour to install from the Lifecycle Services page, your user account probably lacks permission to install solutions in the Dataverse environment. Follow these steps to resolve the issue:

1. Cancel the Inventory Visibility Add-in installation process from the Lifecycle Services page.
2. Install the Inventory Visibility Add-in from the Power Platform admin center.
3. After the installation is completed, go back to the Lifecycle Services page, and check the status of the Inventory Visibility Add-in.

Install the Inventory Visibility Add-in from the Power Platform admin center

After you register an application and add a client secret to Microsoft Entra ID, you must install the Inventory Visibility Add-in.

Important

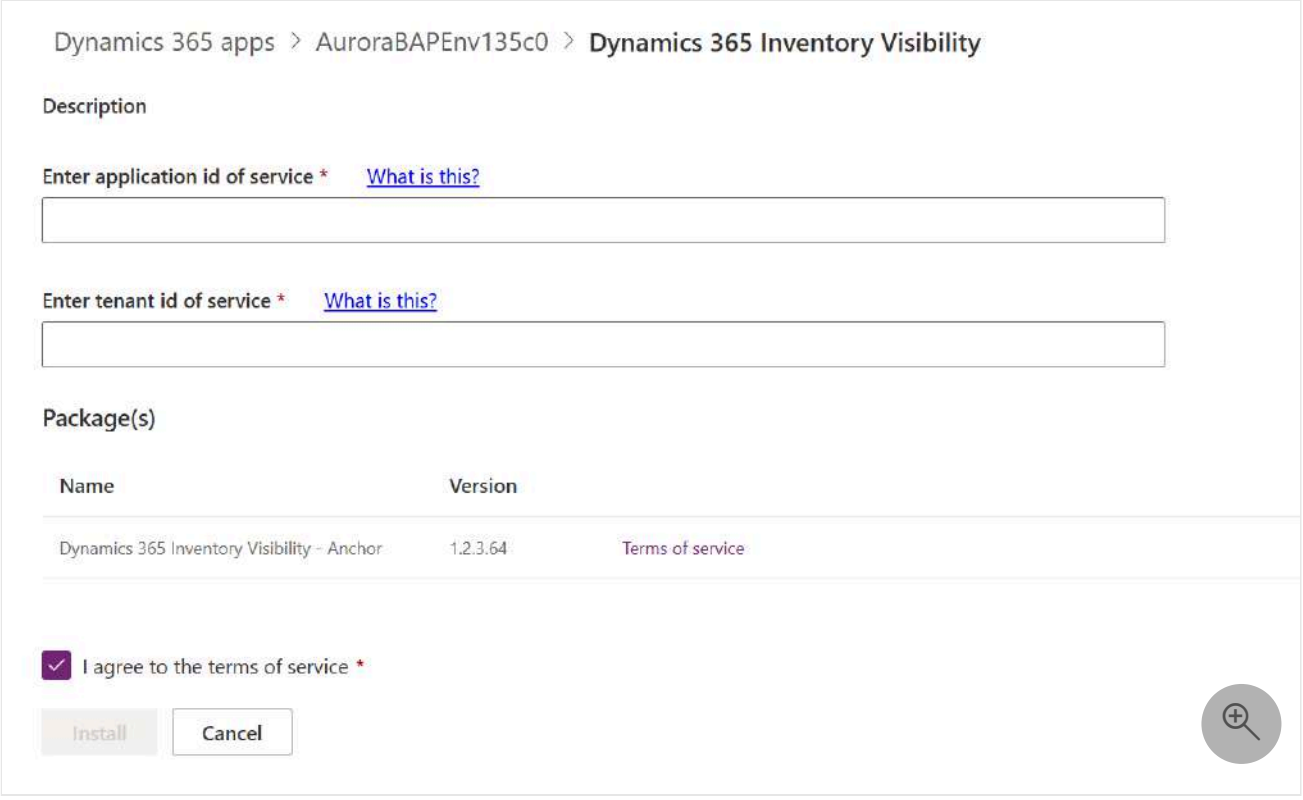
Unless you're using the unified developer experience, check for linking mismatch warnings in Lifecycle Services before you install Inventory Visibility. To do so, open the details page for your environment in Lifecycle Services and look for a warning that resembles the following example: "Microsoft has detected that your environment is linked via dual-write to a different destination than specified in Power Platform Integration, which isn't recommended." If you see this warning, it's possible that your dual-write environment is linked to a Dataverse instance, but Lifecycle Services isn't set up for Power Platform integration. This linking mismatch can cause unexpected behavior. For information about how to fix this issue, see Linking mismatch. After you fix the linking mismatch, you can install Inventory Visibility.

Follow these steps to install the Inventory Visibility Add-in.

1. Sign in to the Microsoft 365 admin center.
2. Make sure that the *Dynamics 365 Unified Operations Plan* license is assigned to the user account that you want to use to install the add-in. If the license isn't assigned, assign it.
3. Sign in to the Power Platform admin center by using the same user account.
4. Select the environment where you want to install the add-in.

5. On the top bar, select **Resources** > **Dynamics 365 apps**.
6. Select **Install App**.
7. Select the *Dynamics 365 Finance and Operations Platform Tools* app. This app provides platform support for registering Inventory Visibility. If you have more than one license that provides access to this app, there might be multiple entries for it. In this case, select any entry that has a status of **Enabled**. Then select **Next**.
8. Agree to the terms of service and select **Install**. Wait for the installation to finish before you move on to the next step. When the installation finishes, the value in the **Status** column changes from *Installing* to *Installed*.
9. Select **Install App**.
10. Find *Dynamics 365 Inventory Visibility* in the list, and select **Next**.
11. Set the following fields for your environment:

    - **Enter application id of service** – Enter the Microsoft Entra application ID that you created and made a note of earlier.
    - **Enter tenant id of service** – Enter the tenant ID that you made a note of earlier.



12. Agree to the terms of service.

13. Select **Install**. During the installation, the status of the add-in is shown as *Installing*. After the installation is completed, refresh the page. The status should change to *Installed*.

Set up Inventory Visibility in Supply Chain Management

Set up Inventory Visibility integration

After you install the add-in, prepare your Supply Chain Management system to work with it by following these steps.

1. In Supply Chain Management, open the **Feature management** workspace, and turn on the *Inventory Visibility integration* feature. As of Supply Chain Management version 10.0.45, it's mandatory and can't be turned off.

2. Go to **Inventory Management** > **Set up** > **Inventory Visibility Integration parameters**.

3. Open the **General** tab and make the following settings:

    - **Inventory Visibility endpoint** – Enter the URL of the environment where you're running Inventory Visibility. Learn more in Find your service endpoint and read the configuration.
    - **Maximum number of records in a single request** – Set the maximum number of records to include in a single request. Enter a positive integer that's less than or equal to 1,000. The default value is 512. Keep the default value unless Microsoft Support advises you to change it or you're otherwise certain that you need to change it.

4. The following optional features enhance the functionality of Inventory Visibility. Decide whether you want to use any of these features. If you do, set them up. (You can also set them up later.)

    - **Soft reservations and offsets** – Soft reservations help organizations achieve a single source of truth for available inventory, especially during the order fulfillment process. For information about how to enable and set up this feature, see Inventory Visibility reservations.
    - **Support for warehouse management processes (WMS) items** – This feature lets you use WMS items with Inventory Visibility. For information about how to enable and set up this feature, see Inventory Visibility support for WMS items.
    - **Inventory summary** – This feature provides an inventory summary for products together with all dimensions. For information about how to enable and set up this feature, see Inventory summary.

- **Preload a streamlined on-hand query** – This feature provides an aggregated inventory summary for products by configured dimensions. For information about how to enable and set up this feature, see Preload a streamlined on-hand query.
- **Track time-series inventory in Inventory Visibility** – This feature enables Supply Chain Management to send its on-hand change schedule to Inventory Visibility to support available-to-promise (ATP) calculations. For information about how to enable and set up this feature, see Track time-series inventory in Inventory Visibility.

5. After you finish setting up the optional features you selected, go to **Inventory Management** > **Periodic tasks** > **Inventory Visibility Integration**, and enable the job. All inventory change events from Supply Chain Management are now posted to Inventory Visibility.

> **Note**
>
> When you enable the Inventory Visibility integration job, if you receive an error that states that you must update the partition schema, see the **Update partition schema to two if you get an error when enabling the Inventory Visibility integration job** section of this article for instructions.

Inventory Visibility integration hotfixes

For a list of the latest hotfixes available for Inventory Visibility integration features in Supply Chain Management, see Recent Dynamics 365 SCM hotfixes.

Update partition schema to two if you get an error when enabling the Inventory Visibility integration job

When you try to enable the Inventory Visibility integration batch job from Supply Chain Management, you might receive the following error:

> Cannot sync more than 500000 records in the same warehouse. To mitigate this issue, update partition schema to 2 in Inventory Visibility Add-in.

If you receive this error, follow these steps to update your partition schema to help prevent out-of-memory issues. If you don't receive this error, you can skip this procedure.

1. In Power Apps, delete all inventory data.
2. Set up a system to send API requests to Inventory Visibility.

3. After data is deleted, call the `Get` API with a body of `none` to get all partition IDs (by using `/api/environment/{environmentId}/allpartitionids`). Review the response to confirm that data is completely cleared. The result should be empty.
4. Call the `Post` API with a body of `none` to change the partition schema (by using `/api/environment/{environmentId}/updatePartitionSchema?newversion=2`).
5. In Power Apps, enable the advanced warehouse inventory feature, and update the configuration.
6. In Power Apps, review the runtime configuration. The `CachePartitonIdVersion` field should show a value of `ByLocationAndProductIdMod64`.
7. In Supply Chain Management, go to **Inventory Management** > **Periodic tasks** > **Inventory Visibility Integration**, and enable the job.

Update the Inventory Visibility Add-in

To update an installed version of the Inventory Visibility Add-in to the latest version, follow these steps:

1. Sign in to the Power Platform admin center.
2. On the navigation pane, select **Environments**.
3. Open the environment where you want to update the Inventory Visibility Add-in.
4. In the **Resources** section, select **Dynamics 365 apps**.
5. In the list, find the row where **Name** is *Dynamics 365 Inventory Visibility*. Check the value in the **Status** column for this row.
    - If the **Status** is *Installed*, then you're already running the latest version so you can skip the rest of this procedure.
    - If the **Status** is *Update available*, then an update is available. Continue with the next step.

    > **Tip**
    >
    > To see which version of the add-in you're currently running, regardless of the status, open the **More application actions** menu (ellipsis button), and then select **Details**.

6. If the page shows that an update is available, open the **More application actions** menu (ellipsis button), and then select **Update**.

7. In the dialog box that appears, select the **I agree to the terms of service** checkbox, and then select **Update**.

8. A pop-up window prompts you to confirm the action. To proceed, enter the name of your environment in the field provided, and then select **Update**.

Uninstall the Inventory Visibility Add-in

To uninstall the Inventory Visibility Add-in, follow these steps:

1. Sign in to Supply Chain Management.

2. Go to **Inventory Management** > **Periodic tasks** > **Inventory Visibility Integration**, and disable the job.

3. Go to Lifecycle Services and open the page for the environment where you want to uninstall the add-in (see also the Install the Inventory Visibility Add-in section).

4. Select **Uninstall**.

> ⓘ **Note**
>
> Uninstalling from the Power Platform admin center isn't officially supported.

The uninstall process terminates the Inventory Visibility Add-in, unregisters the add-in from Lifecycle Services, and deletes any temporary data that's stored in the Inventory Visibility Add-in data cache. Primary inventory data that was synced to your Dataverse subscription isn't deleted.

> ⓘ **Important**
>
> If you want to continue to use Inventory Visibility in the current environment, don't delete all solutions from your Dataverse environment. If you want to delete all inventory data in Dataverse, use the delete all inventory data option.

> ⓘ **Note**
>
> If you must delete all solutions that are related to Inventory Visibility, follow these steps:
>
> 1. Open Power Apps.

2. Use the environment picker in the upper right to select the target Dataverse environment.

3. Go to **Solutions**, and delete solutions in the following order:
   a. Dynamics 365 Inventory Visibility – Anchor
   b. Dynamics 365 Inventory Visibility – Copilot
   c. Dynamics 365 Inventory Visibility – Application
   d. Dynamics 365 Inventory Visibility – Controls
   e. Dynamics 365 Inventory Visibility – Plugins
   f. Dynamics 365 Inventory Visibility – Base
   g. Dynamics 365 Product Search - Anchor
   h. Dynamics 365 Product Search Core

Inventory Visibility data that's stored in tables is deleted together with the solutions.

Move data between Supply Chain Management databases and Dataverse environments

You can migrate data between Supply Chain Management databases and Dataverse environments in several ways. For instructions, see the Inventory Visibility GitHub repo.

Last updated on 04/24/2026

Set up product search for Inventory Visibility

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

The *product search* feature lets users search for products and on-hand inventory information based on specific attributes, such as size and color. It provides the following benefits:

- **Time savings** – Users can quickly find products that meet their exact requirements without having to do extensive browsing. Therefore, the feature helps improve the user experience and save valuable time.
- **Higher confidence** – Attribute-based search ensures that users get exactly what they're looking for. Therefore, user confidence increases.
- **Increased product visibility** – Attribute-based search can highlight products that are less often searched for but still potentially relevant. Therefore, users are exposed to a broader range of options that they might not have considered.

This feature is accessible both through the Inventory Visibility app in Microsoft Power Apps and through the API.

Prerequisites

Before you can use the product search service, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management 10.0.36 or later.
- Inventory Visibility version 1.2.2.54 or later must be installed and set up as described in Install and set up Inventory Visibility.

Install the product search service

New Inventory Visibility installations

The product search service is included in the current version of the Inventory Visibility Add-in. Therefore, you don't have to install it separately if you haven't already installed the add-in for your environment. For information about how to install Inventory Visibility for the first time, see Install and set up Inventory Visibility.

Existing Inventory Visibility installations

If your system is running Inventory Visibility version 1.2.2.53 or earlier, you must update Inventory Visibility in your Microsoft Dynamics Lifecycle Services project and then install the newest version. Learn more in Install and set up Inventory Visibility.

Configure the product search service

Follow these steps to configure product search for Inventory Visibility.

1. If dual-write isn't already enabled for your Lifecycle Services project, enable it as described in Dual-write setup from Lifecycle Services.
2. In Supply Chain Management, go to **System administration** > **Workspaces** > **Data management**, and select the **Framework parameters** tile.
3. On the **Data import/export framework parameters** page, on the **Entity settings** tab, select **Refresh entity list**.
4. The *Refresh entity list* job is added to the batch queue. Wait for the job to be completed.
5. Go back to **System administration** > **Workspaces** > **Data management**, and select the **Dual-write** tile.
6. On the Action Pane, select **Apply solution**.
7. In the **Apply solution** dialog box, select **Dynamics 365 Product Search Core**, and then select **Apply**.
8. Wait for the process to be completed. It might take several minutes. When it's completed, the **Dual-write** page is updated and shows all the table maps that were added. Select all the table maps, and then select **Run** on the Action Pane.
9. The **Initial writes and related table map(s)** dialog box appears. For each table map, select the **Initial sync** checkbox, and set the **Master for initial sync** field to *Finance and*

Operations apps. Then select **Run**.

Troubleshooting

If some table maps fail to do an initial synchronization because of a permission issue, follow these steps to check your team roles.

1. Sign in to your Dataverse environment, go to **Settings** > **Security**, and select **Teams**.
2. Open your team, and select **Manage Roles**.
3. Make sure that the following role is assigned to your team:

    - *System Administrator*

For more information about how to set up dual-write security roles, see Set up dual-write security roles and permissions.

Search for products using the API

After you set up the product search service, you can search for products either by using the Inventory Visibility app in Power Apps or by using the Inventory Visibility API. For details about how to search for products by using the Inventory Visibility API, see Query with product search.

Search for products using the Inventory Visibility app in Power Apps

After you set up the product search service, you can search for products either by using the Inventory Visibility app in Power Apps or by using the Inventory Visibility API. For details about how to search for products by using the Inventory Visibility app in Power Apps, see Search for products using the Inventory Visibility app.

Last updated on 03/10/2026

Enable Inventory Visibility for Commerce

This article describes how to set up Inventory Visibility for Microsoft Dynamics 365 Commerce Scale Units (CSUs) so that e-commerce, point of sale (POS), or other applications on CSUs can use it for real-time inventory lookup, reservation, and adjustments.

Inventory Visibility for Commerce

Accurate inventory information is crucial for any retailer that runs an omnichannel business. Discrepancies between system-recorded inventory and physical inventory can lead to lost revenue or reduced customer satisfaction. Retailers require an accurate, holistic view of inventory across all channels and locations, and throughout all sales and supply chain processes. They must be able to query, reserve, and adjust on-hand inventory in near-real time.

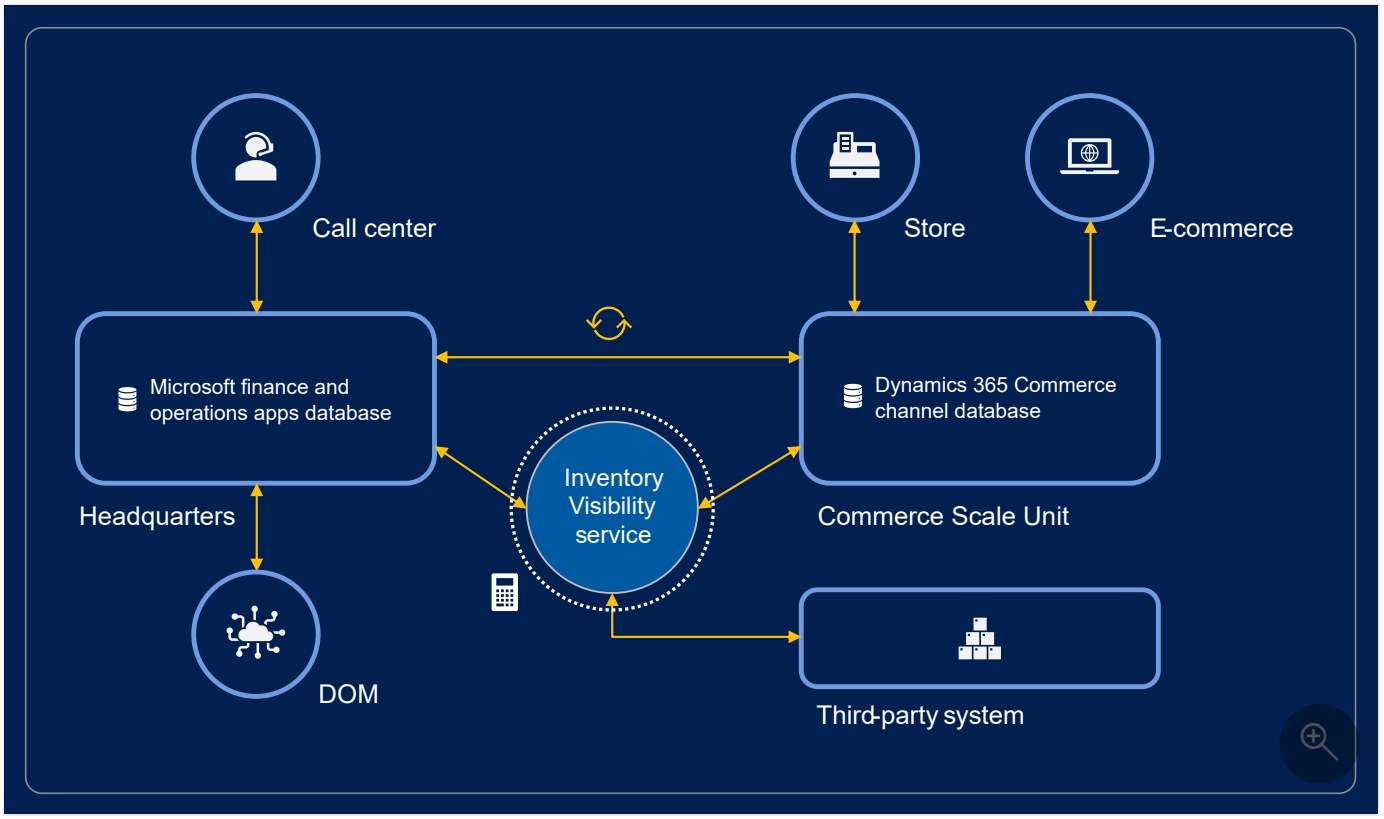
Dynamics 365 Commerce works asynchronously with inventory data. It relies on scheduled back-end jobs to exchange data between Commerce headquarters and CSU components. Because the inventory data that's available to headquarters at any given time might not reflect the latest inventory snapshot, it can't reliably support omnichannel scenarios.

If your company runs any or all of the following scenarios, we recommend that you enable Inventory Visibility so that Commerce can provide real-time inventory data across all channels:

- You're running both Dynamics 365 Commerce and Dynamics 365 Supply Chain Management.
- Your POS or e-commerce business runs on multiple CSUs.
- You have third-party systems that also update inventory.

The Inventory Visibility add-in (also referred to as the *Inventory Visibility service*) for Supply Chain Management is an independent and highly scalable microservice for posting on-hand inventory changes and tracking inventory across data sources in real time. It provides a set of RESTful APIs and includes native support for integrating with Commerce directly out of the box. Whenever a Commerce channel requests real-time inventory data, Inventory Visibility provides it in real time. Whenever an event that changes inventory occurs in the system (such as order creation, editing, cancellation, return, or fulfillment), the event is immediately posted to Inventory Visibility to update its inventory snapshot.

The following illustration shows a system that implements most of the previously mentioned integration scenarios (but just a single CSU).



For more information about Inventory Visibility, see Inventory Visibility Add-in overview.

Prerequisites

Before you can use the features that this article describes, your system must meet the following requirements:

- You must have licenses for both Supply Chain Management and Commerce.
- You must be running Supply Chain Management version 10.0.36 or later.
- The Inventory Visibility add-in must be installed and set up as described in Install and set up Inventory Visibility. Be sure to make a note of the following information from the setup, because you'll need it when you set up the integration with Commerce:
  - Application ID (Client ID)
  - Client secret
  - Tenant ID
- The following features must be turned on in Feature management:
  - *Inventory Visibility integration with inventory adjustment posting* (As of Supply Chain Management version 10.0.45, this feature is mandatory and can't be turned off.)
  - *Inventory Visibility integration with inventory adjustment offset* (As of Supply Chain Management version 10.0.45, this feature is mandatory and can't be turned off.)

- *Enable warehouse items in Inventory Visibility* (As of Supply Chain Management version 10.0.45, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.)

- Inventory Visibility support for WMS items must be enabled for your environment.

Enable adjustments in Supply Chain Management

To enable adjustments in Supply Chain Management, follow these steps:

1. Sign in to Supply Chain Management.
2. Go to **Inventory Management** > **Inventory Visibility** > **Inventory Visibility integration**.
3. On the **Adjustment** tab, on the **Manage inventory adjustment posting and retry** FastTab, on the toolbar, select **Enable adjustment**.

Create a key vault to hold the client secret for Inventory Visibility

To create a key vault to hold the client secret for Inventory Visibility, follow these steps:

1. Sign in to the Azure portal.
2. Search for or select **Key vaults** to open the **Key vaults** page.
3. On the toolbar, select **Create**.
4. Work through the **Create key vault** wizard to create a key vault.
5. Open the key vault that you just created.
6. On the navigation pane, select **Overview**.
7. Copy the **Vault URI** value, and paste it into a temporary text file. Label the value so that you can identify it later.
8. While your key vault is still open, on the navigation pane, select **Object** > **Secrets**.
9. On the toolbar, select **Generate/Import**.
10. On the **Create a secret** page, set the following fields:

    - **Upload options** – Select *Manual*.

- **Name** – Enter a name for the secret (for example, *commerce-iv-01-secret*). Copy the value, paste it into your temporary text file, and label it.
- **Secret value** – Enter the client secret value that you used when you installed the Inventory Visibility add-in (as mentioned in the Prerequisites section).
- **Content type** – This field is optional. The suggested value is *application/vnd.ms-StorageConnectionString*.
- **Set activation date** – Select this checkbox, and then enter the first date when this secret should be valid.
- **Set expiration date** – Select this checkbox, and then enter the last date when this secret should be valid.
- **Enabled** – Set this option to *Yes*.

> **Important**
>
> If the client secret value that you used when you installed the Inventory Visibility add-in expires, then the **Secret value** in this step also becomes invalid. If this happens, then you must create a new client secret value for the Inventory Visibility application registration and update the **Secret value** mentioned in this step accordingly.

11. Select **Create** to create and save the secret.

Register a Microsoft Entra application to provide access to the key vault

To register a Microsoft Entra application to enable Supply Chain Management and Commerce to access the key vault, follow these steps:

1. In the Azure portal, return to the home page.
2. Search for or select **App registrations** to open the **App registrations** page.
3. On the toolbar, select **New registration**.
4. On the **Register an application** page, set the following fields:

    - **Name** – Enter a name.
    - **Supported account types** – Specify who can use the new application. For more information about this setting, see Identity and account types for single- and multitenant apps.

- **Redirect URI** – Leave this field blank.

5. Select **Register** to register the application.
6. The new app registration is opened. On the navigation pane, select **Manage** > **Certificates & secrets**.
7. On the **Client secrets** tab, select **New client secret**.
8. In the **Add a client secret** dialog box, select an expiration date that meets your needs, and then select **Add**.
9. The **Certificates & secrets** page now shows details about the new client secret. These details are shown only once and will be hidden when the page is reloaded. Therefore, you must copy them now. Copy the **Value** value, paste it into your temporary text file, and label it.
10. On the navigation pane, select **Overview**. Copy the following values, paste them into your temporary text file, and label them:

    - **Application (client) ID**
    - **Directory (tenant) ID**

After the application is registered, create an access policy to enable it to access the key vault.

1. Open the key vault you created in Create a key vault to hold the client secret for Inventory Visibility.
2. On the navigation pane, select **Access policies**.
3. On the toolbar, select **Create** to open the **Create an access policy** page.
4. On the **Permissions** tab, under **Secret permissions** > **Secret Management Operations**, Select the **Get** checkbox. A template isn't required.
5. Select **Next**.
6. On the **Principal** tab, select the application you just registered in this section.
7. Select **Next**.
8. The **Application** tab is optional.
9. Select **Next**.
10. On the **Review + create** tab, review your settings and then select **Create**.

Set up Supply Chain Management to access the new key vault

To set up Supply Chain Management to access the new key vault through the new application, follow these steps:

1. Sign in to Supply Chain Management.
2. Use the company picker to select the same company (legal entity ID) as your Dynamics 365 Commerce channel.
3. Go to **System administration** > **Setup** > **Key Vault parameters**.
4. On the Action Pane, select **New** to add a record. Set the following fields for it:
    - **Name** – Enter a name for the Key Vault record.
    - **Description** – Enter a short description.
    - **Key Vault URL** – Paste the **Vault URI** value that you copied when you created the key vault. (See step 7 in the Create a key vault to hold the client secret for Inventory Visibility section.)
    - **Key Vault client** – Enter the **Application (client) ID** value that you copied when you created the app registration. (See step 10 in the Register a Microsoft Entra application to provide access to the key vault section.)
    - **Key Vault secret key** – Enter the secret **Value** value that you copied when you created the app registration. (See step 9 in the Register a Microsoft Entra application to provide access to the key vault section.)
5. On the **Secrets** FastTab, on the toolbar, select **Add** to add a row to the grid. Set the following fields for it:
    - **Name** – Enter a name to identify the secret. You'll need this value later. Therefore, copy it, paste it into your temporary text file, and label it.
    - **Description** – Enter a short description.
    - **Secret** – Enter a value using the following format:

      vault://<KeyVaultName>/<SecretName>/<SecretVersion>

      The *<KeyVaultName>* and *<SecretVersion>* attributes are optional. However, the *<SecretName>* attribute is required, and is the **Name** value that you copied when you created the secret for the key vault (see step 10 in the Create a key vault to hold the client secret for Inventory Visibility section). If the secret version isn't defined in the Key Vault secret key, the system retrieves the active secret that has the latest expiration date. In most cases, you can define a Key Vault secret key in the following format:

vault://<SecretName>

For example, if the name of your key vault secret is *commerce-iv-01-secret*, enter *vault://commerce-iv-01-secret*.

- **Secret type** – Select *Manual*.

After filling in all the necessary fields, you can click the **Validate** button to verify your secret.

6. On the **Secrets** FastTab, on the toolbar, select **Validate**. If you correctly entered all the values, you receive the following informational message: "Validation successful." If you receive an error message, double-check your settings, and try again.

7. On the Action Pane, select **Save**. Be patient, because the save operation takes about two minutes.

For more information about the settings that are mentioned in this procedure, see Set up the Azure Key Vault client.

Set up the integration between Inventory Visibility and Commerce

To set up the integration between Inventory Visibility and Commerce, follow these steps:

1. Sign in to Supply Chain Management.

2. Go to **Retail and Commerce** > **Headquarters setup** > **Parameters** > **Commerce shared parameters**.

3. On the **Inventory** tab, set the following fields:

   - **Microsoft Entra tenant ID** – Enter the **Directory (tenant) ID** value that you used when you installed the Inventory Visibility add-in (as mentioned in the Prerequisites section).
   - **Microsoft Entra app ID** – Enter the **Application (client) ID** value that you used when you installed the Inventory Visibility add-in (as mentioned in the Prerequisites section).
   - **Microsoft Entra client secret** – Select the **Name** value that you copied when you specified the secret for the key vault on the **Key Vault parameters** page in Supply Chain Management. (See step 5 in the Set up Supply Chain Management to access the new key vault section.)
   - **Use Inventory Visibility as inventory data provider** – Set this option to *Yes*.

4. On the Action Pane, select **Save**.

Run the global configuration job

To run the global configuration job, follow these steps:

1. Go to **Retail and Commerce** > **Retail and Commerce IT** > **Distribution schedule**.
2. In the list pane, select the record that's named *1110 Global configuration*.
3. On the Action Pane, select **Run now**.

Last updated on 07/27/2026

Inventory Visibility app user interface version

The Inventory Visibility app runs in Microsoft Power Apps. The current version of the app supports two versions of the user interface. You can choose which version you prefer to use. The following versions are available:

- User interface version 2, which is referred to as *UI version 2* in this article
- User interface version 1, which is referred to as *UI version 1* in this article

This article describes the differences between the two versions, the new operations that are supported in UI version 2, and how to migrate from UI version 1 to UI version 2.

> **Note**
>
> User interface version 1 will be deprecated in a future release. Users are suggested to switch to UI version 2 when possible.

Features and configurations supported by each version

The following settings and features are accessible only from the configuration page for UI version 1:

- Cleanup of preloaded on-hand results
- Operations that are related to inventory allocation

> **Note**
>
> Most of the features for setting up and viewing preloaded on-hand results are supported by both versions of the user interface. Only the cleanup of existing preloaded on-hand results requires UI version 1.

The following settings and features are accessible only from the pages for UI version 2:

- Support for 180-day available-to-promise (ATP) calculations
- Inventory adjustments via the user interface

The following settings are accessible on the pages for both UI version 1 and UI version 2. Changes are synced between versions:

- Index hierarchy setup

All other features and settings can be accessed in both versions of the user interface. However, changes aren't synced between versions.

Switch between UI versions 1 and UI version 2

New installations of Inventory Visibility use UI version 2 by default. If you first installed an older version, the app continues to use UI version 1 by default, even after updates. However, you can switch to UI version 2 when you're ready.

Follow these steps to switch between UI versions.

1. In Power Apps, open the **Inventory Visibility** app.
2. On the navigation pane, select **UI Version Control**.
3. Select **Configuration entity version**.
4. Set the **Entity version** field to *2* or *1*, depending on the version of the UI (and Dataverse configuration entities) that you want to use.

If you're switching to UI version 2, you'll probably want to migrate your UI version 1 settings as described in the next section.

Migrate configuration settings from UI version 1 to UI version 2

When you switch from UI version 1 to UI version 2, Inventory Visibility is updated and uses only configuration values from the Dataverse entities that correspond to UI version 2. You can still read and update the old (legacy) Dataverse entities by using the pages from UI version 1. However, changes to the settings have no effect unless you switch the whole app back to UI version 1. While UI version 2 is active, only the settings that you configure on the UI version 2 pages are used. To view and edit the legacy settings from UI version 1, open the Inventory Visibility menu at the bottom of the navigation pane, and then select **Legacy UI**.

After you switch to UI version 2, you can migrate your previous UI version 1 settings to the new Dataverse entities that are used by UI version 2. Although you can switch back to UI version 1 at any time, you can't migrate settings from UI version 2 to UI version 1. If you switch back to UI version 2 after you've been working in UI version 1 for a while, you can remigrate your UI version

1 settings. However, only newly created records are migrated. Updated and deleted records aren't replicated. Therefore, all your previous UI version 2 settings are kept.

To migrate your settings from UI version 1 to UI version 2, follow these steps:

1. In Power Apps, open the **Inventory Visibility** app.
2. On the navigation pane, select **Admin settings**.
3. On the **Migrate entities to V2** tile, select **Manage**.
4. In the dialog box that appears, enter the credentials that you used when you installed the Inventory Visibility service, and then select **Login**.
5. The system copies your existing settings from UI version 1 to UI version 2.

For a more detailed migration guide with screenshots and troubleshooting tips, see Migrate Configuration Settings from UI Version 1 to UI Version 2 in the Inventory Visibility Add-in Examples repo on GitHub.

Last updated on 12/30/2025

Use the Inventory Visibility app UI version 2

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to use the Inventory Visibility app, which runs in Microsoft Power Apps.

The Inventory Visibility app offers two versions of a model-driven user experience for visualization. Users can now choose between the new user interface (referred to as *UI version 2* in this article) and the old (legacy) user interface (referred to as *UI version 1* in this article).

Prerequisites

To use UI version 2 with the Inventory Visibility app, complete the following prerequisites:

- Install and set up Inventory Visibility.
- Turn on Inventory Visibility UI version 2.

Authenticate with the Inventory Visibility service

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Admin settings**.
3. On the **Set tokens** tile, select **Manage**.
4. In the dialog box, enter the **Client ID**, **Tenant ID**, and **Client secret** values. These values were defined when the Inventory Visibility Add-in was installed.
5. Select **Login**. The system generates a new bearer token for your sessions. This token expires after one hour.

Switch between user interface versions

Regardless of which version of the user interface is active, you can switch to the other version at any time. However, only the settings that you apply in the active version of the UI have any effect. To switch between the two versions, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select one of the following options:
    - **Inventory Visibility** – The navigation pane shows the pages for UI version 2.
    - **Legacy UI** – The navigation pane shows the pages for UI version 1.

Find your service endpoint and read the configuration

The Inventory Visibility service is deployed on Azure Service Fabric in multiple geographies and multiple regions. Currently, there's no central endpoint to automatically redirect your requests to the appropriate geography and region. Therefore, the service endpoint uses the following pattern, which includes the required information:

`https://inventoryservice.<RegionShortName>-il<IsLandNumber>.gateway.prod.island.powerapps.com`

To obtain your service endpoint and runtime configuration, follow these steps:

1. On the UI version 2 navigation pane, select **Admin settings**.
2. On the **Show service details** tile, select **Manage**.
3. In the dialog box, find your service endpoint and configuration details.

Update the configuration

Any time that you modify the configuration and save your changes, the system saves the new configuration settings only temporarily. The changes don't take effect until you commit the new settings to the service. Follow these steps to commit new configuration settings to the service.

1. On the UI version 2 navigation pane, select **Admin settings**.
2. On the **Update Configuration** tile, select **Manage**.
3. Review your modifications in the dialog box.

> **ⓘ Important**
>
> Be sure to verify all the important modifications that are about to be made to your data sources, physical measures, and dimension mappings.

4. Select **Confirm Update** to apply your configuration change.

Data partition rule

Inventory Visibility can distribute and store inventory data in either of the following ways:

- *By location* – Choose this option if you always know site and warehouse information when you make on-hand queries, inventory adjustments, inventory reservations or inventory allocations through Inventory Visibility.
- *By product ID* – Choose this option if you often don't know site or warehouse information when calling Inventory Visibility. For example, when making e-commerce in-basket reservations, the fulfillment warehouse might be unknown when online orders are initially placed. In this case, you need to call Inventory Visibility to query on-hand inventory and make reservations without providing warehouse information.

To change the data partition rule, follow these steps:

1. Open the Inventory Visibility app
2. Select **Data partition rule** and choose the rule you want to use.
3. Clear all inventory data.
4. Update the configuration to apply your changes. (The update-configuration operation fails if you don't clear the data first.)

The data partition rule controls how data is distributed. Operations that are performed inside the same partition provide better performance, at lower cost, than operations that cross partitions. Use the *By product ID* option if you often query across different locations. However, use the *By location* option if you more often query for multiple products at the same location.

When using the *By location* partition rule, the rule in the following table is included in the solution by default and can't be changed (this becomes set 0 of the index hierarchy). When using *By product ID*, the rule has no effect on Inventory Visibility APIs.

Expand table

Base dimension	Hierarchy
SiteId	1
LocationId	2

Important

Don't customize the partition configuration. If you delete or change the configuration without official guidance, you're likely to encounter an unexpected error.

On-hand index configuration

In many cases, on-hand inventory is queried not only at the most detailed level but also at some aggregated levels, based on the inventory dimensions.

For frequently used query patterns, you can help improve query performance by setting up *indexes*.

An index is composed of a *set number*, *dimensions*, and a *hierarchy*.

Expand table

Name	Description
Set number	Dimensions under the same set number belong to the same set (index) and are grouped together.
Dimensions	The dimensions are base dimensions that query results are aggregated on.
Hierarchy	The hierarchy determines the specific combinations of dimensions that can take advantage of an index query. For example, you set up an index that has the following dimensions and hierarchy: *(ColorId, SizeId, StyleId)*. In this case, only queries that use the following four dimension combinations for their filter and group-by fields are enhanced: • **First combination:** Empty • **Second combination:** *(ColorId)* • **Third combination:** *(ColorId, SizeId)* • **Fourth combination:** *(ColorId, SizeId, StyleId)* (The filter field isn't order-restricted.) The index doesn't speed up queries that use other dimension combinations.

The default on-hand index configuration includes a standard set of indexes.

Tip

These tips can help when you set up your on-hand index configuration:

- Base dimensions that are reserved in the partition configuration (set number *0*) shouldn't be included in other on-hand index configurations.

- If you aren't interested in querying specific dimension combinations, set up an index that has only one base dimension, *Empty*.

Set up the on-hand index configuration

1. On the UI version 2 navigation pane, select **Index Configuration**.
2. Add and remove lines in the grid as you require to define your index. For more information, see the introduction to this section and the example in the next subsection.
3. Update the configuration to apply your changes.

On-hand index configuration example

This section provides an example that shows how an on-hand index works.

The following table shows a list of available inventory.

Expand table

Item	ColorId	SizeId	StyleId	Quantity
D0002	Black	Small	Wide	1
D0002	Black	Small	Regular	2
D0002	Black	Large	Wide	3
D0002	Black	Large	Regular	4
D0002	Red	Small	Wide	5
D0002	Red	Small	Regular	6
D0002	Red	Large	Regular	7

The following table shows an on-hand index configuration.

Expand table

Set Number	Dimension	Hierarchy
1	*ColorId*	1
1	*SizeId*	2
1	*StyleId*	3

This index helps improve the performance of the following on-hand inventory queries:

- *()* – Grouped by all
  - D0002, 28
- *(ColorId)* – Grouped by *ColorId*
  - D0002, Black, 10
  - D0002, Red, 18
- *(ColorId, SizeId)* – Grouped by the combination of *ColorId* and *SizeId*
  - D0002, Black, Small, 3
  - D0002, Black, Large, 7
  - D0002, Red, Small, 11
  - D0002, Red, Large, 7
- *(ColorId, SizeId, StyleId)* – Grouped by the combination of *ColorId*, *SizeId*, and *StyleId*
  - D0002, Black, Small, Wide, 1
  - D0002, Black, Small, Regular, 2
  - D0002, Black, Large, Wide, 3
  - D0002, Black, Large, Regular, 4
  - D0002, Red, Small, Wide, 5
  - D0002, Red, Small, Regular, 6
  - D0002, Red, Large, Regular, 7

Feature management

The following table shows the features that Inventory Visibility offers. Each of these features can be turned on or off for your environment.

Expand table

Feature name	Description
Available to promise	Track your available to promise (ATP) across all data sources and channels.
Inventory allocation	Allocate your valuable on-hand stock to your most important channels, customers, or predefined groups, and track the consumption of each allocated pool.
Advanced warehouse inventory	Sync and view advanced warehouse item inventory by using warehouse hierarchies.

Feature name	Description
Soft reservation	Post your omnichannel reservations to Inventory Visibility soft reservations for real-time inventory availability checks and updates.
Inventory log history	Enable Inventory Visibility to store your successful transaction logs. You can query the log history for details about the organization, product, date range, site, and warehouse.
Inventory summary	Periodically sync your raw inventory summary from the cache to Dataverse. This feature isn't compatible with advanced warehouse items. Enable either this feature or the *Preloaded on-hand* feature. Don't enable both features.
Preload on-hand	Periodically preload the inventory summary to Dataverse from a query result. The query result can be configured with the dimensions that are most relevant to your business. This feature is compatible with advanced warehouse items. Enable either this feature or the *Inventory summary* feature. Don't enable both features.

By default, all these features are disabled. To turn a feature on or off, and to make related configuration settings, follow these steps:

1. On the UI version 2 navigation pane, select **Feature management**.
2. Find the tile for the feature that you want to turn on or off, and then select **Manage**.
3. Turn on and set up the feature as you require. Each feature provides different settings. For more information, see the documentation for the feature that you're setting up.
4. Update the configuration to commit your changes.

Delete all configurations

If necessary, you can delete all configurations except those in the *fno* and *@iv* Inventory Visibility data sources. Deleted configurations can't be restored.

1. On the UI version 2 navigation pane, select **Admin Settings**.
2. On the **Delete all configurations** tile, select **Manage**.
3. You're prompted to confirm the deletion. Select **OK** to continue.

Delete all inventory data

If necessary, you can delete all Inventory Visibility data, except configurations, in both the cache and Dataverse. Deleted data can't be restored, and users are blocked until deletion completes.

1. On the UI version 2 navigation pane, select **Admin Settings**.
2. On the **Delete all inventory data** tile, select **Manage**.
3. You're prompted to confirm the deletion. Select **OK** to continue.

Inventory queries and updates

On the navigation pane, the **Operational visibility** group provides access to pages where you can perform real-time on-hand inventory queries and updates. When the soft reservation feature is enabled, you can also post reservation requests to the API. For more information about API requests, see Inventory Visibility public APIs.

The following elements are common to all the pages:

- A **Post** or **Query** button on the toolbar lets you send API requests to the Inventory Visibility service. This button becomes available only after you enter all the content that's required for the API call on the page.
- A **Product** section lets you enter a product ID, an organization ID, and dimension values for the API request. Select the **Edit Dimensions** button in this section to choose which of the available dimensions to include in the request.
- A **Settings** section (the exact name varies) lets you enter API-specific settings, such as whether to query ATP contents for an on-hand query.
- A **Developer reference** button on the toolbar lets you view the raw contents of your API request (read-only).
- A **Settings** button on the toolbar lets you view and edit the authentication details that are used to access the Inventory Visibility service.

The following subsections provide details about how to set up and use each type of operation.

On-hand query

Use the **On hand query** page in the **Operational visibility** group to query real-time on-hand inventory. Follow these steps to set up and run a query.

1. On the UI version 2 navigation pane, select **On hand query**.

2. In the **Product** section, enter the **Organization ID**, **Site ID**, and **Location ID** values of the products that you want to find.

3. In the **Product ID** field, enter one or more product IDs to get an exact match for your query. Alternatively, leave the field blank to include all products at the specified site and location.

4. Select **Edit Dimensions** to choose which dimensions to include in the query body. Then enter values for the selected dimensions in the **Product** section.

> **Note**
>
> If you plan to make an ATP query, ensure that every dimension that is included in the ATP index is also listed in the **Product** section. For example, if your ATP index includes `ColorId` and `SizeId`, both those dimensions must be listed in the **Product** section. If any dimensions are missing, select **Edit Dimensions** to add them. In the search form, you can leave some of the dimension values blank. However, the dimensions must be included in the query body if they're included in the ATP index.

5. For each field that you want to include in the query, but without filtering on any specific value, select **Use all values**.

6. In the **Query Settings** section, set the following options:

    - **Query ATP** – Select this checkbox to include ATP information.
    - **Include negative quantities** – Select this checkbox to include negative quantities for calculated measure results.

7. Select **Query** on the toolbar to send the request.

Inventory adjustment

Use the **Inventory Adjustment** page in the **Operational visibility** group to make real-time updates to inventory. Follow these steps to set up and submit an update.

1. On the UI version 2 navigation pane, select **Inventory Adjustment**.
2. In the **Product** section, enter the **Product ID**, **Organization ID**, **Site ID**, and **Location ID** values of the products that you want to update.
3. Select **Edit Dimensions** to choose which dimensions to include in the request body. Then enter values for the selected dimensions in the **Product** section.
4. In the **Measures to update** section, select **Add** to add a row for the measure that you want to update. For the new row, select a data source, physical measure, and quantity to update. You must specify at least one measure. You can add multiple measures.
5. Select **Post** on the toolbar to send the request.

Soft reserve

Use the **Soft reserve** page in the **Operational visibility** group to make soft reservations of inventory.

> **Important**
>
> Use the capability to make soft reservations through the user interface only to test the feature. Every soft reservation request must be associated with a transaction order line change (create, modify, delete, and so on). Therefore, make only soft reservations that are linked to back-end orders. Learn more in Inventory Visibility soft reservations.

Follow these steps to set up and submit a soft reservation.

1. On the UI version 2 navigation pane, select **Soft reserve**.
2. In the **Product** section, enter the **Product ID**, **Organization ID**, **Site ID**, and **Location ID** values of the products that you want to update.
3. Select **Edit Dimensions** to choose which dimensions to include in the request body. Then enter values for the selected dimensions in the **Product** section.
4. In the **Reservation settings** section, set the following fields:
    - **Enable negative inventory to support oversell** – Select this checkbox to skip the available-for-reservation check. Clear it to enforce the check.
    - **Reserve on measure** – Select the data source and physical measure to perform the soft reservation on.
    - **Quantity** – Specify the quantity to reserve.
5. Select **Post** on the toolbar to send the request.

Change a schedule

Use the **Post on hand change schedule** page in the **Operational visibility** group to post inventory changes with dates to the Inventory Visibility service. Follow these steps to set up and submit a schedule change.

1. On the UI version 2 navigation pane, select **Change schedule**.
2. In the **Product** section, enter the dimension values of the products that you want to update.
3. In the **Change schedule measure, quantity and date** section, select **Add** to specify the date, data source, physical measure, and quantity for the change.
4. Select **Post** on the toolbar to send the request.

Search for products in the Inventory Visibility app

The *product search* feature lets users search for products and on-hand inventory information based on specific attributes, such as size and color. For details about how to set up this feature, see Set up product search for Inventory Visibility. For details about how to use it in the Inventory Visibility app, see Search for products using the Inventory Visibility app.

Last updated on 06/22/2026

Use the Inventory Visibility app UI version 1

Article • 01/02/2024

This article describes how to use the Inventory Visibility app, which runs in Microsoft Power Apps.

The Inventory Visibility app offers two versions of a model-driven user experience for visualization. Users can now choose between the new user interface (referred to as *UI version 2* in this article) and the old (legacy) user interface (referred to as *UI version 1* in this article).

This article describes how to work with UI version 1. For information about how to work with UI version 2, see Use the Inventory Visibility app UI version 2.

Prerequisites

Before you can use UI version 1 to work with the Inventory Visibility app, the following prerequisites must be in place:

- Install and set up Inventory Visibility.
- Turn on Inventory Visibility UI version 1.

Authenticate with the Inventory Visibility service

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Operational Visibility**.
4. Select **Settings** at the top of the page.
5. In the dialog box, enter the **Client ID**, **Tenant ID**, and **Client secret** values. You should have found these values when you installed the Inventory Visibility Add-in.
6. Select the **Refresh** button on the right side of the **Bearer Token** field. The system generates a new bearer token for your sessions. This token expires after one hour.

Switch between user interface versions

Regardless of which version of the user interface is active, you can switch to the other version at any time. However, only the settings that you apply in the active version of the UI have any effect. To switch between the two versions, follow these steps.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select one of the following options:
    - **Inventory Visibility** – The navigation pane shows the pages for UI version 2.
    - **Legacy UI** – The navigation pane shows the pages for UI version 1.

Find your service endpoint and read the configuration

The Inventory Visibility service is deployed on Azure Service Fabric in multiple geographies and multiple regions. Currently, there's no central endpoint to automatically redirect your requests to the appropriate geography and region. Therefore, the service endpoint uses the following pattern, which includes the required information:

```
https://inventoryservice.<RegionShortName>-
il<IsLandNumber>.gateway.prod.island.powerapps.com
```

To obtain your service endpoint and runtime configuration, follow these steps.

1. On the UI version 1 navigation pane, select **Configuration**.
2. Select **Show Service Details** in the upper-right corner.
3. In the dialog box, you can find your service endpoint and configuration details.

Update the configuration

Any time that you modify the configuration and save your changes, the system saves the new configuration settings only temporarily. The changes don't take effect until you commit the new settings to the service. Follow these steps to commit new configuration settings to the service.

1. On the UI version 1 navigation pane, select **Configuration**.
2. On the navigation pane, select **Configuration**.
3. Select **Update Configuration** in the upper-right corner.

4. In the dialog box, set the **Client ID**, **Tenant ID**, and **Client secret** fields. You should have found the values for these fields when you installed the Inventory Visibility Add-in.

5. Select **Login**.

6. Review your modifications in the dialog box.

   > **Important**
   >
   > Be sure to verify all the important modifications that are about to be made to your data sources, physical measures, and dimension mappings.

7. Select **Confirm Update** to apply your configuration changes.

Partition configuration

Partition configuration controls how data is distributed. Operations that are performed inside the same partition provide better performance, at lower cost, than operations that cross partitions. The following partition configuration is included in the solution by default. Therefore, *you don't have to set it up yourself*.

Expand table

Base dimension	Hierarchy
SiteId	1
LocationId	2

> **Important**
>
> Don't customize the partition configuration. If you delete or change the configuration without official guidance, you're likely to encounter an unexpected error.

Set up the on-hand index configuration

The on-hand index configuration has the same purpose and functionality, regardless of which version of the user interface is active. For an overview of the on-hand index configuration and examples that show how it works, see On-hand index configuration.

This section describes only how to find the on-hand index configuration settings in UI version 1.

1. On the UI version 1 navigation pane, select **Configuration**.
2. On the **Product Index Hierarchy** tab, add sets and dimensions as you require. Add dimensions in order, according to the hierarchy that you want to create. For information about what the settings mean and how they work, including some examples, see On-hand index configuration. Set and dimensions numbers are automatically generated as you make modifications.
3. Update the configuration to apply your changes.

Feature management

The following table shows the features that Inventory Visibility offers. Each of these features can be turned on or off for your environment.

Expand table

Feature name	Description
Available to promise	Track your available to promise (ATP) across all data sources and channels.
Inventory allocation	Allocate your valuable on-hand stock to your most important channels, customers, or predefined groups, and track the consumption of each allocated pool.
Advanced warehouse inventory	Sync and view advanced warehouse item inventory by using warehouse hierarchies.
Soft reservation	Post your omnichannel reservations to Inventory Visibility soft reservations for real-time inventory availability checks and updates.
Inventory log history	Enable Inventory Visibility to store your successful transaction logs. You can query the log history for details about the organization, product, date range, site, and warehouse.
Inventory summary	Periodically sync your raw inventory summary from the cache to Dataverse. This feature isn't compatible with advanced warehouse items. Enable either this feature or the *Preloaded on-hand* feature. Don't enable both features.
Preload on-hand	Periodically preload the inventory summary to Dataverse from a query result. The query result can be configured with the dimensions that are most relevant to your business. This feature is compatible with advanced warehouse items. Enable either this feature or the *Inventory summary* feature. Don't enable both features.

By default, all these features are disabled. To turn a feature on or off, and to make related configuration settings, follow these steps.

1. On the UI version 1 navigation pane, select **Configuration**.
2. Find the tile for the feature that you want to turn on or off, and then select **Manage**.
3. On the **Feature management & settings** tab, use the option for each feature that's listed to turn it on or off. More fields might be added to the page when some features are turned on. For more information, see the documentation for the feature that you're setting up.
4. Update the configuration to commit your changes so that they take effect.

Delete all configurations

If necessary, you can delete all configurations except those in the *fno* and *@iv* Inventory Visibility data sources. Deleted configurations can't be restored.

1. On the UI version 1 navigation pane, select **Configuration**.
2. Select the **More** menu (three dots) in the upper-right corner, and then select **Delete all configurations**.

Delete all inventory data

If necessary, you can delete all Inventory Visibility data, except configurations, in both the cache and Dataverse. Deleted data can't be restored, and users are blocked until deletion is completed.

1. On the UI version 1 navigation pane, select **Configuration**.
2. Select the **More** menu (three dots) in the upper-right corner, and then select **Delete all inventory data**.

Inventory queries and updates

The **Operational visibility** page provides tabs where you can perform real-time on-hand inventory queries and updates. When the soft reservation or inventory allocation feature is turned on, you can also post reservation or allocation requests. For more information about API requests, see Inventory Visibility public APIs.

The following subsections provide details about how to set up and use each tab.

On-hand query

Use the **Onhand query** tab of the **Operational visibility** page to query real-time on-hand inventory. Follow these steps to set up and run a query.

1. On the UI version 1 navigation pane, select **Operational visibility**.
2. On the **Onhand query** tab, enter the **Organization ID**, **Site ID**, and **Location ID** values of the products that you want to find.
3. In the **Product ID** field, enter one or more product IDs to get an exact match for your query. Alternatively, leave the field blank to include all products in the specified organization, site, and location.
4. To get a more granular result (for example, a view that's grouped by dimensions such as color and size), select the dimensions to group by in the **Group result by** field.
5. To filter items that have a specific dimension value (for example, color = red), select the dimension in the **Filter dimensions** field, and then enter the value.
6. To include ATP information, select the **Query ATP** checkbox.
7. Select **Query** to send the request.

Reservation posting

Use the **Reservation posting** tab of the **Operational visibility** page to make soft reservations of inventory.

> **Important**
>
> The capability to make soft reservations through the user interface should be used only to test the feature. Every soft reservation request should be associated with a transaction order line change (create, modify, delete, and so on). Therefore, we recommend that you make only soft reservations that are linked to back-end orders. Learn more in Inventory Visibility soft reservations.

Follow these steps to set up and submit a soft reservation.

1. On the UI version 1 navigation pane, select **Operational visibility**.
2. On the **Reservation posting** tab, in the **Quantity** field, enter the quantity that you want to soft reserve.
3. Select the **Enable negative inventory to support oversell** checkbox to skip the available-for-reservation check, or clear it to enforce the check.
4. In the **Operator** field, select the data source and physical measure to make the soft reservation on.
5. Enter the **Organization ID**, **Site ID**, **Location ID**, and **Product ID** values that you want to soft reserve.

6. In the **Dimension** field, select a data source. Then add dimensions from that data source, and enter values for each of the selected dimensions.
7. Select **Post** to send the request.
8. Select **Show details** to view a detailed response, including the `reservationId` value.

Inventory allocation

The inventory allocation feature lets sales planners or key account managers manage and preallocate important stock across allocation groups (such as channels, regions, and customer groups). It also supports real-time tracking, adjustment, and analytics of consumption against allocated quantities, to ensure that replenishment or reallocation can be done on time. The ability to have real-time visibility into allocation, consumption, and allocation balance is especially important at fast-sale or promotion events.

For more information about inventory allocation, including information about how to work with it in the Inventory Visibility app, see Inventory Visibility inventory allocation

Search for products in the Inventory Visibility app

The *product search* feature lets users search for products and on-hand inventory information based on specific attributes, such as size and color. For information about how to set up this feature, see Set up product search for Inventory Visibility. For information about how to use it in the Inventory Visibility app, see Search for products using the Inventory Visibility app.

Feedback

Was this page helpful? Yes No

Provide product feedback

Search for products using the Inventory Visibility app

The *product search* feature lets users search for products and on-hand inventory information based on specific attributes, such as size and color. It provides the following benefits:

- **Time savings** – Users can quickly find products that meet their exact requirements without having to do extensive browsing. Therefore, the feature helps improve the user experience and save valuable time.
- **Higher confidence** – Attribute-based search ensures that users get exactly what they're looking for. Therefore, user confidence increases.
- **Increased product visibility** – Attribute-based search can highlight products that are less often searched for but still potentially relevant. Therefore, users are exposed to a broader range of options that they might not have considered.

This feature is accessible both through the Inventory Visibility app in Microsoft Power Apps and through the API. This article describes how to search for products by using the Inventory Visibility app in Power Apps.

Prerequisites

Before you can search for products in the Inventory Visibility app, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management 10.0.36 or later.
- Inventory Visibility version 1.2.2.54 or later must be installed and set up as described in Install and set up Inventory Visibility.
- The Inventory Visibility search service must be installed and set up as described in Set up product search for Inventory Visibility.

Search for products using attribute filters

When you search by using attribute filters, you select one or more specific product attributes (such as product number, color, and brand). You also specify the value that you're looking for in each attribute. The search results include all products that match all the specified values.

To search for products by using attribute filters in the Inventory Visibility app, follow these steps:

1. Sign in to your Power Apps environment.
2. Open the **Inventory Visibility** app.
3. Open the **Product search** page from the left pane.
4. In the **Attribute filters** pane, find and select the attribute that you want to filter by. A **Find attribute** field is provided to help you find the attribute that you're looking for.
5. A dropdown dialog box appears, where you can select a logical operator (such as *Is exactly*, *Contains*, or *Begins with*), and then enter or select the value that you're looking for. When you've finished, select **OK**.
6. Your new criterion is added to the **Filtered by** list at the top of the **Attribute filters** pane. You can remove any filter that's listed here by selecting its **Close** (**X**) button.
7. Repeat steps 4 through 6 to add more filters as you require. If you include more than one criterion in the filter, the search results include only products that match *all* the criteria. (The criteria are combined by using an *AND* operator.)
8. When you're ready to run the search, select **Apply** at the bottom of the **Attribute filters** pane.
9. The main part of the page lists all matching products. For each product, the list shows the product name, a short description, and the total quantity that's available in inventory.
10. To view more inventory details about a product, select its checkbox in the list. You can select more than one product. When you've finished selecting products, select **View inventory**.

> ⓘ **Note**
>
> You can't combine attribute search criteria with quick search criteria. If you do one type of search and then do the other type of search, the first search results are cleared.

Search for products using quick search

Quick search provides a single search field that matches products based only on their product number or product name. It doesn't support searching by other attributes. The search results include all products that include the specified value in their product number, their product name, or both.

To search for products by using only a product number or product name, follow these steps:

1. Sign in to your Power Apps environment.
2. Open the **Inventory Visibility** app.
3. Open the **Product search** page from the left pane.

4. In the main part of the page, in the field under the **What do you want to search?** heading, enter a product name or number. Then select the search button next to the field.
5. The main part of the page lists all matching products. For each product, the list shows the product name, a short description, and the total quantity that's available in inventory.
6. To view more inventory details about a product, select its checkbox in the list. You can select more than one product. When you've finished selecting products, select **View inventory**.

> **Note**
>
> You can't combine attribute search criteria with quick search criteria. If you do one type of search and then do the other type of search, the first search results are cleared.

Related information

- Use the Inventory Visibility app
- Set up product search for Inventory Visibility

Last updated on 12/30/2025

Configure Inventory Visibility data sources

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article offers a comprehensive overview of data source configuration for Inventory Visibility. It covers the essential steps for setting up and optimizing data source configurations to enhance your inventory management processes.

Each data source configuration includes the following components:

- Data source name
- Dimensions (dimension mapping)
- Physical measures
- Calculated measures

> **Note**
>
> If you recently transitioned from UI version 1 to UI version 2, see the Differences between Inventory Visibility UI version 1 and UI version 2 section of this article to learn what has changed.
>
> For new installations, use UI version 2 from the start. If you're still using UI version 1, test the new version, and then upgrade to UI version 2 as soon as possible.

Prerequisites

Before you begin, install and set up the Inventory Visibility Add-in as described in Install and set up Inventory Visibility.

Differences between Inventory Visibility UI version 1 and UI version 2

The Inventory Visibility app in Microsoft Power Apps supports two versions of the user interface: UI version 1 and UI version 2. Each version uses a different and independent collection of configuration settings. Therefore, you must choose which version to use.

When you're setting up data sources, UI version 2 differs from UI version 1 in the following ways:

- **Physical measures** – In UI version 2, you no longer have to specify the data source for a physical measure. By default, UI version 2 uses the current data source.
- **Calculated measures** – In UI version 2, you specify calculated measures in two steps:
    1. Set up calculated measure metadata to specify the name and data source for a calculated measure.
    2. Add calculated measure lines. Each line consists of a physical measure name and an operator (addition or subtraction) that's used to combine the line with the other lines during the calculation.

Add data sources (data source names)

Each data source represents a system that provides your data. Example data source names include *fno* (which corresponds to Dynamics 365 Supply Chain Management) and *pos* (which stands for "point of sale"). By default, Supply Chain Management is set up as a default data source (*fno*) in Inventory Visibility.

> ⓘ **Note**
>
> The *fno* data source is reserved for Supply Chain Management. If your Inventory Visibility Add-in is integrated with a Supply Chain Management environment, don't delete configurations related to *fno* in the data source.

> ⓘ **Important**
>
> When you add a data source, be sure to validate the data source name, physical measures, and dimension mappings before you update the configuration for the Inventory Visibility service. You can't modify these settings after you select **Update Configuration**.

Add a data source in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

To add a data source in UI version 2, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Data source settings** tile, select **Manage**.
4. On the **Data source settings** page, select **New** on the toolbar to add a data source. Enter a name for the new data source (for example, *ecommerce* or another meaningful data source ID).
5. Select **Save**.

Add a data source in UI version 1

This section applies when you're using Inventory Visibility UI version 1.

To add a data source in UI version 1, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Configuration**.
4. On the **Data Source** tab, select **New data source** to add a data source. Enter a name for the new data source (for example, *ecommerce* or another meaningful data source ID).

> **Note**
>
> When you add a data source, be sure to validate the data source name, physical measures, and dimension mappings before you update the configuration for the Inventory Visibility service. You can't modify these settings after you select **Update Configuration**.

Dimensions (dimension mapping)

Dimension configuration standardizes multi-system integration for posting events and queries based on dimension combinations. Inventory Visibility provides a list of base dimensions that can be mapped from the dimensions of your data source. Thirty-three dimensions are available for mapping.

- If you're using Supply Chain Management as one of your data sources, 13 dimensions are already mapped to the Supply Chain Management standard dimensions by default. The other 12 dimensions (*inventDimension1* through *inventDimension12*) are also mapped to custom dimensions in Supply Chain Management. The remaining eight dimensions

(*ExtendedDimension1* through *ExtendedDimension8*) are extended dimensions that you can map to external data sources.

- If you don't use Supply Chain Management as one of your data sources, you can freely map the dimensions. The following table shows the full list of available dimensions.

> **Note**
>
> If you use Supply Chain Management, and you change the default dimension mappings between Supply Chain Management and Inventory Visibility, the changed dimension doesn't sync data. Therefore, if your dimension isn't on the default dimension list and you're using an external data source, use *ExtendedDimension1* through *ExtendedDimension8* for the mapping.

Expand table

Dimension type	Base dimension
Product	*ColorId*
Product	*SizeId*
Product	*StyleId*
Product	*ConfigId*
Tracking	*BatchId*
Tracking	*SerialId*
Location	*LocationId*
Location	*SiteId*
Inventory status	*StatusId*
Warehouse specific	*WMSLocationId*
Warehouse specific	*WMSPalletId*
Warehouse specific	*LicensePlateId*
Others	*VersionId*
Inventory (custom)	*InventDimension1* through *InventDimension12*
Extension	*ExtendedDimension1* through *ExtendedDimension8*

Dimension type	Base dimension
System	*Empty*

> **Note**
>
> The dimension types that are listed in the preceding table are for your reference only. You don't have to define them in Inventory Visibility.
>
> The inventory (custom) dimensions might be reserved for Supply Chain Management. In that case, use the extended dimensions instead.

External systems can access Inventory Visibility through its RESTful APIs. For the integration, Inventory Visibility lets you configure the *external data source* and the mapping from the *external dimensions* to the *base dimensions*. Here's an example of a dimension mapping table.

Expand table

External dimension	Base dimension
MyColorId	*ColorId*
MySizeId	*SizeId*
MyStyleId	*StyleId*
MyDimension1	*ExtendedDimension1*
MyDimension2	*ExtendedDimension2*

By configuring a dimension mapping, you can send the external dimensions directly to Inventory Visibility. Inventory Visibility then automatically converts external dimensions to base dimensions.

Add dimension mappings in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

To add a dimension mapping in UI version 2, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Data source settings** tile, select **Manage**.

4. On the **Data source settings** page, select the data source where you want to set up a dimension mapping.

5. In the **Dimension mappings** section, select **New Dimension Mapping** on the toolbar.

6. On the **New Dimension Mapping** page, set the following fields:

    - **Customer dimension** – Enter the name of the source dimension.
    - **Base Dimension** – Select the dimension in Inventory Visibility that you want to map to.

    For example, you've already created a data source that's named *ecommerce*, and it includes a product color dimension. In this case, to do the mapping, first add *ProductColor* to the **Dimension Name** field for the *ecommerce* data source. Then select *ColorId* in the **To Base Dimension** field.

7. On the toolbar, select **Save**.

Add dimension mappings in UI version 1

To add a dimension mapping in UI version 1, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.

2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.

3. On the navigation pane, select **Configuration**.

4. On the **Data source** tab, expand the data source where you want to set up a dimension mapping.

5. In the **Dimension Mappings** section, select **Add** to add a dimension mapping row. Then set the following fields for the new row:

    - **Dimension Name** – Enter the name of the source dimension.
    - **To Base Dimension** – Select the dimension in Inventory Visibility that you want to map to.

6. Select the **Save** link for the new row.

Physical measures

When a data source posts an inventory change to Inventory Visibility, it posts that change by using *physical measures*. Physical measures modify the quantity and reflect the inventory status.

You can define your own physical measures based on your requirements. Queries can be based on the physical measures.

Inventory Visibility provides a list of default physical measures that are mapped to Supply Chain Management (the *fno* data source). These default physical measures are taken from the inventory transaction statuses on the **On-hand list** page in Supply Chain Management (**Inventory Management** > **Inquiries and Report** > **On-hand list**). The following table provides an example of physical measures.

Expand table

Physical measure name	Description
NotSpecified	Not specified
Arrived	Arrived
AvailOrdered	Available ordered
AvailPhysical	Available physical
Deducted	Deducted
OnOrder	OnOrder
Ordered	Ordered
PhysicalInvent	Physical inventory
Picked	Picked
PostedQty	Posted quantity
QuotationIssue	Quotation issue
QuotationReceipt	Quotation receipt
Received	Received
Registered	Registered
ReservOrdered	Ordered reserved
ReservPhysical	Physical reserved
OrderedSum	Ordered in total

If your data source is Supply Chain Management, you don't have to recreate the default physical measures. However, for external data sources, you can create new physical measures as you

require.

Add physical measures in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

To add a physical measure in UI version 2, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Data source settings** tile, select **Manage**.
4. On the **Data source settings** page, select the data source where you want to do the dimension mapping. Then, in the **Physical Measure** section, select **New physical measure** on the toolbar.
5. On the **New physical measure** page, set the following fields for the new physical measure:
    - **Display name** – Enter a name that's unique not only within the new measure's own data source but also across all data sources. The recommended format is *<Data source>.<Physical measure name>* (for example, *ecommerce.returned*). This format ensures that the display name is distinct and unique across all data sources.
    - **Physical measure name** – Enter the name of the physical measure.
6. On the toolbar, select **Save**.

Add physical measures in UI version 1

This section applies when you're using Inventory Visibility UI version 1.

To add a physical measure in UI version 1, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Configuration**.
4. On the **Data source** tab, expand the data source where you want to set up a physical measure.
5. In the **Physical measure** section, select **Add** to add a physical measure row. Then, in the **Measure name** field for the new row, enter the name of the measure. For example, enter *Returned* if you want to record returned quantities in this data source for Inventory Visibility.

6. Select the **Save** link for the new row.

Extended dimensions

If you're using external data sources, you can take advantage of the extensibility that the solution offers by creating class extensions for the `InventOnHandChangeEventDimensionSet` and `InventInventoryDataServiceBatchJobTask` classes.

Synchronize with the database after creating the extensions so that the custom fields are added to the `InventSum` table. You can then refer to the "Dimensions" section earlier in this article, to map your custom dimensions to any of the eight extended dimensions in `BaseDimensions` in Inventory.

> **Note**
>
> For additional details about creating extensions, see Extensibility home page.

Calculated measures

Inventory Visibility lets you query both inventory physical measures and *custom calculated measures*. Calculated measures provide a customized computation formula that consists of a combination of physical measures. This functionality lets you define a set of physical measures that are added, and/or a set of physical measures that are subtracted, to form the customized measurement.

> **Important**
>
> A calculated measure is a composition of physical measures. Its formula can include only physical measures without duplicates, not calculated measures.

The configuration lets you define a set of calculated measure formulas that includes modifiers of addition or subtraction to get the total aggregated output quantity.

Add calculated measures in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

To add a calculated measure in UI version 2, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Data source settings** tile, select **Manage**.
4. On the **Data source settings** page, select the data source where you want to do the dimension mapping. Then, in the **Calculated measures** section, select **New calculated measure** on the toolbar.
5. On the **New calculated measure** page, set the following fields for the new calculated measure:
    - **Calculated measure display name** – Enter a name that's unique not only within the new measure's own data source but also across all data sources. The recommended format is *<Data source>.<Calculated measure name>* (for example, *iv.TotalAvailable*). This format ensures that the display name is distinct and unique across all data sources.
    - **Calculated measure name** – Enter the name of the calculated measure.
6. On the toolbar, select **Save & close**. The page is refreshed and now includes the **Calculated measure details** section.
7. In the **Calculated measure details** section, select **New calculated measure details V2** on the toolbar to add a modifier row for the calculated measure.
8. On the **New calculated measure detail** page, set the following fields for the new modifier:
    - **Calculated measure formula description** – Enter a detailed explanation of the significance and purpose of the formula in the calculated measure.
    - **+/- to result** – Select the modifier type (*Addition* or *Subtraction*).
    - **Physical measure name reference** – Select the name of the measure (from the selected data source) that provides the value for the modifier.
9. On the toolbar, select **Save & close**.
10. Repeat steps 7 through 9 until you've added all the modifiers that are required to complete the formula for your calculated measure.
11. On the toolbar, select **Save**.

Add calculated measures in UI version 1

This section applies when you're using Inventory Visibility UI version 1.

To add a calculated measure in UI version 1, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Configuration**.
4. On the **Calculated measure** tab, select **New Calculate Measure** to add a calculated measure.
5. Set the following fields for the new calculated measure:
    - **New calculated measure name** – Enter the name of the calculated measure.
    - **Data source** – Select the data source to include the new calculated measure in. The querying system is a data source.
6. A new, blank modifier line is automatically included for the new calculated measure. Set the following fields for it:
    - **Modifier** – Select the modifier type (*Addition* or *Subtraction*).
    - **Data source** – Select the data source where the measure that provides the modifier value is found.
    - **Measure** – Select the name of the measure (from the selected data source) that provides the value for the modifier.
7. Select **Save measure** to save your settings so far.
8. If you must add another modifier line, select **Add** to add a row, and then set the fields as you did for the first row. When you finish, select the **Save** link for the row.
9. Repeat the previous step until you've added all the required modifiers and completed the formula for your calculated measure.
10. Select **Save**.

Work with calculated measures in Inventory Visibility

For example, a fashion company operates across three data sources:

- `pos` – Corresponds to the store channel.

- `fno` – Corresponds to Supply Chain Management.
- `ecommerce` – Corresponds to your web channel.

Without calculated measures, when you query for product D0002 (Cabinet) under site 1, warehouse 11, and a `ColorID` dimension value of `Red`, you might get the following query result, which shows inventory quantities under each preconfigured physical measure. However, you don't have visibility into the total available for reservation quantities across your data sources.

JSON


```
[
    {
        "productId": "D0002",
        "dimensions": {
            "SiteId": "1",
            "LocationId": "11",
            "ColorId": "Red"
        },
        "quantities": {
            "pos": {
                "inbound": 80.0,
                "outbound": 20.0
            },
            "fno": {
                "availphysical": 100.0,
                "orderedintotal": 50.0,
                "orderedreserved": 10.0
            },
            "ecommerce": {
                "received": 90.0,
                "scheduled": 30.0,
                "issued": 60.0,
                "reserved": 40.0
            }
        }
    }
]
```


You then configure a calculated measure that's named `MyCustomAvailableforReservation`, as shown in the following table. This calculated measure is consumed by the consumption system.

Expand table

Consumption system	Calculated measure	Data source	Physical measure	Calculation type
`CrossChannel`	`MyCustomAvailableforReservation`	`fno`	`availphysical`	`Addition`

Consumption system	Calculated measure	Data source	Physical measure	Calculation type
`CrossChannel`	`MyCustomAvailableforReservation`	`fno`	`orderedintotal`	`Addition`
`CrossChannel`	`MyCustomAvailableforReservation`	`fno`	`orderedreserved`	`Subtraction`
`CrossChannel`	`MyCustomAvailableforReservation`	`pos`	`inbound`	`Addition`
`CrossChannel`	`MyCustomAvailableforReservation`	`pos`	`outbound`	`Subtraction`
`CrossChannel`	`MyCustomAvailableforReservation`	`ecommerce`	`received`	`Addition`
`CrossChannel`	`MyCustomAvailableforReservation`	`ecommerce`	`scheduled`	`Addition`
`CrossChannel`	`MyCustomAvailableforReservation`	`ecommerce`	`issued`	`Subtraction`
`CrossChannel`	`MyCustomAvailableforReservation`	`ecommerce`	`reserved`	`Subtraction`

When you use this computation formula, the new query result includes the customized measurement.

JSON


```
[
    {
        "productId": "D0002",
        "dimensions": {
            "SiteId": "1",
            "LocationId": "11",
            "ColorId": "Red"
        },
        "quantities": {
            "pos": {
                "inbound": 80.0,
                "outbound": 20.0
            },
            "fno": {
                "availphysical": 100.0,
                "orderedintotal": 50.0,
                "orderedreserved": 10.0
            },
            "ecommerce": {
                "received": 90.0,
                "scheduled": 30.0,
                "issued": 60.0,
                "reserved": 40.0
            },
            "CrossChannel": {
                "MyCustomAvailableforReservation": 220.0
            }
        }
```

```
    }
]
```

The `MyCustomAvailableforReservation` output, based on the calculation setting in the custom measurements, is 100 + 50 – 10 + 80 – 20 + 90 + 30 – 60 – 40 = 220.

The default data source configuration

This section provides details about the default data source configuration that's set up when you install Inventory Visibility. You can modify this configuration as needed.

> **ⓘ Important**
>
> The default configuration has evolved through various version iterations. It's possible that your sandbox environment was initially set up with an outdated default configuration, while your production environment was initialized with the latest version of the default configuration. If you've customized your third-party system based on an outdated default configuration, it might encounter issues when your production environment goes live, especially if you haven't reviewed and adjusted the configuration. To prevent this scenario, thoroughly review and update your draft and runtime configurations before transitioning your production environment.

Data source configuration

Configuration of the iv data source

This section describes how the *iv* data source is configured.

Physical measures configured for the "iv" data source

The following physical measures are configured for the *iv* data source:

- *Ordered*
- *Softreserved*

AvailableToReserve calculated measure

The *AvailableToReserve* calculated measure is configured for the *iv* data source as shown in the following table.

Expand table

Calculation type	Data source	Physical measure
Addition	*fno*	*PhysicalInvent*
Addition	*fno*	*Ordered*
Addition	*fno*	*Arrived*
Addition	*pos*	*Inbound*
Addition	*iv*	*Ordered*
Subtraction	*fno*	*ReservPhysical*
Subtraction	*iv*	*Softreserved*
Subtraction	*pos*	*Outbound*
Subtraction	*fno*	*Softreserved*

TotalAvailable calculated measure

The *TotalAvailable* calculated measure is configured for the *iv* data source as shown in the following table.

Expand table

Calculation type	Data source	Physical measure
Addition	*fno*	*AvailOrdered*
Subtraction	*iv*	*Softreserved*
Subtraction	*@iv*	*@allocated*

TotalOnHand calculated measure

The *TotalOnHand* calculated measure is configured for the *iv* data source as shown in the following table.

Expand table

Calculation type	Data source	Physical measure
Addition	*fno*	*PhysicalInvent*

Configuration of the "fno" data source

This section describes how the *fno* data source is configured.

Dimension mappings for the "fno" data source

The dimension mappings that are listed in the following table are configured for the *fno* data source.

Expand table

External dimension	Base dimension
InventBatchId	*BatchId*
InventColorId	*ColorId*
InventLocationId	*LocationId*
InventSerialId	*SerialId*
InventSiteId	*SiteId*
InventSizeId	*SizeId*
InventStatusId	*StatusId*
InventStyleId	*StyleId*
LicensePlateId	*LicensePlateId*
WMSLocationId	*WMSLocationId*
WMSPalletId	*WMSPalletId*
ConfigId	*ConfigId*
InventVersionId	*VersionId*
InventDimension1	*CustomDimension1*
InventDimension2	*CustomDimension2*
InventDimension3	*CustomDimension3*
InventDimension4	*CustomDimension4*
InventDimension5	*CustomDimension5*
InventDimension6	*CustomDimension6*

External dimension	Base dimension
InventDimension7	*CustomDimension7*
InventDimension8	*CustomDimension8*
InventDimension9	*CustomDimension9*
InventDimension10	*CustomDimension10*
InventDimension11	*CustomDimension11*
InventDimension12	*CustomDimension12*

Physical measures configured for the "fno" data source

The following physical measures are configured for the *fno* data source:

- *Arrived*
- *PhysicalInvent*
- *ReservPhysical*
- *onorder*
- *notspecified*
- *availordered*
- *availphysical*
- *picked*
- *postedqty*
- *quotationreceipt*
- *received*
- *ordered*
- *ReservOrdered*
- *OrderedSum*
- *SoftReserved*

Configuration of the "pos" data source

This section describes how the data source *pos* is configured.

Physical measures for the "pos" data source

The following physical measures are configured for the *pos* data source:

- *Inbound*

- *Outbound*

AvailQuantity calculated measure

The *AvailQuantity* calculated measure is configured for the *pos* data source as shown in the following table.

Expand table

Calculation type	Data source	Physical measure
Addition	*fno*	*AvailPhysical*
Addition	*pos*	*Inbound*
Subtraction	*pos*	*Outbound*

Configuration of the "iom" data source

The following physical measures are configured for the *iom* (Intelligent Order Management) data source:

- *OnOrder*
- *OnHand*

Configuration of the "erp" data source

The following physical measures are configured for the *erp* (enterprise resource planning) data source:

- *Unrestricted*
- *QualityInspection*

Last updated on 06/22/2026

Inventory Visibility support for WMS items

This article describes Inventory Visibility support for items that are enabled for warehouse management processes (WMS). The feature that adds this capability to Inventory Visibility is named *Advanced WMS*.

WMS items

A WMS item is an item that's enabled for WMS and processed at a warehouse that's also WMS enabled.

For more information about WMS and the **Warehouse management** module in Microsoft Dynamics 365 Supply Chain Management, see Warehouse management overview.

Scope of the feature

The Advanced WMS feature for Inventory Visibility focuses on on-hand quantity calculations that are based on on-hand queries. The feature isn't intended to let you use Inventory Visibility to manage warehouse processing activities in general. Inventory Visibility doesn't expose warehouse-specific concepts to its users. Here are some examples of these concepts:

- Reservation hierarchy
- Whether an item or warehouse is WMS enabled
- Unit sequence group ID
- Warehouse-specific processes, such as shipments, loads, waves, and work

Inventory Visibility infers this information, based on the data that's sent from Supply Chain Management. The WMS-specific data (in other words, data from the `WHSInventReserve` table) isn't visible to users.

When you use the Advanced WMS feature for Inventory Visibility, all query results will be identical to the results from queries that are made directly in Supply Chain Management. However, they won't be identical to the results from queries that are made by using the Warehouse Management mobile app, because the mobile app uses slightly different calculation logic.

When to use the feature

We recommend that you use the WMS feature for Inventory Visibility in scenarios where all the following conditions are met:

- You're syncing Supply Chain Management data with Inventory Visibility.
- You're using WMS in Supply Chain Management.
- Users make reservations for WMS items at levels below the warehouse level (for example, at the license plate level because you're processing warehouse work).

In other scenarios, on-hand query results will be the same, regardless of whether the Advanced WMS feature for Inventory Visibility is enabled. Additionally, performance will be better if you don't enable the feature in these scenarios, because there are fewer calculations and less overhead.

Enable the WMS feature for Inventory Visibility

To enable the WMS feature for Inventory Visibility, follow these steps:

1. Sign in to Supply Chain Management as an admin.

2. Open the Feature management workspace, and enable the following features in this order:
   a. *Inventory Visibility integration* (As of Supply Chain Management version 10.0.45, this feature is turned on by default.)
   b. *Enable warehouse items in Inventory Visibility* (As of Supply Chain Management version 10.0.45, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.)

3. Go to **Inventory Management** > **Setup** > **Inventory Visibility integration parameters**.

4. On the **Enable WMS items** tab, set the **Enable WMS items sync** option to *Yes*. Then save the setup.

5. Select **Sync WMS data** to sync the WMS data to Inventory Visibility. This step is required because, when you first enable the synchronization of WMS items, only subsequent changes to on-hand quantities of WMS items will be synced to Inventory Visibility. When you select **Sync WMS data**, the system syncs all available WMS data to Inventory Visibility. This synchronization might take a long time, depending on your data volume. Therefore, we recommend that you complete this step during off-peak hours.

If you're using the new user interface for the Inventory Visibility app (UI version 2), the WMS feature is turned on by default. Therefore, you're ready to query on-hand quantities of WMS items. If you're using the old user interface (UI version 1), you must turn on the *Advanced*

warehouse inventory feature in the Inventory Visibility app in Power Apps before you can query for WMS items. Follow these steps to turn on the feature.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Configuration**.
4. On the **Feature management & settings** tab, turn on the *Advanced warehouse inventory* feature.

Choose whether to truncate unused dimensions

This section applies when you're using Inventory Visibility UI version 2. If you want to use the feature that's described here, but you're still using UI version 1, consider updating to UI version 2.

When you use the WMS feature, you can choose to truncate dimensions that you aren't using. The system can then accept queries that exclude specific dimensions. This functionality helps increase query performance and reduce the amount of storage that's required for the feature. Unused dimensions are truncated by default when you sync WMS data from Supply Chain Management, unless one or more of the following dimensions are included in your index hierarchy:

- Batch ID
- Serial ID
- License plate ID
- WMS Location ID

If you're using any of the preceding dimensions, but you haven't added them to your index hierarchy, you should disable the **Truncate unused dimensions** option by following these steps.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Advanced warehouse inventory** tile, select **Manage**.
4. Set the **Truncate unused dimensions** option to *No*.
5. On the toolbar, select **Save & close**.
6. On the navigation pane, select **Admin settings**.
7. On the **Update configuration** tile, select **Manage**.

8. Review your modifications in the dialog box.

> **Important**
>
> Be sure to verify all the important modifications that are about to be made to your data sources, physical measures, and dimension mappings.

9. Select **Confirm Update** to apply your configuration changes.

> **Important**
>
> If you turn off the **Truncate unused dimensions** option after you've already synced some WMS item data from Supply Chain Management, you must sync WMS data to ensure that all the previously truncated dimension data is re-synced to Inventory Visibility.

Query on-hand quantities of WMS items

To query for WMS items, you use the same application programming interface (API) and message syntax that you use for non-WMS items. You don't have to specify whether an item is a WMS item or a non-WMS item. Inventory Visibility automatically distinguishes items, based on the data that's stored.

The results from queries for WMS items are essentially the same as the results for non-WMS items. The only difference is that the following physical measures from the `fno` data source are calculated based on WMS logic in Supply Chain Management:

- `AvailOrdered`
- `AvailPhysical`
- `ReservOrdered`
- `ReservPhysical`

All other physical measures are calculated just as they are when the WMS feature for Inventory Visibility is disabled.

You can find detailed information about how on-hand calculations for WMS items work in Reservations in Warehouse management.

On-hand list view and data entity for WMS items

The **Preload the Inventory Visibility Summary** page provides a view for the *On-hand Index Query Preload Results* entity. Unlike the *Inventory summary* entity, the *On-hand Index Query Preload Results* entity provides an on-hand inventory list for products together with selected dimensions. Inventory Visibility syncs the preloaded summary data every 15 minutes.

If you use Inventory Visibility with WMS items and want to view the on-hand list for WMS items, we recommend that you enable the *Preload the Inventory Visibility Summary* feature. (Learn more in Preload a streamlined on-hand query.) A corresponding data entity in Dataverse stores your preloaded query result, which is updated every 15 minutes. The name of the data entity is *Onhand Index Query Preload Result*.

> **Important**
>
> The Dataverse entity is read-only. You can view and export the data in the Inventory Visibility entities, but **don't modify it**.

Changes to WMS item quantities that are stored in the Supply Chain Management data source (`fno`) are prohibited. This behavior matches the behavior of other features of Inventory Visibility. This restriction is enforced to help prevent conflicts.

WMS item compatibility for other functions in Inventory Visibility

Soft reservations and inventory allocation of WMS items are supported. You can include WMS-related physical measures in soft reservation and allocation calculations.

Calculate available-to-promise quantities

The solution fully supports available to promise (ATP) for WMS items. You can define ATP calculations without worrying about WMS-specific details.

Last updated on 07/27/2026

Inventory Visibility inventory summary

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Inventory summary** page is a customized view for the *Inventory OnHand Sum* entity. It provides an inventory summary for products together with all dimensions. Inventory summary data is periodically synced from Inventory Visibility.

The **Inventory summary** feature isn't compatible with items enabled for warehouse management processes (WMS).

> **Note**
>
> The inventory summary feature tracks only on-hand inventory changes that occur after you turn on the feature. Data for products that haven't changed since you turned on the feature isn't synced from the Inventory Visibility cache to the Microsoft Dataverse environment.
>
> If you change the settings for a calculated measure, data on the **Inventory summary** page isn't automatically updated until the related product data is modified.
>
> If the **Inventory summary** page doesn't show all the on-hand information that you expect, open Dynamics 365 Supply Chain Management, go to **Inventory Management** > **Periodic tasks** > **Inventory Visibility integration**, disable the batch job, and then reenable it to do the initial push. All data is synced to the *Inventory OnHand Sum* entity immediately after the initial push. If you want to use the inventory summary feature, turn it on before you create any on-hand changes and enable the **Inventory Visibility integration** batch job.

Turn on the inventory summary feature in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

To enable the **Inventory summary** page and set the synchronization frequency in UI version 2, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Inventory summary** tile, select **Manage**.
4. Set the following fields:
    - **Enable feature** – Set this option to *True*.
    - **Synchronize frequency** – Set the frequency (in minutes) that inventory summary data should be synced at. The minimum frequency is five minutes.
5. On the toolbar, select **Save & close**.
6. On the navigation pane, select **Admin Settings**.
7. On the **Update configuration** tile, select **Manage**.
8. Review your modifications in the dialog box.

    > ⓘ **Important**
    >
    > Be sure to verify all the important modifications that are about to be made to your data sources, physical measures, and dimension mappings.

9. Select **Confirm Update** to apply your configuration changes.

Turn on the inventory summary feature in UI version 1

To enable the **Inventory summary** page and set the synchronization frequency in UI version 1, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Configuration**.
4. On the **Feature management & settings** tab, turn on the option for the *Inventory summary* feature.
5. After the feature is enabled, the **Service Configuration** section becomes available. It includes a row that you can use to configure the *OnHandMostSpecificBackgroundService*

feature. To set the frequency that inventory summary data is synced at, use the **Up** and **Down** buttons in the **Value** field to change the interval between synchronizations. The minimum interval is five minutes. When you finish, select **Save**.

6. Select **Update configuration** to save and apply your new settings.

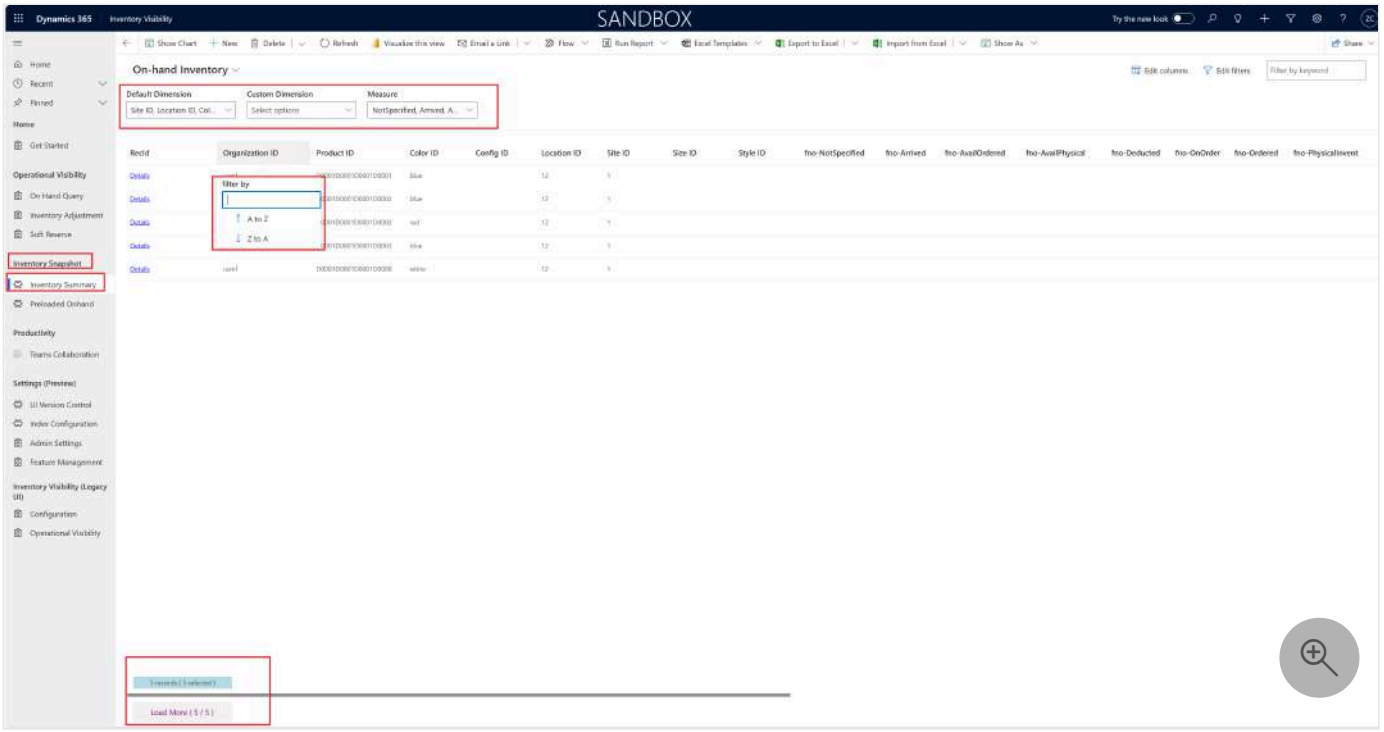
Filter and browse the inventory summaries

Follow these steps to open the **Inventory summary** page, where you can filter and browse your inventory summaries.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Inventory Visibility**.
3. On the navigation pane, select **Inventory summary**.

    - On the **Inventory summary** page, there are three fields above the grid: **Default dimension**, **Custom dimension**, and **Measure**. Use these fields to control which columns are visible. You can also select any column header in the grid to filter or sort the current result by that column.
    - Use the **Edit filters** button in the upper right of the page to create a personal view that shows the rows that are important to you. The filter options let you create a wide range of views, from simple to complex. They also let you add grouped and nested conditions to the filters. For more information about how to use the advanced filter, see Edit or create personal views using advanced grid filters.
    - The lower left of the page shows information such as *2 records (2 selected)* or *50 records*. This information refers to the records that are currently loaded from the **Edit filters** result. The text *2 selected* refers to the number of records that have been selected by using the column header filters for the loaded records. There's a **Load More** button that you can use to load more records from Dataverse. By default, 50 records are loaded. When you select **Load More**, the next 1,000 available records are loaded into the view. The numbers on the **Load More** button indicate the number of currently loaded records and the total number of records that the **Edit filters** result found. For example, the text *(5/5)* indicates that all five of the records from the **Edit filters** result are currently loaded into the view.

The following illustration highlights the dimension, filtering, result count, and "load more" fields on the **Inventory summary** page.



Last updated on 06/22/2026

Preload a streamlined on-hand query

06/17/2025

Microsoft Dynamics 365 Supply Chain Management stores lots of information about your current on-hand inventory and makes it available for a wide range of purposes. However, many everyday operations and third-party integrations require just a small subset of the stored details. If the system is queried for all of them, the result can be large data sets that take time to assemble and transfer. Therefore, the Inventory Visibility service can periodically fetch and store a streamlined set of on-hand inventory data to make that optimized information continuously available. The stored on-hand inventory details are filtered based on configurable business criteria, to ensure that only the most relevant information is included. Because the filtered on-hand inventory lists are stored locally in the Inventory Visibility service and regularly updated, they support quick access, on-demand data exports, and streamlined integration with external systems.

Preloaded on-hand queries provide the following benefits:

- A cleaner view that stores an inventory summary that includes only the dimensions that are relevant to your daily business
- An inventory summary that's compatible with items that are enabled for warehouse management processes (WMS)

> **Important**
>
> We recommend that you use either the *Preloaded onhand* feature or the *Inventory summary* feature, but not both. If you enable both features, performance will be affected.

The **Onhand index query preload results** page provides a view for the *On-hand Index Query Preload Results* entity. Unlike the *Inventory summary* entity, the *On-hand Index Query Preload Results* entity provides an on-hand inventory list for products together with selected dimensions. Inventory Visibility syncs the preloaded summary data every 15 minutes.

> **Note**
>
> Like the *Inventory summary* feature, the *Preloaded onhand* feature tracks only on-hand inventory changes that occur after you turn on the feature. Data for products that haven't changed since you turned on the feature isn't synced from the inventory service cache to the Dataverse environment.
>
> If you change the settings for a calculated measure, data on the **Onhand index query preload results** page isn't automatically updated until the related product data is

modified.

If the **Onhand index query preload results** page doesn't show all the on-hand information that you expect, go to **Inventory management** > **Periodic tasks** > **Inventory Visibility integration**, disable the batch job, and then reenable it to do the initial push. All data will be synced to the *On-hand Index Query Preload Results* entity within 15 minutes. If you want to use this feature, we recommend that you turn it on before you create any on-hand changes and enable the **Inventory Visibility integration** batch job.

Turn on and configure preloaded on-hand queries in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

To turn on and configure preloaded on-hand queries in UI version 2, follow these steps.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Preloaded onhand** tile, select **Manage**.
4. Set the **Enable feature** option to *True*.
5. In the **Preload group by dimensions** section, add a row for each dimension that you want to use for preloaded on-hand queries. The dimensions that you add here are used to group the preloaded on-hand data.
    - To add a row, select **New onhand index query preload configuration** on the toolbar. Set the **Group by value** field to the dimension that you want to group by, and then assign an order. Preloaded on-hand queries use only dimensions from set number *0* as group-by values. Other sets are ignored. When you've finished, select **Save & Close** on the toolbar.
    - To modify a row, select the **Preload set number** link to open the row settings, edit the fields as you require, and then select **Save & Close** on the toolbar.
    - To remove a row, select the **Preload set number** link to open the row settings, and then select **Delete** on the toolbar.
6. If you changed the group-by configuration, select **Admin settings** on the navigation pane, and then, on the **Delete preloaded onhand data** tile, select **Manage**. This step cleans up the database and makes it ready to accept your new group-by settings.

7. On the navigation pane, select **Admin settings**. Then, on the **Update configuration** tile, select **Manage** to update the configuration and activate your changes.

Turn on and configure preloaded on-hand queries in UI version 1

This section applies when you're using Inventory Visibility UI version 1.

To turn on and configure preloaded on-hand queries in UI version 1, follow these steps.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Configuration**.
4. On the **Feature Management & Settings** tab, if the *Preloaded onhand query results* feature is already enabled, we recommend that you use the option to turn it off for now, because the cleanup process might take a long time. You'll turn the feature on again later in this procedure.
5. On the **Preload Setting** tab, in the **Step 1: Clean up preload storage** section, select **Clean** to clean up the database and make it ready to accept your new group-by settings.
6. In the **Step 2: Set up group by values** section, in the **Group result by** field, enter a comma-separated list of field names to group your query results by. After you have data in the preload storage database, you won't be able to change this setting until you clean the database as described in the previous step.
7. On the **Feature Management & Settings** tab, use the option to turn on the *Preload on-hand query results* feature.
8. Select **Update configuration** in the upper-right corner of the page to commit your changes.

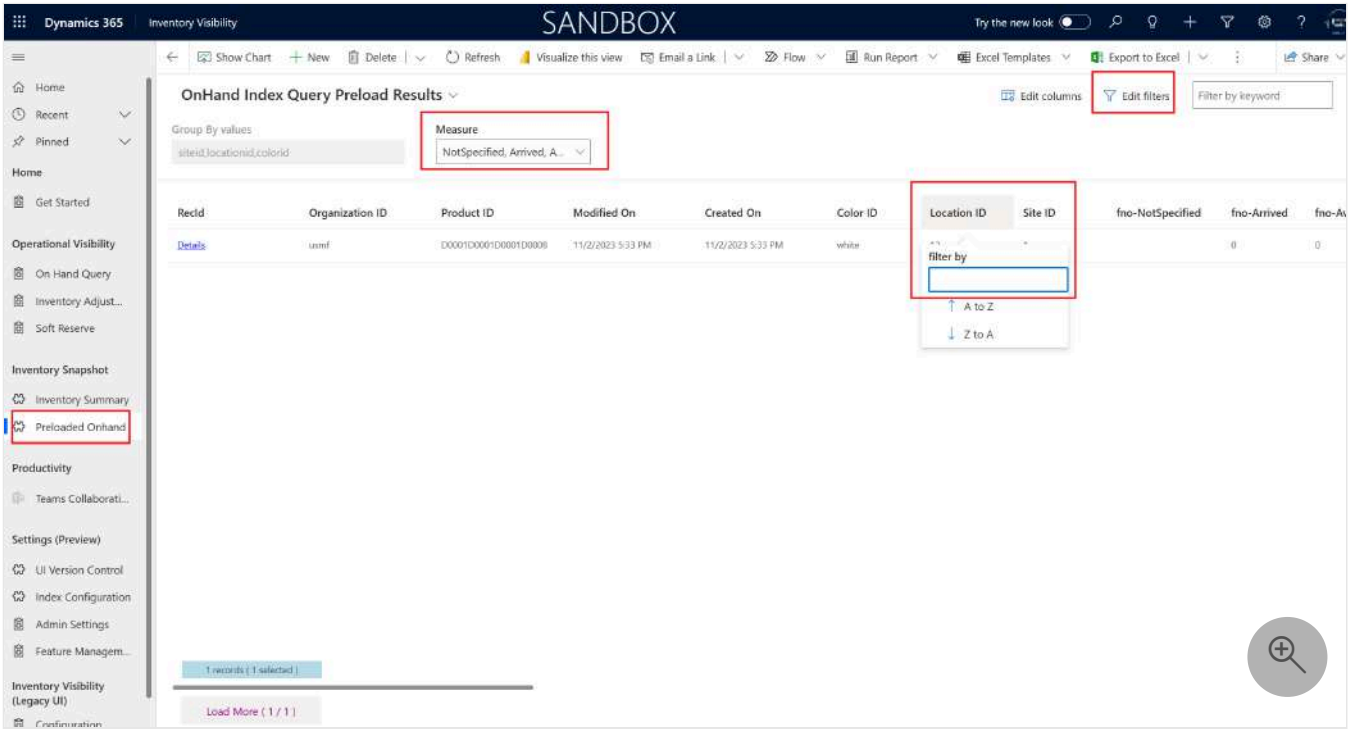
Filter and browse the preloaded query

To filter and browser the preloaded query, follow these steps.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Inventory visibility**.
3. On the navigation pane, select **Preloaded onhand**.
4. On the **Onhand index query preload results** page, you can filter and browse your preloaded summary.

- Use the **Edit filters** button to create a personal view that shows the rows that are important to you. These filter options let you create a wide range of views, from simple to complex. They also let you add grouped and nested conditions to the filters. For more information about how to use the advanced filter, see Edit or create personal views using advanced grid filters.
- Because you've predefined the dimensions that are used to load summary data, the **Onhand index query preload results** page shows dimension-related columns. *The dimensions aren't customizable. The system supports only site and location dimensions for preloaded on-hand lists.* The page provides filters that are similar to the filters on the **Inventory summary** page. However, the dimensions are already selected.
- The bottom of the page shows information such as *50 records (29 selected)* or *50 records*. This information refers to the records that are currently loaded from the **Edit filters** result. The text *29 selected* refers to the number of records that have been selected by using the column header filters for the loaded records. There's a **Load More** button that you can use to load more records from Dataverse. By default, 50 records are loaded. When you select **Load More**, the next 1,000 available records are loaded into the view. The numbers on the **Load More** button indicate the number of currently loaded records and the total number of records that the **Edit filters** result found.

The following illustration highlights the filtering fields that are available on the **Onhand index query preload results** page.



Run on-hand consistency checks while Inventory Visibility is enabled

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft Dynamics 365 Supply Chain Management provides an on-hand consistency check tool that you can use to recalculate inventory on-hand list data based on inventory transactions for selected items or all items. The tool analyzes the inventory transactions (`InventTrans`) table and then reconstructs the `InventSum` and `WHSInventReserve` tables.

> **Note**
>
> This article describes only the additional preparation that's required when the *Inventory Visibility integration* batch job is enabled. For general guidance about when and how to run an on-hand consistency check, see Run an on-hand consistency check.

If you run an on-hand consistency check on *all items* while the *Inventory Visibility integration* batch job is enabled, the following conditions apply:

- The tool usually takes longer to finish running than it does when the *Inventory Visibility integration* batch job isn't enabled.
- The tool uses API calls to sync the changes in the `InventSum` and `WHSInventReserve` tables to the Inventory Visibility service. Therefore, if the *Inventory Visibility integration* batch job is enabled, the consistency check might fail if the external calls result in unexpected errors.

For these reasons, disable the *Inventory Visibility integration* batch job before you run an on-hand consistency check on all items.

Prepare for and run an on-hand consistency check on all items

Before you run an on-hand consistency check on all items, follow these steps:

1. Go to **Inventory Management** > **Periodic Tasks** > **Inventory Visibility integration**.
2. Check the **Status** value to determine whether the job is enabled.
    - If the **Status** value is *Disabled*, you can skip the rest of this procedure and just run the on-hand consistency check.
    - If the **Status** value is *Enabled*, disable the job by selecting **Disable** on the Action Pane.
3. Run the on-hand consistency check.
4. When the check is completed, go to **Inventory Management** > **Setup** > **Inventory Visibility integration parameters**.
5. On the **General** tab, set the **Resync before initial push** option to *Yes*.
6. On the Action Pane, select **Save**.
7. Go to **Inventory Management** > **Periodic Tasks** > **Inventory Visibility integration**.
8. On the Action Pane, select **Enable** to reenable the batch job.

The initial push with the `Resync` action first cleans up legacy Supply Chain Management data in Inventory Visibility service. It then resyncs the `InventSum` and `WHSInventReserve` tables to the Inventory Visibility Service.

> **Important**
>
> The Inventory Visibility service resumes delivering correct query results after the reenabled *Inventory Visibility integration* batch job completes its initial push work.

Prepare for and run an on-hand consistency check on selected items

If you want to run an on-hand consistency check on only a few selected items, you don't need to complete the extra steps that are described in the previous section. Instead, just run the check.

Last updated on 06/04/2026

Inventory Visibility adjustment offset

08/13/2025

This article describes how to offset inventory adjustments that were updated through Inventory Visibility and then also updated in Microsoft Dynamics 365 Supply Chain Management. To prevent inventory quantities from being updated twice, you must offset the previously adjusted inventory from Supply Chain Management to Inventory Visibility.

For example, you have an external sales system, such as a point of sale (POS) or e-commerce system, that makes external sales transactions. You have a data source that's named *POS* in Inventory Visibility. You also have a physical measure, *Sold*, that captures externally sold quantities. Your external system registers a sale of 10 units of item ID A0001, including all relevant dimensions. When you re-create the transaction in Supply Chain Management, if the re-created transaction is a sales order, you must set up sales order offset mappings. If you use an inventory adjustment journal to post inventory changes that are registered through your POS systems, you must set up inventory adjustment journal offset mappings.

When you sync or import external sales orders and inventory adjustment journals from Inventory Visibility into Supply Chain Management, notice that the relevant tables include two new columns: `OFFSETDATASOURCE` and `OFFSETPHYSICALMEASURE`. Be sure to fill in these two columns with the data source and physical measure values that you set up in your inventory offset configuration (as described later in this article). When sales orders and inventory adjustment journals are created in Supply Chain Management, you can view the data source and physical measure mappings on the sales order or journal lines.

> **Note**
>
> Supply Chain Management transaction lines that must be offset are added to the regular data synchronization batch job between Supply Chain Management and Inventory Visibility. Therefore, you should expect a delay of about one minute before you can view the data being offset in Inventory Visibility.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.39 or later.
- The feature that's named *Inventory Visibility integration with inventory adjustment offset* must be turned on in Feature management. As of Supply Chain Management version

10.0.45, this feature is mandatory and can't be turned off.

- Inventory Visibility must be installed, and the basic integration between Supply Chain Management and Inventory Visibility must be set up, as described in Install and set up Inventory Visibility.
- You must be using the **/api/environment/{environmentId}/onhand** and/or **/api/environment/{environmentId}/onhand/bulk** API to make inventory adjustments and post on-hand change events to Inventory Visibility. For information about these two APIs, see Create one on-hand change event and Create multiple change events.

Enable and set up the feature

Follow these steps to enable and set up the Inventory Visibility adjustment offset feature.

1. Go to **Inventory management** > **Inventory Visibility** > **Inventory Visibility integration parameters**.

2. The **Inventory adjustment offset** tab includes two FastTabs: **Sales order** and **Inventory adjustment journal**. Each FastTab represents a type of transaction that you can set up automatic adjustment offsets for. On the appropriate FastTab, set the following fields:

    - **Inventory Visibility offset data source** – Specify the data source that you use Inventory Visibility to post inventory adjustments to (for example, *POS*).

    - **Inventory Visibility offset physical measure** – Specify the physical measure that you use to record the adjustment quantity changes in Inventory Visibility (for example, *Sold*).

    - **Offset trigger** – Select the Supply Chain Management record status that should trigger the offset. For sales orders, you can select *Create* or *Invoice*. For inventory adjustment journals, you can select *Create* or *Post*.
        - **Example 1:** You have a POS system that registers external sales order transactions by using the *POS* data source and the *Sold* physical measure. Each transaction represents inventory that was consumed and paid for in the store. Therefore, you trigger the offset when the sales order reaches *Invoiced* status in Supply Chain Management.
        - **Example 2:** You have an e-commerce website that posts inventory adjustments to Inventory Visibility by using the *Ecommerce* data source and the *On order* physical measure. When you create a new sales order in Supply Chain Management, you trigger the offset at the *Created* status.

    - **Site**, **Warehouse**, **Inventory status**, **Location**, **License plate**, **Batch number**, and **Serial number** – Each of these checkboxes represents a storage or tracking

dimension. Select the checkbox for each dimension that the system should match when it offsets back to Inventory Visibility. Clear the checkbox for each dimension that should be *excluded* when the system matches records. For product and inventory dimensions that aren't listed here (such as color, size, and style), the system exactly matches the values that are specified on the sales order or journal line when it offsets back to Inventory Visibility. *Make sure that only the relevant storage and tracking dimensions are selected*.

- **Example 1:** An external system makes inventory adjustments or on-hand changes through Inventory Visibility by specifying the item ID, color, size, site, and warehouse. Later, when you create a matching sales order in Supply Chain Management, you enter matching item and inventory dimensions, but *you also add a batch number*, which isn't known to the external system. In this case, you should clear the **Batch number** and **Serial number** checkboxes, because batch number and serial number values aren't submitted by the external system and shouldn't be matched when the offset is done.
- **Example 2:** Your process allows site or warehouse information to be changed when an inventory adjustment transaction is re-created in Supply Chain Management. An external system adjusts inventory for item A0001 in site 1 without specifying a warehouse. (In e-commerce, the fulfillment warehouse often isn't known when an order is first submitted.) The fulfillment warehouse (warehouse 13) is specified later in Supply Chain Management. To enable this type of offset to work correctly, you should clear the **Warehouse** checkbox for the appropriate line.

3. If you must set up more offsets for sales order or inventory adjustment journals, select **Add line** on the toolbar of the appropriate FastTab, and repeat the previous steps for each new line.

Inventory Visibility reservations

This article describes a typical use case for soft reservations and explains how to set them up in Inventory Visibility. It includes information about how to create soft reservations, offset them upon physical consumption, and adjust or unreserve specified inventory quantities.

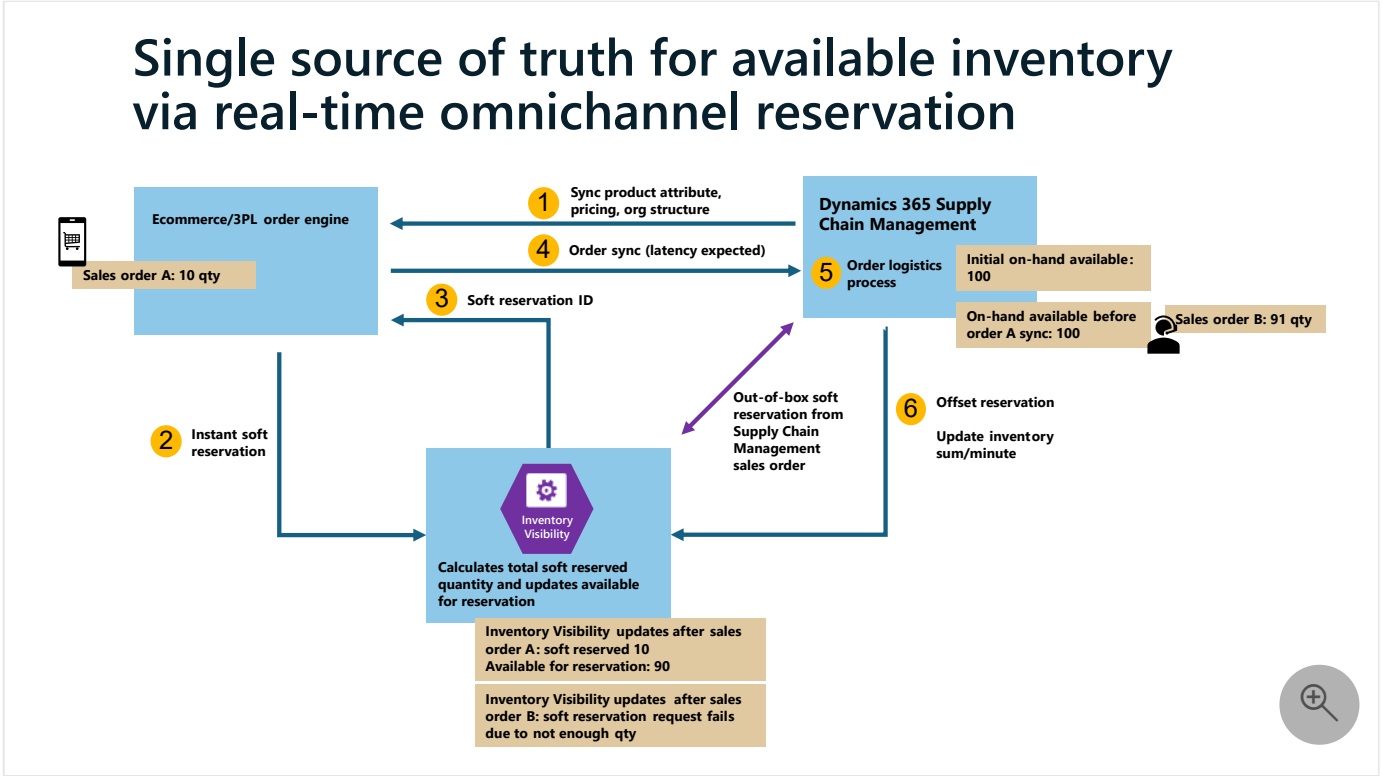
In Microsoft Dynamics 365 Supply Chain Management version 10.0.33 or later, you can make soft reservations from sales orders. When you soft reserve from sales orders, the reservation requests are posted to Inventory Visibility, and the available-for-reservation quantities are validated in Inventory Visibility.

> **Important**
>
> The default soft reservation configuration has evolved through various version iterations. Therefore, it's possible that your sandbox environment was initially set up with an outdated default configuration, whereas your production environment was initialized with the latest version of the default configuration. If you've customized your third-party system based on an outdated default configuration, it might encounter issues when your production environment goes live, especially if you haven't reviewed and adjusted the configuration. To help prevent this scenario, thoroughly review and update your draft and runtime configurations before you transition your production environment.

Sample use case for soft reservation

Soft reservations help organizations achieve a single source of truth for available inventory, especially during the order fulfillment process. This functionality is useful for organizations where the following conditions exist:

- The organization has at least two different systems that are directly taking outbound orders.
- The organization is very strict and wants to prevent double-booking of product inventory, which can happen if multiple systems are able to overbook the last piece of stock. This situation is prevented when all order systems can make instant soft reservation API calls to Inventory Visibility, which provides a single source of truth for inventory availability.



The previous illustration shows how soft reservation works and highlights the following operations:

- Your initial inventory level is synced to Inventory Visibility from Microsoft Dynamics 365 Supply Chain Management.
- You have two order-taking systems running in parallel (an e-commerce system and Supply Chain Management). In an actual case, you could have more systems.
- Soft reservations are posted from each of your order channels or systems to Inventory Visibility. Inventory Visibility validates inventory availability and tries to make a soft reservation. If soft reservation succeeds, Inventory Visibility adds to the soft reserved quantity, deducts from the available for reservation (AFR) quantity, and responds with a soft reservation ID.
- When sales order A is soft reserved in Inventory Visibility, your physical inventory quantity remains the same in Supply Chain Management because sales order A hasn't been synced to Supply Chain Management.
- Now, the call center enters sales order B directly in Supply Chain Management, which makes a soft reservation call to Inventory Visibility. Even though Supply Chain Management still shows 100 pieces available, Inventory Visibility knows there's only 90 pieces available, so the soft reservation request fails. Nevertheless, the call center user can still choose to block or continue processing sales order B in Supply Chain Management (if overselling is allowed).
- Then sync either single or aggregated soft-reserved orders (order lines) into Supply Chain Management to make hard reservations and release to the warehouse or update the final inventory quantity.

- You can set the system to offset soft reservations when physical inventory is updated in Supply Chain Management.

Soft reservations are usually created, consumed, and canceled by using API calls to the Inventory Visibility service.

> **Note**
>
> You can optionally set up Supply Chain Management (and other third-party systems) to automatically offset the quantity that has been reserved by using Inventory Visibility. The offset quantity is deleted from the reservation records in Inventory Visibility.
>
> By default, the offset function is automatically turned on when you enable the soft reservation feature.

Turn on and set up the reservation feature in UI version 2

This section and its subsections apply when you're using Inventory Visibility UI version 2.

Turn on the reservation feature

Follow these steps to turn on the reservation feature in UI version 2.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature Management**.
3. On the **Soft Reservation** tile, select **Manage**.
4. Set the **Enable feature** option to *True*.
5. The **Filter unconfigured dimensions** option controls how the system behaves if a user submits reservation requests that include dimensions that aren't specified on the **Reservation dimensions** FastTab. (Learn more in the Configure reservation mappings and dimensions section of this article.) Select one of the following values:
    - *True* – Reservation requests that include dimensions that aren't defined on the **Reservation dimensions** FastTab succeed, but undefined dimensions are ignored.
    - *False* – Reservation requests that include dimensions that aren't defined on the **Reservation dimensions** FastTab fail.

> **Important**
>
> If you've enabled the *Inventory Visibility integration with soft reservation on sales order lines* feature in Supply Chain Management, you must set the **Filter unconfigured dimensions** option to *True*. Learn more in the Integrate soft reservations and offsets with Supply Chain Management section.

6. On the toolbar, select **Save**.

7. If you changed the setting of the **Filter unconfigured dimensions** option, you must reverse all existing reservations to preserve data consistency. Use the clean up reservation data API to clean up existing soft reservation records.

Configure reservation mappings and dimensions

When you make a reservation, you might want to know whether on-hand inventory is currently available for reservation. The validation is linked to a calculated measure that represents a computation formula of a combination of physical measures.

By setting up the mapping from a physical measure to a calculated measure, you enable the Inventory Visibility service to automatically validate reservation availability based on the physical measure.

> **Important**
>
> All the physical measures and calculated measures that you plan to map for each data source must already be defined for your system. For information about how to configure measures and data sources, see Configure Inventory Visibility.

Default reservation mappings are provided out of the box. You can review and use these mappings directly. You can also add your own mappings by following these steps.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.

2. Define all the data sources, physical measures, and calculated measures that you plan to use with the soft reservations feature. For instructions, see Configure Inventory Visibility. Here are some guidelines for the setup of typically required measures:

- Define the physical measure that serves as the soft reservation measure (for example, *SoftReservPhysical*).
- Define the *available for reservation* calculated measure that contains the available-for-reservation computation formula that you map to the physical measure. For example, you might set up *AvailableToReserve* (available for reservation) so that it's mapped to the previously defined *SoftReservPhysical* physical measure. This setup lets you find which quantities that have the *SoftReservPhysical* inventory status are available for reservation.

The following table shows the available-for-reservation computation formula.

Expand table

Calculation type	Data source	Physical measure
Addition	*fno*	*AvailPhysical*
Addition	*pos*	*Inbound*
Subtraction	*pos*	*Outbound*
Subtraction	*iv*	*SoftReservPhysical*

Set up the physical measure that the reservation measure is based on as a component of the calculated measure. In this way, the calculated measure quantity is affected by the reservation quantity. For example, use the *SoftReservPhysical* physical measure of the *iv* data source as a component of the *AvailableToReserve* calculated measure of the *iv* data source.

3. In the **Reservation mapping** section, set up the mapping from the physical measure to the calculated measure. For the previous example, you might use the following settings to map *AvailableToReserve* to the previously defined *SoftReservPhysical* physical measure:

   - **Physical measure data source:** *iv*
   - **Physical measure:** *SoftReservPhysical*
   - **Available for reservation data source:** *iv*
   - **Available for reservation calculated measure:** *AvailableToReserve*

   Now, when you reserve *SoftReservPhysical*, Inventory Visibility automatically finds *AvailableToReserve* and its related computation formula to validate the reservation.

4. In the **Reservation dimensions** section, set up all the product dimensions that might be specified when a reservation is made. *Each reservation dimension must contain a partition*

configuration.

Here's an example of reservation dimensions.

Expand table

Key	Hierarchy
SiteId	1
LocationId	2
ColorId	3
SizeId	4
StyleId	5

In this example, the partition schema is *ByLocation*, and the basic partition dimension is *(SiteId, LocationId)*.

5. On the toolbar, select **Save**.

Example available-for-reservation calculation

The following example shows how the Inventory Visibility API might report on-hand inventory information.

JSON

```
{
    "productId": "D0002",
    "dimensions": {
        "SiteId": "1",
        "LocationId": "11",
        "ColorId": "Red"
    },
    "quantities": {
        "iv": {
            "SoftReservPhysical": 90
        },
        "fno": {
            "availphysical": 70.0,
        },
        "pos": {
            "inbound": 50.0,
            "outbound": 20.0
        }
```

```
    }
}
```

In this case, the following available-for-reservation calculation applies:

`AvailableToReserve` = `fno.availphysical` + `pos.inbound` – `pos.outbound` – `iv.SoftReservPhysical`
= 70 + 50 – 20 – 90
= 10

Therefore, if you try to make reservations on `iv.SoftReservPhysical`, and the quantity is less than or equal to `AvailableToReserve` (*10*), the soft reservation request succeeds.

Turn on and set up the reservation feature in UI version 1

This section applies when you're using Inventory Visibility UI version 1.

To turn on and set up the reservation feature UI version 1, follow these steps:

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Configuration**.
4. On the **Feature management & settings** tab, use the option to turn on the *Soft reservation* feature.
5. Select the **Soft reservation** tab.
6. Open the **Configuration** page.
7. If you plan to receive soft reservation data from Supply Chain Management, select the **Use soft reservation for sales order in FinOps** checkbox.
8. On the **Soft Reservation** tab, review the default soft reservation hierarchy. Add new dimensions to the hierarchy as you require. For more information about how to work with the soft reservation hierarchy, see the Configure reservation mappings and dimensions section.
9. In the **Set Soft Reservation Mapping** section, review the default settings. For more information about soft reservation mappings and the required measures, see the Configure reservation mappings and dimensions section.

Create soft reservations using the Inventory Visibility API

When you call the reservation API, the system marks the reservation of the specified goods and quantities.

For example, the company Contoso sells product D0002 (Cabinet) from its e-commerce website. A customer places a sales order for a small red cabinet via the website. Contoso decides to fulfill this order by using the following dimensions:

- Organization ID = usmf
- Site = 1
- Warehouse = 11
- Product = D0002
- Color = red
- Size = small

Contoso has already set up an API connection to Inventory Visibility from its own e-commerce system. When the order is received, the system instantly triggers an API call to make a soft reservation for the cabinet in Inventory Visibility.

Reservations are made in the Inventory Visibility service by submitting a POST request to the service's URL, such as `/api/environment/{environmentId}/onhand/reserve`.

For a reservation, the request body must contain an organization ID, a product ID, reserved quantities, and dimensions.

When you call the reservation API, you can control the reservation validation by specifying the Boolean `ifCheckAvailForReserv` parameter in the request body. A value of `True` means that the validation is required, whereas a value of `False` means that the validation isn't required. The default value is `True`.

If you want to cancel a reservation or unreserve specified inventory quantities, set the quantity to a negative value, and set the `ifCheckAvailForReserv` parameter to `False` to skip the validation.

Here's an example of the request body that references the sales order in the previous context.

JSON

```
# Url

# Replace {endpoint} with your system endpoint.
    {endpoint}/api/environment/{environmentId}/onhand/reserve

# Method
Post
```

```
# Header
# replace {access_token} with the one get from security service
Api-version: "1.0"
Content-Type: "application/json"
Authorization: "Bearer {access_token}"

# Body
{
    "id": "Testrequest",
    "organizationId": "usmf",
    "productId": "D0002",
    "dimensions": {
        "SiteId": "1",
        "LocationId": "11",
        "ColorId": "red",
        "SizeId": "small"
    },
    "quantityDataSource": "iv",
    "modifier": "softreserved",
    "quantity": 1,
    "ifCheckAvailForReserv": true
}
```

All dimensions must be defined in your reservation dimensions configuration and must contain the partition for the product. Here are examples of feasible dimension combinations that use the default configuration:

- `(SiteId, LocationId)`
- `(SiteId, LocationId, SizeId)`
- `(SiteId, LocationId, ColorId, SizeId)`
- `(SiteId, LocationId, ColorId, SizeId, StyleId)`

> **Note**
>
> Reservations on different *dimension levels* are independent. For example, your defined reservation dimensions are `(SiteId, LocationId, ColorId, SizeId)`. If you reserve some quantity for one product at dimension level `(SiteId, LocationId, ColorId)`, reservations at other levels, such as `(Site , Location , Color , Size)` or `(Site , Location)`, aren't affected.

A successful soft reservation request returns a *soft reservation ID* for each reservation record. The soft reservation ID isn't a unique identifier for an individual soft reservation record, but a combination of the product ID and dimension values that are associated with the soft reservation request. You can record the soft reservation ID on the order line when you sync the successfully reserved orders to Supply Chain Management or another ERP system for offset.

Integrate soft reservations and offsets with Supply Chain Management

You can trigger a soft reservation from a Supply Chain Management sales order and then make an offset back to Inventory Visibility when the order lines are hard reserved (that is, changed to a status of *Reserve physical*, *Reserve ordered*, *Picked*, and so on). The process differs slightly, depending on the soft reservation feature that you're using (*Inventory Visibility integration with reservation offset* or *Inventory Visibility integration with soft reservation on sales order lines*).

- *Inventory Visibility integration with soft reservation on sales order lines* – This feature lets you avoid double booking by making a soft reservation after you create a sales order line in Supply Chain Management. The soft reservation automatically offsets when this line reaches a status of *Reserve ordered*, *Reserve physical*, *Pick*, and so on.

- *Inventory Visibility integration with reservation offset* – This feature lets you offset soft reservations that were made outside of Supply Chain Management. You can soft reserve directly from the soft reservation API (or another customization channel) and then enter the reservation ID into the corresponding sales order line in Supply Chain Management. This feature automatically offsets the soft reservation once the line status fulfills the conditions established by your settings.

Turn on the feature in Supply Chain Management

Sign in to Supply Chain Management and make sure that you've enabled the following two features in Feature management.

- *Inventory Visibility integration with reservation offset* – This feature requires Supply Chain Management version 10.0.22 or later. As of Supply Chain Management version 10.0.45, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.
- *Inventory Visibility integration with soft reservation on sales order lines* – This feature is recommended if you're running Supply Chain Management version 10.0.33 or later. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

These features are compatible with each other as of build 10.0.1591.98 of Supply Chain Management version 10.0.34 and build 10.0.1627.63 of Supply Chain Management version 10.0.35 (and all builds of versions 10.0.36 and higher). Prior to these versions, the features aren't compatible with each other, so enable only the one you want to use. Upgrade to a newer version if you require both features.

Configure additional settings if the "Inventory Visibility integration with soft reservation on sales order lines" feature is enabled

If you enabled the *Inventory Visibility integration with soft reservation on sales order lines* feature, follow these steps to configure additional settings.

1. In Supply Chain Management, go to **Inventory Management** > **Setup** > **Inventory Visibility integration parameters**.

2. On the **Enable soft reservation** tab, in the **Default soft reserve block level** field, select one of the following values to specify how the system should behave if you try to process sales order lines that no soft reservation exists for in Inventory Visibility:

    - *Block* – You can't process a sales order line to a hard reservation (reserve physical) or further unless Inventory Visibility has already recorded a successful soft reservation for the line.
    - *Warning* – You receive a warning message if you try to process a sales order line to a hard reservation (reserve physical) but no matching soft reservation has been made in Inventory Visibility.
    - *Ignore* – The system doesn't check for a soft reservation in Inventory Visibility when you try to process order lines in Supply Chain Management.

> **Tip**
>
> You don't have to enable or set up the offset modifier, because offset for soft reservation is always enabled and triggers when a sales line proceeds to hard reservation status (or further, if the hard reservation step is skipped).

Configure additional settings if the "Inventory Visibility integration with reservation offset" feature is enabled

If you enabled the *Inventory Visibility integration with reservation offset* feature, follow these steps to configure additional settings.

1. In Supply Chain Management, go to **Inventory Management** > **Setup** > **Inventory Visibility integration parameters**.

2. On the **Reservation offset** tab, set the following fields:

- **Enable reservation offset** – Set this option to *Yes* to enable the functionality.
- **Reservation offset modifier** – Select one of the following values to specify the inventory transaction status that offsets reservations that are made in Inventory Visibility. This setting determines the order processing stage that triggers offsets. The stage is traced by the order's inventory transaction status.
  - *On order* – Orders that have a status of *On order* send an offset request when they're created. The offset quantity is the quantity of the created order (line).
  - *Reserve* – Orders that have a status of *Reserve* send an offset request when they're either order reserved or physically reserved. When you offset at the *Reserve* status, the order sends an offset request at any new inventory status that's closest to reserved picked (for example, pick, packing slip posted, or invoiced). This behavior occurs even if you skip the reservation in Supply Chain Management and continue to another inventory status (for example, if you skip from release to warehouse to pick and pack). The request is triggered only once. If it's triggered at pick, the offset isn't duplicated when a packing slip is posted. The offset quantity is the same as the quantity of the inventory transaction status when the offset was triggered (in other words, *Reserved ordered/Reserve Physical*, or a later status, on the corresponding order line).

Create soft reservations and offsets when you use the Inventory Visibility integration with soft reservation on sales order lines feature

When you use the *Inventory Visibility integration with soft reservation on sales order lines* feature, you can post soft reservations and also offset soft reservations from sales order lines in Supply Chain Management. The offset capabilities in this feature support both *internally* and *externally* created sales lines.

1. Sign in to Supply Chain Management
2. Go to **Sales and marketing** > **Sales Orders** > **All Sales Orders**.
3. On the Action Pane, select **New** to create a sales order.
4. Fill in the **Create sales order** dialog box in the usual way, and then select **OK** to finish creating the new order.
5. The new order is opened. Enter a sales line. Be sure to specify **Product ID**, **Site**, **Warehouse**, and **Quantity** values. Specify any other inventory dimension values that are applicable.

6. There are two ways to make a soft reservation from a sales order. Follow one of these steps:

    - *To soft reserve the whole order, including all lines*, select **Inventory Visibility integration** > **Soft reservation** on the Action Pane. Then select one of the following options:
        - **Reserve entire order directly** – Make an instant soft reservation API call to Inventory Visibility.
        - **Reserve entire order by batch** – Add the reservation requests to a batch queue that's associated with a batch job that syncs Supply Chain Management with Inventory Visibility about once per minute.

    - *To soft reserve a specific order line*, select the sales line, and then select **Inventory** > **Inventory Visibility integration** > **Soft reserve** on the toolbar of the FastTab toolbar to open the **Inventory service reservation details** dialog box. Then select either to reserve directly or to add the reservation as a batch job. The following information is shown:
        - **Unreserved** – The quantity that wasn't soft reserved.
        - **Soft reserve success** – The quantity that was successfully soft reserved.
        - **Batch reserve in progress** – The quantity that was added to the soft reservation batch queue.
        - **Direct reserve in progress** – The quantity that triggered an instant soft reservation API call to Inventory Visibility.
        - **Failed - not enough stock** – The quantity that couldn't be soft reserved because of unavailable stock.
        - **Failed - other reason** – The quantity that couldn't be soft reserved for another reason, such as a failed API call, connection issues, or other issues.
        - **Offset quantity** – The total offset quantity, including both offset success and in-progress quantities.
        - **Pending to offset quantity** – The quantity that skipped soft reservation and proceeded directly to hard reservation or further physical inventory consumption.

7. To view and edit the soft reservation blocking level of a sales line, select the relevant line on the **Sales order lines** FastTab. Then, expand the **Line details** FastTab and open its **General** tab. The **Soft reservation blocking level** field indicates the blocking level for the selected sales line. The default setting for all new order lines is set on the **Inventory visibility integration parameters** page, but you can edit it for individual sales lines as needed. The following values are possible:

    - *Ignore* – You aren't blocked or given a warning if you reserve physical or reserve ordered without a soft reservation.

- *Warning* – You receive a warning, but you aren't blocked, if you reserve physical or reserve ordered without a soft reservation.
- *Blocking* – You must have a soft reservation before you can reserve physical or reserve ordered.

If a line is set to block or warn when a soft reservation couldn't be made, you might see a block notice here. To help avoid the risk of overselling, don't choose to override the soft reservation validation unless overrides are permitted in your business.

When a soft reservation is successfully made, a soft reservation ID is automatically returned and recorded for each sales line.

By default, the soft reservation offset is triggered when the line reaches a hard reservation status (*Reserve physical* or *Reserve ordered*) or later. Sales lines that show a valid soft reservation ID and a qualifying trigger status are automatically added to the offset batch queue.

8. To view the soft reservation status of a sales order line, select the relevant line on the **Sales order lines** FastTab. Then, expand the **Line details** FastTab and open its **General** tab. The **Soft reservation status** field shows the stage of the soft reservation process for the selected sales order line. The following values are possible:

    - *Not started* – The soft reservation process is waiting to start.
    - *Unfinished* – A soft reservation was attempted, but not all quantities were successfully posted. Some or all quantities might have failed to post, or some quantities might have been successfully soft reserved with other reservations still waiting to finish.
    - *Succeeded* – All quantities were successfully soft reserved.

> ⓘ **Note**
>
> If you must reverse a successful soft reservation, open the relevant sales order, and then select **Revert reservation directly** or **Revert reservation by batch** at the sales order or sales line level.

Import sales orders that have been soft reserved externally

It's likely that your company has other order channels that also post soft reservation to Inventory Visibility. Sales orders that have been soft reserved externally might need to be imported into

Supply Chain Management if you're using it as your system of record. Follow these steps to import sales orders that have been soft reserved externally.

1. Sign in to Supply Chain Management.
2. Go to **System Administration** > **Work spaces** > **Data management**.
3. Select the **Framework parameters** tile.
4. On the **Data import/export framework parameters** page, on the **Entity settings** tab, select **Refresh entity list**.
5. Go back to **System Administration** > **Work spaces** > **Data management**.
6. Select the **Export** tile.
7. On the **New Record** page, in the **Group name** field, enter a name.
8. On the **Selected entities** FastTab, select **Add entity** on the toolbar, and then, in the drop-down dialog box, set the following values:
    - **Entity name** – Select *Sales order lines V2*.
    - **Target data format** – Select *Excel*.
    - **Select fields** – Select *Importable fields*.
9. Select **Add** to apply your settings, and then select **Close** to close the dialog box.
10. A new line is added to the grid. On the new line, select the button in the **Filter** column.
11. In the **Inquiry** dialog box, the grid on the **Range** tab should already have one row. Edit the row so that it has the following values:
    - **Table** – Select *Sales order lines V2*.
    - **Derived table** – Select *Sales order lines V2*.
    - **Field** – Select *Sales order*.
    - **Criteria** – Select any sales order number.
12. Select **OK** to apply your settings and close the dialog box.
13. On the Action Pane, select **Export**.
14. On the **Execution summary** page, on the **Entity processing status** FastTab, select **Download file** on the toolbar.

15. Open the downloaded file in Excel, and remove all the data rows. Leave only the header row.

16. Add new rows that include the sales order line details that you want to import into Supply Chain Management.

17. For each row, in addition to the usual order data, be sure to include soft reservation details in the following columns:

    - **INVENTORYSERVICERESERVATIONID** – Enter the soft reservation ID from the external reservation.
    - **ISSOFTRESERVEDEXTERNALLY** – Enter *Yes*.

18. Import the completed Excel file. You can check import status after completion.

19. After the file is successfully imported, you can confirm that the new sales orders have soft reservation details by opening those orders in Supply Chain Management.

> ⓘ **Note**
>
> After you import sales orders that include soft reservation details, you can't modify the soft reservation or revert it from Supply Chain Management. If the data that you imported isn't correct, delete the affected sales line, and then reimport it.

Create soft reservations and offsets when you use the Inventory Visibility integration with reservation offset feature

When you use the *Inventory Visibility integration with reservation offset* feature, you can't trigger soft reservation directly from Supply Chain Management. Instead, this feature supports only the offset of sales order lines that were created *externally* from Supply Chain Management. When externally created soft-reserved sales lines are replicated into Supply Chain Management, offsets might be required from Supply Chain Management to Inventory Visibility.

You can offset a soft-reserved quantity after the quantity on an order is physically deducted in Supply Chain Management or another ERP system. Inventory Visibility offers out-of-box soft reservation offset integration with Supply Chain Management.

Follow these steps to offset a soft reservation.

1. Sign in to Supply Chain Management.

2. Go to **Sales and marketing** > **Sales Orders** > **All Sales Orders**.

3. On the Action Pane, select **New**. Create a new order and a new sales line.

4. Re-create the external sales order, and add a sales line that uses the same product ID, organization, site, warehouse, and dimensions values.

5. On the **Sales order lines** FastTab, select the sales line that you just entered, and then, on the toolbar, select **Inventory** > **Reservation ID**.

6. Follow one of these steps:

    - Copy the soft reservation ID in your soft reservation request response, and paste it into the **Reservation ID** field.
    - Leave the **Reservation ID** field blank, but select the **Inventory service auto offset** checkbox. The system automatically determines which product and product dimensions to offset, based on the item ID and dimension values entered on the selected line.

7. Select **OK**.

8. While the same sales line is still selected, physically reserve the ordered quantity by selecting **Inventory** > **Reservation** on the toolbar of the **Sales order lines** FastTab.

9. If you've previously set the **Reservation offset modifier** field to *Reserved*, the offset is triggered when the order line has a status of *Reserve physical* or *Reserve ordered*. A batch job runs once a minute to sync offset requests from Supply Chain Management to Inventory Visibility.

> ⓘ **Note**
>
> For transaction statuses that include a specified reserve offset modifier, the transaction update offsets the corresponding reservation record when all the following conditions are met:
>
> - The reservation ID on the inventory transaction matches the reservation ID of the reservation record in Inventory Visibility.
> - The dimensions of the inventory transaction match the dimensions of the reservation record in Inventory Visibility.
> - Changes in inventory transaction status trigger offsets for reservations when the inventory transaction status reflects the fact that an order process has been completed

or skipped.

Offset quantities follow the inventory quantities that are specified on the relevant inventory transactions. An offset takes effect only if reserved quantity remains in Inventory Visibility.

Check for failed reservation offsets

To check for failed reservation offsets, go to **Inventory Management** > **Periodic tasks** > **Inventory Visibility integration**. A failed offset might be caused by an incorrect soft reservation ID, an internet issue, a broken system connection, and so on.

Cancel or revert a soft reservation

If an original order line is canceled or deleted, and you must revert the corresponding soft reservation, take one of the following actions:

- Post a soft reservation event with a negative quantity that has the same exact information in your API query body.
- Post an unreserve request with the same quantity and reservation ID. Learn more in Reverse reservation events.

Last updated on 07/27/2026

Inventory Visibility inventory allocation

08/13/2025

Business background and purpose

Organizations often have to preallocate their on-hand stock to their most important sales channels, customer groups, regions, and promotional events to ensure that the preallocated stock is protected against any other use and can be consumed only via sales transactions that are relevant to the allocation. Inventory allocation in Inventory Visibility is a component of the sales operational planning process, and is done before any actual sales activities occur or a sales order is created.

For example, a company that's named Contoso produces a popular bike. Unfortunately, because a recent supply chain disruption has affected all in-transit stock of that bike, Contoso has only limited on-hand stock and must make best use of it. Contoso operates both online and in-store sales. In each sales channel, the company has a few important corporate partners (marketplaces and large retailers) that demand that a specific portion of the bike's available inventory be saved for them. Therefore, the bicycle company must be able to balance stock distribution across channels and also manage the expectations of its VIP partners. The best way to achieve both goals is to use inventory allocation, so that each channel and retailer can receive specific allocated quantities that can be sold to consumers later.

Inventory allocation has two basic business purposes:

- **Inventory protection (ring fencing)** – Organizations want to preallocate restricted or limited stock to prioritized channels, regions, VIP customers, and subsidiary companies. The Inventory Visibility allocation feature aims to protect the allocated inventory, so that the other allocations, reservations, or other sales demands won't affect the previously allocated inventory.
- **Oversell control** – The Inventory Visibility allocation feature aims to put a restriction on the previously allocated quantities, so that the receiving party (for example, a channel or customer group) won't over-consume them when the actual sales transaction that's based on a soft reservation goes into effect.

Allocation definition in Inventory Visibility Service

Allocation virtual pool

Although the allocation feature in Inventory Visibility doesn't set aside physical inventory quantities, it does refer to the available physical inventory quantity to define its initial *available to allocate* virtual pool quantity. Inventory allocation in Inventory Visibility is a soft allocation. It's done before actual sales transactions occur and doesn't depend on sales orders. For example, you can allocate stock to your most important sales channels or large corporate retailers before any end customers visit the sales channel or retail store to purchase it.

Difference between inventory allocation and soft reservation

Soft reservations are often linked to actual sales transactions (sales order lines). Both allocation and soft reservation can be used independently, but if you want to use them together, soft reservation should be done after allocation. We recommend that you do inventory allocation first and then soft reserve against the allocated quantities to achieve near-real-time consumption against allocation. Learn more in the Consume as a soft reservation section of this article.

The inventory allocation feature lets sales planners or key account managers manage and preallocate important stock across allocation groups (such as channels, regions, and customer groups). It also supports real-time tracking, adjustment, and analytics of consumption against allocated quantities, to ensure that replenishment or reallocation can be done on time. This ability to have real-time visibility into allocation, consumption, and allocation balance is especially important at fast-sale or promotion events.

Terminology

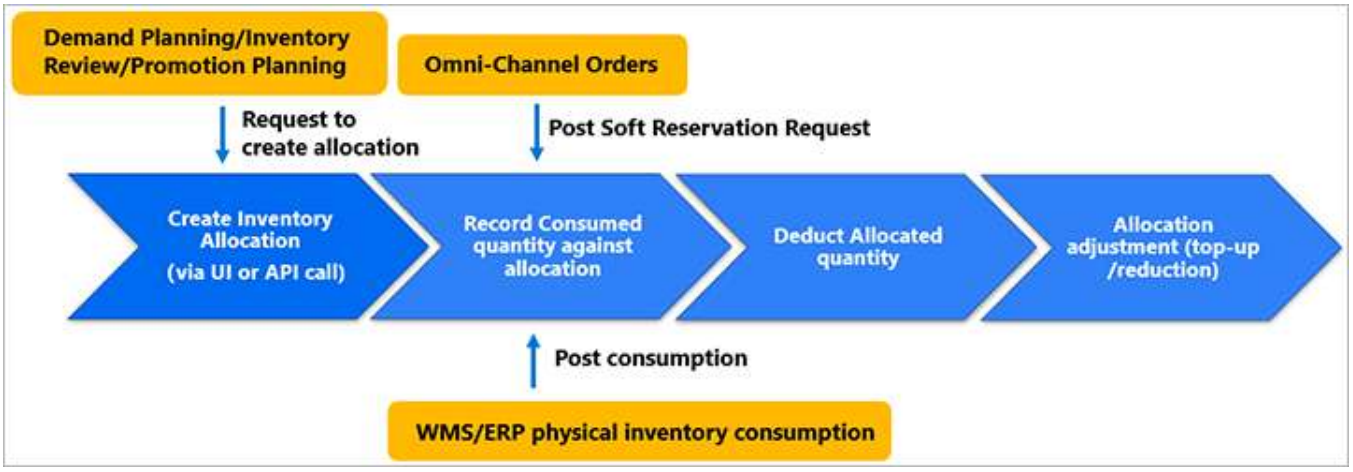
The following terms and concepts are useful in discussions of inventory allocation:

- **Allocation group** – The group that owns the allocation, such as a sales channel, customer group, or order type.
- **Allocation group name** – The name of each allocation group. For example, *web* or *store* might be the name of the sales channel allocation group, whereas *VIP* or *normal* might be the name of the customer allocation group.
- **Allocation hierarchy** – A means to combine allocation groups in a hierarchical manner. A maximum of eight levels of hierarchy are supported. For example, you could define *Channel* as hierarchy level 0, *Region* as level 1, and *Customer group* as level 2. During inventory allocation, you must follow the allocation hierarchy sequence when you specify the value of the allocation group. For example, you could create an allocation of 200 red bikes to the *Web* channel, the *London* region, and the *VIP* customer group.
- **Available to allocate** – The *virtual common pool* that indicates the quantity that's available for further allocation. It's a calculated measure that you can freely define by

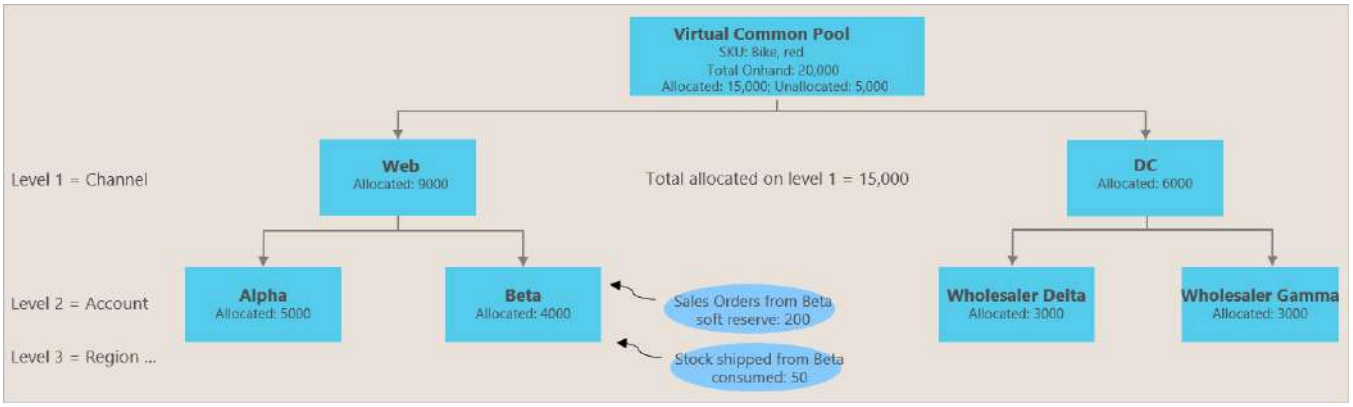
using your own formula. If you're also using the soft reservation feature, we recommend that you use the same formula to calculate available-to-allocate and available-to-reserve.

- **Allocated** – A physical measure that shows the allocated quota that can be consumed by the allocation groups. It's deducted at the same time that the consumed quantity is added.
- **Consumed** – A physical measure that indicates the quantities that have been consumed against the original allocated quantity. As numbers are added to this physical measure, the Allocated physical measure is automatically reduced.

The following illustration shows the business workflow for inventory allocation.



The following illustration shows the allocation hierarchy and allocation groups. The *virtual common pool* shown here is the available-to-allocate quantity.



Turn on and set up inventory allocation in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

To set up inventory allocation, you must complete the following tasks, which are described in the subsections of this section:

- Enable the allocation feature, and update the configuration to initialize it.

- Set up the *available to allocate* calculated measure (and include the *allocated* physical measure in the calculation).
- Set up your allocation groups.
- Update the configuration to activate your new settings.

Enable the inventory allocation feature

Follow these steps to enable the inventory allocation feature.

1. In Power Apps, open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature Management**.
3. On the **Inventory allocation** tile, select **Manage**.
4. Set the **Enable feature** option to *Yes*.
5. On the toolbar, select **Save**.
6. On the navigation pane, select **Admin Settings**.
7. On the **Update configuration** tile, select **Manage**.
8. Review your modifications in the dialog box.

   > **ⓘ Important**
   >
   > Be sure to verify all the important modifications that are about to be made to your data sources, physical measures, and dimension mappings.

9. Select **Confirm Update** to apply your configuration change.

Set up the available to allocate calculated measure

When you enable the inventory allocation feature and update the configuration as described in the previous section, Inventory Visibility creates one predefined data source and several initial measures.

The data source is named *@iv*. It includes the following set of default physical measures:

- *@allocated*
- *@cumulative_allocated*
- *@consumed*
- *@cumulative_consumed*

To use allocation, you must correctly set up the formula for the *available to allocate* calculated measure (*@iv.@available_to_allocate*). For example, you have the *onordered* physical measure in the *fno* data source and the *inbound* physical measure in the *pos* data source. You can then allocate on-hand stock for the sum of *fno.onordered* and *pos.inbound*. In this case, *@iv.@available_to_allocate* should contain *pos.inbound* and *fno.onordered* in the formula. Here's an example calculation:

@iv.@available_to_allocate = fno.onordered + pos.inbound – @iv.@allocated

> **Note**
>
> The *@iv* data source is a predefined data source, and the physical measures in *@iv* that have an at sign (@) as a prefix are predefined measures. These measures are the predefined configuration for the allocation feature. Therefore, don't change or delete them. Otherwise, you're likely to encounter unexpected errors when you use the allocation feature.
>
> You can add new physical measures to the predefined *@iv.@available_to_allocate* calculated measure. However, you must not change its name.

Follow these steps to set up the *available to allocate* calculated measure:

1. In the Inventory Visibility app, select **Feature Management** on the navigation pane.
2. On the **Data source settings** tile, select **Manage**.
3. Open the data source that's named *@iv*.
4. In the **Calculated Measures** section, open the record that's named *@available_to_allocate* if it exists. Otherwise, create a record where the **Calculated measure name** value is *@available_to_allocate*, and **Save** the new record.
5. In the **Calculated measure details** section, add the physical measures that you want to use to calculate your available-to-allocate quantities. Be sure to include the *@iv.@allocated* physical measure in the formula.

Set up your allocation groups and hierarchy

Every allocation that's created in Inventory Visibility must be assigned to a specific *allocation group*. You can create up to eight allocation groups, each of which has a name and a level. A level is required to establish a group hierarchy.

Each level is defined as an integer from *0* (zero) through *7*, representing the order in which the groups are used in the allocation hierarchy. The group that's assigned to level *0* is the highest level in the hierarchy, and the group that's assigned to level *7* is the lowest level in the hierarchy.

When you create an allocation, you must specify hierarchies in order from the highest level to the lowest level. For example, your configuration has *Country/region* for group level 0, *State* for group level 1, and *City* for group level 2. In this case, both *Country/region* and *State* are required when *City* is specified. However, an allocation can be created by using only *Country/region* and *State*, or by using only *Country/region*.

Follow these steps to set up your allocation groups and hierarchy.

1. In the Inventory Visibility app, select **Feature Management** on the navigation pane.
2. On the **Inventory allocation** tile, select **Manage**.
3. In the **Allocation group** section, use the **New allocation group** button on the toolbar to add a row for each allocation group that you need. Assign a level between *0* and *7*. To edit an existing group, select its name in the **Allocation group name** column.
4. On the toolbar at the top of the page, select **Save**.

> **Important**
>
> Be careful when you delete or change the allocation hierarchy. For guidance, see the Tips for using allocation section.

Update the configuration to activate your new settings

After you've finished changing your configuration, you must apply the changes to activate them.

1. In the Inventory Visibility app, select **Admin Settings** on the navigation pane.
2. On the **Update configuration** tile, select **Manage**.
3. Review your modifications in the dialog box.

   > **Important**
   >
   > Be sure to verify all the important modifications that are about to be made to your data sources, physical measures, and dimension mappings.

4. Select **Confirm Update** to apply your configuration changes.

Turn on and set up inventory allocation in UI version 1

This section applies when you're using Inventory Visibility UI version 1.

Follow these steps to enable inventory allocation and set up allocation groups if you're using UI version 1.

1. Enable the inventory allocation feature.
   a. Go to **Legacy UI** > **Configuration**.
   b. On the **Feature Management & Settings** tab, turn on the feature that's named *Inventory Allocation*.
   c. Select **Update configuration** in the upper-right corner to apply the new setting.

2. Configure the *available to allocate* calculated measure and the *allocated* physical measure.
   a. On the **Calculated Measure** tab, review the initial calculated measure, which is named *@iv.@available_to_allocate*.
   b. Edit the formula to meet your business needs by adding and removing physical measures. Be sure to include the *@iv.@allocated* physical measure in the formula.

3. Set up your allocation groups and hierarchy.
   a. Select the **Allocation** tab.
   b. In the default allocation configuration, there are four levels of hierarchy. These levels, from highest to lowest, are *Channel* (group level 0), *customerGroup* (group level 1), *Region* (group level 2), and *OrderType* (group level 3). You can edit the groups by working in the **Edit configuration** field. To remove an existing allocation group, select the **X** next to its name. To add an allocation group, enter the name directly in the **Edit configuration** field.
   c. When you've finished editing the groups, select **Save**.

4. Update the configuration to activate your new settings.

   - When you've finished configuring the allocation group and hierarchy settings, select **Save**, and then select **Update Configuration** in the upper-right corner.

The values of the configured allocation groups will be updated when you create an allocation by using either the user interface or API POST (`/api/environment/{environmentId}/allocation/allocate`). Details about both approaches are provided later in this article. If you use four group names and set them to [`channel`, `customerGroup`, `region`, `orderType`], these names will be valid for allocation-related requests after you call the configuration update API.

> ⓘ **Note**

Settings that are applied for the enabled status of the allocation feature and for allocation groups in UI version 1 don't affect the corresponding settings in UI version 2. Likewise, settings for the enabled status and allocation groups in UI version 2 don't affect the corresponding settings in UI version 1. Only those settings that are applied in the **active UI version** apply.

> **Important**
>
> Be careful when you delete or change the allocation hierarchy. For guidance, see the Tips for using allocation section.

Tips for using allocation

- For every product, the allocation function must use the same *dimension level* according to the on-hand index configuration that's set up in the on-hand index configuration. For example, your index hierarchy is [`Site`, `Location`, `Color`, `Size`]. You allocate some quantity for one product at the [`Site`, `Location`, `Color`] dimension level. In this case, the next time that you want to allocate the same product, you must allocate at the same level. If you use the [`Site`, `Location`, `Color`, `Size`] level or the [`Site`, `Location`] level, the data will be inconsistent.
- When you modify allocation groups and the hierarchy, if allocation data already exists in the system, deletion of existing allocation groups or a shift in the allocation group hierarchy will corrupt the existing mapping between the allocation groups. Therefore, be sure to use the `Unallocate` API to remove all old data before you update the configuration. However, you don't have to clean up the data if you're just adding new allocation groups to the lowest hierarchy.
- Allocation will succeed only for products that have a positive `available_to_allocate` quantity.
- To allocate products from a high allocation hierarchy group to a subgroup, use the `Reallocate` API. For example, your allocation group hierarchy is [`channel`, `customerGroup`, `region`, `orderType`], and you want to allocate some product from the [`Online`, `VIP`] allocation group to the [`Online`, `VIP`, `EU`] allocation subgroup. In this case, use the `Reallocate` API to move the quantity. When you use the `Allocate` API, it will allocate the quantity from the virtual common pool.
- To view overall product availability (the common pool), use the query on-hand API to request the inventory amount that's *available to allocate*. You can then make allocation decisions based on this information.

Use the allocation APIs

The following table lists the allocation APIs that are available.

Expand table

Method	API	Description
POST	`/api/environment/{environmentId}/allocation/allocate`	Create an allocation
POST	`/api/environment/{environmentId}/allocation/unallocate`	Revert or remove allocated records
POST	`/api/environment/{environmentId}/allocation/reallocate`	Move allocated quantities from an existing allocation to other allocation groups
POST	`/api/environment/{environmentId}/allocation/consume`	Deduct (use) the allocated quantity
POST	`/api/environment/{environmentId}/allocation/query`	Check existing allocation records against the allocation groups and hierarchy

Allocate

Call the `Allocate` API to allocate a product that has specific dimensions. Here's the schema for the request body.

JSON

```
{
    "id": "string",
    "productId": "string",
    "dimensionDataSource": "string",
    "groups": {
        "groupA": "string",
        "groupB": "string",
        "groupC": "string"
    },
    "quantity": decimal,
    "organizationId": "string",
    "dimensions": {
        "dimension1": "string",
        "dimension2": "string",
        "dimension3": "string"
    }
}
```

For example, you want to allocate a quantity of 10 for product *Bike*, site *1*, location *11*, color *red*, channel *Online*, customer group *VIP*, and region *US*. To do this allocation, you can make a call that has the following body content.

JSON

```
{
    "id": "test101",
    "productId": "Bike",
    "groups": {
        "channel": "Online",
        "customerGroup": "VIP",
        "region": "US"
    },
    "quantity": 10,
    "organizationId": "usmf",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "colorId": "red"
    }
}
```

The quantity must always be more than 0 (zero).

Unallocate

Use the `Unallocate` API to reverse the `Allocate` operation. Negative quantity isn't allowed in an `Allocate` operation. The body of `Unallocate` is identical to the body of `Allocate`.

Reallocate

Use the `Reallocate` API to move some allocated quantity to another group combination. Here's the schema for the request body.

JSON

```
{
    "id": "string",
    "productId": "string",
    "dimensionDataSource": "string",
    "sourceGroups": {
        "groupA": "string",
        "groupB": "string",
        "groupC": "string"
    },
    "groups": {
        "groupD": "string",
```

```
        "groupE": "string",
        "groupF": "string"
    },
    "quantity": decimal,
    "organizationId": "string",
    "dimensions": {
        "dimension1": "string",
        "dimension2": "string",
        "dimension3": "string"
    }
}
```

For example, you can move two bikes that have the dimensions [site=1, location=11, color=red] from allocation group [Online, VIP, US] to allocation group [Online, VIP, EU] by calling the `Reallocate` API and providing the following body text.

JSON

```
{
    "id": "test102",
    "productId": "Bike",
    "sourceGroups": {
        "channel": "Online",
        "customerGroup": "VIP",
        "region": "US"
    },
    "groups": {
        "channel": "Online",
        "customerGroup": "VIP",
        "region": "EU"
    },
    "quantity": 2,
    "organizationId": "usmf",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "colorId": "red"
    }
}
```

Consume

Use the `Consume` API to post the consumption quantity against allocation. For example, you could use this API to move allocated quantity to some real measures. Here's the schema for the request body.

JSON

```
{
    "id": "string",
    "productId": "string",
    "dimensionDataSource": "string",
    "groups": {
        "groupA": "string",
        "groupB": "string",
        "groupC": "string"
    },
    "quantity": decimal,
    "organizationId": "string",
    "dimensions": {
        "dimension1": "string",
        "dimension2": "string",
        "dimension3": "string"
    },
    "physicalMeasures": {
        "datasource1": {
            "measure": "string" // Addition or Subtraction
        }
    }
}
```

For example, there are eight allocated bikes that have the dimensions [site=1, location=11, color=red] for allocation group [Online, VIP, US]. The following available-to-allocate formula is used:

`@iv.@available_to_allocate` = `fno.onordered` + `pos.inbound` − `@iv.@allocated`

The eight bikes are allocated from the `pos.inbound` measure.

Now, three bikes are sold, and they're taken from the allocation pool. To register this move, you can make a call that has the following request body.

JSON

```
{
    "id": "test103",
    "organizationId": "usmf",
    "productId": "Bike",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "colorId": "red"
    },
    "groups": {
        "channel": "Online",
        "customerGroup": "VIP",
        "region": "US"
    },
```

```
    "quantity": 3,
    "physicalMeasures": {
        "pos": {
            "inbound": "Subtraction"
        }
    }
}
```

After this call, the allocated quantity for the product will be reduced by 3. In addition, Inventory Visibility will generate an on-hand change event where `pos.inbound` = *-3*. Alternatively, you can keep the `pos.inbound` value as is, and just consume the allocated quantity. However, in this case, you must either create another physical measure to keep the consumed quantities or use the predefined measure `@iv.@consumed`.

In this request, notice that the physical measure you use in the consume request body should use the opposite modifier type (Addition or Subtraction), compared with the modifier type used in the calculated measure. So in this consume body, `iv.inbound` has the value `Subtraction`, not `Addition`.

The `fno` data source can't be used in the consume body as we always claimed that Inventory Visibility can't change any data for the `fno` data source. The data flow is one-way, which means that all quantity changes for the `fno` data source must come from your Supply Chain Management environment.

Consume as a soft reservation

The `Consume` API can also consume the allocated quantity as a soft reservation. In this case, the `Consume` operation will reduce the allocated quantity and then create a soft reservation for that quantity. To use this approach, you must also be using the soft reservation feature of Inventory Visibility.

For example, you've set a soft reservation physical measure as `iv.softreserved`. The following formula is used for the available-to-reserve calculated measure:

`iv.available_to_reserve` = `fno.onordered` + `pos.inbound` – `iv.softreserved`

To use this setup with the allocation feature, add `@iv.@allocated` to `iv.available_to_reserve` to produce the following formula:

`iv.available_to_reserve` = `fno.onordered` + `pos.inbound` – `iv.softreserved` – `@iv.@allocated`

Then update `@iv.@available_to_allocate` to the same value.

When you want to consume a quantity of 3 and directly reserve this quantity, you can make a call that has the following request body.

JSON

```
{
    "id": "???",
    "organizationId": "usmf",
    "productId": "Bike",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "colorId": "red"
    },
    "groups": {
        "channel": "Online",
        "customerGroup": "VIP",
        "region": "US"
    },
    "quantity": 3,
    "physicalMeasures": {
        "iv": {
            "softreserved": "Addition"
        }
    }
}
```

In this request, notice that `iv.softreserved` has the value `Addition`, not `Subtraction`.

Query

Use the `Query` API to retrieve allocation related information for some products. You can use dimension filters and allocation group filters to narrow down the results. *The dimensions must match exactly the one you want to retrieve*, for example, [site=1, location=11] will have unrelated results compared with [site=1, location=11, color=red].

JSON

```
{
    "productId": "string",
    "organizationId": "string",
    "dimensions": {
        "dimension1": "string",
        "dimension2": "string",
        "dimension3": "string"
    },
    "groups": {
        "additionalProp1": "string",
        "additionalProp2": "string",
```

```
        "additionalProp3": "string"
    },
}
```

For example, use [site=1, location=11, color=red] and empty groups field to get all allocation records:

JSON

```
{
    "organizationId": "usmf",
    "productId": "Bike",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "colorId": "red"
    },
    "groups": {},
}
```

Use [site=1, location=11, color=red] and groups [channel=Online, customerGroup=VIP, region=US] to get allocation records for this group:

JSON

```
{
    "organizationId": "usmf",
    "productId": "Bike",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "colorId": "red"
    },
    "groups": {
        "channel": "Online",
        "customerGroup": "VIP",
        "region": "US"
    },
}
```

Use the allocation user interface

You can manually manage allocations by using the Inventory Visibility app in Power Apps.

> ⓘ **Important**

In the current version of the Inventory Visibility app, you can manage allocations only when you're using Inventory Visibility UI version 1. If you're using UI version 2, you must use the APIs to manage allocations. For more information about the two UI versions and how to switch between them, see Inventory Visibility app user interface versions.

Open the allocation user interface

The **Allocation** tab on the **Operational visibility** page serves as the user interface where you can create, consume, reallocate, and query allocations, as described in later subsections.

To open the allocation user interface, follow these steps.

1. Sign in to your Power Apps environment, and open the **Inventory Visibility** app.
2. On the **Change area** menu at the bottom of the navigation pane, select **Legacy UI**.
3. On the navigation pane, select **Operational visibility**.
4. Select the **Allocation** tab.

Create an allocation

Follow these steps to create an allocation in the Inventory Visibility app.

1. On the toolbar of the **Allocation** tab, select **Allocate**.
2. Set the **Base fields**, **Dimensions**, and **Target allocation groups** values. When you're selecting values in the **Dimensions** section, first select a data source, then select a dimension from that data source, and then enter dimension values in the fields that appear each time you add a dimension.
3. Select **Submit**.

Consume an allocation

On the toolbar of the **Allocation** tab, select **Consume** to consume an allocation. To ensure that you consume within the correct allocation group and hierarchy, enter the same sets of organization and dimension details that you specified when you created the allocation.

Reallocate an allocation

On the toolbar of the **Allocation** tab, select **Reallocate** to move existing allocated quantity from one set of allocation groups to another.

Query existing allocations

On the toolbar of the **Allocation** tab, select **Query**, and then enter product, organization, dimension, and allocation group values to obtain query results of existing allocations.

Inventory Visibility on-hand change schedules and ATP

06/17/2025

This article describes how to set up the *On-hand change schedule* feature to schedule future on-hand changes and calculate available-to-promise (ATP) quantities. ATP is the quantity of an item that's available and can be promised to a customer in the next period. Use of this calculation can greatly increase your order fulfillment capability.

For many manufacturers, retailers, or sellers, it isn't enough just to know what's currently on hand. They must have full visibility into future availability. This future availability should consider future supply, future demand, and ATP.

Calculated measures for ATP quantities

The *ATP calculated measure* is a predefined calculated measure that's typically used to find the on-hand quantity that's currently available. The *supply quantity* is the sum of quantities for those physical measures that have a modifier type of *addition*, and the *demand quantity* is the sum of quantities for those physical measures that have a modifier type of *subtraction*.

You can add multiple calculated measures to calculate multiple ATP quantities. However, the total number of distinct physical measures across all ATP calculated measures should be less than nine.

> **Important**
>
> A calculated measure is a composition of physical measures. Its formula can include only physical measures without duplicates, not calculated measures.

For example, you set up the following calculated measure:

On-hand-available = (*PhysicalInvent* + *OnHand* + *Unrestricted* + *QualityInspection* + *Inbound*) – (*ReservPhysical* + *SoftReservePhysical* + *Outbound*)

The sum (*PhysicalInvent* + *OnHand* + *Unrestricted* + *QualityInspection* + *Inbound*) represents supply, and the sum (*ReservPhysical* + *SoftReservePhysical* + *Outbound*) represents demand. Therefore, the calculated measure can be understood in the following way:

On-hand-available = *Supply – Demand*

You can add another calculated measure to calculate the **On-hand-physical** ATP quantity.

On-hand-physical = (*PhysicalInvent* + *OnHand* + *Unrestricted* + *QualityInspection* + *Inbound*) – (*Outbound*)

There are eight distinct physical measures across those two ATP calculated measures: *PhysicalInvent*, *OnHand*, *Unrestricted*, *QualityInspection*, *Inbound*, *ReservPhysical*, *SoftReservePhysical*, and *Outbound*.

For more information about calculated measures, see Calculated measures.

Turn on and set up on-hand change scheduling and ATP in UI version 2

This section applies when you're using Inventory Visibility UI version 2.

Before you can use ATP, you must set up one or more calculated measures to calculate the ATP quantities. You must also turn on the feature and configure ATP settings in Microsoft Power Apps.

Follow these steps to turn on the *On-hand change schedule* feature in Power Apps and configure the ATP settings.

1. Sign in to Power Apps, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Feature management**.
3. On the **Available to promise** tile, select **Manage**.
4. Set the **Enable feature** option to *True* to turn on the ATP feature.
5. Set the **Max schedule period (days)** field to the number of days that users can view and submit scheduled on-hand changes. Users who query for stock information will get the on-hand quantity, scheduled on-hand changes, and ATP for each day in the defined period, starting from the current date. The maximum value for this field is 180 days. By default, it's set to 30 days. Therefore, you can schedule changes for up to 30 days from today.

   > **ⓘ Important**
   >
   > The schedule period includes the current date. Therefore, users can schedule on-hand changes to occur any time from the current date (the day when the change is submitted) through (schedule period – 1) days in the future.

6. In the **Schedule Measures** section, set up schedule measures. You can use existing calculated measures as schedule measures, or you can create new ones. When you query Inventory Visibility, the ATP value is provided for defined calculated measures, based on the scheduled changes of component physical measures. In the **Schedule Measures** section, select **New OnHand Change Schedule Configuration V2** on the toolbar to add a new calculated measure binding for ATP. The calculated measure is what you want to use to find the currently available on-hand quantity. For information about how to create a calculated measure, see Calculated measures.

   > **Important**
   >
   > The default ATP calculated formula is for reference. You can modify and add other data sources and physical measures to set up the correct ATP calculation for your business.

7. In the **ATP Index Set Configuration** section, set up your ATP index. The ATP index resembles the *product index hierarchy* that lets you group query results by specific dimensions. For example, if you set *ColorId* and *SizeId* as your ATP index set, query results will be grouped by color and size. You can have multiple index sets.

   > **Important**
   >
   > The default *ColorId* and *SizeId* index is for reference. You can remove dimensions and add other dimensions.

8. Select **Save**.

9. When you've finished configuring all the required settings, select **Update Configuration** under **Admin Settings** on the navigation pane.

Learn more in Complete and update the configuration.

How the on-hand change schedule and ATP calculations work

An *on-hand change schedule* establishes expected dates and quantities of scheduled on-hand changes. You can submit an on-hand change schedule to Inventory Visibility, provided that the dates are within the period that's defined by the **Schedule period** setting (see the Enable and set up the features section of this article). Users who query for stock information will get the on-hand quantity, scheduled on-hand changes, and ATP for each day in that period.

Scheduled changes are initially uncommitted and therefore don't affect your actual on-hand quantities in the system. To commit the changes, you must submit an *on-hand change event*, which updates the actual available on-hand quantity. You must then revert the scheduled change by submitting an on-hand change schedule for a matching negative quantity.

For example, you place an order for 10 bikes and expect it to arrive tomorrow. Therefore, you submit an on-hand change schedule that has an inbound quantity of 10 and is dated for tomorrow. When the order arrives the next day, you add the bikes to your physical on-hand inventory. You must then commit the change to your system to update the actual on-hand quantity. To commit the change, you submit an on-hand change event that has an inbound quantity of 10. You then revert the scheduled change by submitting an on-hand change schedule that has an inbound quantity of -10.

When you query Inventory Visibility for on-hand and ATP quantities, it returns the following information for each day in the schedule period:

- **Date** – The date that the result applies to. The time zone is Coordinated Universal Time (UTC).
- **On-hand quantity** – The actual on-hand quantity for the specified date. This calculation is made according to the ATP calculated measure that's configured for Inventory Visibility.
- **Scheduled supply** – The sum of all scheduled inbound quantities that haven't become physically available for immediate consumption or shipment as of the specified date.
- **Scheduled demand** – The sum of all scheduled outbound quantities that haven't been consumed or shipped as of the specified date.
- **ATP quantity** – The minimum projected on-hand quantity that's available from the specified date through the end of the schedule period. This quantity includes all scheduled quantity adjustments. It's the maximum quantity that can be promised on the current date for delivery or consumption on that day.

For example, if the current date is February 1, 2022, and the schedule period is 7, users can submit scheduled on-hand changes that are expected to occur from February 1 through February 7, 2022. In this case, the ATP quantity for February 3, for example, is calculated based on the on-hand quantity for that day and the scheduled quantities from February 3 through February 7.

Example

The following example shows how a series of scheduled quantity changes affects the on-hand and ATP quantities that Inventory Visibility reports. It also shows how to commit a scheduled change, how a committed schedule change affects the results, and what can occur if you don't commit a scheduled change.

The results in this example show a *projected on-hand* value. This value incorporates all scheduled updates for illustration purposes but isn't actually reported when you query Inventory Visibility.

1. The following settings are configured for your system on the **ATP setting** page of the Inventory Visibility app in Power Apps:

    - **Schedule measures** – A calculated measure that's named *On-hand* is added here. It's calculated as *On-hand = Supply – Demand*.
    - **Max schedule period (days)** – The value is set to *7*.
    - **ATP index set configuration** – *ColorId* and *SizeId* are added here.

2. The following conditions also apply:

    - The current date is February 1, 2022.
    - The current on-hand quantity is 20.

3. For the current date (February 1, 2022), you submit a scheduled demand quantity of 3 to Inventory Visibility. Therefore, the projected on-hand quantity is 17. The following table shows the result.

Expand table

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-01	20		3	17	17
2022-02-02	20			17	17
2022-02-03	20			17	17
2022-02-04	20			17	17
2022-02-05	20			17	17
2022-02-06	20			17	17
2022-02-07	20			17	17

4. On the current date (February 1, 2022), you submit a scheduled supply quantity of 10 for February 3, 2022. The following table shows the result.

Expand table

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-01	20		3	17	17

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-02	20			17	17
2022-02-03	20	10		27	27
2022-02-04	20			27	27
2022-02-05	20			27	27
2022-02-06	20			27	27
2022-02-07	20			27	27

5. On the current date (February 1, 2022), you submit the following scheduled quantity changes:

   - Demand quantity of 15 for February 4, 2022
   - Supply quantity of 1 for February 5, 2022
   - Supply quantity of 3 for February 6, 2022

   The following table shows the result.

Expand table

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-01	20		3	17	12
2022-02-02	20			17	12
2022-02-03	20	10		27	12
2022-02-04	20		15	12	12
2022-02-05	20	1		13	13
2022-02-06	20	3		16	16
2022-02-07	20			16	16

6. On the current date (February 1, 2022), you ship the scheduled demand quantity of 3. Therefore, you must commit this change so that it's reflected in your actual on-hand quantity. To commit the change, you submit an on-hand change event that has an outbound quantity of 3. You then revert the scheduled change by submitting an on-hand change schedule that has an outbound quantity of -3. The following table shows the result.

Expand table

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-01	17		0	17	12
2022-02-02	17			17	12
2022-02-03	17	10		27	12
2022-02-04	17		15	12	12
2022-02-05	17	1		13	13
2022-02-06	17	3		16	16
2022-02-07	17			16	16

7. On the next day (February 2, 2022), the schedule period shifts forward by one day. The following table shows the result.

Expand table

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-02	17			17	12
2022-02-03	17	10		27	12
2022-02-04	17		15	12	12
2022-02-05	17	1		13	13
2022-02-06	17	3		16	16
2022-02-07	17			16	16
2022-02-08	17			16	16

8. However, two days later (February 4, 2022), the supply quantity of 10 that was scheduled for February 3 still hasn't arrived. The following table shows the result.

Expand table

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-04	17		15	2	2
2022-02-05	17	1		3	3

Date	On-hand	Scheduled supply	Scheduled demand	Projected on-hand	ATP
2022-02-06	17	3		6	6
2022-02-07	17			6	6
2022-02-08	17			6	6
2022-02-09	17			6	6
2022-02-10	17			6	6

As you see, the scheduled (but not committed) on-hand changes don't affect the actual on-hand quantity.

Submit change schedules, change events, and ATP queries through the API

You can use the following application programming interface (API) URLs to submit on-hand change schedules, change events, and queries.

Expand table

Path	Method	Description
`/api/environment/{environmentId}/onhand/changeschedule`	`POST`	Create one scheduled on-hand change.
`/api/environment/{environmentId}/onhand/changeschedule/bulk`	`POST`	Create multiple scheduled on-hand changes.
`/api/environment/{environmentId}/onhand`	`POST`	Create one on-hand change event.
`/api/environment/{environmentId}/onhand/bulk`	`POST`	Create multiple change events.
`/api/environment/{environmentId}/onhand/indexquery`	`POST`	Query by using the `POST` method.
`/api/environment/{environmentId}/onhand`	`GET`	Query by using the `GET` method.
`/api/environment/{environmentId}/onhand/exactquery`	`POST`	Exact query by using the `POST` method.

Learn more in Inventory Visibility public APIs.

Create one on-hand change schedule

An on-hand change schedule is created by submitting a `POST` request to the relevant Inventory Visibility service URL. You can also submit bulk requests.

An on-hand change schedule can be created only if the scheduled date is between the current date and the end of the current schedule period. The datetime format should be *year-month-day* (for example, **2022-02-01**). The time format must be accurate only to the day.

This API creates a single on-hand change schedule.

txt

```
Path:
    /api/environment/{environmentId}/onhand/changeschedule
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        id: string,
        organizationId: string,
        productId: string,
        dimensionDataSource: string, # optional
        dimensions: {
            [key:string]: string,
        },
        quantitiesByDate: {
            [datetime:datetime]: {
                [dataSourceName:string]: {
                    [key:string]: number,
                },
            },
        },
    }
```

The following example shows sample body content without `dimensionDataSource`.

JSON

```
{
    "id": "id-bike-0001",
    "organizationId": "usmf",
    "productId": "Bike",
    "dimensions": {
        "SiteId": "1",
```

```
        "LocationId": "11",
        "ColorId": "Red",
        "SizeId": "Small"
    },
    "quantitiesByDate": {
        "2022-02-01": {
            "pos": {
                "inbound": 10
            }
        }
    }
}
```

Create multiple on-hand change schedules

This API can create multiple records at the same time. The only differences between this API and the single-event API are the `Path` and `Body` values. For this API, `Body` provides an array of records. The maximum number of records is 512. Therefore, the on-hand change schedule bulk API can support up to 512 scheduled changes at a time.

txt

```
Path:
    /api/environment/{environmentId}/onhand/changeschedule/bulk
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    [
        {
            id: string,
            organizationId: string,
            productId: string,
            dimensionDataSource: string,
            dimensions: {
                [key:string]: string,
            },
            quantityDataSource: string, # optional
            quantitiesByDate: {
                [datetime:datetime]: {
                    [dataSourceName:string]: {
                        [key:string]: number,
                    },
                },
            },
        },
```

```
        ...
    ]
```

The following example shows sample body content.

JSON

```
[
    {
        "id": "id-bike-0001",
        "organizationId": "usmf",
        "productId": "Bike",
        "dimensions": {
            "SiteId": "1",
            "LocationId": "11",
            "ColorId": "Red",
            "SizeId": "Small"
        },
        "quantitiesByDate": {
            "2022-02-01": {
                "pos": {
                    "inbound": 10
                }
            }
        }
    },
    {
        "id": "id-car-0002",
        "organizationId": "usmf",
        "productId": "Car",
        "dimensions": {
            "SiteId": "1",
            "LocationId": "11",
            "ColorId": "Red",
            "SizeId": "Small"
        },
        "quantitiesByDate": {
            "2022-02-05": {
                "pos": {
                    "outbound": 10
                }
            }
        }
    }
]
```

Create on-hand change events

On-hand change events are made by submitting a `POST` request to the relevant Inventory Visibility service URL (see the start of the Submit change schedules, change events, and ATP

queries through the API section).

> **Note**
>
> On-hand change events aren't unique to the ATP functionality but are part of the standard Inventory Visibility API. This example has been included because events are relevant when you work with ATP. On-hand change events resemble on-hand change reservations, but event messages must be sent to a different API URL, and events use `quantities` instead of `quantityByDate` in the message body. For more information about the on-hand change events and other features of the Inventory Visibility API, see Inventory Visibility public APIs.

The following example shows a request body that contains a single on-hand change event.

JSON

```
{
    "id": "id-bike-0001",
    "organizationId": "usmf",
    "productId": "Bike",
    "dimensions": {
        "SiteId": "1",
        "LocationId": "11",
        "SizeId": "Big",
        "ColorId": "Red"
    },
    "quantities": {
        "pos": {
            "inbound": 10.0
        }
    }
}
```

Query the scheduled on-hand changes and ATP results

You can query scheduled on-hand changes and ATP results by submitting either a `POST` request or a `GET` request to the appropriate API URL (see the Submit change schedules, change events, and ATP queries through the API section).

In your request, set `QueryATP` to *true* if you want to query scheduled on-hand changes and ATP results. By default, the query returns all ATP-related data from today. You can specify `ATPFromDate` and `ATPToDate` to narrow the results. (The "to" and "from" dates just filter the

result. They don't affect how ATP is calculated.) The value of the `groupByValues` parameter must exactly match one of your ATP index sets.

- If you're submitting the request by using the `GET` method, set these parameters in the URL.
- If you're submitting the request by using the `POST` method, set these parameters in the request body.

> **Note**
>
> Regardless of whether the `returnNegative` parameter is set to *true* or *false* in the request body, the result will include negative values when you query for scheduled on-hand changes and ATP results. These negative values will be included because, if only demand orders are scheduled, or if supply quantities are less than demand quantities, the scheduled on-hand change quantities will be negative. If negative values weren't included, the results would be confusing. For more information about this option and how it works for other types of queries, see Inventory Visibility public APIs.

> **Important**
>
> Scheduled on-hand changes are stored exclusively in Dataverse, which means that queries for scheduled on-hand changes and ATP results are subject to Dataverse's performance and data access limits. To avoid exceeding these limits and ensure optimal performance, apply targeted filters to your queries to minimize the volume of data retrieved.

Query by using the POST method

txt

```
Path:
    /api/environment/{environmentId}/onhand/indexquery
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        dimensionDataSource: string, # Optional
        filters: {
            organizationId: string[],
```

```
            productId: string[],
            siteId: string[],
            locationId: string[],
            [dimensionKey:string]: string[],
        },
        groupByValues: string[],
        returnNegative: boolean,
    }
```

The following example shows how to create an index query request body that can be submitted to Inventory Visibility by using the `POST` method.

JSON

```
{
    // OnHand Index Query fields
    "filters": {
        "organizationId": ["usmf"],
        "productId": ["Bike"],
        "SiteId": ["1"],
        "LocationId": ["11"]
    },
    "groupByValues": ["ColorId", "SizeId"],
    "returnNegative": true,

    // ATP related fields
    "QueryATP":true,
    "ATPFromDate": "2022-02-01",
    "ATPToDate": "2022-02-10",
}
```

Query by using the GET method

txt

```
Path:
    /api/environment/{environmentId}/onhand
Method:
    Get
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Query(Url Parameters):
    groupBy
    returnNegative
    [Filters]
```

The following example shows how to create an index query request URL as a `GET` request.

txt

```
https://inventoryservice.{RegionShortName}-
il301.gateway.prod.island.powerapps.com/api/environment/{EnvironmentId}/onhand?
organizationId=usmf&productId=Bike&SiteId=1&LocationId=11&groupBy=ColorId,SizeId&r
eturnNegative=true&QueryATP=true&ATPToDate=2022-02-01&ATPToDate=2022-02-10
```

The result of this `GET` request is exactly the same as the result of the `POST` request in the previous example.

Exact query by using the POST method

To set up an exact query, add ATP-related fields to the query body. Learn more in Exact query by using the post method.

JSON

```
{
    // Exact query fields
    // ...

    // ATP related fields
    "QueryATP":true,
    "ATPFromDate": "2022-02-01",
    "ATPToDate": "2022-02-10",
}
```

Query result example

Any of the previous query examples might produce the following reply. For this example, the system is configured with the following settings:

- **ATP calculated measure:** *iv.onhand = pos.inbound – pos.outbound*
- **Schedule period:** *7*

Here's an example of the reply body.

JSON

```
[
    {
        "quantitiesByDate": {
            "2022-02-02T00:00:00": {
                "pos": {
```

```
                    "outbound": 5,
                    "inbound": 0,
                },
                "iv": {
                    "onhand": -5,
                },
            },
            "2022-02-06T00:00:00": {
                "pos": {
                    "inbound": 7,
                    "outbound": 0,
                },
                "iv": {
                    "onhand": 7,
                },
            }
        },
        "atpQuantities": {
            "2022-02-01T00:00:00Z": {
                "iv": {
                    "onhand": 5.0
                }
            },
            "2022-02-02T00:00:00Z": {
                "iv": {
                    "onhand": 5.0
                }
            },
            "2022-02-03T00:00:00Z": {
                "iv": {
                    "onhand": 5.0
                }
            },
            "2022-02-04T00:00:00Z": {
                "iv": {
                    "onhand": 5.0
                }
            },
            "2022-02-05T00:00:00Z": {
                "iv": {
                    "onhand": 5.0
                }
            },
            "2022-02-06T00:00:00Z": {
                "iv": {
                    "onhand": 12.0
                }
            },
            "2022-02-07T00:00:00Z": {
                "iv": {
                    "onhand": 12.0
                }
            }
        },
        "productId": "Bike ",
```

```
        "dimensions": {
            "ColorId": "Red",
            "SizeId": "Big",
            "siteid": "1",
            "locationid": "11"
        },
        "quantities": {
            "pos": {
                "inbound": 10.0,
                "outbound": 0,
            },
            "iv": {
                "onhand": 10.0,
            }
        }
    }
]
```

Track time-series inventory in Inventory Visibility

The Inventory Visibility service enables external systems to query Microsoft Dynamics 365 Supply Chain Management for inventory changes that will occur up to 180 days in the future. In this way, users who work in an external system can view future inventory availability details and make accurate projections about when orders can be delivered.

The Inventory Visibility service can import information about planned inbound and outbound inventory changes together with the related dates. Available-to-promise (ATP) functionality in Inventory Visibility can then query those time-series based inventory changes and calculate your omnichannel ATP. External systems can query and retrieve this information from Inventory Visibility in near-real time by calling its API.

The *Inventory Visibility integration with ATP* feature that is described in this article adds support for out-of-box ATP integration between Supply Chain Management and Inventory Visibility. Therefore, you can easily bring expected inventory issue and receipts together with time-series information from Supply Chain Management into Inventory Visibility. You can also post updated scheduling information back to Supply Chain Management.

Prerequisites

Before you can use the feature that is described in this article, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.41 or later.
- The feature named *Inventory Visibility integration with ATP* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.
- You must be running the latest version of Inventory Visibility. Learn how to install Inventory Visibility, view version information, and check for updates in Install and set up Inventory Visibility.

Enable Inventory Visibility integration with ATP in Supply Chain Management

To configure Supply Chain Management to exchange ATP information with Inventory Visibility, follow these steps:

1. Sign in to Supply Chain Management.
2. Go to **Inventory Management** > **Inventory Visibility** > **Inventory Visibility integration with ATP**.
3. On the Action Pane, select **Enable** to enable the integration. (If you must ever disable the integration, you can select **Disable** here.)

Default calculated measure for ATP in Supply Chain Management

Inventory Visibility includes a default calculated measure that is named `@iv.@atp`. This calculated measure includes all expected issues and receipts from Supply Chain Management. You typically use it to query Supply Chain Management ATP information in Inventory Visibility. You can customize it as you require. Learn how to work with calculated measures in Calculated measures.

Set up ATP in Inventory Visibility

When ATP in Inventory Visibility is integrated with Supply Chain Management, it works just as it does when it's integrated with other systems. Learn how to set up and use ATP in Inventory Visibility in Inventory Visibility on-hand change schedules and available to promise.

Work with ATP data in the Inventory Visibility app in Power Apps

When you use the Inventory Visibility app in Power Apps, you can query and post ATP information.

- To query ATP information, go to **Operational visibility** > **On hand query**. Be sure to enable the **Query ATP** option.
- To post ATP information, go to **Operational visibility** > **Change schedule**. There, you can specify and submit updated schedule information.

Learn more in Use the Inventory Visibility app UI version 2.

Query and post ATP data using the Inventory Visibility API

You can use the Inventory Visibility API to query and post ATP information. Learn more in Submit change schedules, change events, and ATP queries through the API.

Last updated on 07/27/2026

Inventory Visibility Power BI dashboard

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft provides a sample Microsoft Power BI dashboard file that you can download and connect to your Inventory Visibility instance and other data sources. In this way, you can visualize your inventory levels across regions, entities, and channels. Use the dashboard to track inventory movement, compare supply and demand portions, view the most popular and least popular products, and drive decisions about inventory risk mitigation.

The dashboard lets you perform the following tasks:

- Filter on inventory dimensions to focus on the regions and products that matter most to you.
- Identify out-of-stock, understocked, and overstocked products for timely inventory replenishment or reduction.
- View inventory key performance indicators (KPIs), such as on-hand quantity, supply, and demand, to quickly monitor your inventory status.
- Tailor the sample dashboard to your organization's business needs.

License requirements

Because the dashboard is a standard Power BI file (.pbix file), you must have a Power BI license to use it. Learn more in Power BI licensing guide.

Download the dashboard and connect it to Inventory Visibility

1. Download the Inventory Visibility dashboard file from Inventory Visibility samples folder on GitHub.

2. For all the data that you plan to view by using the Power BI dashboard, configure *preloaded on-hand queries* in Inventory Visibility. For instructions, see Preload a streamlined on-hand query.

3. Connect the Power BI report to your Dataverse environment by following these steps:
   a. Open the downloaded report in Power BI desktop.
   b. Expand the **Data** pane. Select and hold (or right-click) the **Query Preload Results** table, and then select **Edit query** to open the Power Query editor.
   c. Replace the value of the `EnvironmentURL` parameter with the base URL of your Dataverse environment (for example, `example.crm.dynamics.com`).
   d. Refresh the query to update the results.

4. To load data from your entities, create a new data source, and name it *pbi* to indicate that it's your Power BI data source. (Learn more in Configure Inventory Visibility.) To reduce the size of the response body, information from this data source is excluded from results.

Default KPIs

To use the visuals and KPIs provided in the sample dashboard, define the following calculated measures in the `pbi` data source. Add the relevant physical measures from your data sources to these calculated measures.

- `Totalonhand` – The total physical inventory stock that you have in your warehouses.
- `Totalavailablephysical` – The total available physical inventory quantities.
- `Totalavailable` – The total quantities that are available. This total includes both available physical inventory and available ordered inventory, minus the on-order quantity that hasn't been promised.
- `Totalordered` – The total quantity that has been purchased or produced, or that's otherwise inbound.
- `totalonorder` – The total quantities that are on order.
- `Hardreserved` The total hard-reserved quantities. This total includes both reserve physical quantities and reserve ordered quantities.
- `Returned` – The total returned quantities. Make sure that one of your data sources includes a physical measure for returns, and that this physical measure is added to the `Returned` calculated measure.
- `Totalsold` – The total quantities that have been sold.
- `Othersupply` – Supply quantities from other user-specified sources that aren't included in the previously mentioned measures.

- `Otherdemand` – Demand quantities from other user-specified sources that aren't included in the previously mentioned measures.
- `Totalsoftreserved` – The total soft-reserved quantities.
- `Received` – The total inventory quantity received into your warehouses.

These sample calculated measures are included in the following formulas on the Power BI report:

- On the **Supply by Product** dashboard: *Total supply* = `Totalonhand` + `Totalordered` + `Othersupply`
- On the **Demand by Product** dashboard: *Total demand* = `Totalonorder` + `Totalsoftreserved` + `Totalreserved` + `Otherdemand`

Customize the dashboard

Customize the sample dashboard by directly editing the visuals. You can also edit the query to show different measures from different data sources.

To add a new data source, follow these steps:

1. Open the dashboard in Power BI desktop, and expand the **Data** pane. Select and hold (or right-click) the **Query Preload Results** table, and then select **Edit query** to open the Power Query editor.
2. On the ribbon of the Power Query editor, on the **Home** tab, select **Refresh Preview**.
3. In the **Query Settings** pane, in the **Applied Steps** section, select and hold (or right-click) the **Expanded is_quantities** step, and then select **Edit settings**.
4. Search for or select the data source that you want to add to the report.

To add, modify, or delete a column for an existing data source, follow these steps:

1. Open the dashboard in Power BI desktop, and expand the **Data** pane. Select and hold (or right-click) the **Query Preload Results** table, and then select **Edit query** to open the Power Query editor.
2. On the ribbon of the Power Query editor, on the **Home** tab, select **Refresh Preview**.
3. In the **Query Settings** pane, in the **Applied Steps** section, find the **Expanded is_quantities.<data-source>** step, where **<data-source>** is the data source that the new column belongs to.
4. Select and hold (or right-click) the step, and then select **Edit settings**.
5. Search for or select the name of the column that you want to add to the report.

You can also define new columns, based on the retrieved entity. In the sample report, the columns for total supply and total demand are calculated columns that are defined in Power BI.

If you're familiar with Power Query, directly edit the query by using the Power Query language. On the ribbon of the Power Query editor, on the **Home** tab, select **Advanced Editor** to open the advanced editor.

Last updated on 06/22/2026

Inventory Visibility public APIs

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes the public APIs that are provided by Inventory Visibility.

The public REST API of the Inventory Visibility Add-in presents several specific endpoints for integration. It supports four main interaction types:

- Posting on-hand inventory changes to the add-in from an external system
- Setting or overriding on-hand inventory quantities in the add-in from an external system
- Posting reservation events to the add-in from an external system
- Querying current on-hand quantities from an external system

The following table lists the APIs that are currently available:

Expand table

Path	Method	Description
`/api/environment/{environmentId}/onhand`	Post	Create one on-hand change event
`/api/environment/{environmentId}/onhand/bulk`	Post	Create multiple change events
`/api/environment/{environmentId}/setonhand/{inventorySystem}/bulk`	Post	Set/override on-hand quantities

Path	Method	Description
`/api/environment/{environmentId}/onhand/reserve`	Post	Create one soft reservation event
`/api/environment/{environmentId}/onhand/reserve/bulk`	Post	Create multiple soft reservation events
`/api/environment/{environmentId}/onhand/unreserve`	Post	Reverse one soft reservation event
`/api/environment/{environmentId}/onhand/unreserve/bulk`	Post	Reverse multiple soft reservation events
`/api/environment/{environmentId}/onhand/reserve/resyncjob`	Post	Clean up reservation data
`/api/environment/{environmentId}/getJobProgress`	Get	Get job execution progress
`/api/environment/{environmentId}/onhand/changeschedule`	Post	Create one scheduled on-hand change
`/api/environment/{environmentId}/onhand/changeschedule/bulk`	Post	Create multiple on-hand changes with dates
`/api/environment/{environmentId}/onhand/indexquery`	Post	Query by using the post method (recommended)
`/api/environment/{environmentId}/onhand`	Get	Query by using the get method
`/api/environment/{environmentId}/onhand/exactquery`	Post	Exact query by using the post method
`/api/environment/{environmentId}/allocation/allocate`	Post	Create one allocate event
`/api/environment/{environmentId}/allocation/unallocate`	Post	Create one unallocate event
`/api/environment/{environmentId}/allocation/reallocate`	Post	Create one reallocate event
`/api/environment/{environmentId}/allocation/consume`	Post	Create one consume event
`/api/environment/{environmentId}/allocation/query`	Post	Query allocation result

Path	Method	Description
`/api/environment/{environmentId}/onhand/productsearch/indexquery`	Post	Post index query with product search
`/api/environment/{environmentId}/onhand/productsearch/exactquery`	Post	Post exact query with product search
`/api/environment/{environmentId}/transaction/adjustment/bulk`	Post	Sync external inventory changes through Inventory Visibility

> **Note**
>
> The {environmentId} part of the path is the environment ID of Microsoft Dynamics 365 Supply Chain Management. This ID is the one stated for Supply Chain Management in Microsoft Dynamics Lifecycle Services, not the ID of the Power Platform environment that is linked to the Supply Chain Management environment.
>
> The bulk API can return a maximum of 512 records for each request.

Authentication

The platform security token is used to call the Inventory Visibility public API. Therefore, you must generate a *Microsoft Entra token* by using your Microsoft Entra application. You must then use the Microsoft Entra token to get the *access token* from the security service.

To get a security service token, follow these steps:

1. Sign in to the Azure portal, and use it to find the `clientId` and `clientSecret` values for your Dynamics 365 Supply Chain Management app.
2. Fetch a Microsoft Entra token (`aadToken`) by submitting an HTTP request that has the following properties:
   - **URL:** `https://login.microsoftonline.com/${aadTenantId}/oauth2/v2.0/token`
   - **Method:** `GET`
   - **Body content (form data):**

Expand table

Key	Value
client_id	${aadAppId}
client_secret	${aadAppSecret}
grant_type	client_credentials
scope	0cdb527f-a8d1-4bf8-9436-b352c68682b2/.default

You should receive a Microsoft Entra token (`aadToken`) in response. It should resemble the following example.

JSON

```
{
    "token_type": "Bearer",
    "expires_in": "3599",
    "ext_expires_in": "3599",
    "access_token": "eyJ0eX...8WQ"
}
```

3. Formulate a JavaScript Object Notation (JSON) request that resembles the following example.

JSON

```
{
    "grant_type": "client_credentials",
    "client_assertion_type": "aad_app",
    "client_assertion": "${Your_Microsoft_EntraToken}",
    "scope": "https://inventoryservice.operations365.dynamics.com/.default",
    "context": "${fno_environment_id}",
    "context_type": "finops-env"
}
```

Note the following points:

- The `client_assertion` value must be the Microsoft Entra token (`aadToken`) that you received in the previous step.
- The `context` value must be the Supply Chain Management environment ID where you want to deploy the add-in.
- Set all the other values as shown in the example.

4. Fetch an access token (`access_token`) by submitting an HTTP request that has the following properties:

- **URL:** `https://securityservice.operations365.dynamics.com/token`
- **Method:** `POST`
- **HTTP header:**

Expand table

Key	Value
Api-Version	1.0
Content-Type	application/json

- **Body content:** Include the JSON request that you created in the previous step.

You should receive an access token (`access_token`) in response. You must use this token as a bearer token to call the Inventory Visibility API. Here's an example.

JSON

```
{
    "access_token": "${Returned_Token}",
    "token_type": "bearer",
    "expires_in": 3600
}
```

> **Note**
>
> If you receive a response with status code 307, use the value of the `Location` header to resend the token request to the new URL. Most HTTP libraries handle redirects automatically.

Create on-hand change events

There are two APIs for creating on-hand change events:

- Create one record: `/api/environment/{environmentId}/onhand`
- Create multiple records: `/api/environment/{environmentId}/onhand/bulk`

The following table summarizes the meaning of each field in the JSON body.

Expand table

Field ID	Description
`id`	A unique ID for the specific change event. If a resubmission occurs due to a service failure, this ID is used to ensure the same event won't be counted twice in the system.
`organizationId`	The identifier of the organization that's linked to the event. This value is mapped to an organization or data area ID in Supply Chain Management.
`productId`	The identifier of the product.
`quantities`	The quantity that the on-hand quantity must be changed by. For example, if 10 new books are added to a shelf, this value will be `quantities:{ shelf:{ received: 10 }}`. If three books are removed from the shelf or sold, this value will be `quantities:{ shelf:{ sold: 3 }}`.
`dimensionDataSource`	The data source of the dimensions that are used in the posting change event and query. If you specify the data source, you can use the custom dimensions from the specified data source. Inventory Visibility can use the dimension configuration to map the custom dimensions to the general default dimensions. If no `dimensionDataSource` value is specified, you can use only the general base dimensions in your queries.
`dimensions`	A dynamic key-value pair. The values are mapped to some of the dimensions in Supply Chain Management. However, you can also add custom dimensions (for example, *Source*) to indicate whether the event is coming from Supply Chain Management or an external system.

> **Note**
>
> If your data partition rule is set to *By product ID*, `siteId` and `locationId` are optional dimensions. Otherwise, they're required dimensions. This rule also applies to the allocation, soft reserve, and change schedule APIs.

The following subsections provide examples that show how to use these APIs.

Create one on-hand change event

This API creates a single on-hand change event.

txt

```
Path:
    /api/environment/{environmentId}/onhand
Method:
    Post
Headers:
```

```
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        id: string,
        organizationId: string,
        productId: string,
        dimensionDataSource: string, # Optional
        dimensions: {
            [key:string]: string,
        },
        quantities: {
            [dataSourceName:string]: {
                [key:string]: number,
            },
        },
    }
```

The following example shows sample body content. In this example, the company has a point-of-sale (POS) system that processes in-store transactions and therefore inventory changes. The customer has returned a red T-shirt to your store. To reflect the change, you post a single change event for the *T-shirt* product. This event will increase the quantity of the *T-shirt* product by 1.

JSON

```
{
    "id": "Test201",
    "organizationId": "usmf",
    "productId": "T-shirt",
    "dimensionDataSource": "pos",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "posMachineId": "0001",
        "colorId": "red"
    },
    "quantities": {
        "pos": {
            "inbound": 1
        }
    }
}
```

The following example shows sample body content without `dimensionDataSource`. In this case, `dimensions` will be the base dimensions. If `dimensionDataSource` is set, `dimensions` can be either the data source dimensions or the base dimensions.

JSON

```
{
    "id": "Test202",
    "organizationId": "usmf",
    "productId": "T-shirt",
    "dimensions": {
        "siteId": "1",
        "locationId": "11",
        "colorId": "red"
    },
    "quantities": {
        "pos": {
            "inbound": 1
        }
    }
}
```

Create multiple change events

This API can create change events, just as the single-event API can. The only difference is that this API can create multiple records at the same time. Therefore, the `Path` and `Body` values differ. For this API, `Body` provides an array of records. The maximum number of records is 512. Therefore, the on-hand change bulk API can support up to 512 change events at a time.

For example, a retail store POS machine processed the following two transactions:

- One return order of one red T-shirt
- One sales transaction of three black T-shirts

In this case, you can include both inventory updates in one API call.

txt

```
Path:
    /api/environment/{environmentId}/onhand/bulk
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    [
        {
            id: string,
            organizationId: string,
            productId: string,
            dimensionDataSource: string, # Optional
```

```
            dimensions: {
                [key:string]: string,
            },
            quantities: {
                [dataSourceName:string]: {
                    [key:string]: number,
                },
            },
        },
        ...
    ]
```

The following example shows sample body content.

JSON

```
[
    {
        "id": "Test203",
        "organizationId": "usmf",
        "productId": "T-shirt",
        "dimensionDataSource": "pos",
        "dimensions": {
            "SiteId": "Site1",
            "LocationId": "11",
            "posMachineId": "0001"
            "colorId": "red"
        },
        "quantities": {
            "pos": { "inbound": 1 }
        }
    },
    {
        "id": "Test204",
        "organizationId": "usmf",
        "productId": "T-shirt",
        "dimensions": {
            "siteId": "1",
            "locationId": "11",
            "colorId": "black"
        },
        "quantities": {
            "pos": { "outbound": 3 }
        }
    }
]
```

Set/override on-hand quantities

The *Set on-hand* API overrides the current data for the specified product. This functionality is typically used to do inventory counting updates. For example, during its daily inventory counting, a store might find that the actual on-hand inventory for a red T-shirt is 100. Therefore, the POS inbound quantity must be updated to 100, regardless of what the previous quantity was. You can use this API to override the existing value.

txt

```
Path:
    /api/environment/{environmentId}/setonhand/{inventorySystem}/bulk
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    [
        {
            id: string,
            organizationId: string,
            productId: string,
            dimensionDataSource: string, # Optional
            dimensions: {
                [key:string]: string,
            },
            quantities: {
                [dataSourceName:string]: {
                    [key:string]: number,
                },
            },
            modifiedDateTimeUTC: datetime,
        },
        ...
    ]
```

The following example shows sample body content. The behavior of this API differs from the behavior of the APIs that are described in the Create on-hand change events section earlier in this article. In this sample, the quantity of the *T-shirt* product will be set to 1.

JSON

```
[
    {
        "id": "Test204",
        "organizationId": "usmf",
        "productId": "T-shirt",
        "dimensionDataSource": "pos",
        "dimensions": {
```

```
            "SiteId": "1",
            "LocationId": "11",
            "posMachineId": "0001"
            "colorId": "red"
        },
        "quantities": {
            "pos": {
                "inbound": 100
            }
        }
    }
]
```

Create reservation events

To use the *Reserve* API, you must turn on the reservation feature and complete the reservation configuration. For more information (including a dataflow and sample scenario), see Inventory Visibility reservations.

Create one reservation event

A reservation can be made against different data source settings. To configure this type of reservation, first specify the data source in the `dimensionDataSource` parameter. Then, in the `dimensions` parameter, specify dimensions according to the dimension settings in the target data source.

When you call the reservation API, you can control the reservation validation by specifying the Boolean `ifCheckAvailForReserv` parameter in the request body. A value of `True` means that the validation is required, whereas a value of `False` means that the validation isn't required. The default value is `True`.

If you want to reverse a reservation or unreserve specified inventory quantities, set the quantity to a negative value, and set the `ifCheckAvailForReserv` parameter to `False` to skip the validation. There's also a dedicated unreserve API to do the same. The difference is only in the way the two APIs are called. It's easier to reverse a specific reservation event by using `reservationId` with the *unreserve* API. Learn more in Unreserve one reservation event section.

txt

```
Path:
    /api/environment/{environmentId}/onhand/reserve
Method:
    Post
Headers:
    Api-Version="1.0"
```

```
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        id: string,
        organizationId: string,
        productId: string,
        dimensionDataSource: string,
        dimensions: {
            [key:string]: string,
        },
        quantityDataSource: string, # optional
        quantities: {
            [dataSourceName:string]: {
                [key:string]: number,
            },
        },
        modifier: string,
        quantity: number,
        ifCheckAvailForReserv: boolean,
    }
```

The following example shows sample body content.

JSON

```
{
    "id": "reserve-0",
    "organizationId": "SCM_IV",
    "productId": "iv_contoso_product",
    "quantity": 1,
    "quantityDataSource": "iv",
    "modifier": "softReservOrdered",
    "ifCheckAvailForReserv": true,
    "dimensions": {
        "siteId": "iv_contoso_site",
        "locationId": "iv_contoso_location",
        "colorId": "red",
        "sizeId": "small"
    }
}
```

The following example shows a successful response.

JSON

```
{
    "reservationId": "RESERVATION_ID",
    "id": "ohre~id-822-232959-524",
    "processingStatus": "success",
    "message": "",
```

```
    "statusCode": 200
}
```

Create multiple reservation events

This API is a bulk version of the single-event API.

txt

```
Path:
    /api/environment/{environmentId}/onhand/reserve/bulk
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    [
        {
            id: string,
            organizationId: string,
            productId: string,
            dimensionDataSource: string,
            dimensions: {
                [key:string]: string,
            },
            quantityDataSource: string, # optional
            quantities: {
                [dataSourceName:string]: {
                    [key:string]: number,
                },
            },
            modifier: string,
            quantity: number,
            ifCheckAvailForReserv: boolean,
        },
        ...
    ]
```

Reverse reservation events

The *Unreserve* API serves as the reverse operation for *Reservation* events. It provides a way to reverse a reservation event specified by `reservationId` or to decrease the reservation quantity.

Reverse one reservation event

When a reservation is created, a `reservationId` will be included in the response body. You must provide the same `reservationId` to cancel the reservation, and include the same `organizationId`, `productId`, and `dimensions` used for the reservation API call. Finally, specify an `OffsetQty` value that represents the number of items to be freed from the previous reservation. A reservation can either be fully or partially reversed depending on the specified `OffsetQty`. For example, if *100* units of items were reserved, you can specify `OffsetQty: 10` to unreserve *10* of the initial reserved amount.

txt

```
Path:
    /api/environment/{environmentId}/onhand/unreserve
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        id: string,
        organizationId: string,
        productId: string,
        reservationId: string,
        dimensions: {
            [key:string]: string,
        },
        OffsetQty: number
    }
```

The following code shows an example of body content.

JSON

```
{
    "id": "unreserve-0",
    "organizationId": "SCM_IV",
    "productId": "iv_contoso_product",
    "reservationId": "RESERVATION_ID",
    "dimensions": {
        "siteid":"iv_contoso_site",
        "locationid":"iv_contoso_location",
        "ColorId": "red",
        "SizeId": "small"
    },
    "OffsetQty": 1
}
```

The following code shows an example of a successful response body.

JSON

```
{
    "reservationId": "RESERVATION_ID",
    "totalInvalidOffsetQtyByReservId": 0,
    "id": "ohoe~id-823-11744-883",
    "processingStatus": "success",
    "message": "",
    "statusCode": 200
}
```

> **Note**
>
> In the response body, when `OffsetQty` is less than or equal to the reservation quantity, `processingStatus` will be "*success*" and `totalInvalidOffsetQtyByReservId` will be *0*.
>
> If `OffsetQty` is greater than the reserved amount, `processingStatus` will be "*partialSuccess*" and `totalInvalidOffsetQtyByReservId` will be the difference between `OffsetQty` and the reserved amount.
>
> For example, if the reservation has a quantity of *10*, and `OffsetQty` has a value of *12*, `totalInvalidOffsetQtyByReservId` would be *2*.

Reverse multiple reservation events

This API is a bulk version of the single-event API.

txt

```
Path:
    /api/environment/{environmentId}/onhand/unreserve/bulk
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    [
        {
            id: string,
            organizationId: string,
            productId: string,
```

```
            reservationId: string,
            dimensions: {
                [key:string]: string,
            },
            OffsetQty: number
        }
        ...
    ]
```

Clean up reservation data

The *clean up reservation data* API is used to clean up historical reservation data. The body should be a list of data sources. If the list is empty, all data sources will be cleaned up.

txt

```
Path:
    /api/environment/{environmentId}/onhand/reserve/resyncjob
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    [
        "iv",
        "pos"
    ]
```

If the cleanup job is successfully created, a job ID will be returned in the response, which can be used to get job execution progress.

Get job execution progress

Use the *get job execution progress* API to obtain the execution progress of a job. A job ID is required to identify the job.

txt

```
Path:
    /api/environment/{environmentId}/getJobProgress
Method:
    Get
Headers:
    Api-Version="1.0"
```

```
    Authorization="Bearer $access_token"
ContentType:
    application/json
Query(Url Parameters):
    jobId
```

Here's a sample get URL.

txt

```
/api/environment/{environmentId}/getJobProgress?jobId=SomeJob:xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
```

Query on-hand

Use the *Query on-hand* API to fetch current on-hand inventory data for your products. You can use this API whenever you must know the stock, such as when you want to review product stock levels on your e-commerce website, or when you want to check product availability across regions or in nearby stores and warehouses. The API currently supports querying up to 5,000 individual items by `productID` value. Multiple `siteID` and `locationID` values can also be specified in each query. When your data partition rule is set to *By location*, the maximum limit is defined by the following equation:

NumOf(SiteID) × NumOf(LocationID) <= 10,000.

Query by using the post method

The query by post API is available in two versions. The following table outlines the differences.

Expand table

API version 1.0	API version 2.0
Can only query one organization ID.	Can query multiple organization IDs.
Can query up to 10,000 combinations of sites and warehouses.	Can query more than 10,000 combinations of organization IDs, sites, and warehouses. Can return results in multiple pages.

The following subsections show how to use each API version.

Query by post API version 1.0

txt

```
Path:
    /api/environment/{environmentId}/onhand/indexquery
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        dimensionDataSource: string, # Optional
        filters: {
            organizationId: string[],
            productId: string[],
            siteId: string[],
            locationId: string[],
            [dimensionKey:string]: string[],
        },
        groupByValues: string[],
        returnNegative: boolean,
    }
```

In the body part of this request, `dimensionDataSource` is an optional parameter. If it isn't set, `filters` will be treated as *base dimensions*.

The `returnNegative` parameter controls whether the results contain negative entries.

Query data stored by location

This section applies when your data partition rule is set to *By location*.

- `organizationId` should be an array containing exactly one value.
- `productId` can contain one or more values. If it's an empty array, the system will return all products for the specified sites and locations. In this case, `siteId` and `locationId` shouldn't be empty.
- `siteId` and `locationId` are used for partitioning. You can specify more than one `siteId` and `locationId` value in a *Query on-hand* request. If both arrays are empty, the system will return all sites and locations for the specified products. In this case, `productId` shouldn't be empty.

We recommend to use the `groupByValues` parameter in a way that's consistent with your index configuration. Learn more in On-hand index configuration.

Query data stored by product ID

This section applies when your data partition rule is set to *By product ID*. In this case, two `filters` fields are required: `organizationId`, `productId`.

- `organizationId` should be an array containing exactly one value.
- `productId` should be an array with at least one value.

Unlike when storing data by location, if you don't specify values for `siteId` and `locationId`, inventory information for each product ID will be aggregated across all sites and/or locations.

> **Note**
>
> If you've enabled the on-hand change schedule and available-to-promise (ATP) features, your query can also include the `QueryATP` Boolean parameter, which controls whether the query results include ATP information. For more information and examples, see Inventory Visibility on-hand change schedules and available to promise.

The following example shows sample body content. It shows that you can query the on-hand inventory from multiple locations (warehouses).

JSON

```
{
    "dimensionDataSource": "pos",
    "filters": {
        "organizationId": ["usmf"],
        "productId": ["T-shirt"],
        "siteId": ["1"],
        "locationId": ["11","12","13"],
        "colorId": ["red"]
    },
    "groupByValues": ["colorId", "sizeId"],
    "returnNegative": true
}
```

The following example shows how to query all products in a specific site and location.

JSON

```
{
    "filters": {
        "organizationId": ["usmf"],
        "productId": [],
        "siteId": ["1"],
        "locationId": ["11"],
```

```
    },
    "groupByValues": ["colorId", "sizeId"],
    "returnNegative": true
}
```

Query by post API version 2.0

txt

```
Path:
    /api/environment/{environmentId}/onhand/indexquery?pageNumber=
{pageNumber}&pageSize={pageSize}
Method:
    Post
Headers:
    Api-Version="2.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    # Same as version 1.0
```

The request format for API version 2.0 is similar to that of version 1.0, but also supports two optional parameters: `pageNumber` and `pageSize`, which allow the system to split a single large result into several smaller documents. The results are sorted by warehouse (`locationId`), and the parameters are used as follows to split results into pages:

- `pageSize` establishes the number of warehouses (`locationId` values) that are returned in each page.
- `pageNumber` establishes the page number that's returned.

A request of this format returns on-hand inventory data starting from warehouse number *({pageNumber} − 1) × {pageSize}* and includes data for the next *{pageSize}* warehouses.

API version 2.0 responds with a document that uses the following structure:

txt

```
{
    Value: { # Response same as Api-Version=1.0 }
    nextLink: onhand/indexquery?pageNumber={pageNumber+1}&pageSize={pageSize}
}
```

When the request reaches the last warehouse (`locationId`), the `nextLink` value is an empty string.

API version 2.0 also lets you specify more than one organization ID in your request. To do so, include a comma-separated list of organization IDs in the `organizationId` filter of your request document. For example, `"organizationId": ["org1", "org2", "org3"]`.

Query by using the get method

txt

```
Path:
    /api/environment/{environmentId}/onhand
Method:
    Get
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Query(Url Parameters):
    groupBy
    returnNegative
    [Filters]
```

Here's a sample get URL. This get request is exactly the same as the post sample that was provided earlier.

txt

```
/api/environment/{environmentId}/onhand?
organizationId=SCM_IV&productId=iv_contoso_product&siteId=iv_contoso_site&locationI
d=iv_contoso_location&colorId=red&groupBy=colorId,sizeId&returnNegative=true
```

The system doesn't support querying inventory over multiple organization IDs with the GET method.

On-hand exact query

On-hand exact queries resemble regular on-hand queries, but they let you specify a mapping hierarchy between a site and a location. For example, you have the following two sites:

- Site 1, which is mapped to location A
- Site 2, which is mapped to location B

For a regular on-hand query, if you specify `"siteId": ["1","2"]` and `"locationId": ["A","B"]`, Inventory Visibility will automatically query the result for the following sites and locations:

- Site 1, location A
- Site 1, location B
- Site 2, location A
- Site 2, location B

As you see, the regular on-hand query doesn't recognize that location A exists only in site 1, and location B exists only in site 2. Therefore, it makes redundant queries. To accommodate this hierarchical mapping, you can use an on-hand exact query and specify the location mappings in the query body. In this case, you'll query and receive results for only site 1, location A and site 2, location B.

On-hand exact query query using the post method

The on-hand exact query by post API is available in two versions. The following table outlines the differences.

Expand table

API version 1.0	API version 2.0
Can only query one organization ID.	Can query multiple organization IDs.

On-hand exact query by post API version 1.0

txt

```
Path:
    /api/environment/{environmentId}/onhand/exactquery
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        dimensionDataSource: string, # Optional
        filters: {
            organizationId: string[],
            productId: string[],
            dimensions: string[],
            values: string[][],
        },
        groupByValues: string[],
```

```
        returnNegative: boolean,
    }
```

In the body part of this request, `dimensionDataSource` is an optional parameter. If it isn't set, `dimensions` in `filters` will be treated as *base dimensions*. There are four required fields for `filters`: `organizationId`, `productId`, `dimensions`, and `values`.

- `organizationId` should contain only one value, but it's still an array.
- `productId` could contain one or more values. If it's an empty array, all products will be returned.
- In the `dimensions` array, `siteId` and `locationId` are required if and only if your data partition rule is set to *By location*. In this case, they could appear with other elements in any order.
- `values` could contain one or more distinct tuples of values corresponding to `dimensions`.

`dimensions` in `filters` will be automatically added to `groupByValues`.

The `returnNegative` parameter controls whether the results contain negative entries.

The following example shows sample body content.

JSON

```
{
    "dimensionDataSource": "pos",
    "filters": {
        "organizationId": ["SCM_IV"],
        "productId": ["iv_contoso_product"],
        "dimensions": ["siteId", "locationId", "colorId"],
        "values" : [
            ["iv_contoso_site", "iv_contoso_location", "red"],
            ["iv_contoso_site", "iv_contoso_location", "blue"],
        ]
    },
    "groupByValues": ["colorId", "sizeId"],
    "returnNegative": true
}
```

The following example shows how to query all products in multiple sites and locations.

JSON

```
{
    "filters": {
        "organizationId": ["SCM_IV"],
        "productId": [],
        "dimensions": ["siteId", "locationId"],
```

```
        "values" : [
            ["iv_contoso_site_1", "iv_contoso_location_1"],
            ["iv_contoso_site_2", "iv_contoso_location_2"],
        ]
    },
    "groupByValues": ["colorId", "sizeId"],
    "returnNegative": true
}
```

On-hand exact query by post API version 2.0

txt

```
Path:
    /api/environment/{environmentId}/onhand/exactquery
Method:
    Post
Headers:
    Api-Version="2.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        dimensionDataSource: string, # Optional
        filters: {
            productId: string[],
            keys: string[],
            values: string[][],
        },
        groupByValues: string[],
        returnNegative: boolean,
    }
```

API version 2.0 differs from version 1.0 in the following ways:

- The `filters` section now has a `keys` field instead of a `dimensions` field. The `keys` field works like the `dimensions` field in version 1.0, but can now also include `organizationId`. You can specify the keys in any order.
- The `filters` section no longer supports the `organizationId` field. Instead, you can include `organizationId` among the dimensions in the `keys` field (for example, `keys: ["organizationId", "siteId", "locationId"]`) and define organization ID values at the matching position in the `values` field (for example, `values: ["SCM_IV", "iv_contoso_site_1", "iv_contoso_location_1"]`).

Other fields are identical to API version 1.0.

Query with product search

The following on-hand query APIs are enhanced to support product search:

- Query by using the post method
- Exact query by using the post method

> **Note**
>
> When you post an Inventory Visibility query that uses product search, use the `productSearch` request parameter (with a `ProductAttributeQuery` object inside) to find or filter by product ID. The newer APIs no longer support the older `productid` request parameter in the request body.

Prerequisites

Before you can start to use the product search APIs, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management 10.0.36 or later.
- Inventory Visibility version 1.2.2.54 or later must be installed and set up as described in Install and set up Inventory Visibility.
- The Inventory Visibility search service must be installed and set up as described in Set up product search for Inventory Visibility.

Product search contract

The product search contract defines the rules for communicating with the product search APIs. It provides a standardized way to describe the capabilities and behavior of the product search capabilities. Therefore, users can more easily understand, interact with, and build applications that consume the Inventory Visibility APIs.

The following example shows a sample contract.

JSON

```
{
    "productFilter": {
        "logicalOperator": "And",
        "conditions": [
            {
                "conditionOperator": "Contains",
                "productName": [
```

```
                    "Deluxe"
                ],
            },
        ],
        "subFilters": [
            {
                "conditions": [
                    {
                        "conditionOperator": "IsExactly",
                        "productType": [
                            "Item"
                        ]
                    }
                ]
            }
        ]
    },
    "attributeFilter": {
        "logicalOperator": "Or",
        "conditions": [
            {
                "attributeName": "Weight Limit",
                "attributeTypeName":"PoundDomain",
                "attributeArea": " ProductAttribute",
                "attributeValues": [
                    "370"
                ],
                "conditionOperator": "GreaterEqual"
            }
        ],
        "subFilters": [
            {
                "conditions": [
                    {
                        "attributeName": "Weight Limit",
                        "attributeTypeName":"PoundDomain",
                        "attributeArea": " ProductAttribute",
                        "attributeValues": [
                            "330"
                        ],
                        "conditionOperator": "LessEqual"
                    }
                ]
            }
        ]
    },
}
```

The following table describes the fields that are used in the contract.

Expand table

Field ID	Description
`logicalOperator`	The possible values are `And` and `Or`. Use this field to connect multiple conditions or conditions and sub-filters. Note that `subFilters` is actually a `productFilter` or `attributeFilter` object. Therefore, you can have `subFilters` inside `subFilters`.
`conditionOperator`	The possible values are `IsExactly`, `IsNot`, `Contains`, `DoesNotContain`, `BeginsWith`, `IsOneOf`, `GreaterEqual`, `LessEqual`, and `Between`.
`ProductFilter`	Use this field to filter products by product-related information. For example, you can change `productName` in the contract to `Company`, `itemNumber`, `productSearchName`, `productType`, `productName`, `productDescription`, `inventoryUnitSymbol`, `salesUnitSymbol`, or `purchaseUnitSymbol` to fit your business needs.
`AttributeFilter`	Use this field to filter products by attribute-related information.
`attributeArea`	The possible values are `ProductAttribute`, `DimensionAttribute`, and `BatchAttribute`.

Query with product search

txt

```
Path:
    /api/environment/{environmentId}/onhand/productsearch/indexquery
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        productSearch: {ProductAttributeQuery contract object inherited from
Product Search}
            dimensionDataSource: string, # Optional
            filters: {
                organizationId: string[],
                siteId: string[],
                locationId: string[],
                [dimensionKey:string]: string[],
            },
            groupByValues: string[],
            returnNegative: boolean,
    }
```

The following example shows sample body content.

JSON

```
{
    "productSearch": {
        "productFilter": {
            "conditions": [
                {
                    "conditionOperator": "contains",
                    "productName": [
                        "speaker cable"
                    ],
                },
            ],
        },
    },
    "returnNegative": true,
    "filters":
    {
        "organizationId": ["usmf"],
        "siteId": ["1"],
        "locationId": ["13"],
    },
    "groupByValues": ["colorid"],
}
```

The following example shows a successful response.

JSON

```
[
    {
        "productId": "M0030",
        "dimensions": {
            "ColorId": "White",
            "siteid": "1",
            "locationid": "13"
        },
        "quantities": {
            "fno": {
                "arrived": 0,
                "availordered": 20,
                "onorder": 5,
                "ordered": 20,
                "physicalinvent": 0,
                "reservordered": 0,
                "reservphysical": 0,
                "orderedsum": 20,
                "softreserved": 0
            },
            "iv": {
                "ordered": 0,
                "softreserved": 0,
                "softreservphysical": 0,
                "softreservordered": 0,
                "total ordered": 20,
```

```
                "total on order": 5,
                "availabletoreserve": 20,
                "totalavailable": 20,
                "totalordered": 20,
                "totalonorder": 5
            },
            "pos": {
                "inbound": 0,
                "outbound": 0
            },
            "@iv": {
                "@allocated": 0
            }
        }
    },
    {
        "productId": "M0030",
        "dimensions": {
            "ColorId": "Black",
            "siteid": "1",
            "locationid": "13"
        },
        "quantities": {
            "fno": {
                "arrived": 0,
                "availordered": 3,
                "ordered": 3,
                "physicalinvent": 0,
                "reservordered": 0,
                "reservphysical": 0,
                "orderedsum": 3,
                "softreserved": 0
            },
            "iv": {
                "ordered": 0,
                "softreserved": 0,
                "softreservphysical": 0,
                "softreservordered": 0,
                "total ordered": 3,
                "availabletoreserve": 3,
                "totalavailable": 3,
                "totalordered": 3
            },
            "pos": {
                "inbound": 0,
                "outbound": 0
            },
            "@iv": {
                "@allocated": 0
            }
        }
    }
]
```

Exact query with product search

txt

```
Path:
    /api/environment/{environmentId}/onhand/productsearch/exactquery
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Body:
    {
        productSearch: {ProductAttributeQuery contract object inherited from
Product Search}
            dimensionDataSource: string, # Optional
            filters: {
                organizationId: string[],
                dimensions: string[],
                values: string[][],
            },
            groupByValues: string[],
            returnNegative: boolean,
    }
```

The following example shows sample body content.

JSON

```
{
    "productSearch": {
        "productFilter": {
            "conditions": [
                {
                    "conditionOperator": "contains",
                    "productName": [
                        "speaker cable"
                    ],
                },
            ],
        },
    },
    "filters": {
        "organizationId": ["usmf"],
        "dimensions": ["siteId", "locationId", "colorid"],
        "values" : [
            ["1", "13", "Black"],
        ]
    },
    "groupByValues": [],
```

```
    "returnNegative": true
}
```

The following example shows a successful response.

JSON


```
[
    {
        "productId": "M0030",
        "dimensions": {
            "ColorId": "Black",
            "siteid": "1",
            "locationid": "13"
        },
        "quantities": {
            "fno": {
                "arrived": 0,
                "availordered": 3,
                "ordered": 3,
                "physicalinvent": 0,
                "reservordered": 0,
                "reservphysical": 0,
                "orderedsum": 3,
                "softreserved": 0
            },
            "iv": {
                "ordered": 0,
                "softreserved": 0,
                "softreservphysical": 0,
                "softreservordered": 0,
                "total ordered": 3,
                "availabletoreserve": 3,
                "totalavailable": 3,
                "totalordered": 3
            },
            "pos": {
                "inbound": 0,
                "outbound": 0
            },
            "@iv": {
                "@allocated": 0
            }
        }
    }
]
```


Available to promise

You can set up Inventory Visibility to let you schedule future on-hand changes and calculate ATP quantities. ATP is the quantity of an item that's available and can be promised to a customer in the next period. Use of the ATP calculation can greatly increase your order fulfillment capability. For information about how to enable this feature, and how to interact with Inventory Visibility through its API after the feature is enabled, see Inventory Visibility on-hand change schedules and available to promise.

Allocation

Allocation related APIs are located in Inventory Visibility allocation.

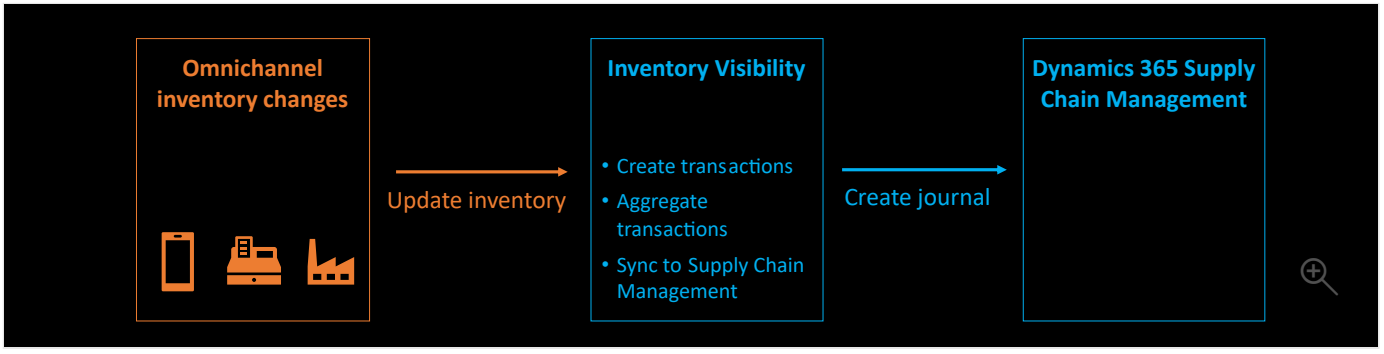
Last updated on 03/10/2026

Sync external inventory adjustments through Inventory Visibility

This article explains how to set up the system to sync inventory adjustment transactions registered in an external system to Microsoft Dynamics 365 Supply Chain Management and Inventory Visibility through the Inventory Visibility service business layer.

Some organizations have multiple sources or channels of inventory operations that work outside Supply Chain Management. By using this feature, those organizations can use Inventory Visibility to store their omnichannel inventory transactions. Based on these transactions, Inventory Visibility can create and post inventory adjustment journals to Supply Chain Management. These capabilities provide the following key business benefits:

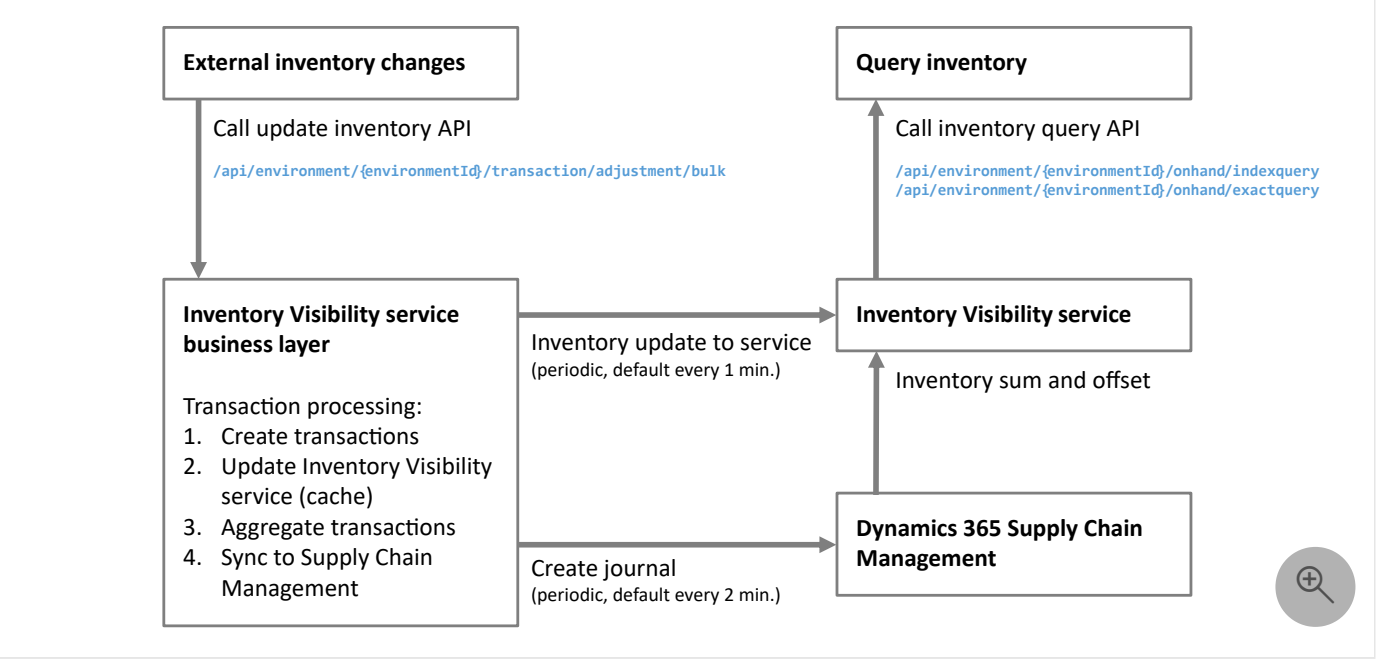
- **Manage all omnichannel inventory adjustments in one place.** You can post inventory adjustments across all channels and data sources to Inventory Visibility. Inventory Visibility then handles the aggregation and posting to Supply Chain Management.
- **Track inventory flow from end to end.** You can track inventory flows from source transactions and documents all the way to Supply Chain Management inventory and financial records. The system stores source document types and reference numbers when it posts inventory adjustments to Inventory Visibility. Inventory Visibility also links Supply Chain Management journal numbers to their original inventory transaction.



Architecture and data flow

Inventory Visibility introduces the concept of the *business layer*, which sits on top of the Inventory Visibility service. The Inventory Visibility service processes and updates data in the cache, but it doesn't directly write data back to Supply Chain Management. The business layer feature adds the following elements and capabilities:

- An inventory adjustment API (`/api/environment/{environmentId}/transaction/adjustment/bulk`) that is designed to create inventory adjustment transactions in the Inventory Visibility business layer. The business layer has columns that are used to integrate with Supply Chain Management journals and track source transactions and documents.
- An entity that stores omnichannel inventory transactions details that are posted to the Inventory Visibility business layer by using the new API.
- A configurable mechanism that automatically updates posted inventory adjustment transactions to the Inventory Visibility service (in-memory cache service) at a regular interval.
- A configurable mechanism that aggregates the detailed inventory adjustment transactions and syncs to Supply Chain Management at a regular interval and according to defined rules.
- A new user interface (UI) page for the Inventory Visibility app in Power Apps. This page lets you manually post inventory adjustment transactions to the Inventory Visibility business layer. It's an alternative to using the API. The UI also lets you view transactions in a detailed or aggregated format.



Prerequisites

Before you can use the features described in this article, ensure your system meets the following requirements:

- You're running Dynamics 365 Supply Chain Management version 10.0.41 or later.
- You're running Inventory Visibility service version 1.2.3.88 or later. For more information about how to install Inventory Visibility, check its version number, and update it as required,

see Install and set up Inventory Visibility.

- The following features are turned on in Feature management:
  - *Inventory Visibility integration with inventory adjustment offset* (This feature is a prerequisite. As of Supply Chain Management version 10.0.45, this feature is mandatory and can't be turned off.)
  - *Inventory Visibility integration* (This feature is a prerequisite. As of Supply Chain Management version 10.0.45, this feature is turned on by default.)
  - *Inventory Visibility transaction integration* (This feature adds the features that are described in this article. As of Supply Chain Management version 10.0.45, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.)

Configure inventory transaction synchronization

Set up the feature for the first time

To set up the feature for the first time, follow these steps:

1. Sign in to Supply Chain Management.
2. Go to **Inventory Management** > **Setup** > **Inventory Visibility** > **Inventory Visibility integration parameters**.
3. On the **Transaction** tab, set the **Enable adjustment journal sync** option to *Yes*.
4. Select **Generate configurations** to run an automatic deployment script that initializes the configuration in Supply Chain Management and the Inventory Visibility service.
5. Select **Enable posting job** to open the **Inventory Visibility posting job** dialog box.
6. Each time the *Inventory Visibility posting job* runs, it automatically posts the inventory adjustment journals submitted from the Inventory Visibility business layer. Schedule this job to run as often as is appropriate for your business. Depending on the volume of incoming data, this job can be resource intensive, so schedule it to run during off-peak hours. Select the **Recurrence** link to open the **Define recurrence** dialog, use it to set up your schedule, and then select **OK** to save your settings.
7. Select **OK** to close the dialog box.

Manage the feature configuration

To manage the feature configuration, follow these steps:

1. Sign in to Power Apps, and open the **Inventory Visibility** app.

2. On the navigation pane, select **Settings** > **Feature management**.

3. On the **Inventory transactions** tile, select **Manage**.

4. On the **Transaction** page, in the upper section, set the following fields:

    - **Enable feature** – Set this option to *True* to enable the feature. Set it to *False* to disable the feature. The feature is automatically turned on when you initialize the configuration as described in the previous section.
    - **Periodically synchronize transaction to IV** – Set this option to *True* if you want the system to automatically sync inventory updates from the Inventory Visibility business layer to the Inventory Visibility service at regular intervals. Set it to *False* to disable this functionality.
    - **Inventory Visibility synchronization frequency** – Enter an integer to specify how often (in minutes) the system syncs inventory updates from the Inventory Visibility business layer to Supply Chain Management (if you enabled this functionality). The default value is *5*.

5. In the **Fno Synchronization Settings** section, set the following fields:

    - **Enable Sync Job** – Set this option to *True* if you want the system to automatically aggregate and sync inventory updates from the Inventory Visibility business layer to Supply Chain Management at regular intervals. When this option is set to *False*, omnichannel transactions are still stored in the Inventory Visibility business layer. However, they aren't synced to Supply Chain Management.
    - **Job settings** – Select the setting value to open the detailed settings page, where you can specify the start and end time, recurrence interval, and unit for synchronizing inventory updates from the Inventory Visibility business layer to the Inventory Visibility service (if this functionality is enabled). The job is automatically scheduled with default settings when you install Inventory Visibility.
    - **Fno sync max line count in adjustment journal** – Enter an integer to specify the maximum number of inventory adjustment lines that are allowed per adjustment journal during synchronization back to Supply Chain Management. The default value is *10,000*.
    - **Fno sync max journal count in package** – Enter an integer to specify the maximum number of inventory adjustment journals that are allowed per data package during synchronization back to Supply Chain Management. The default value is *1*.

6. Use the **Fno integration mappings** section to control how data is mapped from the Inventory Visibility business layer to Supply Chain Management. Use the buttons on the

toolbar to add, edit, or delete mappings as required. For each mapping, set the following fields:

- **Physical measure name** – Select the physical measure that provides the quantity of the inventory.
- **Target FnO Transaction type** – Specify the type of inventory journal that is created in Supply Chain Management. The default value is *Inventory adjustment journal*.
- **Target FnO Transaction Status** – Select one of the following values to specify the status of the inventory journal that is created in Supply Chain Management:
  - *None* – The system doesn't create transactions in Supply Chain Management.
  - *Created* – The system creates transactions but doesn't automatically post them in Supply Chain Management.
  - *Posted* – The system creates transactions and financially posts the journal in Supply Chain Management.

7. If you set the **Periodically synchronize transaction to Fno** option to *True*, you must ensure that your system in configured to offset inventory adjustments to prevent inventory quantities from being updated twice. Learn more in Inventory Visibility adjustment offset.

Manually enter inventory adjustments in the Inventory Visibility app

Use the Inventory Visibility app in Power Apps to manually enter inventory adjustments directly in the Inventory Visibility business layer.

1. Sign in to Power Apps, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Inquiries and reports** > **Inventory transactions**.
3. On the toolbar, select **Create transaction** > **Inventory adjustment**.
4. On **Create inventory adjustment**, on the **Product and dimension information** FastTab, specify the product that you want to adjust inventory for. If some of the product dimensions that you need (such as color or style) aren't shown, select **Edit dimensions** to add them.
5. On the **Adjustment quantity** FastTab, select the data source, physical measure, and quantity of the inventory adjustment. To immediately update inventory adjustments to the Inventory Visibility service cache from the staging area, select the **Instant update to IV service** checkbox.

6. On the **Additional information** FastTab, set the following optional fields:

    - **External reference category** – Specify the source transaction type for the inventory adjustment, such as sales order, inventory counting delta, or POS transaction.
    - **External reference ID** – Enter the source document ID from the external system.
    - **Supply Chain Management journal name** – Specify the inventory adjustment journal to use in Supply Chain Management. If you leave this field blank, the default journal name that is set in Supply Chain Management is used.
    - **Item cost price** – Specify the item cost price to create in Supply Chain Management. If you leave this field blank, the journal that is created in Supply Chain Management uses the default cost calculation method that is set in Supply Chain Management to calculate the cost price.
    - **Trans date** – The transaction date to set on the journal line when it's created in Supply Chain Management.

7. To preview what your inventory adjustment looks like when it's posted to the Inventory Visibility business layer through the API, select **Developer reference**. This action shows the JavaScript Object Notation (JSON) body that's sent to the Inventory Visibility business layer when you post the transaction.

8. On the toolbar, select **Post**.

Submit inventory adjustments to the Inventory Visibility API

External systems typically submit inventory adjustments to the Inventory Visibility business layer through the Inventory adjustment API. The following example shows the API and body content details.

txt

```
Path:
    /api/environment/{environmentId}/transaction/adjustment/bulk
Method:
    Post
Headers:
    Api-Version="1.0"
    Authorization="Bearer $access_token"
ContentType:
    application/json
Query(Url Parameters):
    createWithIVSync # bool, optional
Body:
```

```
[
    {
        "id": string
        "organizationId": string,
        "productId": string,
        "quantityDataSource": string,
        "physicalMeasure": string,
        "dimensions": {
            "siteId": string,
            "locationId": string,
            [key:string]: string, # optional
        },
        "quantity": number,
        "externalReferenceCategory": string, # optional
        "externalReferenceId": string, # optional
        "fnoJournalNameId": string, # optional
        "costPrice": number, # optional
        "transDate": date, # optional
    },
]
```

Learn more about how to work with the Inventory Visibility API, including how to authenticate and get an access token, in Inventory Visibility public APIs.

View inventory transactions in the Inventory Visibility app

The Inventory Visibility app in Power Apps provides a UI where you can view detailed and aggregated inventory transactions.

- Detailed inventory transactions include all individual transactions together with all the original details.
- Aggregated inventory transactions combine lines that have matching dimension values (such as site, warehouse, and color). Some information might be omitted. Instead of showing every individual transaction, the UI shows totals or averages over the specified dimensions. Only the aggregated transactions sync with Supply Chain Management.

Detailed and aggregated transactions differ in the following ways:

- Granularity:
  - Detailed transactions show individual original inventory adjustment transactions and provide fine-grained data.
  - Aggregated transactions show summarized data and provide a higher-level overview.
- Data volume:

- Detailed transaction data sets are typically larger because they include every transaction.
- Aggregated transaction data sets are typically smaller because they condense multiple similar transactions into single records.

View detailed inventory transactions

To view detailed inventory transactions, follow these steps:

1. Sign in to Power Apps, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Inquiries and reports** > **Inventory transactions**.
3. In the row where the **Type** field is set to *Detail*, in the **Action** column, select **View**.

   On the **Transaction details** page, a grid lists details for each successful transaction. Most of the columns are self-explanatory, but here are some important columns to note:

   - **Line record ID** – The globally unique identifier (GUID) that is automatically generated for each original inventory adjustment transaction. Select the link to view more details about the transaction.
   - **Source line status** – The current status of the transaction. The following values are used:
     - *Created* – The adjustment was created in the Inventory Visibility business layer and is waiting to be posted. This status is the initial status for the inventory adjustment records.
     - *IVPosted* – The adjustment was successfully updated in the Inventory Visibility cache from the staging area. This status is the prerequisite status for aggregation.
     - *Aggregated* – Detailed lines were successfully aggregated based on matching dimension values. This status is the prerequisite status for synchronization to Supply Chain Management.
   - **Aggregated line ID** – Select the link to view details about the linked aggregated line. After the aggregated lines are successfully created and/or posted in Supply Chain Management, the journal ID and journal line ID that Supply Chain Management returns are added to the aggregated transaction details.

View aggregated inventory transactions

To view aggregated inventory transactions, follow these steps:

1. Sign in to Power Apps, and open the **Inventory Visibility** app.
2. On the navigation pane, select **Inquiries and reports** > **Inventory transactions**.
3. In the row where the **Type** field is set to *Aggregated*, in the **Action** column, select **View**.

   On the **Transaction aggregated** page, a grid lists details for each successful aggregated transaction. Most of the columns are self-explanatory, but here are some important columns to note:

   - **Line record ID** – The GUID that's automatically generated for each aggregated inventory adjustment transaction. Select the link to view more details about the aggregated transaction. After the aggregated lines are successfully created and/or posted in Supply Chain Management, the journal ID and journal line ID that Supply Chain Management returns are included in the aggregated transaction details.
   - **Aggregated line status** – The current status of the aggregated transaction. The following values are used:
     - *Created* – The aggregated transaction record was successfully created. This status is the initial status for the aggregated transaction record.
     - *DoNotSync* – The line isn't synced to Supply Chain Management. This status typically occurs because the inventory change quantity of the aggregated line is 0 (zero). (Because the inventory quantity wasn't changed, there's no need to sync it.)
     - *FnOSyncEnqueued* – Inventory Visibility sent a request to sync the line with Supply Chain Management. This status indicates that the record was synced in Supply Chain Management and is awaiting further processing.
     - *FnOTransCreated* – Supply Chain Management successfully processed the synchronization request and created the inventory journal.
     - *FnOTransCreateError* – An error occurred during the attempt to create the inventory journal in Supply Chain Management. Use the message ID to check the error. Go to **Supply Chain Management** > **System administration workspace** > **Data Management IT** > **Recurring data jobs**.
     - *FnOTransPosted* – The inventory journal was successfully posted.
     - *FnOTransPostError* – Inventory journal posting failed.

Export transaction details

To export transaction details, you must access them from the Dataverse entity and export them from there. The entity names are *Adjustment Line* and *Aggregated Adjustment Line*.

Check inventory journals posted to Supply Chain Management

To view the aggregated lines that are successfully posted or created in Supply Chain Management, follow these steps:

1. Sign in to Supply Chain Management.
2. Go to **Inventory Management** > **Journal entries** > **Items** > **Inventory adjustment**.
3. Change the filter for the **Posted** column as required to view posted journals, unposted journals, or all journals.

> **Note**
>
> Some of the optional fields that you can submit to the Inventory Visibility business layer don't exist as standard fields in Supply Chain Management. These fields include **External reference category**, **External reference ID**, and any other external dimensions. These fields aren't synced to Supply Chain Management. However, they are stored in the business layer and can be read by using the Inventory Visibility app, as described earlier in this article.

Last updated on 07/27/2026

Log and view successful API posts

08/14/2025

This article describes how to set up and use the *inventory log history* feature for Inventory Visibility. This feature creates a log of successfully updated inventory API posts. The posts include a timestamp and specify the API type.

Set up inventory log history

To start to keep a log of successfully updated inventory API posts, you must enable the feature in Microsoft Power Apps.

1. Sign in to Power Apps, and go to **Inventory Visibility** > **Settings** > **Feature Management**.
2. Enable the *Inventory log history* feature.
3. Go to **Admin settings**, and select **Update configuration**.

View inventory log history

After the feature is enabled, you can view the log entries either by using the Inventory Visibility app in Power Apps or by calling the API.

View log entries by using the Inventory Visibility app in Power Apps

Follow these steps to view the log by using the Inventory Visibility app in Power Apps.

1. Sign in to Power Apps, and go to **Inventory Visibility** > **Operational visibility** > **Inventory log history**.
2. To retrieve and view the log details, enter a product ID, organization ID, site ID, warehouse ID, and date range.

Retrieve log entries by calling the API

Follow these steps to retrieve log entries by calling the Inventory Visibility API.

1. Run a tool that lets you call APIs. Use the tool to access the `{inventoryVisibilityEndpoint}/api/environment/{yourEnvironmentId}/logTransactionDetails` API by using the `Post` method.

2. Set up and submit a request body to retrieve the log data that you're interested in. The following example shows a request body that asks the system to prepare and retrieve the log. This step is necessary because the log is stored in Azure Data Lake and is transferred to Dataverse only on request. The API returns a log history job ID that you can use to check the job completion status.

```txt
Path:

    /api/environment/{environmentId}/logTransactionDetails

Method:

    Post

Headers:

    Api-Version="1.0"

    Authorization="Bearer $access_token"

ContentType:

    application/json

Body:

    {
    "id": "id-contoso-{{datetime}}",
    "organizationId": "usmf",
    "UtcFromDate": "2023/8/23",
    "UtcToDate": "2023/8/23",
    "productId": "D0001",
    "dimensions": {
        "SiteId": "1",
        "LocationId": "13"
    }
}
```

3. Use the `{inventoryVisibilityEndpoint}/api/environment/{yourEnvironmentId}/getJobProgress?jobId={jobIdFromPreviousStep}` API to track the status of the inventory log history job.

4. When the API reports that the job is successfully completed, you can review the logs in Dataverse by opening the transaction logging report table (`is_transactionloggingreport`).

Inventory Visibility diagnostic tool

08/13/2025

This article describes how to set up and use the Inventory Visibility diagnostic tool. This tool helps identify and fix discrepancies between on-hand inventory records in Microsoft Dynamics 365 Supply Chain Management and on-hand inventory records in the `fno` data source in Inventory Visibility.

Set up the Inventory Visibility diagnostic tool

The Inventory Visibility diagnostic tool is turned on by default, and no extra steps are required to make it available. However, the tool requires that Inventory Visibility is set up for your system and the data synchronization batch job is running. Learn more in Set up Inventory Visibility in Supply Chain Management.

Use the Inventory Visibility diagnostic tool

1. Sign in to your Supply Chain Management environment.
2. Go to **Inventory Management** > **Set up** > **Inventory Visibility diagnostic tool**.
3. On the **Inventory Visibility diagnostic tool** page, select **Run diagnostic**.
4. In the **Inventory Visibility on-hand diagnostic process** dialog box, set the following options:

    - **Fix inventory discrepancy** – Set this option to *Yes* to immediately force synchronization and address mismatches. Set it to *No* to check for and view discrepancies, but without immediately fixing or syncing them. (In this case, you can fix previously found discrepancies later by rerunning the tool and setting this option to *Yes*.) The synchronization process takes about a minute.
    - **Only check warehouse items** – Set this option to *Yes* if you want to review only items that are enabled for warehouse management processes (WMS). Set it to *No* to review all items.
    - **Only check previous mismatched data** – Set this option to *Yes* to verify the effectiveness of the forced synchronization, confirm the resolution of previous discrepancies, and also find any new discrepancies. If the **Fix inventory discrepancy** option is also set to *Yes*, the system syncs any remaining mismatches.
    - **Batch Processing** – You must leave this option set to *Yes*, because the tool can run only in batch mode.

5. Select the **Filter** link to open a dialog box where you can set up a filter to limit the set of items that's checked. By default, the **No open quantities** option for the filter is set to *No*. Therefore, the tool checks only for items that have open quantities. You can change the setting of this option as you require.

6. Select **OK** to apply your settings and return to the **Inventory Visibility diagnostic tool** page.

7. You receive the following message: "The Inventory Visibility on-hand diagnostic process job is added to the batch queue." To monitor the progress of the batch job, go to **System Administrator** > **Inquiries** > **Batch Jobs**.

8. When the job has finished running, the Inventory Visibility diagnostic tool shows the results of the test. If you chose not to fix discrepancies, you can fix them now. Rerun the diagnostic, and set the **Fix inventory discrepancy** option and/or the **Only check previous mismatched data** option to *Yes*.

Inventory Visibility tips

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

Here are a few tips that you should consider when you set up and use the Inventory Visibility Add-in:

- Unless you using the unified developer experience, you should check for linking mismatch warnings in Microsoft Dynamics Lifecycle Services before you install Inventory Visibility. To do so, open the details page for your environment in Lifecycle Services and look for a warning that resembles the following example: "Microsoft has detected that your environment is linked via dual-write to a different destination than specified in Power Platform Integration, which isn't recommended." If you see this warning, it's possible that your dual-write environment is linked to a Dataverse instance, but Lifecycle Services isn't set up for Power Platform integration. This linking mismatch can cause unexpected behavior. For information about how to fix this issue, see Linking mismatch. After the linking mismatch is fixed, you can install Inventory Visibility.
- If you have more than one Supply Chain Management environment, create a different Microsoft Entra application for each environment. If you use the same application ID and tenant ID to install the Inventory Visibility Add-in for different environments, a token issue will occur for older environments. Only the latest installation of the Inventory Visibility Add-in will be valid.
- Inventory Visibility isn't currently supported for cloud-hosted environments. It's supported only for Tier-2+ environments.

- The application programming interface (API) allows multiple `SiteID` and `LocationID` values to be specified in each query. The maximum limit is defined by the following equation: `NumOf(SiteID)` × `NumOf(LocationID)` ≤ 100.
- The `fno` data source is reserved for Supply Chain Management. If your Inventory Visibility Add-in is integrated with a Supply Chain Management environment, we recommend that you not delete configurations that are related to `fno` in the data source.
- Inventory Visibility can't change any data for the `fno` data source. The data flow is one-way, which means that all quantity changes for the `fno` data source must come from your Supply Chain Management environment. Therefore, you can't use the API to send on-hand change or reservation requests for the `fno` data source.
- If you add one or more new measures to your Supply Chain Management environment, you should also add them in Inventory Visibility. However, all quantity changes for new measures must come from your Supply Chain Management environment.
- A partition configuration controls how data is distributed. Operations that are performed inside the same partition provide better performance, at lower cost, than operations that cross partitions. By default, partition configuration is automatically set up and shouldn't be customized.
- Base dimensions that are reserved in a partition configuration (set number *0*) shouldn't be included in other on-hand index configurations.
- Your on-hand index configuration must include at least one on-hand index in addition to the partition configuration (set number *0*). If you aren't interested in querying specific dimension combinations, you can set up an index that has only one base dimension, `Empty`. Otherwise, queries will fail, and you'll receive the following error: "No index hierarchy has been set."
- Data source `@iv` is a predefined data source and the physical measures defined in `@iv` with prefix `@` are predefined measures. These measures are a predefined configuration for the allocation feature and the soft reservation feature. Therefore, don't change or delete them. Otherwise, you're likely to encounter unexpected errors when you use those two features.
- You can add new physical measures to the predefined calculated measure `@iv.@available_to_allocate`, but you must not change its name.
- If you restore a Supply Chain Management database or copy a Dataverse environment, additional steps are required to run Inventory Visibility in your target environment. For details, see Move data between Supply Chain Management Database and Dataverse.

Last updated on 03/10/2026

Sensor Data Intelligence home page (preview)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

Sensor Data Intelligence for Microsoft Dynamics 365 Supply Chain Management enables organizations to drive business processes in Supply Chain Management, based on Internet of Things (IoT) signals from machines and equipment on the production floor. It's an updated, renamed version of the *IoT Intelligence* feature that was previously available for Supply Chain Management.

Sensor Data Intelligence lets you perform the following tasks:

- Collect details from machines and equipment to update maintenance asset counter values in Supply Chain Management and drive predictive maintenance.
- Set up the solution by using a simple onboarding wizard instead of manually installing and configuring components in Microsoft Dynamics Lifecycle Services.
- Deploy components on your own subscription, so that you have more flexibility to manage Azure components.

- Configure, scale, and extend the solution as business logic that runs on Azure components. Those components are now managed on your own subscription. Therefore, you can customize them as required to meet your unique business needs.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Business scenarios

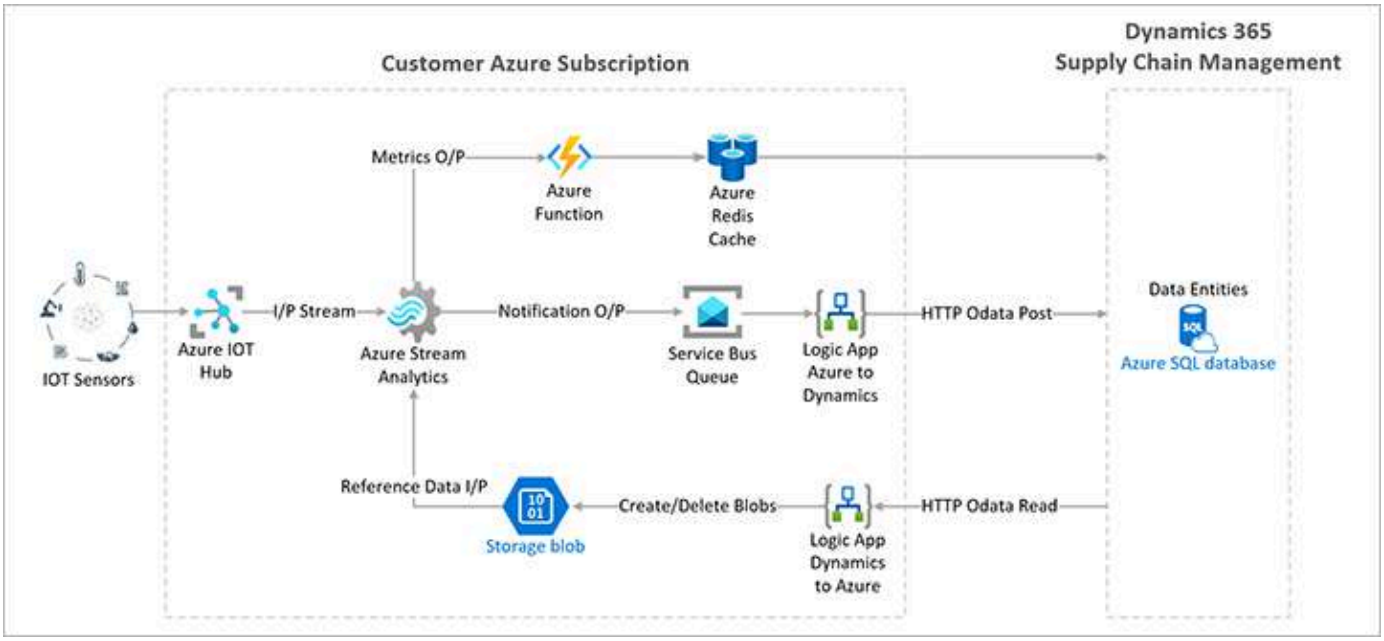
Sensor Data Intelligence enables several types of functionality, each represented by a specific *scenario* in the system. Each scenario provides a specialized set of configuration options, as detailed in the following table.

Expand table

Scenario	Description
Asset downtime	Accurately track the efficiency of machine assets by using sensor data to track machine downtime.
Asset maintenance	Minimize maintenance costs and extend asset life by improving maintenance plans based on sensor readings of critical control points for machine assets.
Machine status	Ensure operation efficiency by using sensor readings to notify planners about machine outages and provide options for mitigating potential delays.
Product quality	Ensure the quality of product batches by comparing sensor readings for actual properties of each product batch, such as moisture, temperature, or custom-defined quality metrics. The system notifies users when deviations occur.
Production auto report	Automatically report finished quantities when a machine generates enough sensor signals to reach a defined output threshold.
Production delays	Use sensor readings to compare actual cycle time to planned cycle time, and notify supervisors when production isn't on schedule. Supervisors can then intervene as required to ensure maximum operation efficiency.

Architecture

The following architectural diagram provides an overview of the solution and its components.



> **Note**
>
> For more information about how to connect sensors to the Azure IoT Hub, see Azure industrial IoT analytics guidance.

Last updated on 05/27/2026

Turn on Sensor Data Intelligence for your system (preview)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

This article explains how to turn on Sensor Data Intelligence for your system.

Video instructions

The following video shows how to turn on the Sensor Data Intelligence feature and deploy the required Azure resources. The other section in this article provides the same instructions in a text-based format.

https://learn-video.azurefd.net/vod/player?id=90804485-5a57-475a-aa9e-753df28dd0d4&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fsensor-data-intelligence%2Fsdi-enable-feature

Procedure

Before you can use Sensor Data Intelligence, it must be turned on for your system.

1. Go to **System administration** > **Workspaces** > **Feature management**. (Learn more in Feature management overview.)
2. On the **All** tab, use the **Filter** field to search for the feature named *Sensor Data Intelligence*.
3. If the *Sensor Data Intelligence* feature is enabled on your system, select it in the list, and then select **Disable** to disable it. You can't use this older version of the feature together with the new preview version that you will enable in the next step.
4. Use the **Filter** field to search for the feature named *(Preview) Sensor Data Intelligence*.
5. Select the feature in the list, and then select **Enable now** to enable it.
6. If you'd also like to use the production auto report scenario (optional), use the **Filter** field to search for the feature named *(Preview) Sensor Data Intelligence - Auto report progress*, and then select **Enable now** to enable it.

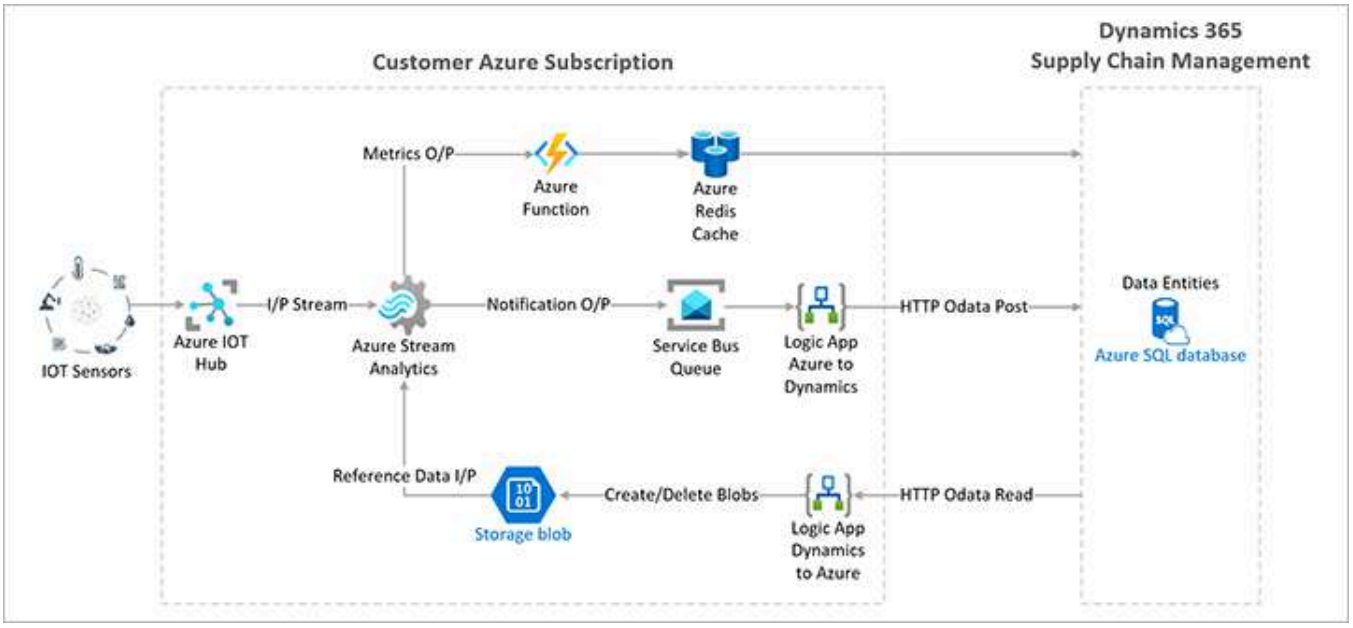
Last updated on 03/25/2026

Deploy an IoT solution on Azure (preview)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

Sensor Data Intelligence uses data from sensors that are connected to Microsoft Azure. To enable Azure to retrieve data from your sensors and share it with Dynamics 365 Supply Chain Management, you must deploy an Internet of Things (IoT) solution on your Azure subscription. The following architectural diagram provides an overview of the solution and its components.



Prerequisites

To deploy and administer Azure components for Sensor Data Intelligence, your Azure environment and user account must meet the following requirements:

1. You must sign in using an Azure organizational account that is tied to your Supply Chain Management environment. You can't use a personal Azure account (such as one registered to an Outlook or Gmail address).
2. Your Azure account must have one of the following security roles:

    - Owner
    - Co-administrator

- Service administrator
- Account administrator

3. The Power Apps component framework must be enabled for your Azure environment. For instructions, see Enable the Power Apps component framework.

Video instructions

The following video shows how to turn on the Sensor Data Intelligence feature and deploy the required Azure resources. The other section in this article provides the same instructions in a text-based format.

https://learn-video.azurefd.net/vod/player?id=90804485-5a57-475a-aa9e-753df28dd0d4&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fsensor-data-intelligence%2Fsdi-deploy-iot-solution-on-azure

Procedure

> **Important**
>
> Before you start this procedure, make sure your Azure environment and user account fulfil the prerequisites listed earlier in this article.

Follow these steps to deploy the required resources on Azure.

1. Sign in to Supply Chain Management as an admin.
2. Go to **System administration** > **Setup** > **Sensor Data Intelligence** > **Deploy and connect Azure resources** to open the deployment wizard.
3. On the **Welcome** page, read the information, and then select **Next**.
4. On the **Deploy the sample IoT solution to Azure** page, read the information, and then, in the **Instructions** section, select **Deploy**.
5. A new browser tab is opened, and you're taken to the Azure portal so that you can deploy the Azure resources. If you're prompted, sign in by using your credentials for your Azure subscription.
6. On the **Custom deployment** page, in the **Subscription** field, select your subscription.
7. Under the **Resource group** field, select **Create new** to create a resource group for the Azure components that you will deploy.

8. In the drop-down dialog box that appears, in the **Name** field, enter a name for the new resource group (for example, *IoT-demo*). Then select **OK**.

9. Set the following fields:

    - **Resource group** – Select the resource group that you just created.
    - **Region** – Select a region, ideally the region where your Supply Chain Management environment is deployed. Keep in mind that Azure regions have different pricing. You can view estimated costs for your region by using the Sensor Data Intelligence price calculator.
    - **Supply Chain Management environment URL** – Enter the URL for your Supply Chain Management environment.
    - **Reuse existing Azure IoT Hub** – Leave this checkbox cleared.

10. Select **Next: Review + Create**.

11. On the **Custom deployment** page, verify that the validation has passed, and then select **Create**.

12. The **Deployment is in progress** page tracks the progress of your deployment. The deployment process can take up to 30 minutes. When the **Deployment is in progress** page indicates that the deployment is completed, select the link for the resource group name to open a page that shows the list of resources that are deployed in the group.

13. In the resource list, find the record where the **Type** field is set to *Managed Identity*. In the **Name** column, select the name to open the details page for the resource.

14. Copy the value in the **Client ID** field (for example, by selecting the **Copy to clipboard** button).

15. Go back to the browser tab where Supply Chain Management is running, *but don't close the tab for the Azure portal*. The **Deploy the sample IoT solution to Azure** wizard page should still be open.

16. Select **Next**.

17. On the **Connect Azure resources** page, in the **Microsoft Entra Application client ID** field, paste the **Client ID** value that you copied.

18. Go back to the browser tab where the Azure portal is open, *but don't close the tab for Supply Chain Management*. The details page for the resource should still be open.

19. Select the browser's **Back** button to return to the list of resources in the new resource group.

20. In the resource list, find the record where the **Type** field is set to *Azure Cache for Redis*. In the **Name** column, select the name to open the details page for the resource.
21. In the left navigation pane, select **Access keys**.
22. On the **Access keys** page, copy the value that is shown for **Primary connection string (StackExchange.Redis)** (for example, by selecting the **Copy to clipboard** button).
23. Go back to the browser tab where Supply Chain Management is running. The **Connect Azure resources** page should still be open.
24. In the **Redis metric store connection string** field, paste the **Primary connection string (StackExchange.Redis)** value that you copied.
25. Select **Finish**.

Azure resources for Sensor Data Intelligence are now deployed on your Azure subscription.

> **Note**
>
> At any time, you can view or edit the connection information that is saved in Supply Chain Management by opening the **Sensor Data Intelligence parameters** page. Learn more in Sensor Data Intelligence parameters.

Last updated on 05/07/2025

Sensor Data Intelligence parameters (preview)

09/26/2025

[This article is prerelease documentation and is subject to change.]

The **Sensor Data Intelligence parameters** page provides a few settings that you can use to configure the feature. These settings include Azure connection parameters and a parameter for the lifetime of alert messages that are sent to users in response to sensor measurements.

Read and change connection details for your Azure IoT solution

Typically, you will set up your Azure IoT solution and connect it to Microsoft Dynamics 365 Supply Chain Management from the **Deploy and connect Azure resources** page, which will guide you through both procedures. (Learn more in Deploy an IoT solution on Azure.) You can also view and edit the connection details at any time in Supply Chain Management by following these steps.

1. Go to **System administration** > **Setup** > **Sensor Data Intelligence** > **Sensor Data Intelligence parameters**.
2. On the **Time series** tab, in the **Redis metric store connection string** field, enter the **Primary connection string (StackExchange.Redis)** value for the Azure IoT solution that you want to connect to. For more information about how to find this value, see Deploy an IoT solution on Azure.
3. On the **Integrations** tab, in the **Microsoft Entra application client ID** field, enter the **Client ID** value for the Azure IoT solution you want to connect to. For more information about how to find this value, see Deploy an IoT solution on Azure.

Set the lifetime of alert messages

Each time that Sensor Data Intelligence detects a situation that requires user attention, it sends a notification to the relevant user. To prevent these notifications from piling up in the system, you should set them to expire after a few days.

1. Go to **System administration** > **Setup** > **Sensor Data Intelligence** > **Sensor Data Intelligence parameters**.
2. On the **Notifications** tab, in the **Notification days to live** field, enter the number of days to keep a notification. After the specified amount of time passes, the notification will be

deleted.

Set up a simulated sensor for testing (preview)

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

If you want to test Sensor Data Intelligence without installing any physical sensors, you can use the *Raspberry PI Azure IoT Online Simulator* service to emulate sensor signals and send them to your Internet of Things (IoT) solution on Microsoft Azure. For more information about the simulator, see Connect Raspberry Pi online simulator to Azure IoT Hub (Node.js).

Video instructions

The following video shows how to set up a simulated sensor for testing. The remaining sections in this article provide the same instructions in a text-based format.

https://learn-video.azurefd.net/vod/player?id=73e8b915-2b43-4285-9100-9bd2a4ed1be6&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fsensor-data-intelligence%2Fsdi-set-up-simulated-sensor

Create a device in Azure IoT Hub

You must first set up a device to authenticate the sensor signals to the Azure IoT Hub.

1. In Azure, go to the list of resources for the resource group that you created for use with Sensor Data Intelligence. (Learn more in Deploy an IoT solution on Azure.)
2. In the resource list, find the record where the **Type** field is set to *IoT Hub*. In the **Name** column, select the name to open the details page for the resource.
3. In the left navigation pane, select **Devices**.
4. On the **Devices** page, select **Add device**.

5. On the **Create a device** page, set the following fields:

    - **Device ID** – Enter a name for the new device (for example, *My-IoT-Device*).
    - **Authentication type** – Select *Symmetric keys*.
    - **Auto-generate keys** – Select this checkbox.
    - **Connect this device to an IoT hub** – Select *Enable*.

6. Select **Save** to return to the **Devices** page.

7. Find the new device in the list. In the **Device ID** column, select the name to open the details page for the device. If you don't see the new device in the list, refresh the page.

8. Copy the **Primary connection string** value (for example, by selecting the **Copy to clipboard** button). You will need this value later when you set up the Raspberry Pi IoT simulator to emulate sensor signals. Therefore, consider pasting it into a text file for now.

Add the Azure connection string to the Raspberry Pi IoT simulator

Follow these steps to add the connection string from the device in Azure IoT Hub to the script in the Raspberry service.

1. Open the Raspberry Pi IoT simulator.

2. In the code editor pane, find the line that contains the following command.

    ```
    const connectionString = '[Your IoT hub device connection string]';
    ```

3. Replace the help text, including the brackets, with the **Primary connection string** value that you copied in the previous section. The result should resemble the following example.

    ```
    const connectionString = 'HostName=XXX;DeviceId=YYY;SharedAccessKey=ZZZ';
    ```

Add sensor IDs and values to the payload in the Raspberry Pi IoT simulator

You must now set up the Raspberry Pi IoT simulator with simulated sensors and the values that they will send as payloads.

- In the code editor of the Raspberry Pi IoT simulator, find the `getMessage` function, and edit it so that it matches the following code. (The sensors are set up in the `cb()` lines.)

C++

```
function getMessage(cb) {
    messageId++;
    sensor.readSensorData()
    .then(function (data) {
        cb(JSON.stringify({ value: 1, sensorId: 'MachineStatus' }), false);
        cb(JSON.stringify({ value: 70, sensorId: 'Quality' }), false);
        cb(JSON.stringify({ value: 1, sensorId: 'AssetMaintenance' }), false);
        cb(JSON.stringify({ value: 1, sensorId: 'ProductionDelay' }), false);
        cb(JSON.stringify({ value: 20, sensorId: 'AssetDowntime' }), false);
    })
    .catch(function (err) {
        console.error('Failed to read out sensor data: ' + err);
    });
}
```

> **ⓘ Important**
>
> The sensors IDs that are defined in the code editor for the Raspberry Pi IoT simulator must be identical to the sensor IDs that you will specify later for the scenarios in Supply Chain Management. The preceding example code uses human-readable sensor IDs. However, in an actual scenario, the sensor IDs will be globally unique identifier (GUID) values that are provided by the sensor manufacturer. The human-readable sensor IDs that are used in this example code are also used in the examples for the product quality scenario, asset maintenance scenario, production delays scenario, asset downtime scenario, and machine status scenario). Therefore, use this code if you will work through those scenarios.

Edit the interval for sending sensor signals

You must now set the interval at which the Raspberry Pi IoT simulator should send the emulated sensor signals.

1. In the code editor of the Raspberry Pi IoT simulator, find the following function invocation.

   ```
   setInterval(sendMessage, 2000);
   ```

2. By default, the Raspberry Pi IoT simulator sends a sensor signal every 2,000 milliseconds (two seconds). You can adjust the value as you require.

Run the Raspberry Pi IoT simulator

- Select **Run** to start the simulator and begin to send simulated sensor data.

Last updated on 02/25/2026

Asset downtime scenario (preview)

Article • 03/19/2024

[This article is prerelease documentation and is subject to change.]

The asset downtime scenario generates a maintenance downtime record if no signal is received from a machine within a defined time threshold since the last signal was received. The scenario requires that you fit your machine with a sensor that periodically sends a signal to your Azure IoT Hub while the machine is operating, but doesn't send a signal when the machine isn't operating.

Set up the asset downtime scenario

Follow these steps to set up the *asset downtime* scenario in Supply Chain Management.

1. Go to **Asset Management** > **Setup** > **Sensor Data Intelligence** > **Scenarios** to open the **Scenarios** page.

2. In the **Asset downtime** scenario box, select **Configure** to open the setup wizard for this scenario.

3. On the **Sensors** page, select **New** to add a sensor to the grid. Then set the following fields for it:

    - **Sensor ID** – Enter the ID of the sensor that you're using. (If you're using the Raspberry PI Azure IoT Online Simulator and have set it up as described in Set up a simulated sensor for testing, enter *AssetDownTime*.)
    - **Sensor description** – Enter a description of the sensor.

4. Repeat the previous step for each additional sensor that you want to add now. You can come back and add more sensors at any time.

5. Select **Next**.

6. On the **Business record mapping** page, in the **Sensors** section, select the record for one of the sensors that you just added.

7. In the **Business record mapping** section, select **New** to add a row to the grid.

8. On the new row, set the **Business record** field to the resource that you're using the selected sensor to monitor. (If you're using the demo data, select *AK-101, Air Knife for Line 1*.)

9. Select **Next**.

10. On the **asset downtime threshold** page, define how long after the last signal the system should wait before send a asset downtime notification. There are two ways to define the threshold:

    - **Default threshold (minutes)** – Set this field to define the default threshold. This value applies to all resources where the **Notification threshold (minutes)** field is set to two minutes or less. The minimum value is *2* (minutes).
    - **Threshold to determine machine not responding (minutes)** – For each resource in the grid where don't want to use the default threshold, enter an override value in this field. Resources that are set to use a threshold of two minutes or less will use the **Default threshold (minutes)** setting instead.

11. Select **Next**.

12. On the **Activate sensors** page, in the grid, select the sensor that you set up, and then select **Activate**. For each activated sensor in the grid, a check mark appears in the **Active** column.

13. Select **Finish**.

Work with the asset downtime scenario

If no sensor signal is received from an asset within the threshold defined in the scenario configuration, the system will create a maintenance downtime record for that asset. Managers should periodically review the downtime records (for example, once during each working shift) and apply appropriate reason codes and end times for each downtime registration. Follow these steps to view the maintenance downtime records for an asset:

1. Go to **Asset management** > **Assets** > **All assets**.
2. Find and select the asset you want to inspect. (If you're using the demo data, select *AK-101*.)
3. On the Action Pane, open the **Asset** tab and, from the **Work order** group, select **Maintenance downtime** to open the page for maintenance downtime records for the selected asset.

Feedback

Was this page helpful? Yes No

Provide product feedback

Asset maintenance scenario (preview)

Article • 03/19/2024

[This article is prerelease documentation and is subject to change.]

The *asset maintenance* scenario lets you use sensor data to create counter records. Counter records track the use of a machine asset and are used as input to generate the maintenance schedule for machine assets.

Video instructions

The following video shows how to set up and try out the asset maintenance scenario using standard demo data. The remaining sections in this article provide the same instructions in a text-based format.
https://learn-video.azurefd.net/vod/player?id=24d48e9f-b79f-4b78-832a-3ee9a6958cad&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fsensor-data-intelligence%2Fsdi-scenario-asset-maintenance

Prepare demo data for the asset maintenance scenario

If you want to use a demo system to test the *asset maintenance* scenario, use a system where the demo data is installed, select the *USMF* legal entity (company), and prepare the additional demo data as described in this section. If you're using your own sensors and data, you can skip this section.

In this section, you will set up the *AK-101, Air knife* asset in demo data as an example. You will then see how upcoming maintenance work can be predicted based on sensor signals that measure the number of hours that the air knife has been running. You will also set up a maintenance plan where the air knife must be maintained every 10,000 hours.

Set up a sensor simulator

If you want to try this scenario without using a physical sensor, you can set up a simulator to generate the required signals. Learn more in Set up a simulated sensor for testing.

Create an asset counter to track production hours

Follow these steps to create an asset counter to track production hours.

1. Go to **Asset management** > **Setup** > **Asset types** > **Counters**.
2. On the Action Pane, select **New** to create a counter.
3. On the header, set the following values:
    - **Counter:** *ProductionHr*
    - **Name:** *Production hours*
4. On the **General** FastTab, set the following values:
    - **Unit:** *hr*
    - **Update:** *Manual*
    - **Total aggregate:** *Sum*
5. On the **Asset types** FastTab, select **Add line**.
6. On the new line, set the **Asset type** field to *Air knife*.

Create a maintenance plan for the asset

Follow these steps to create a maintenance plan for the asset.

1. Go to **Asset management** > **Setup** > **Preventive maintenance** > **Maintenance plan**.
2. On the Action Pane, select **New** to create a maintenance plan.
3. On the header, set the following values:
    - **Maintenance plan:** *AirKnife*
    - **Name:** *Plan for air knives*
4. On the **Details** FastTab, set the following values:
    - **Plan date:** Enter today's date.
    - **Active:** *Yes*
5. On the **Lines** FastTab, select **Add asset counter line** to add a line to the grid. Then set the following values for it:
    - **Work order description:** *Maintenance for air knife*
    - **Maintenance job type:** *Preventive*
    - **Interval type:** *Repeated from last work order*

- **Period frequency:** *10000*
- **Counter:** *ProductionHr*

Set up the asset maintenance scenario

Follow these steps to set up the *asset maintenance* scenario in Supply Chain Management.

1. Go to **Asset management** > **Setup** > **Sensor Data Intelligence** > **Scenarios**.
2. In the **Asset maintenance** scenario box, select **Configure** to open the setup wizard for this scenario.
3. On the **Sensors** page, select **New** to add a sensor to the grid. Then set the following fields for it:
    - **Sensor ID** – Enter the ID of the sensor that you're using. (If you're using the Raspberry PI Azure IoT Online Simulator and have set it up as described in Set up a simulated sensor for testing, enter *AssetMaintenance*.)
    - **Sensor description** – Enter a description of the sensor.
4. Repeat the previous step for each additional sensor that you want to add now. You can come back and add more sensors at any time.
5. Select **Next**.
6. On the **Business record mapping** page, in the **Sensors** section, select the record for one of the sensors that you just added.
7. In the **Business record mapping** section, select **New** to add a row to the grid.
8. On the new row, the **Business record type** field should automatically be set to *Assets(EntAssetObjectTable)*. Set the **Business record** field to the resource that you're using the selected sensor to monitor. (If you're using the demo data that you created earlier in this article, set it to *AK-101, AK-101 Air Knife for Line 1*.)
9. Immediately after you select a business record type for the row that you added in the previous step, a second row is automatically added to the grid. On this row, the **Business record type** field should be set to *Counters(EntAssetCounterType)*. Set the **Business record** field to the asset counter that you're updating based on signals from the selected sensor. (If you're using the demo data that you created earlier in this article, set it to *ProductionHr, Production hours*.)
10. Select **Next**.

11. On the **Activate sensors** page, in the grid, select the sensor that you added, and then select **Activate**. For each activated sensor in the grid, a check mark appears in the **Active** column.

12. Select **Finish**.

Work with the asset maintenance scenario

View counter values

After the data is prepared, and the *asset maintenance* scenario is configured and activated, you can see how records for an asset counter are inserted based on sensor data.

1. Go to **Asset management** > **Assets** > **All assets**.
2. Find and select the asset that you want to inspect. (If you're using the demo data that you created earlier in this article, select *AK-101*.)
3. On the Action Pane, on the **Asset** tab, in the **Preventive** group, select **Counters** to open the page for counter records for asset *AK-101*.

> **Note**
>
> The counter records are configured by default to be inserted every three hours, which means sensor data will be aggregated at that interval. You can change the interval by editing the query in the Azure Stream Analytics component.

Generate maintenance work orders

After you enable the *asset maintenance* scenario and set up the maintenance plan, you can run the maintenance schedule to generate maintenance work orders. For more information about how to work with preventive maintenance, see Preventive maintenance overview.

Feedback

Was this page helpful? Yes No

Provide product feedback

Machine status scenario (preview)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

The *machine status* scenario lets you use sensor data to monitor the availability of your equipment. If you set up a sensor that sends a signal when a production job on a machine resource produces output, but no sensor signal is received within a specified interval, a notification is shown on the supervisor's dashboard.

Scenario dependencies

The *machine status* scenario has the following dependencies:

- A notification can be triggered only if a production job is in progress on a mapped machine.
- A signal that represents a *part-out* signal must be sent to the IoT hub.

Prepare demo data for the machine status scenario

If you want to use a demo system to test the *machine status* scenario, use a system where the demo data is installed, select the *USMF* legal entity (company), and prepare the additional demo data as described in this section. If you're using your own sensors and data, you can skip this section.

Set up a sensor simulator

If you want to try this scenario without using a physical sensor, you can set up a simulator to generate the required signals. Learn more in Set up a simulated sensor for testing.

Verify that resource 3118 is used for product P0111

A production order will be scheduled and released. Therefore, a production job is released to resource *3118* (*Foam cutting machine*). Follow these steps to verify that resource *3118* is used

for product *P0111* in your demo data.

1. Go to **Product information management** > **Products** > **Released products**.
2. Find and select the product where the **Item number** field is set to *P0111*.
3. On the Action Pane, on the **Engineer** tab, in the **View** group, select **Route**.
4. On the **Route** page, on the **Overview** tab at the bottom of the page, select the line where the **Oper. No.** field is set to *30*.
5. On the **Resource requirements** tab at the bottom of the page, make sure that resource *3118* (*Foam cutting machine*) is associated with the operation.

Create and release a production order for product P0111

Follow these steps to create and release a production order for product *P0111*.

1. Go to **Production control** > **Production orders** > **All production orders**.
2. On the **All production orders** page, on the Action Pane, select **New batch order**.
3. In the **Create batch** dialog box, set the following values:
    - **Item number:** *P0111*
    - **Quantity:** *10*
4. Select **Create** to create the order and return to the **All production orders** page.
5. Use the **Filter** field to search for production orders where the **Item number** field is set to *P0111*. Then find and select the production order that you just created.
6. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Estimate**.
7. In the **Estimate** dialog box, select **OK** to run the estimate.
8. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Release**.
9. In the **Release** dialog box, make a note of the number of the batch order that you just created.
10. Select **OK** to release the order.

Configure the production floor execution interface

You will use the production floor execution interface to start the job that was scheduled and released for item number *P0111* in the previous section. Follow these steps to configure the production floor execution interface.

1. Go to **Production control** > **Manufacturing execution** > **Production floor execution**.
2. If you haven't previously set up the interface, a sign-in page appears. Enter your credentials.
3. On the welcome page, select **Configure** to open the **Configure device** wizard.
4. On the **Configure device - Step 1 - Select configuration** page, select the *Default* configuration.
5. Select **Next**.
6. On the **Configure device - Step 2 - Define the production floor area** page, set the **Resource** field to *3118*.
7. Select **OK**.

Enable the search option in the production floor execution interface

To make it easier to find the production job for the production order that was released earlier, follow these steps to enable the search option in the production floor execution interface.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Configure production floor execution**.
2. Select the *Default* configuration.
3. On the Action Pane, select **Edit**.
4. On the **General** FastTab, set the **Enable search** option to *Yes*.
5. Close the page.

Start the first job in the batch order

Follow these steps to start the job scheduled on resource *3118*.

1. Go to **Production control** > **Manufacturing execution** > **Production floor execution**.
2. In the **Badge ID** field, enter *123*. Then select **Sign in**.
3. If you're prompted for an absence reason, select one of the cards for absence, and then select **OK**.
4. In the **Search** field, enter the batch order number that you previously made a note of. Then select the **Return** key.
5. Select the order, and then select **Start job**.
6. In the **Start job** dialog box, select **Start**.

Set up the machine status scenario

Follow these steps to set up the *machine status* scenario in Supply Chain Management.

1. Go to **Production control** > **Setup** > **Sensor Data Intelligence** > **Scenarios**.
2. In the **Machine status** scenario box, select **Configure** to open the setup wizard for this scenario.
3. On the **Sensors** page, select **New** to add a sensor to the grid. Then set the following fields for it:
    - **Sensor ID** – Enter the ID of the sensor that you're using. (If you're using the Raspberry PI Azure IoT Online Simulator and have set it up as described in Set up a simulated sensor for testing, enter *MachineStatus*.)
    - **Sensor description** – Enter a description of the sensor.
4. Repeat the previous step for each additional sensor that you want to add now. You can come back and add more sensors at any time.
5. Select **Next**.
6. On the **Business record mapping** page, in the **Sensors** section, select the record for one of the sensors that you just added.
7. In the **Business record mapping** section, select **New** to add a row to the grid.
8. On the new row, set the **Business record** field to the resource that you're using the selected sensor to monitor. (If you're using the demo data that you created earlier in this article, set the field to *3118, Foam cutting machine*.)
9. Select **Next**.
10. On the **Machine status threshold** page, define how long after the last *part-out* signal the system should send a machine status notification. There are two ways to define the threshold:
    - **Default threshold (minutes)** – Set this field to define the default threshold. The value will then apply to all resources where the **Threshold to determine machine not responding (minutes)** field is set to two minutes or less. The minimum value is *2* (minutes).
    - **Threshold to determine machine not responding (minutes)** – For each resource in the grid that you don't want to use the default threshold for, enter an override value in this field. Resources that are set to use a threshold of two minutes or less will use the **Default threshold (minutes)** setting instead.

    > ⓘ **Note**

> Typically, you will use a *part-out* signal to monitor machine status. Therefore, you should verify that the threshold for each machine resource is longer than the machine requires to produce each part.

11. Select **Next**.
12. On the **Activate sensors** page, in the grid, select the sensor that you added, and then select **Activate**. For each activated sensor in the grid, a check mark appears in the **Active** column.
13. Select **Finish**.

Work with the machine status scenario

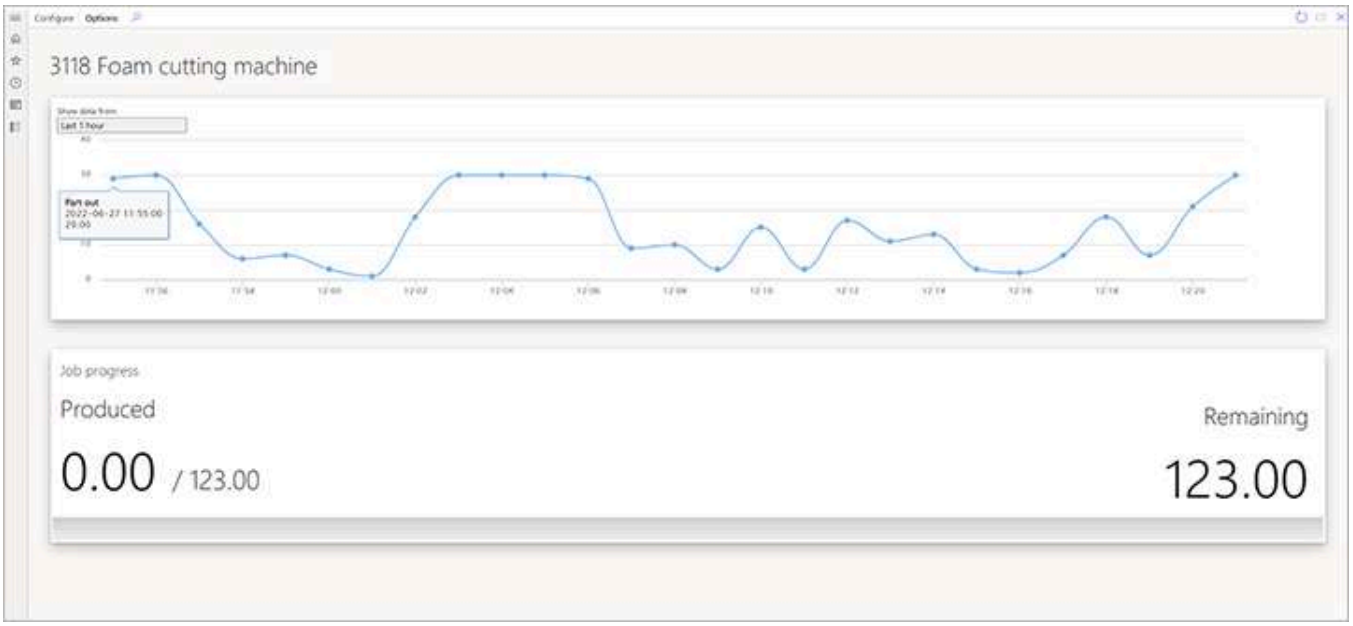
After you install your sensors and configure the scenario, you can view machine status events in Supply Chain Management. This section describes where and how to view this information.

View machine status data on the Resource status page

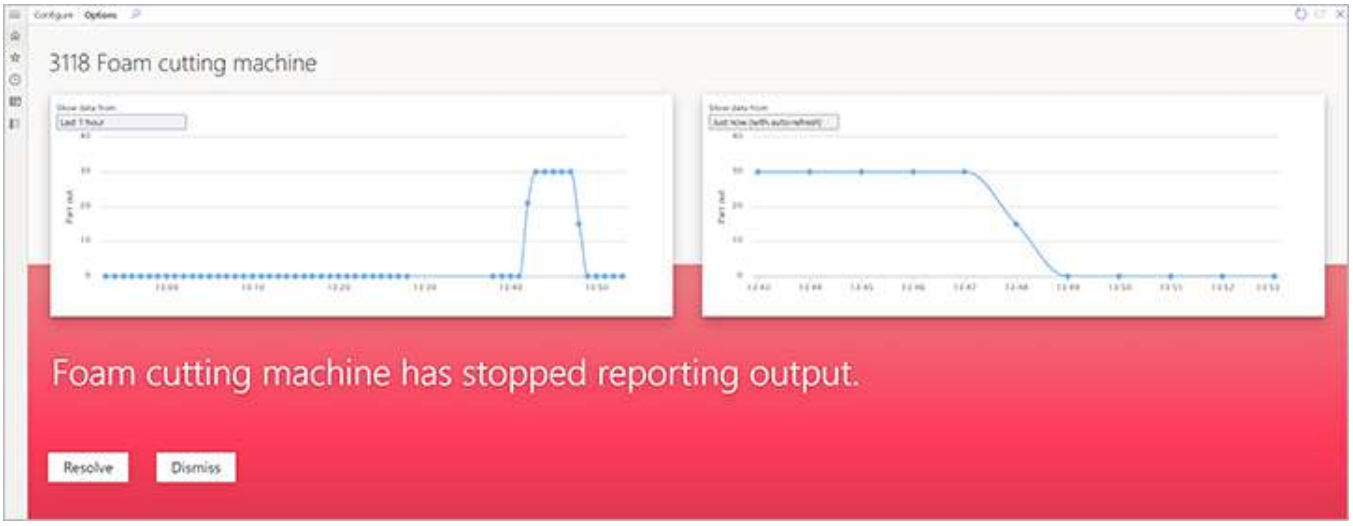
On the **Resource status** page, supervisors can monitor a timeline of the *part-out* signal that is received from the sensors that are mapped to each machine resource. Follow these steps to configure the timeline.

1. Go to **Production control** > **Manufacturing execution** > **Resource status**.
2. In the **Configure** dialog box, set the following fields:
    - **Resource** – Select the resources that you want to monitor. (If you're working with the demo data, select *3118*.)
    - **Time series 1** – Select the record (metric key) that has the following format for its name: *MachineReportingStatus:<Sensor>*
    - **Display name** – Enter *Parts out signal*.

The following illustration shows an example of machine status data on the **Resource status** page during standard operation.



The following illustration shows an example of machine status data when downtime is detected.

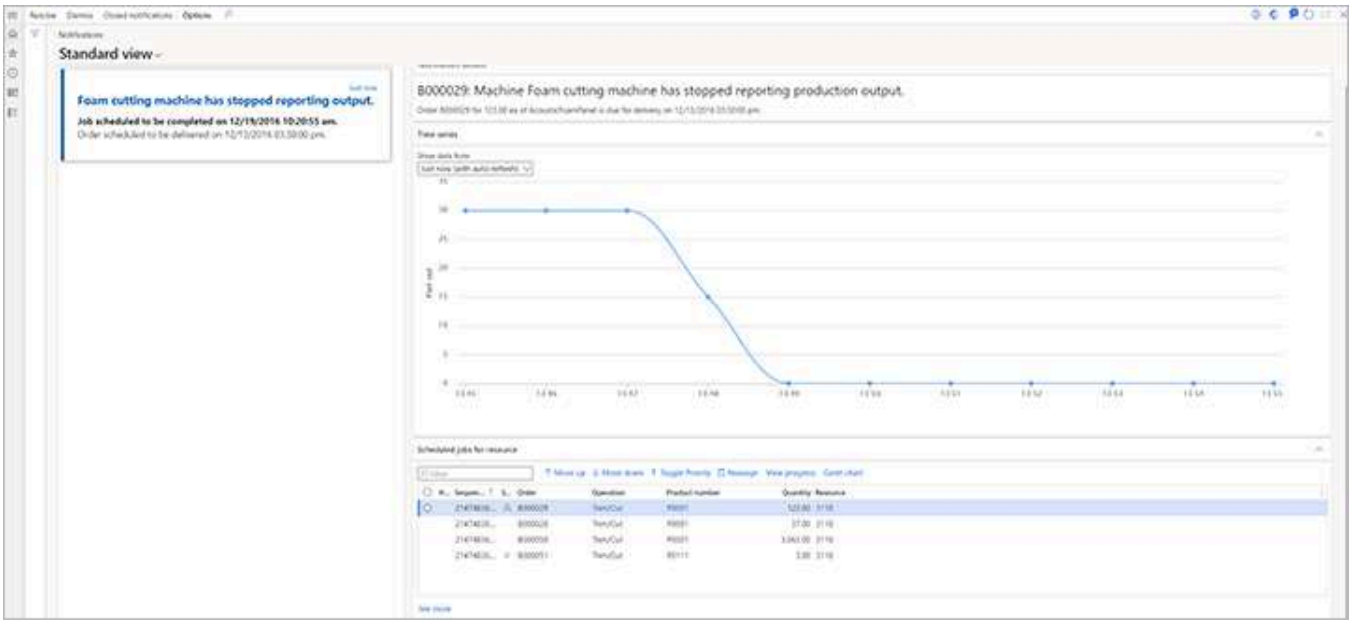


View machine status on the Notifications page

On the **Notifications** page, supervisors can view the notifications that are generated when too much time has passed since the sensor last delivered a *part-out* signal. Each notification provides an overview of the production job that is affected by the outage and gives the option to reassign the affected job to another resource.

To open the **Notification** page, go to **Production control** > **Inquiries and reports** > **Sensor Data Intelligence** > **Notifications**.

The following illustration shows an example of a machine status notification.



Last updated on 05/07/2025

Product quality scenario (preview)

Article • 10/18/2022

[This article is prerelease documentation and is subject to change.]

In the *product quality* scenario, a sensor is set up to measure the quality of a product batch on the shop floor. If a measurement falls outside a defined threshold for the product, a notification is shown on the supervisor's dashboard. For example, a sensor is measuring the moisture of a food product that comes out of the production line. If the measurement is outside the allowed minimum or maximum value for moisture for the product, a notification is generated.

Scenario dependencies

The *product quality* scenario has the following dependencies:

- An alert can be triggered only if a production order is running on a mapped machine, and if that machine is producing a product that has a mapped batch attribute.
- A signal that represents the batch attribute must be sent to the IoT hub, and a unique property name must be included.

Prepare demo data for the product quality scenario

If you want to use a demo system to test the *product quality* scenario, use a system where the demo data is installed, select the *USMF* legal entity (company), and prepare the additional demo data as described in this section. If you're using your own sensors and data, you can skip this section.

In this section, you will set up the demo data so that you can use released product *P0111* (*Composite Case*) with the *product quality* scenario. In this scenario, the system tracks whether a batch attribute value that is measured by a sensor is outside the defined threshold for a batch attribute that is associated with the product.

Set up a sensor simulator

If you want to try this scenario without using a physical sensor, you can set up a simulator to generate the required signals. Learn more in Set up a simulated sensor for

testing.

Associate a batch attribute and resource with product P0111

Follow these steps to associate a batch attribute with product *P0111* (*Composite Case*) and verify that resource *3118* (*Foam cutting machine*) is used for it.

1. Go to **Product information management** > **Products** > **Released products**.
2. Find and select the product where the **Item number** field is set to *P0111*.
3. On the Action Pane, on the **Manage inventory** tab, in the **Batch attributes** group, select **Product specific**.
4. On the **Product specific** page, select **New** to create a product-specific batch attribute.
5. On the header, set the following fields:
    - **Attribute code** – Select the scope of attributes that you will monitor (*Table*, *Group*, or *All*). For this scenario, select *Table*, because you will always monitor a single attribute.
    - **Attribute relation** – Select the attribute that you will use the *product quality* scenario to monitor the value of. For this example, select *Concentration*, which is an attribute that is included in the standard demo data.
6. On the **Values** FastTab, in the **Minimum** and **Maximum** fields, define the range of acceptable values that the attribute must report to pass the quality check. If the sensor reports a value outside this range, the system will identify it as a quality violation. The other fields on this FastTab aren't relevant to the *product quality* scenario. The default values that you currently see come from the demo data. Therefore, leave them as they are, and close the **Product specific** page to return to the **Released product details** page for item *P0111*.
7. On the Action Pane, on the **Engineer** tab, in the **View** group, select **Route**.
8. On the **Route** page, on the **Overview** tab at the bottom of the page, select the line where the **Oper. No.** field is set to *30*.
9. On the **Resource requirements** tab at the bottom of the page, make sure that resource *3118* (*Foam cutting machine*) is associated with the operation.

Create and release a production order for product P0111

Follow these steps to create and release a production order for product *P0111*.

1. Go to **Production control** > **Production orders** > **All production orders**.
2. On the **All production orders** page, on the Action Pane, select **New batch order**.
3. In the **Create batch** dialog box, set the following values:
    - **Item number:** *P0111*
    - **Quantity:** *10*
4. Select **Create** to create the order and return to the **All production orders** page.
5. Use the **Filter** field to search for production orders where the **Item number** field is set to *P0111*. Then find and select the production order that you just created.
6. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Estimate**.
7. In the **Estimate** dialog box, select **OK** to run the estimate.
8. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Release**.
9. In the **Release** dialog box, make a note of the number of the batch order that you just created.
10. Select **OK** to release the order.

Configure the production floor execution interface

If you haven't already done so, configure the production floor execution interface to show jobs that are assigned to resource *3118* (*Foam cutting machine*). For instructions, see the following sections:

- Configure the production floor execution interface
- Enable search option on the production floor execution interface

Start the first job in the batch order

Follow these steps to start the first job in the batch order.

1. Go to **Production control** > **Manufacturing execution** > **Production floor execution**.
2. In the **Badge ID** field, enter *123*. Then select **Sign in**.

3. If you're prompted for an absence reason, select one of the cards for absence, and then select **OK**.
4. In the **Search** field, enter the batch order number that you previously made a note of. Then select the **Return** key.
5. Select the order, and then select **Start job**.
6. In the **Start job** dialog box, select **Start**.

Set up the product quality scenario

Follow these steps to set up the *product quality* scenario in Supply Chain Management.

1. Go to **Production control** > **Setup** > **Sensor Data Intelligence** > **Scenarios**.

2. In the **Product quality** scenario box, select **Configure** to open the setup wizard for this scenario.

3. On the **Sensors** page, select **New** to add a sensor to the grid. Then set the following fields for it:

    - **Sensor ID** – Enter the ID of the sensor that you're using. (If you're using the Raspberry PI Azure IoT Online Simulator and have set it up as described in Set up a simulated sensor for testing, enter *Quality*.)
    - **Sensor description** – Enter a description of the sensor.

4. Repeat the previous step for each additional sensor that you want to add now. You can come back and add more sensors at any time.

5. Select **Next**.

6. On the **Business record mapping** page, in the **Sensors** section, select the record for one of the sensors that you just added.

7. In the **Business record mapping** section, select **New** to add a row to the grid.

8. On the new row, the **Business record type** field should automatically be set to *Resources(WrkCtrTable)*. Set the **Business record** field to the resource that you're using the selected sensor to monitor. (If you're using the demo data that you created earlier in this article, set it to *3118, Foam cutting machine*.)

9. Immediately after you select a business record type for the row that you added in the previous step, a second row is automatically added to the grid. On this row, the **Business record type** field should be set to *Batch attributes(PdsBatchAttrib)*. Set the **Business record** field to the batch attribute that you're using the selected sensor to

monitor. (If you're using the demo data that you created earlier in this article, set it to *Concentration, Concentration Percentage*.)

10. Select **Next**.

11. On the **Activate sensors** page, in the grid, select the sensor that you added, and then select **Activate**. For each activated sensor in the grid, a check mark appears in the **Active** column.

12. Select **Finish**.

Work with the product quality scenario

View product quality data on the Resource status page

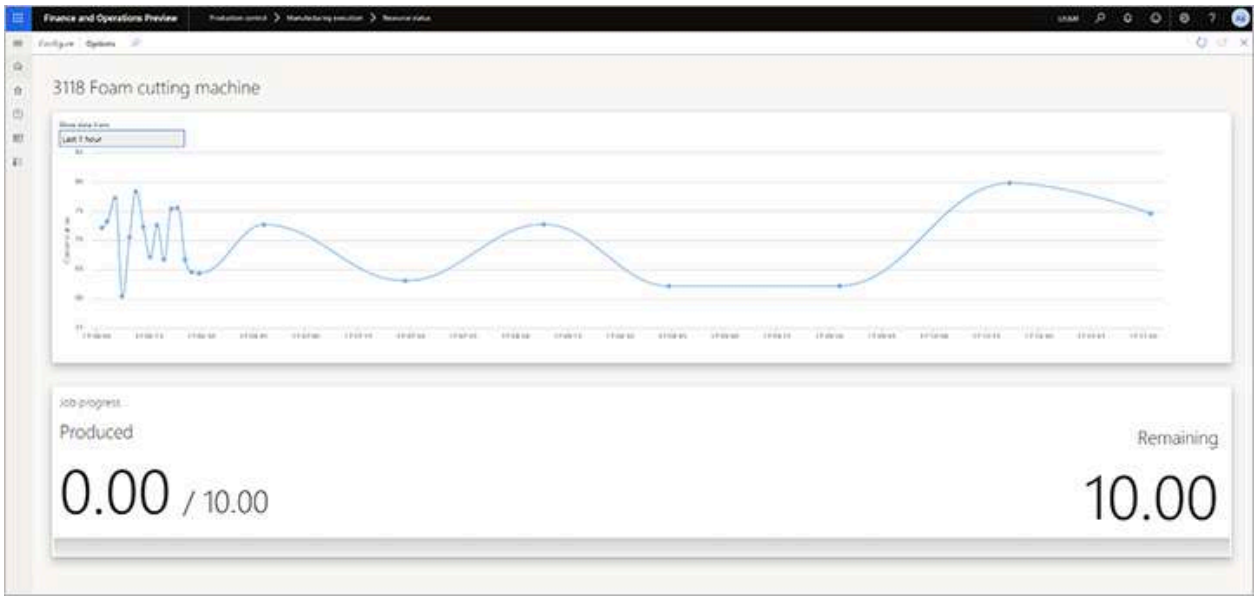
On the **Resource status** page, supervisors can monitor a timeline of the product quality signal that is received from the sensors that are mapped to each machine resource. Follow these steps to configure the timeline.

1. Go to **Production control** > **Manufacturing execution** > **Resource status**.

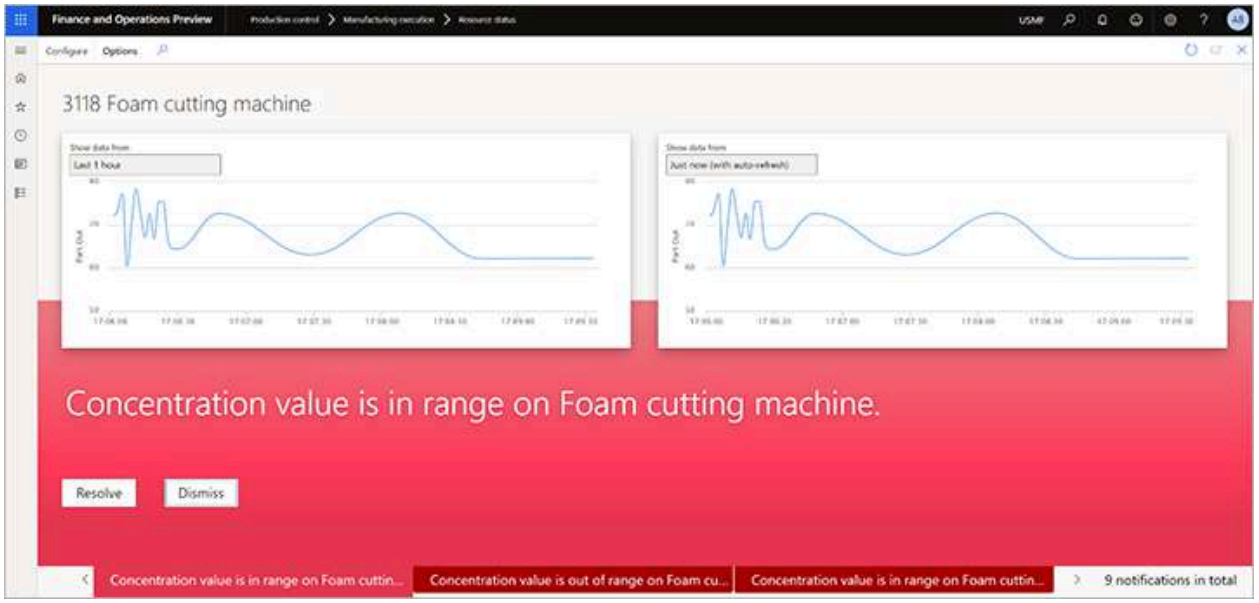
2. In the **Configure** dialog box, set the following fields:

    - **Resource** – Select the resources that you want to monitor. (If you're working with the demo data, select *3118*.)
    - **Time series 1** – Select the record (metric key) that has the following format for its name: *ProductQuality:<JobId>:<AttributeName>*
    - **Display name** – Enter *Batch attribute values*.

The following illustration shows an example of product quality data on the **Resource status** page when the value is in the range of acceptable values.



The following illustration shows an example of product quality data when an out-of-range value is detected.

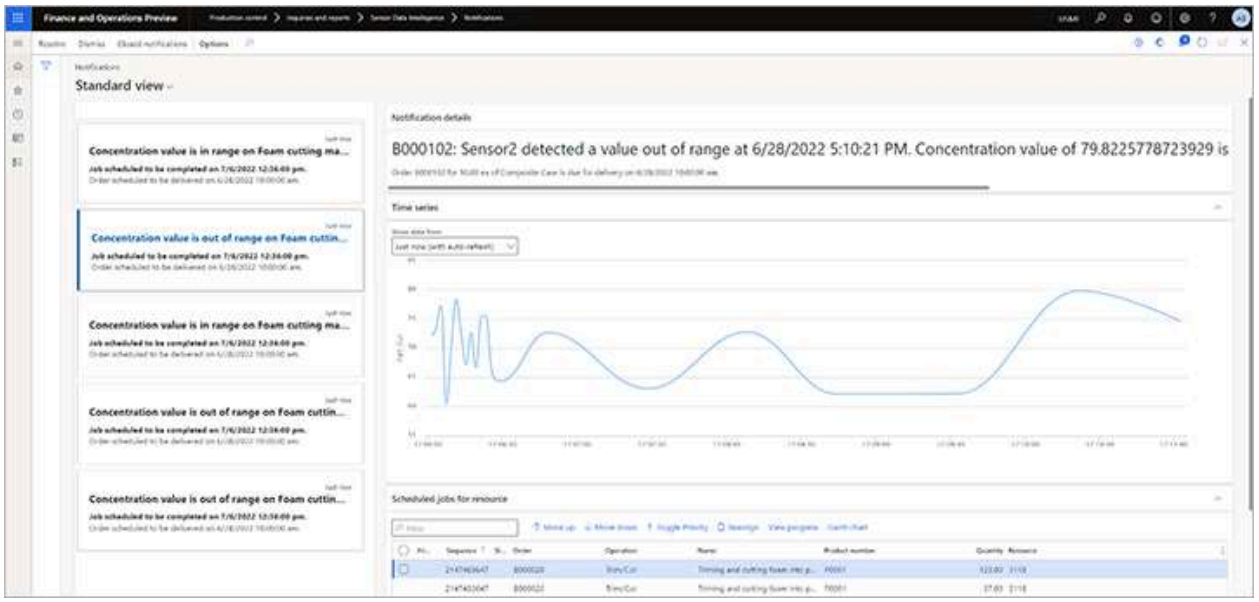


View product quality data on the Notifications page

On the **Notifications** page, supervisors can view the notification that are generated when the sensor measures a batch attribute value that falls outside the defined threshold for the batch attribute. Each notification provides an overview of the production job that is affected by the outage and gives the option to reassign the affected job to another resource.

To open the **Notification** page, go to **Production control** > **Inquiries and reports** > **Sensor Data Intelligence** > **Notifications**.

The following illustration shows an example of a product quality notification.



Feedback

Was this page helpful? Yes No

Provide product feedback

Production auto report scenario (preview)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

The *production auto report* scenario automatically reports finished quantities when a machine generates enough sensor signals to reach a defined output threshold. The scenario requires fitting the machine with a sensor that emits a signal for each production cycle, and mapping that sensor to the corresponding resource and operation in Supply Chain Management. By defining how many signals correspond to one finished unit, the system can automatically determine when progress should be reported and update the production order and inventory accordingly.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.47 or later.
- The following features must be turned on in feature management.
  - *(Preview) Sensor Data Intelligence*
  - *(Preview) Sensor Data Intelligence - Auto report progress*

Set up the production auto report scenario

To set up the production auto report scenario, you need to define the sensors that are installed on the machines, map those sensors to the corresponding resources and operations in Supply Chain Management, and define thresholds for how many sensor signals correspond to finished units for each operation. Follow these steps to set up the scenario:

1. Go to **Production control** > **Setup** > **Sensor Data Intelligence** > **Scenarios** to open the **Scenarios** page.

2. In the **Production auto report** box, select **Configure** to open the setup wizard for this scenario.

3. On the **Sensors** page, select **New** to add a sensor to the grid. Then set the following fields for it:

    - **Sensor ID** – Enter the ID of the sensor that you're using. (If you're using the Raspberry PI Azure IoT Online Simulator and have set it up as described in Set up a simulated sensor for testing, enter *AssetDownTime*.)
    - **Sensor description** – Enter a description of the sensor.

4. Repeat the previous step for each sensor that you want to add now. You can come back and add more sensors at any time.

5. Select **Next**.

6. On the **Business record mapping** page, in the **Sensors** section, select the record for one of the sensors that you just added.

7. In the **Business record mapping** section, select **New** to add a row to the grid.

8. On the new row, set the **Business record** field to the resource that you're using the selected sensor to monitor.

9. Repeat the previous step for each sensor that you want to set up now. You can come back and set up more sensors at any time.

10. Select **Next**.

11. On the **Set auto-reporting threshold** page, select the operation relation you want your configuration to apply to. You can find a specific operation relation by following these steps:

    - Select the **Item relation** column heading to open a drop-down dialog box that includes search filters for the column. Enter the product that the operation relation is created for.
    - Select the **Route relation** column heading to open a drop-down dialog box that includes search filters for the column. Enter the route number that the operation relation is created for.

    > **Tip**
    >
    > Each operation relation defines the detailed properties of an operation—such as setup time, run time, cost categories, and resource requirements—for a specific route or product. It lets each operation have several different properties depending on where it's used.

12. When you have found the applicable operation relation, enter values in the following columns for it:

    - **Pulses per cycle** – Number of pulses the sensor emits in a production cycle.
    - **Units per cycle** – Number of parts produced by the operation in a production cycle.

    > **Tip**
    >
    > A production cycle is the smallest repeatable unit of work performed by a machine that produces a measurable output.

13. Set up more operation relations with their corresponding thresholds as needed. You can come back and set up more operation relations at any time.

14. Select **Next**.

15. On the **Activate sensors** page, in the grid, select the sensor that you set up, and then select **Activate**. For each activated sensor in the grid, a check mark appears in the **Active** column.

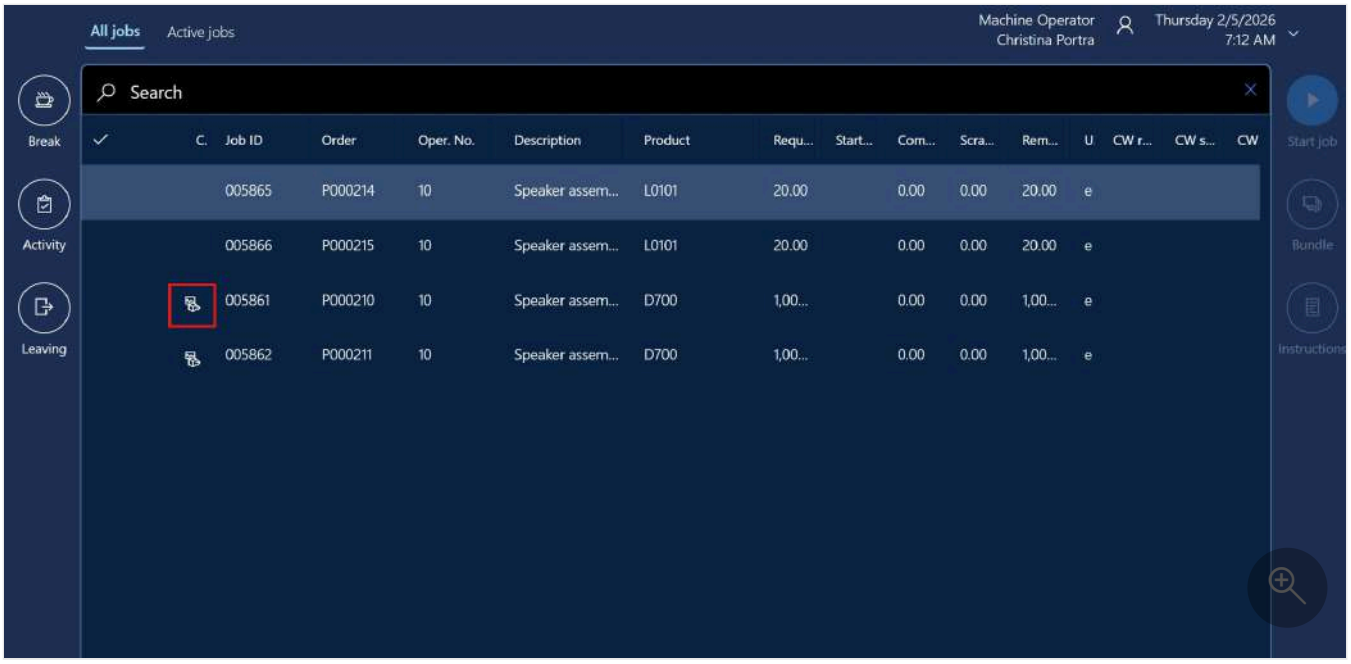
16. Select **Finish**.

Add counter displays to the production floor execution interface

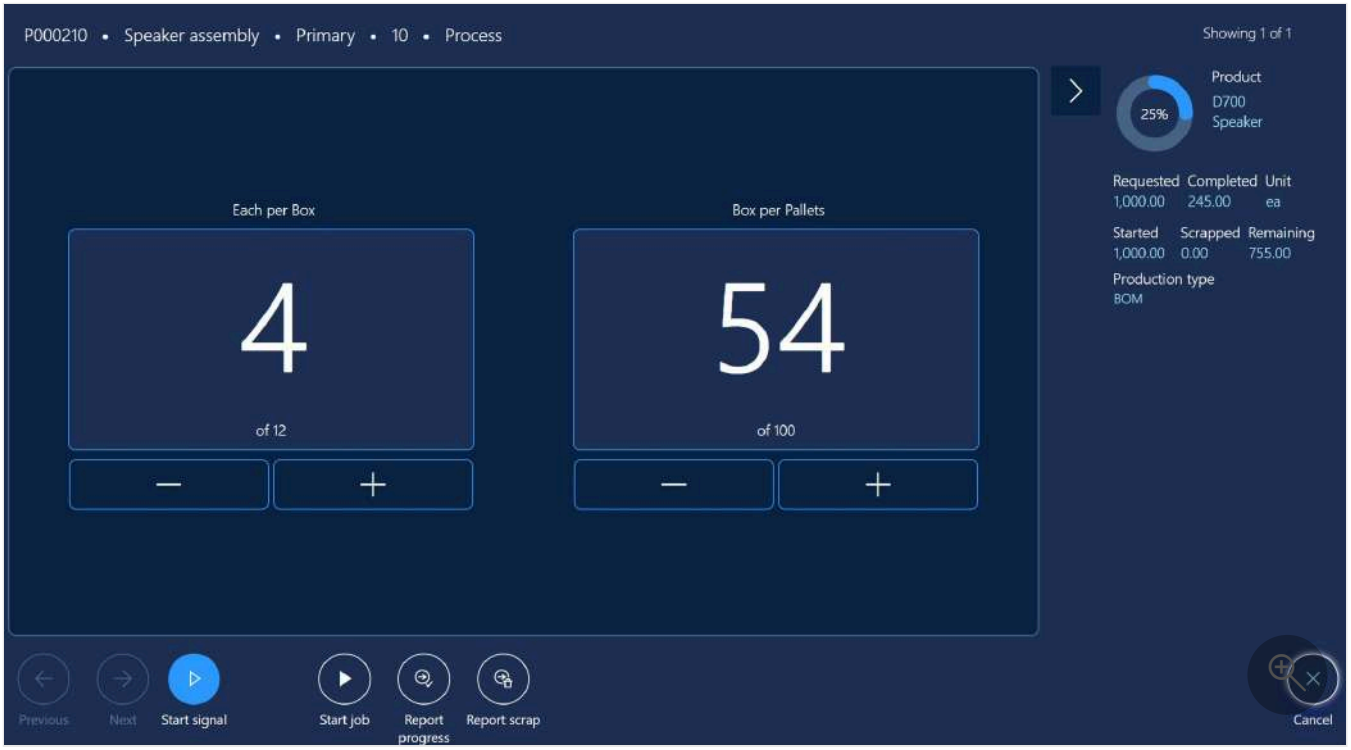
To make counter information available to workers, add a button to the relevant toolbars on the production floor execution interface. Follow the procedure provided in Design the production floor execution interface to insert the action called *View counter* at suitable locations of the production floor execution interface UI.

View sensor counters in the production floor execution interface

Jobs configured for auto reporting through a sensor signal show a dedicated icon in the job list on the production floor execution interface, as indicated in the following image.



After you start a sensor-controlled job, you can open the **View counter** dialog by selecting **View counter** from the toolbar in the job list. As shown in the following image, the **View counter** dialog displays the current progress counters for the job.



Counters are shown in one or two levels. The number of levels depends on the **Unit sequence group ID** defined on the finished product. Learn more about how to set up unit sequence groups in: Unit of measure and stocking policies.

Example of counters in a two level setup

In the example **View counter** dialog shown in the previous image, the first-level (left-hand) counter indicates how many pieces have been added to the box currently being filled. Each

time a sensor signal is received, the level-1 counter (*Pieces per box*) increases. When the threshold of 12 pieces per box is reached, the level-1 counter resets to zero and the level-2 counter (*Boxes per pallet*) increases by one. When the level-2 threshold of 100 boxes per pallet is reached, the level-2 counter also resets to zero, indicating that a new pallet is ready to be filled. At this point, the system automatically reports a pallet as finished. The report as finished posting is processed through the message queue as an asynchronous (deferred) operation. Learn more about deferred posting in Make finished goods physically available before posting to journals.

Pausing signals and adjusting counters

If something unexpected happens, such as a dropped counter signal or a counted item that needs to be scrapped, you can pause the sensor signal and adjust the counters and report quantities as needed. Follow these steps to pause the signal, adjust counters, or adjust reported quantities:

1. Open the counter dialog by selecting **View counter** from the toolbar in the job list.
2. In the counter dialog, select the **Pause signal** button to pause the sensor signal. This allows you to report scrap, report progress, and adjust the counters without new signals coming in.
    - If you need to adjust the counter values, use the + and – buttons below each counter as needed.
    - If you need to report some counted quantity as scrap, select the **Report scrap** button. This action opens the **Reporting as scrap** dialog, where you can enter the quantity to be scrapped and the reason for scrapping.
    - If you need to manually report progress (for example, because a sensor signal was missed), select the **Report progress** button. This action opens the **Reporting progress** dialog, where you can enter the quantity to report as finished.
3. Select the **Start signal** button to restart the sensor signal.

Navigating between multiple sensor enabled jobs

If multiple jobs in the job list are enabled for sensor controlled reporting, use the **Previous** and **Next** buttons of the **View counter** dialog to switch between jobs without closing the dialog.

Last updated on 02/10/2026

Production delays scenario (preview)

Article • 10/18/2022

[This article is prerelease documentation and is subject to change.]

The *production delays* scenario generates a notification if production throughput falls below a specific threshold value. In this scenario, a *part-out* signal is sent to Microsoft Azure IoT Hub for each item that is produced. In Dynamics 365 Supply Chain Management, the order delay is calculated based on the amount of time that the production order is scheduled to run, the number of items that should be produced, the amount of time that the job has been running, and the number of *part-out* signals that have been received. A delay notification is generated if the number of *part-out* signals for the job falls below the threshold value.

Scenario dependencies

The *production delays* scenario has the following dependencies:

- An alert can be triggered only if a production order is running on a mapped machine.
- A signal that represents a mapped machine's *part-out* signal must be sent to IoT Hub, and a unique property name must be included.
- A UNIX timestamp property, where the value is expressed in milliseconds (ms), must be present in the Azure IoT Hub message.

Prepare demo data for the product delays scenario

If you want to use a demo system to test the *production delays* scenario, use a system where the demo data is installed, select the *USMF* legal entity (company), and prepare the additional demo data as described in this section. If you're using your own sensors and data, you can skip this section.

Set up sensor simulator

If you want to try this scenario without using a physical sensor, you can set up a simulator to generate the required signals. Learn more in Set up a simulated sensor for testing.

Verify that resource 3118 is used for product P0111

A production order will be scheduled and released. Therefore, a production job is released to resource *3118* (*Foam cutting machine*). Follow these steps to verify that resource *3118* is used for product *P0111* in your demo data.

1. Go to **Product information management** > **Products** > **Released products**.
2. Find and select the product where the **Item number** field is set to *P0111*.
3. On the Action Pane, on the **Engineer** tab, in the **View** group, select **Route**.
4. On the **Route** page, on the **Overview** tab at the bottom of the page, select the line where the **Oper. No.** field is set to *30*.
5. On the **Resource requirements** tab at the bottom of the page, make sure that resource *3118* (*Foam cutting machine*) is associated with the operation.

Create and release a production order for product P0111

Follow these steps to create and release a production order for product *P0111*.

1. Go to **Production control** > **Production orders** > **All production orders**.
2. On the **All production orders** page, on the Action Pane, select **New batch order**.
3. In the **Create batch** dialog box, set the following values:
    - **Item number:** *P0111*
    - **Quantity:** *10*
4. Select **Create** to create the order and return to the **All production orders** page.
5. Use the **Filter** field to search for production orders where the **Item number** field is set to *P0111*. Then find and select the production order that you just created.
6. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Estimate**.
7. In the **Estimate** dialog box, select **OK** to run the estimate.
8. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Release**.
9. In the **Release** dialog box, make a note of the number of the batch order that you just created.
10. Select **OK** to release the order.

Configure the production floor execution interface

If you haven't already done so, configure the production floor execution interface to show jobs that are assigned to resource *3118* (*Foam cutting machine*). For instructions, see the following sections:

- Configure the production floor execution interface
- Enable search option on the production floor execution interface

Start the first job in the batch order

Follow these steps to start the first job in the batch order.

1. Go to **Production control** > **Manufacturing execution** > **Production floor execution**.
2. In the **Badge ID** field, enter *123*. Then select **Sign in**.
3. If you're prompted for an absence reason, select one of the cards for absence, and then select **OK**.
4. In the **Search** field, enter the batch order number that you previously made a note of. Then select the **Return** key.
5. Select the order, and then select **Start job**.
6. In the **Start job** dialog box, select **Start**.

Set up the production delays scenario

Follow these steps to set up the *production delays* scenario in Supply Chain Management.

1. Go to **Production control** > **Setup** > **Sensor Data Intelligence** > **Scenarios**.

2. In the **Production delays** scenario box, select **Configure** to open the setup wizard for this scenario.

3. On the **Sensors** page, select **New** to add a sensor to the grid. Then set the following fields for it:

    - **Sensor ID** – Enter the ID of the sensor that you're using. (If you're using the Raspberry PI Azure IoT Online Simulator and have set it up as described in Set up a simulated sensor for testing, enter *ProductionDelay*.)
    - **Sensor description** – Enter a description of the sensor.

4. Repeat the previous step for each additional sensor that you want to add now. You can come back and add more sensors at any time.

5. Select **Next**.
6. On the **Business record mapping** page, in the **Sensors** section, select the record for one of the sensors that you just added.
7. In the **Business record mapping** section, select **New** to add a row to the grid.
8. On the new row, set **Business record** to the resource that you're using the selected sensor to monitor. (If you're using the demo data that you created earlier in this article, set it to *3118, Foam cutting machine*.)
9. Select **Next**.
10. On the **Production delay deviation threshold** page, if you're using the demo data that you created earlier in this article, follow these steps:
    a. Select the **Item relation** column heading to open a drop-down dialog box that includes search filters for the column. Enter *P0111* in the search box, and then select **Apply**.
    b. Select the line where the **Operation** field is set to *Trim/Cut*. Set the **Notification threshold for delay (%)** field to the threshold (as a percentage) that a notification should be sent at.
11. Select **Next**.
12. On the **Activate sensors** page, in the grid, select the sensor that you added, and then select **Activate**. For each activated sensor in the grid, a check mark appears in the **Active** column.
13. Select **Finish**.

Work with the production delays scenario

View production delay data on the Resource status page

On the **Resource status** page, supervisors can monitor a timeline of signals that are received from the sensors that are mapped to each machine resource. Follow these steps to configure the timeline.

1. Go to **Production control** > **Manufacturing execution** > **Resource status**.
2. In the **Configure** dialog box, set the following fields:

    - **Resource** – Select the resources that you want to monitor. (If you're working with the demo data, select *3118*.)

- **Time series 1** – Select the record (metric key) that has the following format for its name: *ProductionJobDelayed:ActualQuantity:<JobId>*
- **Display name** – Enter *Parts out signal*.

Feedback

Was this page helpful? Yes No

Provide product feedback

IoT Intelligence now replaced by Sensor Data Intelligence

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The IoT Intelligence feature has been updated to version 2.0 and, at the same time, renamed to *Sensor Data Intelligence*. The new version is currently available as a preview feature for Supply Chain Management version 10.0.30 or later.

Sensor Data Intelligence introduces a new deployment model where the required Azure components are deployed on your own Azure subscription rather than being managed by Microsoft. This new model enhances your ability to customize the feature and to integrate it with third-party systems. The new version also introduces support for two new internet of things (IoT) scenarios for Asset management.

Because of these upcoming changes, Microsoft recommends against starting any new projects based on the existing IoT Intelligence feature. If you are interested in adding IoT capabilities to your system, we recommend that you get started by working with the preview version of Sensor Data Intelligence instead. For more information about Sensor Data Intelligence and how to work with it, see the Sensor Data Intelligence home page.

Last updated on 05/27/2026

Master planning home page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

At its core, master planning helps you determine and balance the future need for raw materials and capacity to meet your goals. Master planning assesses the following questions:

- What raw materials and capacities are currently available?
- What raw materials and capacities are required to complete production? For example, what must be manufactured, purchased, transferred, or set aside as safety stock before you can complete production.

Master planning uses this information to calculate the requirements and generate planned orders.

The three main planning processes are:

- **Master planning** - The master plan calculates net requirements. It's based on actual current orders and enables you to control inventory replenishment on a short-term, day-to-day basis. Another term to describe it is the *Net requirements plan*. Learn more in Master plans overview.
- **Forecast planning** - The forecast schedule calculates gross requirements. It's based on future projections (or forecasts), and enables you to conduct long-term planning of materials and capacity. Learn more in Demand forecasting overview.
- **Intercompany master planning** - The intercompany master plan calculates net requirements across legal entities. It connects demand and supply between companies not only for short term, but also for long-term, planned (not yet firmed) demand and supply. Learn more in Intercompany planning.

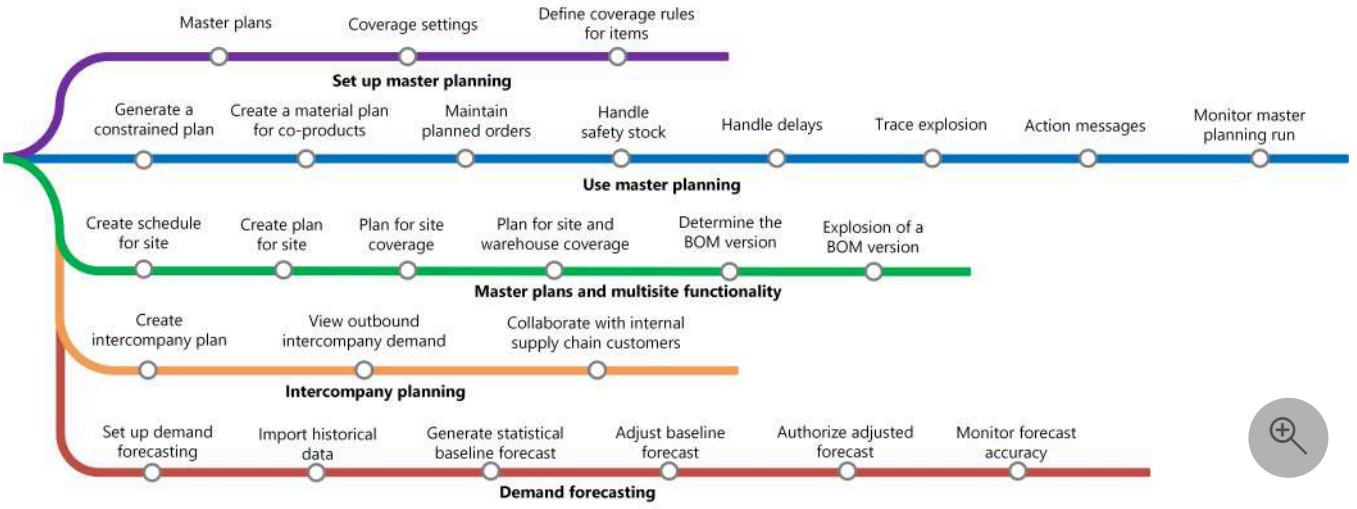
You can change the output of the plan.

Master scheduling plans typically involve the short term, which can be anywhere from one week to six months. The Master planning module determines the supply (materials) and capacity (resources) needs that will meet current demand (the net requirements). In most companies, this planning process extends to include the longest cumulative lead time among the products to be received.

> **Note**
>
> With Planning Optimization, all plans are regenerative, whereas the deprecated master planning engine had a concept of net change and regenerative planning run. Regenerative plans update all requirements, whereas net change plans only update the plan for items with new requirements that were added since the last scheduling run.

Learning map

The following learning map shows the major concepts and tasks that make up the framework of the Master planning module.



Related information

Roadmaps

Go to the Dynamics 365 and Microsoft Power Platform release planner to see what new features are released and what new features are in development.

Blogs

Find opinions, news, and other information about Master planning and other solutions on the Dynamics AX Manufacturing R&D Team blog and Microsoft Dynamics 365 Blog.

Webinars

Use Azure Machine Learning for demand forecasting

Tech conference recordings

- Planning Optimization service
- Get started with Planning Optimization
- Demand forecasting series
- Extend the demand forecasting functionality

Last updated on 03/26/2026

Get started with master planning

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

Master planning in Supply Chain Management is provided by an external service called the Planning Optimization Add-in for Dynamics 365 Supply Chain Management. This article explains how to obtain and set up that service.

Availability

Planning Optimization is currently available in the following Azure geographies: United States, United States government cloud, Canada, Brazil, Europe, France, United Kingdom, Norway, Switzerland, Australia, Asia Pacific, Japan, United Arab Emirates, South Africa, and India. If you try to install the add-in from another geographic region, Microsoft Dynamics Lifecycle Services shows a message that this geographic isn't supported. For more information about Azure geographies and the related regions, see Azure geographies.

> **Note**
>
> Planning Optimization doesn't support on-premises deployments of Dynamics 365 Supply Chain Management.

Licensing

You can run master planning using your current Supply Chain Management licenses. Planning Optimization is the master planning engine for Supply Chain Management and is licensed as part of the standard licenses for the system. There are no extra costs associated with using Planning Optimization.

Install and enable Planning Optimization

To use Planning Optimization, make sure your system has all the prerequisites in place. Then, enable its configuration key and install the Planning Optimization Add-in for Dynamics 365 Supply Chain Management.

Prerequisites

Before you install the Planning Optimization Add-in, make sure the following prerequisites are in place:

- You must be running Supply Chain Management on a Lifecycle Services enabled high-availability environment, tier 2 or higher (not a OneBox environment), with Dynamics 365 Supply Chain Management version 10.0.23 or later. If you try to install the add-in on a OneBox environment, the installation won't complete and you'll need to cancel the installation.

  > **ⓘ Important**
  >
  > Planning Optimization can't be installed on a development (OneBox) environment. It requires a Lifecycle Services–enabled, high-availability environment, tier 2 or higher. If you need to troubleshoot installation issues, see **Troubleshoot Planning Optimization**.

- Your system must be set up for Power Platform integration. Learn more in Microsoft Power Platform integration with finance and operations apps.

- Your Microsoft Entra account must have a Supply Chain Management licensed assigned to it. As mentioned previously, you don't need an extra license for Planning Optimization, but you do need a Supply Chain Management license to install it. If you get an error message regarding licensing while installing the Planning Optimization Add-in, check your licenses as described in Assign licenses to users.

- You must sign in to your Power Platform environment using an account with administrator privileges and an access mode of *Read-Write*. If you get an error message regarding missing user permissions while installing the Planning Optimization Add-in, follow these steps:

    1. Go to the Power Platform admin center.
    2. Open the environment where you want to install the add-in.
    3. Go to **Settings** > **Users** and select your user account from the list to see its details.
    4. From your user details page, select the **Client Access License (CAL) information** link.
    5. On the **Client Access License (CAL) information** page, make sure that **Access Mode** is set to *Read-Write*.

    For more information about licenses and access modes for the Power Platform, see Create a Power Platform admin account.

Enable the Planning Optimization configuration key

To use Planning Optimization, you must enable its configuration key. follow these steps:

1. Put your system into maintenance mode, as described in Maintenance mode.
2. Go to **System administration** > **Setup** > **License configuration**.
3. On the **Configuration keys** tab, select the check box for **Planning Optimization**.
4. Turn off maintenance mode, as described in Maintenance mode.

Install the Planning Optimization Add-in

You must install the add-in from your Lifecycle Services project and then turn on the Planning Optimization functionality from the Supply Chain Management user interface.

To install the Planning Optimization Add-in:

1. Sign in to Lifecycle Services, and open the desired environment.
2. Go to **Full details**.
3. Scroll down to the **Environment add-ins** FastTab.
4. Select **Install a new add-in**.
5. Select **Planning Optimization**.
6. Follow the installation guide, and agree to the terms and conditions.
7. Select **Install**.
8. On the **Environment add-ins** FastTab, you should see that Planning Optimization is installing.

9. After a few minutes, **Installing** changes to **Installed** (you might need to refresh the page). When installed, you're ready to activate Planning Optimization in Dynamics 365 Supply Chain Management.

The main purpose of installing the Planning Optimization Add-in is to connect the service and the environment. Therefore, you must install the add-in separately on each environment where you use Planning Optimization, regardless of any code moved between the environments.

Turn on Planning Optimization for your environment

After you install the Planning Optimization Add-in for your environment, you must enable it in Supply Chain Management before you can start using it.

> **Note**
>
> You can continue to use the deprecated master planning engine for one or more previously created companies (until you're ready to migrate them) even while using Planning Optimization for the others. For instructions on how to set a company to use the deprecated master planning engine, see Continue to use deprecated master planning with existing companies.

To configure your system to use the Planning Optimization Add-in for master planning, follow these steps:

1. Sign in to Supply Chain Management.
2. Go to **Master planning** > **Setup** > **Planning Optimization parameters**.
3. Open the **General** tab.
4. Check the **Connection status**. It shows one of the values listed in the following table.

Expand table

Connection status	Description	Can Planning Optimization be used?
Connected	A connection is established between the Planning Optimization service and Supply Chain Management.	Yes

Connection status	Description	Can Planning Optimization be used?
Enabling connection	A request to turn on the connection to the Planning Optimization service is currently in progress.	No
Disconnected	There's no connection to the Planning Optimization service. You can turn on the connection from Lifecycle Services, as described earlier in this article.	No
Disabling connection	A request to turn off the connection to the Planning Optimization service is currently in progress.	No
Getting status	The system is waiting for status information from the Planning Optimization service.	No

5. If the **Connection status** is *Connected*, the **Use Planning Optimization** option is read-only and set to *Yes*. In most cases, Planning Optimization is now the only planning engine available for Supply Chain Management, so you can't turn it off. However, in some rare cases, Microsoft Support might make the deprecated planning available for one or more specific companies (learn more in Migration to Planning Optimization for master planning).

> **ⓘ Important**
>
> If existing planning batch jobs that were created for the deprecated master planning engine are triggered while the **Use Planning Optimization** option is set to *Yes*, those jobs will fail.

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Last updated on 05/05/2026

Master planning system architecture

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

In Supply Chain Management, the Planning Optimization Add-in for Microsoft Dynamics 365 Supply Chain Management manages master planning. This add-in enables master planning calculation to occur outside Dynamics 365 Supply Chain Management and the related SQL database. The benefits of the Planning Optimization functionality include improved performance and minimal impact on the SQL database during master planning runs. You can run quick planning runs even during office hours, so that planners can immediately react to demand or parameter changes.

To use Planning Optimization, install the Planning Optimization Add-in from your project in Microsoft Dynamics Lifecycle Services and turn on the Planning Optimization functionality in Supply Chain Management. Learn more in Get started with master planning.

The following illustration shows the advantage of running Planning Optimization during office hours.

Performance

You can use Planning Optimization in scenarios that involve long-running master plans. It's specifically designed for very fast calculations that involve very large volumes of data. Because it's built as a hyper-scalable multitenant service, multiple instances can work together simultaneously to calculate the plan. Additionally, the planning service removes the load of master planning from your system and works with a data stream that minimizes the server load.

Planning Optimization can help you achieve the following goals:

- Improve planning performance through a shorter runtime.
- Reduce the impact on other processes during the master planning run.
- Do more frequent planning runs. (You're not limited to daily runs.)
- Be confident that future business growth won't overload the planning system.

Architecture and data flow

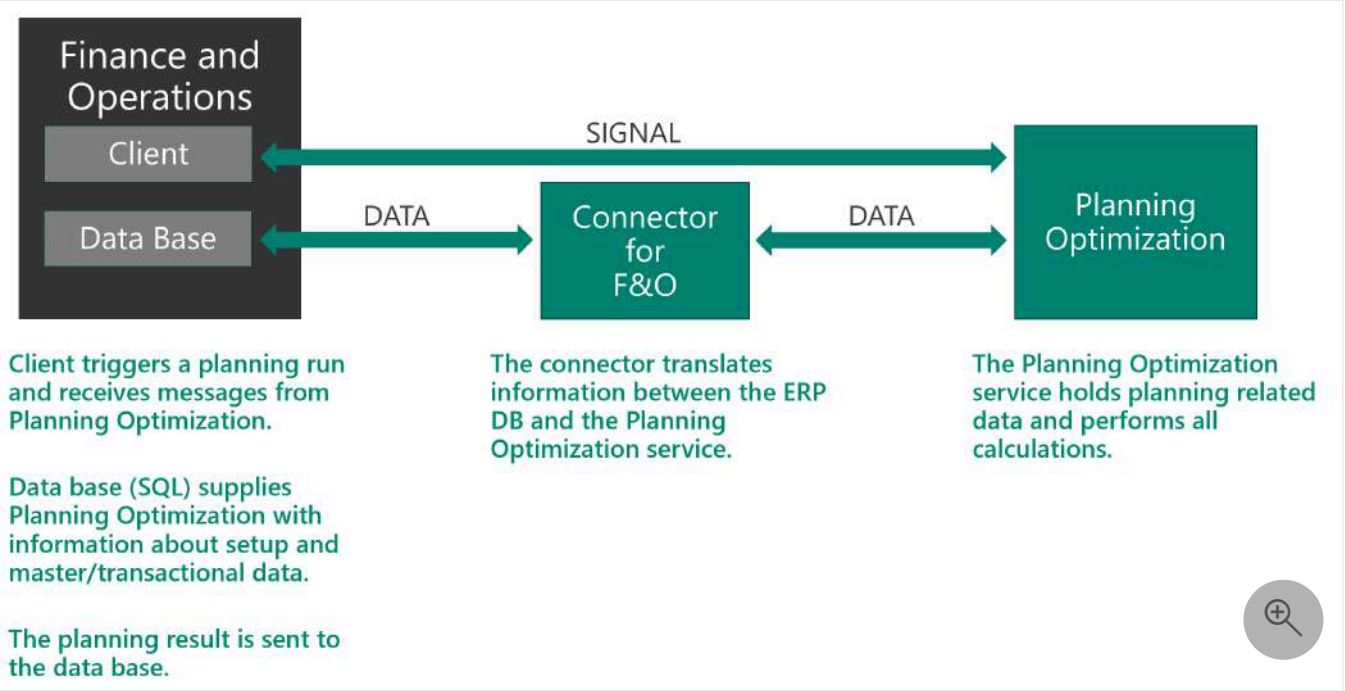
When you install the Planning Optimization Add-in from Lifecycle Services, you establish a secure connection to the Planning Optimization service. The service is located in the same data center country or region as the related Supply Chain Management instance. After you set up Planning Optimization, when you run master planning, Supply Chain Management sends master data and transactional data to the Planning Optimization service.

If you uninstall the Planning Optimization Add-in, the system removes all related data in the Planning Optimization service.

High-level data flow for regeneration runs

1. The Supply Chain Management client sends a signal to request a planning run from Planning Optimization.
2. Planning Optimization requests the required data through the integrated connector.
3. The SQL database sends the requested information about setup, master, and transactional data to Planning Optimization through the connector. The connector translates information between Supply Chain Management and the Planning Optimization service.
4. The Planning Optimization service holds planning-related data in memory and performs the required calculations.
5. The planning result is sent to the Supply Chain Management database through the connector. The results include information such as planned orders and pegging information. Planning Optimization sends a signal to Supply Chain Management to indicate that the planning run is complete. It also sends any relevant messages and warnings.

The following illustration shows the data flow.



Last updated on 03/25/2026

Planning Optimization release process and release history

Microsoft updates Planning Optimization monthly. However, based on business requirements, the product team occasionally releases other updates between the scheduled releases.

Each release is published to the individual regions where Planning Optimization is available. The process typically takes three days.

While Planning Optimization is updating, master planning might run a bit more slowly than usual.

All environments that use Planning Optimization automatically receive the latest release regardless of which version of Supply Chain Management you're using. Therefore, you don't need to backport updates. No user action is required: the service is automatically updated. However, the release process never automatically pushes breaking-change functionality to environments. By default, any changes that are considered breaking are turned off and must be explicitly turned on by using feature management. Therefore, those changes don't appear in environments until you choose to enable them.

Because notifications aren't shown when Planning Optimization is updated in your environment, you can review the release notes in the following table to determine when changes were released and what functionality they introduced. This table shows the changes that were released for Planning Optimization, whether those changes are associated with a feature from feature management, and the date of the release.

Expand table

Changes	Feature management details	Release dates
General performance, quality, and stability improvements. Customer reported bug fixes: • 1144004 – Planning Optimization is not correctly managing vendor calendars when calculating dates for planned purchase orders. • 1145533 – Planning optimization uses Sales Order Requested Ship Date instead of Requested Receipt Date to calculate Sellable Days.	No feature management required	August 18, 2026

Changes	Feature management details	Release dates
• 1146505 – A planned purchase order for service item is generated before a planned production order (FG) is approved, but no planned purchase order is regenerated after approval. • 1147129 – Planned batch order cannot be firmed after BOM explosion fails. • 1159839 – Planning Optimization fails to plan item during earliest replenishment rescan when dynamic positive days is enabled. • 1161762 – Scheduling incorrectly reports "Capacity could not be found" when overlapping slot fetches reload reservations.		
General performance, quality, and stability improvements. Customer reported bug fixes: • 1123148 – Planning results inconsistency in replenishment generation • 1126454 – The requirement date of the expired batch does not match the expiration date • 1133251 – Inconsistent standard formula consumption between BOM calculation and master planning due to midpoint rounding default difference • 1147342 – Transfer order lead time is calculated using source warehouse calendar	No feature management required	July 7, 2026
General performance, quality, and stability improvements. Customer reported bug fixes: • 1121678 – Planning Optimization incorrectly assigns expired supply to future demand despite best-before date limits • 1129635 – WRKCTRACTIVITYRESOURCEREQUIREMENT may cause errors during planning due to 'dat' as ResourceDataAreaId • 1134181 – FORCESEEK hint may select partner-extension index instead of Microsoft index, leading to SQL error 362 after partner index changes	No feature management required	June 1, 2026
General performance, quality, and stability improvements. Customer reported bug fixes:	No feature management required	April 28, 2026

Changes	Feature management details	Release dates
• 1097597 – Planning Optimization does not assign BOM/route to planned production orders when pegged requirement date equals BOM/route To date • 1107603 – The calendars considered by the planned transfer order are inconsistent when calculating the reorder margin • 1119156 – Cleanup process may accidentally remove current records, preventing post-planning actions from running • 1122974 – Data analyzer does not group finished goods and raw materials together when raw materials is set as a configuration group		
General performance, quality, and stability improvements. Customer reported bug fixes: • 1084691 – Planning Optimization incorrect derived requirements for predefined variant • 1084495 – Property value not transferred in Planning Optimization for planned batch orders • 1087477 – Production start dates perpetually push out when using vendor BOM lines, finite capacity, and active change management	No feature management required	January 19, 2026
General performance, quality, and stability improvements. Customer reported bug fixes: • 1027305 – Strict safety stock pegging and batch expiry are not supported by Planning Optimization • 1050426 – Infinite scheduling with primary operation having fractional resource load and 24h calendar resource availability • 987667 – Production start dates perpetually push out when using vendor BOM lines, finite capacity, and active change management • 1037977 – Incorrect requirement date for intercompany demand from demand forecast with transport days on net requirements form • 1068793 – Route Number Not Used in Planning Optimization for Quotations • 1030103 – Planning service is loading capacity reservation for resources that are not finite capacity leading to the performance issue	No feature management required	December 8, 2025

Changes	Feature management details	Release dates
• 1067693 – Optimization planning generates an unnecessary gap between operations for planned production order when one of the materials is linked with the end of the operation • 1028513 – Unnecessary delay given when material is consumed at operation end and next operation starts at non-working day • 1028853 – Issue with the order date in the planned purchase order related to the configuration of purchase transport days. Start date and end dates are disregarding working days and order day is pushed forward • 1063642 – Demand forecast is reduced with transfer issues • 1072379 – In PLOP, there's double reduction in the forecast due to the container reference • 1076273 – Duplicate replenishment order for purchase requisition over multiple planning runs with partial pegging		
General performance, quality and stability improvements. Internal infrastructure changes. Customer reported bug fixes: • 1003359 - Planning Optimization issue with minimum keys • 1019384 - Reference of quarantine order show as "inventory journal" in net requirement when using Planning Optimization • 1055013 - Unhandled error in min/max scenario with short positive days and demand in the future • 1050881 - Inconsistent BOM formula consumption between BOM calculation and master planning • 1020039 - We have identified an issue with how quantities are classified as "Qualified demand" in the net flow calculation for DDMRP-managed parts • 1033476 - Planned transport days are not properly considered when evaluating working calendars marked as "Closed for pickup" in Planning Optimization	No feature management required	October 6, 2025
General performance, quality, and stability improvements. Customer reported bug fixes: • 1015507 – Incorrect raw material estimation in planned batch orders (PLOP) when using yield and rounding	No feature management required	August 21, 2025

Changes	Feature management details	Release dates
• 1020065 – Forecast lines with different required BOM/route for the same date are grouped together • 976755 – Step consumption does not support negative BOM quantity		
General performance, quality, and stability improvements. Customer reported bug fixes: • 987667 – Production start dates perpetually push out when using vendor BOM lines, finite capacity, and active change management • 1029247 – Incorrect start date for planned production order • 987448 – Operation scheduling with overlap and calendar efficiency can lead to error "Internal error in scheduling engine" • 1028099 – Operation scheduling fails with an exception error	No feature management required	July 30, 2025
General performance, quality and stability improvements. Internal infrastructure changes. Customer reported bug fixes: • 1009010 – Planning Optimization is inconsistently considering on-hand that is due to expire • 986770 – Route scrap not considered when firming planned production orders • 977625 – Job scheduling is failing with error "Production order could not be scheduled. Not enough capacity could be found" when route contains two operations with hard links branching into the next operation • 987448 – Operation scheduling with overlap and calendar efficiency can lead to error "Internal error in scheduling engine" • 1034160 – When CX is running Planning Optimization, the system shows the error message "Planning Optimization client retry count exceeded" • 1028099 – Operation scheduling fails with an exception error	No feature management required	July 14, 2025
General performance, quality and stability improvements. Internal infrastructure changes.	No feature management required	May 15, 2025

Changes	Feature management details	Release dates
Customer reported bug fixes: • 987836 – Finished good scheduling to start at 12:00AM prior to an approved planned production order completion time that is being consumed • 976713 – Planning Optimization generates planned orders for inventory which has been blocked • 991562 – Formula line with negative quantity is disappearing from derived requirements form on planned orders upon running planning for different item which contains the same formula line item • 991826 – Lead time does not update to planned order by running Planning Optimization • 968759 – Ship calendar of vendor and lead time is not respected on purchase order		
General performance-, quality- and stability improvements. Customer reported bug fixes: • 1001064 – Planned supply order is considering demand beyond the coverage period • 1003359 – Planning Optimization issue with minimum keys • 1004623 – When disabling strict safety stock feature and then enabling it but not using it, the number of planned orders that have condition "Cancel" is different • 988317 – Forecast reduction not correct when order date is on the first of the reduction period and the forecast is later • 1012963 – Allow batch mixing if no `reqItemTable` record is specified is not considered • 978372 – Forecast is reduced with counting transactions when reduce forecast by Orders	No feature management required	April 11, 2025
General performance, quality, and stability improvements. New feature added: Planning by site/warehouse. This feature will be enabled in version 10.0.44 Customer reported bug fixes: • 974508 – The form "Item requirement statistics" shows wrong period when planned orders are created from Planning Optimization • 976713 – Planning Optimization generates planned orders for inventory which has been blocked	No feature management required	February 26, 2025

Changes	Feature management details	Release dates
• 972843 – Planning generates excess of demand by ignoring available on-hand in some cases of shelf life		
General performance, quality, and stability improvements. Customer reported bug fixes: • 946720 – Strict safety stock pegging feature ignores pegging priority (not chronologically) and negative days (planned purchase orders proposed before firmed purchase orders)	No feature management required	February 12, 2025
General performance, quality, and stability improvements. Support for step consumption Support for catch weight quantity calculations Customer reported bug fixes: • 942711 – Identical demands with required BOM are not grouped according to period coverage • 969100 – [Planning Optimization] Different between pegging quantity and requirement quantity in planned order • 943781 – Derived requirements are not coverage calculated if the planned batch order is inside the freezing time fence	Support for step consumption requires the *Support for material step consumption for Planning Optimization* feature to be enabled in feature management. Support for catch weight quantity calculations requires the *Calculate catch weight quantities for catch weight enabled products in Planning Optimization* feature to be enabled in feature management.	January 13, 2025
General performance, quality, and stability improvements. New features for Planning Optimization: • Support for dimension-based product configuration • Support for reducing demand and supply forecast in the current period by invoiced and delivered orders using "Transactions reduction key" Customer reported bug fixes: • 945437 – Planning Optimization doesn't update the confirmed delivery and receipt date for sales order lines with CTP for Planning Optimization status = Not ready and Delivery date control = CTP for Planning Optimization • 942038 – Excessive planned orders created for expired batch	No feature management required	November 6, 2024

Changes	Feature management details	Release dates
• 962570 – When firming a planned batch order with co-products, error about maximum decimals for stock keeping unit		
General performance, quality, and stability improvements. Customer reported bug fixes included: • 954429 – When running Planning Optimization, unit conversion is not working when conversion is set up at product variant level • 942716 – Job scheduling throws `NullReferenceException` when route group of secondary operation has job type not activated on primary operation • 867912 – Circularity detected for transfer with receipt pegged against shipment • 950963 – Expired batch inventory considered available and pegged despite shelf life setting • 934157 – Forecast plan time fence overrides coverage time fence • 936056 – Planning Optimization fails to meet demand in scenario with planning item, co-product, formula item, and forecast • 947035 – Planned production order route doesn't show link types	No feature management required	September 23, 2024
General performance, quality, and stability improvements. Customer reported bug fixes included: • 942649 – Scheduling engine returns `InternalError` for overlap and finite capacity cases • 947020 – Planned production order may not include BOM line with a validTo date • 942726 – Period coverage can give wrong results when different time zones are used • 942275 – Required BOM/route not passed through transfer orders when using period grouping for the transfer warehouse • 951751 – Coverage by inventory status leads to unpredictable scheduling/planning results • 854350 – Transfer requirement does not consider sellable days from demand sales order or forecast	No feature management required	August 22, 2024

Changes	Feature management details	Release dates
General performance, quality, and stability improvements. Customer bugs fixed: • 944132 – Inefficient finite capacity scheduling during conflict retry • 906505 – Wrong batch expiration date on on-hand inventory when legal entity time zone is changed • 896583 – Quarantine warehouse item is included in on-hand calculation in net requirements • 906576 – Supplier requested and confirmed shipment dates feature does not consider ship calendar set on vendor • Uptake of scheduling engine improvements made for production scheduling: • 883492 – Scheduling optimization timeout and many possible resources can cause misalignment of parallel jobs • 879271 – If delays occur in production orders in certain scenarios, scheduling is not done correctly	No feature management required	July 2, 2024
General performance, quality, and stability improvements. Customer bugs fixed: • 842762 – Planning Optimization does not consider default issue location from warehouse setup • 859663 – Duplicate planned orders created if the plan include purchase requisition but not inventory transactions • 922016 – Resolving location of parent transaction is not respecting active storage dimensions	No feature management required	June 11, 2024
General performance, quality, and stability improvements. Customer bugs fixed: • 842762 – Planning Optimization does not consider default issue location from warehouse setup • 859663 – Duplicate planned orders created if the plan include purchase requisition but not inventory transactions • 922016 – Resolving location of parent transaction is not respecting active storage dimensions	No feature management required	May 14, 2024
General performance, quality, and stability improvements.	No feature management required	March 15–20, 2024

Changes	Feature management details	Release dates
General performance, quality, and stability improvements.	No feature management required	February 10–15, 2024
General performance, quality, and stability improvements.	No feature management required	January 19–25, 2024
Planning items support. General performance, quality, and stability improvements.	No feature management required	December 10–15, 2023
General performance, quality, and stability improvements.	No feature management required	October 12–19, 2023
Positive days support. General performance, quality, and stability improvements.	No feature management required	October 2–6, 2023
General performance, quality, and stability improvements.	No feature management required	September 20–22, 2023
General performance, quality, and stability improvements.	No feature management required	August 15–20, 2023
Custom inventory dimensions. Transport days. General performance, quality, and stability improvements.	No feature management required	July 3–7, 2023
General performance, quality, and stability improvements.	No feature management required	June 15–20, 2023
BOM lines with rounding or multiple setups. BOM/formula lines with negative quantity. General performance, quality, and stability improvements.	No feature management required	May 22–31, 2023
General performance, quality, and stability improvements.	No feature management required	May 4–8, 2023
Process manufacturing support for Planning Optimization. This includes support for creating planned batch orders for formula products, co-products, and by-products, formula measurement, and formula versions with yield, but doesn't include support for planning items or item substitutions (plan groups). General performance, quality, and stability improvements.	No feature management required	April 17–21, 2023

Changes	Feature management details	Release dates
General performance, quality, and stability improvements.	No feature management required	March 27–30, 2023
General performance, quality, and stability improvements.	No feature management required	March 13–17, 2023
General performance, quality, and stability improvements.	No feature management required	February 15–24, 2023
Bills of material with constant scrap or variable scrap defined Demand with specific BOM or route requirements defined Calculated delays: Add the calculated delay to the requirement date settings on master plans	No feature management required	January 10–15, 2023
Batch disposition codes Include on-hand inventory and inventory transaction parameters on master plans Released production orders that aren't started, where scheduled start date is earlier than today General performance, quality, and stability improvements	No feature management required	October 10–14, 2022
Resource scheduling with finite capacity General performance, quality, and stability improvements	No feature management required	September 19–23, 2022
Master planning for products with limited shelf life General performance, quality, and stability improvements	No feature management required	August 29 to September 3, 2022
Centralized calendar maintenance Suggestions to optimize existing supply Support for subcontracting General performance, quality, and stability improvements	No feature management required	March 7–11, 2022
Planning priority support for production orders	Available with version 10.0.25 as part of the feature named *Priority driven MRP support for Planning Optimization*.	November 12–18, 2021

Changes	Feature management details	Release dates
General performance, quality, and stability improvements	No feature management required	November 12–18, 2021
Support for process time calculation formulas, production route with overlap, and production operation number on requirement transactions Enhanced error messages for production scheduling related to timeout, capacity couldn't be found, and cyclic route Improved consistency when calculating receipt dates and issue dates on both planned orders and firmed orders General performance, quality, and stability improvements	No feature management required as of Supply Chain Management version 10.0.43.	October 22–27, 2021
Support for considering scrap percentage in processing time calculation Support for operation number and materials usage during scheduling	No feature management required as of Supply Chain Management version 10.0.43.	October 5–7, 2021
Support for production route job types: **Queue before**, **Queue after**, and **Transport time** General performance, quality, and stability improvements	No feature management required as of Supply Chain Management version 10.0.43.	September 25–30, 2021
Support for master plans with **Scheduling method** set to *Operations scheduling* On the **Route groups** page, respect settings for the **Activation**, **Working time**, and **Capacity** check boxes for rows with a **Route/job type** of *Setup* or *Process* General performance, quality, and stability improvements	No feature management required as of Supply Chain Management version 10.0.43.	September 9–17, 2021
General performance, quality, and stability improvements	No feature management required	August 25–30, 2021
Added **Lead time** field to planned orders. General performance, quality, and stability improvements.	No feature management required	August 12–17, 2021
Added resource type requirements for infinite capacity scheduling Improved resource efficiency and calendar efficiency for infinite capacity scheduling Learn more in Scheduling with infinite capacity	No feature management required as of Supply Chain Management version 10.0.43.	July 6–12, 2021

Changes	Feature management details	Release dates
General quality improvements	No feature management required	July 6–12, 2021

Last updated on 09/08/2026

Improve scheduling engine performance

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The resource scheduling engine schedules routes for planned and released production orders.

The job shop scheduling problem is an extremely complex combinatorial problem where solution time grows exponentially with the number of decision variables. Customers often set up production routes and related data in ways that create scheduling problems that aren't solvable in reasonable time, even on modern hardware. This article explains the scheduling engine and how a specific setup influences performance.

Improve scheduling performance by reducing the complexity of the problem the engine solves. Some of the main factors that can affect performance include:

- Routes with many operations
- Routes with parallel operations
- Operations with more than one resource
- Operations with many applicable resources
- Use of hard links
- Use of finite capacity
- Number of different calendars used
- Number of working time slots per day in the calendar
- Total duration of the route
- Running multiple scheduling engines in parallel

Overview of basic scheduling flow

To understand how a given setup affects performance, you need to know how the process flows inside the engine and in the X++ code that surrounds it.

The basic process of scheduling an order consists of three main steps:

- **Loading data** – X++ data models transform into the engine's internal data model as jobs and constraints.
- **Scheduling** – The engine processes the given model and constraints to generate a result. It requests working time information and existing capacity reservations from X++ as needed.
- **Save data** – X++ code processes the engine result, in the form of job capacity reservation slots, to save capacity reservations and update the start and end times of the jobs, operation, or order.

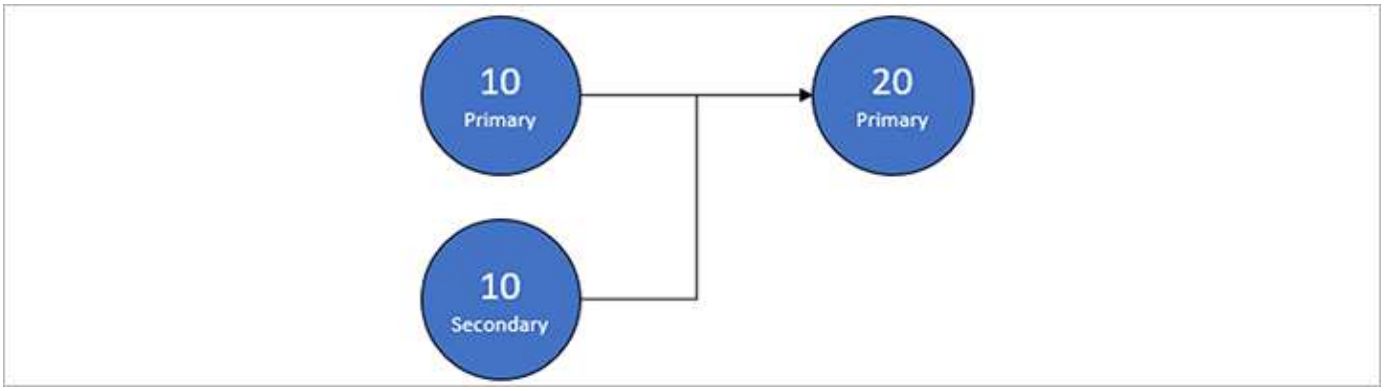
Load data into the engine

The scheduling engine uses a more abstract data model than the Supply Chain Management database because it's a generic engine that handles multiple data sources. The concepts of route, secondary operations, and run time are translated into the generic job and constraint model that the engine exposes. The logic for building the model includes significant business logic and varies depending on the source data. The responsible X++ class is `WrkCtrScheduler`, which includes derived classes for planned production orders, released production orders, and project forecasts.

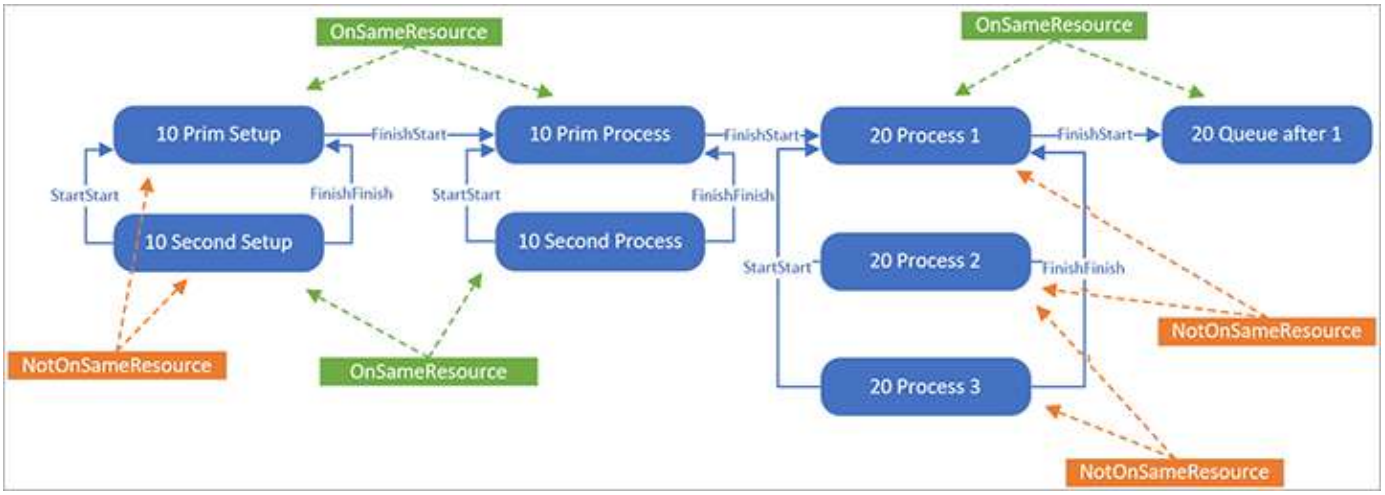
For example, consider a route shown in the following table and image, which is relatively simple.

Expand table

Oper. No.	Priority	Setup time	Run time	Queue time after	Quantity of resources	Next
10	Primary	1.00	2.00		1	20
10	Secondary 1				1	20
20	Primary		3.00	1.00	3	0



When you send this route to the engine, it splits the route into eight jobs, as shown in the following illustration (select the image to enlarge it).



The standard link between two jobs is `FinishStart`, which means the end time of one job must occur before the start time of another. The setup must be done by the same resource that later handles the process, so there are `OnSameResource` constraints between them. Between the jobs for the primary and secondary operation for 10, there are `StartStart` and `FinishFinish` links, which mean the jobs must both start and end at the same time. Additionally, there are `NotOnSameResource` constraints, which prevent the same resource from being used for both primary and secondary operations.

For operation 20, where the quantity of resources is set to 3, the process job is split into three distinct jobs where all the jobs must run at the exact same time. In this case, the route group is set up to not reserve capacity for queue after times, which is why there's only a single job for the queue after.

The scheduling engine understands only the concepts of jobs and doesn't recognize operations. This limitation means that during operation scheduling, the system splits operations into jobs, although these jobs aren't persisted in the database.

For each job, you define what the job capacity requirement is (the number of seconds required). Depending on how the resource requirements are defined, you might also send, for each job, a list of all the potential applicable resources that the job could run on and what the capacity requirement is for that specific resource. Although you send the list of applicable resources when building the model, the engine ensures the resource assignment stays valid for the entire job duration.

Scheduling engine internals

Scheduling engine interface

To understand how the engine works internally, review the functionality it exposes externally. In X++, the main interface is `WrkCtrSchedulerEngineInterface`. It has the methods described in the following subsections.

General engine

Expand table

Method	Purpose
`run`	Schedules all loaded jobs and returns the error code.
`getJobSchedulingSequenceResult`	Gets the scheduling result and the first error job for the sequence identified by a specific job.
`validateJobCapacityReservations`	Validates the capacity reservations for all the jobs stored by the engine.
`setReservationsTimeStamp`	Sends a timestamp to the engine set on all new capacity reservations for the scheduled jobs in the cache of the engine.
`addPropertyToGroupAggregation`	Adds a property prefix to the set of properties used when capacity is aggregated.
`addResource`	Adds a resource to the scheduling engine resource pool.
`addResourceGroup`	Adds a resource group to the scheduling engine resource group pool.
`addResourceGroupMembership`	Adds a resource as a member to a resource group.
`addOptimizationGoal`	Adds a scheduling optimization goal (duration or priority).

Individual jobs

Expand table

Method	Purpose
`addJobInfo`	Adds a job information record that informs the engine about a job that should be scheduled.
`addConstraintJobEndsAt`	Adds a constraint that a job should end at a specified date and time.
`addConstraintJobStartsAt`	Adds a constraint that a job should start at a specified date and time.
`addConstraintMaxJobDays`	Defines the constraint that a job can span over a specified maximal number of days.
`addConstraintResourceRequirement`	Adds the constraint that the job must be scheduled on a specific resource.

Method	Purpose
`addJobBindPriority`	Adds a job bind priority for a (job, constraint level) pair. A higher priority value means the job variables are bound earlier. The job is processed before jobs with lower priority value in the same sequence.
`addJobCapacity`	Adds capacity load information for a job (like the required job runtime) independent on which resource the job runs on.
`addJobResourceCapacity`	Adds a resource to the set of resources that can be used to perform a job, and states the capacity required when running on that resource.
`addJobGoal`	Adds job goal information for a specific constraint level (earliest end time or latest start time).
`addJobResourcePriority`	Adds the priority to use when a job is scheduled on a resource.
`addJobResourceRuntime`	Specifies a job time that is dependent of the resource the job is scheduled on.
`addJobRuntime`	Specifies a job time that is independent of the resource on which the job is scheduled.
`scheduleJobOnResourceGroup`	Marks a job for scheduling on the resource group level.
`setJobResourcePreemptionAllowed`	Sets whether preemption is allowed for a job on a resource (if engine is allowed to schedule the job in noncontiguous capacity slots).
`setRequiredNumberOfResources`	Sets the number of resources required to schedule a job (only for operations scheduling).

Constraints between jobs

Expand table

Method	Purpose
`addJobLink`	Adds a link (such as finish>start) between two jobs.
`addConstraintEndsDelayed`	Defines the constraint that a job can't end before another jobs end plus some delay time.
`addConstraintJobListWorkingTimeIntersect`	Adds a constraint that the capacity slots reserved for the jobs must be on the intersecting working times for the two resources used by the jobs.
`addConstraintJobOverlap`	Adds a constraint that defines how jobs are sequenced when a given quantity of an item can be moved between two resources while the first resource is still not finished processing, so that the second resource can start processing.

Method	Purpose
`addConstraintNotOnSameResource`	Adds a constraint that two jobs shouldn't be scheduled on the same resource.
`addConstraintOnSameResource`	Adds a constraint that two jobs must be using the same resource.
`addJobSameReservations`	Adds a constraint that a job must end up having capacity reservations for the same time slots as the primary job.
`setPrimaryParallelJob`	Adds information about what job is the primary job in a set of parallel jobs.

Solver

The engine is a specialized constraint solver with custom heuristics. The solver is based on two main elements: variables and constraints.

Variable

A variable represents a domain of possible values. The scheduling engine has two types of variables:

- **DateTime variable** - Has a domain of all dates and times. You can restrict the domain by moving the lower and upper bounds for the time of the variable closer to each other.
- **Resource variable** - Has a domain of applicable resources. You can restrict the domain by eliminating resources from the list.

Constraint

A constraint acts on variables by restricting their domains. It also depends on variables, so it gets activated when variables change. The process of "constraint propagation" is when a constraint performs its main function and reports back to the main logic if successful.

A variable is considered bound when it can't be restricted further. For a DateTime variable, this condition means that the upper and lower bounds are the same. For the Resource variable, it means that it has only a single applicable resource. When all variables are bound, a solution is found.

Constraint levels

When scheduling is part of the material requirements planning (MRP) coverage phase, the system schedules orders backward from the requirement date. However, if the system can't find a schedule that starts today or later and ends before the requirement date, it changes the scheduling direction to forward from today.

The system organizes the constraints in levels to handle this main business rule. If the system doesn't find a solution when using the constraints on the highest level, it drops all the constraints on that level and tries the lower level. In practice, this process means that for backward scheduling, the model contains a level 1 with job goals of latest start time given a maximum end time constraint (the requirement date), and a level 0 with job goals of earliest end time and given a minimum start time constraint of today.

Algorithm

The main steps of the engine algorithm are:

1. Find sequences (job chains) that can be solved separately.
2. Try to find an initial solution for the sequence at the highest constraint level.
    a. Sort the jobs in the sequence based on job goal and priorities, so the system can find a start job.
    b. Loop through the jobs in the following sequence:
        i. Find all constraints that need to be propagated and run propagation.
        ii. If all variables for the job are bound, then a solution for that job is found.
        iii. If one of the variables can't be bound without violating the constraints, roll back the variable binding, try a different value in the domain (for resource variable), and rerun the constraint propagation.
3. If no solution is found, remove all constraints on the current constraint level, lower the constraint level (if any lower levels are available), and retry solution search with the new set of constraint.
4. If a feasible solution is found, start the optimization phase, which tries to find a better solution until the optimization timeout is reached or all resource combinations are exhausted.

The constraint solver doesn't account for the specifics of the scheduling algorithm. The "magic" happens in the definition and combination of the various constraints.

Determining working times

A large part of the (internal) constraints in the engine controls the working time and capacity of a resource. Essentially, the task is to traverse the working time slots for a resource from a given point in a given direction, and find a long enough interval in which the job's required capacity (time) can fit.

To do this, the engine needs to know the working times of a resource. Unlike the main model data, the working times are *lazy loaded*, which means that they're loaded into the engine as needed. The reason for this approach is that there are often working times in Supply Chain Management for a calendar for a very long period and typically many calendars exist so the data would be quite large to preload.

The engine requests calendar information in chunks by invoking the X++ class method `WrkCtrSchedulingInteropDataProvider.getWorkingTimes`. The request is for a specific calendar ID in a specific time interval. Depending on the state of the server cache in Supply Chain Management, each of these requests could end up in several database calls, which takes a long time (relative to the pure computational time). Also, if the calendar contains very elaborate working time definitions with many working time intervals per day, this complexity adds to the time the loading takes.

When the scheduling engine loads the working-time data, it retains this data in its internal cache for the specific calendar. This retention means that if any other jobs or resources use the same calendar, the next lookups can be performed quickly from memory. One common cause of bad performance is if a separate calendar ID is used for each resource, because data is then requested for each calendar, even though the content of the calendars might be the same.

Finite capacity

When you use finite capacity, the system splits and reduces the working time slots from the calendar based on the existing capacity reservations. The same `WrkCtrSchedulingInteropDataProvider` class fetches these reservations along with the calendars, but it uses the `getCapacityReservations` method. When scheduling during master planning, the system considers the reservations for the specific master plan. If you enable the setting on the **Master planning parameters** page, the system also includes reservations from firmed production orders. Similarly, when scheduling a production order, you can choose to include reservations from existing planned orders, although this sequence isn't as common as the other way around.

Scheduling takes longer when you use finite capacity for several reasons:

- Fetching the capacity information from the database is a slow operation. The server-side caching of capacity information is typically not as good as for working times because they

aren't shared among resources like calendars typically are.
- The number of working time slots to traverse increases due to the splits. The system typically needs to investigate slots for a longer time period before it can find a solution.
- After scheduling is complete, the system must check for conflicting reservations (see the "Running scheduling engines in parallel" section for details).

Examining the resource combinations

If the job sequence only contains the standard `FinishStart` links, meaning it forms a simple chain without any branches, the system can achieve an optimal result (seen from the single order, not across orders) by finding the best solution for the first job and then moving on to find the best solution for the next job. The best solution for a job is to find the resource that can get the from and to date of the job closest to the job goal (in forward scheduling this means getting the end date of the job as early as possible) while still respecting the constraints.

When there are parallel jobs, finding a solution might involve examining different combinations of resources. The number of possible resource combinations is the product of the number of applicable resources for the connected parallel jobs. Especially when scheduling an order backwards from a requirement date, it can take quite a while for the logic to realize that there's no solution to the problem that will make the parallel jobs fit before today's date. The logic needs to check all the combinations because there could be some resources that have a higher efficiency or a different calendar that might give a result. This condition means that if you don't set a timeout limit, the process runs for a long time before changing the direction to forward.

This combinatorial logic also means that adding more applicable resources might make the engine run slower. If performance problems occur when you have parallel operations and scheduling with infinite capacity, you can partly fix the problem by having the route designer take a decision on which resource should be used and then assign the resource directly on the operation (because the engine in most cases always ends up picking the same resource, so the end result is the same).

Hard links

When you set the link type between two jobs to hard, you ensure that there's no time gap between the finish of one job and the start of the next one. This condition is useful in scenarios where metal is heated in one job and processed in the next, and cooling between jobs isn't desirable.

By using standard soft links and forward scheduling, if the route forms a simple chain without any branches, you can achieve a result by finding a solution for the first job that satisfies its own constraints and then moving on through the chain propagating the end time from the previous job to the next job. If the current job can't find any capacity, the system moves the start time out further, without any consequence for the previous jobs. This action potentially creates gaps between the jobs. However, by using hard links (especially in connection with finite capacity) for the same scenario, the fact that one job later in the chain can't find capacity, means that all previous scheduled jobs must be "dragged" along one by one and thereby rescheduled several times. Especially in scenarios with high load for multiple resources, the hard links can cause a chain reaction where the jobs affect each other and several iterations must be performed before the result stabilizes into a feasible schedule.

Running scheduling engines in parallel

When you perform scheduling as part of a master planning run where you use helpers, each of the master planning helper threads can also pick up production order scheduling tasks. This approach means that multiple scheduling engines can run at the same time. While multithreading provides a significant performance benefit, it also introduces some functional downsides for scheduling.

In MRP, the system schedules all production orders for a given bill of materials (BOM) level in requirement date sequence. This sequence means that the system schedules those orders with the earliest requirement date first, so they have the highest chance of getting the available resource capacity. However, when multiple engines pick from the list of unscheduled orders, the sequence isn't ensured because one engine might complete faster than the other.

When you schedule with finite capacity and multiple engine instances, a race condition can occur if you try to schedule orders that use the same resources at the same time. The **Scheduling conflicts** field on the master plans history page records the number of race conditions. The conflict resolution logic follows these steps:

1. Schedule an order (lock-free) and get capacity reservations.
2. Take the lock.
3. Check if newer capacity reservations exist for the scheduled resources in the time span.

    - If no, write the capacity and release the lock.
    - If yes, release the lock and reschedule the order from the beginning.

When scheduling with multiple engine instances, the result isn't fully deterministic because it depends on the exact timing of each thread.

Operation scheduling performance

Operation scheduling, also known as rough-cut capacity planning, can be harder to solve from an engine standpoint if finite capacity is used because more data is needed to determine feasibility.

The capacity of a resource group depends on which and how many resources are members of the resource group. A resource group itself doesn't have any capacity—only when resources are members of the group does it have capacity. Because the resource group membership can vary over time, capacity must be evaluated per day.

In operations scheduling, the resource group's calendar is used to determine the start and end times for each operation. This calendar places a limit on how much time you can schedule operations for one operation on one day in one resource group. Unlike the calendar for specific resources, the efficiency data of the resource group's calendar is ignored because it only denotes opening hours, not actual capacity.

For example, if the working time for a resource group on one specific date is from 8:00 to 16:00, one operation can't put more load on the resource group than what can fit into 8 hours, no matter how much capacity the resource group has available in total on that day. However, the available capacity can limit the load further.

Job scheduling load on all resources in the resource group for a given day is considered when calculating the available capacity for that day. For each date, the calculation is:

Available resource group capacity = Capacity for resources in the group based on their calendar − Job scheduled load on the resources in the group − Operations scheduled load on the resources in the group − Operations scheduled load on the resource group

On the **Resource requirements** tab on the route operation, you can specify the resource requirements using either a specific resource (in which case the operation is scheduled using that resource), a resource group, a resource type, or one or more capabilities, skill, course, or certificate. Using all these options provides flexibility in route design but complicates scheduling for the engine because capacity must be accounted for per 'property' (an abstract name used in the engine for capability, skills, and so on).

The resource group's capacity for a capability is the sum of the capacity for all resources in the resource group that have the capability in question. If a resource in the group has a capability,

the engine considers it no matter what level of the capacity is required.

In operations scheduling, the available capacity for a certain capability for a resource group is reduced when it's loaded with an operation that requires the capability in question. If the operation requires more than one capability, the capacity is reduced for all required capabilities.

For each date, the required calculation is:

Available capacity for a capability = Capacity for the capability − Job scheduled load on the resources with the specific capability, included in the resource group − Operations scheduled load on the resources with the specific capability, included in the resource group − Operations scheduled load on the resource group itself that require the specific capability

If a specific resource has a load, the engine considers it when calculating the resource group's available capacity per capability because the load reduces its contribution to the group's capacity, regardless of whether the load is for that specific capability. If there's load on the resource group level, the engine considers it in the calculation of the resource group's available capacity per capability only if the load is from an operation that requires the specific capability.

This logic applies to each type of property, so using operations scheduling with finite capacity requires loading a significant amount of data.

Improve MRP performance

The following tech conference video provides tips to improve master planning performance when you use MRP with the deprecated master planning engine: Help! MRP is slow!

Viewing scheduling engine input and output

To get specific details about the input and output of the scheduling process, enable logging by going to **Organization administration** > **Setup** > **Scheduling** > **Scheduling tracing cockpit**.

On this page, select **Enable logging** on the Action Pane, and then run the scheduling for the production order. When complete, return to the **Scheduling tracing cockpit** page and select **Disable logging** on the Action Pane. Refresh the page, and a new line appears in the grid. Select the new line, and then select **Download** on the Action Pane. This action gives you a .zip compressed folder containing the following files:

- **Log.txt** – This log file describes the steps the engine goes through. It's detailed and can be overwhelming, but when you experiment with the route setup to resolve performance

problems, look for the time difference between the first and last line. This difference shows the exact time the scheduler spent.

- **XmlModel.xml** – This file contains the model that is built in X++ and that the engine operates on. The `JobId` used in the file correlates to the `RecId` from the source table containing the jobs (`ReqRouteJob` or `ProdRouteJob`). In this file, check that the dates in `ConstraintJobStartsAt` and `ConstraintJobEndsAt` are as expected, the `JobGoal` property is set correctly, and the jobs are related through the `JobLink` constraints.
- **XmlSlots.xml** – This file contains all the working times and capacity reservations that the engine requests. The calendar working times and reservations are only requested by the engine for the time periods where it tries to place the jobs (and an extra buffer), so if the file contains times very far in the future, it might be an indication of a problem with the setup. The `ResourceProperty` nodes display the resource group and capabilities associated with each resource, along with the relevant time periods.
- **Result.xml** – This file contains the result of the scheduling run.

The tracing functionality adds significant performance overhead, so use it only to investigate scheduling specific orders in a controlled manner. If you turn it on during a master planning run, it quickly reaches its size limit and stops.

Troubleshooting performance

As you can see from the previous sections, some pitfalls in the setup and usage of the scheduling engine can lead to performance problems. The following checklist can help you troubleshoot these problems. Look at all the points because it's often a combination of multiple factors that leads to problems.

> **Tip**
>
> The following table summarizes the most common job scheduling performance problems and their recommended solutions. Use the links to jump to the detailed guidance for each area.
>
> Expand table

Problem	Recommendation	Details
Scheduling not needed during MRP	Set capacity time fence to zero or use a standard production lead time	Scheduling as part of MRP

Problem	Recommendation	Details
Too many applicable resources	Assign a specific resource to the operation at route design time	Many applicable resources
Parallel operations slow scheduling	Model pairs as "virtual" resources or exclude non-bottleneck operations	Parallel operations
Resource quantity greater than one	Review whether multiple resources are truly required for the operation	Resource quantity
Finite capacity overhead	Use the bottleneck option and separate capacity time fence for bottleneck resources	Finite capacity
Hard links complicate scheduling	Use hard links only when strictly necessary	Hard links
Separate calendar per resource	Reuse calendars among resources as much as possible	Calendars
Many time slots per calendar day	Minimize working time slots (for example, skip five-minute break modeling)	Time slots
Missing or large scheduling timeouts	Enable both timeout settings; set maximum scheduling time to about 30 seconds	Timeouts

Performing scheduling as part of MRP when it isn't needed

Although routes are used for production control purposes like costing and reporting, they might not need to be considered during MRP. In some cases, having a standard production lead time specified for the item will be sufficient for planning. Set the capacity time fence to zero to turn off route scheduling. If scheduling is needed, carefully set the capacity time fence because routes might not need to be considered for the full extent of the MRP's coverage time fence.

If the order isn't scheduled during MRP, then it will instead need to be scheduled when the planned order is firmed. This means that the firming process will take longer, so depending on how many of the suggested planned orders get firmed the performance gain during MRP might be lost at firming.

Route with unnecessary operations

When you design the route, avoid trying to model the real world exactly with all the steps the production process goes through. While this approach can be useful in some cases, it isn't good for performance because the engine's model grows larger (in terms of jobs and constraints), and more SQL statements are executed to insert and update jobs and capacity reservations. Also,

there's the downstream effect of having to eventually report progress on the jobs, which automatic postings can mitigate. If the data isn't used for anything, it creates unnecessary load.

Create only the operations strictly needed for scheduling (typically bottleneck resources) or costing purposes. Instead, group many smaller distinct operations into one larger operation that represents a greater part of the process.

Many applicable resources for an operation

Resource requirements set on the operation relation determine the number of applicable resources for an operation. The requirement can either be for a specific resource or it can be based on the resource's membership in a resource group or capability.

If you don't use finite capacity for scheduling and all the applicable resources have the same calendar and efficiency, the scheduling engine always picks the same resource for an operation, but only after it tries all the applicable resources to check if there's one that is better than the others. In this case, you can greatly reduce the load of the scheduling by always assigning a specific resource to the operation at the route design time.

Route with parallel operations

While parallel operations (primary/secondary) are a powerful tool to model scenarios like when a machine and an operator are both needed to perform a specific task, they're also the source of many performance problems. If you assign a requirement for a specific individual resource to both the primary and secondary operation, it's typically not a problem. But if there are many possible resources for each of the operations, then it adds significant computational complexity to the scheduling.

An alternative to using parallel operations is either to model the pairs as "virtual" resources (which then represent the team that always goes together for the operation) or to simply not model one of the operations if it doesn't represent a bottleneck.

Route with quantity of resources higher than one

If the quantity of resources needed for an operation is greater than one, the result is effectively the same as using primary/secondary operations because multiple parallel jobs are sent to the engine. However, for this case, you can't use specific resource assignments because a quantity higher than one requires more than one resource to be applicable for the operation.

A secondary operation that has a resource load quantity greater than one means that the specified quantity of secondary resources is needed for each resource of the primary operation. For example, if a primary operation has its quantity of resources set to two and its secondary operation has its resource quantity set to three, then a total of six resources is needed for the secondary operation.

Excessive use of finite capacity

Using finite capacity requires the engine to load capacity information from a database, which can add computational overhead, especially in environments where resources are booked close to maximum capacity. As a result, it's important to carefully evaluate if a resource really needs to use finite capacity or they can be overbooked. Because there might be a difference among finite capacity resources in how important they are to not overbook, use the bottleneck option on a resource in combination with a separate value on the plan in "Capacity time fence for bottleneck resources." Using the bottleneck concept can enable lowering the general finite capacity time fence.

Setting hard links

The standard link type of the route is *soft*, which allows a time gap between the finishing time of one operation and the start of the next. If you allow this gap, it can cause production to be idle if materials or capacity aren't available for one of the operations. This situation can lead to an increase in work in progress. This problem doesn't happen with hard links because the finish and start must align perfectly. However, setting hard links makes the scheduling problem more difficult because the system must calculate working time and capacity intersections for the two resources of the operations. If the schedule also involves parallel operations, this requirement adds significant computational time. If the resources of the two operations have different calendars that don't overlap at all, the problem is unsolvable.

Use hard links only when strictly necessary, and carefully consider if it's necessary for each operation of the route.

To reduce work in progress without applying hard links, schedule the order twice with changing to the opposite direction for the second pass. If the first schedule was done backward from delivery date, then the second schedule should be done forward from the scheduled start date. This approach compresses the jobs as much as possible so that the work in progress is minimized.

Separate calendar for each resource

One of the main sources of data for the scheduling engine is calendar information, which can be expensive to load from the database. Because calendars are generated based on templates, it might seem tempting to generate a calendar for each resource and then adjust the information in this calendar when the resource has downtime and other issues. However, this approach severely limits the engine's ability to cache the calendar data because it needs to request new data for each resource. This approach can cause performance problems. Reuse calendars as much as possible between resources, and control downtime changes by assigning a different calendar ID for a period.

High number of working time slots per calendar day

Because the engine works by examining time slots one-by-one for capacity, it's beneficial to minimize the number of time slots per calendar day. For example, consider whether it's important for the resulting schedule to reflect that workers take a five-minute break every hour.

Large (or none) scheduling timeouts

You can optimize scheduling engine performance by using parameters found on the **Scheduling parameters** page. Always set the **Scheduling timeout enabled** and **Scheduling optimization timeout enabled** settings to *Yes*. If you set them to *No*, scheduling can potentially run infinitely if an unfeasible route with many options is created.

The value for **Maximum scheduling time per sequence** controls how many seconds can, at most, be spent trying to find a solution for a single sequence (in most cases a sequence corresponds to a single order). The value to use here highly depends on the complexity of the route and settings like finite capacity, but a maximum of about 30 seconds is a good starting point.

The value for **Optimization attempts timeout** controls how many seconds can at most be used to find a better solution than the one originally found. This value only influences routes that use parallel operations as these operations make it necessary to test different combinations.

> **Note**
>
> The values you set for the timeouts apply both to scheduling of released production orders and to planned orders as part of MRP. As a result, setting very high values can significantly add to the run time of MRP when running for a plan with many planned production orders.

Common job scheduling errors

The following table lists known job scheduling errors with links to information that can help you resolve them.

Expand table

Error message	Resolution
Planned production order must be scheduled before it can be firmed	Planned production order must be scheduled before it can be firmed
Production scheduling doesn't consider the safety margins	Production scheduling doesn't consider the safety margins
Master planning is scheduling more than the available capacity	Master planning is scheduling more than the available capacity
Not enough capacity could be found	Not enough capacity could be found Fix the "Not enough capacity could be found" scheduling engine error
The delay value isn't updated when you reschedule a planned order	The delay value isn't updated when you reschedule a planned order

Related information

- Calendars and master planning
- Calculate requested ship dates for purchase orders
- Date and time parameters used by Planning Optimization
- Safety margins

ⓘ **Note:** The author created this article with assistance from AI. Learn more

Last updated on 05/04/2026

Troubleshoot Planning Optimization

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes how to fix common issues that you might encounter while working with Planning Optimization.

Installation of the Planning Optimization Add-in isn't completed or fails with an error

If you're having trouble installing the Planning Optimization Add-in, consider the advice provided in the following subsections.

Make sure your environment meets the prerequisites

Planning Optimization requires a Microsoft Dynamics Lifecycle Services–enabled, high-availability environment, tier 2 or higher (not a OneBox environment), with Dynamics 365 Supply Chain Management version 10.0.23 or later. If you try to install the add-in on a OneBox environment, the installation won't complete.

Fix: Cancel the installation and use a high-availability environment, tier 2 or higher (not a OneBox environment).

Make sure your user account meets the prerequisites

The Planning Optimization Add-in requires a successfully linked Power Platform environment. To install the add-in, you must sign in to your Microsoft Power Platform environment using an account with administrator privileges and an access mode of *Read-Write*. If you try to install the add-in using an account with insufficient permissions, you might see one of the following error messages:

> Current user does not have enough permission on Power platform environment to complete installation for Planning optimization. Consider changing user Access Mode to Read-write on Power platform admin center.

> Current user is not exist in Power platform environment. Please contact with your administrator to assign the user via Power platform admin center.

> Current user does not have assigned license to complete installation for Planning optimization. Consider assigning license on Microsoft admin center.

Fix: Set up your user account as described under *Prerequisites* in Get started with master planning.

Review the error details

An error in your Power Platform environment might prevent the installation of the Planning Optimization Add-in.

Fix: Check the error details and modify your Power Platform organization settings accordingly. It might be a temporary error, so try to install the add-in again later. If the issue persists, contact Microsoft Support.

Planning Optimization job times out

Planning Optimization has a set timeout of 60 minutes. Therefore, if it runs for more than 60 minutes, the planning job stops because of a timeout.

> **Tip**
>
> The following table summarizes the most effective approaches for improving Planning Optimization performance. Try them in order of impact.

Expand table

Approach	When to use	Impact
Reduce time fences	Time fences are larger than your business requires	High—coverage time fence has the largest effect
Plan only needed products	Many items are included that don't need planning	High
Split large jobs	A single job plans too many items for the 60-minute window	High
Reduce scheduling time	Finite capacity scheduling is enabled and adds significant time	Medium
Reduce resource group size	Large resource groups slow down scheduling calculations	Medium
Review item coverage	Multiple coverage lines apply the same settings for all warehouses	Low to medium

If your Planning Optimization jobs frequently time out, consider implementing one or more of the options that are described in the following subsections.

Review your setup to remove time fences and options that you don't need

Follow these steps to review your setup and remove time fences and other options that you don't need:

1. Adjust the time fences to make them as small as possible while still fulfilling your business needs.

    a. Check the following settings for each of the coverage groups on the **Coverage groups** page:

    - **Coverage time fence (days)**
    - **BOM explosion time fence (days)**
    - **Capacity scheduling time fence (days)**
    - **Forecast plan time fence**
    - **Action time fence**
    - **Calculated delays time fence**

b. On the **Master plans** page, check whether the time fences have been overwritten and consider whether the values could be smaller while still fulfilling your business needs for the various time fences. Check the following settings for each plan:

- **Coverage**
- **Explosion**
- **Forecast plan**
- **Capacity**
- **Action message**
- **Calculated delays**
- **Approved requisitions time fence (days)**

> **Important**
>
> The coverage time fence has the largest impact on the time that planning takes to run. Adjust it to the lowest possible value that still fulfills your business needs.

2. On the **Master plans** page, review the following settings for each plan. Set each option to *No* unless it applies to your master plan. Don't leave any option set to *Yes* if it isn't needed for your business processes.

   - **Use shelf life dates**
   - **Finite capacity**
   - **Include on-hand inventory**
   - **Include inventory transactions**
   - **Include requests for quotations**
   - **Include sales quotations**
   - **Include supply forecast**
   - **Include demand forecast**
   - **Include requisitions**

3. Use finite capacity only when it's necessary. For plans that don't need it, set the **Finite capacity** option to *No* on the **Master plans** page.

Reduce scheduling time

Review your scheduling configuration to optimize run times. To check whether scheduling is the main reason that Planning Optimization is timing out, try disabling finite capacity in the plan settings and then rerun planning to see if the issue is solved.

For more information about how to reduce scheduling times, see Improve scheduling engine performance.

Plan only for the products you need

Review the following settings to make sure that you're planning only for the products that you need:

- Use the **Product lifecycle state** field to indicate products or variants that don't need to be fulfilled by master planning. For each such product, select a product lifecycle state where the **Is active for planning** option is set to *No*. Use the **Change lifecycle state for obsolete products** page to identify products that aren't used in any transactions for a while. These products might now be obsolete. Therefore, you can remove them from your planning.
- For plans that should only apply for a certain set of items, set up a plan filter to limit the run to just those items. Learn more in Run planning for a subset of items.
- Set item coverage to manual for each warehouse that doesn't need to be supplied by master planning. For each such warehouse listed on the **Warehouses** page, expand the **Master Planning** FastTab and, in the **Item Coverage** field group, set **Manual** to *Yes*.

Split large planning jobs into several smaller jobs

If you have a large planning job that frequently times out, you might be able to prevent the timeout by splitting the job into several smaller jobs.

Option 1: Run the same master plan but only for a subset of products

For example, you have a master plan named *PlanA*. It runs nightly as a batch job for 1,000 items that have item numbers ranging from *A0001* through *A1000*. If this job often times out after 60 minutes, you can split it into three jobs, each of which runs for a third of the items. You run *PlanA* for the first third (*A0001* through *A0333*), then for the second third (*A0334* through *A0666*), and then for the last third (*A0667* through *A1000*). In this way, each smaller job has the full 60-minute timeout window. You're not trying to use the same 60 minutes to plan all 1,000 items.

To split a large job into several jobs, follow these steps:

1. Go to **System administration** > **Inquiries** > **Batch jobs**.
2. In the grid, find the recurring planning job that's timing out. Then select the value in the **Job ID** column to open the job details.
3. On the Action Pane, select **Change status**.

4. In the **Select new status** dialog box, select *Withhold*, and then select **OK**.

5. On the **Batch tasks** FastTab, the grid should include a single row for Planning Optimization, where the **Class name** field is set to *MpsMasterPlanningRunnerRegen*. Select this task, and then select **Parameters** on the FastTab toolbar.

6. A dialog box appears, where you can set values for the task. On the **Records to include** FastTab, select **Filter** to open a standard query editor dialog box.

7. On the **Range** tab, add a row, and set the following fields for it:

    - **Table** – Select *Items*.
    - **Derived table** – Select *Items*.
    - **Field** – Select *Item number*.
    - **Criteria** – Specify the range of item numbers that you want to include in the first of the smaller jobs. Separate the first and last item numbers in the range with three periods. For example, enter *A0001...A0333*.

8. Select **OK** to close the query editor. Then select **OK** to close the task settings dialog.

9. On the **Batch tasks** FastTab, select **Add** on the toolbar to add a task.

10. Set the following fields for the new task:

    - **Task description** – Enter a description of the task (for example, *PlanA part 2*).
    - **Class name** – Select *MpsMasterPlanningRunnerRegen*.
    - **Company** – Select the same company that's selected for the original task.

11. Set the filter for the new task to find the second third of the items (for example, item numbers in the range *A0334...A0666*).

12. Repeat the previous three steps to add a third task and set the filter to find the last third of the items (for example, item numbers in the range *A0667...A1000*).

13. Select the **Task ID** value for the first task, and copy it to the clipboard (by selecting **Ctrl+C**).

14. Select the second task. Then, on the **Batch task details** FastTab, on the **Constraints** tab, select **New** on the toolbar to add a row to the grid.

15. Set the following fields for the new row. These settings cause the second task to run after the first task finishes running or encounters an error.

    - **Task ID** – Paste the value that you copied to the clipboard.

- **Expected status** – Select *Ended or error*.

16. Repeat the previous three steps to set the third task to run after the second task finishes running or encounters an error.
17. On the Action Pane, select **Change status**.
18. In the **Select new status** dialog box, select *Waiting*, and then select **OK**.
19. On the Action Pane, select **Save**.

> **Tip**
>
> This procedure shows just one way to split a large job into several smaller jobs and set them to run in series. You can split the job into even more jobs and filter on different criteria as needed.

Option 2: Different master plans, each for a subset of products

If your products have different characteristics that affect planning, consider running different master plans, each for a subset of products.

For example, you have a master plan for purchasing items that have a long lead time (such as a year) but that are used to produce manufactured products that have a short manufacturing lead time (such as a week). In this case, you can make one master plan for purchased products (*PlanPurch*) that has a coverage time fence of 365 days. Then make another plan for manufactured items (*PlanManuf*) that has a coverage time fence of 30 days. Because each set of products is in a different master plan, you can run both master plan jobs in parallel. When you run different plans in different batch tasks, each batch task can run in parallel. They don't have to run sequentially.

To implement this strategy, follow these steps:

1. Open your existing master plan, and modify it so that it covers only a subset of the original items (for example, purchased items). You can make this modification by adding a filter as described in Applying a plan filter.
2. Create another master plan to cover the remaining items. Again, set up a plan filter to include only the items that you want to include in this plan (for example, manufactured items). This plan can be a copy of the original plan that you modify to filter for a different set of items.

3. Go to **System administration** > **Inquiries** > **Batch jobs**.
4. In the grid, find the recurring planning job that's timing out. Then select the value in the **Job ID** column to open the job details.
5. On the Action Pane, select **Change status**.
6. In the **Select new status** dialog box, select *Withhold*, and then select **OK**.
7. On the **Batch tasks** FastTab, the grid should include a single row for Planning Optimization, where the **Class name** field is set to *MpsMasterPlanningRunnerRegen*. On the FastTab toolbar, select **Add** to add a task.
8. Set the following fields for the new task:
    - **Task description** – Enter a description of the new task.
    - **Class name** – Select *MpsMasterPlanningRunnerRegen*.
    - **Company** – Select the same company that's selected for the original task.
9. Select the new task, and then select **Parameters** on the FastTab toolbar.
10. A dialog box appears, where you can set values for the task. On the **Parameters** FastTab, set the **Master plan** field to the name of the new plan that you created.
11. On the Action Pane, select **Change status**.
12. In the **Select new status** dialog box, select *Waiting*, and then select **OK**.
13. On the Action Pane, select **Save**.

Because you set each batch task to run a different master plan, both batch tasks run in parallel.

Review your item coverage settings

- Review your item coverage settings. For items that use multiple item coverage lines to apply the same settings for all warehouses at the same site, replace those lines with a single line for the site (with the **Warehouse** column blank). That setting then applies to all warehouses at that site.

Consider reducing the size of your resource groups

Consider whether you can split large resource groups (with many resources) into several smaller groups (each with fewer resources). For example, you might be able to split a large resource

group called *AssemblyStations* (which contains 20 resources) into two smaller resource groups based on location (such as *AssemblyStationsNorth* and *AssemblyStationsSouth*), where each contains just 10 assembly stations. After splitting the group, reassign each new resource group to the right routes based on whether a product is produced in the northern or southern area of the plant.

Data export timeout

The following message appears if a data export times out for Planning Optimization:

> Master planning job timed out when exporting the data to perform the calculation. This condition can be temporary—try running the job again later. If you see this message often, review your setup to limit the amount of data used for planning, as indicated in (this page).

If you receive this message, try one or both of the approaches that are described in the following subsections.

Review your setup for time fences and options that you don't need

Follow these steps to review your setup for time fences and options that you don't need.

1. Go to **Master Planning** > **Setup** > **Coverage** > **Coverage groups**.
2. Review the following settings for each coverage group on the **Coverage groups** page. Consider whether some or all of the values can be smaller but still fulfill your business needs for the different time fences.
    - **Coverage time fence (days)**
    - **BOM explosion time fence (days)**
    - **Capacity scheduling time fence (days)**
    - **Forecast plan time fence**
    - **Action time fence**
    - **Calculated delays time fence**
3. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
4. On the **Master plans** page, check whether the time fences have been overwritten, and consider whether the values can be smaller but still fulfill your business needs for the different time fences. Review the following settings for each plan:

- **Coverage**
- **Explosion**
- **Forecast plan**
- **Capacity**
- **Action message**
- **Calculated delays**
- **Approved requisitions time fence (days)**

5. If your plan times out because it generates a large number of orders, consider changing your business strategy for replenishing items. Here are some examples:

    - If you use coverage groups where the **Coverage code** field is set to *Requirement*, the system creates a specific supply for it each time that there's a demand. Consider whether a **Coverage code** value of *Period* works for your business. In this case, the system groups all demand for a selected number of days into a single supply order that covers that period. This approach also makes planned orders easier to manage. Alternatively, consider using a **Coverage code** value of *Min/Max*. In this case, the system creates a planned order only when the on-hand inventory falls below the minimum value. The on-hand inventory is then replenished to its maximum value.
    - Consider whether you can purchase or produce items in larger amounts. If you can, increase the **Max. order quantity** value on **Default order settings** page for each item that you're ordering. The higher the value, the fewer orders you're likely to generate for that item.

Plan only for the products that you need

Your data export might complete more quickly if you reduce the number of products that the system considers for each planning run. Consider using one or both of the following strategies:

- Identify products and variants that don't need to be fulfilled by master planning, and set their **Product lifecycle state** value to a state where the **Is active for planning** option is set to *No*. (Learn more in Exclude products that have specific product lifecycle states). The **Change lifecycle state for obsolete products** page can help you identify products that haven't been used in any transactions for a while. These products might now be obsolete. Therefore, you can remove them from your planning.
- Use a plan filter to remove unneeded items from your plan.

No planned orders are created

If master planning runs but doesn't create any orders, check the following settings:

- Make sure the items you're expecting to generate supply for are set up with a lifecycle state where **Exclude from master planning** is set to *No*.
- If you're running a filtered plan, make sure that there aren't any typos in your filter values. (Learn more in Run planning for a subset of items.)
- Make sure that there's demand for the items you're expecting master planning to create supply for.

Planning of batch jobs fails when Planning Optimization is enabled

When you enable Planning Optimization, the deprecated master planning engine is automatically disabled. Master planning batch jobs that were created for the deprecated master planning engine fail if they're triggered while Planning Optimization is enabled. You might receive an error message such as *This operation triggered master planning that isn't supported when Planning Optimization is enabled*.

Fix: Cancel all master planning batch jobs that were created for the deprecated master planning engine.

Planning Optimization results are different from earlier results

Planning Optimization differs from the deprecated master planning engine design in some areas. This difference can also be caused by pending features.

Fix: Run Planning Optimization fit analysis and then analyze the results while referring to the related documentation to understand the impact. Learn more in Planning Optimization fit analysis.

Can't disable Planning Optimization

The **Use Planning Optimization** option is set to *Yes* and is grayed out, so you can't turn it off.

This behavior is by design. Planning Optimization is now the only planning engine available for Supply Chain Management, so you can't turn it off. However, in some rare cases, Microsoft Support might make the deprecated planning available for one or more specific companies. Learn more in Migration to Planning Optimization for master planning.

Error message is shown during CTP

If you try to run capable to promise (CTP) from a sales order when Planning Optimization is enabled, you receive the following error message: *This operation triggered master planning that isn't supported when the Planning Optimization is enabled*.

This error message is related to a pending feature that is planned as part of the support for production orders.

Fix: Avoid CTP calculations when Planning Optimization is enabled until CTP support is available.

Error message about active planning dimensions not matching

After running a master plan, you might receive the following error message:

> Supply setting with id: <SettingID> for MinMax on product <ProductID> doesn't match the active planning attributes for this product and was ignored.

If you see this error, check the tracking and coverage dimensions for the specified product. For example, if the product is tracked by serial number, you can't use the serial number dimension as a coverage dimension because the system doesn't know which serial numbers it should supply. Either deselect the serial number as a coverage dimension or change the item coverage group to a group that doesn't use serial number tracking (if serial tracking isn't needed for the specified product).

Planning Optimization client request retry count exceeded

After running master planning, you receive one of the following error messages:

> Planning Optimization client request retry count exceeded.

or

> Planning Optimization run failed.

This error can occur for different reasons. Check the following causes and fixes.

Cause 1: Transient service failure

Planning Optimization is a cloud-based service. Occasionally, transient service failures can occur that are specific to a particular region. These temporary outages can prevent Planning Optimization from completing a run.

Fix: Wait a short time and then retry the master planning run. If the issue persists for more than a few hours, contact Microsoft Support because the issue might require action from the service engineering team.

Cause 2: Connection lost after a database refresh

If you recently performed a database refresh on your environment, the Planning Optimization connection might be broken. The connection status on the **Planning Optimization parameters** page (**Master planning** > **Setup** > **Planning Optimization parameters**) might show *Not connected* even though the add-in appears as installed.

A database refresh is environment-specific, and it resets the Planning Optimization service binding. This is a known behavior.

Fix: Uninstall and reinstall the Planning Optimization Add-in for your environment from the admin center. After reinstallation, verify that the connection status shows *Connected* on the **Planning Optimization parameters** page. Learn more in Get started with master planning.

Cause 3: Temporary connectivity issue

Intermittent network or service connectivity problems between your environment and the Planning Optimization service can cause this error.

Fix: Retry the master planning run. If the error occurs repeatedly, check the **Planning Optimization parameters** page to verify the connection status. If the connection status shows *Not connected*, try reinstalling the add-in. If the status shows *Connected* and the error persists, contact Microsoft Support.

Company information tab is blank on the Planning Optimization parameters page

When you open the **Planning Optimization parameters** page and select the **Company information** tab, the tab appears blank. The **Exclude company from running Planning Optimization** toggle might also appear greyed out and disabled.

This behavior is by design. The **Company information** tab and toggle are only relevant for companies that have been granted an exception to continue using the deprecated master planning engine. Starting from Supply Chain Management version 10.0.41, Planning Optimization is the standard master planning engine for all new deployments and legal entities. If your company wasn't enabled to use the deprecated planning engine, the **Company information** tab is intentionally empty and the toggle is disabled because there's no need to configure an exclusion.

If your company received an exception to use the deprecated planning engine and you still see the tab as blank, create a support ticket stating the reason for the exception.

Learn more in Continue to use deprecated master planning with existing companies and Migration to Planning Optimization for master planning.

Related information

- Get started with master planning
- Planning Optimization fit analysis
- Differences between Planning Optimization and the deprecated master planning engine
- Continue to use deprecated master planning with existing companies

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Last updated on 05/05/2026

Fix the "Not enough capacity could be found" scheduling engine error

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you run scheduling, you might receive the following error message:

> Production order %1 could not be scheduled. Not enough capacity could be found.

There are several reasons why the scheduling engine might fail and issue this error message. This article provides guidelines that help you find the root cause of the error and then mitigate it.

Review the applicable resources

The error can occur if no applicable resources satisfy the operation requirements at the production order site.

To review the applicable resources, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**, and open or select the production order that can't be scheduled.
2. On the Action Pane, on the **Production order** tab, in the **Production details** group, select **Route**.
3. On the **Production route** page, select the operation, and then, on the Action Pane, select **Applicable resources**.
4. In the **Applicable resources** dialog box, set the **Use requirements for** field to *Operations scheduling* or *Job scheduling*, depending on the type of scheduling that you're using.
5. Make sure that there are applicable resources at the production order site.

Review the calendars associated with resources

The error can occur if no calendar is associated with the resource or resource group, or if the associated calendar isn't correctly set up (for example, it has no working times, or its efficiency as a percentage is 0 [zero]).

To verify that a calendar is associated with the resource or resource group, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**, and open or select the production order that can't be scheduled.
2. On the Action Pane, on the **Production order** tab, in the **Production details** group, select **Route**.
3. On the **Production route** page, select the operation, and then, on the Action Pane, select **Applicable resources**.
4. In the **Applicable resources** dialog box, select the resource, and then select **View resource**. Alternatively, select and hold (or right-click) in the **Resource group** field, and then select **View details**.
5. On the **Resources** page or the **Resource groups** page, on the **Calendars** FastTab, make sure that a calendar is associated with the resource or resource group.

> ⓘ **Note**
>
> If the error occurs during operation scheduling, you must make sure that a calendar is associated with all applicable resource groups.

To review the setup of the calendar, follow these steps:

1. Go to **Organization administration** > **Setup** > **Calendars** > **Calendars**, and select the calendar that is associated with the resource or resource group.
2. On the Action Pane, select **Working times**.
3. On the **Working times** page, make sure that the page isn't blank. Additionally, for the days that you're interested in, make sure that the **Control** field isn't set to *Closed*, working times exist, and the **Efficiency is percentage** field isn't set to *0* (zero).

If the calendar is associated with the base calendar, you must also review the setup of the base calendar.

> ⓘ **Note**
>
> In operation scheduling, the resource group's calendar is used to determine the start and end times and dates for each operation. It also limits the amount of time that the system can

schedule for one specific operation for one specific day in one specific resource group. For example, if the working times for a resource group on one specific day are from 8:00 AM to 4:00 PM, the load that one operation puts on the resource group can't exceed the load that can be fit into eight hours, regardless of the amount of available capacity that the resource group has on that day. However, the available capacity can further limit the load.

Review the scheduling parameters

To ensure good performance, the scheduling engine searches for a resource only for a specific amount of time and a specific number of scheduling conflicts.

To review the scheduling parameters, follow these steps:

1. Go to **Organization administration** > **Setup** > **Scheduling parameters**.
2. Follow one of these steps:
    - If the **Scheduling timeout enabled** option is set to *No*, move on to step 3.
    - If the **Scheduling timeout enabled** option is set to *Yes*, increase the value of the **Maximum scheduling time per sequence** field to give processing more time to finish.
3. Follow one of these steps:
    - If the **Scheduling optimization timeout enabled** option is set to *No*, move on to step 4.
    - If the **Scheduling optimization timeout enabled** option is set to *Yes*, increase the value of the **Optimization attempts timeout** field to give processing more time to finish.
4. If you changed any of the settings on the page, reschedule the order to try again.

Review capacity

The error can occur when you perform finite scheduling, but there's no free capacity.

To perform infinite capacity scheduling, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**, and select or open the production order that can't be scheduled.
2. On the Action Pane, on the **Schedule** tab, in the **Production order** group, select **Schedule operations** or **Schedule jobs**.

3. In the **Operations scheduling** or **Jobs scheduling** dialog box, set the **Finite capacity** option to *No*.
4. Select **OK** to schedule the order.

To review the available capacity on the resource, follow these steps:

1. Go to **Organization administration** > **Resources** > **Resources**, and select a resource that is applicable to the order that can't be scheduled.
2. On the Action Pane, on the **Resource** tab, in the **View** group, select **Capacity load** or **Capacity load, graphically**, and make sure that there's available capacity.

To review the available capacity on the resource group, follow these steps:

1. Go to **Organization administration** > **Resources** > **Resource groups**, and select a resource group that is applicable to the order that can't be scheduled.
2. On the Action Pane, on the **Resource group** tab, in the **View** group, select **Capacity load** or **Capacity load, graphically**, and make sure there's available capacity.

Master planning books a resource when the resource calendar is closed

When using operations scheduling, master planning plans capacity according to the calendar of the primary resource group. It books the secondary operation at the same time as the primary operation and doesn't take into account the calendars or capacity of the secondary operation. This can result in the production order being scheduled on a closed calendar or at a time when the secondary operation isn't available (calendar closed, no capacity).

When using job scheduling, master planning takes into account the capacity and calendar of both the primary and secondary operation when scheduling the order. For the order to be scheduled, calendars for the resources of both operations must be open and have available capacity.

Maximum job lead time is too short

The scheduling engine can't schedule an order if the **Maximum job lead time** set for your site is less than the lead time specified for an item in its default order settings or coverage settings.

To view or edit the **Maximum job lead time** setting for your site, go to **Production control** > **Setup** > **Production control parameters** and open the **General** tab.

To view or edit the default order settings for an item, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Find and select the relevant product in the list.
3. On the Action Pane, open the **Manage inventory** tab and select **Default order settings**.
4. Expand the **Inventory** FastTab and view or edit the **Inventory lead time** setting as needed.

To view or edit the coverage settings for an item, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Find and select the relevant product in the list.
3. On the Action Pane, open the **Plan** tab and select **Item coverage**.
4. Open the **Lead time** tab and view or edit the **Production time** value as needed.

Excessive quantity of required resources

During scheduling, the engine tries to match the required resource quantity set for a route operation to the applicable resources according to the operation resource requirements. Setting the resource quantity too high can make a route unfeasible and produce a scheduling error.

Use the following procedure to check both the specified quantity and applicable resources for a selected product, route, and route operation:

1. Go to **Product information management** > **Products** > **Released products**.
2. Find and select the relevant product in the grid.
3. On the Action Pane, open the **Engineer** tab and select **Route**.
4. Find and select the relevant route in the grid.
5. Open to the **Overview** tab at the bottom of the page.
6. Select an operation from the list of the selected route operations.
7. Select **Applicable resources** to open a dialog where you can view the applicable resources for the selected route operation.
8. Open the **Resource load** tab. The **Quantity** field here shows the resource quantity required for the selected route operation. View and/or edit it as needed.

Last updated on 06/22/2026

Cancel a planning job

In Microsoft Dynamics 365 Supply Chain Management, you can cancel an active planning job that uses the Planning optimization functionality. When you select **Cancel** in the dialog box when a Planning optimization job is triggered directly from the user interface (not in the background), this will not cancel the Planning optimization job. Even if you receive a warning such as "Operation canceled", you will still need to use the following steps to cancel a planning job with Planning optimization.

To cancel an active planning job, follow these steps:

> ⓘ **Note**
>
> Only active jobs can be canceled.

1. Go to **Master planning** > **Setup** > **Plans**.
2. Select an appropriate plan for the planning run.
3. Select **History**.
4. Select the planning job to cancel.
5. Select **Cancel**.

The job status will be **Canceling** until the Planning Optimization service confirms that the job has been canceled. The status will then be changed to **Canceled**.

> ⓘ **Note**
>
> To see status changes, you must refresh the page by selecting the **Refresh** button.

Last updated on 12/30/2025

Lean manufacturing for Planning Optimization

Lean manufacturing is a production philosophy that maximizes value for customers while minimizing waste. It comes from the Toyota production system and focuses on continuous improvement and operational efficiency. The *Lean manufacturing for Planning Optimization* feature allows Planning Optimization to consider demand and supply that originates from lean-manufacturing models (such as kanbans) when generating plans in Supply Chain Management. Learn more in Lean manufacturing overview.

To use lean manufacturing processes with Planning Optimization, your system must meet the following requirements:

- To use the preview release of this feature, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.45.
- To use the general availability release of this feature, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- The feature that is named *Lean manufacturing for Planning Optimization* must be turned on in feature management.

Last updated on 12/25/2025

Unified developer experience for planning optimization (preview)

[This article is prerelease documentation and is subject to change.]

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft Power Platform provides a unified developer experience that includes tools and environments for writing and debugging extensions for the Planning Optimization Add-in for Microsoft Dynamics 365 Supply Chain Management. This topic provides an overview of the process for setting up a development environment and testing your extensions.

Learn more in Unified developer experience for finance and operations apps.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Extensibility scenarios

Developers can extend Planning Optimization to add support for scenarios that apply after master planning runs. Developers can test their extensions by connecting the Planning Optimization service to their development environment.

The most common extensibility scenario for Planning Optimization involves the application of custom processing after a plan is updated. Microsoft Power Platform supports low-code development that enables new solutions to be created quickly. At the same time, developers can make sure that their extensions work correctly by using the real-time results that Planning Optimization produces.

Set up an environment for testing your Planning Optimization extensions

Provision a new environment

To provision an environment that you can use for Planning Optimization, follow the instructions in Tutorial: Provision a new environment with an ERP-based template.

> **Important**
>
> When you provision the environment, you must set the template ID to *D365_FinOps_SCM*.

After you complete the steps in the Step-by-step provisioning guide, your new environment will be prepared in Power Platform admin center.

Confirm that the Planning Optimization Add-in is installed

To confirm that you successfully added the Planning Optimization Add-in to your environment, follow these steps:

1. Sign in to Power Platform admin center.
2. On the navigation pane, select **Environments**, and then open the environment where you installed the add-in.
3. In the **Resources** section, select **Dynamics 365 apps**.
4. In the list, find the row where the **Name** field is set to *Planning optimization add-in for D365 SCM*. Confirm that the **Status** field is set to *Installed*.

Turn on Planning Optimization for your Supply Chain Management environment

To turn on Planning Optimization for your Supply Chain Management environment, follow these steps:

1. Sign in to your new Supply Chain Management environment. (You can find your Supply Chain Management URL on the environment page in Power Platform admin center.)
2. Follow the instructions in Turn on Planning Optimization for your environment.

Related information

- Unified developer experience for finance and operations apps
- Unified admin experience for finance and operations apps
- Install and configure development tools
- Provision a new environment with an ERP-based template
- Write, deploy, and debug X++ code

Last updated on 01/09/2026

Planning Optimization extensibility

07/31/2025

This article describes the extensibility scenarios that are related to master planning and supported in Planning Optimization.

> **Tip**
>
> If you've finished developing an extension and want to set up an end-to-end testing environment, see Unified developer experience for planning optimization (preview).

Custom processing when master planning is completed

The most common extensibility scenario for Planning Optimization is to apply custom processing after the plan has been updated. For example, you might add a column to the planned orders table (`ReqPO`), or you might want to derive some statistical information from the generated plan. The main extensibility point that enables this scenario is the `jobCompletedSuccessfully` method in the `MpsMasterPlanningEvents` class. To apply processing after a job fails, use the `jobFailed` method in the same class.

> **Important**
>
> Post-processing actions that use the current extensibility point will only apply to items directly specified in the filters (if used) and not to their subcomponents. Therefore, additional items included as a result of enabling the **Include all BOM levels** and/or **BOM levels to include** options in the **Planning Optimization** run dialog won't be affected by these post-processing actions. For more information about how to use the filters, see Run planning for a subset of items

The job completed successfully method

This section shows how to use the `jobCompletedSuccessfully` method.

```
X++
```

```
public static void
jobCompletedSuccessfully(MpsMasterPlanningJobCompletedSuccessfullyEventArgs _args)
```

Here's an example of an extension that sets a custom `CstmOrderPriority` field on the planned order.

```
X++
```

```
[ExtensionOf(classStr(MpsMasterPlanningEvents))]
public static final class MpsMasterPlanningEvents_Cstm_Extension
{
    public static void
jobCompletedSuccessfully(MpsMasterPlanningJobCompletedSuccessfullyEventArgs _args)
    {
        ttsbegin;

        var affectedPlannedOrdersQuery =
_args.parmPostProcessingQueryBuilder().buildAffectedPlannedOrdersQuery();
        var affectedPlannedOrdersQueryRun = new
QueryRun(affectedPlannedOrdersQuery);

        while (affectedPlannedOrdersQueryRun.next())
        {
            ReqPO reqPO = affectedPlannedOrdersQueryRun.get(tableNum(ReqPO));
            reqPO.selectForUpdate(true);
            reqPO.CstmOrderPriority = reqPO.ReqDate - systemDateGet() < 7 ?
CstmPlannedOrderPriority::Urgent : CstmPlannedOrderPriority::Regular;
            reqPO.doUpdate();
        }

        ttscommit;

        next jobCompletedSuccessfully(_args);
    }

}
```

When you add custom logic, consider the following constraints and best practices:

- The `jobCompletedSuccessfully` method is called outside the transaction scope. Therefore, the plan is already visible to the user when the custom logic starts to run. If your customization sets other fields on planned orders, it's important that you let users know that the values of those fields will eventually be consistent (in other words, they might be out of date immediately after a planning job is completed).
- Another master planning job can start when the `jobCompletedSuccessfully` method is called. The new job might affect the full plan or part of the plan. The new job might update or delete the same planned orders or requirement transactions that were created or updated as part of the planning job that was handled in `jobCompletedSuccessfully`. Update conflict exceptions must be handled when this event is extended.
- Avoid using single transaction scope when you extend this method. A single master planning run can produce millions of requirement transactions and hundreds of

thousands of planned orders. If all these transactions and planned orders are processed in the scope of a single transaction, many SQL locks will occur and block other planning processes. Additionally, if the transaction is held for a long time, "ghost records" will be created in the SQL database. Ghost records will negatively affect every process in the system.

- The query included in `MpsMasterPlanningJobCompletedSuccessfullyEventArgs` includes all the items that were included in the original filters, including those that had errors.
- The query included for `jobCompletedSuccessfully` won't include items added as a result of enabling the **Include all BOM levels** and/or **BOM levels to include** options in the **Planning Optimization** run dialog.

The job failed method

When a Planning Optimization job fails, it might be necessary to start another process (such as a retry operation). For this, you can use the `jobFailed` method, which is supported in Supply Chain Management version 10.0.31 and higher.

When adding custom logic, keep in mind that the query included for the `jobFailed` method won't include items added as a result of enabling the **Include all BOM levels** and/or **BOM levels to include** options in the **Planning Optimization** run dialog.

View plan history and planning logs

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to view the history of planning jobs in Microsoft Dynamics 365 Supply Chain Management.

Each time you run a new master planning calculation, the system adds a new history record.

To view the history for a plan, open the plan by going to **Master planning** > **Setup** > **Plans** > **Master plans** and selecting **History**. The history lists all the jobs for the selected plan. The list includes completed and active jobs.

In addition to seeing the start time and status of jobs, you can view the log for a specific job. The log includes additional information and warnings. Not all jobs have a log. To view the log for a job, select **Log**. Log entries are only stored for 60 days after the date the job finished, after that they are automatically deleted.

If the **Batch processing** option on the **Run in the background** FastTab was enabled when master planning processing was set up, the batch job log shows more information about any warnings and errors that were generated during the master planning run. For example, auto-firming errors are captured only in the batch job log. They aren't shown in logs on the **History** page.

To view auto-firming errors and other warnings or errors that occurred during a master planning run, follow these steps:

1. Go to **System administration** > **Inquiries** > **Batch jobs**.
2. Find and select the record that represents the master planning run that you're interested in. (For example, the value of the **Job description** field might start with *Master planning*.)
3. Follow one of these steps, depending on whether you're using the *enhanced form* or the *legacy (unenhanced) form* for the **Batch jobs** page:
    - If you're using the enhanced form: On the Action Pane, select **Batch job history**. Then, on the **Batch job history** page, on the Action Pane, select **Log**.

- If you're using the legacy form: On the Action Pane, on the **Batch job** tab, select **Log**.

4. Select **Message details** to open the **Message details** pane, where you can view all warnings and errors that were captured during processing.

Last updated on 02/26/2026

Date and time parameters used by Planning Optimization

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about the date and time parameters that Planning Optimization uses during its operation.

Whereas the deprecated master planning engine uses transaction dates in all calculations, Planning Optimization works with date and time values that are converted to dates. This difference in behavior can lead to situations where, for example, forecast transactions that are created at midnight on the day when master planning is run aren't included because Planning Optimization considers that they were created before the current date.

Parameters for issue and demand transactions

The following table lists the parameters that Planning Optimization uses when it processes issue and demand transactions.

Expand table

Parameter	Parameter name in Planning Optimization	Description	Equivalent field in Microsoft Dynamics 365 Supply Chain Management (in the ReqTrans table)
Planned issue time	`PlannedIssueTime`	The date that the issue is currently planned for.	**To date** (`FuturesDate`) and **Delayed to time** (`FuturesTime`)
Requested issue time	`RequestedIssueTime`	The date of issue that the user requests and is set in Supply Chain Management. This parameter is applicable only for	**Requested date** (`ReqDateDlvOrig`)

Parameter	Parameter name in Planning Optimization	Description	Equivalent field in Microsoft Dynamics 365 Supply Chain Management (in the ReqTrans table)
		released or approved planned orders. For planned orders, it's blank by default.	
Required issue time	`RequiredIssueTime`	The required issue date that Planning Optimization adjusts. If this time falls in the past when Planning Optimization runs, it is moved to the first open day that is no earlier than today. If the time is marked as blocked in the calendar, it is instead adjusted to the first open day before that date.	**Requirement date** (`ReqDate`) and **Requirement time** (`ReqTime`)
Issue time delay	`IssueTimeDelay`	The time difference between the planned issue time and either the requested issue time for approved and released orders or the required issue time.	**Delay (in days)** (`FuturesDays`)

Parameters for receipt and supply transactions

The following table lists the parameters that Planning Optimization uses when it processes receipt and supply transactions.

Expand table

Parameter	Parameter name in Planning Optimization	Description	Equivalent field in Supply Chain Management (in the ReqTrans or ReqPO table)
Planned availability time	`PlannedAvailabilityTime`	The date when the receipt is planned to be available.	**Requirement date** (`ReqDate`) and **Requirement time** (`ReqTime`)
Planned receipt time	`PlannedReceiptTime`	The date when the receipt arrives at the location.	**To date** (`FuturesDate`), **Delayed to time** (`FuturesTime`), and **Delivery date** (`ReqDateDlv`) or **Requested date** (`ReqDateDlvOrig`) if the order isn't released yet.
Required availability	`RequiredAvailabilityTime`	The required availability date that Planning Optimization	**Requirement date** (`ReqDate`) and **Requirement time**

Parameter	Parameter name in Planning Optimization	Description	Equivalent field in Supply Chain Management (in the ReqTrans or ReqPO table)
time		adjusts.	(`ReqTime`)
Expected receipt time	`ExpectedReceiptTime`	The expected receipt date for a released receipt. The user sets the value in Supply Chain Management. THe value isn't adjusted in Planning Optimization. This parameter applies only to released receipts.	**Requested date** (`ReqDateDlvOrig`)
Required receipt time	`RequiredReceiptTime`	The required receipt date that Planning Optimization adjusts.	**Requirement date** (`ReqDate`) and **Requirement time** (`ReqTime`)
Planned ordering time	`PlannedOrderingTime`	The ordering date that Planning Optimization calculates.	**Order date** (`ReqDateOrder`) and **Order time** (`ReqTimeOrder`)
Planned activity start time	`PlannedActivityStartTime`	The date when the activity for this receipt should start.	**Start date** (`SchedFromDate`)
Receipt time delay	`ReceiptTimeDelay`	The time difference between the planned receipt time and the required receipt time.	**Delay (days)** (`FuturesDays`) and **Delayed to time** (`FuturesTime`)

Examples of date parameter use by Planning Optimization

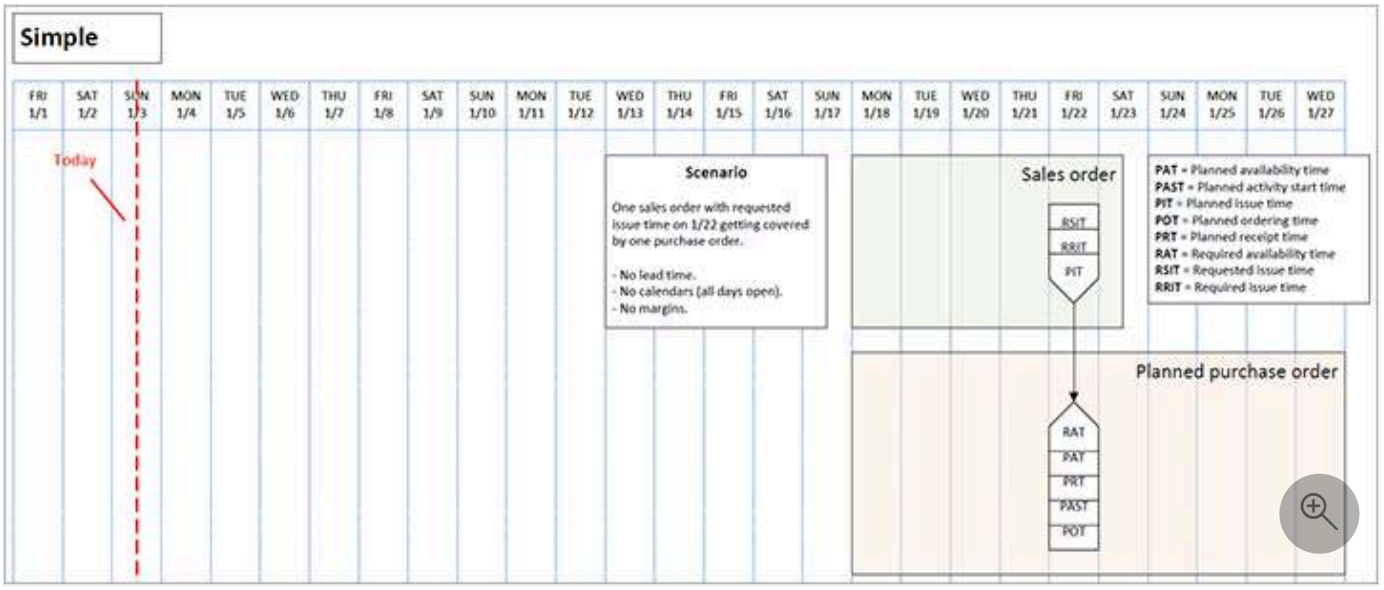
The plans in the following illustrations are at the day level, but Planning Optimization runs at a more detailed level. For example, because margins can be in hours, the planning ordering time can be January 22, 2021, at 11:35, and so on.

Example 1: Simple scenario

One sales order that has a requested issue time on January 22 is covered by one purchase order. The following settings are used:

- No lead time
- No calendars (All days are open.)
- No margins

The following illustration shows this scenario. (Select the illustration to open a larger version.)

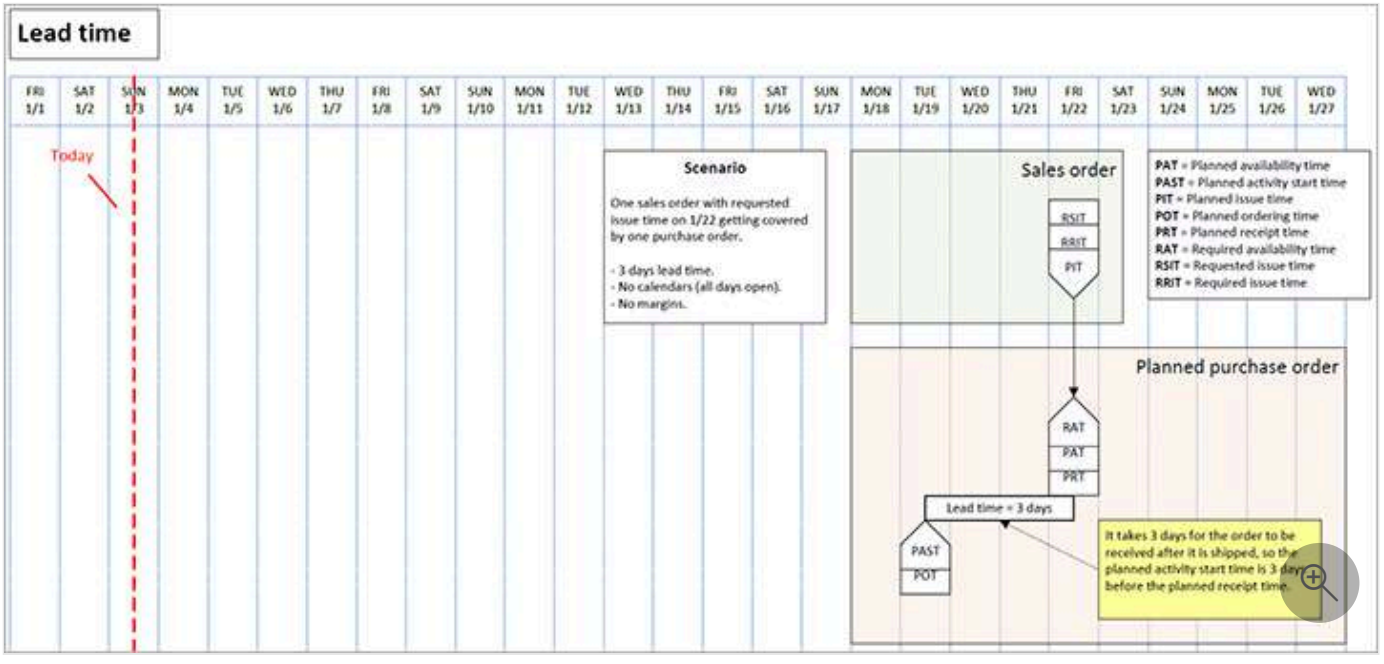


Example 2: Lead time scenario

One sales order that has a requested issue time on January 22 is covered by one purchase order. The following settings are used:

- Three days of lead time
- No calendars (All days are open.)
- No margins

The following illustration shows this scenario. (Select the illustration to open a larger version.)

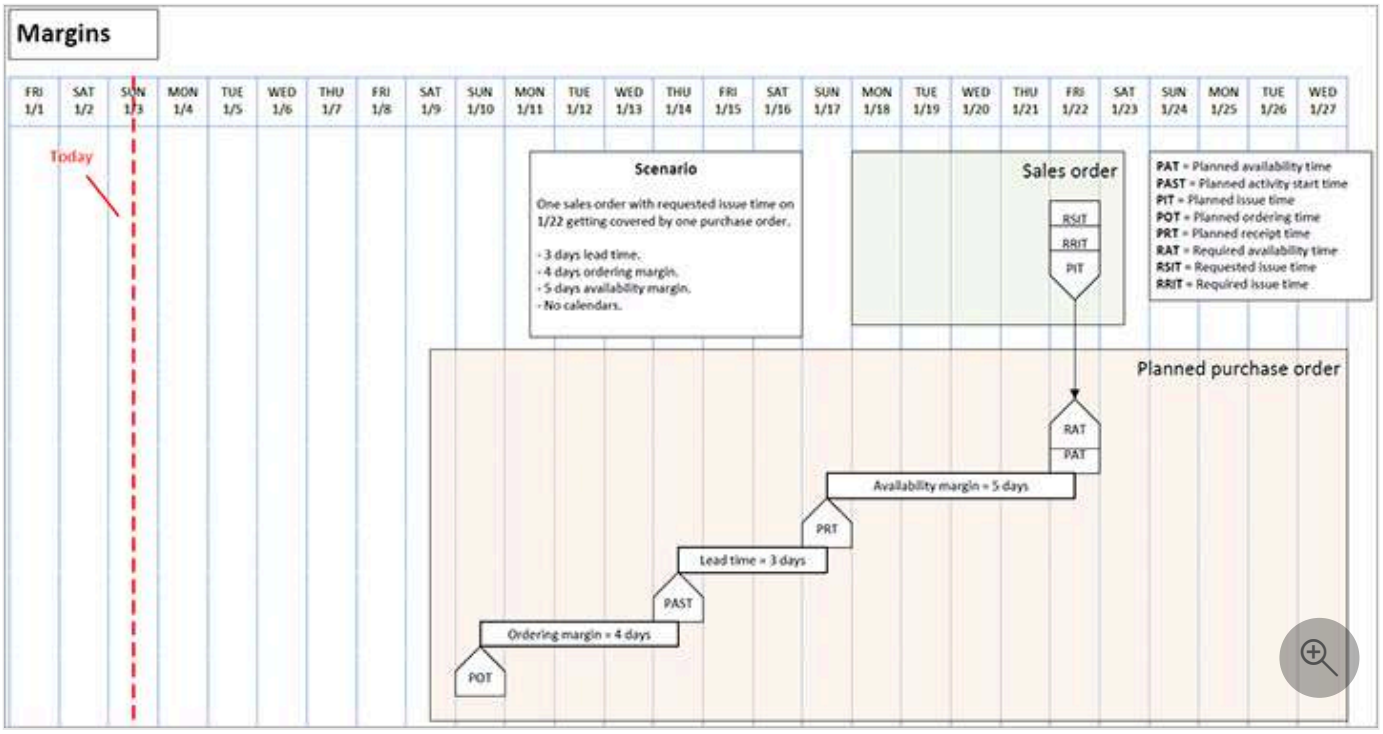


Example 3: Margin scenario

One sales order that has a requested issue time on January 22 is covered by one purchase order. The following settings are used:

- Three days of lead time
- Four-day ordering margin
- Five-day availability margin
- No calendars (All days are open.)

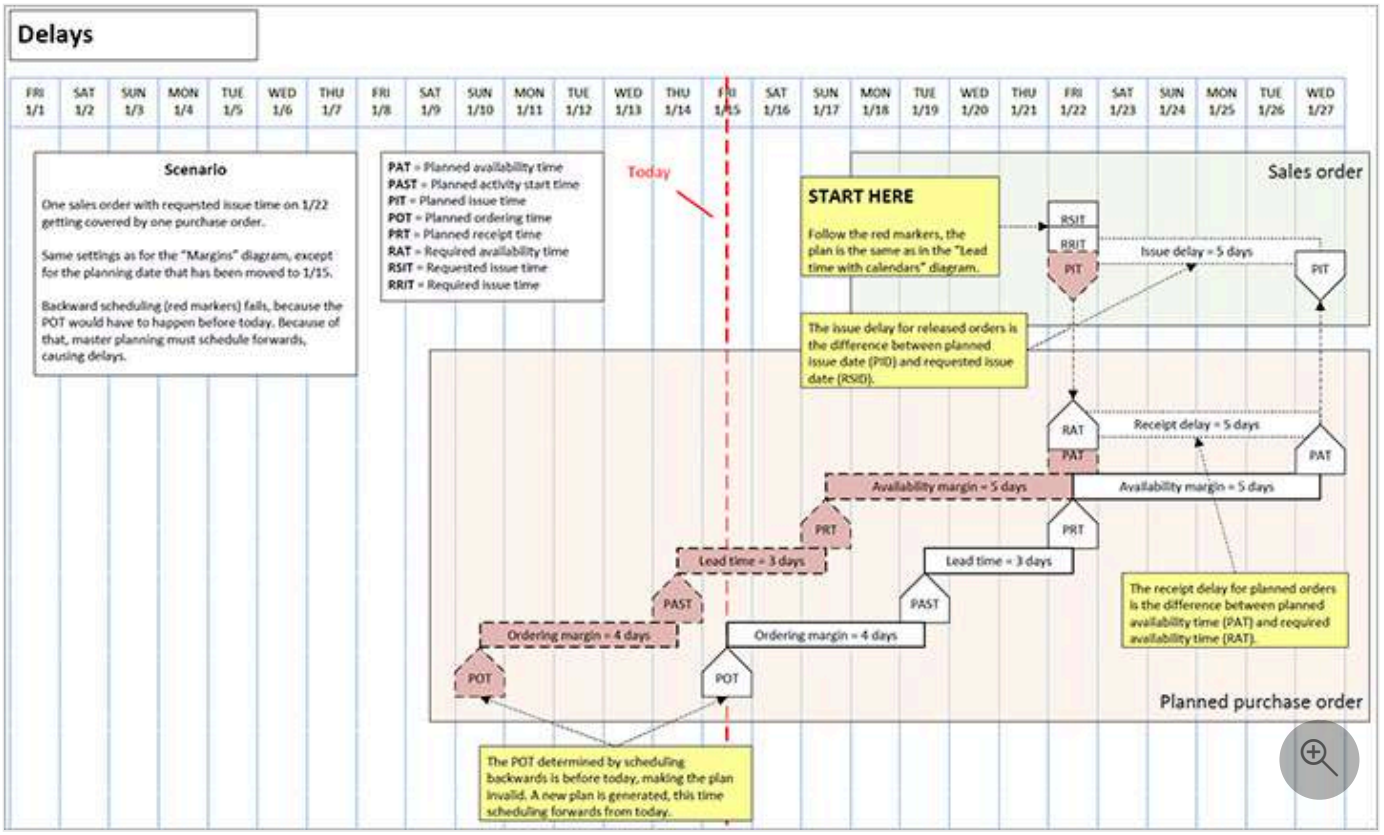
The following illustration shows this scenario. (Select the illustration to open a larger version.)



Example 4: Delay scenario

One sales order that has requested issue time on January 22 is covered by one purchase order. This example uses the same settings as example 3, but the planning date is moved to January 15. Backward scheduling (red markers) fails because the planned ordering time would have to be before today's date. Therefore, master planning must schedule forward, and delays occur.

The following illustration shows this scenario. (Select the illustration to open a larger version.)

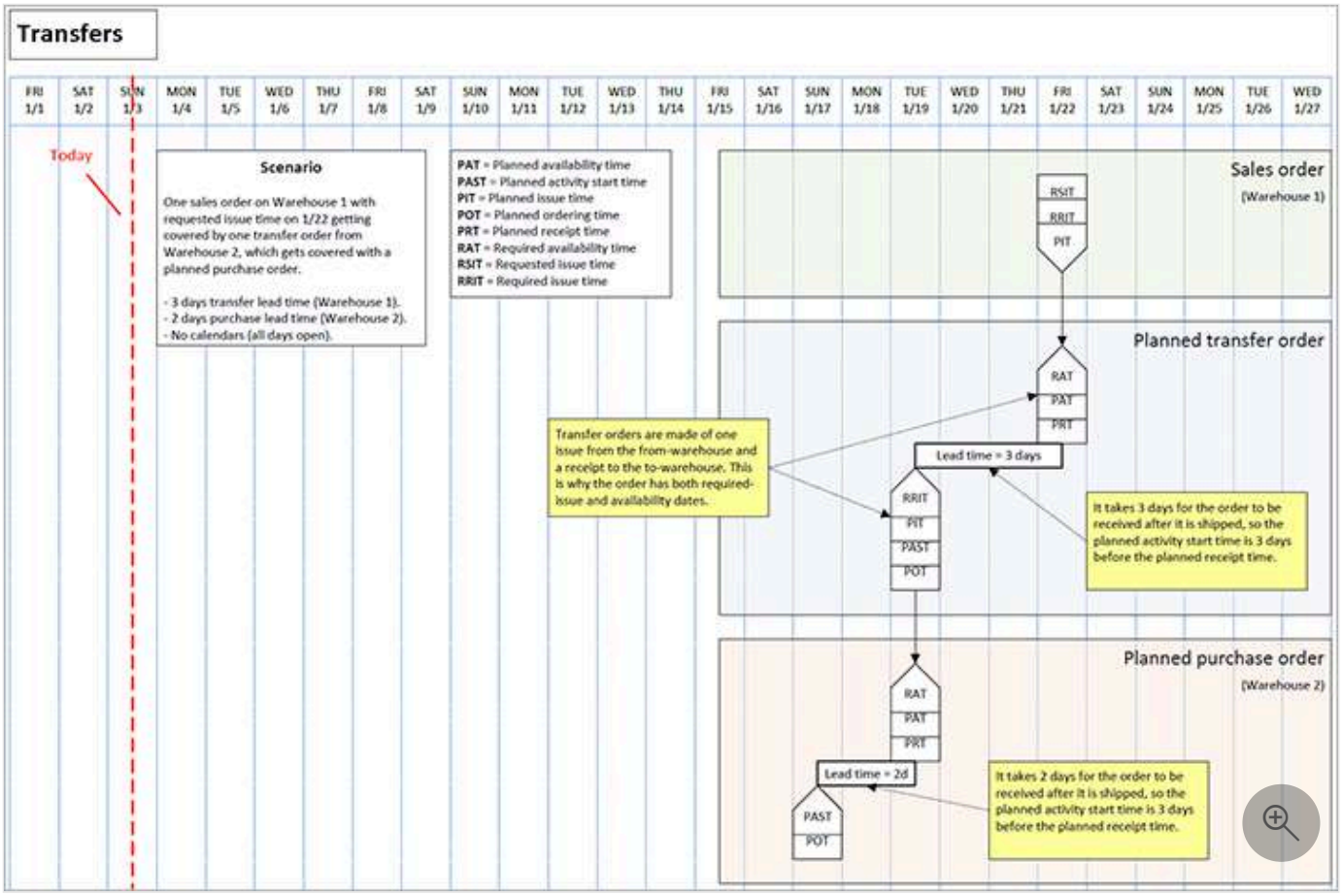


Example 5: Transfer scenario

One sales order from warehouse 1 that has a requested issue time on January 22 is covered by one transfer order from warehouse 2 that is covered by a planned purchase order. The following settings are used:

- Three days of transfer lead time (warehouse 1)
- Two days of purchase lead time (warehouse 2)
- No calendars (All days are open.)

The following illustration shows this scenario. (Select the illustration to open a larger version.)

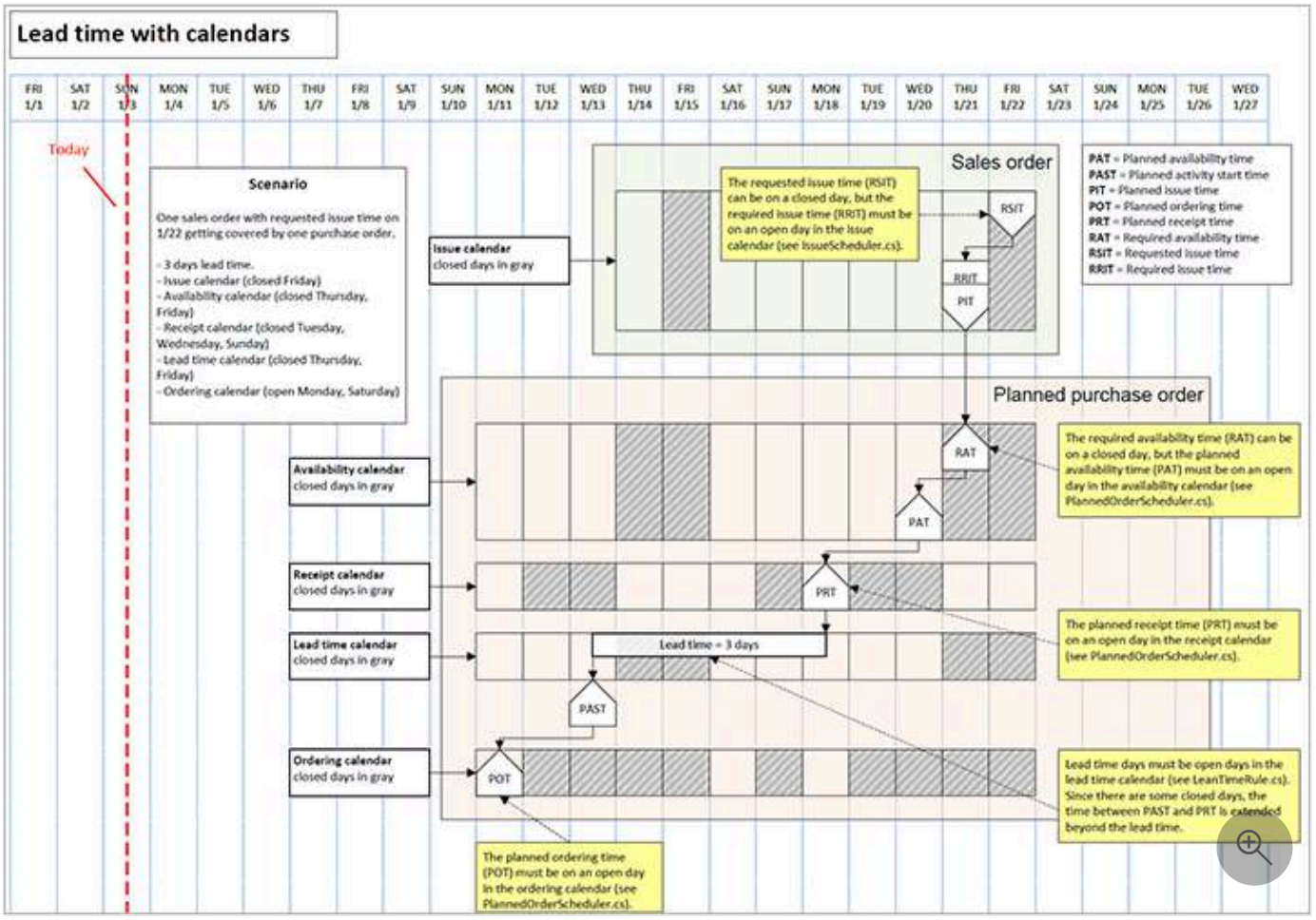


Example 6: Lead time with calendars scenario

One sales order that has a requested issue time on January 22 is covered by one purchase order. The following settings are used:

- Three days of lead time
- Issue calendar (closed on Friday)
- Availability calendar (closed on Thursday and Friday)
- Receipt calendar (closed on Tuesday, Wednesday, and Sunday)
- Lead time calendar (closed on Thursday and Friday)
- Ordering calendar (open on Monday and Saturday)

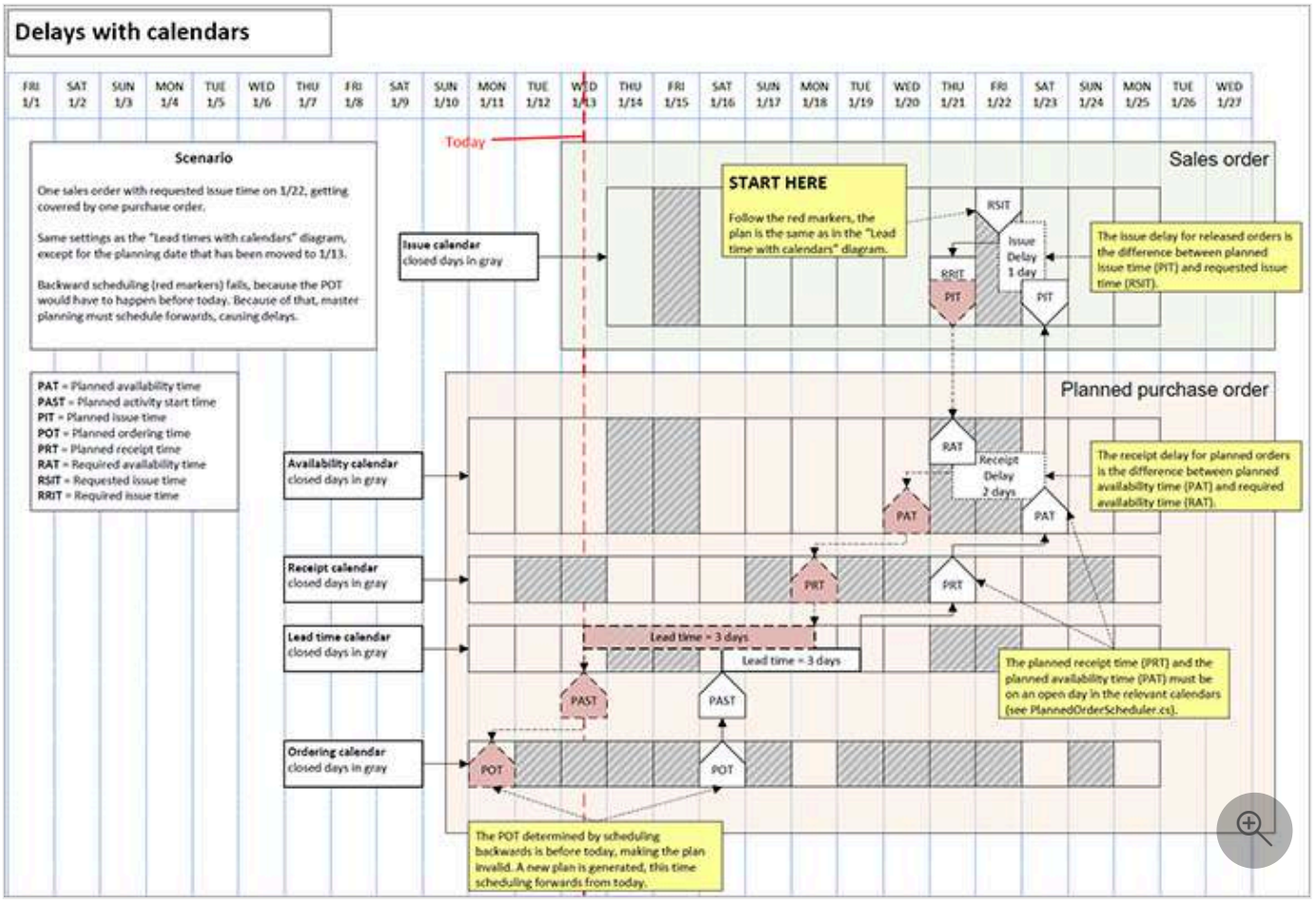
The following illustration shows this scenario. (Select the illustration to open a larger version.)



Example 7: Delay with calendars scenario

One sales order that has a requested issue time on January 22 is covered by one purchase order. This example uses the same settings as example 6, but the planning date is moved to January 13. Backward scheduling (red markers) fails because the planned ordering time would have to be before today's date. Therefore, master planning must schedule forward, and delays occur.

The following illustration shows this scenario. (Select the illustration to open a larger version.)



Related information

- Calendars and master planning
- Calculate requested ship dates for purchase orders
- Improve scheduling engine performance
- Safety margins

Last updated on 05/06/2026

Delays

08/25/2025

This article provides information about delayed dates in master planning. A delayed date is a realistic due date that a transaction receives if the earliest fulfillment date that master planning calculates is later than the requested date.

Master planning can calculate the earliest fulfillment date for a transaction, based on lead times, material availability, capacity availability, and various planning parameters.

If master planning calculates an order date that precedes the current date, the order can't be fulfilled on time. Therefore, the order is delayed. In this case, master planning forward-plans the order from the current date and includes lead times. These lead times start with any lower-level component items. The order then receives a delayed date. A delayed date is a realistic due date, based on the current data. Master planning also calculates the number of delay days.

In some situations, you might choose not to calculate delays, such as when users know that they can expedite lead times by selecting alternative modes of delivery.

You can configure how delays are calculated for a coverage group. You can then attach the coverage group to an item later.

On the **Master planning parameters** page, you can set the start time for the calculation of delays. If an order is fulfilled after this time, a delay of one day is added to the delay date of the order.

> **Note**
>
> In earlier versions, calculated delays were known as *futures messages*, the delayed date was known as the *futures date*, and a delayed transaction was referred to as *a transaction that was future set*.

Limited delays in production setup with multiple BOM levels

When you work with delays in a production setup that has multiple bill of material (BOM) levels, it's important to note that only the items directly above the item (in the BOM structure) causing the delay, will be updated with a delay as part of the master planning run. Other items in the BOM structure won't get the delay applied until the first master planning run, when the planned order for the top level is approved or firmed.

To work around this known limitation, the production orders on the top of the BOM structure with delays can be approved (or firmed) before the next master planning run. This way the delay from the delayed approved planned production order will be kept and all underlying components will be updated accordingly.

Action messages can also be used to identify planned orders that can be moved to a later date, due to other delays in the BOM structure.

Desired date

On the **Planned order** page, under the **Delays** tab is the **Desired date** for the planned order. The desired date of a planned order is the base date for delays, which is a computed date that equals the **Requested date** calculated from the **Net Requirement**. If the planned order is a BOM line, production line, or kanban line, the desired date is based on the **Requirement date** and the desired date won't be shown on the **Planned order** page.

Related information

- Coverage settings

Use tracing for explosion

This article explains how you can use tracing to explore the causes behind the outcome of an order explosion.

By enabling tracing, you can view information about the factors that contributed to the outcome of the explosion of a particular order. The following examples show how you can use the tracing information:

- View relations between the actions on planned orders to optimize the supply chain and inventory reservations.
- View relations to orders that are already approved. You can focus on automatically firming derived requirements and then prioritize orders more accurately.
- Simulate planning results to determine whether the planning parameters are optimal.
- Identify how information such as production dates, quantities, and priorities for an order were determined.

You can view details about futures and actions for a selected order. On the **Explosion** page, tracing information is available on the **Explanation** tab in the upper pane. Tracing occurs when you explode an order. To start tracing for the order, click **Update**, and then select the **Enable trace** check box. You can use the **Find text** field to search the log for specific information. Search results are highlighted in the tree.

Related information

- Master plans overview

Last updated on 07/01/2026

Make-to-order supply automation

08/22/2025

The *Make-to-order supply automation* feature adds several enhancements to Microsoft Dynamics 365 Supply Chain Management. These enhancements let you perform the following tasks:

- View the resource capacity load for a user-defined period, and therefore enable long-term evaluation of the capacity load.
- Improve flexibility by controlling the delay tolerance (negative days) for each master plan.
- Keep products available for last-minute orders and optimize the use of existing supply by pegging the latest possible supply to a demand instead of using the first possible supply.
- Keep component assignment flexible for production orders after firming, so that the system can optimize for last-minute demand changes. In other words, limit marking to one level.
- Control the fulfillment policy for each sales order. Default settings are taken from the customer setup.
- Enhance intercompany information flow. Purchase orders are updated so that they include fields for the mode of delivery, delivery terms, and external item number. This change ensures that detailed demand information can flow to the supplying company.

This article describes how to set up and use each enhancement.

Set the number of days to show on Capacity load page

The **Capacity load** page lets you review a resource's available capacity. The *Make-to-order supply automation* feature enhances the **Capacity load** page by adding a **Number of days** field. You can use this field to limit the number of days that the **Overview** grid shows. The value that you set is saved in memory. Therefore, if you leave the page and then return later, the **Overview** grid will still show the number of days that you last specified.

To open the **Capacity load** page so that you can review the available capacity for an individual resource, follow these steps.

1. Go to **Organization administration** > **Resources** > **Resources**.
2. Select a resource to inspect.
3. On the Action Pane, on the **Resource** tab, in the **View** group, select **Capacity load**.
4. Use the **Number of days** filter to limit the number of days that the **Overview** grid shows.

To open the **Capacity load** page so that you can review the available capacity for a resource group, follow these steps.

1. Go to **Organization administration** > **Resources** > **Resource groups**.
2. Select a resource group to inspect.
3. On the Action Pane, on the **Resource group** tab, in the **View** group, select **Capacity load**.
4. Use the **Number of days** filter to limit the number of days that the **Overview** grid shows.

Apply a single level of marking when you firm planned orders

Marking is used to link supply and demand. It resembles *pegging*, which indicates how master planning expects to cover demand. However, marking is more permanent than pegging. The *Make-to-order supply automation* feature adds the ability to limit inventory marking to a single level when planned orders are firmed. It adds the following new options to the **Update marking** field in the **Firming** dialog box when you're firming a planned order:

- *Single level standard* – Single-level marking is used. Single-level marking marks only the main item, not its bill of materials (BOM) components. Therefore, you can keep component assignment for production orders flexible after firming. Single-level marking enables the system to optimize for last-minute demand changes. In *standard* single-level marking, requirement orders are marked against their fulfillment orders, but fulfillment orders aren't marked if they have remaining quantity.
- *Single level extended* – Single-level marking is used. In *extended* single-level marking, requirement orders are marked against their fulfillment orders, and fulfillment orders are always marked, regardless of whether any quantity remains.

These options are also available in the **Update marking** field on the **Standard update** tab of the **Master planning parameters** page, where you define the default selection for the **Firming** dialog box.

Learn more in Inventory marking.

Set delay tolerance (negative days) at the master plan level

The *Make-to-order supply automation* feature adds the ability to configure delay tolerance (negative days) at the master plan level. The setting is also available at the item coverage and coverage group levels. If negative days are assigned at more than one level, the system applies the following hierarchy to decide which setting to use:

- If negative days are configured on the **Master plans** page, that setting overrides all other negative days settings when the plan runs.
- If negative days are configured on the **Item coverage** page, that setting overrides the coverage group setting.
- Negative days that are configured on the **Coverage groups** page apply only if negative days haven't been configured for a relevant item or master plan.

To set negative days for a master plan, follow these steps.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select a master plan in the list pane, or create a new one.
3. On the **Time fences in days** FastTab, set the **Negative days** option to *Yes*.
4. In the nearby field, enter the number of negative days to allow.

For more information about how to use negative days, see Delay tolerance (negative days).

Control the pegging sequence used during master planning

Master planning uses a *pegging sequence* to determine which supply will cover which demand. The *Make-to-order supply automation* feature adds a new **Use latest possible supply** option that gives you more control over the pegging sequence. The new option lets you keep products available for last-minute orders and optimize the use of existing supply by pegging the latest possible supply to a demand instead of using the first possible supply.

You can set the pegging sequence at the master plan, item coverage, or coverage group level. If a pegging sequence is set up at more than one level, the system applies the following hierarchy to decide which setting to use:

- If a pegging sequence is set up on the **Master plans** page, that setting overrides all other pegging sequence settings when the plan runs.
- If a pegging sequence is set up on the **Item coverage** page, that setting overrides the coverage group setting.
- The pegging sequence that is set up on the **Coverage groups** page applies only if pegging sequence settings haven't been configured for a relevant item or master plan.

To set up pegging at the coverage group level, follow these steps.

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. Select a group in the list pane, or create a new one.
3. On the **General** FastTab, in the **Pegging sequence** section, use the **Consume on-hand inventory** and **Use latest supply** fields to configure your pegging sequence. The table

later in this section shows how these settings are combined to define your pegging sequence.

To set up pegging at the item coverage level, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a product in the grid, or create a new one.
3. On the Action Pane, on the **Plan** tab, select **Item coverage**.
4. The **Item coverage** page provides rows that let you define coverage rules that apply to the item at each warehouse. Select an existing row in the grid, or create a new one.
5. On the **General** tab, select the **Override pegging sequence** checkbox.
6. Use the **Consume on-hand inventory** and **Use latest supply** fields to configure your pegging sequence. The table later in this section shows how these settings are combined to define your pegging sequence.

To set up pegging at the master plan level, follow these steps.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select a master plan in the list pane, or create a new one.
3. On the **General** FastTab, set the **Override pegging sequence** option to *Yes*.
4. Use the **Consume on-hand inventory** and **Use latest supply** fields to configure your pegging sequence. The table later in this section shows how these settings are combined to define your pegging sequence.

The following table shows how the **Consume on-hand inventory** and **Use latest supply** settings are combined to establish your pegging sequence.

Expand table

	Use latest supply = Yes	**Use latest supply = No**
Consume on-hand inventory = *Before all supply*	1. On-hand 2. Existing supply that can meet the demand date 3. Existing supply that can't meet the demand date, but still within negative days 4. Create new supply (planned order)	1. On-hand 2. Existing supply that can meet the demand date 3. Existing supply that can't meet the demand date, but still within negative days 4. Create new supply (planned order)
Consume on-hand inventory = *After all supply*	1. Existing supply that can meet the demand date 2. On-hand	1. Existing supply that can meet the demand date 2. Existing supply that can't meet the demand date, but still within

	Use latest supply = Yes	Use latest supply = No
	3. Existing supply that can't meet the demand date, but still within negative days 4. Create new supply (planned order)	negative days 3. On-hand 4. Create new supply (planned order)

Review and set the fulfillment policy that applies to each sales order

Fulfillment policies control the percentage of the total order price or quantity that must be physically reserved before you can release a sales order to the warehouse. You can set a global default fulfillment policy and then override it for specific customers as you require. The *Make-to-order supply automation* feature adds the ability to view what default policy actually applies to any order and apply an order-specific override as required.

- To set the global fulfillment policy default for sales orders, go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**. Then, on the **Warehouse management** tab, set the **Sales order fulfillment policy** field to the name of the policy that you want to use.
- To set a customer-specific fulfillment policy for sales orders, go to **Accounts receivable** > **Customers** > **All customers**. Then, on the **Warehouse** tab, set the **Fulfillment policy** field to the name of the policy that you want to use.

To view the default policy that applies to any order and apply an order-specific override, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Open the sales order that you want to inspect or edit.
3. Select the **Header** view.
4. On the **Warehouse** FastTab, review and edit the following fields as required:
    - **Order fulfillment policy** – Select the fulfillment policy to use for the selected order. The default policy will be overridden. To use the default fulfillment policy that is shown in the **Default fulfillment policy** field, leave this field blank.
    - **Default fulfillment policy** – This field shows the default fulfillment policy that applies if the **Order fulfillment policy** field is blank. This field is read-only. If it's blank, no customer-specific or global default fulfillment policy is defined.

If both the **Order fulfillment policy** field and the **Default fulfillment policy** field are blank, no fulfillment policy applies. However, other restrictions might be in place, and might require that reservations or other conditions be met before the sales order can be released.

Set the delivery mode and terms on individual purchase order lines

All purchase orders include **Delivery terms** and **Mode of delivery** settings on the order header. The *Make-to-order supply automation* feature adds these settings to each order line. Therefore, you can define overrides of the **Delivery terms** and/or **Mode of delivery** value for any or all order lines as required. For lines where no override is defined, the value from the header is used. You can specify whether a change to the value of one or both of these fields on the order header should also update all lines.

To specify what should happen to line settings if a user changes the **Delivery terms** and/or **Mode of delivery** value on an order header, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **General** tab, on the **Default values and parameters** FastTab, select the **Update order lines** link.
3. In the **Update order lines** dialog box, in the **Updating Delivery terms** and **Update Mode of delivery** fields, select one of the following values:
    - *Never* – Never update the order lines after the settings are changed on the order header.
    - *Always* – Always update all order lines after the settings are changed on the order header.
    - *Prompt* – If a user changes one or both settings on the order header, open a dialog bot that asks whether the user wants to update all lines so that they match the change to one or both settings.

To set delivery information on a purchase order header, follow these steps.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Open the purchase order that you want to edit.
3. Select the **Header** view.
4. On the **Delivery** FastTab, set the **Mode of delivery** and **Delivery terms** fields as required.
5. Depending on the settings on the **Procurement and sourcing parameters** page, you might be asked whether you want to apply your changes to all order lines.

To set delivery information overrides for a purchase order line, follow these steps.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Open the purchase order that you want to edit.
3. Select the **Lines** view.
4. On the **Purchase order lines** FastTab, select the line to edit.
5. On the **Line details** FastTab, on the **Delivery** tab, set the **Mode of delivery** and **Delivery terms** fields as required. Clear one or both fields to use the matching settings on the header.

For intercompany orders, values for the **Mode of delivery** and **Delivery terms** fields on each purchase order line will be synced between the purchase order and its related sales order.

Sync external item IDs and descriptions for intercompany orders

For intercompany orders, the *Make-to-order supply automation* feature enables external item IDs and text descriptions on purchase order lines to be synced with the related intercompany sales order lines, regardless of whether the purchase order is for direct delivery. The feature also adds the ability to specify the final external customer account on an intercompany purchase order. The system will then automatically take external item IDs and text descriptions from the external customer record instead of the (internal) intercompany vendor record.

To set up an intercompany vendor to sync external item IDs and item names between intercompany purchase orders and their related intercompany sales orders, follow these steps.

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Select the intercompany vendor that you want to set up.
3. On the Action Pane, on the **General** tab, in the **Set up** group, select **Intercompany**.
4. On the **Intercompany** page, on the **Purchase order polices** tab, in the **Intercompany purchase order -> Intercompany sales order** section, select the **External item ID and item name** checkbox.

To specify the final external customer account for an intercompany purchase order, follow these steps. The **Customer account** field applies only to intercompany purchase orders. Use it to specify the account number of the customer who will eventually receive the goods. When this field is set, purchase order lines will take external item descriptions and numbers from the specified customer account instead of the intercompany vendor that is specified on the purchase order. If you change the customer account later, external item descriptions and IDs will be updated so that they match the values for the new account. External item descriptions and IDs for each line will also be synced to the matching intercompany sales order.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Open the purchase order that you want to set up, or create a new one.
3. Select the **Header** view.
4. In the **Reference** section, set the **Customer account** field to the relevant external customer account.

To specify external item IDs and text descriptions for a purchase order line for an intercompany order, follow these steps. The values that you set will be synced with the related intercompany sales order lines, regardless of whether the purchase order is for direct delivery.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Open the purchase order that you want to set up, or create a new one.
3. Select the **Lines** view.
4. On the **Purchase order lines** FastTab, select the line to edit.
5. On the **Line details** FastTab, on the **General** tab, in the **Order line** section, in the **Text** field, provide an external description of the item that is specified on the selected order line. This value will be synced with the related intercompany sales order line.
6. In the **Reference** section, in the **External** field, provide an external item ID for the item that is specified on the selected order line. This value will be synced with the related intercompany sales order line.

Learn more in Create and invoice an intercompany sales order for an external customer.

Negative days and dynamic negative days

08/25/2025

This article provides information about negative days and dynamic negative days, and how you can use them to help your business. The *negative days time fence* represents the number of days that you're willing to wait before you order new replenishment when you have negative inventory.

In this article, you'll learn the following information:

- How planned orders are created
- The correlation between the negative days time fence and the item's lead time
- How the dynamic negative days time fence is calculated, and how the item's lead time is factored into the calculation
- How to interpret the suggestions for improving the running time for material requirements planning (MRP) (master planning) that are related to negative days

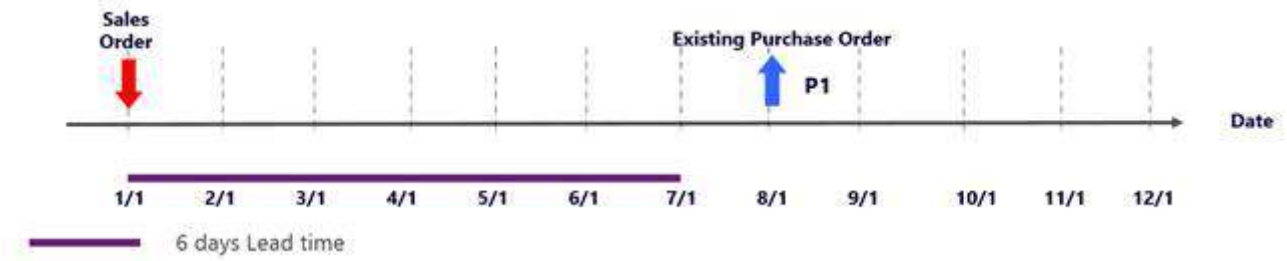
This article uses three hypothetical scenarios to help you understand this information. The difference between the scenarios is the point at which you get demand: before, during, or after the item's lead time period.

Scenario 1: You get demand before the item's lead time period

You might get demand either relatively early in your item's lead time or just before the lead time period begins. Here's an example of this scenario:

- The DemoProduct item has a six-day purchase lead time.
- On day zero (January 1), the inventory level for the DemoProduct item is 0 (zero).
- On day zero (January 1), you get a sales order for a quantity of 10 of the DemoProduct item.
- On day seven (January 8), there's an existing purchase order for a quantity of 10 of the DemoProduct item.

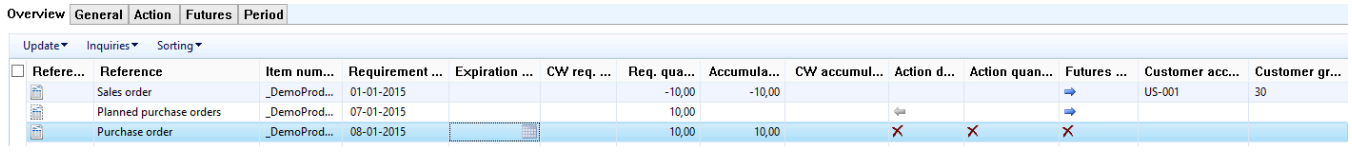
The following illustration shows a graphical view of this scenario.



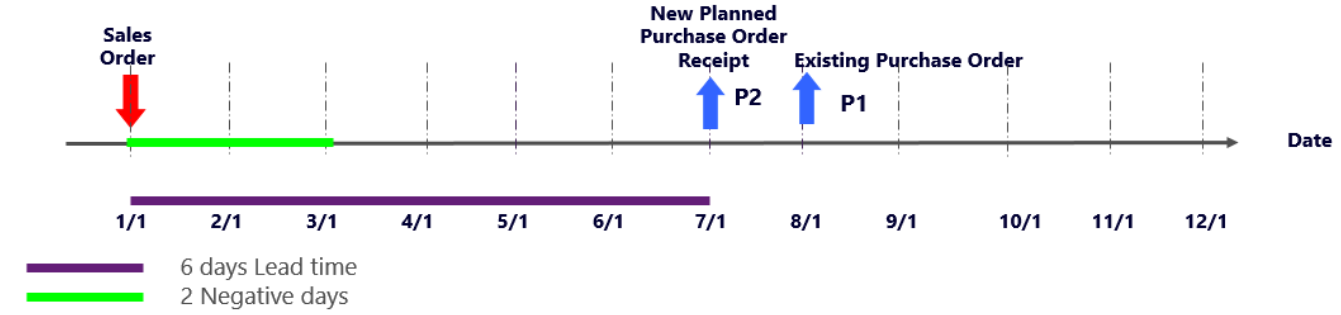
Case 1A: Negative days are less than the item's lead time

If you set the negative days to a number that is less than the item's lead time, MRP looks for receipts for the DemoProduct item inside the negative days time fence. Because it doesn't find any receipts, MRP creates a new planned purchase order. This planned order is immediately delayed by six days (the lead time). Therefore, it will arrive on January 7. The existing purchase order gets a **Cancel** action message, because the creation of the new planned purchase order has made it redundant.

The following illustration shows a screenshot of this case.



The following illustration shows a graphical view of what occurs in this case.



If you consider MRP performance and plan nervousness, this case doesn't perform well. MRP must create a new planned order, and must calculate delays and actions. These tasks are time-consuming. This case also adds two more transactions to your plan. On the other hand, the sales order is delayed by only six days, not seven days.

Case 1B: Negative days are more than the item's lead time

To help improve MRP performance, you can set the negative days to a number that is more than the item's lead time. Because you can't get the supply inside the lead time in this scenario,

you can search for receipts for as long as this search makes sense. Although the running time for MRP will be shorter, you should watch out for additional delays to the orders.

Case 1C: Automatically correlate the item's lead time to the negative days time fence

To automatically correlate the item's lead time to the negative days time fence, use dynamic negative days. To use dynamic negative days, go to **Master planning** > **Setup** > **Master planning parameters**, and then, on the **General** tab, in the **Coverage** section, set the **Use dynamic negative days** option to *Yes*. MRP then looks for receipts inside the dynamic negative days time fence. This time fence is calculated by using the following formula:

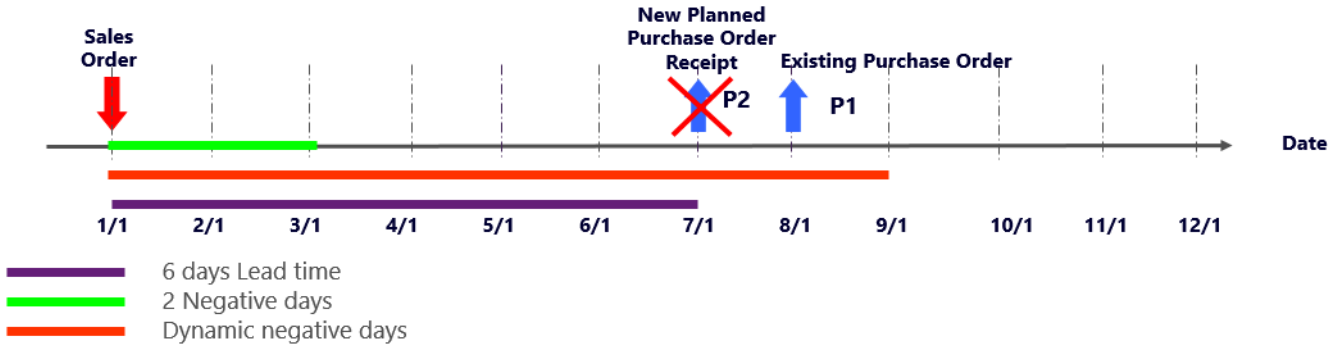
Dynamic negative days time fence = Purchase lead time + Negative days time fence + (Current date – Requirement date)

(If the default order type of the DemoProduct item is *Production* or *Transfer*, the lead time is the *inventory* lead time.)

When dynamic negative days are used, the time fence that MRP looks at for receipts is now 6 + 2 + 0 = 8 days. MRP finds the existing purchase order and pegs the sales order against it. No new planned orders are created. Therefore, the running time for MRP is shorter. The following illustration shows the net requirements for the DemoProduct item.

Reference	Reference	Number	Item number	Requireme...	Expiration date	CW req. qty	Req. quantity	Accumulated	CW accumulated	Action date	Action quantity	Futures date
	Sales order	000787	_DemoProd...	01-Jan-15			-10.00	-10.00				
	Purchase order	000044	_DemoProd...	08-Jan-15			10.00					

The following illustration shows a graphical view of what occurs in this case.

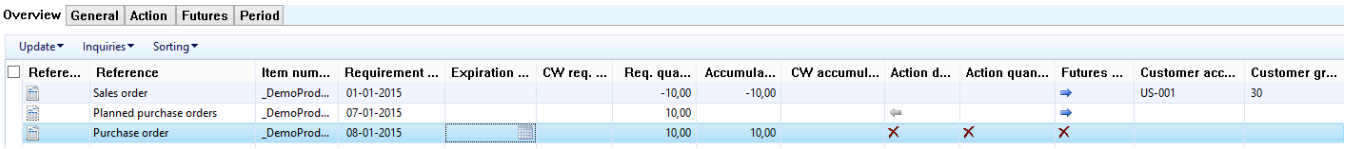


Case 1D: Use only dynamic negative days

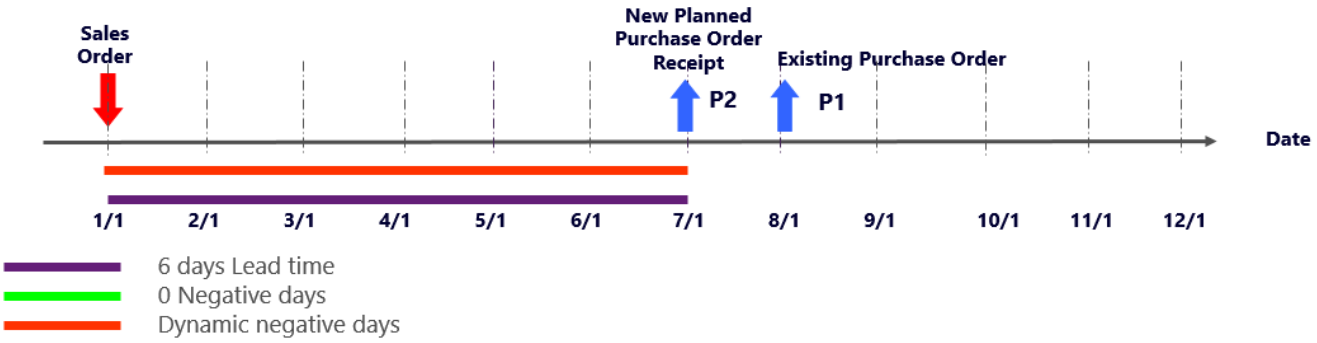
If you set the negative days to *0* (zero) and use only the dynamic negative days time fence, the dynamic negative days time fence is 6 + 0 + 0 = 6 days. In this case, the result is the same as the result in case A for this scenario. MRP must create a new planned order, and must calculate delays and actions. These tasks are time-consuming and can also be frustrating. You also have

two more transactions to process. Because the demand can't be fulfilled on time for the item to arrive, this case adds unnecessary complications to your plan.

The following illustration shows a screenshot for this case.



The following illustration shows a graphical view of what occurs in this case.



Case 1E: Use both negative days that are more than the item's lead time and the dynamic negative days time fence

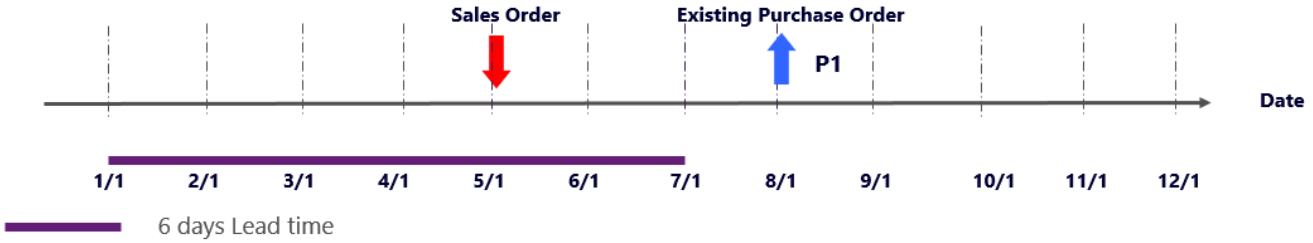
If you set the negative days to a number that is more than the item's lead time, and if you also use the dynamic negative days time fence, the dynamic negative days time fence is 6 + 6 + 0 = 12 days. This approach might produce a very long time fence that MRP must search for results in. For information about how case E is related to a situation where you set the negative days to a long time fence, see the Conclusion section of this article.

Scenario 2: You get demand during the item's lead time period

You might get demand sometime during your item's lead time. Here's an example of this scenario:

- The DemoProduct item has a six-day purchase lead time.
- On day zero (January 1), the inventory level for the DemoProduct item is 0 (zero).
- On day four (January 5), which is inside the item's lead time, you get a sales order for a quantity of 10 of the DemoProduct item.
- On day seven (January 8), there's a purchase order for a quantity of 10 of the DemoProduct item.

The following illustration shows a graphical view of this scenario.



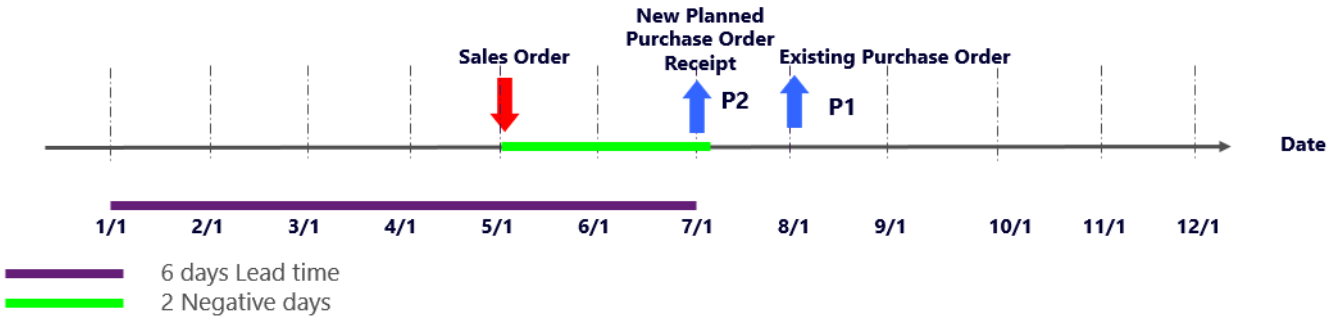
Case 2A: Negative days are less than the item's lead time

If you set the negative days to a number that is less than the item's lead time, MRP looks for receipts for the DemoProduct item inside the negative days time fence. Because it doesn't find any receipts, MRP creates a new planned purchase order that uses the current date as the order date. This planned order is immediately delayed by six days (the lead time). Therefore, it will arrive on January 7. The existing purchase order gets a **Cancel** action message, because the creation of the new planned purchase order has made it redundant.

The following illustration shows a screenshot for this case.

Refere...	Reference	Numb...	Item num...	Requirem...	Expiration ...	CW req. ...	Req. qua...	Accumula...	CW accumul...	Action d...	Action quan...	Futures ...
	Sales order	000787	_DemoProd...	05-Jan-15			-10.00	-10.00				
	Planned purchase orders	013283	_DemoProd...	07-Jan-15			10.00					
	Purchase order	000044	_DemoProd...	08-Jan-15			10.00	10.00				

The following illustration shows a graphical view of what occurs in this case.



Case 2B: Negative days are more than the item's lead time

This case resembles case B for scenario 1. If you set the negative days to a number that is more than the item's lead time, you don't get a new planned order. The sales order is attached to the existing purchase order.

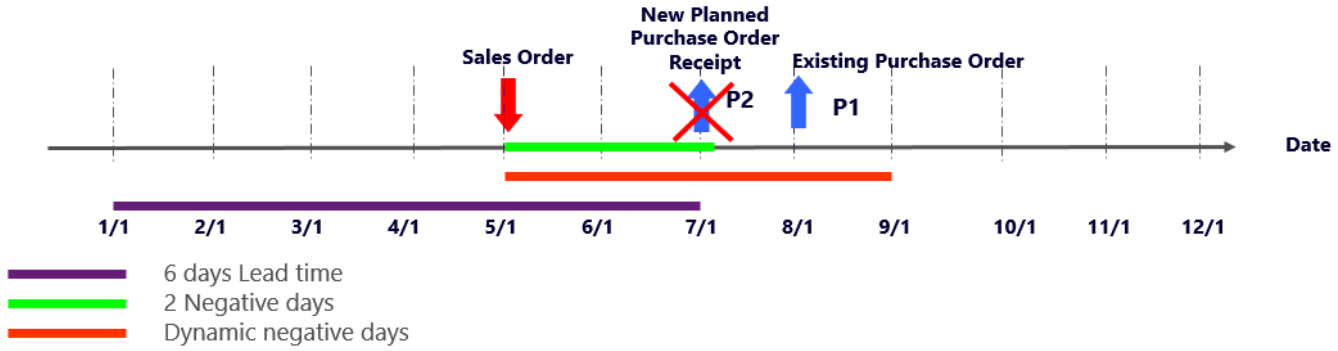
Case 2C: Automatically correlate the item's lead time to the negative days time fence

This case resembles case C for scenario 1, because dynamic negative days work just as well as they do in that case. The dynamic negative days time fence is now 6 + 2 – 4 = 4 days. If you use this approach, MRP finds the existing purchase order and attaches the sales order to it. Because no new planned orders are created, the running time for MRP is shorter.

The following illustration shows a screenshot of this case.

Reference	Reference	Number	Item number	Requirement ...	Expiration date	CW req. qty	Req. quantity	Accumulated	CW accumulated	Action date	Action quantity	Futures date
	Sales order	000787	_DemoProd...	05-Jan-15			-10.00	-10.00				
	Purchase order	000044	_DemoProd...	08-Jan-15			10.00					

The following illustration shows a graphical view of what occurs in this case.



Case 2D: Use only dynamic negative days

If you set the negative days to *0* (zero) and use only the dynamic negative days time fence, the dynamic negative days time fence is now 6 + 0 – 4 = 2 days. In this case, the result is the same as the result in case A for this scenario. For a graphical view of what occurs, see case A for this scenario.

Case 2E: Use both negative days that are more than the item's lead time and the dynamic negative days time fence

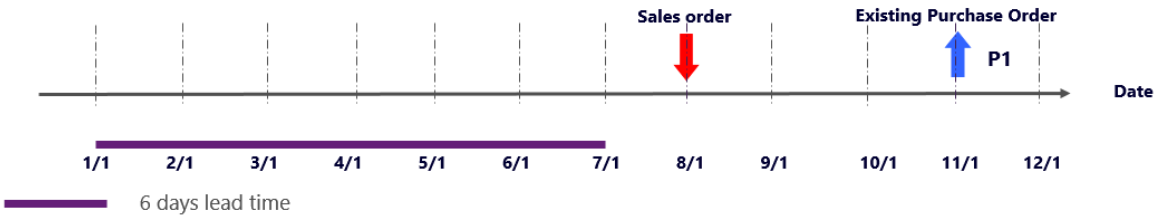
If you set the negative days to a number that is more than the item's lead time, and if you also use the dynamic negative days time fence, the dynamic negative days time fence is 6 + 6 – 4 = 8 days. This approach might produce a very long time fence that MRP must search for results in. For information about how case E is related to a situation where you set the negative days to a long time fence, see the Conclusion section of this article.

Scenario 3: You get demand after the item's lead time period

You might get demand after the item's lead time. Here's an example of this scenario:

- The DemoProduct item has a six-day purchase lead time.
- On day zero (January 1), the inventory for the DemoProduct item is 0 (zero).
- On day seven (January 8), which is outside the item's lead time, you get a sales order for a quantity of 10 of the DemoProduct item.
- On day ten (January 11), there's a purchase order for a quantity of 10 of the DemoProduct item.

The following illustration shows a graphical view of this scenario.



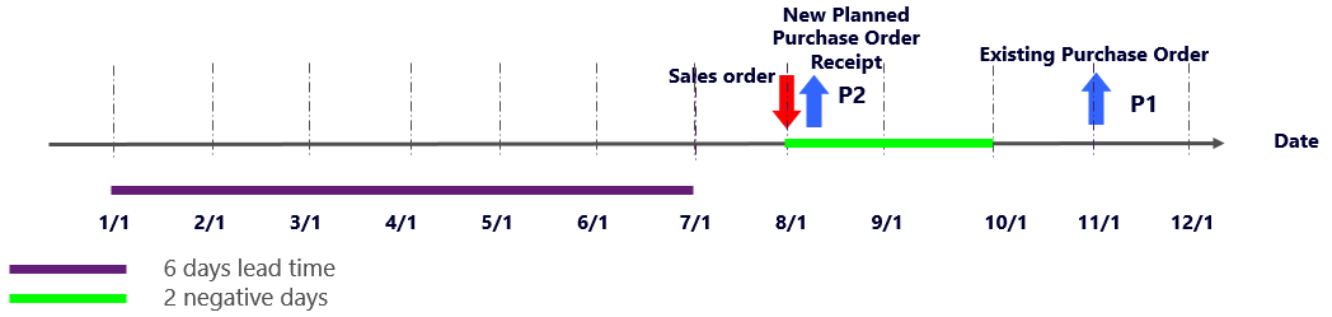
Case 3A: Negative days are less than the item's lead time

If you set the negative days to a number that is less than the item's lead time, MRP looks two days ahead of the sales order's requirement date. Because it doesn't find anything, MRP creates a planned purchase order on January 2. This planned purchase order will be shipped just in time to fulfill the sales order demand. The existing purchase order gets a **Cancel** action message, because it isn't required.

The following illustration shows a screenshot of this case.

Reference	Reference	Number	Item number	Requirement ...	Expiration date	CW req. qty	Req. quantity	Accumulated	CW accumulated	Action date	Action quantity	Futures date
	Planned purchase orders	013695	_DemoProd...	08-Jan-15			10.00	10.00				
	Sales order	000787	_DemoProd...	08-Jan-15			-10.00					
	Purchase order	000044	_DemoProd...	12-Jan-15			10.00	10.00		✕	✕	✕

The following illustration shows a graphical view of what occurs in this case.



> **Note**
>
> In the preceding screenshot, the purchase order requirement date is January 12. Because that screenshot was taken in 2015, when January 11 was a Sunday, MRP moved the

requirement date to the next working day, which was Monday, January 12. Nevertheless, the purchase order has a delivery date of January 11.

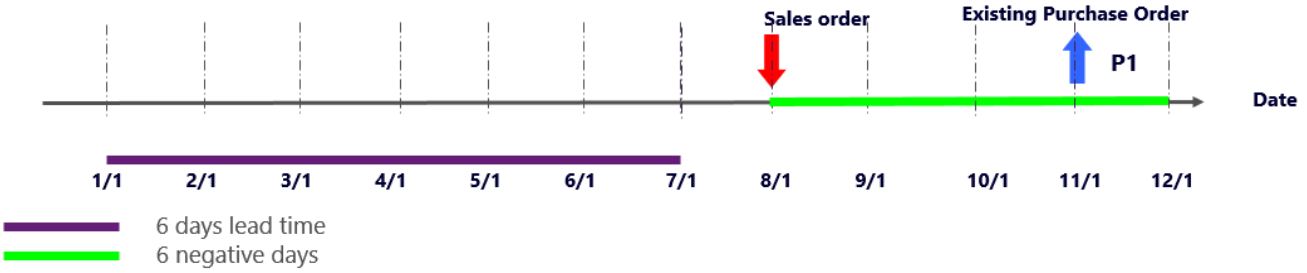
Case 3B: Negative days are more than the item's lead time

If you set the negative days to a number that is more than the item's lead time, you don't get a new planned order. The sales order is pegged against the existing purchase order. Therefore, the sales order is delayed. If you create a planned order, you can ship the sales order on time.

The following illustration shows a screenshot of this case.

Reference	Reference	Number	Item number	Requirement ...	Expiration date	CW req. qty	Req. quantity	Accumulated	CW accumulated	Action date	Action quantity	Futures date
	Sales order	000787	_DemoProd...	08-Jan-15			-10.00	-10.00				
	Purchase order	000044	_DemoProd...	12-Jan-15			10.00					

The following illustration shows a graphical view of what occurs in this case.

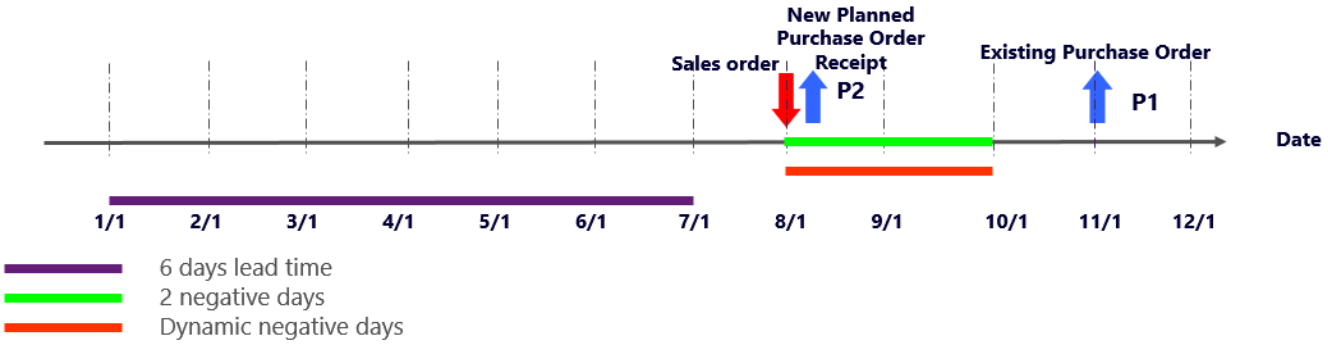


Case 3C: Automatically correlate the item's lead time to the negative days time fence

This case resembles case C for scenario 1, because dynamic negative days work just as well as, if not better than, they work in case B for this scenario.

The dynamic negative days time fence is 6 + 2 – 7 = 1 day. However, in this case, the system still considers the negative days lead time (2), because MRP considers the maximum value between the negative days lead time and the dynamic negative days lead time. Therefore, the result in this case is the same as the result in case A for this scenario.

The following illustration shows a graphical view of what occurs in this case.



Case 3D: Use only dynamic negative days

If you set the negative days to *0* (zero) and use only the dynamic negative days time fence, the dynamic negative days time fence is now 6 + 0 – 7 = -1 day. In this case, the system still considers the negative days lead time (2). Therefore, the result in this case is the same as the result in case A for this scenario and has all the same drawbacks. For a graphical view of what occurs, see case A for scenario 2.

Case 3E: Use both negative days that are more than the item's lead time and the dynamic negative days time fence

This case is the same as case E for scenarios 1 and 2. It has basically the same benefits and drawbacks.

Conclusion

As the three scenarios in this article show, it's a good idea to set the negative days to a number that is more than the lead time of the items in the coverage group. It's also a good idea to use only dynamic negative days, and to set the negative days to the number of days that you're willing to wait before you order new replenishment when you have negative inventory (in other words, the number of days that you're willing to further delay demand). Additionally, items in the same coverage group should have similar lead times.

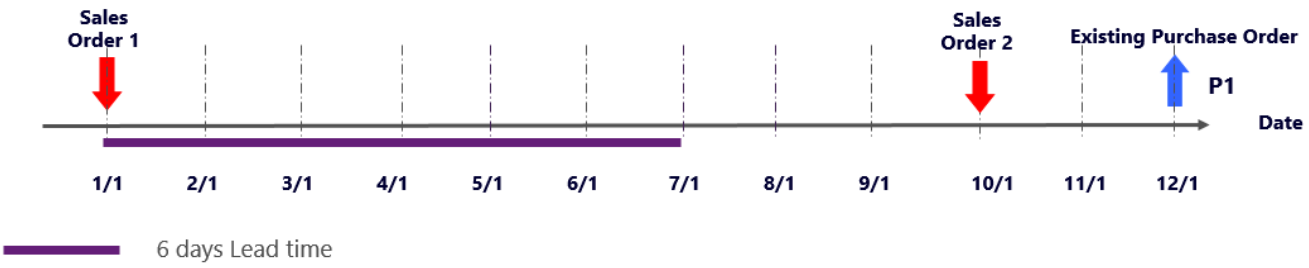
If you set the negative days to *0* (zero) and don't use dynamic negative days, MRP always creates a new planned order to fulfill demand. In this situation, it's important that you work with the action messages to make sure that you don't pile up inventory.

You might want to set the negative days to a long time fence and then work with the action messages. This approach produces good planning results, but it's also a bit slower. It might also be more difficult to analyze, because you must analyze and apply the action messages. Here's an example:

- The DemoProduct item has a six-day purchase lead time.

- On day zero (January 1), the inventory for the DemoProduct item is 0 (zero).
- On day zero (January 1), you get a sales order for a quantity of 10 of the DemoProduct item.
- On day nine (January 10), you get a sales order for a quantity of 10 of the DemoProduct item.
- On day eleven (January 12), there's a purchase order for a quantity of 10 of the DemoProduct item.
- Negative days are set to *20*, which is much more than the item's lead time.

The following illustration shows a graphical view of what occurs.



MRP produces the following results.

Reference	Reference	Number	Item number	Requirement ...	Expiration date	CW req. qty	Req. quantity	Accumulated	CW accumulated	Action date	Action quantity	Futures date
	Sales order	000787	_DemoProd...	01-Jan-15			-10.00	-10.00				
	Planned purchase orders	012459	_DemoProd...	09-Jan-15			10.00					
	Sales order	000788	_DemoProd...	09-Jan-15			-10.00	-10.00				
	Purchase order	000044	_DemoProd...	12-Jan-15			10.00					

In the preceding screenshot, the sales order requirement date is January 9 instead of January 10. Because that screenshot was taken in 2015, when January 10 was a Saturday, the requirement date of the order should be the previous working day, which was Friday, January 9.

MRP creates a planned purchase order to fulfill the demand that is requested by the first sales order, but then it also recommends that you cancel the planned order, because you can advance the existing purchase order and increase the quantity on it.

The results aren't wrong, but the running time for MRP might be longer, because MRP must create all the delays and suggestions. Additionally, the planner might require more time to understand the MRP results. Most importantly, in this case, it's essential that the planner understands and uses the action messages.

If you reduce the negative days to a number that's closer to the item's lead time, and you use dynamic negative days, MRP produces the following results.

Reference	Reference	Number	Item number	Requirement ...	Expiration date	CW req. qty	Req. quantity	Accumulated	CW accumulated	Action date	Action quantity	Futures date
	Sales order	000787	_DemoProd...	01-Jan-15			-10.00	-10.00				
	Planned purchase orders	013066	_DemoProd...	07-Jan-15			10.00					
	Sales order	000788	_DemoProd...	09-Jan-15			-10.00	-10.00				
	Purchase order	000044	_DemoProd...	12-Jan-15			10.00					

MRP creates a planned order that is attached to the first sales order. Then, as is expected, the second sales order is pegged against the existing purchase order, based on the negative days setting. This planning result is also correct, and the running time for MRP might be shorter. In this case, it isn't essential that you understand and know how to work with the action messages.

To help guarantee that the correct values are entered for your business, you must think in terms of both functionality and MRP running time. Therefore, it can take a little trial and error to determine the optimal values.

Related information

For more discussion, see the original More about (dynamic) negative days blog post.

Dynamic positive days for last-minute orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Dynamic positive days help increase your sales order fulfillment rate by using on-hand inventory and planned receipts for last-minute orders. When you use dynamic positive days, master planning applies the following rules to determine whether to create a new planned order or use existing supply:

- If an item is needed outside its lead time, master planning suggests that a new planned order should be created.
- If an item is needed within its lead time, master planning suggests that existing supply should be used.

This strategy enables sales orders that are created later than, but requested earlier than, another sales order to use existing purchase orders. It lets you fulfill all demand and, at the same time, better manage your supply and delivery dates (including when you use capable-to-promise).

When the system determines which supply to peg, it doesn't just consider the lead time. It also considers the positive days. The number of positive days that apply to a given item are set on that item's item coverage group and can be overwritten at the master plan level. This strategy enables some master plans to have a higher number of positive days than others. Therefore, the plans that are used for capable-to-promise can differ from the plans that are used for day-to-day business.

Prerequisites

Before you can use dynamic positive days, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management 10.0.38 or later.
- The feature that's named *Dynamic positive days for Planning Optimization* must be turned on in Feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default.

Set up your system to use dynamic positive days

Follow these steps to set up your system to use dynamic positive days.

1. Go to **Master planning** > **Setup** > **Master planning parameters**.
2. On the **General** tab, set the **Use dynamic positive days** option to *Yes*.
3. On the Action Pane, select **Save**.

> **Note**
>
> The **Use dynamic positive days** setting applies to all legal entities (companies) and master plans in your Supply Chain Management environment.

Assign the number of positive days to an item coverage group

Follow these steps to assign the number of positive days to an item coverage group.

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. Select an existing coverage group, or create a new one.
3. On the **General** FastTab, set the **Positive days** field to the number of positive days that you want to use for the coverage group.

Example scenarios

This section provides some example scenarios to help you understand how dynamic positive days work.

Example scenario 1: Demand within the lead time

The following scenario shows how the system handles demand that's requested for a date that's within the lead time.

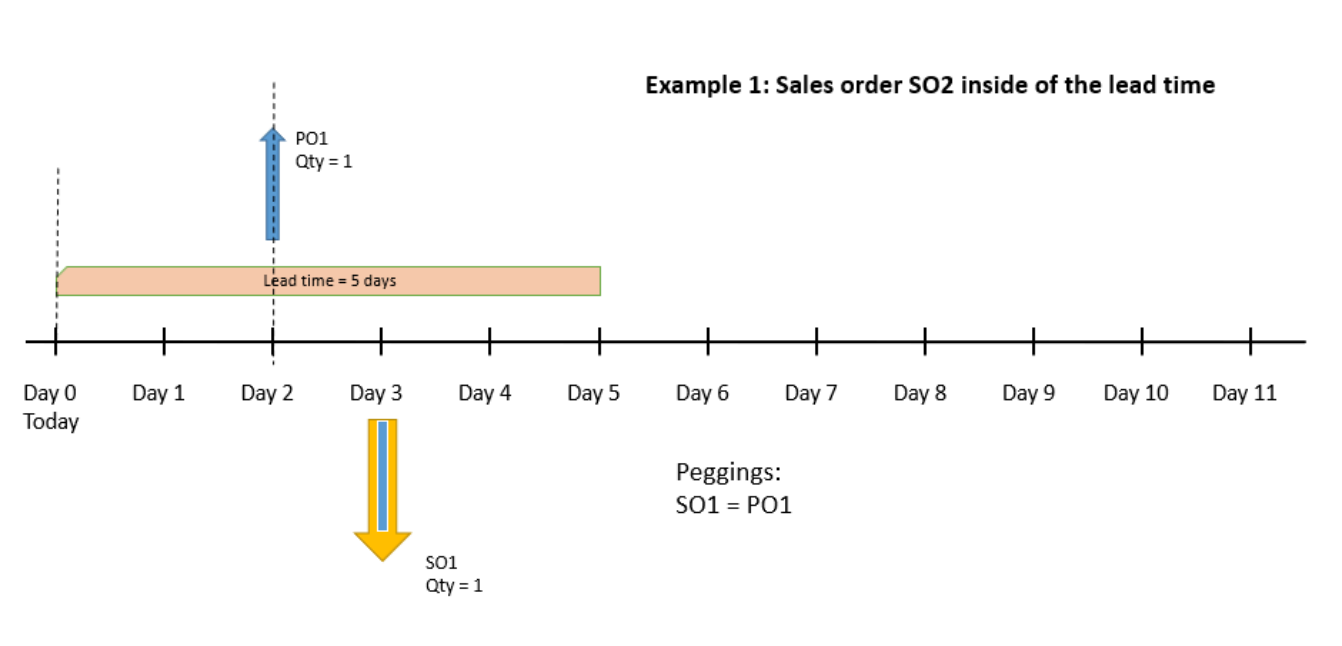
1. Your system is set up in the following way:

    - You have an item that belongs to a coverage group that's set up with three positive days and three negative days.
    - The lead time of the item is five days.
    - Dynamic positive days are enabled.
    - Your master plans don't include any positive day overrides.

- You have a purchase order for the item. It's scheduled to be delivered on day 2.

2. A new sales order is created for the item. The requested ship date is day 3.

3. Because the requested ship date is within the lead time, the system pegs the new sales order to the existing purchase order.



Example scenario 2: Demand outside the lead time

The following scenario shows how the system handles demand that's requested for a date that's outside the lead time.

1. Your system is set up in the following way:

   - You have an item that belongs to a coverage group that's set up with three positive days and three negative days.
   - The lead time of the item is five days.
   - Dynamic positive days are enabled.
   - Your master plans don't include any positive day overrides.
   - You have a purchase order (PO1) for the item. It's scheduled to be delivered on day 2.
   - You have another purchase order (PO2) for the item. It's scheduled to be delivered on day 6.

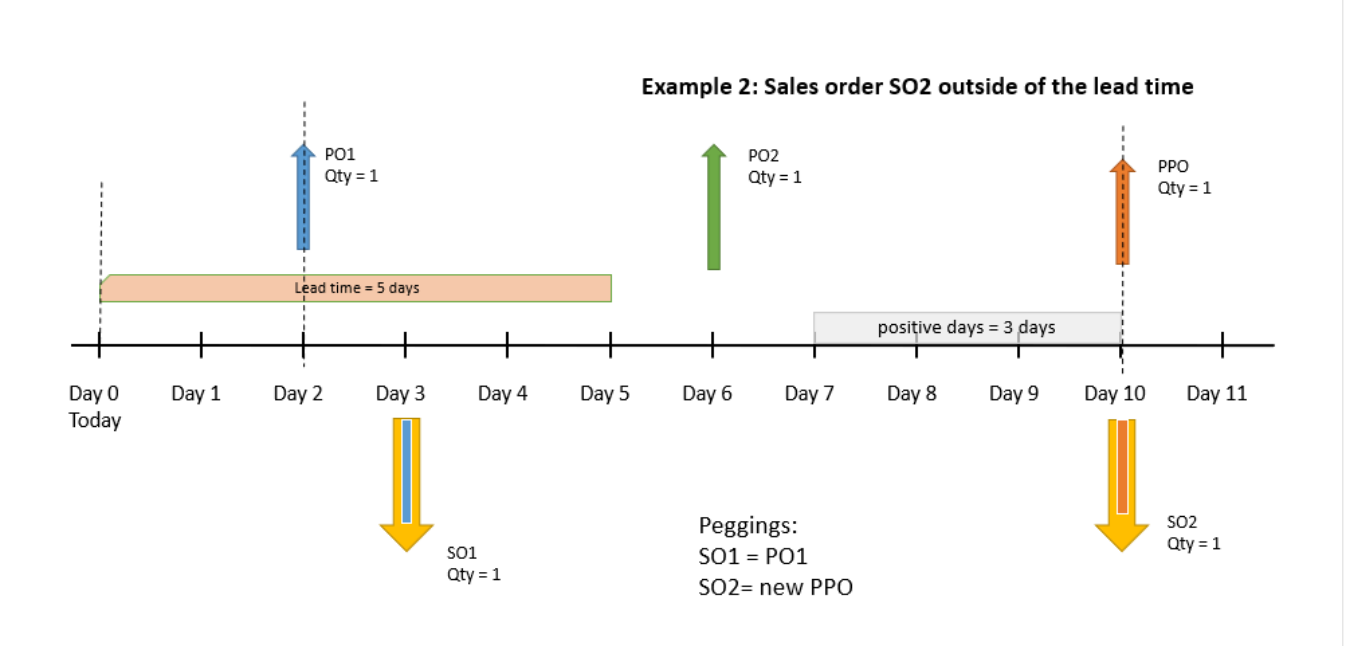
2. The following sales orders are created for the item:

   - Sales order 1 (SO1) has a requested ship date of day 3.
   - Sales order 2 (SO2) has a requested ship date of day 10.

3. The system takes the following actions:

    - Peg SO1 against existing PO1.
    - Create a new planned purchase order, and peg SO2 against it.

    Because SO2 is outside the lead time, the system creates a new planned order to supply it. (There's no supply within the positive days.)



Example scenario 3: Sales order within the positive days but outside the lead time

The following scenario shows how the system handles demand that's requested for a date that's within the positive days but outside the lead time.

1. Your system is set up in the following way:

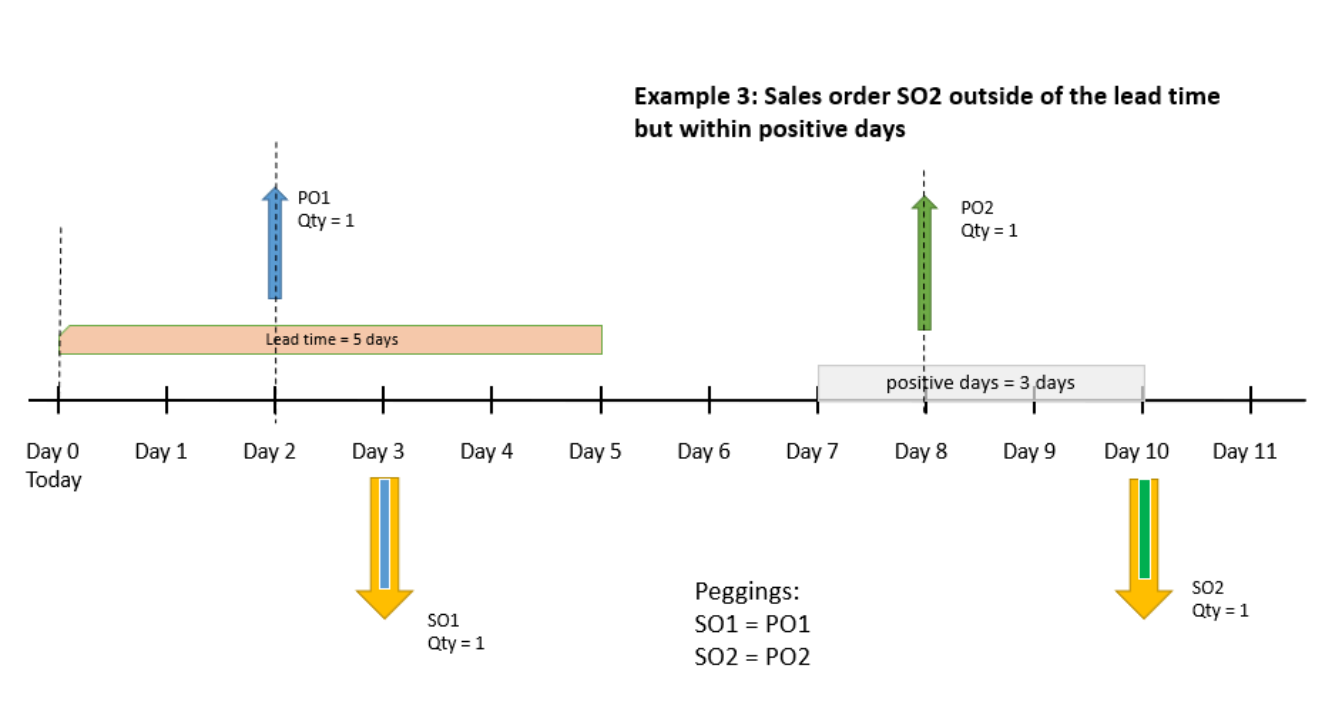
    - You have an item that belongs to a coverage group that's set up with three positive days and three negative days.
    - The lead time of the item is five days.
    - Dynamic positive days are enabled.
    - Your master plans don't include any positive day overrides.
    - You have a purchase order (PO1) for the item. It's scheduled to be delivered on day 2.
    - You have another purchase order (PO2) for the item. It's scheduled to be delivered on day 8. (Therefore, this example differs from the previous example in that PO2 is scheduled for delivery within the range of positive days.)

2. The following sales orders are created for the item:

- Sales order 1 (SO1) has a requested ship date of day 3.
- Sales order 2 (SO2) has a requested ship date of day 10.

3. The system takes the following actions:

- Peg SO1 against existing PO1.
- Peg SO2 against existing PO2.



Last updated on 06/17/2025

Calendars and master planning

This article provides an overview of supply chain calendars and how they affect master planning. It explains the different calendars that the master planning engine uses, including how they affect the shipping and receiving dates in the planned orders. It also provides recommendations for assigning, using, and updating calendars.

Definition of a calendar

To define a calendar for your organization, go to **Organization administration** > **Setup** > **Calendars** > **Calendars**. Use the buttons on the Action Pane to add, remove, or edit calendars as needed.

To manage the details of a selected calendar, select **Working times** on the Action Pane. Each date entry in a calendar can be open or closed, or it can inherit its open or closed status and working times from the base calendar. You specify the value in the **Control** column on the **Working times** page. For each date, set the **Column** field to one of the following values:

- *Open* – Work is performed on the selected day. The calendar is updated according to the working time template.
- *Closed* – Work isn't performed on the selected day.
- *Base calendar* – The specific date inherits its working times and open or closed status from the base calendar. Therefore, when you update the base calendar, you update the selected calendar.

For any closed dates, the **Closed for pickup** checkbox is automatically selected. For open dates, you can manually select the **Closed for pickup** checkbox. By selecting this option, you indicate that work is performed on the date, but shipping isn't performed.

Calendars that affect master planning

Calendar for a coverage group

A coverage group defines a common set of parameters that are used for replenishment of all items that belong to the group.

To add a calendar for a coverage group, go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**. Find the coverage group that you want to assign the calendar to, and then select it in the

Calendar field.

You can assign the coverage group on different pages:

- On the **Released product details** page of an item. To view the coverage group for an item, go to **Product information management** > **Products** > **Released products**, and select the item to open the **Released product details** page. The **Plan** FastTab shows the item coverage group.
- On the **Item coverage** page. On the **Released product details** page, select **Item coverage** to open the **Item coverage** page. The **Overview** tab shows different parameters for replenishment, depending on the site, warehouse, and product dimensions. The coverage group for each item is inherited from the coverage group on the **Released product details** page. However, you can override it by selecting **Use specific settings** or **Override group setting** on the **General** tab.
- On the **Master planning parameters** page. If no coverage group is assigned to an item on the previously mentioned pages, master planning uses the general coverage group that's set in the **General coverage group** field on the **Master planning parameters** page (**Master planning** > **Setup** > **Master planning parameters**).

> ⓘ **Note**
>
> If you don't define a calendar for the coverage group of an item, Planning Optimization uses the calendar of the default coverage group defined on the **Master Planning Parameters** page.

Define the lead time for a purchased item

Specify the purchase lead time for an item (and whether only working days should be considered) on the **Default order settings** page for the product. To open this page, go to **Product information management** > **Products** > **Released products**, and select **Default order settings**.

> ⓘ **Note**
>
> The **Working days** checkbox for purchase lead time indicates working days as counted on the coverage group calendar. For example, a coverage group calendar set to allow delivery only on Tuesdays, with a lead time of 10 days and the **Working days** checkbox selected, indicates that it would take 10 weeks (10 Tuesdays) for the item to be delivered. Therefore, in most cases, don't select **Working days** for purchase order lead times.

Calendars for a vendor

Vendor working days and shipping days

You can set up calendars that define each vendor's standard operating hours, such as the days when the vendor can receive purchase orders, and the days when the vendor can ship goods. The system uses these calendars when it calculates lead times, delivery times, and other dates. Follow these steps to set up vendor calendars.

1. Go to **Organization administration** > **Calendars** > **Calendars**, and either set up or identify the calendars that you want to use for each purpose.
2. Go to **Accounts payable** > **Vendors** > **All vendors**, and select the vendor that you want to assign the calendars to.
3. To set the standard operating hours for the vendor, on the **Purchase order defaults** FastTab, in the **Purchase calendar** field, select a calendar.
4. The days when the vendor can ship orders might differ from the standard operating hours. To set the shipping days, on the **Invoice and delivery** FastTab, in the **Ship calendar** field, select a calendar. For more information about this field and how it's used, see Calculate requested ship dates for purchase orders.

Define lead times from the trade agreements page

You can set up master planning to include all trade agreements for vendors. Trade agreements are fixed prices or discount agreements that you set up for one or more customers or vendors for the sale or purchase of individual or multiple products. To include the trade agreements during planning, go to **Master planning** > **Setup** > **Master planning parameters**, and then, on the **Planned orders** tab, select **Find trade agreements**. Master planning can select the vendor that has either the minimum lead time or the lowest unit price.

Calendar for a warehouse

Assign a calendar to a warehouse to indicate the open dates for receiving and shipping. If you don't assign a calendar to a warehouse, the system assumes that the warehouse is open every day.

> **Note**
>
> Assigning a calendar to a transit warehouse doesn't have any impact. Transit warehouses support transfer orders. The open days within the "from" warehouse and the "to" warehouse,

and the mode of delivery calendar, define the applicable shipping or receipt dates for the orders.

Closed for pickup policy

To indicate that a warehouse is open for receiving but pickup isn't possible, set the **Closed for pickup policy** field in the warehouse calendar. This setting also applies to customer pickups.

Transport calendar

To indicate the dates when transfer orders can be shipped, set up transport calendars for each mode of delivery or for each combination of mode of delivery and "from" warehouse.

To set up a transport calendar, go to **Sales and marketing** > **Setup** > **Distribution** > **Modes of delivery**, select a mode of delivery, and then select **Transport calendar**.

You can create a line for each warehouse and mode of delivery. The **Transport calendar** column specifies the transport calendar that's applied when goods are shipped from the warehouse that uses the mode of delivery.

To apply a transport calendar to all shipments that use a specific mode of delivery, regardless of the warehouse, create a line where you don't specify a warehouse.

If you don't assign a transport calendar, the system assumes that all days are open for transport.

Closed for pickup policy on the transport calendar

By default, Planning Optimization considers the **Closed for pickup** setting only on the warehouse calendar of the issuing warehouse when it schedules transfer order shipments. When the *Consider open for pickup on transport calendar for planned transfer orders with Planning Optimization* feature is enabled, Planning Optimization also considers the **Closed for pickup** setting on the transport calendar. The issue calendar becomes the intersection of both calendars. Therefore, the shipping date must be open for pickup on both the warehouse calendar and the transport calendar.

This feature also changes how Planning Optimization selects the receipt calendar for transfer orders. When the feature is enabled, the warehouse calendar of the receiving warehouse takes precedence over the coverage group calendar, if you specify a warehouse calendar. Without the feature, the coverage group calendar takes precedence. For more information, see Receipt date of a planned transfer order later in this article.

Prerequisites

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.
- The feature named *Consider open for pickup on transport calendar for planned transfer orders with Planning Optimization* must be turned on in feature management.

Example

Consider the following setup:

- Warehouse *WH11* uses a calendar that's open every weekday. Wednesdays are marked as **Closed for pickup**.
- The transport calendar for the *Truck* mode of delivery is open every weekday. Fridays are marked as **Closed for pickup**.
- A planned transfer order ships items from *WH11* by using the *Truck* mode of delivery.

When the feature is disabled, Planning Optimization can schedule the shipment on any weekday except Wednesday, because only the **Closed for pickup** setting on the warehouse calendar is considered.

When the feature is enabled, Planning Optimization can schedule the shipment only on Monday, Tuesday, or Thursday. Both the warehouse calendar (Wednesday is closed for pickup) and the transport calendar (Friday is closed for pickup) must be open for pickup on the shipping date.

Calendar for a customer

Assign a receipt calendar to the customer to indicate the dates when a customer can accept deliveries. If you don't assign a calendar to a customer, the system assumes that the customer can receive orders every day.

This setup affects the receipt date on sales order lines. If the date that you select on the sales order lines isn't open in the customer calendar, the system indicates that the receipt date isn't valid.

You can assign only one calendar to each customer. If a customer has multiple addresses, and you must include a calendar for each address, create one customer per address and then assign the appropriate calendar to each customer.

The customer calendar and the delivery date control method affect the requested receipt date on the sales order lines. Learn more about how the earliest delivery date is calculated in Order Promising.

Shipping calendar for a legal entity

Set up a shipping calendar for a legal entity to indicate the dates when the legal entity can ship goods. Go to **Organization administration** > **Organizations** > **Legal entities**, select the legal entity, and then, on the **Foreign trade and logistics** tab, in the **Shipping calendar** field, add the calendar. The shipping calendar acts as a source of default values for all warehouse calendars in the legal entity.

How calendars affect dates in planning

Order date of a planned purchase order

The order date of a planned purchase order indicates the date when you place the order. It must be an open date in both the vendor calendar and the coverage group calendar. Sometimes, vendors need a margin of a few days between the date when they receive the purchase order and the date when they can ship the goods. The vendor's margin days indicate these dates. However, if the item that you purchase is assigned to a coverage group that has margin days, those margin days override the vendor's margin days.

Delivery date of a planned purchase order

The receipt date of a purchase indicates the date when you expect to receive the goods. It's an open date in the calendar. To determine which days you can receive the purchase orders, the system considers the following calendars, in order from highest to lowest priority:

1. Vendor's calendar
2. Coverage group calendar
3. Warehouse calendar for the receiving warehouse

You can set the coverage group calendar on different pages. The system prioritizes the various page settings in the following order:

1. Item coverage group on the **Item coverage** page
2. Item coverage group on the **Released products details** page
3. Default item coverage group on the **Master planning parameters** page

Shipping date of a planned transfer order

When you create a transfer order between two warehouses, you specify the shipping date and the receipt date in the transfer order header, along with the "from" warehouse and "to" warehouse. The difference between these two dates is the expected transportation time (in days) between the warehouses.

The shipping date of a planned transfer order indicates the date when the goods are shipped from the "from" warehouse. The calendars that are used to specify the available shipping date are listed here in order of priority:

1. Warehouse calendar of the "from" warehouse
2. Coverage group calendar (See the fallback order for this calendar earlier in this article.)

If you set a warehouse calendar, the system sets a shipping date that's open in that calendar. If you don't set a warehouse calendar, the system uses the coverage group calendar.

Receipt date of a planned transfer order

The receipt date for a transfer order indicates the date that the goods are received in the "to" warehouse.

The calendars that specify the receipt date are listed here in order of priority:

1. Coverage group calendar
2. Warehouse calendar of the "to" warehouse

If you set a coverage calendar, the system sets a receipt date that's open in that calendar. If you don't set a coverage group calendar, the system uses the warehouse calendar.

When the *Consider open for pickup on transport calendar for planned transfer orders with Planning Optimization* feature is enabled, this priority order is reversed. The warehouse calendar of the "to" warehouse takes precedence over the coverage group calendar, if you specify a warehouse calendar. If you don't set a warehouse calendar, the system uses the coverage group calendar. For more information, see Closed for pickup policy on the transport calendar earlier in this article.

To determine the shipping and receipt dates for the planned transfer, the system also considers the margins that you define for shipping and receiving.

Using calendars in master planning

Assign calendars

Set the calendars to identify the working days of the company. Best practice is to set a calendar for each element that has different working days. In other words, set all external calendars (customer and vendor) and all internal calendars (warehouse, coverage group, and mode of delivery) that are related to the company.

Recommendation for warehouse calendars

We recommend that you use one calendar for each warehouse, to include the specific changes that affect only that warehouse. For example, two or more warehouses have the same working days but a different pickup policy. In this case, the best practice is to use a separate calendar for each warehouse and its pickup policy. By using this approach, the system can include the best dates for planned purchase, transfer, and production orders.

If you don't set warehouse calendars, the legal entity calendar provides default values for the warehouse calendar.

Recommendation for coverage group calendars

A coverage group calendar overrides receipt dates in master planning. Use coverage group calendars cautiously. They're especially useful in cases where replenishment must happen on specific days of the week.

Update calendars

Although it's important that all relevant calendars are assigned in their appropriate place (vendor, customer, warehouse, mode of delivery, or coverage group), it's equally important to update them to reflect changes. The system defines the production, transfer, purchase, and sales order dates based on the combination of assigned calendars.

Clarify who is responsible for assigning and updating calendars in their appropriate place. In the event of a breakdown or any other unusual change in the working days, update the calendars according to that change. Rerun all tasks that depend on calendars, such as master planning and production scheduling, when you update calendars.

Related information

- Calculate requested ship dates for purchase orders
- Improve scheduling engine performance

- Date and time parameters used by Planning Optimization
- Safety margins

Last updated on 07/23/2026

Calculate requested ship dates for purchase orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Purchase orders in Microsoft Dynamics 365 Supply Chain Management can include both a *requested receipt date*, which specifies when the order should be received at your company's location, and a *requested ship date*, which specifies when the vendor should ship from their location. The system calculates requested ship dates based on the requested receipt date, lead time, the transport days setup, the vendor ship calendar, and other settings. The transport days setup lets you define the number of days that are required to transport items between pairs of addresses. Vendor ship calendars define the days when each vendor can ship.

If any of your vendors don't have specific delivery terms (such as delivery duty paid, delivery at place, or delivery at place unloaded), you must tell them when they should ship.

When master planning suggests a date for placing a planned order, it also considers the transport days setup. This approach helps ensure that the goods arrive on time.

Prerequisites

To calculate requested shipment dates for purchase orders, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management version 10.0.40 or later.
- You must be using Planning Optimization, not the deprecated master planning engine.

Enable supplier requested and confirmed shipment dates for a company

To enable supplier requested and confirmed shipment dates for a legal entity (company), follow these steps:

1. Use the company picker to select the legal entity where you want to calculate requested shipment dates for purchase orders.
2. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
3. Open the **Delivery** tab.
4. Set **Supplier requested and confirmed shipment dates** to *Yes* to make the feature available to the current legal entity. Set to *No* to hide the feature for the current legal entity.
5. Repeat this procedure for each legal entity where you want to turn on (or turn off) the option.

> ⓘ **Note**
>
> If you use intercompany trade, then each of the legal entities involved must have this option turned on to allow requested shipment dates to be calculated for intercompany purchase orders.

Specify whether requested ship dates can be in the past

Sometimes, because of your requested receipt date, plus the lead time, transport days, vendor ship calendar, and other settings, the system might calculate a requested ship date that's in the past. To specify how the system handles this situation, follow these steps:

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Delivery** tab, in the **Requested ship date in the past** field, select one of the following values:
    - *None* – Always allow requested ship dates to be in the past, without showing a warning.
    - *Allow with warning* – Allow requested ship dates to be in the past, but show a warning message to users when this situation occurs.
    - *Disallow with warning* – Always find the next earliest ship date that's after the current date, and show a warning message to users when this situation occurs.

Set up purchase transport days

To define the number of days that are required to transfer goods between addresses, follow these steps:

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.

2. Follow one of these steps:

    - To use an existing record, use the filters at the top of the grid to find it. To edit the record that you find, select the link in the **Shipping country/region** column. To delete the record, select it, and then select **Delete** on the Action Pane.
    - To create a new record, on the Action Pane, select **New**.

3. On the details page for the new or selected record, on the **General** tab, use the fields to define the shipping address, the receiving address, and the number of days that are required to transport goods between them. For more information about each field, hover over its label to open a tooltip.

    > **Tip**
    >
    > - When you must enter an address, first enter a value in the **Shipping ZIP/postal code** field. Often, the system can automatically fill in the rest of the address fields for you, based on the postal code.
    > - To calculate the ship date, the system uses the most specific purchase transport days record that it can find. For example, you have two purchase transport days records: one that specifies 30 days from Spain to the United States and another that specifies 20 days from Madrid to New York. In this case, a purchase order that's shipped from Madrid to New York uses 20 days, whereas an order from Valencia to New York uses 30 days.
    > - If the shipping address and receiving address have equal priority, the shipping address takes precedence.

4. On the **Transport days per mode of delivery** FastTab, you can set specific transport days for any number of delivery modes (such as road, rail, and air). Use the buttons on the toolbar to add and remove rows in the grid as you require.

    - For each row, select a mode of delivery, and then specify the number of transport days for that mode.
    - If you want one of the modes of delivery to define the default transport days that apply to any modes that aren't specified in the grid, select the **Default** checkbox for

that mode's row.
    - If the **Default** checkbox isn't selected for any row, the **Transport in days** value that's specified on the **General** FastTab is used as the default travel time.

5. On the Action Pane, select **Save**.

Vendor settings for calculating requested ship dates

To calculate requested ship dates, you must set up an address for each vendor. To help make the calculation more accurate, you can also set up vendor ship calendars that define specific days when each vendor can ship.

Set up vendor addresses

To set up addresses for a vendor, follow these steps:

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Open the vendor record that you want to work with.
3. On the **Addresses** FastTab, add and remove addresses as you require. If more than one address is specified on the **Addresses** FastTab, when the system uses the purchase transport days setup to calculate the shipping date, it uses the address where the **Primary** option is set to *Yes* as the shipping point.

> **Tip**
>
> - The vendor address is applied to the purchase order and used as the shipping point for purchase transport days.
> - The receiving point for purchase transport days is the second factor that's used to calculate the shipping date. The delivery address depends on the setup, and a default value can be taken from the company, site, or warehouse. Any update to the site or warehouse for a purchase order header or line triggers a recalculation of the shipping date, because the new site or warehouse might have a different delivery address.

How calendars affect ship and receipt date calculations

The following calendars are relevant for ship and receipt date calculations:

- **Purchase calendar** – This calendar defines when a vendor is open. For example, a vendor might accept orders on workdays but not weekends.

- **Vendor ship calendar** – This calendar defines when purchase orders can be shipped from the vendor.
- **Warehouse calendar** – This calendar defines when purchase orders can be received at the site or warehouse.

The purchase and vendor ship calendars are considered when the ship date is calculated. The warehouse (item coverage group) calendar is used when the receipt date is calculated.

> **Note**
>
> If no coverage group is assigned to an item on the **Released product details**, **Item coverage**, or **Master planning parameters** page, then master planning uses the *general coverage group* (which is set in the **General coverage group** field on the **Master planning parameters** page) and calculates the working days based on the calendar assigned to that group.

Learn more in Calendars and master planning.

The following examples show how calendars affect the calculation of requested ship and receipt dates for purchase orders.

Calculation example 1

This example shows how requested ship and receipt dates are calculated for a product that has five days of lead time and two purchase transport days.

The following table shows the calendars that apply.

Expand table

Calendar	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Purchase calendar	Open	Open	Open	Open	Open	Open	Open	Open	Open
Vendor ship calendar	Open	Closed	Closed	Closed	Open	Open	Open	Open	Open
Warehouse calendar	Open	Open	Open	Open	Closed	Closed	Closed	Closed	Open

The following table shows the calculated dates.

Expand table

Date type	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Order date	X								
Ship date						X			
Receipt date									X

The calculation results are based on the following information:

- The ship date is Saturday, which is the first open day that's available in the vendor ship calendar after the five days of lead time.
- The receipt date is Tuesday, which is the first open day that's available in the warehouse calendar after the two transport days.

Calculation example 2

This example shows how requested ship and receipt dates are calculated for a product that has three days of lead time and two purchase transport days.

The following table shows the calendars that apply.

Expand table

Calendar	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Purchase calendar	Open	Open	Open	Open	Open	Open	Open	Open	Open
Vendor ship calendar	Open	Closed	Closed	Closed	Open	Open	Open	Open	Open
Warehouse calendar	Open	Open	Open	Open	Closed	Closed	Closed	Closed	Open

The following table shows the calculated dates.

Expand table

Calendar	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Order date	X								
Ship date					X				
Receipt date									X

Compared to the previous example, the results are calculated in the following way:

- The ship date is moved one day, because the vendor ship calendar shows that the vendor is closed on Thursday.
- The receipt date is moved two days, because the warehouse calendar shows that the warehouse is closed on Sunday and Monday.

Assign vendor ship calendars

For each vendor, you can specify a ship calendar that indicates the dates when the vendor can ship. For example, some vendors might be able to ship during all opening hours, whereas others can ship only on Mondays. You can define as many specific ship calendars as you require and then apply them to each vendor as appropriate.

If you have goods that must be expedited, and the relevant vendor makes an exception on their vendor ship calendar, you can change or remove the vendor ship calendar for the relevant purchase order lines.

For information about how to set up a ship calendar and define the available ship dates and times for it, see Calendars and master planning.

To assign a ship calendar to a vendor, follow these steps:

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Open the vendor record that you want to assign a ship calendar to.
3. On the **Invoice and delivery** FastTab, in the **Ship calendar** field, select the ship calendar that applies to the vendor.

Date calculations and recalculations

This section describes how and when the system calculates and recalculates requested ship dates.

Requested ship date

The requested ship date calculation considers lead time (lead time per vendor if it's set up on trade agreements), the mode of delivery, and the vendor ship calendar. The lead time is the time that the vendor takes to produce the item. It doesn't include transport days (from vendor to company).

You can configure how the system handles requested ship dates that are in the past. Learn more in the Specify whether requested ship dates can be in the past section.

Order date of planned purchase orders

When a master planning run indicates that a planned order is needed, the order date is calculated backward from the requirement date. The calculation considers margins, lead time, and the purchase transport days setup. The requested ship date is calculated according to the same rules that are used for purchase orders. However, for planned orders, the requested ship date isn't saved as a field value or shown on the **Planned purchase order** details page.

The requested ship date is also calculated for manually created planned purchase orders.

Recalculations for updated orders

When you edit a purchase order where the requested and/or confirmed ship date has already been calculated, the system does the following recalculations:

- If the **Requested ship date** value is changed, the **Requested receipt date** value is recalculated.
- If the **Requested receipt date** value is changed, the **Requested ship date** value is recalculated.
- If the **Confirmed ship date** value is changed, the **Confirmed receipt date** value is recalculated.
- If the **Confirmed receipt date** value is changed, the **Confirmed ship date** value is recalculated.
- If any of the following information is changed, the relevant ship and receipt dates are recalculated:
  - Address on the header
  - Mode of delivery (on the header or on a line)
  - Warehouse on a line

If you change the **Requested ship date** or **Confirmed ship date** value on the header of a purchase order, the system asks you whether the relevant ship and receipt dates should also be updated for each line.

> ⓘ **Note**
>
> If any subsequent changes are made at the line level (such as modifying the unit, warehouse quantity, or site), the line **Requested ship date** and **Requested receipt date** are recalculated based on the current date rather than the header date. To ensure accurate date calculations,

we recommend that you complete all necessary updates at the line level before updating the header **Requested ship date** or **Confirmed ship date**.

Purchase order ship and receipt date calculation logic

This section explains how the **Requested ship date** and **Requested receipt date** are calculated for purchase orders. These dates are influenced by lead times, transport days, vendor calendars, and other settings.

Key terms and concepts

The following table describes key terms and concepts related to purchase order ship and receipt date calculations.

Expand table

Term	Description
Requested ship date	The date on which you want the vendor to ship the goods.
Requested receipt date	The date on which you want to receive the goods.
Confirmed ship date	The date on which the vendor ships the goods, as confirmed by the vendor.
Confirmed receipt date	The date on which you receive the goods, as confirmed by the vendor.
Lead time	The number of days the vendor requires to prepare an item.
Transport days	The number of days needed to ship goods from vendor to warehouse.
Purchase calendar	The calendar that defines when a vendor is open for business.
Vendor ship calendar	The calendar that defines when purchase orders can be shipped from the vendor.
Warehouse calendar	The calendar that defines when purchase orders can be received at the site or warehouse
Item coverage calendar	An optional calendar used to adjust receipt dates.

Calculation logic

The following table summarizes the logic used to calculate purchase order ship and receipt dates.

Expand table

Calculation	Description
Requested ship date	Calculated as today's date plus lead time. Adjusted to the next working day if the assigned calendar is *Active*.
Requested receipt date	Calculated as the requested ship date plus transport days. Adjusted to the next working day if the assigned calendar is *Active*.
Lead time (on the **Line details** tab)	Populated only if **Calculate PO delivery date** is set to *Yes* and the purchase order originates from a purchase requisition, purchase agreement, or request for quotation.
Transport days	Applied based either on the selected mode of delivery at the purchase order line level or the default mode of delivery configured for the warehouse master.
Confirmed dates	Manually entered by the vendor and not adjusted by any calendar.
Trade agreement working days	When enabled, lead time is calculated using only open working days.

For details about how to update the **Requested ship date** and **Requested receipt date** in the header, see Recalculations for updated orders.

Example date calculation scenarios

This section provides examples that show how requested ship and receipt dates for purchase orders are calculated in various scenarios.

Example scenario prerequisites

These scenarios have the following prerequisites:

- Your system must meet all the feature prerequisites that are listed in the Prerequisites section of this article.
- The **Purchase transport days** page must be configured with at least one receiving address, shipping address, and number of transport days. The addresses must match the vendors and receiving warehouses that you use for these scenarios.
- Unless otherwise specified, all calendars must have all days open.
- On the **Procurement and sourcing parameters** page, on the **Delivery** tab, the **Requested ship date in the past** field must be set to *None*.

Example scenario 1: Standard purchase order

In this scenario, there are two purchase transport days from the vendor shipping address to the site/warehouse receiving address. Dates are calculated forward, and the purchase transport days are added to the requested ship date to determine the requested receipt date.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
4. On the Action Pane, select **New** to create a purchase order.
5. On the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase order.
6. On the **Purchase order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
7. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:

    - **Requested ship date** – The value is the current date (unless you specified another date for the order header).
    - **Requested receipt date** – The value is a date that's two days later than the **Requested ship date** value for the purchase order. The system calculates this date based on your setup on the **Purchase transport days** page. The following settings can also affect the calculation:
        - If the product that's ordered has lead time, the system adds that lead time to the transport time when it calculates the receipt date. (Therefore, the formula is *Requested ship date = Requested ship date + Product lead time*.)
        - The vendor ship calendar and warehouse calendar can also affect the date calculation by allowing for closing times at one or both locations. (Learn more in the How calendars affect ship and receipt date calculations section.)

    Any update to the **Requested ship date** or **Requested receipt date** value causes the system to recalculate the dates. An update to the **Requested ship date** value triggers a forward

calculation of the **Requested receipt date** value, and an update to the **Requested receipt date** value triggers a backward calculation of the **Requested ship date** value.

Example scenario 2: Purchase order with dates in the past

This scenario shows how the system handles requested ship dates that are in the past, based on the **Requested ship date in the past** setting. (Learn more in the Specify whether requested ship dates can be in the past section.)

The default value of the **Requested ship date in the past** field is *None*. In this case, requested ship dates in the past are always allowed, and no warning is shown. This scenario shows what happens if you change the value to *Disallow with warning*. In this case, a warning message is always shown if requested ship dates are in the past. The system then finds the next earliest ship date that's after the current date.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Delivery** tab, set the **Requested ship date in the past** field to *Disallow with warning*. Then select **Save**.
3. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
4. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
5. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
6. On the Action Pane, select **New** to create a purchase order.
7. On the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase order.
8. On the **Purchase order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
9. On the **Line details** FastTab, on the **Delivery** tab, set the **Requested receipt date** field to a date that doesn't allow for enough time for transport. (For example, select today's date.)

The system shows a warning message such as "The requested ship date 1/18/2024 is not valid because it is before today. Earliest possible requested ship date is 1/22/2024 with requested receipt date 1/26/2024." The requested date fields are updated to match the warning message.

If you try to directly enter a requested ship or receipt date that's in the past, you receive a warning that doesn't include any suggested dates (for example, "The requested ship date 1/8/2024 is not valid because it is before today"). In this case, you can't save the order until you select a valid date.

Example scenario 3: Purchase order created from a sales order

In this scenario, you create a purchase order from a related sales order and observe how the requested ship and receipt dates are calculated for both orders.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
4. On the Action Pane, select **New** to create a sales order.
5. On the **Customer** FastTab, set the **Customer account** field to any customer account (it doesn't matter which one). Then select **OK** to create and open the sales order.
6. On the **Sales order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
7. On the **Line details** FastTab, on the **Delivery** tab, set the **Delivery date control** field to *None*.
8. Set both the **Requested ship date** field and the **Requested receipt date** field to a date that's one week from now.
9. On the Action Pane, on the **Sales order** tab, in the **New** group, select **Purchase order**.
10. On the **Lines** FastTab, you see a line that matches the line from your sales order. For this line, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page.

11. Select the **Include** checkbox for the line.
12. Select **OK** to create the purchase order and return to your sales order.
13. On the Action Pane, on the **General** tab, in the **Related information** group, select **Purchase order**.
14. The related purchase order opens. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value matches the sales order's requested ship date.
    - **Requested ship date** – The value is calculated backward from the receipt date, based on the number of purchase transport days, to the first available open date.

Example scenario 4: Direct delivery created from a sales order

This scenario is similar to the previous one, but you create a direct delivery order from a sales order. Because you're using direct delivery for this scenario, the shipment never arrives at your warehouse. Therefore, transport time is calculated based only on the *vendor's shipping address* and the *customer's receiving address*.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the receiving address of the *customer* that you plan to ship to. Set the **Transport days** field to *2* for this record.
3. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
4. On the Action Pane, select **New** to create a sales order.
5. On the **Customer** FastTab, set the **Customer account** field to the customer whose receiving address you set up on the **Purchase transport days** page. Then select **OK** to create and open the sales order.
6. On the **Sales order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields if they're required. However, these fields don't affect the delivery time calculation, because you're using direct delivery.
7. On the **Line details** FastTab, on the **Delivery** tab, set the **Delivery date control** field to *None*.
8. Set both the **Requested ship date** field and the **Requested receipt date** field to a date that's one week from now.

9. On the Action Pane, on the **Sales order** tab, in the **New** group, select **Direct delivery**.
10. On the **Lines** FastTab, you see a line that matches the line from your sales order. For this line, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page.
11. Select the **Include** checkbox for the line.
12. Select **OK** to create the purchase order and return to your sales order.
13. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value is the requested ship date that you entered earlier.
    - **Requested ship date** – The value is calculated backward from the receipt date, based on the number of purchase transport days, to the first available open date.
14. On the Action Pane, on the **General** tab, in the **Related information** group, select **Purchase order**.
15. The related purchase order is opened. On the **Line details** FastTab, on the **Delivery** tab, notice that the **Requested receipt date** and **Requested ship date** values match the values of the sales order. These dates are synced because this order is a direct delivery order.

Example scenario 5: Intercompany order

For intercompany orders, the requested ship and receipt dates must be synced for both sales orders and purchase orders. The purchase transport days, lead time, delivery date control, and calendar availability must also be considered.

To complete this scenario, first set up an intercompany relationship between the customer and vendor that you use.

1. Use the company picker to select the company that's purchasing an item. This company is the one where you must set up the purchase transport days and create the purchase order.
2. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
3. Make sure that you have a record for delivery from the vendor warehouse to the customer warehouse that you're using for this scenario. Set the **Transport days** field to *2* for this record.
4. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.

5. On the Action Pane, select **New** to create a purchase order.

6. On the **Vendor** FastTab, set the **Vendor account** field to the intercompany vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase order.

7. On the **Purchase order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.

8. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:

    - **Requested ship date** – The value is the current date (unless you specified another date for the order header).
    - **Requested receipt date** – The value is a date that's five days later than the **Requested ship date** value of the purchase order. The system calculated this date based on your purchase transport days setup.

9. On the Action Pane, on the **Manage** tab, in the **Intercompany tracing** group, select **Intercompany sales order**. The requested dates on the sales order line should match the dates that you saw on the purchase order line.

> **Tip**
>
> The sourcing order determines whether and how the purchase order transport days are calculated.
>
> - If dates are updated on the sales order, lead time and customer transport days take precedence, and purchase order transport days are calculated. These dates are editable only when you source from the sales order.
> - If dates are updated on the purchase order, purchase transport days and lead time are calculated, and requested and confirmed dates are synced back to the sales order.

Example scenario 6: Purchase agreement

This scenario shows how requested ship and receipt dates are calculated when you create a purchase order from a purchase agreement.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.

2. Make sure that you have a record for delivery from the vendor shipping address that you plan to create a purchase agreement for to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Purchase agreements** > **Purchase agreements**.
4. On the Action Pane, select **New** to create a purchase agreement.
5. On the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase agreement.
6. On the **Purchase agreement lines** FastTab, select **Add line**.
7. In the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
8. On the Action Pane, on the **Purchase agreement** tab, in the **Generate** group, select **Confirmation**.
9. In the **Confirm purchase agreement** dialog box, select **OK**.
10. On the Action Pane, on the **Purchase agreement** tab, in the **New** group, select **Release order**.
11. In the **Create release order** dialog box, in the **Purchase quantity** field for the line, enter a value. Then set the **Requested receipt date** field to one week from today.
12. Select **Create** to create the purchase order.
13. Make a note of the purchase order number that's shown in the message at the top of the page.
14. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
15. Open the purchase order that you just created. Use the purchase order number that you made a note of to find it.
16. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value is the date that you entered when you released the purchase order.

- **Requested ship date** – The value is calculated backward from the receipt date, based on the number of purchase transport days, to the first available open date.

Example scenario 7: Purchase requisition

This scenario shows how requested ship and receipt dates are calculated when you create a purchase order from a purchase requisition.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to create a purchase requisition for to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Purchase requisitions** > **All purchase requisitions**.
4. On the Action Pane, select **New** to create a purchase requisition.
5. In the **Name** field, enter a name for the requisition. Then select **OK** to create and open the purchase requisition.
6. On the **Purchase requisition lines** FastTab, on the toolbar, select **Add line**.
7. In the **Item number** field, enter a value. Then, in the **Quantity** field, specify a quantity.
8. On the **Line details** FastTab, on the **Inventory dimensions** tab, set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
9. On the Action Pane, select **Workflow** to open a dropdown dialog box. Select **Submit** to open a dialog box where you can enter a comment. Then select **Submit** in this dialog box to submit the requisition for approval.
10. Generate a purchase order for the purchase requisition through the **Submit requisition for approval** workflow. The resulting purchase order should have the following requested dates:

    - **Requested receipt date** – The value is the **Requested date** value from the purchase requisition.
    - **Requested ship date** – The value is calculated backward two purchase transport days from the requested receipt date of the purchase order.

Example scenario 8: Purchase request for quotation

This scenario shows how requested ship and receipt dates are calculated when you create a purchase order from a request for quotation (RFQ) reply.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to create a purchase requisition for to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
4. On the **Request for quotation** tab, set the **Purchaser can edit vendors bid** option to *Yes*.
5. Go to **Procurement and sourcing** > **Requests for quotations** > **All requests for quotations**.
6. On the Action Pane, select **New** to create an RFQ.
7. Set the **Requested receipt date** field to one week from today. Then enter values in any other required fields, and select **OK** to create the RFQ.
8. On the **Request for quotation lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
9. On the **Header** tab, on the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page.
10. On the Action Pane, select **Save**.
11. On the Action Pane, on the **Quotation** tab, in the **Process** group, select **Send**.
12. In the **Sending request for quotation** dialog box, select **OK**.
13. On the Action Pane, on the **Quotation** tab, in the **Replies** group, select **Manage replies**.
14. On the Action Pane, select **Edit** > **Edit RFQ reply**.
15. On the Action Pane, select **Submit**.
16. In the **Submit** dialog box, select **OK**.
17. On the Action Pane, on the **Reply** tab, in the **Process** group, select **Accept**.

18. In the **Accepting request for quotation** dialog box, select **OK** to create the purchase order.
19. On the Action Pane, on the **General** tab, in the **Other information** group, select **Purchase order**.
20. Open the new purchase order.
21. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value matches the **Requested receipt date** value from the RFQ.
    - **Requested ship date** – The value is calculated backward from the requested receipt date, based on the purchase transport days setup.

Related information

- Calendars and master planning
- Improve scheduling engine performance
- Date and time parameters used by Planning Optimization
- Safety margins

Last updated on 07/01/2026

Create a schedule for a site

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to schedule production orders that are not yet started for a site. The demo data company USMF is used to complete this procedure.

Identify production orders that are not started

1. Go to Production control > Production orders > All production orders.
2. Use the Quick Filter to find records. For example, filter on the Site field with a value of '1'.
    - 1 represents a site in USMF. If you are not using USMF, select a site from your own company.
3. Open the Status column filter.
4. Apply a filter on the "Status" field, with a value of "Scheduled", using the "is exactly" filter operator.

Create a schedule

1. In the list, mark or unmark all rows.
2. On the Action Pane, click Schedule.
3. Click Schedule jobs.
4. In the Scheduling direction field, select 'Backward from delivery date'.
5. Select No in the Finite capacity field.
6. Select No in the Finite material field.
7. Click OK.
    - This may take a while.

View the result of scheduled production orders

1. In the list, mark the selected row.

- You can mark any row.

2. On the Action Pane, click Production order.
3. Click All jobs.

    - On this page, you can see the list of jobs. On the Scheduling tab, you can see the Start date and End date for a job.

4. Click Materials.

    - On this page, you can see the estimated material consumption for the operations on the production order and the current available inventory.

Last updated on 06/17/2025

Master planning with demand forecasts

Use a demand forecast to account for expected demand in your master planning. You can manually create a demand forecast, import it, or generate it by using the demand forecasting functionality in Microsoft Dynamics 365 Supply Chain Management. For more information about demand forecasting, see Demand forecasting overview.

> **Note**
>
> Planning Optimization doesn't support separate forecast planning. Therefore, the **Current forecast plan** setting on the **Master planning parameters** page has no effect when you use Planning Optimization.

Set up a master plan to include a demand forecast

To configure a master plan so that it includes a demand forecast, follow these steps:

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select an existing plan, or create a new plan.
3. On the **General** FastTab, set the following fields:
    - **Forecast model** – Select the forecast model to apply. This model is considered when a supply suggestion is generated for the current master plan.
    - **Include demand forecast** – Set this option to *Yes* to include the demand forecast in the current master plan. If you set it to *No*, demand forecast transactions aren't included in the master plan.
    - **Method used to reduce forecast requirements** – Select the method that should be used to reduce forecast requirements. Learn more in the Forecast reduction keys section later in this article.
4. On the **Time fence in days** FastTab, set the following fields to specify the period that the demand forecast is included during:
    - **Forecast plan** – Set this option to *Yes* to override the forecast plan time fence that originates from the individual coverage groups. Set it to *No* to use the values from the individual coverage groups for the current master plan.

- **Forecast time period** – If you set the **Forecast plan** option to *Yes*, specify the number of days (from today's date) that the demand forecast should be applied.

> **Important**
>
> Planning Optimization doesn't support the **Forecast plan** setting.

Set up a coverage group to include a demand forecast

To configure a coverage group to include a demand forecast, follow these steps:

1. Go to **Master planning** > **Setup** > **Plans** > **Coverage groups**.
2. Select an existing coverage group, or create a new group.
3. On the **Other** FastTab, set the following fields:
   - **Forecast plan time fence** – Enter the number of days (from today's date) that the demand forecast should apply to. You can override this value by using the **Forecast plan** option on the master plan, as described in the previous section.
   - **Reduction key** – Select the reduction key to apply. Learn more in the Create and set up a forecast reduction key and Use a reduction key sections later in this article.
   - **Reduce forecast by** – For master plans where the **Method used to reduce forecast requirements** field is set to *Transactions - reduction key* or *Transactions - dynamic period*, specify which transactions should reduce the forecast. Select one of the following values:
     - **All transactions** – All transactions reduce the forecast.
     - **Orders** – Only sales orders reduce the forecast.

     > **Note**
     >
     > If you select *All transactions*, the system considers transactions that have both demand and supply in the same inventory dimensions neutral and it ignores them during the forecast reduction. For example, if the planning dimension is set to site only, not warehouse, the system ignores a transfer order between site 1, warehouse 11, and site 1, warehouse 13, and it doesn't reduce the remaining demand forecast.

- **Include intercompany orders** – Set this option to *Yes* if intercompany orders should be included when the forecast is reduced. Otherwise, set it to *No*.

- **Include customer forecast in the demand forecast** – Specify whether a customer forecast should be included in the overall forecast. This option determines how actual demand reduces the forecasted demand. Use it to ensure that master planning covers the supply of items that specific customers purchase.
  - Set this option to *Yes* to include a customer forecast in the overall forecast. In this case, actual customer demand reduces both the customer forecast and the overall forecast. Master planning generates planned orders to cover only the overall forecast quantity.
  - Set this option to *No* if you don't want to include a customer forecast in the overall forecast. In this case, actual customer demand reduces only the customer forecast. Master planning generates planned orders to cover both the overall forecast quantity and the forecast for each customer quantity.

Forecast reduction keys

This section provides information about the different methods that reduce forecast requirements. It includes examples of the results of each method. It also explains how to create, set up, and use a forecast reduction key. Some methods use a forecast reduction key to reduce forecast requirements.

Methods to reduce forecast requirements

When you include a forecast in a master plan, select how to reduce the forecast requirements when actual demand is included. Master planning excludes forecast requirements from the past, which means all forecast requirements before today's date.

To include a forecast in a master plan and select the method to reduce forecast requirements, go to **Master planning** > **Setup** > **Plans** > **Master plans**. In the **Forecast model** field, select a forecast model. In the **Method used to reduce forecast requirements** field, select a method. The following options are available:

- None
- Percent – reduction key
- Transactions – reduction key
- Transactions – dynamic period

The following sections provide more information about each option.

None

If you select *None*, the forecast requirements aren't reduced during master scheduling. In this case, master planning creates planned orders to supply the forecasted demand (forecast requirements). These planned orders maintain the suggested quantity, regardless of other types of demand. For example, if sales orders are placed, master planning creates additional planned orders to supply the sales orders. The quantity of the forecast requirements isn't reduced.

Percent – reduction key

If you select *Percent - reduction key*, the system reduces the forecast requirements according to the percentages and periods that the reduction key defines. In this case, master planning creates planned orders where the quantity is calculated as forecasted quantity × reduction key in each period. If other types of demand exist, master planning also creates planned orders to supply that demand.

Example: Percent – reduction key

This example shows how a reduction key reduces demand forecast requirements according to the percentages and periods that the reduction key defines.

For this example, you include the following demand forecast in a master plan.

Expand table

Month	Demand forecast
January	1,000
February	1,000
March	1,000
April	1,000

On the **Reduction keys** page, you set up the following lines.

Expand table

Change	Unit	Percent
1	Month	100

Change	Unit	Percent
2	Month	75
3	Month	50
4	Month	25

You assign the reduction key to the item's coverage group. Then, on the **Master plans** page, in the **Method used to reduce forecast requirements** field, you select *Percent - reduction key*.

In this case, if you run forecast scheduling on January 1, the system consumes the demand forecast requirements according to the percentages that you set up on the **Reduction keys** page. The following requirement quantities are transferred to the master plan.

Expand table

Month	Planned order quantity	Calculation
January	0	= 0% × 1,000
February	250	= 25% × 1,000
March	500	= 50% × 1,000
April	750	= 75% × 1,000
May through December	1,000	= 100% × 1,000

Transactions – reduction key

If you set the **Method used to reduce forecast requirements** field to *Transactions - reduction key*, the system reduces the forecast requirements by the qualified demand transactions that occur during the periods that the reduction key defines.

The **Reduce forecast by** field on the **Coverage groups** page defines the qualified demand. If you set the **Reduce forecast by** field to *Orders*, only sales order transactions are considered qualified demand. If you set it to *All transactions*, any non-intercompany issue inventory transactions are considered qualified demand. If intercompany sales orders should also be considered qualified demand, set the **Include intercompany orders** option to *Yes*.

Forecast reduction starts with the first (earliest) demand forecast record in the reduction key period. If the quantity of qualified inventory transactions is more than the quantity of demand forecast lines in the same reduction key period, the balance of inventory transactions quantity is

used to reduce the demand forecast quantity in the previous period (if there's unconsumed forecast).

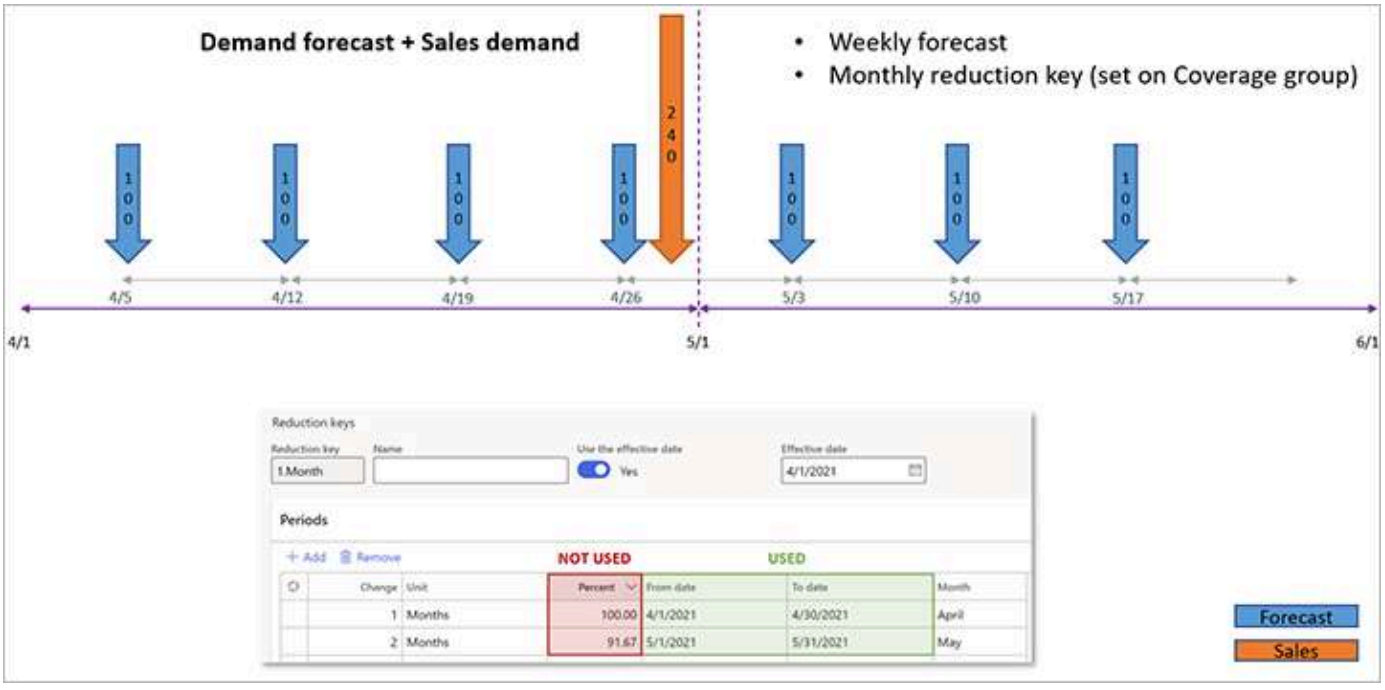
If no unconsumed forecast remains in the previous reduction key period, the balance of inventory transactions quantity reduces the forecast quantity in the next month (if there's unconsumed forecast).

The value of the **Percent** field on the reduction key lines isn't used when the **Method used to reduce forecast requirements** field is set to *Transactions - reduction key*. Only the dates define the reduction key period.

> **Note**
>
> The system ignores any forecast that you post on or before today's date and doesn't use it to create planned orders. For example, if you generate your demand forecast for the month on January 1, and you run master planning that includes demand forecasting on January 2, the calculation ignores the demand forecast line that is dated January 1.

Example: Transactions – reduction key

This example shows how actual orders that occur during the periods that the reduction key defines reduce demand forecast requirements.



For this example, you select *Transactions - reduction key* in the **Method used to reduce forecast requirements** field on the **Master plans** page.

The following demand forecast lines exist on April 1.

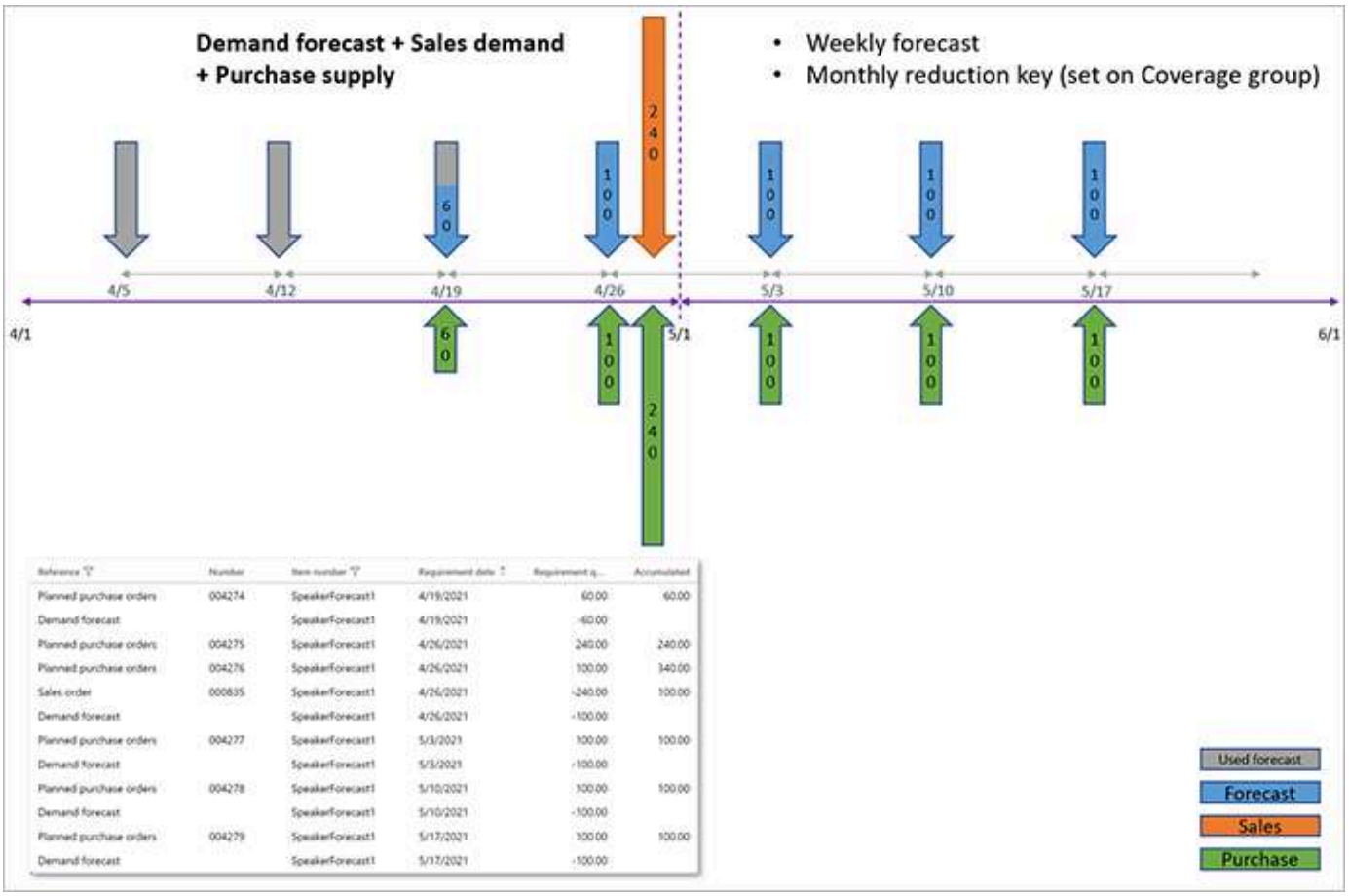
Expand table

Date	Number of pieces forecasted
April 5	100
April 12	100
April 19	100
April 26	100
May 3	100
May 10	100
May 17	100

The following sales order lines exist in April.

Expand table

Date	Number of pieces requested
April 27	240



The following requirement quantities are transferred to the master plan when master planning is run on April 1. As you see, the April forecast transactions were reduced by the demand quantity of 240 in a sequence, starting from the first of those transactions.

Expand table

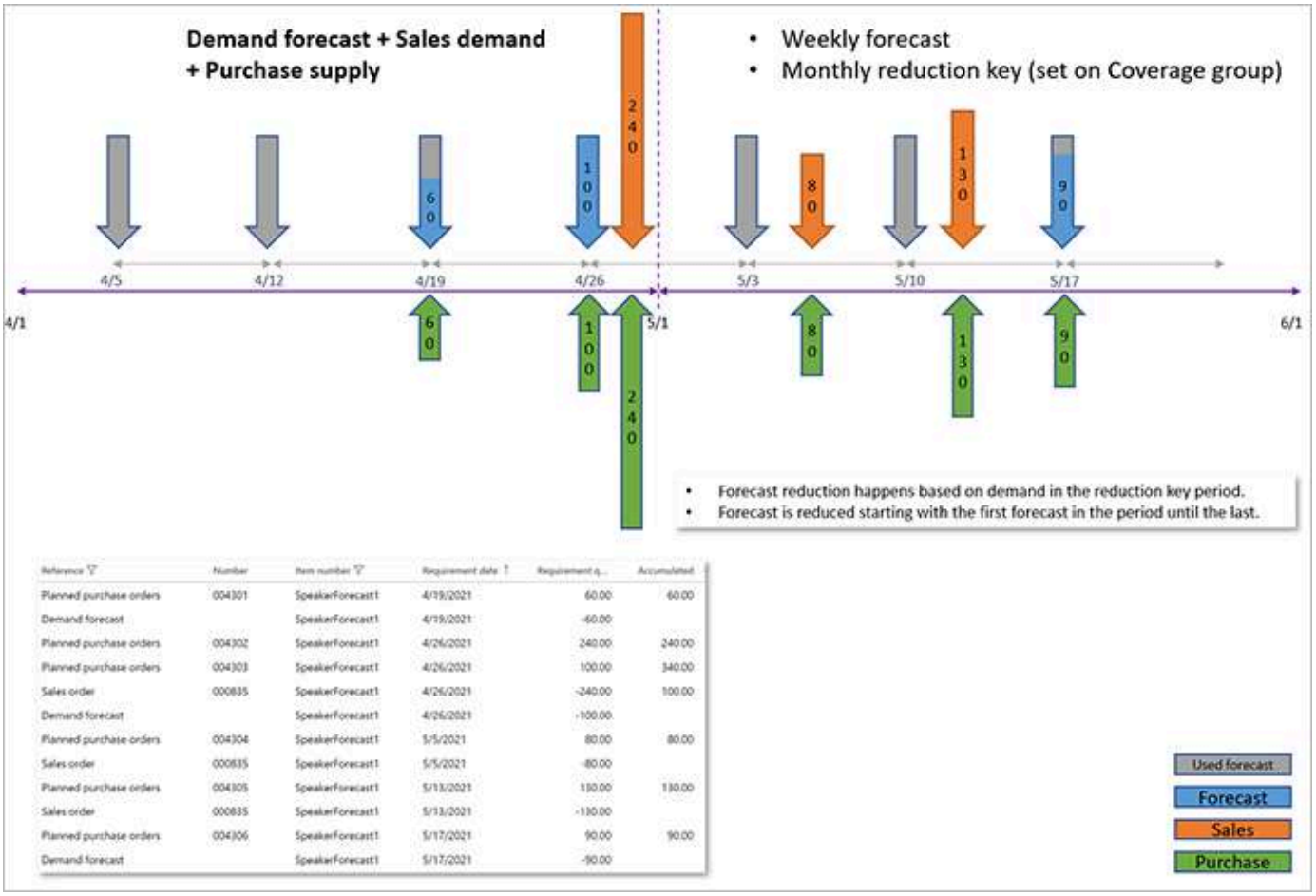
Date	Number of pieces required
April 5	0
April 12	0
April 19	60
April 26	100
April 27	240
May 3	100
May 10	100
May 17	100

Now, assume that you import new orders for the period of May.

The following sales order lines exist in May.

Expand table

Date	Number of pieces requested
May 4	80
May 11	130



When you run master planning on April 1, the system transfers the following requirement quantities to the master plan. As you see, the demand quantity of 240 reduces the April forecast transactions in a sequence, starting from the first of those transactions. However, the demand quantity of 210 reduces the May forecast transactions, starting from the first demand forecast transaction in May. The totals per period are preserved (400 in April and 300 in May).

Expand table

Date	Number of pieces required
April 5	0
April 12	0

Date	Number of pieces required
April 19	60
April 26	100
April 27	240
May 3	0
May 4	80
May 10	0
May 11	130
May 17	90

Transactions – dynamic period

If you select *Transactions - dynamic period*, actual order transactions that occur during the dynamic period reduce the forecast requirements. The dynamic period covers the current forecast dates and ends at the start of the next forecast. In this case, master planning creates planned orders to supply the forecasted demand (forecast requirements). However, when you place actual order transactions, they reduce the forecast requirements. The actual transactions consume part of the forecasted requirements.

When you use this option, the following behavior occurs:

- Reduction keys aren't required or used.
- If the forecast is completely reduced, the forecast requirements for the current forecast become 0 (zero).
- If there's no future forecast, the system reduces forecast requirements from the last forecast that you entered.
- Time fences are included in the forecast reduction calculation.
- Positive days are included in the forecast reduction calculation.
- If actual order transactions exceed the forecasted requirements, the remaining transactions aren't forwarded to the next forecast period.

Example 1: Transactions – dynamic period

Here's a simple example that shows how the *Transactions - dynamic period* method works.

For this example, you include the following demand forecast in a master plan.

Expand table

Date	Demand forecast
January 1	1,000
February 1	1,000

You also create the following sales orders.

Expand table

Date	Sales order quantity
January 15	200
February 15	400

In this case, the system creates the following planned orders.

Expand table

Demand forecast date	Quantity	Explanation
January 1	800	Forecast requirements (= 1,000 – 200)
January 15	200	Sales orders requirements
February 1	600	Forecast requirements (= 1,000 – 400)
February 15	400	Sales orders requirements

Example 2: Transactions – dynamic period

In most cases, systems are set up so that transactions reduce demand forecast in specific forecast periods: weeks, months, and so on. These periods are defined in the reduction key. However, the time between two demand forecast lines can also *imply* a period.

For this example, you create a demand forecast for the following dates and quantities.

Expand table

Date	Demand forecast
January 1	1,000
January 5	500

Date	Demand forecast
January 12	1,000

Notice that, in this forecast, there isn't a clear period between the forecast dates. Between the first and second dates, there's a four-day span, and between the second and third dates there's a seven-day span. These spans are the dynamic periods.

You also create the following sales order lines.

Expand table

Date	Sales order quantity
December 15 in the previous year	500
January 3	100
January 10	200

The sales orders reduce the forecast in the following ways:

- Because the first sales order isn't in any period, it doesn't reduce any forecast.
- Because the second sales order is between January 1 and January 5, it reduces the forecast for January 1 by 100.
- Because the third sales order is between January 5 and January 12, it reduces the forecast for January 5 by 200.

Therefore, the system creates the following planned orders.

Expand table

Demand forecast date	Quantity	Explanation
December 15 in the previous year	500	Sales order requirements
January 1	900	Forecast requirements period January 1 to January 5 (= 1,000 – 100)
January 3	100	Sales order requirements
January 5	300	Forecast requirements period January 5 to January 10 (= 500 – 200)
January 12	1,000	Forecast requirements period January 12 to end

Create and set up a forecast reduction key

A forecast reduction key is used in the *Transactions - reduction key* and *Percent- reduction key* methods for reducing forecast requirements. Follow these steps to create and set up a reduction key.

1. Go to **Master planning** > **Setup** > **Coverage** > **Reduction keys**.
2. Select **New** to create a reduction key.
3. In the **Reduction key** field, enter a unique identifier for the forecast reduction key. Then, in the **Name** field, enter a name.
4. Define the periods and the reduction key percentage in each period:
    - The **Effective date** field indicates the date when creation of the periods starts. When the **Use the effective date** option is set to *Yes*, the periods start on the effective date. When it's set to *No*, the periods start on the date when master planning is run.
    - Define the periods that the forecast reduction should occur during.
    - For a specific period, specify the percentages that the forecast requirements should be reduced by. Enter positive values to decrease requirements or negative values to increase requirements.

Use a reduction key

A forecast reduction key must be assigned to the coverage group of the item. Follow these steps to assign a reduction key to an item's coverage group.

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. On the **Other** FastTab, in the **Reduction key** field, select the reduction key to assign to the coverage group. The reduction key then applies to all items that belong to the coverage group.
3. To use a reduction key to calculate forecast reduction during master scheduling, you must define this setting in the setup of the forecast plan or the master plan. Go to one of the following locations:
    - **Master planning** > **Setup** > **Plans** > **Forecast plans**
    - **Master planning** > **Setup** > **Plans** > **Master plans**

4. On the **Forecast plans** or **Master plans** page, on the **General** FastTab, in the **Method used to reduce forecast requirements** field, select either *Percent - reduction key* or *Transactions - reduction key*.

Reduce a forecast by transactions

When you select *Transactions - reduction key* or *Transactions - dynamic period* as the method for reducing forecast requirements, you can specify which transactions reduce the forecast. On the **Coverage groups** page, on the **Other** FastTab, in the **Reduce forecast by** field, select *All transactions* if all transactions should reduce the forecast or *Orders* if only sales orders should reduce the forecast.

Consider customer, customer group, required BOM, and required Route in demand forecast reduction

Each demand forecast line can specify a customer, a customer group, a *required BOM*, and a *required Route*. When you turn on the *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* feature in Feature management, Planning Optimization respects all four of these dimensions when it decides whether an existing demand transaction (such as a sales order line) reduces a forecast line.

Prerequisites

Before you can use the BOM and route forecast reduction behavior described in this section, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.
- The feature named *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* must be turned on in Feature management.

Matching principle

The behavior follows a single principle:

> A transaction reduces a forecast line only when, for every dimension the forecast line specifies, the transaction either matches that value or doesn't specify a value for that dimension. A transaction that specifies a different value than the forecast line for any dimension can't reduce that forecast line.

In other words, a transaction can be at the same specificity as the forecast line or less specific, but it can't be specific in a *different* way.

Specificity rules for demand forecast lines

The principle leads to the following rules. In each rule, *transaction* refers to a sales order line (or, when the coverage group is set to **All transactions**, any qualifying issue inventory transaction) that would otherwise reduce the forecast.

- A forecast line that specifies both a required BOM and a required Route is reduced only by transactions that specify the same BOM and the same route, or that leave one or both of those values blank.
- A forecast line that specifies a required BOM but no required Route is reduced by transactions that specify the same BOM (regardless of route) and by transactions that specify no BOM.
- A forecast line that specifies a required Route but no required BOM is reduced by transactions that specify the same route (regardless of BOM) and by transactions that specify no route.
- A forecast line that specifies neither a required BOM nor a required Route is reduced by any otherwise-qualifying transaction.
- A transaction that specifies neither a BOM nor a route can reduce any forecast line, regardless of how specific that forecast line is.

The same principle applies to the customer and customer group dimensions on the forecast line: a sales order for a specific customer can reduce a forecast line that specifies the same customer, or that specifies only the customer group that the customer belongs to, or that specifies no customer or customer group. A sales order for a different customer can't reduce a customer-specific forecast line.

The following table summarizes the matching behavior for the BOM and route dimensions. A *Yes* indicates that the transaction reduces the forecast line. A *No* indicates that the transaction doesn't reduce the forecast line because it specifies a conflicting value.

Expand table

Forecast line specifies	**Tx (B1, R1)**	**Tx (B1, R2)**	**Tx (B2, R1)**	**Tx (B1, –)**	**Tx (–, R1)**	**Tx (–, –)**
required BOM = B1, required Route = R1	Yes	No	No	Yes	Yes	Yes

Forecast line specifies	Tx (B1, R1)	Tx (B1, R2)	Tx (B2, R1)	Tx (B1, –)	Tx (–, R1)	Tx (–, –)
required BOM = B1 only	Yes	Yes	No	Yes	Yes	Yes
required Route = R1 only	Yes	No	Yes	Yes	Yes	Yes
Neither specified	Yes	Yes	Yes	Yes	Yes	Yes

> **Note**
>
> This specificity logic is independent of the **Include customer forecast in the demand forecast** option on the coverage group. That option still controls whether a customer-specific forecast also reduces (and contributes to) the overall forecast. The rules in this section determine which transactions are *eligible* to reduce a given forecast line in the first place.

Example: Two demand forecast lines with different required BOMs

You have an item that has the following demand forecast lines.

Expand table

Model	Date	Quantity	Unit	Customer	Customer group	Required BOM	Required Route	Site	Warehouse
CurrentF	10/10/22	10	ea			B1		1	11
CurrentF	10/10/22	10	ea			B2		1	11

A sales order line exists for a quantity of *15 ea* with BOM *B2*, dated within the same forecast period.

The system behaves differently based on whether the *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* feature is turned on:

- **When the feature is turned *off*** – The sales order reduces the demand forecast in the order that the lines are encountered, without checking the BOM. The first line is reduced to *0*, the second line is reduced to *5*, and Planning Optimization creates a planned order to cover the residual *5 ea* of forecast on the second line.

- **When the feature is turned *on*** – Only the second line matches the sales order's BOM. The sales order reduces the second line to *0* and leaves the first line at *10*. Planning Optimization then creates a planned order to cover the *10 ea* of remaining demand with BOM *B1*, plus a planned order to cover the *15 ea* of sales-order demand with BOM *B2*.

Example: Combined customer, customer group, required BOM, and required Route

You have an item that has the following demand forecast lines. Customer *Cust-1* belongs to customer group *CG-1*. Customer *Cust-2* doesn't belong to customer group *CG-1*.

Expand table

Line	Quantity	Customer	Customer group	Required BOM	Required Route
L1	10	Cust-1	CG-1	B1	R1
L2	10		CG-1	B1	
L3	10				R1
L4	10				

The following sales order lines exist for the same item, site, and warehouse within the same forecast period.

Expand table

Sales order	Quantity	Customer	BOM	Route
SO-A	5	Cust-1	B1	R1
SO-B	5	Cust-1	B1	
SO-C	5	Cust-2	B1	R1
SO-D	5			

With the feature turned on, Planning Optimization evaluates each sales order against each forecast line. When a sales order can reduce more than one forecast line, Planning Optimization applies the reduction to the most specific matching line first, so that less-specific forecast lines remain available to cover other demand.

- *SO-A* (Cust-1, B1, R1) matches every dimension of *L1*. It also matches *L2* (the customer belongs to *CG-1*, the BOM matches, and *L2* doesn't constrain the route), but *L1* is more specific, so it reduces *L1* by *5*.
- *SO-B* (Cust-1, B1, no route) matches the customer and BOM of *L1* and is less specific on route, so it also reduces *L1* by *5*. *L1* is now fully consumed.
- *SO-C* (Cust-2, B1, R1) can't reduce *L1* or *L2*, because *Cust-2* doesn't match *L1*'s customer and doesn't belong to *L2*'s customer group. It does match *L3* on route (and is allowed to be more specific than the forecast line on the other dimensions), so it reduces *L3* by *5*.
- *SO-D* (no customer, no BOM, no route) is fully unspecified, so it can reduce any remaining forecast line. Planning Optimization applies it to the most specific line that still has open quantity, which is *L2*, reducing *L2* by *5*.

The residual demand forecast quantities are *L1* = 0, *L2* = 5, *L3* = 5, and *L4* = 10. Planning Optimization creates planned orders to cover this residual demand together with the sales-order demand.

> **Note**
>
> When the feature is turned on, the same BOM and route matching also applies to demand transactions that have already been processed within the forecast period–for example, invoiced sales orders. The BOM and route that were specified on the original sales order are honored when the order reduces a forecast line, just as they are for open sales orders.
>
> Reduction by already-processed transactions is itself controlled by a separate feature management feature: *Include invoiced and delivered orders during supply and demand forecast reduction for planning optimization*. That feature must also be turned on for processed orders to participate in forecast reduction at all. The *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* feature then governs how the BOM and route on those processed orders are matched.

The same rules apply to supply forecasts. Learn more in Consider sub-BOM and sub-Route in supply forecast reduction.

Forecast models and submodels

This section describes how to create forecast models and how to combine multiple forecast models by setting up submodels.

A *forecast model* names and identifies a specific forecast. After you create the forecast model, you can add forecast lines to it. To add forecast lines for multiple items, use the **Demand forecast lines** page. To add forecast lines for a specific selected item, use the **Released products** page.

A forecast model can include forecasts from other forecast models. To achieve this result, add other forecast models as *submodels* of a parent forecast model. You must create each relevant model before you can add it as a submodel of a parent forecast model.

The resulting structure gives you a powerful way to control forecasts, because it lets you combine (aggregate) the input from multiple individual forecasts. Therefore, from a planning point of view, it's easy to combine forecasts for simulations. For example, you might set up a simulation that is based on the combination of a regular forecast with the forecast for a spring promotion.

Submodel levels

You can add unlimited submodels to a parent forecast model. However, the structure can be only one level deep. In other words, a forecast model that is a submodel of another forecast model can't have its own submodels. When you add submodels to a forecast model, the system checks whether that forecast model is already a submodel of another forecast model.

If master planning encounters a submodel that has its own submodels, you receive an error message.

Submodel levels example

Forecast model A has forecast model B as a submodel. Therefore, forecast model B can't have its own submodels. If you try to add a submodel to forecast model B, you receive the following error message: "Forecast model B is a submodel for model A."

Aggregating forecasts across forecast models

The system aggregates forecast lines that occur on the same day across their forecast model and its submodels.

Aggregation example

Forecast model A has forecast models B and C as submodels.

- Forecast model A includes a demand forecast for 2 pieces (pcs) on June 15.
- Forecast model B includes a demand forecast for 3 pcs on June 15.
- Forecast model C includes a demand forecast for 4 pcs on June 15.

The resulting demand forecast is a single demand for 9 pcs (2 + 3 + 4) on June 15.

> ⓘ **Note**
>
> Each submodel uses its own parameters, not the parameters of the parent forecast model.

Create a forecast model

To create a forecast model, follow these steps:

1. Go to **Master planning** > **Setup** > **Demand forecasting** > **Forecast models**.
2. On the Action Pane, select **New**.
3. Set the following fields for the new forecast model:
    - **Model** – Enter a unique identifier for the model.
    - **Name** – Enter a descriptive name for the model.
    - **Stopped** – Usually, you should set this option to *No*. Set it to *Yes* only if you want to prevent editing of all forecast lines that are assigned to the model.

    > ⓘ **Note**
    >
    > The **Include in cash flow forecasts** field and the fields on the **Project** FastTab aren't related to master planning. Therefore, you can ignore them in this context. Consider them only when you work with forecasts for the **Project management and accounting** module.

Assign submodels to a forecast model

To assign submodels to a forecast model, follow these steps:

1. Go to **Inventory management** > **Setup** > **Forecast** > **Forecast models**.
2. In the list pane, select the forecast model to set up a submodel for.
3. On the **Submodel** FastTab, select **Add** to add a row to the grid.
4. In the new row, set the following fields:

- **Submodel** – Select the forecast model to add as a submodel. This forecast model must already exist, and it must not have any submodels of its own.
- **Name** – Enter a descriptive name for the submodel. For example, this name might indicate the submodel's relation to the parent forecast model.

Last updated on 07/23/2026

Master planning with supply forecasts

Supply forecasts let you specify the supply that you expect to need during a future period. Typically, you base the estimate on the previous year's sales history and then use the forecast for budgeting purposes. You can also set up your master plans to consider forecasts during planning.

Set up a master plan to consider supply forecasts

You can specify whether each master plan should consider forecasts when it runs. Use the following procedure to set forecasting options for each plan.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select an existing master plan in the list pane, or select **New** on the Action Pane to create a new one.
3. On the **General** FastTab, set the following fields:
    - **Forecast model** – Select the model that contains the supply and demand forecasts that the master plan should consider.
    - **Include supply forecast** – Set this option to *Yes* if the master plan should consider supply forecasts.
    - **Include demand forecast** – Set this option to *Yes* if the master plan should consider demand forecasts. Although this setting is independent of the **Include supply forecast** setting, some of the other settings on the page apply to both supply forecasts and demand forecasts. For more information about planning that considers demand forecasts, see Master planning with demand forecasts.
    - **Method used to reduce forecast requirements** – Select the method to use to reduce forecast requirements during master planning:
        - *None* – Forecast requirements aren't reduced during master planning.
        - *Percent - reduction key* – Forecast requirements are reduced according to the percentages and time periods that you define in the reduction key.
        - *Transactions – reduction key* – Forecast requirements are reduced by the transactions that occur during the time periods that you define in the reduction key.

- *Transactions – dynamic period* – Forecast requirements are reduced by the order transactions that occur during the dynamic period. The dynamic period covers the current forecast dates, and it ends when the next forecast begins. The *Transactions – dynamic period* method doesn't use or require a reduction key. When you use it, the following conditions apply:
  - If the forecast is completely reduced, the forecast requirements for the current forecast become 0 (zero).
  - If there's no future forecast, forecast requirements from the last forecast that you entered are reduced.
  - Time fences are included in the forecast reduction calculation.
  - Positive days are included in the forecast reduction calculation.
  - If actual order transactions exceed the forecasted requirements, the remaining transactions aren't forwarded to the next forecast period.

> ⓘ **Note**
>
> The **Method used to reduce forecast requirements** setting also applies to demand forecasts if you enable them for the master plan, and it affects them in a similar way. Learn more in Master planning with demand forecasts.

Set reduction options for coverage groups

To set options that control how a coverage group reduces its forecast requirements for master plans that use transaction-based reduction, follow these steps:

1. Go to **Master planning** > **Setup** > **Plans** > **Coverage groups**.
2. Select an existing coverage group in the list pane, or select **New** on the Action Pane to create a new one.
3. On the **Other** FastTab, in the **Reduce forecast by** field, specify how supply forecast requirements should be reduced for items in the coverage group, for master plans where the **Method used to reduce forecast requirements** field is set to *Transactions - reduction key* or *Transactions - dynamic period*. Select one of the following values:
   - *All transactions* – All transactions reduce the forecast.
   - *Orders* – Only orders of the same type reduce the forecast.

For example, the default order settings for an item indicate that it should be produced. There's a supply forecast line for a quantity of 50, and there's an existing purchase order for a quantity of 20. If you set the **Reduce forecast by** field to *Orders*, a planned production order is created for a quantity of 50. If you set it to *All transactions*, the planned production order is reduced to a quantity of 30.

This setting also applies to demand forecasting reduction. Learn more in Master planning with demand forecasts.

Enter a supply forecast for a product

To enter a supply forecast for a product, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product that you want to enter a forecast for.
3. On the Action Pane, on the **Plan** tab, select **Supply forecast**.
4. On the **Supply forecast** page, on the Action Pane, select **New** to add a forecast to the grid.
5. Fill out the new line to specify your forecast.

Plan for an item with supply forecast lines

When you run a master plan that includes an item that has a supply forecast, the system creates a purchase order for the supply lines that you entered. The system respects the default order settings for the item. If those default order settings specify a **Default order type** value of *Purchase order*, and you don't specify a vendor account on the supply forecast line, the system uses the default vendor for the item.

Consider sub-BOM and sub-route in supply forecast reduction

Each supply forecast line can specify a *sub-BOM* and a *sub-route*. When you turn on the *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* feature in Feature management, Planning Optimization respects these values when it decides whether an existing transaction (such as a released purchase order, a released production order, or a released transfer order) reduces a forecast line.

Prerequisites

Before you can use the BOM and route forecast reduction behavior described in this section, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.
- The feature named *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* must be turned on in Feature management.

Matching principle

The behavior follows a single principle:

> A transaction reduces a forecast line only when, for every dimension the forecast line specifies, the transaction either matches that value or doesn't specify a value for that dimension. A transaction that specifies a value that's different from the forecast line value for any dimension can't reduce that forecast line.

In other words, a transaction can be at the same specificity as the forecast line or less specific, but it can't be specific in a *different* way.

Specificity rules for sub-BOM and sub-route

The principle leads to the following rules. In each rule, *transaction* refers to a released purchase order, production order, or transfer order that would otherwise qualify to reduce the forecast.

- A forecast line that specifies both a sub-BOM and a sub-route is reduced only by transactions that specify the same BOM and the same route, or that leave one or both of those values blank.
- A forecast line that specifies a sub-BOM but no sub-route is reduced by transactions that specify the same BOM (regardless of route) and by transactions that specify no BOM.
- A forecast line that specifies a sub-route but no sub-BOM is reduced by transactions that specify the same route (regardless of BOM) and by transactions that specify no route.
- A forecast line that specifies neither a sub-BOM nor a sub-route is reduced by any otherwise-qualifying transaction.
- A transaction that specifies neither a BOM nor a route can reduce any forecast line, regardless of how specific that forecast line is.

The following table summarizes the matching behavior. A *Yes* indicates that the transaction reduces the forecast line. A *No* indicates that the transaction doesn't reduce the forecast line

because it specifies a conflicting value.

Expand table

Forecast line specifies	Tx (B1, R1)	Tx (B1, R2)	Tx (B2, R1)	Tx (B1, –)	Tx (–, R1)	Tx (–, –)
sub-BOM = B1, sub-route = R1	Yes	No	No	Yes	Yes	Yes
sub-BOM = B1 only	Yes	Yes	No	Yes	Yes	Yes
sub-route = R1 only	Yes	No	Yes	Yes	Yes	Yes
Neither specified	Yes	Yes	Yes	Yes	Yes	Yes

This specificity logic is layered on top of the dimension matching that already applies. For purchase orders, the vendor on the transaction must still match the vendor on the forecast line, as described in Reduction of forecasts by matching vendors only. For production orders and transfer orders, the vendor is ignored.

Example: Two supply forecast lines with different sub-BOMs

You have an item where the default order type is *Production*. It has the following supply forecast lines.

Expand table

Model	Date	Quantity	Unit	sub-BOM	sub-route	Site	Warehouse
CurrentF	10/10/22	10	ea	B1		1	11
CurrentF	10/10/22	10	ea	B2		1	11

A released production order exists for a quantity of *15 ea* with BOM *B2*.

The system behaves differently based on whether the *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* feature is turned on:

- **When the feature is turned *off*** – The production order reduces the supply forecast in the order that the lines are encountered, without checking the BOM. The first line is reduced to *0*, the second line is reduced to *5*, and Planning Optimization creates a planned production order for a quantity of *5* with BOM *B2*.

- **When the feature is turned *on*** – Only the second line matches the production order's BOM. The production order reduces the second line to *0* and leaves the first line at *10*. Planning Optimization then creates a planned production order for a quantity of *10* with BOM *B1*, which is the supply that is still needed to cover the unmatched forecast.

Example: Combined sub-BOM and sub-route on the forecast line

You have an item where the default order type is *Production*. It has the following supply forecast line.

Expand table

Model	Date	Quantity	Unit	sub-BOM	sub-route	Site	Warehouse
CurrentF	10/10/22	20	ea	B1	R1	1	11

The following released production orders also exist for the same item, site, and warehouse, all dated within the same forecast period.

Expand table

Production order	Quantity	BOM	Route
PO-A	5	B1	R1
PO-B	5	B1	R2
PO-C	5	B2	R1
PO-D	5		

With the feature turned on, Planning Optimization evaluates each production order against the forecast line.

- *PO-A* (B1, R1) matches both dimensions and reduces the forecast by *5*.
- *PO-B* (B1, R2) specifies a different route than the forecast line, so it doesn't reduce the forecast line. Planning Optimization treats it as separate supply.
- *PO-C* (B2, R1) specifies a different BOM than the forecast line, so it doesn't reduce the forecast line. Planning Optimization treats it as separate supply.
- *PO-D* (no BOM, no route) is less specific than the forecast line, so it reduces the forecast by *5*.

The forecast line is reduced from *20* to *10* (= 20 − 5 − 5), and Planning Optimization creates a planned production order for a quantity of *10 ea* with BOM *B1* and route *R1* to cover the remaining requirement.

> **Note**
>
> When the feature is turned on, the same BOM and route matching also applies to transactions that were already processed for the forecast period–for example, received purchase orders and ended production orders. The BOM and route that were specified on the original order are honored when the order reduces a forecast line, just as they are for open (released) orders.
>
> Reduction by already-processed transactions is itself controlled by a separate feature management feature: *Include invoiced and delivered orders during supply and demand forecast reduction for planning optimization*. You must also turn on that feature for processed orders to participate in forecast reduction at all. The *Consider BOM and route in supply- and demand forecast reduction with Planning Optimization* feature then governs how the BOM and route on those processed orders are matched.

The same rules apply to demand forecasts, with the additional dimensions of customer and customer group. Learn more in Consider customer, customer group, required BOM, and required Route in demand forecast reduction.

Check whether a planned order is a supply forecast order

Use the following procedure to check whether a planned order was created as a result of a supply forecast.

1. Go to **Master planning** > **Master planning** > **Planned orders**.
2. Open the planned order that you want to inspect.
3. On the **General** FastTab, check the value of the **Supply forecast** field. If the order was created to cover a supply forecast, this field is set to *Yes*.

Examples of master planning with supply forecasts

This section provides several examples that show how supply forecasting affects master planning.

Example 1: Supply forecast for an item

You have an item where the default order type is *Purchase order* and the default vendor is *US-002*. It has the following supply forecast line.

Expand table

Model	Date	Vendor account	Vendor group	Quantity	Unit	Site	Warehouse
CurrentF	10/10/22			35	ea	1	11

When you run master planning, a planned purchase order is created for *35 ea* from vendor *US-002*.

Example 2: Several supply forecast lines, with and without a vendor

You have an item where the default order type is *Purchase order* and the default vendor is *US-002*. It has the following supply forecast lines.

Expand table

Model	Date	Vendor account	Vendor group	Quantity	Unit	Site	Warehouse
CurrentF	10/10/22			35	ea	1	11
CurrentF	10/10/22	US-101		25	ea	1	11

In this case, the system treats the line that doesn't specify a vendor as a generic forecast, and it assumes that the line that does specify a vendor should be subtracted from the generic forecast. Therefore, master planning creates the following planned purchase orders for the item:

- A planned purchase order for a quantity of *25 ea* from vendor *US-101* (The vendor and quantity that are specified on the forecast line are used.)
- A planned purchase order for a quantity of *10 ea* from vendor *US-002* (Because no vendor was specified on the other forecast line, the default vendor is used, and the quantity of this forecast line is reduced by the quantity of the vendor-specific forecast line.)

Example 3: Plans that use the Transactions - dynamic period reduction method in a single forecast period

For master plans that use *Transactions - dynamic period* as the forecast reduction method, master planning reduces forecast quantities if existing transactions match those supply characteristics.

For example, you have an item where the default order type is *Purchase order*. It has the following supply forecast line.

Expand table

Model	Date	Vendor account	Vendor group	Quantity	Unit	Site	Warehouse
CurrentF	10/10/22	US-101		25	ea	1	11

Because there's only one supply forecast line, there's only one forecast period.

When you run a master plan that uses *Transactions – dynamic period* as the reduction method, one of the following results can occur:

- If a purchase order exists for vendor *US-101* and a quantity of *10 ea*, and the **Supply forecast** field is set to *Yes*, master planning creates a new planned purchase order for the remaining quantity of *15 ea*. The forecast line is reduced, because the vendor matches the existing purchase order.
- If a purchase order exists for vendor *US-102*, site *1*, warehouse *11*, and a quantity of *10 ea*, and the **Supply forecast** field is set to *Yes*, master planning creates a new planned purchase order for the full quantity of *25 ea*. The forecast line can't be reduced, because it has a different vendor than the existing purchase order.

> **Note**
>
> This situation can occur for planned purchase orders, because a vendor is associated with them. However, for transfer orders and production orders, the supply forecast amounts are always reduced by existing orders, because no vendor is specified for these types of orders.

Example 4: Plans that use the Transactions - dynamic period reduction method over several forecast periods

You have an item where the default order type is *Purchase order*. It has the following supply forecast lines.

Expand table

Model	Date	Vendor account	Vendor group	Quantity	Unit	Site	Warehouse
CurrentF	10/10/22	US-101		25	ea	1	11
CurrentF	10/15/22	US-101		25	ea	1	11

In this example, two forecast lines each have a different date. Therefore, the dates establish the boundaries of the forecast periods. The first period is from October 10 through October 14, 2022 (10/10/22–10/14/22). The second is from October 15, 2022 (10/15/22) onward.

A purchase order exists for vendor *US-101*, a quantity of *10 ea*, and the date *10/12/22* (October 12, 2022). When you run a master plan that is set up to use *Transactions – dynamic period* as the reduction method, it creates the following orders:

- A planned order for a quantity of *15 ea* and the date *10/10/22* (The quantity is reduced by the existing purchase order that is dated during the same forecast period.)
- A planned order for a quantity of *25 ea* and the date *10/15/22* (The quantity is the full quantity of the forecast.)

Example 5: Plans that use no reduction

When you run a plan where the forecast reduction method is *None*, master planning always creates planned supply for all supply forecast lines. After you approve that planned supply, it reduces future planned orders. However, purchase orders don't reduce the supply forecast lines.

For example, you have an item where the default order type is *Purchase order*. It has the following supply forecast line.

Expand table

Model	Date	Vendor account	Vendor group	Quantity	Unit	Site	Warehouse
CurrentF	10/10/22	US-101		25	ea	1	11

There's an existing purchase order for vendor *US-101*, site *1*, warehouse *11*, a quantity of *25 ea*, and the date *10/10/22*.

When you run a master plan that uses *None* as the reduction method, it creates a planned purchase order for vendor *US-101*, site *1*, warehouse *11*, a quantity of *25 ea*, and the date *10/10/22*. In other words, the existing purchase order isn't reduced, because the forecast reduction method is *None*.

You edit the planned purchase order that was created after the last planning run, and change the quantity to *15 ea*. You then approve the order. The next time that you run the master plan, it creates a planned purchase order for vendor *US-101*, site *1*, warehouse *11*, a quantity of *10 ea*, and the date *10/10/22*. This time, the quantity is reduced to reflect the quantity of the existing approved order from the previous planning run.

Differences between Planning Optimization and the deprecated master planning engine

Supply forecasts work slightly differently, depending on the planning engine that you're using (Planning Optimization or the deprecated master planning engine). This section describes the differences.

Vendor groups

When you add a forecasted line, you can specify a vendor and a vendor group. In the deprecated master planning engine, planned orders that you create are grouped by the combination of the vendor and vendor group values. In Planning Optimization, planned orders are grouped by vendor.

The following table provides some examples of supply forecast lines for an item.

Expand table

Model	Date	Vendor account	Vendor group	Quantity	Unit	Site	Warehouse
CurrentF	10/10/22		VendorGroupA	5	ea	1	11
CurrentF	10/10/22		VendorGroupA	6	ea	1	11
CurrentF	10/10/22			7	ea	1	11

Vendor *VendorA* is the default vendor for vendor group *VendorGroupA*. It's also the default vendor for the item.

The deprecated master planning engine creates the following orders:

- A planned purchase order for vendor *VendorA*, vendor group *VendorGroupA*, and a quantity of *11*
- A planned purchase order for vendor *VendorA* and a quantity of *7*

Planning Optimization creates just one order:

- A planned purchase order for vendor *VendorA*, vendor group *VendorGroupA*, and a quantity of *18*

Reduction of general forecasts by more specific forecasts

In the deprecated master planning engine, the result is unpredictable if some forecasts have a vendor but others don't.

In Planning Optimization, more specific forecasts always reduce general forecasts, as the following example shows.

The following table provides some examples of supply forecast lines for an item.

Expand table

Date	Quantity	Vendor	Vendor group
02/11/2022	5.00	Vendor-A	VendorGroup-A
02/11/2022	6.00	Vendor-A	VendorGroup-A
02/11/2022	15.00		

The more specific forecasts (for 5.00 + 6.00 = 11.00 pieces) reduce the general forecast (for 15.00 pieces). The remainder is assigned to the default vendor. The following table shows the planned orders.

Expand table

Date	Quantity	Vendor	Vendor group
02/11/2022	11.00	Vendor-A	VendorGroup-A
02/11/2022	4.00	Vendor-A	VendorGroup-A

Respect for default order settings when planned orders are generated

Each item can have default order settings, such as a minimum purchase order quantity. The deprecated master planning engine ignores these settings, and therefore translates forecasts into planned orders that have the same quantity. Planning Optimization respects these settings when planned orders are generated from supply forecasts.

Aggregation of planned orders as a result of reduction by approved orders

The deprecated master planning engine assumes that only one order reduces the existing supply forecast. Therefore, if several orders match a supply forecast line, only the first order reduces it. In Planning Optimization, all orders that match the supply forecast line reduce it.

Reduction of forecasts by matching vendors only

When the deprecated master planning engine reduces a forecast by existing released purchase orders, it doesn't ensure that the vendor on the purchase order matches the vendor from the forecast. Planning Optimization reduces forecasts only by purchase orders that have a matching value in the vendor field.

For transfer and production orders, the vendor field is always ignored, because it isn't relevant for those order types.

Reduction of forecasts by transfer orders

If the default order type for an item is *Transfer*, forecasts can be reduced only by existing planned transfer orders. However, for production orders and purchase orders, only released orders reduce the supply forecast.

The deprecated master planning engine reduces forecasts for all transfer order states, whereas Planning Optimization reduces forecasts only by transfer orders that are in the *Released* state.

Last updated on 07/23/2026

Supply schedule

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Supply schedule** page shows a comprehensive overview of supply and demand for a product or product family. You can filter the information by location, master plan, and periods. Use the page to create new orders, modify existing planned orders, and run master planning.

Open the Supply schedule page

Open the **Supply schedule** page in any of the following ways:

- Go to **Master planning** > **Master planning** > **Supply schedule**. In the **Modify filter** dialog box, enter filter values in the fields to specify the plan and product that you're looking for. (In the rest of this article, the collection of filter values that you enter is referred to as "the filter" or "the current filter.") When you finish, select **OK**.
- Go to **Product information management** > **Products** > **Released products**. Select or open a product. Then, on the Action Pane, on the **Plan** tab, in the **View** group, select **Supply schedule**.
- Go to **Master planning** > **Setup** > **Demand forecasting** > **Item allocation keys**. Select an item allocation key that's used as a product family. Then, on the Action Pane, select **Supply schedule**.
- Go to **Master planning** > **Master planning** > **Planned orders**. Select or open a planned order. Then, on the Action Pane, on the **View** tab, in the **View** group, select **Supply schedule**.

> **Note**
>
> When you open the **Supply schedule** page from a product, product family, or planned order, you don't have to enter filter values. The system uses the default period template.

Use the Supply schedule page

The **Supply schedule** page consists of an upper section, the **Period end inventory** FastTab, an additional FastTab that becomes visible based on the line selected in the upper section, and the FactBox pane. To open the FactBox pane and view a FactBox, select **Related information** on the right edge of the page.

Upper section

The upper section of the **Supply schedule** page contains a grid that shows the following data for a product or product family. This data is broken down by periods defined by the **Period template** value from the filter.

- **Period start inventory** – This line shows the expected inventory balance at the start of the period if all orders in the system occur.
- **Period end inventory** – This line shows the expected inventory balance at the end of the period if all orders in the system occur.
- **Period end pegged inventory** – This line shows the amount of inventory at the end of the period that is pegged against future demand.
- **Period net supply** – This line shows the difference between supply and demand in the period.
- **Minimum inventory** – This node shows safety stock for a product or product family. To expand or collapse this node, select it, and then select **Expand** or **Collapse** on the toolbar. This node appears only when there's safety stock for the product or product family.
- **Demand** – This node shows demand for a product or product family. The demand is grouped by transaction type. To expand or collapse this node, select it, and then select **Expand** or **Collapse** on the toolbar. Demand transaction types include sales, transfers, and inventory journals. This node appears only when there's demand for the product or product family.
- **Supply** – This node shows supply for a product or product family. The supply is grouped by transaction type. To expand or collapse this node, select it, and then select **Expand** or **Collapse** on the toolbar. Supply transaction types include production, purchase, and transfers. This node appears only when there's supply for the product or product family.

Many of the available toolbar buttons, FactBox displays, and FastTab displays depend on your selections in the upper section. Typically, you choose a transaction type by selecting one of the rows under the **Supply** or **Demand** node. You then choose a period by selecting a specific column for the selected row.

The toolbar above the grid in the upper section of the **Supply schedule** page provides the following buttons:

- **Expand/Collapse** – Expand or collapse a selected node, such as the **Demand** node, the **Supply** node, and their subnodes. (Nodes show a **[+]** or **[-]** prefix to indicate whether they're currently collapsed or expanded.)
- **New** – Open a menu where each command, in turn, opens a dialog box or page that lets you add a specific type of item for the selection. Available commands include **Forecast supply**, **Planned order**, **Scheduled kanban**, **New production order**, **Purchase order**, and **Transfer order**.
- **Master planning** – Run master planning. A dialog box appears, where you can specify the plan to run. By default, the **Master plan** field is set to match the current filter.
- **Max. report as finished** – Open the **Max. report as finished** page for the product defined in the current filter.
- **Update planned orders** – Open the **Planned orders** page, which shows planned orders for the product or product family defined in the current filter.
- **Level** – Spread planned orders according to the settings from the dialog box that appears.
- **Material plan policy by location** – Open the **Item coverage** page for the product defined in the current filter.
- **Kanban rule** – Open the **Kanban rules** page for the product defined in the current filter.

FastTabs

The **Supply schedule** page can contain the following FastTabs:

- **Period end inventory** – This FastTab shows the period end inventory data in a graphical format.
- **Supply profile** – This FastTab shows the supply data in a graphical format. It becomes visible when you select a **Supply** node or a line below it in the upper section.
- **Summary** – This FastTab shows summary details that are related to the transaction type that you selected in the upper section. For example, if you select **Sales** under the **Demand** node, this FastTab shows fields that are related to sales orders, such as **Requested sales order lines** or **Total amount**. If you select **Production orders** under the **Production** subnode of the **Supply** node, this FastTab shows fields that are related to production orders, such as **Production orders scheduled** or **Total scheduled**. Field values depend on the period that you selected in the upper section.
- **[Transaction type] - [Period]** – This FastTab shows orders for the transaction type and period that you selected in the upper section. The name of the FastTab reflects those

selections. For example, it might be named **Sales orders - Backlog** or **Production orders - Week 37**.

Action Pane

The following buttons are available on the Action Pane:

- **Modify filter** – Open the **Modify filter** dialog box, where you can update filter values and then reopen the **Supply schedule** page, which reflects the updated filter settings.
- **Show delays** – Highlight the relevant lines in the upper section if there's a delayed order of that transaction type.

FactBox pane

To open the FactBox pane and view a FactBox, select **Related information** on the right edge of the page. The following FactBoxes are available on the **Supply schedule** page:

- **Item details** – This FactBox shows basic information about the product defined in the current filter. It's blank if you defined a product family in the filter.
- **Actions** – This FactBox shows actions for the orders of the transaction type that you selected in the upper section.
- **Delays** – This FactBox shows delays for the orders of the transaction type that you selected in the upper section.

Last updated on 05/29/2026

Run planning for a subset of items

Article • 09/29/2023

Master planning lets you set up filters to limit the set of items that are included in a planning run, so that not all items are planned.

You can use two types of filters to limit the set of items that is planned:

- *Plan filters*, which are set up on the master plan itself
- *Runtime filters*, where you use options in the **Planning Optimization** run dialog box to select the items and bill of materials (BOM) levels to include in the run

Apply a plan filter

Plan filters are useful when you want to limit a plan to a specific group of items and ensure that no other items are included as part of the resulting master planning. When a plan filter is defined, it's always applied during a master planning run that uses the plan.

> ⓘ **Important**
>
> Plan filters are available only in systems where Planning Optimization is enabled.

Follow these steps to set up a plan filter.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select a plan in the list pane, or create a new one.
3. On the Action Pane, select **Plan filter**.
4. A standard query editor dialog box appears. There, you can enter settings to limit the set of products that are included when the current plan runs. These settings define the plan filter.
5. When you've finished setting up your filter, select **OK**.

> ⓘ **Note**
>
> If you set up a plan filter on the plan that is selected as the **Current dynamic master plan** on the **Master planning parameters** page, that filter will be applied each time that the dynamic plan runs. For example, if the net requirements are updated for an item that isn't part of the plan filter, no result will be generated.

Apply a runtime filter and set the BOM levels to include

You set up the runtime filter for a job from the **Planning Optimization** run dialog box. The runtime filter applies only to the job that you start or schedule from the dialog box. If you both a plan filter and a runtime filter are set up, intersection logic is used to combine them. (See the example later in this article.)

You also use the **Planning Optimization** run dialog box to control the number of levels of BOM components and subcomponents that are included for manufactured items that pass the combined filters.

For a recurring job, the runtime filter and BOM levels that you define when you set up the job are applied each time that it runs.

Follow these steps to set up the runtime filter and BOM levels for a master planning batch job.

1. Go to **Master planning** > **Master planning** > **Run**.
2. In the **Planning Optimization** dialog box, on the **Parameters** FastTab, set the following settings fields:
    - **Master plan** – Select the master plan to run.
    - **Comment** – Add a comment to describe the purpose of the job.
3. On the **Records to include** FastTab, set the following fields:
    - **Filter** – Select this link to open a standard query editor dialog box where you can enter settings to limit the set of products that are included in the current planning job. These settings define the runtime filter.
    - **Include all BOM levels** – For manufactured items that have a BOM, it's often convenient to plan all BOM components together with the main item, to ensure that all the required components are available when production begins. Set this option to *Yes* to include all BOM levels for items that pass both the runtime filter and the plan filter. Set it to *No* to limit the number of BOM levels that are included. Then use the **BOM levels to include** field to specify the maximum number of BOM levels to include.
    - **BOM levels to include** – If **Include all BOM levels** is set to *No*, use this field to specify the number of BOM levels to include. Here are some examples:
        - Set this field to *0* (zero) to ignore all BOM components.
        - Set this field to *1* to include supply for BOM components.

- Set this field to *2* to include supply for BOM components and their derived BOM components.

4. On the **Run in the background** FastTab, set recurrence and other batch options in the usual way. The fields work just as they do for other types of background jobs in Microsoft Dynamics 365 Supply Chain Management.

> **Note**
>
> The following limitations apply when you use a runtime filter with BOM levels:
>
> - If you run auto-firming or another post-processing job using **the current extensibility point**, then the system will only apply the auto-firming and other actions to the items specified directly on the filter (not to the subcomponents).
> - Delays from subcomponents are not propagated to their respective parent items.
> - The requirement date of a subcomponent may fall on a date closed by the calendar of the subcomponent. To update the date to an open date on the subcomponent calendar, you must make a full or filtered run that includes the subcomponent directly on the filter.
>
> If any of these issues are critical for you, then add the relevant subcomponents to the filter manually.

Combine plan filters and runtime filters

If you apply both a plan filter and a runtime filter during a master planning run, only the intersection of the two filters will be included in the run.

To see how the plan and runtime filters are combined, consider this example. You have a plan where a plan filter is set up to include items A, B, and C. You then run master planning run several times. Each run uses the same plan but a different runtime filter. The results are as follows:

- **Run 1: The runtime filter includes item D** – No items are planned, because there is no intersection between the plan filter and the runtime filter.
- **Run 2: The runtime filter includes items A and D** – Only item A is included, because item D isn't part of the plan filter.

- **Run 3: The runtime filter includes item B** – Only item B is included, and the previous planning output for item A is kept.
- **Run 4: The runtime filter includes all items (blank filter)** – Items A, B, and C are included, and the previous planning output for items A and B is overwritten.

The system will also include all the relevant BOM components and subcomponents for the manufactured products that are found by the combined filters, as specified by the BOM level settings that you entered in the **Planning Optimization** run dialog box when you set up the job.

Feedback

Was this page helpful? Yes No

Provide product feedback

Master planning with purchase trade agreements

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how master planning can find the vendor and/or lead time for a planned order, based on the best price or lead time found among all purchase trade agreements that are specified for a given product.

Prepare your system to evaluate purchase trade agreements during master planning

Follow these steps to configure your system to apply master planning that evaluates purchase trade agreements.

1. Go to **Master planning** > **Setup** > **Master planning parameters**. On the **Planned orders** tab, in the **Vendor** section, set the following values:
    - **Find trade agreement** – Set this option to **Yes** to include purchase trade agreements in master planning.
    - **Search criterion** – Select the factor that you want to prioritize for each purchase trade agreement: **Minimum lead time** or **Lowest unit price**.
2. Go to **Procurement and sourcing** > **Setup** > **Prices and discounts** > **Activate price/discount**, and make sure that the **Vendor** option is set to **Yes**.
3. Go to **Product information management** > **Setup** > **Dimension and variant groups** > **Storage dimension groups**, and select a storage dimension group that applies to products that master planning should evaluate purchase trade agreements for. Make sure that each relevant storage dimension in this group has a check mark in the **For purchase prices** column. Repeat this step for every other relevant storage dimension group.

Prepare a released product to evaluate purchase trade agreements during master planning

After your system is prepared as described in the previous section, follow these steps to make sure that each product you want to use with this feature is correctly set up.

1. Go to **Product information management** > **Products** > **Released products**, and open a target product.

2. On the **Purchase** FastTab, make sure that no vendor is assigned in the **Vendor** field.

3. On the Action Pane, on the **Plan** tab, in the **Coverage** group, select **Item coverage** to open the **Item coverage** page for the selected product. Verify the following settings:

    - On the **General** tab, set up vendor overrides. If you want master planning to use purchase trade agreements to select a vendor, prevent vendor overrides by clearing the **Use specific setting** check box.
    - On the **Lead time** tab, set up lead-time overrides. If you want master planning to use purchase trade agreements to select lead times, prevent lead-time overrides. Clear the check box for each type of lead time that you want to select by using purchase trade agreements (**Purchase**, **Production**, and/or **Transfer**).

4. Close the **Item coverage** page to return to the details page for the selected product.

5. On the Action Pane, on the **Plan** tab, in the **Forecast** group, select **Supply forecast** to open the **Supply forecast** page. Make sure that no row that is shown here has a value in the **Vendor account** column.

6. Close the **Supply forecast** page to return to the details page for the selected product.

7. On the Action Pane, on the **Purchase** tab, in the **Trade agreements** group, select **View trade agreements**. Make sure that all the relevant purchase trade agreements are listed. Also make sure that the **Disregard lead time** option is set to **No** for each agreement where you want master planning to use the lead time that is specified for that agreement.

8. On the Action Pane, on the **Plan** tab, in the **Order settings** group, select **Default order settings** to open the **Default order settings** page for the selected product. On the **Purchase order** FastTab, view the value of the **Purchase lead time** field. If no item coverage lead-time override is defined, master planning uses this value when it selects trade agreements where the **Disregard lead time** option is set to **Yes**. Therefore, adjust this value as you require.

9. Repeat this procedure for each relevant product.

> **Note**
>
> Master planning supports multi-currency purchase trade agreements. When searching for a trade agreement by using the **Lowest unit price** option, the system considers purchase trade agreement lines with different currencies, provided an exchange rate is defined between the trade agreement line currency and the accounting currency of the legal entity. Otherwise, the the trade agreement line is ignored, and you see an error during master planning. Therefore, master planning includes information from all relevant purchase trade agreement lines where prices can be converted to the accounting currency. It's important to note that rounding rules aren't taken into account during the trade agreement line price conversion.

Examples of how master planning finds vendor and lead times

The following table provides examples that show how various settings for a released product and its associated purchase trade agreements affect the values that are found for the resulting planned purchase order. The **bold** values in the two rightmost columns are the values selected by master planning. The ***bold and italic*** values in the other columns are the settings that produce those resulting values for each row.

Expand table

Released product: Vendor	Default order settings: Lead time	Item coverage: Override vendor	Item coverage: Override lead time	Trade agreement: Vendor	Trade agreement: Lead time	Trade agreement: Disregard lead time	Resulting vendor	Resulting lead time
US001	***1***	No	No	US003	3	No	**US001**	**1**
US001	1	***Yes: US002***	***Yes: 2***	US003	3	No	**US002**	**2**
(Blank)	1	No	No	***US003***	***3***	No	**US003**	**3**
(Blank)	***1***	No	No	***US003***	3	Yes	**US003**	**1**
(Blank)	***1***	***Yes: US002***	No	US003	3	No	**US002**	**1**
(Blank)	***1***	***Yes: US002***	No	US003	3	No	**US002**	**1**
(Blank)	1	No	Yes: 2	***US003***	***3***	No	**US003**	**3**

Released product: Vendor	Default order settings: Lead time	Item coverage: Override vendor	Item coverage: Override lead time	Trade agreement: Vendor	Trade agreement: Lead time	Trade agreement: Disregard lead time	Resulting vendor	Resulting lead time
(Blank)	1	No	***Yes: 2***	***US003***	3	Yes	**US003**	**2**

Related information

- Purchase agreements

Last updated on 05/29/2026

Plan based on quotations and RFQs

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Quotations and requests for quotations (RFQs) represent possible or even likely future orders. Therefore, it makes sense to consider them during master planning, so that extra supply can be planned based on existing RFQs and the likelihood that each quotation will become an actual order. This article describes how to set up master planning to consider quotations and RFQs when it generates planned orders.

Set up master planning to consider sales quotations and/or RFQs

Use the following procedure to set up master planning to consider sales quotations and/or RFQs.

1. Go to **Master Planning** > **Setup** > **Plans** > **Master plans**.
2. Select an existing plan in the list pane, or select **New** on the Action Pane to create a new one.
3. On the **General** FastTab, set the following fields:
    - **Include sales quotations** – Set this option to *Yes* to consider sales quotations when the current plan is run. Set it to *No* to ignore them.
    - **Probability** % – Set the minimum level of confidence that is required for a quotation to be included in master planning. The master planning calculation will include all quotations that were created from opportunities that have this probability percentage or higher. To include all quotations, including quotations that have no probability assigned to them or no opportunity associated with them, set this field to *0* (zero). For more information about probabilities for quotations, see the next section.
    - **Include requests for quotations** – Set this option to *Yes* to consider RFQs when the current plan is run. Set it to *No* to ignore them. If you choose to consider RFQs, the system will create planned purchase orders for them, but it won't specify the vendor

until the RFQ is resolved. Planned purchase orders that are created for the RFQs might affect the quantities of other planned orders.

4. Continue to set up your master plan in the usual way.

Assign and view probabilities for quotations

As was mentioned in the previous section, a master plan will consider only quotations that meet or exceed the probability threshold that is defined for the plan (if a threshold is defined). However, the probability isn't set directly on each quotation. Instead, it's inherited from the opportunity that was used to generate the quotation. Therefore, quotations that are created directly on the **All quotations** page won't have a probability assigned to them or an opportunity associated with them, and they won't ever be considered by master planning (unless the **Probability %** field is set to *0* [zero]). To have a probability assigned to it, a quotation must be generated from an opportunity.

Create a quotation from an opportunity

Use the following procedure to assign a probability to an opportunity and then create a quotation from that opportunity.

1. Go to **Sales and marketing** > **Relationships** > **Opportunities** > **All opportunities**.
2. Select an existing opportunity, or select **New** on the Action Pane to create a new one.
3. Fill in the opportunity page to identify the opportunity as you want. Be sure to set the **Probability** field to an appropriate value. Only master plans that are set up to look for probabilities of this value or higher will consider quotations that are generated from this opportunity.
4. On the Action Pane, select **Save**.
5. On the Action Pane, on the **Opportunity** tab, in the **New** group, select **Sales quotation** or **Project quotation**, depending on the type of quotation that you want to create.
6. In the **Create quotation** dialog box, set the fields as you require, and then select **OK**.
7. A new quotation is created and opened. On the **Lines** FastTab, use the toolbar to add sales lines or project lines as required to define the content of the quotation.
8. On the Action Pane, select **Save**. Then close the quotation.

View the probability that is assigned to a quotation

The only way to view the probability that is assigned to a quotation is to open the opportunity that was used to create that quotation, as described in the following procedure.

1. Go to **Sales and marketing** > **Sales quotations** > **All quotations**.

2. If the **Opportunity ID** column isn't shown (it's hidden in a default installation), follow these steps to temporarily show it. (Alternatively, to keep the **Opportunity ID** column available, create a saved view that includes it.)
   a. Select and hold (or right-click) in the grid, and then select **Insert columns**.
   b. In the **Insert columns** dialog box, find the row where the **Field** field is set to *Opportunity*, and select the **Select** checkbox for that row.
   c. Select **Update** to add the selected column to the **All quotations** page.

3. In the grid, find the row for the relevant quotation. Then, in the **Opportunity ID** column, select the value for that row.

   > **Note**
   >
   > If you're looking for a project quotation, you might have to select the **Quotation type** column header and clear its filter. In a default installation, this filter is set so that only sales quotations are shown.

4. The related opportunity is opened. You can now view and edit the **Probability** value as you require.

Last updated on 06/17/2025

Safety margins

This article describes how safety margins work during master planning.

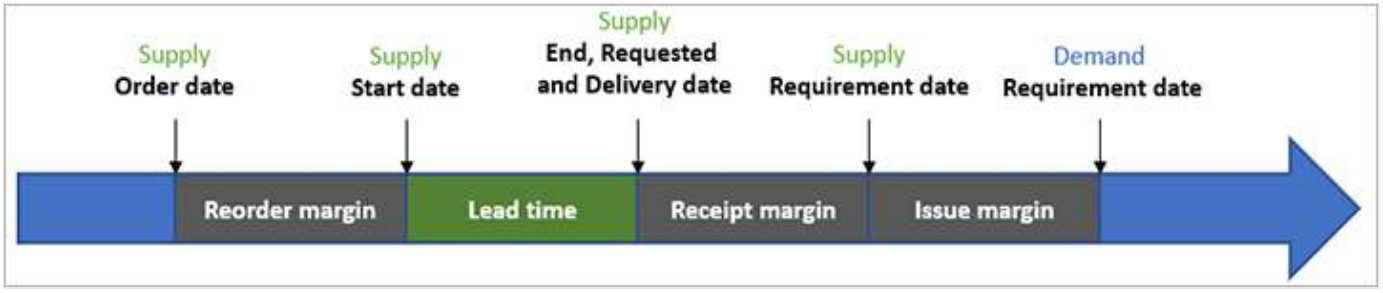
Safety margins overview

Safety margins provide buffer time beyond the normal lead time. For example, you might need to unpack or inspect material after it arrives from the vendor. You can't just add the extra time to the purchase lead time, because that approach gives extra buffer time to the supplier. Instead, use the receipt margin to ensure that the supplier delivers earlier. The receipt margin provides buffer time so that you can handle the goods internally.

Three types of safety margins exist:

- **Reorder margin** – The buffer time for placing the supply order
- **Receipt margin** – The buffer time for handling incoming supply
- **Issue margin** – The buffer time for handling shipments

The following illustration shows how these safety margins apply over time.

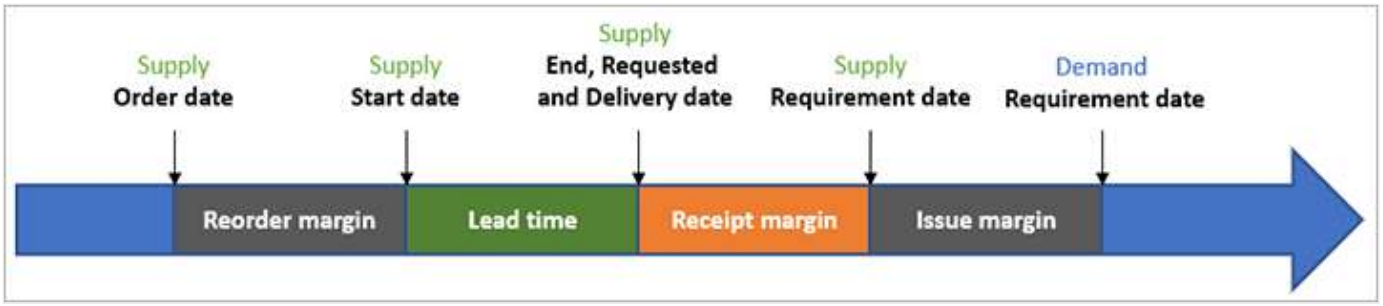


All margins are defined in days. The default value, *0* (zero), indicates that no margin is applied. If you set up multiple margins, they all add to the total time from the supply *order date* to the demand *requirement date*. For example, a setup has no lead time, and all three margin types are set to one day. In this case, three days exist between the supply order date and the demand requirement date, so if the order date is July 1, the requirement date is July 4.

Receipt margin

The receipt margin is probably the most commonly used of the three safety margins. The system applies it to the *delivery date* and calculates backward from the *requirement date*. In other words, you should receive the products the specified number of receipt margin days before you need them.

The following illustration highlights the receipt margin.



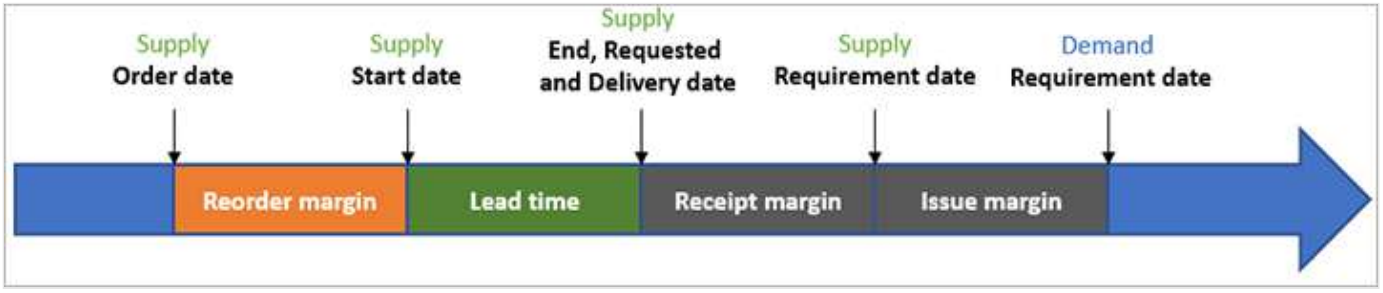
The receipt margin typically acts as a buffer to ensure time for warehouse registration or other time-consuming processes that the system doesn't capture as part of the general lead time. For purchases, one benefit is that the *delivery date* of the purchase order moves forward accordingly. If you increase the lead time instead of using a safety margin, the vendor still needs to deliver at the last minute.

The receipt margin doesn't change the *requirement date* of the supply. Therefore, you don't directly see the receipt margin when you compare requirement dates for demand and supply (for example, on the **Net requirements** page). For example, say the receipt margin is four days and a purchase order line is planned for receipt on the 15th of the month. Master planning calculates the adjusted receipt date as the 19th of the month.

A receipt margin isn't applied when on-hand inventory is used as the supply. All on-hand inventory is assumed to be available immediately, regardless of when you actually received it.

Reorder margin

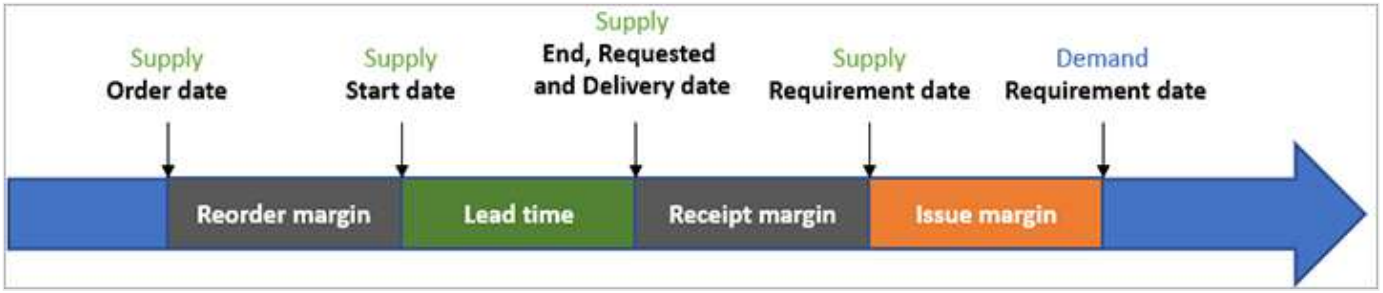
The following illustration highlights the reorder margin.



The reorder margin adds time before the item lead time for all planned orders during master planning. Therefore, it ensures extra time for placing a supply order. This margin typically acts as a buffer to ensure time for approval processes or other internal processes that are required during the creation of supply orders. The reorder margin goes between the supply *order date* and *start date*.

Issue margin

The following illustration highlights the issue margin.



During master planning, the system deducts the issue margin from the demand requirement date. This deduction helps ensure that you have time to react to and ship incoming demand orders. This margin acts as a buffer to ensure time for shipment and related outbound warehouse processes.

When you apply an issue margin, the related supply and demand requirement dates don't match. Instead, they differ by the issue margin, because the issue margin is added between the supply *requirement date* and the demand *requirement date*.

Set up safety margins

Define safety margins

Set safety margins on both the *coverage group* and the *master plan*. The system adds these margins together. For example, if you set a receipt margin of two days on the coverage group and three days on the master plan, the effective receipt margin is five days.

Set the margin on the master plan when you want to simulate longer lead times or uncertainty for a specific plan, but without affecting the daily planning.

Coverage group safety margins

To apply a safety margin to a coverage group, follow these steps:

1. Go to **Master planning** > **Setup** > **Coverage groups**.
2. In the list pane, select the coverage group.
3. On the **Other** FastTab, in the **Safety margins in days** section, use the following fields to set the required safety margins (in days):

- Receipt margin added to requirement date
- Issue margin deducted from requirement date
- Reorder margin added to item lead time

Master plan safety margins

To apply a safety margin to a master plan, follow these steps:

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. In the list pane, select the master plan.
3. On the **Safety margins in days** FastTab, use the following fields to set the required safety margins (in days):
    - Receipt margin added to requirement date
    - Issue margin deducted from requirement date
    - Reorder margin added to item lead time

Define whether calculations are based on calendar days or workdays

Set all safety margins to calculate based on either calendar days or workdays.

1. Go to **Master planning** > **Setup** > **Master planning parameters**.
2. On the **General** tab, in the **Safety margins in days** section, set the **Working days** option to *Yes* to calculate margins based on working days. Set the option to *No* to calculate margins based on calendar days.

For example, a calendar is open from Monday through Friday and closed from Saturday through Sunday. If there's a receipt margin of one day, a requirement date on a Monday produces a delivery date on the previous Friday, because Saturday and Sunday aren't working days.

The calendar that determines the working days depends on the setup and the supply type. The calendars of the coverage group, the warehouse, and the vendor control which days count as working days.

> ⓘ **Note**
>
> If *warehouse* isn't part of the coverage dimension (in other words, planning is based only on *site*), the warehouse calendar isn't used.

The system can handle a setup where one or more calendars are defined. The following subsections describe the possible combinations that you can use to control the result.

Calendar that is used for the duration

The defined calendars control the actual total lead time in calendar days, from the supply order date to the demand requirement date. The system uses the following calendar prioritization:

- **Purchase lead time** – Only the coverage group calendar is considered.
- **Receipt margin** – If a coverage group calendar is defined, the system uses it. Otherwise, the system uses the warehouse calendar.
- **Issue margin** – If a coverage group calendar is defined, the system uses it. Otherwise, the system uses the warehouse calendar.
- **Order margin** – Only the coverage group calendar is considered.

Calendar that is used for the final date

The following rules determine whether the planning engine can use a given date for a given date type:

- **Purchase receipt date** – If a vendor calendar is defined, the system uses it. Otherwise, if a coverage group calendar is defined, the system uses that calendar. If neither calendar is defined, the system uses the warehouse calendar.
- **Transfer receipt date** – If a coverage group calendar is defined, the system uses it. Otherwise, the system uses the warehouse calendar.
- **Production receipt date** – If a coverage group calendar is defined, the system uses it. Otherwise, the system uses the warehouse calendar.
- **Demand issue open day** – If a warehouse calendar is defined, the system uses it. Otherwise, the system uses the coverage group calendar.
- **Order open day** – The system uses a combination (intersection) of the coverage group calendar and the vendor calendar. Both calendars must be open to use the date. If only one calendar is defined, the system uses that calendar alone.

Calendar setup overview matrix

The following illustration presents a matrix that summarizes which calendars apply when safety margins are calculated. (Select the image to open a high-resolution version of it.) The following abbreviations and colors indicate where each type of calendar is specified:

- **Coverage group (CG):** Green

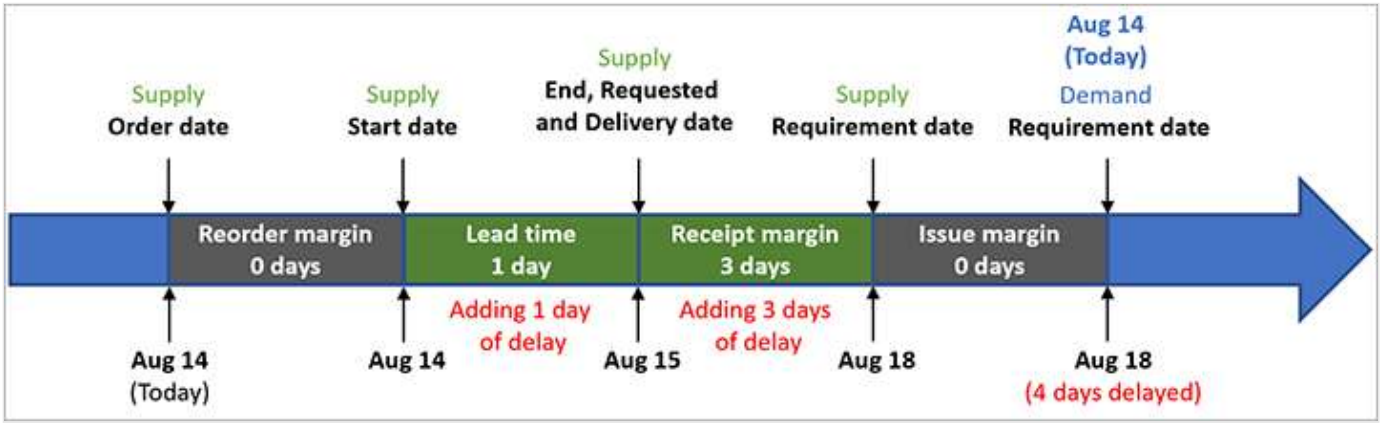
- **Warehouse (WH):** Yellow
- **Vendor (V):** Blue

Calendar is specified on			Calendar used for duration Only counting open days				Calendar used for final date Is the date open or closed				
Coverage group (CG)	Warehouse (WH)	Vendor (V)	Lead time	Receipt margin	Issue margin	Order margin	Purchase requirement date	Transfer requirement date	Production requirement date	Demand requirement date	Supply Order date
Yes	Yes	Yes	CG	CG	CG	CG	V	CG	CG	WH	CG+V
No	Yes	Yes	None	WH	WH	None	V	WH	WH	WH	V
Yes	No	Yes	CG	CG	CG	CG	V	CG	CG	CG	CG+V
Yes	Yes	No	CG	CG	CG	CG	CG	CG	CG	WH	CG
Yes	No	No	CG	CG	CG	CG	CG	CG	CG	CG	CG
No	Yes	No	None	WH	WH	None	WH	WH	WH	WH	None
No	No	Yes	None	None	None	None	V	None	None	None	V
No	No	No	None	None	None	None	None	None	None	None	None

Calculating delays

The system includes all three types of safety margins when it determines whether an order is delayed.

For example, an item has a lead time of one day and a receipt margin of three days. A sales order for this item is set as required today. In this case, the delay is calculated as *lead time* + *receipt margin* = four days. Therefore, if today is August 14, the four days of delay produces a delivery on August 18. The following illustration shows this example.



Issue margin and on-hand

Sometimes, the system doesn't apply the issue margin to an order when on-hand supply exists for an item. The reason for this behavior is that the on-hand supply doesn't have a date, so the system can't apply the issue margin.

This situation usually occurs when you sell an item that has an issue margin from a warehouse such as WH11. A transfer order from another warehouse such as WH13 replenishes the selling warehouse. In this case, one of the following situations could apply:

- **If on-hand supply exists in WH13** – On WH11, the system applies the margin between the sales order date and the transfer order receipt date. However, on WH13, the date of the transfer order shipment is the same as the receipt date because there's on-hand supply, so the system doesn't apply any issue margin.

- **If on-hand supply doesn't exist in WH13** – If there's no on-hand supply, and WH13 is to be replenished by other means (such as a purchase order), then the system applies the issue margin between the transfer order receipt date and the purchase order receipt date.

Soft issue margin

Soft issue margin causes the system to treat the issue margin as a preferred buffer rather than a hard requirement. Instead of always applying the full issue margin, a soft margin lets the system reduce it as needed to avoid pushing the requirement date into the future. Without soft issue margin, the system always applies the full margin, even when doing so delays a demand order.

The soft issue margin works as follows:

- *If the full issue margin can be applied without causing a delay* – The system applies the full issue margin, just as it does with the standard behavior. No change occurs.
- *If the full issue margin would cause a delay* – The system reduces the issue margin to the largest value that doesn't push the planned requirement date past the demand requirement date. The margin can be reduced to any value between zero days and its full configured value minus one day.
- *If the order is already delayed (the demand requirement date is in the past)* – The system ignores the issue margin entirely. The issue margin isn't included in the delay calculation. Instead, the difference between the demand requirement date and the planned order date is surfaced to indicate that the order can't be fulfilled on time.

This behavior applies independently at each level in the supply chain. If a bill of materials (BOM) structure includes both a finished good and a raw material, the system evaluates and adjusts the issue margin separately for each level.

Soft issue margin prerequisites

To use a soft issue margin, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.48 build 10.39.1837 or later.

- The feature named *Soft issue margins with Planning Optimization* must be turned on in feature management.

Enable soft issue margin for a coverage group

You enable soft issue margin at the coverage group level. To enable the soft issue margin for a coverage group, follow these steps:

1. Go to **Master planning** > **Setup** > **Coverage groups**.
2. Select the coverage group.
3. On the **Other** FastTab, set the **Soft issue margin** option to *Yes*.

Example 1: Issue margin is reduced to avoid delay

The following conditions apply in this example:

- The issue margin is 10 days on all items.
- Soft issue margin is enabled.
- Today's date is October 1.
- The **Add the calculated delay to the requirement date for production orders** option is disabled.

The demand requirement date is October 7. Without soft issue margin, the system applies the full 10-day issue margin at both the finished good level and the raw material level. Because only six days are available between today's date and the requirement date, the full margin causes a delay at both levels.

With the soft issue margin enabled, the system reduces the issue margin at both levels to the largest values that don't push the planned dates past the requirement date. The orders are planned without delay.

Example 2: Order already delayed, issue margin is ignored

The same conditions apply as in the previous example.

The demand requirement date is September 24. Without the soft issue margin, the system applies the full 10-day issue margin at both the finished good level and the raw material level, which further increases the delay.

With soft issue margin enabled, the system ignores the issue margin entirely at both levels because the demand requirement date is already in the past. The delay is surfaced as the

difference between the demand requirement date (September 24) and the earliest possible planned order date (October 1). The issue margin doesn't add to the delay.

Greedy application across multilevel supply chains

The soft issue margin is applied *greedily* along the supply chain, starting from the demand side. The first level in the chain (closest to the original demand) consumes as much of the available issue margin as possible. Subsequent levels (further from the demand, closer to the supply source) receive whatever margin remains.

For example, consider a supply chain where warehouse *WH12* fulfills a sales order at warehouse *WH11* through a transfer, and a purchase order replenishes *WH12*. If you configure the soft issue margin for the item at both warehouses, the system first allocates as much issue margin as possible to the issue from *WH11*, the warehouse closest to the customer demand. The issue from *WH12*, which is further upstream in the supply chain, then receives whatever margin is still available without causing a delay.

Greedy allocation means that downstream operations (closer to the customer) are prioritized for margin, while upstream operations absorb the reduction.

Known limitations of soft issue margin

The greedy application of soft issue margins can lead to situations where the requirement dates set by the engine are too optimistic. When the engine applies the soft issue margin date, it doesn't yet know about potential delays that could occur further up the supply chain. Once the system encounters upstream delays and factors them into the plan, requirement dates aren't updated to account for delays compressing the soft issue margin.

Greedy application of soft issue margins can lead to situations where the requirement date of a planned supply is set too early. After the system accounts for delays, it shows delay days even though only the soft issue margin was compressed instead of delaying the demand.

Example of known limitation

The following conditions apply in this example:

- Today's date is *May 18*.
- An item *P1* exists that is replenished by production orders.
- To produce item *P1*, a raw material *R1* is required, which is replenished by purchase orders with a purchase lead time of three days.

- The soft issue margin is set to two days on *P1* and *R1*.

A sales order for item *P1* has a requirement date of *May 22*. A soft issue margin of two days is configured on *P1*. The system greedily applies the full issue margin and schedules a production order with a requirement date of *May 20*. The production order requires raw material *R1*, which also has a two-day issue margin and is scheduled with a requirement date of *May 18*.

However, the purchase order for *R1* has a purchase lead time of three days. This lead time causes a three-day delay on the purchase order, which adjusts the planned date to *May 21*. The delay propagates up to the production order, which also adjusts to *May 21*. Because soft issue margin is enabled, the issue margin at the production level is compressed and doesn't add further delay. The delay also propagates to the sales order level. However, the system compresses the soft issue margin from two days to one day, so the sales order isn't delayed beyond *May 22*.

After delays are resolved, a discrepancy exists between the requirement dates and the planned dates. The requirement dates that were set before the delays were known are now too optimistic. The system doesn't retroactively adjust these requirement dates, which can result in delay days being shown even though the supply isn't delayed according to the soft issue margin. This behavior is by design and a known limitation of the soft issue margin feature.

Impact of known limitation

If delays occur and the system doesn't update the requirement dates, several other features that rely on requirement dates and delays might also be affected. For example:

1. Delay days
2. Actions
3. BOM/route selection
4. Trade agreements

Make sure to test this feature thoroughly to ensure that your intended use of soft issue margins is not affected by the known limitations.

Related information

- Calendars and master planning
- Calculate requested ship dates for purchase orders
- Improve scheduling engine performance
- Date and time parameters used by Planning Optimization

Last updated on 07/27/2026

Delay tolerance (negative days)

10/03/2025

The delay tolerance functionality enables Planning Optimization to consider the **Negative days** value that is set for coverage groups, item coverage, and/or master plans. It's used to extend the delay tolerance period that is applied during master planning. In this way, you can avoid creating new supply orders if existing supply will be able to cover the demand after a short delay. The purpose of the functionality is to determine whether it makes sense to create a new supply order for a given demand.

Delay tolerance in Planning Optimization

Delay tolerance represents the number of days beyond the lead time that you're willing to wait before you order new replenishment when existing supply is already planned. Delay tolerance is defined by using calendar days, not business days.

At the time of master planning, when the system calculates the delay tolerance, it considers the **Negative days** setting. You can set the **Negative days** value on the **Coverage groups** page, the **Item coverage** page, or the **Master plans** page. If negative days are assigned at more than one level, the system applies the following hierarchy to decide which setting to use:

- If negative days are enabled on the **Master plans** page, that setting overrides all other negative days settings when the plan runs.
- If negative days are configured on the **Item coverage** page, that setting overrides the coverage group setting.
- Negative days that are configured on the **Coverage groups** page apply only if negative days haven't been configured for a relevant item or plan.

The system links the delay tolerance calculation to the *earliest replenishment date*, which equals today's date plus the lead time. The delay tolerance is calculated by using the following formula, where *max()* finds the larger of two values:

max(Earliest replenishment date, Demand due date) – Demand due date + Negative days

This formula ensures that master planning doesn't create new supply orders when enough supply exists during the product lead time.

> ⓘ **Note**
>
> The delay tolerance calculation in Planning Optimization always uses the dynamic negative days calculation from the deprecated master planning engine. The **Use dynamic**

negative days setting on the **Master planning parameters** page has no effect on this behavior.

If the existing supply implies a demand delay that is less than or equal to the calculated delay tolerance, Planning Optimization pegs existing supply with the demand. In some cases, it's better to delay the demand than to end up with oversupply.

The following subsections provide examples that show how delay tolerance affects the creation of planned orders in Planning Optimization.

Example 1

A product has the following configuration:

- **Lead time:** *10*
- **Negative days:** *2*

The following supply and demand exist for the product:

- **Demand for today:** A sales order for a quantity of 10
- **Supply in 12 days:** A purchase order for a quantity of 10

Existing supply can cover the demand within 12 days, and that period equals the delay tolerance. Therefore, when master planning runs, no planned orders are created.

Example 2

A product has the following configuration:

- **Lead time:** *10*
- **Negative days:** *0*

The following supply and demand exist for the product:

- **Demand for today:** A sales order for a quantity of 10
- **Supply in 12 days:** A purchase order for a quantity of 10

Existing supply can cover the demand only after 12 days. However, the delay tolerance is 10 days. Therefore, when master planning runs, a planned order for a quantity of 10 is created.

Example 3

A product has the following configuration:

- **Lead time:** *10*
- **Negative days:** *2*

The following supply and demand exist for the product:

- **Demand in 11 days:** A sales order for a quantity of 10
- **Supply in 14 days:** A purchase order for a quantity of 10

Existing supply can cover the demand only after three days. However, the delay tolerance is two days. (In this case, the delay tolerance includes only the two negative days. The demand date isn't included because it's after the product lead time.) Therefore, when master planning runs, a planned order for a quantity of 10 is created.

Purchase requisitions

Master planning can replenish approved purchase requisitions. Therefore, to cover purchase requisitions, users don't have to use a workflow to create purchase orders. Instead, purchase requisitions can be covered by master planning. Because of this functionality, a purchase requisition can produce a purchase order, a transfer order, or a production order, depending on the **Planned order type** value that is set for the related product.

Enable master plans to include requisitions

To include requisitions during the coverage calculation for a master plan, follow these steps:

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Create or select a master plan.
3. On the **General** FastTab, set the **Include requisitions** option to *Yes*.
4. Repeat steps 2 and 3 for each additional master plan where you want to include requisitions.

Approved requisitions time fence

The *approved requisitions time fence* establishes how far back (in days) a master plan will include demand from approved replenishment requisitions. You can set an approved requisitions time fence at both the coverage group level and the master plan level.

Set the approved requisitions time fence for a coverage group

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. Create or select a coverage group.
3. On the **Other** FastTab, set the **Approved requisitions time fence (days)** field to the number of days to include in the time fence.
4. Repeat steps 2 and 3 for each additional coverage group where you want to set an approved requisitions time fence.

Set the approved requisitions time fence for individual master plans

When you set an approved requisitions time fence for an individual master plan, the setting overrides the time fence setting for any applicable coverage group.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Create or select a master plan.
3. On the **TIme fences in days** FastTab, set the **Approved requisitions time fence (days)** field to the number of days to include in the time fence.
4. Repeat steps 2 and 3 for each additional master plan where you want to set an approved requisitions time fence.

> ⓘ **Important**
>
> Approved requisitions time fences aren't supported for Planning Optimization. Until they are supported, all values that you enter in the **Approved requisitions time fence (days)** field will be ignored.

Independent supply, regardless of coverage code

Purchase requisitions are always covered by independent planned orders, regardless of the coverage code. This behavior ensures clear traceability and workflows between purchase requisitions and replenishment orders.

Example 1

A product is set up so that it has a **Coverage code** value of *Min/max*. It has the following inventory and requisition statuses:

- Inventory on-hand quantity = 10.
- Minimum inventory quantity = 15.
- Maximum inventory quantity = 20.
- A purchase requisition for one piece exists. It has a requested date of today.

When master planning runs, two planned orders are created: one for 10 pieces to replenish inventory to the maximum quantity, and one for one piece to replenish the purchase requisition.

Example 2

A product is set up so that it has a **Coverage code** value of *Min/max*. It has the following inventory and requisition statuses:

- Inventory on-hand quantity = 17.
- Minimum inventory quantity = 15.
- Maximum inventory quantity = 20.
- A purchase requisition for one piece exists. It has a requested date of today.

When master planning runs, one planned order for one piece is created to replenish the purchase requisition.

Example 3

A product is set up so that it has a **Coverage code** value of *Period* and a period length of seven days. It has the following inventory, sales order, and requisition statuses:

- Inventory on-hand quantity = 0.
- A sales order for five pieces exists. It has an expected ship date of today plus one day.
- A purchase requisition for three pieces exists. It has a requested date of today plus three days.

When master planning runs, two planned orders are created: one for three pieces to replenish the purchase requisition and one for five pieces to replenish sales order demand.

> **Note**
>
> After a planned order that is pegged to a purchase requisition is firmed, the planning engine keeps the pegging to the purchase requisition. If the firmed order is later found to be missing some quantity that is required to fulfill the purchase requisition, the system will create a new planned order for the difference.

For more information about purchase requisitions, see Purchase requisition overview.

Last updated on 12/30/2025

Production planning

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The following video provides a short introduction to some of the concepts discussed in this article: Dynamics 365 Supply Chain Management: Planning Optimization enhancements.

Planned production orders

When master planning creates planned orders to fulfill requirements, the value of the **Planned order type** field determines the order type. If you set the **Planned order type** field to *Production*, the system creates planned production orders. These planned production orders include information about the active bill of materials (BOM) and the route ID from the related production setup.

Requirements from BOMs

Master planning uses BOM information. The plan output includes material supply to cover related material demand for production.

During master planning, the system uses the current, active BOM to determine the materials required for production. The system checks all levels of the BOM structure that relate to the required production order. The system fulfills material requirements by using available on-hand inventory, existing on-order supply, and approved planned orders. If the system finds that additional material is required, it creates a planned order to cover the demand.

When the system searches for an appropriate BOM version to use for an item in a coverage group that has the **Use the specified BOM or formula version** option turned off, it takes several parameters into account. These parameters include the validity period of the BOM version, whether the BOM version is active or approved, and the applicable quantity. The system doesn't reconsider the chosen BOM version after a receipt is scheduled in cases where a planned production order is delayed or scheduled to start earlier or later.

When you replenish a BOM component by using a production or batch order, the validity date that the system uses to search for the applicable BOM or formula version is determined after the system schedules its parent order. The validity date is based on the scheduled start or end date of an operation from the parent planned production order schedule (by default, the start date of the first operation). The system determines the scheduled start and end dates after it schedules the parent production order. If the system delays the parent planned production order later in the planning process and must push forward its scheduled start or end date times because a subcomponent order is delayed, the system doesn't reconsider the BOM version for the subcomponent. The BOM version is still based on the original scheduling attempt (backward in most cases) of the parent planned production order.

Individual BOM lines also have validity periods. The system evaluates these periods after it schedules a planned production order, so it uses the corresponding operation start or end dates to check for validity. Even though a BOM version and its lines might have the same validity periods, the system uses different dates to evaluate them. For a BOM version, the system uses the demand requirement date. For a BOM line, the system uses the scheduled start of the planned production order for the finished good.

The system takes the following actions:

1. Create a planned production order to replenish demand and select a BOM version to use.
2. Schedule the order without reevaluating the BOM version, even if the date shifts as a result of scheduling.
3. Assess the BOM lines for the selected BOM version to ensure they're valid for the scheduled dates.
4. Create BOM line requirements for the valid BOM lines only.
5. If required by the planning settings, cover BOM line requirements with supply down to the specific BOM level.
6. Evaluate whether supply of the lower BOM levels affects the delivery schedule for any parent production orders and adjust the schedules accordingly.

Scheduling during firming

Planned production orders include the route ID that is required for production scheduling. However, scheduling support during the planning run for planned orders is pending. The route ID is used to schedule planned production orders during firming. Therefore, the lead time on planned production orders can differ from the lead time on related scheduled, firmed production orders that are generated from them.

- **Planned production order** – The lead time is based on the static lead time from the released product.
- **Firmed production order** – The lead time is based on scheduling that uses route information and related resource constraints.

Delays

If the lead time for required material is longer than the period between today's date and the material requirement date, the planned order for the required material and the related production order are delayed. For planned orders, the system calculates the delay (in days) based on the lead time from the released product. The system then propagates the delay information through all levels of the BOM structure. Therefore, you can follow the impact of delayed raw material all the way to the customer sales order.

Modifying planned orders

When you change information on a planned order, you receive the following message: "Note that the impact of manual changes on planned orders won't be reflected in the rest of the plan until the next master planning run."

If you want to change information on a planned order and see the impact on the related material requirements, follow these steps:

1. Update the planned order.
2. Approve the planned order.
3. Run master planning.

> ⓘ **Note**
>
> If you change the delivery date of the planned order to a later date, the demand might peg against a new planned order. This behavior occurs when the new supply date causes a delay for the pegged demand but, according to the lead time settings, the delay can be avoided.

Explosion page

Use the **Explosion** page to analyze the demand that is required for a specific production order or planned production order, the related coverage, and pegging information. The system updates

information on the **Explosion** page during master planning. You can't update the information directly from the **Explosion** page.

Filters

Planning Optimization offers advanced filtering options that can apply at the plan level, runtime level, or both. To learn more about the various filter options, see Run planning for a subset of items.

Last updated on 05/06/2026

Item substitution for formulas and bills of materials

07/21/2025

You can use substitute materials during planning for items that use formulas or bills of materials (BOMs). When you set up item substitution for a formula or BOM, you can specify a group of items that can serve as substitutes for a particular material. You can also specify the priority of each substitute material in the group. When master planning runs, it uses the preferred material if it's available. If the preferred material isn't available, master planning uses the substitute material that has the highest priority and is available. If no substitute materials are available, master planning creates a planned order for the preferred material.

Prerequisites

Before you can use substitution in planning for formulas, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.39 or later.
- The feature that's named *Item substitution (Plan group) support for Planning Optimization* must be turned on in Feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default.

Before you can use substitution in planning for BOMs, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.40 or later.
- The feature that's named *Item substitution for bill of materials in Planning optimization* must be turned on in Feature management.

Set up plan groups

Before you can set up item substitution for formulas or BOMs, you must set up the plan groups that you need.

A plan group is a group of items that can be substituted for a particular material in a formula or BOM. Each plan group record establishes a name and a description, but it doesn't include any items. You assign items and their relative priorities to each plan group when you set up item substitution for a specific formula or BOM. You can reuse plan groups across multiple formulas and/or BOMs.

To set up plan groups, follow these steps.

1. Go to **Product information management** > **Setup** > **Bills of materials and formulas** > **Plan groups**.
2. Use the buttons on the Action Pane to add, remove, or edit plan groups as required. For each plan group, set the following fields:
    - **Plan group** – Specify a unique name for the plan group.
    - **Description** – Enter a short description of the plan group.
3. Continue to work until you finish setting up all the plan groups that you need.
4. On the Action Pane, select **Save**.

Item substitution for formulas

Example scenario: Substituting ingredients in a formula

A food manufacturing company is making an item from a formula that has 20 ingredients. One of the ingredients in the formula can be replaced by one of two other ingredients. However, because those two ingredients are more expensive than the preferred ingredient, substitution is used only when the preferred ingredient is out of stock.

For this scenario, the ingredient that can be replaced (the preferred ingredient) is named *ingredient A*. The two materials that can replace it are named *ingredient B* and *ingredient C*.

To set up the substitution, create a formula line for each of the three ingredients, and associate all three lines with the same plan group. Because ingredient A is the preferred ingredient, assign it the highest priority (lowest number). Assign each of the other ingredients (ingredient B and ingredient C) a priority that matches its relative level of preference. (The higher the number, the lower the preference.)

If you have a demand for the finished item, master planning determines whether the demand for ingredient A can be covered. If it can't, master planning looks at ingredients B and C in order of priority. If one of these ingredients is on hand, it's used (according to its priority) when a batch order is firmed for the formula. If none of the ingredients are on hand, master planning creates a planned order for the preferred ingredient (ingredient A).

Set up item substitution for formulas

Follow these steps to set up item substitution for a formula:

1. Go to **Product information management** > **Bills of materials and formulas** > **Formulas**.
2. Open the formula that you want to set up substitution for.
3. Find the formula line for the material that you want to be able to substitute. Set the following fields for it:
    - **Plan group** – Select the plan group name that you want to use to identify the group of items that can serve as substitutes for this material.
    - **Priority** – Enter a number that indicates the priority of this material in the plan group. The lower the number, the higher the priority. Therefore, the most preferred material should have the lowest number.
    - **Quantity** – Specify the quantity of the preferred material that's required in the formula. This quantity applies to whichever material from the plan group is used.
4. Find or add a formula line for the lower-priority substitute material. Set the following fields for it:
    - **Plan group** – Select the same plan group name that you selected for the preferred material.
    - **Priority** – Enter a number that indicates the priority of this material in the plan group. The lower the number, the higher the priority. Each material in the plan group must have a unique priority number.
    - **Quantity** – Enter *0*. This field isn't used for substitutes. The quantity of the preferred material (that is, the one that has the lowest priority) defines the quantity for all the materials in the plan group.
5. Repeat the previous step for each additional substitute material.
6. Repeat this procedure for each plan group that you want to set up for the current formula.

Shelf life and item substitution for formulas

When material substitution is used with products that have a limited shelf life, master planning always ensures that the shelf life of all materials is respected.

For example, a formula item is set up in the following way:

- **Material A:**
    - **Quantity:** 10
    - **Plan group:** Organic oils
    - **Priority:** 1
    - **Available on-hand:** 4, with an expiry date of today + five days

    - **Lead time:** One day

- **Material B (Substitute for material A):**
    - **Quantity:** 0
    - **Plan group:** Organic oils
    - **Priority:** 2
    - **Available on-hand:** 0
    - **Existing purchase order:** Qty 6, with an expected receipt date of today + six days, with 10 negative days

Master planning first selects material A, which has an available quantity of *4*. Because the required quantity is *10*, there isn't enough of material A. Therefore, master planning determines whether it can use the existing purchase order for the substitute (material B) for the remaining quantity. However, because the available receipt date of the substitute is later than the expiry date of the on-hand inventory of material A, the substitute isn't used. Instead, master planning creates a planned order for a quantity of *6* of material A to fulfill the remaining material that's required.

Item substitution for BOMs

Example scenario: Substituting components in a BOM

A manufacturing company is making an item from a BOM that has 20 components. One of the components in the BOM can be replaced by other components. It can even be replaced by the same item under a different item number.

For this scenario, the material that can be replaced (the preferred component) is named *material A*. The two materials that can replace it are named *material B* and *material C*.

To set up the substitution, create a BOM line for each of the three materials, and associate all three lines with the same plan group. Because material A is the preferred material, assign it the highest priority (lowest number). Assign each of the other materials (material B and material C) a priority that matches its relative level of preference. (The higher the number, the lower the preference.)

If you have a demand for the finished item, master planning determines whether the demand for material A can be covered. If it can't, master planning looks at materials B and C in order of priority. If one of these materials is on hand, it's used (according to its priority) when a batch order is firmed for the BOM. If none of the materials are on hand, master planning creates a planned order for the preferred material (material A).

Set up item substitution for BOMs

Follow these steps to set up item substitution for a BOM.

1. Go to **Product information management** > **Bills of materials and formulas** > **Bill of materials**.

2. Open the BOM that you want to set up substitution for.

3. Find the BOM line for the material that you want to be able to substitute. Set the following fields for it:

    - **Plan group** – Select the plan group name that you want to use to identify the group of items that can serve as substitutes for this material.
    - **Priority** – Enter a number that indicates the priority of this material in the plan group. The lower the number, the higher the priority. Therefore, the most preferred material should have the lowest number.
    - **Quantity** – Specify the quantity of the preferred material that's required in the BOM. This quantity applies to whichever material from the plan group is used.

4. Find or add a BOM line for the lower-priority substitute material. Set the following fields for it:

    - **Plan group** – Select the same plan group name that you selected for the preferred material.
    - **Priority** – Enter a number that indicates the priority of this material in the plan group. The lower the number, the higher the priority. Each material in the plan group must have a unique priority number.
    - **Quantity** – Enter *0*. This field isn't used for substitutes. The quantity of the preferred material (that is, the one that has the lowest priority) defines the quantity for all the materials in the plan group.

5. Repeat the previous step for each additional substitute material.

6. Repeat this procedure for each plan group that you want to set up for the current BOM.

Set up scrap to calculate raw material requirements

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to set up your bills of materials (BOMs), formulas, and route operations to record the quantity of raw material that's scrapped during manufacturing. The master planning engine can then use this information to calculate the quantity of raw materials that must be purchased to produce the desired quantity of the final product.

Scrap is the quantity of raw material that's left over and can't be used after an item is manufactured. You can set up scrap in the following places:

- Formula or BOM line
- Route operations

Scrap for route operations

Route scrap is the scrap that's produced during an operation in a route.

Set up scrap for route operations

To view or set the amount of scrap that an existing route operation produces, follow these steps:

1. Go to **Production control** > **All routes**.
2. In the list pane, select the route that you want to work with.
3. On the Action Pane, on the **Route** tab, select **Route details**.
4. The **Route details** page appears. The top grid lists each operation in the route. For each operation, you can view or set the following values that are related to the amount of raw material consumed as scrap for the operation:

- **Scrap percentage** – Specify the percentage of the input raw material consumed as scrap during the operation.

- **Accumulated** – The accumulated scrap percentage for the operation in the route. This value is read-only. The system calculates it in the following way:

  Accumulated = (*Accumulated percentage for the following operation* × 100) ÷ (100 – *Scrap percentage*)

Calculate scrap for route operations

This section describes an example where you're producing item *FG*, which uses 100 pieces of raw material *RM*. To produce FG, you use a route that has three operations: *Operation-1*, *Operation-2*, and *Operation-3*.

If the BOM line for RM doesn't specify which operation consumes the raw material, the system assumes that the first operation (Operation-1) does. Therefore, it calculates the quantity of RM that must be available at the beginning of Operation-1 to produce the required quantity of FG.

First, consider a scenario where the route operations require the following input:

- Operation-1 requires *X* liters of RM as input.
- Operation-2 requires the output of Operation-1 as input.
- Operation-3 requires the output of Operation-2 as input.

If each operation has a scrap percentage of 10 percent, the system does the following calculations to determine how many liters of RM are required as initial input to produce 100 liters of FG:

- **Operation-1:** *X* × (1 – 10%) = *Y*
- **Operation-2:** *Y* × (1 – 10%) = *Z*
- **Operation-3:** *Z* × (1 – 10%) = 100 liters of FG

When you solve for *X*, you get the following result:

X = 100 ÷ (0.9 × 0.9 × 0.9) = 137 liters of RM as input

Next, consider a scenario where the raw material starts to be consumed only later in the route, at Operation-2. In this case, the **Oper. Nr.** value on the BOM line must be set to the operation number of that operation. (For more information about how to set this value, see the next section.) If each operation has a scrap percentage of 10 percent, the system does the following

calculations to determine how many pieces of RM are required as initial input to produce 100 liters of FG:

- **Operation-1:** $X = Y$
- **Operation-2:** $Y \times (1 - 10\%) = Z$
- **Operation-3:** $Z \times (1 - 10\%) = 100$ liters of FG

When you solve for X, you get the following result:

$X = Y = 100 \div (0.9 \times 0.9) = 123$ liters of RM as input

Scrap for BOMs and formulas

BOM scrap and *formula scrap* are types of scrap that relate to a specific BOM line or formula line, depending on the type of product you're manufacturing. BOMs and formulas can include both constant and variable scrap.

Set up scrap for BOMs and formulas

To view or set the amount of scrap that a BOM or formula produces for a released product, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. In the grid, find and select the product that you want to work with.
3. On the Action Pane, on the **Engineer** tab, select either **BOM versions** or **Formula versions**, depending on the type of product.
4. The **BOM versions** or **Formula versions** page appears. In the list pane, select the version that you want to work with.
5. On the **Bill of materials lines** or **Formula lines** FastTab, select the line that you want to set up scrap for.
6. On the toolbar of the FastTab, select **Edit**.
7. The **Edit BOM line** or **Edit Formula line** dialog appears. On the **General** tab, in the **Oper. No.** field, if the current line is consumed only during a specific route operation, enter the operation number of that route operation. The value must be an integer, and it must match the related operation number that's shown on the **Route details** page. If you leave this field blank, the system assumes that the line starts to be consumed during the first operation.

8. On the **Consumption calculation** tab, set the following fields as required:

    - **Constant scrap** – Enter the quantity of scrap that's always the same for a single production run, regardless of the quantity of finished goods that are manufactured. Specify the scrap in the same unit that's used on the BOM or formula line.
    - **Variable scrap** – Variable scrap grows in proportion to the quantity of finished goods that are produced. Specify the amount of scrap that's created as a percentage of the raw material that's required as input for the route. For example, if the variable scrap for a raw material is 20 percent, and the route requires 100 liters of that material as input for a given production run, the required purchase quantity is 120 liters (excluding any constant scrap).

9. Select **OK** to apply your settings and close the dialog box.

You can also configure the previously described settings directly on the BOM or formula instead of on the **Released products** page.

1. Follow one of these steps:

    - Go to **Product information management** > **Bills of materials and formulas** > **Bills of materials**.
    - Go to **Product information management** > **Bills of materials and formulas** > **Formulas**.

2. Open a BOM or formula, and then configure settings like those that are described in the previous procedure.

Calculate scrap for BOMs and formulas

Based on the configured constant and variable BOM or formula scrap values, the system does the following calculation to determine the quantity of raw material that must be purchased to meet your actual raw material requirement:

Purchased material = (*Required raw material* × [1 + *Variable BOM or formula scrap*]) + *Constant BOM or formula scrap*

For example, you have the following requirements and scrap values:

- Constant scrap = 10 liters
- Variable scrap = 20 percent
- Required raw material = 100 liters

In this case, to meet your material requirements, you must purchase the following quantity of raw material:

Purchased material = (100 liters × 1.20) + 10 liters = 130 liters

Calculate raw material requirements based on the total scrap

If both route scrap and BOM or formula scrap are set up, the system automatically performs the following calculation to determine how much raw material is required to produce a specific quantity of finished goods:

Purchased raw material = *Total start quantity according to the route calculation* × (1 + *Variable BOM or formula scrap*) × *Batch size* + *Constant BOM or formula scrap*

In this calculation, *Batch size* is the number of pieces of the final good that's produced.

Last updated on 05/28/2026

Master planning for products with limited shelf life

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Shelf life is the amount of time that a product can be stored until it can no longer be used or sold. For products that have a limited shelf life, you'll probably use a first-expire, first-out (FEFO) warehouse strategy, which prioritizes the consumption and sale of items based on their remaining shelf life. This warehouse strategy is relevant for food, medicines, and other goods that are characterized by a short storage time. According to FEFO, items in the warehouse are stored like goods on a supermarket shelf: products that have a long shelf life are placed deep into the shelves, so that products that have a shorter remaining shelf life are shipped first.

Using shelf life in master planning

This section explains how master planning suggests supply for shelf-life items.

When you run a master plan, it generates suggested planned orders (supply) that fulfill your demand and also minimize delays. If your plan includes items with a limited shelf life, planning calculations become more complex. The plan must not only minimize delays but also use existing supply before it expires. The plan tries to use supply that's closest to its expiration date before supply that expires later. Therefore, master planning seeks to achieve the following goals, in this order:

1. Minimize the sum of delays.
2. Maximize the sum of FEFO supply.
3. Minimize the replenishment of inventory.

In some cases, there might be a conflict between the first two goals, and a choice must be made: do you want to delay a shipment, or do you want to use supply that expires later instead of supply that expires sooner? To resolve this conflict during master planning, the system prioritizes minimizing delays over using up soon-to-expire supply. In general, this type of conflict occurs when there might be delays and coverage by period. Therefore, use a coverage period that's shorter than the shelf life of an item. Other types of coverage (such as requirement) are unlikely to encounter this type of conflict.

Set up shelf life

Configure each master plan to consider shelf life

By default, master plans don't consider shelf life. Use the following procedure to enable shelf-life calculations for each master plan that requires them.

1. Go to **Master planning** > **setup** > **plans** > **Master plans**.
2. Either select an existing plan in the list pane or create a new one.
3. On the **General** FastTab, set the **Use shelf life dates** option to *Yes*.

Configure tracking dimension groups to track the batch dimension

You can track the shelf life of an item only if you track the item at the batch dimension. In other words, you must record the batch reference and the required dates when you receive or manufacture the item, and through every inventory transaction of the item. To manage this option, set up one or more tracking dimension groups to do the required tracking, and then assign the relevant items to these groups as required.

Use the following procedure to set up a tracking dimension group to track the batch dimension.

1. Go to **Product information management** > **Setup** > **Dimension and variant groups** > **Tracking dimension groups**.

2. Follow one of these steps:

    - On the Action Pane, select **New** to create a new tracking dimension group. Enter a name and description, then select **Save** on the Action Pane.
    - In the list pane, select the existing tracking dimension group that you want to set up to track the batch dimension.

3. On the **Tracking dimension** FastTab, in the **Batch number** row, select the checkboxes in the **Active** and **Physical inventory** columns.

Set up shelf life for a product

Use the following procedure to set up shelf life for a product.

1. Go to **Product information management** > **Products** > **Released products**.

2. Create or open the product that you want to set up.

3. To use the shelf-life settings, on the **General** FastTab, set the **Tracking dimension group** field to a tracking dimension group that tracks the batch dimension. You can set this field only when you're first creating a product. You can't change the value for existing products.

4. On the **Manage inventory** FastTab, set the following fields:

    - **Shelf advice period in days** – Specify the period (in days) by which to check a batch of this product to ensure that it's suitable for consumption or resale. The value of this field is added to a batch's *manufacturing date* to determine its *shelf advice date*. You can configure the system to generate quality orders when a batch approaches its shelf advice date.
    - **Shelf life period in days** – Specify the number of days before a batch of this product expires. This value is added to the *date of manufacture* to determine the *expiration date*. The batch is considered unusable after this date.
    - **Best before period in days** – Specify the period (in days) after which a batch of this product is deemed still sellable but can no longer retain some of its original properties. This value is added to the *date of manufacture* to determine the *best-before date*. You can run reports to identify inventory that's past its best-before date.

Set a sellable days rule for each customer

Sellable days functionality ensures that products from a batch that soon expires aren't sent to customers. Moreover, it ensures that when products are sent to a customer, an adequate number of sellable days still remain after delivery.

To use the sellable days functionality, you must define the number of sellable days that applies for each product (or group of products) for each customer. You must manually complete this process, because there's no data entity for it.

Use the following procedure to set up sellable days for each product for each customer.

1. Go to **Sales and marketing** > **Customers** > **All customers**.

2. Find and select the customer that you want to set up.

3. On the Action Pane, on the **Sell** tab, in the **Set up** group, select **Sell** > **Sellable days**.

4. On the **Sellable days for customer** page, the grid lists existing sellable days rules for each product or group of products. Use buttons on the Action Pane to add or edit rows in the grid as required. A **Filter** is provided to help you find existing rows.

5. For each row, set the following fields:

- **Item code** – Select one of the following values to specify the scope of items that are affected:
  - *Table* – The row applies to a specific item.
  - *Group* – The row applies to a specific item group.
  - *All* – The row applies to all items.

- **Item relation** – If you set the **Item code** field to *Table*, select a specific item. If you set the **Item code** field to *Group*, select an item group. If you set the **Item code** field to *All*, this field is unavailable.

- **Sellable days** – Enter the minimum number of days that the customer must have to sell matching products before the batch expires. The sellable days value is based on the requested receipt date (or the confirmed receipt date, if defined) for the matching products on the sales order.

- *(Other product dimensions)* – To further limit the scope of a row, specify other dimension values (such as **Size** and **Color**) as required. To control which dimensions are shown in the grid, select **Display dimensions** on the Action Pane.

Set up all relevant products so that they're FEFO date controlled

For sellable days to work, each relevant item must belong to an item model group where the **FEFO date-controlled** checkbox is selected.

Use the following procedure to set up an item model group so that it supports sellable days functionality.

1. Go to **Inventory management** > **Setup** > **Inventory** > **Item model groups**.
2. Select an existing group in the list pane or create a new one by selecting **New** on the Action Pane.
3. On the **Inventory policies** FastTab, select the **FEFO date-controlled** checkbox.
4. Set other fields for the group as required.

Use the following procedure to view or set the item model group that a product belongs to.

1. Go to **Product information management** > **Products** > **Released products**.
2. Open the product that you want to inspect or edit.
3. On the **General** FastTab, set the **Item model group** field to a group where the **FEFO date-controlled** checkbox is selected.

Example 1: Simple FEFO, 10-day period, zero days of lead time

This example shows a basic example of shelf life, where pegging between the supply orders and the demand satisfies the following goals of the system:

- Minimize the sum of delays.
- Maximize the sum of FEFO supply.
- Minimize the replenishment of inventory.

The system uses the following item and master plan settings:

- **Coverage code (replenishment strategy):** Period
- **Coverage period:** 10 days (equal to the shelf life)
- **Shelf life:** 10 days
- **Sellable days:** 0 days
- **Lead time:** 0 days
- **Negative days:** 0 days
- **Type of planned order (default order settings of the item):** Purchase order

The system has the following sales orders for the item:

- **SO1:** Quantity (qty) = 2, requested delivery date = today + 1 day
- **SO2:** Qty = 1, requested delivery date = today + 4 days
- **SO3:** Qty = 1, requested delivery date = today + 5 days

All these sales orders create demand for the item.

The system has the following supply for the item:

- **On-hand inventory:** Qty = 1, expiration date = today + 5 days
- **Purchase order 1 (PO1):** Receipt date = today + 2 days, qty = 1, expiration date = today + 4 days

The system creates a list of supply that can cover this demand, and it sorts the list by expiration date (by using FEFO).

Master planning creates the required pegging between supply and demand. It also creates any required demand based on the supply list (by using FEFO) and considers the availability date.

- SO1 can be fulfilled by the on-hand quantity, but it can't be fulfilled by PO1, because the availability date for PO1 is one day later than SO1 requires. Therefore, SO1 generates demand for one unit of goods.

- SO2 can be covered by PO1, because PO1 arrives by the requested time and the expiration date is still valid. Therefore, the SO2 requirement is fully covered by PO1.
- SO3 isn't covered, because resources aren't available. Therefore, SO3 generates demand for one unit of goods.

To cover all the remaining requirements, the system must create the following planned purchase order:

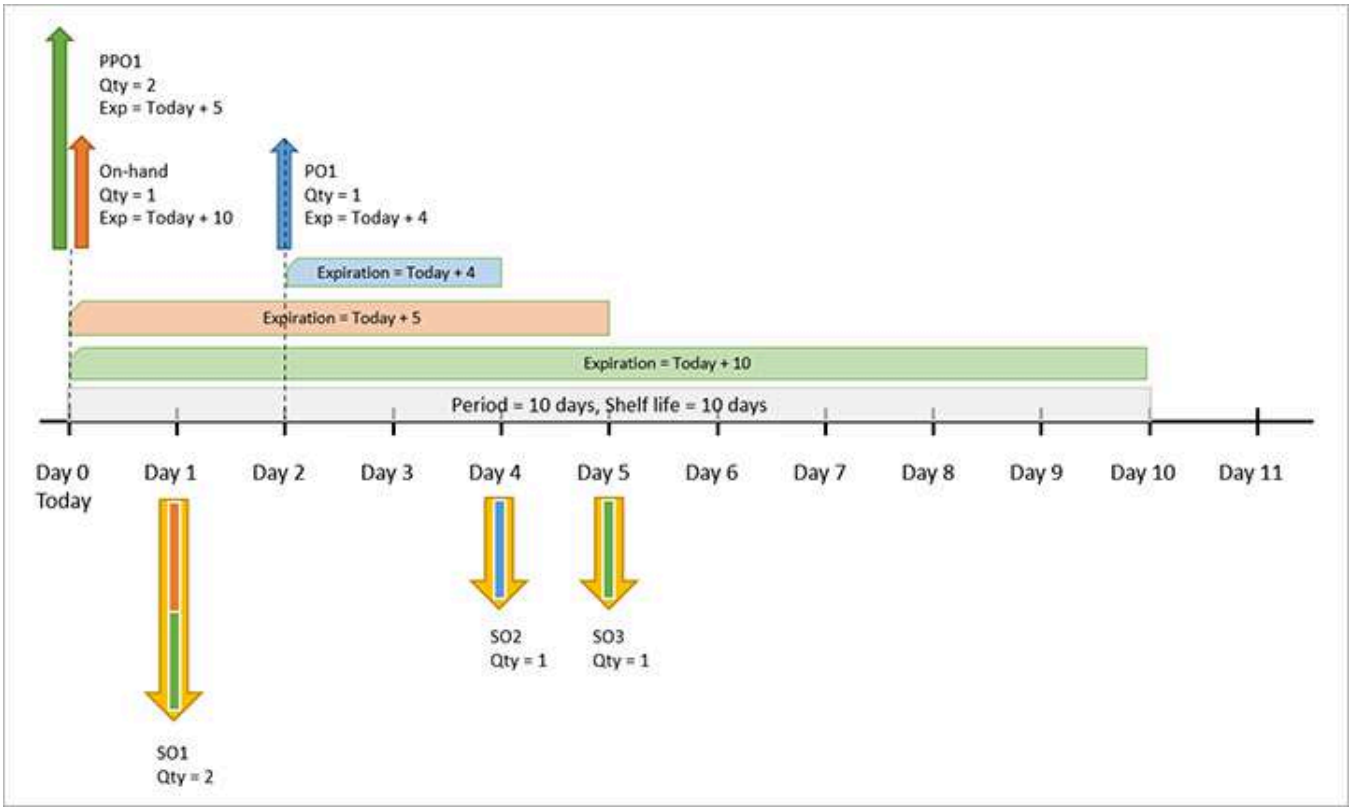
- **PPO1:** Receipt date = today, qty = 2, expiration date = today + 10 days

The following table summarizes the result.

Expand table

Demand	Pegging
SO1: Delivery date = today + 1 day, qty = 2	**On-hand:** Qty = 1, expiration date = today + 5 days **PPO1:** Receipt date = today, qty = 1, expiration date = today + 10 days
SO2: Delivery date = today + 4 days, qty = 1	**PO1:** Receipt date = today + 2 days, 1 qty, expiration date = today + 4 days
SO3: Delivery date = today + 5 days, qty = 1	**PPO1:** Receipt date = today, qty = 2, expiration date = today + 10 days

The following illustration shows the timeline for this example.



Example 2: Simple FEFO, requirement, three days of lead time

This example shows how the system's attempt to minimize delays can sometimes cause overordering.

The system uses the following item and master plan settings:

- **Coverage code (replenishment strategy):** Requirement
- **Shelf life:** 10 days
- **Sellable days:** 0 days
- **Lead time:** Established by the following vendor trade agreements:
  - **Trade agreement 1:** If qty = 1, lead time = 4
  - **Trade agreement 2:** If qty = 2, lead time = 3
- **Negative days:** 0 days
- **Type of planned order (default order settings of the item):** Purchase order

The system has the following sales order:

- **SO1:** Qty = 2, requested delivery date = today + 3 days

Existing supply and a confirmed purchase order cover this demand:

- **On-hand inventory:** Available = today, qty = 1, expiration date = today + 2 days
- **PO1:** Receipt date = today + 3 days, qty = 1, expiration date = today + 4 days

SO1 can't be fulfilled by on-hand inventory, because the inventory expiration date is before the shipment date. PO1 can cover the SO1 requirement with a quantity of only 1. Therefore, SO1 generates demand for one unit of goods. To cover this requirement, the system creates a planned purchase order (PPO1).

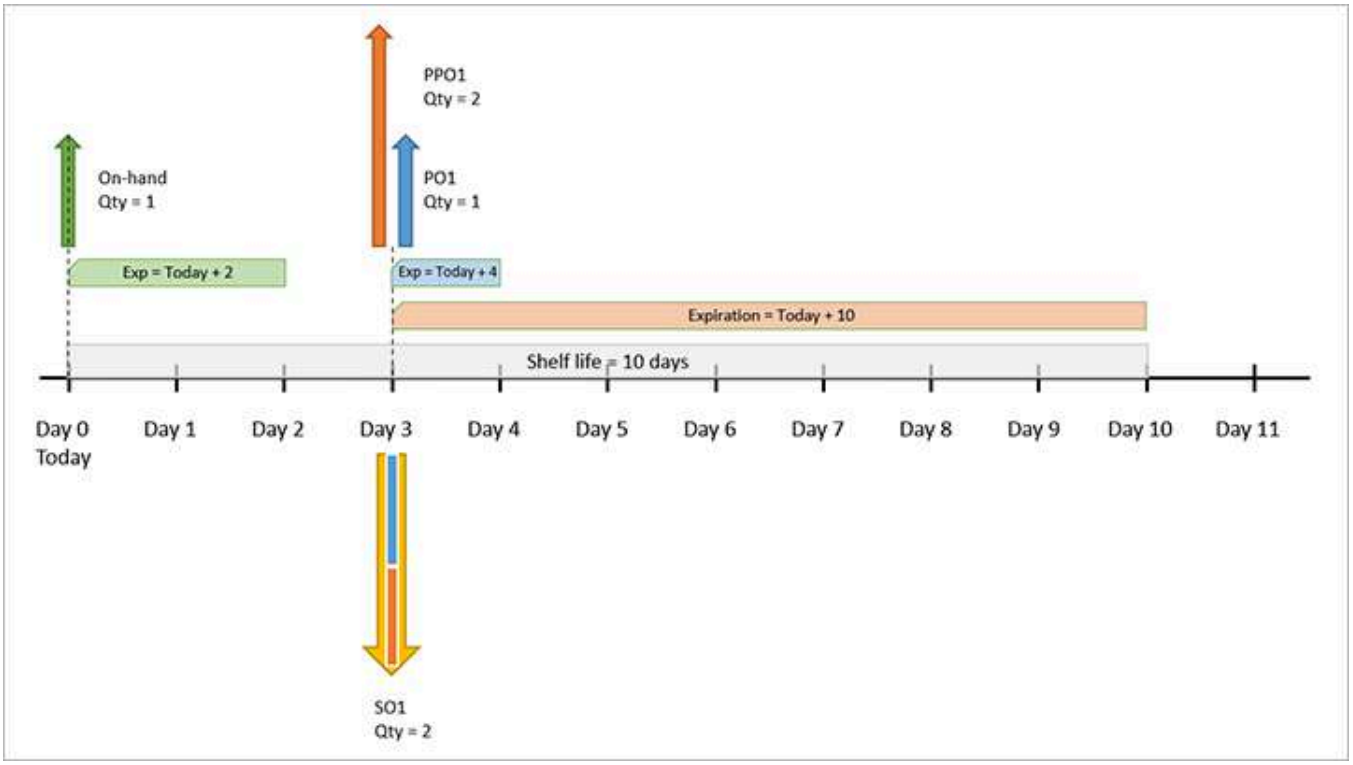
The system has two trade agreements (one for qty = 1, lead time = 4 days, and one for qty = 2, lead time = 3 days). Therefore, the system tries to minimize delays by creating a planned purchase order (PPO1) that meets the second trade agreement. The result is an overdelivery (qty = 2, expiration date = today + 10 days).

The following table summarizes the result.

Expand table

Demand	Pegging
SO1: Delivery date = today + 3 days, qty = 2	**PO1:** Receipt date = today + 3 days, qty = 1, expiration date = today + 4 days **PPO1:** Receipt date = today + 3 days, qty = 1, expiration date = today + 10 days

The following illustration shows the timeline for this example.



Example 3: Simple FEFO, requirement, three days of lead time, five sellable days

This example shows how shelf life works when you add sellable days for an item.

The system uses the following item and master plan settings:

- **Coverage code (replenishment strategy):** Requirement
- **Shelf life:** 10 days
- **Sellable days:** 5 days
- **Lead time:** 5 days
- **Negative days:** 0 days
- **Type of planned order (default order settings of the item):** Purchase order

The system has the following sales orders:

- **SO1:** Qty = 2, requested delivery date = today + 2 days

- **SO2:** Qty = 1, requested delivery date = today + 3 days
- **SO3:** Qty = 1, requested delivery date = today + 5 days

Existing supply and a confirmed purchase order cover this demand:

- **On-hand inventory:** Available = today, qty = 1, expiration date = Today + 6 days
- **PO1:** Receipt date = today + 2 days, qty = 3, expiration date = today + 10 days

The system creates a list of pegging candidates based on the supply (FEFO) list and availability dates. Therefore, the on-hand inventory can't fulfill SO1, because that inventory expires before the end of the sellable days that the customer requires (requested receipt date + 5 days). PO1 can cover the SO1 requirement with two units and the SO2 requirement with one unit. Therefore, only SO3 still has uncovered demand for one unit of goods. To cover this requirement, the system creates the following planned purchase order:

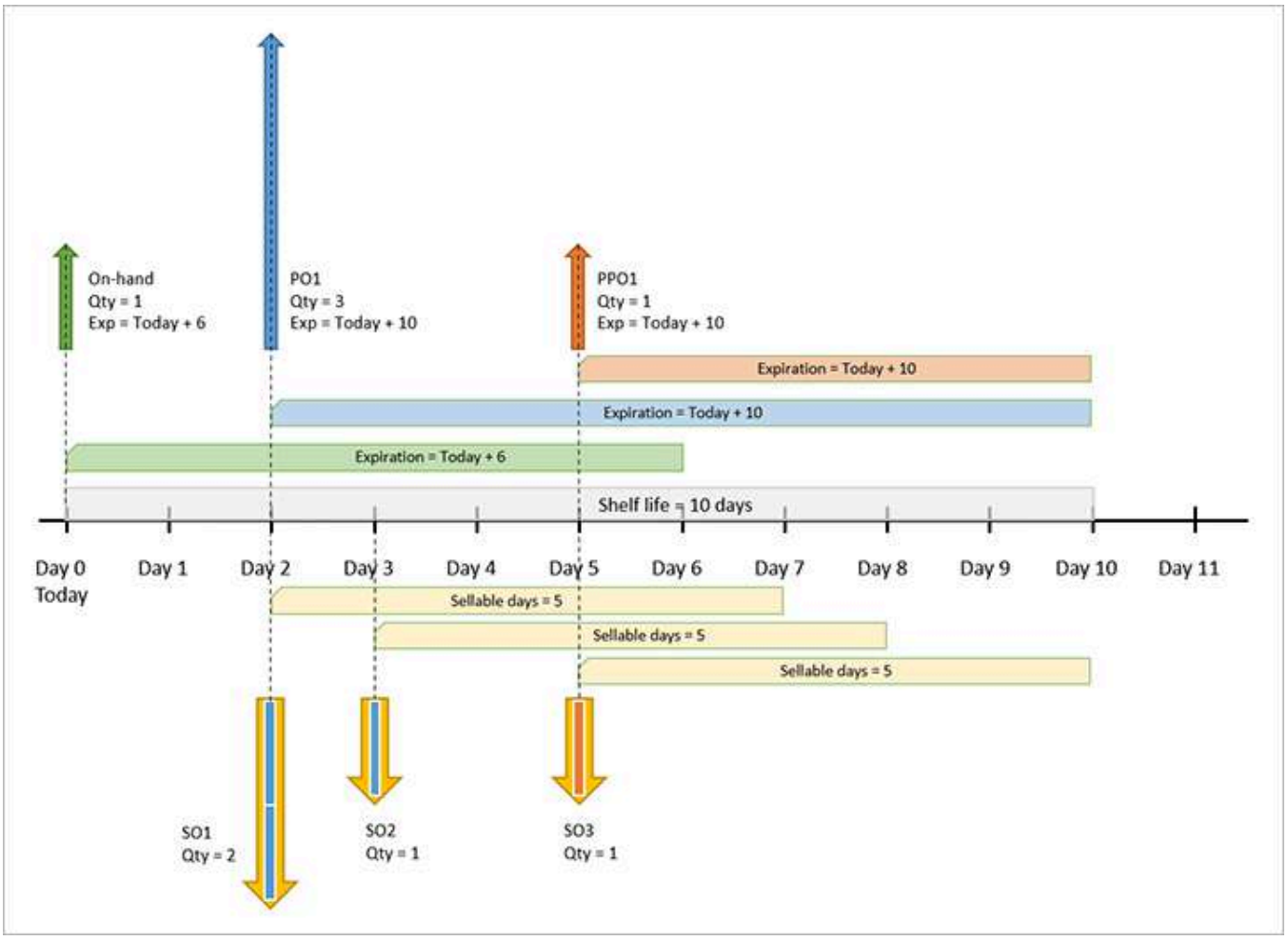
- **PP01:** Receipt date = today + 5 days, qty = 1, expiration date = today + 10 days

The following table summarizes the result.

Expand table

Demand	Pegging
SO1: Delivery date = today + 2 days, qty = 2	**PO1:** Receipt date = today + 2 days, qty = 2, expiration date = today + 10 days
SO2: Delivery date = today + 3 days, qty = 1	**PO1:** Receipt date = today + 2 days, qty = 1, expiration date = today + 10 days
SO3: Delivery date = today + 5 days, qty = 1	**PPO1:** Receipt date = today + 5 days, qty = 1, expiration date = today + 10 days

The following illustration shows the timeline for this example.



Example 4: Simple FEFO, period, lead time depends on the quantity

This example shows how the system's attempt to minimize delays can sometimes cause overordering.

The system uses the following item and master plan settings:

- **Coverage code (replenishment strategy):** Period
- **Coverage period:** 10 days (equal to the shelf life)
- **Shelf life:** 10 days
- **Sellable days:** 0 days
- **Lead time:** Established by the following vendor trade agreements:
  - **Trade agreement 1:** If qty = 1, lead time = 5
  - **Trade agreement 2:** If qty = 2, lead time = 0
- **Negative days:** 0 days

- **Type of planned order (default order settings of the item):** Purchase order

The system has the following sales orders:

- **SO1:** Qty = 1, requested delivery date = today
- **SO2:** Qty = 1, requested delivery date = today + 6 days

The system can partially cover this demand with existing supply from the following confirmed purchase orders:

- **PO1:** Receipt date = today + 1 days, qty = 1, expiration date = today + 2 days
- **PO2:** Receipt date = today + 3 days, qty = 1, expiration date = today + 7 days

The system has two trade agreements (one for qty = 1, lead time = 5 days, and one for qty = 2, lead time = 0 days). Therefore, the system tries to minimize delay by creating the following planned purchase order that meets the second trade agreement:

- **PP01:** Receipt date = today, qty = 2, expiration date = today + 10 days

SO1 is covered by one unit from PPO1. PO2 covers SO2, because PO2 expires sooner than PPO1.

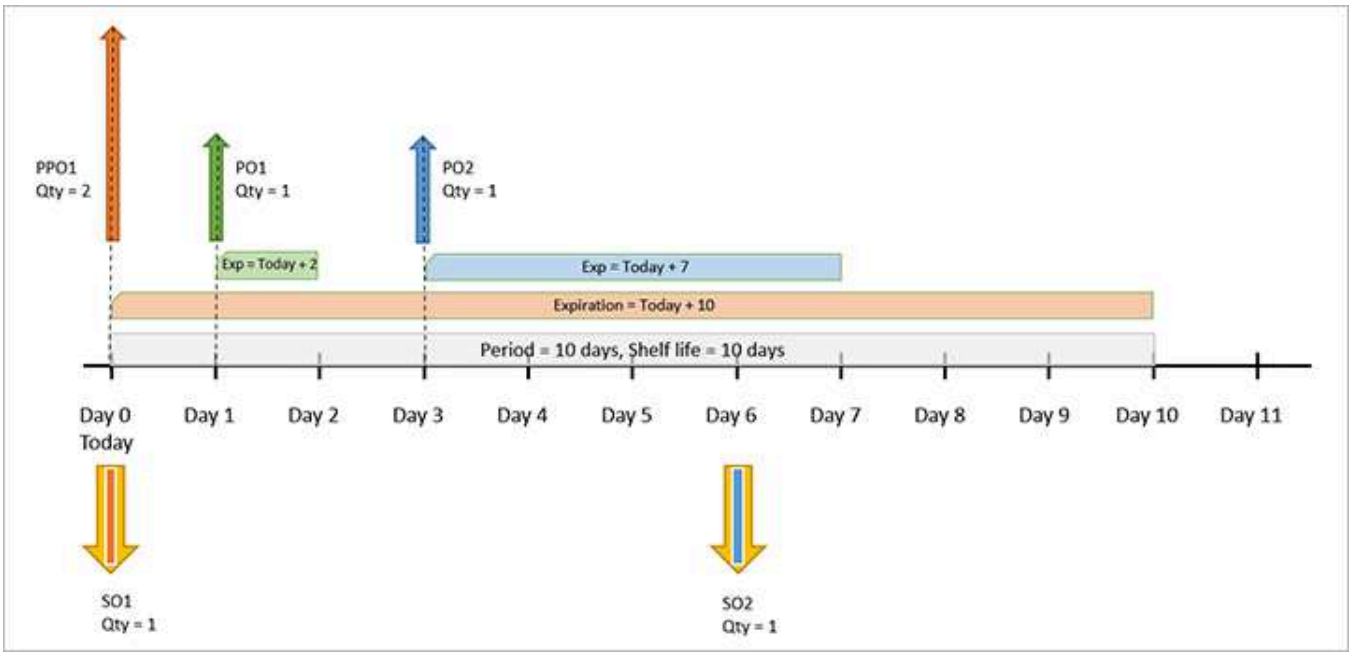
The following table summarizes the result.

Expand table

Demand	Pegging
SO1: Delivery date = today, qty = 1	**PPO1:** Receipt date = today, qty = 1, expiration date = today + 10 days
SO2: Delivery date = today + 6 days, qty = 1	**PO2:** Receipt date = today + 3 days, qty = 1, expiration date = today + 7 days

> **Note**
>
> The system doesn't use PO1, because it arrives too late for S01 and expires before S02 is delivered. PPO1 overorders by one unit to enable the lead time to be 0 (zero), per trade agreement 2.

The following illustration shows the timeline for this example.



Example 5: Simple FEFO, requirement, 10 negative days

This example shows how shelf life works when you add a large number of negative days for an item. Negative days are the number of days that you're willing to wait before you order replenishment of an item that has negative inventory. The system doesn't create supply unless the number of negative days is exceeded.

The system uses the following item and master plan settings:

- **Coverage code (replenishment strategy):** Requirement
- **Lead time:** 0 days
- **Negative days:** 10 days
- **Type of planned order (default order settings of the item):** Purchase order

The system has the following sales order:

- **SO1:** Qty = 1, requested delivery date = today

This demand can be covered by existing supply from the following confirmed purchase order:

- **PO1:** Receipt date = today + 3 days, qty = 1, expiration date = today + 5 days

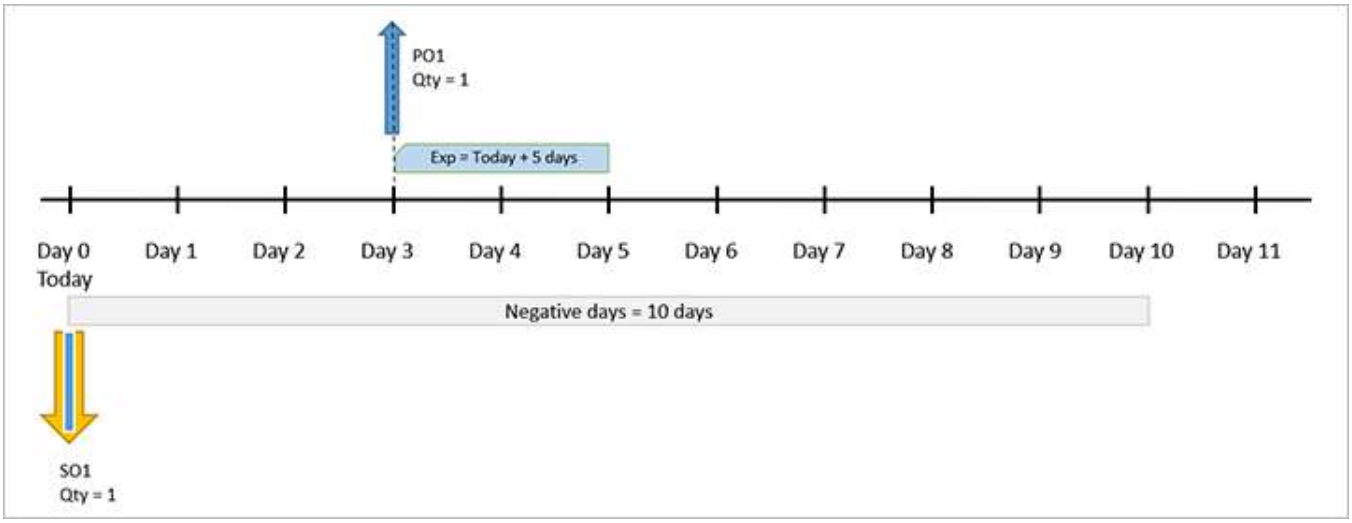
Because the system is configured to allow 10 negative days, it covers the demand of SO1 by using PO1, even though the result is a delay of three days because SO1 creates negative inventory until PO1 arrives. The system doesn't create a planned purchase order, even though the lead time is 0 (zero) and the creation of a planned purchase order would reduce delays.

The following table summarizes the result.

Expand table

Demand	Pegging
SO1: Delivery date = today, qty = 1	**PO1:** Receipt date = today + 3 days, qty = 1, expiration date = today + 5 days

The following illustration shows the timeline for this example.



Example 6: Simple FEFO, requirement, five negative days

This example shows how shelf life works when the number of negative days for an item is less than its shelf-life period.

The system uses the following item and master plan settings:

- **Coverage code (replenishment strategy):** Requirement
- **Sellable days:** 0 days
- **Lead time:** 0 days
- **Negative days:** 5 days
- **Type of planned order (default order settings of the item):** Purchase order

The system has the following sales order:

- **SO1:** Qty = 2, requested delivery date = today

The system can cover this demand with existing supply from the following confirmed purchase orders:

- **PO1:** Receipt date = today, qty = 1, expiration date = today + 1 day
- **PO2:** Receipt date = today + 2 days, qty = 1, expiration date = today + 3 days

However, the system must respect the restriction that shipped items can't be expired at the time of shipment. Therefore, the system can't use both PO2 and PO1 for SO1, because PO1 expires before PO2 arrives. The system creates the following planned purchase order to finish covering the demand for SO1:

- **PPO1:** Receipt date = today, qty = 1, expiration date = today + 10 days

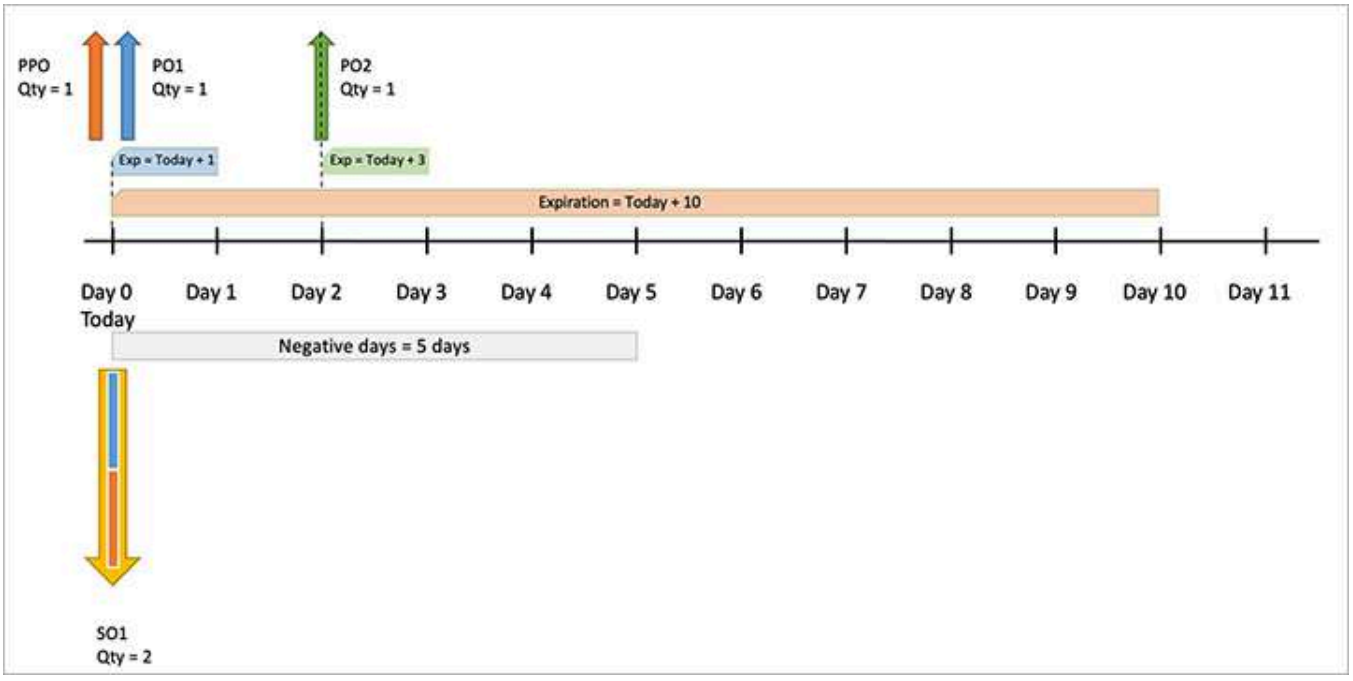
The system can take advantage of the five negative days and use PO2 and PPO1 to cover SO1. However, this approach causes delivery to be delayed until PO2 arrives, and PO1 expires in the meantime. Therefore, the system covers SO1 by using PPO1 and PO1.

The following table summarizes the result.

Expand table

Demand	Pegging
SO1: Delivery date = today, qty = 2	**PO1:** Receipt date = today, qty = 1, expiration date = today + 1 day **PPO1:** Receipt date = today, qty = 1, expiration date = today + 10 days

The following illustration shows the timeline for this example.



Last updated on 11/19/2025

Scheduling with infinite capacity

10/03/2025

The *Infinite capacity scheduling for Planning Optimization* feature introduces scheduling that is based on route information. It lets you schedule jobs based on a wide range of route setups. Scheduling covers frequently used route settings, including the route operation sequence or requirements for route operation resources.

Added functionality

The *Infinite capacity scheduling for Planning Optimization* feature enables job scheduling that is based on route information. Therefore, a route setup can be used to schedule production processes. Although this feature has some limitations that the deprecated master planning engine doesn't have, it supports the most common functionality that is required for manufacturing scenarios.

The feature considers both *simple routes* and *route networks*. By using the **Next** field on a route operation, you can set up complex routes that have multiple starting points and multiple operations that run in parallel. The system will consider complex route structures of this type during scheduling.

Additionally, the feature supports *parallel operations* in routes. By using the *Primary* and *Secondary* options in the **Priority** field on route operations, you can define a route structure where one route operation is the primary operation and another operation is secondary. In this case, the system will consider the route structure during scheduling.

During the scheduling process, the system also considers the *resource requirements* that are specified for an operation. The system uses resource requirements to determine which resources are required to perform the operation. Currently, the *Infinite capacity scheduling for Planning Optimization* feature supports following types of resource requirements:

- Resource type
- Resource
- Resource group
- Capability (Learn more in Scheduling with resource selection based on capability.)

> **Note**
>
> - If the resource and/or the resource group are set to infinite capacity, master planning will consider them as infinite capacity.

- Requirements that are related to human resources, such as skills or certificate requirements, aren't yet supported.

The feature also supports the **Setup time** and **Run time** operational properties. When you set these properties on a route operation, the scheduling process will create the appropriate setup and process jobs.

In summary, scheduling supports the most frequently used scenarios. You can create the route, add primary and secondary operations, define next operations, add resource requirements, and add setup time and run time. The system will then consider this information during scheduling.

Limitations

The following limitations apply when you use the *Infinite capacity scheduling for Planning Optimization* feature:

- The feature supports only infinite capacity.
- The feature doesn't support resource load functionality.
- The feature doesn't consider route scrap.
- The feature supports *Duration* only as the primary resource selection.

Related information

- Scheduling with resource selection based on capability

Finite capacity planning and scheduling

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Finite capacity is an approach that helps you understand how much work can be produced during a specific period when limitations on different resources are taken into consideration. The purpose of finite capacity scheduling is to ensure that work proceeds at an even and efficient pace throughout the plant.

Finite capacity planning and scheduling creates a more realistic schedule for the production processes than the infinite loading approach creates. If there isn't enough capacity on the resources, the delivery date is pushed out, and the job is scheduled when there's enough capacity.

> **Note**
>
> Finite capacity planning and scheduling works in nearly the same way, regardless of whether you use Planning Optimization or the deprecated master planning engine. However, Planning Optimization doesn't use the **Bottleneck time** fence parameter. When you use Planning Optimization, bottleneck resources are always scheduled by using the same time fence as non-bottleneck resources (as indicated by the finite capacity time fence).

Set up finite capacity functionality

To use finite capacity functionality, you must enable capacity planning globally, and you must enable finite capacity planning both for the master plan where you want to use it and for each resource where it applies. For plans and resources where finite capacity is enabled, scheduling of planned production orders considers capacity that is already reserved. Planned production orders are backward-scheduled from the requirement date. If capacity isn't available to meet the optimal schedule, the system tries to require component items on an earlier date. If your capacity can change as requirements change (for example, when you're working with shifts), you shouldn't use finite capacity functionality, because the calculated processing times won't be correct. Scheduling considers only capacity that is already reserved for resources where

finite capacity is enabled. By enabling finite capacity for a resource, you make it possible to modify the finite capacity time fence.

Activate Master planning parameters

To use finite capacity functionality, you must enable capacity planning on the **Master planning parameters** page.

1. Go to **Master planning** > **Setup** > **Master planning parameters**.
2. On the **Planned orders** tab, in the **Capacity planning** section, set the **Production** option to *Yes*.

Activate a master plan

You must enable finite capacity planning and scheduling for each master plan where you want to use it.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. In the list pane, select a master plan that you want to set up to use finite capacity planning and scheduling.
3. On the **General** FastTab, in the **Planned production orders** section, set the **Finite capacity** option to *Yes*.
4. Repeat steps 2 and 3 for each additional master plan that you want to set up.

Activate resources

You must enable finite capacity planning and scheduling for each resource where you want to use it.

1. Go to **Production control** > **Setup** > **Resources** > **Resources**.
2. In the list pane, select a resource that you want to set up to use finite capacity planning and scheduling.
3. On the **Operation** FastTab, in the **Capacity button** section, set the **Finite capacity** option to *Yes*.
4. Repeat steps 2 and 3 for each additional resource that you want to set up.

Examples

This section provides the following examples that show how to work with both infinite and finite capacity planning and scheduling:

- Example 1 – Infinite capacity planning
- Example 2 – Finite capacity planning with a time fence of one day
- Example 3 – Finite capacity planning with a time fence of two days

Preconditions

All three examples assume the preconditions that are described in this section.

Product *Product1* has a route that contains the following operations.

Expand table

Operation no.	Operation name	Resource	Run time	Process qty.	Next
10	Welding	Welding line	1	2	20
20	Assembling	Assembling line	1	4	

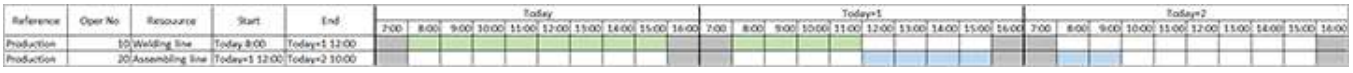
Workers at your company work in one shift for eight hours (8:00–16:00).

There's a scheduled production order for *24 pcs* of *Product1*. It has a delivery date of *Today + 3 days*.

As a result of planning, the system loads the resources in the following way:

- **Welding line:** This resource is loaded from *Today at 8:00* until *Today + 1 at 12:00*.
- **Assembling line:** This resource is loaded from *Today + 1 at 12:00* until *Today + 2 at 10:00*.

The following illustration shows the resulting Gantt chart (select it for a larger view).



Example 1 – Infinite capacity planning

This example shows the planning results when you use infinite capacity planning instead of finite capacity planning.

The master plan has the following relevant setting, which disables finite capacity planning for the plan:

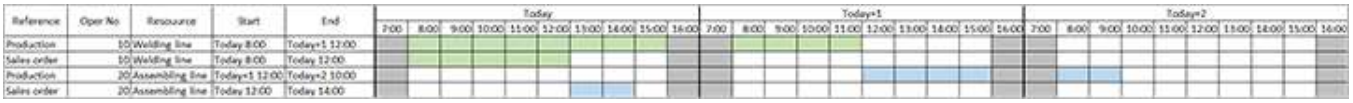
- **Finite capacity:** *No*

The **Finite capacity** option is also set to *No* for both the relevant resources to disable finite capacity planning for them:

- Welding line
- Assembling line

You now add a new sales order for *8 pcs* of *Product1* and run the plan. As a result, the system loads the welding line from *Today at 8:00* until *Today at 12:00*. After operations on the welding line are completed, the system will load the assembling line from *Today at 12:00* until *Today at 14:00*.

The following illustration shows the resulting Gantt chart (select it for a larger view).



Example 2 – Finite capacity planning with a time fence of one day

This example shows planning results when you use finite capacity planning and a time fence of one day.

The master plan has the following relevant settings, which enable finite capacity planning and set a time fence for the plan:

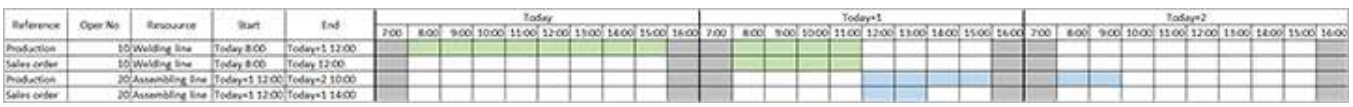
- **Finite capacity:** *Yes*
- **Finite capacity time fence:** *1*

The **Finite capacity** option is also set to *Yes* for both the relevant resources to enable finite capacity planning for them:

- Welding line
- Assembling line

You now add a new sales order for *8 pcs* of *Product1* and run the plan. As a result, the system loads the welding line from *Today + 1 at 8:00* until *Today + 1 at 12:00*. After operations on the welding line are completed, the system will load the assembling line from *Today + 1 at 12:00* until *Today + 1 at 14:00*. The system considers the finite capacity for only one day.

The following illustration shows the resulting Gantt chart (select it for a larger view).



Example 3 – Finite capacity planning with a time fence of two days

This example shows planning results when you use finite capacity planning and a time fence of two days.

The master plan has the following relevant settings, which enable finite capacity planning and set a time fence for the plan:

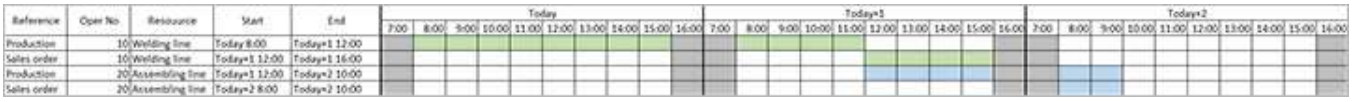
- **Finite capacity:** *Yes*
- **Finite capacity time fence:** *2*

The **Finite capacity** option is also set to *Yes* for both the relevant resources to enable finite capacity planning for them:

- Welding line
- Assembling line

You now add a new sales order for *8 pcs* of *Product1* and run the plan. As a result, the system loads the welding line from *Today + 1 at 12:00* until *Today + 1 at 16:00*. After operations on the welding line are completed, the system will load the assembling line from *Today + 2 at 8:00* until *Today + 2 at 10:00*. The system considers the finite capacity for only two days.

The following illustration shows the resulting Gantt chart (select it for a larger view).



> **ⓘ Important**
>
> You should always set the finite capacity time fence as required to fit your business needs. The examples that are provided in this article merely illustrate the functionality. In reality, a single-day time fence is probably too low for most manufacturers that use finite capacity planning.

Last updated on 08/05/2025

Scheduling with resource selection based on capability

10/03/2025

By specifying resource requirements for an operation of a production route, you define what is required to perform that operation. For example, an operation might require a specific resource or a resource group, or a combination of skills or capabilities. This article describes resource selection during infinite capacity scheduling when you specify capabilities as resource requirements for an operation.

Capability-based scheduling

A capability is the ability of an operation resource to perform a specific activity. More than one capability can be assigned to a single operation resource, and a single capability can be assigned to more than one resource. Capabilities can be assigned to all types of resources, such as tools, vendors, machines, locations, facilities, and human resources.

When you specify capabilities as resource requirements, you can defer resource allocation until orders are scheduled. Instead of assigning specific resources or resource groups to a route operation, you can define the capabilities that are required for each route operation. Then, during scheduling, the system matches required capabilities with the capabilities that are defined for the resources. The system selects only resources that satisfy the requirements.

To assign capabilities to an operation resource, use the **Capabilities** FastTab of the **Resources** page. To assign resources to a capability, use the **Resources** FastTab of the **Resource capabilities** page. Both pages are accessible under **Production control** > **Setup** > **Resources** on the navigation pane. Both FastTabs include a grid that lists the resources that are associated with a selected resource or capability. Both FastTabs also include a toolbar that you can use to add, remove, and edit rows in the grid. The grid includes the following columns:

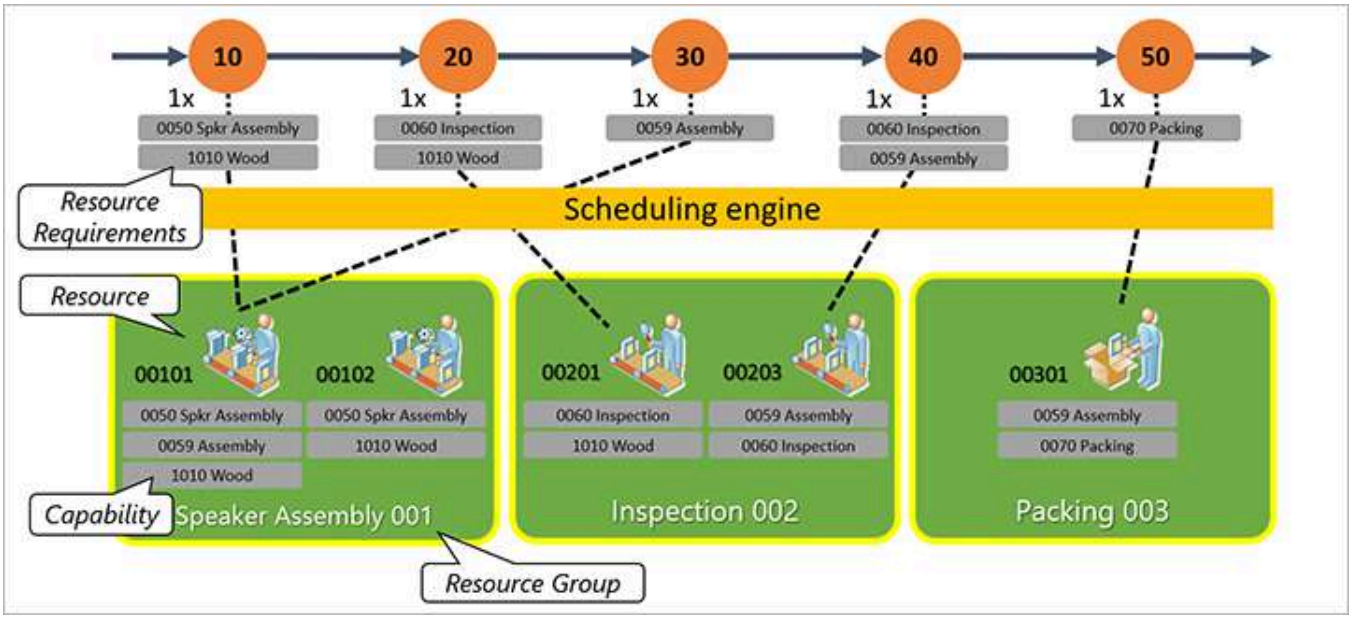
- **Resource** or **Capability** – Select the resource or capability that is being assigned by the row.
- **Description** – Enter a short description of the resource or capability.
- **Effective** – Specify the first date when the resource or capability assignment applies. During scheduling, the system won't use a resource or capability that has an expired capability assignment, even if that resource otherwise satisfies the requirements.
- **Expiration** – Specify the last date when the resource or capability assignment applies. During scheduling, the system won't use a resource or capability that has an expired capability assignment, even if that resource otherwise satisfies the requirements.

- **Level** – Specify the level of proficiency that the resource must have for the capability. Then, if you specify a **Minimum level needed** value for the resource or capability requirement, the scheduling engine considers the level of proficiency during resource selection. The system selects only resources that have the required capability at a level that equals or exceeds the minimum level that is specified in the resource requirement.
- **Priority** – Set a priority for the resource or capability. Then, if *Priority* is selected in the **Primary resource selection** field on the **Scheduling parameters** page, the system selects the available resource with the highest priority (that is, the lowest numeric value in the **Priority** field) during scheduling. The engine only evaluates priority among those resources that are both applicable and available, which means that during backward scheduling, if a resource with a high priority isn't available (for example because its calendar is closed) then another resource would be chosen.

Example

This example shows how the scheduling engine selects resources, based on capability.

A production route has five operations: *10*, *20*, *30*, *40*, and *50*. Each route operation requires a resource that has different capabilities. For example, route operation *10* requires a resource that has the capability to assemble a speaker (*0050 Spkr Assembly*) and the capability do woodworking (*1010 Wood capabilities*). Route operation *50* requires a resource that has the capability to pack a product (*0070 Packing capability*).



During scheduling, the engine looks for resources that satisfy the operation requirements. For example, the scheduling engine selects resource *00101* to perform operation *10*, because this resource is the first resource that is found that has both the required capabilities. The scheduling engine selects resource *00301* to perform operation *50*, because this resource is the only resource that has the packing capability.

Next, consider how this example is affected when proficiency levels are used:

- Operation *10* requires a minimum proficiency level of *7* for the *0059 Assembly* capability.
- Resource *00101* has a proficiency level of *5* for the *0059 Assembly* capability.
- Resource *00102* has a proficiency level of *10* for the *0059 Assembly* capability.

In this case, during infinite capacity scheduling, the scheduling engine selects resource *00102* to perform operation *10*, because this resource, unlike the resource *00101*, has the required proficiency level for the capability.

Related information

- Scheduling with infinite capacity

Inventory marking

10/03/2025

This article provides information about the options that are available for marking inventory in firmed orders.

Marking is used to link supply and demand. It resembles *pegging*, which indicates how master planning expects to cover demand. From a planning point of view, the main difference is that marking is more permanent than pegging.

Marking was introduced to support special costing scenarios for first in, first out (FIFO) and last in, first out (LIFO). However, it's now also used for some non-costing scenarios. Marking between supply and demand is optional and almost permanent. Marking can be removed manually by a user, or it can be removed by running a sales order line explosion where the **Remove marking** option is selected.

Pegging is controlled by master planning during coverage to link demand with the required supply. Pegging can be updated for each planning run to optimize the supply that is required to cover demand. When master planning updates pegging information, it respects any existing marking.

Pegging starts by including relevant marking, on-hand reservations, and on-order reservations, in the following sequence:

1. Marking between demand and supply
2. On-hand reservations
3. On-order reservations

When you firm a planned order, the **Firming** dialog box provides an **Update marking** field that you use to set marking options for the orders that are created during firming. Select one of the following values:

- *No* – No inventory marking is applied.
- *Standard* – Inventory marking is updated according to the pegging. A requirement order (demand) is marked against a fulfillment order (supply). If some quantity remains on the fulfillment order, it isn't marked, and the reference information is left blank. For example, if a sales order for 100 ea is pegged against a purchase order for 150 ea, reference information will be assigned only to the sales order.
- *Extended* – Both the requirement order (demand) and the fulfillment order (supply) are marked, regardless of any quantity that remains on the fulfillment order. For example, if a sales order for 100 ea is pegged against a purchase order for 150 ea, reference information will be assigned to both the sales order and the purchase order.

- *Single level standard* – Single-level marking is used. Single-level marking marks only the main item, not its bill of materials (BOM) components. Therefore, you can keep component assignment for production orders flexible after firming. Single-level marking enables the system to optimize for last-minute demand changes. In *standard* single-level marking, requirement orders are marked against their fulfillment orders, but fulfillment orders aren't marked if they have remaining quantity.
- *Single level extended* – Single-level marking is used. In *extended* single-level marking, requirement orders are marked against their fulfillment orders, and fulfillment orders are always marked, regardless of whether any quantity remains.

To set the default marking option for your system, go to **Master planning** > **Setup** > **Master planning parameters**. Then, on the **Standard update** tab, set the **Update marking** field to your preferred option.

Intercompany planning

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

For some organizations, logistics operations depend on other legal entities (companies) in the organization. Handle these operations by using intercompany sales and purchases because each legal entity has a separate chart of accounts.

This article describes intercompany planning and explains how to configure intercompany planning in Microsoft Dynamics 365 Supply Chain Management.

This article uses the following important intercompany terms:

- **Upstream** – A relative reference in a firm or supply chain. It indicates movement in the direction of the raw material supplier.
- **Downstream** – A relative reference in a firm or supply chain. It indicates movement in the direction of the customer.
- **Planned intercompany demand** – Planned demand for a product in a company, based on planned demand for the product from a downstream company.

In master planning, a plan in one company can include planned intercompany demand that relates to planned orders from a plan in another company. This capability is useful, because it provides full visibility into planned orders across companies. It also ensures that all required planned supply orders are created, but without requiring that planned orders be firmed for the intercompany demand.

If you run master planning from a master plan that includes planned downstream demand, planned purchase orders from the related intercompany vendors are included in the plan as demand.

Required setup

To use intercompany planning, prepare your system in the following way:

1. Release the relevant products in all the relevant companies. Learn more in Configure and use intercompany trade in Dynamics 365 Supply Chain Management.

2. Cover downstream demand by purchasing from a vendor that has an intercompany relation to the upstream company and relevant default inventory dimensions (site and warehouse) on the customer. Learn more in Configure and use intercompany trade in Dynamics 365 Supply Chain Management.
3. Include planned downstream demand in the master plan in the upstream company, and specify the relevant company and master plan in the downstream plans.

Include planned downstream demand

Follow these steps to configure your master plan so that it includes planned downstream demand.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select or create a master plan.
3. On the **Intercompany planning** FastTab, set the following fields:
    - **Include planned downstream demand** – Set this option to *Yes* to enable intercompany planning for the master plan.
    - **Downstream plans** – If you set the **Include planned downstream demand** option to *Yes*, use the toolbar and grid to add the desired master plans from other companies.

Peg across companies by using multilevel pegging

In multilevel pegging, you can view pegging across companies to see the initial source of demand that's covered by a supply.

To view multilevel pegging information, follow these steps.

1. Go to **Master planning** > **Master planning** > **Planned orders**.
2. Select or open a planned order.
3. On the Action Pane, on the **View** tab, in the **Requirements** group, select **Multilevel pegging**.

Intercompany example that involves two companies

For this example, a planned production order is created in the USMF company to cover a sales order in the DEMF company. In USMF, the direct demand is planned intercompany demand. To make this demand appear in USMF, run master planning first in DEMF and then in USMF.

The following illustration shows how this example might appear on the **Multilevel pegging** page for the planned production order.

Intercompany example that involves three companies

In this example, you create a planned purchase order in the USMF company to cover a sales order in the FRRT company. In the DEMF and USMF companies, the direct demand is planned intercompany demand. To make this demand appear in USMF, run master planning first in FRRT, then in DEMF, and finally in USMF.

The following illustration shows how this example might appear on the **Multilevel pegging** page for the planned production order.

Last updated on 11/19/2025

Exclude products that have specific product lifecycle states

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Released products and released product versions include a **Product lifecycle state** field. This field lets you control, among other things, which products are included during master planning. You can add, remove, and edit lifecycle states as required by going to **Product information management** > **Setup** > **Product lifecycle state**. For each product lifecycle state, set the **Is active for planning** option to *Yes* if products that have that state should be included during master planning. When the option is set to *No*, the associated products and variants will be excluded from all master planning and all calculations at the bill of materials (BOM) level.

Released products and variants where the **Product lifecycle state** field is left blank are treated as though they have a product lifecycle state where the **Is active for planning** option is set to *Yes*. In other words, they will be included during master planning.

> **Note**
>
> To help avoid unnecessary supply suggestions, we strongly recommend that you associate all obsolete released products and variants with a product lifecycle state where the **Is active for planning** option is set to *No*. This approach is especially important when you work with product configuration variants that aren't reusable, because it will help prevent waste.

Related information

To learn more about product lifecycle states, including how to exclude products from master planning and BOM-level calculations, go to Product lifecycle states.

Last updated on 01/09/2026

Coverage time fences

This article describes how to set up *coverage time fences*. Planners can define the planning horizon (the coverage time fence in days), and exclude supply and demand that falls beyond that horizon. Therefore, coverage time fences help prevent "noise" caused by supply suggestions that don't require attention for months. Examples include next year's forecast and customer orders that are placed far beyond the normal lead time.

A coverage time fence is the number of days after today's date (or, more precisely, the date when you do the planning run) that supply and demand are excluded. To help avoid delays, you must ensure that the coverage time fence is longer than the total lead time. The default system value is 100 days.

You can specify a coverage time fence at each of the following levels:

- **Coverage group** – You can set a default coverage time fence for each coverage group.
- **Item coverage** – You can override the coverage time fence that is inherited from the coverage group that is assigned to an item.
- **Master plan** – You can override the coverage time fences that are inherited from the coverage group and item coverage settings.

The following sections explain how to specify a coverage group at each level.

Set a coverage time fence for a coverage group

When you specify a coverage time fence for a coverage group, the setting applies to all items (products) that belong to that group. However, you can override the setting for specific items or specific master plans.

To specify a coverage time fence for a coverage group, follow these steps:

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. Select an existing coverage group in the list, or create a new coverage group.
3. On the **General** FastTab, set the **Coverage time fence (days)** field to the number of days that you want to use as the coverage time fence for the coverage group.

Set a coverage time fence for a specific item

Every item (product) belongs to a coverage group. If you don't explicitly assign a coverage group to an item, a default coverage group applies. Every item inherits a coverage time fence from its coverage group. However, you can override this time fence for specific items.

To specify a coverage time fence for a specific item, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a product in the grid.
3. On the Action Pane, on the **Plan** tab, in the **Coverage** group, select **Item coverage**.
4. On the **Item coverage** page, on the **Overview** tab, select or create a row for the site where you want to set a coverage time fence.
5. Select the **General** tab to open the settings for the selected site.
6. Select the **Override coverage group settings** check box.
7. Set the **Coverage time fence (days)** field to the number of days that you want to use as the coverage time fence for the item.

Set a coverage time fence for a specific master plan

You can specify a coverage time fence at the master plan level. By using this setting, you define the number of days that the master planning calculation covers for a master plan. This setting overrides any coverage time settings that you define for each relevant item and coverage group.

To specify a coverage time fence for a specific master plan, follow these steps:

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select an existing master plan in the list, or create a new master plan.
3. On the **Time fences in days** FastTab, set the **Coverage** option to *Yes*. Then, in the field under the option, enter the number of days that you want to use as the coverage time fence for the master plan.

Considerations for coverage time fences

As you set up coverage time fences, consider the following points:

- Coverage time fences affect only input data for master planning. If delays occur, the resulting planned orders might have a date that is after today's date plus the coverage time fence.
- Coverage time fences are specified in calendar days. Calendars that use working days don't affect the time fence calculation. For example, a week is always considered seven days, even if weekends are set up as closed days in the working time calendar.

- The system doesn't generate requirement transactions for any supply and demand that falls outside the coverage time fence.
- If any approved supply and demand fall outside the coverage time fence, the system doesn't load them into the engine. Therefore, they don't trigger any replenishment, and the system doesn't calculate delays. Nevertheless, don't remove this supply and demand from the system.
- The system ignores variations in safety stock quantities (from minimum keys) if they fall outside the coverage time fence.
- The system ignores intercompany demand if the requested ship date that is calculated isn't inside the coverage time fence. For the deprecated master planning engine, the coverage time fence doesn't limit intercompany demand.
- The system ignores demand forecasts if the budget date isn't inside the coverage time fence. For the deprecated master planning engine, the coverage time fence doesn't limit demand forecasts.
- Planning Optimization is time zone–aware. It considers the time zone at the supply and demand sites, and the time of the planning run. For example, master planning is triggered at 11 AM on October 15 from a site in Denmark (GMT+1 time zone), and a coverage time fence of ten days is used. In this case, supply and demand from a site in Seattle (GMT-8 time zone) is included until 2 AM on October 25 (which is ten 24-hour days after master planning was triggered, minus the time zone difference of nine hours). The deprecated master planning engine considers only the date of the time fence. Therefore, the result can differ.

Filter derived requirements by coverage time fence with Planning Optimization

In Supply Chain Management version 10.0.49 and later, the *Filter derived requirements by coverage time fence with Planning Optimization* feature is available in the Feature management workspace. In Supply Chain Management version 10.0.48 and earlier, this feature isn't available, and Planning Optimization doesn't filter derived requirements by coverage time fence.

When you turn on the *Filter derived requirements by coverage time fence with Planning Optimization* feature, Planning Optimization doesn't replenish derived requirements that fall outside the coverage time fence that applies to them. Examples include:

- Planned transfer issues, where the replenishing warehouse has different coverage time fence settings.

- Planned BOM component issues, where the raw material has different coverage time fence settings.

When the feature is on, derived demand follows the same coverage time fence boundaries that apply to a sales order. The *coverage time fence boundary* refers to the coverage time fence that applies to the inventory dimensions of the demand. Planning Optimization first checks the item coverage line, then the coverage group, and finally the global coverage time fence.

Examples of derived and internally created requirements include:

- Planned transfer issue
- Planned BOM/formula component issue
- Planned BOM line type service issue
- Safety stock issue
- Expired supply issue
- Kanban component issue

This behavior applies to all coverage codes: *Period*, *Requirement*, *Min/Max*, *Priority*, and *Decoupling point*.

Last updated on 07/23/2026

Net requirements and pegging information

07/21/2025

When you run master planning, it's important that you understand its output, how existing supply covers the demand, and why specific supply was generated. You can use the **Net requirements** page to better understand the calculated requirements that master planning produces.

The **Net requirements** page shows the net requirements that master planning calculated for the product. It also shows the coverage settings that were applied during the master planning run, a breakdown of the requirements totals by transaction type, and pegging information.

Open the Net requirements page

You can open the **Net requirements** page in any of the following ways:

- Go to **Product information management** > **Products** > **Released products**. Select or open a product. Then, on the Action Pane, on the **Plan** tab, in the **Requirement** group, select **Net requirements**.
- Go to **Sales and marketing** > **Sales orders** > **All sales orders**. Open a sales order. Then, on the **Sales order lines** FastTab, on the toolbar, select **Product and supply** > **Net requirements**.
- Go to **Master planning** > **Master planning** > **Planned orders**. Select or open a planned order. Then, on the Action Pane, on the **View** tab, in the **Requirements** group, select **Requirement profile**.

Use the Net requirements page

The **Net requirements** page consists of upper and lower sections. The Action Pane on this page includes an **Update** button.

Update the Net requirements page to reflect new supply and demand

Before new supply or demand will appear among the net requirements of an item, you must do one of the following:

- Run a master plan that includes the item.
- Open the **Net requirements** page for the item and then, from the Action Pane, select **Update** > **Master planning**. This action runs master planning for the selected item.

For example, if you manually create a planned order (supply), change an existing planned order (supply), or create sales order (demand), the changes won't appear on an item's **Net requirements** page until after master planning runs for that item.

Select a master plan to view

Before you review the net requirements of the product, be sure to select the relevant master plan in the **Plan** field at the top of the page.

Upper section

The upper section of the page provides the following tabs:

- **Overview** – View the item requirements of the product dimensions.
- **Item coverage** – View the coverage settings that were used during the calculation of requirements.
- **Summary** – View a breakdown of the requirements totals by transaction type.
- **Period** – View the receipts, issues, and projected available inventory for each period that is defined by the period template. You can also get a graphical view of the projected available inventory.

Lower section

The lower section of the page provides the following tabs:

- **Overview** – View the list of product requirements that were calculated during the master planning run, together with issue and receipt transactions that correspond to the requirements. By default, the list is sorted by requirement date. When you select a requirement, the **Pegging** FastTab in the lower section shows either the source of the requirement or the transactions that fulfill the requirement.
- **General** – View detailed information about the selected requirement.
- **Action** – View action messages for the requirements.

The Action Pane

The following commands are available on the Action Pane:

- **Update** > **Master planning** – Run master planning directly from the **Net requirements** page.
- **Update** > **Forecast planning** – Run forecast planning directly from the **Net requirements** page. Planning Optimization doesn't support this operation.

- **Update** > **Continuity scheduling** – Run continuity scheduling directly from the **Net requirements** page. Planning Optimization doesn't support this operation.

Example scenario

This example shows how pegging information is presented on the **Net requirements** page.

Prerequisites

Before you work through the scenario, prepare the following prerequisites:

1. You must work on a system where the standard sample data is available, and you must set the legal entity to *USMF*.
2. This example uses product *1000*, which is part of the USMF sample data. Make sure that this product is available, and that it's set up in the following way:
    - **Default order type:** *Purchase order*
    - **Inventory on-hand:** *10.00*
3. Create a sales order for a quantity of *25.00* of product *1000*. Use the storage dimensions where on-hand inventory is located.
4. Run master planning for the *DynPlan* master plan.

Review the calculated requirements

Next, you'll open the **Net requirements** page for product *1000* to review how calculated requirements correspond to each other.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product that has an **Item number** value of *1000*.
3. On the Action Pane, on the **Plan** tab, in the **Requirement** group, select **Net requirements**.
4. On the **Net requirements** page, set the **Plan** field to *DynPlan*.
5. On the **Overview** tab in the lower part of the page, verify that the following requirements are listed as rows in the grid.

Expand table

Reference	Requirement quantity	Accumulated
On-hand	10.00	10.00
Planned purchase orders	15.00	25.00
Sales order	-25.00	(Blank)

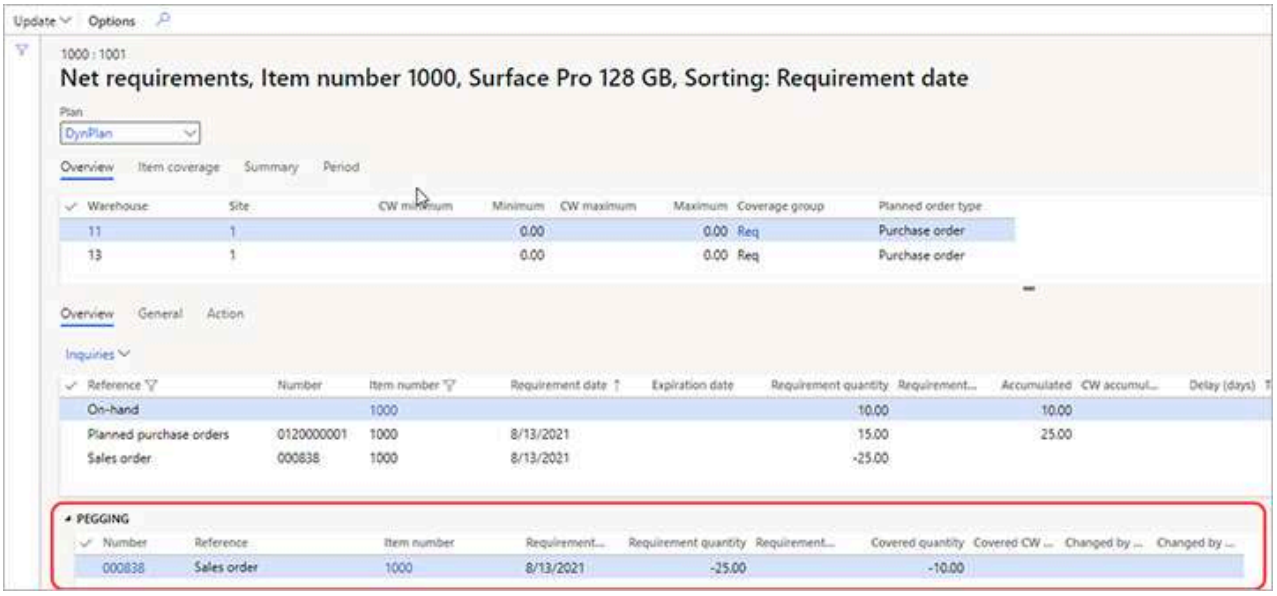
> **Note**
>
> The **Requirement quantity** field represents the total quantity that the requirement either requires (if the value is negative) or supplies (if the value is positive). The **Accumulated** field represents the total receipt and issue quantities that accumulated up through the selected period.

6. Select the *On-hand* requirement line, and then, on the **Pegging** FastTab, review the requirements that are covered by this supply. The following row should appear there.

Expand table

Reference	Requirement quantity	Covered quantity
Sales order	-25.00	-10.00

The existing on-hand inventory partially covers the demand that comes from the sales order.

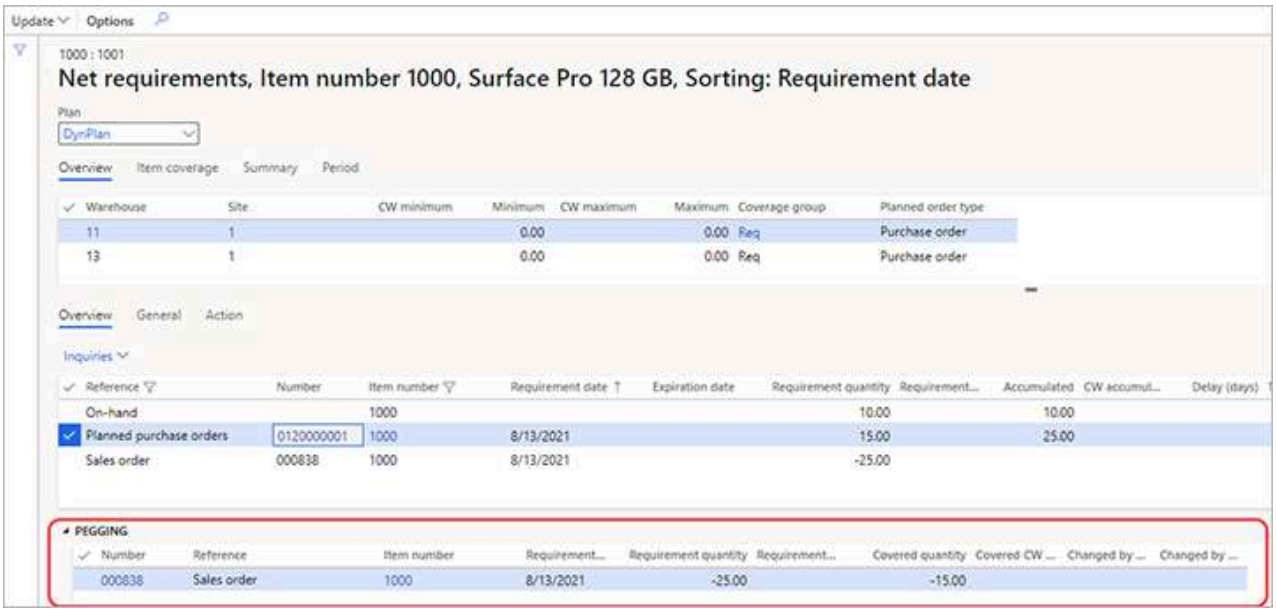


7. Select the *Planned purchase orders* requirement line, and then, on the **Pegging** FastTab, review the requirements that are covered by this supply. The following row should appear there.

Expand table

Reference	Requirement quantity	Covered quantity
Sales order	-25.00	-15.00

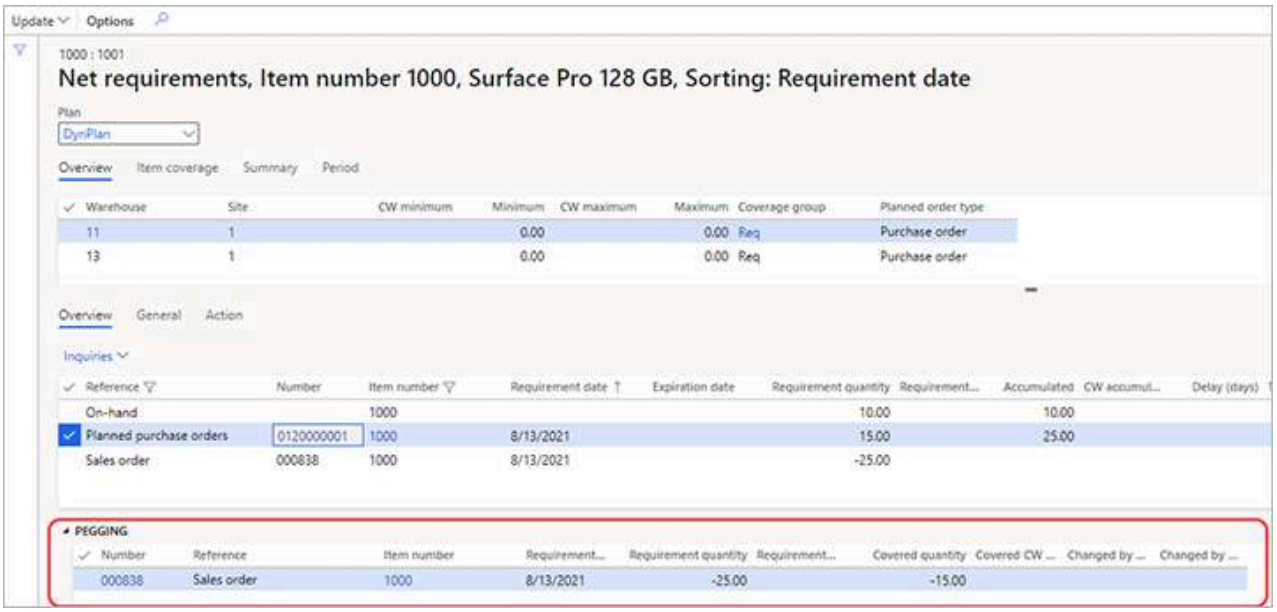
Because the sales order is already partially covered, the system creates a planned purchase order for the remaining uncovered quantity.



8. Select the *Sales order* requirement line, and then, on the **Pegging** FastTab, review the requirements that cover this demand. The following rows should appear there.

Expand table

Reference	Requirement quantity	Covered quantity
On-hand	10.00	10.00
Planned purchase orders	15.00	15.00



> **Note**
>
> The *Safety stock* requirement isn't included on the **Net requirements** page.

Planning with negative on-hand quantities

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

If the system shows a negative aggregate on-hand quantity, the planning engine treats the quantity as 0 to help avoid over-supply. Here's how this functionality works:

1. Master planning aggregates on-hand quantities at the lowest level of coverage dimensions. For example, if *location* isn't a coverage dimension, master planning aggregates on-hand quantities at the *warehouse* level.
2. If the aggregate on-hand quantity at the lowest level of coverage dimensions is negative, the system assumes that the on-hand quantity is really 0.

> **Important**
>
> The planning system is only as precise as the input data. If the input data is inaccurate, negative on-hand records indicate that the inventory information in Microsoft Dynamics 365 Supply Chain Management is out of sync with the real world. Therefore, the planning result is flawed. To get a precise planning result, minimize the number of records that show a negative on-hand quantity.

Example 1

Warehouse 13 is configured in the following manner:

- **Coverage code:** Min./Max.
- **Minimum:** 15 pieces (pcs.)
- **Maximum:** 25 pcs.

The lowest level of coverage dimensions is *warehouse*, and the following on-hand quantities are recorded at the *location* level:

- **Site 1, warehouse 13, location 1:** 20 pcs.
- **Site 1 warehouse 13, location 2:** −8 pcs.

Therefore, the aggregate on-hand quantity for warehouse 13 is 12 pcs. (= 20 pcs. − 8 pcs.).

In this case, the planning engine uses an aggregate on-hand quantity of 12 pcs. for warehouse 13.

The result is a planned order of 13 pcs. (= 25 pcs. − 12 pcs.) to refill warehouse 13 from 12 pcs. to 25 pcs.

Example 2

Warehouse 13 is configured in the following manner:

- **Coverage code:** Min./Max.
- **Minimum:** 15 pcs.
- **Maximum:** 25 pcs.

The lowest level of coverage dimensions is *warehouse*, and the following on-hand quantities are recorded at the *location* level:

- **Site 1, warehouse 13, location 1:** 4 pcs.
- **Site 1 warehouse 13, location 2:** −8 pcs.

Therefore, the aggregate on-hand quantity for warehouse 13 is −4 pcs. (= 4 pcs. − 8 pcs.). In other words, it's less than 0 (zero).

In this case, the planning engine assumes that the on-hand quantity for warehouse 13 is 0 pcs. instead of −4 pcs.

The result is a planned order of 25 pcs. (= 25 pcs. − 0 pcs.) to refill warehouse 13 from 0 pcs. to 25 pcs.

Planning when there is a reservation against negative on-hand inventory

If you adjust inventory while physical reservations exist, you can cause a situation where an order is physically reserved against negative inventory. In this case, because a physical reservation exists, you need to have supply to cover the reserved quantity. Therefore, replenishment is

required, so the system will either create a planned order to replenish the quantity that couldn't be covered by the existing on-hand inventory, or cover it with an existing order for the item.

The following example illustrates this scenario.

Example

The system is configured in the following way:

- Product *FG* exists and has *10* pcs. of on-hand inventory.
- The product configuration allows for physical negative inventory.
- A sales order exists for a quantity of *10* pcs. of product *FG*.
- The sales order quantity is physically reserved against existing on-hand inventory.

You then adjust the quantity of product *FG* so that the on-hand inventory becomes 5. Because the on-hand product inventory is 5, the sales order quantity is now reserved against quantity that isn't available on-hand (it would be similar if on-hand were 0, in which case the sales order would be reserved against negative inventory). If you run master planning now, a planned order of quantity 5 for *FG* is created to supply the sales order, because master planning always uses existing supply or creates a new planned order to supply the physical reservation.

Handle negative on-hand with coverage groups

Because Planning Optimization treats negative aggregate on-hand as 0, you might need to configure your coverage groups to ensure that inventory is replenished to the levels your business requires. To control how the system responds to negative on-hand scenarios, define minimum and maximum quantities in item coverage groups.

To configure coverage groups for handling negative on-hand, follow these steps:

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. Select or create a coverage group, and then set the **Coverage code** field to *Min/Max*.
3. Go to **Master planning** > **Master planning** > **Setup** > **Coverage** > **Item coverage** for the relevant items.
4. Set the **Minimum** field to the lowest inventory level that you want to maintain.
5. Set the **Maximum** field to the target inventory level for replenishment.

When you run master planning, the system creates planned orders to bring inventory back up to the maximum level whenever the projected on-hand falls below the minimum level. This

approach helps maintain appropriate inventory levels even when the system encounters negative on-hand quantities.

Learn more about coverage codes and how each replenishment method works in Coverage settings.

Related information

- Differences between Planning Optimization and the deprecated master planning engine
- Coverage settings
- Safety stock fulfillment for items

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Last updated on 05/05/2026

Calculate sales order delivery dates using CTP

Capable-to-promise (CTP) functionality lets you give customers realistic dates for when you can promise specific goods. For each sales line, you can provide a date that takes account of existing on-hand inventory, production capacity, and transportation times.

CTP extends available-to-promise (ATP) functionality by considering capacity information. Whereas ATP considers material availability and assumes infinite capacity resources, CTP considers availability of both materials and capacity.

CTP works slightly differently, depending on the master planning engine that you're using (Planning Optimization or the deprecated master planning engine). This article describes how to set it up for each engine.

How CTP compares to ATP

Use ATP when you sell from stock and capacity isn't a constraint. Use CTP when you assemble or make-to-order and capacity constraints must be taken into account. In such cases, CTP can often provide a more accurate result, as the following example shows.

Item A is an item that is composed of items B and C, and the on-hand quantity of item A is 0 (zero). If you do an ATP check, the ATP quantity is also 0 (zero). However, if you do a CTP check, the system checks the availability of items B and C, checks the resources that are required to make them into item A, and calculates how many of item A can be made. In addition, the CTP calculation can check the resources and materials that are required to make more of items B and C, and to use them to make more of item A.

A CTP calculation that considers both materials and resources might show a larger quantity than a calculation that checks only materials, particularly when the item that's being checked is an assemble or make-to-order item. In other words, CTP functionality is based on the explosion function and can be run for a selected sales order line to calculate the expected delivery date.

Near real-time CTP

Near real-time CTP enables the system to calculate CTP confirmed dates in the background, without blocking user interface (UI) interactions and without requiring that you run planning to

update the dates. *Near real-time CTP* also lets you use standard CTP delivery date control with Planning Optimization. This approach removes some of the limitations that apply when you use CTP with either planning engine. Without *Near real-time CTP*, you must use *Batch CTP* instead of *CTP* delivery date control if you're using Planning Optimization.

> **Note**
>
> - Because both Planning Optimization and the deprecated master planning engine can now use standard *CTP* delivery date control, CTP for Planning Optimization delivery date control is renamed in Supply Chain Management version 10.0.41. It's now named *Batch CTP* instead of *CTP for Planning Optimization* to better describe the difference.
> - Because CTP considers material and capacity availability and lead times for the bill of material and route for the item sold, the CTP calculation isn't as fast as the ATP calculation. When deciding which order promising policy to apply for an item, base the policy on whether the item is produced to and sold from stock (ATP, ATP + Issue Margin, or Sales Lead Time) or if it's an assemble or make-to-order item (CTP).

Enable Near real-time CTP

Before you can use *Near real-time CTP*, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.41 or later.
- The following features must be turned on in Feature management (in this order):
    1. *Improve Planning Optimization performance by merging and queueing plan regeneration jobs*
    2. *Near real-time CTP*

Queueing and merging plan regeneration jobs

The prerequisite feature, *Improve Planning Optimization performance by merging and queueing plan regeneration jobs*, enables the system to queue and merge plan regeneration requests. This capability eliminates the "Another job for the same plan is currently in progress" error and minimizes the average time that is required to complete a planning run when multiple users are working at the same time.

Here's how queueing and merging works:

- By default, the system can merge up to 10 planning requests into a single job. When the system reaches that limit, it adds new planning requests to the queue as standalone jobs. The system can also merge those jobs into another new job. If you need to change the maximum number of planning requests, contact Microsoft Support for assistance.

- You can merge planning requests into a single job only if they meet the following requirements:
  - All merged requests must belong to the same organization and master plan.
  - All merged requests must use the same filter attribute (such as `ItemId`).
  - All merged requests must have the same UI language.

- Before the system adds a planning request to the queue as a standalone job, it checks for other existing jobs that aren't yet started and can be merged with it.

- By default, the queue can hold up to 20 jobs. When the queue reaches that limit, all users who submit planning runs receive an error until more space becomes available in the queue. Each job in the queue can include several merged planning jobs. If you need to change the maximum number of jobs that are allowed in the queue, contact Microsoft Support for assistance.

Near real-time CTP functionality

Near real-time CTP provides the following functionality:

- Multiple sellers can create sales lines at the same time and immediately get confirmed delivery dates.
- When you mass-import sales lines, the system automatically calculates confirmed delivery dates.
- You can use the **Calculate confirmed delivery dates** dialog box to make the system recalculate all confirmed dates for a sales order. If you set the **Allow multiple deliveries** option to *No*, the system automatically sets all sales order lines so that they use the same delivery date.

How CTP differs, depending on the master planning engine and delivery date control that you use

The following table summarizes how CTP differs, depending on the master planning engine and delivery date control that you use.

Expand table

Element	Batch CTP	CTP
Planning engine support	Planning Optimization only	Planning Optimization and the deprecated master planning engine
Delivery date control setting for orders, order lines, and products	*Batch CTP*	*CTP*
Calculation time	The calculation is triggered when a dynamic plan is run as a scheduled task.	The calculation is immediately triggered each time that you enter or update a sales order line.
Batch CTP status field value	A value of *Not ready* is shown for orders and order lines where the dynamic plan hasn't been run since the orders and lines were created or last updated. A value of *Ready* is shown for orders and lines where CTP was used to calculate confirmed delivery dates by running the dynamic plan.	A value of *Ready* is always shown.
CTP status field value	The field isn't shown.	A value is shown only if you're using Planning Optimization. The value represents the status of the delivery date calculation.

Set default delivery date control methods

The system can use any of several methods to calculate delivery date estimates for each order and order line. Set the delivery date control method that you want to use most often as the global default method. You can also set individual overrides for each product. Later, you can override the default methods for each order and/or order line as you require.

Set the global default delivery date control

The default delivery date control method applies to all new order lines where an override doesn't apply. To select a method, follow these steps:

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.

2. On the **Shipments** tab, on the **Delivery control** FastTab, in the **Delivery date control** field, select the method that you want to use as the default method for your company:

    - *None* – Don't calculate delivery dates.
    - *Sales lead time* – Sales lead time is the time between creation of the sales order and shipment of the items. The delivery date calculation is based on a default number of days, and doesn't consider stock availability, known demand, or planned supply.
    - *ATP* – ATP is the quantity of an item that's available and can be promised to a customer on a specific date. The ATP calculation includes uncommitted inventory, lead times, planned receipts, and issues.
    - *ATP + Issue margin* – The shipping date equals the ATP date plus the issue margin for the item. The issue margin is the time that is required to prepare the items for shipment.
    - *CTP* – Use the standard CTP calculation, which provides near-real-time calculations. To use this option with Planning Optimization, you must enable *Near real-time CTP* for your system.
    - *Batch CTP* – Use the non-real-time CTP calculation that's available for Planning Optimization. This setting has no effect if you're using the deprecated master planning engine.

Set delivery date control overrides for individual products

Assign overrides for specific products when you want to use a delivery date control method other than the one that you set as your global default method.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product that you want to set up.
3. On the Action Pane, on the **Manage inventory** tab, in the **Order settings** group, select **Default order settings**.
4. On the **Default order settings** page, on the **Sales order** FastTab, set the **Override delivery control** option to *Yes*.
5. In the **Delivery date control** field, select the method to use for the selected product. The same options that are described in the Set the global default delivery date control section are available.

Schedule Batch CTP calculations

When you use Batch CTP, you must run a dynamic plan to trigger the system to do the CTP calculations and then set the confirmed ship and receipt dates for all relevant orders. The plan

must include all items that confirmed ship and receipt dates are required for. (When you use standard CTP delivery date control, the CTP calculations are immediately done locally. Therefore, you don't have to run a dynamic plan to see the CTP results.)

To ensure that the dates are available in good time for all users, set up batch jobs to run the relevant plans on a recurring basis. For example, a batch job that is set up to run a dynamic plan every 30 minutes sets the confirmed ship and receive dates every 30 minutes. Therefore, users who enter and import orders wait a maximum of 30 minutes to get the confirmed ship and receive dates.

To set up a batch job to run a dynamic plan on a regular schedule, follow these steps:

1. Go to **Master planning** > **Master planning** > **Run** > **Master planning**.
2. In the **Master planning** dialog box, on the **Parameters** FastTab, set the **Master plan** field to the dynamic plan that you want to run.
3. On the **Run in the background** FastTab, set the **Batch processing** option to *Yes*.
4. Select **Recurrence**, and set up the schedule as required.
5. Select **OK** to save the schedule.
6. Select **OK** to create the batch job and close the dialog box.

Use standard CTP delivery date control

To use standard CTP delivery date control, you must either use the deprecated master planning engine or use Planning Optimization with *Near real-time CTP* enabled.

Create a new order that uses standard CTP delivery date control

Each time that you add a new sales order or order line, the system assigns a default delivery date control method to it. The order header always starts with the global default method. If you assign an override to an ordered item, the new order line uses that override. Otherwise, the new order line also uses the global default method. Therefore, set your default methods so that they match the delivery date control method that you most often use. After you create an order, you can override the default method at the order header and/or order line level as you require. Learn more in Set default delivery date control methods and Change existing sales orders to use CTP.

View confirmed delivery dates when you use standard CTP delivery date control

CTP calculations apply to orders and order lines where the **Delivery date control** field is set to *CTP*.

For sales lines that use standard CTP delivery date control, the system automatically sets the **Confirmed ship date** and **Confirmed receipt date** fields each time that you save a sales line. If you make a relevant change to a sales line later (for example, by changing its quantity or site), the system immediately recalculates the dates.

- To view the confirmed delivery dates for a sales order line, open the sales order, and select the sales line. Then, on the **Line details** FastTab, on the **Delivery** tab, review the **Confirmed ship date** and **Confirmed receipt date** values.
- To view the confirmed delivery dates for an entire order, open the sales order, and select the **Header** view. Then, on the **Delivery** FastTab, review the **Confirmed ship date** and **Confirmed receipt date** values.

Use Batch CTP

Create a new order that uses Batch CTP

Each time that you add a new sales order or order line, the system assigns a default delivery date control method to it. The order header always starts with the global default method. If you assign an override to an ordered item, the new order line uses that override. Otherwise, the new order line also uses the global default method. Therefore, set your default methods so that they match the delivery date control method that you most often use. After you create an order, you can override the default method at the order header and/or order line level as you require. Learn more in Set default delivery date control methods and Change existing sales orders to use CTP.

View confirmed delivery dates when you use Batch CTP

Batch CTP calculations apply to orders and order lines where the **Delivery date control** field is set to *Batch CTP*.

For sales lines that use Batch CTP, the **Confirmed ship date** and **Confirmed receipt date** fields stay blank until you run the appropriate dynamic plan (typically by using a periodic batch job). The system then sets these fields automatically. Learn more in Schedule Batch CTP calculations.

The **Batch CTP status** field shows whether confirmed dates are calculated for each line that uses Batch CTP. The field is set to *Not ready* for lines and orders where the confirmed delivery dates aren't calculated yet or are no longer valid because of other changes. The field is set to *Ready* for lines and orders that are calculated. You can view the status for each line and for the entire order.

- To view the status for a sales order line, open the sales order, and select the sales line. Then, on the **Line details** FastTab, on the **Delivery** tab, review the **Batch CTP status** value. The **Delivery** FastTab also shows the **Confirmed ship date** and **Confirmed receipt date** values for the line after they're calculated.
- To view the status for an entire order, open the sales order, and select the **Header** view. Then, on the **Delivery** FastTab, review the **Batch CTP status** value. The **Delivery** FastTab also shows the **Confirmed ship date** and **Confirmed receipt date** values for the order after they're calculated.

> ⓘ **Note**
>
> - If you update a sales order line after Batch CTP calculates confirmed dates for it, the system clears those dates until the next time that the appropriate dynamic plan runs.
> - If you edit a related setting that might affect existing confirmed dates (for example, by changing lead times, reservations, or markings), clear the confirmed dates for the relevant existing orders. This action changes the status of each relevant line to *Not ready*. This change causes the system to recalculate the confirmed dates the next time that it runs the dynamic plan.
> - Batch CTP only calculates confirmed dates for sales lines with status *Not Ready*, so subsequent Planning Optimization runs that affect planned supply for sales lines with status *Ready* might cause the confirmed dates to become out of sync with the latest plan result. To prevent this, consider firming planned orders early to make sure that generated supply remains unaffected by future Planning Optimization runs.

Change existing sales orders so that they use CTP

You can change the **Delivery date control** value for any open order at any time. You can change the value at the header level and/or for each line as you require.

Change to CTP at the order header level

To change an order so that it uses CTP at the order header level, follow these steps:

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Open the sales order that you want to set up, or create a new one.
3. Select **Header** to open the header information on the **Sales order details** page.

4. On the **Delivery** FastTab, set the **Delivery date control** field to one of the following values, depending on the planning engine that you're using:

    - *CTP* – Use the standard CTP calculation, which provides near-real-time calculations. To use this option with Planning Optimization, you must enable *Near real-time CTP* for your system.
    - *Batch CTP* – Use the non-real-time CTP calculation that's available for Planning Optimization. This setting has no effect if you're using the deprecated master planning engine.

5. Select **OK** to apply your changes.

Change to CTP at the order line level

If you create an order line by using a different delivery date control method, you can change to CTP at any time. Changes that you make at the line level don't affect any other lines. However, they might cause the overall order delivery dates to move forward or backward, depending on how each updated line calculation changes.

To change an order so that it uses CTP at the line level, follow these steps:

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.

2. Open the sales order that you want to set up, or create a new one.

3. On the **Sales order details** page, on the **Sales order line** FastTab, select the sales order line that you want to set up.

4. On the **Line details** FastTab, on the **Delivery** tab, set the **Delivery date control** field to one of the following values, depending on the planning engine that you're using:

    - *CTP* – Use the standard CTP calculation, which provides near-real-time calculations. To use this option with Planning Optimization, you must enable *Near real-time CTP* for your system.
    - *Batch CTP* – Use the non-real-time CTP calculation that's available for Planning Optimization. This setting has no effect if you're using the deprecated master planning engine.

    The **Available ship and receipt dates** dialog box appears, and shows the available ship and receipt dates. This dialog box works the same way for order lines as it does for the order header, as described in the previous section.

5. On the Action Pane, select **Save**.

Additional resources

- Keep supply for confirmed demand in Planning Optimization
- Optimize confirmed dates for CTP line changes

Last updated on 07/27/2026

Keep supply for confirmed demand in Planning Optimization

The *keep supply for confirmed demand* feature ensures that customer commitments provided by capable-to-promise (CTP) functionality are strictly preserved between planning runs. This functionality is applied across all BOM levels, meaning that all derived requirements originating from a confirmed sales line are preserved. Such requirements are referred to as *confirmed requirements*.

What qualifies as confirmed demand

A sales order line is considered confirmed demand when it meets the following criteria:

- The sales order line has a **Line status** equal to *Open order*.
- The **Confirmed ship date** field is populated. The CTP calculation typically sets this date.

What data is preserved

When the system identifies confirmed demand, it preserves the following data across planning runs:

- Planned orders
- Pegging information for planned orders, released orders, and on-hand inventory
- Routes
- Route jobs
- Capacity reservations

> **Important**
>
> The following behavior applies to this feature:
>
> - Action dates and futures are updated for all requirements, including confirmed requirements.
> - This feature doesn't respect the *Period* coverage code for confirmed supplies created between planning runs.
> - This feature doesn't reevaluate product substitution after requirements are confirmed. For example, planning doesn't substitute a confirmed requirement with an alternate

item, even if the alternate item becomes available later.
- Sales order demands for co-products and by-products aren't treated as confirmed demand, even if they have a confirmed ship date.

Prerequisites

Before you can use the *keep supply for confirmed demand* feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.48 build 10.0.2645.33 or later.
- You must use Planning Optimization.
- The *Keep supply for confirmed demand in Planning Optimization* feature must be turned on in Feature management.

> **Note**
>
> This feature works together with CTP delivery date control. You can use either *Near real-time CTP* or *Batch CTP* delivery date control. The supply chain is retained if you later change the delivery date control to other values, as long as the confirmed ship date remains populated.

Set a master plan to keep supply for confirmed demand

You can choose to keep supply for confirmed demand at the master plan level. This setting controls whether the system preserves confirmed requirements for the selected master plan. The value applies to both non-interactive scenarios (such as CTP calculation, BOM explosion, and scheduling with finite capacity) and interactive scenarios (such as the master planning run dialog, where it serves as the default value).

To set a master plan to keep supply for confirmed demand:

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select or create a master plan.
3. On the **General** FastTab, set the **Keep supply for confirmed demand** option to one of the following values:

    - *Yes* – The system identifies confirmed sales lines and preserves the full supply chain linked to them during planning runs. Planned orders that are part of a confirmed

supply chain aren't replanned.

- *No* – The system doesn't consider confirmed requirements. All planned orders that aren't approved and fall outside the freezing time fence are replanned during each planning run. This is the default behavior.

Keep supply for confirmed demand when running a master plan interactively

You can choose to keep supply for confirmed demand when you run master planning interactively. The planning run dialog provides the **Keep supply for confirmed demand** option, and the selected master plan automatically sets the default value.

To keep supply for confirmed demand when running a master plan interactively:

1. Go to **Master planning** > **Master planning** > **Run** > **Master planning**.
2. In the **Master planning** dialog box, on the **Parameters** FastTab, set the **Keep supply for confirmed demand** option to *Yes* or *No*. The behavior is identical to that described in the previous section.

Check master planning history to see whether supply was kept for confirmed demand

After a planning run completes, you can verify whether the **Keep supply for confirmed demand** parameter was active for that run by reviewing the master planning history.

To view the history of a master planning run and check whether supply was kept for confirmed demand:

1. Go to **Master planning** > **Master planning (Workspace)** > **History**.
2. Select the planning run you want to inspect.
3. On the **Request** FastTab, locate the **Keep supply for confirmed demand** field. The value shows whether confirmed requirements were preserved during that specific run.

> **Tip**
>
> Reviewing the master planning history is useful when troubleshooting scenarios where confirmed delivery dates appear to have changed unexpectedly. If the parameter shows *No*

for a given run, that run may have replanned supply that was previously backing a confirmed sales order line.

Control how on-hand inventory is pegged to confirmed demand

By default, the *keep supply for confirmed demand* feature preserves planned orders, pegging, routes, route jobs, and capacity reservations that are part of a confirmed supply chain. However, the pegging from confirmed demand to *received supply* (on-hand inventory) is still subject to the *Positive days* coverage group setting. This condition applies when a confirmed sales line is initially pegged to a purchase order (or other supply) that you later receive and add to on-hand inventory. As a result, if the confirmed sales line falls outside positive days, the system can reallocate the on-hand inventory to an earlier demand during the next planning run, even if the original sales line still has a confirmed ship date.

To override this behavior, use the **Keep received supply outside positive days** parameter on the coverage group.

> ⓘ **Note**
>
> This parameter is visible only when both Planning Optimization and the *Keep supply for confirmed demand in Planning Optimization* feature are enabled. The parameter is used by Planning Optimization and takes effect only for confirmed demand (sales orders with a confirmed ship date) when you run master planning with the **Keep supply for confirmed demand** parameter enabled.

To set the **Keep received supply outside positive days** parameter on a coverage group:

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. Select or create a coverage group.
3. On the **General** FastTab, set the **Keep received supply outside positive days** option to one of the following values:

    - *No* – Confirmed demand is pegged to on-hand inventory only when the demand falls within the positive days. This setting allows on-hand inventory that was previously backing a confirmed sales line to be reallocated to earlier orders if the supply arrives earlier than required. This is the default setting.

- *Yes* – Confirmed demand is always pegged to on-hand inventory, regardless of the positive days setting. The pegging from a confirmed sales line to on-hand inventory is preserved across planning runs even when the demand date is outside positive days.

Interaction with approved planned orders and the freezing time fence

The *keep supply for confirmed demand* feature works alongside approved planned orders and the existing freezing time fence functionality. These features preserve planned orders from being replanned during planning runs, but they use different criteria:

Expand table

Feature	Criteria for preservation
Approved planned orders	Retains approved planned orders with the status of *Approved*.
Freezing time fence	Retains planned orders with a requirement date within the freezing time fence.
Keep supply for confirmed demand	Retains planned orders, pegging information, and other elements that are part of a confirmed supply chain, regardless of their requirement date and status.

Example scenario 1: Period grouping

This example shows how the *keep supply for confirmed demand* feature identifies and preserves confirmed requirements across a multilevel BOM, including scenarios where components are shared (grouped) between multiple sales orders.

Example scenario 1 setup

A manufacturer produces a finished good, *ItemFG1*, with the following BOM structure:

txt

```
ItemFG1 (finished good, produced)
└── ItemSFG1 (semi-finished, produced, coverage by period)
    └── ItemSFG2 (semi-finished, produced)
        └── ItemRM1 (raw material, purchased)
        └── ItemRM2 (raw material, purchased)
        └── ItemRM3 (raw material, purchased)
```

In this example, *ItemSFG1* is a component that uses period grouping, meaning its planned production order is grouped and shared across multiple sales orders.

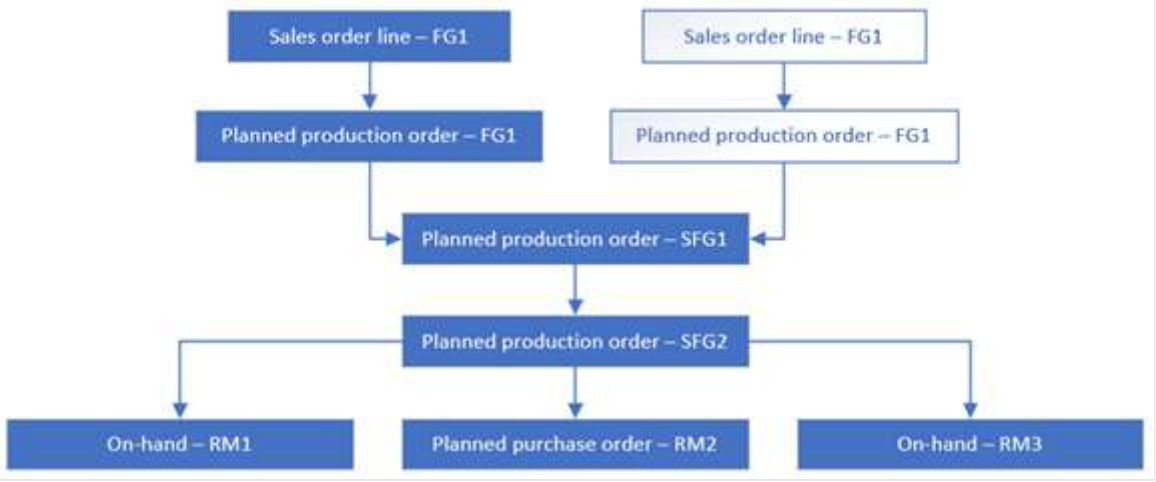
Identify confirmed requirements for example scenario 1

The following orders are entered:

- *Customer 1* places a sales order for *5 pcs* of *ItemFG1* and uses CTP delivery date control.
- *Customer 2* places a sales order for *5 pcs* of *ItemFG1* but doesn't use CTP delivery date control. No confirmed shipping date is set.

When planning runs, it groups the demand for *ItemSFG1* from both orders into a single planned production order. Because the sales line for *Customer 1* has a confirmed shipping date, the system traces the entire supply chain from that sales line and identifies all related requirements as confirmed, including the shared *ItemSFG1* branch and all its derived requirements.

The following illustration shows how the system identifies confirmed requirements for both customers. The shared (grouped) *ItemSFG1* planned production order and its entire subtree are marked as confirmed (highlighted in blue).



Remove a confirmed sales line

If the sales order for *Customer 2* is later canceled, the system removes only the confirmed requirements that are explicitly linked to that sales order line. As the following illustration shows, only the planned production order for *ItemFG1* that was created for *Customer 2* is removed.

The grouped *ItemSFG1* branch and all its subcomponents remain intact. Their quantities aren't reduced either, because they remain linked to demand from *Customer 1* and are therefore considered confirmed.

Example scenario 2: Coverage by requirement

This example shows how the *keep supply for confirmed demand* feature identifies and preserves confirmed requirements when components are planned with the *Requirement* coverage code, including scenarios where part of the supply is already firmed into a purchase order.

Example scenario 2 setup

A manufacturer produces a finished good, *ItemFG1*, with the following BOM structure:

```txt
ItemFG1 (finished good, produced)
└── ItemRM1 (raw material, purchased, coverage by requirement)
```

Because *ItemRM1* uses the *Requirement* coverage code, each demand for *ItemFG1* generates its own dedicated supply for *ItemRM1*.

Identify confirmed requirements for example scenario 2

The following orders are entered:

- *Customer 1* places a sales order for *5 pcs* of *ItemFG1* and uses CTP delivery date control.
- *Customer 2* places a sales order for *5 pcs* of *ItemFG1* and uses CTP delivery date control.

When planning runs, the *Requirement* coverage code creates a separate supply branch for each sales line. Both sales lines have a confirmed ship date, so the system identifies both branches as confirmed. The supply for *ItemRM1* differs between them: the demand for *Customer 1* is covered by a planned purchase order, while the demand for *Customer 2* is covered by a released purchase order that you firmed previously.

The following illustration shows the two confirmed branches. Both branches and their pegging are marked as confirmed (highlighted in blue).

Remove a confirmed sales line backed by a firmed purchase order

If you later cancel the sales order for *Customer 2*, the system removes only the confirmed planned requirements that are explicitly linked to that sales line—in this case, the planned production order for *ItemFG1* that you created for *Customer 2*. The released purchase order for *ItemRM1* that was pegged to that planned production order isn't removed.

The pegging from the planned purchase order is also preserved as-is. Planning doesn't re-peg the released purchase order to other confirmed demand, even though the sales line for *Customer 2* is no longer present. The released purchase order remains in the system and can be used by any other demand that isn't yet confirmed. If the released purchase order remains unpegged, planning assigns it a *Cancel* action, provided that actions are enabled.

Performance considerations

The *keep supply for confirmed demand* feature adds processing time to each planning run, because the system must identify confirmed sales lines, trace the full supply chain across all BOM levels, and preserve the related requirements. The impact depends on the number of open sales lines that have confirmed ship dates and the depth of the BOM structures involved.

To minimize performance impact, consider the following recommendations:

- **Use this feature with dynamic plans used by CTP** – Dynamic plans with a shorter coverage period are the best fit for this feature.
- **Avoid using this feature with static plans or long coverage periods** – For static plans or plans with long coverage time fences, the number of confirmed requirements can grow significantly, which increases processing time and might result in a timeout.

Additional resources

- Calculate sales order delivery dates using CTP
- Optimize confirmed dates for CTP line changes

Last updated on 07/23/2026

Optimize confirmed dates for CTP line changes

The *optimize confirmed dates for CTP line changes* feature ensures that confirmed ship and receipt dates on sales order lines stay accurate when you modify fields that affect transport days. When you use capable-to-promise (CTP) with Planning Optimization, this feature automatically adjusts the confirmed dates to optimize delivery without requiring user action.

When the feature applies

This feature applies when you modify a sales order line in a way that changes the calculated transport days. Changes to any of the following fields can affect transport days:

- **Mode of delivery**
- **Warehouse**
- **Delivery address**

Transport days can either increase (for example, when the delivery mode adds more days, or when the destination or warehouse changes to a more distant location) or decrease (for example, when the delivery mode is faster).

How the feature works

When you modify a field on a sales order line that impacts transport days, the system automatically recalculates the requested and confirmed dates to reflect the new transport days.

The system optimizes the confirmed dates as follows:

- If the confirmed receipt date equals the requested receipt date and no past shipment is needed, the ship date moves closer to the receipt date.
- If the confirmed receipt date is later than the requested receipt date, the system optimizes toward the requested date within shipping constraints.
- If the change would cause the ship date to fall in the past, the confirmed receipt date is scheduled forward from the first feasible ship date.

Prerequisites

Before you can use the *optimize confirmed dates for CTP line changes* feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.48 build 10.0.2645.33 or later.
- You must use Planning Optimization.
- You must use capable-to-promise (CTP) delivery date control on the sales order line. You can use either *Near real-time CTP* or *Batch CTP*.
- The *Optimize confirmed dates for CTP line changes* feature must be turned on in Feature management.

Example scenario 1: Changing to a faster mode of delivery

This example shows how the feature adjusts confirmed dates when you change the mode of delivery to one with fewer transport days.

Example scenario 1 setup

The following modes of delivery are configured:

- *Air* – 1 transport day
- *Truck* – 10 transport days

Initial sales order line with truck

The user creates a sales order line with **Mode of delivery** set to *Truck*. CTP returns the following dates:

- **Requested ship date** – *November 11, 2025*
- **Confirmed ship date** – *November 11, 2025*
- **Requested receipt date** – *November 10, 2025*
- **Confirmed receipt date** – *November 10, 2025*

Change the mode of delivery to air

The user changes **Mode of delivery** to *Air*. The *optimize confirmed dates for CTP line changes* feature automatically adjusts the dates to reflect the shorter transport time. The result is:

- **Requested ship date** – *November 9, 2025*

- **Confirmed ship date** – *November 9, 2025*
- **Requested receipt date** – *November 10, 2025*
- **Confirmed receipt date** – *November 10, 2025*

Without this feature, the requested and confirmed ship dates stay *November 1, 2025* even though only one transport day is required.

Example scenario 2: Changing to a slower mode of delivery

This example shows how the *optimize confirmed dates for CTP line changes* feature adjusts confirmed dates when you change the mode of delivery to one with more transport days.

Example scenario 2 setup

The following modes of delivery are configured:

- *Air* – 1 transport day
- *Truck* – 10 transport days

Initial sales order line with air

You create a sales order line with **Mode of delivery** set to *Air*. CTP returns the following dates:

- **Requested ship date** – *November 9, 2025*
- **Confirmed ship date** – *November 9, 2025*
- **Requested receipt date** – *November 10, 2025*
- **Confirmed receipt date** – *November 10, 2025*

Change the mode of delivery to truck

You change **Mode of delivery** to *Truck*. The *optimize confirmed dates for CTP line changes* feature automatically adjusts the dates to reflect the longer transport time. Assuming there's no delay and the item can be shipped on *November 1, 2025*, the result is:

- **Requested ship date** – *November 1, 2025*
- **Confirmed ship date** – *November 1, 2025*
- **Requested receipt date** – *November 10, 2025*
- **Confirmed receipt date** – *November 10, 2025*

Without this feature, the requested and confirmed receipt dates would shift to *November 19, 2025* because the original ship date of *November 9, 2025* plus 10 transport days would be used, even though the item is available earlier.

Additional resources

- Calculate sales order delivery dates using CTP
- Keep supply for confirmed demand in Planning Optimization

Last updated on 07/23/2026

Master plans overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use various master plans to support your company's daily working operations, simulate different planning strategies that you want to monitor, and implement a company policy, such as a policy about internal performance or customer satisfaction.

This article describes various important strategies and parameters that you use to set up master planning. It also describes the main parameters that affect a plan and explains how those parameters influence the planned orders that are suggested.

You can configure master plans on the **Master plans** page.

Using master plans

Because of the speed at which the system processes master planning, you can set up as many plans as you like and run them as often as needed to fit your business requirements. For example, you can run a plan twice a day to always keep up with the latest demand and supply and to have other plans for different forecasts or simulations.

If you use capable to promise (CTP) for Planning Optimization, it runs the plan specified as the **Current dynamic master plan** fields on the **Master planning parameters** page (**Master planning** > **Setup** > **Master planning parameters**).

Types of scheduling methods

For each plan, on the **General** FastTab of the **Master plans** page (**Master planning** > **Setup** > **Plans** > **Master plans**), select the scheduling method that is used for production orders. You can schedule production at the operation level and the job level.

Operations scheduling

Use operations scheduling to provide a general estimate of the production process over time. Operations scheduling doesn't explode the operations for the production route into jobs. For

more information about operations scheduling, see Operations scheduling.

Job scheduling

Job scheduling is a more detailed scheduling method, where each operation is divided into its individual tasks or jobs. Job scheduling includes information about capacity. Use it to schedule individual jobs on the shop floor for an immediate or short-term time frame. For more information about job scheduling, see Job scheduling.

Time fences in days

For each plan, you can select how far in the future master planning calculates various requirements and other considerations. This period is known as a *time fence*. For the best performance in master planning, adjust the time fences to meet your business requirements. For each plan, find the time fences on the **Time fences in days** FastTab of the **Master plans** page (**Master planning** > **Setup** > **Plans** > **Master plans**).

> **Note**
>
> The time fences you select indicate how far in the future master planning calculates various requirements and other considerations. The time fences you select on this page override the time fences defined in the coverage group. This means setting a time fence option to *Yes* and defining the days overrides the time fence defined in the coverage group. When you set the option to *No*, the time fence is defined in the coverage group. Finally, if you don't want or need to use an option (for example, you don't want to use action messages), set it to *Yes*, and then set the time fence to **0** (zero) days.

Coverage

The coverage time fence represents the scheduling period, or how far out the demand should be included. In other words, it indicates your planning horizon.

By setting the **Coverage** option to *Yes*, you can override the coverage time fence that is defined for the item during master scheduling. Enter the number of days that the master scheduling calculation should cover requirements. The coverage time fence is calculated forward from the current date. Master planning always processes requirements that occur before the current date.

> **Note**

For the best performance in master planning, adjust the coverage time fence to your planning horizon.

Freeze

The freeze time fence represents the period when existing planned orders aren't changed when a new master schedule runs. The planned orders are frozen, and no new planned orders are suggested.

By setting the **Freeze** option to *Yes*, you can override the freeze time fence that is defined for the item during master scheduling. Enter the number of days that planning activity should be frozen. Remember that, during this period, no new planned orders are generated, and existing planned orders can't be changed.

Firming

The firming time fence indicates the time horizon in which planned orders are automatically converted to production and purchase orders. This process is also known as *automatic firming of planned orders*.

By setting the **Firming** option to *Yes*, you can override the firming time fence that is defined for the item during master scheduling. Enter the number of days that planned purchase orders and planned production orders should be automatically firmed. The firming time fence is calculated forward from the master scheduling date. Automatic firming of a planned purchase order can occur only if the item is associated with a vendor.

Forecast plan

The forecast plan time fence indicates how far in the future master planning creates planned orders for items where forecasted demand exists.

Set the **Forecast plan** option to *Yes* to override the forecast plan time fence that is defined for the item during master scheduling. Enter the number of days that the sales forecast from the forecast plan should be included in master planning.

Capacity

The capacity time fence indicates how far in the future the system considers the maximum capacity of your resources when it schedules orders. In other words, the plan schedules the production orders by using the production route for the items, and it considers the production route's resources and the maximum capacity of each resource.

Set the **Capacity** option to *Yes* to override the capacity time fence that is defined for the item during master scheduling. Enter the number of days that capacity should be planned for planned production orders. Master scheduling uses the active production route for the item and schedules backward from the requirement date. If the requirement date for a planned production order is outside the capacity time fence, the lead time is determined by the delivery time of the item. The capacity time fence is calculated forward from the current date.

Action message

Action messages suggest changes that you can make to the existing supply order to help optimize the supply plan. For example, they might recommend that you advance or postpone orders, or that you increase or decrease the order quantities.

Set the **Action message** option to *Yes* to override the action message time fence that is defined for the item during master scheduling. Enter the number of days that master scheduling should generate action messages for requirements. The action message time fence is calculated forward from the current date.

For more information about action messages, see Action messages.

> **Note**
>
> The calculation of action messages causes a longer running time for master planning. If you don't regularly analyze and apply action messages (daily, weekly, and so on), consider turning off the calculation during the master planning run. To turn off the calculation, on the **Master plans** page, set the **Action message** time fence to *0* (zero) for the master plan that you're running. Also make sure that the **Action message** setting is turned off for all the coverage groups.

Calculated delays

You can specify how far in the future possible delays in planned orders are detected and reported. In this way, you can schedule possible (delayed) delivery dates.

If a planned order can't be fulfilled for the requested date, the system plans the order for the earliest fulfillment date for a transaction, based on lead times, and material and capacity availability.

Approved requisitions time fence

You can set up master planning to create planned orders for requisition demand. Set the **Include requisitions** option to *Yes* on the **General** FastTab of the **Master plans** page. Then, when the purpose of an approved requisition is replenishment, master planning automatically creates a corresponding planned order to fulfill it. The replenishment method is determined by the supply policies that you set up for the items in your organization. After you create and approve the replenishment requisition, no additional user action is required.

By setting the **Approved requisitions time fence** option to *Yes* on the **Time fences in days** FastTab, you can override the approved requisitions time fence that is defined for the item during master scheduling. In this case, enter the number of days in the past that demand from approved requisitions with the replenishment purpose should be included in master scheduling. For example, you can specify that only unfulfilled, past-due demand from approved requisitions that were created in the last 10 days should be considered and planned for.

Sequencing

Sequencing enables you to arrange planned orders based on sequencing attributes that are associated with the finished product. It's often used to prepare production orders for packaging. For example, you can use it to pack boxes in a specific sequence, based on color and size.

By setting the **Sequencing** option to *Yes*, you can specify how far in the future the operations or jobs should be sequenced. Keep in mind that the longer the time fence is, the longer it takes for master planning to run.

Calculated delays FastTab

Delay options help guarantee that the orders have feasible planned dates. The following options are available on the **Calculated delays** FastTab of the **Master plans** page:

- **Ensure that the planned orders are not created prior to master planning run date** – Set this option to *Yes* to help guarantee that orders can't be scheduled for dates in the past.
- **Add the calculated delay to the requirement date** (under **Planned purchase orders**) – Set this option to *Yes* to add the calculated delay to the requirements.
- **Add the calculated delay to the requirement date** (under **Planned production orders**) – Set this option to *Yes* to add the calculated delay to the requirements.
- **Add the calculated delay to the requirement date** (under **Planned transfer**) – Set this option to *Yes* to add the calculated delay to the requirements.
- **Add the calculated delay to the requirement date** (under **Planned kanban**) – Set this option to *Yes* to add the calculated delay to the requirements.

When you set the **Add the calculated delay to the requirement date** options to *Yes* to add the delays to the requirements, the system considers the capacity of the resources and creates feasible planned orders. Recalculation of the planned order dates increases the running time for master planning. Therefore, if you don't need to use the delays, set the options to *No*.

Positive and negative days

Positive and negative days affect how master planning suggests planned orders and actions. Set positive and negative days on the item coverage group of the item. You can define the various coverage groups and set their parameters on the **Coverage groups** page (**Master planning** > **Setup** > **Coverage** > **Coverage groups**).

Positive days

Positive days indicate how far in the future master planning considers the current inventory or receipts to fulfill a future demand. For example, if you set the positive days to *100*, the current inventory can fulfill demand for the next 100 days. If there's an order 150 days from the current date, master planning creates a planned order to satisfy that demand, even though the on-hand inventory for the item can satisfy the order. For fast-moving items that have a short lead time, you might not want to use the on-hand inventory for an order that's far in the future. In this fast-moving case, the current on-hand inventory goes quickly, and more orders can be placed in the future to fulfill a future demand on time, which is possible due to the short lead time of the item.

The positive days also affect the action messages. For example, the system might recommend that you increase a planned purchase order so that it includes a demand that is within the number of positive days in the future. If you set the positive days to *100*, and if there's demand for an item in 30 days from the current date, the system creates a planned order to satisfy that demand. If there's demand for the same item in 90 days from the current date, the system recommends that you increase the quantity of the order in 30 days from the current date, so that the order also covers the demand in 90 days. However, if there's demand for the item in 150 days from the current date, the system doesn't recommend that you increase the quantity of the order that you already planned. Instead, a new planned order is created.

As a rule, set the positive days to a number that's between the longest lead time of the items and the coverage time fence. Assign items that you regularly procure or produce to a coverage group where the positive days equal the item's lead time.

Setting zero positive days in Planning optimization means that you always generate new supply for demand because the demand isn't allowed to be pegged against any existing supply.

Dynamic positive days

Dynamic positive days work like positive days, but they also consider the lead time. Learn more in Dynamic positive days for last-minute orders.

Negative days

Negative days indicate how late item receipts are allowed. They represent the number of days that you're willing to wait before you order new replenishment when you have negative inventory or don't have enough inventory. Negative days answer the question, Should we create a new purchase order for the item, or should we use an existing purchase, even though we know that the item will be late?

For example, you have a sales order for an item 15 days from the current date. You also have a purchase order for the same item. This purchase order is scheduled to be received in 20 days from the current date. Do you want the system to create a purchase order for that sales order, or do you want to use the existing order, even though you can't fulfill the sales order on time? If you set the negative days to less than *5* to indicate the item can be delayed a maximum of five days, the system creates a new planned purchase order to satisfy the sales order. If you set the negative days to greater than *5*, the system uses the existing order for the item.

The negative days also affect the performance of master planning. If you set the negative days to a high number, the system generates lots of action messages.

Set the negative days to a number that's less than the lead time of the item.

Dynamic negative days

Dynamic negative days consider the item's lead time and the negative days that you specify. The system creates a new planned purchase order based on the negative days time fence, which it calculates by using the following formula:

Lead time + Negative days + Current date – Requirement date

The system uses only the planned supply orders that are within this time fence, and it creates a new planned order outside it. The advantage of using dynamic negative days is that the system includes individual product lead times and reuses existing orders. It therefore avoids creating new planned orders that the system would push to a later day because of delays caused by lead time.

Learn more in Negative days and dynamic negative days.

Last updated on 03/25/2026

Master planning setup wizard

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides a guide for the **Master planning setup wizard**. It explains how parameter suggestions are calculated and also provides examples that show how different companies set up master planning, based on their business needs.

https://learn-video.azurefd.net/vod/player?id=e574dd27-fecc-4163-a4e8-e3eaa5687574&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fmaster-planning%2Fmaster-planning-setup-wizard

The Master planning setup wizard in Dynamics 365 Supply Chain Management video (shown above) is included in the finance and operations playlist available on YouTube.

Specific requirements of your company

The first page of the wizard asks about the specific requirements of your company. Your answers to these questions don't have to be exact, but you should be able to provide a rough approximation of the number of items and planned orders that there exist for the legal entity. Your answers are used to configure parameters that apply to your legal entity, not just to the master plan that you selected. The following sections describe the parameters that are calculated and the formulas that are used.

Number of threads

- **If you manufacture any of the items:** Number of threads = Number of planned orders ÷ 1,000
- **If you don't manufacture any of the items:** Number of threads = Number of items ÷ 1,000

If the calculated number of threads exceeds 75 percent of the available number of threads, it's capped at 75 percent of the number of threads available for each customer. (The number of

available threads is determined for each customer.)

Learn more in Number of threads.

Bundle size

The bundle size is set to **1**. This value is often the best value, because it helps improve the performance of master planning.

Learn more in Number of tasks in helper task bundle.

Firming bundle size

- **If you manufacture any of the items:** The firming bundle size is the larger of these two values: **10** and the bundle calculation.
- **If you don't manufacture any of the items:** The firming bundle size is the larger of these two values: **50** and the bundle calculation.

Bundle calculation = (Number of planned orders × (Firming time fence ÷ Coverage time fence) ÷ Number of threads) ÷ 10

Cache size

The cache size is set to **Maximum**. This value is often the best value because it helps improve the performance of master planning.

Learn more in Allocate time to jobs in a job bundle.

Manufacturing setup

If you manufacture items, a **Manufacturing setup** page appears later in the wizard.

Scope of the current plan

On the **Scope of the current plan** page of the wizard, answer questions related to how far in the future various requirements are considered and calculated in master planning. Each question asks whether you want to use a feature and how you want to configure it.

For example, for the forecast plan feature, the wizard asks, "Do you want to use a forecast plan in master planning so that planned orders are suggested to fulfill the forecasted demand?"

The following options are available:

- **No** – Master planning doesn't suggest planned orders to fulfill a forecast. On the **Time fences** tab of the **Master plans** page (**Master planning** > **Setup** > **Plans** > **Master plans**), the wizard sets the **Forecast plan (time fence)** option to **Yes** and sets the number of days to **0**. This setup overrides the time fence that is specified in the coverage group. Because the number of days is set to **0**, the feature isn't used.
- **Yes, as defined in this master plan** – A field becomes available, where you can enter the number of days that master planning suggests planned orders to fulfill the forecasted demand. The wizard sets the **Forecast plan (time fence)** option to **Yes** and sets the number of days to the number of days entered in the **Forecast plan** field on the **Time fences** tab of the **Master plans** page. This setup overrides the values that are set in the coverage groups.
- **Yes, as defined in the coverage** – The wizard sets the **Forecast plan (time fence)** option to **No**. The time fences that are specified in the coverage group are used to specify how long you plan for the forecast.

The remaining questions on this page and their answers follow the same schema:

- **No** – The **Forecast plan (time fence)** option is set to **Yes**, and the number of days is set to **0**.
- **Yes, as defined in this master plan** – The **Forecast plan (time fence)** option is set to **Yes**. The number of days that you enter is used and overrides the values that are set in the coverage groups.
- **Yes, as defined in the coverage group** – The **Forecast plan (time fence)** option is set to **No**.

Learn more in Job scheduling.

Scheduling options

The **Scheduling options** page appears only if you answer **Yes** to the "Do you manufacture any of the items planned?" question on the first page of the wizard.

Your answer to the first question on this page ("Do you need to schedule operations divided into individual jobs?") determines the scheduling method on the **General** tab of the **Master plans** page.

- **Yes** – Job scheduling will be used.
- **No** – Operations scheduling is used.

For more information, see Operations scheduling and Job scheduling.

Updates of demand and supply

The questions on the **Updates of demand and supply** page relate to firming, action messages, and delays.

The master planning setup is updated based on your answers, according to the same schema described in the previous section:

- **No** – The **Time fence** option on the **Master plans** page is set to **Yes**, and the number of days is set to **0**.
- **Yes, as defined in this master plan** – The **Time fence** option is set to **Yes**. The number of days that you enter overrides the values that are set in the coverage groups.
- **Yes, as defined in the coverage group** – The **Time fence** option is set to **No**.

For calculated delays, your answers to the questions in the wizard update the corresponding parameters on the **Calculated delays** tab of the **Master plans** page.

Summary of your changes

The last page of the wizard shows the changes that are recommended based on your responses. It shows both the value in your setup (**Current setup**) and the value that the wizard recommends (**New configuration**).

You can also modify the parameters in the new configuration. The parameters are separated into parameters that apply to your legal entity and parameters that apply only to the master plan that you selected. To view all the setup parameters that you can change by using the wizard, select **Show all parameters** at the bottom of the page. You can then change the parameters.

When you select **Finish**, the new configuration is applied. If you select **Cancel**, none of the changes are applied.

Examples

This section describes the setup of two fictional companies to show how the setup can change according to the needs of each business.

Example 1: Contoso Manufacturer

Contoso Manufacturer is a manufacturing company that produces speakers. It purchases the various raw materials and components that are used for the final speakers from various suppliers. Here are some of the characteristics of its supply and manufacturing:

- The final items that the company manufactures have a bill of materials (BOM) structure.

- All final items and components are planned by master planning. Manual planning isn't done.
- A route of operations is defined for the production of each final item.
- The manufacturing plant produces the final items. It has a defined number of milling and drilling machines that are used to process the components. The various components must be processed by these machines.
- There are many suppliers. The average lead time for items is one week. A group of items from the same supplier has a lead time of seven weeks.

In the wizard, the following values are entered for Contoso Manufacturer:

- **Coverage:**
  - **Question:** "Do you want to specify the number of days of your planning horizon?"
  - **Answer:** "Yes, as defined in the coverage groups."

  Because the lead time for items is very different, Contoso doesn't have to plan all the items for the same period in the future. Coverage groups for the items are created. Items that have a similar lead time are assigned to the same coverage group. The planning horizon for each coverage group (that is, the coverage time fence) is approximately the lead time plus a margin of one week. Master planning then makes sure that the items are planned in advance, based on their lead time.

  Therefore, two coverage groups are created for this example. One coverage group has a coverage time fence of two weeks, and the other has a coverage time fence of eight weeks.

  If **Yes, as defined in this master plan**, is selected as the answer, the number of days entered should exceed the longest lead time of all the items. However, because many items don't have to be planned so far in advance, many planned orders are calculated but never used. Therefore, the master planning runtime increases.

- **Scheduling:**
  - **Question:** "Do you need to schedule operations divided into individual jobs?"
  - **Answer:** "Yes."

  Contoso Manufacturing must plan and schedule the individual jobs that it performs on the shop floor. Therefore, it uses job scheduling.

- **Capacity:**
  - **Question:** "Do you want to schedule using the capacity of resources?"
  - **Answer:** "Yes, as defined in this master plan." **10 days** is entered.

The number of milling and drilling machines is limited. Production planning must take this limitation into account and arrange the jobs in time according to the capacity of the resources. In other words, the scheduled jobs are replanned based on the limitations of the resources. Planning uses the lead time in addition to the period that is defined. (Planned production orders can overlap.) Because the production lead time for the items is seven days, the capacity of the resources for master planning is considered during 10 days.

- **Sequencing:**
  - **Question:** "Do you want to sequence planned orders using the product's sequence values?"
  - **Answer:** "Yes, as defined in the coverage groups."

  A route is defined for the production of each final item. Therefore, master planning schedules the orders in time according to the defined routes.

- **Explosion:**
  - **Question:** "Do you want to plan orders for all the elements in a Bill of Materials (plan for the parent and all children items)?"
  - **Answer:** "Yes, as defined in the coverage groups."

  All the items that are used for the production must be planned. Because the items have very different lead times, master planning has better performance when it uses the coverage groups. Again, a margin of one week can be entered, and explosion can be done for the same time as the coverage.

Example 2: Contoso Retailer

Contoso Retailer is a distribution company in the fashion industry. It uses master planning to calculate when to place purchase orders, based on its forecasted sales. Here are some of its characteristics:

- Contoso Retailer uses a demand forecast to predict sales. Purchase orders are planned according to the forecast.
- Stores use requisitions for replenishment.
- The lead time from the main warehouse to each store is approximately two weeks for all items.

In the wizard, the following values are entered for Contoso Retailer:

- **Forecast demand:**

- **Question:** "Do you want to use a forecast plan in master planning so that planned orders are suggested to fulfill the forecasted demand?"
- **Answer:** "Yes, as defined in this master plan."

Contoso includes a demand forecast to predict its sales. Therefore, master planning must recommend planned orders to fulfill the forecast.

- **Firming:**
  - **Question:** "Do you want master planning to automatically firm planned orders into order documents, for example production or purchase orders?"
  - **Answer:** "Yes, as defined in this master plan." **1 day** is entered.

  Because Contoso Retailer creates purchase orders directly from the planned purchase orders, it's useful if the planned purchase orders are automatically firmed. Because the company runs master planning every day, a firming time fence of one day automatically firms all the orders that are required for the next day.

- **Approved requisitions:**
  - **Question:** "Do you want to include demand from approved requisitions to replenish retail stores?"
  - **Answer:** "Yes, as defined in this master plan." **1 day** is entered.

  Contoso uses the approved requisitions from its stores to create planned purchase orders to replenish those stores. Because master planning runs every day, the requisitions from the last day are included in the planning.

Last updated on 05/29/2026

Exclude specific transactions or transaction types from master planning

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to exclude specific transactions or transaction types from master planning calculations.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.36 or later.
- Depending on the functionality that you require, one or both of the following features must be turned on in Feature management:
  - *Ignore specific transaction types using Planning Optimization* – This feature lets you select which transaction types should be excluded from master planning calculations.
  - *Exclude specific sales orders or sales order lines in Planning Optimization* – This feature lets you choose not to create supply for a specific sales order or sales order line when you use Planning Optimization. In this way, you can keep an order unplanned. This capability can be useful for orders that require a review or approval process. As of Supply Chain Management version 10.0.43, this feature is turned on by default.

Set a master plan to ignore specific transaction types

When the *Ignore specific transaction types using Planning Optimization* feature is enabled, you can set up each master plan to exclude specific transaction types.

1. Go to **Master planning** > **Setup** > **Plans**.

2. Select an existing master plan, or create a new one.

3. Set the following fields for the new or selected master plan:

    - **Include sales orders** – Set this option to *No* to ignore all existing sales orders. Set it to *Yes* to plan supply for sales orders.
    - **Include transfer orders** – Set this option to *No* to ignore all existing transfer orders. In this case, master planning replans transfer orders for requirements that must be transferred. (This behavior might be useful for simulation scenarios.) Set this option to *Yes* to include transfer orders as supply.
    - **Include purchase returns** – When a purchase order is scheduled to be returned at a future date, it can be picked up for another customer order. This capability is useful for high-value items such as jewelry or watches. In these cases, the return of an order must be considered to avoid over-purchasing the high-value item.
    - **Include retail sales orders** – Set this option to *No* to ignore orders that originate from the **Commerce** module. Set it to *Yes* to include orders that originate from the **Commerce** module as supply.

Exclude specific sales orders or sales order lines from master planning

When the *Exclude specific sales orders or sales order lines in Planning Optimization* feature is enabled, you can set each sales order and/or sales order line so that it's included in master planning calculations. This capability might be useful when you use some approval processes (such as the approval process for high-value items or high customer prices). It might also be useful in other confirmation workflows where supply shouldn't be planned until an order is cleared.

1. Go to **Master planning** > **Setup** > **Plans**.

2. Select an existing master plan, or create a new one.

3. Set the **Exclude specific transactions** option to *Yes*. If you set this option to *No*, the settings in the next steps have no effect.

4. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

5. Select an existing sales order, or create a new one.

6. To exclude the whole order from master planning, on the **Header** tab, on the **Master planning** FastTab, set the **Exclude from master planning** option to *Yes*. This setting defines the default value for new lines, but it doesn't override the value on existing lines

unless the **Updating Exclude from master planning** option is set to *Always* or *Prompt* on the **Accounts receivable parameters** page (as described in the next section).

7. To exclude one or more order lines from master planning, follow these steps:
   a. On the **Lines** tab, on the **Sales order lines** FastTab, select the order line that you want to exclude.
   b. On the **Line details** FastTab, on the **Sourcing** tab, set the **Exclude from master planning** option to *Yes*.

Configure whether sales lines are updated after changing exclude settings on the sales order header

To configure whether and how sales lines are updated after changing the **Exclude from master planning** option on the sales order header, follow these steps:

1. Go to **Accounts receivable** > **Setup** > **Parameters** > **Accounts receivable parameters**.
2. Open the **Updates** tab.
3. On the **Order lines** FastTab toolbar, select **Update order lines** to open the **Update order lines** dialog box.
4. Set the **Updating Exclude from master planning** option to one of the following values:

   - *Prompt* – When you change the **Exclude from master planning** option on the sales order header, a dialog box appears that asks whether you want to apply the same setting to all lines. If you select **Yes**, the system applies the same setting to all lines. If you select **No**, the system doesn't change the setting for any of the lines.
   - *Always* – When you change the **Exclude from master planning** option on the sales order header, the system automatically applies that setting to all lines without prompting.
   - *Never* – When you change the **Exclude from master planning** option on the sales order header, the system doesn't change the setting for any of the lines and doesn't prompt you.

Last updated on 02/10/2026

Generate a constrained plan

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to create a plan that takes into account both material and capacity constraints. The plan ensures that manufacturing doesn't start before materials are available and resources are not overbooked.

The demo data company used to create this procedure is USMF. This procedure is intended for the production planner.

Set up a constrained plan

1. In the home page, select the **Master planning** workspace.
2. Select **Master plans** in the list of links on the far right side of the workspace.
3. In the list, find and select the desired record. Example: **StaticPlan**
4. Select **Yes** in the **Finite capacity** field.
5. In the **Finite capacity time fence** field, enter `30`.
6. Expand the **Time fences in days** section.
7. Select **Yes** in the **Capacity** field.
8. In the **Capacity scheduling time fence (days)** field, enter a number. Example: `60`
9. Select **Yes** in the **Calculated delays** field.
10. In the **Calculate delays time fence (days)** field, enter a number. Example: `60`
11. Expand the **Calculated delays** section.
12. Select **Yes** in all **Add the calculated delay to the requirement date** fields.
13. Close the page.

Create a constrained plan

1. Select **Run**.
2. In the **Master plan** field, enter or select the plan for which you have set up constraints.
3. Select **OK**.
4. Select **Planned orders**.

Last updated on 06/17/2025

Coverage settings

> **Note**
>
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Master scheduling uses coverage settings to calculate item requirements.

You can specify coverage settings in several ways:

- Specify coverage settings for a coverage group.

  Create a coverage group that contains settings for all products linked to the coverage group. To create a coverage group, go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**. You can link a coverage group to a product. If the link is specific to a site, warehouse, or product dimension, use the **Coverage group** field on the **Item coverage** page. If the link is generic, regardless of the product dimensions, use the **Coverage group** field on the **Plan** FastTab of the **Product details** page. By default, if you don't link a coverage group to a product, master planning uses the general coverage group that you specify on the **Master planning parameters** page.

- Specify coverage settings for a product.

  Create coverage settings for a specific product. Go to **Product information management** > **Products** > **Released products**. Select the product, and then, on the Action Pane, on the **Plan** tab, in the **Coverage** group, select **Item coverage** to open the **Item coverage** page. If the product is already linked to a coverage group, you can override the coverage group settings by using the **Override** field. The coverage settings on the **Item coverage** page take precedence over the settings on the **Coverage group** page.

- Specify coverage settings for a product by using a wizard.

  The wizard guides you step by step through the process of setting up the primary item coverage parameters. On the **Item coverage** page, on the Action Pane, select **Wizard** to open the **Item Coverage Wizard**.

- Specify coverage settings for a dimension group.

  Go to **Product information management** > **Products** > **Released products**. On the **Released product details** page, on the **General** FastTab, in the **Administration** section,

select the link in the **Storage dimension group** field. On the **Storage dimension groups** page, select the **Coverage plan by dimension** check box to create the coverage settings for a dimension in the storage dimension group. You must select the **Coverage plan by dimension** field for all product dimensions, such as configuration, color, size, and style.

Coverage codes

You can configure master planning to use different replenishment methods. The system uses these replenishment methods, or lot-sizing methods, to determine the batch size for purchased or produced items.

Assign each replenishment method one of the following coverage codes:

- **Manual** – The lot-sizing method where the system doesn't suggest purchase, transfer, or production orders for the item. The planner for the item is responsible for creating the required orders to replenish the item.
- **Per requirement** – The lot-sizing method where the system creates a planned purchase, transfer, or production order for each requirement of the item. Use this method for expensive items with intermittent demand.
- **Per period** – The lot-sizing method that combines all the demand for a period into one order for the item. The order is planned for the first day of the period, and its quantity fulfills the net requirements during the established period. The period starts with the first demand of the item and covers the defined length in time. The next period starts with the next requirements of the item.
- **Min/Max** – The lot-sizing method that replenishes inventory up to a certain level when the predicted on-hand quantity is below a threshold. The replenishment quantity is the difference between the maximum level and the predicted on-hand level.
- **Priority** – A strategy that replenishes buffers for a product according to its minimum, reorder point, and maximum stock quantities. Whenever the projected on-hand inventory falls below the reorder point quantity, the system replenishes it to its maximum quantity. For replenishing, priority (not date) is considered, which ensures that the most urgent orders are replenished first. Learn more in Priority-based planning.
- **Decoupling point** – The coverage code that identifies a product as a decoupling point (buffer) according to the Demand Driven Material Requirements Planning (DDMRP) methodology. DDMRP is a specific planning and execution methodology. To specify that a product is a decoupling point, open its **Item coverage** page and, under **Use specific settings**, set **Coverage code** to a coverage group that has coverage code of *Decoupling point*. For more information about the DDMRP methodology, see Demand Driven Material Requirements Planning (DDMRP) overview.

Demand-specified BOM or formula versions and routes

For make-to-order businesses, it can be useful to configure coverage groups to consider BOM or formula versions and routes specified in demand lines (such as sales order lines) when creating supply (such as planned production orders). For demand where no BOM or formula version or route is specified, the system chooses the currently active ones.

Prerequisites

To use the **Prioritize existing supply over required BOM or formula version or route version** setting described in the next section, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management 10.0.38 or later.
- The feature named *Prioritize existing supply over required BOM/route in Planning Optimization* must be turned on in feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default.

The **Use the specified BOM or formula version** and **Use the specified route version** settings described in the next section don't require these prerequisites. However, without the *Prioritize existing supply over required BOM/route in Planning Optimization* feature, the system always works as though **Prioritize existing supply over required BOM or formula version or route version** is set to *No*.

Configure a coverage group to use demand-specified BOM or formula versions and routes

To configure a coverage group to use the BOM or formula versions and routes specified in demand lines, follow these steps:

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. From the list pane, select the group you want to set up.
3. Make the following settings on the **General** FastTab for your selected group:

    - **Use the specified BOM or formula version** – Set to *Yes* to allow demand lines to identify a specific BOM or formula version when creating planned production orders for items that belong to the current coverage group. Set to *No* to always use the currently active BOM or formula version.
    - **Use the specified route version** – Set to *Yes* to allow demand lines to identify a specific route version when creating planned production orders for items that

belong to the current coverage group. Set to *No* to always use the currently active route version.

- **Prioritize existing supply over required BOM or formula version or route version** – This setting only applies when **Use the specified BOM or formula version** and/or **Use the specified route version** are set to *Yes*. Set this setting to *Yes* to prioritize existing supply (such as on-hand inventory, if available) over demand-specified BOM or formula versions and routes. Set to *No* if demand-specified BOM or formula versions or routes should always result in creating new supply (such as planned production orders), even if existing supply is available.

Related information

- Master plans overview

Last updated on 03/25/2026

Define coverage rules for items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to create coverage rules and override coverage settings for a specific item. It also shows how to specify default inventory settings. This procedure shows sample values that you can use with the USMF demo data company, but you can use your own values if you prefer.

Create a coverage group

Create a coverage group by following these steps:

1. Go to **Master planning** > **Setup** > **Coverage groups**.
2. Select **New**.
3. In the **Coverage group** field, enter a value.
4. In the **Name** field, enter a value.
5. In the **Calendar** field, enter a value. Choose the calendar that master planning uses to create replenishment suggestions for items in this group.
6. In the **Coverage code** field, select an option. Select *Requirement* for this procedure.
7. In the **Coverage time fence (days) field**, enter *90*. For items in this group, master planning will create replenishment suggestions for up to 90 days in the future.
8. In the **Negative days** field, enter *1*.
9. In the **Positive days** field, enter *1*.
10. Expand or collapse the **Other** section.
11. Under the **Safety margins in days** section, in the **Receipt margin added to requirement date** field, enter *1*. For example, if the receipt margin is set to one day, and a purchase order line is scheduled for receipt on May 15, master planning calculates the adjusted receipt date as May 16.
12. In the **Issue margin deducted from requirement date** field, enter *1*. For example, if the safety margin is set to one day, and a sales order line is scheduled for delivery on May 15, master scheduling calculates the adjusted delivery date as May 14.
13. In the **Reorder margin added to item lead time** field, enter *1*.
14. Select **Save**.

Create a new product

Create a new product by following these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select **New**.
3. In the **Product number** field, enter a value.
4. In the **Product name** field, enter a value.
5. In the **Item model group** field, select the drop-down button to open the lookup.
6. In the list, find and select the desired record.
7. In the list, select the link in the selected row.
8. In the **Item group** field, select the drop-down button to open the lookup.
9. In the list, find and select the desired record.
10. In the list, select the link in the selected row.
11. In the **Storage dimension group** field, select the drop-down button to open the lookup.
12. In the list, find and select the desired record.
13. In the list, select the link in the selected row.
14. In the **Tracking dimension group** field, select the drop-down button to open the lookup.
15. In the list, find and select the desired record.
16. In the list, select the link in the selected row.
17. Select **OK**.

Set up default order settings

Set up default order settings by following these steps:

1. On the Action Pane, select **Plan**.
2. Under **Order settings**, select **Default order settings**.
3. Under **Purchase order**, in the **Default site** field, enter the site used as default when purchase orders are created.
4. In the **Default warehouse** field, enter the site where the item is stored.
5. Expand or collapse the **Inventory** section.
6. In the **Multiple** field, enter *10*.
7. In the **Min. order quantity** field, enter *10*.
8. In the **Max. order quantity** field, enter *100*.
9. In the **Standard order quantity**, enter *10*.
10. In the **Purchase lead time** field, enter a number.
11. Select or clear the **Working days** check box.
12. Select **Save**.
13. In the **Default order type** field, select *Purchase order*.
14. Select **Save**.

15. Close the page. Close the Default order settings page.

Add an item to a coverage group

Add an item to a coverage group by following these steps:

1. Expand or collapse the **Plan** section.
2. In the **Coverage group** field, select the drop-down button to open the lookup.
3. In the list, find the **Coverage group** you have created.
4. In the list, select the link in the selected row.

Create item coverage rules

Create item coverage rules by following these steps:

1. On the Action Pane, select **Plan**.
2. Under **Coverage**, select **Item coverage**.
3. Select **New**.
4. Select the **General** tab.
5. Check the box on the header of **Override coverage group** settings.
6. In the **Coverage time fence (days)** field, enter *60*. Although items in coverage group *Requirement* are planned 90 days ahead, this item is planned 60 days ahead.
7. In the **Negative days** field, enter *2*.
8. In the **Positive days** field, enter *2*.
9. Select the **Lead time** tab.
10. Check the box on the header of **Purchase**.
11. In the **Purchase time** field, enter *5*.
12. Select **Save**.

> ⓘ **Note**
>
> For manufactured items, the **Production lead time** is used if there's no route for the item. If an active route is associated to the item, then master planning schedules the order and calculates its dates according to the route times and capacity of the resources (if applicable).

Last updated on 08/05/2025

Source products and materials from multiple vendors

08/22/2025

Diversification of the supply network helps businesses be agile and respond to changes. One of the most common ways to diversify is to divide the supply of a specific product among different vendors by assigning a supply percentage to each of them. For example, if one of your products has both a main vendor and a secondary vendor, the usual split might be 80/20 or 70/30. Alternatively, you might divide equally among three different vendors in a 33/33/34 split.

This feature enables the system to automatically suggest a vendor for each planned purchase order so that, over time, the specified percentages for each vendor are respected. However, the system won't split an existing planned order into several smaller orders.

The following table shows a simple example where each planned order quantity is assigned to a single vendor, without splitting the order quantity and in such a way that the overall percentages are respected over time.

Expand table

Demand	Vendor A quantity (target: 50%)	Vendor B quantity (target: 50%)
100	100	0
100	0	100
200	200	0
100	0	100
100	0	100

Automatic vendor selection that's based on predefined splits helps improve supplier resilience. Because the system tracks actual purchases, not just planned purchases, it will also consider purchases that are made outside the planning system when it calculates the percentage of orders to assign to each supplier.

Prerequisites

Before you can use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management 10.0.31 or later.
- The feature that's named *Source products and materials from multiple vendors using Planning Optimization* must be turned on in feature management. As of Supply Chain Management version 10.0.36, this feature is turned on by default.

Define multisource policies

Multisource policies let you establish the sourcing percentages for each vendor. You can create as many policies as you require to set up different percentage splits among different combinations of vendors. Later, you'll assign a policy to each product as needed.

Follow these steps to create a multisource policy.

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Multisource policies**.
2. Follow one of these steps:
    - To edit an existing policy, select it in the list pane, and then select **Edit** on the Action Pane. If the **Active** option is set to *Yes* for the policy, you must set it to *No* and save your change before you can edit or delete the policy. You can deactivate and edit only policies that aren't assigned to any products.
    - To create a new policy, select **New** on the Action Pane.
3. On the header of the new or selected policy, set the following fields:
    - **Name** – Enter a name for the policy. (You can't edit this field after the policy has been saved.)
    - **Description** – Enter a short description of the policy.
4. On the **General** FastTab, set the following fields:
    - **Active** – Specify whether the policy should be active. Only active policies can be assigned to products. You can edit or delete only policies that were last saved while this option was set to *No*. You can't deactivate policies that are currently used (that is, assigned to one or more products for the current period or a future period).
    - **Balance period (days)** – Specify the number of days that master planning should look into the past when it allocates orders to vendors. In other words, when master planning is trying to find a vendor for a planned purchase order, it will look this number of days into the past. It then selects the vendor that will result in an actual allocation that's closest to the preferred allocation that is established by the multisource policy. We recommend that you specify a balance period that will include purchases from the relevant vendors but without going too far back. The default period is 180 days.
5. On the **Policy rules** FastTab, add a line for each vendor that's part of the policy, and assign a target allocation to each vendor. The sum of all allocations that are specified in this grid must be 100 percent. Use the buttons on the toolbar to add lines to the grid or remove them. For each line, set the following fields:
    - **Vendor account** – Select the vendor that the line applies to.
    - **Target allocation (%)** – Specify the percentage of planned orders that should be assigned to this vendor (relative to the others) over time.
6. Follow one of these steps:
    - If you're ready to activate the policy, set the **Active** option to *Yes*, and then select **Save** on the Action Pane. The system will validate your settings (for example, to make sure that all percentages add up to 100 percent) and will proceed with the save only if the validation is passed. If the validation isn't passed, you'll receive an error that you must address before you can save.
    - If you want to prevent the policy from being used, or if you want to save it before you've finished setting up all the allocations, set the **Active** option to *No*, and then select **Save** on the Action Pane. This configuration will allow you to save an invalid policy, but you won't be able to assign the policy to any products.

Define minimum order quantities per vendor

If vendor agreements that you have in place establish a minimum quantity per order for some items, you can add rules for those conditions to the system. These minimum order settings apply to all multisource policies and only to multisource policies. (In other words, rules for minimum orders per vendor apply only to planned purchase orders that are created by master planning. They don't apply to quantities from manually created purchase orders.)

To set up minimum order per vendor rules, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Multisource policies**.
2. On the Action Pane, select **Order quantity per vendor**.
3. Follow one of these steps:
    - To edit an existing rule, find and select it in the grid, and then select **Edit** on the Action Pane.
    - To create a new rule, select **New** on the Action Pane.
4. Set the following fields for the new or selected rule:
    - **Vendor account** – Select the vendor that the rule applies to.
    - **Item number** – Select the item that the rule applies to.
    - **Minimum** – Enter the minimum purchase per order for the specified combination of a vendor and an item.
    - **Unit** – Shows the purchase unit of measure for the selected item (read only). It applies to the **Minimum** value.
5. On the Action Pane, select **Save**.

Assign multisource policies to products

To set up a product so that it can be purchased from multiple vendors, you must specify which multisource policy applies to it.

To specify the multisource policy that applies to a product, follow these steps.

1. Follow one of these steps:
    - Go to **Procurement and sourcing** > **Setup** > **Policies** > **Multisource policy assignments**.
    - If you're already on the **Multisource policies** page, select **Policy assignments** on the Action Pane.
2. Follow one of these steps:
    - To edit an existing assignment, find and select it in the grid, and then select **Edit** on the Action Pane.
    - To create a new assignment, select **New** on the Action Pane.
3. Set the following fields for the new or selected assignment:
    - **Item number** – Select the item to assign a policy to.
    - **Site** – Select the site where the policy should apply. If the assignment should apply to all sites, leave this field blank.
    - **Warehouse** – Select the warehouse where the policy should apply. If the assignment should apply to all warehouses, leave this field blank.
    - **From date** – Select the first date that the assignment should apply.

- **To date** – Select the last date that the assignment should apply.
- **Policy** – Select the policy that should apply to the assignment's combination of an item, site, warehouse, and date range.

4. On the Action Pane, select **Save**.

Review target vs. actual vendor allocations

For each multisource policy assignment, you can view your actual purchases and vendor allocations for the current period, and compare them against your target allocations.

To view and compare target and actual vendor allocations, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Multisource policy assignments**.
2. Select the assignment to inspect.
3. On the Action Pane, select **Current allocation**.
4. The **Current allocation** page appears and shows information about the selected assignment. For each vendor, you can view the total quantity that was purchased, the current allocation, the target allocation, and the amount by which each actual allocation deviates from the target. Each quantity and allocation value counts all purchases that were made during the *current period*. The current period extends backwards the number of days that is defined by the **Balance period (days)** setting for the assigned policy. However, it can't extend earlier than the first date that the assignment is valid (the **From date** value). It extends into the future until the last date that the assignment is valid (the **To date** value).

This feature is useful if, for example, you want to determine whether your Purchasing department is firming orders as master planning suggests or making changes. (If purchasers are following master planning's suggestions, the actual allocation will be close to the current allocation. However, if they're making changes, actual allocations might be quite different than the current allocation.)

Example of target vs. actual vendor allocations

The following table shows an example of how a product that has a multisource policy associated with it could be set up on the **Multisource policy assignments** page.

Expand table

Vendor account	Total quantity	Current allocation (%)	Target allocation (%)	Allocation deviation (%)
1001	350	63.64	70	-9.09
1002	200	36.36	30	21.21

Where:

- **Total quantity** – Represents the vendor's total sourced quantity.
- **Current allocation (%)** – Is equal to (Vendor's total sourced quantity ÷ Total accumulated quantity) × 100%. This value indicates how much of all the sourced quantity has been purchased from this vendor as a percentage of the total.
- **Target allocation (%)** – Is the target allocation as defined by the multisource policy for the vendor.
- **Allocation deviation (%)** – Is equal to ((Current allocation − Target allocation) ÷ Target allocation) × 100%.

How master planning suggests a vendor for each planned order

Each time that master planning suggests a vendor for a planned purchase order, it selects the vendor that will bring the total vendor allocations for the current period closest to the target allocation, as defined by the relevant multisource policy assignment and minimum vendor quantities.

How the system selects which vendor to assign for a new planned order

The system uses the following rules to select the vendor for a new planned order:

- The system will apply a multisource policy according to the following rules:
  - The planned order must match the product and all the dimensions that are specified for a multisource policy assignment.
  - The planned order date must occur during the *current period* for a matching multisource policy assignment. The current period extends backwards the number of days that is defined by the **Balance period (days)** setting for the assigned policy. However, it can't extend earlier than the first date that the assignment is valid (the **From date** value). It extends into the future until the last date that the assignment is valid (the **To date** value).
  - If more than one policy assignment matches the planned order, the most specific assignment is used. For example, if there's demand for item A0001, site 1, and warehouse 11, and the policies that are shown in the following table exist, policy Y will be used, because it's the more specific one.

Expand table

Item	Site	Warehouse	From date	To date	Policy
A0001	1		October 24, 2022		X
A0001	1	11	October 24, 2022		Y

- Classic vendor selection that is based on cost/price isn't used with trade agreements. If a multisource policy is established, the system will try to meet its vendor allocation targets.
- If a vendor lead time is specified in a trade agreement, it will be respected.
- The approved vendor list will be respected if it's available. If the system is set up to disallow vendors besides the approved vendors (that is, if the **Approved vendor check method** field is set to *Not allowed*), that setting will be respected. A vendor that is specified in the multisource policy will be selected only if it's approved. The default **Approved vendor check method** setting is defined at the item model group level but can be overridden at the released product level.
- Default order settings will be respected as follows:
  - The minimum, multiple, and maximum values that are defined in the default order settings will be respected if they're specified.

- If the multisource policy includes a relevant minimum order quantity for the vendor, that minimum quantity will also be applied. Therefore, the higher of the two minimum order values (the default order settings and the minimum vendor quantity) will take effect.

> **Important**
>
> Vendor calendars aren't taken into account when you use multi-vendor sourcing. In some cases, this can result in an order arriving later than it could if you chose a vendor manually.
>
> For example, suppose it's Monday and you have demand for Wednesday for an item provided by two vendors. One vendor is open all days while the second vendor is only open on Fridays. If the sourcing algorithm determines that the second vendor should supply (based on the target percentages), then this vendor will be chosen even though the demand won't arrive until Friday. Therefore, you might want to assign a vendor manually when fulfilling short-term demand.

Example: How vendors are selected based on a multisource policy

The table that follows shows how the system will assign orders to each of two different vendors, based on the target allocations that are defined for the relevant multisource policy. For each new order, the system will select the vendor that results in the least deviation from the target allocation, based on the quantities that are accumulated during the current period, as defined by the relevant policy and policy assignment.

Here's an explanation of the table:

- The "Demand" column shows the planned purchase order quantity.
- The "Vendor A quantity" and "Vendor B quantity" columns show the target allocations, and indicate the actual resulting quantities that are ordered from each vendor for each planned order.
- The "If vendor A is selected" and "If vendor B is selected" columns show the calculations that the system does to determine which vendor it should use. The system selects the vendor that results in the least deviation. The calculation uses the following values:
  - **Qty A** = Accumulated quantity for vendor A
  - **Qty B** = Accumulated quantity for vendor B
  - **A%** = (*Qty A* ÷ *Total accumulated quantity*) × 100%
  - **B%** = (*Qty B* ÷ *Total accumulated quantity*) × 100%
  - **Deviation** = (| *A%* − *Vendor A target* | + | *B%* – *Vendor B target* |) ÷ 2
- The "Total accumulate quantity" column shows the total quantity that is ordered after each new purchase order.
- All planned orders in the table occur during the same period, as defined by the relevant policy and policy assignment.

For vendor A, the target allocation is 80 percent. For vendor B, it's 20 percent.

Expand table

Demand	Vendor A quantity	Vendor B quantity	If vendor A is selected					If vendor B is selected					Total accumulated quantity
			Qty A	Qty B	A%	B%	Deviation	Qty A	Qty B	A%	B%	Deviation	
100	100	0	100	0	100%	0%	20%	0	100	0%	100%	80%	100
100	100	0	200*	0	100%	0%	20%**	100	100	50%	50%	30%	200
100	0	100	300	0	100%	0%	20%	200	100	66%	33%	13%	300

* 100 accumulated + 100 current demand

** (| 100% − 80% | + | 20% − 0% |) ÷ 2 = 20%

Priority-based planning

06/17/2025

This article describes the priority-based planning feature of Microsoft Dynamics 365 Supply Chain Management. The feature adds support for demand-driven planning, which is one step of Demand Driven Material Requirements Planning (DDMRP). Priority-based planning enables the system to generate planned orders that are driven by planning priorities instead of requirement dates.

Priority-based planning lets you prioritize replenishment orders to ensure that urgent demand is prioritized over less important demand. For example, a stockout replenishment order will be prioritized over a standard refill replenishment order. The system can automatically split larger orders into separate smaller orders where order lines are grouped by priority. It can then process all high-priority orders first.

To get a quick overview of this feature, see the following video: Planning optimization support for priority-based planning in Dynamics 365 Supply Chain Management.

Where and how planning priorities are assigned

Planning priority information about supply and demand is the backbone of priority-based planning. Planning priority defines the importance of a demand or supply line. Master planning uses it when the **Coverage code** field is set to *Priority*.

The planning priority is usually a number between 0 (zero) and 100, where 0 represents the highest importance. It's shown and set in the **Planning priority** field. You can find this field on the following pages: **Demand forecast lines**, **Sales order details**, **Purchase order details**, **Transfer order details**, and **Planned order details**.

When the **Coverage code** field for the relevant item or coverage group is set to *Priority*, master planning balances supply with demand by using a demand-driven approach as it calculates the planning priority and, for each released product, considers the values that are set for the **Minimum**, **Reorder point**, and **Maximum** fields on the **Item coverage** page.

> **Note**
>
> The *Priority* value is available for the **Coverage code** field only when Planning Optimization is enabled.

The related *planning priority models* control the planning priority and split planned orders by priority range. They also let you set default planning priority values for each supply or demand

type and define the priority calculation method.

Types of priority calculation methods

Each planning priority model has a **Priority calculation method** setting that controls how master planning applies priority to planned orders. The available values are *Percent of maximum inventory quantity* and *Priority ranges*. *Priority ranges* represents a more advanced version of the *Percent of maximum inventory quantity* method.

Percent of maximum inventory quantity

In the *Percent of maximum inventory quantity* calculation method, the supply priority calculation finds the current *total available inventory* (net flow) as a percentage of the **Maximum inventory quantity** value that is set for an item. A single planned order is then created per item and vendor (unless the maximum order quantity is used to force a split). The planning priority of the order is calculated as a percentage of the maximum.

The following formula is used:

Percentage of maximum = (*Net flow position* × 100) ÷ *Maximum inventory quantity value from the item coverage*

In this formula, *Net flow position* is calculated in the following way:

Net flow position = On-hand + On-order – Qualified demand

- *On-order* is the expected supply.
- *Qualified demand* represents the net requirements that have the requirement date within the planning time fence.

During the master planning run, new planned orders are created when the net flow position is less than the reorder point quantity for an item. The planned order quantity is the difference between the **Maximum inventory quantity** value that is set at the item level and the net flow position. The planned order priority is calculated as a *net flow position* percentage of the **Maximum inventory quantity** value.

> **Note**
>
> The calculated priority can't be negative, even if demand exceeds total supply. If demand exceeds total supply, the calculated priority is set to 0 (zero).

Priority ranges

The *Priority ranges* calculation method is more advanced than the *Percent of maximum inventory quantity* method and is configured at the level of the planning priority model. Several new planned supply orders can be created to fulfill the demand per item. The priorities of the planned supply follow the values that are defined in the **Planning priority ranges** grid on the **Planning priority models** page.

The following additional rules become effective when the **Priority calculation method** field is set to *Priority ranges*:

- If the **Consider demand priority** option for the planned priority model is set to *Yes*, priority that is set on each demand line will limit the priority range bucket. The priority of any new planned order for supply will be no lower than the demand's priority. The range's upper value is considered a threshold that the demand's priority value is compared against. If the demand's priority is exactly in the middle between the upper threshold values of two ranges, the range that has the highest priority (that is, the lowest priority value) will be selected.
- If the **Planned order creation** field for the planned priority model is set to *Single supply with most important priority*, only one supply will be created to fulfill all the way up to the maximum. The priority will be set to the priority of the first range that will trigger a supply.
- If there's no on-hand inventory, no supply, and no demand, the line in the **Planning priority ranges** grid where the **From quantity** field is set to *Zero* will be used.
- If there is demand, but no on-hand inventory or expected supply, the line in the **Planning priority ranges** grid where the **From quantity** field is set to *Zero or less* will be used.
- When the range that the demand is part of is evaluated, the setting of the **Consider demand priority** option will still have an effect.

Differences between traditional timeline calculations and priority-based planning

Priority-based planning differs from traditional timeline calculations in the following ways:

- All the regular preplanning processors are still in effect. These preprocessors include pegging of approved planned orders against sales demand, purchase requisition mapping, and reservation logic. Only demand that isn't fulfilled by these preprocessors is processed.
- During pegging, all supply is considered, regardless of its priority. This supply includes on-hand inventory, released supply, and the non-pegged portion of approved planned orders.

- No "negative days" demand can be pegged against supply during the whole coverage time.
- When supply is pegged against demand, the supply that has the highest priority (that is, the lowest priority value) is used up first. On-hand supply is considered to have a priority value of 0 (zero). Therefore, it will be consumed as the very first source.
- New planned supply lines will be created according to the regular rules for minimum order size, maximum order size, quantity multiple, and so on.

Planning priority models

Planning priority models are assigned to coverage groups and control the planning priority for planned orders. They define the logic that determines how the planning priority value is calculated for each planned order, and how priority is assigned to planned orders, supply lines, and demand lines.

To work with the planning priority models, go to **Master planning** > **Setup** > **Planning priority models**. As was previously discussed, one of a model's most important settings is the **Priority calculation method** value. This setting controls the calculation method that is used when master planning assigns a priority value to planned orders.

> ⓘ **Note**
>
> Planning priority models apply organization-wide, across all legal entities.

Coverage group

Set up a new coverage group that you plan to use for priority-based planning, as described in Define coverage rules for items. After you've created the coverage group, set the following additional fields:

- **Coverage code** – Select *Priority* if the coverage group will use priority-based planning.
- **Planning priority model** – Select any organization-wide planning priority model.

Item coverage

Set up item coverage settings as described in Coverage settings. By default, the **Coverage code** value that is selected for the coverage group is copied to the item coverage settings. However, you can override the default value as you require. In some cases, the **Coverage code** field for an item coverage record is set to *Planning*, but no planning priority model is defined for the related coverage group. In these cases, any model where the **Priority calculation method** field

is set to *Percent of maximum inventory quantity* and the **Planned order creation** field is set to *Single supply with most important priority* will be applied by default.

Set the **Coverage code** field to *Priority* to make the **Reorder point** field in the item coverage settings available. In this field, enter the reorder point quantity that the system should use while it's determining when to place planned orders that have a **Coverage code** value of *Priority*.

The reorder point quantity is often calculated as the demand during the lead time plus a minimum value (safety stock). It must be a value between the **Minimum** and **Maximum** values.

For example, you might set the fields in the following way:

- **Minimum:** *10*
- **Reorder point:** *45*
- **Maximum:** *60*

For this example, the reorder point quantity is based on a lead time of seven days and an average daily usage of 5. Therefore, the demand during the lead time is 35. The minimum value of 10 (safety stock) is then added to get a reorder point of 45. When this setup is used, supply will be suggested when the projected on-hand level is below 45. The order priority will be based on the planning priority setup.

Manage planning priority models

To work with your planning priority models. follow these steps.

1. Go to **Master planning** > **Setup** > **Planning priority models**.
2. Either select an existing model in the list pane, or select **New** on the Action Pane to create a new model.
3. On the header of the record, set the following fields:
    - **Name** – Enter a name for the model. The name must be unique across all legal entities in your organization.
    - **Description** – Enter a description of the model.
    - **Priority calculation method** – Select one of the following values:
        - *Priority ranges* – When you select this value, the **Planning priority ranges** grid becomes available. There, you must establish several priority ranges to define the planning priority value.
        - *Percent of Maximum inventory quantity* – Calculate the planning priority value as a percentage, based on the projected inventory level over the maximum

inventory quantity.

- **Planned order creation** – This field is available when the **Priority calculation method** field is set to *Priority ranges*. Select one of the following values:
  - *Single supply with most important priority* – Don't split planned orders based on the priority range. The planning priority for a planned order is based on the most important priority range (that is, the lowest planning priority value).
  - *Split according to priority ranges* – Split the demand into multiple planned orders, based on the planning priority ranges. The planning priority for individual planned orders is defined by the planning priority of the related planning priority range.

- **Consider demand priority** – Set this option to *Yes* to limit the priority of a new planned order that is created for supply. (The priority will be no lower than the related demand's priority.) If you set this option to *No*, the demand order's priority won't be considered when the supply order's priority is calculated.

4. If you set the **Priority calculation method** field to *Priority ranges*, use the **Add** and **Remove** buttons on the toolbar of the **Planning priority ranges** FastTab to add or remove priority range rows as you require. If multiple rows exist, and you insert a new row, the planning priority will automatically be set to the average of the selected row and the row above it. For each row, set the following fields:

   - **Planning priority** – Enter any value between 0.00 and 100.00. This value represents the planning priority that is used for the row. The lowest priority value represents the highest priority. A default value is assigned, but you can change it as you require. The same **Planning priority** value can't be used for more than one planning priority range in the same planning priority model.

   - **Description** – Enter a description of the planning priority range (such as *Reorder point to Maximum*).

   - **From quantity** – The lower limit of the planning priority range. This value is read-only, and is based on the **To quantity** and **Percentage of to quantity** values for the previous planning priority range.

   - **To quantity** – Select the field from the item coverage that should be used to define the range's upper limit. The following values are supported and will influence the **From quantity** value of the next range:
     - *Zero* – This value represents a negative to zero range (*Zero or less* to *Zero*). For rows where this value is selected, the **Percent of to quantity** field is read-only and is always set to *100* percent.

- *Minimum inventory quantity* – This value represents the **Minimum** value for an item on the **Item coverage** page. For rows where this value is selected, the **Percent of to quantity** field is editable and is used to set the **From quantity** value of the next range (for example, to *80% of Minimum inventory quantity*).
- *Reorder point* – This value represents the **Reorder point** value for an item on the **Item coverage** page. For rows where this value is selected, the **Percent of to quantity** field is editable and is used to set the **From quantity** value for the next range (for example, to *80% of Reorder point*).
- *Maximum inventory quantity* – This value represents the **Maximum** value for an item on the **Item coverage** page. For rows where this value is selected, the **Percent of to quantity** field is editable and is used to set the **From quantity** of the next range (for example, to *80% of Minimum inventory quantity*).
- *Infinite* – This value represents an infinite upper level of the range (*Infinite or less* to *Infinite*). For rows where this value is selected, the **Percent of to quantity** field is read-only and is always set to *100* percent.

- **Percent of to quantity** – Enter a percentage value that is used to calculate the upper limit of the planning priority range, based on the value that is selected in the **To quantity** field. For example, if the **To quantity** field is set to *Minimum inventory quantity*, and the **Percent of to quantity** field is set to *50*, the upper limit will be 50 percent of the minimum inventory quantity from the related item coverage.

5. On the **Planning priority defaults** FastTab, set the fields as you require to define default planning priorities for each type of supply or demand line (sales order, purchase order, transfer order, or demand forecast). Only positive values can be entered.

View and maintain planning priority

The planning priority is shown and set in the **Planning priority** field. You can find this field on the pages that are listed in the following table. The priority is set as a number between 0 (zero) and 100, where 0 represents the highest importance.

Expand table

Page	Field location	Value source
Demand forecast lines	**Item** tab (Select a line in the upper section, and then select the **Item** tab.)	Default value or value that is manually set
Sales order details	**Delivery** tab on the **Line details** FastTab	Default value, value from intercompany, or value that is manually set

Page	Field location	Value source
	(Select a line on the **Sales order lines** FastTab, and then, on the **Line details** FastTab, select the **Delivery** tab.)	
Purchase order details	**Delivery** tab on the **Line details** FastTab (Select a line on the **Purchase order lines** FastTab, and then, on the **Line details** FastTab, select the **Delivery** tab.)	Value that is set during firming from planned orders, value from intercompany, or value that is manually set
Transfer order details	**Delivery** tab on the **Line details** FastTab (Select a line on the **Transfer order lines** FastTab, and then, on the **Line details** FastTab, select the **Delivery** tab.)	Value that is set during firming from planned orders or value that is manually set
Planned order details	**General** FastTab	Value that is calculated during master planning or value that is manually set

Intercompany trade

The **Planning priority** value on intercompany supply and demand lines is shared between the linked entities. Modification on either side will be reflected on the linked order line.

Here are some examples:

- A user changes the planning priority for an intercompany sales order line from 20 to 30. This change is reflected on the linked intercompany purchase order line.
- A user changes the planning priority for an intercompany purchase order line from 40 to 50. This change is reflected on the linked intercompany sales order line.

Action messages

The system doesn't generate action messages for coverage codes with priority-based planning.

Replenishment methods and quantity modification

This article provides information about replenishment methods. It also explains how the multiple order quantity for a product affects the result.

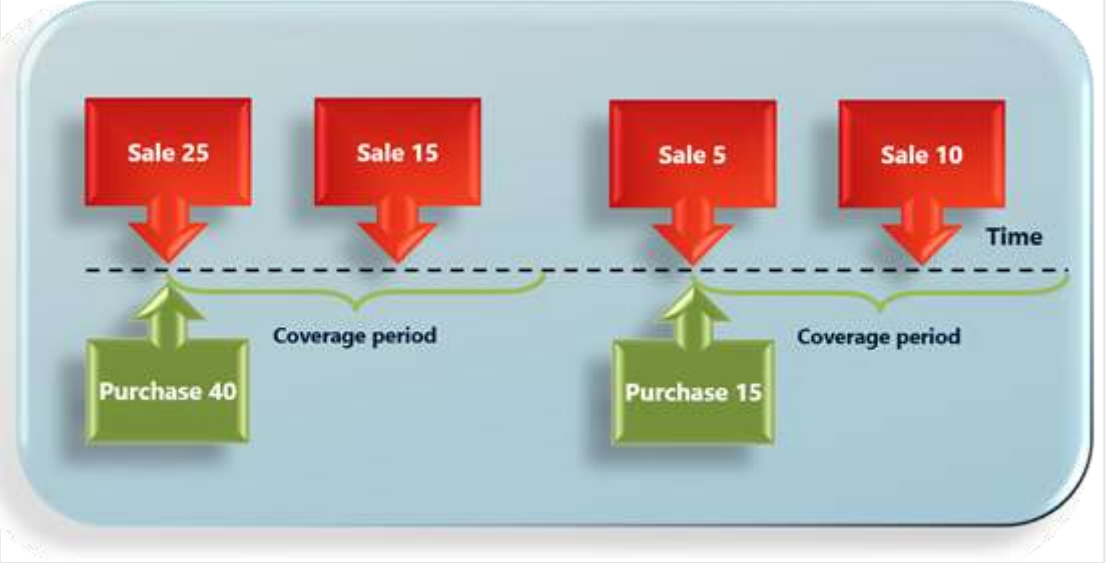
Replenishment methods are also known as coverage methods and lot-sizing methods.

Coverage codes

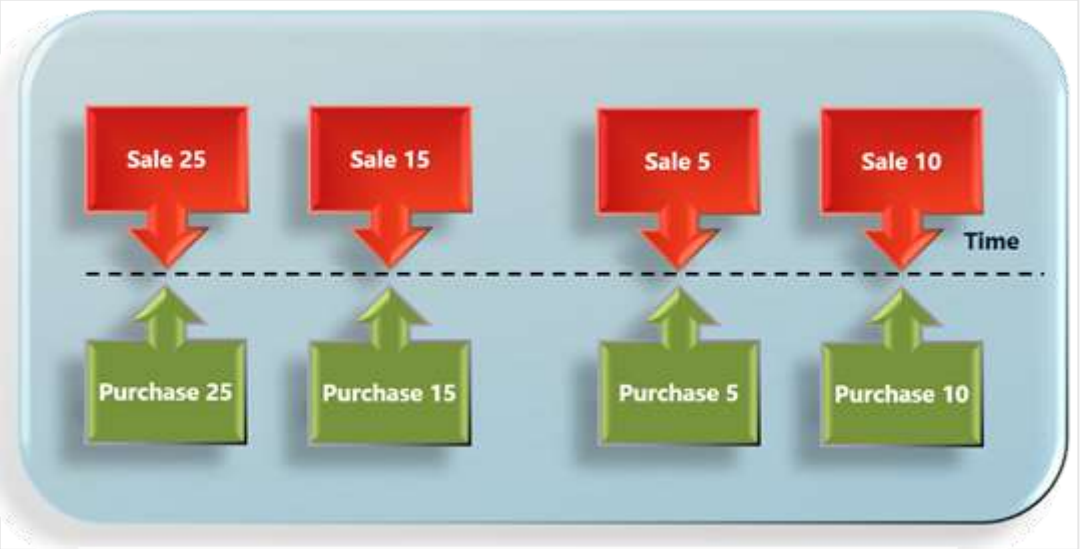
You can configure master planning to use different replenishment methods. Replenishment methods are the techniques that the system uses to calculate requirements for a product. Replenishment methods are defined by coverage codes that you can set up on either the coverage group or the product.

Use the following coverage codes:

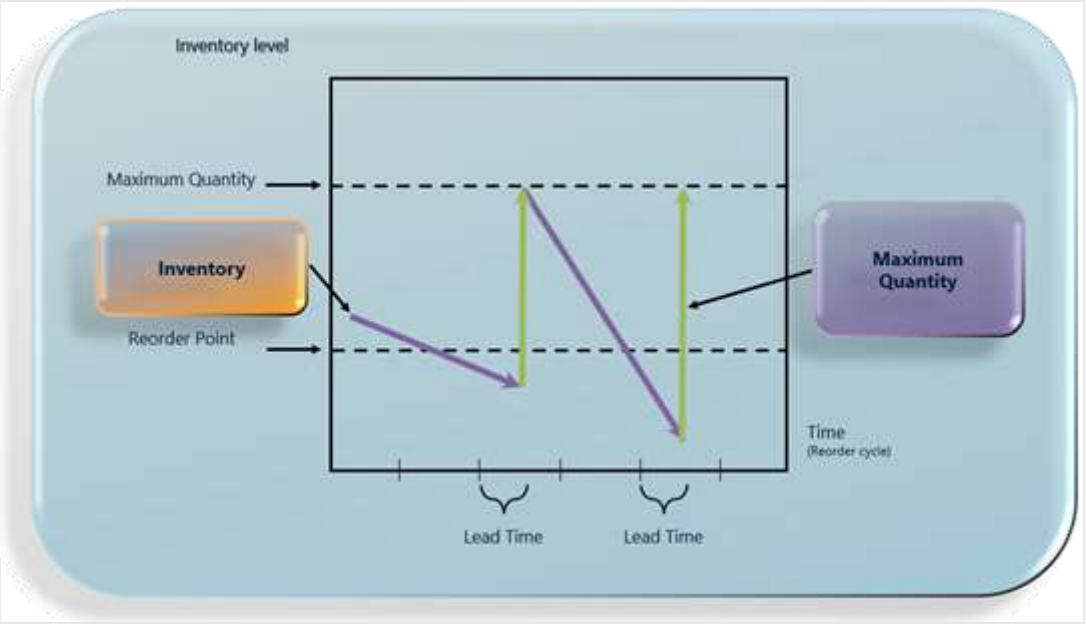
- **Period** – The replenishment method combines all the demand for a period into one order for the product. The order is planned for the first day of the period, and its quantity fulfills the net requirements during the established period. The period starts with the first demand of the product and covers the defined length of time. The next period starts with the next requirements of the product. Use the *Period* coverage code for non-predictable inventory draw, season-influenced products, or high-cost products. The following illustration shows an example.



- **Requirement** – In the replenishment method, the system creates a planned purchase, transfer, or production order per requirement for the product. Use this method for expensive products that have intermittent demand. The *Requirement* coverage code is often used for configurable products or make-to-order scenarios. The following illustration shows an example.



- **Min./Max.** – The replenishment method is based on the inventory level. It defines the replenishment of inventory up to a specific level when the predicted on-hand level is below a specific threshold. The replenishment quantity is the difference between the maximum level and the predicted on-hand level. Use the *Min./Max.* coverage code for predictable inventory draw, high runners, or less expensive products. The following illustration shows an example.



- **Manual** – In the replenishment method, the system doesn't suggest purchase, transfer, or production orders for the product. Instead, the planner for the product is responsible for

creating the required orders for the replenishment of the product. Use the *Manual* coverage code for products for which you don't want the system to generate planned orders.

Impact of the order quantity from default order settings

On the **Default order setting** page for a released product, you can specify each of the following quantity settings on the **Purchase order**, **Inventory**, and **Sales order** FastTabs. (The **Inventory** FastTab is used for both transfer orders and production orders.)

- **Multiple** – Planned orders are a multiple of this quantity.

  For example, if the **Multiple** field is set to *5*, an order can be for a quantity of 5, 10, 15, 20, and so on.

- **Min. order quantity** – Planned orders aren't less than the specified value.

  For example, if the **Min. order quantity** field is set to *10*, a planned order for a quantity of 10 is created, even if only four are required to fulfill the demand.

- **Max. order quantity** – Planned orders don't exceed the specified value. If the demand is more than the **Max. order quantity** value, multiple planned orders are created to cover it.

  For example, if the **Max. order quantity** field is set to *100*, and the demand is 450, four planned orders for a quantity of 100 and one planned order for a quantity of 50 are created.

Examples of replenishment that use the Min./Max. coverage code

If you don't specify a value in the **Multiple** field for a product on the **Default order setting** page, and if you're using the *Min./Max.* replenishment method, master planning replenishes the inventory up to a specific level when the predicted on-hand level is below a specific threshold.

If you define a multiple quantity for a product, the *Min./Max.* replenishment method changes its behavior and considers the **Multiple** value.

In other words, master planning still replenishes the inventory up to the defined maximum level when the predicted on-hand level is less than the defined minimum level. However, the replenishment quantity must be a multiple of the **Multiple** value.

If the replenishment quantity (the difference between the maximum level and the predicted on-hand level) isn't a multiple of the defined multiple quantity, master planning uses the first possible value that, together with predicted on-hand level, is below the maximum level. If the sum is less than the minimum level, master planning uses the first value that, together with predicted on-hand, is above the maximum level.

The following subsections provide some examples that show how the multiple order quantity for a product affects the result of the *Min./Max.* replenishment method.

Example 1

A product has the following configuration:

- **Coverage code:** *Min./Max.*
- **Minimum:** *15*
- **Maximum:** *22*
- **Multiple:** *0*

The product has 10 pieces of on-hand inventory, and there's no other demand or supply.

When master planning runs, it creates a planned order for 12 pieces to replenish inventory to the maximum quantity.

Example 2

A product has the following configuration:

- **Coverage code:** *Min./Max.*
- **Minimum:** *15*
- **Maximum:** *22*
- **Multiple:** *5*

The product has 10 pieces of on-hand inventory, and there's no other demand or supply.

When master planning runs, it creates a planned order for 10 pieces (because 15 pieces of replenishment plus 10 pieces of on-hand inventory exceed the maximum quantity).

Example 3

A product has the following configuration:

- **Coverage code:** *Min./Max.*

- **Minimum:** *21*
- **Maximum:** *24*
- **Multiple:** *5*

The product has 10 pieces of on-hand inventory, and there's no other demand or supply.

When master planning runs, it creates the planned order for 15 pieces (because 10 pieces of replenishment plus 10 pieces of on-hand inventory are less than the minimum quantity).

Last updated on 07/01/2026

Master planning and multisite functionality overview

Master planning takes the settings of the site and warehouse inventory dimensions into account.

The site dimension is mandatory, and you can set the warehouse dimension to be mandatory.

When a dimension is mandatory, a dimension value must be entered on all inventory transactions. Therefore, during master planning, the site and the warehouse for the initial demand are known. The site dimension is also consistent so that during the explosion of lower-level demand, the site value does not change.

When the warehouse is not set to mandatory, it may not be known from the initial demand. The planning engine must determine which warehouse to use based on the settings that are defined for the item, individual warehouses, and the details of the order line.

The following articles describe how the planning engine works, when different settings are defined, to determine the warehouse to use.

Master planning for site and warehouse coverage, warehouse mandatory

Master planning for site coverage, mandatory warehouse

Master planning for site and warehouse coverage, warehouse not mandatory

Master planning for site coverage, warehouse not mandatory

Determine the BOM version

Last updated on 07/01/2026

Master planning for site coverage, mandatory warehouse

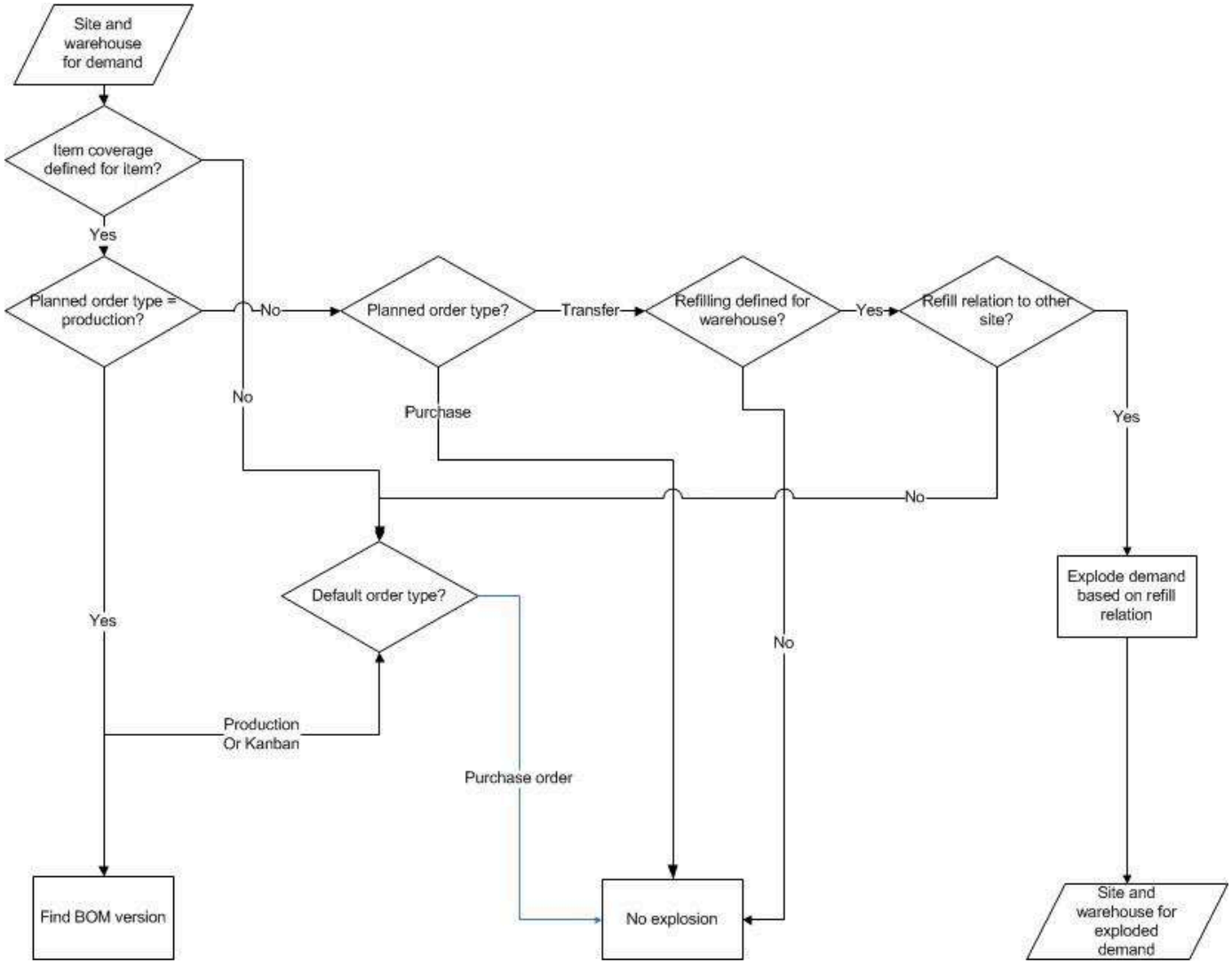
This article describes how an item that has the site as coverage dimension is planned. Warehouse is a mandatory dimension.

This master planning scenario involves the following conditions:

- The site dimension is set to mandatory and must be entered on the demand transaction. This setting can't be modified.
- The warehouse dimension is set to mandatory and must be entered on the demand transaction.
- The site dimension is set for coverage planning. Other dimensions may be set for coverage planning also. However, they are not affected by the multisite functionality.
- The warehouse dimension is not set for coverage planning. Therefore, supply and demand are aggregated by site and, perhaps, other coverage-planned dimensions also.

The following graphic illustrates how master planning proceeds. The parameters that are referred to in the graphic, and their locations, are as follows:

- Item coverage is defined for the item. Click **Product information management** > **Products**> **Released products**. Select the item, and then click **Plan** > **Item coverage**.
- Refill relations are defined for the warehouse. Click **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**. On the **Master planning** tab, see the **Main warehouse** field group.
- The default order type is set to Production, Purchase order, or Kanban. Click **Product information management** > **Products**> **Released products**. Select the item, and then click **Plan** > **Default order settings**. In the **Default order settings** form, see the **Default order type**.



Related information

- Master planning and multisite functionality overview
- Master planning for site and warehouse coverage, warehouse mandatory
- Master planning for site coverage, mandatory warehouse
- Master planning for site and warehouse coverage, warehouse not mandatory
- Determine the BOM version

Last updated on 07/01/2026

Master planning for site coverage, warehouse not mandatory

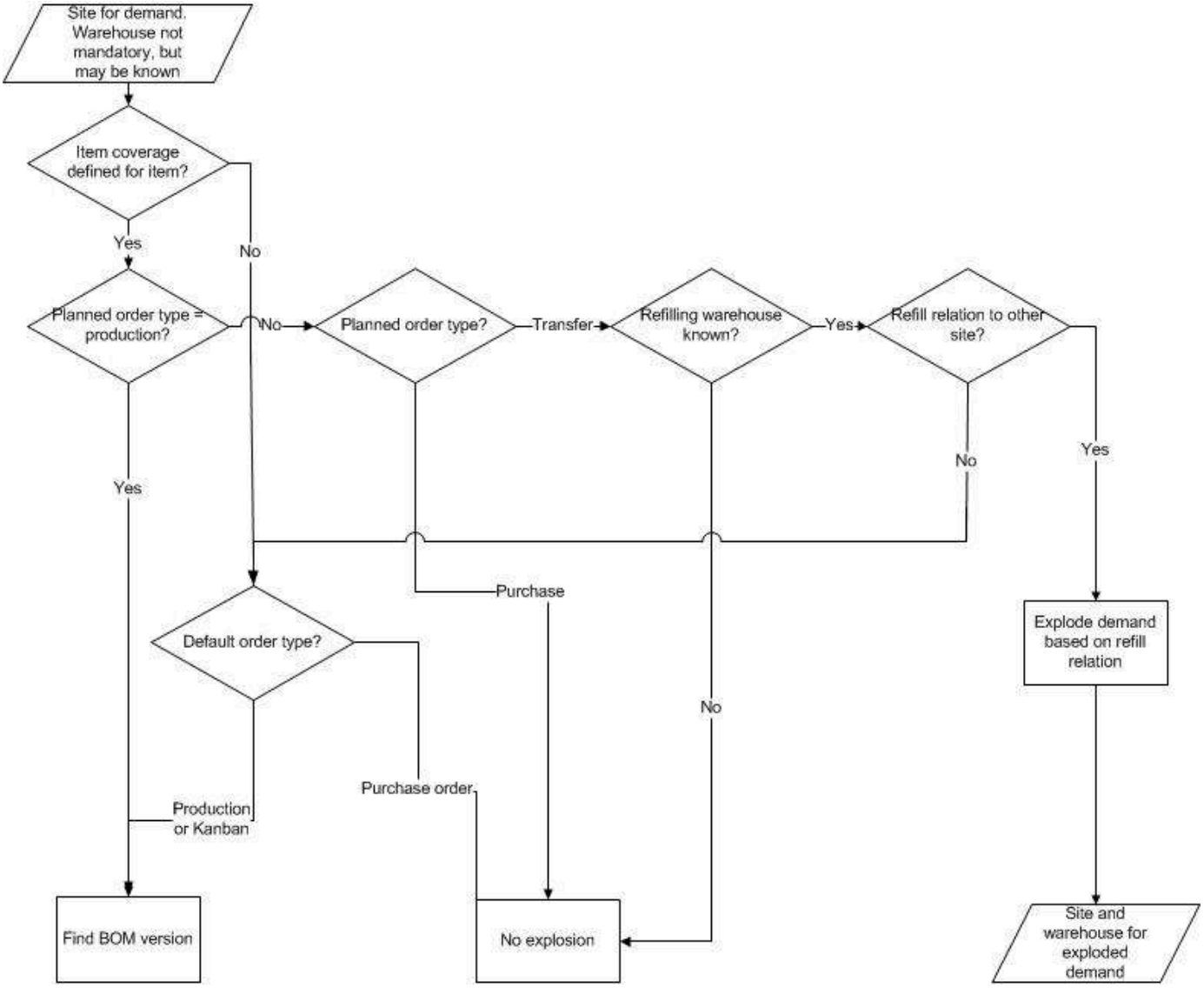
This article describes how an item that has the site dimension set for coverage is planned.

This master planning scenario involves the following conditions:

- The site dimension is set to mandatory and must be entered on the demand transaction.
- The warehouse dimension is not set to mandatory. The warehouse may be known, but it is not used in the master planning calculation.
- The site dimension is set for coverage planning.
- The warehouse dimension is not set for coverage planning. Therefore, supply and demand are aggregated by site and, perhaps, other coverage-planned dimensions also.

The following graphic illustrates how master planning proceeds. The parameters that are referred to in the graphic, and their locations, are as follows:

- Item coverage is defined for the item. Click **Product information management** > **Products**> **Released products**. Select the item, and then click **Plan** > **Item coverage**.
- Refill relations are defined for the warehouse. Click **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**. On the **Master planning** tab, see the **Main warehouse** field group.
- The default order type is set to Production, Purchase order or Kanban. Click **Product information management** > **Products**> **Released products**. Select the item, and then click **Plan** > **Default order settings**. In the **Default order settings** form, see the **Default order type** field.



Related information

- Master planning and multisite functionality overview
- Master planning for site and warehouse coverage, warehouse mandatory
- Master planning for site coverage, mandatory warehouse
- Master planning for site coverage, warehouse not mandatory
- Determine the BOM version

Last updated on 07/01/2026

Master planning for site and warehouse coverage, warehouse mandatory

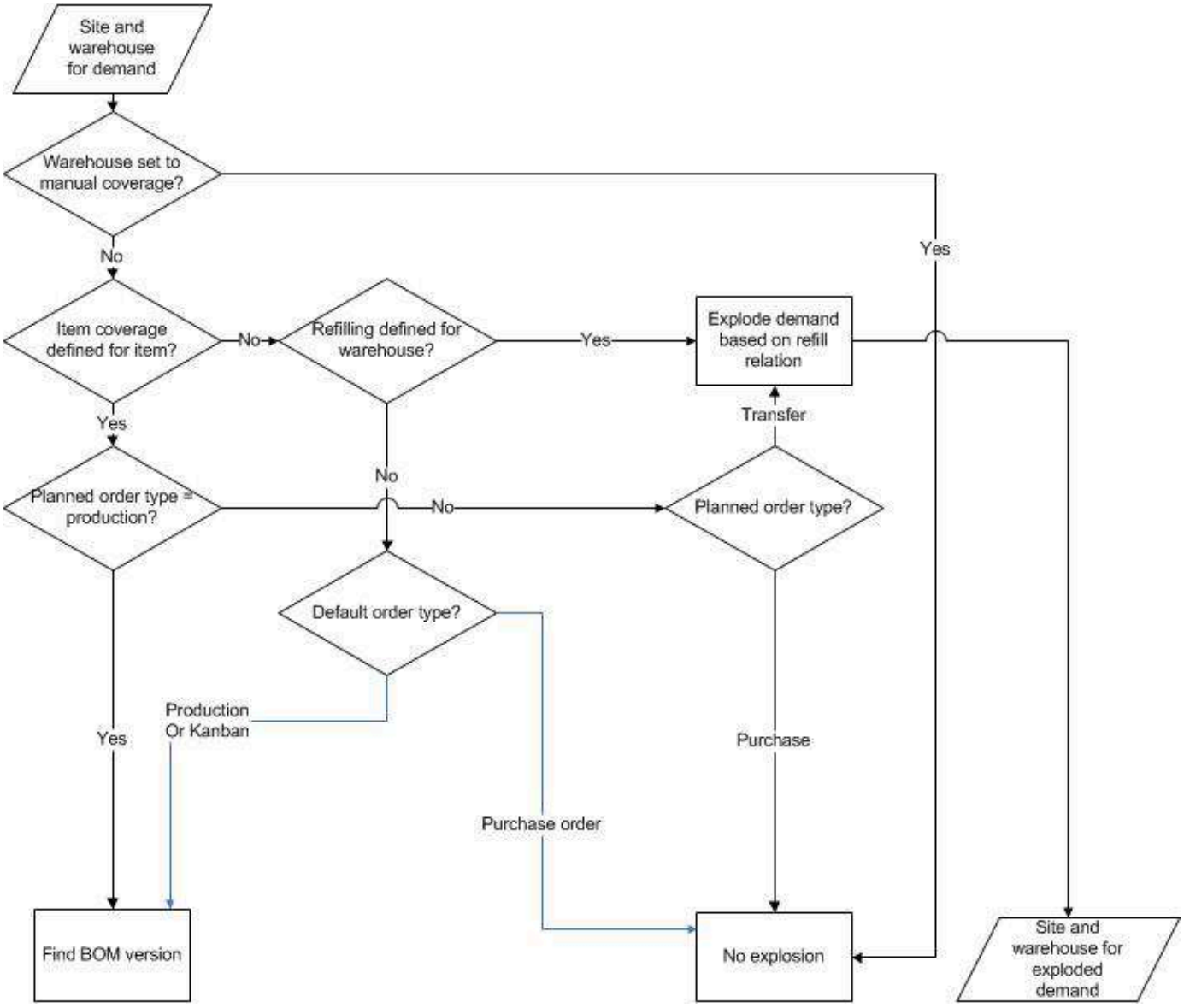
This article describes how an item that has site and warehouse as coverage dimensions is planned. The warehouse dimension is mandatory.

This master planning scenario involves the following conditions:

- The site dimension is set to mandatory and must be entered on the demand transaction.
- The warehouse dimension is set to mandatory and must be entered on the demand transaction.
- The site and warehouse dimensions are set for coverage planning. Other dimensions may be set for coverage planning also. However, they are not affected by the multisite functionality.

The following graphic illustrates how master planning proceeds. The parameters that are referred to in the graphic, and their locations, are as follows:

- The warehouse is set to **Manual**. Click **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**. On the **Master planning** FastTab, see the **Manual** field.
- Item coverage is defined for the item. Click **Product information management** > **Products**> **Released products**. Select the item, and then, on the Action Pane, on the **Plan** tab, click **Item coverage**.
- Refill relations are defined for the warehouse. Click **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**. On the **Master planning** FastTab, see the **Main warehouse** field group.
- The default order type is set to Production, Purchase order, or Kanban. Click **Product information management** > **Products**> **Released products**. Select the item, and then, on the Action Pane, on the **Plan** tab, click **Default order settings**. In the **Default order settings** form, see the **Default order type**.



Related information

- Master planning and multisite functionality overview
- Master planning for site coverage, mandatory warehouse
- Master planning for site coverage, warehouse not mandatory
- Master planning for site and warehouse coverage, warehouse not mandatory
- Determine the BOM version

Last updated on 07/01/2026

Master planning for site and warehouse coverage, warehouse not mandatory

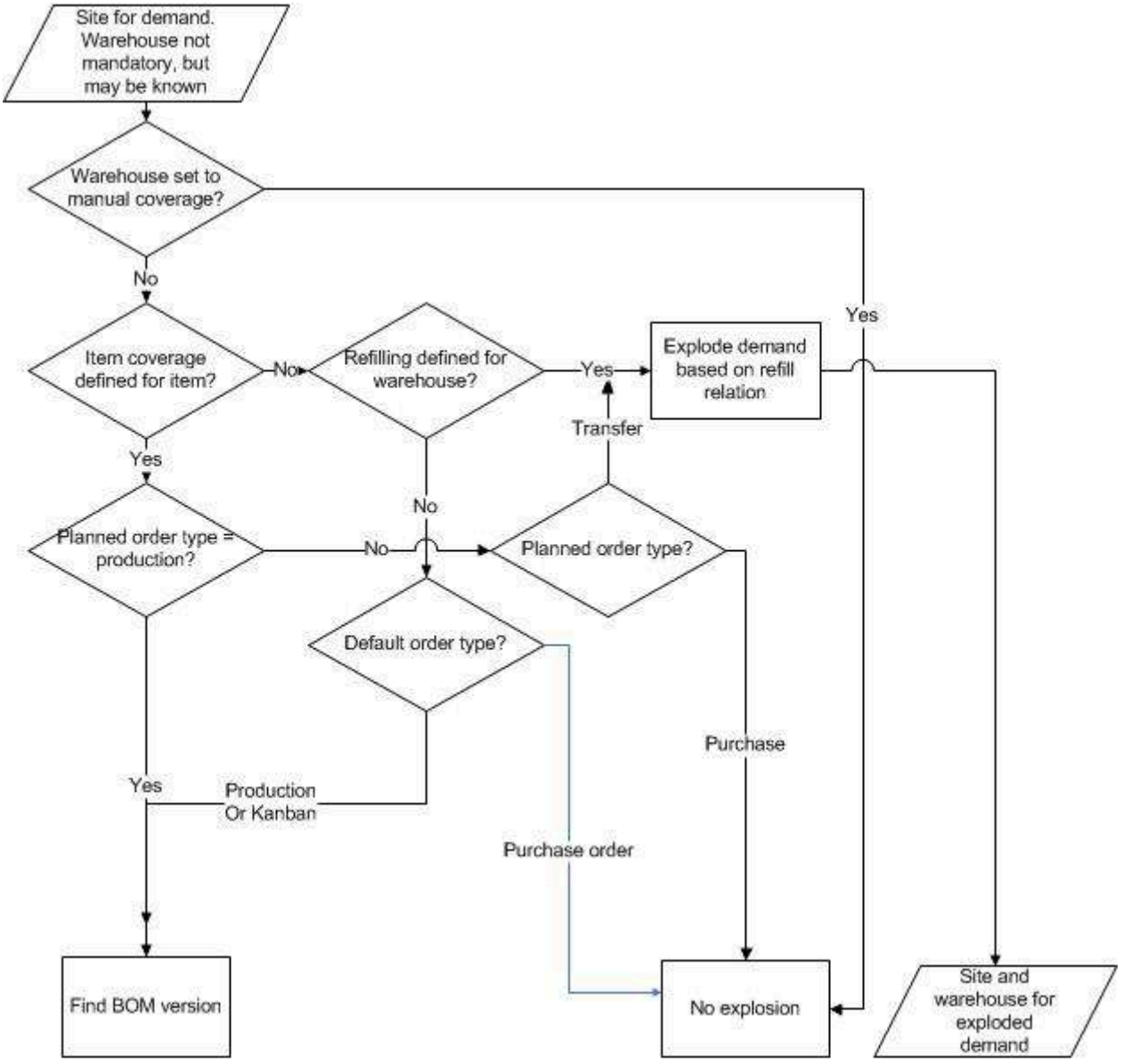
This article describes how an item that has site and warehouse as coverage dimensions is planned. The warehouse dimension is not mandatory.

This master planning scenario involves the following conditions:

- The site dimension is set to mandatory and must be entered on the demand transaction. This setting can't be modified.
- The warehouse dimension is not set to mandatory. The warehouse may be known, but it is not used in the master planning calculation.
- The site and warehouse dimensions are set for coverage planning. Other dimensions may be set for coverage planning also. However, they are not affected by the multisite functionality.

The following graphic illustrates how master planning proceeds. The parameters that are referred to in the graphic, and their locations, are as follows:

- The warehouse is set to Manual. Click **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**. On the **Master planning** FastTab, see the **Manual** field.
- Item coverage is defined for the item. Click **Product information management** > **Products**> **Released products**. Select the item, and then, on the Action Pane, on the **Plan** tab, click **Item coverage**.
- Refill relations are defined for the warehouse. Click **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**. On the **Master planning** FastTab, see the **Main warehouse** field group.
- The default order type is set to Production, Purchase order, or Kanban. Click **Product information management** > **Products**> **Released products**. Select the item, and then, on the Action Pane, on the **Plan** tab, click **Default order settings**. In the **Default order settings** form, see the **Default order type**.



Related information

- Master planning and multisite functionality overview
- Master planning for site coverage, mandatory warehouse
- Master planning for site and warehouse coverage, warehouse mandatory
- Master planning for site and warehouse coverage, warehouse not mandatory
- Determine the BOM version

Last updated on 07/01/2026

Determine the BOM version

During a demand explosion, if an item has a default order type of Production, the planning engine finds a valid BOM version based on the site.

The site dimension is always known and is stated on the demand transaction. The following process is used to determine the BOM version to use:

- If there is a BOM version defined for the item at the demand site, the site-specific BOM is used.
- If there is no site-specific BOM version defined for an item at the demand site, a general BOM is used. A general BOM does not state a site, and it is valid for multiple sites. If there is a general BOM, it is used.
- If there is no general BOM version to use, the demand explosion stops at this point.

A valid BOM version, whether site-specific or general, must meet the required criteria for date and quantity.

Last updated on 07/01/2026

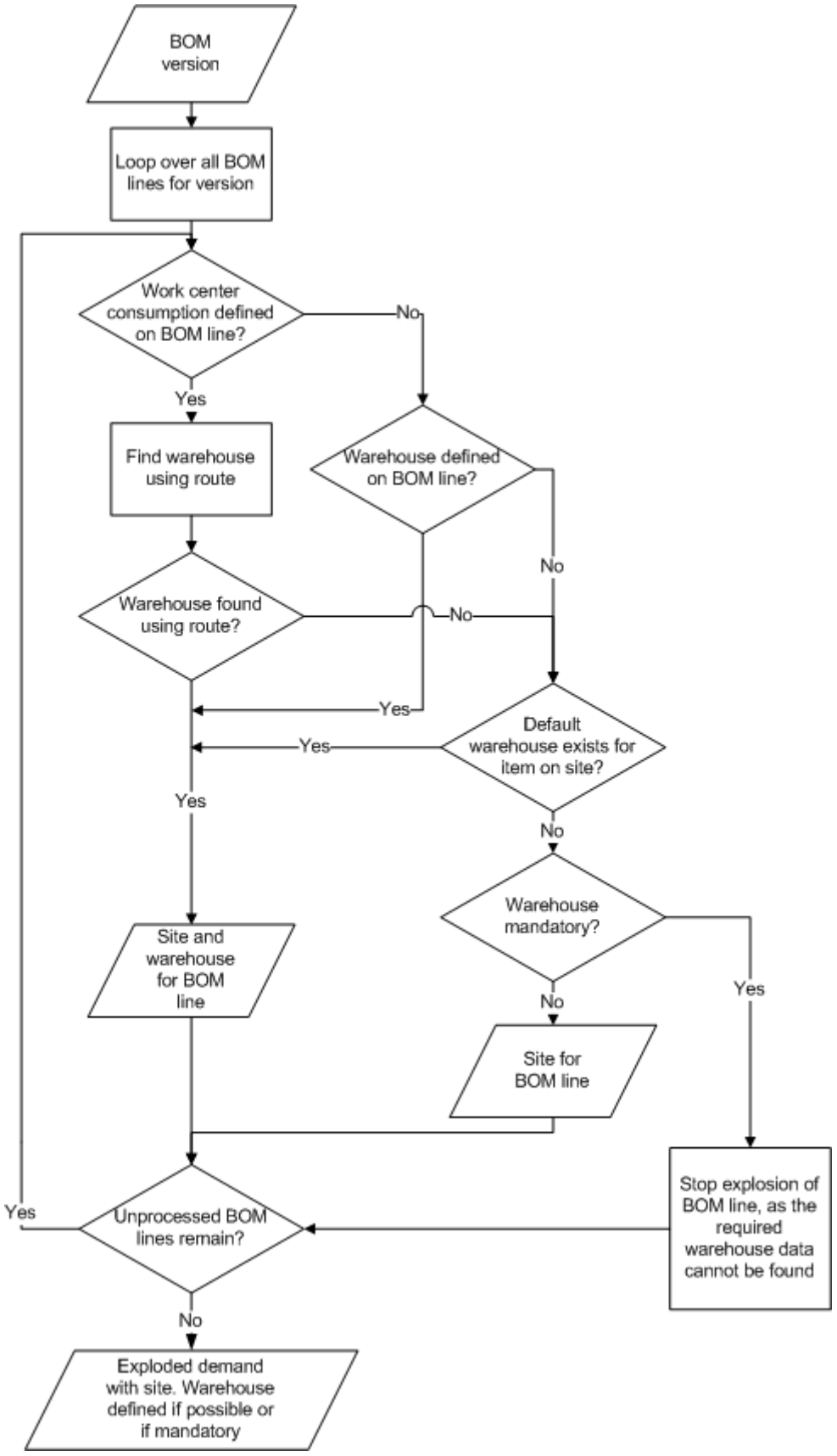
Explosion of a BOM version

This article explains a master planning scenario that involves explosion of a bill of materials (BOM) version.

A demand explosion of a bill of materials (BOM) version creates a demand for each BOM line item at a specific site and, possibly, at a specific warehouse. In a site-specific BOM, a specific warehouse can be defined for each BOM line. Additionally, for each BOM line, the item's dimension settings determine whether the warehouse is required. The resulting demand for each BOM line item then becomes the starting point for additional demand explosion. This master planning scenario involves the following conditions:

- The site dimension is mandatory and must be entered on the demand transaction.
- The site dimension is consistent. Therefore, the site for lower-level demand is the same as the site on the initial demand transaction.

The following illustration shows how the process for master planning demand explosion.



Related information

- Determine the BOM version
- Master planning and multisite functionality overview

Last updated on 07/01/2026

Safety stock fulfillment for items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Safety stock is an item quantity that you hold in inventory to reduce the risk that the item runs out of stock. Use safety stock when the demand from sales orders is more than planned for final goods, and when a supplier can't deliver additional units in the expected time.

The system always tries to prevent the accumulated quantity of an item from falling below its safety stock limit. Whenever the master planning engine detects that the accumulated on-hand inventory for an item will fall below its minimum, it creates a planned order to replenish the item and schedules the order to arrive before the minimum threshold is crossed. Therefore, the system fulfills safety stock on *Today's date* + *Procurement time*.

During planning, if there's actual demand, the system pegs it against a planned order generated for safety stock provided it enables the demand to be fulfilled on time and strict safety stock pegging isn't enforced. Because the system always tries to keep the accumulated on-hand inventory level above the safety stock level, it creates a new planned order to cover the safety stock that the actual demand claimed. When strict safety stock pegging is enforced, actual demand isn't pegged against a planned order generated for safety stock.

> **Note**
>
> Because safety stock isn't really demand, any demand is prioritized over the safety stock when strict safety stock pegging is off. Therefore, the system can create a planned order to fulfill safety stock, but if actual demand arrives later, that demand can claim the safety stock quantity. (The new demand is pegged against the original planned order.)

Set the safety stock level for an item

To define safety stock for an item at a specific location, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the relevant product in the grid.
3. On the Action Pane, on the **Plan** tab, select **Item coverage**.

4. While you're on the **Overview** tab, select **New** on the Action Pane to add a line to the grid.
5. For the new line, specify the relevant product dimensions (**Site**, **Warehouse**, and other dimensions as required, such as **Color** or **Style**).
6. In the **Minimum** field, enter the safety stock value. The master planning engine always generates planned orders to prevent the accumulated inventory level from falling below this limit. The value is expressed in inventory units. If you leave the field blank, a default value of **0** (zero) is used.

> **Note**
>
> The **General** tab of the **Item coverage** page includes a **Fulfill minimum** field. When you use Planning Optimization, the setting of this field is ignored. (Instead, the system always behaves as though **Fulfill minimum** is set to *Today's date + procurement time*.) For information about how the **Fulfill minimum** setting works when you use the deprecated master planning engine, see Safety stock fulfillment with the deprecated planning engine.

Example: Safety stock

Set up your system as follows:

- The warehouse has 30 units of total on-hand inventory of item Z0001.
- A demand forecast shows that 2 units of item Z0001 are consumed every day.
- The safety stock for item Z0001 is set to 20 units.
- The lead time for item Z0001 is five days.

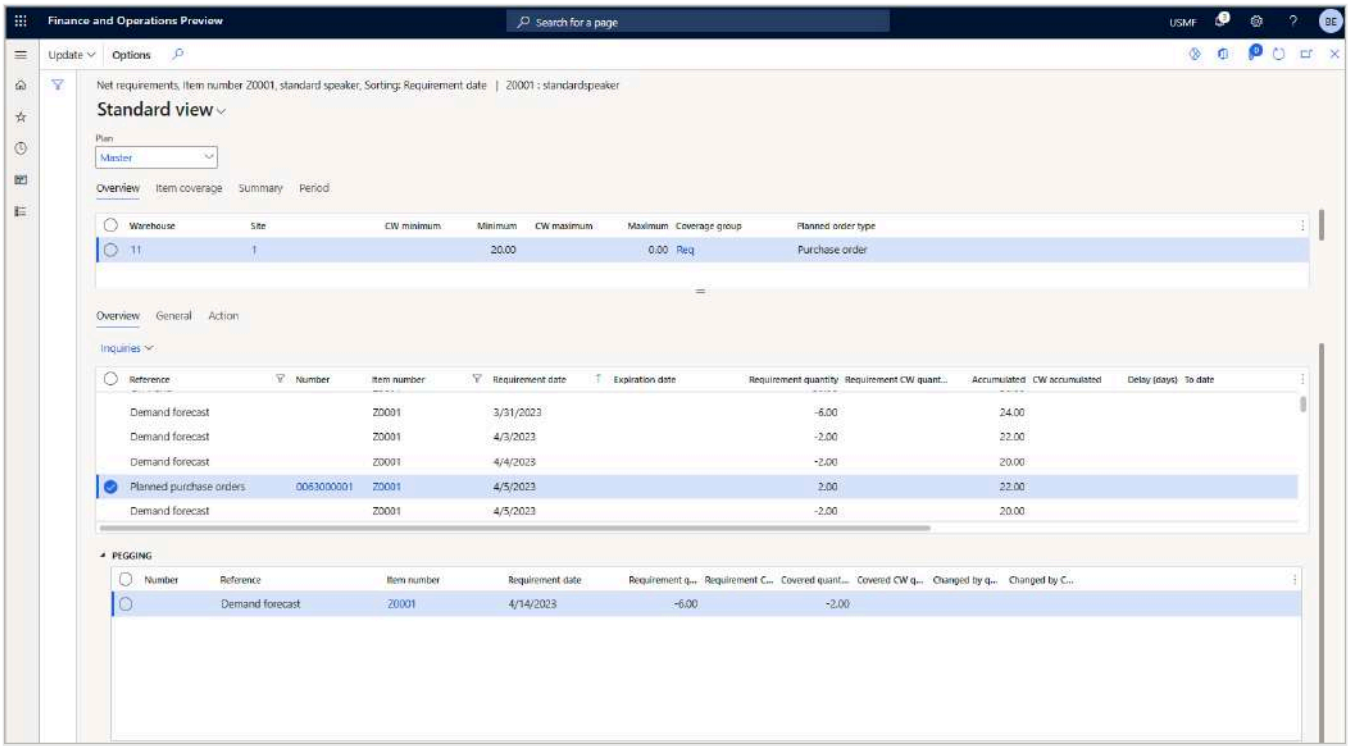
When you run master planning, it creates several orders to ensure that the accumulated on-hand inventory stays above the safety stock threshold. The following table provides an example.

Expand table

Reference	Item number	Requirement date	Requirement quantity	Accumulated
Demand forecast	Z0001	3/31/2023	-6.00	24
Demand forecast	Z0001	4/3/2023	-2.00	22
Demand forecast	Z0001	4/4/2023	-2.00	20
Planned purchase orders	Z0001	4/5/2023	2.00	22

Reference	Item number	Requirement date	Requirement quantity	Accumulated
Demand forecast	Z0001	4/5/2023	-2.00	20

Therefore, the **Net requirements** page for item Z0001 might resemble the following example.



Example: Minimum key

Use minimum keys to handle seasonal fluctuations in demand. For example, you can decrease the minimum inventory level for an item during the off-season and then gradually increase it during the other months.

To create a minimum key, go to **Master planning** > **Setup** > **Coverage** > **Minimum/maximum keys**. You can then specify the minimum key to adjust the safety stock level by seasonality by setting the **Minimum key** field on the **General** tab of the **Item coverage** page.

If you're using minimum keys, set the **Minimum periods** option to *Yes* to fulfill the minimum inventory level for all the periods that are set up in the minimum key. If you set the option to *No*, the minimum inventory is fulfilled for the current period only.

The following procedure shows an example of how to set up a minimum key that accounts for increased seasonal demand during the spring and summer months.

1. Go to **Master planning** > **Setup** > **Coverage** > **Minimum/maximum keys**.
2. Select **New** to create a minimum/maximum key.

3. In the **Minimum or maximum key** field, enter an identifier for the key. In the **Name** field, enter a name for the key.

4. Set the **Use the effective date** option to *Yes* and leave the **Effective date** field blank to make the key valid from the first day of the current year.

   > **Note**
   >
   > The combination of the **Use the effective date** and **Effective date** settings defines the date that the key is valid from.
   >
   > - When the **Use the effective date** option is set to *No*, the key is valid from the current date or system date.
   > - When the **Use the effective date** option is set to *Yes*, the key is valid from the date defined in the **Effective date** field.

5. In the **Periods** section, create 12 lines, and set the following values for them:

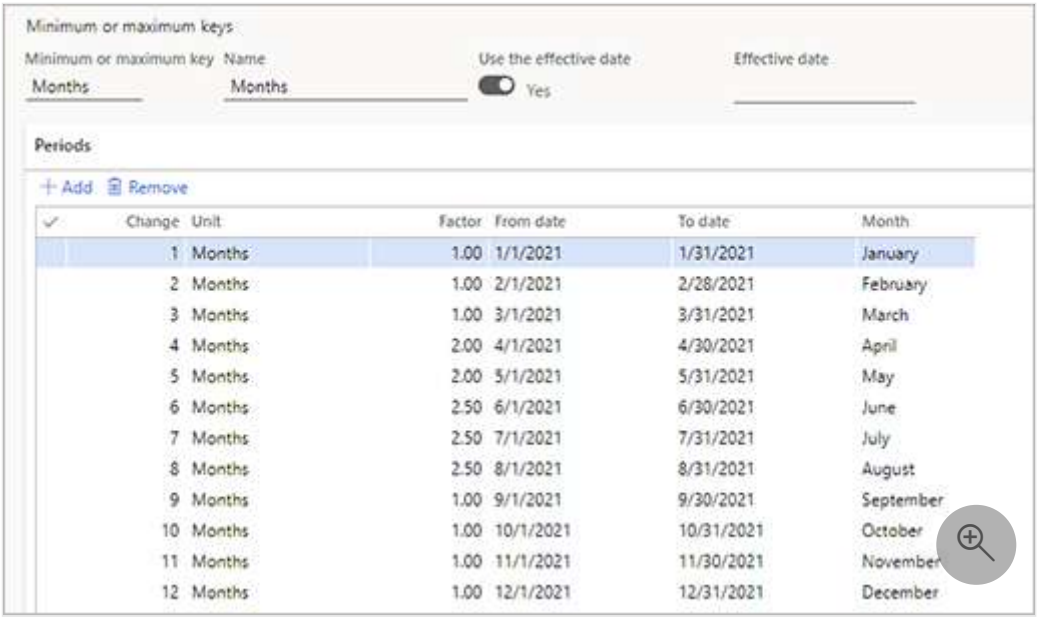
   - **Change** – Assign each line a unique number from 1 through 12. This field indicates the incremental change in the unit of time that is defined by the **Unit** field.
   - **Unit** – Select *Months* for every line.
   - **From date**, **To date**, and **Month** – These fields are automatically set, based on the **Change** and **Unit** settings. Monthly periods start from the first day of the current year.
   - **Factor** – Enter the values that are described in the following table. This field defines the factor that you want to multiply the minimum inventory by.

   Expand table

Line (Change)	Factor	Result
1–3	1	Minimum inventory is based on the setting for January through March on the **Item coverage** page.
4–5	2	Minimum inventory is multiplied by a factor of 2 for April through May.
6–8	2.5	Minimum inventory is multiplied by a factor of 2.5 for June through August.

Line (Change)	Factor	Result
9–12	1	Minimum inventory reverts to the setting for September through December on the **Item coverage** page.

Your settings now resemble the settings in the following illustration.



> **Note**
>
> You can also use a wizard to create a minimum/maximum key. On the **Minimum or maximum keys** page, on the Action Pane, select **Wizard** to open the **Minimum/Maximum Keys** wizard. The wizard guides you step by step through the process of creating and setting up the minimum/maximum key.

If the coverage code is *Min/Max*, you can also specify the maximum inventory quantity that you want to maintain for an item. The value is also expressed in inventory units. If the projected available inventory falls below the minimum quantity, master planning generates a planned order to fulfill all open requirements and brings the available inventory up to the specified maximum quantity. Just as when you set up the minimum inventory quantity, you must define all other planned coverage dimensions before you can set the **Maximum** field.

Example: Min/max coverage code

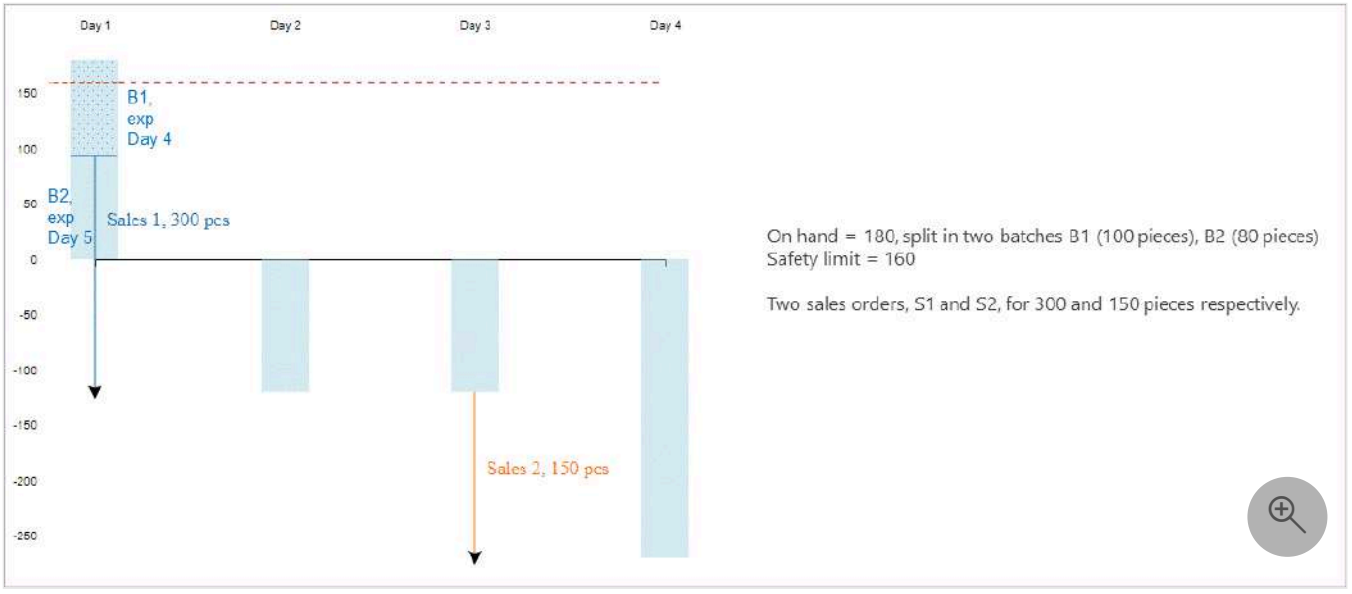
The minimum quantity is 10, and the maximum quantity is 15. Current on-hand inventory is 4. This situation gives a minimum quantity requirement of 6. However, because the maximum quantity is 15, master planning generates a planned order for 11 items.

For items that follow seasonal demands, you might need to maintain different maximum levels. To set up these levels, define **Maximum keys** by going to **Master planning** > **Setup** > **Coverage** > **Minimum/maximum keys**. Fill in the **Maximum key** field on the **Item coverage** page. You can view the information about the safety stock levels, defined through minimum keys on the **Min/Max** tab, on the **Item coverage** page. Make sure that, for a certain period, the minimum and the maximum values stay in sync.

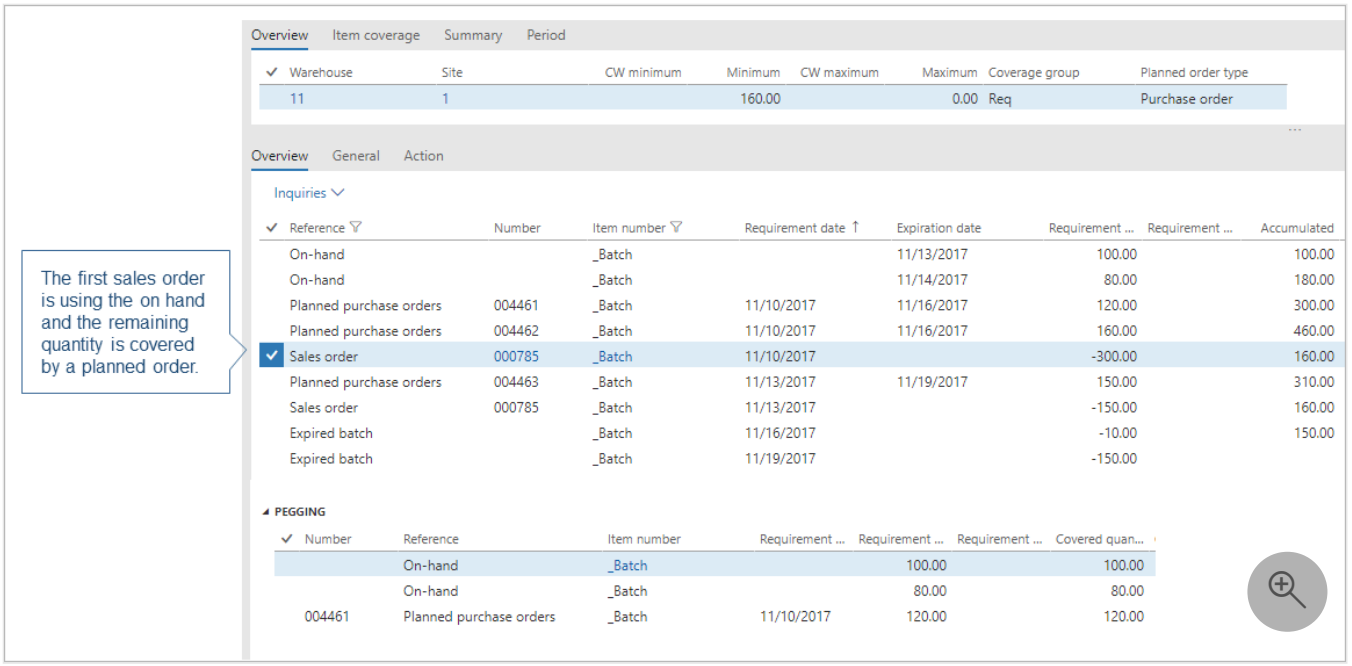
Plan safety stock replenishment for FEFO items

At any point in time, the inventory receipt with the latest expiry date is used for safety stock. This approach ensures that real demand, such as sales lines or BOM lines, is fulfilled in the first expired, first out (FEFO) order.

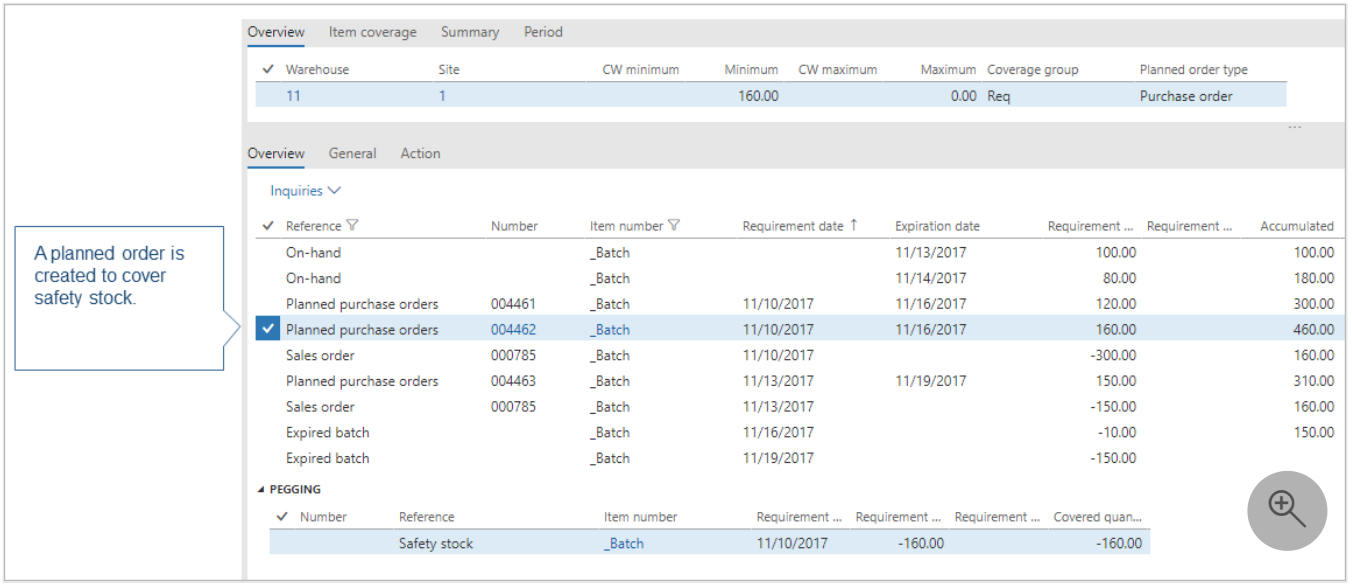
To see how this process works, consider the following scenario:



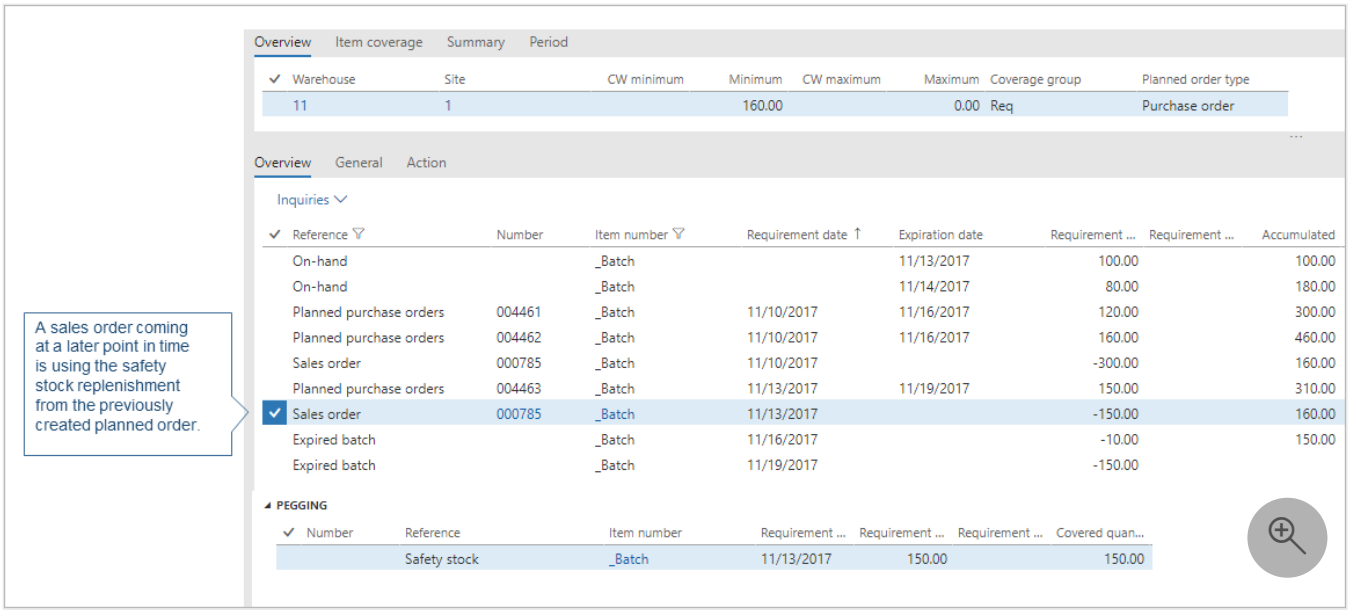
When you run planning, it covers the first sales order from the existing on-hand inventory and an additional purchase order for the remaining quantity.



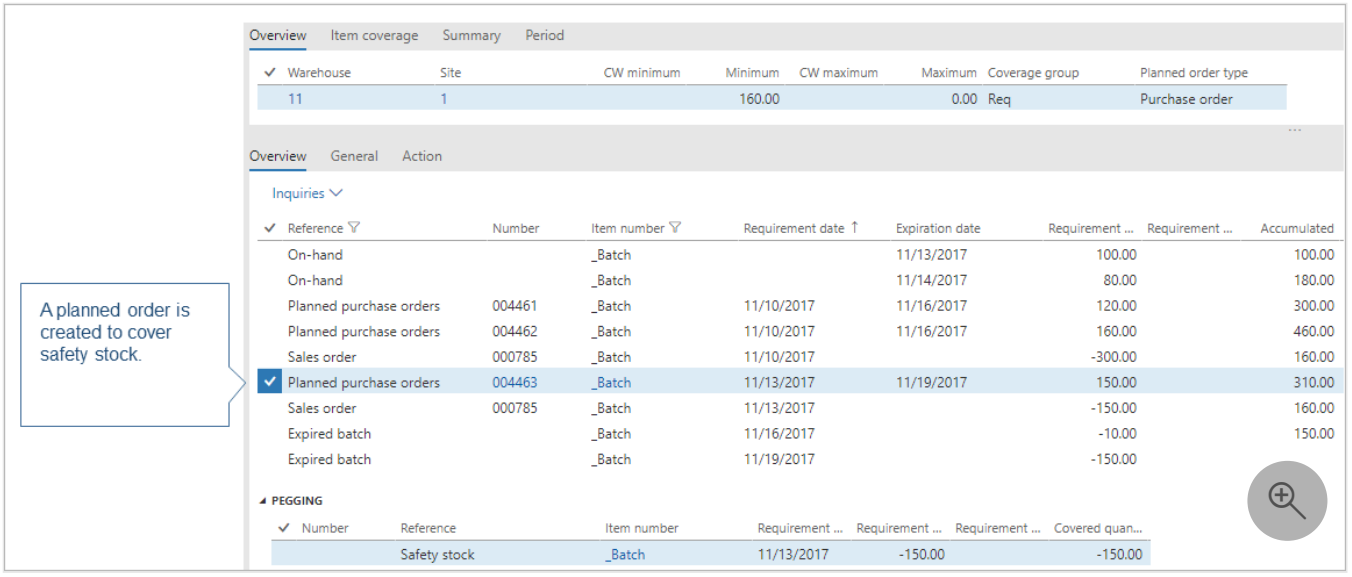
A planned order is created to bring the available inventory back to the safety limit.



When you plan the second sales order, the previously created planned order that covers the safety stock is used to cover this quantity. Hence, the safety stock is constantly rolling.



Finally, another planned order is created to cover the safety stock.



All the batches expire accordingly, and planned orders are created to refill the safety stock after it expires.

How master planning handles the safety stock constraint

The system tracks safety stock as a requirement type, just like sales lines or BOM requirements. You can see the safety stock requirement line on the **Net requirements** page if you remove the default filter on the **Requirement type** column.

If fulfilling the safety stock requirement causes delays in the fulfillment of real demand, such as sales lines, BOM lines, transfer requirements, or demand forecast lines, the system deprioritizes it. Otherwise, making sure that the available inventory is above the safety stock quantity has

the same priority as any other demand types. This approach ensures no delays for real transactions and helps prevent over-replenishment and early replenishment of safety stock.

During the coverage phase of master planning, the system no longer deprioritizes safety stock replenishment. The system can use on-hand inventory before any other demand types. During the delay calculation, new logic goes over the delayed sales lines, BOM line requirements, and all the other demand types to determine whether they can be delivered on time, provided that the safety stock is used. If the system identifies that it can minimize delays by using safety stock, the sales lines or BOM lines replace their initial coverage with the safety stock, and the system triggers the replenishment for the safety stock instead.

If you don't set up the plan or the item for delayed calculation, the safety stock constraint has the same priority as any other demand types. This priority means there's a reserve of on-hand and other available inventory before other demand types.

Related information

- Use the safety stock journal to update minimum coverage for items

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Last updated on 03/26/2026

Safety stock pegging options

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Most industries prioritize customer demand over safety stock. Therefore, when you must both replenish safety stock and fulfill a customer order, the system uses existing planned orders to cover the customer demand instead of leaving those planned orders pegged against safety stock.

However, some businesses prefer to allow the system to keep existing planned orders pegged against safety stock, even if customer demand can't be fulfilled as a result. This approach is known as *strict safety stock pegging*.

For each coverage group, you can specify how strictly the system should peg safety stock as demand against the planned order that's created for it.

Prerequisites

Before you can use the features described in this article, ensure your system meets the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.36 or later.
- The feature that's named *Strict safety stock pegging for Planning Optimization* must be turned on in Feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default.

Set safety stock pegging options

Safety stock pegging options are set at the coverage group level. Follow these steps to set safety stock pegging options for a selected coverage group.

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.

2. Follow one of these steps:

    - To edit an existing group, select it in the list pane, and then select **Edit** on the Action Pane.
    - To create a new group, select **New** on the Action Pane.

3. On the **General** FastTab, in the **Strict safety stock pegging** field, select one of the following options:

    - *No* – The system doesn't consider safety stock as actual demand, and it prioritizes any actual demand over safety stock. Although you can create a planned order to fulfill safety stock, any actual demand that arrives later can claim the planned safety stock quantity. (The actual demand is pegged against the planned order.) This option lets you provide a higher level of service for your customers.
    - *Yes* – The system applies strict safety stock pegging. Any planned order that you create to fulfill safety stock remains pegged against safety stock. If you select this option, whenever the system calculates that a planned order is needed to fulfill safety stock, the system prioritizes that safety stock over real demand. This option can provide better visibility in some scenarios, such as when you're using a large number of negative days.

> ⓘ **Important**
>
> When you enable strict safety stock pegging, the system ignores pegging priority for safety stock planned orders. This behavior can lead to unreliable net requirements pegging and the creation of unwanted planned orders, because the safety stock demand is always prioritized regardless of any pegging priority rules you configured. Before you enable this option, carefully evaluate whether the trade-off between safety stock visibility and potential over-ordering is acceptable for your planning scenario. Learn more about how safety stock behavior differs between the deprecated master planning engine and Planning Optimization in Safety stock fulfillment with the deprecated master planning engine.

Related information

- Safety stock fulfillment for items
- Safety stock fulfillment with the deprecated master planning engine

ⓘ **Note:** The author created this article with assistance from AI. Learn more

Last updated on 05/05/2026

Use the safety stock journal to update minimum coverage for items

08/22/2025

Safety stock is an additional quantity of an item held in inventory to reduce the risk of it going out of stock. It acts as a buffer if sales orders arrive and the supplier can't meet the customer's requested ship date.

This article explains how to use the safety stock journal to calculate minimum coverage proposals from historical transactions and update item coverage with those proposals.

Overview of minimum coverage usage

Safety stock is set up on the **Item coverage** page for each item. Different types of replenishment are represented by different coverage codes. Learn more in Safety stock fulfillment for items. All coverage codes use the value set in the **Minimum** field on the **Item coverage** page for each item. There are two major approaches for using the **Minimum** field:

- **Minimum quantity for min/max purposes** – This approach uses the minimum quantity together with min/max logic. It applies when the **Coverage code** field is set to *Min/Max* for the relevant item or coverage group. The **Minimum** quantity represents the average daily usage multiplied by the item's lead time.
- **Minimum quantity for inventory plan purposes** – This approach uses the minimum quantity to represent an inventory plan in combination with demand forecasts. It applies when the **Coverage code** field is set to *Period* or *Requirement* for the relevant item or coverage group. The **Minimum** quantity represents an inventory plan that reflects the desired customer service level to help reduce stock-outs, partial shipments, and delivery lead times. The minimum quantity reflects a percentage of forecast accuracy for a given item. It doesn't represent a desired inventory position.

You can set the **Minimum** value in three ways:

- Manually on the **Item coverage** page.
- By the Data Management framework (*Item coverage* data entities).
- By the safety stock calculation done by safety stock journals.

Calculate minimum coverage based on historical usage

Safety stock journals calculate a proposed minimum quantity based on an item's historical usage, either for min/max or inventory planning purposes. Historical usage represents all issue transactions during a specified period. These issue transactions include sales order transactions and inventory adjustments. The calculations show the impact of the proposed minimum quantity on inventory value and its change compared to current minimum quantities.

Each safety stock journal line represents an item and its coverage dimensions. These journal lines are created and shown on the **Safety stock journal lines** page (**Master planning** > **Master planning** > **Run** > **Safety stock calculation**). The business process for using the safety stock journals to calculate the proposed minimum quantities is described later in this article.

The planner uses a safety stock journal to calculate proposed minimum quantities for selected items based on historical usage during specific periods. The proposed minimums can be manually overridden as required, and you can review the potential impact of the proposed minimums on inventory value. When the journal is posted, the associated minimum quantities in the item coverage are automatically updated.

Create a new safety stock journal name

Create at least one safety stock journal name before generating this type of journal. You might typically use several journal names to help separate your safety stock calculations.

1. Go to **Master planning** > **Setup** > **Safety stock journal names**.
2. On the Action Pane, select **New**.
3. On the new line, set the following fields:
    - **Name** – Enter a short name for the journal.
    - **Description** – Enter a description of the journal.
    - **Private for user group** – To limit the audience for the current journal name, select a user group.
    - **Delete lines after posting** – Select this checkbox to clean up journal lines by default as you post them. Clear it to keep all posted lines.

    The **Journal type** field is read-only and is preset to *Safety stock*.
4. Close the page.

Create a safety stock journal and lines

This step creates a journal and automatically adds lines to it. Each line identifies an item, its coverage dimensions, and several calculated quantities from the usage history. The calculated

quantities include average issues per item lead time, average issues per month, and the monthly standard deviation.

Automatically generate journal lines

You can automatically generate journal lines only if no lines exist for the current journal.

Use these steps to automatically generate journal lines.

1. Go to **Master planning** > **Master planning** > **Run** > **Safety stock calculation**.
2. On the Action Pane, select **New** to create a new safety stock journal.
3. On the **Journal header details** FastTab, set the following fields:

    - **Name** – Select the safety stock journal name to add the line to.
    - **Description** – The default value is the description of the journal name that you selected. However, you can edit the value as you require.
    - **Delete lines after posting** – Select this checkbox to clean up journal lines as you post them. Clear it to keep all posted lines. The setting is inherited from the selected journal name.

    The **Journal** field is read-only and shows the ID number of the journal that you're creating and adding lines to.
4. On the **Journal lines** FastTab toolbar, select **Create lines**.
5. In the **Create journal lines for proposed minimum inventory levels** dialog box, set the following fields:

    - **From date** – Select the start date of the period that issues should be included in the calculation for.
    - **To date** – Select the end date of the period that issues should be included in this calculation for. There must be at least two months between the start and end dates.
    - **Calculate standard deviation** – Set this option to *Yes* to calculate the standard deviation. You must set this option to *Yes* to use the **Use service level** option when you calculate the proposal (as described later in this article).
6. On the **Records to include** FastTab, you can set up filters and constraints to define which items are included. (For example, you can filter by **Coverage group** value.) Select **Filter** to open a standard query editor dialog box, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Microsoft Dynamics 365 Supply Chain Management.

7. On the **Run in the background** FastTab, select whether to run the job in batch mode, and/or set up a recurrent schedule. The fields work just as they do for other types of background jobs in Supply Chain Management.

8. Select **OK**. Lines are created for the dimensions that have inventory transactions.

Manually add or remove journal lines

You can manually add and/or remove journal lines at any time (either after or instead of automatically generating lines). Use the following buttons on the toolbar of the **Journal lines** FastTab:

- **New** – Add a new line. Then enter a value in each column to define the product to calculate and apply new minimums for. Because proposed minimum calculations aren't available for manually added lines, no new **Calculated minimum quantity** values are shown for them. Therefore, you must manually enter the **New minimum quantity** values. However, you can still view the potential impact of the **New minimum quantity** value on inventory value and the potential change in inventory compared to the currently specified minimum.
- **Delete** – Delete the selected line.
- **Delete journal lines** – Delete all lines in the journal.

Calculate a proposal

This step calculates a proposed minimum for each journal line and the line's potential impact on inventory value. (The proposed minimum is shown as the **Calculated minimum quantity** value.) You can do the calculation multiple times, as you require. The calculation updates the **Calculated minimum quantity** value that is shown for all journal lines.

The calculations shown don't affect the actual minimum quantity values for each product until you select **Post** on the Action Pane. At that time, the **New minimum quantity** values will be applied to each product.

1. Go to **Master planning** > **Master planning** > **Run** > **Safety stock calculation**.

2. Open the journal to calculate a proposal for. Alternatively, create a new journal as described earlier in this article.

3. On the **Journal lines** FastTab, select **Calculate proposal** on the toolbar. (You don't have to select any lines.)

4. In the **Calculate proposal for minimum inventory level** dialog box, set the following fields:

- **Use average issue during lead time** – Select this option to generate **Calculated minimum quantity** values based on the average issue during the specified period. Then, in the **Multiplication factor** field, enter a value to adjust the result by as required. For example, enter *1.0* to use the exact calculated average or *1.1* to add an extra buffer of 10 percent.
- **Use service level** – Select this option to calculate a proposed minimum based on the desired service level. Then, in the **Service level** field, select your preferred service level.
- **Lead time margin** – Enter a value to extend the normal lead time by (for example, to allow for administration).
- **Use the calculated minimum quantity as the new minimum quantity** – Set this option to *Yes* to automatically copy values from the **Calculated minimum quantity** column to the **New minimum quantity** column for all lines after the calculation is completed. If you set this option to *No*, the **Current minimum quantity** value will be copied to the **New minimum quantity** column for all lines. You'll then have to manually edit the **New minimum quantity** values as required.

5. On the **Run in the background** FastTab, select whether to run the job in batch mode, and/or set up a recurrent schedule. The fields work just as they do for other types of background jobs in Supply Chain Management.

6. Select **OK**. The results of the calculation are shown in the **Calculated minimum quantity** column on the **Journal lines** FastTab. For now, the values are only proposed values that haven't yet been applied to your products.

Update minimum quantity

Select any line in a safety stock journal to manually override the value of its **New minimum quantity** field.

1. Go to **Master planning** > **Master planning** > **Run** > **Safety stock calculation**.
2. Open the journal to edit. Alternatively, create a new journal as described earlier.
3. On the **Journal lines** FastTab, find the line to adjust, and then edit the value in the **New minimum quantity** column. For example, you might set the **New minimum quantity** value so that it matches the **Calculated minimum quantity** value. If the **Calculated minimum quantity** value is *0* (zero), you can enter the desired future value.

Post the new minimum quantity and validate the result

Follow these steps to post the new minimum quantity and validate the result.

1. Go to **Master planning** > **Master planning** > **Run** > **Safety stock calculation**.

2. Open the journal to post new minimum quantities for. Alternatively, create a new journal as described earlier.

3. On the Action Pane, select **Post**.

4. In the **Post journal** dialog box, set the **Transfer all posting errors to a new journal** field as required. Then select **OK** to post the journal.

5. If you changed the **New minimum quantity** value for a line on the **Journal lines** FastTab, you can confirm the change by following these steps:
   a. For the line that you edited, select the link in the **Item number** column.
   b. In the **Product information** dialog box, select the link in the **Item number** field to open the related released product record.
   c. On the Action Pane, on the **Plan** tab, in the **Coverage** group, select **Item coverage**.
   d. Notice that the **Minimum** value for the site and warehouse that you adjusted by using the posted safety stock journal has now been updated to match your edit.

Related information

- Safety stock fulfillment for items

Maintain planned orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about how to manage planned orders. It describes how you can update the status of planned orders, firm them, and filter for planned orders that have the same status as a selected planned order.

You can manage planned orders from the **Master planning** workspace, the **Planned order** list, or the **Planned production orders**, **Planned purchase orders**, and **Planned transfer** lists.

Planned order status

You can use the **Status** field to help track your progress. The following values are used:

- When master planning generates planned orders, the planned orders have a status of *Unprocessed*.
- If you decide not to firm a planned order, you can give it a status of *Completed*.
- If you want to firm a planned order, you can change the status to *Approved*. Planned orders with *Approved* status are respected by master planning, so they aren't modified or deleted during a later master planning run. To achieve this, the planning logic copies the *Approved* planned orders from the old plan version to the new plan version during master planning.

Firming planned orders

When planned orders are firmed, real orders are created. These are also known as *released* or *open orders*. When a planned order is firmed, it's moved to the orders section of the relevant module.

You can select two firming options from the **Planned orders** page:

- **Firm** – Firms one or multiple selected planned orders.
- **Firm all** – Firms all planned orders in the filter. Using **Firm all** you don’t have to select the planned order, all planned orders within the filter ar firmed. This option can be useful if you're firming a high number of planned orders.

> **Note**
>
> You can track a planned order that was firmed from **Firming history** under **Planned orders form** > **View** > **Firming history**.

Parallelize firming

If you're planning to firm many orders at the same time, parallelizing the run can improve the run time or performance. This option is available when firming multiple planned orders with either **Firm** or **Firm all**. The following parameters are available:

- **Parallelize firming** – If *Yes*, the firming process is parallelized with the number of threads defined in **Number of threads**.
- **Number of threads** – Controls the number of threads used to parallelize the firming process. The parameter is only shown when **Parallelize firming** is set to *Yes*.

> **Note**
>
> The option for **Parallelize firming** is only shown when you have more than one planned order selected for firming.

Related information

- Master plans overview

Last updated on 08/05/2025

View, manage, and approve planned orders

This article explains how to view, manage, and approve planned orders.

View and manage planned orders

You can view and manage planned orders on any planned orders list page. Go to one of the following places, depending on the type of planned orders that you want to work with:

- **Master planning** > **Workspaces** > **Master planning**
- **Master planning** > **Master planning** > **Planned orders**
- **Production control** > **Production orders** > **Planned production orders**
- **Procurement and sourcing** > **Purchase orders** > **Planned purchase orders**
- **Inventory management** > **Inbound orders** > **Planned transfers**
- **Inventory management** > **Outbound orders** > **Planned transfers**

View and edit the status of planned orders

Use the **Status** field of each planned order to track your progress or change how a planned order is processed. The following **Status** values are available:

- *Unprocessed* – When master planning generates planned orders, it assigns this status. Planned orders with this status are deleted during the next planning run.
- *Completed* – This status indicates that the planned order is completed. If you decide not to firm a planned order, you can manually change its status to *Completed*. The system treats the *Unprocessed* and *Completed* statuses in the same way.
- *Approved* – This status indicates that the planned order is approved for firming. If you want to firm a planned order, change its status to *Approved*. If you want to keep the edits that you made to a planned order, or if you're planning to firm a planned order, change its status to *Approved*. Planned orders with a status of *Approved* are considered fixed and expected supply by master planning. Therefore, they're not modified or deleted during later master planning runs. To achieve this behavior, the planning logic copies planned orders with a status of *Approved* from the old plan version to the new plan version during master planning. Planned orders with a status of *Approved* are considered supply only within the specific master plan.

To change the status of a single planned order, open any planned orders list page, open the order, and then follow one of these steps:

- On the **General** FastTab, change the value of the **Status** field.
- On the Action Pane, on the **Planned order** tab, in the **Process** group, select **Change status**.
- On the Action Pane, select **Approve** to mark the order as approved.

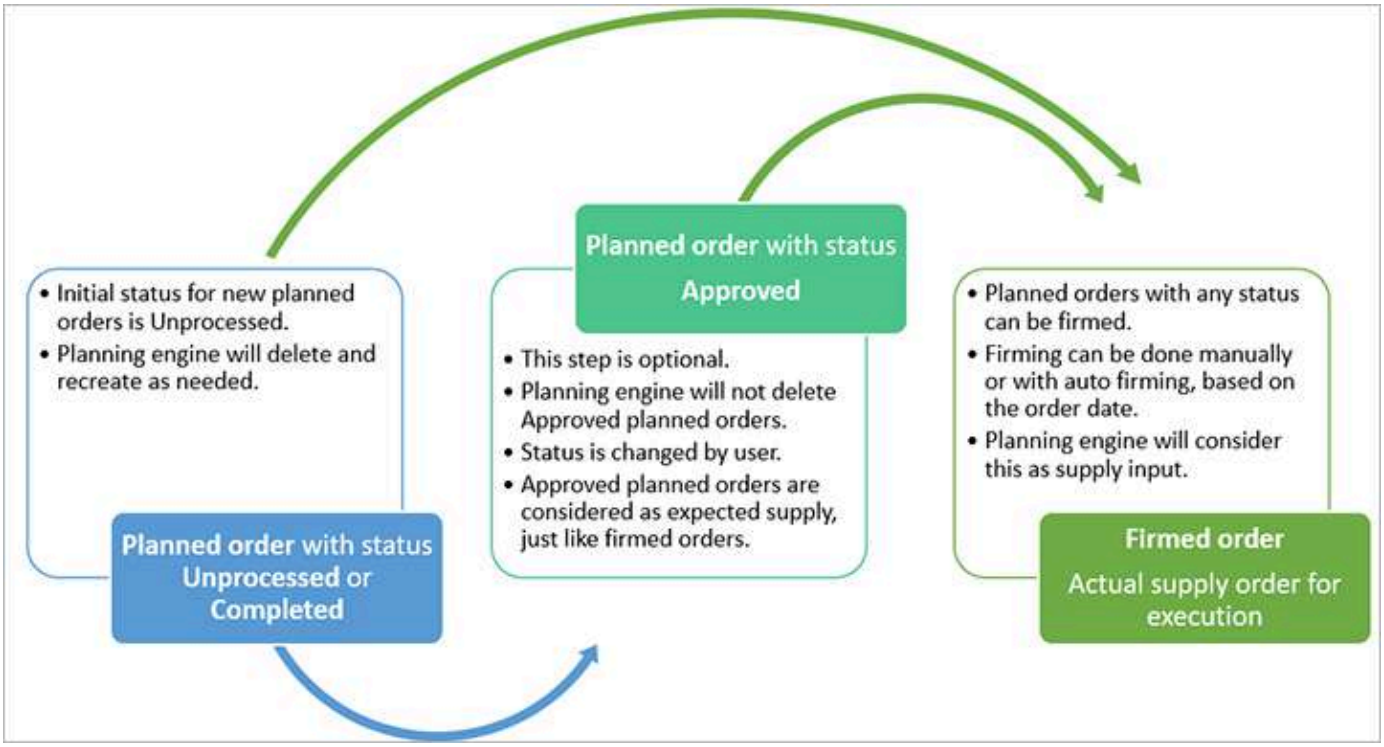
To change the status of several planned orders at the same time, open any planned orders list page, select the check box for each order that you want to change, and then follow one of these steps:

- On the Action Pane, on the **Planned order** tab, in the **Process** group, select **Change status**.
- On the Action Pane, select **Approve** to mark the orders as approved.

Approve planned orders

Approving planned orders is an optional step in the process of creating a firmed order from a planned order.

The following illustration shows how you can use the **Status** value that is assigned to each planned order to implement an approval workflow. To implement an approval process, manually adjust the **Status** value for each planned order as described in the previous section.



> **Tip**

We recommend that you approve all modified planned orders because the next planning run will ignore and overwrite all unapproved edits.

Avoid approving or editing planned orders during a planning run

The preceding text and illustration explain how approving planned orders prevents the system from ignoring and overwriting edits to those orders. However, an exception to this behavior might occur if you make changes to planned orders while the system is executing a full regenerative planning run.

This exception applies to all types of planned orders (production, purchase, and transfer) and can result in the system not preserving your changes, including split orders and field values (quantities, prices, dates, and so on). Even if you approve a planned order and therefore expect to find it after the next regenerative planning run, the order might not be available if you approved it in the middle of a planning run.

This exception occurs because, early in the planning run, the system creates a snapshot in the database during the data export stage. Then, at the end of the planning run, the system creates a new plan version, re-inserts the exported approved planned orders, and clears the previous plan version. As a result, if you approve planned orders or make changes to already approved planned orders during the exact period between the data export and re-insertion, the system overwrites those approved orders and ignores any changes to already approved planned orders. However, the demand still exists, and the planning run still considers it.

Tip

To completely avoid the risk of badly timed edits being ignored and overwritten, run master planning outside normal business hours.

Related information

- Firm planned orders

Last updated on 09/02/2026

Firm planned orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

As part of the master planning process, you must *firm* (release) planned orders. When you firm planned orders, they become actual purchase orders, transfer orders, or production orders. These orders are also known as *released orders* or *open orders*.

Use one of three methods to firm planned orders:

- **Manual firming** – Select specific planned orders in a list, and then manually start the process.
- **Auto-firming** – Define a default firming time fence for coverage groups, individual items, and combinations of items and master plans. Then, during master planning runs, the system automatically firms planned orders if the order date is within the specified time fence for firming.
- **Query-based firming** – Define a query to select planned orders based on their properties. You can set up a batch job to run the query and firm matching orders on a regular schedule.

This article describes each method in detail.

Manually firm planned orders

To manually firm planned orders, find and select the planned orders that you want to firm, and then manually start the firming process.

1. Open any planned orders list page.
2. Use the **Filter** field, **Plan** field, and column headings to filter and sort the list so that you can find the planned orders that you're looking for.
3. Select the check box for each planned order that you want to firm. If you want to firm all the planned orders that are currently listed on the page (based on the filters that you applied), skip this step.
4. On the Action Pane, select one of the following buttons:

- **Firm** – Firm only the selected planned orders.
- **Firm all** – Firm all the planned orders that are currently listed on the page (based on the filters that you applied), regardless of any check boxes that are selected. This option can be useful if you're firming many planned orders.

5. In the **Firming** dialog box, on the **Parameters** FastTab, set the following fields. (Many of these fields take their default values from the **Standard update** tab on the **Master planning parameters** page.)

   - **Update marking** – Select the inventory marking policy to use when planned orders are firmed.

   - **Stop firming if an error occurs** – Set this option to *Yes* to stop firming all selected planned orders if an error occurs in one of them. Set this option to *No* if the **Parallelize firming** option is set to *Yes*.

   - **Parallelize firming** – This option is available only if you select two or more planned orders for firming. Set it to *Yes* to run the firming processes in parallel. Parallel firming can help improve performance.

   - **Number of threads** – This option is available only if you set the **Parallelize firming** option to *Yes*. Enter the number of threads to use to parallelize the firming process. For advice about how to use this option in master planning, see Improve master planning performance.

     > ⓘ **Note**
     >
     > A value of *0* (zero) for the **Number of threads** field increases the running time of master planning. Therefore, always set this field to a value that's more than 0.

   - **Group by vendor** – Set this option to *Yes* to group planned purchase orders and create one purchase order per vendor during firming. Alternatively, you can create one purchase order that has one line for each planned order.

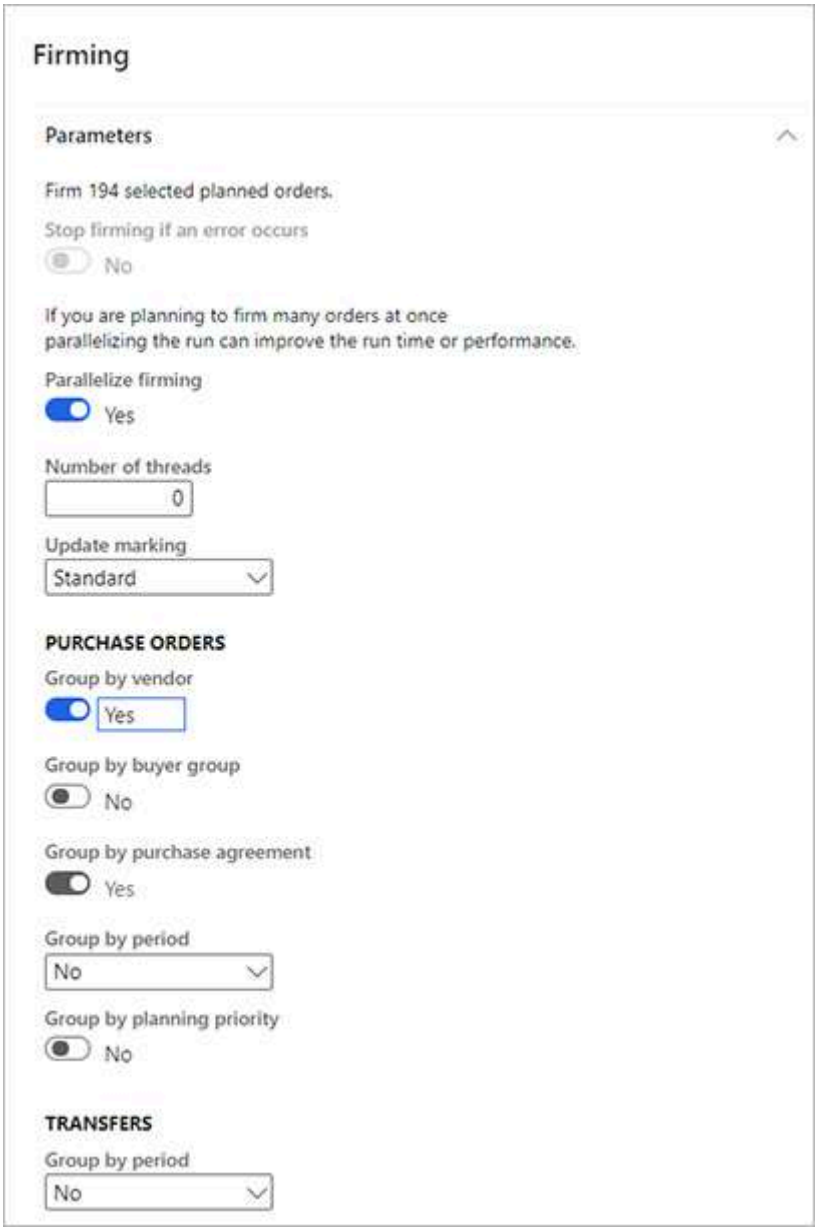
   - **Group by buyer group** – Set this option to *Yes* to group planned purchase orders and create one purchase order that combines the vendor and buyer group. To use this option, you must also set the **Group by vendor** option to *Yes*.

   - **Group by purchase agreement** – Set this option to *Yes* to group planned purchase orders that have the same vendor as existing purchase agreements and create one purchase order per purchase agreement. This option is automatically enabled when

Group by vendor is enabled. To use **Group by purchase agreement**, **Find purchase agreement** must be set to *Yes* on the **Master planning parameters** page.

- **Group by period** (in the **Purchase orders** section) – Select the period to group planned purchase orders by. To use this option, you must also select the **Group by vendor** option.
- **Group by period** (in the **Transfers** section) – Select the period to group planned transfer orders by. The orders are grouped based on **From warehouse** and **To warehouse** values.

> **Note**
>
> Each of the "Group by" options causes the system to convert each planned order to a line in the single purchase order that results from the grouping.



6. On the **Run in the background** FastTab, set up the job so that it runs in batch mode. However, it doesn't make sense to set up a recurrent schedule when you're doing manual firming. The fields work just as they work for other types of background jobs in Supply Chain Management. However, for manual firming, the batch job processes only the currently selected planned orders. It doesn't process any orders that fit the filters that are currently applied on the page.

7. Select **OK** to apply your settings and generate the firmed orders.

Auto-firm planned orders

Automatic firming lets you firm planned orders as part of the master planning process. You can define a firming time fence for coverage groups, individual items, and combinations of items and master plans. Then, during master planning runs, the system automatically firms planned orders if the order date is within the specified time fence for firming. The system handles the order date (start date) differently based on whether the planned order was generated by Planning Optimization or by the deprecated planning engine.

> **Note**
>
> You can auto-firm planned purchase orders only for items that are associated with a vendor.
>
> If you turn on change tracking, derived orders (that is, subcontract purchase orders) that the system firms have a status of *In-review*.

Auto-firming with Planning Optimization vs. the deprecated master planning engine

You can use both Planning Optimization and the deprecated master planning engine to auto-firm planned orders. However, some important differences exist. For example, Planning Optimization uses the order date (that is, the start date) to determine which planned orders to firm, whereas the deprecated master planning engine uses the requirement date (that is, the end date). The following table summarizes the differences.

Expand table

Feature	Planning Optimization	Deprecated master planning engine
Date basis	Auto-firming is based on the order date (start date).	Auto-firming is based on the requirement date (end date).

Feature	Planning Optimization	Deprecated master planning engine
Lead time	Because the order date (start date) triggers the firming, you don't have to consider the lead time as part of the firming time fence.	To help guarantee that the system firms orders quickly, the firming time fence must be longer than the lead time.
Orders for the current week	To firm all orders that must start during the current week, the firming time fence must be one week.	To firm all orders that must start during the current week, the firming time fence must be the lead time plus one week.

Set up auto-firming and the firming time fence

Turn on auto-firming by assigning an automated firming time fence to the relevant coverage setup, as described later in this section. If you don't turn on auto-firming for any coverage setup, or if you start planning from a specific page, such as the **Net requirements** page for a released product, the auto-firming process is skipped.

Grouping and marking options for auto-firming take their values from the **Standard update** tab on the **Master planning parameters** page.

You define the auto-firming time fence by the number of days that you enter for the relevant coverage setup. You can turn on auto-firming and control the firming time fence in the following ways:

- To define the default firming time fence for a coverage group, go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**, and select a coverage group. Then, on the **Other** FastTab, in the **Automatic firming time fence (days)** field, enter the number of days.
- To overwrite the firming time fence that you defined for the coverage group for a specific item, go to **Product information management** > **Released products**. On the Action Pane, select **Plan**, and then select **Item coverage**. On the **General** tab, select **Override time fence**, and then, in the **Automatic firming time fence (days)** field, enter the number of days.
- To overwrite the firming time fence that you defined for the coverage group and item coverage for a specific master plan, go to **Master planning** > **Setup** > **Master plans**, and select a master plan. Then, on the **Time fence in days** FastTab, set the **Firming** option to *Yes*, and enter the number of days.

If you set all the previously mentioned time fences to *0* (zero), you effectively disable auto-firming for the relevant covered items.

Firm planned orders by using a query

Query-based firming lets you plan to firm based on criteria that you define in advance. Unlike auto-firming, query-based firming lets you automate firming for different subsets of orders at different points in time. Additionally, you can use either manual or automated operations to firm different types of planned orders. You can also preview which firmed orders are selected based on your settings. Therefore, you can confirm that the selection fits your expectations.

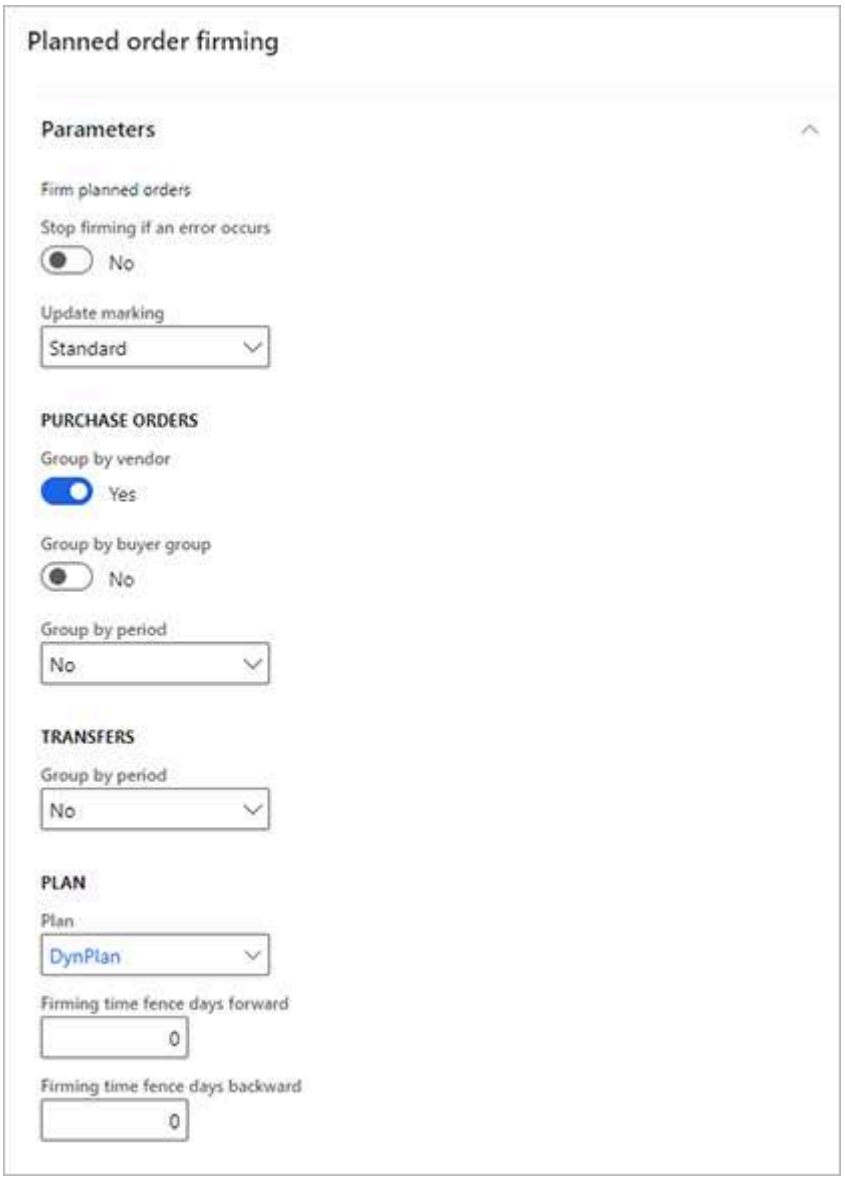
You can combine auto-firming with query-based firming. For example, a query-based firming job has a forward time fence that is longer than the time fence for a matching auto-firming coverage configuration. Therefore, the query-based firming job processes its planned orders before the auto-firming is triggered. You can take advantage of this behavior to schedule orders for specific vendors differently than orders for similar products from other vendors.

To firm a planned order by using the query-based firming process, follow these steps:

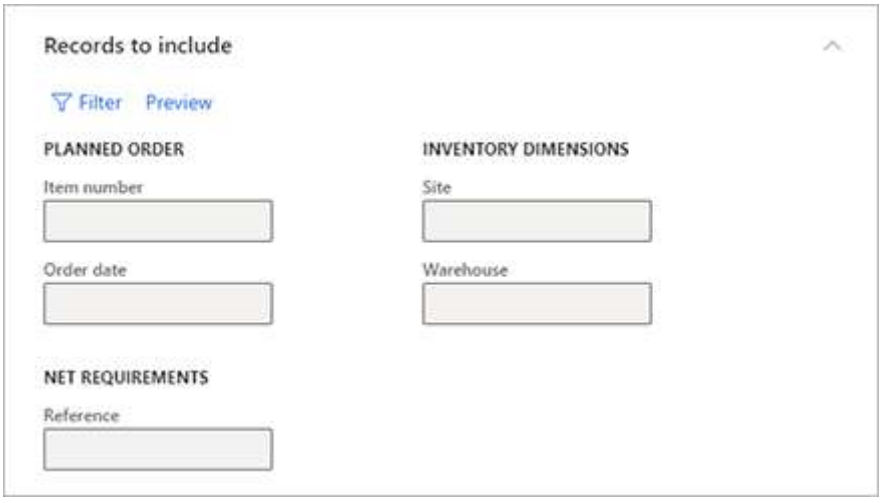
1. Go to **Master-planning** > **Master planning** > **Run** > **Planned order firming**.

2. In the **Planned order firming** dialog box, on the **Parameters** FastTab, set the basic processing, marking, and grouping options. These options work just as they do in the **Firming** dialog box. (See the previous section for descriptions.) Then, in the **Plan** section, set the following fields that are unique to the **Planned order firming** dialog box:

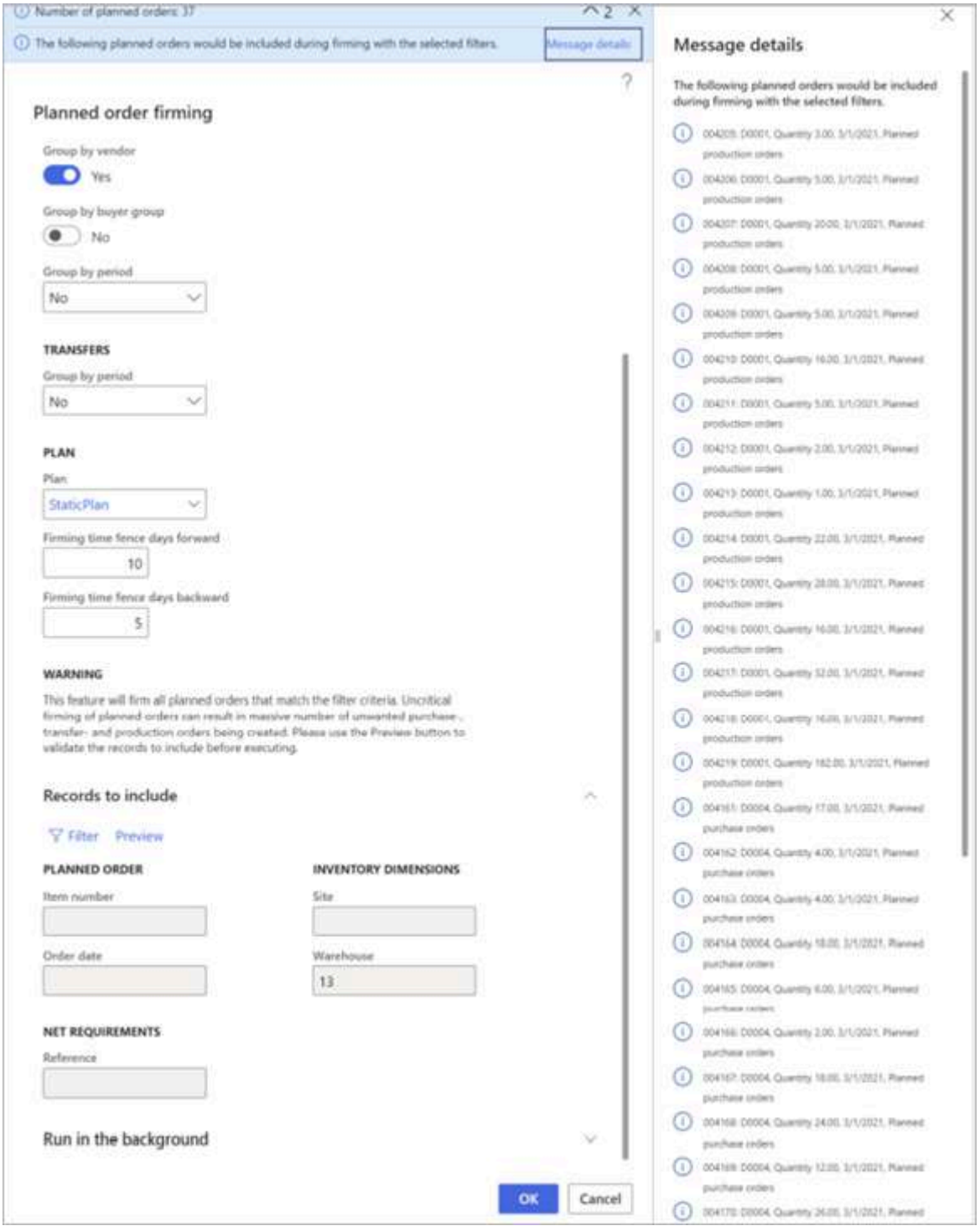
    - **Plan** – Select the master plan that should be applied during firming of the planned orders that this query finds.
    - **Firming time fence days forward** – Select how far in the future the various requirements and other considerations must be calculated by master planning.
    - **Firming time fence days backward** – Select how far in the past the various requirements and other considerations must be calculated by master planning.



3. To specify which records to include in the order, select the **Filter** button on the **Records to include** FastTab. A standard query dialog appears, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Supply Chain Management. The fields are read-only and show values that are related to your query.



4. Select **Preview** to preview the content of your firmed order, based on your settings so far. The list of planned orders that the process firms is shown as a message. You can then adjust your settings as required until the preview shows the firmed order as you intend it.



> **⚠ Warning**
>
> This feature firms all planned orders that match the filter criteria. Uncritical firming of planned orders can cause massive numbers of unwanted purchase, transfer, and production orders to be created. Before you continue, always use the **Preview** button to validate the records that the process includes.

5. On the **Run in the background** FastTab, set up the job so that it runs in batch mode, and/or set up a recurrent schedule. The fields work just as they do for other types of background jobs in Supply Chain Management.

6. Select **OK** to apply your settings and generate the firmed orders.

Track firmed orders

To track planned orders that you firmed, go to **Master-planning** > **Inquiries and reports** > **Master planning** > **Firming history**.

> **Note**
>
> You can also access firming history from any of the planned order pages such as **Master-planning** > **Master planning** > **Planned orders**. The system creates firming history data for planned orders that you firm. Therefore, the firming history you see is unrelated to any planned order you select. The firming history you see is similar to the firming history you see when you access **Master-planning** > **Inquiries and reports** > **Master planning** > **Firming history**.

Last updated on 03/25/2026

Planned orders simplified

10/03/2025

This article describes how you can view, manage, and firm planned orders by using the **Planned orders simplified** page.

The **Planned orders simplified** page is a simplified version of the **Planned orders** page. It provides fewer options but is faster to work with when all you have to do is review, approve, and firm your planned orders.

View, manage, and firm planned orders

The **Planned orders simplified** page is designed primarily for firming functionality. It shows all planned orders that have been generated for the selected plan, just as the standard **Planned orders** page does. Therefore, it can be used as an optimized alternative for viewing the results of a master planning run.

To open the **Planned orders simplified** page, go to **Master planning** > **Master planning** > **Planned orders simplified**. You can use this page to complete the following tasks:

- View all planned orders for a specific plan.
- Create new planned orders.
- Delete single or multiple planned orders.
- Firm single or multiple planned orders.
- Change the planned order status for single or multiple planned orders.
- Seamlessly open planned order details for an individual planned order to view full details of the order.
- Create custom views by using saved views.

Additional scenarios, such as split, are supported by the standard **Planned orders** page.

Action messages

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

An action message is a system-generated suggestion to change an existing planned, approved, or firmed order.

The master planning calculation generates action messages in response to changed requirements. For example, when you change the ship date or quantity on a sales order after you already create a purchase order to fulfill the demand for that sales order. In this case, the master planning calculation generates one or more action messages that suggest that you update the purchase order. You decide whether to make the suggested changes.

You can set up how action messages are calculated for a coverage group that you attach to an item.

> **Note**
>
> The system only creates action messages for unapproved planned orders while they're within the freeze time fence. Outside of the freeze time fence, the system only creates action messages for approved planned orders and firmed orders.

Select action messages

On the **Coverage groups** page, select the action messages that you want the system to generate, and the coverage groups or items that the messages apply to. The following table lists the action messages that you can select.

Expand table

Message	Description
Advance	The system generates action messages, as needed, to move orders to an earlier date. In the **Advance margin** field, specify the maximum number of days that can pass between a receipt and an issue without an advance action.

Message	Description
Postpone	The system generates action messages, as needed, to move orders to a later date. In the **Postpone margin** field, specify the maximum number of days that can pass between a receipt and an issue without a postpone action.
Decrease	The system generates action messages when production orders, purchase orders, and other receipt transactions should be decreased to prevent excess inventory levels.
Increase	The system generates action messages when production orders, purchase orders, and other receipt transactions should be increased to prevent shortages in inventory.
Derived actions	The system propagates action messages that it creates for receipt transactions to any derived requirements, and it generates the necessary action messages for the receipt transactions that satisfy those requirements.

The following sections provide a few detailed scenarios.

Increase and decrease actions with product default order quantities

On the **Default order settings** page for an item, you can set up a minimum order quantity, maximum order quantity, and multiple values. The system takes these settings into account when it suggests actions, to ensure that the suggestions never cause undersupply.

For example, you have an item that has the following settings on its **Default order settings** page:

- **Minimum order quantity:** *0*
- **Maximum order quantity:** *90*
- **Multiple:** *20*

If there's demand for a quantity of 60 of this item, master planning creates a planned purchase order for a quantity of 60. If you increase the demand by 30, master planning suggests an increase of 40, because it respects the multiple of 20 and never causes undersupply.

Action messages for orders related to safety stock

Action messages for orders that supply safety stock never suggest decreasing the quantity below the quantity that's needed for the safety stock. In other words, if there's an order that supplies safety stock and customer demand, and you decrease or cancel the demand, master planning suggests decreasing the planned order by the decreased demand. However, it never suggests a value that is lower than the safety stock that is needed.

The system doesn't suggest postponing actions for supplying safety stock, because it's assumed that the safety stock must be supplied at the required time and on the required date.

Advance and postpone actions related to BOMs

You must apply actions related to components of bills of material (BOMs) before the actions of their parent items. If you don't, further orders that are related to higher-level BOMs might be affected. Then, run master planning again to recalculate and suggest appropriate actions.

For example, consider the following situation:

- Final good *FG* of type *production* has a BOM that includes raw material *RM*.
- Today's date is January 21.
- An existing, released production order for *FG* is scheduled for January 25.
- To support the existing production order, master planning creates a planned purchase order for the required raw material *RM*. This order has a requirement date of January 25.
- A new sales order for *FG* is created today. It has a requirement date of today (January 21).
- January 21 is closed for delivery on the *RM* calendar, but January 22 is open.

When you run master planning, it generates advance action messages that suggest moving up the previously scheduled production so that you can fulfill today's order.

- To meet the new demand, it suggests moving the production order for *FG* up to January 21. (It makes this suggestion without considering the closed date for *RM*.)
- Because *RM* is still required for the production order, it suggests moving up the planned purchase order too. However, this time, it checks the *RM* calendar. Therefore, it suggests moving the planned purchase order for *RM* to January 22 (because January 21 is closed).

As you can see, the required raw material *RM* arrives too late for the scheduled production of *FG*. Therefore, you must first apply the advance action to the planned purchase order for *RM* and then run master planning again. At that point, master planning generates a new action message that suggests moving the production order for *FG* to January 22.

Last updated on 03/26/2026

Inventory forecasts

08/22/2025

> **Tip**
>
> This article describes demand forecasting functionality that's built into Microsoft Dynamics 365 Supply Chain Management. For an even better planning and forecasting experience, we recommend that you try *Demand planning in Microsoft Dynamics 365 Supply Chain Management*, which is Microsoft's next-generation collaborative demand planning solution. For details, see the **Demand planning home page**.

This article describes how to view and create inventory forecasts. You can create and view supply and demand forecast lines for items, item groups, item allocation keys, customer accounts, customer groups, vendor accounts, and vendor groups.

For each forecast line, you can select the forecast model that is used. You can then specify the item or item group, plus the quantity or the transaction amount. You can also set up a time schedule for allocating the forecast quantity.

The supply and demand forecast lines produce an inventory forecast for the same combination of an item and a model. This inventory forecast represents a balance between the supply and demand that were entered for the same item. It can't be edited. The inventory forecast helps the inventory management team review expected changes to the on-hand inventory of an item during the upcoming period that has been forecasted.

After you enter your demand and/or supply, you can run forecast planning to calculate gross requirements for materials and capacity, and to generate planned orders.

View and manually enter forecast lines

Use this procedure to manually create forecast lines for products.

There are also other ways to create forecast lines:

- Generate a statistical baseline forecast.
- Import historical data for demand forecasts.
- Generate the forecast by using a Microsoft Azure Machine Learning web service.
- Import demand or supply forecast lines by using the data management framework (ForecastDemandForecastEntryStaging and ForecastSupplyForecastEntryStaging data entities).

As the table in step 1 shows, there are different ways to access the pages that are used.

1. Depending on the type of entity that you want to create a forecast for, and the type of forecast that you want to create, open a supply, demand, or inventory forecast page as described in the following table.

Expand table

Entity	Instructions
Released products	a. Go to **Product information management** > **Products** > **Released products**. b. Select the product to create a forecast for. c. On the Action Pane, on the **Plan** tab, in the **Forecast** group, select **Demand forecast**, **Supply forecast**, or **Inventory forecast**, depending on the type of forecast that you want to work with.
Released product variants	a. Go to **Product information management** > **Products** > **Released product variants**. b. Select the variant to create a forecast for. c. On the Action Pane, on the **Plan** tab, in the **Forecast** group, select **Demand forecast**, **Supply forecast**, or **Inventory forecast**, depending on the type of forecast that you want to work with.
Item groups	a. Go to **Inventory management** > **Setup** > **Inventory** > **Item groups**. b. Select the item group to create a forecast for. c. On the Action Pane, select **Forecasting** > **Demand**, **Forecasting** > **Supply**, or **Forecasting** > **Inventory forecast**, depending on the type of forecast that you want to work with.
Item allocation keys	a. Go to **Inventory management** > **Setup** > **Forecast**. b. Select the item allocation key to create a forecast for. c. On the Action Pane, select **Demand forecast**.
Customers	a. Go to **Master planning** > **Forecasting** > **Manual forecast entry** > **Customers**. b. Select the customer to create a forecast for. c. On the Action Pane, select **Define demand forecast**.
Customer groups	a. Go to **Master planning** > **Forecasting** > **Manual forecast entry** > **Customer groups**. b. Select the customer group to create a forecast for. c. On the Action Pane, select **Define demand forecast**.
Vendors	a. Go to **Master planning** > **Forecasting** > **Manual forecast entry** > **Vendors**. b. Select the vendor to create a forecast for. c. On the Action Pane, select **Entry** to open the **Supply forecast** page.
Vendor groups	a. Go to **Master planning** > **Forecasting** > **Manual forecast entry** > **Vendor groups**. b. Select the vendor group to create a forecast for. c. On the Action Pane, select **Entry** to open the **Supply forecast** page.

Entity	Instructions
All lines	• Go to **Master planning** > **Forecasting** > **Demand forecast lines** or **Master planning** > **Forecasting** > **Supply forecast lines**, depending on the type of forecast that you want to work with.

Depending on your selection, the **Supply forecast** or **Demand forecast** page appears. It shows any existing forecast lines for the record that you selected before you opened the page.

2. On the Action Pane, select **New** to add a forecast line to the grid in the upper part of the page.

3. On the new line, in the **Model** field, select the forecast model to use. Then enter other details as required, such as the item, item group, customer, or vendor account or group, item quantity, or total transaction amount. For complete details about the fields that are available on the **Supply forecast** and **Demand forecast** pages, see the later sections in this article.

4. To distribute the forecast over the period, on the **Overview** tab, select **Allocate forecast** on the toolbar.

5. In the **Allocation** grid, review the time horizon and the time intervals that are used to distribute the forecast quantities.

Supply forecast lines

The supply forecast lets you create a plan for items that must be purchased. It tells procurement and sourcing clerks what they're expected to order.

You can enter a supply forecast by item, item group, item allocation key, vendor, and vendor group. For information about all the ways to open the **Supply forecast** page for various entities and records, see the View and manually enter forecast lines section earlier in this article.

The upper part of the **Supply forecast** page provides a grid of supply forecast lines and a set of tabs that you can use to view and set more information about a selected forecast line. The lower part of the page provides an **Allocation** grid.

The following subsections describe all the fields that are available on each tab, and all the commands that are available on the page's Action Pane and on the toolbar on the **Overview** tab.

Action Pane commands on the Supply forecast page

The following table describes the commands that are available on the Action Pane of the **Supply forecast** page.

Expand table

Command	Description
Save	Save your current settings.
Edit	Enable settings on the page to be edited.
New	Add a forecast line to the upper grid.
Delete	Remove the selected forecast line from the upper grid.
Forecast balances	View forecast balances that have been calculated for the selected line's model ID for the current fiscal year. The balances are split by period (month).
Cash flow forecasts	View forecast transactions that have been allocated to the general ledger. Learn more in Cash flow forecasting.
Inventory > Display dimensions	Select the inventory dimensions that should be shown in the grid on the **Overview** tab.

Toolbar commands on the Overview tab of the Supply forecast page

The following table describes the commands that are available on the toolbar on the **Overview** tab of the **Supply forecast** page.

Expand table

Command	Description
Allocate forecast	If you're using an allocation method, generate the individual schedule lines for the forecast transaction. The line's quantity is then distributed by date (according to the selected time intervals), quantity, and amount for the whole time horizon. (See the Allocate forecast section later in this article.)
Bulk update	Open the **Edit forecast transactions** page. (See the Bulk update forecast transactions section later in this article.)
Inventory forecast	Open a view of the **Inventory forecast** page that is filtered for the selected item/model combination. (See the Inventory forecast section later in this article.)
Create item requirement	Open a dialog box where you can create item requirements, and sales order or item journal lines for project-related forecast transactions. Although this command is

Command	Description
	available for both supply forecast lines and demand forecast lines, it can't be used on the **Supply forecast** page.

The Overview tab on the Supply forecast page

The following table describes the fields on the **Overview** tab of the **Supply forecast** page.

Expand table

Field	Description
Model	The forecast model that the transaction is associated with.
Date	The start date of the transaction.
Vendor account	The vendor account that the transaction is associated with.
Vendor group	The vendor group that the transaction is associated with.
Item number	The identifier of the item.
Item group	The item group that the transaction is associated with.
Item allocation key	The item allocation key that is associated with the forecast.
CW quantity	The forecast quantity in the specified catch-weight unit.
CW unit	The catch-weight unit that the item is purchased in. The value is automatically retrieved from the item setup.
Quantity	The forecast quantity in the specified purchase unit.
Unit	The unit that the item is purchased in. The value is automatically retrieved from the item setup. However, you can modify it if unit conversions are set up.
Amount	The gross amount, minus discounts, that the forecast transaction contributes to the forecast.
Currency	The currency that is used for the forecast transaction. The default currency is automatically entered, but you can select a different currency.
(Other dimensions)	Additional dimensions can be shown as columns in the grid. To select the additional dimensions that are shown, select **Display dimensions** on the Action Pane.

The General tab on the Supply forecast page

The **General** tab shows more information about the line that is currently selected on the **Overview** tab. Some of this information is repeated from the **Overview** tab. The following table describes the fields that are unique to the **General** tab.

Expand table

Field	Description
Report	Set this option to *Yes* to include the transaction in reporting.
Comments	Enter any comments that you have about the forecast transaction.
Active	Select this checkbox to include the transaction in budget reporting.
Include in cash flow forecasts	Select this checkbox to allocate the forecast transaction to the general ledger. The setting of this checkbox can't be modified for reporting transactions.
Sales tax group	The tax group that is used to specify tax for the forecast transaction.
Item sales tax group	The item tax group that is used to specify tax for the forecast transaction.
Method	Select the method that is used to allocate the forecast transaction: • **None** – No allocation occurs. • **Period** – Forecast the same quantity for each period. If you select this value, specify a quantity in the **Per** field and a unit of time in the **Unit** field. • **Key** – Allocate the forecast according to the period allocation key that you specify in the **Period key** field. You can use this method when you want seasonal variation to be considered.
Per	Enter the number of time intervals into the future that the forecast extends. This field is available only if you select *Period* in the **Method** field. For example, you select *Period* in the **Method** field, enter *1* in the **Per** field, and select *Months* in the **Unit** field. In the **End** field, you specify an end date that extends one year into the future. In this case, one forecast line will be created for each month of the upcoming year, based on the item and quantity that are specified on the header line.
Unit	Select the unit of the time interval: *Days*, *Months*, or *Years*. Allocation then corresponds to the number of days, months, or years that you specify in the **Per** field.
Period key	Specify the period allocation key.
End	Specify the end date when you use the **Per** and **Unit** fields.

The Item tab on the Supply forecast page

The **Item** tab shows more item information about the line that is currently selected on the **Overview** tab.

The grid at the top of the **Item** tab repeats item information that is also shown on the **Overview** tab. In addition, it includes the **Unit price** field, which shows the purchase price for the number of units that is specified in the **Unit** field. The unit price is automatically retrieved from the item setup, but you can modify it.

The fields below the grid provide more item details. Some of this information is repeated from the **Overview** tab. The following table describes the fields that are unique to the **Item** tab.

Expand table

Field	Description
Price unit	The number of units that the purchase price applies to. The value is automatically retrieved from the Items table, but you can modify it.
Charges on purchases	Fixed charges on the purchase price. The charges are independent of the quantity.
Discount percentage	The discount as a percentage.
Discount amount	The discount as an amount per sales unit.
Gross amount	The amount when no discounts are applied.
Quantity	The transaction quantity in the item's inventory unit.
Sub-BOM	The bill of materials (BOM) number of a specific sub-BOM.
Subroute	The route number of a specific sub-route.

The Financial dimensions tab on the Supply forecast page

The **Financial dimensions** tab shows all the financial dimension values for the line that is currently selected on the **Overview** tab.

The Inventory dimensions tab on the Supply forecast page

The **Inventory dimensions** tab shows all the inventory dimension values for the line that is currently selected on the **Overview** tab.

The Allocation grid on the Supply forecast page

If you're using an item allocation key, or if you've entered an item forecast for one or more future periods, you can allocate the forecast by selecting **Allocate forecast** on the toolbar on the **Overview** tab. The quantity is then distributed in the manner that is indicated by the lines in the **Allocation** grid.

Demand forecast lines

The demand forecast lets you enter or generate demand for a customer. It helps sales and marketing clerks inform master planning clerks about the expected demand during the upcoming forecast period.

You can enter a demand forecast by item, item group, item allocation key, customer, and customer group. For information about all the ways to open the **Demand forecast** page for various entities and records, see the View and manually enter forecast lines section earlier in this article.

The upper part of the **Demand forecast** page provides a grid of demand forecast lines and a set of tabs that you can use to view and set more information about a selected forecast line. The lower part of the page provides an **Allocation** grid.

The following subsections describe all the fields that are available on each tab, and all the commands that are available on the page's Action Pane and on the toolbar on the **Overview** tab.

Action Pane commands on the Demand forecast page

The following table describes the commands that are available on the Action Pane of the **Demand forecast** page.

Expand table

Command	Description
Save	Save your current settings.
Edit	Enable settings on the page to be edited.
New	Add a forecast line to the upper grid.
Delete	Remove the selected forecast line from the upper grid.
Forecast balances	View forecast balances that have been calculated for the selected line's model ID for the current fiscal year. The balances are split by period (month).

Command	Description
Cash flow forecast	View forecast transactions that must be allocated to the general ledger. Learn more in Cash flow forecasting.
Display dimensions	Select the product, storage, and tracking dimensions that should be shown in the grid on the **Overview** tab.
General ledger preview	View the general ledger entries for the selected transaction.
Transfer quotation lines	Transfer quotation lines to the selected project.

Toolbar commands on the Overview tab of the Demand forecast page

The following table describes the commands that are available on the toolbar on the **Overview** tab of the **Demand forecast** page.

Expand table

Command	Description
Allocate forecast	If you're using an allocation method, generate the individual schedule lines for the forecast transaction. The line's quantity is then distributed by date (according to the selected time intervals), quantity, and amount for the whole time horizon. (See the Allocate forecast section later in this article.)
Bulk update	Open the **Edit forecast transactions** page. (See the Bulk update forecast transactions section later in this article.)
Inventory forecast	Open a view of the **Inventory forecast** page that is filtered for the selected item/model combination. (See the Inventory forecast section later in this article.)
Create item requirement	Open a dialog box where you can create item requirements, and sales order or item journal lines for project-related forecast transactions.

The Overview tab on the Demand forecast page

The following table describes the fields on the **Overview** tab of the **Demand forecast** page.

Expand table

Field	Description
Task name	The name of the batch task that was used to create the forecast line.
Model	The forecast model that the transaction is associated with.
Date	The start date of the transaction.
Customer account	The customer account that the transaction is associated with.
Customer group	The customer group that the transaction is associated with.
Item number	The identifier of the item.
Product name	The description of the item.
Item allocation key	The item allocation key that is used during inventory forecast allocation.
Sales quantity	The transaction quantity in the specified sales unit.
Unit	The unit that the item is sold in.
Amount	The gross amount, excluding discounts, that the forecast transaction contributes to the forecast.
Sales price	The sales price for the number of units that is specified in the **Price unit** field. The value is retrieved from the item setup, but you can modify it.
Sales currency	The currency that is used for the forecast transaction. The default currency is automatically entered, but you can select a different currency.
CW quantity	The forecast quantity in the specified catch-weight unit.
CW unit	The catch-weight unit that the item is sold in. The value is automatically retrieved from the item setup.
(Other dimensions)	Additional product, storage, and tracking dimensions can be shown as columns in the grid. To select the additional dimensions that are shown, select **Display dimensions** on the Action Pane.

The General tab on the Demand forecast page

The **General** tab shows more information about the line that is currently selected on the **Overview** tab. Some of this information is repeated from the **Overview** tab. The following table describes the fields that are unique to the **General** tab.

Expand table

Field	Description
Demand forecast sequence number	The unique number of the demand forecast line. This number is generated by using the number sequence that is selected for demand forecasts on the **Master planning parameters** page.
Item group	The item group that the transaction is associated with.
Report	Set this option to *Yes* to include the transaction in reporting.
Comments	Enter any comments that you have about the forecast transaction.
Active	Select this checkbox to include the transaction in budget reporting. The setting of this checkbox can't be modified for reporting transactions.
Include in cash flow forecasts	Select this checkbox to allocate the forecast transaction to the general ledger. The setting of this checkbox can't be modified for reporting transactions. Learn more in Cash flow forecasting.
Sales tax group	The tax group that is used to specify tax for the forecast transaction.
Item sales tax group	The item tax group that is used to specify tax for the forecast transaction.
Method	Select the method that is used to allocate the forecast transaction: • **None** – No allocation occurs. • **Period** – Forecast the same quantity for each period. If you select this value, specify a quantity in the **Per** field and a unit of time in the **Unit** field. • **Key** – Allocate the forecast according to the period allocation key that you specify in the **Period key** field. You can use this method when you want seasonal variation to be considered.
Per	Enter the number of time intervals into the future that the forecast extends. This field is available only if you select *Period* in the **Method** field. For example, you select *Period* in the **Method** field, enter *1* in the **Per** field, and select *Months* in the **Unit** field. In the **End** field, you specify an end date that extends one year into the future. In this case, one forecast line will be created for each month of the upcoming year, based on the item and quantity that are specified on the header line.
Unit	Select the unit of the time interval: *Days*, *Months*, or *Years*. Allocation then corresponds to the number of days, months, or years that you specify in the **Per** field.
Period key	Specify the period allocation key that is used to allocate the forecast. Learn more in Budget planning data allocation.
End	Specify the end date when you use the **Per** and **Unit** fields.

The Item tab on the Demand forecast page

The **Item** tab shows more item information about the line that is currently selected on the **Overview** tab. Some of this information is repeated from the **Overview** tab. The following table describes the fields that are unique to the **Item** tab.

Expand table

Field	Description
Price unit	The number of sales units that the sales price applies to. The value is automatically retrieved from the Items table, but you can modify it.
Sales charges	Fixed charges on the sales price. The charges are independent of the quantity.
Cost price	The cost for the current forecast transaction per inventory unit. The value is retrieved from the Items table, but you can modify it.
Discount percentage	The discount as a percentage.
Discount amount	The discount as an amount per sales unit.
Gross amount	The amount when no discounts are applied.
Inventory quantity	The transaction quantity in the item's inventory unit.
Use specific BOM	The BOM number of a specific BOM.
Use specific route	The route number of a specific sub-route.

The Project tab on the Demand forecast page

The **Project** tab shows more project information about the line that is currently selected on the **Overview** tab. Some of this information is repeated from the **Overview**, **General**, or **Item** tab. The following table describes the fields that are unique to the **Project** tab.

Expand table

Field	Description
Project date	The date of the demand forecast transaction.

Field	Description
Project ID	The identifier of the project that is referenced by the current transaction.
Funding source	The funding source that the demand forecast is associated with.
Activity number	The activity that is associated with the demand forecast transaction.
Category	The cost category of the current transaction.
Line property	The status that the transaction is attached to.
Transaction ID	The system identifier for the demand forecast transaction.
Cost amount	The amount of the demand forecast transaction.
Unit price	The price per unit.
Modified By	The identifier of the worker who last modified the selected transaction.
Invoice date	Enter the expected invoice date.
Elimination date	View or modify the elimination date. When the forecast is created, the elimination date is set to the end date of the project. If the project has no end date, the project date is applied. This field applies to fixed-price projects and investment projects.
Cost payment date	Enter the expected date of the purchase payment.
Sales payment date	Enter the expected date of the sales payment.

The Financial dimensions tab on the Demand forecast page

The **Financial dimensions** tab shows all the financial dimension values for the line that is currently selected on the **Overview** tab.

The Inventory dimensions tab on the Demand forecast page

The **Inventory dimensions** tab shows all the inventory dimension values for the line that is currently selected on the **Overview** tab.

The Allocation grid on the Demand forecast page

If you're using an item allocation key, or if you've entered an item forecast for one or more future periods, you can allocate the forecast by selecting **Allocate forecast** on the toolbar on the **Overview** tab. The quantity is then distributed in the manner that is indicated by the lines in the **Allocation** grid. (See the Allocate forecast section later in this article.)

Inventory forecast

The supply and demand forecast lines pages have an **Inventory forecast** button that opens a view of the **Inventory forecast** page that is filtered for the same item/model combination. The inventory forecast represents a balance between the supply and the demand that were entered for the same item. It can't be edited. The inventory forecast helps the inventory management team review expected changes to the on-hand inventory of an item during the upcoming period that has been forecasted.

The Action Pane on the Inventory forecast page

The following table describes the commands that are available on the Action Pane of the **Inventory forecast** page.

Expand table

Action	Description
Supply forecast	Open a view of the **Supply forecast** page that is filtered for the same item/model/date combination.
Demand forecast	Open a view of the **Demand forecast** page that is filtered for the same item/model/date combination.
Inventory > Display dimensions	Select the product, storage, and tracking dimensions that should be shown in the **Overview** grid.

The grid on the Inventory forecast page

The following table describes the fields in the grid on the **Inventory forecast** page.

Expand table

Field	Description
Model	The forecast model that the transaction is associated with.
Item number	The identifier of the item.

Field	Description
Budget date	The date of the forecast transaction.
Forecast type	The source of the transaction (*Demand forecast* or *Supply forecast*).
CW quantity	The forecast quantity in the catch-weight unit.
Quantity	The forecast quantity in the inventory unit.
CW Accumulated	The accumulated forecast quantity in the catch-weight unit when multiple forecast lines have been accumulated for the same item.
Sub-BOM	The BOM number of a specific sub-BOM.
Subroute	The route number of a specific sub-route.
(Other dimensions)	Additional dimensions can be shown as columns in the grid. To select the additional dimensions that are shown, select **Inventory** > **Display dimensions** on the Action Pane.

Allocate forecast

Use the following procedure to process selected forecast transaction lines. When you allocate a forecast, the quantity is then distributed as indicated by the lines in the **Allocation** grid.

1. Depending on the type of entity that you're creating a forecast for, and the type of forecast that you want to create, open a supply or demand forecast page as described in View and manually enter forecast lines.

2. On the supply or demand forecast lines page, select a forecast line, and then, on the **Overview** tab, select **Allocate forecast** on the toolbar.

3. In the **Allocate forecast** dialog box, set the fields that are described in the following table. (The value that you select in the **Method** field determines that other fields that are available.)

Expand table

Field	Description
Method	Select the method that is used to allocate the forecast transaction: • **None** – No allocation occurs. • **Period** – Forecast the same quantity for each period. If you select this value, specify a quantity in the **Per** field and a unit of time in the **Unit** field. • **Key** – Allocate the forecast according to the period allocation key that you specify in the **Period key** field. You can use this method when you want seasonal variations to be considered.
Per	Enter the number of time intervals into the future that the forecast extends. This field is available only if you select *Period* in the **Method** field. For example, you select *Period* in the **Method** field, enter *1* in the **Per** field, and select *Months* in the **Unit** field. Then, in the **End** field, you specify an end date one year into the future. In this case, one forecast line will be created for each month of the upcoming year, based on the item and quantity that are specified on the header line.
Unit	Select the unit of the time interval: *Days*, *Months*, or *Years*. Allocation then corresponds to the number of days, months, or years that you specify in the **Per** field.
Period key	Specify the period allocation key that is used to allocate the forecast. Learn more in Budget planning data allocation.
End	Specify the end date that applies to your settings in the **Per** and **Unit** fields.

4. Select **OK** to confirm your settings.
5. You can review the results on the **Allocation** tab for the same line.

Bulk update forecast transactions

Use this procedure to process existing forecast transaction lines. You can copy, edit, and delete forecast transaction lines.

1. On the supply or demand forecast lines page, select **Bulk update**.
2. In the dialog box, select **Filter**.
3. On the **Range** tab, select the criteria that identify the source forecast data that you want to copy, edit, or delete. This data might include the forecast model, item number, and customer account.
4. Select **OK** to confirm your selection.

After the data is selected, you can use the **Edit forecast transactions** dialog box to define how you want to copy, edit, or delete it.

> ⓘ **Note**
>
> The filtering step is a prerequisite for using the **Bulk edit** functionality. If you skip it, the program selects and updates **all** forecast lines. Therefore, you might unintentionally cause duplication or removal of transactions.

5. In the **Management** section, select whether you want to copy, update, or delete the selected forecast transactions.

6. Use the **Modifications in field** section to change the parameters of the forecast. Select the checkbox for a parameter, and then select among the available options. For example, select the **Model** checkbox, and then select a forecast model number. Existing forecast lines are copied to the selected forecast model.

7. Use the **Change of period** section to move the start and end dates of the forecast forward or backward. Select the checkbox, and then use the quantity and period fields to define how the forecast dates should be moved. You can enter a negative quantity to move the start date forward (that is, make it earlier).

   For example, to delay the forecast transaction start date by six months, select the checkbox, enter *6* as the quantity, and select *Months* as the period.

8. Use the **Correct field** section to update the actual forecast data. In the **Field** field, select the criterion that you want to change. In the **Factor** field, enter a multiplication factor to apply to the selected criterion, or enter a constant to add or subtract. For example, select *Quantity* as the criterion, and enter a factor of *1.5* to multiply the item quantity by 1.5. Alternatively, enter a constant of *-25* to decrease the item quantity by 25 units.

9. Use the **Financial dimensions** section to update the financial dimensions of forecast lines. Select the financial dimensions that you want to change, and then enter a value to apply to the selected dimensions.

10. Select **OK** to apply your changes.

Use forecasts with master planning

After you enter your demand forecast and/or supply forecast, you can include the forecasts during master planning to account for expected demand and/or supply in your master

planning run. When forecasts are included in master planning, gross requirements for materials and capacity are calculated, and planned orders are generated.

Set up a master plan to include an inventory forecast

To set up a master plan so that it includes an inventory forecast, follow these steps.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**.
2. Select an existing plan, or create a new plan.
3. On the **General** FastTab, set the following fields:
    - **Forecast model** – Select the forecast model to apply. This model will be considered when a supply suggestion is generated for the current master plan.
    - **Include supply forecast** – Set this option to *Yes* to include the supply forecast in the current master plan. If you set it to *No*, supply forecast transactions won't be included in the master plan.
    - **Include demand forecast** – Set this option to *Yes* to include the demand forecast in the current master plan. If you set it to *No*, demand forecast transactions won't be included in the master plan.
    - **Method used to reduce forecast requirements** – Select the method that should be used to reduce forecast requirements. Learn more in Forecast reduction keys.
4. On the **Time fences in days** FastTab, you can set the following fields to specify the period that the forecast is included during:
    - **Forecast plan** – Set this option to *Yes* to override the forecast plan time fence that originates from the individual coverage groups. Set it to *No* to use the values from the individual coverage groups for the current master plan.
    - **Forecast time period** – If you set the **Forecast plan** option to *Yes*, specify the number of days (from today's date) that the demand forecast should be applied.

    > **Important**
    >
    > The **Forecast plan** option isn't supported with Planning Optimization.

Run a master plan that includes an inventory forecast

To run a master plan that includes an inventory forecast, follow these steps.

1. Go to **Master planning** > **Workspaces** > **Master planning**.

2. In the **Master plan** field, enter or select the master plan that you set up in the previous procedure.
3. On the **Master planning** tile, select **Run**.
4. In the **Master planning** dialog box, set the **Track processing time** option to *Yes*.
5. In the **Number of threads** field, enter a number.
6. On the **Records to include** FastTab, select **Filter**.
7. A standard query editor dialog box appears. On the **Range** tab, select the row where the **Field** field is set to *Item number*.
8. In the **Criteria** field, select the item number to include in the plan.
9. Select **OK**.

To view the requirements that are calculated, open the **Gross requirement** page. For example, on the **Released products** page, on the Action Pane, on the **Plan** tab, in the **Requirements** group, select **Gross requirement**.

To view the planned orders that are generated, go to **Master planning** > **Common** > **Planned orders**, and select the appropriate forecast plan.

Related information

- Demand forecasting overview
- Demand forecasting setup
- Generate a statistical baseline forecast
- Make manual adjustments to the baseline forecast
- Master planning with demand forecasts

Guide: Modify a demand forecast manually

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Tip**
>
> This article describes demand forecasting functionality that's built into Microsoft Dynamics 365 Supply Chain Management. For an even better planning and forecasting experience, we recommend that you try *Demand planning in Microsoft Dynamics 365 Supply Chain Management*, which is Microsoft's next-generation collaborative demand planning solution. For details, see the Demand planning home page.

This procedure shows how to modify the forecast for an item. This procedure is intended for the production planner.

Modify the forecast for a selected item

To modify the forecast for a selected item:

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, find and select the desired record. Select the item for which you want to modify the forecast.
3. On the Action Pane, open the **Plan** tab and select **Demand forecast**.
4. In the list, select a row. If there are no forecast lines, create a new line by selecting **New** on the Action Pane.
5. In the **Sales quantity** field, enter a positive number. This number represents the forecasted quantity for the item. An error will be shown if you enter a negative number.
6. Fill out the other fields as needed.
7. Select **Save** on the Action Pane.

Modify the forecast for one or more items with Microsoft Excel

To modify the forecast for one or more items with Microsoft Excel:

1. Do one of the following:

    - Open the **Demand forecast** page for any item (it doesn't matter which one) as described in the previous section.
    - Go to **Master planning** > **Forecasting** > **Manual forecast entry** > **Demand forecast lines**.

2. On the Action Pane, select **Open in Microsoft Office** > **Demand forecast entries**.
3. Select a download location, save, and then open the downloaded file in Excel.
4. If you see a warning, choose to **Enable editing**.
5. In Excel, sign in to Supply Chain Management using the Microsoft Dynamics task pane. You must sign in with the **Keep me signed in** option enabled and you must trust the data connection app.
6. The Excel spreadsheet now shows all the current demand forecast lines for your company. Add, delete, and edit demand forecast lines as needed.
7. Select **Publish** on the Microsoft Dynamics task pane to upload your changes back to Supply Chain Management.

Last updated on 06/17/2025

Demand forecasting overview

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Use demand forecasting to predict independent demand from sales orders and dependent demand at any decoupling point for customer orders. The enhanced demand forecast reduction rules provide an ideal solution for mass customization.

To generate the baseline forecast, a summary of historical transactions is passed to Microsoft Azure Machine Learning hosted on Azure. Because this service isn't shared among users, you can easily customize it to meet industry-specific requirements. Use Supply Chain Management to visualize the forecast, adjust the forecast, and view key performance indicators (KPIs) about forecast accuracy.

> **Note**
>
> Microsoft Azure Machine Learning Studio (classic) is required for forecast generation with machine learning. As of December 1, 2021, you can't create new Machine Learning Studio (classic) resources. However, you can continue to use your existing Machine Learning Studio (classic) resources until August 31, 2024. For updated information, see Azure Machine Learning Studio.

Dynamics 365 Supply Chain Management version 10.0.23 and later support the new Azure Machine Learning Studio.

Key features of demand forecasting

Here are some of the main features of demand forecasting:

- Generate a statistical baseline forecast that's based on historical data.
- Use a dynamic set of forecast dimensions.
- Visualize demand trends, confidence intervals, and adjustments of the forecast.
- Authorize the adjusted forecast for use in planning processes.
- Remove outliers.
- Create measurements of forecast accuracy.

Major themes in demand forecasting

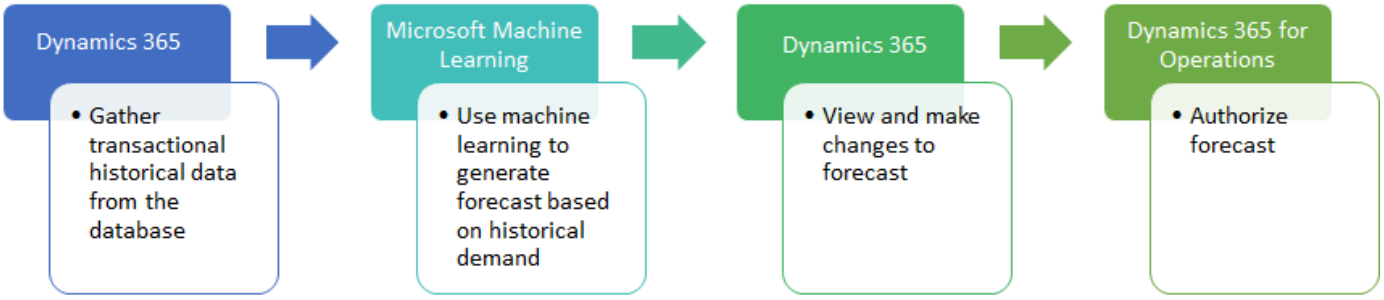
Demand forecasting implements three major themes:

- **Modularity** – Demand forecasting is modular and easy to configure. You can turn the functionality on and off by changing the configuration key at **Trade** > **Inventory forecast** > **Demand forecasting**.
- **Reuse of the Microsoft stack** – Machine Learning lets you quickly and easily create predictive analysis experiments, such as demand estimation experiments, by using algorithms R or Python programming languages and a simple drag-and-drop interface.
  - You can download the Demand forecasting experiments, change them to meet your business requirements, publish them as a web service on Azure, and use them to generate demand forecasts. You can download the experiments if you purchase a Supply Chain Management subscription for a production planner as enterprise-level user.
  - You can create your own experiments in Microsoft Azure Machine Learning studio (classic), publish them as services on Azure, and use them to generate demand forecasts.
  - If you don't require high performance, or if you don't require that a large amount of data be processed, you can use the Machine Learning free tier. Always start from this tier, especially during implementation and testing phases. If you require higher performance and additional storage, you can use the Machine Learning standard tier. This tier requires an Azure subscription and involves additional costs. For details about Machine Learning pricing, see Machine Learning Studio pricing .
- **Forecast reduction at any decoupling point** – Demand forecasting builds on this functionality, which lets you forecast both dependent and independent demand at any

decoupling point.

Basic flow in demand forecasting

The following diagram shows the basic flow in demand forecasting.



Demand forecast generation starts in Supply Chain Management. Historical transactional data from the Supply Chain Management transactional database is gathered and populates a staging table. This staging table is later fed to a Machine Learning service. By performing minimal customization, you can plug various data sources into the staging table. The data sources can include Microsoft Excel files, comma-separated value (CSV) files, and data from Microsoft Dynamics AX 2009 and Microsoft Dynamics AX 2012. Therefore, you can generate demand forecasts that consider historical data that is spread among multiple systems. However, the master data, such as item names and units of measure, must be the same across the various data sources.

If you use the Demand forecasting Machine Learning experiments, they look for a best fit among five time series forecasting methods to calculate a baseline forecast. The parameters for these forecasting methods are managed in Supply Chain Management.

The forecasts, historical data, and any changes that were made to the demand forecasts in previous iterations are then available in Supply Chain Management.

You can use Supply Chain Management to visualize and modify the baseline forecasts. Manual adjustments must be authorized before the forecasts can be used for planning.

Limitations

Demand forecasting is a tool that helps customers in the manufacturing industry create forecasting processes. It offers the core functionality of a demand forecasting solution and is designed so that it can easily be extended. Demand forecasting might not be the best fit for

customers in industries such as commerce, wholesale, warehousing, transportation, or other professional services.

Demand forecast variant conversion limitation

Unit of measure (UOM) per variant conversion isn't fully supported when generating demand forecast if inventory UOM is different from the demand forecast UOM.

Generating forecast (**Inventory UOM** > **Demand forecast UOM**) uses product UOM conversion. When loading historical data for the demand forecast generation, the product level UOM conversion is always used when converting from inventory UOM to the demand forecast UOM, even if there are conversions defined on the variant level.

The first part of authorizing forecast (**Demand forecast UOM** > **Inventory UOM**) uses product UOM conversion. The second part of authorizing forecast (**Inventory UOM** > **Sales UOM**) uses the variant UOM conversion. When the generated demand forecast is authorized, the conversion to inventory UOM from demand forecast UOM uses product level UOM conversion. At the same time, conversion between the inventory unit and the sales UOM respects the variant level defined conversions.

> **Note**
>
> The demand forecast UOM doesn't have to have any specific meaning. It can be defined as *Demand forecast unit*. For each of the products, you can define the conversion to be 1:1 with the inventory UOM.

Troubleshoot demand forecasting

If you experience issues with demand forecasting, review the following table for guidance on common tasks and scenarios before you contact support.

Expand table

Area	Description	Learn more
Setup and configuration	Configure demand forecasting parameters, allocation keys, Machine Learning connections, and intercompany planning groups.	Demand forecasting setup
Baseline forecast generation	Generate a statistical baseline forecast from historical transaction data using Azure Machine	Generate a statistical baseline forecast

Area	Description	Learn more
	Learning.	
Forecast adjustments	Make manual adjustments to a baseline forecast before authorizing it for use in master planning.	Make manual adjustments to the baseline forecast
Forecast authorization	Authorize adjusted demand forecasts so the master planning process can consume them.	Authorize an adjusted forecast
Forecast accuracy	Monitor and measure the accuracy of demand forecasts to improve future predictions.	Monitor forecast accuracy
Outlier removal	Identify and remove outliers from historical transaction data to improve forecast quality.	Remove outliers from historical transaction data when calculating a demand forecast
Forecast reduction	Understand how demand forecasts are reduced by actual transactions during master planning, including differences between Planning Optimization and the deprecated engine.	Master planning with demand forecasts
Business process guidance	Review the end-to-end business process for demand forecasting within the forecast to plan scenario.	Predict future demand for products and services with demand forecasting to drive business decisions

Related information

- Demand forecasting setup
- Generate a statistical baseline forecast
- Make manual adjustments to the baseline forecast
- Authorize an adjusted forecast
- Monitor forecast accuracy
- Remove outliers from historical transaction data when calculating a demand forecast
- Intercompany planning groups for demand forecasting
- Master planning with demand forecasts
- Predict future demand for products and services with demand forecasting to drive business decisions
- Video: Extend the demand forecasting functionality
- Webinar: Demand Forecasting with Azure Machine Learning Series

ⓘ **Note:** The author created this article with assistance from AI. Learn more

Last updated on 05/06/2026

Demand forecasting setup

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This article describes how to set up demand forecasting.

Item allocation keys

Item allocation keys establish groups of items. The system calculates a demand forecast for an item and its dimensions only if the item is part of an item allocation key. This rule groups large numbers of items so that the system can create demand forecasts more quickly. The system creates forecasts based on historical data only.

An item and its dimensions must be part of only one item allocation key if you use the item allocation key during forecast creation.

To create item allocation keys and add a stock-keeping unit (SKU) to them, follow these steps:

1. Go to **Master planning** > **Setup** > **Demand forecasting** > **Item allocation keys**.
2. Either select an item allocation key in the list pane, or select **New** on the Action Pane to create a new one. On the header for the new or selected key, set the following fields:
    - **Item allocation key** – Enter a unique name for the key.

- **Name** – Enter a descriptive name for the key.

3. Follow one of these steps to add items to the selected item allocation key or remove items:

    - On the **Item allocation** FastTab, use the **New** and **Delete** buttons on the toolbar to add or remove items as needed. For each row, select the item number, and then assign dimension values in the other columns as you require. Select **Display dimensions** on the toolbar to change the set of dimension columns that is shown in the grid. (The value in the **Percent** column is ignored when demand forecasts are generated.)
    - If you want to add a large number of items to the key, select **Assign items** on the Action Pane to open a page where you can find and assign multiple items to the selected key.

> ⓘ **Important**
>
> Be careful to include only relevant items in each item allocation key. Unnecessary items might cause increased costs when you use Microsoft Azure Machine Learning.

Intercompany planning groups

Demand forecasting can generate cross-company forecasts. In Dynamics 365 Supply Chain Management, group companies that you plan together into the same intercompany planning group. To specify, per company, which item allocation keys to consider for demand forecasting, associate an item allocation key with the intercompany planning group member.

> ⓘ **Important**
>
> Planning Optimization doesn't currently support intercompany planning groups. To do intercompany planning that uses Planning Optimization, set up master planning batch jobs that include master plans for all the relevant companies.

To set up your intercompany planning groups, follow these steps:

1. Go to **Master planning** > **Setup** > **Intercompany planning groups**.

2. Either select a planning group in the list pane, or select **New** on the Action Pane to create a new one. On the header for the new or selected group, set the following fields:

    - **Name** – Enter a unique name for the planning group.

- **Description** – Enter a short description of the planning group.

3. On the **Intercompany planning group members** FastTab, use the buttons on the toolbar to add a row for each company (legal entity) that should be part of the group. For each row, set the following fields:

    - **Legal entity** – Select the name of a company (legal entity) that is a member of the selected group.
    - **Scheduling sequence** – Assign the order that the company should be processed in relative to other companies. Low values are processed first. This order can be important when demand for one company affects other companies. In these cases, the company that supplies the demand is processed last.
    - **Master plan** – Select the master plan to trigger for the current company.
    - **Automatic copy to static plan** – Select this checkbox to copy the result of the plan to the static plan.
    - **Automatic copy to dynamic plan** – Select this checkbox to copy the result of the plan to the dynamic plan.

4. By default, if you don't assign item allocation keys to intercompany planning group members, the system calculates a demand forecast for all items that are assigned to all item allocation keys from all companies. You can find additional filtering options for companies and item allocation keys in the **Generate statistical baseline forecast** dialog box (**Master planning** > **Forecasting** > **Demand forecasting** > **Generate statistical baseline forecast**). To assign item allocation keys to a company in the selected intercompany planning group, select the company, and then, on the **Intercompany planning group members** FastTab, select **Item allocation keys** on the toolbar.

Learn more in Intercompany planning groups for demand forecasting.

> **Important**
>
> Include only relevant item allocation keys in each intercompany planning group. Unnecessary items might cause increased costs when you use Azure Machine Learning.

Set up demand forecasting parameters

Use the **Demand forecasting parameters** page to set up options that control how demand forecasting works in your system.

Open the Demand forecasting parameters page

To set up demand forecasting parameters, go to **Master planning** > **Setup** > **Demand forecasting** > **Demand forecasting parameters**. Because demand forecasting runs cross-company, the setup is global. It applies to all legal entities (companies).

General settings

Use the **General** tab of the **Demand forecasting parameters** page to define general settings for demand forecasting.

Demand forecast unit

Demand forecasting generates the forecast in quantities. Therefore, you must specify the unit of measure for the quantity in the **Demand forecast unit** field. This field defines the unit that the system uses for all demand forecasts, regardless of the usual inventory units for each product. By using a consistent forecast unit, you help ensure that the aggregation and percentage distribution make sense. For more information about aggregation and percentage distribution, see Make manual adjustments to the baseline forecast.

For every unit of measure that you use for SKUs included in demand forecasting, make sure that there's a conversion rule for the unit of measure and the general forecasting unit of measure that you select here. When you generate a forecast, the system logs the list of items that don't have a unit of measure conversion. Therefore, you can easily correct the setup. For more information about units of measure and how to convert among them, see Manage units of measure.

Transaction types

Use the fields on the **Transaction types** FastTab to select the transaction types that the system uses when it generates the statistical baseline forecast.

You can use demand forecasting to forecast both dependent demand and independent demand. For example, if you set only the **Sales order** option to *Yes*, and all the items that you consider for demand forecasting are items that are sold, the system calculates independent demand. However, you can add critical subcomponents to item allocation keys and include them in demand forecasting. In this case, if you set the **Production line** option to *Yes*, the system calculates a dependent forecast.

You can override transaction types for one or more specific item allocation keys by using the **Item allocation keys** tab. That tab provides similar fields.

Choose how to create the baseline forecast

Use the **Forecast generation strategy** field to select the method that the system uses to create a baseline forecast. Three methods are available:

- *Copy over historical demand* – Create forecasts by just copying historical data.
- *Azure Machine Learning Service* – Use a forecast model that uses the Azure Machine Learning Service. The Azure Machine Learning Service is the current machine learning solution for Azure. Therefore, use it if you want to use a forecast model.
- *Azure Machine Learning* – Use a forecast model that uses Azure Machine Learning Studio (classic). Azure Machine Learning Studio (classic) is deprecated and will soon be removed from Azure. Therefore, select *Azure Machine Learning Service* if you're setting up demand forecasting for the first time. If you're currently using Azure Machine Learning Studio (classic), you should plan to switch to the Azure Machine Learning Service as soon as possible.

You can override the forecast generation method for one or more specific item allocation keys by using the **Item allocation keys** tab. That tab provides similar fields.

Override default forecast algorithm parameters globally

Default forecast algorithm parameters and values are assigned on the **Demand forecasting parameters** page (**Master Planning** > **Setup** > **Demand forecasting** > **Demand forecasting parameters**). However, you can override them globally by using the **Forecast algorithm parameters** FastTab on the **General** tab of the **Demand forecasting parameters** page. (You can also override them for specific allocation keys by using the **Item allocation keys** tab on the **Demand forecasting parameters** page.)

Use the **Add** and **Remove** buttons on the toolbar to establish the required collection of parameter overrides. For each parameter in the list, select a value in the **Name** field, and then enter an appropriate value in the **Value** field. All parameters that aren't listed here take their values from the settings on the **Demand forecasting parameters** page. For more information about how to use the standard set of parameters and select values for them, see the Default parameters and values for demand forecasting models section.

Set forecast dimensions

A forecast dimension shows the level of detail for the forecast. Company, site, and item allocation key are required forecast dimensions. You can also generate forecasts at the warehouse, inventory status, customer group, customer account, country/region, state, and item level, and at

all item dimension levels. Use the **Forecast dimensions** tab on the **Demand forecasting parameters** page to select the set of forecast dimensions that is used when the demand forecast is generated.

At any time, you can add forecast dimensions to the list of dimensions that are used for demand forecasting. You can also remove forecast dimensions from the list. However, manual adjustments are lost if you add or remove a forecast dimension.

Set up overrides for specific item allocation keys

Not all items work in the same way from a demand forecasting perspective. Therefore, you can establish allocation key–specific overrides for most of the settings available on the **General** tab. The exception is the demand forecast unit. To set up overrides for a specific item allocation key, follow these steps:

1. On the **Demand forecasting parameters** page, on the **Item allocation keys** tab, use the toolbar buttons to add item allocation keys to the grid on the left, or remove them, as you require. Then select the allocation key that you want to set up overrides for.
2. On the **Transaction types** FastTab, enable the types of transactions that you want to use to generate the statistical baseline forecast for products that belong to the selected allocation key. The settings work just as they work on the **General** tab, but they apply only to the selected item allocation key. All the settings here (both the *Yes* values and the *No* values) override all the **Transaction types** settings on the **General** tab.
3. On the **Forecast algorithm parameters** FastTab, select the forecast generation strategy and forecast algorithm parameter overrides for products that belong to the selected allocation key. These settings work just as they work on the **General** tab, but they apply only to the selected item allocation key. Use the **Add** and **Remove** buttons on the toolbar to define the required collection of parameter overrides. For each parameter in the list, select a value in the **Name** field, and then enter an appropriate value in the **Value** field. For more information about how to use the standard set of parameters and select values for them, see the Default parameters and values for demand forecasting models section.

Set up the connection to the Azure Machine Learning Service

Use the **Azure Machine Learning Service** tab to set up the connection to the Azure Machine Learning Service. This solution is one of the options for creating the baseline forecast. These settings on this tab take effect only when the **Forecast generation strategy** field is set to *Azure Machine Learning Service*.

For more information about how to set up the Azure Machine Learning Service and then use the settings here to connect to it, see the Set up the Azure Machine Learning Service section.

Set up the connection to Azure Machine Learning Studio (classic)

> **Important**
>
> Azure Machine Learning Studio (classic) is deprecated. Therefore, you can't create new workspaces for it in Azure. Azure Machine Learning Studio (classic) is replaced by the Azure Machine Learning service, which provides similar functionality and more. If you're not already using Azure Machine Learning, start using the Azure Machine Learning service. If you already have a workspace that you created for Azure Machine Learning Studio (classic), you can continue to use it until the feature is removed from Azure. However, update to the Azure Machine Learning service as soon as possible. Although the application continues to warn you that Azure Machine Learning Studio (classic) is deprecated, the forecasting result isn't affected. For more information about the new Azure Machine Learning service and how to set it up, go to the Set up the Azure Machine Learning Service section.
>
> You can freely switch between using the new and old machine learning solutions with Supply Chain Management for as long as your old Azure Machine Learning Studio (classic) workspace remains available.

If you already have an available Azure Machine Learning Studio (classic) workspace, you can use it to generate forecasts by connecting it to Supply Chain Management. You can establish this connection by using the **Azure Machine Learning** tab on the **Demand forecasting parameters** page. (The settings on this tab take effect only when the **Forecast generation strategy** field is set to *Azure Machine Learning*.) Enter the following details for your Azure Machine Learning Studio (classic) workspace:

- Web service application programming interface (API) key
- Web service endpoint URL
- Azure storage account name
- Azure storage account key

> **Note**

You need the Azure storage account name and key only if you use a custom storage account. If you deploy the on-premises version, you must have a custom storage account in Azure, so that Machine Learning can access the historical data.

Default parameters and values for demand forecasting models

When you use machine learning to generate your forecast planning models, you control machine learning options by setting values for *forecasting algorithm parameters*. Supply Chain Management sends these values to Azure Machine Learning. Use the **Forecasting algorithm parameters** page to control which types of values to provide and which values each should have.

To set up the default parameters and values for demand forecasting models, go to **Master Planning** > **Setup** > **Demand forecasting** > **Forecasting algorithm parameters**. A standard set of parameters is provided. Each parameter has the following fields:

- **Name** – The name of the parameter, as used by Azure. Usually, you shouldn't change this name unless you've customized the experiment in Azure Machine Learning.
- **Description** – A common name for the parameter. Use this name to identify the parameter in other places in the system (for example, on the **Demand forecasting parameters** page).
- **Value** – The default value for the parameter. The value that you enter depends on the parameter that you're editing.
- **Explanation** – A short description of the parameter and how to use it. This description typically includes advice about valid values for the **Value** field.

The following parameters are provided by default. (To revert to this standard list, select **Restore** on the Action Pane.)

- **Confidence level percentage** – A confidence interval consists of a range of values that act as good estimates for the demand forecast. A 95% confidence level percentage indicates there's a 5% risk that the future demand falls outside the confidence interval range.

- **Force seasonality** – Specifies whether to force the model to use a specific type of seasonality. This parameter applies to ARIMA and ETS only. Options: *AUTO (default)*, *NONE*, *ADDITIVE*, *MULTIPLICATIVE*.

- **Forecasting model** – Specifies which forecasting model to use. Options: *ARIMA*, *ETS*, *STL*, *ETS+ARIMA*, *ETS+STL*, *ALL*. To select the best-fit model, use *ALL*.

- **Maximum forecasted value** – Specifies the maximum value to use for predictions. Format: +1E[n] or numeric constant.
- **Minimum forecasted value** – Specifies the minimum value to use for predictions. Format: -1E[n] or numeric constant.
- **Missing value substitution** – Specifies how to fill gaps in historical data. Options: *(numeric value)*, *MEAN*, *PREVIOUS*, *INTERPOLATE LINEAR*, *INTERPOLATE POLYNOMIAL*.
- **Missing value substitution scope** – Specifies whether the value substitution applies only to the date range of each individual granularity attribute or to the entire dataset. The following options are available for establishing the date range that the system uses when filling in gaps in historical data:
  - *GLOBAL* – The system uses the full range of dates of all granularity attributes.
  - *HISTORY_DATE_RANGE* – The system uses a specific date range defined by **From date** and **To date** fields in the **Historical horizon** section in the **Generate statistical baseline forecast** dialog.
  - *GRANULARITY_ATTRIBUTE* – The system uses the date range of the currently processed granularity attribute.

  > ⓘ **Note**
  >
  > A granularity attribute is a combination of forecast dimensions against which the forecast is done. You can define forecast dimensions on the **Demand forecasting parameters** page.

- **Seasonality hint** – For seasonal data, provide a hint to the forecasting model to improve forecast accuracy. Format: integer number that represents the number of buckets that a demand pattern repeats itself for. For example, enter *6* for data that repeats itself every six months.
- **Test set size percentage** – Percentage of historical data to use as a test set for forecast accuracy calculation.

Override the values for these parameters by going to **Master Planning** > **Setup** > **Demand forecasting** > **Demand forecasting parameters**. On the **Demand forecasting parameters** page, you can override the parameters in the following ways:

- Use the **General** tab to override the parameters globally.

- Use the **Item allocation keys** tab to override the parameters for specific item allocation keys. Parameters that you override for a specific item allocation key affect only the forecast of items that are associated with that item allocation key.

> ⓘ Note
>
> On the **Forecasting algorithm parameters** page, use the buttons on the Action Pane to add parameters to the list or remove parameters from the list. However, don't use this approach unless you customize the experiment in Azure Machine Learning.

Set up the Azure Machine Learning Service

Supply Chain Management calculates demand forecasts by using the Azure Machine Learning Service. You must set up and run this service on your own Azure subscription. This section describes how to set up the Azure Machine Learning Service in Azure and then connect it to your Supply Chain Management environment.

Enable the Azure Machine Learning Service in Feature management

To use this feature, it must be turned on for your system. As of Supply Chain Management version 10.0.32, it's turned on by default. As of Supply Chain Management version 10.0.36, the feature is mandatory and can't be turned off. If you're running a version older than 10.0.36, then admins can turn this functionality on or off by searching for the *Azure Machine Learning Service for demand forecasting* feature in the Feature management workspace.

Set up machine learning in Azure

> ⓘ Important
>
> Azure Machine Learning SDK v1 (`azureml-core`) is retired, so the demand forecasting template is updated to use Azure Machine Learning SDK v2 (`azure-ai-ml`). Because the Supply Chain Management connector calls the Azure Machine Learning v1 published-pipeline REST contract, which SDK v2 doesn't expose, the updated template adds a small *relay* (an Azure Function) that receives those calls and submits the SDK v2 pipeline job. The `quick_setup.ps1` script provisions and deploys the relay for you. Setups that still use the v1

template continue to work, but switch to the updated template as soon as it's convenient. Learn more in **About the relay**.

To enable Azure to use machine learning to process your forecasts, you must set up an Azure Machine Learning workspace for this purpose. You have two options:

- To set up the workspace by running a script that Microsoft provides, follow the instructions in the Option 1: Run a script to automatically set up your machine learning workspace section, and then skip ahead to the Set up Azure Machine Learning Service connection parameters in Supply Chain Management section.
- To manually set up your workspace, follow the instructions in the Option 2: Manually set up your machine learning workspace section, and then skip ahead to the Set up Azure Machine Learning Service connection parameters in Supply Chain Management section. This option takes more time, but it gives you more control.

About the relay

The Supply Chain Management demand forecasting connector communicates with Azure Machine Learning through the v1 published-pipeline REST contract: it submits a run, polls the run status, cancels a run, and queries active runs. Azure Machine Learning SDK v2 doesn't expose that contract, so the updated template includes a *relay*—a lightweight Azure Function—that accepts those four calls and submits the equivalent SDK v2 pipeline job. Input and output data continue to flow through the *demplan-azureml* blob container, exactly as in the v1 template.

The relay authenticates each caller by validating the service principal's Microsoft Entra token, so that only your Supply Chain Management environment can trigger forecasts. The `quick_setup.ps1` script creates the Function app, assigns it a managed identity that has access to the workspace and storage account, configures it, and deploys the relay code. The **Pipeline endpoint address** that you enter in Supply Chain Management is the relay's URL, which the script prints when it finishes.

Option 1: Run a script to automatically set up your machine learning workspace

This section describes how to set up your machine learning workspace by using an automated script that Microsoft provides. The script provisions the Azure Machine Learning resources, creates the service principal, and provisions and deploys the relay. If you prefer, you can manually set up all the resources by following the instructions in the Option 2: Manually set up your machine learning workspace section. You don't have to complete both options.

The script requires the Azure CLI and the Azure CLI ml extension (v2).

1. On GitHub, open the Templates for Dynamics 365 Supply Chain Management demand forecasting with Azure Machine Learning repository (repo). Either clone the repo or download it as a .zip file, so that you have the complete folder structure, including these items:

   - `quick_setup.ps1`
   - `sampleInput.csv`
   - `src/parameters.py`
   - `src/api_trigger.py`
   - `src/run.py`
   - `src/REntryScript/entry.py`
   - `src/REntryScript/forecast_entry.r`
   - `src/REntryScript/forecast.r`
   - The `relay` folder

2. Open a PowerShell window in the root of the repo, and run the `quick_setup.ps1` script. Provide the parameters that are described in the following table. Follow the on-screen instructions, including the Azure sign-in prompt.

Expand table

Parameter	Description
`-subscriptionId`	The Azure subscription to create the resources in.
`-resourceGroupName`	The resource group to create or reuse.
`-location`	The Azure region. Select the region that matches your Supply Chain Management deployment.
`-storageAccountName`	The storage account that is dedicated to demand forecasting.
`-workspaceName`	The Azure Machine Learning workspace to create or reuse.
`-AADApplicationName`	The name of the Microsoft Entra application (service principal) to create or reuse.
`-functionAppName`	The name of the Azure Function app that hosts the relay.
`-functionStorageAccountName`	The storage account that the Function app uses for its runtime.
`-d365TenantId` (optional)	The tenant that the connector authenticates from. If you omit this

Parameter	Description
	parameter, the script uses the current tenant.
`-d365AppId` (optional)	The connector's service principal application ID. If you omit this parameter, the script uses the application that it created.

The script is idempotent. If you run it again, it reuses existing resources. It does the following work:

- Creates (or reuses) the workspace, the storage account, the *demplan-azureml* blob container, the *workspaceblobdemplan* datastore, a compute instance, and the *e2ecpucluster* compute cluster.
- Creates (or reuses) the Microsoft Entra application and service principal, and assigns it the Contributor role on the workspace and storage account, and the Storage Blob Data Contributor role on the storage account.
- Provisions the Function app for the relay, assigns it a managed identity that has the required roles, configures it, and deploys the relay code.

3. In the **Certificates & secrets** section of the Microsoft Entra application that the script created, create a client secret. For instructions, see Add a client secret. Make a note of the secret value. You'll enter it in Supply Chain Management later.

4. When the script finishes, it prints the demand forecasting parameters that you enter in Supply Chain Management, including the **Pipeline endpoint address** (the relay URL). Make a note of these values.

5. Your workspace and relay are now set up. Skip ahead to the Set up Azure Machine Learning Service connection parameters in Supply Chain Management section.

Option 2: Manually set up your machine learning workspace

This section describes how to manually set up your machine learning workspace. Complete the procedures in this section only if you decided not to run the automated setup script, as described in the Option 1: Run a script to set up your machine learning workspace section.

Step 1: Create a new workspace

Use the following procedure to create a new machine learning workspace.

1. Sign in to your Azure portal.

2. Open the **Machine learning** service.

3. Select **Create** on the toolbar to open the **Create** wizard.

4. Complete the wizard by following the on-screen instructions. Keep the following points in mind as you work:

    - Use default settings unless other points in this list recommend different settings.
    - Be sure to select the geographic region that matches the region where your instance of Supply Chain Management is deployed. Otherwise, some of your data might pass through region boundaries. Learn more in the privacy notice later in this article.
    - Use dedicated resources, such as resource groups, storage accounts, container registries, Azure Key Vaults, and networking resources.
    - On the **Set up Azure Machine Learning Service connection parameters** page of the wizard, you must provide a storage account name. Use an account that's dedicated to demand forecasting. Demand forecasting input and output data are stored in this storage account.

Learn more in Create the workspace.

Step 2: Configure storage

Use the following procedure to set up your storage.

1. On GitHub, open the Templates for Dynamics 365 Supply Chain Management demand forecasting with Azure Machine Learning repo, and download the `sampleInput.csv` file.
2. Open the storage account that you created in the Step 1: Create a new workspace section.
3. Follow the instructions in Create a container to create a container that's named *demplan-azureml*.
4. Upload the `sampleInput.csv` file that you downloaded in step 1 to the container that you just created. You need this file to publish the pipeline and generate a test forecast. For instructions, see Upload a block blob.

Step 3: Configure a default datastore

To set up your default datastore, follow these steps:

1. In Azure Machine Learning studio, select **Datastores** in the navigator.
2. Create a new datastore of the *Azure Blob Storage* type that points to the *demplan-azureml* Blob storage container that you created in the Step 2: Configure storage section. If the authentication type of the new datastore is *Account key*, provide an account key for the created storage account. For instructions, see Manage storage account access keys.

3. Make your new datastore the default datastore by opening its details and selecting **Set as default datastore**.

Step 4: Configure compute resources

To set up a compute resource in Azure Machine Learning studio to run your forecasting generation scripts, follow these steps:

1. Open the details page for the machine learning workspace that you created in the Step 1: Create a new workspace section. Find the **Studio web URL** value, and select the link to open it.

2. Select **Compute** on the navigation pane.

3. On the **Compute instances** tab, select **New** to open a wizard that helps you create a new compute instance. Follow the on-screen instructions. Use the default settings. The compute instance is used to create the demand forecasting pipeline. You can delete it after the pipeline is published.

4. On the **Compute clusters** tab, select **New** to open a wizard that helps you create a new compute cluster. Follow the on-screen instructions. The compute cluster generates demand forecasts. Its settings affect performance and the maximum level of parallelization of the run. Set the following fields, but use the default settings for all other fields:

    - **Name** – Enter *e2ecpucluster*.
    - **Virtual machine size** – Adjust this setting according to the volume of data that you expect to use as input for demand forecasting. The node count shouldn't exceed 11, because one node is required to trigger demand forecast generation and the maximum number of nodes that can then be used to generate a forecast is 10. You'll also set the node count in the `parameters.py` file in the Step 5: Deploy the relay section. On each node, several worker processes run forecasting scripts in parallel. The total number of worker processes in your job is *Number of cores that a node has* × *Node count*. For example, if your compute cluster has a type of *Standard_D4* (eight cores) and a maximum of 11 nodes, and if the `nodes_count` value is set to *10* in the `parameters.py` file, the effective level of parallelism is 80.

Step 5: Deploy the relay

In SDK v2, the connector reaches Azure Machine Learning through the relay instead of through a v1 published pipeline. Follow these steps to deploy the relay manually. These steps use the Azure CLI and the Azure CLI ml extension (v2).

> **Tip**
>
> The `quick_setup.ps1` script in **Option 1** does all of these steps for you. You can run it with the `-RelayOnly` switch to deploy only the relay against a workspace that you set up manually.

1. On GitHub, open the Templates for Dynamics 365 Supply Chain Management demand forecasting with Azure Machine Learning repo, and clone it or download it as a .zip file, so that you have the `src` and `relay` folders.

2. Open the `src/parameters.py` file. Make sure that the `nodes_count` value isn't greater than the number of nodes that you configured for the compute cluster in the Step 4: Configure compute resources section. If the `nodes_count` value is greater than the number of nodes that are available in the cluster, the run might be queued until more resources become available.

3. Create a storage account for the Function app's runtime. This account is separate from the demand forecasting storage account.

   Azure CLI

   ```
   az storage account create --name <functionStorageAccountName> --resource-group
   <resourceGroupName> --location <location> --sku Standard_LRS
   ```

4. Create a Linux Function app that runs Python 3.11 on a Consumption plan, and enable the remote build.

   Azure CLI

   ```
   az functionapp create --name <functionAppName> --resource-group
   <resourceGroupName> --storage-account <functionStorageAccountName> --consumption-
   plan-location <location> --os-type Linux --runtime python --runtime-version 3.11
   --functions-version 4
   az functionapp config appsettings set --name <functionAppName> --resource-group
   <resourceGroupName> --settings SCM_DO_BUILD_DURING_DEPLOYMENT=1
   ENABLE_ORYX_BUILD=true
   ```

5. Assign the Function app a system-assigned managed identity.

   Azure CLI

   ```
   az functionapp identity assign --name <functionAppName> --resource-group
   ```

```
<resourceGroupName>
```

Assign the returned principal ID the *Contributor* role on the Azure Machine Learning workspace, and the *Storage Blob Data Contributor* role on the demand forecasting storage account. For instructions, see Assign Azure roles using the Azure portal.

6. Configure the relay's app settings. Set `RELAY_EXPECTED_TENANT_ID` and `RELAY_EXPECTED_APP_ID` to the tenant ID and the service principal application ID from the Set up a new Microsoft Entra application section. The relay rejects any caller that doesn't present a token from this tenant and application.

Azure CLI

```
az functionapp config appsettings set --name <functionAppName> --resource-group
<resourceGroupName> --settings AML_SUBSCRIPTION_ID=<subscriptionId>
AML_RESOURCE_GROUP=<resourceGroupName> AML_WORKSPACE_NAME=<workspaceName>
RELAY_DATASTORE=workspaceblobdemplan RELAY_EXPECTED_TENANT_ID=<tenantId>
RELAY_EXPECTED_APP_ID=<servicePrincipalApplicationId>
```

7. Copy the `src` folder into the `relay` folder (the relay imports the pipeline builders from it), package the relay, and deploy it with a remote build.

PowerShell

```
Copy-Item .\src .\relay\src -Recurse -Force
Compress-Archive -Path .\relay\* -DestinationPath .\relay.zip -Force
az functionapp deployment source config-zip --name <functionAppName> --resource-
group <resourceGroupName> --src .\relay.zip --build-remote true
```

> **ⓘ Important**
>
> The `--build-remote true` option is required. Without it, the packages in `relay/requirements.txt` aren't installed, and the relay fails at runtime.

8. The relay is now deployed. Its pipeline endpoint address has the following form. You enter this URL as the **Pipeline endpoint address** in Supply Chain Management. Only the host is used by the connector; the path segments supply the subscription, resource group, and workspace.

text

```
https://<functionAppName>.azurewebsites.net/pipelines/v1.0/subscriptions/<subscri
ptionId>/resourceGroups/<resourceGroupName>/providers/Microsoft.MachineLearningSe
```

```
rvices/workspaces/<workspaceName>/PipelineRuns/PipelineEndpointSubmit/Id/relay
```

Set up a new Microsoft Entra application

> **Note**
>
> If you set up your workspace by running the script (Option 1), the script already created this Microsoft Entra application and assigned the required roles. In that case, you only need to create a client secret (step 5 in this section). Complete the other steps in this section only if you set up your workspace manually (Option 2).

To authenticate by using a service principal with the resources dedicated to demand forecasting, you need a Microsoft Entra application. Therefore, the application should have the lowest level of privilege that is required to generate the forecast.

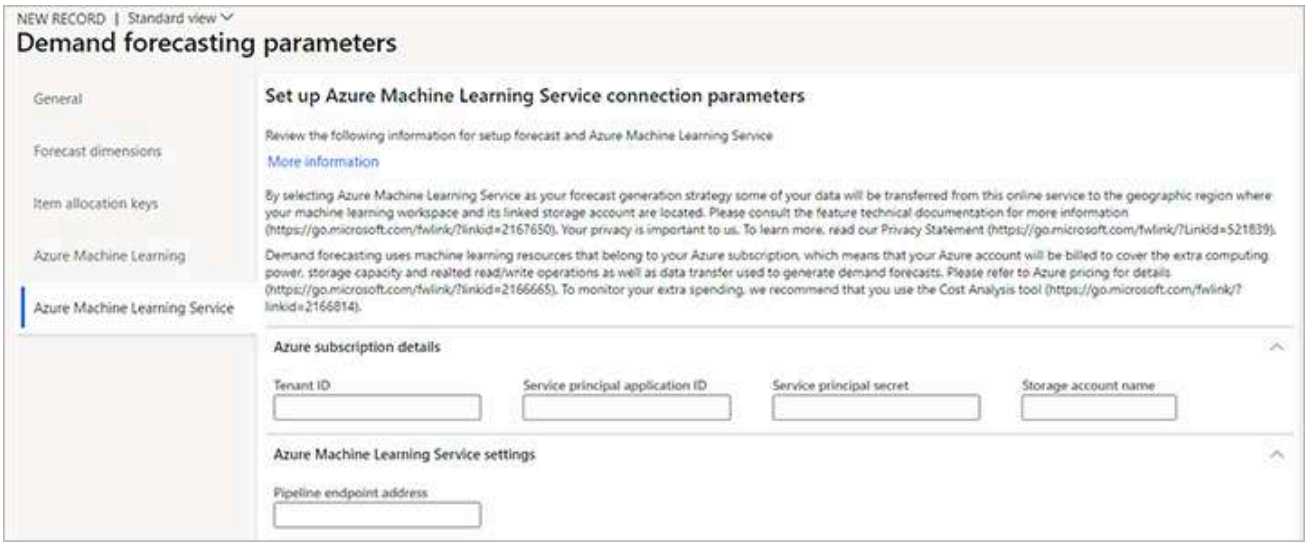
1. Sign in to your Azure portal.
2. Register a new application in the tenant's Microsoft Entra ID. For instructions, see Use the portal to create a Microsoft Entra application and service principal that can access resources.
3. Follow the on-screen instructions as you complete the wizard. Use the default settings.
4. Give your new Microsoft Entra application access to the following resources that you created in the Set up machine learning in Azure section. For instructions, see Assign Azure roles using the Azure portal. This step enables the system to import and export forecasting data, and to trigger machine learning pipeline runs from Supply Chain Management.
    - Contributor role to the machine learning workspace
    - Contributor role to the dedicated storage account
    - Storage Blob Data Contributor role to the dedicated storage account
5. In the **Certificates & secrets** section of the application that you created, create a secret for the application. For instructions, see Add a client secret.
6. Make a note of the application ID and its secret. You'll need the details of this application later, when you set up the **Demand forecasting parameters** page in Supply Chain Management.

Set up Azure Machine Learning Service connection parameters in Supply Chain Management

Use the following procedure to connect your Supply Chain Management environment to the machine learning service that you set up in Azure.

1. Sign in to Supply Chain Management.
2. Go to **Master planning** > **Setup** > **Demand forecasting** > **Demand forecasting parameters**.
3. On the **General** tab, make sure that the **Forecast generation strategy** field is set to *Azure Machine Learning Service*.
4. On the **Item allocation keys** tab, make sure that the **Forecast generation strategy** field is set to *Azure Machine Learning Service* for each allocation key that should use the Azure Machine Learning Service for demand forecasting.
5. On the **Azure Machine Learning Service** tab, set the following fields:

    - **Tenant ID** – Enter the ID for your Azure tenant. Supply Chain Management uses this ID to authenticate with the Azure Machine Learning Service. You can find your tenant ID on the **Overview** page for Microsoft Entra ID in the Azure portal.
    - **Service principal application ID** – Enter the application ID for the application that you created in the Active Directory Application section. Supply Chain Management uses this value to authorize API requests to Azure Machine Learning Service.
    - **Service principal secret** – Enter the service principal application secret for the application that you created in the Active Directory Application section. Supply Chain Management uses this value to acquire the access token for the security principal that you created to perform authorized operations against Azure Storage and the Azure Machine Language workspace.
    - **Storage account name** – Enter the Azure storage account name that you specified when you ran the setup wizard in your Azure workspace. (Learn more in the Set up machine learning in Azure section.)
    - **Pipeline endpoint address** – Enter the URL of the relay that you deployed when you set up machine learning in Azure. If you used the script (Option 1), this URL is the **Pipeline endpoint address** that the `quick_setup.ps1` script prints when it finishes. If you set up the relay manually (Option 2), construct the URL as described in the Step 5: Deploy the relay section.



Privacy notice

When you select *Azure Machine Learning Service* as your forecast generation strategy, Supply Chain Management automatically sends your customer data for historical demand, such as aggregated quantities, product names and their product dimensions, shipping and receiving locations, customer identifiers, and also forecast parameters, to the geographic region where your machine learning workspace and its linked storage account are located, for the purpose of forecasting future demands. The Azure Machine Learning Service might be in a different geographic region than the geographic region where Supply Chain Management is deployed. Some users can control whether this functionality is enabled by selecting the forecast generation strategy on the **Demand forecasting parameters** page.

Related information

- Demand forecasting overview
- Generate a statistical baseline forecast
- Make manual adjustments to the baseline forecast
- Webinar: Demand Forecasting with Azure Machine Learning Series

Last updated on 06/30/2026

Intercompany planning groups for demand forecasting

> **Tip**
>
> This article describes demand forecasting functionality that's built into Microsoft Dynamics 365 Supply Chain Management. For an even better planning and forecasting experience, we recommend that you try *Demand planning in Microsoft Dynamics 365 Supply Chain Management*, which is Microsoft's next-generation collaborative demand planning solution. For details, see the **Demand planning home page**.

This article describes how you can use intercompany planning groups during the demand forecasting process.

Collaboration is a vital aspect for many large-scale projects. When working with other legal entities (companies) during the development of a product or service, you can create intercompany planning groups to assist you during the demand forecasting process.

This gives all companies involved a greater understanding of the anticipated revenue that the product or service will generate. It also gives them the information necessary to devise and conduct business plans accordingly.

Many organizations utilize other companies within their organization to oversee various facets within collaborations and projects. Logistics operations that depend on other companies are handled by using intercompany sales and purchases since each legal entity has a separate chart of accounts.

Intercompany planning groups are created during the demand forecasting process after creating forecast models and removing outliers. For demand forecasts, set up downstream companies first, then set up upstream companies afterwards.

To further understand intercompany planning groups for demand forecasting, understand the following terms:

- **Demand forecast** – Allows your company to create "what if" scenarios and efficiently and cost-effectively plan for and meet demand. Accurate forecasts can make a critical difference in customer satisfaction levels about order promising dates and on-time delivery. Allows

businesses to predict future sales demand and use that data to drive certain business decisions.

- **Upstream** – A relative reference in a firm or supply chain. It indicates movement in the direction of the raw material supplier.
- **Downstream** – A relative reference in a firm or supply chain. It indicates movement in the direction of the customer.
- **Planned intercompany demand** – Planned demand for a product in a company, based on planned demand for the product from a downstream company.

A plan in one company can include planned intercompany demand that is related to orders from another company's plan. This capability is useful, because it provides full visibility into planned orders across companies. It also ensures that all required planned supply orders are created, but without requiring that planned orders be firmed for the intercompany demand.

To propagate the demand throughout the intercompany chain, you must set parameters to ensure that planned purchase orders are automatically firmed; that is, orders can't be changed in terms of time or quantity. Set up the **Firming time fence** on the coverage group, or the **Firming time fence** in the master plan.

If no **Firming time fence** is set up, no intercompany purchase orders are created automatically. Only the first execution of the master scheduling results in planned orders; however, because the intercompany purchase order is not actually created, no intercompany sales order is created, and, therefore, no additional intercompany purchase orders are created, and so on.

Learn more in Intercompany planning groups.

Related information

- Demand forecasting overview
- Master planning with demand forecasts

Last updated on 01/09/2026

Generate a statistical baseline forecast

08/22/2025

> **Tip**
>
> This article describes demand forecasting functionality that's built into Microsoft Dynamics 365 Supply Chain Management. For an even better planning and forecasting experience, we recommend that you try *Demand planning in Microsoft Dynamics 365 Supply Chain Management*, which is Microsoft's next-generation collaborative demand planning solution. For details, see the **Demand planning home page**.

This article provides information about the parameters and filters that are used in the calculation of demand forecasting.

When you create a baseline forecast, you must first specify the parameters and filters that are used in the calculation. For example, you can create a baseline forecast that estimates demand based on transaction data from the past year for a specific company, for the coming month, and for a selected group of items.

To generate a demand forecast, go to **Master planning** > **Forecasting** > **Demand forecasting** > **Generate statistical baseline forecast**.

The forecast bucket can be selected at forecast generation time. The available values are Day, Week, and Month.

The number of buckets to generate a forecast for is set in the **Forecast horizon** field.

When the forecast strategy is set to **Copy over historical demand**, the end of the historical horizon is ignored. The system copies the number of buckets specified in the **Forecast horizon** field to the forecast demand, starting from the date set in the **From date** field under **Historical horizon**. By copying historical demand from a certain date forward, production planners can make the plan for the next quarter in two ways:

- By copying the demand from the same quarter last year.
- By copying the demand from the previous quarter.

To prevent confusion in the production plans, a certain number of forecast buckets can be frozen. This number is set in the **Freeze time fence** field. On the **Adjusted demand forecast** page, the cells for the frozen buckets are disabled, to give a visual indication that those values shouldn't be changed.

The start date for the baseline demand forecast doesn't have to be the current date or a date in the future. To set a different start date, use the **Baseline forecast start date - From date** field. For example, in June, users can generate a forecast for the next year. Because the forecast buckets between the end of historical demand and the start of the baseline are missing, the predictions might not be accurate. If you're using the Demand forecasting service, there are four ways in which you can fill in the missing gaps. You can choose the method that you want by setting the MISSING_VALUE_SUBSTITUTION parameter on the **Demand forecasting parameters** page.

> **Note**
>
> Missing value substitution only works for the gaps in data in between the start and end dates for historical data. It will not fill in data before or after the last physical data point, it only acts as extrapolation between actual existing data points.

The **Baseline forecast start date** - **From date** field has to be set to the beginning of a forecast bucket, for example, in the United States, a Sunday if the forecasting bucket is the week. The system automatically adjusts the **Baseline forecast start date** - **From date** field to match the beginning of a forecast bucket.

The **Baseline forecast start date** - **From date** field can be set to a date in the past. In other words, it's possible to generate a demand forecast in the past. This is useful, because it lets users adjust the forecast service parameters so that the statistical forecast generated in the past matches the actual historical demand. Users can then continue using these parameter settings to generate a statistical baseline forecast for the future.

Manual adjustments made in previous demand forecasting iterations can be automatically applied to the new baseline forecast if the **Transfer manual adjustments to the demand forecast** checkbox is selected. If the checkbox is cleared, the manual adjustments aren't added to the baseline forecast – but they aren't deleted. Manual adjustments made to a forecast can be deleted only at forecast import time, by clearing the **Save the manual adjustments made to the baseline demand forecast** checkbox. Manual adjustments are saved at authorization time. Therefore, if a user makes manual adjustments to the forecast, but doesn't authorize the forecast back to Supply Chain Management, the changes are lost. For more information about manual adjustments and how they work, see Authorize an adjusted forecast.

A demand forecast generation can have a name and comments to help users identify the forecast that was generated. These values are visible in forecast generation history on the **Statistical baseline forecast generation history** page.

The intercompany planning group, item allocation keys, and other filters can be applied at forecast generation time. These can be used to improve performance or to split the data into

manageable chunks. However, a demand forecast isn't generated for the members of any item allocation key that isn't associated with an intercompany planning group, even if the item allocation key is selected in the query.

> **Tip**
>
> Sometimes users might receive errors while generating a demand forecast, or forecast generation is completed with no session log. This can happen because of leftover data in the query that was previously used for forecast generation. To resolve this issue, select **Select** to open the **Query** page, select **Reset**, and then regenerate the baseline forecast.

If the forecast isn't generated for a large set of items, but, for example, for one item or one item allocation key at a time, then in order to get better performance, you can select the **Use request response mode** checkbox on the **Master planning - Setup - Demand forecasting - Demand forecasting parameters - Azure Machine Learning** tab.

> **Note**
>
> A potentially flat looking forecast can be due to the historical data that has to be of a longer historical timeframe (a minimum of three time periods in order to pick out patterns, such as three years with monthly forecast). To get a better result, you can try changing the granularity of the time range or increase the time range.

Related information

- Demand forecasting setup
- Make manual adjustments to the baseline forecast
- Authorize an adjusted forecast
- Webinar: Demand Forecasting with Azure Machine Learning Series

Make manual adjustments to the baseline forecast

10/03/2025

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This article explains how you can make manual adjustments to a baseline forecast and view details of the forecast.

Before you make manual adjustments, it's important that you understand a few concepts on various pages.

Grid on the Adjusted demand forecast page

The **Adjusted demand forecast** page includes a grid that has the following structure:

- The first column shows the items, item allocation keys, companies, and so on, that the forecast has been generated for. The subtitle of the page provides a description of the current forecast dimensions that are shown in the grid. For example, if the subtitle of the page is **Company / Site / Item allocation key**, and one of the row headers in the grid is **USMF / 1 / D_Alloc**, that row shows the forecast for the USMF company, site 1, and the **D_Alloc** item allocation key.
- Subsequent columns represent the forecast buckets that the forecast has been generated for. Each column header is the first date of the forecast bucket that the column shows.
- The values in the cells represent the forecast for one item, item allocation key, and so on, for that specific forecast bucket.

Forecast aggregation and de-aggregation

The subtitle of the page shows the level of forecast aggregation.

For example, if the subtitle of the page is **Company / Site / Allocation key / Item number / Color / Size / Configuration / Style**, there is no forecast aggregation, and the forecast is shown

at the level of the item and its dimensions. To change the aggregation, use the **Change forecast dimensions** page, which you can open from the application menu.

To modify the forecast, click in any of the cells that are available, and type the adjusted forecast value. The edited cell immediately becomes bold to indicate that the forecast that it shows isn't the forecast that the demand forecasting service created, but has been manually adjusted.

If you change the aggregation to make the page show more aggregated data, you can use the **Demand forecast lines** page to see the individual forecast lines that make up the aggregated forecast.

For example, you've generated the forecast at the item level, but you know that the demand for this item will increase across all sites because of a promotion or another similar event. In this case, you can set the aggregation to **Company / Item allocation key / Item** on the **Change forecast dimensions** page. You can adjust the global forecast for the item across all sites in the **Adjusted demand forecast** grid. To see the effect of your change across all sites, open the **Demand forecast lines** page. On this page, you will see one line for the item for each site, the adjusted forecast quantity, and the original forecast quantity.

When the adjustment of the forecasted quantity is made at an aggregated level, the system uses weighted allocation to distribute the change among the lines that create the aggregation.

You can also make manual adjustments on the **Demand forecast lines** page, by modifying either the **Total quantity** value or the **Quantity** cells in the de-aggregation grid.

Viewing details of the forecast

You can open the **Demand forecast details** page to view more information about the forecast.

The **Demand forecast details** page shows the following information in graphical and tabular formats:

- The historical demand that the forecast predictions are based on.
- The current forecast that is used by Master planning.
- The new demand forecast values and the amounts they have been manually adjusted by.
- The confidence interval for the forecasted values.
- The forecast model that was used to generate the forecast. If you're viewing aggregated data, you will see the list of all the methods that were used for all the aggregated time series.
- The internal model accuracy (MAPE). For more information about forecast accuracy, see Monitor forecast accuracy.

> **Note**

- The confidence interval that appears in the **Forecast** section of the page represents the difference between the confidence interval upper limit and the confidence interval lower limit. To see the values for the upper and lower limits, hover over the chart in the **Historical demand and forecast graphically** section.
- If you use the Demand forecasting Microsoft Azure Machine Learning, you can specify the confidence level percentage that the forecast that is generated should have. A confidence interval consists of a range of values that act as good estimates for the demand forecast. A 95-percent confidence level percentage indicates that there is a 5-percent risk that the demand forecast falls outside the confidence interval range.

You can also make manual adjustments to the forecast on the **Demand forecast details** page, by modifying the values in the **Forecast** row in the **Forecast** section.

Related information

- Monitor forecast accuracy
- Generate a statistical baseline forecast

Authorize an adjusted forecast

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Not all forecast data must be authorized immediately. This article explains how you can specify the period that a forecast is authorized for. It also explains how you can authorize the forecast for specific companies and forecast models.

Not all forecast data must be authorized immediately. You can specify the start and end dates of the period that the forecast is authorized for. This functionality lets you freeze specific buckets.

The start and end dates that you specify must correspond to the start and end dates of the bucket that the forecast is generated in. The system enforces this restriction and automatically adjusts the dates, if adjustment is required.

On the **Details** tab of the **Authorization** page, you can view details about the forecast that was most recently generated.

You can select the companies and the forecast models to authorize the forecast for use. By default, the grid includes all the companies that forecast demand has been created for. For each company, the forecast model that corresponds to the current forecast plan that is set up in the master planning parameters is prefilled. However, you can change this forecast model to any forecast model that belongs to that company. If no forecast demand data has been generated for a selected company, you receive a warning message at import time.

It's very important that you understand how the **Save the manual adjustments made to the baseline demand forecast** checkbox works. If you've made manual adjustments to the

statistical baseline forecast, the adjusted values are authorized for use, even if this checkbox is cleared. However, the changes are discarded after the authorization. Therefore, the next time that a forecast is generated, that forecast is only a statistical forecast and doesn't have any manual overrides, even if **Transfer manual adjustments to the demand forecast** is selected. Therefore, you can consider the **Save the manual adjustments made to the baseline demand forecast** checkbox a mechanism that lets you keep or discard all manual changes.

Related information

- Make manual adjustments to the baseline forecast
- Monitor forecast accuracy

Last updated on 06/17/2025

Import historical data for demand forecasts

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To help guarantee the accuracy of demand forecasts, you must have as much historical demand data as you can get per item or item allocation key. If the historical demand data isn't already imported, use the **Historical external demand** (ReqDemPlanHistoricalExternalDemandEntity) data entity in Dynamics 365 Supply Chain Management to import it.

In the **Data management** workspace, you can see an overview of all the fields in the entity.

1. Open the **Data management** workspace.
2. Select the **Data entities** tile.
3. Search the entity list for **Historical external demand**.
4. Select **Target fields**. The following entity fields are mandatory: site (**DeliveringSiteId**), date (**DemandDate**), quantity (**DemandQuantity**), and either item number (**ItemNumber**) or item allocation key (**ProductAllocationKeyId**).

To use the data entity, you must have a Microsoft Excel file or comma-separated values (CSV) file that contains the historical demand data. The following example shows how to import the data from a CSV file.

For more information about how to import data, including how to clean up data after an import, see Data import and export jobs overview and its related articles.

Learn more in Generate a statistical baseline forecast.

Last updated on 06/17/2025

Remove outliers from historical transaction data when calculating a demand forecast

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This article describes how to exclude outliers from the historical data that is used to calculate a demand forecast. By excluding outliers, you can improve forecast accuracy.

You can exclude outliers to improve forecast accuracy. This is an optional task. Here is an overview of the process:

1. Go to **Master planning** > **Setup** > **Demand forecasting** > **Outlier removal** to open the **Outlier removal** page, where you can use a query to select the transactions to exclude.
2. Select the company that the query applies to, and then enter a name and description. The **Query date** field is automatically set to the current date.
3. Select the **Active** checkbox to exclude the transactions that the query finds from the historical data. This setting will take effect when you create a baseline forecast.
4. On the **Outlier removal query** page, you can add, remove, and select the criteria that define which transactions will be excluded when the baseline forecast is calculated. For example, select a specific item or order transaction to exclude.
5. Select **Display transactions**. The **Outlier transactions** page lists the transactions that meet the criteria that you defined in the query, and that will be excluded from the historical data when the demand forecast is calculated.

> **Note**

You can also create a query that is based on an existing query. Select the query to copy, and then select **Duplicate**. The **Query date** field identifies the version. You can use the query as it is, or you can select **Edit query** to modify the criteria. You can optionally modify the name and description of the new query.

Related information

- Demand forecasting overview
- Monitor forecast accuracy

Last updated on 06/17/2025

Monitor forecast accuracy

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This article describes the types of forecast accuracy that Microsoft Dynamics 365 Supply Chain Management calculates, and explains how you can view the accuracy values.

Supply Chain Management calculates the following types of forecast accuracy:

- Historical forecast accuracy, by comparing the historical forecast that Master Planning uses with the historical demand. To view the values (both absolute values and percentage values) for historical forecast accuracy, select **Show accuracy** on the **Demand forecast details** page.
- The estimated accuracy of the forecasting model that is used to generate the predictions. You can view the accuracy percentage under **Model details - MAPE** on the **Demand forecast details** page.

> **Note**
>
> If you use the Demand forecasting Microsoft Azure Machine Learning, the calculation of internal model accuracy is based on the test data set. To specify the size of the test data set, set the **TEST_SET_SIZE_PERCENT** parameter on the **Demand forecasting parameters** page. For example, if you set the value to **20**, the last 20 percent of the historical data will be used to calculate the internal model accuracy.

Related information

- Authorize an adjusted forecast
- Remove outliers from historical transaction data when calculating a demand forecast

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Last updated on 06/17/2025

Guide: Create a baseline forecast

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A production planner can create a baseline forecast either by using time series forecast models or by copying the historical demand. This procedure shows how to copy the historical demand to create a baseline forecast for all products using one item allocation key.

Set up an item allocation key

1. Go to **Master planning** > **Setup** > **Intercompany planning groups**.
2. Use the Quick Filter to find records. For example, filter on the *Name* field with a value of *10*.
    - Demand forecasting runs across legal entities. That's why you need to set up all the companies for which you want to generate forecasts in one intercompany planning group.
3. In the list, find and select the desired record.
4. Select **Item allocation keys**.
    - Select all the item allocation keys for which you want to create forecasts.
5. In the list, mark the selected row.
    - Select D_Aloc item allocation key.
6. Select >.

7. Close the page.
8. Close the page.

Set up the demand forecasting parameters

1. Go to **Master planning** > **Setup** > **Demand forecasting** > **Demand forecasting parameters**.
2. Expand the **Forecast algorithm parameters** section.
3. In the **Forecast generation strategy** field, select **Copy over historical demand**.
4. Select **Save**.

Create a baseline forecast

1. Go to **Master planning** > **Forecasting** > **Demand forecasting** > **Generate statistical baseline forecast**.
2. In the **From date** field, enter a date.

    - If you have sales orders starting from January 1, 2015, enter this date. If you don't, enter the earliest date of your sales orders.

3. In the **To date** field, enter a date.

    - Enter the last date of your sales orders, for example *2015-03-31*.

4. In the **From date** field, enter a date.

    - Enter *2015-04-01*. This date will be automatically calculated as the start date of the next forecasting bucket.

5. Expand the **Records to include** section.
6. Select **Filter**.
7. In the list, mark the selected row.

    - Mark the row where **Field** = *Intercompany planning group*.

8. In the **Criteria** field, type a value.

    - Type the intercompany planning group (for example, *10*) that you used in the first task.

9. In the list, find and select the desired record.

    - Select the row where **Field** = *Item allocation key*.

10. In the **Criteria** field, type a value.
11. Select **OK**.
12. Expand the **Advanced parameters** section.
13. In the **Forecast bucket** field, select *Month*.
14. In the **Forecast horizon** field, enter *3*.
15. In the **Freeze time fence** field, enter *1*.
16. Select **OK**.

Visualize the demand forecast

1. Go to **Master planning** > **Forecasting** > **Demand forecasting** > **Adjusted demand forecast**.
2. In the aggregated view table, select the cell in row 1, column 2. This is the second month for which you have created a forecast.
3. Set **QtyCell** to *400*.

    - In the cell, enter a different number than the one that was forecasted, for example, 400.

4. You have made a manual adjustment to the forecast. Notice the graphical indication in the next step.
5. Select **Forecast line details**.

    - In this page, you can see the accuracy values, historical demand, and forecast. You can make changes to the forecast as well.

Last updated on 06/17/2025

Forecast reduction keys

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This article provides information about the different methods that you can use to reduce forecast requirements. It includes examples of the results of each method. It also explains how to create, set up, and use a forecast reduction key. Some methods use a forecast reduction key to reduce forecast requirements.

Methods to reduce forecast requirements

When you include a forecast in a master plan, you can select how the forecast requirements are reduced when actual demand is included. Master planning excludes forecast requirements from the past, which means all forecast requirements before today's date.

To include a forecast in a master plan and select the method to reduce forecast requirements, go to **Master planning** > **Setup** > **Plans** > **Master plans**. In the **Forecast model** field, select a forecast model. In the **Method used to reduce forecast requirements** field, select a method. The following options are available:

- *None*
- *Percent - reduction key*
- *Transactions - reduction key*
- *Transactions - dynamic period*

The following sections provide more information about each option.

None

If you select *None*, the forecast requirements aren't reduced during master scheduling. In this case, master planning creates planned orders to supply the forecasted demand (forecast requirements). These planned orders maintain the suggested quantity, regardless of other types of demand. For example, if sales orders are placed, master planning creates additional planned orders to supply the sales orders. The quantity of the forecast requirements isn't reduced.

Percent - reduction key

If you select *Percent - reduction key*, the system reduces the forecast requirements according to the percentages and periods that the reduction key defines. In this case, master planning creates planned orders where the quantity is calculated as forecasted quantity × reduction key in each period. If other types of demand exist, master planning also creates planned orders to supply that demand.

Example: Percent - reduction key

This example shows how a reduction key reduces demand forecast requirements according to the percentages and periods that the reduction key defines.

For this example, you include the following demand forecast in a master plan.

Expand table

Month	Demand forecast
January	1,000
February	1,000
March	1,000
April	1,000

On the **Reduction keys** page, you set up the following lines.

Expand table

Change	Unit	Percent
1	Month	100

Change	Unit	Percent
2	Month	75
3	Month	50
4	Month	25

You assign the reduction key to the item's coverage group. Then, on the **Master plans** page, in the **Method used to reduce forecast requirements** field, you select *Percent - reduction key*.

In this case, if you run forecast scheduling on January 1, the system consumes the demand forecast requirements according to the percentages that you set up on the **Reduction keys** page. The following requirement quantities are transferred to the master plan.

Expand table

Month	Planned order quantity	Calculation
January	0	= 0% × 1,000
February	250	= 25% × 1,000
March	500	= 50% × 1,000
April	750	= 75% × 1,000
May through December	1,000	= 100% × 1,000

Transactions - reduction key

If you set the **Method used to reduce forecast requirements** field to *Transactions - reduction key*, the system reduces the forecast requirements by the qualified demand transactions that occur during the periods defined by the reduction key.

The **Reduce forecast by** field on the **Coverage groups** page defines the qualified demand. If you set the **Reduce forecast by** field to *Orders*, only sales order transactions are considered qualified demand. If you set it to *All transactions*, any non-intercompany issue inventory transactions are considered qualified demand. If intercompany sales orders should also be considered qualified demand, set the **Include intercompany orders** option to *Yes*.

Forecast reduction starts with the first (earliest) demand forecast record in the reduction key period. If the quantity of qualified inventory transactions is more than the quantity of demand forecast lines in the same reduction key period, the balance of inventory transactions quantity doesn't reduce previous or future periods.

The value of the **Percent** field on the reduction key lines isn't used when the **Method used to reduce forecast requirements** field is set to *Transactions - reduction key*. Only the dates are used to define the reduction key period.

> **Note**
>
> The system ignores any forecast that you post on or before today's date and doesn't use it to create planned orders. For example, if you generate a demand forecast for the month on January 1, and you run master planning that includes demand forecasting on January 2, the calculation ignores the demand forecast line dated January 1.

Example: Transactions - reduction key

This example shows how actual orders that occur during the periods defined by the reduction key reduce demand forecast requirements.

For this example, you select *Transactions - reduction key* in the **Method used to reduce forecast requirements** field on the **Master plans** page.

The following sales orders exist on January 1.

Expand table

Month	Number of pieces ordered
January	956
February	1,176
March	451
April	119

If you use the same demand forecast of 1,000 pieces per month that was used in the previous example, the following requirement quantities are transferred to the master plan.

Expand table

Month	Number of pieces required
January	44
February	0
March	549

Month	Number of pieces required
April	881
May through December	1,000

Transactions - dynamic period

If you select *Transactions - dynamic period*, the system reduces the forecast requirements by the actual order transactions that occur during the dynamic period. The dynamic period covers the current forecast dates and ends at the start of the next forecast. In this case, master planning creates planned orders to supply the forecasted demand (forecast requirements). However, when actual order transactions are placed, the forecast requirements are reduced. The actual transactions consume part of the forecasted requirements.

When you use this option, the following behavior occurs:

- The system doesn't require or use reduction keys.
- If the forecast is completely reduced, the forecast requirements for the current forecast become 0 (zero).
- If there's no future forecast, the system reduces forecast requirements from the last forecast that you entered.
- The demand forecast reduction time fence isn't included in the forecast reduction calculation. Instead, the coverage group time fence is used for forecast reduction.
- Positive days are included in the forecast reduction calculation.
- If actual order transactions exceed the forecasted requirements, the system doesn't forward the remaining transactions to the next forecast period.

Example 1: Transactions - dynamic period

Here's a simple example that shows how the *Transactions - dynamic period* method works.

For this example, you include the following demand forecast in a master plan.

Expand table

Date	Demand forecast
January 1	1,000
February 1	1,000

You also create the following sales orders.

Expand table

Date	Sales order quantity
January 15	200
February 15	400

In this case, the system creates the following planned orders.

Expand table

Demand forecast date	Quantity	Explanation
January 1	800	Forecast requirements (= 1,000 – 200)
January 15	200	Sales orders requirements
February 1	600	Forecast requirements (= 1,000 – 400)
February 15	400	Sales orders requirements

Example 2: Transactions - dynamic period

In most cases, systems are set up so that transactions reduce demand forecasts in specific forecast periods: weeks, months, and so on. These periods are defined in the reduction key. However, the time between two demand forecast lines can also *imply* a period.

For this example, you create a demand forecast for the following dates and quantities.

Expand table

Date	Demand forecast
January 1	1,000
January 5	500
January 12	1,000

Notice that, in this forecast, there isn't a clear period between the forecast dates. Between the first and second dates there's a four-day span, and between the second and third dates there's a seven-day span. These spans are the dynamic periods.

You also create the following sales order lines.

Expand table

Date	Sales order quantity
December 15 in the previous year	500
January 3	100
January 10	200

In this case, the forecast is reduced in the following manner:

- The first sales order isn't in any period, so it doesn't reduce any forecast.
- The second sales order is between January 1 and January 5, so it reduces the forecast for January 1 by 100.
- The third sales order is between January 5 and January 12, so it reduces the forecast for January 5 by 200.

Therefore, the following planned orders are created.

Expand table

Demand forecast date	Quantity	Explanation
December 15 in the previous year	500	Sales order requirements
January 1	900	Forecast requirements period January 1 to January 5 (= 1,000 – 100)
January 3	100	Sales order requirements
January 5	300	Forecast requirements period January 5 to January 10 (= 500 – 200)
January 12	1,000	Forecast requirements period January 12 to end

Create and set up a forecast reduction key

Use a forecast reduction key with the *Transactions - reduction key* and *Percent - reduction key* methods to reduce forecast requirements. Follow these steps to create and set up a reduction key.

1. Go to **Master planning** > **Setup** > **Coverage** > **Reduction keys**.
2. Select **New** to create a reduction key.

3. In the **Reduction key** field, enter a unique identifier for the forecast reduction key. Then, in the **Name** field, enter a name.

4. Define the periods and the reduction key percentage in each period:

    - The **Effective date** field indicates the date when creation of the periods starts. When the **Use the effective date** option is set to *Yes*, the periods start on the effective date. When it's set to *No*, the periods start on the date when master planning is run.
    - Define the periods that the forecast reduction should occur during.
    - For a specific period, specify the percentages that the forecast requirements should be reduced by. Enter positive values to decrease requirements or negative values to increase requirements.

Use a reduction key

You must assign a forecast reduction key to the coverage group of the item. Follow these steps to assign a reduction key to an item's coverage group.

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.

2. On the **Other** FastTab, in the **Reduction key** field, select the reduction key to assign to the coverage group. The reduction key then applies to all items that belong to the coverage group.

3. To use a reduction key to calculate forecast reduction during master scheduling, you must define this setting in the setup of the forecast plan or the master plan. Go to one of the following locations:

    - **Master planning** > **Setup** > **Plans** > **Forecast plans**
    - **Master planning** > **Setup** > **Plans** > **Master plans**

4. On the **Forecast plans** or **Master plans** page, on the **General** FastTab, in the **Method used to reduce forecast requirements** field, select either *Percent - reduction key* or *Transactions - reduction key*.

Reduce a forecast by transactions

When you select *Transactions - reduction key* or *Transactions - dynamic period* as the method for reducing forecast requirements, you can specify which transactions reduce the forecast. On the **Coverage groups** page, on the **Other** FastTab, in the **Reduce forecast by** field, select **All transactions** if all transactions should reduce the forecast or **Orders** if only sales orders should reduce the forecast.

Related information

- Master plans overview

Last updated on 11/19/2025

Demand Driven Material Requirements Planning (DDMRP) overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

For years, companies have used material requirements planning (MRP) as a system for calculating the materials and components that are required to manufacture a product. However, supply chains have changed. Parts have longer lead times because companies increasingly source them from overseas. Therefore, many companies experience stockouts or overstocks because they don't know how much inventory to stock. There's also more market fluctuation (sometimes inaccurately forecast), and customers are demanding products at a short lead time. Therefore, there are supply chain shortages all over the world. In addition, MRP tools often give planners thousands of actions to do. Therefore, it's hard to know what to focus on. Often, the solution to many of these issues is to switch to Demand Driven Material Requirements Planning (DDMRP).

DDMRP is a planning methodology that's based on the decoupling of supply and demand. You achieve this decoupling by setting up decoupling point items. For those items, you maintain buffers to ensure that you keep the correct amount of stock. The decoupling of supply and demand helps prevent the "bullwhip effect" because variability isn't passed through the chain. (The *bullwhip effect* refers to how small fluctuations in demand at the retail level can cause progressively larger fluctuations in demand at the wholesale, distributor, manufacturer, and raw material supplier levels.) Each buffer is intended to cover the average use of a part and can also be adjusted to cover demand spikes.

DDMRP is a valuable planning methodology for variable environments where customer tolerance times are shorter than cumulative lead times.

The five components of DDMRP

DDMRP has five sequential components (steps). The first three components essentially define the initial and evolving configuration of a demand-driven material requirements planning model. The last two components define the day-to-day operation of the methodology.

1. **Strategic inventory positioning** – Identify decoupling points in the supply chain network. Decoupling points are specific points of your supply chain where you put an inventory buffer that you monitor and replenish.
2. **Buffer profiles and levels** – For each decoupling point, identify the buffer sizes (minimum quantity, maximum quantity, and reorder point) and the reorder quantity.
3. **Dynamic buffer adjustments** – Adjust buffer levels, based on varying operating parameters or planned future events.
4. **Demand-driven planning** – Generate supply orders as they're required. These supply orders include manufacturing orders, purchase orders, and stock transfer orders.
5. **Highly collaborative and visible execution** – Run the supply orders with the help of visualization.

Manufacturers typically use DDMRP with a multilevel bill of materials (BOM). However, you can also apply it to distribution and retail networks.

DDMRP in Dynamics 365 Supply Chain Management

Microsoft Dynamics 365 Supply Chain Management includes DDMRP with no additional license fees. In Supply Chain Management, developers added DDMRP functionality to the existing **Master planning** module. However, it requires that you use the Planning Optimization Add-in.

DDMRP integrates with the existing planning setups in Supply Chain Management. Use it together with those setups to arrive at the correct planning configuration for your business. A new coverage code controls it. This code is completely different from period, min/max, requirement, and so on. DDMRP isn't a new module, and it doesn't replace existing planning functionality. However, it does give you more functionality to use.

Last updated on 03/25/2026

Turn on the DDMRP feature for your system

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the **Request access to Finance and Operations Viva Engage Community** form and choose the community you want to join.

To use Demand Driven Material Requirements Planning (DDMRP) functionality, it must be enabled for your system. As of version 10.0.36, this feature is turned on by default. As of Supply Chain Management version 10.0.43, it's mandatory and can't be turned off. If you're running a version older than 10.0.43, then admins can turn this functionality on or off by searching for the *DDMRP for Planning Optimization* feature in the Feature management workspace.

Learn more in Priority-based planning.

Last updated on 06/17/2025

Inventory positioning

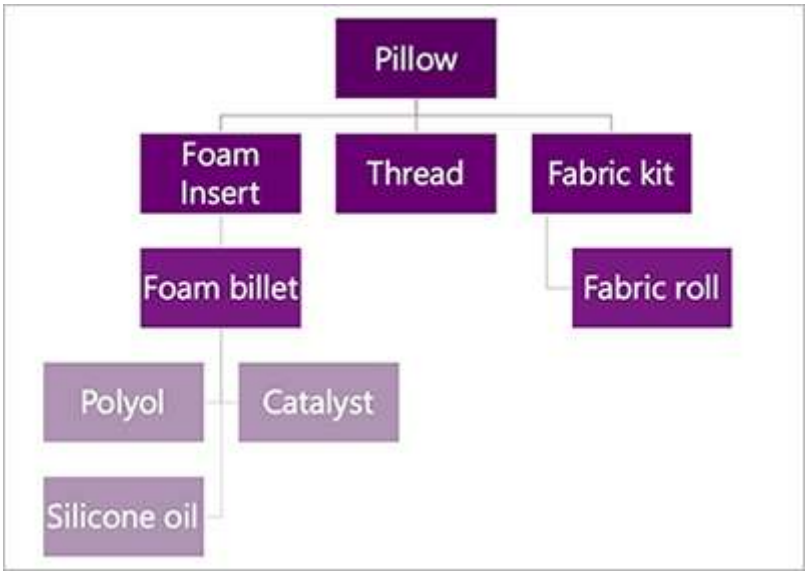
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Strategic inventory positioning involves identifying decoupling points in your supply chain where you can build up on-hand inventory. Use this approach to help compress lead times and absorb shocks to your supply chain. It helps you mitigate the "bullwhip effect," because demand variability doesn't pass all the way down the supply chain. (The *bullwhip effect* refers to how small fluctuations in demand at the retail level can cause progressively larger fluctuations in demand at the wholesale, distributor, manufacturer, and raw material supplier levels.)

Inventory positioning is the first step of Demand Driven Materials Resource Planning (DDMRP).

Inventory positioning for manufacturing

This section provides an example that shows how to make inventory positioning decisions if you manufacture a typical pillow product. The pillow has a multilevel bill of materials (BOM), as shown in the following illustration.



Choose your decoupling points

When you choose where to put your decoupling points, consider all the following aspects of each item in the BOM as criteria:

- External variability
- Inventory leverage and flexibility
- Critical operation protection
- Customer tolerance time
- Sales order visibility horizon
- Market potential lead time

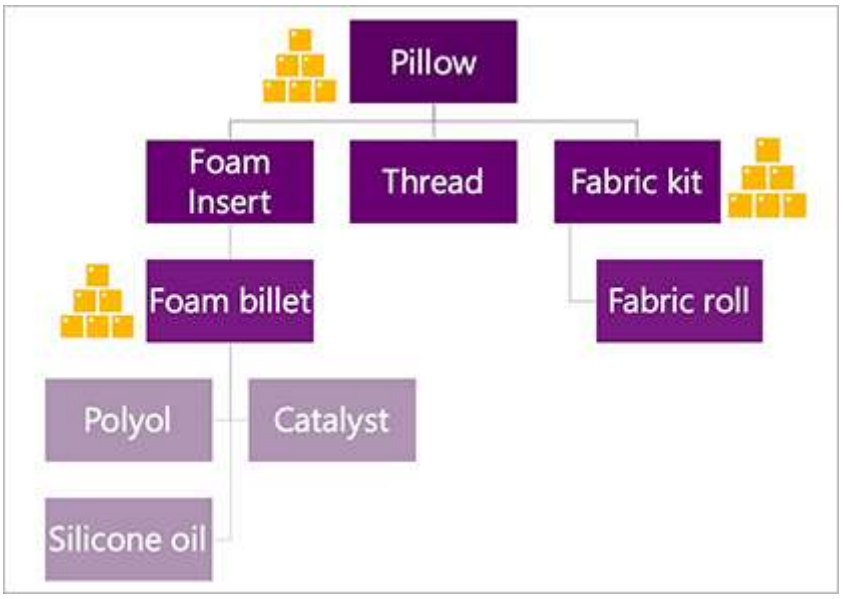
In the pillow example, you might put your first decoupling point at the *foam billets* for the following reasons:

- It's difficult to source the materials that are used to make the foam billets, and availability is volatile. Therefore, the *external variability* criterion is met.
- The foam billets can be cut into many different shapes and sizes to create foam inserts for other products that you manufacture, in addition to the pillow. Therefore, the *inventory leverage and flexibility* criterion is met.

You might then put your next decoupling point at the *fabric kit*, which is precut pillow fabric. You might choose this point because you have only one fabric cutting machine. Therefore, the *critical operation protection* criterion is met.

Finally, you might put your last decoupling point at the finished good pillow item. You might choose this point because you have a very low *customer tolerance time* on sales, and because your *sales order visibility horizon* is fairly short. Therefore, you want to ensure that you have the on-hand inventory to meet incoming orders. You can also set a higher price by keeping the lead time this short, which is what the *market potential lead time* criterion refers to.

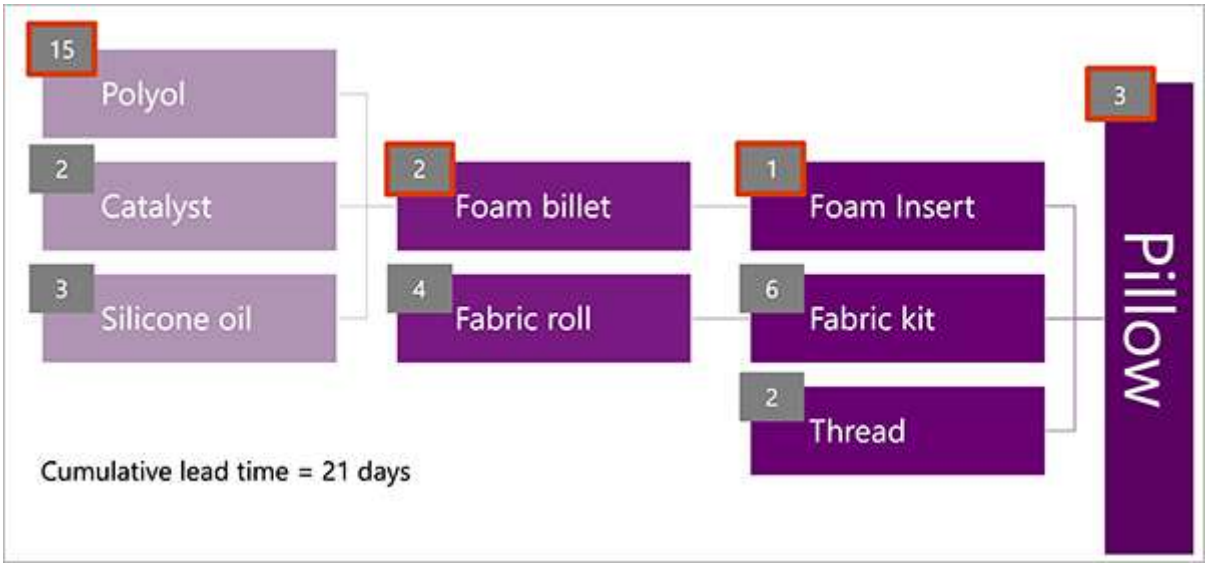
Based on this analysis, the following illustration shows what the pillow BOM looks like. Yellow inventory symbols highlight the decoupling points.



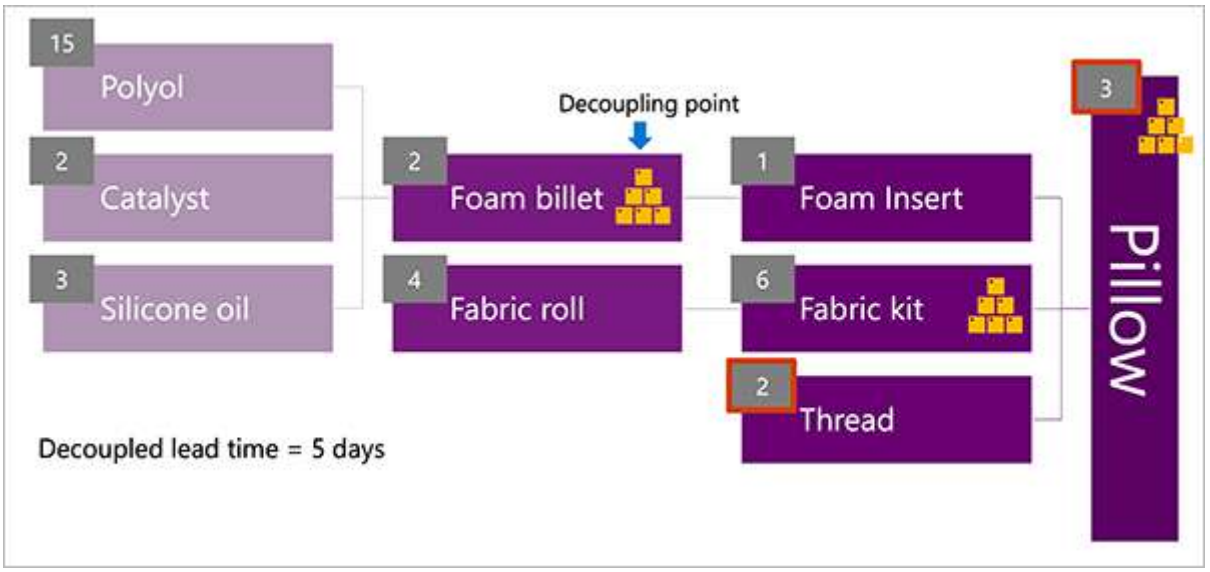
Calculate your decoupled lead time

This section shows how to calculate your new lead times after you introduce decoupling points.

In the following illustration for the pillow example that was started in the previous section, lead times appear in gray boxes at the upper left of each BOM component. Boxes that have a red outline indicate items that drive the cumulative lead time (the sum of the longest lead times at each level of the BOM). This lead time is 21 days when you start from scratch.



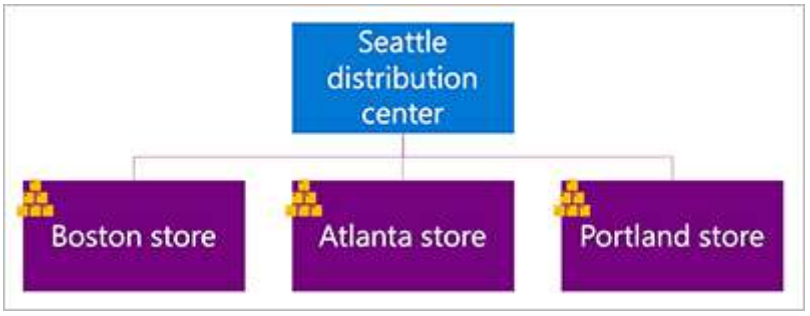
However, if you apply the decoupling points that you previously chose, the decoupled items are always in stock. Therefore, they have a lead time of 0 (zero). The new lead time for the pillow is now just five days: two days to purchase the thread and three days to produce the pillow. This lead time is known as the *decoupled lead time*.



Strategic inventory positioning in a retail model

Because retailers stock only finished products, BOMs aren't an issue. However, retailers can still use DDMRP by setting strategic inventory positioning and buffer levels based on storage locations in the distribution network.

The following illustration shows an example of a company that has a distribution center in Seattle, and stores in Boston, Atlanta, and Portland.



You might decide that the transfer time to move a blanket product between the distribution center and the stores violates your *customer tolerance time*, because your customers expect the blanket to be in stock when they visit. In this case, you set up a decoupling point for the blanket item at each of the three stores. Each store has different buffer levels, based on its lead times, demand patterns, and so on.

Implement inventory positioning in Dynamics 365 Supply Chain Management

This section describes how to implement your inventory positioning strategy in Microsoft Dynamics 365 Supply Chain Management.

Set up item coverage groups that create decoupling points

Items become decoupling points when they belong to a coverage group that you configure with a **Coverage code** value of *Decoupling point*. Therefore, the first step in the process of setting up DDMRP is to decide which coverage groups you must implement for your DDMRP strategy, and then create them by following these steps.

1. Go to **Master planning** > **Setup** > **Coverage** > **Coverage groups**.
2. On the Action Pane, select **New** to create a coverage group.
3. Enter information that identifies the coverage group, and then select the calendar to use.
4. On the **General** tab, set the **Coverage code** field to *Decoupling point*. This setting causes all items that belong to this coverage group to be treated as decoupling points for DDMRP. It also enables all DDMRP settings for this group, as described later in this procedure.
5. On the **Other** tab, in the **DDMRP parameters** section, set the following fields:
    - **Lead time factor** – Specify a factor (as a decimal value between 0 and 1) to control the impact that lead time should have when the minimum and maximum stock levels are calculated for items in this coverage group. In general, the longer the lead time an item has, the lower its lead time factor should be. A lower lead time factor produces lower minimum and maximum stock levels, and therefore causes smaller and more frequent orders. DDMRP methodology recommends a value between 0.20 and 0.40 for items that have long lead times, between 0.41 and 0.60 for items that have medium lead times, and between 0.61 and 1.00 for items that have short lead times. Learn more in Buffer profile and levels.
    - **Variability factor** – Specify a factor (as a decimal value between 0 and 1) to control the impact that varying demand should have when the minimum stock level is calculated for items in this coverage group. In general, the more variable an item's demand is, the higher its variability factor should be. A higher variability factor produces a higher minimum stock level. DDMRP methodology recommends a value between 0.00 and 0.40 for items that have low variability, between 0.41 and 0.60 for items that have medium variability, and between 0.61 and 1.00 for items that have high variability. Learn more in Buffer profile and levels.
    - **Min, max, and re-order point period** – Specify how often to calculate buffer values (*Daily* or *Weekly*).
6. On the **Other** tab, in the **Average daily usage** section, set the following fields:
    - **Average daily usage based on** – Select which time periods the calculation of the average daily usage (ADU) should be based on. Select one of the following values:

- *Past* – Look only at past usage for the number of days that you specify in the **Past period (days)** field. The ADU is calculated as the total demand for an item during the calculation period (in inventory units) divided by the number of days in the calculation period.
- *Forward* – Look only at projected future usage (including forecasts) for the number of days that you specify in the **Forward period (days)** field. The ADU is calculated as the total demand for an item during the calculation period (in inventory units) divided by the number of days in the calculation period.
- *Blended* – Look at both the past and future usage. Settings for the **Past period (days)** field, the **Forward period (days)** field, and blending options all apply.

  Blended ADU = ([*Past weighting* × *Past ADU*] + [*Forward weighting* × *Forward ADU*]) ÷ (*Past weighting* + *Forward weighting*)

- **Past period (days)** – Enter the number of past days (up to and including today) that the system should consider when it calculates the ADU of items in this coverage group. This setting applies only when the **Average daily usage based on** field is set to *Past* or *Blended*.
- **Forward period (days)** – Enter the number of future days (from today and up to the specified day) that the system should consider when it calculates the ADU of items in this coverage group. This setting applies only when the **Average daily usage based on** field is set to *Forward* or *Blended*.
- **Relative weight of past period for blended average daily usage** – Enter the weight (as a percentage) to apply to the past period when the blended ADU is calculated. This setting applies only when the **Average daily usage based on** field is set to *Blended*.
- **Relative weight of forward period for blended average daily usage** – Enter the weight (as a percentage) to apply to the forward period when the blended ADU is calculated. This setting applies only when the **Average daily usage based on** field is set to *Blended*.

7. For all other tabs and fields, enter the detailed settings that are used to calculate requirements for the items that are linked to this coverage group.

Set an item as a decoupling point

To set an item as a decoupling point, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.

2. Find and select a released item that you want to set up for DDMRP.
3. On the Action Pane, on the **Plan** tab, select **Item coverage**.
4. On the **Item coverage** page, you might see several item coverage records, each of which applies to a different combination of storage and product dimensions. You can select an existing item coverage record that applies to the dimensions where you want to create a decoupling point. Alternatively, you can select **New** on the Action Pane to create a new item coverage record.
5. Set up the item coverage record as usual. At a minimum, you must specify the site and warehouse where the decoupling point applies.
6. While the appropriate record is still selected, select the **General** tab.
7. Select the **Use specific settings** checkbox.
8. Set the **Coverage group** field to a coverage group that is set up to create decoupling points (as described in the previous section).
9. The item is now configured as a decoupling point. Usually, when you use DDMRP, you also configure settings here that affect buffer sizes and the reorder quantity. However, you can complete that configuration later. For more information about the settings, see Set up buffers for a decoupling point item.

> **Note**
>
> To plan items that aren't decoupling points, follow the same steps that you follow when standard material requirements planning (MRP) is used.

Last updated on 11/19/2025

Buffer profile and levels

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

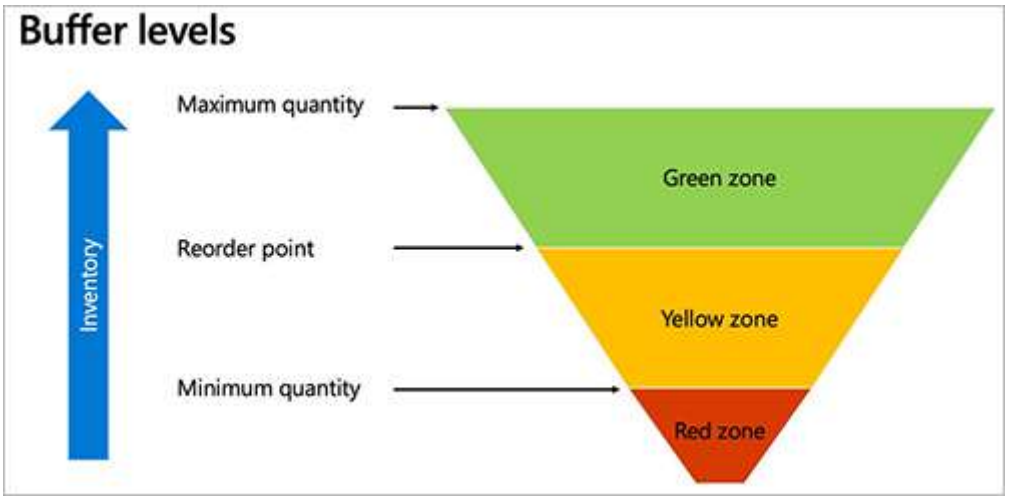
After you identify your decoupling points (key items that you strategically keep in stock), decide how much stock (buffer) to keep at each one. This task is the second step of Demand Driven Materials Resource Planning (DDMRP).

Buffer levels and zones

In DDMRP, each stock buffer is defined by using three values: the minimum quantity, the maximum quantity, and the reorder point. These values establish three difference zones, which are identified by the following color codes:

- **Red zone** – The area below the minimum quantity. The minimum quantity is also referred to as "top of red," and your planning strategy should be designed to ensure that stock levels are always above this point.
- **Yellow zone** – The area between the minimum quantity and the reorder point. The reorder point is also referred to as "top of yellow." When this point is reached, the system should reorder.
- **Green zone** – The area between the reorder point and the maximum quantity. The maximum quantity is also referred to as "top of green." This point is the maximum level that the stock is replenished to.

The following illustration shows the three colored zones and how they relate to the minimum quantity, maximum quantity, and reorder point.



Calculating the buffer zones

This section explains how the height of each buffer zone is calculated.

The yellow zone is typically calculated first. This zone represents the quantity that you consume from the moment when you order until the order arrives. In other words, it's the expected consumption during the replenishment lead time. It's calculated by using the following equation:

- **Yellow zone** = *Average daily usage (ADU)* × *Decoupled lead time*

The *decoupled lead time* represents the time that is required to produce or receive an item if the decoupling points are always in stock. It's usually much shorter than the *cumulative lead time* that is traditionally used in master planning. Correct buffer settings are key to ensuring that decoupling points are always actually in stock but not overstocked.

The red zone is calculated by using the following equations:

- **Red base** = *ADU* × *Decoupled lead time* × *Lead time factor*
- **Red safety** = *Red base* × *Variability factor*
- **Red zone** = *Red base* + *Red safety*

The green zone is calculated as the maximum result of the following three equations:

- *Minimum order quantity*
- *ADU* × *Order cycle*
- *ADU* × *Decoupled lead time* × *Lead time factor*

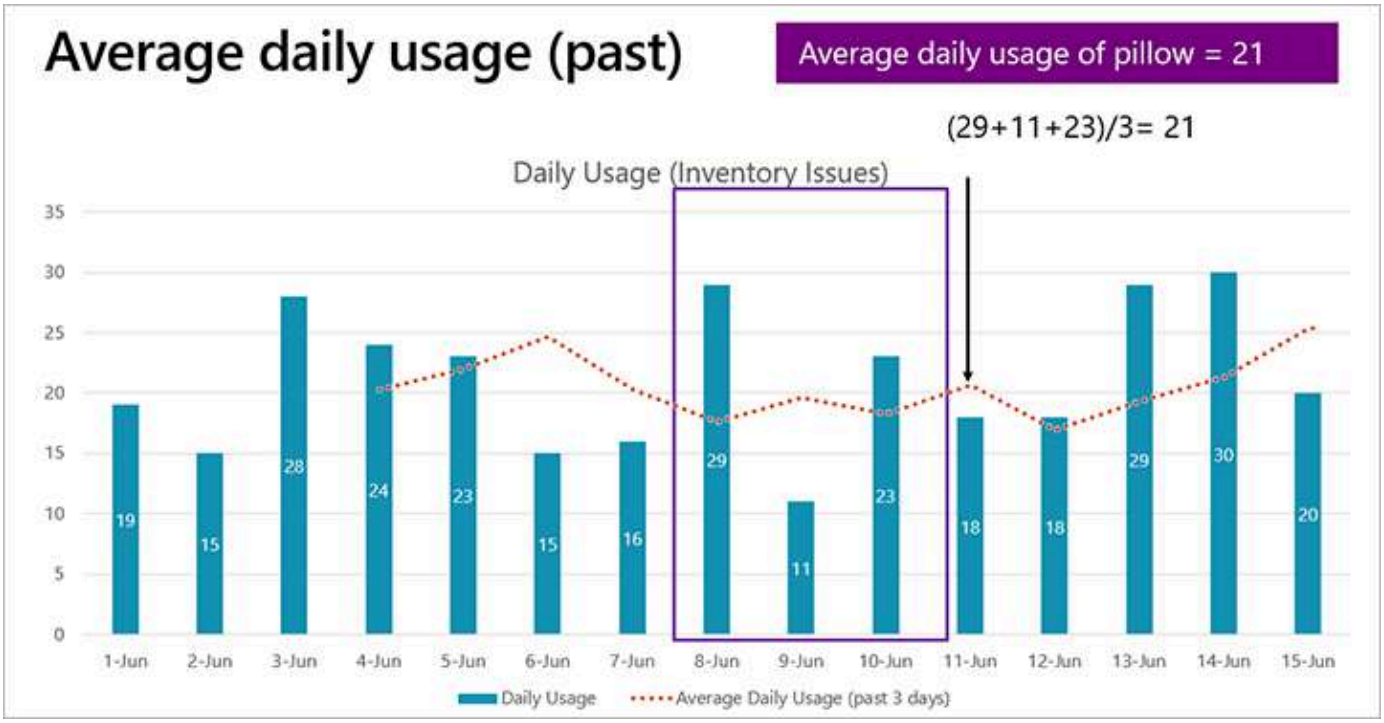
Calculating average daily usage

The system uses one of three approaches to calculate the amount that you consume per day:

- **Average daily usage (past)** – This approach is based on actual past consumption.
- **Average daily usage (forward)** – This approach is based on the forecasted future consumption.
- **Average daily usage (blended)** – This approach is based on a weighted mix of past and forecasted consumption.

Average daily usage (past)

Past ADU is calculated as an average by adding the quantities that are used each day for a specified number of past days and then dividing the total by the number of days. The following illustration shows how this approach works when the calculation looks three days into the past.



In the previous illustration, if today is the morning of June 11, the ADU for the previous three days (June 8, 9, and 10) is 21.

- **ADU (past)** = (29 + 11 + 23) ÷ 3 = 21

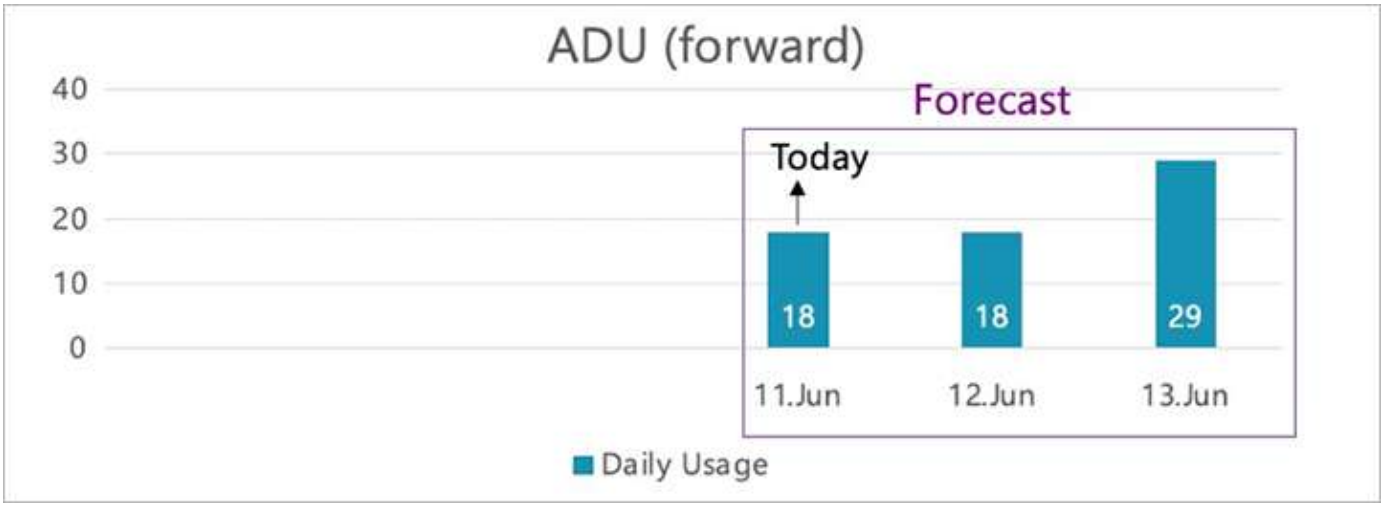
The following transactions are taken into account for the average daily usage (past) calculation:

- Transactions that diminish the quantity of the item (in the `inventtrans` table where quantity is less than zero)
- Transactions with a status of *On order*, *Reserved ordered*, *Reserved physical*, *Picked*, *Deducted*, or *Sold*
- Transactions dated within the chosen backwards period (the average daily usage past period)

- Transactions other than warehouse work, quarantine, sales quotations, or statements (`WHSWork`, `WHSQuarantine`, `SalesQuotation`, or `Statement`)
- Transactions other than transfer journals that are within the same coverage dimension

Average daily usage (forward)

For a new product, you might not have any past usage data. Therefore, you might instead use the projected ADU going forward (for example, based on forecasted demand). The following illustration shows how this approach works when the calculation looks three days into the future (including today).



In the previous illustration, if today is the morning of June 11, the ADU for the next three days (June 11, 12, and 13) is 21.66.

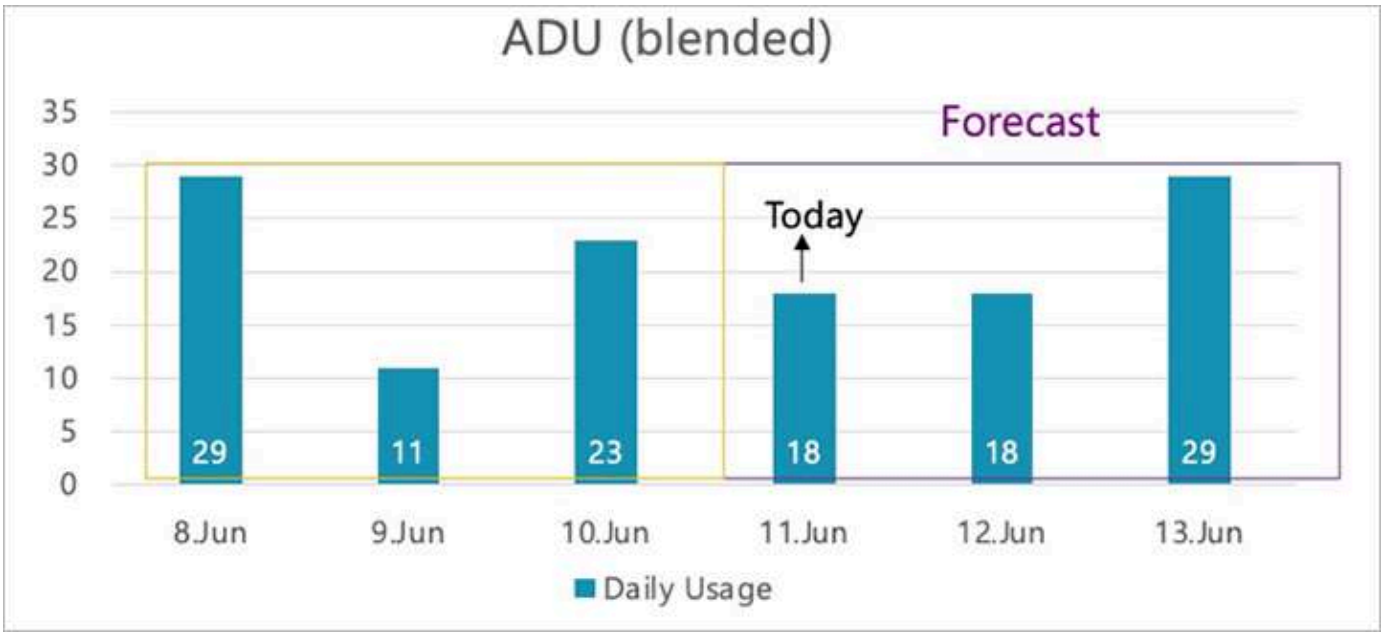
- **ADU (forward)** = (18 + 18 + 29) ÷ 3 = 21.66

The following transactions are taken into account for the average daily usage (forward) calculation:

- Forecast transactions for the item where the forecast is selected on the master plan
- Transactions dated within the chosen forward period (the average daily usage forward period)

Average daily usage (blended)

The blended ADU combines the average past usage and average forward usage, as shown in the following illustration.

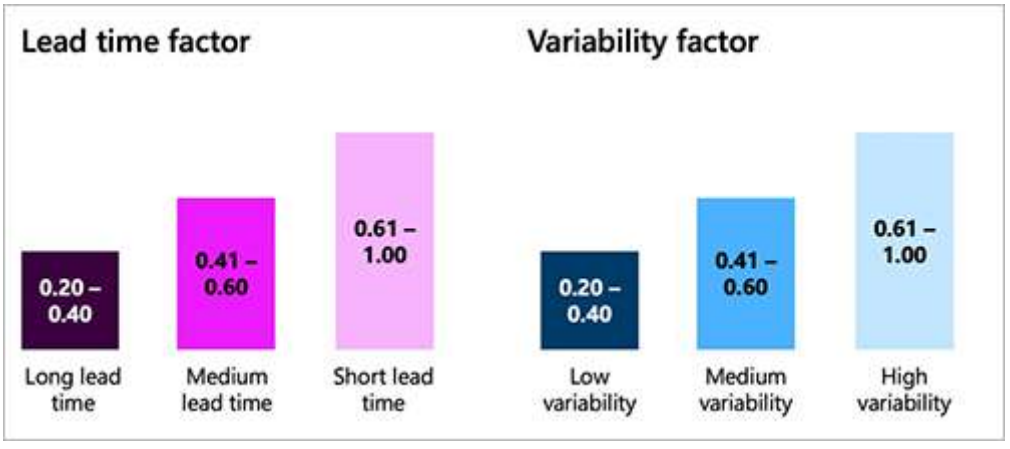


In the previous illustration, if today is the morning of June 11, the blended ADU for the previous and next three days (June 8 to 13) is 21.33.

- **ADU blended** = (*ADU past* + *ADU forward*) ÷ 2
  = (21 + 21.66) ÷ 2
  = 21.33

Buffer calculation factors

For each item, you can define two factors to adjust how large the red and green zones should be. In this way, you can compensate for the expected lead time and demand variability.



The first factor is the *lead time factor*. The value is a decimal value from 0 to 1. The longer the lead time is, the lower the value should be. The Demand Driven Institute recommends the following ranges:

- **Long lead time:** 0.20–0.40
- **Medium lead time:** 0.41–0.60

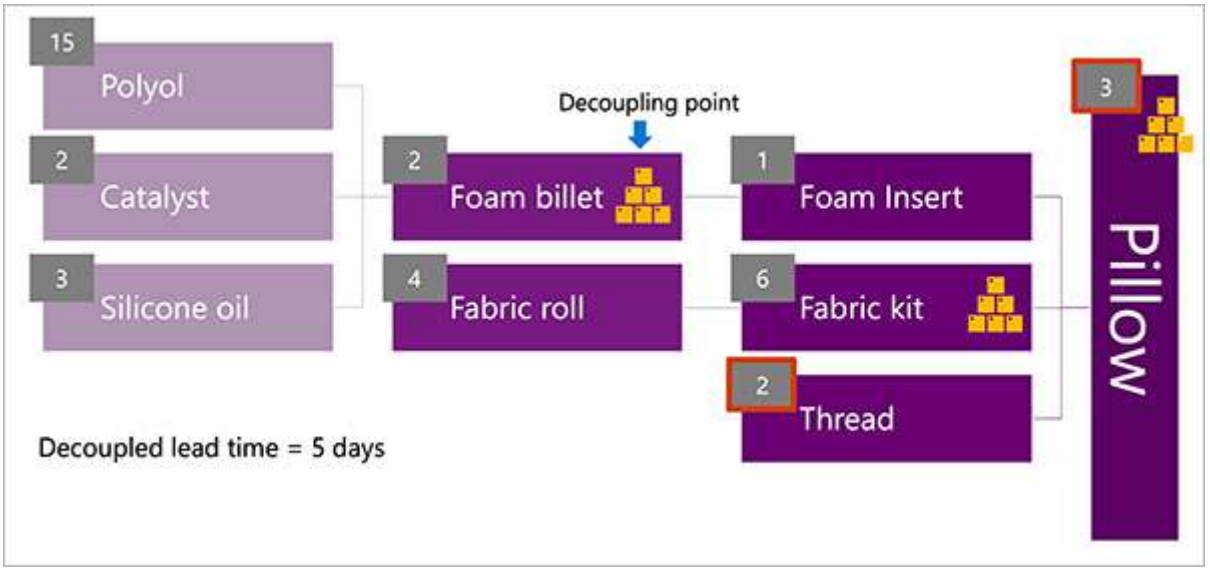
- **Short lead time:** 0.61–1.00

The second factor is the *variability factor*. The value is a decimal value from 0 to 1. The higher the demand variability is, the higher the value should be. The Demand Driven Institute recommends the following ranges:

- **Low variability:** 0.20–0.40
- **Medium variability:** 0.41–0.60
- **High variability:** 0.61–1.00

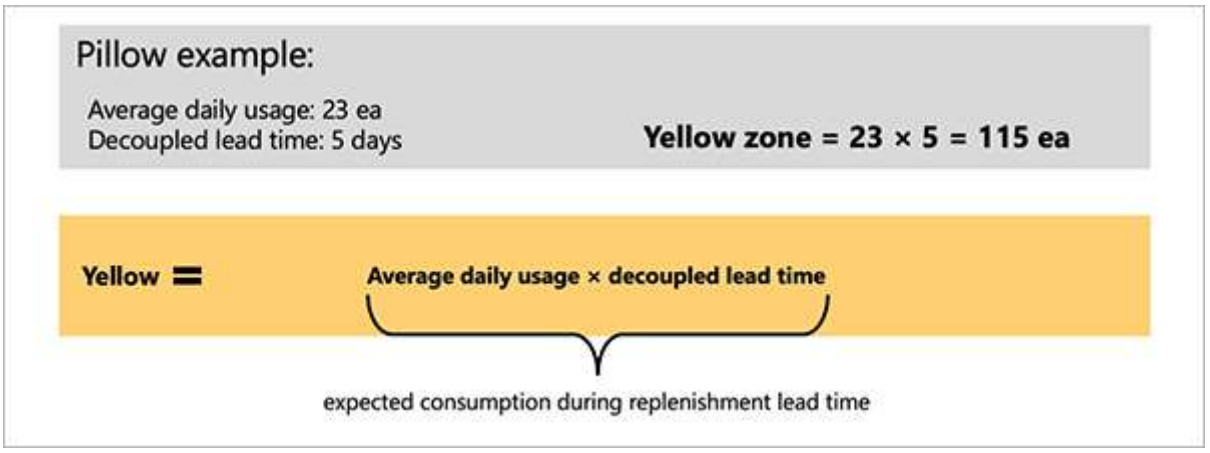
Buffer calculation examples

This example continues the pillow production example that is provided in Inventory positioning. In that example, you selected decoupling points that reduced the lead time from 21 days to five days, as shown in the following illustration.



For this example, assume that the ADU has been calculated as 23 pieces and, as shown in the previous illustration, the decoupled lead time is five days. By using these values, you can calculate the yellow zone by using the following equation:

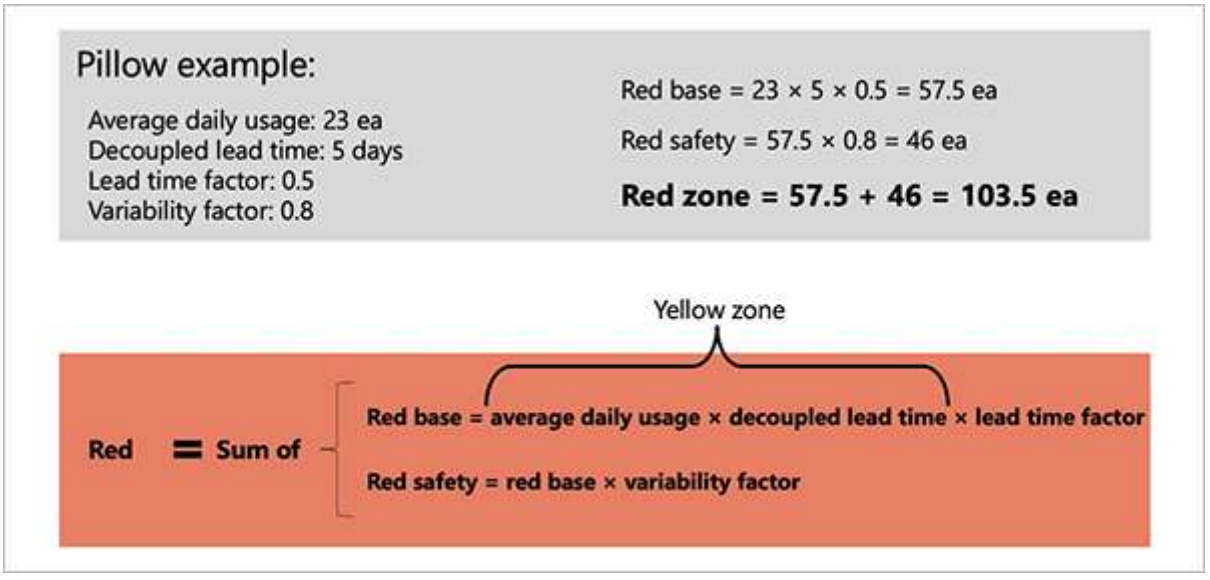
- **Yellow zone** = *ADU* × *Decoupled lead time* = 115



The red zone calculation resembles the yellow zone calculation, but it's padded for variability and lead time. For this example, assume that you've observed a medium lead time (factor = 0.50) and high demand variability (factor = 0.8). By using these values together with the components from the yellow zone equation, you can calculate the red zone by using the following equations:

- **Red base** = *ADU* × *Decoupled lead time* × *Lead time factor* = 57.5
- **Red safety** = *Red base* × *Variability factor* = 46
- **Red zone** = *Red base* + *Red safety* = 103.5

The system will round the red zone to 104 pieces (ea), because pieces are counted in whole numbers.



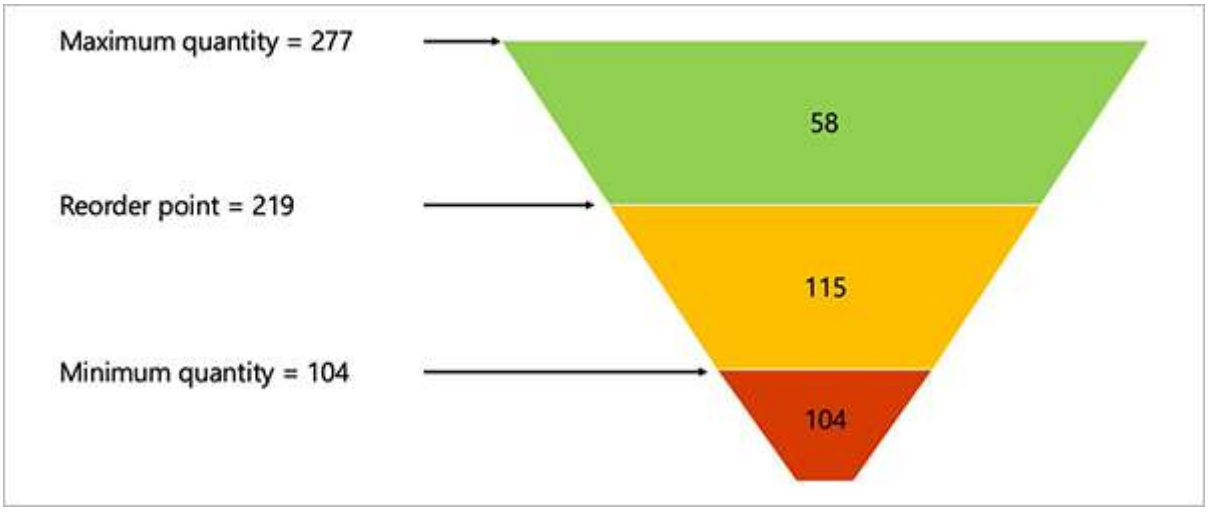
The green zone calculation also includes the components from the yellow zone equation, but it allows for a minimum order size, order cycle, and lead time factor. For this example, assume that there's no order cycle (therefore, you don't have any time constraints about how frequently you order), and the minimum order quantity is 10 pieces. The green zone is then calculated as the maximum result of the following three equations:

- *Minimum order quantity* = 10

- *ADU* × *Order cycle* = 0
- *ADU* × *Decoupled lead time* × *Lead time factor* = 57.5

The system will round the green zone to 58 pieces (ea), because pieces are counted in whole numbers.

The following illustration summarizes these zone calculation results by using the funnel graphic that is often used in DDMRP.



Dynamic adjustments

Dynamic adjustments let you apply a *demand adjustment factor* during periods of high or low demand. This factor multiplies the ADU in all calculations for the selected period. The buffer zones are then modified in turn. You usually apply this factor after you generate your initial buffer values so that you can fine-tune them over time and in response to changing conditions. This task is the third step of DDMRP.

For example, there might be more demand for a pillow product in August as people head out on vacation. Therefore, sales are expected to be higher. In this case, you can change the **Demand adjustment factor** value for the product to *1.5* for all the weeks in August.

In this way, you can calculate buffer values over time and then adjust them based on more than just the information that the system has. In a full DDMRP implementation, you calculate new buffer values every day through a batch job and automatically accept them. You then run planning as a batch job and review the planned orders every day to refill the buffers.

Implement buffers in Supply Chain Management

This section describes how to implement your buffer zone strategy in Microsoft Dynamics 365 Supply Chain Management. It assumes that you've already done the analyses and calculations that are outlined in the first half of this article.

Set up buffers for a decoupling point item

Follow these steps to set up buffer values for a decoupling point.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a released item that is set up as a decoupling point. (Learn more in Inventory positioning.)
3. On the Action Pane, on the **Plan** tab, select **Item coverage**.
4. On the **Item coverage** page, select an item coverage record that creates a decoupling point. (This record shows the name of a coverage group that is set up to create decoupling points.)
5. Select the **General** tab.
6. If you want the system to recalculate buffer values every day or every week, based on your sales history, forecasts, and coverage group settings, follow these steps:
   a. Set **Buffer values over time** option to *Yes*.
   b. A message box notifies you that your manual buffer settings (**Minimum**, **Reorder point**, and **Maximum**) will be reset if you continue. Select **Yes** to keep the new setting.

   Alternatively, if you prefer to calculate or enter your buffer settings just once, follow these steps:
   a. Set **Buffer values over time** option to *No*.

b. Set the **Minimum**, **Reorder point**, and **Maximum** fields to the buffer values that you've calculated for the item, as described earlier in this article.

7. Set the following fields to finish setting up the DDMRP calculations for the item:

   - **Order cycle** – Specify the number of days that must pass between purchase orders for the item. Set the value to *0* (zero) if no order restrictions exist. This field affects the maximum quantity buffer, as previously discussed in this article.
   - **Average daily usage** – You can optionally enter an estimated ADU for the item. This field is for informational purposes only. Usually, the value is automatically calculated as part of the buffer calculations.
   - **Order spike threshold** – Specify the minimum number of daily sales of the item that qualify as a sales spike (unusually high demand). The system uses this field to increase the reorder quantity of planned orders in periods of high demand. Learn more in Demand-driven planning.

Calculate or enter decoupled lead times

For items where you choose to allow the system to calculate your buffer zones automatically, follow these steps to calculate or enter decoupled lead times for a decoupling point item.

1. Open the **Item coverage** page for your decoupling point item. (Learn more in Set up buffers for a decoupling point item.)

2. Select an item coverage record that creates a decoupling point.

3. Select the **Buffer values** tab.

4. If no time periods are shown in the grid, on the Action Pane, on the **Buffer values** tab, select **Add time periods**. The system fills the grid with rows for each daily or weekly time period, depending on whether the **Min, max, and re-order point period** field for the coverage group is set to *Daily* or *Weekly*. The system adds enough rows to reach the time fence that is specified for the coverage group that is assigned to the item.

5. Select the time period where you want to calculate the decoupled lead time. (Usually, this time period is the period that includes today's date.)

6. On the Action Pane, on the **Buffer values** tab, select **Calculate decoupled lead time**.

7. In the **Calculate decoupled lead time** dialog box, set the following fields:

   - **BOM** – Select the bill of materials (BOM) that you want to run the calculation on.

- **Date** – Select the date that you want to run the calculation on. The set of available BOMs is filtered so that only BOMs that are active for the selected date are shown.
- **Quantity** – Enter the quantity that you want to run the calculation for. The set of available BOMs is filtered so that only BOMs that apply to the specified quantity are shown.

8. Select **OK** to run the calculation and close the **Calculate decoupled lead time** dialog box. The **Decoupled lead time** column for your selected time period now shows the calculated value.

Calculate or enter average daily usage

For items where you choose to allow the system to calculate your buffer zones automatically, follow these steps to calculate or enter the ADU for a decoupling point item.

1. Open the **Item coverage** page for your decoupling point item. (Learn more in Set up buffers for a decoupling point item.)

2. Select an item coverage record that creates a decoupling point.

3. Select the **Buffer values** tab.

4. If no time periods are shown in the grid, on the Action Pane, on the **Buffer values** tab, select **Add time periods**. The system fills the grid with rows for each daily or weekly time period, depending on whether the **Min, max, and re-order point period** field for the coverage group is set to *Daily* or *Weekly*. In addition, default values for the **Lead time factor** and **Variability factor** fields are taken from the coverage group. You can edit these values for each row as required.

5. Select a time period where you want to calculate ADU.

6. On the Action Pane, on the **Buffer values** tab, select **Calculate average daily usage**. The system tries to collect data that is required for the ADU calculation, as defined for the coverage group.

7. Follow one of these steps:

    - If the required data is available, calculation results are added to the **Average daily usage** column. In this case, no action is required.
    - If the required data isn't available, no values are automatically added. In this case, manually enter estimated values for each row where you're planning to calculate

buffer values.

Calculate and apply buffer values

For items where you choose to allow the system to calculate your buffer zones automatically, follow these steps to manually trigger the calculation of buffer values.

1. For the relevant decoupling point item, configure the buffer calculation, calculate or enter decoupled lead times, and calculate or enter average daily usage for all relevant time periods, as previously described in this article.

2. Open the **Item coverage** page for your decoupling point item.

3. Select the **Buffer values** tab, which should already show a list of time periods.

4. Select the time period where you want to calculate buffer values. (Usually, this time period is the period that includes today.) The row that you select must already have non-zero values in the **Average daily usage** and **Decoupled lead time** columns.

5. Edit the **Demand adjustment factor** field for one or more rows as required. The system applies this factor to the **Average daily usage** value in all buffer calculations where that value is used. This factor lets you adjust for the way that demand fluctuates by date (for example, for holidays or seasonal items).

6. On the Action Pane, on the **Buffer values** tab, select **Calculate min, max and reorder point quantities**. The system calculates and fills in the **Calculated min**, **Calculated reorder point**, and **Calculated max** columns in the grid on the **Item coverage** page.

7. When you finish reviewing the calculated values, you must apply them. Otherwise, they have no effect. When you apply a calculation for one or more rows, values from the **Calculated min**, **Calculated reorder**, and **Calculated max** fields are copied to the **Min**, **Reorder point**, and **Max** columns, respectively. On the Action Pane, on the **Buffer values** tab, in the **Take action** group, select one of the following buttons:

    - **Accept all calculations** – Apply all calculated values in the grid.
    - **Accept calculations for selected rows** – Apply calculated values for the selected rows only.
    - **Discard all calculations** – Discard all calculated values for minimum quantities, maximum quantities, and reorder points in the grid.
    - **Discard calculations for selected rows** – Discard all calculated values for minimum quantities, maximum quantities, and reorder points for the selected rows.

Schedule automatic buffer value calculations

After you fully set up your DDMRP settings and confirm that they work as expected, set up a batch job to periodically recalculate ADU and related buffer values as required, based on actual consumption data and/or updated forecasts. This procedure applies only to items where you choose to allow the system to automatically calculate your buffer zones.

Follow these steps to schedule automatic buffer value calculations.

1. Go to **Master planning** > **Master planning** > **DDMRP** > **Calculate buffer values**.

2. In the **Calculate buffer values** dialog box, set the following fields:

    - **Calculate average daily usage** – Set this option to *Yes* to recalculate the ADU of decoupling point items each time that the job runs. Set it to *No* to skip this calculation. Usually, set this option to *Yes*.
    - **Calculate decoupled lead time** – Set this option to *Yes* to recalculate the decoupled lead times each time that the job runs. Set it to *No* to skip this calculation. Usually, set this option to *Yes*.
    - **Calculate buffer values** – Set this option to *Yes* to recalculate buffer values each time that the job runs. Set it to *No* to skip this calculation. Usually, set this option to *Yes*.
    - **Accept calculation for min, max and reorder point** – Set this option to *Yes* to automatically approve and apply the recalculated buffer values each time that the job runs. Set it to *No* to leave the recalculated values unapplied. In this case, the recalculated values don't take effect unless somebody manually applies them later from each product's **Item coverage** page. Usually, set this option to *Yes*.
    - **Master plan** – Select a master plan that includes the items that are affected by the calculation. The calculation applies to all the items in the plan filter, which is further limited by the **Filter** settings in this dialog box.

3. To limit the set of records that this batch job should run on, on the **Records to include** FastTab, select **Filter** to open the **Inquiry** dialog box. This dialog box works just as it does for other types of background jobs in Supply Chain Management.

4. On the **Run in the background** FastTab, specify how, when, and how often the selected calculations should be done for the selected items. The fields work just as they do for other types of background jobs in Supply Chain Management.

5. Select **OK** to add the new job to a batch queue for execution.

Review and recalculate decoupled lead times for all items

Follow these steps to review and recalculate all decoupled lead times that are available in your legal entity (company).

1. Go to **Master planning** > **Master planning** > **DDMRP** > **Decoupled lead time**.
2. On the **Decoupled lead time** page, browse and filter the list as required to find the information that you're looking for. To view even more information for an item, select its link in the **Item number** column.
3. If you want to recalculate the decoupled lead time for any item, select the item, and then select **Calculate decoupled lead time** on the Action Pane. The **Calculate decoupled lead time** dialog box appears. This dialog box works just as it does when you calculate decoupled lead times for the same item on the **Item coverage** page.

Last updated on 07/01/2026

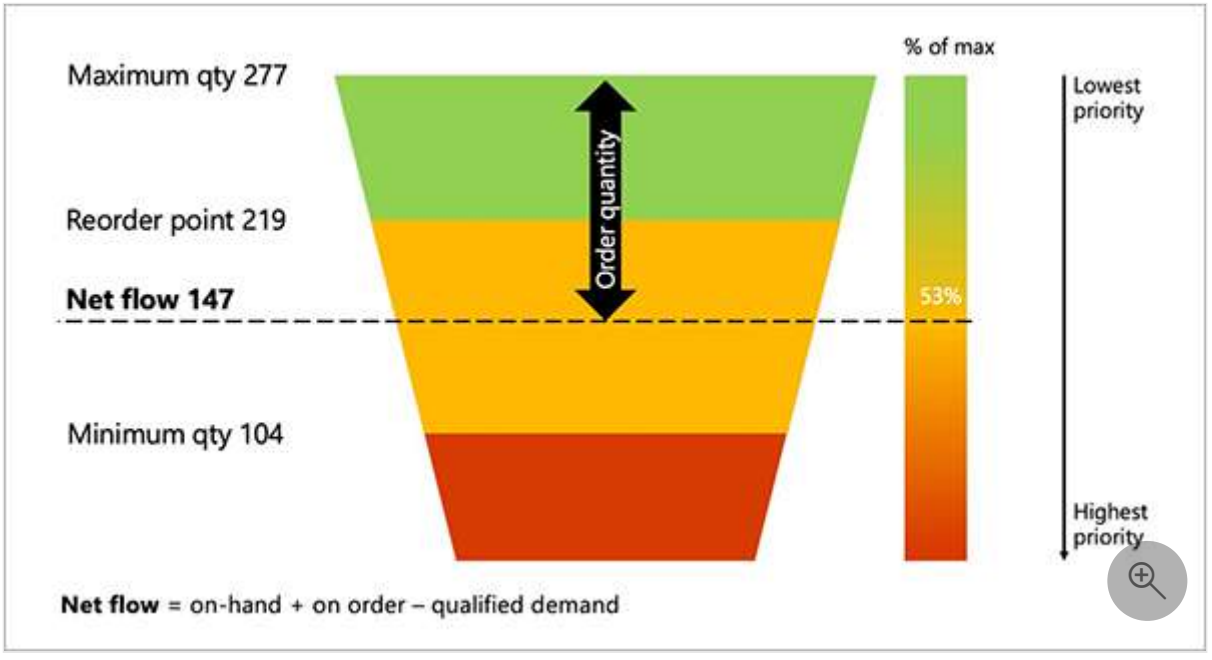
Demand-driven planning

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to generate planned orders for items that are set up as decoupling points.

Net flow and qualified demand

During master planning, the system uses the concept of *net flow* to calculate the effective on-hand quantity. The calculation starts with the actual on-hand inventory, adds inventory that is on order (existing supply orders that aren't yet received), and subtracts what is referred to as *qualified demand* (qualified upcoming sales), as shown in the following illustration. When the system determines which buffer zone an item belongs in and what the ordered quantity should be, it evaluates the net flow, not the actual on-hand inventory.



When a planned order is triggered during a planning run, the ordered quantity is the maximum level minus the net flow. To assign an order priority, the system uses priority-based planning functionality instead of requirement dates. Demand Driven Material Requirements Planning (DDMRP) assigns the priority of a planned order based on the net flow quantity as a percentage of maximum inventory. In the previous illustration, the net flow quantity is 53

percent of the maximum quantity. Therefore, the order priority for replenishment is 53. (For context, 0 is the highest priority, and 100 is the lowest priority.)

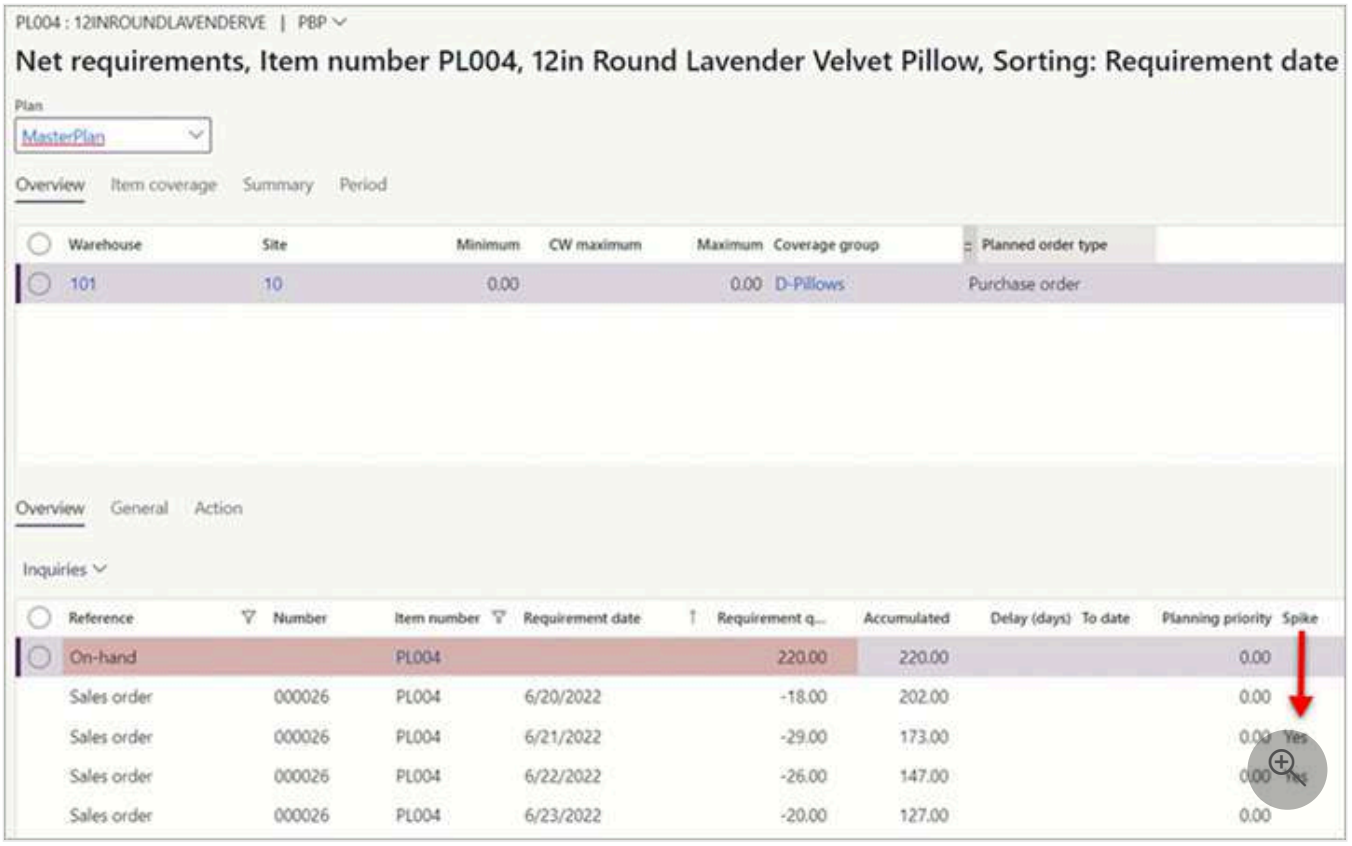
Qualified demand is the past-due demand, plus today's demand, plus qualified order spikes in the future. The following illustration shows an example of demand levels for today (June 12) and the next three days. DDMRP lets you set an order spike threshold to identify demand that exceeds normal expectations. If the threshold is set at 25 pieces, two of the future dates that are shown in the illustration qualify as order spikes. You must assign an order spike threshold for each product individually by using its **Item coverage** page, as described in Set up buffers for a decoupling point item.

If there's no past-due demand, you can add today's demand (18) to the quantities of the two order spikes (29 and 26) to get a qualified demand of 73. Even though there's demand for June 23, notice that it isn't included in the net flow calculation, because DDMRP triggers planned order differently than traditional planning coverage groups.

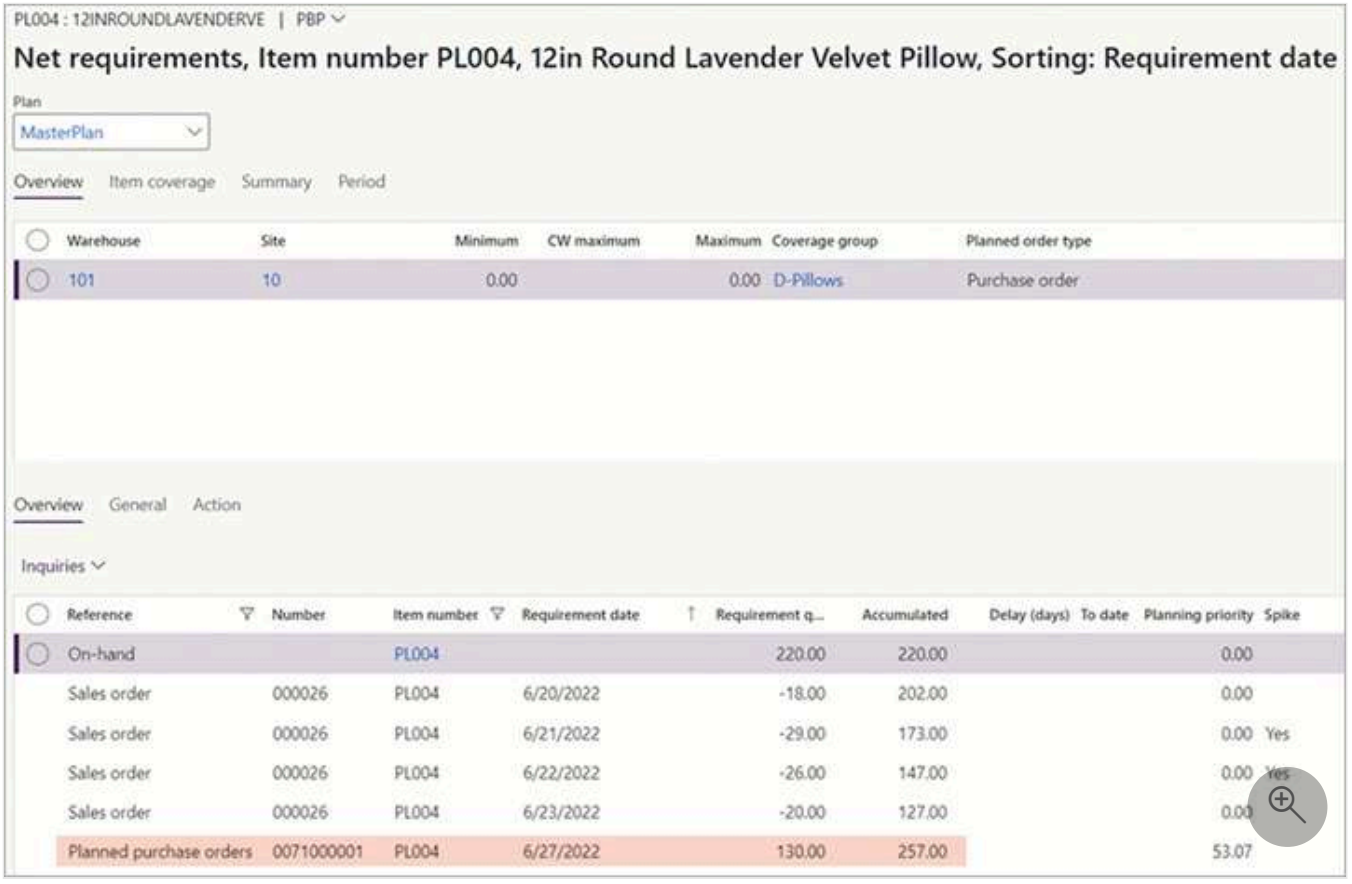
Generating planned orders with DDMRP

During a planning run, the system creates a new planned order if the net flow for an item drops below the reorder point. When you use DDMRP, you usually run a planning run every day. Therefore, you check your inventory levels once a day to determine which items you must replenish.

The following illustration shows an example of a situation where you have an order for 18 pieces on June 20, 29 pieces on June 21, 26 pieces on June 22, and 20 pieces on June 23. Because the spike threshold is set at 25 pieces, two of these orders are flagged as spikes. You have 220 pieces of this item on hand.



If you run master planning now, it generates a planned order if the net flow drops below the reorder point (219 pieces in this example).



This example produces a planned purchase order for a quantity of 130, which equals the maximum level minus the net flow. The planned order gets a priority of 53.07, based on its percentage of the maximum quantity. Because these values were found on June 20, the system

creates a planned order that is dated June 20 plus the decoupled lead time for the item (five business days in this example). Therefore, because five business days are one week from today, the planned order is dated June 27.

> **Note**
>
> Master planning calculates only decoupled items by using DDMRP. All other items are calculated by using standard material requirements planning (MRP).

Last updated on 03/25/2026

Visual and collaborative execution

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to monitor your Demand Driven Material Requirements Planning (DDMRP) decoupling points, buffer zones, planned orders, and history.

Visually track buffers and quantities for a selected item

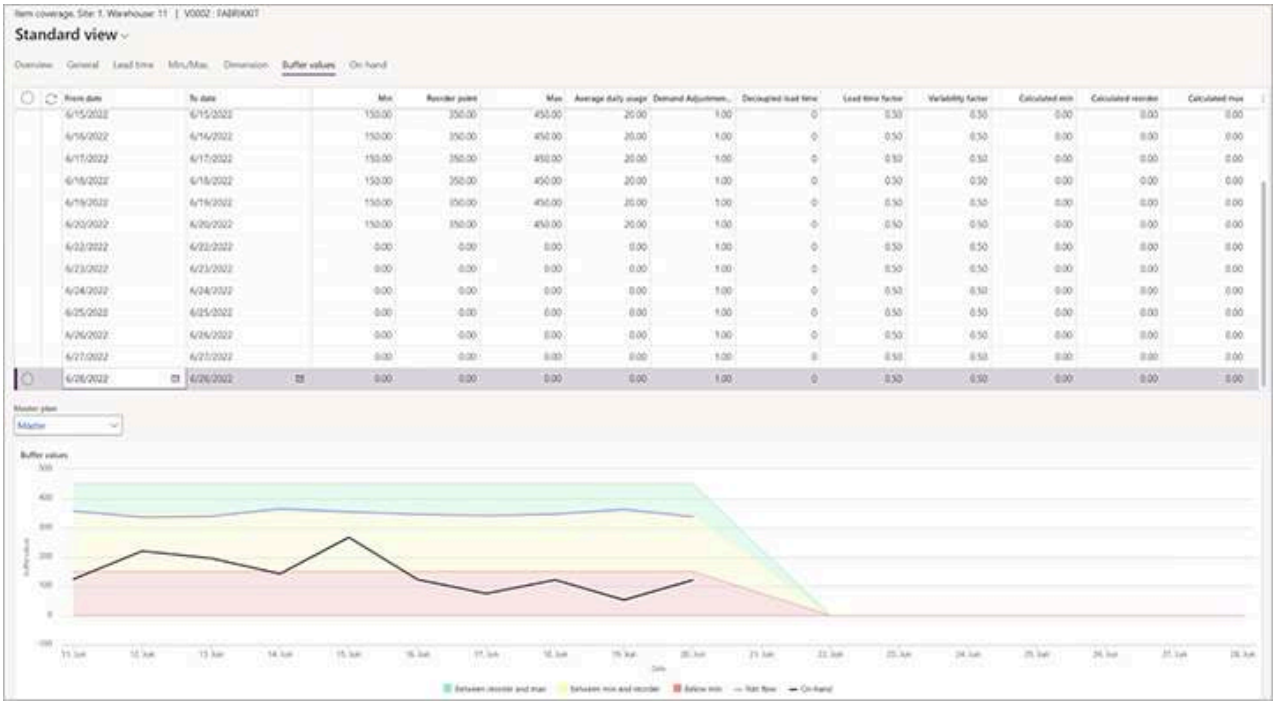
In Microsoft Dynamics 365 Supply Chain Management, you can visually track how buffers, on-hand quantities, and net-flow levels change over time for any selected released product. Follow these steps to open and review the charts.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a released item that is set up as a decoupling point. (Learn more in Inventory positioning.)
3. On the Action Pane, on the **Plan** tab, select **Item coverage**.
4. On the **Item coverage** page, select an item coverage record that creates a decoupling point. (This record will show the name of a coverage group that is set up to create decoupling points.)
5. Select the **On-hand** tab. This tab includes a chart that shows how on-hand quantities changed over time, together with the value of the on-hand level that was recorded for a specific period every time that master planning is run. The tab also includes a table that shows which of the following categories each recorded on-hand level falls into:

    - **Critically low** – Less than half the minimum for the period.
    - **Low** – Between half the minimum and the minimum.
    - **Average on-hand** – Between the minimum and the minimum plus the green zone.
    - **Higher than average** – Higher than average.

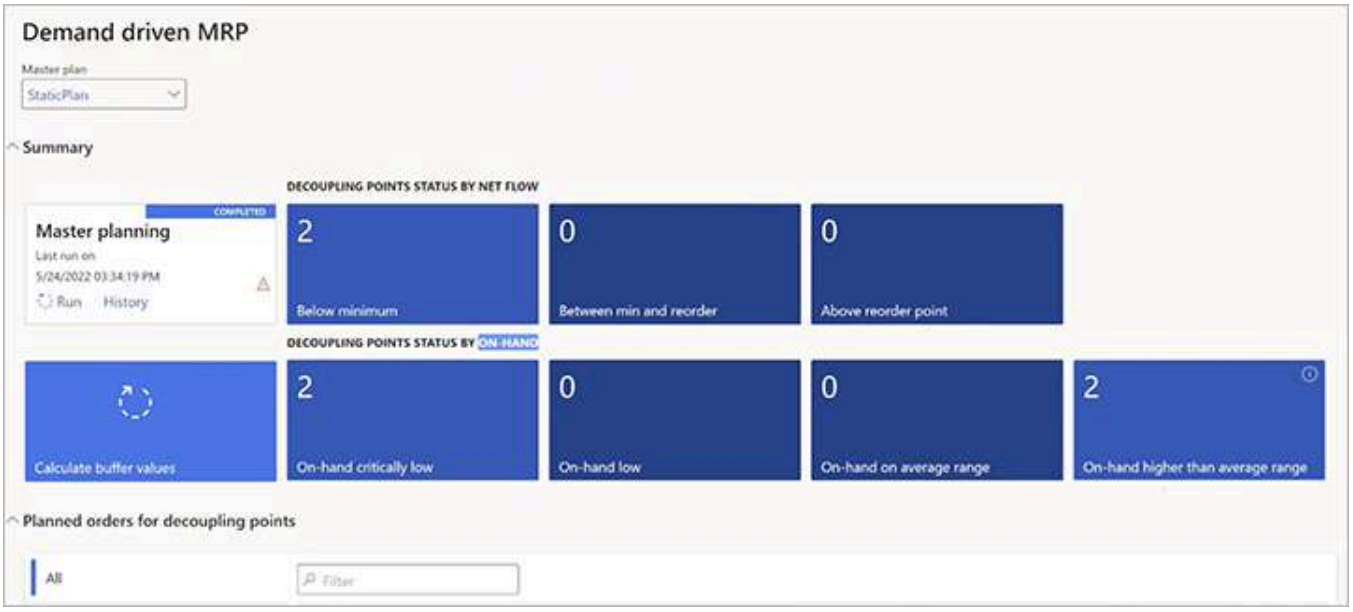
> **Note**
>
> The colors that are used on the **On-hand** tab represent different ranges than the similar colors used on the **Buffer values** tab.

6. To view how your buffer values changed over time, and how they compare to on-hand and net-flow levels, select the **Buffer values** tab.



Track the status and planned orders for all decoupling points

The **Demand driven MRP** workspace provides several tools, together with key performance indicators (KPIs) and overviews of the status of your decoupling point items and related planned orders. To open it, go to **Master planning** > **Workspaces** > **Demand driven MRP**.



Get overviews of decoupling points and planned orders

In addition to the **Demand driven MRP** workspace, Supply Chain Management provides several pages where you can view information about your DDMRP setup, decoupling points, and planned orders. Some of these pages also provide convenient buttons on the Action Pane that let you manage buffers, run master planning, and open other related views.

- To view decoupling point status by net flow level, go to **Master planning** > **Master planning** > **DDMRP** > **Decoupling points status by net flow**.
- To view decoupling point status by on-hand inventory level, go to **Master planning** > **Master planning** > **DDMRP** > **Decoupling points status by on-hand**.
- To view planned orders for decoupling points, go to **Master planning** > **Master planning** > **DDMRP** > **Planned orders for decoupling points**.
- To view decoupled lead times, go to **Master planning** > **Master planning** > **DDMRP** > **Decoupled lead time**.

Clean up decoupling point buffer values

Over time, your system will build up a large amount of historical data that is related to ongoing buffer calculations. This historical data will cause your data volume to increase and can eventually affect the performance of your system. Therefore, we recommend that you occasionally clean up the old decoupling point buffer values.

> **⚠ Warning**
>
> The cleanup job will remove historical buffer value settings and historical on-hand/net-flow information. It isn't reversible.

Follow these steps to clean up your decoupling point buffer values.

1. Go to **Master planning** > **Master planning** > **DDMRP** > **Clean up decoupling point buffer values**.

2. In the **Clean up decoupling point buffer values** dialog box, set the following fields:

    - **Delete records older than (days)** – Specify the age of the youngest records to keep, in days. All decoupling point buffer value records that are older than this value will be deleted.
    - **Master plan** – Select a master plan that includes items that should be affected by this cleanup. The cleanup will apply to all the items in the plan after the **Filter** settings that you specify in this dialog box are applied.

3. To limit the set of records that the batch job should run on, on the **Records to include** FastTab, select **Filter** to open the **Inquiry** dialog box. This dialog box works just as it does for other types of background jobs in Supply Chain Management.

4. On the **Run in the background** FastTab, specify how, when, and how often the cleanup should be done. The fields work just as they do for other types of background jobs in Supply Chain Management.

5. Select **OK** to add the new job to a batch queue for execution.

Last updated on 06/17/2025

DDMRP FAQ

Article • 01/06/2023

This topic provides answers to frequently asked questions (FAQ) about Demand Driven Material Requirements Planning (DDMRP) in Microsoft Dynamics 365 Supply Chain Management.

General

Does DDMRP require Planning Optimization?

Yes, DDMRP functionality requires the Planning Optimization Add-in for Dynamics 365 Supply Chain Management. It isn't available for the deprecated master planning engine. For more information about Planning Optimization, see Get started with master planning.

Can priority-based planning be used together with DDMRP?

DDMRP uses priority-based planning functionality to set the priority of planned orders. It calculates the priority of each order by determining the net flow based on the maximum inventory level. Master planning then uses the order priority (instead of the requirement date) to drive supply allocation.

DDMRP seems to be an extension of the min/max coverage code functionality. Are there any scenarios where I might still use min/max functionality instead of DDMRP when I produce or procure stock?

One of the key differences between min/max and DDMRP is that DDMRP doesn't consider all future net requirements when it's determining whether to reorder. DDMRP is run daily and looks only at past-due demand (today's demand) and order spikes to determine whether an order must be placed immediately. By contrast, min/max looks at all demand in the coverage time fence. It will propose planned orders farther in the future, based on when net requirements will deplete inventory below the minimum point.

Can I run the existing material requirements planning (MRP) functionality together with DDMRP (for example, when I use coverage groups)?

Yes, we expect that companies will run some combination of decoupling points and other coverage methods (such as min/max, period, or requirement). Decoupling points should be put at strategic points in the supply chain but probably aren't the best answer for every product. Therefore, it's likely that companies will run DDMRP together with other coverage methods.

Can I simulate DDMRP by using the past two years of data in Supply Chain Management?

Unfortunately not. However, you can easily create an Excel worksheet that contains the formulas, and then use that worksheet to determine what the system would have created as the buffer values.

Can Supply Chain Management analyze whether an item is in the correct coverage group (for example, to test whether the factors that I set up for variability, and so on, will still meet my item consumption)?

No automated functionality for this purpose currently exists in Supply Chain Management. However, you can manually review the performance of your net flow and on-hand inventory by using the buffer values chart. This chart will help you determine whether your actual net flow and on-hand inventory values are in the zones that you expect them to be in.

Is there a data entity for importing usage history from legacy enterprise resource planning (ERP) systems?

Yes, if you come from another system and want to keep your past buffer value history, there's a data entity for updating that history.

Does DDMRP work with Supply Chain Management process manufacturing?

Yes, DDMRP works with any functionality that the deprecated master planning engine works with, provided that you're using Planning Optimization.

Can DDMRP work with projects?

Yes, DDMRP works with any functionality that the deprecated master planning engine works with, provided that you're using Planning Optimization.

Does DDMRP work in an intercompany scenario?

Yes, DDMRP works with any functionality that the deprecated master planning engine works with, provided that you're using Planning Optimization.

The Demand Driven Institute

Is the Demand Driven Institute part of Microsoft?

No, the Demand Driven Institute is a separate organization that developed the DDMRP methodology. Microsoft developed the functionality that supports the DDMRP methodology in Supply Chain Management, and our solution has been certified by the Demand Driven Institute. Learn more at the Demand Driven institute website.

Which Demand Driven Institute certifications are recommended for consulting teams?

The recommended certification for consulting teams is Demand Driven Planner (DDP). This certification includes all the DDMRP material and provides a deep understanding of DDMRP principles, to help consultants successfully explain and implement DDRMP in Supply Chain Management.

Decoupling points

How can I decide where to define decoupling points?

When you're choosing where to put decoupling points in your supply chain, consider the following factors:

- **External variability** – Add buffers to mitigate effects that can arise from supply variability that's out of your control.
- **Inventory leverage and flexibility** – Add buffers to prioritize items that are used as components in multiple products.

- **Critical operation protection** – Add buffers before critical operations or bottlenecks, to ensure that operations don't sit idle because of a lack of raw materials.
- **Customer tolerance time** – Add buffers at stages where customers expect a short lead time for the item.
- **Sales order visibility horizon** – Add buffers at points where you must quickly react to meet new demand.
- **Market potential lead time** – Add buffers at points where you can sell items at a higher price if the lead time is shorter.

Learn more in Inventory positioning.

Are negative and positive days considered for coverage code decoupling points?

No, the DDMRP model doesn't consider negative and positive days.

Decoupling points can be added only at the item coverage level. Should I continue to indicate the coverage group at the item level, so that it can be considered for other locations that aren't specified in the item coverage?

Yes, it's possible that not all locations for a given item must be set up as decoupling points. Therefore, you can choose to have a coverage group for an item that uses min/max, period, requirement, and so on, and then indicate the specific decoupling points on the item coverage page. The settings on the item coverage page override the item-level coverage group.

Is the new coverage code compatible with all product and storage dimensions?

Yes, the *decoupling point* coverage code works with all products and storage dimensions. You must specify all the dimensions for a product that's a decoupling point.

Do I have to create item coverage records for every variant of a product, or can the calculation be run on product masters?

Decoupling points must be set at the product variant level because each variant will have its own demand pattern, and planned orders are generated at the variant level.

Planned order generation with DDMRP

Does DDMRP plan based on forecast values unless qualified demand (orders) for the time frame exceeds the forecast?

DDMRP doesn't use forecasts to generate planned orders. Planned orders are created when the net flow is below the reorder point.

Forecasts are used when the system is set to calculate average daily usage (ADU) based on a *Forward* or *Blended* time period. ADU is one of the factors that affect the buffer values (minimum, reorder point, and maximum). Therefore, it indirectly affects how much you reorder.

How is production scheduling done when a manufactured part uses DDMRP?

Planned production orders are generated and assigned a priority value that's calculated by dividing the net flow by the maximum inventory level. Master planning then uses the order priority (instead of the requirement date) to schedule production.

If I set up a maximum reorder quantity that's lower than the reorder quantity, will DDMRP generate multiple planned orders that have different priority values?

The default order settings are respected when planned orders are created, and the same priority will be assigned to all planned orders.

If different priorities must be assigned to different parts, use planning priority models. Planning priority models let you split a replenishment order into multiple orders that have different priorities. For example, you can assign the highest priority (for example, *10*) to replenish until the minimum, a lower priority (for example, *30*) to replenish until the reorder point, and an even lower priority (for example, *50*) to replenish until the maximum.

In cases where a planned purchase order was generated, how will Supply Chain Management peg the purchase order quantity?

One key concept of DDMRP is that it doesn't use pegging when it generates planned orders. Decoupling points are buffers where supply and demand are decoupled. In other words, there's no specific supply for a demand. Therefore, there's no pegging. All replenishment orders replenish the buffer, which is on-hand inventory.

Lead time and variability factors

Does the variability factor refer to lead time or demand?

The variability factor refers to demand variability (in other words, how much demand fluctuates or how consistent it is).

How can we classify a lead time as long, medium, or short?

Lead time classification differs for every company. You should start by looking at the lead times for all your items, and group them based on whether their lead time is short, medium, or long. After you assign a lead time factor to each item, you can test and adjust the factor to identify the optimal value.

Qualified demand

What is spike demand?

Spikes refer to dates in the future when the total amount of demand exceeds the specified order spike threshold. This situation indicates that the demand for a date exceeds what's considered normal and will be included in the net flow calculation as part of the qualified demand.

How is the order spike threshold determined? Is it calculated or manually defined?

The order spike threshold is manually set on the item coverage record.

Is there a time fence for qualified demand (looking forward)?

The time fence for qualified demand is determined by the coverage time fence setting.

Feedback

Was this page helpful? Yes No

Provide product feedback

Deprecated master planning overview

> **Note**
>
> Although we strongly recommend that you migrate to Planning Optimization as soon as possible for the reasons outlined in this article, you can continue to use the deprecated master planning engine while you prepare to migrate without requesting an exception from Microsoft. For more information about how to do so, see **Continue to use deprecated master planning with existing companies**.

As previously announced, the built-in master planning engine is deprecated. The engine has been deprecated for a long time and receives no meaningful support from Microsoft. Microsoft no longer invests in it (no new features are released).

Planning Optimization is the master planning engine for Supply Chain Management and completely replaces the built-in master planning engine.

> **⚠ Warning**
>
> If you're still using the deprecated master planning engine, you should strongly reconsider doing so. The engine receives no support and has no active development. We strongly encourage you to complete the migration to Planning Optimization as soon as possible so that you can take advantage of the many performance improvements and other new capabilities provided by the new service.

Work with a partner to evaluate and plan the migration to Planning Optimization.

Before you switch to Planning Optimization, evaluate the results of the Planning Optimization fit analysis. Learn more in Planning Optimization fit analysis.

When will the deprecated planning engine be removed?

There's currently no timeline for the full removal of the deprecated master planning engine from Supply Chain Management. Microsoft isn't currently planning to remove it.

> **Important**
>
> If Microsoft decides to remove the deprecated master planning engine, the company will announce those plans at least 12 months before the removal date on the Removed or deprecated features in Dynamics 365 Supply Chain Management page. Monitor that page and the What's new or changed in Dynamics 365 Supply Chain Management page for the latest announcements.

Although the deprecated master planning engine isn't being removed right now, it's important to understand what *deprecated* means in practice:

- **No new features** – Microsoft isn't investing in new capabilities for the deprecated engine.
- **No support** – The deprecated master planning engine has been deprecated for a long time and receives no support from Microsoft. There are no bug fixes, no new features, and no investment in the engine going forward. Customers who continue to use it do so entirely at their own risk.
- **Version restrictions** – Starting in Supply Chain Management version 10.0.41, the deprecated master planning engine is blocked for all new deployments. Existing deployments can continue using it on a per-company basis.

We strongly recommend that you begin planning your migration to Planning Optimization now, even though there's no removal date. Learn more in Migration to Planning Optimization for master planning.

Note: The author created this article with assistance from AI. Learn more

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Last updated on 05/01/2026

Migration to Planning Optimization for master planning

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

The built-in master planning engine is deprecated. It's replaced by the Planning Optimization Add-in for Microsoft Dynamics 365 Supply Chain Management. This article provides information about the impact on new and existing deployments. It includes information about required actions.

Planning Optimization enables master planning calculations to occur outside Supply Chain Management and its Azure SQL database. The benefits that are associated with Planning Optimization include improved performance and minimized impact on the SQL database during master planning runs. Because quick planning runs can be done even during office hours, planners can immediately react to demand or parameter changes.

For more information about Planning Optimization, see Master planning system architecture.

Deprecation of the legacy master planning engine

Microsoft deprecated the built-in master planning engine in favor of Planning Optimization. This change affects all cloud environments. On-premises installations aren't affected.

For more information about the deprecated master planning engine, see the announcements in Removed or deprecated features in Dynamics 365 Supply Chain Management.

A deprecated feature is a feature that's no longer in active development and might be removed in a future release. For the deprecated master planning engine, Microsoft won't add new features. Support is provided only for blocking problems (where master planning doesn't create any planned orders and/or continuously fails) and for regressions in the functionality. In March 2023, Microsoft adopted a strict policy against providing support for the deprecated master planning engine, except for blocking problems or regressions. These conditions apply to all customers, including the following types:

- Customers whose needs aren't yet fully supported by Planning Optimization.
- Customers who received an exception so that they can continue to use the deprecated planning engine.
- All on-premises customers.

There's currently no timeline for full removal of the deprecated planning engine from Supply Chain Management. If Microsoft makes plans to remove it, the company announces those plans 12 months before the removal date.

New deployments

Planning Optimization is now the standard master planning engine and you must use it for all new cloud deployments. Starting with Supply Chain Management version 10.0.32, the first time you enable planning processes for any legal entity, the system requires you to install and enable Planning Optimization if you didn't already do so.

Starting in Supply Chain Management version 10.0.41, the deprecated master planning engine is blocked for all new deployments. There's no manual way to enable the deprecated master planning engine for these environments.

Adding a new legal entity to an existing environment

When you add a new legal entity (company) to an existing environment, that company must use Planning Optimization. Starting with Supply Chain Management version 10.0.32, the first time you enable planning processes for any newly added legal entity, the system requires you to install and enable Planning Optimization if you didn't already do so.

You can continue to use the deprecated master planning engine for one or more previously created companies (until you're ready to migrate them) even while using Planning Optimization for the others. For instructions on how to set a company to use the deprecated master planning engine, see Continue to use deprecated master planning with existing companies.

> **Note**
>
> Remember that you must still move each legal entity to Planning Optimization once they're supported.

If you use kanban or are going live very soon and need time to test Planning Optimization

All new deployments and legal entities must use Planning Optimization from the beginning. However, if you've been preparing a new deployment for some time, and the functionality you needed wasn't yet supported by Planning Optimization while you were developing it, you can still go live by using the deprecated master planning engine for that deployment until you have enough time to install and test Planning Optimization on it.

Planning Optimization doesn't yet support kanban, so if you need to use kanban, you can go live by using the deprecated planning engine for now. You should still plan to move to Planning Optimization as soon as kanban is supported.

Existing deployments

Owners of existing cloud-based deployments that depend on master planning must plan to migrate to Planning Optimization. If your implementation depends on functionality that Planning Optimization doesn't currently support, you must request an exception to continue to use the deprecated master planning engine.

Starting in Supply Chain Management version 10.0.32, you can allow some companies (legal entities) to run Planning Optimization while others continue to use the deprecated master planning engine until they're ready to be migrated. Therefore, Microsoft grants exceptions on a per-company basis. The exception only applies to existing companies—starting with version 10.0.32, all new companies that you add to your existing environment must use Planning Optimization. For instructions on how to set a company to use the deprecated master planning engine, see Continue to use deprecated master planning with existing companies.

Migrate companies to Planning Optimization one at a time, as soon as they're supported.

Migration recommendations

Several differences exist between the deprecated master planning engine and Planning Optimization.

For distribution companies, the two planning engines provide very similar feature sets. Based on our experience helping other customers migrate, we recommend that distribution companies enable and test Planning Optimization in a test environment. When testing is successful, enable it in a production environment.

Manufacturing companies might be affected by some of the minor architectural differences that exist between Planning Optimization and the deprecated planning engine. Based on our experience helping other customers migrate, set up a test environment and proceed in the following way:

1. Create two testing plans, one for Planning Optimization and the other for the deprecated planning engine. Use the same settings for both plans.
2. While the deprecated planning engine is enabled, run the plan that you created for it.
3. Enable Planning Optimization, and run the plan that you created for it.
4. For each plan, export the planned orders to an Excel file.
5. For each plan, sum the planned order quantities for each of several regular periods (for example, every month).
6. Compare the quantities for each plan to make sure that the result is the same (or very similar). Some variation can be expected for orders that occur at the start or end of a period.
7. If your test is successful, continue testing on the test environment.
8. If all of your tests are successful, enable Planning Optimization in your production system.

Exception process for version 10.0.32 and later

As of Supply Chain Management version 10.0.32, the process of evaluating your system and moving to Planning Optimization is fully automated. The system analyzes your setup and automatically shows you the correct instructions for your situation and for each company (legal entity). The following subsections provide details about the possible cases.

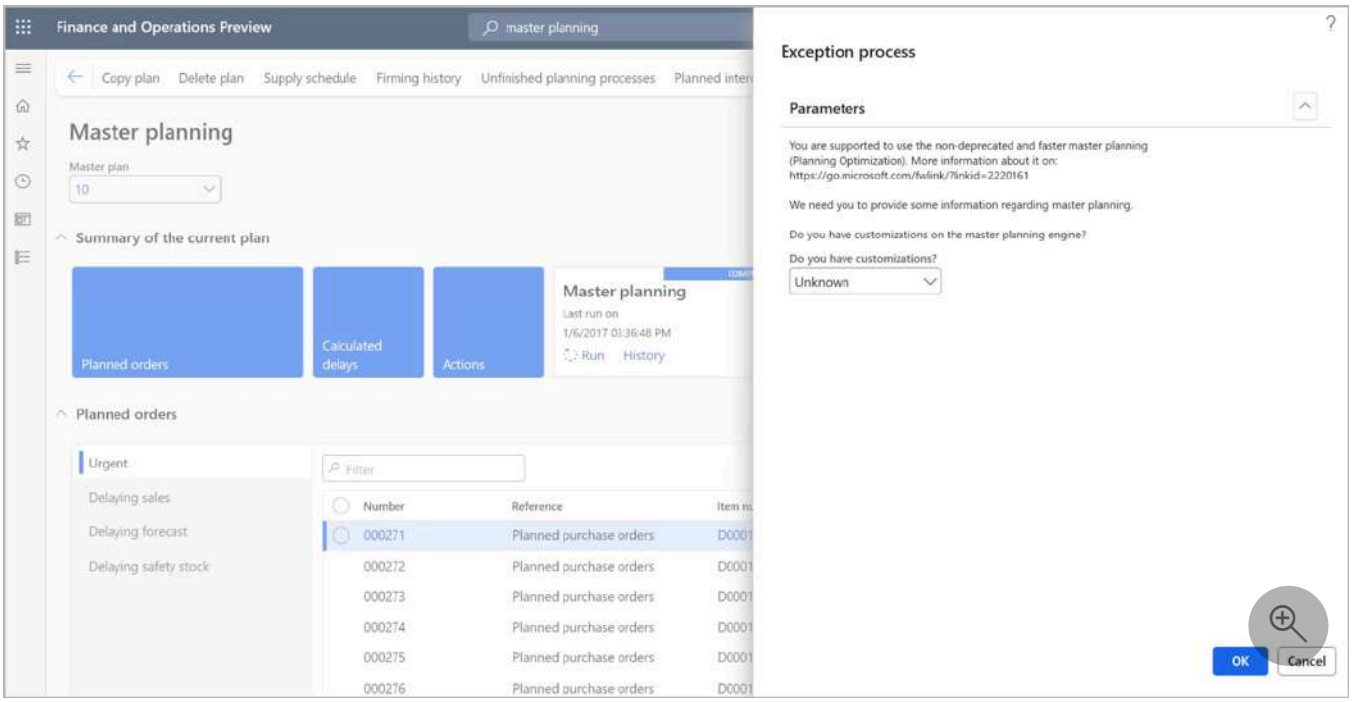
Deployments where Planning Optimization supports all required features

If the system detects that all the relevant features you're using are supported by Planning Optimization, but you're still running the deprecated planning engine, it prompts you to migrate. The next time that you manually run master planning, the system shows the following message:

> You can use the non-deprecated and faster master planning (Planning Optimization).
>
> We need you to provide some information regarding master planning.

> Do you have customizations on the master planning engine?

The following screenshot shows how the message looks.



If you didn't customize the master planning engine for this deployment, you must migrate to Planning Optimization. If you need more time to test and migrate, select the time that you need. The system automatically applies an exception for the selected time.

If you do have customizations, you must move them to the existing extensibility point. Learn more in Planning Optimization extensibility.

Deployments requiring features not yet supported by Planning Optimization

If the system detects that you're using features that aren't supported by Planning Optimization, it shows the following message the next time that you manually run master planning:

> You are not supported yet to use the non-deprecated master planning (Planning Optimization). We expect to be supporting you in the coming future. When you are supported, you will be asked to move to Planning Optimization. If you have customizations on the master planning engine, you can already start evaluating them and preparing to move them to the Planning Optimization extensibility point.

This message informs you that you should start planning to move to Planning Optimization as soon as it supports the features that you're using. Therefore, you should evaluate any customizations that you made to the deprecated planning engine, plan to move them to the

existing extensibility point (see Planning Optimization extensibility), and take other actions to prepare for the migration (for example, by contacting your Microsoft partner or consultant).

For information about which features are already supported and estimates about when each feature will become available for Planning Optimization, see Planning Optimization fit analysis.

If you already received an exception, it remains in place until Planning Optimization supports the features that you require.

Environments that don't support Planning Optimization

Regardless of the features that you're using, to use Planning Optimization, you must be running Supply Chain Management version 10.0.7 or later in a tier 2 or higher high-availability environment that's Microsoft Dynamics Lifecycle Services–enabled. The environment must **not** be a OneBox environment. If you try to install the add-in in a OneBox environment, the installation doesn't complete, so you must cancel it.

If your environment doesn't support Planning Optimization, you receive the following message:

> You can only run deprecated master planning in this environment. If you would like to get an environment that supports non-deprecated planning (Planning Optimization) please follow the instructions: Get started with master planning

If you're a Microsoft partner or independent software vendor (ISV), you can get a nonproduction environment that supports Planning Optimization at a reduced price. It includes Microsoft business applications and demo data. These environments are available only to partners and ISVs. They can be used only on partner tenants, never on customer tenants. Use the environment that you get to learn how Planning Optimization works, test your solutions while you're using it, and deliver end-to-end customer demos. To request a license, go to the partner sandbox request page.

Frequently asked questions about migration

What impact does Planning Optimization have on current planning users?

Users still work in the same module and use the same pages to initiate planning and review planned orders. The only visible difference is that the processing window shows **Run Planning Optimization**.

If I'm running classic planning in a batch job, do I have to update this job when I enable Planning Optimization?

Yes, you must set up a new batch job to run Planning Optimization instead of classic master planning.

If I'm running deprecated planning in some companies and Planning Optimization in others, can I use intercompany master planning?

Yes. Although the intercompany master planning page isn't supported, you can achieve the same effect by scheduling sequential batch jobs to run planning for the different companies in their desired order. Use the Planning Optimization batch job for the companies that are running Planning Optimization (*Planning Optimization* task) and the deprecated engine for the others (*Master planning* task).

To use intercompany master planning for different companies that use Planning Optimization, the process is the same: schedule Planning Optimization tasks as batch jobs the run sequentially.

How much will my performance improve when I move to Planning Optimization?

There's no specific rule about how much your performance improves. In general, companies that run with large datasets and are already experiencing performance problems with deprecated master planning see the biggest improvement. The best way to find out is to test Planning Optimization.

Does it cost extra to use Planning Optimization?

No, it's included in your Supply Chain Management license. There are no extra costs.

Can I run Planning Optimization in a Government Community Cloud (GCC) environment?

Yes, Planning Optimization is supported in Government Community Cloud (GCC) environments now that GCC is compatible with the Microsoft Dynamics Lifecycle Services microservices framework.

Can I run Planning Optimization in a tier-1 environment?

No, Planning Optimization runs through a Lifecycle Services add-in that you can install only in tier-2 or higher environments.

What data center does Planning Optimization run on?

Planning Optimization runs on the same data center as your Supply Chain Management environment.

I'm a partner, and I want to demo or test Planning Optimization. How can I do that?

For partners and independent software vendors (ISVs), Microsoft offers a special license for accessing tier-2 environments. For details, see [ISV] Request License.

I'm going live with version 10.0.32 or higher, but Planning Optimization doesn't yet have all the features I need. Can I go live by using the deprecated planning engine for now?

Yes. When you start using the planning features, the system asks you to install and enable Planning Optimization because this requirement has been mandatory since Supply Chain Management version 10.0.32. However, if one or more of your companies isn't yet ready to use Planning Optimization, you can exclude them from running Planning Optimization by going to the **Planning Optimization parameters** page. For instructions, see Continue to use deprecated master planning with existing companies.

How do I request an exception to continue using the deprecated planning engine on version 10.0.32 or higher?

The system automatically gives you an opportunity to request an exception. Fill out the dialogs when prompted. The exception is applied automatically in the background.

Why do I get an error message when running the deprecated master planning engine?

You might sometimes get an error message when running the deprecated master planning engine. It indicates that you must move to Planning Optimization and provides the following information:

> The built-in master planning engine is deprecated. This means that it is not supported (unless a blocking issue) and it will no longer be invested in. It's being replaced by the Planning Optimization Add-in for Microsoft Dynamics 365 Supply Chain Management.

For information about how to migrate to Planning Optimization, see the earlier sections of this article.

You can continue using the deprecated master planning engine for one or more of your companies if necessary. For instructions, see Continue to use deprecated master planning with existing companies.

If you see this error while running on a sandbox and you'd like to remove it, follow the instructions provided in Can I use the deprecated master planning engine on my sandbox environment?

Can I use the deprecated master planning engine on my sandbox environment?

Yes. Even if you receive the error message described in the answer to the previous question on a sandbox environment, the deprecated master planning engine still runs successfully. However, if the error message disturbs you, you can disable it on an IaaS (not Service Fabric) sandbox environment by running the following query on your database:

SQL

```
-- Insert or update an enabled flight:
DECLARE @flightName NVARCHAR(100) = 'ReqPlanningOptimizationExceptionToggle';
IF NOT EXISTS (SELECT TOP 1 1 FROM SysFlighting WHERE flightName = @flightName)
    INSERT INTO SYSFLIGHTING(FLIGHTNAME,ENABLED, FLIGHTSERVICEID,PARTITION)
    SELECT @flightName, 1, 12719367,RECID FROM DBO.[PARTITIONS];
ELSE
    UPDATE SysFlighting SET enabled = 1, flightServiceId = 12719367 WHERE
flightName = @flightName;
```

My environment is on-premises. Do I need an exception to continue using the deprecated master planning engine?

No. An exception isn't required for on-premises environments. You can continue to use the deprecated master planning engine. Your environment admin is informed if any action is required.

We use planned production orders, but I'm concerned about what will happen when we upgrade to version 10.0.16. Should I take any action?

You shouldn't be concerned. You can continue to use the deprecated master planning engine in version 10.0.16. However, evaluate whether migration to Planning Optimization can start with the current functionality, and stay informed about new functionality.

I'm getting an error message when running master planning. Is master planning blocked?

If you're running version 10.0.16 or later, you might receive the following error message when you run master planning:

> You receive this error message because the deprecated master planning engine was used for scenarios supported by Planning Optimization. You should migrate to Planning Optimization now, as the built-in master planning engine has been deprecated. Note that this master planning run did complete successfully.
>
> In case your migration has strong dependencies on pending features, an exception to continued usage of the deprecated master planning engine can be requested.
>
> Please complete the following questionnaire to get started and if relevant request exception from migration to Planning Optimization.

Master planning isn't blocked. Your master planning run was successfully completed, and you can use the result in the usual way. However, to avoid receiving this error message during future master planning runs, you must either migrate to Planning Optimization immediately or request an exception by using the link in the error message.

Last updated on 03/26/2026

Continue to use deprecated master planning with existing companies

> **Note**
>
> This article applies to **existing deployments** where one or more companies (legal entities) already use the deprecated master planning engine. If you're setting up a new deployment or a new tenant, Planning Optimization is the required engine starting with Supply Chain Management version 10.0.41. For more information, see Migration to Planning Optimization for master planning.

Starting in Supply Chain Management version 10.0.41, the deprecated master planning engine is blocked for all new legal entities (companies) added to existing deployments. There's no manual way to enable the deprecated master planning engine for these companies. However, existing companies can continue to use the deprecated master planning engine until they're ready to migrate.

> **Important**
>
> The deprecated master planning engine is only available to existing companies that already use it. Existing companies that use the deprecated engine aren't affected when you upgrade to a new version of Supply Chain Management. However, any new companies added to the deployment must use Planning Optimization.

Companies where planning processes are disabled

Starting with release 10.0.32, you must install the Planning Optimization Add-in before you can enable planning processes, even if you plan to continue using the deprecated master planning engine. If the company where you want to use deprecated master planning has planning processes disabled, follow the steps in this section to enable them.

When you try to set **Disable all planning processes** to *No* on the **Master planning parameters** page without the Planning Optimization Add-in installed, you receive one of the following error messages:

> To enable all planning processes, first follow the instructions provided in Get started with master planning. Then return to this page and enable the parameter.

or

> All planning processes are disabled. Please enable planning processes to continue.

To resolve this error and enable planning processes, follow these steps:

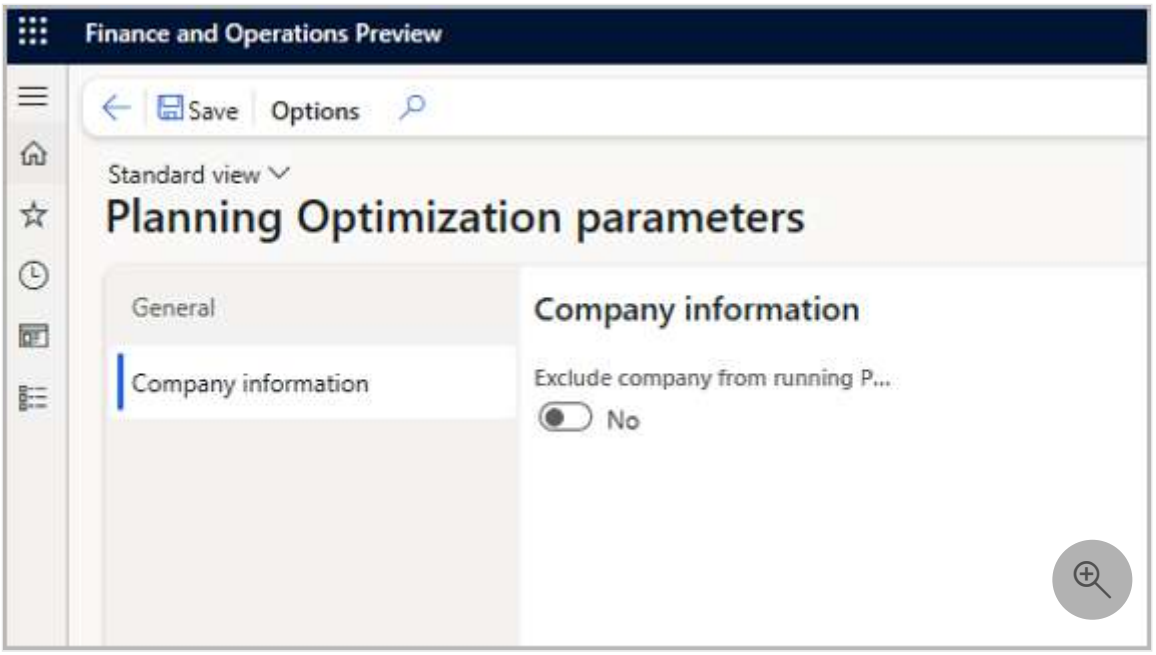
1. Make sure Planning Optimization is installed and enabled as described in Get started with master planning. This step is mandatory starting with release 10.0.32, even if you don't intend to use Planning Optimization.
2. Make sure Planning Optimization is enabled on the **Planning Optimization parameters** page.
3. Enable planning processes for the relevant company by setting **Disable all planning processes** to *No*, on the **Master planning parameters** page.

> **Note**
>
> Starting in Supply Chain Management version 10.0.41, the deprecated master planning engine is blocked for all new legal entities (companies) added to existing deployments. If you're setting up a new company, you must use Planning Optimization. To continue using the deprecated engine for existing companies, see the next section.

Exclude companies from using Planning Optimization and continue to use deprecated master planning

Follow these steps to set a company to continue to use the deprecated master planning engine on an environment that is otherwise enabled to use Planning Optimization:

1. Use the company picker to choose the company (legal entity) that you want to set up.
2. Go to **Master planning** > **Setup** > **Planning Optimization parameters**.
3. Open the **Company information** tab.



4. Set **Exclude company from running Planning Optimization** to one of the following values

    - *Yes* – Use the deprecated master planning engine for this company.
    - *No* – Use Planning Optimization for this company.

5. Select **Save** on the Action Pane.

6. Repeat this procedure for each company you want to set up.

> ⓘ **Note**
>
> If the **Company information** tab appears blank and the **Exclude company from running Planning Optimization** toggle is greyed out, this behavior is expected. The tab and toggle are only relevant for companies that have been granted an exception to use the deprecated master planning engine. Starting from Supply Chain Management version 10.0.41, Planning Optimization is the standard engine for all new deployments and legal entities. Companies that aren't enabled for the deprecated engine don't need this configuration, so the tab remains empty by design. Learn more in Migration to Planning Optimization for master planning.

Unsupported features and exceptions

If your company requires features that aren't yet supported by Planning Optimization, you can continue to use the deprecated engine for the affected companies. Supply Chain Management includes an automated exception process that evaluates your system and shows appropriate

instructions for each legal entity. Learn more about the exception process and the features supported by Planning Optimization in Migration to Planning Optimization for master planning.

Next steps

- Migration to Planning Optimization for master planning
- Planning Optimization fit analysis
- Deprecated master planning overview

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Parameters not used by Planning Optimization

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article lists the parameters that Planning Optimization doesn't currently consider during its operation. The planning service might skip a parameter because, for example, it doesn't yet support related functionality. Alternatively, functional changes might make the parameter obsolete.

The following sections list the parameters that Planning Optimization doesn't use on specific pages. They also explain why each parameter isn't used.

Master planning parameters page

Planning Optimization doesn't use the following parameters or options on the **Master planning parameters** page:

- **General** tab:
  - **Current forecast plan** – Planning Optimization doesn't consider this parameter. Instead, create a master plan that specifies a forecast model.
  - **Copy the complete and updated static master plan to the dynamic master plan** – Planning Optimization doesn't support this option. You can run each plan independently, but results aren't copied between them.
  - **Start time for calculated delays** – Planning Optimization doesn't support this option.
  - **Use dynamic negative days** – Planning Optimization always uses the *Dynamic negative days* approach.
  - **Today's date calendar** – Planning Optimization doesn't support this option. Planning Optimization always uses current time rounded to the next whole hour.
  - **Use of cache** – Planning Optimization doesn't consider this parameter because it manages performance and scaling.

- **Number of tasks in helper task bundle** – Planning Optimization doesn't consider this parameter because it manages performance and scaling.
- **Pre-processing: Automatically filter by items with direct demand** – Planning Optimization doesn't support this option.
- **Post-processing: Automatically filter by items with direct demand** – Planning Optimization doesn't support this option.
- **Number of threads** – Planning Optimization doesn't consider this parameter because it manages performance and scaling.
- **Planning process timeout in minutes** – In Planning Optimization, there's a fixed timeout of 1 hour.
- **Scheduling start time** – Planning Optimization doesn't consider this parameter. With Planning Optimization, an explosion run is a run of the plan for the selected items.

- **Planned orders** tab:
  - **Receipt time** – Planning Optimization doesn't support this option. The receipt time exactly matches the demand time in Planning Optimization.

- **Number sequences** tab:
  - **Master scheduling planned order number** – Planning Optimization uses its own proprietary number sequence for all generated planned order numbers, and there's no method available for modifying this number sequence. The system ignores this parameter entirely. Learn more about how Planning Optimization handles number sequences in Differences between Planning Optimization and the deprecated master planning engine.

Coverage groups page

Planning Optimization doesn't use the following parameters or options on the **Coverage groups** page:

- **Other** FastTab:
  - **Freeze time fence (days)** – Freeze time fence is supported in Supply Chain Management version 10.0.40 and later. To use it, turn on the feature named *Freezing time fence for Planning optimization* in feature management.
  - **Forecast plan time fence** – Planning Optimization doesn't support forecast plans. However, it does consider this value when consuming forecast data within a master plan.

- **Delays** FastTab:

- **Calculated delays** – Planning Optimization doesn't consider this parameter because it always calculates delays for the coverage group.
- **Calculate delays time fence (days)** – Planning Optimization doesn't consider this parameter because it calculates delays for the **Coverage time fence (days)**.

Item coverage page

Planning Optimization doesn't use the following parameters or options on the **Item coverage** page:

- **General** tab:
  - **Planned order type** – Planning Optimization supports all order types, including kanban. To use kanban, enable the *Lean manufacturing for Planning Optimization* feature in feature management. This is a preview feature in Supply Chain Management version 10.0.45 and is generally available as of version 10.0.46.
  - **Freeze time fence (days)** – Freeze time fence is supported in Supply Chain Management version 10.0.40 and later. To use it, turn on the feature named *Freezing time fence for Planning optimization* in feature management.
  - **Fulfill minimum** – Planning Optimization doesn't support the *Today's date*, *First issue*, and *Coverage time fence* options. It always uses the *Today's date + procurement time* option.
  - **Minimum periods** – Planning Optimization doesn't support this parameter.

Master plans page

Planning Optimization doesn't use the following parameters or options on the **Master plans** page:

- **General** FastTab:
  - **Include continuity plan** – Planning Optimization doesn't support this parameter.
  - **Capacity time fence for bottleneck resources** – Planning Optimization doesn't support this parameter because customers didn't use it.
  - **Planned orders** – Planning Optimization uses its own proprietary number sequences that you can't modify.
  - **Session** – Planning Optimization uses its own proprietary number sequences that you can't modify.
  - **Continuity plan** – Planning Optimization doesn't support this parameter.

- **Time fences in days** FastTab:

- **Freeze** – Freeze time fence is supported in Supply Chain Management version 10.0.40 and later. To use it, turn on the *Freezing time fence for Planning optimization* feature in feature management.
- **Continuity plan** – Planning Optimization doesn't support this parameter.
- **Calculated delays** – Planning Optimization doesn't consider this parameter. Calculated delays are created for the **coverage time fence (days)**.

- **Calculated delays** FastTab:
  - **Ensure that the planned orders are not created prior to the master planning run date** – Planning Optimization doesn't support this parameter. Planning Optimization never creates planned orders in the past.
  - **Add the calculated delay to the requirement date** (in the **Planned kanban** section) – To use kanban, enable the *Lean manufacturing for Planning Optimization* feature in feature management. This is a preview feature in Supply Chain Management version 10.0.45 and is generally available as of version 10.0.46.

- **Action message** FastTab:
  - **Update postponed date as requirement date** – Planning Optimization discontinued this parameter.

Scheduling parameters page

Planning Optimization doesn't use the following parameters or options on the **Scheduling parameters** page:

- **Schedule timeout enabled** – In Planning Optimization, schedule timeout is always enabled.
- **Maximum scheduling time per sequence** – Planning Optimization doesn't support this parameter.
- **Schedule optimization timeout enabled** – Planning Optimization doesn't support this parameter.
- **Optimization attempts timeout** – Planning Optimization doesn't support this parameter.
- **Keep production unit** – Planning Optimization doesn't support this parameter.
- **Keep warehouse from resource** – Planning Optimization doesn't support this parameter.

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Last updated on 05/01/2026

Differences between Planning Optimization and the deprecated master planning engine

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you transition from the deprecated built-in master planning engine to Planning Optimization, you might notice differences in planned orders, quantities, and dates. Use this article and the related resources to identify whether an issue you're experiencing is due to an expected difference, an unsupported feature, or a parameter change.

Identify the type of issue

Use the following table to find the right resource for your situation:

Expand table

Issue	Resource
Planning Optimization shows different planned order results, dates, quantities, or behavior	Expected differences (this article)
A feature doesn't work or is missing in Planning Optimization	Planning Optimization fit analysis
Date or time calculations produce different results	Date and time parameters used by Planning Optimization
A master planning parameter seems to have no effect	Parameters not used by Planning Optimization
You have questions about when the deprecated engine will be removed	Deprecated master planning overview
You need guidance on the full migration process	Migration to Planning Optimization

Expected differences

The following table lists specific differences between Planning Optimization and the deprecated master planning engine that aren't listed on the **Planning Optimization fit analysis** page.

Expand table

Feature	How Planning Optimization differs from deprecated master planning
Forecasts for today's date	Planning Optimization doesn't consider forecasts for today's date, though the deprecated master planning engine considered them.
Filtered production runs	For details, see Production planning - Filters.
Forecast planning	Forecast planning isn't supported. Use master planning where a forecast model is assigned to the master plan.
Number sequences for planned orders	Number sequences for planned orders aren't supported. Planning Optimization uses its own proprietary number sequence for all generated planned order numbers, and there's no method available for modifying this number sequence. The **Master scheduling planned order number** setting on the **Number sequences** tab of the **Master planning parameters** page has no effect when you use Planning Optimization. The planned order number normally shows 14 digits, but the sequence is actually built on 20 characters, with 6 digits allocated for the planning run count and 10 for the planned orders count.
Plan copy, plan delete, and plan version cleanup	Under **Master planning** > **Master planning** > **Maintain plans** in the navigation pane, the following items are disabled: • Plan copy • Delete plan • Plan version cleanup Plan version cleanup is automatic with Planning Optimization, so you don't need to do it manually.
Return orders	Planning Optimization doesn't consider return orders.
Scheduling related features	For details, see Scheduling with infinite capacity.
Safety stock fulfillment	Planning Optimization always uses the *Today's date + procurement time* option for the **Fulfill minimum** field on the **Item coverage** page. This choice helps prevent unwanted planned orders and other issues. If the procurement time isn't included for safety stock, planned orders that are created for low on-hand inventory are always delayed because of the lead time. Learn more in Safety stock fulfillment with the deprecated master planning engine.
Safety stock pegging and net requirements	The *Safety stock* requirement type isn't included and isn't displayed on the **Net requirements** page. Safety stock doesn't represent demand and doesn't have a requirement date associated with it. Instead, it sets a constraint on how much inventory must be present at all times. However, the **Minimum** field value is still

Feature	How Planning Optimization differs from deprecated master planning
	taken into account when calculating planned orders during master planning. Inspect the **Accumulated quantity** column on the **Net requirements** page to see that this value was considered. Because the pegging is different, different actions might be suggested. Learn more in Safety stock fulfillment with the deprecated master planning engine.
Transport calendars	The value in the **Transport calendar** column on the **Modes of delivery** page is ignored.
Min/max coverage code with no values	With the deprecated master planning engine, when you use a min/max coverage code where no minimum or maximum values are set, the planning engine treats the coverage code as a requirement and creates one order for each requirement. With Planning Optimization, the system creates one order per day to cover the full amount for that day.
Net requirements and manually created planned orders	With the deprecated master planning engine, manually created supply orders for an item automatically appear among the net requirements for that item. For example, when you create a purchase order from a sales order, the purchase order appears on the **Net requirements** page without requiring any prior actions. This behavior occurs because the deprecated master planning engine logs inventory transactions in the `inventLogTTS` table and shows changes on the **Net requirements** page for dynamic plans. However, with Planning Optimization, manually created orders don't appear among the net requirements of an item until you run Planning Optimization (using a plan that includes the item), or until you select **Update** > **Master planning** on the Action Pane on the **Net requirements** page, which runs master planning for the item. For more information about how to work with the **Net requirements** page, see Net requirements and pegging information.
Net requirements and any changes to supply or demand	With the deprecated master planning engine, any changes to supply or demand for an item automatically appear among the net requirements for that item. For example, when you increase the quantity of a planned order, the purchase order appears on the **Net requirements** page without requiring any prior actions. This behavior occurs because the deprecated master planning engine logs inventory transactions in the `inventLogTTS` table and shows changes on the **Net requirements** page for dynamic plans. However, with Planning Optimization, any changes don't appear among the net requirements of an item until you run Planning Optimization (using a plan that includes the item), or until you select **Update** > **Master planning** on the Action Pane on the **Net requirements** page, which runs master planning for the item. Learn more about how to work with the **Net requirements** page in Net requirements and pegging information.
Resource assignment	When working with infinite capacity, the deprecated master planning engine assigns all planned orders to the same resource on a given resource group. Planning Optimization improves on this behavior by selecting resources at random so different production orders can use different resources. If you want to use the same resource for all planned orders, specify that resource in the route.
Extended data types	Planning Optimization doesn't support changes to the precision of EDTs. This

Feature	How Planning Optimization differs from deprecated master planning
(EDTs)	process isn't officially supported and isn't guaranteed to work.
Safety stock fulfillment	Planning Optimization always uses a **Fulfill minimum** of *Today's date + procurement time*. This choice prepares the system to use a simplified planning setup in the future, and provides an actionable result. If the procurement time isn't included for safety stock, planned orders that are created for low on-hand inventory are always delayed because of the lead time. This behavior can cause significant noise and unwanted planned orders. The best practice is to change the setting to use *Today's date + procurement time*. Update master data to avoid warnings. Learn more in Safety stock fulfillment with the deprecated master planning engine.
Purchase order fulfillment	When fulfilling a purchase order, the system always applies a lead time and never disregards it, even if the **Add delay to calculated requirement date** parameter is set to *No*. This behavior differs from that of the deprecated master planning engine, which would set an unfeasible date in this case. For example, if today is March 1, 2024, lead time is 30 days, and **Add delay to calculated requirement date** is set to *No*, the deprecated engine creates a purchase order with date March 1, 2024, while Planning Optimization creates it with date April 1, 2024 (today plus the 30-day lead time).
Copy static to dynamic plan	Planning Optimization doesn't copy static plans to dynamic plans, regardless of the setting on the **Master planning parameters** page. In general, this operation is less relevant because of the speed and complete regeneration that Planning Optimization provides. If you use two or more plans, trigger master planning for each plan.
Negative days	Planning Optimization always uses dynamic negative days, regardless of the setting of **Use dynamic negative days** on the **Master planning parameters**. The **Use dynamic negative days** setting on the **Master planning parameters** page has no effect on this behavior. For more information on negative days, see Delay tolerance (negative days)
Today's date calendar	Planning Optimization doesn't consider the **Today's date calendar** setting on the **Master planning parameters** page. Planning Optimization always uses the current time rounded to the next whole hour. In the deprecated master planning engine, this setting defined the start of a freezing time fence. However, the system still uses the **Today's date calendar** parameter to determine whether a manually created production order is scheduled within the freezing time fence, even when you're using Planning Optimization.
Planned orders in the past	Planning Optimization never plans orders in the past, regardless of the **Requested date** setting. If you set a **Requested date** that's in the past, the system keeps your setting, but sets the **Required date** to *Today*.
Definition/meaning of safety stock	In Planning Optimization, safety stock isn't actual demand. For each coverage group, you can specify how strictly the system should peg safety stock as demand against the planned order that's created for it. (Learn more in Safety stock pegging options.) Coverage groups that use strict safety stock pegging work just like all

Feature	How Planning Optimization differs from deprecated master planning
	groups in the deprecated engine where the **Fulfill minimum** field is set to *Today's date + procurement time*. In the deprecated master planning engine, safety stock is demand, the same as other demand types. You can specify when safety stock will be fulfilled.
When safety stock is fulfilled	In Planning Optimization, safety stock is always fulfilled on *today's date + procurement time*, regardless of the item coverage setting. In the deprecated master planning engine, item coverage settings include a **Fulfill minimum** field that defines when the safety stock must be fulfilled. (Learn more in Safety stock fulfillment with the deprecated master planning engine.)
Forecast plan time fence	When deciding which demand and supply forecast entries to take into consideration, Planning Optimization always takes the lower of the following two values: *coverage time fence* or *forecast plan time fence*. The deprecated planning engine always uses the *forecast plan time fence*. Planning Optimization therefore guarantees that the supply covers all demand forecast entries that are included in a plan.
Issue margin	When the issue margin can be met, Planning Optimization applies its value to the *receipt requirement date*, not the *issue requirement date*; if the margin can't be met, Planning Optimization pushes the issue requirement date forward. The deprecated planning engine always adjusts the *issue requirement date* by the value of the issue margin. Planning Optimization therefore treats the issue margin more correctly, as a shipping-time buffer that affects how far in advance receipt is required. Learn more in Safety margins.
Vendor calendar lead time	Planning Optimization doesn't consider closed days from the vendor's calendar when calculating lead times for delivery dates. It only considers the coverage group calendar when calculating lead times. Learn more in Calendar matrix for Planning Optimization.
Zero positive days	Setting positive days to one in Planning Optimization provides the same behavior as setting positive days to zero in the deprecated master planning engine.
Planning with negative on-hand quantities	If the system shows a negative aggregate on-hand quantity, Planning Optimization treats it as quantity 0 (zero) to help avoid over supply. Learn more in Planning with negative on-hand quantities. The deprecated master planning engine instead replenished the negative quantity.
Firming manual planned production orders	When firming planned production orders that were created manually, Planning Optimization doesn't automatically trigger the explosion of items. You must manually trigger item explosions as needed.
CTP with lean manufacturing	The *event* kanban rule replenishment strategy with event type *sales event* set to *Automatic with CTP* doesn't automatically generate and add kanbans to the schedule for sales order order lines that use the *CTP* delivery date control method. Instead, replenish through a planning kanban and apply auto-firming.

Related information

- Migration to Planning Optimization for master planning
- Deprecated master planning overview
- Planning Optimization fit analysis
- Parameters not used by Planning Optimization
- Date and time parameters used by Planning Optimization

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Last updated on 05/04/2026

Planning Optimization fit analysis

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Planning Optimization fit analysis helps you identify where the result might differ between the deprecated master planning engine and Planning Optimization. The system runs this analysis based on your current setup and data.

You should analyze the result from the Planning Optimization fit analysis as part of the migration process. The scope of Planning Optimization isn't equal to the deprecated master planning engine functionality. Work with your partner and read the documentation to prepare for the migration.

> **Note**
>
> - The Planning Optimization fit analysis can't identify some inconsistencies. For more information, see Differences between classic master planning and Planning Optimization.
> - If the analysis finds inconsistencies, you can still use Planning Optimization. The results of the fit analysis just show places where the planning service doesn't honor your current setup. In other words, they show places where some processes might be ignored or aren't supported.

Run Planning Optimization fit analysis

To run the Planning Optimization fit analysis and view the results, follow these steps:

1. Select a company (legal entity) from the company picker in the navigation bar.
2. Go to **Master planning** > **Setup** > **Planning Optimization fit analysis**.
3. On the Action Pane, select **Run analysis**.
4. The system runs the analysis and then shows the results. For each result that shown, consult the table later in this article to see whether that feature is currently supported for Planning Optimization and, if it isn't yet supported, when support is expected to become

available. If no issues exist for your selected company, the result list is blank, and you receive a "No issues found" message.

5. Repeat this procedure for each company in your organization.

Possible results from the fit analysis

The following table shows the various results that can appear after a fit analysis. Number signs (#) are replaced with a number that indicates the number of records that have the listed issue.

> **Important**
>
> For features that aren't yet supported, the following table provides an availability estimate based on the current roadmap. These estimates are subject to change without notice.

Expand table

Feature	Listed issue	Explanation	Expected availability
Actions	Coverage groups with Actions calculation enabled	This feature is now supported.	Supported
Base calendars	Calendars using base calendar	This feature is now supported.	Supported
Batch disposition codes	Non-nettable batch disposition masters	This feature is now supported. Learn more in Use batch disposition codes to mark batches as available or unavailable	Supported
Capable to promise (CTP)	Default order settings with delivery date control set to CTP	In Supply Chain Management 10.0.28 and newer, a process called *CTP for Planning Optimization* makes confirmed ship and receipt dates available after the dynamic plan runs. For older versions of Supply Chain Management, the legacy CTP setting is ignored when Planning Optimization is enabled.	Supported
Firming	Coverage groups with auto firming time fence set	In version 10.0.7 and later, firming is supported as a separate firming batch job after master planning is completed. Auto firming for Planning Optimization is based on the order date (start date), not the requirement date (end date). This behavior ensures that firming of planned orders occurs in due time, without	Supported

Feature	Listed issue	Explanation	Expected availability
		having to include lead time in the firming time fence.	
Firming	Item coverage records with auto firming set	In version 10.0.7 and later, auto firming is supported as a separate firming batch job after master planning is completed. Auto firming for Planning Optimization is based on the order date (start date), not the requirement date (end date). This behavior ensures that firming of planned orders occurs in due time, without having to include lead time in the firming time fence.	Supported
Firming	Master plans with auto firming set	In version 10.0.7 and later, auto firming is supported as a separate firming batch job after master planning is completed. Auto firming for Planning Optimization is based on the order date (start date), not the requirement date (end date). This behavior ensures that firming of planned orders occurs in due time, without having to include lead time in the firming time fence.	Supported
Planning items	Planning Items	This feature is now supported for all versions of Supply Chain Management.	Supported
Forecast	Coverage groups with "Include intercompany orders" disabled	This feature is now supported. Learn more in Intercompany planning	Supported
Forecast	Coverage groups with "Reduce forecast by" setting set to a value different than "Orders"	This feature is now supported. Learn more in Master planning with demand forecasts	Supported
Forecast	Forecast models with sub models	This feature is now supported. Learn more in Master planning with demand forecasts	Supported
Forecast	Master plans with "Include supply forecast" enabled	This feature is now supported. Learn more in Master planning with supply forecasts	Supported
Freeze time fence	Coverage groups with freeze time fence set	This feature is now supported. To use it, enable the *Freezing time fence for Planning optimization* feature in Feature management. As of Supply	Supported

Feature	Listed issue	Explanation	Expected availability
		Chain Management version 10.0.43, this feature is turned on by default.	
Freeze time fence	Item coverage records with freeze time fence set	This feature is pending. Currently, the freeze time fence setup is ignored when Planning Optimization is enabled, regardless of this setting.	Supported
Freeze time fence	Master plans with freeze time fence set	This feature is now supported. To use it, enable the *Freezing time fence for Planning optimization* feature in Feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default.	Supported
Intercompany	Master plans including planned downstream demand	This feature is now supported. Learn more in Intercompany planning	Supported
Kanban	Item coverage records with planned order type kanban	This feature is now supported. To use it, enable the *Lean manufacturing for Planning Optimization* feature in feature management. This is a preview feature in Supply Chain Management version 10.0.45 and is generally available as of version 10.0.46.	Supported
Kanban	Items with default order type kanban	This feature is now supported. To use it, enable the *Lean manufacturing for Planning Optimization* feature in feature management. This is a preview feature in Supply Chain Management version 10.0.45 and is generally available as of version 10.0.46.	Supported
Product lifecycle state	Product lifecycle states not active for planning	This feature is now supported. Learn more in Exclude products that have specific product lifecycle states	Supported
Production	BOM lines with rounding or multiple setup	This feature is supported as of June 1, 2023. No feature management is required.	Supported
Production	BOM/formula lines with formula measurement	This feature is supported in version 10.0.33 and higher.	Supported
Production	BOM/formula lines with item substitution (plan groups): #	This feature is now supported. To use it, enable the *Item substitution (Plan group) support for Planning Optimization* and/or *Item substitution for bill of materials in Planning optimization*	Supported

Feature	Listed issue	Explanation	Expected availability
		features in Feature management (on by default as of Supply Chain Management version 10.0.43).	
Production	BOM/formula lines with negative quantity: #	This feature is supported as of June 1, 2023. No feature management is required.	Supported
Production	BOM/formula lines with resource consumption: #	This feature is now supported.	Supported
Production	BOM/formula lines with step consumption: #	This feature is pending. Currently, step consumption is ignored on BOM and formula lines when Planning Optimization is enabled.	Supported
Production	BOMs with constant scrap or variable scrap defined: #	This feature is now supported.	Supported
Production	BOMs with subcontracting: #	This feature is now supported.	Supported
Production	BOMs without a site: #	This feature is now supported. Learn more in Production planning	Supported
Production	Demand with specific BOM or route requirements defined: #	This feature is now supported.	Supported
Production	Formula versions with Co/By products: #	This feature is supported in version 10.0.33 and higher.	Supported
Production	Formula versions with yield: #	This feature is supported in version 10.0.33 and higher.	Supported
Production	Plans including sequencing: #	This feature is now supported.	Supported
Production	Released production orders that aren't started, where scheduled start is earlier than today: #	This feature is now supported.	Supported
Production	Resources scheduled with finite capacity: #	This feature is now supported.	Supported

Feature	Listed issue	Explanation	Expected availability
Production	Routes used in planning: #	This feature is supported.	Supported
Production	Sales line reservation using explosion: #	This scenario isn't yet supported. Sales line reservations aren't automatically made during explosion.	Future wave
Production	Scheduling with explosion of production orders: #	Scheduling that uses explosion of production orders is supported starting in version 10.0.32, with a private preview available for version 10.0.31 (contact Microsoft Support to request access to the private preview feature on version 10.0.31). A Planning Optimization run is performed for the items contained in the production order and its respective components in its BOM.	Supported
Request for quotations	Master plans with request for quotations enabled: #	This feature is supported, see Plan based on quotations and RFQs	Supported
Requisitions	Master plans with requisitions enabled: #	This feature is now supported. Learn more in Purchase requisitions	Supported
Safety margins	Coverage groups with safety margin: #	This feature is now supported. Learn more in Safety margins	Supported
Safety margins	Master plans with safety margin: #	This feature is now supported. Learn more in Safety margins	Supported
Sales quotations	Master plans with sales quotations enabled: #	This feature is now supported, see Plan based on quotations and RFQs	Supported
Shelf life	Master plans with shelf life enabled: #	This feature is now supported.	Supported
Custom inventory dimensions	Tracking dimension groups with one or more custom inventory dimensions: #	This feature is now supported for all versions of Supply Chain Management. No set up is required.	Supported
Intercompany Planning	Setup Intercompany planning execution with planning optimization: #	This scenario isn't yet supported. It is not yet possible to setup Planning Optimization to execute automatically across legal entities according to the Intercompany planning group	Future wave

Feature	Listed issue	Explanation	Expected availability
		setup and using the run menu item to invoke the actual planning execution. It is possible to run the cross legal entity planning execution manually or by setting up sequentially scheduled batch jobs for each legal entity planning execution	
Production	Planning Optimization support for requirement types skills, courses, certificates, and titles.	This scenario isn't yet supported.	Future wave

Related information

- Master planning system architecture
- Get started with master planning
- Differences between classic master planning and Planning Optimization
- Parameters not used by Planning Optimization
- View plan history and planning logs
- Run planning for a subset of items
- Cancel a planning job

Last updated on 03/26/2026

Improve master planning performance

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains the various options that can help you improve the performance of master planning or troubleshoot issues. It includes information about parameters and settings, as well as recommended configurations and actions. Also included is a summary of all the important parameters that you should consider when you have long-running master planning jobs.

This article is intended for system admins or IT users who have the capability to troubleshoot. It's also intended for production or supply planners because it includes information about parameters that are related to business planning requirements.

Parameters related to master planning performance

Various parameters influence the running time of master planning and should be considered.

Number of threads

The **Number of threads** parameter lets you adjust the master scheduling process to help improve performance on the specific data set. This parameter specifies the total number of threads used to run master planning. It causes parallelization of the master planning run, which helps decrease the running time.

Set the **Number of threads** parameter in the **Master planning run** dialog box. To open this dialog box, go to **Master planning** > **Master planning** > **Run** > **Master planning**, or select **Run** in the **Master planning** workspace. To determine the best value for this parameter, use a trial-and-error process. However, you can use the following formulas to calculate an initial value:

- **If your industry is manufacturing:** (Number of threads) = (Number of planned orders ÷ 1,000)
- **Otherwise:** (Number of threads) = (Number of items ÷ 1,000)

The number of helpers used during master planning must be less than or equal to the maximum number of threads that the batch server allows. If the number of helpers exceeds the number of threads on the batch server, the extra threads don't do any work.

> **Note**
>
> Setting the **Number of threads** parameter to **0** (zero) increases the master planning running time. Therefore, always set a value that's more than 0.

Number of tasks in helper task bundle

Change the **Number of tasks in task bundle** setting (that is, the bundle size) to potentially decrease the running time. This setting controls the number of items that a single helper plans together.

Set the **Number of tasks in task bundle** parameter in the **Performance** section on the **General** tab of the **Master planning parameters** page (**Master planning** > **Setup** > **Master planning parameters**). The best value for this parameter depends on your data. Therefore, start with a value of **1**, and then use a trial-and-error process to determine the best value for your setup.

In general, increase the number of tasks when the number of items is very large (in the hundreds of thousands). Otherwise, decrease the number of tasks. For the following specific industries, consider these recommendations:

- In the retail and distribution industries where there are many independent items, use many helpers because there's no dependency between items.
- In the manufacturing industry where there are many bills of materials (BOMs) and shared subcomponents, use fewer helpers because dependencies between items might cause waiting times.

> **Tip**
>
> If you have performance issues, reduce the **Number of helpers in task bundle** setting to **1**. Then, start the trial-and-error process to find the best value for your setup. In general, performance issues occur when one item takes longer to process than the remaining items. In this case, you see that two subsequent tasks that have a status of **Coverage** in the master planning run take significantly different amounts of time to run. In extreme cases, this

difference might be as much as 30 minutes. You can infer the amount of time that tasks take to run by looking at the duration of each task.

Use of cache

The **Use of cache** parameter lets you adjust the master scheduling process to help it perform better on the specific data set. For example, you can adjust it to achieve the following results:

- If more caching is done, more data is collected in memory. The expectation is that the data can be used again later. If the data is in memory, you might save some database requests. However, if more caching is done, memory requirements increase.
- If less caching is done, the same data might have to be fetched from the database more often. Additionally, Application Object Server (AOS) stores little data in memory throughout the process.

It's difficult to predict which option is better because each case depends not only on the data but also on the hardware. For example, because less caching causes additional load on the database server, it probably isn't a good idea to use that option if your database server is already overloaded.

Set the **Use of cache** parameter in the **Performance** section on the **General** tab of the **Master planning parameters** page (**Master planning** > **Setup** > **Master planning parameters**). The effectiveness of caching depends heavily on the customer data. For example, if cached data is never needed, you just waste memory if you store data until the end of the scheduling process. In this case, if you configure less caching, performance might increase because AOS requires less memory and server resources are freed up for other tasks.

> **Tip**
>
> In general, set the **Use of cache** parameter to **Maximum**, because the parameter is intended as a performance enhancing feature. Set the parameter to **Minimum** if you run on-premises and have limited memory (approximately 2 gigabytes [GB]).

Number of orders in firming bundle

The **Number of orders in firming bundle** parameter specifies the total number of orders that each thread or batch processes at a time. It causes parallelization of the auto-firming process.

Set the **Number of orders in firming bundle** parameter in the **Performance** section on the **General** tab of the **Master planning parameters** page (**Master planning** > **Setup** > **Master planning parameters**). Parallelization of the auto-firming process is based on the orders that must be processed together. For example, if this parameter is set to **50**, each thread or batch task picks up 50 orders at a time and processes them together. Use a trial-and-error process to find the best value. However, you can use the following formula to calculate an initial value:

(Number of orders per bundle) = (Number of demand items ÷ Number of threads)

> **Note**
>
> If you set the **Number of orders in firming bundle** parameter to **0**, the auto-firming process isn't parallelized. The whole process runs on a single batch task and has a cumulative running time. Therefore, the running time of your master planning increases. For this reason, set this parameter to a value that is more than **0**.

Time fences

Time fences specify how far in the future master planning must calculate requirements. The larger the time fence is, the longer it takes master planning to run. Therefore, set the time fences according to your business requirements. For more information about time fences, see Master plans overview.

Actions

Among the time fences, you can also find the **Action message** parameter. The calculation of action messages causes a longer running time for master planning. If you don't regularly analyze and apply action messages (daily, weekly, and so on), consider turning off the calculation during the master planning run. To turn off the calculation, on the **Master plans** page (**Master planning** > **Setup** > **Plans** > **Master plans**), set the **Action message** time fence to **0**. Also make sure that the **Action message** setting is turned off for all the coverage groups.

Futures

The calculation of futures causes a longer running time for master planning. If you aren't planning BOMs, or if delays don't have to be propagated from supply to demand during planning, consider turning off futures calculations during master planning. To turn off the

calculations, set the **Futures** time fence to **0** for the master plan that you're running. Also make sure that the **Futures** setting is turned off for all the coverage groups.

One heavy routine at a time

When you schedule master planning, don't schedule any other batch job at the same time. Be especially careful that you don't schedule any other heavy routines, such as inventory close, at the same time.

Review the session log

The system can collect extra information about the tasks that run during master planning. To have the system collect this information, turn on the **Track processing time** setting in the **Master planning run** dialog box. The information that the system collects can help you find bottlenecks in the run. For example, when the **Number of tasks in helper task bundle** parameter is set to **1**, you can identify the item that has the longest running time. You can also compare the running times for the various threads that have a status of **Coverage** and compare the duration for each task.

To review the master planning runs of your system, follow one of these steps:

- In the **Master planning** workspace, select a master plan in the drop-down field. On the **Master planning** tile, select **History**. Select a job, select **Inquiries** on the FastTab, and then select **Process task duration**.
- On the **Master plans** page, select a plan in the left pane. Select **History** on the FastTab. Select a job, select **Inquiries** on the FastTab, and then select **Process task duration**.

When you review the session log, consider the following points:

- **Update** shouldn't take a long time (in general, it should take up to 30 minutes), but it's single-threaded.
- **Copy plan** shouldn't take a long time (it should take about one minute).
- **Auto firming** usually takes about 30 minutes. However, it can take up to multiple hours, depending on the number of orders and the complexity of the items.
- **Auto firming** should take less time than **Coverage**.
- **Coverage** should take the longest time relative to the rest.
- **Action** and **Future message** can take a long time, depending on the data and the number of BOMs.

Filtering of items

Filters applied in the **Master planning run** dialog box affect the duration of the master planning run. Go to **Master planning** > **Master planning** > **Run** > **Master planning**, or select **Run** in the **Master planning** workspace. To exclude items from the run, filter by the lifecycle state of the item (not by item numbers). When you filter by lifecycle state, the update process takes less time than when you filter by item numbers.

Automatically filter by items with direct demand

To improve the master planning run time, you can choose to only include items with direct demand. You can use this filter only for a complete master planning run with no other filters applied in the **Records to include** field. A master planning run with filters ignores the **Automatically filter by items with direct demand** setting.

You can find the **Automatically filter by items with direct demand** field on the **Master planning parameters** page. Use it with both pre-processing and post-processing settings.

Pre-processing

The **Pre-processing: Automatically filter by items with direct demand** parameter ensures that the pre-processing phase of master planning only includes items that fulfill at least one of the following conditions:

- The item has an expected receipt or issue, such as a purchase order, sales order, quote, transfer order, or production order.
- The item has item coverage with safety stock (minimum on-hand inventory).
- Forecast demand after today exists for the item.
- Forecast supply after today exists for the item.
- The item includes any continuity lines from the call center module yet to be created.

> **Note**
>
> An item that has physically available on-hand inventory doesn't show a requirement transaction because there's no demand for the item.

Post-processing

The **Post-processing: Automatically filter by items with direct demand** option is only relevant if you set **BOM version requirement** in your coverage groups. Otherwise, you don't need to enable the parameter.

Before the coverage step starts, there's a pre-coverage step during which items with the coverage setting **BOM version requirement** enabled are reprocessed. This step ensures that items from the required BOM version are planned. Items that are considered to have demand during pre-processing no longer have any demand and therefore should be excluded from the planning run.

Performance checklist summary

- **Number of threads** – Set to a value that's more than **0** (zero).
- **Number of tasks in helper task bundle** – Set to a value that's more than **0** (zero). (Use the formulas that are given earlier in this article.)
- **Use of cache** – Set to **Maximum** unless your system is low on memory.
- **Number of orders in firming bundle** – Set to a value that's more than **0** (zero). (Use the formula that is given earlier in this article.)
- **Time fences** – Adjust to your business needs.
- **Actions and futures** – Disable actions and future if you don't use them.
- **One heavy routine at a time** – Don't run master planning together with any other heavy routine.
- **Review the session log.**
- **Filtering of items** – Use the lifecycle state to exclude items from the master planning run. (Don't use the item numbers.)

Last updated on 05/29/2026

Set up master plans for the deprecated planning engine

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes various important strategies and parameters that are used to set up master plans when you are using the deprecated planning engine. It includes an overview of the types of plans that are used by master planning and explains which plan strategy you should use, depending on your business requirements. It also describes the main parameters that affect the plan and explains how those parameters influence the planned orders that are suggested.

Types of master plans

The deprecated master planning engine uses two types of plans: static and dynamic. Each type is designed for a different use. For the best performance, we recommend that you use the appropriate plan for your scenario. You can configure master plans on the **Master plans** page.

- **Static plan** – The master planning calculation uses the current data to generate a net requirements plan. This plan remains unchanged until the next time that you run master planning or manually change the plan. This is an operating plan that various company personnel, such as a purchaser or production planner, can use to base their decisions on and perform their daily activities.
- **Dynamic plan** – This plan starts with the same net requirements plan that was generated by master planning. However, you can update the dynamic plan every time that the master data changes. This could be when you create a new sales order, for example. This enables you to monitor the changing order network and item availability without disturbing the static plan that others are using for their work processes.

A company may choose to work with just a dynamic plan or it may use both static and dynamic plans. In addition, you can configure any master plan to reflect a particular strategy or address an issue. Examples are as follows:

- Set higher inventory levels to guarantee against stockouts.
- Set longer safety margins to protect against unreliable vendors.

You can also set up the starting dynamic plan so that it is updated with the new requirements plan every time that you run master planning. You can specify these settings on the **Master planning parameters** page.

Static plan

The static plan remains unchanged, regardless of any changes in the supply and demand, until the next time that master planning is run.

The static plan is used to approve and firm the orders that are suggested. It's an operating plan that various company personnel (such as a purchaser or production planner) can base their decisions on and use to perform their daily tasks and activities.

The static plan also supports regeneration. A totally new optimized plan is calculated every time that master planning is run, and existing orders that haven't been approved are deleted.

Dynamic plan

The dynamic plan usually starts as a copy of the static plan, but it can be updated every time that master data changes. For example, if the data changes, a new sales order is created.

The dynamic plan is used for ad-hoc planning. It lets the company monitor the changing order network and item availability without disturbing the static plan that other people are using for their work processes. It's especially used for capable to promise (CTP). The dynamic plan is the default plan when net requirements are opened from order pages (such as sales order, purchase order, or production order pages).

The dynamic plan uses net change. Therefore, it helps guarantee a faster running time.

Strategies for using master plans

You can use either one plan or two plans in master planning. The strategy that you use depends on your business requirements.

One-plan strategy

For the one-plan strategy, you use the same master plan for the static plan and the dynamic plan. This strategy is used for make-to-stock scenarios, where you don't usually have to simulate a plan that fulfills an order.

The one-plan strategy is typically used in retail and distribution industries, or where sales delivery dates are confirmed based on service level agreements (SLAs) or lead times. For the

plan, delivery date control can be used without CTP.

If you must do a simulation, the plan can be run again for the items that require the simulation. The planned orders that order simulations produce are usually firmed.

Two-plan strategy

For the two-plan strategy, you use a static plan and a different dynamic plan. The two-plan strategy is typically used for configure-to-order and make-to-order scenarios, where you must do sales order simulations and calculate exact delivery dates for sales orders, but the calculations must not affect everyday operations. These simulations are always done in the dynamic plan. The two-plan strategy is useful in the automotive and original equipment manufacturer (OEM) industries, for example.

For the two-plan strategy, delivery date control CTP can be used. When CTP is used, it automatically triggers the run in the dynamic plan.

Setting up the plans

You can create plans on the **Master plans** page (**Master planning** > **Setup** > **Plans** > **Master plans**).

You can specify which plans will be used for the static plan and the dynamic plan by setting the **Current static master plan** and **Current dynamic master plan** fields on the **Master planning parameters** page (**Master planning** > **Setup** > **Master planning parameters**). To use a one-plan strategy, select the same plan in the **Current static master plan** and **Current dynamic master plan** fields.

Types of planning methods

Three calculation methods can be used to run master planning: regeneration and net change. Each method produces a different plan when it's run.

You specify the planning method in the **Master planning run** dialog box. To open this dialog box, go to **Master planning** > **Master planning** > **Run** > **Master planning**, or select **Run** in the **Master planning** workspace.

Regeneration

The regeneration planning method deletes existing planned orders, unless they are firmed. It generates new planned orders, based on all the requirements. Regeneration is the only

planning method that is available for static plans.

- Changes in supply are considered. These changes include changes in the forecast.
- This method respects the **Period** coverage code.
- This method supports product substitution functionality (PI).

Net change

The net change planning method generates planned orders to cover the requirements that have been created or changed since the last master planning run. Changes in supply aren't considered when this method is run. The system doesn't consider any new supplies, and previously created planned orders aren't deleted if they are no longer required. The net change planning method runs faster than the regeneration method. It's available only for dynamic plans.

- Action dates and futures are updated for all requirements.
- This method doesn't respect the **Period** coverage code.
- This method doesn't fulfill the forecast, even if it's selected on the plan.
- This method doesn't support product substitution functionality (PI).

Net change minimized

The net change minimized planning method generates planned orders to cover only the requirements that have been created or changed since the last master planning scheduling run. For this method, unlike the net change method, action dates and future dates are updated only for new or changed requirements. This planning method is available only for dynamic plans.

Scheduling methods, time fences, positive days, and negative days

Other aspects of setting up master plans (including Scheduling methods, time fences, positive days, and negative days) work the same way when using the deprecated planning engine as they do when using Planning Optimization. For more information about these topics, see Master plans overview.

Related information

- Coverage settings
- Separating tactical and operative planning for master scheduling
- Master Planning: Use a static and dynamic Master Plan or use one plan?

Last updated on 06/17/2025

Create an intercompany plan

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to create an intercompany plan. The demo data company used to create this procedure is USMF.

Set up an intercompany planning group

1. Go to **Master planning** > **Setup** > **Intercompany planning groups**.
2. Use the Quick Filter to find records. For example, filter on the Name field with a value of '10'.
3. In the list, mark the selected row.
4. Select **Delete**. This step is necessary in order to shorten the intercompany planning run. Intercompany planning will run master planning in all the companies in a planning group, starting from the lowest scheduling sequence.
5. Select **Yes**.
6. Close the page.

Create an intercompany master plan

1. Go to **Master planning** > **Workspaces** > **Master planning**.
2. Select **Intercompany master planning**.
3. In the **Intercompany planning group** field, select the drop-down button to open the lookup.
4. In the list, select the link in the selected row. Select intercompany planning group 10.
5. In the **Number of intercompany planning iterations** field, enter '2'. Intercompany planning group 10 has two members. In order to propagate the delays from the source company (USMF) to the customer company (DEMF), you will need to run intercompany in both companies two times. The first iteration will propagate the demand and identify the delays in the source company (USMF). The second iteration will propagate the delays from USMF to DEMF.
6. In the **First iteration** field, select 'Regeneration'.
7. In the **Subsequent iterations** field, select 'Regeneration'.

8. In the **Number of threads** field, enter a number. This represents the number of parallel threads used for planning.
9. Select **OK**.

View the result of the plan

1. In the **Plan** field, select the drop-down button to open the lookup.
2. In the list, select the link in the selected row. Select the link for StaticPlan. You need to be in company USMF.
3. Select **Planned orders**.

Last updated on 06/17/2025

View outbound planned intercompany demand

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to view all the planned orders that will be fulfilled by an intercompany vendor. The demo data company used to create this procedure is DEMF.

1. Select **Master planning**.
2. In the **Plan** field, enter or select a value.
    - In the **Plan** field, select plan *10*.
3. Select *Run*.
4. In the **Number of threads** field, enter a number.
    - This represents the number of parallel threads to be used for master planning.
5. Select **OK**.
    - This may take a while.
6. Select **Planned intercompany demand**.
7. Select **Outbound planned intercompany demand**.
    - This page provides an overview of all the planned demand that will be fulfilled by an internal supply chain vendor.
8. Expand the **Upstream demand details** section.
    - In this section, you can see the details about how the demand will be fulfilled. You may need to wait for master planning to be run in the supply company before you can see additional information here.

Last updated on 06/17/2025

Monitor a master planning run

10/03/2025

Use a Gantt chart to visualize master planning progress

From the **View master planning progress** page, you can view details of historical master planning runs as a Gantt chart. This functionality can help you understand the time that is spent on the various phases of master planning. For a current active planning job, the **View master planning progress** page can be used to track progress and view the estimated remaining time.

Use the Master plan progress visualization feature

The **View master planning progress** page can display both historical planning jobs and active planning jobs.

To view historical planning jobs, there are two options.

1. Go to **Master planning** > **Setup** > **Plans** > **Master plans**, and then, on the Action Pane, select **History**. With the desired job selected, select **Inquiries**, and then select **View progress**
2. Go to **Master planning** > **Workspaces** > **Master planning**, on the Master planning tile click **History**. With the desired job selected, select **Inquiries**, and then select **View progress**

To view active planning jobs, there are two options.

1. Go to **Master planning** > **Workspaces** > **Master planning**, on the Action Pane, select **Unfinished planning process**. With the desired job selected, select **Inquiries**, and then select **View progress**.
2. Go to **Master planning** > **Workspaces** > **Master planning**, on the Master planning tile click **View progress**. With the desired job selected, select **Inquiries**, and then select **View progress**

Note you can view active jobs only when a planning job is processing.

Analyze a master planning job

In the Gantt chart, you can expand each of the following planning processes to view additional details about the time that is spent:

- Initializing
- Deleting and inserting data
- Coverage planning
- Delays
- Action messages
- Finalization
- Auto-firming

The Gantt chart is a useful tool if you want to view the impact of using action messages.

Navigation in the Gantt chart

- To expand the selected group and show the details, select the plus sign (**+**) in the tree view.
- To collapse the selected group, select the minus sign (**–**) in the tree view.
- You can use your keyboard for navigation. Use the **Up arrow** and **Down arrow** keys to move between rows. Use the **Right arrow** and **Left arrow** keys to expand and collapse groups.
- To open or close all levels in the Gantt chart, select **Expand all** or **Collapse all**.
- To view the related processing time, hover over a task. (Tasks are the lowest level in the Gantt chart.)

View an additional master planning run to compare jobs

By selecting a master planning job on field **Show additional master planning run**, you can view an additional master planning run in the Gantt chart and compare the two jobs.

BOM-level display

Bill of materials (BOM) levels are shown differently for coverage planning, delays, actions, and firming.

- **Coverage planning** – BOM levels are shown as expected. They are calculated from the top down.

  Example: BOM level 0, 1, 2

- **Delays** – BOM levels are shown as the coverage planning BOM levels multiplied by –1. (In other words, they have a negative sign.)

Example: BOM level –2, –1, 0

- **Action message** – BOM levels are shown as expected. They are calculated from the top down.

  Example: BOM level 0, 1, 2

- **Auto-firming** – BOM levels are shown as 999 minus the coverage planning BOM level.

  Example: BOM level 999, 998, 997

The following table summarizes the behavior.

Expand table

BOM level that is shown	End item	Subcomponent	Raw material
Coverage planning	0	1	2
Delays	0	–1	–2
Action message	0	1	2
Auto-firming	999	998	997

Visualize progress

If you view a master planning job that is currently running, progress is shown through colors in the Gantt chart. The following colors apply to the blue theme. For other color themes, the colors will differ.

- **Dark blue** – Completed planning tasks.
- **Orange** – The task that is currently in progress.
- **Light blue** – The estimate for remaining tasks.

The color is shown only on the lowest level in the Gantt chart. Select **Expand all** to view all tasks in the master planning job. The estimate of remaining tasks is based on historical master planning jobs.

Run master planning and track processing time

1. On the default dashboard, select **Master planning**.
2. In the **Plan** field, enter or select a value.
3. Select **Run**.
4. Set the **Track processing time** option to **Yes**.

5. In the **Number of threads** field, enter a number.
6. On the **Records to include** FastTab, select **Filter**.
7. In the grid, select the row where the **Field** field is set to **Item number**.
8. In the **Criteria** field, enter a value.
9. Select **OK**.

Net requirements and pegging information with the deprecated master planning engine

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you run master planning, it's important that you understand its output, how existing supply covers the demand, and why specific supply was generated. You can use the **Net requirements** page to better understand the calculated requirements that master planning produces.

The **Net requirements** page shows the net requirements that are calculated for the product during master planning. It also shows the coverage settings that were applied during the master planning run, a breakdown of the requirement totals by transaction type, and pegging information.

The **Net requirements** page behaves slightly differently based on whether you're using Planning Optimization or the deprecated master planning engine. For more information about this difference, see Differences between Planning Optimization and the deprecated master planning engine

For more information about how to work with this page, see Net requirements and pegging information.

Last updated on 06/17/2025

Safety stock fulfillment with the deprecated master planning engine

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how safety stock functionality differs when you use the deprecated master planning engine compared to when you use Planning Optimization. For more information about safety stock, including information that applies to both planning engines, see Safety stock fulfillment for items.

Differences in safety stock functionality

The following table summarizes the differences in safety stock functionality between the deprecated master planning engine and Planning Optimization.

Expand table

	Planning Optimization	Deprecated master planning engine
Meaning of safety stock	Safety stock ensures that the accumulated on-hand inventory is always above the minimum value that's defined in the item coverage settings for each item. Safety stock isn't actual demand. Demand is prioritized over the safety stock.	Safety stock is a demand just like other types of demand. You can choose when safety stock will be fulfilled.
When safety stock is fulfilled	Safety stock is always fulfilled on *Today's date* + *Procurement time*, regardless of the setting for the item coverage.	On the **Item coverage** page for an item, on the **General** tab, the **Fulfill minimum** field defines when the safety stock must be fulfilled. The options are *Today's date*, *Today's date*

	Planning Optimization	Deprecated master planning engine
		+ procurement time, *First issue*, and *Coverage time fence*.
Safety stock pegging and net requirements	The *Safety stock* requirement type isn't included and isn't shown on the **Net requirements** page. Safety stock doesn't represent demand, and no requirement date is associated with it. Instead, it sets a constraint on the amount of inventory that must be present at all times. However, the value of the **Minimum** field is still considered when planned orders are calculated during master planning. We recommend that you inspect the **Accumulated quantity** column on the **Net requirements** page to confirm that this value was considered. Because the pegging differs, different actions might be suggested.	Safety stock is a demand type, similar to a sales order line, forecast line, or bill of materials (BOM) line. Master planning always fulfills it on the date that's specified in the **Fulfill minimum** field. In some cases, safety stock might take precedence over actual demand fulfillment.

> ⓘ Note
>
> When strict safety stock pegging is enabled, the system ignores pegging priority for safety stock planned orders. This condition can result in unreliable net requirements pegging and the creation of unwanted planned orders. Learn more in Safety stock pegging options.

Safety stock fulfillment date options

The deprecated master planning engine respects the value of the **Fulfill minimum** field on the **General** tab of the **Item coverage** page. This field lets you select the date or the period during which the inventory level must meet the quantity that you specified in the **Minimum** field. The **Fulfill minimum** field is available when you select **Period**, **Requirement**, or **Min/Max** in the **Coverage code** list.

If you use minimum keys, set the **Minimum periods** option to *Yes* to fulfill the minimum inventory level for all the periods that are set up in the minimum key. If you set the option to *No*, the minimum inventory is fulfilled for the current period only.

The following scenario shows how the **Fulfill minimum** setting works and how the values differ.

> ⓘ Note
>
> For all the illustrations in this article, the x-axis represents inventory, the y-axis represents days, the bars represent the inventory level, and the arrows represent transactions, such as

sales order lines, purchase order lines, or planned orders.

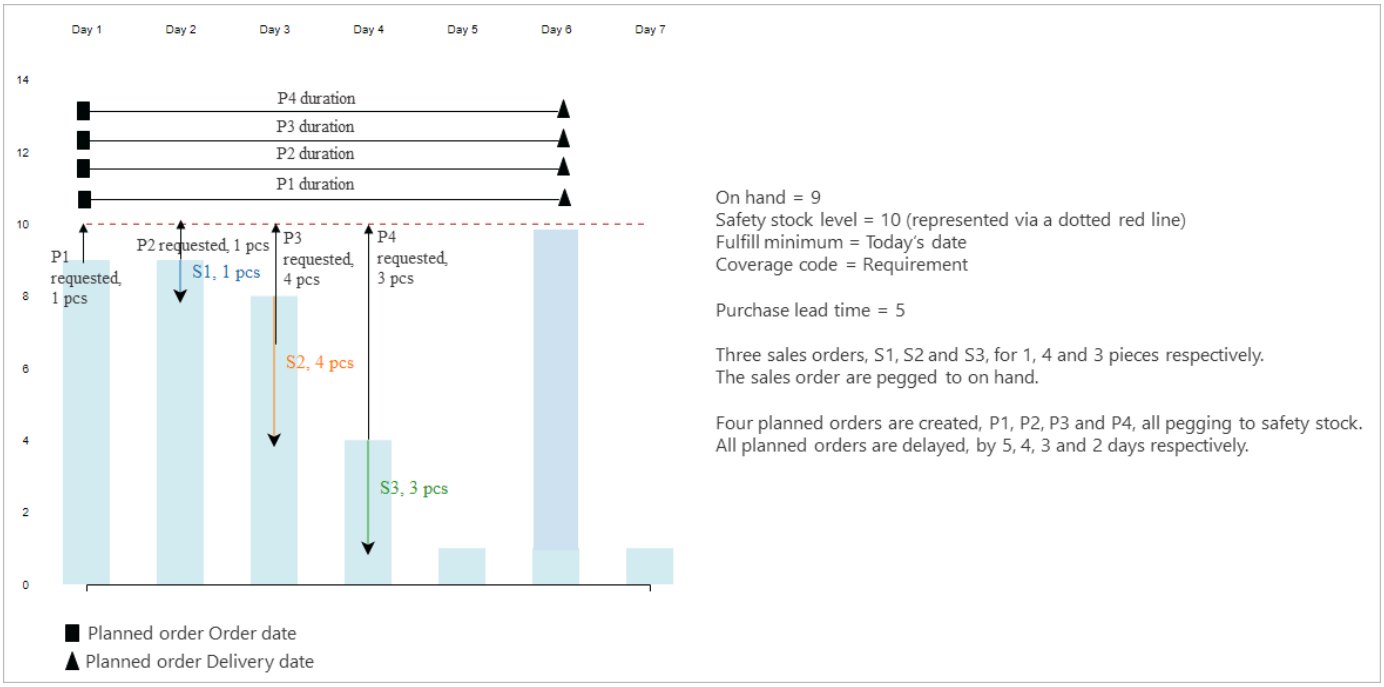
The **Fulfill minimum** field can have the following values:

- *Today's date*
- *Today's date + procurement time*
- *First issue*
- *Coverage time fence*

The following subsections describe how each value works.

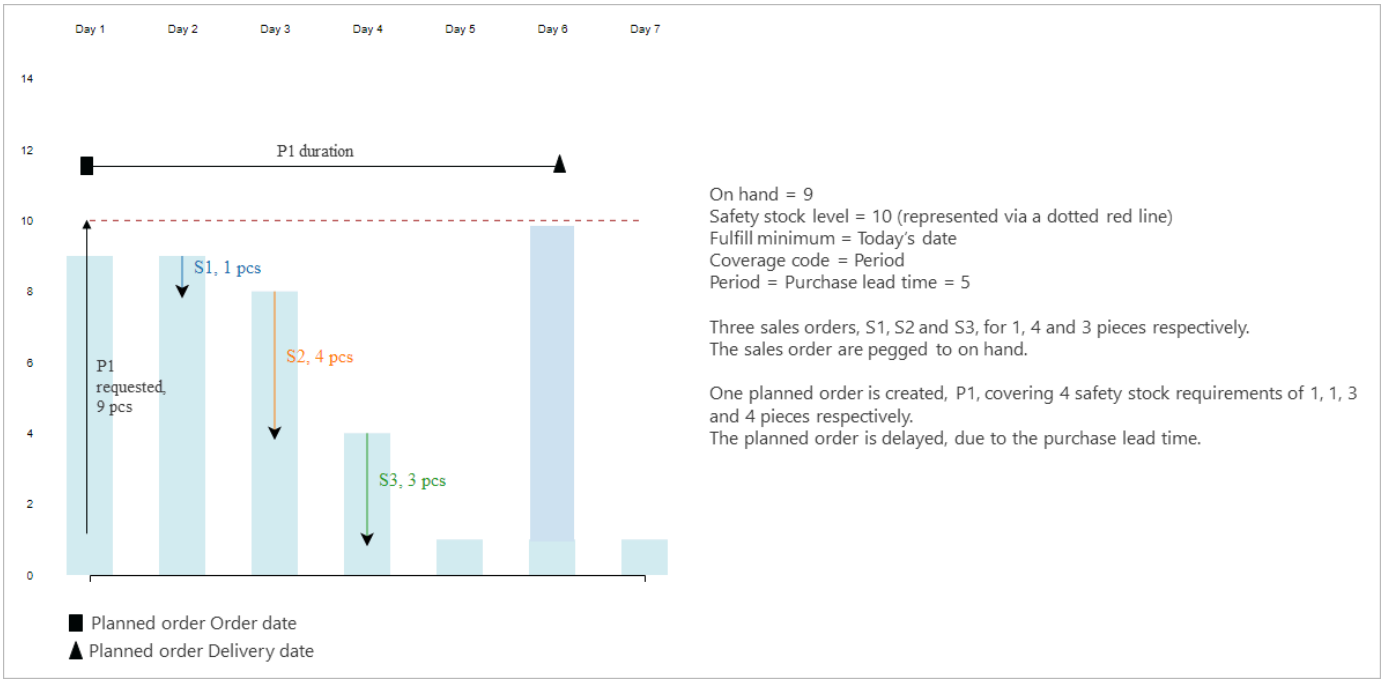
Today's date

The specified minimum quantity is met on the date when master planning runs. The system tries to fulfill the safety stock limit as soon as possible, even though the attempt might be unrealistic because of the lead time.

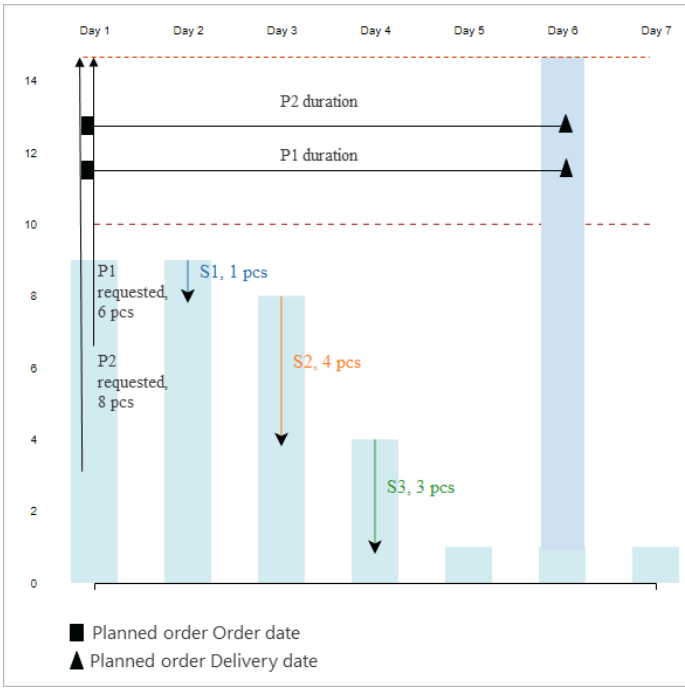


Planned order P1 is created for today's date to bring the available inventory above the safety stock level on this date. The sales order lines S1 through S3 continue to lower the inventory level. Master planning generates planned orders P2 through P4 to bring the inventory level back to the safety limit after each sales order requirement.

When you use the **Requirement** coverage code, the system creates multiple planned orders. Always use either **Period** or **Min/Max** coverage for items and materials that are in frequent demand. In that way, the replenishment is bundled. The following illustration shows an example for the **Period** coverage code.



The following illustration shows an example for the **Min/Max** coverage code.



On hand = 9
Safety stock level = 10 (represented via a dotted red line)
Fulfill minimum = Today's date
Coverage code = Min./Max.
Maximum level = 15

Three sales orders, S1, S2 and S3, for 1, 4 and 3 pieces respectively.
The sales order are pegged to on hand.

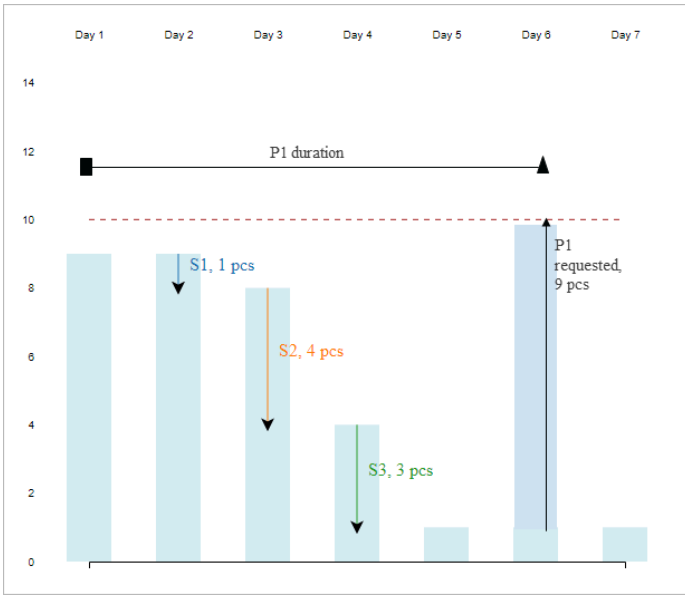
Two planned orders are created, P1 and P2.
P1 is for 6 pcs and is requested on Day 1, because the current on hand is below the safety limit.
P2 is for 8 pcs and is also requested on Day 1, due to the sales orders that are coming in and that would bring the inventory below the safety limit.
Both planned orders are delayed, due to the purchase lead time.

Today's date + procurement time

The specified minimum quantity is met on the date when master planning runs, plus the purchase or production lead time. This time includes any safety margins. If the item carries a trade agreement, and you select the **Find trade agreements** checkbox on the **Master planning parameters** page, the delivery lead time from the trade agreement isn't considered. Instead, lead times are taken from either the item coverage settings or the item.

This fulfillment mode creates plans that have fewer delays and fewer planned orders, regardless of the coverage group that's set up on the item.

The following illustration shows the outcome of the plan if the coverage code is **Requirement** or **Period**.



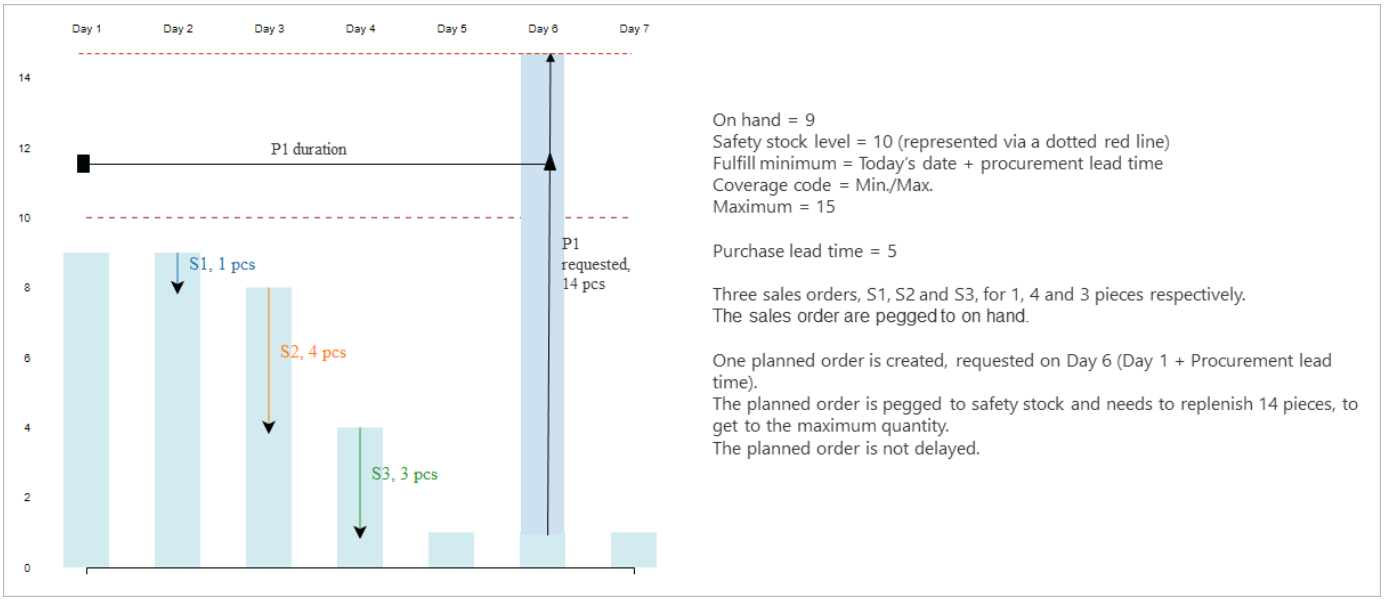
On hand = 9
Safety stock level = 10 (represented via a dotted red line)
Fulfill minimum = Today's date + procurement lead time
Coverage code = Requirement or Period (Period = 5)

Purchase lead time = 5

Three sales orders, S1, S2 and S3, for 1, 4 and 3 pieces respectively.
The sales order are pegged to on hand.

One planned order is created, requested on Day 6 (Day 1 + Procurement lead time).
The planned order is pegged to safety stock and needs to replenish 9 pieces.
The planned order is not delayed.

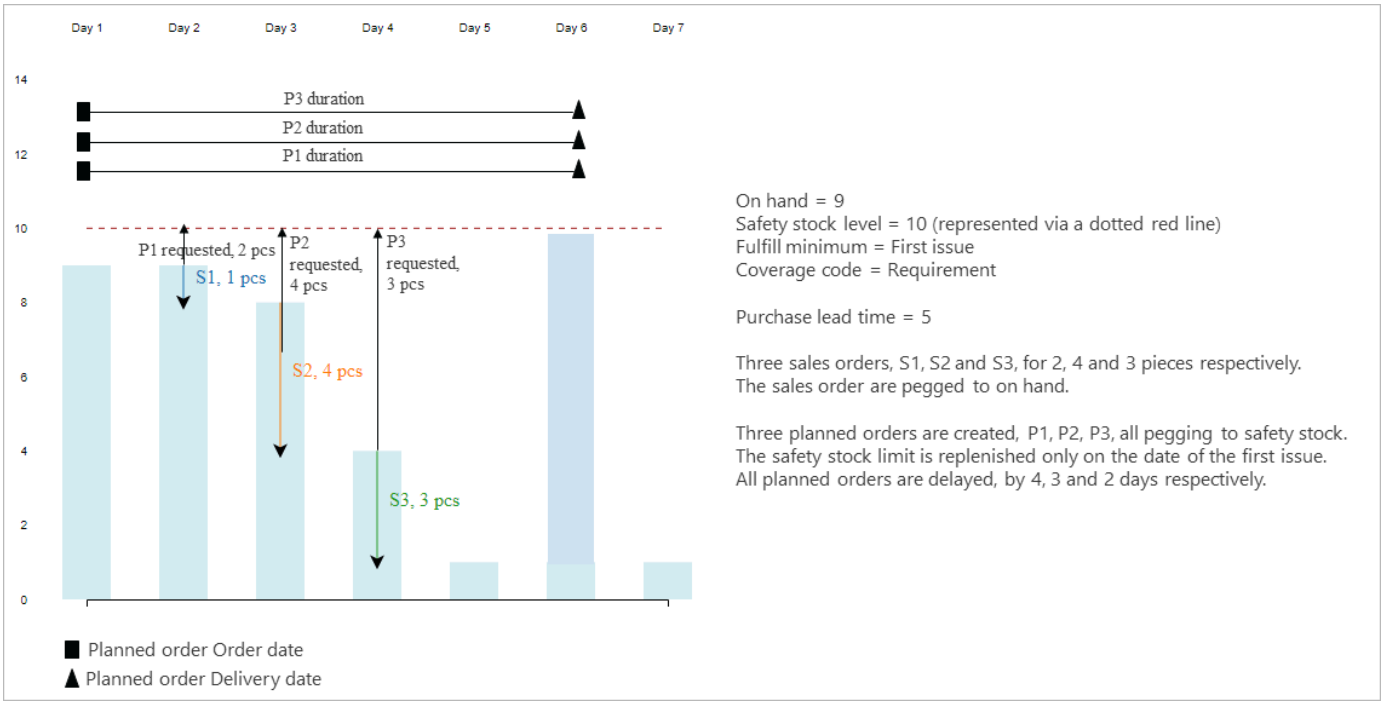
The following illustration shows the outcome of the plan if the coverage code is **Min/Max**.



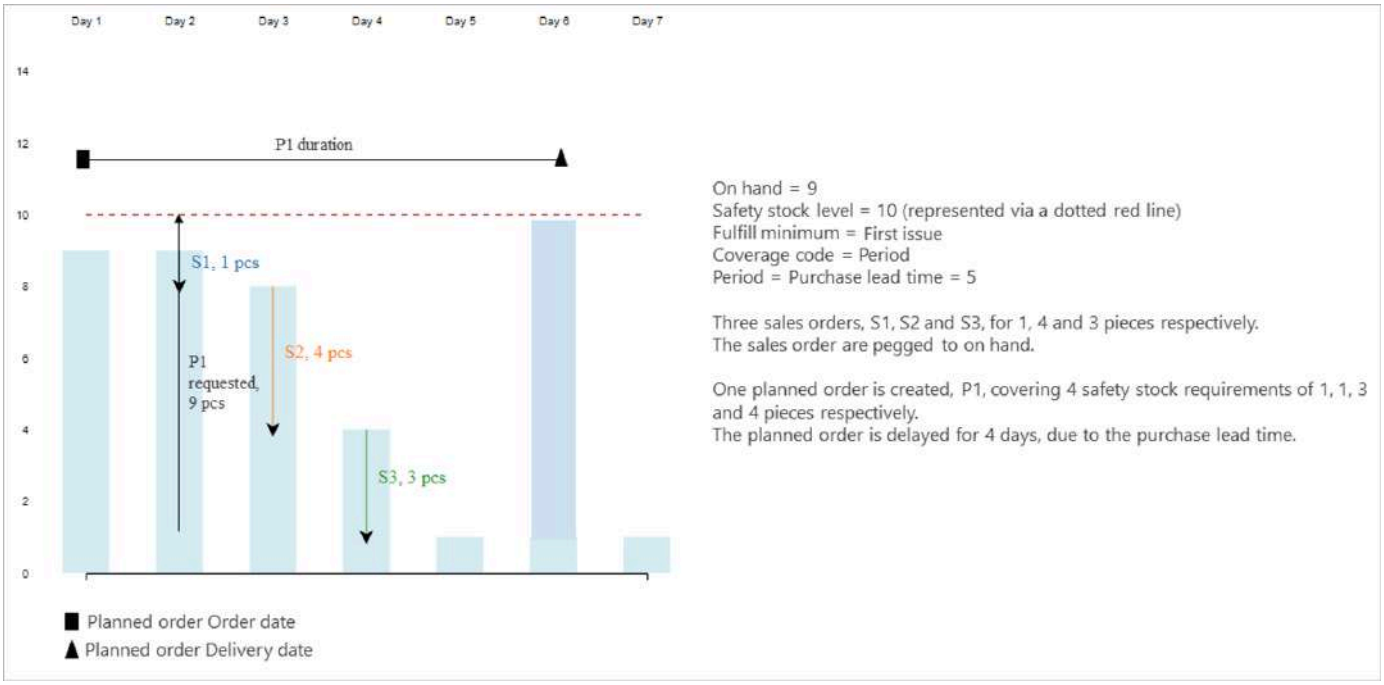
First issue

The specified minimum quantity is met on the date when the available inventory goes below the minimum level. Even if the available inventory is below the minimum level on the date when master planning runs, this fulfillment mode doesn't try to cover it until the next requirement comes in.

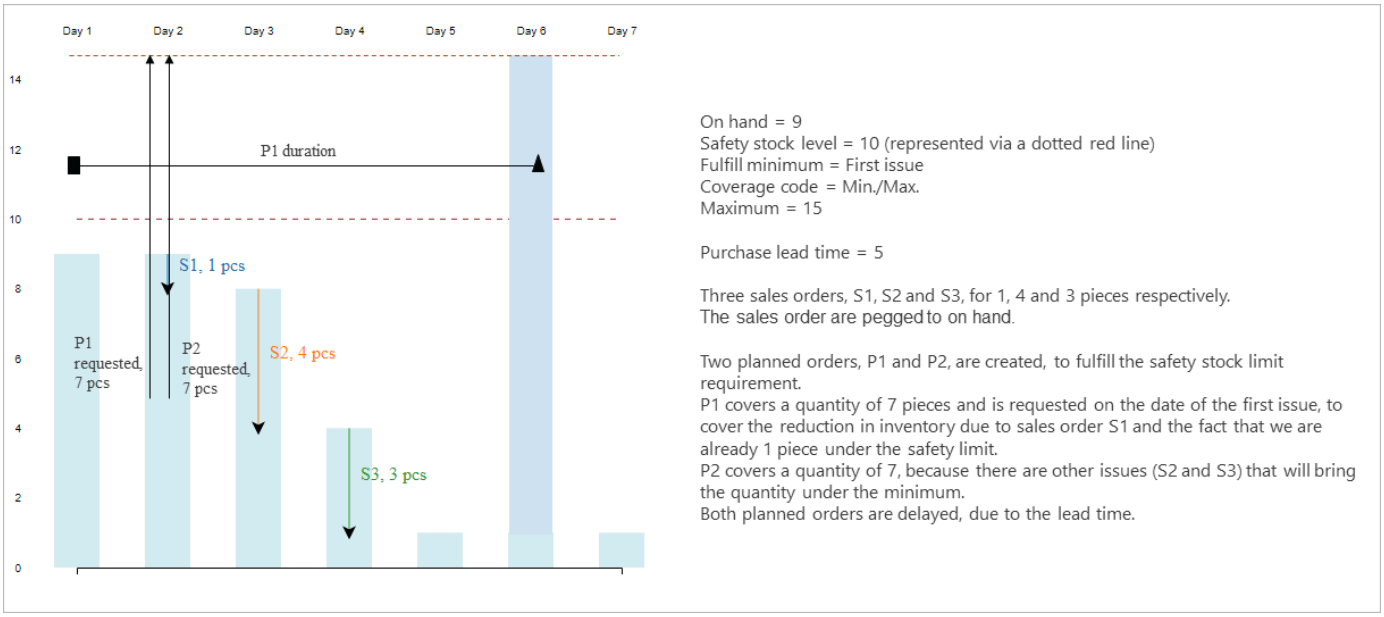
The following illustration shows an example for the **Requirement** coverage code.



The following illustration shows an example for the **Period** coverage code.

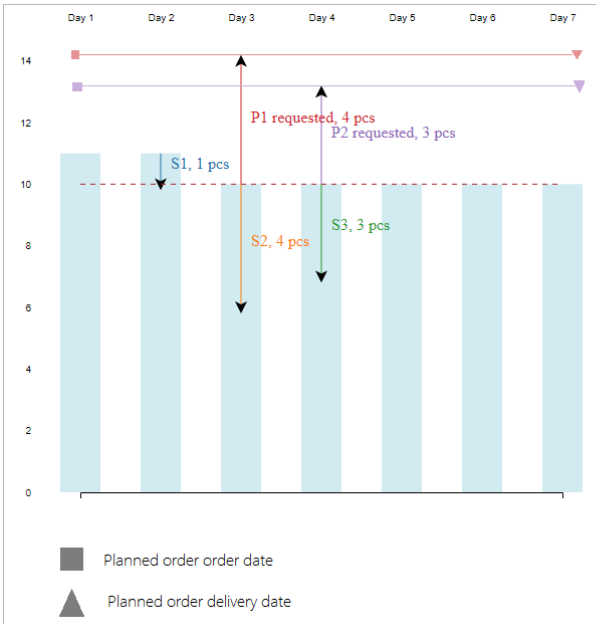


The following illustration shows an example for the **Min/Max** coverage code.



On the date when master planning runs, if the available inventory is already under the safety stock limit, both the **Today's date** fulfillment mode and the **Today's date + procurement time** fulfillment mode trigger the replenishment immediately. However, the **First issue** fulfillment mode waits until there's another issue transaction for the item, such as a sales order and BOM line requirement. It then triggers the replenishment on the date of that transaction.

On the date when master planning runs, if the available inventory isn't under the safety stock limit, the **Today's date** and **First issue** fulfillment modes provide the same result, as shown in the following illustration.



Coverage code = Requirement
Safety stock = 10
On hand = 11
Fulfill minimum = First issue

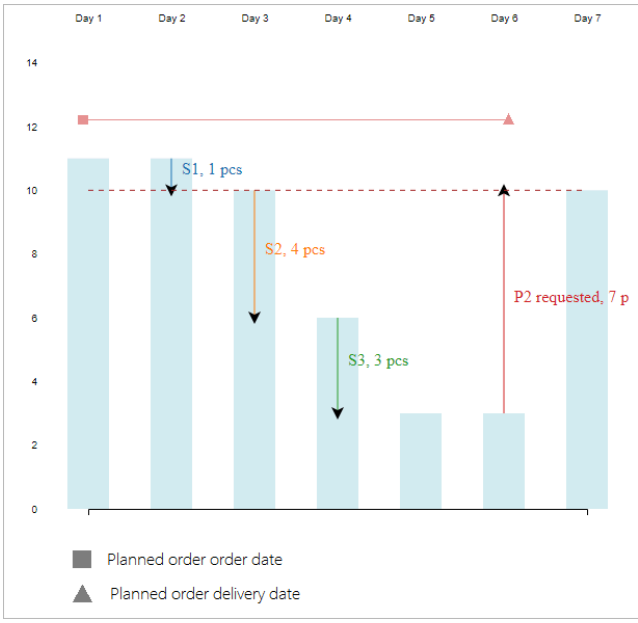
Purchase lead time = 5

Given three sales orders, S1, S2 and S3, master planning will create two planned orders, P1 and P2, as follows:

P1
Quantity = 4
Requested date = Day 3
Order date = Day 1
Delivery date = Day 1 + purchase lead time
Pegging to safety stock requirement, coverage quantity 4
Delay date = Day1 + purchase lead time
Delay = 3 days
Requirement date = Delay date (per plan setup)

P2
Quantity = 3
Requested date = Day 4
Order date = Day 1
Delivery date = Day 1 + purchase lead time
Pegging to safety stock requirement, coverage quantity 3
Delay date = Day 1 + purchase lead time
Delay = 2 days
Requirement date = Delay date (per plan setup)

On the date when master planning runs, if the available inventory isn't under the safety stock limit, the **Today's date + procurement time** fulfillment mode provides the following result, because it postpones the fulfillment until the end of the procurement lead time.



Coverage code = Requirement
Safety stock = 10
On hand = 11
Fulfill minimum = Today's date + procurement time

Purchase lead time = 5

Given three sales orders, S1, S2 and S3, master planning will create one planned order, P2, as follows:

P2
Quantity = 7
Requested date = Day 1 + purchase lead time
Order date = Day 1
Delivery date = Day 1 + purchase lead time
Pegging to safety stock, covered quantity = 7
There are no delays

The sales orders are ll pegged to on hand.

Coverage time fence

The specified minimum quantity is met during the period that's specified by the **Coverage time fence** field. This option is useful when master planning doesn't allow available inventory to be used for real orders, such as sales or transfers, because it's trying to maintain the safety level. However, in a future release, this mode of replenishment won't be needed, and this option will be deprecated.

ⓘ **Note:** The author created this article with assistance from AI. Learn more

Last updated on 05/05/2026

Create a material plan for co products

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The production planner plans the material requirements for items that are formula co-products. The demo data company used to create this procedure is USP2.

Create requirement for a co-product

1. Go to **Sales and marketing** > **Workspaces** > **Sales order processing and inquiry**.
2. Select **New**.
3. Select **Sales order**.
4. In the **Customer account** field, type a value.
    - Example: US-001
5. Select **OK**.
6. In the **Item number** field, type a value.
    - Example: P6003
7. In the **Quantity** field, enter a number.
    - Example: 50000
8. Select **Save**.

Create a material plan for co-products

1. Close the page.
2. Close the page.
3. Select **Master planning**.
4. In the **Plan** field, select the drop-down button to open the lookup.
5. In the list, select the link in the selected row.
    - Example: MasterPlan

6. Select **Run**.
7. Expand or collapse the **Records to include** section.
8. Select **Filter**.
9. In the list, select the row for **Field** = *Item number*.
10. In the **Criteria** field, type a value.

    - Example: P6003

11. Select **OK**.
12. Select **OK**.
13. Select **Planned orders**.
14. Use the Quick Filter to find records. For example, filter on the **Item number** field with a value of 'P6000'.

    - Filter by the formula item that has as co-product of the item that you created a sales order for.

15. In the list, mark the selected row.

    - Select any of the rows returned by the filter.

16. In the list, select the link in the selected row.
17. Expand the **Pegging** section.
18. In the list, select the link in the selected row.

    - The planned order is pegged to the sales order for the co-product.

19. Close the page.

Create a second requirement for a co-product

1. Go to **Sales and marketing** > **Workspaces** > **Sales order processing and inquiry**.
2. Select **New**.
3. Select **Sales order**.
4. In the **Customer account** field, type a value.

    - Example: US-001

5. Select **OK**.
6. In the **Item number** field, type a value.

    - Example: P6003

7. In the **Quantity** field, enter a number.

- Example: 50000

8. Select **Save**.

Create a second material plan for co-products

1. In the **Plan** field, select the drop-down button to open the lookup.
2. In the list, select the link in the selected row.

    - Example: MasterPlan

3. Select **Run**.
4. Expand or collapse the **Records to include** section.
5. Select **Filter**.
6. In the list, select the row for **Field** = *Item number*.
7. In the *Criteria* field, type a value.

    - Example: P6003

8. Select **OK**.
9. Select **OK**.
10. Select **Planned orders**.
11. Use the Quick Filter to find records. For example, filter on the **Item number** field with a value of 'P6000'.

    - Filter by the formula item that has as co-product of the item that you created a sales order for.

12. In the list, mark the selected row.

    - Select any of the rows returned by the filter.

13. In the list, select the link in the selected row.
14. Expand or collapse the **Pegging** section.
15. In the list, select the link in the selected row.

    - The planned order is pegged to the sales order for the co-product.

16. Close the page.
17. Select **Master planning**.
18. Go to **Master planning** > **Setup** > **Master planning parameters**.
19. Select *No* in the **Disable all planning processes** field.
20. Close the page.

Last updated on 06/17/2025

Cancel a job that was created using the deprecated master planning engine

06/17/2025

There are multiple options for canceling a job that was created using the deprecated master planning engine. For example, you might want to cancel a master planning job if it was started by mistake or is running longer than expected and you want to end it.

The best way to cancel a planning job is from the **Unfinished planning processes** page. Alternative options from the **Batch jobs** and **Batch jobs enhanced** pages should only be used if canceling the master planning job from the **Unfinished planning processes** page didn't complete within a few minutes.

Preferred cancel option

Cancel master planning job from the Unfinished planning processes page

1. Go to **Master planning** > **Inquiries and reports** > **Master planning** > **Unfinished planning processes**.
2. Select the line with the planning process that you want to cancel.
3. Select **Cancel**.

Additional cancel options

These should only be used if canceling the master planning job from the **Unfinished planning processes** page didn't complete within a few minutes.

Delete master planning job from the Batch jobs page

1. Go to **System administration** > **Inquiries** > **Batch jobs**.
2. Select the line with the planning job that you want to delete.
3. Select **Delete**.

Abort master planning job task from the Batch jobs enhanced page

1. Go to **System administration** > **Inquiries** > **Batch jobs**.

2. If the job ID isn't shown in the list, select **Switch to enhanced form**, otherwise proceed with the next step.
3. Open the batch job. Select the **Job ID** for the batch job with tasks that you want to end.
4. In **Batch tasks**, select the tasks to end.
5. Select **Change status**, choose **Canceling** and select **OK**.
6. On the **Batch tasks** FastTab, select **Abort**.

Demand planning home page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Inaccurate forecasts and demand plans can lead to lost revenue and inefficiency in the supply chain. However, forecasts that are infused with intelligence lead to more accurate and effective demand plans. You can measure the direct impact in improved revenue and fewer stockouts. Operating expenses go down in supply chains that require fewer fire drills.

Demand planning in Microsoft Dynamics 365 Supply Chain Management is Microsoft's next-generation collaborative demand planning solution. This app is powered by best-in-class forecasting algorithms and models, and offers immersive user experiences, intelligent reports, and analytics. It empowers organizations to build an agile, resilient, and sustainable demand planning practice that's fueled by intelligence and collaboration.

Demand planning provides the following capabilities:

- **A no-code approach** to demand modeling and planning configuration. Flexible building blocks enable the vast majority (over 85 percent) of demand planners who aren't data scientists to do what-if planning and analyze, optimize, and compare scenarios in minutes.
- **Seamless, on-the-fly aggregation and disaggregation**. Planners can edit forecasts at the corporate or product group level, and then zoom in and instantly see the impact at the regional and stock-keeping unit (SKU) levels.
- **Improved forecast accuracy** with automatic AI parameter tuning to help ensure accurate forecasting and preprocessing. External signals enable superior forecast accuracy by considering promotions or stockouts.
- **Disruption readiness** with interactive and fast what-if analysis. Version history enables tracking, evaluating forecast changes, and using the lessons learned to improve the decision making process.
- **Effective collaboration throughout the planning cycle**. This capability is enabled through Microsoft Teams in-context communication, in-product commenting, and restorable

versions of forecast values throughout the planning process.

- **Increased agility through integrated planning and execution flow** with native integration with Supply Chain Management, customizable worksheets, and exception-based planning.

The demand planning process

Demand planning provides functionality for the complete demand planning process. This process has the following steps:

1. **Import data** – Import your historical data, products, sites, warehouses, prices, and other information into the app.

   https://learn-video.azurefd.net/vod/player?id=62832cd0-1bd9-456b-9dd1-3dc33a101b0b&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fdemand-planning%2Fdemand-planning-home-page

2. **Create transformation** – Transform imported data from tables into time series by identifying data columns, choosing time buckets, and shifting dates. For example, you can shift historical data from last year to next year so that you can use it as a basis for the forecast. Alternatively, you can apply multiplications or combine data from different systems.

3. **Create forecasts** – Use different forecasting models (including your own Azure Machine Learning model) to create a forecast, or let AI determine which forecast model works best.

   https://learn-video.azurefd.net/vod/player?id=9d178a98-0a58-4765-9885-b73ee2ad17ef&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fdemand-planning%2Fdemand-planning-home-page

4. **Review and adjust forecast** – Work on the forecast, adjust values, and get different perspectives about how pricing, weather, promotional events, and other factors might affect the forecast. Collaborate to reach the most accurate forecast.

   https://learn-video.azurefd.net/vod/player?id=14bf0230-5088-49c6-9b14-685f322f2376&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fdemand-planning%2Fdemand-planning-home-page

5. **Export data** – After your forecast is done, you can export it to any external system that can consume forecasts.

Deep dive video

For a deep dive into the capabilities of Demand planning, watch the following video on YouTube: Deep dive into Demand Planning for Supply Chain Management | Dynamics 365 TechTalk.

Licensing

To use Demand planning in a production environment, each relevant user must have a license for it. Learn more in Demand planning license requirements.

Last updated on 07/01/2026

What's new or changed in Demand planning

This article lists new features, fixes, improvements, and known issues for each released version of Demand planning in Microsoft Dynamics 365 Supply Chain Management.

Version 1.4

New features introduced in version 1.4

Demand planning version 1.4 introduces the following new features:

- **Best fit model - version 4** – The *Best fit model - version 4* forecasting algorithm introduces symmetric mean absolute percentage error (SMAPE) as the new metric for selecting between forecast models. This metric improves model selection for products with periods where no historical sales occurred.
- **Forecast calculation accuracy analysis** – The new calculation accuracy page for forecast time series improves on the explainability display previously available in the forecast profile job run history. For each planning object in the forecast, it shows which demand forecasting algorithm was used and the resulting SMAPE value. New data visualizations include model distribution and an average SMAPE indicator that responds to filters. It improves the overall user experience and provides deeper insights.
- **Detailed forecast explainability** – The calculation accuracy page also lets you drill down into individual planning objects (such as product and store) for an even more detailed explainability view. Here, you can see historical data alongside each calculated forecast and get a natural-language explanation of the factors driving it.
- **View forecast with full component breakdown** – View each forecast across the full horizon with a detailed component breakdown. Demand planning splits each forecast into *baseline*, *signals impact*, and *manual adjustments* to provide clearer insight into manual changes and the influence of signals.
- **Signals and fill missing values** – New **Fill missing value** options make it easier to create continuous data signals based on input originating from external sources, including Supply Chain Management and third parties. New options include *Forward fill*, *Backward fill*, and *Interpolation*. Learn more in Signal fill strategies.
- **New calculation options for transformations** – Previously, Demand planning provided a single calculation method for data transformations: Summarize. This version adds several

new options, including *Average*, *Max*, *Min*, and *Last value*, which can be better suited for signals such as inflation, prices, or weather. Learn more in Aggregation methods.

- **Engineering enhancements for upcoming Dataverse storage alignment** – This release includes behind-the-scenes enhancements that prepare the app to support Dataverse storage for timeseries data. As a result of these preparations, both new installations and updates to version 1.4 might require significantly more processing time than previous versions.

New fixes and improvements in version 1.4

Demand planning no longer resets unsaved edits to their start value when you double-click on other cells.

Version 1.2

New features introduced in version 1.2

This release improves data consumption management by allowing you to delete obsolete time series versions or entire time series that you no longer need. The system provides an overview of dependencies between time series, time series versions, and the profiles that use them, so you can easily see what you need to clean up to allow a time series or version to be deleted. Learn more in Clean up time series data.

New fixes and improvements in version 1.2

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Added a new API that you can use to export time series programmatically. Learn more in Export data using custom APIs.
- Improved performance when exporting forecasts to Supply Chain Management. This improvement is made possible by a new index introduced in Supply Chain Management version 10.0.48.
- Improved forecast calculation performance.

Version 1.1

New features introduced in version 1.1

This version of Demand planning enhances basic forecast calculations by adding two new features:

- The new *Best fit model - version 3 (preview)* forecast algorithm now includes the *Croston's method* forecasting model. It's designed specifically for intermittent demand, which is demand data with many zero-demand periods and occasional non-zero demands. Learn more in Croston's method forecasting (preview).
- Forecasting with signals by using the *XGBoost* forecast model now supports up to five signals. Learn more in Forecast with signals (preview).

New fixes and improvements in version 1.1

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Improved performance when exporting forecast to Supply Chain Management with data aggregation.
- Improved forecast calculation performance.

Version 1.0.0.3424

New feature introduced in version 1.0.0.3424

This version of Demand planning enhances basic forecast calculations by adding two new features:

- The new *Best fit model - version 2 (preview)* forecast algorithm now applies naive forecasting when an input time series contains a low number of data points. This approach helps improve statistics forecasting results when input data is limited, such as for newly added products. Learn more at Naive forecasting.
- Time freeze can now preserve all manual forecast adjustments. This option ensures that all manually adjusted cell values remain at the adjusted value even after you recalculate an existing forecast. No freeze rules are required. Learn more at Limit automatic time series updates with time freezes.

New fixes and improvements in version 1.0.0.3424

This version of Demand planning includes the following fixes and improvements:

- Increased stability.

- Improvements to Generative insights.
- Improved forecast calculation performance.

Version 1.0.0.3336

New feature introduced in version 1.0.0.3336

This version of Demand planning enhances the *Copilot grid cursor* feature so that it now shows the impact of the signal input separately from the baseline for a selected forecasted cell. Learn more in Copilot grid cursor.

New fixes and improvements in version 1.0.0.3336

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Auto-detect seasonality is improved to account for certain data compositions.
- Forecast calculations now respect the decimal precision set in the forecast profile.
- Improved the performance of forecast calculations.

Version 1.0.0.2999

New feature introduced in version 1.0.0.2999

This version of Demand planning marks the public preview of *Generative insights*, which offers detailed insights calculated using AI models. The production-ready preview supports two metrics: *seasonality* and *signal correlation*. The system clusters forecast data based on distinct patterns found for each metric. For each cluster, the system provides insights such as its relative size and confidence level. You can also choose the major contributor across various dimensions in the forecast. Learn more in Generative insights for Demand planning (production-ready preview).

New fixes and improvements in version 1.0.0.2999

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Improvements in demo data. Features like row level security, time fence, and time freeze now work seamlessly when using demo data.

Version 1.0.0.2794

New features introduced in version 1.0.0.2794

This version of Demand planning adds several new features, as described in the following subsections.

Copilot grid cursor

The *Copilot grid cursor* offers detailed insights into a selected cell's value, including its original value, manual adjustments, and full adjustment history. User comments are also shown, to help make the changes easier to understand. Learn more in Copilot grid cursor.

New operator in rules: Select all

To improve efficiency and make queries easier to formulate, Demand planning now provides a *select all* operator. The new operator is available in the following policies: *time fence*, *time freeze*, and *row level access*. Learn more in Using the select all operator.

Multiple rules for time freezes

Demand planning now lets you assign multiple time freeze rules to *Forecast* and *Forecast with signals* steps. Each rule can have a different time freeze horizon. In this way, updates to a continuous forecast calculation are prevented during the selected periods. Learn more in Limit automatic time series updates with time freezes.

New fixes and improvements in version 1.0.0.2794

This version of Demand planning includes the following fixes and improvements:

- Increased stability
- New demo data with seasonal clusters

Version 1.0.0.2502

New features introduced in version 1.0.0.2502

This version of Demand planning adds several new features, as described in the following subsections.

Autodetection of seasonality patterns is now available for all forecast algorithms (preview)

Automatic detection of seasonality patterns was introduced in Demand planning version 1.0.01700, but it was previously available only for the ARIMA forecast algorithm. In this version, it's now available for all forecast algorithms.

Time freeze rules

Time freeze rules let demand planning managers establish policies that prevent forecast calculations from recalculating and overwriting manual adjustments in an existing forecast during a specified date range. Time freeze rules are based on the dimensions that are available in each forecast, and they're easy to maintain. Learn more in Limit automatic time series updates with time freezes.

Generative insights into seasonality patterns (production-ready preview)

Generative insights provide AI-generated insights into your data. The initial preview of the feature provides generative insights into your seasonality patterns. Visual displays make it easy for planners to understand seasonality data and make business decisions based on it. Information is clustered (grouped) according to the detected seasonality patterns. For each cluster, generative insights provide a confidence score and describe the seasonality pattern in natural language. The system also indicates the proportion of planning items that follow the pattern. Learn more in Generative insights for Demand planning (production-ready preview).

Simplified security role dependencies

The *Basic user* security role no longer needs to be applied to all users of Demand planning. Instead, Demand planning now incorporates all privileges from the *App Opener* security role into the *Demand planning contributor* and *Demand planning manager* security roles. Learn more in Security roles and row-level security in Demand planning.

New fixes and improvements in version 1.0.0.2502

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- New demo data that includes seasonal clusters.

Version 1.0.0.1700

New features introduced in version 1.0.01700

This version of Demand planning adds several new features, as described in the following subsections.

Select input for forecasts and calculations at the model level

In previous versions of Demand planning, you set the input time series used by forecast and calculation profiles at the *profile* level. This version makes the setting part of the *model* configuration, which improves usability while adding flexibility to your model designs (including the ability to add a signal input). As before, each forecast or calculation model starts with an *Input* step, but now it includes a setting where you choose the time series to use as input for the model. You can also choose a time series version and add filters based on dates and dimensions, such as products and locations. As with all steps, access these settings by opening the **Actions** menu (ellipsis button) on the card. Learn more in Design calculation models and Design forecast models.

Apply input filters in calculation models

You can now filter the input used in calculation models. Create filters based on dates and dimensions, such as products and locations. You could already apply filters in forecast models. Learn more in Design calculation models.

Analyze demand plans with Copilot

Copilot cursor prompts in Demand planning let you explore specific data points or data ranges in a forecast or time series. Each prompt presents a set of predefined questions that you can ask Copilot, which then returns insights into notable shifts, trends, anomalies, or deviations across multiple dimensions. Copilot replies using natural-language summaries and visuals, which make it easy for you to digest the information and use it to make informed decisions. Learn more in Analyze demand plans with Copilot.

Forecast with signals (preview)

One way to improve the accuracy of a forecast is to include input signal data beyond just historical sales. This version of Demand planning adds a new *Signal* step, which lets planners include any signal, such as inflation or weather data, as input to their forecast models. The current release supports up to one signal, though support for more signals might be added in a future release. The new *Forecast with signals* step lets you combine the signal input and main input to create the forecast. In the current version, *Forecast with signals* always uses the XGBoost demand

forecasting algorithm, but support for other algorithms might be added in a future release. Learn more in Forecast with signals.

Autodetect seasonality patterns (preview)

Seasonality patterns typically vary for different products and different locations, so using forecast models that try to apply the same pattern everywhere can result in inaccurate forecasts. This version of Demand planning adds an algorithm that automatically detects seasonality patterns for each combination of location and product and applies the result to its forecast calculations to help improve forecast accuracy. Automatic seasonality detection is only available when you use the ARIMA forecast algorithm. Learn more in Design forecast models.

New fixes and improvements in version 1.0.0.1700

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Improved forecast error messages.
- Improved the *best fit model* forecasting algorithm for datasets that contain low inventories of phased-out products.

Version 1.0.0.1281

New feature introduced in version 1.0.0.1281

This version adds support for *time fences*, which allow demand planning managers to establish policies that prevent planners from editing certain date ranges of a forecast. Time fences are both flexible and simple to maintain. Managers create time fence rules based on the dimensions available in each plan. For example, a single product could be set up with different time fences for each store or geographical location. Time fence rules can also apply based on each user's role. For example, a role-based time fence rule could allow managers to edit a forecast in a period that can't be edited by planners. Learn more in Limit manual time series edits with time fences.

New fixes and improvements in version 1.0.0.1281

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Improved forecast error messages.

Version 1.0.0.1232

New features introduced in version 1.0.0.1232

Forecast calculation job-run records now provide an **Explainability** tab, which lists each combination of dimensions (such as item variant, warehouse location, and so on) the forecast was calculated for. For each combination, it shows the forecast algorithm that was used and the mean average percentage error (MAPE) for the calculation. A lower MAPE value indicates greater accuracy. Learn more in Work with forecast profiles.

New fixes and improvements in version 1.0.0.1232

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Transformation jobs that include the **Country/Region** column are now processed correctly. Previously, these jobs failed with an error.
- Improved the **Microsoft finance and operations apps** export data provider.
- Improved error messages shown during transformation, calculation, forecast, export, and import operations.
- Improved translation coverage in supported languages.

Version 1.0.0.1182

New features introduced in version 1.0.0.1182

The Microsoft Dynamics 365 Finance and Operations data provider now lets you choose which legal entities to import from. This feature applies to all data entities that contain a data area ID, including custom-built data entities. Learn more in Import data into Demand planning.

New fixes and improvements in version 1.0.0.1182

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Improved the ARIMA forecast model. The model now provides better error messages if it fails due to a data error.
- Improved the best fit forecast model. Calculations now succeed provided at least one of the models provides a result.

- Each organization instance can now run up to five forecasts in parallel. Newly created forecasts run in parallel only if all existing forecast jobs already started (in the *Executing* state). If one or more existing forecast jobs are still waiting to start (in the *Created* state) when a new forecast is created, then all forecasts execute sequentially.

Version 1.0.0.1132

New features introduced in version 1.0.0.1132

This version of Demand planning focuses on quality and stability improvements and doesn't include any new features.

New fixes and improvements in version 1.0.0.1132

This version of Demand planning includes the following fixes and improvements:

- Increased stability.
- Fixed an issue with data transformations that caused zero values to appear in output.
- Improved the *best fit* forecast model.

Version 1.0.0.1067

Version 1.0.0.1067 is the first general availability (GA) release of Demand planning in Microsoft Dynamics 365 Supply Chain Management.

Last updated on 08/18/2026

Demand planning license requirements

To use Demand planning in Microsoft Dynamics 365 Supply Chain Management in a production environment, each relevant user must have a license for it.

- For an overview of Supply Chain Management and Demand planning licensing plans and prices, see Supply Chain Management pricing.
- For complete licensing details for all Dynamics 365 products, including Supply Chain Management and Demand planning, see the Dynamics 365 Licensing Guide.
- For links to licensing and pricing details for all Dynamics 365 applications, see the Dynamics 365 pricing overview.

After purchasing the required licenses, an administrator must assign one to each user that needs it. For details about how to assign licenses to users, see Assign licenses and Assign or unassign licenses for users in the Microsoft 365 admin center.

Last updated on 07/01/2026

Install, enable, and set up Demand planning

This article describes how to install, enable, and set up Demand planning in Microsoft Dynamics 365 Supply Chain Management. It explains how to use Power Platform admin center to install Demand planning in your tenant and set up the feature in Supply Chain Management.

Prerequisites

To use Demand planning with Supply Chain Management, you must be running one of the following versions of Supply Chain Management:

- Supply Chain Management version 10.0.36, build 10.0.1695.83 or later
- Supply Chain Management version 10.0.37, build 10.0.1725.60 or later
- Supply Chain Management version 10.0.38 (any build) or later

To use Demand planning in a production environment, each relevant user must have a license for it. Learn more in Demand planning license requirements.

> **Important**
>
> **Demand planning no longer supports Supply Chain Management *Cloud hosted* environments.**
>
> As of February 28, 2024, Demand planning no longer supports Supply Chain Management environments of the *Cloud hosted* type. After that date, you can't import, export, or sync data between Demand planning and Supply Chain Management environments of this type.
>
> If you're running a Supply Chain Management *Cloud hosted* environment and want to continue to use Demand planning, you must either switch to a Supply Chain Management environment of the *Tier-2* type or higher, or set up a unified development environment. You must then create new import and export profiles in Demand planning and configure them to connect to your new Supply Chain Management environment.

Install Demand planning in Power Platform admin center

> **Important**

You must install Demand planning on the same tenant as your Supply Chain Management environment to ensure that the built-in import and export profiles can be used to import and export data.

Due to a current technical limitation, you can't install Demand planning on your tenant's default environment.

Follow these steps to install Demand planning in Power Platform admin center.

1. Sign in to Power Platform admin center.
2. Select **Resources** > **Dynamics 365 apps** on the left navigation pane.
3. Search for and select the app named *Dynamics 365 Demand Planning Application*.
4. Select **Install** on the top toolbar.
5. Select an environment, review the packages that will be installed, and select the **I agree to the terms of service** checkbox.
6. Select **Install**.

Enable and configure Demand planning in Supply Chain Management

Follow these steps to enable and configure Demand planning in Supply Chain Management.

1. In the **Feature management** workspace, turn on the *Demand Planning* feature. (As of Supply Chain Management version 10.0.40, it's turned on by default.)

2. Go to **System administration** > **Setup** > **Demand planning app parameters**.

3. On the **General** tab, in the **Demand planning app URL** field, enter the URL where Demand planning is installed.

4. Go to **Master Planning** > **Setup** > **Master planning parameters**.

5. On the **Number sequences** tab, in the grid, find the row where the **Reference** field is set to *Demand forecast sequence number*. Then select the link in the **Number sequence** column for that row.

6. On the details page for the selected number sequence, on the **Performance** FastTab, set the following field:

    - **Preallocation** – Set this option to *Yes*.
    - **Quantity of numbers** – Set the value to *500*.

7. On the Action Pane, select **Save**.

8. You must now refresh the data entity list in Supply Chain Management to ensure that data imports and exports between Supply Chain Management and Demand planning work correctly. Follow these steps:
   a. Go to **System administration** > **Workspaces** > **Data management**.
   b. Select the **Framework parameters** tile.
   c. Open the **Entity settings** tab.
   d. Select **Refresh entity list**.

Last updated on 07/01/2026

Clean up staging data

08/18/2025

When you import data from Demand planning into Supply Chain Management, the data is stored in staging tables in the Supply Chain Management database. After the import is complete, the staging data is no longer needed, so it's deleted automatically. However, if an import operation fails, its staging data is typically left behind in the staging table, which might eventually grow to be very large and therefore reduce database performance and add to data storage charges. To mitigate this problem, you can periodically clean up the staging data tables. The cleanup process deletes all data from the staging tables that is older than a specified number of days.

Manually clean up staging data

You can manually clean up the staging data tables at any time by following these steps:

1. Sign in to Supply Chain Management.
2. Go to **System administration** > **Periodic tasks** > **Setup** > **Demand planning app parameters**.
3. Open the **Periodic tasks** tab.
4. In the **Delete data from jobs executed before** field, enter the newest date for which you want to keep staging data. All data that is older than this date will be deleted.
5. Select **Run clean up job**.

Automatically clean up staging data periodically

To automatically clean up staging data periodically, you can set up a batch job that runs the cleanup process once a month. To do this, follow these steps:

1. Sign in to Supply Chain Management.
2. Go to **System administration** > **Periodic tasks** > **Setup** > **Demand planning app parameters**.
3. Open the **General** tab.
4. Set **Clean up staging data** to *Yes*.

Security roles and row-level security in Demand planning

07/30/2025

This article describes how to set up users, security roles, and access rules for Demand planning. It explains how to configure security roles and duties in Supply Chain Management and how to assign security roles for accessing the app and its various features on Power Platform. It also explains how to set up row-level security within Demand planning.

Configure security roles and duties in Supply Chain Management

Demand planning features in Supply Chain Management are available only to users who have the required security roles and related duties. When you enable Demand planning in Supply Chain Management, the feature adds a privilege named *Demand planning informational tile*. This privilege grants users access to a button that opens Demand planning from the home page of Supply Chain Management. The privilege is added to the out-of-box roles and duties that are listed in the following table.

Expand table

Role name	Duty
Production planner	Maintain forecasts
Production planner	Maintain firming of planned orders
Production manager	Enable forecast process
Production planner	Maintain forecast planning
Production planner	Maintain demand forecasts
Sales manager	Maintain demand forecast
Production manager	Enable the demand planning process

If you're using custom roles in Supply Chain Management, you must manually assign the *Demand planning informational tile* privilege to them as needed after you enable Demand planning in Supply Chain Management.

Assign security roles for accessing Demand planning on Power Platform

When you install Demand planning on Power Platform, it creates and updates several security roles to control access to its various features. To be able to access Demand planning, users must sign in a Power Platform user account that has the security roles for the features they need.

The following security roles are used by Demand planning. Some of them are standard for Power Platform, while others are specific to Demand planning.

- **Demand Planning Contributor**
  - This role is added when you install Demand planning.
  - Grants access to Demand planing.
  - View worksheets (see shared worksheets and save their own views).
  - Controlled by row-level access and can edit data they have access to.
  - Collaborate using Microsoft Teams and in-app comments.

- **Demand Planning Manager**
  - This role is added when you install Demand planning.
  - Grants access to Demand planning and all features that are available to the *Demand Planning Contributor* role.
  - Configure demand planning (role-level security, time fences, worksheets).
  - View and create planning data (forecasts and calculations).
  - View and create tables and import data from excel.
  - View and create transformations.
  - View and create worksheets.
  - Export plans when they're ready to share with Supply Chain Management.

- **Basic User**
  - This is a standard security role that's included with Power Platform.
  - This role isn't required to access Demand planning, nor does it grant access to Demand planning.
  - For users that also have the *Demand Planning Manager* and/or *Demand Planning Contributor* roles, the *Basic User* role adds the following privileges in Demand planning:
    - Export any data table to Microsoft Excel using buttons in the Demand planning user interface.
    - Access core business tables such as accounts, contacts, and so on.

- **System Administrator**
  - This is a standard security role that's included with Power Platform.

    - Grants access to Demand planning and all features that are available to the *Demand Planning Manager*, *Demand Planning Contributor*, and *Basic User* roles.
    - Grants access to admin features within Demand planning.
    - Install the app, add users, assign roles, manage teams, and so on.
    - View, create, and manage tables, relations, and data.
    - Create and manage imports and exports.
    - View and create transformations based on requests from the demand planning manager.
    - Monitor jobs (imports, calculations, forecasts, transformations, and exports).

- **Demand Planning Service Role**
    - This role is added when you install Demand planning.
    - An internal role required by Demand planning, which uses this role for internal operations.
    - You should never assign this role to a human user.

Use the Power Platform admin center to set up user accounts and assign security roles to them as needed. For more instructions, see Assign a security role to a user.

Set up row-level security for accessing data within Demand planning

Users who have the *Demand Planning Contributor* role (but not *Demand Planning Manager* or *System Administrator*) can access Demand planning, but they can only view and edit data that they're granted access to. You control this access using *row-level security*, which is set up in Demand planning. For example, a user might only be granted access to the specific products or product families they manage; or they might be restricted to only viewing data for a specific site or warehouse. By default, demand planning contributors can't see any data until they're granted access to it by a row-level access rule.

To set up row-level security, follow these steps:

1. Sign in to Demand planning using a system admin account.
2. On the navigation pane, select **Configuration** > **Row level access**.
3. Do one of the following steps:

    - To add a new row-level access rule, select **New** from the top toolbar. This launches a setup wizard, where each page of the wizard maps to a tab in the edit dialog.
    - To edit an existing row-level access rule, select the rule in the list, and then select **Edit** from the grid toolbar.

4. On the **Create rule** tab or wizard page, make the following settings.

- **Name** – Enter a name for the rule.
- **Description** – Enter a description for the rule.

5. On the **Conditions** tab or wizard page, set up the logic for which data tables and columns the rule grants access to. Use the buttons to add or remove conditions as needed. The following logic applies.

    - Each condition line is based on a data table and column, which you select in the **Table** and **Column** fields. The values in the column are compared to the values that you enter in the **Value** field. The **Operator** field determines how the comparison is made. For example, you can select *equals*, *greater than*, or *less than*. There is also a *Select all* operator, which matches all values in the column (in which chase the **Value** field is disabled); learn more in Using the select all operator.
    - If you have more than one condition line, then the lines are combined with an AND operator, which means that users will only have access to data rows where all of the condition lines are true.
    - You can select more than one value in the **Value** field for each condition. These values are combined with an OR operator, which means that the condition line evaluates to true for data rows where any of the values are true.

6. On the **Users** tab or wizard page, add each user for whom the current row level access rule applies. If you assign the same user to multiple rules, then the user will have access to data rows that are granted by any of the rules (an OR operator is applies across rules).
7. On the **Activation** tab or wizard page, set **Activation** to *Active* or *Inactive*, as needed. Only active rules take effect, but you might set a rule to inactive if you're still working on it or if you want to temporarily disable it without deleting it.

Convert a Demand planning environment from sandbox to production

Article • 03/21/2025

You can convert a Demand planning environment from sandbox to production. You might want to do this conversion if, for example, you use a sandbox environment to set up models and check integrations before you use them with production data.

Conversion process

The process for converting a Demand planning environment from sandbox to production is the same as the process for converting an environment for any other Microsoft Power Platform–based application. Follow the instructions in Change environment type.

Special considerations

Although the process for environment conversion is the same for every Microsoft Power Platform–based application, conversion of a Demand planning environment involves some special considerations, as described in the following subsections.

Remove sandbox connections

After you convert a Demand planning environment from sandbox to production, the environment can no longer connect to its sandbox sources of data. (Those sources include the previous instance of Dynamics 365 Supply Chain Management and Power Query tables.) This behavior makes sense from a security standpoint, because it prevents accidental leakage of production data. However, one result is that some of your configurations, such as import or export profiles, no longer work as you expect. Therefore, you should delete all existing import and export profiles before you convert the environment to production. Then re-create them after the conversion is completed. Configure the new profiles so that they point to production sources of data.

To delete import and export profiles, follow these steps.

1. Go to **Data management** > **Import** or **Data management** > **Export**.
2. Select the profile to delete.

3. Select **Delete**.

Learn more about how to create new profiles in the following articles:

- **Import profiles:** Import data into Demand planning
- **Export profiles:** Export and download data

Consider removing test data

After you convert a Demand planning environment from sandbox to production, you will probably start to use production data. To avoid mixing test data and production data, consider going through tables and time series, and deleting any that contain test data.

To delete tables and time series, follow these steps.

1. Go to **Data management** > **Tables and data** or **Planning data** > **All**.
2. Select the table or time series to delete.
3. Select **Delete**.

> ⓘ **Note**
>
> The system might block you from deleting some tables or time series (for example, if a time series is used as an input in a forecast or calculation profile). In this case, read the error messages that you receive to learn how to address the issue. For example, you might first have to delete dependent records.

Review user access roles

After you convert a Demand planning environment from sandbox to production, the environment uses production data and generates production forecasts. Therefore, you should review who can access Demand planning, and what kind of data they can view or edit. Learn more in the following articles:

- **Security roles and row-level security:** Security roles and row-level security in Demand planning
- **Time fence functionality:** Limit time series edits with time fences

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View and customize tables for holding imported data

Article • 02/21/2024

Data that's imported into Demand planning must be loaded into tables that are set with the fields and relations that are required to accommodate the incoming data. Demand planning includes a set of predefined tables that support the most useful standard data that you'll probably want to import from Microsoft Dynamics 365 Supply Chain Management. If you've customized your tables in Supply Chain Management, you can use the tools that are provided to reproduce the custom fields. You can also create new custom tables to store data from Supply Chain Management and other sources.

View tables

To view the full list of tables that are available to Demand planning, select **Data management** > **Tables and data** on the navigation pane. The tables are listed by name.

- The **Name** column shows the name of each table. Select a table name to view and edit details about that table, including column definitions, relationships, and data.
- The **Is System** column indicates whether each table is a standard table that's provided by the system or a custom table that was added by a user. It shows one of the following values:
  - *Yes* – The table is a system table. You can't delete these tables.
  - *No* – The table is a custom table that was added by a user. You can delete these tables if you no longer need them.

> **Note**
>
> Tables that hold prices aren't marked as systems tables.

Standard tables

Demand planning includes a set of predefined *system tables* that support the most useful standard data that you'll probably want to import from Supply Chain Management. Most system tables represent important data that's imported from Supply Chain Management. However, many of these tables are slightly modified in Demand

planning through minor name changes and the addition of extra columns. This section lists these important tables and explains how they're modified in Demand planning.

Site Location table

Sites and warehouses are implemented as a hierarchy. The site is the parent, and each site can have one or more warehouses. This concept is inherited from Supply Chain Management.

To import sites from Supply Chain Management, export an Excel file from the **Sites V2** data entity in Supply Chain Management, and then import that file into Demand planning.

In the current release, only one-level relations are supported in transformations. Therefore, the **Site location** table and its data can't be used in transformations. Site ID values can still be obtained from the **Warehouse Location** table.

Warehouse Location table

The **Location** table from Supply Chain Management is renamed **Warehouse Location** in Demand planning. Local columns that are named **Operation Site ID** and **Site Location ID** have been added to the **Warehouse Location** table in Demand planning. These extra columns let you generate forecasts based on the warehouse and/or site location ID.

You can derive site IDs by applying a warehouse location ID to the historical demand.

Customer Accounts table

Customer Accounts is a new table that lets you import customers and related details such as the customer group and address information (for example, country/region). This table also lets you generate forecasts based on the customer, the customer group, and/or the customer's country/region.

Legal Entity table

The **Legal Entity** table stores information about legal entities and related details. Usually, you'll import this data from Supply Chain Management.

This table lets you generate forecasts across legal entities. It also lets you slice forecasts according to the owning legal entity.

> **Note**
>
> When you export forecasts back into Supply Chain Management so that they can be used in supply planning, you must include the **Legal Entity ID** column, because supply planning is performed per legal entity in Supply Chain Management.

Product table

The **Product** table stores product information and related details. Usually, you'll import this data from Supply Chain Management.

The **Product** table supports both *Product* and *Service* product types.

You can forecast on products and product variants as a hierarchy. In this way, you can get an aggregate of product variants.

You can add columns for color ID, size ID, style ID, and configuration ID to the forecast. In this way, you can slice the forecast across products variants. For example, you can forecast demand for all silver LCD TVs across sizes.

Historical Demand table

The **Historical Demand** table stores information about historical demand in the form of orders. Each order includes a buyer and a seller, to cater to both external sales and intercompany trade. An order can also be internal. (In this case, the buyer and the seller are the same.) Usually, you'll import this data from Supply Chain Management.

The **Historical Demand** table includes an **Order Type** column that classifies the origin of the demand (*Sales*, *Transfer*, or *Production*). Sales returns are classified as sales that have a negative quantity.

Create or edit a table and its columns and relationships

You can create your own non-system tables, or you can extend system tables by adding extra non-system columns. Custom tables and columns can be used in Demand planning just as system tables and columns can. For example, you can add important signals, such as weather data, inflation statistics, or other macroeconomic data.

To create or edit a table, follow these steps.

1. On the navigation pane, select **Data management** > **Tables and data**.

2. Follow one of these steps:

    - To add a new non-system table, select **New** on the Action Pane.
    - To edit an existing table, select the link for it in the **Name** column.
    - To delete one or more existing non-system tables, select the checkbox next to the name of each table, and then select **Delete** on the Action Pane.

3. On the **General** tab, set the following fields:

    - **Name** – For new tables, enter the table name. For existing tables, this field is read-only.
    - **Owner** – Select the owner of the table (user account, master data manager, or a similar role).
    - **Key** – Specify one or more columns whose combined values will uniquely identify each row in the table. If you're creating a new table, you can't set this field until you add the columns that you'll use as keys. This limitation is important because it helps prevent duplicate records from being created at each import.

4. On the Action Pane, select **Save**.

5. On the **Columns** tab, you can view the name, data type, and system status of each column. Use the buttons on the toolbar to add new columns or delete existing non-system columns. Each column must have a **Name** value and a **Data Type** value.

    > ⓘ **Note**
    >
    > To transform a table and its data into a time series, the table must include at least three columns:
    >
    > - One *timestamp* column (*DateTime* data type)
    > - One *measure* column (*Decimal* or *Integer* data type)
    > - One *dimension* column (*String* data type)

6. If you didn't specify key columns in step 3, go back to the **General** tab, and specify them.

7. Select the **Relationships** tab.

Relationships between tables let you construct a data model that's suitable for your organization's needs. As for tables and columns, predefined system relationships are required. Therefore, the **Is System** field is set to *Yes* for them. You can also add custom relationships as you require. (Set the **Is System** field to *No* for these custom relationships.) System relationships can't be deleted or changed.

Each relationship links two tables together. This link lets you match related information that's stored in different tables. Only columns from related tables can be selected in transformations. You can set up both *many-to-one* (n:1) and *many-to-many* (n:n) relationships. For *many-to-many* relationships, the system shows the first matching record.

Relationships are currently *directional*. Therefore, it's important that you pay attention to which table is the "from" table and which is the "to" table. Often, the "from" table is a transactional table such as **Historical Demand**.

8. To add a new relationship, select **New Relationship** on the toolbar, and then set the following fields in the **Quick Create: Relationship** dialog box:

    - **From Table** – This table is the current table. It's usually a table that holds transaction data.
    - **From Column** – Select the field in the "from" table that holds values that should be matched in the related table.
    - **To Table** – Select the target table that contains the data that you want to relate to.
    - **To Column** – Select the field in the "to" table that holds data values that should be matched to the value that's found in the "from" column.

9. Select **Save and Close**.

    By default, all relationships are active. Therefore, columns from related tables can be selected in transformations. You can change the activation status of a relationship or completely delete the relationship. Select the relationship on the **Relationship** tab, and then select the appropriate button on the toolbar.

10. On the **Data** tab, inspect the data that your table contains.

11. Select **Save** on the Action Pane to save your changes.

Delete all records in a table

As required, you can delete all the records from a selected table but keep the table schema and its relationships. This approach can be useful if you must correct data that

was incorrectly imported.

To delete all the records in a table, follow these steps.

1. On the navigation pane, select **Data management** > **Tables and data**.
2. In the **Name** column, select the name of the target table.
3. On the Action Pane, select **Delete Data**.
4. Select **OK** to confirm the deletion.

Import data from Supply Chain Management

You can easily import data from Supply Chain Management into the standard tables by using the Finance and Operations connector. By default, all system (standard) entities are selected, and their fields are automatically mapped.

You can edit any relationship, unmap fields, or add custom field mappings as you require. Before you can import a custom field, you must add it in the standard table.

When you create an import job, an export data project for the required data entities is created in Supply Chain Management.

Security configuration for custom entities

To allow data to be read from custom entities, you must configure their security settings in Dynamics 365 Supply Chain Management. To do so, follow these steps:

1. Go to **System Administration** > **Security** > **Security Configuration**.
2. Open the **Privileges** tab.
3. Select **Create new** from the toolbar.
4. In the dialog, add a **Name** for your new privilege, and then select **OK**.
5. Your new privilege is added to the list and selected. In the middle column, select **Entities**.
6. On the toolbar, select **Add references**.
7. In the dialog, find and select your custom entity. Select the access properties you want to grant and then select **OK**.
8. In the middle column, select **Duties**.
9. On the toolbar, select **Add references**.

10. In the dialog, find and select the duty named *View Document entity data for data management* or *Create data management project and details using entity*. Choose the duty based on the level of access you require.

    - *View Document entity data for data management* is commonly associated with entities used in the Data Management Framework in Supply Chain Management. It's typically assigned to the role *Data management migration user*, which is a subordinate role to *Demand planning app role*.
    - *Create data management project and details using entity* is part of the *Demand planning app role* role. *Demand planning app role* is assigned to *DemandPlanAppUser*, which is a user role often employed for integrating the Demand Planning Service with Supply Chain Management.

11. Select **OK** to add the selected duty to your new privilege.

12. Open the **Unpublished objects** tab. On the toolbar, select **Publish all**.

Feedback

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Import data into Demand planning

You can import data from a range of sources and file types. For example, you can import data directly from Microsoft Dynamics 365 Supply Chain Management or by importing text files in Excel or comma-separated values (CSV) format. You can also import data that's stored in an Azure data lake. Before you import the data into Demand planning, you can use Power Query to transform it as you require.

Demand planning lets you build a collection of *import profiles*. Each profile imports data from a specific external source into one or more specific tables in Demand planning.

Typically, a manager or system administrator creates the initial collection of required profiles. Forecasters and other users can then run the profiles to update the data as they require.

In addition, you can use the standard user interface (UI) to import your custom data entities into custom tables or add custom fields that you've extended in the standard tables. No developer is required.

View and run existing data import profiles

You only need to import data as often as the relevant data changes in the external systems. Some profiles require occasional runs, whereas others require almost every time that a user works with the app.

To update your data by running an existing data import profile, follow these steps:

1. On the navigation pane, select **Data management** > **Import**.
2. Find the profile for the type of import that you want to run, and select the link for it in the **Name** column.

   The details page for the selected profile appears. It contains the following tabs:

   - **Summary** – This tab provides basic information about the profile. You can edit the name and description to make the profile easier to identify and work with.
   - **Configure provider** – This tab lets you view and edit the settings that are specific to the data provider that the profile uses. For information about how to work with the settings on this tab, see the Create and manage data import profiles section.

- **Run schedule** – This tab lets you set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
- **Jobs** – This tab shows a list of every run of the profile. It includes date information, the job status, the provider that was used, the table that was updated, and the number of records that were imported. Links to more information are provided.

3. To run the profile, select **Run** on the Action Pane. This command adds a new row to the grid on the **Jobs** tab. There, you can follow the status of the new import. The page isn't automatically refreshed. To update the status information, you must select **Refresh** on the grid toolbar.

Create and manage data import profiles

Each time your organization needs to run a new type of data import, a manager or admin must create a new data import profile. After the profile is created, it becomes available to users, who can run it as often as they require.

Each data import profile uses a *data provider* that's optimized to import data from a specific type of external data source or file type. The following types of data providers are currently available:

- **Power Query providers** – Each of these providers imports from a specific type of file (CSV, Excel, or Azure data lake storage) through Power Query. Therefore, you can transform and map the data for the relevant table in Demand planning.
- **Microsoft finance and operations apps provider** – This provider connects directly to Supply Chain Management (or another finance and operations app) and imports data that one or more data entities provide into the relevant tables in Demand planning.

Create an import profile for importing directly from Supply Chain Management

To import directly from Supply Chain Management (and other finance and operations apps), create an import profile that uses the *Microsoft finance and operations apps* data provider.

1. On the navigation pane, select **Data management** > **Import**.
2. On the Action Pane, select **New**.
3. On **Select data provider**, select the **Microsoft finance and operations apps** tile.
4. A setup wizard opens. On **Get started**, enter a name and description for the new profile. Then select **Next**.

5. On **Configure connection**, in the **Connection URL** field, enter the URL of your Supply Chain Management environment. Then select **Next**.
6. The **Entity selection** page lets you map **Source** data entities from Supply Chain Management to **Target** tables in Demand planning. It initially lists all of the mappings that the solution supports out of the box. You can edit the list, remap fields, and add more mappings as needed. You can make the following settings:

    - For each listed **Source** entity that you want to import, set **Synchronize** to *Yes*. If you set **Synchronize** to *No*, the profile ignores that entity when it runs.
    - To delete all data from a listed **Target** table each time the profile runs, set **Delete data** to *Yes*.
    - Source entities that include company-specific data provide a drop-down dialog box in the **Company** column. To limit the import to only include data from one or more specific companies (data entities), select those companies. This feature supports all data entities that contain a *Data area ID* field, including custom-built data entities. If none of the listed source entities include company-specific data, the **Company** column is hidden.
    - To view or edit the field mappings for any listed entity, select that entity and then select **Edit** from the toolbar. Then use the **Add mapping** dialog to map source entity fields to target table fields.
    - To add an unlisted entity, select **Add** from the toolbar. Use the **Add mapping** dialog to identify the **Source** entity and **Target** table. Then map the required source entity fields to their matching target table fields. The target table and fields must already exist in Demand planning (learn more in View and customize tables for holding imported data).

7. Select **Next**.
8. On **Set run schedule**, you can choose to set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
9. Select **Next**.
10. On **Review and finish**, review the summary of settings that you configured, and then select **Review and finish** to create the new profile.
11. You're returned to the **Active Import Data Profiles** page, which now shows the new profile in the list. The profile is now available, but it didn't yet run. To run it, follow the instructions in the View and run existing data import profiles section.

Create an import profile for importing data through Power Query

To import data from a text or workbook file exported from an external system, create an import profile that uses one of the *Power Query* data providers. Use the provider that matches the format of the exported file. The procedure is the same for all providers.

1. On the navigation pane, select **Data management** > **Import**.
2. On the Action Pane, select **New**.
3. On **Select data provider**, under **Power Query providers**, select the tile that matches the format of the exported file that you want to import.
4. A setup wizard opens. On **Get started**, enter a name and description for the new profile. Then select **Next**.
5. On **Select target table**, in the **Select table** field, select the name of the table in Demand planning that you want to import data into. Then select **Next**.
6. On **Configure provider**, select the file that you want to import and then send to Power Query, where you can transform the data and map columns from the source document to the selected Demand planning table. Power Query provides the connection, transformation, and mapping features. The procedure varies slightly, depending on the Power Query provider that you're using:
    - **Excel** – For information about how to select the target file, connect to Power Query, and work with data, see Excel in the Power Query documentation.
    - **CSV** – For information about how to select the target file, connect to Power Query, and work with data, see Text/CSV in the Power Query documentation.
        - **Datalake** – For information about how to select the target file, connect to Power Query, and work with data, see Datalake in the Power Query documentation. Set up the data lake as described in Datalake setup.
7. When you finish transforming and mapping your data by using Power Query, select **Next**.
8. On **Refresh settings**, select or clear the **Delete all data for the chosen table before import** checkbox to specify what should happen to the data that's currently in the target table in Demand planning:

- *Selected* – Delete all the data that's currently in the target table. Use this option if, for example, the incoming data includes all the required records that are already in your table, and you don't select a key field for the table. This option ensures that you don't have any duplicate records, but it also deletes all the data that's currently in the target table.
- *Cleared* – Keep all the data that's currently in the target table. Use this option if, for example, the incoming data is an incremental export that contains only new records, and you selected key fields for the table. Incoming records that have key field values that match existing records update those records, whereas incoming records that have unique key field values create new records. If you don't configure and map your key fields correctly, this option might create duplicate records.

9. Select **Next**.

10. On **Set run schedule**, you can choose to set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.

11. Select **Next**.

12. On **Review and finish**, review the summary of settings that you configured, and then select **Review and finish** to create the new profile.

13. You're returned to the **Active Import Data Profiles** page, which now shows the new profile in the list. The profile is now available, but it didn't yet run. To run it, follow the instructions in the View and run existing data import profiles section.

Troubleshoot data import issues

To troubleshoot data import issues in Demand planning, see the Demand Planning Import Profile Troubleshooting Guide.

Last updated on 08/20/2026

Transform data

Transformations let you transform the imported data in a selected table into time series. You can then use the time series to create a forecast, do a comparison to a forecast, or do calculations. For example, you might want to shift the historical demand time from last year by 12 months, so that it can be used as the basis for next year's forecast. Alternatively, you might want to combine daily historical demand data into monthly buckets, because you run planning on a monthly basis.

Demand planning lets you build a collection of *transformation profiles*. Each profile transforms data from a specific data table that's already set up in the app into a time series that you can analyze by using the app.

Each transformation profile takes a selected main table as its input. That main table can have other tables that are related to it.

- The main table typically represents historical demand, such as sales orders.
- The main table must include a *timestamp* column. This column will become the x-axis of the output time series and typically represents the order date or delivery date.
- The main table must also include a *measure* column. This column will become the y-axis of the output time series and typically represents units that are sold.
- Either the main table or one of its related tables must include at least one dimension column. Often, you'll select more than one dimension, so that you can analyze the data in multiple ways. For example, if you're looking at T-shirts, you might include columns for the color, size, style, and warehouse dimensions. You can then view the output time series to see, for example, the historical demand for red, extra-large T-shirts that are sold from the Chicago warehouse. You can also compare that demand with the overall sales of all T-shirts from all warehouses.

Typically, a manager or system administrator creates the initial collection of required profiles. Forecasters and other users can then run the profiles to update the time series as they require.

View and run existing transformation profiles

You have to run a transformation profile only as often as you import new data into the relevant tables. Some profiles must be run only occasionally, whereas others must be run almost every time that a user works with the app.

To update your time series by running an existing transformation profile, follow these steps:

1. On the navigation pane, select **Operations** > **Transformations**.
2. Find the profile for the type of transformation that you want to run, and select the link for it in the **Name** column.

   The details page for the selected profile appears. It contains the following tabs:

   - **Summary** – This tab provides basic information about the profile. You can edit the name and/or description to make the profile easier to identify and work with. Although you can also change the owner and adjust the time bucket, you typically shouldn't change these details for existing profiles. For information about how to work with the settings on this tab, see the Create and manage transformation profiles section.
   - **Transformations** – This tab shows details about the transformation that the profile will process. For information about how to work with the settings on this tab, see the Create and manage transformation profiles section.
   - **Run schedule** – This tab lets you set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
   - **Jobs** – This tab shows a list of every run of the profile. It includes date information, the job status, and the time series that was updated. Select a link in the **Time Series** column to open the time series.

3. To run the profile, select **Run** on the Action Pane. This command adds a new row to the grid on the **Jobs** tab. There, you can follow the status of the new transformation. The page isn't automatically refreshed. To update the status information, you must select **Refresh** on the grid toolbar.

Create and manage transformation profiles

Each time that your organization requires a new type of data transformation, a manager or admin must create a new transformation profile. After the profile is created, it becomes available to users, who can run it as often as they require.

To create or edit a transformation profile, follow these steps:

1. On the navigation pane, select **Operations** > **Transformations**.
2. On the Action Pane, select **New**.
3. A setup wizard is opened. On the **Get started** page, set the following fields:

- **Name** – Enter a name for the new profile.
- **Description** – Enter a short description of the profile.
- **Owner** – Select the user account that owns the profile.
- **Time Bucket** – Specify the time span to group source records by in the time series that this profile generates as output: *Daily*, *Weekly*, or *Monthly*. The shorter the time bucket, the more data points there are on the x-axis. Select the time bucket that gives the required level of granularity.
- **Precision** – Specify the rounding precision for the output values. For example, select *0.01* to keep two decimal places. The default value is *0.01*.
- **Aggregation Method** – Select how the profile combines multiple source records that have the same dimension values and belong to the same time bucket. The default method is *Sum*. Learn more about the options available here and how they work in Aggregation methods.

4. Select **Next**.

5. On the **Configure transformation** page, select a table, and then map the columns of that table and its related tables to transform the data into a timeline.

   - The grid has a row for each column that the transformation generates as output.
   - You can use the buttons on the toolbar to add columns to the grid, remove selected columns from the grid, or completely reset the grid.
   - You can select only one main table at a time. To change the main table, you must reset the grid.

6. To select the main table, and/or add or remove columns for the main table and related tables, select **Add column** on the toolbar.

7. In the **Add columns to your transformation** dialog box, if no main table has been selected yet, select one in the **Tables** field. (If you have to change the main table, you must first reset the whole transformation by selecting **Reset**.)

8. To add a data column to the transformation, select a table in the left column of the dialog box. Then, in the middle column, select the name of the data column in that table that you want to add. For information about how the different types of columns are used, see the beginning of this article.

   - Under **Timestamp**, you must select exactly one column. This column will become the x-axis of the output time series and typically comes from the main table.

- Under **Measure**, you select exactly one column. This column will become the y-axis of the output time series and typically comes from the main table.
- Under **Dimension**, you must select at least one dimension column. Dimensions provide interactive parameters that let you choose which values to view in the output time series. These columns can come from the main table and/or any related tables.

> ⓘ **Note**
>
> The order that you add the dimension columns in defines how the output time series aggregates the data.

Each time that you add a data column, it's added to the list in the right column of the dialog box. Therefore, you can view your previous selections while you explore the different tables. Under **Dimension**, the order reflects the order that you selected the dimensions in.

9. When you've finished selecting data columns, select **OK**.
10. Select **Next**.
11. On the **Set run schedule** page, you can choose to set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
12. Select **Next**.
13. On the **Review and finish** page, review the summary of settings that you've configured, and then select **Review and finish** to create the new profile.
14. The profile is now saved, but it hasn't yet run. If you're ready to create a time series now, select **Run** on the Action Pane.

Aggregation methods

Each transformation profile has an **Aggregation method** setting that determines how the profile combines multiple source records that have the same dimension values and belong to the same time bucket. It groups the relevant records in each bucket and applies the selected aggregation method to the measure column to produce one output value for the group. The profile applies its **Precision** setting after it calculates the value.

For example, suppose that the following records belong to the same product dimension and monthly time bucket:

Expand table

Date	Measure value
January 1	40
January 10	20
January 20	80
January 28	60

The following table shows the result of each aggregation method.

Expand table

Aggregation method	How the output value is calculated	Result in the example	Common use
Sum	Adds all measure values in the bucket.	200	Additive values, such as units sold or ordered.
Minimum	Uses the smallest measure value in the bucket.	20	The lowest value recorded during a period.
Maximum	Uses the largest measure value in the bucket.	80	The highest or peak value recorded during a period.
Average	Calculates the arithmetic mean of the measure values in the bucket.	50	Rates, prices, or measurements where a total wouldn't be meaningful.
First	Uses the measure value from the record with the earliest timestamp in the bucket.	40	The opening value or balance for a period.
Last	Uses the measure value from the record with the latest timestamp in the bucket.	60	The closing value or balance for a period.
Count	Counts the source records in the bucket. The measure values aren't used in the calculation.	4	The number of transactions, orders, or events.

Note

Sum is the default method and preserves the behavior of transformation profiles that were created before other aggregation methods were available. *First* and *Last* are based on the timestamp of each record, not the order in which records were imported or displayed.

Last updated on 08/10/2026

Work with calculation profiles

Demand planning in Microsoft Dynamics 365 Supply Chain Management lets you build a collection of *calculation profiles*. Each profile takes one or more existing time series as input, and applies a set of predefined calculations to generate a new time series as output.

Here are some examples of purposes that you might use calculations for:

- Increase all forecasts by 10 percent.
- Combine two forecasts, so that the result is the average of the two.
- Remove outliers from the forecast by removing some values that are completely out of range.

Typically, a manager or system administrator creates the initial collection of required profiles. Forecasters and other users can then run the profiles to generate new calculated time series as they require.

View and run existing calculation profiles

You have to run a calculation profile only as often as you import new data into the relevant tables. Some profiles must be run only occasionally, whereas others must be run almost every time that a user works with the app.

To generate a new calculated time series by running an existing calculation profile, follow these steps:

1. On the navigation pane, select **Operations** > **Calculations**.

2. Find the profile for the type of calculation that you want to run, and select the link for it in the **Name** column.

   The details page for the selected profile appears. It contains the following tabs:

   - **Summary** – This tab provides basic information about the profile. You can edit the name and/or description to make the profile easier to identify and work with.
   - **Calculation model** – This tab shows the calculation that the profile does. It uses a flowchart of interconnected steps. Each step does a specific type of operation and has settings that let you define how that operation works. For information about how to

work with the settings on this tab, see the Create and manage calculation profiles section.

- **Run schedule** – This tab lets you set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
- **Jobs** – This tab shows a list of every run of the profile. It includes date information, the job status, and the time series that was generated. Select a link in the **Time Series** column to open the time series.

3. To run the profile, select **Run** on the Action Pane.

4. In the dialog box that appears, set the **Output option** field to one of the following values to define the output of the job:

   - *Create a new time series* – The job will create a new time series. If you select this value, enter a name for the new series in the **Time series name** field.
   - *Use an existing time series* – The job will overwrite an existing series or create a new version of it. If you select this value, use the fields that are provided to select the target series, specify whether you want to overwrite the existing version or create a new version, and, if you want to create a new version, specify the name of the new version.

5. Select **Save and close**.

6. The new job is added to the grid on the **Jobs** tab. There, you can follow the status of the new calculation. To update the status information, select **Refresh** on the grid toolbar.

Create and manage calculation profiles

Each time that your organization requires a new type of calculation, a manager or admin must create a new calculation profile. After the profile is created, it becomes available to users, who can run it as often as they require.

To create or edit a calculation profile, follow these steps:

1. On the navigation pane, select **Operations** > **Calculations**.

2. On the Action Pane, select **New**.

3. A setup wizard is opened. On the **Get started** page, set the following fields:

   - **Name** – Enter a name for the new profile.
   - **Description** – Enter a short description of the profile.
   - **Owner** – Select the user account that owns the profile.

- **Precision** – Specify number of decimal places that the profile should show for calculation results and output.
- **Category** – Select the category of the output time series (*Forecast*, *Demand*, *Financials*, or *Miscellaneous*). The category affects the location of the output time series under the **Planning data** heading on the navigation pane.

4. Select **Next**.

5. On the **Select and configure calculation model** page, you can select a calculation model preset to use with your current profile. Browse the presets that are listed under **Available model presets** to preview the calculation that each preset does. You can configure settings and customize the calculation model as you require after you save the profile. Therefore, select the preset that's closest to what you're looking for, and then select **Next**. For information about how to configure settings and customize the calculation model that a profile uses, and how to create new presets, see Design calculation models.

6. On the **Set run schedule** page, you can choose to set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.

7. Select **Next**.

8. On the **Review and finish** page, review the summary of settings that you've configured, and then select **Review and finish** to create the new profile.

9. Your profile is now saved, but it hasn't yet run. If you're ready to run the calculation now, select **Run** on the Action Pane.

Last updated on 12/30/2025

Design calculation models

Article • 01/10/2025

Calculation models let you arrange and configure steps to define the calculation that's done by a calculation profile. Each model presents a flowchart that graphically represents the calculation that the model does.

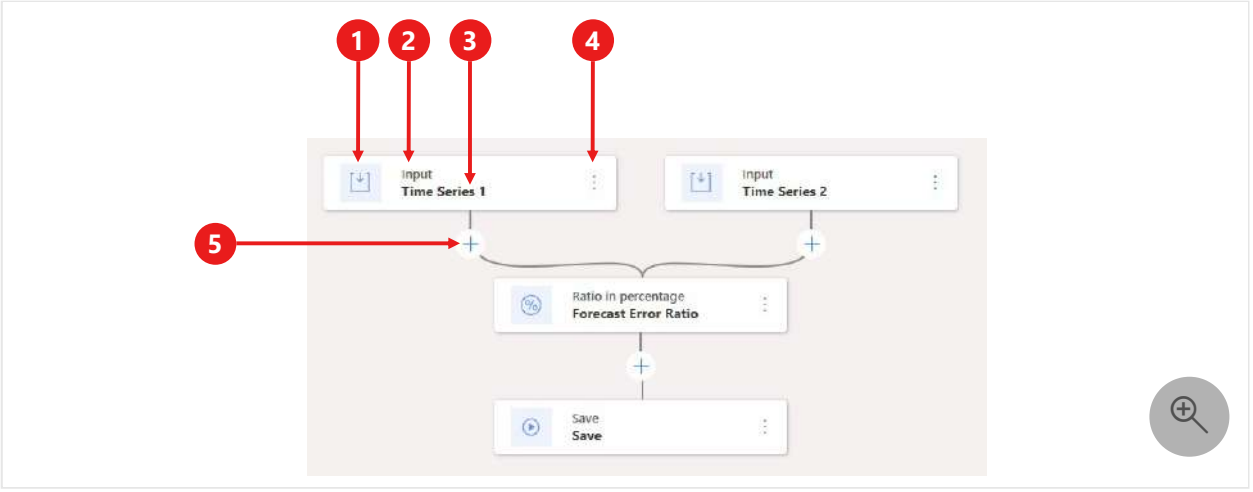
Create and customize a calculation model

To create and customize a calculation model, you must first open an existing calculation profile. (Learn more in Work with calculation profiles.) You can then fully customize the model that the selected profile uses by adding, removing, and arranging steps, and configuring settings for each of them.

Follow these steps to create and customize a calculation model.

1. On the navigation pane, select **Operations** > **Calculations**.
2. Select the calculation profile that you want to create or customize a calculation model for.
3. On the **Calculation Model** tab, there will always be at least one step (of the *Input* type) at the top of the flow chart. The model is processed from top to bottom, and the last step must be a step of the *Save** type. Add, remove, and arrange steps as you require, and configure settings for each of them. For guidelines, see the illustration after this procedure.
4. When you've finished designing your calculation model, select the **Validate** button in the upper-right corner. The system runs a few tests to validate that your model will work, and then provides feedback. Fix any issues that the validation test reports.
5. Continue to work until your model is ready. Then, on the Action Pane, select **Save**.
6. If you want to save your calculation model as a preset, so that it's available when you and other users create a new calculation profile, select the **Save as model template** button in the upper-right corner.

The following illustration highlights the information and controls that are available for steps in a calculation model.



Legend:

1. **Step icon** – A symbol that represents the role of the step.

2. **Step type** – The type of step. This text usually describes the type of calculation or other action that the step represents.

3. **Step name** – The name that's applied to the step. Usually, you can manually enter this text in each step's settings. However, some types of steps have a predefined value here.

4. **Step actions** – Open a menu of actions that you can take on the step. Although some of these actions are specific to the step type, many are common to all steps. If any actions appear dimmed, they can't be used because of the step's current position or for some other contextual reason. Here are some common actions that are available:

    - **Settings** – Open a dialog box where you can configure settings for the step.
    - **Remove** – Remove the step.
    - **Move Up** and **Move Down** – Reposition the step in the flowchart.
    - **Set to 'Pass Through'** – Temporarily disable a currently enabled step without deleting it or its settings.
    - **Unset 'Pass Through'** – Reenable a currently disabled step.
    - **Connect with** – For steps that support multiple inputs, specify the second input. The first input is automatically set based on where you put the step.

5. **Add a step** – Add a new step at the selected location. (If you insert a new *Input* step, that step will always be put at the top of a new flow, not at the button location.)

Calculation step types

This section describes the purpose of each type of calculation step. It also explains how to use and configure each type.

Input steps

Each *Input* step represents a time series that provides input to the calculation model.

Input steps have the following options that you can set:

- **Time series** – The time series that is added by the step. When you first set up an *Input* step, the **Configure step** dialog box provides a button that lets you select the time series to add. After you add a time series, the dialog box shows the name of the time series and provides an **Edit** button that lets you change the time series. It also provides a **Filter** button that lets you set up filtering criteria based on fixed dates, relative dates, and/or field values.
- **Time series version** – If the selected time series includes more than one version, you can select a specific version. By default, the latest (current) version is used.

Arithmetic operator steps

Arithmetic operator steps do a calculation that combines values from two input time series. You might use this type of step, for example, to calculate the expected margins of a forecast by subtracting the cost of goods sold (COGS) from the expected revenue.

Arithmetic operator steps implement the following calculation:

f(x) = a(x) <*operator*> b(x)

In this formula, <*operator*> is one of the following arithmetic operators: +, -, *, or /.

You can use this type of step only if your calculation model has at least two input time series in parallel branches. The first series is at the top of the branch where you create the step. The second series must start with an *Input* step. It can have any number of steps in it, but it must still be open. (In other words, it must not end with a *Save* step.) To specify the second series, open the step's **Action** menu, and select **Connect with**. The flowchart is then updated so that it shows the two branches combining at the *Arithmetic operator* step.

Arithmetic operator steps have the following fields that you can set:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.

- **Created by** – The user who created the step.
- **Value 1** – The first series in the calculation (*a(x)*), which is on the left side of the operator. Select *Input 1* or *Input 2*.
- **Operator** – The operator that's applied between the first series (**Value 1**) and the second series (**Value 2**). Select whether to add (+), subtract (-), multiply (*), or divide (/).
- **Value 2** – The second series in the calculation (*b(x)*), which is on the right side of the operator. Select *Input 1* or *Input 2*.

Constant arithmetic operator steps

Constant arithmetic operator steps do a calculation that applies a constant arithmetic operation to each value in an input time series.

Constant arithmetic operator steps implement the following calculation:

f(x) = a(x) <*operator*> C

In this formula:

- <*Operator*> is one of the following arithmetic operators: +, -, *, or /.
- *C* is a constant value.

Constant arithmetic operator steps have the following fields that you can set:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Operator** – The operator that's applied. Select whether to add (+), subtract (-), multiply (*), or divide (/).
- **Constant** – The constant value that's applied to each time series value by using the selected operator.

Date offset steps

Date offset steps shift the input time series by a fixed value along the x-axis. You might use this type of step, for example, to shift last year's demand one year forward, so that you can compare it to next year's forecast by overlaying the time-shifted demand graph on the forecast graph.

Date offset steps implement the following calculation:

$f(x) = a(x - t)$

In this formula, *t* is an integer that represents the number of days, months, or years.

Date offset steps have the following fields that you can set:

- **Step name** – The specific name of the step. The name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Time value** – The amount of time to shift the input time series forward by. The amount is in the unit that's specified in the **Time unit** field.
- **Time unit** – The unit that applies to the time value.

Logical operator steps

Logical operator steps test whether a value exceeds a constant (threshold) at each point along a time series. As output, it returns a *1* for true and a *0* (zero) for false. You might use this type of step, for example, to evaluate when the forecast error is above a specific threshold in either direction, so that you can focus on important products.

Logical operator steps implement the following calculation:

$f(x) = ABS[a(x)]$ <*operator*> C

In this formula:

- <*Operator*> is one of the following logical operators: >, >=, <, or <=.
- *C* is a constant.
- *ABS* is optional. If it's enabled, the absolute value (that is, the distance from zero) is used instead of the literal value.

Logical operator steps have the following fields that you can set:

- **Step name** – The specific name of the step. The name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Apply ABS** – Set this option to *Yes* to use the absolute value (distance from zero) of the input series. Set it to *No* to use the literal value (including +/- signs) of the input series.
- **Operator** – The logical operator that's applied for the comparison against the constant value.
- **Constant** – The constant value to compare against the input time series.

Merge steps

Merge steps merge two time series into a new time series by applying the selected merge logic. You can merge the time series by minimum value, maximum value, average, or sum. You might use this type of step, for example, to calculate a merged result of two forecast methods.

You can use this type of step only if your calculation model has at least two input time series in parallel branches. The first series is at the top of the branch where you create the step. The second series must start with an *Input* step. It can have any number of steps in it, but it must still be open. (In other words, it must not end with a *Save* step.) To specify the second series, open the step's **Action** menu, and select **Connect with**. The flowchart is then updated so that it shows the two branches combining at the *Merge* step.

Merge steps have just one field that you can set: **Merge policy**. Select one of the following values:

- *Min value* – Take the lowest of the two input values.
- *Max value* – Take the largest of the two input values.
- *Average* – Calculate the average of the two input values.
- *Sum* – Add the two input values together.

Monetary value steps

Monetary value steps convert a forecast or historical demand time series that's based on sales units into a series of monetary values that are based on a selected price list. The calculation first tries to match the cost or price from a primary price list. It then tries a fallback price list if one is selected. You might use this type of step, for example, to convert a forecast (quantity) into expected revenue (monetary value).

Monetary value steps implement the following calculation:

$f(x) = a(x) \times \text{Price(product-ID)}$

Monetary value steps have the following fields that you can set:

- **Step name** – The specific name of the step. The name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Primary price list** – Select the primary data table that contains your price list. The list includes all the tables that are currently defined for use with Demand planning.

(Learn more in Set up tables.) The price list must include columns for product IDs and prices.
- **Fallback price list** – If the system can't find a matching price in the primary price list, it checks any price list that you select here.

Phase in/out steps

Phase in/out steps modify the values of a data column in a time series to simulate the gradual phasing in of a new element (such as a new product or warehouse) or phasing out of an old element. The phase in/out calculation lasts for a specific period and uses values that are drawn from the same time series (from either the same data column that is being adjusted or another data column that represents a similar element).

Phase in/out steps have the following fields that you can set:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Rule group** – The name of the rule group that defines the calculation that the step does.

For more information about phase in/out functionality, including details about how to set up your phase in/out rule groups, see Use phase in/out functionality to simulate planned changes.

Ratio in percentage steps

Ratio in percentage steps calculate a ratio in percentage, based on two input time series. You might use this type of step, for example, to calculate the error percentage of a previous forecast, based on a historical demand and forecast time series.

Ratio in percentage steps implement the following calculation:

$$f(x) = \Sigma \{[a(x) - b(x)] \div a(x)\} \times 100\ \%$$

You can use this type of step only if your calculation model has at least two input time series in parallel branches. The first series is at the top of the branch where you create the step. The second series must start with an *Input* step. It can have any number of steps in it, but it must still be open. (In other words, it must not end with a *Save* step). To specify the second series, open the step's **Action** menu, and select **Connect with**. The flowchart is then updated so that it shows the two branches combining at the *Ratio in percentage* step.

Ratio in percentage steps have the following fields that you can set:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Value 1** – The first series in the calculation (*a(x)*). Select *Input 1* or *Input 2*.
- **Value 2** – The second series in the calculation (*b(x)*). Select *Input 1* or *Input 2*.

Save steps

Save steps save the result of the calculation model as a new or updated series. All calculation models must end with a single *Save* step.

The calculated series will be saved according to the settings that you configure each time that you run a calculation job as described in Work with calculation profiles.

Feedback

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Provide product feedback

Work with forecast profiles

Demand planning lets you build a collection of *forecast profiles*. Each profile takes an existing time series as input, prepares the data, and then runs forecasting algorithms to create a new time series that predicts demand over a coming period.

Typically, a manager or system administrator creates the initial collection of required profiles. Forecasters and other users can then run the profiles to generate new calculated time series as they require.

View and run existing forecast profiles

To generate a new forecast by running an existing forecast profile, follow these steps:

1. On the navigation pane, select **Operations** > **Forecast profiles**.

2. Find the profile for the type of forecast that you want to run, and select the link for it in the **Name** column.

   The details page for the selected profile appears. It contains the following tabs:

   - **Summary** – This tab provides basic information about the profile. You can edit the name and/or description to make the profile easier to identify and work with.
   - **Horizon and time buckets** – This tab shows the granularity (bucket size) that the forecast will be made at (daily, weekly, or monthly) and the number of buckets that will be included in the forecast. You can modify the selections as you require. For information about how to work with the settings on this tab, see the Create and manage forecast profiles section.
   - **Forecast model** – This tab shows the forecast calculation that the profile does. It uses a flowchart of interconnected steps. Each step does a specific type of operation and has settings that let you define how that operation works. For information about how to work with the settings on this tab, see the Create and manage forecast profiles section.
   - **Run schedule** – This tab lets you set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
   - **Jobs** – This tab shows a list of every run of the profile. It includes date information, the job status, and the time series that was generated. Select a link in the **Time Series** column to open the time series.

3. To run the profile, select **Run** on the Action Pane.

4. In the dialog box that appears, set the **Output option** field to one of the following values to define the output of the job:

    - *Create a new time series* – The job will create a new time series. If you select this value, enter a name for the new series in the **Time series name** field.
    - *Use an existing time series* – The job will overwrite an existing series or create a new version of it. If you select this value, use the fields that are provided to select the target series, specify whether you want to overwrite the existing version or create a new version, and, if you want to create a new version, specify the name of the new version.

5. Select **Save and close**.

6. The new job is added to the grid on the **Jobs** tab. There, you can follow the status of the new calculation. To update the status information, select **Refresh** on the grid toolbar.

Review forecast job run history

Each time you run a forecast profile, the system creates a record of the job run. To review the details of a job run, follow these steps:

1. On the navigation pane, select **Operations** > **Forecast profiles**.

2. Select the forecast profile that you want to inspect.

3. Open the **Jobs** tab, which shows a grid with a row for each time the profile ran.

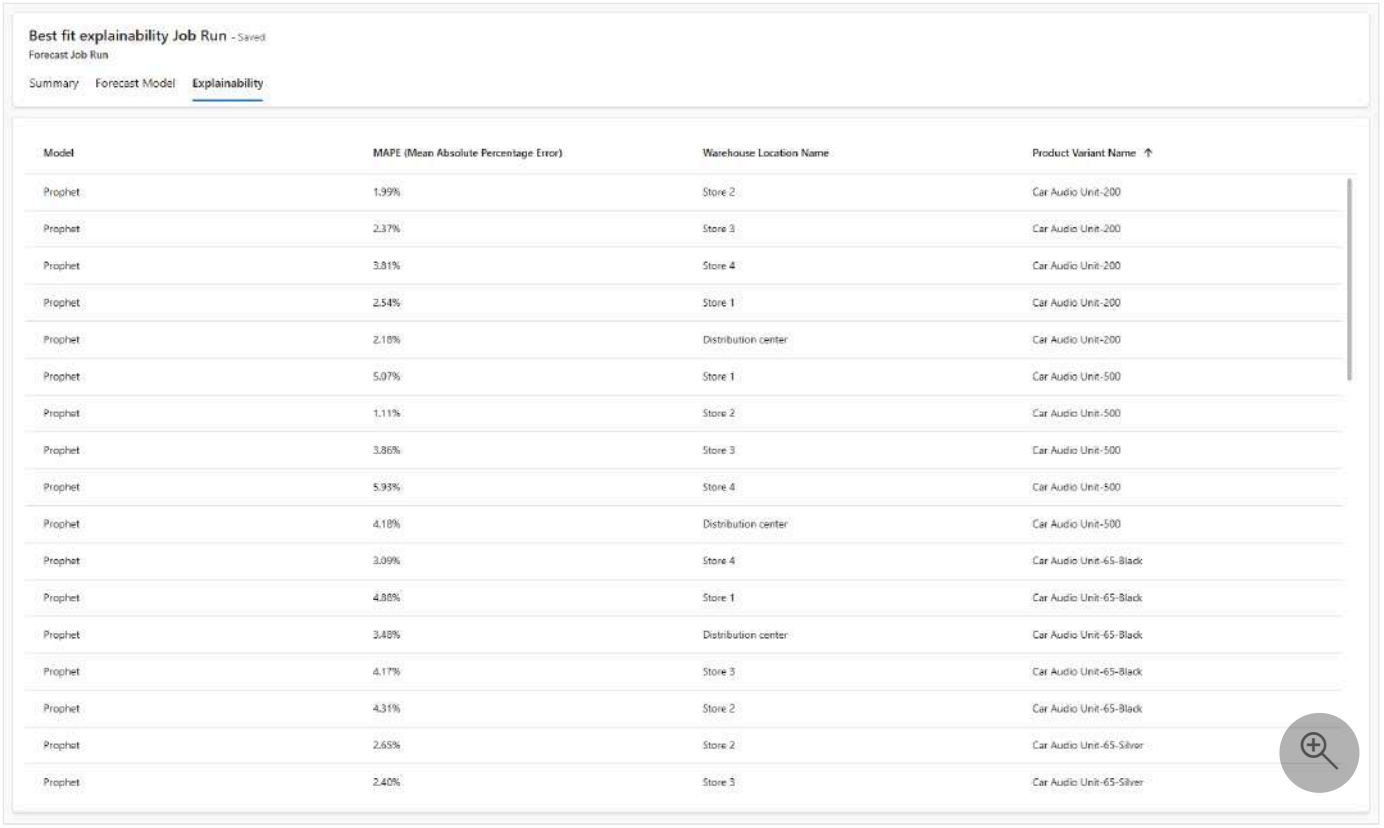
4. Find the job run that you want to inspect and select the link in the **Job run name** column for that run.

5. On the **Forecast job run** page, browse through the following tabs to review the details of the job run:

    - **Summary** – This tab provides basic information about the job run, including who ran it, when it was run, and a link to the time series that was generated or updated. You can also see the job status and any error messages that occurred.
    - **Forecast model** – This tab shows the forecast model run by the job.
    - **Explainability** – This tab lists each combination of dimensions (such as item variant, warehouse location, and so on) that the forecast was calculated for. For each combination, it shows the demand forecasting algorithm that was used and the mean average percentage error (MAPE) for the calculation. A lower MAPE value indicates

greater accuracy. You can sort the grid by selecting the column heading you want to sort by. To resize, filter, or search in a column, select the menu button on the appropriate column heading.

The following image shows an example of the **Explainability** tab of the **Forecast job run** page. This job ran using the *best fit* forecast algorithm, which calculates results for each of the standard demand forecasting algorithms and then picks the one with the lowest MAPE for each row. This is why forecasts for the same product variant (such as *Car Audio Unit-200*) used varying algorithms depending on which warehouse was analyzed.



Create and manage forecast profiles

Each time that your organization requires a new type of forecast, a manager or admin must create a new forecast profile. After the profile is created, it becomes available to users, who can run it as often as they require.

To create or edit a forecast profile, follow these steps:

1. On the navigation pane, select **Operations** > **Forecast profiles**.

2. On the Action Pane, select **New**.

3. A setup wizard is opened. On the **Get started** page, set the following fields:

- **Name** – Enter a name for the new profile.
- **Description** – Enter a short description of the profile.
- **Owner** – Select the user account that owns the profile.
- **Precision** – Specify number of decimal places that the profile should show for calculation results and output.
- **Category** – Select the category of the output time series (*Forecast*, *Demand*, *Financials*, or *Miscellaneous*). The category affects the location of the output time series under the **Planning data** heading on the navigation pane.

4. Select **Next**.

5. On the **Set horizon and time buckets** page, set the following fields:

   - **Time bucket** – Select the granularity (bucket size) that the forecast should be made at (daily, weekly, or monthly).
   - **Horizon in time buckets** – Enter the number of buckets to include in the forecast.

6. Select **Next**.

7. On the **Select a forecasting model preset** page, select a forecast model preset to use with your current profile. Browse the presets that are listed under **Available model presets** to preview the forecast that each preset makes. You can configure settings and customize the forecast model as you require after you save the profile. Therefore, select the preset that's closest to what you're looking for, and then select **Next**. For information about how to configure settings and customize the forecast model that a profile uses, and how to create new presets, see Design forecast models.

8. On the **Set run schedule** page, you can choose to set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.

9. Select **Next**.

10. On the **Review and finish** page, review the summary of settings that you've configured, and then select **Review and finish** to create the new profile.

11. The profile is now saved, but it hasn't yet run. If you're ready to run the forecast now, select **Run** on the Action Pane.

Last updated on 12/30/2025

Design forecast models

Forecast models let you arrange and configure steps to define the forecast that a forecast profile makes. Each model presents a flowchart that graphically represents the calculation that the model does.

For a presentation about how forecast models and outlier removal methods work in Demand planning, watch the following video on YouTube (starting at 16:53): Deep dive into Demand Planning for Supply Chain Management | Dynamics 365 TechTalk.

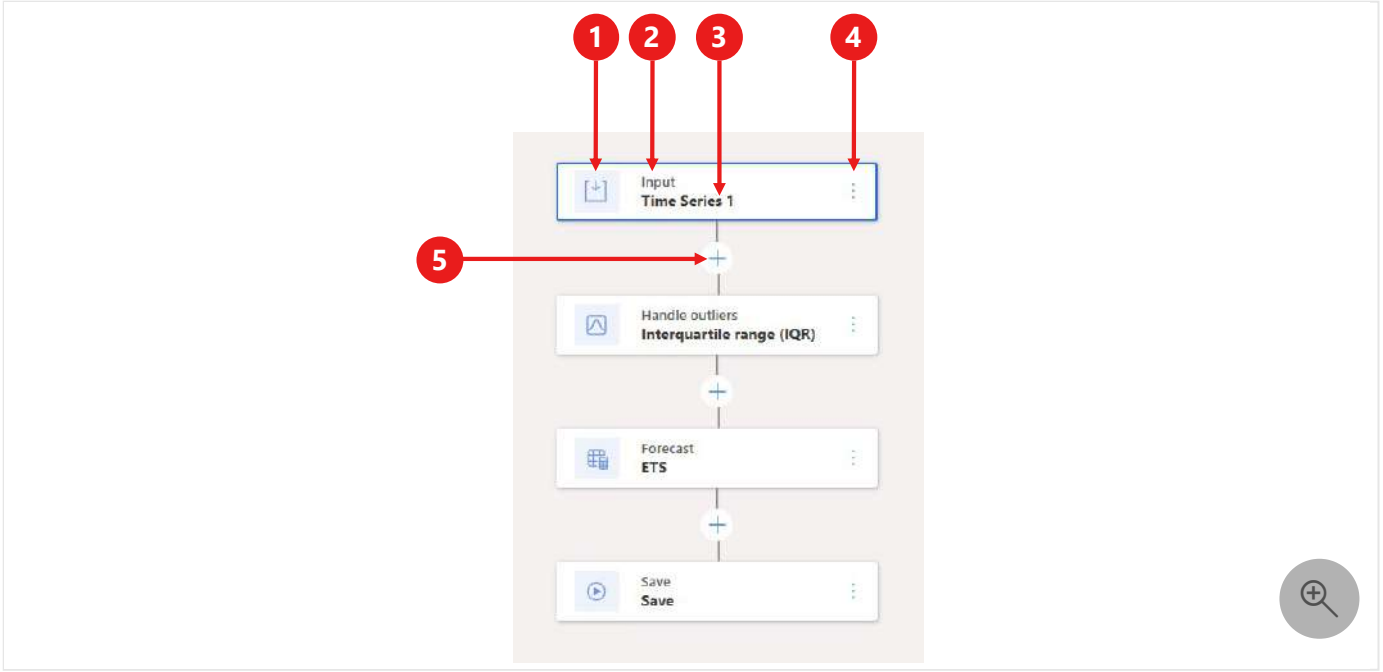
Create and customize a forecast model

To create and customize a forecast model, you must first open an existing forecast profile. (Learn more in Work with forecast profiles.) You can then fully customize the model that the selected profile uses by adding, removing, and arranging steps, and configuring settings for each of them.

Follow these steps to create and customize a forecast model.

1. On the navigation pane, select **Operations** > **Forecast profiles**.
2. Select the forecast profile that you want to create or customize a forecast model for.
3. On the **Forecast model** tab, there's always at least one step of the *Input* type at the top of the flow chart. The model is processed from top to bottom, and the last step must be a step of the *Save* type. Add, remove, and arrange steps as you require, and configure settings for each of them. The illustration after this procedure provides guidelines.
4. After you finish designing your forecast model, select the **Validate** button in the upper-right corner. The system runs a few tests to validate that your model will work, and then provides feedback. Fix any issues that the validation test reports.
5. Continue to work until your model is ready. Then, on the Action Pane, select **Save**.
6. If you want to save your forecast model as a preset, so that it's available when you and other users create a new forecast profile, select the **Save as model template** button in the upper-right corner.

The following illustration shows the information and controls that are available for steps in a forecast model.



Legend:

1. **Step symbol** – A representation of the purpose of the step.
2. **Step type** – The type of step. This text usually describes the type of roles, calculations, or other actions that the step represents.
3. **Step name** – The name that is applied to the step. Sometimes, you can manually enter this text in the step's settings. However, it usually indicates the value of one of the settings that were configured for the step.
4. **Step actions** – Open a menu of actions that you can take on the step. Although some of these actions are specific to the step type, most are common to all steps. If any actions appear dimmed, they can't be used because of the step's current position or for some other contextual reason. Here are some common actions that are available:
    - **Settings** – Open a dialog box where you can configure settings for the step.
    - **Remove** – Remove the step.
    - **Move Up** and **Move Down** – Reposition the step in the flowchart.
    - **Set to 'Pass Through'** – Temporarily disable a currently enabled step without deleting it or its settings.
    - **Unset 'Pass Through'** – Reenable a currently disabled step.
    - **Connect with** – For steps that support multiple inputs, specify the second input. The first input is automatically set based on where you put the step.
5. **Add a step** – Add a new step at the selected location.

Forecast step types

This section describes the purpose of each type of forecast step. It also explains how to use and configure each type.

Input steps

Input steps represent a time series that provides input to the forecast model. All forecast models must start with an *Input* step.

Input steps have the following settings:

- **Step name** – The specific name of the step. This name automatically matches the name of the time series.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Time series** – The time series that is added by the step. When you first set up an *Input* step, the **Configure step** dialog box provides a button that lets you select the time series to add. After you add a time series, the dialog box shows the name of the time series and provides an **Edit** button that lets you change the time series. It also provides a **Filter** button that lets you set up filtering criteria based on fixed dates, relative dates, and/or field values.
- **Time series version** – If the selected time series includes more than one version, you can select a specific version. By default, the latest (current) version is used.
- **Fill missing values** – Specify whether the system should fill in placeholder values for time buckets if the input doesn't provide values. In most cases, you should enable this feature, because it helps produce better forecasts.
- **Fill missing values strategy** – Select the strategy that the system should use to enter values in empty time buckets. In the current version, the only available strategy is *Zeros*. In this strategy, the system sets all missing values to *0* (zero).

Signal steps (preview)

[This section is prerelease documentation and is subject to change.]

Signal steps represent additional time series that can be combined with the primary time series that the *Input* step provides. For example, the primary input that the *Input* step provides might represent historical sales. A *Signal* step might then provide input that represents inflation or weather data.

When you add a *Signal* step, the system automatically creates a parallel branch in your model and puts the step at the top of that branch. The initial branch always has an *Input* step at the top of it. Each branch can include any number of steps, but you must eventually combine the branches by using a *Forecast with signals* step.

Learn more in Forecast with signals (preview).

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Signal step settings

Signal steps have the following settings:

- **Step name** – The specific name of the step. This name automatically matches the name of the time series.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Time series** – The time series that is added by the step. When you first set up a *Signal* step, the **Configure step** dialog box provides a button that lets you select the time series to add. After you add a time series, the dialog box shows the name of the time series and provides an **Edit** button that lets you change the time series. It also provides a **Filter** button that lets you set up filtering criteria based on fixed dates, relative dates, and/or field values.
- **Time series version** – If the selected time series includes more than one version, you can select a specific version. By default, the latest (current) version is used.
- **Fill missing values** – Specify whether the system should fill in placeholder values for time buckets if the input doesn't provide values. In most cases, you should enable this feature, because it helps produce better forecasts. In the current version, all missing values are set to *0* (zero).
- **Fill missing values strategy** – Select the strategy that the system should use to enter values in empty time buckets. The next section describes the options available here and how they work.

Signal fill strategies

When the forecast is generated, Demand planning prepares each signal so that it has one value for each time bucket through the end of the forecast period. If a signal has no row for a time bucket, the system adds a temporary value according to the **Fill missing values strategy** setting on the *Signal* step.

A missing value in this context means that a row is absent for a specific time bucket and combination of dimensions. For example, if a monthly signal has rows for January and March for the same product and warehouse, but no row for February, then February is a missing time bucket. This feature doesn't replace a blank value in a row that already exists.

The system fills missing values separately for each combination of dimensions. The filled values are used only for the current forecast run. The source signal time series isn't changed.

The following table describes the available strategies for filling missing values, how they work, and when to use them:

Expand table

Strategy	How missing time buckets are filled	Example use
Zeros	Enter 0 (zero). This strategy is the default.	Use when a missing bucket means that no signal activity occurred, such as no promotion.
Forward fill	Repeat the previous known value until a new value is available.	Use when the latest known value remains in effect, such as a price or rate.
Backward fill	Use the next known value for earlier missing buckets. If no later value is available, repeat the last known value.	Use when the next observation best represents the preceding gap.
Linear interpolation	Calculate evenly spaced values between the previous and next known values. If no later value is available, repeat the last known value. Linear interpolation treats each time bucket as one equal step; for example, each month is one step even though calendar months have different numbers of days.	Use for a continuous signal that normally changes gradually, such as temperature or an economic index.

> **Note**

The available strategies depend on the version of Demand planning you're using. Earlier versions provide only the *Zeros* strategy.

For example, suppose that a monthly signal has a value of 10 in January, no row in February, a value of 30 in March, and no row in April. The forecast period ends in April, and no later signal value is available. Each strategy prepares the signal as follows:

Expand table

Strategy	January	February	March	April
Zeros	10	0	30	0
Forward fill	10	10	30	30
Backward fill	10	30	30	30
Linear interpolation	10	20	30	30

The system applies the following rules when it fills a signal:

- Filling starts with the first available value for each combination of dimensions. The system doesn't add values before that point.
- Filling continues through the end of the forecast period. Therefore, the stored signal doesn't need to contain a row for every time bucket or an explicit row at the end of the forecast horizon.
- Every combination of shared dimension values in the primary input must appear at least once in each signal. The system can't create an entirely missing combination of signal dimensions, and the forecast fails if a required combination has no signal data.
- Each signal must contain at least one value on or before the end of the forecast period. If the entire signal starts after the forecast period ends, the forecast fails.
- *Backward fill* and *Linear interpolation* can use the next known signal value even if that value occurs after the end of the forecast period. The later value helps calculate missing values through the forecast end date, but isn't included beyond the forecast horizon.

Handle outliers steps

Handle outliers steps identify and compensate for outlier data points in the input. These data points are considered anomalies that should be ignored or smoothed out to prevent them from throwing off the forecast calculation.

Settings for Handle outlier steps

Handle outliers steps have the following settings:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Handle outliers** – Select one of the following options. Learn more about these options in the How the handle outliers options work section.
  - *Interquartile range (IQR)*
  - *Seasonal and trend decomposition using Loess (STL)*
- **Interquartile range multiplier** – This field is available only when the **Handle outliers** field is set to *IQR*. It affects the number of data points that are removed as outliers. Learn more about this setting in the How the handle outliers options work section.
- **Correction methods** – This field is available only when the **Handle outliers** field is set to *IQR*. It affects how outliers are replaced in the data after they're removed. The available options are *Median smoothing* and *Mean smoothing*.
- **Select seasonality detection setting** – This field is available only when the **Handle outliers** field is set to *STL*. It influences the size of the window that is used for locally estimated scatterplot smoothing (LOESS) when the seasonal component is estimated. Learn more about this setting in the Seasonality in forecasts section.

How to choose the method for handling outliers

The following table provides examples that will help you choose which method to use to handle outliers in your time series data, based on your scenario.

Expand table

Scenario	IQR	STL
Data with seasonal or trend patterns	Not recommended	Recommended
Robust handling of outliers	Recommended	Not recommended
Sensitivity to extreme values	Recommended	Recommended

How the handle outliers options work

Interquartile range

The *interquartile range* (IQR) is the range between the first quartile (Q1) and the third quartile (Q3). It measures the spread of the middle 50% of the data. Any data outside the IQR is identified as an outlier that should be smoothed out later. The IQR method applies the following formula to identify outliers:

IQR = Q3 − Q1

Key:

- *Q1* – A representation of the twenty-fifth percentile (that is, the value that 25% of the data lies below).
- *Q3* – A representation of the seventy-fifth percentile (that is, the value that 75% of the data lies below).

When you use the IQR method to handle outliers, you must set an **Interquartile range multiplier** value. This value affects the number of data points that are removed as outliers. It's calculated in the following way:

- Lower bound = Q1 − (Multiplier × IQR)
- Upper bound = Q3 + (Multiplier × IQR)

A *high* multiplier increases the range within which data points are considered non-outliers. High multipliers lead to softer boundaries and include more points. Therefore, the result is less sensitive to variations in the time series.

A *low* multiplier decreases the range within which data points are considered non-outliers. Low multipliers lead to stricter boundaries and remove more data points. Therefore, the result is more sensitive to variations in the time series.

When you use the IQR method to handle outliers, you must also set a **Correction method** value. This value affects how outliers are replaced in the data after they're removed. The available options are *Median smoothing* and *Mean smoothing*.

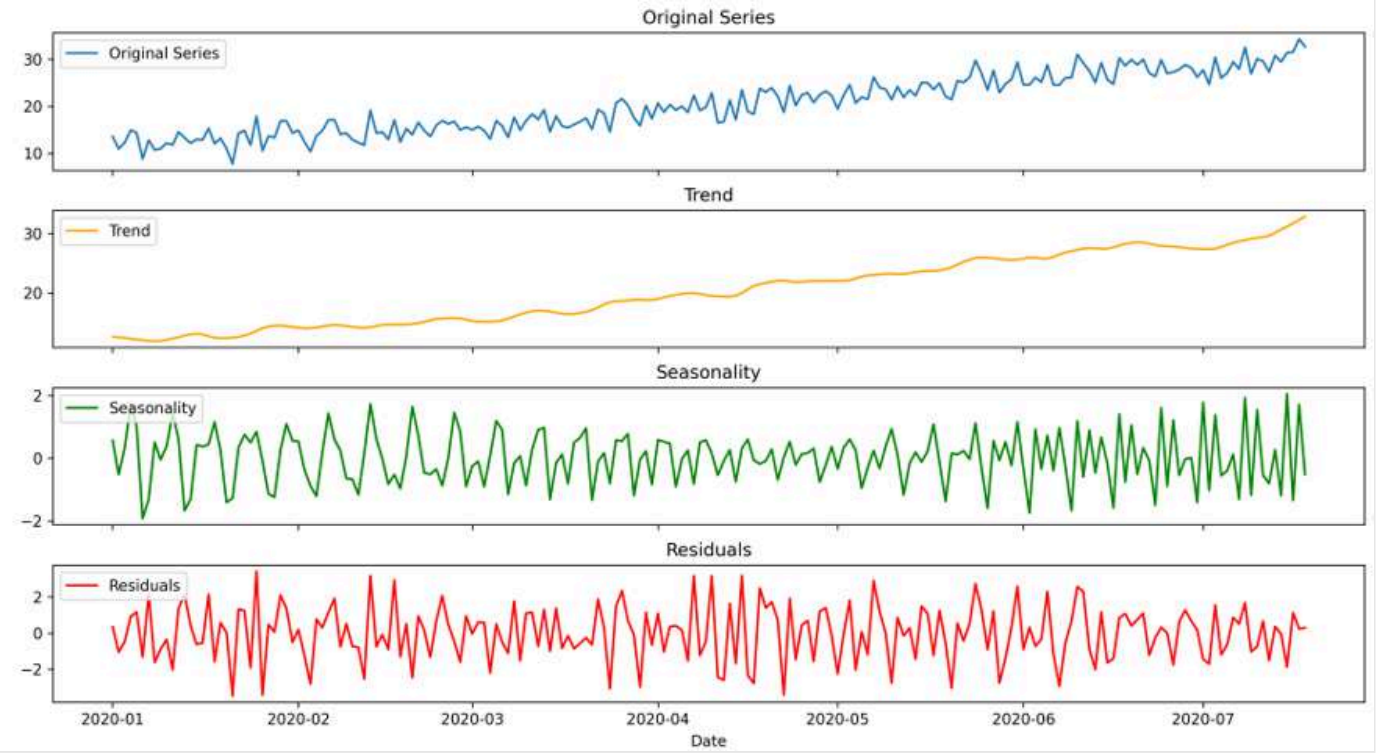
Seasonal and trend decomposition using LOESS

Seasonal and trend decomposition using LOESS (STL) starts by decomposing the time series into three components:

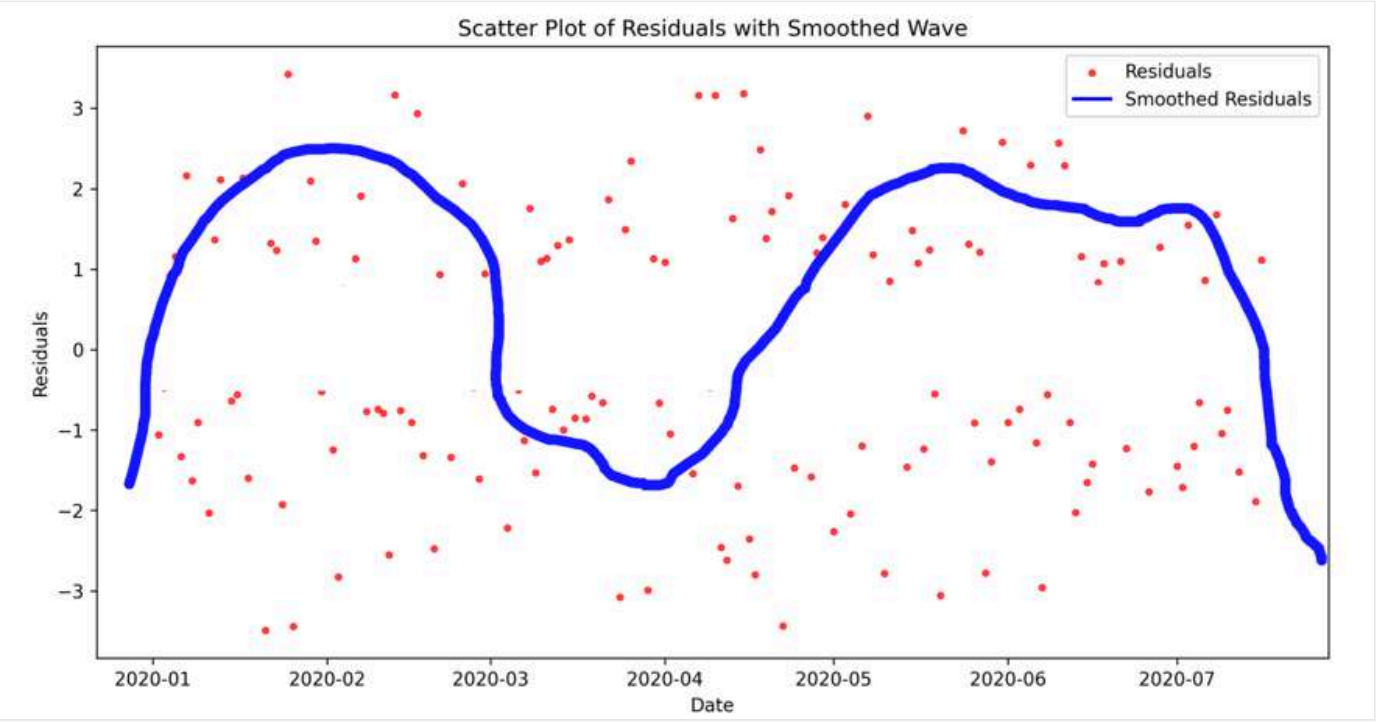
- *Trend* – The overall trend in the data.

- *Seasonality* – Regular and predictable patterns that are driven by factors such as weather, holidays, or economic cycles.
- *Residuals* – The time series after trend and seasonality are removed.

The following illustration shows examples of an original time series, the trend that is decomposed from the time series, the seasonality spikes that are found in the time series, and the residuals after the trend and seasonality are removed.



After STL decomposes the time series, it tries to smooth out the outliers by using LOESS. LOESS is a generalization of the moving average and polynomial regression. It tries to fit a chart through scattered data points and therefore smooth out the residuals, as shown in the following illustration.



After STL calculates the mean and standard deviation, it uses them to eliminate the outliers and mitigate anomalies.

Finally, STL reintroduces the seasonality and trend to the newly smoothed residual.

When you use the STL method to handle outliers, you must also set the **Select seasonality detection setting** option. This setting influences the size of the period (in time buckets) that is used when the seasonality component is estimated. It helps divide the data into segments to ensure that the algorithm correctly extracts repeating patterns. Learn more in Seasonality in forecasts.

Forecast steps

Forecast steps apply a selected forecast algorithm to the input time series to create a forecast time series.

Forecast steps have the following settings:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Model type** – Select the forecast algorithm to use. Learn about each of the available algorithms in Demand forecasting algorithms. The following algorithms are available:
  - *Best fit model - version 1*

- *Best fit model - version 2 (preview)*
- *Best fit model - version 3 (preview)*
- *ARIMA* (auto regressive integrated moving average)
- *ETS* (error, trend, seasonality)
- *Prophet*

- **Select seasonality detection setting** – Choose how the forecast should identify seasonality patterns. Seasonality refers to a pattern of demand that fluctuates according to a regular, recurring schedule. Learn more about seasonality, seasonality auto detection, and seasonality hints in the Seasonality in forecasts section.

- **Enable time freeze** – Select this check box if you want to enable time freeze functionality for the forecasts created by this step. Then select one of the following options:
  - **Freeze manual adjustments** – Select this option to prevent the system from automatically updating any time series cells that have been manually adjusted by a user.
  - **Use time freeze rules** – Select this option to apply one or more time freeze rules to the forecasts created by this step. Then use the **Time freeze rules** field to search for and select the relevant time freeze rules. Time freeze rules are defined as logical expressions and prevent the system from automatically updating time series cells that match that expression. For example, you could create a time freeze rule that lasts two months and applies to product ID *K0001* in color *Red*. Learn more about how to work with time freeze rules at Limit automatic time series updates with time freezes.

Forecast with signals steps (preview)

[This section is prerelease documentation and is subject to change.]

Forecast with signals steps generate a forecast based on multiple input time series. Learn more about forecasting with signals in Forecast with signals (preview).

Forecast with signals steps always use the XGBoost demand forecasting algorithm. Learn about this algorithm in Demand forecasting algorithms.

> **Important**
>
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> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to **supplemental terms of use**, and are

available before an official release so that customers can get early access and provide feedback.

You can use this type of step only if your forecast model has at least two parallel branches: one that starts with an *Input* step and up to five others that start with a *Signal* step. The first branch is the one where you create the step. To specify a second branch, open the **Action** menu for the *Forecast with signals* step, select **Connect with**, and then select the step to connect to. The flowchart is then updated so that it shows the branches combining at the *Forecast with signals* step. You can add up to five signal branches.

Forecast with signals steps have the following settings:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Model type** – Select the forecast algorithm to use. In the current version, only the *XGBoost* algorithm is available.
- **Enable time freeze** – Select this check box if you want to enable time freeze functionality for the forecasts created by this step. Then select one of the following options:
  - **Freeze manual adjustments** – Select this option to prevent the system from automatically updating any time series cells that have been manually adjusted by a user.
  - **Use time freeze rules** – Select this option to apply one or more time freeze rules to the forecasts created by this step. Then use the **Time freeze rules** field to search for and select the relevant time freeze rules. Time freeze rules are defined as logical expressions and prevent the system from automatically updating time series cells that match that expression. For example, you could create a time freeze rule that lasts two months and applies to product ID *K0001* in color *Red*. Learn more about how to work with time freeze rules at Limit automatic time series updates with time freezes.

Finance and operations steps

You might already be using custom Azure Machine Learning algorithms that you created for use with the legacy Supply Chain Management forecasting functionality, as described in Demand forecasting overview. To use them with Demand planning, you must set up the connection to Azure and then add the forecast calculation to your forecast model using a *Finance and operations* step instead of a *Forecast* step.

Learn how to set up Demand planning to connect to and use your Azure Machine Learning forecast algorithms in Use your own custom Azure Machine Learning algorithms in Demand

planning.

Custom steps

It's possible for you to design your own custom Microsoft Azure Machine Learning algorithms for demand forecasting, even if you've never used them with Dynamics 365 Supply Chain Management before. To use them with Demand planning, you must set up the connection to Azure and then add the forecast calculation to your forecast model using a *Custom* step instead of a *Forecast* step.

Learn how to set up Demand planning to connect to and use your Azure Machine Learning forecast algorithms in Use your own custom Azure Machine Learning algorithms in Demand planning.

Phase in/out steps

Phase in/out steps modify the values of a data column in a time series to simulate the gradual process of phasing in a new element or phasing out an old element. For example, a *Phase in/out* step might simulate the process of phasing in a new product or warehouse. The phase in/out calculation lasts for a specific period and uses values that are drawn from the same time series. (The values come from either the same data column that is being adjusted or another data column that represents a similar element.)

Phase in/out steps have the following settings:

- **Step name** – The specific name of the step. This name is also shown in the flowchart.
- **Description** – A short description of the step.
- **Created by** – The user who created the step.
- **Rule group** – The name of the rule group that defines the calculation that the step does.

When you set up your forecast model, the position of the *Phase in/out* step affects the calculation result.

- To apply the phase in/out calculation to the historical sales numbers, put the *Phase in/out* step before the *Forecast* step, as shown on the left side of the following illustration.
- To apply the phase in/out calculation to the forecasted result, put the *Phase in/out* step after the *Forecast* step, as shown on the right side of the illustration.

Learn more about phase in/out functionality, including details about how to set up your phase in/out rule groups, in Use phase in/out functionality to simulate planned changes.

Save steps

Save steps save the result of the forecast model as a new or updated series. All forecast models must end with a single *Save* step.

The forecast time series is saved according to the settings that you configure each time that you run a forecast job. Learn more in Work with forecast profiles.

Seasonality in forecasts

Seasonality refers to regular and predictable patterns in a time series that occur at fixed intervals because of recurring events or influences. These patterns are often driven by factors such as weather, holidays, or economic cycles. Examples include increased retail sales every December because of holiday shopping or higher electricity demand during summer months because of air conditioning.

Demand planning identifies and compensates for seasonality patterns both when it handles outliers (in a *Handle outliers* step that is set to use STL) and when it generates a forecast (in any *Forecast* step).

Forecast with signals steps (which use the XGBoost algorithm) detect seasonality patterns by analyzing the time series, but they use a different approach than is used by *Forecast* and *Handle outliers* steps and don't provide any related settings.

How seasonality differs from cycles

Although both seasonality and cycles involve repeating patterns, there are key differences, as summarized in the following table.

Expand table

Aspect	Seasonality	Cycles
Frequency	Fixed, regular intervals (such as daily, weekly, or yearly)	Variable, often irregular, and influenced by external factors
Duration	Short-term and predictable (such as seasonal sales)	Long-term and unpredictable (such as business cycles)
Cause	Calendar-based events or natural patterns	Economic, social, or structural forces
Examples	Summer travel peaks or holiday shopping trends	Economic recession cycles or stock market booms and busts

Autodetect seasonality patterns (preview)

[This section is prerelease documentation and is subject to change.]

Automatic seasonality detection uses an algorithm that finds seasonality patterns for each combination of a location and a product, and applies the result to forecast calculations. Seasonality patterns typically vary for different products and different locations. Therefore, auto detection often works better than applying the same pattern everywhere.

To set a *Forecast* or *Handle outliers* step to use seasonality auto detection, edit the step's settings, and then set **Select seasonality detection setting** to *Auto detection*.

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available before an official release so that customers can get early access and provide feedback.

Detect seasonality using a hint

The system can detect seasonality by using a *seasonality hint* that you choose. This option works best when you know the seasonality pattern in advance. The **Seasonality hint** setting influences the size of the period (in time buckets) that is used in the identification of seasonality patterns. It helps divide the data into segments to ensure that the algorithm correctly extracts repeating patterns. For example, if you notice a weekly seasonality pattern where most customers shop on Saturdays, and you're using buckets in days, enter *7* as the **Seasonality hint** value.

A well-defined seasonal period captures the true periodicity of the data. It enables the algorithm to successfully isolate the seasonality. As a result, the trend and residuals are more interpretable. By contrast, an incorrect seasonal period leads to poor seasonal extraction. As a result, trend and residual components are inaccurate.

If the seasonal period that you use is too short (that is, shorter than the actual cycle), you risk encountering the following issues:

- **Missed patterns** – Important parts of the seasonal pattern might be excluded, so that seasonality extraction is incomplete.
- **Trend contamination** – Some seasonal variations might be mistaken for the trend component, so that the trend estimation is distorted.
- **Residual issues** – Residuals might show unexplained periodicity. This situation indicates that the algorithm failed to capture seasonality correctly.

For example, you use a seasonal period of *6 months* for yearly retail sales data. In this case, only part of the yearly cycle is captured. As a result, critical holiday season peaks might be missing.

If the seasonal period that you use is too long (that is, longer than the actual cycle), you risk encountering the following issues:

- **Overfitting** – Noise or random fluctuations might be falsely interpreted as seasonality.
- **Distorted decomposition** – Artificially long seasonal cycles might obscure the true trend or residual components.
- **Increased complexity** – The model might become unnecessarily complex and therefore harder to interpret and validate.

For example, you use a seasonal period of *24 months* for data that has an annual cycle. The result is overfitting, where small irregularities in the data are treated as seasonal patterns.

To set a *Forecast* or *Handle outliers* step to use a seasonality hint, edit the step's settings, and set **Select seasonality detection setting** to *Detection using hint*. Then, enter the number of time buckets in the **Seasonality hint (in periods of time buckets)** field.

Last updated on 08/10/2026

Forecast with signals (preview)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

One way to improve the accuracy of a forecast is to include input signal data beyond just historical sales. You can add signals in your forecast models by including a *Signal* step. This type of step lets you include any signal as input data. You can then use the new *Forecast with signals* step to combine the signal input and main input to create the forecast.

This article explains how to set up a forecast model that includes both a primary input (such as historical sales) and up to five signal inputs (such as inflation or weather data).

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>
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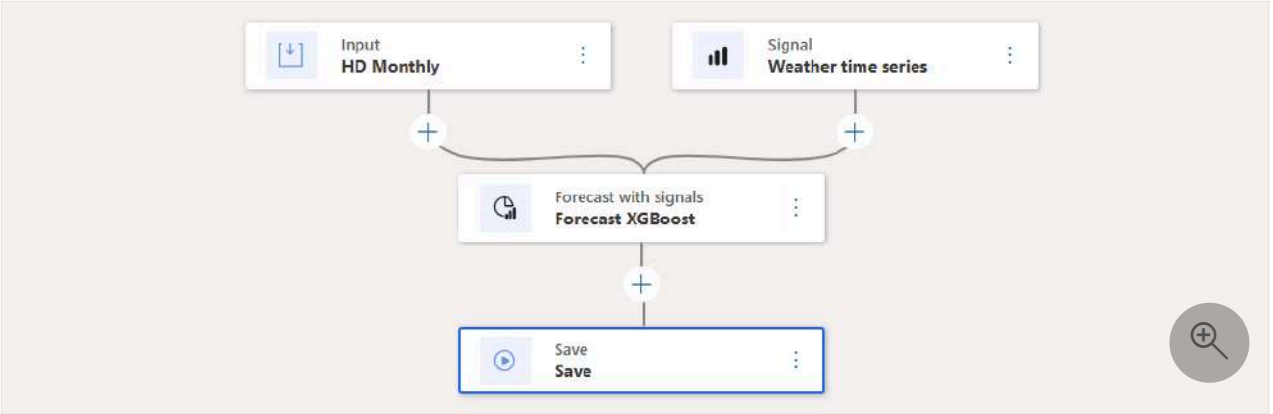
Set up a forecast model that includes both input and signals

Follow these steps to set up a forecast that includes both input and up to five signal inputs.

1. Create a new forecast profile as described in Create and manage forecast profiles.
2. On the **Select a forecasting model preset** page, select *None*.
3. After you create and save the profile, on the **Forecast model** tab, set up your model in the following way. (Learn more in Design forecast models.)
   a. Use an *Input* step to set up your primary time series.
   b. Add a *Signal* step to set up a signal time series in a parallel branch.

c. Add other steps as required to condition the data in each branch.
d. Add a *Forecast with signals* step to combine the branches.
e. Complete the model by adding a *Save* step.



4. On the Action Pane, select **Save**.

Align input and signal time series

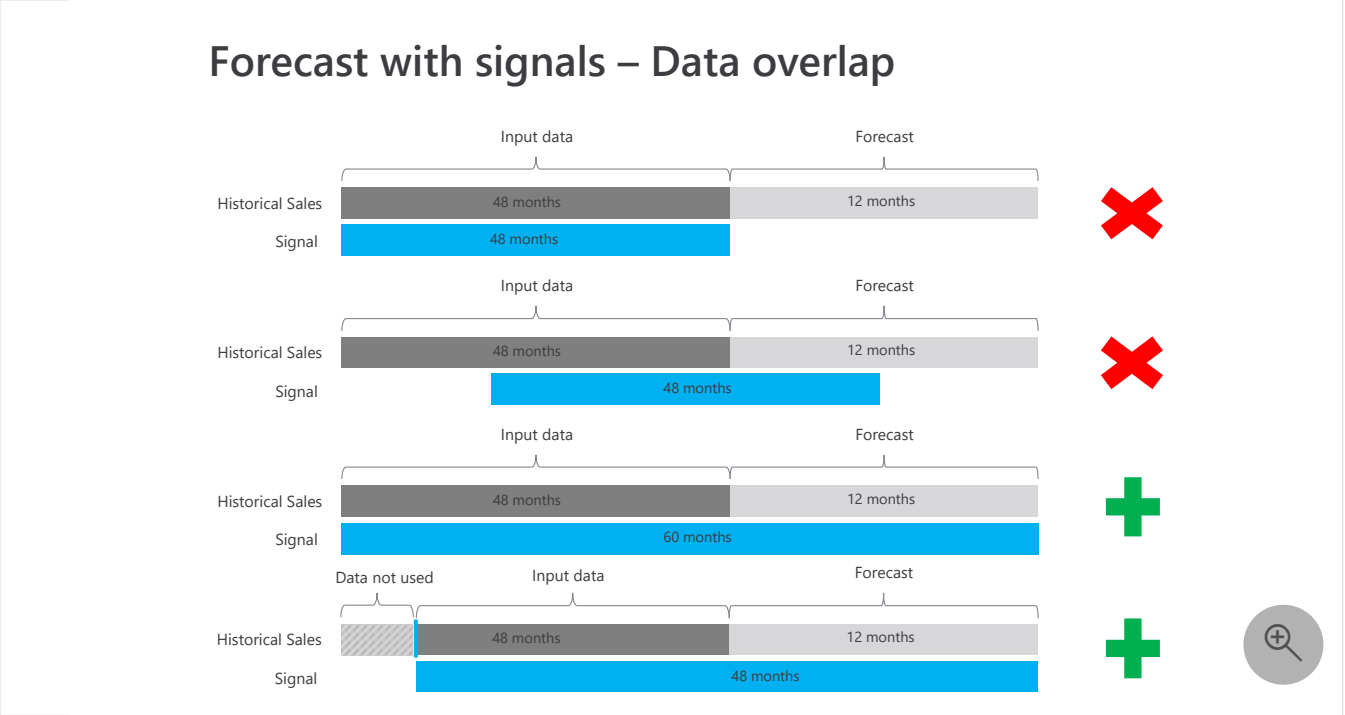
To calculate a forecast with signals, the input and signal time series must be aligned. This means that they must have compatible time bucket sizes, dimensions, and time spans.

Time span requirements

The time span of the signal time series must cover all of the historical input data that you want to consider and extend to the end of the forecast time horizon. If the input time series extends further into the past than the signal time series, then the forecast calculation will ignore all input data that is older than the oldest signal data. If the forecast time horizon extends further into the future than the signal time series, then the forecast will fail and you'll see an error message.

The following illustration provides examples of successful and unsuccessful time span alignments.

- The first example won't work because the signal time series doesn't include any data for the forecast horizon.
- The second example won't work because the signal time series doesn't include enough data to cover the full forecast horizon.
- The third example is perfectly aligned.
- The fourth example works, but the forecast will ignore the input data that is older than the oldest signal data.



Dimension requirements

The input and signal time series must have compatible dimensions. For example, if the input time series has dimensions for *Country* and *Month*, then the signal time series must also have *Country* and *Month* dimensions. The system uses these dimensions to align the two time series. If the dimensions don't match, the forecast calculation will fail and you'll see an error message.

The following illustration provides examples of successful and unsuccessful dimension alignments.

- The first example won't work because the signal time series has a *Country* dimension, but the input time series doesn't.
- The second example works because both time series include dimensions for both the *Country* and the *Month*.

Forecast with signals – Dimension matching

Historical Sales:

Product	Month	Quantity
Product A	Jan	10
Product A	Feb	12

Signal:

Country	Month	Inflation%
USA	Jan	5.00
USA	Feb	5.50

Dimension match

Product	Country	Month	Quantity
Product A	USA	Jan	10
Product A	USA	Feb	12

Country	Month	Inflation%
USA	Jan	5.00
USA	Feb	5.50

Last updated on 11/07/2025

Demand forecasting algorithms

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Demand planning in Microsoft Dynamics 365 Supply Chain Management includes four popular demand forecasting algorithms: *auto-ARIMA*, *ETS*, *Prophet*, and *XGBoost*.

- Auto-ARIMA works well with *stationary data*. Stationary data is data that has a constant mean, constant standard deviation, and no seasonality.
- Error, trend, and seasonality (ETS) works well if your business case is simple and the data has various patterns, such as linear or exponential trends, *or* if you want the forecast to give more weight to the most recent data.
- Prophet works best with complex, real-world data.
- eXtreme Gradient Boosting (XGBoost) can generate a forecast based on multiple inputs.

In addition, Demand planning provides a *best fit model* algorithm, which automatically selects the best of the available algorithms for each product and dimension combination. Demand planning also lets you develop and use your own custom algorithms.

In Demand planning, you choose a forecast algorithm when you place and configure a *Forecast* or *Forecast with signals* step in a forecast model. You then use that forecast model in a forecast profile to generate a forecast.

This article describes how each algorithm works and its suitability for different types of historical demand data.

When to use each forecasting algorithm

The demand forecasting algorithm that you use depends on the specific characteristics of your historical data. The following table shows which single-input forecasting algorithms are best suited to each of several different business scenarios. XGBoost is excluded from this table, because it's always used for multi-input forecasting. For most other scenarios, use the best fit

model algorithm, because it automatically selects the correct forecasting algorithm for each product and dimension combination.

Expand table

Scenario	Auto-ARIMA	ETS	Prophet
Simple business case	Acceptable	Recommended	Acceptable
The time series has a different (linear/exponential) trend and several seasonality types.	Not recommended	Recommended	Recommended
The time series shows a clear linear trend.	Recommended	Recommended	Acceptable
Data is stationary.	Recommended	Not recommended	Acceptable
Data is nonstationary.	Not recommended	Acceptable	Recommended
Quick forecasting is required.	Not recommended	Recommended	Acceptable
Forecasting is focused on a recent period.	Acceptable	Recommended	Acceptable

Best fit model algorithm

The best fit model algorithm automatically determines which of the other available single-input algorithms (auto-ARIMA, ETS, or Prophet) best fits your data for each product and dimension combination. In this way, you can use different models for different products. In most cases, use the best fit model, because it combines the strengths of all the other standard models. The following example shows how.

Example of how the best fit model algorithm works

For this example, you have historical demand time series data that includes the following dimension combinations.

Expand table

Product	Store
A	1
A	2

Product	Store
B	1
B	2

When you run a forecast calculation by using the Prophet model, you get the following results. In this example, the system always uses the Prophet model, regardless of the calculated mean absolute percentage error (MAPE) for each product and dimension combination.

Expand table

Product	Store	Forecast model	MAPE
A	1	Prophet	0.12
A	2	Prophet	0.56
B	1	Prophet	0.65
B	2	Prophet	0.09

When you run a forecast calculation by using the ETS model, you get the following results. In this example, the system always uses the ETS model, regardless of the calculated MAPE for each product and dimension combination.

Expand table

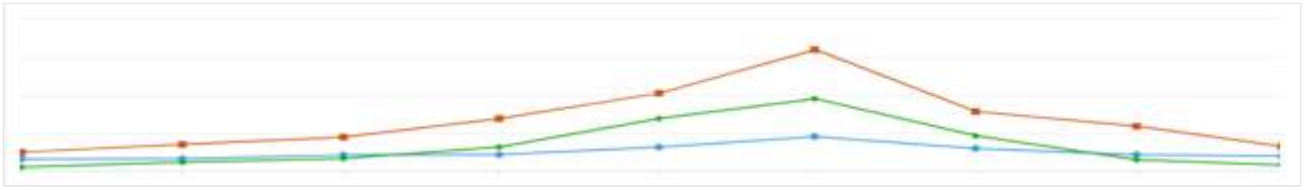
Product	Store	Forecast model	MAPE
A	1	ETS	0.18
A	2	ETS	0.15
B	1	ETS	0.21
B	2	ETS	0.31

When you run a forecast calculation by using the best fit model, the system optimizes the model selection for each product and dimension combination. The selection changes based on patterns that are found in the historical sales data.

Expand table

Product	Store	Prophet MAPE	Auto-ARIMA MAPE	ETS MAPE	Best fit forecast model	Best fit MAPE
A	1	0.12	0.34	0.18	Prophet	0.12
A	2	0.56	0.23	0.15	ETS	0.15
B	1	0.65	0.09	0.21	Auto-ARIMA	0.09
B	2	0.10	0.27	0.31	Prophet	0.10

The following chart shows the overall sales forecast across all dimensions (all products in all stores) over the next nine months, found by using three different forecast models. The green line represents the best fit model. Because the best fit model selects the best forecast model for each product and dimension combination, it avoids the outliers that can occur if a single model is used for all dimension combinations. As a result, the overall best fit forecast resembles an average of the single-model forecasts.



Legend:

- **Red** – Prophet only.
- **Blue** – ETS only.
- **Green** – Best fit.

Best fit model versions

The best fit model algorithm is available in several versions, as described in the following table. Usually, you should use the newest version that's available. However, to ensure that all of your existing forecast models continue to work, the older versions remain available and supported until further notice. To use one of these algorithms, select the appropriate version in the configuration settings for the *Forecast* step in your forecast model.

Expand table

Name	Version required	Description
Best fit model - version 1	Demand planning version 1.0.0.1067 or higher	Works as described in this section

Name	Version required	Description
Best fit model - version 2 (preview)	Demand planning version 1.0.0.3424 or higher	Same as version 1, but with the following changes: • Adds support for Naive forecasting as a way to handle low-data scenarios. • Data used for training and testing models is limited to values from before the forecast start date.
Best fit model - version 3 (preview)	Demand planning version 1.1.0.4 or higher	Same as version 2, but adds support for the Croston's method for forecasting based on intermittent demand (which is demand data with many zero-demand periods and occasional non-zero demands).

> **Important**
>
> - *Best fit model - version 2* and *Best fit model - version 3* are preview features.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to **supplemental terms of use**. They're available before an official release so that customers can get early access and provide feedback.

Auto-ARIMA: The time traveler's delight

The auto-ARIMA algorithm is like a time machine. It takes you on a journey through past demand patterns so that you can make informed predictions about the future.

Auto-ARIMA uses a technique that's known as ARIMA. The name ARIMA is an abbreviation for the three key components that the technique combines:

- *AR* is short for "auto regressive." This component regresses the time series on its own previous values. It captures the influence of past values on the current value.
- *I* is short for "integrated." This component, which is also known as *differencing*, is a step that the model takes to morph a nonstationary time series into stationary data.
- *MA* is short for "moving average." This component accounts for past forecast errors and improves the model's accuracy by smoothing out the noise.

Therefore, the ARIMA technique combines autoregression and moving averages after it differences the data. The final prediction combines the influence of past values and the adjustments from past errors.

The auto-ARIMA algorithm automatically identifies the best combination of the three components to create a forecast model that suits your data. It follows these steps to generate forecasts:

1. Run differencing on the data if it isn't stationary.
2. Find correlation between lagged data points.
3. Calculate moving average error.

Auto-ARIMA works especially well with time series data that shows a stable pattern over time, such as seasonal fluctuations or trends. If your historical demand follows a reasonably consistent path, you might prefer to use auto-ARIMA as your forecasting method.

Auto-ARIMA algorithm equations

Auto regressive calculation

The AR component uses the following equation:

$$Y(t) = c + \phi 1Y(t-1) + \phi 2Y(t-2) + \ldots + \phi pY(t-p) + \epsilon(t)$$

Key:

- *Y(t)* – The value at time *t*.
- *c* – A constant.
- *ϕ1, ϕ2, … ϕp* – Coefficients of the model.
- *ϵ(t)* – The white noise error term.

Moving average calculation

The MA component uses the following equation:

$$Y(t) = c + \epsilon(t) + \Theta 1\epsilon(t-1) + \Theta 2\epsilon(t-2) + \ldots + \Theta q\epsilon(t-q)$$

Key:

- *Y(t)* – The value at time *t*.
- *c* – A constant.
- *ϵ(t), ϵ(t−1), … ϵ(t−q)* – Error terms at time *t*, *t*−1, … *t*−*q*.
- *Θ1, Θ2, … Θq* – Coefficients of the model.

ARIMA calculation

The auto-ARIMA algorithm combines the AR and MA components by using the following equation:

ARIMA = AR + MA (after differencing the time series)

ETS: The shape-shifter

ETS is a versatile demand forecasting algorithm that adapts to the shape of your data. It changes its approach based on the characteristics of your historical demand. Therefore, it's suitable for a wide range of scenarios.

The name ETS is an abbreviation for the three essential components that the algorithm decomposes the time series data into:

- *E* is short for "error." This component captures the random noise or irregular fluctuations.
- *T* is short for "trend." This component represents the overall direction of the data over time: increasing, decreasing, or constant.
- *S* is short for "seasonality." This component reflects repetitive patterns or cycles in the data (for example, yearly or monthly).

By understanding and modeling these components, ETS generates forecasts that capture the underlying patterns in your data.

ETS forecasts future data points by applying varying weights to different observations. More recent data points carry more weight than older ones. ETS can also decompose the time series into error, trend, and seasonality components. (The error comes from noise and fluctuations in the time series.) ETS uses the seasonal period parameter that you set as a seasonal index, estimates the trend in the upcoming horizon, and tries multiple values to determine what fits. Finally, it forecasts the error and combines it with the estimated trend and seasonality components.

Demand planning in Supply Chain Management determines which "flavor" of ETS is most suitable for each time series and applies it accordingly.

Here's a step-by-step explanation of the algorithm:

1. **Decompose components.** Break down the time series into the three components: error (E), trend (T), and seasonality (S).

2. **Select models for the components.** Each component follows an *additive* model:

   ETS(A,A,A) – Additive error, additive trend, additive seasonality.

3. **Specify the initial states.** Calculate initial values for the *level*, *trend*, and *seasonality* states of the model to start the recursive update process. The *level* is the baseline forecast, which the model updates as it trains.

4. **Update the states.** As new data points arrive, update the states of the model (*level*, *trend*, and *seasonality*) by using weighted smoothing equations.

5. **Forecast.** Predict future values by combining the most recent estimates of the *level*, *trend*, and *seasonality*.

ETS algorithm equation

The ETS algorithm uses the following equation:

$F(t+1) = \alpha A(t) + [1-\alpha]F(t)$

Key:

- *F(t+1)* – The forecasted value.
- *F(t)* – The previous forecasted value.
- *A(t)* – The actual historical value.
- α – A smoothing constant ($0 \leq \alpha \leq 1$).

Prophet: The visionary forecasting guru

Facebook's research team developed Prophet. It's a modern and flexible forecasting algorithm that can handle the challenges of real-world data. It's especially effective at handling missing values, outliers, and complex patterns. It performs best with seasonal data, accounts for holidays during forecasting, and doesn't need much preprocessing.

Prophet works by decomposing the time series data into several components, such as trend, seasonality, and holidays, and then fitting a model to each component. This approach enables Prophet to accurately capture the nuances in your data and produce reliable forecasts. Prophet is ideal for businesses that have irregular demand patterns or frequent outliers. It's also ideal for businesses that are affected by special events such as holidays or promotions.

The Prophet algorithm follows these steps to generate forecasts:

1. Decompose the time series into trend, seasonality, and holiday components.
2. Detect and handle change points for trend shifts.
3. Use Fourier series for seasonal patterns.

4. Add holiday regressors for irregular events.
5. Fit the model parameters by using Bayesian optimization.
6. Generate predictions and uncertainty intervals.

Prophet algorithm equation

The Prophet algorithm uses the following equation:

$y(t) = g(t) + s(t) + h(t) + \epsilon(t)$

Key:

- *g(t)* – A value that captures the nonperiodic trend changes over time. The algorithm calculates this value by using a piecewise linear trend equation.
- *s(t)* – A value that represents recurring seasonality patterns, such as daily, weekly, or yearly patterns. The algorithm models this value by using Fourier series.
- *h(t)* – A value that accounts for known, irregular effects caused by holidays or special events. The algorithm treats these effects as additional regressors, which provide flexibility in modeling special events.
- *ϵ(t)* – Random noise or unexplained variability.

XGBoost

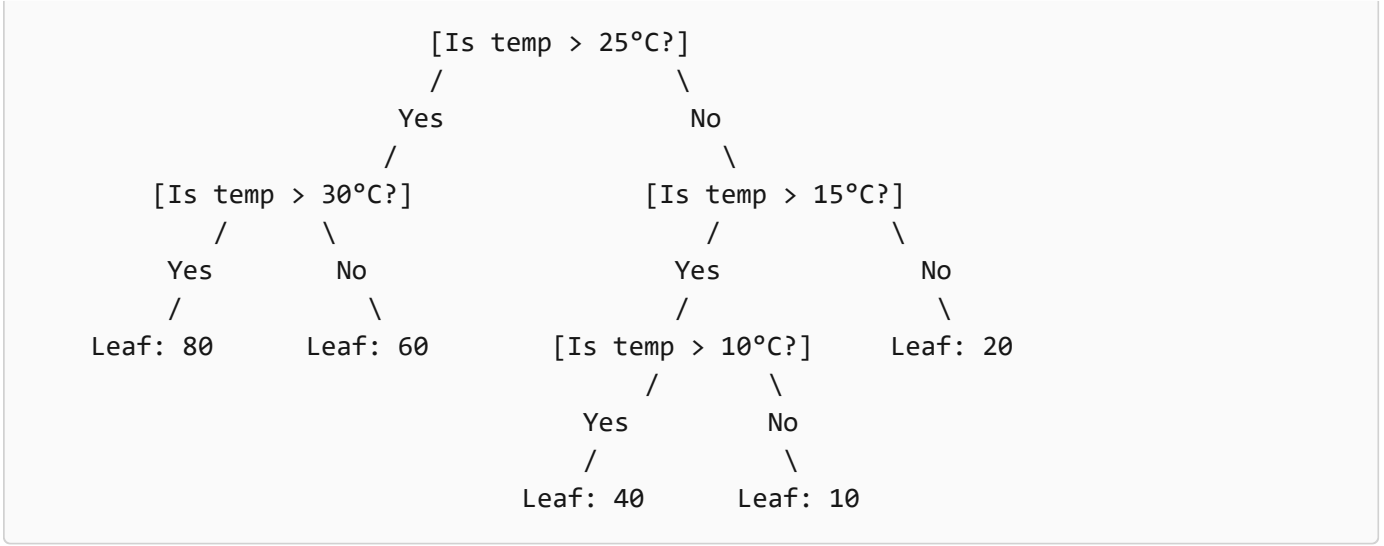
Unlike the other algorithms described in this article, eXtreme Gradient Boosting (XGBoost) generates a forecast based on multiple inputs. It's currently the only algorithm you can use with the *Forecast with signals* forecast model step in Demand planning. In addition, only that type of step supports it. Learn more about how to set up forecast models that use XGBoost and signals input in Forecast with signals.

XGBoost is a highly efficient and scalable implementation of gradient boosting. It builds an ensemble of decision trees to make predictions. The following subsections break down each of the components.

Decision trees

A *decision tree* is a machine learning model that splits data into subsets based on signal values (also known as *dimensions* or *features*) and forms a tree-like structure. The following example shows sales based on weather data.

```
plaintext
```



This decision tree progresses in the following way:

1. **Root node** – The tree splits based on whether the temperature exceeds 25°C:
    - *Yes* – Go to the left subtree.
    - *No* – Go to the right subtree.
2. **Left subtree (temp > 25°C)** – The tree further splits based on whether the temperature exceeds 30°C:
    - *Yes* – Predict *80* sales.
    - *No* – Predict *60* sales.
3. **Right subtree (temp ≤ 25°C)** – The tree splits based on whether the temperature exceeds 15°C:
    - *Yes* – The tree further splits based on whether the temperature exceeds 10°C:
        - *Yes* – Predict *40* sales.
        - *No* – Predict *10* sales.
    - *No* – Predict *20* sales.

Ensemble learning

Ensemble learning is a machine learning approach that combines multiple models (often called *weak learners*) to make predictions. The combined output of many models is often more accurate and robust than any single model.

One type of ensemble learning is known as *boosting*. In this approach, models are built sequentially, and each model corrects the errors of the previous one.

Gradient boosting

Gradient boosting is a powerful machine learning technique that you can use for both regression (which is the case here) and classification problems. It builds an ensemble of weak models (typically decision trees) sequentially, and each model focuses on reducing the errors (*residuals*) that the previous models made.

Gradient boosting effectively captures complex relationships between signals (also known as *exogenous variables*) and the input data (also known as the *target variable*). It also provides better predictive performance than other methods.

How the XGBoost algorithm works

XGBoost is a highly efficient and scalable implementation of gradient boosting. It builds an ensemble of decision trees to make predictions. Here's a step-by-step explanation of how it works:

1. Initialize predictions.

    - **Task** – Start by predicting a base value for all instances.
    - **Purpose** – The base value is typically the mean of the target variable for regression or the log odds for classification.

2. Calculate residuals (gradients).

    - **Task** – Compute the residuals or gradients, which represent the difference between the predicted and actual values.
    - **Purpose** – These residuals serve as the error signal that the model tries to minimize.

3. Fit a decision tree.

    - **Task** – Train a new decision tree by using the residuals (gradients) as the target values.
    - **Purpose** – The tree predicts adjustments to the previous model's predictions.
    - **Key details**
        - XGBoost uses a greedy algorithm to split the data.
        - Splits are selected based on the gain in the objective function, which is regularized to avoid overfitting.

4. Regularize tree growth.

- **Task** – Apply constraints to prevent overfitting.
- **Purpose** – Regularization helps generalize the model and maintain performance on unseen data.
- **Techniques**
  - *Tree depth* – Limit the maximum depth of trees.
  - *Leaf weights* – Penalize overly complex trees by adding regularization terms.
  - *Minimum split gain* – Allow a split only if a minimum improvement occurs in the loss function.

5. Update predictions.

- **Task** – Adjust predictions by adding the outputs of the new tree.
- **Purpose** – This step reduces the error progressively.

6. Repeat the process.

- **Task** – Repeat steps 2 through 5 to sequentially add more trees.
- **Purpose** – Each tree reduces the residuals and therefore gradually improves the model.
- **Stopping criteria**
  - A fixed number of trees are implemented in the algorithm of the Demand planning app.
  - There's no significant improvement in the loss function (*convergence*).

7. Combine trees for the final prediction.

- **Task** – Aggregate the outputs of all trees to produce the final prediction.
- **Purpose** – Each tree contributes to the final result. Therefore, an ensemble effect is created.

Custom Azure Machine Learning algorithm

If you have a custom Azure Machine Learning algorithm that you want to use with your forecasting models, you can use it in Demand planning.

Last updated on 04/22/2026

Naive forecasting (preview)

[This article is prerelease documentation and is subject to change.]

Forecasting models are useful for creating accurate forecasts when they have access to large amounts of data. However, in many cases, such data isn't available, leading to inaccurate results like overstocking or understocking. Naive forecasting is a method for handling small datasets that ensures forecasts are predictable and tied to historical data.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

The idea behind naive forecasting

Models and data

Forecasting models use the training data you provide, and their ability to make useful predictions usually increases as the training dataset size grows. However, sometimes you can't provide enough data. For example, new products typically have only a few data points available to train a model. When that happens, regular forecasting models don't have enough information to identify trends and seasonality patterns in the data, which can lead to inaccurate forecasts.

Consequences of insufficient data on models

When there's not enough data to train a model, the model's behavior can be difficult to predict. Sometimes, the model stops trying to find the right approach and instead selects a constant value as its output, usually the last encountered value or zero. In other cases, the model might identify and extend a trend based on limited data, resulting in forecasts that differ significantly from historical data. These inaccurate forecasts might result in overstocking or understocking.

Naive forecasting as a fallback option

When confronted with unpredictable outcomes, demand planning can use naive forecasting as a fallback approach. Demand planning applies this method when the available data doesn't meet the requirements for standard forecasting models. Naive forecasting uses straightforward rules to produce approximate forecasts, reducing the uncertainty that arises from using regular models trained on limited datasets. Naive forecasting is only a fallback, not a replacement for regular forecasting models. Using naive forecasting within a dimension combination indicates that the data you provided might be insufficient.

Conditions and thresholds for using naive forecasting

Naive forecasting applies when both of the following conditions are met:

1. You must use the *Best fit model - version 2 (preview)* (or newer) forecasting algorithm. Naive forecasting isn't available for other forecast algorithms or as a separate algorithm.
2. Your dataset size is below the defined threshold, with no seasonality automatically detected. The following rules apply:

    - If auto seasonality detection is on but no seasonality pattern is detected, Demand planning uses a threshold of 14 days, 12 weeks, or 12 months (depending on the size of the time buckets in your input time series). If the number of data points is less than double the threshold, it's considered low data, and naive forecasting applies.
    - If auto seasonality detection is on and a seasonality pattern is detected, Demand planning applies a regular forecasting model.
    - If auto seasonality detection is off, Demand planning uses the seasonality hint you provide instead of applying naive forecasting.

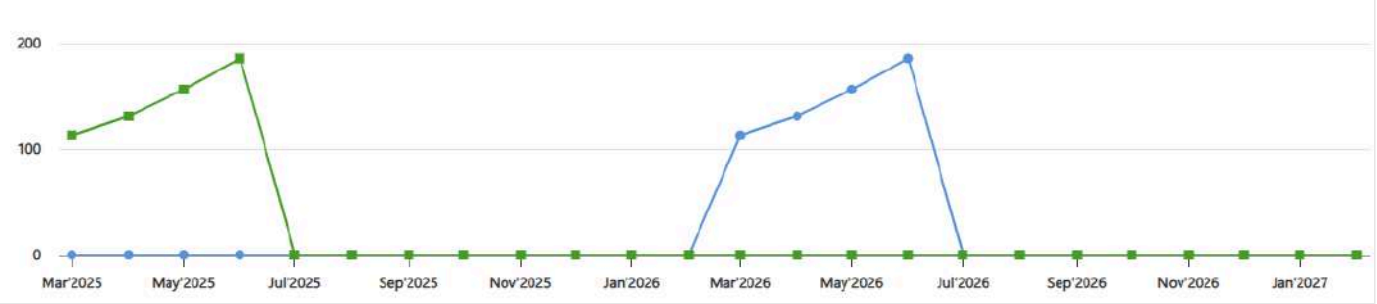
How naive forecasting works

The behavior of the naive forecasting algorithm depends on the size of the input data and the threshold that triggers it. The algorithm uses one of three methods, as described in the following subsections.

Copy and fill with zeros

The *copy and fill with zeros* method is used when the input dataset size is less than half of the threshold. Because only a small amount of data is available, naive forecasting copies the provided data and fills the remaining data points with zeros up to the threshold. This data is then repeated until predictions for all requested dates are found.

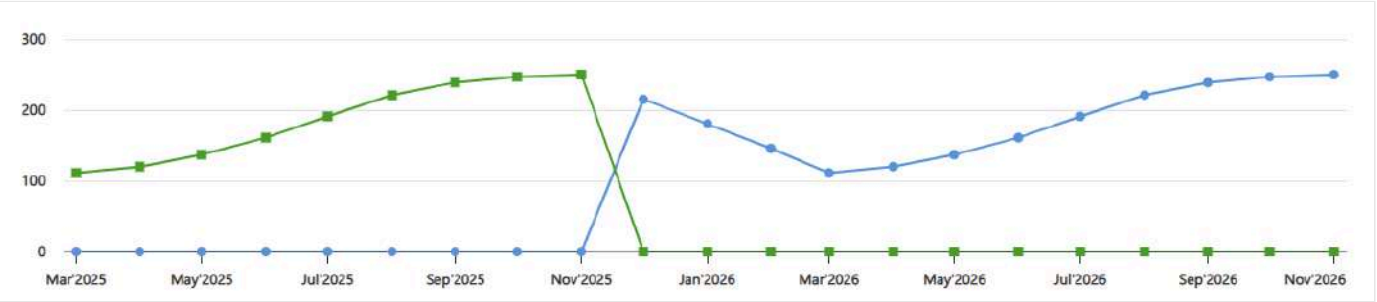
The following image shows an example of the *copy and fill with zeros* method. Input data appears in green, and forecasted data appears in blue. Because there are only 4 input data points, and the threshold is 12, the algorithm applies the *copy and fill with zeros* method, which results in the input values for March to June 2025 being copied to March to June 2026, with zero values inserted in between.



Copy and linear interpolation

When the input dataset size is more than half the threshold but still less than the threshold, naive forecasting uses the *copy and linear interpolation* method. This method fills gaps in the data by using linear interpolation between the first and last data points, ensuring that filled data is closer to copied data compared to the *copy and fill with zeros* method.

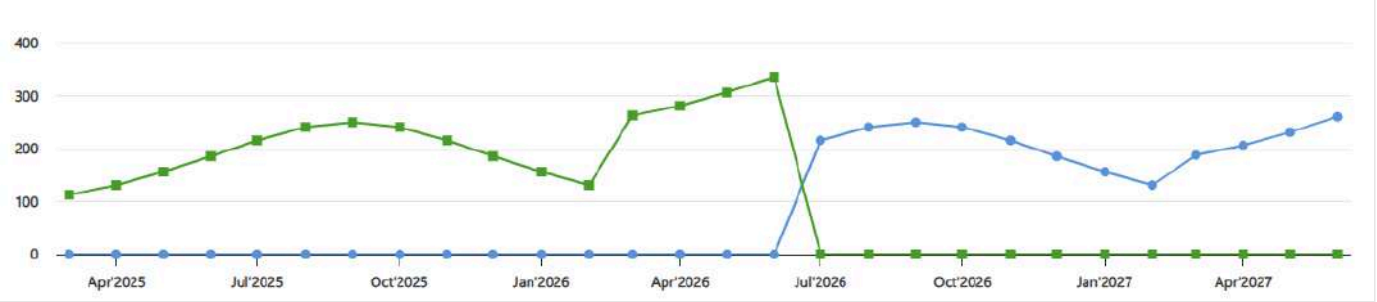
The following image shows an example of the *copy and linear interpolation* method. Input data is shown in green and forecasted data is shown in blue. Because there are 8 input data points, and the threshold is 12, the algorithm applies the *copy and linear interpolation* method, which results in the input values for March–June 2025 being copied to March–June 2026, with linearly interpolated values inserted in between.



Averaged copy

When the input dataset is larger than the threshold but smaller than double the threshold, naive forecasting uses the *averaged copy* method. This method takes an average of values seen before and after the threshold and then uses these values to calculate predictions. This method ensures that calculated values reflect changes in the data and follow the seasonality determined by the threshold.

The following image shows an example of the *averaged copy* method. Input data is shown in green and forecasted data is shown in blue. Because there are 16 input data points, and the threshold is 12, the algorithm applies the *averaged copy* method, which results in the input values for July 2025 to February 2026 being copied to July 2026 to February 2027, with values for March to June 2027 being an average of values for March to June 2025 and March to June 2026.



Limitations and caveats

When you use naive forecasting, the following limitations and caveats apply:

- Naive forecasting isn't a forecasting model but a fallback method for low-data scenarios. Check the explainability tables for your forecast jobs to see which planning objects use the naive method, and consider improving the input data quality where possible.
- Naive forecasting assumes seasonality of the data based on a manual hint or predetermined thresholds. If there's a mismatch between the assumed seasonality and the actual seasonality of the data (for example, if the time series has different seasonality or no seasonality at all), naive forecasting might lead to suboptimal results.
- Zero values are considered when determining whether to use naive forecasting. For planning objects with data that extends far into the past (compared with the rest of the time series), naive forecasting isn't used. To find out how naive forecasting works on this type of data, consider changing the forecast start date to be closer to where data is present. Be careful, because this approach also impacts the accuracy of other models as it limits their training data.

Last updated on 11/07/2025

Croston's method forecasting (preview)

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

[This article is prerelease documentation and is subject to change.]

Forecasting models are useful for creating accurate forecasts when they have access to large amounts of data. However, in some cases, such data isn't available, leading to inaccurate results that can cause overstocking or understocking. *Croston's method* is a forecasting algorithm designed specifically for *intermittent demand*, which is demand data that includes many zero-demand periods with occasional nonzero demands.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Models and data

Forecasting models typically rely on consistent, detailed historical data. As the dataset grows, models become better at identifying trends, seasonality, and patterns, which improves forecast accuracy. However, for some products, the data behaves differently; there are long stretches with no demand at all, followed by occasional spikes. In these cases, regular forecasting models might misinterpret the data and produce misleading predictions.

Challenges with intermittent demand

Standard models struggle when the data includes many zero-demand periods. They might overreact to small changes, forecast zero across the board, or attempt to create a trend where

none exists. These inaccurate outputs can lead to poor inventory decisions, such as overstocking low-usage items or understocking critical spare parts.

Croston's method as a fallback option

As with naive forecasting, Croston's method is a fallback option. Demand planning applies this method when the available data doesn't meet the requirements for standard forecasting models. When the system chooses to use Croston's method within a dimension combination, it indicates that the data you provided might be insufficient.

Conditions and thresholds for using Croston's method

Croston's method forecasting applies when all of the following conditions are met:

- You're using the *Best fit model - version 3 (preview)* (or newer) forecasting algorithm. Croston's method isn't available for other forecast algorithms or as a separate algorithm.
- Your dataset doesn't fulfill the conditions for using the naive forecasting algorithm.
- Your dataset doesn't have a clear seasonal pattern.
- Over 80% of values in the input time series are equal to zero.

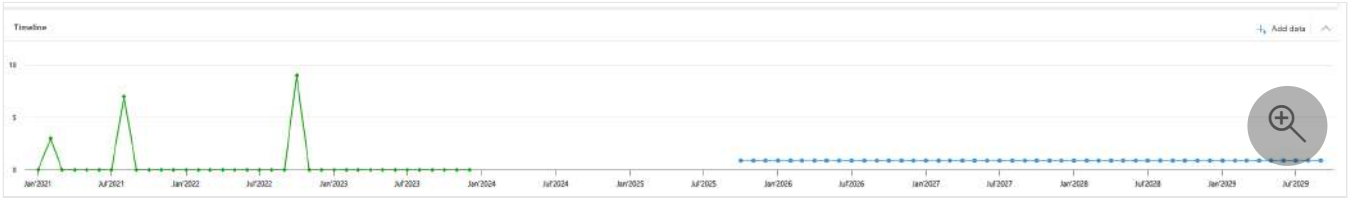
How Croston's method works

Croston's method doesn't try to guess exactly *when* the next demand will happen. Instead, it provides a steady, average forecast over time that reflects:

- How frequently the item is needed.
- How large the average demand is when it occurs.

The forecast stays stable and realistic, without reacting to random fluctuations or zero-demand periods. This approach gives planners a clearer view of what to expect over time.

The following image shows an example time series where Croston's method is a suitable forecasting algorithm.



In this example, the demand history (green line) shows that most periods have zero orders, with occasional spikes that don't follow a pattern. This example is a typical case where Croston's method is appropriate. Instead of chasing the noise, Croston's method produces a flat forecast line (the blue line), which reflects the average demand and its frequency weighted toward recent data.

Limitations and caveats

When you use Croston's method, the following limitations and caveats apply:

- Croston's method isn't a forecasting model that you can choose. The system uses it as a fallback for datasets when most data points are zero. Check the explainability tables for your forecast jobs to see which planning objects use Croston's method and consider improving the input data quality where possible.
- Adding a seasonal hint prevents the best fit algorithm from selecting Croston's method, forcing it to choose an alternative.
- The method predicts only the average interval between demands—not *when* the next demand will occur—and doesn't provide a probability distribution of sizes. Instead, it produces a flat line for a stable, realistic forecast.
- The model gives more weight to recent demand activity, allowing the forecast to better reflect current usage volume.

Last updated on 11/07/2025

Use your own custom Azure Machine Learning algorithms in Demand planning

08/18/2025

In addition to the standard forecasting algorithms provided with Demand planning, you can also create your own custom algorithms and use them in your Demand planning forecast models. To do so, you must create and publish your algorithms using Microsoft Azure Machine Learning, set up a connection from Demand planning to Azure, and place one of the following types of steps to your forecast models as needed (instead of using the standard *Forecast* step):

- *Custom* step – Lets you use custom algorithms that you've created using Azure Machine Learning. Use the *Custom* step with custom algorithms that you haven't used with the legacy Supply Chain Management forecasting functionality. This type of step communicates with Azure through a *batch* endpoint and allows you to pass any custom parameters your algorithm requires, though no preconfigured parameters are set up by default.
- *Finance and operations* step – Lets you use custom Azure Machine Learning algorithms that you created for use with the legacy Supply Chain Management forecasting functionality, as described in Demand forecasting overview. This type of step communicates with Azure through a *pipeline* endpoint and is preconfigured to pass the parameters supported by legacy Supply Chain Management forecasting functionality.

This article describes the setup that is required to enable Demand planning to connect to your Azure Machine Learning workspace, including how to set up your Azure Machine Learning workspace, how to connect to that workspace in Demand planning, and how to set up your forecast models to run your custom algorithms.

Download sample code and other resources

To download sample code and other resources that demonstrate how to use custom Azure Machine Learning algorithms in Demand planning, go to the Custom-ML-in-DemandPlanning repository on GitHub. It provides insights into how to read input data, handle standard and custom parameters, and generate output that aligns with the application's consumption requirements. The repository consists of a readme file, a scoring script, and demonstrative videos.

Set up your Azure Machine Learning resources

You must develop your custom forecast algorithms to run in Azure Machine Learning and then set up Azure to make them available to Supply Chain Management. The following subsections explain how. If you've already been using your custom Azure Machine Learning algorithms with the legacy Supply Chain Management forecasting functionality, then these elements might already be set up. In that case, you can skip to the section Connect to the Azure Machine Learning service from Demand planning.

Set up a new Microsoft Entra application

Follow the steps in this section to create a new Microsoft Entra application in your Azure Machine Learning workspace. This resource in the Azure portal holds your algorithms. The Microsoft Entra application is an enterprise application that enables Demand planning to connect to your Azure Machine Learning algorithms. Learn more about how to set up a Microsoft Entra application in Register an application.

1. Sign in to the Azure portal by using an account that has at least *Cloud Application Administrator* privileges.
2. Register a new Microsoft Entra application as described in Create a Microsoft Entra application and service principal that can access resources.
3. Follow the on-screen instructions to complete the wizard. Use the default settings.
4. In the **Certificates & secrets** section of the new Microsoft Entra application, create a secret for the application as described in Add a client secret.
5. Make a note of the application ID and its secret. You'll need these details later.

Grant access for the new Microsoft Entra application to the Azure Machine Learning workspace and Azure Machine Learning workspace storage account

Follow these steps to allow your new Microsoft Entra application to access the Azure Machine Learning workspace.

1. In the Azure portal, go to the resource group that contains your Azure Machine Learning workspace.
2. On the left navigation pane, select **Access control**.
3. On the **Role assignments** tab, select **Add** to add a new role assignment.
4. On the **Privileged administrator roles** tab, select *Contributor*.
5. Select **Next**.
6. Select the **User, group, or service principal** option.
7. Select **Select members**. Use the filter on the menu on the right to find the name of the Microsoft Entra application that you created, and then select it.
8. The application now appears in the **Members** list. Select **Next**.

9. On the **Review + assign** tab, select **Next**.

Follow these steps to allow your new Microsoft Entra application to access the storage account that Azure Machine Learning workspace is connected to.

1. In the Azure portal, go to the resource group that contains your storage account (the storage account that your Azure Machine Learning workspace is using).
2. On the left navigation pane, select **Access control**.
3. On the **Role assignments** tab, select **Add** to add a new role assignment.
4. On the **Job function roles** tab, select *Storage account contributor* and *Storage blob data contributor*. To find these roles quickly, enter *storage contributor* into the **Search** field.
5. Select **Next**.
6. Select the **User, group, or service principal** option.
7. Select **Select members**. Use the filter on the menu on the right to find the name of the Microsoft Entra application that you created, and then select it.
8. The application now appears in the **Members** list. Select **Next**.
9. On the **Review + assign** tab, select **Next**.

The application is now listed in the **All** section of the **Role assignments** tab on both the Azure Machine Learning workspace and the storage account.

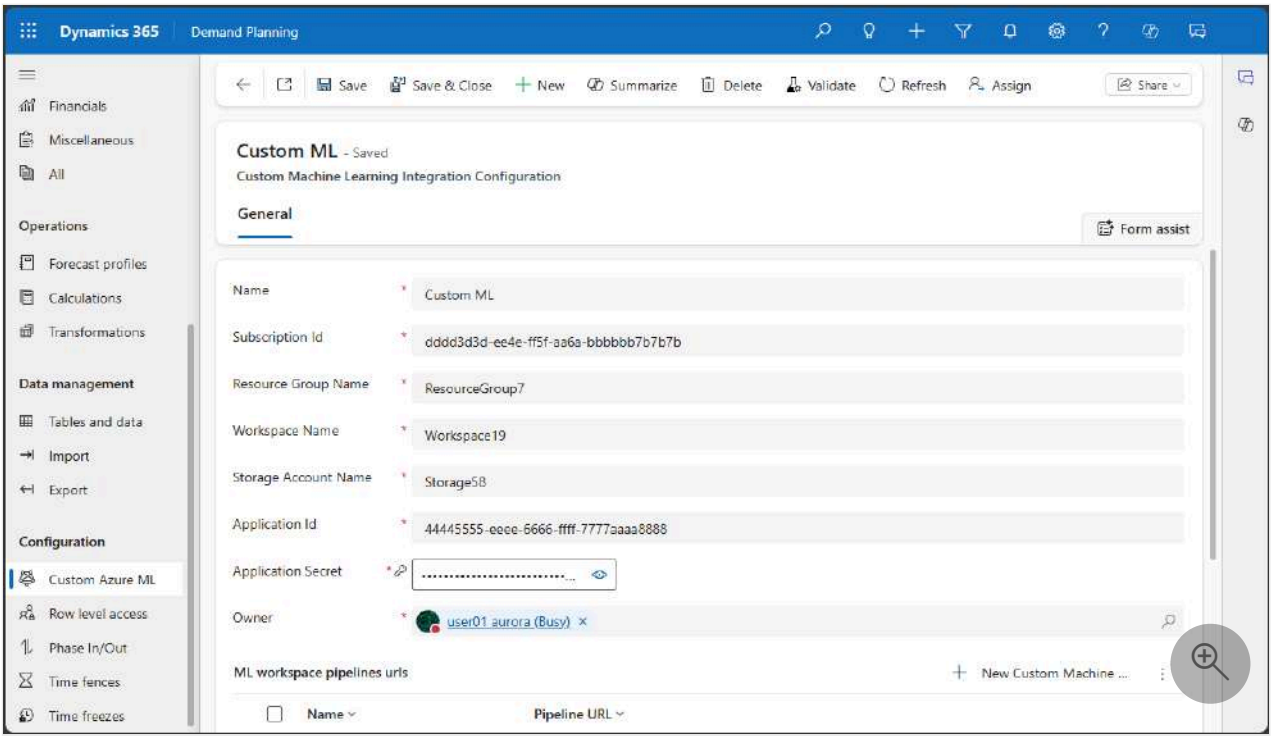
Connect to the Azure Machine Learning service from Demand planning

Follow these steps to set up the Azure Machine Learning service connection in Demand planning.

1. Sign in to Demand planning.

2. On the left navigation pane, select **Custom Azure ML**.

3. In the toolbar, select **New** to create a new connection.



4. Make the following settings for the new connection:

    - **Name** – Enter a name for the connection.
    - **Subscription ID** – Enter the ID for your Azure subscription.
    - **Resource group name** – Enter the name of the resource group that contains your Azure Machine Learning workspace.
    - **Workspace name** – Enter the name of your Azure Machine Learning workspace.
    - **Storage Account Name** – Enter the name of the Azure storage account that you specified when you ran the setup wizard in your Azure workspace.
    - **Application ID** – Enter the application ID of the Microsoft Entra application that you created. This value is used to authorize API requests to the Azure Machine Learning service.
    - **Application secret** – Enter the service principal application secret for the Microsoft Entra application that you created. This value is used to acquire the access token for the security principal that you created to perform authorized operations against Azure Storage and the Azure Machine Language workspace.

5. On the Action Pane, select **Save**.

6. On the **ML workspace pipelines URLs** FastTab, add a row for each Azure Machine Learning endpoint that you want to make available to Demand planning. Each endpoint requires a unique **Name** (which is only used internally in Demand planning) and a **Pipeline URL**, which is the full URL that Demand planning must use to access the endpoint.

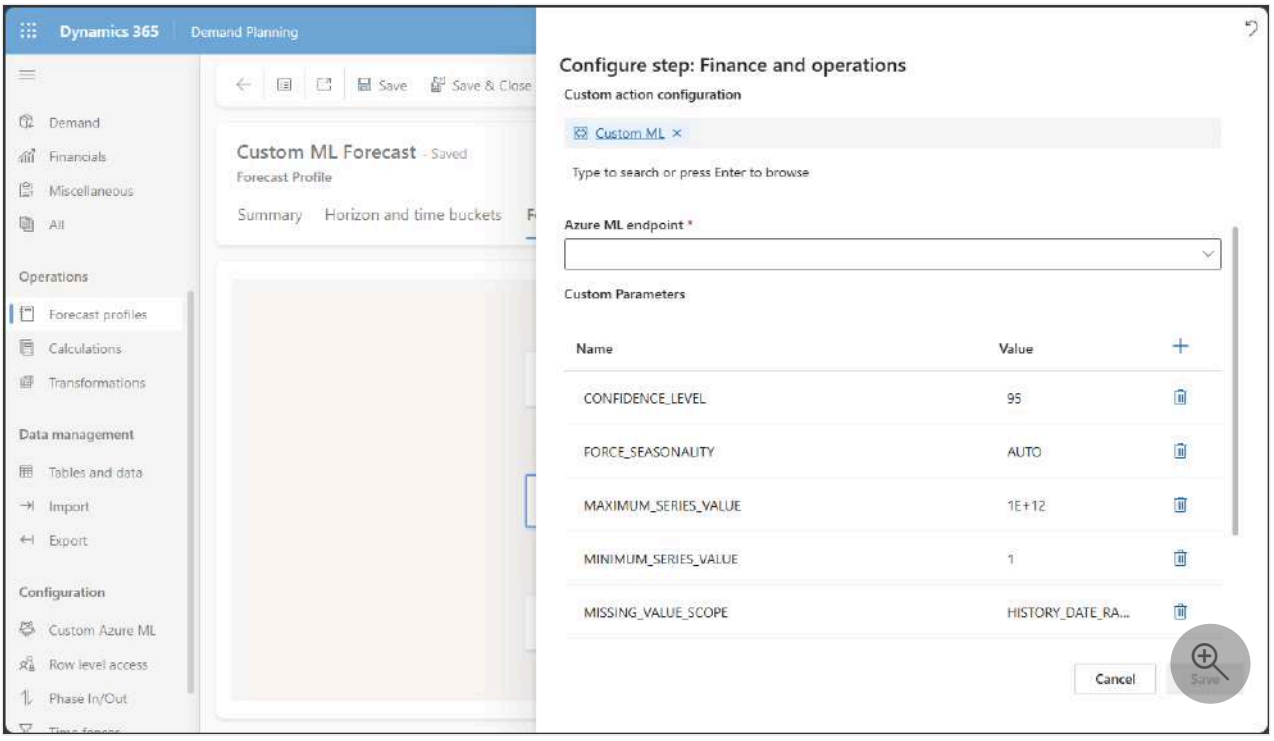
    - For *Finance and operations* steps, these must be Azure Machine Learning *pipeline* endpoints.

- For *Custom* steps, these must be Azure Machine Learning *batch* endpoints. Learn more in Batch endpoints.

Set up a forecast model that uses a custom Azure Machine Learning algorithm

Follow these steps to set up a forecast model that uses a custom Azure Machine Learning algorithm. This procedure applies both to new custom algorithms and to custom algorithms that you previously used with Supply Chain Management; the only difference is the type of forecast step you should use in your model (*Custom* or *Finance and operations*).

1. Create a new forecast profile as described in Create and manage forecast profiles.
2. On the **Select a forecasting model preset** page, select *None*.
3. After you've created and saved the profile, select the **Forecast model** tab. (Learn more in Design forecast models.)
4. Set up your model. Add a *Finance and operations* or *Custom* step in the position where you want to run your algorithm.
    - Use a *Custom* step if you want to connect to an Azure Machine Learning algorithm that you haven't used with the legacy Supply Chain Management forecasting functionality.
    - Use a *Finance and operations* step if you want to connect to an Azure Machine Learning algorithm that you created for use with the legacy Supply Chain Management forecasting functionality.
5. Open the **Actions** menu for the *Finance and operations* or *Custom* step and then select **Settings**.



6. In the **Configure step** dialog, make the following settings for the step:

   - **Step name** – Enter a name for the step.
   - **Description** – Enter a description for the step.
   - **Custom action configuration** – Select the Custom Azure ML profile that you created to connect to your Azure Machine Learning resource.
   - **Azure ML endpoint** – Select the Azure Machine Learning endpoint that triggers the machine learning algorithm that you want to use. The available endpoints must be defined as part of the Custom Azure ML profile that you created.
   - **Custom parameters** – If you've designed your Azure Machine Learning forecast algorithm to accept input parameters, then add them as key/value pairs here as required.
     - For *Custom* steps, no default parameters are provided. You must manually add each one that you require following the specifications of your algorithm.
     - For *Finance and operations* steps, a collection of default parameter names are provided, so all you need to do is adjust the values as needed. These are the parameters used by the legacy Supply Chain Management forecasting functionality. For details about how to use these settings, see Default parameters and values for demand forecasting models.

> **⊗ Caution**
>
> You must only include parameters that your forecast algorithm is designed to accept. If you include parameters that your algorithm doesn't recognize, the forecast will fail when it runs.

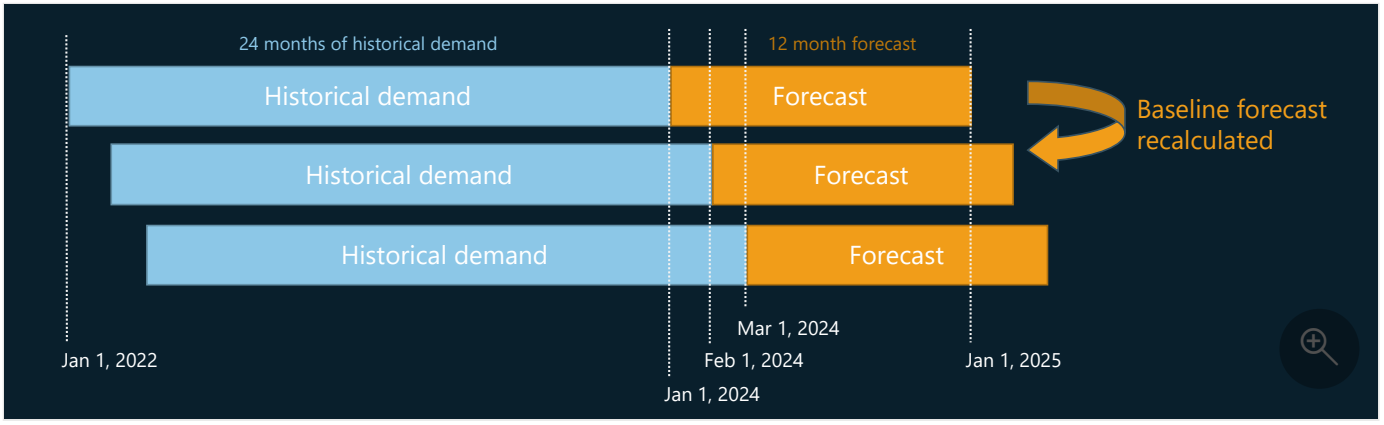
7. Select **Save** to close the dialog.

8. Complete the model by adding a *Save* step.

Rolling forecasts

Rolling forecasts let you establish a regular schedule that your time series follow to automatically update and extend their forecast horizon based on the latest data and assumptions. Rolling forecasts help align the demand planning process with the changing business environment. They also help improve the accuracy and reliability of the forecasts.

The following illustration shows an example of a rolling forecast that recurs on the first of each month.



> **Note**
>
> On a continuous basis, when a forecast is fully recalculated, its baseline forecast is also automatically recalculated. Edits that were made during the previous period aren't kept.

How to use rolling forecasts

After you set up a rolling forecast, you can view different versions of the same time series to compare the historical demand, the current forecast, and the previous forecast. You can also modify, review, and edit the forecast as you require. Rolling forecasts automatically update and extend the forecast horizon each time that they run.

Rolling forecasts can help you monitor the performance of your demand planning process, identify and address issues by collaborating on a forecast, and adjust your plans accordingly. They can also help you communicate and collaborate with other stakeholders (such as sales, marketing, and finance), and align your demand planning with the strategic goals of your organization.

Set up rolling forecasts

You can fully automate a rolling forecast by scheduling each of its processes (importing data, running a transformation, running a calculation, running a forecast, and exporting the forecast).

To schedule a process, follow these steps:

1. Go to one of the following pages, depending on the type of process that you want to schedule:

    - **Data management** > **Import**
    - **Data management** > **Export**
    - **Operations** > **Forecast profile**
    - **Operations** > **Calculations**
    - **Operations** > **Transformations**

2. Select the profile that you want to schedule, or create a new profile.

    - If you're creating a new profile, work through the creation wizard in the usual way. When you reach the **Set run schedule** page of the wizard, you can set up a schedule by configuring the settings that are described in the rest of this procedure. However, to fully automate most types of processes, you must have an existing time series. Therefore, we recommend that you manually run the profile for the first time to create the required time series. Then edit the profile to set up the schedule by using that time series.
    - If you're editing an existing profile, select the **Run Schedule** tab.

3. Set the **Run schedule** field to one of the following values:

    - **None** – The process isn't scheduled to run automatically. You must manually run it.
    - **Recurring** – The process is triggered at a specific date and time, according to the configured schedule (daily, weekly, or monthly).

4. In the **Set run date recurrence** section, define when and how often the process should run, and set the start and end dates that it should run during.

5. In the **Output settings** section, follow one of these steps to specify how you want to save the output of the process:

    - To create a new time series after each run, set the **Save output as** field to *Create new time series*. Then, in the **Time series name** field, enter a base name for the new time

series. In the **Append to name** field, select a dynamic suffix to add to the base name, so that each time series has a unique name. You can use the run date, the run month, or the run month and year as the suffix.

- To add a new version to an existing time series after each run, set the **Save output as** field to *New version of the same time series*. Then set the **Use existing time series** field to the target time series. (The target time series must already exist.) We recommend that you also select the **Save current as a version before overwriting** checkbox to maintain the traceability of the whole process and ensure that the system doesn't accidentally overwrite any version work that's in progress. For more information about how time series versions work, see Time series, worksheets, and planning data.

Last updated on 12/30/2025

Use phase in/out functionality to simulate planned changes

Article • 01/10/2025

Phase in/out functionality lets you set up forecasts and calculations that simulate changes that you plan to make. Here are some examples:

- Phase out a product that's no longer popular or that's leaving the market.
- Phase in a new product.
- Replace an older product with a new, similar one.
- Open a new store or warehouse that should be similar to an existing one.

To apply calculations of this type, add a *Phase in/out* step to the relevant calculation model or forecast model. The *Phase in/out* step uses a *phase in/out rule group* to modify a selected data column based on values that are found in the same column or another column of the time series. Each phase in/out rule group defines selection criteria that identify input and output columns, a period to apply the calculation during, and a scaling factor to apply during that period.

Example: Phase in a new product that replaces an existing product

You're introducing a new product (Speaker V2) that replaces a similar existing product (Speaker V1). You expect that customers will increasingly choose the new speaker. However, you also expect that the total number of speakers that are sold each month will remain about the same as last year and follow a familiar seasonal sales pattern. Therefore, you add the following phase in/out calculations to next year's forecast:

- For the first six months of next year, you'll offer Speaker V2 at a premium price. Therefore, you predict that the quantity of Speaker V2 that you sell will be 40 percent of the quantity of Speaker V1 that you sold during the same month last year. Speaker V1 will make up the difference, and you'll sell 60 percent of the quantity that you sold last year.
- After six months, you'll slightly lower the price of Speaker V2. Therefore, for the last half of next year, you predict that the quantity of Speaker V2 that you sell will be 75 percent of the quantity of Speaker V1 that you sold last year. The quantity of Speaker V1 that you sell will be 25 percent of the quantity that you sold last year.
- After a year, you expect that sales of Speaker V2 will completely replace sales of Speaker V1.

To add this calculation to a forecast model, you set up a phase in/out rule group that has the following rules:

- **Phase in rule 1** – This rule applies from January through June. For each month, select the forecast quantity based on last year's sales of Speaker V1, multiply it by 40 percent, and add the result to the forecast quantity for Speaker V2. (The forecast quantity for Speaker V2 is currently 0 [zero], because it's a new product.)
- **Phase out rule 1** – This rule applies from January through June. For each month, select the forecast quantity based on last year's sales of Speaker V1, multiply it by 60 percent, and set the result as the new forecast quantity for Speaker V1.
- **Phase in rule 2** – This rule applies from July through December. For each month, select the forecast quantity based on last year's sales of Speaker V1, multiply it by 75 percent, and add the result to the forecast quantity for Speaker V2.
- **Phase out rule 2** – This rule applies from July through December. For each month, select the forecast quantity based on last year's sales of Speaker V1, multiply it by 25 percent, and set the result as the new forecast quantity for Speaker V1.

The following table shows an example of forecasted sales before the phase in and phase out calculations are applied.

Expand table

Product	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Speaker V1	100	100	100	100	100	100	100	100	100	100	100	100

The following table shows an example of forecasted sales after the phase in and phase out calculations are applied.

Expand table

Product	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Speaker V1	60	60	60	60	60	60	25	25	25	25	25	25
Speaker V2	40	40	40	40	40	40	75	75	75	75	75	75

Manage phase in/out rule groups

Use phase in/out rule groups to define the phase in/out calculations that are applied in your forecast and calculation models. To create, view, or edit a rule group, follow these steps.

1. On the navigation pane, select **Configuration** > **Phase in/out**.

2. Follow one of these steps:

    - To create a new rule group, select **New** on the Action Pane. A wizard guides you through the steps for creating the group.
    - To view or edit an existing rule group, select the link for it in the **Name** column of the grid to open its details page.

3. The details page and wizard pages provide nearly identical fields that you can set:

    - **Name** – Enter a name for the rule group.
    - **Description** – Enter a short description of the rule group.
    - **Owner** – Select the user account that owns the rule group. (This field is available only in the wizard.)
    - **Active** – Set this option to *Yes* or *No*. Only active rules groups can be assigned to the phase in/out steps of your forecast or calculation models.
    - **Rules** – Use the fields on this tab to define the phase in/out calculations that the group does. For information about how to use these fields, see the next section.

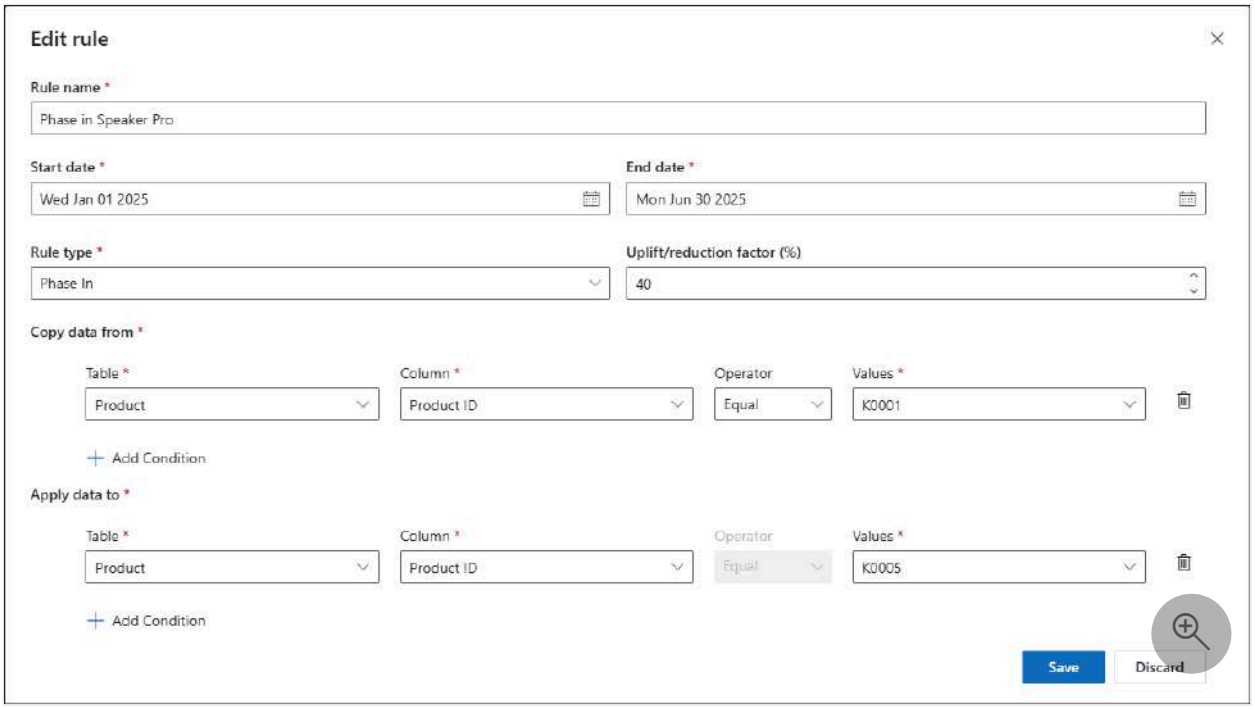
4. Follow one of these steps:

    - If you're creating a new group, select **Review and finish** on the last wizard page to save your rule group and open its details page.
    - If you're editing an existing group, save your settings by selecting **Save** or **Save & close** on the Action Pane.

Set up rules

Each rule group must have at least one rule and might have several rules. You set up rules on the **Rules** tab of either the details page or the **Add rules** page of the new group wizard. The process is the same in both places.

Use the buttons on the toolbar to add, remove, and edit rules. You can include as many rules as you need for each group. Each time that you add or edit a rule, the **Edit rule** dialog box appears.



In the **Edit rule** dialog box, set the following fields for each rule:

- **Rule name** – Enter a name for the rule. Choose a name that makes the rule easy to identify in the rule list.
- **Start date** – Enter the first date when the rule should apply.
- **End date** – Enter the last date when the rule should apply. For a phase in rule, data will stop being copied on this date. (By this date, you should have actual sales data to base your forecasts on.) For a phase out rule, time series values will fall to 0 (zero) on this date unless another phase out rule applies immediately after this rule.
- **Rule type** – Select one of the following values to specify the type of rule:
  - *Phase in* – The rule takes a value from the data column that you identify in the **Copy data from** section, multiplies it by the **Uplift/reduction factor (%)** value, and adds the result to the data column that you identify in the **Apply data to** section.
  - *Phase out* – The rule takes a value from the data column that you identify in the **Copy data from** section, multiplies it by the **Uplift/reduction factor (%)** value, and sets the result as the new value for the same data column that you identify in the **Copy data from** section.
- **Uplift/reduction factor (%)** – Specify a factor (as a percentage) to apply to the input time series values.
- **Copy data from** – Use this section to specify criteria for selecting input values for the calculation. You can use the **Add condition** button to add as many rows as you

need. The data tables that are listed are the data tables that are defined in Demand planning. (Learn more in View and customize tables for holding imported data.) You typically use this section to identify a product from the product table that you want to phase in or out.

- **Apply data to** – This section applies only to phase in rules. (Phase out rules always apply to the same data columns that are identified in the **Copy data from** section.) Specify criteria for selecting the data column that the phase in rule applies to. You typically use this section to identify a new product that you're phasing in. You can use the **Add condition** button to add as many rows as you need.

Assign rule groups to phase in/out steps

After you finish setting up your rule groups, you can use them by putting *Phase in/out* steps at the appropriate positions in your calculation models and forecast models. Learn more in Design calculation models and Design forecast models.

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Time series, worksheets, and planning data

08/18/2025

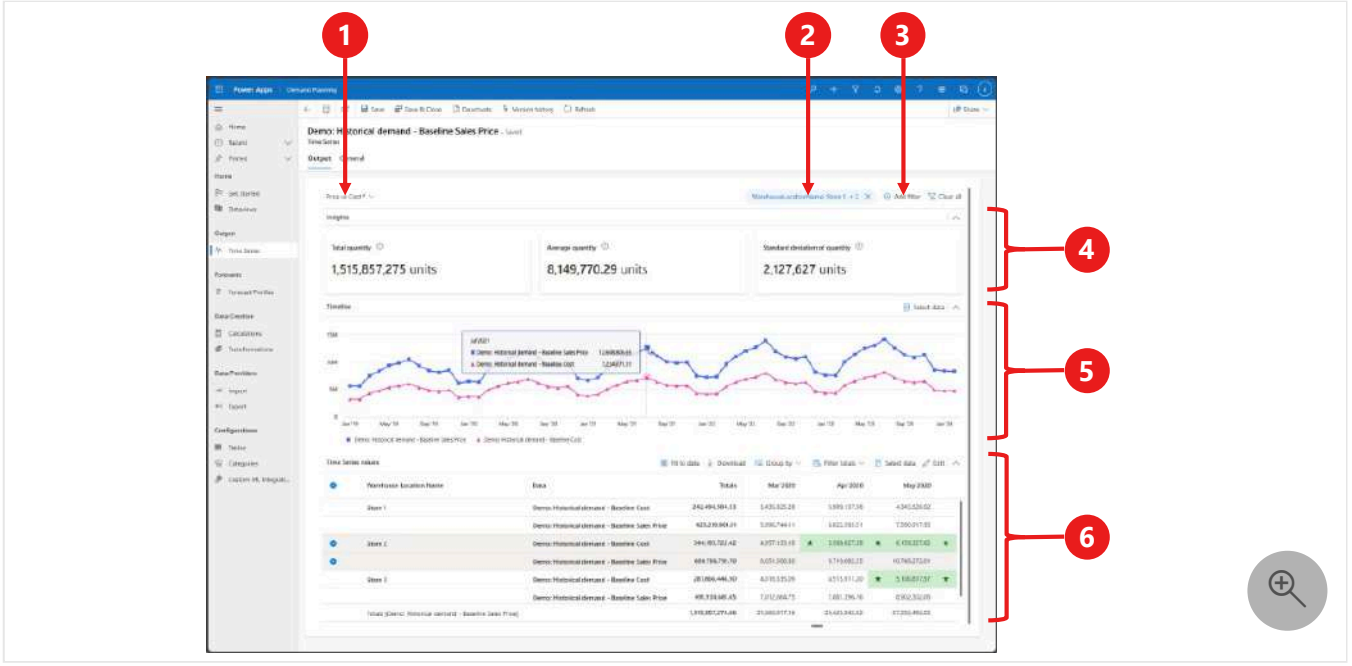
Time series, worksheets, and planning data represent the output from forecasts, calculations, and transformations in daily, weekly, or monthly time buckets.

- *Time series* is a term that describes the type of data that forecasts, calculations, and transformations consume and create. Each time series is a collection of data values that are plotted against a series at regular time intervals. Both planning data and worksheets contain time series data.
- *Planning data* is the processed and categorized output that forecasts, calculations, and transformations create. Demand planning presents planning data by using both a chart and a grid of values. It also provides tools for organizing, analyzing, editing, combining, and presenting this data. Every planning data record is categorized as containing *Forecast*, *Demand*, *Financials*, or *Miscellaneous* data, based on how it was created.
- *Worksheets* let you view, modify, and combine various types, versions, and views of planning data. You can create and switch between any number of worksheets for each planning data record. You can also share worksheets with other users or keep them private for yourself.

Explore planning data and worksheets

To view all your planning data, on the navigation pane, select **Planning data** > **All**. Alternatively, select an appropriate planning data category under **Planning data**. To view shared worksheets, on the navigation pane, select **Home** > **Worksheets**.

To open a worksheet or planning data record, select its name in the **Name** column. The following illustration highlights the features on the **Output** tab of the time series details page.



Legend:

1. **Create and select worksheets** – You can use this menu to save the current view (including all filters, overlaid series, and other selections) as a worksheet in the current planning data record. If you've already saved one or more worksheets, you can use this menu to load them. Each worksheet that's created from this menu is also listed on the **Worksheets** page (available at **Home** > **Worksheets** on the navigation pane). Each worksheet has a share setting that lets you share it with other users or keep it private to yourself. You can also use this menu to delete worksheets.

2. **Current filters** – If you've applied any filters to the data, they're listed next to the **Add filter** button as colored blocks. You can remove any listed filter by selecting its **Close** button (**X**).

3. **Add filter** – Open a menu where you can choose to filter the time series by any combination of date or dimension values. This filtering affects the key performance indicators (KPIs) that are shown on the **Insights** FastTab, the chart that's shown on the **Timeline** FastTab, and the values that are available on the **Time series values** FastTab. Filters that you add from here are added to the current filters list (2).

4. **Insights** – This FastTab shows KPIs for the current time series, including total quantity, average quantity, and standard deviation of quantity. These values reflect the filter settings.

5. **Timeline chart** – This chart is a graphical representation of the selected time series. It provides the following capabilities:

    - **The chart reflects time series filters.** Chart values are consistent with the filter settings that are configured at the top of the page.

- **The chart reflects row selections.** Chart values reflect any rows that are selected on the **Time series values** FastTab.
- **View precise data values.** You can view precise data values at any point by hovering your mouse pointer over the chart.
- **Set chart options.** You can adjust chart options (such as color, style, and grid options) by using the **Select data** button.
- **Overlay another time series.** You can overlay another time series for comparison by using the **Select data** button. In this illustration, a second time series (pink) is overlaid on the current time series (blue).

6. **Time series values** – This FastTab shows tabular data for the selected time series. The following features and toolbar buttons are available:

    - **Row selection** – Select a row to see only that row reflected in the timeline chart. To select multiple rows, select the **Shift** key while you tap or click. A blue check mark indicates each row that's selected. Select the check mark to clear the selection of a row. Select the check mark at the top of the table to clear the selection of all rows.
    - **Add a cell comment** – Enter a comment for the selected cell. The comment is saved with the time series and can be viewed by other users. Cells that have comments show a comment symbol (purple speech bubble). To view the comment for a cell, select the comment symbol (or use the Copilot grid cursor).
    - **Fit to data** – Adjust the width of visible columns to ensure that the data isn't clipped.
    - **Downloads** – Download the data as a comma-separated values (CSV) file where the current filter and group options are applied.
    - **Group by** – Select which of the available dimensions to use to group the data. For example, if you have dimensions for warehouses and product variants, you can group by both warehouse and variant (to show each warehouse, with each product variant listed under it), group by variant only (to show the variants but not the warehouses), or group by warehouse only (to show the warehouses but not the variants). Totals values are aggregated to match the group selection.
    - **Filter totals** – Set up a logical expression to filter the selection of rows that are shown in the table. For example, you can view only rows that have totals above 100,000 units. This setting doesn't affect the chart or KPIs.
    - **Select data** – Add another time series to the chart and/or data table, so that you can compare them. This button opens the same dialog box that the **Select data** button above the chart opens. Therefore, you can also adjust chart options (such as color, style, and grid options).
    - **Edit** – Edit values directly in the data table. If an edited value is aggregated because of the **Group by** settings, the value of your edit will be divided proportionally among the aggregated data rows. If a cell shows a lock symbol, it's currently locked

by a time fence and therefore can't be edited. Learn more about time fences in Limit manual time series edits with time fences.

Select the **General** tab of the time series details page to view and adjust the following basic settings. Many of these settings affect the way that the time series is shown on the list page.

- **Name** – The name of the time series.
- **Description** – A short description of the time series.
- **Category** – The category of the time series. This setting controls which navigation entry, in addition to **All**, the time series is listed in under **Planning data** on the navigation pane. Select the **Search** button (magnifying glass symbol) to select a category.
- **Time buckets** – The time buckets that are used by the time series (daily, monthly, or yearly).
- **Allow manual edits** – A value that indicates whether the series allows users to manually edit data values by using the data table.

> **Tip**
>
> In addition to the features described in this article, time series also provide *generative insights* and a *Copilot grid cursor*, each of which help you analyze and explore time series data with the help of AI. Learn more in Generative insights for Demand planning (production ready preview) and Copilot grid cursor.

Customize time series chart and data table layouts

Demand planning provides many options that help you analyze your time series. For example, you can customize the chart layout (such as style, color, and size), overlay the chart or data table with another time series so that you can easily make comparisons, and assign logic to highlight specific cells in the data table so that they stand out.

To configure these options, open a time series, and then select **Select data** on either the **Timeline** FastTab or the **Time series values** FastTab. The **Select data** dialog box appears.

To add a new time series to the chart and/or data table, select **Add**. You're prompted to select an existing time series. The time series that you select is then added as a new row in the **Select data** dialog box. To remove a time series from the chart and data table, select its **Close** box on the right side of the row. The top row of the **Select data** dialog box lists the current time series. You can customize this time series, but you can't remove it.

The row for each time series that's listed in the **Select data** dialog box provides the following information and options:

- **Name** – The name of the time series.
- **Version** – For time series that have more than one version, select the version to use.
- **Description** – A short description of the time series.
- **Show in chart** – Enable this option for each time series that you want to include in the chart.
- **Chart styling** – Select the **Customize** link in this column to open a dialog box where you can customize the way that the selected row appears in the chart. You can select whether the series is presented as a line or a series of bars, and you can select the color. You can also add a secondary axis. A secondary axis is useful if the selected series use very different values (for example, when you compare sales in the millions of units against prices in the hundreds of dollars). If you add a secondary axis, a new set of values is shown on the right side of the chart to indicate the scale of the second time series.
- **Show in grid** – Enable this option for each time series that you want to include in the data table.
- **Grid formatting** – Select the **Customize** link in this column to open a dialog box where you can specify logic to highlight cells in the data table. You can highlight cells for the selected series according to either their own values or values from another time series for the same time bucket. (If you want to compare the selected series to another series, set the **Use values from another time series** option to *Yes*, and then specify a series and version.) You can add as many rules as you need. For each rule, specify the logic (for example, find values that are less than 4,000,000) and the style (color and symbol) that are applied to cells that match that rule.

After you've spent time customizing and formatting your time series, you'll probably want to save your settings as a new data view that you can return to later. To save the current settings as a data view, select **Save as new dataview** on the dropdown menu in the upper left of the page. (Learn more in the Explore time series section.) You can save as many data views as you need for each time series.

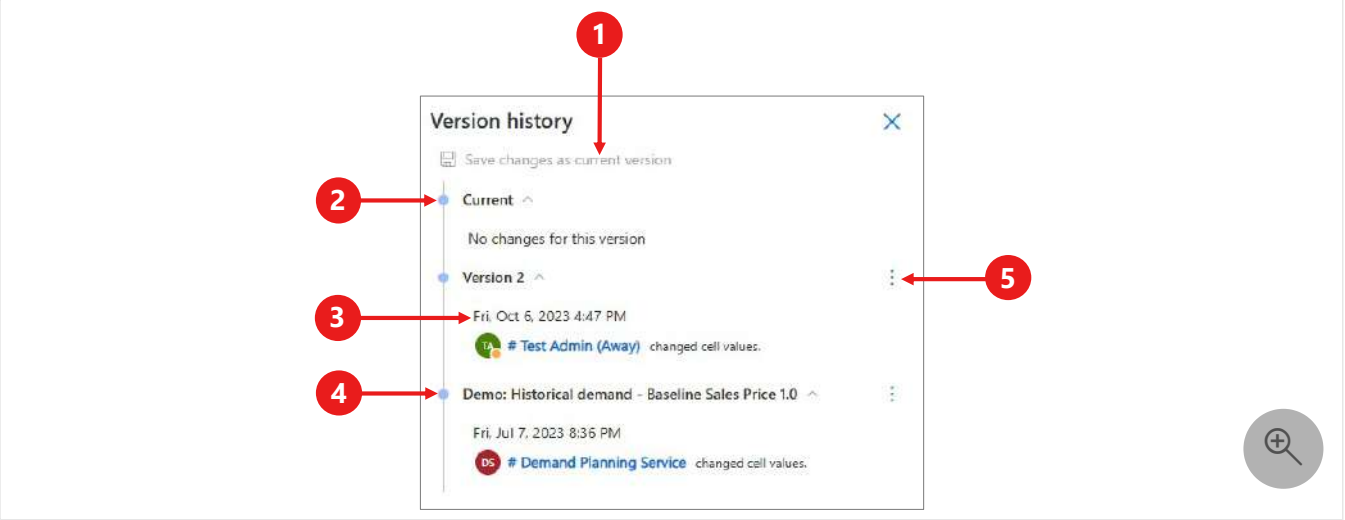
Time series versions, version control, and change log

Demand planning lets you edit cell values directly. Therefore, you can modify the underlying data for any time series. Each time that you edit cell values, the change is noted in the change log, where all users can see when the change was made and who made it. At any time, you can save the edits that you've made so far (since the last version) as a new version and provide a name for it. You can view any version by selecting it in the version history, and you can restore any previous version to the current version.

Sometimes, when you're selecting a time series, you can select which version to use. For example, when you use **Select data** to overlay one time series on the current time series, you can choose which version to use. You can even overlay a previous version of the same time series to quickly see what changed between versions.

To view the change log and version history for a time series, open the time series, and then select **Version history** on the Action Pane. The **Version history** pane that appears shows a log entry for each relevant edit event. Log entries are grouped under the version that they apply to.

The following illustration highlights the features of the **Version history** pane.



Legend:

1. **Save changes as current version** – Save a new version that includes all the changes that have been made since the last saved version. When you select this button, you're prompted to provide a name for the version.

2. **Time series version** – Each saved version shows its version name next to a blue dot on the timeline. Select a version to view it in the chart and data table. You can collapse or expand each version to hide or view the log of edits that were made to create that version. The top version is always the current version. This version might include edits that have been made since the last saved version.

3. **Change log entry** – Change log entries are nested below the version where they were made. Each entry shows the date and time when the change was made, and the user who made it.

4. **Initial version** – The initial version and its change log entry represent the time series creation event. It's always saved as a version. Therefore, you can easily view or restore it as you require.

5. **Version edit button** – Open a menu where you can select any of the following actions:

- **Rename** – Enter a new name for the version.
- **Restore** – Make the selected version the current version.
- **Deactivate** – Deactivate the selected version. The values of newer versions aren't affected. Deactivated versions aren't shown in version selectors.
- **Activate** – Reactivate an inactive version.

Commenting and collaboration

You can add comments to a time series to communicate with other users. For example, you can explain why you chose to edit specific time series values or how to understand a saved data view. Users can also reply to each other's comments, edit or delete comments, and mark comment threads as resolved.

To view and work with comments, open a time series, and select the **Comments** button in the Action Pane at the top of the page. The commenting pane that appears shows all the comments that have been made so far for the current series. From here, you can perform the following actions:

- Select **New** to add a new comment.
- For exiting comments, select the **More thread actions** button ··· to open a menu where you can choose to edit or delete the selected comment or mark the selected thread as resolved.
- Reply to an existing comment by entering text in its **Reply** field.

As mentioned previously, you can also add comments to specific cells in the data table. To do this, select a cell and select the **Add a cell comment** button in the toolbar above the data grid. Cells that have comments show a comment symbol (purple speech bubble). To view the comment for a cell, select its comment symbol (or use the Copilot grid cursor). When you select the **Comments** button in the Action Pane at the top of the page, you'll see both the overall time series comments and all cell-specific comments.

Copilot grid cursor

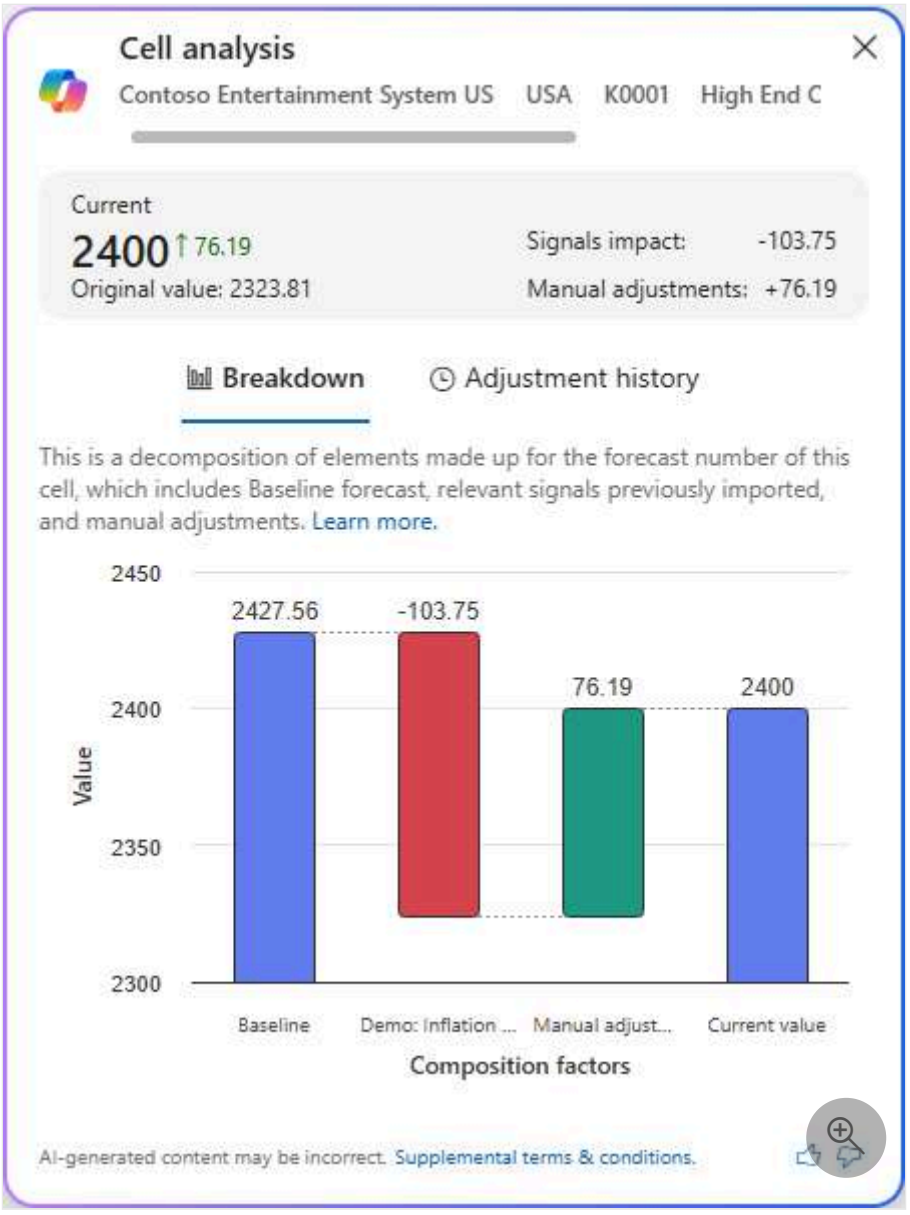
09/09/2025

The *Copilot grid cursor* offers detailed insights into a selected time-series cell's value, including its original value, manual adjustments, and full adjustment history. User comments are also shown, to help make the changes easier to understand.

To use the Copilot grid cursor, follow these steps:

1. Open the time series that you want to analyze. For example, on the navigation pane, select **Planning data** > **Forecasts**, and then select the time series.
2. In the **Time series values section**, find and select the cell that you want to analyze. After you select the cell, the Copilot grid cursor icon appears to the right of the cell and points to it.

3. Select the Copilot grid cursor icon to open the **Cell analysis** dialog for your selected cell.



4. Use the following features of the **Cell analysis** dialog to analyze the selected cell:

    - **Summary** – At the top of the dialog, you can see a summary of the cell's value and its history.
    - **Breakdown** – Select this button to see a graph that shows how the cell's value has been adjusted over time. The previous screenshot shows an example that indicates (from left to right) the baseline value (primary input), the signal impact, manual adjustments, and the current final value of a selected cell.
    - **Adjustment history** – Select this button to see a timeline that lists each time the cell was edited. To read the comments associated with a listed event, select the **Show details** button for the relevant row.

5. When you're done exploring the cell history, select the **Close** button in the upper-right corner of the dialog.

Generative insights for Demand planning (production-ready preview)

[This article is prerelease documentation and is subject to change.]

Generative insights offers detailed insights calculated using AI models. It currently supports two metrics: *seasonality* and *signal correlation*. The system clusters forecast data based on distinct patterns found for each metric.

Visual displays make it easy for planners to understand time series data and make business decisions based on it. Information is clustered (grouped) according to the detected patterns. For each cluster, generative insights provides a confidence score and describes the pattern in natural language. The system also indicates the proportion of planning items that follow the pattern.

> **Important**
>
> - This is a production-ready preview feature.
> - Production-ready previews are subject to supplemental terms of use.

To view generative insights, follow these steps:

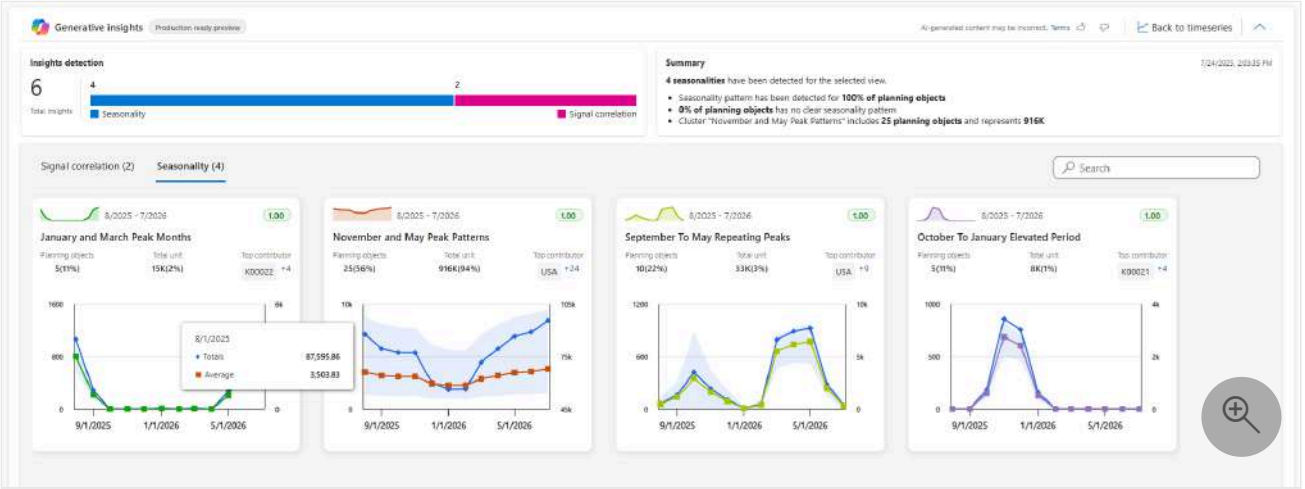
1. Open the time series that you want to analyze. For example, on the navigation pane, select **Planning data** > **Forecasts**, and then select a time series.
2. The **Generative insights** section appears near the top of the page. Do one of the following steps:
    - If you've never generated insights for the current time series, then select the **Generate insights** button available here. The system then runs the AI model to generate insights for the current time series. The process might take a few minutes, depending on the size of the time series.
    - If you have generated insights for the current time series, then summarized results are already shown. There's also a **Regenerate** button to regenerate the insights, which is useful if you've updated the time series since the last time you generated insights for it.

3. When insights are available, they are summarized in the **Generative insights** section (you might need to expand the section to see them). There are two tiles here:

    - **Insights detection** – Shows how many of each type of insight was detected in the time series.
    - **Summary** – Summarizes the insights that were detected.

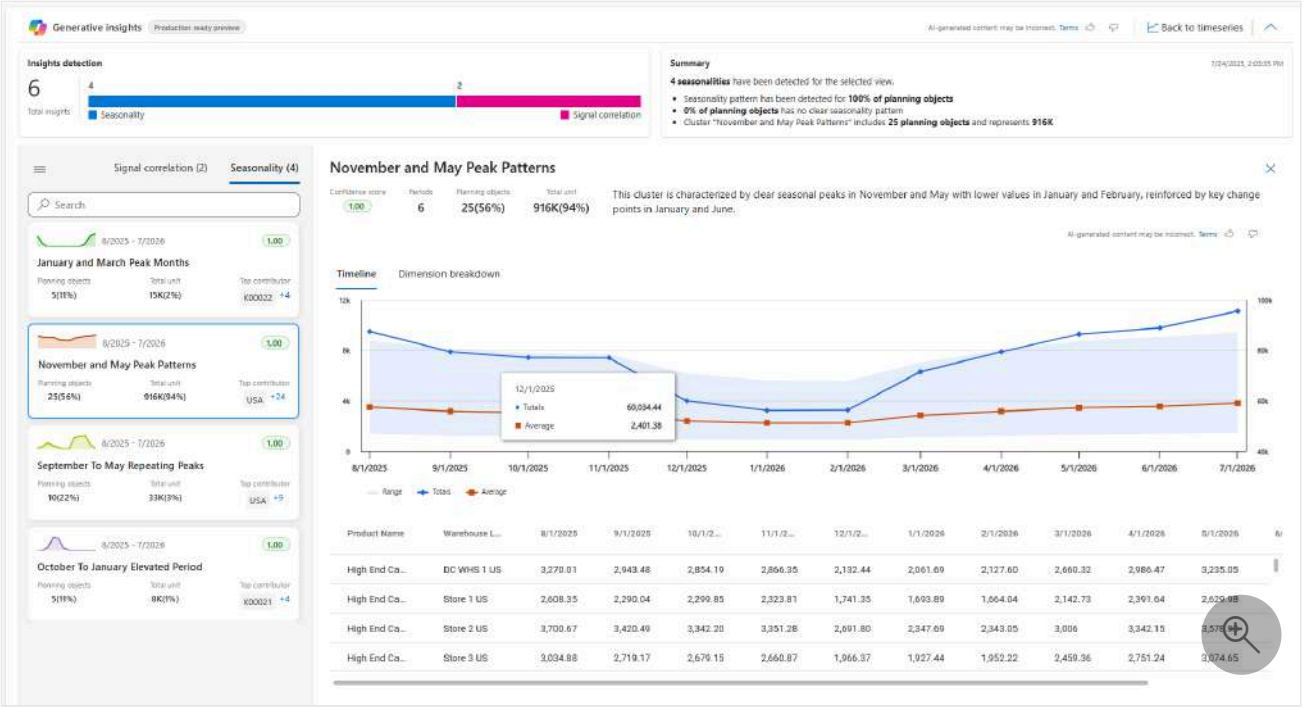
4. In the **Insight detection** tile, select **View insights** to see the details. Depending on what was detected, the system shows one or more of the following:

    - Several graphs of relevant data, with tabs that organize the data by insight type (**Signal correlation** and **Seasonality**). Select a tab to view insights of the indicated type.
    - A confidence score that indicates how confident the system is about each detected insight.
    - Various relevant KPI values for each insight.
    - A description of each insight, which is written in natural language. This description typically explains the detected pattern and its significance.



5. Select a graph on the **Signal correlation** or **Seasonality** tab to view even more details about it.

    - Some detailed graphs provide both a timeline view and data grid. Hover the mouse pointer to read timeline values. Select a row in the grid to add that row to the graph.
    - Some detailed graphs provide a dimension breakdown that lets you review the major contributors to the detected insights. For example, you can see which products or locations contributed most to the detected seasonality or signal correlation.



6. When you're done exploring the insights, select **Back to time series** to return to the time series.

Related information

- Responsible AI FAQ for Generative insights for Demand planning (production-ready preview)

Last updated on 05/13/2026

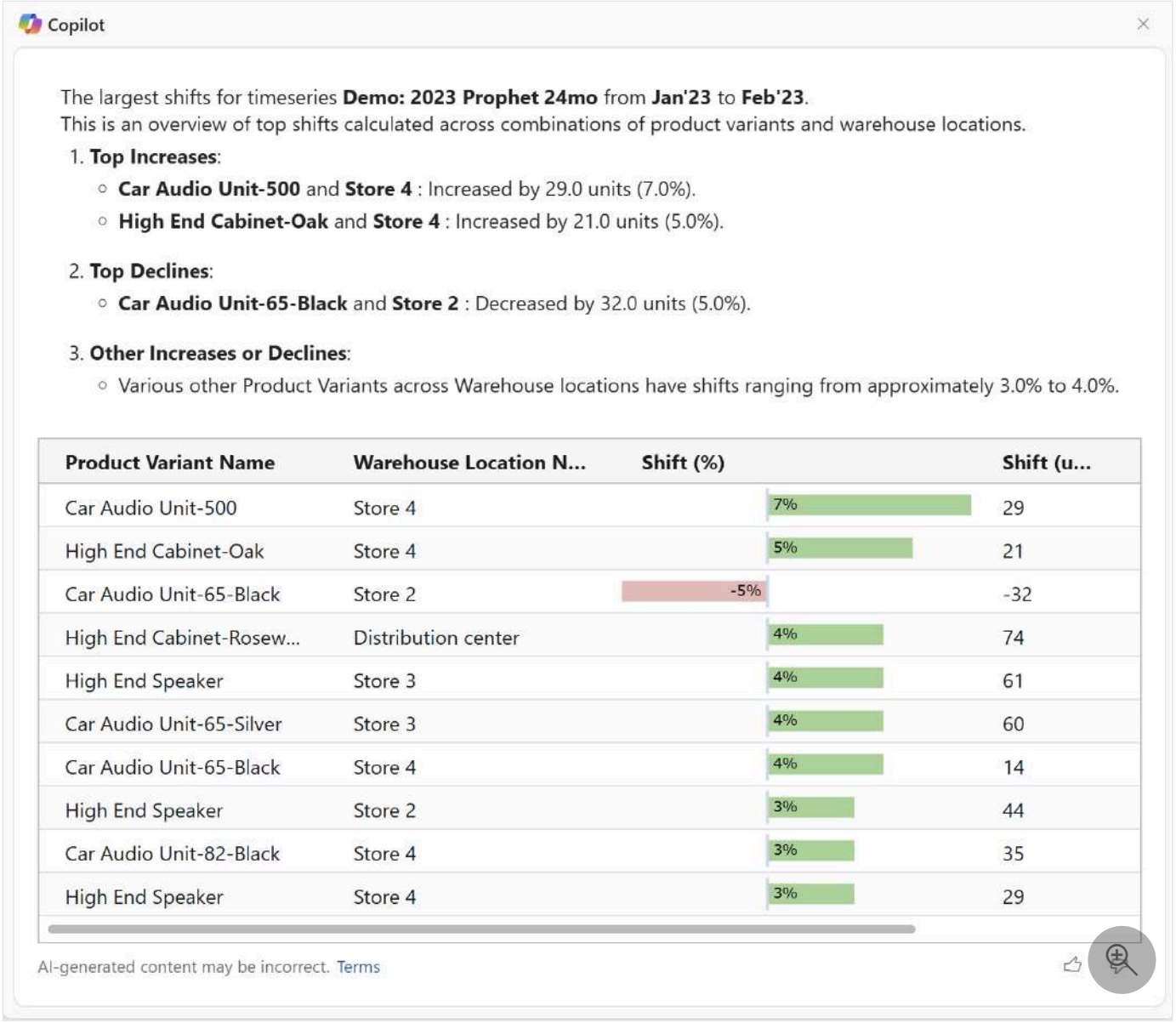
Analyze demand plans with Copilot

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Demand planning in Microsoft Dynamics 365 Supply Chain Management lets you use Copilot to analyze your demand plans. You can select from a set of predefined questions. The app then provides answers in real time by using natural language.

By using Copilot, you can quickly gain valuable insights across multiple dimensions. You can use these insights to improve operational efficiency, increase profitability, and improve customer satisfaction.

For example, a demand planner is interested in learning the most significant factors that contributed to a notable shift between two consecutive periods. The planner opens the relevant worksheet, selects a point of interest on the chart, and then selects a predefined question to ask Copilot. Copilot's response might resemble the following example.



Prerequisites

To use Copilot to analyze demand plans, you must be running the general availability (GA) version of Demand planning in Microsoft Dynamics 365 Supply Chain Management.

Country/region availability and language support

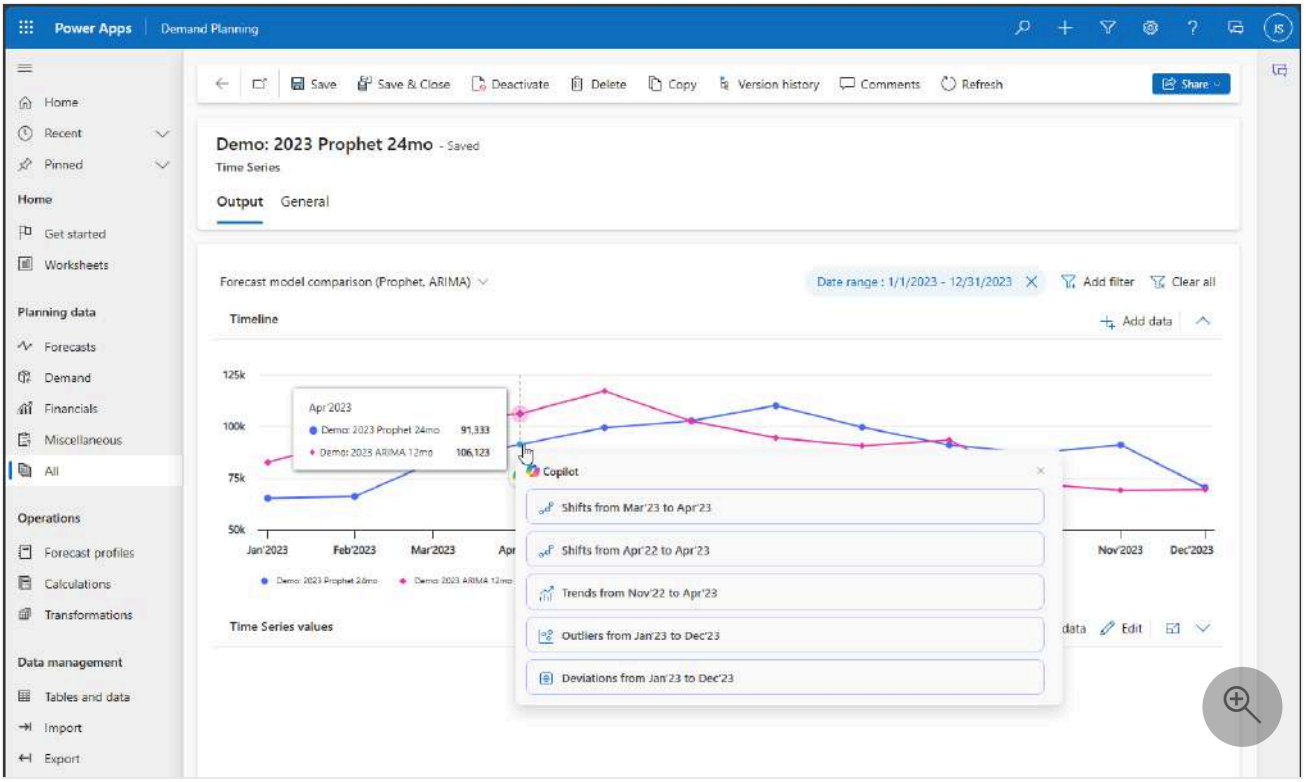
The *Analyze demand plans with Copilot* feature is available in all regions.

As of January 2025, the *Analyze demand plans with Copilot* feature is available in all languages supported by Demand planning.

Ask Copilot to analyze your data

To ask Copilot to analyze your data, follow these steps:

1. On the navigation pane, select **Worksheets**.
2. Open the worksheet that you want to analyze.
3. In the **Timeline** section, on the chart, select the point that represents the period that you want to analyze.
4. The **Copilot** menu shows one or more questions that you can ask about the selected point. Select the question that you want to ask. (For more information about the types of questions that can be shown, see the next section.)



5. Copilot analyzes the data and provides its answer.

Chat with Copilot

The following table provides a quick reference to the predefined questions that you can select on the **Copilot** menu.

Expand table

Example question	Description
Shifts from Apr '24 to May '24	Compare two consecutive periods in the same year. The length of the period (day, week, or month) depends on the setup of the time series in your worksheet. The example question that's shown here indicates that you selected a point that

Example question	Description
	represents the one-month period of May 2024. Copilot will compare that period to the previous one-month period (April 2024).
Shifts from May '23 to May '24	Compare a selected period to the same period one year earlier. The length of the period (day, week, or month) depends on the setup of the time series in your worksheet. The example question that's shown here indicates that you selected a point that represents the one-month period of May 2024. Copilot will compare that period to the one-month period of May 2023.
Trends from Nov '23 to Apr '24	Identify trends that occurred over the previous six time periods. The length of the period (day, week, or month) depends on the setup of the time series in your worksheet. The example question that's shown here indicates that you selected a point that represents the one-month period of April 2024. Copilot will analyze data across all available dimensions for the previous six months (November 2023 to April 2024) and highlight the most significant trends.
Outliers from Jan '24 to Dec '24	Analyze the full calendar year that includes the selected point and apply *interquartile range* (IQR) methodology to identify outlying values that occurred during that year. The example question that's shown here indicates that you selected a point that represents a period from 2024. Copilot will analyze data across all available dimensions to find outliers that occurred in the selected time series during the calendar year 2024.
Deviations from Jan '24 to Dec '24	Compare the selected time series to a second time series to identify the largest deviations that occurred between them during the selected calendar year. On choosing this question, Copilot asks you to choose a second time series included in the same worksheet. This question is only available if you're working with a worksheet that includes at least two time series. Copilot applies *mean average percentage error* (MAPE) methodology to identify the largest deviations occurring between the two selected time series over the selected calendar year.

Turn the Analyze demand plans with Copilot feature on or off

The *Analyze demand plans with Copilot* feature is on by default. Follow these steps to turn the feature on or off for your system. A system administrator of the environment must complete this procedure.

1. Sign in to the Power Apps maker portal.
2. Use the selector in the upper right of the page to select the environment where you want to turn the feature on or off.

3. Go to **Solutions**.
4. In the *Default Solution* row, select the link in the **Default name** column.
5. In the **Objects** list, select **Settings**.
6. In the *Copilot for demand planning* row, select the link in the **Display name** column.
7. In the **Edit Copilot for demand planning** dialog box, in the **Setting environment value** section, select **Add existing value**.
8. In the dropdown list, select *Yes* to enable the feature for the current environment, or select *No* to disable it.
9. Select **Save**.

Related information

- Responsible AI FAQ for Analyze demand plans with Copilot

Last updated on 05/07/2026

Limit manual time series edits with time fences

08/18/2025

Time fences let demand planning managers define rules that prevent users from *manually* editing time series values that are associated with a specified time span. They ensure that agreed-upon plans remain intact and unchanged during specified periods. Time fences are similar to time freezes, which are used to prevent the system from *automatically* editing certain existing time series values when you use rolling forecasts or when you manually rerun a forecast to update an existing time series. However, time fences only prevent *manual* updates.

For example, users might be prevented from editing specific time series values that fall within the current month. However, they can still edit values for the previous month or the next month. Every time fence uses the bucket size of the time series to establish a time span that starts in the current period and extends a fixed number of periods into the future.

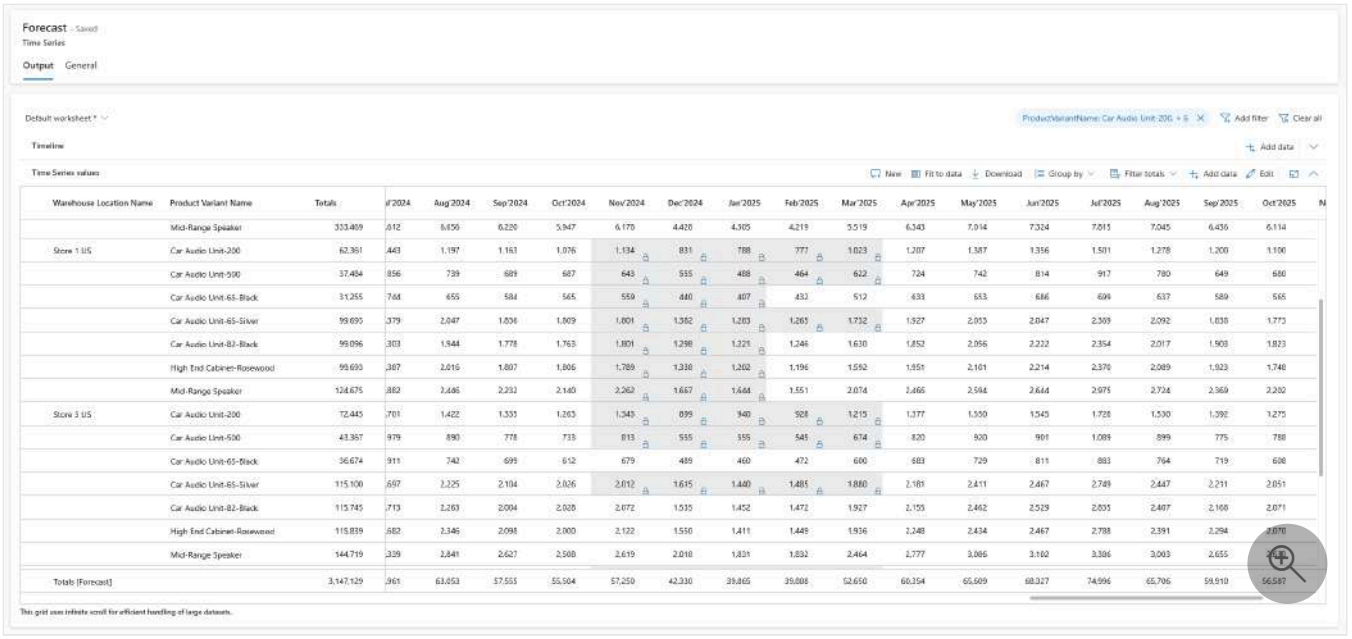
Time fences are defined as logical expressions and apply to all time series cells where the expression is true. For example, you could create a time fence that lasts two months and applies to product ID *K0001* in color *Red* for all time series that use *Monthly* time buckets.

Time fences are both flexible and easy to maintain. Managers create time fence rules based on the dimensions that are available in each plan. For example, a single product can be set up so that different time fences apply to each store or geographical location. Time fence rules can also apply based on each user's role. For example, a role-based time fence rule might allow managers to edit a forecast in a period that planners can't edit.

Time fence example

The following screenshot shows an example of a time series where two time fences apply. The time fences are represented by cells that show a lock symbol. This symbol indicates that the cells are locked. Therefore, the current user can't edit the values in them. However, the user can still edit the values in all unlocked cells. The following time fences apply here:

- Warehouse location name (**Warehouse location**) = *Store 1 US*; Time bucket = *Monthly*; Size = *Current period + 2 Time buckets*
- Product variant name (**Product**) = *Car Audio Unit-200, Car Audio Unit-500, Car Audio 65-Silver*; Time bucket = *Monthly*; Size = *Current period + 4 time buckets*



Manage time fences

To view, create, edit, or delete a time fence, follow these steps.

1. On the navigation pane, select **Configuration** > **Time fences**.

   The **Active time fence rules** page shows a list of active time fences. For each time fence, you can review the name, conditions, role, and other settings.

2. Follow one or more of these steps as required:

   - To create a new time fence, select **New** on the Action Pane.
   - To edit an existing time fence, select the link in the **Name** column.
   - To delete a time fence, select the row for it, and then, on the Action Pane, select **Delete**.
   - To view and edit inactive time fences, select the **Active time fence rules** heading, and then select **Inactive time fence rules** in the dropdown list.

3. If you chose to edit a time fence in the previous step, a tabbed window appears. If you choose to create a new time freeze, then a wizard launches, which offers similar settings. On the **Summary** edit tab (or **Get started** wizard page), set the following fields:

   - **Name** – Enter a name for the time fence.
   - **Description** – Enter a short description of the time fence.
   - **Owner** – Select the user account that owns the time fence record.
   - **Active** – Select whether the time fence should be active. (This option is only shown on the **Summary** edit tab. In the wizard, you can set this field on the **Review and finish** page.)

4. On the **Conditions** edit tab (or **Add conditions** wizard page), define the conditions under which the time fence applies. Use the **Add condition** and **Delete** (trashcan) buttons to add or remove rows as required. The rows are combined by using an *AND* operator. Therefore, the time fence locks only cells where the condition in *every* row is true. For each row, set the following fields:

    - **Table** – Select the data table that provides the cell value to compare.
    - **Column** – For the selected table, select the column that provides the cell value to compare.
    - **Operator** – Select the logic to apply to test the cell value against the row value. For example, select *equals*, *greater than*, or *less than*. There is also a *Select all* operator, which matches all values in the column; learn more in Using the select all operator.
    - **Value** – Enter a comma-separated list of values to compare the cell value to. You can either select among available values in a dropdown list or enter custom values. If you specify more than one value, the values are combined by using an *OR* operator. Therefore, the condition in the row is true for every time series cell that matches any one of the specified values. The **Value** field is disabled when you use the *Select all* operator.

5. On the **Time fence horizon** edit tab or wizard page, define the time span that the time fence applies to. The time span always starts in the time bucket that includes the current date. It then extends a fixed number of time buckets into the future. Set the following fields:

    - **Time buckets** – Specify the size of the time buckets that you want to use to define the time span. The time fence applies only to time series that also use this time bucket size. The system doesn't convert between time bucket sizes. For example, if you select *Monthly* here, your rule doesn't apply to time series that use *Weekly* time buckets (even though a month is about four weeks long).
    - **Current period +** – Specify the total number of time buckets, after the current one, to include in the time fence. The time fence always applies to the current time bucket. Therefore, if you want to include only the current time bucket in the time fence, set this field to *0* (zero). To include only the current time bucket and the next one, set this field to *1*.

    The **Example** area shows the time fence that results from your settings on this page.

6. On the **Role** edit tab (or **Select roles** wizard page), select the user security roles that the time fence applies to.

7. On the **Review and finish** wizard page (which isn't shown when you edit an existing time fence), review the settings that you made in the previous steps and set the **Active** option as needed. If everything looks correct, select **Review and Finish** to create the time fence.

Using the select all operator

To improve efficiency and make queries easier to formulate, Demand planning provides a *select all* operator for creating rules for time fences, time freezes, and row level access. The *Select all* operator matches all values in a selected column and doesn't require a value to compare to.

The following table provides an example that shows how the *select all* operator can be used in a time fence rule. As a result of the rules, all products except *Product A* have a two-month time fence. *Product A* has a three-month time fence.

Expand table

Conditions				**Ruling**			
Table	**Column**	**Operator**	**Value**	**Time bucket**	**Time bucket size**	**Role**	**Status**
Product	Product name	Select all	Not applicable	Monthly	2 months	All	Active
Product	Product name	Equals	Product A	Monthly	3 months	All	Active

Overlapping time fences

In situations where time fences overlap, the more specific condition applies. Here are some examples:

- A time fence for *all users* prevents a specific cell from being edited, but the time fence for the *manager* user role allows the cell to be edited. In this case, a user who has the *manager* role is allowed to edit the cell.
- A time fence of four months is set for *all products*, but a time fence of two months is set for *product 00001*. In this case, the time fence for *product 0001* is two months, but is four months for all other products that don't have more specific rules defined for them.

Limit automatic time series updates with time freezes

09/09/2025

Time freezes let planners define rules that prevent the system from automatically updating selected cells in existing time series when a forecast gets recalculated (such as when you use rolling forecasts or when you manually rerun a forecast to update an existing time series). Time freezes are similar to time fences, which are used to prevent users from *manually* editing certain time series values. However, time freezes only prevent *automated* updates that would otherwise occur when you rerun a forecast; users can still edit those values manually in the time series.

Unlike time fences (which apply to all time series that match the time fence logic), you must explicitly configure each relevant *Forecast* and *Forecast with signals* step to use the time freezes that should apply to it. You can assign any number of time freezes to each step, and you can also assign no time freezes at all. If you don't assign any time freezes to a step, then the step won't use any time freeze rules.

For example, the system might be prevented from updating specific forecast values that fall within the next three months. However, the system can still update values for the previous month or for four months from now. Time freezes use the bucket size of the time series to establish a time span that starts in the current period and extends a fixed number of periods into the future.

There are two types of time freezes:

- **Time freeze rules** – Time freeze rules are defined as logical expressions. For example, you could create a time freeze rule that lasts two months and applies to product ID *K0001* in color *Red*. The rule would then apply for all *Forecast* and *Forecast with signals* steps that have that time freeze assigned.
- **Freeze manual adjustments** – This type of time freeze is a simple on/off setting that you can enable for each *Forecast* and *Forecast with signals* step. When it's enabled, the system prevents automatic updates to any time series cells that have been manually adjusted by a user. No rules are required.

Manage time freeze rules

To view, create, edit, or delete a time freeze rule, follow these steps.

1. On the navigation pane, select **Configuration** > **Time freezes**.

The **Active time freeze rules** page shows a list of active rules. For each rule, you can review the name, conditions, and other settings.

2. Follow one or more of these steps as required:

   - To create a new time freeze rule, select **New** on the Action Pane.
   - To edit an existing time freeze rule, select the link in the **Name** column.
   - To delete a time freeze rule, select the row for it, and then, on the Action Pane, select **Delete**.
   - To view and edit deactivated time freeze rules, select the **Active time freeze rules** heading, and then select **Inactive time freeze rules** in the dropdown list. Other views are also available here, including views for viewing only daily, monthly, or weekly time freeze rules.

3. If you chose to edit a time freeze rule in the previous step, a tabbed window appears. If you choose to create a new time freeze, then a wizard launches, which offers similar settings. On the **Summary** edit tab (or **Get started** wizard page), set the following fields:

   - **Name** – Enter a name for the rule.
   - **Description** – Enter a short description of the rule.
   - **Owner** – Select the user account that owns the time freeze rule record.
   - **Active** – Select whether the rule should be active. Inactive rules have no effect, even when explicitly assigned to step. (This option is only shown on the **Summary** edit tab. In the wizard, you can set this option on the **Review and finish** page.)

4. On the **Conditions** edit tab (or **Add conditions** wizard page), define the conditions under which the rule applies. Use the **Add condition** and **Delete** (trashcan) buttons to add or remove rows as required. The rows are combined by using an *AND* operator. Therefore, the rule locks only cells where the condition in *every* row is true. For each row, set the following fields:

   - **Table** – Select the data table that provides the cell value to compare.
   - **Column** – For the selected table, select the column that provides the cell value to compare.
   - **Operator** – Select the logic to apply to test the cell value against the row value. For example, select *equals*, *greater than*, or *less than*. There's also a *Select all* operator, which matches all values in the column (learn more in Using the select all operator).
   - **Value** – Enter a comma-separated list of values to compare the cell value to. You can either select among available values in a dropdown list or enter custom values. If you specify more than one value, the values are combined by using an *OR* operator. Therefore, the condition in the row is true for every time series cell that matches any one of the specified values. The **Value** field is disabled when you use the *Select all* operator.

5. On the **Time freeze horizon** edit tab or wizard page, define the time span that the time freeze rule applies to. The time span always starts in the time bucket that includes the current date. It then extends a fixed number of time buckets into the future. Set the following fields:

    - **Time buckets** – Specify the size of the time buckets that you want to use to define the time span. The time freeze rule can only be used with forecast profiles that also use this time bucket size. The system doesn't convert between time bucket sizes. For example, if you select *Monthly* here, your rule won't be available to forecast models in profiles that use *Weekly* time buckets (even though a month is about four weeks long).
    - **Current period +** – Specify the total number of time buckets, after the current one, to include in the time freeze rule. The rule always applies to the current time bucket. Therefore, if you want to include only the current time bucket in the rule, set this field to *0* (zero). To include only the current time bucket and the next one, set this field to *1*.

    The **Example** area shows the time freeze rule that results from your settings on this page.

6. On the **Review and finish** wizard page (which isn't shown when you edit an existing rule), review the settings that you made in the previous steps and set the **Active** option as needed. If everything looks correct, select **Review and Finish** to create the rule.

Add a time freeze rule to a forecast profile

As mentioned, you must explicitly configure each relevant *Forecast* and *Forecast with signals* step to use the time freeze rules that should apply to it. To do this, follow these steps:

1. Create the time freeze records that you want to use, as described in the previous sections of this article.
2. Create or open a relevant forecast profile as described in Work with forecast profiles.
3. On the **Forecast model** tab, crate or select the step that you want to add a time freeze rule to. The step must be a *Forecast* or *Forecast with signals* step. (Learn more in Design forecast models.) If you're editing an existing step, open the **Actions** menu for the step and then select **Settings**.
4. In the Freeze values section, select the **Enable time freeze** check box.
5. Select the **Use time freeze** rules radio button.
6. Use the **Time freeze rules** field to search for and select the time freeze rules that you want to apply to the step.

Freeze manual adjustments for a selected forecast profile

To prevent the system from automatically updating any time series cells that have been manually adjusted by a user, use the following procedure. You don't need to create any time freeze rules to use this feature.

1. Create or open a relevant forecast profile as described in Work with forecast profiles.
2. On the **Forecast model** tab, crate or select the step that you want to add a time freeze rule to. The step must be a *Forecast* or *Forecast with signals* step. (Learn more in Design forecast models.) If you're editing an existing step, open the **Actions** menu for the step and then select **Settings**.
3. In the Freeze values section, select the **Enable time freeze** check box.
4. Select the **Freeze manual adjustments** rules radio button.

Clean up time series data

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Almost every operation that changes a time series in Demand planning—such as running a transformation, calculation, or forecast—creates a new version. This version history is valuable, but over time, unused versions and entire time series that you no longer need accumulate and consume storage.

The *data cleanup* feature lets you review everything that's stored and permanently delete the versions and time series you no longer need. The feature shows you exactly what you're removing, including its size and anything that still references it, so you can make informed decisions before deleting.

> **Important**
>
> Deletion is permanent and can't be undone. Always review the details shown in the confirmation dialog before proceeding.

Required security role

Only users with the *Demand planning manager* role can access data cleanup. This restriction ensures that only authorized users can permanently remove data.

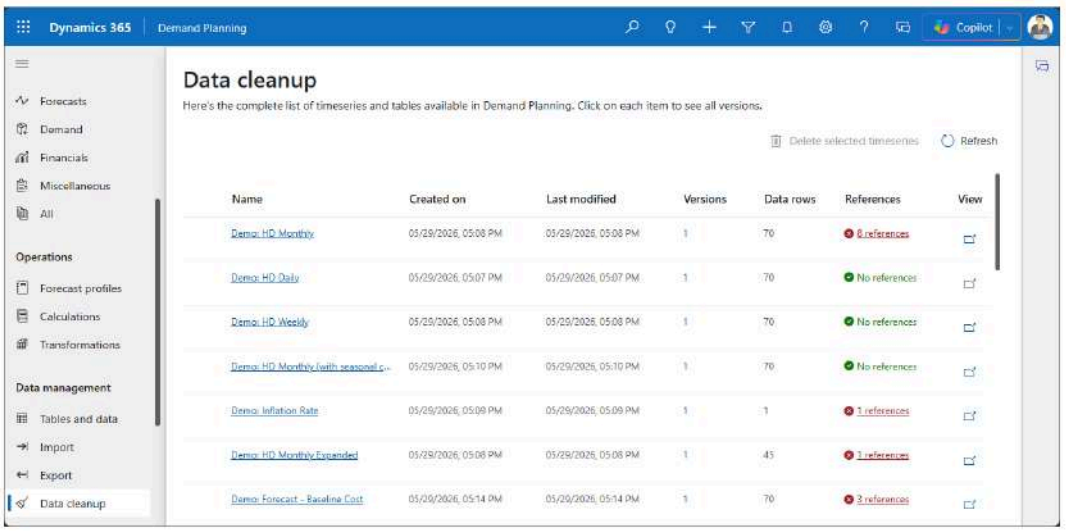
Open the data cleanup page

To review and clean up time series data, open the Demand planning app, go to the navigation pane, and select **Data management** > **Data cleanup**.

The **Data cleanup** page shows a list of all time series in the system. For each time series, you can see the number of versions, the number of data rows, and whether any references to the time

series or its versions still exist.

- Select a time series name to see its versions.
- Select the link in the **References** column to open a list of profiles that are using the time series as input.
- Select the link in the **View** column to open the time series itself.

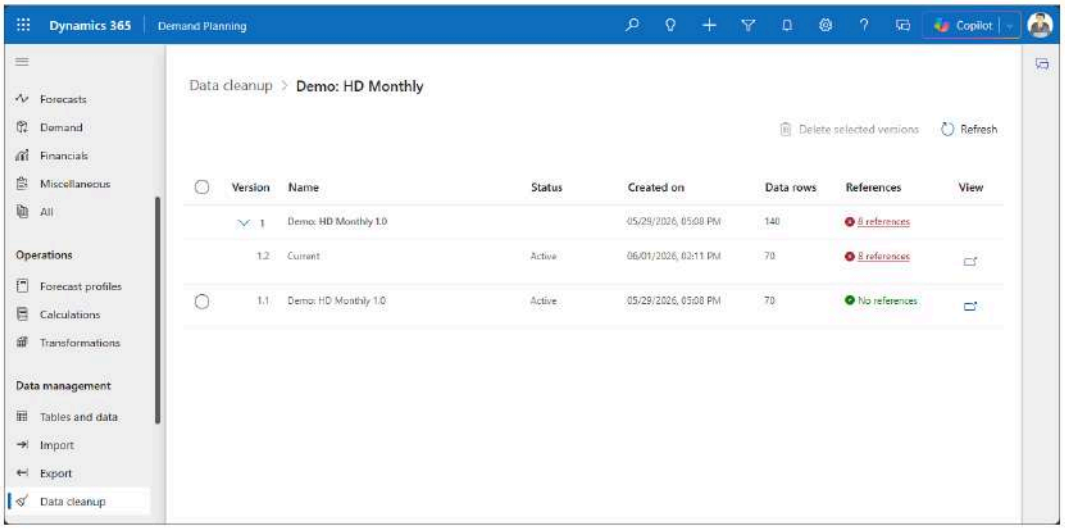


View time series versions

To view the versions of a time series, open the the **Data cleanup** page and select the link in the **Name** column for the target time series.

The versions view shows a list of all major and minor versions of the selected time series, along with details such as their status, creation date, number of data rows, and references.

- The newest versions appear first. The version currently in use is named *Current*.
- Expand or collapse major versions to show or hide their minor versions.
- Select the link in the **References** column to open a list of profiles that use the time series version as input.
- Select the link in the **View** column to open the version itself.



Understand time series versions

Versions are organized in two levels:

- **Major versions** – Created by a job or action such as a transformation, calculation, forecast, revert, or copy. They're numbered sequentially (1, 2, 3, and so on).
- **Minor versions** – Individual data versions within a major version. They're numbered hierarchically (1.1, 1.2, and so on). Users name minor versions and they typically represent saved edits of the major version.

Learn more about how to create and manage versions in Time series, worksheets, and planning data.

Understand version statuses

Each version listed on the version view of the **Data cleanup** page shows one of the following status values:

- *Active* – The version exists normally and is available for use.
- *Marked for deletion* – You confirmed the deletion of this version. A background cleanup process will soon remove the version. This is a normal intermediate state, not an error.
- *Failed deletion* – A cleanup attempt didn't finish, typically because of a temporary issue. The system retries these automatically, so a version might briefly show this status before the retry succeeds.

Understand references

When you use a time series or time series version as input for a calculation or forecast profile, the system creates a reference between the time series and the profile. To preserve data integrity, the

system doesn't allow you to delete a time series or version that is referenced by a profile. You must remove all references to a time series or version before you can delete it. The **Data cleanup** page shows the number of references for each version and provides tools for viewing and removing them, as described later in this article.

Delete time series versions

To delete one or more versions of a time series, follow these steps:

1. On the navigation pane, select **Data management** > **Data cleanup**.
2. Find the time series that has the versions you want to delete and select the link in the **Name** column to open the versions view for that time series.
3. Find a version you want to delete and check the **References** column for that time series version. If it shows that references exist, do the following steps:

    - Select the link in the **References** column to open a dialog that lists the profiles that use your target time series version.
    - Open each listed profile and either delete it or change its settings to use a different input.
    - When you've removed all references, close the dialog to return to the **Data cleanup** page.

4. Repeat the previous step as needed for each version you want to delete.
5. In the versions view, select the check box for each version you want to delete. Selecting a major version automatically selects all of its eligible minor versions, and clearing it clears them. You can't select the *Current* version or any version that has references.
6. On the toolbar, select **Delete selected versions**.
7. You're asked to confirm the deletion. Select **Confirm** to proceed, or **Cancel** to back out.

> ⓘ **Note**
>
> Deleting versions of a time series also affects worksheets that include them. If the deleted version is the main version for a worksheet, the system also deletes the worksheet. If a deleted version isn't the main version, then the system just deletes its reference from the worksheet, which means that the worksheet still appears but without the deleted version.

Delete an entire time series

To delete one or more complete time series, including all of their versions and data, follow these steps.

1. On the navigation pane, select **Data management** > **Data cleanup**.
2. Find the time series you want to delete.
3. Check the **References** column for your selected time series. If it shows that references exist, do the following steps:

    - Select the link in the **References** column to open a dialog that lists the profiles that use your target time series.
    - Open each listed profile and either delete it or change its settings to use a different input.
    - When you've removed all references, close the dialog to return to the **Data cleanup** page.

4. For your target time series, select the checkbox in the left column.
5. On the toolbar, select **Delete selected timeseries**.
6. You're asked to confirm the deletion. Select **Confirm** to proceed, or **Cancel** to back out.

> ⓘ **Note**
>
> Deleting time series also affects worksheets that include them. If the deleted time series is the main time series for a worksheet, the system also deletes the worksheet. If a deleted time series isn't the main time series, then the system just deletes its reference from the worksheet, which means that the worksheet still appears but without the deleted time series.

Related information

- Time series, worksheets, and planning data
- Work with calculation profiles
- Work with forecast profiles

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Last updated on 06/03/2026

Export and download data

Article • 01/10/2025

When you've finished creating, analyzing, and editing a forecast, you can export it back to Microsoft Dynamics 365 Supply Chain Management. Alternatively, you can download the data as a comma-separated values (CSV) file that can be viewed in Excel.

Export to Supply Chain Management

View and run existing data export profiles to export to Supply Chain Management

Demand planning lets you build a collection of *export profiles*. Each Supply Chain Management export profile exports data to a specific Supply Chain Management instance. Typically, a manager or system administrator creates the initial collection of required profiles. Forecasters and other users can then run the profiles to export as they require.

To run an existing data export profile, follow these steps.

1. On the navigation pane, select **Data management** > **Export**.
2. Find the profile for the type of export that you want to run, and select the link for it in the **Name** column.

   The details page for the selected profile appears. Supply Chain Management export profiles contain the following tabs:

   - **Get started** – This tab provides basic information about the profile. You can edit the name and/or description to make the profile easier to identify and work with.
   - **Map columns** – This tab shows how the selected profile maps columns in Demand planning to columns in the target system. You can edit the mappings from here as you require.
   - **Define data export rules** – This tab shows any export rules that have been defined for the profile. You can edit the settings from here as you require.
   - **Run schedule** – This tab lets you set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
   - **Jobs** – This tab shows a list of every run of the profile.

3. To run the profile, select **Run** on the Action Pane. This command adds a new row to the grid on the **Jobs** tab. There, you can follow the status of the new export. The page isn't automatically refreshed. To update the status information, you must select **Refresh** on the grid toolbar.

Create and manage profiles for export to Supply Chain Management

Each time that your organization has to run a new type of data export, a manager or admin must create a new data export profile. After the profile is created, it becomes available to users, who can run it as often as they require. Each profile exports to either Supply Chain Management or a CSV file, depending on how you set it up. This section describes how to create a profile that exports to your selected Supply Chain Management instance.

1. On the navigation pane, select **Data management** > **Export**.
2. On the Action Pane, select **New**.
3. On the **Select data provider** page, select the **Microsoft finance and operations apps** tile.
4. A setup wizard is opened. On the **Get started** page, enter a name and description for the new profile. Then select **Next**.
5. On the **Configure data provider** page, in the **Connection URL** field, enter the URL of your Supply Chain Management environment. Then select **Next**.
6. On the **Select output data** page, select the time series to export to Supply Chain Management. You must select exactly one time series.

    - The **Available** tab shows a list of available time series. Select the time series to export in the grid, and then select **Include data source** on the toolbar. By default, the most recent version of the selected time series is exported. However, you can select older versions on the **Included** tab.

    - The **Included** tab shows the time series that's selected for export.
        - If more than one version of the time series is available, select the version that you want to use in the **Output version** field.
        - To remove a time series, select it, and then select **Remove** on the toolbar.
        - To set up one or more filter rules to apply to the exported data, select **Filter data source** on the toolbar.

7. When you've finished selecting and setting up the time series to export, select **Next**.

8. On the **Map columns** page, use the dropdown lists to map each column in the selected time series to the appropriate column in the target Supply Chain Management data entity. When you've finished, select **Next**.

9. On the **Define data export rules** page, specify the target company (legal entity) and forecast model ID to export the data to. You can select **All** to export all the data to the same company and forecast model, or you can select **Custom** to set up rules for splitting the export between different forecast models.

10. Select **Next**.

11. On the **Set run schedule** page, you can choose to set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.

12. Select **Next**.

13. On the **Review and finish** page, review the summary of settings that you've configured, and then select **Review and finish** to create the new profile.

14. The profile is now available, but it hasn't yet run. To run it, follow the instructions in the View and run existing data export profiles to export to Supply Chain Management section.

Export to a CSV file

Download a time series values grid as a CSV file

You can download the **Time series values** grid from any selected time series as a CSV file. The CSV file presents the data just as it's presented in the **Time series values** grid for the time series. In other words, it has a row for each combination of dimensions and a column for each dimension, the grand total value, and the value for each available date.

If you require a file that has a row for each date instead, you should create a CSV export profile and run it as described in the Create and manage profiles for export to CSV section. This type of CSV file might be more useful for importing data into other systems.

To export the **Time series values** grid from any selected time series to a CSV file, follow these steps.

1. Open the time series that you want to export from.
2. On the **Time series values** FastTab, select **Download** on the toolbar.

The file is saved as a CSV file in your local download folder.

View and run existing data export profiles to export to CSV

Demand planning lets you build a collection of *export profiles*. Each CSV export profile exports data from a selected time series to a CSV file that's saved on your local computer. The CSV file includes a row for each date and columns for the date, value, and each dimension.

Typically, a manager or system administrator creates the initial collection of required profiles. Forecasters and other users can then run the profiles to export as they require.

To run an existing data export profile, follow these steps.

1. On the navigation pane, select **Data management** > **Export**.

2. Find the profile for the type of export that you want to run, and select the link for it in the **Name** column.

   The details page for the selected profile appears. CSV export profiles contain the following tabs:

   - **Get started** – This tab provides basic information about the profile. You can edit the name and/or description to make the profile easier to identify and work with.
   - **Run schedule** – This tab lets you set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
   - **Jobs** – This tab shows a list of every run of the profile.

3. To run the profile, select **Run** on the Action Pane. This command adds a new row to the grid on the **Jobs** tab. There, you can follow the status of the new export. The page isn't automatically refreshed. To update the status information, you must select **Refresh** on the grid toolbar.

4. When the job is finished, select **Download** on the Action Pane to download the data as a CSV file and save it in your local download folder.

Create and manage profiles for export to CSV

Each time that your organization has to run a new type of data export, a manager or admin must create a new data export profile. After the profile is created, it becomes available to users, who can run it as often as they require. Each profile exports to either Supply Chain Management or a CSV file, depending on how you set it up. This section describes how to create a profile that exports to CSV.

1. On the navigation pane, select **Data management** > **Export**.
2. On the Action Pane, select **New**.
3. On the **Select data provider** page, select the **CSV** tile.
4. A setup wizard is opened. On the **Get started** page, enter a name and description for the new profile. Then select **Next**.
5. On the **Configure data provider** page, in the **Connection URL** field, enter the URL of your Supply Chain Management environment. Then select **Next**.
6. On the **Select output data** page, select the time series to export to CSV. You must select exactly one time series.
    - The **Available** tab shows a list of available time series. Select the time series to export in the grid, and then select **Include data source** on the toolbar. By default, the most recent version of the selected time series is exported. However, you can select older versions on the **Included** tab.
    - The **Included** tab shows the time series that's selected for export.
        - If more than one version of the time series is available, select the version that you want to use in the **Output version** field.
        - To remove a time series, select it, and then select **Remove** on the toolbar.
        - To set up one or more filter rules to apply to the exported data, select **Filter data source** on the toolbar.
7. When you've finished selecting and setting up the time series to export, select **Next**.
8. On the **Set run schedule** page, you can set up a schedule for the profile to run automatically. For details about this functionality and how to configure it, see Rolling forecasts.
9. Select **Next**.

10. On the **Review and finish** page, review the summary of settings that you've configured, and then select **Review and finish** to create the new profile.

11. The profile is now available, but it hasn't yet run. To run it, follow the instructions in the View and run existing data export profiles to export to CSV section.

Improve performance by clearing unneeded staging data

If you're experiencing performance issues, it might help to clear staging data from the *Forecast Sales Import Entity* entity in Supply Chain Management. To do this, take one of the following actions:

- Manually clear the staging data as needed.
- Set up an automatic cleanup process to clear the staging data on a regular basis.

Feedback

Was this page helpful? Yes No

Provide product feedback

Export data using custom APIs

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Overview

You can programmatically export Demand planning data (forecasts and time series) to external systems using Dataverse custom API actions. This approach enables you to automate export workflows without needing to manually interact with the Demand planning interface. The custom APIs support retrieving exported data as comma-separated values (CSV) files through secure shared access signature (SAS) URLs.

Dataverse custom API actions provide the public interface for export operations. You can call them using:

- **Dataverse Web API** – OData HTTP requests for external applications and services
- **Power Platform tools** – Native integration with Power Automate, custom connectors, and Dataverse plugins

> **Note**
>
> This article describes programmatic export using Dataverse custom APIs. Learn how to export data using the Demand planning interface in Export and download data.

Prerequisites

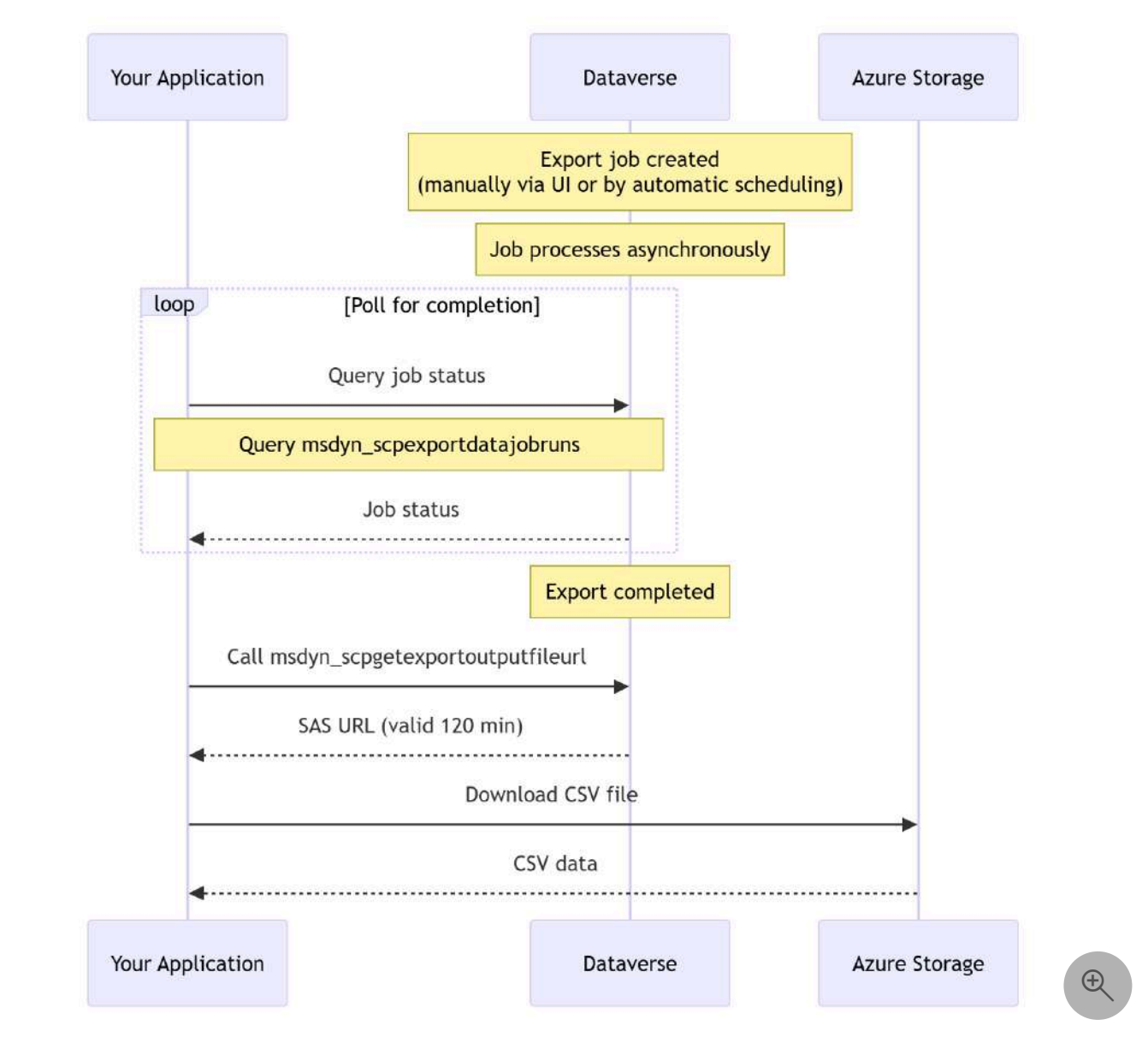
Before you use the export custom APIs, make sure you have:

- An export profile configured in Demand planning. Learn how to create export profiles in Export and download data.
- Access to the Dataverse environment for your Demand planning instance.

> **Tip**
>
> To find your export profile ID, open the export profile in the Demand planning interface and copy the GUID from the browser URL. Alternatively, query the `msdyn_scpexportdataprofile` entity using the Dataverse Web API to retrieve profile IDs programmatically.

Export data programmatically

The following diagram shows the workflow for exporting data using Dataverse custom APIs:



To export Demand planning data using Dataverse custom APIs, follow these steps:

1. **Create an export job** – Create an export job using the Demand planning interface. You can create jobs manually or configure scheduled jobs within the Demand planning

application. The job creation process runs asynchronously, so the export job is created immediately and then processes in the background.

2. **Monitor job status** – Poll the Dataverse **Export Job Run** entity (`msdyn_scpexportdatajobruns`) to track job progress. Query the entity using Dataverse Web API OData queries to check the `msdyn_jobrunstatus` field. Learn more in Monitor job status.

3. **Retrieve the file URL** – When the job status shows as completed (status value 192350002), call the `msdyn_scpgetexportoutputfileurl` custom API action to retrieve a SAS URL for downloading the exported file. The URL is valid for 120 minutes. Learn more in Power Platform integration.

4. **Download the file** – Use the returned SAS URL to download the CSV file. The file contains your exported Demand planning data in the format described in the Export file format section.

Monitor job status

After scheduling an export job, monitor its progress by querying the Dataverse **Export Job Run** entity. Export jobs run asynchronously, so you need to poll for status updates until the job completes.

Entity details

Query the following Dataverse entity to check export job status:

Expand table

Property	Value
Entity name	`msdyn_scpexportdatajobrun`
Display name	Export Job Run

The following table lists the key fields in the `msdyn_scpexportdatajobrun` entity that are relevant for monitoring export jobs:

Expand table

Field name	Description
`msdyn_jobrunstatus`	Current job status (see status values in the next section)
`msdyn_name`	Job name provided when scheduling
`msdyn_starttime`	Job start timestamp
`msdyn_endtime`	Job completion timestamp
`msdyn_errormessage`	Error details if the job failed
`_msdyn_profile_value`	Associated export profile ID (lookup field)

Job status values

The `msdyn_jobrunstatus` field uses the following integer values:

Expand table

Status	Value	Description
Created	192350000	Job is created but not started yet
Executing	192350001	Job is currently running
Completed	192350002	Job finished successfully
Failed	192350003	Job encountered an error
Canceling	192350004	Job cancellation in progress
Canceled	192350005	Job was cancelled

Query examples

Use Dataverse Web API OData queries to retrieve job status. The base query pattern is:

OData

```
GET [Organization URI]/api/data/v9.2/msdyn_scpexportdatajobruns?$filter={filter expression}&$select={columns}
```

The following examples show common query patterns for monitoring export jobs.

Filter by completed status:

OData

```
GET [Organization URI]/api/data/v9.2/msdyn_scpexportdatajobruns?
$filter=msdyn_jobrunstatus eq 192350002&$select=msdyn_name,msdyn_endtime
```

Filter by export profile:

OData

```
GET [Organization URI]/api/data/v9.2/msdyn_scpexportdatajobruns?
$filter=_msdyn_profile_value eq 11111111-1111-1111-1111-
111111111111&$select=msdyn_name,msdyn_jobrunstatus
```

Combine multiple filters:

OData

```
GET [Organization URI]/api/data/v9.2/msdyn_scpexportdatajobruns?
$filter=msdyn_jobrunstatus eq 192350002 and _msdyn_profile_value eq 11111111-1111-
1111-1111-111111111111
```

> **Note**
>
> When calling Dataverse Web API endpoints, authenticate by using one of the supported methods. Options include OAuth tokens, Azure managed identities, Azure DefaultAzureCredential, or service principal authentication. For detailed guidance on authentication options and implementation, see Authenticate with Microsoft Dataverse web services.

> **Tip**
>
> For Dataverse-native integrations, you can also use the `msdyn_scpgetexportoutputfileurl` custom API action to retrieve the output file URL directly from Dataverse.

For more information about Dataverse Web API queries, see Query data using the Web API.

Power Platform integration

For Power Platform scenarios such as Power Automate flows, custom connectors, or Dataverse plugins, use Dataverse custom API actions. These actions provide native integration with the Power Platform ecosystem.

Available custom API actions

The following custom API action is available for export operations:

Expand table

Action name	Description	Use case
`msdyn_scpgetexportoutputfileurl`	Retrieves the SAS URL for a completed export file	Downloading exported data in Power Automate flows or custom code

Use the custom API action

The `msdyn_scpgetexportoutputfileurl` action retrieves a SAS URL for downloading the exported CSV file from a completed export job.

The following table lists the input parameters required to call the `msdyn_scpgetexportoutputfileurl` custom API action:

Expand table

Parameter name	Type	Required	Description
`msdyn_exportprofileid`	String (GUID)	Yes	The ID of the export profile

The following table describes the output returned by the `msdyn_scpgetexportoutputfileurl` custom API action:

Expand table

Property name	Type	Description
`output`	String	The SAS URL for downloading the exported CSV file (valid for 120 minutes)

Use the `msdyn_scpgetexportoutputfileurl` custom API action in the following scenarios:

- **Power Automate flows** – Use the *Perform an unbound action* step to call the custom API action natively.
- **Custom connectors** – Invoke the custom API action through Dataverse connectors.
- **External applications** – Call the custom API action via Dataverse Web API (OData HTTP requests).
- **Dataverse plugins** – Execute the custom API action from within plugin code.

Example: Power Automate flow

To retrieve an export file URL in Power Automate:

1. Add the *Perform an unbound action* step.
2. Select the action name `msdyn_scpgetexportoutputfileurl`.
3. Enter the export profile ID as `msdyn_exportprofileid`.
4. Use the `output` property to download the file or pass the URL to subsequent steps.

> **Note**
>
> The custom API action requires Dataverse authentication and appropriate permissions. Make sure your application or Power Platform connection has access to the Dataverse organization where Demand planning is deployed.

For more information about using custom API actions in Power Platform, see Use custom APIs.

Walkthrough: Automatically download exports to SharePoint

This walkthrough shows how to create a Power Automate flow that automatically downloads completed export files to a SharePoint folder. The flow monitors export job completions and retrieves the CSV file by using the custom API action.

Scenario

When an export job completes in Demand planning, the flow automatically:

1. Detects the completed job
2. Retrieves the export file URL by using the custom API action
3. Downloads the CSV file
4. Saves it to a designated SharePoint folder

Scenario prerequisites

To implement this scenario, you need the following prerequisites:

- Access to Power Automate with a Dataverse connection
- SharePoint site and folder for storing export files
- Export profile configured in Demand planning

Create the flow

To create the flow, follow these steps:

1. **Add trigger** – Use the *When a row is added, modified or deleted* trigger

    - Set **Change type** – Added, Modified, or Deleted
    - Set **Table name** – Export Data Job Runs (`msdyn_scpexportdatajobruns`)
    - Set **Filter rows** – `msdyn_jobrunstatus eq 192350002` (filters for completed jobs only)
    - Optional: Add `_msdyn_profile_value eq {your-profile-guid}` to filter for specific export profiles

2. **Get export file URL** – Add the *Perform an unbound action in selected environment* step

    - Set **Environment** – Your Dataverse environment
    - Set **Action name** – `msdyn_scpgetexportoutputfileurl`
    - Set **msdyn_exportprofileid** – The known export profile ID, or use the profile ID from the trigger output

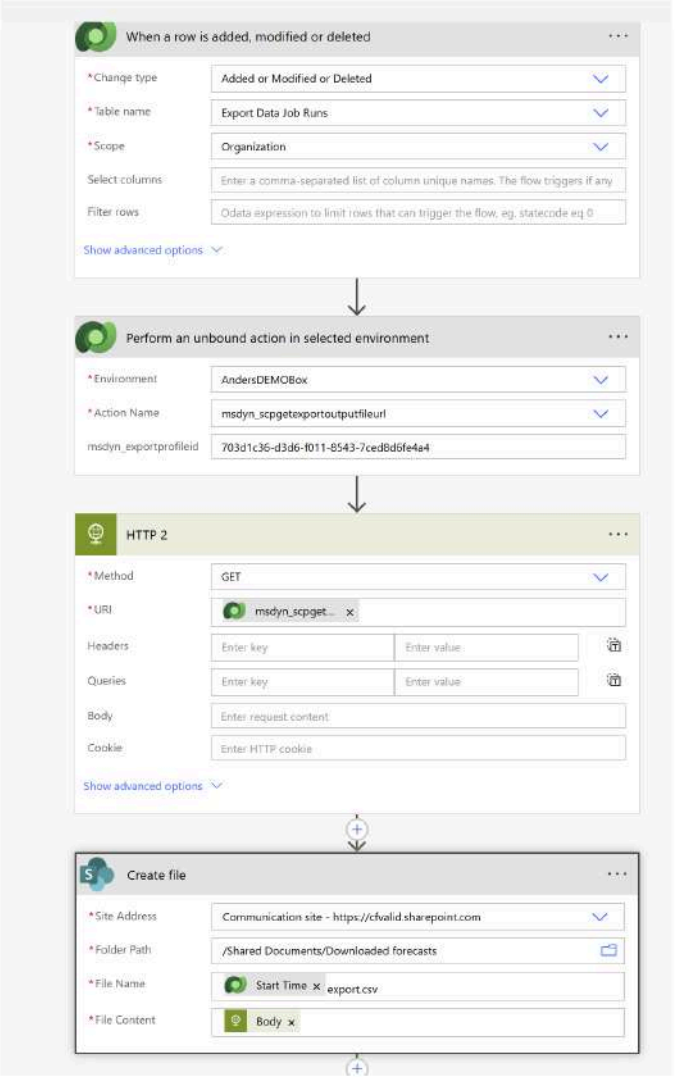
3. **Download file** – Add the *HTTP 2* action

    - Set **Method** – GET
    - Set **URI** – Use the `output` from the previous step (the SAS URL)

4. **Save to SharePoint** – Add the *Create file* action (SharePoint)

    - Set **Site address** – Your SharePoint site
    - Set **Folder path** – Your target folder
    - Set **File name** – Use the job name from trigger or create a naming pattern
    - Set **File content** – Use the body output from the HTTP download step

The following image shows the final created flow:



> **Tip**
>
> To download exports from multiple profiles, remove the profile filter from the trigger. The flow processes all completed export jobs. You can add condition logic to route files to different SharePoint folders based on the profile ID.

Export file format

The exported files use a standard comma-separated values (CSV) format with configurable columns.

CSV structure

Exported files have the following standard columns:

- **Time** – Timestamp of the data point (format configurable via organization settings)
- **Value** – The numeric value (forecast, demand, and so on)

- **Dimension columns** – Additional columns based on the time series dimensions (such as Product, Location, Customer)

Here's an example of CSV file content:

```csv
Time,Value,Product,Location
2024-01-01,1500.50,PROD001,WH-EAST
2024-01-02,1623.75,PROD001,WH-EAST
2024-01-01,892.25,PROD002,WH-WEST
2024-01-02,945.00,PROD002,WH-WEST
```

File storage

Files are stored as follows:

- **Location** – Azure Data Lake Storage Gen2
- **Path** – `ExportFiles/{exportProfileId}.csv`
- **Format** – UTF-8 encoded CSV
- **Access** – SAS URL valid for 120 minutes (2 hours)

Best practices

Follow these best practices to ensure successful exports:

- **SAS URL expiration** – The SAS URL expires after 120 minutes. Download the file promptly or request a new URL if needed.
- **Job naming** – Use descriptive job names that include context (such as date range and purpose) for easier tracking and auditing.
- **Export profile management** – Configure export profiles in advance with appropriate filters, date ranges, and dimension selections.
- **Error handling** – Implement retry logic for transient failures and proper error handling for authentication and authorization issues when calling Dataverse custom APIs.
- **File size** – Be mindful of large exports. Consider filtering data by date range or dimensions to manage file sizes.

Error responses

When an error occurs calling Dataverse custom APIs, the API returns standard Dataverse error responses. For details on error handling, see Web API error handling.

Here are some common error scenarios you might encounter when calling the export custom API action:

- **401 Unauthorized** – Authentication token is missing, expired, or invalid.
- **403 Forbidden** – The authenticated user lacks permissions to access the Dataverse organization or execute custom API actions.
- **404 Not Found** – The specified export profile or export file doesn't exist.
- **400 Bad Request** – Invalid request parameters exist, such as malformed GUIDs or missing required fields.

Data classification

> **Important**
>
> Exported data might contain customer content and system metadata. The `msdyn_scpgetexportoutputfileurl` custom API action returns SAS URLs to access customer content. Make sure you follow appropriate data handling and storage practices according to your organization's security policies.

Additional resources

- Export and download data
- Rolling forecasts

Last updated on 04/10/2026

Office integration overview

Article • 01/03/2025

Applies to these Dynamics 365 apps:
Commerce, Finance, Supply Chain Management

This article reviews Microsoft Office integration concepts and features. The integration depends on several technologies:

- Working in Microsoft Azure
- Working with Microsoft Entra ID
- Running a web client in multiple browsers

The Microsoft Office integration capabilities provide users with a productive environment that helps them get the job done by using Office products.

Excel Data Connector add-in

Microsoft Excel can change and quickly analyze data. The Excel Data Connector app interacts with Excel workbooks and OData services that are created for publicly exposed data entities. The Excel Data Connector add-in enables Excel to become a seamless part of the user experience. The Excel Data Connector add-in is built by using the Office Web add-ins framework. The add-in runs in a task pane. Office Web Add-ins are web applications that run inside an embedded Microsoft Edge browser window.

Dynamics AX 2012 architecture vs. Finance and operations architecture

There are several differences between versions. For both, we built lightweight add-ins that run in Excel and use services to connect to the application.

Dynamics AX 2012

Excel > VSTO (.NET) Add-in > Windows Communication foundation (WCF) > Authentication through Active Directory (AD) > AIF SOAP services on the AOS > AX Services and Tables > AX query engine > Database

Finance and operations

Excel > Office Web Add-in (JS + HTML) > JavaScript OData API (Olingo) > Authentication through Microsoft Entra ID (Microsoft Entra ID) > AX OData services on the AOS > AX Entities > AX LINQ provider > AX Database

Office Add-in explained

The Excel Data Connector app is located in a task pane on the right side of a workbook. The following table describes the parts of the add-in.

Expand table

Name	Description
Add-in primary title	The title of the add-in that is provided to the Office Web Add-ins framework.
Add-in secondary title	The title of the add-in that is provided by the add-in.
Source name	The label of the entity that provides data for the selected data table. You can hover over the label to see the corresponding name.
Field name	The label of the field that provides data for the selected data table column. You can hover over the label to see the corresponding name and type.
Refresh button	Refresh the data in the workbook.
Publish button	Publish the data changes in the workbook.
Design button	Open the design-time experience.
Status bar	The status bar provides brief temporary information alerts. Information that is appears in the status bar also appears in the **Messages** dialog box.
Options button	Open the **Options** dialog box.
Messages button	Open the **Messages** dialog box, which displays the information messages, warnings, and errors that the program provides to the user. A number sometimes appears next to the **Messages** button to provide a count of the warnings or errors that the user might be interested in.
Excel data table containing data	The filter and sort controls in the columns headers can be used on this data. The filters must be removed before data changes are published.
Office Web Add-ins menu	The Office Web Add-ins menu button provides several standard links. The most important of the links is used to reload the add-in. When the add-in is reloaded, it updates all the data for the workbook that is contained in tables that are associated with the add-in.

Authentication

OData sits on the same authentication stack as the server. The add-in uses OAuth to facilitate authentication.

Lookups and drop-down lists

When you click in a table cell, any lookup, enumeration drop-down list, or date picker that is associated with that cell will be shown inside the add-in, underneath the source and field information. Any value that you select inside the add-in is put into the currently selected table cell.

Adding and deleting records

To add a record, either start typing in a row directly below a table, or use the Tab key to tab away from the last cell of the last row in the table. To delete a record, select the row label (1, 2, 3, and so on), and then delete all the cells in that row. To publish the changes, select **Publish**. The **Messages** dialog box shows how many records were added, edited, and deleted.

Workbook Designer

You can use the **Workbook Designer** page to design an editable custom export workbook that contains an entity and a set of fields. To open the **Workbook Designer** (**ExportToExcelWorkbookDesigner**) page, select **Common** > **Common** > **Office Integration** > **Excel workbook designer**. Before you can publish data edits, all the key fields of the entity must be in the Excel table. Key fields have a key symbol next to them. To successfully create or update a record, it must have all the mandatory fields in the Excel table. Mandatory fields have an asterisk (*) next to them.

To retrieve the resulting workbook, select **Create workbook** in the app bar.

Click **View related form** to see the data that the entity exposes. This button is only enabled for entities that have a **FormRef** property value.

Document management

Document management supports saving record attachments in Azure Blob storage and SharePoint Online. Database storage is deprecated. Azure Blob storage is equivalent to storage in the database since documents can only be accessed through the application and it provides the added benefit of providing storage that doesn't negatively affect the

performance of the database. Azure blob storage is the default and works immediately. SharePoint storage will work immediately if you have an O365 license since we auto-discover the SharePoint tenant e.g. a user on the TenantA.onmicrosoft.com O365/Microsoft Entra tenant gets TenantA.sharepoint.com as the SharePoint site. If document management has been turned off by the user, turn it on by selecting **Options** > **General** > **Miscellaneous** and setting **Document handling active** to **Yes**.

On any page that has data, an **Attach** button is available in the upper-right corner.

The **Attachments** page provides a view of the attachments (documents) that are associated with the record that was selected on the previous page. You can add new attachments to the record by selecting the **New** button (+) in the app bar. For the **File** and **Image** document types, you will be prompted to provide the associated file.

Document preview

A preview for supported file types is provided on the **Preview** FastTab. Basic document types, such as PNG images and text files, are supported by default. Office document types, such as Microsoft Word, Excel, and PowerPoint files, must use a production Office Web Apps Server, which might not be available in a OneBox configuration.

Frequently asked questions

Office Licensing

What Microsoft 365 licenses are available?

There are lots of Microsoft 365 license options. You should select the license that makes sense for your organization.

After purchasing a Microsoft 365 license, what needs to be done to set up SharePoint storage for attachments?

Open the Document Parameters form and ensure that the SharePoint server has been automatically discovered and set. Now open or create a Document Type, set the Document Type's location to "SharePoint" and select the folder that the files should be stored in.

Additional resources

Office integration tutorial

Troubleshoot the Office integration

Application stack and server architecture

Feedback

Was this page helpful? Yes No

Provide product feedback

Office integration tutorial

Applies to these Dynamics 365 apps:
Commerce, Finance, Human Resources, Supply Chain Management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In this tutorial, you will use and build Office integration experiences that involve Excel, Word, document management, and email.

Overview

In this tutorial, you will use and build Microsoft Office integration experiences that involve Microsoft Excel, Microsoft Word, the document management capabilities of finance and operations apps, and email. You will see how Excel and Word use data entities as an entry point into the system, how Excel can become a core part of the user experience, and how Excel and Word can be used for ad-hoc lightweight reporting. You will also see how files can be stored and shared by using the document management and email capabilities.

Key concepts

- **Entities and OData** – You will use the Microsoft Dynamics Excel Data Connector App (Excel App) to create, read, update, and delete. The connector uses OData services that are created for any entity that is left in the default state of "public" (**DataEntity.Public**=**Yes**).
- **Apps for Office** – The Excel App is built by using the Apps for Office framework (which is also known as the Office Web API). The Excel App is web-based, and therefore shares technology with the client and will run inside both on-premises Excel instances and Microsoft Excel Online (Microsoft 365). The app runs inside Excel in a task pane.
- **Microsoft Office 2016** – The Excel and Word Apps use advances in the Apps for Office framework that were introduced in Office 2016. Therefore, Office 2016 is required in order to run the Excel and Word Apps.
- **Authentication** – The Excel and Word apps run in a browser window inside Excel and Word. For details about which browser will be used for your configuration, see Browsers used by Office Add-ins. The specified browser is used even if the user is running the client

in an InPrivate browsing session in Microsoft Edge, or in a different browser such as Google Chrome. Authentication is facilitated by OAuth, and the user can select accounts and sign in within the app. The browser will first try to automatically sign the user in, so if you aren't signed in as the correct user or if you have trouble signing in, you might need to force a sign-out from the app by using the sign-out link. After signing out, try to sign in again to the app.

- **Excel App** – In addition to facilitating refresh and publish data operations, the Excel App also provides source and field information, lookups, filtering, error messaging, and a design experience for adding or removing fields, table columns, or labels from entity data sources.

Setup

Load the Fleet data set

During this tutorial, we will mainly use forms, entities, and data in the Fleet Management model. Therefore, we must first load the Fleet data set.

1. Go to **Fleet Management** > **Setup** > **Fleet setup**.
2. Select **Create**.

Static Export to Excel experiences

Static Export to Excel

Static Export to Excel provides a quick mechanism for getting data into grids on a page. The standard mechanism for triggering Export to Excel is the **Open in Microsoft Office** menu. Static Export to Excel is also available via a shortcut menu on the grid.

1. Go to **Fleet Management** > **Customers** > **Customer**.
2. Select **Open in Microsoft Office** > **Export to Excel** > **Customers**.
3. Download and open the workbook that is generated. Note that the columns in the workbook match the columns in the grid.
4. Select ("mark") the first two rows by clicking in the left edge of the row, below the "Select all" check mark.
5. Right-click the grid header to open the shortcut menu. Note that both **Export all rows** and **Export marked rows** are available as commands.
6. Select **Export marked rows**. Note that the columns in the workbook match the columns in the grid, and that the rows that are exported match the rows that you marked.

Modify the static Export to Excel experience

You can suppress the static Export to Excel mechanism for a grid or change the label that appears on the **Open in Microsoft Office** menu.

1. Start Visual Studio. Make sure that it's running as an administrator.
2. Select **View** > **Application Explorer** (or press Ctrl+E, Ctrl+E).
3. Go to **AOT** > **User Interface** > **Forms** > **FMCustomer**.
4. Right-click **FMCustomer**, and then click **Add to new project**.
5. In Solution Explorer, double-click the **FMCustomer** form to open the designer view.
6. Select **FMCustomerDesignTab(Tab)TabPageGrid(TabPage)MainGrid(Grid)**.
7. In the **Properties** window, find the **Export Label** property.
8. Set the **Export Label** property to **Fleet Customers**.
9. Save the form. If you're asked whether you want to overwrite the existing form or save it as a new form, click **Overwrite**.
10. Build the solution (press Ctrl+Shift+B).
11. In the browser, go to **Fleet Management** > **Customers** > **Customer**.
12. Select **Open in Microsoft Office**. Note that the **Customers** option has changed to **Fleet Customers**.

Generated Open in Excel experiences

Generated Open in Excel

Generated Open in Excel options are automatically added to forms when the system finds data entities that have the same root data source as the form. The workbook that is generated will contain a single table data source where the data from that entity is loaded. The Open in Excel experiences are listed on the **Open in Microsoft Office** menu. (When an entity has the same root data source as a form, it's added as an option in the **Open in Excel** section of the **Open in Microsoft Office** menu. This option is referred to as a “generated” option.)

1. Go to **Fleet Management** > **Customers** > **Customer**.
2. Select **Open in Microsoft Office** > **Open in Excel** > **Fleet Management Customers (unfiltered)**.
3. Download and open the workbook that is generated. This workbook contains the Excel Data Connector App, a binding to the **Fleet Management Customer** entity, and a pointer to the server that the workbook was generated from.
4. Select **Enable editing** to enable the Excel Data Connector App to load. Customer data is read from the OData service on the server and added to the table.
5. In the browser on the **Customer** page, click **Edit** (or press F2), and change the email address of one of the customers.

6. In the Excel App, click **Refresh**. Note that the new email address is shown in Excel.
7. In Excel, change the email address of one of the customers.
8. In the Excel App, click **Publish**.
9. In the browser, click **Refresh** in the upper right of the page (or press Shift+F5). Note that the new email address is shown on the **Customer** page.
10. In Excel, click the **Settings** (gear) button in the lower-right corner of the Excel App. You can use the dialog box that appears to adjust the settings in the current workbook. Note that the **Server URL** value matches the start of the URL that is shown in the browser. Also note that the data refresh and data publish operations are listed.
11. Select **Cancel** to close the **Settings** dialog box.
12. Select the **Message Center** (flag) button in the lower-right corner of the Excel App. The message center dialog box that appears provides information about what is occurring in the Excel App.

Add and remove table columns from an existing table data source in the Excel App

The Excel App has a design experience that lets users add and edit bindings to entity data sources and labels. To add and remove fields from an existing binding, you use the edit experience that is outlined in the following steps.

1. Get a workbook that has an existing table data source:
   a. Go to **Fleet Management** > **Customers** > **Customer**.
   b. Select **Open in Microsoft Office** > **Open in Excel** > **Fleet Management Customers (unfiltered)**.
   c. Download and open the workbook that is generated. This workbook contains the Excel Data Connector App, a binding to the Fleet Management Customer entity, and a pointer to the server that the workbook was generated from.
   d. Select **Enable editing** to enable the Excel Data Connector App to load. Customer data is read from the OData service on the server and added to the table.

2. Open the data source for editing:
   a. In Excel, in the Excel App, click **Design**. A list of table and field data sources appears.
   b. Select the **Edit** (pencil) button next to the existing table data source. The data source details are shown.

3. Remove fields. In the **Selected** list, double-click a field. Alternatively, click a field, and then click **Remove**. To select multiple fields, keep the Ctrl key held down while you click them. To select all fields, press Ctrl+A.

4. Add fields. In the **Available** list, double-click a field. Alternatively, click a field, and then click **Add**. To select multiple fields, keep the Ctrl key held down while you click them. To

select all fields, press Ctrl+A.

5. Change the field order. In the **Selected** list, click a field, and then click **Up** or **Down**.
6. Change a field label. In the **Selected** list, click a field, and then click in the **Column label** field below the list. You can change the label to either a static string or a label identifier that will be translated to the active language (for example, **@SYS129977**).
7. Apply the changes that you made to data source fields:
   a. Select **Update** to return to the data source list.
   b. Select **Refresh** to make sure that any new fields are filled with data.

Change an automatically generated Open in Excel experience

The automatically generated Open in Excel experiences that are created for entities have a single table binding. The list of fields that are added to that table binding is defined by the **AutoReport** field group if the table binding contains fields. Otherwise, the key and mandatory fields for the entity are automatically added. The order of fields in the **AutoReport** group determines the order of fields in the table binding.

1. Start Visual Studio. Make sure that it's running as an administrator.
2. Select **View** > **Application Explorer** (or press Ctrl+E, Ctrl+E).
3. Go to **AOT** > **Data Model** > **Data Entities** > **FMCustomerEntity**.
4. Right-click **FMCustomerEntity**, and then click **Add to project**.
5. Expand **FMCustomerEntity** > **Field Groups** > **AutoReport**.
6. Reverse the order of the **First name** and **Last name** fields by clicking the **Last name** field and moving it up (press Alt+Up arrow key).
7. Save the entity. If you're asked whether you want to overwrite the existing entity or save it as a new entity, click **Overwrite**.
8. Build the solution (press Ctrl+Shift+B).
9. In the browser, go to **Fleet Management** > **Customers** > **Customer**.
10. Select **Open in Microsoft Office** > **Open in Excel** > **Fleet Management Customers**.
11. Open the workbook that is generated.
12. Select **Enable editing** to enable the Excel Data Connector App to load. Note that the **Last name** column appears before the **First name** column.

Open in Excel Online

The Excel App is built by using a new Apps for Office framework. This framework provides a JavaScript-based web application programming interface (API) that enables apps to communicate with Office applications. The biggest advantage of this new framework is that

apps can run in on-premises Excel instances (Win32), Excel Online (Microsoft 365), and Excel on the Apple iPad. They will also be able to run in other Excel apps in the future.

1. Go to **Fleet Management** > **Customers** > **Customer**.
2. Select **Open in Microsoft Office** > **Open in Excel** > **Fleet Management Customers**.
3. Select **SharePoint**.
4. Browse to the desired Microsoft SharePoint folder.
5. Select **Save**. The default behavior is to open the file after it's saved. Note that the workbook opens in Excel Online. In Excel Online, capabilities of the Excel App, such as refresh and publish, and the design experience, should work just as they work in on-premises Excel instances.

Template Open in Excel experiences

Template Open in Excel

Template options resemble the generated Open in Excel options. They are automatically added to forms when the system finds templates that have the same first data source as the root data source in the form. A workbook template can have multiple data sources. It can also have unbound content. The Open in Excel experiences are listed on the **Open in Microsoft Office** menu. The **Excel workbook designer** page provides an easy way to get a generated Open in Excel experience for an entity. It also provides a mechanism getting a blank workbook that contains just the Excel App and a pointer to the server.

1. Go to **Common** > **Common** > **Office integration** > **Excel workbook designer**.
2. Select the **FleetCustomer** entity.
3. Add all fields in the list of available fields to the list of selected fields.
4. Select **Create workbook**.
5. Open the workbook that is generated. This workbook contains the Excel Data Connector App, a binding to the **Fleet Management Customer** entity, and a pointer to the server that the workbook was generated from.
6. Select **Enable editing** to enable the Excel Data Connector App to load. Customer data is read from the OData service on the server and added to the table.
7. Insert a blank row above the table, and enter **Fleet Customers** as the title.
8. Rename the worksheet **FleetCustomers**.
9. Rearrange some of the fields in the table. Select **Design** to open the design experience.
10. Next to the **FleetCustomer** data source, there are buttons for editing and deleting the data source. Select **Edit** to see the field list.
11. Select fields, and move them as you require. Set the order for the first three fields to **FirstName**, **LastName**, and **DriverLicense**.

12. Select **Update**. Note that the field order is changed.
13. Select **Done**.
14. Select the **Settings** (gear) button.
15. Select **Clear binding data** so that the workbook contains no bound data.
16. Select **OK**.
17. Save the workbook as **FleetCustomersBasic.xlsx**.
18. In the browser, go to **Common** > **Common** > **Office integration** > **Document templates**.
19. Select **New**.
20. Browse to the file that you just saved.
21. Select **OK**. The template is added as a line in the templates table.
22. In the **FleetCustomersBasic** row, clear the **Apply current record filter** check box, so that an unfiltered list of customers will be loaded after the template is opened.
23. Change the **Template display name** value to **Fleet Customers Basic**.
24. Go to **Fleet Management** > **Customers** > **Customer**.
25. Select **Open in Microsoft Office**. Note that **Fleet Customers Basic** is now an option in the **Open in Excel** section. Select that option.
26. Open the workbook that is generated.
27. Select **Enable editing** to enable the Excel Data Connector App to load. Customer data is read from the OData service on the server and added to the table binding that you created.

Register a template as a system-defined template

Templates that are registered as system-defined templates are loaded at deployment. This behavior is useful for independent software vendors (ISVs) and partners that want to package templates together with other model artifacts.

1. Start Visual Studio by opening the previously created project where the model is set to **Fleet Management**, or create a new project.

2. Right-click the project, and then click **Add** > **New item**.

3. Select the **Resource** item type.

4. Set the name to **FleetCustomersBasicTemplate**.

5. Make sure that the FleetCustomersBasic.xlsx file is closed.

6. Select **Add**.

7. Select the **FleetCustomersBasic.xlsx** file. Note that the resource is added to the project.

8. Select **View** > **Application Explorer** (or press Ctrl+E, Ctrl+E).

9. Go to **AOT** > **Classes** > **Code** > **FMTemplateRegistrations**.
10. Right-click **FMTemplateRegistrations**, and then click **Add to project**.
11. Open **FMTemplateRegistrations**. The FMTemplateRegistrations.xpp code file should be shown.
12. Copy one of the existing lines, and change it by providing the template name, resource name, description, display name, and **Apply current record filter** and **List in Open in Office menu** values. The display name is the text that appears as an Open in Excel option. The description appears when the user holds the pointer over that item. The display name and description can be either labels or static strings. The code should resemble the following example.

    X++

    ```
    this.addTemplate(
        OfficeAppApplicationType::Excel,
        resourceStr(FleetCustomersBasicTemplate),
        resourceStr(FleetCustomersBasicTemplate),
        "Template for fleet customers", "Fleet customers basic", NoYes::No,
    NoYes::Yes);
    ```

13. Save the code. If you're asked whether you want to overwrite the existing code or save it as a new file, click **Overwrite**.
14. Build the solution (press Ctrl+Shift+B).
15. Verify that the change was successful. In the browser, go to **Common** > **Common** > **Office integration** > **Document templates**.
16. Select **Reload system templates**.
17. Select **Yes** to confirm that you want to reload the system templates.
18. Verify that the new system-defined template is loaded, and that the template name is **FleetCustomersBasicTemplate**.

Journal Entry in Excel experience powered by a template

1. Go to **General ledger** > **Journal entries** > **General journals**.
2. Make sure that you're in company **USMF**.
3. Create a new journal by clicking **New**.
4. Set the name to **GenJrn**.
5. Select **Open lines in Excel**.

6. Open the workbook that is generated, and enable editing as required. Note that header fields are filled with data.
7. Enter a new line, and set the **MainAccount** field to **110110**. Enter a description, a currency, and a debit amount. Note that lookups are provided for the company and currency fields, because those relationships are defined for this entity.
8. Select **Publish**. Note that the line is updated with the current date and a debit amount of 0 (zero).
9. In the browser, click **Lines**. Note that line that you entered in Excel is shown.

Lookups in Excel experiences

Lookups in the Excel App

To facilitate data entry, the Excel App provides lookups and data assistance. Date fields provide a date picker, enumeration (enum) fields provide an enum list, and relationships provide a relationship lookup.

1. Go to **Fleet Management** > **Rentals** > **Rental**.
2. Select **Open in Microsoft Office** > **Open in Excel** > **Fleet Management Rentals**.
3. Open the workbook that is generated.
4. Select **Enable editing** to enable the Excel Data Connector App to load and read in data.
5. Select a **Drivers license** value. Note that a relationship lookup now appears in the Excel App and shows a list of customers. Because relationship lookups are in their first generation, no filtering or sorting is currently supported.
6. Select another customer in the lookup, and note that the **Drivers license** value changes. Because this field is part of the key, click the original **Drivers license** value to reset it. Note that the **Drivers license**, **First name**, and **Last name** fields form a multi-part key, but the Excel App doesn't immediately change all parts of the multi-part key.
7. Select a **Start date** value. Note that a date picker now appears in the Excel App.
8. Select another date to change the **Start date** value.
9. Select **Design**, and edit the FleetRental data source by adding the **Status** field as a column in the table binding.
10. When you've finished adding the **Status** column, click a **Status** value. Note that an enum list now appears in the Excel App.
11. While focus is in the **Status** column, move up and down the list of rentals to see how quickly the enum list changes to reflect the current value. The whole enum list is shown, so that the user can quickly see all the available values.
12. Select a different **Status** value to see how an enum value can be changed by using a single click.

Create a relationship lookup

When relationships exist between entities, a relationship lookup is shown.

1. Start Visual Studio by opening the previously created project where the model is set to **Fleet Management**, or create a new project.
2. Select **View** > **Application Explorer** (or press Ctrl+E, Ctrl+E).
3. Go to **AOT** > **Data Model** > **Tables** > **FMCustGroup**.
4. Right-click, and then click **Open designer**.
5. In the designer, right-click **FMCustGroup**, and then click **Add-ins** > **Create data entity**. Artifacts are added to the project.
6. Open the designer view for **FMCustGroupEntity**.
7. In the property sheet for **FMCustGroupEntity**, set **Public Collection Name** to **FleetCustomerGroups** and **Public Entity Name** to **FleetCustomerGroup**.
8. Add the **CustGroup** and **Description** fields to the **AutoLookup** field group.
9. If **FMCustomerEntity** isn't already in the project, add it.
10. Open the designer view for **FMCustomerEntity**.
11. Right-click **Relations**, and then click **New** > **Relation**.
12. On the new relation, set **Name** to **CustomerGroup**, **Cardinality** to **ZeroMore**, **RelatedDataEntity** to **FMCustGroupEntity**, **RelatedDataEntityCardinality** to **ZeroOne**, **RelationshipType** to **Association**, **Role** to **CustomerGroupSource**, and **RelatedDataEntityRole** to **CustomerGroupTarget**.
13. Build the solution (press Ctrl+Shift+B).
14. Verify that the change was successful. In the browser, go to **Fleet Management** > **Customers** > **Customer**.
15. Select **Open in Microsoft Office** > **Open in Excel** > **Fleet Management Customers**.
16. Open the workbook that is generated.
17. Select a **Customer group** value.
18. Change the **Customer group** value for a customer.
19. Publish the change.
20. Change the value back, and publish that change.

Create a custom lookup

You can create custom lookups to show data options when an enum or relationship isn't sufficient. The main use case is when data must be retrieved from an external service and presented in real time.

1. Start Visual Studio by opening the previously created project where the model is set to **Fleet Management**, or create a new project.

2. Open the designer view for **FMCustomerEntity**.

3. Right-click **Methods**, and then click **New Method**.

4. Add the **lookup_Country** code from the following example.

```
X++
```

```
  public class FMCustomerEntity extends common
{
    [SysODataAction("FMCustomerEntityCountryCustomLookup", false), //Name in
$metadata
    SysODataCollection("_fields", Types::String), //Types in context
    SysODataFieldLookup(fieldStr(FMCustomerEntity, Country))] //Name of field
    public static str lookup_Country(Array _fields)
    {
        OfficeAppCustomLookupListResult result = new
OfficeAppCustomLookupListResult();

        result.items().value(1, "US");
        result.items().value(2, "AU");
        result.items().value(3, "FR");
        result.items().value(4, "GR");
        result.items().value(5, "NZ");

        return result.serialize();
    }
  }
```

5. Save the code. If you're asked whether you want to overwrite the existing code or save is as a new file, click **Overwrite**.

6. Build the solution (press Ctrl+Shift+B).

7. Verify that the change was successful. In the browser, go to **Fleet Management** > **Customers** > **Customer**.

8. Select **Open in Microsoft Office** > **Open in Excel** > **Fleet Management Customers**.

9. Open the workbook that is generated.

10. Select a **Country/region** value.

11. Change the **Country/region** value for a customer.

12. Publish the change.

13. Change the value back, and publish that change.

Export to Word experiences

Export to Word

Export to Word experiences can be used for lightweight reporting. They are powered by pre-built templates. The Export to Word experiences are listed on the **Open in Microsoft Office** menu. Let's look at an example experience that has been created for Fleet Management Customers.

1. Go to **Fleet Management** > **Customers** > **Customer**.
2. Select **Open in Microsoft Office** > **Export to Word** > **Customer information Fleet Management Customers (unfiltered)**.
3. Download and open the document that is generated. The document contains data from the record that is currently selected.

Create a Word template

The Microsoft Dynamics App for Office can be run in Word to enable the creation of templates that can then be used for document generation.

1. Create a blank document
   a. Go to **Common** > **Common** > **Excel workbook designer**.
   b. Select **Create blank document**.
   c. Download and open the document that is generated.

2. Sign in to the app:
   a. Select **Sign In**. The Microsoft Entra sign-in screen should provide a list of credentials. If you encounter an error, force a sign-out (by using the sign-out link), and then sign in again.
   b. Select the appropriate account, or click **Use another account**.
   c. Enter the credentials for that environment, and then click **Sign in**.

3. Follow one of these steps:

   - Add a fields data source:
     a. In the app, click **Design**.
     b. Select **Add fields**.
     c. Select **FleetCustomer**.
     d. Select **Next** to go to the field selection page.
     e. In the document, add a title and/or some blank lines at the top of the document.
     f. In the app, in the **Available** list, select the **FirstName** field.
     g. Select **Add label** to add a content control that references the "First name" label.
     h. In the document, click to put focus into the document, click again to put focus at the end of the label, and then press the Right arrow key until the cursor is outside

the content control (the control box will disappear).
    i. Add a separator, such as space+hyphen+space (" - ") or space+colon+space (" : ").
    j. In the app, click **Add value** to add a content control that references the **FirstName** field.
    k. Repeat the process for the **LastName** field label and value.
    l. Continue to add fields as desired.
- Add a table data source:
    a. In the app, click **Design**.
    b. Select **Add table**.
    c. Select **FleetCustomer**.
    d. Select **Next** to go to the field selection page.
    e. In the document, add a title and/or some blank lines at the top of the document.
    f. In the app, in the **Available fields** list, add the **FirstName**, **LastName**, and **City** fields.
    g. Select **Done**.

4. In Word, save the template document.

Create a Word template and use it for document generation

After you've built a Word template, you can upload it to create an Export to Word experience.

1. Upload a template:
    a. Go to **Common** > **Common** > **Office integration** > **Document templates**. Alternatively, search for the page.
    b. Select **New**.
    c. Select **Browse**.
    d. In the dialog box, select a previously created template, and then click **Open**. Note that the Root data entity is obtained from the template and appears near the bottom of the dialog box.
    e. Select **OK**.
    f. Scroll down the list of templates to confirm that the template was added.
    g. Optional: If the template should not be filtered to the user's current record, clear the **Apply current record filter** check box.
    h. Optional: If the template should not be filtered to the user's current company, clear the **Apply company filter** check box.

2. Use the uploaded template for document generation:
    a. Go to a page that shares the same root data source as the template's root data entity. For **FleetCustomer** (**FMCustomerEntity**), that page is **Fleet Management** > **Customers** > **Customer**.

b. Select **Open in Microsoft Office** > **Export to Word**, and click the template.
c. Download and open the document that is generated.

Document Management and SharePoint experiences

Add a SharePoint document type

The Document Management subsystem can be used to attach files to records. Most non-executable file types are supported as attachments. A document preview is provided for Office document files and PDFs. Administrators create document types to indicate where attachments should be stored. When administrators use SharePoint as the storage location, they must provide a specific folder that the files should be put in. Security of that SharePoint folder is a separate administration responsibility.

1. Go to **Organization administration** > **Document management** > **Document management parameters**.
2. Select **SharePoint**.
3. Make sure that the **Default SharePoint server** field is set to a default value for the tenant, such as **contosoax7.sharepoint.com**.
4. Select **Test SharePoint connection**. Note a successful connection.
5. Select **Save**.
6. Go to **Organization administration** > **Document management** > **Document types**.
7. Select **New**.
8. Set **Type** to **SharePointDoc**.
9. Set **Name** to **SharePointDoc**.
10. Set **Location** to **SharePoint**.
11. Select the **Edit** (pencil) button next to the **SharePoint Address** field.
12. On the left side of the dialog box, select a site. For the contosoax7 tenant, select **ContosoAX Team Site**.
13. On the right side of the dialog box, select a folder. For the contosoax7 tenant, select **ContosoAX Team Site** > **Documents** > **OfficeIntegration** > **Attachments**.
14. Select **OK**.
15. Select **Save**.
16. Select the **Browse** (globe) button next to the **SharePoint Address** field. Note that a new browser tab that shows the selected folder appears.
17. Use Windows Explorer to create a Word document in the Documents folder, and enter a few words in the document.
18. In the browser, go to **Fleet Management** > **Customers** > **Customer**.

19. Put focus on the first customer, and then click the **Attach** (paperclip) button in the upper-right corner of the page.
20. Select **New** > **SharePointDoc**.
21. Select **Browse**, and select the Word document that you created.
22. Expand the **Preview** FastTab to see a preview of the Word document.
23. Expand the **Attachment** FastTab to see the file location of the Word document.
24. In the browser, use the previously opened tab that shows the SharePoint folder to double-check that the file has been placed appropriately.

Email experiences

Send mail via a local mail client

Email workflows that are enabled via the SysEmail framework can generate email messages (.eml files) that contain attachments. You can then send these messages via Microsoft Outlook or another email client.

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select **US-008 Sparrow Retail**.
3. Select **Collect** > **Customer balances** > **Collections** to open the **Collections** page.
4. Select **Communicate** > **Email** > **Statements to contact**.
5. Select **OK** to accept the default values in the dialog box.
6. If you're prompted for the mail option to use, clear the **Do not ask again** check box (you can change this option from the user options page), select **Use an email app, such as Outlook**, and then click **OK**.
7. If you're running the browser on your laptop, open the email (.eml) file that is generated. If you're running the browser on the VM, copy the file to your laptop, and open it there.
8. Note the email address in the **To** field and the generated workbook attachment.

Send mail via SMTP

Email workflows that are enabled via the SysEmail framework can also be created in a simple email dialog box and then sent via Simple Mail Transfer Protocol (SMTP).

1. Go to **System administration** > **Setup** > **Email** > **Email parameters**.
2. Select **SMTP settings**.
3. Set the **Outgoing mail server** to the desired SMTP server:
    - For Microsoft 365 production (including *.onmicrosoft.com accounts): smtp.office365.com (Find this setting via outlook.office.com, at **Settings** > **Mail** >

POP and IMAP.)
- For Outlook/Hotmail: smtp-mail.outlook.com

4. Set the user name and password to an appropriate email account and password.
5. Leave **SSLRequired** turned on, and leave **SMTP port number** set to **587**.
6. Select **Save**.
7. In the browser, go to **Accounts receivable** > **Customers** > **All customers**.
8. Select **US-008 Sparrow Retail**.
9. Select **Collect** > **Customer balances** > **Collections** to open the **Collections** page.
10. Select **Communicate** > **Email** > **Statements to contact**.
11. Select **OK** to accept the default values in the dialog box.
12. If you're prompted for the mail option to use, select **Use the Microsoft Dynamics 365 Finance email client**, and then click **OK**.
13. To receive the test message, change the To address to your email address.
14. Enter a subject and body for the message.
15. Select **Send**. The message should be delivered in one to five minutes. Note that the message will appear to be sent from the email account that is set on the **Email parameters** page. If that email account is given "Send As" (or "Send email from this mailbox") permissions for the From address that is used in the **Send email** dialog box, messages will appear to come from that address.
    - You can configure "Send As" permissions in the Microsoft 365 admin center (portal.office.com/Admin), at **Users** > **Active users** > **User** > **Edit mailbox permissions** > **Send email from this mailbox**. For more information, see Enable sending email from another user's mailbox in Microsoft 365.
    - Before users can send email messages, "Send As" permissions for each user email account in the client must be given to the email account that is set on the **Email parameters** page. For more information, see How to set up a multifunction device or application to send email using Microsoft 365.
16. Email that is sent directly from the server, without user interaction, is sent via a batch process and requires that the **Email distributor batch** process be started. Follow these steps to start the process:
    a. Go to **System administration** > **Periodic tasks** > **Email processing** > **Batch**.
    b. Turn on **Batch processing**.

Additional resources

- Office integration overview
- Troubleshoot the Office integration

Last updated on 03/17/2026

View and update entity data with Excel

Summarize this article for me

Applies to these Dynamics 365 apps:
Commerce, Finance, Supply Chain Management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to open entity data in Microsoft Excel, and then view, update, and edit the data by using the Microsoft Dynamics Excel add-in. To open entity data, you can start from either Excel or finance and operations apps.

By opening entity data in Excel, you can quickly and easily view and edit the data by using the Excel add-in. This add-in requires Microsoft Excel 2016 or later.

> **Note**
>
> If your Microsoft Entra ID (Azure AD) tenant is configured to use Active Directory Federation Services (AD FS), you must make sure that the May 2016 update for Office is applied, so that the Excel add-in can correctly sign you in.

To learn more about how to use the Excel add-in, watch the short Create an Excel template for header and line patterns video.

Open entity data in Excel when you start from a finance and operations app

1. On a page in a finance and operations app, select **Open in Microsoft Office**.

   If the root data source (table) for the page is the same as the root data source for any entities, default **Open in Excel** options are generated for the page. You can find **Open in Excel** options on frequently used pages, such as **All vendors** and **All customers**.

2. Select an **Open in Excel** option, and open the workbook that is generated. This workbook has binding information for the entity, a pointer to your environment, and a pointer to

the Excel add-in.

3. In Excel, select **Enable editing** to enable the Excel add-in to run. The Excel add-in runs in a pane on the right side of the Excel window.

4. If you're running the Excel add-in for the first time, select **Trust this Add-in**.

5. If you're prompted to sign in, select **Sign in**, and then sign in by using the same credentials that you used to sign in to the finance and operations app. The Excel add-in uses a previous sign-in context from the browser and automatically signs you in, if it can. (For information about the browser that is used based on the operating system, see Browsers used by Office add-ins. To ensure that sign-in was successful, verify the user name in the upper-right corner of the Excel add-in.

The Excel add-in automatically reads the data for the entity that you selected. The workbook contains no data until the Excel add-in reads it in.

Open entity data in Excel when you start from Excel

1. In Excel, on the **Insert** tab, in the **Add-ins** group, select **Store** to open the Office Store.

2. In the Office Store, search on the keyword **Dynamics**, and then select **Add** next to **Microsoft Dynamics Office Add-in** (the Excel add-in).

3. If you're running the Excel add-in for the first time, select **Trust this Add-in** to enable the Excel add-in to run. The Excel add-in runs in a pane on the right side of the Excel window.

4. Select **Add server information** to open the **Options** pane.

5. In your browser, copy the URL of your target finance and operations app instance, paste it into the **Server URL** field, and then delete everything after the host name. The resulting URL should have only the host name.

   For example, if the URL is `https://xxx.dynamics.com/?cmp=usmf&mi=CustTableListPage`, delete everything except `https://xxx.dynamics.com`.

6. Select **OK**, and then select **Yes** to confirm the change. The Excel add-in restarts and loads metadata.

   The **Design** button is now available. If the Excel add-in has a **Load applets** link, you likely aren't signed in as the correct user. For more information on addressing this issue, see the Load applets troubleshooting entry.

7. Select **Design**. The Excel add-in retrieves entity metadata.
8. Select **Add table**. A list of entities appears. The entities are listed in "Name - Label" format.
9. Select an entity in the list, such as **Customer - Customers**, and then select **Next**.
10. To add a field from the **Available fields** list to the **Selected fields** list, select the field, and then select **Add**. Alternatively, double-click the field in the **Available fields** list.
11. After you finish adding fields to the **Selected fields** list, make sure that the cursor is in the correct place in the worksheet (for example, cell A1), and then select **Done**. Then select **Done** to exit the designer.
12. Select **Refresh** to pull in a set of data.

View and update entity data in Excel

After the Excel add-in reads entity data into the workbook, you can update the data at any time by selecting **Refresh** in the Excel add-in.

Edit entity data in Excel

You can change entity data as needed and then publish it back to finance and operations apps by selecting **Publish** in the Excel add-in. To edit a record, select a cell in the worksheet, and then change the cell value. To add a new record, follow one of these steps:

- Click anywhere in the data sources table, and then select **New** in the Excel add-in.
- Click anywhere in the last row of the data sources table, and then press the Tab key until the cursor moves out of the last column of that row and a new row is created.
- Click anywhere in the row immediately below the data sources table, and start to enter data in a cell. When you move the focus out of that cell, the table expands to include the new row.
- For field bindings of header records, select one of the fields, and then select **New** in the Excel add-in.

You can create a new record only if the worksheet binds all the key and mandatory fields, or if default values are filled in by using the filter condition.

To delete a record, follow one of these steps:

- Right-click the row number next to the worksheet row that you want to delete, and then select **Delete**.

- Right-click anywhere in the worksheet row that you want to delete, and then select **Delete** > **Table Rows**.

If you add data sources as related data sources, the header publishes before the lines. If dependencies exist between other data sources, you might need to change the default publishing order. To change the publishing order, in the Excel add-in, select the **Options** button (the gear symbol). On the **Data Connector** FastTab, select **Configure publish order**.

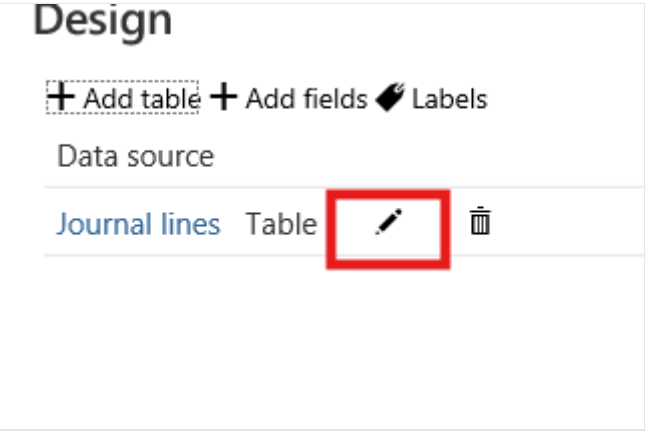
Add or remove columns

Use the designer to adjust the columns that the worksheet automatically adds.

> **Note**
>
> If the **Design** button doesn't appear below the **Filter** button in the Excel add-in, you must enable the data source designer. Select the **Options** button (the gear symbol), and then select the **Enable design** check box.

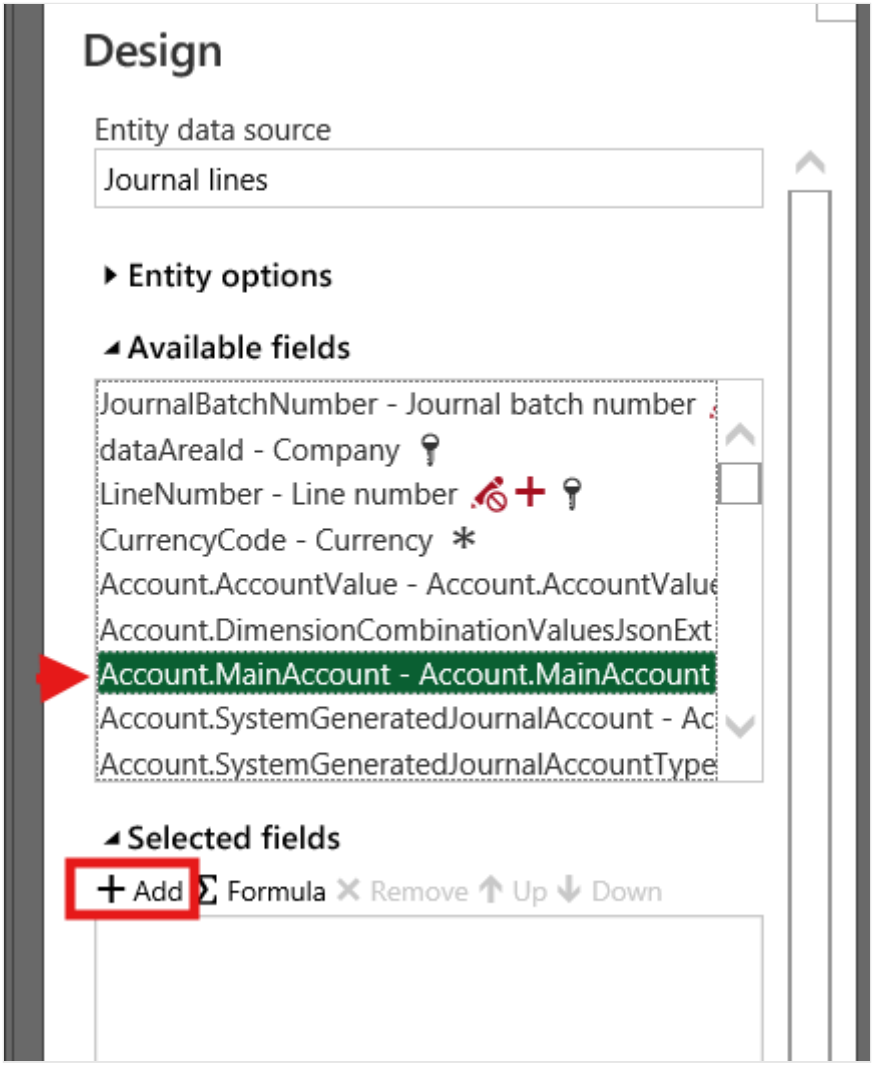
1. In the Excel add-in, select **Design**. All the data sources are listed.

1. Next to the data source, select the **Edit** button (the pencil symbol).



1. In the **Selected fields** list, adjust the list of fields as you require:

    - To add a field from the **Available fields** list to the **Selected fields** list, select the field, and then select **Add**. Alternatively, double-click the field in the **Available fields** list.
    - To remove a field from the **Selected fields** list, select the field, and then select **Remove**. Alternatively, double-click the field.
    - To change the order of fields in the **Selected fields** list, select a field, and then select **Up** or **Down**.



1. To apply your changes to the data source, select **Update**. Then select **Done** to exit the designer.
2. If you added a field (column), select **Refresh** to pull in an updated set of data.

Change the publish batch size

When users publish changes to data records by using the Excel add-in, the updates are submitted in batches. The default and maximum publish batch size is 100 rows. However, the **Allow configuration of the publish batch size in the Excel add-in** feature gives you flexibility in lowering the publish batch size, especially if you see time outs when attempting to publish updates from Excel.

System administrators can specify a system-wide limit on the publish batch size for "Open in Excel" workbooks by setting the **Publish batch limit** field in the **App parameters** section of the **Office app parameters** page.

You can also change the publish batch size for an individual workbook by using the Excel add-in.

1. Open the workbook in Excel.
2. Select the **Option** (gear) button in the upper right of the Excel add-in.
3. Set the **Publish batch size** field as desired. The value that you set must be less than the system-wide publish batch limit.
4. Select **OK**.
5. Save the workbook. If you don't save the workbook after you make changes to the add-in settings, those changes don't persist when the workbook is reopened.

Excel workbook template authors can use the same procedure to set the publish batch size for templates before they upload them into the system.

Copy environment data

The data that is read into the workbook from one environment can be copied to another environment. However, you can't just change the connection URL, because the data cache in the workbook will continue to treat the data as existing data. Instead, you must use the Copy Environment Data functionality to publish the data to a new environment as new data.

1. Select the **Options** button (the gear symbol). On the **Data Connector** FastTab, select **Copy Environment Data**.

2. Enter the server URL for the new environment.

3. Select **OK**, and then select **Yes** to confirm the action. The Excel add-in restarts and connects to the new environment. The existing data in the workbook is treated as new data.

4. After the Excel add-in restarts, a message box states that the workbook is in Environment copy mode.

5. To copy the data into the new environment as new data, select **Publish**. To cancel the environment copy operation and review the existing data in the new environment, select **Refresh**.

Troubleshooting

There are a few issues that can be resolved through some easy steps.

- **The "Load applets" link appears** – For more information about this problem, see Load applets.
- **You receive a "Forbidden" message** – If you receive a "Forbidden" message while the Excel add-in is loading metadata, the account that is signed in to the Excel add-in doesn't have permission to use the targeted service, instance, or database. To resolve this problem, verify that the correct user name appears in the upper-right corner of the Excel add-in. If an incorrect user name appears, select it, sign out, and then sign back in.
- **A blank webpage appears over Excel** – If a blank webpage opens during the sign-in process, the account requires AD FS, but the version of Excel that is running the Excel add-in isn't recent enough to load the sign-in dialog box. To resolve this problem, update the version of Excel that you're using. To update the version of Excel when you're in an enterprise that is on the deferred channel, use the Office deployment tool to move from the deferred channel to the current channel.
- **You receive a timeout while you publish data changes** – If you receive timeout messages while you're trying to publish data changes to an entity, consider reducing the publish batch size for the affected workbook. Entities that trigger larger amounts of logic on record changes might require updates to be sent in smaller batches to help prevent timeouts.

Last updated on 02/26/2026

Open in Excel experiences FAQ

Article • 01/23/2025

Applies to these Dynamics 365 apps:
Commerce, Finance, Human Resources, Supply Chain Management

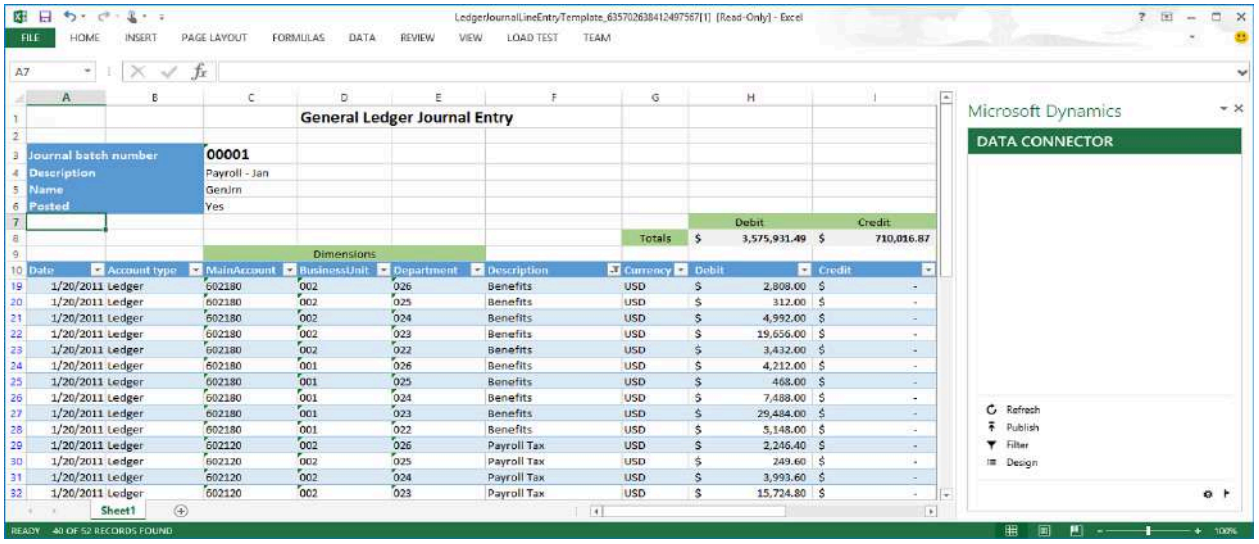
Learn about creating Open in Office experiences for Excel and Word.

What are Open in Excel experiences?

Open in Excel experiences are:

- Based on entities and the OData services that they create.
- Dynamically-generated or based on a pre-defined template.
- Editable and refreshable via the Excel Add-in.

The following image shows the **Excel Add-in** being used for Journal entry.



Where are the Open in Excel experiences?

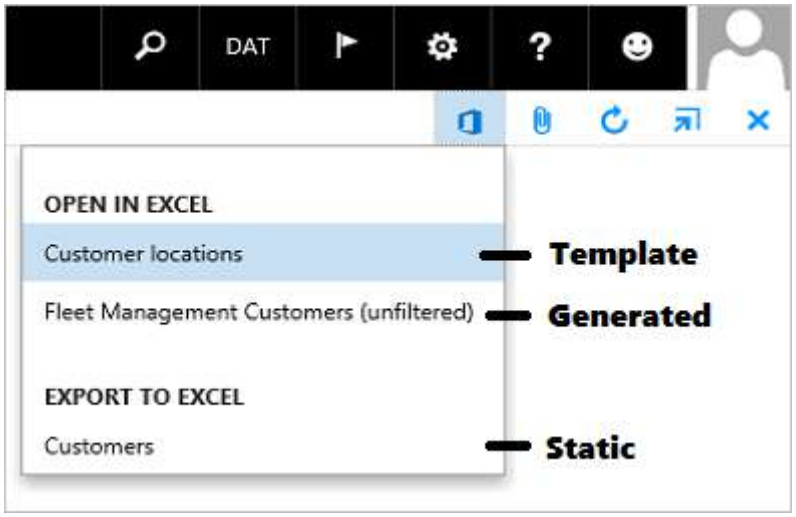
Open in Excel experiences are found under in the Open in Excel section of the Open in Microsoft Office menu, but an explicit button can be added for these experiences.

What's the difference between Export to Excel and Open in Excel?

The Export to Excel options and experiences are both found in the Open in Microsoft Office menu:

- The Export to Excel options are static exports of grid data. Each one corresponds to a visible grid. All the grid data for the current filter is placed into a workbook.
- The Open in Excel experiences utilize the Excel Add-in to facilitate refresh and publish.

The following image shows the **Open in Microsoft Office** menu on the **Fleet Customers** form with a template **Open in Excel** option, a generated **Open in Excel** option, and a static **Export to Excel** option.



When will an entity show as an Open in Excel option?

When an entity has the same root datasource (table) as a form, it will be added as an option in the Open in Excel section of the Open in Microsoft Office menu. This is referred to as a "generated" option.

> ⓘ Note
>
> The user must have security access to the underyling entity for that option to appear in the Open in Excel menu. See Security and data entities for more details.

What fields will be shown in the workbook?

The default fields that will be added into the workbook are the key and mandatory fields of the entity. If a different set of fields should be provided by default, then those fields can be added into the **AutoReport field group** on the entity. The following image shows the Visual Studio view of the AutoReport field group for the FMCustomerEntity.

What fields will be shown when an entity is the target of a lookup?

When a relationship is defined between two entities, if the identifier for one entity is shown on the other entity, then the fields that will be shown in that lookup are either the key fields, or the fields in the **AutoLookup field group** if it isn't empty. Relationship lookups aren't currently supported, but they will eventually display in the app in a similar way to the enumeration lookups.

What should be done to make an entity ready for use in Excel?

Define the AutoReport and AutoLookup field groups and test them using the Excel App design experience.

Why does an automatically added entity option have “(unfiltered)” after the entity name?

Currently, a filter isn't added to these options, hence the term “(unfiltered)”. In the future, an attempt will be made to apply the filter from the form to these options. For example, if a list of Customers was filtered to just Customers in the state of California, then, in the future, the entity will be scanned for the state field and if it's found then a filter would be added automatically.

How can an entity be added as an Open in Excel option on a form that doesn’t share the same root datasource?

A generated Open in Excel option can be added on any form by implementing the OfficeIGeneratedWorkbookCustomExporter interface. When adding a generated option programmatically, the set of fields can be explicitly specified. For more information, see Modifying the Open in Office menu through interfaces.

What are the region-specific considerations for defining entities?

The Open in Excel generated experiences can be made region-specific by adding region-specific fields into the AutoLookup group. These region-specific fields will then be included in the generated workbook.

How can I create a custom lookup for an entity field in Excel?

A custom lookup can be shown for an Entity field.

- Name - The method needs to have a name that is “lookup_<fieldname>” e.g. a field “MyField” could have a lookup method “lookup_MyField”.
- Attributes – Attributes need to be added to the method:
  - SysODataActionAttribute(str <name>, Boolean <isInstanceMethod>)
  - SysODataCollectionAttribute(str <name>, Types <type>, “Value”)
- Return – The method should return a list of strings.

Example

X++

```
public class ExportToExcel_SimpleEntity extends common
{
    [SysODataActionAttribute("Lookup_StringLookupField", true),
    SysODataCollectionAttribute("return", Types::String, "Value")]
    public List lookup_StringLookupField()
    {
        List lookupList = new List(Types::String);
        const int items = 5;

        for (int item = 0; item < items; item++)
        {
            lookupList.addEnd(strfmt('%1 - %2 (%3)', this.StringField,
this.IntField, item));
        }

        return lookupList;
    }
}
```

How does the app get injected into a workbook to start building a template?

The Excel Add-in is injected into a workbook when a generated Open in Excel experience is triggered or when a workbook is created using the **Common** > **Common**

> **Office integration** > **Excel workbook designer** form.

- The **Create workbook** button will add the selected entity and fields, a pointer to the server, and the app into a workbook.
- The **Create blank workbook** button adds a pointer to the server and the app into a workbook.
- The **View related** form will navigate to the form relating to the currently selected entity to more easily review the effect of data changes made in Excel.
- The **Get entity record count** button will show the record count for the currently selected entity. The Excel Add-in will handle large sets of data within the memory limits of a user's machine. By default, the Excel Add-in has a data governor that restricts the data size to one million cells but, depending on the performance abilities of the user's machine, this can usually be extended to around 2.5 million cells.

After obtaining a workbook containing the Excel Add-in, additional datasources can be added using the **Design** button. Currently, datasources can't be removed.

When will a template show as an Open in Excel option?

When a template listed in the **Common** > **Common** > **Office integration** > **Document templates** form (DocuTemplate) has ShowInOpenInOfficeMenu set to Yes and has the same root datasource (table) as the current form, it will be added as an option in the Open in Excel section of the Open in Microsoft Office menu.

Will a filter be added to the template?

In the **Document Templates** form, the standard filter for “current record” can be turned on and off. If the filter is on, when the template is invoked as an Open in Excel option, then a filter for the current record will be added to the workbook. The filter will be the key fields and their values.

How can templates be defined in metadata and code and loaded automatically?

When adding a template into the **Document Templates** form, it's added for that instance and is referred to as a “user-defined” template. Templates can also be defined in metadata and code and loaded automatically, thus making them “system-defined”

templates. To create a system-defined template using metadata and code, you need to do the following:

- Define the template.
- Create a new resource in a project.
- Define a new class that extends the DocuTemplateRegistrationBase class and add an implementation of the registerTemplates method.

The LedgerJournalLineEntryTemplateRegistration and FMTemplateRegistrations classes are good examples of template registrations defined in code. The LedgerJournalLineEntryTemplate and FMTemplateCustomersWithLocations resources are the corresponding templates stored in metadata as resources. When a template has a registration class, it will be loaded when the **Reload system templates** button is clicked in the **Document Templates** form.

How do templates get loaded into a fresh deployment?

To load system defined templates, click the **Reload system templates** button in the **Common** > **Common** > **Office integration** > **Document templates** form.

How do I decide if I should create a template?

A template is an artifact that needs to be maintained and versioned. If you can avoid defining a template without sacrificing much from the user experience, then you probably should use a template. Create a template if:

- You need additional content or formatting in the template.
- You want to combine multiple entities/datasources in the same workbook.

Don’t create a template if:

- You can just specify a set of fields to show in a table binding.

What are the region-specific considerations for templates?

When creating a template for an entity that has region-specific fields, you should leave those region-specific fields out of the template, otherwise all users will see the region-specific fields. Templates should cater to the majority of users by default and region-

specific users can add those fields using the easy-to-use design experience of the Excel Add-in. The region-specific fields and columns can be added by users as needed. That template can be either saved to local computer for reuse by a single user or uploaded via the Document Templates form for reuse by any user of that instance. A couple of other considerations:

- If a region has a region-specific entity, then a region-specific template could be created.
- If a region is important enough, then you could define a region-specific template as well as a region-generic template.

How do I add an explicit button for a template Open in Excel option?

An explicit button can be added for Open in Excel experiences. The label shown on the button should usually be "Open target in Excel" where target is the name of the target data like "lines" or "catalog". The code behind such a button will:

- Obtain the template to be used.
- Add a filter.
- Pass the template to the user.

An example of this code can be found on the **LedgerJournalTable** form (**General ledger** > **Journals** > **General journal**) in the **Clicked** method on the **OpenLinesInExcel** button.

X++

```
[Control("Button")]
class OpenLinesInExcel
{

    /// <summary>
    /// Opens the current journal in Excel for line entry and editing
    /// </summary>
    public void clicked()
    {
        super();

        const str templateName =
resourceStr(LedgerJournalLineEntryTemplate);
                DocuTemplate template =
DocuTemplate::findTemplate(OfficeAppApplicationType::Excel, templateName);

        // Ensure the template was present
        if (template && template.TemplateID == templateName)
        {
```

```
            Map filtersToApply = new Map(Types::String, Types::String);

            // Create lines filter
            ExportToExcelFilterBuilder filterBuilder = new
ExportToExcelFilterBuilder(tablestr(LedgerJournalLineEntity));
            str filterString =
filterBuilder.areEqual(fieldstr(LedgerJournalLineEntity,
JournalBatchNumber), LedgerJournalTable.JournalNum);
            filtersToApply.insert(tablestr(LedgerJournalLineEntity),
filterString);

            // Create header filter
            filterBuilder = new
ExportToExcelFilterBuilder(tablestr(LedgerJournalHeaderEntity));
            filterString =
filterBuilder.areEqual(fieldstr(LedgerJournalHeaderEntity,
JournalBatchNumber), LedgerJournalTable.JournalNum);
            filtersToApply.insert(tablestr(LedgerJournalHeaderEntity),
filterString);

            // Generate the workbook using the template and filters
            DocuTemplateRender renderer = new DocuTemplateRender();
            str documentUrl = renderer.renderTemplateToStorage(template,
filtersToApply);

            // Pass the workbook to the user
            if (documentUrl)
            {
                Browser b = new Browser();
                b.navigate(documentUrl, false, false);
            }
            else
            {

error(strFmt("@ApplicationFoundation:DocuTemplateGenerationFailed",
templateName));
            }
        }
        else
        {
            warning(strFmt("@ApplicationFoundation:DocuTemplateNotFound",
templateName));
        }
    }

}
```

To programmatically add generated and template Open in Excel options, Open in Excel options can be added by implementing the ExportToExcelIGeneratedCustomExport and ExportToExcelITemplateCustomExport interfaces. This allows the addition of options to forms where the entity or template doesn't have the same table as the root datasource. An example of when you would use this capability is on forms without a datasource,

potentially containing only a collection of form parts. The following example adds generated and template Open in Excel options programmatically to the **FMRental** form.

X++

```
[Form]
public class FMRental extends FormRun implements
ExportToExcelIGeneratedCustomExport, ExportToExcelITemplateCustomExport
{
...

    public List getExportOptions()
    {
        List exportOptions = new List(Types::Class);

        ExportToExcelExportOption exportOption =
ExportToExcelExportOption::construct(ExportToExcelExportType::CustomGenerate
d, int2str(1));
        exportOption.setDisplayNameWithDataEntity(tablestr(FMRentalEntity));
        exportOptions.addEnd(exportOption);

        ExportToExcelExportOption exportOption2 =
ExportToExcelExportOption::construct(ExportToExcelExportType::CustomTemplate
, int2str(2));
        exportOption2.displayName("Analyze rentals");
        exportOptions.addEnd(exportOption2);

        return exportOptions;
    }

    public ExportToExcelDataEntityContext
getDataEntityContext(ExportToExcelExportOption _exportOption)
    {
        ExportToExcelDataEntityContext context = null;

        if (_exportOption.id() == int2str(1))
        {
            context =
ExportToExcelDataEntityContext::construct(tablestr(FMRentalEntity),
tablefieldgroupstr(FMRentalEntity, AutoReport));
        }

        return context;
    }

    public System.IO.Stream getTemplate(ExportToExcelExportOption
_exportOption)
    {
        System.IO.Stream stream = null;

        if (_exportOption.id() == int2str(2))
        {
            stream =
Microsoft.Dynamics.Ax.Xpp.MetadataSupport::GetResourceContentStream(resource
```

```
str(FMRentalEditableExportTemplate));
        }

        return stream;
    }

    public void updateTemplateSettings(ExportToExcelExportOption
_exportOption,
Microsoft.Dynamics.Platform.Integration.Office.ExportToExcelHelper.SettingsE
ditor _settingsEditor)
    {
    }
...
```

How do I add a filter for a programmatically-added template Open in Excel option?

A template Open in Excel option can be programmatically added by implementing the ExportToExcelITemplateCustomExport interface and providing a template in the getTemplate method. A filter for that option can be programmatically added by using the ExportToExcelFilterBuilder API in the updateTemplateSettings method.

X++

```
public void updateTemplateSettings(ExportToExcelExportOption _exportOption,
Microsoft.Dynamics.Platform.Integration.Office.ExportToExcelHelper.SettingsE
ditor _settingsEditor)

{

_settingsEditor.SetFilterExpression(tableStr(RetailTmpBulkProductAttributeVa
lueEntity), element.getExportToExcelFilterExpression());

DictDataEntity dictDataEntity = new
DictDataEntity(tableNum(RetailTmpBulkProductAttributeValueEntity));

_settingsEditor.SetFilterExpressionByPublicName(dictDataEntity.publicEntityN
ame(), element.getExportToExcelFilterExpression());

}
```

After a filter has been added programmatically, the resulting filter can be viewed in the Excel Add-in using the **Filter** button.

How do I enable relationship lookups in Excel?

To enable relationship lookups in the Excel Data Connector, you must ensure that the following metadata is set.

- The Role and Related Data Entity Role defined on the relationship need to be unique among all relationships on both the source and target entity. Also, the relation role properties must be unique across all entities. This is particularly important for relationships involving entities with many relationships, such as DimensionCombinationEntity. If you're not seeing an expected lookup, try changing the role names to the following format:
    - **Role**: [this entity's public name] + [target entity's public name] + [target entity field] + "Source"
    - **Related Data Entity Role**: [this entity's public name] + [target entity's public name] + [target entity field] + "Target"
    - The Cardinality and Related Data Entity Cardinality need to be set appropriately.
    - At least one constraint must be added to the relationship. With the exception of dimension relationships, which are a special case, the properties constrained must both be public.

How can I enable users to create new header records as well as lines in a workbook?

To enable creation of header records and related lines, the header data source must be added as a set of "fields" and the lines data source must be added as a related table. This pattern can work well for document data entry scenarios such as Journal entry.

To learn more about header records and related lines, watch the short Create an Excel template for header and line patterns in Dynamics 365 Finance video.

To design a workbook with header fields and a lines table that enables header creation:

1. In the Excel Add-in, click **Design** to open the Designer. Select **Add fields** to add a header data source.
2. Select the header fields that you want to use. Be sure to include all the key fields or the **New** button won't be enabled.
3. For all of the string header value fields, manually apply "Text" format for that cell using **Excel ribbon** > **Home tab** > **Number group** > **set "Number"** in the format drop-down menu. If the Text format isn't manually set on a string field and there's a string value with leading zeros like "00045", then Excel will automatically change it to "45" and an error will be shown like: "*Unable to change the value of*

PurchaseOrderHeader's PurchaseOrderNumber field as it's a key field". Currently, the API doesn't allow for automatically applying the text formatting on individual cells (versus table columns).

4. In the Designer, on the header data source, click the **Add related table** button represented by a double plus icon.
5. Select the line fields that you want to use.

Here's an example of a header data source with a related table data source.

- PurchaseOrderHeader (Fields)
  - dataAreaId
  - PurchaseOrderNumber
  - PurchaseOrderName
  - OrderVendorAccountNumber
- PurchaseOrderLine (Table - related)
  - LineNumber
  - ItemNumber
  - LineDescription
  - OrderedPurchaseQuantity
  - LineAmount

To use a header and lines workbook to create a new header and lines, follow these steps:

1. In the workbook, move the focus to a cell with a header value.
2. In the Excel Add-in, click **New**.
3. Enter header values and lines as needed.
4. Select **Publish**.

How can fields be added, removed, or moved within an existing template workbook?

Fields can be added into an existing template workbook by editing the workbook stored in **Document Templates**.

1. Get the original template workbook.
   a. Open the **Document Templates** form.
   b. Find the existing template workbook.
   c. Download the workbook.
   d. Open the workbook and enable editing so that the Excel Add-in runs.
2. Make changes to the template.
   a. In the Excel Add-in, select **Design**.

b. Click the **Edit** button (pencil icon) next to the datasource that you want to add a field into.
c. Add fields by moving them from the **Available fields** list into the **Selected fields** list. Double-clicking a field will move it. Remove fields by moving them from the **Selected fields** list into the **Available fields** list.Move fields using the **Up** and **Down** buttons.
d. After changes are complete, select **Update**, select **Yes** to confirm, and then select **Done** to exit the Designer (if appropriate, select **Refresh** to verify that the data is correctly populated).
e. Clear the data from the template before upload by clicking **Options** (gear icon), expand the **Data Connector** section, then click the **Clear binding data** button.
f. Use **Save As** to store the template somewhere temporarily.

3. Upload the changed template.
   a. Return to the **Document Templates** form and upload the changed template.
   b. Click **New** and browse to find the changed template.
   c. Select the saved template file and click **Open**.
   d. In the **Upload template** dialog box, remove the underscore and trailing random number from the name For example, "CustInvoiceJournalTemplate_636564840743000567" becomes "CustInvoiceJournalTemplate".
   e. A confirmation dialog box should show that "A template with this name already exists...", click **Yes** to confirm replacement of the previous template. Note that if this confirmation isn't shown, then the template name is different and it's being uploaded as a new template.
4. Open the form that the template is used on and use the changed template.

Troubleshooting

If you aren't seeing an expected lookup, validate relationship metadata by checking the metadata feed available at [YourSiteURL]/data/$metadata. Search the $metadat feed for the public name of your entity to find its EntityType element, then make sure there's a child NavigationProperty element with a name equal to the Role value of the relationship. If the navigation property exists, it will be used by the Excel Data Connector to show a relationship lookup. Lookups aren't shown under the following conditions:

- All of the entity's key fields are included as constraints in the relationship.
- The selected field is a key and the selected record isn't new.
- The authenticated user does not have permission to access the entity targeted by the lookup.

How do dimensions work?

The easiest way to set up dimension metadata on data entities is to use the data entity creation wizard, which will automatically create the private relationships and public display value fields exactly as the dimensions framework needs them. If you want to customize your dimensions setup, see Add dimensions to Excel templates. Lookups, are only generated automatically for non-ledger dimensions. Custom dimensions aren't supported currently. If you want to enable lookups for ledger dimensions (MainAccount, Department, CostCenter, etc.), see Add dimensions to Excel templates for guidance on creating relationships on DimensionCombationEntity and DimensionSetEntity fields. When those relationships are present, relationship lookups will be displayed in the Excel Data Connector. The Excel Data Connector supports two types of dimension data entry: editing the display value directly or editing each attribute of the display value in a separate column. If both the display value column and the individual attribute columns are bound, they can both be edited and published separately. If both the display value and an individual attribute are edited in the same row, the individual attribute change overrides the display value change.

How do I create formula table columns?

If a formula is needed in a table, then add a formula column. When in the field selection page for a table binding, click the **Formula** button above the Selected fields list to add a new formula column. The label and value for the formula are entered in the fields immediately below the Selected fields list. After adding a new formula column, leave the value empty and click **Update**. After the field has been added to the table, use standard Excel capabilities to create a formula, then copy the formula and paste it into the formula column value field. When defining a formula, make sure there's more than one row in the table, otherwise the formula that Excel provides may be for ALL rows instead of THAT row. To specify just the current row, the at sign (@) is needed. For example, sum of four columns for all rows "=SUM(Table1[[ColumnA]:[ColumnD]])" versus sum of four columns for the current row "=SUM(Table1[@[ColumnA]:[ColumnD]])".

Known issues

Error triggered when selecting custom Excel templates

This issue occurs when the user selects an Excel template from an **Open in Excel** button and receives the following error: `Record for Id {guid} not found`

The error occurs when a custom Excel template can't be found. One scenario that could cause this error is moving the finance and operations database between environments, but not copying the Excel templates.

To resolve this issue, delete the database entry for the template from the **Document templates** page (**Common** > **Common** > **Office integration** > **Document templates**) and then reload the template.

Refresh doesn’t automatically occur in old templates

The ability to control “refresh on open” was added as a setting. To add this to the default behavior, existing templates and workbooks need to have the **Refresh on open** check box selected in **Options** > **Data Connector** > **Refresh Options**.

Error finding entity

The reference to entities changed from using the Private Entity Name (DataEntity.Name) to Public Entity Name (DataEntity.PublicEntityName). If the public and private names for an entity were different and that entity was used in an Excel template or workbook, then this will cause the following error to be displayed in the Excel App: “Error Finding Entity. Details: Entity "<DataEntity.Name>" not found”.

To resolve this, change the binding information in the affected template so that it points to DataEntity.PublicEntityName instead of DataEntity.Name.

1. For the DataEntity.Name that needs to be replaced, determine the DataEntity.PublicEntityName, for example replace FMCustomerEntity with FleetCustomer.
2. Find the affected template.
3. Change the file extension on the template from .xlsx to .zip.
4. The file to be changed will be one of the webextension*.xml files in the xlwebextensions directory, such as 2015-05-25-FleetCustomersWithLocations.zipxlwebextensionswebextension2.xml.
5. Open the file to ensure that you have the correct location.
6. Find the DataEntity.Name, such as FMCustomerEntity.
7. Extract the zip file.
8. Open the webextension xml file.
9. Replace the DataEntity.Name with the corresponding DataEntity.PublicEntityName.
10. Save the webextension .xml file changes.
11. Rename the old zip file, for example, add “.old” to the name.

12. Create a new zip file of all the previously extracted content. This usually involves highlighting the content inside the archive/zip folder and creating a zipped folder using that content.
13. Verify that the zip file has the “_rels”, “docProps”, and “xl” folders in the root of the zip file.
14. Rename the zip file as needed, for example rename the file 2015-05-25-FleetCustomersWithLocations.zip.
15. Change the zip file extension to .xlsx.
16. Re-publish the workbook .xlsx file, if needed.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add templates to the Open lines in Excel menu

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how you can promote a template to the **Open lines in Excel** menu that is available on journal pages.

Some of the most frequently used templates are the journal templates. You can promote these journal templates so they appear on the **Open lines in Excel** menu by default. However, when you add a new template to the system, you must promote it so it's available on both the **Open in Office** menu and the **Open lines in Excel** menu. To promote a new template, follow these steps:

1. Create a Microsoft Excel template, and save it locally. For more information, see Create Open in Excel experiences.

2. In Microsoft Visual Studio, create a new project for a model that has a reference to the ApplicationSuite model.

3. Create a new class, implement the **LedgerIJournalExcelTemplate** interface, and extend **DocuTemplateRegistrationBase**. Your implementation (supported journal type, and so on) defines the context that your template will be available as an option for in the Open in Excel experience. This example uses LedgerJournalHeaderEntity and LedgerJournalLineEntity, but you're not limited to these entities. You can define your own entities, as long as they follow the journal header/line entity pattern. Here's an example from the **LedgerDailyJournalExcelTemplate** class.

   X++

   ```
   using Microsoft.Dynamics.Platform.Integration.Office;
   public class TestNewTemplate extends DocuTemplateRegistrationBase implements
   LedgerIJournalExcelTemplate
   {
       private const DocuTemplateName ExcelTemplateName =
   resourceStr(TestNewTemplate);
       private const EntityName LineEntityName =
   tableStr(LedgerJournalLineEntity);
   ```

```
    private const FieldName LineEntityJournalNum =
fieldStr(LedgerJournalLineEntity, JournalBatchNumber);
    private const FieldName LineEntityDataAreaId =
fieldStr(LedgerJournalLineEntity, dataAreaId);
    private const FieldName HeaderEntityName =
tableStr(LedgerJournalHeaderEntity);
    private const FieldName HeaderEntityJournalNum =
fieldStr(LedgerJournalHeaderEntity, JournalBatchNumber);
    private const FieldName HeaderEntityDataAreaId =
fieldStr(LedgerJournalHeaderEntity, dataAreaId);
    /// <summary>
    /// A boolean value which indicates whether the journal type is supported
for the Excel template.
    /// </summary>
    /// <param name = "_ledgerJournalType">The ledger journal type.</param>
    /// <returns>True if the journal type is supported; otherwise, false.
</returns>
    public boolean isJournalTypeSupported(LedgerJournalType
_ledgerJournalType)
    {
        return _ledgerJournalType == LedgerJournalType::Daily;
    }
    /// <summary>
    /// Gets the document template name.
    /// </summary>
    /// <returns>The document template name</returns>
    public DocuTemplateName documentTemplateName()
    {
        return ExcelTemplateName;
    }
    /// <summary>
    /// Gets a collection of the supported account types for the entity.
    /// </summary>
    /// <returns>A collection of <c>LedgerJournalACType</c> values.</returns>
    public Set supportedAccountTypes()
    {
        Set accountTypeSet = new Set(Types::Integer);
        accountTypeSet.add(LedgerJournalACType::Ledger);
        return accountTypeSet;
    }
    /// <summary>
    /// Gets a collection of the supported offset account types for the
entity.
    /// </summary>
    /// <returns>A collection of <c>LedgerJournalACType</c> values.</returns>
    public Set supportedOffsetAccountTypes()
    {
        Set offsetAccountTypeSet = new Set(Types::Integer);
        offsetAccountTypeSet.add(LedgerJournalACType::Ledger);
        return offsetAccountTypeSet;
    }
    /// <summary>
    /// Validates the journal is valid for the template.
    /// </summary>
    /// <param name = "_ledgerJournalTable">The <c>LedgerJournalTable</c>
```

```
record.</param>
    /// <returns>True if the journal is valid for the template; otherwise,
false.</returns>
    public boolean validateJournalForTemplate(LedgerJournalTable
_ledgerJournalTable)
    {
        return
LedgerJournalExcelTemplate::validateJournalForTemplate(_ledgerJournalTable,
this);
    }
    public void registerTemplates()
    {
        this.addTemplate(
            OfficeAppApplicationType::Excel,
            ExcelTemplateName,
            ExcelTemplateName,
            'Test new template',
            'Test new template',
            NoYes::No,
            NoYes::No,
            NoYes::No);
    }
    /// <summary>
    /// The resource name of the header entity.
    /// </summary>
    /// <returns>The resource name of the header entity.</returns>
    public EntityName headerEntityName()
    {
        return HeaderEntityName;
    }
    /// <summary>
    /// The resource name of the line entity.
    /// </summary>
    /// <returns>The resource name of the line entity.</returns>
    public EntityName lineEntityName()
    {
        return LineEntityName;
    }
    /// <summary>
    /// The field name for the header journal batch number.
    /// </summary>
    /// <returns>The field name for the header journal batch number.</returns>
    public FieldName headerJournalBatchNumberFieldName()
    {
        return HeaderEntityJournalNum;
    }
    /// <summary>
    /// The field name for the header data area.
    /// </summary>
    /// <returns>The field name for the header data area.</returns>
    public FieldName headerDataAreaFieldName()
    {
        return HeaderEntityDataAreaId;
    }
    /// <summary>
```

```
    /// The field name for the line journal batch number.
    /// </summary>
    /// <returns>The field name for the line journal batch number.</returns>
    public FieldName lineJournalBatchNumberFieldName()
    {
        return LineEntityJournalNum;
    }
    /// <summary>
    /// The field name for the line data area.
    /// </summary>
    /// <returns>The field name for the line data area.</returns>
    public FieldName lineDataAreaFieldName()
    {
        return LineEntityDataAreaId;
    }
    /// <summary>
    /// Append additional filter to the default filtering behavior.
    /// </summary>
    /// <returns>The original filter with new filter(s) appended; Otherwise,
the original filter</returns>
    public FilterCollectionNode appendHeaderEntityFilters(FilterCollectionNode
_headerFilter, ExportToExcelFilterTreeBuilder _headerFilterBuilder)
    {
        return _headerFilter;
    }
    /// <summary>
    /// Append additional filter to the default filtering behavior.
    /// </summary>
    /// <returns>The original filter with new filter(s) appended; Otherwise,
the original filter</returns>
    public FilterCollectionNode appendLineEntityFilters(FilterCollectionNode
_lineFilter, ExportToExcelFilterTreeBuilder _lineFilterBuilder)
    {
        FilterCollectionNode lineFilter = _lineFilterBuilder.and(
            _lineFilterBuilder.areEqual(fieldStr(LedgerJournalLineEntity,
AccountType), LedgerJournalACType::Ledger),
            _lineFilterBuilder.areEqual(fieldStr(LedgerJournalLineEntity,
OffsetAccountType), LedgerJournalACType::Ledger));
        return _lineFilterBuilder.and(_lineFilter, lineFilter);
    }
}
```

4. Build the project/model that has the new resources. You should have one new resource and one new class.

5. In the client, go to **Common** > **Common** > **Office integration** > **Document templates** > **Reload system templates**. You see the new template in the list. If you open the journal page that you added the template to, you also see that template on the **Open lines in Excel** menu.

Additional resources

Create Open in Excel experiences

Last updated on 03/14/2026

Customize the Open in Microsoft Office menu

Article • 12/31/2024

Applies to these Dynamics 365 apps:
Commerce, Finance, Supply Chain Management

Most pages include an Open in Microsoft Office menu. This topics provides information about the Open in Office menu, and explains how customize it by adding, removing, and changing options.

Overview

The **Open in Microsoft Office** menu button (**Open in Office** menu) is a system-defined button that appears on pages. The **Open in Office** menu contains menu items that let you export data to various Office products, such as Microsoft Excel and Microsoft Word. The following table describes the menu items on the **Open in Office** menu.

Expand table

Menu item	Description
Export to Excel	The data is exported to an Excel workbook. The workbook contains no references back to finance and operations, and the data can't be refreshed.
Export to Word	The data is exported to a Word document. The document contains no references back to finance and operations, and the data can't be refreshed.
Open in Excel	A workbook is created that contains the Microsoft Dynamics Office add-in. The workbook contains a reference back to finance and operations, and the data can be refreshed, updated, and published from the Data Connector that is hosted in the add-in.

How menu items are added to the Open in Office menu

The export options are added to the **Open in Office** menu in the following manner:

1. In the **Export to Excel** group, a menu item is added for each visible grid.

2. For each root data source on the page, the set of data entities that have the same root data source is determined. For each of these data entities, the following menu items are added:

    - In the **Open in Excel** group, a menu item is added for a default export of the data entity.
    - In the **Open in Excel** group, a menu item is added for each Document Template record of the **Excel** type that has the same root data entity and is marked for inclusion on the **Open in Office** menu.
    - In the **Export to Word** group, a menu item is added for each Document Template record of the **Word** type that has the same root data entity and is marked for inclusion on the **Open in Office** menu.

Default exports

The **Open in Office** menu provides a default export for each data entity. This export includes all the fields in the **AutoReport** group on the data entity. If **SaveDataPerCompany** is set to **Yes** for the data entity, a filter is applied to limit the data to the current company.

Document templates

Document templates can be added from the **Document templates** page. Several fields that are associated with each Document Template record control the behavior of that template on the **Open in Office** menu.

Expand table

Field	Description
Root data entity	The root data entity of the template. The root data entity is used to determine which pages the template can be included on.
List in Office menu	If this field is selected, the template will be included on the **Open in Office** menu on applicable pages. (The applicable pages depend on the root data entity).
Apply record filter	If this field is selected, the data will be filtered, based on the record that is currently selected on the page.
Apply company filter	If this field is selected, the data will be filtered, based on the current company.

Trimmed template columns and fields

For Excel templates that are included on the **Open in Office** menu, columns and fields will be trimmed from the workbook, based on the configuration keys that are applied to the system and the applicable country/region context. If a configuration key is associated with a column or field in the workbook, the column or field will be removed if the configuration key is disabled. If a set of country/region codes is associated with the column or field in the workbook, the column or field will be removed if the country/region code isn't in scope.

Open in Office menu customization

There are several methods for customizing the content that appears on the **Open in Office** menu on a given page. For example, you can customize the content statically through metadata properties on model element and code attributes. However, customization via code gives you the finest level of control. In code, you can either implement one or more interfaces on a page, or use extensions and event subscriptions. The following sections describe the customization scenarios that are most often used.

Modifying the Open in Office menu through interfaces

If you must modify a page that you own, interfaces are the most appropriate customization method, because they give access to all private members of the page and allow for deeper customization. You can apply the following interfaces to the code for a page.

Expand table

Interface	Description
OfficeIMenuCustomizer	Use this interface to modify the set of data entities that is considered for a page and add custom menu items.
OfficeIGeneratedWorkbookCustomExporter	Use this interface to do a custom export of a workbook that is generated at run time.
OfficeITemplateCustomExporter	Use this interface to do a custom export that is based on a Document Template record.

Modifying the Open in Office menu through extensions and event subscriptions

If you must modify a page that you don't own, you should avoid using interfaces, because that approach will require over-layering. Instead, you should do the customization through extensions and event subscriptions. To use this approach, implement an extension class that subscribes to the **OnIntializing** event of the page that you're customizing. From this event handler, get the **OfficeFormRunHelper** for the page, and subscribe to its **OfficeMenuInitializing** event. The following example shows sample code for this approach.

X++

```
public static class MyForm_Extension
{
    [FormEventHandler(formStr(MyForm), FormEventType::Initializing)]
    public static void
ExportToExcel_DataEntityCustom_OnInitializing(xFormRun sender, FormEventArgs
e)
    {
        FormRun formRun = sender as FormRun;
        if (formRun)
        {
            OfficeFormRunHelper officeHelper = formRun.officeHelper();
            if (officeHelper)
            {
                officeHelper.OfficeMenuInitializing +=
eventhandler(MyForm_Extension::officeMenuInitializingHandler);
            }
        }
    }
    private static void officeMenuInitializingHandler(FormRun _formRun,
OfficeMenuEventArgs _eventArgs)
    {
        // Modify the OfficeMenuOptions available on the
OfficeMenuEventArgs.menuOptions() as necessary.
    }
}
```

Office menu best practices

Avoid APIs that could initialize the metadata cache during form load

The following classes could initialize the metadata cache, which can be extremely time consuming. For this reason, these classes should never be used in methods that are executed during form load (such as `init()` and `run()`), as this can cause a form to take a long time to load.

- ExportToExcelMetadataCache
- ExportToExcelDataEntityContext
- OfficeDataEntityExportMenuItem
- ExportToExcelDataEntityHelper
- ExportToExcelFilterTreeBuilder

Typical customization scenarios

The following examples assume that the **_menuOptions** variable contains the **OfficeMenuOptions** instance that you're customizing.

Modifying the set of data entities that is considered for a page

Many of the menu items on the **Open in Office** menu are added automatically, based on the data entities that are considered for the page. However, in some cases, the algorithm that is used to determine the set of data entities might not determine the correct set. To modify the set of data entities that is considered for the page, you can use the **OfficeMenuOptions** that is available from either the **OfficeIMenuCustomizer.customizeMenuOptions** method or the **OfficeFormRunHelper.OfficeMenuInitializing** delegate.

> ⚠ **Warning**
>
> Do not use this code in form init() or run(). This can negatively impact form load speed, especially upon system reboot.

X++

```
// Add an entity to the list
OfficeMenuDataEntityOptions entityOptions  =
OfficeMenuDataEntityOptions::construct(tableStr(MyEntity));
_menuOptions.dataEntityOptions().addEnd(entityOptions);
// Remove an entity from the list
ListIterator dataEntityOptionsIterator = new
ListIterator(_menuOptions.dataEntityOptions());
while (dataEntityOptionsIterator.more())
{
    OfficeMenuDataEntityOptions dataEntityOptions =
dataEntityOptionsIterator.value();
    if (dataEntityOptions.dataEntityName() == tableStr(MyOtherEntity))
    {
        dataEntityOptionsIterator.delete();
    }
```

```
    else
    {
        dataEntityOptionsIterator.next();
    }
}
```

Specifying the default data entity–related options that are included

The **OfficeMenuDataEntityOptions** class lets you specify whether to include a menu item for a default export or a menu item that is related to a document template.

X++

```
// Find the entity options if they were included by default.
OfficeMenuDataEntityOptions entityOptions =
_menuOptions.getOptionsForEntity(tableStr(MyEntity));
if (!entityOptions)
{
    // The entity options were not included. Add them.
    entityOptions =
OfficeMenuDataEntityOptions::construct(tableStr(MyEntity);
    _menuOptions.dataEntityOptions().addEnd(entityOptions);
}
entityOptions.includeDefault(false); // Don't include the default export
menu item.
entityOptions.includeTemplates(false); // Don’t include Document Template
related menu items.
```

Adding a custom export menu item – Generating a workbook

To explicitly add a menu item, you must add it to the **OfficeMenuOptions.customMenuItems()** list. To add a menu item that corresponds to a workbook that is generated at run time, use an **OfficeGeneratedExportMenuItem**.

X++

```
OfficeGeneratedExportMenuItem menuItem =
OfficeGeneratedExportMenuItem::construct(tableStr(MyEntity),
"MyCustomGeneratedExportId");
menuItem.setDisplayNameWithDataEntity();
_menuOptions.customMenuItems().addEnd(menuItem);
```

To define what is actually exported, use an **ExportToExcelDataEntityContext**. The method for specifying the **ExportToExcelDataEntityContext** depends on whether you're using interfaces or extensions and event subscriptions to customize the **Open in Office** menu.

Using interfaces

If you're using interfaces, you must implement the **OfficeIGeneratedWorkbookCustomExporter.getDataEntityContext()** method.

```
X++
```

```
public ExportToExcelDataEntityContext
getDataEntityContext(OfficeGeneratedExportMenuItem _menuItem)
{
    ExportToExcelDataEntityContext context = null;
    if (_menuItem.id() == "MyCustomGeneratedExportId")
    {
        context =
ExportToExcelDataEntityContext::construct(tableStr(MyEntity),
tableFieldGroupStr(MyEntity, MyFieldGroup);
    }
    return context;
}
```

Using extensions and event subscriptions

If you're using extensions and event subscriptions, the **OfficeGeneratedExportMenuItem.getDataEntityContext** delegate should be subscribed to before you add the menu item to the **OfficeMenuOptions.customMenuItems()** list. The code for the event handler should resemble the preceding code for the interface. The following example shows how to do the event subscription.

```
X++
```

```
menuItem.getDataEntityContext +=
eventhandler(MyForm_Extension::getDataEntityContextHandler);
```

Adding a custom export menu item – Specifying a document template

To explicitly add a menu item, you must add it to the **OfficeMenuOptions.customMenuItems()** list. To add a menu item that corresponds to a

Document Template record, use an **OfficeTemplateExportMenuItem**.

```
X++
```

```
OfficeTemplateExportMenuItem menuItem =
OfficeTemplateExportMenuItem::construct(OfficeAppApplicationType::Excel,
"MyTemplateId", "MyCustomTemplateExportId");
_menuOptions.customMenuItems().addEnd(menuItem);
```

To modify the template at run time, you can supply a set of initial filters. These filters will replace any filters in the template for the specified data entities. Additionally, you can modify filters and specify many settings by using **WorkbookSettingsManager**. The following sections show examples.

Using interfaces

If you're using interfaces, you must implement the **OfficeITemplateCustomExporter.getInitialTemplateFilters()** and **OfficeITemplateCustomExporter.updateTemplateSettings()** methods.

```
X++
```

```
public Map getInitialTemplateFilters(OfficeTemplateExportMenuItem _menuItem)
{
    Map initialFilters = new Map(Types::String, Types::Class);
    if (_menuItem.id() == "MyCustomTemplateExportId")
    {
        // Add an initial filter.
        ExportToExcelFilterTreeBuilder bldr = new
ExportToExcelFilterTreeBuilder(_menuItem.dataEntityName());
        FilterNode filter = // create the filter…
        initialFilters.insert(_menuItem.dataEntityName(), filter);
    }
    return initialFilters;
}
public void updateTemplateSettings(OfficeTemplateExportMenuItem _menuItem,
Microsoft.Dynamics.Platform.Integration.Office.SettingsManager
_settingsManager)
{
    if (_menuItem.id() == "MyCustomTemplateExportId")
    {
        // Set a new filter.
        ExportToExcelFilterTreeBuilder bldr = new
ExportToExcelFilterTreeBuilder(_menuItem.dataEntityName());
        FilterNode filter = // create the filter…
        Excel.WorkbookSettingsManager workbookSettingsManager =
_settingsManager as Excel.WorkbookSettingsManager;

workbookSettingsManager.SetEntityFilter(entityMetadata.PublicEntityName,
```

```
filter);
        // Adjust settings.
        DataConnectorAppletSettings settings =
settingsManager.DataConnectorSettings;
        DataConnectorAppletUserOptions options = settings.DataOptions;
        options.RefreshOnOpen = true;
        options.EnableDesign = false;
        workbookSettingsManager.DataConnectorSettings = settings;
    }
}
```

Using extensions and event subscriptions

If you're using extensions and event subscriptions, the **OfficeTemplateExportMenuItem.getInitialTemplateFilters** and **OfficeTemplateExportMenuItem.updateTemplateSettings** delegates should be subscribed to before you add the menu item to the **OfficeMenuOptions.customMenuItems()** list. The code for the event handlers should resemble the preceding code for the interface. The following example shows how to do the event subscription.

```
X++
```

```
menuItem.getInitialTemplateFilters +=
eventhandler(MyForm_Extension::getInitialTemplateFiltersHander);
menuItem.updateTemplateSettings +=
eventhandler(MyForm_Extension::updateTemplateSettingsHandler);
```

Additional customizations

The following customizations let you modify the contents of the **Open in Office** menu for a page without using interfaces or extensions and event handlers.

Removing an Export to Excel menu item for a grid

On the **Open in Office** menu, the **Export to Excel** group will contain a menu item for each visible grid. To remove a grid from the **Open in Office** menu, set the **ExportAllowed** metadata property on the grid to **No**. After you make this change, users won't be able to export the grid by using the shortcut menu for the grid.

Renaming an Export to Excel menu item for a grid

To rename the **Export to Excel** menu item that is related to a grid, set the **ExportLabel** metadata property on the grid to the appropriate label.

Removing all menu items for a data entity from all pages

Integration scenarios require that some data entities be publicly available via the OData Service. However, it isn't always appropriate that these data entities appear on the **Open in Office** menu by default. In this scenario, you can add the **OfficeMenuOmit** code attribute to the entity declaration.

```
X++
```

```
[OfficeMenuOmit]
public class MyEntity extends common
{
    // Entity code…
}
```

After you make this change, by default, the entity won't appear on the **Open in Office** menu on pages that have a matching root data source. However, if the entity should be added to a specific page, you can use other customization mechanisms to add it.

Adding a menu item button to a page that corresponds to an Open in Office menu entry

Sometimes, it's appropriate that the Action Pane of a page have a menu item button that corresponds to a custom menu item on the **Open in Office** menu. In this case, you can model a menu item button that has the following properties:

- **Menu Item Type: Action**
- **Menu Item Name: ExportToExcelExportForm**
- **Parameters:** The ID of the menu item

Then, a mouse click of this menu item button is equivalent to a mouse click of the corresponding menu item on the **Open in Office** menu.

Feedback

Was this page helpful? Yes No

Provide product feedback

Configure and send email

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The behavior of the email subsystem is influenced by a combination of administrator configuration, user configuration, and user choices. This article is divided into sections for administrators and users to make it easy to find relevant information.

Both administrators and users set the behavior of the email subsystem.

[Administrator] Email parameters page

Email configuration for the environment

On the **Email parameters** page, administrators can configure the high-level email behavior for the environment from the **Configuration** tab.

Expand table

Field	Description
Batch email provider	Specifies which email provider will be used to send emails that are sent by processes in a batch or non-interactive manner. The Microsoft Graph and Exchange providers will use the account that's associated with the batch process. Note: Microsoft Graph is available in Dynamics 365 Finance version 10.0.38 and later.
Attachment size limit	Specifies the maximum size of a single email that can be sent via the email subsystem.
Email expiration in days	Specifies the number of days before the status of an unsent message is set to **Expired**. A value of **0** means that the default period of 30 days is in effect.

The **Email history** section serves two purposes. First, it provides an entry point to the **Email history** page, where administrators can review all sent emails and also any errors that prevented an email from being sent. Second, it lets you configure how long email history is maintained. By default, the last 30 days of email history are retained. You can adjust this period

by changing the value of the **Number of days to retain email history** field to a non-zero number. If you set the value to **0** (zero), the default number and behavior are used.

The **Email throttling** section enables non-interactive email providers (such as the batch email provider) to adhere to a per-minute sending limit. This feature can help prevent some errors if the system tries to send more emails than the provider allows. Specifically, if an email can't originally be sent because the per-minute sending limit has been reached, the send attempt for the email will be deferred for up to one minute. After ten deferrals, the system will try to send the email regardless. You can remove the per-minute sending limit from a provider by resetting the **per-minute email sending limit** field to **0**. The sending limits for the Office 365 SMTP email provider is automatically set according to the Exchange Online sending limits. Manual configuration is required for all other email providers.

> **Note**
>
> Use of the **Email throttling** feature isn't recommended for the Microsoft Graph and Exchange email providers, because they have their own throttling mechanisms. Therefore, you should ensure that the **per-minute email sending limit** field is set to **0** for both those providers.

Send email with Microsoft Graph

For customers who use Office 365, Microsoft Graph is the recommended email provider for Dynamics 365 finance and operations apps. This email provider replaces the deprecated Exchange email provider.

You must have the following permissions to set up the Microsoft Graph integration:

- You must be a system administrator in the application to link the Dynamics 365 finance and operations environment to Microsoft Graph.
- You must be an administrator for your Microsoft Entra ID account. If you aren't the administrator, an administrative user must perform the first part of the configuration for you.

One-time registration process

To create an app, follow these steps:

1. Sign in to the Azure portal by using an Azure tenant admin account.

   > **Note**

> The user who completes this procedure must have Admin rights for the tenant to register applications.

2. Go to **Microsoft Entra ID** > **App registrations** > **New application**.

3. Enter the following values:

   - **Name** – Enter the name of your app.
   - **Supported account types** – Enter only accounts that are directly in this organization (single tenant).

4. Select **Register**.

5. Make a note of the **Application (client) ID** value. You will use this value to connect to the Microsoft Graph service from your finance and operations environment.

   > ⓘ **Important**
   >
   > Be sure to capture the **Application (client) ID** value before you continue.

6. To add permissions, follow these steps:

   a. Select **Manage** > **API permissions** > **Add a permission** > **Microsoft APIs** > **Microsoft Graph**.

   b. Select **Application permissions**, and enable **Mail.Send**.

      > ⓘ **Note**
      >
      > By default, the app should include the **User.Read** delegated permission for Microsoft Graph. If that permission is missing, you must add it from the delegated permissions.

   c. Select **Add permissions**.

   d. Select **Grant admin consent for** to allow emails to be sent.

7. To create a client secret, follow these steps:
   a. Select **Manage** > **Certificates and secrets**.
   b. On the **Client secrets** tab, select **New client secret**.
   c. Enter a value in the **Description** and **Expires** fields, and then select **Add**.
   d. Make a note of the **Secret value** value. You will use this value to connect to the Microsoft Graph service from your Dynamics 365 finance and operations

environments.

> **Important**
>
> Be sure to capture the **Secret value** value before you continue.

Restrict mailboxes

Administrators can modify mailbox access. For more information, see Limiting application permissions to mailboxes.

Configure Microsoft Graph in Dynamics 365 finance and operations apps

1. In Dynamics 365 finance and operations apps, open the **Email parameters** page.
2. On the **Microsoft Graph settings** tab, in the **Application ID** field, enter the **Application ID** value that you captured when you created the new app in the previous procedure.
3. In the **Application secret** field, enter the **Secret value** value that you captured when you created a client secret in the previous procedure.
4. Select **Save**.

Exchange email provider (Deprecated)

On the **Email parameters** page, an administrator can select **Exchange** as an interactive email provider and as the Batch email provider. This mail provider will use the current user's Exchange Online account to send emails. When it's used as the Batch email provider, the batch account will be used. No additional configuration is required.

If troubleshooting is required, ensure that it's possible to sign in to the current user's account, and that emails can be sent from that account to the intended recipients.

Note that the Exchange mail provider has the following limitations:

- It isn't supported for external users, because those users won't have Exchange accounts on the system tenant.
- It's available only in Microsoft-managed environments.

Send email by using SMTP

On the **Email parameters** page, note the following settings on the **SMTP settings** tab.

Server information

Expand table

Field	Description
Outgoing mail server	The host name of the desired Simple Mail Transfer Protocol (SMTP) server. For Microsoft 365 production (including *.onmicrosoft.com accounts), use smtp.office365.com. (You can find this setting at outlook.office.com at **Settings** > **Mail** > **POP and IMAP**.) For Outlook/Hotmail, use smtp-mail.outlook.com.
SMTP port number	Typically, the port number should be set to 587 for secure transport.
SSL required	Determines whether secure transport is used. Typically, this is **Yes**, except for internal or troubleshooting scenarios.

> **Note**
>
> If you are running into issues from too many emails being sent in a short period of time, it is recommended to utilize the **Email throttling** feature mentioned in the **Configuration tab** section above. If that doesn't resolve the issue, you may consider adding the appropriate IP addresses from the outbound IP safe list to your DNS SPF record authorizing Dynamics 365 finance and operations to send emails from your domain. For more information, see Outbound IP safe list.

Authentication

Expand table

Field	Description
Authentication required	Determines whether a user name and password are needed to send emails.
User name and **Password**	If authentication is required, specify the appropriate mail account to send email from. All users need to provide the SMTP account **Send As** and **Send On Behalf Of** permissions to enable the ability to send SMTP mail. You can configure Send As permissions in the Microsoft 365 admin center (portal.office.com/Admin) at **Users** > **Active users** > **User** > **Edit mailbox permissions** > **Send email from this mailbox**. For more information, see Enable sending email from another user's mailbox in Microsoft 365.

> **Note**

Finance and operations apps don't support multifactor authentication or Modern auth (OAuth 2.0) for SMTP. The Microsoft Graph email provider can be used if a more modern integration is desired. Administrators might have to re-enable Basic authentication to allow for SMTP AUTH. For more information, see Enable or disable SMTP AUTH. Note that the deprecation of Basic authentication for Exchange online only impacts the use of Office 365 SMTP servers. Other SMTP servers that still support basic authentication won't be impacted by this Office 365 deprecation.

[Administrator] Email distributor batch process

Email that is sent directly from the server, without user interaction, via SMTP is sent by the **Email distributor batch** process. That batch process must be started to process the email queue. To start the process, open the **Email distributor batch** pane (**System administration** > **Periodic tasks** > **Email processing** > **Batch**) and turn on **Batch processing**.

If the Microsoft Graph or Exchange provider is used, then the user account associated with the batch process (usually admin) will be the sender.

[Administrator] User email

The default **send from** address for each user is pulled from the **Email** field on the **Users** page (**System administration** > **Users** > **Users**). Administrators can override this **send from** default if needed using the **Sender email** field on the **Options** page.

[User] Email provider selection section on the Options page

The **Options** page can be opened via **Settings** > **User options**. The **Email provider selection** section is on the **Account** tab.

Expand table

Field	Description
Email provider ID	Allows the user to select the email provider that should be used when sending an email. Selecting an option here is the equivalent of selecting **Do not ask again** in the **How would you like to send email** dialog box. Selecting the blank option **Prompt for which email provider to use** will cause the **How would you like to send email** dialog box to display when an email is going to be sent.

Field	Description
Sender email	Allows the administrator to provide an email address override for the user in the **From** field of the email. By default, the email alias that is associated with the user account is used as the **From** field in new emails, but this user option email address will override that. When sending email via SMTP, the user needs to have appropriate **Send As** and **Send On Behalf Of** permissions configured in Exchange or on the SMTP server. **NOTE:** You can configure **Send As** and **Send On Behalf Of** permissions in the Microsoft 365 admin center (portal.office.com/Admin) at **Users** > **Active users** > **User** > **Edit mailbox permissions** > **Send email from this mailbox**. For more information, see Enable sending email from another user's mailbox in Microsoft 365.

[User] How would you like to send email dialog box (optional)

When an email is going to be sent, the user will see the **How would you like to send email** dialog box that will list the available options for sending email.

Expand table

Field	Description
Use an email app, such as Outlook	Provides the user with a generated email (.eml) file.
Use Exchange email server	Uses the Exchange Online server associated with the tenant. The **Exchange** mail provider is currently not supported outside the public cloud, including on-premises environments.
Use the system email client	Opens the **Send email** composition dialog box and then sends the resulting email via SMTP.
Do not ask again	If this field is not selected, the next time an email is sent the most recently selected option will be used and the dialog box will not open.

[User] Send email dialog box (optional)

The **Send email** dialog box is opened to allow the user to edit the contents of the email that will be sent. Some of the following fields will be pre-populated in this window.

Expand table

Field	Description
From	Populated from the **Email** field on the **Options** page.
To, **Cc**, **Bcc**, **Subject**, and **Body**	Populated with values specified by the process that initiated the sending of the email. These fields can be edited as needed by the user.
Attachments list	May be populated with attachments specified by the process that initiated the sending of the email. This list can be edited as needed by the user.

When the email is ready to be sent, the **Send** button will cause the email to be sent via SMTP.

Usage scenarios to verify if email is configured correctly

Send mail via a local mail client

Email workflows that are enabled via the SysEmail framework can generate email messages (.eml files) that contain attachments. You can then send these messages via Microsoft Outlook or another email client.

1. In Microsoft Edge, navigate to **Accounts receivable** > **Customers** > **All customers**.
2. Select **US-008 Sparrow Retail**.
3. Click **Collect** > **Customer balances** > **Collections** to open the **Collections** page.
4. Click **Communicate** > **Email** > **Statements to contact**.
5. Click **OK** to accept the default values in the dialog box.
6. If you're prompted for the mail option to use, clear the **Do not ask again** check box (you can change this option from the **User options** page), select **Use an email app, such as Outlook**, and then click **OK**.
7. If you're using Microsoft Edge on your computer, open the email (.eml) file that is generated. If you're using Microsoft Edge on the VM, copy the file to your computer, and open it there.
8. Note the email address in the **To** field and the generated workbook attachment.

Send mail via SMTP

Email workflows that are enabled via the SysEmail framework can also be created in a simple email dialog box and then sent via Simple Mail Transfer Protocol (SMTP).

1. Go to the **Email parameters** page.
2. Click **SMTP settings**.

3. Set the **Outgoing mail server** to the desired SMTP server:

    - For Microsoft 365 production (including *.onmicrosoft.com accounts) use smtp.office365.com. (Find this setting via outlook.office.com, at **Settings** > **Mail** > **POP and IMAP**.)
    - For Outlook/Hotmail use smtp-mail.outlook.com.

4. Set the user name and password to an appropriate email account and password.
5. Leave **SSLRequired** turned on, and leave **SMTP port number** set to **587**.
6. Click **Save**.
7. In Microsoft Edge, navigate to **Accounts receivable** > **Customers** > **All customers**.
8. Select **US-008 Sparrow Retail**.
9. Click **Collect** > **Customer balances** > **Collections** to open the **Collections** page.
10. Click **Communicate** > **Email** > **Statements to contact**.
11. Click **OK** to accept the default values in the dialog box.
12. If you're prompted for the mail option to use, select **Use the Microsoft Dynamics 365 Finance email client**, and then click **OK**.
13. To receive the test message, change the **To address** to your email address.

    Ensure that the account specified in the SMTP settings is able to **Send As** and **Send On Behalf Of** your email account. The easiest way to ensure this is to use your email account in the SMTP settings.

14. Enter a subject and body for the message.
15. Click **Send**. The message should be delivered in one to five minutes.

[Administrator] Workflow email notifications

Workflow email configuration is a collection of related settings that work in conjunction.

Workflow email notification setup

1. Verify email settings:
    a. Go to **System administration** > **Setup** > **Email** > **Email parameters**.
    b. Verify that SMTP is enabled.

c. Set the SMTP mail server settings.

2. Verify that the email batch process is running:

   a. Go to **System administration** > **Periodic tasks** > **Email processing** > **Email distributor batch**.

   b. Enable the **Batch processing** option.

   c. Optionally, adjust the recurrence of the email process:
      i. Select **No end date** to adjust all recurrences of the email batch process.
      ii. Adjust the count.
      iii. Adjust to run every minute if needed.

3. Verify workflow notification system email templates:
   a. Go to **System administration** > **Setup** > **Email** > **System email templates** (for system-wide templates).
   b. Verify that the **Sender email** field is set and valid.

4. Verify workflow notification organization email templates:
   a. Go to **Organization administration** > **Setup** > **Organization email templates** (for organization-specific templates).
   b. Verify that the **Sender email** field is set and valid.

5. Verify that the user can receive email notifications:

   a. Go to **Settings** > **User options**.

   b. Go to the **Account** tab.
      i. Set the **Email provider ID** (for example, SMTP).
      ii. Optionally, set a **Sender email** override if the default **send from** address should not be used for the current user.

   c. Navigate to the **Workflow** tab. Set the option to send notifications in email to **Yes**.

6. Verify that the workflow system will send email notifications:

   For each workflow that should have a notification, open the workflow properties in the workflow editor.

   a. Click **Basic settings**. Adjust the email template for the workflow notifications.

   b. Click **Notifications**.
      i. Enable the events for which a user should be notified.
      ii. Set the recipient of the workflow notification for each event notification that is enabled.

c. On a workflow approval element for which a user should be notified:
   i. Go to **Properties**.
   ii. Enable the events for which a user should be notified.
   iii. Set the recipient of the workflow notification for each event notification that is enabled.

Workflow email notification testing

The testing for email notifications is to simply trigger the notification and then check for it.

1. Submit a workflow that has been set up for email notifications.
2. Check the workflow history to make sure that a workflow work item was assigned to the expected user.
3. Check the status of the pending email notification in **System administration** > **Periodic tasks** > **Email processing** > **Batch email sending status**.

   If the email fails to be sent, make sure that the SMTP mail account can be opened.
4. Check for the email notification in the appropriate inbox.

Troubleshooting

Common issues with sending email

There are some standard processes that can help you troubleshoot the configuration of email settings.

Expand table

Verify email settings, and send a test email.	1. Go to **System administration** > **Setup** > **Email** > **Email parameters**. 1. Verify that SMTP is enabled. 1. Verify the settings of the SMTP mail server. 1. Sign in to the SMTP account in a separate window to make sure that the account and password are correct. 1. Send a test email by going to **System administration** > **Setup** > **Email** > **Email parameters** > **Test email**.
Verify that the email batch process is running.	1. Go to **System administration** > **Periodic tasks** > **Email processing** > **Batch**. 1. Make sure that the **Batch processing** option is set to **Yes**. 1. Review the recurrence of the email process: 1. Select **No end date** to adjust all recurrences of the email batch process. 1. Adjust the count as needed.

Review the status of batch emails.	1. Go to **System administration** > **Periodic tasks** > **Email processing** > **Batch email sending status**. 1. Verify that emails are being sent from the correct account. If the account is incorrect, you must adjust settings such as user options, system templates, or organization templates, as needed. 1. Verify that all email user accounts have been granted **Send As** permission for the configured SMTP account (see Verify that all email accounts have appropriate Send As permissions later in this table).
Review any errors on the Email history page.	1. Go to **System administration** > **Setup** > **Email** > **Email history**. The **Email history** page lets administrators review all sent emails and also any errors that prevented an email from being sent. It shows both interactive and non-interactive/batch emails. 1. For any emails where the **Email status** value is **Failed**, review the error message on the **Failure details** tab, and determine whether corrective actions should be taken. Some of these errors are covered later in this article.
Verify that all email accounts have appropriate Send as permissions.	In the Microsoft 365 admin center, verify that all user mail accounts that will be used to send emails have **Send As** and **Send On Behalf Of** permissions for the configured SMTP account. For more information, see Give mailbox permissions to another user in Microsoft 365.
Verify user mailboxes.	Consider signing in to the affected (or all) user mailboxes to verify that they are valid and can be accessed by using sign-in.

Specific Microsoft Graph/Exchange email issues

- **"(401) Unauthorized" or "(403) Forbidden" error when email is sent via Exchange**

Expand table

Explanation	This issue might indicate an invalid or incorrectly set up mailbox in Office 365.
Fix	To fix this issue, make sure that the specified user exists in Office 365 and has appropriate permissions. To find the affected users, follow these steps: 1. Open the **Email history** page. 1. Add a filter for **Failed** emails to the **Email status** field. 1. Make a note of the value in the **Email sender** field. This field shows the user that Exchange is indicating isn't a valid or correctly permissioned user in Office 365.

- **"(404) Not found" error when email is sent via Exchange**

Expand table

Explanation	This issue indicates that no mailbox exists for the user account in Exchange.
Fix	To fix this issue, make sure that the specified user exists in Office 365 and has appropriate permissions, or use alternate user accounts that have valid Exchange mailboxes. To find the affected users, follow these steps: 1. Open the **Email history** page. 1. Add a filter for **Failed** emails to the **Email status** field. 1. Make a note of the value in the **Email sender** field. This field shows the user that doesn't have a mailbox in Exchange.

Specific SMTP email issues

If you continue to experience issues when email is sent via SMTP, you may be running into one of the specific errors below. If not, consider entering the SMTP account information in a tool such as SMTPer.net to verify that the SMTP server and account are valid and working correctly.

- **SMTP emails fail to be sent with "Recipient addresses in single label domains not accepted"**

Expand table

Explanation	An email failed to be sent because a recipient is using a single-label domain, but Office 365 doesn't support single-label domains. Single-label domains are Domain Name System (DNS) names that don't contain a suffix such as .com, .corp, .net, or .org. For example, contoso is a single-label domain. However, contoso.com, contoso.net, and contoso.local aren't single-label domains.
Fix	Specify alternate addresses for email recipients that aren't single-label domains.

- **SMTP emails fail to be sent with "Mailbox full"**

Expand table

Explanation	An email failed to be sent because at least one recipient's mailbox is full. For more information, see Error (554 5.2.2 mailbox full) when sending email to mail-enabled public folders in Office 365.
Fix	Use an alternate address for the recipient, or contact the recipient by using alternate means before you try again.

- **SMTP emails fail to be sent with "Authentication unsuccessful, the request did not meet the criteria to be authenticated successfully"**

Expand table

Explanation	An email failed to be sent because additional criteria are required to successfully authenticate the SMTP user account. These additional criteria might be required because multifactor authentication (MFA) is configured. However, finance and operations apps don't currently support MFA.
Fix	Adjust the user account configuration as appropriate in Office 365 before you try again. For more information, see Fix issues with printers, scanners, and LOB applications that send email using Microsoft 365 or Office 365.

- **SMTP emails fail to be sent with "Authentication unsuccessful, SmtpClientAuthentication is disabled for the Mailbox"**

Expand table

Explanation	An email failed to be sent because SMTP client authentication is disabled for the SMTP user account.
Fix	Enable SMTP client authentication for the user account in Office 365 before you try again. For more information, see Enable or disable authenticated client SMTP submission.

- **SMTP emails fail to be sent with "If your SMTP server doesn't support authentication, please clear the SMTP user name and password"**

Expand table

Explanation	This issue is related to migration from the .NET SMTP mail client (which is now obsolete) to MailKit per Microsoft recommendations. This migration causes a change in the way that SMTP user names and passwords are handled in situations where the mail server didn't support authentication. Previously, if an SMTP user name was provided, but the server didn't support authentication, the .NET SMTP mail client ignored the provided user name and password, and continued without authentication. This behavior led some customers to incorrectly believe that they were using an authenticated mail server. In MailKit, authentication is required if a user name is provided. This requirement will trigger an error for mail servers that don't support authentication, and emails will fail to be sent.

Fix	The user must go to the **Email parameters** page (**System administration** > **Setup** > **Email**) and clear the **User name** and **Password** fields on the **SMTP settings** tab.

- **SMTP emails fail to be sent with an "5.7.57 SMTP" error, or an indication that either you aren't authenticated or authentication is required**

Expand table

Explanation	The mail server is indicating that the SMTP user credentials aren't valid.
Fix	Verify that the SMTP user credentials are correct.

- **SMTP emails fail to be sent with "Microsoft.Dynamics.Ax.Xpp.Security.CryptoEncryptionException: Encryption error occurred with exception"**

Expand table

Explanation	The SMTP password is encrypted by using a key that is specific to a single environment. This error typically occurs when the password can no longer be decrypted after database migration between environments.
Fix	Clear and reenter the SMTP password on the **SMTP settings** tab of the **Email parameters** page.

- **SMTP emails fail to be sent with "Client does not have permissions to send as this sender"**

Expand table

Explanation	This issue is usually caused by incorrect setup of the **Send As** permissions for the email account.
Fix	Ensure that the email account has appropriate **Send As** permissions. For more information, see Verify that all email accounts have appropriate Send As permissions earlier in this article.

Frequently asked questions

Where do workflow email templates come from?

The email templates will be sourced from either system email templates or organization email templates depending on whether the workflow is a system-level (not company specific) or organization-level (company specific) workflow.

What email limits apply when using Microsoft Graph, Exchange, or SMTP?

The system communicates with Microsoft Graph, Exchange, or an SMTP server like a typical email client, so standard behavior and limits apply. For example, standard Exchange Online receiving and sending limits apply.

Additional resources

Troubleshoot the Office integration

Office integration tutorial

Configure email functionality in Microsoft Dynamics AX [AX 2012]

Last updated on 03/17/2026

Develop email experiences by using the SysMailer framework

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Sending emails

The SysMailer framework is a new, extensible way to send email. It replaces all previous application programming interfaces (APIs) for mail, such as CDO.Messaging (SysMailer), MAPI (SysINetMail), and Outlook COM (SmmOutlookEmail). Those older mail APIs don't work correctly in finance and operations applications. By taking advantage of the SysPlugin framework and several .NET technologies, SysMailer provides a configurable experience for users and enables the application consumers to remain agnostic to the email option that users use to send email.

The SysMailer framework consists of a factory class that you use to retrieve an email provider, a set of email providers that send messages, a message builder that builds the messages, and the forms that are related to configuring and interacting with the email providers. To consume the SysMailer framework, an application developer primarily uses the **SysMailerFactory** and **SysMailerMessageBuilder** classes. The email provider factory is used to retrieve interactive or non-interactive email providers, so that you can send multiple messages at the same time, or so that you can send a message directly. The email providers expect the messages that they send to be encapsulated in .NET **System.Net.Mail.MailMessage** objects. Use the message builder class to build the .NET object that you pass to the email provider.

Scenarios

This article describes three scenarios:

- Sending an interactive message
- Sending a non-interactive (batch) message

- Sending multiple non-interactive (batch) messages

Sending an interactive message

The following example is taken from the **CustCollectionsEmail** class. It demonstrates multiple features of the framework, such as the ability to chain message builder calls, conditionally set the sender address ("from" address), and add attachments.

X++

```
using (System.IO.Stream attachmentStream = this.generateAttachment())
{
    var messageBuilder = new SysMailerMessageBuilder();
    messageBuilder.addTo(context.parmEmailAddress())
    .setSubject(emailSubject)
    .setBody(SysEmailMessage::stringExpand(emailBody,
    SysEmailTable::htmlEncodeParameters(templateTokens)));
    if (emailSenderAddr)
    {
        messageBuilder.setFrom(emailSenderAddr, emailSenderName);
    }
    else if (custParameters.CollectionsOMTeam)
    {
        var collectionsEmail =
OMTeam::find(custParameters.CollectionsOMTeam).primaryEmail();
        if (strLen(collectionsEmail) > 0)
        {
            messageBuilder.setFrom(collectionsEmail);
        }
    }
    if (attachmentStream != null)
    {
        messageBuilder.addAttachment(
            attachmentStream,
            strFmt('%1%2', strReplace(DateTimeUtil::toStr(DateTimeUtil::utcNow()),
':', ''), '.xlsx'));
    }
    SysMailerFactory::sendInteractive(messageBuilder.getMessage());
}
```

Sending a non-interactive (batch) message

The following example is taken from the **VendRequestCompanyWorkflowManager** class. It shows how you can send a single message non-interactively.

X++

```
// The vendor <vendor name> has been approved and has been added to the vendor
master.
```

```
messageText = strFmt("@SYS134393",
DirPartyTable::findRec(vendRequestCompany.VendParty).Name);
// Request
var messageBuilder = new SysMailerMessageBuilder();
messageBuilder.setFrom(senderEmail)
    .addTo(recipientEmail)
    .setSubject("@SYS130372")
    .setBody(messageText);
SysMailerFactory::sendNonInteractive(messageBuilder.getMessage());
```

Sending multiple non-interactive (batch) messages

The following example is taken from the **UserAdAddManager** class. It shows how you can get an instance of a batch email provider before you iterate over a list of emails to send.

X++

```
var mail = SysMailerFactory::getNonInteractiveMailer();
var messageBuilder = new SysMailerMessageBuilder();
for (i = 1; i <= conLen(_notifyCon); i++)
{
    notifyEmailsStr = conPeek(_notifyCon, i);
    select firstonly RecId, Email from sysUser where sysUser.Id == notifyEmailsStr;
    if (sysUser.RecId && sysUser.Email != '')
    {
        messageBuilder.reset()
            .setFrom(_emailFrom)
            .addTo(sysUser.Email)
            .setSubject("@SYS129183")
            .setBody(errorStr);
        mail.sendNonInteractive(messageBuilder.getMessage());
        delete_from tmpUserErrorNotification;
    }
}
```

Important considerations

- When you send messages to an email provider, you need a sender address ("from" address). When you send messages non-interactively, you need a receiver address ("to" address). If you don't meet these conditions, the framework throws an exception. If you call **getMessage** on the message builder before you call **setFrom**, the builder tries to set the sender address to the current user's email address or network alias.

- When you send messages, how the sender address ("from" address) is used depends on the provider:
    - **EML provider:** The sender address is removed from the message before the message is opened in the user's email client. Therefore, the email client can set the sender

address to the default account that is configured for sending mail.
    - **SMTP provider:** You must configure the Simple Mail Transfer Protocol (SMTP) server to allow messages to be sent by using the sender address. In other words, the SMTP server must allow the impersonation of emails that are sent from it. Otherwise, the SMTP server might prevent the messages from being sent, flag them as spam, and so on.

- When you send messages, the framework returns a Boolean value that indicates whether the operation to send the message was successful. However, it doesn't report any messages to the Action Center when the operation is successful. You decide whether messages are shown in the Action Center.

- By default, the body of all messages that you send is in rich-text (HTML) format. If an application scenario requires that plain text be used to maintain newline spacing, you can pass **false** to the optional **_isHtml** parameter of the **setBody** method on the message builder.

Adding a new email provider

You can add email providers by using the pluggable framework. When you use the factory class and interfaces, new email providers immediately become available for use across all relevant application scenarios. You can find examples of email providers in the existing provider implementations, SysMailerEML and SysMailerSMTP, and also in an existing tutorial implementation, Tutorial_SysMailerMailTo. The following examples are excerpts from the implementation of the SysMailerEML email provider.

To implement an email provider, create an implementation class that has the following properties:

- The class must have the appropriate **Export** attributes.
- The class must implement the base **SysIMailer** methods, **getId** and **getDescription**.
- The class must implement the **SysImailerInteractive** interface, the **SysIMailerNonInteractive** interface, or both interfaces.

Specify attributes for the implementation class

First, specify attributes for the implementation class. The class must have two attributes:

- **ExportAttribute** – The framework uses this attribute to discover the plug-in. The attribute specifies that the plug-in is an implementation of **SysIMailer** and therefore part of the

SysMailer framework.

- **ExportMetadataAttribute** – The framework uses this attribute to find specific instances of a plug-in that have specific metadata. The attribute specifies the ID of the email provider, so that a single provider can be discovered quickly. **No two email providers should ever have the same ID.**

X++

```
using System.IO;
using System.Net.Mail;
using System.Text.RegularExpressions;
#define.SysMailerEML_ID('EML')
/// <summary>
/// The <c>SysMailerEML</c> class is an interactive email provider implementation
that sends messages by generating
/// an EML file, uploading it to Azure temporary blob storage, and then redirecting
the user's browser to
/// the file to save or open for sending using their default email client.
/// </summary>
// This is a framework class. Customizing this class may cause problems with future
upgrades to the software.
[System.ComponentModel.Composition.ExportAttribute(identifierStr(Dynamics.AX.Applic
ation.SysIMailer)),
System.ComponentModel.Composition.ExportMetadataAttribute(extendedTypeStr(SysMailer
Id), #SysMailerEML_ID)]
public class SysMailerEML implements SysIMailerInteractive
{
```

Implement the SysIMailer interface

The **SysIMailer** interface includes identifiable information about an email provider, regardless of the capabilities of the email provider itself. To implement this interface, you must create two methods:

- **getId** – This method returns the same ID that you specify in the **ExportMetadataAttribute** attribute.
- **getDescription** – This method returns a non-technical description that indicates how the email is sent.

X++

```
public SysMailerId getId()
{
    return #SysMailerEML_ID;
}
public SysMailerDescription getDescription()
{
```

```
    // Use an email app, such as Outlook
    return "@ApplicationFoundation:EmailProviderEMLDescription";
}
```

Implement a combination of the SysIMailerInteractive and SysIMailerNonInteractive interfaces

The **SysIMailerInteractive** and **SysIMailerNonInteractive** interfaces are very simple. They implement the **sendInteractive** and **sendNonInteractive** methods, respectively. An email provider might implement one or both methods, depending on its capabilities. Each method that you implement takes a .NET **System.Net.Mail.MailMessage** object that you use to construct and send the email.

X++

```
public boolean sendInteractive(System.Net.Mail.MailMessage _message)
{
    using (var emlStream = this.generateEmlFile(_message))
    {
        Dynamics.AX.Application.File::SendFileToUser(emlStream, 'message.eml');
    }
    return true;
}
```

The **System.Net.Mail.MailMessage** object contains a large amount of advanced functionality that's related to MIME messages. You can build a relatively complex message object and pass it to an email provider. If there's specific functionality that an email provider doesn't support, the email provider is expected to handle the functionality actively (by modifying the message), passively (by making an internal call to another email provider), or not at all (by throwing an error).

Migration from Microsoft Dynamics AX 2012 to finance and operations applications

Migration involves using the **SysMailerMessageBuilder** object to build a message and then using the **SysMailerFactory** to send it, as outlined in the examples in this article.

Last updated on 04/01/2026

Office integration FAQ

09/23/2025

Applies to these Dynamics 365 apps:
Commerce, Finance, Human Resources, Supply Chain Management

This article provides answers to frequently asked questions about the capabilities of the Microsoft Office integration. The questions and issues that are discussed range across user, administration, and development scenarios.

Frequently asked questions

What platforms do the Office add-ins support?

The Microsoft Excel add-in and Microsoft Word Add-in are built by using the Office Web/JavaScript add-in framework. This framework was originally released for Microsoft Office 2013 but received significant updates in Microsoft Office 2016. For more information, see Office Add-in host and platform availability. The Excel add-in requires ExcelAPI 1.2. Therefore, use the Office Add-in host and platform availability matrix to determine which platforms support the Excel add-in. For many users, the phrase "Excel 2016 with the latest updates" is sufficient.

Are the Office Add-ins safe?

In an age of malware, full connectivity, and compliance risks, nothing is completely secure. However, the web add-ins, like other websites, are basically a web application that interacts with the Office client products via a limited application programming interface (API). For more information, see Office add-ins platform overview

Does the Excel add-in support Office for Mac?

No. The Office JavaScript (JS) APIs work differently in Apple Safari and Microsoft Edge/Internet Explorer, especially in respect to authentication. For details about platform support for the Office JS APIs, see Office Add-in host and platform availability.

What version of Office is required for the Excel add-in to support AD FS?

For more information, see the "Troubleshooting issue" section later in this article.

How can I force an update of Office?

If your Office build isn't updated, you might be on the deferred track (Overview of update channels for Microsoft 365 Apps). In this case, you can use the Office Deployment Tool to move to the Current channel or sign up for the Office Insider program to help guarantee that you have the latest updates. The easiest method is to use the Office Deployment Tool to switch to the current channel. In this case, the latest updates are installed immediately.

Why can't you tell me what version of Office or Excel a particular issue is fixed in?

Office has many releases. These releases receive updates at different times and have different version numbers that don't correspond. Some frequently used Office versions and update methods are Click to Run (C2R) Current channel, C2R Deferred, C2R First Update Deferred, Office Insider Fast, Office Insider Slow, and MSI/MSO (install from DVD). For more information about Office versions, see the Release information for updates to Microsoft 365 Apps page.

Why am I having trouble signing into the Excel add-in?

The browser used to authenticate the Excel add-in (or any Office add-in) is dependent on the Office and operation system versions. For more information about the specific browser used to sign in for your configuration, see Browsers used by Office add-ins. By default, the Excel add-in picks up stored credentials from the browser, and the browser provides the current Microsoft Windows credentials if there are no stored credentials. Make sure that you're using the correct credentials to sign in. In the Excel add-in, explicitly sign out, and then sign in to ensure that the correct credentials are used.

The Excel add-in seems to be slow when it publishes records. How can I learn more about what is occurring?

Most of the work that the Excel add-in does should occur on the server. To learn where the time is being spent, you can use Fiddler (a free download) to make sure that the Excel Add-in works as you expect.

The Excel add-in sends the published records as a request. When those records are processed, the response is sent back from the server. The Excel add-in then creates another message that contains the next set of records to publish, and sends that request. Five to 10 seconds of processing time in the Excel add-in should be required between the previous response from the server and the next request to the server.

To check processing time in the Excel add-in versus the server/service, follow these steps.

1. Start Fiddler.
2. Publish a few records to test the process.
3. Make sure that you can view that request and response in Fiddler (make sure that HTTPS traffic is being decrypted).
4. Publish a larger number of records.
5. In Fiddler, watch the time that is required from a request to its response, and from a response to the next request.
    - If the time from a request to its response is large, the bottleneck is the server/service.
    - If the time from a response to the next request is large, the bottleneck is the Excel add-in (that is, the client).

Is Export to Excel the best way to get data out of the system?

The best way for end users to get data out of the system is to use the Excel add-in. The add-in relies on the Open Data Protocol (OData) service to retrieve data and takes advantage of the security that data entities provide. The import and export capabilities in the Data management framework (DMF) and Data import/export framework (DIXF) can also be used. However, DMF and DIXF are often limited to administrators. Additionally, you might consider exploring the available reporting and analytic features in the product.

When the previously mentioned options can't be used, you might consider the Export to Excel functionality, which can be useful for retrieving data from calculated columns (display methods), columns that have formatted values, and data from temporary tables. Because the export process uses the data from the active form to retrieve data, Export to Excel returns the data exactly as it's shown in the grid. Additionally, it only returns the data that matches the current set of filters.

Export to Excel does have its own set of limitations. First, this mechanism takes more time to export than the options that go through the OData service, and the time to export can be significant for larger exports. Secondly, the export of data to Excel is a memory-intensive process that can cause out-of-memory errors to occur and servers to stop responding, especially if the organization increases the export row limit beyond the default value. Export to Excel isn't intended for auditing purposes. When very large datasets must be exported from the system, data management is recommended.

> ⚠ **Warning**

Although organizations can increase the default Export to Excel row limit (50,000 records), we **strongly recommend** that you **not** raise this limit above 100,000 records. Depending on the characteristics of the environment and the specific grid that is being exported, even that limit might be too high and might cause memory issues.

Because an export from Excel can take some time, it's recommended that the export is done with the Chrome or Edge browsers, with the automatic download option enabled. The automatic download option ensures that the browser downloads the file as soon as the export is complete to ensure that the download link is used within the 15-minute time limit.

Why is the Publish button in the Excel add-in unavailable?

Hover over the **Publish** button to get more information as to why publishing currently isn't supported. Typically when this occurs, it's because all key and required fields must be present to publish data back to the entity, in which case you should open the designer and make sure all the key fields are bound.

Why are the Excel add-in, the Word add-in, and the Open in Excel options only available when the Internet is available?

All environments including on-premises, the Excel and Word add-ins, and the libraries they use, are loaded from multiple Internet locations and only run when the Internet is available. For on-premises environments, when the Internet isn't available, the Open in Excel options are hidden because the Excel Add-in doesn't run without access to the Content Delivery Network (CDN) that houses the Excel add-in.

Can the Excel add-in and Word add-in be made available to users using centralized deployment?

Yes, centralized deployment is supported. For more information, see Centralized Deployment.

To use Centralized Deployment, on the **App parameters** tab on the **Office App Parameters** page change the **App ID**, **Store**, and **Store Type**:

- **App ID**: 61bcc63f-b860-4280-8280-3e4fb5ea7726
- **Store**: EXCatalog
- **Store Type**: Centralized Deployment

In case a change back to Office Store is needed, the standard values are:

- **App ID**: WA104379629

- **Store**: en-US
- **Store Type**: Office Store

> ⓘ Note
>
> - **Name**, **Version**, and **Notes** are values that provide information but they aren't needed to run the Excel add-in.
> - These same values are also used for the Word add-in when it's run from the Document Templates form.

If you encounter issues with centralized deployment for some users, it could be one of these problems:

- One or more users are members in a group that is more restrictive than others
- The user referenced is on a different Microsoft 365 account (such as a personal account)

What is the cell limit for the Excel add-in?

The default Excel add-in cell limit is about half the limit of what the Excel Add-in can handle on a reasonably fast machine. The speed of the machine is the limitation. If problems are encountered, then the cell limit should be reduced and/or the filter should be adjusted to reduce the data set. A common workaround is to use a filter to manage the data in smaller pieces instead of all at once.

How do I make an entity available in the Excel add-in and/or as an Open in Excel option?

If the entity is marked as “IsPublic=Yes” and has unique PublicEntityName and PublicCollectionName values, then it will be available via the OData service. Check that there aren’t any existing entities with the same PublicEntityName and PublicCollectionName values by looking at the $metadata feed for the environment (preferably in Google Chrome): https://*SomeFullEnvironmentURL*.dynamics.com/data/$metadata

Why are date and time values in UTC in the Excel add-in?

The Excel add-In, Data Management Framework, and Power BI reporting are all designed to interact with data directly on the database level. Because there's no client to adjust Date and Time data to the time zone of the user, all Date and Time values are in UTC.

Troubleshooting issues

Issue: The Excel add-in loads, but instead of showing data, it displays "Load applets" in the task pane

Issue: The Excel add-in loads, but instead of showing data, it displays "Load applets" in the task pane.

Explanation: This issue usually occurs due to one of the following reasons:

- Incorrect sign in
- Add-in registration data hasn't been initialized in the environment
- OData issues

Fix:

- **Incorrect sign in**: The most likely cause of this issue is that the user is signed in as the wrong user. This can happen if the user has multiple accounts and the browser uses the wrong user context. Before trying any other fix, you should sign out of the add-in using the user menu in the upper-right corner and then sign back in to the add-in. If the issue persists, ensure registration data is properly initialized, as noted in the following fix.
- **Registration data not initialized**: If the issue wasn't addressed by signing back in to the add-in, another potential cause is that the environment doesn't yet have the add-in data initialized. To check this, the admin can navigate to the **Office app parameters** page. For each of the **App parameters**, **Registered applets**, and **Registered resources** tabs on that page, verify that there's data populated in each tab. If any tab has an empty grid, select the appropriate **Initialize** button on that tab.
- **OData issue**: If the issue persists after attempting the previous two fixes, then the final cause of this issue could be that the OData service, through which the add-in communicates with finance and operations, is unable to return the registration data to the add-in. Without that data, the add-in fails to load applets. At this stage, you need to contact Microsoft Support with information from the **Application correlation ID** from the Excel add-in with the failed session. You can find this field under **Options**.

Issue: During sign-in to the Excel add-in, users receive an error message saying they "can't access the application '2bc50526-cdc3-4e36-a970-c284c34cbd6e' in that tenant"

Issue: During sign in to the Excel add-in, a user receives an error similar to the following:

- "Microsoft EntraSTS50020: User account 'XXX' from identity provider 'https://sts.windows.net/XXX' doesn't exist in tenant 'XXX' and can't access the application '2bc50526-cdc3-4e36-a970-c284c34cbd6e'(Microsoft Business Office Add-in) in that tenant."
- "Selected user account doesn't exist in tenant 'XXX' and can't access the application '2bc50526-cdc3-4e36-a970-c284c34cbd6e' in that tenant."

Explanation: This issue is caused by a change made to Microsoft Entra ID in April 2021 in regard to external users. Because this change wasn't made to the finance and operations apps, it can affect customers on any version of the platform or application.

Fix: All external users need to be invited to the tenant through Microsoft Entra ID. For more information, see Invite users with Microsoft Entra B2B collaboration.

[Fixed] Issue: During sign-in to the Excel add-in, I receive the following error message: "Microsoft EntraSTS65001: The user or administrator hasn't consented to use the application with ID XYZ"

Issue: During sign in to the Excel add-in, you receive the following error message: "Microsoft EntraSTS65001: The user or administrator hasn't consented to use the application with ID XYZ."

Explanation: Typically, this issue occurs because Microsoft Entra ID can't find the Microsoft Entra application that represents the Excel add-in. That issue occurs because, during the configuration of Microsoft Power BI, a Microsoft Entra application was added that has the App ID URI set to the environment URL.

Fix: Make sure that no Microsoft Entra apps have the App ID URI set to the environment URL. App ID URIs should be fabricated, unique URIs, such as `https://contosoAXPowerBI`.

[Fixed] Issue: During sign-in to the Excel add-in, I receive the following error message: "Microsoft EntraSTS50001: The application named ABC wasn't found in the tenant named XYZ"

Issue: During sign-in to the Excel add-in, you receive the following error message: "Microsoft EntraSTS50001: The application named ABC wasn't found in the tenant named XYZ."

Explanation: This issue probably occurs because an error in the deployment system caused the environment to get a URL that wasn't added to the configured list of service principals for the tenant.

Fix: File a support issue for your environment, so that the problem can be investigated and the configuration can be adjusted.

[Fixed] Issue: After the Excel add-in starts and updates data, I receive the following error message: "An error occurred while writing to the data cache"

Issue: After the Excel add-in starts and updates data, you receive the following error message: "An error occurred while writing to the data cache." The details of the error state, "The argument is invalid or missing or has an incorrect format."

Explanation: You receive this error message if the client is open in Internet Explorer, and the user clicks **Open** immediately after selecting the **Open in Excel** option. The way that Internet Explorer handles temporary Internet files causes an issue in Excel. This issue, in turn, causes API calls to fail.

Workaround: The recommendation is to use Microsoft Edge or another modern browser. These browsers tend to save files to a Downloads folder by default, which mitigates the issue. If you're using Internet Explorer, which is now deprecated, when you open a workbook, select **Save** first, and then select **Open**. The file will then be opened from your Downloads folder.

Long-term fix: We are working with the Office team to understand this issue so that it can be fixed in Excel.

Issue: VPN users see a blank authentication screen when trying to sign into the Excel add-in

Issue: Users using a virtual private network (VPN) aren't able to sign in to the add-in because the authentication dialog appears blank or is missing text.

Fix: The issue must be solved from the customer side, as the VPN software or browser settings when running under VPN are preventing their machines from retrieving resources and/or interpreting those resources correctly.

Here are some common solutions to VPN issues:

- Verify that the user has access to the locations listed in: Issue: The Excel Add-in doesn't correctly run or enable sign-in.
- Verify that your browser isn't set to "Display intranet sites in Compatibility view" for Microsoft Edge (or Internet Explorer 11). This can be configured by organization policy.
- Have the affected user compare their VPN and Edge/IE browser settings with other users who aren't impacted.

[Fixed] Issue: The Excel add-in doesn't correctly run or enable sign-in

Issue: When users try to sign into the Excel add-in, a blank authentication dialog box appears, or an error message is shown instead of the authentication page. The user can't sign in.

Explanation: The Excel add-in relies on the Office Web JS Add-in platform and uses Microsoft Entra ID authentication. If a proxy is used, several URLs must be accessible for users to run and sign in to the Excel Add-in. Additionally, if AD FS is used, the AD FS URL must use HTTPS.

Solution: Because this issue is a customer-specific network issue, it requires a customer-specific resolution. If AD FS is used, make sure that the AD FS URL uses HTTPS. Additionally, make sure that all the following URLs are accessible from the user's computer.

The following URLs are accessed for loading.

- `https://az689774.vo.msecnd.net:443`
- `https://az689774.vo.msecnd.net`
- `https://appsforoffice.microsoft.com:443`
- `https://appsforoffice.microsoft.com`
- `https://secure.aadcdn.microsoftonline-p.com:443`
- `https://secure.aadcdn.microsoftonline-p.com`
- `https://az416426.vo.msecnd.net:443`
- `https://az416426.vo.msecnd.net`
- `https://telemetryservice.firstpartyapps.oaspapps.com:443`
- `https://telemetryservice.firstpartyapps.oaspapps.com`
- `https://nexus.officeapps.live.com:443`
- `https://nexus.officeapps.live.com`
- `https://browser.pipe.aria.microsoft.com:443`
- `https://browser.pipe.aria.microsoft.com`
- `http://schemas.microsoft.com`

The following URLs are accessed for authentication.

- `https://login.windows.net:443`
- `https://login.windows.net`
- `https://login.microsoftonline.com:443`
- `https://login.microsoftonline.com`

Alternate solution: If the user is running an older version of Windows 10 (earlier than version 1909), consider upgrading Windows 10 to version 1909 or later. Version 1909 may need KB5003635. For more information, see May 2021 resolved issues.

Issue: The Excel add-in needs an explicit sign out after encountering a Microsoft EntraSTS50058 "silent sign in failed" error

Issue: When users try to sign in to the Excel add-in after some period of inactivity, the user encounters the Microsoft EntraSTS50058 "silent sign in failed" error and is forced to sign out before signing back in.

Explanation: The Excel add-in uses Microsoft Entra ID authentication. When authentication occurs, a token is created for the user. That token has an expiration period. After the token has expired, a Microsoft EntraSTS50058 error occurs indicating that "silent sign in failed".

Solution: The user needs to sign out and sign back in. We will improve this behavior in the future by automatically signing the user out to enable faster sign in.

Issue: When trying to use a document template with Open in Excel a "Record for id GUID not found" error displays

Issue: The "Record for id GUID not found" error can display when copying a database from one environment to another.

Explanation: Copying the database is problematic for document templates, record attachments, and other files that are stored in Azure blob storage. When the database is copied from one environment to another, the files aren't copied along with the records, so the files that the application tries to access aren't found.

Solution: For document templates, the solution is to identify the templates that are needed and load a copy of those template files into the target environment.

Additional resources

Office integration

Office integration tutorial

Configuring Power BI integration

Organization administration home page

This article points to content that helps power users and administrators configure the system to work smoothly and effectively for your organization and business.

Much of the content listed here applies to features in the **Organizational administration** module. However, you can perform a couple of tasks, such as creating and using a record template, in any module to help your organization run more efficiently.

Number sequences

Use number sequences to generate readable, unique identifiers for master data records and transaction records that require identifiers. A master data record or transaction record that requires an identifier is a *reference*. Before you can create new records for a reference, you must set up a number sequence and associate it with the reference.

- Number sequences overview
- Set up number sequences using a wizard (Task guide)
- Set up number sequences on an individual basis (Task guide)

Organizations

An organization is a group of people who work together to carry out a business process or achieve a goal. Organizational hierarchies represent the relationships between the organizations that make up your business.

Before you set up organizations and organization hierarchies, plan how your business is modeled. The organization model has a significant effect on implementation and business processes.

- Organizations and organizational hierarchies overview
- Plan your organizational hierarchy
- Create an organization hierarchy (Task guide)
- Create a legal entity (Task guide)
- Create an operating unit (Task guide)

Address books

The global address book is a centralized repository for master data that you must store for all internal and external persons and organizations that your company interacts with. The data associated with party records includes the party's name, address, and contact information.

After you create the global address book, you can create more address books as you require, such as a separate address book for each company in your organization or for each line of business.

- Global address book overview
- Plan for the global address book and other address books
- Configure the global address book
- Address books FAQ

Workflow

Workflow is a system that you can use to create individual workflows or business processes. When you create a workflow, you specify how a document flows or moves through the system by showing who must complete a task, make a decision, or approve a document.

- Workflow system overview
- Workflow elements
- Actions in workflow approval processes
- Create workflows overview

Electronic signatures

An electronic signature confirms the identity of a person who is about to start or approve a computing process. In some industries, an electronic signature is as legally binding as a handwritten signature. Electronic signatures are a regulations compliance requirement for several regulated industries, such as pharmaceuticals, food and beverage, and aerospace and defense.

You can use electronic signatures for critical business processes. Some processes have built-in electronic signature capabilities. You can also create custom signature requirements for any database table and field.

- Electronic signatures overview
- Set up electronic signatures (Task guide)

Case management

By planning, tracking, and analyzing cases, you can develop efficient resolutions that you can use for similar issues. For example, when customer service representatives or Human Resources generalists create cases, they can find information in knowledge articles to help them work with or resolve a case more efficiently.

- Case management overview
- Plan case category security, case processes, and case categories

Record templates

Record templates can help you create records more quickly. You can create a record template so that you don't have to enter field values that you use often for each new record.

- Record templates overview
- Create a record template to facilitate data entry (Task guide)
- Use record template to create a new record (Task guide)

General organization administration

- Change the banner or logo (Task guide)
- Configure document management
- Configure and send email
- Date/time data and time zones

Last updated on 06/26/2026

Number sequences overview

Summarize this article for me

> **Note**
>
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Use number sequences to generate readable, unique identifiers for master data records and transaction records that need identifiers. A master data record or transaction record that needs an identifier is called a *reference*.

Before you can create new records for a reference, set up a number sequence and associate it with the reference. Use the pages in **Organization administration** to set up number sequences. If you need module-specific settings, use the parameters page in a module to specify number sequences for the references in that module. For example, in **Accounts receivable** and **Accounts payable**, you can set up number sequence groups to allocate specific number sequences to specific customers or vendors.

When you set up a number sequence, specify a scope to define which organization uses the number sequence. The scope can be **Shared**, **Company**, **Legal entity**, or **Operating unit**. You can combine **Legal entity** and **Company** scopes with **Fiscal calendar period** to create even more specific number sequences.

Number sequence formats consist of segments. Number sequences with a scope other than **Shared** can contain segments that correspond to the scope. For example, a number sequence with a scope of **Legal entity** can contain a legal entity segment. By including a scope segment in the number sequence format, you can identify the scope of a particular record by looking at its number.

In addition to segments that correspond to scopes, number sequence formats can contain **Constant** and **Alphanumeric segments**. A **Constant** segment contains a set of letters, numbers, or symbols that doesn't change. An **Alphanumeric** segment contains a set of letters or numbers that increment each time you use a number. Use a number sign (#) to represent incrementing numbers and an ampersand (&) to represent incrementing letters. For example, the format #####_2017 creates the sequence 00001_2017, 00002_2017, and so on.

Number sequence examples

The following examples show how to use segments to create number sequence formats. In particular, the examples demonstrate the effects of using scope segments.

Expense report numbers

In the following example, you set up expense report numbers for the legal entity that is titled **CS**.

- **Area:** Travel and expense
- **Reference:** Expense report number
- **Scope:** Legal entity
- **Legal entity:** CS

Expand table

Segments	Segment type	Value
Segment 1	Legal entity	CS
Segment 2	Constant	-EXPENSE-
Segment 3	Alphanumeric	####

Example of formatted number: CS-EXPENSE-0039

You can set up a similar number sequence format for other legal entities. For example, for a legal entity that is named **RW**, if you change only the value of the legal entity segment, the formatted number is RW-EXPENSE-0039. You can also change the whole number sequence format for other legal entities. For example, you can omit the legal entity scope segment to create a formatted number such as Exp-0001.

Sales order numbers

In the following example, you set up sales order numbers for the company ID **CEU**.

- **Area:** Sales
- **Reference:** Sales order
- **Scope:** Company
- **Company:** CEU

Expand table

Segments	Segment type	Value
Segment 1	Constant	SO-
Segment 2	Alphanumeric	####

Example of formatted number: SO-0029

Even though the format doesn't include a scope segment, numbering restarts for each company ID. If you use the same format for all company IDs, each company uses the same numbers. For example, sales order number SO-0029 is used in each company. You can also change the whole number sequence format for other company IDs.

Purchase requisition numbers

In the following example, purchase requisition numbers are organization-wide.

- **Area:** Purchase
- **Reference:** Purchase requisition
- **Scope:** Shared

Expand table

Segments	Segment type	Value
Segment 1	Constant	Req
Segment 2	Alphanumeric	####

Example of formatted number: Req0052

Because the scope is **Shared**, the number sequence format is used across the organization. You can't set up different number sequence formats for different parts of the organization.

Performance considerations for number sequences

Consider the following information about how the configuration of number sequences can affect system performance before you set up number sequences.

Continuous and noncontinuous number sequences

Number sequences can be continuous or noncontinuous. A continuous number sequence doesn't skip any numbers, but you might not use the numbers sequentially. You use numbers

from a noncontinuous number sequence sequentially, but the number sequence might skip numbers.

- Continuous Number Sequence (CNS)
  - Doesn't skip any numbers
  - You might not use numbers sequentially
  - Example: If a user cancels a transaction, the system generates a number but later recycles (reuses) it
- Noncontinuous Number Sequence (Non-CNS)
  - Might skip numbers
  - You use numbers sequentially (based on caching)
  - Example: If a user cancels a transaction, the system generates a number but doesn't use it

> **Note**
>
> The **Enable continuous number sequence performance improvements** feature provides enterprise readiness with continuous number sequences, which was challenging in finance and operations apps. The feature is in public preview in version 10.0.34, and generally available in version 10.0.36.
>
> This feature improves performance with continuous number sequences by preallocating a number in the sequence for each request. By default, the system allocates five numbers in a sequence, but you can adjust this value. If there's an unexpected termination of any number, the clean-up job is improved.

For continuous and noncontinuous number sequences, you can enable **Preallocation** on the **Performance** FastTab of the **Number sequences** page. When you specify a quantity of numbers to preallocate, the system selects those numbers and stores them in memory for noncontinuous number sequences and in the database for continuous number sequences.

If you use a noncontinuous number sequence, you can enable **Preallocation** on the **Performance** FastTab of the **Number sequences** page. When you specify a quantity of numbers to preallocate, the system selects those numbers and stores them in memory. The system requests new numbers from the database only after it uses the preallocated quantity.

Unless there's a regulatory or legal compliance requirement that you use continuous number sequences, use noncontinuous number sequences.

Automatic cleanup of number sequences

If an application error, crash, or other unexpected failure occurs, the system tries to automatically recycle numbers for continuous number sequences. You can run the cleanup process manually or automatically to recover lost numbers.

Carefully consider server usage when you plan the cleanup process. Perform the cleanup as a batch job during nonpeak hours.

Last updated on 03/11/2026

Set up number sequences on an individual basis

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up number sequences on an individual basis. Use number sequences to generate readable, unique identifiers for master data records and transaction records that need them. A master data or transaction record that needs an identifier is called a reference. Before you can create new records for a reference, you must set up a number sequence and associate it with the reference. You can set up all required number sequences at the same time by using the **Set up number sequences** wizard, or you can create or modify individual number sequences by using the **Number sequences** page.

1. Go to **Navigation pane** > **Modules** > **Organization administration** > **Number sequences** > **Number sequences**.
2. Select **Number sequence**.
3. In the **Number sequence code** field, enter a value.
4. In the **Name** field, enter a value.
5. On the **Scope parameters** FastTab, select a scope for the number sequence and select scope values from the drop-down list. The scope defines which organizations use the number sequence. In addition, number sequences that have a scope other than **Shared** can have segments that correspond to their scope. For example, a number sequence with a scope of **Legal entity** can have a legal entity segment. For more information about scopes, see Number sequence overview.
6. Expand the **Segments** section.

    - Define the format for the number sequence by adding, removing, and rearranging segments.
    - Number sequences of all scopes can contain *Constant segments* and *Alphanumeric segments*. Constant segments contain a set of alphanumeric characters that don't change. Use this segment type to add a hyphen or other separators between number sequence segments. Alphanumeric segments contain a combination of number signs (#) and ampersands (&). These characters represent letters and

numbers that increment each time you use a number from the sequence. Use a number sign (#) to indicate incrementing numbers and an ampersand (&) to indicate incrementing letters. For example, the format `#####_2014` creates the sequence `00001_2014`, `00002_2014`, and so on. At least one alphanumeric segment must be present. Scope segments, such as company or legal entity, aren't required. However, if you don't include scope segments in the format, the system still generates numbers for the selected reference per scope.

7. Expand the **References** section. Select the document type or record to assign this number sequence to. This step is optional for sequences that you define for special application usage patterns. In these scenarios, the system generates a new number by using the value of a number sequence code or ID, without using a reference. An example of a special application usage pattern is a voucher series that is used for specific journal names. However, don't use such patterns.
8. Expand the **General** section. On the General FastTab, specify whether the number sequence is manual, and continuous or non-continuous. In addition, enter the lowest and highest numbers that can be used in the number sequence. Don't change a non-continuous number sequence to a continuous number sequence. The number sequence won't be truly continuous. This change might also cause duplicate key violations in the database. In addition, continuous number sequences have a larger effect on performance.
9. Select **Save**.

Last updated on 03/11/2026

Set up number sequences by using a wizard

Summarize this article for me

> **Note**
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Number sequences generate readable, unique identifiers for master data records and transaction records that need them. A master data or transaction record that needs an identifier is called a reference. Before you can create new records for a reference, you need to set up a number sequence and associate it with the reference. This article explains how to set up all required number sequences at the same time by using a wizard. The demo data company used to create this procedure is USMF.

1. Go to **Navigation** > **Modules** > **Organization administration** > **Number sequences** > **Number sequences**.
2. Select **Generate**.
3. Select **Next**.
    - On this page, you can modify the identification code, the lowest value, and the highest value. You can also specify whether the number sequence must be continuous.
    - Don't select the **Continuous** option if you need to preallocate numbers for the number sequence. To add a scope segment to the format of a number sequence, select the format in the list, and then select **Include scope in format**. To remove a scope segment from the format of a number sequence, select the format in the list, and then select **Remove scope from format**. To exclude a number sequence from automatic generation, select the number sequence in the list, and then select **Delete**.
4. Select **Next**.
5. Select **Finish**.

Last updated on 03/11/2026

Configure number sequences for warehouse flows

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The *Number sequence extensions* feature adds a new configuration page for setting up number sequences. It enables flexible configuration of GS1-regulated IDs, including GS1 prefixes and check digits (modulo 10), and enforces a length limit on existing number sequences.

Standard number sequence segments aren't suitable for GS1 implementation, because no check digit is calculated, and the company's GS1 prefix must be updated manually.

This feature adds the following functionality:

- Bill of lading (BOL) IDs can be generated in advance.
- A unique number sequence can be generated for serial shipping container code (SSCC) numbers.
- GS1-compliant number sequences can be created for BOL and SSCC numbers. The feature adds out-of-box support for license plate IDs, container IDs, wave label IDs, and BOL IDs.
- Configuration of license plate ID numbers is flexible. For example, you can include or exclude application identifiers (AI), such as leading zeros (00).

This functionality makes it more efficient to support labeling of cartons and to adjust new numbers that are generated by the system.

Set up the feature

To set up number sequence extensions in your system, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, in the **GS1 company prefix** field, and enter your company's GS1 prefix. This value will affect all number sequences where the GS1 prefix is included as a segment.

3. If you want to generate BOL numbers for wave labels, on the **Reports** tab, select the **Generate BOL number when printing wave labels** checkbox.

   > ⓘ **Note**
   >
   > This checkbox is available only if the functionality for wave label printing is turned on.

4. Go to **Warehouse management** > **Setup** > **Number sequence extensions**

5. On the Action Pane, select **Create default setup**. A GS1-compliant BOL number sequence and three types of SSCC number sequences are created. All these number sequences take the length of your company's GS1 prefix into account.

   For more information about how to customize these default number sequences and/or add new sequences, see the next section. You can also remove any of these sequences if you don't need them.

6. Go back to **Warehouse management** > **Setup** > **Warehouse management parameters**.

7. On the **Number sequences** tab, select a relevant number sequence extension to use to generate numbers for your license plate IDs, wave label IDs, container IDs (in this case, select the appropriate **SSCC-18 number** sequence), and/or BOL IDs (in this case, select the **BOL** sequence). By default, number sequence extensions are supported only for these four types of IDs.

The next time that a new number is generated for one of these number sequences, the new logic will be used.

> ⓘ **Note**
>
> If you don't select a number sequence extension for license plate IDs, the current rules for generating license plate IDs will be used. Otherwise, your selected number sequence will be used. The other IDs will use a plain number sequence until you apply a number sequence extension for them.

Create and edit number sequences

In the previous section, you generated a default set of number sequences. This section explains how each number sequence is defined. It also explains how to create custom sequences, and how to edit the default or custom sequences.

To create and edit number sequences, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Number sequence extensions**.
2. On the Action Pane, select **New**.
3. In the **Number sequence extension** field, enter a name for the new sequence. In the **Description** field, enter a description.
4. On the **Segments** FastTab, use the buttons on the toolbar to assemble your numbering format by adding, deleting, and arranging segments. In the **Segment** field for each row, assign a segment type to define the purpose and content of that segment. The following table describes the types of segments that are available.

Expand table

Segment type	Description
Constant	This segment type adds the same constant text for each generated number in the sequence. In the **Value** field, enter the required text. The **Length** field is automatically updated to the length of the text that you entered in the **Value** field.
Number sequence	In the **Value** field, enter a number sign (#) for each character that should be shown in the generated sequence. The number sequence itself might generate longer numbers, but only the right-most characters will be shown. The **Length** field is automatically updated to the number of number signs that you entered in the **Value** field. To comply with GS1 requirements for SSCC-18 numbers, make sure that the length of this segment is 16 minus the length of your GS1 prefix.
GS1 prefix	This segment type adds the value that is set in the **GS1 company prefix** field on the **Warehouse management parameters** page. The **Value** field shows the value that is set on the **Warehouse management parameters** page, and the **Length** field shows the number of characters in the value. Both the **Value** field and the **Length** field are read-only.
Application identifier	In the **Value** field, enter an application identifier, as specified by the relevant bar code data policy for this type of number sequence. For example, enter *00* for SSCC or *420* for BOL. The **Length** field is automatically updated to the length of the identifier that you entered in the **Value** field.
Packing type	For items that can be clearly identified, this segment type adds a field value from the relevant unit sequence group (from the **Unit sequence groups** page). (This behavior matches the existing logic for license plate IDs.) For license plates that include multiple stock keeping units (SKUs), this segment type adds *0* (zero) by default. For this segment type, the **Value** field is always set to *P*, and the **Length** field is always set to *1*.

Segment type	Description
Check digit	This segment type adds a check digit, which is a modulo 10 calculation. (This behavior matches the existing logic for license plate IDs.) For this segment type, the **Value** field is always set to a caret (^), and the **Length** field is always set to *1*.

5. To view an example of your final number format, inspect the **Format** field at the bottom of the **Segments** FastTab.

Last updated on 12/30/2025

Organizations and organizational hierarchies overview

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>
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An organization is a group of people who work together to carry out a business process or achieve a goal. Organizational hierarchies show the relationships between the organizations that make up your business.

Organizations

You can define the following types of internal organizations: legal entities, operating units, and teams.

All internal organizations are types of the **Party** entity. Therefore, these organizations use the address book to store address and contact information. A party, which can be either a person or an organization, can belong to one or more address books.

Legal entities

A legal entity is an organization that has a registered or legislated legal structure. Legal entities can enter into legal contracts and must prepare statements that report on their performance.

A company is a type of legal entity. Currently, companies are the only kind of legal entity that you can create, and every legal entity is associated with a company ID. This association exists because some functional areas in the program use a company ID, or DataAreaId, in their data models. In these functional areas, companies are used as a boundary for data security. Users can access data only for the company that they're currently signed in to.

Operating units

An operating unit is an organization that divides the control of economic resources and operational processes in a business. People in an operating unit have a duty to maximize the use of scarce resources, improve processes, and account for their performance.

The types of operating units include cost centers, business units, value streams, departments, and commerce channels. The following table provides more information about each type of operating unit.

Expand table

Operating unit type	Description	Purpose
Cost center	An operating unit in which managers are accountable for budgeted and actual expenditures.	Used for the management and operational control of business processes that span legal entities.
Business unit	A semi-autonomous operating unit that is created to meet strategic business objectives.	Used for financial reporting that is based on industries or product lines that the organization serves independently of legal entities.
Value stream	An operating unit that controls one or more production flows.	Commonly used in lean manufacturing to control the activities and flows that are required to supply a product or service to consumers.
Department	An operating unit that represents a category or functional part of an organization that performs a specific task, such as sales or accounting.	Used to report on functional areas. A department might have profit and loss responsibility, and might consist of a group of cost centers.
Retail channel	An operating unit that represents a brick-and-mortar store, an online store, or call center.	Used for the management and operational control of one or more stores within or across legal entities.

Establishments

> **Note**
>
> **Availability**: Establishments support in Dynamics 365 Finance is available starting with version **10.0.48**. **Establishments** functionality in Finance is controlled by **Feature management**. To use **Establishments** in business documents and enable establishment-level **Registration ID** validation and immutable storage on invoices, enable the **Establishment and Registration ID governance on invoices** feature. When the feature is turned off,

establishment-specific behavior (such as establishment selection, defaulting, and validation on invoices) isn't applied.

In Finance, an establishment represents an operational unit of a legal entity that carries out economic activity on a stable basis and might require its own regulatory identifiers, such as registration numbers used on invoices and regulatory reports.

You model establishments by using existing **Operating units**, combined with an **Organizational hierarchy** assigned with *Enterprise establishment structure* **Organizational hierarchy purpose**, to support invoicing and compliance scenarios.

A single legal entity can have one or multiple establishments, while remaining the sole legal and accounting entity. All establishments share the same legal identity of the company but might have distinct operational identities for invoicing and regulatory purposes.

Use **Operating units** to represent **Establishments** when a business requires:

- Identification of the issuing or receiving unit on invoices
- Separate regulatory identifiers per operational location
- Consistent establishment assignment across customer, vendor, project invoices

The system treats only **Operating units** that you explicitly include in a dedicated **Organizational hierarchy** assigned with **Enterprise establishment structure** purpose as **Establishments**. You can select these establishments in the **Establishment** field in the following documents where you can post an invoice:

- Free text invoices
- Sales orders
- Purchase orders
- Vendor invoices
- Pending vendor invoices
- Project invoice proposals
- Intercompany customer invoices
- General journals

Associate **Establishments** with inventory **Sites** (**Inventory management** > **Setup** > **Inventory breakdown** > **Sites**) to support transactional defaulting.

When you link an inventory **Site** to an **Establishment**, the system can derive the issuing or receiving establishment of the legal entity for customer, vendor, or project invoice that reference

that site. This association ensures consistent establishment identification across business documents that result in invoice posting.

Associate **Establishments** with **Financial dimension values** to support transactional defaulting. When a **Financial dimension value** linked to an **Operating unit** included into an **Organizational hierarchy** assigned with **Enterprise establishment structure** purpose, is used on a business document, the system can derive and default the applicable **Establishments** for customer, vendor, or project invoices created from that document. This association complements site-based defaulting and helps ensure consistent establishment identification across invoice-generating processes.

Teams

A team is an organization in which the members share a common responsibility, interest, or objective. Teams can't be used in organizational hierarchies.

Organizational hierarchies

Set up organizational hierarchies to view and report on your business from different perspectives. For example, you can set up a hierarchy of legal entities for tax, legal, or statutory reporting. Set up a hierarchy that is based on operating units to report financial information that isn't legally required, but that is used for internal control. For example, you can create a purchasing hierarchy to control purchasing policies, rules, and business processes.

> **Note**
>
> After an operating unit is added to a hierarchy, the operating unit can't be deleted.

Each hierarchy is assigned to a purpose. The purpose of a hierarchy determines the types of organizations that can be included in the hierarchy. The purpose also determines which application scenarios a hierarchy can be used in.

Organizations in a hierarchy can share parameters, policies, and transactions. An organization can inherit or override the parameters of its parent organization. However, shared master data, such as products and address books, applies to the whole organization and can't be overridden for individual organizations. Creating organizations and hierarchies requires careful planning. Learn more in Plan your organizational hierarchy.

Establishment hierarchy

> **Note**
>
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Organizational hierarchies can be used to define which **Operating units** represent valid **Establishments** within a legal entity.

To support establishment scenarios, Finance introduces a dedicated organization hierarchy purpose: **Enterprise establishment structure**. This hierarchy purpose determines which operating units are considered establishments and participate in **Registration ID** validation and immutability on documents such as invoices.

Using Organizational Hierarchies in Financial Reporting

Organizational Hierarchies can be selected as a **Tree type** option in Financial Reporting. Financial reporting selects the current effective hierarch as noted by hierarchy.

Additional resources

- Plan your organizational hierarchy
- Create an organization hierarchy
- Create a legal entity
- Create an operating unit

Last updated on 04/24/2026

Plan your organizational hierarchy

> **Note**
>
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Before you set up organizations and organization hierarchies, plan how your business will be modeled. The organization model significantly affects the implementation and business processes.

Organizational hierarchies represent the relationships between the organizations that make up a business. Therefore, the most important consideration when you model organizations is the structure of your business. Define organization structures based on feedback from executives and senior managers from functional areas, such as finance and accounting, human resources, operations, purchasing, and sales and marketing.

When you plan hierarchies, consider the relationship between the organizational hierarchy and financial dimensions. You can set up multiple organizational hierarchies to represent different views of your business. By using financial dimensions, you can create reports based on these views. Work with your partner to create hierarchies that address both organizational and statutory reporting needs.

> **Note**
>
> Although you can use financial dimensions to represent legal entities without creating the legal entities, financial dimensions aren't designed to address the operational or business needs of legal entities. The interunit accounting functionality is designed to address only the accounting entries that are created by each transaction.

> **Important**
>
> You shouldn't decide how to model organizations based only on the information in this article. This documentation is a guide. You can work with your partner for additional

guidance. Your partner has gained experience in various industries and across the customer base.

Decide whether to model internal organizations as legal entities or operating units

You must have at least one legal entity to represent your business. A legal entity can enter legal contracts and is required to prepare financial statements that report on its performance.

Use legal entities for transactional business or for consolidation. This structure means that a legal entity in finance and operations doesn't necessarily represent a real entity in your business. For example, a company that participates in transactions can own subsidiary legal entities. In this scenario, a legal entity is required for transactions, and a virtual legal entity is required to consolidate the results and balances of the subsidiary legal entities.

Internal organizations in your business, such as regional offices, can be represented as additional legal entities, or as operating units of the main legal entity. An operating unit isn't required to be a legally defined organization. Operating units are used to control economic resources and operational processes in the business. For example, departments and cost centers are operating units.

Some functionality works differently depending on whether the organization is a legal entity or an operating unit. Carefully consider the functionality described in the following section as you make your decision.

Master data

If you model the organization as a legal entity

Some master data, such as customers, payment terms, tax authorities, and site-specific stock ordering, must be set up for each legal entity. Some master data, such as users, products, and most human resources data, is shared among all legal entities.

If you model the organization as an operating unit

Master data is shared among operating units.

Module parameters

If you model the organization as a legal entity

Set module parameters, such as Accounts receivable, Accounts payable, and Cash and bank management, per legal entity. Because the module setup for legal entities is separate, each subsidiary can comply with local statutory requirements and business practices. For example, a professional services legal entity and a manufacturing legal entity can have different module parameters even though they report to the same parent company.

If you model the organization as an operating unit

Module parameters are shared among operating units.

Data security

If you model the organization as a legal entity

Most data is automatically secured by company ID. A company ID is a unique identifier for the data that is associated with a legal entity. A company can be associated with only one legal entity, and a legal entity can be associated with only one company. Users can access data only for the companies that they have access to. You don't need to customize to secure data by company ID.

If you model the organization as an operating unit

Data can be secured per operating unit by creating customized data security policies. Data security policies are used to limit access to data. For example, assume that a user is allowed to create purchase orders only in a particular operating unit. Data security policies can be created to prevent the user from accessing purchase order data from any other operating unit. The volume of transactions and the number of security policies can affect performance. When you design security policies, keep performance in mind.

Ledgers

If you model the organization as a legal entity

Each legal entity requires a ledger that provides a chart of accounts, accounting currency, reporting currency, and fiscal calendar. A balance sheet can be created only for a legal entity. Main accounts, dimensions, account structures, charts of accounts, and account rules can be used by more than one legal entity.

If you model the organization as an operating unit

An operating unit can't have its own ledger information. If your internal organizations don't require unique ledgers, model them as operating units. Set up ledger information for the parent legal entity in the hierarchy. You can create income statements for operating units within a legal entity or for the parent legal entity.

Fiscal calendars

If you model the organization as a legal entity

Each legal entity has its own fiscal calendar. If your internal organizations use different fiscal years and fiscal calendars, model the organizations as legal entities.

If you model the organization as an operating unit

Operating units must share a fiscal calendar. If your internal organizations can use the same fiscal years and fiscal calendars, model the organizations as operating units.

Consolidation

If you model the organization as a legal entity

You must consolidate the financial results for regional offices into a single, consolidated company to prepare financial statements.

If you model the organization as an operating unit

Consolidation isn't required, because data is already shared among operating units.

Centralized payments

If you model the organization as a legal entity

Set up centralized payments so you can pay invoices for all child legal entities to or from a single parent legal entity.

If you model the organization as an operating unit

You don't need centralized payments because you record all invoices in a single legal entity.

Intercompany transactions

If you model the organization as a legal entity

You can apply intercompany sales orders, purchase orders, payments, or receipts to one another. You don't need to use journal vouchers. You can view intercompany transactions at the subledger level, such as Accounts receivable and Accounts payable. The following examples illustrate how the system handles intercompany transactions.

Example 1: Headquarters provides services to regional offices and must charge the costs of those services to the regional offices

If you model the regional office as a legal entity, you have the following options:

- Headquarters creates a journal entry to cross-charge the regional office for the expense. The transactions can't be aged.
- Headquarters sends a purchase order for the services to the regional office. A sales order is automatically created in the legal entity for the regional office, with intercompany subledger transactions.

Example 2: Headquarters procures and pays for service that is delivered to a regional office

If you model the regional office as a legal entity, you have the following options:

- The invoice and payment follow the regulatory requirements of headquarters. Headquarters can create a journal entry to cross-charge the regional office for the expense. The transactions can't be aged.
- The invoice and payment follow the regulatory requirements of headquarters. Headquarters can create an intercompany subledger transaction.

If you model the organization as an operating unit

Intercompany transactions among operating units are supported only through journal vouchers. An operating unit can't issue or receive a purchase order, sales order, or invoice from another operating unit in the same legal entity. You can't view intercompany transactions at the subledger level (Accounts receivable, Accounts payable). The following examples illustrate how intercompany transactions are handled.

Example 1: Headquarters provides services to regional offices and must charge the costs of those services to the regional offices

If you model the regional office as an operating unit, headquarters enters an expense transaction and codes it to the regional office.

Example 2: Headquarters procures and pays for service that is delivered to a regional office

If you model the regional office as an operating unit, the invoice and payment follow the regulatory requirements of headquarters. The invoice can be coded to the regional office. On the income statement, use a balancing financial dimension to report costs for the regional office.

Local tax requirements

If you model the organization as a legal entity

A legal entity is subject to the tax laws of the tax authority in the country/region where the legal entity is registered. For example, a legal entity that is registered in Denmark is subject to Danish tax laws and regulations. A legal entity can belong to only one country/region. The country/region that you select for the primary address of the legal entity controls the country/region-specific features that are available to that legal entity. For example, if the primary address of the legal entity is in Denmark, features that are related to Danish tax laws and regulations become available. Therefore, if your organizations are in different countries/regions and require different local tax options, you must set up the organizations as separate legal entities.

If you model the organization as an operating unit

Operating units use the country context of the parent legal entity. Operating units in the same legal entity can't have different country/region-specific requirements. If your organizations are in the same country/region and use the same tax options, you can set them up as operating units.

Statutory reporting for a country/region

If you model the organization as a legal entity

For countries/regions that are supported, most statutory reports can be created.

> **Note**
>
> A posting layer in the general ledger allows you to make adjusting entries to a parent company that uses a different accounting standard than the child company. For example, for a company that uses generally accepted accounting practices in the United Kingdom (UK GAAP), you can make adjusting entries in the posting layer. These entries can be

consolidated into a parent company that uses generally accepted accounting principles (GAAP) in the United States. The adjusting entries don't affect UK GAAP reporting.

If you model the organization as an operating unit

Statutory reports must be created by using another application. You must ensure that data is captured in finance and operations apps to support the requirements of each operating unit, where they differ from the requirements of headquarters.

Currency

If you model the organization as a legal entity

If your organizations must use different functional currencies, you must model the organizations as legal entities. Functional currencies are set up per legal entity. However, you can enter transactions in multiple currencies.

If you model the organization as an operating unit

If your organizations can use a single functional currency, you can model the organizations as operating units. Operating units must share a functional currency. However, you can enter transactions and create reports in multiple currencies.

Year-end closing

If you model the organization as a legal entity

Model the organizations as legal entities if laws and accounting practices differ among the countries or regions where your organizations are located. Each legal entity has its own year-end procedures.

If you model the organization as an operating unit

Model the organizations as operating units if laws and accounting practices are the same among the countries or regions where your organizations are located. All operating units must use the same year-end closing procedure.

Number sequences

If you model the organization as a legal entity

Set up number sequences for some references per legal entity. Share some number sequences.

If you model the organization as an operating unit

Number sequences for some references can be set up per operating unit. Some number sequences can be shared.

Products

If you model the organization as a legal entity

Product definitions are shared, and they must be released to individual legal entities before they can be included in transactions. Each legal entity has its own set of released products that can be included in transaction documents. If your internal organizations must use different sets of products, you must model the organizations as legal entities.

> **Note**
>
> Even though product definitions are shared, in each legal entity where a product has been released, you can specify different sales, purchase, and stocking parameters for the item at each inventory site.

If you model the organization as an operating unit

All operating units share the same set of products. If your internal organizations can share the same set of products, you can model the organizations as operating units.

Inquiry and reporting

If you model the organization as a legal entity

You must manually change companies to enter transactions and perform inquiries in multiple legal entities. Because of data security boundaries, consolidated inquiry and reporting can be resource intensive and time-consuming.

If you model the organization as an operating unit

You don't need to change companies to access data from multiple operating units. Consolidated inquiry and reporting and individual regional inquiry is easier and faster.

Multi-company processing for hierarchical legal entity selection

To enable users who have access to multiple companies to execute transactions, generate reports, and perform inquiries across legal entities, assign the **Multi-company processing** purpose to one or more organization hierarchies. This capability was introduced in Dynamics 365 Finance version 10.0.44. Starting with Dynamics 365 Finance version 10.0.47, the feature is enabled by default.

To enable the feature manually, follow these steps:

1. Go to **Feature management**.
2. Enable the **Multi-company purpose** feature.

The feature activates a hierarchical parent-child legal entity selection experience, replacing the previous flat multiselection company picker. This enhancement provides a more scalable and structured approach for cross-company operations and improves usability for organizations managing complex legal entity structures.

You can search, sort, and select multiple legal entities in:

- General ledger
  - Currency revaluation
  - Year-end close
- Fixed assets
  - Fixed assets depreciation proposal
- Asset leasing
  - Asset leasing batch journal creation
- Cash and bank
  - Currency revaluation
- Accounts payable
  - Pending vendor invoice
  - Vendor invoice automation
  - Payment workspace (Vendor)
  - Global transaction
- Accounts receivable
  - Payment workspace (Customer)
  - Global transaction
  - Payment workspace

Best practices for modeling organizations and hierarchies

Consider the following best practices when you implement an organization hierarchy:

- Create a department to model the intersection between a legal entity and a business unit. You can then roll up data from a department to a legal entity for statutory reporting, and from a department to a business unit for internal reporting. Departments can serve as profit centers. If you use departments, you don't have to use legal entities and business units as dimensions in the account structure. You can use just departments as a dimension. However, you must use both cost centers and departments as dimensions in the account structure if cost centers are used only as cost accumulators, and departments are used for revenue recognition.
- Model multiple hierarchies for operating units if you have complex requirements for reporting profit and loss.
- In a single legal entity, don't model multiple hierarchies for the same hierarchy purpose.
- Don't create a hierarchy for every purpose. Usually, you can use one hierarchy for multiple purposes. For example, one hierarchy of operating units can be assigned to all policy-related purposes.
- Create balanced hierarchies. In a hierarchy, all nodes that are the same distance from the root node are defined as a level. In a balanced hierarchy, only one type of operating unit can occur at each level, and the distance from the root node to each level is consistent. If there are intermediate levels between a department and a legal entity or a business unit, placeholder organizations may be required to create a balanced hierarchy.
- Don't model a separate hierarchy of operating units if the structure for legal entities is also your operating structure. A mixed hierarchy of legal entities and operating units may serve both purposes.
- Before you model major restructuring scenarios, use the hierarchy's effective dates to perform an impact analysis and a validation test.
- Use draft mode to change a hierarchy before you publish a new version in a production environment.
- Limit the number of people who have permissions to add or remove organizations from a hierarchy in a production environment. A smaller number reduces the chance that costly mistakes can occur and corrections must be made.

Last updated on 06/05/2026

Create an organization hierarchy

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the following procedure to create an organizational hierarchy. You can use organizational hierarchies to view and report on your business from various perspectives. For example, you can set up one hierarchy for tax, legal, or statutory reporting. You can then set up another hierarchy to report financial information that isn't legally required, but that is used for internal reporting.

Starting with Dynamics 365 Finance version 10.0.48, organization hierarchies can also be used to model Establishments. Establishments represent operational units of a legal entity that may require their own regulatory identifiers on invoices. Establishments are modeled using existing operating units and are identified through a dedicated organization hierarchy purpose, **Enterprise establishment structure**. This hierarchy determines which operating units are treated as establishments in invoicing, while the legal entity remains the single legal and accounting entity.

Before you create an organizational hierarchy, you must create organizations. For more information, see the "Create a legal entity" or "Create an operating unit" tasks. You can also create departments and teams.

The demo data company used to create this procedure is USMF.

Create a hierarchy

1. Go to **Navigation pane** > **Modules** > **Organization administration** > **Organizations** > **Organization hierarchies**.
2. On the **Action pane**, select **New**.
3. In the **Name** field, enter a value.
4. In the **Purpose** section, select **Assign purpose**.
5. In the list, find and select the desired record. Select a purpose to assign to your organization hierarchy.

6. In the **Assigned hierarchies** section, select **Add**.
7. In the list, select the row. Find the hierarchy you just created.
8. Select **OK**.

Add organizations to the hierarchy

1. In the list, find and select the desired record. Select your hierarchy.
2. In the **Assigned hierarchies** section, select **View hierarchy**.

    - Add organizations, as necessary.
    - To add an organization, select **Edit** and then **Insert** to add the organization. When you're done making changes, you can **Save** a draft and/or **Publish** the changes.

Last updated on 04/24/2026

Create a legal entity

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

A legal entity is an organization that a legal authority identifies through registration. Legal entities can enter into contracts and must prepare statements that report on their performance. The following procedure explains how to create a legal entity. The demo data company used to create this procedure is USMF.

1. Go to **Organization administration** > **Organizations** > **Legal entities**.
2. Select **New**.
3. In the **Name** field, enter a value.
4. In the **Company** field, enter a value.
5. In the **Country/region** field, enter or select a value.
6. Select **OK**. In the **General** section, provide the following general information about the legal entity: Enter a search name, if a search name is required. A search name is an alternate name that you can use to search for this legal entity. Select whether this legal entity is being used as a consolidation company or an elimination company.
7. Expand the **Addresses** section. In the **Addresses** section, select **Edit** to enter address information, such as the street name and number, postal code, and city.
8. Expand the **Contact information** section and enter information about methods of communication, such as email addresses, URLs, and telephone numbers.
9. Expand the **Statutory reporting** section and enter the registration numbers that are used for statutory reporting.
10. Expand the **Registration numbers** section and enter any information required by the legal entity.
11. Expand the **Bank account information** section and enter bank accounts and routing numbers for the legal entity.
12. Expand the **Foreign trade and logistics** section and enter shipping information for the legal entity.
13. Expand the **Number sequences** section and view the number sequences that are associated with the legal entity.
14. Expand the **Images** section and view or change the logo and dashboard image that are associated with the legal entity.

15. Expand the **Tax registration** section and enter the registration numbers that are used to report to tax authorities.
16. Expand the **Tax 1099** section and enter 1099 information for the legal entity.
17. Select **Save**.

Last updated on 03/11/2026

Create an operating unit

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

An operating unit is an organization that is used to divide the control of economic resources and operational processes in a business. People in an operating unit have a duty to maximize the use of scarce resources, improve processes, and account for their performance. The types of operating units include cost centers, business units, departments, and value streams.

Starting with Dynamics 365 Finance version 10.0.48, operating units can be used to represent Establishments. An establishment represents an operational unit of a legal entity that carries out economic activity and may require its own regulatory identifiers on business documents, such as invoices. Establishments are modeled by using existing operating units and identified through an organization hierarchy that's assigned the **Enterprise establishment structure** purpose. Only operating units that are included in such a hierarchy are treated as establishments. The legal entity remains the single legal and accounting entity, while establishments are used for operational and regulatory purposes.

Use the following procedure to create an operating unit. The demo data company used to create this procedure is USMF.

1. Go to **Navigation pane** > **Modules** > **Organization administration** > **Organizations** > **Operating units**.
2. Select **New** to open the dialog.
3. Find and select the desired record. Select the type of operating unit you want to create.
4. Select the link in the selected row.
5. In the **Name** field, enter a value.

    - Expand the **General** section, if necessary.
    - Provide general information about the operating unit, such as an identification number, DUNS number, and manager.
    - Expand the **Addresses** section, if necessary.

- Enter address information, such as the street name and number, postal code, and city. Select **Add** to enter a new address record, or select **Edit** to modify an existing address record.
- Expand the **Contact information** section, if necessary.
- Enter information about methods of communication, such as email addresses, URLs, and telephone numbers. To enter a new communication record, select **New**. To modify an existing communication record, select **More options** > **Advanced**.

6. Optionally, change the **Operating unit number** as needed. This number is a unique identifier for the corresponding **Party** record and can't be the same as any other operating unit.
7. Select **Save**.

Last updated on 04/24/2026

Global address book overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The global address book is a centralized repository for master data that you must store for all internal and external persons and organizations that your company interacts with. The data associated with party records includes the party's name, address, and contact information. Other details vary, depending on whether the party is a person or an organization. You assign each party record to a party, and each party can be associated with one or more party roles in a company. Party roles include customer, prospect, worker, user, vendor, competitor, applicant, and contact. For example, the organization party First Up Consultants, can be associated with customer, business relation, and vendor roles in the CEE company, and can also be associated with the vendor role in the CEU company. Here are some of the benefits of this shared data:

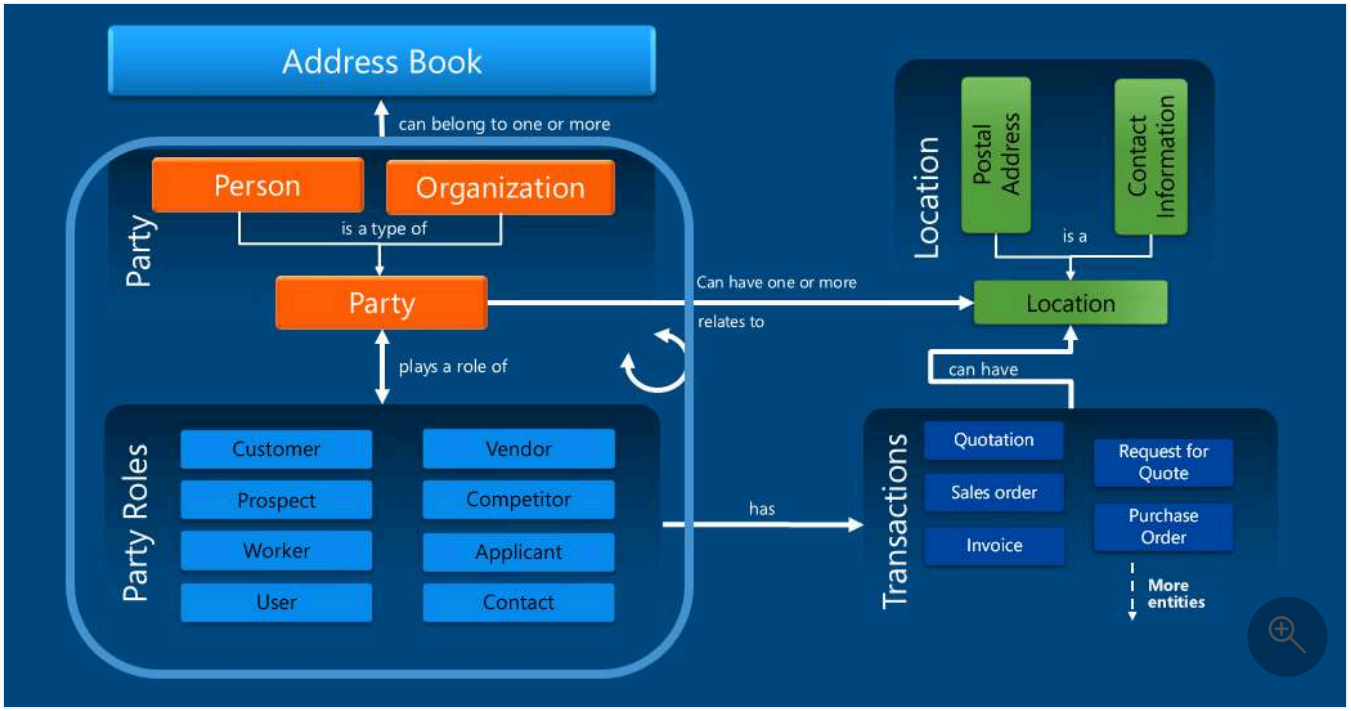
- The data shows the relationships that people and organizations have with other areas of the company. The relationship between two organizations changes when one organization has multiple roles, such as vendor and customer. Communication between the two organizations also changes. There might be special agreements that you can negotiate to encourage a closer partnership with the other organization.
- Setup and maintenance are easier. For example, when an address changes, you need to update the address in only one place. All the other associated records are updated automatically.

You can't delete any party in the global address book that's a legal entity. The **Delete** button is unavailable for these parties.

How the global address book works

The following illustration shows how party records, party roles, locations, and transactions interact and relate to an address book. As the illustration shows, a party record can belong to one or more address books. Each party record can store one or more locations, or addresses, and is assigned a party role. The role that is assigned to the party record can have specific transactions types associated with it. The following sections provide more information about party roles, locations, and transaction types. The following image is a graphical representation

of the ways that parties, party roles, locations, and transactions interact in relation to the global address book.



Party roles

You refer to roles that are associated with party records as party roles. You can assign several party roles to both party types, person and organization. Here are the definitions for each party role:

- **Customer** – Individuals, companies, or other entities who purchase goods and services that are produced by other individuals, companies, or entities.
- **Prospect** – A party that might provide a service or benefit to a legal entity.
- **Worker** – A person who assumes the role of an employee or a contractor, and who is paid in exchange for services.
- **User** – A person who is a user of the system.
- **Vendor** – A party that supplies products to one or more legal entities in exchange for payment.
- **Competitor** – A person or organization that provides goods or services that are similar to the goods or services that your business provides.
- **Applicant** – A person who makes a formal written or electronic request to work for or fill an open position in an organization.
- **Contact** – A person, either inside or outside your organization, that you create an entry for. In this entry, you can save information such as the person's street and email addresses, telephone and fax numbers, and webpage URLs.

Creating new party records

Enter party records in the global address book in two ways:

- **Create a party record when you don't know the role** – Create a party record in the global address book when you don't know the role type (for example, you don't know whether the party is a customer or an opportunity). You can select the role type later.
- **Create a party record when you know the role** – If you know the role type for the party, create a record on the appropriate page for that type. For example, if the party is a customer, create a record on the **Customer** page. When you create and save a record by using the page for the party's role type, you automatically create the record in the global address book.

Party roles and transactions

For transactions that are a part of the business processes, multiple parties might be associated with each transaction. An example is a customer that you need to reference on project quotations.

Parties locations, addresses, and contact information

Each party record's addresses, locations, and contact information are shared across all the party roles that are associated with that party. Therefore, when you change any of this information, all other associated records are updated accordingly.

Locations and transactions

When you include a party role in a transaction, you can access the location, address, or contact information of the party when you enter transaction details.

Last updated on 03/18/2026

Plan for the global address book and other address books

This article describes the considerations and decisions that you must make during the planning process before you set up and configure the global address book and any additional address books. Some of the decisions require that you confirm the decisions made for other areas of the product, such as the organization hierarchy.

Global address book

Before you begin to work with the global address book, determine the default values for it. Use these default values for any additional address books that you create.

Decisions

- What sequence should names be displayed in for party records of the **Person** type? For example, one sequence is last name, middle name, first name.
- Should the system delete party records from the address book when the system deletes the role record? For example, if the system deletes a customer record, should the system also delete the party record?
- When a user creates a new record, should the system notify the user if a duplicate record is found in the global address book?
- Should the Data Universal Numbering System (DUNS) number be included in a party record's information?
- If the system includes the DUNS number in a party record, should the system check the uniqueness of the number?
- When a user creates a party record in the global address book, do you want a default party type, person, or organization?
- Which user roles should have access to the private addresses and contact information of party records?

Additional address books

After you create the global address book, you can create additional address books as you require, such as a separate address book for each company in your organization or for each line of business. For example, Fabrikam is an international organization that has multiple companies and

multiple lines of business. Fabrikam plans to create an address book for each line of business. For lines of business that occur in more than one location, such as the pneumatic tools business, Fabrikam plans to create an address book for each location. Chris, the IT manager at Fabrikam, created the following list of address books that are required. This list also describes the party records that each address book must include.

- **Public Sector Contracts (PubSC)** – Party records for all parties that are involved in the public sector contracts that Fabrikam holds.
- **Private Sector Contracts (PriSC)** – Party records for all parties that are involved in the private sector contracts that Fabrikam holds.
- **Electronic Tools (ET)** – Party records for all parties that are involved in the purchase or sale of electronic tools, or that otherwise interact with the electronic tools that are provided by or purchased for Fabrikam in the Fabrikam-Japan company.
- **Pneumatic Tools (PTJPN)** – Party records for all parties that are involved in the purchase or sale of pneumatic tools, or that otherwise interact with the pneumatic tools that are provided by or purchased for Fabrikam in the Fabrikam-Japan company.
- **Pneumatic Tools (PTUSA)** – Party records for all parties that are involved in the purchase or sale of pneumatic tools, or that otherwise interact with the pneumatic tools that are provided by or purchased for Fabrikam in the Fabrikam-US company.

Decision:

- How many additional address books will you create?

Last updated on 06/26/2026

Configure the global address book

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use this procedure to set the default values and security policies for the global address book.

The demo data company used to create this task is USMF. This task is intended for the Planning and configuration team.

1. In the navigation pane, go to **Modules** > **Organization administration** > **Global address book** > **Global address book parameters**.
2. In the **Name sequence** field, select how names should be shown.
3. In **Delete parties with no roles**, select whether to delete parties that don't have an assigned role.
4. In **Use duplicate check**, select whether to check for duplicate records.
5. In **Display DUNS number on addresses**, select whether to display the DUNS number on addresses.
6. In **Check for unique DUNS number**, select whether to check for unique DUNS numbers.
7. In the **Party** field, select an option.
8. In the **Customer** field, select an option.
9. In the **Vendor** field, select an option.
10. In the **Prospect** field, select an option.
11. In the **Competitor** field, select an option.
12. Select the **Private location security** tab.
13. In the list, find and select the desired record. Press the **Shift** key to select multiple roles to add to the **Selected roles** pane, and then select the arrow to add the selected roles.
14. Select **Save**.

Last updated on 03/11/2026

Address setup

> **Note**
>
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Global regions where your organization operates might have different address format types. You must be able to adjust to the appropriate address format when a postal address is shown for each region. You can use the **Address setup** page to set up the postal address formats that are required for your organization.

Define address parameters

The **Parameters** tab of the **Address setup** page defines which fields must be validated when addresses are created. You can select options to require validation for **ZIP/postal code**, **District**, and **City** values.

If the option that corresponds to a field isn't selected, no validation is done for that field when new addresses are created in the global address book. In this case, the field is a free-text field where users can enter any value. If the option that corresponds to a field is selected, validation is done for that field when new addresses are created in the global address book. In this case, the value that is entered in the field must have been previously created in the address setup.

For example, the **City** option is selected in the **Validate on creating addresses** section. In this case, when a new address is created in the global address book, the **City** value for the new address must be selected in the list of cities that is set up in the **City** section of the **Address setup** page.

Set up address formats

You can set up address component information and formatting on the **Address format** tab. As you enter address information for a selected address type, the **Preview** field in the upper-right pane shows how the address will appear.

1. On the **Address setup** page, on the **Address format** tab, select **New** to add a new address format type.

2. Enter the name of the address format type and a brief description. For example, you might enter **NO** in the **Address format** field and **Norway** in the **Description** field.

3. In the lower pane, select **New** to configure a new address component for the address format type that is selected in the upper pane.

4. In the **Address application object** field, select the first object that must appear when an address in this format is shown. For example, select **Street**.

5. In the **Separator** field, enter the character that is used to separate this object from the next object when an address in this format is shown. For example, enter a comma (**,**).

6. Set the following checkboxes as appropriate:

    - **New line** – Select this checkbox if the object should appear on a new line.
    - **Data entry only** – Select this checkbox if the object should be shown only when the address is entered. If this checkbox is selected, the object won't be shown when the address is printed.
    - **Not active** – Select this checkbox if the object should not be used in an address of the selected format type.
    - **Expand** – Select this checkbox to show the full name of the country/region, state/province, or county instead of the code when the address is printed.
    - **Special** – Select this checkbox if the field value contains the character that you specified as a separator.

Set up country/region information

Use the **Country/region** tab to define the country/region information for the address setup.

1. On the **Address setup** page, on the **Country/region** tab, select **New** to create a new line.
2. Enter the country/region code, a short name for country/region, and the full name. For example, enter **DEU**, **Germany**, and **Federal Republic of Germany**, respectively.
3. In the **Address format** field, select an address format to use for the country/region. The address formats that are defined on the **Address format** tab of the page are available for selection.
4. In the **Time zone** field, select the time zone of the country/region.
5. Enter the International Organization for Standardization (ISO) code for the country/region.
6. In the **BACEN code**, **IOR facility ID**, and **Code by OKSM** fields, enter values if they are required for your financial, purchasing, or sales transactions. Also select whether the country/region is a member of the customs union.

7. To enable only valid five-digit or nine-digit ZIP/postal codes, select the **Use ZIP + 4 postal code validation rules** option.

Set up state/province information

Use the **State/province** tab to define the state/province information for the address setup.

1. On the **Address setup** page, on the **State/province** tab, in the **Country/region** filter, select the country/region that you're adding the state/province information to.
2. Select **New** to create a new line.
3. Enter the state/province code and the name of the state/province.
4. Select the time zone for the state/province.
5. Enter the IT state code for the state/province, and any additional codes that are required for transactional information.
6. To define the state/province as the default state/province for the selected country/region, select the **Default state/province** option. The default state/province will be entered by default in state/province fields when a new county or city record is created.

Set up county information

Use the following procedure to add county information to any existing countries/regions or states/provinces.

1. On the **Address setup** page, on the **County** tab, select **New** to create a new line.
2. In the **Country/region** field, select the country/region where the county is located.
3. In the **State or province** field, select the state/province where the county is located.
4. Enter the name of the county and a brief description.

Set up city information

Use the following procedure to add city information to any existing countries/regions or states/provinces.

1. On the **Address setup** page, on the **City** tab, select **New** to create a new line.
2. In the **Country/region** field, select the country/region that you're adding the city information to.
3. Enter the name of the city and a brief description.
4. Select the state/province or county where the city is located.

Set up district information

1. On the **Address setup** page, on the **District** tab, use the filters to select the country/region, state/province, county, and city, as appropriate, where the district is located.
2. Select **New** to create a new line.
3. Enter the name of the district and a brief description.

Set up ZIP/postal code information

1. On the **Address setup** page, on the **ZIP/postal codes** tab, use the filter to select the country/region for the ZIP/postal code.
2. Select **New** to create a new line.
3. Enter the ZIP/postal code, and select the corresponding city.
4. Enter the street name that the ZIP/postal code applies to.
5. Enter the lowest and highest street numbers that use the ZIP/postal code, and enter the number types. For example, the ZIP/postal code might apply only to odd-numbered street addresses.
6. Select the state/province, county, district, and time zone for the ZIP/postal code.

Last updated on 03/17/2026

Address books FAQ

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

How do I check for duplicate records?

You can check for duplicate records directly from the **Global address book** list page. On the Action Pane, on the **Party** tab, in the **Maintain** group, select **Check for duplicates**. Then select the values to include in the check for duplicates.

Can I bulk add or delete party records from an address book?

Yes, you can add multiple party records to an address book and also remove multiple party records.

- To add multiple party records to an address book, on the **Global address book** list page, select the parties in the list. Then, on the Action Pane, on the **Party** tab, in the **Maintain** group, select **Assign parties**. Select the address books to add the selected party records to, and then select **OK**. All the selected party records are added to the address books that you selected.
- To remove multiple party records from an address book, on the **Global address book** list page, select the parties in the list. Then, on the Action Pane, on the **Party** tab, in the **Maintain** group, select **Remove parties**. Select the address books to remove the parties from, and then select **OK**. All the selected party records are removed from the address books that you selected.

Can I delete historical addresses?

As of Microsoft Dynamics 365 Finance version 10.0.32, an administrator can use the following procedure to enable the advanced address maintenance feature so that historical addresses can be deleted.

1. Go to **Feature Management** > **Check for updates**.
2. Search for **Advanced address maintenance**. Advanced address maintenance is disabled by default.
3. Select **Enable**.

After the **Advanced address maintenance** feature is enabled by an administrator, follow these steps to delete historical addresses.

1. Open the **Global address book** list page.
2. Select the party record to remove historical addresses for.
3. On the **Addresses** tab, select **More options** > **Advanced**.
4. On the **History** tab, select one or more addresses.
5. Select **Delete** to delete the selected addresses.

If an address that you're trying to delete is linked to another record, you receive an error message.

Alternatively, you can set up a SysOperation framework batch job to delete multiple historical addresses.

1. Go to **Organization administration** > **Periodic** > **Delete deactivated addresses**.
2. Select **OK**.

You're notified whether the batch job succeeded or failed.

You can also delete inactive postal addresses through the Microsoft Office add-in.

1. Open the **Global address book** list page.
2. Select the party record to remove historical addresses for.
3. On the **Addresses** tab, select **More options** > **Advanced**.
4. Select the Microsoft Office logo in the upper-right corner.
5. Select **Inactive Logistics Postal Address**.
6. Sign in to the add-in by using your credentials. You can use the add-in to delete only inactive addresses.

> ⓘ **Note**
>
> You can't use this procedure to update records or insert new records. For example, if you try to update the **ValidFrom** date, the record will be deleted instead.

Can I change the party type of a record, or do I have to delete the old record and create a new

one?

Occasionally, you might have to change the party type of a record from person to organization or from organization to person. For example, Nancy is a member of the sales team for Fabrikam U.K. At a trade show in London, Nancy meets six new prospects, and creates a prospect party record for each prospect. When Nancy saves the records, each record is also created in the global address book. Fabrikam has set the default party type to organization, but two of the new prospects should have a record type of person. Therefore, on returning from the trade show, Nancy must change the party type of the two prospect records. To change a party record from one party type to another, you must first create a new party record of the correct type in the global address book. You then associate the old party record with this new record. After you have made the new party association, delete the original party record that has the incorrect record type.

How do I change the name or address of a party record?

You can update the name of a party record, and the addresses that are associated with that record, at any time.

- To update the name of a party record, open the party record, and then, on the Action Pane, select **Edit**. On the **General** FastTab, enter the new name for the party, and then save the record.
- To update an address for a party record, open the party record, and then, on the **Addresses** FastTab, select the address to update. Select **Edit**, and then, on the **Edit address** page, make the required changes to the address or address parameters.

Can I merge two or more party records into one record?

Occasionally, you might want to merge two or more party records into a single record. This can occur if you create one or more duplicate party records, either on purpose or unintentionally. When you merge party records, you select one record to keep. The information from the other records is then merged into this record. For example, you discover that information about Fabrikam is stored in three party records: A, B, and C. You decide to keep party record A. Therefore, the information that is stored in party records B and C will be merged into party record A. There are some situations where you can't merge party records:

- You can't merge party records that are associated with the same party role, such as customer or vendor, in the same legal entity. For example, party A is associated with a

customer in legal entity 123, and party B is associated with a different customer in legal entity 123. These party records can't be merged, because if they were merged, the merged party record would be associated with multiple customers in the same legal entity, and this isn't allowed. However, the records can be merged if party B is associated with a vendor in legal entity 123 or with a customer in a different legal entity.
- You can't merge internal party organization records in the same legal entity, team, or operating unit.

Should I create a party record in the global address book or in another place, such as the Customer or Vendor page?

You can enter party records either in the global address book or on the appropriate entity page. When you add a record in one location, the same record is always added in the other location. For example, if you add a party record for a customer in the global address book, the record is also added on the **Customer** page. Likewise, if you add a party record for a customer on the **Customer** page, the record is also added in the global address book. Use the following guidelines to decide where you should enter new party records:

- **Creating a party record when you don't know the entity type** – If you must create a party record but don't know the entity type (for example, you don't know whether the entity is a customer or an opportunity), create the record in the global address book. You can select the entity type later.
- **Creating a party record when you know the entity type** – If you know the entity type for the party, you can create a record on the applicable page for that type. For example, create a record for a customer on the **Customer** page. When you create and save a record by using the appropriate entity page, the record is automatically created in the global address book.

Can I translate address information for party records?

You can set up translations of address information, so that the information appears in your user language (system language) in your program, but in another language on documents such as sales orders. You can enter translations for country/region names, address purposes, and name sequences. For example, your system language is Danish, and you create a sales order for a customer in France. In this case, you can view the customer record in Danish in the program but display the address information in French on the printed sales order. When you set up translations, you should enter a translation for every item in the list. Any item that you don't

enter a translation for will appear in the system language. For example, your system language is Danish, and you send a document to a customer in Spain. If you haven't entered Spanish (ESP) translations for the address information, that information will appear in Danish both in the program and on the printed document.

After I import addresses, why can't I edit the records?

When you import addresses, there is a field that is named **IsLocationOwner**. This field indicates whether the party that is associated with the location (address) is the owner of the address. If the party is the owner of the address, the address can be edited when the party is used in the global address book or from the master record page (such as the customer, vendor, or worker). If the party isn't the owner of the address, the record can't be edited.

When you import addresses, the **IsLocationOwner** field should be set to **Yes** if you want the address to be editable by using the associated party. If this field is incorrectly imported, the location owner can be updated in the global address book.

For more information about how to change the location owner of an imported address, see Managing location owners.

If the party type is Legal entity or Team, will the global address book synchronize the customer record?

No, the global address book won't synchronize the customer. Because Customer's V3 mapping supports only the **Person** party type for contacts and **Organization** party type for accounts, the record doesn't reach Dataverse, and no errors are displayed. A separate party record of the **Person** or **Organization** types must be created.

> **Note**
>
> Data can't be deleted using excel due to dependency on the platform. Users can remove addresses using the **Manage addresses** page.

Last updated on 03/09/2026

Configure address books

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use this procedure and the decisions you made when planning the configuration of the global address book and additional address books to set up additional address books for your organization.

The demo data company used to create this task is USMF.

Configure address books

1. In the **Navigation pane**, go to **Modules** > **Organization administration** > **Global address book** > **Address books**.
2. Select **New**.
3. Enter a value in the **Name** field.
4. Enter a value in the **Description** field.
5. Select **Save**.
6. In the list, find and select the record you want.
7. Select the arrow to add the selected available teams to the address book.
8. Select **Save**.

Last updated on 03/11/2026

Workflow system overview

Summarize this article for me

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>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the workflow system.

What is workflow?

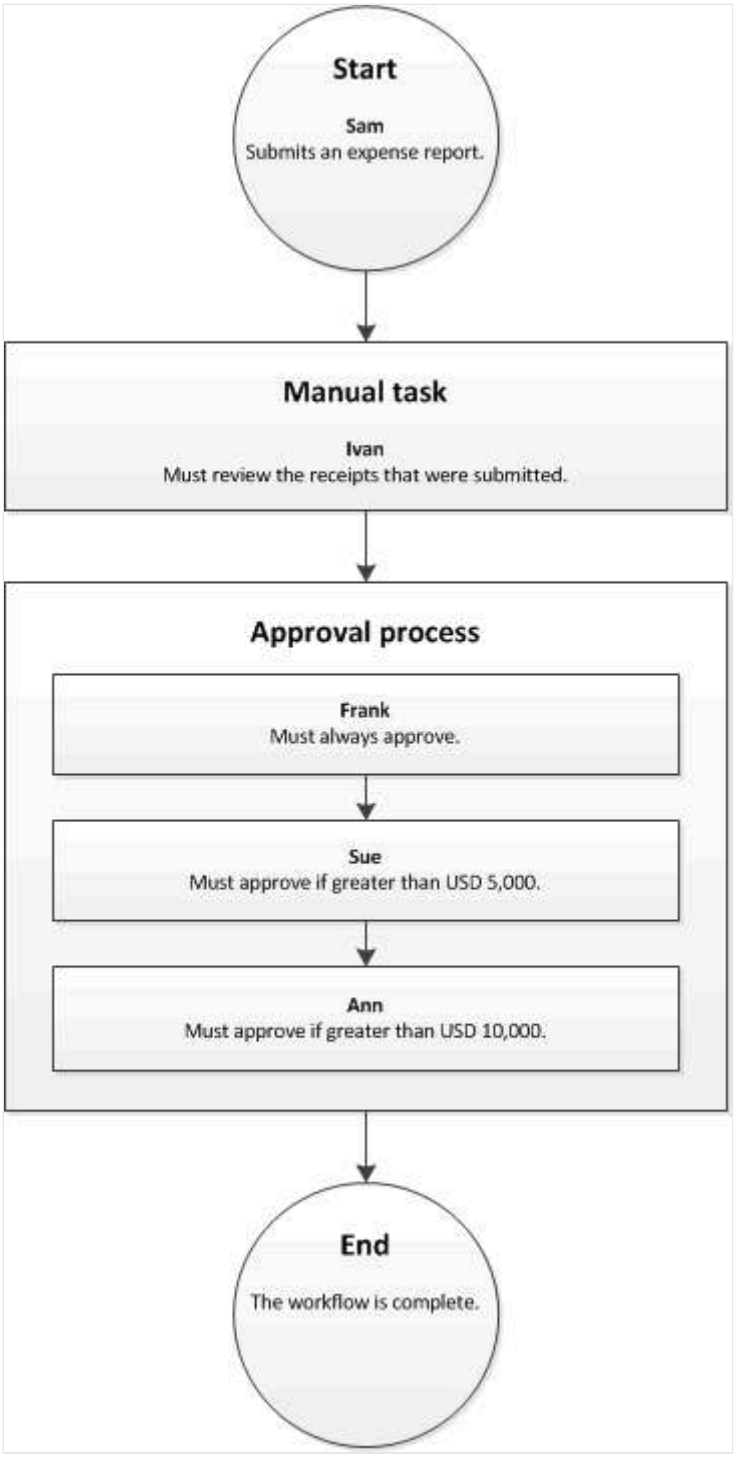
You can define the term *workflow* in two ways: as a system and as a business process.

Workflow is a system

Workflow is a system that runs on the Application Object Server (AOS). The workflow system provides functionality that you can use to create individual workflows, or business processes.

Workflow is a business process

A workflow represents a business process. It defines how a document flows, or moves, through the system by showing who must complete a task, make a decision, or approve a document. For example, the following illustration shows a workflow for expense reports.



To better understand this workflow, suppose that Sam submits an expense report for USD 7,000. In this scenario, Ivan must review the receipts that Sam submitted. Then Frank and Sue must approve the expense report. Now suppose that Sam submits an expense report for USD 11,000. In this scenario, Ivan must review the receipts, and Frank, Sue, and Ann must approve the expense report.

Benefits of using the workflow system

Use the workflow system in your organization to get these benefits:

- **Consistent processes** – Define how to process specific documents, such as purchase requisitions and expense reports. By using the workflow system, you ensure that documents are processed and approved in a consistent and efficient manner.
- **Process visibility** – Track the status, history, and performance metrics of workflow instances. This information helps you determine whether you need to make changes to the workflow to improve efficiency.
- **Centralized work list** – View a centralized work list that displays the workflow tasks and approvals assigned to you.

Workflow content

- Workflow system architecture
- Workflow elements
- Actions in workflow approval processes
- Create workflows overview
- Configure workflow properties
- Configure manual tasks in a workflow
- Configure automated tasks in a workflow
- Configure approval processes in a workflow
- Configure approval steps in a workflow
- Configure manual decisions in a workflow
- Configure conditional decisions in a workflow
- Configure parallel activities in a workflow
- Configure parallel branches in a workflow
- Configure line-item workflows
- Workflow FAQ

Last updated on 03/11/2026

Workflow system architecture

This article describes the architecture of the workflow system.

The workflow infrastructure consists of two components that AOS hosts: the X++ workflow runtime and the managed workflow runtime.

The X++ workflow runtime consists of the following components:

- Workflow runtime application programming interface (API)
- A messaging batch job
- A message queue

Either the messaging batch job or the workflow runtime API can invoke the application code, if it's required. The X++ workflow runtime compiles into the Common Intermediate Language (CIL) of the Microsoft .NET Framework.

The managed workflow runtime consists of the Windows Workflow Foundation and finance and operations apps extensions.

Logically, the workflow infrastructure is an extension and is transparent to users. Physically, AOS hosts both the X++ workflow and the managed workflow runtimes. The workflow infrastructure uses batch processing on AOS and .NET Interop to integrate both subsystems, and to pass messages from one subsystem to the other. The X++ code that runs in the batch processor compiles to .NET CIL. The batch processing runs in the .NET common language runtime (CLR).

Users can use workflow pages and controls to participate in business processes.

Developers can create workflows for the objects that they add. The following table describes the workflow steps that occur when a user submits an expense report to the workflow system for approval.

Expand table

Step	Runtime	Activity
1	X++ workflow runtime	A user submits an expense report by clicking the **Submit** button on one of the workflow controls. This action causes X++ code to activate a workflow instance by calling the workflow runtime API. The workflow runtime API posts a message to the message queue. The messaging batch job reads the message and sends a workflow activation request to the managed workflow runtime. > [!NOTE]

The messaging batch job processes the message queue at one-minute intervals. | | 2 | Managed workflow runtime | .NET Interop from X++ receives the message and starts a new workflow instance via Windows Workflow Foundation. This workflow instance performs a callback to the X++ workflow runtime API via .NET Interop to X++ CIL and posts a message that the workflow started.

After the message is posted, the managed workflow runtime saves the idle workflow instance to the database. The runtime then removes the workflow instance from memory. When the managed workflow runtime receives another message from the X++ workflow runtime for this workflow instance, it restores the workflow instance to memory and resumes it.

Each workflow instance is unique. If two users submit their expense reports for approval, two workflow instances start.
| | 3 | X++ workflow runtime | The messaging batch job reads the **workflow started** message from the message queue and invokes the application event handler to process a **workflow started** event. The batch job then posts an acknowledgment message that the event was processed. | | 4 | Both | This same messaging pattern repeats, as required, throughout the lifecycle of the workflow instance. |

The workflow architecture helps provide a reliable and durable messaging system, and also helps guarantee that the state of the workflow is always synchronized with the state of the application. If an unexpected hardware or software failure occurs, the workflow instance state returns to its last known saved point, and the message stays in the queue. Therefore, from an architecture perspective, the recovery model is to fix the problem and resume the workflow.

Last updated on 06/26/2026

Workflow elements

This article describes the various elements that make up a workflow.

A workflow consists of elements. The sections that follow describe each type of element.

Tasks

A *task* is a unit of work that must be performed. You can add two types of tasks to a workflow: manual tasks and automated tasks.

Manual task

A *manual task* is a unit of work that a user must perform. For example, an expense report workflow can have manual tasks that require the assigned users to complete the following actions:

- Review the receipts that are submitted together with an expense report.
- Call an employee's manager.

Automated task

An *automated task* is a unit of work that the system performs. No human interaction is required. For example, a sales order workflow can have automated tasks that require the system to complete the following actions:

- Perform a credit check.
- Create a customer record for the customer, if a record doesn't already exist.

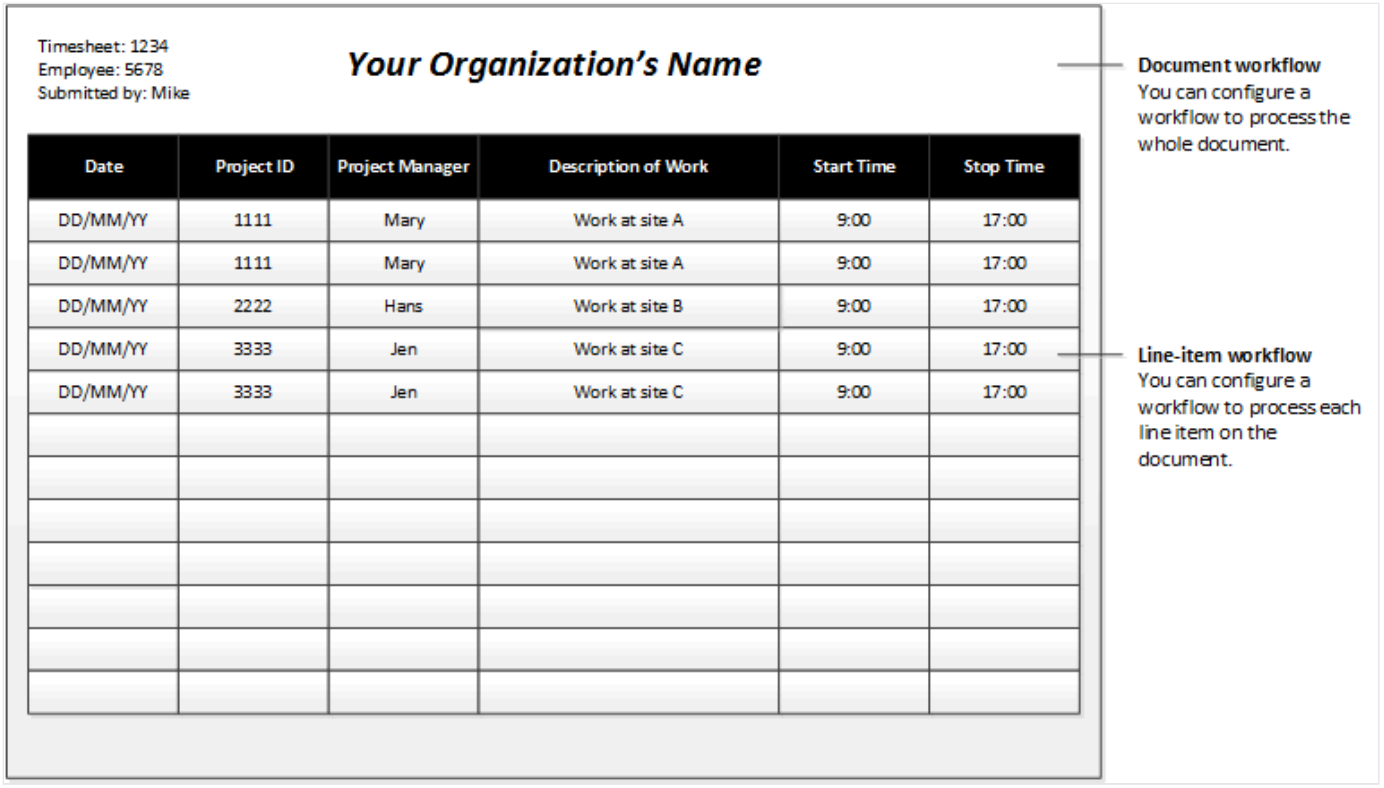
Approval processes

An *approval process* is a process that consists of separate steps. At each approval step, the user can perform the following actions:

- Approve the document.
- Reject the document.
- Request a change to the document.
- Assign the document to another user for approval.

Line-item workflow elements

You can create a workflow to process either documents or the line items on a document. For example, you create an approval workflow for timesheets. (Refer to this workflow as the *document workflow*.) You can add a *line-item workflow* element to that document workflow. When the line-item element runs, each line item on the document is submitted for processing. You might want all the line items to be processed by the same line-item workflow, or you might want each line item to be processed by a different line-item workflow. Imagine that an employee submits a timesheet that resembles the following figure.



In this scenario, you might want to create the following line-item workflows:

- **Line-item workflow 1** – Use this workflow to process line items where the project ID is 1111.
- **Line-item workflow 2** – Use this workflow to process line items where the project ID is 2222.
- **Line-item workflow 3** – Use this workflow to process line items where the project ID is 3333.

Flow-control elements

By using the following elements, you can design workflows that have alternate branches or branches that run at the same time.

Manual decision

A *manual decision* is a point where a workflow divides into two branches. A user must make a decision, and this decision determines which branch to use to process the submitted document.

Conditional decision

A *conditional decision* is a point where a workflow divides into two branches. The system decides which branch to use for processing the submitted document. To make this decision, the system evaluates the document to determine whether it meets specified conditions.

Parallel activity

A *parallel activity* is a workflow element that includes two or more workflow branches that run simultaneously.

Subworkflow

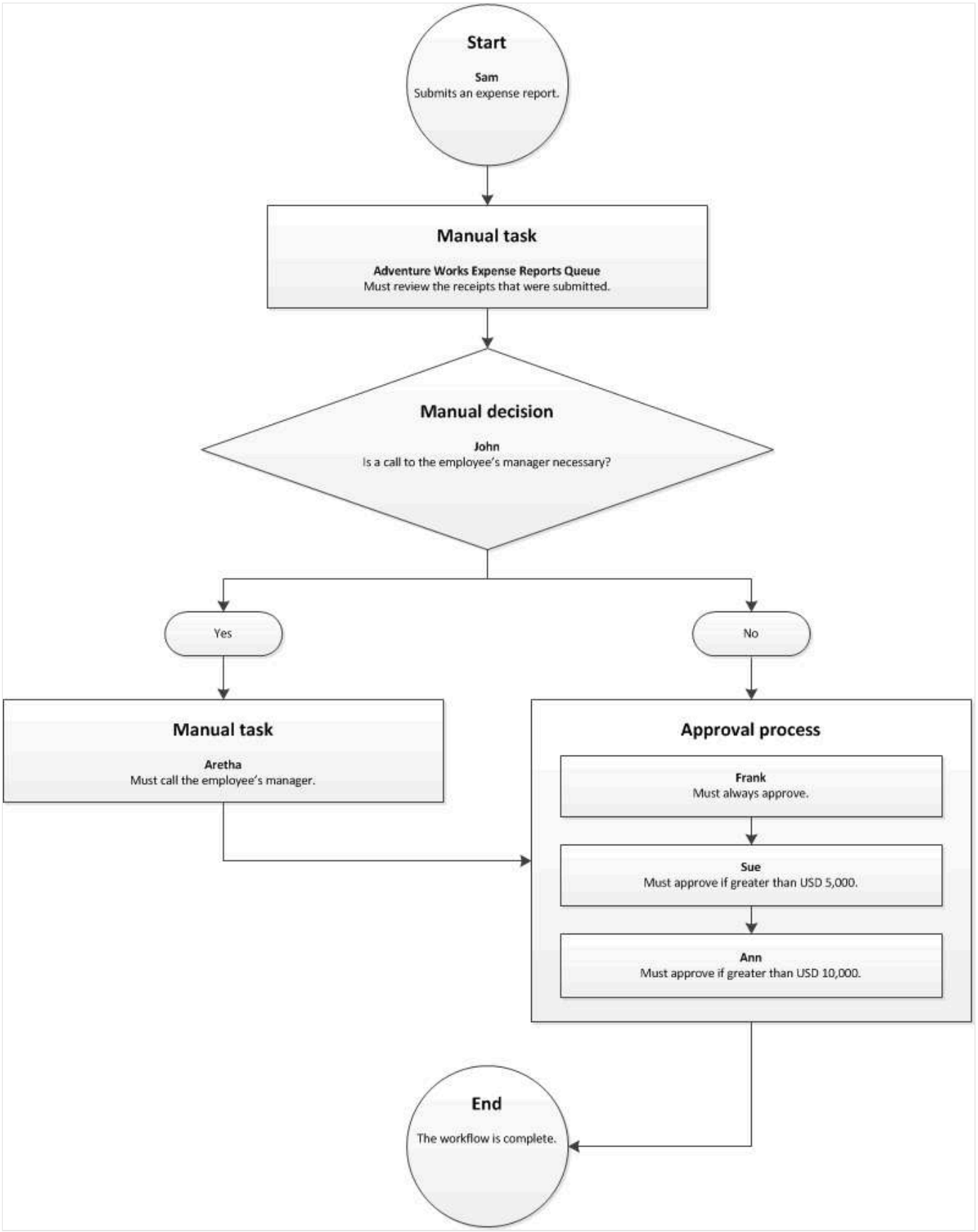
A *subworkflow* is a workflow that runs in the context of another workflow.

Last updated on 06/26/2026

Actions in workflow approval processes

This article explains the actions that each participant in a workflow approval process can take.

A workflow can involve several groups of people: the originator, task assignees, decision makers, and approvers. For example, in the following expense report workflow, Sam is the originator, the members of the queue are task assignees, John is a decision maker, and Frank, Sue, and Ann are approvers.



The following sections explain the workflow actions that each group can perform.

Actions that an originator can perform

The originator starts a workflow instance by submitting a document for processing. For example, Sam must select the **Submit** button on the **Expense report** page to submit the expense report.

Actions that a task assignee can perform

You can assign a task to multiple people or to a work item queue that several people monitor. However, only one person can complete a task. For example, Sam submits an expense report and routes the receipts to the organization's Expense Reports department for review.

The members of the Adventure Works Expense Reports department monitor the queue. Julie, a member of that department, accepts the task of reviewing Sam's expense report and receipts. Julie can now perform one of the following actions: complete, reject, delegate, request change, reassign, or release.

> **Note**
>
> The available actions vary, depending on how the software developer designed the task.

Complete

When a user completes a task, the system assigns the document submitted for processing to the next user in the workflow, if there's a next user. If no additional processing is required, the workflow process ends.

For example, Julie, a member of the Adventure Works Expense Reports department, accepts the task of reviewing Sam's expense report and receipts. After Julie completes the review, the system assigns the document to John.

Reject

When a user rejects a document, the workflow process ends.

For example, Julie, a member of the Adventure Works Expense Reports department, accepts the task of reviewing Sam's expense report and receipts. If Julie rejects the expense report, the workflow process ends.

Sam can then resubmit the expense report. Sam can make changes first, or resubmit the original version. If Sam resubmits the expense report, the workflow process starts at the manual review task.

Delegate

When a user delegates a task, the system assigns the task to another user.

For example, Julie, a member of the Adventure Works Expense Reports department, accepts the task of reviewing Sam's expense report and receipts. Julie delegates this task to Tim, who is Julie's assistant.

Tim acts on behalf of Julie. Therefore, when Tim completes the review, the system assigns the expense report to John, just as if Julie had completed the task.

Request change

When a user requests a change to a document that was submitted, the system sends the document back to the originator.

For example, Julie, a member of the Adventure Works Expense Reports department, accepts the task of reviewing Sam's expense report and receipts. Julie notices some errors on the expense report and requests changes. The system sends the expense report back to Sam.

Sam can resubmit the expense report. Sam can make the requested changes first, or resubmit the original version. If Sam resubmits the expense report, a member of the work item queue must review the expense report and the receipts again.

Reassign

Members of a work item queue can reassign documents in that queue to another queue.

For example, Julie is a member of the Adventure Works Expense Reports department and monitors the queue. To help balance the workload, Julie can reassign the expense report and the included receipts to another queue.

Release

Occasionally, a member of a work item queue accepts a task but then decides they can't complete it. In this case, the person who accepted the task can release the document back to the work item queue.

For example, Julie is a member of the Adventure Works Expense Reports department and accepts the task of reviewing Sam's expense report and receipts. If Julie can't complete the task, she can

release the document. The expense report returns to the queue so that other members of the Adventure Works Expense Reports department can complete the task.

Actions that a decision maker can perform

Typically, you assign a document to a decision maker because there's a question that the decision maker must answer. The answer to the question is typically **Yes** or **No**, or **True** or **False**. If the decision maker doesn't select one of those choices, they can delegate the decision.

[Choice 1] or [Choice 2]

A decision maker must answer a question that's related to the document. The answer to the question is typically **Yes** or **No**, or **True** or **False**. The answer that the decision maker selects determines the workflow branch that's used to process the document.

For example, you assign Sam's expense report to John. John must decide whether the information in the document requires a call to Sam's manager. If John decides that a call is required, the expense report is assigned to Aretha, who must then call Sam's manager. If John decides that a call isn't required, the expense report is assigned to Frank for approval.

Delegate

When a decision maker delegates a decision, they assign the document to another user who must make the decision.

For example, you assign Sam's expense report to John. John delegates the decision to Maria, who is John's assistant.

Maria acts on behalf of John. If Maria decides that a call to Sam's manager is required, she assigns the expense report to Aretha, who must then call Sam's manager. If Maria decides that a call isn't required, she assigns the expense report to Frank for approval.

Actions that an approver can perform

When you assign a document to an approver, the approver can perform one of the following actions: approve, reject, delegate, or request change.

Approve

When an approver approves a document, the document is assigned to the next user in the workflow, if there's a next user. If no additional processing is required, the workflow process ends.

For example, Sam submits an expense report for USD 6,000, and you assign this document to Frank. When Frank approves the document, it's assigned to Sue for approval. When Sue approves the expense report, the workflow process ends.

Reject

When an approver rejects a document, the workflow process ends.

For example, Sam submits an expense report for USD 12,000, and you assign this document to Sue. If Sue rejects the expense report, the workflow process ends.

Sam can resubmit the expense report. Sam can make changes first, or resubmit the original version of the expense report. If Sam resubmits the expense report, the workflow process starts at the approval process.

Delegate

When an approver delegates a document, the document is assigned to another user for approval.

For example, Sam submits an expense report for USD 12,000, and you assign this document to Frank. Frank delegates the expense report to Ann.

Ann then acts on behalf of Frank. Therefore, when Ann approves the document, it's assigned to Sue for approval, just as if Frank had approved it. After Sue approves the document, it's sent to Ann for approval.

Request change

When an approver requests a change to a document, the system sends the document back to the originator.

For example, Sam submits an expense report for USD 12,000, and the system assigns the document to Sue. If Sue requests a change, the system sends the expense report back to Sam.

Sam can resubmit the expense report. Sam can make the requested changes first, or resubmit the original version of the expense report. If Sam resubmits the expense report, the system sends it to Frank for approval, because Frank is the first approver in the approval process.

Last updated on 06/26/2026

Create workflows overview

Summarize this article for me

> **Note**
>
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This article explains how to create a workflow.

Open the workflow editor

The module that you're working in determines the types of workflows that you can create. To select the type of workflow to create a new workflow by using the workflow editor, follow these steps:

1. Open the module that you want to create a new workflow for. For example, to create a workflow for purchase requisitions, select **Procurement and sourcing**.
2. Select **Setup** > **[Module name] workflows**.
3. On the list page that appears, on the action pane, select **New**.
4. On the **Create workflow** page, select the type of workflow to create, and then select **Create workflow**. The workflow editor launches.
5. Select or sign in to the account to use with the workflow editor. If prompted to **Stay signed in to all your apps**, select **OK** to allow the account to automatically sign in to the workflow editor and other apps in the future, or select **No, sign in to this app only** if you prefer to only sign in to the workflow editor for this session.

Drag workflow elements onto the canvas

The **Workflow elements** area of the workflow editor contains the elements that you can add to your workflow. To add elements to the workflow, drag them onto the canvas.

Connect the elements

To connect one workflow element to another, hold the pointer over an element until connection points appear. Select a connection point, and then drag it to another element. Be sure to connect all the elements.

Configure the properties of the workflow

Follow these steps to configure the properties of the workflow.

1. Select the canvas to ensure that no workflow element is selected.
2. Select **Properties** to open the **Properties** page for the workflow.
3. Follow the procedures in the Configure workflow properties article.

Configure the elements of the workflow

Configure each element that you dragged onto the canvas. For information about how to configure each workflow element, see the following topics:

- Configure manual tasks in a workflow
- Configure automated tasks in a workflow
- Configure approval processes in a workflow
- Configure approval steps in a workflow
- Configure manual decisions in a workflow
- Configure conditional decisions in a workflow
- Configure parallel branches in a workflow
- Configure a parallel branch
- Configure line-item workflows.

Resolve any errors or warnings

The **Errors and warnings** pane at the bottom of the workflow editor shows messages that the workflow generates. To find the element where an error or warning occurred, double-click the error or warning message. You must resolve all errors and warnings before you can make the workflow active.

Save and activate the workflow

When you're ready to save and activate the workflow, follow these steps:

1. Select **Save and close** to close the workflow editor and open the **Save workflow** page.
2. Enter comments about the changes that you made to the workflow, and then select **OK**.
3. If you resolved all errors and warnings, the **Activate workflow** page appears. Select one of the following options:

- To activate this version of the workflow, select **Activate the new version**. When a workflow is active, users can submit documents to it for processing.
- If you don't want to activate this version, select **Do not activate the new version**. You can activate the workflow later.

Last updated on 03/10/2026

Configure workflow properties

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure the various properties of a workflow.

To configure the properties of a workflow, open the workflow in the workflow editor. Select the canvas of the workflow editor, and then select **Properties** to open the **Properties** page. Use the following procedures to configure the various properties of the workflow.

Name the workflow

Follow these steps to enter a name for the workflow.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the workflow. For example, if you create a purchase requisition workflow for each country or region that you operate in, you can name the purchase requisition workflow **Purchase Requisitions Denmark** or **Purchase Requisitions Spain**.

Specify the workflow owner

The workflow owner is the person who manages and maintains the workflow. Follow these steps to specify the workflow owner.

1. In the left pane, select **Basic Settings**.
2. In the **Owner** list, select the name of the person who will manage the workflow.

Select an email template

Follow these steps to select the email template that is used to generate notification messages about the workflow.

1. In the left pane, select **Basic Settings**.

2. In the **Email template for workflow notifications** list, select the template.

Enter instructions for users

You can provide instructions to users who submit documents for processing and approval. These users are also referred to as *originators*. For example, you create a purchase requisition workflow, and you enter instructions. Users who enter purchase requisitions on the **Purchase requisitions** page can view those instructions. To view instructions, the originator selects the icon in the workflow message bar. Follow these steps to enter instructions for users.

1. In the left pane, select **Basic Settings**.
2. In the **Submission instructions** field, enter the instructions.
3. To personalize the instructions, insert placeholders. Placeholders are replaced with the appropriate data when the instructions are shown to users. To insert a placeholder, follow these steps:
   a. Select the **Submission instructions** field to specify where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.
4. To add translations of the instructions, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the text.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders. For instructions about how to enter a placeholder, see step 3.
   f. Select **Close**.

> ⓘ **Note**
>
> You can't add placeholders by using copy and paste because the target information isn't pasted in correctly. Use the interface to add placeholders.

Specify when to use this workflow through activation conditions

You can create multiple workflows that use the same workflow type. When you have multiple workflows that use the same type, you must specify when to use each workflow through activation conditions. If activation conditions aren't met, the default workflow is used. Similarly, if you define only one workflow configuration for a workflow type, the workflow configuration is used regardless of the activation conditions.

For example, you can create a purchase requisition workflow for each country or region that you operate in, such as Purchase Requisitions Denmark and Purchase Requisitions Spain, with the following conditions:

- Use Purchase Requisitions Denmark when: country/region = DK
- Use Purchase Requisitions Spain when: country/region = ES

Follow these steps to specify when to use the workflow that you're configuring.

1. In the left pane, select **Activation**.
2. Select the **Set the conditions for running this workflow** check box.
3. Select **Add condition**.
4. Enter a condition.
5. Enter any additional conditions that you need.
6. Run through the workflow with some target records to verify that the condition correctly includes and excludes records.

Specify when to send notifications

When you submit a document for processing, you create a workflow instance. You can send notifications to users when workflow instances that are based on the workflow start, complete, terminate, or stop because of an error. Follow these steps to specify when to send notifications.

1. In the left pane, select **Notifications**.

2. Select the check box for each event that should trigger notifications:

    - **Started** – Send notifications when a workflow instance starts.
    - **Stopped** – Send notifications when a workflow instance stops because of an error.
    - **Completed** – Send notifications when a workflow instance completes.
    - **Unrecoverable** – Send notifications when a workflow instance stops because of an unrecoverable error.
    - **Terminated** – Send notifications when a workflow instance terminates.

3. Select the row for an event that you selected in step 2.

4. On the **Notification text** tab, enter the text of the notification.

5. To personalize the text, insert placeholders. Placeholders are replaced with the appropriate data when the text is shown to users. To insert a placeholder, follow these steps:
   a. Select the field to specify where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.
   e. A common **Notification text** placeholder to include is "Last Notes: %Workflow.Last note%", which displays any comments from the previous step.

6. To add translations of the text, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the text.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders. For instructions about how to enter a placeholder, see step 5.
   f. Select **Close**.

7. On the **Recipient** tab, use the following options to specify who should receive the notifications.

Expand table

Option	Notifications are sent to these users	To send notifications, follow these steps
Participant	Users who are assigned to a specific group or role	1. On the **Recipient** tab, select **Participant**. 1. On the **Role based** tab, in the **Type of participant** list, select the type of group or role to send notifications to. 1. In the **Participant** list, select the group or role to send notifications to.
Workflow user	Users who are participants in this workflow	1. On the **Recipient** tab, select **Workflow user**. 1. On the **Workflow user** tab, in the **Workflow user** list, select a participant in this workflow.
User	Specific users	1. On the **Recipient** tab, select **User**. 1. On the **User** tab, the **Available users** list includes all users. Select the users to send notifications to, and move those users into the **Selected users** list.

1. Repeat steps 3 through 7 for each event that you selected in step 2.

Enter comments about the changes that you made to the workflow

To enter comments about the changes that you made to the workflow, follow these steps:

1. In the left pane, select **Notes**.
2. In the **Enter comments about the workflow** field, enter your comments.
3. Review your comments. After you add comments, you can't modify them.
4. Select **Add** to add your comments to the **Comment history** area.

Last updated on 03/11/2026

Configure manual tasks in a workflow

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure the properties for a manual task.

To configure a manual task in the workflow editor, right-click the task, and then select **Properties** to open the **Properties** page. Then use the following procedures to configure the properties for the manual task.

Name the task

Follow these steps to enter a name for the manual task.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the task.

Enter a subject line and instructions

You must provide a subject line and instructions for users assigned to the task. For example, if you configure a task for purchase requisitions, the user assigned to the task sees the subject line and instructions on the **Purchase requisitions** page. The subject line appears in a message bar on the page. The user can then select the icon in the message bar to view the instructions. Follow these steps to enter a subject line and instructions.

1. In the left pane, select **Basic Settings**.
2. In the **Work item subject** field, enter the subject line.
3. To personalize the subject line, insert placeholders. Placeholders are replaced with appropriate data when the subject line is shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.

d. Select **Insert**.

4. To add translations of the subject line, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the translated text.
   e. To personalize the text, insert placeholders as described in step 3.
   f. Select **Close**.

5. In the **Work item instructions** field, enter the instructions.

6. To personalize the instructions, insert placeholders. Placeholders are replaced with appropriate data when the instructions are shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.

7. To add translations of the instructions, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the translated text.
   e. To personalize the text, insert placeholders as described in step 6.
   f. Select **Close**.

Assign the task

Follow these steps to specify who should receive the manual task.

1. In the left pane, click **Assignment**.

2. On the **Assignment type** tab, select one of the options in the following table. Follow the additional steps for that option before you go to step 3.

Expand table

Option	Users that the task is assigned to	Additional steps
Participant	Users who are assigned to a specific group or role	1. After you select **Participant**, on the **Role based** tab, in the **Type of participant** list, select the type of group or role to assign the task to. 1. In the **Participant** list, select the group or role to assign the task to.
Hierarchy	Users in a specific organizational hierarchy	1. After you select **Hierarchy**, on the **Hierarchy selection** tab, in the **Hierarchy type** list, select the type of hierarchy to assign the task to. 1. The system retrieves a range of user names from the hierarchy. These names represent users that the task can be assigned to. Follow these steps to specify the starting point and ending point of the range of user names that the system retrieves: 1. To specify the starting point, select a person in the **Start from** list. 1. To specify the ending point, select **Add condition** and then enter a condition that determines where in the hierarchy the system stops retrieving names. 1. On the **Hierarchy options** tab, specify which users in the range the task should be assigned to: **Assign to all users retrieved** – The task is assigned to all users in the range. **Assign only to last user retrieved** – The task is assigned to only the last user in the range. **Exclude users with the following condition** – The task isn't assigned to users in the range who meet a specific condition. Select **Add condition** to specify the condition.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to assign the task to, and then move those users to the **Selected users** list.
Queue	A work item queue	1. After you select **Queue**, select the **Queue based** tab. 1. To assign the task to a specific queue, follow these steps: 1. In the **Queue type** list, select **Work item queues**. 1. In the **Queue name** list, select the queue. 1. If a specific condition should determine which queue the task is assigned to, follow these steps: 1. In the **Queue type** list, select **Conditional work item queues**. 1. In the **Queue name** list, select **Conditional queue**. **NOTE:** This option is used for only a few workflows, such as Case management.

3. On the **Time limit** tab, in the **Duration** field, specify how much time the user has to complete the task. Select one of the following options:

    - **Hours** – Enter the number of hours that the user has to complete the task. Then select the calendar that your organization uses, and enter information about your organization's work week.

- **Days** – Enter the number of days that the user has to complete the task. Then select the calendar that your organization uses, and enter information about your organization's work week.
- **Weeks** – Enter the number of weeks that the user has to complete the task.
- **Months** – Select the day and week that the user must complete the task by. For example, you might want the user to complete the task by Friday of the third week of the month.
- **Years** – Select the day, week, and month that the user must complete the task by. For example, you might want the user to complete the task by Friday of the third week of December.

If the user doesn't complete the task in the allotted time, the task is overdue. You can escalate an overdue task based on the options that you select in the **Escalation** area of the page.

Specify what happens when the task is overdue

If a user doesn't complete the manual task in the allotted time, the task is overdue. You can escalate an overdue task or automatically assign it to another user. Follow these steps to escalate the task if it's overdue.

1. In the left pane, select **Escalation**.
2. Select the **Use escalation path** check box to create an escalation path. The system automatically assigns the task to the users who are listed in the escalation path. For example, the following table represents an escalation path.

Expand table

Sequence	Escalation path
1	Assign to: Donna
2	Assign to: Erin
3	Final action: Reject

In this example, the system assigns the overdue task to Donna. If Donna doesn't complete the task in the allotted time, the system assigns the task to Erin. If Erin doesn't complete the task in the allotted time, the system rejects the document that was submitted for processing.

3. To add a user to the escalation path, select **Add escalation**. On the **Assignment type** tab, select one of the options in the following table, and then follow the additional steps for

that option before you go to step 4.

Expand table

Option	Users that the task is escalated to	Additional steps
Hierarchy	Users in a specific organizational hierarchy	1. After you select **Hierarchy**, on the **Hierarchy selection** tab, in the **Hierarchy type** list, select the type of hierarchy to escalate the task to. 1. The system must retrieve a range of user names from the hierarchy. These names represent users that the task can be escalated to. Follow these steps to specify the starting point and ending point of the range of user names that the system retrieves: 1. To specify the starting point, select a person in the **Start from** list. 1. To specify the ending point, select **Add condition**. Then enter a condition that determines where in the hierarchy the system stops retrieving names. 1. On the **Hierarchy options** tab, specify which users in the range the task should be escalated to: **Assign to all users retrieved** – The task is escalated to all users in the range. **Assign only to last user retrieved** – The task is escalated to only the last user in the range. **Exclude users with the following condition** – This task isn't escalated to users in the range who meet a specific condition. Select **Add condition** to specify the condition.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to escalate the task to, and then move those users to the **Selected users** list.

4. On the **Time limit** tab, in the **Duration** field, specify how much time the user has to complete the task. Select one of the following options:

   - **Hours** – Enter the number of hours that the user has to complete the task. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Days** – Enter the number of days that the user has to complete the task. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Weeks** – Enter the number of weeks that the user has to complete the task.
   - **Months** – Select the day and week that the user must complete the task by. For example, you might want the user to complete the task by Friday of the third week of the month.

- **Years** – Select the day, week, and month that the user must complete the task by. For example, you might want the user to complete the task by Friday of the third week of December.

5. Repeat steps 3 through 4 for each user that you want to add to the escalation path. You can change the order of the users.

6. If the users in the escalation path don't complete the task in the allotted time, the system takes action on the task. To specify the action that the system takes, select the **Action** row, and then, on the **End action** tab, select an action.

Specify when the system automatically acts on the task

You can configure the system to take action on the manual task if specific conditions are met. For example, a task requires that a member of the Expense reports department review the receipts that are submitted together with an expense report. According to company policy, this task must be performed if the total amount of the expense report is more than $100. In this scenario, you can configure the system to automatically mark the task as **Complete** when the total amount is less than 100. Follow these steps to specify when the system takes action on the manual task.

1. In the left pane, select **Automatic actions**.

2. Select the **Enable automatic actions** check box.

3. Select **Add condition**.

4. Enter a condition.

5. Enter any additional conditions that are required.

6. To verify that the conditions you entered are configured correctly, follow these steps:
   a. Select **Test**.
   b. On **Test workflow condition**, in **Validate condition**, select a record.
   c. Select **Test**. The system evaluates the record to determine whether it meets the conditions that you defined.
   d. Select **OK** or **Cancel** to return to **Properties**.

7. In **Auto complete action**, select the action that the system should take on the task.

Specify when to send notifications

You can send notifications to people when you delegate, escalate, complete, or reject a manual task, or when you request a change. Follow these steps to specify when to send notifications and who to send them to.

1. In the left pane, select **Notifications**.
2. Select the check box next to the events that you want to send notifications for:
   - **Delegate** – You assigned the task to another user.
   - **Escalate** – The assigned user didn't complete the task in the allotted time.
   - **Complete** – The assigned user completed the task.
   - **Reject** – The assigned user rejected the document that was submitted.
   - **Request change** – The assigned user requested a change to the document that was submitted.
3. Select the row for an event that you selected in step 2.
4. On the **Notification text** tab, enter the text of the notification.
5. To personalize the notification, insert placeholders. Placeholders are replaced with appropriate information when the notification is shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.
6. To add translations of the notification, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the translated text.
   e. To personalize the text, insert placeholders as described in step 5.
   f. Select **Close**.
7. On the **Recipient** tab, specify who to send the notifications to. Select one of the options in the following table, and then follow the additional steps for that option before you go to step 8.

Expand table

Option	Notification recipients	Additional steps
Participant	Users who are assigned to a specific group or role	1. After you select **Participant**, on the **Role based** tab, in the **Type of participant** list, select the type of group or role to send notifications to. 1. In the **Participant** list, select the group or role to send notifications to.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to send notifications to, and then move those users to the **Selected users** list.

8. Repeat steps 3 through 7 for each event that you selected in step 2.

Set a time limit

Follow these steps if you need to complete the manual task within a specific time.

> ⓘ **Note**
>
> The options that you select in this procedure override the options that you selected in the **Assignment** and **Escalation** areas of the page.

1. In the left pane, select **Advanced settings**.
2. Select the **Set a time limit for the workflow element** check box.
3. In the **Duration** field, specify when the task must be completed. Select one of the following options:

   - **Hours** – Enter the number of hours for task completion. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Days** – Enter the number of days for task completion. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Weeks** – Enter the number of weeks for task completion.
   - **Months** – Select the day and week for task completion. For example, you might want the task to be completed by Friday of the third week of the month.

- **Years** – Select the day, week, and month for task completion. For example, you might want the task to be completed by Friday of the third week of December.

4. If the time limit is exceeded, the system takes action on the task. In the **Action** list, select the action that the system should take.

Specify which actions are available to the user

When you assign the manual task to a user, the user must take action on the task. Follow these steps to specify which actions the user can take on the task.

> **Note**
>
> The available actions vary, depending on the design of the task.

1. In the left pane, select **Advanced settings**.
2. Select the **Complete** check box if the user should be able to mark the task as **Complete**.
3. Select the **Reject** check box if the user should be able to reject the document that was submitted.
4. Select the **Request change** check box if the user should be able to request changes to the document that was submitted.
5. Select the **Delegate** check box if the user should be able to assign the task to another user.
6. Select the **Reassign** check box if the user should be able to reassign the task to another user in the work item queue.
7. Select the **Release** check box if the user should be able to reassign the task to the work item queue. Another user can then complete the task.

Last updated on 03/11/2026

Configure automated tasks in a workflow

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure the properties for an automated task.

To configure an automated task in the workflow editor, right-click the task, and then select **Properties** to open the **Properties** page. Then use the following procedures to configure the properties for the automated task.

Name the task

Follow these steps to enter a name for the automated task.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the task.

Specify when to send notifications

You can send notifications to people when an automated task runs or cancels. Follow these steps to specify when to send notifications and who receives them.

1. In the left pane, select **Notifications**.
2. Select the check box next to the events to send notifications for:
   - **Execution** – Send notifications when the task runs.
   - **Canceled** – Send notifications when the task cancels.
3. Select the row for an event that you selected in step 2.
4. On the **Notification text** tab, enter the text of the notification.
5. To personalize the notification, insert placeholders. Placeholders are replaced with appropriate data when the notification appears to users. Follow these steps to insert a placeholder:

a. In the text box, select where the placeholder should appear.
b. Select **Insert placeholder**.
c. In the list that appears, select the placeholder to insert.
d. Select **Insert**.

6. To add translations of the notification, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the text.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders as described in step 5.
   f. Select **Close**.

7. On the **Recipient** tab, specify who receives the notifications. Select one of the options in the following table, and then follow the additional steps for that option before you go to step 8.

Expand table

Option	Notification recipients	Additional steps
Participant	Users assigned to a specific group or role	1. After you select **Participant**, on the **Role based** tab, in the **Type of participant** list, select the type of group or role to send notifications to. 1. In the **Participant** list, select the group or role to send notifications to.
Workflow user	Users who participate in the current workflow	- After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to send notifications to, and then move those users to the **Selected users** list.

1. Repeat steps 3 through 7 for each event that you selected in step 2.

Last updated on 03/11/2026

Configure approval processes in a workflow

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the following procedure to configure the properties of the approval process.

To configure an approval process, in the workflow editor, right-click the approval element, and then select **Properties** to open the **Properties** form.

Name the approval process

Follow these steps to enter a name for the approval process.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the approval process.

Specify when the system automatically acts on the document

You can configure the system to automatically act on the document if specific conditions are met. For example, the system can approve expense reports that have total amounts that are less than USD 100. Follow these steps to specify when the system acts on the document:

1. In the left pane, select **Automatic actions**.
2. Select the **Enable automatic actions** check box.
3. Select **Add condition**.
4. Enter a condition.
5. Enter additional conditions, if necessary.
6. To verify that the conditions you entered are configured correctly, complete the following steps:

a. Select **Test** to open the **Test workflow condition** form.
b. Select a record in the **Validate condition** area of the form.
c. Select **Test**. The system evaluates the record to determine whether it meets the conditions that you defined.
d. Select **OK** or **Cancel** to return to the **Properties** form.

7. In the **Auto complete action** list, select the action that the system should take on the document.

Specify when to send notifications

You can send notifications to people when a document is approved, rejected, delegated, or escalated, or when a change is requested. Follow these steps to specify when to send notifications and who receives them.

1. In the left pane, select **Notifications**.

2. Select the check box next to the events to send notifications for:

   - **Delegate** – When a document is assigned to another user for approval.
   - **Escalate** – When the assigned user doesn't act on a document in the allotted time.
   - **Approve** – When a document is approved.
   - **Reject** – When a document is rejected.
   - **Request change** – When the assigned user requests a change to a submitted document.

3. Select the row for an event that you selected in step 2.

4. Select the **Notification text** tab.

5. In the text box, enter the text for the notification.

6. To personalize the text, insert placeholders, which are replaced with the appropriate data when they're displayed to users. To insert a placeholder, follow these steps:
   a. Select the text box at the location where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.

7. To add translations of the notification, select **Translations**. In the form that appears, follow these steps:
   a. Select **Add**.
   b. In the list that appears, select the language for the text.
   c. In the **Translated text** text box, enter the text.

d. To personalize the text, insert placeholders.
e. Select **Close**.

8. Select the **Recipient** tab.

9. Specify who receives the notifications. Select one of the options in the following table, and then follow the additional steps for the option before you go to step 10.

Expand table

Option	Notification recipients	Additional steps
Participant	Users who are assigned to a specific group or role	1. After you select **Participant**, select the **Role based** tab. 1. In the **Type of participant** list, select the type of group or role to send notifications to. 1. In the **Participant** list, select the group or role to send notifications to.
Workflow user	Users who participate in the current workflow	1. After you select **Workflow user**, select the **Workflow user** tab. 1. In the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. Select the users to send notifications to, and then move these users to the **Selected users** list.

1. Repeat steps 3 through 9 for each event that you selected in step 2.

Specify a final approver

To require extra approval when the user who submitted the document also approves it, specify a final approver for the approval step.

1. In the workflow editor, right-click the approval element, and then select **Properties** to open the **Properties** form.
2. In the left pane, select **Advanced settings**.
3. Select the **Use final approver** check box.
4. In the list, select a user to be the final approver.

Prevent the submitter from approving steps in the workflow

To prevent users who submit documents for approval from approving the documents themselves, follow these steps:

1. Go to **System administration** > **Workflow** > **Workflow parameters** > **General** > **Approver**.
2. Update the **Disallow approval by submitter** setting on the workflow to **Yes**.

By default, this setting is **No**, and users can approve the document if they're included in the approval step's assignment settings.

If you set the workflow to **Disallow approval by submitter** and include approval steps with a final approver, make sure the final approver isn't a user who typically submits documents to workflow, since they can't complete the approval.

Set a time limit

Follow these steps if you need to complete the approval process within a specific time.

> ⓘ **Note**
>
> The options that you select in these steps override the options that you selected in the **Assignment** and **Escalation** areas of each approval step.

1. In the left pane, select **Advanced settings**.
2. Select the **Set a time limit for the workflow element** check box.
3. In the **Duration** field, specify when the approval process must be completed. Select one of the following options:

    - **Hours** – Enter the number of hours in which the approval process must be completed. Then select the calendar that your organization uses, and enter information about your organization's work week.
    - **Days** – Enter the number of days in which the approval process must be completed. Then select the calendar that your organization uses, and enter information about your organization's work week.
    - **Weeks** – Enter the number of weeks in which the approval process must be completed.
    - **Months** – Select the day and week by which the approval process must be completed. For example, you might want the approval process to be completed by Friday of the third week of the month.
    - **Years** – Select the day, week, and month by which the approval process must be completed. For example, you might want the approval process to be completed by Friday of the third week of December.

4. If the time limit is exceeded, the system acts on the document. In the **Action** list, select the action that the system should take.

Specify which actions are available to the user

When you assign a document to a user for approval, the user must act on the document. Follow these steps to specify which actions the user can take on the submitted document.

1. In the left pane, select **Advanced settings**.
2. Select the **Approve** check box if the user can approve the document.
3. Select the **Reject** check box if the user can reject the document.
4. Select the **Request change** check box if the user can request changes to the document.
5. Select the **Delegate** check box if the user can assign the document to another user for approval.

> **Note**
>
> The **Enable actions from the work list in Enterprise Portal** check box is deprecated.

Configure the approval steps

An approval process consists of approval steps. Complete the following procedure to add steps to the approval process and configure the steps.

1. In the workflow editor, double-click the approval process. The workflow editor displays the steps of the approval process.
2. To add an approval step, drag the step from the **Workflow elements** area to the canvas.
3. To configure an approval step, see Configure approval steps in a workflow.

Last updated on 03/11/2026

Configure approval steps in a workflow

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure the properties of an approval step.

To configure an approval step in the workflow editor, right-click the approval step, and then select **Properties** to open the **Properties** page. Use the following procedures to configure the properties of the approval step.

Name the step

Follow these steps to enter a name for the approval step.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the approval step.

Enter a subject line and instructions

You must provide a subject line and instructions for users assigned to the approval step. For example, if you configure an approval step for purchase requisitions, the user assigned to the step sees the subject line and instructions on the **Purchase requisitions** page. The subject line appears in a message bar on the page. The user can then select the icon in the message bar to see the instructions. Follow these steps to enter a subject line and instructions.

1. In the left pane, select **Basic Settings**.
2. In the **Work item subject** field, enter the subject line.
3. To personalize the subject line, insert placeholders. Placeholders are replaced with appropriate data when the subject line is shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.

d. Select **Insert**.

4. To add translations of the subject line, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders as described in step 3.
   f. Select **Close**.

5. In the **Work item instructions** field, enter the instructions.

6. To personalize the instructions, insert placeholders. Placeholders are replaced with appropriate data when the instructions are shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.

7. To add translations of the instructions, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders as described in step 6.
   f. Select **Close**.

Assign the approval step

Follow these steps to specify who the approval step should be assigned to.

1. In the left pane, select **Assignment**.

2. On the **Assignment type** tab, select one of the options in the following table, and then follow the additional steps for that option before you go to step 3.

Expand table

Option	Users that the approval step is assigned to	Additional steps
Participant	Users who are assigned to a specific group or role	1. After you select **Participant**, on the **Role based** tab, in the **Type of participant** list, select the type of group or role to assign the step to. 1. In the **Participant** list, select the group or role to assign the step to.
Hierarchy	Users in a specific organizational hierarchy	1. After you select **Hierarchy**, on the **Hierarchy selection** tab, in the **Hierarchy type** list, select the type of hierarchy to assign the step to. 1. The system must retrieve a range of user names from the hierarchy. These names represent users that the step can be assigned to. Follow these steps to specify the starting point and ending point of the range of user names that the system retrieves: 1. To specify the starting point, select a person in the **Start from** list. 1. To specify the ending point, select **Add condition**. Then enter a condition that determines where in the hierarchy the system stops retrieving names. 1. On the **Hierarchy options** tab, specify which users in the range the step should be assigned to: **Assign to all users retrieved** – The step is assigned to all users in the range. **Assign only to last user retrieved** – The step is assigned to only the last user in the range. **Exclude users with the following condition** – The step isn't assigned to any users in the range who meet a specific condition. Select **Add condition** to specify the condition.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all system users. Select the users to assign the step to, and then move those users to the **Selected users** list.

3. On the **Time limit** tab, in the **Duration** field, specify how much time the user has to take action on, or respond to, documents that reach the approval step. Select one of the following options:

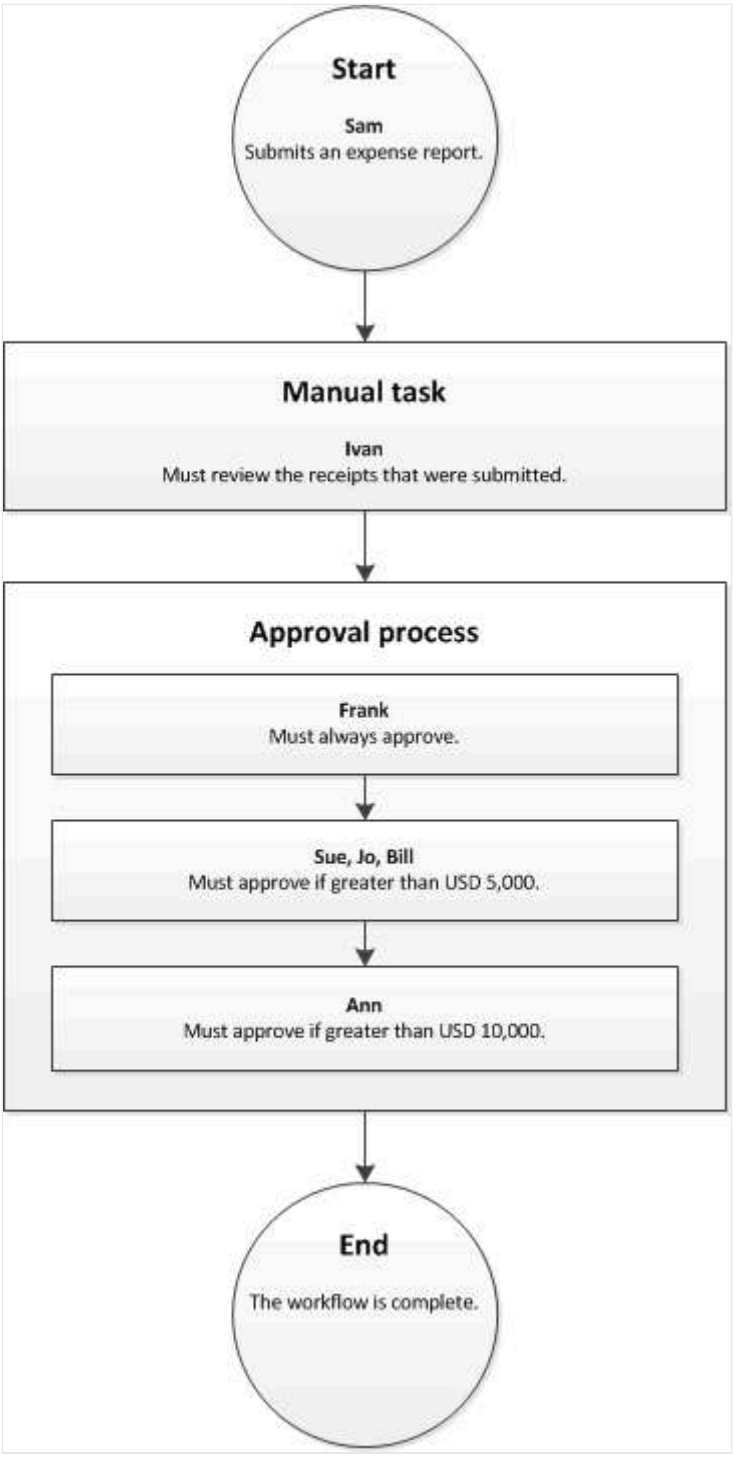
   - **Hours** – Enter the number of hours that the user has to respond. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Days** – Enter the number of days that the user has to respond. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Weeks** – Enter the number of weeks that the user has to respond.
   - **Months** – Select the day and week that the user must respond by. For example, you might want the user to respond by Friday of the third week of the month.

- **Years** – Select the day, week, and month that the user must respond by. For example, you might want the user to respond by Friday of the third week of December.

If the user doesn't take action on the document in the allotted time, the document is overdue. You can escalate an overdue document by selecting options in the **Escalation** area of the page.

4. If you assigned the approval step to multiple users or a group of users, on the **Completion policy** tab, select one of the following options:

    - **Single approver** – The action that is applied to the document is determined by the first person who responds. For example, Sam submits an expense report for USD 15,000. The expense report is currently assigned to Sue, Jo, and Bill. If Sue is the first person who responds to the document, the action that Sue takes is applied to the document. If Sue rejects the document, it's rejected and sent back to Sam. If Sue approves the document, it's sent to Ann for approval.



- **Majority of approvers** – The action that is applied to the document is determined when most of the approvers respond. For example, Sam submits an expense report for USD 15,000. The expense report is currently assigned to Sue, Jo, and Bill. If Sue and Jo are the first two approvers who respond, the action that they take is applied to the document.
  - If Sue approves the document, but Jo rejects it, the document is rejected and sent back to Sam.
  - If both Sue and Jo approve the document, it's sent to Ann for approval.

- **Percentage of approvers** – The action that is applied to the document is determined when a specific percentage of the approvers respond. For example, Sam submits an expense report for USD 15,000. The expense report is currently assigned to Sue, Jo,

and Bill, and you entered **50** as the percentage. If Sue and Jo are the first two approvers who respond, the action that they take is applied to the document, because they meet the requirement for 50 percent of approvers.
    - If Sue approves the document, but Jo rejects it, the document is rejected and sent back to Sam.
    - If both Sue and Jo approve the document, it's sent to Ann for approval.

- **All approvers** – All the approvers must approve the document. Otherwise, the workflow can't continue. For example, Sam submits an expense report for USD 15,000. The expense report is currently assigned to Sue, Jo, and Bill. If Sue and Joe approve the document, but Bill rejects it, the document is rejected and sent back to Sam. If Sue, Jo, and Bill all approve the document, it's sent to Ann for approval.

Specify when the approval step is required

You can specify when the approval step is required. The approval step can always be required, or it can be required only if specific conditions are met.

Always require the approval step

Follow these steps if the approval step is always required.

1. In the left pane, select **Condition**.
2. Select the **Always run this step** option.

Require the approval step in specific conditions

The approval step that you're configuring might be required only if specific conditions are met. For example, if you're configuring an approval step for a purchase requisition workflow, you might want the approval step to occur only if the amount of the purchase requisition is more than USD 10,000. Follow these steps to specify when the approval step is required.

1. In the left pane, select **Condition**.
2. Select the **Run this step only when the following condition is met** option.
3. Enter a condition.
4. Enter any additional conditions that are required.
5. To verify that the conditions that you entered are configured correctly, follow these steps:
    a. Select **Test**.
    b. On the **Test workflow condition** page, in the **Validate condition** area, select a record.

c. Select **Test**. The system evaluates the record to determine whether it meets the conditions that you defined.
d. Select **OK** or **Cancel** to return to the **Properties** page.

Specify what happens when the document is overdue

If a user doesn't take action on a document in the allotted time, the document is overdue. You can escalate an overdue document or automatically assign it to another user for approval. Follow these steps to escalate the document if it's overdue.

1. In the left pane, select **Escalation**.

2. Select the **Use escalation path** check box to create an escalation path. The system automatically assigns the document to the users who are listed in the escalation path. For example, the following table represents an escalation path.

   Expand table

Sequence	Escalation path
1	Assign to: Donna
2	Assign to: Erin
3	Final action: Reject

   In this example, the system assigns the overdue document to Donna. If Donna doesn't respond in the allotted time, the system assigns the document to Erin. If Erin doesn't respond in the allotted time, the system rejects the document.

3. To add a user to the escalation path, select **Add escalation**. On the **Assignment type** tab, select one of the options in the following table, and then follow the additional steps for that option before you go to step 4.

   Expand table

Option	Users that the document is escalated to	Additional steps
Hierarchy	Users in a specific organizational hierarchy	1. After you select **Hierarchy**, on the **Hierarchy selection** tab, in the **Hierarchy type** list, select the type of hierarchy to escalate the document to. 1. The system must retrieve a range of user names from the hierarchy. These names represent users that the document can be escalated to. Follow these steps to specify the starting point and ending point of the range of user names that the system retrieves: 1. To specify the starting point, select a person in the **Start from** list. 1. To specify the ending point, select **Add condition**. Then enter a condition that determines where in the hierarchy the system stops retrieving names. 1. On the **Hierarchy options** tab, specify which users in the range the document should be escalated to: **Assign to all users retrieved** – The document is escalated to all users in the range. **Assign only to last user retrieved** – The document is escalated to only the last user in the range. **Exclude users with the following condition** – The document isn't escalated to any users in the range who meet a specific condition. Select **Add condition** to specify the condition.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to escalate the document to, and then move those users to the **Selected users** list.

4. On the **Time limit** tab, in the **Duration** field, specify how much time the user has to take action on, or respond to, documents. Select one of the following options:

   - **Hours** – Enter the number of hours that the user has to respond. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Days** – Enter the number of days that the user has to respond. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Weeks** – Enter the number of weeks that the user has to respond.
   - **Months** – Select the day and week that the user must respond by. For example, you might want the user to respond by Friday of the third week of the month.
   - **Years** – Select the day, week, and month that the user must respond by. For example, you might want the user to respond by Friday of the third week of December.

5. Repeat steps 3 through 4 for each user that you want to add to the escalation path. You can change the order of the users.

6. If the users in the escalation path don't respond in the allotted time, the system automatically takes action on the document. To specify the action that the system takes, select the **Action** row, and then, on the **End action** tab, select an action.

Last updated on 03/11/2026

Configure expenditure reviewers

Article • 06/19/2024

You can set up dynamic expenditure reviewers to route expenditures for review, based on either the user who is assigned to a project role or the financial dimension where the expenditure is being charged. The workflow process uses the specified project role or financial dimension owner to determine who the expenditure should be routed to.

You can define one or more expenditure reviewer configurations, and then select a configuration when you create a workflow. You can configure the expenditure reviewer values for each legal entity in your organization. After you define the expenditure reviewer configurations, you assign the configuration. Expenditure reviewers can be assigned to workflow tasks, approvals, and manual decisions.

> **Note**
>
> Although expenditure reviewers aren't mandatory, they can help simplify the configuration of your workflow. For example, you don't have to create one conditional decision to check for each cost center dimension, and then create another element to assign the approval or task to a specific user or groups of users. Instead, you can configure the owner of the cost center dimension and use an expenditure reviewer to dynamically select the correct user. In this way, when the ownership or assignment of cost centers changes, you just have to update the **Owner** field for the financial dimension.

Types of expenditure reviewers

Not all workflows support the concept of expenditure reviewers. By default, you can configure expenditure reviewers for purchase requisitions, purchase orders, vendor invoices, and expense reports. Each of these workflow types requires that you configure the expenditure reviewers on a separate page that is specific to the feature and module. For each supported document, expenditure reviewers can be used with either header-level workflows or line-level workflows.

The following table shows where you go to configure an expenditure reviewer for each supported document.

Expand table

Document	Navigation path
Expense reports	Expense management > Setup > Policies > Expenditure reviewers
Purchase orders	Procurement and sourcing > Setup > Policies > Purchase order expenditure reviewers
Purchase requisitions	Procurement and sourcing > Setup > Policies > Purchase requisition expenditure reviewers
Vendor invoices	Accounts payable > Policy setup > Vendor invoice expenditure reviewers

Expenditure reviewer assignments

Two types of assignment are available for each expenditure reviewer: *project distributions* and *organization distributions*. For a project distribution, you can select between project authorities and financial dimensions. For an organization distribution, you can select financial dimensions.

Project authorities include the project manager, project controller, and project sales manager. You define these authorities directly on your project, by selecting a worker in the appropriate fields.

Financial dimensions are controlled by the account structures in each legal entity. For each legal entity, you can select which financial dimensions will be used with the expenditure reviewer. The dimensions can come from either the project (in this case, you select the dimensions on the **Project distributions** tab) or the document (in this case, you select the dimensions on the **Organization distributions** tab). In the **Owner** field on the **Financial dimensions values** page, you can select the worker for each financial dimension.

Example 1: Expenditure reviewers based on organization distributions

You work for Contoso Appliances, and your organization has six departments and 10 cost centers. When a new purchase requisition is submitted, approval must come first from the department manager and then from the cost center manager.

For this example, you configure two *purchase requisition expenditure reviewers*:

- For the first reviewer, you name the department and then select the **Department** financial dimension on the **Organization distributions** tab.

- For the second reviewer, you name the cost center and then select the **Cost Center** financial dimension on the **Organization distributions** tab.

When you configure the workflow, you create two approval steps. The first is for the department, and the second is for the cost center. For each approval element, you select **Participant** in the **Assignment type** field on the **Role-based** tab, select **Expenditure participants** in the **Type of participant** field, and then select the appropriate participant in the **Participant** field.

When the purchase requisition is created, the department and cost center financial dimensions are assigned to the lines of the purchase requisition. When the workflow is submitted, the system first evaluates the department on the purchase requisition and assigns the approval element to the user who is related to the worker that you selected for the department. After the first approval step is completed, the system evaluates the second approval step and assigns the second approval element to the user who is related to the worker that you selected for the cost center.

Example 2: Expenditure reviewers based on project distributions

You work for the services division of Contoso Appliances. The organization requires that the project manager for each purchase order must approve the expense. In addition, the cost center manager for the project must approve it. The approvals can be done at the same time. In any case, both users must approve the purchase order before the workflow can proceed.

For this example, you create one *purchase order expenditure reviewer* that is named **PM And Cost Center**. You select the **Project manager** checkbox and set the **Cost center dimension** option to **Yes** on the **Project distributions** tab of the **Purchase order expenditure reviewer** page. As a part of the configuration, you must ensure that the **Project manager** field is set for all projects, and that an owner is specified for all cost centers on the **Financial dimension values** page.

When you configure the workflow, you need one approval element. You select **Participant** in the **Assignment type** field. Then, on the **Role-based** tab, you select **Expenditure participants** in the **Type of participant** field, and select the project manager and cost center in the **Participant** field. To ensure that the workflow can't proceed until both the project manager and the cost center owner approve the workflow, you set the **Completion policy** field to **All approvers**.

When the purchase order is created, the **Project** field must be specified. If you don't require a project for all your purchase orders, you should add a conditional decision to

your workflow to check for a project before the approval step is run. The system then evaluates the project that is specified for the purchase order and assigns the approval element to the user who is related to the worker in the **Project manager** field. Additionally, the system creates a task and assigns it to the user who is related to the worker that you select in the **Owner** field for the cost center from the project. Note that this example uses the cost center of the project, not the cost center of the purchase order.

Set up expenditure reviewers

This example shows how to configure a purchase requisition expenditure reviewer. To configure other types of expenditure reviewers, replace the navigation path in step 1 with the appropriate path from the table in the Types of expenditure reviewers section, earlier this article.

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Purchase requisition expenditure reviewers**.
2. On the **Purchase requisition expenditure reviewers** page, select **New**.
3. In the **Name** field, enter a name for the expenditure reviewer configuration, such as **cost center**.
4. On the **Expenditure reviewers by legal entity** FastTab, select the legal entity to configure.
5. For a project distribution, select the checkbox for each project authority, and select **Yes** for each financial dimension that you want to enable.
6. For an organization distribution, on the **Organization distributions** tab, select **Yes** for each financial dimension that you want to enable.
7. Repeat steps 4 through 6 for each additional legal entity.

Assign owners to financial dimensions

To assign an owner to a financial dimension, follow these steps.

1. Go to **General ledger** > **Chart of accounts** > **Dimensions** > **Financial dimensions**.
2. In the list, select the financial dimension to assign owners to.
3. Select **Dimension values**.
4. Select **Edit** to make changes to the dimension values.
5. In the list, select the dimension value to assign an owner to.
6. In the **Owner** field, select the worker to assign the dimension value to.

Configure project distributions

To assign project authorities to a project, follow these steps.

1. Go to **Project management and accounting** > **Projects** > **All projects**.
2. Select the project to assign authorities to.
3. Select **Edit** to make changes to the project.
4. On the **General** FastTab, in the **Responsible** section, select a worker for the **Sales manager**, **Project manager**, and **Project controller** fields as required.
5. On the **Financial dimensions** FastTab, select the required financial dimensions for your project.

Assign expenditure reviewers to a workflow task

This example shows how to configure a purchase requisition workflow so that it uses a purchase requisition expenditure reviewer. To configure other types of workflows, the workflow page that you must open in step 1 might be in the **Expense management** or **Accounts payable** module instead of the **Procurement and sourcing** module.

To assign purchase requisition expenditure reviewers to a workflow task, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing workflows**.
2. Double-tap (or double-click) an existing workflow, or create a new workflow.
3. In the workflow editor, in the **Workflow** pane, select the task to assign the expenditure reviewer configuration to. Then, on the **Action Pane**, in the **Show** group, select **Properties**.
4. In the left pane of the **Properties** page, select **Assignment**.
5. On the **Assignment type** tab, select **Participant**.
6. On the **Role-based** tab, in the **Type of participant** field, select **Expenditure participants**. Then, in the **Participant** field, select the expenditure reviewer configuration to use for the workflow task.

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Configure manual decisions in a workflow

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure the properties of a manual decision.

To configure a manual decision in the workflow editor, right-click the manual decision, and then select **Properties** to open the **Properties** page. Then use the following procedures to configure the properties of the manual decision.

Name the decision

Follow these steps to enter a name for the manual decision.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the manual decision.

Enter a subject line and instructions

You must provide a subject line and instructions for users assigned to the manual decision. For example, if you're configuring a decision for purchase requisitions, the user assigned to the decision sees the subject line and instructions on the **Purchase requisitions** page. The subject line appears in a message bar on the page. The user can then select the icon in the message bar to view the instructions. Follow these steps to enter a subject line and instructions.

1. In the left pane, select **Basic Settings**.
2. On the **Instructions** tab, in the **Work item subject** field, enter the subject line.
3. To personalize the subject line, insert placeholders. Placeholders are replaced with appropriate data when the subject line is shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.

d. Select **Insert**.

4. To add translations of the subject line, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders as described in step 3.
   f. Select **Close**.

5. In the **Work item instructions** field, enter the instructions.

6. To personalize the instructions, insert placeholders. Placeholders are replaced with appropriate data when the instructions are shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.

7. To add translations of the instructions, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders as described in step 6.
   f. Select **Close**.

Specify the possible outcomes of a decision

Typically, when you assign a document to a decision maker, you ask a question. The answer to this question is usually **Yes** or **No**, or **True** or **False**. Follow these steps to specify the possible outcomes of the manual decision.

1. In the left pane, select **Basic Settings**.

2. On the **Outcomes** tab, in the **Outcome 1** field, enter the name of the outcome or the option.

3. To add translations of the outcome, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.

d. In the **Translated text** field, enter the text.
e. Select **Close**.

4. In the **Outcome 2** field, enter the name of the outcome or the option.

5. To add translations of the outcome, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the text.
   e. Select **Close**.

Specify when to send notifications

You can send notifications to people when a decision is made, delegated, or escalated. Follow these steps to specify when to send notifications and who receives them.

1. In the left pane, select **Notifications**.

2. Select the check box next to the events that send notifications:

   - **[Choice 1]** – The assigned user selects **[Choice 1]**.
   - **[Choice 2]** – The assigned user selects **[Choice 2]**.
   - **Delegate** – The assigned user assigns the decision to another user.
   - **Escalate** – The assigned user doesn't make the decision in the allotted time.

3. Select the row for an event that you selected in step 2.

4. On the **Notification text** tab, enter the text of the notification.

5. To personalize the notification, insert placeholders. Placeholders are replaced with appropriate data when the notification is shown to users. Follow these steps to insert a placeholder:
   a. In the text box, select where the placeholder should appear.
   b. Select **Insert placeholder**.
   c. In the list that appears, select the placeholder to insert.
   d. Select **Insert**.

6. To add translations of the notification, follow these steps:
   a. Select **Translations**.
   b. On the page that appears, select **Add**.
   c. In the list that appears, select the language for the translation.
   d. In the **Translated text** field, enter the text.
   e. To personalize the text, insert placeholders as described in step 5.

f. Select **Close**.

7. On the **Recipient** tab, specify who receives the notifications. Select one of the options in the following table, and then follow the additional steps for that option before you go to step 8.

Expand table

Option	Notification recipients	Additional steps
Participant	Users who are assigned to a specific group or role	1. After you select **Participant**, on the **Role based** tab, in the **Type of participant** list, select the type of group or role to send notifications to. 1. In the **Participant** list, select the group or to send notifications to.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to send notifications to, and then move those users to the **Selected users** list.

1. Repeat steps 3 through 7 for each event that you selected in step 2.

Assign a decision

Follow these steps to specify who should receive a manual decision.

1. In the left pane, select **Assignment**.
2. On the **Assignment type** tab, select one of the options in the following table. Follow the additional steps for that option before you go to step 3.

Expand table

Option	Users that the decision is assigned to	Additional steps
Participant	Users assigned to a specific group or role	1. After you select **Participant**, on the **Role based** tab, in the **Type of participant** list, select the type of group or role to assign the decision to. 1. In the **Participant** list, select the group or role to assign the decision to.

Option	Users that the decision is assigned to	Additional steps
Hierarchy	Users in a specific organizational hierarchy	1. After you select **Hierarchy**, on the **Hierarchy selection** tab, in the **Hierarchy type** list, select the type of hierarchy to assign the decision to. 1. The system retrieves a range of user names from the hierarchy. These names represent users that the decision can be assigned to. Follow these steps to specify the starting point and ending point of the range of user names that the system retrieves: 1. To specify the starting point, select a person in the **Start from** list. 1. To specify the ending point, select **Add condition**. Then enter a condition that determines where in the hierarchy the system stops retrieving names. 1. On the **Hierarchy options** tab, specify which users in the range the decision should be assigned to: **Assign to all users retrieved** – The decision is assigned to all users in the range. **Assign only to last user retrieved** – The decision is assigned to only the last user in the range. **Exclude users with the following condition** – The decision isn't assigned to any users in the range who meet a specific condition. Select **Add condition** to specify the condition.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to assign the decision to, and then move those users to the **Selected users** list.

1. On the **Time limit** tab, in the **Duration** field, specify how much time the user has to make the decision. Select one of the following options:

    - **Hours** – Enter the number of hours that the user has to make the decision. Then select the calendar that your organization uses, and enter information about your organization's work week.
    - **Days** – Enter the number of days that the user has to make the decision. Then select the calendar that your organization uses, and enter information about your organization's work week.
    - **Weeks** – Enter the number of weeks that the user has to make the decision.
    - **Months** – Select the day and week that the user must make the decision by. For example, you might want the user to make the decision by Friday of the third week of the month.
    - **Years** – Select the day, week, and month that the user must make the decision by. For example, you might want the user to make the decision by Friday of the third week of December.

If the user doesn't make the decision in the allotted time, the decision is overdue. You can escalate an overdue decision based on the options that you select in the **Escalation** area of the page.

Specify what happens when a decision is overdue

If a user doesn't make the decision in the allotted time, the decision is overdue. You can escalate an overdue decision or automatically assign it to another user. Follow these steps to escalate the decision if it's overdue.

1. In the left pane, select **Escalation**.
2. Select the **Use escalation path** check box to create an escalation path. The system automatically assigns the decision to the users who are listed in the escalation path. For example, the following table represents an escalation path.

   Expand table

Sequence	Escalation path
1	Assign to: Donna
2	Assign to: Erin
3	Final action: [Choice 1]

   In this example, the system assigns the overdue decision to Donna. If Donna doesn't make the decision in the allotted time, the system assigns the decision to Erin. If Erin doesn't make the decision in the allotted time, the system selects **[Choice 1]** as the decision.
3. To add a user to the escalation path, select **Add escalation**. Select one of the options in the following table, and then follow the additional steps for that option before you go to step 4.

   Expand table

Option	Users that the decision escalates to	Additional steps
Hierarchy	Users in a specific organizational hierarchy	1. After you select **Hierarchy**, on the **Hierarchy selection** tab, in the **Hierarchy type** list, select the type of hierarchy to escalate the decision to. 1. The system retrieves a range of user names from the hierarchy. These names represent users that the decision can escalate to. Follow these steps

Option	Users that the decision escalates to	Additional steps
		to specify the starting point and ending point of the range of user names that the system retrieves: 1. To specify the starting point, select a person in the **Start from** list. 1. To specify the ending point, select **Add condition**. Then enter a condition that determines where in the hierarchy the system stops retrieving names. 1. On the **Hierarchy options** tab, specify which users in the range the decision should escalate to: **Assign to all users retrieved** – The decision escalates to all users in the range. **Assign only to last user retrieved** – The decision escalates to only the last user in the range. **Exclude users with the following condition:** – The decision doesn't escalate to any users in the range who meet a specific condition. Select **Add condition** to specify the condition.
Workflow user	Users in the current workflow	After you select **Workflow user**, on the **Workflow user** tab, in the **Workflow user** list, select a user who participates in the workflow.
User	Specific users	1. After you select **User**, select the **User** tab. 1. The **Available users** list includes all users. Select the users to escalate the decision to, and then move those users to the **Selected users** list.

1. On the **Time limit** tab, in the **Duration** field, specify how much time the user has to make the decision. Select one of the following options:

    - **Hours** – Enter the number of hours that the user has to make the decision. Then select the calendar that your organization uses, and enter information about your organization's work week.
    - **Days** – Enter the number of days that the user has to make the decision. Then select the calendar that your organization uses, and enter information about your organization's work week.
    - **Weeks** – Enter the number of weeks that the user has to make the decision.
    - **Months** – Select the day and week that the user must make the decision by. For example, you might want the user to make the decision by Friday of the third week of the month.
    - **Years** – Select the day, week, and month that the user must make the decision by. For example, you might want the user to make the decision by Friday of the third week of December.

2. Repeat steps 3 through 4 for each user that you want to add to the escalation path. You can change the order of the users.

3. If the users in the escalation path don't make the decision in the allotted time, the system makes the decision. To specify the option that the system selects, select the **Action** row,

and then, on the **End action** tab, select an option.

Set a time limit

Follow these steps if you need to make the decision by a specific time.

> **Note**
>
> The options that you select in this procedure override the options that you selected in the **Assignment** and **Escalation** areas of the page.

1. In the left pane, select **Advanced settings**.
2. Select the **Set a time limit for the workflow element** check box.
3. In the **Duration** field, specify when the decision must be made. Select one of the following options:
   - **Hours** – Enter the number of hours. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Days** – Enter the number of days. Then select the calendar that your organization uses, and enter information about your organization's work week.
   - **Weeks** – Enter the number of weeks.
   - **Months** – Select the day and week that the decision must be made by. For example, you might want the decision to be made by Friday of the third week of the month.
   - **Years** – Select the day, week, and month that the decision must be made by. For example, you might want the decision to be made by Friday of the third week of December.
4. If the time limit is exceeded, the system makes the decision. In the **Action** list, select the option that the system should choose.

Last updated on 03/11/2026

Configure conditional decisions in a workflow

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the following procedure to configure the properties of a conditional decision.

A conditional decision is a point where a workflow divides into two branches. To configure a conditional decision, right-click the conditional decision in the workflow editor, and then select **Properties** to open the **Properties** form.

Name a decision

Follow these steps to enter a name for a conditional decision.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the conditional decision.

Set conditions

The system determines which branch to use by evaluating the submitted document to see if it meets specific conditions.

1. In the left pane, select **Basic Settings**.
2. Select **Add condition**.
3. Enter a condition.
4. Enter additional conditions, if needed.
5. To verify that the conditions you entered are configured correctly, complete the following steps:
   a. Select **Test** to open the **Test workflow condition** form.
   b. Select a record in the **Validate condition** area of the form.

c. Select **Test**. The system evaluates the record to see if it meets the conditions you defined.
d. Select **OK** or **Cancel** to return to the **Properties** form.

Last updated on 03/11/2026

Configure parallel activities in a workflow

Summarize this article for me

> **Note**
>
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To configure a parallel activity, use the workflow editor.

A parallel activity consists of workflow branches that run at the same time.

Name a parallel activity

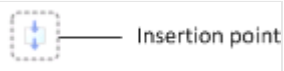
Follow these steps to enter a name for a parallel activity.

1. Right-click the parallel activity, and then select **Properties** to open the **Properties** form.
2. In the left pane, select **Basic Settings**.
3. In the **Name** field, enter a unique name for the parallel activity.
4. Select **Close**.

Configure the branches of a parallel activity

Follow these steps to add and configure the branches of this parallel activity.

1. Double-click the parallel activity to display the branches of the parallel activity.
2. To add a branch, drag the **Branch** element from the **Workflow elements** area to an insertion point on the canvas. The following figure shows an insertion point.

   > **Note**
   >
   > The order of the branches doesn't matter because all the branches of a parallel activity run at the same time.

3. To configure each branch, see Configure parallel branches in a workflow.

Last updated on 03/11/2026

Configure parallel branches in a workflow

Summarize this article for me

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To configure a parallel branch, use the workflow editor.

A parallel branch is a workflow that runs in the context of a parent workflow.

Name a branch

Follow these steps to enter a name for a parallel branch.

1. Right-click the parallel branch, and then select **Properties**. The **Properties** form appears.
2. In the left pane, select **Basic Settings**.
3. In the **Name** field, enter a unique name for the parallel branch.
4. Select **Close**.

Design and configure the elements of a branch

Follow these steps to design and configure the elements of a parallel branch.

1. Double-click the parallel branch.
2. Drag workflow elements onto the canvas, and then configure the elements, just as you would to create any other workflow. For more information, see Create workflows overview.

Additional resources

Create workflows overview

Last updated on 03/11/2026

Configure line item workflows

Summarize this article for me

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>
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This article explains how to configure a line item workflow element.

To configure a line item workflow element, in the workflow editor, right-click the element, and then select **Properties** to open the **Properties** page. Then use the following procedures to configure the properties of the line item workflow element.

Name the line item workflow element

Follow these steps to enter a name for the line item workflow element.

1. In the left pane, select **Basic Settings**.
2. In the **Name** field, enter a unique name for the line item workflow element.

Specify whether the same workflow processes all line items

Follow these steps to specify whether the same workflow processes all the line items on a document.

1. In the left pane, select **Basic Settings**.
2. If the same workflow should process all the line items on a document, select **Invoke a single workflow for all line-items**. Then select the workflow to use to process the line items.
3. If a specific workflow should process line items that meet a specific set of conditions, select **Invoke a workflow for each line-item**. Then follow these steps to define the set of conditions:
   a. Select **Add**.
   b. Select the condition in the table.

c. On the **Condition name** tab, enter a name for the set of conditions that you're defining.
d. Select **Add condition** to enter a condition.
e. Enter any additional conditions that are required.
f. To verify that the set of conditions that you entered is configured correctly, select **Test**. On the **Test workflow condition** page, in the **Validate condition** area, select a record, and then select **Test**. The system evaluates the record to determine whether it meets the conditions that you defined. Select **OK** or **Cancel** to return to the **Properties** page.

On the **Workflow** tab, select the workflow to use to process line items that meet the set of conditions that you defined.

Last updated on 03/11/2026

Configure the Workflow message processing batch job as critical

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The workflow system uses various batch jobs. **Workflow message processing** is an important batch job that processes workflow messages. If workflow is a key component of your organization, consider configuring the **Workflow message processing** batch job as critical.

When you configure the **Workflow message processing** batch job as critical, the system actively tracks its status. When a critical batch job fails, the support team can better monitor failures and take action to resolve any problems.

Follow these steps to configure the **Workflow message processing** batch job as critical:

1. Go to the **Batch jobs** page.
2. Search for **Workflow message processing** by using the quick filter.
3. Select the **Workflow message processing** batch job.
4. Select **Edit** in the action pane.
5. Select the **Critical Job** check box.
6. Select **Save** in the action pane.

Last updated on 03/18/2026

Delegate work items in a workflow

Summarize this article for me

> **Note**
>
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Manually delegate a work item

To delegate an individual work item, select the **Delegate** option in the **Workflow** menu. Enter the user to delegate to along with a comment. This action reassigns the work item to that user so they can complete it.

Manually delegate multiple work items

You can delegate multiple work items together from the **Work items assigned to me** page. The following workflow types are eligible for mass delegation: Purchase agreement approval workflow, Purchase order workflow, Purchase requisition review, and Vendor invoice workflow. The **Delegate multiple work items** feature is disabled by default. You can enable it in **Workspaces** > **Feature management**. Contact your system administrator for help with enabling this feature.

1. Go to **Common** > **Common** > **Work items** > **Work items assigned to me**.
2. Select the work items that you want to delegate.
3. Select the **Delegate work items** menu.
4. In the **User** field, select the user to delegate the work items to.
5. In the **Comment** field, enter a comment that explains why you're delegating the work items.
6. Select the **Delegate work items** button to complete the work item delegation.

Automatically delegate work items

If you plan to be out of the office or otherwise unavailable to act on work items for a period of time, you can automatically delegate new work items to other users by using the **User options** page.

Set up automatic delegation

1. Go to **Common** > **Setup** > **User options**.
2. Select the **Workflow** tab. Make sure the Delegation section is expanded. To configure the system to automatically delegate your work items to other users, create delegation rules that specify when to delegate certain types of work items. Follow these steps to create a delegation rule.
3. Select **Add**.
4. In the **Scope** field, select an option:

    - All – Delegate all work items that are assigned to you.
    - Module – Delegate only the work items that are related to a specific type of workflow. If you select this option, you must select the type of workflow in the **Name** field.
    - Workflow – Delegate only the work items that are related to a specific workflow. If you select this option, you must select the workflow in the **Name** field.

5. In the **Name** field:

    - For **Module** scope, select the target module.
    - For **Workflow** scope, select the target workflow.

6. In the **Delegate** field, select the user to delegate the work items to. Use the **Start date/time** and **End date/time** fields to specify when you want the work items to be automatically delegated.
7. In the **Start date/time** field, enter a date and time.
8. In the **End date/time** field, enter a date and time.
9. Select the **Enabled** check box to activate the delegation rule.
10. In the **Comment** field, enter a comment that explains why you're delegating the work items.

Last updated on 03/11/2026

View workflow history

Summarize this article for me

> **Note**
>
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This article describes the steps to view the status of a document that you submitted to the workflow system for processing and approval. The demo data company used to create this procedure is USMF.

1. Go to **Navigation pane** > **Modules** > **Common** > **Inquiries** > **Workflow** > **Workflow history**.

    - Use this form to view the status of a document that you submitted to the workflow system for processing and approval.
    - The **Instance ID** is the identification code of the workflow instance that is processing, or has processed the document.
    - The **Status** is the workflow status of the document.
    - The **Workflow ID** is the identification code of the workflow that is processing, or has processed the document.
    - The **Document** is the identification code of the document.
    - The **Document type** is the type of document that you submitted for processing.
    - The **Workflow** is the name of the workflow that is processing, or has processed the document.
    - The **Version** is the version number of the workflow that is processing, or has processed the document.
    - The **Created date and time** is the date and time that you submitted the document.
    - The **Elapsed time** is the time that passed since you submitted the document.
    - The **Resume** button allows you to resume the workflow process for the selected document.
    - The **Recall** button allows you to recall the selected document so that it isn't processed.

2. In the list, select the link in the desired row.

    - Make sure the **Work items** section is expanded. In this section, you can view the work items that are associated with the selected document. For example, a task

might need to be completed, or the document might need to be approved.

- The **Reassign** button opens a dialog box where you can reassign a work item to another user.
- Make sure the **Tracking details** section is expanded. In this section, you can view the workflow history of the selected document.

Last updated on 03/11/2026

Workflow history summary

Summarize this article for me

This article explains how to enable the optional **Summary by Copilot** field on any **Workflow history** page in Microsoft Dynamics 365 Finance and Dynamics 365 Supply Chain Management.

Turn on Copilot support for the Workflow history page

The **Workflow history summary** feature is enabled by default in Dynamics 365 Finance version 10.0.39.

> **Note**
>
> As of version 10.0.40, and in the proactive quality updates (PQUs) for version 10.0.39, the **Workflow history summary** feature is turned on by default in Feature management.

You can use workflow history either with or without its AI-powered copilot functionality. To use the copilot functionality in workflow history, you must install the **Copilot in Microsoft Dynamics 365 Finance** app in your Dataverse environment. For more information, see Copilot prerequisites.

Use this feature to get an AI-generated summary of the workflow history that you're viewing on the **Workflow history** page. The submitter, submission date, current status, approval or rejection, request change actions, due dates, and comments are summarized for you. This feature is powered by Azure OpenAI Service's large language model. It's designed to reduce the time that you spend searching the workflow history for the most important details, so that you can approve documents quickly and efficiently.

For more information, see Workflow history summary: Transparency FAQ.

This feature has two purposes:

- Provide a concise workflow history summary.
- Help you make quicker decisions when you approve workflow documents.

Country/region and language availability

For information about the languages that Microsoft validated this Copilot feature for, see Explore Copilot features by geography and languages. Although you can use the feature in

other languages, it might not work as intended. Language quality might vary, based on the user's interactions or system settings, and might therefore affect accuracy and the user experience.

Version requirements

Workflow history summary requires the latest PQU on Dynamics 365 Finance versions 10.0.39 and 10.0.40.

Role requirements

To use this feature, you must be able to view workflow history in Dynamics 365 Finance or Dynamics 365 Supply Chain Management.

Summary by Copilot

The **Summary by Copilot** field appears at the top of any **Workflow history** page if you submit the workflow. The first line shows the submitter, submitted date, current status, and comments. The next lines show the most recent workflow actions. Workflow actions include approvals, delegations, rejections, and change requests. The workflow action dates and the user who performed each action are shown together with any comments that were entered. The comments might be summarized if they exceed the space that was allocated for comments.

Last updated on 03/11/2026

Enable users to receive workflow-related email messages

Summarize this article for me

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>
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You can configure the system to send email messages to users when workflow-related events occur. For example, you can send email messages to users when you assign documents to them for approval. The demo data company used to create this procedure is USMF.

1. Go to **Navigation pane** > **Modules** > **System administration** > **Users** > **Users**.
2. In the list, find and select the desired record.
3. On the **Action pane**, select **User options**.
4. Select the **Workflow** tab. Make sure that the **Notifications** section is expanded. In the **Notifications** section, specify how you want the user to be notified about workflow-related events.
5. In the **Line-item workflow notification type** field, select an option.
    - Grouped – Notifications for line items are grouped into a single email message.
    - Individual – An email message is sent for each line item.
    - If you want the user to receive notifications in the client, select the **Send notifications in email** check box.
6. Select **Save**.
7. Close the page.

> **Note**
>
> The workflow email templates come from either system email templates or organization email templates. The source depends on whether the workflow is a system-level (not company specific) or organization-level (company specific) workflow.

Last updated on 03/11/2026

Create a workflow type

> **Note**
>
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To add workflow support for a document, you must create a workflow type. After you create a workflow type, you can use it to create workflow configurations for the document. This article provides links to the procedures for creating a workflow type.

You create workflow types to define the following elements:

- The workflow document to assign the workflow to
- A category that defines the module that the workflow type is available in
- Tasks, automated tasks, and approvals that the workflow supports
- The workflow started, completed, configuration data change, and canceled event handlers
- A **SubmitToWorkflowMenuItem** menu item

In this section

- Workflow type checklist
- Create a workflow category
- Create a query for a workflow type
- Create a new workflow type
- Create a workflow document class
- Create a SubmitToWorkflow class
- Associate a workflow document class with a workflow type

Last updated on 03/18/2026

Workflow type checklist

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>
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This article describes the steps that are required to create a new workflow type. Use workflow types to create configurations for a workflow.

Workflow type checklist

1. If an existing category isn't appropriate, create a new workflow category to use for the workflow type. For more information, see Create a workflow category.
2. Create a query that accesses the document that the new workflow type is for. For more information, see Create a query for a workflow type.
3. Create a workflow type in Application Explorer. Typically, you complete this step by using the **Workflow** wizard. For more information, see Create a new workflow type.

Use the **Workflow** wizard to create a workflow type. The wizard performs the following tasks. In some cases, you must add more code for your workflow type. Links are provided so that you can see the details of the actions that the wizard performs and the additional steps that you must perform to complete your workflow type.

1. Create a workflow document class, and then bind the query that the workflow uses to the class. For more information, see Create a workflow document class.
2. Bind the workflow document class to the workflow type. For more information, see Associate a workflow document class with a workflow type.
3. Create workflow event handlers for started, completed, configuration change, and canceled events, and then bind event handlers to the workflow type. For more information, see Create a workflow event handler.
4. Create a class for **SubmitToWorkflowMenuItem**. Optionally create a class for **CancelMenuItem**. Bind the classes to the action menu items that you create in the next step. For more information, see Classes and methods.
5. Create an action menu item for the **SubmitToWorkflowMenuItem** workflow property. Optionally create an action menu item for the **CancelMenuItem** property. Bind the actions to the workflow type. For more information, see Create a new workflow type.

Next steps

After you create a workflow type, you can add tasks, automated tasks, and approvals.

See also

Configure workflow properties

Configure manual tasks in a workflow

Configure automated tasks in a workflow

Configure approval processes in a workflow

Last updated on 03/17/2026

Create a workflow category

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you create a workflow type, you must assign it to a workflow category. The workflow category determines whether the workflow type is available in a specific module. If an appropriate workflow category doesn't already exist, you must create it.

For example, a workflow type for a customer invoice shouldn't be available in the **Master planning** module. To make the workflow type available only in the **Customer** module, select the **Customer** module when you create a workflow category.

This article describes how to create a workflow category.

Create a workflow category

1. In Application Explorer, expand the **Business Process and Workflow** node.
2. Follow one of these steps to create the workflow category:
   - Right-click the **Workflow Categories** node, and then select **New Workflow Category**. A new workflow category group appears under the **Workflow Categories** node. Right-click the new workflow category, and then select **Properties**.
   - On the **Project** menu, select **Add new item**. In the **Add new item** dialog box, select **Workflow category**. Enter a name for the category, and then select **Create**.
3. On the **Properties** sheet, set the following properties, as required.

Expand table

Property	Value
Name	Specify the name that you use to reference the workflow category.
Label	Specify the label for the workflow category in the user interface (UI).
Help Text	Specify the description that's shown for the workflow category in the workflow configuration UI.

Property	Value
Module	Specify the module that the workflow is available in. The default module is **Ledger**.

After you create a workflow category, you can bind a workflow type to it. Typically, the **Workflow** wizard completes this step. However, the following procedure explains how to manually bind a workflow type to a workflow category.

Bind a workflow type to a workflow category

1. In Application Explorer, expand the **Business Process and Workflow** node, and then expand the **Workflow Types** node.
2. Right-click the workflow type that you want to bind to a workflow category, and then select **Properties**.
3. On the **Properties** sheet, set the **Category** property to the workflow category that you created in the previous procedure.

See also

Create a workflow type.

Create a new workflow type.

Last updated on 03/18/2026

Create a query for a workflow type

> **Note**
>
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Before you create a workflow type, create a query that accesses the table fields for the workflow document. This article describes how to create a query for a workflow type.

Create a query for a workflow type

1. Follow one of these steps to create a new query:
    - In Application Explorer, right-click the **Queries** node, and then select **New Query**. A query group appears under the **Queries** node. Right-click the new query, and then select **Rename**. Enter a name for the query.
    - On the **Project** menu, select **Add new item**. In the **Add new item** dialog box, select **Query**. Enter a name for the query, and then select **Create**.
2. Expand the new query, right-click the **Data Sources** node, and then select **New Data Source**. A data source group appears under the **Data Sources** node.
3. Right-click the new data source group, and then select **Properties**.
4. On the **Properties** sheet, set the **Table** property to the main table for the document type that you're defining a workflow for.
5. In the Application Object Tree (AOT), right-click the new query, and then select **Save**.

For more information about how to create queries, see Create queries by using the AOT.

Last updated on 03/18/2026

Create a new workflow type

> **Note**
>
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To make the workflow process available for a workflow document, you must create the workflow types that are used in the workflow configuration user interface (UI). This article describes how to create a workflow type in Application Explorer.

A workflow type defines the following information:

- The document that the workflow supports
- Workflow categories, which you use to assign a workflow type to a specific module
- Tasks, automated tasks, approvals, and line item workflows that you can configure
- Menu items and event handlers

Create a new workflow type

1. Follow one of these steps to open the **Workflow** wizard. The wizard helps you create a new workflow type.

    - In **Application Explorer**, right-click the **Business Process and Workflow** node, and then select **Add-Ins** > **Workflow type wizard**.
    - On the **Project** menu, select **Add new item**. In the **Add new item** dialog box, select **Workflow type**. Enter a name for the query, and then select **Create**.

2. Set the following values for the wizard.

Expand table

Value	Description
Name	Specify the name for the workflow type.
Category	Select the workflow category for the workflow type. The category determines the module that the workflow type is available in. You can use an existing category or create a new category. For more information, see Create a workflow category.
Query	Select the query that accesses the table fields for the workflow document. For

Value	Description
	more information, see Create a query for a workflow type.
Document menu item	Select the menu item that points to the main page that shows the document for the workflow type.
Document web menu item	Select the web menu item that points to the Enterprise Portal page that shows the document for the workflow type.

3. Specify the types of menu items that you want to create. You can create menu items, web menu items, or both.
4. Select **Next**. A list of all the resources that the wizard creates for the workflow type is shown.
5. Select **Finish** to create the resources. The wizard creates classes, menu items, web menu items, the workflow type, and a project that contains all the items.
6. When the dialog box appears that indicates the status, select **OK**. The portal shows the project that contains the workflow type resources.

After you create a workflow type, the next step is to create a workflow document class to expose document data for conditions. For more information, see Create a workflow document class.

See also

Create a workflow type

Create a workflow category

Last updated on 03/18/2026

Create a workflow document class

> **Note**
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You define table fields in a query to create workflow conditions. In a typical scenario, you use calculated fields to determine the behavior of a workflow. For example, you can use a dynamic sales total of all records in a table as a workflow condition to determine whether the step should be used. However, a limitation of queries is that you can't define calculated fields in the queries themselves. To overcome this query limitation, you must use a workflow document class. This article describes how to create a workflow document class.

The workflow document class that you create defines table fields for conditions in two ways: the Application Explorer query and parameter methods. You must override the **getQueryName** method of the WorkflowDocument class to return the name of the query. You can optionally add calculated fields by adding parameter methods that have a specific signature on the class. For more information about workflow conditions, see Configure workflow properties and Configure conditional decisions in a workflow.

The following procedures explain how to create a workflow document class that includes a parameter method for a calculated field. Before you begin, you must create a query that specifies the data that the workflow accesses. For more information about workflow queries, see Create a query for a workflow type.

> **Note**
>
> If you use the **Workflow** wizard to create the workflow type, the wizard already creates the workflow document class.

Create a workflow document class

1. Follow one of these steps to create a new query:

    - In **Application Explorer**, expand the **Classes** node. Right-click the **Classes** node, and then select **New Class**. A class group appears under the **Classes** node. Right-click

the new class, select **Rename**, and then enter a name for the workflow document class.
- On the **Project** menu, select **Add new item**. In the **Add new item** dialog box, select **Class**. Enter a name for the class, and then select **Create**.

2. Expand the new class, select **classDeclaration**, right-click the class declaration, and then select **Edit**.

3. Enter the following code in the class declaration.

```
X++
```

```
class <MyWorkflowDocumentClassName> extends WorkflowDocument
{
}
```

4. Close the **Editor** window, and select **Yes** to save your changes.

5. Right-click the new class, point to **Override Method**, and then select **getQueryName**. A method node that is named **getQueryName** appears under the workflow document class node, and the **Editor** window appears.

> ⓘ Note
>
> Be sure to select **getQueryName** as the method to override. The WorkflowDocument.getQuery method is used only internally to retrieve the actual query, based on the string that is returned by the WorkflowDocument.getQueryName method.

6. In the **Editor** window, enter the following code for the query method.

```
X++
```

```
QueryName getQueryName()
{
    return querystr(<MyWorkflowDocumentQueryName>);
}
```

After you create the workflow document class, you can bind it to the workflow type. For more information, see Associate a workflow document class with a workflow type.

You can add a calculated field to the workflow document class only if it meets these conditions:

- You name it **parm <name>**.

- It defines the **CompanyId**, **TableId**, and **RecId** parameters.
- It returns an extended data type (EDT) that's included in the list of fields that define conditions or notification messages. The label for the EDT must uniquely identify the value.

Add a calculated field to the workflow document class

1. In the workflow document class where you want to add a calculated field, right-click the class, and then select **New Method**. A new method node appears under the **Classes** node.

2. Right-click the new method node, and then select **Edit**.

3. Enter code in the format shown in the following example. This example shows how to add a calculated field to determine the total credit amount for a journal.

```
X++
```

```
public TotalJournalCreditAmount parmTotalJournalCreditAmount(CompanyId
_companyId, TableId _tableId, RecId _recId)
{
    // The calculateAmounts method contains business and validation logic
    this.calculateAmounts(_companyId, _tableId, _recId);
    return totalJournalCreditAmount;
}
```

See also

Associate a workflow document class with a workflow type

Last updated on 03/18/2026

Create a SubmitToWorkflow class

> **Note**
>
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A workflow starts when the user selects the **Submit** button on the workflow toolbar. The **Submit** button connects to an action menu item that calls the **main** method of a class you create to activate a workflow. This article describes how to create a **SubmitToWorkflow** class and use the name of the workflow type to activate the workflow.

You can also activate a workflow by using the workflow configuration ID or the workflow sequence number. The basic procedure is the same. For more information, see Activating a workflow.

> **Note**
>
> If you use the **Workflow** wizard to create the workflow type, the wizard already creates a workflow submission manager class. You just need to add code to it.

Create a SubmitToWorkflow class

1. In Application Explorer, expand the **Classes** node.
2. Right-click the **Classes** node, and then select **New Class**. A class group appears under the **Classes** node.
3. Right-click the new class, and then select **New Method**. A new method node appears under the **Classes** node.
4. Right-click the new method, and then select **Edit**.
5. Enter the following code for the **main** method to use the name of the workflow type to activate the workflow.

   > **Note**

This example applies to workflow submissions. For an example that also works with Enterprise Portal, see Adding enterprise portal support for workflow submission.

X++

```
public static void main(Args args)
{
    // Variable declaration.
    recId _recId = args.record().RecId;
    WorkflowCorrelationId _workflowCorrelationId;
    // Hardcoded workflow type name.
    WorkflowTypeName _workflowTypeName = workflowtypestr("MyWorkflowType");
    // Initial note is the information that users enter when they
    // submit the document for workflow.
    WorkflowComment _initialNote = "";
    WorkflowSubmitDialog workflowSubmitDialog;
    // Opens the submit to workflow dialog box for user comments.
    workflowSubmitDialog =
WorkflowSubmitDialog::construct(args.caller().getActiveWorkflowConfiguration()
);
    workflowSubmitDialog.run();
    if (workflowSubmitDialog.parmIsClosedOK())
    {
        _recId = args.record().RecId;
        // Get user comments from the submit to workflow dialog box.
        _initialNote = workflowSubmitDialog.parmWorkflowComment();
        try
        {
            ttsbegin;
            // Activate the workflow from a template.
            _workflowCorrelationId =
Workflow::activateFromWorkflowType(_workflowTypeName, _recId, _initialNote,
NoYes::No);
            ttscommit;
            // Updates the workflow button to display Actions instead of
Submit.
            args.caller().updateWorkflowControls();
        }
        catch(exception::Error)
        {
            // ToDo Insert your error code here.
        }
    }
}
```

6. Close the **Editor** window, and select **Yes** to save your changes.

> ⓘ Note

When you save this code, you receive an "Empty compound statement" warning message in the **Compiler Output** window unless you add valid code in the **catch(exception::Error)** block.

See also

Activating a workflow

Create a new workflow type

Workflow::activateFromWorkflowType method

Workflow::activateFromWorkflowSequenceNumber method

Workflow::activateFromWorkflowConfigurationId method

Last updated on 03/18/2026

Associate a workflow document class with a workflow type

> **Note**
>
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To create a workflow, you must bind a workflow document class to the workflow type. The workflow document class contains references to the table data fields that the workflow uses. This article describes how to bind a workflow document class to a workflow type.

> **Note**
>
> If you use the **Workflow** wizard to create the workflow type, the wizard already bound the workflow document class to the workflow type.

Before you begin the following procedure, create a workflow document class to expose the table fields that are used for conditions in the configuration user interface (UI). For more information, see Create a workflow document class.

Bind a workflow document class to a workflow type

1. In Application Explorer, expand the **Workflow** node.
2. In the **Workflow Types** node, right-click the workflow type that you want to bind a workflow document class to, and then select **Properties**.
3. On the **Properties** sheet, set the **Document** property to the workflow document class that defines the workflow document.

See also

Create a new workflow type

Create a workflow document class

Last updated on 03/18/2026

Workflow types report

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This article points you to a report that lists the available workflow types.

To create a workflow, select the *type* of workflow that you want to create. For example, you might want to create a **Purchase requisition line review** workflow to define who must approve line items on purchase requisitions. For more information about creating a workflow, see Create workflows overview.

View the report

The **Workflow types report**, included with the Technical reference reports, lists each type of available workflow. The report also describes what each type of workflow is used for and indicates whether the workflows of each type are associated with a specific company in the organization or with the whole organization.

Last updated on 03/18/2026

Workflow FAQ

Summarize this article for me

> **Note**
>
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This article answers frequently asked questions about the workflow system.

Why do I receive multiple notifications when a work item is rejected?

When you reject a work item, the system completes that work item as rejected. It also creates and assigns a new work item to the originator. The system sends a notification to the originator for the rejected work item, and it sends a separate notification to the user assigned to the new "change requested" work item.

Each notification corresponds to a different work item, but their similarity can cause confusion.

Why are my workflow exports failing?

The workflow export feature currently doesn't support workflow names longer than 48 characters. If you use a name that's longer than 48 characters, you might see a "Server failed to authenticate the request" error. You might also be unable to export a file with a file type. For more information, see Workflow export troubleshooting.

Can the submitter of a workflow approve the workflow?

Yes, a submitter of a workflow can approve the workflow unless the setting **Disallow approval by submitter** is enabled for the workflow. To enable this setting, go to **System administration** > **Workflow** > **Workflow parameters** > **General** > **Approver**.

Can I add alerts to workflows to provide notifications to users?

Here are a few key areas to note about adding alerts to workflows to provide notifications:

- Alerts versus workflow notification mechanisms
  - You can set up alerts for record changes. Workflows change records, so you can set up an alert related to a record change caused by a workflow. However, because workflows have different built-in notification options, using alerts isn't ideal.
- Standard notifications from workflows
  - Workflows have built-in email notifications. You can configure the notifications so users are sent emails. Those notifications don't result in Action Center messages for users.
  - In a future update, an Action Center message will be added so a user is assigned a workflow work item.
- Adding notifications to workflows
  - You can create Action Center messages for specific users, such as a message created from a workflow in X++.
  - Workflows have business events that you can use to trigger Flows have the notifications that they're looking for.

If a user doesn't get the proper notification from the Action Center when they're assigned a workflow work item, Workflow business events with Microsoft Power Automate can provide additional or different notifications.

There's some overlap between workflow and Microsoft Power Automate.

- Workflows that you create in Dynamics 365 Finance run inside Dynamics 365 Finance.
  - Use Dynamics 365 Finance to set up workflows that are specific to your Dynamics 365 Finance.
- Power Automate allows you to set up workflows that run outside of Dynamics 365 Finance. For more information, see Power Automate.
  - Use Power Automate to set up flows that involve information that flows through your company.

Why can't the workflow editor start under AD FS?

When running under Active Directory Federation Services (AD FS) in an upgraded environment, the workflow editor might have trouble starting. If it does, make sure that you add the URL `<https://dynamicsaxworkfloweditor/>` to the property **Microsoft Dynamics 365 for Operations On-premises - Workflow - Native application** in the ADFS settings.

Why am I getting SQL deadlocks on workflow processing?

The default field value for the **Number of workflow items per batch** on the **Workflow parameters** page is 0. A value of 0 sets the default to 20 items per batch. Be careful when adjusting this value because a high number of items per batch (> 40) can cause SQL deadlocks.

What is the Workflow Enhanced Error feature?

The Workflow Enhanced Error feature in version 10.0.13 adds error codes that differentiate various classes of workflow errors. The feature also provides error messages that are mostly similar but include minor differences to make them clearer.

What should I do if I encounter error code CAAC000E when signing in to the workflow editor?

You might encounter error code CAAC000E ("**Something went wrong. User is not eligible to enroll a device. Reason: DeviceCapReached**") when you sign in to the workflow editor if your account reaches the Microsoft Intune device registration cap that your organization configured. To ignore this error, select **Continue** to proceed with using the workflow editor to create or edit workflows.

To avoid this error in the future, select **No, sign in to this app only** the next time you sign in to the workflow editor, or contact your IT administrator to raise your device cap.

Last updated on 03/11/2026

Alerts overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

About alerts

Alerts from a notification system for critical events in the system. Use alerts to stay informed about events that you want to track during the workday. You can easily create your own set of alert rules so that you're alerted about deliveries that are overdue, orders that are deleted, prices that change, or other events that you must respond to.

In enterprise resource planning (ERP), there are several typical scenarios where you can use the alerts feature. Here are some examples.

Scenario 1: Create an alert rule for new sales orders

To create an alert rule for new sales orders, follow these steps:

1. Open the **All sales orders** page.
2. On the Action Pane, on the **Options** tab, in the **Share** group, select **Create a custom alert**.
3. In the **Create alert rule** dialog box, on the **Alert me when** FastTab, in the **Event** field, select **Record has been created**.

Scenario 2: Create an alert rule for postponement of a delivery date

To create an alert rule for new sales orders, follow these steps:

1. Open the **All purchase orders** page.
2. Select a purchase order ID to access the purchase order details.
3. Expand the **Purchase order header** FastTab.
4. On the Action Pane, on the **Options** tab, in the **Share** group, select **Create a custom alert**.
5. In the **Create alert rule** dialog box, on the **Alert me when** FastTab, in the **Field** field, select **Delivery date**.
6. In the **Event** field, select **has been postponed**.

After you close the **Create alert rule** dialog box, your rule appears on the **Manage alert rules** page. You can use the **Manage alert rules** page to update your existing alert rules. For example, you can modify event triggers, update event notifications, and update expiration dates. To open the **Manage alert rules** page, use the **Alert me** button on the **Options** tab of the Action Pane.

What happens when you create an alert rule?

When you create alert rules, you can associate a predefined event with a specific field. For example, the event can be the date that you specify in the field, or the contents of the field change. Alternatively, you can associate an event with the records on a specific page. For example, a record is created, or a record is deleted.

When the selected event occurs for the field or for a record on the page, an alert is sent to you. For example, you create a rule where you associate the **Delivery date** field on a specific purchase order line with the **was due this amount of time ago** event. You set the time frame to five days. In this case, an alert is sent five days after the delivery date of that purchase order line.

Additionally, you can refine alert rules by setting conditions. For example, you can be alerted about new purchase orders that are created for specific vendor accounts.

Preparing for an alert

Before you set up an alert rule, decide when or in what situations you want to receive alerts. When you know which event you want to be notified about, find the page where the data that causes that event appears. The event can be a date that arrives or a specific change that occurs. Therefore, you must find the page where the date is specified, or where the field that changes or the new record that is created appears. After you have this information, you can create the alert rule.

Components of an alert rule

An alert rule has five components:

- **Event** – The event that triggers an alert rule can be a date that arrives or a specific change that occurs. You define events on the **Send email alerts for job status changes** FastTab of the **Create alert rule** dialog box.
- **Condition** – On the **Alert me for** FastTab of the **Create alert rule** dialog box, you can select the scope of the condition, to control when you're alerted about events. You can

apply the rule either to the current record only or to all visible records on the page. If the rule applies across legal entities, you can set the **Organization-wide** option to **Yes**.

- **Expiry of rule** – On the **Alert me until** FastTab of the **Create alert rule** dialog box, you can specify how long the alert rule should be active.
- **Contents** – On the **Alert me with** FastTab of the **Create alert rule** dialog box, you can specify the subject text and message text that the alert messages should use.
- **User** – On the **Alert who** FastTab of the **Create alert rule** dialog box, you can specify which user should receive the alert messages. By default, your user ID is selected.

> **Note**
>
> This option is restricted to organization administrators.

Videos

How to use alerts to monitor filtered data

https://learn-video.azurefd.net/vod/player?id=f9d42474-b1ab-40a1-a938-06175dfa9554&locale=en-us&embedUrl=%2Fdynamics365%2Ffin-ops-core%2Ffin-ops%2Fget-started%2Falerts-overview

The How to use alerts to monitor filtered data video (shown preceding) is part of the finance and operations playlist on YouTube.

Alert rule options

https://learn-video.azurefd.net/vod/player?id=24a11730-58ba-4054-9b20-9b708073999a&locale=en-us&embedUrl=%2Fdynamics365%2Ffin-ops-core%2Ffin-ops%2Fget-started%2Falerts-overview

The Alert rule options video (shown preceding) is part of the finance and operations playlist on YouTube.

Last updated on 11/26/2025

Create alert rules

> **Note**
>
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Getting started

Before you set up an alert rule, decide when or in what situations you want to receive alerts. When you know which event you want to be notified about, find the page where the data that causes that event appears. The event can be a date that arrives or a specific change that occurs. Therefore, you must find the page where the date is specified, or where the field that changes or the new record that is created appears. After you have this information, you can create the alert rule.

When you create an alert rule, you define the criteria that must be met before an alert is triggered. Criteria are basically the match between the occurrence of an event and the fulfillment of specific conditions. When an event occurs, the system starts to perform a check according to the conditions that are set up.

Ensure the alert batch jobs are running

The batch jobs for data change and due date alerts need to be running for the alert conditions to be processed and the notifications to be sent. To run batch jobs, go to **System administration** > **Periodic tasks** > **Alerts** and add a new batch job for **Change based alerts** and/or **Due date alerts**. If a long and frequently running batch job is needed, select **Recurrence** and set **No end date** with a **Recurrence pattern** of **Minutes** and a **Count** of **1**.

Events

The event that triggers an alert rule can be a date that arrives or a specific change that occurs. You define triggers for events on the **Alert me when** FastTab of the **Create alert rule** dialog box. The events that are available for a particular field depend on the trigger that you select.

For example, if you're setting up an alert rule for the **Start date** field, due date events are appropriate. Therefore, the `is due in` event type is available for that field. However, for a field

such as **Cost center**, a due date event isn't appropriate. Therefore, the `is due in` event type isn't available. Instead, the `has changed` event type is available.

Event types

Three types of events can occur:

- **Create-type and delete-type events** – These events trigger an alert when a record is created or deleted.
- **Update-type events** – These events trigger an alert when the data in a specific field changes.
- **Due date-type events** – These events trigger an alert when a date arrives.

Changes that occur can be initiated by a user. For example, a user changes the delivery date of a purchase order. Alternatively, changes can occur as part of a process. For example, the **Status** field on a page changes to reflect the life cycle of various processes in the system.

Conditions

On the **Alert me for** FastTab in the **Create alert rule** dialog box, you can use conditions to control when you're alerted about events.

For example, you can specify that the system should alert you when the status of purchase orders changes, but only if the status matches a specific set of conditions. Specifically, you want to be alerted when the status of a purchase order is set to **Received**. This change in status is the event that triggers the alert.

Next, you must decide which purchase orders you want to be alerted about. For example, you can select one of the following options. These options define the conditions for the alert rule.

- **Current selected record** – You receive an alert when the status of a specific purchase order changes to **Received**.
- **All records** – You receive an alert when the status of a purchase order is changed for an item in the active page view. You can use the advanced filtering that's available on the page to create rules for a specific set of records. For example, you can create an alert that is triggered for all purchase orders for the customers in a specific customer group.

Expiry of rule

On the **Alert me until** FastTab of the **Create alert rule** dialog box, you can specify how long the alert rule should be active.

Alert contents

On the **Alert me with** FastTab of the **Create alert rule** dialog box, you can specify the subject text and message text that the alert messages should use.

User ID

On the **Alert me with** FastTab of the **Create alert rule** dialog box, you can specify which user should receive the alert messages. By default, your user ID is selected. The ability to change the user receiving the alert is restricted to organization administrators.

Alerts as business events

You can send alerts externally using the business events framework. When creating an alert, set **Organization-wide** to **No** and set **Send externally** to **Yes**. After you have the alert triggering the business event, you can trigger a flow built in Power Automate using the **When a business event occurs** trigger on the finance and operations connector, or explicitly send the event to a business events endpoint via the **Business events catalog**.

Create an alert rule

To create an alert rule, follow these steps:

1. Ensure the alert batch jobs are running (see preceding).
2. Open the page that contains the data to monitor.
3. On the Action Pane, on the **Options** tab, in the **Share** group, select **Create alert rule**.
4. In the **Create alert rule** dialog box, in the **Field** field, select the field to monitor.
5. In the **Event** field, select the type of event.
6. On the **Alert me for** FastTab, select the desired option. If you want to send the alert as a business event, set the **Organization-wide** value to **No**.
7. If the alert rule should become inactive on a specific date, on the **Alert me until** FastTab, select an end date.
8. On the **Alert me with** FastTab, in the **Subject** field, accept the default subject heading for the email message, or enter a new subject. The text becomes the subject heading for the email message that you receive when an alert is triggered. If you want to send the alert as a business event, set **Send externally** to **Yes**.
9. In the **Message** field, enter an optional message. The text becomes the message that you receive when an alert is triggered.
10. Select **OK** to save the settings and create the alert rule.

Limitations and workarounds

Workaround for creating alerts for the secondary data sources of a form

You can't create alerts for some secondary data sources on forms. For example, when creating alerts on the customer or vendor posting profiles form, only the fields on the header (CustLedger or VendLedger) are available and not the dimension accounts. The workaround for this limitation is to use **SysTableBrowser** to open that table as primary data source.

1. Open the table in the **SysTableBrowser** form.

   ```
   https://<EnvironmentURL>/?cmp=USMF&mi=SysTableBrowser&TableName=<TableName>
   ```

2. Create an alert from the SysTableBrowser form.

Change based alerts don't work for batch status changes

Change based Alerts doesn't work with batch status changes because it's turned off for performance reasons. Instead, you should set up the **Batch alerts** capability. For more information, see Set up alerts for batch enhanced forms.

Last updated on 11/20/2025

Batch processing of alerts

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Batch processing functionality processes alerts. You must set up batch processing before the process and deliver alerts.

Batch processing functionality supports two types of events:

- Events triggered by change-based events. These events are also referred to as create, delete, and update events.
- Events triggered by due dates.

You can set up batch processes for each type of event.

Batch processing for change-based events

The system reads all change-based events that occur since batch processing last ran. Change-based events include updates to fields, the deletion of records, and the creation of records. The system compares these events with the conditions that you set up in alert rules. When an event matches the conditions in a rule, the batch process generates an alert.

Frequency for change-based events

For change-based events, you can set up a batch job that triggers the processing of an event soon after the system logs the event. If you set up the batch job to recur more often, users receive their alerts sooner after changes occur. However, a high frequency for batch processing might adversely affect system performance.

On the other hand, a batch job that recurs less often, and that you schedule for times when the system load is low, might help improve system performance. However, a low frequency for batch processing might not meet the users' demands for timely alerts.

Therefore, when you set up the frequency of batch processing for change-based events, consider the balance between the timeliness of alerts and the performance of the whole system. These considerations become more relevant as the number of users who create alert

rules increases. The frequency doesn't affect the number of events that the system processes. However, if more users create rules, the process runs more checks. This type of data exchange might affect system performance.

The risks of low batch frequency

If you set up a low frequency for batch processing for change-based events, data that's relevant to the conditions in alert rules might change before processing. Therefore, you might lose alerts.

For example, you create an alert to trigger when the event is **customer contact changes** and the condition is **customer = BB**. In other words, when the customer contact for customer BB changes, the process logs the event. However, the batch processing system is set up so that batch processing occurs less often than data entry. If the customer name changes from **BB** to **AA** before the event is processed, the data in the database no longer matches the condition in the rule, **customer = BB**. Therefore, when the event is finally processed, no alert is generated.

Set up processing for change-based alerts

To set up processing for change-based alerts, follow these steps:

1. Go to **System administration** > **Periodic tasks** > **Alerts** > **Change based alerts**.
2. In the **Change based alerts** dialog box, enter the appropriate information.

Batch processing for due-date events

The system detects all events that are caused by due dates, and these events are compared with the conditions that are set up in alert rules. The batch process generates an alert when an event matches the conditions in a rule.

Frequency for due-date events

For due-date events, you might want to set up batch jobs that are run during the night or at specific times of the day, to balance the system load. We recommend that you set up the batch job so that it's run at least one time per day. If alerts should be sent as early as possible, set up the batch processing to occur immediately after the system date changes. If you want to generate alerts for due-date events that occur after a batch job has already processed alerts, you can run the batch job again on the same day.

For example, a batch job has been run on a particular day. You then create a purchase order that has a due date that should trigger an alert on that same day. To receive the alert on that

day, you must run the batch job again after the purchase order is created. However, if you don't run the batch job again on that day, the next day's batch job detects any due-date events that weren't processed on previous days.

> **Note**
>
> Even when the batch process is run more than one time per day, alerts aren't duplicated for the same due-date event and conditions. Alerts are generated only for dates that have become due because of changes that occurred in the system after the last batch job was run.

Batch processing window

The processing of alert rules in a company can be stopped for several reasons. These reasons include vacations, system errors, or other issues that prevent the batch jobs from being run for some time.

To prevent due-date alerts from becoming obsolete because the batch job hasn't been run for several days, you can set up a batch processing window. A batch processing window can be used to prevent a batch job from being run for a specified number of days.

If you set up a batch processing window, an alert is sent when the alert rule is processed, even if the alert exceeds the time limit that is defined in the due-date criteria. An alert continues to be sent for as long as the period that is defined by this time limit plus the batch processing window isn't exceeded. However, when the period exceeds the value defined by the time limit plus the batch processing window, an alert is no longer sent.

Set up processing for due-date alerts

To set up processing for due-date alerts, follow these steps:

1. Go to **System administration** > **Periodic tasks** > **Alerts** > **Due date alerts**.
2. In the **Due date alerts** dialog box, enter the appropriate information.

Last updated on 11/20/2025

Client alert notifications by email

You can define custom alert rules that monitor filtered views of data and automatically send email notifications when predefined events occur. The option to send email notifications is available for all supported alert types, and you can also turn on email notifications for existing alert rules.

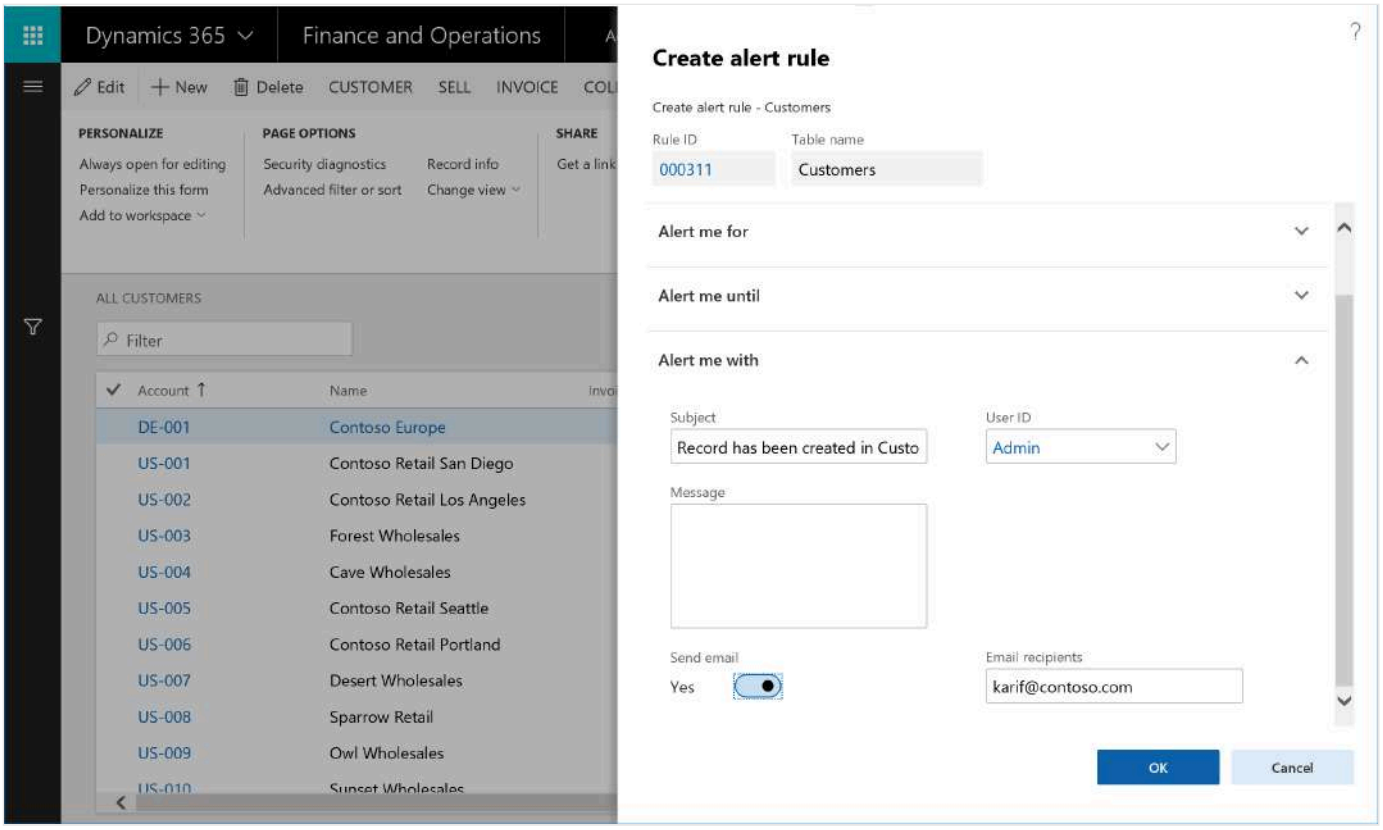
You can use built-in controls to create alert rules that monitor the filtered views of system batch jobs. By monitoring the value of the **Status** field, you can also configure alert rules that send email when a batch job fails. After you create these alert rules, you no longer have to check reports for changes to business data. Instead, you can let the intelligent change detection service do the monitoring for you.

Client alerts depend on the email subsystem that Microsoft Office integration provides. We recommend that you use the Simple Mail Transfer Protocol (SMTP) provider, so that email distribution doesn't have to rely on a local mail client.

To send notifications by email, customers must configure integrated email services. Email notifications are sent to recipients on behalf of alert owners.

For more information about how to configure email, see Configure and send email.

The following image shows the **Create alert rule** dialog box, which now includes a **Send email** option.

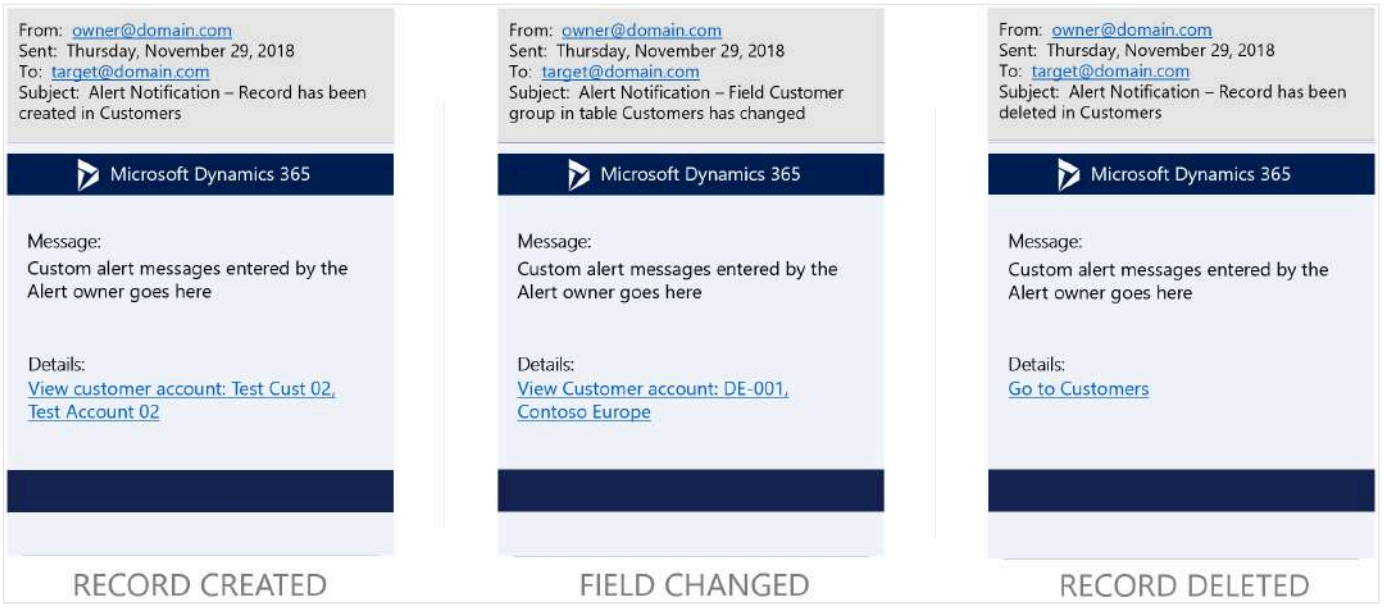


> **Note**
>
> When the **Send email** option is set to **Yes**, alert notifications continue to be delivered from the Action Center.

Alert notification email templates

The service sends email notifications by using predefined email templates that deliver the basic details of the alert notification.

The following image shows the structure of the alert notifications when they're received by email.



Last updated on 06/26/2026

Electronic signatures overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an overview of electronic signatures and describes how you can use them.

What is an electronic signature?

An electronic signature confirms the identity of a person who starts or approves a computing process. In some industries, an electronic signature is as legally binding as a handwritten signature.

Electronic signatures are a regulations compliance requirement for several regulated industries, such as pharmaceuticals, food and beverage, and aerospace and defense. They're also required for compliance with regulations in 21 CFR Part 11 that the Food and Drug Administration (FDA) in the United States issued.

> **Note**
>
> An electronic signature by itself isn't the same as a digital signature. An electronic signature is just a substitute for a handwritten signature, whereas a digital signature provides other security measures. A digital signature can help identify whether another user or process has tampered with the data. A digital signature can also be verified, and this verification can't be refuted by the owner of the certificate that was used to sign the data. As described in the following section, electronic signatures have built-in digital signature functionality.

Electronic signatures

You can use electronic signatures for critical business processes. Some processes have built-in electronic signature capabilities. You can also create custom signature requirements for any database table and field.

Electronic signatures have built-in digital signature functionality. Every user who signs documents must obtain a valid cryptographic certificate. When a user signs a document, the private key that is associated with that certificate is validated. The system records electronic signature information in a log to provide an audit trail. To set up electronic signatures, see Set up electronic signatures.

Users who require access to electronic signatures

Three kinds of users typically require security access to electronic signatures: electronic signature administrators, signers, and electronic signature auditors.

Electronic signature administrator

The electronic signature administrator sets up signature requirements, general parameters, and approvers, and receives alerts when signatures can't be verified. By default, a user who belongs to the **Information technology manager** security role has permission to administer electronic signatures.

Signer

A signer provides electronic signatures for documents and processes that require signatures. By default, a user who belongs to the **System user** security role has permission to sign documents electronically.

> **Note**
>
> The signer might need extra permissions to access data related to the document or process that they're signing. A user who changes data and then signs for those changes must have permission to change the data. A user who signs on behalf of another user might not need access to the data. An example of this kind of user is a supervisor who signs for an employee's changes.

Electronic signature auditor

The electronic signature auditor reviews the database log and the signature review log that's available from the database log. By default, a user who belongs to the **Information technology manager** security role has permission to audit electronic signatures.

If you use a role other than **Information technology manager**, make sure that the role is assigned to the following privileges:

- View electronic signature failures
- View database log

Signing documents electronically

Get a certificate

Before you sign documents electronically, you must request a certificate.

> ⓘ **Note**
>
> Microsoft SQL Server features are used to create certificates and enable electronic signing. You don't need any other certificate or public key infrastructure (PKI).

When you request a certificate, the system creates a public key and a private key for you. You encrypt the private key by using a password that only you know. When you sign a document electronically, you verify your identity by entering the password.

To request a certificate, on the **Options** page, on the **Accounts** tab, select **Get certificate**.

You must enter and confirm the password that you use for signing. The password protects your private key and authorizes the use of your certificate. The database doesn't store this password, and no one else has access to it, not even the administrator.

If you forget the password that connects with your certificate, you must reset the certificate. Resetting the certificate doesn't affect documents that you signed by using the previous certificate. To reset the certificate, on the **Options** page, select **Reset certificate**.

Sign a document electronically

The **Sign document** page appears when you make a change that requires an electronic signature.

1. On the **Sign document** page, select the **Document** tab to review the changes to the document.
2. On the **Signature** tab, select a reason code.
3. Enter a comment, if a comment is required.
4. If your user ID doesn't appear in the **Signer** field, select it in the list.
5. Enter your location, if this information is required.
6. Select **OK**.

Sign for another user's changes

Occasionally, you might want a user to sign for another user's changes. For example, a supervisor might be required to sign for changes that an employee makes to a bill of materials (BOM). Use this procedure to designate a user as a signer for another user.

> **Note**
>
> When one user signs for another user's change, the signature must be provided at the workstation of the user who made the change. The user can't save the change until the signature is provided.

To designate approvers, follow these steps.

1. On the **Options** page, on the **Accounts** tab, select **Designate approver**.
2. In the **Approver user ID** field, select the ID of the user who must sign for another user's changes.
3. In the **Sign for user ID** field, select the ID of the user whose changes must be signed for.

Last updated on 11/26/2025

Set up electronic signatures

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

An electronic signature confirms the identity of a person who is about to start or approve a computing process. In some industries, an electronic signature is as legally binding as a handwritten one.

Electronic signatures are a regulations compliance requirement for several regulated industries, such as pharmaceuticals, food and beverage, aerospace, and defense. They're also necessary for compliance with regulations that the United States Food and Drug Administration (FDA) issues in part 11 of Title 21 of the Code of Federal Regulations (21 CFR Part 11).

Electronic signatures aren't the same as digital signatures. An electronic signature is just a substitute for a handwritten signature, whereas a digital signature provides additional security measures. A digital signature can help identify whether another user or process tampered with the data. A digital signature can also be verified, and this verification can't be refuted by the owner of the certificate that was used to sign the data. Although electronic signatures and digital signatures differ, they can work together. For example, in finance and operations apps, electronic signatures have built-in digital signature functionality.

You can use electronic signatures for critical business processes. Some processes have built-in electronic signature capabilities. You can also create custom signature requirements for any database table and field. As was mentioned earlier, electronic signatures in finance and operations apps have built-in digital signature functionality. Each user who signs documents must obtain a valid cryptographic certificate. When a document is signed, the private key that's associated with that certificate is validated.

Enable the Electronic signature configuration key

1. Go to **System administration** > **Setup** > **License configuration**.
2. In the tree, expand **Administration**, and verify that the **Electronic signature** checkbox is selected. If the checkbox is cleared, you must enable maintenance mode before you can select it. You can enable maintenance mode in the environment by running a

maintenance job from Microsoft Dynamics 365 Lifecycle Services. Alternatively, you can use the Deployment.Setup tool locally. After you select the **Electronic signature** checkbox, turn off maintenance mode.

3. Close the page.

Set up electronic signature parameters

Prerequisites for passphrase constraints (preview)

[This section is prerelease documentation and is subject to change.]

Most electronic signature features are standard for all current versions of Microsoft Dynamics 365 Supply Chain Management. However, passphrase constraints and the ability to require comments are recent enhancements. Before you can use these features, your system must meet the following requirements:

- You're running Supply Chain Management version 10.0.44 or later.
- You turn on the **(Preview) Electronic signature improvements** feature in Feature management.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Set up the electronic signature parameters

1. Go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature parameters**.
2. Select **Edit**.
3. Set the following fields:
    - **Notice** – Enter the notice that is shown to signers on the **Sign document** page. You can enter any text. Typically, this notice informs the user what it means to sign a

document electronically, including legal implications. If you want to enter the notice text in more languages, select **Translations** on the Action Pane.

- **Require comments** – Set this option to *Yes* if all signers must enter a comment when they sign.
- **Signature timeout** – Enter a timeout period (in seconds). If np signature is provided within this period, the signing process fails.
- **Key expiry** – Enter the number of days that a digital key remains valid after it's created.
- **Signature alert recipient** – Select the user who should receive an alert email if signature validation fails. Validation fails if the system determines that the certificate and private key that a user provided show signs of tampering.
- **Minimum length** – Enter the required minimum length of the passphrase. We recommend a minimum of eight characters.
- **Minimum alpha** – Enter the required number of alphabetical characters in the passphrase.
- **Minimum numerals** – Enter the required number of numeric characters in the passphrase.
- **Minimum special** – Enter the required number of special characters in the passphrase. Special characters include the following symbols:
  - *!* (exclamation point)
  - @ (at sign)
  - # (number sign)
  - *$* (dollar sign)
  - *%* (percent sign)
  - ^ (circumflex accent)
  - *&* (ampersand)
  - * (asterisk)
  - *(* (opening parenthesis)
  - *)* (closing parenthesis)
  - *;* (semicolon)
  - *:* (colon)
  - *?* (question mark)
  - < (less than sign)
  - > (greater than sign)
  - - (hyphen)
  - + (plus sign)

- = (equal sign)
- ~ (tilde)

4. Select **Save**.
5. Close the page.

> **Note**
>
> All the parameters that are related to passphrase constraints work in conjunction with the Windows policy requirements. Although the passphrase constraints can be more complex than the Windows policy, they can't be less complex. For example, the Windows policy might require a minimum length of eight characters, including more than one numeric character and more than one special character. By contrast, the passphrase constraints might just require six characters. In this case, the password is rejected when the certificate is generated.

Set up reason codes for electronic signatures

Before electronic signatures can be used, you must set up reason codes. A valid reason code is required whenever a document is signed. The reason code indicates the purpose of the electronic signature. For example, a reason code can indicate legal approval.

1. Go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature reason codes**.
2. Select **New**.
3. In the **Reason code** field, enter a value.
4. In the **Description** field, enter a value.
5. Repeat these steps to add more reason codes as required.
6. Select **Save**.
7. Close the page.

Require electronic signatures for existing processes

1. Go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature requirements**.
2. In the list, find and select the desired record.
3. Select a process that requires electronic signatures.
4. Select or clear the **Signature required** checkbox.
5. Repeat these steps for every other process that requires electronic signatures.

6. Select **Save**.

Create a custom requirement for electronic signatures

1. Go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature requirements**.
2. Select **New**.
3. Select or clear the **Signature required** check box.
4. In the **Name** field, enter a name for the process that requires electronic signatures.
5. In the **Table name** field, select the dropdown button to open the lookup. Find and select the table that store the data that must be signed. Then select the link in the selected row.
6. In the **Field name** field, select the dropdown button to open the lookup. Find and select the table field that you want to monitor. Then select the link in the selected row.
7. Select one of the following values to specify when a signature is required:
    - *Always* – Always require a signature when the data in the field is changed.
    - *Only* – Require a signature only under certain conditions. If you select *Only*, you must also select one of the following options: *When a record is inserted*, *When a record is updated*, or *When a record is deleted*.
8. Select **Save**.
9. Close the page.

Last updated on 03/05/2026

Case management overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

By planning, tracking, and analyzing cases, you can develop efficient resolutions that can be used for similar issues. For example, when customer service representatives or Human Resources generalists create cases, they can find information in knowledge articles to help them work more efficiently. The following examples show how cases can be used for different situations in an organization.

Example: How Fabrikam uses cases for customers in the private sector

Lisa, a customer service representative at Fabrikam, receives a telephone call from Lionel, a Fabrikam customer. Lionel is having trouble setting the correct volume level on the new sound system that Fabrikam installed in Lionel's music store. Lisa creates a case for Lionel and assigns the **Volume** category to the case. Lisa then elevates the priority and assigns a one-day service level agreement (SLA) to the case.

Lisa also enters the case details in the case log. Lisa notices that several knowledge articles are associated with the **Volume** category, and that three of them are marked as helpful for resolving cases. Lisa opens each article and discusses the resolution steps with Lionel, but none of the solutions solve the issue that Lionel is having with the new sound system. Lisa tells Lionel that an audio technician calls within 24 hours and work to solve the issue.

Lisa activates the case, and a set of activities is created. Lisa assigns the activities to Terrence, a member of the audio engineering team. Terrence sees that new activities are assigned, opens the case and reads the case log to learn more about the case. Terrence encountered the same issue the day before, and developed a solution. Terrence contacts Lionel and offers this solution for the issue. Terrence also enters it in the case details.

Because the solution is successful, Terrence decides to document it, so that other people can use it if they encounter the same issue. Terrence adds the document to the **Knowledge article** page, assigns the document to the **Volume** category, and manually elevates the ranking, so that other Fabrikam employees know that it's a successful solution. Terrence then elevates the

case to the next level. By elevating the case, Terrence creates a new activity for Marie, who is a quality assurance representative in the customer service department.

Marie sees that a new activity is assigned, and opens the case that is associated with the activity. Marie reviews the case and the case details to make sure that the correct process was followed for the case. Marie verifies that the actual case time didn't exceed the time frame that was estimated in the SLA. Marie notes that Terrence contacted the customer, and that the issue was resolved. Marie is satisfied with the treatment that the customer received and the results of the case. Marie resolves the case as closed. On closing the case, the open activity that is assigned to Marie is also closed.

Example: How City Power & Light uses cases for customers in the public sector

Annie, a customer service representative with City Power & Light, receives a telephone call from a resident of the city that City Power & Light serves. Annie records the call as an activity and takes notes of the conversation. The resident tells Annie that their house has no power. Annie informs the resident that City Power & Light will investigate, find, and resolve the issue as quickly as possible.

Annie then creates a case, associates the telephone call with the case, and creates a service order. Annie knows that other residents are likely to call to report a power outage. Therefore, to avoid overwhelming the customer service center, and to save time, Annie sends a group instant message (IM) to inform the other representatives about the issue, and to tell them that a case and service order have been created. Annie includes the case number and service order number in the IM. Then, if City Power & Light receives more telephone calls about the power outage, the customer service representatives can create an activity for each telephone call and assign it to the existing case.

Example: How Fabrikam uses cases for employees

The following scenarios show how Fabrikam Human Resources generalists in different locations can use case management when they address issues for employees.

In England

Christine, the Human Resources generalist for the English division of Fabrikam, receives a telephone call from Peter, a Fabrikam employee. Christine is informed that nine weeks ago, immediately after the birth of Peter's son, Peter changed the number of dependents on the tax withholdings. Peter wants to know why the changes haven't become effective. Christine creates

a case for Peter and reviews Peter's tax information. Christine learns that, although Peter entered new dependent information, a start date for the new tax withholdings wasn't selected. Christine informs Peter to select a start date and resubmit the changes. Peter replies to Christine's message informing that Peter has now selected a start date and resubmitted the changes. Christine attaches the email message from Peter to the case record, verifies that the correct changes were made and submitted, and closes the case.

In the United States

Luke, the Human Resources generalist for the United States division of Fabrikam, receives an email message from Shannon, a Fabrikam employee. Shannon is a machine operator who was injured on the job six months ago, and has been working with Humongous Insurance to have the medical expenses paid. Because Shannon contacted Luke about this issue four weeks ago, a case has already been created. Shannon's new email message explains that Humongous Insurance is still not returning the telephone calls. Luke opens the existing case, adds Shannon's email message as a document, and reviews the case log. When creating this case, Luke assigned the **Insurance** category to it but now sees that there's a new knowledge article that is associated with the **Insurance** category. Luke reads the knowledge article and learns that all phones at Humongous Insurance are down while the company's telephone system is being updated. The article states that the insurance company sent an email message to all its customers, but that several customers didn't receive the message because of a problem with the company's email system. All customers who have active insurance claims are asked to send their inquiries to Humongous Insurance by email or paper mail. Luke sends Shannon an email message that explains what Shannon must do to have the insurance claim settled. Luke also ranks the knowledge article as a helpful source of information. Luke creates another activity to follow up with both Shannon and Humongous Insurance in four weeks and make sure that the claim has been resolved. After four weeks, Luke contacts Shannon and learns that Humongous Insurance has paid the claims, and that Shannon is happy with the resolution. Luke changes the status of the case to **Closed**.

Last updated on 12/02/2025

Plan case category security, case processes, and case categories

This article describes the considerations and decisions that you must make during the planning process, before you begin to configure cases.

You can use the case functionality to resolve both issues that are external to your organization and internal issues.

Case category security by role

Only appropriate employees in an organization should have access to cases and related information. To control which employees have access to view, create, and update different types of cases, assign security roles to case category types. Determine which security roles should have access to the various case category types.

Decision: Determine which security roles should have access to the following case category types. Your organization might not use all these category types, so make decisions only for those categories that are appropriate.

- General
- Sales
- Purchase
- Service
- Project
- Production
- Collections
- Audit
- Web
- Human Resources
- Product change
- FMLA

Case processes

Set up processes that employees must follow for the cases they open in your organization. Processes help guarantee consistency for the people involved in cases, and they help employees

resolve cases faster and more efficiently. Set up a process for each case category that you assign cases to. Although planning a separate process for each case type takes time, case resolution goes much more smoothly if you plan out the processes.

Decisions: For each case process, make the following decisions:

- What are the name and description of the process?
- Is the process active, and should employees use it when they handle a case that the process is assigned to?
- Who in the organization is responsible for applying the process to a case? For example, cost accounting or human resources might be responsible for some case processes. Multiple areas can be responsible for completing one process.

Case categories

The case category hierarchy provides a list of categories that you can assign cases to (see the "Case category security by role" section). Each top-level category includes subcategories, so that you can create more specific categories for the cases that your organization works with. Review the list of existing categories and subcategories to determine whether you need to create more. If you need to create more categories and subcategories, make the following decisions for each addition.

Decisions:

- Are you creating a new top-level category?
  - What is the name of the category?

- Are you creating a new subcategory?
  - What is the parent category of the subcategory?
  - What is the name of the subcategory?
  - Which worker owns the subcategory?
  - What department is the subcategory assigned to?
  - What case process is assigned to this subcategory?
  - What is the default service level agreement (SLA) that is assigned to this subcategory?
  - Should an activity be created when a case that this subcategory is assigned to is opened?
  - If so, what is the activity category, type, purpose, and phase?
  - Should a follow-up activity for the case be created?
  - If so, what is the follow-up activity category, type, purpose, and phase?

Last updated on 06/26/2026

Record templates overview

This article introduces the concept of record templates and explains how you can use them to create records that share information.

Record templates can help you create records more quickly. However, you can only create record templates for some record types.

For example, imagine you're entering rental information for a car rental business that's located in San Francisco. Since most of the customers are likely to be from the San Francisco area, it would be helpful if you could automatically fill in the values for the **State**, **Country/Region**, and **City** fields on the rental form.

> **Note**
>
> You can apply templates only in areas that you have access to. However, you can see all template titles when you create a new record. Other users can also see all template titles if you create templates that are available to all users. Be sure to consider this visibility when naming templates. For example, avoid using names that include words such as "commission" if it's confidential that some employees in the company have commission-based salaries.

When one or more templates that you have access to exist for a specific form and you attempt to create a new record in the form, the **Select a template for** page is displayed. When you select a template from the list, the new record is created and contains default information based on the template that you selected. If you don't want to use templates when you create new records, select the **Do not ask again** check box in the **Select a template for** page. To display the template selection dialog box again, right-click any record, select **Record info**, and then select **Show template selection**.

Last updated on 06/26/2026

Create a record template to facilitate data entry

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows how to create a record template so that you don't have to enter field values that you use often for each new record. In this procedure, you create a new record for new laptops that you add to your fixed assets. This procedure uses the USMF sample company.

To create a record template to facilitate data entry, follow these steps:

1. In the navigation pane, go to **Modules** > **Fixed assets** > **Fixed assets** > **Fixed assets**.
2. Select **New**.
3. In the **Fixed asset group** field, enter or select a value.
4. In the **Name** field, type a value. For example, enter **Corporate lead laptop**.
5. In the **Search name** field, type a value. For example, enter **laptop**.
6. Expand the **Technical information** section.
7. In the **Make**, **Model**, and **Model year** fields, type values.
8. On the Action Pane, select **Options**.
9. Select **Record info**.
10. Select **User template**.
11. In the **Name** field, type a value.
12. In the **Description** field, type a value.
13. Select **OK**.
14. Select **Close**.

Last updated on 11/26/2025

Use record template to create a new record

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to use a previously defined record template to create a new record. To complete this procedure, you must first complete the "Create a record template to facilitate data entry" procedure.

This procedure uses the USMF company.

1. In the **Navigation pane**, go to **Fixed assets** > **Fixed assets** > **Fixed assets**.
2. Select **New**. You're prompted to select a template. Select the one that corresponds to your business need.
3. In the list, find and select the desired record.
4. Select **OK**.

Last updated on 11/26/2025

Configure document management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure document management (document handling) so that it stores file attachments and notes for records. It includes information about the concepts and features that are involved in this functionality.

To learn more about document management, watch the short Document Management video.

Configure document types

Use document types to categorize the documents that you attach to records or the templates that you create. You can store each document type in a unique location.

The system provides a default set of document types. Use these document types to categorize an attachment as a file, image, note, or URL. The **File** and **Image** default document types are configured to use **Azure storage** as the location.

To create a new document type, follow these steps:

1. Go to the **Document types** page.
2. Select **New**.
3. In the **Type** field, enter a short name for the new document type, such as **SharePoint** or **HR Docs**.
4. In the **Name** field, enter a longer name, such as **SharePoint files** or **HR Docs**.
5. In the **Class** field, specify a class to define the behavior for the document type:
    - **Attach file** – The user is prompted for a file.
    - **Attach URL** – The user can enter a URL in the **Notes** field, such as `https://www.microsoft.com`. The **Open** button on the **Attachments** page opens the URL on a browser tab.
    - **Simple note** – The user can add a note in the **Notes** field.

6. If you specify **Attach file** in the **Class** field, use the **Location** field to specify the storage mechanism.

7. If you specify **SharePoint** in the **Location** field, select the **Edit** button (pencil symbol) and use the **Folder selection** dialog box to specify the Microsoft SharePoint address in the **SharePoint Address** field.

Configure SharePoint storage

SharePoint Online is a supported storage location for attachments.

> ⓘ **Important**
>
> - SharePoint storage is only available in Microsoft-managed environments.
> - On-premises SharePoint (a local SharePoint server) isn't currently supported.
> - SharePoint managed device policies aren't compatible with an integration to finance and operations apps.

One-time registration process

In Dynamics 365 finance and operations version 10.0.40 and later, when you enable the **SharePoint user authentication** feature, your organization needs to perform a one-time setup to use the SharePoint integration for non-interactive batch scenarios. This setup is necessary because Microsoft deprecated the Microsoft-managed high-trust connection between the finance and operations environment and SharePoint.

With this updated SharePoint authentication mechanism, batch connections use application access. As this access isn't granted by default for tenants, an Entra ID tenant administrator needs to grant access manually one time for the tenant. To provide the required application consent for finance and operations batch scenarios to connect to SharePoint, run the following PowerShell commands.

PowerShell

```
Import-Module Microsoft.Graph.Applications

# The parameter for TenantId needs to be changed
Connect-MgGraph -TenantId microsoft.onmicrosoft.com -Scopes
'Application.ReadWrite.All'

# These AppIds do not change as they are the first party application IDs
```

```
$erpServicePrincipal = Get-MgServicePrincipal -Filter "AppId eq '00000015-0000-0000-c000-000000000000'"
$sharePointServicePrincipal = Get-MgServicePrincipal -Filter "AppId eq '00000003-0000-0ff1-ce00-000000000000'"
$spAppRole = $sharePointServicePrincipal.AppRoles | where {$_.Value -eq 'Sites.ReadWrite.All'}

# Assign the SharePoint 'Sites.ReadWrite.All' permission to the Microsoft Dynamics 365 finance and operations application
New-MgServicePrincipalAppRoleAssignedTo -ServicePrincipalId $erpServicePrincipal.Id -PrincipalId $erpServicePrincipal.Id -ResourceId $sharePointServicePrincipal.Id -AppRoleId $spAppRole.Id
```

> **ⓘ Important**
>
> Interactive connections use the logged-in user's context.
>
> - As conditional access can pass through the finance and operations sign-in to SharePoint, ensure that conditional access settings for SharePoint are applied to finance and operations.
> - Calling SharePoint as a user that isn't the currently authenticated user is no longer supported.

Setting up SharePoint inside your finance and operations environment

To configure SharePoint storage, follow these steps:

1. Go to the **Document management parameters** page.
2. On the **SharePoint** tab, in the **Default SharePoint server** field, review the host name that the system automatically detects for the SharePoint site, such as contosoax7.sharepoint.com. Typically, the SharePoint host name is in the form `tenantname.sharepoint.com`, and accounts on that tenant are in the form `user1@tenantname.onmicrosoft.com`.

   Typically, if you don't specify a default SharePoint server, either there's no SharePoint site for the tenant, or a valid Microsoft 365 license isn't associated with the current user (the admin).
3. Optional: Test the SharePoint connection

   - If you disable the **SharePoint user authentication** feature, select **Test SharePoint connection** to test the specified SharePoint host name. This action verifies that the

security and license are working correctly.
    - If you enable the **SharePoint user authentication** feature, there are two SharePoint connection types to test. Use the **Test interactive SharePoint connection** action to test interactive scenarios by using the logged-in user's context. Use the **Test batch SharePoint connection** action to test batch scenarios by using the granted application access.

4. Optional: Select **Open SharePoint** to open the specified SharePoint host name in a browser. This action doesn't verify security. It just opens the SharePoint path on a browser tab for easy exploration.

5. Optional: On the **General** tab, turn on **Open attachments in new window**. For more information, see the Other configuration section later in this article.

To use SharePoint storage, set the **Location** field for a document type to **SharePoint**. Then, in the **SharePoint Address** field, enter a valid SharePoint address.

Troubleshooting SharePoint communication

SharePoint communication works for the current user only if the following conditions are met:

- A Microsoft 365 license is associated with the user's account.
- The user is a typical user on the tenant, not an external user (for example, a user from another tenant).
- There's a SharePoint site for the tenant (for example, Contoso.SharePoint.com).
- The SharePoint site is configured to **Allow this site to appear in search results**.
- The SharePoint site doesn't use managed device policies.

  If you enable managed device policies on the SharePoint instance, the finance and operations SharePoint integration stops working. Therefore, users can't download, view, or create documents that are stored in SharePoint from finance and operations apps.
- The user has access to the folder that the document is stored in.

If you can't open documents stored in SharePoint or they don't appear in preview, follow these steps to troubleshoot the issue.

1. Verify that the admin account has an associated email account. You can verify or change the associated email account on the **User** page. If you don't set up an associated email account, you must add the email account and provider via the OData Excel add-in. By

default, the email address isn't present in the Excel design. You must edit the Excel design, add all fields, apply the change, and refresh. You can then update the admin account.
2. After the admin account has an associated email account, sign in to Dynamics 365 as the admin.
3. Open an attachment that's stored in SharePoint.
4. Sign in by using a different user account that has read access to the attachments page and the configured SharePoint folder. Verify that this user account can open and preview the attachment.

Configure file types

By modifying the list of allowed file extensions, you can control the types of files that users can attach to records.

To specify file types, follow these steps:

1. Go to the **Document management parameters** page.
2. On the **File types** tab, review the default file types.
3. Remove any file types that users shouldn't attach to records. Add any file types that users should attach to records.

Configure document preview

The attachments preview uses the Web app Open Platform Interface (WOPI) that Office Online Server provides. On the **Document management parameters** page, on the **General** tab, in the **Office Web Apps Server** field, specify the Office Online Server instance to use for attachment previews. The default value is `https://onenote.officeapps.live.com`, which points to the cloud-based WOPI server.

> ⓘ **Note**
>
> For the following situations, you must adjust the **Office Web Apps Server** value:
>
> - For environments in China, use `https://onenote.partner.officewebapps.cn`.
> - For environments in the Government Community Cloud (GCC), use `https://gb4-onenote.officeapps.live.com`.

For a Dynamics 365 Finance + Operations (on-premises) environment

The default cloud-based WOPI server in Finance + Operations (on-premises) can't read the attachment file to provide a preview. If you need previews, you must install an on-premises Office Online Server instance and configure it inside the environment. Set the **Office Web Apps Server** field to the host name of the installed Office Online Server instance, and then select **Save**.

If you don't need previews, set the **Office Web Apps Server** field to `https://localhost`. The preview then shows the message "No preview available" instead of an error message.

Document preview (WOPI) doesn't work in environments where an IP allow list is enabled

Document preview (WOPI) doesn't work in environments where an IP allow list is enabled, because the WOPI service that provides the preview can't connect back to the file service to retrieve the file for rendering.

Other configuration

Consider these configuration options:

- On the **Document management parameters** page, on the **General** tab, use the **Use active document tables** option to enable the **Active document tables** allow list. If you set this option to **Yes**, you disable attachments on all other tables. Turn on this option only when it's required.
- On the **Document management parameters** page, on the **General** tab, use the **Maximum file size in megabytes** field to set the maximum file size for attachments. When you use SharePoint as a document type, users can upload only documents that have a maximum file size of 262 megabytes (MB).
- On the **Document management parameters** page, on the **General** tab, use the **Open attachments in new window** option to determine whether attachments are opened in place, or in a new window or on a new tab. Consider turning on this option especially if you use SharePoint to store attachments, because this option prevents the finance and operations user session from being reset when attachments are opened. This option is available as of version 10.0.23.
- On the **Options** page (**Settings** > **User options**), on the **Preferences** tab, use the **Enable document handling** option to disable document handling (document management).

Accessing document management attachments

Document management appears to users as the **Attach** button at the top of most pages that contain data. When you select the **Attach** button (or when you use the corresponding keyboard shortcut, **Ctrl**+**Shift**+**A**), you open the **Attachments** page in the context of the data source of the control that's currently selected on the page. This page shows all the attachments that are related to the corresponding data source.

The **Attach** button also shows a count of the attachments for the currently selected record. Therefore, you can determine whether there are attachments for the current record without having to open the **Attachments** page. The button shows exact counts for zero through nine attachments. If there are more than nine attachments, the button shows **9+** as the count. In this way, the performance impact and visual noise that exact larger counts might cause are reduced.

Document attachment history

As of version 10.0.16/Platform update 40, a history mechanism is available for record attachments. Therefore, your organization can maintain an audit of actions that are related to individual attachments. You can see when an attachment was created, marked for pending deletion, restored, deleted, or moved. You can also see who performed the action. Attachment history isn't maintained before version 10.0.16/Platform update 40. Therefore, any actions that were performed on attachments in earlier versions aren't available.

Configuration of document attachment history

To enable or disable document attachment history, go to **Document management parameters** > **General** > **History** > **Enable document history**. The default setting of the **Number of days to retain history** field is 180 days, but you can change it as you require.

Viewing an attachment's history

There are two entry points for viewing the history of a record attachment:

- When you're looking at the attachments for a specific record (see the Accessing document management attachments section), you can view the history for the current set of attachments on the **Attachments** page by selecting **Document history** on the Action Pane.
- Administrators can select the **Document history** button in the **History** section of the **Document management parameters** page. This action opens the **Document history** page, which shows a list of all attachments in the system. You can then drill into any record to view the detailed history for the selected attachment.

Attachment recovery

Platform update 29 introduced an attachment recovery feature. This feature provides a recycle bin for record attachments, so you can recover them within a configured period.

Configuration of attachment recovery

To enable attachment recovery, go to **Document management parameters** > **General** > **Deferred deletion** > **Deferred deletion enabled**. The default setting for the **Number of days to defer deletion** field is 30 days, but you can change it as you require. If you set the **Number of days to defer deletion** value to **0** (zero), deleted attachments are recoverable for an indefinite period.

After you enable attachment recovery, you create a batch job named **Scans for deleted references which have reached the end of their retention period**. This batch job uses the **Number of days to defer deletion** value to determine how long a deleted attachment should be retained, based on the **Deleted date and time** value.

Deleting attachments when attachment recovery is active

When you delete an attachment, the system adds a notification to the Message Center to confirm the deletion. The notification also provides an option that lets you undo the deletion if it was unintended.

Table extension support is built in. Therefore, the system retains any extension or custom field values on the DocuRef or DocuValue table to enable their recovery.

Recovering attachments

When you enable attachment recovery, you can recover attachments in one of three ways:

- Immediately after deletion, use the undo link in the **Attachment deleted** notification.
- On the **Attachments** page, use the **Deleted attachments** button to access the list of deleted attachments that you can recover for a particular record. You can open the deleted attachments for review, permanently delete them, or restore them.
- At **System administration** > **Inquiries**, use the **Deleted attachments** page to access the list of deleted attachments that you can recover for any record. You can open the deleted attachments for review, permanently delete them, or restore them.

Scanning attachments for viruses and malicious code

When you work with attachments, you might want to scan the files for viruses and malicious code. Therefore, in version 10.0.12 and later, extension points are available so that you can integrate with the file scanning software of your choice when you work with attachments. A similar extension point is also available for file upload. For more information, see File upload control.

> **ⓘ Important**
>
> Out of the box, finance and operations apps don't scan files for viruses and malicious code, and they don't recommend specific software for file scanning. Instead, you're responsible for choosing your own file scanning software, and for adding the appropriate code to the delegate handlers so that you can use the software or service of your choice to scan files.

The **Docu** class exposes the following two delegates. You can implement handlers for these delegates for document scanning purposes:

- **Docu.delegateScanDocument()** – This delegate applies the file scanning logic when you upload a new document attachment, or when a user tries to preview or download an existing attachment. The corresponding action fails if the scanning service determines that the file is malicious.
- **Docu.delegateScanDeletedDocument()** – This delegate applies the file scanning logic to documents in the attachment recycle bin when a user tries to preview or download a file. The corresponding action fails if the scanning service determines that the file is malicious.

Implementation details

The following example of the **ScanDocuments** class shows boilerplate code for the two handlers. For general information about how to implement handlers for delegates, see EventHandlerResult classes in request or response scenarios.

X++

```
public final class ScanDocuments
{

    [SubscribesTo(classStr(Docu), staticDelegateStr(Docu, delegateScanDocument))]
    public static void Docu_delegateScanDocument(DocuRef _docuRef,
EventHandlerRejectResult _validationResult)
```

```
    {
        if (!ScanDocuments::scanDocument(_docuRef))
        {
            _validationResult.reject();
        }
    }

    [SubscribesTo(classStr(Docu), staticDelegateStr(Docu,
delegateScanDeletedDocument))]
    public static void Docu_delegateScanDeletedDocument(DocuDeletedRef
_docuDeletedRef, EventHandlerRejectResult _validationResult)
    {
        if (!ScanDocuments::scanDeletedDocument(_docuDeletedRef))
        {
            _validationResult.reject();
        }
    }

    private static boolean scanDocument(DocuRef _docuRef)
    {
        /*
        Custom implementation required for connecting to a scanning service
        If document scanning process found an issue, return false; otherwise,
return true;
        */
        return true;
    }

    private static boolean scanDeletedDocument(DocuDeletedRef _docuDeletedRef)
    {
        /*
        Custom implementation required for connecting to a scanning service
        If document scanning process found an issue, return false; otherwise,
return true;
        */
        return true;
    }

}
```

[Developer] Specifying valid content types when attaching documents programmatically

The following APIs from the `DocumentManagement` class let you specify the file content type (MIME type) of the file you're attaching:

- `attachFileToCommon()`
- `attachFile()`
- `attachFileToDocuRef()`

If you don't specify the file content type correctly, the attached document might not behave as expected. Therefore, if you use these APIs, consider one of the following actions:

- Pass **null** for the `_fileContentType` parameter in any of the preceding APIs. This action enables the correct content type to be inferred from the file name.
- Switch to using one of the following methods that don't include a `_fileContentType` parameter. This action helps prevent the possibility that incorrect file content types are passed.
  - **attachFileForRecord()**, which replaces **attachFileToCommon()**
  - **attachFileForReference()**, which replaces **attachFile()**
  - **attachFileForDocuRefRecord()**, which replaces **attachFileToDocuRef()**

Export attachments

In version 10.0.40 and later, Document management includes the **Export attachments** feature. Use this feature to export files and metadata that are attached to records of tables in finance and operations apps.

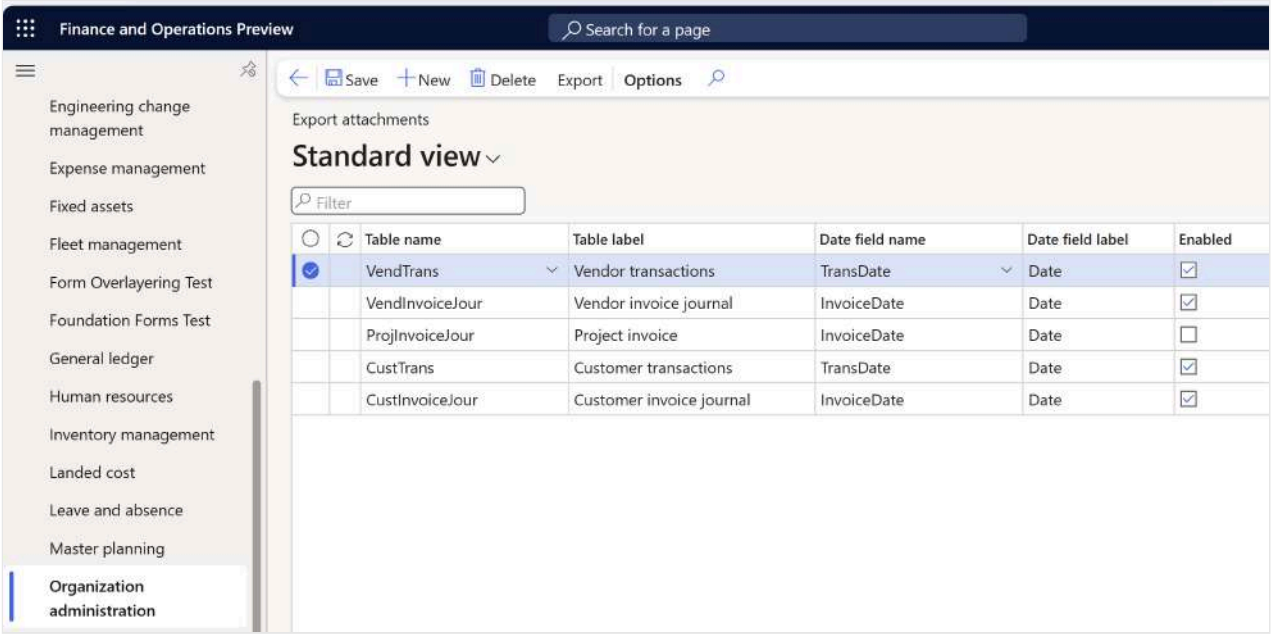
To export attachments, follow these steps:

1. Go to **Organization administration** > **Document management parameters** > **Export attachments**.
2. On the **Export attachments** page, on the Action Pane, select **New** to add a row for the first table that you want to export attachments for. Repeat this step until all the tables that you need attachments for are listed. The following table describes the fields that are available on the **Export attachments** page.

Expand table

Field name	Description	Example value
Table name	The name of the table that you want to export attachments for.	CustInvoiceJour
Table label	The label that's assigned to the selected table. Note that multiple tables can share the same label.	Customer invoice journal
Date field name	A date field of the selected table. The export job uses this field as a criterion to apply the **Date from** and **Date to** range that you specify to export attachments.	InvoiceDate

Field name	Description	Example value
Date field label	The label that's assigned to the selected date field.	Date
Enabled	When this checkbox is selected, the system includes the selected table with the next export job.	Selected



3. On the Action Pane, select **Export** to open the **Export attachments** dialog box. The following table describes the fields that are available in the **Export attachments** dialog box.

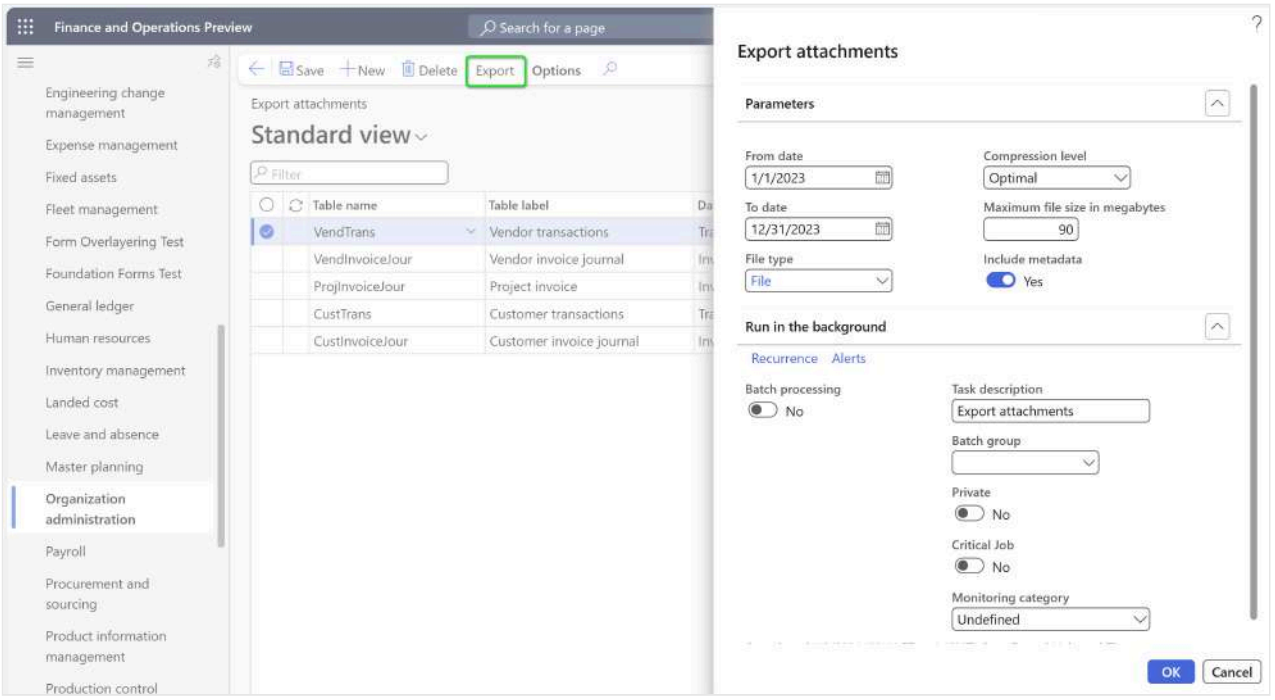
Expand table

Field name	Description	Example value
From date	Specify the first date of the period that you want to export attachments for. The export job applies this period to the **Date field name** value of the selected tables that are configured to export attachments.	1/1/2023
To date	Specify the last date of the period that you want to export attachments for. The export job applies this period to the **Date field name** value of the selected tables that are configured to export attachments.	12/31/2023
File type	Specify the file type that's used to create an archive of exported attachments. Set up this file type as described in the Configure file types section. Select **File** in the list of file types. Alternatively, use a	File

Field name	Description	Example value
	similar custom file type that you set up with an **Attach file** value in the **Class** property of the document type.	
Compression level	This parameter determines how much the data in an archive is compressed. Specifically, it defines whether a compression operation emphasizes speed or compression size. Select one of the following options: • **Fastest** – The compression operation should be completed as quickly as possible, even if the resulting file isn't optimally compressed. • **No compression** – No compression should be done on the file. • **Optimal** (default) – The compression operation should optimally balance compression speed and output size.	Optimal
Maximum file size in megabytes	Use this parameter to limit the size of the created archive. The value redefines the value that's set for the **Maximum file size in megabytes** parameter on the **Document management parameters** page for a specific execution of the **Export attachments** job. If the size of the archive file exceeds the value that you set here, the process creates and attaches multiple files. A large **Maximum file size in megabytes** value might affect the performance of job execution.	90
Include metadata	If you set this option to **Yes**, the system includes **Note** and **URL** type attachments and the metadata of **File** type attachments in the export job. Exported metadata is exported as separate JavaScript Object Notation (JSON) files. If you leave this option set to **No** (the default value), only attachments of the **File** type are exported without metadata.	Yes

4. On the **Run in the background** FastTab, set the **Batch processing** option to **Yes**, and then specify the necessary parameters to run the **Export attachments** job in the background. The job exports attachments for the legal entity where you run the job.

5. Select **OK** to start running the **Export attachments** job.



When the **Export attachments** job finishes, you can find the exported attachments in the archives that are attached for each enabled table. Depending on the volume of the exported attachments in the specified period, and the value of the **Maximum file size in megabytes** parameter in the **Export attachments** dialog box, the process might automatically split the archive into several files.

The name of each attached archive is in the format *LE_TableName_DateFieldName_Period_SeqNum.zip* and contains the following information:

- **LE** – The ID of the legal entity.
- **TableName** – The name of the referenced table.
- **DateFieldName** – The date field that's selected as a criterion.
- **Period** – The values of the **Date from** and **Date to** parameters.
- **SeqNum** – The sequential number of the archive, in case the process creates several archive files.

The name of each attachment that's included in an archive is in the format *RecID_AttachmentName_AttachmentDate_SeqNum.xxx* and contains the following information:

- **RecID** – The ID of the referenced record in the selected table.
- **AttachmentName** – The original file name of the attachment.
- **AttachmentDate** – The date when the attachment was attached to the referenced record of the table.
- **SeqNum** – The sequential number, in case there are simultaneous duplicate attachments.
- **xxx** – The attachment file extension.

Frequently asked questions

What is the difference between document handling and document management?

There's no difference between document handling and document management. Both terms refer to the same functionality. Different terms are used in different versions of the product.

What is the difference between document management and print management?

Document management lets you add notes, documents, and other files to records.

Print management lets you control print settings for selected reports. Print settings include the number of copies, the printer destination, and the multilanguage text that can be included on the report. For more information, see Document Reporting Services.

What is the difference between document types and file types?

Use document types to categorize the documents that you attach to records or the templates that you create. You can store each document type in a unique location. The table for document types is named DocuType.

File types include Word documents and images. A file type is denoted by the extension of the file, such as .txt, .png, .doc, .xlsx, or .pdf.

Does document management integrate with Microsoft 365?

Yes. The system supports SharePoint storage natively, and you can select it as the storage location for a document type. In addition, you can make any URL addressable file an attachment by using the **URL** document type.

What is the default storage location for attachments in Finance + Operations environments?

By default, the system saves attachments in Azure Blob storage automatically as part of the product cloud offering.

If I accidentally delete an attachment stored in Azure Blob Storage, can I restore it?

When you delete attachments that are stored in Azure Blob Storage, you permanently delete them. The system also deletes the references to those attachments. Therefore, if you accidentally delete an attachment, you can't restore or recover it.

Is the database information about attachments stored separately from the attachments themselves?

The system stores record attachment information in the **DocuRef** and **DocuValue** tables. The **DocuRef** table contains the record that represents the attachment. Each **DocuRef** record points to both the record it's attaching to and a **DocuValue** record. Each **DocuValue** record points to the file for the attachment. The system stores files outside the database. Therefore, database operations, such as restorations from a backup, affect only the database information about the attachment, not the attachment file itself.

Can the system store attachments in the database?

No. By default, the system stores attachments in Azure Blob storage.

What are the main differences between Azure Blob storage and database storage?

Database storage uses Azure SQL Database. File storage uses Azure Blob storage. Azure Blob storage is simpler and much less expensive.

How much storage do I get for Azure Blob storage?

See the licensing guide. Currently, you get 40 gigabytes (GB) of storage.

What is the cost for additional storage?

The cost for additional storage varies, but it's similar to the standard Azure costs for storage. In other words, the cost is about $0.05 per GB.

How do I know how much storage is already used?

The system provides proactive communication when you're approaching your database and file storage limits. However, Microsoft Dynamics Lifecycle Services provides some information, and you can log support requests for additional information.

Is there an option to export all document attachments from the system?

Although you can export attachments, this capability isn't a standard capability because the system doesn't include a standard attachment entity. You must build entities that provide attachments for a specific business document or record.

How can you extract attachments from the system?

To extract attachments, you must build an Attachments entity for a specific business document or record. There's no standard attachment entity because the identity for each record type differs. To learn how to build an Attachments entity, review the examples in Application explorer. To find these examples, search for "Attachment" under the **AOT** > **Data Model** > **Data Entities** node.

How does the document preview work for attachments stored in SharePoint?

The WOPI service retrieves the files from SharePoint by using the current user permissions. The service then renders the files in HTML to provide a document preview. The current user must have access to the files to preview or open them.

Troubleshooting problems

Problem: When users interact with Document management or Electronic reporting, they receive an error similar to "Invalid length for a Base-64 char array or string."

Explanation: Typically, this problem occurs because the token for the Office Web Apps Server is no longer valid.

Fix: The admin needs to select the **Token refresh** button to the right of the **Office Web Apps Server** field on the **Document management parameters** page under the **General** tab.

Last updated on 03/18/2026

Configure and send email

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The behavior of the email subsystem is influenced by a combination of administrator configuration, user configuration, and user choices. This article is divided into sections for administrators and users to make it easy to find relevant information.

Both administrators and users set the behavior of the email subsystem.

[Administrator] Email parameters page

Email configuration for the environment

On the **Email parameters** page, administrators can configure the high-level email behavior for the environment from the **Configuration** tab.

Expand table

Field	Description
Batch email provider	Specifies which email provider will be used to send emails that are sent by processes in a batch or non-interactive manner. The Microsoft Graph and Exchange providers will use the account that's associated with the batch process. Note: Microsoft Graph is available in Dynamics 365 Finance version 10.0.38 and later.
Attachment size limit	Specifies the maximum size of a single email that can be sent via the email subsystem.
Email expiration in days	Specifies the number of days before the status of an unsent message is set to **Expired**. A value of **0** means that the default period of 30 days is in effect.

The **Email history** section serves two purposes. First, it provides an entry point to the **Email history** page, where administrators can review all sent emails and also any errors that prevented an email from being sent. Second, it lets you configure how long email history is maintained. By default, the last 30 days of email history are retained. You can adjust this period

by changing the value of the **Number of days to retain email history** field to a non-zero number. If you set the value to **0** (zero), the default number and behavior are used.

The **Email throttling** section enables non-interactive email providers (such as the batch email provider) to adhere to a per-minute sending limit. This feature can help prevent some errors if the system tries to send more emails than the provider allows. Specifically, if an email can't originally be sent because the per-minute sending limit has been reached, the send attempt for the email will be deferred for up to one minute. After ten deferrals, the system will try to send the email regardless. You can remove the per-minute sending limit from a provider by resetting the **per-minute email sending limit** field to **0**. The sending limits for the Office 365 SMTP email provider is automatically set according to the Exchange Online sending limits. Manual configuration is required for all other email providers.

> **Note**
>
> Use of the **Email throttling** feature isn't recommended for the Microsoft Graph and Exchange email providers, because they have their own throttling mechanisms. Therefore, you should ensure that the **per-minute email sending limit** field is set to **0** for both those providers.

Send email with Microsoft Graph

For customers who use Office 365, Microsoft Graph is the recommended email provider for Dynamics 365 finance and operations apps. This email provider replaces the deprecated Exchange email provider.

You must have the following permissions to set up the Microsoft Graph integration:

- You must be a system administrator in the application to link the Dynamics 365 finance and operations environment to Microsoft Graph.
- You must be an administrator for your Microsoft Entra ID account. If you aren't the administrator, an administrative user must perform the first part of the configuration for you.

One-time registration process

To create an app, follow these steps:

1. Sign in to the Azure portal by using an Azure tenant admin account.

   > **Note**

> The user who completes this procedure must have Admin rights for the tenant to register applications.

2. Go to **Microsoft Entra ID** > **App registrations** > **New application**.

3. Enter the following values:

   - **Name** – Enter the name of your app.
   - **Supported account types** – Enter only accounts that are directly in this organization (single tenant).

4. Select **Register**.

5. Make a note of the **Application (client) ID** value. You will use this value to connect to the Microsoft Graph service from your finance and operations environment.

   > ⓘ **Important**
   >
   > Be sure to capture the **Application (client) ID** value before you continue.

6. To add permissions, follow these steps:

   a. Select **Manage** > **API permissions** > **Add a permission** > **Microsoft APIs** > **Microsoft Graph**.

   b. Select **Application permissions**, and enable **Mail.Send**.

      > ⓘ **Note**
      >
      > By default, the app should include the **User.Read** delegated permission for Microsoft Graph. If that permission is missing, you must add it from the delegated permissions.

   c. Select **Add permissions**.

   d. Select **Grant admin consent for** to allow emails to be sent.

7. To create a client secret, follow these steps:
   a. Select **Manage** > **Certificates and secrets**.
   b. On the **Client secrets** tab, select **New client secret**.
   c. Enter a value in the **Description** and **Expires** fields, and then select **Add**.
   d. Make a note of the **Secret value** value. You will use this value to connect to the Microsoft Graph service from your Dynamics 365 finance and operations

environments.

> **Important**
>
> Be sure to capture the **Secret value** value before you continue.

Restrict mailboxes

Administrators can modify mailbox access. For more information, see Limiting application permissions to mailboxes.

Configure Microsoft Graph in Dynamics 365 finance and operations apps

1. In Dynamics 365 finance and operations apps, open the **Email parameters** page.
2. On the **Microsoft Graph settings** tab, in the **Application ID** field, enter the **Application ID** value that you captured when you created the new app in the previous procedure.
3. In the **Application secret** field, enter the **Secret value** value that you captured when you created a client secret in the previous procedure.
4. Select **Save**.

Exchange email provider (Deprecated)

On the **Email parameters** page, an administrator can select **Exchange** as an interactive email provider and as the Batch email provider. This mail provider will use the current user's Exchange Online account to send emails. When it's used as the Batch email provider, the batch account will be used. No additional configuration is required.

If troubleshooting is required, ensure that it's possible to sign in to the current user's account, and that emails can be sent from that account to the intended recipients.

Note that the Exchange mail provider has the following limitations:

- It isn't supported for external users, because those users won't have Exchange accounts on the system tenant.
- It's available only in Microsoft-managed environments.

Send email by using SMTP

On the **Email parameters** page, note the following settings on the **SMTP settings** tab.

Server information

Expand table

Field	Description
Outgoing mail server	The host name of the desired Simple Mail Transfer Protocol (SMTP) server. For Microsoft 365 production (including *.onmicrosoft.com accounts), use smtp.office365.com. (You can find this setting at outlook.office.com at **Settings** > **Mail** > **POP and IMAP**.) For Outlook/Hotmail, use smtp-mail.outlook.com.
SMTP port number	Typically, the port number should be set to 587 for secure transport.
SSL required	Determines whether secure transport is used. Typically, this is **Yes**, except for internal or troubleshooting scenarios.

> **Note**
>
> If you are running into issues from too many emails being sent in a short period of time, it is recommended to utilize the **Email throttling** feature mentioned in the **Configuration tab** section above. If that doesn't resolve the issue, you may consider adding the appropriate IP addresses from the outbound IP safe list to your DNS SPF record authorizing Dynamics 365 finance and operations to send emails from your domain. For more information, see Outbound IP safe list.

Authentication

Expand table

Field	Description
Authentication required	Determines whether a user name and password are needed to send emails.
User name and **Password**	If authentication is required, specify the appropriate mail account to send email from. All users need to provide the SMTP account **Send As** and **Send On Behalf Of** permissions to enable the ability to send SMTP mail. You can configure Send As permissions in the Microsoft 365 admin center (portal.office.com/Admin) at **Users** > **Active users** > **User** > **Edit mailbox permissions** > **Send email from this mailbox**. For more information, see Enable sending email from another user's mailbox in Microsoft 365.

> **Note**

Finance and operations apps don't support multifactor authentication or Modern auth (OAuth 2.0) for SMTP. The Microsoft Graph email provider can be used if a more modern integration is desired. Administrators might have to re-enable Basic authentication to allow for SMTP AUTH. For more information, see Enable or disable SMTP AUTH. Note that the deprecation of Basic authentication for Exchange online only impacts the use of Office 365 SMTP servers. Other SMTP servers that still support basic authentication won't be impacted by this Office 365 deprecation.

[Administrator] Email distributor batch process

Email that is sent directly from the server, without user interaction, via SMTP is sent by the **Email distributor batch** process. That batch process must be started to process the email queue. To start the process, open the **Email distributor batch** pane (**System administration** > **Periodic tasks** > **Email processing** > **Batch**) and turn on **Batch processing**.

If the Microsoft Graph or Exchange provider is used, then the user account associated with the batch process (usually admin) will be the sender.

[Administrator] User email

The default **send from** address for each user is pulled from the **Email** field on the **Users** page (**System administration** > **Users** > **Users**). Administrators can override this **send from** default if needed using the **Sender email** field on the **Options** page.

[User] Email provider selection section on the Options page

The **Options** page can be opened via **Settings** > **User options**. The **Email provider selection** section is on the **Account** tab.

Expand table

Field	Description
Email provider ID	Allows the user to select the email provider that should be used when sending an email. Selecting an option here is the equivalent of selecting **Do not ask again** in the **How would you like to send email** dialog box. Selecting the blank option **Prompt for which email provider to use** will cause the **How would you like to send email** dialog box to display when an email is going to be sent.

Field	Description
Sender email	Allows the administrator to provide an email address override for the user in the **From** field of the email. By default, the email alias that is associated with the user account is used as the **From** field in new emails, but this user option email address will override that. When sending email via SMTP, the user needs to have appropriate **Send As** and **Send On Behalf Of** permissions configured in Exchange or on the SMTP server. **NOTE:** You can configure **Send As** and **Send On Behalf Of** permissions in the Microsoft 365 admin center (portal.office.com/Admin) at **Users** > **Active users** > **User** > **Edit mailbox permissions** > **Send email from this mailbox**. For more information, see Enable sending email from another user's mailbox in Microsoft 365.

[User] How would you like to send email dialog box (optional)

When an email is going to be sent, the user will see the **How would you like to send email** dialog box that will list the available options for sending email.

Expand table

Field	Description
Use an email app, such as Outlook	Provides the user with a generated email (.eml) file.
Use Exchange email server	Uses the Exchange Online server associated with the tenant. The **Exchange** mail provider is currently not supported outside the public cloud, including on-premises environments.
Use the system email client	Opens the **Send email** composition dialog box and then sends the resulting email via SMTP.
Do not ask again	If this field is not selected, the next time an email is sent the most recently selected option will be used and the dialog box will not open.

[User] Send email dialog box (optional)

The **Send email** dialog box is opened to allow the user to edit the contents of the email that will be sent. Some of the following fields will be pre-populated in this window.

Expand table

Field	Description
From	Populated from the **Email** field on the **Options** page.
To, **Cc**, **Bcc**, **Subject**, and **Body**	Populated with values specified by the process that initiated the sending of the email. These fields can be edited as needed by the user.
Attachments list	May be populated with attachments specified by the process that initiated the sending of the email. This list can be edited as needed by the user.

When the email is ready to be sent, the **Send** button will cause the email to be sent via SMTP.

Usage scenarios to verify if email is configured correctly

Send mail via a local mail client

Email workflows that are enabled via the SysEmail framework can generate email messages (.eml files) that contain attachments. You can then send these messages via Microsoft Outlook or another email client.

1. In Microsoft Edge, navigate to **Accounts receivable** > **Customers** > **All customers**.
2. Select **US-008 Sparrow Retail**.
3. Click **Collect** > **Customer balances** > **Collections** to open the **Collections** page.
4. Click **Communicate** > **Email** > **Statements to contact**.
5. Click **OK** to accept the default values in the dialog box.
6. If you're prompted for the mail option to use, clear the **Do not ask again** check box (you can change this option from the **User options** page), select **Use an email app, such as Outlook**, and then click **OK**.
7. If you're using Microsoft Edge on your computer, open the email (.eml) file that is generated. If you're using Microsoft Edge on the VM, copy the file to your computer, and open it there.
8. Note the email address in the **To** field and the generated workbook attachment.

Send mail via SMTP

Email workflows that are enabled via the SysEmail framework can also be created in a simple email dialog box and then sent via Simple Mail Transfer Protocol (SMTP).

1. Go to the **Email parameters** page.
2. Click **SMTP settings**.

3. Set the **Outgoing mail server** to the desired SMTP server:

    - For Microsoft 365 production (including *.onmicrosoft.com accounts) use smtp.office365.com. (Find this setting via outlook.office.com, at **Settings** > **Mail** > **POP and IMAP**.)
    - For Outlook/Hotmail use smtp-mail.outlook.com.

4. Set the user name and password to an appropriate email account and password.
5. Leave **SSLRequired** turned on, and leave **SMTP port number** set to **587**.
6. Click **Save**.
7. In Microsoft Edge, navigate to **Accounts receivable** > **Customers** > **All customers**.
8. Select **US-008 Sparrow Retail**.
9. Click **Collect** > **Customer balances** > **Collections** to open the **Collections** page.
10. Click **Communicate** > **Email** > **Statements to contact**.
11. Click **OK** to accept the default values in the dialog box.
12. If you're prompted for the mail option to use, select **Use the Microsoft Dynamics 365 Finance email client**, and then click **OK**.
13. To receive the test message, change the **To address** to your email address.

    Ensure that the account specified in the SMTP settings is able to **Send As** and **Send On Behalf Of** your email account. The easiest way to ensure this is to use your email account in the SMTP settings.

14. Enter a subject and body for the message.
15. Click **Send**. The message should be delivered in one to five minutes.

[Administrator] Workflow email notifications

Workflow email configuration is a collection of related settings that work in conjunction.

Workflow email notification setup

1. Verify email settings:
    a. Go to **System administration** > **Setup** > **Email** > **Email parameters**.
    b. Verify that SMTP is enabled.

c. Set the SMTP mail server settings.

2. Verify that the email batch process is running:

   a. Go to **System administration** > **Periodic tasks** > **Email processing** > **Email distributor batch**.

   b. Enable the **Batch processing** option.

   c. Optionally, adjust the recurrence of the email process:
      i. Select **No end date** to adjust all recurrences of the email batch process.
      ii. Adjust the count.
      iii. Adjust to run every minute if needed.

3. Verify workflow notification system email templates:
   a. Go to **System administration** > **Setup** > **Email** > **System email templates** (for system-wide templates).
   b. Verify that the **Sender email** field is set and valid.

4. Verify workflow notification organization email templates:
   a. Go to **Organization administration** > **Setup** > **Organization email templates** (for organization-specific templates).
   b. Verify that the **Sender email** field is set and valid.

5. Verify that the user can receive email notifications:

   a. Go to **Settings** > **User options**.

   b. Go to the **Account** tab.
      i. Set the **Email provider ID** (for example, SMTP).
      ii. Optionally, set a **Sender email** override if the default **send from** address should not be used for the current user.

   c. Navigate to the **Workflow** tab. Set the option to send notifications in email to **Yes**.

6. Verify that the workflow system will send email notifications:

   For each workflow that should have a notification, open the workflow properties in the workflow editor.

   a. Click **Basic settings**. Adjust the email template for the workflow notifications.

   b. Click **Notifications**.
      i. Enable the events for which a user should be notified.
      ii. Set the recipient of the workflow notification for each event notification that is enabled.

c. On a workflow approval element for which a user should be notified:
   i. Go to **Properties**.
   ii. Enable the events for which a user should be notified.
   iii. Set the recipient of the workflow notification for each event notification that is enabled.

Workflow email notification testing

The testing for email notifications is to simply trigger the notification and then check for it.

1. Submit a workflow that has been set up for email notifications.
2. Check the workflow history to make sure that a workflow work item was assigned to the expected user.
3. Check the status of the pending email notification in **System administration** > **Periodic tasks** > **Email processing** > **Batch email sending status**.

   If the email fails to be sent, make sure that the SMTP mail account can be opened.
4. Check for the email notification in the appropriate inbox.

Troubleshooting

Common issues with sending email

There are some standard processes that can help you troubleshoot the configuration of email settings.

Expand table

Verify email settings, and send a test email.	1. Go to **System administration** > **Setup** > **Email** > **Email parameters**. 1. Verify that SMTP is enabled. 1. Verify the settings of the SMTP mail server. 1. Sign in to the SMTP account in a separate window to make sure that the account and password are correct. 1. Send a test email by going to **System administration** > **Setup** > **Email** > **Email parameters** > **Test email**.
Verify that the email batch process is running.	1. Go to **System administration** > **Periodic tasks** > **Email processing** > **Batch**. 1. Make sure that the **Batch processing** option is set to **Yes**. 1. Review the recurrence of the email process: 1. Select **No end date** to adjust all recurrences of the email batch process. 1. Adjust the count as needed.

Review the status of batch emails.	1. Go to **System administration** > **Periodic tasks** > **Email processing** > **Batch email sending status**. 1. Verify that emails are being sent from the correct account. If the account is incorrect, you must adjust settings such as user options, system templates, or organization templates, as needed. 1. Verify that all email user accounts have been granted **Send As** permission for the configured SMTP account (see Verify that all email accounts have appropriate Send As permissions later in this table).
Review any errors on the Email history page.	1. Go to **System administration** > **Setup** > **Email** > **Email history**. The **Email history** page lets administrators review all sent emails and also any errors that prevented an email from being sent. It shows both interactive and non-interactive/batch emails. 1. For any emails where the **Email status** value is **Failed**, review the error message on the **Failure details** tab, and determine whether corrective actions should be taken. Some of these errors are covered later in this article.
Verify that all email accounts have appropriate Send as permissions.	In the Microsoft 365 admin center, verify that all user mail accounts that will be used to send emails have **Send As** and **Send On Behalf Of** permissions for the configured SMTP account. For more information, see Give mailbox permissions to another user in Microsoft 365.
Verify user mailboxes.	Consider signing in to the affected (or all) user mailboxes to verify that they are valid and can be accessed by using sign-in.

Specific Microsoft Graph/Exchange email issues

- **"(401) Unauthorized" or "(403) Forbidden" error when email is sent via Exchange**

Expand table

Explanation	This issue might indicate an invalid or incorrectly set up mailbox in Office 365.
Fix	To fix this issue, make sure that the specified user exists in Office 365 and has appropriate permissions. To find the affected users, follow these steps: 1. Open the **Email history** page. 1. Add a filter for **Failed** emails to the **Email status** field. 1. Make a note of the value in the **Email sender** field. This field shows the user that Exchange is indicating isn't a valid or correctly permissioned user in Office 365.

- **"(404) Not found" error when email is sent via Exchange**

Expand table

Explanation	This issue indicates that no mailbox exists for the user account in Exchange.
Fix	To fix this issue, make sure that the specified user exists in Office 365 and has appropriate permissions, or use alternate user accounts that have valid Exchange mailboxes. To find the affected users, follow these steps: 1. Open the **Email history** page. 1. Add a filter for **Failed** emails to the **Email status** field. 1. Make a note of the value in the **Email sender** field. This field shows the user that doesn't have a mailbox in Exchange.

Specific SMTP email issues

If you continue to experience issues when email is sent via SMTP, you may be running into one of the specific errors below. If not, consider entering the SMTP account information in a tool such as SMTPer.net to verify that the SMTP server and account are valid and working correctly.

- **SMTP emails fail to be sent with "Recipient addresses in single label domains not accepted"**

Expand table

Explanation	An email failed to be sent because a recipient is using a single-label domain, but Office 365 doesn't support single-label domains. Single-label domains are Domain Name System (DNS) names that don't contain a suffix such as .com, .corp, .net, or .org. For example, contoso is a single-label domain. However, contoso.com, contoso.net, and contoso.local aren't single-label domains.
Fix	Specify alternate addresses for email recipients that aren't single-label domains.

- **SMTP emails fail to be sent with "Mailbox full"**

Expand table

Explanation	An email failed to be sent because at least one recipient's mailbox is full. For more information, see Error (554 5.2.2 mailbox full) when sending email to mail-enabled public folders in Office 365.
Fix	Use an alternate address for the recipient, or contact the recipient by using alternate means before you try again.

- **SMTP emails fail to be sent with "Authentication unsuccessful, the request did not meet the criteria to be authenticated successfully"**

Expand table

Explanation	An email failed to be sent because additional criteria are required to successfully authenticate the SMTP user account. These additional criteria might be required because multifactor authentication (MFA) is configured. However, finance and operations apps don't currently support MFA.
Fix	Adjust the user account configuration as appropriate in Office 365 before you try again. For more information, see Fix issues with printers, scanners, and LOB applications that send email using Microsoft 365 or Office 365.

- **SMTP emails fail to be sent with "Authentication unsuccessful, SmtpClientAuthentication is disabled for the Mailbox"**

Expand table

Explanation	An email failed to be sent because SMTP client authentication is disabled for the SMTP user account.
Fix	Enable SMTP client authentication for the user account in Office 365 before you try again. For more information, see Enable or disable authenticated client SMTP submission.

- **SMTP emails fail to be sent with "If your SMTP server doesn't support authentication, please clear the SMTP user name and password"**

Expand table

Explanation	This issue is related to migration from the .NET SMTP mail client (which is now obsolete) to MailKit per Microsoft recommendations. This migration causes a change in the way that SMTP user names and passwords are handled in situations where the mail server didn't support authentication. Previously, if an SMTP user name was provided, but the server didn't support authentication, the .NET SMTP mail client ignored the provided user name and password, and continued without authentication. This behavior led some customers to incorrectly believe that they were using an authenticated mail server. In MailKit, authentication is required if a user name is provided. This requirement will trigger an error for mail servers that don't support authentication, and emails will fail to be sent.

Fix	The user must go to the **Email parameters** page (**System administration** > **Setup** > **Email**) and clear the **User name** and **Password** fields on the **SMTP settings** tab.

- **SMTP emails fail to be sent with an "5.7.57 SMTP" error, or an indication that either you aren't authenticated or authentication is required**

Expand table

Explanation	The mail server is indicating that the SMTP user credentials aren't valid.
Fix	Verify that the SMTP user credentials are correct.

- **SMTP emails fail to be sent with "Microsoft.Dynamics.Ax.Xpp.Security.CryptoEncryptionException: Encryption error occurred with exception"**

Expand table

Explanation	The SMTP password is encrypted by using a key that is specific to a single environment. This error typically occurs when the password can no longer be decrypted after database migration between environments.
Fix	Clear and reenter the SMTP password on the **SMTP settings** tab of the **Email parameters** page.

- **SMTP emails fail to be sent with "Client does not have permissions to send as this sender"**

Expand table

Explanation	This issue is usually caused by incorrect setup of the **Send As** permissions for the email account.
Fix	Ensure that the email account has appropriate **Send As** permissions. For more information, see Verify that all email accounts have appropriate Send As permissions earlier in this article.

Frequently asked questions

Where do workflow email templates come from?

The email templates will be sourced from either system email templates or organization email templates depending on whether the workflow is a system-level (not company specific) or organization-level (company specific) workflow.

What email limits apply when using Microsoft Graph, Exchange, or SMTP?

The system communicates with Microsoft Graph, Exchange, or an SMTP server like a typical email client, so standard behavior and limits apply. For example, standard Exchange Online receiving and sending limits apply.

Additional resources

Troubleshoot the Office integration

Office integration tutorial

Configure email functionality in Microsoft Dynamics AX [AX 2012]

Last updated on 03/17/2026

Date and time data and time zones

This article provides information about date and time fields, and time zones.

Date and time fields

Three types of date and time fields correspond to different data types in the database:

- **Combined date/time fields** – Use these fields to enter date and time data. The **utcdatetime** data type stores time and date data in a single field in Coordinated Universal Time (UTC). UTC is the primary time standard by which the world regulates clocks and time. It's, within about 1 second, mean solar time at 0° longitude. It doesn't observe daylight saving time. Time zones around the world are expressed using positive or negative offsets from UTC. For most purposes, UTC is considered interchangeable with Greenwich Mean Time (GMT). The current version of UTC is defined by International Telecommunications Union Recommendation (ITU-R TF.460-6).
- **Date fields** – Use these fields to enter dates only. The **date** data type stores a day, month, and year. However, these values aren't stored in UTC and can't be associated with a time zone.
- **Time fields** – Use these fields to display the number of seconds that have elapsed since midnight on the current date. The **timeofDay** data type stores an integer value. Time values aren't stored in UTC.

Time zones

To express UTC times in the local time, you must provide a time zone. The time zone controls the offset from UTC that is the equivalent of the local time. For example, the offset for Moscow is UTC+3. The Windows locale of your computer first sets your preferred time zone, although an administrator might change it. Use your preferred time zone only when displaying combined dates and times. To set the preferred time zone for a user, go to the **Users** page. The page shows the list of users of the system. Select the user that you want to set the preferred time zone for, and select **User options**. On the **Language and region** tab, select the preferred time zone.

Last updated on 06/26/2026

Unified pricing management module overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an overview of the **Unified pricing management** module for Microsoft Dynamics 365 Commerce and Dynamics 365 Supply Chain Management. For a video overview, see Pricing Management Overview in Dynamics 365 Supply Chain Management - TechTalk.

Increasingly, traditional business-to-business (B2B) companies are considering converting to omnichannel sales and selling directly to customers, so that they can gain more control over pricing and margins. An omnichannel conversion causes massive changes to pricing models and rules. To help B2B companies that are converting to omnichannel pricing, Unified pricing management enables attribute-based pricing rules that use attributes and data from customers, products, and order segments.

Here are some of the main challenges:

- **Pricing silos** – There's a lack of transparency across channels. There's also a lack of centralized pricing management.
- **Pricing relevancy** – All pricing factors must be considered when B2B and business-to-consumer (B2C) pricing rules are defined.
- **Pricing complexity** – There can be difficulties during the conversion to B2B and B2C pricing logic without validation.
- **Real-time execution** – The omnichannel customer experience requires a real-time pricing calculation response.

Microsoft Dynamics 365 Commerce and Dynamics 365 Supply Chain Management use Commerce Scale Unit (CSU) Core to help traditional B2B companies embrace omnichannel pricing. The CSU Core function provides the following capabilities:

- Offer, manage, and calculate pricing, and provide transparent pricing across channels. This capability is essential for the alignment of pricing strategies across multiple channels.
- Quickly determine prices while you also consider a range of commercial aspects, such as general base prices, sales trade agreements, long-term discount agreements, short-term promotion discounts, and retrospective rebate calculations for each sales order.
- Create pricing data models that are based on price attributes. Price attributes can be based on categorized product pricing differentiators, customer groups, and order types.
- Manage complex price structures by using price component breakdowns, and define margin component price adjustments on top of item base prices. When you create an order, the pricing details can provide a list of price component breakdowns for future in-depth analysis.
- Easily manage sophisticated pricing rules when you convert from B2B pricing to B2B and B2C pricing that considers discount concurrency, bundle sales, mandatory sales items, and bonus free item pricing rules.
- Simulate prices, and show detailed price calculations.
- Use enhanced discount budget controls to help avoid margin leakage from fund consumption.
- Use the CSU-based pricing API for Dynamics 365 Commerce to enable pricing calculations to be retrieved by a native Commerce point of sale (POS) system and/or consumed by external applications to support different pricing scenarios.

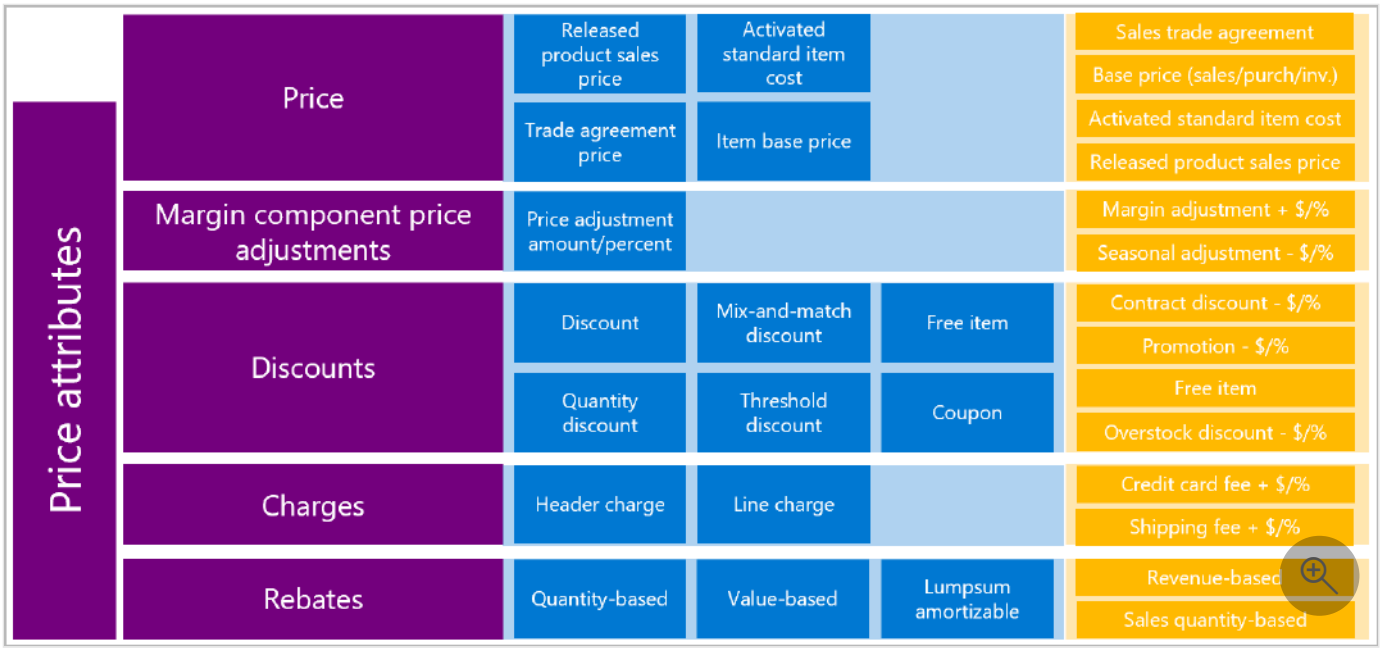
Pricing components

Unified pricing management applies the following key elements to determine pricing:

- **Price attributes** – Price attributes give you a flexible way to define your pricing factors. They use information about customers, products, sales order headers, and sales order lines. Because price attributes work with the customer and product attribute framework, they're highly configurable. You can extend them through available extension points.
- **Price component codes** – Price component codes group together price attributes. They represent the building blocks of your price structure. When you create a price and discount rule record, you also assign that record to a price component code.
- **Price structures** – Price structures help you understand the sequence of your price component codes. You can use one price structure for each company, or you can use a different price structure for each order attribute in a company. Price determination logic that's embedded in the price structure ensures that the base price can be determined. The base price is the price before any price adjustments are made (*Base price* + *Price adjustment* = *Selling price*).

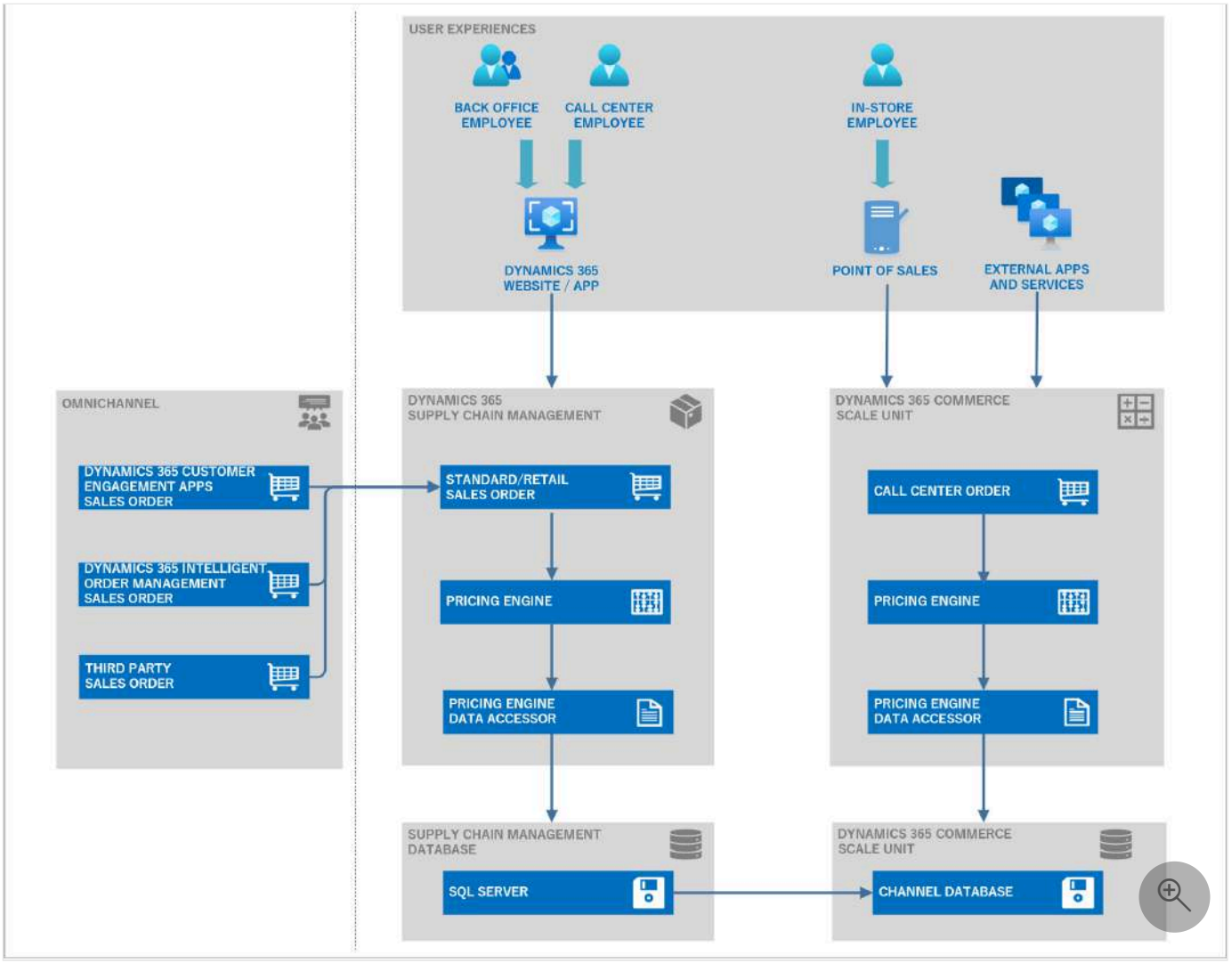
- **Concurrency modes** – Concurrency modes control how the final price is calculated in situations where multiple pricing rules are associated with the same price component code.

The following illustration shows many of the elements that affect price calculations in Unified pricing management.



Architecture overview

The following illustration shows the architecture of the **Unified pricing management** module.



> **Note**
>
> - You can create and maintain the pricing rules in Microsoft Dynamics 365 Commerce and Dynamics 365 Supply Chain Management.
> - Omnichannel orders can come from a range of front-end ordering systems.
> - For orders that are created in Microsoft Dynamics 365 Commerce or Dynamics 365 Supply Chain Management, through integration, or through dual-write (such as orders from Dynamics 365 customer engagement apps or Dynamics 365 Intelligent Order Management), the pricing engine can provide calculated pricing and pricing details for each sales order.
> - For orders that are created in a native Commerce POS or a third-party system, calculated prices can be retrieved via the CSU-based pricing API.

Next steps

- Price attributes overview

- Price structure overview
- Vendor list price and price adjustments overview
- Sales trade agreement prices
- Pricing rules for discounts and margin price adjustments

Last updated on 04/20/2026

Turn on the Unified pricing management module for your system

Prerequisites

To use the **Unified pricing management** module, you must be running version 10.0.41 or later of Microsoft Dynamics 365 Commerce and/or Dynamics 365 Supply Chain Management.

Turn on the Unified pricing management feature in Feature management

Before you can use the **Unified pricing management** module, you must turn it on for your system. Admins can use the Feature management settings to review the status of the feature and turn it on as required. In the **Feature management** workspace, the feature appears as follows:

- **Module:** *Sales and marketing*
- **Feature name:** *Unified pricing management*

Turn on the optional Unified pricing management pricing rule performance enhancement feature

The optional *Unified pricing management pricing rule performance enhancement* feature improves performance and provides enhanced pricing data import and export entities. To use this feature, turn on the following feature in Feature management:

- **Module:** *Sales and marketing*
- **Feature name:** *Unified pricing management pricing rule performance enhancement*

Learn more in Pricing data import and export entities.

Last updated on 07/01/2026

Migrate to Unified pricing management (preview)

[This article is prerelease documentation and is subject to change.]

This article explains how to migrate to Unified pricing management in Microsoft Dynamics 365 Commerce version 10.0.46 to simplify pricing strategies with centralized control and flexible rules.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Unified pricing management migration helps Dynamics 365 Commerce customers transition from the legacy Commerce pricing engine to the Unified pricing management framework without extensive configuration or disruption.

This enhancement simplifies adoption, so you can quickly experience the benefits of centralized and flexible pricing management. It lays the groundwork for you to later extend your pricing strategies by incorporating more pricing attributes for greater precision and control.

Unified pricing management migration is structured into three incremental phases from Commerce versions 10.0.46 to 10.0.48, so you can move at your own pace and address various user scenarios and existing Commerce pricing engine configurations.

The following table describes whether various rules are supported for pricing features.

Expand table

Feature	Current nonattribute rule	New nonattribute rules	New attribute-based rules
Trade agreement price	Supported	Can also migrate	Supported
Discount	Migrate	Not supported	Supported

Feature	Current nonattribute rule	New nonattribute rules	New attribute-based rules
Shipping discount	Not supported	Not supported	Supported
Tender discount	Not supported	Not supported	Not supported
Charges	Supported	No migration (migrating existing charges rules is planned)	Supported
Rebate management	Supported	Supported	Supported
Trade agreement discount	Supported	No migration	Not supported
Price group	Migrate to attributes	Not supported	Supported
Price adjustments	Migration isn't supported.	Not supported	Supported

Prerequisites

You must be running Commerce version 10.0.46 to use Unified pricing management.

Migrate to Unified pricing management

To migrate to Unified pricing management, follow these steps:

1. In Dynamics 365 Commerce headquarters, go to the **Feature Management** workspace at **System administration** > **Workspaces** > **Feature management**.
2. Search for the **Unified pricing management pricing rule performance and enhancement** feature.
3. Select the feature, then select **Enable**.

   > ⓘ **Note**
   >
   > When you enable the feature, you unlock the migration options to migrate from the Commerce pricing engine to Unified pricing management.

4. Go to **Retail and Commerce** > **Migration** > **Set up Price group mapping for migration**. You see the following:

- The system automatically converts all existing Commerce price groups to Unified pricing management price groups.
- Each new group keeps the same price group conditions as its legacy configuration.

5. If duplicate mappings exist, the system prompts you to resolve them manually. After resolving any duplicates, select **Validate** before proceeding.

6. Go to **Retail and Commerce** > **Migration** > **Migrate Commerce Pricing Setup**. This page lets you run the migration job that transitions your current Commerce pricing engine data into the Unified pricing management framework.

   > **Note**
   >
   > In Commerce version 10.0.46, price adjustments aren't included. Price adjustments migration support is included in Commerce version 10.0.47.

7. Use the **Effective after date** field to select pricing rules that are still active and should be migrated.

8. Choose whether or not to include disabled discounts and skip trade agreements.

9. If you want to continue applying existing trade agreements without attributes, configure the corresponding parameters accordingly.

10. Select **X** to start the migration.

When the migration is complete, your data and configurations are successfully migrated to Unified pricing management, providing a foundation for future enhancements to flexible pricing strategy.

Last updated on 08/24/2026

Price attributes overview

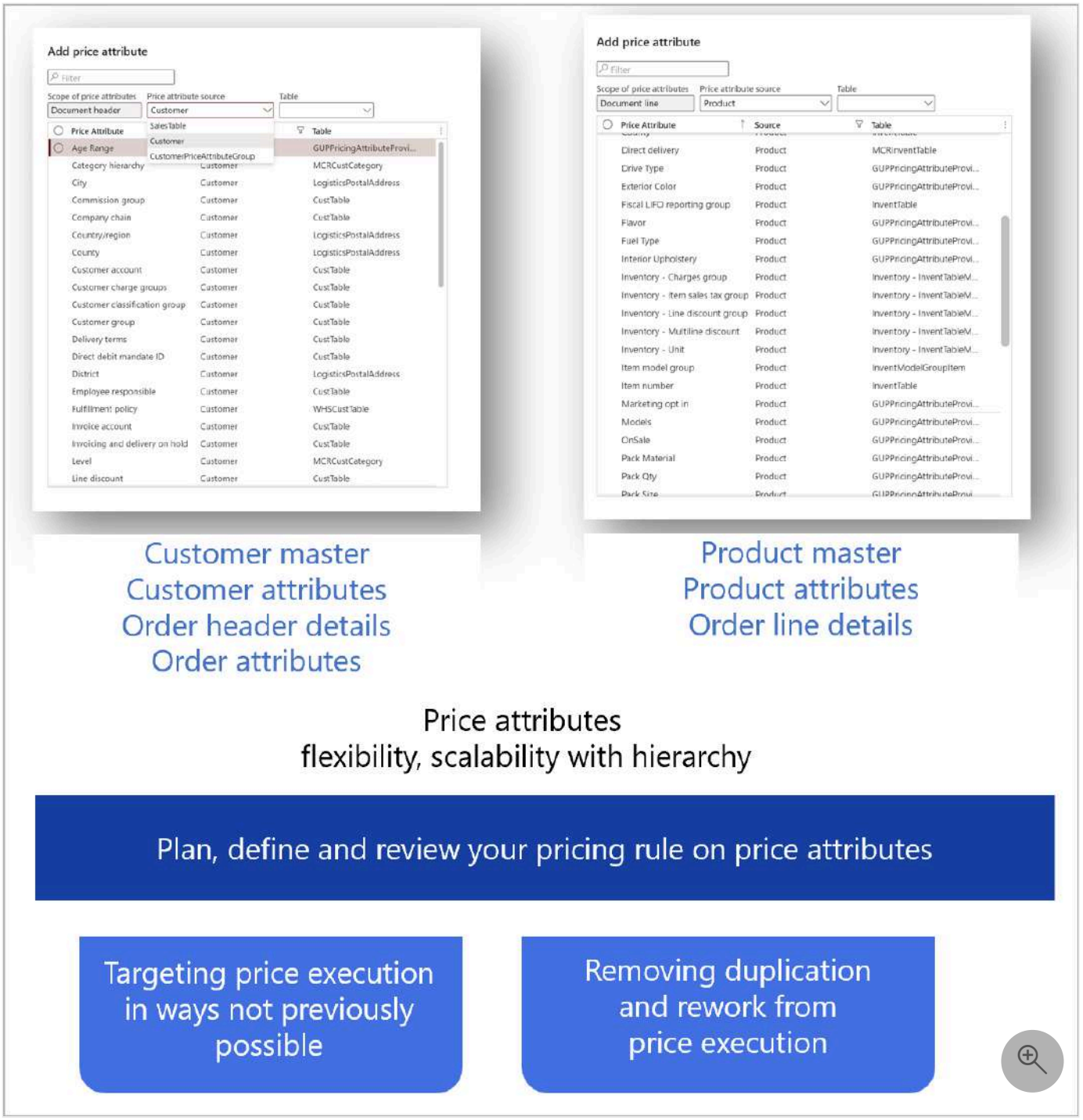
One of the key functions of a price manager is to work together with the product manager to accomplish the following tasks:

- Identify and classify products, customers, and order modes (for example, by region, pack size, brand, or age) to reflect each product's differentiating features, customer segments, and price sensitivity.
- Set up pricing rules that consider any combination of the customer, product, and order attributes to set up a flexible pricing architecture.

The Unified pricing management module provides *price attributes*. These attributes let price managers and sales promotion managers mark and group price differentiators defined for *products*, *customers*, and *sales order information*, and set up the pricing rules. Here are some examples of pricing rules:

- Sales trade agreement prices
- Margin component price adjustments
- Sales discounts
- Customer rebates on sales

Define these pricing rules by using combinations of price attributes to define the rule criteria.



Price attribute sources

Microsoft Dynamics 365 Commerce and Dynamics 365 Supply Chain Management can maintain prices based on price groups, discount groups, product categories, product variants, storage dimensions, and tracking dimensions. These pricing factors are fixed, and can't be extended or changed to accommodate additional pricing criteria. Examples of possible product-based pricing variables include product types, brands, flavors, pack types, and pack sizes.

Customer groups don't always provide enough control over the setup of customer-based pricing. Therefore, customer-based pricing variables can also include customer area, payment method, payment terms, and loyalty program membership.

Other sales order details (such as the ordering channel, ordering site, ordering campaign, and mode of delivery) can also include pricing conditions.

Unified pricing management offers three price attribute sources that can serve as a foundation for the setup of pricing rules and conditions. For each source, it also provides out-of-box price attributes. You can build price attribute groups (header or line scope) and combine them to create price attribute combinations.

Expand table

Attribute group	Source	Source table	Price attributes
Header attribute group	Order (header)	Order table	The attributes provide out-of-box fields from the order header. You can extend them to add more fields.
Header attribute group	Order (header)	Order attributes; Attributes of the attribute group that are defined as the order attribute group in parameters	Configurable
Header attribute group	Customer	Customer master	The attributes provide out-of-box fields from customer masters. You can extend them to add more fields.
Header attribute group	Customer	Attributes of the customer attribute group that are defined as the customer attribute group in parameters	Configurable
Line attribute group	Product	Product master; Released product master	The attributes provide out-of-box fields from product masters. You can extend them to add more fields.
Line attribute group	Product	Associated product attributes that are defined as price attributes and assigned to a product	Configurable
Line attribute group	Order (line)	Order line	The attributes provide out-of-box fields from the order line. You can extend them to add more fields.

Each pricing rule allows for the combination of one header attribute group and one line attribute group.

Note

The number of attributes can affect the performance of the pricing engine. For pricing rule data entities (such as sales trade agreement prices, discounts, and rebates), up to 14 price attributes can be imported into Unified pricing management. Each pricing rule allows for the combination of one header attribute group (which can have up to seven header price attributes) and one line attribute group (which can have up to seven line price attributes).

Next steps

- Price attributes for products, customers, and orders

Last updated on 07/01/2026

Price attributes for products, customers, and orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Unified pricing management provides a set of prefilled price attributes from fields in related tables that are associated with products, customers, orders, and order lines. It also lets you create custom price attributes and associate them with orders, customers, and items. You can use those price attributes when you create pricing rules and conditions. When an order is placed, the pricing engine uses the associations to determine the appropriate price.

This article explains how to set up the configurable price attributes and associate them with products, customers, and orders.

Set up attribute types

When you create an attribute for any purpose, you must assign an *attribute type*. The attribute type defines the data type and the range of values that can be entered for attributes of that type.

To maintain your attribute types, go to **Product information management** > **Setup** > **Categories and attributes** > **Attribute types**.

Set up and assign product price attributes

You can create pricing rules that consider the attribute values assigned to each item in an order. *Attributes* provide a way to describe products and their characteristics through user-defined fields. Examples of attributes include brand, flavor, memory size, and commodity group.

When you create new attributes and assign them to a category, products in that category inherit the attributes and their default values. Attributes can be associated both with a released product

and with a product master. Default attribute values can be overridden at the level of the individual product. Only one value per attribute can be assigned to each product.

> **Note**
>
> Attributes apply to each *product*, not to each *item of inventory*. They serve as a broad classification of a product and aren't connected to inventory dimensions.

Set up product price attributes

To use attributes when you design Unified pricing management pricing rules, mark them as *price attributes* by following the procedure described in this section.

1. Go to **Product information management** > **Setup** > **Categories and attributes** > **Attributes**.
2. Select an attribute in the list pane, or create a new one.
3. Set the **Attribute type** field to the type that you identified or selected earlier.
4. Set the **Can be used as price attribute** option to *Yes*.
5. In the **Default value** field, assign a default attribute value as required.
6. On the Action Pane, select **Save**.

Assign product price attributes to products

Before attributes can have an effect, assign them to the relevant products by using one of the following methods:

- Assign product attributes to products by using the procurement category hierarchy.
- Group attributes into price attribute groups, and assign attribute groups to Commerce categories. If products are assigned to categories that are associated with attribute groups, they inherit the attributes that are included in those attribute groups.

For more information about attribute management and association, see Manage attributes and attribute groups.

Set up and assign customer price attributes

You can create Unified pricing management pricing rules that consider the attribute values assigned to the customer who places an order. To set up customer price attributes, first create an attribute group that includes the attributes that you need. Then set up that attribute group as the

customer attribute group. You can have only one customer attribute group, and it applies to all customers. However, each customer can have different values for each attribute.

Set up customer price attributes

Follow these steps to set up your customer price attributes.

1. Go to **Product information management** > **Setup** > **Categories and attributes** > **Attributes**, and set up all the customer price attributes that you need. For each attribute, make sure that the **Can be used as price attribute** option is set to *Yes*. (Learn more in the Set up product price attributes section of this article.) In the **Default value** field, assign a default attribute value as required.
2. Go to **Product information management** > **Setup** > **Categories and attributes** > **Attribute groups**, and create a customer price attribute group. Give it an appropriate name, such as *Customer price attributes*, and add each relevant attribute on the **Attributes** FastTab.
3. Go to **Pricing management** > **Setup** > **Pricing management parameters**. On the **Price attribute** tab, set the **Customer attribute group** field to the attribute group that you created to hold your customer price attributes.

Assign customer price attribute values to customers

Customer price attributes are initially assigned the default value set up for the attribute. If you need to assign a specific value for a customer, follow these steps:

1. Go to **Sales and marketing** > **Customers** > **All customers**.
2. Select a customer in the grid.
3. On the Action Pane, on the **Price** tab, select **Customer attributes**.
4. Select a customer attribute in the list pane, and then, on the **Value** FastTab, edit the attribute value for the customer as required.
5. Repeat the previous step until you assign all the attribute values that are required for the customer.

Set up and assign sales order price attributes

You can create Unified pricing management pricing rules that consider attribute values assigned to each sales order. For example, you might set up sales order price attributes for *Campaign event*, *Order type*, and *Order channel*. To set up sales order price attributes, first create an attribute group that includes the attributes that you need. Then set up that attribute group as the

sales order attribute group. You can have only one sales order attribute group, and it applies to all sales orders. However, each sales order can have different values for each attribute.

Set up sales order price attributes

Follow these steps to set up your sales order price attributes.

1. Go to **Product information management** > **Setup** > **Categories and attributes** > **Attributes**, and set up all the sales order price attributes that you need. For each attribute, make sure that the **Can be used as price attribute** option is set to *Yes*. (Learn more in the Set up product price attributes section of this article.) In the **Default value** field, assign a default value as required.
2. Go to **Product information management** > **Setup** > **Categories and attributes** > **Attribute groups**, and create a sales order price attributes group. Give it an appropriate name, such as *Order price attributes*, and add each relevant attribute on the **Attributes** FastTab.
3. Go to **Pricing management** > **Setup** > **Pricing management parameters**. On the **Price attribute** tab, set the **Sales order attribute group** field to the attribute group that you created to hold your sales order price attributes.

Assign sales order price attribute values to sales orders

Sales order price attributes are initially assigned the default value set up for the attribute. If you need to assign a specific value for an order, follow these steps:

1. Go to **Sales and marketing** > **Orders** > **All sales orders**.
2. Select an order in the grid, or create a new one.
3. On the Action Pane, on the **Price** tab, select **Sales order attributes**.
4. Select a sales order price attribute in the list pane. On the **Value** FastTab, edit the attribute value for the customer as required.
5. Repeat the previous step until you assign all the attribute values that are required for the order.

Last updated on 05/04/2026

Price attribute groups

Each *price attribute group* is a collection of price attributes. When you define a pricing rule, select a price attribute group and then specify values for each of the associated attributes. In this way, you define the match criteria used to select the sales orders and order lines that the rule applies to. This article explains how to configure price attribute groups for the Unified pricing management module.

Price attribute scopes

Price attribute groups have two scopes: *header* and *line*.

- **Header scope** – These attributes are on *sales order headers*. They describe the overall sales order. Examples include the sold-to customer and the invoice account.
- **Line scope** – These attributes are on *sales order lines*. They describe each individual order line. Examples include the product ID, the quantity, and delivery details.

When you build a pricing rule, you can combine one header price attribute group and one line price attribute group to set up your condition.

Expand table

Attribute group	Source	Source table	Price attributes
Header attribute group	Order (header)	Order table	The attributes provide out-of-box fields from the order header. You can extend them to add more fields.
Header attribute group	Order (header)	Order attributes - Attributes of the attribute group that are defined as the order attribute group in parameters	Configurable
Header attribute group	Customer	Customer master	The attributes provide out-of-box fields from customer masters. You can extend them to add more fields.
Header attribute group	Customer	Attributes of the customer attribute group that are defined as the customer attribute group in parameters	Configurable

Attribute group	Source	Source table	Price attributes
Line attribute group	Product	Product master - Released product master	The attributes provide out-of-box fields from product masters. You can extend them to add more fields.
Line attribute group	Product	Associated product attributes that are defined as price attributes and assigned to a product	Configurable
Line attribute group	Order (line)	Order line	The attributes provide out-of-box fields from the order line. You can extend them to add more fields.

Manage price attribute groups

To add, edit, or delete a price attribute group, follow these steps:

1. Go to **Pricing management** > **Setup** > **Price attribute groups** > **Price attribute groups**.
2. Follow one of these steps:
   - To create a new group, select **New** on the Action Pane.
   - To edit an existing group, select it in the list pane, and then select **Edit** on the Action Pane.
   - To delete an existing group, select it in the list pane, and then select **Delete** on the Action Pane.
3. On the header of the new or selected group, review the settings of the following fields, and edit them as required:
   - **Price attribute group** – If you're creating a new group, enter a unique name for the group. This field becomes read-only after you save the group.
   - **Validated** – This option indicates whether the group is validated. Only validated groups are active and can be used in pricing rules. You can't edit this option directly. When you create a new group or edit a previously validated group, this option is set to *No*. To validate a new or edited group, select **Validation** on the Action Pane to trigger a validation check. If the check passes, the system sets this option to *Yes*.
4. On the **General** FastTab, set the following fields:
   - **Friendly name** – Enter a descriptive or common name for the group. For new groups, the value initially matches the value that you entered in the **Price attribute group** field,

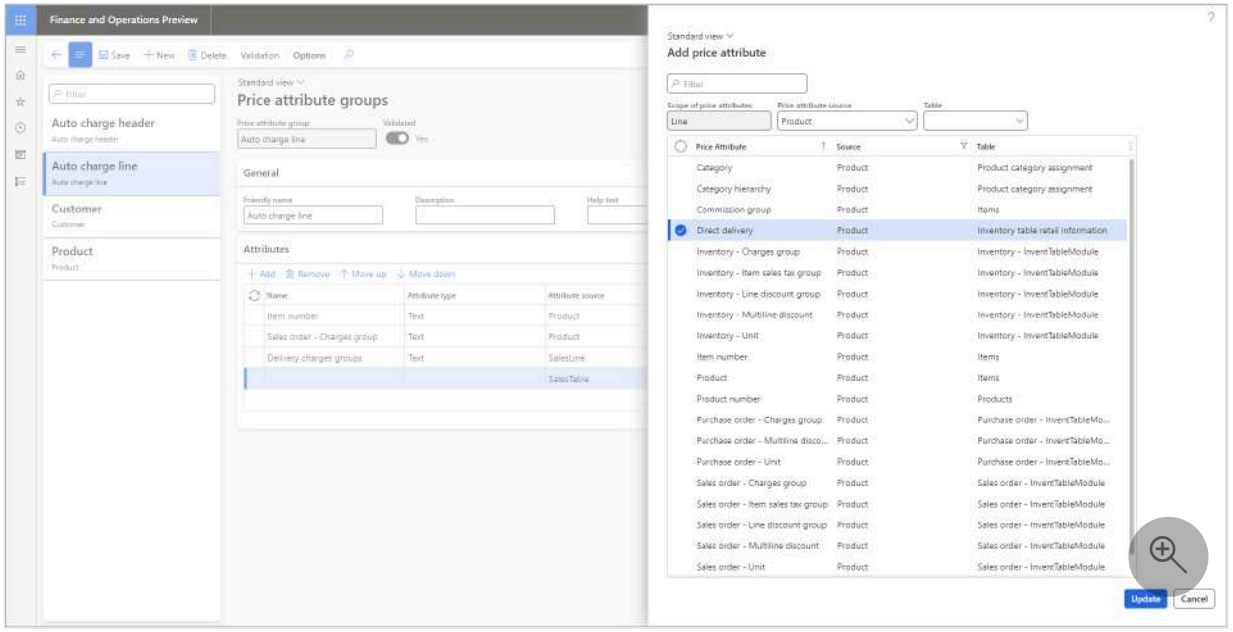
but you can change it.

- **Description** – Enter a short description of the group.
- **Help text** – Enter help text that describes the group.
- **Sales order matching scope** – If you're creating a new group, select the scope (*Header* or *Line*) where this group applies. This field becomes read-only after you save the group.

5. On the Action Pane, select **Save**.

6. On the **Attributes** FastTab, review the attributes that belong to the group, and edit them as required. Use the following toolbar buttons to work with the list:

   - **Add** – Open the **Add price attribute** dialog box, where you can add one or more attributes to the list. This dialog box shows attributes that match the selected scope and provides filtering controls to help you find the attributes that you're looking for. Select the checkbox for each attribute that you want to add, and then select **Update** to add those attributes to the group.



   - **Remove** – Remove a selected attribute from the list.

   - **Move up** and **Move down** – Move a selected attribute up or down in rank. For more information about how to use these buttons, see the next section.

> ⓘ **Note**
>
> You can define groups of products or customers, and then add attributes here that let you specify those groups by name in your pricing rules. For information about how to

create and use those groups, see Customer and product price attribute groups.

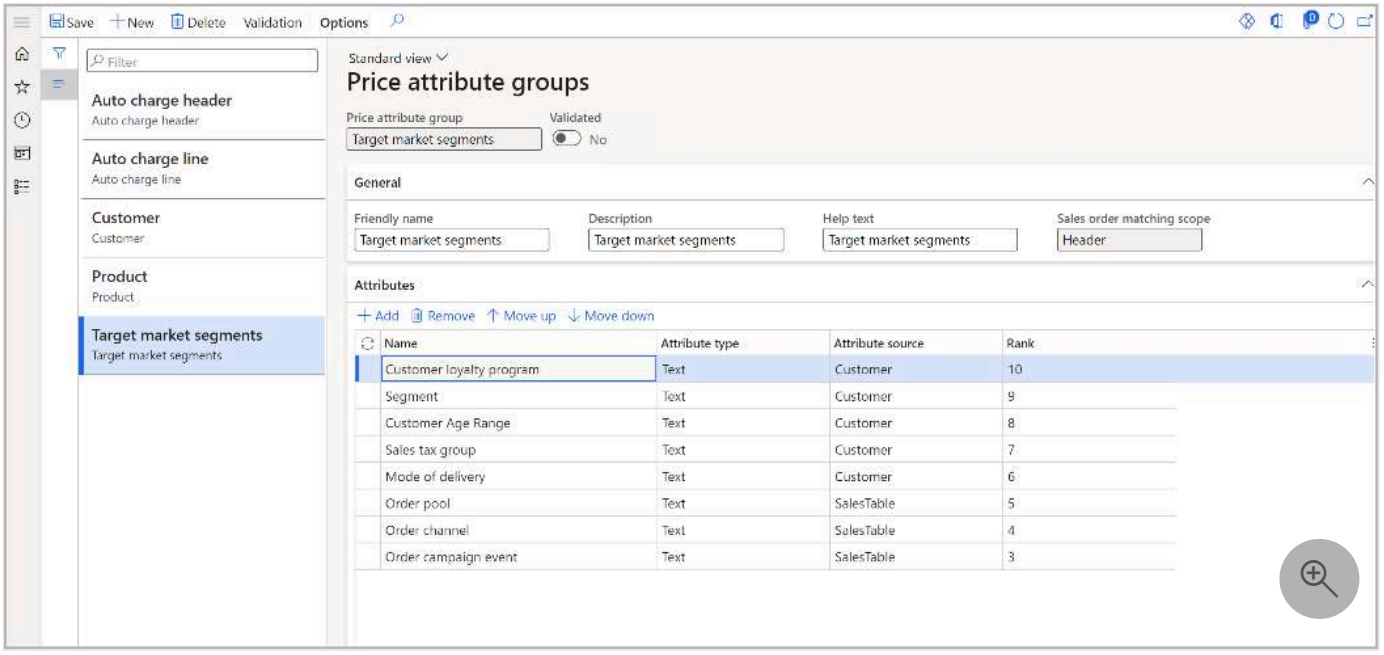
7. When you finish adding and arranging attributes, select **Validation** on the Action Pane to trigger a validation check. If the check passes, the system sets the **Validated** option to *Yes*. This setting indicates that the price attribute group is activated and can be used in pricing rules.

Price attribute ranks

A rank is assigned to each attribute in a price attribute group. The rank determines which rule applies if multiple pricing rules match for a given sales order. In this scenario, only the pricing rule that specifies a value for the highest-ranked attribute applies.

For example, a price attribute group includes the following two attributes:

- **Price attribute 1:**
  - **Name** = *Customer loyalty program*
  - **Rank** = *10*
- **Price attribute 2:**
  - **Name** = *Segment*
  - **Rank** = *9*



The following two pricing rules use this price attribute group:

- **Pricing rule 1:** If **Customer loyalty program** = *Platinum* and **Segment** = blank (any segment), **Price** = *$100*.

- **Pricing rule 2:** If **Customer loyalty program** = blank (any loyalty program) and **Segment** = *20*, **Price** = *$150*.

Provided that no other pricing rules apply, the price is $100, because that price is associated with a specific loyalty program, and the **Customer loyalty program** attribute has a higher rank than the **Segment** attribute.

Last updated on 07/01/2026

Enable non-attribute based rebate management deals in Unified pricing management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the Rebate management module to create contracts, deals, or agreements between a business and its customers or vendors, so that you can calculate rebates, deductions, and royalties. When you enable Unified pricing, you can determine rebates based on pricing attributes.

Starting in Microsoft Dynamics 365 Supply Chain version 10.0.48, the **Non-attribute based rebate management deals** feature can be included in Unified pricing management scenarios. When you enable this feature, the pricing evaluation process considers existing rebate agreements without requiring changes to their configuration.

This capability allows organizations to continue using existing rebate management deals while adopting Unified pricing. Customers can also create new non-attribute based rebate deals, facilitating the migration to Unified pricing for those customers who rely on the legacy behavior.

Prerequisites

To enable the features, follow these steps:

1. Go to the Feature management workspace.
2. Enable the **Unified pricing management** and **Enable non-attribute based rebate management deals for Unified pricing scenario** features.

> **Note**

There's a dependency between the listed features. You can't enable the **Enable non-attribute based rebate management deals for Unified pricing scenario** feature if the **Unified pricing management** feature is disabled. You can't disable the **Unified pricing management** feature if the **Enable non-attribute based rebate management deals for Unified pricing scenario** feature is enabled.

Enable non-attribute based rebate management deals

1. Go to **Rebate management** to see all pages related to rebate management deals. Two new pages manage non-attribute based rebate deals:

    - **Non-attribute based customer rebate deals**
    - **Non-attribute based customer royalty deals**

2. Go to the **All rebate management deals** page. On this page, you can view all rebate deals. The **Attribute-based rebate** column shows the status of each rebate deal. From this page, you can manage any existing rebate deals or create new rebate deals.

3. To create new rebate deals, in the **Type of rebate management deal** field, select attribute or non-attribute rebate deal types.

To view and manage attribute-based customer rebate deals or royalty deals, see the following pages:

- **Customer rebate deals**
- **Customer royalty deals**

To view and manage non-attribute based customer rebate deals or royalty deals, see the following pages:

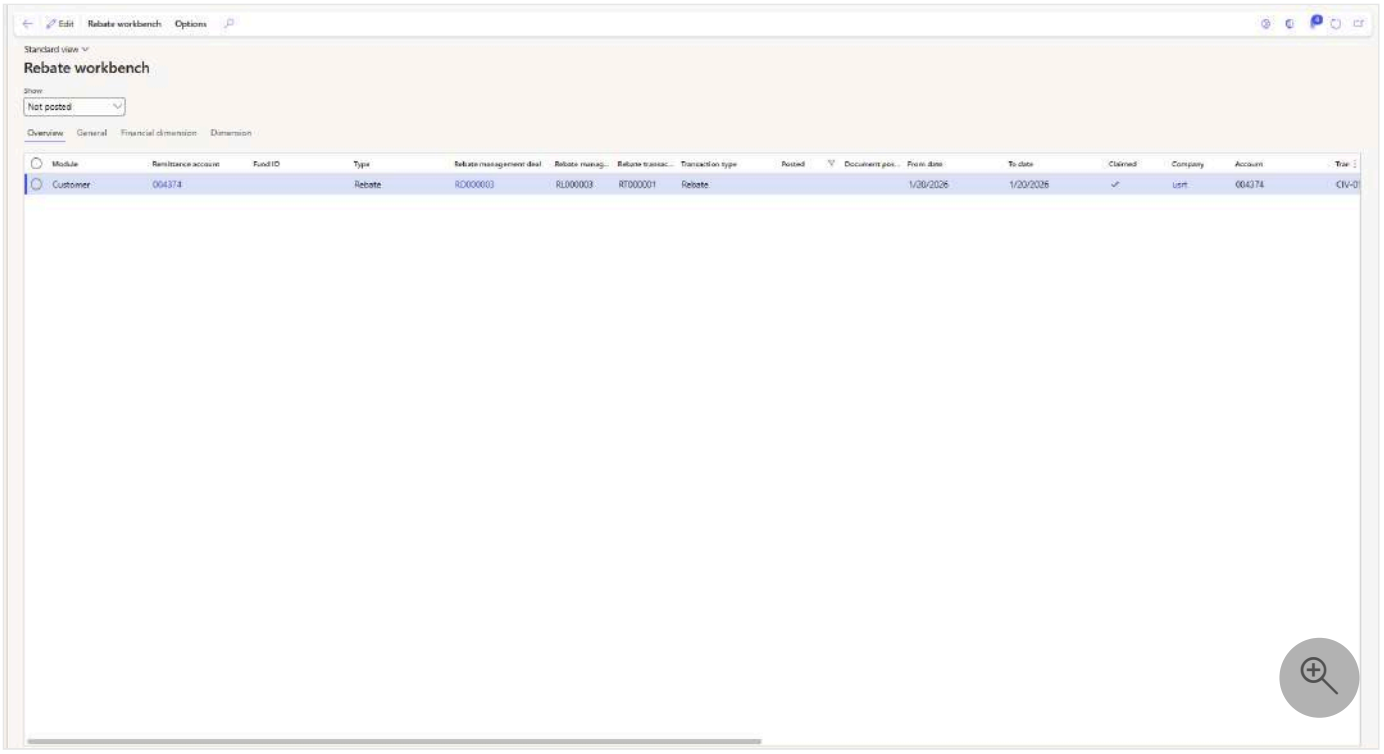
- **Non-attribute based customer rebate deals**
- **Non-attribute based customer royalty deals**

All vendor rebate deals are non-attribute based and be accessed using the **Vendor rebate deals** page.

To process, post, or cancel multiple rebate management deals, follow these steps:

1. Go to **Rebate management**.
2. Expand **Periodic tasks**. You can process, post, or cancel multiple rebate management deals by using the commands.

To view and manage all rebate management deals, see the **Rebate workbench** page.



Troubleshooting

To get more information about errors or unexpected behavior, contact Microsoft Support and open a support ticket.

Last updated on 05/22/2026

Customer and product price attribute groups

Article • 03/13/2025

Customer price attribute groups and *product price attribute groups* let you create logic to define groups of customers and products, respectively. The membership of each group is based on the values of attributes that you assign to that group. These groups can make it easier to set up collections of customers and products that you can target in your pricing rules.

Usage examples

The following examples show how you can simplify your price attribute groups and pricing rules by setting up a customer price attribute group. Although these examples are focused on customer price attribute groups, product price attribute groups work in a similar way (but define groups of products instead of groups of prices).

The more often you have to use a given customer price attribute group in your price attribute groups and pricing rules, the more useful that group will be.

Customer segment for the examples

For these examples, you often target a customer segment that consists of American female wholesale and retail customers aged 20–45. The following table shows the attribute values that you can use to define this segment.

Expand table

Price attribute	Value
Gender	Woman
Customer group	30, 10
Customer age range	20-45
Country/region	USA

Example 1: Price attribute group that doesn't use a customer price attribute group

To set up price attribute groups that can target the preceding segment without using a customer price attribute group, you must ensure that each of your price attribute groups includes all four attributes. Each pricing rule that's designed to address that segment must specify all four values, plus any additional values.

Example 2: Price attribute group that uses a customer price attribute group

If you want to simplify the price attribute groups and pricing rules that you use to target the segment, you can create a customer price attribute group that has the name and attribute values that are shown in the following table.

Expand table

Customer attribute group	Attribute	Value	Rank
US-F-20-45-RE&WH	Gender	Girls, Women	4
	Customer group	10, 30	3
	Customer age range	20-45	2
	Country/region	USA	1

Each price attribute group can now include just the *Customer Price Attribute Group* attribute, plus whatever other attributes are required for each individual group. The related pricing rules can just specify that a price applies when **Customer Price Attribute Group** = *US-F-20-45-RE&WH*.

Manage customer and product price attribute groups

To add, edit, or delete a price attribute group, follow these steps.

1. Follow one of these steps, depending on the type of group that you want to create:

   - Go to **Pricing management** > **Setup** > **Price attribute groups** > **Customer price attribute groups**.
   - Go to **Pricing management** > **Setup** > **Price attribute groups** > **Product price attribute groups**.

2. Follow one of these steps:

- To create a new group, select **New** on the Action Pane.
- To edit an existing group, select it in the list pane, and then select **Edit** on the Action Pane.
- To delete an existing group, select it in the list pane, and then select **Delete** on the Action Pane.

3. On the header of the new or selected group, review the settings of the following fields, and edit them as required:

    - **Price attribute group** – If you're creating a new group, enter a unique name for the group. This field becomes read-only after you save the group.
    - **Validated** – This option indicates whether the group has been validated. Only validated groups are active and can be used in pricing rules. You can't edit this option directly. When you create a new group or edit a previously validated group, this option is set to *No*. To validate a new or edited group, you must select **Validation** on the Action Pane to trigger a validation check. If the check passes, the system sets this option to *Yes*.

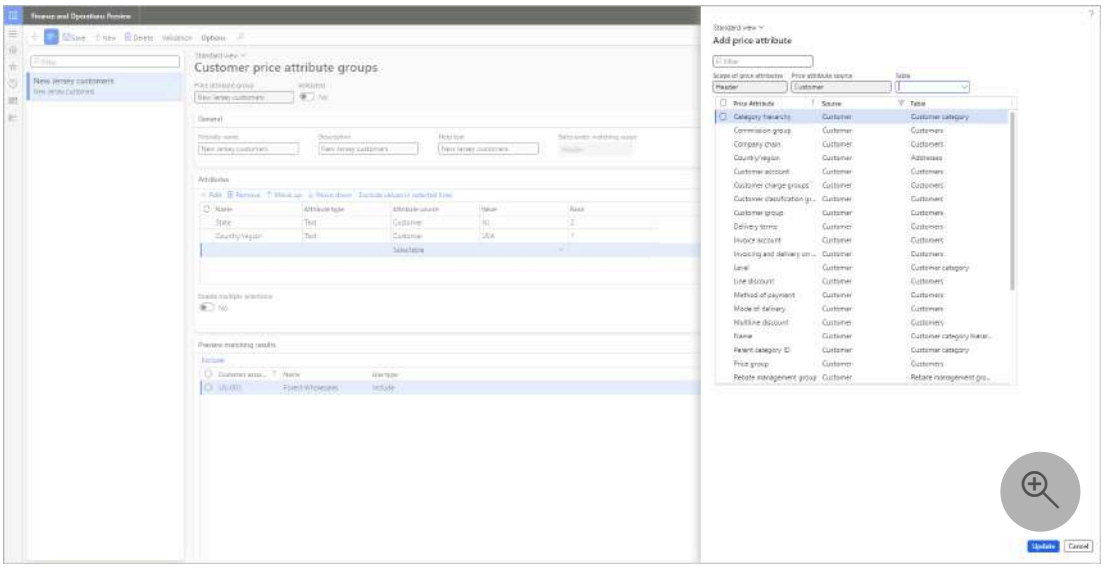
4. On the **General** FastTab, set the following fields:

    - **Friendly name** – Enter a descriptive or common name for the group. For new groups, the value initially matches the value that you entered in the **Price attribute group** field, but you can change it.
    - **Description** – Enter a short description of the group.
    - **Help text** – Enter help text that will be shown to describe the group.
    - **Sales order matching scope** – This read-only field indicates whether the group applies to the overall order header values or to individual order line values. If you're creating a customer price attribute group, the value is always *Header*. If you're creating a product price attribute group, the value is always *Line*. For more information about order matching scopes, see Price attribute groups.

5. On the Action Pane, select **Save**.

6. On the **Attributes** FastTab, review the attributes that belong to the group, and edit them as required. Use the following toolbar buttons to work with the list:

    - **Add** – Open the **Add price attribute** dialog box, where you can add one or more attributes to the list. This dialog box shows attributes that match the selected group type (product or customer) and provides filtering controls to help you find the attributes that you're looking for. Select the checkbox for each attribute that you want to add, and then select **Update** to add those attributes to the group.



- **Remove** – Remove a selected attribute from the list.
- **Move up** and **Move down** – Move a selected attribute up or down in rank.

7. For each row on the **Attributes** FastTab, enter or select one or more values in the **Value** column. All customers or products that match the specified attribute values will become a member of the current group. The following rules apply:

    - All rows are combined by using a logical AND operator. Therefore, only those companies or products that have matching values for *all* rows will be included in the group.
    - If you want to create one or more rows that include multiple values, set the **Enable multiple selections** option to *Yes*. You can then add a comma-separated list of values in the **Values** column for each row. These values are combined by using an OR operator. Therefore, the row will find products or customers that have *any* of the values in the list.
    - You can specify values to exclude by adding an exclamation mark (!) before the value. This exclamation mark is known as an exclusion prefix. For example, to find all companies that are located outside the United States, set the value of the **Country/region** attribute to *!USA*. You can automatically add the exclusion prefix to the values of any row by selecting the row and then selecting **Exclude values in selected lines** on the FastTab toolbar.

8. As you set up attributes and values on the **Attributes** FastTab, the **Preview matching results** FastTab is updated to show the resulting group members. Therefore, you can verify that your settings are giving you the result that you expect.

    - If the list includes any rows that you want to exclude, select the target rows, and select **Exclude** on the toolbar.
    - The **Line type** column indicates which rows you've selected to exclude.

- To re-include a previously excluded row, select it, and then select **Include** on the toolbar.
- If you're working with a product group, to view, exclude, and/or re-include specific item variants, select **All variants** on the toolbar. While you're viewing the variants, you can select **Product master** on the toolbar to go back to the standard item list.

9. When you've finished adding and arranging attributes, select **Validation** on the Action Pane to trigger a validation check. If the check passes, the system sets the **Validated** option to *Yes*. This setting indicates that the price attribute group is activated and can be used in pricing rules.

Use customer and product price attribute groups in price attribute groups

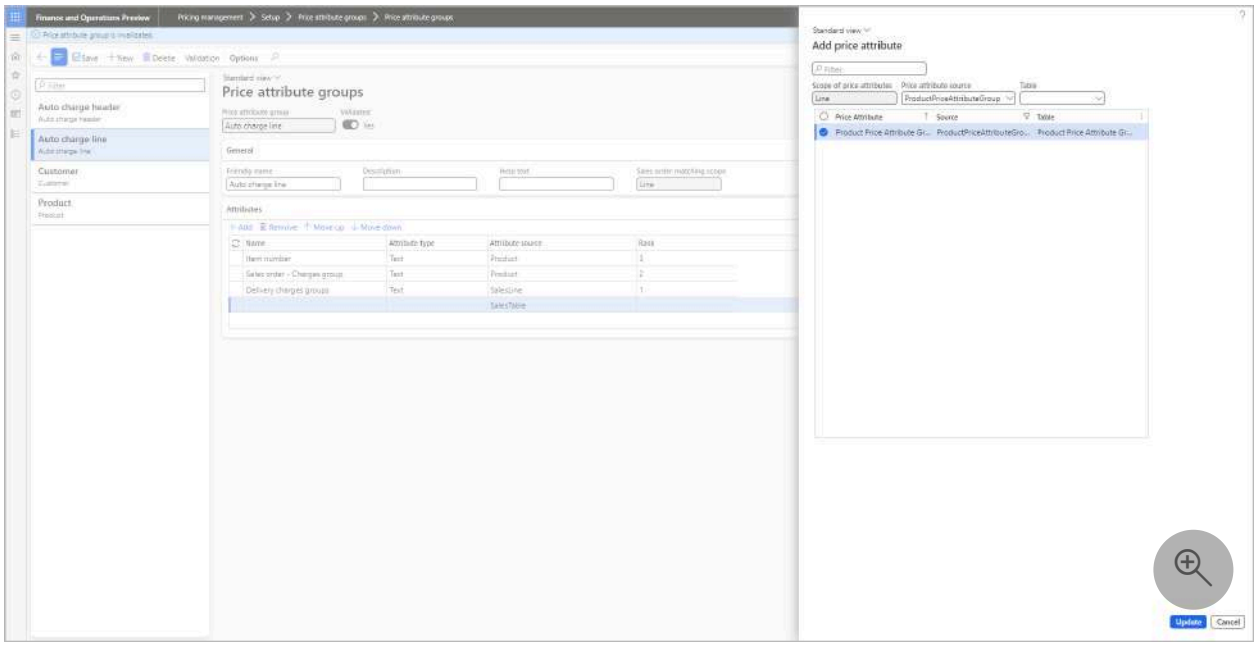
To use a customer or product price attribute group in a price attribute group, add the group as an attribute. You add these attributes just as you add other types of attributes.

Customer price attribute groups can be used only in price attribute groups where the **Sales order matching scope** field is set to *Header*. *Product* price attribute groups can be used only in price attribute groups where the **Sales order matching scope** field is set to *Line*.

You can use the **Price attribute source** filter in the **Add price attribute** dialog box to quickly find the value, as summarized in the following table.

Expand table

Price attribute group scope	Attribute name	Source	Table
Header	Customer Price Attribute Group	CustomerPriceAttributeGroup	Customer Price Attribute Group
Line	Product Price Attribute Group	ProductPriceAttributeGroup	Product Price Attribute Group



For more information about how to set up price attribute groups, see Price attribute groups.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add custom pricing attributes for Dynamics 365 Commerce Point of Sale

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Unified pricing management lets you configure pricing rules that can also consider values of custom pricing attributes for products, customers, orders, and order lines. It provides the following two components that you can use to implement your own custom request handler and develop your logic: `GetCustomizedPricingPropertiesRequest` and `GetCustomizedPricingPropertiesResponse`.

Custom attributes are marked in the `GUPPRICINGATTRIBUTELINK` table using a specific identifier, which indicates that they require special logic in Commerce Scale Unit (CSU) to be supported in Dynamics 365 Commerce Point of Sale (POS). To mark a pricing attribute as custom, the `GUPPRICINGATTRIBUTELINK.TypeName` column must be set to `Customization`.

In CSU logic, if a custom pricing attribute is validated as `GUPPRICINGATTRIBUTELINK.TypeName = 'Customization'`, then the system calls the custom request handler.

Prerequisites

To use the features described in this article, you must be running version 10.0.46 or later of your Microsoft finance and operations apps.

First steps

Start by setting up your environment to implement custom pricing attributes, as described in the following procedure:

1. Implement a custom request handler to process custom pricing attributes. Learn more in Example custom request handler code.
2. Register the output library in the `CommerceRuntime.Ext.config` file. Here's an example of a custom request handler registration (if the output library were

`Contoso.Commerce.Runtime.Services`).

```xml
<add source="assembly" value="Contoso.Commerce.Runtime.Services" />
```

3. Build the solution and deploy it to your CSU.

Add custom pricing attributes

To implement custom pricing attributes for products, customers, orders, and order lines, follow these steps:

1. Create a class to represent each new custom pricing attribute. The class should extend `GUPAbstractPricingAttribute` and implement `GUPIPricingAttribute`. Some methods should also be overwritten, such as `getValueOfAttribute()`, `getName()`, and `getDataType()`. Describe this class with three system attributes:

   - [`GUPPricingMetadataDiscovery`] – Describe all pricing attributes.
   - [`GUPPricingAttributeSourceDiscovery(, )`] – Identify the attribute source this attribute applies to.
   - [`GUPPricingAttributeFieldsDiscovery(fieldStr(, ))`] – Identify the table and field this attribute is related to.

   You must add a new class for each custom pricing attribute you want to add.

   - Here's an example of code that creates a header-level custom pricing attribute (for `Customer` `GUPPricingAttributeSource`):

     ```x++
     /// <summary>
     /// PricingAttribute of table <c>CustTable</c> field StatisticsGroup.
     /// </summary>
     [GUPPricingMetadataDiscovery]
     [GUPPricingAttributeSourceDiscovery(GUPPricingAttributeSource::Customer,
     GUPPricingAttributeSourceLevel::Header)]
     [GUPPricingAttributeFieldsDiscovery(fieldStr(CustTable, StatisticsGroup))]
     internal class GUPPricingAttributeCustTableStatisticsGroup extends
     GUPPricingAttributeCustTable implements GUPIPricingAttribute
     {
         private Name typeName = 'Customization';

         public anytype getValueOfAttribute(Common _pricingObject)
         {
             return this.getTableRecord(_pricingObject).StatisticsGroup;
     ```

```
    }

    public str getName()
    {
        return this.getNameFromField();
    }

    public AttributeDataType getDataType()
    {
        return AttributeDataType::Text;
    }

    public Name getAttributeType()
    {
        return this.typeName;
    }

}
```

- Here's an example of code that creates a line-level custom pricing attribute (for `SalesLine` `GUPPricingAttributeSource`):

X++

```
/// <summary>
/// PricingAttribute of table <c>SalesLine</c> field LineNum.
/// </summary>
[GUPPricingMetadataDiscovery]
[GUPPricingAttributeSourceDiscovery(GUPPricingAttributeSource::SalesLine,
GUPPricingAttributeSourceLevel::Line)]
[GUPPricingAttributeFieldsDiscovery(fieldStr(SalesLine, LineNum))]
internal final class GUPPricingAttributeSalesLineLineNum extends
GUPPricingAttributeSalesLine implements GUPIPricingAttribute
{
    private Name typeName = 'Customization';

    public anytype getValueOfAttribute(Common _pricingObject)
    {
        return this.getTableRecord(_pricingObject).LineNum;
    }

    public str getName()
    {
        return this.getNameFromField();
    }

    public AttributeDataType getDataType()
    {
        return AttributeDataType::Integer;
    }

    public FieldId getField()
    {
```

```
        return fieldNum(SalesLine, LineNum);
    }

    public Name getAttributeType()
    {
        return this.typeName;
    }

}
```

2. For *new* custom pricing attributes, you must programmatically set the `TypeName` column of the `GUPPRICINGATTRIBUTELINK` table to `Customization`. Create an extension to `GUPPricingAttributeRepository` and add a statement to the `toPriceAttributeLink()` method that programmatically sets the `TypeName` column for each new custom pricing attribute. Here's an example of how to do this:

X++

```
[ExtensionOf(classStr(GUPPricingAttributeRepository))]
public static final class GUPPricingAttributeRepository_Extension
{
    public static GUPPricingAttributeLink
toPriceAttributeLink(GUPIPricingAttribute _pricingAttr)
    {
        GUPPricingAttributeLink link = next toPriceAttributeLink(_pricingAttr);

        if (link.AttributeName == "StatisticsGroup")
        {
            link.TypeName = (_pricingAttr as
GUPPricingAttributeCustTableStatisticsGroup).getAttributeType();
        }

        return link;
    }

}
```

3. Build the relevant models, Global Unified Pricing (GUP) or extension, and restart Internet Information Services (IIS). Then, clear the cache for your Microsoft finance and operations apps.

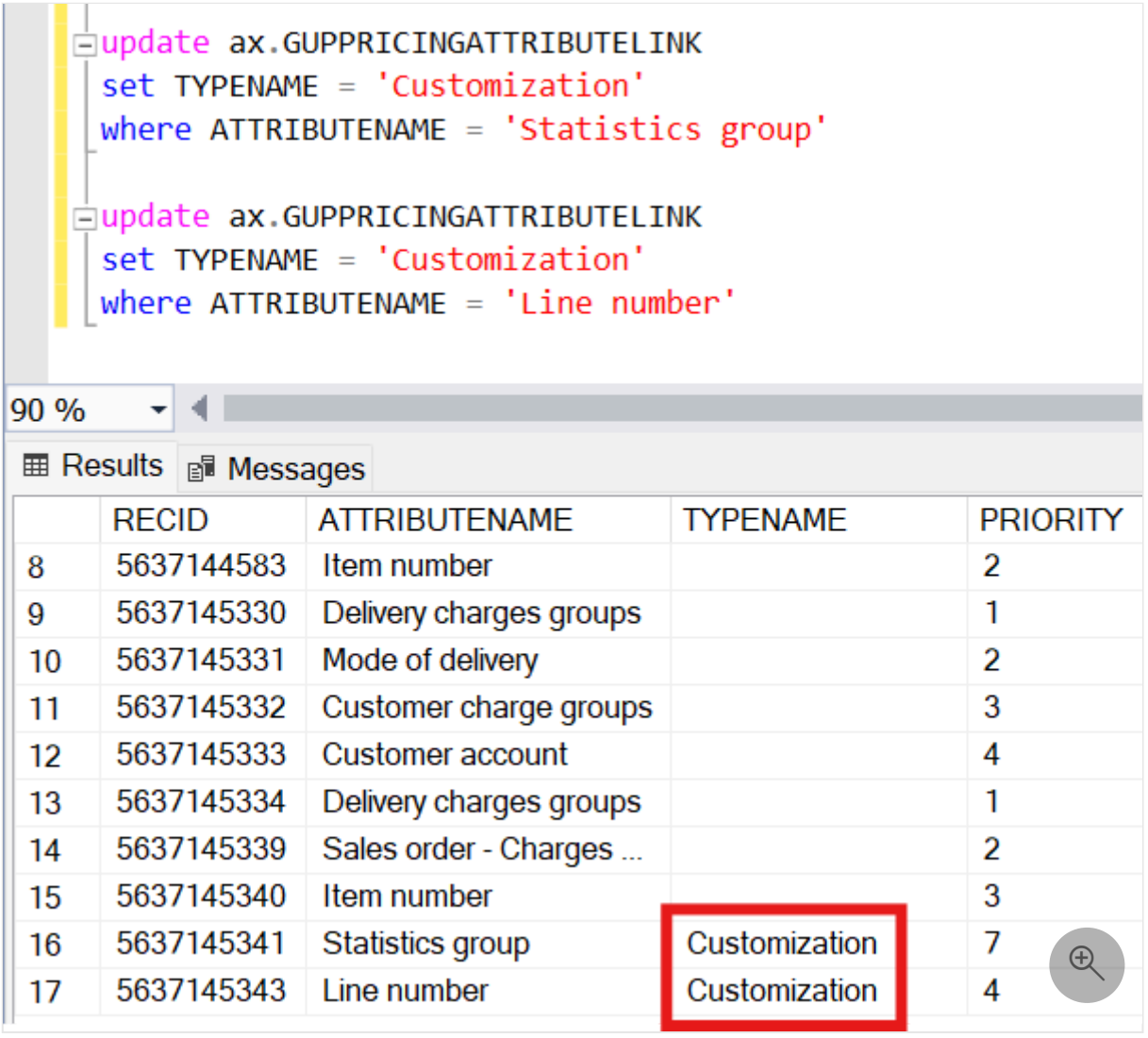
4. Open your Microsoft finance and operations app and go to **Pricing management** > **Setup** > **Price attribute groups** > **Price attribute groups**. Select the price attribute group that you want to customize and then use the **Attributes** FastTab to add your custom pricing attributes to it. Update each group as needed.

5. Go to **Retail and Commerce** > **Retail and Commerce IT** > **Distribution Schedule** and run the *1210 Pricing management* job to sync changes to the Channel database.

6. For *existing* custom pricing attributes, manually set the `TypeName` column of the `GUPPRICINGATTRIBUTELINK` table to `Customization` in SQL Server Management Studio (SSMS). Here's an example of how to set the `Customization` identifier:

```
SQL
update dbo.GUPPRICINGATTRIBUTELINK
set TYPENAME = 'Customization'
where ATTRIBUTENAME = 'Custom attribute name'
```

   The following image shows an example of how entries marked as custom are shown in the `GUPPRICINGATTRIBUTELINK` table.



	RECID	ATTRIBUTENAME	TYPENAME	PRIORITY
8	5637144583	Item number		2
9	5637145330	Delivery charges groups		1
10	5637145331	Mode of delivery		2
11	5637145332	Customer charge groups		3
12	5637145333	Customer account		4
13	5637145334	Delivery charges groups		1
14	5637145339	Sales order - Charges ...		2
15	5637145340	Item number		3
16	5637145341	Statistics group	Customization	7
17	5637145343	Line number	Customization	4

7. For `Customer` or `Product`, create a trade agreement journal that tests the new custom pricing attribute. For `SalesTable` or `SalesLine`, create an auto charge that tests the new custom pricing attribute. In both cases, make sure to set a specific value.

8. Go to **Retail and Commerce** > **Retail and Commerce IT** > **Distribution Schedule** and run the *9999 All jobs* job to sync changes to the Channel database.

9. In POS, you should see the value configured in the trade agreement journal or that an auto charge was correctly applied.

Troubleshooting

You receive a POS notification that a custom request handler isn't implemented

While creating a transaction in POS that is associated with one or more custom pricing attributes, you might receive an error message similar to the following message:

> The request GetCustomizedPricingPropertiesRequest is not implemented in customized code. Please contact your system administrator.

To address this issue, follow these steps:

1. Verify that a custom request handler is implemented and included in the build.
2. Ensure that the custom request handler is correctly registered in the `CommerceRuntime.Ext.config` file. Keep in mind that this registration can be overwritten during the build process, so it's important for you to confirm the registration after the build completes.

You receive a POS notification that custom pricing attributes are detected among out-of-box attributes

While creating a transaction in POS that is associated with one or more custom pricing attributes, you might receive one of the following error messages:

> A custom product pricing attribute has been detected. Please contact your system administrator to verify the setup of custom attributes.

Or:

> A custom customer pricing attribute has been detected. Please contact your system administrator to verify the setup of custom attributes.

To address this issue, follow these steps:

1. Verify that the CSU flight `UPSupportCustomPricingAttributesFlight` is enabled.

2. Confirm that the new custom pricing attributes are identified as custom. The `GUPPRICINGATTRIBUTELINK.TypeName` column should be set to `Customization` for all custom pricing attributes.

Further troubleshooting

To find more information about errors or exceptions, or to review CSU and POS logs, contact Microsoft Support and open a support ticket.

Last updated on 10/30/2025

Example custom request handler code

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides sample code for the `GetCustomizedPricingPropertiesRequestHandler` class, which is used to implement custom pricing attributes. This handler enables you to retrieve and process custom pricing properties for products, customers, orders, and order lines.

The code demonstrates how to extend the pricing framework by defining custom attributes and linking them to specific fields in the database. It also shows how to process requests programmatically to fetch and merge these attributes into the pricing logic. For more details about adding custom pricing attributes, see Add custom pricing attributes for Dynamics 365 Commerce Point of Sale.

X++

```
/**
 * SAMPLE CODE NOTICE
 *
 * THIS SAMPLE CODE IS MADE AVAILABLE AS IS.  MICROSOFT MAKES NO WARRANTIES, WHETHER
EXPRESS OR IMPLIED,
 * OF FITNESS FOR A PARTICULAR PURPOSE, OF ACCURACY OR COMPLETENESS OF RESPONSES, OF
RESULTS, OR CONDITIONS OF MERCHANTABILITY.
 * THE ENTIRE RISK OF THE USE OR THE RESULTS FROM THE USE OF THIS SAMPLE CODE
REMAINS WITH THE USER.
 * NO TECHNICAL SUPPORT IS PROVIDED.  YOU MAY NOT DISTRIBUTE THIS CODE UNLESS YOU
HAVE A LICENSE AGREEMENT WITH MICROSOFT THAT ALLOWS YOU TO DO SO.
 */

namespace Microsoft.Dynamics.Commerce.Runtime.DataServices.Messages
{
    using System;
    using System.Collections;
    using System.Collections.Concurrent;
    using System.Collections.Generic;
    using System.Collections.ObjectModel;
    using System.Diagnostics.CodeAnalysis;
    using System.Linq;
    using System.Runtime.Serialization;
    using System.Text;
    using System.Text.RegularExpressions;
    using System.Threading.Tasks;
    using Microsoft.Dynamics.Commerce.Runtime;
```

```
    using Microsoft.Dynamics.Commerce.Runtime.Data;
    using Microsoft.Dynamics.Commerce.Runtime.Data.Types;
    using Microsoft.Dynamics.Commerce.Runtime.DataModel;
    using Microsoft.Dynamics.Commerce.Runtime.DataServices.Messages;
    using
Microsoft.Dynamics.Commerce.Runtime.DataServices.Messages.Pricing.AttributePricing;
    using Microsoft.Dynamics.Commerce.Runtime.Handlers;
    using Microsoft.Dynamics.Commerce.Runtime.Messages;
    using Microsoft.Dynamics.Retail.Diagnostics;

    /// <summary>
    /// The request handler for <c>GetCustomizedPricingPropertiesRequest</c> class.
    /// </summary>
    public sealed class GetCustomizedPricingPropertiesRequestHandler :
SingleAsyncRequestHandler<GetCustomizedPricingPropertiesRequest>
    {
        private const string PricingAttributeLinkFieldPathNotExists =
"PATH_NOT_EXIST";

        private IPricingAttribute SalesOrderInclTax
        {
            get
            {
                return new PricingAttributeLink()
                {
                    AttributeName = "Prices include sales tax",
                    TypeName = "Customization",
                    SourceName = "SalesTable",
                    FieldName = "InclTax",
                    FieldPath = "SalesTable#InclTax",
                    DataType = ExtensibleAttributeDataType.Boolean,
                };
            }
        }

        private IPricingAttribute SalesLineLineNumber
        {
            get
            {
                return new PricingAttributeLink()
                {
                    AttributeName = "Line number",
                    TypeName = "Customization",
                    SourceName = "SalesLine",
                    FieldName = "LineNum",
                    FieldPath = "SalesLine#LineNum",
                    DataType = ExtensibleAttributeDataType.Integer,
                };
            }
        }

        /// <summary>
        /// Processes the <c>GetCustomizedPricingPropertiesRequest</c>.
        /// The request should not be null.
        /// </summary>
```

```
        /// <param name="request">The request parameter.</param>
        /// <returns>The <c>GetCustomizedPricingPropertiesResponse</c>.</returns>
        protected override async Task<Response>
Process(GetCustomizedPricingPropertiesRequest request)
        {
            ThrowIf.Null(request, nameof(request));
            ThrowIf.Null(request.RequestContext, nameof(request.RequestContext));

            var databaseContext = new DatabaseContext(request.RequestContext);

            foreach (PricingAttributeLink link in request.PricingAttributeLinks)
            {
                if (request.SourceEntity is Customer customer)
                {
                    if (string.Equals(link.TypeName, "Customization",
StringComparison.OrdinalIgnoreCase)
                            && string.Equals(link.FieldName, "StatisticsGroup",
StringComparison.OrdinalIgnoreCase))
                    {
                        await this.GetCustomizedCustomerPricingPropertiesFromTable(
                                    request,
                                    databaseContext,
                                    request.PricingAttributeLinks,
                                    customer.AccountNumber)
                                .ConfigureAwait(false);
                    }
                }

                if (request.SourceEntity is Item item)
                {
                    if (string.Equals(link.TypeName, "Customization",
StringComparison.OrdinalIgnoreCase)
                            && string.Equals(link.FieldName, "NameAlias",
StringComparison.OrdinalIgnoreCase))
                    {
                        await this.GetCustomizedItemPricingPropertiesFromTable(
                                    request,
                                    databaseContext,
                                    request.PricingAttributeLinks,
                                    item.ItemId)
                                .ConfigureAwait(false);
                    }
                }

                if (request.SourceEntity is SalesLine salesLine)
                {
                    if (salesLine.LineNumber != 0)
                    {

request.AttributeSource.PricingProperties[this.SalesLineLineNumber] =
salesLine.LineNumber;
                    }
                }

                if (request.SourceEntity is SalesTransaction salesTransaction)
```

```
                {

request.AttributeSource.PricingProperties[this.SalesOrderInclTax] =
salesTransaction.IsTaxIncludedInPrice;
                }
            }

            return new
GetCustomizedPricingPropertiesResponse(request.AttributeSource.PricingProperties);
        }

        private async Task<Response>
GetCustomizedCustomerPricingPropertiesFromTable(
            GetCustomizedPricingPropertiesRequest request,
            DatabaseContext databaseContext,
            IEnumerable<PricingAttributeLink> pricingAttributes,
            string accountNumber)
        {
            pricingAttributes = pricingAttributes
                .Where(attr =>
attr.Source.Equals(PricingAttributeSourceType.Customer))
                .Where(attr =>
                    !string.IsNullOrWhiteSpace(attr.FieldPath)
                    &&
!attr.FieldPath.Equals(PricingAttributeLinkFieldPathNotExists,
StringComparison.OrdinalIgnoreCase));

            if (pricingAttributes.Any())
            {
                var query = new SqlPagedQuery(QueryResultSettings.AllRecords)
                {
                    DatabaseSchema = "ax",
                    Select = new ColumnSet("StatisticsGroup"),
                    From = "CustTable",
                    Where = "ACCOUNTNUM = @accountNumber and DATAAREAID =
@dataAreaId",
                    Parameters =
                    {
                        ["@accountNumber"] = accountNumber,
                        ["@dataAreaId"] =
request.RequestContext.GetChannelConfiguration().InventLocationDataAreaId,
                    },
                };

                var selectedPricingProperties = await
this.ExecuteSelectedPricingAttributesQueryAndResolveResult(databaseContext, query,
pricingAttributes, request).ConfigureAwait(false);
                request.AttributeSource.Merge(selectedPricingProperties);
            }

            return NullResponse.Instance;
        }

        private async Task<Response> GetCustomizedItemPricingPropertiesFromTable(
            GetCustomizedPricingPropertiesRequest request,
```

```
            DatabaseContext databaseContext,
            IEnumerable<PricingAttributeLink> pricingAttributes,
            string itemId)
        {
            pricingAttributes = pricingAttributes
                .Where(attr =>
attr.Source.Equals(PricingAttributeSourceType.Product))
                .Where(attr =>
                    !string.IsNullOrWhiteSpace(attr.FieldPath)
                    &&
!attr.FieldPath.Equals(PricingAttributeLinkFieldPathNotExists,
StringComparison.OrdinalIgnoreCase));

            if (pricingAttributes.Any())
            {
                var query = new SqlPagedQuery(QueryResultSettings.AllRecords)
                {
                    DatabaseSchema = "ax",
                    Select = new ColumnSet("NameAlias"),
                    From = "InventTable",
                    Where = "ITEMID = @itemId and DATAAREAID = @dataAreaId",
                    Parameters =
                    {
                        ["@itemId"] = itemId,
                        ["@dataAreaId"] =
request.RequestContext.GetChannelConfiguration().InventLocationDataAreaId,
                    },
                };

                var selectedPricingProperties = await
this.ExecuteSelectedPricingAttributesQueryAndResolveResult(databaseContext, query,
pricingAttributes, request).ConfigureAwait(false);
                request.AttributeSource.Merge(selectedPricingProperties);
            }

            return NullResponse.Instance;
        }

        private async Task<SortedDictionary<IPricingAttribute, object>>
ExecuteSelectedPricingAttributesQueryAndResolveResult(
            DatabaseContext databaseContext,
            SqlPagedQuery query,
            IEnumerable<PricingAttributeLink> pricingAttributes,
            Request request)
        {
            IDictionary<string, ISet<object>> dbResults = new Dictionary<string,
ISet<object>>();

            await databaseContext
                .ExecuteQueryWithRetryAsync(
                    query,
                    (result) =>
                    {
                        for (int i = 0; i < result.FieldCount; ++i)
                        {
```

```
                            var columnName = result.GetName(i);
                            var columnValue = result.GetValue<object>(i);

                            if (columnValue is not null)
                            {
                                if (!dbResults.ContainsKey(columnName))
                                {
                                    dbResults[columnName] = new HashSet<object>();
                                }

                                dbResults[columnName].Add(columnValue);
                            }
                        }

                        return Task.CompletedTask;
                    })
                .ConfigureAwait(false);

            var selectedPricingProperties = new SortedDictionary<IPricingAttribute,
object>(PricingComparer.Instance);

            foreach (var dbResult in dbResults)
            {
                var attributeLink = pricingAttributes.FirstOrDefault(attr =>
attr.FieldName.Equals(dbResult.Key, StringComparison.OrdinalIgnoreCase));

                if (attributeLink != default)
                {
                    if (dbResult.Value.Count > 1)
                    {
                        selectedPricingProperties[attributeLink] = string.Join(",",
dbResult.Value);
                    }
                    else
                    {
                        selectedPricingProperties[attributeLink] =
dbResult.Value.FirstOrDefault();
                    }
                }
            }

            return selectedPricingProperties;
        }
    }
}
```

Last updated on 10/30/2025

Price groups overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

A *price group* is a flexible tool that enables businesses to manage and adjust their pricing strategies across a wide range of attributes. These attributes include sales channels, customer groups, order types, and delivery methods. The power of a price group lies in its ability to logically bundle these attributes together. Businesses can consistently apply pricing rules and promotions across multiple entities without having to modify individual trade agreements, discount setups, or pricing components. This flexibility is valuable in both Microsoft Dynamics 365 Commerce and Dynamics 365 Supply Chain Management, where pricing must adapt quickly to changing business conditions but also maintain consistency across various settings.

In Commerce, price groups offer a crucial level of flexibility for businesses that must run promotions or adjust pricing for specific customer segments or sales channels. For example, a retail company wants to launch a promotion that targets specific stores or e-commerce channels. By creating a price group, the retailer can easily apply the promotion across those stores and channels. In addition, the company can add stores to the promotion, or remove stores from it, without changing the underlying trade agreements or discount configurations. This ease of adjustment ensures that the company can quickly react to market trends or customer behavior, and that it can maintain the efficiency of its pricing strategy without having to do complex reconfiguration.

In Supply Chain Management, price groups can be just as valuable. For example, a company offers different pricing based on *order types*. Priority orders have a higher price, because delivery is expedited. Standard orders receive discounts for longer delivery times. By grouping these order types into a price group, the company can adjust pricing for the different categories without having to rework the underlying agreements or discount structures. If the company later decides to introduce a new order type or retire an existing one, it can add the new order type to the price group, or remove the existing order type from it. Therefore, the management of pricing strategies is streamlined.

Because of the flexibility that price groups provide, companies can make quick, broad changes across a set of trade agreements or discounts without disrupting the underlying setup. This capability is especially beneficial for companies that use Commerce, because they are already accustomed to using price groups to efficiently manage promotions and customer-specific pricing.

Price groups in Microsoft finance and operations apps

Price groups classify products or services based on their price level. They give companies a flexible way to manage and adjust pricing strategies. Retailers, wholesalers, and manufacturers can segment their target markets and define different prices for different customers.

Price groups are used to calculate prices in Supply Chain Management and Commerce, and attribute-based pricing management.

The following table shows how price groups fit into the various modules and feature areas in Supply Chain Management and Commerce.

Expand table

Feature area	Association type	Context type	Description
Sales and marketing module pricing	Customer or product	• Trade agreement	Price groups that are associated with customers and products can be selected as group criteria during the maintenance of trade agreement lines.
Commerce pricing	Channel, affiliation, loyalty program, or catalog	• Trade agreement • Adjustment • Discounts	Price groups can be associated with channels, affiliations, loyalty programs, and catalogs in trade agreements, adjustments, and discounts.
Pricing management (deprecated) pricing	As an attribute	• Trade agreement • Adjustment • Discounts	Price groups can be used as an attribute in trade agreements, adjustments, and discounts.
Unified pricing management pricing	As an attribute or with attribute assignment	• Trade agreement • Adjustment • Discounts	Price groups can be used as an attribute in trade agreements, adjustments, and discounts.

You can associate price groups with attributes from customers, sales order headers, and other entities, including channels, loyalty programs, and affiliations.

Price groups are optional in Unified pricing management.

Price groups in Unified pricing management

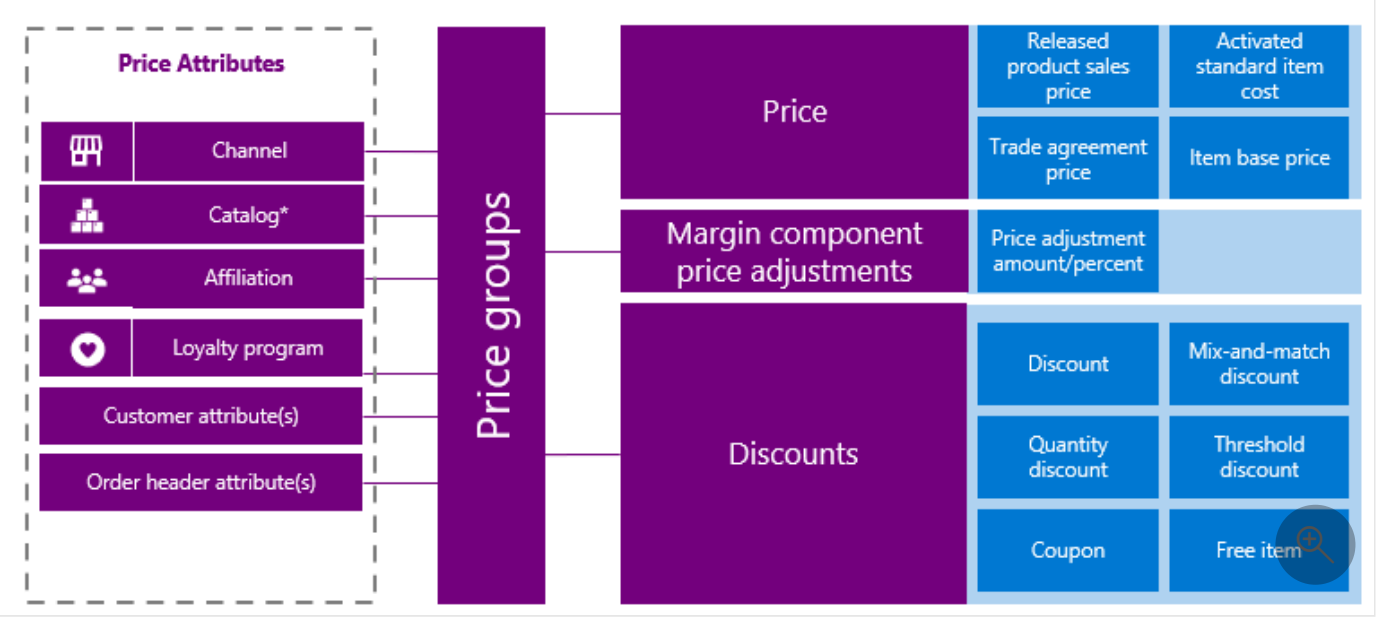
In the **Unified pricing management** module, price groups can be used to implement price discrimination strategies. For example, different prices can be charged for the same product, depending on the time of purchase, the location of the customer, or the demand conditions.

> **Note**
>
> Price groups are typically used in Commerce because they suit the way that Commerce users are accustomed to working. However, they're completely optional. Depending on the pricing scenarios that you want to implement, you might decide not to use price groups, especially if you use only Supply Chain Management.

Price groups can be defined in the following ways:

- Price groups can be used to define strategies that are associated with attributes.
- Only price groups that have no attributes associated with them can be used as attributes in header attribute groups.

The following illustration shows how price groups relate price attributes to prices, margin adjustments, and discounts. The price group is centrally positioned in pricing and discount management. The left side of the illustration shows the attributes and commerce entities that are used to manage differential prices (channels, affiliations, and loyalty programs) and discounts. The right side shows the areas where price groups can be configured in price and discount records.



In Unified pricing management, price groups provide the following functionality:

- Create logical groups of multiple pricing rules.
- Define shared context for price segmentation.
- Assign prices and discounts to Commerce entities (including channels, affiliations, and loyalty programs), and attributes from customers and sales order headers.
- Establish priorities. After price groups are configured with attribute association, ranks that are assigned to the price groups take priority over ranks that are assigned in other places.

> ⓘ Note
>
> You can't currently use price groups with Commerce catalogs.

Benefits of price groups

Price groups offer the following benefits:

- **Flexibility** – Price groups provide a flexible way to manage pricing strategies across various attributes, such as customer segments, channels, order types, and more. Therefore, businesses can quickly adapt without having to modify individual trade agreements or discount setups. Currently, only pricing header attributes are supported. However, we plan to extend support to pricing line attributes.
- **Unified nature** – Price groups offer significant value for Commerce customers. They enable efficient management of dynamic promotions, channel-specific pricing strategies, affiliations programs, and loyalty programs. In Supply Chain Management, price groups bring similar benefits by simplifying pricing strategies for key attributes, such as customer

segments, order types, and delivery methods. Therefore, they provide the flexibility to make adjustments without having to reconfigure underlying pricing structures.

- **Versatility** – Price attribute groups are linked to specific price components, and their configuration isn't easily changed after they're enabled. By contrast, price groups provide a versatile and flexible way to dynamically adjust prices on top of the established setup. Additionally, price groups aren't mandatory. Therefore, businesses can decide whether to use them, based on their specific needs and complexity.
- **Capability to make dynamic adjustments** – As business needs evolve, pricing attributes can be modified to reflect changes in strategy. For example, new channels can be introduced, or loyalty programs can be expanded. Unified pricing management simplifies the implementation of these changes without disrupting the overall pricing strategies.
- **Consistency across platforms** – The unified approach ensures that pricing strategies are consistently applied across all touch points in both Supply Chain Management and Commerce.

Examples of price group use

The following subsections provide examples that show how price groups can be used in various business-to-business (B2B) and business-to-consumer (B2C) scenarios.

Example 1: Channels

A wholesaler, retailer, or distributor wants to offer different prices for a product, depending on whether it's sold online, in a physical store, or through a mobile app. Therefore, it defines the following price groups by using the *Channels* attribute:

- Online store pricing
- Physical store pricing
- Mobile app pricing

The product is priced at £100 online, £105 in physical stores, and £98 through the mobile app. These prices are created as trade agreements and linked to channels via the price group association. The final step is to attach the relevant price groups to the appropriate stores to ensure that the correct pricing is applied across all sales channels.

Example 2: Affiliations

A company offers a special discount to all its employees. Therefore, it creates a price group that is named *Employee benefits* by using the *Affiliation* attribute. Employees receive a 15% discount on

all products. This discount is automatically applied at checkout when the system recognizes an employee's affiliation.

Example 3: Loyalty programs

The Fabrikam company wants to reward members of the *Fabrikam rewards* loyalty program by offering exclusive pricing. Therefore, it sets up a price group that is named *Fabrikam rewards pricing*. Members of this loyalty program receive a 10% discount on selected products, and this pricing is linked directly to their loyalty program membership.

Example 4: Customer

A company wants to configure a specific range of customers for some discounts. Therefore, it sets up price groups that are named *US customers*, *European customers*, and *Asian customers*. The company can then assign different discounts, based on geographic region.

Next steps

- Set up price groups for Unified pricing management
- Assign price groups
- Price group priorities

Last updated on 05/18/2026

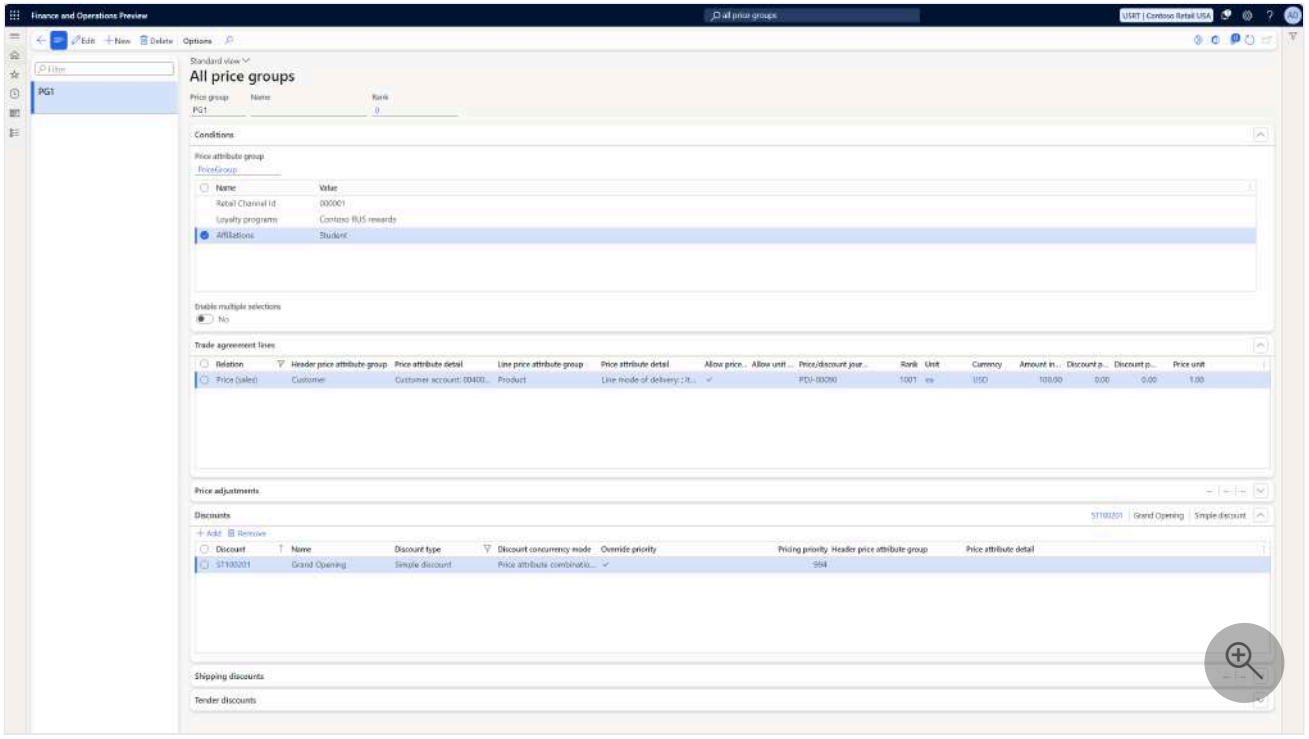
Set up price groups for Unified pricing management

To create and manage price groups that you can associate with attributes, follow these steps:

1. Go to **Pricing management** > **During-sales pricing** > **Price groups** > **All price groups**.

2. Follow one of these steps:

    - To create a new price group, select **New** on the Action Pane.
    - To view an existing price group, select it in the list pane. Then, if you want to edit the price group, select **Edit** on the Action Pane.
    - To delete an existing price group, select it in the list pane, and then select **Delete** on the Action Pane.



3. On the header of the new record, set the following fields:

    - **Price group** – Enter a unique identifier for the price group. You can edit this field only when you create a new record.
    - **Name** – Enter a descriptive name for the price group.
    - **Rank** – Select a rank for the price group. In situations where multiple price groups apply, the system uses the price group that has the highest priority. The larger the value is, the higher the priority.

4. On the **Conditions** FastTab, set the following fields to define membership rules that control which market segments are part of the price group:

    - Set the **Price attribute group** field to the price attribute group that you want to use with the price group. The grid is then updated to show the attributes that belong to the selected price attribute group.
    - For each attribute in the grid, select an appropriate value in the **Value** field.
    - Set the **Enable multiple selections** option to *Yes* to allow multiple values for each row. If you define more than one value for a row, separate the values with commas. The system combines the values by using an *OR* operator.

5. To view the list of trade agreement lines that the current price group was assigned to, select the **Trade agreement lines** FastTab.

6. To view and edit the price adjustments that apply to the current price group, select the **Price adjustments** FastTab. Then use the buttons on the toolbar to add or remove price adjustments as you require.

7. To view and edit the discounts that apply to the current price group, select the **Discounts** FastTab. Then use the buttons on the toolbar to add or remove discounts as you require.

> **Note**
>
> In the current version, the **Shipping discounts** and **Tender discounts** FastTabs on the **All price groups** page have no effect.

Last updated on 07/01/2026

Assign price groups

Assign price groups to trade agreements

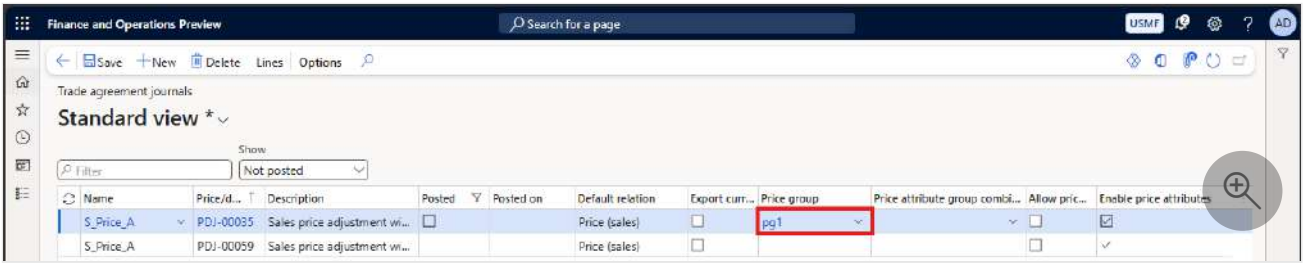
To assign a price group to a trade agreement, follow these steps:

1. Go to **Pricing management** > **During-sales pricing** > **Sales trade agreement price** > **Trade agreements journals**.

2. You can assign price groups only to journals that haven't yet been posted. Therefore, set the **Show** filter to *Not posted*.

3. Find and select the trade agreement journal that you want to assign a price group to.

4. In the **Price group** column, specify the price group that should be inherited by all lines that are added to the journal.



5. On the Action Pane, select **Lines**.

6. The **Journal lines, trade agreement** page shows the lines that belong to the selected journal. You can perform the following actions as required:

    - To add a line, select **New** on the Action Pane. Each new line inherits the price group that you assigned to the journal in step 4.
    - Edit the price group for each line.

Assign price groups to price adjustments or discounts

To assign price groups to a price adjustment or discount, follow these steps:

1. Follow one of these steps:

    - To work with a price adjustment, go to **Pricing management** > **During-sales pricing** > **Price adjustments** > **Margin component price adjustments**.

- To work with a discount, go to **Pricing management** > **During-sales pricing** > **Discounts** > **All discounts**.

2. In the list pane, select the price adjustment or discount that you want to assign a price group to. The price adjustment or discount that you select must have a status of *Disabled*. (You can view the status on the **General** FastTab.)

3. On the Action Pane, select **Price groups**.

4. On the **Price groups** page, on the Action Pane, select **New** to add a price group.

5. On the header of the new record, in the **Price group** field, specify a name for the price group.

6. On the Action Pane, select **Save**. The following fields on the page are updated:

   - **Pricing priority** – The value is updated to the **Rank** value of the selected price group.
   - **Price attribute group** – The value is updated to the **Price attribute group** value of the selected price group. The grid under this field is updated to show the conditions for the selected price group.

7. On the Action Pane, select the **Back** button to return to the price adjustment or discount. On the **General** FastTab, you can view the pricing priority that applies to that adjustment or discount. By default, the highest priority among the price groups that you assigned is shown. However, you can override the value by setting the **Override priority** option to *Yes*.

8. Repeat this procedure to add more price groups as required.

Last updated on 12/30/2025

Price group priorities

In situations where more than one price calculation can apply, the system evaluates the priority of each group to determine which one should apply. This article explains how the priority is evaluated.

Priority configuration

You can assign ranks for pricing calculations in several places. Records that you assign a higher rank to have a higher priority when the system searches for a calculation.

The following table lists the pages where you can assign ranks. It also shows how the ranks that are assigned on each page are used to calculate priorities.

Expand table

Element	Page	Field name	Calculated priority
Price attribute	Price attribute groups (**Pricing management** > **Setup** > **Price attribute groups** > **Price attribute groups**)	Rank	Same value (No conversion is done.)
Price group	All price groups (**Pricing management** > **During-sales pricing** > **Price groups** > **All price groups**)	Rank	1,000 + *Rank*
Price attribute group combination	Price component codes (**Pricing management** > **Setup** > **Price component codes** > **Price component codes**)	Combination rank	Same value (No conversion is done.)
Price component code	Price trees (**Pricing management** > **Setup** > **Price component codes** > **Price trees**)	Pricing sequence	1,000 – *Sequence*

Priority application sequence

When the system calculates prices, it checks for priorities for trade agreements, adjustments, and discounts in a specific sequence. The following table outlines the sequence that the system uses to check pricing rules during price calculations. Each sequence is numbered from 1 through 4. These numbers indicate the order of checks.

Expand table

Pricing rule	Check sequence	Field name	Visible in the pricing rule	Configured from
Trade agreement	1	Rank	No	Price group
	2	Pricing sequence	No	Price tree
	3	Combination rank	Yes	Price component code
	4	Attribute rank	No	Price attribute group
Adjustment	1	Rank	Yes	Price group
	2	Pricing sequence	Yes	Price tree
	3	Combination rank	Yes	Price component code
	4	Attribute rank	No	Price attribute group
Discount	1	Rank	Yes	Price group
	2	Pricing sequence	Yes	Price tree
	3	Combination rank	Yes	Price component code
	4	Attribute rank	No	Price attribute group

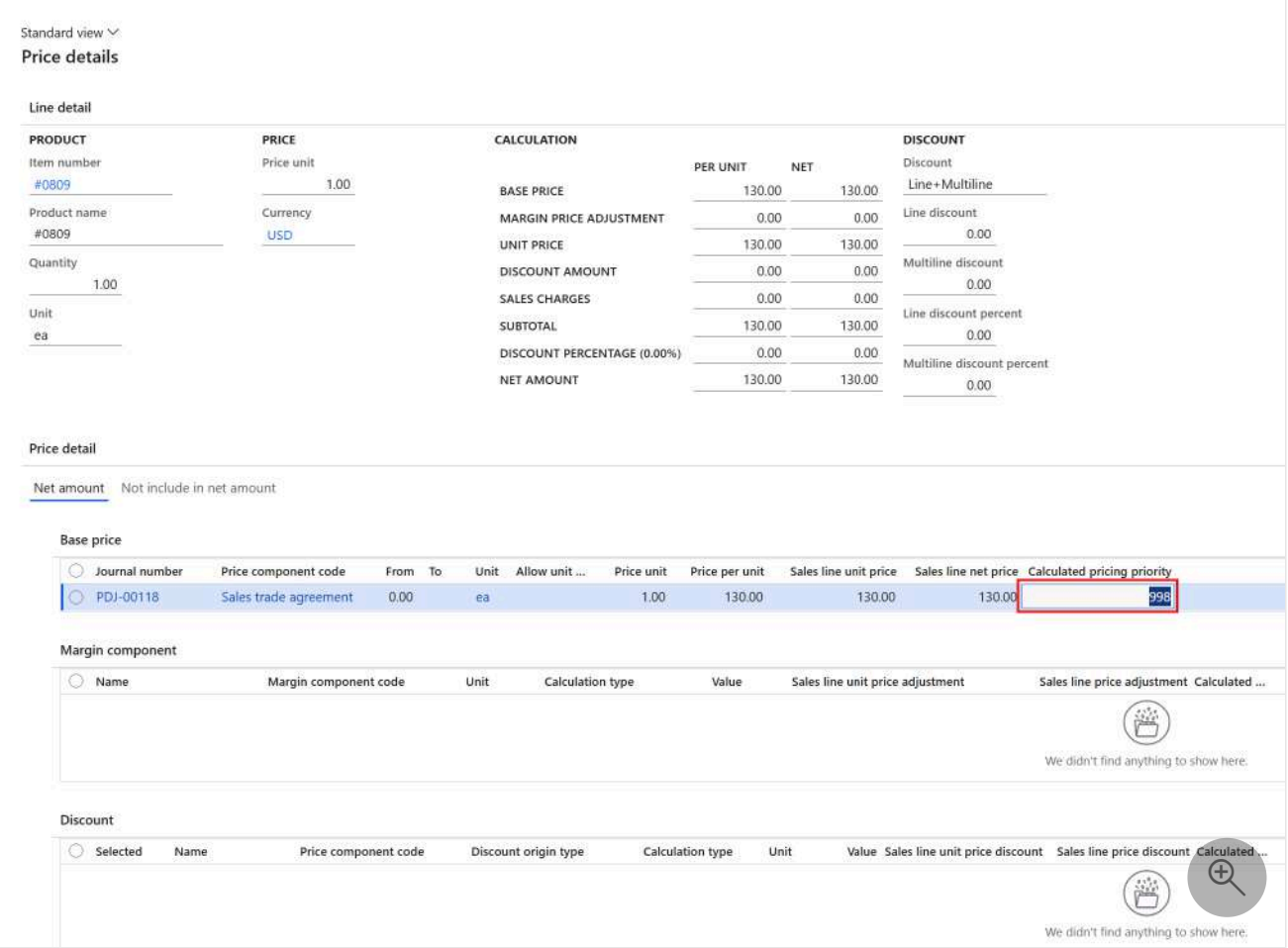
Review calculated priorities

After a calculation, the calculated priorities are shown on sales order lines for the applied pricing rules. To review the calculated priority values, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Find and open the sales order that you want to check.
3. On the **Sales order lines** FastTab, select the line that you want to check.
4. On the toolbar of the **Sales order lines** FastTab, select **Sales order line** > **View** > **Price details**.

   On the **Price details** page, each grid (**Base price**, **Margin component**, and **Discount lines**) includes a **Calculated pricing priority** column. This column shows the calculated priority of

each line in the grid.



Related information

- Price attribute groups
- Set up price groups for Unified pricing management
- Price component codes
- Arrange price component codes into a price structure

Last updated on 12/30/2025

Price component codes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The goal of making pricing decisions is to increase profitability. Good decision making requires a thorough understanding of the different elements that determine the price. By providing a list of pricing building blocks, Unified pricing management helps you create a price structure for pricing at the macro level. The price component code is the basic unit of the price structure. Set up your price structure by creating one or more price trees in Unified pricing management.

Expand table

Name	Description

Price components	*Price components* are the main price elements of the price structure. Here are some examples:
	• Price:
	  ◦ Base price
	  ◦ Sales trade agreement price
	  ◦ Margin component
	• Discounts
	• Rebate management
	• Auto charges

Price component codes	*Price component codes* are defined and structured based on the function and business use of your pricing elements. • Price component codes can be built according to the price sequence. • Price component codes can define default values for posting and the discount concurrency mode. • Each pricing and discount record is assigned to a specific price component code.
Price component groups	Price component codes can be grouped into *price component groups*.

Name	Description
Price attribute groups	Each *price attribute group* contains several price attributes. Each price component code can have one *header price attribute group* and one *line price attribute group*. Use these groups to define pricing rules based on attribute values that are defined in *pricing component codes*.
Combination rank	The *combination rank* for a price attribute group lets you define what occurs if multiple rules from the same price component code apply to a given order. If the concurrency mode is based on the *price attribute combination rank*, the rule that has highest rank applies.

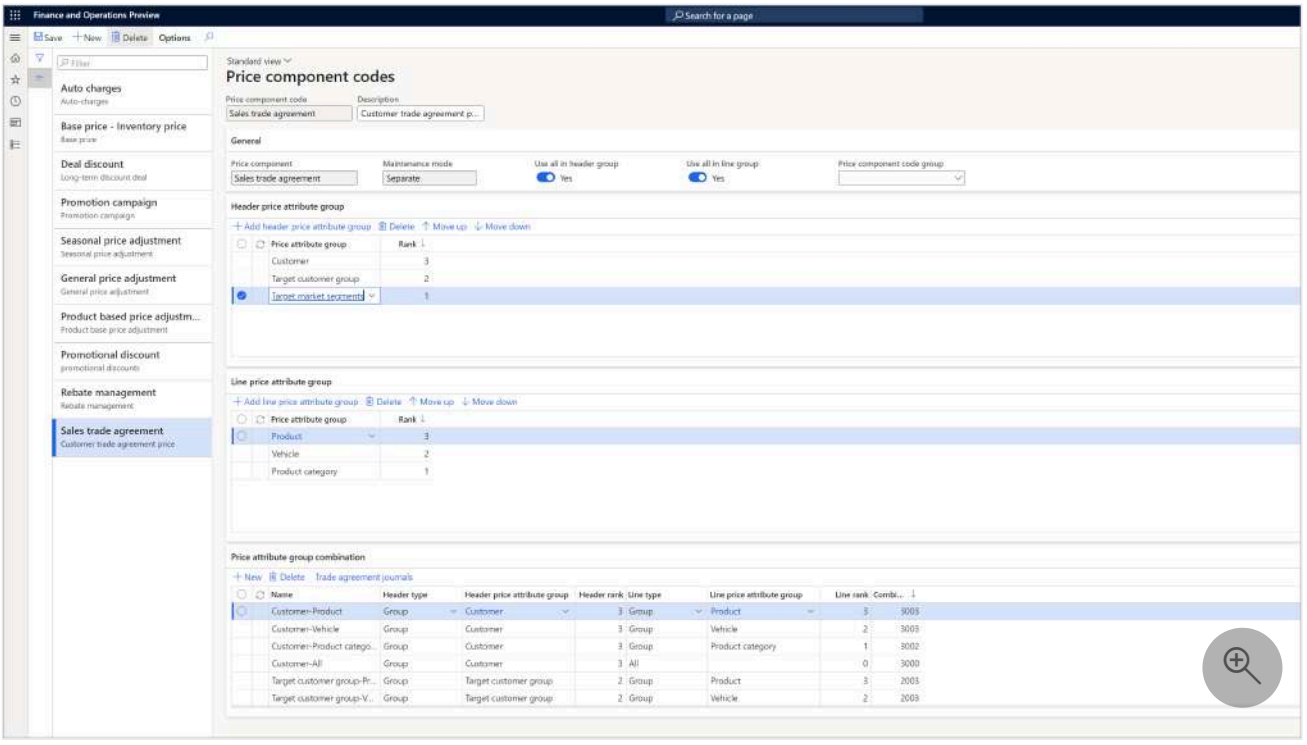
> **Note**
>
> You can have a maximum of one price component code record for each of the following price components:
>
> - Base price - inventory price
> - Base price - purchase price
> - Base price - sales price
> - Sales trade agreement
>
> You can have any number of price component code records for each of the remaining price components.

Within each price component code, you can define multiple pricing rule records that have different combinations of price attribute groups.

Set up price component codes

Follow these steps to set up your price component codes.

1. Go to **Pricing management** > **Setup** > **Price component codes** > **Price component codes**.



2. Follow one of these steps:

   - To create a new price component code, select **New** on the Action Pane.
   - To edit an existing price component code, select it in the list pane, and then select **Edit** on the Action Pane.
   - To delete an existing price component code, select it in the list pane, and then select **Delete** on the Action Pane.

3. On the header of the new or selected price component code, set the following fields:

   - **Price component code** – Enter a name for the code. You can edit this field only for new records. (It becomes read-only when you save the record.)
   - **Description** – Enter more details that are related to the price component code that you're creating.

4. On the **General** FastTab, set the following fields:

   - **Price component** – Select the type of price component that you're creating. You can edit this field only for new records. (It becomes read-only when you save the record.) For more information about price component types, see Price structure overview.

   - **Maintenance mode** – Select one of the following values to specify how you'll create or generate the different combinations of price attribute groups that the price component code supports when you create pricing rules for it. You can edit this field only for new records. (It becomes read-only when you save the record.)

    - *Separate* – You can assign a rank to each individual header and line attribute group. The system then automatically generates each possible combination of header and line attribute groups, and assigns a combined rank to each combination, based on your individual rankings.
    - *Combined* – You can define each relevant combination of header and line attributes, and manually assign a combination rank to each of them.

- **Use all in header group** – Set this option to *Yes* if you have the combination where header attributes have a value of *All*, so that you can create pricing rules that apply to all customers.

- **Use all in line group** – Set this option to *Yes* if you have the combination where line attributes have a value of *All*, so that you can create pricing rules that apply to all products.

- **Default auto charge concurrency mode** – Select the default concurrency rule to use for auto charges that are associated with the price component code. For information about how this type of concurrency works and the effects of each value, see Resolve concurrency within price component codes. This field is available only when the **Price component** field is set to *Auto charges*.

- **Default discount concurrency mode** – Select the default concurrency rule to use for discount pricing rules that are associated with the price component code. For information about how this type of concurrency works and the effects of each value, see Resolve concurrency within price component codes. This setting is available only when the **Price component** field is set to *Margin component* or *Discount*.

- **Price component code group** – Select a price component code group. Set up and maintain your component code groups on the **Price component groups** page (**Pricing management** > **Setup** > **Price component codes** > **Price component groups**).

5. If you set the **Maintenance mode** field to *Separate*, on the **Header price attribute group** FastTab, add each header attribute group that you want to use with the price component code. Use the buttons on the toolbar to add, remove, and rearrange header price attribute groups as required.

6. If you set the **Maintenance mode** field to *Separate*, on the **Line price attribute group** FastTab, add each line attribute group that you want to use with the price component code. Use the buttons on the toolbar to add, remove, and rearrange line price attribute groups as required.

7. If you set the **Maintenance mode** field to *Combined*, use these guidelines to set up the **Price attribute group combination** FastTab:

    - Use the buttons on the toolbar to add and remove price attribute groups as required.

    - For each attribute group, set the following fields:
        - **Name** – Enter a descriptive name for the line.
        - **Header type** – Select the type of criteria that should be available for specifying header attributes in pricing rules that use the price attribute group combination. Select *All* to create rules that apply to all customers. Select *Group* to select a group of price attributes that the pricing rules can specify values for to identify specific customers.
        - **Header price attribute group** – If you set the **Header type** field to *Group*, specify the attribute group that the price attribute group combination provides so that header attributes can be specified in pricing rules.
        - **Line type** – Select the type of criteria that should be available for specifying line attributes in pricing rules that use the price attribute group combination. Select *All* to create rules that apply to all products. Select *Group* to select a group of price attributes that the pricing rules can specify values for to identify specific products.
        - **Line price attribute group** – If you set the **Line type** field to *Group*, specify the attribute group that the price attribute group combination provides so that line attributes can be specified in pricing rules.
        - **Combination rank** – Assign a rank to the price attribute group combination. The rank is used to resolve concurrency if more than one pricing rule applies to the price component code. Learn more in the Price attribute combination rank section of this article.

    - For price component codes where **Price component** is *Sales trade agreement*, the toolbar on the **Price attribute group combination** FastTab includes a **Trade agreement journals** button. Select a row, and then select this button to create a new trade agreement journal for the selected row. The selected row is the default **Price attribute group combination** field in the trade agreement journal. Learn more in Sales trade agreement prices.

8. If you set the **Maintenance mode** field to *Separate*, use these guidelines to set up the **Price attribute group combination** FastTab:

    - The system automatically generates a row for each possible combination of header attributes (as listed on the **Header price attribute group** FastTab, plus *All* if the **Use all**

in header group option is enabled) and line attributes (as listed on the **Line price attribute group** FastTab, plus *All* if the **Use all in line group** option is enabled).

- The system automatically assigns a **Combination rank** value to each row, based on the **Rank** setting of individual header and line attribute groups.
- Use the buttons on the toolbar to add and remove price attribute groups as required.
- For each attribute group, set the following fields:
  - **Name** – For automatically generated rows, a name is generated to indicate the header and line attribute that were combined to create the row.
  - **Header type** – Select the type of criteria that should be available for specifying header attributes in pricing rules that use the price attribute group combination. Select *All* to create rules that apply to all customers. Select *Group* to select a group of price attributes that the pricing rules can specify values for to identify specific customers.
  - **Header price attribute group** – If you set the **Header type** field to *Group*, specify the attribute group that the price attribute group combination provides so that header attributes can be specified in pricing rules.
  - **Line type** – Select the type of criteria that should be available for specifying line attributes in pricing rules that use the price attribute group combination. Select *All* to create rules that apply to all products. Select *Group* to select a group of price attributes that your pricing rules can specify values for to identify specific products.
  - **Line price attribute group** – If you set the **Line type** field to *Group*, specify the attribute group that the price attribute group combination provides so that line attribute can be specified in pricing rules.
  - **Combination rank** – Assign a rank to the price attribute group combination. The rank resolves concurrency if more than one pricing rule applies to the price component code. For more information, see the Price attribute combination rank section.
- For price component codes where the **Price component** field is set to *Sales trade agreement*, the toolbar on the **Price attribute group combination** FastTab includes a **Trade agreement journals** button. Select a row, and then select this button to create a new trade agreement journal for the selected row. The selected row is the default **Price attribute group combination** field in the trade agreement journal. For more information, see Sales trade agreement prices.

9. On the Action Pane, select **Save**.

Price attribute combination rank

Rankings enable the system to determine which pricing rule to use if an order qualifies for more than one rule. In general, specific pricing rules (that is, rules that apply to a specific customer account and product number) take priority over more general rules (that is, rules that apply to a group of customers and/or a group of products). However, if two equally general rules apply (for example, a rule that targets a specific customer group and a rule that targets a specific region), it can be difficult to determine which rule should have priority. Therefore, Unified pricing management lets you specify a rank for each attribute group and each attribute group combination.

The price engine considers the **Combination rank** value if both of the following conditions are met:

- Multiple pricing rules in the same price component code apply to the same order line.
- For all the qualifying rules, the **Concurrency mode** field is set to *Price attribute combination rank*.

In this situation, the rule that matches the row with the highest **Combination rank** value applies.

For example, you have a price component code where the **Price component** field is set to *Trade agreement* and the **Maintenance mode** field is set to *Separate*. The price component code uses the attribute group settings that are shown in the following table.

Expand table

Price attribute group	Group scope	Rank
Target customer group	Header	2
Target market segment	Header	1
Product group	Line	2
Product category	Line	1

The following table shows the attribute group combinations and combination ranks that the system automatically generates based on these settings.

Expand table

Combination	Combination rank
Target customer group-Product group	2002 (highest rank)
Target customer group-Product category	2001
Target market segment-Product group	1002
Target market segment-Product category	1001

The system includes two sales trade agreements that can apply to the same sales order line, as shown in the following table.

Expand table

Rule no.	Concurrency mode	Criteria with price attributes	Price	Combination Rank
RD001	Price attribute combination rank	**Customer group** = *20* **Product group** = *Standard*	$20	2002
RD002	Price attribute combination rank	**Target market group** = *Online* **Product category** = *Electronic*	$30	1001

In this case, the pricing engine selects rule *RD001*, because that rule has the highest combination rank. Therefore, the price is $20.

Last updated on 05/29/2026

Price component posting

Article • 10/28/2024

Unified pricing management lets you set up a different sales order posting ledger for each of the following types of price component codes:

- Base price
- Sales trade agreement price
- Margin component price adjustment
- Discounts

This capability enables the system to keep ledger accounts that distinguish the many layers of your pricing composition at the price component level.

For more information about how to set up your price structures, including how to assign posting profiles for each relevant price component code in a structure, see Arrange price component codes into a price structure.

Set up price component posting

Follow these steps to prepare your system to support price component posting.

1. Go to **Pricing management** > **Setup** > **Pricing management parameters**.

2. On the **Posting** tab, on the **Sales base price** FastTab, set the following fields:

    - **Post sales base price** – Set this option to *Yes* to enable *base prices* and *sales agreement prices* to be posted to a specific ledger. When you post a sales invoice, the system will use the ledger account that's specified by the posting profile that's assigned to each price component code line.
    - **Sales base price account** – Specify a fallback ledger account to use for sales base prices. This account will be used as required for price component code lines where no posting profile is specified.

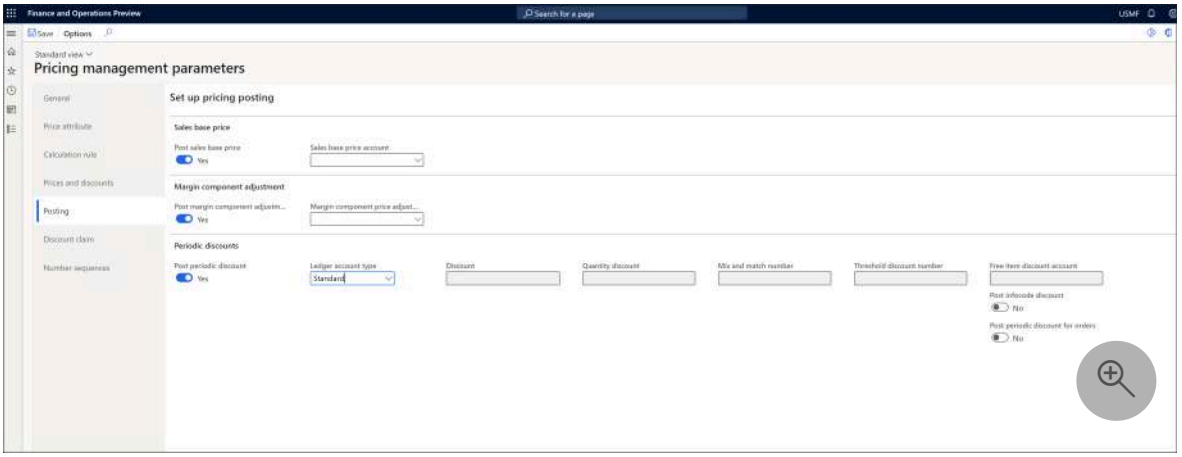
3. On the **Margin component adjustment** FastTab, set the following fields:

    - **Post margin component adjustments** – Set this option to *Yes* to enable *margin component price adjustments* to be posted a specific ledger. When you post a sales invoice, the system will use the account that's specified by the posting profile that's assigned to each price component code line.
    - **Margin component price adjustment account** – Specify a fallback ledger account to use for margin component price adjustments. This account will be

used as required for price component code lines where no posting profile is specified.

4. On the **Periodic discounts** FastTab, set the following fields:

    - **Post periodic discount** – Set this option to *Yes* to enable *periodic discounts* to be posted to a specific ledger. When you post a sales invoice, the system will first determine whether there's a specific discount account in the applied rule record. If there isn't, the system will instead use the account that's specified by the posting profile that's assigned to each price component code line. Periodic discounts include mix-and-match discounts, quantity discounts, and discount offers.
    - **Ledger account type** – Set this field to *Periodic* to set up fallback ledger accounts by using the other fields on the FastTab.



5. On the Action Pane, select **Save**.

Configure price component posting profile

Follow these steps to set up your price component posting profiles.

1. Go to **Pricing management** > **Setup** > **Posting** > **Pricing component posting**.

2. Follow one of these steps:

    - To create a new profile, select **New** on the Action Pane.
    - To edit an existing profile, select it in the list pane.
    - To delete an existing profile, select it in the list pane, and then select **Delete** on the Action Pane.

3. On the header of the new or selected record, set the following fields:

    - **Posting profile** – Enter a unique name.
    - **Description** – Enter a short description.

- **Price component** – Select the price component type that the posting profile applies to.

4. On the **Lines** FastTab, set up the rules that the profile will use to assign ledger accounts when price component codes of the selected price component type are posted. Use the buttons on the toolbar to add or remove lines as required. For each line, set the following fields:

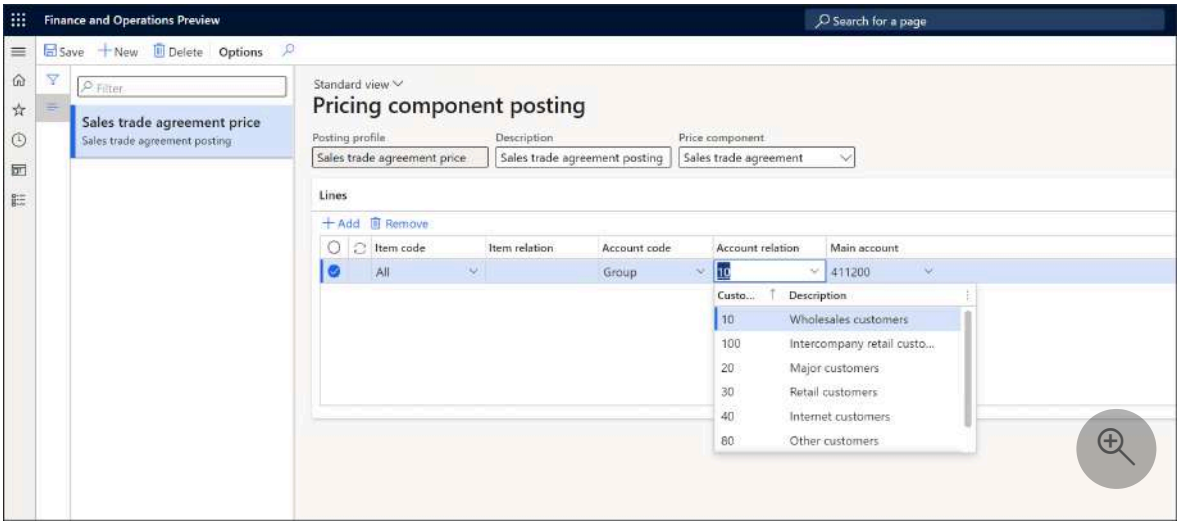
    - **Item code** – Select one of the following values to specify the scope of items where the line will apply:
        - *Table* – Assign an account for a specific item.
        - *Group* – Assign an account for an item group.
        - *All* – Assign an account for all items.

    - **Item relation** – If you set the **Item code** field to *Table*, select a specific item. If you set the **Item code** field to *Group*, select an item group.

    - **Account code** – Select one of the following values to specify the scope of customer accounts where the line will apply:
        - *Table* – Assign an account for a specific account.
        - *Group* – Assign an account for an account group.
        - *All* – Assign an account for all accounts.

    - **Account relation** – If you set the **Account code** field to *Table*, select a specific account. If you set the **Item code** field to *Group*, select an account group.

    - **Main account** – Select the account to use when the conditions that are defined by the line are met.

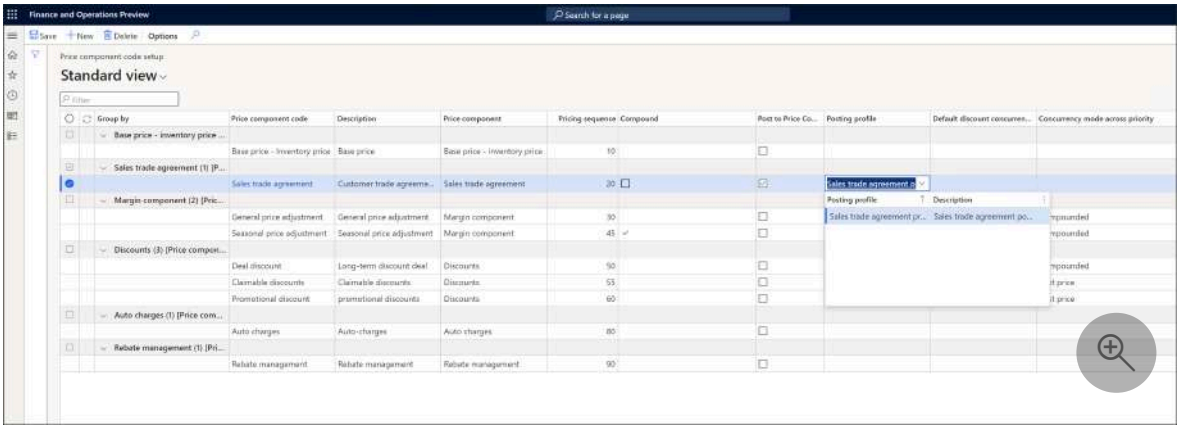


5. When you've set up all the lines that you need for the current profile, select **Save** on the Action Pane.

Associate price component posting with the price structure

Follow these steps to set up your price structures to use a specific posting profile for each relevant price component code.

1. Go to **Pricing management** > **Setup** > **Price component codes** > **Price trees**
2. Find the price component code line that you want to assign posting for, and select its **Post to price component code** checkbox. Then set the **Posting profile** field to the profile that you want to use for the line.



3. Repeat the previous step until you've assigned a posting profile to each relevant line.
4. On the Action Pane, select **Save**.
5. If you're using multiple pricing trees, repeat the previous steps until you've set up each tree.

Feedback

Was this page helpful? Yes No

Provide product feedback

Price structure overview

When a sales order is placed, you can review the detailed price breakdown, including the price component. The breakdown corresponds to the price structure that you configured. The price details provide an audit record that shows how the price was determined. They can serve as a starting point for future price investigation.

You can set up your system to use either a single price structure or multiple price structures.

Price component types

The following table summarizes the main types of components that can be included in a price structure.

Expand table

Price component	Price position	Details
Base price	Unit Price	The *base price* represents the common price for general use. It's at the level of the stock keeping unit (SKU).

Price component	Price position	Details
		You can have only *one* price component code of this type in each price structure. That price component code can be of any of the following base price types: • Base price - inventory price • Base price - purchase price • Base price - sales price The base price comes from the **Item base price** page. it can be either calculated or manually entered. If you set up the base price by using a price structure (price trees), and no matching record exists on the **Item base price** page, the system determines whether the item is a standard cost item. If it's a standard cost item, the system uses the active item standard cost as the source for the base price.
Sales trade agreement price	Unit Price	If the *sales trade agreement price* is available, it takes precedence over the base price, because it represents a special price arrangement with a specific group of consumers for a specific set of products. You can have only *one* price component code of this type. However, any number of sales trade agreement prices can apply to that code in different situations.
Margin component price adjustment	Unit Price	*Margin component price adjustments* move item prices up or down from the base price. Margin component price adjustments can be associated with many types of agreements, promotions, and events. They let you easily adjust prices without changing the base price. You can have *multiple* price component codes of this type, and you can associate any number of pricing rules with each one. Price adjustments are compounded across price component codes and add up to the total price adjustment.
Discounts	On-invoice discounts	*Discounts* let you adjust prices based on a wide range of calculation methods that are designed to encourage the purchase of specific products, volumes, packages, and so on. You can have *multiple* price component codes of this type, and you can associate any number of pricing rules with each one.
Rebate management	Off-invoice discounts	You can have *multiple* price component codes of this type.

Concurrency rules

If your price structure has multiple price rules that apply to the same price component code and/or to multiple price component codes of the same type, the system applies concurrency rules to resolve them.

- If more than one pricing rule applies to the same price component code in a price structure, *within-price-component-code concurrency* rules are applied to determine which rules apply and how they're combined. For more information, see Resolve concurrency within price component codes.
- If your price structure includes more than one price component code of the same type, *across-price-component-code concurrency* rules are applied to determine how they're combined. For more information, see Resolve concurrency across price component codes.

Next steps

- Set up a company to use multiple price structures
- Set up a company to use a single price structure
- Arrange price component codes into a price structure

Last updated on 07/01/2026

Set up a company to use a single price structure

This article explains how to configure a company to use a single price structure, and how to set up that structure. The purpose of a price structure is to define the order that the system calculates each type of price adjustment in, and to define other options for each price component code, such as concurrency and compounding rules.

In this scenario, companies and price structures have a one-to-one (1:1) relationship. The pricing engine matches each sales order with the price component codes, based on the pricing sequence that's defined in the single structure.

Configure a company to use a single price structure

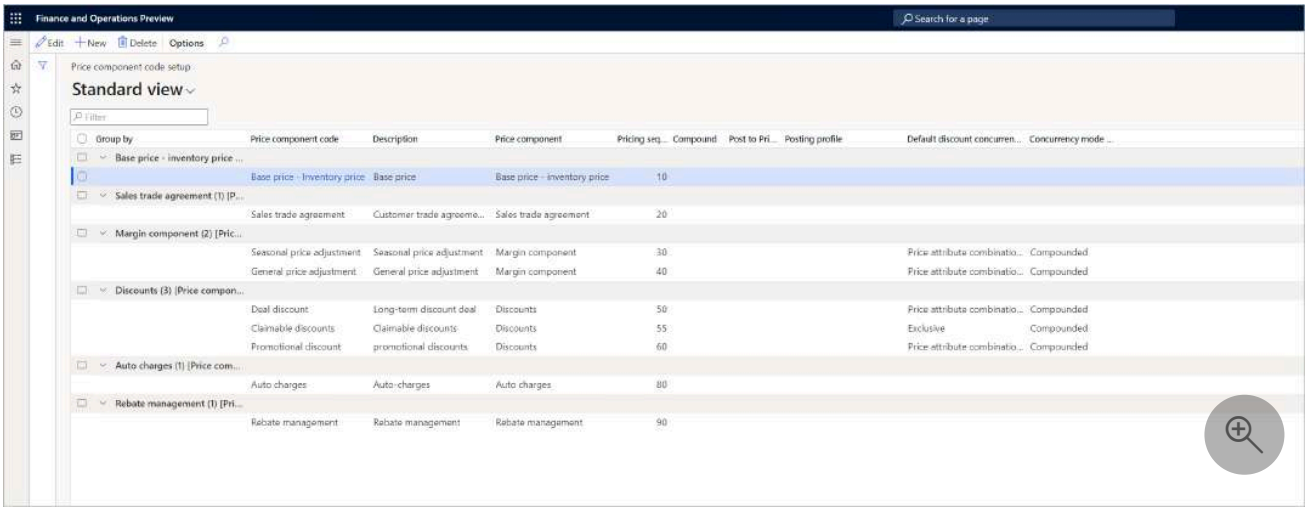
Follow these steps to use a single price structure for a company.

1. Select the company in the company picker.
2. Go to **Pricing management** > **Setup** > **Pricing management parameters**.
3. On the **Price attribute** tab, set the **Enable multiple price trees** option to *No*.
4. On the Action Pane, select **Save**.

Configure the single price structure

To set up the price structure for a company that uses a single structure, follow these steps:

1. Go to **Pricing management** > **Setup** > **Price component codes** > **Price trees**. The **Price trees** page shows your current price structure (if you have one), and lets you add or remove price component codes in the structure. The components are grouped according to the **Price component** value that's set for each price component code in the list.



2. Set up your price structure by adding and removing lines to define a price sequence. Set the fields for each price component code that you want to include. These fields work just as they do on the **Price component code list** FastTab of the **Price trees** page. For information about how to set these fields, and examples that show how they work, see Arrange price component codes into a price structure.

3. When you've set up all lines as required, select **Save** on the Action Pane.

Auto charges in single and multiple price structures

Price component codes of the *Auto charges* type are handled differently, based on whether you're using a single price structure or multiple price structures.

- *If you're using a single price structure*, you can add the price component code for auto charges in the price trees.
- *If you're using multiple price structures*, you can't add auto charges in the price tree structures. Instead, the system will apply the standard auto charge logic to determine which auto charges apply to the sales order.

Last updated on 12/30/2025

Set up a company to use multiple price structures

This article explains how to set up multiple price structures in a company. A price structure defines the order in which the system calculates each type of price adjustment. It also defines other options for each price component code, such as concurrency and compounding rules.

In this scenario, companies and price structures have a one-to-many (1:N) relationship. In addition, there are multiple price structures. The pricing engine selects a price structure based on the specified **Price tree attribute**, which is one of the order attributes. After it determines the applicable price structure, the price engine matches the sales order with the price component codes according to the pricing sequence that's defined in the applicable price tree.

Configure a company to use multiple price structures

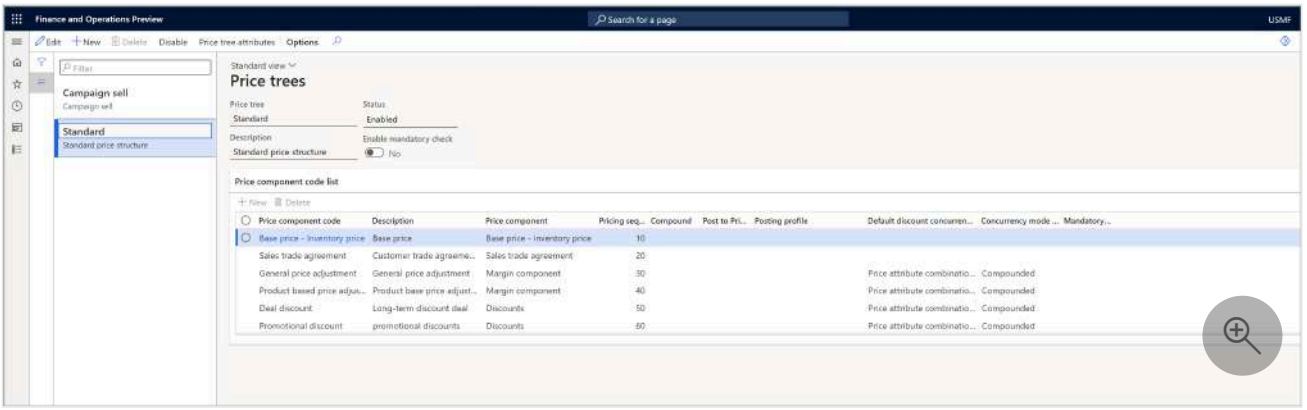
Follow these steps to use multiple price structures for a company.

1. Select the company in the company picker.
2. Go to **Pricing management** > **Setup** > **Pricing management parameters**.
3. On the **Price attribute** tab, set the **Enable multiple price trees** option to *Yes*.
4. Use the value of one of the order attributes to select which price tree to use. Set the **Price tree attribute** field to the order attribute that you want to use.

Configure the price structures

Follow these steps to create and configure a price structure.

1. Go to **Pricing management** > **Setup** > **Price component codes** > **Price trees**. The **Price trees** page shows your current price structures (if you have any), and lets you add, remove, and configure price trees.



2. Follow one of these steps:

    - To add a new price tree, select **New** on the Action Pane.
    - To edit an existing price tree, select it in the list pane, and then select **Edit** on the Action Pane.
    - To delete a price tree, select it in the list pane, and then select **Delete** on the Action Pane.

3. On the header of the new or selected price tree, set the following fields:

    - **Price tree** – Enter a name for the price tree.
    - **Status** – This field shows the current status of the price tree. Only price trees that have a **Status** of *Enabled* affect your price calculations. To change the status, select **Disable** or **Enable** on the Action Pane.
    - **Description** – Enter a short description of the price tree.
    - **Enable mandatory check** – If you want to mark one or more price component codes as mandatory, set this option to *Yes*.

4. On the **Price component code list** FastTab, add each price component code that should be part of the price structure for the current price tree. Use the buttons on the toolbar to add and remove lines as required. For information about how to set these fields, and examples that show how the settings work, see Arrange price component codes into a price structure.

5. When you set up all lines as required, select **Save** on the Action Pane.

6. On the Action Pane, select **Price tree attribute** to specify the value of the price tree attribute that causes this price tree to apply to a given order.

7. On the Action Pane, select **Enable** to enable the current price tree.

Auto charges in single and multiple price structures

Price component codes of the *Auto charges* type work differently depending on whether you use a single price structure or multiple price structures.

- *If you use a single price structure*, you can add the price component code for auto charges in the price tree.
- *If you use multiple price structures*, you can't add auto charges in the price tree structures. Instead, the system applies the standard auto charge logic to determine which auto charges apply to the sales order.

Last updated on 07/01/2026

Arrange price component codes into a price structure

Article • 10/28/2024

Regardless of whether you're using a single price structure or multiple price structures, you use the same basic procedure to arrange your price component codes into a price structure. The only difference is that, if you use multiple price structures, you also define a condition to select the structure that's used in each situation.

- For information about how to create and manage a single or multiple price structures, see Set up a company to use multiple price structures. Then, when you're ready to add, arrange, and configure the price component codes in each structure, return to this article.

Configure a price structure

Follow these steps to create and configure a price structure.

1. Create the pricing tree or trees that you need as described in Set up a company to use multiple price structures or Set up a company to use a single price structure.

2. Go to **Pricing management** > **Setup** > **Price component codes** > **Price trees**, create or select a price tree, and expand the **Price component code list** FastTab.

3. Add each price component code that should be part of the current price structure. Use the buttons on the toolbar to add and remove lines as required. The following fields are available for each line:

    - **Group by** – In the grid, the price component codes are grouped by type. The type is shown in the **Group by** column, and the relevant price component codes are indented under their group name. When you first create a new code, a default group is selected for it. However, the default group will automatically be replaced when you select a value in the **Price component code** field. You can't edit this field directly.

    - **Price component code** – If you're adding a new line, select an existing price component code for it. (You can't edit this field for existing lines.) Each price component code can appear only once in the list. Therefore, you'll receive an error if you try to select a code that's already listed. For information about how to create and configure price component codes, see Price component codes.

- **Description** – Enter a short description.
- **Price component** – This field shows the type of the price component code that was selected for the line. The value comes from the **Price component** setting of the selected price component code and establishes the group that the line was put into.
- **Pricing sequence** – Enter an integer to define a calculation sequence for the different lines. The calculation proceeds from the lowest sequence number to the highest number. We recommend that you leave some space between numbers, so that there's room for new pricing components to be added between existing lines later. For example, use *10*, *20*, *30*, *40*, and so on, instead of *1*, *2*, *3*, *4*, and so on. The sequence can have a significant effect on the final unit price, especially if you use percentage-based calculations and compounding.
- **Compound** – This checkbox applies only to price component codes where the **Price component** field is set to *Margin component*. Select the checkbox for lines where you want to calculate the price adjustment based on the compound price (running total). Clear it for lines where you want to calculate the price adjustment based on the original base price. For discounts, the compounding behavior is controlled instead by the **Discount compound behavior** setting on the **Pricing management parameters** page. (Learn more in Resolve concurrency across price component codes.)
- **Post to price component code** and **Posting profile** – To post to a price component code, select the **Post to price component code** checkbox, and then select the posting profile that you want to use. Learn more in Price component posting.
- **Default concurrency mode** – This field shows the default concurrency mode that was set for the selected price component code. For information about how to set the default concurrency mode and interpret its value, see Price component codes.
- **Concurrency mode across priority** – This field applies only to lines where the **Price component** field is set to either *Margin component* or *Discounts*. Select one of the following values:
  - *Compounded* – The system will first compute the price adjustment or discount for each line in the structure. It will then calculate the final discount or adjustment by adding the value for every line. This value is available only for lines where the **Price component** field is set to *Discounts*

or *Margin component*. (It's the only available value for margin components.)
    - *Best price* – If the price structure includes more than one price component code that has this value, only one of those lines (the one that has the largest discount) will contribute to the final unit price. This value is available only for lines where the **Price component** field is set to *Discounts*.

  This field affects discount calculations only when the **Best price and compound concurrency control model** field on the **Pricing management parameters** page is set to *Best price and compound within priority, best price and compound across priority*. Learn more in Resolve concurrency across price component codes.

- **Mandatory** – For price trees where the **Enable mandatory check** option is set to *Yes*, use this checkbox to mark each price component code that's mandatory in the sales order. If a price component code isn't applicable, an error will be shown.

> **Note**
>
> If you use multiple price structures, they can't include price component codes where the **Price component** field is set to *Auto charge*. Instead, the system will apply the standard auto charge logic. In other words, if an auto charge rule record is applicable, the auto charge will apply to the sales order.

4. When you've set up all the lines that you need in your price structure, select **Save** on the Action Pane.

Feedback

Was this page helpful? Yes No

Provide product feedback

Vendor list prices and price adjustments

In some cases, Unified pricing management can derive a product's sales price from the vendor's list price or procurement price. For example, this functionality can be used for the trade of items in the retail and distribution industry, where products are procured from vendors and sold to customers.

Vendor list prices come from a vendor price catalogue and can be used as the foundation for calculating the sales price of an item.

The vendor list price features have three core elements:

- **Vendor list price** – The vendor list price lets you record the general vendor price catalog and specify a valid date period. The vendor list price is on the *item and inventory dimensions* level.
- **Vendor price term codes** – Vendor price term codes establish a list of price adjustment codes that you can use when you define vendor price term agreements. For calculations where the **Compound** option is set to *Yes*, pricing calculations apply price term codes together with the pricing sequence.
- **Vendor price term agreements** – Vendor price term agreements keep track of the contracts that you have with vendors for price adjustments for each site. The valid periods, vendors, calculation methods for price adjustments, and values are all recorded. You can use price attributes to specify the applicable items and sales order lines, because vendor price term agreements use the line price attribute group.

Unified pricing management lets you define the selling price as *Item base price ± Margin component price adjustments*.

For the items that you sell, the system can determine the item base price based on the vendor list price. The item base price is calculated as *Vendor list price + Vendor price term agreement price*. During the calculation, the system evaluates each vendor's item base price for each site. Because the item base price has just one unique active value per site, if multiple calculated prices exist, the system uses only the lowest calculated price.

Next steps

- Base price determination rules
- Base price versions

- Sales trade agreement prices

Last updated on 07/01/2026

Sales order base price determination rules

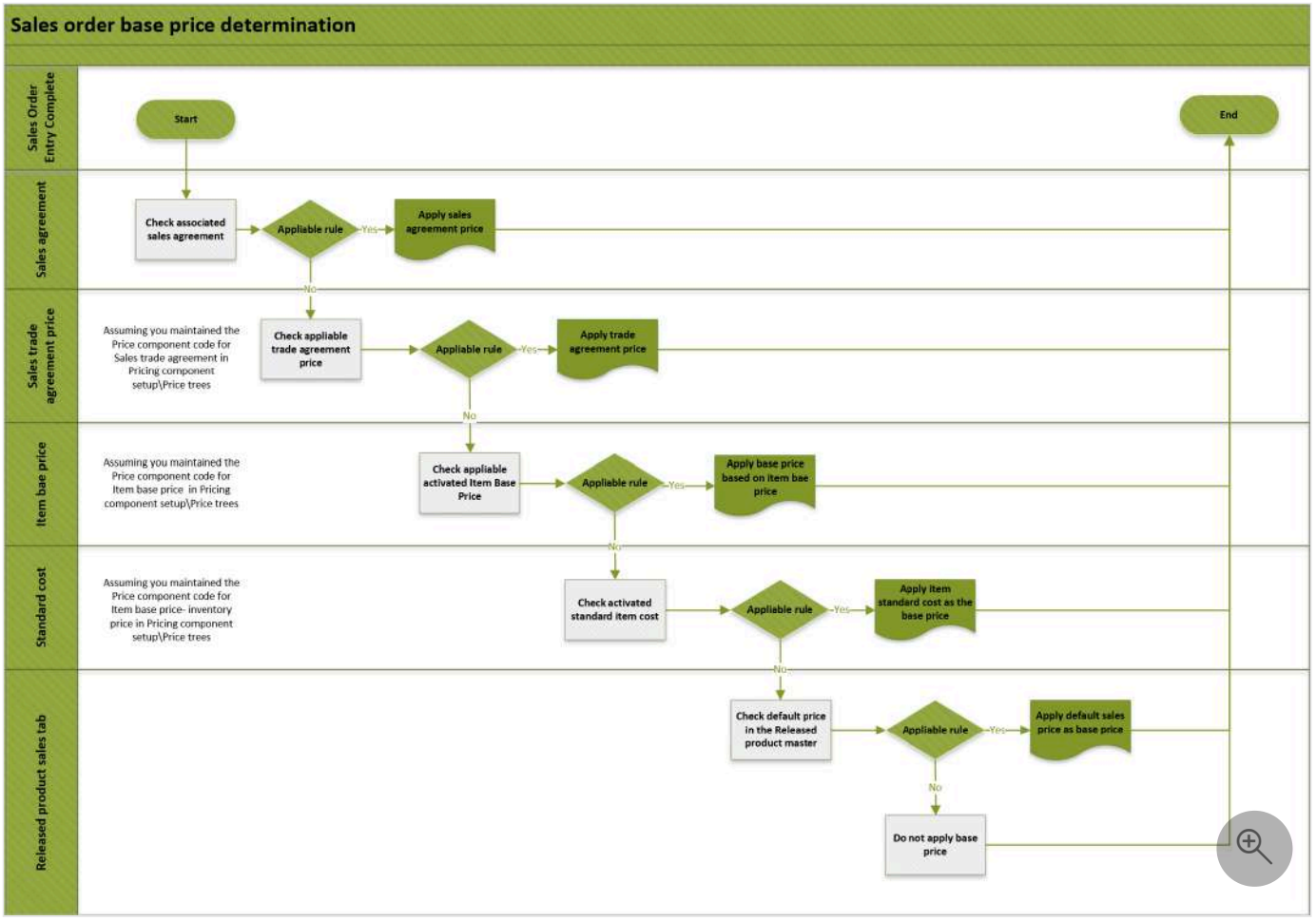
This article describes the price determination rules for calculating the sales order base price.

Expand table

Terms	Definition, usage, and notes	Price component type
Unit price	The price calculated for each unit in a sales order. It's calculated by using the following formula: *Unit price = Base price ± Margin component price adjustments*.	
Base price	The basis for price adjustments. It's intended for all customers and is the standard rate for general purposes. • If there's an applicable sales trade agreement price, the base price is the *sales trade agreement price*. • If there's no applicable sales trade agreement price, the base price is the *item base price*.	
Sales trade agreement price	The price that reflects a negotiated pricing strategy for a collection of customers and a group of specific products. You can configure the system so that it uses one of the following formulas to calculate the unit price based on the sales trade agreement price: • *Unit price = Sales trade agreement price* (if no margin component price adjustment applies) • *Unit price = Sales trade agreement price ± Margin component price adjustments*	Sales trade agreement
Item base price	The **Item base price** page lets you set prices for each item by using either calculated values or manually entered values. Calculated prices use the following formula: *Item base price = Vendor list price ± Vendor term agreements*.	Any of the following: • Base price - purchase price • Base price - inventory price • Base price - sales price
Standard cost	The cost version of an item that's calculated by using the standard cost model.	Base price - inventory price

Terms	Definition, usage, and notes	Price component type
Margin component price adjustments	You can set up layers of margin components to adjust the base price up (positive) or down (negative). Margin component price adjustments have a calculation sequence and can be compounded to get the total adjustment price.	Margin component
Sales agreement	Although the Unified pricing management module respects sales agreement rules, the sales agreement feature isn't otherwise integrated with the Unified pricing management module.	

The following flow chart shows the sales order base price determination rules.



Next steps

- Base price versions
- Sales trade agreement prices

Last updated on 07/01/2026

Base price versions

This article describes how base price versions work in Unified pricing management. The primary purpose of a base price version is to maintain a list of *item base prices* for a specific period. The base price is a common price at the level of the stock keeping unit (SKU). Unified pricing management lets you build selling prices by using *cost-plus pricing*. When you use cost-plus pricing, you first build a price structure that starts with the base price.

The following table shows an example of a price structure that you can use to build final selling prices based on base prices.

Expand table

Pricing sequence	Price component code	Price component type	Description	Price details on the sales order
10	BAS01	Base price	Price component code *BAS01* must be associated with one of the following price component code types: • Base price – inventory price • Base price – purchase price • Base price – cost price	Base price
20	MAC01	Margin component	Adjust the price up or down based on a fixed amount or percentage.	Price adjustments
25	MAC02	Margin component	Adjust the price up or down based on a fixed amount or percentage.	Price adjustments
30	MAC03	Margin component	Adjust the price up or down based on a fixed amount or percentage.	Price adjustments

The selling price is calculated by using the following formula: *Selling price* = *Base price* ± *Margin component price adjustments*.

Determining the base price

The cost-plus pricing model works by determining the product cost and then adding a markup to determine the selling price.

- In retail and distribution industries, many products are items that are procured and then sold to customers. For these industries, the base price is typically the procurement price for a product.
- In the manufacturing industries, the base price is typically based on the calculated total cost of the bill of materials (BOM) for a product.

Unified pricing management derives base prices from one of the following two sources:

- The primary source is the item base price that's listed for each product on the **Item base price** page. To open this page, go to **Pricing management** > **Pre-sales pricing** > **Base price versions** to open the **Base price versions** page, which lists the base price versions that you created and shows the dates that each version is valid for. Select a version in the list, and then select **Price** > **Base price** on the Action Pane to view and edit the prices that apply for that version.
- If an item doesn't have an active item base price, the system determines whether it has an active cost price in a costing version. A costing version can support a standard cost inventory model for items, where the costing version contains a set of standard cost records about the items and their manufacturing processes.

The following table shows the rules for determining the base price.

Expand table

Product type	Industries	How base prices are determined
Traded items that you procure and then sell	Retail, distribution	• Unified pricing management calculates item base prices based on prices found on the **Base price versions** page. • IBase prices are calculated by using the following formula: *Base price = Vendor list price ± Vendor price term agreement*. • If several calculated prices apply to the same item, the price engine always selects the *lowest* price as the item base price. • You can activate a calculated item base price. The price engine then uses that calculated price as the base price.
Manufactured items (with BOMs) that use the standard cost model	Manufacturing	• Microsoft Dynamics 365 Commerce and Dynamics 365 Supply Chain Management can calculate and activate item cost prices based on costing versions. • You don't have to maintain item base prices on the **Base price versions** page.

Product type	Industries	How base prices are determined
		• The price engine uses the active item cost price as the base price. • Item selling prices are calculated by using the following formula: *Selling price = Active calculated item cost (base price) ± Margin component price adjustments*.
Manufactured items (with BOMs) that don't use the standard cost model	Manufacturing	• Item base prices are defined on the **Base price versions** page. • The price engine uses the active item base price as the base price.

Base price determination exception

As mentioned earlier, the base price is a common price at the SKU level. A price differentiation strategy is often applied when a price manager negotiates with a customer to set up a special price for a specific group of products for a specific date range. Record these prices as *sales trade agreement prices*. Sales trade agreement prices take priority over base prices.

The following table shows how sales trade agreements relate to selling prices.

Expand table

Do applicable margin component price adjustments exist?	Sales price calculation
No	*Selling price = Sales trade agreement price*
Yes	*Selling price = Sales trade agreement price ± Margin component price adjustments*

> **Note**
>
> To use sales trade agreement prices, your price structure must include a price component code where the **Price component** field is set to *Sales trade agreement*.

The following table shows an example of a price structure that includes a sales trade agreement.

Expand table

Pricing sequence	Price component code	Price component type	Description	Price details on the sales order
10	BAS01	Base price	Price component code *BAS01* must be associated with one of the following price component code types: • Base price – inventory price • Base price – purchase price • Base price – cost price	Base price or Sales trade agreement price
15	TAM	Sales trade agreement	Sales trade agreement	Base price or Sales trade agreement price
20	MAC01	Margin component	Adjust the price up or down based on a fixed amount or percentage.	Price adjustments
25	MAC02	Margin component	Adjust the price up or down based on a fixed amount or percentage.	Price adjustments
30	MAC03	Margin component	Adjust the price up or down based on a fixed amount or percentage.	Price adjustments

For example, when this price structure is used, the price engine might find that the following pricing rules are applicable to item *EV0001* on a sales order line:

- Activated item base price (BAS01) = $5
- Posted sales trade agreement journal price (TAM) = $8
- Margin component price adjustment (MAC01) = $3
- Margin component price adjustment (MAC02) = $2
- Margin component price adjustment (MAC03) = -$1

Therefore, the selling price for the sales order line is 8 + 3 + 2 – 1 = $12.

Next steps

- Sales trade agreement prices

Last updated on 07/01/2026

Sales trade agreement prices

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Sales trade agreement prices are negotiated prices for specific products that apply to specific customers. For sales where a sales trade agreement applies, this price takes priority over the item base price.

Unified pricing management uses the standard *Trade agreement price - sales price* side, but it's enhanced with price attributes.

> **Note**
>
> Unified pricing management respects standard *sales agreements*, which differ from the *sales trade agreement prices* that are described in this article. For order lines where a *sales agreement* applies, Unified pricing management uses the sales agreement. If no sales agreement applies, Unified pricing management determines whether an applicable *sales trade agreement price* exists. The discounts that are included in sales trade agreements (line discounts, multiple discounts, and total discounts) are outside the purview of Unified pricing management. Instead, Unified pricing management provides a new approach to defining discount rules.

Net pricing

To support negotiated final prices in trade agreements, you can define sales prices directly with or without price adjustments and explicitly prevent other discount applications. This approach ensures accurate discount-exempt pricing aligned with contractual or wholesale pricing needs. This function supports sales trade agreement pricing by allowing pricing administrators to define:

- A final exclusive net price that reflects the outcome of negotiation
- Whether price adjustments (such as percentage increases or decreases) are included

- Whether promotional or system-applied discounts should be prevented from further affecting the transaction price

This capability is especially relevant in business-to-business (B2B), channel sales, or regulated pricing models, where discount logic shouldn't be allowed to alter the agreed price. It can also be important for long-term contracts that require strict adherence to fixed pricing that excludes further discounts.

To define a trade agreement sales price as a *net* (final, exclusive) price that isn't subject to further discounting, set **Allow Price Adjustment** to *No* and **Prevent Discount** to *Yes* for the relevant sales trade agreement lines. Learn more in Manage pricing rules for a sales trade agreement journal.

Configure sales trade agreements, including concurrence rules

Several configuration settings affect the way that sales trade agreements work in Unified pricing management. Before you start to create any sales trade agreement pricing rules, follow these steps to set up your system.

1. Go to **Pricing management** > **Setup** > **Pricing management parameters**.
2. On the **Price and discounts** tab, on the **Trade agreements** FastTab, set the **Enable default find next** option to one of the following values to specify how the system should resolve concurrent sales trade agreement rules:
    - *Yes* – The price engine checks all applicable trade agreement prices and applies the lowest price.
    - *No* – The price engine uses the *price attribute combination rank* to determine the price, by applying the following rules:
        a. Apply the record that has the highest *price attribute combination rank*.
        b. If two or more records have the same highest *price attribute combination rank*, check the *header price attribute*, and apply the highest-ranked record price group.
        c. If two or more records have the same highest *header price attribute*, check the *line price attribute*, and apply the highest-ranked record in the price attribute group.
        d. If multiple price records have the same rank, apply the lowest price.
3. Set the **Apply existing trade agreement** option to *Yes* if the price engine should consider posted sales trade agreement pricing rule records where the **Default relation** field is set to

Price (sales). (These records are available at **Sales and Marketing** > **Prices and discounts** > **Trade agreement journals**.) The price engine applies the following rules:

a. Apply the sales trade agreement price that meets the criteria.

b. If no sales trade agreement price applies, use the qualifying rules that are posted to the trade agreement journals.

4. On the **General** tab, set the **Date type** field to the type of date that you use when you set up criteria for matching pricing rule records. The available values are *Today*, *Requested ship date*, *Requested receipt date*, and *Created date*.

> **Note**
>
> If your organization uses only Unified pricing management and doesn't use traditional trade agreements (created via **Sales and marketing** > **Prices and discounts** > **Trade agreement journals**), the parameters on the **Activate price/discount** page (**Sales and marketing** > **Setup** > **Prices and discounts** > **Activate price/discount**) aren't required. You can safely set all toggles on this page to *No*. These parameters control which combinations of customer/item dimensions are available for traditional trade agreement journal lines, and they have no effect on UPM pricing rules, price structures, or price component codes.

Concurrency example

The following table shows an example of combinations of price component code attributes.

Expand table

Price attribute combination	Price attribute combination	Header price attribute	Rank	Line price attribute	Rank
Target customer segments + Vehicle product	2003	Customer account	4	Interior Upholstery	4
Target customer segments + Vehicle product	2003	Customer group	3	Exterior color	3
Target customer segments + Vehicle product	2003	Price group	2	Fuel type	2
Target customer segments + Vehicle product	2003	Sales group	1	Drive type	1

The following two records are posted for sales trade agreement price lines that have the same date range. Because the price attribute combinations are the same, the ranks of the price attribute combinations are also the same. Therefore, the price engine applies the RID0002 price.

Expand table

ID	Price attribute combination	Header price attribute criteria	Line price attribute criteria	Price	Applicable
RID0001	Target customer segments + Product segments	Price group = 01	Interior Upholstery = Package B	$1,500	No
RID0002	Target customer segments + Product segments	Customer account = US-003	Interior Upholstery = Package B	$1,550	Yes

Set up your price component codes and price structures

To use sales trade agreements, you must have the following configurations in your system:

- You must have exactly one price component code where the **Price component** field is set to *Sales trade agreement*. This setting lets you include sales trade agreements as part of your price structures. Learn more in Price component codes.
- Each price structure where you want to consider sales trade agreement prices must include the previously mentioned *Sales trade agreement* price component code. Learn more in Arrange price component codes into a price structure.

Manage trade agreement journal names

Follow these steps to create and manage your trade agreement journal names.

1. Go to **Pricing management** > **Setup** > **Trade agreement prices** > **Trade agreement journal names**.

2. Follow one of these steps:

    - To create a new journal name, select **New** on the Action Pane.
    - To edit an existing journal name, select it in the grid, and then select **Edit** on the Action Pane.
    - To delete an existing journal name, select it in the grid, and then select **Delete** on the Action Pane.

3. Set the following fields for the new or selected record:

    - **Name** – Enter a name for the trade agreement journal name.
    - **Description** – Enter a short description of the trade agreement journal name.
    - **Relation** – This read-only field always shows a value of *Price (sales)*.
    - **Enable price attributes** – Select this checkbox to create sales trade agreement journals that have the price attributes.

4. On the Action Pane, select **Save**.

5. To view and manage all the trade agreement journals that use a selected trade agreement journal name, select **Trade agreement journals** on the Action Pane.

Manage sales trade agreement journals

Follow these steps to create and manage your sales trade agreement journals.

1. Go to **Pricing management** > **During-sales pricing** > **Sales trade agreement price** > **Trade agreement journals**.

2. Follow one of these steps:

    - To create a new journal, make sure that the **Show** field is set to *All* or *Not posted*, and then select **New** on the Action Pane.
    - To edit an existing journal, select it in the grid.
    - To delete an existing journal, select it in the grid, and then select **Delete** on the Action Pane.

    > **Tip**
    >
    > If you're looking for an existing journal to edit or delete, use the filter and the **Show** field to find it.

3. Set the following fields for the new or selected journal:

    - **Name** – Select an existing trade agreement journal name. For the name that you select, set the **Enable price attributes** option to *Yes* and set the **Relation** field to *Price (sales)* on the **Trade agreement journal names** page.
    - **Price/discount journal number** – A value is automatically generated the first time that you save a new journal. The field then becomes read-only.

- **Description** – Enter a short description of the journal. For new journals, this field takes its default value from the selected trade agreement journal name.
- **Posted** – This read-only field shows whether the journal is posted.
- **Posted on** – If the journal is posted, this read-only field shows the date when it was posted.
- **Enable price attributes** – This read-only field takes its default value from the selected trade agreement journal name.

4. On the Action Pane, select **Save**.

Manage pricing rules for a sales trade agreement journal

Sales trade agreement journal pricing rules control which prices apply to which combinations of customers and products. Each rule exists as a *line* that belongs to the relevant sales trade agreement journal.

Follow these steps to add, view, and manage pricing rules for a sales trade agreement journal.

1. Go to **Pricing management** > **During-sales pricing** > **Sales trade agreement price** > **Trade agreement journals**.
2. Create or select the journal that you want to work with, as described in the previous section.
3. On the Action Pane, select **Lines**.
4. The **Journal lines, trade agreement** page appears. If you want to add a new pricing rule, select **New** on the Action Pane. If you want to work with existing rules only, skip ahead to step 10.
5. In the **Edit price attributes** dialog box, set up the logic for selecting the customers and products that the line applies to. On the **General** FastTab, in the **Price attribute group combination** field, select one of the price attribute combinations that's associated with your *Sales trade agreement price* price component code. The value that you select controls which attributes are available to define the customer and products that the line applies to.
6. If the selected price attribute group combination considers header values (that is, if it doesn't apply to *all* customers), the **Header price attribute group** FastTab is available. In this case, for each row on the **Header price attribute group** FastTab, enter or select one or more

values in the **Condition** column to define the rules for selecting customers. The following rules apply:

- All rows are combined by using a logical AND operator. Therefore, only those customers that have matching values for *all* rows are included.
- If you want to create one or more rows that include multiple values, set the **Enable multiple selections** option to *Yes*. You can then add a comma-separated list of values in the **Values** column for each row. These values are combined by using an OR operator. Therefore, the row finds customers that have *any* of the values in the list.
- You can specify values to exclude by adding an exclamation mark (!) before the value. This exclamation mark is known as an exclusion prefix. For example, to find all customer accounts except *DE-001*, set the **Condition** field to *!DE-001* for the attribute where the **Attribute** field is set to *Customer account*. You can automatically add the exclusion prefix to the values of any row by selecting the row and then selecting **Exclude values in selected lines**.

7. To preview the results of your settings on the **Header price attribute group** FastTab, select **Preview matching results**. A dialog box shows a preview of customers that match the conditions that you set up so far.

    - If the list includes any customers that you want to exclude, select the target rows, and then select **Exclude** on the toolbar.
    - The **Line type** column indicates which customers you selected to exclude.
    - To reinclude a previously excluded customer, select it, and then select **Include** on the toolbar.

8. If the selected price attribute group combination considers line values (that is, if it doesn't apply to *all* products), the **Line price attribute group** FastTab is available. Use this FastTab to define rules for selecting products. The fields and preview options work just as they do on the **Header price attribute group** FastTab.

9. Select **OK** to add the new line to the grid on the **Overview** FastTab of the **Journal lines, trade agreement** page.

10. On the **Overview** FastTab, for each line in the grid, set the following fields as required:

    - **Header price attribute group** – This field shows which header price attribute group was selected for the line. For lines that apply to all customers, this field is blank. To change the value, select **Edit price attribute** on the toolbar of the FastTab.

- **Price attribute detail** – This field summarizes the header attribute values that identify the customers that the line applies to. For lines that apply to all customers, this field is blank. To change the values, select **Edit price attribute** on the toolbar.
- **Line price attribute group** – This field shows which line price attribute group was selected for this line. For lines that apply to all products, this field is blank. To change the value, select **Edit price attribute** on the toolbar.
- **Price attribute detail** – This field summarizes the line attribute values that identify the products that the line applies to. For lines that apply to all products, this field is blank. To change the values, select **Edit price attribute** on the toolbar.
- **Price attribute combination rank** – This field shows the rank that's assigned to the price attribute group combination that was selected for the current line in the **Edit price attribute** dialog box. The value affects how concurrent rules are resolved when more than one rule can apply to the same order line.
- **Allow price adjustment** – Select this checkbox if your sales trade agreement price isn't your final unit price, but other margin component price adjustments are permitted. For an example that shows how this setting can affect the final unit price, see the table after this procedure.
- **Prevent Discount** – Select this checkbox to prevent any other discounts from being applied for this line. To define a trade agreement sales price as a *net* (final, exclusive) price that isn't subject to further discounting, set **Allow Price Adjustment** to *No* and **Prevent Discount** to *Yes*. Learn more in Net pricing.
- **Allow Price Adjustment** – Select this checkbox to apply adjustments (such as markup or markdown) to include in the sales price. To define a trade agreement sales price as a *net* (final, exclusive) price that isn't subject to further discounting, set **Allow Price Adjustment** to *No* and **Prevent Discount** to *Yes*. Learn more in Net pricing.
- **Allow unit conversion** – Select this checkbox to allow the price that you specified for the selected unit to be converted proportionately for sales lines that specify other units. The conversion is based on the available unit conversion factors. This feature lets you maintain a single record that can apply to sales in different units.
- **From** – Enter a minimum quantity for the line. This value defines the minimum quantity that a customer must order to qualify for the agreement price.
- **To** – Enter a maximum quantity for the line. This value defines the maximum quantity that a customer can order and still qualify for the agreement price.
- **Unit** – Select the unit that the specified price applies to.
- **Amount in currency** – Enter the price that applies to the line.
- **Currency** – Select the currency that you used to specify the price in the **Amount in currency** field.

- *Inventory dimensions* – Use columns that show inventory dimensions (such as **Color**, **Warehouse**, and **Serial number**) to further refine the conditions for a line. To select which dimensions are shown in the grid, select **Inventory** > **Dimensions** on the toolbar.

> **Tip**
>
> If you offer prices based on multiple quantity breaks, specify each quantity bracket as a row that has the relevant **From** and **To** values. For the sales trade agreement price, the tiered quantity range is per sales order line. This behavior differs from the behavior for discounts, where quantity discount tiers are per sales order line.

11. Select each line that you added, and then, on the **Details** tab, set the following fields for it:

    - **From date** – Enter the first date that the selected line will be valid.
    - **To date** – Enter the last date that the selected line is valid.

12. Continue to add lines as required. Use the following buttons on the toolbar to modify or copy existing lines:

    - **Select** – Copy existing lines from posted journals to the current journal. For information about how to use this button, see the Create a new journal based on posted journal lines section of this article.
    - **Edit price attributes** – Open the **Edit price attributes** dialog box for the selected line, so that you can edit its settings as described earlier in this procedure.
    - **Copy and revise** – Create a copy of the selected line, and open a dialog box where you can pre-edit settings for the new line.
    - **Copy line** – Create a copy of the selected line.
    - **Clear journal** – Delete *all* the lines in the journal. (To delete a single existing line, select the line, and select **Delete** on the Action Pane.)

13. On the Action Pane, select **Save**.

14. To validate all the lines, select **Validate** > **Validate all lines** on the Action Pane. To validate only the selected lines, select **Validate** > **Validate selected lines**.

15. In the **Price/discount Journal posting** dialog box, select **OK** to run the validation.

The following table shows an example of a price structure that the **Allow price adjustment** setting affects for a sales trade agreement.

Expand table

Price component code	Price component	Price sequence	Value
TAM01	Sales trade agreement price	10	$200
MAC01	Price adjustment 01	20	$10
MAC02	Price adjustment 02	30	$20

The **Allow price adjustment** setting affects the final price in the following way:

- If you select the **Allow price adjustment** checkbox, the final unit price is $230.
- If you don't select the **Allow price adjustment** checkbox, the final unit price is $200.

Post a sales trade agreement journal

Follow these steps to post a sales trade agreement journal.

1. Go to **Pricing management** > **During-sales pricing** > **Sales trade agreement price** > **Trade agreement journals**.
2. Select the journal that you want to post, and then select **Lines** on the Action Pane.
3. On the **Journal lines, trade agreement** page, on the Action Pane, select **Post**.
4. In the **Price/discount Journal posting** dialog box, select **OK** to post the journal.

Create a new journal based on posted journal lines

If you must create a new journal that includes lines that are similar to the lines in a previously posted journal, you can start by adding copies of the posted journal lines to your new journal.

1. Go to **Pricing management** > **During-sales pricing** > **Sales trade agreement price** > **Trade agreement journals**.
2. On the Action Pane, select **New**.
3. On the Action Pane, select **Lines**.
4. On the **Overview** FastTab, select **Select** on the toolbar.
5. The **Select** dialog box appears. Use the fields to build a query that finds the matching lines from posted journals that you want to copy to your new journal.
6. Set the **Match value only** option to control how the system filters the records. For example, you're looking for a journal line that matches price attribute **Customer group** = *A*, and you

have customer *US-001* that belongs to customer group A. In this case, the **Match value only** settings work in the following way:

- *Yes* – When you select customer account *US-001*, the system doesn't match the line, because there's no record that specifies the customer account, even though the rule applies to customer account US-001.
- *No* – When you select customer account *US-001*, the system matches the line, because the rule applies to customer account US-001.

7. After you set up your query, select **Select** to copy the matching posted trade agreement lines to a new journal.

8. Edit the copied lines as required.

9. Validate and post the journal.

View posted sales trade agreement prices for products and customers

Follow these steps to view posted sales trade agreement prices that apply to a selected product or customer.

1. Follow one of these steps:

   - To find trade agreement prices for products, go to **Product information management** > **Products** > **Released products**.
   - To find trade agreement prices for customers, go to **Sales and marketing** > **Customers** > **All customers**.

2. Select the product or customer that you want to inspect.

3. On the Action Pane, on the **Price** tab, select **Trade agreements**.

4. The **Trade agreements** dialog box appears. Criteria for finding lines for the selected product or customer are already included. Set the **Posted** option to *Yes* to find only posted lines.

5. Set the **Match value only** option to *No*.

6. Select **OK**. The trade agreement lines that are found are shown.

Last updated on 04/27/2026

Margin price adjustments

This article describes how to set up and use margin price adjustments.

Unified pricing management lets you use *margin price adjustments* to move item prices up or down from the base price. Set up these adjustments by including one or more price component codes of the *Margin component price adjustment* type in your price structures. Price adjustments are compounded across price component codes and add up to the total price adjustment.

In scenarios where you build your selling price based on the inventory standard cost or purchase price, it represents the layers of the margin components that add up to those base prices.

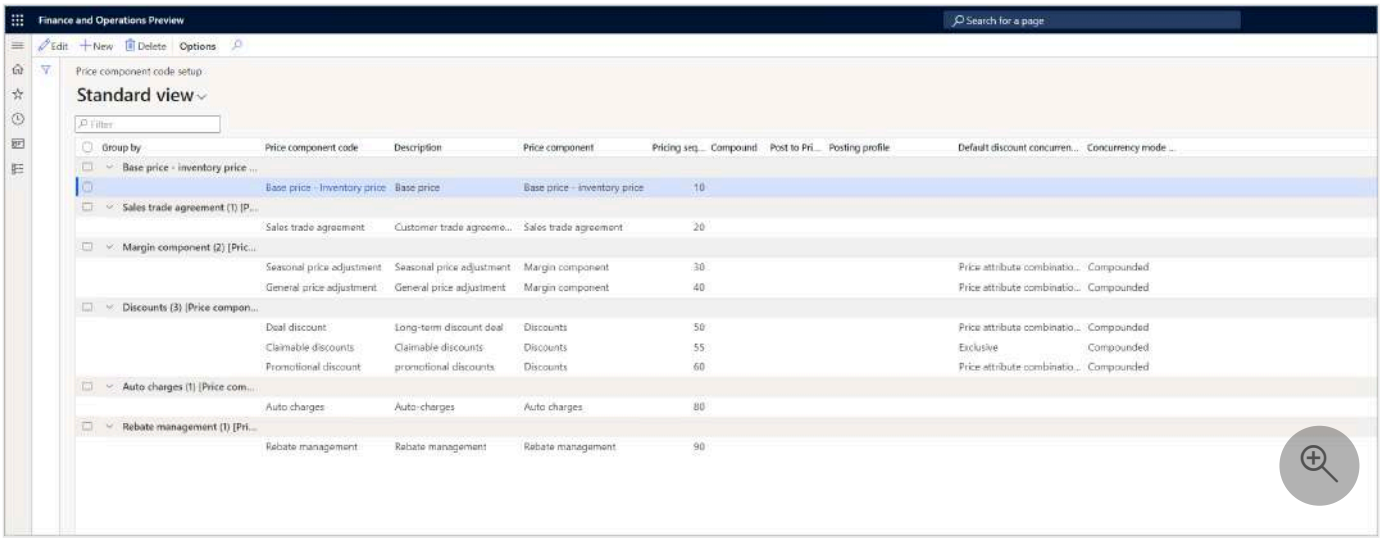
You can associate margin component price adjustments with many types of agreements, promotions, and events. They let you easily adjust prices without changing the base price.

To use margin component price adjustments, complete the following configuration steps:

- Create one or more price component codes to set up the different types of margin price adjustments that you can include in your price structures.
- Create one or more price structures to define how your margin price adjustments combine with other price elements (such as base prices and discounts) to determine the final unit price.
- Set up one or more margin price adjustment pricing rules to configure the margin price adjustments that you need, and to define which customers and items they apply to, and how they're calculated. Associate each price adjustment with a specific price component code and then define the details of the calculation.

For information about how to create pricing rules for each margin price adjustment (and discount), see Pricing rules for discounts and margin price adjustments.

For example, the following illustration shows a price structure that contains two sequential margin component price adjustments (*General price adjustments* and *Seasonal price adjustments*).



Last updated on 07/01/2026

Discount types

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article introduces the different types of discounts that you can set up by using Unified pricing management.

To use discounts, complete the following setup steps:

- Create one or more price component codes to set up the different types of discounts that you can include in your price structures.
- Create one or more price structures to define how your discounts combine with other price elements (such as the base price and margin price adjustments) to determine the final unit price.
- Set up one or more discount pricing rules to configure each discount that you need, and define which customers and items they apply to, and how they're calculated. For each rule, select the type of discount and associate it with a specific price component code. Then, define the discount by using a combination of settings that are common to all discounts and settings that are specific to your selected discount type.

For information about how to create pricing rules for each type of discount (and for margin price adjustments), see Pricing rules for discounts and margin price adjustments.

> **Note**
>
> The *pricing sequence* of your price structure isn't associated with discount types.

Simple discounts

Simple discounts reduce the product price by a percentage or amount. Here are some examples:

- Buy items of brand A, and get 5 percent off.

- Buy items from package group B, and get $10 off.

Quantity discounts

Quantity discounts are given to customers when they purchase a specific quantity of a product. The quantity tier applies to each order, not each line. Here are some examples:

- Buy two or more items of brand A, and get 5 percent off.
- Buy five or more items of brand A, and get 7 percent off.
- Buy 10 or more items of brand A, and get 9 percent off.
- Buy 15 or more items of brand A, and get 12 percent off.

Threshold discounts

Threshold discounts are given to customers when the total for a transaction reaches one or more specified amounts. The amount threshold applies to each order, not each line. Here are some examples:

- Buy a total of $1,000 or more of items of brand A, and get $100 off.
- Buy a total of $2,000 or more of items of brand A, and get 10 percent off.

Threshold discounts are applied after other discount types, because the total order value determines their eligibility. Create a price component code for threshold discounts, and avoid mixing threshold discounts with other types in the same price component code. Set up the price component code for threshold discounts after the other discount-related price component codes that have a higher pricing sequence.

Mix-and-match discounts

Mix-and-match discounts are given to customers when they purchase a specific combination of products. Here's an example:

- Buy 10 or more items of brand A, and get 5 percent off items from package group B.

> **Note**
>
> The discount claim feature isn't available for mix-and-match discounts.
>
> For mix-and-match least-expensive discounts, the number of least-expensive products must be more than one and less than the number of products that are required to trigger the

discount.

For mix-and-match line spec discounts, the **Calculation type** and **Discount** values must be the same for all discount lines in the same line group.

Set up line groups for mix-and-match discounts

To define the different combinations of item groups and related purchase thresholds that qualify for a discount, set up *line groups* and assign them to lines in your mix-and-match pricing rules. Follow these steps to set up and apply mix-and-match line groups.

1. Go to **Pricing management** > **During-sales pricing** > **Discounts** > **Mix and match line group setup**.

2. Review the existing line groups, and create any that you need. Use the buttons on the Action Pane to add or remove line groups as required. For each row that you add to the grid, set the following fields:

    - **Line group** – Enter a unique identifier for the line group.
    - **Number of products needed** – Enter the quantity of a product that a customer must purchase to qualify for the discount. This field defines a default value that is used every time a user selects the line group for a specific discount pricing rule. However, you can override the value for each specific rule as needed.

3. Go to **Pricing management** > **During-sales pricing** > **Discounts** > **Mix and match line groups**.

4. Review the line group assignments for any or all existing mix-and-match discount rules, and create any that you need. Use the buttons on the Action Pane to add or remove line groups as required. For each row that you add to the grid, set the following fields:

    - **Discount** – Select the existing mix-and-match discount pricing rule to add a line group to.
    - **Line group** – Select the existing line group to add to the selected pricing rule.
    - **Number of products needed** – Enter the quantity of a product that a customer must purchase to qualify the discount. For new rows, this field initially shows the default value that you previously set for the selected line group on the **Mix and match line group setup** page. However, you can override the default value.
    - **Line color** – Select a color for the line group. This color appears as the background color for lines that are assigned to the group on the **Lines** FastTab when you view or

edit a mix-and-match discount pricing rule.

5. To use your line groups in a pricing rule, open the **All discounts** or **Mix and match discounts** page, and select or create your rule. Then follow these steps:

    - On the **Price/discount** FastTab, define the terms of the mix-and-match discount.
    - On the **Lines** FastTab, add a line for each item (or collection of items) that's part of the mix-and-match discount. Then use the **Line group** column to assign a line group to each line.
    - If you need to change the selection or configuration of the line groups that are available for the current pricing rule (including the number of products that are required), select **Mix and match line groups** on the Action Pane. The same fields that were available on the **Mix and match line group setup** page are available. However, only the lines for the current pricing rule are shown.

Mix-and-match example 1: Discount for bundle sales

A car dealer offers a bundle where, if a customer purchases any **Interior option** item from package *B*, they get 10% off a dashboard camera (item number *DAC0001*). The dealer sets up a mix-and-match discount pricing rule and sets the following value on the **Price/discount** FastTab:

- **Calculation type:** *Line spec*

The dealer also sets the following values on the **Lines** FastTab.

Expand table

Line no.	Price attribute	Price attribute value	Number of products needed	Unit	Discount value	Calculation type	Line group
1	Interior option	Package B	1	Ea	0.00	Percentage off	A
2	Item number	DAC0001	1	Ea	10.00	Percentage off	B

Mix-and-match example 2: Discount for any second purchase

A car dealer offers a discount where, if a customer purchases any two **Interior option** items from package *C*, they get 15 percent off the less expensive of the two items. Therefore, the dealer sets

up a mix-and-match discount pricing rule and sets the following values on the **Price/discount** FastTab:

- **Calculation type:** *Least expensive*
- **Number of least expensive lines:** *1*
- **Percentage off:** *15%*

The dealer also sets the following values on the **Lines** FastTab.

Expand table

Line no.	Price attributes	Price attribute value	Number of products needed	Unit	Line group
Line 1	Interior option	Package C	1	Ea	A
Line 2	Interior option	Package C	1	Ea	B

The following table shows the items that are available to customers.

Expand table

Item no.	Interior option	Price
EV004	Package C	$100
EV005	Package C	$120

Therefore, the customer gets 15 percent off the less expensive item (*EV004*) and gets both items for $205.

Mix-and-match example 3: Mandatory item to trigger the discount

A car dealer offers a discount where, if a customer purchases any ten **Interior option** items from package *D*, including at least one of item *EV007* (which is also an item in package *D*), they get a 12-percent discount on the whole package of interior options. Therefore, the dealer sets up a mix-and-match discount pricing rule and sets the following values on the **Price/discount** FastTab:

- **Calculation type:** *Percentage off*
- **Percentage off:** *12%*

The dealer also sets the following values on the **Lines** FastTab.

Expand table

Line No.	Price attributes	Price attribute value	Mandatory	Number of products needed	Unit	Line group
1	Interior option	Package D	No	10	Ea	A
2	Item number	EV007	Yes	1	Ea	B

The following table shows the items that are available to customers.

Expand table

Item no.	Interior option	Price
EV009	Package D	$100
EV007	Package D	$100

If a customer now orders five of item *EV009* and five of item *EV007*, they've ordered 10 items from package *D* and have also purchased the required quantity of item *EV007*. Therefore, the customer gets the 12-percent discount and pays $880 for the package.

However, if the customer orders ten of item *EV009*, they still order 10 items from package *D*, but they didn't purchase the required quantity of item *EV007*. Therefore, the customer doesn't get the discount and pays $1,000 for the package.

"Always apply" discounts

Always apply isn't a discount type. It's a concurrency mode that's available for all discount types. All discounts created by using the *Always apply* mode are applied to the appropriate items after all the existing discounts are applied. For more information, see Resolve concurrency within price component codes.

Next steps

- Create price component codes.
- Create price structures.
- Configure discount pricing rules for each discount.

Last updated on 05/29/2026

Pricing rules for discounts and margin price adjustments

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure pricing rules for margin component price adjustments, simple discounts, quantity discounts, mix-and-match discounts, threshold discounts, and free-item discounts.

Create and manage pricing rules

1. Depending on the type of pricing rule that you want to work with, follow one of these steps:

    - To view, edit, and create margin price adjustments, go to **Pricing management** > **During-sales pricing** > **Price adjustments** > **Margin component price adjustments**.
    - To view, edit, and create all types of discounts (including all the other discount types in this list), go to **Pricing management** > **During-sales pricing** > **Discounts** > **All discounts**.
    - To view, edit, and create simple discounts only, go to **Pricing management** > **During-sales pricing** > **Discounts** > **Discounts**.
    - To view, edit, and create quantity discounts only, go to **Pricing management** > **During-sales pricing** > **Discounts** > **Quantity discounts**.
    - To view, edit, and create mix-and-match discounts only, go to **Pricing management** > **During-sales pricing** > **Discounts** > **Mix and match discounts**.
    - To view, edit, and create threshold discounts only, go to **Pricing management** > **During-sales pricing** > **Discounts** > **Threshold discounts**.
    - To view, edit, and create free-item discounts only, go to **Pricing management** > **During-sales pricing** > **Discounts** > **Free item**.

2. Follow one of these steps:

- To add a new record, select **New** on the Action Pane, and then, on the menu, select the type of record that you want to create.
- To create a copy of an existing record, select it in the list pane, and then select **Copy** on the Action Pane.
- To edit an existing record, select it in the list pane, and then, on the **General** FastTab, set the **Status** field to *Disabled*. (All rule records are read-only while they're enabled.)
- To delete a record, select it in the list pane, and then select **Delete** on the Action Pane.

3. On the header of the new or selected rule record, set the following fields:

   - **Discount** or **Margin component adjustment** – Enter a unique identifier for the rule.
   - **Name** – Enter a common name for the rule.
   - **Discount type** – If you're working with a discount rule, this field shows the type of discount and is read-only. This field isn't shown for margin price components.
   - **Publish status** – This field is initially set to *Draft*. After you complete the record, you can activate the rule.

4. Continue to set up the pricing rule as described in the remaining sections of this article. Many settings are shared by all types of pricing rules. However, some settings are specific to margin component price adjustments, all discounts, or specific types of discounts.

Enter information on the General FastTab

The **General** FastTab provides basic settings for the pricing rule. Set the fields described in the following subsections, depending on the type of pricing rule that you're setting up.

General settings that apply to both margin price adjustments and discounts

Set the following fields on the **General** FastTab for all types of pricing rules:

- **Status** – Set this field to *Enabled* or *Disabled*. Only enabled records are available for calculations. New records are set to *Disabled* and can be enabled only after you finish setting all the required fields. All fields are read-only for enabled rules. Therefore, you must disable a rule before you can edit it.
- **Currency** – Select the currency that the rule applies to.
- **Price component** – This read-only field shows the type of pricing rule (*Discounts* or *Margin component*).

- **Price component code** – Select the price component code that the rule applies to. Only existing price component codes of the applicable type are listed.
- **Header price attribute group type** – This read-only field indicates how customers that the rule applies to are selected. It shows *All* if the rule applies to all customers. It shows *Group* if the rule applies to only a set of customers, as defined by values that are assigned to attributes that are part of an attribute group. You can set the value for the current rule by selecting **Header price attribute group** on the Action Pane. For more information, see the Set up header price attribute values section of this article.
- **Price attribute group** – If the **Header price attribute group type** field is set to *Group*, this read-only field shows the attribute group that's used to define the set of customers that the current rule applies to. For more information, see the Set up header price attribute values section.

General settings that apply only to margin price adjustments

If you're setting up a pricing rule for a margin component price adjustment, set the following fields on the **General** FastTab:

- **Concurrency model** – This read-only field indicates how the system handles situations where more than one price adjustment applies to the same price component code in a price structure. This type of situation is known as concurrency. The value *Price component code rank* means that the system selects a price adjustment based on its price attribute combination rank. If multiple price adjustments have the same highest rank, they all apply to the sales order.
- **Quantity tiers** – Set this option to *Yes* to enable the **Quantity tiers** FastTab, where you can define price adjustments that vary based on the quantity on the sales order line.

> **Note**
>
> Margin component price adjustment quantity tiers differ from quantity discounts. Quantity discounts tiers are based on the item quantity per order. Margin component price adjustment tiers are based on the item quantity per line.

General settings that apply only to discounts

If you're setting up a pricing rule for a discount, set the following fields on the **General** FastTab. Not all these fields are available for every type of discount.

- **Discount account** – If you want to post the discount amount for the discount record to a separate general ledger account, specify that account. This field overrides the similar field that's available for price structures (price trees). Learn more in Price component posting.

- **Discount concurrency mode** – Select one of the following values to specify how the system should handle situations where more than one discount rule applies to the same price component code in a price structure. (This type of situation is known as concurrency.)
  - *Exclusive* – The discount is an exclusive discount. Therefore, it can't be combined with other discounts. If there are multiple exclusive discounts, the price engine applies only the one that gives the largest discount.
  - *Best price* – Discounts of this concurrency mode compete for the largest discount (lowest price).
  - *Compounded* – The discount combines with other discounts according to the pricing sequence that's defined in the price structure. Settings on the **Pricing management parameters** page control whether the compounded discount calculation is based on the original price or the compounded amount. (Learn more in Resolve concurrency within price component codes.)
  - *Always apply* – The discount always applies. Always-apply discounts are applied last. Therefore, they represent universal discounts that apply to all sales orders.
  - *Price attribute combination rank* – Discounts of this concurrency mode don't compete based on price. Instead, they compete based on which discount has the highest price attribute combination rank. If multiple discounts have the same highest rank, they all apply to the sales order.

- **Coupon code required** – Set this option to *Yes* if a coupon code is required to qualify for the discount. The discount must be linked to the coupon. The discount triggers only if a valid coupon code is entered together with the order. When you enable this option, the **Status**, **Effective date**, and **Expiration date** fields become unavailable. The settings of those fields depend on the coupon configuration. To set up coupons, go to **Pricing management** > **During-sales pricing** > **Coupons**.

- **Claimable** – Set this option to *Yes* to make the discount claimable. (In other words, the discount is sponsored by a vendor and can therefore be charged back to that vendor.)

- **Discount claim group** – If you set the **Claimable** option to *Yes*, select the applicable claim group.

- **Claim offset posting** – If you set the **Claimable** option to *Yes*, select the offset account that you want to post the claim to.

- **Max criteria type** – If a maximum discount per order applies, select the type of limit that you want to apply:
  - *Amount*
  - *Quantity*
- **Max** – If a maximum discount per order applies, select the value of the limit.
- **Max quantity unit** – If a maximum discount per order applies, and the **Max criteria type** field is set to *Quantity*, select the unit that applies to the value that's selected in the **Max** field.

Enter information on the Details FastTab

Use the **Details** FastTab to enter more information about the pricing rule. It also provides extra information about the pricing rule. You can set and review the following fields:

- **Description** – Enter a description of the pricing rule.
- **Disclaimer** – Enter a disclaimer for the pricing rule.
- **Text for fiscal receipt** – Enter text that appears on the related fiscal receipt.
- **Header price attribute details** – This read-only field summarizes the header price attribute values for the pricing rule. (You can access the header price attribute values by selecting **Header price attribute group** on the Action Pane.) For more information, see the Set up header price attribute values section of this article.

Enter information on the Price/discount FastTab

The **Price/discount** FastTab is available only for simple, threshold, and mix-and-match discounts. The settings here combine with the settings on the **Lines** FastTab to define the discount values for different combinations of customers and items.

Price/discount settings for simple discounts

If you're setting up a simple discount, set the following fields on the **Price/discount** FastTab:

- **Percentage off** – Enter the default discount percentage for new lines added on the **Lines** FastTab. As you add each line to the **Lines** FastTab, you can change the percentage, or change the percentage off to an amount off, as described later in this article.
- **Quantity limit** – Enter the maximum order quantity that the discount applies to. This limit applies to all lines on the **Lines** FastTab. Although the value is repeated for each line on the

Lines FastTab, you can edit it only on the **Price/discount** FastTab. If you edit the value, all existing lines are updated. You can't edit the value separately for each line.

Price/discount settings for mix-and-match discounts

When you set up a mix-and-match discount, set the following fields on the **Price/discount** FastTab:

- **Percentage off** – Enter the default discount percentage for new lines added on the **Lines** FastTab. As you add each line to the **Lines** FastTab, you can change the percentage, or change the percentage off to an amount off, as described later in this article.
- **Quantity limit** – Enter the maximum order quantity that the discount applies to. This limit applies to all lines on the **Lines** FastTab. Although the value is repeated for each line on the **Lines** FastTab, you can edit it only on the **Price/discount** FastTab. If you edit the value, all existing lines are updated. You can't edit the value separately for each line.
- **Calculation type** – Select the type of calculation that the current pricing rule applies. The other fields that are available on the **Price/discount** FastTab vary based on the value that you select.
    - *Deal price* – Use a specific final price instead of a calculated discount. After you select this value, enter the price in the **Deal price** field.
    - *Percentage off* – Calculate the discount as a percentage of the item price. After you select this value, enter the percentage in the **Percentage off** field. For example, the usual price of item A is 40.00, the usual price of item B is 60.00, and a mix-and-match discount of 30 percent applies to these items. If a customer buys item A and item B together, the cost is 70.00 instead of 100.00.
    - *Amount off* – Define the discount as a fixed value. After you select this value, enter the discount in the **Amount off** field.
    - *Least expensive* – After you select this value, enter the discount price in the **Deal price**, **Percentage off**, or **Amount off** field, depending on the method that you want to use to calculate the price. Then, in the **Number of least expensive lines** field, enter the number of least-expensive items that you want to apply the discount to. Finally, set the **Multiple occurrences mode** field to one of the following values:
        - *Favor customer*
        - *Favor retailer*

- *Line spec* – Use the discount that's specified for each line on the **Lines** FastTab.

- **Count non-discountable products** – Set this option to *Yes* to count non-discountable items toward the quantity.

> ⓘ **Note**
>
> For mix-and-match and least-expensive discounts, the number of least-expensive products must be more than one and less than the number of products.

Price/discount settings for threshold discounts

When you set up a threshold discount, set the following field on the **Price/discount** FastTab:

- **Count non-discountable items toward threshold** – Set this option to *Yes* to count non-discountable items toward the quantity.

Enter information on the Quantity discount configuration FastTab

The **Quantity discount configuration** FastTab is available only for quantity discounts. Use it to set up the discounts that apply to each quantity threshold that you set up.

Quantity discount settings

On the **Quantity discount** FastTab, set the following fields, which affect how the quantity tiers work:

- **Calculation type** – Select one of the following options to specify how the discount is calculated based on the table values:
  - *Percentage off* – The discount is calculated as a percentage of the price.
  - *Amount off* – A specific value is subtracted from the price.
  - *Unit price* – A new, specific unit price is applied.

- **Interval** – This option applies only when the **Calculation type** field is set to *Amount off*. Set it to *Yes* to use a stepped calculation for each quantity tier. See the next section for an example that shows how this setting affects the discount calculation.

The grid on the **Quantity discount configuration** FastTab defines a set of quantity tiers and the discount that applies to each tier. Use the buttons on the toolbar to add and remove rows from

the grid. For each row, set the following fields:

- **Minimum quantity** – Enter the minimum quantity of an item that a customer must order to get the discount specified on the line.
- **Value** – Enter the value of the discount for the line (based on the selected calculation type).

View information on the Calculation FastTab

The **Calculation** FastTab provides information about how the price component code selected for the pricing rule is set up in your price structure.

The following information is shown:

- **Pricing sequence** – This read-only field shows the position of the selected price component code in your price structure.
- **Compound** – This read-only field is shown only for margin price adjustments, not for discounts. It indicates whether the selected price component code is set up to make compound calculations in your price structure.

Enter information on the Validation period FastTab

Use the fields on the **Validation period** FastTab to control when the pricing rule applies. The following conditions apply:

- The date range is defined at the header level.
- The date, which is used when a pricing rule is validated, is based on the value of the **Date type** field on the **General** tab of the **Pricing management parameters** page. You can set the **Date type** field to *Today*, *Requested ship date*, *Requested receipt date*, or *Created date*.

Set up the validation period by using either standard rules or advanced rules.

Set up standard date validation rules

Follow these steps to set up the validation period by using the standard validation rules.

1. Set the **Date validation type** field to *Standard* to enable the fields that are required to set up standard date validation.

2. Set the following fields:

    - **Effective date** – Select the first date that the pricing rule should apply.
    - **Expiration date** – Select the last date that the pricing rule should apply.

Set up advanced date validation rules

Advanced validation rules let you set the validation period by using a discount period record instead of entering simple start and end dates.

Discount period records let you set up detailed date and time limits that can include a start date and time, an end date and time, and specific times of day for each day of the week. Set up discount period records by going to **Retail and Commerce** > **Pricing and discounts** > **Discount periods**.

Follow these steps to set up the validation period by using the advanced validation rules.

1. Set the **Date validation type** field to *Advanced* to enable the fields that are required to set up advanced date validation.
2. In the **Discount period number** field, select the discount period record that defines the validation period that you want to use. The read-only **Description**, **Start date**, and **End date** fields are updated to show values from the selected discount period record.

Enter information on the Lines and Line details FastTabs

Use the **Lines** FastTab to control which discounts and margin price adjustments apply to each item. All the lines in a pricing rule apply to the same collection of customers. You configure this collection in the header price attribute group settings, which you can access by selecting **Header price attribute group** on the Action Pane.

Add a line and configure the set of items that it applies to

Each line assigns calculation rules that apply to a specific collection of items. When you add a line, use a dialog box to set up the items that the line affects and then add the line to the grid. Later, set up the calculation rules for the line by using the columns in the grid. Follow these steps to add a line.

1. On the **Lines** FastTab, select **New** on the toolbar.

2. The **Edit price attributes** dialog box appears. This dialog box lets you set up the logic for finding the items that the current line applies to. On the **Header price attribute group** FastTab, review the logic that's set up for selecting the customers that the current pricing rule applies to. The values apply to all lines in the current pricing rule and are read-only here. (You can change them in the header price attribute group settings.)

3. In the **Price attribute group combination** field, select the combination of price attributes that you use to define the items for the current line. The combinations that are available for selection are defined for the price component code that's selected for the current pricing rule. However, only the right side of the combination name (the line price attribute group name) is relevant, because the header attributes are common to all lines for the pricing rule, as mentioned in the previous step. Select the combination where the right side of the name matches the way that you want to define the collection of items that the line applies to.

4. If the selected price attribute group combination considers item (line) attribute values (that is, if it doesn't apply to *all* items), the **Line price attribute group** FastTab is available. In this case, for each row on the **Line price attribute group** FastTab, enter or select one or more values in the **Value** column to define the rules for selecting items. The collection of attributes that appears comes from the price attribute group combination that you selected. The following rules apply:

    - All rows are combined by using a logical AND operator. Therefore, only those items that have matching values for *all* rows are included.
    - If you want to create one or more rows that include multiple values, set the **Enable multiple selections** option to *Yes*. You can then add a comma-separated list of values in the **Value** column for each row. These values are combined by using an OR operator. Therefore, the row finds items that match *any* of the values in the list.
    - You can specify values to exclude by adding an exclamation mark (!) before the value. This exclamation mark is known as an exclusion prefix. For example, to find all items except *A0001*, set the **Value** field to *!A0001* for the attribute where the **Attribute** field is set to *Item number*. You can automatically add the exclusion prefix to the values of any row by selecting the row and then selecting **Exclude values in selected lines**.

5. On the **Preview matching results (Products)** FastTab, view the preview of items that match the conditions set up so far.

    - If the list includes any items that you want to exclude, select the target rows, and then select **Exclude** on the toolbar.
    - The **Line type** column indicates which items you selected to exclude.
    - To re-include a previously excluded item, select it, and then select **Include** on the toolbar.
    - To view, exclude, and/or re-include specific item variants, select **All variants** on the toolbar. While you're viewing the variants, you can select **Product master** on the toolbar to go back to the standard item list.

6. Select **OK** to add the new line to the grid on the **Lines** FastTab.

7. Set up the calculation rules for the new line by using the columns in the grid, as described in the next section.

If you need to return to the **Edit price attributes** dialog box to change the set of items that a line applies to, select the line, and then select **Edit line price attribute** on the toolbar.

Set line values to define calculation rules and further refine the item selection

The settings for each line on the **Lines** FastTab vary based on the type of pricing rule that you're working with. The following fields and information might be available:

- **Line group type** – This field shows whether the line applies to a specific group of items or to all items.
- **Line price attribute group** – If the **Line group type** field is set to *Group*, this field shows the name of the selected line price attribute group.
- **Price attribute detail** – This field summarizes the values set for the line in the **Edit price attributes** dialog box. To change these values, select **Edit price attribute** on the FastTab toolbar.
- **Combination rank** – This field shows the combination rank that was generated for the price attribute group combination that you selected for the line in the **Edit price attributes** dialog box. Rankings enable the system to determine which pricing rule is used if an order or order line qualifies for more than one rule. Learn more in Price component codes.
- **Required line quantity** – For free-item discounts where the **Criteria type** field is set to *Quantity*, enter the minimum quantity that must be ordered to qualify for the discount.
- **Required line amount** – For free-item discounts where the **Criteria type** field is set to *Amount*, enter the minimum amount that must be ordered to qualify for the discount.
- **Allow unit conversion** – Select this checkbox to convert the selling unit to the unit that's used for the pricing rule as required.
- **Line type** – Select one of the following values:
  - *Include*
  - *Exclude*

- **Calculation type** – Select one of the following values to specify how the discount or price adjustment for the current line should be calculated:
  - *Percentage off* or *Percent* – Calculate the discount or price adjustment value as a percentage of the price.
  - *Amount off* or *Amount* – Calculate the discount or price adjustment by using a fixed value that's added to or subtracted from the price.
  - *Unit price* – Set a new, fixed unit price.
- **Percent** or **Percentage off** – If you're using a percentage-based calculation type, enter the percentage.
- **Amount** or **Amount off** – If you're using an amount-based calculation type, enter the amount.
- **Unit price** – If you're using the *Unit price* calculation type, enter the new unit price.
- **Quantity limit** – The field shows the maximum order quantity that the discount applies to. This limit applies to all lines on the **Lines** FastTab. You can't edit this value separately for each line, but you can edit it for all lines at the same time by using the **Quantity limit** field on the **Price/discount** FastTab.
- **Number of products needed** – This field shows the number of items that the customer must buy in this line group to qualify for the discount. For mix-and-match discounts, the value is read-only and is taken from the group that's selected in the **Line group** field.
- **Line group** – If you're setting up a mix-and-match discount, select the line group that applies to the line. Before you can select a value, you must set up your line groups, add them to the current pricing rule, and configure a color and item threshold for them, as described in Mix-and-match discounts. The line is highlighted in the color that's configured for the selected line group (if any color was configured), and the **Number of products needed** value is updated to match the selected line group configuration.
- *Other dimensions* – The grid might include additional columns that show item, storage, and tracking dimensions (such as **Color**, **Warehouse**, and **Serial number**). To select which dimensions are shown, select **Display dimensions** on the toolbar. Enter values in these columns to limit the line so that it applies only to items that match the specified dimension values.

Enter information on the Lines details FastTab

To add a note that provides more information about a line on the **Lines** FastTab, select the relevant line. Enter your note in the **Description** field on the **Line details** FastTab.

Set up header price attribute values

Use header price attributes to control which customers the current pricing rule applies to. The limits apply to all lines and items that the rule applies to. (Learn more in the Enter information on the Lines and Line details FastTabs section of this article.) Follow these steps to set the header price attributes for the current pricing rule.

> ⓘ **Note**
>
> Before you can set header price attributes, you must assign a price component code on the **General** FastTab.

1. On the Action Pane, select **Header price attribute group**.
2. The **Edit price attributes** dialog box appears. This dialog box lets you set up the logic for finding the customers that the current pricing rule applies to. On the **Header price attribute group** FastTab, review and set the following fields:
    - **Scope of price attributes** – This read-only field indicates the type of attributes that your settings apply to. It always shows *Header*.
    - **Group type** – Select the type of customer selection that you want to set up. The available values depend on the price attribute combinations that are available for your selected price component code. The following values might be available:
        - *Group* – If you select this value, the dialog box provides a group of header attributes that you can assign values for, so that you can limit the set of customers that the pricing rule applies to.
        - *All* – The pricing rule applies to all customers. If you select this value, no further steps are required. Therefore, you can skip the rest of this procedure and just select **OK** to apply your settings.
    - **Price attribute group** – If you set the **Group type** field to *All*, select a price attribute group. The available values depend on the attribute groups that you enabled for the selected price component code. The value that you select affects the set of attributes that you can work with in the dialog box.

3. If you selected a price attribute group, the attributes from it appear on the **Header price attribute group** FastTab. For each row, enter or select one or more values in the **Value** column to define the rules for selecting customers. The following rules apply:

    - All rows are combined by using a logical AND operator. Therefore, only those customers that have matching values for *all* rows are included.
    - If you want to create one or more rows that include multiple values, set the **Enable multiple selections** option to *Yes*. You can then add a comma-separated list of values in the **Value** column for each row. These values are combined by using an OR operator. Therefore, the row finds customers that match *any* of the values in the list.
    - You can specify values to exclude by adding an exclamation mark (!) before the value. This exclamation mark is known as an exclusion prefix. For example, to find all customers except *US-001*, set the **Value** field to *!US-001* for the attribute where the **Attribute** field is set to *Customer account*. You can automatically add the exclusion prefix to the values of any row by selecting the row and then selecting **Exclude values in selected lines**.

4. On the **Preview matching results (Customers)** FastTab, view the preview of customers that match the conditions that you set up so far.

    - If the list includes any customers that you want to exclude, select the target rows, and then select **Exclude** on the toolbar.
    - The **Line type** column indicates which customers you selected to exclude.
    - To re-include a previously excluded customer, select it, and then select **Include** on the toolbar.

5. Select **OK** to save your settings and close the dialog box.

Set up mix-and-match line groups

If you're working with a mix-and-match discount pricing rule, select **Mix and match line groups** to select which mix-and-match line groups can be assigned on the **Lines** FastTab. Specify the number of products that must be bought from each group to qualify for the discount. For information about how to set up mix-and-match line groups, see Mix-and-match discounts.

Last updated on 04/27/2026

Resolve concurrency within price component codes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Concurrency rules let you define what happens if multiple pricing rules apply to the same order and/or order line. You can set up concurrency rules to control whether the customer receives just one of the matching rules (and if so, which one they receive), or whether the rules are combined (and if so, how they're combined). There are two types of concurrency:

- **Across-price-component-code concurrency** – This type of concurrency controls how the different price component codes that are included in a price structure combine with each other. A price structure defines a collection of different types of price component codes (including base price, sales agreement price, margin components, discounts, and/or charges), the order that the codes are calculated in, and the way that the price adjustments are calculated and combined to determine the final price.
- **Within-price-component-code concurrency** – Any number of pricing rules can be associated with each price component code that's included in a price structure. This principle is especially true of discounts. This type of concurrency occurs when an order or order line qualifies for more than one pricing rule that's associated with the same component code. For example, if you define a price component code that's named *Seasonal promotion events*, and you associate multiple discount rules with it, multiple pricing rules might apply to the same order line. Therefore, in the *Seasonal promotion events* price component code, you'll set up concurrency rules to control whether the customer receives just one of the matching discounts (and if so, which one they receive), or whether the discounts are combined (and if so, how they're combined).

This article explains how to manage *within-price-component-code concurrency*. For information about how to manage *across-price-component-code concurrency*, see Resolve concurrency across price component codes.

Types of within-price-component-code concurrency modes

The following types of within-price-component-code concurrency modes are available, depending on the type of price component code:

- **Exclusive** – The pricing rule can't be combined with other rules that are associated with the same price component code. If more than one of these rules are set up as exclusive, the price engine applies the rule that has the largest discount.
- **Best price** – Pricing rules that use this concurrency mode will compete for the largest discount (lowest price).
- **Compounded** – All applicable pricing rules are combined. On the **Pricing management parameters** page, you can configure the system so that each calculation is based on either the original price or a running total of all adjustments so far. Learn more in the System settings for resolving discount concurrency within a price component code section of this article.
- **Always apply** – The pricing rule always applies. This calculation is always applied last within a price component code.
- **Price attribute combination rank** – Pricing rules that use this concurrency mode don't compete for prices. Instead, they compete based on which rule has the highest *price attribute combination rank*. If multiple rules have the same highest rank in the same price component code, they're combined.

Configure within-price-component-code concurrency modes

You can view and/or set within-price-component-code concurrency modes in several places in the system:

- **Price component code level** – Each relevant price component code has a default within-price-component-code concurrency setting. The available modes vary, based on the type of price component code. This setting can be overwritten in any or all pricing rules as required. Learn more in Price component codes.
- **Price structure level** – Each price component code that's included in a price structure shows its default concurrency mode, but you can't edit it here. Learn more in Arrange price component codes into a price structure.
- **Pricing rule level for discounts and margin price adjustments** – The concurrency mode that's assigned to a pricing rule overrides the default mode that's assigned by its associated price component code. Learn more in Pricing rules for discounts and margin price adjustments.
- **Company level for sales trade agreements** – Sales trade agreements are set up by using trade agreement journals, where the pricing rules are defined by the related journal lines. Concurrency rules for sales trade agreements are defined on the **Pricing management**

parameters page for each company. For information about the options that are available and how to use them, see Sales trade agreement prices.

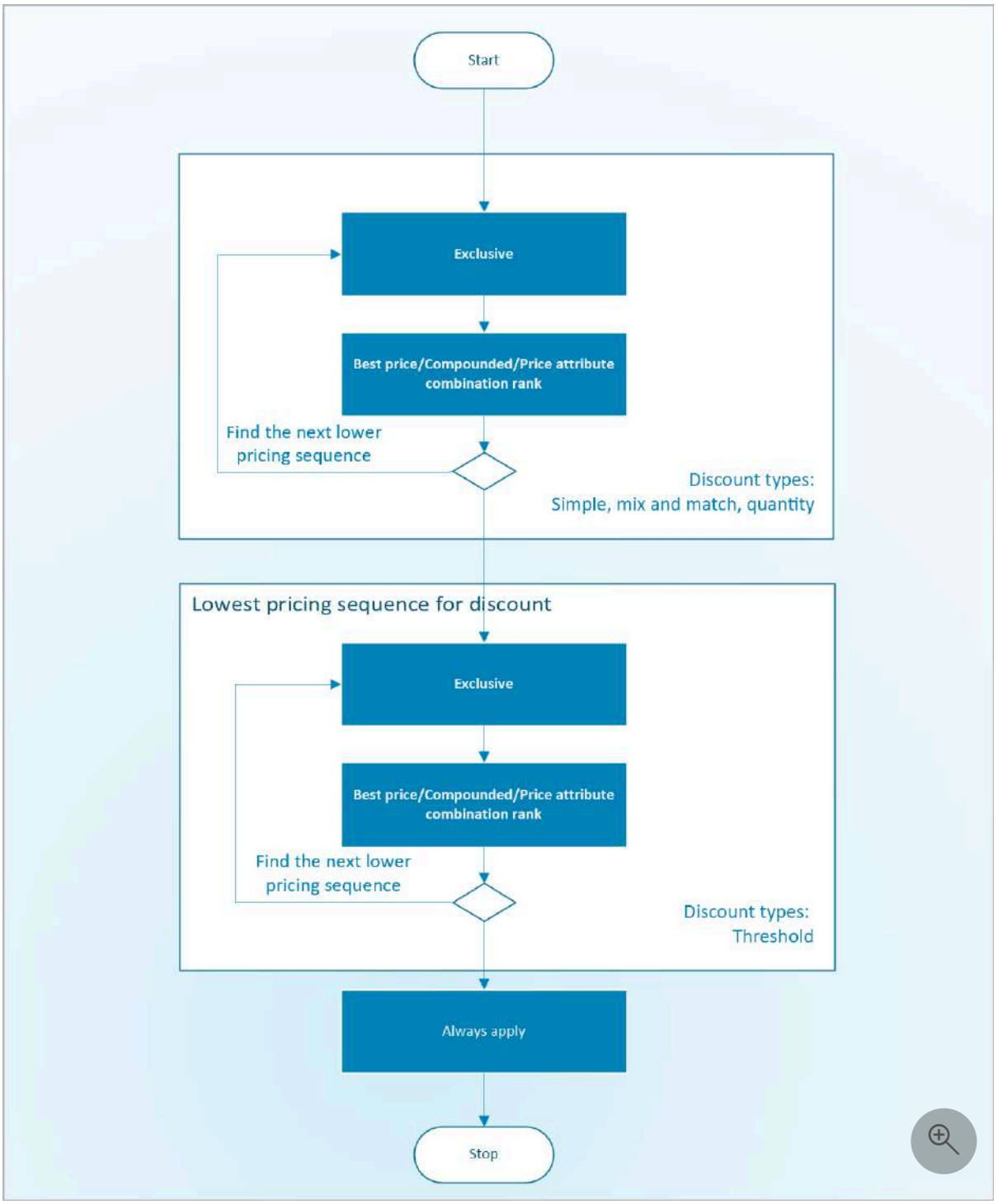
System settings for resolving discount concurrency within a price component code

A few settings on the **Pricing management parameters** page affect the way that concurrent discounts are calculated. The calculation method can, in turn, affect the way that some of your price structure settings work as they apply to discounts. Follow these steps to set up the calculation.

1. Go to **Pricing management** > **Setup** > **Pricing management parameters**.
2. On the **Prices and discounts** tab, on the **Discount concurrency control** FastTab, under **Best price and compound concurrency control model**, select one of the following options:
    - *Best price and compound within priority, never compound across priorities*
    - *Best price only within priority, always compound across priorities*
    - *Best price and compound within priority, best price and compound across priority*
3. Under **Discount compound behavior**, select one of the following options:
    - *Compound*
    - *Compound on the original price*
4. Set the **Enable competition between exclusive threshold and other periodic discounts** option to *Yes* if exclusive threshold discounts should compete with exclusive non-threshold discounts, but priority should still apply. The behavior of best-price and compounded threshold discounts will remain unchanged (that is, they won't compete with the non-threshold discounts).

Discount determination flow

The following flow chart shows how pricing rules, the price structure, and concurrency models work together to resolve within-price-component-code concurrency.



Example

Here's an example that shows how concurrency determination works.

Item *BT023* has an item price of *$1,565.00*, the concurrency mode is *Compounded*, and your system includes the discount pricing rules that are shown in the following table. The pricing sequence comes from your price structure.

Expand table

Price component code	Pricing sequence	Pricing rule ID	Concurrency mode	Calculation method	Value	Discount type
DIS 01	30	DIS01-01	Best price	Percentage	10	Simple
DIS 01	30	DIS01-02	Always apply	Percentage	5	Simple
DIS 02	40	DIS02-01	Compounded	Percentage	10	Simple
DIS 02	40	DIS02-02	Always apply	Percentage	8	Simple
DIS 03	50	DIS03-01	Compounded	Amount	20	Threshold
DIS 03	50	DIS03-02	Always apply	Amount	10	Threshold

An order is placed for a quantity of *1* of item *BT023*. Therefore, the system applies the rules in the previous table to calculate the discount that's shown in the following table. The result is a final price of $1,080.45.

Expand table

Concurrency mode	Pricing sequence	Discount price component code	Pricing rule ID	Calculation method	Value	Discount amount	New unit price
Best price	30	DIS01	DIS01-01	Percentage	10	156.50	1,408.50
Compounded	40	DIS02	DIS02-01	Percentage	10	140.85	1,267.65
Threshold	50	DIS03	DIS03-01	Percentage	20	20.00	1,247.65
Always apply	30	DIS01	DIS01-02	Percentage	5	62.38	1,185.27
Always apply	40	DIS02	DIS02-02	Amount	8	94.82	1,090.45
Always apply	50	DIS 03	DIS03-02	Amount	10	10.00	1,080.45

Last updated on 05/07/2025

Resolve concurrency across price component codes

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Concurrency rules let you define what happens if multiple pricing rules apply to the same order and/or order line. You can set up concurrency rules to control whether the customer receives just one of the matching rules (and if so, which one they receive), or whether the rules are combined (and if so, how they're combined). There are two types of concurrency:

- **Across-price-component-code concurrency** – This type of concurrency controls how the different price component codes that are included in a price structure combine with each other. A price structure defines a collection of different types of price component codes (including base price, sales agreement price, margin components, discounts, and/or charges), the order that the codes are calculated in, and the way that the price adjustments are calculated and combined to determine the final price.
- **Within-price-component-code concurrency** – Any number of pricing rules can be associated with each price component code that's included in a price structure. This principle is especially true of discounts. This type of concurrency occurs when an order or order line qualifies for more than one pricing rule that's associated with the same component code. For example, if you define a price component code that's named *Seasonal promotion events*, and you associate multiple discount rules with it, multiple pricing rules might apply to the same order line. Therefore, in the *Seasonal promotion events* price component code, you'll set up concurrency rules to control whether the customer receives just one of the matching discounts (and if so, which one they receive), or whether the discounts are combined (and if so, how they're combined).

This article explains how to manage *across-price-component-code concurrency*. For information about how to manage *within-price-component-code concurrency*, see Resolve concurrency within price component codes.

Set across-price-component-code concurrency options for each price structure

Concurrency across price components codes affects only discounts and margin price adjustments. You configure the setting at the price structure level, by using the

Concurrency mode across priority field on the **Price trees** page. The following values are available:

- *Compounded* – The system first computes the price adjustment or discount for each line in the price structure. It then calculates the final discount or adjustment by adding the value of every line. This value is available only for lines where the **Price component** field is set to *Discounts* or *Margin component*. (For margin components, this value is the only one that's available.)
- *Best price* – If the price structure includes more than one price component code that has this value, only one of the lines (the one that has the largest discount) will contribute to the final unit price. This value is available only for lines where the **Price component** field is set to *Discounts*.

The **Concurrency mode across priority** setting affects discount calculations only when the *Best price and compound within priority, best price and compound across priority* option is selected under **Best price and compound concurrency control model** on the **Pricing management parameters** page. Learn more in the System settings for resolving discount concurrency across price component codes section of this article.

As was mentioned, margin components are always compounded across components (in other words, the **Concurrency mode across priority** field is always set to *Compounded* for margin components in your price structures). However, you must still set up each price structure to control whether each adjustment is calculated by using the base price or the compound price (that is, the running total). The option that you choose can have a significant effect on your final prices for margin price adjustments that are calculated as a percentage. For each margin component in your price structures, you can set the option by using the **Compound** checkbox. For an example that shows how the setting of this checkbox affects your final prices, see the Example price calculation using a mix of compound and non-compound margin components section of this article.

For information about how to set up price structures, see Arrange price component codes into a price structure.

System settings for resolving discount concurrency across price component codes

A setting on the **Pricing management parameters** page affects the way that concurrent discounts are calculated. The calculation method can, in turn, affect the way that some of your price structure settings work as they apply to discounts. Follow these steps to set up the calculation.

1. Go to **Pricing management** > **Setup** > **Pricing management parameters**.

2. On the **Prices and discounts** tab, on the **Discount concurrency control** FastTab, under **Best price and compound concurrency control model**, select one of the following options:

    - *Best price and compound within priority, never compound across priorities*
    - *Best price only within priority, always compound across priorities*
    - *Best price and compound within priority, best price and compound across priority*

Example: Price calculation that uses a mix of compound and non-compound margin components

If your price structure includes multiple margin and/or discount components, the order that the components are calculated in and the method that's used to combine them can significantly affect the final price.

The following table shows an example of a single price structure setup and its resulting calculations.

Expand table

Price component code	Description	Pricing sequence	Calculation method	Value	Compounded	Calculated line value	New unit price
Base	Base price - inventory Cost	10				$1,000.00	$1,000.00
MC01	Ullage	20	Percent	5	No	$50.00	$1,050.00
MC02	Freight	30	Percent	-2	Yes	-$21.00	$1,029.00
MC03	Group Margin price adjustment	40	Amount	10	No	$10.00	$1,039.00
MC04	Contract Margin price adjustment	50	Percent	5	Yes	$51.95	$1,090.95
MC05	Pickup fee	60	Amount	2	Yes	$2.00	$1,092.95

Price component code	Description	Pricing sequence	Calculation method	Value	Compounded	Calculated line value	New unit price
MC06	Other cost to sell	70	Percent	5	Yes	$54.65	$1,147.60

The system applies the following rules to calculate the final unit price:

- A new unit price is calculated for each line, according to the pricing sequence.
- For non-compounded lines, the line value is calculated based on the original base price.
- For compound lines, the line value is calculated based on the previous line's new unit price.
- Each line's new unit price is found by adding its calculated line value to the new unit price from the previous line.
- Calculation continues until all lines have been processed. The result is the final unit price.

Example: Price calculation that uses a mix of compound and best-price discount components

The following table shows an example of a price structure that uses a mix of best-price and compounded components.

Expand table

Price component code	Price sequence	Concurrency mode across priority	Price component	Computed line value
MAC01	10	Compounded	Margin component	$50
MAC02	20	Compounded	Margin component	$20
DIS 01	30	Best price	Discounts	$10
DIS 02	40	Best price	Discounts	**$20**
DIS 03	50	Compounded	Discounts	$30

In addition, on the **Pricing management parameters** page, the *Best price and compound within priority, best price and compound across priority* option under **Best price and compound concurrency control model** is selected. In the name of this option, *priority* refers to each price component code in the sequence. (This setting affects the way that the discount lines are interpreted.) Therefore, the following unit price modifications apply:

- The total price adjustment is the sum of the margin component lines: **$70**.
- The best price (largest discount) for the best-price lines is $20. The compounded discount is then added to this price. Therefore, the total discount is **$50**.

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Calculate prices and discounts without creating sales orders

10/24/2025

Unified pricing management in Dynamics 365 Supply Chain Management can calculate customer and volume-specific prices and discounts without requiring you to create a sales order. This capability is useful both when creating quotes and when integrating with external selling systems where you need to calculate prices without creating a sales order. To trigger pricing and discount calculations programmatically, you just need to provide product, quantity, and customer details.

This feature exposes key calculation methods and provides sample code that you can use to implement your own custom pricing logic. If you don't have a Commerce Scale Unit (CSU) API license but still want to integrate pricing with third-party systems, use this capability to build custom services to expose pricing via APIs. You can leverage this extensibility to connect pricing seamlessly with external applications.

By carefully preparing transaction data, managing attributes consistently, and leveraging the pricing engine's capabilities, you can achieve accurate, dynamic pricing tailored to your business needs.

This article demonstrates a practical method for integrating external systems with the Unified pricing management pricing engine by using custom logic without requiring a Commerce Scale Unit. This approach enables flexible, attribute-driven pricing logic that you can embed into your applications.

Learn more about custom service development in Custom service development.

Step 1: Gather input data

The first step in the pricing workflow is to collect all relevant input data required for accurate and context-aware pricing calculations. These inputs serve as the foundation for building the transaction and determining which pricing rules apply. You might get these inputs from various sources such as user input, query strings in URLs, external APIs, or other integrated systems. Gathering input data ensures that the pricing engine receives all necessary context to evaluate and apply the correct pricing logic. Missing or incomplete inputs can lead to inaccurate pricing or skipped discount rules.

The following list outlines the most common data elements used in pricing scenarios. Use these inputs to populate transaction objects and influence pricing calculations.

- **Item numbers** – Unique identifiers for the products that customers purchase. These identifiers are mandatory.
- **Customer account numbers** *(optional)* – Identify the buyer and apply customer-specific pricing, discounts, or loyalty-based rules.
- **Quantities** *(optional)* – Specifies the number of units per item. Quantity can affect volume-based pricing or tiered discount structures.
- **Attributes** *(optional)* – Custom fields that provide additional context, such as:
  - **Site** – Indicates the location or warehouse, which might influence regional pricing.
  - **Channel ID** – Helps apply channel-specific pricing rules.
  - **Delivery mode**, **Tax group**, or **Promotion code** – Other optional fields that might trigger specific pricing logic.

X++

```
// Sample Code: Extracting Input Parameters from URL
// This X++ snippet demonstrates how to retrieve parameter values for item numbers
and Customer account numbers from the URL input. This is typically used in
runnable classes.
str itemid = urlUtil.getQueryParamValue('itemid');
str custAccount = urlUtil.getQueryParamValue('custAccount');
```

Step 2: Prepare transaction objects

After you collect the input data, construct structured transaction objects that represent the sales order and its individual lines. These objects serve as the core data model for the pricing engine. They're essential for passing all relevant information—such as item details, quantities, customer context, and unit of measure—into the pricing logic. This step transforms raw input parameters into structured Commerce Runtime (CRT) objects that the pricing engine can process. These objects encapsulate the full context of the transaction, enabling accurate and rule-based price calculations.

The pricing engine expects the transaction to be represented by two primary object types:

- **Sales transaction** – A container for the overall order. It includes customer information, currency, channel context, and a collection of sales lines.
- **Sales lines** – Each line represents a single item in the order. It includes item-specific data such as product ID, quantity, unit of measure, and line identifiers.

The following X++ code demonstrates how to initialize a sales transaction and populate it with one or more sales lines by using the input data. You need this setup before invoking the pricing engine.

X++

```
// This code initializes the transaction and populates it with sales lines using
the input data. It sets up the necessary fields for each item, preparing the data
for pricing calculations.
using Microsoft.Dynamics.Commerce.Runtime.Services.PricingEngine;
using CrtSalesTransaction =
Microsoft.Dynamics.Commerce.Runtime.DataModel.SalesTransaction;
using CrtSalesLine = Microsoft.Dynamics.Commerce.Runtime.DataModel.SalesLine;
using CrtPriceAndDiscountCalculationParameters =
Microsoft.Dynamics.Commerce.Runtime.Services.PricingEngine.PriceAndDiscountCalcula
tionParameters;
int salesLinesOrderField = 1;
str lineId = strRFix(int2str(salesLinesOrderField), 3, '0');
InventTable invTbl = InventTable::find(iItemId);
CrtSalesTransaction crtSalesTransaction = new CrtSalesTransaction();
crtSalesTransaction.set_CustomerId(custAccount);
CLRObject clrSalesLines = crtSalesTransaction.get_SalesLines();
CrtSalesLine crtSalesLine = new CrtSalesLine();
crtSalesLine.set_ItemId(itemId);
crtSalesLine.set_Quantity(qty);
crtSalesLine.set_SalesOrderUnitOfMeasure(invTbl.inventTableModuleSales().UnitId);
crtSalesLine.set_LineId(lineId);
crtSalesLine.set_LineNumber(salesLinesOrderField);
clrSalesLines.Add(crtSalesLine);
```

Step 3: Handle pricing attributes

To ensure accurate and context-aware pricing, the pricing engine relies on attribute-based logic. These attributes can exist at both the header level (order context) and the line level (item context). Calculating hash keys for these attributes is essential, as they serve as identifiers that help the pricing engine match the correct pricing rules. It prepares the pricing context by generating and storing attribute hash keys, enabling the pricing engine to match and apply the correct pricing rules based on attributes. This step enables the pricing engine to apply differentiated pricing rules based on custom attributes such as site, channel, product category, or customer type. By generating hash keys, you create a consistent and secure way to reference attribute values during pricing evaluation.

The following X++ sample code demonstrates how to process pricing attributes and generate hash keys for both the transaction header and each sales line that the pricing engine uses to evaluate applicable pricing rules.

X++

```
// Put the attribute value into the corresponding field in the temporary sales
table. The sales table record don't need to be inserted into the database
SalesTable salesTable;
salesTable.CustAccount = custTable.AccountNum; // Put the customer account to make
sure the customer attribute can be matched
```

```
SalesTable.InventSiteId = 'CENTRAL'; // Put the sales table attribute if that is
provided in parameter
RetailTempOrderItem tempOrderItem;

// Put the attribute value into the corresponding field in the temporary sales
line. The sale lines record don't need to be inserted into the database
SalesLine salesLine;
salesLine.ItemId = itemId; // Put the itemid to make sure the product attribute
can be matched
salesLine.InventTransId = lineId;
Set headerHashSet = new Set(Types::String);
Map lineHashMap = new Map(Types::String, Types::Class);
Set lineHashSet = new Set(Types::String);
Set hashKeySet = GUPHashHelper::getPropertyHashKeySetForPricingObject(salesTable,
GUPPricingAttributeSourceLevel::Header);
headerHashSet = GUPHashHelper::calculateHashValueForHashKeySet(hashKeySet);
hashKeySet = GUPHashHelper::getPropertyHashKeySetForPricingObject(salesLine,
GUPPricingAttributeSourceLevel::Line);
Set lineHashSetOneLine =
GUPHashHelper::calculateHashValueForHashKeySet(hashKeySet);
lineHashSet = Set::union(lineHashSet, lineHashSetOneLine);
lineHashMap.add(salesLine.InventTransId, lineHashSetOneLine);
ttsbegin;
tempOrderItem.itemId = itemId;
tempOrderItem.LineNum = salesLinesOrderField;
tempOrderItem.UnitOfMeasure =
UnitOfMeasure::findBySymbol(invTbl.inventTableModuleSales().UnitId).RecId;
tempOrderItem.insert();

// Create pricing manager
GUPRetailPricingDataManagerV4 gupPricingManager =
GUPRetailPricingDataManager::constructDefault(0, transId, tempOrderItem, true);
gupPricingManager.parmHeaderHashSet(headerHashSet);
gupPricingManager.parmLineHashMap(lineHashMap);

/* For version 10.0.45.
GUPPropertyHashTmp orderAttributeValueHeaderHash, orderAttributeValueLineHash;
GUPHashHelper::insertHashTempDB(orderAttributeValueHeaderHash, headerHashSet);
GUPHashHelper::insertHashTempDB(orderAttributeValueLineHash, lineHashSet);
gupPricingManager.parmOrderAttributeValueHeaderHash(orderAttributeValueHeaderHash)
;
gupPricingManager.parmOrderAttributeValueLineHash(orderAttributeValueLineHash);
*/

// For version 10.0.46.
gupPricingManager.parmLineHashSet(headerHashSet);
gupPricingManager.processHash();

ttscommit;
```

If your input attribute values include custom attributes (such as a customer's loyalty tier), use specialized helper methods to generate, hash, and register these attributes before passing them to the pricing engine. These attributes aren't part of the standard transaction schema, so

you need to explicitly handle them to ensure the pricing logic recognizes and evaluates them correctly.

To support these scenarios, you must:

- Identify the relevant attributes at either the header (order-level) or line (item-level).
- Map them to pricing attribute definitions by using `GUPPricingAttributeLink`.
- Generate hash keys by using `GUPHashHelper::getPropertyHashKeySetForPricingObject()`.

X++

```
Map salesTableAttributeValues = new Map(Types::String, Types::AnyType);
GUPPricingAttributeLink attrLink; // find the attribute that need to be specified
salesTableAttributeValues.add(attrLink.pricingAttribute().GetHashString(),
&lt;particular attribute value&gt;);
Set hashKeySet = GUPHashHelper::getPropertyHashKeySetForPricingObject(salesTable,
GUPPricingAttributeSourceLevel::Header, null, null, null,
salesTableAttributeValues);
Map salesLineAttributeValues = new Map(Types::String, Types::AnyType);
GUPPricingAttributeLink attrLink; // find the attribute that need to be specified

// Put the sale table attribute value on the map
salesLineAttributeValues.add(attrLink.pricingAttribute().GetHashString(),
&lt;particular attribute value&gt;);
Set hashKeySet = GUPHashHelper::getPropertyHashKeySetForPricingObject(salesLine,
GUPPricingAttributeSourceLevel::Line, null, salesLineAttributeValues);
```

Step 4: Specify price tree name

If your pricing logic depends on a specific pricing structure with different price components, you must define the price tree name. This configuration determines which pricing tree the engine uses to evaluate and apply pricing rules. It allows for differentiated pricing structures across scenarios. The price tree acts as a logical framework that organizes pricing rules. By specifying the correct tree, you ensure that the pricing engine applies the appropriate pricing component structure.

- **Set the Price Tree Name from input parameters** – Use a predefined value passed into your service or class to assign the price tree name directly. Use this approach when the pricing structure is known and consistent across transactions.
- **Derive the Price Tree Name dynamically based on input attributes** – In more flexible scenarios, you can compute the tree name at runtime by using product or customer attributes. For example, you might derive the tree name from an input attribute value by using code such as `unifiedPricingParameter.PriceTreeName = GUPPricingTree::getPriceTreeByEcoResValue(<EcoResValue>).Name;`.

This sample code demonstrates how to set a static tree name directly.

```
X++
```

```
Microsoft.Dynamics.Commerce.Runtime.DataModel.UnifiedPricingParameter
unifiedPricingParameter = gupPricingManager.GetGlobalUnifiedPricingParameter();
unifiedPricingParameter.PriceTreeName = 'Inheritance';
```

Step 5: Invoke pricing engine methods

With all input data, transaction objects, and pricing attributes prepared, the next step is to invoke the pricing engine to calculate prices and discounts. This step is the core execution phase of the pricing workflow. The engine applies business rules, evaluates attribute-based conditions, and returns computed pricing results. This step activates the pricing logic by passing the prepared transaction and attribute data to the pricing engine. It ensures that all relevant pricing rules—such as discounts, price groups, and attribute-based conditions—are applied consistently and accurately.

The pricing engine processes the transaction object, which includes sales lines, customer information, and pricing attributes. It evaluates applicable pricing rules based on the configured price tree, attribute hash keys, and business logic. The engine returns calculated unit prices and discount amounts for each sales line.

The following X++ code demonstrates how to invoke the pricing engine methods to calculate prices and discounts:

```
X++
```

```
RetailCurrencyOperations currencyAndRoundingHelper = new
RetailCurrencyOperations(CompanyInfoHelper::standardCurrency());
System.DateTimeOffset dateTimeOffset =
RetailPricingEngineHelper::getSessionDateTimeInChannelTimeZone(0, today());
CrtPriceAndDiscountCalculationParameters calculationParameters = new
CrtPriceAndDiscountCalculationParameters();
CurrencyCode transactionCurrency = 'USD';

// Calculate price
PricingEngine::CalculatePricesForTransaction(
    crtSalesTransaction,
    gupPricingManager,
    currencyAndRoundingHelper,
    custTable.PriceGroup,
    transactionCurrency,
    dateTimeOffset);

// Calculate discount
PricingEngine::CalculateDiscountsForLines(
```

```
    gupPricingManager,
    crtSalesTransaction,
    currencyAndRoundingHelper,
    transactionCurrency,
    custTable.LineDisc,
    custTable.MultiLineDisc,
    custTable.EndDisc,
    false, // calculateSimpleDiscountOnly
    dateTimeOffset,
    calculationParameters);
```

Step 6: Retrieve and display results

After the pricing engine completes its calculations, the final step is to extract and utilize the computed prices and discounts from the transaction object. This step transforms the pricing engine's output into actionable data that you can present to users or integrate into downstream business processes such as order creation or analytics.

The pricing engine populates each sales line in the transaction object with calculated values. These values include the final unit price, discount amount, and potentially other pricing-related metadata. You can iterate through the sales lines to extract these values and format them for display or further processing. This step is critical for ensuring that the pricing logic is correctly applied and that the results are visible and usable. The following X++ code demonstrates how to loop through each sales line in the transaction and retrieve the final price and discount amount. Typically, you place this code at the end of your pricing service or runnable class, after the pricing engine has been invoked.

X++

```
clrSalesLines = crtSalesTransaction.get_SalesLines();
CLRObject enumeratorSalesLine = clrSalesLines.GetEnumerator();
while (enumeratorSalesLine.MoveNext())
{
    crtSalesLine = enumeratorSalesLine.get_Current();
    Price price = crtSalesLine.get_Price();
    Price discount = crtSalesLine.get_DiscountAmount();
}
```

Calculate prices for external systems through the pricing calculation API

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The pricing calculation API enables external applications to retrieve accurate, real-time pricing and discount calculation results directly from Microsoft Dynamics 365 Supply Chain Management. By providing key input data, such as product and customer details, external systems can programmatically access calculated prices without creating sales orders. This approach helps ensure pricing consistency across your sales channels and simplifies quoting and integration workflows.

> **Important**
>
> This API is designed for low- to moderate-frequency pricing queries from system-to-system integrations. It isn't intended to replace the Dynamics 365 Commerce Scale Unit (CSU) pricing APIs. For high-scale, high-performance pricing scenarios—such as e-commerce storefronts, point-of-sale (POS), or product catalog browsing—Commerce Scale Unit APIs remain the recommended approach. Learn more at Commerce pricing APIs.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.47 or later.
- Unified pricing management must be enabled in your environment.

Business scenarios

This API is a good fit when external systems need to validate or retrieve prices on demand, and the request volume is low or moderate. Typical scenarios include:

- **Quoting or contract validation tools** – Retrieve calculated prices to validate quotes against Supply Chain Management pricing rules.
- **Partner or ISV integrations** – Allow third-party solutions to access pricing logic without direct access to the Supply Chain Management UI.
- **Backend workflow services** – Trigger pricing lookups from automated workflows or batch processes.
- **Internal applications** – Give internal tools pricing insight without requiring order creation.

When to use Commerce Scale Unit pricing APIs instead

For high-performance pricing workloads, use the Dynamics 365 Commerce Scale Unit (CSU) pricing APIs instead of the pricing calculation API. CSU pricing APIs are purpose-built for scenarios that include:

- E-commerce and web shopping experiences
- Product detail page (PDP) pricing
- Cart and basket pricing
- Anonymous and authenticated consumer scenarios
- High concurrency and scalability requirements

If your scenario involves online storefronts, high-traffic B2B or B2C commerce, frequent or burst pricing requests, or customer-facing digital experiences, use Commerce pricing APIs instead.

Functional scope and limitations

The following table summarizes what the pricing calculation API supports and doesn't support.

Expand table

Supported	Not supported
Single-line price calculation per product	Multiline or cart-level discounts
Simple discounts	Basket pricing or promotion concurrency
Supply Chain Management pricing rules and pricing attributes	High-frequency or high-volume pricing calls
Quantity-based pricing (default quantity is 1)	E-commerce or storefront scenarios
Variant price ranges for product masters	

Authentication

The pricing calculation API uses OAuth 2.0 with Microsoft Entra ID (formerly Azure Active Directory). To authenticate, you need the following information:

- **Directory ID** (Tenant ID) – The identifier for your Microsoft Entra tenant.
- **Client ID** – The application (client) ID of your registered app.
- **Client Secret** – A secret generated for your registered app.
- **Scope** – Set to `{D365_URL}/.default`, where `{D365_URL}` is the base URL of your Supply Chain Management environment.

Register the external application

Before you can call the API, register your external application in both Microsoft Entra ID and Supply Chain Management. The following procedure describes how to set up the registration.

1. Go to the Azure portal and sign in with an account that has permissions to manage app registrations.

2. Go to **Microsoft Entra ID** > **App registrations** and either select an existing registration or create a new one. Note the **Application (client) ID** value.

3. Under the same app registration, go to **Certificates & secrets** and select **New client secret**. Enter a description, set the expiration period, and then select **Add**.

   > **Important**
   >
   > Copy the secret value immediately after it's generated. The value is only shown once and can't be retrieved later.

4. Sign in to Supply Chain Management and go to **System administration** > **Setup** > **Microsoft Entra ID applications**.

5. Select **New** to add a record to the grid. Then set the following fields for it:

   - **Client ID** – Enter the application (client) ID that you noted earlier.
   - **Name** – Enter a descriptive name for the application.
   - **User ID** – Select a user account that has appropriate permissions.

6. Select **Save**.

The following resources provide more information about how to register an application in Microsoft Entra ID:

- For instructions that show how to use Windows PowerShell to register an application in Microsoft Entra ID, see Use Azure PowerShell to create a service principal with a certificate.
- For complete details about how to manually register an application in Microsoft Entra ID, see the following articles:
  - Register an application in Microsoft Entra ID
  - Register a Microsoft Entra app and create a service principal

API endpoint

Send pricing requests to the following endpoint by using an HTTP POST method:

HTTP

```
POST
{D365_URL}/api/services/GUPPricingServiceGroup/GUPPricingService/getActivePrices
```

Replace `{D365_URL}` with the base URL of your Supply Chain Management environment.

Request structure

Wrap all input parameters inside a single request object. Parameter names are case-sensitive.

Request parameters

The following table describes the top-level parameters that you can include in the request body.

Expand table

Name	Type	Required/Optional	Description
`dataAreaId`	string	Required if `ChannelId` isn't provided	The company identifier (for example, `"usrt"`).
`activeDate`	DateTimeOffset	Optional (defaults to the current system date and time)	The date when prices and discounts are calculated.
`productIds`	array of long	Required if `LineContexts` isn't provided	A list of product record IDs to calculate prices for.

Name	Type	Required/Optional	Description
`priceLookupContext`	object	Required if `productIds` isn't provided	An object that provides additional context for filtering prices and discounts during lookup.
`calculateSimpleDiscountOnly`	boolean	Optional (defaults to `true`)	If set to `true`, only simple discounts are calculated. If set to `false`, the input is treated as a transaction for full calculation.
`includeVariantPriceRange`	boolean	Optional (defaults to `false`)	If set to `true`, the response returns the minimum and maximum prices among all variants for a product master.

> **Note**
>
> You must provide either `productIds` or `priceLookupContext` with `LineContexts` in each request, but don't provide both in the same request.

PriceLookupContext structure

The `priceLookupContext` parameter contains a `HeaderContext` object and an optional `LineContexts` array. Together, these elements provide the pricing engine with the information it needs to calculate accurate prices.

HeaderContext

The `HeaderContext` object provides order-level details such as the customer, channel, and inventory dimensions. The following table describes the fields that make up this object.

Expand table

Name	Type	Required/Optional	Description
`CustomerAccount`	string	Optional	The customer account number.
`ChannelId`	long	Required if `dataAreaId` isn't provided	The record ID of the channel where prices and discounts are calculated.

Name	Type	Required/Optional	Description
			Either `dataAreaId` or `ChannelId` must be provided.
`InventorySiteId`	string	Optional	The inventory site. If you don't specify this value, the system takes the default from the channel. If the channel isn't provided, the system takes the default from the customer account. To override defaults and leave the value blank, specify an empty string (`""`).
`InventoryLocationId`	string	Optional	The inventory location (warehouse). The same defaulting logic applies as for `InventorySiteId`.
`AffiliationLoyaltyTierLines`	array of objects	Optional	An array of affiliation and loyalty tier IDs. Each object contains an `AffiliationId` (long) and a `LoyaltyTierId` (long).
`SalesOrderProperties`	array of objects	Optional	Key-value pairs for additional order-level attributes. Each object contains a `Name` (string), an optional `TypeName` (string), and a `Value` (string).

> **Tip**
>
> The system automatically applies affiliation and loyalty tier information if the customer account links to the affiliation or loyalty tier. You don't need to include this information explicitly in the request unless you want to specify additional affiliations.

The following example shows a `HeaderContext` object that specifies a customer, channel, inventory dimensions, affiliation and loyalty tier data, and sales order properties.

JSON

```
"HeaderContext": {
    "CustomerAccount": "2001",
    "ChannelId": 5637144592,
    "InventorySiteId": "CENTRAL",
    "InventoryLocationId": "DC-CENTRAL",
    "AffiliationLoyaltyTierLines": [
        {
            "AffiliationId": 5637144579,
            "LoyaltyTierId": 5637144578
        },
```

```
        {
            "AffiliationId": 5637144577,
            "LoyaltyTierId": 0
        }
    ],
    "SalesOrderProperties": [
        {
            "Name": "order type",
            "Value": "3"
        },
        {
            "Name": "AW Material",
            "TypeName": "AW Text Type",
            "Value": "aluminium"
        }
    ]
}
```

LineContexts

The `LineContexts` array provides line-level details for each product. Use `LineContexts` when you need to specify attributes such as quantity, unit of measure, or delivery mode for individual products.

> **Important**
>
> Don't provide both `productIds` and `LineContexts` in the same request. Doing so can lead to conflicts or unexpected results.

The following table describes the fields available for each object in the `LineContexts` array.

Expand table

Name	Type	Required/Optional	Description
`ProductRecordId`	long	Required	The unique record ID of the product.
`UnitOfMeasureSymbol`	string	Optional	The unit of measure for the product (for example, `"ea"` or `"kg"`). If omitted, the product's base unit is used.
`InventorySiteId`	string	Optional	The inventory site for this line. The same defaulting logic applies as described for `HeaderContext`.
`InventoryLocationId`	string	Optional	The inventory location for this line. The same defaulting logic applies as described for

Name	Type	Required/Optional	Description
			`HeaderContext`.
`DeliveryMode`	string	Optional	The delivery mode for the sales line (for example, `"99"` for a specific mode).
`Quantity`	real	Optional (defaults to 1)	The quantity of the product.
`SalesLineProperties`	array of objects	Optional	Key-value pairs for additional line-level attributes. Each object contains a `Name` (string), an optional `TypeName` (string), and a `Value` (string).

The following example shows a `LineContexts` array with one product line that specifies a quantity, delivery mode, and a custom line property.

JSON

```
"LineContexts": [
    {
        "ProductRecordId": 22565421974,
        "UnitOfMeasureSymbol": "ea",
        "InventorySiteId": "",
        "InventoryLocationId": "",
        "DeliveryMode": "99",
        "Quantity": 3,
        "SalesLineProperties": [
            {
                "Name": "aw Material",
                "Value": "brass"
            }
        ]
    }
]
```

Sample request

The following example shows a complete request that uses `priceLookupContext` with both a `HeaderContext` and `LineContexts` array.

JSON

```
{
    "request": {
        "dataAreaId": "usrt",
        "activeDate": "2025-11-06T14:25:26.847-08:00",
        "priceLookupContext": {
```

```
            "HeaderContext": {
                "CustomerAccount": "004009",
                "ChannelId": 5637145360,
                "InventorySiteId": "CENTRAL",
                "InventoryLocationId": "DC-CENTRAL",
                "SalesOrderProperties": [
                    {
                        "Name": "order type",
                        "Value": "3"
                    }
                ]
            },
            "LineContexts": [
                {
                    "ProductRecordId": 22565421974,
                    "UnitOfMeasureSymbol": "ea",
                    "Quantity": 3,
                    "SalesLineProperties": [
                        {
                            "Name": "aw Material",
                            "Value": "brass"
                        }
                    ]
                }
            ]
        },
        "calculateSimpleDiscountOnly": true,
        "includeVariantPriceRange": false
    }
}
```

Response structure

The API returns a JSON response that contains transaction-level information and an array of item-level pricing results. Each item result includes the calculated price, discount details, and price source information.

Key response fields

The following table describes the key fields that the API returns for each item in the `ItemResults` array.

Expand table

Field	Type	Description
`ProductId`	long	The record ID of the product.

Field	Type	Description
`ItemId`	string	The item number of the product.
`BasePrice`	decimal	The base price of the product as defined in the product master.
`TradeAgreementPrice`	decimal	The price determined by applicable trade agreements.
`AdjustedPrice`	decimal	The price after any price adjustments are applied.
`CustomerContextualPrice`	decimal	The final calculated price for the customer, including applicable discounts.
`DiscountAmount`	decimal	The total discount amount applied to the product.
`CurrencyCode`	string	The currency code for the calculated price.
`UnitOfMeasure`	string	The unit of measure for the product.
`Quantity`	decimal	The quantity used in the calculation.
`MaxVariantPrice`	decimal	The maximum price among all variants (returned when `includeVariantPriceRange` is `true`).
`MinVariantPrice`	decimal	The minimum price among all variants (returned when `includeVariantPriceRange` is `true`).
`AttainablePriceLines`	array	An array of price lines that shows how the final price was determined, including the price method and origin.
`DiscountLines`	array	An array of discount lines that shows the discounts applied, including the offer name, percentage, and effective amount.

Sample response

The following example shows a response for a single product that has a trade agreement price of 30.00 USD and a 25% simple discount applied.

JSON


```
{
    "$id": "1",
    "Transaction": {
        "$id": "2",
        "InventorySiteId": "",
        "InventoryLocationId": "",
        "LoyaltyCardId": ""
    },
    "ItemResults": [
        {
```

```
            "$id": "3",
            "ProductId": 22565421967,
            "ListingId": 0,
            "BasePrice": 7.99,
            "TradeAgreementPrice": 30.0,
            "AdjustedPrice": 30.0,
            "MaxVariantPrice": 0.0,
            "MinVariantPrice": 0.0,
            "CustomerContextualPrice": 22.500,
            "DiscountAmount": 7.500,
            "CurrencyCode": "USD",
            "ItemId": "0005",
            "InventoryDimensionId": "AllBlank",
            "UnitOfMeasure": "Ea",
            "ValidFrom": "2025-12-13T01:07:43Z",
            "ProductLookupId": 0,
            "ChannelId": 0,
            "CatalogId": 0,
            "SalesAgreementPrice": 0.0,
            "PriceSourceTypeValue": 2,
            "DeliveryMode": "",
            "Quantity": 1.0,
            "SalesLineProperties": null,
            "AttainablePriceLines": [
                {
                    "$id": "4",
                    "RecordId": 0,
                    "Value": 7.99,
                    "PriceMethod": 1,
                    "OriginId": "0005",
                    "PriceChangedByExtensions": false,
                    "SaleLineNumber": 0.0,
                    "ExtensionProperties": []
                },
                {
                    "$id": "5",
                    "CanApplyPriceAdjustments": false,
                    "PricePreventDiscount": false,
                    "PricingPriorityNumber": 998,
                    "RecordId": 68719503366,
                    "Value": 30.0,
                    "PriceMethod": 1,
                    "OriginId": "68719503366",
                    "PriceChangedByExtensions": false,
                    "SaleLineNumber": 0.0,
                    "ExtensionProperties": []
                }
            ],
            "DiscountLines": [
                {
                    "$id": "6",
                    "SaleLineNumber": 1.0,
                    "OfferId": "ST100201",
                    "OfferName": "Grand Opening",
                    "OfferDescription": "",
```

```
                    "Amount": 0.0,
                    "DiscountCost": 0.0,
                    "EffectiveAmount": 7.500,
                    "EffectivePercentage": 25.00,
                    "LineNumber": 1.0,
                    "RecordId": 0,
                    "SavedEffectiveAmount": null,
                    "Percentage": 25.0,
                    "DealPrice": 0.0,
                    "DiscountLineTypeValue": 2,
                    "ManualDiscountTypeValue": 0,
                    "CustomerDiscountTypeValue": 0,
                    "PeriodicDiscountTypeValue": 2,
                    "DiscountApplicationGroup": "0005",
                    "ConcurrencyModeValue": 4,
                    "IsCompoundable": true,
                    "DiscountCode": "",
                    "PricingPriorityNumber": 1994,
                    "PricingAttributeCombinationPriority": 1.0,
                    "IsDiscountCodeRequired": false,
                    "ThresholdAmountRequired": 0.0,
                    "BundleId": 0,
                    "ValidFrom": "1900-01-01T08:00:00+00:00",
                    "ValidTo": "2154-12-31T08:00:00+00:00",
                    "ExtensionProperties": []
                }
            ]
        }
    ],
    "FreeItemLines": null
}
```


Understand the sample pricing calculation

The response provides several price-related values that help you understand how the final customer price was determined. The following walkthrough explains the relationship between these values using the sample response shown earlier.

1. **Base price** – The system starts with the product's base price of $7.99, which comes from the released product definition.
2. **Trade agreement price** – Because a trade agreement applies to this product, the system evaluates it and determines a trade agreement price of $30.00.
3. **Adjusted price** – No price adjustments apply in this case, so the adjusted price remains $30.00.
4. **Discount calculation** – The system identifies an applicable simple discount ("Grand Opening") that provides a 25% discount. The effective discount amount is $7.50.
5. **Customer contextual price** – The final price for the customer is calculated as $30.00 – $7.50 = $22.50. This value is returned in the `CustomerContextualPrice` field.

Best practices

Keep the following best practices in mind when you work with the pricing calculation API:

- **Case sensitivity** – All parameter names are case-sensitive. Make sure you use the exact casing shown in this article when you construct requests.
- **Site and location defaults** – If `InventorySiteId` or `InventoryLocationId` isn't specified, the system takes defaults from the channel configuration. If the channel isn't provided, the system takes defaults from the customer account. To override defaults and leave the values blank, specify empty strings (`""`).
- **Choosing between productIds and LineContexts** – Use `LineContexts` for detailed line-level pricing scenarios, such as when you need to specify multiple products, attributes, or specific quantities. Don't mix `LineContexts` with `productIds` in the same request.
- **Affiliation and loyalty tier** – The system automatically applies affiliation and loyalty tier information if it's linked to the customer account. You don't need to include it explicitly in the request unless you want to specify additional affiliations.
- **Multiple attribute values** – The `SalesOrderProperties` and `SalesLineProperties` fields don't support multiple values for the same attribute in a single request. If you need results for different attribute values, split them into separate requests.

Extend the API

If you need to add custom parameters to the input or output of the pricing calculation API, use the extension points available in the `GUPPricingServiceClass` class. You can extend the following methods:

- `initHeaderPricingObjectHash` – Extend to add custom logic when the header-level pricing object is initialized.
- `initLinePricingObjectHash` – Extend to add custom logic when the line-level pricing object is initialized.
- `setProductPrice` – Extend to add custom fields or logic to the product price output.

Extension example

The following example shows how to extend the `initLinePricingObjectHash` method to read a custom property from the `SalesLineProperties` array and apply custom logic.

X++

```
using CrtSalesLine = Microsoft.Dynamics.Commerce.Runtime.DataModel.SalesLine;
```

```
[ExtensionOf(classStr(GUPPricingServiceClass))]
final class GUPPricingServiceClass_Extension
{
    protected void initLinePricingObjectHash(
        CrtSalesLine _crtSalesLine,
        SalesLine _pricingObject,
        Map _salesLineAttributeValues,
        GUPAPIPriceLookupLineContextContract _lineContext)
    {
        if (_lineContext.parmSalesLineProperties())
        {
            ListEnumerator enumerator =
_lineContext.parmSalesLineProperties().getEnumerator();
            str extensionInputName, extensionInputValue;
            while (enumerator.moveNext())
            {
                Newtonsoft.Json.Linq.JObject jsonObject = enumerator.current();
                extensionInputName = jsonObject.SelectToken('ExtensionInput')
                    ? jsonObject.SelectToken('ExtensionInput').ToString()
                    : '';
                extensionInputValue = jsonObject.SelectToken('ExtensionInputValue')
                    ? jsonObject.SelectToken('ExtensionInputValue').ToString()
                    : '';
                if (extensionInputName == "Test")
                {
                    // Implement logic to assign the value to the SalesLine
                    // or set the appropriate attribute.
                }
            }
        }
        next initLinePricingObjectHash(
            _crtSalesLine,
            _pricingObject,
            _salesLineAttributeValues,
            _lineContext);
    }
}
```

> **Note**
>
> The extension class must use the `ExtensionOf` attribute and call `next` to ensure that the standard logic continues to execute after your custom logic.

Related information

- Commerce pricing APIs
- Unified pricing management module overview

Last updated on 03/27/2026

Pricing data import and export entities

Microsoft Dynamics 365 Supply Chain Management and Dynamics 365 Commerce use the data management framework (DMF) for import and export. The following articles provide more information about it:

- Data management overview
- Data import and export jobs overview

Unified pricing management provides an enhanced set of data entities designed specifically for importing and exporting pricing rules. These improvements strengthen DMF mapping accuracy, data validation, and staging performance.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- The feature named *Unified pricing management pricing rule performance enhancement* must be turned on in feature management.

Available entities

Unified pricing management includes an enhanced set of data entities designed for importing and exporting pricing rules. The set comprises the following entities:

- `GUPMarginComponentPriceAdjustmentLineV2Entity`
- `GUPMarginComponentPriceAdjustmentV2Entity`
- `GUPOpenTradeAgreementJournalLineV2Entity`
- `GUPPriceDiscPriceAttributeCustomerGroupV2Entity`
- `GUPPriceTermJournalTransV2Entity`
- `GUPRetailDiscountLineV2Entity`
- `GUPRetailDiscountV2Entity`
- `GUPSalesAutomaticSalesDocumentHeaderChargeTableV2Entity`
- `GUPSalesAutomaticSalesDocumentLineChargeTableV2Entity`
- `GUPSalesAutomaticSalesDocumentLineChargeV2Entity`
- `GUPTAMRebateAndDeductionsAgreementHeaderV2Entity`

Last updated on 01/27/2026

Procurement and sourcing overview

07/21/2025

This article gives an overview of the functionality that's available in the Procurement and sourcing module.

Procurement and sourcing covers all the steps from identifying a need for product and services through procuring the product, receipt, invoicing, and processing of payment with vendors. Procurement processes can be configured toward specific business needs by defining purchasing policies and workflows.

Identifying a need for product and services

The need for products or services can arise from *requisitions*, for example, when an employee requires a product. *Product catalogs* can be set up to guide the selection of available products to select from, or requests can be made for products that aren't yet made available in a catalog, allowing the purchasing department to consider how the product can be supplied.

Spending limits can be used to constrain requisition spending, and the *purchasing workflow* adds the option of requiring approval before ordering happen. It's also possible to specify budget fund allocation if necessary.

The procurement department identifies suppliers for required products and services, and this can involve a *request for quotation* being sent out to multiple potential suppliers. It's possible to share the specifications of the product that's being requested and potential vendors can view these to see if they can deliver a product that conforms with them. Vendors return their bids, which the procurement department then reviews before they select the supplier that they want to procure from.

Purchase orders include an option to send out a *purchase inquiry* to the vendor as an alternative to a more comprehensive request for quotation process. The purchase inquiry can be used to help establish terms like prices, discounts, and receipt date for the order. If vendors are set up to use the **Vendor** portal, purchase inquiry functionality is disabled. Instead the order is shared on the **Vendor** portal, and when a *confirmation request* is sent, the vendor can directly confirm the order.

Vendor catalogs can be used to collect information on the product assortment that vendors can supply. Vendors can publish their own catalog, so it's easier to keep the catalog up to date. It's possible to attach an *approved vendor list* to a product, and this can help guide vendor selection when new purchase orders are opened, and prevent the use of unintended vendors.

Procurement

Purchase orders can be created in several different ways including:

- As an outcome of a master planning fun that identified a demand that requires a purchase. This process generates planned purchase orders, and when these are released, purchase orders are generated.
- Through the processing of purchase requisitions that result in procurement.
- Through the processing of purchase agreements, where purchase orders are created as released orders from the agreements. This is commonly used when purchase agreements are used to represent blanket orders.
- Manually, when the purchase order that's created isn't based on another document.

Purchase orders that are configured with *purchase approval workflows* require approval before they're recorded as approved, and this is required before the order can be processed further.

Purchase orders are *confirmed* to represent that an agreement has been established with the vendor. The purchase order will then gradually progress through different states until ultimately being invoiced or canceled.

When you create a purchase order, many of the fields are prepopulated with values that default from the information stored about the vendor in the **Vendors** page. This means that there are a limited number of fields that you need to fill in on the purchase order, although you can choose to override the default values.

Prices and discounts

Prices and discounts include information about the prices, discounts, and rebate terms that they offer. Prices and discounts can be represented as *trade agreements*. Trade agreements represent vendor price lists with prices or discounts, and have a specific set of dates for which the agreement is valid. Prices and discounts can be negotiated and represented through *purchase agreements* with conditions like commitments to buy certain volumes or monetary amounts as a precondition for the negotiated terms. *Rebate agreements* can be created with vendors where the procurement of specific products or groups of products might trigger a rebate from the vendor depending on the purchase amount or volume.

Delivery options

There are different options for the delivery process associated with a purchase order. Ordered products can be split into *delivery* schedules, where parts of the ordered quantity can be planned for delivery on different dates. Delivery can also include *direct delivery* initiated from a sales order, which automates the generation of the packing slip on the sales order at the same

time as the product receipt is recorded on the purchase order. Purchase orders can also be part an *intercompany order* chain, also referred to as intercompany purchase orders, where products are ordered from a matching intercompany sales order. In this situation, some steps are automated across the two related intercompany orders.

Supplementary items

Products can be set up to include *supplementary items*. This is to propose products that are related to the product that's being ordered. The additional products might be required, or might be optional. In some cases, supplementary items might be added as free products that accompany the purchase of other products.

Purchase order charges

Charges can be assigned to the purchase order. This can happen automatically through setup of automatic charges or by adding the charges manually. Charges can be assigned to the order at the header level, or at the order line level. Accounting of charges can be set up in different ways. For example, you can set up a charge to be accounted as a product cost. If you do this, the charges must be assigned at the order line level before the order can be confirmed. There's an option that can help allocate charges from the order header to the lines.

Product receipt and invoicing

Purchase orders that include physical products commonly require *arrival registration* to happen within a warehouse, and after this a *product receipt* is registered for the order.

Some purchase orders include products that are services or other nonphysical products where receipt in a warehouse isn't needed. Products can be created as services or *procurement categories* can be used directly on the purchase order for such orders. With these orders, accounting of product receipt is sometimes skipped and the order is invoiced directly, or alternatively product receipt registration is done on the purchase order without any prior arrival registration.

Receipt of products can result in automatic consumption for a specific purpose. This includes implied consumption with direct delivery, consumption towards a project, or accounting the product as a fixed asset.

When *vendor invoices* arrive from the vendor, they can first be recorded in the *invoice register* independent of the purchase order, and then later approved as a record against the purchase order. Recording the vendor invoice with the purchase order includes matching of the product receipt toward the invoice.

Accounting distributions can be specified on the purchase order to describe how accounting should be done within the ledger, and can also define how budget fund allocation is obtained when this is included in your configuration.

Invoiced purchase orders will record the liability into the vendor account within accounts payable, from where the *vendor payment* can be processed.

Vendor performance

Performance and review of purchasing is supported through *procurement and account payable reports*, which include spend analysis and vendor performance analysis.

Related information

Documentation resources

- Monitor consignment inventory using vendor collaboration
- Create a purchase order from a sales order
- Create a consignment replenishment order
- Overview of vendor invoices
- Vendor posting profiles
- Importing vendor catalogs (blog post)

White papers

The following white papers explore various aspects of procurement in Dynamics 365 Supply Chain Management:

- Defining business process workflows for purchase requisitions
- Inbound consignment inventory demo script
- Defining business process workflows for purchase requisitions
- Trade agreements

Procurement and sourcing workflows

08/13/2025

Some organizations require that purchase requisitions and purchase orders are approved by a user other than the person who entered the transaction. To set up an approval process, you can create a workflow.

A workflow represents a business process. It defines how a document flows through the system and indicates who must complete a task or approve a document. There are several benefits of using the workflow system in your organization:

- **Consistent processes** – You can define the approval process for specific documents, such as purchase requisitions and expense reports. Using the workflow system helps to ensure that documents are processed and approved in a consistent and efficient manner.
- **Process visibility** – You can track the status, history, and performance metrics of a specific workflow instance. This helps you determine whether changes should be made to the workflow to improve efficiency.
- **Centralized work list** – Users can view a centralized work list to view the workflow tasks and approvals assigned to them across all workflows they participate in. This is available on the **Work items** page.

The types of workflows that you can create

The following workflow types are available for Procurement and sourcing.

Expand table

Type	Use this type to
Purchase requisition review	Create review and approval workflows for purchase requisitions.
Purchase requisition line review	Create review and approval workflows for purchase requisition lines.
Purchase order workflow	Create review and approval workflows for purchase orders.
Purchase order line workflow	Create review and approve workflows for purchase order lines.
Vendor add application workflow	Create review and approval workflows for adding new vendors via vendor requests.

> **Important**

When you're adding a new workflow, you might also see the following obsolete workflows listed in the **Create workflow** dialog box. These are related to the *confirmation of receipt* functionality that was available in Dynamics AX 2012, but which has now been deprecated. These workflows are currently unsupported.

- Delivery due date notification workflow
- Invoice received notification workflow
- Product receipt failed notification workflow
- Unconfirmed product receipt rejection notification workflow

Creating a workflow

To create a workflow, go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing workflows** and create a new workflow by selecting the type of workflow you want to create.

In the workflow canvas, you can drag workflow elements into the designer and link the elements into a flow. The workflow elements should be configured. For approval and task workflow elements, you can configure which participant should take action.

Types of participants

You can assign an approval step to the following groups of participants.

Expand table

User group	Description
Participant	Assign the approval step to members of a group or role.
Hierarchy	Assign the approval step to users in a specific organizational hierarchy.
Workflow user	Assign the approval step to users of this workflow.
Queue	Assign the approval step to a work item queue.
User	Assign the approval step to specific users.

Related information

- Defining business process workflows for purchase requisitions
- Purchase requisition workflow
- Onboard vendors

Procurement catalogs overview

This article describes, at a high level, how purchasing professionals can set up and maintain procurement catalogs. Procurement catalogs define the items and services that company employees can order for internal use.

Purchasing professionals can create and maintain catalogs of the items and services that can be purchased for internal use in an organization. After catalogs are set up, company employees can create purchase requisitions to order from them. The catalogs can be used to enforce purchasing policies, so that employees can order only the items and services that are allowed for their buying legal entity. When you create a procurement catalog, consider the following tasks:

- Configure your procurement category hierarchy before you create the catalog.
- Determine which products you want your employees to be able to order. You can show or hide specific products in a catalog node, or you can show or hide all the products in a node.
- Determine how many procurement catalogs you require. Access to a procurement catalog is determined by the catalog policy rule that you configure for the legal entity and operating unit that an employee is assigned to.

Several factors determine the products that employees can order and the procurement categories that they can use when they create purchase requisitions:

- The category access policy rule in the purchasing policy determines which procurement categories are available in the purchase requisition. If no rule is defined, all procurement categories are available.
- Employees can order a product only if it's in the active procurement catalog for your organization, and if it's in an enabled node and not hidden. The product must also be in a category that a particular employee has access to according to the category access policy rule.
- The catalog policy rule specifies the catalog that is used. If no catalog policy rule is defined, the category access rule alone determines the products that an employee can order on the requisition.

Prerequisites

The following table describes the tasks that must be completed before a purchasing professional can create a procurement catalog.

Expand table

Task	Role	Description
Set up a procurement category hierarchy.	Purchasing manager	Procurement category hierarchies classify items or transactions for reporting and analysis. By using a procurement category hierarchy, companies can strategically manage categories, products, vendors, and other procurement factors from a central location. One procurement category hierarchy is defined for a whole organization. The catalog is based on the procurement category hierarchy: the categories in the hierarchy become nodes in the catalog. The vendors and products are included in your catalog.
Add vendors and products to procurement categories.	Purchasing manager	When the procurement category hierarchy is created for your organization, each procurement category can be associated with specific vendors, products, and so on. These associations are copied automatically to the catalog.

Set up a catalog

After meeting the prerequisites, you can set up catalogs. You can create one catalog for your whole organization or multiple catalogs for different divisions within your organization. If you create one catalog for the entire organization, your purchasing policy rules control access to the catalog.

The catalog defines which products are available when purchase requisitions are created, but you can use category access policies rules to apply additional restrictions. Because the nodes in a catalog are procurement categories, they can be suppressed by a category access policy rule. In this case, the products in that category aren't available for employees to use on requisitions. You define category access policy rules on the **Purchasing policies** page. The following table describes the tasks that must be completed to set up a catalog.

Expand table

Task	Role	Description
Create a new catalog.	Purchasing agent	When you create a catalog, specify a name and description for the catalog. Also, define whether the catalog is updated manually or automatically, and specify the catalog owner.
Control whether products are available in the catalog.	Purchasing agent	Because the products are inherited from the procurement categories, they all appear in the appropriate catalog nodes. You can control whether all products in a node are hidden or shown when the catalog is used in a purchase requisition. You can also control whether individual products in a node are hidden or shown.

Task	Role	Description
Publish the catalog.	Purchasing agent	Before employees can use a catalog in a requisition, you must define a catalog policy rule for the catalog, set the catalog's status to *Active*, and publish the catalog. You can deactivate catalogs that you no longer want to be available to your users.

Updates are published either automatically or manually, depending on the option that you select for the catalog in the **Default update type** field on the **Catalogs** page. The following default update types are available for catalogs:

- *Dynamic* – The catalog is automatically updated whenever it's changed.
- *Static* – The catalogs must be manually updated.
- *Both* – If the catalog includes product categories that have a default update type of *Static*, it must be manually updated when these categories are updated. If the catalog includes product categories that have a default update type of *Dynamic*, it's automatically updated whenever it's changed.

Related information

- Set up a procurement category hierarchy

Last updated on 07/01/2026

Create a procurement catalog

Article • 01/24/2025

This article explains how to create a procurement catalog. This task would typically be carried out by a procurement professional. You'll also learn how employees can use the catalog when they create a requisition. Before you can create a catalog, there must be a procurement category hierarchy in your system. The hierarchy is inherited by the new catalog, along with all the products that are in the hierarchy. You can use this guide in demo data company USMF where the procurement category hierarchy and other sample data are available.

Ensure that a procurement category hierarchy exists

1. Go to **Procurement and sourcing** > **Procurement categories**. A procurement category hierarchy is available in the USMF demo data company and products have been added to the **Office machines/Computers** category. If you're running this procedure as a task guide, you'll need to unlock the guide if you want to browse through the category. If a hierarchy isn't available, you can create it by selecting **New** (this action can only be done once).
2. Close the page.

Create a catalog

1. Go to **Procurement and sourcing** > **Catalogs** > **Procurement catalogs**.
2. Select **New procurement catalog** to open the drop dialog.
3. In the **Name** field, type a value.
4. Select **OK**.
5. In the tree, expand **CORP PROCUREMENT CATEGORIES**.
6. In the tree, expand **OFFICE MACHINES**.
7. In the tree, select **Computers**.
    - The products from the procurement category are displayed in the list. If you want to add a product to the category you need to do this on the **Procurement category hierarchy** page or on the **Item details** page.

- The **Default** update type determines whether new products that have been added to the procurement category hierarchy are immediately visible in the catalog. If the update type is set to *Dynamic*, changes are visible immediately. If the update type is *Static*, new products are only visible to people using the catalog after the catalog has been republished. The **Publish** action is available on the Action Pane at the top of the page. If products are removed from the procurement category hierarchy, the change is immediately visible, regardless of the value in the **Default** update type field.

8. On the Action Pane, select **Category navigation** and ensure that **Enable** is selected.
9. Select **Activate catalog**.
10. Close the page.

Make the catalog visible

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Purchasing policies**.
2. Select **Procurement Policy USMF**. You need to select the purchasing policy for the legal entity that the worker connected to your user profile is allowed to order products in. In the USMF demo data, the Admin user is connected to the worker called *Julia Funderburk*, and orders products in USMF by default.
3. Select the catalog that you just created.
4. Select **OK**.

Use the catalog

1. Go to **Procurement and sourcing** > **Purchase requisitions** > **All purchase requisitions**.
2. Select **New**.
3. In the **Name** field, type a value.
4. Select **OK**.
5. Select **Add products**.
6. In the list, find and select the desired record. You can use the category hierarchy on the left or the filter at the top of the list to filter the products.
7. Select **Add to lines**.
8. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up a procurement category hierarchy

08/13/2025

This procedure shows you how to create new nodes in a procurement category hierarchy and how to configure a procurement category to be used in a procurement process. These tasks would typically be carried out by a Purchasing manager. Before you can start this procedure, there must be a category hierarchy of type *Procurement*.

Add a new procurement category

1. Go to **Procurement and sourcing** > **Procurement categories**.
2. On the Action Pane, select **Edit category hierarchy**. The current procurement category hierarchy is displayed in the left side of the page. You're about to modify the hierarchy.
3. On the Action Pane, select **New category node**. The system selects the top node by default.
4. In the **Name** field, type a value.
5. In the **Description** field, type a value.
6. In the **Friendly name** field, type a value. The friendly name is optional. It's displayed in category lookups together with the category name.
7. Select **Save**.

Add products to your new procurement category

1. Go to **Procurement and sourcing** > **Procurement categories**. Select the node you just added.
2. Toggle the expansion of the **Products** section.
3. Select **Add** to associate products with the procurement category.
4. Select the products you want to add to the procurement category.
5. Select the arrow to add the products to the **Selected** table.
6. Select **OK**.

Purchase requisition overview

This article describes the purchase requisition workflow and the different statuses that a purchase requisition can have.

Depending on the setup of your organization, you can create purchase requisitions for products that your organization uses. A purchase requisition is an internal document that authorizes the Purchasing department to buy items or services.

After a purchase requisition is approved, you can use it to generate a purchase order. Purchase orders are the external documents that the Purchasing department submits to vendors.

Creating purchase requisitions

Create a purchase requisition on the **My purchase requisitions** page, and select the items and services that you require. You can select items from a procurement catalog that your organization created, or you can request items that aren't found in a catalog by selecting a procurement category and entering the product details.

Before you can submit a purchase requisition for review, you must configure the required workflows. Use a workflow to move a purchase requisition through the review process, from an initial status of *Draft* to a final status of *Approved*.

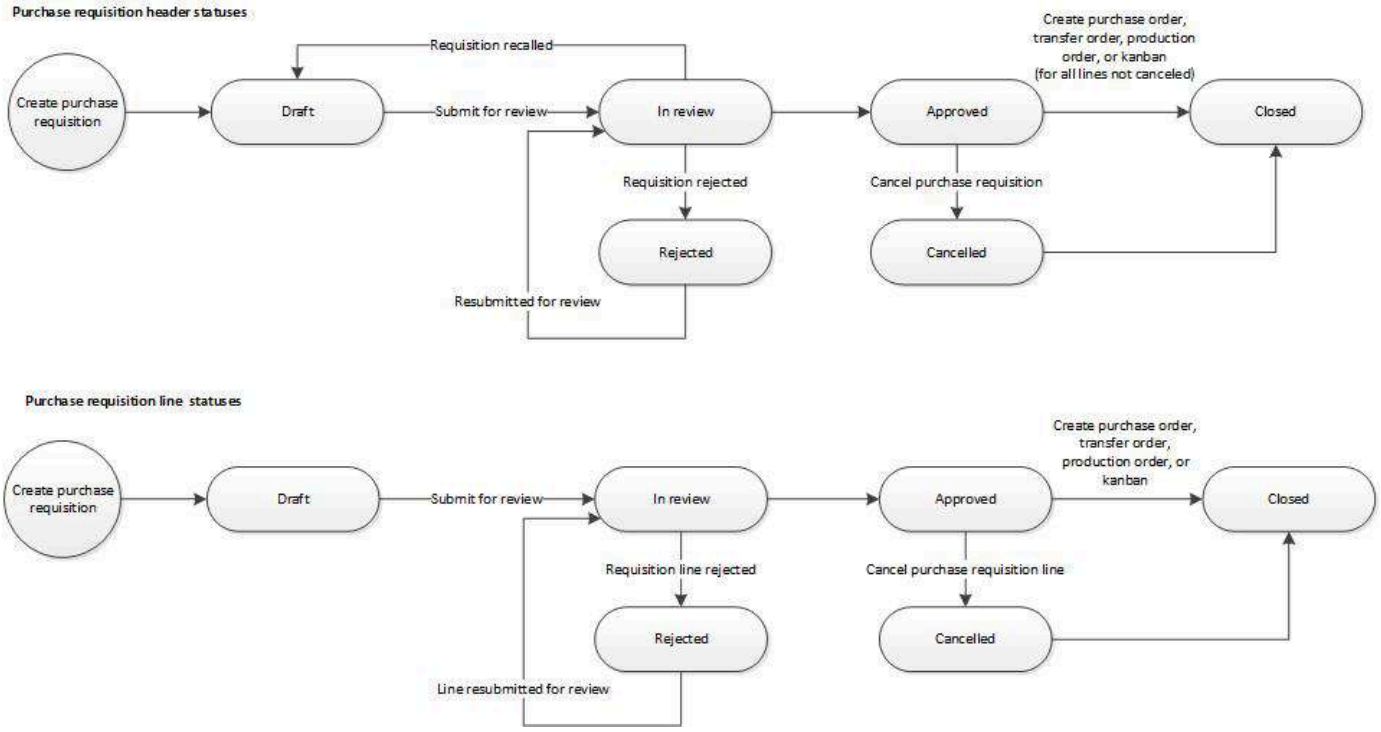
Purchase requisition statuses

When you create a purchase requisition, the system assigns a status to it. The system also assigns a status to every line that you add to a purchase requisition. When you submit a purchase requisition to a workflow for review, the system updates the status of the purchase requisition and the status of each line as the lines move through the workflow process.

You can configure the purchase requisition workflow process to route a purchase requisition through the review process as a single document. Alternatively, you can route the lines on a purchase requisition individually to the appropriate reviewers. If the purchase requisition lines are reviewed individually, the system can update the status of each purchase requisition line as the line moves through the review process. When all lines complete the review process and no review steps remain for the purchase requisition, the system updates the status of the whole purchase requisition.

Purchase requisition workflow

The following diagram shows the statuses that the system assigns to a purchase requisition and a purchase requisition line as they move through the workflow process.



Purchase requisition header and line status relationships

The overall status of a purchase requisition depends on the status of the purchase requisition lines. Therefore, you must complete the review process for all purchase requisition lines before you can complete the review process for the whole purchase requisition. The following table describes the statuses that are assigned to a purchase requisition header and lines as the purchase requisition moves through the workflow process.

Expand table

Purchase requisition status	Purchase requisition line status	Description
Draft	Draft	The purchase requisition and purchase requisition line have been created, but they haven't been submitted for review. You can modify purchase requisitions and purchase requisition lines that have a status of *Draft*. A purchase requisition or purchase requisition line also has a status of *Draft* if you recalled it but didn't resubmit it for review.

Purchase requisition status	Purchase requisition line status	Description
		Note: You can submit or recall a purchase requisition at the document level. However, you can't submit or recall a single purchase requisition line.
In review	• In review • Rejected	If the workflow is configured to route purchase requisition lines to individual reviewers, each line can have a status of *In review* or *Rejected*. The purchase requisition status updates when the review process is completed for all purchase requisition lines and no review steps remain for the purchase requisition. • *In review* – The purchase requisition lines are submitted for review. When the workflow process is completed for a purchase requisition line, the status of that line remains *In review* until all remaining purchase requisition lines are reviewed. • *Rejected* – A purchase requisition line is rejected. You can modify and resubmit purchase requisition lines that are rejected. If you resubmit a purchase requisition line that was rejected, the review process starts over for all lines in the purchase requisition that are still in review. **Note:** You can recall a purchase requisition that you already submitted. When you recall a purchase requisition, you also recall all other purchase requisition lines. You can delete purchase requisition lines that are recalled.
Rejected	Rejected	The purchase requisition and all purchase requisition lines have been rejected. You can resubmit purchase requisitions and purchase requisition lines that have been rejected.

Approved	• Approved
	• Cancelled
	• Closed
		All purchase requisition lines have completed the review process, and there are no more review steps for the purchase requisition.
		
		• *Approved* – The review process for a purchase requisition line is complete, and the line is approved.
		• *Cancelled* – The purchase requisition line was approved, but it's canceled because it's no longer required. Only purchase requisition lines that are approved can be canceled.
		• *Closed* – The purchase requisition line was approved, and documents are generated, depending on the requisition purpose.
		  ◦ If the requisition purpose is consumption, a purchase order is generated for the purchase requisition line.
		  ◦ If the requisition purpose is replenishment, one or more fulfillment documents are generated.

Cancelled	Cancelled	The purchase requisition and all purchase requisition lines have been canceled.

Purchase requisition status	Purchase requisition line status	Description
		Note: If you no longer require an item that is on a purchase requisition line, you must cancel the purchase requisition line if it's already approved. Only purchase requisition lines that are approved can be canceled. If any purchase requisition lines are in review, the purchase requisition has a status of **In review**. In this case, you can recall the purchase requisition and delete the appropriate purchase requisition line.

Closed	• Closed
	• Cancelled
		The purchase requisition is closed, and one or more fulfillment documents have been generated.
		
		• *Closed* – The purchase requisition line was approved, and documents are generated, depending on the requisition purpose.
		  ◦ If the requisition purpose is consumption, a purchase order is generated for the purchase requisition line.
		  ◦ If the requisition purpose is replenishment, one or more fulfillment documents are generated.
		• *Cancelled* – The purchase requisition line was approved, but it's canceled because it's no longer required. Only purchase requisition lines that are approved can be canceled.
		
		Note: If you no longer require an item on a purchase requisition line that is closed, you must cancel the line on the fulfillment document that was generated for the purchase requisition line.

Distributing costs to multiple financial accounts

You can distribute the cost of a product that is included in a purchase requisition to multiple financial accounts. If your organization uses dimensions, such as cost centers and departments, you can distribute the cost of a product to dimensions for financial accounts.

Requisition purposes

Requisition purposes make the process of fulfilling requisition demand more flexible. When you create a requisition, you can assign one of two purposes to it: consumption or replenishment. Depending on the requisition purpose and the setup of your organization, requisition demand can be fulfilled by a purchase order, transfer order, production order, or kanban.

In the procurement policies, you can control the requisition purposes that are available when a requisition is created for your organization.

Requisitions that have a purpose of consumption

A requisition that has a purpose of consumption represents demand for items or services that your organization uses internally. The demand that this kind of requisition creates is always fulfilled by a purchase order. If Supply Chain Management is set up to automatically generate purchase orders, purchase orders are created after the purchase requisition is approved.

Requisitions that have a purpose of replenishment

A requisition that has a purpose of replenishment represents demand to replenish inventory. For example, you create a requisition to replenish items so that you can sell them at a specific retail location at a specific time. The demand that this kind of requisition creates can be fulfilled by a purchase order, transfer order, production order, or kanban.

When the requisition purpose is replenishment, demand is expressed as a quantity instead of a monetary amount. Therefore, encumbrance accounting, budgetary control, business rules for fixed asset determination (BRAD), project accounting, and any related rules don't apply. Only products that are stocked and released to the specified legal entity can fulfill replenishment requisition demand. To define the products that are available when the requisition purpose is replenishment, use the **Replenishment category access policy rule** page.

To use purchase requisitions that have a purpose of replenishment, set up master scheduling to include requisition demand. The system automatically determines the fulfillment method for the demand that this kind of requisition creates, based on the supply policies that you set up for the items in your organization and planned by using master scheduling.

Purchase requisitions and requests for quotation

In some cases, you must start a request for quotation (RFQ) process to identify the vendor and price for products that a purchase requisition requests. You can generate an RFQ when the purchase requisition is in review. When you accept a bid, the system transfers information about the vendor, price, and other details to the requisition.

You can control whether the system should update fixed asset fields when you accept an RFQ bid for purchase requisitions that have business rules for fixed asset determination (BRAD). To set this option, follow these steps (requires Supply Chain Management version 10.0.49 or later):

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. Open the **Purchase requisition** tab.
3. Set **RFQ bid–driven fixed asset field updates** to one of the following values:

- *Yes* – Reevaluate asset classification and accounting values based on the accepted RFQ bid. This value is the default setting.
- *No* – Only transfer the RFQ price to the requisition. Preserve existing accounting values and budget control reservations.

You can put a purchase requisition on hold by selecting the **On hold** check box on the **Purchase requisition details** page. Processing of the purchase requisition can continue only after you remove the hold by clearing the check box.

> ⓘ **Note**
>
> In e-procurement, the RFQ for your purchase requisition might allow vendors to add alternate lines. In this case, your purchase requisition reflects approved alternates.

Demand consolidation

When you consolidate purchase requisition lines from multiple purchase requisitions, you increase your negotiating power with vendors. You can get better pricing, lower shipping and handling costs, and reduced overhead costs.

Purchase requisition lines are eligible for demand consolidation only if the following statements are true:

- The purchase requisition is approved.
- The purchase requisition meets the purchasing policy rule criteria for manual processing and demand consolidation.

The **Release approved purchase requisitions** page lists approved purchase requisition lines that meet the criteria for manual processing. If a purchase requisition line also meets the criteria for demand consolidation, you can add the line to a consolidation opportunity.

A consolidation opportunity is a set of purchase requisition lines that you group together so that the purchasing professional can negotiate the best deal with vendors. Purchase requisition lines that you select for a consolidation opportunity appear on the **Purchase requisition consolidation** page. You can modify the lines on this page, if changes are required. You can also add new lines to the consolidation opportunity or remove existing lines.

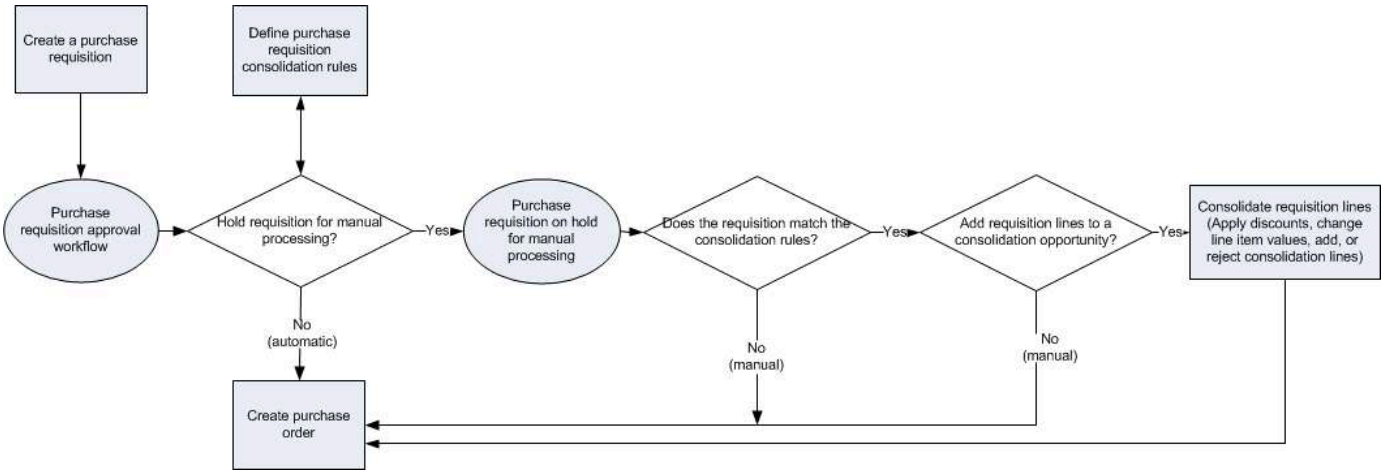
After you add requisition lines to a consolidation opportunity and make any changes that you require, you can create a purchase order for the consolidated purchase requisition lines.

> **Note**
>
> Changes that you make to a purchase requisition line on the **Purchase requisition consolidation** page are reflected on the purchase order that you create. However, the line remains unchanged in the purchase requisition, so that its history is preserved.

To create a purchase order for purchase requisition lines that aren't eligible for demand consolidation or aren't selected for a consolidation opportunity, you must process the lines manually.

Consolidating purchase requisition lines

The process for demand consolidation starts when a purchase requisition is approved in a workflow and, if budget control is configured for your organization, when the budget reservations and pre-encumbrances have been recorded. The following diagram shows the process flow for demand consolidation.



To consolidate approved purchase requisition lines, follow these steps:

1. Review approved requisition lines that have been held for manual processing, and that are eligible for demand consolidation.
2. Select lines to add to a consolidation opportunity.
3. Create a new consolidation opportunity, or add requisition lines to an existing consolidation opportunity.
4. Make any required changes to the requisition lines, and remove requisition line items that you no longer want to include in the consolidation opportunity.
5. Create purchase orders for consolidated requisition lines or for purchase requisition lines in a consolidation opportunity.

Next steps

- Create a requisition for consumption
- Purchase requisition workflow

Last updated on 07/23/2026

Purchase requisition workflow

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The workflow process moves purchase requisitions through the review process, from an initial status of Draft to a final status of Approved. When a purchase requisition is submitted for review, the workflow process is started. After a purchase requisition is approved, a purchase order can be generated for the purchase requisition lines and submitted to the vendor for order fulfillment.

Before a purchase requisition can be submitted for review, you must configure a workflow. The workflow process can include one or more review steps, in any order. The workflow process can also be configured to skip the review tasks and automatically approve the purchase requisition. You can configure the workflow to route the purchase requisition as a single document, or you can route individual purchase requisition lines to the appropriate reviewers. You can also create a scenario where the purchase requisition is routed as a single document to some reviewers and selected purchase requisition lines are routed to other reviewers.

If purchase requisition lines are reviewed individually, the review process must be completed for all purchase requisition lines before the workflow process can move to the next step, and before the review process for the purchase requisition as a whole can be completed. When the review process has been completed for the purchase requisition and all its lines, the overall status of the purchase requisition is updated to *Approved*.

You can configure your workflow to represent the business process for purchase requisitions in your organization. When you configure your purchase requisition workflow process, consider the following questions:

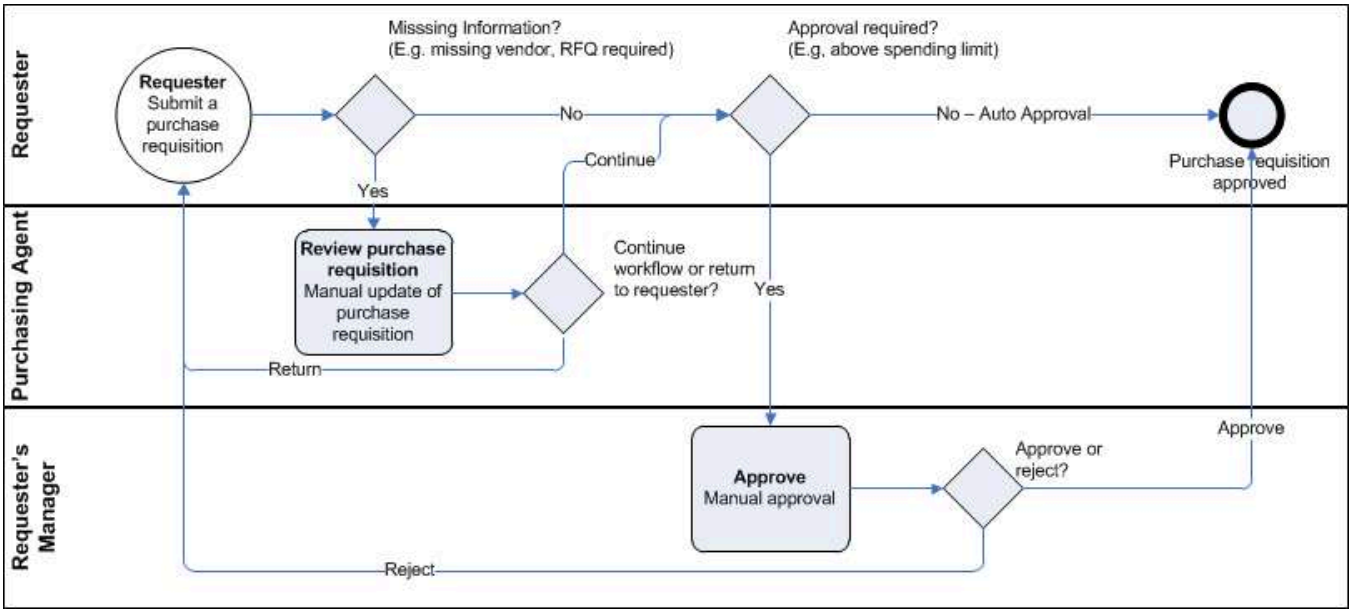
- What expenditures must be reviewed?
- What expenditures can be automatically approved?
- Who is required to review and approve expenditure requests? What role are these users assigned to?
- What process must be followed if a reviewer isn't available?

The following examples illustrate two ways that you can configure a workflow for purchase requisitions.

Example 1: Route a purchase requisition as a single document for review

The following illustration shows how a purchase requisition can flow through the workflow review process as a single document. The lines on the purchase requisition aren't routed individually. The following roles are included in the workflow process for this example:

- *Requester* – The user who requests the items or services. The requester can prepare the purchase requisition, or another worker can prepare the purchase requisition on the requester's behalf. This worker is the preparer. The preparer is responsible for managing the purchase requisition throughout the review process. Only the preparer of the purchase requisition can modify it. A worker must be granted the appropriate permissions to create a purchase requisition on behalf of someone else. Use the **Purchase requisition permission** page to set up these permissions.
- *Purchasing agent* – The user who performs a procurement review and can approve the document.
- *The requester's manager* – The user who performs a managerial review and can approve the document.



In this example, the workflow process for the purchase requisition includes the following steps:

1. The preparer submits a purchase requisition for review.
2. The purchasing agent receives a notification. The notification requests that the purchasing agent verify the information in the purchase requisition. If required information is missing, the purchasing agent can either add it or return the purchase requisition to the preparer to add it. When all the required information has been filled in, the purchase requisition can move to the next step in the review process.
3. The requester's manager reviews the purchase requisition. The purchase requisition might be routed to the requester's manager if, for example, the amount of the purchase

requisition exceeds the requester's spending limit for purchase requisitions. The requester's manager can approve or reject the purchase requisition, or return it to the preparer for changes.

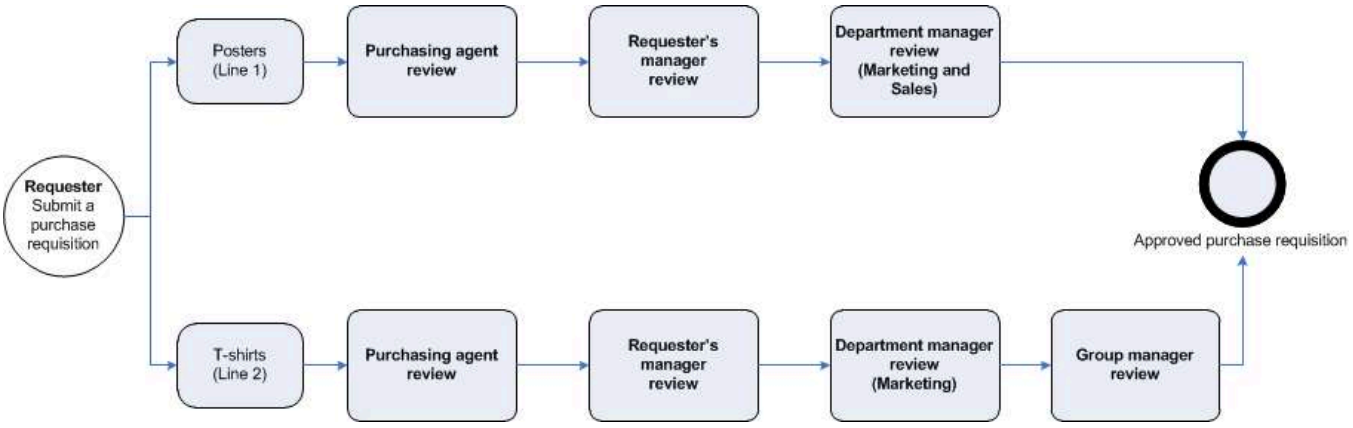
Example 2: Route the individual purchase requisition lines for review

The following illustration shows how the individual purchase requisition lines can be routed through a workflow. In general, the process for each line is the same as the process for a purchase requisition that is reviewed as a single document. However, each line must complete the workflow process individually before the workflow can be completed for the whole purchase requisition.

In this example, a worker enters a request for posters and T-shirts for a marketing campaign. The cost of the posters is split between the Marketing department and the Sales department. If the cost of the posters or T-shirts exceeds the signing limit authority for department managers, the purchase requisition must also be reviewed by the group manager.

The following roles are included in the workflow process for this example:

- *Requester* – The user who requests the items or services. The requester can prepare the purchase requisition, or another worker can prepare the purchase requisition on the requester's behalf. This worker is the preparer. The preparer is responsible for managing the purchase requisition throughout the review process. Only the preparer of the purchase requisition can modify it. A worker must be granted the appropriate permissions to create a purchase requisition on behalf of someone else. Use the **Purchase requisition permission** page to set up these permissions.
- *Purchasing agent* – The user who performs a procurement review and can approve the document.
- *The requester's manager* – The user who performs a managerial review and can approve the document.
- *Department manager* – The user who performs an expenditure review and can approve the document.
- *Group manager* – The user who performs a signature authority review and can approve the document.



In this example, the workflow process for the purchase requisition lines includes the following steps:

1. The preparer submits a purchase requisition for review. Each line is routed to the reviewer who is configured to receive it in the workflow process.
2. The purchasing agent receives a notification. The notification requests that the purchasing agent verify the information in the purchase requisition and on the purchase requisition lines. When the purchase requisition is opened by the purchasing agent all the lines are visible, but a visual indicator shows which lines have been sent to the purchasing agent for review. If required information is missing, the purchasing agent can either add it or return a purchase requisition line to the preparer to add it. When all the required information has been filled in, the purchase requisition line can move to the next step in the review process. Purchase requisition lines can continue through the review process independently of each other.
3. The requester's line manager reviews and approves the purchase requisition lines. The approval might be routed to the requester's manager if, for example, the amount on a purchase requisition line exceeds the requester's spending limit for purchase requisition lines. The manager can approve or reject one or both of the purchase requisition lines.
4. The department manager for the Marketing department reviews the purchase requisition lines for both the posters and the T-shirts. The Sales department manager reviews the purchase requisition line only for the posters, because that is the only cost that is being charged to the Sales department.
5. The group manager reviews and approves the purchase requisition line for the T-shirts only if group manager approval is required because, for example, the amount on the purchase requisition line exceeds the department manager's approval limit. The group manager doesn't have to approve the purchase requisition line for the posters.

> **Note**
>
> The system currency must be set if the header workflow for a purchase requisition requires approvals related to signing limits.

Configuring a workflow for purchase requisitions

To route a purchase requisition for review, you must configure the purchase requisition workflow processes. The workflow process that you define controls the interaction between the user who requested the items (the requester) and the reviewer and approver in the workflow. The routing of the purchase requisition depends on the conditions that are specified in the workflow configuration. For example, these conditions determine when the purchase requisition should be routed, the user or role that it should be routed to, and the actions that users can take.

The examples in this article show how a purchase requisition can be routed through a workflow as a single document or as individual purchase requisition lines. You can also configure a workflow for purchase requisitions that reflects the internal control review of purchase requisitions that is defined for your organization.

The participants or the reviewers that a task is assigned to in a workflow can be members of a particular user group, users who have a particular security role, users who are associated with the submitter in a managerial hierarchy, or named users or users who have specific expenditure responsibilities.

Purchase requisition expenditure reviewers

Expenditure reviewer configurations let you dynamically route expenditures for review, based on the user who is assigned to a project role or a financial dimension where the expenditure is being charged. The workflow process uses the specified project role or financial dimension owner to determine who the expenditure should be routed to.

You can define one or more expenditure reviewer configurations and then select a configuration when you create a workflow. You can configure the expenditure reviewer values for every legal entity in your organization. After you define the expenditure reviewer configurations, you assign a configuration to your workflow task.

You don't have to define expenditure reviewer configurations. Instead, you can assign specific users or user groups as reviewers when you define your workflow. However, if you have a complex organization, expenditure reviewers can increase the efficiency of your approval process. In addition, if you set up expenditure reviewers, you don't have to update workflow reviewer assignments every time that a reviewer changes job roles.

You can set up the expenditure reviewers on the **Purchase requisition expenditure reviewers** page. Create an expenditure reviewer configuration, and enter values for each legal entity in your organization. For requisitions that are assigned to a project, you can specify the role that is responsible for reviewing the requisitions: Project manager, Project controller, or Project

sales manager. Expenditures will be routed to the user who is assigned to the specified role. You can also route the expenditure to the financial dimension owner by selecting the appropriate financial dimension option on the **Organization distributions** tab.

To use one of the expenditure reviewers that you set up in a workflow, you must set the **Type of participant** option to **Expenditure participants** in the **Assignment** properties for the relevant workflow element.

Related information

- Create a requisition for consumption
- Defining business process workflows for purchase requisitions
- Procurement and sourcing workflows
- Purchase requisition overview

Last updated on 08/05/2025

Create a requisition for consumption

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

his article describes the process of creating a requisition. It shows you different ways to search for products in your procurement catalog and how to add a product that isn't in your catalog. Before you start this procedure, you must have a purchasing policy set up with Consumption as the default type of requisition. You can walk through this procedure in demo data company USMF, or using your own data. The procedure can only be carried out by a user profile that is set up as worker. This task is normally be carried out by an employee. The **Employee** employ security role allows you to carry out the tasks, or if you're using USMF, you can sign in as *Alicia*.

Create a new requisition

1. Go to **Procurement and sourcing** > **Purchase requisitions** > **Purchase requisitions prepared by me**.
2. Select **New**.
3. In the **Name** field, enter a name for the requisition.
4. In the **Requested date** field, enter a date. By default, the requested date and accounting date are copied to the purchase requisition lines. You can change them at the line level. The requested date is the requested receipt date.
5. In the **Accounting date** field, enter a date. The accounting date records the accounting entry in the general ledger, and validates whether budget funds are available.
6. Select **OK**.
7. In the **Reason** field, select an option from the drop-down menu. By default, the business justification reason that you select appears for the purchase requisition lines, but you can change it at the line level.
8. Select the reason.
9. In the **Details** field, enter a more descriptive justification for the requisition.

Add a line to the requisition

1. Select **Add line**. There are two ways to add lines to the purchase requisition. If you already know the product number or you already know that you're requesting a product that isn't in the product catalog, add the line directly by using **Add line**. The other way is to use **Add products** where you can use searching and filtering to find items in the product catalog.
2. Select the row you just created.

    - The requester is the worker that requests the requisition.
    - By default, the person preparing the requisition is the worker who requests it. You must be given permission to prepare a requisition line on behalf of another worker. If you have such permissions, the other workers appear in this lookup.

3. In the **Item number** field, type a value. The items that you can choose from are limited by the category access policy and the procurement catalog for the buying legal entity.
4. In the **Quantity** field, enter a number.

Add more products to the requisition

1. Select **Add products**. This option lets you search for products in the product catalog.
2. In the **Find procurement category node** field, enter the first part of the name of the category that you're looking for, and then select **Enter**. For example, enter *comput*.
3. Use the **InvokeDefaultButton** shortcut.
4. Use the **Filter** to filter the list of products in the selected category.
5. Select the product card that you want to add to the requisition.
6. Select **Add to lines**.
7. In the **Quantity** field, enter a number.
8. In the **Find procurement category node** field, type the first part of the name of the category that you're looking for, and then select **Enter**. For example, enter *High* (highlighters).
9. Use the **InvokeDefaultButton** shortcut.
10. Select **Add unlisted product to lines** to add a product that's not listed in the procurement catalog.
11. In the **Product name** field, type a value.
12. In the **Unit** field, type a value.
13. Select **OK**.
14. In the **Item description** field, add a description of the product.
15. In the **Quantity** field, enter a number.

16. In the **Unit price** field, enter a number. If you know the price for a particular vendor (that you select in the vendor account field) enter that price.
17. In the **Vendor account** field, open the drop-down list to select an option. The vendors that are available in this field depend on the purchasing policies and the status that the vendor has for the current procurement category. As an alternative to selecting a vendor here, you can select **Suggest vendor**.
18. In the list, select the vendor you want to use.
19. In the **External item number** field, type a value. This value is a reference number for the product that the vendor knows. For example, this value could be the item number of the product in the vendor's own catalog.
20. Select **OK**.

Distribute amounts

1. Select **Financials**.
2. Select **Distribute amounts**. This process shows you how to distribute the cost for the first line between two accounts. You can also distribute amounts later when the requisition is in review.
3. Select **Split** to create a new distribution line.
4. In the **Ledger account** field, select the first cost center that should take part of the cost.
5. Select the other distribution line.
6. In the **Ledger account** field, specify the other cost center.
7. Select **Distribute equally**.
8. Close the page.

View line details

1. Toggle the expansion of the **Line details** section.

Submit the requisition

1. Select **Workflow** to open the drop dialog.
2. Select **Submit**.
3. Close the page.
4. In the **Comment** field, type a note for the approver of the requisition.
5. Select **Submit**.
6. Close the page.
7. Refresh the page.

Last updated on 05/29/2026

Create a requisition that uses an RFQ

06/17/2025

This article explains how to add price and vendor information to a purchase requisition from a request for quotation (RFQ) process. The example shown in this guide can be used in the USMF demo data company, and you must be logged in as an Admin to complete all the steps. The tasks in this guide would typically be done by procurement professionals.

Create a requisition

1. Go to **Procurement and sourcing** > **Purchase requisitions** > **Purchase requisitions prepared by me**.
2. Select **New**.
3. In the **Name** field, type a value.
4. In the **Requested date** field, enter a date.
5. In the **Accounting date** field, enter a date.
6. Select **OK**.
7. In the **Reason** field, enter or select a value.
8. Select **Add line**.
9. In the **Procurement category** field, select a category in the tree, and then select **OK**.
10. In the **Product name** field, type a value.
11. In the **Quantity** field, enter a number.
12. In the **Unit** field, enter or select a value.
13. Select **Save**.
14. Select **Workflow** to open the drop dialog.
15. Select **Submit**.
16. Close the page.
17. Select **Submit**.

Reassign a workflow task

The next task is to create an RFQ to get bids from vendors for the product. In USMF demo data, the requisition workflow is set up with a rule so that if a vendor isn't selected, or the unit price is 0 for a line, a task is assigned to a specific worker to create an RFQ. To continue with this guide, you need to reassign that task to another user (yourself). You can only do this if you're logged in as an Admin.

1. Select **Workflow** to open the drop dialog.
2. Select **View history**.
3. Refresh the page.

4. Expand the **Tracking details** section.
5. In the tree, select the line that starts with *Line workflow activated on*.
6. Select **View workflow details**.
7. Expand the **Work items** section.
8. Select **Reassign**.
9. In the **User** field, select *Admin*.
10. Select **Reassign**.
11. Close the two pages.

Create an RFQ

1. Refresh the page.
2. Select **Request for quotation**.
3. In the **Buying legal entity** field, select *USMF*. You must select the same legal entity that's on the requisition line.
4. In the list, mark the selected row. If you had multiple lines on your purchase requisition, select all the lines that you want to add to the RFQ.
5. Select **OK**.
6. Refresh the page.
7. Ensure that the FactBox is open, then expand the **Related documents** section.
8. Select the link in the **Request for quotation** field to open the RFQ that you created.
9. Select **Header**.
10. Select **Add**.
11. In the **Vendor account** field, enter or select a value.
12. Select **Add**.
13. In the **Vendor account** field, enter or select a value.
14. Select **Send**.
15. Select **OK**.
16. Select **Enter reply**.
17. On the Action Pane, select **Reply**.
18. Select **Copy data to reply**. This copies data, such as the quantity and dates, from the RFQ to the reply.
19. In the **Unit price** field, enter a number. This is the price that you've received from the vendor. You might also want to enter additional information from the vendor.
20. Select **Accept**.
21. Select **OK**.

Verify that vendor and price have been transferred to the requisition

1. Close the page.
2. Select **Lines**.
3. Select **Related information**.
4. Select **Purchase requisition**.
5. Select the line that was transferred to the RFQ. Verify that the price and vendor have been copied to the requisition.
6. Select **Workflow** to open the drop dialog.
7. Select **Complete**.
8. Select the page.
9. Select **Complete**.

Requests for quotation (RFQs) overview

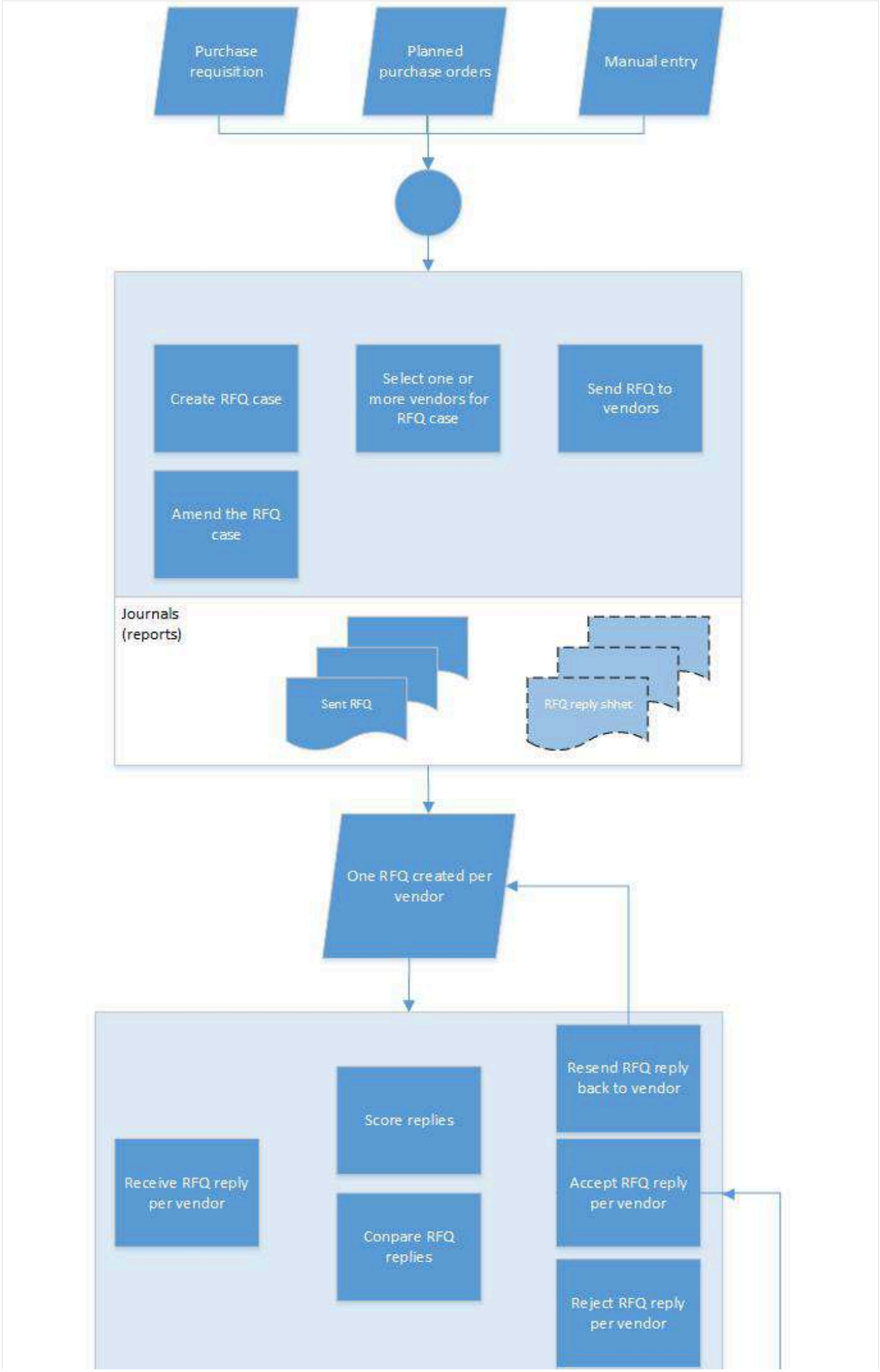
07/21/2025

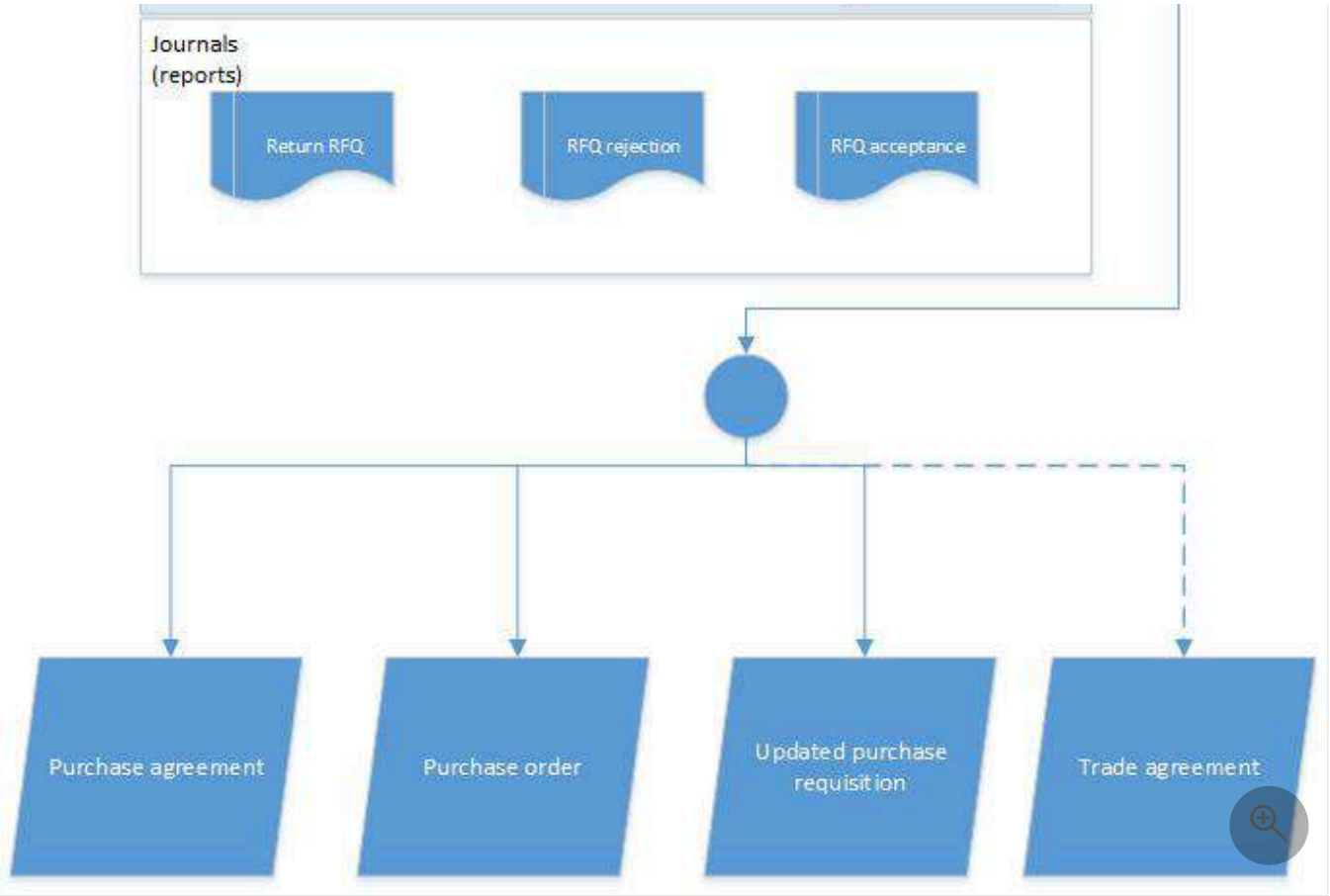
This article provides an overview of requests for quotation (RFQs). Organizations issue RFQs when they want to receive competitive offers from several vendors for the items or services that they must purchase. In an RFQ, you ask vendors to provide the prices and delivery times for the item quantities that you specify. You can also ask vendors to specify whether there are any incidental charges, such as shipping costs, or any discounts for large orders or early payment of the vendor invoice.

The RFQ process consists of the following tasks:

1. Create and send an RFQ to one or more vendors.
2. Receive and register bids (RFQ replies).
3. Transfer bids that you accept to a purchase order, purchase agreement, or purchase requisition.

The following illustration shows an overview of the RFQ process.


Purchase requisition
Planned purchase orders
Manual entry
Create RFQ case
Select one or more vendors for RFQ case
Send RFQ to vendors
Amend the RFQ case
Journals (reports)
Sent RFQ
RFQ reply sheet
One RFQ created per vendor
Resend RFQ reply back to vendor
Score replies
Receive RFQ reply per vendor
Accept RFQ reply per vendor
Conpare RFQ replies
Reject RFQ reply per vendor



You can create an RFQ case from planned orders, from a purchase requisition, or by manual entry. The RFQ case is the base document that you use to issue an RFQ to each vendor.

After you prepare the RFQ case and add vendors, select **Send** (**Send and publish** for public sector) on the RFQ case. An RFQ journal is generated for each vendor that you sent the RFQ to. You can configure the printing options for the **Send** action so that it either prints a report for each vendor to an archive or sends a report to each vendor's email address. In addition, you can use the RFQ journal for each vendor to generate a report that you can send or resend to the vendor later. You can also configure the **Send** action so that it generates a reply sheet that the vendor can fill in.

This article covers the process for handling RFQs when vendor collaboration isn't used. If your system is set up for vendor collaboration, vendors can enter bids directly in Supply Chain Management. Learn more in Vendor collaboration with customers and Vendor collaboration with external vendors.

If you must amend an RFQ after you send it, you can resend the RFQ to vendors when you've finished by using the two amendment actions: Create and Finalize. When you receive bids by email, you can handle these bids from the **Request for quotations** page.

If a second iteration of a reply from a vendor is required, select **Return** on the **Request for quotation** page. The Return action generates a new journal and a report that will be printed, archived, and sent according to your print settings.

If you added scoring criteria to your RFQ case, the RFQ will have a scoring panel where you can enter the scores. The total scores will appear on the RFQ and when you compare the replies on the **Compare replies** page. On the **Compare replies** page, you can also compare other reply data, such as the line price, receipt date, and total price.

After you choose a bid or bid lines, you can accept all or some lines and reject the rest. Acceptance journals, rejection journals, and corresponding reports are generated, and will be printed, archived, and sent according to your print settings. When you accept a bid or specific lines in a bid, either a purchase agreement or a purchase order is generated, or a purchase requisition is updated, depending on the purchase type of the RFQ. You can create a trade agreement that you can use later for any of the replies, regardless of whether you accepted or rejected them.

An RFQ case has two statuses: lowest and highest, you can view the status on the list page for **All request for quotations**. The lowest status is the least advanced stage of any line in the RFQ case, and the highest status is the most advanced stage of any line in the RFQ case. For example, say that an RFQ case with three lines is sent to two vendors, so there are two RFQs each with three lines. All lines are *Sent*. Now a bid is entered from one of the vendors and the RFQ lines get the status *Received*. This means that out of the three lines on the RFQ case, all of them are *Sent* for one RFQ and *Received* for another RFQ. The lowest status will then be *Sent*, and the highest status is *Received*.

These statuses will be described in more detail later in this article.

Setting up RFQ functionality

Before you can create an RFQ case, you must set up RFQ information on the **Procurement and sourcing parameters** page. When you create an RFQ case, you can specify default values that are copied to the RFQ. You can specify the following default values:

- The purchase type of new RFQs: **Purchase order** or **Purchase agreement**
- The expiration date and time offset from the day the RFQ case is created.
- Solicitation type, which might default a specific scoring method to the RFQ case.
- Delivery information and payment terms.

You can override these values for a specific RFQ case.

You should also configure the amendment process. As part of this configuration, you can turn on field locking. When field locking is turned on, a procurement professional who wants to amend an RFQ must first select **Create** in the **Amendment** section of the **Quotation** tab on the RFQ case. Then, after the RFQ case has been updated with the amendment, the procurement

professional must complete the process by selecting **Finalize**. The Finalize action generates an email that notifies the vendors about the amended RFQ.

On the **Procurement and sourcing parameters** page, you select the template to use for the email notification that is sent to vendors. When a template is created in **Email templates**, it can contain the following replacement tokens:

- %RFQ case%
- %Reason for bid return%
- %Reason for amendment%
- %Amendment prepared by%
- %Company%
- %RFQ case name%
- %Expiry Date Time%
- %Date%

The %Reason for bid return% and %Reason for amendment% tokens are replaced by text that the procurement professional can enter when they complete the amendment in the **Amendment** wizard. The values for the %Amendment prepared by% and %Company% tokens are automatically taken from the RFQ. The %Date% token is replaced by the current date.

If you want to cancel an RFQ after it's been sent, you can do that from the RFQ case. For the cancellation, an email template is required to send the cancellation notification to the vendor's contact persons. The template must be selected on the **Procurement and sourcing parameters** page. When the template is created, it can contain the following replacements tokens:

- %Reason for cancellation%
- %RFQ case%
- %RFQ canceled by%
- %Company%
- %RFQ case name%
- %Date%

The %Reason for cancellation% token is replaced by text that the procurement professional can enter in the **Cancellation** wizard. The %Date% token is replaced by the current date.

If you want to use reason codes on a bid to indicate why it was rejected or accepted, you must set up reason codes on the **Vendor reasons** page.

On the **Form setup** page in Procurement and sourcing, you can configure the appearance of your printed or stored RFQ documents.

> ⓘ **Note**

For a public-sector configuration, you must use the amendment process to change an RFQ that has already been sent. When an RFQ is sent, fields are locked. Therefore, to make changes to the RFQ, you must select **Create** to start the amendment process, as described earlier. The locking behavior is controlled by the **Lock RFQ when they are sent** option on the **Procurement and sourcing parameters** page. By default, this parameter is set to *Yes*, and for a public-sector configuration, the default setting can't be changed. Therefore, although the amendment process can be handled manually in a non-public-sector configuration, it must be used for a public-sector configuration.

When you create an RFQ case of the type Purchase order and add an inventory item to the RFQ, an inventory transaction is generated that has a receipt status of Quotation receipt. Only RFQ case lines that have this status are considered when you use a master plan to calculate supplies. If you want the master plan to include RFQ case lines as an expected receipt, you must configure this behavior in the setup of master planning.

A purchasing manager or agent can create and maintain solicitation types to suit the organization's procurement requirements. Each solicitation type can be associated with a scoring method. Scoring methods consist of a set of criteria that can be used when you score bids. You must set up solicitation types, scoring methods, and scoring criteria on the **Solicitation type** and **Scoring method** pages.

Choose default fields to include in vendor RFQ reply forms

You can specify specific types of information that you want to receive from vendors when they reply to (bid on) a request for quotation (RFQ). Fields that you mark as default are included on the online form provided for vendor collaboration. To make these settings:

1. If you haven't already done so, use the Feature management page to enable the *Select RFQ fields to include in vendor RFQ reply forms* feature.
2. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
3. Open the **Request for quotation** tab.
4. Select the **Default requests for quotation** reply fields link under the **Set up default values for requests for quotations** heading.
5. The **Default request for quotation reply fields** dialog box opens.
6. The **RFQ fields included in vendor RFQ reply forms** section includes a slider for each field that is available for use in RFQ reply forms. Fields set to *Yes* in this section will be included (together with their values) in RFQ reply forms. Set the slider to *No* for each field where you want to prevent vendors from seeing data when reviewing bids. This allows you to

enter estimated or expected values during RFQ entry for internal purposes without the vendor seeing what has been entered.

You can override these settings for individual RFQ cases as needed.

Creating and sending an RFQ

You create an RFQ case, select the vendors that you want to bid on the RFQ case, and then send RFQs to the vendors. You can use Printing settings to route the RFQ report and reply sheet reports to your preferred destination.

You can manually create an RFQ case of either the **Purchase order** purchase type or the **Purchase agreement** purchase type.

If the RFQ case is of the **Purchase order** type, the following behavior occurs that deviates from other types of RFQ cases:

- When RFQ case lines are created, inventory transactions are generated that have a receipt status of **Quotation receipt**.
- When you accept a bid, a purchase order is generated.

If the RFQ is of the **Purchase agreement** type, the following behavior occurs that deviates from other RFQ cases:

- The RFQ case is used for an agreement to purchase a specific quantity or value of product over time. You must select the date range that applies to the purchase agreement and the name of the person who manages the purchase agreement.
- When you accept a bid, a purchase agreement is generated.

If the RFQ case is generated from a purchase requisition, the **Purchase requisition** type is automatically assigned. You can't manually create an RFQ case of the **Purchase requisition** type.

You can create an RFQ case from a purchase requisition only if the status of the purchase requisition is **In review** and you're assigned to do the next workflow task. The lines in the purchase requisition are automatically updated as you accept lines from bids (RFQ replies) that you received from vendors. You can't complete, reject, approve, or perform any other actions on the purchase requisition until the requisition line is updated with an accepted RFQ line or the RFQ case is canceled.

When you create an RFQ case, you can select a solicitation type. The solicitation type determines the set of scoring criteria that is used to score RFQ replies to the RFQ case.

You can add a questionnaire to an RFQ case. This questionnaire then appears on all RFQ replies after you send the RFQ. The completion of the questionnaire is a mandatory task before the bid can be submitted.

Although defaults are provided, you can change the **RFQ fields included in vendor RFQ reply forms** settings for each individual RFQ case as needed. To do so, create or open an RFQ case. Then, on the Action Pane, open the **Quotation** tab and, from the **Replies** section, select **Set RFQ reply defaults**. The **Default request for quotation reply fields** dialog box opens, which works the same as it does when setting the defaults for vendor RFQ reply forms, except your changes here will only affect the current RFQ case. For details about how to enable this functionality and how it works, see Choose default fields to include in vendor RFQ reply forms.

There are three ways to select the vendors to add to an RFQ case:

- Add the vendors one by one.
- Search for all vendors that meet specific criteria.
- Automatically add all vendors that are approved for the procurement categories that are used on the RFQ case lines.

When the RFQ case is ready, select **Send**. The Send action generates journals and reports that will be printed, archived, and sent according to your printing settings.

If you set **Use vendor for recalculating prices** and **Use vendor specific item information** to *Yes* on the **Sending request for quotation** page when you sent the RFQ to a vendor, some vendor-specific information is automatically entered in the RFQ for that vendor.

Amending an RFQ case

Occasionally, you must change an RFQ case after you send it. You might have to change an RFQ case if, for example, the receipt dates have changed, or you want additional products or different quantities of products. You can configure the amendment process so that it's either more restrictive or less restrictive.

If you configure the amendment process so that it's more restrictive, before you can modify the fields on an RFQ case that has already been sent, you must select **Create** on the RFQ case to start an amendment. After you've completed your changes, you must select **Finalize**. You're then guided through the process of adding information for the email that is sent to notify vendors about the amendment. The updated RFQ report, which includes an amendment note, is automatically attached to the email.

If you configure the amendment process so that it's less restrictive, you don't have to select **Create** before you can modify the fields on an RFQ case that has already been sent. However, you must manually add an amendment note on the RFQ and send the case again. This

approach can be used only if none of the replies (bids) have been edited. If you've entered a reply and it's in a *Received* state, the **Send** button is unavailable. In this case, you must select **Create** and then **Finalize**, as you must do in the more restrictive process. The reply is then reset to reflect the changes to the RFQ case.

If vendors use the vendor collaboration interface to enter bids, you must always use the amendment process to notify vendors about changes to the RFQ case. This process helps prevent the situation where vendors bid on an outdated RFQ case while their bid is in progress. For more information about vendor collaboration, see Vendor collaboration with external vendors.

If you want to invite additional vendors to bid, and no changes have been made to the RFQ case, you can use the **Send** button. The vendors that you added will appear on the **Send** page and will receive the email invitation.

Receiving and registering RFQ replies

When you send an RFQ, a reply sheet is automatically generated. As you receive bids on an RFQ, you must enter them via the **Request for quotation** page by selecting the action **Edit RFQ reply.** This will allow you to enter the bid information in a dedicated bid form. Initially, the **Reply progress** is *Not started*. When you select **Edit RFQ reply,** the progress status is *Purchaser is updating* until the bid is submitted. Select **Submit** when you have entered the bid information. The **Reply progress** status changes to *Submitted by purchaser*. Similarly, with vendor collaboration enabled, the **Reply progress** will update as the vendor interacts with the bid. The status then changes from *Vendor is updating* to *Submitted by vendor*. When the bid is submitted, a journal is created as *Received*. The reply (bid) has to be submitted in order to be registered as received, and only then it can be further processed as accepted or rejected.

If you need to update the bid, you should go through the same process as above and submit again.

You can only edit the **Request for quotation** form with information that relates to processing the bid, not for entering the bid. To enter or modify the bid, select **Edit RFQ reply.**

When you enter the bid information, and if the RFQ case allows for alternate lines, you can add alternate lines for lines that only have a procurement category and no catalog item specified. Select **Add alternate** to add alternate lines.

If you've entered a reply but require a new offer from the vendor, you can return the RFQ. A new journal and a report are generated, that can be sent to the vendor.

You can see an overview of all RFQs and their statuses: *Sent*, *Received*, *Accepted*, *Rejected*, *Canceled*, *Declined* on the **Request for quotation follow-up** page.

Accepting and rejecting bids, and transferring accepted bids to downstream documents

After you've identified the best bid, such as the bid that offers the best total price, you accept the bid. You can accept some lines in a bid and reject others. You can also accept lines from different vendors. If you accept some lines, you're prompted to reject all the other lines. Therefore, if you want to accept other lines, you must select **Cancel** when you're prompted. The status of the RFQ reply for each vendor that you accept bids or lines from is updated to **Accepted**.

If you, while you prepare the purchase order or purchase agreement, need to add an additional line to the RFQ, you can do so by selecting **Add line** on the **Request for quotation** page line grid. You can only view and edit this line in **Request for quotation** page. It will be visible on the bid page when it's accepted.

When you accept a bid or one or more lines in a bid, a purchase order or a purchase agreement is automatically generated. You can then reject the bids from all the other vendors.

On the reply, you can add a reason code to explain why you accepted or rejected a bid.

When you accept a bid of the **Purchase requisition** type, the purchase requisition lines will be updated with the following information that reflects the information of the accepted bid:

- Unit price
- Discount percentage
- Discount amount
- Purchase charges
- Line charges
- Vendor
- External number
- External description

The following table shows how the RFQ status changes as you accept and reject bids from vendors.

Statuses – highest and lowest

On the Vendor tab of the RFQ case, you can see the lines with the highest and the lowest status for a particular vendor. When the vendor is added, and no lines have yet been sent, both the lowest and the highest status is *Created*. When the RFQ is sent to the vendor with all lines, the status of the two lines will be *Sent*. If some lines in a bid from a vendor are accepted and

others are rejected, the rejected lines will get the lowest status, which is *Rejected*, and the accepted lines will get the highest status, which is *Accepted*.

On the RFQ case lines, you can see the highest and the lowest status per line across all vendors. If you have sent a line to all the vendors in the RFQ case and no one have responded yet both the lowest and the highest status is *Sent*. When at least one vendor responds, the highest state will change to *Received*. If you add a new vendor to the case, the lowest status will change to *Created*

The highest and the lowest status on the RFQ case is an aggregation of the status on the **Vendor** tab and the **Lines** tab.

The statuses are ranked in the following way from lowest to highest: *Created*, *Sent*, *Received*, *Rejected*, *Accepted*, *Declined*, *Canceled*.

The following table shows how the RFQ case status changes when you create an RFQ case with lines and then send it to vendors.

Expand table

Action	Lowest RFQ case status	Highest RFQ case status	Lowest RFQ case line status	Highest RFQ case line status
Create the RFQ case header and line.	Created	Created	Created	Created
Send RFQs to all vendors in the RFQ case.	Sent	Sent	Sent	Sent
Add another vendor.	Created	Sent	Created	Sent
Send the RFQ to the second vendor.	Sent	Sent	Sent	Sent

All the lines on the RFQs that are related to the RFQ case will be in *Sent* state.

The following table shows how the RFQ status changes as you receive bids and register the information on the RFQ reply sheet.

Expand table

Action	Lowest status across all lines of all RFQs	Highest status across all lines of all RFQs	Lowest RFQ case header status	Highest RFQ case header status	Lowest RFQ case line status	Highest RFQ case line status
Register one vendor's bid to an RFQ, and save it.	Sent	Received	Sent	Received	Sent	Received
Register the second vendor's bid to an RFQ, and save it.	Received	Received	Received	Received	Received	Received

In the example below you can see the highest and lowest status on the RFQ case where one bid has been received and the other bid has been accepted. When a received bid is rejected, the lowest status will change from received to rejected on the RFQ case header and line.

Expand table

Action	Lowest status across all lines of all RFQs	Highest status across all lines of all RFQs	Lowest RFQ case header status	Highest RFQ case header status	Lowest RFQ case line status	Highest RFQ case line status
Accept one of the bids (or at least one line).	Received	Accepted	Received	Accepted	Received	Accepted
Reject all the other bids.	Rejected	Accepted	Rejected	Accepted	Rejected	Accepted

Create a request for quotation

10/17/2025

This article shows you how to create a request for quotation. A purchasing agent typically performs this task. You can use this procedure in demo data company USMF or on your own data. You need to set up solicitation types before you start. After you complete this task and create and send a request for quotation (RFQ), you can enter the replies per vendor, compare them, and award the contract.

Prepare a new RFQ

1. Go to **Procurement and sourcing** > **Requests for quotations** > **All requests for quotations**.
2. Select **New**. The following purchase types are available:

    - *Purchase order* – A document that confirms the offer to buy products, or the acceptance of an offer to sell products in exchange for payment. This type is the default.
    - *Purchase requisition* – This type is automatically selected if you create an RFQ directly from a purchase requisition. If you manually select this option, you get an error.
    - *Purchase agreement* – An agreement to purchase a specific quantity or value of product over time. If you select this option, you must select the date range that applies to the purchase agreement.

3. In the **Document title** field, type a value.
4. In the **Solicitation type** field, enter or select a value.

    - If a scoring method is associated with the solicitation type, this method is the default scoring method for the RFQ that you're creating. You can change the scoring method later.
    - In the **Receipt date** field, enter a date.
    - Select the date by which you want to receive the items.
    - In the **Expiration date and time** field, enter a date and time.
    - Specify the date and time by which vendors must respond to the RFQ.

5. In the **Warehouse field**, enter or select a value. The delivery address defaults to the warehouse address.
6. Select **OK**.

Add lines

After you specify the basic information about your RFQ, specify the goods or services that you want vendors to bid on. Item is the default line type.

1. In the **Item number** field, enter or select a value. If you're using USMF, you can select *T0020*.
2. In the **Quantity** field, enter a number.
3. Select **Add line**.
4. In the **Line type** field, select *Category*. Use the *Category* line type to create RFQs for noninventory goods or services. Then select the type of goods or services from a hierarchy of procurement categories.
5. In the **Procurement category** field, enter or select a value.
6. In the **Product name** field, type a value.
7. In the **Quantity** field, enter a number.
8. In the **Unit** field, enter or select a value.

Add vendors

1. Select **Header** to change from the lines view to the header view.
2. Expand the **Vendor** section.
3. Select **Auto-add vendors**. You can add vendors to the RFQ automatically, based on the procurement category of the items requested. If there are no vendors approved for the categories included in the lines, you can add vendors manually.
4. Select **Add**.
5. In the **Vendor account** field, enter or select a value.
6. Select **Add**.
7. In the **Vendor account** field, enter or select a value. When you select a vendor, the status is *Created*. This status means that the vendor information is saved in the RFQ, but you didn't yet send the RFQ to the vendor. You can add a vendor to an RFQ regardless of the vendor status.

Send the RFQ to vendors

1. On the Action Pane, select **Send**. On the **Sending request for quotation** page, check that the vendors in the list are the ones that you want to receive the RFQ.
2. Select **Print**. This dialog allows you to print the RFQ. If you choose to print a reply sheet, the contents of this sheet are defined in **Procurement and sourcing parameters**. To choose how to print reply sheets, select **Advanced printing options** in the **Print** dialog.

One RFQ is printed for each vendor containing the lines that have the status of *Created* or *Sent*. Canceled lines and lines with registered replies aren't printed.

3. Select **Cancel**.
4. Select **OK**.
5. Close the page.
6. Close the page.

Send several RFQs at once

To send several RFQs to vendors all at once, follow these steps:

1. Select the checkbox for each RFQ that you'd like to send.
2. On the Action Pane, open the **Quotation** tab and select **Send all lines**.
3. On the **Sending request for quotation** dialog, make sure that the vendors shown are the ones that you want to send the RFQ to.
4. Select **OK**.

You can configure the system to filter the set of vendors listed on the **Sending request for quotation** dialog based on their status. To set this option, follow these steps:

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**
2. Open the **Request for quotation** tab.
3. Set **Send all lines default status** to one of the following values:
    - *All* – List all vendors regardless of their status.
    - *Created* – Only list vendors that have **Lowest status** or **Highest status** set to *Created*.
    - *Sent* – Only list vendors that have **Lowest status** or **Highest status** set to *Sent*.
4. On the Action Pane, select **Save**.

View the RFQ journal

1. Go to **Procurement and sourcing > Requests for quotations > Request for quotations follow-up > Request for quotation journals**.
2. Select **Preview/Print**.
3. Select **Original preview**.
4. Close the page.
5. Close the page.

Sealed bidding for RFQs

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Sealed bidding keeps vendor bid replies secret until they're unsealed by purchasing personnel. All bids that are related to a request for quotation (RFQ) are first available to be unsealed upon bid expiry. Before a bid is unsealed, only users with dedicated user roles and who represent the vendor can access it.

> **Important**
>
> Modifying or extending forms, or creating new forms or business logic can defeat the protection Microsoft provides for sealed bidding. You bear the risk of using any modifications, extensions, new forms or business logic, or the inability to use the sealed bidding feature due to such modifications, extensions, new forms, or business logic. Microsoft isn't responsible for any damages arising from any modifications or extensions to forms, or any new forms or business logic that you create, or any third party creates for you. Microsoft doesn't provide technical or other support for any modifications, extensions, new forms, or business logic made by you or any third party on your behalf. You are solely responsible for complying with all applicable laws or regulations concerning sealed bidding.

How bids are kept secure

Sealed bidding uses asymmetric encryption to encrypt the bid encryption key (KEK) and symmetric encryption to encrypt the actual bid. The system integrates with Microsoft Azure Key Vault to generate and manage the encryption keys that are used to encrypt sealed bids. Each bid has its own encryption key. This encryption is kept safe in a key vault that is owned by the organization that runs Dynamics 365 Supply Chain Management. The system accesses the key vault on demand, whenever encryption and decryption are required.

Setup and configuration

This section describes the prerequisites that must be in place before you can send out RFQs that require sealed bidding.

Step 1: Set up user access to sealed bidding

When you use sealed bidding, only users who are set as the contact person for a vendor can view, edit, and submit bids for that vendor until the bidding period expires. These users must have the **Vendor (external)** role, and they must be set as a contact person for the vendor account. The contact person must also have permission to do vendor collaboration, as described in Set up and maintain vendor collaboration.

Because users who have appropriate permissions and who are set up as vendor contacts can access the sealed bids for a vendor, it's important to track who those users are. The system admin who sets up users and security roles is responsible for limiting access to sealed bids to those users who are allowed to view them.

Step 2: Set up Azure Key Vault

Supply Chain Management uses encryption keys to protect all sealed bids and keep them secret until the appropriate time. It takes advantage of the capabilities of Key Vault to generate and manage the required keys. Therefore, you must set up a connection from Supply Chain Management to a key vault to enable the system.

> **Important**
>
> The key vaults that you use for sealed bidding must meet the following requirements:
>
> - If you use a sandbox for development and testing, then you must have one dedicated key vault for the sandbox and a separate one for production.
> - Each key vault must be created in an Azure subscription that is owned by your organization (not the subscription where you're running Supply Chain Management).
> - Each key vault must be used exclusively for sealed bidding. You must not use your sealed-bidding key vaults for any other purpose.

Every bid retrieves its own secret key. This key is used every time that a user views, updates, or unseals the bid.

Key Vault generates the key that is used to retrieve the secret when the vendor selects **Bid** in the vendor collaboration interface. The key then expires after a fixed time that the procurement manager sets on the **Procurement and Sourcing Parameters** page in Supply Chain Management. After the key expires, nobody will be able to view, edit, or unseal the bid.

Therefore, it's important to configure key expiration so that there's enough time for the bidding process to be completed.

In the next three steps, you'll set up the connection to Key Vault. First, you'll set up a key vault to use with sealed bidding. Next, you'll configure Supply Chain Management to communicate with the key vault. Finally, you'll set the expiration time for the key.

Step 3: Set up a key vault to use with sealed bidding

To set up your key vault, follow these steps. The order of the steps is important.

1. If you haven't already set up an Azure subscription that is separate from the subscription where you're running Supply Chain Management, set it up.
2. Set up a key vault in your separate Azure storage. Learn more in Maintaining Azure Key Vault storage.
3. Set up your Supply Chain Management app to work as a client for your key vault. Learn more in Setting up Azure Key Vault Client.

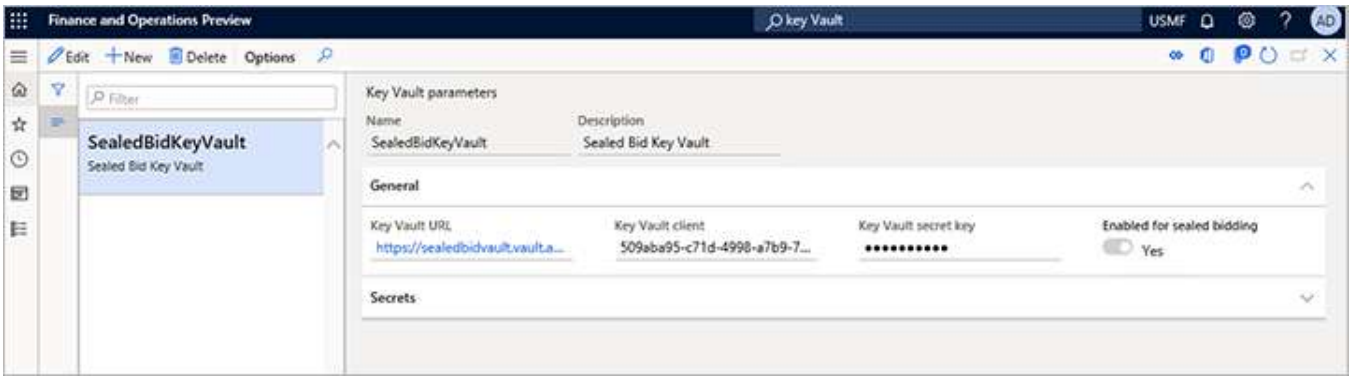
Step 4: Configure Key Vault parameters in Supply Chain Management

To configure Supply Chain Management to communicate with the key vault during sealed bidding, follow these steps.

1. Sign in to Supply Chain Management, and go to **System administration** > **Setup** > **Key Vault parameters**.

2. Select **New** to create a record, and set the following fields for it:

    - **Name** – Enter a name.
    - **Description** – Enter a description.
    - **Key Vault URL** – Enter the default URL for your key vault.
    - **Key Vault client** – Enter the interactive client ID of the Active Directory application that is associated with a key vault for authentication.
    - **Key Vault secret key** – Enter the secret reference for the certificate.
    - **Enabled for sealed bidding** – Set this option to *Yes*.



> **Note**
>
> You can enable only one key vault configuration for sealed bidding at a time. Before you disable an existing key vault configuration, you must make sure that all sealed bids that use it are unsealed. Inspect every RFQ case of the *Sealed* type, and make sure that all replies for it are unsealed.

Step 5: Set the key expiration time

To set the expiration time that is applied to the key that is generated for every new bid, follow these steps.

1. Go to **Procurement and sourcing parameters** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Request for quotation** tab, in the **Days offset** field, enter the number of days that the RFQ period should last. When an RFQ is created, the number of days that you enter here is added to the system date to define the default expiration date for the RFQ.
3. In the **Encryption key expiration day offset** field, enter the number of days that every encryption key should be valid after it's issued. After the encryption key expires, nobody will be able to view, edit, or unseal the sealed bid that uses it.

> **Tip**
>
> The value of the **Encryption key expiration day offset** field shouldn't be less than the value of the **Days offset** field.

Step 7: Set the default bid type

Every RFQ case has a bid type. The bid type defines whether that RFQ case provides open or sealed bidding.

RFQ cases that don't have a solicitation type

To set the default bid type that is assigned to new RFQ cases that aren't assigned a solicitation type when they're created, follow these steps.

1. Go to **Procurement and sourcing parameters** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Request for quotation** tab, set the **Bid type** field to *Sealed*.

RFQ cases that have a solicitation type

To set the default bid type that is assigned to new RFQ cases that are assigned a solicitation type when they're created, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Request for Quotation** > **Solicitation type**.
2. Create a new solicitation type, or select an existing solicitation type where you want to use a bid type of *Sealed*.
3. Set the **Bid type** field to *Sealed*.
4. Repeat these steps for every additional solicitation type where you want to implement sealed bidding.

> **Tip**
>
> A solicitation type doesn't have to be assigned when a new RFQ is created. If a solicitation type is assigned, the default bid type of an RFQ can retrieve it. Otherwise, the default solicitation type that is defined in Procurement and sourcing parameters can be used.

The sealed bidding process

Sealed bidding follows almost the same process that is described in Requests for quotation (RFQs) overview. The main difference is that the bid data and its attachments are kept encrypted until the bid is unsealed.

> **Important**
>
> Questionnaires aren't encrypted and shouldn't be used to collect sensitive information

Here's an outline of the process:

1. You create and send an RFQ to one or more vendors.
2. Vendors respond by submitting their sealed bid.

3. You unseal the bids after the bid entry expiration time.
4. The bids become visible, and you can evaluate and compare them.
5. After a bid is accepted, you generate a purchase order, generate a purchase agreement, or update a purchase requisition.

Create an RFQ case that uses sealed bidding

The process of creating an RFQ case for sealed bidding is almost the same as the process for non-sealed bidding. For information about how to create both types of RFQ cases, see Create a request for quotation. This section highlights a few important considerations when you create an RFQ for sealed bidding.

RFQ cases for sealed bidding must have a **Bid type** value of *Sealed*. There are three ways to assign this value to an RFQ case:

- Set the value directly on the RFQ case after you create it.
- Define sealed bidding as the default bid type for all RFQ cases in Procurement and sourcing parameters. (See the Set the default bid type section earlier in this article.)
- When you create a new RFQ case, select a solicitation type that is set up for sealed bidding. (See the Set the default bid type section.)

For sealed bidding, the RFQ case's **Expiration date and time** value establishes when the submitted bids can be unsealed. The **Expiration date and time** value on each line will match the value on the header.

You can't change the bid type after an RFQ case is sent.

Vendors respond to an RFQ

Vendors that respond to a sealed bid use the same procedure that they use to respond to open bids, as outlined in Working with RFQs in the Vendor bidding workspace. For detailed instructions about how to work with both open bids and sealed bids, see Enter and compare RFQ bids and award contracts. The only difference between processing for sealed bids and processing for open bids is that, until the bidding period has expired, procurement professionals can't open a vendor's sealed bid. Only a contact person for the vendor can open that vendor's sealed bid.

> ⓘ **Important**
>
> For sealed bidding, vendors can upload attachments only in PDF format. No other formats are accepted.

After a registered vendor user selects **Bid** on a sealed-bid RFQ, they can enter their bid data, and that data will be kept secure. Vendors can save their work as they prepare a bid, return to it as often as required, and then submit it when it's ready. Vendors can also view their bid after they submit it. If vendors must change their bid after they submit it, they can recall it, update it, and resubmit it at any time until the bidding period expires.

The following conditions apply during sealed bidding:

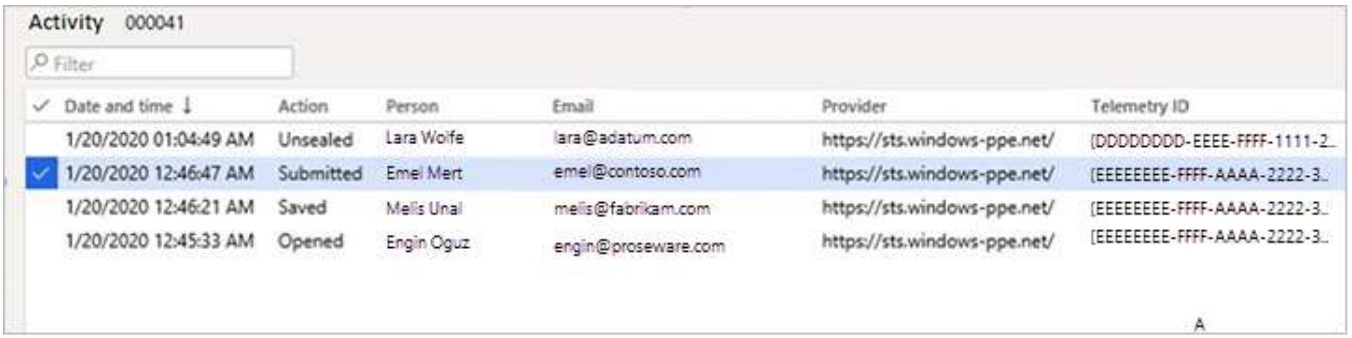
- During sealed bidding, the system creates a *Request for quotation* journal.
- When a vendor submits a bid, RFQ journals without lines are created that have a sealed price.
- After the case is unsealed, RFQ journals are created that have a price and amount. You can access this journal by selecting **Request for quotation journals** on the **All requests for quotations** page.
- The system stores a log of each action that users perform on a sealed bid. These actions include viewing, editing, and saving the bid. This log is visible both to the vendor and to procurement professionals who have access to the bid.
- As the bid progresses, procurement professionals can view its status. The status will be either *Vendor is updating* or *Submitted by vendor*.
- The status of the lines in a sealed bid remains *Sent* until the bid is unsealed.
- If the vendor chooses to decline a bid, the bid will have a **Reply Progress** status of *Declined by vendor*. This status will be visible to procurement personnel.
- Answers in questionnaires are **not** stored in encrypted form in the database. Therefore, they aren't sealed. However, they won't be visible in the RFQ user interface until the case is unsealed.

All sealed bid activities are logged

A detailed log records all user activities and actions for a bid. The activity log enables organizations to determine who updated a bid during its lifetime, and what the updates were. It also enables organizations to confirm that only authorized people accessed a sealed bid. The activity log is available on every bid's **Activity** page.

Review RFQ activity

Every interaction with the bid is logged and can be viewed on the **Activity** page. The following illustration shows an example.



Vendors can access the **Activity** page by selecting **Activity** on the **Request for Quotation Sealed bid** page. Procurement personnel can access it by selecting **Activity** on the **Request for Quotation** page. The activity log provides full visibility for vendors and for the procurement personnel who have accessed the bid. Therefore, it can reveal any unauthorized access.

Unseal sealed bids

When the **Expiration date and time** value that was configured for a sealed-bid RFQ case has passed, the **Unseal** button becomes available. Select **Unseal** to unseal all the bids for the selected RFQ case. All the bid data and attachments then become visible so that replies to the case can be managed. Additionally, journals are created that contain the submitted bid data.

The unsealing event is logged and can be viewed on the **Activity** page.

Process accepted bids

The process of comparing and approving previously sealed bids is the same as the process for open bids. For information about how to score, compare, reject, and accept unsealed bids, see Enter and compare RFQ bids and award contracts.

The RFQ activity log can never be deleted

Nobody, not even administrators and Microsoft Support, can edit or delete the RFQ activity log. This restriction helps ensure the fairness of the sealed bidding process. It also helps ensure that an accurate audit trail is maintained.

Last updated on 10/03/2025

Create a requisition that uses an RFQ

06/17/2025

This article explains how to add price and vendor information to a purchase requisition from a request for quotation (RFQ) process. The example shown in this guide can be used in the USMF demo data company, and you must be logged in as an Admin to complete all the steps. The tasks in this guide would typically be done by procurement professionals.

Create a requisition

1. Go to **Procurement and sourcing** > **Purchase requisitions** > **Purchase requisitions prepared by me**.
2. Select **New**.
3. In the **Name** field, type a value.
4. In the **Requested date** field, enter a date.
5. In the **Accounting date** field, enter a date.
6. Select **OK**.
7. In the **Reason** field, enter or select a value.
8. Select **Add line**.
9. In the **Procurement category** field, select a category in the tree, and then select **OK**.
10. In the **Product name** field, type a value.
11. In the **Quantity** field, enter a number.
12. In the **Unit** field, enter or select a value.
13. Select **Save**.
14. Select **Workflow** to open the drop dialog.
15. Select **Submit**.
16. Close the page.
17. Select **Submit**.

Reassign a workflow task

The next task is to create an RFQ to get bids from vendors for the product. In USMF demo data, the requisition workflow is set up with a rule so that if a vendor isn't selected, or the unit price is 0 for a line, a task is assigned to a specific worker to create an RFQ. To continue with this guide, you need to reassign that task to another user (yourself). You can only do this if you're logged in as an Admin.

1. Select **Workflow** to open the drop dialog.
2. Select **View history**.
3. Refresh the page.

4. Expand the **Tracking details** section.
5. In the tree, select the line that starts with *Line workflow activated on*.
6. Select **View workflow details**.
7. Expand the **Work items** section.
8. Select **Reassign**.
9. In the **User** field, select *Admin*.
10. Select **Reassign**.
11. Close the two pages.

Create an RFQ

1. Refresh the page.
2. Select **Request for quotation**.
3. In the **Buying legal entity** field, select *USMF*. You must select the same legal entity that's on the requisition line.
4. In the list, mark the selected row. If you had multiple lines on your purchase requisition, select all the lines that you want to add to the RFQ.
5. Select **OK**.
6. Refresh the page.
7. Ensure that the FactBox is open, then expand the **Related documents** section.
8. Select the link in the **Request for quotation** field to open the RFQ that you created.
9. Select **Header**.
10. Select **Add**.
11. In the **Vendor account** field, enter or select a value.
12. Select **Add**.
13. In the **Vendor account** field, enter or select a value.
14. Select **Send**.
15. Select **OK**.
16. Select **Enter reply**.
17. On the Action Pane, select **Reply**.
18. Select **Copy data to reply**. This copies data, such as the quantity and dates, from the RFQ to the reply.
19. In the **Unit price** field, enter a number. This is the price that you've received from the vendor. You might also want to enter additional information from the vendor.
20. Select **Accept**.
21. Select **OK**.

Verify that vendor and price have been transferred to the requisition

1. Close the page.
2. Select **Lines**.
3. Select **Related information**.
4. Select **Purchase requisition**.
5. Select the line that was transferred to the RFQ. Verify that the price and vendor have been copied to the requisition.
6. Select **Workflow** to open the drop dialog.
7. Select **Complete**.
8. Select the page.
9. Select **Complete**.

Create a scoring method for RFQs

Article • 05/02/2022

This procedure shows you how to create a scoring method. A scoring method is a set of criteria that can be used to compare bids that are sent in reply to a request for quotation (RFQ). For example, you might want to rate a vendor on past performance, or rate whether the company is environmentally friendly or a good collaborator, or you might want to compare bids based on price. The scoring method can be associated with a solicitation type as the default scoring method for RFQs of that type. These tasks would typically be carried out by a purchasing manager. You can use this procedure in demo data company USMF or on your own data.

1. Go to Procurement and sourcing > Setup > Request for quotation > Scoring method.
2. Click New.
3. In the Name field, type a value.
4. In the Description field, type a value.
5. Click Save.
6. Click New.
7. In the Name field, type a value.
8. In the Description field, type a value.

    - This description is shown along with the scoring method name when a scoring method is selected for an RFQ.

9. In the Range from field, enter a number.

    - The range limits what the procurement professional can enter as a score. When there are multiple scoring criteria on an RFQ, the scores that have been entered are added to each other and the sum is made available to allow the bids to be compared.

10. In the Range to field, enter a number.
11. Click New.
12. In the Name field, type a value.
13. In the Description field, type a value.
14. In the Range from field, enter a number.
15. In the Range to field, enter a number.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create solicitation types and scoring criteria for RFQs

06/17/2025

This guide shows you how to create a solicitation type and associate this with a scoring method. It also shows how to use the solicitation type on a request for quotation (RFQ) which then sets the default scoring method. These tasks would typically be carried out by a purchasing manager. You can use this procedure in demo data company USMF or on your own data. You need to have a scoring method available before you start.

Create a solicitation type

1. Go to **Procurement and sourcing** > **Setup** > **Request for quotation** > **Solicitation type**.
2. Select **New**.
3. In the **Name** field, type a value.
4. In the **Description** field, type a value.
5. In the **Scoring method** field, select the scoring method that you want to use for this solicitation type.
6. Select **Save**.
7. Close the page.

Use the solicitation type

1. Go to **Procurement and sourcing** > **Requests for quotations** > **All requests for quotations**.
2. Select **New**.
3. In the **Solicitation type** field, select the solicitation type that you created.
4. Select **OK**.
5. Select **Scoring criteria**. The scoring criteria that are shown are the ones from the scoring method that you associated with the solicitation type. You can choose to add or delete criteria on this page. It's also possible to add new criteria by copying them from other scoring methods.
6. Select **Copy criteria**.
7. In the **Scoring method** field, enter or select a value.
8. Select **OK**.
9. Close the page.

Responding to vendor questions on Request for quotations

10/03/2025

When your agency has sent a request for quotation (RFQ), vendors sometimes have questions that pertain to the request. Vendors that have questions related to an RFP can submit their questions and read the answers on **Vendor collaboration** page, when you make that page available to them. When vendor questions are accepted, **Questions and answers** is available on the **Request for quotation bid** page on **Vendor collaboration**, and for your agency through the **Request for quotation** page, **Questions and answers**.

Users can publish answers to vendor questions more than once. Vendors can't no longer post questions after a vendor is selected and the RFQ is awarded, or after the cutoff date for questions is reached.

Allow questions and answers to be used in RFQs

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. Open the **Request for quotation** tab.
3. Set the following options as needed:

    - **Allow vendor questions**: Enables or disables vendor questions for RFQ cases. You must set this to *Yes* to use the features described in this article.
    - **Default direct response**: When you reply to a question, you can choose to reply to all vendors who received the RFQ or to reply only to the specific vendor who submitted the question. You can make choose this option each time you reply, but this setting controls the default. If you usually reply to all vendors, set this to *No*. If you usually reply to individual vendors, set this to *Yes*.

Setting up for vendor questions

When creating a request for quotation, you determine whether vendors can ask questions about the RFQ.

1. Navigate to **Procurement and sourcing** > **Requests for quotations** then click **New** > **Request for quotation**
2. In the **New request for quotation** page, **Header** to set the **Vendor question options** fields to allow questions before a certain date.
3. Set the **Allow vendor question** option to **Yes** so the vendors can enter questions. Users can enter and answer questions and designate commonly asked questions to publish for

vendors when the RFQ is sent to vendors.
4. Optional: In the **Cutoff date** field, define the final date by which questions must be submitted. If no cutoff date is entered, questions are accepted as long as the RFQ is open and accepting bids.
5. Click **Save** to save the RFQ.
6. Click **Send** to send the RFQ to the vendors for bidding.

Entering and replying to vendor questions

Vendors enter questions in the **Vendor Collaboration** > **Request for quotations bid** page, **Vendor questions** FastTab. The question is visible only to the vendor and users.

Entering a vendor question

1. In Vendor collaboration, in the **Request for quotation bid** page, click **Questions and answers**, and then click + **Ask a question**.

   > ⓘ **Note**
   >
   > Alternately, a user can enter questions for a vendor on the **Request for quotation** page, by clicking **Manage replies**, **Edit RFQ reply** then clicking **Questions and asnwers**.

2. On the **Question** field, enter the question text.

3. Click the **Submit**. Repeat steps 1-3 to add a question.

4. When done, click **Save** to save your questions.

Replying to a single vendor

The question and answer are visible only to the vendor and users.

1. On the **Request for quotations** page, click the **Questions and answers** to display the **Questions and answers** page.
2. Click **Edit**.
3. Enter text in the **Answer** field to respond to the vendor's question.
4. Check the **Direct response** box.
5. Click **Save** to save your replies.
6. Click **Send answers** to send the answers to the vendor.

Replying to all vendors

If you receive the same question from multiple vendors, you can group the questions and respond with one answer. All vendors receive notification when the commonly asked questions and answers are published. The vendors and anyone with access to the request for quotation can view the summary of questions and answers.

1. On the **Request for quotations** page, click the **Questions and answers** to display the **Questions and answers** page.
2. Click **Edit**.
3. Choose a code for the common question, such as the letter 'a.'
4. For each line that asks a similar question, enter the code in the **Group code** field. For example, for each line asking about item color, enter 'a.'
5. Choose one of the lines with the code value, and enter the question and answer the way you want them to read in the summary that will be available when the questions and answers are published (**Group question, Group answer** fields).
6. Optional: You can mark the **Direct response** check box to send the answers only to the selected vendors.
7. Click **Save** to save your answers.
8. Optional: You can revert the questions and answers to the previously published values if you want to undo your changes.
9. Click **Send answers** to send your answers to the vendors.

Changing RFQ to allow or disallow questions

You can make changes to allow or disallow questions to RFQs until the RFQ is awarded. You can also extend or shorten the time frame in which vendors can submit questions. For published RFQs, you must modify a request for quotation to allow or disallow vendor questions or adjust the time frame for questions.

> ⓘ **Important**
>
> If you amend an existing RFQ for the purpose of allowing vendor questions, the system will clear all existing responses when you resend the RFQ.

Enter and compare RFQ bids and award contracts

06/17/2025

This article explains how to enter replies to a request for quotation (RFQ), score and compare bids, and then award the contract to one of the vendors. You can use this procedure in the **USMF** demo data company.

Before you start this procedure, you must have an RFQ that has two lines, and that has been sent to at least two vendors. To create this RFQ, complete the Create a request for quotation procedure. Scoring criteria must also be set up before you can complete this procedure.

You can enter the bid as either a vendor or a procurement professional. Learn more in Set up and maintain vendor collaboration.

Enter a reply as a vendor

1. Go to **Vendor collaboration** > **Workspaces** > **Vendor bidding**.
2. In the **New bid invitations** list, find an RFQ that was recently sent. Select the RFQ to review what was requested.
3. Select **RFQ attachments** to review any attachments that have been added.
4. Select **Bid** to make the fields editable. Notice that the **Bid progress** field is set to **Vendor is updating**.
5. On the header and lines, enter the values from the bid reply.
6. If any attachments should be added to the bid, select **Bid attachments**.
7. Select the **Bidding guiding items** FastTab to view whether any documents are required.
8. Select the **Amendments** FastTab to view whether the RFQ has been amended.
9. Select the **Questionnaire** FastTab. Any questionnaires that appear here must be answered.
10. Select the **Line details** FastTab to view extended information about the line.
11. Select **Reset from RFQ** only if you must reset the values that have been entered to the original RFQ values.
12. You can save the bid at any time and do additional processing later, provided that the expiration date and time haven't passed. In this case, you can find the bid in the **Bids in progress** list in the **Vendor bidding** workspace.
13. When the bid is ready to be sent, select **Submit**. If you don't want to bid, select **Decline**. Submitted bids are available in the **Submitted bids** list in the **Vendor bidding** workspace.
14. After the bid is submitted, you can recall it at any time before the expiration date and time. When a bid is recalled, it isn't treated as submitted. When the bid is accepted or rejected by the procurement department, it appears in either the **Awarded bids** or **Lost bids** list in the **Vendor bidding** workspace.

Enter a reply from a vendor as a procurement professional

1. Make sure that the permission to edit vendor bids is set up. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**. On the **Request for quotations** tab, set the **Purchaser can edit vendors bid** option to **Yes**.
2. Go to **Procurement and sourcing** > **Requests for quotations** > **All requests for quotations**.
3. Select an RFQ that has a status of **Sent**, and then select the link in the **Request for quotation case** field.
4. Select **Manage replies**. The page that appears shows an RFQ for each vendor that was invited to bid.
5. Select an RFQ that hasn't been replied to. (The **Reply progress** field should be set to **Not started**.)
6. Select **Edit** > **Edit RFQ reply**. The **RFQ reply** page appears. As a procurement professional, you can now enter the reply on behalf of the vendor. Notice that the **Bid progress** field is set to **Purchaser is updating**.
7. Enter the bid data. When you've finished, select **Submit**.

Score the bids

1. On the **All requests for quotations** page, select the RFQ case that you want to score replies for.
2. Select **Manage replies**.
3. Select the reply to score.
4. Select **Header** so that you can view the scoring for the bid.
5. On the **Bid scoring** FastTab, enter a number in the **Score** field for one of the scoring criteria. If you hover over a scoring criterion, a tooltip shows the range that the score must be within. In this demo, you can enter a number between 1 and 5 for any of the scoring criteria.
6. Repeat step 5 for another scoring criterion.
7. If the RFQ case has a questionnaire that was sent to the vendors, you can enter the vendor responses on the **Questionnaires** FastTab.
8. Close the page.
9. Repeat steps 1 through 8 for all the other bids.

Compare the replies

1. On the Action Pane, on the **General** tab, select **Compare replies**.

2. In the **Rank** field, enter a number.

    - This page shows the bids, together with the header and line information, and also the total score at the header level. You can compare the lines by sorting the grid so that comparable lines are next to each other. The following information is also included:
    - **Quantity** – The quantity that the vendor quoted. This quantity might not equal the quantity that is specified in the RFQ.
    - **Net amount** – The price that the vendor quoted for the items on the line, minus any discounts.
    - **Deviation** – The number of days by which the receipt date on the bid header or line differs from the requested receipt date on the RFQ header or line. You can enter a rank for each bid.

3. Select the header line for the other bid that you want to rank.
4. In the **Rank** field, enter a number.
5. Select **Save**.

Reject a bid

1. Select the header line for the bid that you want to reject. You can accept, reject, or return only one bid or the lines on only one bid at a time.
2. Select the **Mark** check box.

    - If you select the **Mark** check box on the header of the bid, all the lines are also marked. To reject or accept only some of the lines on the bid, you can mark just those lines. Additionally, you can accept one vendor's bid for some lines of an RFQ and then award other RFQ lines to a different vendor. However, you must do one bid at a time.
    - If alternate lines are present, you can accept either the original bid line or its alternate, but not both.

3. Select **Reject**.
4. Select **Parameters**, and then, in the **Reason reject** field, enter or select your reason for rejecting the bid. The reason is stored in the reply.
5. Select **OK**.
6. Select **OK**.

Accept a bid

1. Select the bid to accept, and then select the link in the **Request for quotation** field. If you're on the **Compare request for quotation replies** page, the highlighted bid that has focus is the bid that the system will consider during the Accept action. You can accept lines from only one bid at a time.
2. On the Action Pane, select **Reply**.
3. Select **Accept**. If you marked only specific lines, the Accept action will include only those lines. If you want to accept all the lines on the bid, you don't have to mark the lines.
4. Select **Parameters**, and then, in the **Reason accept** field, enter or select your reason for accepting the bid. The reason is stored in the bid.
5. Select **OK**.
6. Select **OK**. When you select **OK**, a purchase order is generated based on the lines that are included in the RFQ acceptance. If there are other bids that haven't been processed (accepted, rejected, or returned), the system prompts you to reject them.

View the purchase order that is generated

On the Action Pane, on the **General** tab, select **Purchase order**. The page that appears shows the purchase order that was generated when you accepted the bid.

Purchase order overview

This article provides general information about purchase orders (POs) and links to other articles that are related to the various stages that a PO goes through.

A purchase order (PO) is a document that represents an agreement with a vendor to buy goods or services. The document also helps keep track of product receipts that are made toward the order and, later, the accounting of vendor invoices that the vendor bills toward the order.

The **Purchase orders** page contains an overview of the available orders and lets you modify those orders. When you open a PO, you can select the **Header** view, which contains information that is specified only one time for each PO, such as the vendor details. Alternatively, you can select the **Lines** view, where you can modify order lines. Typically, you'll switch between these two views as you modify POs. Charges aren't listed directly on the **Purchase orders** page, but are accessed via menus on the order header and lines.

There are many reports where you can view information about POs, product receipts, and vendor invoices. These reports are found in the **Procurement and sourcing** and **Accounts payable** modules.

The **Purchase order preparation** and **Purchase order receipt and follow-up** workspaces let you view lists of POs in the various states that they've progressed to. They also provide a summary of the actions that must be taken. The **Purchase order preparation** workspace is focused on PO creation and review, processing of the order through approval, and confirmation with the vendor. The **Purchase order receipt and follow-up** workspace is focused on processing the receipt of goods or services against POs. It includes lists that give insight into receipts that are overdue, or that will soon be due for delivery by the supplier. These workspaces aren't used to perform the related receipt activities that are done in the warehouse. Those activities are performed by using pages in the **Inventory management** and **Warehouse management** modules. Processing of vendor invoices should be done by using the **Vendor invoice entry** workspace, and payments should be done by using the **Vendor payments** workspace.

The following articles provide an overview of the various stages that a PO goes through:

- Create purchase orders
- Approve and confirm purchase orders
- Product receipt against purchase orders
- Overview of vendor invoices

Types of purchase orders

There are three types of purchase orders. When you create a purchase order, you must specify the type. Set up a default order type for new orders on the **Procurement and sourcing parameters** page.

Expand table

PO type	Description
Journal	Use this type to create a draft order. This type doesn't affect stock quantities or generate inventory transactions. The PO journal lines aren't included in master scheduling.
Purchase order	Use this type to create purchase orders when you confirm orders with a vendor, and as the orders are processed through receipt and invoicing before payment is made to the vendor. This type of purchase order is the most common.
Returned order	Use this type when you return goods to the vendor. This type of order requires that you specify the return material authorization (RMA) number that the vendor gives you. Specify the RMA number on the **General** tab of the purchase order. The order lines must have negative quantities.

Purchase order data entities

If you integrate purchase orders with external systems through OData or the Data management framework, be aware of which purchase order types each standard data entity covers.

The *Purchase order headers V2* (`PurchPurchaseOrderHeaderV2Entity`) and *Purchase order lines V2* (`PurchPurchaseOrderLineV2Entity`) data entities include only records where the purchase order type is *Purchase order*. By design, they don't return *Returned order* records. Therefore, queries against these entities don't include purchase return orders, and you can't create or update purchase return orders through them.

There's currently no separate standard public data entity for purchase return order headers and lines.

If your integration must read or write purchase return orders, use the extensibility model to create a custom data entity that exposes the header and line data that you require. Before you use a custom entity in production, test its validations, table relations, cross-company behavior, and create and update operations.

> **Important**

Don't modify the generated SQL view behind a standard data entity, and don't change the application database directly. These approaches bypass supported application validation, and your changes can be overwritten when the database is synchronized.

Purchase order statuses

POs include several status fields that indicate the progress of the order. You can see all these fields in the **Header** view of the order. You can also see some of these fields in the grid overview of all orders. The **Purchase order status** field shows the status for quantities on the order. The following values are available:

- *Open order* – Orders are created, and quantities are on order.
- *Received* – You received the full quantity on the order, but you didn't invoice it yet.
- *Invoiced* – You invoiced the full quantity on the order. If an order is partially received or invoiced, *Received* status and *Invoiced* status aren't appropriate. Therefore, the order still has a status of *Open order*.
- *Canceled* – You confirmed an order but later canceled it. Therefore, this status indicates that there are no longer any open quantities on order.

The **Document status** field helps you quickly review the order's progress in terms of documents that are processed. It shows the status of the most recent document that is completed for the order. The following values are available:

- *None* – No document is processed for the order yet.
- *Purchase inquiry* – The system generated a purchase inquiry, and the order is awaiting feedback from the vendor.
- *Purchase order* – The system processed confirmation of the order.
- *Product receipt* – The system processed product receipt on the order.
- *Invoice* – The system accounted an invoice with the order.

Use the **Approval status** field when a PO goes through a review process or workflow. The following values are available:

- *Draft*, *In review*, and *Rejected* – These statuses are used only when an approval workflow is used for the PO.
- *Approved* – This status is assigned to orders that have completed workflow approval. Orders that are created without using an approval workflow receive a status of *Approved* immediately.

- *In external review* – This status is used in scenarios where a purchase inquiry is sent to the vendor, so that the vendor can confirm terms of the PO. This status is also used in the process that is initiated by the **Confirmation request** action. For this process, the vendor is asked to confirm terms of the PO by connecting to your system and registering whether it confirms or rejects the order.
- *Confirmed* – This status is assigned after the order has been confirmed. Typically, this status is the last approval status that is assigned to an order.

Related information

- Create purchase orders
- Approve and confirm purchase orders
- Product receipt against purchase orders
- Overview of vendor invoices

Last updated on 09/08/2026

Approve and confirm purchase orders

This article describes the statuses that a purchase order (PO) goes through after you create it, and the effect of enabling change management on POs.

After you create a PO, it might need to go through an approval process. After the vendor agrees to the order, set the PO to a status of *Confirmed*.

Approval of purchase orders

POs that don't use change management have a status of *Approved* as soon as you create them, whereas POs that use change management have a status of *Draft* when you first create them. A PO that you create by firming a planned order from master planning is always set to a status of *Approved*, regardless of the change management settings. A PO creates inventory transactions only when it reaches the *Approved* status. Therefore, that inventory doesn't appear as available for reservation or marking until the order is accepted.

Enable change management for POs by setting the **Activate change management** option on the **Procurement and sourcing parameters** page. When you enable change management, POs must go through an approval workflow after you complete them. Supply Chain Management has a workflow process editor where you can define a workflow to represent your approval process. This workflow can include rules for automatic approval, rules that determine who will be assigned to approve particular POs, and rules for escalating a workflow that has been waiting for approval for a long time. You can enable the change management process for all vendors or for specific vendors. You can also set up the process so that it can be overridden for individual POs.

When you enable change management, POs move through six approval statuses, from *Draft* to *Finalized*. After an order is approved, users who want to modify it must use the **Request change** action.

Expand table

Approval status	Description	Request change is enabled
Draft	The PO is a draft and hasn't been submitted for approval in the PO workflow.	No
In review	The PO was submitted for approval in the PO workflow.	No

Approval status	Description	Request change is enabled
	Approval is pending.	
Rejected	The PO was rejected during the approval process.	No
Approved	The PO was approved.	Yes
Confirmed	The PO was confirmed. A PO can't be confirmed until it has been approved.	Yes
Finalized	The PO was made final. It's now financially closed and can no longer be changed.	No

Confirming purchase orders

POs that have an approval status of *Approved* can go through additional steps before they're confirmed. For example, you might have to send a purchase inquiry to the vendor to inquire about prices, discounts, or receipt dates. In this case, you can set the PO to the *In external review* status by using the **Purchase inquiry** action.

Vendors that are set up to use the vendor collaboration interface can review orders on the portal, and approve or reject them. During this review process, the PO has a status of *In external review*. The vendor collaboration interface can be configured so that a confirmation from the vendor automatically confirms the order in Supply Chain Management. In this scenario, you must schedule the **Confirm accepted purchase orders from vendor collaboration** batch job, which is responsible for processing and confirming your POs. Alternatively, you can manually confirm a PO after you receive confirmation from the vendor. If a vendor rejects a PO, the rejection is received together with the reason for the rejection and suggestions for changes. In this case, the status of the PO remains *In external review*.

There's also an option to generate a pro-forma confirmation for an order before the actual confirmation has been processed. This option just creates a report that you can share with the vendor. It doesn't create any journal information.

After the vendor has agreed to the order, the next step is to record the PO as committed. You can complete this step by using either the **Confirmation** action or the **Confirm** action. Both these actions set the approval status of the order to *Confirmed*. Confirmation of an order initiates two additional processes:

- A journal is created to store an exact copy of what was confirmed in the system. Sometimes, orders require changes, and additional journals are created after the updated order is

confirmed. These journals let you view the history of the various versions of the order that were confirmed.

- Accounting distributions are created, and order checks and budget checks occur if this functionality has been enabled. If either check fails, you receive an error message that states that changes must be made to the PO before it can be confirmed again.

> **Note**
>
> The purchase order confirmation report includes price and amount information—the **Unit price**, **Discount**, **Discount percent**, and **Amount** columns, together with the totals table—only when the vendor is set up to include that information. This behavior is controlled by the **Purchase order prices/amount** option on the vendor record. The system applies this option when it creates the purchase order confirmation. Therefore, if you change the option on the vendor after an order was already confirmed, the confirmation that was already generated isn't affected. If you change the setting, confirm the purchase order again to generate a new confirmation report. (You don't have to delete the earlier confirmation.) The **Purchase order prices/amount** field on the **General** tab of the **Procurement and sourcing parameters** page sets the default value for newly created vendors only. It doesn't change existing vendors or existing confirmations.

A vendor might request some type of assurance that payment will be provided for a purchase. There are various methods for providing this guarantee within accounts payable processes. For example, the **Prepayment** action reserves funds for the PO, and this prepayment is recorded on the PO.

Changing purchase orders

In some situations, you might have to change a PO after it has reached an approval status of *Approved* or *Confirmed*.

If the PO was created by using a change management process, you can make changes by recalling the order or, if the order has already been approved, by using the **Request change** action. In this case, the approval status is changed back to *Draft*, and you can then modify the order. After you've finished making changes, you might have to submit the PO for reapproval. You can configure the types of changes that require reapproval by using a **Re-approval rule for purchase orders** policy rule on the **Purchasing policies** page.

If part of the ordered quantity for a PO line has been delivered, you can't change the ordered quantity when the purchase order is in *Draft*. However, you can change the **Deliver remainder** quantity on the line for the purchase order that is in *Draft* status.

After an order has been confirmed, you can no longer delete it. However, you can cancel the total quantity or any remaining quantity on an order, provided that the quantity hasn't been received or invoiced. You can then use the **Finalize** action to prevent further processing.

Canceling purchase orders

Use the **Cancel** action on the header to cancel a purchase order (PO).

If the quantity is partially registered, received, or invoiced, you can cancel only the remaining quantity that isn't registered, received, or invoiced. The order quantity is reduced accordingly. When you update the quantity on the line, you also update the line status. For example, the original quantity on the line is 5, and you receive a quantity of three. In this case, you can cancel only two. The line updates to *Received* status.

If you add a delivery remainder to the order line and it exceeds the quantity on the order line, the **Cancel** action doesn't cancel the excess quantity. Instead, the line remains in *Open order* status because it has a remaining quantity. For example, the original quantity on the line is 5, and the delivery remainder is 7. If you cancel the order, you cancel five, and a quantity of two remains, as you can see in the inventory transactions.

To cancel the whole quantity on a PO line, cancel the delivery remainder quantity on the line. The line updates to *Canceled* status.

If a PO is under change management, you must submit any change, such as cancellation of the order or the delivery remainder, to the workflow system for approval before the process can be completed and the inventory transactions can be updated as canceled.

Related information

- Purchase order overview
- Create purchase orders
- Product receipt against purchase orders
- Vendor invoices overview

Last updated on 09/08/2026

Configure purchase order confirmation batch jobs

Use the *Confirm purchase orders* batch job to confirm multiple purchase orders at the same time. For example, you can schedule the job to confirm all approved purchase orders at the end of each day, confirm drop-shipment purchase orders that are generated from sales orders, or confirm time-sensitive purchase orders before supplier cutoff times.

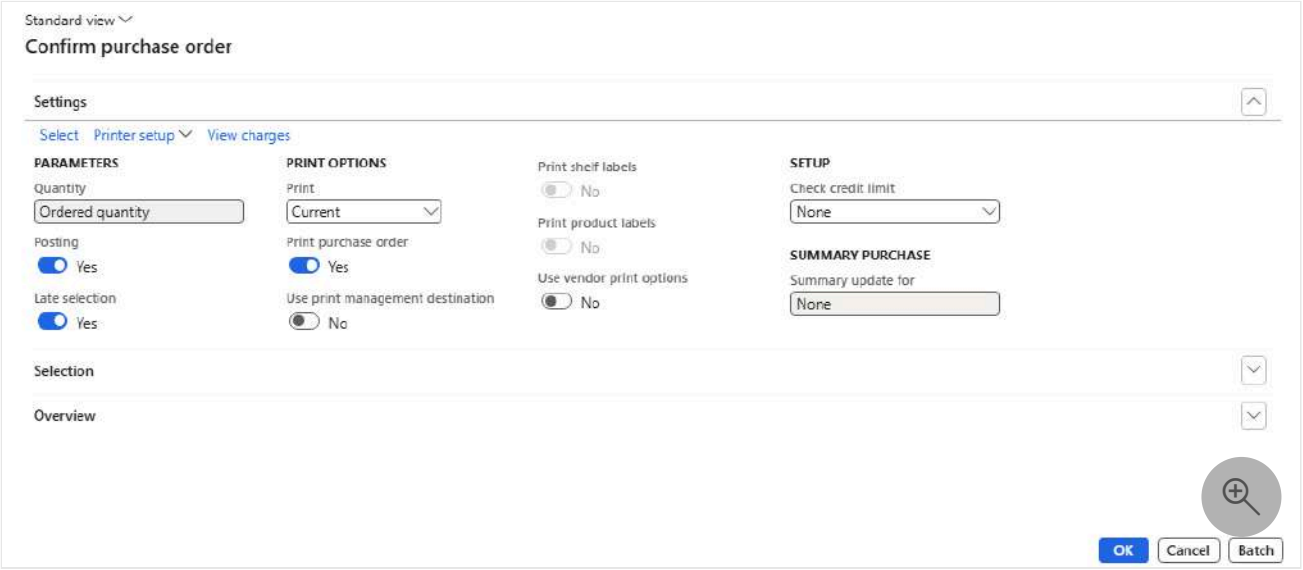
This article describes how to schedule the job, prevent failures when a vendor is on hold, and configure options for printing or emailing purchase order confirmation journals.

Schedule the auto-confirmation batch job

The *Confirm purchase orders* batch job processes purchase orders in bulk. Follow these steps to schedule and configure the batch job:

1. Go to **Procurement and sourcing** > **Purchase orders** > **Purchase order confirmation** > **Confirm purchase orders**.

2. The **Confirm purchase order** dialog opens.



3. On the **Settings** FastTab, set **Print** to one of the following values.

   - *Current* – The system generates and sends a purchase order confirmation journal for each purchase order it confirms while it works. Use this option when each confirmation must be sent even if another purchase order in the same batch fails.

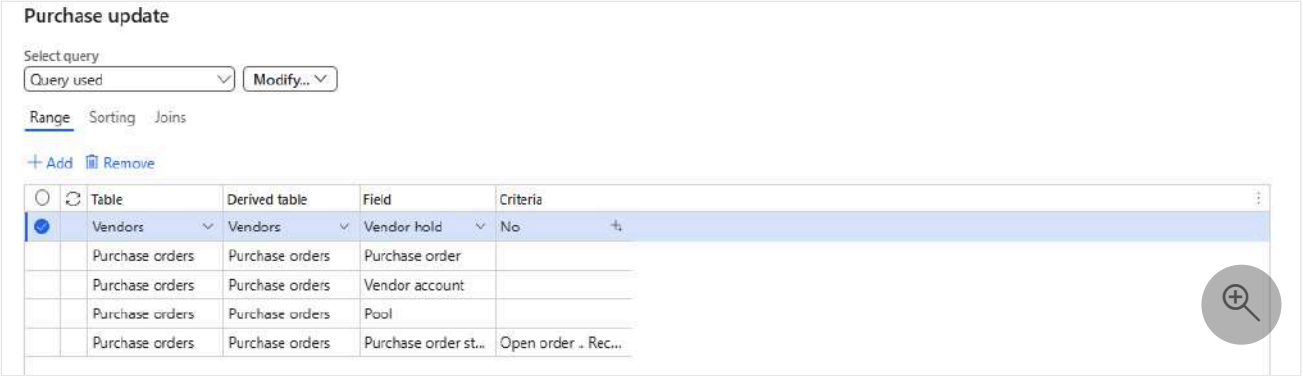
- *After* – The system confirms all purchase orders before it generates and sends any purchase order confirmation journals. With this option, if one purchase order in a batch fails validation (for example, because of a credit management check or because change management is enabled), the system might not send confirmations for any of the purchase orders in the batch, including those that were confirmed successfully. Use this option if your organization needs to use the standard (global) print management destination set on the **Print management setup** page.

4. Set other options on the **Settings** FastTab as needed. Use tooltips to learn more about each setting. The following settings are recommended for most organizations:

    - **Print purchase order** – Set to *Yes* to print or send purchase order confirmation journal after the system confirms a purchase order.
    - **Use print management destination** – Choose how the system decides where to send purchase order confirmation journals. Set to *No* to use the destination specified in the local **Printer setup** dialog (as set on this page). Set to *Yes* to use the standard (global) print management destination set on the **Print management setup** page (available from **Accounts payable** > **Setup** > **Forms**). If you want to deliver confirmations by email, set this option to *No* so you can use the local printer setup to configure the email. See the next section for details. When this option is set to *Yes*, the system always uses the *After* print behavior in batch processing, even if *Current* is selected in the **Print** field.
    - **Late selection** – Set to *Yes* to select the most recent purchase orders for confirmation each time the batch job runs.

5. The batch job fails if it tries to confirm an order or generate a document for a vendor that's on hold, so you should exclude these vendors. To exclude purchase orders for vendors that are on hold, select **Select** on the **Settings** FastTab toolbar and then make the following settings in the **Purchase update** dialog:

    - Go to the **Joins** tab. Expand the **Tables** node and select **Purchase orders** in the tree. On the toolbar, select **Add table join**. Set **Show details** to *Yes* and select the row where **Relation** is *Vendors (Vendor account)* and **Relation source** is *VendTable : InvoiceVendor*. On the toolbar, select **Select**.
    - Go to the **Range** tab. On the toolbar, select **Add**. Add a row where **Table** = *Vendors*, **Field** = *Vendor hold*, and **Criteria** = *No*.
    - Select **OK**.



6. Select the **Batch** button to open the batch job scheduler. Make the following settings on the **Parameters** FastTab:

    - On the toolbar, select **Recurrence**. Set up the run schedule for the batch job, and then select **OK**. Choose a recurrence interval that reflects your order volume, other batch workloads, and vendor lead times. In environments that process large volumes of data or run many batch jobs, don't run this job more often than every 10 minutes.
    - Set **Batch processing** to *Yes*.
    - Set other batch options as needed.
    - Select **OK**.

7. Select **OK** to save the batch job configuration.

> ⓘ **Note**
>
> When **Print management destination** is set to *Yes*, the system always uses the *After* print behavior in batch processing, even if you set **Print** to *Current* (the batch job always saves and runs with **Print** set to *After*). With *After*, if one purchase order in a batch fails validation (for example, because of a credit management check or because change management is enabled), the document sending step might not run for any purchase orders in the same batch, even those that were confirmed successfully.

Where to find confirmation history after a recurring batch runs

The **Confirmations** page that you open from a purchase order shows *posting-parameter* records. These records are staging data for the current batch operation, not the permanent confirmation history. On every recurrence, the *Confirm purchase orders* batch job clears the existing records in its posting-parameter set before it selects the purchase orders for the next run. As a result, purchase orders from a previous run no longer appear on that page. This behavior is by design, and the **Late selection** setting isn't the reason that the previous staging records are cleared.

The permanent confirmation history isn't affected. To review confirmations for audit or historical purposes, go to **Accounts payable** > **Purchase orders** > **Purchase order confirmations**, and then filter by purchase order, vendor, confirmation number, or date. Build reports and integrations from the purchase order confirmation journals rather than from the posting-parameter staging tables.

Configure email delivery of purchase order confirmation documents

To set your system to deliver purchase order confirmation journals by email after auto-confirmation, follow the procedure in the previous section, but be sure to use the following settings in the **Confirm purchase order** dialog:

- Set **Print purchase order** to *Yes*.
- Set **Use print management destination** to *No*.
- Set **Print** to *Current*.
- Set **Late selection** to *Yes*.
- On the toolbar, select **Printer setup** > **Purchase order**. Select **Email** and then set up your email options for delivering purchase order confirmation journals.
- If you want to send or print a copy of the purchase order confirmation journals, select **Printer setup** > **Purchase order (copy)**. Then set up the destination options for the copy.

Related information

- Approve and confirm purchase orders
- Purchase order overview
- Vendor collaboration with external vendors

Last updated on 09/08/2026

Calculate requested ship dates for purchase orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Purchase orders in Microsoft Dynamics 365 Supply Chain Management can include both a *requested receipt date*, which specifies when the order should be received at your company's location, and a *requested ship date*, which specifies when the vendor should ship from their location. The system calculates requested ship dates based on the requested receipt date, lead time, the transport days setup, the vendor ship calendar, and other settings. The transport days setup lets you define the number of days that are required to transport items between pairs of addresses. Vendor ship calendars define the days when each vendor can ship.

If any of your vendors don't have specific delivery terms (such as delivery duty paid, delivery at place, or delivery at place unloaded), you must tell them when they should ship.

When master planning suggests a date for placing a planned order, it also considers the transport days setup. This approach helps ensure that the goods arrive on time.

Prerequisites

To calculate requested shipment dates for purchase orders, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management version 10.0.40 or later.
- You must be using Planning Optimization, not the deprecated master planning engine.

Enable supplier requested and confirmed shipment dates for a company

To enable supplier requested and confirmed shipment dates for a legal entity (company), follow these steps:

1. Use the company picker to select the legal entity where you want to calculate requested shipment dates for purchase orders.
2. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
3. Open the **Delivery** tab.
4. Set **Supplier requested and confirmed shipment dates** to *Yes* to make the feature available to the current legal entity. Set to *No* to hide the feature for the current legal entity.
5. Repeat this procedure for each legal entity where you want to turn on (or turn off) the option.

> ⓘ **Note**
>
> If you use intercompany trade, then each of the legal entities involved must have this option turned on to allow requested shipment dates to be calculated for intercompany purchase orders.

Specify whether requested ship dates can be in the past

Sometimes, because of your requested receipt date, plus the lead time, transport days, vendor ship calendar, and other settings, the system might calculate a requested ship date that's in the past. To specify how the system handles this situation, follow these steps:

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Delivery** tab, in the **Requested ship date in the past** field, select one of the following values:
    - *None* – Always allow requested ship dates to be in the past, without showing a warning.
    - *Allow with warning* – Allow requested ship dates to be in the past, but show a warning message to users when this situation occurs.
    - *Disallow with warning* – Always find the next earliest ship date that's after the current date, and show a warning message to users when this situation occurs.

Set up purchase transport days

To define the number of days that are required to transfer goods between addresses, follow these steps:

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.

2. Follow one of these steps:

    - To use an existing record, use the filters at the top of the grid to find it. To edit the record that you find, select the link in the **Shipping country/region** column. To delete the record, select it, and then select **Delete** on the Action Pane.
    - To create a new record, on the Action Pane, select **New**.

3. On the details page for the new or selected record, on the **General** tab, use the fields to define the shipping address, the receiving address, and the number of days that are required to transport goods between them. For more information about each field, hover over its label to open a tooltip.

    > **Tip**
    >
    > - When you must enter an address, first enter a value in the **Shipping ZIP/postal code** field. Often, the system can automatically fill in the rest of the address fields for you, based on the postal code.
    > - To calculate the ship date, the system uses the most specific purchase transport days record that it can find. For example, you have two purchase transport days records: one that specifies 30 days from Spain to the United States and another that specifies 20 days from Madrid to New York. In this case, a purchase order that's shipped from Madrid to New York uses 20 days, whereas an order from Valencia to New York uses 30 days.
    > - If the shipping address and receiving address have equal priority, the shipping address takes precedence.

4. On the **Transport days per mode of delivery** FastTab, you can set specific transport days for any number of delivery modes (such as road, rail, and air). Use the buttons on the toolbar to add and remove rows in the grid as you require.

    - For each row, select a mode of delivery, and then specify the number of transport days for that mode.
    - If you want one of the modes of delivery to define the default transport days that apply to any modes that aren't specified in the grid, select the **Default** checkbox for

that mode's row.
    - If the **Default** checkbox isn't selected for any row, the **Transport in days** value that's specified on the **General** FastTab is used as the default travel time.

5. On the Action Pane, select **Save**.

Vendor settings for calculating requested ship dates

To calculate requested ship dates, you must set up an address for each vendor. To help make the calculation more accurate, you can also set up vendor ship calendars that define specific days when each vendor can ship.

Set up vendor addresses

To set up addresses for a vendor, follow these steps:

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Open the vendor record that you want to work with.
3. On the **Addresses** FastTab, add and remove addresses as you require. If more than one address is specified on the **Addresses** FastTab, when the system uses the purchase transport days setup to calculate the shipping date, it uses the address where the **Primary** option is set to *Yes* as the shipping point.

> **Tip**
>
> - The vendor address is applied to the purchase order and used as the shipping point for purchase transport days.
> - The receiving point for purchase transport days is the second factor that's used to calculate the shipping date. The delivery address depends on the setup, and a default value can be taken from the company, site, or warehouse. Any update to the site or warehouse for a purchase order header or line triggers a recalculation of the shipping date, because the new site or warehouse might have a different delivery address.

How calendars affect ship and receipt date calculations

The following calendars are relevant for ship and receipt date calculations:

- **Purchase calendar** – This calendar defines when a vendor is open. For example, a vendor might accept orders on workdays but not weekends.

- **Vendor ship calendar** – This calendar defines when purchase orders can be shipped from the vendor.
- **Warehouse calendar** – This calendar defines when purchase orders can be received at the site or warehouse.

The purchase and vendor ship calendars are considered when the ship date is calculated. The warehouse (item coverage group) calendar is used when the receipt date is calculated.

> **Note**
>
> If no coverage group is assigned to an item on the **Released product details**, **Item coverage**, or **Master planning parameters** page, then master planning uses the *general coverage group* (which is set in the **General coverage group** field on the **Master planning parameters** page) and calculates the working days based on the calendar assigned to that group.

Learn more in Calendars and master planning.

The following examples show how calendars affect the calculation of requested ship and receipt dates for purchase orders.

Calculation example 1

This example shows how requested ship and receipt dates are calculated for a product that has five days of lead time and two purchase transport days.

The following table shows the calendars that apply.

Expand table

Calendar	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Purchase calendar	Open	Open	Open	Open	Open	Open	Open	Open	Open
Vendor ship calendar	Open	Closed	Closed	Closed	Open	Open	Open	Open	Open
Warehouse calendar	Open	Open	Open	Open	Closed	Closed	Closed	Closed	Open

The following table shows the calculated dates.

Expand table

Date type	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Order date	X								
Ship date						X			
Receipt date									X

The calculation results are based on the following information:

- The ship date is Saturday, which is the first open day that's available in the vendor ship calendar after the five days of lead time.
- The receipt date is Tuesday, which is the first open day that's available in the warehouse calendar after the two transport days.

Calculation example 2

This example shows how requested ship and receipt dates are calculated for a product that has three days of lead time and two purchase transport days.

The following table shows the calendars that apply.

Expand table

Calendar	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Purchase calendar	Open	Open	Open	Open	Open	Open	Open	Open	Open
Vendor ship calendar	Open	Closed	Closed	Closed	Open	Open	Open	Open	Open
Warehouse calendar	Open	Open	Open	Open	Closed	Closed	Closed	Closed	Open

The following table shows the calculated dates.

Expand table

Calendar	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue
Order date	X								
Ship date					X				
Receipt date									X

Compared to the previous example, the results are calculated in the following way:

- The ship date is moved one day, because the vendor ship calendar shows that the vendor is closed on Thursday.
- The receipt date is moved two days, because the warehouse calendar shows that the warehouse is closed on Sunday and Monday.

Assign vendor ship calendars

For each vendor, you can specify a ship calendar that indicates the dates when the vendor can ship. For example, some vendors might be able to ship during all opening hours, whereas others can ship only on Mondays. You can define as many specific ship calendars as you require and then apply them to each vendor as appropriate.

If you have goods that must be expedited, and the relevant vendor makes an exception on their vendor ship calendar, you can change or remove the vendor ship calendar for the relevant purchase order lines.

For information about how to set up a ship calendar and define the available ship dates and times for it, see Calendars and master planning.

To assign a ship calendar to a vendor, follow these steps:

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Open the vendor record that you want to assign a ship calendar to.
3. On the **Invoice and delivery** FastTab, in the **Ship calendar** field, select the ship calendar that applies to the vendor.

Date calculations and recalculations

This section describes how and when the system calculates and recalculates requested ship dates.

Requested ship date

The requested ship date calculation considers lead time (lead time per vendor if it's set up on trade agreements), the mode of delivery, and the vendor ship calendar. The lead time is the time that the vendor takes to produce the item. It doesn't include transport days (from vendor to company).

You can configure how the system handles requested ship dates that are in the past. Learn more in the Specify whether requested ship dates can be in the past section.

Order date of planned purchase orders

When a master planning run indicates that a planned order is needed, the order date is calculated backward from the requirement date. The calculation considers margins, lead time, and the purchase transport days setup. The requested ship date is calculated according to the same rules that are used for purchase orders. However, for planned orders, the requested ship date isn't saved as a field value or shown on the **Planned purchase order** details page.

The requested ship date is also calculated for manually created planned purchase orders.

Recalculations for updated orders

When you edit a purchase order where the requested and/or confirmed ship date has already been calculated, the system does the following recalculations:

- If the **Requested ship date** value is changed, the **Requested receipt date** value is recalculated.
- If the **Requested receipt date** value is changed, the **Requested ship date** value is recalculated.
- If the **Confirmed ship date** value is changed, the **Confirmed receipt date** value is recalculated.
- If the **Confirmed receipt date** value is changed, the **Confirmed ship date** value is recalculated.
- If any of the following information is changed, the relevant ship and receipt dates are recalculated:
  - Address on the header
  - Mode of delivery (on the header or on a line)
  - Warehouse on a line

If you change the **Requested ship date** or **Confirmed ship date** value on the header of a purchase order, the system asks you whether the relevant ship and receipt dates should also be updated for each line.

> ⓘ **Note**
>
> If any subsequent changes are made at the line level (such as modifying the unit, warehouse quantity, or site), the line **Requested ship date** and **Requested receipt date** are recalculated based on the current date rather than the header date. To ensure accurate date calculations,

we recommend that you complete all necessary updates at the line level before updating the header **Requested ship date** or **Confirmed ship date**.

Purchase order ship and receipt date calculation logic

This section explains how the **Requested ship date** and **Requested receipt date** are calculated for purchase orders. These dates are influenced by lead times, transport days, vendor calendars, and other settings.

Key terms and concepts

The following table describes key terms and concepts related to purchase order ship and receipt date calculations.

Expand table

Term	Description
Requested ship date	The date on which you want the vendor to ship the goods.
Requested receipt date	The date on which you want to receive the goods.
Confirmed ship date	The date on which the vendor ships the goods, as confirmed by the vendor.
Confirmed receipt date	The date on which you receive the goods, as confirmed by the vendor.
Lead time	The number of days the vendor requires to prepare an item.
Transport days	The number of days needed to ship goods from vendor to warehouse.
Purchase calendar	The calendar that defines when a vendor is open for business.
Vendor ship calendar	The calendar that defines when purchase orders can be shipped from the vendor.
Warehouse calendar	The calendar that defines when purchase orders can be received at the site or warehouse
Item coverage calendar	An optional calendar used to adjust receipt dates.

Calculation logic

The following table summarizes the logic used to calculate purchase order ship and receipt dates.

Expand table

Calculation	Description
Requested ship date	Calculated as today's date plus lead time. Adjusted to the next working day if the assigned calendar is *Active*.
Requested receipt date	Calculated as the requested ship date plus transport days. Adjusted to the next working day if the assigned calendar is *Active*.
Lead time (on the **Line details** tab)	Populated only if **Calculate PO delivery date** is set to *Yes* and the purchase order originates from a purchase requisition, purchase agreement, or request for quotation.
Transport days	Applied based either on the selected mode of delivery at the purchase order line level or the default mode of delivery configured for the warehouse master.
Confirmed dates	Manually entered by the vendor and not adjusted by any calendar.
Trade agreement working days	When enabled, lead time is calculated using only open working days.

For details about how to update the **Requested ship date** and **Requested receipt date** in the header, see Recalculations for updated orders.

Example date calculation scenarios

This section provides examples that show how requested ship and receipt dates for purchase orders are calculated in various scenarios.

Example scenario prerequisites

These scenarios have the following prerequisites:

- Your system must meet all the feature prerequisites that are listed in the Prerequisites section of this article.
- The **Purchase transport days** page must be configured with at least one receiving address, shipping address, and number of transport days. The addresses must match the vendors and receiving warehouses that you use for these scenarios.
- Unless otherwise specified, all calendars must have all days open.
- On the **Procurement and sourcing parameters** page, on the **Delivery** tab, the **Requested ship date in the past** field must be set to *None*.

Example scenario 1: Standard purchase order

In this scenario, there are two purchase transport days from the vendor shipping address to the site/warehouse receiving address. Dates are calculated forward, and the purchase transport days are added to the requested ship date to determine the requested receipt date.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
4. On the Action Pane, select **New** to create a purchase order.
5. On the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase order.
6. On the **Purchase order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
7. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:

    - **Requested ship date** – The value is the current date (unless you specified another date for the order header).
    - **Requested receipt date** – The value is a date that's two days later than the **Requested ship date** value for the purchase order. The system calculates this date based on your setup on the **Purchase transport days** page. The following settings can also affect the calculation:
        - If the product that's ordered has lead time, the system adds that lead time to the transport time when it calculates the receipt date. (Therefore, the formula is *Requested ship date = Requested ship date + Product lead time*.)
        - The vendor ship calendar and warehouse calendar can also affect the date calculation by allowing for closing times at one or both locations. (Learn more in the How calendars affect ship and receipt date calculations section.)

    Any update to the **Requested ship date** or **Requested receipt date** value causes the system to recalculate the dates. An update to the **Requested ship date** value triggers a forward

calculation of the **Requested receipt date** value, and an update to the **Requested receipt date** value triggers a backward calculation of the **Requested ship date** value.

Example scenario 2: Purchase order with dates in the past

This scenario shows how the system handles requested ship dates that are in the past, based on the **Requested ship date in the past** setting. (Learn more in the Specify whether requested ship dates can be in the past section.)

The default value of the **Requested ship date in the past** field is *None*. In this case, requested ship dates in the past are always allowed, and no warning is shown. This scenario shows what happens if you change the value to *Disallow with warning*. In this case, a warning message is always shown if requested ship dates are in the past. The system then finds the next earliest ship date that's after the current date.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Delivery** tab, set the **Requested ship date in the past** field to *Disallow with warning*. Then select **Save**.
3. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
4. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
5. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
6. On the Action Pane, select **New** to create a purchase order.
7. On the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase order.
8. On the **Purchase order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
9. On the **Line details** FastTab, on the **Delivery** tab, set the **Requested receipt date** field to a date that doesn't allow for enough time for transport. (For example, select today's date.)

The system shows a warning message such as "The requested ship date 1/18/2024 is not valid because it is before today. Earliest possible requested ship date is 1/22/2024 with requested receipt date 1/26/2024." The requested date fields are updated to match the warning message.

If you try to directly enter a requested ship or receipt date that's in the past, you receive a warning that doesn't include any suggested dates (for example, "The requested ship date 1/8/2024 is not valid because it is before today"). In this case, you can't save the order until you select a valid date.

Example scenario 3: Purchase order created from a sales order

In this scenario, you create a purchase order from a related sales order and observe how the requested ship and receipt dates are calculated for both orders.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
4. On the Action Pane, select **New** to create a sales order.
5. On the **Customer** FastTab, set the **Customer account** field to any customer account (it doesn't matter which one). Then select **OK** to create and open the sales order.
6. On the **Sales order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
7. On the **Line details** FastTab, on the **Delivery** tab, set the **Delivery date control** field to *None*.
8. Set both the **Requested ship date** field and the **Requested receipt date** field to a date that's one week from now.
9. On the Action Pane, on the **Sales order** tab, in the **New** group, select **Purchase order**.
10. On the **Lines** FastTab, you see a line that matches the line from your sales order. For this line, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page.

11. Select the **Include** checkbox for the line.
12. Select **OK** to create the purchase order and return to your sales order.
13. On the Action Pane, on the **General** tab, in the **Related information** group, select **Purchase order**.
14. The related purchase order opens. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value matches the sales order's requested ship date.
    - **Requested ship date** – The value is calculated backward from the receipt date, based on the number of purchase transport days, to the first available open date.

Example scenario 4: Direct delivery created from a sales order

This scenario is similar to the previous one, but you create a direct delivery order from a sales order. Because you're using direct delivery for this scenario, the shipment never arrives at your warehouse. Therefore, transport time is calculated based only on the *vendor's shipping address* and the *customer's receiving address*.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to order from to the receiving address of the *customer* that you plan to ship to. Set the **Transport days** field to *2* for this record.
3. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
4. On the Action Pane, select **New** to create a sales order.
5. On the **Customer** FastTab, set the **Customer account** field to the customer whose receiving address you set up on the **Purchase transport days** page. Then select **OK** to create and open the sales order.
6. On the **Sales order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields if they're required. However, these fields don't affect the delivery time calculation, because you're using direct delivery.
7. On the **Line details** FastTab, on the **Delivery** tab, set the **Delivery date control** field to *None*.
8. Set both the **Requested ship date** field and the **Requested receipt date** field to a date that's one week from now.

9. On the Action Pane, on the **Sales order** tab, in the **New** group, select **Direct delivery**.
10. On the **Lines** FastTab, you see a line that matches the line from your sales order. For this line, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page.
11. Select the **Include** checkbox for the line.
12. Select **OK** to create the purchase order and return to your sales order.
13. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value is the requested ship date that you entered earlier.
    - **Requested ship date** – The value is calculated backward from the receipt date, based on the number of purchase transport days, to the first available open date.
14. On the Action Pane, on the **General** tab, in the **Related information** group, select **Purchase order**.
15. The related purchase order is opened. On the **Line details** FastTab, on the **Delivery** tab, notice that the **Requested receipt date** and **Requested ship date** values match the values of the sales order. These dates are synced because this order is a direct delivery order.

Example scenario 5: Intercompany order

For intercompany orders, the requested ship and receipt dates must be synced for both sales orders and purchase orders. The purchase transport days, lead time, delivery date control, and calendar availability must also be considered.

To complete this scenario, first set up an intercompany relationship between the customer and vendor that you use.

1. Use the company picker to select the company that's purchasing an item. This company is the one where you must set up the purchase transport days and create the purchase order.
2. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
3. Make sure that you have a record for delivery from the vendor warehouse to the customer warehouse that you're using for this scenario. Set the **Transport days** field to *2* for this record.
4. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.

5. On the Action Pane, select **New** to create a purchase order.

6. On the **Vendor** FastTab, set the **Vendor account** field to the intercompany vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase order.

7. On the **Purchase order lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.

8. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:

    - **Requested ship date** – The value is the current date (unless you specified another date for the order header).
    - **Requested receipt date** – The value is a date that's five days later than the **Requested ship date** value of the purchase order. The system calculated this date based on your purchase transport days setup.

9. On the Action Pane, on the **Manage** tab, in the **Intercompany tracing** group, select **Intercompany sales order**. The requested dates on the sales order line should match the dates that you saw on the purchase order line.

> **Tip**
>
> The sourcing order determines whether and how the purchase order transport days are calculated.
>
> - If dates are updated on the sales order, lead time and customer transport days take precedence, and purchase order transport days are calculated. These dates are editable only when you source from the sales order.
> - If dates are updated on the purchase order, purchase transport days and lead time are calculated, and requested and confirmed dates are synced back to the sales order.

Example scenario 6: Purchase agreement

This scenario shows how requested ship and receipt dates are calculated when you create a purchase order from a purchase agreement.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.

2. Make sure that you have a record for delivery from the vendor shipping address that you plan to create a purchase agreement for to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Purchase agreements** > **Purchase agreements**.
4. On the Action Pane, select **New** to create a purchase agreement.
5. On the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page. Then select **OK** to create and open the purchase agreement.
6. On the **Purchase agreement lines** FastTab, select **Add line**.
7. In the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
8. On the Action Pane, on the **Purchase agreement** tab, in the **Generate** group, select **Confirmation**.
9. In the **Confirm purchase agreement** dialog box, select **OK**.
10. On the Action Pane, on the **Purchase agreement** tab, in the **New** group, select **Release order**.
11. In the **Create release order** dialog box, in the **Purchase quantity** field for the line, enter a value. Then set the **Requested receipt date** field to one week from today.
12. Select **Create** to create the purchase order.
13. Make a note of the purchase order number that's shown in the message at the top of the page.
14. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
15. Open the purchase order that you just created. Use the purchase order number that you made a note of to find it.
16. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value is the date that you entered when you released the purchase order.

- **Requested ship date** – The value is calculated backward from the receipt date, based on the number of purchase transport days, to the first available open date.

Example scenario 7: Purchase requisition

This scenario shows how requested ship and receipt dates are calculated when you create a purchase order from a purchase requisition.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to create a purchase requisition for to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Purchase requisitions** > **All purchase requisitions**.
4. On the Action Pane, select **New** to create a purchase requisition.
5. In the **Name** field, enter a name for the requisition. Then select **OK** to create and open the purchase requisition.
6. On the **Purchase requisition lines** FastTab, on the toolbar, select **Add line**.
7. In the **Item number** field, enter a value. Then, in the **Quantity** field, specify a quantity.
8. On the **Line details** FastTab, on the **Inventory dimensions** tab, set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
9. On the Action Pane, select **Workflow** to open a dropdown dialog box. Select **Submit** to open a dialog box where you can enter a comment. Then select **Submit** in this dialog box to submit the requisition for approval.
10. Generate a purchase order for the purchase requisition through the **Submit requisition for approval** workflow. The resulting purchase order should have the following requested dates:

    - **Requested receipt date** – The value is the **Requested date** value from the purchase requisition.
    - **Requested ship date** – The value is calculated backward two purchase transport days from the requested receipt date of the purchase order.

Example scenario 8: Purchase request for quotation

This scenario shows how requested ship and receipt dates are calculated when you create a purchase order from a request for quotation (RFQ) reply.

1. Go to **Procurement and sourcing** > **Setup** > **Distribution** > **Purchase transport days**.
2. Make sure that you have a record for delivery from the vendor shipping address that you plan to create a purchase requisition for to the warehouse receiving address where you plan to receive the order. Set the **Transport days** field to *2* for this record.
3. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
4. On the **Request for quotation** tab, set the **Purchaser can edit vendors bid** option to *Yes*.
5. Go to **Procurement and sourcing** > **Requests for quotations** > **All requests for quotations**.
6. On the Action Pane, select **New** to create an RFQ.
7. Set the **Requested receipt date** field to one week from today. Then enter values in any other required fields, and select **OK** to create the RFQ.
8. On the **Request for quotation lines** FastTab, in the **Item number** field, select an item. Then set the **Site** and **Warehouse** fields to the warehouse that you set up a receiving address for on the **Purchase transport days** page.
9. On the **Header** tab, on the **Vendor** FastTab, set the **Vendor account** field to the vendor whose shipping address you set up on the **Purchase transport days** page.
10. On the Action Pane, select **Save**.
11. On the Action Pane, on the **Quotation** tab, in the **Process** group, select **Send**.
12. In the **Sending request for quotation** dialog box, select **OK**.
13. On the Action Pane, on the **Quotation** tab, in the **Replies** group, select **Manage replies**.
14. On the Action Pane, select **Edit** > **Edit RFQ reply**.
15. On the Action Pane, select **Submit**.
16. In the **Submit** dialog box, select **OK**.
17. On the Action Pane, on the **Reply** tab, in the **Process** group, select **Accept**.

18. In the **Accepting request for quotation** dialog box, select **OK** to create the purchase order.
19. On the Action Pane, on the **General** tab, in the **Other information** group, select **Purchase order**.
20. Open the new purchase order.
21. On the **Line details** FastTab, on the **Delivery** tab, you see the following values:
    - **Requested receipt date** – The value matches the **Requested receipt date** value from the RFQ.
    - **Requested ship date** – The value is calculated backward from the requested receipt date, based on the purchase transport days setup.

Related information

- Calendars and master planning
- Improve scheduling engine performance
- Date and time parameters used by Planning Optimization
- Safety margins

Last updated on 07/01/2026

Review and accept changes to confirmed purchase orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

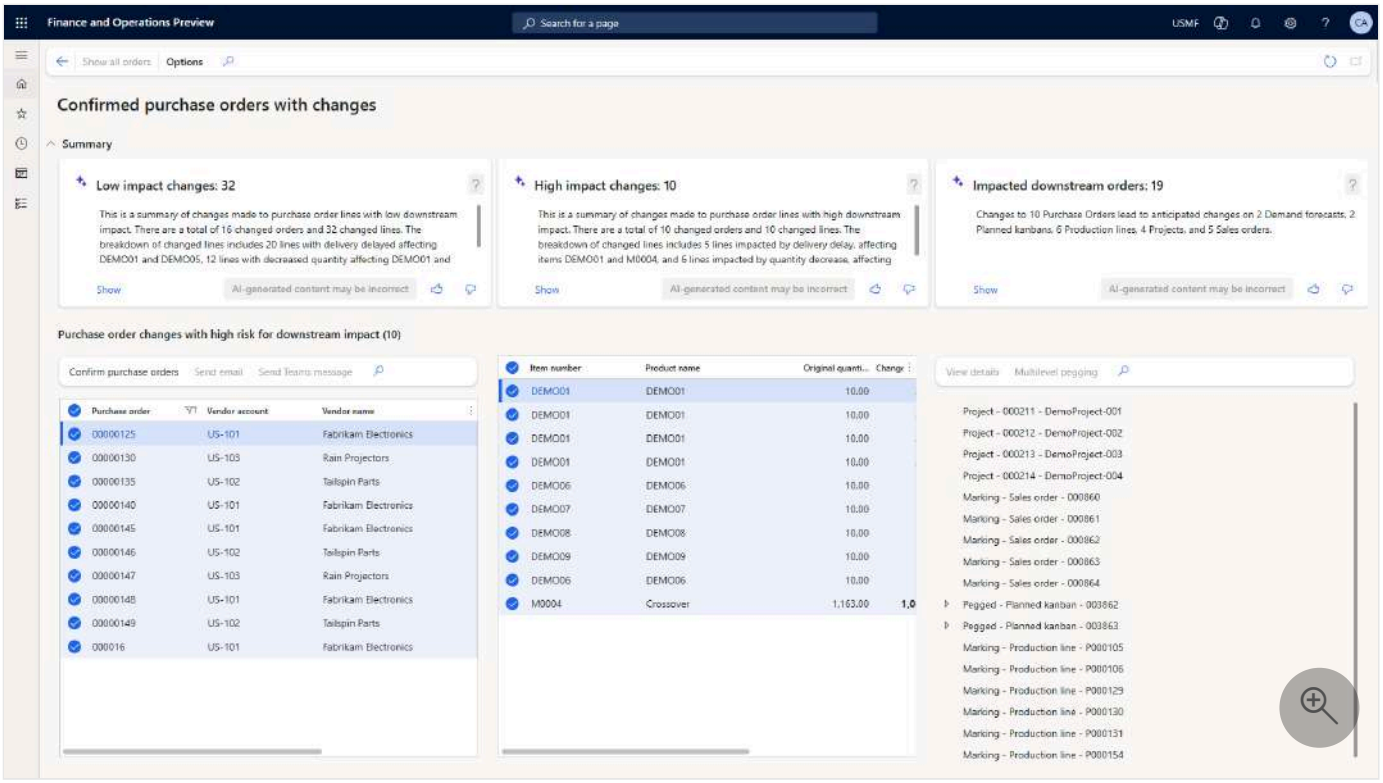
During procurement planning, changes to confirmed purchase orders can significantly affect downstream processes such as planned production, service work, or sales orders. The new **Confirmed purchase orders with changes** workspace makes it fast and easy to identify and reconfirm changes that have only a low risk of downstream impact. Procurement managers can focus on high-impact changes to assess downstream order impacts and communicate directly with vendors.

Prerequisites

The **Confirmed purchase orders with changes** workspace is enabled by default. However, its AI-powered and Copilot functionality must be enabled separately. To take advantage of the Copilot features, see Enable Copilot support for managing changes to confirmed purchase orders for the system requirements and information about how to set them up.

The Confirmed purchase orders with changes workspace

To open the workspace, go to **Procurement and sourcing** > **Workspaces** > **Confirmed purchase orders with changes**.



The workspace lists all previously confirmed supplying purchase orders that are changed since confirmation. It helps during the review process by identifying potential impacts on demand such as production work, service work orders, and sales orders.

The relations between supplying purchase orders and demand can originate from different sources, such as pegging from master panning, inventory markings, and purchase orders created for demand in projects.

At the top of the workspace, three tiles summarize groups of changes to confirmed purchase orders and the related impact on downstream demand:

- **Low impact changes** – This tile highlights changes to purchase orders that have no known impact on downstream demand. No known impact is assumed if changes to the quantity or delivery dates don't affect those requirements from demand. Examples of these changes include an increased quantity or an earlier date. A summary of changes is generated to help you review and validate the purchase orders.
- **High impact changes** – This tile highlights purchase orders that have known downstream impacts on demand. These changes are summarized together with the detected demand impacts, to help you investigate further and decide what action to take.
- **Impacted downstream orders** – This tile summarizes impacts on different kinds of downstream demand orders by total per type of order. This information helps you review potential impacts on demand and identify each originating change.

Each tile provides a **Show** link. When you select this link, the view beneath shows the changes to supplying purchase orders and the affected demand according to the tile's classification and filter.

Review changes to confirmed purchase orders

Purchasers use the workspace to review and accept changes to confirmed purchase orders. They typically follow these steps:

1. Study all changes to confirmed purchase orders and their downstream impact.
2. Focus on changes that have a low risk of downstream impact.
3. Review changes that have a high risk of downstream impact.
4. Review the remaining impact on downstream demand orders.

The following subsections describe these steps in more detail.

Step 1: Study all changes to confirmed purchase orders and their downstream impact

The purchaser first opens the **Confirmed purchase orders with changes** workspace and studies all the changes that vendors submit through the purchase order change management process. Vendors can submit these changes through several channels.

When a purchase order that was previously confirmed is changed, it's moved back to the *Approved* state. The workspace shows all purchase orders changed after confirmation.

The workspace shows two lists: one for the purchase orders and one for the purchase order lines.

The purchase order list shows the purchase order number, the vendor, and the requested receipt date from the purchase order header. To view the related records, select the purchase order or vendor link in the list.

The purchase order lines list shows information about each line that has proposed changes. This information includes the item number, product name, original quantity, new quantity, original confirmed receipt date, and new confirmed receipt date. The workspace also provides a hierarchical view of any downstream impact. The **Reference** field indicates the type of downstream impact, such as a planned kanban for a production order, a sales order, or a maintenance work order.

Step 2: Focus on changes that have a low risk of downstream impact

To take action effectively and efficiently, start by focusing on low-impact changes. These changes have no known downstream impact according to the current plan.

The **Low impact changes** tile at the top of the workspace provides an AI-generated, natural-language summary of the changes that have a low risk of impact. Select the **Show** link to open the corresponding filtered view.

The purchaser reviews the changes and can then accept them by selecting individual or multiple purchase orders and then selecting **Confirm purchase orders** on the toolbar above the list. The procedure for confirming the changes is the same as the procedure for confirming a purchase order for the first time, and it supports batch processing. Any extension that's registered for purchase order confirmation also runs.

After the selected purchase orders are reconfirmed, they're moved back to the *Confirmed* state.

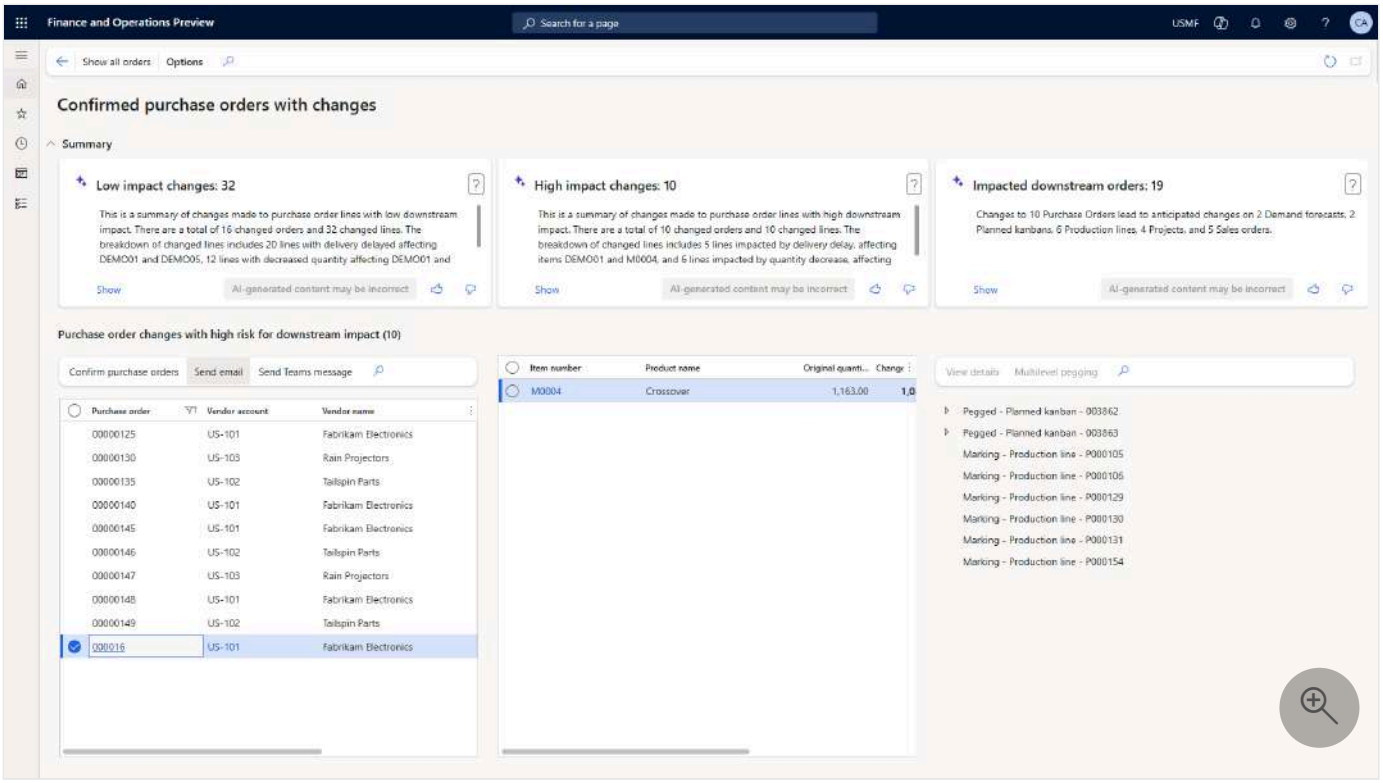
Step 3: Review changes that have a high risk of downstream impact

The purchaser spends more time on changes that have a high risk of downstream impact. These changes have downstream orders allocated to them, and the downstream impact is identified.

The **High impact changes** tile provides an AI-generated, natural-language summary of the changes that have a high risk of impact. Select the **Show** link to open a filtered view where you can focus on those impacts and take action.

> **Note**
>
> Only direct downstream impacts are considered. Indirect downstream impacts, such as product work that depends on other production work, aren't yet considered.

From this view, you can reach out to a vendor by using email or Microsoft Teams. By using the contact details of the vendor contact person, the system generates a proposed message text and adds it to a new email message or Microsoft Teams chat. You can then review, update, and send the message.



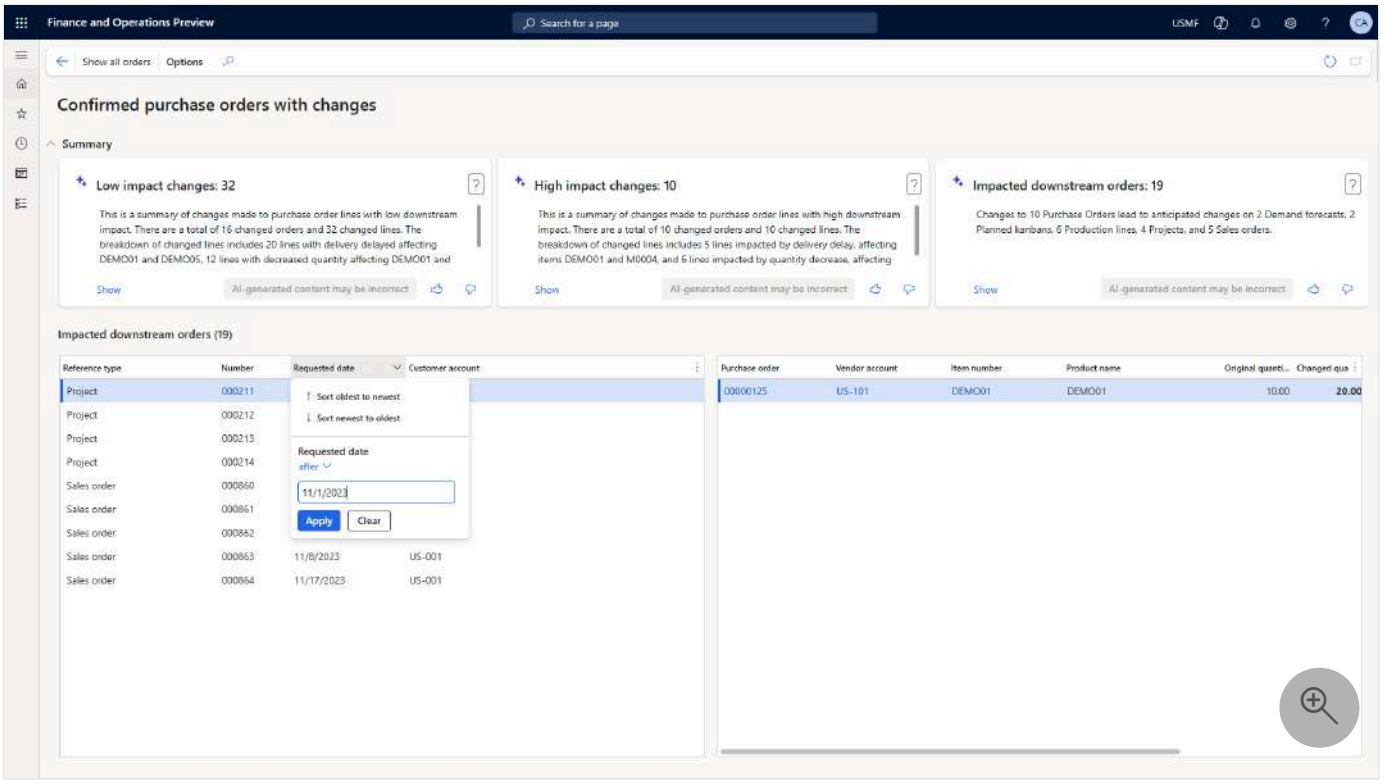
After you communicate with the vendor, you can update the order further and then send it back to the vendor for confirmation. Alternatively, you might cancel the order and plan for alternative supply.

Step 4: Review the remaining impact on downstream demand orders

After addressing most of the order changes, the purchaser can review the overall downstream impact of the remaining open order changes.

The **Impacted downstream orders** tile provides a summary of the downstream impact. Select the **Show** link to open a filtered view where you can focus on those impacts and take action.

In this view, you can analyze the changes from the perspective of downstream impact and identify the purchase order changes that cause the impact. You can then take action on those purchase orders.



A good way to focus on the most pressing downstream impact is to sort and filter the list by the date of impact (the requested date in the downstream order). You can then take action by communicating with downstream stakeholders. Those stakeholders might, in turn, contact affected customers and replan the downstream orders.

Related information

- Responsible AI FAQ for the Confirmed purchase orders with changes workspace
- For administrators: Enable Copilot support for managing changes to confirmed purchase orders

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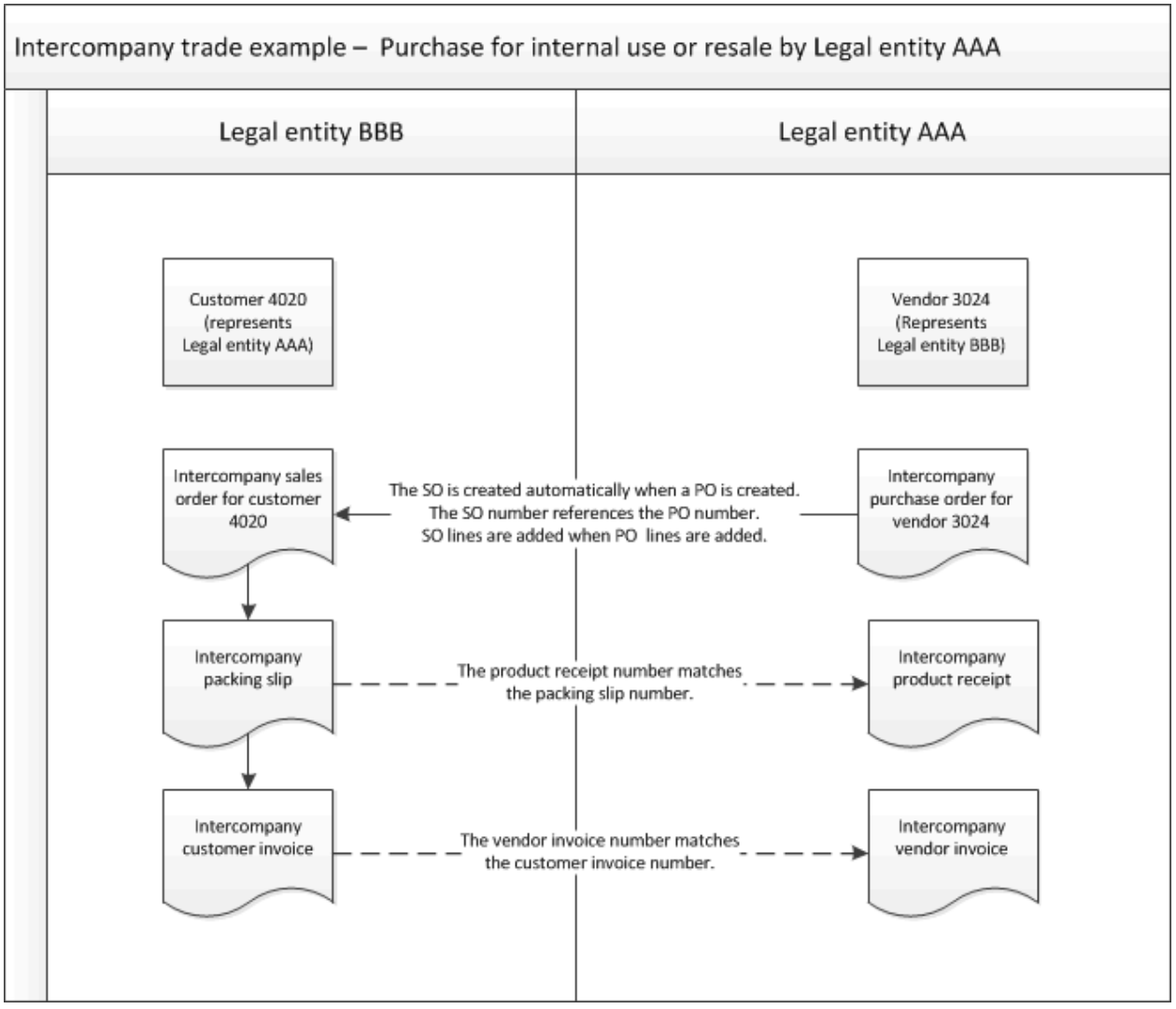
Last updated on 07/01/2026

Create and invoice an intercompany purchase order for internal use

Article • 01/07/2025

You can create an intercompany purchase order for an intercompany vendor. This automatically creates an intercompany sales order at the intercompany vendor.



Create an intercompany purchase order and a corresponding intercompany sales order

Do these steps in legal entity AAA, as shown in the illustration.

1. Select **Accounts payable** > **Purchase orders** > **All purchase orders**.
2. On the **All purchase orders** list page, create a purchase order for an intercompany vendor. The field values are copied from the vendor account to the purchase order.

Because you're working with an intercompany vendor, an intercompany sales order is created in the legal entity that corresponds to the vendor. The number of the intercompany sales order can be the same as the number of the intercompany purchase order, and it can include the ID of the legal entity. The number structure that is used depends on the selection in the **Sales order numbering** field in the **Intercompany** page. For example, if you create purchase order 00029_064 in legal entity AAA, the sales order number in legal entity BBB is AAA00029_64.

An informational message tells you that an intercompany purchase order and intercompany sales order have been created. The message includes the intercompany sales order number, for your information.

3. Add line items to the purchase order. The corresponding line items are added automatically to the intercompany sales order. If an item doesn't exist in the other legal entity, a message is displayed and you can't add the item to the purchase order. To fix this problem, switch to the other legal entity and release the product to that legal entity. The item will be available to be added to sales orders in that legal entity. Then, switch back to the legal entity of the purchase order and continue adding line items.

4. When you have finished entering information for the purchase order, confirm it.

Process the intercompany packing slip and customer invoice

Do these steps in legal entity BBB, as shown in the illustration.

1. Go to **Accounts receivable** > **Sales orders** > **All sales orders**.

2. On the **All sales orders** list page, select the intercompany sales order.

3. On the Action Pane, select the **Pick and pack** tab, and then select **Packing slip**.

4. Select the **Posting** check box.

5. Select **OK**. The packing slip is posted in legal entity BBB.

6. On the **All sales orders** list page, select the intercompany sales order.

7. On the Action Pane, select the **Invoice** tab, and then select **Invoice**.

8. Select the **Posting** check box.

9. Select **OK**.

The customer invoice for the intercompany sales order is posted in legal entity BBB.

Process the intercompany product receipt and vendor invoice

Do these steps in legal entity AAA, as shown in the illustration.

1. Go to **Accounts payable** > **Purchase orders** > **All purchase orders**.
2. On the **All purchase orders** list page, select the intercompany purchase order.
3. On the Action Pane, select **Receive**, and then select **Product receipt**. The product receipt is created. The product receipt number is the same as the intercompany packing slip number.
4. Select the **Posting** check box.
5. Select **OK**.
6. On the **All purchase orders** list page, select the intercompany purchase order.
7. On the Action Pane, select **Invoice**, and then select **Invoice**. The vendor invoice is created. The vendor invoice number is the same as the intercompany customer invoice number.
8. Finish entering the vendor invoice, and then post it.

Feedback

Was this page helpful? Yes No

Provide product feedback

Check intercompany order price discrepancies

Article • 02/09/2024

You can use this procedure to check for price discrepancies on intercompany orders.

1. Go to **Procurement and sourcing** > **Purchase orders** > **Intercompany** > **Check intercompany order price discrepancies**.

2. Set up a batch job, if you need to, and then select **OK** to check the prices and discounts for intercompany sales orders and purchase orders. Otherwise, select **Cancel** to close the page without performing the check.

   > **Tip**
   >
   > If there are any synchronization issues or issues with the prices, these will be listed in an informational message that will be displayed. You can print the informational message for reference.

3. On the informational message, select the relevant line to open the order in the current legal entity. Open the order in the related legal entity by selecting **Intercompany**, and then **Intercompany purchase order** or **Intercompany sales order**.

   > **Note**
   >
   > To allow an update of the intercompany order:
   >
   > a. Go to **Accounts receivable** > **Customers** > **All customers**.
   > b. On the Action Pane, select the **General** tab, and then select **Intercompany**.
   > c. On the **Intercompany** page, select **Purchase order policies** or **Sales order policies**.
   > d. Select **Allow price edit** and **Allow discount edit** in both areas.

4. Open the relevant intercompany order where you want to keep the price.

5. For each order line, change the **Unit price** field for the line, and then change it back to the original value. Change the **Discount** field for the line back and forth, and change the relevant **Charges on purchases** or **Sales charges** fields back and forth. Changing the values back and forth will trigger a synchronization of the

prices, discounts, and charges from this intercompany order line to the referenced intercompany order line, so they will be automatically aligned.

> **Note**
>
> This synchronization is valid only if the intercompany sales order has not been invoiced. If the intercompany sales order has been invoice-posted and a customer invoice journal has been created, changes must be performed directly on the intercompany purchase order lines by setting the prices, discounts, and charges equal to those on the intercompany customer invoice journal. If this is not done, the intercompany purchase order cannot be invoice-posted, because there will be a mismatch in the order totals.

6. Repeat the procedure until all intercompany purchase and sales orders are synchronized.

Feedback

Was this page helpful? Yes No

Provide product feedback

Record the receipt of goods on the purchase order

This article explains how to record receipt of goods directly on a purchase order. You can also register product receipt in the warehouse, and then later record it on the purchase order. A purchasing agent or an accounts payable coordinator typically performs this task. The example shown in this guide uses values from the USMF demo data company and has you prepare an example purchase order to receive against. If you use the procedure on your own data, start at the *Record receipt of goods* section.

Prepare a new purchase order for receipt of goods

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Select **New**.
3. In the **Vendor account** field, enter *US-101*.
4. Select **OK**.
5. In the **Item number** field, enter *M0001*.
6. In the **Quantity** field, enter *5*.
7. On the Action Pane, select **Purchase**.
8. Select **Confirm**.

Record receipt of goods

1. On the Action Pane, select **Receive**.
2. Select **Product receipt**. The **Quantity** field allows you to select different options for the quantity that you want to receive. For example, if a quantity is previously registered in the warehouse, you can select *Registered quantity*. For this example, use the value *Ordered quantity*.
3. Expand the **Overview** section.
4. In the **Product receipt** field, type any value. Enter a reference used as voucher for the product receipt journal.
5. Expand the **Lines** section.
6. Set **Quantity** to *4*. Manually specify the quantity that you're receiving for each line on the order.

7. Select **OK**. The goods are now recorded as received on the purchase order, and a product receipt journal is created as document to reflect this. Use the **Product receipt** action to review the journals created with the purchase order, and see what was received, and when.

Last updated on 07/01/2026

Create purchase orders

08/13/2025

This article describes the process and options when you manually create a purchase order.

When you create a purchase order (PO), general information about the whole order is specified in the PO header, and you then add one or more PO lines. This article describes some of the most frequently used options that are available.

You can also create POs by copying lines from another PO document or a sales order. In this case, you can reverse the sign on the inventory, as you would reverse the sign on an invoice to indicate credit.

Although you can manually create POs, they're more typically generated from other processes. Orders can be automatically created based on other documents, such as requisitions. Alternatively, they can be created as part of the master planning process through planned POs. If you use purchase agreements, POs can be created by the **Release order** action. There are also more advanced methods for automatically creating a PO. For example, orders can be created when you use direct delivery or intercompany order chains.

Creating a purchase order header

When you create a new PO, a dialog box appears, where you can enter the most common information for the PO header. When you select **OK** to close the dialog box, the order is created, and you can then specify additional information in the header.

The first detail that you must consider when you create a PO is the type of order. The **Purchase order** type is used most often. However, if a credit invoice is required, you can use the **Returned order** type.

You must specify the supplier in the **Vendor account** field. For this field, you can search on either the account or the vendor name. If a vendor delivers from multiple locations but uses a single invoice account, you can select that invoice account in the **Invoice account** field and then use it with different vendor accounts. If you must create a PO for products that won't be ordered repeatedly, you can use the **One-time supplier** option. This option automatically creates a new vendor account that is marked as a one-time account, to support a later clean-up process for one-time accounts in the **Accounts payable** module. When you select a vendor account, many fields in the PO header inherit default values from the information that is associated with the vendor account. For example, the default delivery site and warehouse are copied from the vendor information. However, you can override these default values if the purchase is intended for another location.

If the supplier has provided a reference number for the order, you can record this information in the **Vendor reference** field. For returned orders, you must specify a value in the **RMA** field to reference the supplier's authorization for processing the return.

If a purchase agreement is associated with the order, you must specify this information in the **Purchase agreement** field.

The PO header also contains information about charges that apply to the whole order instead of individual lines. Charges can be automatically added to the order if automatic charges have been set up for the vendor or the vendor's charge group. You can also manually add charges to the order header by selecting **Maintain charges** on the Action Pane.

Adding purchase order lines

POs can be for physical products or for services. A setting on the inventory model group determines whether a particular item number applies to a product or a service. Usually, the item that is purchased is specified by an item number. However, if the order is for products or services that are directly consumed, you can also specify the item by using a procurement category.

PO lines contain lots of fields, but many of these fields have a default value or a value that is inherited from the order header. Additional fields are set when you select a product or service. The fields that are most often set manually include the fields for the item number, quantity, and requested receipt date. Information about unit price and discounts is also important, but the values of those fields are often determined by trade agreements or purchase agreements.

When you select a product, you can search on all or part of the product name instead of using the item number. If the product has several variants, such as different sizes, you can see an overview of the available variants by using the **Add lines** function or by using the lookup that is available in the **Variant number** field.

Often, you'll have to specify several dimensions for the item that is selected on each PO line. The dimensions that must be specified depend on the dimension groups that have been assigned to the product master definition. For example, you'll often have to specify a site and warehouse to indicate the location that the product should be delivered to. You identify product variants by specifying a variant number, or by entering values for one or more product dimensions, such as color, size, configuration, or style. Tracking dimensions, such as batch and serial number, let you uniquely identify each inventory lot. After you've created an order, you can capture dimension values on the order by using the **Registration** action. For example, you've ordered a quantity of five pieces of an item. Later, you register that three of those pieces will be black, and two of them will be blue. This approach is an alternative to capturing the dimension information during arrival registration.

You can check the details of the inventory transaction status for stocked products. For example, you might want to check the on-hand inventory to help you decide how much to order. Alternatively, you might want to review the inventory status of an ordered quantity to see whether inbound arrival registration has occurred.

A PO line that is being used to return a product to the vendor will have a negative quantity. You can select a specific lot to return by using the **Reservation** action.

Sometimes, you might want to divide the quantity that you've ordered, so that different parts of it are delivered on different dates. You can set up these deliveries by using the **Delivery schedule** action, which is available on the **Purchase order line** menu in the **Lines** view.

Charges can be automatically added to PO lines, if automatic charges have been set up for the vendor or vendor charge group, and for the item or item charge group. However, more typically, charges are added manually at the order line level. To add a charge, open the **Maintain charges** page by using the **Maintain charges** action on the **Financials** menu in the **Lines** view. The advantage of adding charges directly at the order line level is that the charge can be allocated as an inventory cost. To set up charge codes to account product cost, use the **Item** debit option. These types of charges must be allocated from the PO header to the lines before the order can be confirmed. For example, you might want to allocate charges based on the quantity on each line. The charge category also affects how charges are accounted. For example, fixed charges specify a fixed amount, and percent charges are calculated as a percentage of the net amount for the order line. POs can be assigned to a load, and the load might include an estimate of the expected expense for the transportation cost. You can allocate this expense from the load back to the PO lines.

Purchase order actions

After you've added the header and lines to the PO, you must often complete additional steps before the order is ready to be sent to the vendor. Because so many options are available, you might find it helpful to use Action search to find the relevant menu item.

You can configure products on the order so that they have supplementary items. Supplementary items are products that must or can be bought together with other products. Supplementary products might be added free of charge as accompanying products, or you might be able to decide whether to add them to the order or not. You can review the supplementary items after each order line that is added. However, you'll probably find it more convenient to review and add relevant supplementary items for all the order lines by using the **Supplementary items** page, which you can open from the Action Pane.

Discounts are usually added to lines as they're created. However, a few discounts apply to the whole order:

- The **Total discount** action calculates a total discount percentage that is applied to the full order. Don't confuse this discount with the cash discount percentage. Cash discounts are applied when the invoice is paid, and they depend on payment settlement by a specific date.
- If a multi-line discount applies, you must use the **Multiline discount** action to calculate and assign it to the order. Multi-line discounts are discounts that can be offered if a mix of products on the order exceeds a joint threshold. Only a few companies use this type of discount.

Charges that have a charge code that uses the **Item** debit type must be assigned to the line level before the order can be sent to the vendor. You might find it convenient to assign these charges at the order header level, so that you can specify the total amount of the charge. However, in this case, the charge must then be allocated down to each line before the order can be confirmed. You can use the **Allocate charges** action to split amounts from charges that are assigned at the header level down to the order lines. Charges can be split according to the net amount of each line, according to the quantity that has been ordered, or evenly across the order lines. After you've allocated charges to the lines, the charge is removed from the order header.

POs can be configured to require that budget funds be allocated to the order before it can be processed. In this case, you can use the **Budget checking** action to allocate the budget.

You might have to delay the completion of a PO. For example, you might require additional information about products or services, or you might have to get authorization for the spend. There are several ways to hold back an order. For example, you can wait to send the order. Alternatively, if a change management workflow is being used, don't submit the order for approval. If you must block all orders for a particular vendor, you can also mark the vendor as *On hold* for processing on the vendor master. There are also circumstances that might prevent the order from being processed. For example, processing might be prevented if credit limits have been exceeded, or if the required budget funds aren't available.

Related information

- Purchase order overview
- Approve and confirm purchase orders
- Product receipt against purchase orders
- Vendor invoices overview

Create a purchase order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows you how to create a purchase order manually. The system often creates purchase orders automatically as a result of master planning, direct delivery, and other processes. A purchasing agents also create purchase orders. This procedure includes sample values based on the USMF demo data company.

Create the purchase order header

The purchase order header captures information that applies to the entire order, such as the vendor, delivery location, receipt date, and the person placing the order. Follow these steps to start a new purchase order and set the header values that will act as defaults for the order lines you add later.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Select **New**.
3. Select vendor account *US-101*. When you select a vendor, the system copies details from the vendor record, such as address, invoice account, delivery terms, and delivery mode, as default values into the order header. You can change these values at any time.
4. Expand the **General** section.
    - The **Site** field together with the **Warehouse** field specifies where the procured goods or services must be delivered. The default delivery address is the site. You can populate both fields with values set up for the selected vendor, or you can specify them manually.
    - Use the **Receipt date** field to specify when procured goods and services need to be delivered. You can specify a single receipt date for the order, or you can give the individual order lines unique receipt dates. If the receipt date you specify here can't be

met for specific products or services because they have longer lead times, the system creates those lines with a later receipt date to accommodate for this.

5. Expand the **Administration** section. Use the **Orderer** field to specify who is placing the order. This information might be convenient to share with the vendor in case they need to contact that person. The system might automatically assign the field a value if the current user account is associated with a name on the **Users** page.

6. Select **OK**. The order header is created. When you work with purchase order lines, you see only a summary of the header information. If you need to view the rest of the information, select **Header**.

Add a purchase order line

Purchase order lines specify the individual products, services, or expense categories that you are buying, along with quantities, prices, discounts, and delivery details. Follow these steps to add a line to the order, choose the product dimensions that you want to track, and enter the pricing and quantity information that will be sent to the vendor.

1. Select **Purchase order line**.

2. Select **Dimensions**. Products can be in variants that are differentiated by dimensions, such as color, size, or style. You can also set up products to use storage dimensions, such as site and warehouse. There are also optional tracking dimensions, such as batch and serial numbers. To improve the efficiency of order entry, add the dimension fields that you commonly use directly to the order grid.

3. Select the **Color** check box. If you select the **Save setup** option, the selected dimensions appear by default the next time you open the purchase order page.

4. Select **OK**.

5. In the **Item number** field, select *T0004*.

    - Create order lines for products and services by specifying an item number. Create order lines for expenses by specifying a procurement category.
    - Use the **Procurement** category field for adding lines where procured items are expensed directly, rather than going into inventory. This means that if you need to expense a purchase, you can create a purchase order line that specifies a procurement category, rather than creating a line with an item number. You can also associate items

with a procurement category. In this case, the procurement category is shown as informational only.

6. In the **Color** field, enter or select a value. The **Site** and **Warehouse** fields typically populate with values from the order header, but you can override the fields if some lines need to be delivered to different locations.

7. In the **Quantity** field, enter a number.

    - Select the quantity that you want to purchase. The **Quantity** field automatically populates with the minimum order quantity for the product if this is set up, or with the value of 1.
    - The **Unit** field indicates the unit of measure for the ordered quantity. Typically, the unit is automatically provided from the purchasing unit on the product master data, but you can change this.
    - The **Unit price** field typically contains a value from either a purchase agreement or a trade agreement. You can change the unit price on individual order lines, for example if a unique price is negotiated with the vendor.
    - The **Discount** field represents a discount amount per unit. This discount therefore reduces the unit price by the discount. This discount is commonly supplied automatically from purchase agreements or trade agreements, but you can override on individual lines if unique discounts are negotiated with the vendor.
    - Enter a discount percentage that reduces the net amount for the line accordingly. The discount percent is often supplied automatically from purchase agreements or trade agreements, but you can override on individual lines if a unique discount percentage is negotiated with the vendor.
    - The value in the **Net Amount** field is calculated from other fields on the line including quantity, unit price, discount, and discount percent. You can change the net amount, but then the **Unit Price**, **Discount**, and **Discount percent** fields are blank. When you post toward the line, the amount posted is proportional to the net amount. Typically the **Net Amount** field is only used for displaying the net amount of the line.

8. Expand the **Line details** section.

9. Select the **Delivery** tab. You can assign a unique receipt date to each order line. The date is inherited from the field on the purchase order header, but you can change this.

Review order totals

Before you submit a purchase order, it's a good idea to verify the totals so that you can confirm the overall cost, taxes, charges, and discounts that apply to the order. Follow these steps to open the totals dialog box and review the calculated amounts for the whole order.

1. Select **Totals**.

    - If you don't see the **Totals** action, select the **Purchase Order** tab on the Action Pane.
    - This dialog box shows totals for the whole order.
    - The **Selection** field allows you to change the basis of how totals are calculated. For example, you could choose **Product receipt quantity** to show totals that relate to the amount of the products that are received, or **Ordered quantity** to show the amount of product that you ordered.

2. Select **OK**.

Default financial dimensions for purchase orders created from purchase requisitions

When you create a purchase order from a purchase requisition, the system initializes financial dimensions on the purchase order based on the following rules:

- **Purchase order header dimensions** – The system always initializes the financial dimensions on the purchase order header based on the vendor setup. This setup includes dimensions such as *Business unit*, *Cost center*, and any other mandatory financial dimensions configured on the vendor.
- **Purchase order line dimensions** – If you create the purchase order from a purchase requisition, the system copies line-level financial dimensions from the corresponding purchase requisition lines.

Last updated on 06/12/2026

Create a purchase order governed by budget

06/17/2025

Use this procedure to create a purchase order that is checked for available budget.

Review the budget control configuration

1. Go to **Budgeting** > **Setup** > **Budget control** > **Budget control configuration**.
2. Select the **Budget funds available** tab.
3. Select the **Documents and journals** tab.
4. Select the **Define budget control rules** tab.
5. Select the **Define budget groups** tab.
6. Close the page.

Create a purchase order

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Select **New**.
3. In the **Vendor account** field, enter or select a value.
4. Expand the **General** FastTab.
5. In the **Accounting date** field, set the date.
6. Select **OK** to close the dialog and open your new purchase order.
7. On the **Purchase order lines** FastTab, select **Add line** from the toolbar to add a new line and then fill out the line as needed to add an item to the order.
8. On the **Purchase order lines** FastTab toolbar, select **Financials** > **Distribute amounts**.
9. In the **Ledger account** field, specify an account.
10. Close the page.

Perform budget checking

1. Continue working with the purchase order you just added a line to.
2. On the **Purchase order lines** FastTab toolbar, select **Financials** > **Perform budget checking**.
3. On the **Purchase order lines** FastTab toolbar, select **Financials** > **Budget check errors or warnings**.
4. The **Budget check errors or warnings** dialog opens. Check the results of the check and then select **Close** to close the dialog.

Create a purchase order with a delivery schedule

Article • 01/24/2025

This article demonstrates how to create a delivery schedule for a purchase order. A delivery schedule is used when a quantity on an order or a journal is requested to be delivered in multiple shipments. The example shown in this guide can be used in the USMF demo data company. This procedure would typically be done by a purchasing agent.

Create a delivery schedule

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. On the Action Pane, select **New**.
3. In the **Vendor account** field, enter *US-101*.
4. Select **OK**.
5. In the **Item number** field, enter *M0001*.
6. In the **Quantity** field, enter *10*.
7. Select **Purchase order line**.
8. Select **Delivery schedule**.
    - The **Delivery schedule** page allows you to specify the number of shipments in which the total quantity of the order line will be delivered from the vendor.
    - By default, the system copies the total quantity and other delivery details of the original purchase line into the first delivery schedule line. In this example, we'll create a schedule for two shipments, with the second shipment's date offset by a week from the first shipment.
9. In the **Quantity** field, change the quantity to *4*.
10. Select **New**.
11. In the **Quantity** field, enter *6* as the remaining quantity.
    - In the receipt date field, select a date that's one week after the date on the first delivery line.

- You can keep track of the total quantity that's allocated to the delivery schedule lines by looking at the **Total** and **Remaining** fields. When the remaining quantity is zero, the full quantity from the original line has been allocated to the schedule.

12. Expand the **Charges conversion** section.

    - The options here allow you to control how you want charges to be distributed across the delivery schedule lines. If you select *Copy gross amounts*, the charge amount on the original order line is copied to each delivery line. The *Allocate to delivery lines* option divides the original line charge according to the quantity on each delivery line.

13. Collapse the **Charges conversion** section.

14. Select **OK**.

    - The delivery schedule has now been applied to the order.
    - The original order line, now referred to as a Commercial line, has been converted to an Order line with multiple deliveries. It's marked with a distinct icon and acts as a header for the delivery lines.

15. Select the second order line, which is the first of the two delivery lines.

    - The two new lines, referred to as Delivery lines, make up one delivery schedule. The order will be processed against these lines and not the original line. If documents such as confirmations, product receipt journals, or invoices are printed, only the delivery lines are shown.

Change the delivery schedule

You can change the quantity on delivery lines. If you do this, the commercial line is automatically updated to the total quantity in the delivery lines.

1. In the **Quantity** field of the first delivery line, change the quantity from *4* to *5*.

2. Select the first order line (the commercial line).

    - The quantity on the commercial line has been changed to 11.

Process product receipt using delivery schedules

The purchase order must be confirmed before the product receipt can be processed. In this example, receipt is recorded directly on the purchase order. The receipt could also have been recorded when the goods arrived in the warehouse.

1. On the Action Pane, select **Purchase**.
2. Select **Confirm**.
3. On the Action Pane, select **Receive**.
4. Select **Product receipt**. In the **Product receipt** field, type any value.
    - This field is used to enter a reference that will be used as voucher for the product receipt journal.
    - In the **Quantity** field, select *Ordered quantity*. This option means that the receipt will process for the quantity that the order lines were created with.
    - Make sure that the **Print product receipt** field is set to *No*. Printing isn't needed in this example.
5. Expand the **Lines** section.
    - Notice how the product receipt is created for the two delivery lines and not the original order line. If the receipt had been recorded in the warehouse, it would also have been recorded on the delivery schedule lines.
6. Collapse the **Lines** section.
7. Select **OK** to post the receipt.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a purchase order for a one-time supplier

06/17/2025

This procedure shows you how to create a purchase order for a one-time supplier. The supplier is created automatically with the purchase order, rather than having to create the vendor account manually. Purchase orders are typically created by a purchasing agent. A one-time vendor account must already be set up in the **Account payable parameters** page.

To create a purchase order for a one-time supplier, follow these steps:

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Select **New**.
3. Select *Yes* in the **One-time supplier** field. A vendor account is automatically created and assigned to the purchase order. The vendor account is created based on the template that is specified on the **General** tab in the **Accounts payable parameters** page.
4. In the **Name** field, type a name for the supplier.
5. Select **OK**. The purchase order can now be completed and processed like any other order. There are no special characteristics related to how this is done. The invoice will account a due transaction on the vendor account that was created with the order, and payment will then be processed.

Apply a purchase agreement when creating a purchase order

Article • 01/24/2025

This procedure shows how to use a purchase agreement when you create a purchase order. The purchase agreement must be applied when you create the purchase order because there are general terms that should be copied to the purchase order header. Typically this task would be carried out by a purchasing agent. As a prerequisite for this guide, you must have an effective purchase agreement with a product quantity commitment for a vendor and items. The same procedure can be used if you have a purchase agreement with other types of commitments.

Create a purchase order

1. Go to **Procurement and sourcing** > **Workspaces** > **Purchase order preparation**.
2. On the Action Pane, select **New purchase order**.
3. The **Create purchase order** dialog opens. Select a **Vendor account**. Inspect and adjust the other address fields as needed.
4. Expand the **General** FastTab.
5. In the **Purchase agreement** field, find and select the effective agreement that you want to use. All available agreements for the vendor are listed here.
6. Select **Yes**.
7. Select **OK**.

Add a line

1. In the **Item number** field, type a value. If there are specific inventory or location dimensions on the commitment, you must enter the same values on the purchase order line to make use of the agreement.
2. In the **Site** field, select the drop-down button to open the lookup. The site may already be populated with the default value from the order, or from the vendor. If so, skip this step.
3. In the list, find and select the desired record.
4. In the list, select the link in the selected row.
5. In the **Quantity** field, enter a number. Validate that the price is copied from the commitment.

Look up the commitment

1. Select **Update line**.
2. Select **Attached**. Here you can get details for the purchase agreement. For example, you can see the price and whether the price and discount are fixed, which means that if you change price or discount on the purchase order to a different value than on the commitment, the system will remove the link so the purchase order line doesn't fulfill the commitment. You can also see if the Max is enforced option is selected, which means that the quantity on the commitment can't be exceeded by summing all of the purchases that fulfill the commitment.
3. Close the page.

Look up the purchase agreement

1. On the Action Pane, select **General**.
2. Select **Purchase agreement**.
3. Close the page.
4. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a purchase return order

09/03/2025

Use purchase return orders (return POs) when you need to send goods back to a vendor. Here are a few common scenarios where you might create a return PO:

- You received goods that need to be returned
- You created an invoice but didn't receive the goods
- You need to return some of the goods that you received (partial return of goods)
- You received items that weren't ordered (unplanned receipt)
- Vendor buyback or consignment return

This article explains how to create a return PO by using the **Credit note** action to copy lines from a vendor invoice document to a new purchase order. It also shows you how to confirm the order and process shipment of the goods back to the vendor. A purchasing agent typically carries out this task.

Create a new return PO

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**. The first step is to create a new purchase order to use as the return PO.
2. Select **New**.
3. In the **Vendor account** field, specify the vendor you're returning to.
4. Select **OK**.
5. On the Action Pane, select **Purchase**.
6. Select **Credit note**. This page lets you copy from an existing vendor invoice to your return order. This page is the same page that's used for other copy actions. But because you opened it from the **Credit note** action, the page is configured to support the creation of a return order that offsets vendor invoices.
7. Expand the **Parameters** section.

    - The **Invert sign** option is automatically selected, and you can't change it. This option ensures that the sign is changed for the quantities, and that order lines that you add offset the vendor invoice.
    - The **Copy charges** option is automatically selected, and you can't change it. This option means that charges from the vendor invoice are added to the return PO to offset the original charge. You can modify the changes on the order header and lines later.
    - The **Copy precisely** option is automatically selected, and you can't change it. This option ensures that an exact copy is made of the values in all the fields on the

vendor invoice header and lines. This option means that a return PO is created with values that match all terms used with the vendor invoice document.
    - The **Delete purchase lines** option deletes any purchase order lines that already exist on the purchase order before adding the new lines. In this example, you didn't yet add any lines to the return PO, so this option has no effect. Use this option with caution, as it deletes all existing lines without further warning.
    - The **Copy order header** option is automatically selected, and you can't change it. This option ensures that information is copied from the vendor invoice and applied to the return PO header. This option is useful because it helps to ensure that the return PO offsets the invoice by using similar terms.
8. Collapse the **Parameters** section.
9. Expand the **Invoices** section. You opened the page from the **Credit note** action, so the only option available is to copy information from vendor invoices. This tab shows all the available invoices for the vendor account that you specified on the return PO. The invoices are identified by the invoice voucher or the purchase order IDs.
10. Find the vendor invoice that includes the items you want to return and highlight it by selecting any field in that line.
11. Select the check box for the line. The lines available on the vendor invoice are automatically selected together with the order.
12. Highlight the line for the item you want to return by selecting any field in that line. In the **Quantity** field, enter the quantity that you want to return to the vendor.
13. Clear the check box for each line that you don't want to return items for. Only the lines that you select are copied to your order.
14. Collapse the **Invoices** section.
15. Expand the **Selected lines or header to be copied** section. This view shows a summary of all the documents and lines that you selected to copy to your order.
16. Collapse the **Selected lines or header to be copied section**.
17. Select **OK**. The line that you selected is now copied to your return PO. The **Quantity** field shows a negative value for each item you're returning.
18. In the **Purchase order line** section, select **Inventory**.
19. Select **Marking**. The order line that you created is marked against the inventory transaction from the vendor invoice. This marking ensures that the inventory that's returned to the vendor is the same as the inventory that was received from them earlier. There are some situations where marking doesn't occur, for example, if the inventory is already marked as *Consumed*, or if the product is one that doesn't use marking.
20. Select **OK**.

Confirm and record the shipment of goods

1. Select **Actions** > **Confirm**.

2. On the Action Pane, select **Receive**.
3. Select **Product receipt**.

    - Use this page to record product receipt for purchase orders and to process the return of goods back to the vendor. Order lines with a negative quantity mean that goods are to be returned to the vendor, and the document that you can generate from this page can be used as packing slip for this use.
    - In the **Quantity** field, select *Ordered quantity*. This step ensures that the shipment processes the full ordered quantity that the order lines include.

4. In the **Product receipt** field, type a value. Use this field to enter a reference that acts as a voucher for the product receipt journal.
5. Select **OK**. The goods are now recorded as shipped on the return PO, and a product receipt journal is created. Use the Product receipt action to review the journals created with the purchase order, and see what was received or returned, and when.

Create project purchase order

Article • 05/02/2022

This procedure shows you how to create a project purchase order. This task uses the USSI data set.

1. Go to Project management and accounting > Projects > All projects.
2. In the list, click the link in the selected row.
3. On the Action Pane, click Manage.
4. Click Item task.
5. Click Purchase order.
6. In the Vendor account field, enter or select a value.
7. In the Site field, enter or select a value.

    - These steps aren't required, but they do simplify the purchase order by setting up a default site and warehouse for the purchase order lines.

8. In the Warehouse field, enter or select a value.
9. Click OK.
10. In the list, mark the selected row.
11. In the Item number field, enter or select a value.

    - This can be the item number or a procurement category.

12. Expand the Line details section.
13. Click the Project tab.

    - Verify that the sales and cost prices are available. If they are not available but needed, enter the information.

14. Click Save.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a repeat purchase order

This article shows you how to create a repeat purchase order (PO) by copying lines from an earlier purchase order document to a new PO or to an existing PO. You can create repeat orders by using two methods. Use the actions available at the document level from the Action Pane, or use the line detail actions. The document level actions are intended for creating a new purchase order by adding lines and header information from another order, while the line details action is mainly for adding lines to an existing order. You can use the example shown in this guide in the USMF demo data company. A purchasing agent typically carries out this task.

Create a new repeat purchase order

1. In the navigation pane, go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**. First, try the option for copying information to a new order.
2. Select **New**.
3. In the **Vendor account** field, enter *US-101*.
4. Select **OK**.
5. On the Action Pane, open the **Purchase order** tab and, from the **Copy** group, select **From all**. The **Copy from other documents** dialog opens. From here, you can copy from existing orders to your order. There are different options for how the lines are copied, and different kinds of documents that you can copy from. See the options for how lines are copied first.
6. Expand the **Parameters** FastTab.
    - The **Quantity factor** field is useful if you need to use a quantity that is different from the one on the order that you're copying from. For example, if you want to order twice the quantity that's in the lines that you're copying from, type `2` in this field and then the system adds lines where the original quantity is doubled.
    - The **Invert sign** field also supports changing the ordered quantity by changing the sign of the quantity for order lines that are added. This change might be useful if you need to reverse a transaction, by creating order lines that negate the transaction. This option is automatically selected when the page is opened from the **Create credit note** action.

- The **Copy charges** option allows you to copy charges to your new order from the document that you're copying the order lines from.
- The **Recalculate prices** option uses the current prices and discounts rather than copying these values from the document that you're copying other information from.
- The **Copy precisely** option copies the values of several fields on the order header and lines. If you don't select this option, default values are used for many of the fields relating to the vendor and products, just like when you create a new order manually. For example, if you set **Copy precisely** to *Yes* and copy an order that overrides the default invoice account for the vendor, then that same invoice account is copied to your new order. If you set **Copy precisely** to *No*, the default invoice account for the vendor is used on your new order instead. Values for the following fields are copied when **Copy precisely** is set to *Yes*:
  - **Language**
  - **Terms of payment**
  - **Method of payment**
  - **Payment specification**
  - **Number sequence group**
  - **Cash discount**
  - **Discount percentage**
  - **Currency**
  - **Delivery terms**
  - **Mode of delivery**
  - **Dimension**
  - **Sales tax group**
  - **Override sales tax**
  - **Prices include sales tax**
  - **Transport**
  - **Port**
  - **Statistics procedure**
  - **Template ID**
  - **Delivery name**
  - **Postal address**
  - **Reference**
  - **Reference Table ID**
- The **Delete purchase lines** option deletes all purchase order lines that already exist on the purchase order that you're copying to, before applying the new lines. Use this option with caution, as it deletes all existing lines without further warning.

- If you use the **Copy order header** option, you don't need to manually create the header information on your new order. This option results in default values being used for the fields associated with the vendor. If the document that you're copying from has non-default values that you want to copy, use the **Copy precisely** option as well.
- There are different document sources that you can copy from, and each has a separate section on this page. For example, the **Purchase orders** section allows you to copy from existing purchase orders.

7. In the **Purchase orders** section, select the lines you want to copy to your clipboard. You can select additional purchase order lines from other purchase orders and copy them to your order as well. You can also add lines from other kinds of purchase documents. The next few steps review the different options.

8. Expand the **Confirmation** section. Here you can select purchase order confirmations to copy from. The purchase order confirmations are identified by the associated purchase journal ID or the purchase order ID.

9. Expand the **Product receipts** section. Here you can select product receipt journals to copy from. The product receipt journals are identified by the product receipt voucher or the purchase order ID.

10. Expand the **Invoices** section. Here you can select vendor invoices to copy from. The invoices are identified by the invoice voucher or the purchase order ID.

11. Expand the **Selected lines or header to be copied** section. This view shows a summary of all documents and lines that you selected to copy to your order.

12. Select **OK**. The lines that you selected are copied to your new purchase order.

Copy lines to an existing purchase order

Instead of copying an entire order, it's more common to create a new PO and complete the information on the PO header. Then, copy individual lines from existing orders.

1. Select **Purchase order** line.

2. Select **From all**. The page that opens is identical to the one shown earlier, but different options are selected when you open it from the order lines view. Review the parameters.

3. Expand the **Parameters** section.

- The **Delete purchase lines** option isn't selected. This setting means that you can copy new lines to your order without removing existing lines.
- The **Copy order header** option isn't selected, as you're only adding extra lines to the order.
- For this example, copy lines from an existing purchase order.

4. Select the line for the desired purchase order. Notice that the single order line on this PO is also selected.

5. Select **OK**. The extra order line is added to your purchase order.

Last updated on 07/01/2026

Procurement Agent overview (production-ready preview)

[This article is prerelease documentation and is subject to change.]

The Procurement Agent in Dynamics 365 Supply Chain Management works with procurement teams to manage supplier changes that affect purchase orders and supply commitments. By streamlining routine supplier communication and automating impact evaluation, the Procurement Agent enables purchasers to spend less time on manual updates and analysis, and more time on informed decision-making and exception handling.

> **Important**
>
> - This is a production-ready preview feature.
> - Production-ready previews are subject to supplemental terms of use.

The Procurement Agent includes the following capabilities that work together in the procurement workflow:

- Supplier communications
- Impact analysis

You can choose to enable and use both capabilities together or just one of them.

Supplier communications

Supplier communications features of the Procurement Agent manage inbound and outbound updates tied to purchase orders. With it, teams can quickly coordinate with vendors and keep purchase records aligned with supplier commitments.

The inbound functionality monitors and classifies emails from vendors. For those related to purchase orders, it identifies which messages are confirmations and which are change requests. For change requests, it identifies the changed fields. Purchasers can confirm and apply changes without having to find and open the purchase orders, ensuring purchase orders are up-to-date.

Supplier communications also supports outbound communication, enabling teams to automate outgoing or follow-up communication.

Learn more about supplier communications in Supplier communications features of the Procurement Agent.

Impact analysis

The *Impact analysis* features of the Procurement Agent evaluate how supplier changes affect inventory, production schedules, and customer deliveries. When you receive a supplier change through configured sources, impact analysis automatically assesses the downstream effects on orders and inventory. This process helps procurement teams understand the consequences of a change before deciding whether to accept it or take further action.

Vendors might send change requests through various channels. Impact analysis supports change requests arriving through emails processed by supplier communications and changes submitted through the vendor collaboration interface. It classifies each change as either *Has impact* or *No impact*. Learn more in Review the impact of purchase order changes from vendors.

You can also manually initiate impact analysis for a selected purchase order. Learn more in Impact simulation.

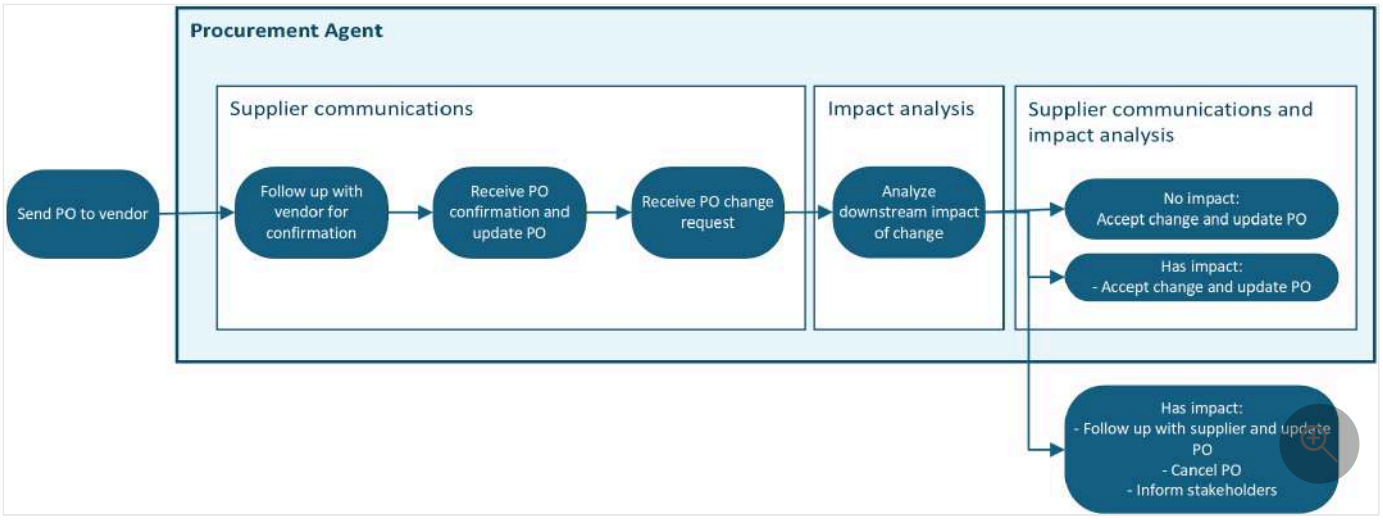
Learn more about impact analysis in Impact analysis overview.

How the Procurement Agent capabilities work together

Procurement Agent capabilities fit into the procurement workflow differently based on whether supplier communications or the vendor collaboration interface is used to communicate with suppliers.

When you process emails through supplier communications together with impact analysis

The following diagram illustrates how the supplier communications and impact analysis capabilities of the Procurement Agent work together when you receive change requests through emails processed by supplier communications. The Procurement Agent detects changes in supplier emails, runs impact analysis for relevant changes, and surfaces the impact analysis result back in the **Emails from vendors** workspace to help purchasers make informed decisions.

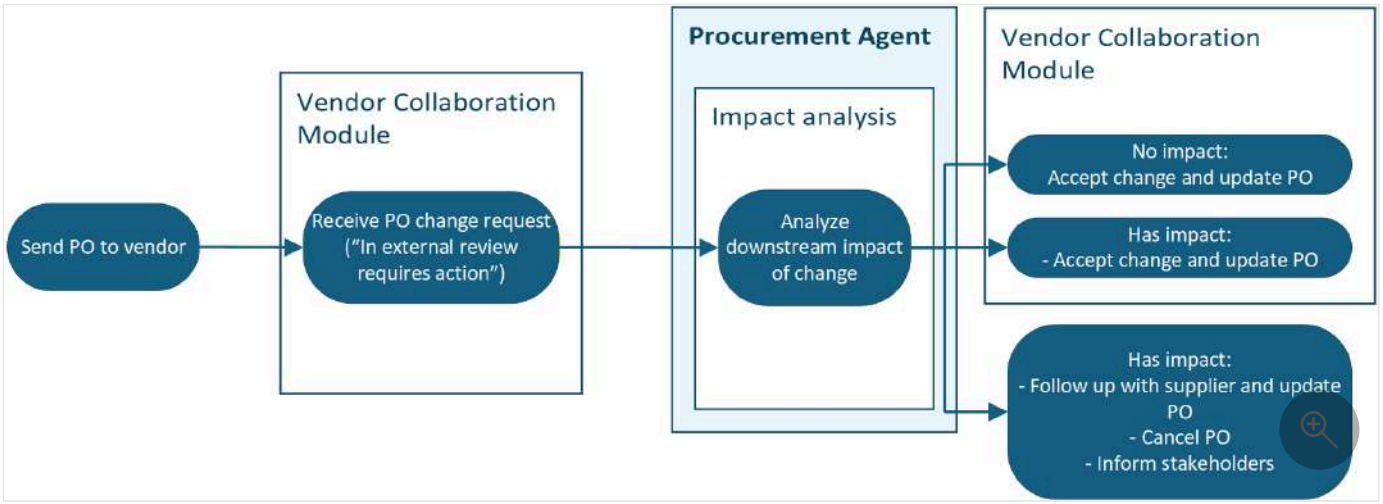


1. Send purchase order to vendor
2. Follow up with vendor for confirmation (supplier communications)
3. Receive purchase order confirmation and update purchase order (supplier communications)
4. Receive purchase order change request and view changed fields in the **Emails from vendors** workspace (supplier communications)
5. Analyze downstream impact of the change and view result in the **Emails from vendors** workspace (impact analysis)
6. Make a decision based on the impact analysis result (supplier communications + impact analysis)

    - *No impact* – The purchaser can accept the change and update the purchase order (supplier communications)
    - *Has impact* – The purchaser can view more details about the impact (impact analysis) and determine how to proceed. This action can be to follow up with the supplier, accept the change and update the purchase order if nothing can be done (supplier communications), cancel the purchase order, and inform stakeholders.

When you use the vendor collaboration interface together with impact analysis

The following diagram shows how the supplier communications and impact analysis capabilities of the Procurement Agent work together when you receive change requests through the vendor collaboration interface. The Procurement Agent runs impact analysis for changes received through the vendor collaboration interface and surfaces the impact analysis result in the **Purchase order preparation** workspace to help purchasers make informed decisions.



1. Send purchase order to vendor.
2. Receive purchase order change request via the vendor collaboration interface's *In external review requires action* tab in the **Purchase order preparation** workspace.
3. Analyze downstream impact of the change (impact analysis).
4. Make a decision based on the impact analysis result (impact analysis).

    - *No impact* – The purchaser can accept the change and update the purchase order.
    - *Has impact* – The purchaser can view more details about the impact (impact analysis) and determine how to proceed. This action can be to follow up with the supplier, accept the change and update the purchase order if nothing can be done, cancel the purchase order, as well as informing stakeholders.

Cost

The Procurement Agent incurs charges based on the number of Microsoft Copilot Studio credits you use when running it. The agent has a fixed cost per run and a variable cost that depends on the resources it consumes. The agent management configurations control this cost.

For information about how charges apply to each capability, see Cost for supplier communications and Cost for impact analysis.

Learn about the billing rates and management for Copilot Studio in Billing rates and management.

Last updated on 08/10/2026

Impact analysis overview (production-ready preview)

[This article is prerelease documentation and is subject to change.]

The Procurement Agent in Dynamics 365 Supply Chain Management works with procurement teams to manage supplier changes that affect purchase orders and supply commitments. Impact analysis evaluates whether and how supplier changes affect inventory levels, production schedules, and customer deliveries. It helps teams react faster and with higher confidence by clearly indicating when purchase order changes proposed by suppliers have downstream impact.

> **Important**
>
> - This is a production-ready preview feature.
> - Production-ready previews are subject to supplemental terms of use.

Instead of spending a lot of time manually tracing pegging, reservations, or planning dependencies across the system for each change communicated by suppliers, teams can quickly determine which changes they can safely accept and which require further action. This process helps prioritize time spent on changes with impact, ultimately protecting service levels and production continuity.

When a supplier proposes a change to a purchase order through a configured source, or when it's manually initiated, impact analysis looks for downstream impact on sales orders, production orders, transfer orders, and inventory levels. It does this by tracing the changed supply across the system to identify orders that are linked to the purchase order through markings and peggings (if available), and checking this tracing against confirmed incoming supply and projected on-hand inventory.

Review the impact of purchase order changes from vendors

Purchasing teams regularly receive change requests from suppliers, including delivery delays, quantity reductions, and cancellations. Each change to a purchase order can potentially affect safety stock, production schedules, internal transfers, and customer commitments. In most cases, purchasers don't want to spend time on changes to purchase orders with no downstream impact.

Impact analysis gives purchasers the confidence, visibility, and ability to quickly identify those changes that can be safely accepted, apply them right away, and move on.

For changes with downstream impact on orders, or changes that cause inventory levels to go below certain thresholds, purchasers need to quickly understand the details and consequences, and decide how to respond. Purchasers often follow up with a supplier to request alternatives, such as expediting production or transportation, resulting in an earlier delivery date, increased quantity delivered, or a split delivery, while also looking for other alternatives such as alternative vendors.

If purchasers must accept a change with impact, they now know straight away which internal stakeholders to inform.

Impact analysis automatically analyzes the downstream impact of changes that arrive through emails (if you're using the supplier communication feature of the Procurement Agent) and through the vendor collaboration interface of Supply Chain Management.

Learn more in Review impact of purchase order changes from vendors.

Simulate whether purchase order changes have impact

The impact simulation capability allows purchasers and planners to test the impact of proposed changes that aren't directly communicated through emails or the vendor collaboration interface. Some key use cases for this simulation include:

- Testing and getting a feel for the impact analysis feature before using it on live supplier change requests.
- Proactively testing potential scenarios for at-risk or high-priority purchase orders. This scenario could be purchase orders for which you didn't receive a reply yet.
- Evaluating whether supplier proposals for expediting or setting new dates or quantities solve the impact. These proposals can be alternatives that result from follow-up conversations with the supplier for changes with impact.
- Evaluating changes received and reviewed in channels that don't automatically trigger an impact analysis.

Learn more in Simulate whether purchase order changes have impact.

Cost

Impact analysis incurs charges based on the number of Microsoft Copilot Studio credits you use when running it. It has a fixed cost per run and a variable cost that depends on the resources it consumes. The agent management configurations control this cost.

Impact analysis runs on changes that vendors communicate through the configured sources. If you set up and configure impact analysis to run for changes coming through emails via supplier communications, you pay credits when the conditions in the configuration of supplier communications are met. For example, if you configure supplier communications to run on incoming emails from all vendors in a shared mailbox, and you configure impact analysis to run on vendor emails, impact analysis consumes credits when any vendor sends an email about purchase order changes.

The fixed cost applies each time the agent runs, and the variable cost depends on the number of line changes communicated.

Learn about the billing rates and management for Copilot Studio in Billing rates and management.

Last updated on 08/10/2026

How impact analysis works (production-ready preview)

[This article is prerelease documentation and is subject to change.]

This article explains how impact analysis determines whether a purchase order line change has downstream impact. It runs automatically based on configured sources, but you can also run it manually.

Impact analysis considers the following purchase order line changes:

- Delayed delivery dates (changes to the *Confirmed delivery date*)
- Quantity decreases
- Cancellations (where quantity is changed to 0)

Supplier communications features of the Procurement Agent detect the following kinds of changes in supplier emails, but these changes don't trigger an impact analysis:

- Unit of measure
- Price
- Quantity increases
- Earlier delivery dates

Planning data used for the impact analysis

Impact analysis uses planning data available in Dynamics 365 Supply Chain Management. It uses any available markings and peggings between supply and demand orders present in the last-run dynamic plan to trace downstream orders impacted by a purchase order line change.

If you use the master planning features of Supply Chain Management, impact analysis follows the supply and demand chain through pegging relationships in the active dynamic plan to find which planned and confirmed orders depend on changed supply. The dynamic plan used for impact analysis is the one set in the **Current dynamic master plan** field on the **Master planning parameters** page (**Master planning** > **Setup** > **Master planning parameters**).

When pegging data is available, impact analysis can trace dependency chains to identify indirectly affected orders, such as those listed on the following pages:

- **Purchase order** > **Production order** > **Production order** > **Sales order**

- **Purchase order** > **Transfer order** > **Sales order**

Pegging enables broader, multilevel impact tracing. When you use the master planning features of Supply Chain Management, the available planning data is richer, which enables deeper and broader impact visibility.

How impact analysis determines impact

Impact analysis evaluates planning and inventory data to place changes into one of the following categories:

- **Has impact** – The change delays downstream orders or causes inventory levels to fall below zero or the minimum level configured for that item.
- **No impact** – The change doesn't delay downstream orders or cause inventory to drop below minimum levels.

Impact analysis considers the following types of downstream orders when determining impact:

- **Sales orders** – Customer orders that might be delayed or can't be fulfilled.
- **Production orders** – Manufacturing orders that depend on the supply.
- **Transfer orders** – Transfers that rely on the item.
- **Forecast orders** – Demand forecast requirements used by the planning system to anticipate and prepare for expected future demand.
- **Safety stock requirements** – Planned requirements generated by the planning system to maintain minimum inventory levels. These requirements appear when the planning system creates planned orders to replenish safety stock, and the purchase order change affects the ability to fulfill those planned requirements.

Has impact scenarios

Impact analysis flags impact when:

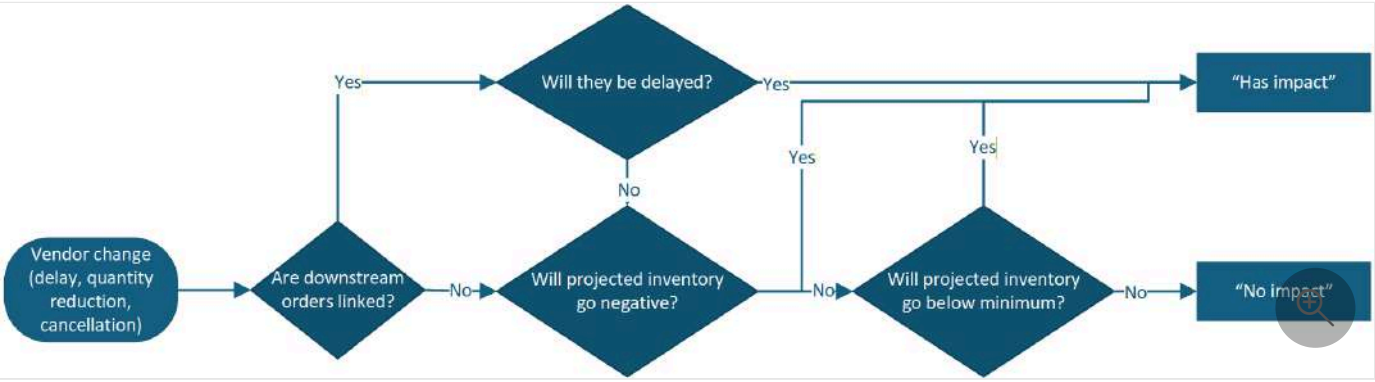
- **Downstream orders are delayed** – The supply date moves after the delivery date required by connected demand.
- **Inventory threshold breaches** – The change is projected to cause inventory levels to drop below one of the following critical thresholds:
  - **Below minimum** – On-hand inventory falls below the configured safety stock level (but remains positive).
  - **Negative** – On-hand inventory goes below zero, resulting in a complete stock-out.

No impact scenarios

A change has no impact when it meets one of the following conditions:

- **No downstream orders found** – The purchase order isn't linked to any dependent demand.
- **Sufficient buffer exists** – Inventory levels stay above the minimum level you set for that item.
- **Change is an improvement** – The purchase order date moves earlier or quantity increases (positive changes don't flag impact).

The following diagram illustrates how impact analysis evaluates whether a change has impact or not based on the downstream orders and inventory levels.



Last updated on 08/24/2026

Set up and configure impact analysis features of the Procurement Agent (production-ready preview)

[This article is prerelease documentation and is subject to change.]

This article explains how system administrators can use the agent deployment wizard from Copilot Hub to set up and configure the impact analysis features of the Procurement Agent, regardless of whether you're also using the supplier communications features of the agent.

If you're already using supplier communications, you can choose to keep the same Entra ID and connections you created in the supplier communications set-up, or you can create new ones.

Prerequisites

To use the impact analysis features of the Procurement Agent, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.48 build 10.39.2117 or later. If you're running 10.0.48, ensure you're running the newest available build.
- The following features must be turned on in feature management. Select **Check for updates** if the features don't appear on your system.
  - *Immersive Home*
  - *Agent management*
  - *(Production-ready preview) Procurement Agent - Impact analysis*

  > **Tip**
  > - If you don't see all of the features you're looking for in the **Feature management** workspace, select **Check for updates** to refresh the list of features.
  > - If you can't enable the *Agent management* feature, ensure that your environment fulfills all of the **prerequisites**, including version requirements and Copilot Studio billing.

- In the Power Platform admin center, ensure you're running the following versions of the following Dynamics 365 Apps in your Supply Chain Management environment. It's important that you install or update them in the following order:
  - First, install *Copilot for finance and operations apps* version 1.0.03048.2 or later. If it's already installed, update it to the latest version.
  - Then, install *Copilot in Microsoft Dynamics 365 Supply Chain Management* version 1.1.03510.1 or later. If it's already installed, update it to the latest version.

- Normally, the Microsoft Copilot Studio agent needed for impact analysis to run is published automatically. But there might be data loss prevention (DLP) policies on your environment that prevent the publishing of this agent. To check if the agents are successfully published, go to Copilot Studio and find your environment. Ensure that the following Microsoft Copilot Studio agent is published in that environment: *Procurement Agent - Impact Analysis*. If the agent isn't published, you can find help in Troubleshoot data policy enforcement for Copilot Studio.

Understand the agent identity user

An *agent identity user* is a dedicated Microsoft Entra ID user account that the Procurement Agent uses to sign in to the systems it connects to. This account isn't a person account for day-to-day business use. Instead, it's a service-style identity that gives the agent the permissions it needs to access Dataverse, Microsoft Copilot Studio, and Supply Chain Management while it performs its work.

You create or select this user during setup because the agent must authenticate to several services and components before it can run impact analysis. The identity user is what allows the agent to read incoming vendor change requests, review the related purchase orders and downstream data, and trigger the processes that evaluate the impact of those changes. The agent deployment wizard uses this identity user as the foundation for the connection and authorization steps you complete to enable impact analysis.

Run the agent deployment wizard

To use the agent deployment wizard to set up the impact analysis features, follow these steps:

1. Open Copilot Hub in Power Platform admin center and select **Dynamics 365 agents** in the **Manage** section.

2. Select **Select environment** and choose your target environment.

3. In the **All agents** section, find the **Procurement agent - Impact analysis (Preview)** tile and select the **Add** button in that tile to launch the agent deployment wizard.

4. The **Overview** page opens, which provides a summary of the agent deployment wizard and its capabilities. Select **Next** to continue.

5. The **Check prerequisites** page opens. This step makes sure your environment meets all of the prerequisites for the agent.

    - The deployment wizard automatically checks most of these prerequisites for you. It shows a green check mark for each fulfilled prerequisite. If any of the features or settings aren't enabled, enable them before you continue. Each section provides links to the Power Platform admin center to help you enable the relevant features or settings. Learn more in Enable or disable Copilot features.
    - In the **Make sure the following apps are up to date with at least the versions noted** section, review this list of required apps and versions. Make sure that the apps listed are installed in your environment and that their versions are equal to or greater than the ones listed. For more information about the required apps and versions, see the prerequisites section.
    - Each time you make changes to meet the prerequisites, go back to the agent deployment wizard and select the **Reload** button at the right side of each section to let the wizard check the status of that section again. When all prerequisites are met, green check marks are shown for all sections.
    - Your environment might include data loss prevention (DLP) policies that prevent it from creating connections for the Procurement Agent. Ensure that the required connections are allowed in your organization. Learn more in Advanced connector policies and Data Loss Prevention Policies.

    When all prerequisites are met, select **Next** to continue.

6. The **Set up agent identity** page opens. Use this page to set up the *agent identity user* account that the agent uses to interact with Dataverse and Microsoft Copilot Studio. To set up your agent identity user, follow links and make settings in the following sections on this page:

    - **Create your agent's Entra user ID** – For security and ease of maintenance, use a dedicated identity for the agent. If you don't already have an eligible user available, select the link provided to open the Microsoft 365 admin center and create a new user that will be the agent identity user. Then select that user in the drop-down list

provided. Select **Set up**. This action triggers the wizard to check the required roles in the next section, **Set up identity in environment**.

- **Set up identity in environment** – This section shows the steps required to configure the agent identity user in your Power Platform environment. Green check marks indicate which steps are completed successfully. A link to the Power Platform admin center environment user settings is provided so that you can complete the required actions manually if needed.
- **Assign required product licenses** – The Procurement Agent uses premium tier connectors, so the agent identity user you selected must have a license that permits those connectors. You must check this license manually in the Power Platform admin center. Learn more in Power Platform licensing FAQs or download the Licensing Guide. Examples of sufficient licenses are listed in this section. Select the link provided to open the Microsoft 365 admin center, where you can review and assign licenses for the agent identity user. Once you verify that the licenses are in place, check **Complete**.
- **Set up identity in Finance and Operations** – Add the agent identity user as a user in Supply Chain Management and assign the security roles listed in this section. You must check this setting manually in Supply Chain Management. Make a note of the roles listed and then select the link provided to open the **User settings** page of Supply Chain Management, where you can review and assign security roles for the agent identity user. Ensure that each of the required roles is assigned to the agent identity user, and then go back and continue to complete wizard. Once you verify that the roles are in place, check **Complete**.

After you confirm all the required settings, select **Next** to continue.

7. The **Connect agent** page opens. The agent identity user you specified on the previous page is used to set up these connections.

   - **Connect the agent** – The agent uses the types of connections listed here. For each type, select an existing connection from the menu if one is available. If no connections are available, select the + button to create a new connection. Select **Connect agent** to use the connections you selected, and wait until the agent is connected.
   - **Activate data flows and processes** – Select **Activate flows** and wait for all of the flows listed to switch to the *Activated* state.

   When all connections and data flows are shown as successful, select **Next** to continue.

8. The **Set up user permissions** page opens. Add users who can interact with the agent and specify if they can configure it or only view results.

    - In the **Add a user** search field, search for a user. In the **Select a role** dropdown, assign the *Agent manager* or *Agent user* role. The *Agent manager* role grants the user permission to enable and edit configurations of the agent, while the *Agent user* role only grants the user permission to view impact analysis results. Users assigned the *Agent manager* role can also view results of the agent.
    - If you need to create new users in your environment to manage the agent configuration or review agent results, see Create users.

    After you assign permissions to users, select **Next** to continue.

9. The **Enable agent** page opens.

    - **Publish Copilot Studio agents** – The wizard automatically publishes the Microsoft Copilot Studio agents needed for the Procurement Agent to run. If this action succeeds, the **Deploy** button is greyed out and shows the help text *All agents are already published*. If the agent isn't automatically published, it might be because data loss prevention (DLP) policies in your environment prevent this action. Learn more in Troubleshoot data policy enforcement for Copilot Studio. Select the **Microsoft Copilot Studio: Agents** link provided here to go to Copilot Studio, where you can check whether these agents are published and publish them if necessary. Learn more in Key concepts - Publish and deploy your agent.
    - **Enable Immersive Home and agent features in Finance and Operations** – This section lists the features that you must turn on in Supply Chain Management. Make a note of the features listed here and then select the link provided to open the **Feature management** workspace, where you can enable each of the features.

    > **Tip**
    >
    > - If you don't see all of the features you're looking for in the **Feature management** workspace, select **Check for updates** to refresh the list of features.
    > - If you can't enable the *Agent management* feature, ensure that your environment fulfills all of the **prerequisites**, including version requirements and Copilot Studio billing.

    Select **Next** to continue.

10. The final page of the wizard opens. Select **Finish** to complete the setup.

Configure sources that automatically trigger impact analysis

To use impact analysis on incoming change requests received through emails or the vendor collaboration interface, first enable the relevant sources. Learn more in Supplier communications features of the Procurement Agent and Vendor collaboration with external vendors.

To enable impact analysis to run based on change requests coming through one or both sources, follow these steps:

1. Sign in to the Supply Chain Management environment as a user who has permissions to manage the agent configuration.
2. Go to **Agents** > **Agents**.
3. Open the **Library** tab.
4. Find the *Show impact of changes in purchase orders - Impact analysis - Procurement Agent* tile and select **Select** for that tile.
5. Open the **Source** dropdown list and select one or both of the following options:

    - *Vendor emails* – Select this option if you're using supplier communications features of the Procurement Agent to receive and classify emails from vendors.
    - *Vendor collaboration module* – Select this option if you're using the vendor collaboration interface to receive and manage vendor responses.

6. Select **Activate**.

If you're using impact analysis with supplier communications, impact analysis only runs on configurations set for the *Updates from vendors (reading vendor emails)* feature (see Configure the agent to track your email). If you set up a configuration for the Procurement Agent to read and review emails from all vendors, impact analysis runs on all change requests received from all vendors. If you set up a configuration for the Procurement Agent to read and review only emails from certain vendors, impact analysis only runs on change requests received from those specific vendors. If your organization sets up multiple configurations, for example to accommodate different purchasers responsible for their own vendors, the impact analysis configuration applies to all of them. Impact analysis runs for all change requests received from all specified vendors in all configurations.

Last updated on 08/26/2026

Review impact of purchase order changes from vendors (production-ready preview)

[This article is prerelease documentation and is subject to change.]

Impact analysis can automatically run based on supplier changes received through emails or the vendor collaboration interface. Whichever options you configure also show the results of the impact analysis for review directly in Dynamics 365 Supply Chain Management, so purchasers can decide how to respond in their existing workflow.

Review changes received via email (supplier communications)

The supplier communications features of the Procurement Agent classify whether emails from vendors are confirmations, change requests, or related to something else (indicated as *Other*). If you enable impact analysis, it runs automatically for all the change requests you receive. If an email that the system classifies as a *Change request* in the list pane is shown as *Has impact*, it means that it includes either a purchase order header or line change that *Has impact*.

Vendors can communicate changes to both purchase order headers and lines. For example, the whole purchase order can be delayed, specific lines can have delays or quantity changes, or a combination of these changes can occur. For this reason, there's an **Impact** column on both the **Purchase order header** and **Purchase order lines** grids.

The system runs the impact analysis for purchase order line changes only. If a delay to an entire purchase order is communicated, that delay applies to the purchase order header, the Procurement Agent propagates this delay to all of the lines in that purchase order, and runs an impact analysis on all lines individually. Any line-level changes always take precedence, so if a delay to the purchase order header is communicated, but one line is even more delayed, the Procurement Agent runs the impact analysis on the line-level delay.

To review if changes have downstream impact, follow these steps:

1. Open the **(Preview) Emails from vendors** page by doing one of the following steps:

    - Go to **Procurement and sourcing** > **Purchase order receipt and follow-up** to open the **Purchase order receipt and follow-up** workspace. Then select the **(Preview)**

Emails from vendors tile.

- Go to **Procurement and sourcing** > **(Preview) Procurement Agent - Supplier communications** > **(Preview) Emails from vendors**.

2. Select an email classified as a *Change request* in the list pane, and then look at the grids on the **Summary** FastTab, where changed field values are highlighted in bold. These changes can be on the purchase order header and/or lines.

3. Check the **Impact** column on both the **Purchase order header** and **Purchase order line** grids. It shows one of the following values based on the impact analysis results: *Has impact* or *No impact*.
   a. If a change to a **Purchase order header** *Has impact*, this condition means that a purchase order header delay causes at least one line to have downstream impact. If there isn't a header change, but instead line-level changes, check the **Impact** column on the **Purchase order lines** grid.
   b. Purchase order line changes can show the following results:

      - *Has impact* - Change affects downstream orders and/or inventory levels.
      - *No impact* - Change can be safely accepted.

4. Expand the **Impact of purchase order line** FastTab to see a brief summary of each line change that *Has impact*.

5. To view the full impact analysis of a change that *Has impact*, select the **View details** button. The **Impact analysis** page opens. Learn more about this page in the Impact analysis details page section later in this article.

Review changes received via vendor collaboration interface

When vendors respond to purchase orders through the vendor collaboration interface, you can review proposed changes and their downstream effects directly from the purchase order preparation workspace. If you enabled impact analysis for the vendor collaboration source, the impact results are available alongside the vendor's response so you can decide how to proceed. To review the impact of changes received through the vendor collaboration interface, follow these steps:

1. Go to **Procurement and sourcing** > **Workspaces** > **Purchase order preparation**.
2. On the **Orders** FastTab, open the **In external review requires action** tab.
3. Find and select a purchase order with a **Vendor response status** of *Accepted with changes*.

4. To view more details about the impact of the proposed change on the selected purchase order, do one of the following steps:

    - On the toolbar, select **View impact analysis** to open the **Impact analysis** page, which is described in the next section.
    - On the toolbar, select **View responses to latest sent order** to open the vendor's proposed changes. In the **Purchase order line impact** section, review the impact summary and select **View details** to see the full impact analysis on the **Impact analysis** page, which is described in the next section.

Impact analysis details page

The **Impact analysis** page opens when you select **View details** from either the email-based workflow or the vendor collaboration workflow. This page shows all the downstream effects of the proposed change in one place. The **Impact analysis** page includes a summary of the impact, a detailed list of impacted orders, and a detailed list of inventory impacts, including a graph that shows projected inventory levels over time. The following subsections describe the various parts of the **Impact analysis** page.

Filter and item information

The **Filter by** section shows key information about the purchase order and item affected by the proposed change, along with a selector to switch between different lines if the purchase order has multiple lines. The information in this section helps you understand which item and purchase order line you're analyzing, and gives you quick access to related information and planning details.

This section provides the following information and features:

- The **Purchase order line** being shown, which provides a drop-down list that lets you select a line for multi-line purchase orders.
- The **Item number** and **Item name** being analyzed.
- An **Open net requirements** link, which opens the detailed planning logic used for the impact calculation if you're using master planning.

Purchase order line impact

The **Purchase order line impact** FastTab provides an AI-generated summary of the downstream impact, including specific mentions of affected orders and inventory problems. The agent

analyzes the full impact details and provides a concise natural language explanation to give you a quick overview before you dive into detailed tables and graphs.

Impacted orders

The **Impacted orders** FastTab shows a detailed table of all downstream orders affected by the purchase order change. It includes sales orders, production orders, transfer orders, safety stock orders, and forecasts that the proposed change impacts. This detailed view helps you understand exactly which orders are affected and how, so you can make informed decisions about accepting the change and communicating with relevant stakeholders.

Each row in the grid on the **Impacted orders** FastTab provides the following information about the impacted order:

Expand table

Column	Description
Order type	Sales order, Production order, Transfer order, Safety stock order, or Forecast
Order number	Specific order ID
Customer/reference	Customer name for sales orders, or production/transfer number
Tracing method	How the order is linked: *Pegging* (from planning) or *Marking* (explicit reservation)

Use the **View order**, **View item**, and **View customer** buttons to open related pages in Supply Chain Management.

If this table is empty, the change has no direct impact on specific orders.

Impacted inventory

The **Impacted inventory** FastTab shows whether the purchase order change causes inventory to drop below safety stock or go negative. The following table summarizes the columns shown on the **Impacted inventory** FastTab:

Expand table

Column	Description
Site	The site where inventory is affected.
Warehouse	The warehouse where inventory is affected.

Column	Description
Expected inventory level	Shows one of the following values to indicate how the change impacts inventory levels: • *Below minimum* – Inventory drops below configured safety stock level but remains positive. You still have inventory, but your safety buffer is consumed. • *Negative* – Inventory goes below zero (complete stockout). You can't fulfill orders from on-hand inventory.
Expected impact date	The date when the inventory breach occurs.

The inventory projection graph on the **Impacted inventory** FastTab visualizes inventory levels over time, showing how the purchase order change affects your projected inventory. The graph includes a legend with two lines:

- **Projected inventory before change** – Shows expected inventory levels based on the original purchase order and plan (displayed as a stepped line).
- **Projected inventory after change** – Shows projected inventory levels if you accept the vendor's proposed change (displayed as a stepped line). If the after-change line drops below the safety stock reference line, this condition is highlighted as a **Below minimum** impact. If it goes below zero, it is highlighted as a **Negative** impact.

The two lines might diverge without resulting in an impact, which means that projected inventory levels after the vendor change might be lower, but still above minimum level.

Additional visual elements:

- **X-axis** – Timeline showing dates
- **Y-axis** – Inventory quantity
- **Horizontal reference lines** – Show the safety stock threshold if available

> **Important**
>
> This projection is based on current planning data. It shows what *would happen* if you accept the change, but the plan isn't updated yet. Once you accept the change and update the purchase order, planning needs to run to create the new official plan.

Accept a change and apply it to the purchase order

Purchasers can accept changes that:

- Have no or low downstream impact.
- Have an impact, but the purchaser decides to accept the change anyway. This decision might be because there's no other option but to accept.

To accept changes without going back to the purchase order, see one of the following articles based on how you receive changes from vendors:

- To accept changes processed by the supplier communications features of the Procurement Agent, see Review and apply purchase order changes received in vendor emails.
- To accept changes when received through the vendor collaboration interface, see Updating a PO when a vendor suggests changes.

When purchasers choose to accept a change that has a downstream impact, they now have a clear understanding of that impact and can communicate it directly to the relevant stakeholders.

Last updated on 08/24/2026

Simulate whether purchase order changes have impact (production-ready preview)

[This article is prerelease documentation and is subject to change.]

The impact simulation capability enables purchasers and planners to test the impact of changes that aren't directly or explicitly communicated through emails or the vendor collaboration interface. This functionality is useful when proactively testing potential changes to critical or at-risk purchase orders, seeing whether changes received through external channels have an impact, or evaluating whether supplier follow-up proposals to increase quantities or deliver earlier solves the impact.

To run an impact simulation, follow these steps:

1. Go to **Procurement and sourcing** > **Purchase orders** > **All Purchase orders** and select a purchase order.
2. On the Action Pane, open the **Purchase Order** tab and, from the **Impact simulation** group, select **Simulate purchase order changes**.
3. Modify the purchase order details for the simulation by completing any or all of the following steps:

    - In the **Purchase order heading** section, select a new **Confirmed receipt date** to change the date for all purchase order lines.
    - In the **Purchase order lines** section, select a new **Confirmed receipt date** to change the date for one or more specific purchase order lines.
    - In the **Purchase order lines** section, select a new **Quantity** to change the quantity for one or more specific purchase order lines.

4. Select **Simulate impact**.
5. The result is displayed as described in Review impact of purchase order changes from vendors.

> **Important**
>
> Impact simulations *don't* change the actual purchase order or affect planning calculations. Simulations use the current planning data at the time of simulation and are for analysis only.

Last updated on 08/10/2026

Test impact analysis features (production-ready preview)

[This article is prerelease documentation and is subject to change.]

Impact analysis helps you understand how vendor change requests affect downstream orders, such as production, sales, and transfer orders, before you take action. This article describes four ways to test impact analysis, depending on whether you want to test with or without supplier communications, and whether you're using a sandbox or production environment:

- **Forward actual vendor emails** to a testing email address in your sandbox environment.
- **Use actual vendor emails** in your production environment.
- **Use the Add vendor messages functionality** in your sandbox or production environment to test without setting up a complete email pipeline.
- **Simulate a purchase order change** in your sandbox or production environment without using supplier communications.

Choose the approach that best matches your current setup and testing goals.

Test impact analysis with supplier communications

Test with supplier communications by forwarding actual vendor emails to a testing email address in your sandbox environment

If you're testing supplier communications in your sandbox environment, you can enable impact analysis to see the impact of forwarded change requests. This testing approach corresponds to the supplier communications testing phases *Phase 1* and *Phase 2* outlined in Test the "Updates from vendors" (reading emails) feature.

Test with supplier communications on actual vendor emails in your production environment

If you're already using supplier communications in production, you can see impact analysis running on live change requests that you receive and use up-to-date planning data. This testing approach corresponds to the supplier communications testing phase *Phase 3* outlined in Test the "Updates from vendors" (reading emails) feature, and is the optimal way to test impact analysis with supplier communications.

Test with supplier communications using the *Add vendor messages* functionality in your sandbox or production environment without a complete email setup

Instead of testing impact analysis on actual change request emails you receive from vendors, use the *Add vendor messages* functionality as described in Try out the *updates from vendors* feature without a complete email setup (production-ready preview). You can write a sample email from one of your vendors or you can copy and paste a real email you received from a vendor. Ensure the message communicates a change request to at least one purchase order line.

When you select **Save**, it takes a few minutes for supplier communications and impact analysis to review the message. Select **Refresh** to update the page, after which the email queue in the left-hand navigation pane of the **(Preview) Emails from vendors** workspace updates to show the new message. The message is labeled a *Change request* and shows a status of *Analyzing impact*. Select **Refresh** again to see the result of the impact analysis, which is shown as *Has impact* or *No impact*. Learn more about navigating and interpreting the impact analysis results in Review impact of purchase order changes from vendors.

Testing impact analysis with supplier communications in sandbox vs production

Impact analysis can only find the affected downstream orders when planning data is current. This requirement can cause problems when you test impact analysis with supplier communications in a sandbox environment. If planning data is outdated, you might not find downstream orders such as production, sales, and transfer orders for the purchase order referenced in the incoming email.

You can avoid the risk of a blank result by testing impact analysis with supplier communications in a production environment. In this case, you can use actual vendor emails and up-to-date planning data to see the impact of incoming change requests. Impact analysis doesn't take any action on your behalf—it only runs on incoming change requests that you configure to trigger it. Learn more in Configure sources that automatically trigger impact analysis.

If you want to test in a sandbox environment, consider refreshing data immediately before testing and verify that the items on the test purchase order are linked to demand orders such as sales, production, or transfer orders. Check this condition on the **Net requirements** page. If you don't want to refresh data, you can create test demand orders and link them to the purchase order you want to test before you run impact analysis. If you use this approach, remember to rerun planning before trying out impact analysis for the test order.

Test impact analysis without supplier communications using the simulation capability in your sandbox or production environment

You can test impact analysis without supplier communications by simulating a date and quantity change to a purchase order. Learn more in Simulate whether purchase order changes have impact.

Ensure that the item on the purchase order for which you're simulating a change is linked to downstream demand in **Net requirements**. If you manually add linked orders such as sales orders, remember to rerun planning before running the simulation.

Last updated on 08/26/2026

Supplier communications features of the Procurement Agent (production-ready preview)

[This article is prerelease documentation and is subject to change.]

Purchasing personnel spend a large part of their day on manual tasks that are related to communicating with vendors, following up with them, and maintaining changes that are related to purchase orders. Often, these tasks have low complexity, but they're manual and repetitive. Therefore, procurement personnel are eager to automate them. They can then spend the time that they save on more complex and meaningful tasks. Automating these tasks can also help companies save on procurement costs.

> **Important**
>
> - This is a production-ready preview feature.
> - Production-ready previews are subject to supplemental terms of use.

Communication between procurement departments and vendors is often unstructured and conducted via email, even for companies that implement electronic data interchange (EDI) and companies that work with recurrent vendors.

The Procurement Agent uses AI to automate many communications with vendors. Therefore, purchasers can focus on tasks that add more value and can be more productive.

Follow up on purchase orders

The Procurement Agent can automatically generate emails that remind vendors to confirm a purchase order or ask why a delivery is late. Therefore, purchasers don't have to spend time finding the orders that require attention and then writing emails for them. The agent can automatically send the emails right away, or it can save them as drafts that require human review before they can be sent.

Learn more in Follow up on purchase orders using the Procurement Agent.

Speed up purchase order communications

The Procurement Agent helps purchasers speed up communications with vendors by enabling Copilot to read emails from some or all vendors.

The agent can determine what each email is about (for example, whether it's a purchase order confirmation or a purchase order change request) and which purchase order numbers it applies to. The agent then matches information that it extracts from the email to fields in the system and indicates any changes. In this way, purchasers can review just that information and update the fields in the system as required. They don't have to manually open the purchase order and do all the work. They can then spend the time that they save on other tasks that add more value.

Learn more in Review and apply purchase order changes received in vendor emails.

Specifically, the agent helps with the following scenarios:

- **Increase or decrease order quantity** – The agent detects a supplier's request to increase or decrease the ordered quantity by reading the email body and any attachments (PDF). It summarizes the requested change, identifies the impacted purchase order and purchase order line, and flags it for approval when needed.
- **Cancel quantity or line** – The agent recognizes that the supplier is asking to cancel (for one line or multiple lines) and matches the message to the correct purchase order and lines. It provides a short summary of what is being reduced.
- **Price increase or decrease** – The agent detects a requested unit price increase or decrease (including supporting rationale if provided) by reading the supplier's email and any attached documentation. It summarizes the old versus new price (where available) and flags it as a commercial change requiring buyer decision or approval.
- **Cancel balance** – When a supplier indicates they won't fulfill the remaining (open) balance of a purchase order, the agent detects this intent and identifies the affected purchase orders and lines. It summarizes what portion is being canceled (if stated) and flags it as an exception that typically needs buyer intervention (alternate sourcing, renegotiation, and so on).
- **Updating confirmed delivery date** – If the supplier confirms or revises the order confirmation date, the agent extracts the date change from the email or attachment and links it to the correct purchase order. It summarizes what changed (previous versus updated date if present) and surfaces it so the buyer can quickly validate and accept.
- **Updating confirmed shipping date (ETA based on ETD)** – When the supplier provides only an estimated time of departure (ETD) or ship date (not an *into warehouse* date), the agent detects which date is provided and indicates the estimated time of arrival (ETA) date. The agent can learn to infer or derive the receiving ETA using agreed transit-time assumptions.

- **Match unit of measure** – If the supplier uses a different unit representation (for example, *packs* versus *each*), the agent identifies the unit mismatch and still extracts the intent of the change. The agent can learn or standardize unit patterns over time and presents a clear summary for the buyer rather than failing the match.
- **Follow up for not confirmed purchase order** – For purchase orders that remain unconfirmed after a defined threshold (such as seven days), the agent identifies the gap and prompts follow-up with the vendor.
- **Follow up on delayed purchase order** – For purchase orders that are delayed beyond a defined threshold, the agent identifies the gap and prompts follow-up with the vendor.

The agent doesn't currently support the following scenarios:

- **Splitting a line into multiple deliveries** – When a supplier proposes splitting delivery (for example, *ship 30 now, 70 later*), the agent should extract the proposed delivery breakdown from the email or attachment and summarize it. It should clearly call out new delivery dates and quantities, associate them to the correct purchase order lines, and split them into multiple deliveries. This scenario isn't yet supported.
- **Changing site for delivery receipt for specific line** – If the supplier requests shipping to a different site or warehouse for a particular line, the agent should extract the new delivery location from the message or attachment and match it to the purchase order line. It should summarize the requested site change and highlight it as a logistics-impacting update requiring review before acceptance. This scenario isn't yet supported.
- **Changing vendor for a specific line** – When a supplier asks to ship from a different operating entity (same supplier, different state or account), the agent should detect that the request implies a vendor account change for processing. It should summarize the reason and proposed vendor ID (if provided) and flag it as an exception scenario for procurement to validate and approve. This scenario isn't yet supported.
- **Changing vendor for the purchase order** – When a supplier asks to ship from a different operating entity (same supplier, different state or account) for the entire purchase order, the agent should detect that the request implies a vendor account change for processing. It should summarize the reason and proposed vendor ID (if provided) and flag it as an exception scenario for procurement to validate and approve. This scenario isn't yet supported.

Cost

The supplier communications features of the Procurement Agent incur charges based on the number of Microsoft Copilot Studio credits you use when running it. The agent has a fixed cost

per run and a variable cost that depends on the resources it consumes.

- For the *follow up on purchase orders* feature, the fixed cost applies each time the agent runs. The variable cost depends on the number of emails the agent writes.
- For the *review and apply purchase order changes* feature, the fixed cost applies each time the agent runs. The variable cost depends on the number of emails the agent reads and the number of attachments those emails have.

Learn about the billing rates and management for Copilot Studio in Billing rates and management.

Related information

- Responsible AI FAQ for the Procurement Agent

Last updated on 08/10/2026

Try out the *updates from vendors* feature without a complete email setup (production-ready preview)

[This article is prerelease documentation and is subject to change.]

This article explains how to try out the *updates from vendors* (reading vendor emails) functionality of the Procurement Agent without having to do a complete email setup first.

The Procurement Agent has two independent supplier communication features:

- *Follow-up with vendors* (writing reminders)
- *Updates from vendors* (reading vendor emails)

The *follow-up with vendors* feature is straightforward to set up and try out, but the *updates from vendors* feature requires a more complex email setup. If you want to try the *updates from vendors* feature without doing a complete email setup, use the approach described in this article. This approach lets you try the feature by creating a few test emails so you can see how the agent works and how it handles the kinds of messages that you receive.

To try out this feature, follow these steps:

1. Enable the *(Production Ready Preview) Procurement Agent - Supplier communications* feature in Feature management.

2. Go to **Procurement and sourcing** > **(Preview) Procurement Agent - Supplier Communications** > **(Preview) Emails from vendors**.

3. When the feature is fully configured, the **Emails from vendors** workspace lists the emails that the agent identified as coming from vendors. However, you can also add messages manually to try out the feature without needing to set up email integration. To do that, select **Add vendor messages** from the Action Pane.

4. Enter the following information into the dialog:

    - **Title** – Enter a title for your test message.
    - **Message** – Enter or paste the body of the email. You could copy this content from the body of an existing email that you want to try or just enter a fictional message.

- **Attachment** – If you'd like to add an attachment, select the **Browse** button here and then select an image or PDF file. You can add up to 10 attachments.

5. Select **Save**.

6. Wait for a few minutes and refresh the page, which then shows your message. Select the email to see how the agent processed it, what information it extracted, and how it classified the email.

Last updated on 08/10/2026

Set up and configure supplier communications features of the Procurement Agent (production-ready preview)

[This article is prerelease documentation and is subject to change.]

This article explains how system administrators can use the agent deployment wizard from Copilot Hub to set up and configure the supplier communications features of the Procurement Agent.

Prerequisites

Before you can use the supplier communications features of the Procurement Agent, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later, with all available quality updates.
- The following features must be turned on in feature management. Select **Check for updates** if the features don't appear on your system.
  - *(Production ready Preview) Immersive Home*
  - *(Production ready preview) Agent management*
  - *(Production ready preview) Procurement Agent - Supplier Communications*
  - Optional: To have the agent send emails automatically, turn on the feature *(Production ready preview) Procurement Agent - Supplier Communications - automatically sending follow-up emails*. We recommend that you turn off this feature for sandbox environments because data (such as purchase orders) might not be up to date, or vendor email addresses might be missing.

> **Tip**
> - If you don't see all of the features you're looking for in the **Feature management** workspace, select **Check for updates** to refresh the list of features.
> - If you can't enable the *Agent management* feature, ensure that your environment fulfills all of the **prerequisites**, including version requirements and Copilot Studio

billing.

- In the Power Platform admin center, ensure you're running the following versions of the following Dynamics 365 Apps in your Supply Chain Management environment. Install or update them in the following order:
  - First, install *Copilot for finance and operations apps* version 1.0.03048.2 or later. If it's already installed, update it to the latest version.
  - Then, install *Copilot in Microsoft Dynamics 365 Supply Chain Management* version 1.1.03510.1 or later. If it's already installed, update it to the latest version.

- Normally, the Microsoft Copilot Studio agents needed by supplier communications features are published automatically. However, data loss prevention (DLP) policies on your environment might prevent publishing these agents. To check if the agents are successfully published, go to Copilot Studio and find your environment. Ensure that the following Microsoft Copilot Studio agents are published in that environment:
  - *Supplier communications Agent - inbound*
  - *Supplier Communications Agent - outbound*.

  If the two agents aren't published, see Troubleshoot data policy enforcement for Copilot Studio.

Video instructions

For video instructions on how to set up supplier communications, watch Supplier Communications: Setup and configure via agent deployment wizard.

The remaining sections in this article provide the same instructions in a text-based format.

Understand the agent identity user

An *agent identity user* is a dedicated Microsoft Entra ID user account that the Procurement Agent uses to sign in to the systems it connects to. This account is distinct from a person who uses the mailbox day to day. Instead, it's a service-style identity that gives the agent the permissions it needs to access Dataverse, Microsoft Copilot Studio, Exchange, and Supply Chain Management while it reads and responds to supplier emails.

You create or select this user during setup because the agent must authenticate to multiple services before it can monitor mailboxes, classify incoming emails, and send follow-up messages. In the context of supplier communications, the identity user is the account that the wizard uses to

connect the agent to the relevant mailboxes and background services. That connection lets the agent process incoming vendor emails and manage related actions in Supply Chain Management without requiring a human to sign in for each step.

Run the agent deployment wizard

To set up the supplier communications features, follow these steps:

1. Open Copilot Hub in Power Platform admin center and select **Dynamics 365**.
2. Select **Select environment** and choose your target environment.
3. In the **All agents** section, find the **Supplier Communications Agent** tile and select the **Add** button in that tile to launch the agent deployment wizard.
4. The **Overview** page opens, which provides a summary of the agent deployment wizard and its capabilities. Select **Next** to continue.
5. The **Check prerequisites** page opens. Make sure your environment meets all of the prerequisites for the agent. The prerequisites are organized into sections on this page, and you must meet all of the prerequisites in each section to continue. Here's how to review the prerequisites:

    - In the **Make sure the following apps are up to date with at least the versions noted** section, review this list of required apps and versions. Make sure that the apps listed are installed in your environment and that their versions are equal to or greater than the ones listed. For more information about the required apps and versions, see the prerequisites section.
    - If any of the apps aren't installed or updated to the required version, install or update them before you continue. A link to the Power Platform admin center is provided to help you check the app versions and do the installations or updates if needed. Mark the **Complete** check after you've confirmed that all of the apps meet the requirements.
    - The remaining sections on this page automatically check whether all other required features and settings are enabled in your environment. If they are, green check marks are shown. If any of the features or settings aren't enabled, enable them before you continue. Links to the Power Platform admin center are provided in each section to help you enable the relevant features or settings if needed. Learn more in Enable or disable Copilot features.
    - Each time you make changes to meet the prerequisites, go back to the agent deployment wizard and select the reload button at the right side of each section to let

the wizard check the status of that section again. When all prerequisites are met, green check marks are shown for all sections.
- Your environment might include data loss prevention (DLP) policies that prevent creating connections for the Procurement Agent. Ensure that your organization allows the required connections. For more information, see Advanced connector policies and Data Loss Prevention Policies.

When all prerequisites are met, select **Next** to continue.

6. The **Set up agent identity** page opens. Use this page to set up the *agent identity user* account that the agent uses to interact with Dataverse and Microsoft Copilot Studio. To set up your agent identity user, follow links and make settings in the following sections on this page:

   - **Create your agent's Entra user ID** – For security and ease of maintenance, use a dedicated identity for the agent. If you don't already have an eligible user available, select the link provided to open the Microsoft 365 admin center and create a new user that will be the agent identity user. Then select that user in the drop-down list provided.
   - **Set up identity in environment** - This section shows the required steps to configure the agent identity user in your Power Platform environment. Green check marks indicate which steps have been completed successfully. A link to the Power Platform admin center environment user settings is provided so that you can complete the required actions manually if needed.
   - **Assign required product licenses** – The Procurement Agent uses premium tier connectors, so the agent identity user you selected must have a license that permits those connectors. Learn more in Power Platform licensing FAQs or download the Licensing Guide. Examples of sufficient licenses are listed in this section. Select the link provided to open the Microsoft 365 admin center, where you can review and assign licenses for the agent identity user.
   - **Set up identity in Finance and Operations** – The agent identity user must be added as a user in Supply Chain Management and assigned the security roles listed in this section. Make a note of the roles listed and then select the link provided to open the **Users** page of Supply Chain Management, where you can review and assign security roles for the agent identity user. Ensure that each of the required roles is assigned to the agent identity user, and then go back and continue to complete wizard.

   After you confirm all the required settings, select **Next** to continue.

7. The **Connect agent** page opens. To set up each of the required connections, make settings in the following sections on this page. The agent identity user you specified on the previous page is used to set up these connections.

   - **Connect the agent** – The agent uses the types of connections listed here. For each type, select an existing connection from the menu if one is available. If no connections are available, select the + button to create a new connection. Select **Connect the agent** to use the connections you selected, and wait until the agent is connected.
   - **Activate data flows and processes** – Select **Activate data flows** and wait for all of the flows listed to switch to the *Activated* state.

   When all connections and data flows are shown as successful, select **Next** to continue.

8. The **Configure mailboxes** page opens – To enable the email analysis and delivery features of the Procurement Agent, you must configure one or more mailboxes and synchronize them with Dataverse using server-side synchronization. You can choose to use shared mailboxes, private mailboxes, or both. At a minimum, you must configure at least one mailbox.

   - To use a shared mailbox, follow these steps:
     a. Select **Set up shared mailbox**.
     b. Enter the shared mailbox email address. Ensure the shared mailbox already exists in Exchange Server. Learn more in Create a shared mailbox.
     c. Select **Search** and then select **Set up**.
     d. Follow the on-screen instructions to approve and enable the configuration for the shared mailbox.

     This process associates the shared mailbox with a team in the Power Platform environment.

     > **Important**
     >
     > Add all users who create agent configurations or review agent results related to this mailbox as members of the associated team.

   - To set up a private mailbox, follow these steps:
     a. Select **Set up private mailbox**.
     b. Search for a user. If the user doesn't appear, see Add users to environment.
     c. Select the mailbox associated with the selected user.
     d. Select **Set up**.

e. Follow the on-screen instructions to approve and enable the configuration. This process enables synchronization between the email server and Dataverse for the selected mailbox.

> **Important**
>
> Only the owner of a private mailbox can create agent configurations and review agent results related to that mailbox. The owner must have **permissions** to manage the agent configuration and review agent results.

Learn more in Set up server-side synchronization of email.

9. The **Enable agent** page opens. To enable the agent, follow links and make settings in the following sections on this page:

   - **Publish Copilot Studio agents** – Normally, the Microsoft Copilot Studio agents needed for the Procurement Agent to run are published automatically. But there might be data loss prevention (DLP) policies in your environment that prevent the publishing of these agents (learn more in Troubleshoot data policy enforcement for Copilot Studio). This section lists the agents that must be published. Select the link provided here to go to Copilot Studio, where you can check whether these agents are published and publish them if necessary. Learn more in Key concepts - Publish and deploy your agent.
   - **Enable agent related features** – This section lists the features that must be turned on in Supply Chain Management. Make a note of the features listed here and then select the link provided to open the Feature feature management workspace, where you can enable each of the features.

> **Tip**
>
> - The *(Production ready preview) Procurement Agent - Supplier communications - automatically sending follow-up emails* feature is optional. It allows the agent to send emails automatically. Turn off this feature for sandbox environments because data there (such as purchase orders) might not be be current, or vendor email addresses might be missing.
> - If you don't see all of the features listed on this page, select **Check for updates** to refresh the list of features.
> - If you can't enable the *Agent management* feature, ensure that your environment fulfills all of the **prerequisites**, including version requirements and Copilot Studio

billing.

After you complete all of the settings on this page, select **Next** to continue.

10. The final page of the wizard opens. Select **Finish** to complete the setup.

Assign user permissions

After you run the setup wizard, set up user permissions for the users who manage the agent configuration and review agent results. The permissions differ for users who manage the agent configuration and users who review agent results. The following sections describe these differences.

If you need to create new users in your environment to manage the agent configuration or review agent results, learn how to Create users.

Required permissions for users who manage the agent configuration

The following permissions are required for users who create and manage the agent configuration:

- Required Dataverse user roles:
  - *Basic User*
  - *Finance and Operations Agent Configuration Manager*
  - *Finance and Operations Basic User*
- Required Supply Chain Management user roles:
  - *System user*
  - *Purchasing manager* and/or *Purchasing agent*

Required permissions for users who review agent results

Users who review the agent results need the following permissions:

- Required Dataverse user roles:
  - *Basic User*
  - *Finance and Operations Basic User*
- Required Supply Chain Management user roles:
  - *System user*

- *Purchasing agent*

Troubleshooting

Issues with setting up supplier communications features of the Procurement Agent

For help with problems that might occur when setting up supplier communications features of the Procurement Agent, go to Solve common problems when setting up supplier communications features.

Issues with server-side synchronization

Learn how to fix common issues that are related to server-side synchronization in Troubleshooting and monitoring.

Last updated on 08/26/2026

Solve common problems when setting up supplier communications features (production-ready preview)

[This article is prerelease documentation and is subject to change.]

This article helps system administrators to solve typical issues when configuring supplier communications features of the Procurement Agent.

The *Dynamics 365 apps* link isn't shown for my environment in the Power Platform admin center

Symptoms

In the Power Platform admin center, the environment is deployed for finance and operations apps, but it doesn't show **Dynamics 365 apps** under **Resources** on the environment details page.

Solution

Open your project in Microsoft Dynamics Lifecycle Services and then open the **Full details** for your finance and operations apps environment. Under **Power Platform Integration**, check whether any errors are reported (such as "LCS Power Platform Integration provisioning fails with timeout"). Then attempt a retry to complete the deployment.

> **Note**
>
> Dual write isn't required for agent deployment.

It isn't possible to install the Copilot apps on my environment in the Power Platform admin center

Symptoms

In the Power Platform admin center, you go to **Manage** > **Dynamics 365 Apps** and try to install the applications *Copilot for finance and operations apps* and *Copilot in Microsoft Dynamics 365*

Supply Chain Management, but your finance and operations apps environment isn't shown in the **Select an environment** drop-down list in the install dialog.

Solution

Open your project in Lifecycle Services and then open the **Full details** for your finance and operations apps environment. Under **Power Platform Integration**, check whether any errors are reported (such as "LCS Power Platform Integration provisioning fails with timeout"). Then attempt a retry to complete the deployment.

I can't enable the *Agent management* feature in feature management

Symptoms

In the **Feature management** workspace for your finance and operations apps environment, when you try to enable the *Agent management* feature, you receive the following error message:

> You cannot enable this Feature. Environment must be linked to Dataverse for agents to function

The error appears even though a Dataverse environment was successfully created in Lifecycle Services.

Solution

If you're running version 10.0.44 of finance and operations apps, then install the latest quality update for that version. This is required because the initial preview release of version 10.0.44 didn't include important changes to the Copilot message capacity, and these changes are required to enable the agent management feature.

The Procurement Agent shows error "Exception has been thrown by target of invocation"

Symptoms

When you're configuring the supplier communications features of the Procurement Agent, and specifying the conditions in which the agent should operate, the following error is shown when you press the **Activate** button:

> Exception has been thrown by target of invocation

Solution 1

It could be that, when you installed or updated the *Copilot for finance and operations app* Dynamics 365 app in the Power Platform admin center, the included cloud flows weren't turned on automatically. To resolve the issue, open your Power App maker environment, select the Dataverse environment associated with your finance and operations app, and select **Solutions**. Under **Managed**, find *Copilot for finance and operations apps*, open the ellipsis (...) menu and select **Edit**. Select the object category **Cloud flows** and make sure that each of the two cloud flows is turned on.

Solution 2

Make sure that the mailbox email address is always identified using lower case only.

1. In Power Platform admin center, open your environment details and then select **Settings** from the top navigation bar.
2. Expand the **Email** heading and select **Mailboxes**. Identify the mailbox that you configured in Dataverse for the Supplier Communication Agent and open the details. In the field **Email Address**, make sure that the email is written in lower case.
3. Back on the **Settings** page, expand the **Business** heading and select **Queues**. Identify the queue you created for your team and open it. In the field **Incoming Email**, make sure that the email is written in lower caps.

Last updated on 08/10/2026

Test supplier communications features (production-ready preview)

[This article is prerelease documentation and is subject to change.]

Learn how to test the supplier communications features of the Procurement Agent in Microsoft Dynamics 365 Supply Chain Management.

The agent has two independent supplier communications features:

- *Follow-up with vendors* (writing reminders)
- *Updates from vendors* (reading vendor emails).

Test the feature you need first. If you plan to use both supplier communications features, start with *Follow-up with vendors* because it's easier to set up, configure, and test.

Test the "Follow-up with vendors" (writing reminders) feature

Use the user interface to set up the feature to fit your needs. Then test to make sure that the emails work as expected.

Test the "Updates from vendors" (reading emails) feature

The agent works by matching incoming vendor emails to existing purchase orders. To try this feature, your sandbox environment must have both of the following items:

- **Purchase orders** – Synchronize recent purchase orders from production to your sandbox environment.
- **Vendor emails** – Forward or send test emails that reference those purchase orders.

To ensure that you can continuously test in sandbox by using your latest purchase orders and emails, update the sandbox by using the following process.

Synchronize data between production and sandbox environments for vendor email matching

When you test in a sandbox environment (Phase 1 or Phase 2 in the following section), the agent needs purchase order data to match against vendor emails. Prepare your sandbox environment by following these steps:

1. Run a full data refresh.

2. Synchronize just the purchase orders to ensure that your sandbox environment reflects any new delta purchase orders as you progress in your testing. Follow the instructions in the following subsections to export purchase orders from your production environment and import them into your sandbox environment by using the Data management framework.

Export purchase orders from the production environment

Follow these steps in the production environment.

1. Open **Data management**.
2. Create a new export project.
3. Select the **Purchase orders composite V3** entity.
4. Select the **XML-Element** format.
5. Create a filter by selecting the filter icon in the **Filter** column.
6. Add a range for the **Created date and time** field.
7. Set the filter value to `(DayRange(-14, 0))`. Modify the first argument (`-14`) to control how many days back to export. A higher value exports more data, which could exceed the import capacity of the sandbox environment.
8. Run the export in batch mode.

Prepare the sandbox environment (one-time setup)

To avoid number sequence conflicts when importing purchase orders, enable manual numbering for purchase orders in the sandbox environment.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. Select the **Number sequences** tab.
3. Find the number sequence that's used for purchase orders and navigate to it.
4. On the **General** FastTab, set **Manual** to *Yes*.

Import purchase orders into the sandbox environment

Repeat these steps regularly (for example, weekly) to keep the sandbox environment up to date.

1. Go to **Data management**.

2. Create a new import project.
3. Select the **Purchase orders composite V3** entity.
4. Start the import in batch mode.

Forward emails

This feature requires more setup permissions, test email addresses, and multiple scenarios. Start small and then expand, as described in the following subsections.

Phase 1: Forward some emails from a vendor email address to a testing email address in a sandbox environment

This phase lets you test the smallest possible scope. Start with a test email address so you can test without risk. It doesn't matter if you plan to use the purchaser's email in production or a common mailbox for the purchasing department.

- The agent only reads emails from vendor domains. If you want to send or forward vendor emails from your own email account, you must add your email address as a vendor contact. To add your email address, follow these steps:

    1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
    2. Create or select a vendor.
    3. On the **Contact information** FastTab, add a row with your own email address (the one you send or forward test messages from).

- Ask your Microsoft Exchange admin to create a test email address and send the messages to that address.

- Connect that email address to the sandbox environment where you do the testing.

- Refresh your data (copy production data to the sandbox) or forward older emails so the system can match them.

- Forward emails in batches—for example, five emails at a time. Check results and note if there's anything unexpected, and continue until you have a good idea of how it performs.

Phase 2: Forward emails from selected vendors

Now that you have a few emails available on your system, start working with messages from those vendors that you communicate with the most or that have the most confirmations that you want the agent to handle.

- Use the same setup as before, but this time autoforward a set of emails from real vendor email addresses to the test address.
- Refresh data or import purchase orders and products before you do the test, especially if your testing should last for a few weeks. The agent can't match an email to a purchase order if the purchase order isn't in the system. Collaborate with one or more of your purchasers if you haven't already.

For this phase, you might want to involve one or more of your purchasers.

Phase 3: Use real email addresses in production

Now you're ready for the final test: testing in production. Set up the agent in production using real email addresses. Stop the agent from working in your sandbox.

Testing in production is safe. The feature doesn't interfere with standard business processes. At best, the agent does the work automatically. At worst, if the agent doesn't understand something correctly, the purchaser still needs to do it manually.

The supplier communications features of the Procurement Agent are production-ready preview features. They have the quality of generally available (GA) features and you can go live with them. The reason They aren't fully GA is that the Procurement Agent is under continuous development. By designating these features as production-ready previews, Microsoft can continue improving them.

For this test, use the following recommendations:

- Have purchasers work with the agent to do their daily tasks.
- Start with just one or a few purchasers.
- Optionally filter messages from a few vendors to start small.

Phase 4: Expand the number of vendors in production

Keep adding more vendors to the configuration so you can gradually increase the value provided by the agent.

Last updated on 08/26/2026

Follow up on purchase orders by using the Procurement Agent (production-ready preview)

[This article is prerelease documentation and is subject to change.]

The Procurement Agent helps you identify and follow up on purchase orders that vendors didn't yet confirm or that delivery is late for. For each order that it finds, the agent automatically generates a draft email that is addressed to the relevant vendor.

The Procurement Agent helps you perform the following actions:

- Create agent tasks to find purchase orders that require follow-up. Each user has unique tasks, so you can create and adjust them to meet your specific business needs. Two default tasks are provided. One task follows up on unconfirmed purchase orders, and the other task follows up on late deliveries. You can modify or delete the default tasks as required to meet your business needs.
- Configure how emails are generated for each task.
- Review the draft emails that the agent generates. After you review each message, you can modify it, and send it. To generate the emails, the agent uses data from the purchase order lines and related tables.

By default, the system provides two tasks that help you find purchase orders that require action:

- *Unconfirmed purchase orders* – This task finds purchase orders assigned to you with a delivery date within the next 60 days that the vendor didn't yet confirm (that is, the confirmed delivery date is blank). The task uses the following criteria:
  - The **Deliver remainder (line)** field is set to a value other than *0* (zero).
  - The **Confirmed receipt date (line)** field is blank.
  - The **Requested receipt date (line)** field is set to a date that is before the current date.
  - The **Orderer (header)** field is set to your user account (the current user).
  - The **Document status (header)** field isn't *None*.

- *Delayed purchase orders* – This task finds purchase orders assigned to you that are delayed (that is, the confirmed delivery date is one or more days ago). The task uses the following criteria:
  - The **Deliver remainder (line)** field is set to a value other than *0* (zero).

- The **Confirmed receipt date** field is set to a date that is before the current date.
- The **Orderer (header)** field is set to your user account (the current user).

Configure task criteria and email settings

The following procedure explains how to set up agent tasks to find purchase orders that require follow-up. It also explains how to configure the way that Copilot generates vendor email content for each task.

1. Open the **(Preview) Follow-up emails** page by following one of these steps:

   - Go to **Procurement and Sourcing** > **(Preview) Procurement Agent - Supplier communications** > **(Preview) Follow-up emails**.
   - Go to **Procurement and sourcing** > **Purchase order receipt and follow-up** to open the **Purchase order receipt and follow-up** workspace. A tile named **(Preview) Follow-up emails** indicates the number of emails that require review. Select the tile.

2. Follow one of these steps:

   - To edit an existing task, select it in the list, and then select **Edit**.
   - To delete an existing task, select it in the list, and then select **Delete**.
   - To create a new task, select **Configure agents**. Then, under **Library**, select **Send follow-up emails to vendors with Supplier Communications Agent**.

3. Edit the name of the task as required. (The default name is *Send follow-up emails to vendors with Supplier Communications Agent*.)

4. Select whether you want the task to find unconfirmed purchase orders or delayed purchase orders.

5. Modify the criteria to define which purchase orders require follow-up. For example, you might want the task to find: unconfirmed orders for the next two months, or orders created more than three days ago, or orders sent but not confirmed.

6. Select the fields that you want to include in the email, such as delivery dates or the address.

7. To add a signature, select **Signature**, and enter the desired text. You can also add an email footer that has text such as "This email was written with the help of AI."

   > **Important**

If your system is set up to send emails automatically, the email footer is mandatory.

8. Select the tone of the emails (*Casual* or *Formal*, *Urgent*, or *Non-urgent*).

Examples

Here are some examples of other tasks that you might set up:

- To generate emails for orders that you created less than three days ago, and that the vendor didn't yet confirm, specify the following criteria:
  - The **Deliver remainder (line)** field on the line is set to a value other than *0* (zero).
  - The **Confirmed receipt date (line)** field on the line is blank.
  - The **Created date** field is set to a date that's between three days before the current date and current date.
  - The **Orderer (header)** field on the header is set to your user account (the current user).
- To generate emails for orders that you created for vendor group A less than three days ago, and that the vendor didn't yet confirm, specify the following criteria:
  - The **Deliver remainder (line)** field on the line is set to a value other than *0* (zero).
  - The **Confirmed receipt date (line)** field on the line is blank.
  - The **Created date** field is set to a date that's between three days before the current date and current date.
  - The **Orderer (header)** field on the header is set to your user account (the current user).
  - The **Vendor (group)** field is set to *A*.
- You might want to send a second follow-up email if a vendor doesn't reply to the first one you sent. However, agent tasks are designed to generate just one email per purchase order unless the purchase order changes. Therefore, to continue following up when no vendor reply is received, create two agent tasks that use the same core criteria but different date ranges that don't overlap. You can use any suitable date field to define the ranges. For example, you can use the delivery date, as illustrated in the following table:

Expand table

Agent task	Example criteria
Recent overdue purchase orders	The delivery date is between seven days ago and the current date.
Extended overdue purchase orders	The delivery date is earlier than eight days ago.

This setup lets the first task generate draft emails for recently overdue purchase orders while a second task generate draft emails for purchase orders that are still overdue after that. Even though no purchase order changes are made, a second draft email is generated when the order moves into the second date range.

> **Important**
>
> Make sure that the date ranges don't overlap. Otherwise, the same purchase order might get evaluated by multiple tasks at the same time.

Configure automatic sending of follow-up emails

To configure automatic sending of follow-up emails without requiring user review, follow these steps:

1. Sign in to Supply Chain Management as an administrator.
2. Go to **System administration** > **Workspaces** > **Feature management**.
3. Make sure the *(Production ready preview) Procurement Agent – Supplier communications - automatically sending follow-up emails* feature is turned on. This feature is optional and is turned off by default. Learn more in Set up and configure supplier communications features of the Procurement Agent.
4. Go to **System administration** > **Setup** > **Email** > **Email parameters**.
5. Open the **Configuration** tab.
6. On the **General** FastTab, set **Batch email provider** to *Graph*. This setting turns on the batch email provider.
7. On the **Enabled interactive email providers** FastTab, make sure that *Graph* is listed in the **Enabled** column.
8. On the **Microsoft Graph settings** tab, make sure that the **Application ID** and **Application secret** fields are filled in. If they aren't, you might need to contact your organization's Azure tenant admin for help. Learn more in Configure Microsoft Graph in Dynamics 365 finance and operations apps.
9. To test whether the email is working, open the **Test email** tab. Set **Email provider** to *Graph* and enter the email address for an account that you can read in the **Send to** field. Then select **Send test email** and check your inbox for the test email.

Configure the email sender

When a user reviews and sends an email that the agent drafts, the email comes from that user's email address.

When the system automatically sends an email without user review, the email comes from the agent identity user email address.

To configure the email address of any user (including the agent identity user):

1. Sign in to Supply Chain Management as an administrator.
2. Go to **System administration** > **Users**.
3. Select the user you want to configure (for example, the agent identity user).
4. On the Action Pane, select **User options**.
5. Open the **Account** tab.
6. On the **Email provider selection** FastTab, make the following settings:
    - **Email provider ID** – Set to *Graph*.
    - **Sender email** – Enter the email address that should appear in the **From** field of the follow-up emails sent by the selected user.
7. On the **Account** FastTab, enter the **Name** that you want to show in the signature of the email body. The emails conclude with a "Best regards" signature that includes the name configured in this field.

> **Tip**
>
> If you're a user without administrator rights in Supply Chain Management, you can configure your own email parameters, but not other users'. To configure your own email parameters, select the settings gear icon at the upper right side of the screen, and choose **User Options** from the menu. Then continue from step 5 in the previous procedure.

Configure the email receiver for each vendor

To choose the recipient of the follow-up emails for a vendor, follow these steps:

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Find and open the target vendor from the list.

3. The **Contact information** FastTab lists various ways that you can contact the selected vendor and lets you edit the list. Use the settings here to configure the email address the agent should use when sending follow-up emails to this vendor. The agent chooses which contact line to use as follows:

    - If a contact line is set up as the priority email address for purchase order communications, the agent always uses that email address for this vendor, regardless of whether the agent task is set up to send emails per purchase order or grouped per vendor. For details about how to set up this address, see Set up the priority purchase order communications email address for a vendor.
    - If no contact line is set up as the priority email address for purchase order communications, the agent applies the following logic to find the email address to use:
        - If the agent task is configured to send emails grouped per vendor, the agent uses the contact line of **Type** *Email address* that has the **Primary** checkbox selected. If no email address is set as **Primary**, then the agent arbitrarily chooses one of the lines of **Type** *Email address*.
        - If the agent task is configured to send emails per purchase order, the agent prioritizes the **Email** field of the purchase order if there is one. If the purchase order doesn't have an email address, the agent uses the vendor's contact information as described in the previous bullet point.

> **Tip**
>
> To see whether an email address listed on the **Contact information** FastTab is set up as the priority email address for purchase order communications, select the row and then select **Edit contact information** from the FastTab toolbar. In the **Edit contact information** dialog, check the value of the **Purpose** field. If it's set to *Purchase Order Communications* (or to another name that was created for this purpose), the email address is set up as the priority email address for purchase order communications. Learn more in **Set up the priority purchase order communications email address for a vendor**.

Set up the priority purchase order communications email address for a vendor

As described in the previous section, the agent always uses the contact line that is set up as the priority email address for purchase order communications if such a line exists for the vendor,

regardless of whether the agent task is set up to group emails per vendor or per purchase order. To set up this priority address, follow these steps:

1. Go to **Organization administration** > **Global address book** > **Address and contact information purpose**.
2. At the top-right corner of the grid, open the **Grid options** menu (ellipsis button) and select **Insert columns**.
3. In the **Insert columns** dialog, select the check box for the row with a **Field value** of *Type*. Then select **Update**. The grid now shows a **Type** column.
4. If a row already exists with a **Type** of *Purchase Order Communications*, note its **Purpose** name and skip this step. Otherwise, add it by selecting **New** on the Action Pane and make the following settings for the new row:
    - **Purpose** – Enter a short name, such as *My Purchase Order Agent Communications*.
    - **Description** – Enter a short description, such as *Priority purchase order communications email address for a vendor*.
    - **Postal address** – Don't select this check box because this purpose is for email addresses, not postal addresses.
    - **Contact information** – Select this check box. This option makes the row available for use with contact information lines, including email addresses.
    - **Type** – Select *Purchase Order Communications*.
5. On the Action Pane, select **Save**.
6. Go to **Accounts payable** > **Vendors** > **All vendors**.
7. Find and open the vendor you want to set up from the list.
8. On the **Contact information** FastTab, select the row with the email address you want to send follow-up messages to. Then select **Edit contact information** from the FastTab toolbar.
9. In the **Edit contact information** dialog, set **Purpose** to the purpose name you found or created earlier in this procedure.
10. Select **OK**.

Review and send drafted emails

To review the emails that were previously created for the various configurations, go to **Procurement and Sourcing** > **(Preview) Procurement Agent - Supplier communications** > **(Preview) Follow-up emails**. The configurations appear on the left, and the emails for each configuration appear on the right.

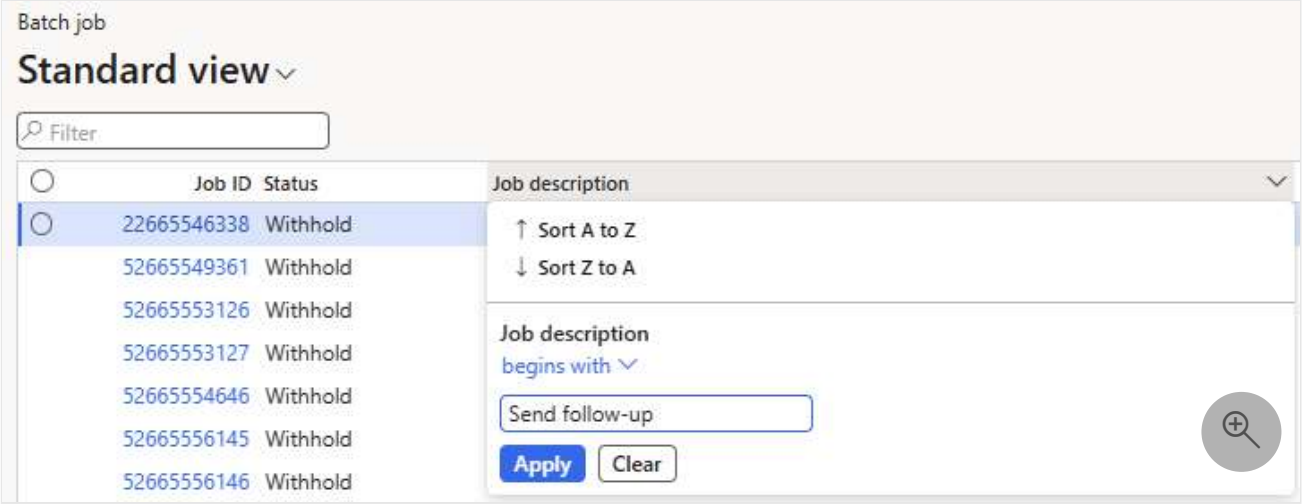
Edit each message as required, and then select **Send** to send it to the vendor.

Set up the periodic agent run schedule and batch job options

The Procurement Agent includes a batch job that runs the agent automatically. To change the default configuration of the job, follow these steps:

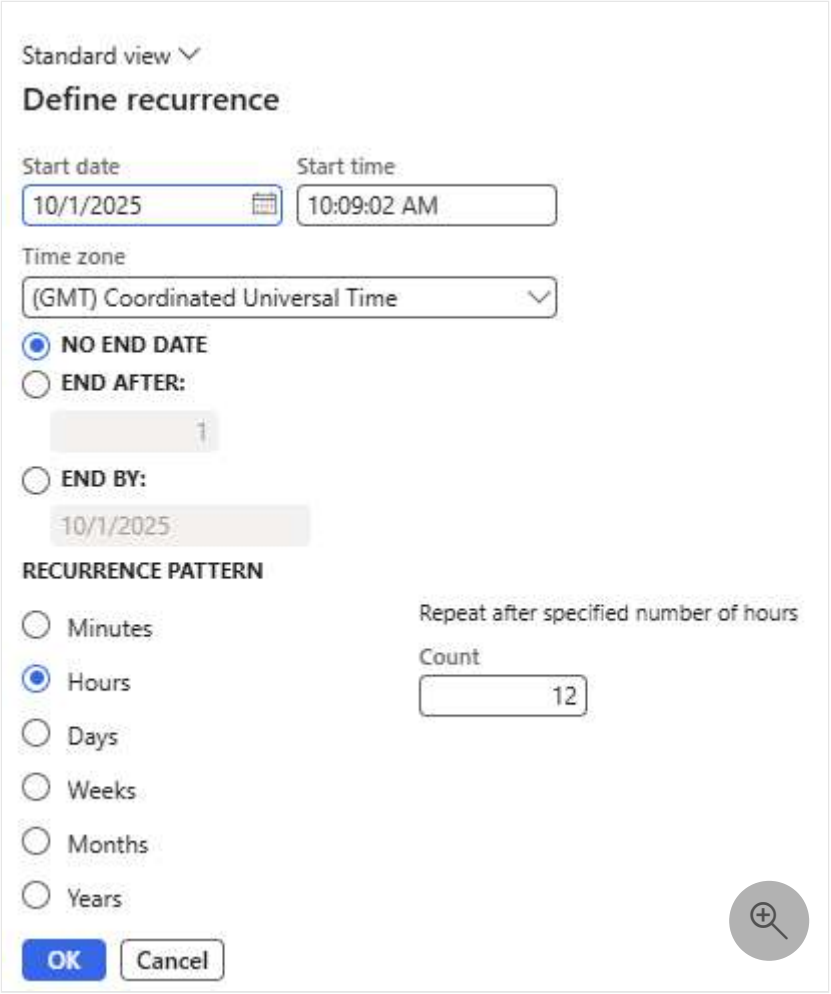
1. Go to **System Administration** > **Inquiries** > **Batch jobs**.

2. Find and select the job that has a **Job description** that matches the name of the agent configuration, which is described in Configure task criteria and email settings. For example, if the agent name is *Send follow-up emails to vendors with Supplier Communications Agent*, you can locate its batch job quickly by using this text in the **Job description** column header filter. The agent name might be different on your system, so check the name before proceeding.



3. Select the job, and then select **Recurrence** from the Action Pane.

4. In the **Define recurrence** dialog, set up the run schedule as desired, and then select **OK**.



5. To get more information about the batch job and adjust other settings, return to the **Batch jobs** page and select the link in the **Job ID** column.

Last updated on 08/26/2026

Review and apply purchase order changes received in vendor emails (production-ready preview)

[This article is prerelease documentation and is subject to change.]

The Procurement Agent helps speed up communications with vendors about purchase orders by enabling Copilot to read emails from some or all vendors.

When Copilot analyzes a vendor email, it determines what the message is about. For example, the agent recognizes whether it's a purchase order confirmation or a purchase order change request. It also identifies which purchase order it applies to. Copilot then matches the information that it extracts from the email to fields in the system and indicates whether there are any changes. Therefore, you just need to review the information that Copilot provides, review the proposed changes, and decide whether to accept those changes. Copilot saves you time, because you don't have to manually find, open, and edit the purchase order in the system.

Copilot classifies each vendor email that it reads into one of the following categories, based on the intent that it detects in the message:

- *Purchase order confirmation* – Copilot detected that the intent of the message is to confirm one or more purchase orders (all the purchase orders that the email mentions).
- *Purchase order change request* – Copilot detected that the message contains at least one requested change to a purchase order. For example, the vendor wrote to tell you that they can't deliver the full quantity that was requested for one or more lines, to tell you that they can't deliver on time, or to reject the entire order.
- *Rejected* – Copilot detected that the vendor can't supply the requested purchase order.
- *Other* – Copilot couldn't identify any of the previous intents.

For the first two categories, Copilot identifies which purchase orders the email is related to and matches the information in the email to information in the system.

> **ⓘ Important**
>
> The agent can read incoming PDF attachments but not Microsoft Word or Excel documents.

When you review incoming changes that are based on an email, the system shows the original email, the current information in the system, and the changes proposed. Therefore, you can more easily understand each change. After you finish reviewing the proposal and making any corrections that are required, you can apply the changes directly to the relevant purchase order.

Configure the agent to track your email

You need to configure the Procurement Agent to monitor a specific email address for incoming messages from vendors. Follow these steps to complete the configuration.

> **Important**
>
> Before you begin, complete the steps in Set up and configure supplier communications features of the Procurement Agent. If you want to send or forward email communications from your own email address for testing, you must also set up your email address as a vendor contact as described in Forward emails.

1. Sign in to the Microsoft Dynamics 365 Supply Chain Management environment as a user who has permissions to manage the agent configuration.
2. Go to **Agents** > **Agents**.
3. On the **Library** tab, for **Speed up updates in purchase orders with Supplier communications agent**, select **Select**.
4. On the **Agent configuration** page, use the dropdown menu to specify which vendors the agent should analyze emails for. Select **Any vendor** to track all vendors or **Specific vendors** to track a list of specific vendors that you provide.
5. Select the mailbox that the agent should monitor. The dropdown menu shows all mailboxes that the sign-in user has access to. If the mailbox that you're looking for isn't in the list, you might need to share and synchronize it with Dataverse. Learn more in Set up and configure supplier communications features of the Procurement Agent.
6. Select **Activate**.

Review and accept changes suggested by the agent

The agent detects changes in the following fields:

- Quantity
- Unit of measure
- Price

- Confirmation
- Delivery date
- Cancellation

To review and accept changes that the agent suggests based on the emails that it reads, follow these steps:

> **ⓘ Important**
>
> Before you begin, confirm that the signed-in user has access to the mailboxes. Learn more in Set up and configure supplier communications features of the Procurement Agent.

1. Go to **Procurement and sourcing** > **Purchase order receipt and follow-up** to open the **Purchase order receipt and follow-up** workspace.

2. The **(Preview) Emails from vendors** tile indicates the number of emails that require review. Select the tile.

   The left side of the **(Preview) Emails from vendors** page lists all the emails that the agent reads. The right side shows summaries that Copilot generates.

3. Follow one of these steps:

   - To apply all the changes that the agent suggests, select **Apply all suggestions**. This action affects all purchase orders that the message mentions.
   - To apply all suggestions to a single selected purchase header and all its lines, select **Apply suggestions** in the **Purchase order header** section.
   - To apply suggestions to specific purchase order lines, select the relevant lines, and then select **Apply suggestions** in the **Purchaser order lines** section.

Teach the agent how to better interpret incoming email content

If a vendor uses acronyms or other language that the system doesn't recognize, the Procurement Agent might not have enough data to interpret it. However, you can teach the agent to better process incoming email content. By teaching the agent, you help improve its accuracy.

Two types of teaching are available:

1. Column mapping teaching (improves recognition of field names)

2. Value teaching (improves recognition of field values)

Teach the agent about column mappings

Agent teaching applies when any of the following situations occur:

- The incoming email from the vendor contains a term or acronym that the agent can't interpret.
- The email contains multiple possible matches for a single field in Supply Chain Management. For example, various vendors might use many different names to refer to the field named **Confirmed receipt date** in Supply Chain Management. Examples include:
  - **Estimated delivery date**
  - **Confirmed delivery date**
  - **Transportation load date**
  - **Ship date**
  - **Ship by**
  - **Approx. delivery date**
  - And more.

If the agent detects ambiguity when trying to identify a field, it shows that "Some columns are mapped with low confidence." In this case, select **Review** to manually choose which of the fields from the vendor's email you want the agent to use. This action opens a dialog that lists possible mappings for the field. After you choose the correct mapping, choose how broadly to apply it: to all vendors from now on, to the current vendor from now on, or just for this one time.

To map more fields, select **Show more**. This action expands the list of fields that are available for mapping. Agent teaching for field mappings is currently supported for the **Confirmed receipt date**, **Unit**, **Quantity**, and **Unit price** fields.

Teach the agent about value mappings

For some fields, a vendor might use values that are synonyms or industry equivalences that aren't saved in your system. You can train the agent to recognize these equivalences. For example, as a value for the **Unit** field, you might use *cartons* while one of your vendors always uses the name *cassettes* for the same unit. When the agent detects the different value as a change, you can clarify the equivalence. When you edit the field, the teaching pane appears. You can then save the equivalence for all vendors, for the current vendor, or not at all.

Locate existing teaching items

If any teaching is previously saved for all vendors or a specific vendor, you can always find it later, review it, or delete it by following these steps:

1. Go to **Procurement and sourcing** > **(Preview) Procurement Agent - Supplier communications** > **(Preview) Emails from vendors**.
2. On the Action Pane, select **Taught items** to view everything that the agent was taught.
3. If you see any lessons that you want to delete, select them, and then select **Delete** on the Action Pane.

Last updated on 08/26/2026

Purchase agreements

This article provides information about purchase agreements. A purchase agreement is a contract that commits an organization to buy a specified quantity or amount by using multiple purchase orders over time. In exchange for this commitment, the buyer receives special prices and discounts.

Purchase agreements can apply to a specific quantity of a product, a specific currency amount of a product, or a specific currency amount of the products in a procurement category. The prices and discounts of the purchase agreement override the prices and discounts that are specified in any trade agreements that exist.

On the **Purchase agreements** page, you can create, apply, and follow up on purchase agreements that exist between your organization and your vendors. For example, after you create a purchase agreement, you can order directly from it. Each purchase agreement has a validity period that is defined by the person who creates the purchase agreement. The receipt date of a purchase must be within the effective dates of this validity period.

After you create a purchase agreement, you must activate it before it becomes effective. To activate a purchase agreement, set the **Mark agreement as effective** option to *Yes*.

To prevent your purchase agreement from being used and confirmed, mark the agreement status as *Closed*. You can still update the status to *Effective** at any time after making this change.

Responsible workers on purchase agreements

You can identify a primary responsible worker and secondary responsible worker on the purchase agreement classification. The resulting purchase agreement inherits these values. You're not required to add responsible workers to the purchase agreement, and you can modify them directly on a case-by-case basis on the purchase agreement itself. You can't specify a secondary responsible worker without a primary responsible worker, although you don't have to have a secondary responsible worker. You can't specify the same worker as both the primary and secondary responsible worker.

Commitment types

Each line in a purchase agreement is a commitment to buy something. You can use lines from multiple purchase orders (POs) to fulfill the commitment. There are four types of commitments:

- **Product quantity commitment** – You purchase a specific quantity of a product.
- **Product value commitment** – You purchase a specific currency amount of a product.
- **Product category value commitment** – You purchase a specific currency amount in a procurement category. The amount can be for a catalog item or a noncatalog item.
- **Value commitment** – You purchase a specific currency amount of any product or products in any procurement category.

Pricing terms for purchase agreements

Pricing terms can vary, depending on the type of commitment. The pricing terms from purchase agreements override any other pricing terms that are set up for trade agreements. The following table describes the price-related fields that are affected by each commitment type. You can update fields that contain *Yes* on an order line.

Expand table

Commitment type	Unit price	Price unit	Discount percent	Cash discount amount
Product quantity commitment	Yes	Yes	Yes	Yes
Product value commitment			Yes	
Product category value commitment			Yes	
Value commitment			Yes	

Policies for purchase agreements

The following policies affect the way that the link between a purchase agreement commitment and the corresponding PO lines works:

- **Max is enforced** – The total quantity or amount for all order lines can't exceed the quantity or amount that is specified on the related commitment.
- **Price and discount is fixed** – The price on an order line and the price on the related commitment must be the same. If the price is changed on the order line, the link to the commitment is broken. If the link is broken, the order line doesn't contribute to the fulfillment of the commitment.
- **Minimum release amount and Maximum release amount** – If an amount is specified, you receive a message if you make any change to an order line that causes the order line to

differ from the related commitment.

Fulfillment calculations for purchase agreements

Fulfillment quantities and amounts are shown on the **Fulfillment** tab on the **Line details** FastTab of the **Purchase agreements** page.

The **Fulfillment** area shows the remaining amount or quantity that is required to fulfill the commitment.

The **Agreement** area shows the total quantity or the total amount that the sales agreement line is valid for.

You can access the PO lines and the invoice lines that contribute to the fulfillment calculation by selecting the **Related information** action on the lines or on the header of a purchase agreement.

Confirmations and version history for purchase agreements

When you confirm a purchase agreement, the current version of the purchase agreement is stored in a history table. If you change the purchase agreement, you can confirm it again to store another version of the purchase agreement in the history. If you don't confirm a purchase agreement, you can still use it to create POs. However, the history information for the purchase agreement isn't stored. You can preview or print all versions of the agreement. You can then share the revisions with your vendor to obtain approval.

Applying purchase agreements in the ordering process

When you create a PO, you can apply a purchase agreement to it. Information from the terms for the agreement, such as the payment terms, delivery terms, and delivery address, is then copied to the header of the PO. If the PO contains one or more lines for products or categories that are covered by the agreement, the prices and discounts from the purchase agreement are used for those lines. The amount or quantity on the order line contributes to the fulfillment of the commitment in the purchase agreement. The same PO can include both lines that aren't related to a purchase agreement and lines that have a commitment for a purchase agreement.

You can select a purchase agreement only when you're creating a PO. You can't select a purchase agreement after the PO has been created.
In some situations where POs are created indirectly, you can control whetherSupply Chain

Management automatically searches for applicable purchase agreements. For example, you might do this when you're automatically firming planned POs or creating POs that are based on sales orders.

Delivery address defaults from a purchase agreement

When you apply a purchase agreement to a PO, you're prompted to transfer the currency and delivery address from the agreement. If you select **Yes**, the agreement values replace the values that are already on the PO. Select **No** if you want to keep the existing PO delivery address.

The transfer uses the default values that are saved on the purchase agreement header. If the agreement doesn't yet have a saved set of header defaults, there's no address for the transfer to apply, so it clears the delivery address on the PO instead of replacing it. This behavior is by design.

To make sure that the agreement address is transferred, follow these steps:

1. Open the purchase agreement and confirm that it's marked as effective and remains valid through the required delivery date.
2. On the agreement header, update at least one field. For example, reselect the **Delivery name** or **Delivery address** value.
3. Save the agreement. Saving the header creates the default values, including the delivery address, that the transfer uses.
4. Return to the PO, select the purchase agreement again, and select **Yes** at the transfer prompt. Verify that the expected delivery address remains on the PO.

Matching policy on purchase agreements

You can define a line matching policy on the header of the purchase agreement. This line matching policy will respect the accounts payable parameters line matching policy when the **Allow matching policy override** field on the **Accounts payable parameters** page (on the **Price and quantity matching** FastTab) is set to *Higher than company policy*. Documents that reference the purchase agreement will use the line matching policy that's defined on the purchase agreement header unless otherwise defined on the corresponding item, item and vendor, or category purchasing policy.

Purchase agreements and intercompany trade

Intercompany trading relationships can be created between vendor accounts and customer accounts that are in different legal entities. When a sales order or PO is created for one of the parties, an intercompany order chain is created. In the order chain, the sales order and PO are created in the appropriate legal entities.

You can create a purchase agreement or sales agreement for one of the intercompany trading parties. You can then generate the corresponding sales agreement or purchase agreement for the other intercompany trading party in the other legal entity.

If you create an intercompany PO that uses the intercompany purchase agreement in one legal entity, the corresponding intercompany sales order uses the corresponding intercompany sales agreement in the other legal entity. The fulfillment of the sales agreement commitments and the fulfillment of the purchase agreements are synchronized, just as the intercompany sales order and the intercompany PO are synchronized.

Financial dimensions on purchase agreements

You can copy financial dimensions to document headers or to individual lines of a purchase agreement. If you change the dimensions in the agreement header or on the agreement line, the change doesn't affect any released orders, but it will be reflected on any new orders.

Related information

- Create a purchase agreement
- Apply a purchase agreement when creating a purchase order

Last updated on 09/08/2026

Create a purchase agreement

This article shows you how to create a purchase agreement. Typically, a purchasing manager completes this procedure. Set up purchase agreement classifications before you start. After you create an agreement, use it when you create a purchase order. This action copies the purchase agreement conditions to the header and to any lines in the order that the agreement affects.

Create a new purchase agreement

1. Go to **Procurement and sourcing** > **Purchase agreements** > **Purchase agreements**.
2. Select **New**.
3. In the **Vendor account** field, select the drop-down menu and select the row of the desired record.
4. In the **Purchase agreement classification** field, select the drop-down menu and select the row of the desired record.
5. Expand the **General** FastTab.
6. In the **Expiration date** field, enter a date. This expiration date is the default for all commitment lines and determines how long each specific commitment is valid.
7. In the **Document title** field, type a name for your purchase agreement.

    - Leave the **Default commitment** field set to **Product quantity commitment** (or change it if it's not set to this).
    - The default commitment value determines your options on the agreement lines. If you need a new commitment type when you're creating the agreement lines, change the default commitment on the header. There are four types of commitments:
        - *Product quantity commitment* – for a specific quantity of a product.
        - *Product value commitment* – for a specific currency amount of a product.
        - *Product category value commitment* – for a specific currency amount in a procurement category where the amount can be for a catalog item or a non-catalog item.
        - *Value commitment* – for a specific currency amount which any product or any procurement category can fulfill.

8. Select **OK**.

Add a commitment

1. Select **Add line**.
2. In the **Item number** field, select the desired record from the drop-down menu.
3. In the **Quantity** field, enter a number. This number is the total quantity that you agree to buy from your vendor.
4. In the **Unit price** field, enter a number.
5. Expand the **Line details** section.
6. Set the **Max is enforced** option to *Yes*. The **Max is enforced** option limits the use of the commitment. You can only purchase up to the quantity specified in the **Quantity** field for the line.

Add header conditions

1. On the Action Pane, select **Options**.
2. Select **Change view**.
3. Select **Header view**.
4. Expand the **Terms** section.
5. In the **Method of payment** field, select the desired record in the drop-down menu. The payment terms from the vendor account appear here by default.
6. In the **Mode of delivery** field, select the desired record in the drop-down menu.
7. In the **Delivery terms** field, select the drop-down button to open the lookup.

Confirm and activate the agreement

1. On the Action Pane, select **Purchase agreement**.
2. Select **Confirmation**. Set the **Mark agreement as effective** option to **Yes**.
3. Select **OK**.
4. On the Action Pane, select **Purchase agreement**.
5. Select **Purchase agreement confirmations**. The **Preview/Print** option generates a document for the purchase agreement that you can print or send to the vendor. If you update the agreement later and reconfirm it, both versions appear here.
6. Close the page.

Last updated on 07/01/2026

Automatic application of charges

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Based on the customer or vendor that you're working with or the item that you're selling, you might want to apply specific additional charges. The *charges* feature in Microsoft Dynamics 365 Supply Chain Management helps you automatically apply charges to purchase orders or sales orders.

Automatic charges (auto charges) are automatically applied when you create a sales order or a purchase order. You can define auto charges for specific vendors, customers, groups of vendors, groups of customers, items, or groups of items. You can also define auto charges that apply to all vendors, customers, or items.

Set up parameters

The **Procurement and sourcing parameters** page has a few settings that are especially relevant when you want to apply charges automatically in the purchase order. To complete this setup, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Prices** tab, on the **Prices** FastTab, set the following fields:
    - **Find auto charges for header** – Set this option to *Yes* if charges should automatically be applied to purchase order headers.
    - **Find auto charges for line** – Set this option to *Yes* if charges should automatically be applied to purchase order lines.

The **Accounts receivable parameters** page also has a few settings that are especially relevant when you want to apply charges automatically in the sales order. To complete this setup, follow these steps.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Prices** tab, on the **Prices** FastTab, set the following fields:

- **Find auto charges for header** – Set this option to *Yes* if charges should automatically be applied to sales quotation and sales order headers.
- **Find auto charges for line** – Set this option to *Yes* if charges should automatically be applied to sales quotation and sales order lines.

Set up charges codes

To apply charges, you must first define charges codes.

1. Follow one of these steps:

   - For purchase orders: Go to **Procurement and sourcing** > **Setup** > **Charges** > **Charges code**.
   - For sales orders: Go to **Accounts receivable** > **Setup** > **Charges** > **Charges code**.

2. On the Action Pane, select **New** to create a charges code.

3. In the header of the new record, set the following fields:

   - **Charges code** – Enter a code for the charges.
   - **Description** – Enter a description of the charges.
   - **Item sales tax group** – Select an item sales tax group, if applicable.
   - **Prorate** – Set this option to *Yes* if you want to prorate your charges. This option is available only for sales orders.
   - **Maximum amount** – Enter the maximum amount that's allowed for the charges code. This field is used to validate charges for vendor invoices. It's available only for purchase orders.

     > ⓘ **Note**
     >
     > To turn on the functionality for validating charges for purchase orders, go to **Accounts payable** > **Setup** > **Accounts payable parameters**. On the **Invoice validation** FastTab, in the **Invoice validation** section, set the **Enable invoice matching validation** option to *Yes*.

4. The **Posting** FastTab includes **Debit** and **Credit** sections. Set the following fields, depending on the ledger that you want to post the charges to:

   - **Type** – Select the type of account that you're posting to (*Ledger*, *Customer*, or *Item*). If you select *Ledger account* as the debit type or credit type, specify a posting type in

the **Posting** field, and specify the main account in the **Account** field.
- **Posting** – Select the type of postings to create (such as *Broker fee* or *Customer settlement*).
- **Account** – Select the account to post the charge for.

5. On the Action Pane, select **Save**.

Create charge groups

Charge groups automatically apply specific charges to a group of customers or vendors. The following subsections describe how to create and assign these charge groups.

Charge groups for purchase orders

To create charge groups for purchase orders, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Charges** > **Vendor charges group**.
2. On the Action Pane, select **New** to add a row to the grid, and then set the following fields:
    - **Charges group** – Enter the name of the charge group.
    - **Description** – Enter a description of the charge group.
3. On the Action Pane, select **Save**.
4. Go to **Procurement and sourcing** > **Vendors** > **All vendors**, and either open an existing vendor or create a new vendor.
5. On the **Purchase order defaults** FastTab, in the **Purchase order** section, set the **Charges group** field to the charge group that you created.

Charge groups for sales orders

To create charge groups for sales orders, follow these steps.

1. Go to **Accounts receivable** > **Setup** > **Charges** > **Customer charge groups**.
2. On the Action Pane, select **New** to add a row to the grid, and then set the following fields:
    - **Charges group** – Enter the name of the charge group.
    - **Description** – Enter a description of the charge group.
3. On the Action Pane, select **Save**.

4. Go to **Accounts receivable** > **Customers** > **All customers**, and either open an existing customer or create a new customer.

5. On the **Sales order defaults** FastTab, in the **Sales order** section, set the **Charges group** field to the charge group that you created.

Define auto charges

After you set up your charge codes, follow these steps to define the auto charges.

1. Follow one of these steps:

    - For purchase orders: Go to **Procurement and sourcing** > **Setup** > **Charges** > **Automatic charges**.
    - For sales orders: Go to **Accounts receivable** > **Setup** > **Charges setup** > **Auto charges**.

2. In the list pane, in the **Level** field, select the level where your auto charge applies:

    - *Header* – Apply charges to the order header.
    - *Line* – Apply charges to the order lines.

3. Select an existing auto charge to edit it, or select **New** to define a new auto charge.

4. In the **Account code** list, select one of the following values to specify the scope of accounts that the auto charge affects:

    - *Table* – Assign charges to a specific customer or vendor.
    - *Group* – Assign charges to a miscellaneous charges group.
    - *All* – Assign charges to all customers or vendors.

5. In the **Customer relation** or **Vendor relation** field, select a specific customer or vendor if you set the **Account code** field to *Table*. If you set the **Account code** field to *Group*, select a customer or vendor charges group.

6. In the **Item code** field, select one of the following values to specify the scope of items that the auto charge affects. You can select an item code only when you define auto charges at the line level.

    - *Table* – Assign charges to a specific item.
    - *Group* – Assign charges to an item charges group.
    - *All* – Assign charges to all items.

7. In the **Item relation** field, select a specific item if you set the **Item code** field to *Table*. If you set the **Item code** field to *Group*, select an item charges group.

8. *For sales orders only:* In the **Mode of delivery code** field, select one of the following values to specify the scope of delivery modes that the auto charge affects:

    - *Table* – Assign charges to a specific mode of delivery.
    - *Group* – Assign charges to a mode of delivery group.
    - *All* – Assign charges to all modes of delivery.

9. *For sales orders only:* In the **Mode of delivery relation** field, select a specific mode of delivery if you set the **Mode of delivery code** field to *Table*. If you set the **Mode of delivery code** field to *Group*, select a mode of delivery group.

10. On the **Lines** FastTab, define the charges and the charge rates that the current auto charge uses. You can use the toolbar on this FastTab to add as many lines as you require. For each line, set the following fields:

    - **Currency** – Select the currency that you want to use to calculate the charge. This field applies to the *Header* and *Line* levels for sales quotation and sales order charges.

    - **Charges code** – Select the code for the charge.

    - **Category** – Select one of the following values:
        - *Fixed* – Enter the charge as a fixed amount on the line. You can use fixed charges on charges both in the order header and on the order lines.
        - *Pcs* – Base the charge on unit with no unit of measure conversion. *Pcs* in this case represents any unit of measure and not the specific *Pcs* unit of measure. You can use these charges only on order lines. They appear when you calculate the order total.
        - *Percent* – Enter the charge as a percentage on the line. You can use percentage charges on charges both in the order header and on the order lines.
        - *Intercompany percent* – Enter the charge as a percentage on the line for intercompany orders. You can use intercompany percentage charges only on order lines.
        - *External* – A third-party service that's associated with one or more shipping carriers calculates the charge.
        - *Specific unit* – Express the charge value in the unit of measure on the charge line. Unit of measure conversion is used to apply it proportionally to the unit of measure on the sales line. This category applies only to the *Line* level.
        - *Specific unit match* – Express the charge value in the unit of measure on the charge line. The unit of measure on the sales line must match this unit of measure for the charge line to be applied. No unit of measure conversion is applied. This category applies only to the *Line* level.

- **Unit** – Enter the unit that applies when the **Category** field is set to *Specific unit* or *Specific unit match*.
- **Charges value** – Enter the charge value, based on the category that you selected.
- **Charges currency code** – Specify a currency for the charge if you want to use a currency other than the currency that you specified in the **Currency** field. You can use a different currency only if the **Debit type** or **Credit type** field is set to either *Ledger account* or *Item* for the selected charges code.
- **From amount** – Specify a starting amount to apply the auto charge to. In this context, the amount refers to the order total.
- **To amount** – Specify the ending amount to apply the auto charge to. In this context, the amount refers to the order total.
- **Sales tax group** – Specify a sales tax group.
- **Site** and **Warehouse** – Specify a site and warehouse if charges should be applied only for a specific site and warehouse.
- **Keep** – Select this checkbox to keep the charges transactions after invoicing is completed, so that the charge is applied every time that you create a new invoice for the selected customer account.

For sales orders only: If you want to calculate tiered charges, see Tiered charges on sales orders for information.

Supply Chain Management provides the following optional functionality for calculating auto charges. You might need to enable and configure these features before you can use them. For more information, follow the links.

- Auto charge compounding and sequencing
- Units of measure for line-level charges

> **Tip**
>
> When configuring auto charges for accounts payable in Dynamics 365 Finance, you might see duplicate charge calculations if you set up auto charges for both specific vendors (*Table*) and all vendors (*All*). This situation occurs because the system applies all relevant auto charge configurations, regardless of hierarchy, whenever you create a new purchase order. For example, if you set up a charge code with account code *All*, the system applies that charge code to every purchase order, even if you also set up other specific vendor or group configurations.

Allocate charges from the header to a line

The following procedure shows how to allocate header-level charges to a line. Before you start this procedure, you should already have a header-level charge of the *fixed amount* type and an order where you apply that charge. Additionally, the order should already include at least one line item.

1. Open the purchase order or charge order.

2. On the Action Pane, follow one of these steps:

    - For purchase orders: On the **Purchase** tab, in the **Charges** group, select **Allocate charges**.
    - For sales orders: On the **Sell** tab, in the **Charges** group, select **Allocate charges**.

3. In the **Allocate charges to order lines** dialog box, set the following fields:

    - **Charges allocation** – Select one of the following values to specify how the charges should be allocated:
        - *Net amount* – Allocate charges according to each line amount relative to the total net amount.
        - *Quantity* – Allocate charges according to the number of units for each line relative to the total number of units.
        - *Per line* – Allocate charges equally among the total number of lines.

    - **Allocate charges to lines** – Select a value to specify whether charges should be allocated to all lines, to positive lines only, or to negative lines only.

    - **Allocate all** – Select this checkbox to allocate charges to order lines even if the charges code has a debit type other than *Item*.

    - **Received** – Select this checkbox to allocate charges only to received order lines.

    - **Stocked** – Select this checkbox to allocate charges to only inventoried order lines.

    - **Show selections and clear specific lines** – Select this checkbox to exclude specific lines from this allocation. When you select this checkbox, the **Choose lines to exclude from allocation** grid opens. This grid includes only lines that match the criteria that are defined by the **Allocate charges to lines** and **Stocked** settings. For example, if you set the **Allocate charges to lines** field to *Positive lines* and select the **Stocked** checkbox, the grid shows only lines that are both positive and inventoried. In addition, the grid automatically filters out any lines that the full quantity is already received for. While the grid is open, clear the **Include** checkbox for each line that you want to exclude from allocation.

> **Important**
>
> When you work with the **Choose lines to exclude from allocation** grid, be sure to leave the grid open until you select **Allocate**. If you close the grid before you select **Allocate**, your settings in the grid are lost. Therefore, charges are allocated based on the criteria that you previously defined.

4. Select **Allocate** to apply your settings and close the dialog box.

Reapply header auto charges

In some cases, the system reapplies header auto charges by deleting and inserting them after you update a header field.

You can set up auto charges to consider vendor charge groups or modes of delivery. Therefore, when you update the **Charges group** or **Mode of delivery** on a purchase order header, the charge might change based on the auto charge setup. When the charge needs to be recalculated, the system deletes the existing auto charge and inserts a new auto charge. The recalculation doesn't affect charges you add manually to the purchase order.

After the system allocates an auto charge from the header to a line, it doesn't track its origin (there's no reference from the line to the auto charge setup), so the charge becomes a line charge. If you then update the **Charges group** or **Mode of delivery** on a purchase order header, the system reapplies the new, recalculated header charge, but it doesn't affect the line charges. This pattern applies to both purchase order headers and sales order headers.

The following examples show how to update header-level charges and select a new **Mode of delivery** on the purchase order header. Each example assumes that you already have a header-level charge of type *fixed amount* and an order where that charge is applied. The order must already include at least one line item.

Example 1: Header auto charge value resets after changing the mode of delivery

This example shows that if you customize the value of a header auto charge and then update the mode of delivery for the purchase order, the system resets the header auto charge to its original value.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders** and open a purchase order that includes a header auto charge of type *fixed amount*.

2. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.
3. For the existing header auto charge, enter a new value in the **Charges value** field.
4. On the Action Pane, select the **Back** button to return to the purchase order.
5. Open the **Header** tab of the purchase order.
6. Expand the **Delivery** FastTab and select a new value for the **Mode of delivery** field.
7. On the Action Pane, select **Save**.
8. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

   Note that the **Charges value** is now reset to its original auto charge value.

Example 2: Header auto charge is reapplied after allocating it to a line and then changing the mode of delivery

This example shows that if you allocate a header auto charge to a line and then update the mode of delivery for the order, the system reapplies the original header auto charge while leaving the allocated charge applied to the line.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders** and open a purchase order that includes a header auto charge of type *fixed amount*.
2. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

   Note the header auto charges listed here.
3. On the Action Pane, select the **Back** button to return to the purchase order.
4. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Allocate charges**.
5. On the **Allocate charges to order lines** dialog, select the **Allocate all** check box. Then select **Allocate** to allocate the header auto charge to the order lines.
6. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

   Note that the header auto charges are no longer listed because you allocated them to the line level.

7. On the Action Pane, select the **Back** button to return to the purchase order.
8. On the **Purchase order lines** FastTab, select a purchase order line.
9. On the **Purchase order lines** FastTab toolbar, select **Financials** > **Maintain charges**.

   Note that the previous header auto charge is now shown here, at the line level.
10. On the Action Pane, select the **Back** button to return to the purchase order.
11. Open the **Header** tab of the purchase order.
12. Expand the **Delivery** FastTab and select a new value for the **Mode of delivery** field.
13. On the Action Pane, select **Save**.
14. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

    Note that the original header auto charge is now reapplied to the order, even though the charge is also listed at the line level.

Last updated on 12/09/2025

Units of measure for line-level charges

All line-level auto charges and line charges include a category that controls how and where the charge applies. The available categories include *Specific unit* and *Specific unit match*. These categories let you set up and apply line charges that are specific to the unit of measure for a line item that's either sold or purchased. They work in the following way:

- *Specific unit* – The auto charge is defined by using a specific unit (such as *ea*). The system then converts proportionally from other units if a conversion factor is available. For example, a sales line is for 2 boxes of an item where a box is defined as containing 10 ea. There's a charge of 2 USD per ea. In this case, the sales line produces a charge of 40 USD (= 2 boxes × 10 ea/box × 2 USD/ea). If no conversion factor is available, the charge isn't applied.
- *Specific unit match* – The auto charge is defined by using a specific unit (such as *ea*). However, the system applies that charge only to lines that use the same unit. For example, you set up a charge so that a line charge is added when the item is sold in smaller units (such as *ea*). However, there's no line charge for orders that are specified in larger units (such as cases).

The *Specific unit* and *Specific unit match* categories are supported for sales quotation lines, sales order lines, purchase requisition lines, request for quotation lines, and purchase order lines. Because both categories apply to both sales orders and purchase orders, they're also supported for intercompany trade.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.38 or later.
- The feature named *Unit of measure for line level charges* must be turned on in feature management. As of Supply Chain Management version 10.0.45, it's mandatory and can't be turned off.

Example scenario: Set up line-level auto charges for specific units and unit matching

This example scenario shows how to set up line-level auto charges that apply to a specific unit, or that are calculated based on unit matching.

Enable demo data

To work through this scenario by using the demo records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin.

Set up the auto charges

Follow these steps to set up a line charge in auto charges.

1. Go to **Accounts receivable** > **Setup** > **Charges** > **Auto charges**. (Similar steps can be completed at **Accounts payable** > **Setup** > **Charges** > **Automatic charges** and **Procurement and sourcing** > **Setup** > **Charges** > **Automatic charges**.)

2. In the list pane, set the **Level** field to *Link*.

3. On the Action Pane, select **New** to create an auto charges record.

4. On the header of the new record, set the following values:

    - **Account code** – Select *All*.
    - **Item code** – Select *All*.
    - **Mode of delivery code** – Select *All*.
    - **Charge description** – Enter a short description.

5. On the Action Pane, select **Save**.

6. The **Lines** FastTab should now include a blank line. If it doesn't, select **Add** on the toolbar to add one. Set the following values for this line:

    - **Currency** – Select *USD*.
    - **Charges code** – Select *Freight*.
    - **Category** – Select *Specific unit*.
    - **Unit** - Select *ea*.
    - **Charges value** – Enter *2*.
    - **Sales tax group** – Leave this field blank.
    - **Site** – Leave this field blank.
    - **Warehouse** – Leave this field blank.

- **Keep** – Leave this checkbox cleared.

7. On the **Lines** FastTab, select **Add** on the toolbar to add a second charge line. Set the following values for it:

    - **Currency** – Select *USD*.
    - **Charges code** – Select *Install*.
    - **Category** – Select *Specific unit match*.
    - **Unit** - Select *dz* (dozen).
    - **Charges value** – Enter *5*.
    - **Sales tax group** – Leave this field blank.
    - **Site** – Leave this field blank.
    - **Warehouse** – Leave this field blank.
    - **Keep** – Leave this checkbox cleared.

8. On the Action Pane, select **Save**.

Use the auto charges on a sales order

The charges can now be automatically applied to a sales quotation line or a sales order line. To apply the charges to a sales order line, follow these steps:

1. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.

2. On the Action Pane, select **New** to create a sales order.

3. In the **Create sales order** dialog box, set the **Customer account** field to *US-004*. Then select **OK** to create the order. The selected customer must use the same currency as the auto charges that you set up (*USD*).

4. The new sales order is opened. On the Action Pane, on the **Sell** tab, in the **Charges** group, select **Maintain charges**.

5. If any charges are listed on the **Maintain charges** page, select them, and then select **Delete** on the Action Pane.

6. Close the **Maintain charges** page by selecting the back button on the Action Pane.

7. You're returned to the order. On the **Sales order lines** FastTab, add an order line, and set the following values for it:

    - **Item number** – Select *A0002*.

- **Quantity** – Enter *2*.
- **Unit** – Select *ea*.
- **Site** – Select *6*.
- **Unit price** – Enter *50*.

The **Net amount** field should show a calculated value of *100*.

8. While the new order line is still selected on the **Sales order lines** FastTab, select **Financials** > **Maintain charges** on the toolbar.

9. Confirm that the **Maintain charges** page shows the two auto charges that you set up earlier. Then close the page by selecting the back button on the Action Pane.

10. On Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.

11. The **Totals** dialog box shows the different calculation results that apply to the current order. Notice that the value that's shown for **Total Charges** is 4 USD. This value is calculated in the following way:

    - There's a *Specific unit* charge of 2 USD per ea. Because the line is for the equivalent of 2 ea, this charge is 2 ea × 2 USD/ea = 4 USD.
    - There's also a *Specific unit match* charge of 5 USD per dz. However, because the unit that you're ordering in is ea, not dz, this charge doesn't apply.
    - The total charge for the line combines both charges. Therefore, the total charge is 4 USD.

12. Select **OK** to return to the sales order.

13. On the sales line, change the **Unit** value to *dz*. When you're prompted to overwrite prices and discounts, select *No*, and then select **OK**. (The system is configured to convert 1 dz to 12 ea.)

14. On Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.

15. In the **Totals** dialog box, notice that the value that's shown for **Total Charges** is now 58 USD. This value is calculated in the following way:

    - The unit is now *dz*, and there's a *Specific unit match* charge of 5 USD per dz. Because you're now ordering two dozens, this charge is 2 dz × 5 USD/dz = 10 USD.
    - There's also a *Specific unit* charge of 2 USD per ea. Because the system is configured to convert 1 dz to 12 ea, and you're now ordering a total of two dozen items, it can

determine that the order line is for 24 ea. Therefore, this charge is 24 ea × 2 USD/ea = 48 USD.

- The total charge for the line combines both charges. Therefore, the total charge is 10 USD + 48 USD = 58 USD.

Pricing management

The features that are described in this article work together with the Unified pricing management module. Pricing management introduces changes to the setup and search of auto charges. Therefore, it also changes some of the behavior that's described in this article. When the Unified pricing management module is enabled, charges that are set up at **Accounts receivable** > **Charges setup** don't apply. Instead, only charges that are set up on the auto charges page that's specific to the **Pricing management** module apply to sales orders and sales quotations. To set up auto charges in Pricing management, go to **Pricing management** > **During-sales pricing** > **Charges setup** > **Auto charges**.

Last updated on 03/25/2026

Vendor rebates

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Vendor rebates help companies better manage their supplier rebate programs by automating tasks required to administer, track, and claim earned rebates.

This article provides an overview of the most common tasks that you might want to perform when you work with vendor rebates. The overview covers the following tasks:

- Review details of a rebate agreement.
- Identify orders that qualify for rebates, and generate rebate claims.
- Review and approve claims.

Audience and purpose

The information in this article is intended for business decision makers in enterprise companies, in positions such as purchase manager, chief financial officer (CFO), and accounting manager, who have the following responsibilities:

- Negotiate vendor price, discount, and rebate agreements.
- Manage staff that processes rebate claims and collects payments.
- Order inventory at the best possible prices.

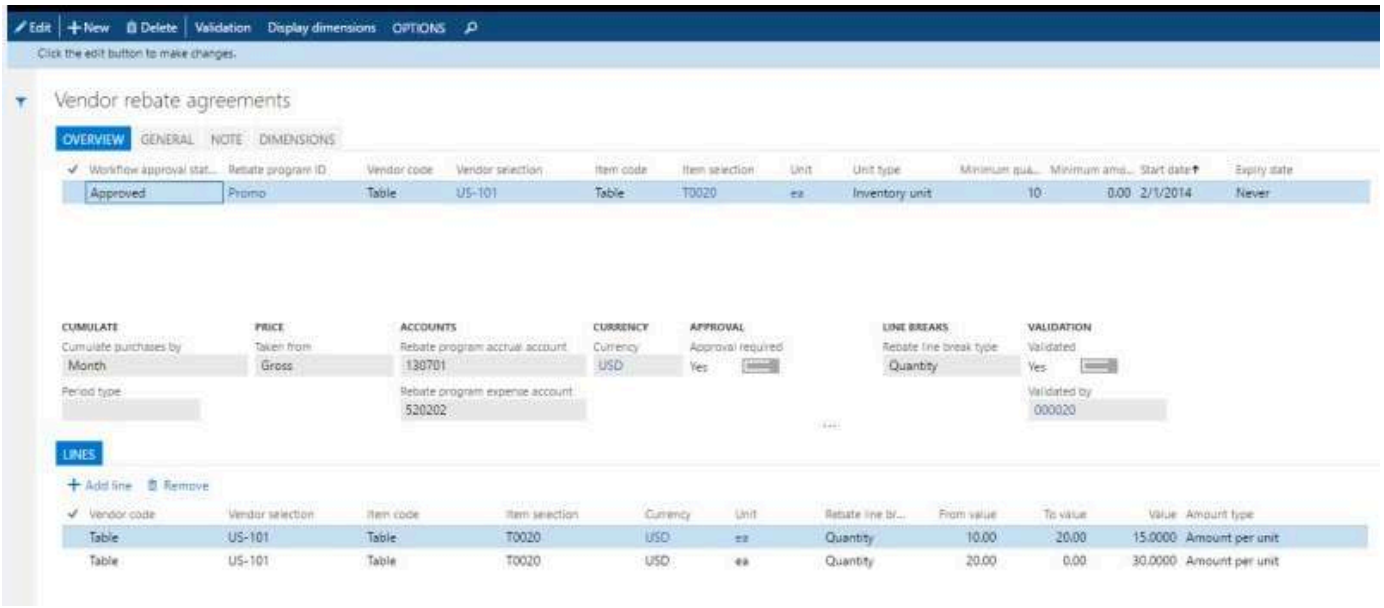
People in these positions are looking for ways to achieve various goals. Here are some examples:

- Flexibly accommodate different types of vendor promotion programs and rebate conditions.
- Reduce the administrative burden and errors associated with monitoring promotion performance and processing claims.
- Improve cash flow forecasts by accruing for future receivables.
- Have a quantified basis for ongoing and future negotiations with vendors about rebates.

Review details of a vendor rebate agreement

A vendor rebate agreement is a record of a contract with a vendor that specifies the negotiated terms and conditions under which the company qualifies for a monetary reward in return for achieving preset purchase targets. Vendor rebate agreements are recorded on the **Rebate agreements** page.

To open the **Vendor rebate agreements** page, select **Procurement and sourcing** > **Vendor rebates** > **Rebate agreements**.



On the **Vendor rebate agreements** page, you can view details about the negotiated conditions of a vendor agreement.

The agreement's header specifies the general conditions that qualify a company for rebates. In effect, the header information specifies that a vendor grants a rebate when a specific product is bought in a specific quantity. On the header, you also specify the unit of measure rebate option and the calculation date type.

- On the **Overview** tab, if you have lines with **Item code** set to *table* to specify the item, then the agreement is for that specific item. If you have lines with **Item code** set to *Group* or *All* to specify the items, then the vendor rebate agreement is individually processed per item qualifying for the item code, not across all items qualifying for the item code.

- On the **General** tab, in the **Unit of measure rebate option** field, you can define whether a unit of measure is a condition for the purchase order line to qualify for a rebate claim.
    - *Convert* – A purchase order line qualifies for a vendor rebate per the rebate agreement. You receive a rebate regardless of the unit of measure applied on the line.

    - *Exact match* – To qualify for a rebate, a purchase line must have the same unit of measure that is specified on the agreement.

- On the **General** tab, in the **Calculation date type** field, select the date used to determine whether the purchase occurs in the validity period of the rebate agreement.
    - *Created* – Use the creation date of the purchase order.
    - *Requested delivery* – Use the requested receipt date.

On the agreement lines, you can specify the vendor rebate agreement in more detail.

- In the **Cumulate purchase by** field, you can set the calculation of the rebate claim. The amount can depend on a period (week, month, year, lifetime, or a customized period). The **Invoice** value indicates that a rebate claim is determined every time that a purchase order line is invoiced.

- In the **Taken from** field, you can specify the basis for the rebate calculation.
    - *Gross* – The rebate is calculated based on the gross price of the item.
    - *Net* – The rebate is calculated based on the net price of the item (that is, the price after other discounts have been applied).

- The **Rebate program accrual account** and **Rebate program expense account** fields specify account numbers that receive accrued rebate amounts during the intermediate stage between approval and processing.

- When the **Approval required** option is set to *Yes*, the rebate claim must be approved before it can be accrued or paid out.

- The **Rebate line break type** field specifies the basis for the rebates.
    - *Quantity* – The rebates are volume-based.
    - *Amount* – The rebates are amount-based.

- On the **Lines** FastTab, you can see how different quantity tiers can be set up to grant different rebates. For example, in the previous illustration, the **From value** and **To value** fields indicate that a product quantity between 10 and 19 units will qualify for a rebate of USD 15 per unit.

    > ⓘ **Note**
    >
    > The **From value** value is inclusive, whereas the **To value** value is exclusive. For example, the **Rebate line break type** field is set to *Quantity*, and you enter *1* in the **From value**

field and *3* in the **To value** field. In this case, the rebate amount applies when you purchase one or two items, but not when you purchase three items.

- In the **Workflow approval status** field, the *Approved* value indicates that the agreement applies to purchase orders that meet the agreement's conditions.

Identify orders that qualify for rebates, and generate rebate claims

When purchase orders are placed with a vendor that the company has a rebate agreement with, the program identifies any future vendor credit payments. If the purchase orders qualify for a rebate, a rebate claim is generated for every order line as soon as a purchase invoice has been posted. This process is automatic. Later, you can review the expected rebates and see the impact of those rebates on the product's cost and profit margin.

View details of rebates applied to a purchase order line per the vendor rebate agreement

1. On the **Purchase order** page, select an order line, and then select **Purchase order line** > **View** > **Price details**.
2. On the **Price details** page, select the **Rebates** FastTab.

The **Vendor rebate** field in the **Margin estimation** section of the **Price details** page also shows the rebate information.

> ⓘ **Note**
>
> On the **Procurement and sourcing parameters** page, on the **Prices** tab, verify that the **Enable price details** option is set to *Yes*. If the option is set to *No*, you can't view the rebates.

Review and approve claims

Generated rebate claims represent the future payments that can be expected from the vendor. Before a credit note is issued to the vendor, the agreement owner typically wants to review and approve the claims. However, the status of a claim determines whether it's ready for the approval process.

The status of claims and the effect on the approval process

When a claim is generated, its status is set to *To be calculated* if the rebate is granted on a cumulative basis or *Calculated* if the rebate is granted per invoice. If the status of a claim is *To be calculated*, the claim must go through a calculation process that the Cumulate function handles. Only claims that have a status of *Calculated* can be included in the approval process.

> ⓘ **Note**
>
> If the **Approval required** option on a vendor rebate agreement is set to *No*, any claims that are generated have a status of *Approved*. Approval is mandatory for claims that are granted on a cumulative basis.

Approve claims, and view postings and invoice details

After claims are approved, Accounts payable (A/P) can process them. A credit memo (vendor invoice) for the rebate claim amount is automatically generated. The credit is then added to the vendor balance, and the A/P team can include it in the regular settlement process.

1. Select **Procurement and sourcing** > **Vendor Rebates** > **Rebate claims** to open a rebate claim.
2. Close the rebate claim.
3. Mark the claim, and then, on the Action Pane, select **Approve**.
4. On the request page, in the **Vendor** field, select the vendor that you're authorized to receive a rebate from, and then select **OK**.

   A Rebate accrual journal is posted for the claim amount. This posting debits the Accrued Vendor Rebates Receivable account for the expected vendor credit and credits the interim Accrued Vendor Rebates Received account for the expected gain.

5. In the rebate list, select the line, and then, on the Action Pane, select **Rebate transactions** to see and navigate to the journal batch number for this rebate accrual posting.

   To move the claims to the regular A/P process, the A/P clerk must now complete the rebate claim handling by running the Process function.

6. On the Action Pane, select **Process**, and then select **Filter**. In the **Criteria** field for the **Vendor account** field, select the vendor to process rebate claims for, select other relevant filters, and then select **OK**.

   The message bars and the status change to *Completed* indicating that the following events have occurred:

   - A Rebate accrual journal posting has reversed the previous interim amounts on the accrual receivable and expense accounts.
   - A vendor invoice (credit note) for the rebate amount has been created.

     > ⓘ **Note**
     >
     > The setting of the **Manual invoice posting** option on the **Rebate program** tab of the **Procurement and sourcing parameters** page determines whether a vendor invoice is posted manually or automatically as part of claim processing.

   - When the vendor invoice is posted, either automatically or manually, the vendor's Payable account has been debited, and the Discounts and Allowances Received account has been credited.

     > ⓘ **Note**
     >
     > The Discounts and Allowances Received account number is specified for the procurement category that is used on the purchase invoice line for the rebate. The procurement category, in turn, is set on the **Rebate program** tab of the **Procurement and sourcing parameters** page.

7. In the rebate list, select the line, and then, on the Action Pane, select **Rebate transactions** to see and navigate to the journal batch number for this rebate accrual posting and also the vendor invoice number.

8. Select the line for the vendor invoice transaction, and then, on the Action Pane, select **Vendor invoice**. If the vendor invoice has been posted, you see the Invoice journal. Otherwise, you see the vendor invoice as a pending vendor invoice that requires manual posting.

   The invoice line specifies the details of the vendor invoice for the **Commissions and Rebates** procurement category.

9. On the **All vendors** page, select the vendor that you receive a rebate from, and then, on the Action Pane, select **Transactions**. Find the line for the invoice. The rebate amount has now been added to the vendor balance.

Summary

The process for handling vendor rebates involves multiple manual tracking tasks that are often tedious. By automating these tasks, the vendor rebate management feature helps you move through the following processes:

- Generating accurate rebate claims
- Accruing the expected receivable and interim gain in the general ledger
- Updating the vendor balance and the income statement with the allowance that is due

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Last updated on 07/01/2026

Purchasing policies overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about purchasing policies. A purchasing policy is a collection of rules that control the requisition process. Purchasing policies help procurement administrators implement their procurement strategy by creating a policy structure that aligns with the organization's strategic purchasing requirements.

A purchasing policy consists of a set of policy rules. When you define a policy rule, you first select a rule type. You then create a rule for the rule type by defining the settings, the start date, and the end date for the rule.

For example, an administrator creates a policy, selects the *Catalog policy* rule type, and then adds a catalog policy rule to the policy. This catalog policy rule specifies that the Adventure catalog must be used for internal procurement. After you associate the purchasing policy with a particular organization, employees of that organization see the Adventure catalog when they create requisitions.

Manage purchasing policies

To manage your purchasing policies, go to **Procurement and sourcing** > **Setup** > **Policies** > **Purchasing policies**. From there, you can create, edit, and delete purchasing policies as needed.

Assigning policies to organizations

Before a policy can take effect, you must associate it with an organization. When you create or edit a policy, use the **Policy organization** FastTab of the **Purchase policy** page to associate it with an organization.

Purchasing policies are associated with the *Procurement internal control* hierarchy purpose. Therefore, purchasing policies apply only to organizations in hierarchies that have a hierarchy

purpose of *Procurement internal control*. You can also select organizations from the flat list of legal entities in the CompanyInfo table. These legal entities are designated in the policy framework as *Companies*.

Determine which rule to apply

Depending on how you configure your purchasing policies, multiple rules can affect the users in an organization. The following examples illustrate different ways that you can configure purchasing policies and specify how policies are applied when a transaction occurs.

Example 1: Simple purchasing policy configuration

Small and less complex organizations can set up purchasing policies by legal entity and use only the *Companies* organization hierarchy.

For Fabrikam, a small business, purchasing requirements vary little across the organization. Purchasing rules vary only among the organization's legal entities. For example, employees of Fabrikam Canada and employees of Fabrikam U.S. purchase goods and services from different catalogs and different vendors. Therefore, Fabrikam sets up its purchasing policies at the legal-entity level.

Fabrikam creates two purchasing policies. *Policy A* applies to its U.S. legal entity, *1111*. *Policy B* applies to its Canadian legal entity, *2222*. When an employee in legal entity *1111* creates a purchase requisition, the policy rules come from *policy A*. For example, the product catalog that the employee sees is specified in the catalog policy rule for *policy A*.

When an employee in legal entity *2222* creates a purchase requisition, the policy rules come from *policy B*.

> **Note**
>
> If an employee of legal entity *1111* purchases an item on behalf of an employee of legal entity *2222*, the policy rules that are specified for legal entity *2222* (that is, the policy rules from *policy B*) are applied.

Example 2: Complex purchasing policy configuration

In the previous example, all purchasing rules were defined in a single organization hierarchy, the *Companies* organization hierarchy. However, a complex organization might define policies for

multiple organization hierarchies.

Contoso is a large company that requires complex purchasing rules to control the requisition process. Contoso has defined rules for two different organization hierarchies: *Department* and *Global purchasing control*.

Policy 123 is defined for the *Department* organization hierarchy for the *Sales UK – Sales* department. In *policy 123*, the purchase requisition control rule specifies that restrictions must be enforced for minimum order quantities. In this rule, the **Enforce minimum order quantity restrictions** option is selected.

Policy 456 is defined for the *Global purchasing control* organization hierarchy for the *Sales and Marketing* department. In *policy 456*, the purchase requisition control rule doesn't specify that restrictions must be enforced for minimum order quantities. In this rule, the **Enforce minimum order quantity restrictions** option is deselected.

Sam works in the *Sales UK – Sales* department in Contoso's United Kingdom office. The policies for both the *Department* and *Global purchasing control* organization hierarchies apply to Sam's department. When Sam creates a purchase requisition, the system must determine which policy to apply. The system administrator set up the purchasing policy parameters to specify that purchasing policies must be applied in the following order of precedence:

1. *Global purchasing control*
2. *Department*
3. *Companies*

Therefore, *policy 456* is applied to the purchase requisition that Sam creates, and no minimum order quantity is required for the purchase requisition.

Policy rules

When you create or edit a policy, add and configure its policy rules by using the **Policy rules** FastTab of the **Purchase policy** page. Each type of rule has its own set of conditions and actions, as described in the following subsections.

Catalog policy rule

The catalog policy rule determines which procurement catalog users see when they create purchase requisitions. If a user has been granted permission to order products on behalf of another user, the requisition uses the catalog policy rule that is defined for the requester's legal

entity and operating unit to determine which catalog to display. Before you can define a catalog policy rule, you must create and publish a procurement catalog.

Category access policy rule

The category access policy rule determines which categories users can access when they create purchase requisitions. If you don't specify a rule, users can add all the procurement categories to the purchase requisition.

1. Select the **Include parent rule** option to apply the category access policy rule of the parent organization to the category.
2. In the **Available categories** pane, select the categories that the rule applies to. When you select a category, all categories that are higher in the hierarchy are also added to the **Selected categories** list.
3. Select the **Include subcategories** option to apply the rule to all subcategories of the selected category.

Category policy rule

The category policy rule defines how users can select vendors for each category. It also defines requirements for the receiving and invoicing processes.

Reapproval rule for purchase orders

The reapproval rule is an optional rule that defines the criteria for requiring reapproval when a purchase order is changed. The selected fields are evaluated in the purchase order workflow when the *Requires purchase order reapproval* condition is set up in the workflow.

> ⓘ **Note**
>
> Accounting distribution is always reset when you change an approved purchase order with change management enabled. To avoid reapproval of a purchase order when certain fields are changed, don't include the field Accounting distribution.changed as a selected field for reapproval.

Purchase requisition RFQ rule

The purchase requisition RFQ rule defines criteria for requiring a request for quotation (RFQ) for a purchase requisition line. If a line meets the conditions, the *informal RFQ* or *formal RFQ* stamp

appears on the requisition line.

Purchase requisition control rule

The purchase requisition control rule for requisitions of type *Consumption* is an optional rule. When you create rules of this type, you can set options on various tabs:

- On the **Workflow submission** tab, configure the fields that must be entered on the requisition line for the requisition to be submitted for approval.
- On the **Order quantities** tab, configure the fields that are required on the purchase requisition under certain conditions. You can also enforce order quantities.
- On the **Dates** tab, configure whether the accounting date is the same as the requested date.
- On the **Address** tab, define whether the user is allowed to create new addresses in the system to apply to the purchase requisition.

Requisition purpose rule

The requisition purpose rule is an optional rule that determines the type of requisition purpose that is allowed for a specific legal entity. Unless another purpose is indicated in this rule, requisitions automatically have a purpose of **Consumption** when they're created.

Replenishment category access policy rule

The replenishment category access policy rule is an optional rule that determines the products that are available to fulfill requisition demand for a specific legal entity when the requisition purpose is *Replenishment*.

Replenishment control rule

The replenishment control rule is an optional rule that defines the fields that must be entered on the requisition line for the requisition to be submitted for approval when the requisition purpose is *Replenishment*.

Purchase order creation and demand consolidation rule

The purchase order creation and demand consolidation rule defines the policy rules to use when generating a purchase order from an approved purchase requisition. When you create rules of this type, set options on various tabs:

- On the **Purchase order split** tab, define criteria for splitting purchase requisition lines onto separate purchase orders.

- On the **Price/discount transfer** tab, define when to recalculate the price agreement when creating a purchase order:
  - **Only if no trade agreement** – Transfer prices and discounts from the purchase requisition only if there's no applicable trade agreement or base price. If a trade agreement or base price exists for the item or vendor, recalculate prices and discounts based on the trade agreement or the base price, and apply them to the purchase order. Unless otherwise specified, this is the default behavior.
  - **Always** – Always transfer prices and discounts from the purchase requisition.

  You can also allow the requester to change the method of price and discount transfer for individual purchase requisition lines, regardless of the price and discount transfer rule that you define. Select the **Allow manual override per purchase requisition line** option to enable this capability.

- On the **Item description transfer** tab, transfer the item description from the requisition when it originates from an RFQ.

- On the **Price Tolerance** tab, define rules to route approved purchase requisitions back through the review process when the price of a procurement catalog item increases. Set the maximum amount that the net amount on a line item on a purchase requisition can increase between the time when the purchase requisition is approved and the time when the purchase order is created. The net amount is calculated by using the following formula: ([Quantity × (Unit price – Discount) ÷ Price unit] + Purchase miscellaneous charges) × (100 – Discount percent) ÷ 100 Purchase requisition lines that exceed the price tolerance that you set are held for manual processing. The rules that you configure on the **Error processing** tab determine how the purchase requisition lines are processed.

- On the **Error processing** tab, configure the processing rule that is applied to a purchase requisition if it fails validation during purchase order creation because of a vendor error or a price tolerance error. Select one of the following options:
  - **No action** – The purchase requisition lines remain on the **Release approved purchase requisitions** page. The status of the purchase requisition lines remains *Approved*. However, you must resolve the errors before a purchase order can be generated for the purchase requisition lines.
  - **Cancel the purchase requisition line** – The purchase requisition lines are canceled. The requester can create a new purchase requisition for the canceled lines if they still want to

request the line items.
    - **Create a new purchase requisition line** – The purchase requisition lines are canceled. New purchase requisitions are then generated that contain only the purchase requisition lines that failed validation. The new purchase requisitions that are generated have a status of *Draft*. These purchase requisitions can be resubmitted for review after the validation errors are resolved. The preparer of the purchase requisition lines is notified that the lines were canceled, and that new purchase requisitions were generated for the purchase requisition lines that failed.

- On the **Manual purchase order creation** tab, define the parameters that determine whether a purchase requisition must be manually processed, or whether it can be automatically converted to a purchase order. The parameters can apply to internal catalog items, external catalog items, or non-catalog items. Select one of the following options:
    - **Manually create purchase orders** – Manually create purchase orders for all purchase requisitions.
    - **Automatically create purchase orders** – Automatically create purchase orders for all approved purchase requisitions. No purchase requisitions are held for manual purchase order creation.
    - **Automatically create purchase orders except under these conditions** – Manually create purchase orders for purchase requisitions that match the criteria that you define. All other purchase requisitions that are approved are automatically converted to purchase orders. If you select **Automatically create purchase orders except under these conditions**, you can add procurement categories and vendors to specify which approved purchase requisition lines are held for manual processing. This option can apply to internal catalog items, external catalog items, and non-catalog items. When you select a procurement category, any subcategories for that procurement category are also selected. Select the **All** option for a specific type of purchase requisition line to hold all lines of that line type for manual processing.

  If you want purchase orders to be generated automatically from approved purchase requisitions when the batch job for purchase order generation runs, select the **Run automatic purchase order creation as a batch job** option. This option applies only to purchase requisitions that don't require manual processing. By running automatic purchase order generation as a batch job, you can schedule this activity at a time when resources are less constrained. If the **Prepayment required** option is selected on the purchase requisition lines, select the **When the requisition is set up for prepayment** option to hold approved purchase requisitions for manual processing. You can filter purchase requisitions that are

held for manual processing so that you can view only those purchase requisition lines that require prepayment.

- On the **Demand consolidation** tab, define the parameters that determine whether purchase requisitions that are manually processed can be considered for purchase requisition consolidation. The parameters can apply to internal catalog items, external catalog items, or non-catalog items. Select one of the following options:
  - **Do not allow demand consolidation** – No approved purchase requisition lines are eligible for demand consolidation. This option is selected by default and applies only to purchase requisition lines that require manual processing for purchase order creation.
  - **Always allow demand consolidation** – All approved purchase requisition lines are eligible for demand consolidation. **Note:** If you select the **Always allow demand consolidation** option on the **Demand consolidation** tab, but you select the **Automatically create purchase orders** option on the **Manual purchase order creation** tab, all purchase requisitions are held for manual processing.
  - **Allow demand consolidation under these conditions** – Define the criteria that determine whether approved purchase requisition lines are eligible for demand consolidation. For each type of purchase requisition line, you can set the criteria by procurement category and vendor. If you select **Allow demand consolidation under these conditions**, you can set the criteria by procurement category and vendor for each type of purchase requisition line. When you select a procurement category, any subcategories for that procurement category are also selected. If you select the **All** option for a specific line type, all purchase requisition lines of that line type are eligible for demand consolidation.

> **ⓘ Important**
>
> If you enable budget control and set purchase requisitions to **Automatically create purchase orders** on approval, also turn on **Run automatic purchase order creation as a batch job** on the **Purchase order creation and demand consolidation** policy rule (**Procurement and sourcing** > **Setup** > **Policies** > **Purchasing policies**). Otherwise, the budget check that runs during requisition release can repeatedly re-trigger the release process, trapping requisition processing in a never-ending loop that degrades performance and can make the system unresponsive. Running automatic purchase order creation as a batch job processes approved requisitions asynchronously and prevents the loop.

Last updated on 06/30/2026

Create purchasing policies

Article • 01/24/2025

This article shows you how to create purchasing policies to align with your business processes for purchasing. Before you can create purchasing policies, you must set up the purchasing policy parameters. It's possible to create, modify, and retire a purchasing policy, but you can't delete a purchasing policy. This procedure would typically be carried out by a purchasing manager. You can use this procedure in demo data company USMF or on your own data.

Set up policy parameters

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Purchasing policies**.
2. On the Action Pane, select **Parameters**.

    - Policy precedence rules apply to different levels in your organization. The organizational units that are shown depend on your organizational hierarchy, and on which levels in the hierarchy have been assigned the purpose of Procurement internal control. For example, your organization might have legal entities, cost centers, regions, and departments, but it might be that only some of these have a hierarchy purpose of Procurement internal control. As a default, the organization of type Company is available.

3. Select the **Policy rule type parameters** tab.
4. In the tree, go to **Purchasing policy** > **Purchase requisition control rule**.

    - You define the order of precedence for policy resolution at the policy level. However, for some policy types, you can override the order of precedence for individual policy rule types. For example, you might define the order of precedence for purchasing policies to be: cost center, department, company. For the catalog policy rule, you might want the order of precedence to be: department, cost center, company. You can change the order of precedence for the Catalog policy rule. When a worker creates a requisition, the catalog that is displayed is determined by the policies that are associated with the worker's department, then their cost center, and then their company.
    - If there's more than one organizational level listed, you can use the Up/Down arrows to set the order of precedence for the Purchase requisition control rule.

5. Close the page.

Create a new policy

1. Select **New**.
2. In the **Name** field, type a value.
3. In the **Description** field, type a value.

    - A single purchasing policy can only apply to one organization hierarchy. For example, you could have one hierarchy called *Geographic* and one called *Department*, and have a different purchasing policy for each.
    - Select an organization that the policy should apply to.

4. Select the arrow to add the selected organization.

You can repeat this process to add more organizations.

Add a policy rule

1. In the **Policy rule type** list, select **Requisition purpose rule**.

    - You'll create a rule that sets the default requisition purpose to type Consumption but allows the Replenishment type to be selected instead.

2. Select **Create policy rule**.
3. Select **Yes** in the **Allow manual override** field.
4. Select **Close**.

Now you can set up other policy rules for the purchasing policy. Note that a Policy rule type cannot have overlapping rules that are active at the same time within a single procurement policy.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up policies for procurement category hierarchies

Use this procedure to set up rules for ordering products in a category. The rules are defined for a specific purchasing policy. The category access rule controls which procurement categories employees can access when they create a requisition. When an employee creates a requisition, the legal entity and the operational unit that the employee belongs to determine the purchasing policy and category access rule that apply. A purchasing manager typically carries out this task.

Find the procurement policy

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Purchasing policies**.
2. Open the policy that you want to add a rule to. It must be an active policy.

Create a category access rule

1. Expand the **Policy rules** FastTab.
2. In the **Policy rule type** list, select the *Category access policy rule*. If the **Create policy rule** button is dimmed, it's because there's already an active policy rule for category access. Check the **Effective** and **Expiration** fields to determine which rule is active. Then select the rule, and select **Retire policy rule**. If the **Create policy rule** button is available, you don't need to do anything.
3. Select **Create policy rule**.
4. In the **Effective date** field, enter a date and time. The time must not overlap with another rule that's already active.
5. Select a category that the rule applies to. Make a note of which category this is – you need it later. When you select a category, its parent category or categories are also added to the **Selected categories** list. If you want the rule to apply to all subcategories of the selected category, select the **Include subcategories** check box.
6. Select the right arrow to add to the **Selected categories** list.
7. Select **OK**. If you set the **Include parent rule** option to *Yes*, the policy rule that you define for a parent category is also assigned to its child categories, if no policy rule exists for the child categories.

Create a category policy rule

1. In the **Policy rule type** list, select **Category policy rule**. If the **Create policy rule** button is dimmed, select the active policy rule, and then select **Retire policy rule**.
2. Select **Create policy rule**.
3. In the **Effective date** field, enter a date and time.
4. Select **Add**.
5. In the **Category** field, select the same category that you used for the **Category access rule**.
6. In the **Vendor selection** field, select an option. Select a rule to control which kind of vendors can be selected for the category when requisitions are created.
7. Select **Close**. The policy rules that you defined apply to requisitions of type Consumption. To define policies for requisitions of type *Replenishment*, create a rule for the **Policy rule type** called *Replenishment category access policy rule*.

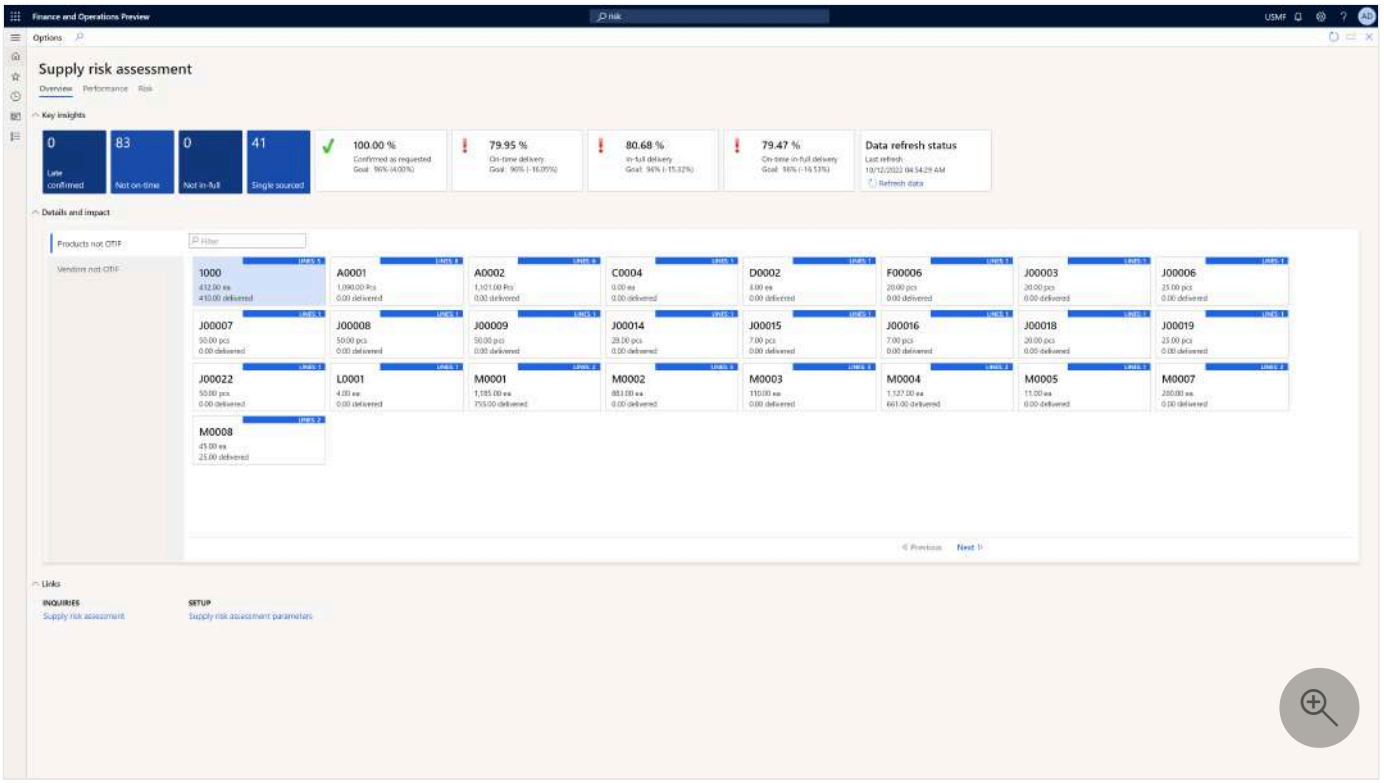
Last updated on 07/01/2026

Supply risk assessment overview

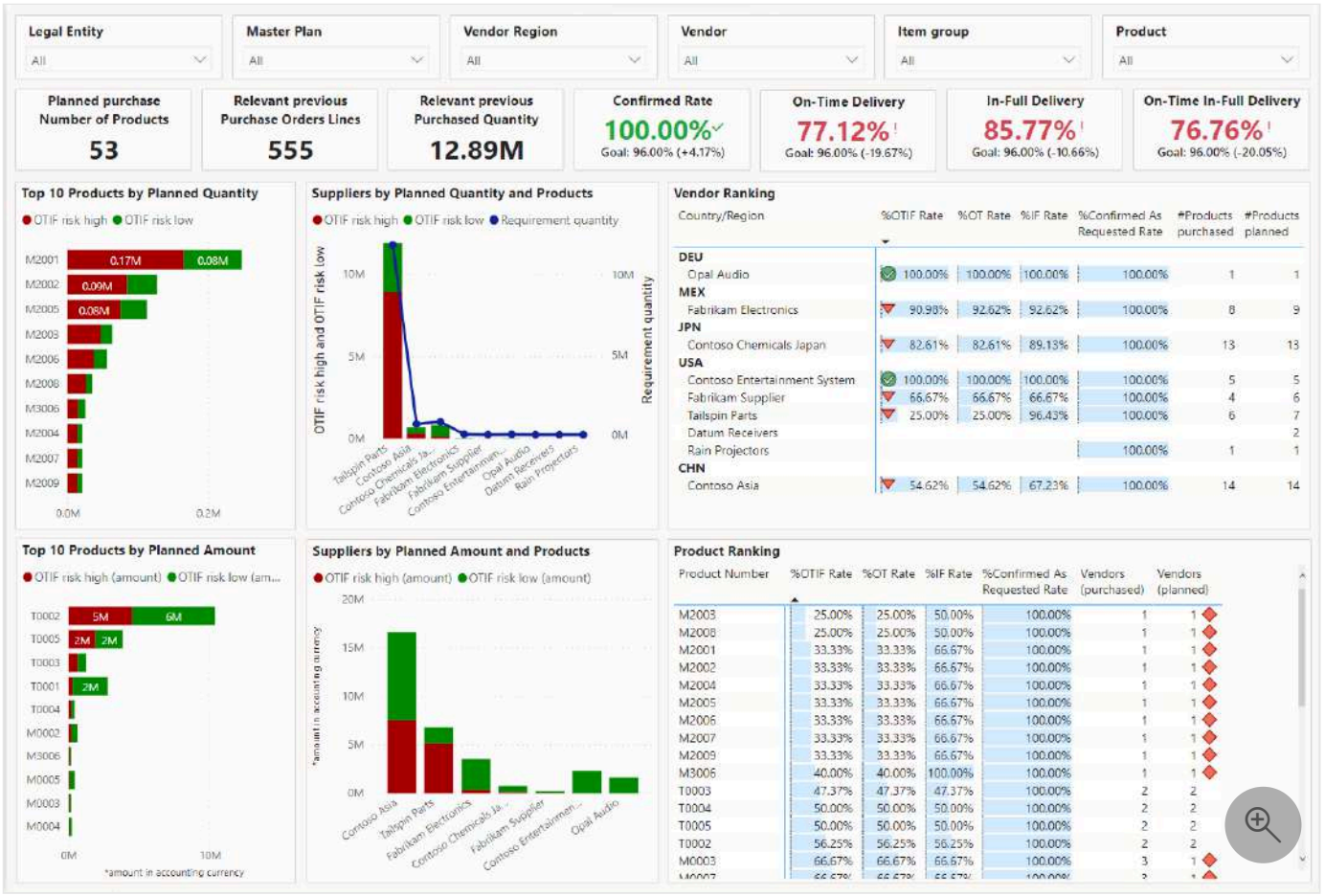
Supply risk assessment for Microsoft Dynamics 365 Supply Chain Management helps you identify supply risks, so you can take action to prevent supply chain disruptions.

The Supply risk assessment workspace, together with the supplier performance and risk analysis reports, helps supply managers understand the risk of encountering sourcing shortages and delays. It enables businesses to take proactive actions to balance cost and resilience when they optimize their supply chains.

The following illustration shows an example of the **Supply risk assessment** workspace.



The following illustration shows an example of the **Risk assessment** report for planned purchases. This report is one of the supplier performance and risk analysis reports.



The first release of this set of capabilities is available in release version 10.0.31. This release focuses on delivery performance by suppliers. It enables ranking and risk assessment based on planned orders.

The capabilities let you perform the following tasks:

- Measure the supply delivery performance of suppliers and products along several dimensions.
- Derive a risk assessment for future planned supply.

This release doesn't include the following capabilities:

- Holistic assessment of vendors along other aspects, such as financial ratings, compliance, security, and other risks
- Risks from other types of supply, such as production

The Supply risk assessment workspace as your point of entry

The **Supply risk assessment** workspace consists of lists of actionable items and embedded Power BI reports. It shows metrics such as on-time in-full (OTIF) ratings for vendors and product

rankings. It also represents the past performance as a risk for future supply. Planned purchase orders can quantify risk in terms of the quantity and amount that are at risk if the same performance and trend continue into the future.

To help mitigate the risks that you discover, you can take different approaches. For example, diversify vendors, or use different shipping methods or sourcing locations for planning. After you update the plan, reassess the supply risks and validate the improvements.

Next steps

- Turn on and set up the feature as described in Configure Supply risk assessment.
- Open the **Supply risk assessment** workspace to start discovering products and vendors that had low ratings in the past, and to access the Power BI reports for performance and supply risk analysis, as described in Actionable workspace for discovering and handling supplier risks.
- Open the Power BI reports for detailed performance and risk analysis.

Last updated on 07/01/2026

Configure supply risk assessment

This article describes how to enable and set up supply risk assessment.

Configure thresholds

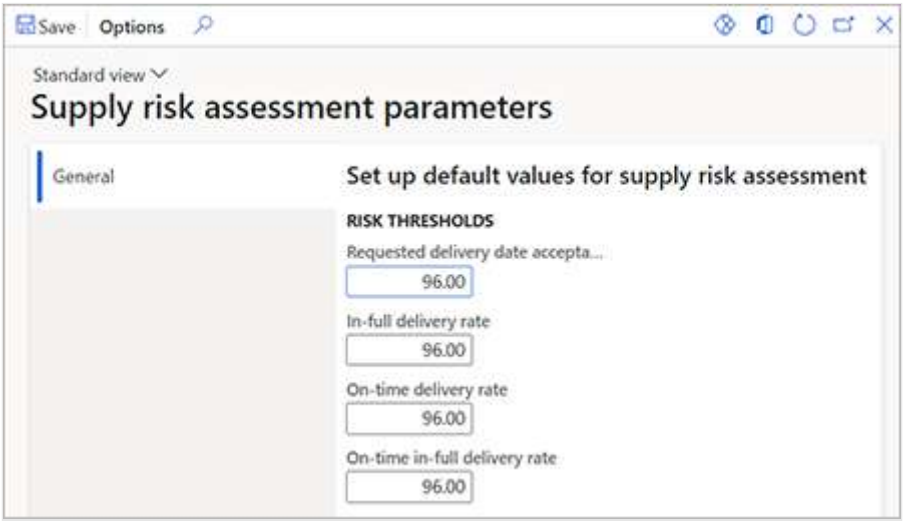
Use the following procedure to configure the feature by selecting the minimum rates that you want the system to apply when it calculates your supply risks.

1. Go to **Procurement and sourcing** > **Setup** > **Supply risk assessment parameters**.

2. On the **General** tab, set the following fields:

    - **Requested receipt date acceptance rate** – Enter the minimum percentage of orders where vendors are expected to return a confirmed receipt date (CRD) that meets your requested receipt date (RRD).
    - **In-full delivery rate** – Enter the minimum percentage of deliveries that are expected to arrive in full (IF).
    - **On-Time delivery rate** – Enter the minimum percentage of deliveries that are expected to arrive on time (OT).
    - **On-time in-full delivery rate** – Enter the minimum percentage of deliveries that are expected to arrive on time and in full (OTIF).

    Each of these metrics is calculated as a rate within the relevant scope of analysis. Set the rates that you consider a risk for your business. By default, each rate threshold is set to 96 percent.



3. On the Action Pane, select **Save**.

Initialize workspace data and configure refresh options

When you first turn on and set up the feature, the **Supply risk assessment** workspace doesn't show any data. However, because the system periodically updates the data, the workspace eventually shows data. Alternatively, you can manually initialize or refresh the data as described in the following procedure.

1. Go to **Procurement and sourcing** > **Workspaces** > **Supply risk assessment**.

   > ⓘ **Note**
   >
   > You don't have to do anything in the **Supply risk assessment** workspace right now. However, you must visit it at least once after you enable the feature, to make the required data cache configuration available.

2. Go to **System administration** > **Setup** > **Data cache** > **Data set cache configuration**.

3. On the Action Pane, select **Edit**.

4. In the grid, select the row where the **Consumer** field is set to *VendSupplyRiskCacheDataSet*, and set the following fields for it:

   - **Enabled** – Select this checkbox to enable the system to keep the data in the **Supply risk assessment** workspace up to date. If you clear this checkbox, the data isn't updated.
   - **Refresh frequency (seconds)** – Specify how often (in seconds) the system should refresh the data in the **Supply risk assessment** workspace.
   - **Manual refresh** – Select this checkbox to enable the **Refresh data** button on the **Data refresh status** tile on the **Supply risk assessment** page. If you clear this checkbox, the button is still shown, but it doesn't do anything.

5. On the Action Pane, select **Save**.

6. If you selected the **Manual refresh** checkbox and want to initialize or refresh the data in the **Supply risk assessment** workspace now, follow these steps:
   a. Go to **Procurement and sourcing** > **Workspaces** > **Supply risk assessment**.
   b. On the **Overview** tab, on the **Data refresh status** tile, select **Refresh data**.

Refresh the analytics entity store data used by Power BI reports

If you didn't use purchase data for Power BI until now, you might need to refresh this data and and/or set it up so that it's automatically updated from now on. If this data isn't refreshed, your Power BI reports for supply risk assessment show errors.

1. Go to **System administration** > **Setup** > **Entity store**.
2. In the list pane, find and select the record named *Purchase cube*.
3. On the Action Pane, select **Edit**.
4. Select **Refresh** to start the refresh process. This process takes about 15 minutes.
5. To set up a scheduled refresh (recommended), set **Automatic refresh enabled** to *Yes*, and then set the **Recurrence** field to the desired refresh interval.

Last updated on 07/01/2026

Actionable workspace for discovering and handling supplier risks

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Supply risk assessment** workspace provides a direct view of the top key insights that are related to supplier performance and related risks. It also provides embedded reports that you can use for detailed performance and risk analysis.

Open the Supply risk assessment workspace

To open the workspace, go to **Procurement and sourcing** > **Workspaces** > **Supply risk assessment**.

The workspace provides three tabs for direct actionable navigation and embedded analytic reporting that uses Microsoft Power BI:

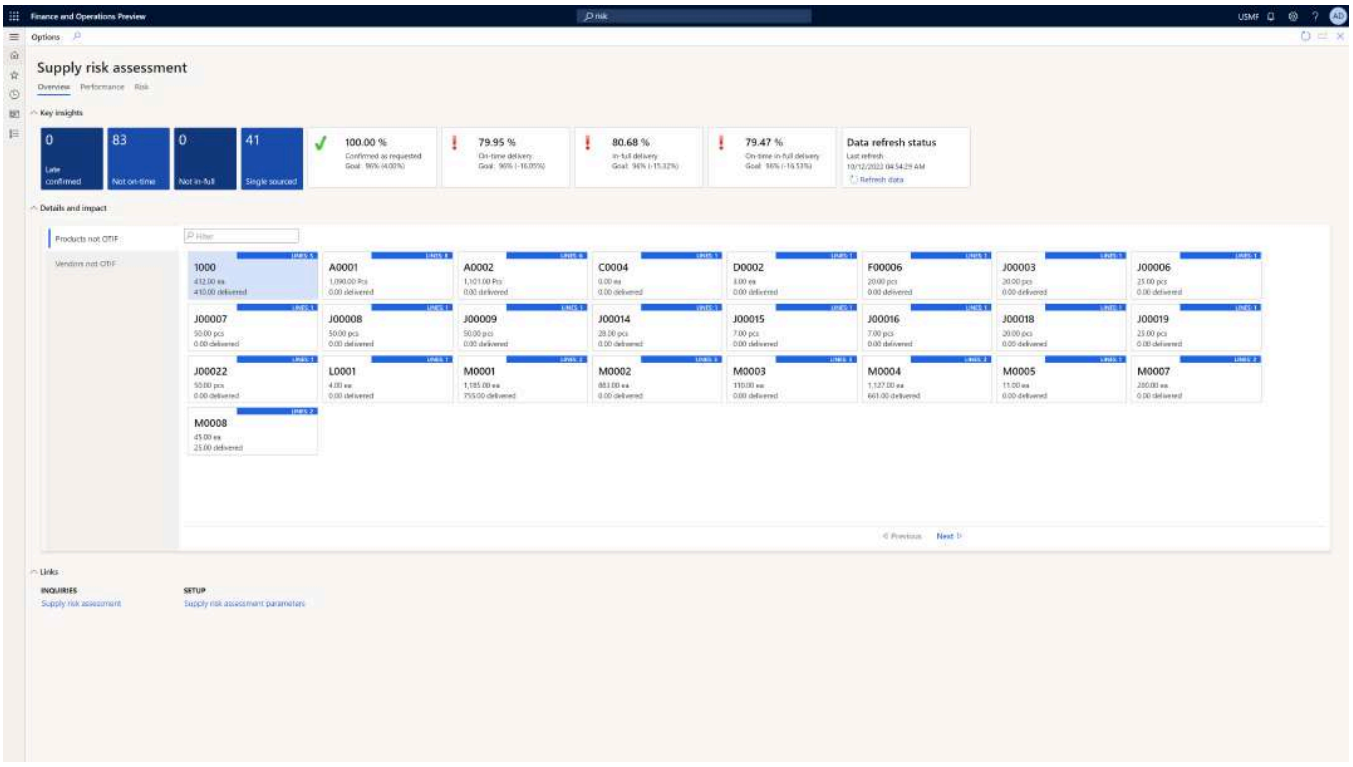
- Overview
- Performance
- Risk

Each tab is described in detail later in this article.

> **Note**
>
> If the workspace is blank or doesn't show the latest data that you expect to see, you might have to initialize or refresh the data. For instructions, see Configure supply risk assessment.

Overview tab

The **Overview** tab shows metrics and actionable content that let you navigate to relevant views and records in Dynamics 365 Supply Chain Management, such as planned orders, products, and vendors. The data on the **Overview** tab is filtered by the currently selected legal entity.

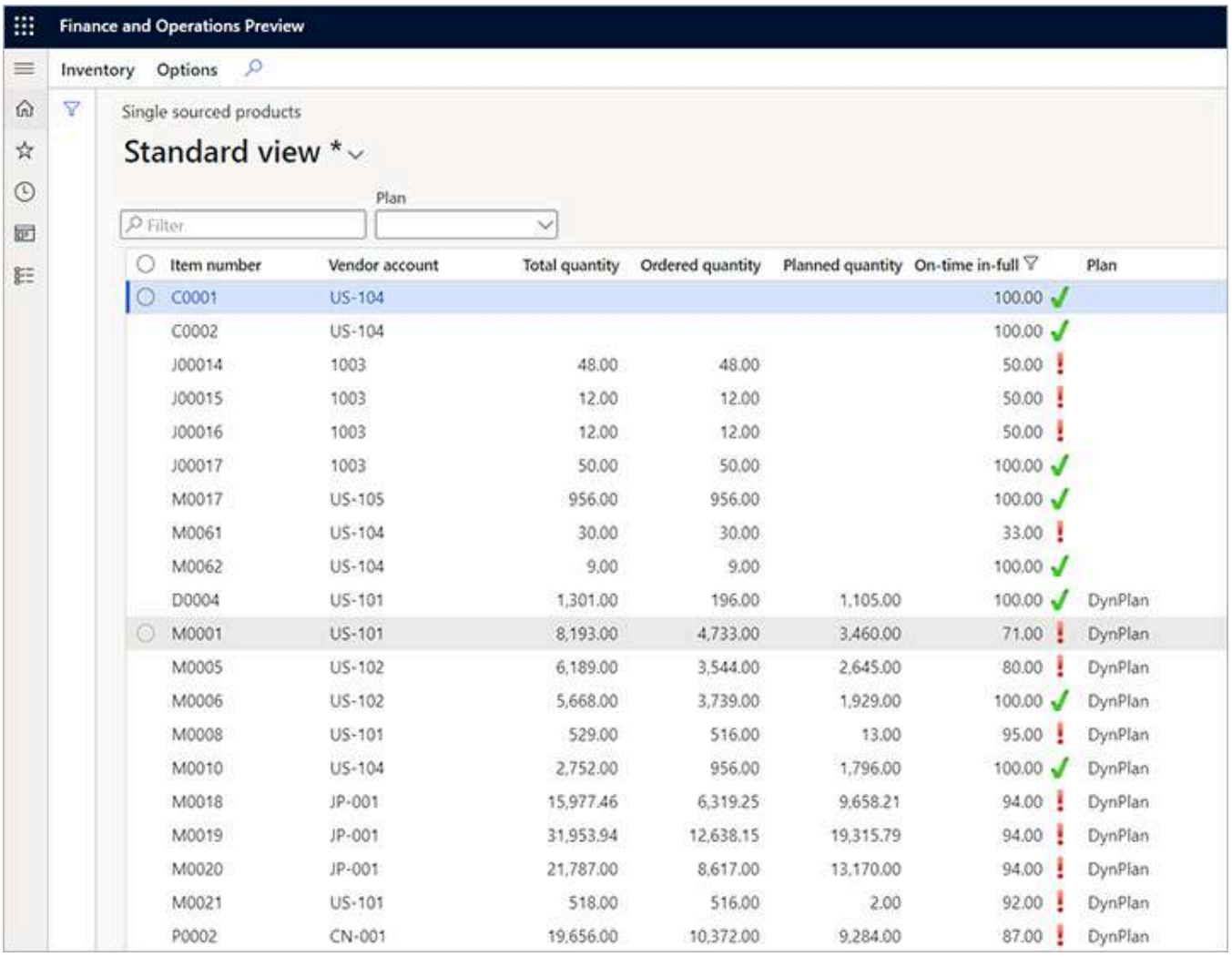


Key insights FastTab

The **Key insights** FastTab of the **Overview** tab shows the following tiles. Each tile reports a metric that is calculated for the scope of the workspace.

- **Late confirmed** – The number of purchase order lines where vendors returned a confirmed receipt date (CRD) that was later than your requested receipt date (RRD).
- **Not on time** – The number of purchase order lines that weren't delivered on time (OT).
- **Not in full** – The number of purchase order lines that weren't delivered in full (IF).
- **Single sourced** – The number of planned order items that only one vendor is assigned to. These items might indicate increased risk.

Each tile summarizes a details view, which you can open by selecting the tile. The following illustration shows the details view that is available from the **Single sourced** tile. It shows on-time in-full (OTIF) ratings and highlights whether your configured threshold was met for each item.



Item number	Vendor account	Total quantity	Ordered quantity	Planned quantity	On-time in-full	Plan
C0001	US-104				100.00	
C0002	US-104				100.00	
J00014	1003	48.00	48.00		50.00	
J00015	1003	12.00	12.00		50.00	
J00016	1003	12.00	12.00		50.00	
J00017	1003	50.00	50.00		100.00	
M0017	US-105	956.00	956.00		100.00	
M0061	US-104	30.00	30.00		33.00	
M0062	US-104	9.00	9.00		100.00	
D0004	US-101	1,301.00	196.00	1,105.00	100.00	DynPlan
M0001	US-101	8,193.00	4,733.00	3,460.00	71.00	DynPlan
M0005	US-102	6,189.00	3,544.00	2,645.00	80.00	DynPlan
M0006	US-102	5,668.00	3,739.00	1,929.00	100.00	DynPlan
M0008	US-101	529.00	516.00	13.00	95.00	DynPlan
M0010	US-104	2,752.00	956.00	1,796.00	100.00	DynPlan
M0018	JP-001	15,977.46	6,319.25	9,658.21	94.00	DynPlan
M0019	JP-001	31,953.94	12,638.15	19,315.79	94.00	DynPlan
M0020	JP-001	21,787.00	8,617.00	13,170.00	94.00	DynPlan
M0021	US-101	518.00	516.00	2.00	92.00	DynPlan
P0002	CN-001	19,656.00	10,372.00	9,284.00	87.00	DynPlan

Details and impact FastTab

The **Details and impact** FastTab has two tabs. The **Products not OTIF** tab shows a tile for each product that didn't meet the OTIF rate expectations, based on past purchase orders, whereas the **Vendors not OTIF** tab shows a tile for each vendor that didn't meet the OTIF rate expectations.

Each tile shows the number of items that were ordered and the number of items that were delivered in the relevant unit of measure. The header of each tile shows the number of relevant order lines for this calculation. The following illustration shows an example of tiles on the **Vendors not OTIF** tab.

Select a tile to view the list of order lines that were counted by that tile.

Links FastTab

The **Links** FastTab provides convenient links to the **Supply risk assessment parameters** page (see Configure supply risk assessment) and to the integrated supply risk assessment report (see Power BI reports for risks analysis and performance ranking).

Performance and Risk tabs

The **Performance** and **Risk** tabs provide in-depth analytics reports of your vendor and supply performance, and the calculated risks for your planned supply orders. For more information about how to work with these reports, see Power BI reports for risks analysis and performance ranking.

Last updated on 07/10/2025

Power BI reports for risks analysis and performance ranking

08/13/2025

Microsoft Dynamics 365 Supply Chain Management provides an integrated risk and performance report, and also two reports that are embedded into the **Supply risk assessment** workspace (supplier performance analysis and risk assessment analysis). All three reports use Power BI and are based on the same metrics as the workspace. Each report offers rich filtering and data slicing options to help you identify the most pressing supplier performance issues and potential risks for planned supply.

The tiles for metrics and key performance indicators (KPIs) at the top of each report page show relevant key figures that summarize the performance in each currently selected filter scope (including the total number of purchase orders and order lines). Some risk assessment reports show information about the planned order volume. Typically, you'll also find performance rates for the percentage of order lines that have delivery data on or before the requested date, based on order line receipts and the rates of order lines that were delivered on time (OT), in full (IF), and on time and in full (OTIF).

The metrics and KPI tiles are updated when you filter or select different vendors, products, regions, or delivery methods. This changing information can help you identify correlations with different factors that affect supply performance.

Integrated risk and performance report

The integrated risk and performance report shows information only about planned orders and is therefore most relevant for risk assessment. It combines elements of the two embedded reports. For information about how to use and interpret the various views of this report, see the later sections of this article.

You can open the integrated risk and performance report in two ways:

- Go to **Procurement and sourcing** > **Inquiries and reports** > **Supply risk assessment** > **Supply risk assessment**.
- Go to **Procurement and sourcing** > **Workspaces** > **Supply risk assessment**, and then, on the **Links** FastTab, select the **Supply risk assessment** link.

Embedded supplier performance analysis

The workspace-embedded supplier performance report shows a holistic performance assessment, not only for planned orders but also for all products that have previously been purchased.

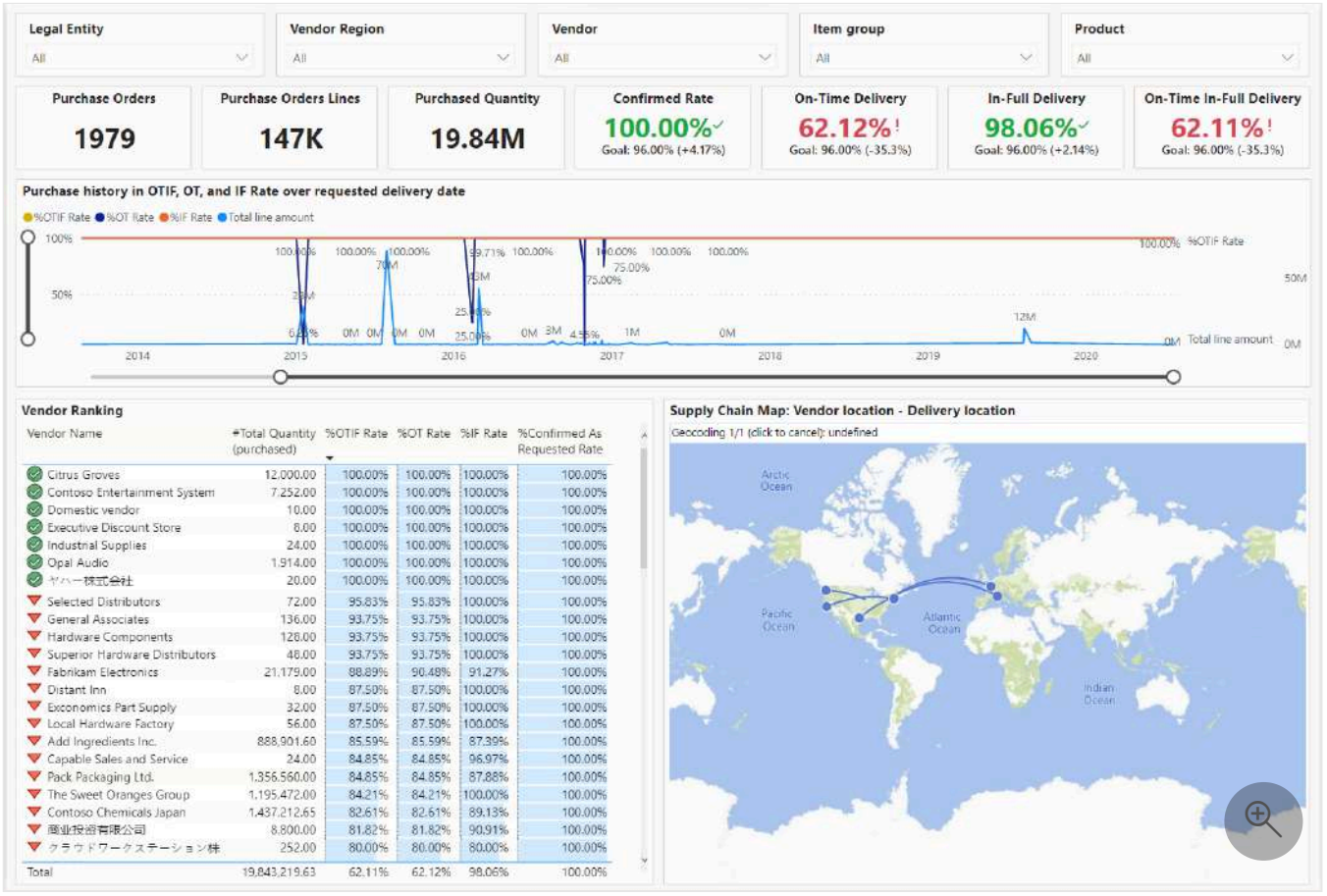
To open the embedded supplier performance analysis report, go to **Procurement and sourcing** > **Workspaces** > **Supply risk assessment**, and then select the **Performance** tab.

All supply performance analysis reports provide a header that includes slicers and filters. To limit your analysis, select products, and then use the slicers and filters to select the relevant legal entities (companies), regions, vendors, and item groups. The filters are carried through the different report pages. Therefore, you don't have to repeatedly reapply the filter settings.

Vendor and product ranking report

Use the two similar reports for product and vendor ranking to study the historic order performance in terms of OTIF deliveries. You can find OTIF trends and outliers in the timeline of deliveries, and see them in relation to the total line amounts.

You can zoom in on the timeline and select individual data points to see the related vendors or products. Alternatively, you can select one or more vendors, based on their ranking, and see the appropriate performance timeline.



Multi perspective OTIF ranking

On this report, the worst OTIF rankings are calculated from various perspectives and shown side-by-side for products, vendors, delivery methods, and sites. You can use individual or combined dimensions to identify bottlenecks that might cause a drop in supply performance.

Embedded supply risk assessment analysis

The workspace-embedded supply risk assessment report shows a holistic performance assessment, not only for planned orders but also for all products that have previously been purchased.

To open the embedded supplier performance analysis report, go to **Procurement and sourcing** > **Workspaces** > **Supply risk assessment**, and then select the **Risk** tab.

The risk assessment report turns the performance observations into a risk calculation for planned orders, based on purchase order receipts.

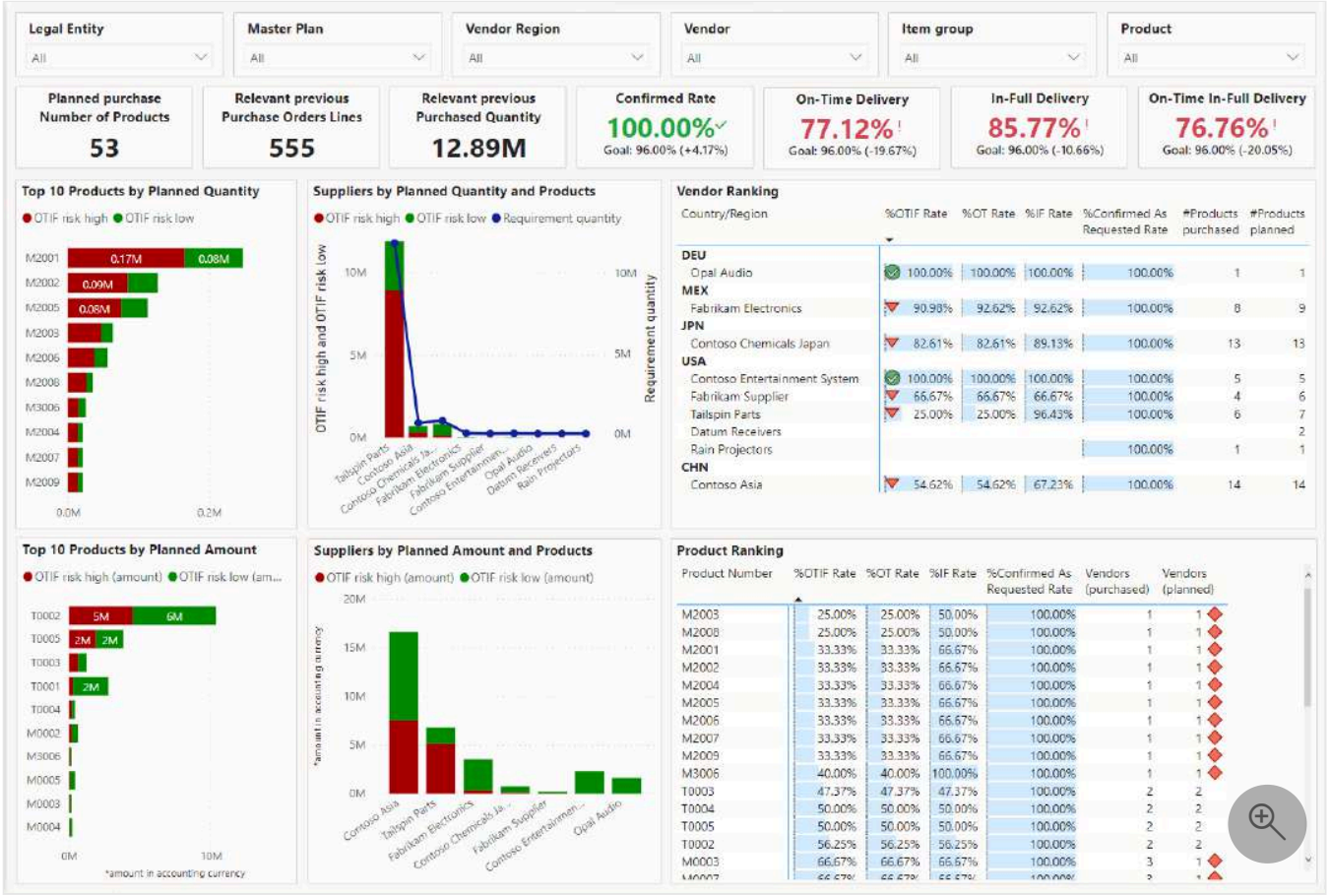
All supply risk analysis reports provide a slicer and filter header that includes a master plan selector. You can use this filter to select a single master plan. In this way, you can help prevent the unwanted double accounting of planned orders from multiple plans.

Risk assessment overview report

This report helps you analyze the calculated risks for your planned purchases. After you filter to the desired scope, you'll see the top-10 products ranked by risk in their aggregated quantity and amount. The high-risk classification is derived from the planned product quantity and amount multiplied by the observed OTIF rate of past orders. The low OTIF risk is the remaining total quantity or amount.

> **Note**
>
> The risk assessment is most statically significant when most of the order lines in the observed time frame are for products where few fundamental changes are expected to surrounding factors.

After you zoom in on a group of planned product supplies, you can study additional performance statistics for the products and vendors that are assigned to the planned orders.

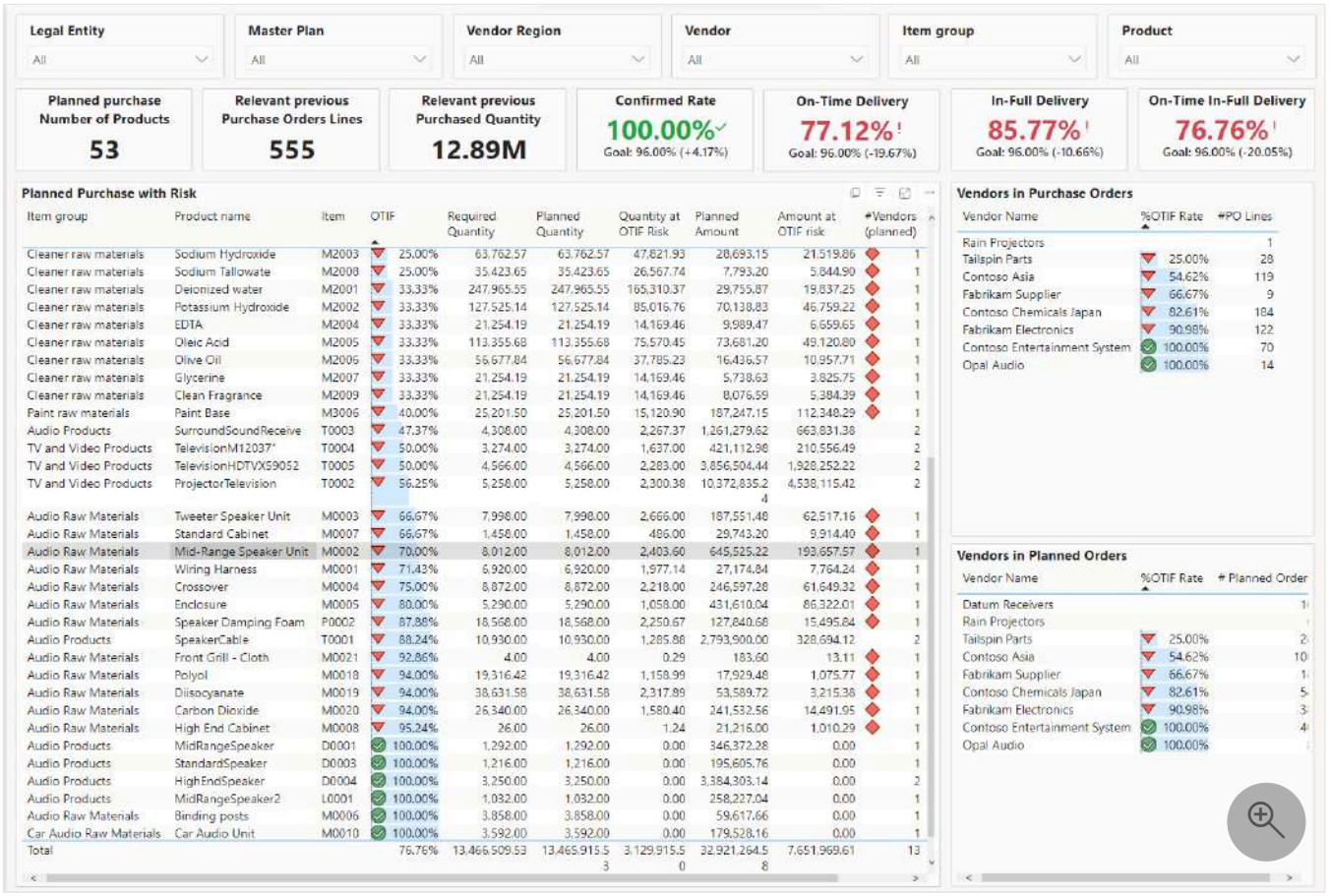


Detailed sourcing risk breakdown

The detailed planned purchase risk breakdown lists show all at-risk products within their item groups, together with the planned quantity and amount, and the corresponding risk amounts.

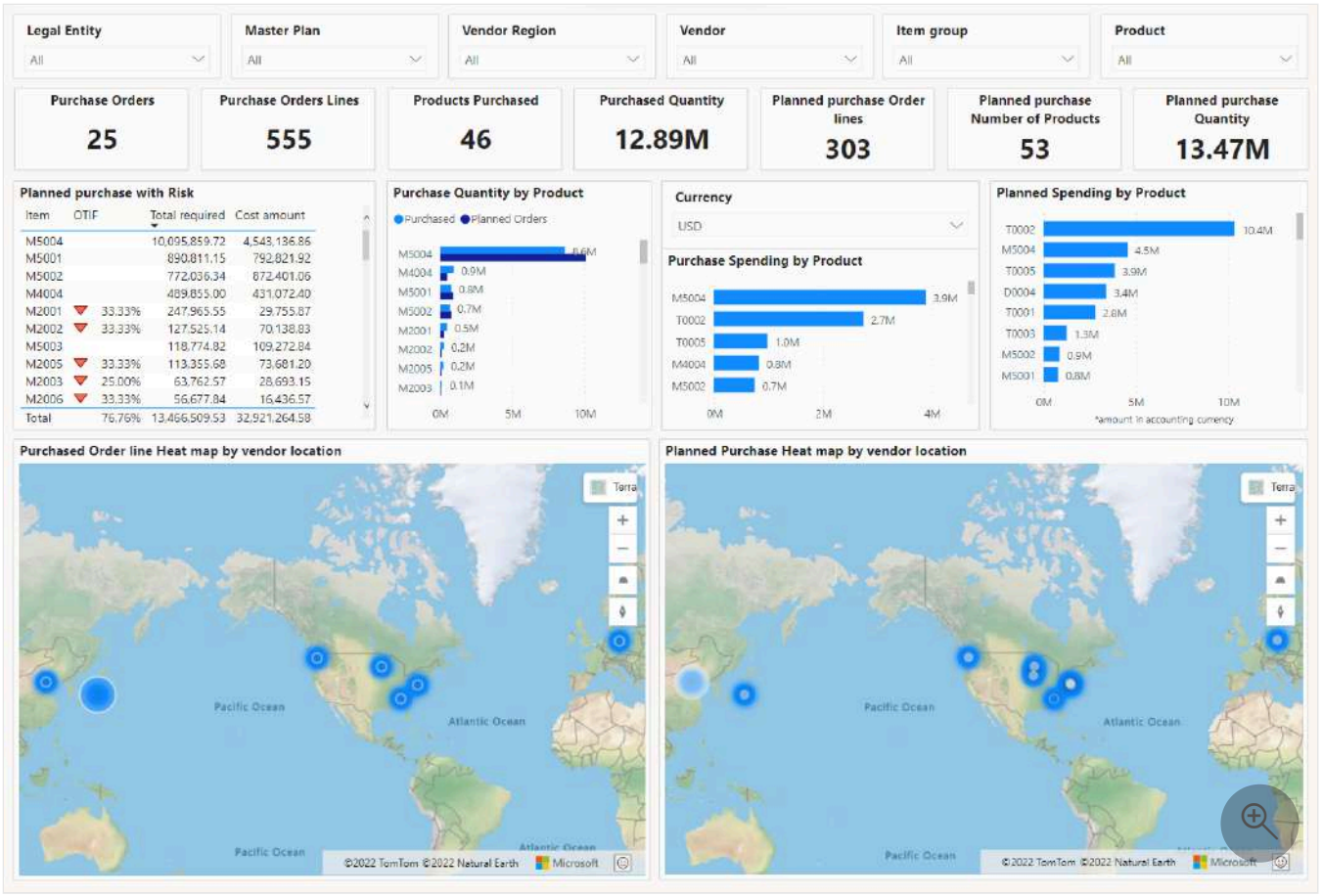
This view can help you study vendors that were used for past orders that led to bad performance. Consider using other or additional vendors to help reduce the risk for future

planned purchases.



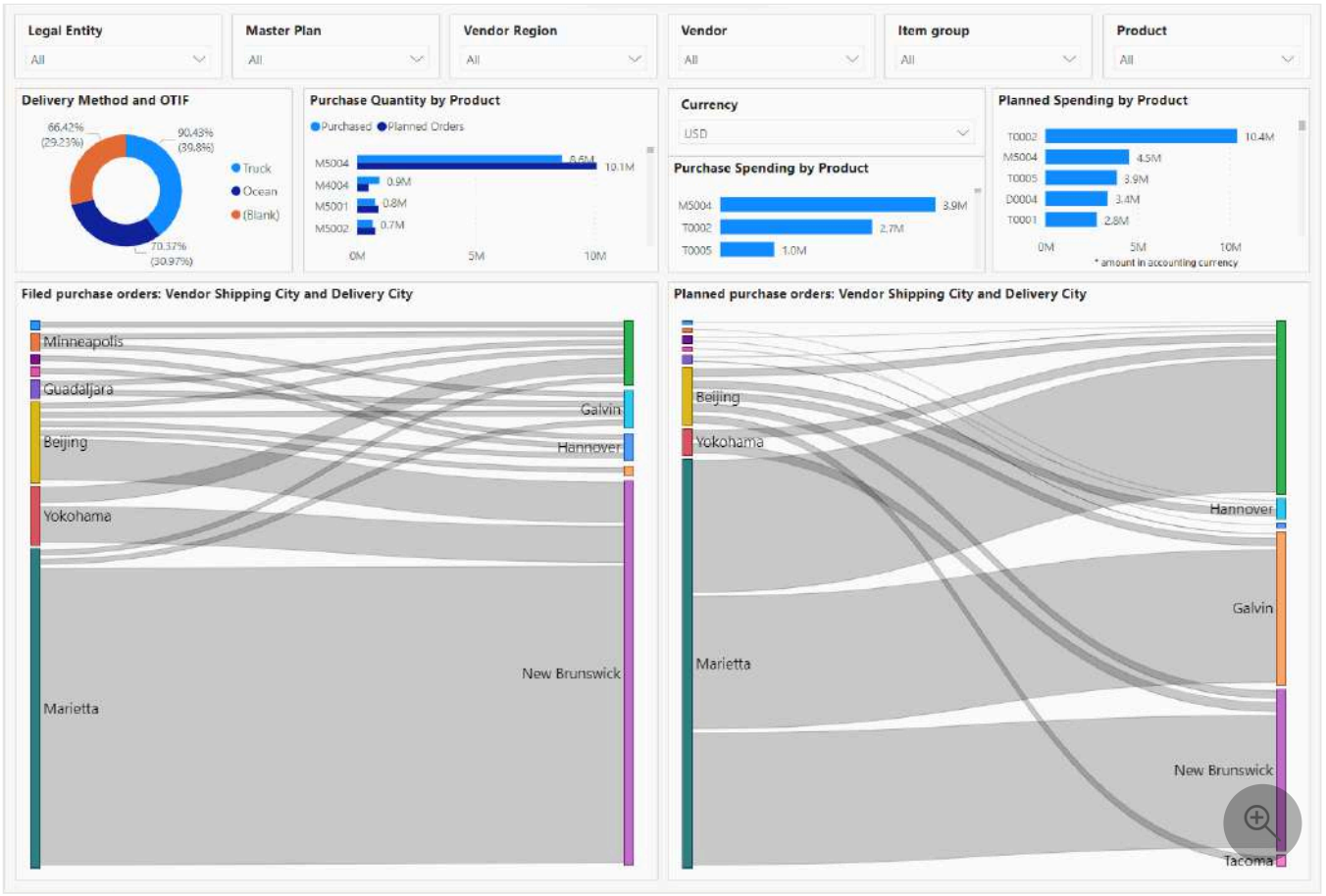
Compare vendor locations between past order and planned orders

To better compare the overall supply-originating location (vendor location), you can study the side-by-side geographic maps. These maps are heat maps that highlight where supply was coming from for purchase orders versus planned supply orders.



Supply flow comparison from the past against planned supply

You can dive deeper into the supply from source to delivery location in comparison to past purchase and planned supply. You can filter by items that have been identified as at risk and by delivery methods. In this way, you can identify potential risks that are imposed by the route and method of delivery.


Legal Entity
All
Master Plan
All
Vendor Region
All
Vendor
All
Item group
All
Product
All
Delivery Method and OTIF
66.42% (29.23%)
90.43% (39.8%)
70.37% (30.97%)
Truck
Ocean
(Blank)
Purchase Quantity by Product
Purchased
Planned Orders
M5004
M4004
M5001
M5002
0.9M
0.8M
0.7M
10.1M
0M
5M
10M
Currency
USD
Purchase Spending by Product
M5004
T0002
T0005
3.9M
2.7M
1.0M
Planned Spending by Product
T0002
M5004
T0005
D0004
T0001
10.4M
4.5M
3.9M
3.4M
2.8M
0M
5M
10M
* amount in accounting currency
Filed purchase orders: Vendor Shipping City and Delivery City
Minneapolis
Guadaljara
Beijing
Yokohama
Marietta
Galvin
Hannover
New Brunswick
Planned purchase orders: Vendor Shipping City and Delivery City
Beijing
Yokohama
Marietta
Hannover
Galvin
New Brunswick
Tacoma

Product receipt against purchase orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the various options for registering products as received.

Product receipt is the process of recording that products you ordered are received so that the purchase order (PO) lines can be processed for invoicing. In some cases, products go through preregistration, where you record additional information from the supplier before receiving the products. When products arrive, you first mark them as *Registered*. You might then send the products through other processes, such as quality management, before you finally mark them as *Received*.

Preregistration (ASN)

Suppliers might share information about products that they'll ship. In this case, you can preregister the products to record this information before receiving the products. By preregistering products, you reduce the amount of work that is required during item registration and receipt. Suppliers can provide product information electronically through an Advance Shipment Notice (ASN) that the system automatically records. The information in the ASN includes the quantity of products that will be shipped and the date when they'll be shipped. The ASN might also include information such as batch or serial numbers. The **Transportation management** module registers the ASN.

Registration

You often register product receipt at the inbound docks in a warehouse. Workers register receipts using a hand-held device or through arrival journals. Alternatively, you can manually register product receipt by using the **Registration** action on the **Purchase order** page. In both cases, you mark the products as *Registered*. You don't yet mark the products as *Received*.

Products that you receive in a warehouse might go through quality inspection before workers put them away into inventory. Either quality orders or quarantine orders can be used to perform quality inspection. If you use quality orders, you can configure the process to temporarily block products through a reservation while they're inspected. If you use quarantine

orders, workers move products to another warehouse for inspection. This warehouse is known as the quarantine warehouse. In both quality inspection processes, workers might scrap some of the goods, either because they don't conform to the quality expectations or because the quality inspection involves destructive testing of a sample of the product.

Product receipt

Use the **Product receipt** action on the **Purchase orders** page to mark products as *Received* on the purchase order. The **Posting product receipt** page has various options for the quantity that you account as received. For example, you can set the **Quantity** field to *Ordered quantity* or *Receive now quantity*. Alternatively, if a warehouse arrival process is used, set this field to *Registered quantity*. You can modify the quantities on each order line that you mark as *Received*, to account for any discrepancies, such as under-delivery and over-delivery. During product receipt, specify a product receipt identifier, which is typically a reference to the packing slip from the supplier. This identifier is required for accounting, because it enables checks or audits of supplier packing slips against what is received, and the accounted inventory or expense.

Create purchase orders for products that aren't intended as inventory but are considered an expense. This category includes order lines where the products are marked as *Not stocked* by their inventory model group, and also lines that use procurement categories. In this case, the items might not go through arrival registration and receipt in the warehouse. Instead, use the **Product receipt** action to record the receipt directly on the purchase order, and the receipt is based on the ordered quantity, not a registered quantity.

Use the **New fixed asset** option for purchase order lines to indicate that the purchase is a fixed asset instead of inventory. In this case, the fixed asset determination rules that you configure determine whether the purchase of the product or category exceeds specific thresholds, and must therefore be accounted for as an asset and go through fixed asset management. You can also make purchases toward an existing fixed asset. In this case, adjust the amount as appropriate.

You can select multiple orders and process receipt on all those orders together. This approach isn't used often, but you might want to use it if a supplier consolidates shipments for you into a single load. During product receipt on the purchase, there's a function for doing summary updates. Summary updates let you post a single packing slip from the supplier for more than one purchase order.

You might create purchase orders from a sales order where the **Direct delivery** option was selected. When you use direct delivery, the products never arrive in your warehouse but are shipped directly from the supplier to the customer. In this case, you usually record the receipt directly on the purchase order. The receipt can be done automatically, such as through electronic data interchange (EDI) integration with the supplier. Alternatively, if the purchase

order is an intercompany order, Supply Chain Management automates the receipt on the intercompany sales order when shipment occurs. When you use direct delivery, you still account products as inventory, even though they don't physically arrive at the warehouse. Therefore, when you register product receipt on the purchase order, the sales order is automatically updated with a packing slip, so that the overall change to inventory is 0 (zero). In direct delivery scenarios, you shouldn't require preregistration. If you're using warehouses that are enabled for warehouse management, you can get around the requirement for license plate registration by specifying a virtual warehouse instead. You specify this warehouse in the **Direct delivery warehouse** field on the product.

After you process the product receipt on the purchase order, the system sets the purchase order status to *Received* to indicate that the invoice can be processed for the order. You can review details about products that are already received by using the **Product receipt journals** page.

You can access this page from the **Receipt** action group on the **Purchase order** page. The information in the journals includes details about the quantities, dates, and dimensions.

Auto posting product receipts

To automatically post product receipts for multiple purchase orders, follow these steps:

1. Go to **Procurement and sourcing** > **Purchase orders** > **Receiving products** > **Post Product receipt**.
2. In the **Posting product receipt** dialog, on the **Settings** FastTab toolbar, select **Select**.
3. In the **Purchase update** dialog, use the **Range** tab to specify selection criteria for finding the purchase orders you want to post.
4. Select **OK** to return to the **Posting product receipt** dialog. The purchase orders that match the criteria you specified are displayed on the **Overview** FastTab.
5. On the **Overview** FastTab, enter the product receipt identifier in the **Product receipt** column for each purchase order in the grid. This information enables the system to post the product receipt.

   You must specify a product receipt identifier when each product is received. It's typically a reference to the packing slip from the supplier. This identifier is required for accounting because it enables supplier packing slips to be checked against what was received and against the accounted inventory or expense.
6. Select **Batch** to open the **Batch processing** dialog, where you can set up the batch job that posts the product receipt.

7. Select **OK** to return to the **Posting product receipt** dialog.
8. Select **OK** to close the **Posting product receipt** dialog.

After the system finishes processing product receipts for all lines in the purchase order, it sets the purchase order status to *Received* to indicate that the invoice can now be processed for the order.

To correct or cancel a product receipt, follow these steps:

1. Go to **Procurement and sourcing** > **Purchase orders** > **Receiving Products** > **Product receipt**. The **Product receipt journal** page opens.
2. On the **Overview** tab, select the product receipt that you want to correct or cancel.
3. On the **Overview** tab toolbar, select **Correct** to make corrections or **Cancel** to cancel the product receipt.

> **Note**
>
> - When correcting a product receipt, you can only reduce the received quantity. To raise the quantity, you must post a new product receipt journal.
> - Late selection isn't supported for product receipt posting. A product receipt number is required when you post a packing slip, but when you use late selection, this number isn't available because no parameter table exists and the system doesn't create an interaction during batch recurrence processing. Therefore, late selection is disabled for packing slip posting to ensure successful batch processing.

Auto post product receipts when using WMS

Auto posting works differently if you're using warehouse management processes (WMS). Learn more in Warehouse handling of inbound loads for purchase and inbound shipment orders.

Related information

- Warehouse handling of inbound loads for purchase and inbound shipment orders
- Purchase order overview
- Create purchase orders
- Approve and confirm purchase orders
- Overview of vendor invoices

Last updated on 01/27/2026

Vendor invoices overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides general information about vendor invoices. Vendor invoices are requests for payment for products and services. Vendor invoices might represent a bill for ongoing services, or they can be based on purchase orders for specific items and services.

Vendor invoices

A vendor invoice from a purchase order is produced when products or services are received according to a purchase order placed with a vendor. The vendor invoice contains a header, and one or more lines for items or services. A vendor invoice completes the cycle from purchase order to product receipt to vendor invoice.

Although some vendor invoices connect to a purchase order, vendor invoices can also contain lines that don't correspond to purchase order lines. You can also create vendor invoices that aren't associated with any purchase order. These vendor invoices might represent ongoing services, such as a utility bill. You don't have to reference a purchase order when you add an ongoing service.

There are several ways to enter a vendor invoice:

- Use the vendor invoice register to quickly enter invoices that don't reference a purchase order, so that you can accrue the expense. By using the vendor invoice approval journal, you can select those invoices and post them to the vendor balance to reverse the accrual.
- Use the vendor invoice journal to quickly enter invoices that don't reference a purchase order, in a single step.
- Together with the vendor invoice pool, use the vendor invoice register to quickly enter invoices to accrue the expense. You can open the associated purchase orders later to post the invoice against the expense account.

- Use the **Open vendor invoices** and **Pending vendor invoices** pages to create vendor invoices from confirmed purchase orders.
- The **Invoice capture solution** automatically creates vendor invoices from digital invoice images. For more information, see Invoice capture.

The following discussion provides more information about how to use the **Open vendor invoices** or **Pending vendor invoices** page to create a vendor invoice from a purchase order.

Understanding invoice line quantities

When you open a vendor invoice from a related purchase order, the system creates invoice lines from the purchase order. By default, the system takes the quantities from the product receipt. However, you can use any of the following default behaviors:

- **Receive now quantity** – Use this option for partial shipments. The default value in the **Quantity** field is set to the quantity specified in the **Receive now** field on the purchase order.
- **Ordered quantity** – Use this option for complete shipments. The default value in the **Quantity** field is set to the quantity specified in the **Ordered** field on the purchase order.
- **Registered quantity** – Use this option if the item requires registration, as specified on the **Item model groups** page. The default value in the **Quantity** field is the physical update quantity that has been registered.
- **Product receipt quantity** – Use this option if a product receipt has already been received for the order. The default value in the **Quantity** field is the total quantity of available product receipts.
- **Registered quantity and services** – Use this option if quantities are registered in arrival journals for stocked items or items that aren't stocked. This option also includes services, regardless of whether they're registered.

If your legal entity uses invoice matching, you can view the results of the quantity matching in the **Product receipt quantity match** column. You can also use the **Matching details** button on the **Review** tab of the Action Pane to view the results of the quantity matching.

Adding a line that wasn't on the purchase order

You can add a line that wasn't on the purchase order to the vendor invoice. You must select an item number or procurement category. You can then add quantities, prices, and amounts to the line. The line is included only in matching policies for invoice totals.

Submitting a vendor invoice for review

Your organization might use workflows to manage the review process for vendor invoices. Workflow review can be required for the invoice header, the invoice line, or both. The workflow controls apply to the header or the line, depending on where the focus is when you select the control. Instead of the **Post** button, a **Submit** button sends the vendor invoice through the review process.

Preventing invoice submission to workflow

To prevent an invoice from being submitted to a workflow, use the following options.

- **Invoice total and the registered total aren't equal.** The user who submitted the invoice receives an alert that the totals aren't equal. This alert gives the user an opportunity to correct the balances before they resubmit the invoice to the workflow system. This feature is available if the **Prohibit submission to workflow when the invoice total and registered invoice total aren't equal** parameter on the **Feature management** page and the **Workflow option when invoice total and registered total aren't equal** parameter on the **Accounts payable parameters** page are turned on.
- **Invoice contains unallocated charges.** The user who submitted the invoice receives an alert that the invoice has unallocated charges. In this way, the user can correct the invoice before they resubmit it to the workflow system. This feature is available if the **Prohibit submission to workflow when there are unallocated charges on a vendor invoice** parameter on the **Feature management** page and the **Workflow option when unallocated charges exist** parameter on the **Accounts payable parameters** page are turned on.
- **Invoice contains the same invoice number as another posted invoice.** The user who submitted the invoice receives an alert that an invoice was found that has a duplicate number. The user can correct the duplicate number before they resubmit the invoice to the workflow system. The alert is shown if the **Check the invoice number used** parameter in Accounts payable is set to **Reject duplicate**. This feature is available if the **Prohibit submission to workflow when the invoice number already exists on a posted invoice, and your system isn't set up to accept duplicate invoice numbers** parameter on the **Feature management** page is turned on.
- **Invoice contains a line where the invoice quantity is less than the matched product receipt quantity.** The user who submits the invoice or tries to post it receives a message that the quantities aren't equal. This message gives the user an opportunity to correct the values before they resubmit the invoice to the workflow system. This feature is available if the **Block posting and submission of vendor invoices to workflow** parameter on the

Feature management page and the **Block posting and submission to workflow** parameter on the **Accounts payable parameters** page are turned on.

Matching vendor invoices to product receipts

Enter and save information for vendor invoices, and match invoice lines to product receipt lines. You can also match partial quantities for a line.

Create a vendor invoice based on the product receipt line items that you received through the current date, even if you didn't receive all the items for a particular purchase order. For example, use this option if a vendor sends one invoice per month to cover all the deliveries that it shipped during that month. Each product receipt represents a partial or complete delivery of the items on the purchase order.

When an invoice is in workflow, the approver can update invoice quantities so that they match the value in the **Product-receipt-quantity-to-match** field. To do so, select the **Update the invoice quantities to match product receipt quantities in workflow** feature in the **Feature management** workspace and select **Enable**. If an approver in the workflow process removes all the matches from all the product receipts from the invoice line, the invoice line is deleted. When this feature isn't enabled, invoice quantities aren't updated for invoices in workflow.

When you post the invoice, the **Invoice remainder** quantity for each item is updated with the total of the received quantities from the selected product receipts. If both the **Invoice remainder** quantity and the **Deliver remainder** quantity for all items on the purchase order are 0, the status of the purchase order changes to **Invoiced**. If the **Invoice remainder** quantity isn't 0, the status of the purchase order remains unchanged, and you can enter additional invoices for it.

This option assumes that you posted at least one product receipt for the purchase order. The vendor invoice is based on these product receipts and reflects the quantities from them. The financial information for the invoice is based on the information that you enter when you post the invoice.

For more information, see Record vendor invoice and match against received quantity.

Configure an automated task for vendor invoice workflow to post the vendor invoice using a batch job

Add an automated posting task to the Vendor invoice workflow so that invoices are processed in a batch. Posting invoices in a batch lets the workflow process continue without having to wait for

the posting to finish, which improves the overall performance of all the tasks submitted to the workflow.

To post a vendor invoice in a batch, on the **Feature management** page, turn on the **Vendor invoice batch posting** parameter. Configure vendor invoice workflows by going to **Accounts payable** > **Setup** > **Accounts payable workflows**.

You can see the **Post the vendor invoice using a batch** task in the workflow editor, regardless of whether the feature parameter, **Vendor invoice batch posting**, is enabled. When the feature parameter isn't enabled, an invoice that contains the **Post the vendor invoice using a batch task** isn't processed in workflow until the parameter is enabled. The **Post the vendor invoice using a batch** task must not be used in the same workflow as the **Post vendor invoices** automated task. Also, the **Post the vendor invoice using a batch** task should be the last element in the workflow configuration.

You can specify the number of invoices to include in the batch, and the number of hours to wait before rescheduling a batch, by going to **Accounts payable** > **Setup** > **Accounts payable parameters** > **Invoice** > **Invoice workflow**.

Working with multiple invoices

You can work with multiple invoices at the same time and post all of them at the same time. If you need to create multiple invoices, use the **Pending vendor invoices** page. If you must post and print multiple vendor invoices, use the **Invoice approval journal**. If you're using the **Invoice approval journal**, you must post at least one product receipt for the purchase order, and you must post an invoice for the purchase order in an invoice register. The financial information for the invoice comes from the invoice that you posted in the register.

Recovering vendor invoices that are in use

While a user is using a vendor invoice, other users can't edit it. However, the state of an invoice might sometimes indicate that the invoice is in use, even though no user is actively editing it. For example, the application might stop responding while the invoice is being edited, or a user might inadvertently leave the invoice open in the application.

Use the **Recover vendor invoices** page to recover or release vendor invoices that are in use for more than four hours, so that you can edit them. You can open this page from the **Periodic task** navigation or a tile on the **Vendor invoice entry** workspace. After you recover an invoice, it's available for editing on the **Vendor invoice** page.

You can access the **Recover vendor invoices** page only if you're assigned the **Recover vendor invoices in use** security duty and privilege. Additionally, the **Allow vendor invoice recovery** parameter on the **Accounts payable parameters** page must be turned on.

Default financial dimension in vendor invoice lines

For vendor invoices that are associated with a purchase order, the default financial dimension comes from the corresponding purchase order lines.

For vendor invoices that aren't associated with any purchase order, the default financial dimension merges the dimensions from the dimension link, invoice header, and item. If the same dimension is defined on the dimension link, invoice header, and item, the dimension on the invoice line follows this priority sequence: dimension link > invoice header > item

Resetting the workflow status for vendor invoices from Unrecoverable to Draft

A workflow instance that stops because of an unrecoverable error has a workflow status of **Unrecoverable**. When the status of a vendor invoice workflow is **Unrecoverable**, you can reset it to **Draft** by selecting **Recall**. You can then edit the vendor invoice. This feature is available if the **Resetting the workflow status for vendor invoices from Unrecoverable to Draft** parameter on the **Feature management** page is turned on.

Use the **Workflow history** page to reset the workflow status to **Draft**. You can open this page from **Vendor invoice** or from the **Common** > **Inquires** > **Workflow** navigation. To reset the workflow status to **Draft**, select **Recall**. You can also reset the workflow status to **Draft** by selecting the **Recall** action on the **Vendor invoice** or **Pending vendor invoices** page. After you reset the workflow status to **Draft**, you can edit the vendor invoice on the **Vendor invoice** page.

Viewing the invoice total on the Pending vendor invoices page

You can view the invoice total on the **Pending vendor invoices** page by enabling the **Display invoice total on pending vendor invoices list** parameter on the **Accounts payable parameters** page.

Vendor open transactions report

The **Vendor open transactions** report provides detailed information about the open transactions for each vendor as of the date that you specify. Use this report during the audit procedure to verify balances between vendor book transactions and ledger account transactions.

For each transaction, the report includes the following details:

- Invoice number
- Transaction date
- Voucher number
- Transaction amount in the transaction currency and accounting currency
- Credit balance in the transaction currency and accounting currency
- Debit balance in the transaction currency and accounting currency
- Subtotal amount in the accounting currency
- Payment due date

Filter the data on the report

When you generate the **Vendor open transactions** report, the following default parameters are available. Use them to filter the data that the report includes.

- **Exclude future settlement** – Select this checkbox to exclude transactions that are settled after the date that you enter in the **Open transactions per** field.
- **Open transactions per** – Enter a date to include transactions that are open as of that date. If you don't enter a date, this field is set to the maximum date. (The maximum date is the latest date that the system accepts, December 31, 2154.) By default, the next time that you run the report, this field is set to the last date that you entered in it.

Use the filters under the **Record to include** field to further limit the transaction data that the report includes.

Extend invoice number length

Starting in Dynamics 365 Finance version 10.0.40, you can enable the **Extend the length of invoice number for vendor invoice** feature to increase the invoice number from 20 characters to 50 characters in the vendor invoice and invoice journal. Before enabling the feature, create a ticket in Lifecycle services to open the **EnableEdtStringDatabaseStringLengthAttributeInComputation** flight.

Additional resources

- Set up vendor invoice policies
- Key invoice data in AP system using vendor invoice
- Key invoice data into accounts payable using an approval journal
- Key invoice data into the AP system using invoice pool
- Record a vendor invoice in the invoice journal

Last updated on 05/06/2026

Set up and maintain vendor collaboration

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The vendor collaboration interface exposes a limited set of information about purchase orders, invoices, and consignment stock to external vendor users. From this interface, a vendor can also reply to requests for quotation (RFQs), and view and edit basic company information.

This article explains how to set up vendor collaboration in Dynamics 365 Supply Chain Management. It also explains how to set up a workflow to provision new vendor collaboration users, and how to manage the security roles for those users.

Set up vendor collaboration security roles

A procurement professional or a vendor that has enough permissions can request that a contact person be provisioned as a user by enabling **Provision vendor user** on the contact person record. During the provisioning process, user permissions are selected for the new external user, and the new vendor user request is submitted. It's important that you correctly set up the user permissions that are available for selection in the vendor user request. Otherwise, vendors might be granted access to information that they shouldn't have access to in Supply Chain Management.

Set up the security roles that are available for selection when a new user request is used for a contact person

1. Select **System administration** > **Security** > **External roles**.
2. Select **New**, and then select a security role and the *Vendor* party role.

You might want to add the *Vendor admin (external)* and *Vendor (external)* roles that are provided in Supply Chain Management. Alternatively, you can use security roles that your company has created.

You should make the *Vendor admin (external)* role available only if vendors should be able to create new contacts, submit vendor collaboration user requests for new users and changes to user information, and handle those requests via a workflow.

If you plan to manually set up vendor contacts and users, you can make just the *Vendor (external) role* available. This role will then be the only role that can be requested through a vendor user request.

> **Note**
>
> The *SystemUser* role is automatically granted when you manually create a new user account. Therefore, you must remove that role and assign the *SystemExternalUser* role. If new user accounts are created via the workflow that is initiated by a vendor user request to provision a new user, one or more of the roles that you've set up for vendor collaboration and the *SystemExternalUser* role will be assigned.

Vendor admin (external) security role

The *Vendor admin (external)* role can be used for external vendors that maintain vendor contact information and make requests to provision new vendor collaboration users. External users who have this security role can perform the following tasks:

- View and modify contact person information, such as the person's title, email address, and telephone number.
- Add a new or existing contact person to the vendor accounts that they're a contact for.
- Delete any contact person that they have created.
- Activate or deactivate the association between a contact person and a vendor account. After the association between a contact person and a vendor account is deactivated, the contact person can't be referred to on new purchase orders or other documents.
- Deny or allow a contact person's access to documents on the vendor collaboration interface that are specific to the vendor account. After the association between a contact person and a vendor account is deactivated, access to documents that are specific to the vendor account is always denied.
- Request a new user account for a contact person by using the **Provision user** action.
- Request that a contact person's user account be deactivated.
- Request that a contact person's user account be modified to add or remove security roles.
- View RFQs.

Vendor (external) security role

The *Vendor (external) role* can be used for external vendors that will work with purchase orders. External users who have this security role can perform the following tasks:

- Respond to and view information about purchase orders.
- Maintain vendor collaboration invoices.

- View consignment inventory.
- View and respond to RFQs.
- View vendor information.

Set up security roles that are used when prospective vendors are onboarded

To onboard vendors that are initiated via a prospective vendor registration request, you must set up an external security role. This role will be assigned to new users during the provisioning process that is controlled by the workflow of the *User request workflow (platform)* type. Learn more in the Set up workflows to process vendor collaboration user requests section later in this article.

For information about how to onboard prospective vendors, see Onboard vendors.

Set up a security role that is used when a new prospective vendor user request is submitted

1. Select **System administration** > **Security** > **External roles**.
2. Select **New**, and then select a security role and the *Prospective vendor* party role.

You should add the *Vendor prospect (external)* role that is provided in Supply Chain Management.

The security role will grant access only to the new vendor registration wizard.

Set up workflows to process vendor collaboration user requests

To help guarantee that all the relevant tasks are completed, and that the appropriate approvals are given, you must set up workflows to handle vendor collaboration user requests.

Vendor collaboration user requests are submitted either by external vendors that have the *Vendor admin (external)* security role or similar permissions, or by procurement professionals in your company. They can also be generated from prospective vendor registration requests during the vendor onboarding process.

There are three types of requests:

- Requests to provision a new user
- Requests to deactivate an existing user

- Requests to modify the security roles of an existing user

For more information about vendor collaboration user requests, see Manage vendor collaboration users.

You must create two or more workflows to process all three types of vendor collaboration user requests. New workflows are created on the **User workflows** page.

Example of a workflow for provisioning new users and modifying security roles

To handle vendor user requests to create new users and modify security roles, you can put a branching condition at the beginning of the workflow. In this way, a different branch of the workflow is used, depending on whether the request is to create a new user or modify an existing user.

To set up this branching, create a new workflow of the *User Request Workflow (Platform)* type. The branches of this workflow might contain the following elements.

Branch to provision new users

1. Assign an approval task to the person who is responsible for approving that new users should be granted access to vendor collaboration information.

2. Assign a task to the person who is responsible for requesting new Microsoft Microsoft Entra user accounts in Azure portal. Use the predefined *Send Azure B2B user invitation* task for this step. B2B users can be automatically exported to Microsoft Entra ID. Use the predefined *Provision Microsoft Entra B2B user*. Learn more in Export B2B users to Microsoft Entra ID.

3. Assign an approval task to the person who uploads to Azure. If an account isn't successfully created, this person rejects the task and ends the workflow. This approval task can be skipped if you've included the step that automatically exports new user accounts to Azure via the B2B application programming interface (API).

4. Add an automated task that provisions a new user. Use the predefined *Automated provision user* task for this step.

5. Add a task that notifies the new user. You might want to send the new user a welcome email that includes a URL for Supply Chain Management. This email can use a template that you create on the **Email messages** page and then select on the **User workflow parameters** page. The template can include the *%portalURL%* tag. When the welcome

email is generated, this tag will be replaced by the URL of the Supply Chain Management tenant.

> **Note**
>
> This workflow can be used in multiple scenarios that involve user onboarding. For example, it can be used when prospective vendors or contact persons require a vendor collaboration account. Therefore, you should phrase the email as a general statement that can be used for multiple purposes.

Branch to modify security roles

1. Assign an approval task to the person who is responsible for approving changes to security roles.
2. Add an automated task that adds or removes the relevant security roles. Use the *Automated provision user* task for this step.

Example of a workflow for inactivating a user

Create a workflow of the *Inactivate user request workflow platform* type, and then add the following tasks.

1. Assign an approval task to the person who is responsible for accepting requests to deactivate users. You can add conditions to automate this approval step.
2. Add an automated task that deactivates the user. Use the *Automated user inactivation* task for this step.
3. Add any clean-up tasks that are required. For example, you can add a task that removes the account from your directory in Azure portal.

Enable vendor collaboration for a specific vendor

Before you create a user account for someone who will use vendor collaboration, you must set up the vendor so that it can use vendor collaboration. For details about how to do this, see Vendor collaboration with external vendors.

Troubleshoot the provisioning of new vendor collaboration users

New vendor collaboration users are provisioned via the workflow that you set up to process vendor collaboration user requests of the **Provision vendor user** type.

If the email address of a new vendor collaboration user belongs to a domain that is registered with Azure as a tenant (that is, if it's a managed domain account), the email address must be an existing Microsoft Entra account. Otherwise, the provisioning process can't be completed.

For more information about the process that is used in the *Send Azure B2B user invitation* task in the workflow for Microsoft Entra account management, see Microsoft Entra B2B collaboration.

Related information

- Vendor collaboration with external vendors
- Watch a short video on the vendor onboarding process: Onboard a new vendor

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Last updated on 09/26/2025

Manage vendor collaboration users

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to request the provisioning of new vendor collaboration users and how to add new vendor collaboration contacts.

The vendor collaboration interface in Dynamics 365 Supply Chain Management exposes information about purchase orders, invoices, and consignment stock to external vendors. You can create new vendor collaboration contacts and request new user provisioning if you're working as an external vendor with the *Vendor admin (external)* security role or similar permissions. You can also perform these tasks if you're working as a procurement professional. In this article, this role refers to a procurement professional who is working within the company that owns the instance of Supply Chain Management. For more information about how to use vendor collaboration if you're an external vendor, see Vendor collaboration with customers.

For more information about how to use vendor collaboration if you're a procurement professional, see Vendor collaboration with external vendors.

Add new vendor collaboration contacts

To access vendor collaboration, you first need to be added as a vendor collaboration contact. You might also want to add contacts for employees in your company who don't use vendor collaboration. For example, they could be the point of contact for other kinds of procurement information. Add new contacts on the **All contacts** page, which you access from the **Vendor collaboration** > **Contacts** menu. To add a new contact:

1. Select **New.**
2. Enter the contact person details.
3. Choose which legal entity they're representing in your company, and which legal entity they'll work with in the company that they'll collaborate with. Select a **Legal entity in my company/Legal entity in customer company** pair.

4. Select **Create.**

If you want to delete a contact, you can only delete the contacts that you created.

Vendor collaboration user requests

Only procurement professionals or external vendor administrators can raise vendor collaboration user requests.

- If you're an external vendor, submit requests from the **All contacts** page within the **Vendor collaboration** module.
- If you're a procurement professional, submit requests from the **View contacts** page. To do this, on the vendor record, in the **Setup** section on the Action Pane, select **Contacts** > **View contacts**.

You can request to provision a user, deactivate a user, or modify security roles. If you're an external vendor administrator, you must be registered as a contact person for the vendor accounts that you want to make user requests for, and you must have access to the vendor collaboration interface for those vendor accounts.

When a request is submitted, it's added to the **Vendor collaboration user requests** list in the **Vendor collaboration** module, and to the **Vendor collaboration user request** list in the **Procurement and sourcing** module (the Procurement and sourcing module isn't accessible to external users).

Provision a user

Before you can request to provision a new user, you need to set up that person as a contact for one or more vendor accounts. To create a request for a new vendor collaboration user:

1. On the **All contacts** page, select **Provision vendor user**.
2. Enter an email address for the user. The user uses this address to sign in to Supply Chain Management. If the email address belongs to a domain registered as a tenant with Microsoft Azure, the email address must be an existing Microsoft Entra ID account for the provisioning process to complete successfully. If the email address doesn't belong to a domain registered with Microsoft Azure, a Microsoft Entra account is created as part of the provisioning process and the new user receives an invitation email. Consumer email addresses with domains such as @hotmail.com, @gmail.com, or @comcast.net can't be used to register a user.

3. Set the **Vendor collaboration access allowed** option to *Yes* for all the legal entities that the user needs access to.
4. In the **Assign user roles** section, select the **Assign** check box for the security roles that the new user should have.
5. Select **Submit**.

When you submit the vendor user request, the **Vendor collaboration access allowed** field is set to *Yes* for the selected vendor account and a user request workflow starts. As part of the workflow, a new user is created and security roles are assigned. In addition, an Azure B2B service is activated which initiates interaction with Azure portal and associates a new or existing Microsoft Entra account with the Supply Chain Management user account. Learn more in What is Microsoft Entra B2B collaboration?.

Inactivate a user

To remove access to vendor collaboration for a user, use one of the following methods:

- On the **Contacts** page for the vendor, set the **Vendor collaboration access allowed** option to *No* for the contact. Set this option individually for each legal entity that the person is a contact for. Only procurement professionals can use this option.
- Inactivate the entire user account by submitting an **Inactivate vendor user** request.

To request that a user is deactivated:

1. On the **All contacts** page, select **Inactivate vendor user**.
2. Enter a comment in the **Business justification** field.
3. Select **Submit**.

Modify security roles

The **Maintain vendor user roles** page is the same as the **Provision vendor user** page except that you can edit the list of security roles.

To request that the security roles are modified for a user:

1. On the **All contacts** page, select **Maintain vendor user roles**.
2. Enter a comment in the **Business justification** field.
3. In the **Maintain user roles** section, select the security roles that you want to assign, or clear the ones that you want to remove.
4. Select **Submit**.

Last updated on 05/06/2026

Set up vendor accounts

07/21/2025

This article describes the types of information that you must specify when you create a new vendor account.

When you create a vendor account, you enter information about the vendor. This information is used to automatically enter data in documents and to track activity that involves the vendor. For example, you can configure the following information for a vendor:

- Assign a vendor group. Every vendor must be assigned to a vendor group. The vendors in a vendor group have parameters that they share. For example, they might have the same terms of payment.
- Configure the vendor for catalog import. Vendors can provide a file that contains the catalog of their items and services. This file can be uploaded so that your workers can order from the vendor.
- Assign the vendor to procurement categories.
- Allow an existing vendor to do business with another legal entity in your organization.
- Put a vendor on hold for specific types of transactions.
- Set up banking information for the vendor, so that you can send payments electronically.
- Set up tax, delivery, invoice, and payment information for the vendor. By default, these settings are copied to new documents that you create for the vendor.
- Set up default financial dimensions that are used to automatically post transactions with the vendor to financial accounts.

To speed up the process of creating vendor accounts, you can create templates. To create a template, on the **Vendor** page, on the Action Pane, select **Options** > **Record info**. Then select **Company accounts template**. Company account templates are shared with other users.

You can also create a user template for your own use. You can't delete a vendor that is associated with other records, such as contacts or products.

Create, edit, and manage vendor accounts

To create, edit, and manage vendor accounts, go to **Procurement and sourcing** > **Vendors** > **All vendors**. Learn more in Set up vendor accounts.

Vendor account numbers

The account number is a unique identifier for a vendor. You can set up account numbers so that they're generated automatically when you create a vendor. You can also configure the

number sequence so that account numbers are entered manually. For example, you might want to use the vendor's telephone number as the identifier.

Vendor organizations and individual vendors

When you create a new vendor account, you must select whether the vendor is an organization or a person. Your selection affects the information that you must fill in for the vendor. For a person, this information includes the first name, family name, and title. For an organization, this information includes the organization number and the number of employees.

Addresses

For each vendor, you can define multiple addresses, each of which is used for a different purpose. For example, you can create an address that has a purpose of *Invoice*. Or, if you'll pay the vendor by using checks, you can set up an address that has a purpose of *Remit-to*. If you must specify an address to use for money transfers to foreign banks, the purpose will be *SWIFT*.

Vendor contacts

You can store contacts for a vendor. These contacts can then be used on documents such as purchase orders or requests for quotation (RFQs).

To add contacts for a vendor, on the **All vendors** page, on the **Vendor** tab, in the **Set up** group, select **Contacts** > **Add contacts**.

You can create vendor contacts from scratch. Alternatively, you can copy details from another person who is already registered in Supply Chain Management, and edit the information as you require.

> **Note**
>
> Adding a contact for a vendor isn't the same as adding contact information for that vendor. Although you might add general contact information for a vendor, you might also have several specific people who are contacts at that company, and who all have their own contact information.

You can't delete a contact person record if the contact is referenced on a document. Instead, you can deactivate the contact.

Vendors in different legal entities

If a vendor is registered for only one legal entity in your organization, and other legal entities must register the same vendor, you can use the **Add vendor to another legal entity** page to configure the vendor to do business with another legal entity. You must select a vendor group, currency, and hold status for the vendor in the selected legal entity.

If multiple legal entities in your organization do business with the same vendor, and each legal entity maintains a separate vendor account for that vendor, you can merge the party IDs for the vendor accounts. In this way, information such as the address and the number of employees can be shared, so that you must update it in only one place.

To merge party IDs, follow these steps.

1. On the **Global address book** page, select the address book records that represent the vendor in each legal entity that should be included in the mapping.
2. On the Action Pane, select **Merge records**.

Agreements

When you set up a vendor account, you might also want to register the agreements that you have with the vendor. You can set up price and discount agreements by using the actions on the vendor record. You can also set up a purchase agreement on the **Purchase agreements** page.

Putting a vendor on hold

You can put a vendor on hold for various transaction types. The following options are available:

- *No* – No holds have been put on the vendor.
- *Invoice* – No invoices can be posted for the vendor.
- *All* – The vendor is on hold for all transaction types. These transaction types include purchase requisitions, invoices, and payments.
- *Payment* – No payments can be generated for the vendor.
- *Requisition* – Purchase requisitions can't be created for the vendor, and requisition lines already created before the vendor was set on hold can't be converted to a purchase order. Requisition lines for the vendor will be canceled if your policy is set to create purchase orders automatically.
- *Never* – The vendor is never put on hold for inactivity.
- *Purchase order* – Purchase orders can't be created for the vendor, but you can still proceed with any open invoices or payments to the vendor.

When you put a vendor on hold, you can also specify a reason and a date when the on-hold status will end. If you don't enter an end date, the vendor's on-hold status lasts indefinitely.

You can bulk update the on-hold status to *All* for vendors based on the selected criteria on the **Vendor inactivation** page, and assign a reason for why the vendor is on-hold.

The following criteria are used to include vendors that have been inactive in a period, include or exclude vendors that are employees, and exclude vendors that are under a grace time before the next hold.

- Based on the number of days that you enter in the **In activity period** field on the **Vendor inactivation** page, the application calculates the latest date where the vendor can have any activity to be considered inactive. That is, the current date minus the number of days that you enter. If one or more invoices exist for the vendor where the date is later than the calculated latest date, the vendor will be excluded from the inactivation. This is also validated if the vendor has payments after that date, open purchase requisitions, open purchase orders, requests for quotations, or replies.
- The number of days in the **Grace time before next hold** field is used to calculate the latest grace date. That is, the current date minus the days that you enter. This only applies to vendors who have previously been deactivated. The application verifies the history of other occurrences of inactivation for the vendor and checks if the latest inactivation occurred before the latest grace date. If so, the vendor will be included in the inactivation process.
- The parameter **Include employees** refers to vendors that are linked to an employee. You can set if you want to include those employees.

This process will always exclude vendors where the value in the **Vendor hold** field is *Never*.

Vendors that pass the validations are put on-hold, which sets the **Vendor hold** field value to *All* and the **Reason** to what has been selected. A record in the on-hold history is created for the vendor.

Vendor invoice account

If more than one vendor has the same billing address, or if a vendor is invoiced through a third party, you can specify an invoice account on the vendor record. The invoice account is the account that the invoice amount is credited to when you create a vendor invoice from a purchase order. If you don't enter an invoice account on the vendor record, the vendor account is used as the invoice account.

Vendor bank accounts

If you must make payments to a vendor bank account, you can enter information about the vendor’s bank and bank accounts on the **Vendor bank accounts** page. You can also enter information about validation and payments for the bank account. For example, you can add prenotes to vendor bank accounts. These prenotes can be used to verify the accuracy of account data, such as routing numbers and account numbers. You must specify a default account for payments to the vendor. However, when you make an actual payment, you can change this account to one of the vendor's other accounts.

Ledger accounts

You can specify the default accounts that automatically appear in vendor invoice journals for the specified vendor. This functionality can be useful if you typically pay for the same types of items or services from the same vendors over time. When you specify a default account, you can quickly and efficiently enter journal entries in the invoice journal. The default accounts that you specify aren't used for purchase orders, or for vendor invoices that are entered on the **Vendor invoice** page.

You select default accounts on the **Default account setup** page, which you can open from the **Invoice** tab on the vendor record. The accounts that you select here appear in the filtered list of accounts for the vendor account when you enter a journal entry. You can set one of the accounts as a default account.

Create a vendor account

This procedure shows how to create a vendor account, and add an address and contact information. The procedure doesn't show how to populate all fields for purchasing and financial purposes. To learn more about those fields, read the field descriptions. You can use this procedure in demo data company USMF or on your own data. Procurement professionals or accounts receivable personnel typically create vendor accounts.

Create a new vendor account

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. On the Action Pane, select **New**.
3. In the **Vendor account** field, enter a value that uniquely identifies the vendor.

    - The value might be populated automatically. If so, you can skip this step.
    - You can create vendor accounts for a person or organization. This choice affects which fields are available. In this example, create a vendor account for an organization.

4. In the **Name** field, enter or select a vendor name. If your vendor is already registered in your system, select it in the dropdown list. The new vendor account inherits the address and contact information that's already registered.
5. In the **Group** field, select a vendor group. Use the vendor group to group vendors that have any of the following parameters in common: terms of payment, settle period, inventory posting ledger accounts - including the sales tax group, default ledger accounts, product filter codes, or supply forecast configuration.
6. In the **Number of employees** field, enter the number of employees.
7. In the **Organization number** field, enter the legal organization number of the vendor.

Add an address

1. Expand the **Addresses** section.
2. Select **Add**.
3. In the **Purpose** field, select a purpose of the address. You can select one or more purposes. These purposes help you select the correct address for a given purpose. For example, if the purpose is *Invoice*, use that address when you send invoices.
4. In the **Name or description** field, enter a name of the address.

5. In the **Country/region** field, select a country/region. Enter the address details. The country/region that you select determines the fields you're presented with, corresponding to the address format for the country/region.
6. Select **OK**.

Add contact information

1. Expand the **Contact information** section.
2. Select **Add**.
3. In the **Description** field, enter a contact name.
4. In the **Type** field, select a contact type. For example, if the contact details include the phone number, select *Phone* option in the **Type** field.
5. In the **Contact number/address** field, type a contact information. You can select the Primary check box if this is the primary contact.
6. Select **Save**.
7. Close the page.
8. Close the page.

Last updated on 07/01/2026

Create a vendor bank account

This procedure shows you how to create a bank account for a vendor. You can use this procedure in demo data company USMF.

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Select the vendor that you want to create a bank account for, and then select the link on the **Vendor account ID** field.
3. On the Action Pane, select **Vendor**.
4. Select **Bank accounts**.
5. On the Action Pane, select **New**.
6. In the **Bank account** field, type a value. This ID identifies the bank account on the vendor record.
7. In the **Name** field, type a value.
8. In the **Bank groups** field, enter or select a value.
9. In the **Routing number type** field, select an option. This type of routing number is used for international payments.
10. In the **Bank account number** field, type a value.
11. In the **SWIFT code** field, type a value.
12. In the **IBAN** field, type a value.

    - The IBAN number must be in the correct format. For example, you could use *DE89370400440532013000*.
    - The status of the bank account is *Active* if the **Active date** is reached, and the **Expiration date** isn't exceeded. It's also active if both the **Active date** and **Expiration date** fields are blank. If the dates in both the **Active date** and **Expiration date** fields are in the future, electronic payments aren't available. Other payment types are available and the bank account is active.

13. Expand the **Setup** section.
14. In the **Text code** field, type a value. This field specifies a code that appears on the bank statement of the recipient.
15. In the **Message to bank** field, type a value.
16. In the **Exchange reference** field, type a value. This is the reference number for any forward-term or fixed-term rate of exchange.
17. In the **Currency** field, enter or select a value. When prenotes are issued, this section provides an overview of their status (pending or approved).

18. Expand the **Address** section.
19. Expand the **Prenotes** section.
20. Expand the **Contact information** section.
21. In the **Telephone** field, type a value.
22. Close the page.
23. Select **Edit**.
24. Expand the **Payment** section.
25. In the **Bank account** field, select the account that you just created.
26. Select **Save**. The address might be inherited from the bank group, if one is specified, or you can add it here.

Last updated on 07/01/2026

Vendor posting profiles

Vendor posting profiles control the posting of vendor transactions to the general ledger.

Vendor posting profiles

Vendor posting profiles enable you to assign general ledger accounts and document settings to all vendors, a group of vendors, or a single vendor. These settings apply when you create purchase orders, vendor invoices, and cash payments. For some transactions, you can select a posting profile that differs from and takes precedence over the posting profiles that you set up on this page. You define the default posting profile on the **Ledger and Sales Tax** FastTab on the **Accounts payable parameters** page. The default posting profile automatically appears on the header of new documents, where you can change it to a different posting profile if needed.

You can also associate posting definitions with transaction posting types on the **Transaction posting definitions** page. Posting definitions control the posting of vendor transactions to the general ledger instead of posting profiles.

Creating a posting profile

Setup

Specify the ledger accounts that are used in the posting of transactions that use the selected posting profile. Select an account code and, whenever possible, an account or group number for the selected posting profile. In the posting process, the system locates the most appropriate posting profile for each transaction by searching for the most specific account code, account number, or group and number combination in the following priority.

Expand table

Account code **field value**	Account/Group number **field value**	**Search priority**
Table	Specific vendor account	1
Group	Vendor group that is assigned to the vendor	2
All	Blank	3

If you want all vendor transactions to use the same posting profile, set up only one posting profile with **All** in the **Account code** field. Specify the following values to set up your posting profile.

Expand table

Field	Description
Posting profile	Enter a code for the posting profile. For example, you could create two posting profiles to obtain one account for vendor balances in the national currency and another for vendor balances in a foreign currency. You could call one account National and the other Foreign.
Description	Enter a description of the posting profile.
Account code	Specify whether the posting profile applies to a specific vendor, a group of vendors, or all vendors: • **Table** – The posting profile applies to a single vendor. Select the vendor account in the **Account/Group number** field. • **Group** – The posting profile applies to a vendor group. Select the vendor group in the **Account/Group number** field. • **All** – The posting profile applies to all vendors. Leave the **Account/Group number** field blank.
Account/Group number	If you select **Table** in the **Account code** field, select the account number of the vendor that is associated with the posting profile. If you select **Group**, select a vendor group. If you select **All**, leave this field blank.
Summary account	Select the ledger account that the system uses as the summary account for the vendors that the posting profile relates to. The **Do not allow manual entry** parameter for this main account is marked. If you subsequently remove this account from the posting profile, validate the **Do not allow manual entry** setting on the **Main accounts** page. **Note:** If you select the **Use posting definitions** option on the **General ledger parameters** page, the transaction posting definition for vendor invoices is used instead of the summary account.
Settle account	Select the liquidity ledger account that is used for cash flow forecasts. This field is only available when cash flow forecasting is enabled.
Sales tax prepayments	Select the account for sales tax payments that are received in advance. **Note:** The posting profile that the system uses when the payment is marked as a prepayment is selected in the **Posting** profile with **Prepayment journal voucher** field in the **Ledger and sales tax** area on the **Accounts payable parameters** page.
Arrival	Select the ledger account that the system posts information about unapproved vendor invoices to. The user enters the information in the **Invoice register journal**. For example, a user enters very basic information about vendor invoices when they are received in the

Field	Description
	invoice register. When the invoice register is posted, the system posts the transactions to the account that is entered here and in the **Offset account** field. When the invoices are approved, the system transfers the debt from the arrival account to the vendor summary account.
Offset account	Select the ledger account that you use to offset unapproved vendor invoices when you update them by using the invoice register. The offset account works as the offset account for transactions that you post to arrival accounts. Therefore, the account contains vendor purchases that aren't yet approved.

Table restrictions

For transactions that have the selected posting profile, specify whether to settle transactions automatically, calculate interest, and send collection letters. You can also select the account to use when closing transactions that have the selected posting profile.

Specify the following values to set up your posting profile

Expand table

Field	Description
Settlement	Select this option to enable automatic settlement of transactions that have this posting profile. If you clear this option, you must manually settle transactions by using the **Settle open transactions** page.
Cancel	Select this option if you want to be able to cancel transactions that have this posting profile.
Close	Select a posting profile to change to when closing transactions that have this posting profile. A transaction is closed when it's fully settled.

Last updated on 08/04/2026

Vendor request configurations

06/17/2025

To complete a vendor request, a vendor contact person must complete the prospective vendor registration wizard.

On the **Vendor request configurations** page, you can create profiles that specify required fields and visible fields in the prospective vendor registration wizard.

The vendor registration wizard starts by asking the prospective vendor user which country/region the vendor is doing business in. This information determines the applicable configuration. If no specific configuration is defined for a country/region, a default configuration is used.

Set up a vendor request configuration

By default, there's a vendor configuration available on the **Vendor request configurations** page.

It isn't possible to select country/regions for the default configuration, so the **Countries/regions** section can't be changed.

1. Go to **Procurement and sourcing** > **Setup** > **Vendors** and then select **Vendor request configurations**.
2. Select the **Fields** tab to set the status of the listed fields.

    - *Hidden* (Not visible)
    - *Displayed* (Visible but not mandatory)
    - *Required* (Visible and mandatory)

3. Select the **Content** tab to specify if text is going to be shown on the wizard and if there should be an acknowledgment that the prospective vendor user must accept this before moving to the next step in the wizard. The acknowledgment is requested for any terms and conditions that the user must accept to continue.

You can also enter a confirmation message that is displayed when the wizard is finalized, and you can add one or more questionnaires.

Create a vendor configuration for a specific country/region

1. Go to **Procurement and sourcing** > **Setup** > **Vendors** and then select **Vendor request configurations**.
2. Select **New** to create a new configuration, and provide a name for the configuration.
3. Select **Save**.
4. Open the **Country/regions** tab to select the country/region that the configuration should be used for.
5. Complete the configuration by following the guidelines for the default configuration.

Onboard vendors

08/13/2025

> **Note**
>
> Azure Active Directory is now Microsoft Entra ID. Learn more

New vendors can be onboarded and registered as vendors in Microsoft Dynamics 365 Supply Chain Management, based on information that is collected from a person who represents the vendor.

The process consists of the following steps, where various roles perform actions in the system.

1. **Data management OData** – Entity import. The initial request is the prospective vendor registration request. Typically, this request comes from a source such as a customer-hosted website that allows anonymous access. Vendors can sign up by providing basic information, such as the vendor name, justification, organization number, and name and email address of the contact person. The requests are imported via the Data management interface.
2. **Prospective vendor registration request list page** – Based on the information that is provided in the prospective vendor registration request, a procurement professional decides whether the vendor should be onboarded. The procurement professional views the incoming request on the **Prospective vendor registration requests** list page.
3. **User provisioning workflow** – When a procurement professional has verified the information in the incoming request and has decided to continue with the onboarding process, the user request workflow provisions the new user and sends an invitation email to accept the contact person as an authenticated user of Microsoft Dynamics 365.
4. **Vendor registration wizard** – The vendor's contact person signs in by using the new user account. They complete a vendor registration wizard to provide information such as addresses, business information, procurement categories, and questionnaire responses.
5. **Approval workflow** – A vendor request that includes the registration information is created. This vendor request is submitted to a workflow, and is routed for review and approval.
6. **Creation of a vendor master and user role modification** – When the vendor request is approved, a vendor record is created. The user account of the vendor's contact person is either granted permission to vendor collaboration or deactivated.

The following table shows the steps and roles that are involved in the process.

Expand table

Role and "process"	Data management OData – Entity import	Prospective vendor registration request list page	User provisioning workflow	Vendor registration wizard	Approval workflow	Creation of a vendor master and user role modification
System	The request for a new vendor is imported.					After the vendor request is accepted, the vendor record is created.
Procurement professional		Start the onboarding process.			Review and either accept or reject the vendor request.	
Administrator			Create a user in Supply Chain Management and Microsoft Azure.			
Vendor contact person			Send email to the contact person.	Register vendor information.		

For a quick demonstration of the vendor onboarding process, watch this short YouTube video about How to onboard a new vendor in finance and operations.

Importing the prospective vendor registration request

The prospective vendor registration request is an entity in Supply Chain Management. You can set up the system to import data via this entity.

The following table shows the information that this entity contains, and that can be imported.

Expand table

Field	Description
Vendor name	The name of the vendor.
Business justification	The reason or reasons for the vendor request.
Organization number	An officially known registration number.
Line of business	The line of business that the vendor is in.
Contact person's first name	The first name of the person who will be invited to register vendor information.
Contact person's middle name	The middle name of the person who will be invited to register vendor information.
Contact person's last name	The last name of the person who will be invited to register vendor information.
Contact person's email	The email address that will be used to create a new user in Supply Chain Management, and that will be registered in the tenant's Microsoft Entra account.
Submitted date	The date when the request was created in an external system.
Legal entity	The legal entity where the vendor is requesting to become a vendor. This value must be a legal entity code that has been registered in Supply Chain Management. If no value is received though the import process, a value from the Procurement and sourcing parameters is applied.
Vendor type	The vendor can be either an organization or a person. The vendor type determines how the vendor is finally created.

After the prospective vendor registration request is imported, it appears on the **Prospective vendor registration request** list page. From this list page, a procurement professional can invite the user. A user request for provisioning the user is sent to a workflow.

Submitting a prospective vendor user request

The purpose of a prospective vendor user request is to provision the person who submitted the initial request, so that they can sign in to Supply Chain Management by using the email account that is provided in the prospective vendor registration request.

The prospective vendor user request is processed by the user request workflow. This workflow communicates through Microsoft Entra B2B collaboration. It creates a user in Supply Chain Management that has the appropriate security settings.

New users that are set up have the following security roles:

- System external user
- Vendor prospective (external)

The new user will receive an email that is generated by the user request workflow. This email invites the user to sign in to the system.

For information about the configuration of the email and the workflow in general, see the description of a user request workflow in Set up and maintain vendor collaboration.

Vendor registration

A prospective vendor user who signs in to Supply Chain Management will see the first page of a vendor registration wizard, where they can enter vendor information.

The wizard reflects the configuration of the vendor request. The country or region where the vendor does business determines what information is requested in the wizard and what information is mandatory.

For more information about the vendor request configuration, see Set up and maintain vendor collaboration. The following table gives an overview of the pages in the wizard and the purpose of each page.

Expand table

Page	Description
Country/region	The country or region determines the vendor request configuration that is applied on the remaining wizard pages. It also determines values in the **Tax state** lookup.
Terms and conditions	This page might be available, depending on the vendor request configuration. If it's available, the user must acknowledge the terms and conditions to continue.
Vendor information	This page contains the vendor name, which is automatically entered from the original prospective vendor registration request. It also contains the organization number, the vendor's telephone number, fax number, and email address, and the vendor's addresses for various purposes.
Contact person information	This page contains the contact person's name, which is automatically entered from the original prospect vendor registration request. It also contains the contact person's telephone number and email address, and the contact person's addresses for various purposes.
Business information	This page contains tax registration numbers (for various countries or regions) and the numbers of employees. It also indicates whether the business is minority owned.

Page	Description
Procurement categories	This page contains the procurement categories that the vendor is requesting approval for. The user can select categories in the procurement category hierarchy. You can configure the number of levels that are shown in the hierarchy at **Procurement and sourcing parameters** > **Vendor collaboration**, under **Procurement and sourcing** > **Setup**.
Questionnaires	The wizard might include a set of questionnaires for the vendor. Questionnaires that appear in the wizard are configured either on the vendor request or per procurement category. If questionnaires are configured per procurement category, the procurement categories that the vendor requests approval for determine the questionnaires that appear in the wizard. On the **Procurement categories** page, you can add a questionnaire under the relevant category and set the activity type to *Vendor onboarding*.

When the prospective vendor user completes the vendor registration wizard, a vendor request is created.

Manually or automatically submit a vendor request

A vendor request can be created as a draft and manually submitted to a workflow. Alternatively, the vendor request can be automatically submitted to a workflow when the vendor registration wizard is completed. A request can be submitted manually if, for example, a procurement professional wants to assess whether the request should be routed through an approval process before it's submitted to the workflow.

To configure the vendor request so that it's submitted automatically to a workflow when the vendor registration wizard is completed, select **Procurement and sourcing parameters** > **Vendor collaboration** and then select **Auto submit prospective vendor registration to workflow**.

Vendor requests

Vendor requests are available on the **Vendor collaboration user requests** page.

A vendor request contains the information that the prospective vendor user entered in the vendor registration wizard.

The request lets you review the vendor information and decide whether the vendor should become a registered vendor.

The vendor request should be submitted to a workflow, and it should be routed to the relevant reviewers and approvers. For basic information about how to set up workflows, see

Procurement and sourcing workflows.

The following table shows the statuses that vendor requests can have.

Expand table

Status	Description
Draft	The vendor request hasn't yet been submitted.
Request submitted	The vendor request has been submitted, and the first step in the workflow is being processed.
Pending review	If there are multiple reviewers in a workflow task, a reviewer can accept the task of reviewing the vendor request and then complete the review. If there's only one reviewer, that participant can complete the review by selecting *Completed* in the workflow action. They don't have to accept the work item first.
Request pending approval	The vendor request has been routed to the participants for approval, and there's an option to request additional information. A request for additional information causes the work item to be routed back to the submitter. The vendor request can also be approved or rejected while it's in this status.
Application change request	Additional information has been requested by the approver, and the vendor request has been routed to the person who submitted the vendor request. The submitter can add required information and then resubmit the vendor request. If a vendor request is resubmitted, the status is changed back to *Request pending approval* status.
Request approved	This status is a final state.
Request rejected	This status is a final state.

Approving a vendor request

When a vendor request is approved, a vendor account is created, and the status *Approved* appears on both the initial prospective vendor registration request and the vendor request.

Before you approve a vendor request, on the **New vendor** page, on the **General** FastTab, select **Vendor group** to select a vendor group.

If the prospective vendor user should have access to Supply Chain Management as a vendor collaboration user who represents the vendor, set the vendor collaboration access permission to *Yes*. To deactivate the user account that the prospective vendor used to register, set this permission to *No*.

If the vendor collaboration access permission is set to *Yes*, when the vendor request is approved, a request is submitted to modify the user's roles so that the user has the roles that are defined for the **Vendor** type in **External roles**. If this permission is set to *No*, when the vendor request is approved, a request is submitted to deactivate the user. In this case, the workflow to deactivate a user request must be set up.

For a vendor account to be created when the vendor request is approved, the number sequence for creating vendors from vendor requests must be set to *Auto*.

For an overview of the access permissions of a vendor collaboration user, see Set up and maintain vendor collaboration.

Rejecting a vendor request

If a vendor request is rejected, a reason for rejection must be selected in the vendor request.

When a vendor request is rejected, a request is submitted to deactivate the user. In this case, the workflow to deactivate a user request must be set up. Learn more in Set up and maintain vendor collaboration.

When a vendor request is rejected, the status *Rejected* appears on both the initial prospective vendor registration request and the vendor request.

Deleting a prospective vendor registration request in various statuses

The various statuses of a prospective vendor registration request give an overview of the request's progress.

By using the **Delete** action on the prospective vendor registration request, you can clean up and remove the chain of records that has been created, and you can deactivate the user account. The result of the **Delete** action varies, depending on the status of the prospective vendor registration request, as shown in the following table.

Expand table

Status	Status description	Result of the Delete action
New	No actions have been taken on the request.	The prospective vendor registration request is deleted.
User requested	When you select **Invite user**, the status is changed to *User requested*, and a	You can't delete a prospective vendor registration request that has this status,

Status	Status description	Result of the Delete action
	prospective user request is created and submitted to a user request workflow.	because the user request workflow hasn't ended.
User invited	The user request workflow is approved, and the user is created.	A request to deactivate the user is created, and the prospective vendor registration request is deleted.
Registration in progress	The new user has signed in and has started the vendor registration wizard.	A request to deactivate the user is created, and the prospective vendor registration request and the data that was entered in the vendor registration wizard are deleted.
Vendor request created	The vendor registration wizard has been completed.	A request to deactivate the user is created, and the prospective vendor registration request, the data that was entered in the vendor registration wizard, and the vendor request are deleted. **NOTE:** You can't use the **Delete** action when the vendor request is in a review process in the workflow.
Approved	The vendor request is approved.	The prospective vendor registration request, the data that was entered in the vendor registration wizard, and the vendor request are deleted.
Rejected	The vendor request is rejected.	The prospective vendor registration request, the data that was entered in the vendor registration wizard, and the vendor request are deleted.

Vendor collaboration with external vendors

07/21/2025

The **Vendor collaboration** module is targeted at vendors who don't have electronic data interchange (EDI) integration with Microsoft Dynamics 365 Supply Chain Management. It lets vendors work with purchase orders (POs), invoices, consignment inventory information, and requests for quotation (RFQs), and also lets them access parts of their vendor master data. This article explains how you can collaborate with external vendors who use the vendor collaboration interface to work with POs, RFQs, and consignment inventory. It also explains how to enable a specific vendor to use vendor collaboration, and how to define the information that all vendors see when they respond to a PO.

For more information about what external vendors can do in the vendor collaboration interface, see Vendor collaboration with customers.

For more information about how vendors can use vendor collaboration in invoicing processes, see Vendor collaboration invoicing workspace. For information about how to provision new vendor collaboration users, see Manage vendor collaboration users.

Defining the information that is shown to vendors when they respond to POs

When vendors respond to a PO that you send them, they see one of three message boxes, where they must confirm that they want to accept the PO, reject it, or accept it with changes. Because the information that must be shown to the vendor at that point might be specific to your business, you can specify the text that appears in each confirmation message. For example, the text can inform the vendor about the next steps in the process, or about terms and conditions.

To define the text that's shown in the PO response, follow these steps

1. On the **Information for vendors responding to POs** page, select the response type, and then select **Edit**.
2. In the **Information message** box, enter the information that should be shown to vendors in the message box.

If you must add messages in more than one language, create separate messages, and specify the appropriate language code for each. The message that is shown to each vendor will be in the language that the vendor uses.

Setting the vendor collaboration options for a specific vendor

An administrator configures the general settings for vendor collaboration, such as the security roles that are available for all vendors that you collaborate with. However, there are also settings that can differ for each vendor account. You should configure these settings.

- Enable vendor collaboration.
- Specify whether the vendor should see price information.

Enabling vendor collaboration

Before user accounts can be created for an external vendor, you must configure the vendor account so that vendor can use vendor collaboration. On the **Vendors** page, on the **General** tab, set the **Collaboration activation** field. The following options are available:

- *Active (PO is auto-confirmed)* – POs are automatically confirmed if the vendor accepts them without changes. If you use this option, be sure to schedule the *Confirm accepted purchase orders from vendor collaboration* batch job, which is responsible for processing the confirmations. For instructions, see the next section.
- *Active (PO is not auto-confirmed)* – Your organization must manually confirm POs after the vendor has accepted them.

Scheduling the auto-confirmation batch job

If you use the *Active (PO is auto-confirmed)* option for one or more of your vendors (as described in the previous section), you must schedule the *Confirm accepted purchase orders from vendor collaboration* batch job, which is responsible for processing and confirming your POs. Otherwise, auto-confirmations will never occur. Use the following procedure to schedule this job.

1. Go to **Procurement and sourcing** > **Purchase orders** > **Purchase order confirmation** > **Confirm accepted purchase orders from vendor collaboration**.

2. In the **Confirm accepted purchase orders from vendor collaboration** dialog box, on the **Run in the background** FastTab, select **Recurrence**.

3. In the **Define recurrence** dialog box, define the schedule that the job should run on. When you're choosing your schedule, consider the following issues:

    - If your system processes a large volume of data and runs many batch jobs, performance might be an issue. In this case, you probably shouldn't run this job

more often than every 10 minutes (depending on your other requirements). If performance isn't an issue for you, you can run it as often as every 1 to 2 minutes if necessary.
    - If your vendors tend to deliver goods quickly (within the day that they agreed to), the recurrence should be frequent (every 10 to 30 minutes or so). In this way, warehouse workers will be able to receive the goods against the confirmed PO after confirmation is done.
    - If your vendors tend to have a long lead time (more than 24 hours), you can set this task to run just once a day or so.

4. Select **OK** to apply your schedule and return to the **Confirm accepted purchase orders from vendor collaboration** dialog box.

5. Set additional background options as needed. The dialog box provides the usual options for setting up batch jobs in Supply Chain Management.

For more information about batch jobs, see Batch processing overview.

Specifying whether the vendor should see price information

To share price information for POs via the vendor collaboration interface, you must set the **Purchase order prices/amount** option on the **Purchase order defaults** tab for the vendor account to *Yes*. This price information includes the unit price, discounts, and charges.

Working with POs when vendor collaboration is used

Sending a PO to a vendor

POs are prepared in Supply Chain Management. When a PO has a status of *Approved*, you send it to the vendor by selecting **Send for confirmation** on the **Purchase order** page. The PO status is then changed to *In external review*. After the PO has been sent, the vendor can see it on the **Purchase orders for review** page in the vendor collaboration interface. The vendor can then accept the PO, reject it, or suggest changes to it. The vendor can also add comments to communicate information such as changes to the PO. If you want to draw the vendor's attention to a new PO, you can also use the Print management system to send the PO by email.

Confirmation and acceptance of a PO by a vendor

After a vendor accepts a PO, the PO can be automatically confirmed, or it might have to be manually confirmed. The behavior depends on whether the **Collaboration activation** field is set to *Active (PO is auto-confirmed)* or *Active (PO is not auto-confirmed)* for the vendor.

The following table shows the typical exchange of information, depending on the vendor's response when you send a PO for confirmation.

Expand table

Vendor response	Result
The vendor accepts the order, and Supply Chain Management is configured to automatically confirm POs that the vendor accepts.	The status of the order is updated to *Confirmed*. If the order can't be updated for some reason, the vendor response is still recorded as *Accepted*, but the status of the PO remains *In External Review*. The PO that was sent to the vendor and that has a status of *In external review* is updated with confirmed receipt dates on the lines. This update initiates a new version that is automatically set to *Confirmed* status. When the PO is confirmed, it appears in the vendor collaboration interface.
The vendor accepts the order, but Supply Chain Management isn't configured to automatically confirm POs that the vendor accepts.	The vendor response is recorded as *Accepted*, but the status of the PO remains *In external review*. The PO that was sent to the vendor and that has a status of *In external review* is updated with confirmed receipt dates on the lines. This update initiates a new version that is automatically set to *In external review* status. You can then manually confirm the PO.
The vendor rejects the order.	The vendor response is recorded as *Rejected*, and the status of the PO remains *In external review*. The rejection is received together with the vendor's note.
The vendor accepts the order with changes. Changes are suggested at the line level. The vendor can accept or reject individual lines. Here are some other changes that the vendor can suggest: • Change dates or quantities. • Split lines for different receipt dates or quantities. • Substitute an item. The vendor can't change price information and charges. However, the vendor can suggest these changes by using notes.	The vendor response is recorded as *Accepted with changes*, and the status of the PO remains *In external review*. The statuses show the types of changes that the vendor has suggested. For information about the automatic consumption of changes, see the "Update the PO when a vendor suggests changes" section later in this article.

You can use the **Purchase order preparation** workspace to monitor which POs the vendor has responded to. This workspace contains two lists that contain POs that have a status of *In external review*:

- In external review requires action
- In external review awaiting vendor response

Changing a PO

To change a PO that a vendor has already responded to, you must send the vendor a new version of the PO. The new PO will have a version suffix to indicate that it's a modified version of a PO that was previously sent. The **Purchase order vendor confirmation history** page lets you and your vendors track the history of each order. The previously confirmed version of a PO remains in the list of confirmed POs until the new PO has been confirmed.

Canceling a PO

When you cancel a PO, the status is changed to *Approved*. You must send the PO back to the vendor, so that the vendor can confirm or reject the cancellation. After the cancellation is confirmed, the PO appears in the vendor's list of confirmed POs as *Canceled*.

Adding attachments to a PO

You can add attachments such as files, images, and notes to the PO by using the document management system. Attachments of the *External* type will be visible to the vendor when you send the PO.

Updating a PO when a vendor suggests changes

If a vendor has responded to a PO and suggested changes, the next step is to process the response.

In the **Purchase order preparation** workspace, in the **In external review requires action** list, you can identify POs that a vendor has accepted with changes. From this list, you can also navigate to the vendor's response.

On a response, a vendor can change the following information on the header:

- Vendor document reference
- Mode of delivery
- Delivery terms

- Confirmed receipt date

The vendor can also add a note or attachment.

On the lines, the vendor can change the quantity and the receipt dates, add notes and attachments, reject a line, substitute a line with another product that is entered as text, and split a line into multiple deliveries. The status of a line varies, depending on the changes that the vendor has suggested:

- *Accepted with changes*
- *Rejected*
- *Substituted* – In this case, an extra line is added that has a status of *Rejected*.
- *Confirmed*
- *Split into schedule* – In this case, extra lines are added that have a status of *Schedule lines*.

If a line has no changes, the line status is *Accepted*.

On the response, the line statuses tell you the types of changes that the vendor made. Additionally, all fields that were changed appear bold to help you identify the changes.

You can update a PO by selecting **Process PO update** on the response or on one line at a time. An **Is PO update processed?** field on the header and the lines indicates whether the system has processed the header or lines to update the PO with changes that originate from the response. You can run the **Process PO update** action only one time per header or line.

Not all suggested changes can be updated on a PO. Only updates on the header, and updates of dates and quantities on lines, can be automatically updated on the PO. For other changes, you must manually update the PO. In this case, the value of the **Is PO update processed?** field is *Manual update*. For example, if a vendor suggests that a line be split into a schedule, this change must be made manually.

Every line that has a status of *Accepted* will have a confirmed receipt date. When you run the **Process PO update** action, this date is updated on the PO. Notes and attachments aren't automatically transferred to the current PO. Additionally, trade agreements aren't reassessed on the PO lines when you update the current PO via the **Process PO update** action.

PO statuses and versions

This section describes the various statuses that a PO can have up to the time when it's confirmed. It also describes when new versions of a PO are made available to the vendor. The behavior varies, depending on whether you use change management for POs.

Versions and statuses if you don't use change management

The following table shows an example of the changes in status and version that a PO might go through.

Expand table

Action	Status and version
The initial version of the PO is created in Supply Chain Management.	The status is *Approved*.
The PO is sent to the vendor.	A version is registered in the vendor collaboration interface, and the status is changed to *In external review*.
The vendor sends an *Accepted with changes* response.	The status is still *In external review*.
You make some changes that the vendor requested.	The state is changed to *Approved*.
You send the new version of the PO to the vendor.	A new version is registered in the vendor collaboration interface, and the status is changed to *In external review*.
The vendor accepts the new version of the PO.	The status is still *In external review*, unless the vendor account is configured to automatically set POs to *Confirmed* status when the vendor accepts them.

Vendors don't have to confirm a PO by using the vendor collaboration interface. They can also send an email or communicate their acceptance of a PO via other channels. You can then manually confirm the order. In this case, you receive a warning that states that the order is being confirmed even though there's no response from the vendor. The PO then appears in the confirmation history as an open confirmed order that doesn't have any responses. At this point, the vendor no longer has the option to confirm or reject the PO.

> **Note**
>
> The version of the PO that is available to other processes in Supply Chain Management is always the latest version, even if that version hasn't yet been registered in the vendor collaboration interface.

Versions and statuses if you use change management

If change management is enabled for POs, POs go through an approval workflow to reach the *Approved* status. This process isn't visible to the vendor.

The following table shows an example of the changes in status and version that a PO might go through when change management is turned on. A version is registered when the PO is approved, not when the PO is sent to the vendor or confirmed.

Expand table

Action	Status and version
The initial version of the PO is created in Supply Chain Management.	The status is *Draft*.
The PO is submitted to the approval process. (The approval process is an internal process that the vendor isn't involved in.)	The status is changed from *Draft* to *In review* to *Approval* if the PO isn't rejected during the approval process. The approved PO is registered as a version.
The PO is sent to the vendor.	The version is registered in the vendor collaboration interface, and the status is changed to *In external review*.
You make some changes that the vendor requested, either manually or by using the **Process PO update** action on the response to update the PO.	The status is changed back to *Draft*.
The PO is submitted to the approval process again.	The status is changed from *Draft* to *In review* to *Approval* if the PO isn't rejected during the approval process. Alternatively, the system can be configured so that specific field changes don't require re-approval. In this case, the status is first changed to *Draft* and is then automatically updated to *Approved*. The approved PO is registered as a new version.
You send the new version of the PO to the vendor.	The new version is registered in the vendor collaboration interface, and the status is changed to *In external review*.
The vendor approves the new version of the PO.	The status is changed to *Confirmed*, either automatically, or when you receive the response from the vendor and then confirm the PO.

Sharing information about consignment inventory

If you're using consignment inventory, vendors can use the vendor collaboration interface to view information on the following pages:

- **Purchase orders consuming consignment inventory** – POs for consignment inventory are generated when the ownership of the inventory is changed from the vendor to your company. A product receipt is posted at the same time. These consignment POs are

shown only on the **Purchase orders consuming consignment inventory** page. They aren't included on the **All confirmed purchase orders** page in the **Vendor collaboration** module.

- **Products received from consignment inventory** – This page lists all the transactions where the ownership of products has been transferred from the vendor to your company. Vendors can use this information to invoice the customer.
- **On-hand consignment inventory** – This page shows the vendor-owned on-hand consignment inventory that has been received at your warehouse.

Working with RFQs when you use vendor collaboration

This section describes the interactions between customers and vendors during the RFQ process. It also explains how information is communicated to the vendors.

For a basic overview of support for the RFQ process, see Requests for quotation (RFQs) overview.

Alternates, attachments, amendments, and returns

- **Alternates** – On the header of an RFQ case, you can specify that alternates are allowed for non-catalog item lines. When alternates are enabled, vendors can add an alternate line for each requested line.

- **Attachments** – Attachments can be added at both the header level and the line level of an RFQ case. Attachments can be classified as either internal or external. Internal attachments can be viewed only on the customer side, whereas vendors can view external attachments after they're sent.

  Vendors can also add attachments on their bid reply. These attachments can be viewed on the customer side after a vendor submits the bid reply. Attachments that vendors add are always classified as external. To access an attachment that a vendor has submitted together with a bid, select **Bid attachments** or **Bid line attachments**.

  To open attachments that were sent together with the RFQ case, use the document handling paper clip symbol on the reply.

- **Amendments** – When an amendment is finalized, the existing bid replies are removed so that they can be replaced by updated values. Information such as the line price and quantity from previous bid replies can be viewed via the journals on the RFQ case.

To enforce amendment processing, on the **Procurement and sourcing parameters** page, on the **Request for quotations** FastTab, set the **Lock RFQs when they are sent** option to **Yes**. (This option is set and required for Public sector.)

- **Returns** – If a vendor has submitted a bid, but more or modified information is required for the RFQ case, the customer can return the bid to the vendor. The data from the bid that was previously submitted is retained, and the vendor can make the requested modifications without having to restart the bid process.

Public sector extensions

For Public sector, the extended functionality enables an RFQ case to be sent to vendors and published. When you publish an RFQ, anyone who requests the information can view the work that complies with most public-sector regulations. All available work is reflected on the **Open published requests for quotations** list page, and the canceled, pending, or awarded RFQs can be viewed on the **Closed published requests for quotations** list page. These documents can also be viewed on a site outside Supply Chain Management through integrations with the following data entities:

- Published requests for quotations
- Published requests for quotations line
- Published requests for quotations header attachments

These entities let people who aren't provisioned users in Supply Chain Management, but who have anonymous access to the external site, view the available and closed work. Additionally, extended functionality in **Send and publish** lets the user who sets up parameters for the RFQ process define an email template. Then, when the procurement professional creates the RFQ case, they must select the email template to send the required information to the vendors on the RFQ case.

The user who sets up parameters for the RFQ process can create multiple email templates. These email templates can contain both static text and the following replacement tokens. The tokens will be replaced with contextual values when an email is created.

- %RFQCase%
- %RFQCaseName%
- %bidType%
- %inviteOnly%
- %expiryDateTime%
- %requester%
- %requestingDepartment%
- %accountnum%

- %todaysdate%
- %createddate%

If an amendment is required and is sent after the RFQ is sent, the RFQ will be resent to all invited vendors. The published document will also be updated on the **Open published requests for quotations** page.

Maintain vendor certification

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the steps that your vendors can use to maintain their certifications using the **Vendor collaboration workspace**. Examples of certifications might include a Woman Business Enterprise (WBE) or a Leadership in Energy and Environment Design (LEED) company. Vendors need to enter certification information in the **Vendor information** workspace. From there, vendors select **More details**, and then select **Certifications**.

Turn on the vendor certification feature

Before you can use this feature, it must be turned on in your system. Administrators can use the Feature management page to check the feature status and enable it if needed. In the **Feature management** workspace, the feature is listed in the following way:

- **Module** - *Accounts payable*
- **Feature name** - *Enable vendor collaboration certification management*

Add a new certification

To add a new certification, select the **Add** button located above the **Certification** grid in the **Vendor information** workspace. Enter the following information:

- Certification number
- Certification type
- Certifying organization
- Certification date
- Liability amount, if applicable
- Effective date
- Expiration date
- Comments, optional

If there are documents related to the specific certification, you can attach them by selecting the **Document** button.

Certifications that are entered by your vendors on this page will be assigned a source of “Vendor”. You can enter certificate information on your vendor's behalf under vendor bank accounts. The information displays here and the source will display as **Customer**.

Vendors can edit or delete their certifications as needed.

Vendor collaboration generated certification records

After certification information has been added by a vendor, the information will be visible on the **Vendor collaboration generated certifications** page. To open the page, go to **Accounts payable** > **Inquiries** > **Vendor reports** > **Vendor collaboration generated certifications**. By default, all new or modified certification records are visible. An Accounts payable clerk can view the changes and validate the information through their confirmation process to validate. When the information has been confirmed, the certification record listed on the page can be selected and marked as reviewed. Marking the record as reviewed will remove it from the default list.

All certification changes are visible on the **Vendor collaboration generated certifications** page. If a change isn't displayed on the page, you can view it by adjusting the filters for the vendor account, effective date range, or choosing whether to include information for certification changes that have been reviewed.

Last updated on 06/15/2026

Category requests from vendors

10/03/2025

The category request process lets vendors request that new procurement categories be associated with their account. Those procurement categories can then be used by the related procurement and sourcing processes. (Learn more in Procurement catalogs overview.)

Category requests are initiated by vendors in the **Vendor information** workspace. They are then submitted to your agency for review and approval. Approved categories are added to the list of procurement categories for the vendor's account.

You can also manually add procurement categories to vendor accounts. Learn more in Approve vendors for specific procurement categories.

Vendor collaboration requirements

Before a vendor can interact with category requests, it must be set up for vendor collaboration.

The vendor must have at least one vendor collaboration user. Only vendor users with the *Vendor admin (external)* security role can create and submit category requests.

Learn more in Set up and maintain vendor collaboration.

Set up the Vendor category request workflow

The *Vendor category request* workflow must be set up in the procurement and sourcing workflows. Vendors will submit new category requests that you can review and approve. Requested procurement categories are added to a vendor account after a category request is approved.

The following example shows how to set up a simple *Vendor category request* workflow that has a single approver. You must review your internal processes to determine the appropriate workflow setup for your agency.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing workflows**.
2. On the Action Pane, select **New**.
3. In the dialog box, select **Vendor category request workflow** to open the workflow editor.
4. Sign in to the workflow editor.
5. On the Action Pane, select **Properties**.
6. On the **Properties** page for the workflow, in the **Submission instructions** field, enter instruction text. The instructions will be visible to vendors when they submit a category

request.
7. Close the **Properties** page.
8. Drag the **Approve new category request** workflow element onto the canvas.
9. Drag a link from the **Start** workflow element to the **Approve new category request** workflow element.
10. Drag a link from the **Approve new category request** workflow element to the **End** workflow element.
11. Double-tap (or double-click) the **Approve new category request** workflow element.
12. Select and hold (or right-click) the **Step 1** workflow element, and then select **Properties**.
13. On the **Properties** page for the workflow element, in the **Work item subject** field, enter subject text. This text will be shown as the value of the **Subject** field on the **Work items assigned to me** page.
14. In the **Work item instructions** field, enter instruction text. The instructions will be visible to approvers when they select **Workflow** on the Action Pane of a category request.
15. In the left pane, select **Assignment**.
16. On the **Assignment type** tab, set **Assign users to this workflow element** to *User*.
17. On the **User** tab, in the **Available users** list, select an approver. For example, select a user in the Procurement department.
18. Select the right arrow button (>) to move the approver to the **Selected users** list.
19. Close the **Properties** page.
20. Select **Save and close**.
21. In the **Version notes** field, enter notes about the workflow.
22. Select **OK** to save the workflow.
23. If you're ready to start to test the workflow, select **Activate the new version**. Otherwise, select **Do not activate the new version**.
24. Select **OK** to complete the workflow setup.

> **Tip**
>
> If your agency doesn't require approval of category requests, you should configure the workflow for automatic approval.

For more information about how to set up workflows, see Workflow system overview.

Create and submit a category request

This section describes how vendors can use the **Vendor information** workspace to create, edit, view, and submit category requests.

Start a new request

To start a new category request, follow these steps.

1. In the **Vendor information** workspace, select the **Category requests** tile.
2. On the **Category requests** page, on the Action Pane, select **New category request**.
3. In the **New category request** dialog box, find the category that you want to apply for by navigating the tree and/or using the filter at the top of the list. Select the checkbox for each relevant category.

   Note the following points:

   - Categories that are currently active on your vendor account are shown as selected and can't be removed.
   - You can request multiple procurement categories in a single category request.
4. Select **OK** to create the draft request.

   The new draft request now appears on the **Category requests** page.
5. Open the new draft request to review and edit it as required.

   - To view the list of categories that are currently included in the request, select the **Requested categories** FastTab.
   - To view your active categories, open the **Active categories** FactBox on the right side of the page.
   - To add a category to the request, select **Add** on the toolbar on the **Requested categories** FastTab.
   - To remove a category from the request, select the category on the **Requested categories** FastTab, and then select **Remove** on the toolbar.
   - To attach a document to the request, select **Attachments** on the Action Pane. Attached documents will be available to approvers when they review the category request.
   - To remove an attached document from the request, select **Attachments** on the Action Pane. Select the document to remove, and then select **Delete** on the Action Pane.
6. If you're ready to submit the request, select **Submit** on the Action Pane. Otherwise, just close the page and skip the remaining steps of this procedure. You can then return to the request later.
7. Read any submission instructions that appear, and then select **Submit**.
8. In the **Comment** box, enter any additional information that is required. Then select **Submit** to complete the request.

Edit a draft or recalled request

To edit a draft or recalled category request, follow these steps.

1. In the **Vendor information** workspace, select the **Category requests** tile.
2. Select the draft or recalled request to open it.
3. Edit the request as required, and then either close it or submit it as described in the previous procedure.

Submit a draft or recalled request

To submit a draft or recalled category request, follow these steps.

1. In the **Vendor information** workspace, select the **Category requests** tile.
2. Select the draft or recalled request that you want to submit.
3. On the Action Pane, select **Submit**.
4. Read any submission instructions that appear, and then select **Submit**.
5. In the **Comment** box, enter any additional information that is required. Then select **Submit** to complete the request.

   The status of the category request is changed to one of the following values:

   - *Pending review* – The request is in the workflow.
   - *Pending approval* – Approval is required.

Recall a submitted request that hasn't yet been approved

To recall a category request that has been submitted but hasn't yet been approved, follow these steps.

1. In the **Vendor information** workspace, select the **Category requests** tile.
2. Select the pending request that you want to recall.
3. On the Action Pane, select **Recall**.
4. In the **Enter a comment** box, enter any additional information that is required. Then select **Submit** to complete the request.

   The status of the category request is changed to *Canceled*. The request will remain in this status until you delete or resubmit it.

Delete a draft or recalled request

To delete a draft or recalled category request, follow these steps.

1. In the **Vendor information** workspace, select the **Category requests** tile.
2. Select the draft or recalled request that you want to delete.
3. On the Action Pane, select **Delete**.
4. Select **Yes** to confirm the deletion.

View completed requests

To view completed requests, open the **Vendor information** workspace and select the **Category requests** tile. Category requests that have been completed will have one of the following statuses:

- *Approved* – The request was approved. To view the newly added categories, go back to the **Vendor information** workspace, open the **More details** drop-down list in the left pane, and then select **Categories**.
- *Rejected* – The request was rejected. You can create a new category request as required.

Review category requests

This section explains how to approve, reject, and delegate category requests that vendors submitted, and how to view completed requests. These workflow actions are for the whole category request.

View category requests

To view category requests, follow these steps.

1. Go to **Procurement and sourcing** > **Vendors** > **Vendor collaboration requests** > **Category requests**.

   The **Category requests** page appears. The default page view shows category requests that have a status of *Pending action*.

2. To view all requests, select *All* in the **Show requests** field.

3. Open a request to review and edit it as required.

   - To view the list of categories that are currently included in the request, select the **Requested categories** FastTab.

- To view the active categories, open the **Active categories** FactBox on the right side of the page.
- If documents were submitted, the **Attachments** (paper clip) button on the Action Pane shows a count of the attached documents. To view attached documents, select the **Attachments** button. Then select the document to view and select **Open** to view it.
- To attach a document to the request, select **Attachments** on the Action Pane. Attached documents will be available to approvers when they review the category request.
- To remove an attached document from the request, select **Attachments** on the Action Pane. Select the document to remove, and then select **Delete** on the Action Pane.
- To view the workflow history, select **Workflow** on the Action Pane. In the workflow options, select **More** and then **Workflow history**. The **Workflow history** page appears.

Approve a pending category request

To approve a pending category request, follow these steps.

1. Go to **Procurement and sourcing** > **Vendors** > **Vendor collaboration requests** > **Category requests**.
2. Select the pending request to approve.
3. Review the category request.
4. Optional: On the **General** FastTab, in the **Reason code** field, select a reason code. Then, in the **Reason comment** field, enter a comment about the reason code.
5. On the Action Pane, select **Workflow**.
6. In the workflow options, select **Approve**.
7. In the **Comment** field, enter any additional information that is required. Then select **Approve** to complete the request.

   The status of the category request is changed to *Approved*, and the procurement categories are added to the vendor account.

Reject a pending category request

To reject a pending category request, follow these steps.

1. Go to **Procurement and sourcing** > **Vendors** > **Vendor collaboration requests** > **Category requests**.
2. Select the pending request to reject.
3. Review the category request.
4. On the Action Pane, select **Edit**.
5. On the **General** FastTab, in the **Reason code** field, select a reason code. Then, in the **Reason comment** field, enter a comment about the reason code.
6. On the Action Pane, select **Save**.
7. On the Action Pane, select **Workflow**.
8. In the workflow options, select **More** and then **Reject**.
9. In the **Comment** field, enter any additional information that is required. Then select **Reject** to complete the request.

   The status of the category request is changed to *Rejected*. At this point, the vendor can create a new category request as required.

Delegate a pending category request

To delegate a pending category request to another user, follow these steps.

1. Go to **Procurement and sourcing** > **Vendors** > **Vendor collaboration requests** > **Category requests**.
2. Select the pending request that you want to approve.
3. On the Action Pane, select **Workflow**.
4. In the workflow options, select **More** and then **Delegate**.
5. In the **User** field, select the user to assign the category request to for review.
6. In the **Comment** field, enter any additional information that is required.
7. Select **Delegate** to complete the delegation. The selected user can now review the category request.

View procurement categories for a vendor

To view procurement categories for a vendor after a category request is approved, follow these steps.

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.

2. On the **All vendors** page, select the vendor that you want to view procurement categories for.

3. On the Action Pane, open the **General** tab and, from the **Set up** group, select **Categories**.

   The **Categories** page appears. The **Procurement** FastTab shows procurement categories that were added through the category request.

4. On the **Procurement** FastTab, you can make changes if needed. For example, you can set the **Vendor category status** field to *Preferred*.

Vendor collaboration with customers

06/17/2025

This article describes how you can use vendor collaboration to work with customers in Microsoft Dynamics 365 Supply Chain Management. Vendors can complete a series of business processes from the following workspaces:

- **Purchase order confirmation** – Monitor and respond to purchase orders (POs).
- **Vendor bidding** – View requests for quotation (RFQs), and respond to them by entering bids.
- **Vendor information** – View and update vendor master data.
- **Invoicing** – Work with invoices. This article doesn't cover the **Invoicing** workspace. For more information about this workspace, see Vendor collaboration invoicing workspace.

Vendors can also monitor information about consignment inventory.

Working with POs in the Purchase order confirmation workspace

The **Purchase order confirmation** workspace lets you respond to the purchase orders (POs) that have been sent to you for review. It also lets you view information about POs that are awaiting action from the customer, and POs that have been confirmed but are still open.

There are three lists in the **Purchase order confirmation** workspace:

- **Purchase orders for review** – This list shows POs that have been sent to you and are awaiting a response from you. After you respond, the PO disappears from the list. If the customer sends you a new version of the PO before you've responded to the previous version, only the latest version is shown.
- **Awaiting customer action** – This list shows all the POs that you've responded to but that the customer hasn't yet confirmed. If you accept a PO, you can monitor it in this list until the status is changed to **Confirmed**. If you reject a PO or accept it with changes, you can monitor it here until the customer sends a new version.
- **Open confirmed purchase orders** – This list shows all the POs for your account that have a status of **Confirmed**. When products or services are fully received against the PO, the PO disappears from the list.

You can use the following pages to work with POs:

- **Purchase orders for review** – This page contains the same information as the **Purchase orders for review** list in the workspace. See the description earlier in this article.

- **Purchase order vendor confirmation history** – This page contains all POs and all versions of POs that have been sent to the vendor. It also contains all the responses that have been returned from the vendor.
- **Open confirmed purchase orders** This page contains the same information as the **Open confirmed purchase order** list in the workspace. See the description earlier in this article.
- **All confirmed purchase orders** – This page contains all the POs that have been confirmed. The POs that are shown on this page include POs where products or services have been received. You can use this list to monitor POs that you can send invoices for.

Responding to POs

The POs that the customer sends you to review appear in the **Purchase order confirmation** workspace and on the **Purchase orders for review** page. After you open a PO, you can accept it, reject it, or accept it with changes. There might be attachments on the PO header or on individual lines. Additionally, you can attach information to your response on the PO header or individual lines. For example, you might suggest a substitute item for one of the lines.

You can preview and print the PO as a PDF file by using the **Preview/Print** option. You can also use the **Display dimensions** action to hide or show the following dimension columns: **Site**, **Warehouse**, **Color**, **Size**, **Style**, and **Configuration**.

If you use the **Accept with changes** option, you can accept or reject individual lines. You can also make the following changes to lines:

- Change dates or quantities. To update the confirmed receipt date on all lines, use the **Update receipt date** option on the PO header.
- Split lines for different receipt dates or quantities.
- Substitute an item. In the **Line details** section, enter an item description and the item number in the **External** field.

You can't change pricing information or charges, but you can use notes to make suggestions for these changes.

If the customer sends you a new version of a PO, it will have a version suffix to indicate that it's a modified version of a PO that was previously sent. The **Purchase order vendor confirmation history** page lets you track the history of each order.

Monitoring consignment inventory

If you're using consignment inventory, you can use the vendor collaboration interface to view information on the following pages:

- **Purchase orders consuming consignment inventory** – POs for consignment inventory are generated when the customer takes ownership of the inventory. These consignment POs are shown only on this page. They aren't included on the **All confirmed purchase orders** page.
- **Products received from consignment inventory** – This page lists all the transactions where the ownership of products has been transferred to the company that is consuming the inventory. You can use this information to invoice the customer.
- **On-hand consignment inventory** – This page shows the on-hand consignment inventory that is owned by your company, but that is on hand at the customer's warehouse.

Working with RFQs in the Vendor bidding workspace

The **Vendor bidding** workspace lets you view the requests for quotation (RFQs) that your company has been invited to respond to. You can also respond to the RFQs.

The workspace also shows all the RFQs that you've lost or won. Additionally, if the system is configured for the public sector, the workspace shows the RFQs that are publicly available.

Viewing RFQs

Open the **Vendor bidding** workspace to access the following information:

- Select **New bid invitations** to see the RFQs that your company has been invited to respond to. From here, you can view an RFQ and start the bidding process. You can also see amended RFQs that a new bid must be submitted for.
- Select **Returned bids** to see the RFQs that the customer has returned to you so that you can provide more information or update the bid.
- Select **Bids in progress** to see the RFQs that you or a contact person who represents your company has been working on but hasn't yet submitted.
- Select **Awarded bids** to see when the customer has awarded at least one line item in your bid.
- Select **Lost bids** to see bids where all lines have been rejected.
- Select the **Request for quotations** link to see a list of all the vendor's RFQ invitations and any bids that have been submitted. The **Request for quotations** page lists all the RFQs that a vendor has been involved in. You can search by status.
- Select the **Declined bids** link to see a list of all the RFQs where a vendor's contact person has declined to bid.

Working with RFQs that are publicly available

People who work in the public sector can see open and expired RFQs that have been made available to the public.

- Select the **Open published requests for quotations** link to see a list of open RFQs that are available to the public. An open RFQ is an RFQ that hasn't yet expired. You can find the expiration date and time on the header of the RFQ.

  If you've been invited to bid, you can find the same RFQ on the **New bid invitations** page. Sometimes, you might want to bid on an open RFQ, but you haven't been invited to bid. In this case, you might be able to invite yourself, provided that the customer has enabled self-invitation for the RFQ case.

  The **New bid invitations** page provides a filter that lets you view the open RFQs and identify those that contain lines that match your approved procurement categories.

- Select the **Closed published requests for quotations** link to see a list of closed RFQs that are available to the public. A closed RFQ is an RFQ that has expired. You can find the expiration date and time on the header of the RFQ.

  A closed RFQ shows all vendor bids down to the line level. As bids are awarded or rejected, this information is reflected in the closed RFQ. Any attachments that are included in the bid are also available.

> **Note**
>
> This functionality is available only if the public sector configuration is turned on.

Bidding

- Select **Bid** to start to bid on an RFQ.

  When the editing is enabled for bid fields on the headers and lines of an RFQ, you can enter your bid directly in the line grid. You must also consider any additional bid information that should be added in the line details.

  When you start to work on a bid, it appears in the **Bids in progress** section.

  At any time before the expiration date, you can save a bid. You can then return later to finish and submit the bid. After you submit a bid, you can recall and update it up until the expiration date.

- Select **Reset from RFQ** to reset the data that you entered for a bid and revert to the original RFQ. You can reset the header or the line.

- Select **Add alternate** or **Remove alternate** in the line grid to work with alternates.

  Some RFQs allow for alternate bids. You can specify alternate bids only for lines of the **Category** type, because specific items can't be added as alternates.

- Select **RFQ attachment** or **RFQ lines attachment** to open any attachment that the customer has added to an RFQ. Select **Bid attachments** or **Bid line attachments** to upload attachments together with the bid.

  There might be questionnaires that you must answer before you're allowed to submit a bid.

- Select **Decline** if you don't want to bid. After you select **Decline**, you can't recall the action and enter a bid.

If an RFQ is amended, you must enter a new bid. You can find information about the amendment on the **Amendments** tab of the RFQ page. Amended RFQs appear on the **New bid invitations** page.

Accessing vendor master data in the Vendor information workspace

As a vendor, you can access part of the information that the customer maintains in the vendor master record. Therefore, you can keep the information up to date. To update the information, you must have a vendor admin (external) role.

The accessible information is the vendor name, addresses, contact information, contact persons and their contact information, identification numbers, tax registration numbers, procurement categories that the vendor is approved to sell to the customer in, and information about certifications.

Related information

- Manage vendor collaboration users

Maintain vendor bank account information

Vendors can use the Vendor collaboration functionality to maintain their bank account information. This article explains how to add and maintain bank information for vendors that you do business with.

After you give access to vendors, they can add information for new bank accounts. You can then review that information and complete the pre-note process, so that the new accounts are used for payments to those vendors.

You can maintain your vendor account in the **Vendor information** workspace. There, vendors select **More details** and then **Bank accounts** from the drop-down list. To add a new bank account, select **Add** above the bank account grid. In the **New** dialog box that appears, enter the following information:

- Bank account
- Bank name
- Bank account number
- Bank routing number
- Effective date
- Expiration date
- Comments (optional)

SWIFT and the International Bank Account Number (IBAN) codes are required for all non-US based companies. You can update the SWIFT and IBAN requirements parameter on the **Accounts payable parameters** page after the feature is enabled.

If there are any documents that are related to the specific certification, you can attach them by selecting **Document**.

Bank information that vendors enter on the page shows **Vendor** as the source. You can also enter bank account information on a vendor's behalf. That information appears here, and **Customer** is shown as the source. For more information, see Create a vendor bank account.

After an account is added, vendors can edit their bank's effective and expiration dates as required.

Vendor collaboration-generated bank changes page

After vendors update their bank information, you can see that information on the new **Vendor collaboration-generated bank changes** page, available under **Accounts payable** > **Inquiries** > **Vendor reports**. By default, the page shows all newly entered or modified bank records. The accounts payable clerk can view the changes and run the account information through the pre-note process to validate it. When this process is complete and the primary payment method is manually updated, you can select and mark the bank account as reviewed on the **Vendor collaboration-generated bank changes** page. This action removes the account from the default list.

To view all changes to a vendor's bank information, change the filters to view the page by vendor account, effective date range, and whether the changes are reviewed.

Last updated on 08/04/2026

Import vendor catalogs

Article • 11/18/2024

Vendor catalogs import

In Dynamics 365 Supply Chain Management, purchasing professionals can create and maintain catalogs for company employees to use when they order items and services for internal use. To create a procurement catalog, you can add the items and services that you want to make available to employees, either by importing the product catalog data or by manually adding the product catalog data to the product master.

You can upload catalog data submitted by a vendor from the Microsoft Dynamics 365 client.

The product data that a vendor submits to you, in the form of a catalog maintenance request (CMR) file, must be in XML file format. The CMR file should contain the details for the products that the vendor supplies to your company.

Import vendor catalog data

To import vendor catalog data, you must complete the following tasks:

1. Set up a project in the Data management workspace where you have defined your data mapping rules. Select **Data management** and then select **Set up roles for data projects**.

2. Set up a procurement category hierarchy, and assign your vendors to procurement categories. If you use commodity codes, add the commodity codes to the procurement categories. Learn more about setting up a procurement category hierarchy in Set up a procurement category hierarchy.

3. Configure the vendor for catalog import. Select a vendor, and then select **Procurement** > **Set up** > **Configure vendor for catalog import**.

4. Configure workflow for catalog import. Create a CMR file template and share this with your vendor.

5. Select **Procurement and sourcing** > **Catalogs** > **Vendor catalogs** to create a vendor catalog. The catalog maintenance request (CMR) files that you receive from your vendor are grouped in this catalog.

6. Upload the CMR file.

7. Review, approve, or reject the products in the vendor catalog. The products are automatically mapped to the procurement categories.

Approved products are added to the product master and are released to the selected legal entities. Only approved products can be added to the procurement catalog.

Generate a catalog import file template

The catalog import file template is an XSD file that you use to create a CMR file for a vendor's products. You can use the CMR file to create a new catalog, replace an existing catalog, or modify an existing catalog.

1. Go to **Procurement and sourcing** > **Catalogs** > **Vendor catalogs** and open the catalog that you want to work with.

2. Download a current catalog import template (XSD file). On the **Update catalog** page, on the Action Pane, on the **Catalogs** tab, in the **Related information** group, select **Generate catalog template** and select **Procurement category**.

    - With the **Procurement category** option, you can generate a catalog template that includes the procurement categories in which the vendor is authorized to provide products.

3. In the **Save as** dialog box, select the location where you want to store the catalog file template and save the file.

For more information and for examples, refer to this blog post: Vendor catalogs in Dynamics AX.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up an external catalog for PunchOut e-procurement

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

By using the external catalog, you can ensure that the product and price information that you subsequently process in Supply Chain Management is accurate and up to date.

The requisition can then be approved and converted to a purchase order and an order can be placed at the vendor.

When the external catalog is set up and an employee prepares a requisition, an option is available to redirect to an external site, the external catalog, and return the shopping basket that was created at the external site.

This communication is based on the cXML protocol and it needs to be set up between the systems of the buying and the selling organization.

To set up the communication, your vendor needs to provide information for you to use in the configuration of the external catalog, such as identity, domain of the buyer's company (for example, "DUNS" and "DUNS number"), credentials, and the URL to reach the vendor's catalog.

Set up an external catalog

Set up the external catalog so that when an employee enters a purchase requisition, they're redirected to an external site to select products.

The products that the employee selects from the external catalog are returned with up-to-date price information and from here, they can be added to the purchase requisition.

The intention isn't to enable employees to place an order on the external site.

When setting up the external catalog, make sure that the purpose of the site that the external catalog accesses is to collect quote information and not to place a real order.

To set up an external vendor catalog, complete the following tasks:

1. Set up a procurement category hierarchy. Learn more in Set up policies for procurement category hierarchies.
2. Register the vendor in Supply Chain Management. Before you can set up configurations to access an external vendor's catalog, set up the vendor and the vendor contact in Microsoft Dynamics 365. Add the external catalog's vendor to the selected procurement category. For more information about registering vendors, see Manage vendor collaboration users. Learn more about how to assign vendors to a procurement category in Approve vendors for specific procurement categories.
3. Make sure that the units of measure and the currency that the vendor uses are set up. Learn more about how to create a unit of measure in Manage units of measure.
4. Configure the external vendor catalog by using the requirements for your vendor's external catalog site. Learn more about this task in Configure the external vendor catalog.
5. Test the vendor's external catalog configurations to verify that the settings are valid and that you can access the vendor's external catalog. Use the **Validate settings** action to validate the request setup message that you've defined. This message should cause the vendors external catalog site to be opened in a browser window. During validation, you can't order items and services from the vendor. To order items and services, you must access the vendor's catalog from a purchase requisition.
6. Activate the external catalog by using the **Activate catalog** button on the **External catalogs** page. Activate the external catalog before employees can use it. You can deactivate the external catalog at any time.

Configure the external vendor catalog

This section gives more details about task 4 in the preceding section.

1. Enter a name and description for the vendor's external catalog. The name that you enter will appear on the cart that represents the external catalog that is shown to employees who creates a requisition. Employees can select on the cart to open the catalog on the vendor's external catalog site.

2. Add an image by using the **External catalog image** action. The image appears on the cart that represents the external catalog. Employees who create a requisition can see the cart. The image's width and height must be equal. Otherwise, the image doesn't display correctly.

3. Select whether the vendor's external catalog website should appear in the same browser window as the one where the employee created the requisition, or if it should open in a new window.

4. Select the vendor for the catalog. In the **Legal entities** list, there's a row for each legal entity where the vendor is set up. To allow users to request products directly from the vendor's catalog in some legal entities but not others, use the **Prevent access** or **Allow access** button for each legal entity where you want the catalog to be or not to be available.

5. In the **Default expiration (Days)** field, enter the number of days that a quotation received from the external catalog is valid and can be used to purchase from the external vendor. When a quotation is created and retrieved from the vendor's external catalog site, the quotation is valid as of the current system date and remains valid for the number of days that you enter in this field.

6. Select the **Add** button to start mapping the procurement categories to the external catalog. Then, in the Category name list, select a category. The list of categories is a superset of procurement categories that the vendor is mapped to in all the legal entities set up for the vendor.

   > ⓘ **Note**
   >
   > Procurement policies are used to allow or restrict access to categories for the buying legal entity or receiving operating unit. Punchout to an external catalog requires that you allow access to at least one of the procurement categories mapped to the catalog.

7. Set up the cXML setup request message that is sent to the vendor. The automatically generated message format is the minimal template that is required in order to start a session. Fill in values for the tags.

At any time, you can reload the system-generated message template by selecting **Restore message format**.

If you restore the message format, the current message is replaced by the automatically generated message format, which has empty tags.

cXML setup message

The following table describes the tags included in the template:

Expand table

Field	Description
< Header >< From >< Credential domain="" >	The domain of the buyer's company.
< Header >< From >< Credential>< Identity >< /Identity >	The identity of the buyer's company.
< Header >< To >< Credential domain="" >	The domain of the vendor's company.
< Header >< To >< Credential>< Identity >< /Identity>	The identity of the vendor's company.
< Header >< Sender >< Credential domain="" >	The domain of the buyer's company.
< Header >< Sender >< Credential >< Identity >< /Identity>	The identity of the buyer's company.
< Header >< Sender >< Credential >< SharedSecret >< /SharedSecret >	The shared secret for the buyer's company.
< Request deploymentMode="" >	The test or production deployment.
< Request >< PunchOutSetupRequest >< SupplierSetup >< URL >< /URL>	The URL of the vendor's PunchOut endpoint.

Extrinsic elements

An extrinsic element is additional information, such as a user name that is based on a user that punches out. The extrinsic element is set when the PunchOut occurs and it can be sent in the request setup message.

Your vendor might require an extrinsic element in the setup request. In that case, add the extrinsic element to the list of extrinsic elements in the **Message format** section of the **External catalog** page.

Specify a name for the extrinsic element that the vendor can recognize and map it to a value. The options for values are: User name, User email, or Random value. For more information about the cXML protocol, see the cXML.org website.

Post back message

The post back message is the message that is received from the vendor when the user checks out from the external site and returns to Supply Chain Management. Post back messages can't be configured. The messages are based on the cXML protocol definition. Here's the information that can be part of the post back message that is received on a requisition line.

Expand table

Message received from vendor	Copied to requisition line
< ItemIn quantity="" >	Quantity
< ItemIn>< ItemID >< SupplierPartID >< /SupplierPartID >	External item ID
< ItemDetail>< UnitPrice >< Money currency="" >	Currency
< ItemDetail >< UnitPrice >< Money >< /Money >	Unit price
< ItemDetail >< Description ShortName="" >	Product name
< ItemDetail >< Description >< /Description >	Included in item description; Product name if ShortName isn't specified.
< ItemDetail >< UnitOfMeasure >< /UnitOfMeasure >	Unit
< ItemDetail >< Classification >< /Classification >	Included in item description
< ItemDetail >< Classification domain="" >	Included in item description

Delete an external catalog

Delete an external catalog by using the **Delete** action on the page.

If a product is requested from the external vendor catalog, the external vendor catalog can't be deleted. Instead, the status of the external vendor catalog is set to inactive. If you want to remove access to the external vendor's catalog site, but not delete it, change the external catalog status to **Inactive**.

Related information

- Purchasing cXML enhancements
- Use external catalogs for PunchOut e-procurement

Last updated on 05/06/2026

Purchasing cXML Enhancements

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

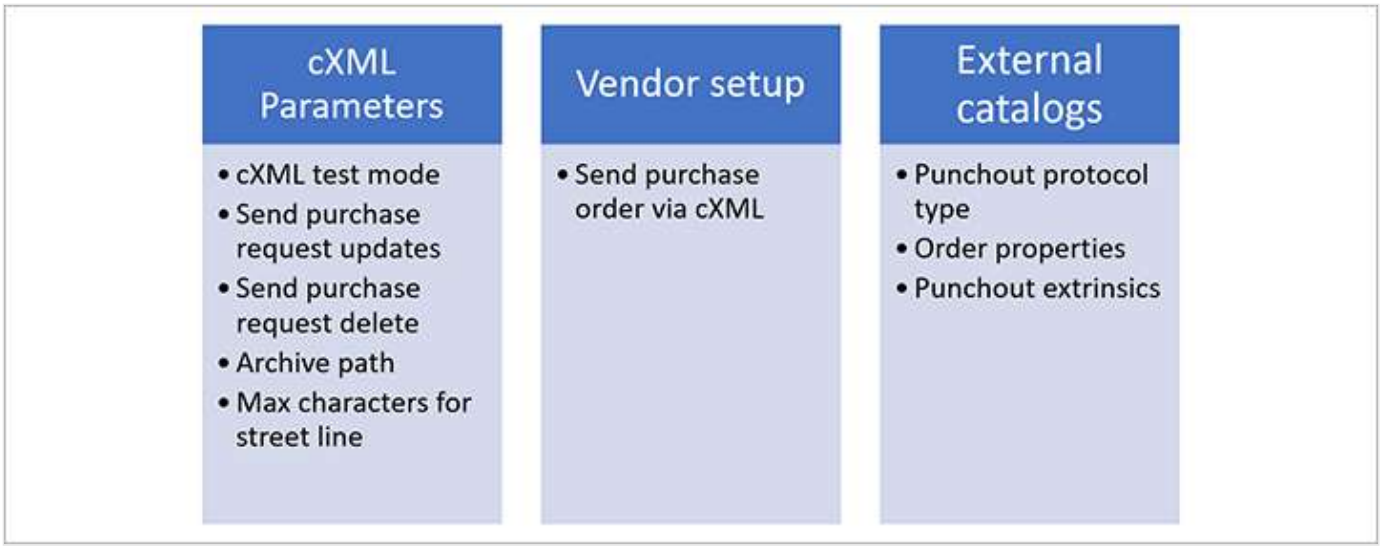
The *Purchasing cXML Enhancements* feature builds on the external catalog functionality used for purchase requisitions. The external catalog functionality is known as *PunchOut*. Although a purchase order doesn't have to originate from a purchase requisition, there must be a connection between the vendor on a purchase order and the parameters that are used to send the purchase order document.

Prepare to use the Purchasing cXML Enhancements feature

Before using this feature for the first time, configure settings in the following three areas:

- **cXML parameters** – Set up global parameters for the functionality for sending purchase orders.
- **Vendor setup** – If you want to use commerce eXtensible Markup Language (cXML) by default for all new purchase orders created for any vendor, set the **Send purchase order via cXML** option to *Yes* for that vendor.
- **External catalogs** – Use the new **Order properties** settings to define the format of the purchase order document and how it's sent.

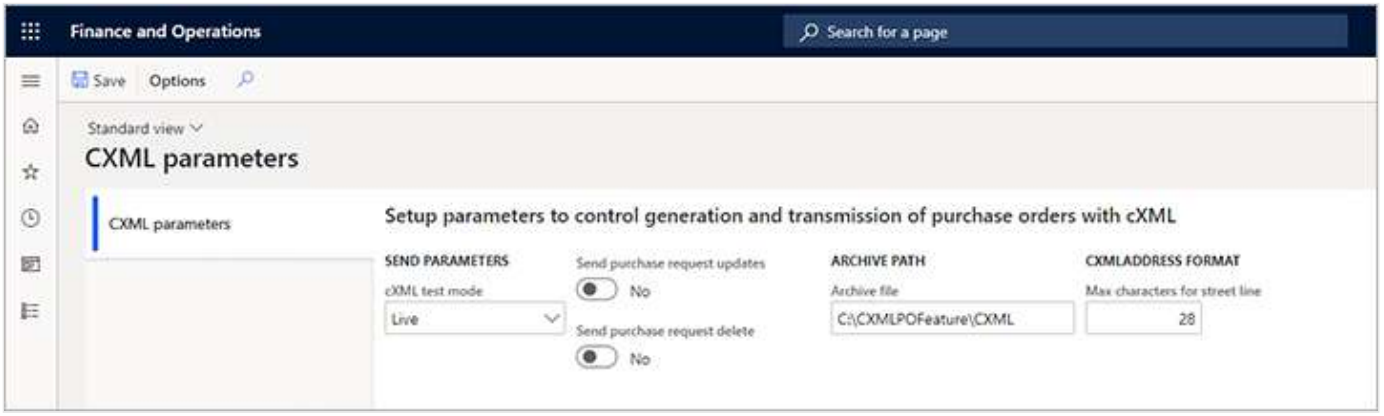
The following illustration summarizes this configuration.



In addition, set up the Purchase order request batch job. This batch job sends the confirmed purchase orders.

Set up global cXML parameters

Use the **cXML parameters** page to make a few global settings that apply to the functionality for sending purchase orders.



Go to **Procurement and sourcing** > **Setup** > **cXML management** > **cXML parameters**, and set the following parameters:

- **cXML test mode** – This global parameter affects the way that purchase orders are physically sent from the batch job. Select one of the following values:
  - **Test** – In this mode, the batch job can be running, and the XML document for the message is generated, but it isn't sent. Instead, it's saved on the purchase order request for review purposes. This mode is helpful when you're in an initial implementation, and you want to see how data is entered in the cXML message. You might also want to generate sample messages that you can send to vendors for initial validation.

    - **Live** – In this mode, the feature uses the external catalog settings to physically transmit each document to the vendor.

- **Send purchase request updates** – Set this option to *Yes* to send an update message for purchase orders. Set it to *No* to prevent the message from being sent. Most vendors prefer not to receive update messages. Instead, they require that customers contact them by phone or email if a purchase order should be changed. This parameter is a global parameter, and no override can be specified on the external catalog for each vendor. A purchase order is marked as updated if you post a second confirmation on a purchase order, but the first confirmation is already sent and acknowledged by the vendor. If there's a second confirmation, but the first confirmation isn't sent, the second confirmation is treated as a new document. You can confirm a purchase order as many times as you want until one confirmation is sent. The next confirmation is then treated as an update message.

- **Send purchase request delete** – Set this option to *Yes* to send a delete message for purchase orders. Set it to *No* to prevent the message from being sent. Most vendors prefer not to receive delete messages. Instead, they require that customers contact them by phone or email if a purchase order was sent by mistake. This parameter is a global parameter, and no override can be specified on the external catalog for each vendor. A purchase order is marked as deleted if you cancel the purchase order in Supply Chain Management.

- **Archive file** – Specify the file path where you want to export and save archived cXML documents. The path is used when you run the purge function from the **Purchase order request** page.

- **Max characters for street line** – Enter the maximum number of characters that can be used in the street field for addresses in the cXML document. This global parameter affects all vendors unless an override is specified on the external catalog properties.

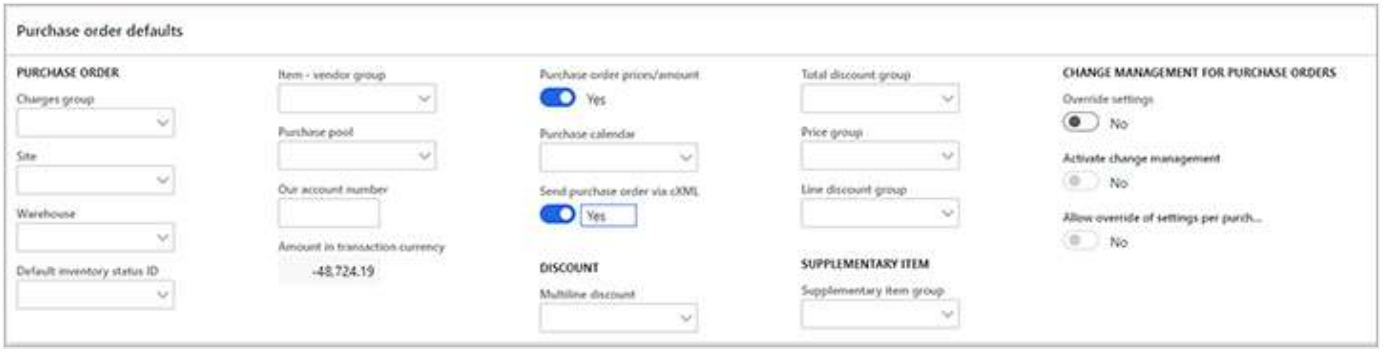
Set up vendor purchase orders to use cXML

Every time that you confirm a purchase order where the **Send purchase order via cXML** option is set to *Yes*, the system automatically generates the cXML message and delivers it to the vendor that is associated with that purchase order. There are two ways to control this option for your purchase orders:

- To set up a vendor so that it automatically uses cXML for all new purchase orders that are created from a requisition, go to **Procurement and sourcing** > **Vendors** > **All vendors**, and select or create a vendor to open its details page. Then, on the **Purchase order defaults**

FastTab, set the **Send purchase order via cXML** option to *Yes*. If cXML should also automatically be used for new purchase orders that are **not** created from a requisition, you must also set the **ENABLEMANUALPO** order property to *True* for the related external catalog, as described in the Set the order properties section later in this article.

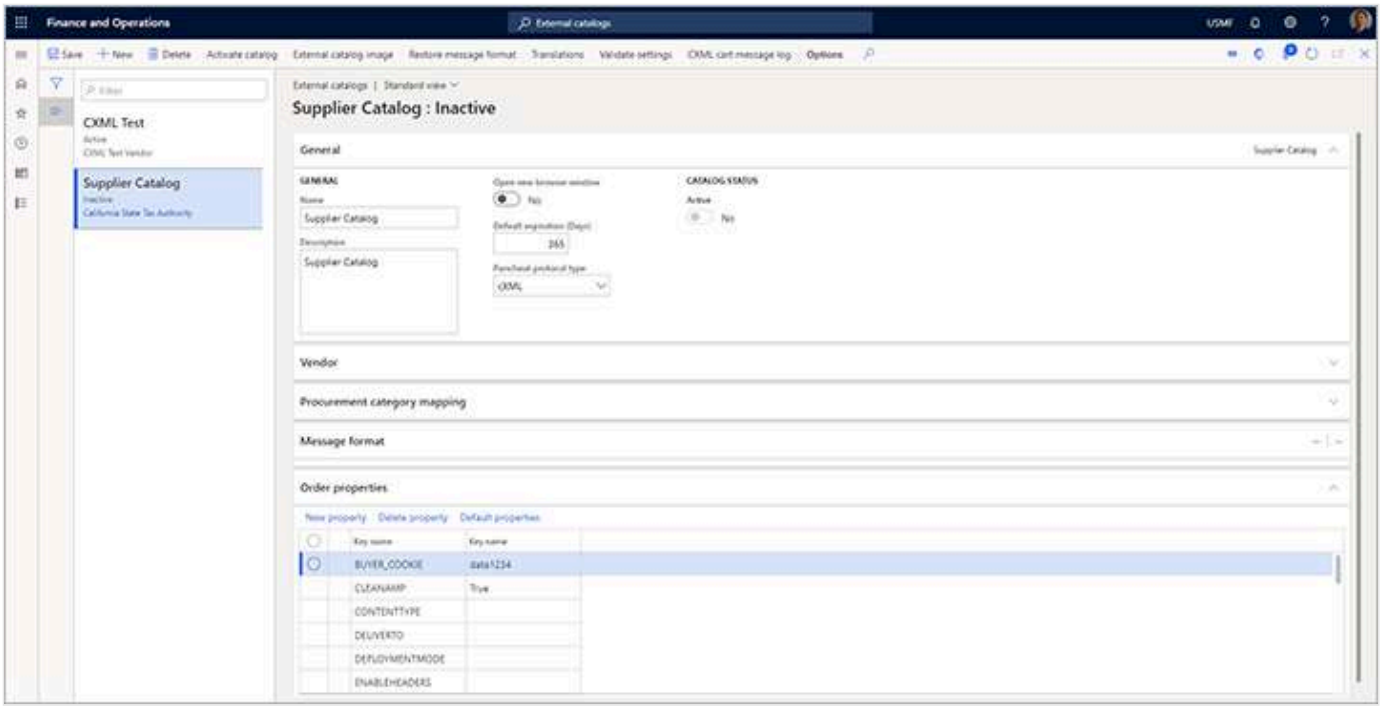
- For individual purchase orders, go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**, and select or create a purchase order to open its details page. Switch to the **Header** view, and then, on the **Setup** FastTab, set the **Send purchase order via cXML** option as required.



Set up an external catalog to use cXML

On the **External catalogs** page, for each of your catalogs, you can set up the PunchOut functionality and the functionality for sending purchase orders. To find the relevant settings, go to **Procurement and sourcing** > **Catalogs** > **External catalogs**. Start by setting up each catalog as usual. This process includes assigning a vendor, selecting the categories that the vendor is allowed to supply, and activating the catalog. Then configure the additional settings that are described in this section.

> **Note**
>
> When you confirm a purchase order that can be sent via cXML, the system looks up the vendor that is associated with the purchase order and then finds the first active external catalog that is associated with that vendor. The system then uses the settings from that external catalog to send the purchase order. If multiple external catalogs are set up, the system uses only the first external catalog that it finds, based on the vendor on the purchase order. Therefore, create just one external catalog for each vendor.



Set the PunchOut protocol type

On the **General** FastTab of the **External catalogs** page, set the **Punchout protocol type** field to *cXML*. This option will be the only available option, unless a partner has extended the functionality and provides an additional option in the extension.

If you use the catalog for PunchOut, set up the message format. The message format establishes the connection to the vendor in the PunchOut transaction from the requisition. When a purchase order is sent, the order properties establish the connection with a vendor.

Set the order properties

The **Order properties** FastTab for external catalogs provides a grid where you can define the order properties. It also provides a toolbar. This toolbar contains the following three buttons that you can use to manage the order properties:

- **Default properties** – When you're setting up a new catalog, select this button to add a predefined collection of commonly used properties to the grid.
- **New** – Add a new property to the grid.
- **Delete** – Delete the currently selected property from the grid. If you accidentally delete a default property, select **Default properties** to restore all the missing properties.

Every time that you add one or more properties to the grid, use the right column to specify a value for each.

Use the default properties in the following way:

- **BUYER_COOKIE** – This tracking field can be used to indicate specific information for your company. Unless you have an agreement with the vendor about how this property is used, it doesn't have very much meaning when sending a purchase order. Therefore, you should set it to a simple value.

- **DELIVERTO** – When the shipping address is entered in the document from the purchase order, the **Attention information** field is used to set the **DeliverTo** field in the XML message. If you require that this value be a requester name, and you'll set the requester field on the purchase order header, enter the value *REQUESTER* for this property, so that the requester name will be entered in the **DeliverTo** field in the XML. In this case, the primary email address and phone number that are used will be from the requester instead of the orderer.

- **DEPLOYMENTMODE** – Set this property as required by the vendor. The values are usually *PRODUCTION* or *TEST*. Set the value based on your communication with the vendor. Usually, it must match the intended system behind the **ORDERCHECKURL** value that the vendor indicates as a test or production system.

- **FIXEDBILLADDRESSID** – When the **addressID** field in the XML message is set, it picks up the location that is specified on the address. If the ID value that you've communicated to the vendor differs from the value on the address location for some reason, you can force an override by specifying the value here. The assumption is that you'll use only one address with the vendor, and that the address is set up in the vendor's system. The billing address is the primary invoice address that is specified for the legal entity in Supply Chain Management.

- **FIXEDSHIPADDRESSID** – When the **addressID** field in the XML message is set, it picks up the location that is specified on the address. If the ID value that you've communicated to the vendor differs from the value on the address location for some reason, you can force an override by specifying the value here. The assumption is that you'll use only one address with the vendor, and that the address is set up in the vendor's system. The shipping address is the address that is specified on the header of the purchase order. Most vendors accept only header addresses, not line addresses. Although there are fields for line addresses in the XML, they'll be set to the header address.

- **FROM_DOMAIN** – Enter the value that is used to send purchase order documents. This value is supplied by your vendor.

- **FROM_IDENTITY** – Enter the value that is used to send purchase order documents. This value is supplied by your vendor.

- **ORDERCHECKURL** – Enter the URL to transmit the purchase order documents to. This URL starts with `https://` and is supplied by your vendor.
- **PAYLOAD_ID** – Enter a prefix value for the payload ID, as required for the business processes that are in place for the current vendor.
- **SENDER_DOMAIN** – Enter the value that is used to send purchase order documents. This value is supplied by your vendor.
- **SENDER_IDENTITY** – Enter the value that is used to send purchase order documents. This value is supplied by your vendor.
- **SHARED_SECRET** – Enter the value that is used to send purchase order documents. This value is supplied by your vendor.
- **STREETLENGTH** – Enter a number that represents the maximum number of characters that the vendor will accept as a street name. If a value is entered here, it overrides the value that is specified on the **cXML parameters** page. The system removes line breaks and spaces to try to fit the standard address in Supply Chain Management into the number of characters that is specified here. Any additional characters are truncated.
- **TO_DOMAIN** – Enter the value that is used to send purchase order documents. This value is supplied by your vendor.
- **TO_IDENTITY** – Enter the value that is used to send purchase order documents. This value is supplied by your vendor.
- **USERAGENT** – Enter a value to identify the system that you're using. For example, enter *Dynamics 365 Supply Chain Management*.
- **VERSION** – Enter a cXML version number, if the vendor requests this information. The default version is *1.2.008*. This version is stable and is accepted by most vendors.
- **RESPONSETEXT** – Enter any custom text that you expect the vendor to return in the cXML response message after the order is sent. In this way, the system can mark the message as *Acknowledged*. If the response doesn't match standard text or the customer text that you enter here, the request is marked as *Error*.
- **RESPONSETEXTSUB** – Set this property to *TRUE* if you want to search the vendor response text for the values that are specified in the **RESPONSETEXT** field. For example, the vendor might return a long string that includes "OK" in the response. In this case, you can enter *OK*

in the **RESPONSETEXT** field and set **RESPONSETESTSUB** to *TRUE* to search for "OK" anywhere in the response. The order can then be set to *Acknowledged*.

- **CONTENTTYPE** – In a typical catalog setup, you don't have to set this property. If you receive a server 500 error from a vendor's system when you send a purchase order, you can do testing by setting this property to *FALSE*. That value changes a setting in the web request and might enable the message to be sent for some platforms.

- **ENABLEHEADERS** – Set this property to *TRUE* to send headers together with the purchase order. You must set this property only if the vendor requires it. If you set this property to *TRUE*, add extra custom properties that are based on the names that the vendor provides, and prefix them with *H_*. Typical examples include **H_USERID**, **H_PASSWORD**, **H_RECEIVERID**, and **H_ACTIONREQUEST**. The following custom properties are included in the default properties:
  - **H_USERID** – If the trading partner requires that you send a user ID as part of the URL to submit a purchase order, enter the value here.
  - **H_PASSWORD** – If the trading partner requires that you send a password as part of the URL to submit a purchase order, enter the value here.

- **ENABLEMANUALPO** – If this property is set to *TRUE*, when users manually create purchase orders (that is, when they don't start from a requisition), those purchase orders inherit the setting of the **Send purchase order via cXML** option from the vendor. If this property isn't set, or if it's set to *FALSE*, the **Send purchase order via cXML** option isn't set on the purchase order header for manually created purchase orders. For purchase orders that are created from a requisition, the setting of the **Send purchase order via cXML** option is always inherited from the vendor, regardless of the setting of this property. Learn more in the Set up vendor purchase orders to use cXML section earlier in this article.

- **PUNCHOUTPOONLY** – If this property is set to *TRUE*, only purchase requisition lines that are created from the PunchOut process set the **Send purchase order via cXML** option on the purchase order header. In addition, the purchase requisition line type of all lines on the purchase order must be *External catalog item*. Otherwise, the cXML purchase order can't be created.

- **PUNCHOUTSHIPTO** – If this property is set to *TRUE*, the default address of the legal entity is added to the PunchOut setup request message when the user opens an external catalog. This address is added as the **ShipTo** address. Vendors use the **ShipTo** address to show pricing based on the company location.

- **TRACEPUNCHOUT** – Set this property to *TRUE* if you receive an error message when you try to browse to an external catalog from the requisition. Trace information is then filled in for the **PunchOutSetupRequest** and **PunchOutResponse** messages that are sent between Supply Chain Management and the vendor site. You can view this information on the **cXML cart message log** page, which you can open from the **External catalog setup** page for the vendor catalog that you're having issues with. Set this property to *TRUE* only if you're troubleshooting, because it creates a large performance overhead on the database for each PunchOut. Learn more in the View the cXML cart message log for external catalog PunchOut section later in this article.

- **TRACESEND** – Set this property to *TRUE* to collect more information about purchase orders sent using cXML. To view the collected information, go to the **Purchase order request** page. Set this property to *TRUE* only while troubleshooting cXML functionality because it generates a large performance overhead on the database for each purchase order that you send.

- **REPLACENEWLINE** – Set this property to *TRUE* if you're having an issue because a vendor's system is sending a **PunchOutResponse** message that includes newline characters (\n). This issue might occur if the vendor's messages are parsed through middleware or a procurement hub. If you're having an issue with a new vendor setup, set the **TRACEPUNCHOUT** property to *TRUE* to view the **PunchOutResponse** message and determine whether the XML tags are broken up by newline characters.

- **POCOMMENTS** – Set this property to *TRUE* if you want the cXML document to include notes that are attached to the purchase order in Supply Chain Management. The attachment text is included in the header comments in the purchase order message. For more information about how the system selects and processes these attachments, see the Attach notes to a purchase order section later in this article.

- **VENDCOMMENTS** – Set this property to *TRUE* if you want the cXML document to include notes that are attached to the purchase order in Supply Chain Management. The attachment text is included in the header comments in the purchase order message. For more information about how the system selects and processes these attachments, see the Attach notes to a purchase order section.

- **CLEANAMP** – Set this property to *TRUE* if you receive an error message when you try to do a PunchOut to a vendor, and the vendor's return URL includes incorrectly encoded ampersands (&).

- **PUNCHOUTTZ** – Set this property to *TRUE* if the vendor that you're working with uses the International Organization for Standardization (ISO) 8601 standard to do a specific validation of the date/time of PunchOut request message. Supply Chain Management uses the Coordinated Universal Time (UTC) date in the **PunchOutSetupRequest** message. Therefore, when you set this property to *TRUE*, *+00:00* is added to the date/time format.

- **WMSADDRESSID** – Set this property to *TRUE* to use the warehouse number on the purchase order header as the **addressID** value in the **ShipTo** address for the purchase order request that is sent to the vendor. If you set a value for the **FIXEDSHIPADDRESSID** property, that value takes precedence over this value. Therefore, use one property or the other to set the **addressID** value in the **ShipTo** address for the document.

- **PUNCHOUTSHIPTOUSER** – This property works together with the **PUNCHOUTSHIPTO** property. If **PUNCHOUTSHIPTO** is set to *TRUE*, the address for the legal entity is picked up. If **PUNCHOUTSHIPTOUSER** is set to *TRUE*, the primary address from the PunchOut user is used instead of the legal entity address.

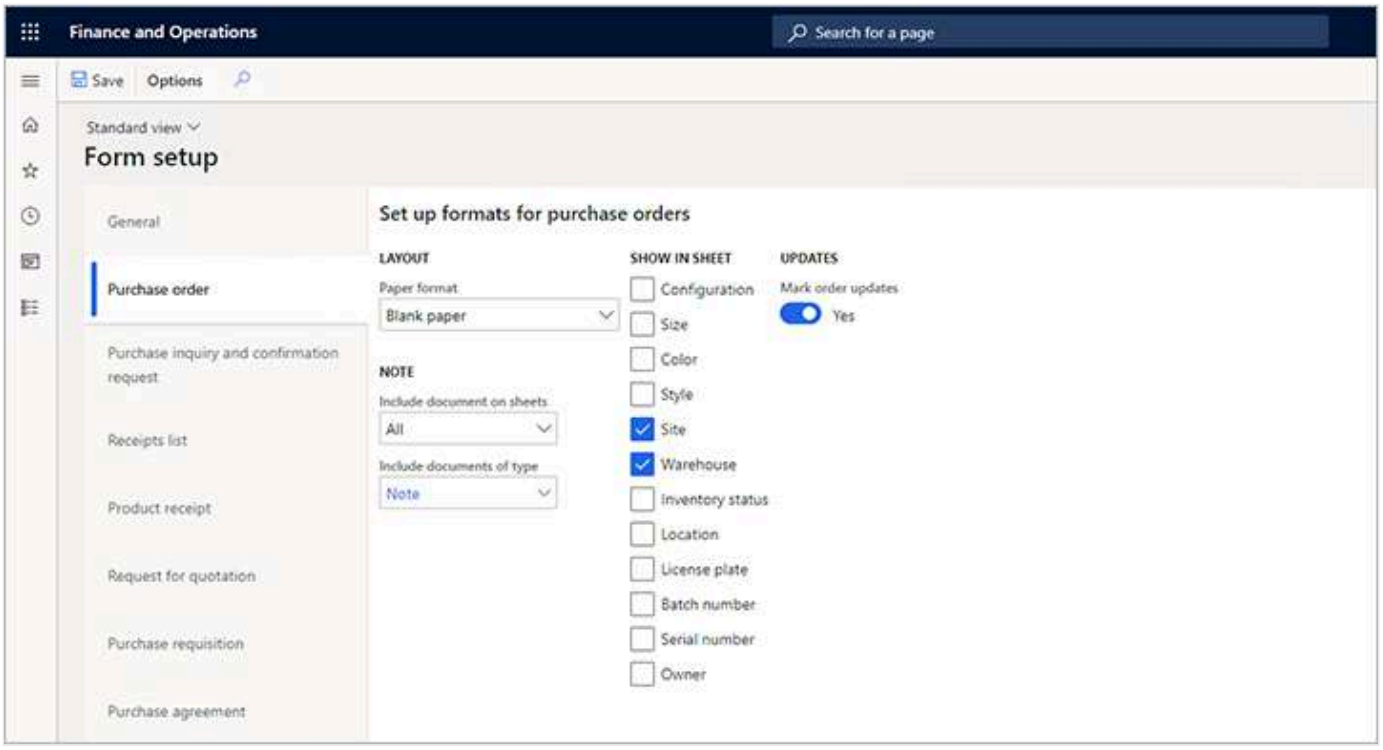
Activate the external catalog

After you set up all the properties and configure other settings for your external catalog, return to the **General** FastTab of the **External catalogs** page, and set the **Active** option to *Yes*.

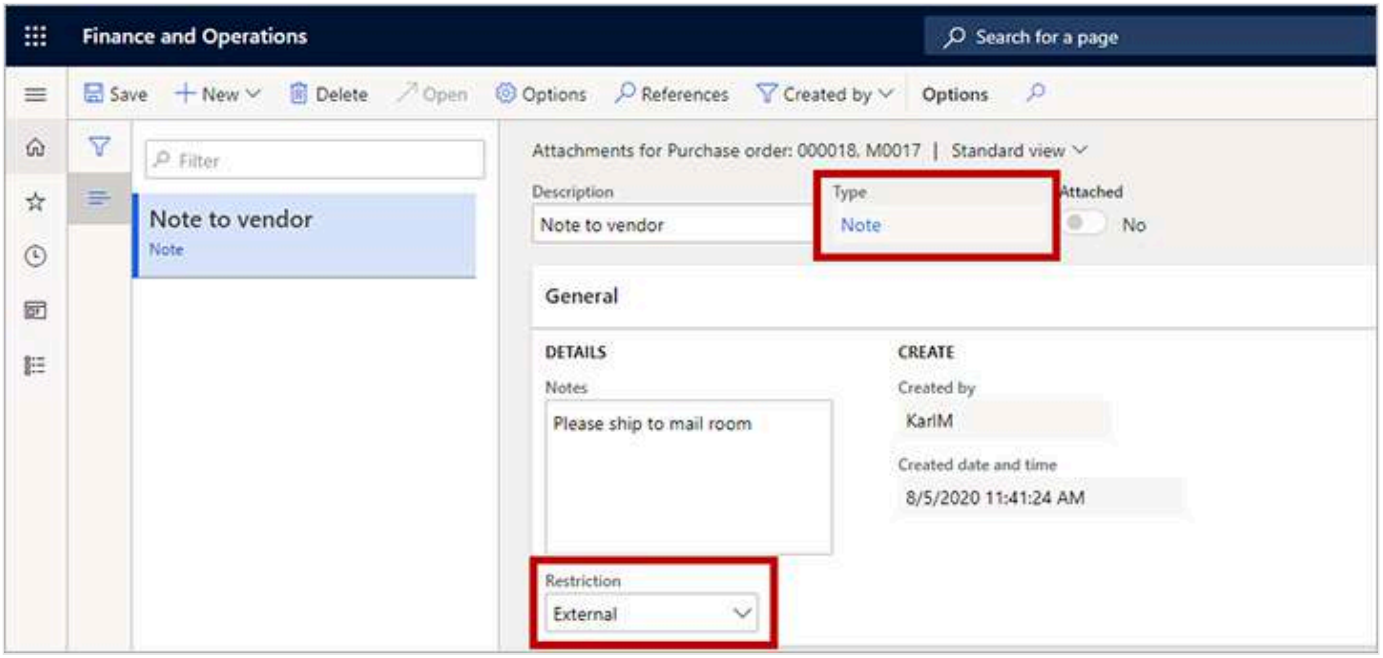
Attach notes to a purchase order

As mentioned in the Set the order properties section, if you want your delivered cXML to include text from notes attached to the relevant purchase order and/or vendor records, set the **POCOMMENTS** and **VENDCOMMENTS** properties to *TRUE* in the external catalog setup. This section provides more detail about how the system selects and processes these attachments.

To set the types of notes that the system looks for, go to **Procurement and sourcing** > **Setup** > **Forms** > **Form setup**. Then, on the **Purchase order** tab, set the **Include documents of type** field to the type of note that you want to include. Only text notes are included, not document attachments.



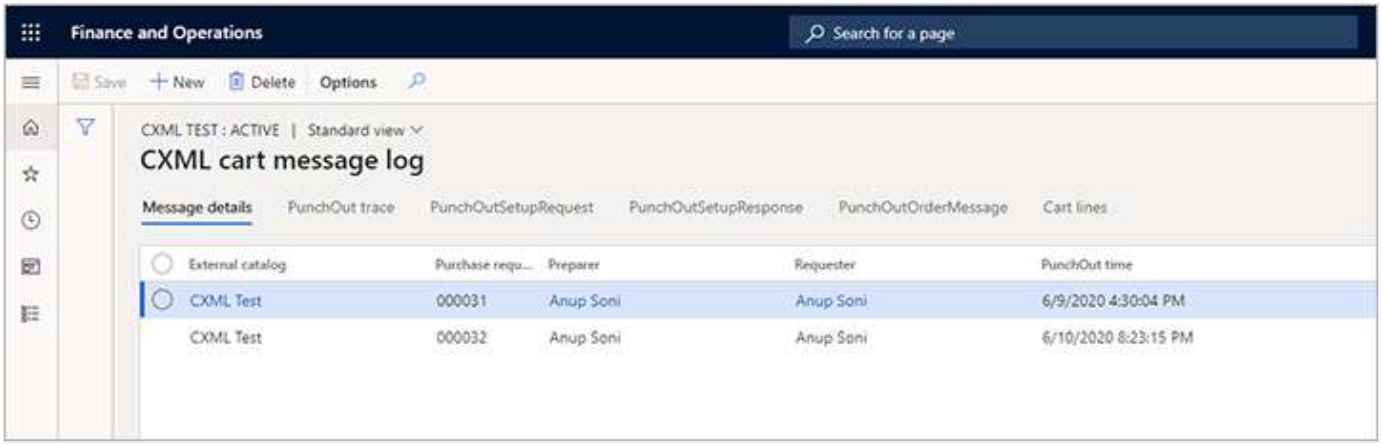
Attachments are included with a purchase order only if their **Type** field matches the value that you select in the **Include documents of type** field, and if their **Restriction** field is set to *External*. To create, view, or edit the attachments for a purchase order, go to **Procurement and sourcing** > **All purchase orders**, select or create a purchase order, and then select the **Attachments** button (paperclip symbol) in the upper-right corner.



View the cXML cart message log for external catalog PunchOut

When you set the **Punchout protocol type** field to *cXML* for an external catalog, the system will capture a message log of the carts that come back from vendors. This log can be used for troubleshooting and other data purposes.

To open the log for an external catalog, select the relevant catalog, and then, on the Action Pane, select **cXML cart message log**. The **cXML cart message log** page shows a list of the carts that are returned, the XML that is related to those carts, and the lines that are created on the related purchase requisition.



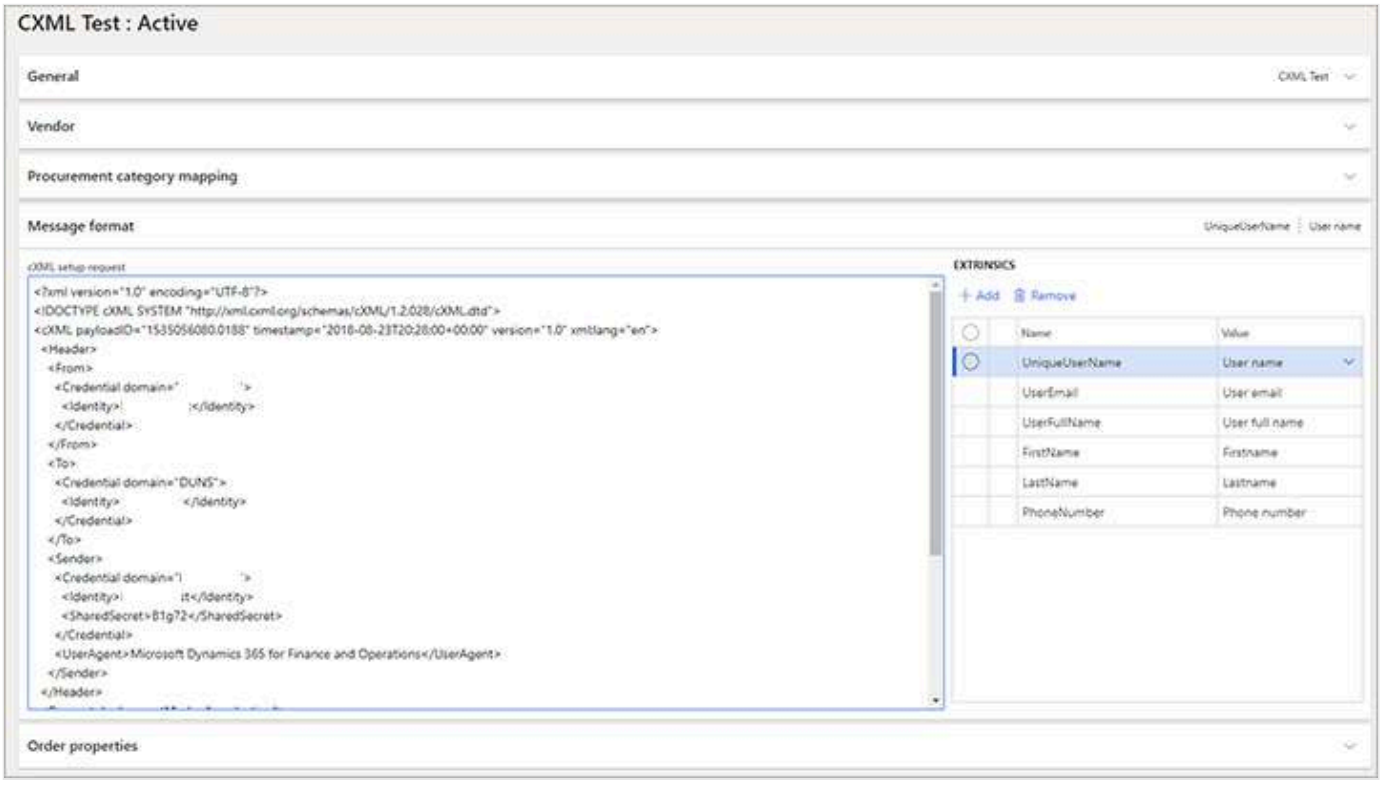
Set the extrinsic elements for external catalog PunchOut

When you set the **Punchout protocol type** field to *cXML* for an external catalog, you can add custom extrinsic elements that are inserted at the correct place in the request XML message.

To add extrinsic elements to an external catalog, follow these steps:

1. Go to **Procurement and sourcing** > **Catalogs** > **External catalogs**.
2. Select the relevant catalog.
3. On the **Message format** FastTab, in the **Extrinsics** section, select **Add** to add a row to the grid for each extrinsic element that you want to include. On each row, set the following fields:
    - **Name** – Enter a name for the element. This value becomes the value of the **name** attribute for the **Extrinsic** XML element that each row creates. Usually, you work with your vendor to agree on the name for each extrinsic element.
    - **Value** – Select a value for the element. This value becomes the value of the XML element that each row creates. The following values are available:

- **User name** – Use the name of the user who is doing the PunchOut. This name is the sign-in name for Supply Chain Management.
- **User email** – Use the email address of the user who is doing the PunchOut. This address is the address that the user sets up on the **Account** tab of the **User options** page.
- **Random value** – Use a unique random value.
- **User full name** – Use the full name of the contact person who is associated with the user who is accessing the external catalog.
- **Firstname** – Use the given name of the contact person who is associated with the user who is accessing the external catalog.
- **Lastname** – Use the family name of the contact person who is associated with the user who is accessing the external catalog.
- **Phone number** – Use the primary phone number of the contact person who is associated with the user who is accessing the external catalog.



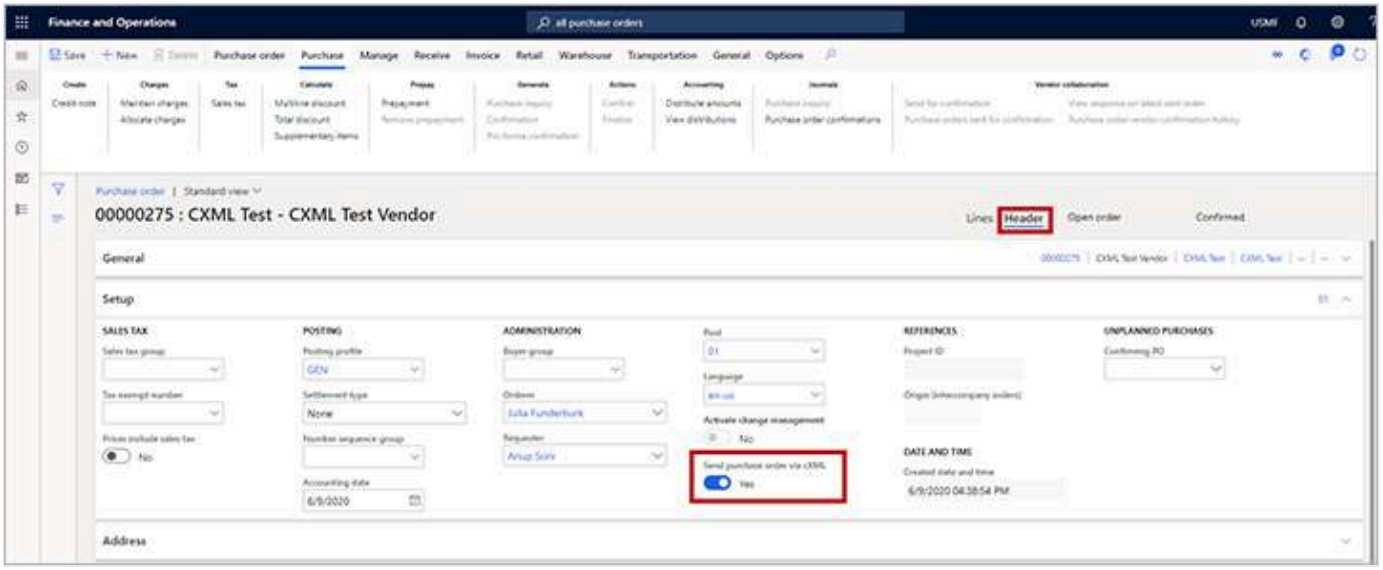
The user or admin doesn't see the extrinsic elements, because they aren't added until the user does a PunchOut. They're automatically inserted between the **BuyerCookie** and **BrowserFromPost** elements in the cXML setup request message. Therefore, you don't have to set them manually in the XML when you set up the external catalog.

```
<Request deploymentMode="production">
  <PunchOutSetupRequest operation="create">
    <BuyerCookie></BuyerCookie>
    <BrowserFormPost>
      <URL></URL>
    </BrowserFormPost>
```

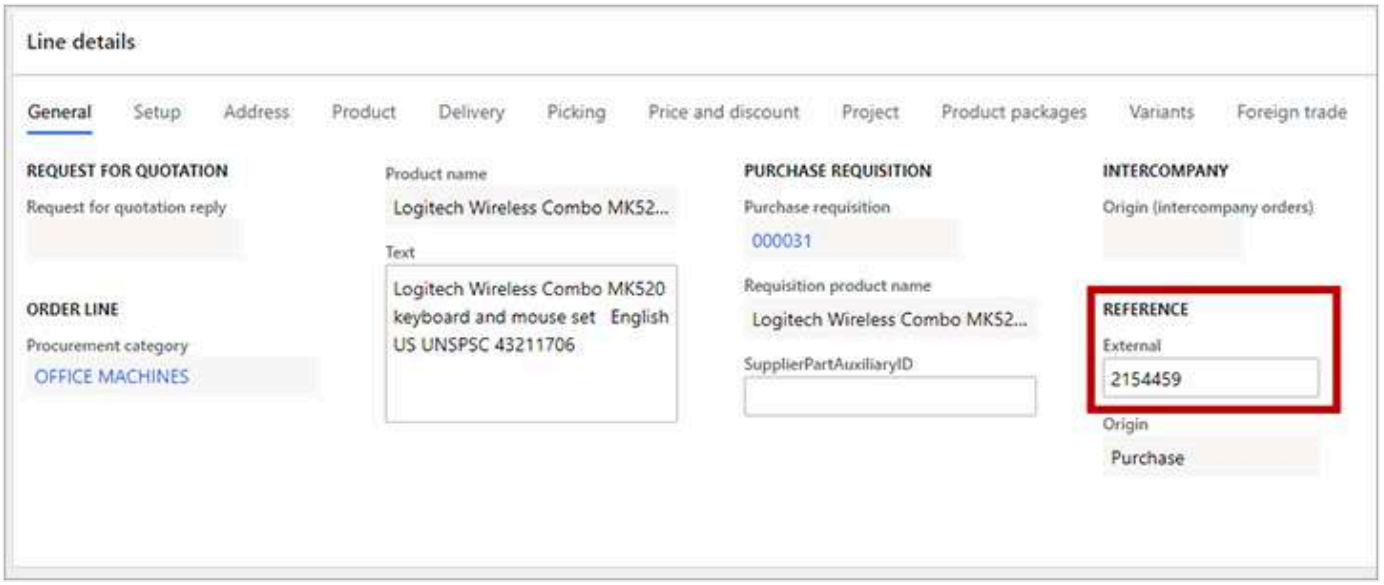
```
<Request deploymentMode="production">
  <PunchOutSetupRequest operation="create">
    <BuyerCookie></BuyerCookie>
    <Extrinsic name="UniqueName"></Extrinsic>
    <Extrinsic name="UserEmail"></Extrinsic>
    <BrowserFormPost>
      <URL></URL>
    </BrowserFormPost>
```

Create and process a purchase order

When you create a purchase order for a vendor, the purchase order inherits the setting for the **Send purchase order via cXML** option from that vendor. However, the setting remains available on the **Setup** FastTab in the **Header** view of the purchase order, so that you can change it later as required.



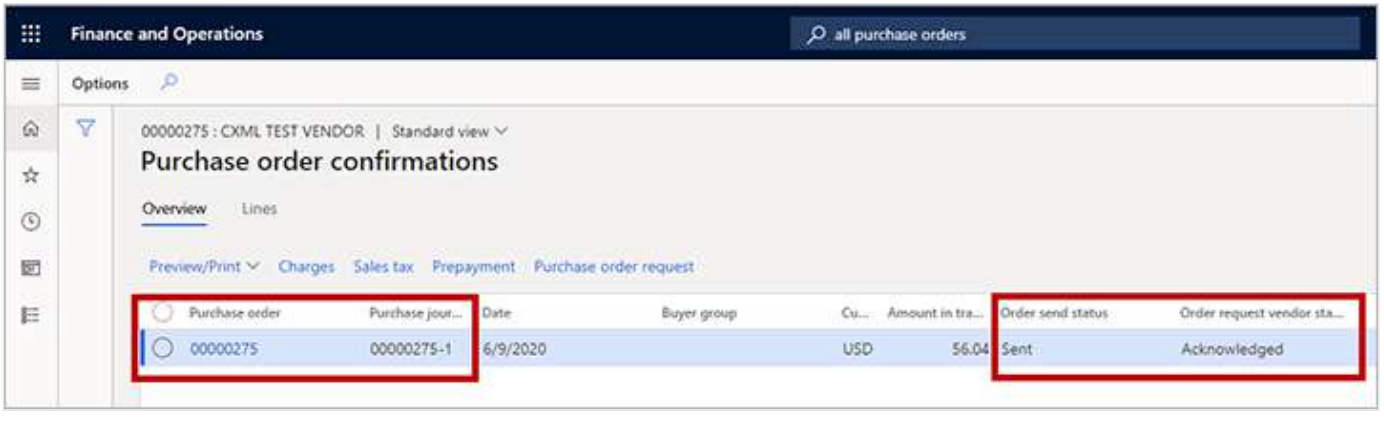
When you create a purchase order from a purchase requisition that came from a PunchOut flow, all the required line details are filled in. You can then manually add purchase order lines or copy them from other purchase orders. Be sure to set all required fields. These required fields include the external reference number, which is the vendor number that used in the cXML message.



When you finish filling in all the details for the purchase order, confirm it. The system doesn't send a message unless you confirm the purchase order. To confirm a purchase order, on the Action Pane, on the **Purchase** tab, in the **Actions** group, select **Confirm**.

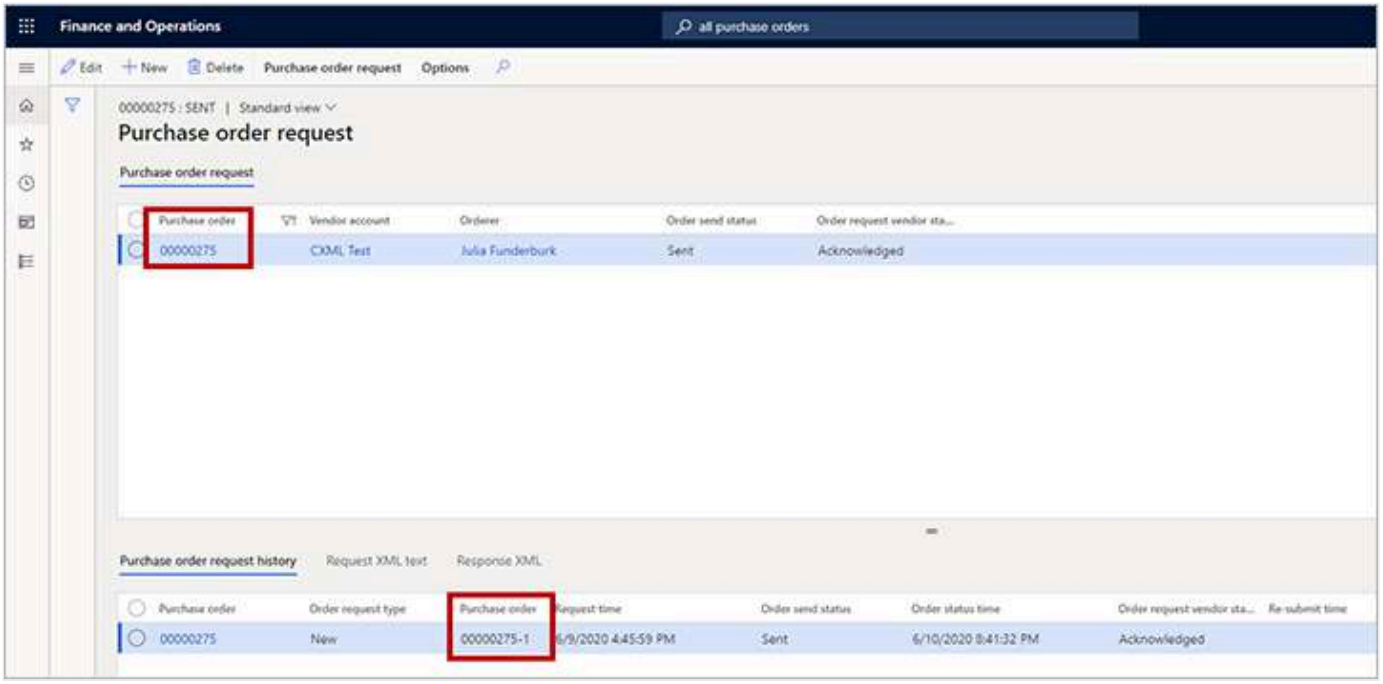
After you confirm the purchase order, you can view the status of the confirmation through the **Purchase order confirmations** journals. On the Action Pane, on the **Purchase** tab, in the **Journals** group, select **Purchase order confirmations**.

Each purchase order can have many confirmations. Each confirmation has an incremental number. In the following illustration, the purchase order is *00000275*, and the confirmation is *00000275-1*. This numbering reflects standard Supply Chain Management functionality, where changes in a purchase order, and therefore the type of cXML message that should be sent to the vendor, are identified based on the confirmation. As the illustration shows, the **Purchase order confirmations** page also includes **Order send status** and **Order request vendor status** fields. For more information about the various status values that you might see on this page, see the Monitor purchase order requests section later in this article.

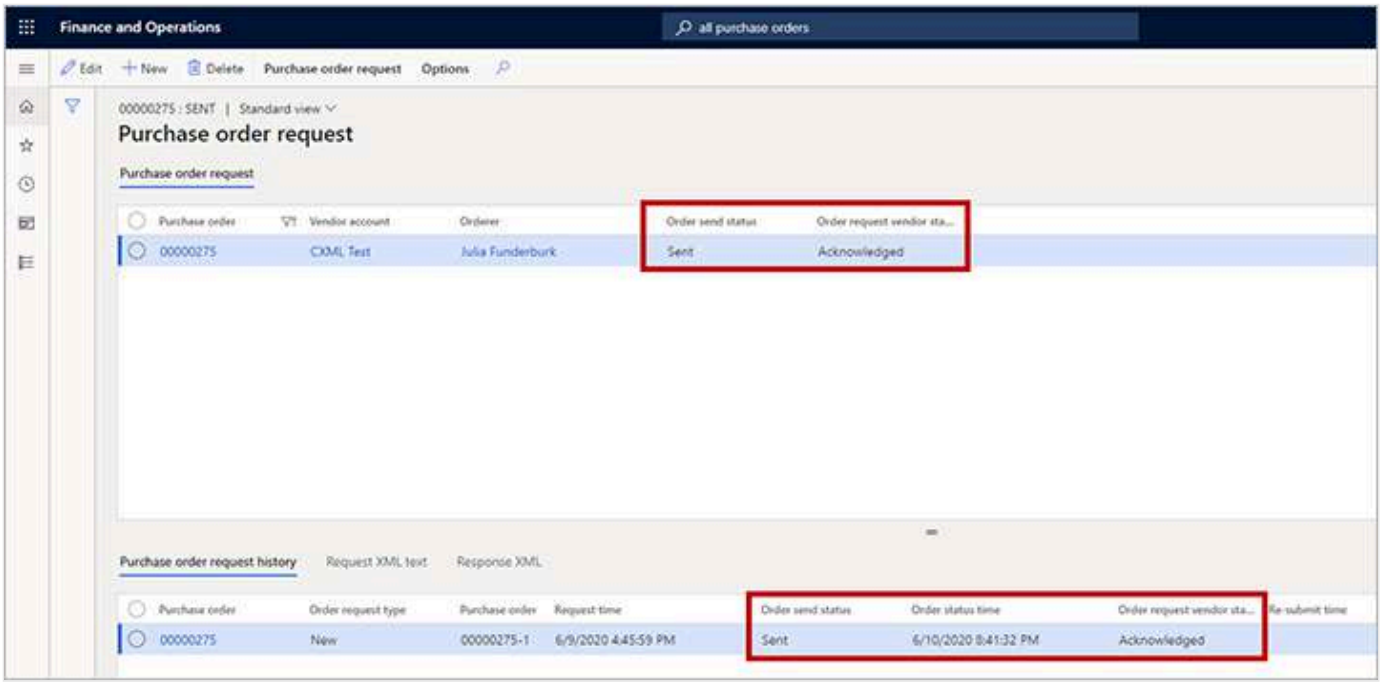


To view more information about the document, select **Purchase order request** above the grid.

The **Purchase order request** page includes two grids. The grid in the upper part of the page has one record for each purchase order that is marked for sending. The grid on the **Purchase order request history** tab in the lower part of the page might have several records for the selected purchase order, to indicate the status of each confirmation. The following illustration shows purchase order 00000275 in the upper grid and document 00000275-1 in the grid on the **Purchase order request history** tab.



If the batch job is set up and running, the document will be sent. You can view the status change after the document is sent. In the following illustration, the **Order send status** field is set to *Sent*. The **Order request vendor status** field is set to *Acknowledged* to indicate that the vendor received the document, and was able to read it and store it in its system. The grid on the **Purchase order request history** tab shows the time when the document was sent. For more information about the various status values that you might see on this page, see the Monitor purchase order requests section.



Schedule the purchase order request batch job

To activate the batch job for sending purchase order requests, go to **Procurement and sourcing** > **Setup** > **cXML Management** > **Purchase order request**. On the Action Pane, on the **Purchase order request** tab, in the **Batch** group, select **Submit job** to open the **Purchase request prepare and send** dialog box. Use this dialog box to set up the recurrence, just as you usually do for batch jobs in Supply Chain Management. Select an interval based on your order volume. Although you can run the batch job every minute, it's probably best to send batches throughout the business day, based on order receipt windows that match your vendors' schedules.

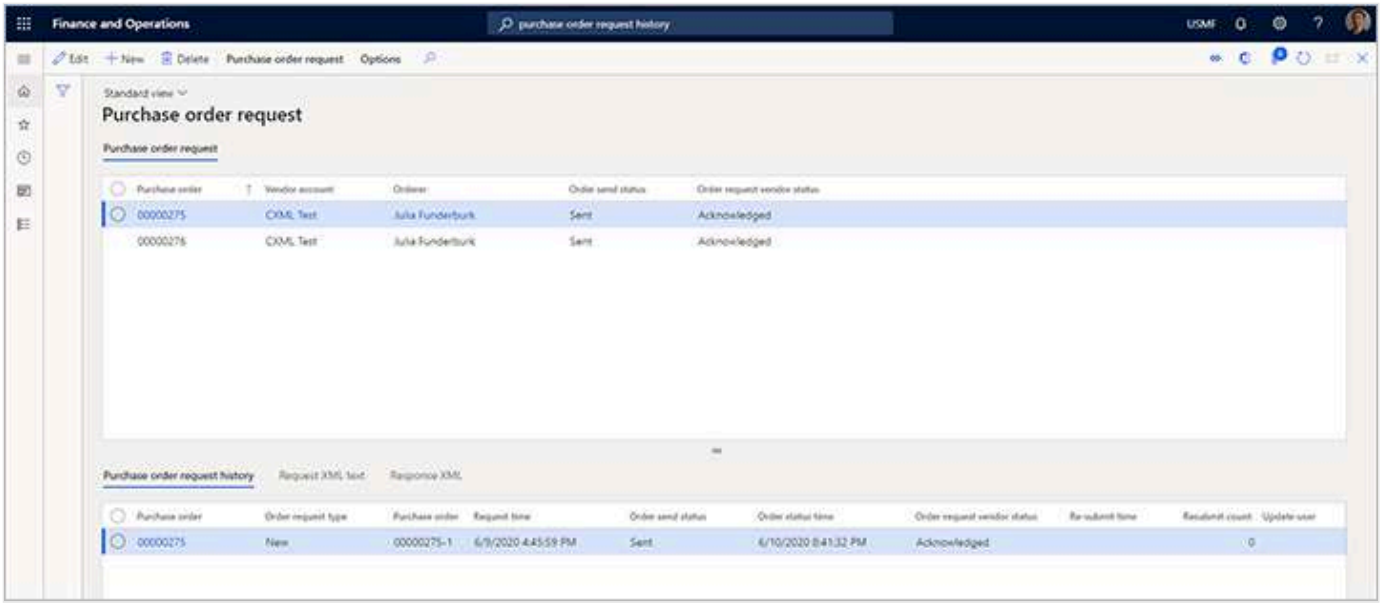
For example, your vendor has a policy that states that all orders that are received by 1 PM are shipped on the same day. In this case, you might run the batch only a few times during the morning to communicate any orders that you have. The remaining orders are sent the next day. This decision is purely a business decision. You can review it and set the parameters for it accordingly.

The process looks for purchase order request documents that have a status of *Waiting*. If you have an order that you must send to a vendor immediately, you can select **Submit job** and set the **Batch processing** option to *No*.

Monitor purchase order requests

View the status of a purchase order

When orders that can be sent via cXML are confirmed, they go into *Waiting* status. As described in the Create and process a purchase order section, you can view the purchase order status on the **Purchase order request** page. Each purchase order request can have one of several statuses, depending on its parameters and data. This section describes the various status types and the values that they can have. This information can help you manage issues and understand the status of your purchase orders.



The grid in the upper part of the **Purchase order request** page might show the following status values:

- **Order send status** – This field can have one of the following values:
  - **Waiting** – The document is waiting to be sent.
  - **Sent** – The document is sent.
  - **Stopped** – The document is marked as *Stopped* before it's sent. To mark a document as *Stopped*, select **Stop** on the Action Pane of the **Purchase request** page.
  - **Archive** – The document is purged. To purge a document, select **Purge** on the Action Pane of the **Purchase request** page.

- **Order request vendor status** – This field can have one of the following values:
  - **Waiting** – The document is waiting to be sent.
  - **Acknowledged** – The vendor acknowledges to have received the document. You can review the detailed XML message returned from the vendor by selecting the **Response XML** tab in the lower part of the page.
  - **Error** – The document is sent to the vendor, but an error occurs. You can review the error message by selecting the **Response XML** tab in the lower part of the page.

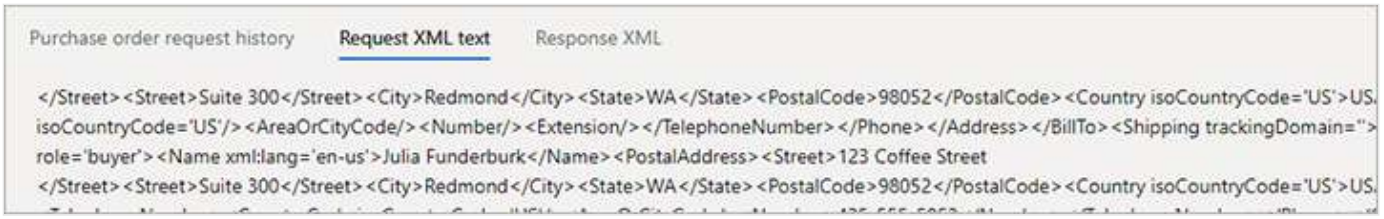
The grid on the **Purchase order request history** tab in the lower part of the **Purchase order request** page might show the following values:

- **Order request type** – This field can have one of the following values:
  - **New** – The line is marked as new immediately after the purchase order is confirmed.
  - **Update** – If a confirmation has is already sent and acknowledged by the vendor, the next confirmation is marked as *Update*. Updates are sent only if the **Send purchase request updates** option is set to *Yes* in the global cXML parameters.
  - **Delete** – If a confirmation is already sent and acknowledged by the vendor, but the purchase order is canceled, the confirmation that's waiting is marked as *Delete*. Deletions are sent only if the **Send purchase request delete** option is set to *Yes* in the global cXML parameters.
- **Request time** – The time of creation of the order confirmation. You can compare the request time with the order status time to determine the time between confirmation and vendor acknowledgment.
- **Order send status** – This field is the same as the **Order send status** field in the upper part of the page. It's repeated here to make it easier to view the status at the confirmation level. If the **Order request type** field is set to *New*, and the purchase order is reconfirmed before a confirmation is sent, the **Order send status** field is set to *Archive*.
- **Order status time** – The time when the purchase order request history record was last updated. (The updates include status changes.)
- **Order request vendor status** – This field is the same as the **Order request vendor status** field in the upper part of the page. It's repeated here to make it easier to view the status at the confirmation level.
- **Re-submit time** – The time when the record was resubmitted.
- **Re-submit count** – The number of times that the record is resubmitted. A high number indicates multiple failures, and therefore might indicate an issue with either your data setup or your vendor's data setup.

View the XML for a purchase order request message

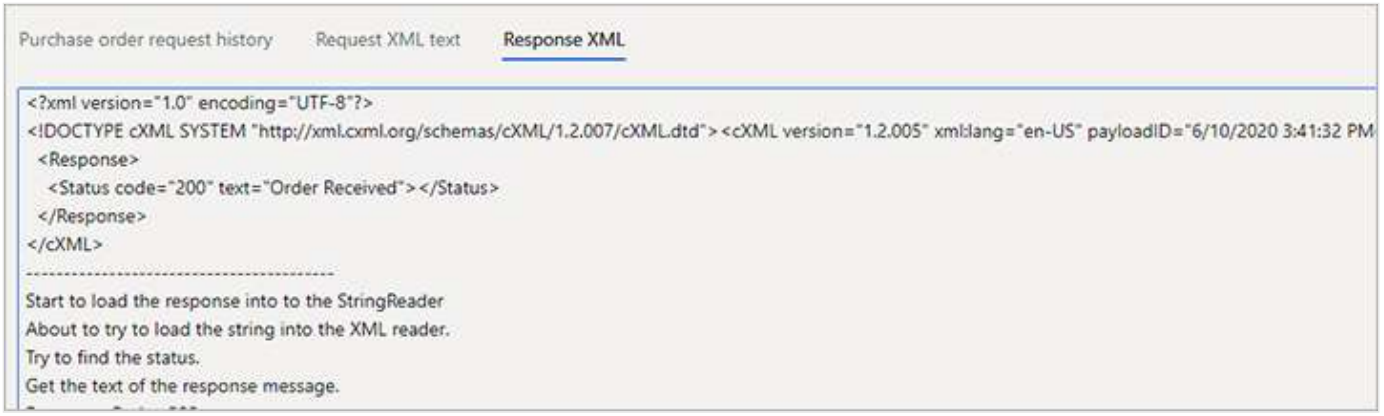
To view the XML for the purchase order request message, select the **Request XML text** tab at the bottom of the **Purchase order request** page. The information on this tab can help during testing

or error validation. To make the information easier to read, you can view it as a formatted message. Copy the contents of the tab to a text file, and then view it in an XML editor.



View the details of the vendor response

To view the content of a vendor acknowledgment or error response, select the **Response XML** tab at the bottom of the **Purchase order request** page.



Related information

- Set up an external catalog for PunchOut e-procurement
- Use external catalogs for PunchOut e-procurement

Last updated on 05/06/2026

Use external catalogs for PunchOut e-procurement

06/17/2025

By using external catalogs for PunchOut e-procurement, you don't have to maintain information about your vendors' products in your own master data. Instead, the shopping cart on a vendor's website is converted to requisition lines that have the correct product information.

You should avoid maintaining the descriptions and prices of your vendors' products in your own product master data. Instead, use external catalogs for PunchOut e-procurement. Then, when employees create requisitions, they can *punch out* to a vendor's external catalog site (in other words, they leave your system and go to the vendor's site). The products that are added to the shopping cart on the vendor's website can then be converted to requisition lines. Therefore, you get the correct product information: product ID, name, price, and so on.

To use external catalogs, an employee must create a requisition on the **My purchase requisitions** page.

The employee who creates a requisition is referred to as the preparer of the requisition. As a preparer, you can complete the following tasks.

Use the **External catalogs** line action to open a page that includes all external catalogs that are available for the specified requester, buying legal entity, and receiving operating unit.

Depending on your permissions, change the requester, buying legal entity, and receiving operating unit. A change in those values might change the list of external catalogs that are available to a requester. The external catalogs that are available depend on the current active purchasing policies for the buying legal entity or the receiving operating unit. These policies can allow or prevent access to specific procurement categories. Therefore, the list of external catalogs that are mapped to these procurement categories can be affected.

For more information about policies, see Purchasing policies overview.

- To find external catalogs for specific procurement categories, enter text in the catalog search field.
- To add products from a vendor's external catalog on the vendor's website, select the external catalog. Then add products to the shopping cart, and check out. The shopping cart lines will be transferred to Microsoft Dynamics 365.

If there are multiple options for procurement categories, select the correct procurement category before you add the lines to the requisition. After lines have been added to a

requisition, you can add more lines without using external catalogs. Alternatively, you can continue to use external catalogs to add lines.

When the requisition is ready, use the **Workflow** > **Submit** action to submit it for approval.

Related information

- Set up an external catalog for PunchOut e-procurement
- Purchasing cXML enhancements

Approve vendors for specific procurement categories

Article • 01/24/2025

This article explains how to approve vendors for specific procurement categories in Dynamics 365 Supply Chain Management. When a purchase requisition is created, there may be a requirement to select an approved or preferred vendor, depending on how the purchasing policies are set up. This procedure shows you how to specify that a vendor is approved or preferred for a specific procurement category. This task would usually be carried out by a procurement professional. You can use this procedure in demo data company USMF.

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Select the vendor that you want to set as an approved or preferred vendor for a category.
3. On the Action Pane, select **General**.
4. Select **Categories**.
5. Select **Add category**.
6. In the **Category** field, select *OFFICE AND DESK ACCESSORIES (OFFICE AND DESK ACCESSORIES)*.
7. In the **Vendor category status** field, select *Preferred*. It's possible to specify more than one preferred vendor for a category.
8. Select **Save**.
9. Go to **Procurement and sourcing** > **Procurement categories**.
10. In the tree, select *CORP PROCUREMENT CATEGORIES\OFFICE AND DESK ACCESSORIES*.
11. Expand the **Vendors** section. Check if the vendor that you selected appears as a preferred vendor for the Office and desk accessories category. If you're running this procedure as a task guide, you may have to select the **Unlock** button to be able to scroll down to the list of vendors. It's also possible to add preferred and approved vendors on this page.
12. In the tree, expand *OFFICE AND DESK ACCESSORIES* and select *Scissors*.
13. Select *No* in the **Inherit vendors from parent category** field.
14. Select *Yes* in the **Inherit vendors from parent category** field.

Feedback

Was this page helpful? Yes No

Provide product feedback

Approve vendors for specific products

This procedure shows you how to approve vendors for specific products. When you approve vendors, you control which vendors can be used when adding the product to a purchase order. A purchasing manager typically carries out this task.

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, find and select the desired record.
3. In the list, select the link in the selected row.
4. Expand the **Purchase** FastTab. If there's a primary vendor shown in the **Vendor** field, add this vendor as an approved vendor in the following steps. Make a note of the vendor number, if one is shown.
5. On the Action Pane, select **Purchase**.
6. Select **Setup**.
7. Select **Add**.
8. In the **Vendor** field, enter or select a value. Select the approved vendor. At least one of the lines has to be the primary vendor if there was one in the product record. If you made a note of the vendor number earlier, select it here.
9. In the **Expiration** field, enter a date. Choose a date that is a few months ahead.
10. Select **Add**.
11. In the **Vendor** field, enter or select a value.
12. In the **Expiration** field, enter a date. Choose a date that is different from the previous expiration date.
13. Close the page.
14. On the Action Pane, select **Approved vendors**.
15. In the **Expiration** field, enter a date. This date acts as a filter so you can see who the approved vendors are, up to a certain date.
16. Close the page.
17. On the Action Pane, select **Effective period**.
18. In the **Show vendors expired by** field, enter a date. Use this page to identify vendors where the approval status expires after a certain date.
19. Close the page.
20. Select **Edit**.
21. In the **Approved vendor check method** field, select an option. This field allows you to select the policy for what should happen if the product is added to a purchase order line where the vendor isn't an approved vendor.

22. Select **Save**.
23. Close the page.
24. Close the page.
25. Go to **Procurement and sourcing** > **Vendors** > **Vendor/item relations** > **Approved vendor list by item**. This page gives you an overview of all products and the approved vendors.
26. Close the page.
27. Go to **Procurement and sourcing** > **Vendors** > **All vendors**. You can also start from a vendor and then go to the list of approved products for that vendor account.
28. In the list, find and select the desired record.
29. On the Action Pane, select **Procurement**.
30. Select **Approved vendor list by vendor**.
31. Close the page.
32. Close the page.

Last updated on 07/01/2026

Search for vendors

06/17/2025

Learn how to search for vendors based on specific criteria. This example shows you how to search for vendors that are approved for a particular procurement category and have their primary address in a specific country/region. You can run this procedure in demo data company USMF, or on your own data. This task would usually be carried out by a procurement professional.

1. Go to **Procurement and sourcing** > **Vendors** > **Vendor search**.
2. Select the plus icon to open the **Procurement category selection** page.
3. In the tree, select *CORP PROCUREMENT CATEGORIES\OFFICE MACHINES*. If you're running this procedure as a task guide, you might need to select the **Unlock** button before you can select the correct node. If you're not using USMF, select one of the categories that you have.
4. Select **Add**. It's possible to select more than one category here. If you do so, the search finds all the vendors that are approved for at least one of the categories.
5. Select **OK**.
6. In the **Country/region** field, type a value.
7. Select **OK**.

Set up permissions for ordering products on behalf of someone else

Article • 01/24/2025

This article explains how to grant workers permission to prepare purchase requisitions on behalf of other workers. In other words, a purchase requisition "preparer" can create a requisition for another "requester." The procedure also shows how to grant a worker permission to order items and services in different legal entities or operating units. Typically, these tasks are performed by a purchasing manager. You can use this procedure either on data for the USMF demo company or on your own data.

Grant permission to enter purchase requisitions on behalf of another worker

1. Go to **Procurement and sourcing** > **Setup** > **Policies** > **Purchase requisition permissions**. Make sure that the **Current view** field is set to *By preparer*. The list in the left pane shows the people who can be granted permission to prepare requisitions on behalf of other people.
2. Select the person to grant permission to (the preparer).
3. Select **Add**.
4. Find and select the person to add as a requester.

    - The requester is the person that the preparer can create requisitions on behalf of.
    - In the **Authorization** field, select *Specific* if the preparer should be able to create purchase requisitions on behalf of the selected worker. Select *Reporting* if the preparer should also be able to create purchase requisitions on behalf of all workers who report to that worker.

5. In the **Effective** field, enter a date.
6. In the **Expiration** field, enter a date.

View preparers who have permission to create purchase requisitions for a selected worker

1. In the **Current view** field, select *By requester*. This view shows a list of preparers who have been granted permission to create purchase requisitions on behalf of a selected worker.

2. Use the Quick Filter to find the worker that you just added as the requester.
3. Select the requester. The Preparer list shows the people who have permission to order items on behalf of the requester who is selected in the left pane. You can add additional preparers here. This view also lets you grant the requester permission to create requisitions in legal entities and operating units that aren't that person's primary legal entity or operating unit.

Feedback

Was this page helpful? Yes No

Provide product feedback

Supplier Engagement overview (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement solution provides a unified framework for managing interactions between your organization and its suppliers. It covers the full supplier lifecycle—from registration and onboarding to managing requests for quotation (RFQs), purchase orders, invoices, and consignment inventory. Suppliers enter and maintain their own data directly in the system, which reduces manual entry and helps keep information consistent across procurement, finance, and supply chain processes.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Supplier Engagement replaces the older vendor collaboration interface with broader capabilities, including global vendor data management, a dedicated Power Pages portal, and lifecycle processes such as qualification and termination. You can use both solutions in parallel during the transition. Learn more in How Supplier Engagement compares to the vendor collaboration interface.

Solution components

Supplier Engagement consists of three components that work together: the Supplier Engagement app, the supplier portal, and Supply Chain Management.

Supplier Engagement app

The Supplier Engagement app is a Microsoft Power Platform model-driven app for internal users such as procurement staff and supplier managers. It provides an overview of all suppliers across legal entities and supports the following activities:

- Registering new suppliers and managing the new-supplier lifecycle from qualification to approval
- Recording supplier capabilities and maintaining certificates
- Reviewing risks and tracking corrective actions
- Managing user accounts that grant access to the supplier portal

The app is connected to Supply Chain Management, which makes supplier information available in related business processes.

Supplier portal

The supplier portal is a Microsoft Power Pages site for external supplier representatives. Through the portal, suppliers can:

- View and respond to requests for quotation
- Review purchase orders
- Submit invoices
- Manage consignment inventory
- Update organization details and maintain certificates
- Provide compliance information when required

The portal serves as the primary channel of communication between suppliers and your organization. Information entered by suppliers is available in Supply Chain Management for procurement, finance, and inventory processes.

Supply Chain Management

Supply Chain Management stores and processes the data that is exchanged with suppliers through the Supplier Engagement app and the supplier portal. Information such as purchase orders, requests for quotation, invoices, and consignment inventory is recorded and maintained in Supply Chain Management.

Global vendors, local vendors, and vendor contacts

Global vendor concept

The Supplier Engagement solution introduces the concept of a *global vendor* in Supply Chain Management. A global vendor represents a single party that can act as a vendor across multiple legal entities. This design differs from vendor collaboration, where you create and manage vendor records separately for each legal entity.

Global vendors help organizations maintain consistent supplier information across legal entities:

- Certifications, contacts, and addresses are entered once and shared.
- Redundant data entry and maintenance are reduced.
- Supplier information can be viewed and managed consistently across all entities.

Learn more in Global vendor management overview and Work with Supplier Engagement data in Supply Chain Management.

Global vendors can be created in three ways:

- Through self-registration from the supplier portal
- By creating the vendor manually in the Supplier Engagement app
- By creating from Supply Chain Management using an existing vendor party

Initially, a new global vendor is considered a *prospect*. After review, it becomes *qualified*. Once qualified, a global vendor can be *released* to a legal entity in Supply Chain Management. Releasing a global vendor creates a *local vendor* account in the selected legal entity. The local vendor is linked to the same party record as the global vendor, so core information such as name, addresses, and contact details is shared.

> **Note**
>
> In Supply Chain Management, a *party* is a shared record that represents a person or an organization. A party can hold multiple roles, such as vendor, customer, or worker. With Supplier Engagement, a global vendor is linked to a party of type *organization*. The same party can hold a vendor role in each company that it supplies.

For example, you might set up a global vendor and its related party and local vendors as follows:

- **Global vendor** – Contoso Office Supplies (Global vendor number: GV-01001-P0H3N7)
- **Party** – Contoso Office Supplies (Party number: 000001745)
- **Local vendors**:
  - Contoso Office Supplies – Vendor number: 1001 – Company: USMF
  - Contoso Office Supplies – Vendor number: 2001 – Company: DEMF

At the global vendor level, you can manage the information shared by the party, including:

- Name
- Addresses
- Contact information (phone, email, website)

- Number of employees
- Organization number
- DUNS number
- Known as
- Phonetic name
- Language

Besides the information shared by the party, the local vendors also share certifications added to the global vendor. Certifications maintained at the global vendor level are marked as global.

You can put a global vendor on hold, and this action propagates to the local vendor accounts. When you set a global vendor on hold, you set all local vendor accounts on hold as well. However, you can manage the hold independently for each local vendor account.

> **Important**
>
> For data to synchronize to Supplier Engagement, a global vendor must exist in Supply Chain Management. The global vendor concept is only available when the Supplier Engagement feature is turned on.

Local vendors

In Supply Chain Management, a local vendor represents a supplier relationship that exists within a specific legal entity (company). In environments where Supplier Engagement isn't turned on, you create and manage all vendor records only at this local level. Each legal entity maintains its own vendor accounts, even if the same supplier works with multiple legal entities.

The global vendor feature, introduced with Supplier Engagement, builds on top of local vendor management. A global vendor represents the supplier as a single party across all legal entities. Use the Supplier Engagement app to release a global vendor to one or more legal entities. Releasing creates a local vendor record in the selected legal entity that's linked to the same party as the global vendor.

You can also create local vendors directly in Supply Chain Management by using the standard vendor creation process from an existing party record. This process creates a vendor that's valid only for the selected legal entity, without the global vendor context.

Global vendor contacts and local vendor contacts

A global vendor contact represents a person who acts on behalf of a supplier at the global level. It's stored as a party of type person and linked to the global vendor's party through a relationship of type global vendor contact. This record is the master reference for the contact across the organization.

A local vendor contact is a representation of a global vendor contact in the context of a specific legal entity. It allows the same person to be recognized as a contact for one or more local vendor accounts under the same global vendor. The global vendor contact is the central record, while each local vendor contact reflects that person's role as a vendor contact for a specific legal entity.

Internal buyers can use the Supplier Engagement app to create and maintain global vendor contacts for any supplier company. External vendor users use the supplier portal to create global vendor contacts for their own companies. You must use the Supplier Engagement app or supplier portal to make updates and deletions.

In Supply Chain Management, you create and manage local vendor contacts by using the **Vendor contacts** page. When you associate a local vendor with a global vendor, creating a local vendor contact automatically establishes the link to the corresponding global vendor contact.

> ⓘ **Note**
>
> Global vendor contact management is restricted to the Supplier Engagement app and the supplier portal.

Invite, onboard, and qualify new vendors

Invite new suppliers

Two scenarios exist for inviting a new supplier, depending on who initiates the process:

- **Self-registration** – A supplier registers themselves through the anonymous registration page on the supplier portal. The registration request is created as a prospect global vendor in the Supplier Engagement app. Users at your organization review the request, and if it's approved, the supplier is invited to the supplier portal to complete the onboarding process. Learn more in Review and approve incoming vendor registrations.
- **Internal registration** – An internal buyer directly creates a prospective global vendor in the Supplier Engagement app. From there, the buyer sends the vendor an invitation to the supplier portal so the vendor can continue with onboarding.

Onboard suppliers

Onboarding is a guided process in the supplier portal where suppliers provide detailed information about their organization. The onboarding form covers contact information, addresses, business profile, certificates, and capabilities. The final step is a questionnaire that captures any additional business information required.

The information entered during onboarding synchronizes with the global vendor profile in the Supplier Engagement app, where internal users can review it.

Qualify a new global vendor

After onboarding is completed, an internal buyer reviews the data submitted by the supplier. The buyer evaluates whether the supplier meets business requirements and required standards.

If the supplier doesn't meet the requirements, the global vendor record is marked as disqualified. If the supplier is accepted, the record is promoted to qualified. Once qualified, the global vendor is available in Supply Chain Management and can be released to one or more legal entities. Releasing creates local vendor accounts in the selected companies, making the supplier available for transactional processes such as purchase orders and RFQs.

Learn more in Global vendor onboarding lifecycle.

Related information

- How Supplier Engagement compares to the vendor collaboration interface
- Synchronize data between Supplier Engagement and Supply Chain Management
- Supplier Engagement deployment overview, prerequisites, and licensing
- Configure the Supplier Engagement app overview
- Manage users of the Supplier Engagement app and supplier portal
- Manage Supplier Engagement app users
- Manage supplier portal users
- Personalize the supplier portal
- Work with the Supplier Engagement app
- Work with Supplier Engagement data in Supply Chain Management

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

How Supplier Engagement compares to the vendor collaboration interface (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement solution provides broader capabilities compared to the older vendor collaboration interface. Both solutions support collaboration with suppliers during purchasing processes, but they differ in scope, architecture, and data management. You can use both solutions in parallel.

Vendor collaboration is embedded directly in Supply Chain Management and focuses on document exchange—purchase orders, invoices, RFQs, and consignment inventory—with vendor data maintained separately in each legal entity. Supplier Engagement is built on the Microsoft Power Platform and introduces global vendor data management, structured lifecycle processes (registration, onboarding, qualification, approval, and termination), supplier capabilities and certifications, risk management, and a dedicated Power Pages portal with anonymous self-registration. This article compares the two solutions across these areas and includes a detailed feature comparison table.

Vendor collaboration

Vendor collaboration is embedded in Supply Chain Management. It provides a portal where suppliers can:

- Exchange documents such as purchase orders, invoices, consignment inventory, and responses to requests for quotation (RFQs).
- Participate in sealed bidding, where RFQ responses remain confidential until the deadline.
- Respond to public RFQs, where invitations are open to a broader supplier base.

You manage vendor organizational and transactional data separately for each legal entity. When a supplier operates across multiple entities, you must maintain the same information multiple times, which can lead to inconsistencies.

Learn more in Vendor collaboration with external vendors.

Contact mapping and user access

In the vendor collaboration interface, you manage vendor contacts for each legal entity. To grant a portal user access to vendor data, you must create the user as a contact for that vendor in the specific legal entity. When the same vendor operates across multiple entities, you often need to create the same user multiple times as a contact, which leads to duplication and extra maintenance.

In Supplier Engagement, you link contacts to the global vendor party rather than to released vendor records in individual legal entities. Each contact only needs to exist once as a global vendor contact, and the contact information is shared across legal entities through the global vendor.

Supplier Engagement also introduces vendor access management, which allows administrators to define which vendors a portal user can access. In Supply Chain Management, these access settings are reflected in the portal user's assigned-organizations configuration.

Supplier Engagement capabilities

Supplier Engagement is built on the Microsoft Power Platform and consists of two applications:

- **Supplier Engagement app** (model-driven app, for internal users):
  - Introduces global vendor data management, where a single supplier record can be used across multiple legal entities
  - Supports lifecycle processes: registration, onboarding, qualification, approval, and termination
  - Includes supplier capabilities, certifications, and risk management
  - Allows internal users to record feedback on suppliers
  - Uses Microsoft Dataverse, making it possible to extend and configure processes such as onboarding questionnaires and qualification workflows

- **Supplier portal** (Power Pages, for external suppliers):
  - Provides access to purchase orders, RFQs, invoices, and consignment inventory
  - Includes anonymous supplier registration, where potential suppliers can submit registration requests without prior setup
  - Offers an improved user interface with clearer navigation and layouts
  - Supports certificate management at a global level

> ⓘ **Note**
>
> The supplier portal doesn't support sealed bidding or public RFQs.

Feature comparison

The following table provides a side-by-side comparison of vendor collaboration and Supplier Engagement features.

Expand table

Category	Feature/capability	Supplier Engagement	Vendor collaboration
Architecture and applications	Platform	Microsoft Power Platform (Dataverse, model-driven app, Power Pages portal)	Embedded in Supply Chain Management
	Applications	Two apps: Supplier Engagement app (internal) and supplier portal (external)	Single vendor collaboration interface
Supplier data management	Vendor master data	Global vendor record shared across multiple legal entities. Users have an overview of the data across all legal entities.	Vendor data maintained separately in each legal entity
	Supplier capabilities	Supported and managed across legal entities	Not available
	Supplier certifications	Managed once at the global vendor level and shared across entities	Managed per vendor record in each legal entity
	Supplier feedback	Internal users can provide structured feedback on suppliers	Not available
Supplier lifecycle	Registration	Supported, including anonymous self-registration	Supported, but requires setup by the buying organization (no anonymous registration)
	Onboarding	Extended onboarding processes supported (qualification, approval, disqualification) via Dataverse workflows	Supported, focused on enabling vendors to access transactional data in Supply Chain Management
	Lifecycle scope	Full lifecycle: registration, onboarding, qualification, approval, termination	Limited scope: mainly registration and onboarding
	User management	Managed in Supplier Engagement app for portal access	Managed in Supply Chain Management per vendor account

Category	Feature/capability	Supplier Engagement	Vendor collaboration
Collaboration features	Aggregated view	Home page and workspaces contain aggregated view across all legal entities	Not available. Workspaces are limited to one legal entity at a time.
	Purchase orders	Supported	Supported
	RFQs	Supported, but no sealed bidding or public RFQs	Supported, including sealed bidding and public RFQs
	Invoices	Supported	Supported
	Consignment inventory	Supported	Supported
Implementation and extensibility	Portal user interface	Power Pages-based, configurable design, responsive layouts, centralized home page for navigation	Standard Supply Chain Management UI, limited customization
	Extensibility	Supported via Dataverse and Power Platform customization	Limited to Supply Chain Management configuration

Related information

- Supplier Engagement overview
- Work with the Supplier Engagement app
- Synchronize data between Supplier Engagement and Supply Chain Management

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

Supplier Engagement deployment overview, prerequisites, and licensing (preview)

[This article is prerelease documentation and is subject to change.]

Supplier Engagement deployment spans both Power Platform and Supply Chain Management. You install the solution in your Power Platform tenant, enable the related functionality in Supply Chain Management, and then finish the setup tasks that connect both experiences. Planning the prerequisites up front helps you avoid setup delays later in the deployment.

Deployment process overview

Deploying Supplier Engagement includes the following tasks:

- Confirm the prerequisites.
- Install the Supplier Engagement app on Power Platform.
- Install the supplier portal in Power Pages.
- Turn on the Supplier Engagement feature in Supply Chain Management.

Power Platform hosts the model-driven app, supplier portal, cloud flows, and supporting Dataverse components. Supply Chain Management provides the business processes, shared supplier data, and feature configuration that the solution depends on.

Prerequisites

Before you start deployment, make sure that your environment and service account meet the following requirements:

- **Dataverse readiness** – Confirm that Microsoft Power Platform is up to date and that the latest release updates are applied.
- **Supply Chain Management version** – Run Microsoft Dynamics 365 Supply Chain Management version 10.0.48 build 10.0.2645.52 or later.
- **Dataverse base language** – In the current public preview release, the Supplier Engagement app is only available in US English (Microsoft locale identifier (LCID) 1033). If you created your Dataverse environment with a base language other than US English, the Supplier Engagement app might display a mixed-language UI, where labels provided by Dataverse

appear in your selected base language while labels provided by the Supplier Engagement app appear in English. You select your Dataverse environment's base language during provisioning and can't change it later, so we recommend that you install and test the preview version of the Supplier Engagement app on a Dataverse environment that was created to use US English as its base language. Support for additional languages will be added for the general availability release.

- **Dataverse connection** – Connect your Supply Chain Management environment to Dataverse. Learn more in Microsoft Power Platform integration with finance and operations apps.

- **Finance and operations virtual entity solutions** – In the Power Apps maker portal, open your environment, go to **Solutions**, open the **Managed** tab, and confirm that the following solutions are installed:
  - *Dynamics 365 Company*
  - *Dynamics Operations Virtual Entity Support*
  - *Microsoft Operations ERP Catalog*
  - *Dynamics 365 ERP Virtual Entities*

  If one or more solutions are missing, install them as described in Configure Dataverse virtual entities.

- **Power Platform Dynamics 365 apps** – Run the latest update of the *Finance and Operations Virtual Entity* app from Power Platform admin center. The app must be version 2.20.3385.8 or later. Learn more in Manage Dynamics 365 apps.

- **Dedicated service account** – Set up a dedicated Microsoft Entra ID service account for installation and configuration in both Dataverse and Supply Chain Management. The account must be a standard user account, not an app registration account, and it must have all required licenses. The service account is used for flows, notifications, and background automation. Grant it the following security privileges:
  - **Dataverse** – *System Administrator* role
  - **Supply Chain Management** – *System Administrator* role
  - **Microsoft Entra ID** – *Power Platform administrator* role

Supplier Engagement licensing

Supplier Engagement is currently a preview feature that is provided for evaluation and testing. The licensing model for Supplier Engagement features in Dynamics 365 Supply Chain

Management and the Supplier Engagement app in Dataverse is still under review. The supplier portal offers a 30-day trial period. Additional information regarding the licensing requirements will be published before the solution is made generally available.

What to do next

After you confirm the prerequisites, install the Supplier Engagement app and supplier portal in Power Platform. When those components are available, configure Power Platform and Supply Chain Management so that onboarding, notifications, synchronization, and workflows can run correctly.

Related information

- Install the Supplier Engagement app and supplier portal on Power Platform
- Onboard using the onboarding guide
- Configure Supplier Engagement elements in Power Platform
- Configure Supplier Engagement features in Supply Chain Management
- Post-deployment validation checklist
- Deployment FAQs

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/17/2026

Install the Supplier Engagement app and supplier portal on Power Platform (preview)

[This article is prerelease documentation and is subject to change.]

Installing Supplier Engagement on Power Platform adds the model-driven app, supporting solutions, and supplier-facing portal components to your environment. You also need to reactivate the supplier portal and prepare sign-in settings so that administrators and suppliers can access the right experience. Completing these tasks in order helps you confirm that the environment is ready before you move on to configuration.

Prerequisites

Before you install the Supplier Engagement app and supplier portal on Power Platform, complete the prerequisites listed in Supplier Engagement deployment overview, prerequisites, and licensing.

Import the Supplier Engagement application

Use the Power Platform admin center to install the Supplier Engagement app in the target environment.

1. Sign in to the Power Platform admin center.
2. Go to **Manage** > **Environments** and open your environment.
3. On the command bar, select **Resources** > **Dynamics 365 apps**.
4. On the command bar, select **Install app**.
5. Search for *Supplier Engagement in Dynamics 365 Supply Chain Management (Preview)* and select it.
6. Select **I agree to the terms of service**, and then select **Install**.

Track installation progress

Use the solution history in Power Apps to monitor the import and confirm when installation finishes.

1. Go to the Power Apps maker portal.
2. On the navigation pane, select **Solutions**.

3. On the command bar, select **See History**.

If the *msdyn_SupplierEngagementVendorEntity* solution fails with the following error, wait two hours and then repeat the import procedure.

> **Exception message** Failed to sync entity metadata for entity 'VRMVENDORENTITY'. Exception details: CreateRequest failed with error: String or binary data would be truncated in table '{0}', column '{1}'. Truncated value: '{2}'...

Activate the supplier portal in Power Pages

After the app is installed, activate the supplier portal site so that you can continue with portal setup and testing.

1. Sign in to the Power Apps maker portal.
2. On the navigation pane, select **Solutions** and open **Supplier Engagement Portal Solution**.
3. Go to **Objects** > **Sites**, find *Microsoft Dynamics 365 Supplier Portal*, open its **Commands** (three dots) menu, and select **Open**.
4. Wait for the browser to redirect you to the Power Pages portals site.
5. Open the **Inactive sites** tab and find *Microsoft Dynamics 365 Supplier Portal*.
6. Select **Reactivate** for the site.

Learn more in Reactivate sites.

Configure identity providers

When you're testing the preview version of Supplier Engagement, we recommend that you use Microsoft Entra ID with the configuration described in this section. For a list of available identity providers, learn more in Overview of authentication in Power Pages. To choose Microsoft Entra ID as the identity provider for the supplier portal, follow these steps.

1. Go to Power Pages portals.
2. Open the *Microsoft Dynamics 365 Supplier Portal* site.
3. On the navigation pane, select **Security**.
4. Under **Manage**, select **Identity providers**.
5. Find *Microsoft Entra ID*. Open its **Commands** (three dots) menu and select **Edit configuration**.
6. Set **Contact mapping with email** to *On*, and then select **Save**.

Configure site visibility

You can keep the portal hidden while you finish setup, testing, and validation. Update site visibility only when you're ready to let suppliers reach the site externally.

To learn more about how to hide and show the supplier portal, go to Site visibility in Power Pages.

Related information

- Supplier Engagement deployment overview, prerequisites, and licensing
- Onboard using the onboarding guide
- Configure Supplier Engagement elements in Power Platform
- Configure Supplier Engagement features in Supply Chain Management
- Post-deployment validation checklist
- Deployment FAQs

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Onboard using the onboarding guide (preview)

[This article is prerelease documentation and is subject to change.]

The onboarding guide gives administrators a single place to track Supplier Engagement setup progress. It brings together the core deployment activities, helps you understand what still needs attention, and makes it easier to return to incomplete tasks later. Use it throughout deployment instead of managing setup with a separate checklist.

Open the onboarding guide

Open the onboarding guide from the Supplier Engagement app whenever you want to review setup status or continue deployment work.

1. Sign in to the Supplier Engagement app by using the service account that you're using for installation.
2. At the bottom of the navigation pane, select the **Configuration** area.
3. Under **Onboarding**, select **Onboarding Guide**.
4. Review the onboarding checklist that opens.

The onboarding guide remains available during the full setup process, so you can return to it at any time and continue where you left off.

Use the onboarding guide

Each onboarding activity points to a specific configuration area, such as Supply Chain Management setup, Power Pages setup, cloud flow activation, or notification setup.

1. Select an onboarding activity in the left pane to review its setup instructions.
2. Use the linked Supplier Engagement documentation for additional context when you need it.
3. Complete the required configuration outside the onboarding guide.
4. Select the checkbox next to the activity after you verify that the task is complete.
5. Clear a checkbox if more work is required.

Progress is saved automatically, and the completion status is retained across sessions.

Best practices

Use the following practices to keep onboarding organized and accurate:

- Complete onboarding activities in the recommended order.
- Mark activities as complete only after you verify the related configuration.
- Review the onboarding guide periodically to confirm that no required tasks were missed.
- If you need more detailed setup guidance for a specific activity, refer to this and other related documentation articles on Microsoft Learn.

Related information

- Supplier Engagement deployment overview, prerequisites, and licensing
- Install the Supplier Engagement app and supplier portal on Power Platform
- Configure Supplier Engagement elements in Power Platform
- Configure Supplier Engagement features in Supply Chain Management
- Post-deployment validation checklist
- Deployment FAQs

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Configure Supplier Engagement elements in Power Platform (preview)

[This article is prerelease documentation and is subject to change.]

After you install the Supplier Engagement components, align the Power Platform environment with your portal, communication, and automation requirements. This article covers the configuration tasks to complete before broad onboarding or production validation, including environment variables, email templates, mailbox activation, cloud flows, and duplicate detection rules. These tasks make sure that users receive the right links, cloud flows can run, and duplicate vendor data is detected before it causes problems.

Prerequisites

Before you configure Supplier Engagement elements in Power Platform, complete the prerequisites listed in Supplier Engagement deployment overview, prerequisites, and licensing.

Customize supplier invitation and notification emails

You can customize the email template text that Supplier Engagement sends to suppliers, including invitation and notification emails. This setup is useful when you want to align system-generated emails with your organization's tone, branding, and onboarding guidance.

> **Important**
>
> The Supplier Engagement invitation email template must include a link to your supplier portal. Update that link before you go live.

> **Tip**
>
> Learn more in **Create templates for email**.

To customize supplier invitation and notification emails, follow these steps.

1. Sign in to the Power Platform admin center as a system administrator and open your environment.

2. Select the link under **Environment URL** to open the environment in a new browser tab.

3. Do one of the following steps:

    - If a list of apps opens, select the **Power Platform Environment Settings** app.
    - If the environment opens directly to an app, select the app name on the app header bar at the top to open the list of published apps, and then select **Power Platform Environment Settings**.

4. On the navigation pane, go to **Templates** > **Email Templates**.

5. Supplier Engagement uses the following templates. Open the template that you want to customize.

    - *Supplier Engagement contact portal access*
    - *Supplier Engagement invitation*
    - *Supplier Engagement registration status*

6. Make the following changes to the selected email template:

    - Update the subject and body text as needed.
    - If the message includes a portal action, verify that the link points to the correct supplier portal URL for your environment. By default, the *Supplier Engagement invitation* email template includes a placeholder for your supplier portal URL and you must update it before going live with the solution.

7. Select **Save**.

8. Inspect and update the other templates as needed.

Enable a mailbox for cloud flow notifications

Cloud flows, which are part of Microsoft Power Automate, automate business processes and ensure timely communication across platforms such as Supply Chain Management and Microsoft Teams. To maximize the effectiveness of these flows—especially those that send notifications—you must enable a mailbox for the flow owner or service account. This setup allows the flow to send email notifications reliably, which ensures that users and stakeholders receive critical updates, alerts, and approvals directly in their inbox.

To enable a mailbox for cloud flow notifications, follow these steps.

1. Open the **Power Platform Environment Settings** app in the Power Platform admin center as described previously.
2. On the navigation pane, go to **Email Configuration** > **Mailboxes**.
3. Select the mailbox to activate.
4. On the command bar, select **Activate**. In the **Confirm Mailbox Activation** dialog, select **Activate**.
5. On the command bar, select **Approve Email**. In the **Approve Primary Email** dialog, select **OK**.
6. On the command bar, select **Test and Enable Mailboxes**. In the **Test Email Configuration** dialog, select **OK**.

> **Note**
>
> These mailboxes are used both for automated notifications and to send updates from cloud flows. Choose mailboxes for valid service accounts or non-personal accounts that are permanent and unaffected by organizational changes. This approach helps maintain stable, long-term automation and avoids unnecessary interruptions in workflow.

> **Note**
>
> Power Automate can also work with on-premises Exchange through the appropriate connectors and gateway configuration. Learn more in Manage an on-premises data gateway in Power Automate.

Enable supporting cloud flows

Cloud flows automate key business processes and ensure integration between the Supplier Engagement app, supplier portal, and Supply Chain Management. These flows streamline notifications, synchronize data, and automate approvals to help reduce manual effort and minimize errors.

Connection references act as secure bridges between your Power Platform environment and external services such as Dataverse, Outlook, Teams, and Power Apps. By establishing these references, you ensure that cloud flows and integrations can authenticate and interact with the necessary data sources and applications. Proper configuration of connection references enables automated workflows that synchronize supplier data and support collaboration across the Supplier Engagement app, supplier portal, and Supply Chain Management.

Although the cloud flows are already present in the system after installation, they're inactive by default and don't run until you enable them. After you configure the necessary connection references, activate the cloud flows in your environment so that all related functionalities become operational.

To create the required connection references and activate the cloud flows, follow these steps:

1. Sign in to the Power Apps maker portal and select your environment.
2. On the navigation pane, select **More** > **Connections**.
3. On the command bar, select **New connection**.
4. A list of services opens. Add connections for the **Microsoft Dataverse** service by using the **+** in the **Actions** column.
5. Add a second connection for the **Microsoft Teams** service.
6. On the navigation pane, select **Solutions**.
7. Find and open the solution that has a **Display name** of *Default solution*.
8. On the navigation pane, go to **Objects** > **Connection References**.
9. Use your browser's on-page search function to find connection references that include the text *SupplierEngagement* (in other words, use Ctrl + F to do an on-page search, don't use the filter provided by the site). There should be three of them:
    - *Microsoft Dataverse MicrosoftSupplierEngagement-77aa1*
    - *Microsoft Dataverse msdyn_SupplierEngagementBase-abf2e*
    - *Microsoft Teams MicrosoftSupplierEngagement-0e21b*
10. For *Microsoft Dataverse MicrosoftSupplierEngagement-77aa1*, open the **Commands** (three dots) menu, select **Edit**, set **Connection** to the *Microsoft Dataverse* connection that you created, and then select **Save**.
11. For *Microsoft Dataverse msdyn_SupplierEngagementBase-abf2e*, open the **Commands** menu, select **Edit**, set **Connection** to the *Microsoft Dataverse* connection that you created, and then select **Save**.
12. For *Microsoft Teams MicrosoftSupplierEngagement-0e21b*, open the **Commands** menu, select **Edit**, set **Connection** to the *Microsoft Teams* connection that you created, and then select **Save**.

13. On the navigation pane, select **Back to solutions**.
14. On the **Solutions** page, open the **Managed** tab.
15. Find and open the solution that has a **Display name** of *Supplier Engagement Core*.
16. On the navigation pane, go to **Objects** > **Cloud flows**.
17. Turn on each cloud flow that shows a **Status** of *Off*. To turn on a flow, select it and then select **Turn on** on the command bar.
18. Repeat the previous steps to turn on all the cloud flows in the *Supplier Engagement Portal* solution.

Enable duplicate detection rules

Duplicate detection rules help maintain clean and consistent vendor data within the Supplier Engagement solution. These rules prevent the creation of duplicate global vendor records by automatically identifying potential matches based on key fields such as name, email address, phone number, and website. By configuring and activating duplicate detection, you ensure data integrity, streamline vendor management processes, and reduce the risk of errors caused by redundant or conflicting records.

Set up rules to detect duplicate global vendor records by making sure that none of the following fields have values that match those of an existing global vendor:

- **Primary phone number** – Use an exact match comparison and include blank values in the evaluation.
- **Email address** – Use an exact match comparison and ignore blank values.
- **Name** – Compare the first five characters by using a same-first-characters matching rule and ignore blank values.
- **Primary URL** – Use an exact match comparison and ignore blank values.

To enable the duplicate detection rules in the Supplier Engagement solution, follow these steps:

1. Open the **Power Platform Environment Settings** app in the Power Platform admin center as described previously.
2. On the navigation pane, go to **Data Management** > **Duplicate Detection Rules**.
3. Select the check box for each of the following rules, and then select **Publish** on the command bar:

- *Global vendor with same primary phone number*
- *Global vendor with same email address*
- *Global vendor with same name*
- *Global vendor with same primary URL*

4. Confirm that each rule now shows a **Status Reason** of *Published*.

Learn more in Set up duplicate detection rules to keep your data clean.

Related information

- Supplier Engagement deployment overview, prerequisites, and licensing
- Install the Supplier Engagement app and supplier portal on Power Platform
- Onboard using the onboarding guide
- Configure Supplier Engagement features in Supply Chain Management
- Post-deployment validation checklist
- Deployment FAQs

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/03/2026

Configure Supplier Engagement features in Supply Chain Management (preview)

[This article is prerelease documentation and is subject to change.]

After you install the Power Platform components, you must configure Supply Chain Management to expose the feature set, support approvals, and keep reporting and workflows running. These tasks connect the operational side of Supplier Engagement to the app and portal experiences. Completing them helps you avoid workflow failures, missing permissions, and incomplete supplier synchronization.

Prerequisites

Before you configure Supply Chain Management for Supplier Engagement, complete the prerequisites listed in Supplier Engagement deployment overview, prerequisites, and licensing.

Enable the Supplier Engagement feature

Turn on the feature in **Feature management** before you try to use Supplier Engagement capabilities in Supply Chain Management.

1. Go to **Workspaces** > **Feature management**.
2. If you don't see *Supplier Engagement*, select **Check for updates**.
3. Select the *Supplier Engagement* feature, and then select **Enable now**.

Learn more in Feature management overview.

Enable batch processing for the user approval workflow

Workflow batch jobs must be active so that approval, notification, and due-date processing can continue in the background.

1. Go to **System administration** > **Inquiries** > **Batch jobs**.
2. Use the filter in the **Job description** column to find rows that contain *Workflow*.
3. Select the check box for each of the following jobs:
    - *Workflow message processing*

- *Workflow line-item notifications*
- *Workflow Maintenance*
- *Workflow due date processing*

4. On the Action Pane, select **Change status**, and then change the status to *Waiting*.

Configure security roles

Supply Chain Management includes two standard security roles that support purchasers: *Purchasing manager* and *Purchasing agent*. The default versions of these roles might not include all the duties you need after you install Supplier Engagement. Affected functionality includes the abilities to:

- Create global vendors from Supply Chain Management and view party details
- Create replenishment orders
- Create inventory ownership change journals
- Create replenishment order product receipts
- Create purchase orders

To make the required functionality available, add the duties specified in the following lists to the respective roles. If you're using custom security roles, add some or all of these duties to your roles as needed.

- Purchasing manager:
  - *Maintain global address book master*
  - *Maintain consignment replenishment orders*
  - *Maintain inventory ownership change journals*
  - *Maintain consignment product receipt*
  - *Maintain purchase orders*

- Purchasing agent:
  - *Maintain consignment replenishment orders*
  - *Maintain inventory ownership change journals*
  - *Maintain consignment product receipt*

Learn more in Role-based security.

Configure the supply risk assessment report

The supply risk assessment report is a Power BI report built on data stores that stage aggregate measurements for reporting. In Supply Chain Management, this data store is called an entity store.

To see the data refreshed on the supply risk assessment report, you must enable the automated refresh of the entity stores that the report uses. To enable the automated entity store refresh, follow these steps.

1. Go to **System administration** > **Set up** > **Entity store**.
2. Use **Filter** to find *PurchaseCube* and *VRMPurchaseCube*.
3. Select **Edit** on the Action Pane. 1.Select the *PurchaseCube* record and expand the **General** FastTab. Set **Automatic refresh enabled** to *Yes* and choose a *Recurrence interval* (for example, *Every hour*). Then select **Refresh** to schedule the batch job.
4. Repeat the previous step for *VRMPurchaseCube*.

Configure B2B collaboration for vendor user requests

To provision supplier portal users as Microsoft Entra B2B guest users from the Supply Chain Management user workflow, first create an application registration in Microsoft Entra ID and then enter that application information in Supply Chain Management. You need this configuration before you can use the *Provision Microsoft Entra B2B user* workflow task.

The one-time setup comprises the following steps.

1. Set up a B2B invitation service application in Microsoft Entra ID.
2. Configure the B2B invitation service settings in Supply Chain Management.

Learn more in Export business-to-business (B2B) users to Microsoft Entra ID.

> ⓘ **Note**
>
> This setup is mandatory when your environment requires cross-domain user access. If cross-domain access isn't required, you can skip it.

Set up the vendor user request workflow

The *Vendor user request (new user or modify user)* workflow is a standard workflow included with Supply Chain Management. You can set it up to automatically approve portal user requests (new

vendor user requests that are routed from the supplier portal), or you can assign the approvals to a particular role or set of users.

Open the Vendor user request (new user or modify user) workflow in the workflow editor

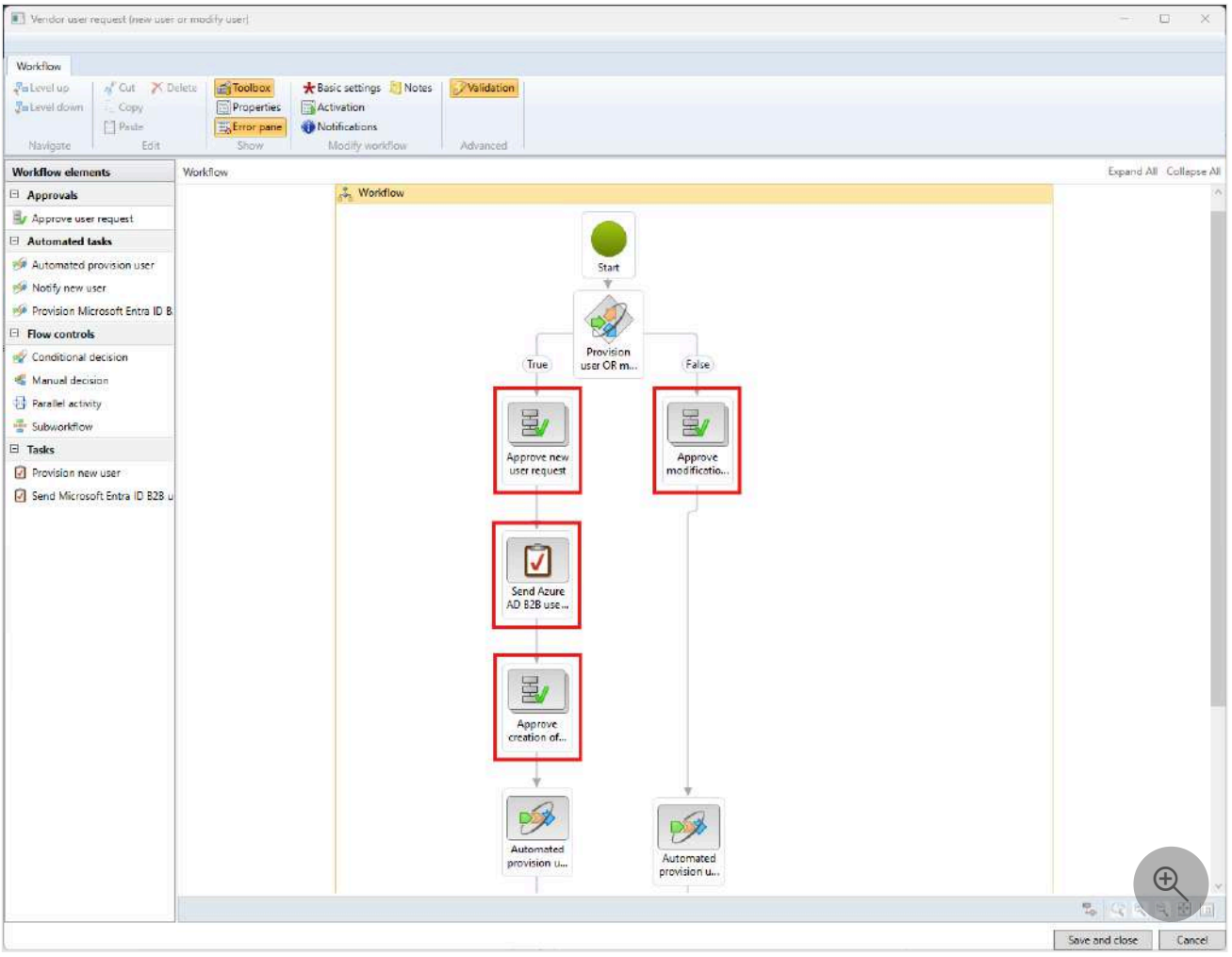
To start editing the workflow, follow these steps:

1. Go to **System administration** > **Workflow** > **User workflows**.
2. Open the *Vendor user request (new user or modify user)* workflow. The system suggests downloading and opening the workflow editor. You must use the Microsoft Edge browser to download and run the app.
3. The app opens, showing the default workflow. You can now edit the workflow to modify its behavior as needed, as described in the following sections. Choose which functionality to enable based on your business needs.

Auto-approve supplier portal user requests (optional)

If you want to approve supplier portal user requests automatically, follow these steps:

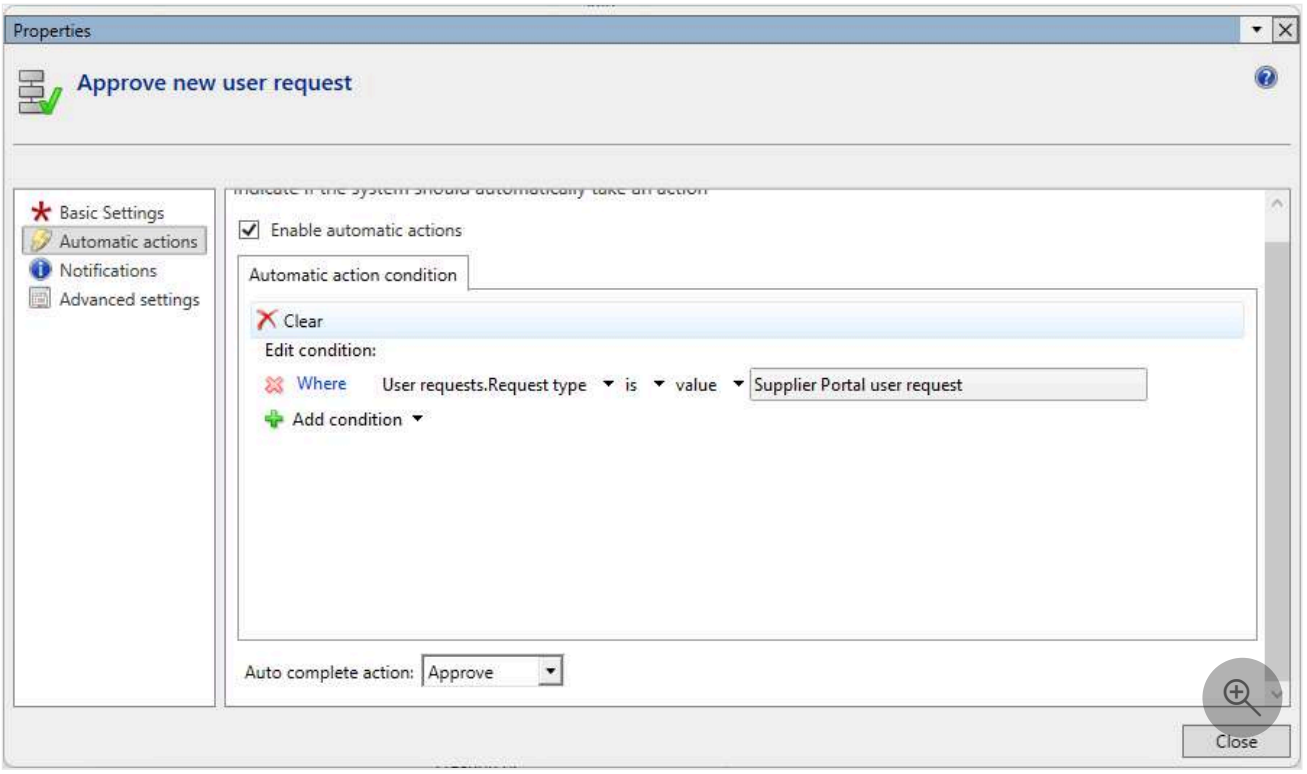
1. Open the *Vendor user request (new user or modify user)* workflow in the workflow editor. In this procedure, you must edit each of the steps highlighted in the following screenshot.



2. Select the first step highlighted in the previous screenshot, and then select **Automatic actions** on the ribbon.

3. Select **Add condition** and set **Condition** to *Where User requests.Request type is Value Supplier Portal user request*.

   The following screenshot shows an example of the automatic actions settings.



4. Set Automatic action to *Approve*.

5. Repeat the previous five steps for each of the other three approval steps in the workflow.

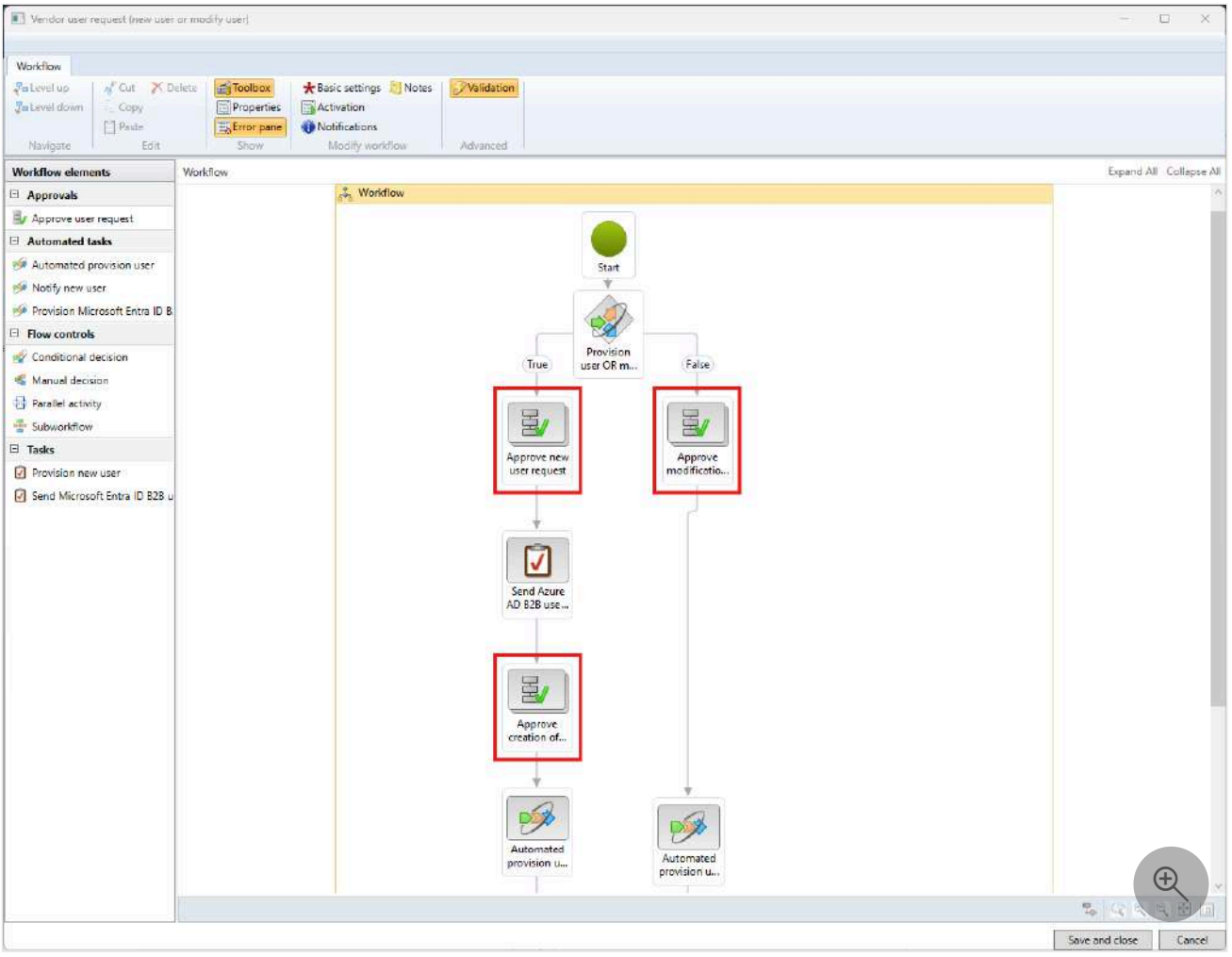
6. Select **Save and close** to save your changes to the workflow as a new version.

7. In Supply Chain Management, go back to **System administration** > **Workflow** > **User workflows** and select the *Vendor user request (new user or modify user)* workflow. On the Action Pane, open the Workflow tab and select **Versions**. Use the **Workflow versions** dialog to set the new version of the workflow as the active version.

Require internal users to approve supplier portal user requests (optional)

If you want to require internal users (such as the purchasing manager) to approve supplier portal user requests, follow these steps:

1. Open the *Vendor user request (new user or modify user)* workflow in the workflow editor. In this procedure, you must edit each of the steps highlighted in the following screenshot.



2. Select the first step highlighted in the preceding screenshot, and then select **Advanced** on the ribbon.

3. Select the **Use final approver** check box.

4. Select the user who should approve the user request.

5. Select **Save and close** to save your changes to the workflow as a new version.

6. In Supply Chain Management, go back to **System administration** > **Workflow** > **User workflows** and select the *Vendor user request (new user or modify user)* workflow. On the Action Pane, open the Workflow tab and select **Versions**. Use the **Workflow versions** dialog to set the new version of the workflow as the active version.

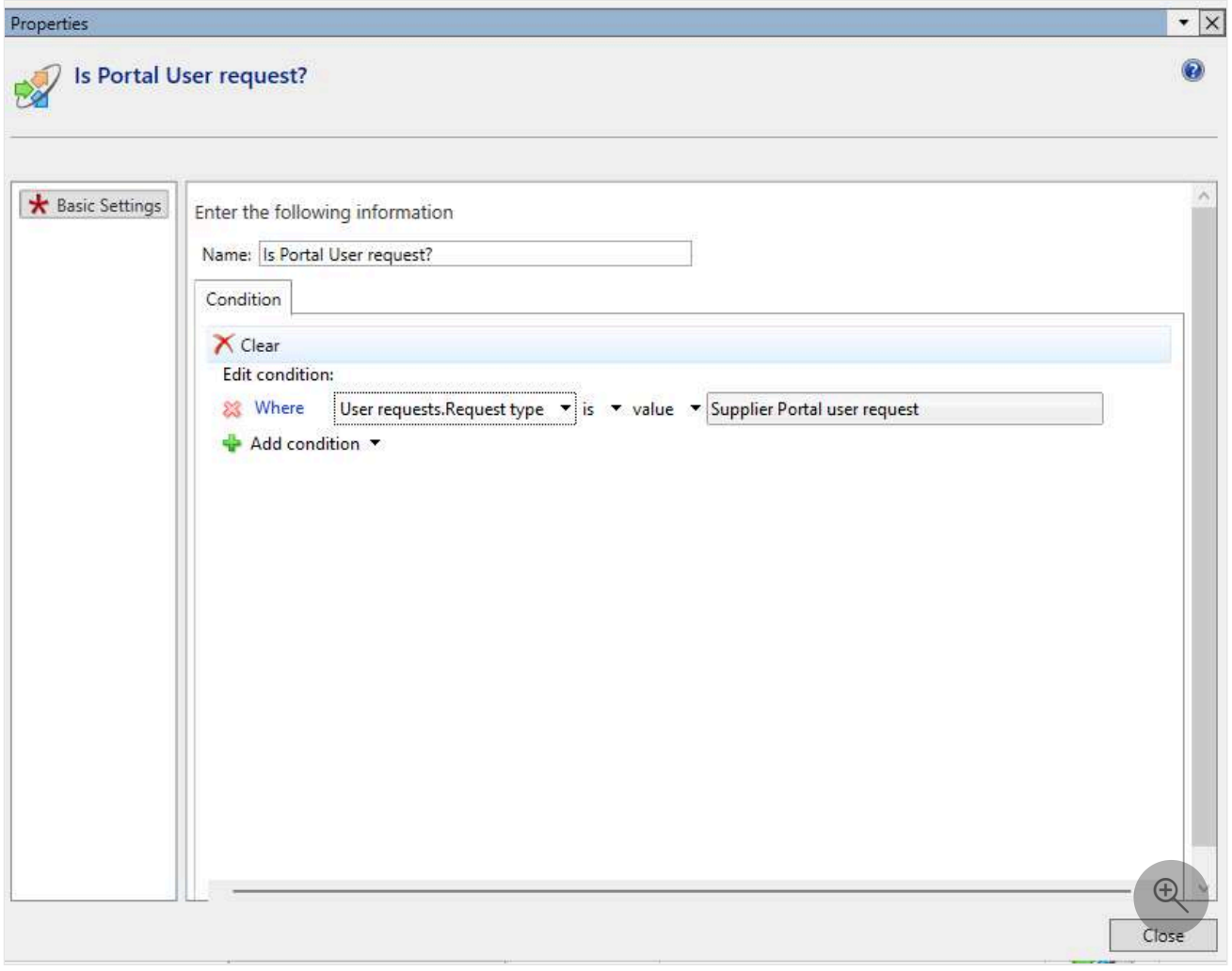
Automate guest supplier user provisioning in Microsoft Entra (optional)

To automate guest supplier user provisioning in Microsoft Entra, add a conditional workflow step that verifies whether the request pertains to a portal user. If so, the workflow bypasses all

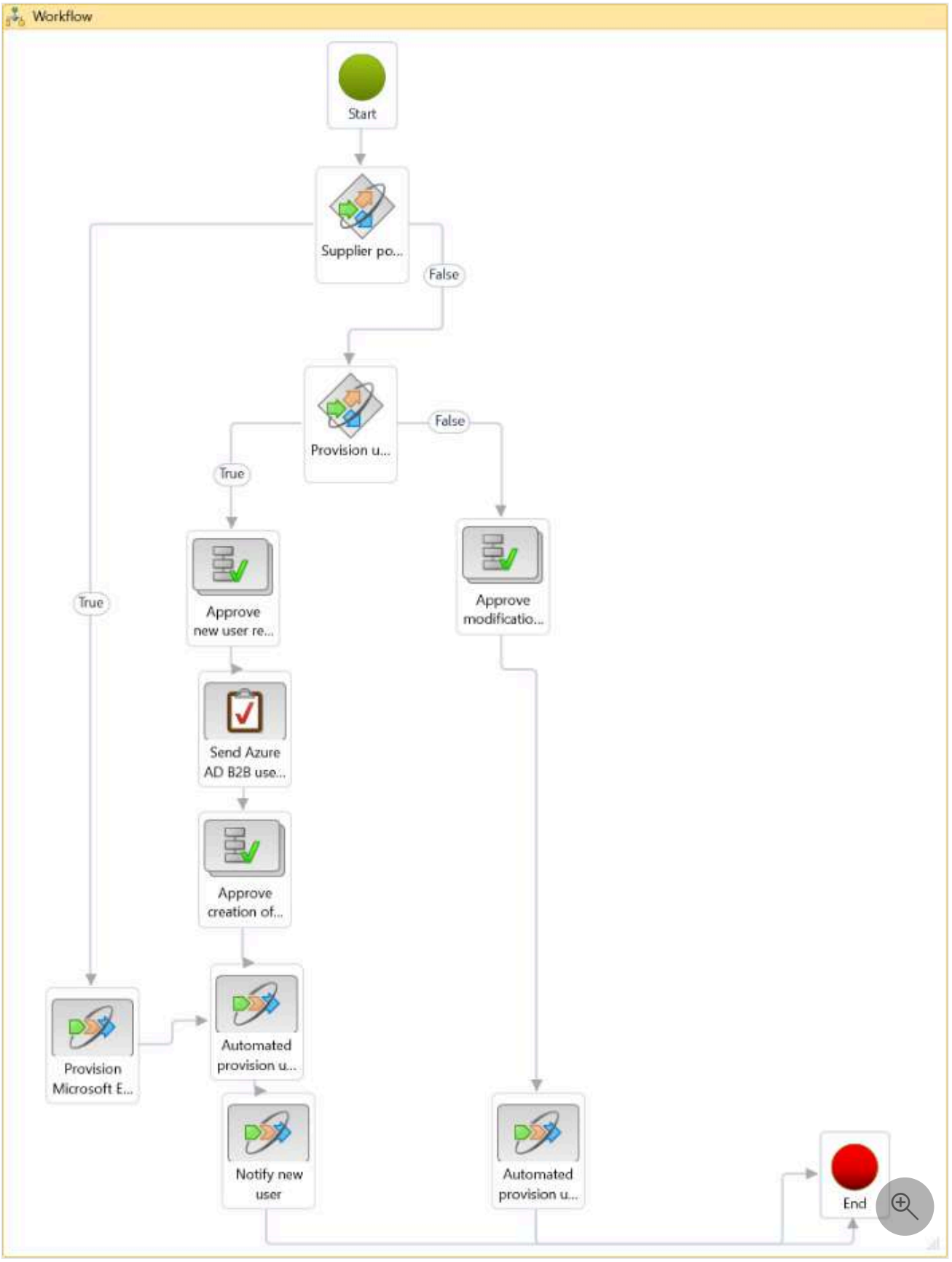
subsequent approval steps and proceeds directly to user creation. If you want to implement this functionality, follow these steps:

1. Open the *Vendor user request (new user or modify user)* workflow in the workflow editor.
2. Add a new *Conditional decision* step immediately after the *Start* step.
    - Move the output of the *Start* step to the input of the new conditional step.
    - Link the **True** outcome of the new conditional step to the *Automated provision user* step.
    - Link the **False** outcome of the new conditional step to the *Provision user OR modify user* step.
3. Select the new conditional step, and then select **Basic settings** on the ribbon.
4. Set **Name** to *Is portal user request?*.
5. Select **Add condition** and set **Condition** to *Where User requests.Request type is Value Supplier Portal user request*.

    The following screenshot shows an example of the conditional step settings.



6. Add the *Provision Microsoft Entra ID B2B user* automated task on the **True** branch before **Automated provision user**. The workflow diagram should now look like this:



7. Select **Save and close** to save your changes to the workflow as a new version.

8. In Supply Chain Management, go back to **System administration** > **Workflow** > **User workflows** and select the *Vendor user request (new user or modify user)* workflow. On the Action Pane, open the Workflow tab and select **Versions**. Use the **Workflow versions** dialog to set the new version of the workflow as the active version.

Learn more in Example of a workflow for provisioning new users and modifying security roles.

Set up the vendor category request workflow

The vendor category request workflow is triggered when a vendor requests a new procurement category.

To create or modify the workflow, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing workflows**.
2. Find or add the *Vendor category request* workflow.
3. Configure the workflow steps and activate the workflow.

Learn more in Set up the Vendor category request workflow.

Set up the vendor invoice workflow

The vendor invoice workflow is triggered when a vendor invoice is submitted.

To create or modify the workflow, follow these steps.

1. Go to **Accounts payable** > **Setup** > **Accounts payable workflows**.
2. Create or update the vendor invoice workflow that you want suppliers to use.
3. Activate the workflow after configuration is complete.

Learn more in Set up options for vendor invoice automation and Vendor invoices overview.

Set up localization metadata

Local laws and regulations vary from place to place, so the fields and options available for each vendor must adapt to that vendor's specific country/region. To prepare your system to provide localized features, you must generate the required entity field metadata. To generate the entity field metadata, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **(Preview) Supplier Engagement** > **Entity field metadata for supplier portal**.

2. On the Action Pane, select **Generate entity field metadata**.

Create global vendors from existing data

To enable interaction with the supplier portal and ensure visibility in Supplier Engagement processes, each of your existing suppliers must be registered as a global vendor. Supply Chain Management can create global vendors based on existing vendor records or parties. The new global vendors are then linked to the existing vendor party.

This capability is particularly useful for organizations transitioning to Supplier Engagement that want to integrate their current vendor base with the supplier portal.

Supply Chain Management provides a global vendor creation wizard that streamlines the registration process by generating global vendors based on existing vendors or global address book (party) records. It can also merge parties, which helps eliminate duplicate records in your global address book.

To run the global vendor creation wizard, follow these steps:

1. Sign in to Supply Chain Management.
2. Start the wizard in one of the following ways:

    - Go to **Procurement and sourcing** > **Vendors** > **All vendors**, select up to 200 vendors that aren't already linked to global vendors. On the Action Pane, open the **Vendor** tab and, from the **Set up** group, select **Global vendor**.
    - Go to **Organization administration** > **Global address book** > **Global address book**, select up to 200 parties that have a vendor role and aren't already linked to global vendors, and then select **Party** > **New** > **Global vendor**.
    - Go to **Procurement and sourcing** > **Vendors** > **Supplier engagement** > **Global vendor creation**.

3. The **Global vendor creation** wizard opens. On the **Welcome** page, select **Next**.
4. If the **Select vendor** page appears, select the vendors that you want to process. Use the **Vendor selection** filter to help find vendors that meet your criteria. Then select **Next**.
5. The **Check for duplicates** page opens.
6. Review each row in **Selected vendors** and evaluate possible matches in **Duplicates search result**.
7. For each vendor, choose the party that should become the global vendor and select **Global vendor**.
8. Select **Merge** for any parties that should be merged into the selected global vendor.

9. Repeat the duplicate review and merge decisions for each selected vendor.
10. Select **Next** to review the **Summary** page.
11. If the summary is correct, select **Finish**.

When the wizard finishes, it creates the new global vendors, merges selected parties, links local vendors to the related global vendor parties, and synchronizes the new records to the Supplier Engagement app.

Review the global vendor creation log

The global vendor creation wizard runs asynchronously, so use the log to confirm results and handle failures.

1. Go to **Procurement and sourcing** > **Vendors** > **Supplier Engagement** > **Global vendor creation log**.
2. Review the creation status and failure reason for each record.
3. If a record has a status of *Error*, select it and then select **Resync**.

Related information

- Supplier Engagement deployment overview, prerequisites, and licensing
- Install the Supplier Engagement app and supplier portal on Power Platform
- Onboard using the onboarding guide
- Configure Supplier Engagement elements in Power Platform
- Post-deployment validation checklist
- Deployment FAQs

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Post-deployment validation checklist (preview)

[This article is prerelease documentation and is subject to change.]

After deployment is complete, validate the environment before you hand it over to business users or suppliers. These checks help you confirm that the expected solutions are available, automation is turned on, and synchronization data was created correctly. Running the checklist early makes troubleshooting much easier if something is missing.

Confirm Power Apps solutions are available

Review the managed solutions in the Power Apps maker portal to make sure that the Supplier Engagement package finished installing.

1. Sign in to the Power Apps maker portal.
2. Go to **Solutions**.
3. Open the **Managed** tab.
4. Confirm that the the following Supplier Engagement solutions are listed:

    - Supplier Engagement Anchor
    - Supplier Engagement Portal
    - Supplier Engagement Plugins
    - Supplier Engagement Core
    - Supplier Engagement Controls
    - Supplier Engagement Consignment Inventory
    - Supplier Engagement Vendor Invoice
    - Supplier Engagement Request for Quotation
    - Supplier Engagement Purchase Order
    - Supplier Engagement Vendor Onboarding
    - Supplier Engagement Contact Person Entity
    - Supplier Engagement Vendor Entity
    - Supplier Engagement Base

> **Note**

If any expected solutions are missing, reinstall the Supplier Engagement package.

Confirm cloud flows are enabled

Check both managed solutions that contain cloud flows so that you can confirm that automation is active.

1. Sign in to the Power Apps maker portal and select your environment.
2. On the navigation pane, select **Solutions**.
3. Find and open the solution that has a **Display name** of *Supplier Engagement Core*.
4. On the navigation pane, select **Objects** > **Cloud flows**.
5. Verify that every flow has a **Status** of *On*.
6. On the navigation pane, select **Back to solutions**.
7. Find and open the solution that has a **Display name** of *Supplier Engagement Portal*.
8. On the navigation pane, select **Objects** > **Cloud flows**.
9. Verify that every flow has a **Status** of *On*.

> **Note**
>
> If any flow status is *Off*, follow the instructions in Enable supporting cloud flows.

Confirm sync configurations are available

The Supplier Engagement app should contain the sync configuration records created when you deployed the solution.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, go to **Data Sync** > **Sync Configs**.
3. Confirm that the sync configuration mappings listed in the following table exist on your system.

Expand table

Source Table	Destination Table
mserp_vrmpersonentity	contact

Source Table	Destination Table
mserp_vrmglobalvendorentity	msdyn_globalvendor
msdyn_vrmpartycontact	mserp_vrmpartycontactentity
msdyn_vrmvendorpostaladdress	mserp_vrmpartylocationpostaladdressentity
contact	mserp_vrmpersonentity
mserp_vrmportaluserrequestentity	msdyn_globalvendorportalaccess
mserp_vrmpartycontactentity	msdyn_vrmpartycontact
msdyn_globalvendor	mserp_vrmglobalvendorentity
mserp_vrmpartylocationpostaladdressentity	msdyn_vrmvendorpostaladdress

4. If sync configuration records are missing or the record count doesn't match expectations, follow the instructions in How can I resolve sync configuration issues?.

Confirm sync config table mappings are current

Use the **Modified On** value to confirm that the sync configuration records were refreshed during the latest deployment.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, go to **Data Sync** > **Sync Configs**.
3. Select **Edit columns**.
4. In the **Edit columns** dialog, select **Add columns**.
5. Select **Modified On**, and then select **Close**.
6. Select **Apply**.
7. Verify that the **Modified On** value for each sync configuration matches the date when you last deployed the Supplier Engagement solution.
8. If the dates aren't correct, follow the instructions in How can I resolve sync configuration issues?.

Related information

- Supplier Engagement deployment overview, prerequisites, and licensing
- Install the Supplier Engagement app and supplier portal on Power Platform
- Onboard using the onboarding guide

- Configure Supplier Engagement elements in Power Platform
- Configure Supplier Engagement features in Supply Chain Management
- Deployment FAQs

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Deployment FAQs (preview)

[This article is prerelease documentation and is subject to change.]

Deployment questions often come up after the initial installation, especially when sync records don't appear as expected or when administrators can't find the supplier portal site. This FAQ collects the most common troubleshooting paths so that you can get deployment back on track quickly. Use it together with the validation checklist when you're confirming a new environment.

How can I resolve sync configuration issues?

If sync configurations are missing or incorrect after deployment, reinstall the anchor solution package components in the environment.

1. Sign in to the Power Apps maker portal using the service account that you used for installation.
2. Go to **Solutions** > **Managed Solutions**.
3. Find and delete the *Supplier Engagement Anchor* solution.
4. Go to the Power Platform admin center.
5. Reinstall the Supplier Engagement package.

> **Note**
>
> Learn more in Install the Supplier Engagement app and supplier portal on Power Platform.

How do I locate the supplier portal site?

Use Power Pages to find the installed supplier portal site when you need to reactivate it, configure security, or review site settings.

1. Go to the Power Pages maker portal.
2. In some environments, you might be redirected to the *Welcome to Power Pages* onboarding experience instead of the site listing page. If that happens, select **Get started**.
3. Go to the **Sites** page. If the **Tell us about yourself** page appears, update the browser address. Replace the path ending in `portals/create#roleAndIndustry` with `environments/<environment-id>/portals/home`.
4. Find *Microsoft Dynamics 365 Supplier Portal* in the site list.

Related information

- Supplier Engagement deployment overview, prerequisites, and licensing
- Install the Supplier Engagement app and supplier portal on Power Platform
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- Configure Supplier Engagement elements in Power Platform
- Configure Supplier Engagement features in Supply Chain Management
- Post-deployment validation checklist

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Configure the Supplier Engagement app overview (preview)

[This article is prerelease documentation and is subject to change.]

Supplier Engagement configuration establishes the reference data that supports supplier onboarding, vendor management, risk assessment, and collaboration. When these entities are set up correctly, users can work with consistent options and complete related processes without missing required values. Planning this setup early helps the rest of the app work as intended.

Why configuration matters

Configuration matters for the following reasons:

- Accurate setup ensures that users see only valid options during onboarding, vendor management, and risk assessment.
- If required configuration is missing, users might not be able to complete related tasks such as assigning certificates or categorizing vendors.
- Features such as onboarding questionnaires, risk management, and compliance tracking depend on these entities being available in advance.

For example, if you want vendors to provide a specific certificate during onboarding, you must first define the certificate type and certifying organization. Likewise, if you want to segment vendors by industry, you must first configure the relevant segments and product categories.

Configuration entities and dependent features

The following table summarizes each configuration entity, its purpose, and the features that depend on it.

Expand table

Configuration entity	Purpose	Dependent features
Certificate types	Defines standard types of certificates vendors can hold.	Vendor certification management, onboarding
Certifying organizations	Stores recognized bodies that issue certificates.	Vendor certification management

Configuration entity	Purpose	Dependent features
Product categories	Classifies vendor offerings and supports a hierarchical structure.	Vendor registration, product and service assignment
Segments	Further classifies vendors for reporting and management.	Vendor segmentation, analytics
Capability types	Categorizes vendor skills and capabilities.	Capability assignment, vendor qualification
Qualities	Defines specific attributes or standards that vendors might have.	Vendor qualification, compliance checks
Risk scenarios	Identifies potential risks that are associated with vendors.	Risk assessment, corrective action planning
Corrective action types	Defines actions to mitigate identified risks.	Risk management, compliance tracking
Self-assessment questions	Evaluates vendor suitability during onboarding.	Vendor onboarding, qualification
Global vendor types	Classifies vendors for reporting and management.	Vendor registration, reporting
Countries/regions	Maintains a list of countries/regions for vendor classification.	Vendor registration, compliance, reporting
Hold reasons	Explains why a vendor is placed on hold.	Vendor management, compliance, audit

> **Note**
>
> You can import all configuration data by using Excel. Learn more in /power-apps/user/import-data.

Related information

- Configure certificate types and organizations
- Configure capabilities
- Configure risk management elements
- Configure self-assessment questions
- Configure Supplier Engagement app settings

Note: The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

Configure certificate types and organizations (preview)

[This article is prerelease documentation and is subject to change.]

Certificates help validate vendor qualifications and compliance requirements. Certificate types standardize the certifications that you track, while certifying organizations identify the recognized bodies that issue them. Configuring both areas first helps ensure accurate reporting and smoother onboarding later.

Understand certificate setup

To prepare your system for use, configure the certificate types and certifying organizations that apply to your business. You'll add individual vendor certificates later during vendor setup and onboarding.

Configure certificate types and certifying organizations

Use the following procedure to create, update, or delete certificate types and certifying organizations in the Supplier Engagement app.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. Do one of the following steps:

    - To configure certificate types, go to **Configuration** > **Certificates** > **Certificate types**.
    - To configure certifying organizations, go to **Configuration** > **Certificates** > **Certifying organizations**.

3. Either open an existing record to edit it or, on the command bar, select **New** to create a record. You can also select **Delete** to remove a selected record.
4. Enter the following information:

    - **Name** – Enter a descriptive name for the certificate type or certifying organization.
    - **Description** – Enter details that help users understand when to use the record.

5. Select **Save**.

Related information

- Configure the Supplier Engagement app overview
- Configure capabilities
- Configure risk management elements
- Configure self-assessment questions
- Configure Supplier Engagement app settings

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Configure capabilities (preview)

[This article is prerelease documentation and is subject to change.]

Capabilities define how vendor offerings, specializations, and standards are organized in Supplier Engagement. By using predefined values, teams can classify vendors consistently and support better sourcing, qualification, and supplier management decisions. These settings also support hierarchical structures that make complex vendor profiles easier to manage.

Capability areas

In the Supplier Engagement app, you can configure four types of capabilities:

- **Product categories** – Classify a vendor's products using a structured list based on industry standards. This classification helps you organize and search for vendors efficiently within the system. Examples include:
  - **Automotive components** – Tires, batteries, radiators, brakes
  - **Industrial machinery** – Cranes, forklifts, pumps, compressors
  - **Food and beverage** – Frozen meals, canned goods, baked goods, dairy products
  - **Construction materials** – Doors, roofing, cement, bricks, tiles

- **Segments** – Represent the broader product or service groups a vendor operates in. This representation helps you categorize vendors by the industries they serve. Examples include:
  - **Medical equipment** – Irradiation equipment, electro-medical devices, hospital furniture
  - **Electronic and electricals** – Lighting equipment, motors, generators, switches, connectors

- **Capability types** – Describe a vendor’s technical skills, manufacturing processes, and production capacity. This information helps you determine whether a vendor can meet operational or project-specific requirements. Examples include:
  - **Metal fabrication** – Bending, punching, stamping, tool manufacturing, progressive die stamping

- **Qualities** – Define the standards, certifications, and testing processes a vendor follows. This definition ensures vendors meet industry, customer, or regulatory requirements. Examples include:
  - **Inspection and testing** – Proof load testing, impact testing
  - **Certifications** – ISO 9001, FDA approval
  - **Processes** – PPAP levels, VDA levels

- **Additional areas** – Compliance, traceability

Capability hierarchies

All capability entities use parent-child hierarchies. For example, a product category such as *Automotive Components* can include child categories such as *Tires* and *Brakes*, while a capability type such as *Metal Fabrication* can include sub-capabilities such as *Stamping* and *Bending*.

These hierarchies use self-referential one-to-many relationships, so each record can link to a parent record of the same type. This approach supports flexible classification and helps users navigate vendor information efficiently.

Create, delete, and edit capabilities

Use the following procedure to manage product categories, segments, capability types, and qualities.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, under the **Capabilities** heading, select the capability entity that you want to set up: **Product categories**, **Segments**, **Capability types**, or **Qualities**.
3. Either open an existing record to edit it or, on the command bar, select **New** to create a record. You can also select **Delete** to remove a selected record.
4. Enter the following information for the new or selected record:

    - **Name** – Enter a descriptive name.
    - **Parent** – If the record belongs under another record of the same type, select the parent record to place it in the hierarchy.

5. Select **Save**.

Related information

- Configure the Supplier Engagement app overview
- Configure certificate types and organizations
- Configure risk management elements
- Configure self-assessment questions
- Configure Supplier Engagement app settings

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

Configure risk management elements (preview)

[This article is prerelease documentation and is subject to change.]

Risk configuration helps purchasers identify, assess, and monitor issues that might affect vendors or supply chain operations. When you define risk scenarios and corrective action types ahead of time, teams can evaluate risks consistently and track mitigation work more effectively. Historical analysis and reporting also become easier because the same structures are used across the supplier network.

Risk scenarios

Risk scenarios define the types of issues that can affect vendors or supply chain operations, such as financial instability, regulatory noncompliance, quality failures, or delivery delays. Configuring these scenarios helps purchasers classify and evaluate risks consistently across vendors.

Corrective action types

Corrective action types define the measures that purchasers take to reduce or eliminate identified risks. Each action type represents a mitigation approach that you can associate with one or more risk scenarios.

Risk hierarchies

Both risk scenarios and corrective action types support parent-child relationships. This structure lets you group related risks and mitigation strategies into multilevel hierarchies that are easier to analyze and manage.

For example, a parent risk scenario such as *Dependence* can include child scenarios, while a corrective action type such as *Mitigation* can include more specific actions beneath it. These relationships use self-referential one-to-many relationships, where each record can link to a parent record of the same type.

Create, delete, and edit risk management elements

Use the following procedure to manage risk scenarios and corrective action types.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, under the **Risks** heading, select the risk entity that you want to set up: **Risk scenarios** or **Corrective action types**.
3. Either open an existing record to edit it or, on the command bar, select **New** to create a record. You can also select **Delete** to remove a selected record.
4. Enter the following information for the new or selected record:

    - **Name** – Enter a descriptive name.
    - **Parent** – If the record belongs under another record of the same type, select the parent record to place it in the hierarchy.

5. Select **Save**.

Related information

- Configure the Supplier Engagement app overview
- Configure certificate types and organizations
- Configure capabilities
- Configure self-assessment questions
- Configure Supplier Engagement app settings

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Configure self-assessment questions (preview)

[This article is prerelease documentation and is subject to change.]

Vendors see self-assessment questions in the supplier portal during onboarding. These questions help you evaluate vendor compliance and capabilities before you make qualification decisions. A well-structured question set also makes assessments more consistent across vendors.

Create, delete, and edit self-assessment questions

Use the following procedure to manage self-assessment questions in the Supplier Engagement app.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, go to **Self-assessment** > **Questions**.
3. Either open an existing record to edit it or, on the command bar, select **New** to create a record. You can also select **Delete** to remove a selected record.
4. Enter the following information for the new or selected record:

    - **Name** – Enter the question exactly as you want potential vendors to see it.
    - **Number** – Control the order in which questions appear on the page. Questions with lower numbers appear first. If two questions have the same number, they're shown in creation order. To keep sorting consistent, use leading zeros for single-digit values, such as 01, 02, 03, and 10.
    - **Area** – Select the operational area that the question applies to. Available areas include *Supply Chain Management*, *Quality management*, *Quality control*, *Human resources*, *Environmental*, *Social*, and *Governance (ESG)*, and *Production*.
    - **Answer type** – Choose whether vendors respond by entering **Text** or selecting **Yes/No**.

5. Select **Save**.

Related information

- Configure the Supplier Engagement app overview
- Configure certificate types and organizations

- Configure capabilities
- Configure risk management elements
- Configure Supplier Engagement app settings

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Configure Supplier Engagement app settings (preview)

[This article is prerelease documentation and is subject to change.]

Portal management setup defines reference data that's used throughout the Supplier Engagement application. Before you onboard vendors, configure the global settings that support consistent classification, compliance alignment, and reporting. These settings include global vendor types, countries and regions, and hold reasons.

Global vendor types

Global vendor types classify vendors based on the nature of their business or the goods and services that they provide. This setup helps organize vendors for management and analysis in the Supplier Engagement app.

Countries/regions

Countries/regions identify a vendor's geographic location in the system. This setup also supports registration requests from anonymous users who don't have direct access to Supply Chain Management.

The list should match the countries/regions that are configured in Supply Chain Management so data stays consistent between systems.

Hold reasons

Hold reasons specify why a vendor is on hold, such as compliance issues, performance concerns, or regulatory requirements. These reasons help users record and communicate why vendor activity is restricted.

Create, delete, and edit settings

Use the following procedure to manage global vendor types, countries/regions, and hold reasons.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, in the **Settings** section, select **Global vendor types**, **Countries/regions**, or **Hold reasons**.
3. Either open an existing record to edit it or, on the command bar, select **New** to create a record. You can also select **Delete** to remove a selected record.
4. Enter the following information for the new or selected record:

    - **Name** – For global vendor types and hold reasons only, enter a descriptive name. Users and vendors see these names as selectable options in Supplier Engagement and the supplier portal, so each name should clearly represent the option.
    - **Description** – For global vendor types and hold reasons only, enter a longer description.
    - **Country/region** – For countries/regions only, enter the three-letter country or region code that's used in Supply Chain Management.
    - **Long name** and **Short name** – For countries/regions only, use **Short name** for a common name such as *United Kingdom* and **Long name** for the full formal name such as *United Kingdom of Great Britain and Northern Ireland*.

5. Select **Save**.

Related information

- Configure the Supplier Engagement app overview
- Configure certificate types and organizations
- Configure capabilities
- Configure risk management elements
- Configure self-assessment questions

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Synchronize data between Supplier Engagement and Supply Chain Management (preview)

[This article is prerelease documentation and is subject to change.]

Data synchronization keeps supplier-related information consistent between Supply Chain Management and the Supplier Engagement app. It uses event-driven processes and configurable sync logic to support full create, read, update, and delete (CRUD) operations on data that resides in Supply Chain Management while maintaining the data in its original location. These synchronization features are available both during initial installation and day-to-day use of the solution.

> **Tip**
>
> We recommend that you always use the Supplier Engagement application to create and maintain aggregated supplier information across legal entities. System synchronization logic ensures that supplier data is synced to the respective legal entities within Dynamics 365 Supply Chain Management.

Prerequisites

Before you can start synchronizing data, your systems must meet the following requirements:

- Your Supply Chain Management environment must be connected to Dataverse. Learn more in Microsoft Power Platform integration with finance and operations apps.
- The Supplier Engagement app must be installed. Learn more in Install the Supplier Engagement app and supplier portal on Power Platform.
- The *Supplier Engagement* feature must be turned on in Supply Chain Management. Learn more in Enable the Supplier Engagement feature.
- You must complete all configuration steps described in the deployment guide. Learn more in Supplier Engagement deployment overview, prerequisites, and licensing.

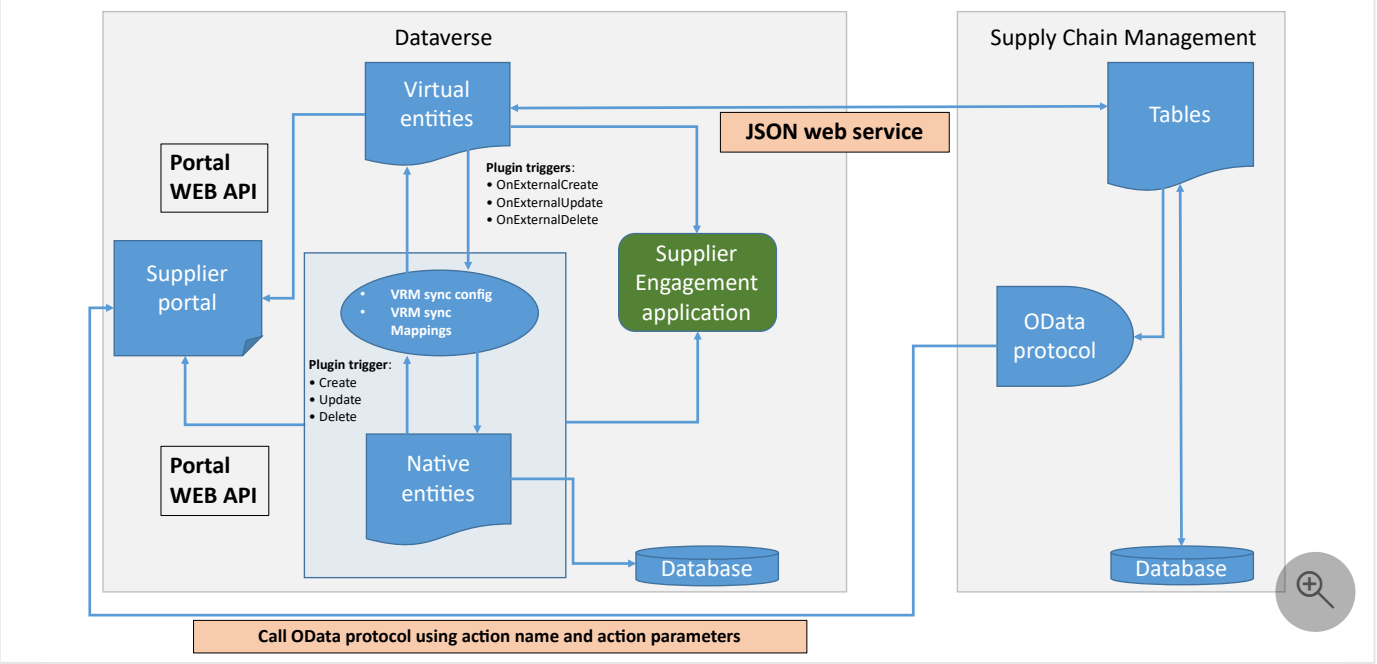
Data synchronization overview

The Supplier Engagement app uses the following components to synchronize data between the systems:

- **Sync configuration manager** – Uses plugin triggers and field mappings to keep Dataverse tables and virtual entities in sync.
- **Virtual entities** – Represent Supply Chain Management data in Dataverse without duplicating it.
- **Supplier portal Web API** – Allows the supplier portal to perform CRUD operations on Dataverse tables and virtual entities.
- **Open Data Protocol (OData)** – Provides standardized web service access to Supply Chain Management data and processes.

The system uses virtual entities from Supply Chain Management and Dataverse tables to maintain data between both systems. The sync configuration manager uses plugin triggers on both virtual entities and Dataverse tables. These plugins use the mappings from the sync configuration manager to synchronize data between both systems.

The following illustration shows the integration architecture between the Dataverse environment and Supply Chain Management.



You can learn more about how synchronization is implemented in Dataverse in the following articles:

- Use plug-ins to extend business processes
- Data synchronization in Dataverse

Sync configuration manager

The sync configuration manager ensures that data remains consistent between Supply Chain Management and the Supplier Engagement app. It uses the data stored in the Supplier Engagement app to determine which records require synchronization and triggers updates on both Supply Chain Management and Supplier Engagement tables. The supplier portal also uses this system to keep data updated between the two platforms.

Key capabilities of the sync configuration manager include:

- **Real-time synchronization** – Keeps data aligned between Supply Chain Management and the Supplier Engagement app, reducing manual updates and errors.
- **Configurable sync logic** – Administrators can define which tables and fields are synchronized, allowing for tailored integration scenarios.
- **Event-driven updates** – Changes in either system trigger synchronization, ensuring timely data updates.

Synchronized entities

The following table lists the entities that are exposed for use by the sync configuration manager. Entities with the *(mserp)* suffix are virtual entities from Supply Chain Management that are exposed in Dataverse. Entities without the suffix are native Dataverse tables. The names shown here are the display names, which you can search for in Power Platform.

Expand table

Dataverse table	Virtual entity (mserp)
Global Vendor	Global Vendors (mserp)
Contact	Person for VRM (mserp)
Contact Method	Contact information for VRM (mserp)
Address	Addresses for Supplier Portal (mserp)
Global Vendor Portal Access	Supplier Portal user requests (mserp)

Data flows

This section describes the data flow for each synchronized entity pair. Each entity supports bidirectional synchronization between the Supplier Engagement app (or supplier portal) and

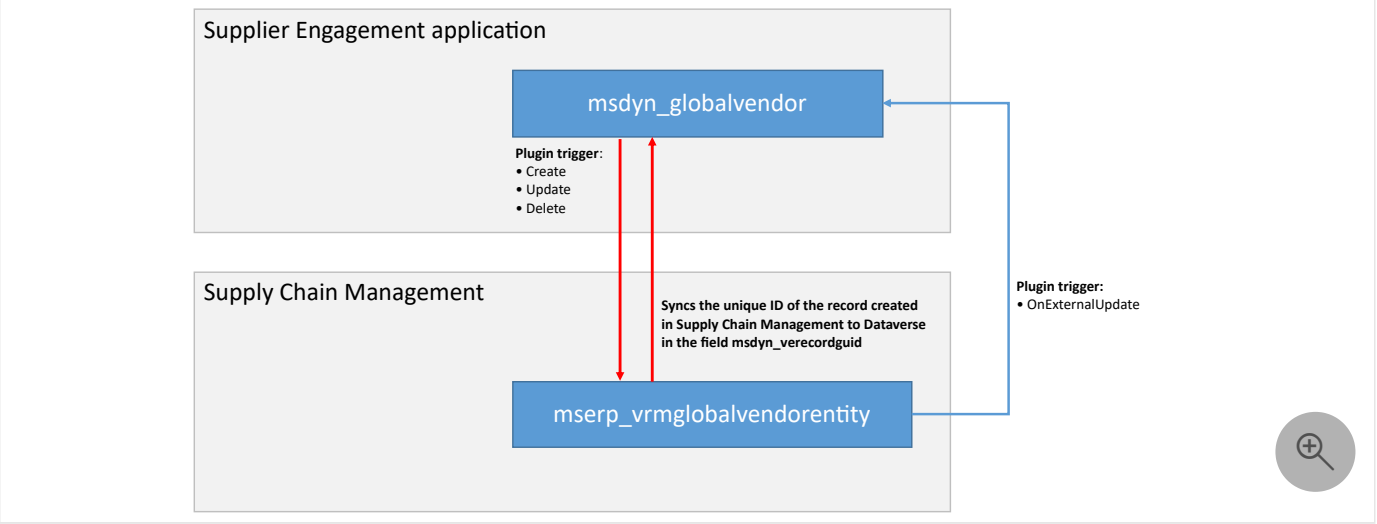
Supply Chain Management, using plugin triggers and the sync configuration manager mappings. Records are linked between systems using the `VERecordGuid` field, which stores the Supply Chain Management record ID in Dataverse.

Global vendor

When you create, update, or delete a global vendor record in the Supplier Engagement app, plugins trigger on the `msdyn_globalvendor` Dataverse table and synchronize the changes to the `mserp_vrmglobalvendorentity` virtual entity in Supply Chain Management.

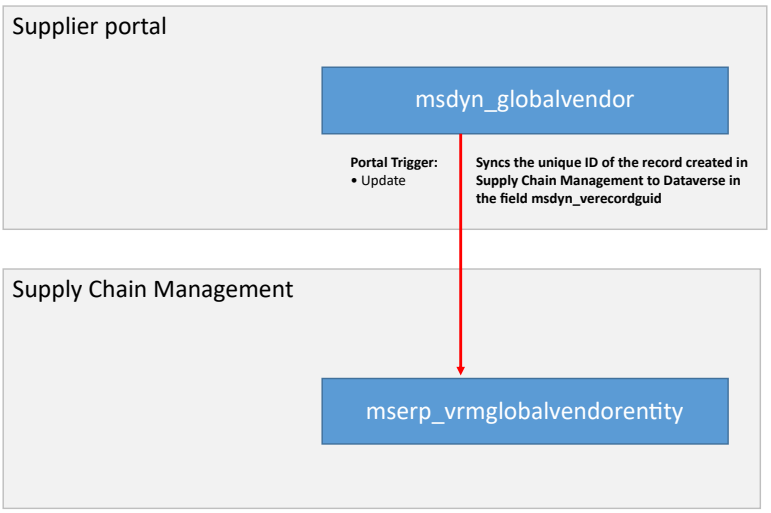
When you update the virtual entity in Supply Chain Management, an `OnExternalUpdate` plugin triggers and synchronizes the changes back to Dataverse.

The following illustration shows the data flow between the Supplier Engagement app and Supply Chain Management for global vendor records.



When you update a global vendor through the supplier portal, the updated data synchronizes to Supply Chain Management using the sync configuration manager mappings via the Web API.

The following illustration shows the data flow from the supplier portal to Supply Chain Management for global vendor records.

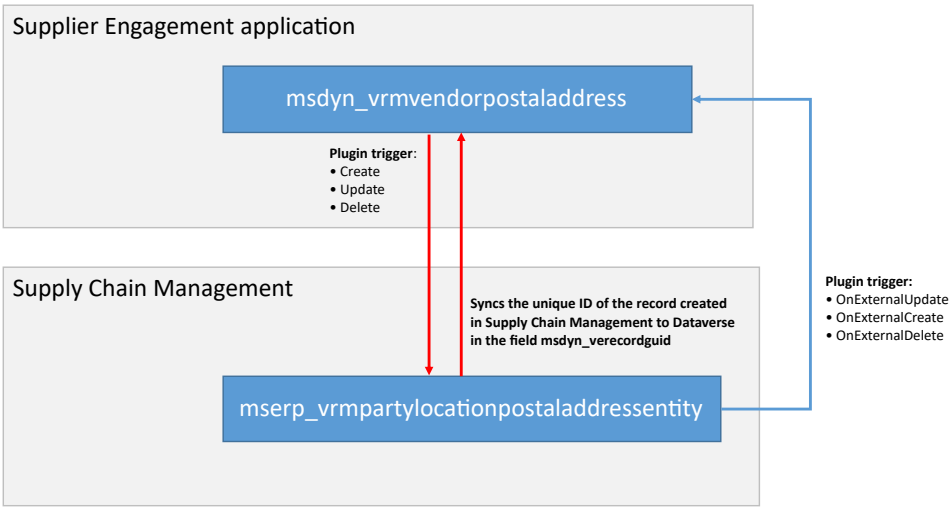


Address

When you create, update, or remove an address for a global vendor that is qualified or approved, plugins trigger on the `msdyn_vrmvendorpostaladdress` Dataverse table and synchronize the changes to the `mserp_vrmpartylocationpostaladdressentity` virtual entity in Supply Chain Management.

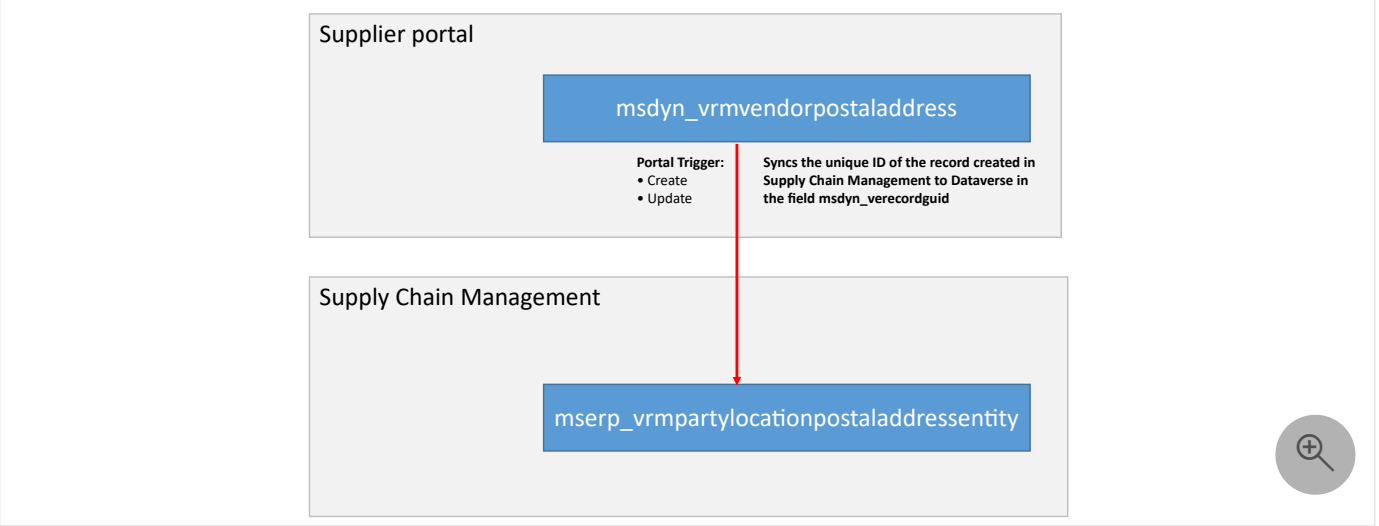
When you create, update, or delete the virtual entity record in Supply Chain Management, the corresponding `OnExternalCreate`, `OnExternalUpdate`, or `OnExternalDelete` plugin triggers and synchronizes the changes back to Dataverse. If you delete a record in Supply Chain Management, the corresponding Dataverse record is also deleted.

The following illustration shows the data flow between the Supplier Engagement app and Supply Chain Management for address records.



You can also create or update addresses through the supplier portal. Changes synchronize to Supply Chain Management using the sync configuration manager mappings via the Web API.

The following illustration shows the data flow from the supplier portal to Supply Chain Management for address records.

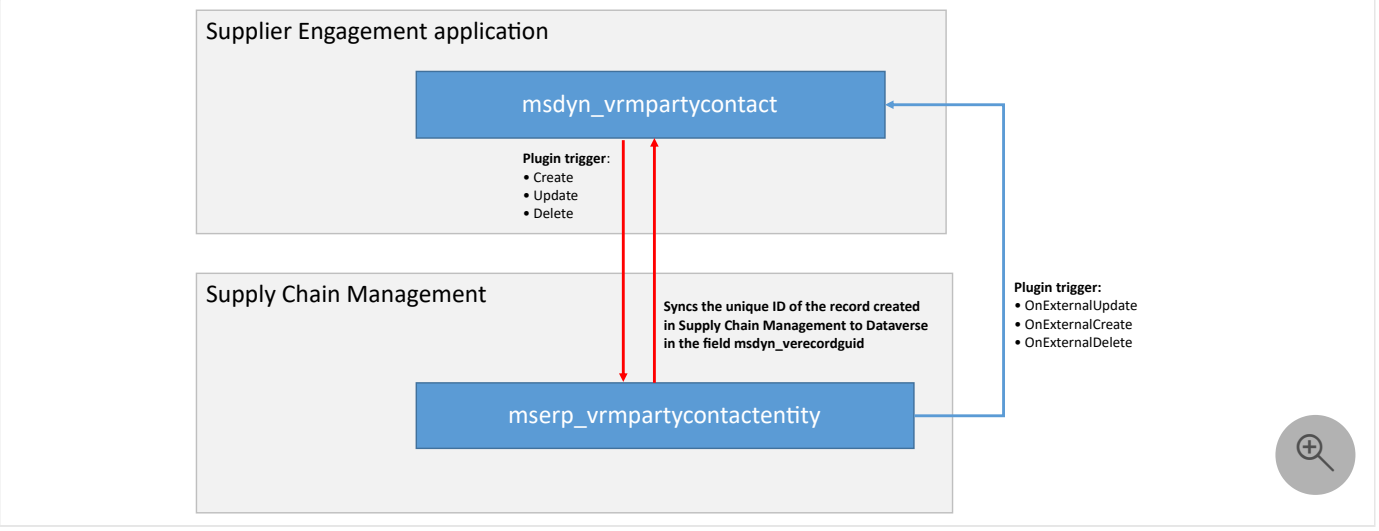


Contact method

When you create, update, or delete a contact method record, plugins trigger on the `msdyn_vrmpartycontact` Dataverse table and synchronize the changes to the `mserp_vrmpartycontactentity` virtual entity in Supply Chain Management.

When you create, update, or delete the virtual entity record in Supply Chain Management, the corresponding plugin triggers and synchronizes the changes back to Dataverse. If you delete a record in Supply Chain Management, the corresponding Dataverse record is also deleted.

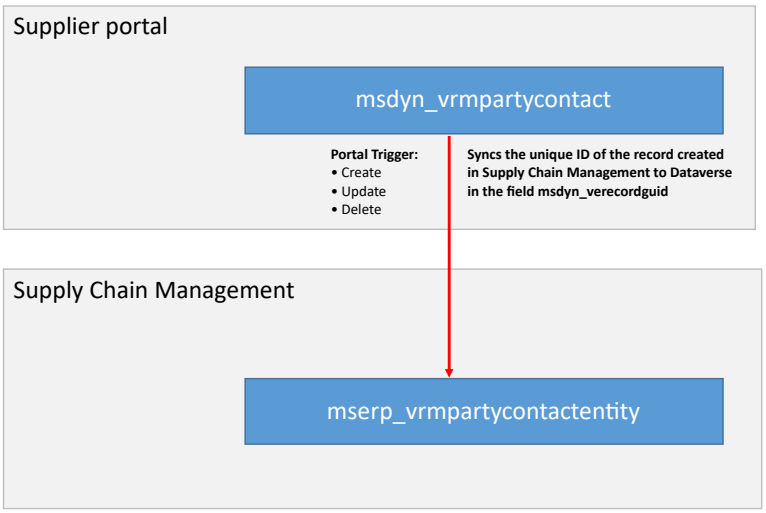
The following illustration shows the data flow between the Supplier Engagement app and Supply Chain Management for contact method records.



You can also create, update, or delete contact method records through the supplier portal. Changes synchronize to Supply Chain Management by using the sync configuration manager

mappings via the Web API.

The following illustration shows the data flow from the supplier portal to Supply Chain Management for contact method records.

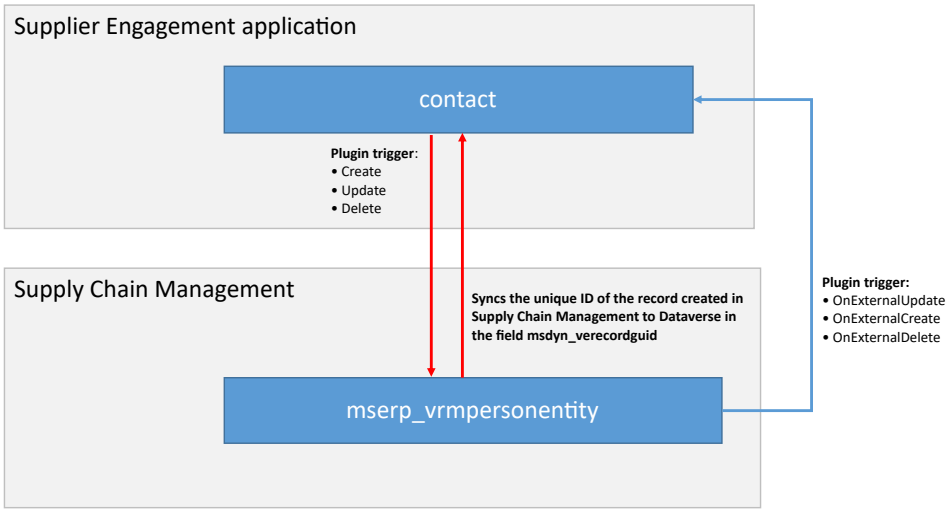


Contact

When you create, update, or delete a contact record, plugins trigger on the `contact` Dataverse table and synchronize the changes to the `mserp_vrmpersonentity` virtual entity in Supply Chain Management.

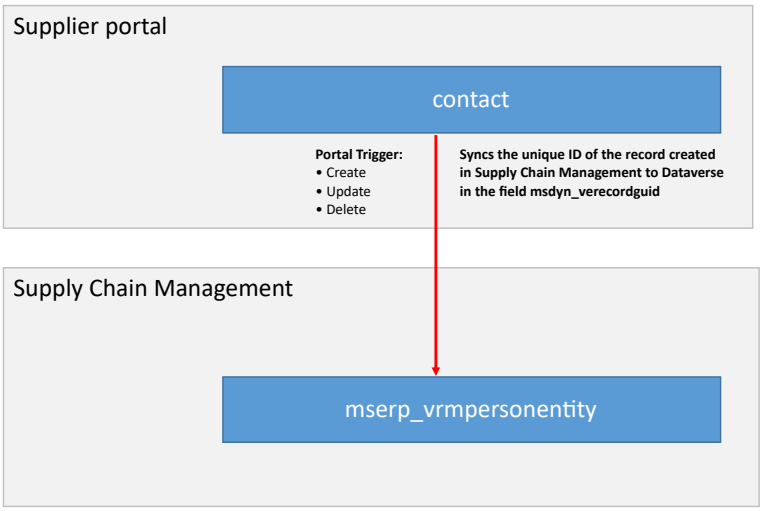
When you create, update, or delete the virtual entity record in Supply Chain Management, the corresponding plugin triggers and synchronizes the changes back to Dataverse. If you delete a record in Supply Chain Management, the corresponding Dataverse record is also deleted.

The following illustration shows the data flow between the Supplier Engagement app and Supply Chain Management for contact records.



You can also create, update, or delete contacts through the supplier portal. Changes are synchronized to Supply Chain Management using the sync configuration manager mappings via the Web API.

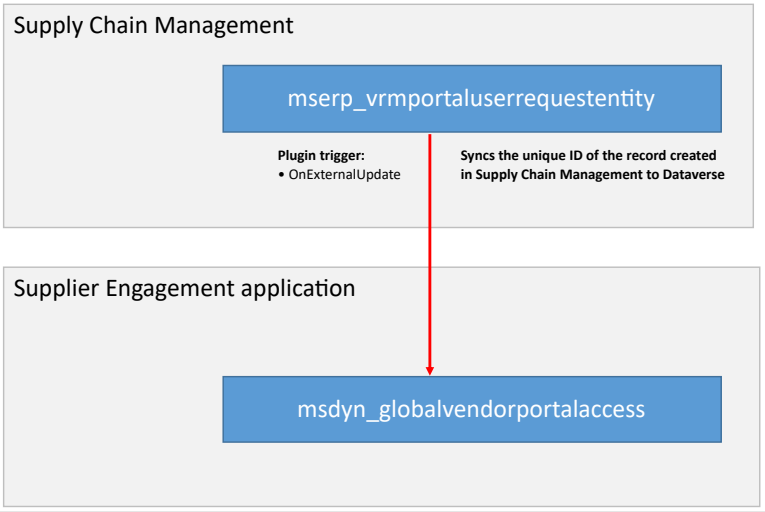
The following illustration shows the data flow from the supplier portal to Supply Chain Management for contact records.



Portal access

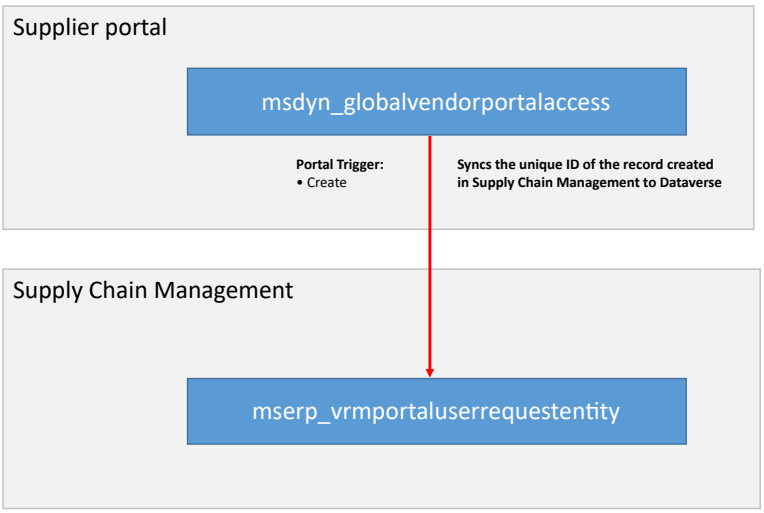
When you update a supplier portal user request record in Supply Chain Management, an `OnExternalUpdate` plugin triggers on the `mserp_vrmportaluserrequestentity` virtual entity and synchronizes the changes to the `msdyn_globalvendorportalaccess` Dataverse table.

The following illustration shows the data flow from Supply Chain Management to the Supplier Engagement app for portal access records.



You can also create global vendor portal access records from the supplier portal. The records are synchronized to Supply Chain Management using the sync configuration manager mappings via the Web API.

The following illustration shows the data flow from the supplier portal to Supply Chain Management for portal access records.



> **Note**
>
> For portal access records, the link between Dataverse and Supply Chain Management uses the Entra ID and User ID fields (rather than `VERecordGuid`).

Mapping components

The sync configuration manager uses two related tables to manage field-level mappings:

- **VRM Sync Config** – The parent table. Each record defines a synchronization configuration that specifies the source table and the destination table.
- **VRM Sync Mappings** – The child table. Each record specifies how individual fields from the source table are mapped to fields in the destination table. Each mapping record is linked to a parent record in the VRM Sync Config table.

In summary, VRM Sync Config defines *what* is synchronized (which tables), and VRM Sync Mappings defines *how* the data is synchronized (which fields are mapped). Each sync configuration can have multiple attribute mappings associated with it.

Configure sync mappings

When you install the Supplier Engagement app, it creates data for the VRM Sync Config and VRM Sync Mappings tables. Only users with the *Supplier Engagement Admin* role can modify these tables. Users with other roles have read-only access.

> **⚠ Warning**
>
> Changes to the sync configuration manager setup affect data synchronization for all configured entities. Only make changes as part of post-deployment steps, and only if required. Only administrators should perform these activities.

To add a new sync configuration mapping, follow these steps:

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, select **Data Sync** > **Sync Config**.
3. On the command bar, select **New**.
4. In the **Source Table** field, enter the logical name of the source entity in text format.
5. In the **Destination Table** field, enter the logical name of the destination entity in text format.
6. On the command bar, select **Save**. A **Field Mappings** area appears on the record.
7. On the toolbar, select **New Field Mappings** to add a field mapping.
8. In the new mapping window, set the following fields:

    - **Source field** – Enter the source field logical name in text format.
    - **Destination field** – Enter the destination field logical name in text format.
    - **Type** – Select the field type. Allowed types are *Text*, *Whole Number*, *Decimal*, *Lookup*, *Choice*, and *Date and Time*.

9. Select **Save**.
10. Go back to the parent record, set **Generate Config** to *Yes*, and select **Save** on the command bar.

The system creates a JSON file on the backend that the sync configuration manager uses to synchronize records between the source and destination tables.

To remove a sync configuration mapping, follow these steps:

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Configuration** area.
2. On the navigation pane, select **Data Sync** > **Sync Config**.
3. Open the record from which you want to remove a mapping.
4. In the **Field Mappings** area, select the mapping that you want to remove.
5. On the toolbar, select **Delete Sync Config Mapping**.

6. Go back to the parent record, set **Generate Config** to *Yes*, and select **Save** on the command bar.

You can also activate or deactivate individual sync config mappings in the same way. Always select **Generate Config and Save** after making changes to update the configuration with the latest settings.

Relation to dual-write

The custom sync logic in Supplier Engagement is distinct from the standard dual-write functionality in Supply Chain Management. While dual-write provides out-of-the-box data integration, the sync configuration manager offers more tailored and granular control over which fields are synchronized.

> ⓘ **Important**
>
> If you enable both dual-write and the sync configuration manager for the same entities, the system might create duplicate records. Don't configure both systems to synchronize the same data.

Virtual entities

You can integrate Supply Chain Management with Microsoft Dataverse as a virtual data source. This integration makes Supply Chain Management entities, such as vendors, purchase orders, and contacts, available in Dataverse as virtual entities. These virtual entities act as a bridge, so users and applications in Dataverse and the broader Microsoft Power Platform can perform CRUD operations on Supply Chain Management data.

Key points about virtual entities:

- **Data location** – The actual data stays in Supply Chain Management. Dataverse only references this data and doesn't store it.
- **Enablement** – Before you can perform CRUD operations from Dataverse, you must expose the relevant Supply Chain Management entities as virtual entities. This step requires configuration. Learn more in Enable Microsoft Dataverse virtual entities.
- **Operations** – Once enabled, you can interact with Supply Chain Management data from Dataverse and Power Platform tools, just as you would with native Dataverse tables.

Learn more about virtual entities in Virtual entities overview.

Open Data Protocol

Open Data Protocol (OData) actions enable external systems and applications to interact with Supply Chain Management data and processes using standard web services. By using OData, you can securely and consistently create, update, or retrieve business data.

The following entities in the Supplier Engagement solution expose OData actions:

- Supplier Portal users
- Vendor-approved procurement categories for Supplier Portal
- Vendor category request headers for Supplier Portal
- Purchase order response headers for Supplier Portal
- Purchase order response lines for Supplier Portal
- RFQ reply headers for Supplier Portal
- Request for quotation reply lines for Supplier Portal
- Pending vendor invoice headers for Supplier Portal
- Request for quotation questionnaires for Supplier Portal

To support OData actions within the Supply Chain Management user security context, each of these entities exposes two key fields:

- `ActionParameters` – Specifies the input data for the OData action.
- `ActionName` – Identifies the specific OData action to execute.

Related information

- Supplier Engagement overview
- Supplier Engagement deployment overview, prerequisites, and licensing
- Configure the Supplier Engagement app overview
- Work with the Supplier Engagement app
- Work with Supplier Engagement data in Supply Chain Management

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Manage users of the Supplier Engagement app and supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement solution uses a multilayered security framework to control access across the Supplier Engagement app, the supplier portal, and Supply Chain Management. Security roles and web roles define what each user can see and do—from managing global vendor data and processing registrations to inviting portal users and accessing transactional documents.

User management is divided into two areas:

- **Supplier Engagement app users** – Internal users such as procurement staff and supplier managers who work in the Supplier Engagement model-driven app. Access is controlled through Dataverse security roles that map to corresponding roles in Supply Chain Management.
- **Supplier portal users** – External supplier representatives who access the portal to collaborate on purchase orders, RFQs, invoices, and consignment inventory. Access is controlled through Power Pages web roles that are synchronized with Supply Chain Management external user roles.

Both areas share a common principle: user permissions are managed centrally in the Supplier Engagement app and automatically synchronized to Supply Chain Management. This approach eliminates the need for separate user administration in each system.

> **Important**
>
> Internal users who are part of the procurement team must have the required security roles in both Supply Chain Management and the Supplier Engagement (Dataverse) app. This dual access is required to allow synchronization and related Supplier Engagement processes to work correctly across both systems.

How user access works

The following table summarizes how access control works at each layer of the solution.

Expand table

Layer	User type	Access mechanism
Supplier Engagement app (Dataverse)	Internal users	Dataverse security roles
Supplier portal (Power Pages)	External supplier contacts	Web roles assigned in the Supplier Engagement app
Supply Chain Management	Both internal and external	Roles synchronized from the Supplier Engagement app

For internal users, Dataverse security roles determine what features are available in the Supplier Engagement app. These roles map to Supply Chain Management roles, such as *Purchasing Manager* and *Purchasing Agent*, so that internal users also have appropriate access to supply chain data.

For external users, web roles control what portal pages and features are visible. Each portal user automatically maps to an external user in Supply Chain Management. Supply Chain Management roles are assigned based on the web roles, and extensible data security (XDS) policies ensure that each user can only access data belonging to their vendor.

Related information

- Manage Supplier Engagement app users
- Manage supplier portal users
- Supplier Engagement overview
- Supplier Engagement deployment overview, prerequisites, and licensing

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

Manage Supplier Engagement app users (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement app defines security roles in Dataverse that control what internal users can do in the app. These roles determine access to features such as configuration management, vendor lifecycle processes, user administration, and feedback. Each Dataverse role maps to a corresponding role in Supply Chain Management to ensure consistent access across both systems.

Supplier Engagement app security roles

The Supplier Engagement app defines four security roles for Dataverse. The following table summarizes each of these roles and their key permissions. In addition, each internal user must also have the *Finance and Operations Basic User Role*.

Expand table

Security role	Description	Key permissions
Supplier Relationship Management System Administrator	Has full access to all Supplier Engagement operations. Manages configurations, master data, user access, and flow history. Oversees the full lifecycle of global vendor records.	Can create configuration data, approve or reject registrations, update records, and initiate legal entity transfers. Can delete all records, including configuration data.
Supplier Portal User Admin	Manages portal user access. Invites users, approves or rejects requests, assigns roles, and manages vendor portal access.	Has access specifically to manage portal user access to vendor records and therefore has limited access to other Supplier Engagement features.
Global Vendor Manager	Oversees global vendor operations, including certificates, feedback, risk assessment, onboarding, and qualification.	Can approve or reject registrations, update records, and initiate legal entity transfers. Can delete records they created.
Global Vendor Professional User	Supports onboarding, maintains vendor data, manages certificates, provides feedback, and participates in qualification and approval workflows for assigned vendors.	Can delete records they created, but not those created by others or configuration data.

Detailed permissions by area

The following table provides a detailed overview of which areas of the Supplier Engagement app each security role has access to.

Expand table

Category	Scenario	*Supplier Relationship Management System Administrator*	*Supplier Portal User Admin*	*Global Vendor Manager*	*Global Vendor Professional User*
Configuration data	Capabilities	Yes	No	No	No
	Risks	Yes	No	No	No
	Self-assessment	Yes	No	No	No
	Settings	Yes	No	No	No
Registration request	View registration requests	Yes	No	Yes	Yes
	Review registration requests	Yes	No	Yes	Yes
	Approve/disapprove registration requests	Yes	No	Yes	Yes
Global vendor	General information	Yes	No	Yes	Yes
	Contacts	Yes	No	Yes	Yes
	Certificate management	Yes	No	Yes	Yes
	Feedback	Yes	No	Yes	Yes
	Risk management	Yes	No	Yes	Yes
	Portal contacts	Yes	Yes	Yes	No
	Lifecycle and vendors	Yes	No	Yes	Yes

Map Supplier Engagement roles to Supply Chain Management roles

Each Supplier Engagement app security role maps to a corresponding role in Supply Chain Management. The following table shows these mappings.

Expand table

Supplier Engagement app role	Supply Chain Management role
Supplier Relationship Management Admin	*Purchasing Manager*
Supplier Portal User Admin	*Purchasing Manager*
Global Vendor Manager	*Purchasing Manager*
Global Vendor Professional User	*Purchasing Agent*

In Supply Chain Management, the *Purchasing Manager* role manages the full procurement lifecycle, including requisitions, orders, vendor interactions, and purchasing policies. The *Purchasing Agent* role supports day-to-day procurement tasks such as creating and managing purchase orders and maintaining vendor details, with more limited access than the manager role.

Related information

- Manage users of the Supplier Engagement app and supplier portal
- Manage supplier portal users
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

Manage supplier portal users (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement solution provides tools for managing external supplier users through the Supplier Engagement app and the supplier portal. You can invite suppliers, add or remove portal users, assign roles, manage access to legal entities, and deactivate portal access when needed. Supplier portal user management ensures that external contacts have the right access to the right data while maintaining centralized control across all legal entities.

How supplier portal user management works

Supplier portal user management is based on an integration between Microsoft Power Pages (Dataverse) and Supply Chain Management. Together, these systems control authentication, role assignments, and access to both portal content and supply chain data.

Power Pages (Dataverse) layer

The following principles apply when managing users at the Power Pages layer:

- Supplier portal users are represented as contacts in Dataverse.
- Each contact must have a unique email address to ensure proper identity mapping.
- Contacts are assigned web roles, which control access to Dataverse tables and portal pages.
- Web roles define what the user can see and do within the supplier portal interface.
- Every contact has an external user mapping in Supply Chain Management. Administrators configure portal roles and vendor assignments in the Supplier Engagement app, while Supply Chain Management enforces access to supply chain transactional data.

Supply Chain Management layer

The following principles apply when managing users at the Supply Chain Management layer:

- Each Power Pages contact maps to an external user in Supply Chain Management.
- The system automatically provisions external users when you invite a contact to the supplier portal.
- Supply Chain Management roles assigned to the external user define what supply chain data the portal user can access.

- Access is exposed through Dataverse virtual entities, which surface Supply Chain Management records to the portal.
- Extensible data security (XDS) policies restrict data visibility, ensuring that users see only data that belongs to their vendor.

Data access boundaries

Each portal user's access is limited to the data of their global vendor. The scope of access depends on the type of role assigned:

- **Global roles** (for example, *Global Vendor Admin* or *Global Vendor User*) – Provide access across all legal entities linked to that vendor.
- **Local roles** (for example, *Local Vendor Admin* or *Local Vendor User*) – Restrict access to specific legal entities where vendor collaboration is turned on.

User roles in the supplier portal

Supplier portal access is defined through web roles, which determine what a user can see and do in the portal. Roles fall into two broad categories: anonymous access and authenticated access.

Anonymous access

Anonymous access is granted to site visitors who aren't signed in. These users are assigned the *Global Vendor Anonymous User* role, which allows them to open the supplier portal landing page and submit new vendor registration requests. Their access is strictly limited—they can't view, edit, or delete any data, including their own submissions.

Authenticated access

Authenticated access begins once you invite and provision a contact as a portal user. Each authenticated user is linked to a global vendor in the Supplier Engagement app. Their web roles are assigned in the app and synchronized with Supply Chain Management.

Administrator and standard-user roles

The supplier portal defines the following user roles:

- *Global Vendor Administrator* – Provides full access to vendor information and user management across all legal entities.

- *Local Vendor Administrator* – Offers the same capabilities as *Global Vendor Administrator* but only within selected legal entities.
- *Global Vendor User* and *Local Vendor User* – Standard user roles that allow participation in vendor collaboration processes like managing RFQs, purchase orders, or invoices, with the scope defined globally or locally.

The following table shows the functions available to each supplier portal user role.

Expand table

Category	**Scenario**	***Global Vendor Administrator***	***Local Vendor Administrator***	***Global Vendor User***	***Local Vendor User***
Global vendor onboarding	Company details	Yes	No	Yes	No
	Contacts, addresses, and contact information	Yes	No	Yes	No
	Certificates and capabilities	Yes	No	Yes	No
	Assessment questionnaire	Yes	No	Yes	No
User management	View users list and user request	Yes	Yes	No	No
	Create a user request	Yes	Yes	No	No
	Manage user roles and access to vendors	Yes	No	No	No
	Deactivate a user	Yes	No	No	No
Global vendor setup	Update company details	Yes	Read only	Read only	Read only
	Contacts, addresses, and contact information	Yes	Read only	Read only	Read only
	Certificates and capabilities	Yes	Read only	Read only	Read only

Category	Scenario	*Global Vendor Administrator*	*Local Vendor Administrator*	*Global Vendor User*	*Local Vendor User*
Local vendor management	View vendors list, update vendor details	Yes	Yes	Yes	Read only
	Manage vendor contacts	Yes	Yes	Yes	Read only
	Bank accounts	Yes	Yes	Yes	Read only
	State tax	Yes	Yes	Yes	Read only
	Procurement category request	Yes	Yes	Yes	Read only
	View approved procurement categories	Yes	Yes	Yes	Read only
	Manage vendor certificates	Yes	Yes	Yes	Read only
Supply Chain Management transactions	Purchase orders	Yes	Yes	Yes	Yes
	Vendor invoices	Yes	Yes	Yes	Yes
	Invoices	Yes	Yes	Yes	Yes
	Consignment inventory	Yes	Yes	Yes	Yes

> **Note**
>
> The *Global Vendor Authenticated User* role is a supporting web role that's also included in the solution. It's the only role with the **Authenticated User Role** setting set to *Yes*. Don't assign this role directly to users. It serves as the base role that enables correct authentication and session management. Other web roles that you assign on top of this authenticated user role control additional access and permissions.

Transactional web roles

In addition to administrator and standard user roles, the supplier portal supports transactional web roles. These roles are designed for scenarios where you want to limit a user's access to specific business processes rather than granting access to the entire application.

The following transactional web roles are available.

Expand table

Web role name	Description
Invoice Management User	Grants access to view, create, and manage invoice-related data and actions only.
Consignment Inventory User	Grants access to consignment inventory and related transactions.
Purchase Order User	Grants access to purchase order collaboration only.
Request for Quotation User	Grants access to RFQ processes such as submitting and managing a bid.

Transactional web roles behave in the same way as the *Local Vendor User* role. Users with these roles don't have administrative access to the local vendor and can only work with vendor data in the specific legal entities where vendor collaboration is turned on.

You can also combine transactional web roles. For example, you can assign a user both the *Purchase Order User* role and the *Request for Quotation User* role if they should only participate in these two processes, without accessing invoices or consignment inventory.

Map web roles to Supply Chain Management roles

When you provision a portal user, the system maps them to an external user in Supply Chain Management. Supply Chain Management assigns roles that align with portal access, such as *Supplier Portal Prospective User*, *Supplier Portal Administrator*, or *Supplier Portal User*. These roles ensure that users can access the appropriate supply chain data entities through virtual entities, always limited by XDS policies and vendor-specific boundaries.

The following table shows how supplier portal web roles map to Supply Chain Management external user roles.

Expand table

Supplier portal web role	Supply Chain Management external user role	Description
Global Vendor Administrator / Local Vendor Administrator	*Supplier Portal Administrator*	Grants administrative access to the supplier portal.
Global Vendor User / Local Vendor User	*Supplier Portal User*	Grants access to view and manage supplier portal data.
(N/A)	*Supplier Portal Prospective User*	Assigned during vendor onboarding, before a vendor is linked to a legal entity. It's assigned to users not yet connected to a person—specifically for prospect global vendors not yet synchronized to Supply Chain Management. This role provides access only to reference data needed for onboarding. After the global vendor is qualified, the security role is automatically updated.
Invoice Management User	*Supplier Portal Invoice Management User*	Grants access to view, create, and manage invoice-related data and actions only.
Consignment Inventory User	*Supplier Portal Consignment Inventory User*	Grants access to consignment inventory and related transactions.
Purchase Order User	*Supplier Portal Purchase Order User*	Grants access to purchase order collaboration only.
Request for Quotation User	*Supplier Portal Request for Quotation User*	Grants access to RFQ processes such as submitting and managing a bid.

Enable a vendor for the supplier portal

Before a vendor can access transactional data in the supplier portal, you must activate the vendor for the portal. To activate a vendor, follow these steps:

1. Open the global vendor record that you want to activate.
2. Open the **Lifecycle** tab.
3. In the **Vendors** section, select the **Vendor account** number for the vendor you want to enable.

4. The vendor opens. On the **General** tab, in the **Vendor collaboration** section, set the **Collaboration activation** field to one of the following values:

    - *Active (PO is not auto-confirmed)* – Purchase orders require manual confirmation.
    - *Active (PO is auto-confirmed)* – Purchase orders are automatically confirmed upon submission.

5. On the command bar, select **Save**.

This activation step turns on supplier portal features and grants the vendor access to the portal based on their assigned roles.

Invite a global vendor to the supplier portal

You can invite a global vendor to join the supplier portal. By default, the global vendor's primary contact is provisioned as a *Global Vendor Administrator*. You can add more users later. To invite a global vendor, follow these steps:

1. Open the global vendor record associated with the user that you want to invite.
2. On the **Summary** tab, verify that a **Primary contact** is assigned and that the contact has a valid email address. If not, assign a primary contact and provide an email address.
3. On the command bar, select **Invite to supplier portal**.

    > ⓘ **Note**
    >
    > The **Invite to supplier portal** button is only available for global vendors that aren't yet invited to the portal.

The system performs the following actions:

- Updates the global vendor portal status to *Invited*.
- Initiates user provisioning for the primary contact.
- Creates a global vendor portal access record in the user provisioning status.
- Assigns the *Global Vendor Administrator* web role to the primary contact.
- Triggers a user provisioning workflow in Supply Chain Management.

An admin user in Supply Chain Management must then review and approve the user request. Learn more in Approve portal user requests in Supply Chain Management.

The contact receives an email invitation to access the supplier portal.

> **Important**
>
> The email invitation requires that the cloud flow *CF - Global Vendor Portal Access - Send Portal Invitation email to Contact* is set up and enabled on your system. Learn more in Enable supporting cloud flows. You must also customize the invitation email template to include the correct portal URL for your environment.

Invite additional users to the supplier portal

In addition to the primary contact, you can invite extra contacts from a supplier organization. To invite a new user, follow these steps:

1. Open the global vendor record associated with the user that you want to invite.
2. Open the **Contacts** tab.
3. Select an existing contact to open it, or select **New contact** on the **Contacts** toolbar to create a new contact and then open it.
4. On the command bar, select **Invite to supplier portal**.
5. In the invitation dialog, set the following fields:
    - **Role** – Select *Admin* and/or *User*.
    - **Access level** – Select *Global* or *Local*.
    - **Legal entity** – If **Access level** is *Local*, select the default legal entity for this user.
6. Select **OK** to confirm the selection.

The system provisions the portal user in Dataverse and creates a portal user request in Supply Chain Management with the assigned roles and legal entities. An admin user in Supply Chain Management must then review and approve the request. Learn more in Approve portal user requests in Supply Chain Management.

Approve portal user requests in Supply Chain Management

When you invite a supplier contact to the portal, the system automatically creates a portal user request in Supply Chain Management. To review portal user requests, follow these steps:

1. Open Supply Chain Management.

2. Go to **Procurement and sourcing** > **Vendors** > **Supplier Engagement** > **Portal user requests**.
3. Review and approve or reject the pending requests.

Portal user requests follow a three-level approval workflow. By default, this workflow controls the review and authorization of new users, but administrators can modify the workflow configuration to meet organizational requirements. Learn more in Set up the vendor user request workflow.

When you approve the request:

- The system creates a new external user account in Supply Chain Management. You can find new portal users at **Procurement and sourcing** > **Vendors** > **(Preview) Supplier Engagement** > **Portal users**.
- The system automatically assigns the requested security roles. The available roles are listed and described in the Map web roles to Supply Chain Management roles section.
- The system grants access either globally (all organizations) or locally (specific legal entities), depending on how you invited the user in the Supplier Engagement app.

As part of the provisioning workflow, the system initiates an Azure B2B service process. This process associates the new Supply Chain Management user account with a Microsoft Entra account. If the supplier already has a Microsoft Entra identity, the process links it; otherwise, it provisions a new one. Learn more in Configure B2B collaboration for vendor user requests.

Manage web roles for a user

To change a user's role assignment, follow these steps:

1. In the Supplier Engagement app, do one of the following steps to find the contact record you want to manage:

    - At the bottom of the navigation pane, select the **Configuration** area, and then select **Portal management** > **Contacts**.
    - At the bottom of the navigation pane, select the **Menu** area, and then select **General** > **Global vendors**. Open the relevant global vendor, and then open the **Contacts** tab.

2. Find and open the contact record you want to work with.
3. Open the **Portal access** tab.
4. In the **Web roles** section, use the toolbar buttons to add or remove roles as needed.

The system synchronizes with Supply Chain Management, granting or revoking rights accordingly.

> ⚠ **Warning**
>
> Follow these guidelines when assigning web roles:
>
> - Don't assign the *Global Vendor Anonymous User* or *Global Vendor Authenticated User* roles. These are system roles that support portal functionality and aren't meant for direct assignment.
> - Don't assign both global and local roles to the same user. For correct behavior, assign roles from *only one* of the following groups:
>   - *Global Vendor Administrator* and/or *Global Vendor User*
>   - *Local Vendor Administrator* and/or *Local Vendor User*
>   - One or more transactional roles (*Purchase Order User*, *Request for Quotation User*, *Invoice Management User*, *Consignment Inventory User*)
> - Although the system technically allows multiple role assignments, mixing global and local roles can lead to inconsistent access.

Manage access to legal entities

Administrators can grant or remove access for users whose access is limited to specific legal entities. To manage legal entity access, follow these steps:

1. In the Supplier Engagement app, do one of the following steps to find the contact record you want to manage:

   - At the bottom of the navigation pane, select the **Configuration** area, and then select **Portal management** > **Contacts**.
   - At the bottom of the navigation pane, select the **Menu** area, and then select **General** > **Global vendors**. Open the relevant global vendor, and then open the **Contacts** tab.

2. Find and open the contact record you want to work with.
3. Open the **Portal access** tab.
4. In the **Vendor access** section, do one of the following steps:

   - To add access to a legal entity, select **New contact portal access vendors** on the toolbar. Then use the dialog to choose the vendor legal entity to add.
   - To remove access to a vendor legal entity, select the check box for one or more vendor legal entities that you want to remove and select **Delete** on the toolbar.

The system synchronizes with Supply Chain Management, granting or revoking rights as needed to ensure the user only sees authorized data.

Deactivate portal access for a single contact

To revoke a user's access to the supplier portal, follow these steps:

1. In the Supplier Engagement app, do one of the following steps to find the contact record you want to manage:
    - At the bottom of the navigation pane, select the **Configuration** area, and then select **Portal management** > **Contacts**.
    - At the bottom of the navigation pane, select the **Menu** area, and then select **General** > **Global vendors**. Open the relevant global vendor, and then open the **Contacts** tab.
2. Find and open the contact record you want to work with.
3. On the command bar, select **Deactivate portal access**.
4. Confirm the action.

The system removes all web roles for the contact, sets the portal access status to *Blocked*, and disables the user in Supply Chain Management.

Deactivate portal access for a global vendor and all users

If you no longer collaborate with a global vendor, you can deactivate portal access for all of the vendor's users at once. To deactivate access for a global vendor, follow these steps:

1. Open the global vendor record you want to deactivate.
2. On the command bar, select **Deactivate portal access**.
3. Confirm the action.

The system updates the global vendor portal status to *Blocked*, which blocks portal access for all supplier portal users linked to the global vendor, and removes all associated roles in both Dataverse and Supply Chain Management.

Reactivate portal access

You can restore access for a global vendor or individual contact that you previously deactivated:

- **Global vendor** – Open the global vendor record, as described earlier, and select **Invite to supplier portal** on the command bar. The primary contact is re-provisioned as a *Global Vendor Administrator*.
- **Contact** – Open the contact record, as described earlier, and select **Invite to supplier portal** on the command bar. Choose the role and access level as in a new invitation.

If the user already exists in Supply Chain Management, their account is reactivated. Otherwise, the provisioning workflow creates a new user.

Work with portal users from the Portal tab

Many of the portal user management tasks described in this article—such as viewing active users, adding new portal contacts, and reviewing pending access requests—can also be performed directly from the **Portal** tab on a global vendor record. The **Portal** tab provides a consolidated view that combines portal status, the portal users grid, pending user access requests, and portal notifications in one place.

This approach is convenient when you're already working with a specific global vendor and want to manage its portal users without navigating to a separate contact record. Learn more in Manage global vendor information.

Related information

- Manage users of the Supplier Engagement app and supplier portal
- Manage Supplier Engagement app users
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Personalize the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

The supplier portal is built on Microsoft Power Pages and serves as the primary interface for external supplier representatives to collaborate with your organization. Through the portal, suppliers register, complete onboarding, respond to requests for quotation, manage purchase orders, submit invoices, and handle consignment inventory.

The portal provides settings that you can use to personalize its appearance to match your organization's branding and communication style. Out of the box, the portal includes placeholder elements for the company name, company logo, and a banner image. You can replace these elements with your own branding, and you can also modify the layout, text, and images on key pages such as the landing page, registration page, and home page.

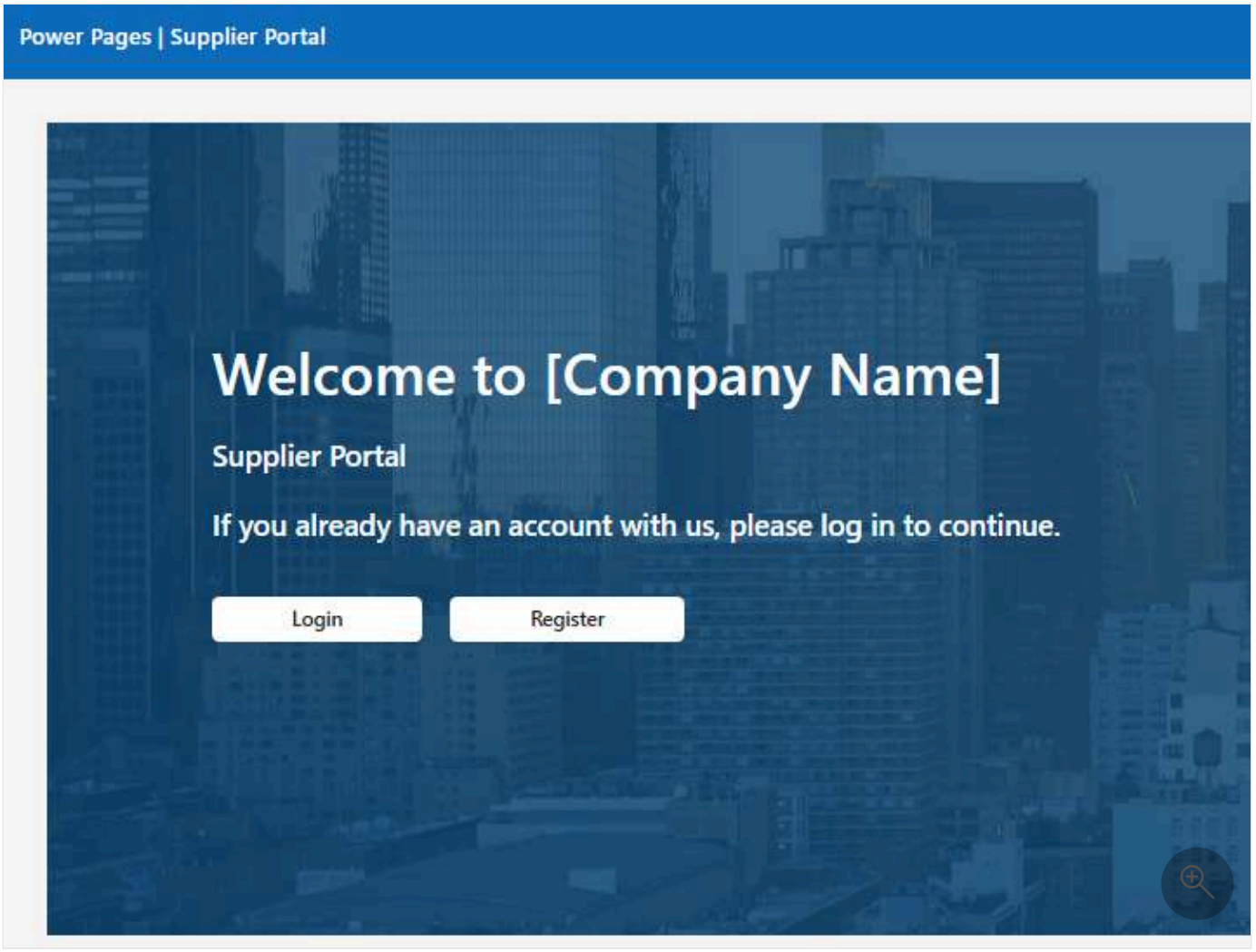
Portal pages that support personalization

The supplier portal has several key pages that visitors and authenticated users interact with:

- **Landing page** – The default page for all visitors. It's publicly accessible (no sign-in required) and introduces the supplier portal to potential suppliers. It includes links to register as a new supplier or sign in with an existing account.
- **Registration page** – An anonymous page where potential suppliers submit their company details, contact information, capabilities, and certifications.
- **Home page** – The main page for authenticated users. It shows a personalized banner with the vendor's name and provides navigation to purchase orders, RFQs, invoices, and other collaboration features.
- **Onboarding pages** – A guided wizard where newly invited suppliers provide detailed business, contact, and compliance information.

Each of these pages includes branding elements—logos, banner images, and text—that you can personalize.

The following screenshot shows the default supplier portal home page with placeholder branding elements.



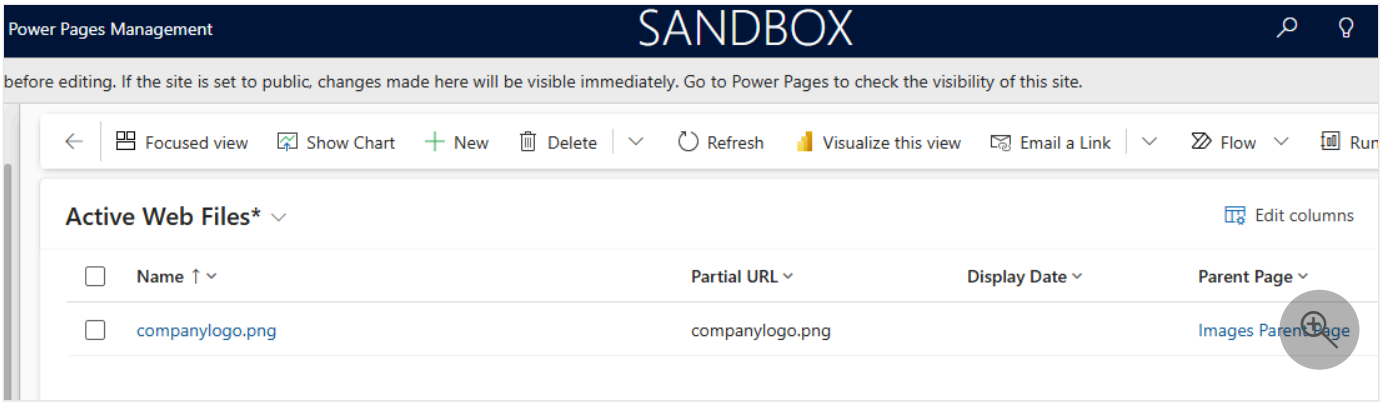
How portal branding works

Two types of Power Pages components control portal branding:

- **Content snippets** – Use these to configure text-based elements such as the company name. Content snippets are stored in Microsoft Dataverse, so you can centrally manage text content without modifying page templates.
- **Web files** – Use these to store and display image assets such as the company logo and the home page banner image. Web files are stored as Dataverse file attachments.

The following screenshot shows the content snippet list in the Power Pages Management app, filtered to show company-related snippets.

The following screenshot shows the web file list in the Power Pages Management app, showing the company logo web file.



Use the Power Pages Management app to manage both content snippets and web files.

Personalization areas

The following table summarizes the branding elements you can personalize and where to find detailed instructions.

Expand table

Branding element	Component type	Name	Learn more
Company logo	Web file and content snippet	*companylogo.png* and *Home/CompanyLogo*	Update the company logo
Company name	Content snippet	*Home/CompanyName*	Update the company name
Home page banner image	Web file	*homepage-banner*	Update the home page banner
Registration and onboarding page text	Content snippets	*Registration/HeaderPageTitle* and *Header/VendorOnboarding*	Update the home page banner

Advanced personalization

Beyond branding, the Power Pages platform supports deeper configuration and customization through CSS, web templates, and page layouts. You control the portal's visual styling through CSS files, such as `dom.css`. Modify these files to adjust colors, fonts, spacing, and responsive behavior. Test any advanced personalization across different screen sizes and browsers to ensure consistent rendering.

> **Important**
>
> When you modify portal components directly in an environment, you create unmanaged layers in Dataverse that can affect solution management. Learn more in Manage unmanaged customizations.

Related information

- Update the company logo on the supplier portal
- Update the company name on the supplier portal
- Update banner texts and images
- Manage unmanaged customizations on the supplier portal
- Supplier Engagement overview

Note: The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Update the company logo on the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

The portal logo is a web file record in Microsoft Dataverse. A content snippet reference renders the logo on the home page. Replace the default placeholder logo with your organization's logo by uploading a new image and ensuring the related content snippet is active.

Prerequisites

To update the company logo on the supplier portal, you must have the following:

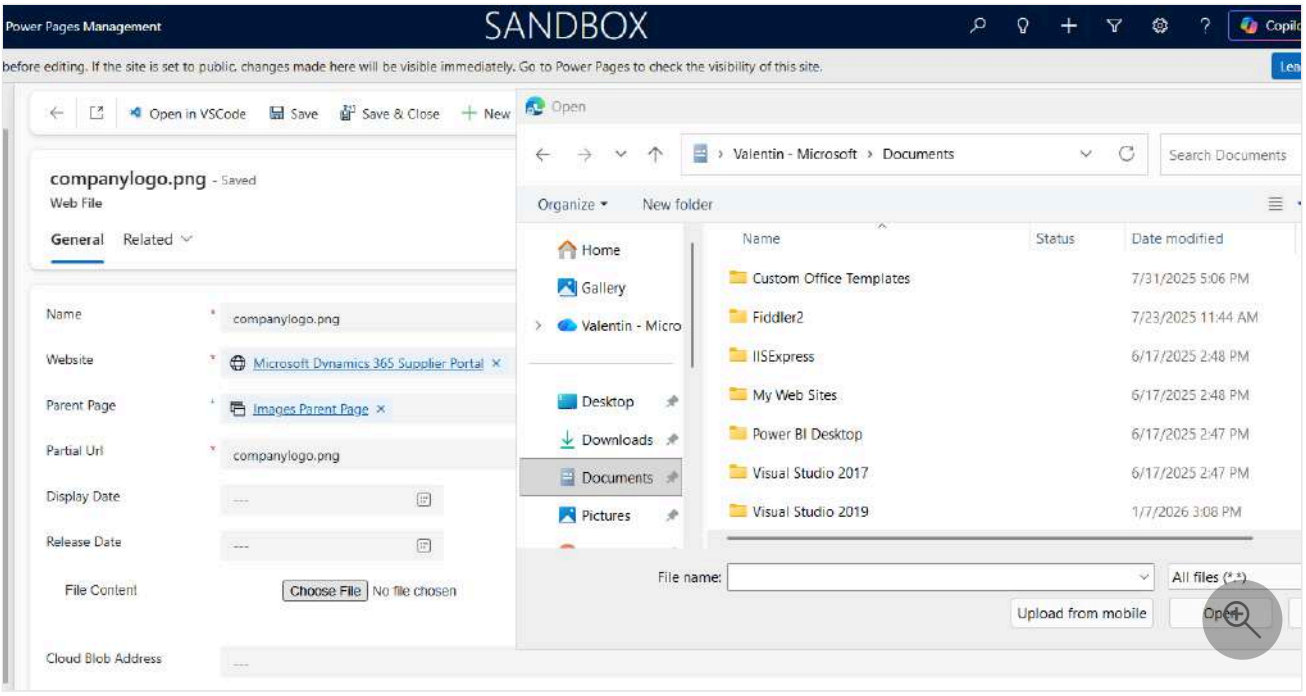
- Access to the Power Pages Management app.
- A logo file in one of the supported formats, such as .png, .jpg, .svg, or .webp.

Upload a new logo

To replace the company logo, follow these steps:

1. Open the Power Pages Management app.
2. Go to **Content** > **Web Files**.
3. Locate and open the web file named *companylogo.png*.
4. Select the **Choose File** button next to the **File content** field.

   The following screenshot shows the *companylogo.png* web file record with the file upload dialog.



5. Select your new logo image.

6. Select **Save & Close**.

Activate the logo snippet

After uploading the logo, verify that the content snippet that references it is active. To check the snippet, follow these steps:

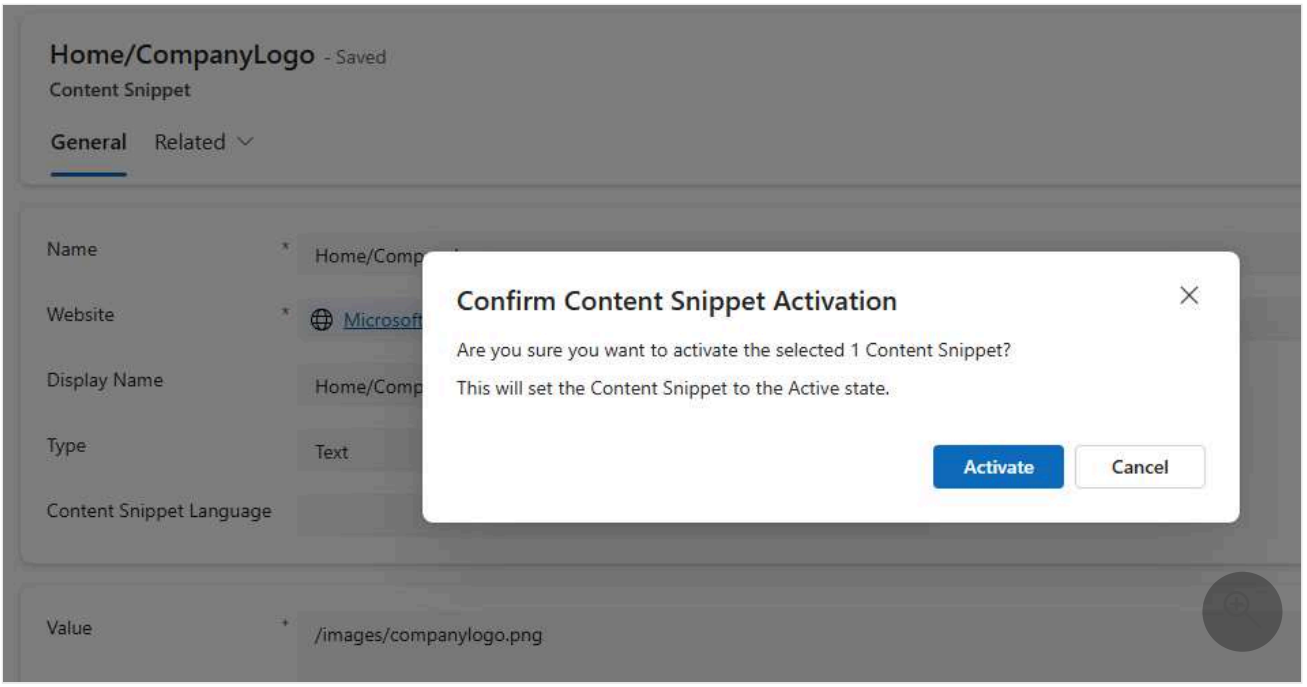
1. Open the Power Pages Management app.

2. Go to **Content** > **Content Snippets**.

3. Set the view selector to **Active Content Snippets** and locate the snippet named *Home/CompanyLogo*. If you don't see the snippet, set the view selector to **Inactive Content Snippets** and try again.

4. Open the *Home/CompanyLogo* snippet. If the snippet is inactive, select **Activate** on the command bar.

   The following screenshot shows the *Home/CompanyLogo* content snippet with the activation confirmation dialog.

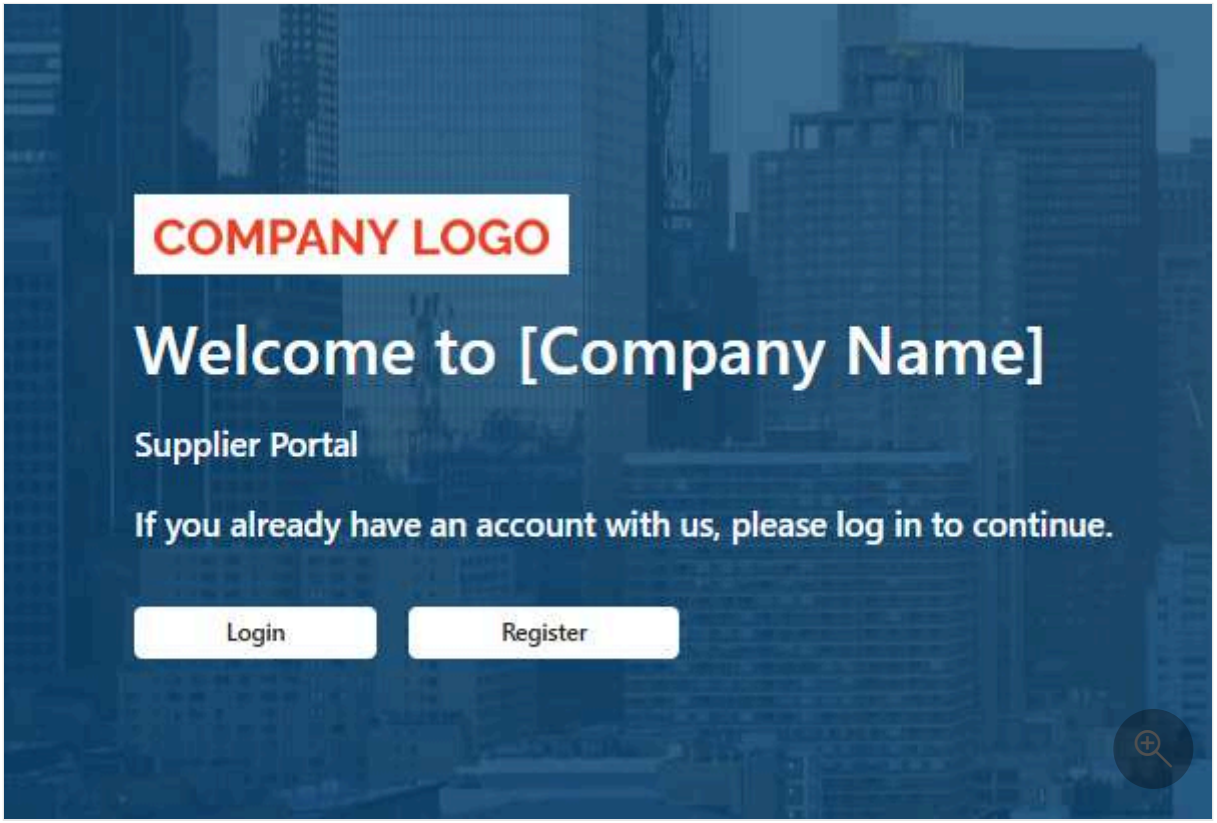


5. Select **Save & Close** if you made any changes.

Verify the changes

Clear your browser cache and refresh the portal home page. The updated logo should now appear.

The following screenshot shows the portal home page with an updated company logo.



> **Tip**
>
> Ensure uploaded logo images are optimized for web use to maintain portal performance. If the updated logo doesn't appear, verify that the *Home/CompanyLogo* content snippet is active and correctly configured.

Related information

- Personalize the supplier portal
- Update the company name on the supplier portal
- Update banner texts and images

Note: The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Update the company name on the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

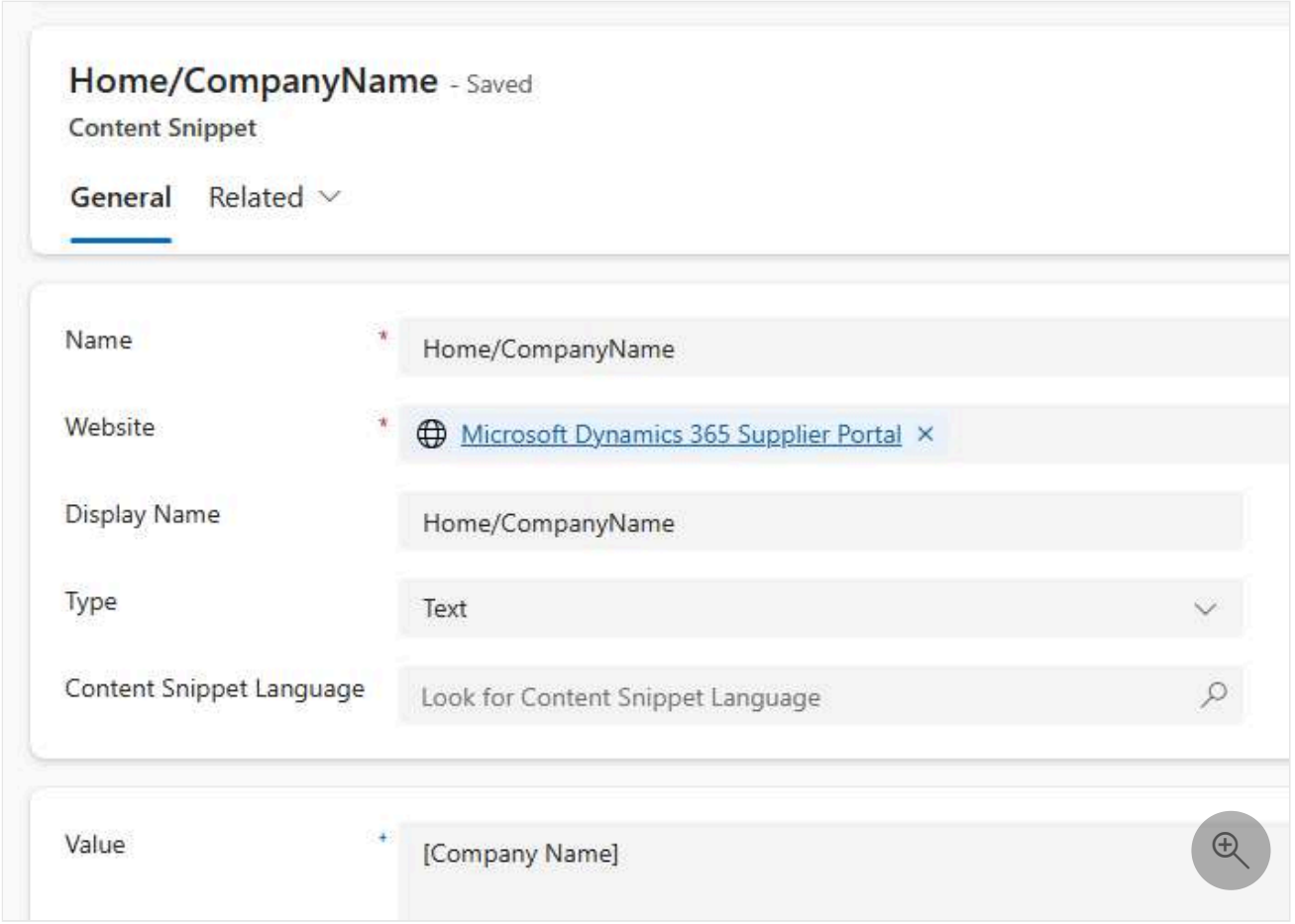
A content snippet in Microsoft Dataverse controls the company name displayed on the supplier portal home page. You can update this value at any time to reflect your organization's name.

Update the company name

To update the company name shown on the portal, follow these steps:

1. Open the Power Pages Management app.
2. Go to **Content** > **Content Snippets**.
3. Locate and open the snippet named *Home/CompanyName*.
4. In the **Value** field, enter your company's name.

   The following screenshot shows the *Home/CompanyName* content snippet with text in the default company name in the **Value** field.

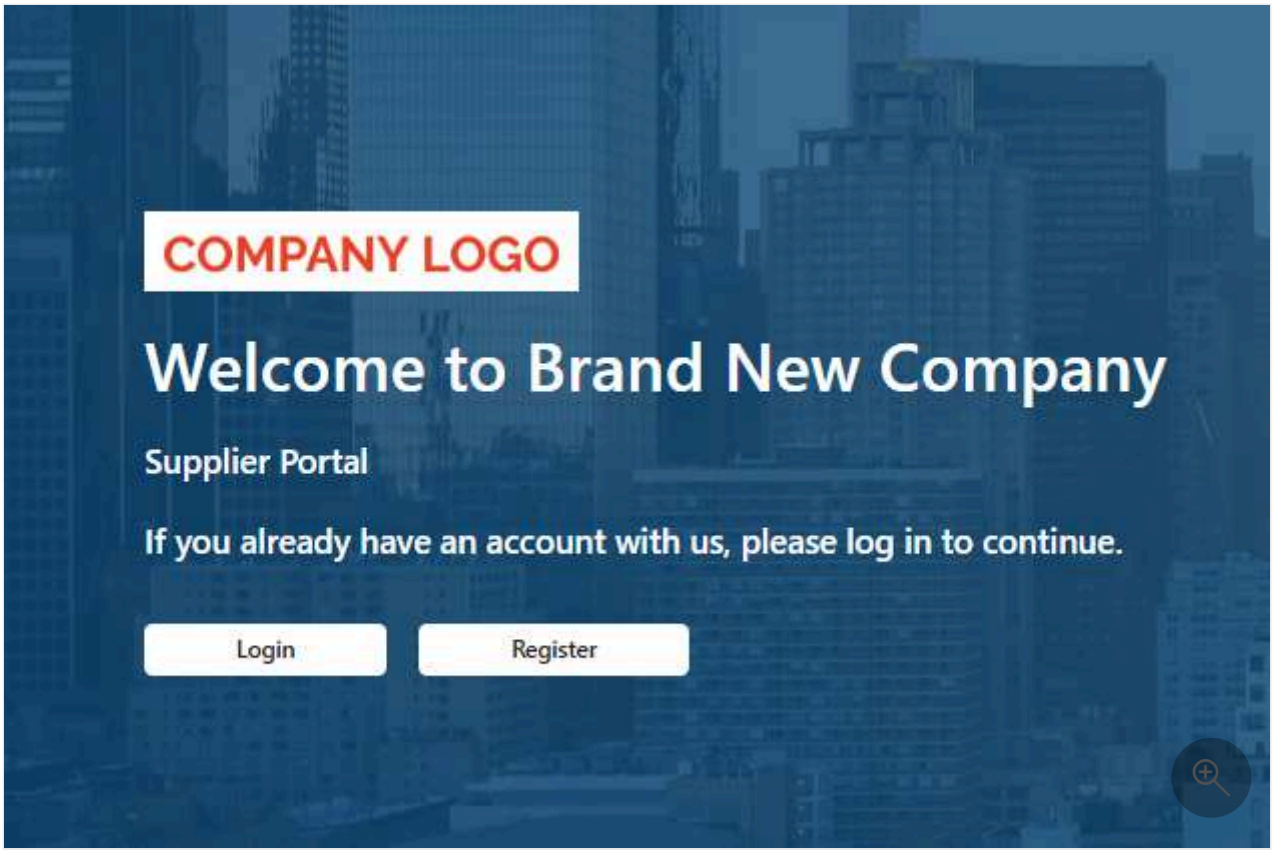


5. Select **Save & Close**.

Verify the changes

Clear your browser cache and refresh the portal. The updated company name is now visible on the home page.

The following screenshot shows the portal home page with an updated company name.



> **Tip**
>
> If the updated company name doesn't appear immediately, verify that the content snippet is active and that your browser cache is cleared.

Related information

- Personalize the supplier portal
- Update the company logo on the supplier portal
- Update banner texts and images

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Update banner texts and images (preview)

[This article is prerelease documentation and is subject to change.]

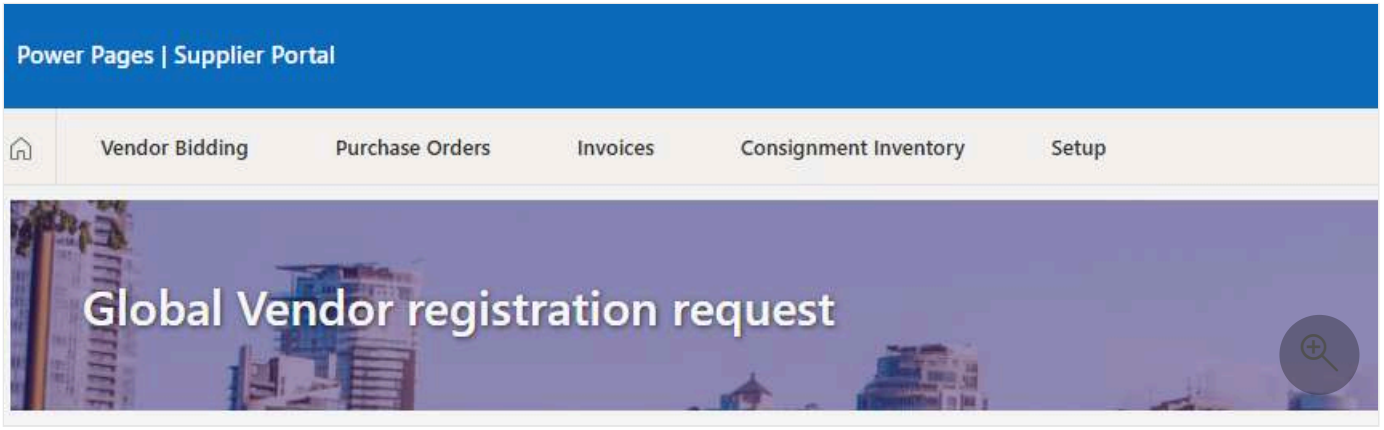
The supplier portal home page displays a banner at the top that includes both an image and text. The registration and onboarding pages also include banner text. You can configure these elements to align the portal with your organization's branding guidelines and communication style.

Banner text

The banner text that appears on the registration and onboarding pages is stored in content snippets. The following snippet names are used:

- *Registration/HeaderPageTitle*
- *Header/VendorOnboarding*

The following screenshot shows the banner text on the registration page.



Update banner text on registration and onboarding pages

To update the banner text on registration and onboarding pages, follow these steps:

1. Open the Power Pages Management app.
2. Go to **Content** > **Content Snippets**.
3. Find the snippet by name (*Registration/HeaderPageTitle* or *Header/VendorOnboarding*) and open it.
4. Update the **Value** field with your desired text.
5. Select **Save**.

Home page banner text

The home page banner doesn't use a content snippet. Instead, it displays the value from the **Global vendor name** field.

The technical maximum length of this field is 850 characters. However, don't exceed 500 characters. Longer values might appear visually crowded, overflow or wrap excessively, reduce readability on smaller screens, and negatively affect the overall banner layout.

Banner image

The banner image is stored as a web file record and rendered on the home page. The image file is saved as a standard attachment in Microsoft Dataverse and associated with the corresponding web file record.

Prerequisites

Before you begin, make sure you have the following:

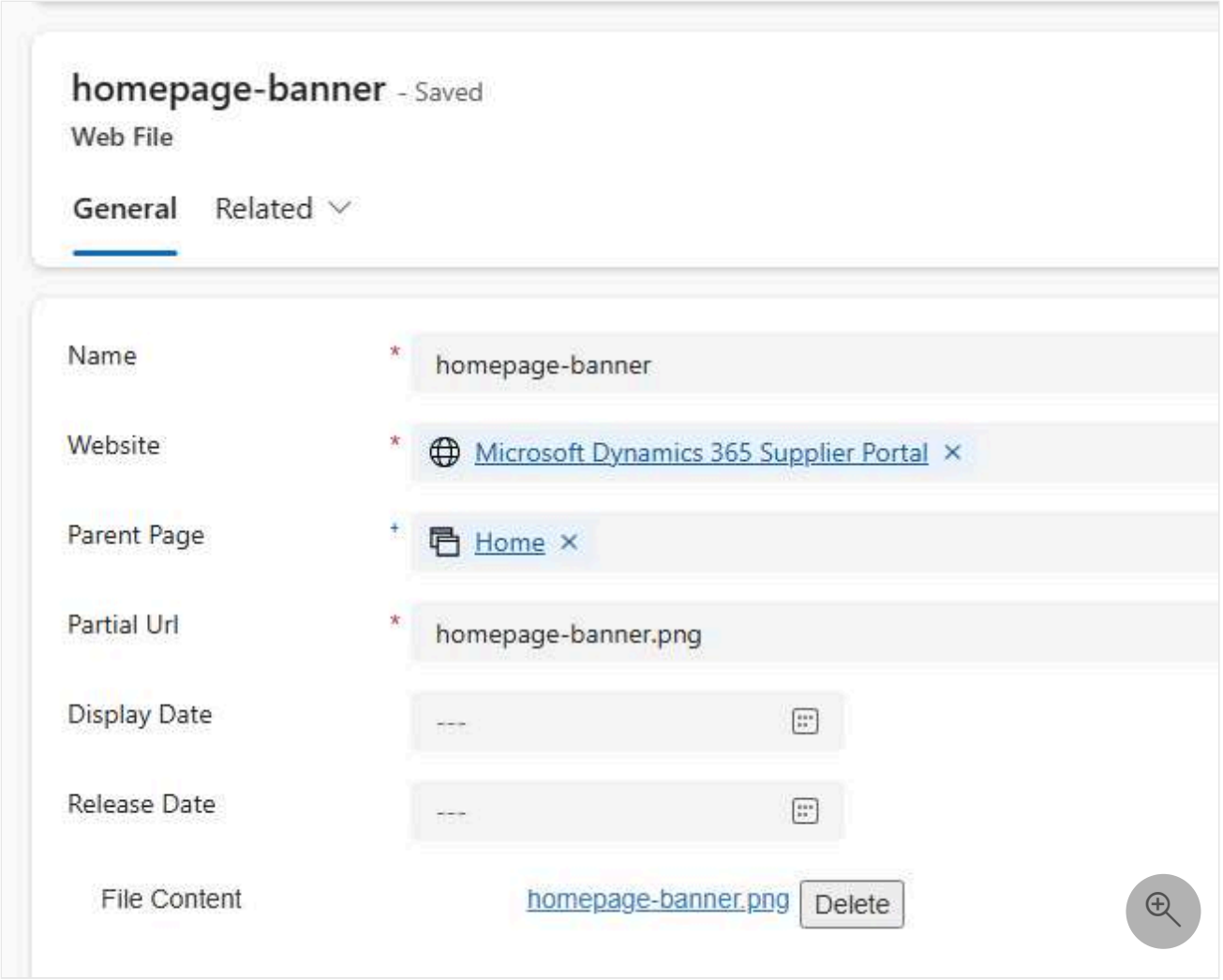
- Access to the Power Pages Management app.
- A banner image file in a supported format, such as .png, .jpg, .svg, or .webp.

Upload a new banner image

To replace the banner image, follow these steps:

1. Open the Power Pages Management app.
2. Go to **Content** > **Web Files**.
3. Locate and open the web file named *homepage-banner*.

   The following screenshot shows the homepage-banner web file record.



4. In the **File Content** field, select **Delete** to remove the existing image (if applicable).
5. In the **File Content** field, select **Choose File** and upload your new banner image.
6. Select **Save & Close**.

Banner image requirements and recommendations

The following table summarizes the requirements and recommendations for the banner image.

Expand table

Property	Requirement
Aspect ratio	12:1 (width to height). If the ratio doesn't match, the image might appear stretched or cropped.
Recommended render size	3,758 × 308 pixels
Supported formats	.png, .jpg, .svg, .webp

Property	Requirement
Maximum file size	Determined by Dataverse environment attachment limit (default is 5 MB). An administrator can increase or decrease this limit. If your image exceeds the configured size limit, either optimize the image or adjust the environment setting.

To ensure optimal quality, upload an image that matches or exceeds the recommended render size while maintaining the 12:1 ratio. Avoid significantly larger images that might affect performance. Optimize images for web use to balance quality and load time.

CSS styling

You can further personalize the banner by using cascading style sheets (CSS). By default, `dom.css` controls the styling through the `.workspace-hero-banner` class. You can adjust properties such as height, background size, positioning, and responsiveness. Test any CSS changes across different screen sizes to ensure consistent rendering.

Related information

- Personalize the supplier portal
- Update the company logo on the supplier portal
- Update the company name on the supplier portal
- Manage unmanaged customizations on the supplier portal

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Manage unmanaged customizations on the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

When you modify portal components such as site settings, content snippets, web files, web templates, or pages, you create unmanaged layers on those components in Microsoft Dataverse. Understanding how unmanaged layers work is essential when you plan to delete individual components or remove an entire solution.

Why unmanaged layers matter

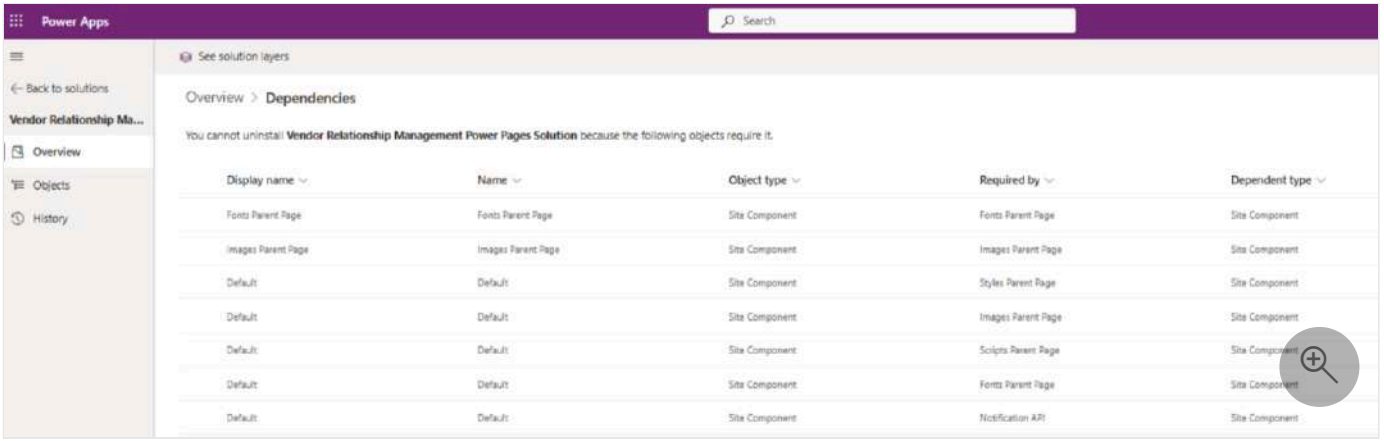
When you include a component in a managed solution and you make changes directly in the environment, you create an unmanaged layer on top of the managed layer. This layer overrides the managed solution configuration.

The presence of unmanaged layers can prevent:

- Deleting individual components
- Deleting the managed solution

If you don't remove unmanaged layers first, deletion might fail due to dependency or layering conflicts.

The following screenshot shows the solution dependencies view in Power Apps. It lists the components that prevent a solution from being uninstalled.



Delete a component

To delete a portal component, first remove any unmanaged layers. To delete a component correctly, follow these steps:

1. Go to the Power Apps maker portal.
2. Locate the component in the appropriate solution.
3. Right-click the component, and then select **Advanced** > **See Solution Layers**.
4. Select and remove the unmanaged layer.
5. After you remove the unmanaged layer, delete the component.

This process ensures the managed solution can properly release the component.

Delete a solution

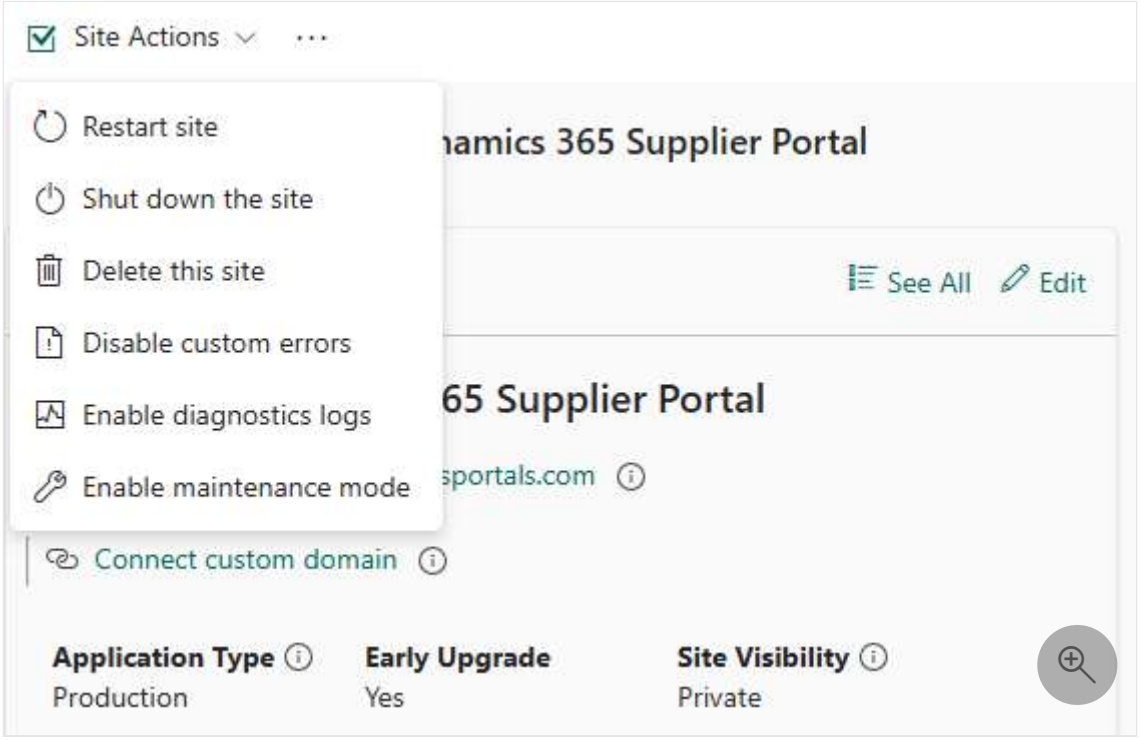
When deleting a solution that contains Power Pages components, first disable the associated site.

Shut down the site

To shut down the Power Pages site, follow these steps:

1. Sign in to the Power Platform admin center.
2. Select **Power Pages**.
3. Select your site.
4. Select **Site Actions** > **Shut down the site**.

The following screenshot shows the Site Actions menu with the **Shut down the site** option.



When you shut down the site, you disable active portal interactions, remove runtime dependencies, prevent additional unmanaged changes, and help ensure unmanaged layers are properly handled.

Delete the solution

After you clear unmanaged layers and shut down the site, you can delete the managed solution. Make sure no dependency errors exist before you complete the deletion.

Best practices

- Avoid making direct changes in production environments. Use development environments for customizations.
- Track solution layering carefully when working with managed solutions.
- Always shut down the Power Pages site before you attempt large-scale deletions.
- Validate dependencies before removing components.

Related information

- Personalize the supplier portal
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Work with the Supplier Engagement app (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement app is a model-driven app for internal users who manage supplier relationships. It's built on Microsoft Dataverse. It gives teams a single place to review supplier registrations, maintain shared vendor data, and coordinate legal entity–specific setups. Changes that you make in the app sync automatically with Supply Chain Management. The app also works with the supplier portal to support vendor self-service.

Supplier Engagement app overview

The Supplier Engagement app organizes supplier work into the following areas:

- **Review and approve vendor registrations** – Process incoming supplier registration requests, review submitted details, and approve or reject requests to start onboarding.
- **Manage global vendors** – Create and maintain global vendor records that act as master profiles across legal entities. You can track vendor information, capabilities, certificates, contacts, lifecycle status, feedback, risks, and activities.
- **Manage local vendors** – Release global vendors to specific legal entities as local vendor accounts, update local vendor details, add contacts and bank accounts, and manage localization-specific data.
- **Vendor holds** – Put global or local vendors on hold to restrict invoices, payments, requisitions, or all transactions, and release the hold when it's no longer needed.
- **Dashboards** – Use role-based dashboards to monitor activities, registration pipelines, risk insights, vendor distributions, and portal access status.

Because the app is connected to Supply Chain Management and the supplier portal, users can work across onboarding and ongoing vendor management without reentering the same information. Global vendor data stays centralized in Dataverse, while transactional readiness and legal entity release stay aligned with Supply Chain Management.

Related information

- Review and approve incoming vendor registrations
- Global vendor management overview

- Manage local vendors
- Global and local vendor holds
- Dashboards
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Global vendor management overview (preview)

[This article is prerelease documentation and is subject to change.]

Global vendors give organizations a centralized way to manage supplier relationships across legal entities in the Supplier Engagement app. Instead of maintaining the same supplier profile separately for each company, you can use one master profile to consolidate shared information and connect it to released vendors in Supply Chain Management. The global vendor record isn't transactional, but it becomes the governing record for supplier identity, contact details, addresses, certifications, agreements, and ownership information.

Understand the global vendor model

A global vendor acts as the shared supplier profile across your organization. It helps you standardize supplier data before you release the vendor to one or more legal entities.

Key characteristics of a global vendor include the following capabilities:

- It serves as the master profile for the supplier.
- It supports data consolidation rather than day-to-day transaction processing.
- It links to local vendors, also known as released vendors, in Supply Chain Management.
- It centralizes company details, contact methods, addresses, agreements, certifications, and ownership profiles.

Because the profile is shared, internal teams can review supplier information once and reuse it across multiple legal entities. This approach reduces duplicate data entry and helps keep supplier records aligned.

Create global vendors

You can create a global vendor in three ways, depending on how the supplier relationship starts:

- **Portal registration** – When a supplier submits a registration request through the supplier portal and your organization approves it, the system automatically creates a prospect global vendor. Records created this way show *Supplier portal* as the origin.
- **Manual creation** – Internal users can create a new global vendor directly in the Supplier Engagement app. Records created this way show *Supplier Engagement* as the origin.

- **Global vendor creation in Supply Chain Management** – Use existing vendor data to create qualified global vendors. Records created this way show *Supply Chain Management* as the origin.

Each method supports a different onboarding pattern, but they all lead to the same centralized vendor profile. Learn more in Create a global vendor.

Manage vendor information

The global vendor record includes multiple tabs that organize supplier data for onboarding, review, and ongoing governance.

Summary and general information

The **Summary** and **General** tabs focus on core company data. They help you maintain the supplier name, type, parent company, origin, primary contact, company profile, ownership profile, contact methods, addresses, certificates, audit details, feedback summary, and rating levels.

Lifecycle and released vendor information

The **Lifecycle** tab tracks the supplier's status and readiness for use across legal entities. It also stores agreement information such as non-disclosure agreements (NDAs), terms and conditions, and code-of-conduct acceptance, alongside related released vendor records.

Capabilities, risks, assessments, portal, and related records

The remaining tabs support broader supplier profiling and governance:

- **Capabilities** – Captures non-transactional attributes such as products, market segments, quality standards, and process capabilities.
- **Risks** – Documents supplier risks and related corrective actions.
- **Assessment** – Stores onboarding questionnaire responses and other evaluation data.
- **Portal** – Tracks onboarding invitations, portal access status, active users, and notifications.
- **Related** – Provides a menu where you can choose to view various types of records that are related to the current vendor.

Learn more in Manage global vendor information.

Detect duplicates

Each time you create or validate a vendor, the system checks to make sure that the vendor doesn't already exist. The Supplier Engagement app applies Dataverse duplicate detection rules during manual creation, and Supply Chain Management validates against existing vendor party records during qualification. These checks don't automatically block you from continuing. Instead, they surface possible matches so you can review the records and decide whether to merge, keep the record anyway, or cancel the action. Learn more in Detect and manage duplicate global vendors.

Profile capabilities and certifications

Global vendor profiling lets organizations evaluate what a supplier can do, not just who the supplier is.

- **Capabilities** – Help teams classify vendors by products, services, market segments, quality standards, and process strengths.
- **Certificates** – Provide centralized certification tracking, expiration monitoring, and synchronization to local vendors when the global vendor is released.

These records support sourcing, qualification, and compliance decisions across legal entities. Learn more in Manage global vendor capabilities and Manage certificates.

Manage contacts and lifecycle

By using global vendor contacts, you can maintain primary and additional contacts at the shared vendor level. These contacts can support portal access and synchronize to related party contacts in Supply Chain Management.

The lifecycle model tracks how a supplier moves through your process:

- *Prospect* – The supplier is newly created and under review.
- *Qualified* – The supplier passed initial validation and can be released to legal entities.
- *Approved* – The supplier is fully approved for business use.
- *Disqualified*, *Disapproved*, or *Terminated* – The supplier didn't meet requirements or is no longer active.

Learn more in Global vendor contacts and Global vendor onboarding lifecycle.

Review feedback, risks, and activities

The Supplier Engagement app lets you track ongoing supplier performance and interactions, including:

- **Feedback** – Records structured star ratings across four domains and calculates a dynamic overall rating.
- **Risk management** – Captures identified risks, evaluates them through likelihood and impact, and tracks corrective actions.
- **Activities** – Records activities such as tasks, appointments, phone calls, and emails to provide a history of vendor engagement.

Together, these features support informed supplier decisions throughout the relationship. Learn more in Global vendor feedback, Risk management and corrective actions, and Manage activities.

Related information

- Create a global vendor
- Manage global vendor information
- Global vendor onboarding lifecycle
- Risk management and corrective actions
- Work with the Supplier Engagement app

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

Create a global vendor (preview)

[This article is prerelease documentation and is subject to change.]

You can create global vendors through supplier self-registration, direct entry by an internal user, or synchronization from existing vendor data in Supply Chain Management. The creation method determines the initial **Origin** value, but each method creates the same centralized vendor profile that you can qualify, approve, and release to legal entities.

Understand the creation methods

The Supplier Engagement solution supports three creation methods for global vendors.

Manual creation in the Supplier Engagement app

An internal user such as a Supplier Engagement manager can create the global vendor directly in the app. The **Origin** field is set to *Supplier Engagement*. Manual creation works well when your organization identifies a supplier first and wants to create the master profile before inviting the supplier to the portal or qualifying the record. This method is described later in this article.

Portal registration and approval

When a supplier submits a registration request through the supplier portal and your organization approves it, the system automatically creates a prospect global vendor. The **Origin** field is set to *Supplier portal* so you can identify how the record entered the system. This method is useful when suppliers begin the relationship through self-service registration and then continue onboarding in the portal. Learn more in Review and approve incoming vendor registrations.

Automated creation through initial sync

During the initial synchronization process, the system can create a qualified global vendor from an existing vendor in Supply Chain Management. The **Origin** field is set to *Supply Chain Management*. This method helps you adopt Supplier Engagement without recreating supplier records manually. Learn more in Create global vendors from existing data.

Create a global vendor manually

Use this procedure when you want to create a new vendor profile directly in the Supplier Engagement app. After you save the record, the system assigns a unique global vendor number, sets the initial status to *Prospect*, and records *Supplier Engagement* as the origin.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Menu** area.
2. On the navigation pane, go to **General** > **Global vendors**.
3. On the command bar, select **New**.
4. Enter the vendor details in the fields provided. Use tooltips to understand the purpose of each field. Required fields are marked with an asterisk (*).
5. On the command bar, select **Save**. Notice the following values, which are generated on save:

    - **Global vendor number** – The system assigns this number automatically.
    - **Status reason** – The initial status is *Prospect*, which indicates that the vendor is not yet qualified for operational use.
    - **Origin** – Set to *Supplier Engagement* to indicate that you created the vendor manually.

Next steps

After the global vendor is created, you can continue onboarding by adding addresses, contact methods, certificates, capabilities, and assessment details. You can then qualify the vendor and release it to one or more legal entities when it's ready for operational use. Learn more in Manage global vendor information.

Related information

- Global vendor management overview
- Manage global vendor information
- Detect and manage duplicate global vendors
- Synchronize data between Supplier Engagement and Supply Chain Management
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Review and approve incoming vendor registrations (preview)

[This article is prerelease documentation and is subject to change.]

Prospective suppliers can use the supplier portal to submit a supplier registration request, even without an invitation. Before a prospective supplier can access the supplier portal, authorized internal users must use the Supplier Engagement app to review and approve the supplier's registration request. During review, you can assess the submitted organization details, move requests into review, and decide whether to approve or reject them. Approval starts global vendor creation and onboarding activities.

View new registration requests in the Supplier Engagement app

Internal users such as global vendor managers and procurement professionals can access registration requests. Each request typically shows key details such as supplier name, contact information, company type, country or region, status, owner, and submission date.

To open the list of submitted requests, follow these steps:

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Menu** area.
2. On the navigation pane, select **Portal registrations** > **Registration requests**.
3. Use the search field, column filters, sorting options, view selector, and other tools to find the request that you want to review. You might often want to use the view selector or filter by **Status Reason** to focus on requests that are in one of the following states:

    - *New* – Recently submitted requests that are waiting for action
    - *In review* – Requests that are currently being assessed
    - *Approved* – Accepted requests that triggered vendor creation and onboarding
    - *Rejected* – Declined requests that won't move forward

4. Open the registration request that you want to inspect.

Start reviewing a request

When a prospective supplier first submits a registration, its **Status Reason** is *New*. Move the request to *In review* when you're ready to begin evaluation.

1. Find and open the registration request that you want to review. It should have a **Status Reason** of *New*.
2. On the command bar, select **Review**.
3. In the confirmation dialog box, select **OK**.

After you complete these steps, the **Status Reason** changes from *New* to *In review*.

Reject a registration request

If the submitted information doesn't meet your organization's requirements, you can reject the request and stop the onboarding process.

1. Find and open the registration request that you want to review. It should have a **Status Reason** of *In review*.
2. On the command bar, select **Reject**.
3. In the confirmation dialog box, select **OK**.

After the request is rejected, the **Status Reason** changes to *Rejected*, and the supplier is notified by email that the application wasn't approved.

Approve a registration request

Approve the request when the supplier should move forward into onboarding and vendor creation.

1. Find and open the registration request that you want to review. It should have a **Status Reason** of *In review*.
2. On the command bar, select **Approve**.
3. In the confirmation dialog box, select **OK**.
4. The system checks for duplicate records to make sure that it doesn't create overlapping supplier data. It looks for existing global vendor records that match the registration by name, website, or primary email. It also checks for existing contact records that use the same primary email address. If the system finds a possible duplicate, it informs you and blocks the approval until you review the existing global vendor data and decide how to proceed.

When approval succeeds, the system completes several actions automatically to start the supplier onboarding flow:

- The registration request **Status Reason** changes to *Approved*.
- The registration contact receives an email that confirms approval.
- A new global vendor record is created with a **Status Reason** of *Prospect*.
- A contact record is created and linked to the global vendor as the primary contact.
- The primary contact's portal access status is set to *Invited*, and prospective user provisioning starts.
- The primary contact is assigned the *Global Vendor Admin* role, and a *Portal User* request is generated in Supply Chain Management.
- After user provisioning finishes, the supplier's primary contact receives onboarding instructions by email.

Related information

- Work with the Supplier Engagement app
- Global vendor management overview
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Global vendor onboarding lifecycle (preview)

[This article is prerelease documentation and is subject to change.]

Global vendor onboarding lifecycle management helps you move suppliers from early prospecting through qualification, approval, and end-of-relationship decisions. Each lifecycle stage reflects the vendor's current readiness, compliance standing, and operational status. The lifecycle also controls when data is synchronized to Supply Chain Management and when portal access is allowed.

Lifecycle stages overview

Global vendors move through lifecycle stages as onboarding and review activities progress.

Expand table

Status reason	Supplier Engagement app	Supply Chain Management
Prospect	Initial vendor record is created manually or through registration and is used for onboarding and data collection.	No vendor record exists in Supply Chain Management. Data is stored only in Dataverse.
Qualified	Vendor has passed initial validation and due diligence and is ready for release to legal entities.	A global vendor party record is created in Supply Chain Management. Contact methods, contact person relationships, and addresses are synchronized.
Approved	Vendor has met all compliance, risk, and performance criteria and is considered a preferred vendor.	Vendor is ready for transactional use and can be released to legal entities as a local vendor.
Disqualified	Vendor failed onboarding or due diligence, and the record is made inactive. Portal access is blocked.	No vendor record is created in Supply Chain Management.
Disapproved	Vendor failed to meet performance or compliance standards after qualification and can be reactivated to qualified status.	The vendor record is marked as disapproved. Portal access is blocked.

Status reason	Supplier Engagement app	Supply Chain Management
Terminated	The vendor relationship is formally ended, the record is blocked, and access is revoked. The status can't be reactivated.	The vendor and related parties are placed on hold. Portal access is blocked.

Status change tracking

The system logs each lifecycle status change so you can review how and why a vendor moved to a new state. To see this information, open the relevant global vendor record and go to the **Lifecycle** tab. Tracked values include the following details:

- **Status reason** – The status that was applied
- **Last updated by** – The user who made the change
- **Last updated on** – The date and time of the change
- **Status change comments** – The comment that explains the change

Qualify or disqualify a global vendor

You can qualify or disqualify a vendor when it's in *Prospect* status. Qualification confirms that the vendor has passed initial validation and due diligence, while disqualification indicates that the vendor failed onboarding or due diligence. When you qualify a vendor, key data synchronizes to Supply Chain Management, as shown in the following table.

Expand table

Data category	Information synchronized to Supply Chain Management
Basic information	Name, status, global vendor hold, and feedback level
Contact methods	Primary email address, primary phone number, primary URL, and additional contact methods
Addresses	Primary address and any additional addresses, including purpose and country/region-specific formatting
Contacts	Contact person party relationships and their corresponding contact methods
Company details	Number of employees, organization number, and DUNS number

Data category	Information synchronized to Supply Chain Management
Portal access	If portal access is enabled, the portal user for the primary contact is mapped to the synchronized contact person in Supply Chain Management, and the user's role changes from *Prospective User* to *Supplier Portal Administrator*

To qualify a prospect, follow these steps.

1. Open the relevant global vendor record, which must have *Prospect* status.
2. On the command bar, select **Qualify**.
3. In the confirmation dialog box, enter a comment.
4. Select **OK**.

When you qualify the vendor, the status changes to *Qualified*, a global vendor party is created in Supply Chain Management, and supported data is transferred.

To disqualify a prospect, follow these steps.

1. Open the relevant global vendor record, which must have *Prospect* status.
2. On the command bar, select **Disqualify**.
3. In the confirmation dialog box, enter a comment.
4. Select **OK**.

When you disqualify the vendor, the status changes to *Disqualified*, the record becomes inactive, portal access is blocked, and all portal roles are removed.

Approve or disapprove a global vendor

You can approve or disapprove a vendor after the vendor reaches *Qualified* status. Approval confirms that the vendor has passed all required assessments. Disapproval blocks further use until the issue is resolved.

To approve a qualified vendor, follow these steps:

1. Open the relevant global vendor record, which must have *Qualified* status.
2. On the command bar, select **Approve** > **Approve**.
3. Confirm the operation and enter a comment to explain why you approved the vendor.
4. Select **OK**.

When you approve the vendor, the status changes to *Approved*, and the vendor becomes eligible for release to legal entities.

To disapprove a qualified vendor, follow these steps:

1. Open the relevant global vendor record, which must have *Qualified* status.
2. On the command bar, select **Approve** > **Disapprove**.
3. Select the required hold type and reason code.
4. Enter any comments that are needed.
5. Confirm the action.

When you disapprove the vendor, the status changes to *Disapproved*, the record becomes inactive, portal access is blocked, a vendor hold is applied, and the update synchronizes to Supply Chain Management.

Request and approve global vendor termination

Termination uses a two-step process where one user requests the action and an authorized user approves it.

Request termination

You can request termination when a vendor is in *Qualified* or *Approved* status.

1. Open the relevant global vendor record, which must have *Qualified* or *Approved* status.
2. On the command bar, select **Terminate Request**.
3. Enter a comment that explains why termination is requested.
4. Confirm the action.

After you submit the request, **Terminate Review** is set to *Initiated*. You can see this information on the **Lifecycle** tab of the global vendor record, in the **Termination details** section. The termination request is sent to an authorized user for review and approval.

Approve termination

An authorized user completes the termination review and applies the final status.

1. Open the relevant global vendor record, which must have a pending termination request.
2. Review the request details and comments.
3. On the command bar, select **Terminate** > **Approve** to approve the termination request. Select **Terminate** > **Disapprove** to reject the request.

When you approve the termination request, **Status reason** updates to *Terminated*, the record becomes inactive, a vendor hold is applied to all local vendors, portal access is blocked, and all

portal roles are removed.

If you disapprove a termination request, **Status reason** remains *Approved*, and the record remains active. On the **Lifecycle** tab, the **Termination details** section shows that the request was disapproved and includes any comments that you entered.

> **Important**
>
> You can't reactivate a terminated vendor. If you want to work with the supplier again, you must start a new onboarding process.

Related information

- Global vendor management overview
- Global vendor contacts
- Global vendor feedback
- Risk management and corrective actions
- Manage local vendors
- Synchronize data between Supplier Engagement and Supply Chain Management

Note: The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Detect and manage duplicate global vendors (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement app includes duplicate detection features that help you identify possible duplicate supplier records before they move too far through onboarding. The validation checks both existing global vendors in Dataverse and vendor party records in Supply Chain Management. Duplicate detection warns you, but doesn't automatically block record creation, so you can review the warning and decide how to proceed.

How duplicate detection works

Duplicate detection can run at two points in the process:

1. During manual global vendor creation, by using Dataverse duplicate detection rules.
2. During qualification, by checking unlinked vendor party records in Supply Chain Management.

These checks help reduce duplicate supplier identities while still allowing you to make an informed decision when a match needs manual review.

Detect duplicates in the Supplier Engagement app

When you create a global vendor manually, the app uses standard Power Platform duplicate detection rules in Dataverse. If the system finds a potential match, it shows a warning so that you can compare the records before continuing.

Because these rules exist in Dataverse, administrators can update the matching logic in the Power Platform admin center to align with organizational policies. The warning is informational rather than blocking, so you can still continue if you can see that the new record is required.

> **Note**
>
> Duplicate detection helps you review possible matches, but it doesn't stop you from creating the record.

Detect duplicates against vendor parties in Supply Chain Management

During qualification, the system checks vendor party records in Supply Chain Management that aren't already linked to a global vendor. This validation helps you avoid qualifying a new global vendor when a matching party already exists.

The following table shows the matching criteria that the system uses during qualification.

Expand table

Field	Matching logic
Company name	Contains
Vendor name	Contains
Primary email	Exact match
Primary phone	Exact match

If the system finds possible matches, a dialog box shows the list of vendor parties so that you can choose the most appropriate action.

Choose an action when duplicates are found

When the system detects possible duplicates during qualification, select one of the following actions.

Expand table

Action	What it does	Typical use
Merge	Associates the global vendor with the existing vendor party. The system merges party information, and links released vendor records that belong to that party to the new global vendor.	Use this option when the vendor already exists in Supply Chain Management and the records represent the same supplier.
Keep anyway	Continues qualification without linking to the matched vendor party. The global vendor receives its own party ID.	Use this option when the records look similar but actually represent different suppliers.
Cancel	Closes the duplicate review without making a change. The vendor remains in *Prospect* status.	Use this option when you need more validation before continuing.

Choosing the right action is important because it determines whether the supplier reuses an existing party structure or creates a new one for downstream processes.

Related information

- Global vendor management overview
- Create a global vendor
- Manage global vendor information
- Synchronize data between Supplier Engagement and Supply Chain Management

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Manage global vendor information (preview)

[This article is prerelease documentation and is subject to change.]

Global vendor records hold the supplier information that your organization uses for onboarding, qualification, compliance, and reporting across legal entities. The Supplier Engagement app organizes that information into tabs so you can maintain a shared company profile, monitor readiness, and review connected records from one place. A well-maintained global vendor profile also helps keep released vendor data accurate when the supplier is used in Supply Chain Management.

Open a global vendor record

To open a global vendor record, follow these steps:

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Menu** area.
2. On the navigation pane, go to **General** > **Global vendors**.
3. On the **Global vendors** page, use the view selector at the upper left to choose which types of global vendors to show, such as **Active global vendors** or **All global vendors**. You can also use the search box and column headings to filter and sort the list.
4. Select the link in the **Name** column for the global vendor that you want to open.

Understand the global vendor tabs

The following table summarizes the main tabs on a global vendor record and the information that each tab contains.

Expand table

Tab	What you can manage
Summary	Basic information such as vendor name, type, parent company, primary contact, origin, activity timeline, last visit date, contact methods, and addresses.
General	Company profile details such as foundation year, employee count, sales turnover, ownership profile, certificates, audit information, feedback summary, and rating levels.

Tab	What you can manage
Lifecycle	Vendor onboarding status, agreement information such as NDA, terms and conditions, and code of conduct, plus related released vendor records.
Contacts	Contact persons associated with the vendor, including the primary contact who serves as the main point of communication and portal administrator.
Capabilities	A hierarchical view of capabilities, products, market segments, quality standards, and other profiling attributes.
Risks	Risk records and corrective actions tied to the vendor.
Assessment	Onboarding questions, answers, and related assessment data.
Portal	Shows information about current portal onboarding status including a list of portal invitations and active users. You can also can create portal notifications for the vendor here.
Related	Provides a menu where you can choose to view various types of records that are related to the current vendor.

Together, these tabs provide a complete picture of the supplier's profile, readiness, and history.

Manage contact methods for a global vendor

Use contact methods to maintain the supplier's phone numbers, email addresses, and URLs in one shared record. You can add multiple entries of each type, but you can mark only one entry of each type as primary at a time.

To create contact methods for a global vendor, follow these steps:

1. Open the vendor that you want to update.
2. Open the **Summary** tab.
3. In the **Contact information** section, select **New contact method** in the toolbar above the grid.
4. Enter the contact method details:
    - **Type** – Select the type of contact method, such as *Phone*, *Email*, or *URL*.
    - **Primary** – Set to *Yes* for the entry that is the primary method to use for the selected **Type**. When you mark a new contact method as primary, the previous primary record of the same type is automatically unmarked. This behavior helps keep the current preferred phone number, email address, and website clear.

Fill in other details as requested for your selected **Type**.

5. Repeat the previous two steps to add additional contact methods as needed.
6. On the command bar, select **Save**.

To edit an existing contact method, double-click it in the grid.

To delete an existing contact method, select it in the grid and then select **Delete contact method** in the toolbar above the grid.

Create and manage global vendor addresses

Addresses follow the global address book structure from Supply Chain Management, so address behavior stays consistent across both applications. Address fields can change by country or region, and address validation follows the address setup that's defined in Supply Chain Management.

To create a new address for a global vendor, follow these steps:

1. Open the vendor that you want to update.
2. Open the **Summary** tab.
3. In the **Addresses** section, select **New address** in the toolbar above the grid.
4. Enter the address details that apply for the selected country or region.
5. If the address should be the main address, set **Primary** to *Yes*.
6. On the command bar, select **Save**.

To edit an existing contact address, double-click it in the grid.

Keep the following address rules in mind:

- Only one primary address can exist for a vendor at a time. If you mark a different address as primary, the previous primary address is automatically unmarked.
- A vendor can't delete the primary address from the supplier portal.
- ZIP or postal code selection might automatically populate the city or state, depending on the country or region configuration.
- Only vendors that have valid address and contact information can be qualified.

Manage vendor contacts

The **Contacts** tab shows the people associated with a global vendor.

Each global vendor can have multiple contacts, but one of them must be designated as the primary contact. The primary contact is the main point of communication for the vendor and is typically the first person to sign in and complete the onboarding process on behalf of the supplier. You identify the primary contact using the **Primary contact** field on the **Summary** page.

You can create additional contacts, set or change the primary contact, and view each contact's relationship to local vendors—all from the **Contacts** tab. Contact persons are synchronized to Supply Chain Management when the global vendor is qualified.

Learn more in Global vendor contacts.

Track lifecycle status and agreements

The **Lifecycle** tab tracks a global vendor's progression from prospect through qualification, approval, and—if necessary—termination. It shows the vendor's current status, the date and comments from the most recent status change, and agreement compliance details such as non-disclosure agreements (NDAs), terms and conditions, and code of conduct.

The tab also shows the released vendor records that have been created from this global vendor in specific legal entities. This gives you a single view of where the supplier is actively used for transactions.

Each status transition (for example, from *Prospect* to *Qualified*, or from *Approved* to *Terminated*) has specific effects on data synchronization, portal access, and vendor holds across legal entities. Learn more in Global vendor onboarding lifecycle.

Manage portal status and users

The **Portal** tab gives you a centralized view of a vendor's portal onboarding progress, active portal users, pending access requests, and portal notifications. From this tab you can:

- View and monitor the **Supplier portal status**, including whether the vendor has been invited, whether onboarding is required, and when onboarding was finalized.
- Review the **Portal users** grid to see which contacts currently have portal access. You can also add new portal contacts directly from this grid.
- Review **User access requests** that are pending approval, including the contact name, business justification, access level, and status.
- View and manage **Portal notifications** that have been sent to the vendor.

Much of the portal user management functionality available on this tab is also accessible from the **Contacts** tab and the command bar. Learn more in Global vendor contacts and Manage supplier portal users.

Maintain profile and compliance data

Use the **General** tab to manage business profile and governance details that support qualification and approval decisions.

Examples of the information that teams commonly maintain include:

- Company profile details such as foundation year, number of employees, and sales turnover
- Ownership profile indicators such as woman-owned, minority-owned, and locally owned classifications
- Certificates and audit information
- Released vendor information for legal entities that already use the supplier

This data helps procurement and compliance teams evaluate suppliers consistently across the organization.

Review operational context

The remaining tabs help teams evaluate suppliers beyond basic profile data:

- **Capabilities** – Records what the supplier offers and where it operates. Learn more in Manage global vendor capabilities
- **Risks** – Tracks issues, severity, and corrective actions. Learn more in Risk management and corrective actions
- **Assessment** – Captures onboarding responses and other review inputs.
- **Related** – Provides a menu where you can choose to view various types of records that are related to the current vendor.

These areas make the global vendor record the main workspace for centralized supplier management.

Related information

- Global vendor management overview
- Create a global vendor
- Manage global vendor capabilities

- Manage certificates
- Global vendor contacts
- Global vendor onboarding lifecycle

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Global vendor contacts (preview)

[This article is prerelease documentation and is subject to change.]

Global vendor contacts help procurement teams manage the people who represent each supplier across onboarding and ongoing collaboration. In the Supplier Engagement app, contacts can be internal records or external users who have portal access. These records stay connected to vendor lifecycle processes, portal access, and synchronization with Supply Chain Management.

Contact roles for global vendors

Each global vendor can include the following types of contacts:

- **Primary contact** – The main point of communication for the global vendor and typically the global vendor administrator.
- **Additional contacts** – Extra contact people who support onboarding, communication, and day-to-day vendor management.
- **Portal-enabled contacts** – Contacts who can sign in to the supplier portal when portal access is enabled.

When a global vendor is qualified, contact people are synchronized to Supply Chain Management as party contacts.

Create a contact person

Create a contact person when you need to add someone who represents the vendor for onboarding, communication, or portal use.

1. Open the relevant global vendor record and go to the **Contacts** tab.
2. In the **Contacts** grid, select **New**.
3. Enter the required details, such as **First Name**, **Last Name**, **Salutation**, **Job Title**, **Email Address**, and **Phone Number**.
4. Select **Save**.

When you save a new contact, the primary contact methods are created automatically. You can add more contact methods later if they're needed.

Set a primary contact

Each global vendor should have a primary contact, who has the following role and responsibilities:

- **Role** – The primary contact is the main contact person for the global vendor.
- **Portal invitation behavior** – When the vendor is invited to the portal, the system creates a user request for the primary contact and assigns the *Global vendor administrator* security role.
- **Onboarding responsibility** – The primary contact is the first person to sign in, start onboarding, and provide initial data and documentation.

To assign or change the primary contact, follow these steps.

1. Open the relevant global vendor record and go to the **Summary** tab.
2. In the **Primary Contact** field, select an existing contact or create a new one.
3. Select **Save & Close**.

Delete a global vendor contact

You can delete a contact only when the contact isn't tied to active downstream processes or protected account relationships.

Contact deletion is blocked in the following cases:

- The contact has an active supplier portal user.
- The contact is associated with a vendor contact in Supply Chain Management.
- The contact is trying to delete their own account through the supplier portal.

If none of these conditions apply, you can delete the contact from the global vendor record. If the contact has a synchronized party contact in Supply Chain Management, the deletion is also synchronized.

Contact relationship in Supply Chain Management

After a global vendor is qualified, its contacts are synchronized to Supply Chain Management as party contacts. You can review these relationships by following these steps:

1. Sign in to Supply Chain Management.
2. Go to **Procurement and sourcing** > **Vendors** > **(Preview) Supplier Engagement** > **All global vendors**.
3. On the list pane, select the relevant global vendor record.
4. Select the link in the **Name** field.

5. Expand the **Relationships** FastTab. The **Relationship A to B** column can show the following types of party relationships:

    - *Global contact for* – Identifies the person as a contact for the global vendor.
    - *Contact for* – Identifies the person as a contact for a local vendor.

View local vendor contact relationships

When a local vendor contact is created either in Supply Chain Management or in the supplier portal, you can use the Supplier Engagement app to see the contact's relationship to a local vendor that is released to a specific legal entity.

To review local vendor relationships for a contact, follow these steps.

1. Open the relevant global vendor record and go to the **Contacts** tab.
2. Open the relevant contact record.
3. If the selected contact is linked to a local vendor, the **Contact for legal entities** section shows information about the local vendor. If the contact isn't linked to any local vendors, this section isn't shown.

Manage contacts from the Portal tab

In addition to the **Contacts** tab, you can also view and manage portal-enabled contacts from the **Portal** tab on the global vendor record. The **Portal** tab provides a focused view that combines portal status information with user management. To access this view, open the relevant global vendor record and go to the **Portal** tab.

From this tab, the **Portal Users** grid shows contacts who currently have portal access. You can select **New Contact** to add a portal user directly from here. The **User Access Requests** section shows any pending approval requests for portal access.

The contact creation and user management functionality on the **Portal** tab works similarly to what's described in the sections above—the main difference is that the **Portal** tab gives you a portal-focused workspace with portal status and notifications visible alongside the user list. Learn more in Manage supplier portal users.

Related information

- Global vendor management overview
- Global vendor onboarding lifecycle

- Global vendor feedback
- Risk management and corrective actions
- Manage local vendors
- Work with the Supplier Engagement app

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Manage global vendor capabilities (preview)

[This article is prerelease documentation and is subject to change.]

Global vendor capabilities help organizations capture what a supplier offers at a shared, cross-entity level. These records support vendor profiling rather than transaction processing, so teams can compare suppliers, evaluate fit, and segment vendors based on strategic needs. Capability data is stored in Dataverse and managed in the Supplier Engagement app.

Understand what capabilities represent

Global vendor capabilities describe non-transactional attributes such as the following examples:

- Products that the vendor specializes in
- Market segments that the vendor serves
- Quality standards that the vendor follows
- Process capabilities that matter in specialized industries

Because you maintain this information at the global vendor level, procurement teams can review a supplier's broader business fit before they decide whether to qualify, approve, or release the vendor to legal entities.

Use capabilities for evaluation and segmentation

Capabilities help internal teams answer business questions that transactional data alone can't answer. For example, you can use capability data to identify suppliers that support a target industry, hold a required quality standard, or provide a specialized manufacturing process.

That broader view supports activities such as:

- Comparing suppliers during sourcing and onboarding
- Grouping suppliers for category or market analysis
- Identifying suppliers that meet regional or technical requirements
- Supporting business reviews with richer profile information

Organize capabilities in hierarchies

Capabilities can be arranged in parent-child relationships so that complex offerings are easier to understand and maintain. A hierarchy lets you group broad business areas under more specific capability records.

For example, a top-level capability such as *Manufacturing services* might contain child capabilities for *Precision machining*, *Injection molding*, and *Assembly*. This structure helps users capture both the supplier's broad service area and the detailed specialties underneath it.

Understand how capabilities differ from procurement categories

Global vendor capabilities are different from procurement categories in Supply Chain Management:

- **Global vendor capabilities** are used for profiling, evaluation, and business alignment.
- **Procurement categories** are used for operational and transactional classification in purchasing processes.

Capability data isn't synchronized to Supply Chain Management. That separation helps organizations keep strategic supplier profiling distinct from the procurement structures that drive transactions.

Assign capabilities to global vendors

To assign capabilities to global vendors, open the relevant global vendor record and go to the **Capabilities** tab. Use the various **Edit** buttons on this page to add or edit capabilities in each capability category (**Products**, **Segments**, **Capabilities**, or **Quality**).

Learn more in Configure capabilities (preview).

Related information

- Global vendor management overview
- Manage global vendor information
- Create a global vendor
- Manage certificates

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

Manage certificates (preview)

[This article is prerelease documentation and is subject to change.]

Global vendor certificate management provides a single place to track supplier certifications across industries and legal entities. Managing certificates at the global vendor level helps you avoid maintaining the same record separately for each local vendor and makes compliance reviews easier. Certificate data synchronizes to related local vendor records in Supply Chain Management.

Create a certificate record

Use this procedure to create a certificate for a global vendor. Create the record first, and then upload the supporting file or add a document URL if the evidence is stored externally.

1. Open the relevant global vendor record and go to the **General** tab.
2. In the **Certificates** section, select **New global vendor certificate**.
3. Enter the certificate details in the fields provided. Use the tooltips to understand the purpose of each field.
4. Select **Save**.

View and edit a certificate record

To view or edit a certificate record, follow these steps.

1. Open the relevant global vendor record and go to the **General** tab.
2. In the **Certificates** section, find the certificate record that you want to add a file to. Scroll to the right and select the **Go to record** button for your selected row.
3. The app displays certificate details, including review details and links to uploaded and external files. Edit the fields as needed and select **Save** to update the record.

Upload a certificate file

After you create a certificate record, you can attach the supporting evidence directly to that record. Each certificate can have one uploaded file, or you can store a document URL that points to an external source instead.

1. Open the certificate record.

2. Do one of the following steps:

    - If the document is stored locally, select **Upload file**, go to the **Certificate file** section, and select **Choose file**. Then select the file to upload. If a file is already uploaded, you must first delete it by selecting **Delete**.
    - If the document is stored externally, go to the **Certificate file** section and enter the URL in the **Document URL**.

3. Select **Save**.

> ⓘ **Note**
>
> You can attach only one file to each certificate record.

Track expired certificates

Each certificate includes an **Expiration date** field so you can monitor validity over time. A Power Automate flow checks whether the expiration date is the current date and then updates the certificate status to **Expired**.

Organizations can surface expiring or expired certificates through dashboards, views, or other reporting mechanisms so that reviewers can act before compliance issues affect supplier readiness.

Synchronize certificates with Supply Chain Management

When you release a global vendor to a legal entity, the system synchronizes its certificate records to the related local vendor and marks them as global vendor certificates. For qualified vendors that already have related local vendors, newly created certificates synchronize automatically to all connected local vendor records.

This behavior helps teams maintain certificates once at the shared supplier level while still making the data available where operational users need it.

Mark a certificate as reviewed

Use this procedure when an internal reviewer validates the certificate content and wants to capture the review result on the record.

1. Open the certificate record that you want to review.
2. On the command bar, select **Mark as reviewed**.
3. Select **OK**.
4. Enter review comments.
5. Select **OK**.

After the action is completed, the system updates the following fields on the certificate:

- **Reviewed?** – Set to **Yes**
- **Reviewer** – Set to the current user
- **Review comment** – Populated with the entered review comments

Related information

- Global vendor management overview
- Manage global vendor information
- Manage global vendor capabilities
- Synchronize data between Supplier Engagement and Supply Chain Management

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Global vendor feedback (preview)

[This article is prerelease documentation and is subject to change.]

Global vendor feedback provides a standardized way to evaluate suppliers and capture performance insights in one place. Ratings are stored centrally and contribute to a dynamic overall score. As you add new feedback, the overall rating updates automatically.

Enter global vendor feedback

Create a feedback entry when you want to record a new supplier evaluation for a global vendor.

1. Open the relevant global vendor record.
2. On the command bar, select **Feedback**.
3. On the feedback form, enter a **Title**.
4. Enter a rating (1–5 stars) for each factor:
    - **Sales policy and flexibility** – How well the supplier supports commercial terms, negotiations, and flexibility in doing business.
    - **Communication** – Responsiveness, clarity, and consistency in supplier communication.
    - **Competitiveness** – Pricing strength and commercial competitiveness.
    - **Product offering and innovation** – Value, breadth, and innovation of the supplier's products or services.
5. Enter a **Comment**.
6. Select **Save and close**.

View feedback

You can submit multiple feedback entries for the same vendor. As you add more entries, the overall rating adjusts automatically. The system calculates each global vendor's overall feedback score as the average of all the submitted rating factors. To view a global vendor's feedback history, open the vendor record, go to the **General** tab, and look in the **Feedback** section.

Feedback score in Supply Chain Management

The latest feedback score syncs to Supply Chain Management, where it is also shown on the global vendor record. This shared visibility lets you review supplier performance from either application.

Related information

- Global vendor management overview
- Global vendor onboarding lifecycle
- Global vendor contacts
- Risk management and corrective actions
- Work with the Supplier Engagement app

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Risk management and corrective actions (preview)

[This article is prerelease documentation and is subject to change.]

Supplier risk management helps organizations identify, assess, and mitigate risks that are associated with global vendors. The feature supports proactive monitoring by combining structured risk evaluations with corrective action tracking. It also preserves historical records so teams can review how risks were handled over time.

Key capabilities

The Supplier Engagement app supports the following risk management capabilities:

- **Risk identification** – Select predefined risk scenarios.
- **Risk evaluation** – Assess likelihood and impact on a 1-to-5 scale.
- **Risk level calculation** – Automatically determine the overall risk level.
- **Corrective action tracking** – Define and monitor mitigation steps.
- **Historical risk records** – Keep prior evaluations and actions for reference.

Prerequisites

Before you can create risk evaluations or corrective actions, complete the following configuration steps:

- **Risk scenarios** – Create and activate risk scenarios in **Configuration** > **Risks** > **Risk Scenarios**.
- **Corrective action types** – Configure corrective action types in **Configuration** > **Risks** > **Corrective Action Types**.

Learn more in Configure risk management elements (preview).

Identify and evaluate risks

Create a risk evaluation when you need to document a vendor risk, score its severity, and capture the required mitigation details.

1. Open the relevant global vendor record and open the **Risks** tab.

2. On the command bar, select **Risk Evaluation**.
3. Fill out the dialog with the details of the risk and a possible corrective action.
4. Select **Save and close**.
5. The risk is now added to the **Active global vendor risks** list, where you and other users can review, open, and eventually delete it if appropriate.

The system calculates the **Risk number** by multiplying **Likelihood** by **Impact**, which produces a value from *1* to *25*. It then assigns a **Risk level** of *Low*, *Medium*, or *High* based on the most recent risk evaluation.

Create corrective actions

Create a corrective action when a risk requires mitigation activities, ownership, and due date tracking.

1. Open the relevant global vendor record and open the **Risks** tab.
2. In the **Active corrective actions** section toolbar, select **New global vendor corrective action**.
3. Fill out the dialog with the details of the corrective action.
4. Select **Save and close**.

Corrective actions link to the related vendor and risk record so you can monitor mitigation work in context.

Close or cancel corrective actions

Close a corrective action when the mitigation work is finished. Cancel it when the action is no longer required.

1. Open the relevant global vendor record and open the **Risks** tab.
2. In the **Active corrective actions** section, locate the corrective action row and select the **Go to record** button at the right side of the row.
3. In the **Measures Implemented** field, enter the actions that were taken to mitigate the risk.
4. On the command bar, select **Deactivate**.
5. In the **Confirm deactivation** dialog, select one of the following options from the **Status reason** drop-down list:

- **Closed** – The corrective action was successfully implemented
- **Cancelled** – The corrective action is no longer required

6. Select **Deactivate** to confirm the status change.

> **Note**
>
> After a corrective action is closed or cancelled, the record becomes read-only. You can view deactivated corrective actions using the drop-down list at the top of this section (change the section name from **Active corrective actions** section to **Inactive corrective actions**).

Related information

- Global vendor management overview
- Global vendor onboarding lifecycle
- Global vendor feedback
- Manage activities
- Work with the Supplier Engagement app

Note: The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Manage local vendors (preview)

[This article is prerelease documentation and is subject to change.]

Local vendor management in the Supplier Engagement app lets you create, update, and control vendor accounts at the legal entity level. You can release a global vendor to a legal entity, maintain local vendor details, and add supporting records such as contacts and bank accounts. All supported updates sync automatically between the Supplier Engagement app and Supply Chain Management.

Release a global vendor to a legal entity

When a supplier is ready for operational use, release the global vendor to a legal entity and create the local vendor account.

1. Open the relevant global vendor record.
2. On the command bar, select **Release vendor**. Then select **OK** in the confirmation dialog.
3. Enter the following values in the **Release vendor** dialog:

    - **Global vendor** – Read-only reference to the selected global vendor
    - **Legal entity** – Select the legal entity (company) to create the local vendor for.
    - **Vendor group** – Select the vendor group for the local account.
    - **Vendor number** – Enter a vendor number for the local account.
    - **Currency code** – Select the currency used by the local vendor. Leave blank to use the default currency for the selected legal entity.
    - **Vendor hold** – If the vendor should be on hold, select the appropriate hold type. Leave blank if the vendor is not on hold. Learn more in Global and local vendor holds (preview).

4. Select **Release**.

View and edit local vendor information

Use the local vendor record when you need to maintain legal entity–specific settings.

1. Open the relevant global vendor record.
2. Open the **Lifecycle** tab.

3. The **Vendors** section lists the currently defined local vendors. To open a local vendor record so you can view and edit its details, select the link in the **Vendor account** column.
4. Update any fields that are required.
5. Select **Save**.

After you save the record, the updated information syncs automatically to Supply Chain Management.

Add a contact for a local vendor

You can add vendor contacts for a local vendor in the Supplier Engagement App. The contact must already exist as a global vendor contact. To add a local vendor contact, follow these steps:

1. Open the relevant global vendor record.
2. Open the **Contacts** tab.
3. Open a contact record by selecting the link in the **Full name** column.
4. On the **Summary** tab, in the **Contact for legal entities** section, select **New Vendor Contact** from the toolbar.
5. Select the **Vendor account** for the local vendor to add the contact to.
6. On the command bar, select **Save & close**.

Maintain bank information for local vendors

To view, add, or edit bank account details for a local vendor, follow these steps:

1. Open the local vendor record.
2. Open the **Bank information** tab. Each existing bank detail record is summarized in the grid. You can do the following actions:

    - To open an existing bank account record, select the **Go to record** button in the right column of the row.
    - To add a new bank account, select **New vendor bank account** in the toolbar.

3. View, enter, and edit bank account details as needed.
4. On the command bar, select **Save & close**.

View approved categories

Approved categories determine which products or services the vendor is authorized to supply. To see which categories are approved for a local vendor, follow these steps:

1. Open the local vendor record.
2. Open the **Approved Categories** tab.
3. Review the categories that are assigned to the vendor.

View certificates

Certificates that are managed at the global vendor level are inherited by local vendors, so you can review them from the local record.

1. Open the local vendor record.
2. Open the **Certifications** tab.
3. Review the inherited certificate records.

Add state tax IDs to local vendors

Use the state tax ID section when you're working with a United States legal entity that requires state-specific tax registration details.

1. Open the local vendor record.
2. Open the **State Tax Id** tab.
3. Select **Add State Tax Id**.
4. Enter the required tax ID details.
5. On the command bar, select **Save & close**.

> **Note**
>
> State tax IDs are available only for supported United States localization scenarios.

Localization

Some local vendor fields and sections appear only for specific countries/regions. The app automatically shows or hides these areas based on the local vendor's legal entity configuration.

Country/region-specific sections can include the following examples:

- **Ownership profile**
- **Business information**
- **State tax ID**

Related information

- Work with the Supplier Engagement app
- Global vendor management overview
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Manage activities (preview)

[This article is prerelease documentation and is subject to change.]

The **Activities** page provides a central place to track interactions, follow-up work, and scheduled events in the Supplier Engagement app. Built-in views help teams monitor open work, recent activities, due dates, and portal-facing timeline items. Users can create tasks, appointments, and other activity types directly from the **Activities** page.

Open the activities page

To open the **Activities** page:

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Menu** area.
2. On the navigation pane, go to **General** > **Activities**.
3. On the **Activities** page, use the drop-down list at the top of the page to choose the types of activities to show. The default view is **My Activities**, which shows activities that are assigned to you. You can also use the search bar to filter activities by keyword, and you can sort the list by selecting the column headers.

Add a new activity

To add a new activity, follow these steps:

1. Open the **Activities** page.
2. On the command bar, select the button for the type of activity that you want to create. For example, create a **Task** when you need to track a work item or follow-up action. Create an **Appointment** when you need to schedule a meeting, review, or other time-based event.
3. Fill out the details for your activity.
4. On the command bar, select **Save & close**.

Work with activities

To work with your assigned activities, follow these steps:

1. Open the **Activities** page.
2. Use the view selector and other filters to find the activities you need.

3. Either select the link in the Subject column to open the activity record, or select the check box for one or more activities.
4. Use the buttons on the command bar to take action. For example, select **Mark as complete** to close an activity, select **Assign** to assign the activity to another user, or select **Delete** to remove an activity. If you opened an activity record, you can also edit the details and select **Save & close** to save your changes.

The command bar also includes the following buttons for working with activities:

- **Focused view** – Opens a view that makes it easy to browse records using a list pane and a details pane. Select a record in the list to view its details. Use the details pane to edit and take action on the record. Select the **Read only grid** button to go back to the standard view.
- **Show chart** – Opens a chart view where you can choose various graphical representations of your and your team's activities, their statuses, completion rates, and so on.

Dashboard views

The **Activities** page includes the following dashboard views.

- **All Activities** – Displays all activity records, regardless of status or owner.
- **Closed Activities** – Shows activities that are completed or closed.
- **Due and Overdue Activities** – Lists activities that are due soon or past the due date.
- **Last Activity** – Displays the most recent activity for each record.
- **My Activities** – Shows activities that are assigned to you. This is the default view.
- **My Closed Activities** – Lists your closed activities.
- **My Team Members' Activities** – Displays activities for your team members.
- **Next Activity** – Shows the next scheduled activity for each record.
- **Open Activities** – Lists activities that are currently open.
- **Portal Timeline View** – Shows a timeline of activities that are visible in the external portal.
- **Scheduled Activities** – Displays activities that are scheduled for a future date.
- **All Tasks** – Shows all task records.
- **My Tasks** – Displays tasks that are assigned to you.
- **All Appointments** – Shows all appointment records.
- **My Appointments** – Displays appointments that are assigned to you.
- **My Completed Appointments** – Lists your completed appointments.
- **My Draft Appointments** – Shows draft appointments that you created.

Related information

- Global vendor management overview
- Global vendor onboarding lifecycle
- Risk management and corrective actions
- Dashboards
- Work with the Supplier Engagement app

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Global and local vendor holds (preview)

[This article is prerelease documentation and is subject to change.]

Vendor holds help you control supplier activity when compliance, risk, or operational conditions require temporary restrictions. You can apply holds at the global vendor level to affect all released vendors, or at the local vendor level for a single legal entity. Hold updates sync with Supply Chain Management so that transaction controls stay aligned.

Understand vendor hold types

Depending on the hold type that you apply, the system blocks specific actions such as invoice posting, payment generation, or requisition creation.

Expand table

Hold type	Effect
No	All transactions are allowed.
Invoice	Invoice creation or posting is blocked.
Payment	New payments can't be generated, although existing payments can still post.
Requisition	Requisition creation is blocked.
All	All transactions are blocked for the vendor.
Never	The vendor can't be put on hold for inactivity.

Place a global or local vendor on hold

To place a global or local vendor on hold, follow these steps:

1. Open the relevant global vendor record.
2. To do one of these steps:

    - To place the currently selected global vendor on hold, stay on this page. When you place a global vendor on hold, all related local vendors are also placed on hold.
    - To place a local vendor on hold, open the **Lifecycle** tab, and in the **Vendors** section, open the target local vendor record.

3. On the command bar, select **On Hold**.
4. Confirm by selecting **OK**.
5. Enter the hold details:

    - **On hold** – Select the hold type to apply
    - **On hold reason** – Select the business reason for the hold.
    - **On hold release date** – Optionally, select a date when the hold should be automatically released. If you leave this field blank, the hold remains in effect until you manually release it.

6. Select **OK**.

After you apply the hold, the vendor status updates and the changes sync to Supply Chain Management.

Release a global or local vendor from hold

To release a global or local vendor from hold, follow these steps:

1. Open the relevant global vendor record.
2. To do one of these steps:

    - To release the currently selected global vendor from hold, stay on this page. When you release a global vendor from hold, all related local vendors are also released.
    - To release a local vendor from hold, open the **Lifecycle** tab, and in the **Vendors** section, open the target local vendor record.

3. On the command bar, select **On Hold Release**.
4. Confirm by selecting **OK**.

After you complete these steps, the system clears the hold status and syncs the updates to Supply Chain Management.

Related information

- Work with the Supplier Engagement app
- Manage local vendors
- Supplier Engagement overview

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 07/31/2026

Dashboards (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement app includes dashboards that provide a role-based view of supplier activity and progress. These dashboards help administrators, managers, and operational users monitor registrations, risks, ongoing work, and portal access from one place. You can switch between dashboards to focus on the metrics and lists that matter for your role.

To open and use the dashboards.

1. Open the Supplier Engagement app, and at the bottom of the navigation pane, select the **Menu** area.
2. On the navigation pane, go to **General** > **Dashboards**.
3. On the **Dashboards** page, use the list at the top of the page to choose a dashboard. The following dashboards are available:

    - **Global vendor activities overview** – This dashboard helps you track follow-up work and keep vendor-related tasks moving. It can include lists of current and overdue appointments, tasks and their status, and corrective actions.
    - **Global vendor registrations** – This dashboard focuses on the onboarding pipeline. It can include lists of new and pending registration requests and charts that group requests by status reason.
    - **Global vendor risks overview** – This dashboard highlights supplier risk information. It can include charts that show vendors by risk level, risk scenario, status, corrective action type, and country or region.
    - **Global vendors overview** – This dashboard shows vendor distribution and trend data across the supplier base. It can include charts for vendor status, vendor type, risk level, business unit, and new vendors by creation month.
    - **Global vendors portal overview** – This dashboard focuses on supplier portal access. It can include charts that show vendors by portal status and lists that identify vendors that already have portal access enabled.

Related information

- Work with the Supplier Engagement app
- Supplier Engagement overview

- Manage users of the Supplier Engagement app and supplier portal

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Work with Supplier Engagement data in Supply Chain Management (preview)

[This article is prerelease documentation and is subject to change.]

The Supplier Engagement solution introduces several changes to Supply Chain Management that support the new global vendor model. These changes include a dedicated global vendors form, global vendor certificates, and enhancements to the supply risk assessment report. This article describes how to work with Supplier Engagement data directly in Supply Chain Management.

Supplier Engagement and vendor collaboration

The Supplier Engagement solution is intended to replace the older vendor collaboration feature. Key differences between vendor collaboration and the new solution include:

- **Global vendor concept** – Supplier Engagement introduces the global vendor concept, where a single master profile represents a vendor across legal entities. Only vendors that are associated with a global vendor are accessible through the Supplier Engagement app or the supplier portal. Learn more in Global vendor management overview.
- **Supplier Engagement app** – A model-driven Power Apps application is available for internal procurement specialists and buyers to manage vendor relationships. Learn more in Work with the Supplier Engagement app.
- **Supplier portal** – The supplier portal is a dedicated Power Pages app that vendors use to collaborate with your organization. Previously, vendors interacted through the Vendor Collaboration module in Supply Chain Management.

Learn more in How Supplier Engagement compares to the vendor collaboration interface.

Global vendors in Supply Chain Management

In Supply Chain Management, you can access global vendors by going to **Procurement and sourcing** > **Vendors** > **Supplier engagement** > **All global vendors**. The **All global vendors** page shows the following data:

- Global vendor name and ID
- Global vendor status
- Global vendor holds

- Released vendors

The global vendor table in Supply Chain Management is shared, with each global vendor represented by a single party in the global address book. When details such as contacts, addresses, or contact persons for a global vendor party are updated in Supply Chain Management, these changes are synchronized with the Supplier Engagement app.

Local vendors

Local vendors that are involved in Supplier Engagement continue to be managed as they were previously. You can create new vendors either from the **All vendors** page or by releasing from the party within the global address book.

Learn more in Create global vendors from existing data.

Manage certificates

Local vendor certificate management remains unchanged. With Supplier Engagement and the global vendor model, certificates can now be added at the global vendor level and automatically apply to all associated local vendors. These certificates are marked with a **Global vendor certification** checkbox on the **Certifications** page and can only be created in the Supplier Engagement app. To open the **Certifications** page, go to **Procurement and sourcing** > **Vendors** > **All vendors**, select a vendor, and then, on the Action Pane, open the **Vendor** tab and select **Certifications**.

Learn more in Manage certificates.

Create a global vendor based on existing vendor data

You can convert existing vendors to global vendors or merge an existing vendor with a global vendor. Learn more in Create global vendors from existing data.

Learn more in Create a global vendor.

Supply risk assessment report

When the Supplier Engagement feature is turned on in Supply Chain Management, the standard **Supply Risk Assessment** report (**Procurement and Sourcing** > **Inquiries and Reports** > **Supply Risk Assessment** > **Supply Risk Assessment**) is enhanced with the following changes:

- A **Date** filter is added to enable time-based analysis.
- A **Global Vendor** filter is introduced as a hierarchy level above vendors.
- Global vendor data is incorporated into report visuals to provide aggregated insights.

Learn more in Risk management and corrective actions.

Related information

- Supplier Engagement overview
- Work with the Supplier Engagement app
- Global vendor management overview
- Manage certificates
- Risk management and corrective actions
- Synchronize data between Supplier Engagement and Supply Chain Management

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 07/31/2026

Supplier portal overview (preview)

[This article is prerelease documentation and is subject to change.]

The supplier portal provides an external-facing web experience where vendors can interact with your organization's procurement processes. Built on Microsoft Power Pages, the portal gives vendors self-service access to manage their profiles, respond to requests for quotation (RFQs), confirm purchase orders, submit invoices, and track consignment inventory—all without needing direct access to Supply Chain Management.

This article provides an overview of the portal's capabilities from the vendor's perspective. The information here is intended for administrators and Microsoft partners who need to understand what vendors can do in the portal and where customization opportunities exist.

Learn more about the supplier portal and how it works with the other components of Supplier Engagement in Supplier Engagement overview.

Portal capabilities

The supplier portal supports the following vendor workflows:

- **Landing page and onboarding** – Vendors can register as new suppliers and complete onboarding steps to provide company information, capabilities, and certifications. Learn more in Landing page and vendor onboarding.
- **Home page** – After signing in, vendors see a personalized home page with workspace tiles, essential links, and notifications. Learn more in Home page.
- **User profile** – Vendors can view and edit their personal profile information. Learn more in User profile.
- **RFQ responses** – Vendors can view, bid on, and manage requests for quotation. Learn more in View and respond to RFQs.
- **Purchase order confirmation** – Vendors can review, confirm, or respond to purchase orders. Learn more in View and confirm purchase orders.
- **Invoicing** – Vendors can create, submit, and track invoices. Learn more in View and create invoices.
- **Consignment inventory** – Vendors can monitor consignment stock levels and consumption. Learn more in View consignment details.

- **Global and local vendor data** – Both global and local vendors can manage their organizational profile, contacts, certificates, and capabilities. Learn more in Vendor details.
- **User access management** – Vendor administrators can manage portal users, assign roles, and control access. Learn more in Vendor details.

Related information

- Supplier Engagement overview
- Manage supplier portal users
- Personalize the supplier portal
- Supplier portal landing page and vendor onboarding
- Supplier portal home page

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

Supplier portal landing page and vendor onboarding (preview)

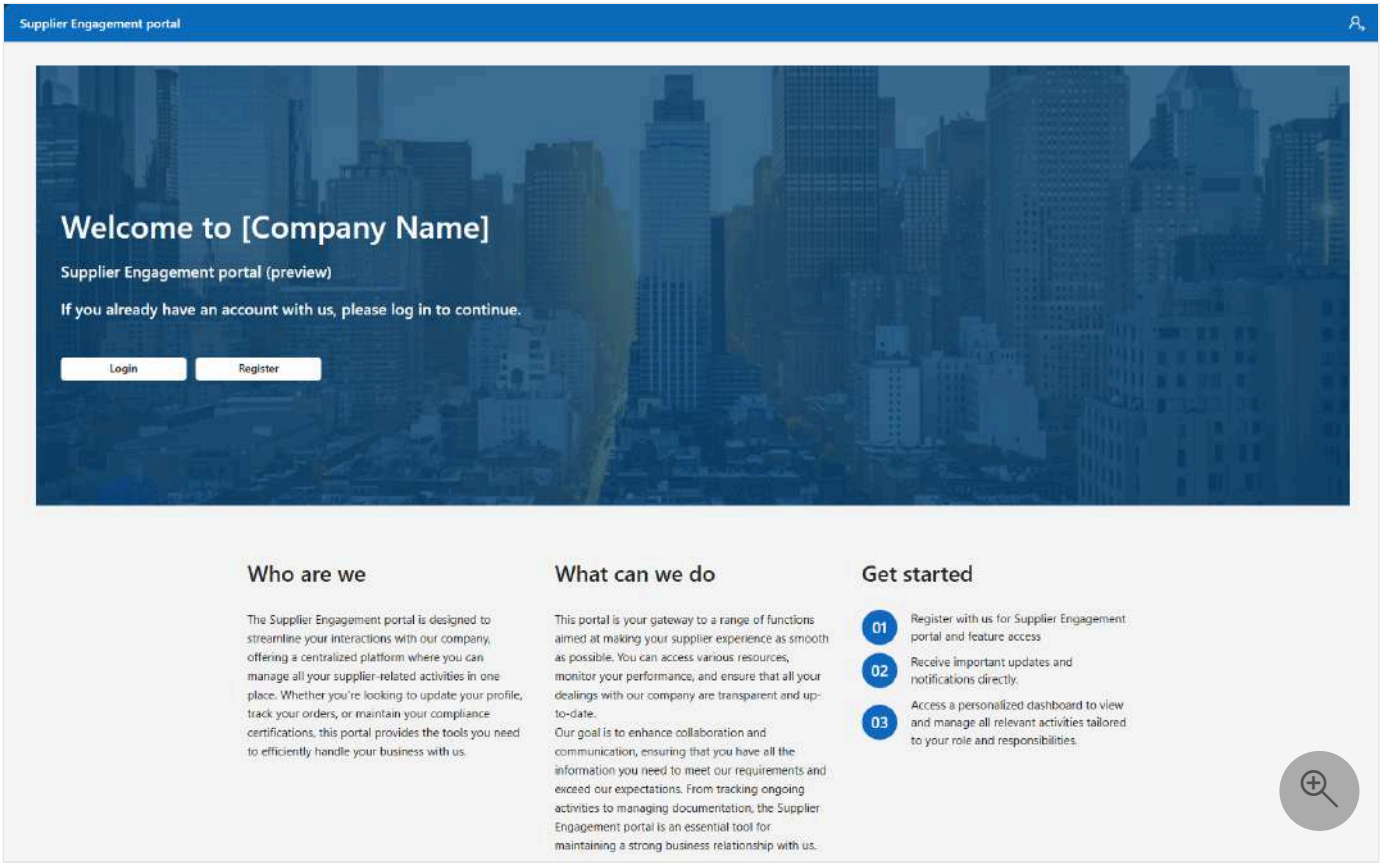
[This article is prerelease documentation and is subject to change.]

The supplier portal provides a landing page that's available to all anonymous visitors and an onboarding wizard that lets new vendors apply to become vendors to your organization. These features let potential vendors introduce themselves to your organization and, once approved, provide the detailed information needed to begin doing business.

Landing page

The supplier portal landing page is the default page for all visitors. This page is available anonymously, which means visitors don't need to sign in to view it.

The landing page introduces the concept of the supplier portal and guides visitors to register as a new vendor or sign in if they already have an account. It provides a brief description of your organization's vendor collaboration approach and links to key entry points of the portal.



You can customize the landing page to align with your organization's visual identity. You can update the logo, colors, and content to reflect your company's style and communication standards.

The supplier portal is built on Microsoft Power Pages, which allows organizations to tailor the look and feel of the landing page. You can modify the layout, text, and images, or add information about your vendor engagement process.

For detailed guidance on branding and customization, go to Personalize the supplier portal overview.

Initial registration

Portal registration enables potential vendors to introduce themselves by submitting their company details through the supplier portal. Anonymous users can register without prior authentication. Once submitted, the request goes to the Supplier Engagement app for review.

To register as a new vendor, follow these steps:

1. Go to the supplier portal and select **Register** on the landing page.
2. In the **Welcome to registration** dialog, review the registration instructions and select **Continue**.
3. Fill out the **Global Vendor registration request** form with the required information, including company details, contact information, capabilities, and certifications.
4. Accept the **Confirmation** check boxes and complete the security captcha.
5. Select **Submit** to finalize the registration request.

After submission, the potential vendor sees a confirmation message and receives an acknowledgment email. The request becomes available in the Supplier Engagement app for internal users to review.

For information on the internal review process, go to Review and approve incoming vendor registrations.

Complete vendor onboarding

After a vendor registration request is approved, the vendor is invited to complete onboarding using the supplier portal. The onboarding process ensures that vendors provide all necessary business, contact, and compliance information before they can be qualified and approved.

Prerequisites

- The vendor must have received an invitation to the supplier portal.
- The vendor must have a user provisioned with the *Global Vendor Admin* role assigned.

Complete onboarding

To complete the onboarding wizard, follow these steps:

1. On the portal home page, select **Login**. Then select **Entra user** and sign in by using your vendor account credentials. Because you didn't yet complete onboarding, a pop-up message opens indicating that you must follow the onboarding process.

2. On the **Contact information** page, review and enter the requested information, including the primary contact details for your organization. Use this information for communication and supplier-related correspondence. Enter at least one address to proceed.

3. On the **Company details** page, provide the following information: - Business profile details, such as founding year and number of employees - Ownership profile details, such as minority-owned or small business

4. On the **Capabilities** page, define your scope of supply by selecting from:

    - Product categories
    - Market segments
    - Capabilities
    - Quality standards

5. On the **Certificates** page, manage certification records:

    - Add certification details by specifying certification type, certifying organization, and effective and expiration dates.
    - Upload supporting documents.
    - Edit, delete, or mark certificates as expired.

6. On the **Assessment** page, complete the questionnaire configured by the buyer. Questions might include both yes/no responses and free-text answers.

After you complete the onboarding wizard, a confirmation page is displayed. In the Supplier Engagement app, internal staff can see that the onboarding process is marked as complete.

What happens after onboarding

When you finalize onboarding:

- The global vendor's **Finalized Date** and **Status** update.
- The vendor record is ready for qualification and approval by procurement staff.
- After completing the onboarding wizard, supplier portal users can sign in to the portal and then go to **Setup** > **Global Vendor Setup** to manage global vendor information whenever needed.

Related information

- Supplier portal overview
- Review and approve incoming vendor registrations
- Personalize the supplier portal

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

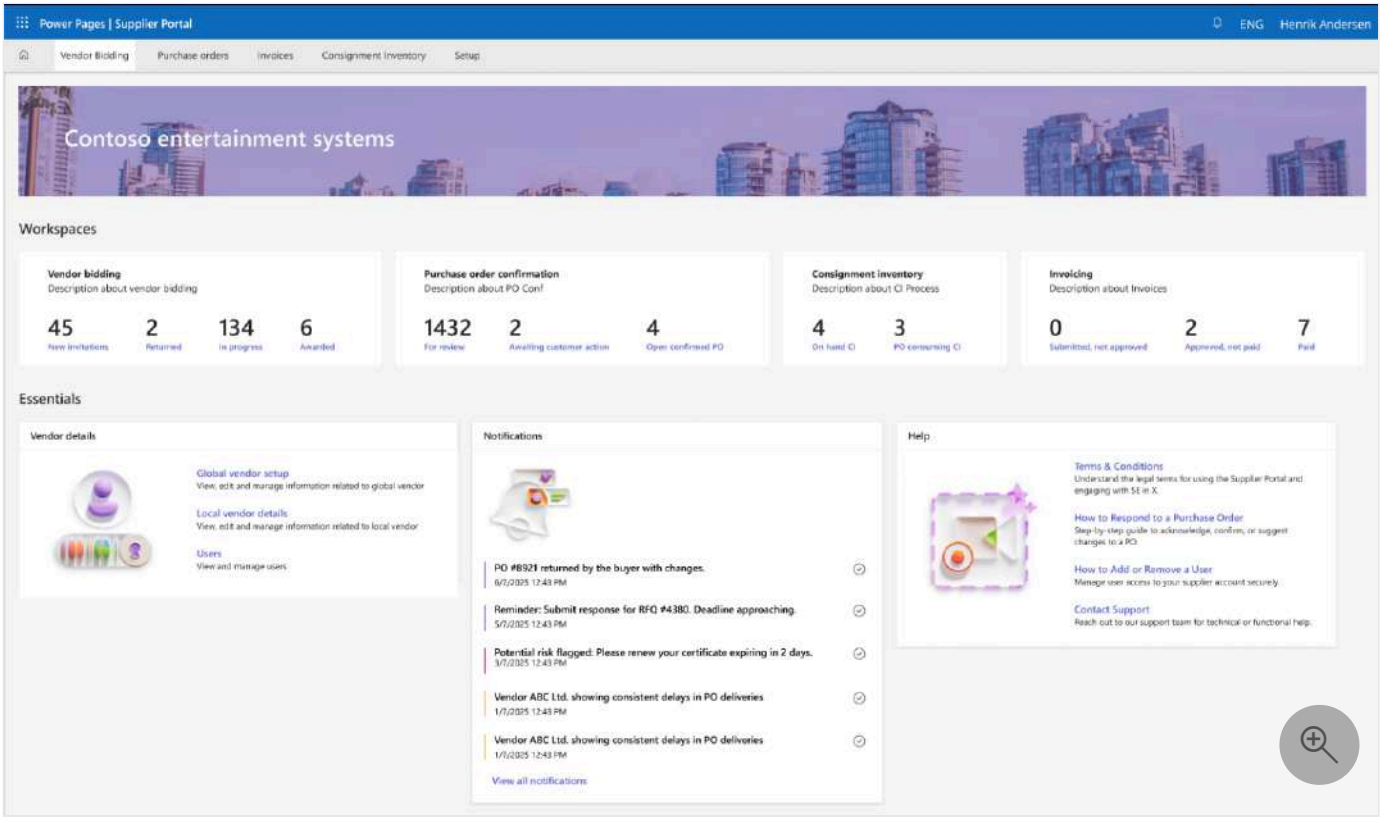
Supplier portal home page (preview)

[This article is prerelease documentation and is subject to change.]

The supplier portal home page is the main entry point for vendor collaboration within the Supplier Engagement solution. It provides vendors with quick access to key business areas such as purchase order confirmations, vendor bidding, consignment inventory, and invoicing. The home page layout is designed for ease of navigation and can be customized to reflect your organization's branding and communication style.

The content displayed on the home page is dynamically personalized based on the vendor's profile and access permissions.

The following screenshot shows an example of the supplier portal home page.



Workspaces section

The **Workspaces** area provides a consolidated view of vendor-related activities. Each workspace panel displays key metrics and links directly to detailed pages.

Expand table

Panel	Description	Example metrics
Vendor bidding	View and respond to requests for quotation (RFQs).	New invitations, Returned, In progress, Awarded
Purchase order confirmation	Manage and confirm purchase orders.	For review, Awaiting customer action, Open confirmed PO
Consignment inventory	Track consignment stock and consumption status.	On hand CI, PO consuming CI
Invoicing	View and manage invoice submissions and approvals.	Submitted, Approved, Paid

> **Note**
>
> Record counts are calculated based on the signed-in user's access rights. The workspaces section is hidden for prospective vendors who aren't yet fully approved.

Essentials section

The **Essentials** section provides quick links to the most frequently used pages and management functions.

Vendor details

The **Vendor details** panel provides quick links to the pages where you maintain your organization's profile, legal entity information, and portal user accounts. The following table describes each link.

Expand table

Link	Description
Global vendor setup	View, edit, and manage information related to the global vendor profile.
Local vendor details	View, edit, and manage information related to local vendor entities.
Users	View and manage user accounts related to the vendor organization.

Notifications

The **Notifications** panel provides updates and alerts about vendor data management activities, such as certificate expiration. Here you can:

- View recent notifications directly on the home page.
- **Dismiss** notifications you already reviewed.
- Select **View all notifications** to open the full list.

Help

The **Help** section offers guidance and support links for portal users. By default, the portal includes example links on the home page:

- **Terms & Conditions** – Legal information for using the supplier portal.
- **How to respond to a Purchase Order** – Step-by-step instructions for confirming or updating POs.
- **How to Add or Remove a User** – Guidance for managing user access.
- **Contact Support** – Direct link to customer support for technical or functional help.

These links are placeholders. Customize them to include your organization's actual help materials.

Home page versions based on global vendor status

The home page layout adapts depending on the global vendor's qualification status:

- **Qualified or Approved global vendor** – Displays the full home page, including **Workspaces**, **Essentials**, **Notifications**, and **Help** sections.
- **Prospective global vendor** – Shows a simplified version with introductory content and limited access. Workspace tiles and transactional data are hidden.

Related information

- Supplier portal overview
- Personalize the supplier portal
- Supplier portal landing page and vendor onboarding

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

Manage vendor details, local vendors, and users in the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

After completing onboarding, vendors can manage their organization's data and user access directly in the supplier portal.

- The **Global vendor setup** page provides a centralized interface for updating company details, contacts, certificates, and capabilities.
- The **Local vendor setup** page allows vendors to manage their records at the legal entity level, including ownership profiles, contact information, approved categories, and bank details.
- The **User management** page allows vendor administrators to control who has access to the portal, what roles they hold, and which vendors they can work with.

Global vendor setup

Use the **Global vendor setup** page to manage your organization's profile, contacts, certificates, and capabilities. The page provides a structured interface for maintaining accurate and up-to-date information that's essential for supplier engagement and compliance. To open the page, complete one of the following steps:

- From the home page, select **Global vendor setup** from the **Vendor details** tile.
- From the navigation menu, go to **Setup** > **Global vendor setup**.

The following table summarizes the settings available in the **Global vendor setup** page.

Expand table

Section	Description
General information	Maintain and update your core company profile, including the key organizational details required for supplier registration and business identification.
Ownership profile	Provide and manage your company ownership and business classification information for supplier diversity and business categorization purposes.
Contact information	Manage the primary company contact details used for communication and supplier-related correspondence.

Section	Description
Contact methods	Maintain the different communication channels associated with your company, including adding, updating, or removing contact methods as required.
Addresses	Manage your company address information by adding new addresses, updating existing address details, or removing outdated address records.
Contacts	Manage the contact persons associated with your organization, including maintaining active contact details and updating company representatives when needed.
Certificates	Manage compliance and certification records by maintaining certificate information, uploading supporting documents, and updating certification details as part of supplier compliance management.
Capabilities	Maintain and update your business capabilities, including areas of expertise, product categories, quality standards, and other capabilities that define your scope of supply. Make selections by using a tree-structured control that the buyer configures.

View and manage local vendors

Use the **Local vendor setup** page to view and manage your records at the legal entity level. The page provides a structured interface for maintaining accurate and up-to-date information that's essential for supplier engagement and compliance. To open the page, follow these steps:

1. Do one of the following steps:

    - From the home page, select **Local Vendor Setup** from the **Vendor details** panel.
    - From the navigation menu, go to **Setup** > **For buyer's legal entities**.

2. Select the local vendor you want to work with.

The following table summarizes the key sections available in the **Local vendor setup** page.

Expand table

Section	Description
General information	Maintain and update the local vendor profile, including the key organizational details required for supplier registration and business identification.
Ownership profile	Provide and manage company ownership and business classification information for supplier diversity and business categorization purposes.
Vendor contacts	Manage the local vendor contact details and add or remove contact persons as needed.

Section	Description
Approved categories	View the product categories that the local vendor is approved by the buyer to supply. These categories are defined by the buyer and used for procurement classification and sourcing purposes.
Category requests	Submit requests to become an approved supplier for additional product categories. These requests are subject to buyer approval and used for procurement classification and sourcing purposes.
Certificates	Manage compliance and certification records by maintaining certificate information, uploading supporting documents, and updating certification details as part of supplier compliance management.
Capabilities	Maintain and update your business capabilities, including areas of expertise, product categories, quality standards, and other capabilities that define your scope of supply. Make selections by using a tree-structured control that the buyer configures.
Bank details	View and manage the bank account information associated with the local vendor for payment processing and financial transactions.

Localization

Some global and local vendor fields and sections are relevant only for specific countries or regions. The system automatically shows or hides these fields and tabs based on the vendor's legal entity. This feature ensures that vendors see only the information required for their local regulatory and business requirements. Country/region-specific sections include:

- **Ownership profile**
- **Business information**
- **State tax IDs**

Manage portal users

The supplier portal provides user management capabilities that vendor administrators can use to control who has access to the portal, what roles they hold, and which vendors they can work with. The **User management** page in the supplier portal provides:

- A list of all current portal users.
- The ability to create new portal user requests.
- A list of submitted access requests with their status.

To open the **User management** page, sign in to the supplier portal and complete one of the following steps: - From the home page, select **Users** from the **Vendor details** panel. - From the

navigation menu, go to **Setup** > **Users**.

Request to add a new user

Vendors can request to add new users by submitting an access request. This process ensures that requests are reviewed and approved before the user receives access. You can grant global or local access, depending on the scope of work.

To create a new user request, follow these steps:

1. Open the **User management** page.
2. In the **Access requests** section, select **Request access**.
3. Fill out the request form and select **Submit**.

Personnel from the purchaser company use the Supplier Engagement app review and approve the portal user access request.

Manage existing users

After you provision a user, you can update their access and roles. The supplier portal supports role changes, vendor assignments, and deactivation to maintain proper access control.

Changes you make in the Supplier Engagement app automatically synchronize with Supply Chain Management, updating user access rights accordingly.

Assign or remove roles or local vendor access

Roles define what a portal user can do. A user can act as an administrator or a standard user. Vendor access controls which vendor records a user can view or manage. This access is useful when a supplier operates across multiple entities but users only need access to certain vendors. Global Vendor Administrators can change roles as needed.

To manage roles and local vendor access, follow these steps:

1. Open the **User management** page.

2. In the **Portal users** section, select a portal user's name to open their details.

3. Manage roles and local vendor access as follows:

    - To add a role, select **Add web role**. Then choose a role and select **Add**.
    - To remove a role, select the role to open a menu, and then select **Remove**.

- To add the users to a local vendor, select **Add vendor access**. Then choose the local vendors to assign and select **Submit**.
- To remove local vendor access, select the local vendor to remove to open a menu, and then select **Remove**.

4. Select **Submit** to save your changes and close the dialog.

Deactivate a user

When a user no longer requires access, administrators can deactivate the user. This action immediately prevents the user from signing in to the supplier portal.

To deactivate a user, follow these steps:

1. Open the **User management** page.
2. In the **Portal users** section, find the portal user you want to deactivate. At the right side of their row, open the **Options** (ellipsis) menu and select **Deactivate**.
3. Confirm the action. The system immediately revokes the user's portal access.

Related information

- Supplier portal overview
- Manage global vendor information
- Manage global vendor capabilities
- Manage certificates
- Manage supplier portal users

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

Supplier portal user profile and sign out (preview)

[This article is prerelease documentation and is subject to change.]

The supplier portal enables vendors to view and update their personal profile information, including contact details and preferences. Each portal user manages their profile settings individually, so changes don't affect other users or the vendor's organizational profile.

Access and edit your user profile

To view or edit your profile, follow these steps:

1. Select your user profile icon in the top-right corner of the portal.
2. Select **My Profile** from the drop-down menu.
3. On the **My Profile** page, view and update your information and preferences as needed.
4. Select **Submit** to apply your changes.

Sign out

To exit the portal, select your user profile icon in the top-right corner of the portal. Then, select **Sign Out** from the drop-down menu.

Related information

- Supplier portal overview
- Supplier portal home page
- Manage supplier portal users

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

View and respond to RFQs in the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

The **Vendor bidding** workspace in the supplier portal lets vendors view the requests for quotation (RFQs) that the buying company invited them to and submit bids. The workspace also shows all RFQs that the vendor lost or won.

Open the vendor bidding workspace

To open the **Vendor bidding** workspace, sign in to the supplier portal and complete any of the following steps:

- On the top navigation bar, select **Vendor bidding** and then select any of the views listed.
- On the home page, select any of the links in the **Vendor bidding** panel.

The workspace provides the following tiles. Each tile represents a category or status of RFQs. Each tile shows the number of RFQs in that category and provides a link to view the related list of RFQs.

- **Bid invites** – RFQs that the vendor is invited to respond to. This category includes RFQs where bidding can begin or where an amended RFQ requires a new submission.
- **Submitted** – RFQs the vendor submitted that are awaiting the purchaser's response.
- **In progress** – RFQs that the vendor or a contact person is working on but hasn't yet submitted.
- **Awarded** – RFQs where the purchaser awarded at least one line item in the vendor's bid.
- **Returned** – RFQs that the purchaser returned so the vendor can provide more information or update the bid.
- **Lost** – Bids where all lines are rejected and the bid is closed as lost.
- **Declined** – RFQs where the vendor declined to participate in bidding.

Bid on an RFQ

Bidding is the procedure through which a vendor submits an offer in response to an RFQ. To submit a bid, follow these steps:

1. Go to **Vendor bidding** > **Bid invites** and open the RFQ.

2. Examine the RFQ details, including the **General** tab for header information, the **Lines** tab for line items, and any amendments or comments provided by the purchaser.

3. Select **Bid** to start bidding on the RFQ.

4. Respond to the RFQ by entering your bid details. Work as follows:

    - When editing is enabled for bidding fields on the headers and lines of an RFQ, you can enter your bid directly in the **Lines** grid. Consider any additional bid information that you need to add to the line details.
    - At any time before the expiration date, save a bid by selecting **Save as draft** on the toolbar. This action lets you return later to finish and submit the bid. While you're working on a bid, and until you submit it, the bid is available in the **Bids in progress** section.
    - To remove all entered bid data on the **Lines** tab and restore the original RFQ details, select **Reset to default** on the toolbar.
    - To reset only one line, select **Reset line** from the line **Actions** column.
    - To work with alternates, select **Add alternate** or **Remove alternate** in the line grid. If the purchaser enables **Allow alternates on response lines** for a specific RFQ, you can only add alternate items for lines of the *Category* type.
    - When you're done reviewing and updating a line, select the check box in the **Done** column to mark it as complete.
    - To mark all lines as complete, select the **Mark as done** button at the top of the **Lines** grid.
    - The **Lines** and **Amendments** tabs include a **Communications** panel that vendors and purchasers can use to communicate.
        - Purchaser-provided information remains read-only. Vendor-added comments and attachments can be modified or removed.
        - Read and provide clarifications or notes in the **Comments** section.
        - Upload documents using the **Add Attachment** button.
    - If the Q&A feature is enabled for the RFQ, use the **Q&A** tab to ask questions and view answers from the purchaser. This tab is available if **Q&A** is set to *Yes* on the RFQ and is visible based on the cutoff date.
    - If any questions are shown on the **Questionnaire** tab, complete all required questions as part of the bid submission. You must complete the questionnaire in full before the system allows you to submit the bid.

5. Select **Submit** on the toolbar to submit the bid.

Recall and resubmit a bid

You can recall and update a submitted bid up until the expiration date.

1. Go to **Vendor bidding** > **Submitted** and open the bid.
2. To bring the bid back to an editable state, select **Recall** on the toolbar.
3. Make changes and resubmit before the RFQ expires.

Respond to a returned bid

If the purchaser returns a submitted bid, the vendor can find it on the **Returned** tile of the **Vendor bidding** workspace.

1. Go to **Vendor bidding** > **Returned** and open the returned bid.
2. Go to the **General** tab and check the **Reason Code** and **Reason comment** on the bid **General** tab.
3. Take the necessary actions and resubmit.

Decline an RFQ

If you don't want to bid on an RFQ, open it and select **Decline** from the toolbar. After you decline, you can't recall the action and enter a bid.

Amended RFQs

If the purchaser amends an RFQ, you must enter a new bid for it. Information about the amendment is on the **Amendments** tab of the RFQ page. Amended RFQs appear on the **Bid invites** page.

Related information

- Supplier portal overview
- Requests for quotation (RFQs) overview
- View and confirm purchase orders in the supplier portal

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

View and confirm purchase orders in the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

The **Purchase order confirmation** workspace in the supplier portal lets vendors review, confirm, or respond to purchase orders (POs) that purchasers send. The portal provides a workspace where vendors can access all relevant POs, check their status, and take appropriate actions such as accepting, rejecting, or suggesting changes.

To open the **Purchase order confirmation** workspace, sign in to the supplier portal and complete any of the following steps:

- On the top navigation bar, select **Purchase orders** and then select any of the views listed.
- On the home page, select any of the links in the **Purchase order confirmation** panel.

The workspace displays tiles for the following statuses:

- **For Review** – POs that require a vendor response (unanswered or edited).
- **Awaiting Customer Action** – POs that await purchaser action after vendor response.
- **Pending delivery** – POs that the purchaser confirmed but aren't yet fully received.

Each tile shows the number of POs in that category and provides a link to view the related list of POs. Select a tile to view the list of POs in that status. From the list, you can open a PO to review its details and communicate with the purchaser as needed.

Related information

- Supplier portal overview
- Purchase order overview
- View and create invoices in the supplier portal

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 08/07/2026

View and create invoices in the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

The supplier portal enables vendors to create, submit, and track invoices through the **Invoice management** workspace. Vendors can create invoices from purchase orders or as cost invoices, add lines and attachments, and monitor invoice status through to payment.

Prerequisites

To use the invoice management features in the supplier portal, the purchasing organization must set up the vendor invoice workflow as described in Configure Supplier Engagement features in Supply Chain Management.

Open the Invoice management workspace

To open the **Invoice management** workspace, sign in to the supplier portal and complete any of the following steps:

- On the top navigation bar, select **Invoices** and then select any of the views listed.
- On the home page, select any of the links in the **Invoicing** panel.

The workspace displays tiles for the following statuses:

The workspace displays a set of tiles, each of which represents an invoice status. Each tile displays a count of invoices in that status. Select a tile to view a list of invoices in that status. The following tiles are available in the **Invoice management** workspace:

- **Draft** – Invoices that the vendor created but didn't submit yet.
- **Submitted** – Invoices that the vendor submitted for approval but aren't yet approved.
- **Approved** – Invoices that are approved and waiting to be paid.
- **Paid** – Invoices that are fully paid.
- **Rejected** – Invoices rejected in the workflow.

Create and submit an invoice

To create an invoice from a purchase order, follow these steps:

1. Open the **Invoice management** workspace.
2. Select **New Invoice**.
3. In the **Create invoice** dialog, select one of the following options:

    - Select **Choose a purchase order** to create an invoice from an existing purchase order.
    - Select **Start from Scratch** to create a cost invoice that isn't associated with a purchase order.

4. Fill out the rest of the dialog to identify the vendor and purchase order, and enter details about the invoice. Then select **Submit**.
5. A detailed invoice form opens where you can add terms and conditions, lines, attachments, and other details.
6. Select **Save as draft** on the toolbar to save the invoice without submitting it. You can find and open your draft invoices by selecting the **Draft** tile of the **Invoice management** workspace.
7. To view details about charges and sales tax, select the **Total charges** or **Total sales tax** information icon at the top of the invoice form.
8. When the invoice is ready, select **Submit** on the toolbar to submit the invoice to the purchaser for approval.

> ⓘ **Note**
>
> The supplier portal only shows procurement categories that the current vendor has access to. Vendors that need additional procurement categories can request them through vendor management.

Recall an invoice

To recall an invoice and make corrections while it's still under review, follow these steps:

1. Open the **Invoice management** workspace and select the **Submitted** tile.
2. Open the invoice you want to recall.
3. On the toolbar, select **Recall**.
4. Confirm the action.
5. The details of the invoice are now editable. Make the necessary corrections and then select **Save as draft** or **Submit** on the toolbar. If you don't save or submit it, the invoice is deleted.

Track invoice status

To check the status of your invoices, open the **Invoice management** workspace and view the tiles for **Draft**, **Submitted**, **Approved**, **Paid**, and **Rejected** invoices.

Related information

- Supplier portal overview
- View and confirm purchase orders in the supplier portal
- Supplier portal home page

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

View consignment details in the supplier portal (preview)

[This article is prerelease documentation and is subject to change.]

The supplier portal lets vendors view information about the stock level of products that they place in consignment with a customer. Vendors can also monitor consumption of their stock when the customer takes ownership of the inventory.

Work with the Consignment inventory workspace

The **Consignment inventory** workspace helps vendors track and manage consignment inventory. To open it, sign in to the supplier portal and complete any of the following steps:

- On the top navigation bar, select **Consignment inventory** and then select any of the views listed.
- On the home page, select any of the links in the **Consignment inventory** panel.

The workspace displays a set of tiles, each of which represents a status of consignment inventory. Each tile displays a count of items in that status. Select a tile to view a list of consignment inventory items in that status. The following tiles are available in the **Consignment inventory** workspace:

- **On-hand** – View the current stock owned by the vendor at the customer's warehouse. The page lists inventory owned by the vendor that is stored at the purchaser's warehouse.
- **Received items** – Track the inventory that the customer consumed after the inventory ownership change journal was posted. The page lists product receipt lines that were generated when ownership of the consignment inventory changed from the vendor to the customer. Select a purchase order number to view the related purchase order.
- **Consigned POs** – View purchase orders generated when consignment inventory is consumed and received. Select a purchase order number to view its details.
- **Invoiced POs** – View consignment purchase orders that were invoiced. Select a purchase order number to view header and line details, including consignment-inventory-specific fields.

Related information

- Supplier portal overview
- Supplier portal home page
- View and confirm purchase orders in the supplier portal

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/07/2026

Supplier Engagement frequently asked questions (preview)

[This article is prerelease documentation and is subject to change.]

Questions about Supplier Engagement often come up as organizations evaluate the solution, plan a rollout, or begin working with the supplier portal. This FAQ answers the questions that customers ask most often, covering global vendor management, how Supplier Engagement relates to the vendor collaboration interface, supported languages, and common sign-in issues.

Why should I adopt global vendors, and how do they differ from previous vendor management features in Supply Chain Management?

Global vendors give your organization a centralized way to manage supplier relationships across legal entities. Instead of maintaining the same supplier profile separately for each company, you use one master profile to consolidate shared information and connect it to released vendors in Supply Chain Management.

Unlike vendor records in Supply Chain Management, the global vendor record isn't transactional. Instead, it's the governing record for the following supplier information:

- Supplier identity and hold status
- Contact details and addresses
- Certifications and agreements
- Risk assessments and feedback
- Audit history and activities
- Ownership, offboarding, and termination details

Can I use the supplier portal and the vendor collaboration portal in parallel?

Yes. Supplier Engagement replaces the older vendor collaboration interface with broader capabilities, including global vendor data management, a dedicated Power Pages portal, and lifecycle processes such as qualification and termination. You can run both solutions in parallel

during the transition. Learn more in How Supplier Engagement compares to the vendor collaboration interface.

What happens to the existing vendor collaboration portal?

Enhancements to the vendor collaboration portal are on hold. Future strategic investment is planned for the supplier portal instead. Follow release communications for updates.

Is sealed bidding or public RFQ supported in Supplier Engagement?

No. The supplier portal doesn't currently support sealed bidding or public requests for quotation (RFQs). To learn more, see How Supplier Engagement compares to the vendor collaboration interface.

Who can use Supplier Engagement, and does it span multiple environments?

Supplier Engagement is only available for Microsoft finance and operations apps, such as Dynamics 365 Supply Chain Management. Organizations that run other enterprise resource planning (ERP) systems can't use the solution.

The solution is scoped to a single Finance and Operations instance. If you run several instances and want to share a vendor across them, you must implement a custom integration because out-of-the-box cross-instance vendor sharing isn't currently available.

Which languages does Supplier Engagement support?

In the current public preview release, the Supplier Engagement app is only available in US English (Microsoft locale identifier (LCID) 1033). If you created your Dataverse environment with a base language other than US English, the Supplier Engagement app might display a mixed-language UI, where labels provided by Dataverse appear in your selected base language while labels provided by the Supplier Engagement app appear in English.

You select your Dataverse environment's base language during provisioning and can't change it later, so we recommend that you install and test the preview version of the Supplier Engagement

app on a Dataverse environment that was created to use US English as its base language. Support for additional languages will be added for the general availability release.

Why do I see an *Invalid sign-in attempt* message when I open the supplier portal from an invitation email?

This message usually appears when your browser already has an active session for a different account, such as a Dataverse or Supply Chain Management administrator account. The supplier portal might automatically use that identity instead of the invited portal user.

To resolve the issue, follow these steps:

1. Copy the supplier portal link from the invitation email.
2. Open a new in-private or incognito browser window.
3. Paste the link into the new window.
4. Sign in with the portal user account that received the invitation.

A private browsing window prevents existing administrator sessions from interfering, and it ensures that the portal authenticates the correct supplier user. If the issue continues, verify that the invited account has the required portal access and roles.

Related information

- Supplier Engagement overview
- How Supplier Engagement compares to the vendor collaboration interface
- Global vendor management overview
- Supplier portal overview
- Deployment FAQs

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 09/03/2026

Contract lifecycle management integration overview

Contract lifecycle management (CLM) helps organizations manage their contracts with suppliers. It's an important aspect of the source-to-pay process.

CLM helps streamline contract creation, negotiation, execution, and renewal processes. In this way, it reduces the time and resources that are required to manage contracts. In addition, CLM helps ensure that contracts comply with organizational policies and regulations. In this way, it reduces the risk of penalties and other negative consequences. CLM also helps improve visibility into contract performance. Therefore, organizations can make more informed decisions about their supplier relationships.

By integrating CLM into the source-to-pay process, organizations can improve their overall supply chain performance and deliver greater value to their customers.

Microsoft Dynamics 365 Supply Chain Management provides built-in capabilities that help you operationalize contracts through purchase agreements, rebate agreements, brokerage agreements, and regular purchase orders. However, to meet the needs of organizations that want more comprehensive CLM, Supply Chain Management also provides a common integration framework that lets you integrate with an external CLM system. By integrating with an external CLM system, you can add support for more advanced CLM processes, such as contract creation, authoring, redlining, negotiation, signing, amendment, and termination.

Although the CLM integration framework lets you integrate CLM with Supply Chain Management, it isn't a CLM solution. To take advantage of this feature, you must integrate it with an external CLM solution.

When you enable and configure the integration with an external CLM solution, the following contract flows are enhanced in Supply Chain Management:

- **Purchase agreements** – The purchase agreement lifecycle can be seamlessly managed between the external CLM system and Supply Chain Management.
- **Non-disclosure agreements (NDAs)** – The NDA lifecycle is managed in the external CLM system and accessible from Supply Chain Management.

The purchase agreement and NDA contract types represent two different types of integration. Purchase agreement contracts are deeply integrated with the purchase agreement source

document and process in Supply Chain Management. By contrast, the integration of NDA contracts is looser and more lightweight. An NDA contract represents a supplier contract from an external CLM system that doesn't directly affect any processes in Supply Chain Management. In Supply Chain Management, NDA contracts exist only as records on the **All contracts** page.

The feature is extensible. Therefore, you can integrate more contract flows and types as you require.

Next steps

- Enable and configure CLM integration
- Work with integrated CLM features

Last updated on 07/01/2026

Work with integrated CLM features

Article • 10/28/2024

This article explains how the addition of contract lifecycle management (CLM) integration affects the way that Microsoft Dynamics 365 Supply Chain Management works. It also explains where you can find CLM elements in the Supply Chain Management user interface (UI).

> **Note**
>
> These changes take effect only when the CLM integration feature is configured for and connected to a CLM solution. Learn how to enable the feature and set up the integration in Enable and configure CLM integration.

View and work with all integrated contracts

To work with integrated contracts in Supply Chain Management, go to **Procurement and sourcing** > **Contracts** > **All contracts**. On the **All contracts** page, you can perform the following actions:

- **View a list all contracts that were created by using your CLM system.** These contracts are synced from the external system and are read-only here. For each contract, you can view the current status, important dates, and other information.
- **Find a specific contract.** Use the **Filter** field and/or the filtering and sorting controls that are available through the column headings.
- **View, edit, or amend a specific contract.** Select the contract in the grid, and then, on the Action Pane, on the **Contract** tab, select **View**, **Edit**, or **Amend**. Each of these buttons opens the selected contract in your external CLM system. The availability of these buttons depends on how you configured the integration.
- **Add a new contract.** On the Action Pane, on the **Contract** tab, select **New** to open the contract creation page in your external CLM system. The availability of the **New** button depends on how you configured the integration.
- **View all purchase agreements that are associated with a specific contract.** Select the contract in the grid, and then, on the Action Pane, on the **Contract** tab, select **Purchase agreement**.

View contracts associated with a selected vendor

While you're working with a vendor record, you can easily identify which contracts are associated with that vendor. To view the contracts that are associated with a specific vendor, follow these steps.

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Find and select the vendor that you want to work with. You must select only one vendor at a time.
3. On the Action Pane, on the **Procurement** tab, in the **Agreements** group, select **Contracts** to open the **All contracts** page. The page is filtered so that it shows only contracts from the selected vendor.

Purchase agreements with external CLM integration

When you're using CLM integration, purchase agreements work slightly differently than they work in a non-integrated Supply Chain Management system.

To view your purchase agreements, go to **Procurement and sourcing** > **Purchase agreements** > **Purchase agreements**. There, the following differences apply when you use CLM integration:

- The **Purchase agreements** list page includes the following new columns:
  - **Contract ID** – This column identifies any externally managed contract that is associated with each purchase agreement. The value is generated by and synced from the external CLM system. A purchase agreement can reference only one external contract.
  - **Contract status** – This column shows the status of contracts that exist in the context of an external contract.

- On the **Purchase agreements** details page, the **Header** tab includes a new **Contract** FastTab that has the following fields:

  - **Contract ID** – This field identifies any externally managed contract that is associated with the selected purchase agreement. The value is generated by and synced from the external CLM system.

  - **Status** – This field shows the status of contracts that exist in the context of an external contract. The value is synced from the external CLM system.

  - **Purchase agreement ownership type** – This field identifies the owner of the purchase agreement. It can have the following values:

    - *Supply Chain Management* – The purchase agreement was created directly in Supply Chain Management. All aspects of it must be managed by using Supply Chain Management instead of your external CLM system.
    - *External contract management* – The purchase agreement was created by a contract in an external CLM. Therefore, processing and editing options are limited in Supply Chain Management. For purchase agreements of this ownership type, the **Contract ID** and **Status** fields on the **Contract** FastTab have values. All edits and lifecycle changes are managed by the external CLM system, which automatically syncs updated values back to Supply Chain Management.

- No approval workflows are applied to purchase agreements where the **Purchase agreement ownership type** field is set to *External contract management*. In an integrated solution, a contract is approved by all relevant parties, signed, and executed by using the external CLM system. Any additional approval workflows that are applied in Supply Chain Management would disrupt the approval process that is already implemented in the external CLM system. Therefore, any workflow configuration in Supply Chain Management is bypassed. Instead, the external CLM system is responsible for updating the status of the purchase agreement status from *On hold* to *Effective*.

- If a purchase agreement that is owned by an external CLM system is selected, the following buttons are disabled on the **Purchase agreement** tab on the Action Pane of both the list page and the details page:

    - In the **Intercompany** group:
        - **Generate sales agreement**
        - **View sales agreement**

    - In the **Maintain** group:
        - **Activities**

    - In the **Generate** group:
        - **Confirmation and proforma confirmation**
        - **Proforma confirmation**

- A purchase agreement that is owned by an external CLM system can't be deleted from Supply Chain Management. Therefore, if a purchase agreement of this type is selected, the **Delete** button on the Action Pane is disabled.

- While a purchase agreement that is owned by an external CLM system is open, most of the fields on the **Header** tab of the **Purchase agreements** details page are read-only. Exceptions to this rule include the following fields:

- **Vendor reference**
- **External document reference**
- **Line matching policy**
- **Delivery address**
- **Payment specification**
- **Charges group**
- **Mode of delivery**
- **Buyer group**
- **Pool**
- **Shipping carrier**
- **Carrier service**
- **Carrier group**
- **Mode**
- **Route plan**
- **Transportation template ID**
- All fields on the **Financial dimensions** FastTab

Known limitations

CLM integration isn't supported for intercompany purchase and sales agreements.

Feedback

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Enable and configure CLM integration

Article • 03/13/2025

This article describes the configuration steps that must be completed to integrate Microsoft Dynamics 365 Supply Chain Management and third-party contract lifecycle management (CLM) providers. The configuration must be done in both Supply Chain Management and your CLM system before you can start to use the integration.

You can always manually configure the settings in Supply Chain Management as described in this article. However, in some cases, your CLM provider can remotely configure most of these settings instead. Learn more in Establish a connection from a CLM system to Supply Chain Management.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.42 or later.
- The feature that is named *Integrate with external contract lifecycle management systems* must be turned on in feature management.

Set up CLM connection parameters

To set up the connection to your external CLM system, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **CLM integration** > **Contract management parameters**.

2. On the **Connection** tab, on the **General** FastTab, set the following fields:

    - **Connection name** – Enter a name to identify the CLM system.
    - **Base URL** – Enter the host name of the external CLM service.
    - **External navigation base URL** – Some CLM providers might require an extra URL, but most don't. Leave this field blank unless your CLM provider or Microsoft Support tells you to set it.

3. On the Action Pane, select **Save**.

Set up external navigation links

External navigation links provide the capability to redirect users to an external CLM system where they can perform various actions. For example, they can create new contracts, edit existing contracts, and view contracts. These actions are optional, and the corresponding buttons appear on the contract page only after you configure the relevant external navigation links.

To configure external navigation links, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **CLM integration** > **Contract management parameters**.

2. On the **External navigation links** tab, on the toolbar, select **New**.

3. On the **New record** page, on the **General** FastTab, set the following fields:

    - **Navigation type** – Select one of the following values to specify the type of navigation link that you're setting up:
        - *New contract* – Settings for this record enable and configure the **New** button on the **All contracts** page.
        - *New contract from purchase agreement* – Settings for this record enable and configure the **New** button on the **Purchase agreements** page.
        - *Edit contract* – Settings for this record enable and configure the **Edit** button on the **All contracts** and **Purchase agreements** pages.
        - *Amend contract* – Settings for this record enable and configure the **Amend** button on the **All contracts** and **Purchase agreements** pages.
        - *View contract* – Settings for this record enable and configure the **View** button on the **All contracts** and **Purchase agreements** pages.

    - **Navigation name** – Enter a descriptive name for the record.

    - **Relative URL** – Enter the relative path from the base URL that is defined on the **Connections** tab to the page that should be opened when a user selects the button for the selected navigation type. You can view the full resulting URL in the **Absolute URL** field on the **Preview** FastTab.

    - **Action** – Select one of the following values to specify where the page should be opened when a user selects the button for the selected navigation type:
        - *Open in new tab* – Open the external page on a new browser tab.
        - *Open in existing tab* – Open the external page on the current browser tab.

4. On the **Navigation link query strings** FastTab, you can add any parameters that the CLM system requires to the URL. Use the toolbar buttons to add parameters to the

grid, or remove them, as you require. For each parameter, set the following fields:

- **Key** – Enter the key name of the parameter.
- **Value** – Enter the value of the parameter. To include dynamic values, you can insert supported placeholders be selecting them in a dropdown list. Placeholders are formatted as `%placeholder_name%`. They are replaced with actual values at runtime. You can construct a value that combines multiple placeholders and static text.

  The placeholders that are available for selection in the dropdown list are determined by the current context. This context depends on the navigation type and is derived from either the contract record or the purchase agreement record.

5. On the Action Pane, select **Save**.
6. Continue to work until you finish setting up all the external navigation links that you need.

Review the connection status

To review the current status of the connection between Supply Chain Management and the CLM system, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **CLM integration** > **Contract management parameters**.
2. On the **Connection** tab, select the **Connection status** FastTab. Here, the connection status is indicated through one of the following values:

   - *Not started* – The system hasn't yet tried to connect.
   - *Connection establishment failed* – The system tried to connect but failed.
   - *Connection established* – The connection was established, and a handshake between the two systems succeeded.
   - *Configuration failed* – The connection is working, but at least one of the external navigation links isn't working.
   - *Ready* – The integration is ready to be used.

The connection status is based on information that your CLM system submits through the Supply Chain Management API. Learn how to set up your CLM system to return a value for this status indicator in Establish a connection from a CLM system to Supply Chain Management.

View and edit contract types

Contract types represent the types of external contracts that are either integrated with or accessible from Supply Chain Management. (Examples of external contracts include purchase contracts and non-disclosure contracts.) Contract types are used to help describe each contract that is listed on the **All contracts** page. In most cases, the values are submitted and maintained exclusively by your CLM provider.

To view the names and descriptions of the various contract types that are integrated with your CLM system, go to **Procurement and Sourcing** > **Setup** > **CLM integration** > **Contract types**. Although admins can add, edit, and remove values on the **Contract types** page, you typically should do so only at the direction of your CLM provider or Microsoft Support.

Feedback

Was this page helpful? Yes No

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Data integration APIs

This article describes the integration patterns that are available in Microsoft Dynamics 365 Supply Chain Management. It also provides guidelines for syncing master data between Supply Chain Management and the contract lifecycle management (CLM) system.

The following table describes the recommended integration patterns for CLM integration.

Expand table

Pattern	Timing	Batch	Documentation
Data management package REST API	Asynchronous	Yes	Data management package REST API
OData	Synchronous	No	Open Data Protocol (OData)

Guidelines

You should select an integration pattern based on your data volume and real-time requirements.

- For high-volume master data, we generally recommend that you use the *Data management package REST API* pattern. For low-volume master data, use either the *Data management package REST API* pattern or the *OData* pattern.
- For integrations that require real-time data synchronization and error handling, we generally recommend that you use the *OData* pattern, provided that your peak data volume isn't excessively high. To handle large volumes of data, we recommend that you use the *Data management package REST API* pattern instead.

Learn more about the integration patterns and scenarios in Integration between finance and operations apps and third-party services.

Enable change tracking

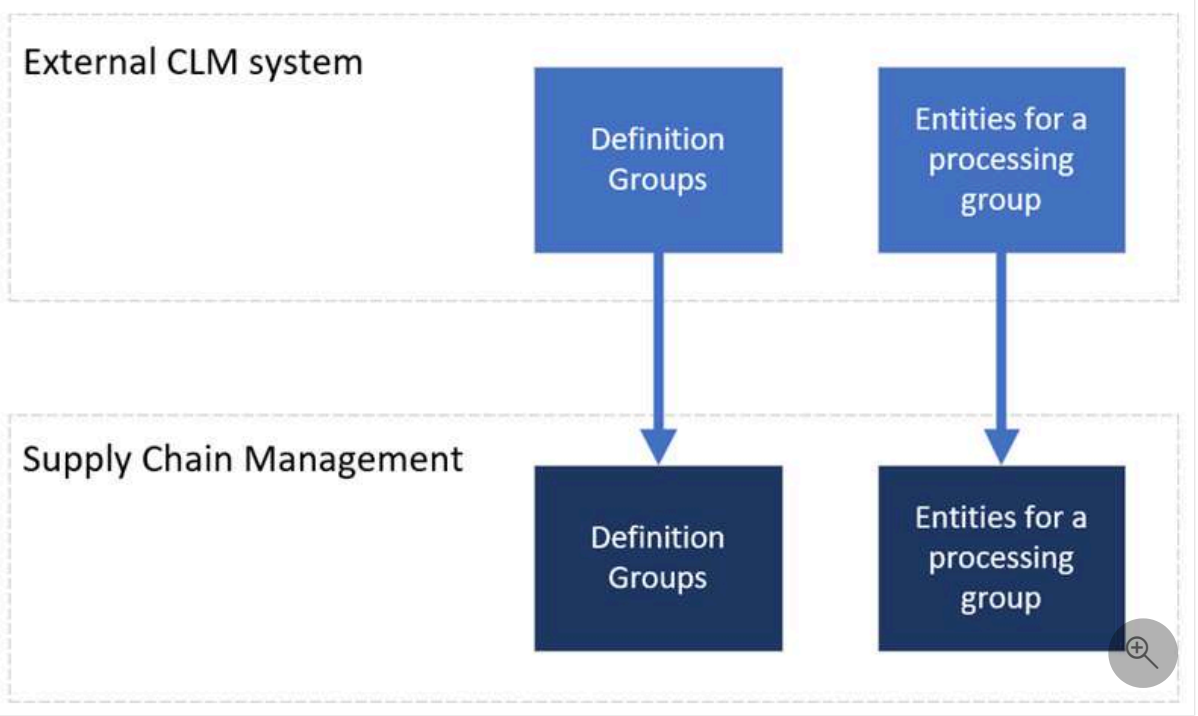
When you use the *Data management package REST API* integration pattern, you can enable change tracking for selected data entities. In this way, you enable *incremental exports*. In an incremental export, only changed records are exported.

Learn how to enable change tracking for an entity in Enable change tracking for entities.

Create data projects

As a prerequisite for using the *Data management package REST API* integration pattern, you must create a *data project* in the **Data management** workspace. You can manually create the data project in Supply Chain Management, or you can create it from the external system by using the *OData* pattern.

The following illustration shows how data projects are integrated and synced.



The following table lists the data entities that are available for the creation of a data project.

Expand table

Entity	Target entity	Public name (OData)	Company-specific	Direct
Definition Groups	`DataManagementDefinitionGroupEntity`	`DataManagementDefinitionGroups`	No	CLM - Supply Manag
Entities for a processing group	`DataManagementDefinitionGroupDetailEntity`	`DataManagementDefinitionGroupDetails`	No	CLM - Supply Manag

Learn more about data management and data import/export projects in Data management overview.

Configure and use service-based authentication

To gain access to the data entities through the API, your CLM system must authenticate with Supply Chain Management. Service-based authentication with Microsoft Entra ID provides a secure way to allow an external CLM system to connect with Supply Chain Management. To set up service-based authentication, follow these steps:

1. Register an application in Microsoft Entra ID. Make a note of the application ID (client ID) for the new application.

2. In Supply Chain Management, go to **System administration** > **Setup** > **Microsoft Entra ID applications**.

3. Add a row, and set the following fields for it:

    - **Client ID** – Enter the application ID (client ID) that you noted when you registered the application in Microsoft Entra ID.
    - **Name** – Enter a descriptive name of the application.
    - **User ID** – Select the service account to run the application under. We recommend that you create a new user and assign the *CLM integration role* role to it.

Learn how to register applications in Microsoft Entra ID and external applications in Supply Chain Management in Service endpoints overview.

After you register the Microsoft Entra ID application, you can use it to set up your external CLM system to authenticate with the Supply Chian Management API. Learn more about how authentication works in Authentication.

Related information

- Establish a connection from a CLM system to Supply Chain Management
- Synchronize master data
- Synchronize contracts

Last updated on 12/30/2025

Establish a connection from a CLM system to Supply Chain Management

Article • 10/28/2024

This article explains how a third-party contract lifecycle management (CLM) provider can simplify the process of establishing connections by efficiently configuring the contract management parameters in Microsoft Dynamics 365 Supply Chain Management and initiating the connection. The process can be initiated from the CLM system. It seamlessly transfers all the required configuration settings to the Supply Chain Management system.

Most of the code examples in this article show how to *fetch* configuration information. However, a CLM provider can also use the same APIs to *post* configuration settings. Therefore, the provider can remotely configure the connection for their Supply Chain Management customers.

Access the data integration APIs

Learn which data integration APIs are available in Data integration APIs. There, you can also learn how to enable change tracking, and how to authenticate with Supply Chain Management so that your CLM system can access the data entities that are described in this article.

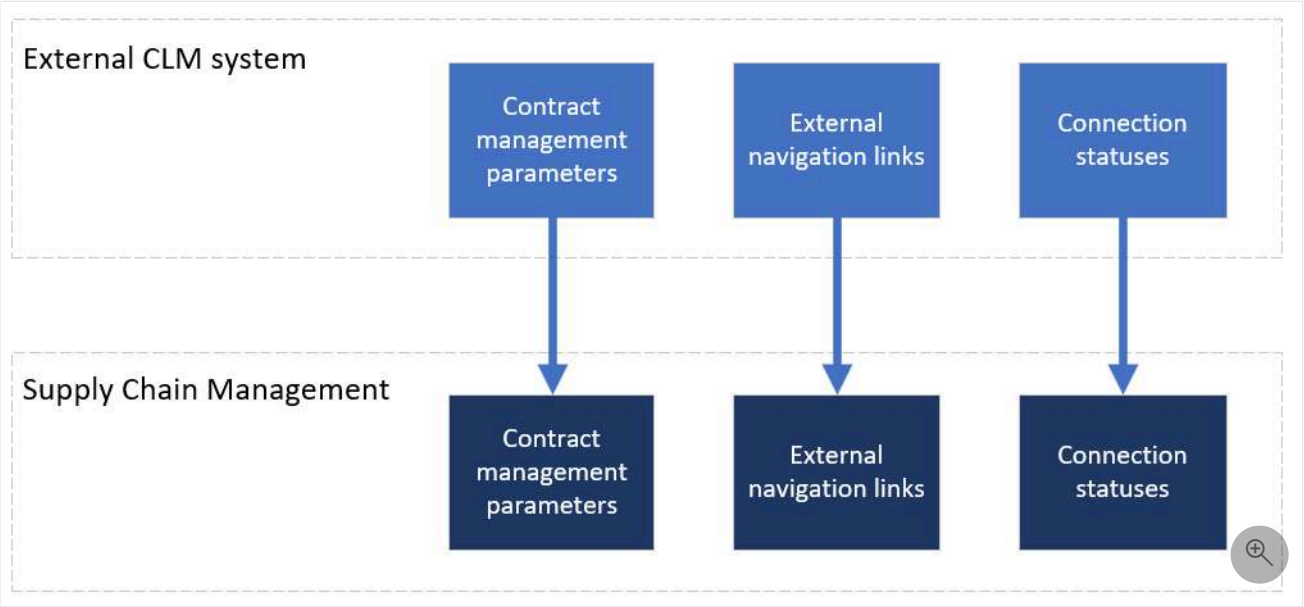
Available data entities

To streamline the connection process, Supply Chain Management offers built-in data entities that enable external CLM systems to automatically configure parameters and establish the connection. The following table describes the available data entities.

Expand table

Entity	Target entity	Public name (OData)	Purpose	Direction
CLM integration service instance	`CLMIntegrationServiceInstanceEntity`	`CLMIntegrationServiceInstances`	Set up contract management parameters.	CLM → Supply Chain Management
CLM integration external navigation links	`CLMIntegrationExternalNavigationLinkEntity`	`CLMIntegrationExternalNavigationLinks`	Set up external navigation links.	CLM → Supply Chain Management
CLM integration external navigation link key-value pairs	`CLMIntegrationExternalNavigationLinkKeyValueEntity`	`CLMIntegrationExternalNavigationLinkKeyValues`	Set up query strings for external navigation links.	CLM → Supply Chain Management

The following illustration shows the integration.



Important

Your CLM system must update the relevant status fields in the *CLM integration service instance* entity so that they reflect the **Connection status** indicator.

CLM integration service instance entity

The *CLM integration service instance* entity provides information about the instance of the CLM integration service that contains contract management parameters.

Expand table

Physical name	Property	Type	Description
`ID` (entity key)	ID	Int	The parameter key. The value should always be set to `0`.
`InstanceName`	Connection name	String	The connection name.
`BaseURL`	Base URL	String	The base URL.
`ExternalNavigationBaseURL`	External navigation base URL	String	The external navigation base URL.

The following example shows a request query for the *CLM integration service instance* entity.

HTTP

```
GET https://[baseURI]/data/CLMIntegrationServiceInstances
```

The following example shows a response to the preceding query.

JSON

```
{
    "ID": 0,
    "InstanceName": "CLM instance",
    "BaseURL": "https://clm-instance.com",
    "ExternalNavigationBaseURL": "https://clm-instance.com"
}
```

The CLM integration external navigation links entity

The *CLM integration external navigation links* entity provides information about external navigation links.

Expand table

Physical name	Property	Type	Description
`ServiceInstanceName`	Connection name	String	The instance name.
`NavigationType` (entity key)	Navigation type	Enum	The navigation type. The possible values are `ViewContract`, `EditContract`, `NewContract`, `NewContractFromPurchAgreement`, and `AmendContract`.
`NavigationName`	Navigation name	String	The navigation name.
`RelativeURL`	Relative URL	String	The relative URL.
`Action`	Action	Enum	The action that is performed. The possible values are `OpenInNewTab` and `OpenInExistingTab`.

The following example shows a request query for the *CLM integration external navigation links* entity.

HTTP

```
GET https://[baseURI]/data/CLMIntegrationExternalNavigationLinks
```

The following example shows a response to the preceding query.

JSON

```
{
    "ServiceInstanceName": "CLM instance",
    "NavigationType": "ViewContract",
    "NavigationName": "View contract",
    "RelativeURL": "/view",
    "Action": "OpenInNewTab"
}
```

The CLM integration external navigation link key-value pairs entity

The *CLM integration external navigation link key-value pairs* entity provides information about the key-value pairs for external navigation links.

Expand table

Physical name	Property	Type	Description
`NavigationType` (entity key)	Navigation type	Enum	The navigation type. The possible values are `ViewContract`, `EditContract`, `NewContract`, `NewContractFromPurchAgreement`, and `AmendContract`.
`KeyValueType` (entity key)	Key-value type	Enum	The key-value type. `QueryString` is the only allowed value. (The other possible value, `Header`, isn't supported.)
`Key` (entity key)	Key	String	The key.
`Value`	Value	String	The value.

The following example shows a request query for the *CLM integration external navigation link key-value pairs* entity.

HTTP

```
GET https://[baseURI]/data/CLMIntegrationExternalNavigationLinkKeyValues
```

The following example shows a response to the preceding query.

JSON

```
{
    "NavigationType": "ViewContract",
    "KeyValueType": "QueryString",
    "Key": "id",
    "Value": "%CLMIntegrationContractTable.ExternalContractId%"
}
```

Update the connection status

To clearly reflect the connection status, the CLM system must use the *CLM integration external navigation links* entity to update the following two statuses:

- `CLMIntegrationServiceInstanceEntity.EstablishmentStatus` – This status indicates whether a connection was established, so that the external CLM system can communicate with Supply Chain Management. The following values are used to report the connection status:
  - `NotStarted` – Establishment of a connection hasn't yet begun.
  - `Success` – A connection was successfully established.
  - `Error` – Establishment of a connection failed.
- `CLMIntegrationServiceInstanceEntity.ConfigurationStatus` – This status indicates whether external navigation links were configured. The following values are used to report the configuration status:
  - `NotStarted` – Configuration of external navigation links hasn't yet begun.
  - `Success` – External navigation links were successfully configured.
  - `Error` – Configuration of external navigation links failed.

Administrators can read status information on the **Contract management parameters** page in Supply Chain Management. Learn more in Enable and configure CLM integration.

The following example shows a request query to update the status on the *CLM integration service instance* entity.

HTTP

```
PATCH https://[baseURI]/data/CLMIntegrationServiceInstances(ID=0)
```

JSON

```
{
    "EstablishmentStatus": "Success",
    "ConfigurationStatus": "Error"
}
```

Related information

- Data integration APIs

Feedback

Was this page helpful? Yes No

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Synchronize master data

Article • 10/28/2024

This article describes the data entities and integration methods that you can use to sync master data between Microsoft Dynamics 365 Supply Chain Management and an external contract lifecycle management (CLM) system. Master data is essential for the CLM system to integrate contracts and related purchase agreements into Supply Chain Management. Supply Chain Management functions as the master system for providing the master data.

Access the data integration APIs

Learn which data integration APIs are available in Data integration APIs. There, you can also learn how to enable change tracking, and how to authenticate with Supply Chain Management so that your CLM system can access the data entities that are described in this article.

Available data entities

Supply Chain Management offers several out-of-box data entities that external CLM systems can use to sync master data. The following table describes the available data entities that are required for the synchronization of contracts and purchase agreements. Other data entities might be required, depending on your business requirements.

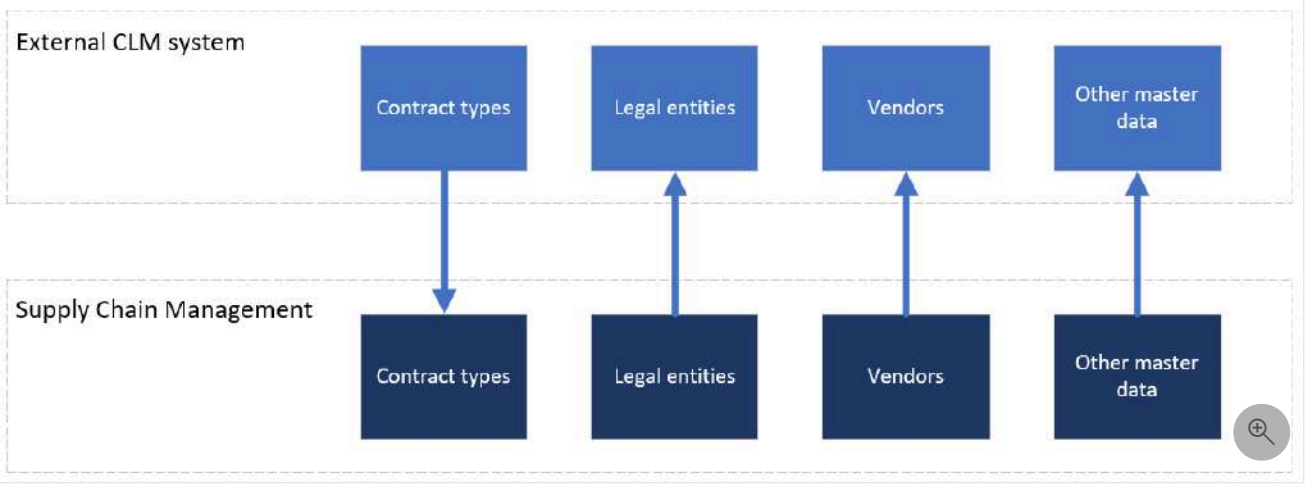
Expand table

Entity	Target entity	Public name (OData)	Company-specific	Required by	Direction
CLM integration contract types	`CLMIntegrationContractTypeEntity`	`CLMIntegrationContractTypes`	No	Contracts	CLM → Supply Chain Management
Legal entities	`OMLegalEntity`	`LegalEntities`	No	Contracts/Purchase agreements	Supply Chain Management → CLM
Vendors V2	`VendVendorV2Entity`	`VendorsV2`	Yes	Contracts/Purchase agreements	Supply Chain Management → CLM
Contacts V2	`smmContactPersonV2Entity`	`ContactPersons`	Yes	Contracts/Purchase agreements	Supply Chain Management → CLM
Purchase agreement classification	`PurchAgreementClassificationEntity`	Not applicable	No	Purchase agreements	Supply Chain Management → CLM
Sites V2	`InventOperationalSiteV2Entity`	`OperationalSitesV2`	Yes	Purchase agreements	Supply Chain Management → CLM
Warehouses	`InventWarehouseEntity`	`Warehouses`	Yes	Purchase agreements	Supply Chain Management → CLM
Procurement product categories	`CatProcurementProductCategoryEntity`	`ProcurementProductCategories`	No	Purchase agreements	Supply Chain Management → CLM
CLM integration released products	`CLMIntegrationReleasedProductEntity`	`CLMIntegrationReleasedProducts`	Yes	Purchase agreements	Supply Chain Management → CLM
Released product	`EcoResReleasedProductVariantV2Entity`	`ReleasedProductVariantsV2`	Yes	Purchase agreements	Supply Chain Management

Entity	Target entity	Public name (OData)	Company-specific	Required by	Direction
variants V2					→ CLM
Units	`UnitOfMeasureEntity`	`UnitsOfMeasure`	No	Purchase agreements	Supply Chain Management → CLM
Terms of payment	`PaymentTermEntity`	`PaymentTerms`	Yes	Purchase agreements	Supply Chain Management → CLM
Currencies	`CurrencyEntity`	`Currencies`	No	Purchase agreements	Supply Chain Management → CLM
Vendor payment method	`VendorPaymentMethodEntity`	`VendorPaymentMethods`	Yes	Purchase agreements	Supply Chain Management → CLM
Payment schedule	`PaymentScheduleEntity`	`PaymentSchedules`	Yes	Purchase agreements	Supply Chain Management → CLM
Terms of delivery	`DeliveryTermsEntity`	`DeliveryTerms`	Yes	Purchase agreements	Supply Chain Management → CLM
Projects	`ProjectEntity`	`Projects`	Yes	Purchase agreements	Supply Chain Management → CLM
Cash discount	`CashDiscountEntity`	`CashDiscounts`	Yes	Purchase agreements	Supply Chain Management → CLM
CLM integration country/regions	`CLMIntegrationAddressCountryRegionEntity`	`CLMIntegrationAddressCountryRegions`	No	Purchase agreements	Supply Chain Management → CLM

> **Note**
>
> Company-specific data entities include the `dataAreaId` field, which indicates the legal entity that a record belongs to. Your CLM system must adopt the same behavior to support both company-specific records and cross-company records. Learn more about cross-company behavior in Cross-company behavior of data entities.

The following illustration shows how master data is integrated and synced.



Related information

- Data integration APIs

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Synchronize contracts

Article • 10/28/2024

This article describes the scenarios, data entities, and integration methods that are available for the synchronization of contracts and related purchase agreements from an external contract lifecycle management (CLM) system to Microsoft Dynamics 365 Supply Chain Management. The integration process establishes a workflow that transfers contracts and related purchase agreements from the CLM system to Supply Chain Management. The CLM system is assumed to function as the master system for managing contracts.

Access the data integration APIs

Learn which data integration APIs are available in Data integration APIs. There, you can also learn how to enable change tracking, and how to authenticate with Supply Chain Management so that your CLM system can access the data entities that are described in this article.

Scenarios

Supply Chain Management supports the following integration scenarios:

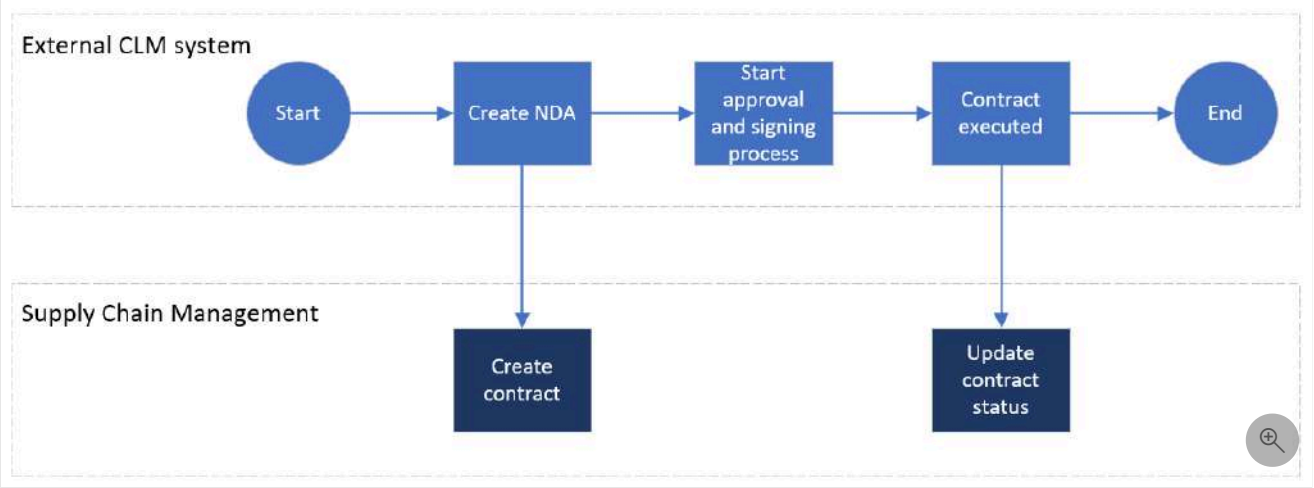
- Create a non-disclosure agreement (NDA)
- Create a purchase contract
- Create a contract amendment

This section outlines each scenario and provides a process diagram for it.

Create an NDA

NDAs are independent contracts that aren't related to any document in Supply Chain Management. The integration points involve creating the contract and updating its status after it's approved and signed.

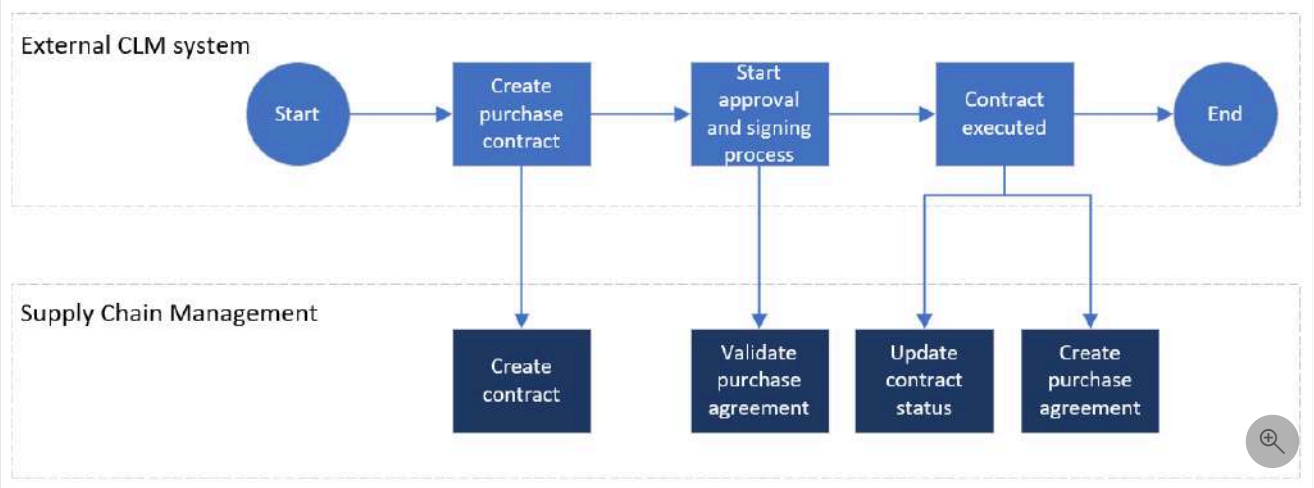
The following illustration shows how NDAs are created and synced between systems.



Create a purchase contract

Purchase contracts are contracts that are associated with a purchase agreement in Supply Chain Management. The integration points include creating the contract, validating the purchase agreement, submitting it for approval and signing, updating the contract's status after it's approved and signed, and eventually creating the purchase agreement.

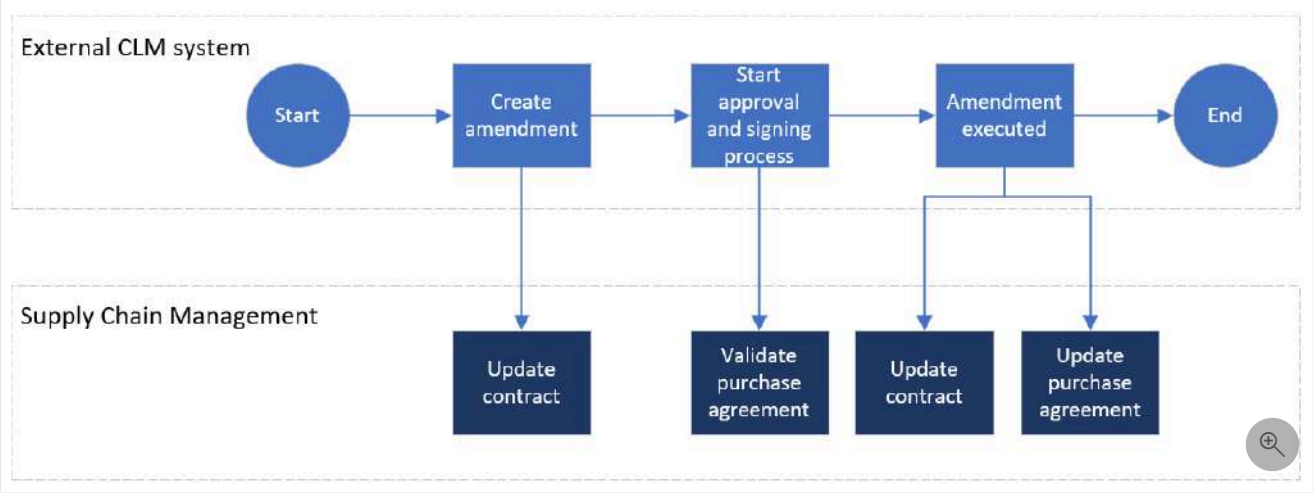
The following illustration shows how purchase contracts are created and synced between systems.



Amend a contract

Purchase contract amendments and NDA amendments are documents that change the terms of an existing agreement. The integration points include updating the contract with amendment information, validating the purchase agreement, submitting it for approval and signing, updating the contract's status after it's approved and signed, and eventually updating the purchase agreement. In Supply Chain Management, the amendment information is updated in the same contract and purchase agreement record.

The following illustration shows how contract amendments are created and synced between systems.



Available data entities for syncing contracts and purchase agreements

Supply Chain Management offers several out-of-box data entities that external CLM systems can use to sync contract data. The following table describes the available data entities that are required for the synchronization of contracts and purchase agreements.

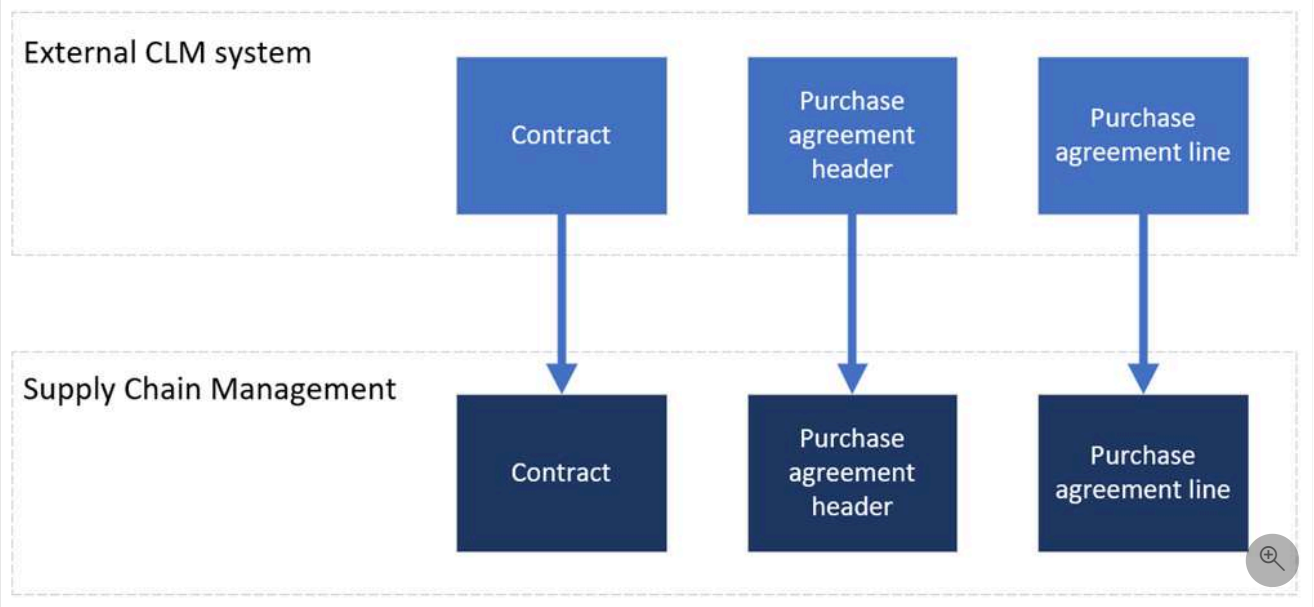
Expand table

Entity	Target entity	Public name (OData)	Company-specific	Direction
CLM integration contract	`CLMIntegrationContractEntity`	`CLMIntegrationContracts`	Yes	CLM → Supply Chain Management
CLM integration purchase agreements	`CLMIntegrationPurchaseAgreementHeaderEntity`	`CLMIntegrationPurchaseAgreements`	Yes	CLM → Supply Chain Management
CLM integration purchase agreement lines	`CLMIntegrationPurchaseAgreementLineEntity`	`CLMIntegrationPurchaseAgreementLines`	Yes	CLM → Supply Chain Management

ⓘ Note

Company-specific data entities include the `dataAreaId` field, which indicates the legal entity that a record belongs to. Your CLM system must adopt the same behavior to support both company-specific records and cross-company records. Learn more about cross-company behavior in Cross-company behavior of data entities.

The following illustration shows how contracts and purchase agreements are integrated and synced.



Contract entity

The *CLM integration contract* entity provides information about the contract. The following table describes its properties.

Expand table

Physical name	Property	Type	Description
`ContractId` (entity key)	Contract ID	String	The contract identifier in the CLM system.
`LegalEntityId` (entity key)	Legal entity ID	String	The legal entity ID.
`ExternalContractId`	External contract ID	String	The internal, system-generated contract identifier in the CLM system.
`ContractType`	Contract type	String	The contract type name reference.
`ContractStatus`	Contract status	Enum	The contract status. The possible values are `Draft`, `WaitingForApproval`, `Approved`, `Rejected`, `WaitingForSignatures`, `WaitingForExternalSignatures`, `WaitingForInternalSignatures`, `Executed`, `Terminated`, `Expired`, `OnHold`, `Cancelled`, `WaitingForReview`, and `Reviewed`.
`ContractName`	Contract title	String	The contract title.
`EffectiveDate`	Effective date	Date	The effective date.
`ExpirationDate`	Expiration date	Date	The expiration date.
`RequesterWorkerName`	Requester name	String	The name of the requester worker. If the specified worker name isn't found in the system, the `RequesterWorkerEmail` field is used to identify the worker.
`RequesterWorkerEmail`	Requester email	String	The email address of the requester worker.
`AccountType`	Account type	Enum	A value that determines the type of contracting party. The possible values are `Vend`, `Cust`, and `Party`.
`AccountRelationId`	Account number	String	The contracting account identifier that corresponds to the specified account type.
`PartyNumber`	Party ID	String	The party number in the global address book.
`ContactPersonId`	Contact person ID	String	The contact person ID.

Physical name	Property	Type	Description
`LatestAmendmentId`	Latest amendment ID	String	The latest amendment identifier in the CLM system.
`LatestExternalAmendmentId`	Latest amendment external ID	String able	The internal, system-generated latest amendment identifier in the CLM system.
`AmendmentInProcess`	Amendment in process	Enum	A value that indicates whether amendment is in progress. The possible values are `Yes` and `No`.

The following example shows a request query for the *CLM integration contract* entity.

HTTP

```
GET https://[baseURI]/data/CLMIntegrationContracts
```

The following example shows a response to the preceding query.

JSON

```
{
    "ContractId": "C00001",
    "LegalEntityId": "USMF",
    "ExternalContractId": "EXTC00001",
    "ContractType": "NDA",
    "ContractStatus": "Draft",
    "ContractName": "Non-disclosure agreement",
    "EffectiveDate": "2024-12-31T12:00:00Z",
    "ExpirationDate": "2024-01-01T12:00:00Z",
    "RequesterWorkerEmail": "julia.funderburk@constoso.com",
    "RequesterWorkerName": "Julia Funderburk",
    "AccountType": "Vend",
    "AccountRelationId": "1001",
    "PartyNumber": "000001745",
    "ContactPersonId": "",
    "LatestAmendmentId": "A00001",
    "LatestExternalAmendmentId": "EXTA00001",
    "AmendmentInProcess": "No"
}
```

Purchase agreement header entity

The *CLM integration agreement header* entity provides information about the purchase agreement header. The following table describes the most common properties of this entity. You might also use other properties, depending on your business requirements.

Expand table

Physical name	Property	Type	Description
`PurchaseAgreementId` (entity key)	Purchase agreement ID	String	The purchase agreement identifier from Supply Chain Management.
`BuyingLegalEntityId` (entity key)	Legal entity ID	String	The legal entity ID.
`ExternalContractId`	External contract ID	String	The internal, system-generated purchase agreement identifier in the CLM system.
`ContractId`	Contract ID	String	The contract identifier reference that is used to associate the purchase agreement with the contract.
`ContractLegalEntity`	Contract legal entity ID	String	The contract legal entity identifier reference that is used to associate the purchase agreement with the contract.
`AgreementOwnership`	Purchase agreement ownership type	Enum	The ownership type of the purchase agreement. The possible values are `SCM` and `CLM`.
`AgreementStatus`	Agreement status	Enum	The purchase agreement status. The possible values are `OnHold`, `Effective`, and `Closed`.
`DocumentTitle`	Document title	String	The document title.
`PurchaseAgreementClassificationName`	Purchase agreement	String	The purchase agreement classification name reference.

Physical name	Property	Type	Description
	classification		
`DefaultEffectiveDate`	Default effective date	Date	The default effective date.
`DefaultExpirationDate`	Default expiration date	Date	The default expiration date.
`AgreementVendorAccountNumber`	Vendor account	String	The agreement vendor account reference.
`InvoiceVendorAccountNumber`	Invoice account	String	The invoicing vendor account reference.
`VendorReference`	Vendor reference	String	The vendor reference.
`PrimaryResponsibleWorkerEmail`	Primary responsible worker email	String	The email address of the primary responsible worker.
`PrimaryResponsibleWorkerName`	Primary responsible worker name	String	The name of the primary responsible worker. If the specified worker name isn't found in the system, the `PrimaryResponsibleWorkerEmail` field is used to identify the worker.
`SecondaryResponsibleWorkerEmail`	Secondary responsible worker email	String	The email address of the secondary responsible worker.
`SecondaryResponsibleWorkerName`	Secondary responsible worker name	String	The name of the secondary responsible worker. If the specified worker name isn't found in the system, the `SecondaryResponsibleWorkerEmail` field is used to identify the worker.
`CashDiscountCode`	Cash discount	String	The cash discount.
`ContactPersonId`	Contact person ID	String	The contact person ID.
`CurrencyCode`	Currency	String	The currency.
`DefaultCommitmentType`	Default commitment type	Enum	The default commitment type. The possible values are `ProductQuantity`, `ProductVolume`, `ProductCategory`, and `ProductRootCategory`.
`BuyerGroupId`	Buyer group ID	String	The buyer group ID.
`ProjectId`	Project ID	String	The project ID.
`PaymentTermsName`	Payment terms	String	The payment terms.

The following example shows a request query for the *CLM integration agreement header* entity.

HTTP

```
GET https://[baseURI]/data/CLMIntegrationPurchaseAgreements
```

The following example shows a response to the preceding query.

JSON


```
{
    "PurchaseAgreementId": "000026",
    "BuyingLegalEntityId": "USMF",
    "ExternalContractId": "EXT000026",
    "ContractId": "C00003",
    "ContractLegalEntityId": "USMF",
    "AgreementOwnership": "CLM",
    "AgreementStatus": "Effective",
    "DocumentTitle": "Purchase agreement title",
    "PurchaseAgreementClassificationName": "General purchases",
    "DefaultEffectiveDate": "2024-01-01T12:00:00Z",
    "DefaultExpirationDate": "2024-12-31T12:00:00Z",
    "AgreementVendorAccountNumber": "1001",
    "InvoiceVendorAccountNumber": "1001",
    "VendorReference": "VENREF00001",
    "PrimaryResponsibleWorkerEmail": "julia.funderburk@constoso.com",
    "PrimaryResponsibleWorkerName": "Julia Funderburk",
    "SecondaryResponsibleWorkerEmail": "charlie.carson@constoso.com",
    "SecondaryResponsibleWorkerName": "Charlie Carson",
    "CashDiscountCode": "0.5%D10",
    "ContactPersonId": "",
    "CurrencyCode": "USD",
    "DefaultCommitmentType": "ProductQuantity",
```

```
    "BuyerGroupId": "",
    "ProjectId": "",
    "PaymentTermsName": "Net30"
}
```

Purchase agreement line entity

The *CLM integration purchase agreement line* entity provides information about the purchase agreement line. The following table describes the most common properties of this entity. You might also use other properties, depending on your business requirements.

Expand table

Physical name	Property	Type	Description
`PurchaseAgreementId` (entity key)	Purchase agreement ID	String	The purchase agreement identifier reference.
`PurchaseAgreementLegalEntityId` (entity key)	Purchase agreement legal entity ID	String	The purchase agreement legal entity identifier reference.
`LineNumber` (entity key)	Line number	Real	The line number.
`ExternalContractId`	External contract ID	String	The internal, system-generated purchase agreement identifier reference in the CLM system. This field can be used instead of the `PurchaseAgreementId` field when a record is created.
`ExternalContractLineId`	External contract line ID	String	The internal, system-generated purchase agreement line identifier in the CLM system.
`CommitmentType`	Commitment type	Enum	The commitment type. The possible values are `ProductQuantity`, `ProductVolume`, `ProductCategory`, and `ProductRootCategory`.
`CommittedAmount`	Committed amount	Real	The committed amount for volume-based commitment types (`ProductVolume`, `ProductCategory`, and `ProductRootCategory`).
`CommittedQuantity`	Committed quantity	Real	The committed quantity for quantity-based commitment types (`ProductQuantity`).
`EffectiveDate`	Effective date	Date	The effective date.
`ExpirationDate`	Expiration date	Date	The expiration date.
`ItemNumber`	Item number	String	The item number. This field is required by the `ProductQuantity` and `ProductVolume` commitment types.
`ProductVariantNumber`	Product variant number	String	The product variant number. This field is relevant if the specified product contains product variants.
`ProcurementProductCategoryName`	Procurement category	String	The procurement category. This field is required by the `ProductCategory` commitment type.
`Price`	Item price	Real	The item price. This field is available for the `ProductQuantity` commitment type.
`PriceQuantity`	Price unit	Real	The item price unit. This field is available for the `ProductQuantity` commitment type.
`LineDiscountAmount`	Line discount amount	Real	The item line discount amount. This field is available for the `ProductQuantity` commitment type.
`LineDiscountPercentage`	Line discount percentage	Real	The line discount percentage. This field is available for all commitment types.
`MinimumReleaseAmount`	Minimum release amount	Real	The minimum release amount.
`MaximumReleaseAmount`	Maximum release amount	Real	The maximum release amount.
`IsCommitmentMaximumEnforced`	Is max enforced	Enum	A value that indicates whether a commitment maximum is enforced. The possible values are `Yes` and `No`.
`IsPriceAndDiscountFixed`	Is price and discount fixed	Enum	A value that indicates whether the price and discount are fixed. The possible values are `Yes` and `No`.
`UnitSymbol`	Unit of measure	String	The unit of measure.
`ProjectId`	Project ID	String	The project ID.

Physical name	Property	Type	Description
`ProjectCategoryId`	Project category ID	String	The project category ID.
`ProjectActivityNumber`	Project activity number	String	The project activity number.
`ReceivingSiteId`	Site	String	The site.
`ReceivingWarehouseId`	Warehouse	String	The warehouse.

The following example shows a request query for the *CLM integration purchase agreement line* entity.

HTTP

```
GET https://[baseURI]/data/CLMIntegrationPurchaseAgreementLines
```

The following example shows a response to the preceding query.

JSON


```
{
    "PurchaseAgreementId": "000026",
    "PurchaseAgreementLegalEntityId": "USMF",
    "LineNumber": 1,
    "ExternalContractId": "EXT000026",
    "ExternalContractLineId": "EXT000026_1",
    "CommitmentType": "ProductVolume",
    "CommittedAmount": 1100,
    "CommittedQuantity": 0,
    "EffectiveDate": "2024-01-01T12:00:00Z",
    "ExpirationDate": "2024-12-31T12:00:00Z",
    "ItemNumber": "1000",
    "ProductVariantNumber": "",
    "ProcurementProductCategoryName": "",
    "Price": 0,
    "PriceQuantity": 0,
    "LineDiscountAmount": 0,
    "LineDiscountPercentage": 10,
    "MinimumReleaseAmount": 0,
    "MaximumReleaseAmount": 0,
    "IsCommitmentMaximumEnforced": "No",
    "IsPriceAndDiscountFixed": "No",
    "UnitSymbol": "",
    "ProjectId": "000057",
    "ProjectCategoryId": "ProjItem",
    "ProjectActivityNumber": "",
    "ReceivingSiteId": "",
    "ReceivingWarehouseId": ""
}
```


Purchase agreement validation entities

In the scenario for creating and amending purchase contracts, Supply Chain Management provides data entities that are designed for the validation of purchase agreements. These validation entities contain the same list of fields as regular entities, but they don't persist data in the database. Instead, they perform validation and return the result as an exception if a validation error occurs. The following table describes the data entities that are available for the validation of purchase agreements.

Expand table

Entity	Target entity	Public name (OData)	Validated operations	Compa specifi
CLM integration validation purchase agreements	`CLMIntegrationValidationPurchaseAgreementHeaderEntity`	`CLMIntegrationValidationPurchaseAgreements`	Update	Yes
CLM integration validate insert	`CLMIntegrationValidateInsertPurchaseAgreementLineEntity`	`CLMIntegrationValidateInsertPurchaseAgreementLines`	Insert	Yes

Entity	Target entity	Public name (OData)	Validated operations	Compa specifi
purchase agreement lines				
CLM integration update, delete validation purchase agreement lines	`CLMIntegrationValidateUpdateDeletePurchaseAgreementLineEntity`	`CLMIntegrationValidateUpdateDeletePurchaseAgreementLines`	Update, Delete	Yes

> **Note**
>
> As the preceding table shows, each validation entity can validate only specific database operations. No entity for validating purchase agreement insert operations is supported. Therefore, data can be validated only by persisting it through the regular `CLMIntegrationPurchaseAgreementHeaderEntity` entity.

Create shareable, secured deep links

The following APIs create shareable, secured URLs that point to specific pages in Supply Chain Management, such as the vendor, contract, or purchase agreement page. These URLs are also known as *deep links*.

The APIs enable scenarios such as embedding links directly in an external CLM system. In this way, users can quickly and easily find the specified pages and records just by selecting the link that is generated.

Learn more about deep links in Create shareable, secured URLs (deep links).

Generate a deep link for a vendor

This API generates a deep link that opens the page for a specific vendor in Supply Chain Management.

The following example shows a request for a deep link for a vendor.

HTTP

```
POST https://[baseURI]/data/VendorsV2/Microsoft.Dynamics.DataEntities.GenerateVendorDetailFormURL
```

JSON

```
{
    "_vendAccount": "1001",
    "_companyId": "USMF"
}
```

The following example shows a response to the preceding request.

JSON

```
{
    "@odata.context": "https://[baseURI]/data/$metadata#Edm.String",
    "value": "https://[baseURI]/?cmp=usmf&prt=initial&mi=display:VendTable&pageType=Details&q=[queryValue]"
}
```

Generate a deep link for a contract by contract ID

This API generates a deep link that opens the page for a specific contract in Supply Chain Management by using the contract ID.

The following example shows a request for a deep link for a contract by contract ID.

HTTP

```
POST
https://[baseURI]/data/CLMIntegrationContracts/Microsoft.Dynamics.DataEntities.GenerateContractDetailFormURLByContractId
```

JSON

```
{
    "_contractId": "C00001",
    "_companyId": "USMF"
}
```

The following example shows a response to the preceding request.

JSON

```
{
    "@odata.context": "https://[baseURI]/data/$metadata#Edm.String",
    "value": "https://[baseURI]/?cmp=usmf&prt=initial&mi=display:CLMIntegrationContractTable&pageType=Details&q=
[queryValue]"
}
```

Generate a deep link for a contract by contract external ID

This API generates a deep link that opens the page for a specific contract in Supply Chain Management by using the contract's external ID.

The following example shows a request for a deep link for a contract by contract external ID.

HTTP

```
POST
https://[baseURI]/data/CLMIntegrationContracts/Microsoft.Dynamics.DataEntities.GenerateContractDetailFormURLByExternalContr
actId
```

JSON

```
{
    "_externalContractId": "EXTC00001",
    "_companyId": "USMF"
}
```

The following example shows a response to the preceding request.

JSON

```
{
    "@odata.context": "https://[baseURI]/data/$metadata#Edm.String",
    "value": "https://[baseURI]/?cmp=usmf&prt=initial&mi=display:CLMIntegrationContractTable&pageType=Details&q=
[queryValue]"
}
```

Generate a deep link for a purchase agreement by agreement ID

This API generates a deep link that opens the page for a specific purchase agreement in Supply Chain Management by using the agreement ID.

The following example shows a request for a deep link for a purchase agreement by agreement ID.

HTTP

```
POST
https://[baseURI]/data/CLMIntegrationPurchaseAgreements/Microsoft.Dynamics.DataEntities.GenerateAgreementFormURLByAgreement
Id
```

JSON

```
{
    "_agreementId": "000026",
```

```
    "_companyId": "USMF"
}
```

The following example shows a response to the preceding request.

JSON

```
{
    "@odata.context": "https://[baseURI]/data/$metadata#Edm.String",
    "value": "https://[baseURI]/?cmp=usmf&prt=initial&mi=display:PurchAgreementDetails&pageType=Details&q=[queryValue]"
}
```

Generate a deep link for a purchase agreement by agreement external ID

This API generates a deep link that opens the page for a specific purchase agreement in Supply Chain Management by using the agreement's external ID.

The following example shows a request for a deep link for a purchase agreement by agreement external ID.

HTTP

```
POST
https://[baseURI]/data/CLMIntegrationPurchaseAgreements/Microsoft.Dynamics.DataEntities.GenerateAgreementFormURLByExternalContractId
```

JSON

```
{
    "_externalContractId": "EXT000026",
    "_companyId": "USMF"
}
```

The following example shows a response to the preceding request.

JSON

```
{
    "@odata.context": "https://[baseURI]/data/$metadata#Edm.String",
    "value": "https://[baseURI]/?cmp=usmf&prt=initial&mi=display:PurchAgreementDetails&pageType=Details&q=[queryValue]"
}
```

Related information

- Data integration APIs

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Integrate with Icertis for Microsoft Dynamics 365 Supply Chain Management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Icertis for Microsoft Dynamics 365 Supply Chain Management is a full-featured third-party contract lifecycle management (CLM) solution that uses the CLM integration capabilities of Microsoft Dynamics 365 Supply Chain Management. This article provides an overview of Icertis for Microsoft Dynamics 365 Supply Chain Management. It summarizes the following aspects of the integrated solution:

- Capabilities
- Key features
- Benefits
- Setup

You can learn more about the solution on the Icertis for Microsoft Dynamics 365 Supply Chain Management page on Microsoft Marketplace.

Capabilities

Supply Chain Management provides functionality that streamlines and automates source-to-pay processes for organizations. Icertis for Microsoft Dynamics 365 Supply Chain Management enhances this functionality by integrating capabilities for contract authoring, assembly, redlining, and analytics. This integration supports the following workflows:

- **Contract authoring** – Create procurement contracts and non-disclosure agreements (NDAs) by using configured attributes, templates, clauses, master data, and rules. Use seeded configurations to get started right away, but tailor configurations to your business needs.
- **Reviews and negotiations** – Use preapproved legal templates and playbooks to review, redline, and approve contracts. After contracts are approved, you can seamlessly send them

for digital signature.

- **Post-execution workflows** – After a contract is executed, an associated purchase agreement is created in Supply Chain Management. The purchase agreement is created in an activated state for operational purposes such as creating release orders and invoicing.
- **Update contracts** – You can amend an executed contract by using the same safeguards that apply when new contracts are drafted. Additionally, contracts can be seamlessly terminated. In this case, the required due processes are followed.
- **Risk Assessment Copilot** – Risk Assessment Copilot significantly accelerates the contract review process. At the same time, it helps reduce risk by using AI to compare contract terms to a company's predefined risk parameters, based on the contract type.
- **Pre-seeded information** – Icertis for Microsoft Dynamics 365 Supply Chain Management includes extensive pre-seeded attributes and entities. Therefore, you can benefit from the integration without having to spend lots of time on implementation.

To learn more about Icertis for Microsoft Dynamics 365 Supply Chain Management, follow these steps:

1. Sign in to your Icertis Contract Intelligence (ICI) instance.
2. Select the **Help** button.
3. Go to Welcome to Icertis for Microsoft Dynamics 365 Supply Chain Management .

Key features

This section describes the key features of Icertis for Microsoft Dynamics 365 Supply Chain Management.

Manage and operationalize procurement contracts

Icertis for Microsoft Dynamics 365 Supply Chain Management offers a seamless source-to-contract workflow that includes an embedded experience in Supply Chain Management. This integration lets Supply Chain Management users take advantage of the robust CLM capabilities of Icertis to create and manage procurement contracts.

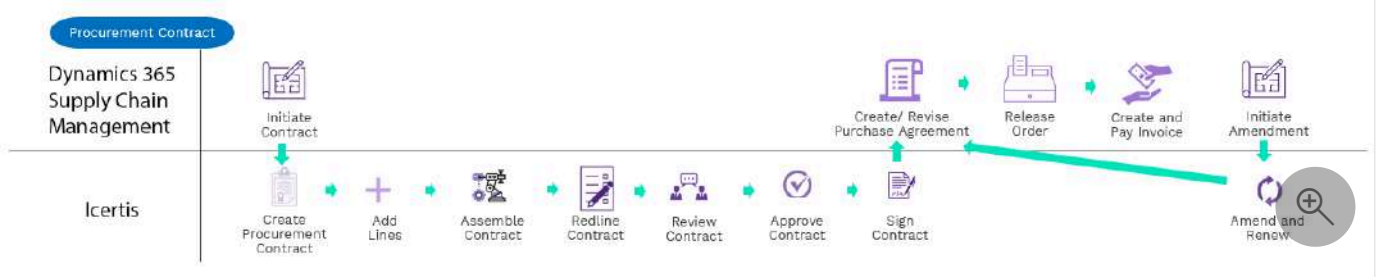
After you conclude a sourcing event and shortlist a vendor, you can initiate the creation of a procurement contract from Supply Chain Management. The embedded Icertis experience lets you perform the following actions:

- Define the terms, conditions, and specifics of the procurement contract with Icertis.
- Add line items for the procurement contract.

- Confirm that the pricing, quantities, and other terms are aligned with your business needs.
- Obtain all the required approvals.
- Validate the contract details to ensure that they comply with Supply Chain Management rules.
- Review and negotiate the contract with all relevant parties.
- Execute the contract.

After the Icertis procurement contract is executed, an effective purchase agreement is created in Supply Chain Management. The statuses of the Icertis procurement contract and the corresponding Supply Chain Management purchase agreements are regularly synced.

The following illustration shows the contract workflow between the two applications.



Create and manage NDAs

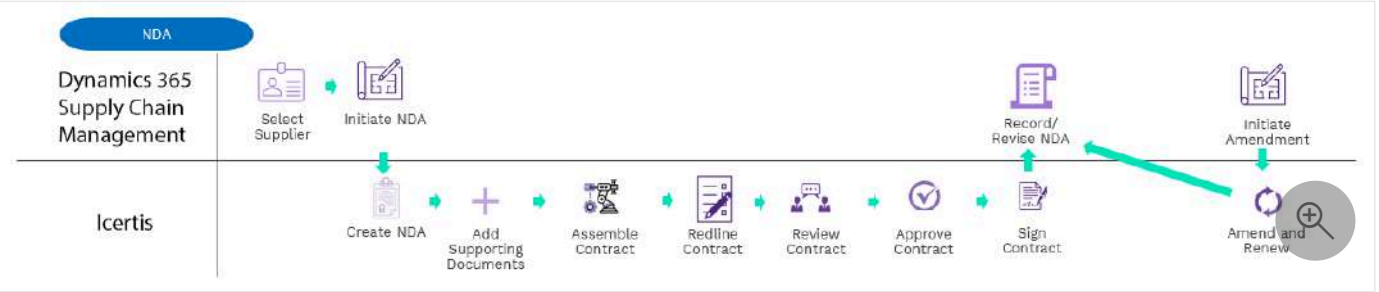
Icertis for Microsoft Dynamics 365 Supply Chain Management offers an embedded experience in Supply Chain Management. This integration lets you take advantage of the robust CLM capabilities of Icertis to create and manage NDAs.

After you decide to transact business with a vendor, you can initiate the creation of an NDA from Supply Chain Management. The embedded Icertis experience lets you perform the following actions:

- Specify effective dates, expiry dates, signature types, signatory details, and other details. The statuses of Icertis procurement contracts and their corresponding Supply Chain Management purchase agreements are regularly synced.
- Use a system-provided template, review details, and create and publish the contract.
- Add supporting documents, team members, and so on.
- Review and negotiate the contract with the vendor.
- After the agreement is approved, send it for signature.

NDAs can be amended, terminated, or updated as required. The process is similar.

The following illustration shows the NDA workflow between the two applications.



Amend and update agreements

Icertis for Microsoft Dynamics 365 Supply Chain Management offers options that let you update or modify executed contracts. You can add line items, modify expiry dates, or terminate contracts. Updates to contracts can trigger one of the following events:

- **Synchronize amendments** – The statuses of an Icertis procurement contract and the corresponding Supply Chain Management purchase agreements are regularly synced. Users can create amendments directly in Supply Chain Management. Contract managers can add an amendment to a signed or executed contract in Icertis. Downstream consumption then uses the revised information in Supply Chain Management.
- **Termination of contract** – If an Icertis procurement contract is terminated in Icertis Contract Intelligence (ICI), the associated purchase agreement in Supply Chain Management system is automatically terminated.
- **Reset expiry** – You can renew agreements or reset expiry dates as you require, based on your business needs.

Integrate master data

The integration of master data that stores less frequently changing data and business information enables all the previously mentioned workflows. The master data entity is based on attributes and rules. It can be reused for the various organizations that you work with.

Comprehensive pre-seeded information for seamless integration

Icertis for Microsoft Dynamics 365 Supply Chain Management includes extensive pre-seeded attributes and entities. This pre-seeded information enables the application and integration to work seamlessly. It also lets users benefit from the integration after only minimal implementation time.

If the pre-seeded attributes and entities require configuration, an administrator or implementation team can configure them.

The pre-seeded information includes the following elements:

- Data models for procurement contracts, NDAs, line items, and associated documents
- Attributes that are part of contract types and master data
- Master data and rules
- Standard reports and saved searches
- Clauses and templates
- eSign configurations
- Role–action mappings
- Entities for notifications, reasons, organizations, cascade teams, and implementation tasks

Risk Assessment Copilot

You can enhance the integration by adding Risk Assessment Copilot, which is powered by the Icertis ExploreAI Service. Risk Assessment Copilot significantly accelerates the contract review process. At the same time, it helps reduce risk by using AI to compare contract terms to a company's predefined risk parameters, based on the contract type.

Benefits

Icertis for Microsoft Dynamics 365 Supply Chain Management provides the following benefits:

- **Accelerate contract turnaround times** – The seamless integration enables organizations to quickly author and negotiate contracts, and operationalize them in Supply Chain Management.
- **Plug cost leakage** – Maximize the value of every contract by using the negotiated commercial terms and enforcing them during procurement and payable processes.
- **Effectively manage legal and business risks** – Manage legal and business risks by using preapproved legal templates and predefined corporate policies, and by constantly monitoring contractual obligations and service-level agreements (SLAs).
- **Effectively collaborate on contracts with all parties** – Reduce rework, improve supplier satisfaction, and help reduce risk exposure for your company through accurate redlining, versioning, and approval workflows. Enable secure communication with suppliers, customers, and partners.
- **Gain deeper insights into purchases and contracts** – Manage agreements across multiple departments to gain deeper insights. Align supplier and customer terms through AI-supported search, digitization, and negotiation.

Set up Icertis for Microsoft Dynamics 365 Supply Chain Management

To set up Icertis for Microsoft Dynamics 365 Supply Chain Management, follow these steps:

1. Purchase a subscription for Icertis for Microsoft Dynamics 365 Supply Chain Management.

   After the subscription is confirmed, the designated administrator receives an email from Icertis. This email includes instructions for downloading and deploying the package.

2. Deploy the package by using Microsoft Dynamics Lifecycle Service.

3. Optional: Configure single sign-on (SSO).

4. Configure ICI with attributes, templates, clauses, master data, rules, and other useful metadata to seamlessly integrate the two products.

To get detailed instructions for deploying the package and setting up an instance, follow these steps:

1. Sign in to your Icertis Contract Intelligence (ICI) instance.
2. Select the **Help** button.
3. Go to Icertis for Microsoft Dynamics 365 SCM Package Installation & Setup.

Last updated on 07/01/2026

Use financial tags in procurement documents

Financial tags let you attach reporting context to procurement transactions. They help you track the business purpose behind purchases throughout the procurement lifecycle. For example, a company could tag all purchases related to an employee onboarding initiative, making it easy to find and analyze the associated spending later—without relying on document descriptions or manual tracking.

You can use financial tags across procurement documents, including purchase requisitions, purchase orders, purchase agreements, requests for quotation (RFQs), and inventory ownership change journals. Because financial tag values flow through the process from one document to the next (for example, from a purchase requisition to a purchase order), you can maintain consistent reporting context across the entire procurement lifecycle.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.
- The feature named *Enable financial tags for purchase order invoicing* must be turned on in feature management.

How financial tags work in procurement

Financial tags in procurement follow the same general principles across all document types. Understanding these common behaviors helps you use tags consistently.

Header and line behavior

You can enter financial tag values on both document headers and document lines. The following rules apply across all procurement document types:

- Line-level tag values are the source of truth. A line-level change doesn't flow back to the header.

- When you add new lines, the header tag value can default to the line. However, a blank header value doesn't overwrite an existing line value—only populated header fields cascade to lines.
- Users can override financial tag values directly on individual lines, which allows different lines within the same document to carry different reporting context.

Tag propagation between documents

When one procurement document creates another document (for example, when you release a purchase requisition to a purchase order), financial tag values flow from the source document's lines to the target document's lines. The following rules apply:

- Tag propagation is always line-to-line. The source document's line-level tags are the primary source.
- If the source document has a single line, the target document's header can also receive the same tag value. If the source has multiple lines with different tag values, the target document's header remains blank to avoid ambiguity.
- Changing financial tags on a source document doesn't change documents that you already created from it. Earlier documents preserve the tag values that existed when they were created.

Editability

The following rules determine when you can edit financial tag values on procurement documents:

- You can edit financial tags while a document is in a draft or editable state.
- When a document enters a workflow for approval, the system typically locks the tags. Depending on the workflow configuration, the approver might be able to update them during review.
- After you post a document, its financial tag values are read-only.
- On purchase orders that you received but didn't post yet, you can edit financial tags in the same way that you can edit financial dimensions.

Create financial tags

You can create up to 20 financial tags as needed to meet your business needs. Set up each tag by assigning a name and a value type. The value type might ask users to select from a list of predefined values or allow them to enter free text. When you're working with a relevant procurement document, you can see the available financial tag names and enter values for them

as needed. The system stores the tag values with the document and uses them for reporting and analysis.

To create, edit, and activate financial tags, go to **General ledger** > **Chart of accounts** > **Financial tags** > **Financial tags**. Learn more about financial tags and how to set them up in Financial tags.

Financial tags on purchase requisitions

Procurement professionals use financial tags on purchase requisitions to capture the reporting context for each requisition line. When you enter tags here, they flow downstream into purchase orders when you release the requisition.

Set financial tag values on purchase requisitions

To set financial tag values for a purchase requisition, follow these steps:

1. Go to **Procurement and sourcing** > **Purchase requisitions** > **All purchase requisitions**.
2. Create or open a purchase requisition.
3. On the **Lines** tab of the requisition details page, select or create a line that you want to assign a financial tag to.
4. On the **Line details** FastTab, open the **Financial tags** tab and set the desired financial tag values.

How financial tags behave on purchase requisitions

The following table describes how financial tags behave in purchase requisition scenarios.

Expand table

Scenario	Behavior
Entry point	Enter and maintain financial tags on requisition lines. Both item-number lines and procurement-category lines can carry financial tag values.
Line-level override	Users can update or add financial tag values directly on requisition lines. A line-level change doesn't update the requisition header and doesn't affect other lines. This behavior allows different requisition lines to carry different reporting context.
Workflow and approval	During review, an approver might be able to update the financial tags. After a requisition is recalled to draft, financial tags become editable again. After final approval, they aren't editable.
Release to purchase order	When you release an approved requisition (manually or automatically), the purchase order lines receive the financial tag values from the originating requisition lines.

Scenario	Behavior
Purchase order header outcome	If the requisition has a single line, the purchase order header can receive the same tag value as the requisition line. If the requisition has multiple lines with different tag values, the purchase order header remains blank.
Copying requisitions	When you copy a requisition or selected requisition lines, the new requisition lines retain the financial tag values from the source lines.
RFQ creation from requisition	When you create an RFQ from a requisition, financial tags are copied to the RFQ header and lines. However, changes made to the RFQ tags don't propagate back to the originating requisition.

Financial tags on purchase orders

Financial tags on purchase orders help you track the business purpose of each purchase throughout the order lifecycle, from creation through receipt and invoicing.

Set financial tag values on purchase orders

To set financial tag values for a purchase order, follow these steps:

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Create or open a purchase order.
3. To assign financial tags to the purchase order header, open the **Header** tab of the purchase order details page. Then expand the **Financial tags** FastTab and set the desired financial tag values.
4. To assign financial tags to a purchase order line, open the **Lines** tab of the purchase order details page. Then select or create the line that you want to assign a financial tag to. On the **Line details** FastTab, open the **Financial tags** tab and set the desired financial tag values.

How financial tags behave on purchase orders

The following table describes how financial tags behave in purchase order scenarios.

Expand table

Scenario	Behavior
Entry point	You can maintain financial tags on both the purchase order header and on purchase order lines.
Header-to-line defaulting	Line-level tag values can differ from the header. A line change doesn't flow back to the header. A blank header value doesn't overwrite an existing line value—only populated

Scenario	Behavior
	header fields cascade.
Existing lines	For existing purchase order lines, header-to-line updates are controlled by defaulting behavior. A blank header value doesn't overwrite an existing line value—only populated header fields cascade.
Line-level override	Users can update or add financial tag values directly on purchase order lines. A line-level change doesn't update the purchase order header and doesn't affect other lines. This behavior allows different purchase order lines to carry different reporting context.
From purchase requisition	Purchase order lines inherit financial tags from the linked purchase requisition lines. The purchase order header is populated from the requisition line only when the source requisition has a single line. For multi-line requisitions, the purchase order header remains blank.
Copying purchase orders	Copy behavior depends on the copy options. With precise line copy, line tags are copied line by line. With header copy, the purchase order header tag can be copied or overwritten. Without precise or header copy, tags can remain blank.
Copying from journals	When copying from confirmation journals, product receipts, or invoices, financial tags follow the same copy-option pattern: precise copy retains line tags, and non-precise header copy can apply header tags to new lines.
Version and history views	A confirmed purchase order version shows the tag values that existed at confirmation. Distribution views show the current financial tag values, which might differ if the purchase order was changed after confirmation.
Editability after receipt	You can edit tags on a received purchase order before posting, in the same way that you can edit financial dimensions at that stage. Tags on posted journals are read-only. Pending vendor invoice line tags are read-only, consistent with financial dimension behavior.

Financial tags on purchase agreements

Financial tags on purchase agreements let you capture agreement-specific reporting context that carries into release orders.

Set financial tag values on purchase agreements

To set financial tag values for a purchase agreement, follow these steps:

1. Go to **Procurement and sourcing** > **Purchase agreements** > **Purchase agreements**.
2. Create or open a purchase agreement.
3. To assign financial tags to the purchase agreement header, open the **Header** tab of the purchase agreement details page. Then expand the **Financial tags** FastTab and set the desired financial tag values.

4. To assign financial tags to a purchase agreement line, open the **Lines** tab of the purchase agreement details page. Then select or create the line that you want to assign a financial tag to. On the **Line details** FastTab, open the **Financial tags** tab and set the desired financial tag values.

How financial tags behave on purchase agreements

The following table describes how financial tags behave in purchase agreement scenarios.

Expand table

Scenario	Behavior
Entry point	You can enter financial tags on purchase agreement headers and lines for different agreement classifications, including bulk purchase, employee purchases, and general purchases.
Header-to-line defaulting	Line-level tag values can differ from the header. A line change doesn't flow back to the header. A blank header value doesn't overwrite an existing line value—only populated header fields cascade.
Existing lines	For existing agreement lines, header-to-line updates are controlled by defaulting behavior. A blank header value doesn't overwrite an existing line value—only populated header fields cascade.
Line-level override	Users can update or add financial tag values directly on agreement lines. A line-level change doesn't update the agreement header and doesn't affect other lines. This behavior allows different agreement lines to carry different reporting context.
Release to purchase order	When a release order creates a purchase order, the purchase order receives the financial tag values from the purchase agreement. The line value must be updated on the agreement line for the released purchase order to reflect it.
Historical purchase orders	Changing financial tags on a purchase agreement doesn't change purchase orders that you already created from that agreement. Earlier purchase orders preserve the tags that existed when they were created.

Financial tags on requests for quotation

Financial tags on requests for quotation (RFQs) help you maintain reporting context while soliciting bids from vendors. Tags flow into the resulting purchase order or purchase agreement when the RFQ is accepted.

Set financial tag values on RFQs

To set financial tag values for an RFQ, follow these steps:

1. Go to **Procurement and sourcing** > **Requests for quotation** > **All requests for quotation**.
2. Create or open an RFQ.
3. To assign financial tags to the RFQ header, open the **Header** tab of the RFQ details page. Then expand the **Financial tags** FastTab and set the desired financial tag values.
4. To assign financial tags to an RFQ line, open the **Lines** tab of the RFQ details page. Then select or create the line that you want to assign a financial tag to. On the **Line details** FastTab, open the **Financial tags** tab and set the desired financial tag values.

How financial tags behave on RFQs

The following table describes how financial tags behave in RFQ scenarios.

Expand table

Scenario	Behavior
Entry point	You can add financial tags on RFQ headers and lines for RFQs created with purchase type *Purchase order* or *Purchase agreement*. Header values can default to lines when you add them.
Header-to-line defaulting	Line-level tag values can differ from the header. A line change doesn't flow back to the header.
Line-level override	Buyers can change or add financial tag values directly on RFQ lines before the RFQ outcome is accepted. Line changes don't flow back to the RFQ header.
Vendor response	Vendors can't edit financial tag values when submitting RFQ responses. Financial tags are buyer-controlled context.
After vendor response	The buyer can continue to update financial tags after the vendor submits a response and before the RFQ is accepted.
RFQ to purchase order	When an RFQ for a purchase order is accepted, the generated purchase order receives the RFQ financial tag values at both header and line level. After creation, the purchase order follows standard purchase order financial tag rules.
RFQ to purchase agreement	When an RFQ for a purchase agreement is accepted, the generated purchase agreement receives the RFQ tag values.

Financial tags on inventory ownership change journals

Use financial tags on inventory ownership change journals to capture reporting context for consignment ownership transfers.

Set financial tag values on inventory ownership change journals

To set financial tag values for an inventory ownership change journal, follow these steps:

1. Go to **Procurement and sourcing** > **Consignment** > **Inventory ownership change**.
2. Create or open an inventory ownership change journal.
3. On the **Lines** tab of the inventory ownership change journal details page, select or create a line that you want to assign a financial tag to.
4. On the **Line details** FastTab, open the **Financial tags** tab and set the desired financial tag values.

How financial tags behave on inventory ownership change journals

The following table describes how financial tags behave in ownership change journal scenarios.

Expand table

Scenario	Behavior
Consignment replenishment	Financial tags aren't required when creating a consignment replenishment order or generating the product receipt for that replenishment order.
Ownership change journal	Enter financial tags on inventory ownership change journal lines. You can update saved journal lines before posting.
After posting	After the inventory ownership change journal is posted, its financial tag values aren't editable.
Auto-created purchase order	Posting the ownership change journal can generate a purchase order. The generated purchase order line receives the financial tag values from the journal line. The purchase order header doesn't receive those tags.
Purchase order editability	On the generated purchase order, users can add or edit financial tag values on fields that weren't populated by the ownership change journal, subject to normal purchase order editability rules.
Product receipt and invoice	Product receipt journal reporting is voucher-oriented rather than requiring editable line-level tags. Invoice line tags are read-only before posting, consistent with financial dimension behavior.

General financial tag behaviors

The following table summarizes a few financial tag behaviors that you might not expect, but that are based on intentional design decisions.

Expand table

Area	Behavior
Product receipt	Product receipt journal carries tags at the voucher level, not per line.
Purchase agreements	On header-to-line cascade, a blank header never overwrites an existing line value—only populated fields cascade.
Purchase agreements	Editing a purchase agreement's tags doesn't change purchase orders that were already created from it (historical values are preserved).
Purchase agreements and RFQs	A line-level tag change never flows back up to the header.
Purchase order copy	Copy behavior depends on the `copyHeader` and `copyPrecisely` options (header-only vs. line-precise).
Purchase order versions	A confirmed purchase order version shows the tag as of confirmation. View distributions shows the current value.
Purchase orders and consignment	Editing tags on a received (pre-posting) purchase order is allowed (same as financial dimensions). Posted-journal tags are read-only.
Purchase Requisition	Financial tags are supported for purchase requisition lines but not for purchase requisition headers.
Purchase requisition to purchase order	The created purchase order header tag equals the purchase requisition line tag, but only when the requisition has a single line. A multiline requisition leaves the purchase order header tag blank.
Purchase requisitions, purchase orders, and purchase agreements	Financial tag header and line behavior mirrors financial dimensions (header cascades to lines; line values are the source of truth).
RFQ to purchase requisition	Updated RFQ tags aren't propagated back to the originating purchase requisition.
RFQs	Vendors can't edit tags. The buyer can edit tags until the bid is accepted.

Related information

- Financial tags
- Purchase requisition overview
- Purchase order overview
- Purchase agreements
- Request for quotations (RFQs) overview

Last updated on 07/29/2026

Procurement FAQ

06/17/2025

This article provides answers to frequently asked questions (FAQs) about the procurement functionality of Supply Chain Management.

Can I show only purchase orders that I created?

This functionality isn't currently available.

Can I reserve inventory and create transactions against registered inventory on a purchase order?

Issue description

Even when items are in a *Registered* state on a purchase order, you can still reserve the inventory. In other words, you can create transactions against the registered inventory.

Reproduce the issue

The following procedure shows one way to reproduce the issue.

1. Create a purchase order.
2. Register the purchase order line.
3. Notice that you can generate reservations or transactions against the registered inventory.

Issue resolution

This behavior is by design. The expectation is that the registered items have physically arrived in the warehouse or inventory, and that they're therefore available for reservation.

Can I find the user who canceled a purchase order?

This information is tracked only if the purchase order is subject to change management. If you use change management, you can see who submitted the change (the cancellation), and who approved it.

Why can't I link a purchase agreement to an existing purchase order?

A purchase agreement must be associated with a purchase order when the purchase order is created. Otherwise, purchase order lines can't be associated with purchase agreement lines.

What check triggers the "Update prices and discounts entered manually or external document" message?

When you change the shipping date, you might receive a message that states, "Update prices and discounts entered manually or external document." You receive this message because, if the shipping date is changed, a different trade or sales agreement might be triggered and cause a price change. A change to the shipping date can also affect warehouse schedules and other related information.

The message is triggered whenever any of the dates or some other parameters are changed. The purpose of the message is to make sure that you're aware of price changes that can occur because of those changes.

The message is the trade agreement evaluation (TAE) prompt. For a full description, see Trade agreement evaluation policies.

Why can't I post more than one invoice for a purchase order line that has category-based items?

A quantity is mandatory if you want to post invoices. Therefore, if the full quantity of a line has been invoiced for only a partial amount, you won't be able to invoice the remaining amount on another invoice.

Product information overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about product information management. Product information management works with a shared product definition, categorization, identifiers, and specific configurations of a product.

Product information is the backbone of supply chain and commerce applications across all industries. It refers to processes and technologies that focus on centrally managing information about products (for example, across supply chains). It's crucial that shared product definitions, documentation, attributes, and identifiers be used. In the various modules of a business solution, product-specific information and configuration are required to manage the business processes that are related to specific products, product families, or product categories.

Product definition

A product is primarily defined by a product number, name, and description. However, other data is also required to describe a product or service:

- Product type: Item or service
- Product subtype: Distinct products or product masters
- Definition of the product variant model:
  - Product dimensions and dimension groups
  - Product nomenclature
  - Product configuration models
- Association of the product with one or more categories
- Definition of the product and category attributes
- Product images
- Attachments
- Units of measure and related conversions

- Translations for all names and descriptions

Distribution, export, and import of product data

Create the product definition in Supply Chain Management. You can also import it from product lifecycle management (PLM), product data management (PDM), or product information management (PIM) systems. When you use more than one instance of Supply Chain Management, one instance typically acts as the master of the product data for all other instances. A large set of data entities supports this approach and enables the export and import of product definition data from one instance to another.

To support the distribution of product data to many instances, Supply Chain Management lets you use Microsoft Dataverse. You can export the product definitions from an instance of Supply Chain Management to Microsoft Dataverse. You can then use the product definitions to provision other business applications, such as Dynamics 365 Sales, with product data.

In dynamic and agile organizations, product information data changes every day. Therefore, maintenance of accurate and actual product data is a critical business process on its own.

Product masters and product variants

In an agile world, where products must be quickly adapted to customer requirements, product definitions specify a set of products instead of distinct products. In Supply Chain Management, those generic products are known as *product masters*. Product masters hold the definition and rules that specify how distinct products are described and behave in business processes. Based on these definitions, distinct products can be generated. These distinct products are known as *product variants*.

A product master is associated with a product dimension group and a configuration technology to specify the business rules. The product dimensions (Color, Size, Style, and Configuration) are a specific set of attributes that can be used throughout the application to define and track specific behaviors of the related products. These dimensions also help users search for and identify the products.

Configuration technologies

Choose among three configuration technologies:

- The predefined variants are defined by predefined product dimensions. The variant definition includes the definition of a specific valid combination of dimensions, such as

Color, Style, and Size. Each combination produces a distinct product variant.

- The dimension-based configuration is typically used in manufacturing scenarios and lets you use the Configuration dimension in the definition of the bills of materials (BOMs). After you select a specific configuration, the system uses the subset of BOM lines that are valid for that configuration for planning and production. This concept is also known as *global BOM*, because one shared BOM is used for all configurations of a product.
- The constraint-based configuration uses a product configuration model to describe all possible attributes and components that are required in order to describe all possible variants of a product in a single model. The constraints of combinations of attributes can be described through regular expressions or table-based constraints. Configuration models and configurators become more important in product information management and are used across all industries.

When you plan the implementation of Supply Chain Management, it's important that you choose the correct configuration technology for a business process. A product can't be converted from one model to another after implementation.

Product variant model definition workspace

The **Product variant model definition** workspace gives an overview of the product masters. It also shows the status of the release of masters and related variants to specific legal entities.

Released products

Released products are products that you release to a specific legal entity. You can release products in bulk to one legal entity or many legal entities at the same time. Because you might need to add various properties and attributes of the products for each legal entity, the **Released product maintenance** workspace helps you monitor and complete the recently released products in each legal entity or in the suborganizations of a legal entity.

Released product maintenance workspace

You can configure the **Released product maintenance** workspace from the **Configure my workspace** menu item. Select a category hierarchy and category by which to filter the workspace. To adjust the relevant product data in the workspace, you can also define, in days, the time fences for **Recently released products** and **Stopped released products**.

The workspace consists of a summary of tiles and two lists. The **Open cases** list shows product change cases that have products in the selected product category hierarchy that aren't

completed and closed. The **Recently released** list shows products that are released within the time fence set in the workspace configuration. For each item in the list, validation is run, and the validation status is shown. This status might indicate that the required configurations for the legal entity aren't completed. From the list, you can directly access the **Released product details**, **Product attribute maintenance**, **Product category maintenance**, **Default order settings**, and **Text translations** pages to complete the required configuration of the product.

Manually creating a new released product

You can manually create a released product in a single run, depending on the organization's business processes and any rules about whether this function should be used. This function creates a new product and automatically releases it to the current legal entity. To create a new product, select **Released products** in the **Released product maintenance** workspace or on the **Released product** list page.

Change managed products (versioned products)

Products where you need to keep track of versions and have an established change management process (especially products with BOMs or formulas) are often best set up as *engineering products*. Engineering products are change managed, include version identifiers, and use change orders to keep track of the changes made between versions. If needed, you can choose to track the version used in each transaction (each version is a product variant).

For more information about engineering change management, see Engineering change management overview.

Last updated on 07/01/2026

Product identifiers

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about the various types of product identifiers and explains how you can add product identifiers in your product data.

When you work with products on the shop floor or in a warehouse in Microsoft Dynamics ERP or Microsoft Dynamics CRM, you must have a good strategy for identifying those products and product variants.

Unique product number/product ID

In Dynamics 365 Supply Chain Management, the primary identifier for a product is the product number (that is, the unique product ID). This number can be automatically generated by a number sequence or manually associated with a product. For product variants, define the numbers through the product nomenclature template.

In many cases, the product number isn't originally created in Dynamics 365 Supply Chain Management. Instead, it's associated with a product in a product lifecycle management (PLM) system or a product data management (PDM) system. In this case, use data entities to import the products and product variants. Supply Chain Management then uses the numbers in all operations.

When you implement Supply Chain Management, give special consideration to your strategy for product numbers. A good numbering system improves logistics flows and helps prevent errors. A good product identifier usually has 20 characters or less, but use fewer than 10 characters and include no more than five classifying characters. You can also use search names to enable quick searches. A search name is an extra name that represents the classifications of a product.

When you use Microsoft Dataverse, the product number in Supply Chain Management is also the product number in Microsoft Dataverse. Product variants are synchronized to Dataverse as

distinct products.

Item number and product dimensions

The item number is the product identifier that a specific legal entity uses. Ideally, the item number should be identical to the product number. If the nomenclature differs per legal entity, it becomes difficult to follow a product throughout the supply chain, and it introduces burdensome relabeling and referencing processes. For compatibility with older versions (that is, with Microsoft Dynamics AX 2009 and earlier), this model is preserved. However, eliminate identifiers that are specific to legal entities whenever you can, and use the unique product number as the primary identifier instead.

Additionally, an item number can't uniquely identify a product variant. It always requires the combination of an item number and all the product dimensions that are defined on the product master. This requirement can become cumbersome and can slow down the identification processes. For this reason, use the unique product number instead of the item number whenever you can.

Many pages still have the item number and product dimensions as the primary identifiers. However, you can use the product numbers for searches. At **Sales and marketing** > **Setup** > **Search** > **Search parameters**, you can change the search lookup so that it uses product numbers instead of item numbers as the primary search strategy. If you set the **Enable lookup for product search** option to *Yes*, the lookup shows not only product masters but product variants. For more information, see Search for products and product variants during order entry.

Additionally, you can search and filter on the product number, the product name and description, and the product dimension IDs of the product variant. When you select a variant, the related item number and all product dimension IDs are selected. Therefore, you can more easily find and select the correct variant. This setting is highly recommended if you use product variants and the unique product number as the primary identifiers for products. The only exception might be the fashion industry, where the business processes often require that you select the master before you select a variant. You should carefully evaluate this option before you implement the numbering system.

> **Note**
>
> The item number for a product can't be changed once one or more transactions exist for that product.

Product name and description

The product name and description are the human-readable identifiers of a product and can be maintained in many languages. By default, the Supply Chain Management client shows all product information in the default company language, not in the user's language. However, translated product names and descriptions are used in all communication with customers and vendors. The translations are based on the language code of the customer and vendor accounts.

For product variants, the product name can be generated through a product nomenclature template. Because there's no requirement that product names be unique, you might find multiple products that have the same name.

Product and item search names

Supply Chain Management offers a secondary search name for products and also for items (released products). This search name doesn't have to be unique, and you can change it after a product or product variant is created. Use the search name to search for products by categories. The search names enable quick searches, especially in sales and purchase processes.

The search name can also contain a customer or vendor product ID, or some other external product ID, if this external ID is the primary search criterion for a product.

External product identifiers (Customer and vendor identifiers)

For released products, you can maintain the item numbers, item names, and item descriptions that the customer or vendor uses. The references appear on external documents, such as sales orders, purchase orders, packing slips, and invoices. In the current version of Supply Chain Management, the external references don't appear on core operations pages. The only exception is the vendor item number. This number appears in the **Product information** dialog box if a default vendor is defined for the released product.

You can maintain the external product identifiers by released product, released product variant, customer, customer group, vendor, or vendor group.

On the **Released products** page, follow one of these steps:

- For customers, on the Action Pane, open the **Sell** tab and, in the **Related information** group, select **External item description**.

- For vendors, on the Action Pane, open the **Purchase** tab and, in the **Related information** group, select **External item description**.

On the **External item descriptions** page, you can associate the customer's or vendor's item number with a released product. You must make this association for each legal entity. You can gather the following information. However, the labels are slightly misleading in the current version of Supply Chain Management. These labels might be changed in a future version.

Expand table

Field	Corresponding customer information	Corresponding vendor information
External item number	The customer's item number	The vendor's item number
Description	The name that the customer associates with the item	The name that the vendor associates with the item
External item text	The customer's item description	The vendor's item description

If many customers or vendors use the same item numbers (as with a purchase association or a commerce group, for example), you can create groups of customers or vendors to simplify the maintenance of external product information.

- For customer groups, go to **Sales** > **Setup** > **Items** > **External item description** to create and maintain the groups and the related item numbers. To associate customers with a group, go to **Accounts receivable** > **Customers** > **All customers**, and then, on the **Sales order defaults** FastTab, specify a value in the **Item - customer group** field.
- For vendor groups, go to **Procurement and sourcing** > **Setup** > **External item description group** to create and maintain the groups and the related item numbers. To associate vendors with a group, go to **Accounts payable** > **Vendors** > **All vendors**, and then, on the **Purchase order defaults** FastTab, specify a value in the **Item - vendor group** field.

Bar codes

To use a bar code scanner to identify products, the product identifier must meet the requirements of the bar code standard that you use. Therefore, bar codes don't typically contain the raw product number but a number that's generated specifically for the selected bar code technology. You can maintain multiple bar codes by bar code type. You can even associate the same bar code with multiple products and then select the actual active association when you scan a bar code.

Before you define bar codes, define one or more bar code setups. The bar code setups can help validate that bar codes follow the required guidelines, and that they can therefore be effectively printed and scanned. You can also maintain special bar codes for specific product quantities.

Use the bar code setup to maintain Global Trade Item Number (GTIN) or International Article Number (EAN) codes.

To maintain bar codes, on the **Released products** page Action Pane, open the **Manage inventory** tab and, in the **Warehouse** group, select **Bar codes**.

GTIN codes

In e-commerce, it's crucial that all parties speak a common language and refer to products by using a common set of identifiers. Therefore, some industries rely on the GTIN, which is a global item number system that's facilitated by GS1.

Maintain the GTIN as a bar code. However, you can also maintain it on the **Item - GTIN** page. To open this page, on the **Released products** page Action Pane, open the **Manage inventory** tab and, in the **Warehouse** group, select **GTIN codes**. The GTIN isn't maintained as a global number. Instead, legal entities maintain it.

In Supply Chain Management, you define packaging variants in the warehouse operations by defining specific units of measure. For example, an item might be stored in pieces, in bundles of six, in trays of 18, or in full pallets. A specific unit of measure is defined for each of these packaging variants. Because the GTIN is typically related to the packaging unit of a product, the **Item - GTIN** page lets you maintain multiple GTIN codes per product and unit of measure. However, you can't use the same GTIN code more than one time for different items or product variants of a legal entity.

To maintain **GTIN codes**, on the **Released products** page Action Pane, open the **Manage inventory** tab and, in the **Warehouse** group, select **GTIN**.

External codes

External codes can be defined for many entities. For example, you can define external codes to identify products and released products. These external codes can be used to associate statistical codes or tax codes with released products and released product variants. External codes are defined by legal entity and code type. They must be unique by legal entity, code type, and table reference.

Unfortunately, there's no standard functionality that lets you search for products by external codes.

Data entities for importing and exporting product identifiers

Expand table

Entity name	Import identifiers	Export identifiers	Comments
Products V2	Product number, product search name, product name, product description	Product number, product search name, product name, product description	Depending on the settings of the entity and the number sequence for the product number, the product number can be automatically created during import.
Product variants	Product number, product search name, product name, product description	Product number, product search name, product name, product description	Depending on the product nomenclature template, the product number can be automatically created during import. However, you can import any unique product number, and that product number doesn't have to follow the structure of the product nomenclature templates.
Product translations	Product name, product description	Product name, product description	This entity overwrites any language. When the name or description of a legal entity's primary language is overwritten, the name and description of the product itself is changed.
Released product creation V2	Item number, product number, item search name	Item number, product number, item search name, product search name, product name	This entity can be a challenge when number sequences are used during the creation of new released products. Both the *Item number* number sequence and the *Product number* number sequence have an influence. However, the *Item number* number sequence is per legal entity, whereas the *Product number* number sequence is global. Therefore, don't use the *Item number* number sequence when you deploy new released products. When the entity is used to release an existing product, the product number must be given in the entity. For more information, see the "Product and item number sequences" section in this article.
Released product variants	Item number, product dimensions,	Product number, product search name, product	Like the **Product variants** entity, this entity can be used to create new products that either follow the

Entity name	Import identifiers	Export identifiers	Comments
	product number	name, product description, product dimensions	product nomenclature template or use their own product numbers for the variant.
External item description for customers	Customer item number, customer item name, customer description, customer account	Customer item number, customer item name, customer description, customer account	Use the **External item description customer groups** entity to aggregate a group of customers (for example, a buyer association) into one group.
External item description for vendors	Vendor item number, vendor item name, vendor description, vendor account	Vendor item number, vendor item name, vendor description, vendor account	Use the **External item description vendor groups** entity to aggregate a group of vendors (for example, a sales association or industry organization) into one group.
Item Barcode	Bar code	Bar code	During import, you must refer to a bar code setup that is defined in the target system. The imported bar code references are validated against that bar code setup and are rejected if the bar codes don't match the requirements that are defined in that setup.
External codes for released products	External code	External code, external code classes, item number	External codes are by legal entity. For import, you must refer to a defined code class. Import the code classes by using the **External code classes for released products** entity.
External codes for released product variants	External code	External code, external code classes, item number, product dimensions	External codes are by legal entity. For import, you must refer to a defined code class. Import the code classes by using the **External code classes for released products** entity. This entity refers to product variants by the item number and product dimensions.
External codes for released product variants by product number identifier	External code	External code, external code classes, product number	External codes are by legal entity. For import, you must refer to a defined code class. Import the code classes by using the **External code classes for released products** entity. This entity refers to product variants by the product number of the variant. (From the next major release)

Entity name	Import identifiers	Export identifiers	Comments
GTIN	Not applicable	Not applicable	Currently, there's no specific entity that is used to import and export GTIN codes. Use the **Item Barcode** entity instead.
Product entity common data service identifier entity	Not applicable	Item number, item search name, product search name, vendor item number, customer item number, external codes, GTIN codes, bar codes	This entity consolidates all identifiers into one data model, so that one interface can be used to easily export all identifiers and their related types. Use the **Product entity identifier code** entity to export the identifier codes and descriptions. Use the **Product entity identifier scope** entity to export additional scope information to an identifier, such as the party, legal entity, quantity, or unit.

Product and item number sequences

You can define two different number sequences:

- The *Product number* number sequence for the global product number
- The *Item number* number sequence for the item number per legal entity

> **Note**
>
> Use the item number as a separate identifier only when you migrate different legal entities from different sources that had different numbering systems. Always try to use a product identifier that is unique across all legal entities. Therefore, set the **Manual** option to *Yes* for the *Item number* number sequence. In this way, the item number follows the product number on creation. If Supply Chain Management isn't the leading system for new product numbers, set the **Manual** option to *Yes* for both the *Item number* and *Product number* number sequences.

When you use the **Released product creation V2** entity to create products, multiple settings can affect how the number sequences are used to create the product number and item number:

- Settings of the *Product number* number sequence
- Settings of the *Item number* number sequence
- The mapping of the item number
- The mapping of the product number

The following table provides an overview of the results of import and manual creation when specific combinations of the number sequence and field mapping settings.

Expand table

Product number number sequence	Item number number sequence	Mapping of the item number	Mapping of the product number	Result of entity import	Result of manual creation	Conclusion
Manual = No	Manual = No	No mapping	No mapping	Product numbers use the *Product number* number sequence. Item numbers use the *Item number* number sequence.	Product numbers use the *Product number* number sequence. Item numbers use the *Item number* number sequence.	With this configuration, product numbers follow the product number sequence and item numbers follow the item number sequence. However, this configuration doesn't work if there's more than one item (row) to be imported.
Manual = No	Manual = Yes	Auto generate	No mapping	Both product numbers and item numbers use the *Item number* number sequence.	Both product numbers and item numbers use the *Product number* number sequence.	Both product numbers and item numbers follow the product number sequence. This approach is the recommended approach to import bulk products with the Released product creation V2 data entity. You can only use this approach when bulk importing items (several rows) and when you aren't creating items through the user interface. If you need both to bulk import and to create products through the user interface, use the procedure in the next row of this table instead. To transition from using a bulk import approach to

Product number number sequence	Item number number sequence	Mapping of the item number	Mapping of the product number	Result of entity import	Result of manual creation	Conclusion
						using the user interface to manually, import and create products, you must manually adjust the **Next number** in the item number number sequence to match the **Next number** in the product number number sequence. Then you can switch to the approach in the next row of this table.
Manual = No	Manual = Yes	No mapping	No mapping	Both product numbers and item numbers use the *Product number* number sequence.	Both product numbers and item numbers use the *Product number* number sequence.	Both product numbers and item numbers use the product number sequence. However, this configuration doesn't work if there's more than one item (row) to be imported. You must use this approach if you need both to import products using the entities (only one row can be imported at a time) and to create products through the user interface.
Manual = Yes	Not applicable	Not applicable	Auto generate	You receive the following error message: "Number sequence can't be detected."	According to the *Item number* number sequence	This setting isn't supported for import.

Product entity identifier (Export all product identifiers)

The product entity identifier model enables version 1.0 of Dataverse to be provisioned with all identifiers that refer to a product. To simplify this task, the model aggregates all identifiers into one global identifier table, so you can export them as one model. This version of Dataverse doesn't use the product identifiers model. Therefore, the **Product entity common data service identifier entity** and this process have limited practical use and might change in the future.

The product identifier table is a global table that's populated from all reference tables of the main legal entity through a recurring batch job. You must select a legal entity and a product category hierarchy as the definition of the global product master scope. Generation of the global product identifier table is limited to products that are released to the selected legal entity and products that are members of the product hierarchy selected for the **Common data service** role in the product category hierarchy.

This process assumes that product master data is primarily maintained in one central legal entity. (However, this legal entity can be a virtual legal entity that's used only to maintain global master data.)

Follow these steps to configure the environment.

1. Select the category hierarchy for Dataverse. On the **Category hierarchy role associations** page, if no hierarchy is associated with the **Common data service** role, create a new association. Select the **Common data service** role, and then associate the category hierarchy that represents the product portfolio that you want to synchronize to Dataverse.

2. Select the legal entity for global product master data. On the **Product information management parameters** page, on the **Product attributes** tab, select the master company where the product and item identifiers are primarily maintained.

3. Define the identifier code types and codes that you want to export. Go to **Product information management** > **Setup** > **Product identifier codes**. To generate the identifier code types, select **Generate codes**. A code type entry is generated for each type of identifier that is found in the selected legal entity.

   For bar codes, a code type is generated for each bar code setup. For external codes, a code type is generated for each external code class.

   You can now maintain the list of code types. You can change the code, name, and description. You can also delete code types. Code types that you delete aren't used to populate the global product entity identifier tables.

4. When you finish defining the product identifier code types, create the identifiers in the global table by starting the **Create product entity identifiers** job on the **Product entity identifier codes** page. Run this job in a batch. Set up this job as a periodic batch job so that the table is populated according to new entries.

You can now use the **Product entity common data service identifier entity**, **Product entity identifier code**, and **Product entity identifier scope** data entities to export the identifiers for any target system.

Related information

- Search for products and product variants during order entry

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Last updated on 06/16/2026

Create a hierarchy of product classification

This procedure shows how to create a new category hierarchy and assign a commodity code hierarchy type. The demo data company used to create this procedure is USMF. This procedure is intended for the category manager.

Create the new category hierarchy

1. Go to **Product information management** > **Setup** > **Categories and attributes** > **Category hierarchies**.
2. Select **New**.
3. In the **Name** field, type a value.
4. In the **Description** field, enter a value.
5. Select **Create**.

Build the hierarchy

1. Select **New** category node.
2. In the **Name** field, type a value.
3. In the **Code** field, type a value.
4. In the **Friendly name** field, type a value.
5. Select **New** category node.
6. In the **Name** field, type a value.
7. In the **Code** field, type a value.
8. In the **Friendly name** field, type a value.
9. Select **New** category node.
10. In the **Name** field, type a value.
11. In the **Code** field, type a value.
12. In the **Friendly name** field, type a value.
13. Select **New** category node.
14. In the **Name** field, type a value.
15. In the **Code** field, type a value.
16. In the **Friendly name** field, type a value.
17. Close the page.

Classify the hierarchy

1. In the list, find and select the desired record.
2. On the Action Pane, select **Category hierarchy**.
3. Select **Associate hierarchy type**.
4. Select **New**.
5. In the **Category hierarchy type** field, select an option. Select the **Commodity code category hierarchy type for product classification**.
6. In the **Category hierarchy** field, select the drop-down button to open the lookup.
7. In the list, find and select the desired record.
8. In the list, select the link in the selected row.
9. Close the page.

Last updated on 07/01/2026

Classify a product using category hierarchies

This article describes how to classify a product using category hierarchies. Use category hierarchies to classify products for export, procurement, sales, or other purposes. Typically, a product manager performs this task. The example uses product number M0009 in demo data company USMF.

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, find and select the desired record.
3. On the Action Pane, select **Product**.
4. Select **Product categories**.
5. Select **New**.
6. In the **Category hierarchy** field, enter or select a value.
7. In the **Category** field, enter or select a value.
8. Select **New**.
9. In the **Category hierarchy** field, enter or select a value.
10. In the **Category** field, enter or select a value.
11. Close the page.

Last updated on 07/01/2026

Nomenclature of product variant numbers and names

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to set up a product number nomenclature to replace the fixed [Product master number - Configuration - Size - Color - Style] format. The new nomenclature has a targeted format that includes the product master number, active product dimensions, and text delimiters of your choice. You can also create a nomenclature for product names. Finally, you can build a nomenclature to identify configurations that the constraint-based product configurator creates. These nomenclatures can contain attributes of your choice.

The new nomenclatures for product variant numbers and product variant names let you include segments in the identifiers for product variants. These segments can include the product master number and name, product dimension IDs and names, number sequences, text constants, and attributes. This functionality lets you quickly find a specific product variant when you create a sales order or a purchase order. You create nomenclatures for both product variant numbers and product variant names by using the **Product nomenclature** page. To open this page, click **Product information management** > **Setup**.

Nomenclature of predefined product variants

Product variants are generated for product masters according to one of three configuration technologies:

- Predefined variants
- Constraint-based
- Dimension-based

Each product variant has a number and a name. The product variant identification nomenclatures let you select the segments to include in each product variant number or name. Select the

following segments on the **Product nomenclature** page:

- Product master number
- Product master name
- Number sequence value
- Text constant
- Product dimensions
  - Configuration ID or name
  - Color ID or name
  - Size ID or name
  - Style ID or name

After you define a product variant identification number nomenclature, associate it with a product dimension group. All product masters that reference this product dimension group are assigned product variant numbers according to the nomenclature. However, product variant name nomenclatures can't be associated with product dimension groups. You can also assign a product variant identification nomenclature directly to a product master. In this case, the product variants that belong to the product master are assigned product variant numbers and names according to the nomenclatures.

Example

A T-shirt (TS1234) is produced in three sizes (S, M, L), four colors (Red, Green, Blue, Yellow), and two styles (Polo, V). Therefore, 24 product variants are possible (= 3 × 4 × 2). You create a product variant number nomenclature that has the following segments:

1. Product master number
2. Text constant: "-"
3. Color
4. Text constant: "-"
5. Size
6. Text constant: "-"
7. Style

In this case, the product variant number for a red, small, polo T-shirt is TS1234-Red-Small-Polo.

Nomenclature of constraint-based configurations

For constraint-based configurations, you can create a dedicated nomenclature for the configuration product dimension. On the **Product nomenclature** page, select the following segments:

- Number sequence value
- Text constant
- Attribute value

Each component in a product configuration model can have its own configuration nomenclature. You can only use attributes that belong to the component. You can't use attributes from subcomponents or user requirements.

Example

A product configuration model has a root component that has two attributes:

- Material (Plastic, Wood, Steel)
- Length (10...100)

You create a configuration nomenclature that has the following segments:

1. Attribute value: Material
2. Text constant: "AAA"
3. Attribute value: Length

In this case, the configuration ID for wood material that has a length of 78 is WoodAAA78.

Nomenclature of dimension-based configurations

For dimension-based configurations, you can create a dedicated nomenclature for the configuration product dimension. You can select the following segments on the **Product nomenclature** page:

- Number sequence value
- Text constant
- Configuration group item

You can define a configuration nomenclature for a bill of materials (BOM).

Example

A BOM has four BOM lines that are divided into two configuration groups:

- BOM line: M0007, Standard cabinet
  - Configuration group: Cabinet
- BOM line: M0008, High end cabinet
  - Configuration group: Cabinet
- BOM line: M0021, Front grill cloth
  - Configuration group: Front grill
- BOM line: M0022, Front grill metal
  - Configuration group: Front grill

You create a configuration nomenclature that has the following segments:

1. Configuration group: Cabinet
2. Text constant: "&"
3. Configuration group: Front grill

In this case, the configuration ID for a standard cabinet that has a cloth front grill is M0007&M0021.

Nomenclature for a combination of product variants and configurations

When you use either the constraint-based configuration technology or the dimension-based configuration technology to configure product variants for a product master, the product variant numbers of the product variants can include the nomenclature from the configuration dimension. Follow these steps to configure variants.

1. On the **Product nomenclature** page, define a product variant number nomenclature that includes the configuration dimension.
2. Assign the nomenclature to a product dimension group that has the configuration dimension.
3. Define a configuration nomenclature for the components or BOMs used to configure the product variants.

You can also create nomenclatures for the product variant names. The product variant names can be configured to include the configuration ID or name.

Example for constraint-based configurations

For this example, use a product variant number nomenclature that includes the following segments:

1. Product master number
2. Text constant "_"
3. Configuration

The configuration nomenclature consists of the following segments:

1. Attribute value: Material
2. Text constant: "AAA"
3. Attribute value: Length

Enter the following values for segments:

- Product master number = **M0099**
- Material = **Plastic**
- Length = **12**

In this case, the product variant number is M0099_PlasticAAA12.

Example for dimension-based configurations

For this example, use a product variant number nomenclature that includes the following segments:

1. Product master number
2. Text constant "//"
3. Configuration

The configuration nomenclature consists of the following segments:

1. Configuration group: Cabinet
2. Text constant: "&"
3. Configuration group: Front grill

Enter the following values for segments:

- Product master number = **D0123**
- Cabinet = **M0008**
- Front grill = **M0022**

In this case, the product variant number is D0123//M0008&M0022.

Numbering conflicts

In some cases, a product variant number nomenclature that you set up might not produce unique product variant numbers. For example, the product variant numbers aren't unique if one active product dimension isn't included in the nomenclature for a product master that uses the predefined variant configuration technology. The way that you handle conflicts varies, depending on the configuration technology.

Predefined variants

An error occurs if you try to manually create or automatically generate product variants, and more than one product variant ends up with the same product variant number. To avoid this scenario, use all active product dimensions in the product dimension group. Alternatively, include a number sequence to help guarantee that the product variant numbers are unique.

Constraint-based configurations

Depending on the nomenclature, the system might try to assign a non-unique product variant number to a configuration. In this case, the system uses the number sequence for the configuration dimension as the product variant number instead, and you receive a warning. To avoid this scenario, include enough attributes in the nomenclature to help guarantee unique product variant numbers. Also, make sure that the **Reuse** option is turned on for the component.

Dimension-based configurations

During one step of the configuration process, the system suggests a configuration value according to the nomenclature. In this step, you can manually change the configuration value. When you save the configuration, the system verifies that the configuration value is unique. If the value that you entered isn't unique, you receive an error message. To save the configuration, you must enter a unique configuration value.

Related information

Create a product number nomenclature for predefined product variants

Create a product number nomenclature for configured product variants

Last updated on 05/18/2026

Create a product number nomenclature for predefined product variants

Article • 06/03/2022

This article explains how to set up a product number nomenclature for predefined product variants, and how you assign it to the appropriate product dimension group. The demo data company used to create this procedure is USMF. The new product number nomenclature is assigned to the Color and Size product dimension group. This task would typically be done by a product designer.

Create a product number nomenclature

1. Go to **Product information management** > **Setup** > **Product nomenclature**.
2. Select **New**.
3. In the **Name** field, enter a nomenclature name that helps to identify the target product dimension group, for example, `ColorSize`.
4. In the **Description** field, type a value.
5. Select **Add**.
6. Select **Product master** number.
7. Select **Add**.
8. Select **Text constant**.
9. In the **Text** field, type a value.
10. Select **Add**.
11. Select **Color**.
12. Select **Add**.
13. Select **Text constant**.
14. In the **Text** field, type a value.
15. Select **Add**.
16. Select **Size**.
17. Close the page.

Assign the nomenclature to a product master

1. Select **Product dimension groups**.
2. Select the **SizeCol product dimension** group.
3. Select **Edit**.
4. Select **Yes** in the **Use nomenclature** field.
5. In the **Product variant number nomenclature** field, enter or select a value.

6. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a product number nomenclature for configured product variants

Article • 09/29/2021

This procedure shows you how to set up a product number nomenclature for configured product variants, and how it can be attached to a configurable product master. This procedure also demonstrates how you can build a configuration nomenclature for a product configuration model component. The demo data company used to create this procedure is USMF. The new product number nomenclature is assigned to the D0004 product master. This task would typically be done by a product designer.

Create a product number nomenclature

1. Go to **Product information management** > **Setup** > **Product nomenclature**.
2. Select **New**.
3. In the **Name** field, type a value.
4. In the **Description** field, type a value.
5. Select **Add**.
6. Select **Product master number**.
7. Select **Add**.
8. Select **Text constant**.
9. In the list, mark the selected row.
10. In the **Text** field, type a value.
11. Select **Add**.
12. Select **Configuration**.
13. Close the page.

Assign the product number nomenclature to a product master

1. Go to **Product information management** > **Products** > **Product masters**.
2. Use the Quick Filter to find records. For example, filter on the **Product number** field with a value of 'D'.
3. In the list, select the link in the selected row.
4. Select **Edit**.
5. Select *Yes* in the **Use nomenclature** field.
6. In the **Product variant number nomenclature** field, enter or select a value.

7. Close the page.
8. Close the page.

Create nomenclature for a product configuration model component

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, find and select the desired record.
3. In the list, select the link in the selected row.
4. Select **Edit**.
5. Select *Yes* in the **Use configuration nomenclature** field.
6. Select **Add**.
7. Select **Attribute value**.
8. In the list, mark the selected row.
9. In the **Attribute** field, enter or select a value.
10. Select **Add**.
11. Select **Text constant**.
12. In the list, mark the selected row.
13. In the **Text** field, type a value.
14. Select **Add**.
15. Select **Attribute value**.
16. In the list, mark the selected row.
17. In the **Attribute** field, enter or select a value.
18. Select **Add**.
19. Select **Text constant**.
20. In the list, mark the selected row.
21. In the **Text** field, type a value.
22. Select **Add**.
23. Select **Attribute value**.
24. In the list, mark the selected row.
25. In the **Attribute** field, enter or select a value.
26. Select **Add**.
27. Select **Text constant**.
28. In the list, mark the selected row.
29. In the **Text** field, type a value.
30. Select **Add**.
31. Select **Attribute value**.
32. In the list, mark the selected row.
33. In the **Attribute** field, enter or select a value.

34. Select **Add**.
35. Select **Text constant**.
36. In the list, mark the selected row.
37. In the **Text** field, type a value.
38. Select **Add**.
39. Select **Number sequence value**.
40. In the list, mark the selected row.
41. In the **Number sequence** field, enter or select a value.
42. Close the page.
43. Close the page.
44. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a product master

06/17/2025

Create a product master for the predefined variants. The demo data company used to create this procedure is USMF. This procedure is intended for the product designer.

Create a new product master

1. Go to **Product information management** > **Products** > **Product masters**.
2. Select **New**.
3. In the **Product number** field, type a value. The number must be unique. A number sequence can be set for the **Product number** field. In this case, the user doesn't have to enter a value.
4. In the **Product name** field, type a value. Enter a descriptive product name. The value defaults to the search name, but this can be changed by the user.
5. In the **Product dimension group** field, select the drop-down button to open the lookup. The product dimension group determines which of the four product dimensions that can be used to create product variants. This example uses a group with color and size.
6. In the list, find and select the desired record.
7. In the list, select the link in the selected row. The default **Configuration technology** is *Predefined variant*. This value is used for this example.
8. Select **OK**.

Select product dimension groups

1. In the **Color group** field, select the drop-down button to open the lookup.
2. In the list, find and select the desired record.
3. In the list, select the link in the selected row.
4. In the **Size group** field, select the drop-down button to open the lookup.
5. In the list, find and select the desired record.
6. In the list, select the link in the selected row.

Add dimension groups

1. On the Action Pane, select **Product**.
2. Select **Dimension groups** to open the drop dialog.
3. In the **Storage dimension group** field, select the drop-down button to open the lookup. The storage dimensions help you control how items are stored and taken from inventory. For example, a storage dimension can include Site and Warehouse.

4. In the list, find and select the desired record.
5. In the list, select the link in the selected row.
6. In the **Tracking dimension group** field, select the drop-down button to open the lookup. The tracking dimension group determines which tracking dimensions you can add to a product. For example, the batch number and serial number are used to track inventory items.
7. In the list, find and select the desired record.
8. In the list, select the link in the selected row.
9. Select **OK**.
10. Select **Save**.
11. Close the page.

Create predefined product variants

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an example scenario that shows how to create predefined product variants for a product master. It also describes how to work with improved variant suggestions and how to assign variant-specific item sales tax groups if needed.

Example scenario: Create predefined product variants

This scenario shows how to create product variants for a product master by using a combination of product dimensions.

Make demo data available

To follow this scenario using the values suggested here, you must have demo data installed, and you must select the *USMF* legal entity.

Step 1: Create a product master

To create a product master:

1. Go to **Product information management** > **Products** > **Product masters**.
2. Select **New**.
3. If the **Product number** field doesn't already show a number, then enter a value. This is only required if no number sequence has been set for this field.
4. Enter a name in the **Product name** field.
5. In the **Product dimension group** field, select the product dimension group *SizeCol* (Size and Color).
6. Select **OK** to create and open the new product master.

Step 2: Add product dimensions

This example shows how to manually enter product dimensions. You can also select a size, color, or style group that includes the product dimension values you want to use.

To add product dimensions:

1. With your new product master still open, select **Product dimensions** on the Action Pane.
2. Open the **Size** tab and select **New** on the toolbar to add a row to the grid. Make the following settings for the new row:
    - **Size** – Select a size value.
    - **Name** – Enter a name for the size.
3. Select **New** on the toolbar and add a second size to the grid with a new **Size** and **Name**.
4. Open the **Colors** tab and select **New** on the toolbar to add a row to the grid. Make the following settings for the new row:
    - **Color** – Select a color value.
    - **Name** – Enter a name for the color.
5. Select **New** on the toolbar and add a second color to the grid with a new **Color** and **Name**.
6. Select **Save**.
7. Close the page to return to your new product master.

Step 3: Generate product variants

To generate product variants based on suggestions, follow these steps:

1. Open or create a product master and add the required product dimensions to it, as described in the previous section.
2. With the product master open, select **Product variants** on the Action Pane.
3. Select **Variant suggestions** on the Action Pane.
4. Select the values that you want to use for each of the dimensions.
5. On the top toolbar, select **Suggest**.
6. The system generates a list with all possible combinations of the sizes and colors you selected. On the **Suggested variants** FastTab, select the check box for each product dimension combination that you want to use, or select **Select all** on the toolbar to select all of them.
7. Select **Create** to add the variants to the current product master.

Set up variant-specific sales tax groups for sales and/or procurement

The *Apply sales tax group for product variants in sales and procurement* feature lets you assign variant-specific item sales tax groups both for sales and for procurement. The feature applies the following rules:

- If a specific item sales tax group is assigned to a released product variant for sales, that group is used by default when a sales quotation or sales order line is created.
- If no specific item sales tax group is assigned to a released product variant for sales, the item sales tax group that's assigned to the released product master is used by default when a sales quotation or sales order line is created.
- In a similar way, for purchase orders, purchase requisitions, and requests for quotation, if a specific item sales tax group is assigned to a released product variant for procurement, that group is used by default. If no specific item sales tax group is assigned to a released product variant, the item sales tax group that's assigned to the product master is used by default.

This functionality supports a setup where most released product variants use the item sales tax groups from the product master setup, but a few variants use an alternative item sales tax group by default. Therefore, this functionality helps eliminate the cost, effort, and risk that can occur when incorrect taxation is applied to an order line for a product variant.

Assign an item sales tax group to a released product variant

To assign variant-specific item sales tax groups, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.

2. Open the product that you want to set up variant-specific item sales tax groups for.

3. On the **Released product details** page, you can view the default item sales tax groups for the master product and edit them as required. These default groups apply to all variants that you don't assign a variant-specific item sales tax group to for sales and/or procurement.

    - To specify the default item sales tax group that's used when product variants for this master are purchased, set the **Item sales tax group** field on the **Purchase** FastTab.
    - To specify the default item sales tax group that's used when product variants for this master are sold, set the **Item sales tax group** field on the **Sell** FastTab.

4. On the Action Pane, on the **Product** tab, select **Released product variants**.

5. On the **Released products variants** page, find the variant that you want to set up, and open it by selecting the link in the **Product number** column.

    - To specify the default item sales tax group that's used when the selected variant is purchased, set the **Item sales tax group** field on the **Purchase** FastTab.
    - To specify the default item sales tax group that's used when the selected variant is sold, set the **Item sales tax group** field on the **Sell** FastTab.

You can now create a sales order line, sales quotation line, purchase requisition line, purchase order line, or request for quotation line. Notice that the default item sales tax group for the product variant is entered from the variant setup, not from the product master setup.

Last updated on 07/01/2026

Show translated product names and descriptions in the UI

For companies that operate worldwide, in multiple countries or regions, it's important that each user can work with product information in their own language. Sales representatives who communicate with customers by telephone see product names in their local language. Purchasers can search for products in the language that they speak. Shop floor workers get product names and instructions in familiar terms. In each case, you can show the information in a language other than the system language.

All users of Microsoft Dynamics 365 Supply Chain Management can select the language that they prefer to see for all user interface (UI) elements, such as field names, button labels, and page names. The system also lets you enter product names and details for each product in as many different languages as you require, and you can set up each printed document to show this product information in any selected language. However, the UI doesn't necessarily show translated product information to users who work online. Instead, it shows product names and descriptions in the system language, unless you configure your system to use translated product names and descriptions. This behavior affects features such as the production floor execution interface and order line entry.

By entering product translations and configuring the system to show them in the UI, each user can see product data (names and descriptions) in their own preferred language while they work online. For products where no name or description is available in the user's preferred language, the system shows the name or description in the system language instead.

The following video provides an introduction to this feature.

https://learn-video.azurefd.net/vod/player?id=4c9d7e56-36f7-49e8-a6de-ed0c8feed1e5&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fpim%2Fshow-translated-product-info

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You're running Microsoft Dynamics 365 Supply Chain Management version 10.0.31 or later.

- The feature named *Display product info in user's language* is turned on in feature management.

Turn the feature on or off in your system

Before you can use this feature, you must turn it on in your system. Admins can use the feature management settings to check the status of the feature and turn it on. In the **Feature management** workspace, the feature is listed in the following way:

- **Module:** *Product information management*
- **Feature name:** *Display product info in user's language*

Configure your system to show translated product information

Follow these steps to set up the system to show translated product names and descriptions in the UI.

1. Go to **Product information management** > **Setup** > **Product information management parameters**.

2. On the **General** tab, in the **Product identification** section, set the **Display product info in user's language** option to one of the following values:

    - *Yes* – Show product names and descriptions in each user's preferred language in the UI. For products where the name or description isn't available in the user's preferred language, the system language is used instead.
    - *No* – Always show product names and descriptions in the system language in the UI, regardless of each user's preferred language setting. Translated product names and descriptions are still used in printed documents.

Enter translated product names and details

Follow these steps to enter translated names and descriptions for products.

1. Go to **Product information management** > **Products** > **Released products**.
2. Find and select the product that you want to work with.
3. On the Action Pane, on the **Product** tab, select **Translations**.
4. On the **Text translations** page, on the Action Pane, select **Add**.

5. In the drop-down dialog box, select the language that you want to enter translations for. Then enter translations in the fields that are provided.
6. To add translations for more languages, switch to each language by using the **Language** drop-down list, and then enter translations as required.

Select your preferred language

Follow these steps to select your preferred language. This language is used for all UI elements and might also be used for product names and descriptions (if they're entered and enabled as described earlier in this article).

1. Select the **Settings** button (gear symbol) in the upper-right corner, and then select **User options** on the menu.
2. On the **Preferences** tab, on the **Language and country/region** FastTab, set the **Language** field to your preferred language.

Last updated on 07/01/2026

Manage units of measure

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to define a unit of measure, provide translations for the unit and its description, and define conversion rules for related units.

Open the Units page

To create and work with the units of measure that are available in your system, go to **Organization administration** > **Setup** > **Units** > **Units**.

The remaining sections of this article describe what you can do on the **Units** page.

Create standard units and conversions

If your system doesn't already include the most commonly used units of measure for the metric system and/or the US customary system (USCS), the unit setup wizard can help you quickly get started with basic unit definitions and conversions. To complete the wizard, select **Unit creation wizard** on the Action Pane, and then follow the on-screen instructions.

Create or edit a unit of measure

To create or edit a unit of measure, follow these steps:

1. Follow one of these steps:

    - To edit an existing unit, select it in the list pane.
    - To create a new unit, select **New** on the Action Pane.

2. On the header of the record, set the following fields:

- **Unit** – Enter the ID or symbol to use to refer to the unit in the system language. This ID or symbol is usually a common abbreviation for the unit, such as *ea* for each or *cm* for centimeter.
- **Description** – Enter a descriptive name for the unit in the system language. This name is usually the full name of the unit, such as *Each* or *Centimeter*.

3. On the **General** FastTab, set the following fields:

    - **Unit class** – Select the property that the unit measures (such as length, area, mass, or quantity).

    - **System of units** – Select the measurement system that the unit belongs to (*Metric units* or *United States customary units*).

    - **Base unit** – Set this option to *Yes* to use the current unit as the base unit for its unit class. In this case, you only have to specify the conversion factor between the base unit and each additional unit in the unit class. The system can then convert between all units in that unit class. Therefore, it's easier to set up conversions.

      For example, if gallon is the base unit for the *Volume* unit class, you only have to set up conversion factors from quart to gallon and from pint to gallon. The system can also convert from quart to pint.

      You can have only one base unit per unit class.

    - **System unit** – Set this option to *Yes* to use the current unit as the assumed unit for all unspecified measurements in its unit class. For example, if a field that is used to enter a quantity doesn't allow a unit to be specified (of if the user doesn't select a unit), the system uses the unit that is set as the system unit for the *Quantity* unit class. You can have only one system unit per unit class.

    - **Decimal precision** – Specify the number of decimal places that values that are specified for the current unit or converted to it should be rounded to.

4. On the Action Pane, select **Save**.

Define unit translations

To define translations for the ID or symbol and the description for a unit of measure, follow these steps:

1. Create or select the unit to create translations for.

2. On the Action Pane, select **Unit texts**.

   The **Unit texts** page appears. You use this page to define translations for the ID or symbol for the selected unit. You can use those translations on external documents in customer-specific or vendor-specific languages.

3. On the Action Pane, select **New**.
4. In the **Language** field, select the language to translate the unit ID or symbol to.
5. In the **Text** field, enter the translation of the unit ID or symbol in the selected language.
6. On the Action Pane, select **Save**.
7. Close the page.
8. On the Action Pane, select **Translated unit descriptions**.

   The **Translated unit descriptions** page appears. You use this page to define language-specific descriptions for the selected unit.

9. On the Action Pane, select **New**.
10. In the **Language** field, select the language to translate the unit description to.
11. In the **Description** field, enter the translation of the unit description in the selected language.
12. On the Action Pane, select **Save**.
13. Close the page.

Define unit conversion rules

To define rules for conversions between units of measure, follow these steps:

1. Create or select the unit to define conversion rules for.
2. On the Action Pane, select **Unit conversions**.

   The **Unit conversions** page appears. You use this page to define rules for converting the selected unit to and from other units in the unit class.

3. Select one of the following tabs, depending on the type of conversion that you want to set up:

- **Standard conversions** – Set up standard conversion rules for all products.
- **Intra-class conversions** – Set up product-specific conversion rules for units in the same unit class.
- **Inter-class conversions** – Set up product-specific conversion rules for units across unit classes.

4. Follow one of these steps:

   - To create a new conversion, select **New** on the toolbar.
   - To edit an existing conversion, select the conversion in the grid, and then select **Edit** on the toolbar.

5. In the drop-down dialog box that appears, set the following fields:

   - **Product** – Select the specific product that the conversion applies to. This field is available only for intra-class and inter-class conversions.
   - **Formula layout** – Leave this field set to *Simple* to specify a simple conversion that has a single factor. Set it to *Advanced* to set up a more complex equation. The format for advanced equations varies, depending on the unit class.
   - **From unit** – This field shows the selected unit. Usually, don't change the value. (If you do change the value, you must open the **Unit conversions** page for the selected unit to view your new conversion after you save it.)
   - **To unit** – Select the unit to convert to.
   - **Rounding** – Select how fractions should be rounded, based on the **Decimal precision** value of the selected unit (*To nearest*, *Up*, or *Down*).
   - **Conversion formula** – Use the remaining fields at the top of the drop-down dialog box to specify the formula for converting between the two units. The available fields vary, depending on the unit class and formula layout that you've selected.

6. Select **OK**.

7. Close the page.

> **Tip**
>
> You can also set up unit conversions per product variant. Learn more in Unit of measure conversion per product variant.

Last updated on 06/22/2026

Unit of measure conversion per product variant

This article explains how to set up unit of measure conversions for various product variants.

Instead of creating multiple individual products that must be maintained, you can use product variants to create variations of a single product. For example, a product variant might be a T-shirt of a given size and color.

Previously, unit conversions could be set up only on the product master. Therefore, all product variants had the same unit conversion rules. However, when the *Unit of measure conversions for product variants* feature is turned on, if your T-shirts are sold in boxes, and the number of T-shirts that can be packed in a box depends on the size of the T-shirts, you can now set up unit conversions between the different shirt sizes and the boxes that are used for packaging.

Turn on the feature in your system

If you don't already see this feature in your system, go to Feature management, and turn on the *Unit of measure conversions for product variants* feature.

Set up a product for unit conversion per variant

You can create product variants only for products that are product masters. For more information, see Create a product master. The *Unit of measure conversions for product variants* feature isn't available for products that are set up for catch-weight processes.

To configure a product master to support unit conversion per variant, follow these steps:

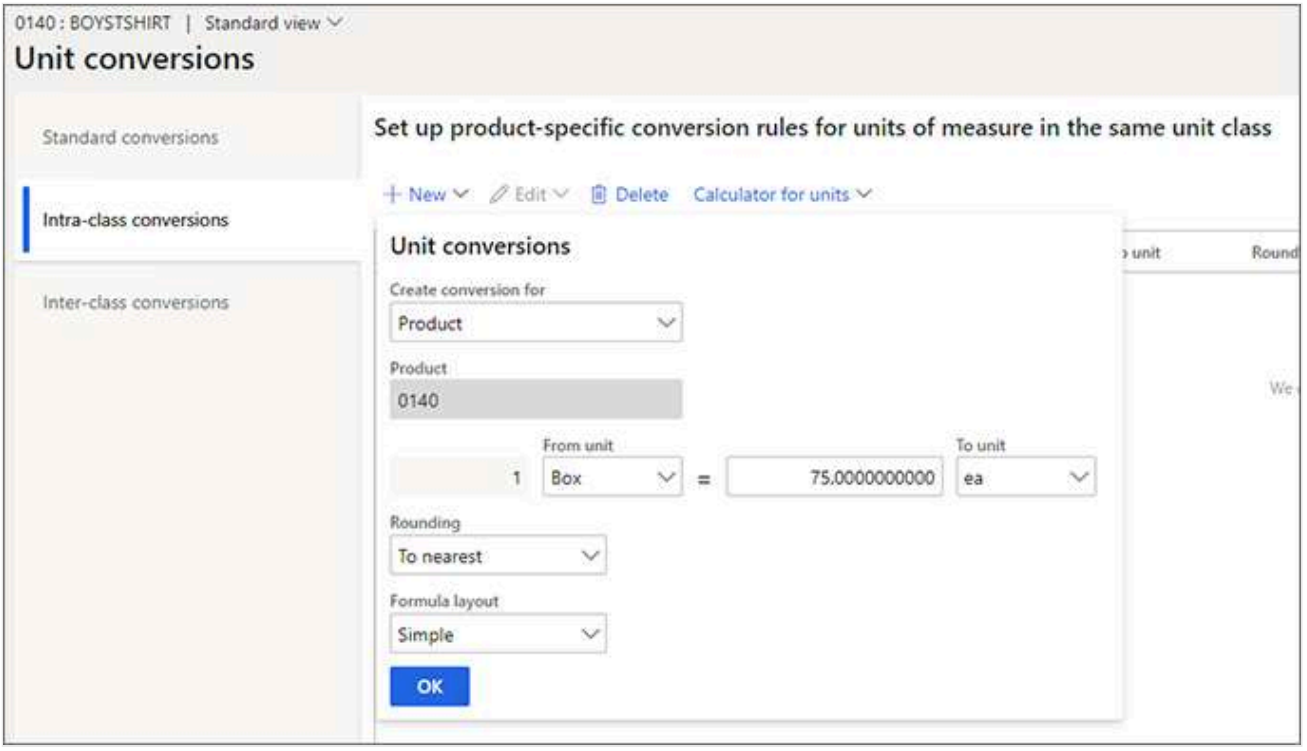
1. Go to **Product information management** > **Products** > **Product masters**.
2. Create or open a product master to go to its **Product details** page.
3. Set the **Enable unit of measure conversions** option to *Yes*.
4. On the Action Pane, on the **Product** tab, in the **Set up** group, select **Unit conversions**.
5. The **Unit conversions** page opens. Select one of the following tabs:

- **Intra-class conversions** – Select this tab to convert between units that belong to the same unit class.
- **Inter-class conversions** – Select this tab to convert between units that belong to different unit classes.

6. Select **New** to add a new unit conversion.

7. Set the **Create conversion for** field to one of the following values:

   - **Product** – If you select this value, you can set up a unit conversion for the product master. That unit conversion acts as a fallback for all product variants when no unit conversion is defined for them.
   - **Product variant** – If you select this value, you can set up a unit conversion for a specific product variant. Use the **Product variant** field to select the variant.



8. Use the other provided fields to set up your unit conversion.

9. Select **OK** to save the new unit conversion.

> **Tip**
>
> You can open the **Unit conversions** page for a product or a product variant from any of the following pages:
>
> - Product details
> - Released products details

- Released product variants

Example scenario

In this scenario, a company sells T-shirts in sizes small, medium, large, and extra-large. The T-shirt is defined as a product, and the different sizes are defined as variants of that product. The shirts are packed in boxes. For sizes small, medium, and large, there can be five shirts in each box. However, for size extra-large, there is space for only four shirts in each box.

The company wants to track the different variants in the *Pieces* unit, but it's selling them in the *Boxes* unit. For sizes small, medium, and large, the conversion between the inventory unit and the sales unit is 1 Box = 5 Pieces. For size extra-large, the conversion is 1 Box = 4 Pieces.

1. From the **Released product details** page for the **T-Shirt** product, open the **Unit conversions** page.

2. On the **Unit conversions** page, set up the following unit conversion for the **X-Large** released product variant.

Expand table

Field	Setting
Create conversion for	Product variant
Product variant	T-Shirt : : X-Large : :
From unit	Boxes
Factor	4
To Unit	Pieces

3. Because the **Small**, **Medium**, and **Large** product variants all have the same unit conversion between the *Box* and *Pieces* units, you can define the following unit conversion for them on the product master.

Expand table

Field	Setting
Create conversion for	Product
Product	T-Shirt

Field	Setting
From unit	Boxes
Factor	5
To Unit	Pieces

Using Excel to update the unit conversions

If a product has many product variants that have different unit conversions, export the unit conversions to a Microsoft Excel workbook, update them, and then publish them back to Dynamics 365 Supply Chain Management.

To export unit conversions to Excel, on the **Unit conversions** page, on the Action Pane, select **Open in Microsoft Office**.

Related information

Manage units of measure

Last updated on 07/01/2026

Create a released product for a single company

This procedure shows how to create a single released product for a single legal unit. When you create the released product, it's available only in that legal unit. You can perform this task for demo data company USMF. A product designer usually performs the task.

Create a released product

1. Go to **Released products**.
2. Select **New**.
3. In the **Product number** field, enter a value.

   If the **Product number** field isn't automatically filled in, enter a unique product number. Perform this step only if you haven't set up any number sequence for product numbers.
4. In the **Product name** field, enter a value.

   The **Product name** defaults to the search name. If needed, select a different search name.
5. In the **Item model group** field, select the dropdown button to open the lookup.

   The item model groups contain settings that determine how items are controlled and handled on item receipts and issues. The settings also determine how the consumption of items is calculated.
6. In the list, find and select the desired record.
7. In the list, select the link in the selected row.
8. In the Item group field, click the drop-down button to open the lookup.

   Use item groups to manage inventory by dividing inventory items into groups based on item characteristics. For example, to manage how information is posted to main accounts, create a series of different item groups that are associated with specific main accounts. This process lets you track the inventory value of items at different stages.
9. In the list, find and select the desired record.

10. In the list, select the link in the selected row.
11. In the **Storage dimension group** field, select the dropdown button to open the lookup.

    The storage dimensions help you control how items are stored and taken from inventory. For example, a storage dimension can be Site and Warehouse.
12. In the list, find and select the desired record.
13. In the list, select the link in the selected row.
14. In the **Tracking dimension group** field, select the dropdown button to open the lookup.

    The tracking dimension group determines which tracking dimensions you can add to a product. For example, the batch number and serial number are used to track inventory items.
15. In the list, find and select the desired record.
16. In the list, select the link in the selected row.
17. In the **Inventory unit** field, select the dropdown button to open the lookup.

    The inventory unit determines how the product is quantified when it's stored in inventory.
18. In the list, find and select the desired record.
19. In the list, select the link in the selected row.
20. In the **Purchase unit** field, select the dropdown button to open the lookup.

    The purchase unit determines how the product is quantified when it's purchased from a vendor.
21. In the list, find and select the desired record.
22. In the list, select the link in the selected row.
23. In the **Sales unit** field, select the dropdown button to open the lookup. The sales unit determines how the product is quantified when it's sold to a customer.
24. In the list, find and select the desired record.
25. In the list, select the link in the selected row.
26. In the **BOM unit** field, select the dropdown button to open the lookup.

The BOM unit determines how the product is quantified when including it in a bill of materials (BOM).

27. In the list, find and select the desired record.
28. In the list, select the link in the selected row.
29. In the **Item sales tax group** field, select the dropdown button to open the lookup.

    The item sales tax group in the **Sales taxation group** determines how sales tax is calculated for each item.

30. In the list, find and select the desired record.
31. In the list, select the link in the selected row.
32. In the **Item sales tax group** field, select the dropdown button to open the lookup.

    The item sales tax group in the **Purchase taxation group** determines how purchase tax is calculated for each item.

33. In the list, find and select the desired record.
34. In the list, select the link in the selected row.
35. Click OK.

Add product characteristics

1. Expand or collapse the **Manage inventory** section.
2. In the **Net weight** field, enter a number.
3. In the **Tare weight** field, enter a number.
4. In the **Gross depth** field, enter a number.
5. In the **Gross width** field, enter a number.
6. In the **Gross height** field, enter a number.
7. In the **Volume** field, enter a number.

Add financial dimensions

1. Expand or collapse the **Financial dimensions** section.
2. In the **BusinessUnit** field, select the dropdown button to open the lookup.
3. In the list, find and select the desired record.

4. In the list, select the link in the selected row.
5. In the **CostCenter** field, select the dropdown button to open the lookup.
6. In the list, find and select the desired record.
7. In the list, select the link in the selected row.
8. In the **Department** field, select the dropdown button to open the lookup.
9. In the list, find and select the desired record.
10. In the list, select the link in the selected row.
11. In the **ItemGroup** field, select the dropdown button to open the lookup.
12. In the list, find and select the desired record.
13. In the list, select the link in the selected row.

Last updated on 07/01/2026

Cross-company product sharing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Organizations that have many companies (legal entities) and a large product portfolio (for example, large sales and distribution networks) often experience a high level of duplicated product data. The cross-company data sharing capabilities of Microsoft Dynamics 365 Supply Chain Management let you share data about released products across multiple companies. In this way, you can reduce the volume of data that must be maintained and simplify product master data maintenance.

Prerequisites

To use the features that this article describes, your system must meet the following requirements:

- Run Microsoft Dynamics 365 Supply Chain Management version 10.0.43 or later.
- Turn on the following features in feature management.
  - *Master company data sharing*
  - *Cross-company data sharing for products*

Get started with cross-company data sharing

Single record sharing and duplicate record sharing work in the following way:

- **Single record sharing** – Just one shared record exists in the database, and all relevant companies can view and edit that record.
- **Duplicate record sharing** – Each company has its own copy of every shared record. Each time that the shared record is edited in any company, the edit is immediately replicated to every other company's copy of the record.

When you share product information across companies, the sharing works in the following way:

- The released products table (`Inventtable`) must use *single record sharing*.
- For related tables (which typically hold policies for handing a product, such as the bar code setup and the cost group), you must decide whether you want to use *single record sharing* or *duplicate record sharing*. In most cases, duplicate record sharing is the better choice for related tables, because single record sharing imposes several restrictions that duplicate record sharing doesn't impose. (Learn more in Cross-company data sharing overview.)

Before you continue, read Cross-company data sharing overview to learn more about how data sharing works in Supply Chain Management.

Prepare your system to enable cross-company data sharing for products

Before you enable cross-company data sharing for products, work through the following checklist:

- **Set up the currency.** Either the currency must be the same across all companies in the data sharing policy, or you must avoid using the base price fields.
- **Align number sequences.** Number sequences must be aligned across all companies in the data sharing policy.
- **Enable policies in the correct order.** Duplicate shared policies must be enabled before single record sharing policies.
- **Specify the default parameters for each company.** Consider aligning default parameters across all companies in the data sharing policy.
- **Consider whether any of your business processes are affected by the single record sharing functionality.** Read the detailed list of limitations and notes that apply for shared products later in this article, and prepare your system as required.

Set up sharing policies for products and product-related information

To share products and product-related information, you must create and set up your sharing policies on the **Configure cross-company data sharing** page. Each sharing policy establishes a set of tables and fields that are shared.

> **Important**

If a field in a table that's being enabled for single record sharing holds a foreign key for a related table, that related table is also shared by using single record sharing, unless a duplicate record sharing policy for it already exists and is enabled. Therefore, all the duplicate record sharing policies that you require must be enabled before you start to set up your single record sharing policies. For example, if you plan to use duplicate record sharing for production pools, you must enable the production pool policy before you enable the products (and the products must use single record sharing).

Set up duplicate record sharing policies

Use duplicate record sharing for tables that handle policies and related information.

To create a duplicate record sharing policy, follow these steps. You can create as many duplicate record sharing policies as you want.

1. Go to **System administration** > **Setup** > **Configure cross-company data sharing**.
2. On the Action Pane, select **New**.
3. In the **Name** field, enter a name for the policy (for example, *Product policies - duplicate sharing*).
4. Leave the **Master company data sharing policy** option set to *No*. (This option doesn't apply to *duplicate record sharing* policies.)
5. On the Action Pane, select **Save**.
6. In the **Companies which share the records in these tables** section, add each company that shares the tables that you add for this policy.
7. In the **Tables and fields to share** section, select **Add** on the toolbar.
8. In the drop-down dialog box, in the **Table name** field, select a table. For more information about which tables you might select, see the Product information tables that can be shared section later in this article.

   ⓘ **Note**

> Not all tables are enabled for record sharing. If the table that you want to share isn't available for selection in the **Table name** field, send a request to Microsoft Support, and we'll consider adding sharing support for the requested table in the future.

9. Select **Add table**. Your table is added.
10. Select which fields (and which related tables, if any) to share by selecting and clearing the different checkboxes.
11. Repeat steps 7 through 10 for each additional table that you want to share by using this policy.
12. On the Action Pane, select **Save**.

Set up single record sharing policies

Use single record sharing for the *Released product* table (`Inventtable`).

To create single record sharing policies for the **Released product** table and other tables, follow these steps:

1. Go to **System administration** > **Setup** > **Configure cross-company data sharing**.
2. On the Action Pane, select **New**.
3. In the **Name** field, enter a name for the policy (for example, *Products - single record sharing*).
4. Set the **Master company data sharing policy** option to *Yes*. This setting is required for *single record sharing* policies.
5. Set the **Master company** field to the company to use as the master. The single record is assigned to this company.
6. On the Action Pane, select **Save**.
7. In the **Child companies** section, add each company that shares the tables that you add for this policy.
8. In the **Tables and fields to share** section, select **Add** on the toolbar.
9. In the drop-down dialog box, in the **Table name** field, select a table (for example, *Inventtable*). Then select **Add table**.
10. Expand the newly added table, and review the status of each of its fields. (A check mark indicates enabled fields.) Take special note of the status of fields that reference another table. (These are labeled with the text "foreign key.") Either make sure that you've included the related table as part of a duplicate record sharing policy or be aware that the related tables are now also shared by using single record sharing.

Product information tables that can be shared

To share products across companies, the following table *must* be shared by using *single record sharing*:

- Released products (`Inventtable`)

The following product-related tables can be shared by using either *duplicate record sharing* or *single record sharing*:

- Barcode setup
- Batch attribute
- Batch disposition master
- Buyer group
- Calculation group
- Charges group
- Commission group
- Cost group
- Counting reason code policy
- Duty rates group
- Fiscal LIFO reporting group
- Freight allocation group
- Inventory level profile
- Item coverage group
- Item group
- Item model group
- Item price tolerance group
- Item rebate group
- Number group
- Packing group
- Price group
- Production group
- Production pool
- Property
- Revenue schedule
- Statistics procedure
- Supplementary item group
- Tax rate type

- Warehouse item setup

Limitations and notes that apply to shared products

This section provides notes and summarizes the limitations that apply when you share product information between companies in different scenarios.

Item templates

You can't apply item templates to child companies, because the `SysRecordTemplateTable` table isn't shared. Therefore, the **Apply template** command is unavailable in child companies.

Currencies and base prices

Base prices (such as the sales price, purchase price, and inventory price) are implicitly expressed in the accounting currency of each company.

The accounting currency for each company is set up in the **Accounting currency** field on the **Ledger** page (**General ledger** > **Ledger setup** > **Ledger**).

For companies that are part of a cross-company policy, the following cases apply:

- **All companies in the policy use the same accounting currency** – There are no special limitations to consider.
- **Companies in the policy use different accounting currencies** – The currency isn't explicitly specified. Instead, it's implicit for each company. Therefore, if one amount (such as the base sales price) is shared across companies, each company assumes a different currency. For example, the USMF company uses US dollars (USD), whereas the JPM company uses Japanese yen (JPY). If the base sales price is 20, the USMF company assumes that the amount is 20 USD, but the JPM company assumes that it's 20 JPY. Because 20 USD and 20 JPY don't represent the same monetary value, one of the companies sells the item at a price that is either too high or too low.

Therefore, only one of the following rules applies:

- **All companies that are part of a cross-company data sharing policy for sharing products must use the same accounting currency.** If you rely on fields that use an implicit currency (such as the base sales price or purchase sales price), you must set the same accounting currency in all companies that are part of the product sharing policy.

- **You must not rely on the value of these fields.** Use trade agreements for purchase and sales prices, and don't use the inventory price or its related processes.

Country/region-specific fields

By default, most country/region-specific fields are disabled in product tables, because they aren't supported. Therefore, if you want to share those fields, you must add them as an extension. Add the country/region-specific tables to the policy, and then select the fields that must be shared.

Default unit of measure

When you create a product, you must specify several different units of measure, each for a specific purpose. For example, you might specify an inventory unit, a purchase unit, and a sales unit. The **New released product** dialog box shows the same default unit for all of them, based on the default setting for the current company. However, you can edit the unit for each unit of measure separately in the dialog box.

To assign the default unit for a company, follow these steps:

1. On the navigation bar, select the company (legal entity) that you want to set defaults for.
2. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
3. On the **General** tab, in the **Default values** section, in the **Unit** field, set the default unit for the current company.

> **Important**
>
> The default unit depends on the company that the product is created in. If you want the default behavior to be aligned, set the same default unit for all companies.

Physical dimensions units

The **Released products** page shows physical dimensions for each released product. These dimensions include weight and height. However, the unit for the measurements isn't shown on the page and is therefore implicit. The unit that is used for each measurement is the system unit in each unit class. Because these system units are cross-company, there are no specific actions or limitations that you must consider in this area.

> **Note**

The system unit for each class is set up on the unit itself, and only one unit in the unit class can be a system unit. If no unit is set up as the system unit, the value of the fields has no meaning, and processes (such as warehousing) that rely on the existence of a system unit can't be completed. Learn more in **Manage units of measure**.

Bills of materials, formulas, and routes

Manage bills of materials (BOMs), formulas, and routes in each company. Therefore, BOMs, formulas, and routes aren't shared in the provided templates.

Coverage groups – calendar

When you share the coverage groups for an item, the calendar on a coverage group isn't shared in the provided templates. If you want to share it, create a new policy for calendar sharing, or add the calendar table to the coverage group policy.

Product configurator

Constraint-based configuration products don't have any limitations. The product is shared when it's created, and the variant that is created on the fly is also shared cross-company.

Number sequences

Number sequences can be used for product numbers and other purposes. Number sequences must be shared if they're used with single record sharing of products.

Dual-write

Dual-write isn't currently supported when you use cross-company data sharing for products.

Dynamics 365 Commerce

Dynamics 365 Commerce isn't supported when you use cross-company data sharing for products. You can't use Commerce and cross-company data sharing in the same system, because shared products aren't synced to the point of sale (POS) system.

Financial dimensions

Fields that reference financial dimensions (for example, the Ledger or Default dimension) can't be shared across companies. This limitation is a restriction of cross-company data sharing. Learn

more in Cross-company data sharing overview.

Vendors

There's no template policy for the vendors table (`vendtable`). Because the vendors table is marked as a single record shared table, you can create a policy for single record sharing and add the table.

Item sales tax group and item purchase tax groups

Item sales tax groups and item purchase tax groups are specified at the product level. The item sales tax group value automatically enters a default tax group on each sales order line, and the item purchase tax group value automatically enters a default tax group on each purchase order line. For convenience when default values are entered, the values are set to duplicate record sharing and are added to the *Products* template for single record sharing. Although the names of the tax groups are shared across companies, their details and settings are company-specific. Therefore, because tax groups are specific to each item in the company context, you must set up those details and settings in the company part of the sharing policy.

Distributed hybrid environment

If you're using a distributed hybrid environment that includes cloud or edge scale units, cross-company data sharing for products isn't supported.

Engineering change management

Some functionality that engineering change management provides might conflict with single record sharing of products or, when it's used in specific ways, might be complementary.

The following engineering change management concepts are affected if you use single record sharing of products:

- **Definition of engineering companies** – Engineering change management lets you set up one or more engineering companies. These engineering companies might be organizations that represent one or more existing companies where your engineering department is located. Alternatively, you might set up a dedicated company that doesn't represent an actual engineering department, but that you use to manage master data and control the release of products for your organization. For more information about engineering companies, see Engineering companies and data ownership rules.

- **Release product structure** – From the engineering company, you can use a controlled process to release a product to other companies. For more information about the release product structure, see Release product structures.

The tables that are related to engineering versions aren't shared, because they're meant to be contained in engineering and to be shared only with a controlled release. Therefore, if you use engineering change management, you might choose one of the following setups:

- **Your engineering company is part of the single record sharing policy** – In this case, most of the engineering-related functionality isn't applicable, especially the controlled release. As soon as the product is created as a released product in the engineering company, it's shared (that is, made available) to the rest of your companies. Therefore, the engineering company loses its engineering character, because it's no longer a space where the engineering team can work on products, and only those products that continue forward are used in the system.

  Because the engineering versions aren't shared, the version information is available only in the engineering company. Therefore, this setup has the following implications:
  - Engineering versions exist only in the engineering company, unless they're shared by using the release product structure.
  - The lifecycle state at the version level works only in those companies where the version information has been shared. Therefore, you can't control the processes that a version of a product is being used in unless the version is shared with its released product structure.
  - Release control is lost, because the product is shared across all companies.
  - The BOM and route are kept locally. You must manage the BOMs and BOM versions (and also the routes and route versions) at the company level. Therefore, you can still use the release product structure to share the BOM and route.

  Don't use this setup, because much of the value of engineering change management is lost.

- **Your engineering company isn't part of the single record sharing policy for products** – You can set up your engineering company so that it isn't part of the shared product policy. You can then make a sharing policy that includes all the companies where product introduction is done at the same time. In this case, you can release a product to one company to make it available in all the companies.

  If you require version management, you must still release the version information with the release product structure. Therefore, your organization must have an engineering company where products are designed and managed. Then, when a product is ready, you do a controlled release to the rest of the companies. (This type of controlled release is the main

goal of the engineering change management functionality.) In this case, you create the product in the engineering company and then release it to all the companies that are part of the sharing policy. The released product is released to all the companies at the same time, because they're sharing the same product record.

Version-related tables (such as engineering versions) aren't shared. Therefore, you must not rely on version management outside the engineering organization. Alternatively, if you require version management in all companies, you must release version-related information with the release product structure.

Sites and warehouses

Sites and warehouses can be shared. To share them, include the `InventLocation` table (for warehouses) and/or the `InventSite` table (for sites) in a sharing policy. If you don't share these tables, all sites and warehouses are company specific. Therefore, you must create them in every company.

Existing products

When you enable single record sharing for products, create and enable the single record sharing policies before you start to import or create your product portfolio.

> **Important**
>
> A company that already has product records can never be made into a child company for single record sharing. Therefore, in most cases, you can set up product sharing only when you're setting up a new system.

If the system already contains products before you enable the single record sharing policies, those products aren't shared.

Modifying policies

It isn't possible to add more fields to a data sharing policy after it has been created and enabled.

More information about cross-company data sharing

To learn more about cross-company data sharing, see Cross-company data sharing overview.

Last updated on 06/22/2026

Product data entities

06/23/2025

To import and export product data, you must use data entities. Learn more at Configuration data and data migration in Dynamics 365 implementation projects.

Product-related data entities

The following table provides details about the product-related data entities and describes the purpose of each. Learn more in Built-in entities.

Expand table

Entity	Application Object Tree (AOT) name (type)	Notes
Products V2	`EcoResProductV2Entity`	This entity is used to import and export shared products-distinct products and product masters. It allows for updates. It doesn't support set-based SQL operations. It's enabled for Open Data Protocol (OData).
Released products V2	`EcoResReleasedProductV2Entity`	This entity is used to import and export released products-distinct products and product masters. It allows for updates. It requires that the shared product is already created. When a new released product is imported, a release of the shared product occurs. There are also separate entities that can be used to import and export released product masters and released distinct variants. This entity doesn't support set-based SQL operations or delete operations. It's enabled for OData.
Released product creation V2	`EcoResReleasedProductCreationV2Entity`	This entity is used to import shared products and released products in one step.

Entity	Application Object Tree (AOT) name (type)	Notes
		Although it supports exports, we don't recommend using it for exports because the purpose of the entity is product creation. It doesn't support updates. It supports a limited set of fields (fields that are available in the product creation dialog box). It doesn't support set-based SQL operations. It isn't exposed through OData.
Product variants	`EcoResProductVariantV2Entity`	This entity is used to import and export shared product variants. It allows for updates. It requires that dimension values are already created. The integration key is the product master plus product dimensions. This entity doesn't support set-based SQL operations. It's enabled for OData. It supports delete operations. It can't be extended through the addition of new product dimensions.
Product variants by product number identification	`EcoResProductNumberIdentifiedProductVariantEntity`	This entity is used to import and export shared product variants. It allows for updates. It requires that dimension values are already created. The integration key is the product number (whereas the integration key for the **Product variants** entity is the product master plus product dimensions).
Released product variants	`EcoResReleasedProductVariantV2Entity`	This entity is used to import and export released product variants. It allows for updates. It requires that shared product variants already be created. When a new released product variant is imported, a

Entity	Application Object Tree (AOT) name (type)	Notes
		release of the shared product variant occurs. This entity doesn't support set-based SQL operations. It's enabled for OData. Although it supports delete operations, doing so currently causes data corruption because of a bug in the current platform. This entity can't be extended through the addition of new product dimensions. This entity doesn't support deletion of global products and variants.
Released product variants by product number identification	`EcoResProductNumberIdentifiedReleasedProductVariantEntity`	This entity resembles the **Released product variants** entity, but the integration key is the product number instead of the product master plus product dimensions. It can be extended through the addition of new product dimensions.
Sellable released products	`EcoResSellableReleasedProductEntity`	This entity is used to export only sellable products. Sellable products are products that have the information that they require in order to be used in a sales order. The same rules apply when a product is validated by using the **Validate** function on the **Released products** page.
Released Distinct products V2	`EcoResDistinctProductEntity`	This entity is used for distinct products. Those distinct products can be products, subtype products, and product variants. This entity is enabled for OData but not for data management.
Released products masters V2	`EcoResProductMasterV2Entity`	This entity is used for product masters. It's enabled for

Entity	Application Object Tree (AOT) name (type)	Notes
		OData but not for data management.
Item - barcodes V3	`EcoResProductBarcodeV3Entity`	This entity is used to export products and bar codes. This entity doesn't allow change tracking, updates, or deletes. To use change tracking, updates, or deletes on bar codes, use the **Item - barcode association** entity.
Item - barcode association	`EcoResProductBarcodeAssociationEntity`	This entity is used to export products and bar codes. It allows change tracking, updates, and deletes. Its entity key is `AssociationID`, which creates the association between the bar code and the product. To add support for this key, the table `InventitemBarcodeAssociation` is populated for existing item bar code data when you turn on the feature. The table is populated using a batch job and if your bar code table has a large number of records, it could take significant time to run the batch job. Therefore, we recommend that you plan to enable the feature (and therefore run the batch job) at a time that fits your business schedule.
Product lifecycle states	`EcoResProductLifecycleStateEntity`	This entity is used to import and export the different product lifecycle states that can be assigned to a product.

> **Note**
>
> You can use the **Released Products V2** data entity to import products into the system only if the shared product has already been created. Otherwise, to import products into the system, you must use the **Product creation** data entity.

Next steps

- Data management overview

Product lifecycle states

Product lifecycle states document the lifecycle state of a released product or variant. They control which business processes (such as purchase orders, production, planning, or forecasts) can use released products or variants assigned to a given lifecycle state.

You can define any number of product lifecycle states to represent the different stages in a product's lifecycle, such as *Prototype*, *Active*, *Obsolete*, or *End-of-life*. Associate these lifecycle states to released products and released product variants to manage their availability and usage within your organization. Select one lifecycle state as the default for new released products. Released product variants inherit their product lifecycle state from their released product master when you create them. When you change the lifecycle state on a released product master, you can choose to update all existing variants that have the same original state.

Create, configure, and manage product lifecycle states

The following procedure describes how to create, configure, and manage product lifecycle states. It explains how to mark a lifecycle state as the default for new released products and how to control which business processes are allowed or blocked for products in a specific lifecycle state.

1. Go to **Product information management** > **Setup** > **Product lifecycle state**.
2. On the Action Pane, select **Refresh processes** to ensure all of the latest business processes are available for use with your product lifecycle states.
3. Follow one of these steps:
    - To create a new lifecycle state, select **New** on the Action Pane.
    - To edit an existing lifecycle state, select it in the list pane, and select **Edit** on the Action Pane.
    - To delete an existing lifecycle state, select it in the list pane, and then select **Delete** on the Action Pane.
4. Make the following settings in the header of your new or selected lifecycle state:
    - **State** – Enter a name for the product lifecycle state (such as *Prototype*, *Active*, *Obsolete*, or *End-of-life*).
    - **Description** – Enter a description of the product lifecycle state.

5. Make the following settings on the **General** FastTab:

   - **Default when released to a legal entity** – Set to *Yes* if this lifecycle state should be applied to all products by default when they're first released. Set it to *No* if this lifecycle state is manually applied later.
   - **Is active for planning** – Set to *Yes* to include products that are in this lifecycle state in calculations at the master planning and bill of materials (BOM) levels. Set it to *No* to exclude products that are in this lifecycle state from the calculations.

   > ⓘ **Important**
   >
   > The **Default when released to a legal entity** setting doesn't apply to engineering products. Learn more in Product lifecycle states and transactions.

6. Use the **Enabled business processes** FastTab to control which of the available business processes can be used with products in the current lifecycle state. For each process that is listed on the FastTab, view and set the following values:

   - **Process** – This read-only field shows the name of an existing business process. The table at the end of this procedure lists and describes the business processes that might appear here.
   - **Process area** – This field shows the name of the process area that the specified business process belongs to. Although this field is editable, don't change it.
   - **Policy** – Select one of the following values to control whether and how the business process is permitted to use products that are in this lifecycle state:
     - *Enabled* – The business process is allowed to use products in this lifecycle state.
     - *Blocked* – The business process isn't allowed to use products in this lifecycle state. If a user tries to use the process on a product that is in this lifecycle state, the system blocks the attempt and shows an error instead. For example, you might block end-of-life products from being purchased.
     - *Enabled with warning* – The process is allowed, but a warning is shown. For example, you might want a prototype product to be put on a production order that is created by the Research and Development department. However, other departments should be aware that they shouldn't produce the product yet.

   > ⚠ **Warning**

Although the **Process area** values are editable, don't change them. For each row, the value shown in the **Process area** column relates directly to the value shown in the **Process** column. These combinations are generated by the underlying framework. Altering them might break system-driven validations or cause incorrect business rule behavior.

> **Tip**
>
> If you don't see all of the business processes that you expect on the **Enabled business processes** FastTab, select **Refresh processes** on the Action Pane to update the list.

7. On the Action Pane, select **Save** to save your new or updated product lifecycle state.

The following table lists and describes the business processes available on the **Enabled business processes** FastTab.

Expand table

Process	Process area	Description
BOM report as finished	Production	Governs the standalone *Report as finished* function for a bill of materials, such as the **Bills of materials** inventory journal.
Inbound order	Warehouse	Enables or blocks inserting or updating the item on an inbound warehouse shipment line.
Inbound order creation	Warehouse	Enables or blocks inserting the item on an inbound warehouse shipment line during creation.
Inventory adjustment	Inventory	Enables or blocks adding the item to an inventory adjustment journal line.
Inventory journal	Inventory	Enables or blocks adding the item to inventory transfer, adjustment, or movement journal lines.
Inventory movement	Inventory	Enables or blocks adding the item to an inventory movement journal line.
Inventory transfer	Inventory	Enables or blocks adding the item to an inventory transfer journal line.
Item forecast	Planning	Enables or blocks creating manual demand forecasts for the item in Master Planning.
Outbound order	Warehouse	Enables or blocks inserting or updating the item on an outbound

Process	Process area	Description
		warehouse shipment line.
Outbound order creation	Warehouse	Enables or blocks inserting the item on an outbound warehouse shipment line during creation.
Production BOM	Production	Enables or blocks creating a production order if the item appears in the BOM.
Production order	Production	Enables or blocks creating a production order for the item.
Production picking list	Production	Enables or blocks adding the item to a production picking list journal line.
Purchase order	Purchasing	Enables or blocks adding the item to a purchase order line, and posting product receipt or invoice.
Purchase order creation	Purchasing	Enables or blocks adding the item to a purchase order line during creation.
Request for quote	Purchasing	Enables or blocks creating a request for quote for the item.
Return order	Sales	Enables or blocks adding the item to a return order line.
Sales order	Sales	Enables or blocks adding the item to a sales order line, and posting packing slip or invoice.
Sales order creation	Sales	Enables or blocks adding the item to a sales order line during creation.
Sales picking list	Sales	Enables or blocks generating a sales picking list for the item in a sales order.
Sales quotation	Sales	Enables or blocks adding the item to a sales quotation line.
WBS estimate for items	Planning	Enables or blocks creating a transaction of type *Item* in a work breakdown structure for a project.

Associate product lifecycle states to released products

You can associate a product lifecycle state to newly released products or product variants in several ways.

- *When you create a new released product* – The system assigns the lifecycle state that has **Default when released to a legal entity** selected to the **Product lifecycle state** field of the new released product.

- *When you release a product to a legal entity* – The system assigns the lifecycle state that has **Default when released to a legal entity** selected to the **Product lifecycle state** field of the new released product.
- *When you release a product variant to a legal entity* – The system assigns the lifecycle state assigned to the released product master in this legal entity to the **Product lifecycle state** field of the new variant.

> **Important**
>
> The **Default when released to a legal entity** setting doesn't apply to engineering products. The engineering change category specifies the lifecycle state of an engineering product version after you create it in the engineering company. When you release the product to an operational company, the lifecycle state of the product is copied. In other words, when an engineering product is released to an operational company, it has the same lifecycle state that it had in the engineering company. You can overwrite the lifecycle state in the operational company. Learn more in Product lifecycle states and transactions.

To manually update the product lifecycle state of an existing released product or product master, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Find the product or product master that you want to update. Product masters show a **Product subtype** of *Product master* on the **Released products** page.
3. Do one of the following steps:
   - If you're working with a standard (non-engineering) product or variant, do one of the following steps:
     - From the list view, select **Edit** on the Action Pane and then change the **Product lifecycle state** field to the desired lifecycle state.
     - Open the record, select **Edit** on the Action Pane, and then, on the **General** FastTab, change the **Product lifecycle state** field to the desired lifecycle state.
   - If you're working with an engineering product, select or open the record. On the Action Pane, open the **Engineer** tab, and from the **Engineering change management** group, select **Change lifecycle state** to open a drop-down dialog. From the dialog, select the desired lifecycle state and then select **OK**.
4. On the Action Pane, select **Save**.

5. If you edited a product master for a standard (non-engineering) product, you're asked to confirm whether you want to update the variants. Select one of the following options:

    - Select **Yes** to update all related released product variants that have the same original status as the released product master.
    - Select **No** to keep the actual state of all variants. Variants that have a different product lifecycle state from the released product master aren't updated.

6. If you chose to update product variants related to a product master, you can view and edit the new lifecycle states of the variants by going to the Action Pane, opening the **Product** tab, and from the **Product master** group, selecting **Released product variants**.

To manually update the product lifecycle state of an existing released product variant, follow these steps:

1. Go to **Product information management** > **Products** > **Released product variants**.
2. Find the product variant that you want to update. Select or open the record.
3. On the Action Pane, open the **Product variant** tab and select **Change lifecycle state** to open a drop-down dialog. From the dialog, select the desired lifecycle state and then select **OK**.

Impact on master planning

Master planning and bill of materials (BOM) level calculations exclude items assigned a product lifecycle state when the **Is active for planning** option is set to *No*.

> **Tip**
>
> For performance reasons, associate all obsolete released products and variants with a product lifecycle state that deactivates master planning. This association is especially important when you work with non-reusable product configuration variants.

Default migration, import, and export

Data entities support product lifecycle states. You can set the lifecycle state to a variable state through either the *released product* data entity or the *released variant* data entity.

Find obsolete products and product variants

The *Change lifecycle state for obsolete products* periodic task makes it easy to find obsolete products or variants and update their product lifecycle status. Learn more in Find obsolete released products or variants.

Related information

- Find obsolete released products or variants
- Product lifecycle states and transactions

Last updated on 08/24/2026

Find obsolete released products or variants

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to find obsolete released products or product variants and how to associate a product lifecycle state to the obsolete products. It also explains how to view the simulation results and assess how many products and product variants will be associated with a new product lifecycle state when running the update without simulation.

When you run the analysis in simulation mode, it displays the products and product variants identified as obsolete. You can easily review these products. The analysis searches for transactions and specific master data to identify products that have no demand within a variable period and no master data that can result in demand. You can exclude new released products within a variable period from the analysis. When the analysis simulation returns the expected result, you can run the analysis and set a new product lifecycle state to all products identified as obsolete by the analysis.

> **Note**
>
> Perform all analyses and updates within the same legal entity.

Find obsolete products and variants and assign a new product lifecycle state to them

To find obsolete product variants and assign a product lifecycle state to them, follow these steps:

1. Go to **Product information management** > **Periodic tasks** > **Change lifecycle state for obsolete products**.
2. In the **New product lifecycle state** field, select the product lifecycle state that your company uses to indicate obsolete products.
3. Set **Run simulation without updating product data** to *Yes*.

4. In the **Exclude products created within this number of days** field, enter a number.
5. In the **Exclude products used in transactions (in number of days)** field, enter a number.
6. If you want to limit the scope of the simulation, expand the **Records to include** FastTab and select **Filter** to open the standard query dialog, where you can define selection criteria. The fields work just as they do for other types of queries in Supply Chain Management. The next section lists a few ways to identify obsolete products and variants that you can use as filter criteria.
7. On the **Run in the background** FastTab, specify how, when, and how often the job runs. The fields work just as they do for other types of background jobs in Supply Chain Management.

   > **Tip**
   >
   > Run the simulation in batch if you expect to search a large number of products. Also, don't run the simulation during the most active working time of the company.

8. Select **OK** to apply your settings and close the dialog.
9. Go to **Product information management** > **Inquiries and reports** > **Product lifecycle state maintenance history**.
10. The **Product lifecycle state maintenance history** page shows the results of the simulation. Review the results and decide whether you want to apply the suggested changes. You might decide to adjust the filter criteria and run the simulation again.
11. When you're satisfied with the simulation results, go to **Product information management** > **Periodic tasks** > **Change lifecycle state for obsolete products**.
12. Set **Run simulation without updating product data** to *No*.
13. Expand the **Records to include** FastTab and confirm that it still shows the filter criteria that you used for the simulation that you want to apply.
14. Depending on how many products and product variants are affected, consider running this job in batch by using the settings on the **Run in the background** FastTab. Make sure that you're not running a large update job during the most active working hours in the company.
15. Select **OK**.

16. Go to **Product information management** > **Inquiries and reports** > **Product lifecycle state maintenance history** and confirm that the changes you expected were applied.

Ways to identify obsolete products and product variants

Here are some signs that a released product or product variant is obsolete:

- The product or product variant was created some days ago based on the number of days that you enter in the selection dialog box.
- There are no open production orders (status less than ended) for the product or product variant.
- There are no open inventory transactions (status issue `ReservPhysical` to `QuotationIssue` or status receipt `Registered` to `QuotationReceipt`) for the product or product variant.
- There are no inventory transactions within the last number of days for the product or product variant.
- There's no future demand or supply forecast for the product or product variant.
- No minimum stock level is set in item coverage for the product or product variant.
- No active fixed quantity kanban rules exist for the product or product variant.
- No service order lines exist for the product or product variant.
- No active or future sales or purchase agreement lines exist for the product or product variant.
- The product or product variant isn't used in a BOM that is associated with a non-expired approved BOM version for a product or variant that's active for planning.

Related information

- Product lifecycle states
- Product lifecycle states and transactions

Last updated on 12/15/2025

Create a bar code for a product

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows how to manually create a bar code by using item number M0001 as an example. The demo data company for this procedure is USMF.

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, find and select the desired record. For this example, select item number *M0001*.
3. On the Action Pane, select **Manage inventory**.
4. Select **Bar codes**.
5. Select **New**.
6. Select the new row in the list.
7. In the **Bar code setup** field, enter or select a value.
8. In the **Bar code** field, enter or select a value.
9. In the **Bar code** field, type a value.
10. In the **Quantity** field, enter a number.
11. Select **Save**.
12. Close the page.

Last updated on 07/01/2026

Enter the GTIN code for a product

A Global Trade Item Number (GTIN) is a unique and internationally recognized identifier for a product. The GS1 organization assigns GTIN codes.

To assign GTIN codes to a product, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a product in the grid.
3. On the Action Pane, open the **Manage inventory** tab and select **GTIN codes**.
4. Use the Action Pane buttons to add or remove rows as needed.
5. Make the following settings for a new or selected row:

    - **Unit** – Select a unit of measure.
    - **GTIN code** – Select the GTIN standard you're using.
    - **GTIN** – Enter the GTIN code for the product.
    - *Other dimensions* – Enter values in the other dimension columns as needed. To control which dimensions are shown, select **Display dimensions** on the Action Pane.

6. On the Action Pane, select **Save**.

Last updated on 07/01/2026

Create batch attributes for a product

Article • 09/29/2021

This procedure shows how to create a batch attribute, assign default value ranges, and include the attribute in a group. The demo data company used to create this procedure is the USP2 Company.

1. Go to Inventory management > Setup > Batch > Batch attributes.
2. Click New.
3. In the Attribute field, type a value.
4. In the Description field, type a value.
5. In the Attribute type field, select 'Fraction'.

    - This procedure uses the Fraction type to enable decimal values. You can select other attribute types. If you select the Enumeration type, you must enter values in the enumeration list before you can enter a value in the Target field.

6. In the Minimum field, enter a number.
7. In the Maximum field, enter a number.
8. In the Increment field, enter a number.
9. In the Target field, type a value.
10. Click Save.
11. Close the page.
12. Go to Inventory management > Setup > Batch > Batch attribute groups.
13. Click New.
14. In the Attribute group field, type a value.
15. In the Description field, type a value.
16. Click Save.
17. Click Group attributes.
18. Click New.
19. In the Attribute field, click the drop-down button to open the lookup.
20. In the list, find and select the desired record.
21. In the list, click the link in the selected row.

    - An attribute can be included in any of the groups.

22. Click Save.
23. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add an image to a product

09/17/2025

This procedure shows how to add an image to a product.

1. Go to **Product information management** > **Products** > **Released products**.
2. In the grid, find and select the product you want to add an image for.
3. On the far-right pane, select the **Related information** button to open the **Related information** pane (also called the FactBox pane).
4. On the **Related information** pane, expand the **Product image** FastTab.
5. Select **Change image** to open the **Product images** dialog, where you can upload an image or select an existing one from the gallery.
6. Select **OK** to apply the image and close the dialog.

> **Tip**
>
> Admins can choose whether and how product images should be shown in the Warehouse Management mobile app by default, and workers can override this setting on their local devices if they wish. Learn more in User settings, color themes, and sound themes and Mobile device user settings.

Country of origin

Many organizations issue certificates to their vendors to ensure that products meet specific certification standards. These certificates often depend on the country/region of origin. The country of origin feature lets you link a product to its country/region of origin and keep track of its product certifications.

> **Important**
>
> The *Country of origin management* feature described in this article helps you keep track of vendor certificates but doesn't modify the standard functionality of invoices, packing slips, or order confirmations to include them. If you want to print certificate numbers on one or more of these reports, you must extend them in your project.

Configure source and destination countries/regions

Before you issue a certificate for a product, you must link the product to its destination country/region and its country/region of origin.

1. Go to **Product information management** > **Setup** > **Product compliance** > **Country of origin** > **Country of origin rules**.
2. Select an existing country/region setup to edit, or select **New** on the Action Pane to create a new country/region setup.
3. Set the following values for the selected or new country/region.

Expand table

Field	Description
Item number	Select the item number of the product.
Destination country	Select the country/region that you're sending the product to.
Origin country	Select the country/region that you're shipping the product from.

This setup helps you generate a bill of materials (BOM) report where you can include the country/region of origin for each part that source and destination countries/regions are specified

for. This report helps you get a holistic picture of where your parts come from and where they're going.

Keep track of vendor certificates

Use the **Country of origin vendor certificates** page to keep track of certificates that you issue to vendors.

Decide which certificate documents you're issuing and how you'll report them to customers. This feature helps you keep track of your certificates. It also lets you choose whether the relevant certificate numbers appear on invoices, packing slips, and/or order confirmations.

To set up your certificate information, follow these steps:

1. Go to **Product information management** > **Setup** > **Product compliance** > **Country of origin** > **Country of origin vendor certificates**.
2. Select an existing certificate setup to edit, or select **New** on the Action Pane to create a new certificate setup.
3. Set the following settings for the selected or new certificate.

Expand table

Field	Description
Vendor account	Select the vendor that you issued the certificate to.
Item number	Select the item that you issued the certificate for.
Country/region	The destination country/region where you must use this certificate.
Certificate number	Enter the identification number of the certificate that you issued.
Effective	Select the first date when the current certificate is valid.
Expiration	Select the last date when the current certificate is valid.
Print on invoice	If you extended your invoice reports to support certificate numbers, select this check box to include them when relevant. The out-of-box invoice report design doesn't provide support for certificate numbers.
Print on packing slip	If you extended your packing slip reports to support certificate numbers, select this check box to include them when relevant. The out-of-box packing slip report deign doesn't provide support for certificate numbers.

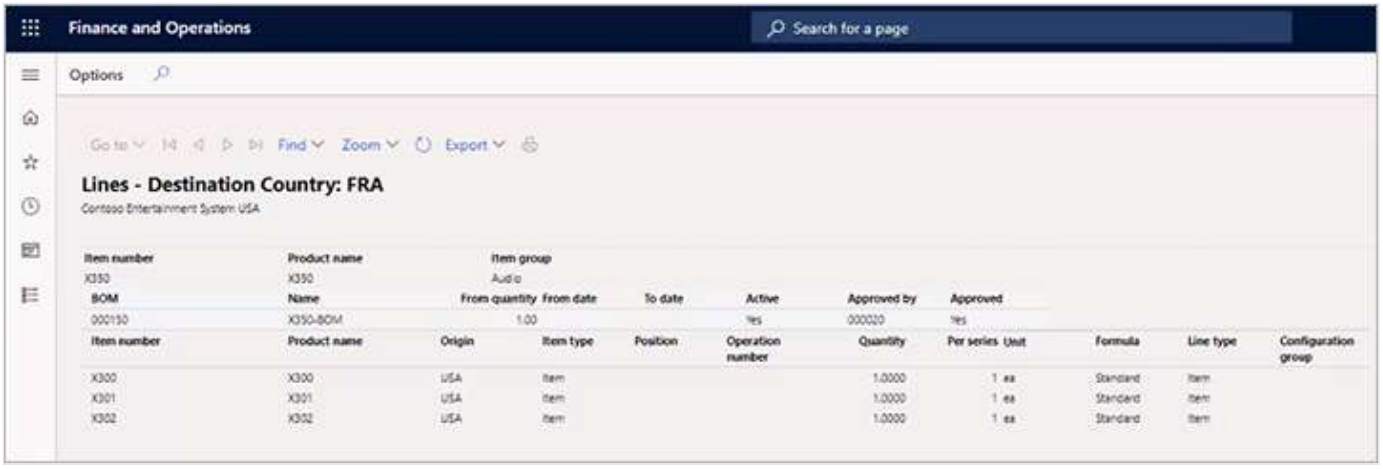
Field	Description
Print on sales order	If you extended your sales order reports to support certificate numbers, select this check box to include them when relevant. The out-of-box sales order report design doesn't provide support for certificate numbers.

Include the country/region of origin on BOM reports

When you generate a BOM report, you can include the country/region of origin for each part that you specify source and destination countries/regions for on the **Country of origin rules** page.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select or create a product to open its **Released product details** page.
3. On the Action Pane, on the **Engineer** tab, in the **BOM** group, select **Designer**.
4. On the page that appears, on the Action Pane, select **BOM** > **Print**.
5. In **Bill of materials lines** dialog box, set the **Destination country** field to the destination country/region that you want to view on your report.
6. Select **OK**.

A report that shows information about the country/region of origin of each part is generated and shown. Here's an example of the report.



Last updated on 07/01/2026

Dual-use goods

Article • 07/14/2023

Dual-use goods are typically items that have both civilian and military applications. For example, a chemical might be used as either a fertilizer or an explosive. Many countries/regions have special regulations that apply to the export, import, and transportation of dual-use goods. Therefore, it's important that companies that are involved in the international trade of dual-use goods keep track of the various policies and certificates.

The dual-use feature helps companies keep track of products that are identified as dual-use goods, stores certificate numbers for each relevant product and destination country/region, and print valid certificate numbers on relevant invoices, packing slips, and/or sales orders. It helps ensure that, when your products are shipped, they always include up-to-date certifications.

Consider the following scenario:

1. The **Dual use country setup** page in your system indicates that shipments to France require a certification.

2. The **Released product details** page for product X-100 indicates that it's a dual-use good. Together, the code, category, group, and regime indicate the export control classification that the product belongs to.

3. The **Dual use certificates** page includes a certificate for product X-100 when it's shipped to France. This certificate expires January 1, 2020.

4. On June 17, 2020, you create a sales order for a customer company that is based in France, and the order includes product X-100.

5. When you confirm the sales order, the system determines the following information:
   a. Does the order include any products that are dual-use goods?
   b. If the order includes dual-use goods, does the destination country/region require dual-use certificates?
   c. If the country/region requires dual-use certificates, does a valid certificate exist for each dual-use good for the destination country/region?

6. The order includes product X-100, the product is being shipped to France, and a French certificate exists for the product. However, the certificate has expired. Therefore, you receive the following warning message: "Dual use certificates for

one or more dual-use items in this sales order aren't valid. Do you want to proceed with the confirmation?"

This article explains how to configure all the settings that are required to set up dual-use goods and support this scenario.

Define dual-use requirements for each relevant country/region

Different countries/regions have different requirements for dual-use goods. You use the **Dual use country setup** page to keep track of the countries/regions that do and don't require a certificate. The information that you specify here is checked when you create sales orders, and you will be reminded to provide the required certifications.

To set up the information about dual-use requirements for different countries/regions, follow these steps.

1. Go to **Product information management** > **Setup** > **Product compliance** > **Dual use products** > **Dual use country setup**.
2. Select an existing country/region setup to edit it, or select **New** on the Action Pane to create a new country/region setup.
3. Set the following values for the selected or new country/region setup.

Expand table

Field	Description
Country/region	Select the country/region that you're tracking requirements for.
Certificate Required	Select this check box for countries/regions that require a certification for dual-use goods. Clear it for countries/regions that don't require this certification.

Create dual-use categories

Dual-use goods must often be categorized according to their export control classification number (ECCN). The ECCN is an alphanumeric code that categorizes items based on factors such as the commodity and technology. The **Dual use categories** page helps you make a list of the categories that you use, for reporting purposes.

To set up dual-use categories, follow these steps.

1. Go to **Product information management** > **Setup** > **Product compliance** > **Dual use products** > **Dual use categories**.
2. Select an existing category to edit it, or select **New** on the Action Pane to create a new category.
3. Set the following values for the selected or new category.

Expand table

Fields	Description
Dual use code	Enter the full ECCN code (for example, *3A001*).
Dual use category	Enter the commerce control list (CCL) category part of the ECCN code. For example, for the ECCN code *3A001*, this value is *3*.
Dual use group	Enter the product group part of the ECCN code. For example, for the ECCN code *3A001*, this value is *A*.
Dual use regime	Enter the regime code for the item. This code identifies the reason why the item is classified as a dual-use good. For example, for the ECCN code *3A001*, this value is *001*.

Apply dual-use categories to products

To identify a product as a dual-use good and apply a dual-use category to it, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select or create a product to open its **Released product details** page.
3. On the **Foreign trade** FastTab, set the **Dual use products** option to **Yes** to identify the current product as a dual-use good.
4. Set the **Dual use code** field to the code that applies to the current product. (You defined this code on the **Dual use categories** page.)

> **Note**
>
> The system makes the following dual-use checks when it generates a sales confirmation:
>
> 1. Does the order include any dual-use goods?
> 2. If so, does the destination country/region require dual-use certificates?

3. If so, do certificates exist for each dual-use good for the destination country/region, and are those certificates valid for the confirmed ship dates?
4. If the answers to questions 1 and 2 are "Yes" and the answer to question 3 is "No", then the system shows a warning to inform the user that dual-use certificates are missing for one or more dual-use goods in the sales order. The user should probably obtain the required certificates and try again, but could instead overrule the warning and proceed with the sales confirmation if they wish.

Set up dual-use certificates

You use the **Dual use certificates** page to set up and manage the required dual-use certificates for each product and country/region. You can track each certificate's details, such as the country/region and the dates of validity. You can also set options to specify where this information should be printed. For example, the information can be printed on the invoice, packing slip, and/or sales order. This setup is checked when you create a sales order.

1. Go to **Product information management** > **Setup** > **Product compliance** > **Dual use products** > **Dual use certificates**.
2. Select an existing certificate to edit it, or select **New** on the Action Pane to create a new certificate.
3. Set the following values for the selected or new certificate.

Expand table

Field	Description
Item number	Select the item number of the dual-use good that this certificate applies to.
Country/region	The destination country or region where you must use this certificate.
Certificate number	The number that appears on the certificate that is issued to the vendor or customer.
Effective	Select the first date when the current certificate is valid.
Expiration	Select the last date when the current certificate is valid.
Print on invoice	Select this check box to print the certificate number on invoices that are addressed to the specified country/region during the specified date

Field	Description
	range.
Print on packing slip	Select this check box to print the certificate number on packing slips that are addressed to the specified country/region during the specified date range.
Print on sales order	Select this check box to print the certificate number on sales orders that are addressed to the specified country/region during the specified date range.

Feedback

Was this page helpful? Yes No

Provide product feedback

Advanced export management overview

06/17/2025

Companies that trade internationally in goods that are subject to external regulations and internal policies must keep track of many rules and policies. The advanced export management solution lets you express these rules, even complex ones, by using formulas that are similar to the formulas in Microsoft Excel. The system then ensures that the rules are honored throughout the sales and fulfillment process.

Dynamics 365 Supply Chain Management lets you manage, track, and verify compliance with export control restrictions before you confirm, pick, pack, ship, and invoice sales orders. It can also check sales quotations. You manage your export control policies by using a native Dataverse solution that interfaces directly with your Supply Chain Management instance. Supply Chain Management then enforces compliance with international trade regulations by consulting your export control policies in real time. Because the architecture is based on Dataverse, many other systems that you use can also access your export control rules through the hundreds of connectors that are available for Dataverse. You can also use the rich features of Microsoft Power Platform, such as the Microsoft Power Fx language, to extend the solution. In this way, your export control administrator can support complex scenarios by using formulas that are similar to the formulas in Excel.

The advanced export management solution provides the foundation for managing, tracking, and verifying export compliance. You can use it both with and without Supply Chain Management. The solution implements five primary concepts:

- Jurisdictions
- Codes and categories
- Restrictions
- Exceptions
- Licenses

Jurisdictions

A jurisdiction is a set of codes, categories, restrictions, exceptions, and licenses. It represents a set of configurations that apply to incoming requests. Each jurisdiction is associated with a set of rules that indicate when the jurisdiction applies. If a rule is configured as an error, any activity that matches the rule is blocked for that jurisdiction, unless an exception exists. A rule is usually a combination of a country/region, a transaction purpose, and a set of codes and categories.

Here are some examples of export control jurisdictions:

- US International Traffic in Arms Regulation (ITAR)
- The Wassenaar Arrangement
- EU Dual Use
- Norway Liste I and II
- US Export Administration Regulations (EAR)
- Controls on exports by an individual company
- Multilateral Export Controls
- Sanctions

Jurisdictions don't have to be based on countries/regions. They can also be configured to control export activities based on your company's own policies.

Codes and categories

The codes that make up a jurisdiction are often referred to as Export Control Classification Numbers (ECCNs). If you use the export control functionality for customs scenarios, Harmonized System (HS) codes might also be used. Codes can be linked to one or more control categories, such as *Missile Technology* or *National Security*. The set of codes and categories is unique for each jurisdiction. Each ECCN can be a member of zero or more control categories, and each control category can contain many ECCNs.

An example of an ECCN is *7A994*, which is defined by the US EAR export control jurisdiction. This ECCN applies to "Other navigation direction finding equipment, airborne communication equipment, all aircraft inertial navigation systems not controlled under 7A003 or 7A103, and other avionic equipment, including parts and components." According to the US EAR, ECCN 7A994 is a part of the *Anti Terrorism (AT)* control category.

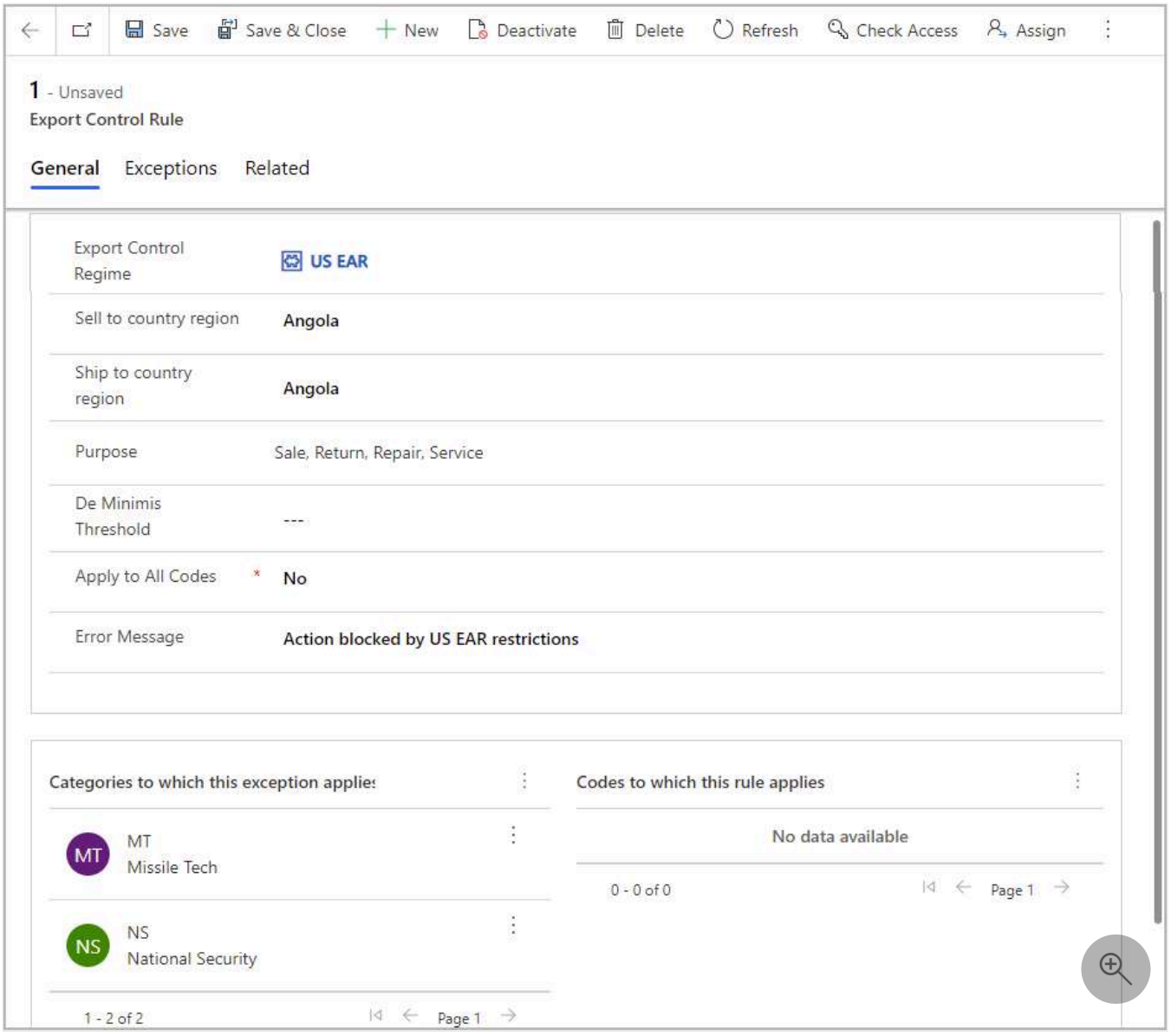
Because ECCNs and control categories are defined by export control jurisdictions, the same ECCN can appear in more than one export control jurisdiction. Therefore, you must always specify which jurisdiction a code references. HS codes, which are used primarily for customs tracking, can also be used to manage export controls that are related to customs.

Restrictions

Each export control jurisdiction defines a set of restrictions under which export actions should be disallowed unless an exception exists. A restriction is a set of ECCNs and/or control categories, together with a country/region, transaction purpose, and other aspects.

Often, restrictions are defined in terms of a commerce country/region chart, as in this downloadable example from the US EAR. In the example in the following illustration, a rule for the US EAR restrictions is defined based on the country/region chart. According to this rule,

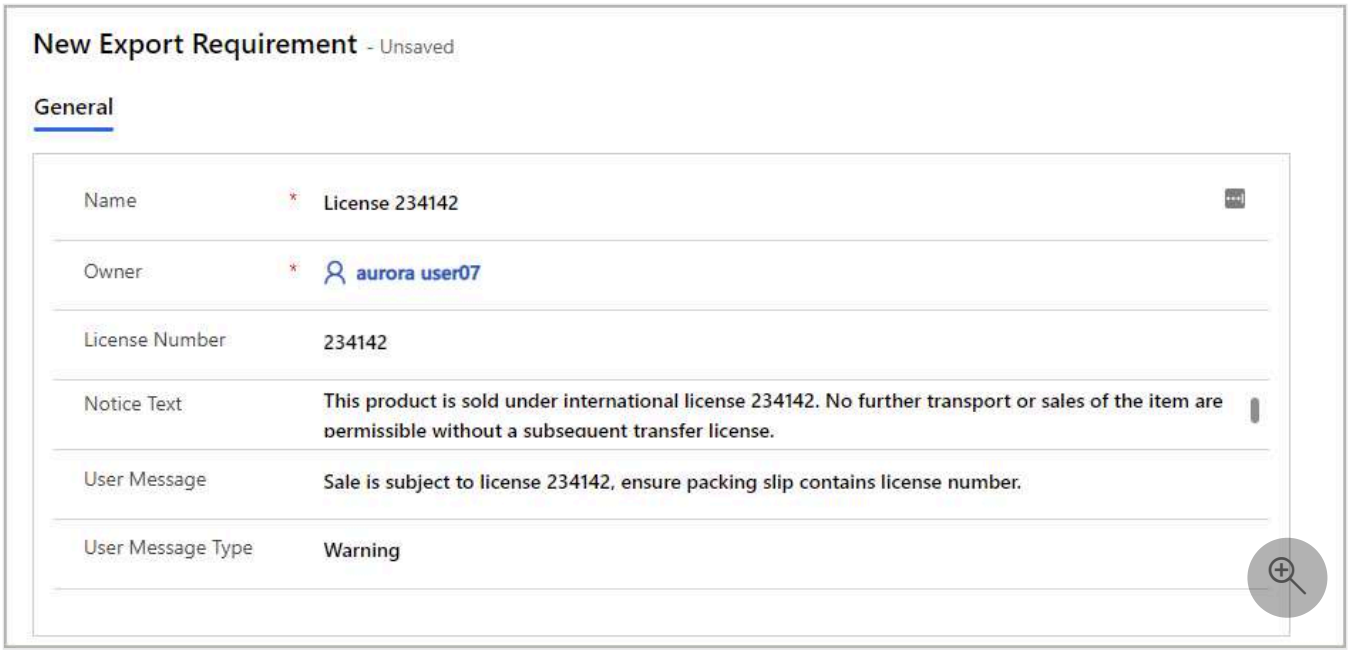
if an order or quotation contains any items that have an ECCN in the *MT* or *NS* control category, the system won't allow it to be sold to Angola. If a user tries to confirm an order or quotation that contains one of these items for shipment to Angola, they receive the following error message: "Action blocked by US EAR restrictions."



Exceptions

Exceptions allow an action that a restriction would otherwise block. Common types of exceptions include blanket exemptions and corporate policies. Exceptions are defined in the same way as restrictions. However, they also provide extra requirements that apply when the exception is used. For example, a message might have to be shown to the user, or text and licenses might have to be printed on documents.

The following illustration shows an example of the setup for an exception.



Licenses

Authorities issue licenses to give a company specific permission to trade in restricted items, or a set of restricted items, in a specific context. It's a common practice to specify the license under which an item can be traded for each transaction.

Next steps

- Enable and configure advanced export management
- Work with advanced export management for products, sales orders, and sales quotations

Enable and configure advanced export management

Prerequisites

Before you can use advanced export management, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management 10.0.36 or later.
- The feature that's named *Advanced export management configuration* must be turned on in feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default. As of version 10.0.45, it's mandatory and can't be turned off.
- Your Supply Chain Management environment must be linked to a Dataverse environment.

Install the export control app

Follow these steps to install the export control app.

1. Sign in to Power Platform admin center.
2. Go to **Environments**, and select your environment.
3. On the page for the selected environment, in the **Resources** section, select **Dynamics 365 apps**.
4. In the list of apps, select **Dynamics 365 Export Management**.
5. Follow the on-screen instructions to install the app.

Authentication and authorization

To perform export control checks, the system uses application user calls. Therefore, the checks can be performed regardless of which user is working with the document in Supply Chain Management. Supply Chain Management users don't have to be added to Dataverse, and they don't have to be given extra permissions. To support the service-to-service (S2S) authentication, a Microsoft Entra application must be created for each environment. You must not share or reuse the Microsoft Entra applications across environments.

Register a new Microsoft Entra application in the Azure portal

Follow these steps to create a new Microsoft Entra application.

1. Sign in to the Azure portal.
2. Go to **Microsoft Entra ID> App registrations**.
3. Select **New registration**.
4. On the **Register an application** page, enter the following information:
    - **Name** – Enter a unique name.
    - **Supported account types** – Select *Any Microsoft Entra directory* (single or multi-tenant).
    - **Redirect URI** – Leave the field blank.
5. Select **Register**.
6. The page that appears shows details about the new app. Copy the **Application (client) ID** value to a temporary text file, because you'll need it later.
7. In the **Manage** list, select **Certificates & secrets**.
8. On the **Certificates & secrets** page, select **New client secret**.
9. In the **Add a client secret** dialog box, enter a description and an expiration date, and then select **Add**.
10. You're returned to the **Certificates & secrets** page, which now includes a row that contains details about your new client secret. Copy the value in the **Value** column to a temporary text file, because you'll need it later, and it will be hidden the next time that you open this page.

Grant the application permissions in Power Platform admin center

Follow these steps to grant the application permissions in Power Platform admin center.

1. Sign in to Power Platform admin center.
2. Go to **Environments**, and select your environment.
3. On the page for the selected environment, in the **Access** section, under **Users**, select **See all**.
4. On the **Users** page, select the **app users list** link.
5. On the **Application users** page, on the toolbar, select **New app user**.

6. In the **Create a new app user** dialog box, select **Add an app**.

7. In the **Add an app from Microsoft Entra** dialog box, in the search field, paste the application (client) ID of your new app. (You should have copied this ID when you registered the app in the Azure portal.) Then select your app in the list or results, and select **Add**.

8. You're returned to the **Create a new app user** dialog box. Select your business unit, and then select **Edit security roles**.

9. In the **Add security roles** dialog box, in the **Role** list, select each of the following roles:

    - *Export Management Application*
    - *Export Management Admin*
    - *Finance and Operations Basic User*

10. Select **Save** to apply the roles.

11. Select **Create** to create the new app user.

Enable the functionality and configure the application in Supply Chain Management

Follow these steps to enable the functionality and configure the application in Supply Chain Management.

1. Sign in to Supply Chain Management.

2. Go to **Product information management** > **Setup** > **Product compliance** > **Country of Origin** > **Advanced export management configuration**.

3. On the **General** tab, review and set the following fields:

    - **Linked Dataverse environment URL** – This read-only field shows the URL of the Dataverse environment that's linked to your Supply Chain Management environment. Your advanced export management jurisdictions, codes, restrictions, exceptions, and licenses are defined in this Dataverse environment.
    - **Application ID** – Paste the application (client) ID of your new app. (You should have copied this ID when you registered the app in the Azure portal.)
    - **Application secret** – Paste the client secret of your new app. (You should have copied this secret when you registered the app in the Azure portal.)

As you enter values into these fields, the system checks whether it can use them to connect to Dataverse and whether the required solution is installed there. The three checkboxes under the **Status** heading automatically update to show a checkmark as each test passes. If one or more tests fail, make sure that the advanced export management solution is successfully installed, and that the specified Microsoft Entra application has all of the following security roles assigned to it in Dataverse: *Export Management Application*, *Export Management Admin*, and *Finance and Operations Basic User*. When all three checkboxes under the **Status** heading show a check, continue to the next step.

4. The following settings appear after all of the tests have passed. Set each option as required.

    - **Advanced export management functionality enabled** – Set this option to *Yes* to turn on advanced export management functionality in your system.
    - **Validate data in Dataverse** – Set this option to *Yes* to ensure that the system checks all sales orders and shipments against the export rules that are defined in the Dataverse app. The check confirms whether your company is allowed to sell or ship each item on an order, according to the rules that are established for the relevant jurisdiction.
    - **Track sales order check history** – Set this option to *Yes* to keep a log of the order check history. Set it to *No* if you don't need this log.

5. Select **Synchronize country/region list** to copy the existing list of country/region codes from Supply Chain Management to the Dataverse solution. This list is used when rules are defined. Each time that you change the country/region code list, you must return to the **Advanced export management configuration** page and select **Synchronize country/region list** again to sync the changes. The language of the user who selects this button is used to sync the names of the countries/regions.

Configure jurisdictions for each legal entity

You must configure the set of jurisdictions that are available for each legal entity (company). For companies where no jurisdictions are enabled, advanced export management features aren't available on sales orders, items, or products. Instead, simpler dual-use goods functionality is provided. If at least one jurisdiction is enabled for a legal entity, the dual-use goods functionality is hidden on pages in that legal entity, and advanced export management functionality is shown instead.

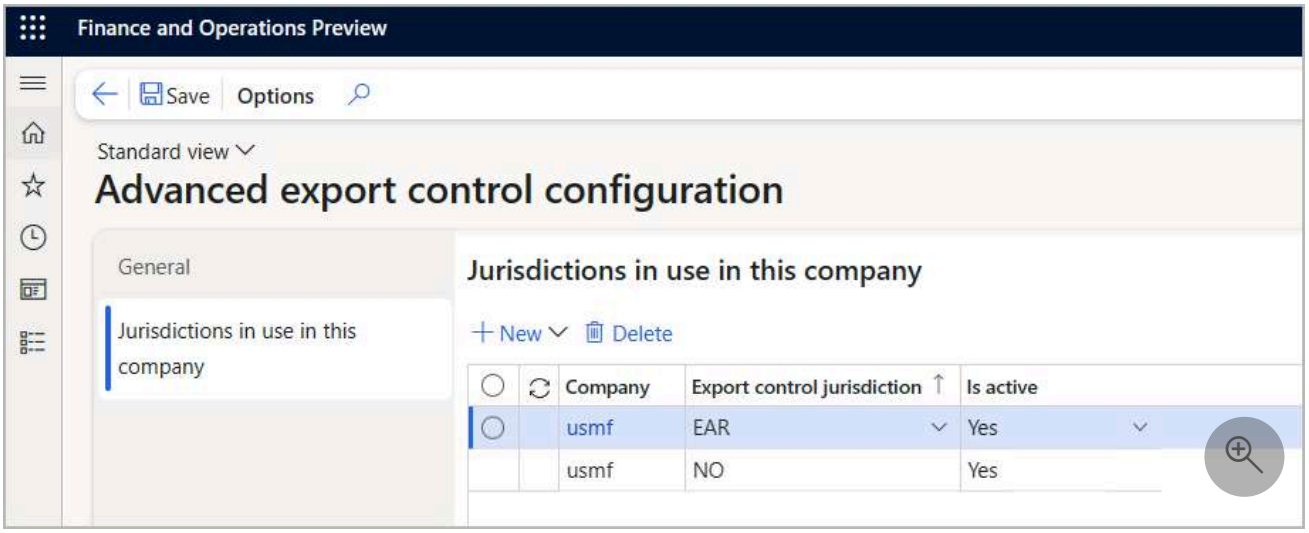
Follow these steps to configure the jurisdictions for a legal entity.

1. Sign in to Supply Chain Management.

2. Use the company picker to select the legal entity that you want to configure.

3. Go to **Product information management** > **Setup** > **Product compliance** > **Country of Origin** > **Advanced export management configuration**.

4. On the **Jurisdictions in use in this company** tab, the grid shows the jurisdictions that are enabled for the current legal entity. Initially, the grid is blank. Use the buttons on the toolbar to add or remove jurisdictions as you require.



5. For each jurisdiction, choose whether that jurisdiction is active and choose whether to skip sales order and/or sales quotation checks for that jurisdiction.

6. On the Action Pane, select **Save**.

Last updated on 12/05/2025

Work with advanced export management for products, sales orders, and sales quotations

Article • 03/27/2024

The advanced export management feature lets you apply export control rules to sales orders and sales quotations. Individual jurisdictions can be enabled or disabled for each legal entity (company). Therefore, each company can have different export control behaviors and rules.

Set up jurisdictions, codes, restrictions, exceptions, and licenses

To define the export controls that apply to the items that you trade in, set up the required *export control jurisdictions* in Dataverse. Then configure the *codes*, *restrictions*, *exceptions*, and *licenses* that apply for each jurisdiction. (For more information about these terms, see Advanced export management overview.)

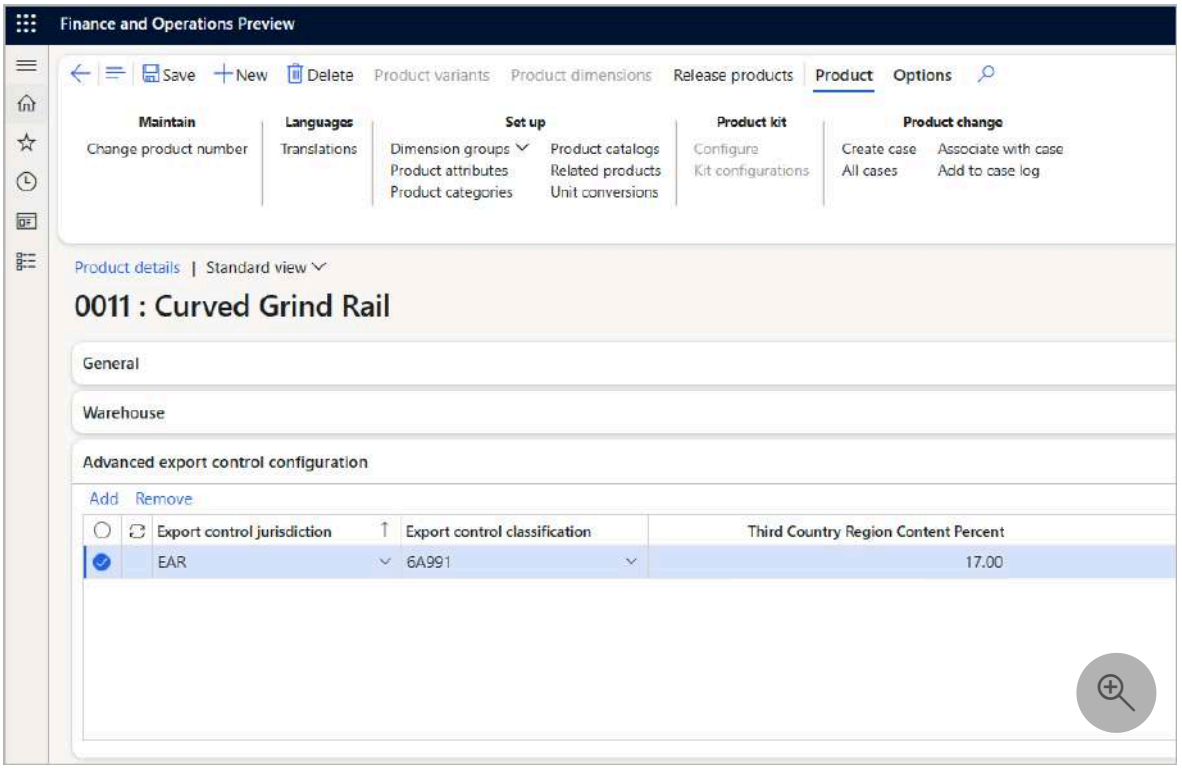
1. Sign in to Power Platform admin center.
2. Go to **Environments**, and select your environment.
3. On the page for the selected environment, in the **Details** section, select the link under **Environment URL**.
4. Your Dataverse environment is opened. On the left navigation pane, select **Export control jurisdictions**.
5. A list of existing export control jurisdictions is shown. Use the buttons on the toolbar to add or remove jurisdictions as you require.
6. Select a jurisdiction to open its details page. Use the jurisdiction details page to define the jurisdiction, and to add or remove codes, restrictions, exceptions, and licenses as you require.

Assign export control classifications for global products

When advanced export management is enabled, it adds a new **Advanced export management configuration** FastTab to the **Product details** page for products and product masters. Use this FastTab to specify the Export Control Classification Number (ECCN) and other export control settings for the product in different jurisdictions.

To set up advanced export management for a product or product master, follow these steps.

1. Sign in to Microsoft Dynamics 365 Supply Chain Management.
2. Go to **Product information management** > **Products** > **All products and product masters**.
3. Find and open the product that you want to work with.
4. On the **Product details** page, select the **Advanced export management configuration** FastTab.



5. The grid lists each export control that applies to the current product. Use the buttons on the toolbar to add and remove rows as you require. For each row, set the following fields:

    - **Export control jurisdiction** – Select the jurisdiction where the current row applies. Because global products can be released to different legal entities, the list shows all available jurisdictions, not just the jurisdictions that are enabled for the currently selected legal entity.
    - **Export control classification** – Select the ECCN that applies to the current row. The ECCN establishes the set of export control rules that apply to the current product for the jurisdiction that's selected in this row.

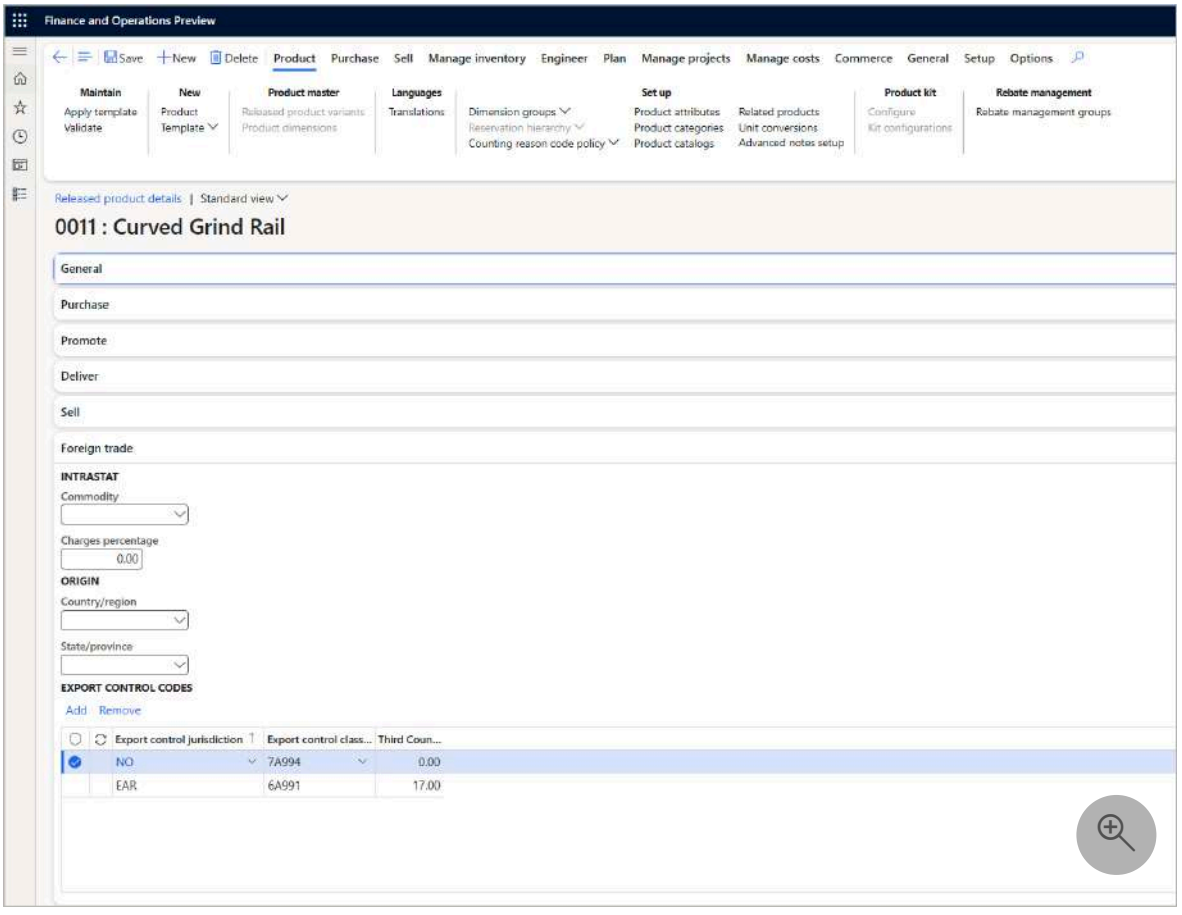
6. On the Action Pane, select **Save**.

Assign export control classifications for released products

When you create a released product that's based on a global product, the **Advanced export management configuration** settings for the global product become default export control values for the released product. If the **Advanced export management configuration** settings for the global product are changed after the product is released, those new settings aren't updated on the related released product unless the product is re-released. Because released products are associated with a specific legal entity, only the ECCNs for jurisdictions that are enabled for that legal entity are inherited from the global product.

To set up advanced export management for a released product, follow these steps.

1. Use the company picker to select the legal entity where you want to work.
2. Go to **Product information management** > **Products** > **Released products**.
3. Find and open the product that you want to work with.
4. On the **Released product details** page, select the **Foreign trade** FastTab.



5. If at least one jurisdiction is enabled for the selected legal entity, the **Foreign trade** FastTab includes an **Export control codes** section. (Otherwise, simpler dual-use

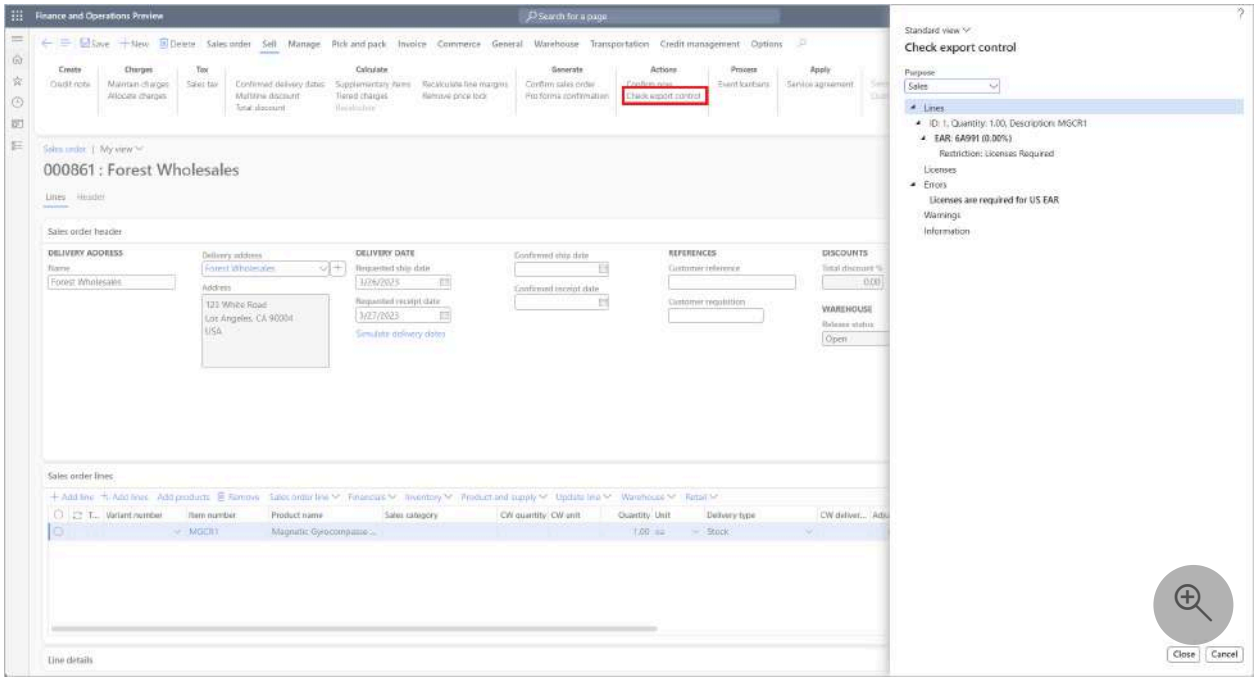
goods functionality is shown instead.) The grid in this section lists each export control that applies to the current released product. Use the buttons on the toolbar to add and remove rows as you require. For each row, set the following fields:

- **Export control jurisdiction** – Select the jurisdiction where the current row applies. This list is filtered so that it shows only the jurisdictions that are used in the currently selected legal entity.
- **Export control classification** – Select the ECCN that applies to the current row. The ECCN establishes the set of export control rules that apply to the current product for the jurisdiction that's selected in this row.

6. On the Action Pane, select **Save**.

Checking and enforcing export control rules for sales orders

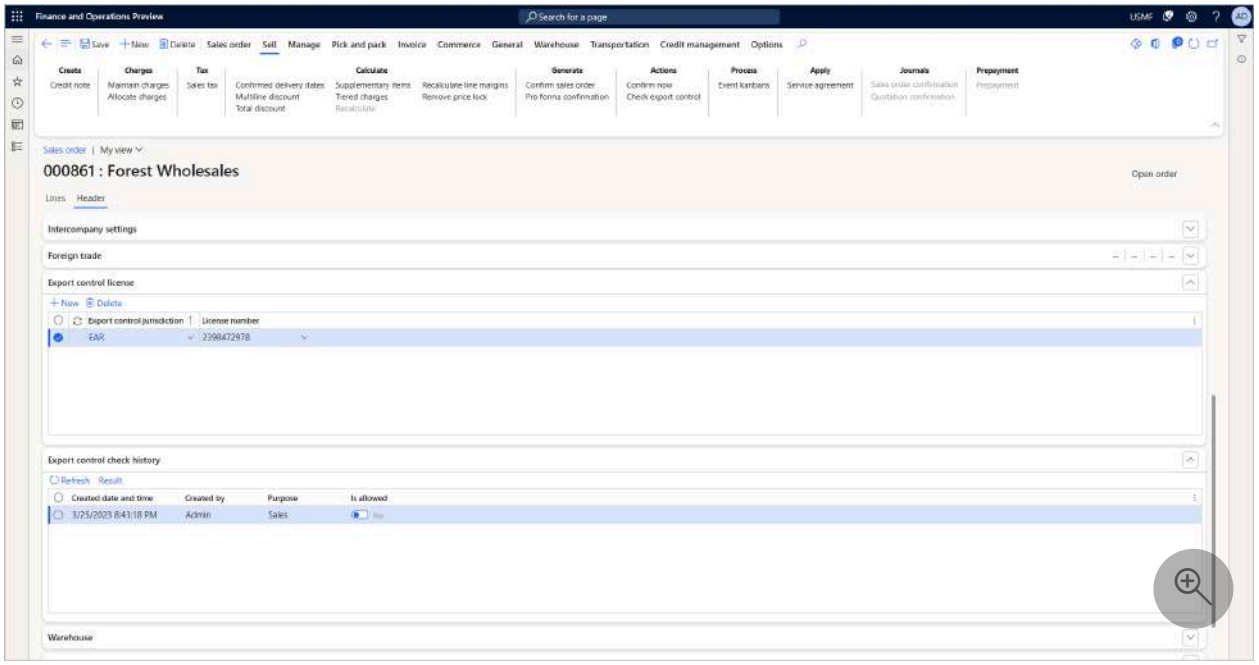
When advanced export management is enabled for a legal entity, a **Check export control** button is provided on the **Sell** tab of the Action Pane for sales orders. Select this button to check the export control status of the current order. The same check is performed during confirmation, picking, packing, shipping, and invoicing. License quantity is consumed and history is tracked only during confirmation and similar steps. Manual checks don't change license consumption and aren't tracked in history.



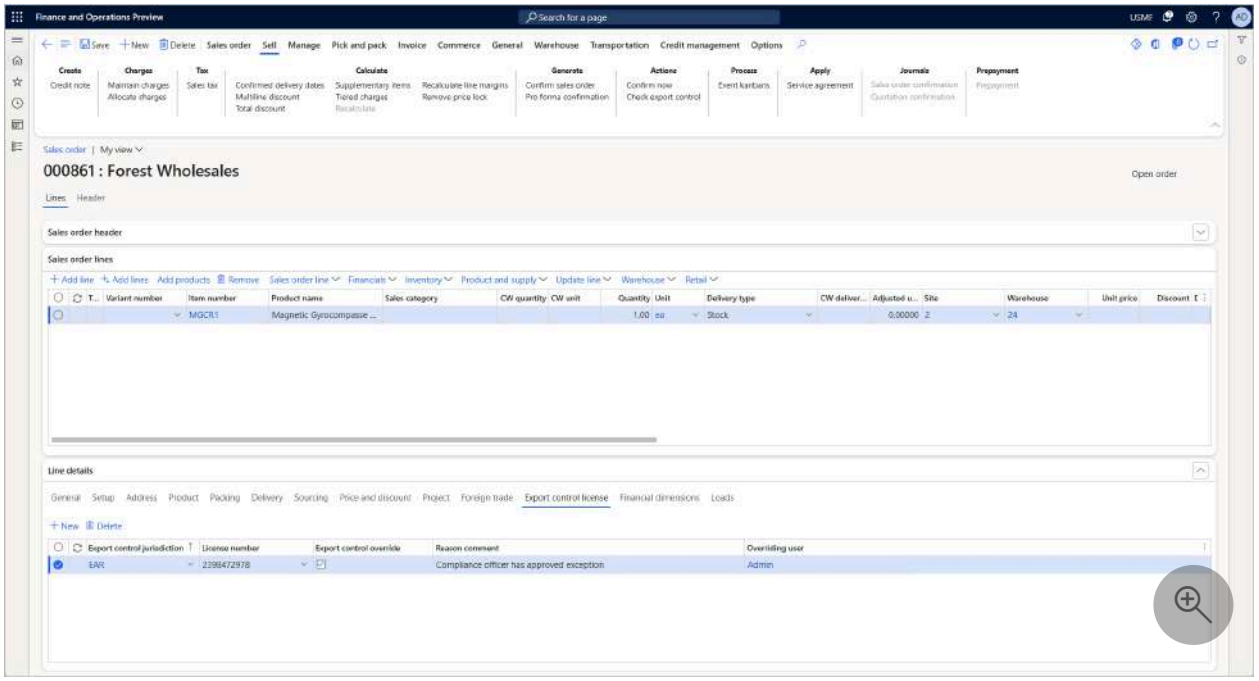
The sales order header has a tab where you can specify licenses per jurisdiction. Licenses that are specified on the header are assumed to be used by default on all lines of the document. However, the licenses can be overridden at the line level. Licenses must pass

export control checks only if restrictions and exceptions require a license. Blanket exemptions or corporate policies don't require licenses.

During confirmation and similar posting steps, the system tracks the history of export control checks. You can view this history from the sales order header by selecting **Result**.



Licenses from the sales order header can be overridden on the sales order lines. In this way, different lines can be related to different licenses. You can also override the export control check at the line level. In this case, the export control check is still performed and tracked, but failures don't block the document processing. The user who edited the override information is automatically tracked on the record. This information can be audited by using the standard logging functionality for finance and operations apps.



Checking and enforcing export control rules for sales quotations

Advanced export management functionality works similarly for sales quotations as it does for sales orders. See the previous section for guidelines.

Feedback

Was this page helpful? Yes No

Provide product feedback

Extend export control sales order functionality

Article • 12/06/2023

This article provides information that's useful for developers who are extending sales order functionality for implementing export controls.

During confirmation and similar posting steps, the system tracks the history of export control checks. You can view this history from the sales order header by selecting **Result**. The details of the check that was performed are shown. This data is available for reference from reports and other processes in the `COOExportControlSalesTableHistory` table. This table also provides helper methods for accessing common information.

The document-level *de minimis* threshold value is **not** computed. By default, it's always *0* (zero). To provide a computation of the document-level *de minimis* threshold value, extend the `COOValidateSalesTable::createRequest()` API.

The `SalesLine.QtyOrdered` field is passed for the quantity, and the `SalesLine.LineAmount` field is passed for the value. No currency conversion is performed on these values. To override and pass different values, extend the `COOValidateSalesTable::createRequest()` method to update the values. The *de minimis* percentage from the associated item is passed for the line, without calculation.

Extended properties

Additional properties can be added to requests from Supply Chain Management to the advanced export management solution. These properties can then be referenced in Power Fx rules on both restrictions and exceptions.

To add additional properties, create a chain of command extension of the `COOValidateSalesTable` class and override the appropriate method for your property.

- **Document properties:** `getExtendedProperties()`
- **Line properties:** `getExtendedLineProperties()`
- **Line code properties:** `getExtendedLineCodeProperties()`

Each of these methods contains a map named `_extendedProperties`. Add the physical name of the property and the value. For a given document part and property name, you must always use the same value type. For example, the property `TestProp` can't be of

type `string` on one call and type `int` on the next call. Here is an example that adds properties to the document, line, and line code:

X++

```
[ExtensionOf(classStr(COOValidateSalesTable))]
final class COOValidateSalesTable_SampleModel_Extension
{
    public void getExtendedProperties(Map _extendedProperties)
    {
        _extendedProperties.add("TestDocString", strFmt("SalesId = %1",
this.parmSalesTable().SalesId));
        _extendedProperties.add("TestDocInt", 42);
        _extendedProperties.add("TestDocReal", 3.14159);
        _extendedProperties.add("TestDocDate", today());
        _extendedProperties.add("TestDocUtcDateTime",
DateTimeUtil::utcNow());
        next getExtendedProperties(_extendedProperties);
    }
    public void getExtendedLineProperties(Map _extendedProperties,
COOValidationRequestLineContract _line)
    {
        _extendedProperties.add("TestLineString", strFmt("LineRecId = %1",
_line.parmRecordId()));
        _extendedProperties.add("TestLineInt", 43);
        _extendedProperties.add("TestLineReal", 3.14159);
        _extendedProperties.add("TestLineDate", today());
        _extendedProperties.add("TestLineUtcDateTime",
DateTimeUtil::utcNow());
        next getExtendedLineProperties(_extendedProperties, _line);
    }
    public void getExtendedLineCodeProperties(Map _extendedProperties,
COOValidationRequestLineContract _line, COOValidationRequestCodeContract
_code)
    {
        _extendedProperties.add("TestLineCodeString", strFmt("Code = %1",
_code.parmCode()));
        _extendedProperties.add("TestLineCodeInt", 44);
        _extendedProperties.add("TestLineCodeReal", 3.14159);
        _extendedProperties.add("TestLineCodeDate", today());
        _extendedProperties.add("TestLineCodeUtcDateTime",
DateTimeUtil::utcNow());
        next getExtendedLineCodeProperties(_extendedProperties, _line,
_code);
    }
}
```

The first time a new property is used in an export control check, that property is added to the `msdyn_ExportControlExtendedProperty` table in the export control solution. After that, the property may be referenced in Power Fx rules. Properties can also be pre-

added to this table prior to performing any checks. Here is an example of a Power Fx rule that makes use of the mentioned extension properties:

```
And(And(Document.TestDocInt=42, Line.TestLineInt=43), LineCode.TestLineCodeInt=44
```

Feedback

Was this page helpful? Yes No

Provide product feedback

Extend export control rules functionality

Article • 08/29/2023

This article provides information that's useful for developers who are extending rules functionality for implementing export controls.

Rules are defined primarily as data in the `msdyn_exportcontroljurisdiction`, `msdyn_exportcontrolrule`, and `msdyn_exportcontrollicense` tables.

High-level flow

Evaluations are performed per document line. The result is a set of matching restrictions, exceptions, and applicable licenses for each line. If no restrictions are found, the set of exceptions and licenses is empty.

No document-level checks are performed. Instead, the document is considered blocked if there are any restrictions that don't have a matching exception. The following pseudo-code shows how the restrictions, exceptions, and licenses are considered.

plaintext

```
foreach (Line in Request)
{
    Line.MatchingRestrictions = FindAnyMatchingRestrictions(Line);

    foreach (MatchingRestriction in Line.MatchingRestrictions)
    {
        Line.MatchingExceptions = FindAnyMatchingExceptions(Line);

        foreach (MatchingException in Line.MatchingExceptions)
        {
            if (MatchingException.RequireLicense)
            {
                Line.ApplicableLincenses = FindApplicableLicenses(Line);
            }
        }
    }
}
```

Evaluation of an individual rule

The `msdyn_exportcontrolrule` table has fields for information such as the sell-to country/region, ship-to country/region, *de minimis* threshold, and transaction purpose.

If a field isn't set, it has no impact on rule evaluation. The values from the line must match *all* rules where a value is set. Otherwise, the rule doesn't apply.

For example, a line has the following values:

- **Sell to country/region:** *France*
- **Ship to country/region:** *Mexico*
- **Transaction purpose:** *Return*
- **De minimis threshold:** *28%*

Rule example 1

A rule has the following values:

- **Sell to country/region:** Blank
- **Ship to country/region:** *Mexico, Canada*
- **Transaction purpose:** Blank
- **De minimis threshold:** *25%*

In this case, only the ship-to country/region and *de minimis* values are evaluated. Because the ship-to country/region, Mexico, is in the list of values that's provided by the rule, and 28 percent is more than the 25 percent threshold that's provided by the rule, the rule is evaluated to *true*.

Rule example 2

A rule has the following values:

- **Sell to country/region:** *Italy*
- **Ship to country/region:** *Mexico, Canada*
- **Transaction purpose:** Blank
- **De minimis threshold:** blank

In this case, only the sell-to country/region and ship-to country/region values are evaluated. Because the sell-to country/region, France, isn't in the list of values that's provided by the rule, the rule is evaluated to *false* .

A rule applies only if the Export Control Classification Number (ECCN) from the line is one of the codes that are explicitly listed for the rule, or if the ECCN is part of one of the categories for the rule. A rule that's marked as *Apply to All Codes* is automatically evaluated to *true*, regardless of the ECCN that's included on the line.

An exception rule that's marked as *Require License* is evaluated to *true* only if one or more licenses are found that match the line.

License evaluation works in the same way. Fields that are left blank are ignored during validation. If a field has a value (that is, it isn't blank), the item that's being tested must match one of the values in the field. The overall evaluation is *true* only if all fields are evaluated to *true*.

Power Fx formulas

Microsoft Power Fx formulas enable complex rules to be expressed for scenarios that aren't simple references to countries/regions or purposes. As is the case for other fields of rules, if a Power Fx formula isn't provided, it has no impact on the rule evaluation. The document object that's being evaluated is passed to the rule and can be used in formulas for comparison.

Here are some examples of Power Fx formulas for export control. Learn more in the Formula reference.

- Documents that are being shipped to one of three countries/regions:

  ```
  CountRows(Filter(["NOR","SWE","FIN"], Value=Document.SellToCountryRegion)) > 0
  ```

- Documents that are being shipped to one of three countries/regions and contain at least one 6A994 product in the US Export Administration Regulations (EAR) jurisdiction:

  ```
  And(CountRows(Filter(["NOR","SWE","FIN"], Value=Document.SellToCountryRegion)) > 0, CountIf(Document.Lines, CountIf(Codes, And(Jurisdiction="EAR",Code="6A994")) > 0) > 0)
  ```

- Documents that are after a specific date and contain no more than five items on any line:

  ```
  And(Document.DocumentDate > Date(2023, 12, 1), CountIf(Document.Lines, Quantity > 5) = 0)
  ```

Overrides

Export control checks can be overridden at the line code level. Therefore, you can override a failing export control check for one jurisdiction on one line of a document but still enforce checks on other lines. To override an export control check, pass `"msdyn_overridden": true` as a property of one of the line codes in the `msdyn_CheckExportControlLineCodes` collection. The export checks are still evaluated, and

any informational or warning messages are still shown. However, any error messages are downgraded to warning messages and don't block the activity.

Feedback

Was this page helpful? Yes No

Provide product feedback

Extend export control license functionality

Article • 12/06/2023

This article provides information that's useful for developers who are extending license functionality for implementing export controls.

License information is stored in the `msdyn_exportcontrollicense` table. Tracking of licenses is provided only as it pertains to export control. It should not be considered a general-purpose license management subsystem.

License evaluation

All license checks occur at the document line level. A single document can contain lines that reference multiple licenses. When the system evaluates a license, the same evaluation that's used for restriction and exception rules is performed. The evaluation considers any field that has a value in the license definition. It ignores fields that don't have a value. As is the case for rules, the license is considered valid only if the value matches the line that's being processed.

The document date is compared against the license's valid-from and valid-to dates, if they're provided. To perform the check based on the current date instead of the document date, either use a Power Fx rule or pass the current date in the document that's being analyzed. The **Expected export date** field of the license has no impact and is provided only for reference. However, it can be referenced from Power Fx rules.

A license can be provided at either the document level or the line level. The assumption is that a document-level license is by default on all lines. It's overridden if a license is provided on a specific line.

License lines and consumption

The Export Control Classification Number (ECCN) from the license line must match the ECCN from the document line that references the license. If no line in the license has a matching ECCN, the check fails, and the license issues collection contains a reference to the missing license.

The following three properties on the export control check call control how license consumption is handled:

- `"msdyn_ExportControlSourceApplication": "Supply Chain"`
- `"msdyn_ExportControlSourceDocument": "SO1347134"`
- `"msdyn_DecrementLicenseQuantity": true`

Consumption is tracked in the `msdyn_exportcontrollicenselineconsumption` table. The `SourceApplication` and `SourceDocument` fields are used to reference which incoming document from an external system has consumed the quantity or value. If you specify *true* for `msdyn_DecrementLicenseQuantity`, rows are written in the `msdyn_exportcontrollicenselineconsumption` table, and future license checks will be blocked if no quantity or value remains for that line. If the same `SourceApplication` and `SourceDocument` values are passed in again with `DecrementLicenseQuantity`, all existing decrements for that document are deleted and then re-created with the updated quantities and values. Therefore, if a sales order is updated, and a different quantity (either larger or smaller) must be reserved, all the existing values can be updated just by sending in the sales order a second time.

The consumed quantity and value are calculated as rollup fields on the license line. If no total quantity or value is provided, export control checks aren't blocked for that license line. Instead, total consumed quantity and total consumed value continue to accrue. The line value is a standard Dataverse currency field. For more information about currency fields and exchange rates, see Using currency columns.

License consumption is automatically computed and updated based on checks when `msdyn_DecrementLicenseQuantity` is set to *True*. However, users who have the appropriate permissions can also create, edit, or delete consumption lines directly from the `msdyn_exportcontrollicenselineconsumption` table as required. Access to view and edit this and all other tables is based on standard Dataverse role-based security.

Unit and currency conversion

If a unit or currency is specified on both a license line and a related incoming document, then the system converts the unit and currency automatically. If either the document being checked or the license line doesn't specify a unit or currency, then it's assumed that the amount provided is in the same unit or currency as the license line.

If the Dynamics 365 Supply Chain Extended and Dynamics 365 Finance Extended dual-write solutions are installed on your environment, then the system converts units and currencies using the `msdyn_unitofmeasureconversion` and `msdyn_currencyexchangerate` values, respectively. If the dual-write solutions aren't installed, then the system applies the default Dataverse conversions. If a conversion for the currency or unit isn't found in existing data, then a license issue is returned and the check is blocked.

Feedback

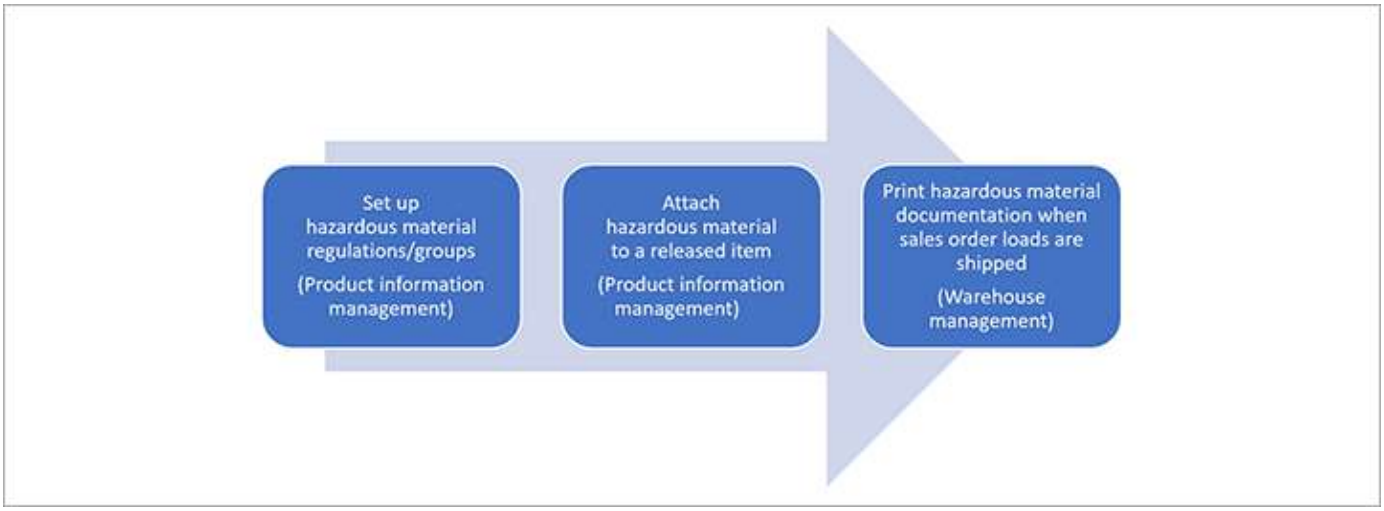
Was this page helpful? Yes No

Provide product feedback

Hazardous materials overview

To remain compliant with shipping and transport regulations, organizations that ship materials classified as dangerous goods must include additional paperwork with their shipments. The hazardous materials feature lets you store information related to released items. Use this information to help prepare shipping documentation. An organization that ships dangerous goods must have its own processes and procedures for managing the shipping process. Microsoft Dynamics 365 Supply Chain Management is just a tool that can help generate the required documents.

The following diagram illustrates the steps needed to set up and use the hazardous materials feature.



Set up the hazardous materials feature in Product information management. You can print the documents through Warehouse management. Therefore, broadly speaking, those areas are the two main areas where you review, set up, and use this feature's functionality:

- **Product information management** – Set up the codes that are applied to a released product.
- **Warehouse management** – Work with additional shipping documents that are printed for shipments.

> ⓘ **Important**
>
> The hazardous materials features in Supply Chain Management provide a collection of useful product information fields and related functionality that can help you record and reference information related to your hazardous products. These features can also help you design

and print shipment documents that include some of the same information about any hazardous materials that you're shipping. However, the system doesn't automatically make you compliant with all applicable regulations in your country or region. Although these tools are intended to help you comply with common regulations, they're neither sufficient in themselves nor guaranteed to be so. Your organization is responsible for being aware of all applicable regulations and for taking all necessary steps to comply with them.

Product information management

Product information management provides a range of setup tables where you can enter reference information for the various dangerous goods lists for road, air, and sea freight.

The following common regulations were referenced when this functionality was developed:

- **ADR** – Regulations related to the international carriage of dangerous goods by road
- **CFR 49** – Regulations in the United States for the carriage of dangerous goods
- **IMDG** – The International Marine Dangerous Goods (IMDG) code
- **IATA** – The International Air Transport Association (IATA) dangerous goods regulations

Each set of regulations provides standardized lists of dangerous goods and reference codes. Therefore, Supply Chain Management provides a reference table for the common codes on those lists. Each list also has some unique codes that you can define.

For more information about how to set up regulations and values for hazardous materials, and how to assign the values to relevant products, see the following articles:

- Set up hazardous materials
- Hazardous materials in products, orders, shipments, and loads

Warehouse management

When you prepare a shipment in Warehouse management, you can print several new reports that use the information you set up in Product information management. For more information about the available reports and how to use them, see Hazardous materials inquiries and reports.

Last updated on 07/01/2026

Set up hazardous materials

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To use hazardous materials functionality, first set up the data required to classify items as hazardous materials. Then, when you create a sales order that includes an item classified as a hazardous material, the system generates hazardous material documentation for that sales order when it ships.

Hazardous material regulations

To use the hazardous materials processes, first create a regulation. Regulations define how the printed shipping text is created for an item. They also define the associated modes of delivery.

Here are some common regulations:

- **ADR** – Regulations related to the international carriage of dangerous goods by road
- **CFR 49** – Regulations in the United States for the carriage of dangerous goods
- **IMDG** – The International Marine Dangerous Goods (IMDG) code
- **IATA** – The International Air Transport Association (IATA) dangerous goods regulations

These regulations help determine what information shows when you print the shipping text for an item. You can define as many regulations as you must comply with.

> **Important**
>
> Setting up information codes that are related to hazardous materials doesn't make your company compliant with regulations. It's only a tool that helps you build processes for your company.

Typically, a regulation is available for a specific mode of transport, such as sea freight, road freight, or air freight. Therefore, you can associate each regulation with a mode of delivery. The

mode of delivery is used when specific documents are printed in Warehouse management. In Warehouse management, each shipment is linked to a shipping carrier and carrier service set up in the **Transportation** module. The mode of delivery is associated with the shipping carrier and carrier service. When you run a report, the mode of delivery is used to find the shipping print text that is associated with a released item.

This setup data isn't specific to each legal entity (company). Therefore, you can have a common set of hazardous material information shared among all your legal entities.

For each regulation, you can define a materials list and use it as a template list associated with released items. For each regulation, you can also define a print setup. A print setup lets you define how the shipping text for an item is constructed. The print setup is used to construct the shipping print text for a released item.

To set up hazardous material regulations, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material regulation**. On the **Hazardous material regulation** page, you can create any number of regulations and configure each by using the fields that are described in the following subsections.

Hazardous material regulation header

Each regulation has a code and a description. The following table describes the fields that are available on the header.

Expand table

Field	Description
Regulation code	Enter a code to identify the hazardous material regulation record.
Description	Enter a description of the hazardous material regulation.

Print setup FastTab

Each regulation can have any number of print setups. You define the print setups on the **Print setup** FastTab. The following table describes the fields that are available for each print setup.

Expand table

Field	Description
Sequence	Define the order in which fields are referenced in the shipping text.

Field	Description
Print field	Select the field to include in the shipping text. Not all fields for the hazardous material are available to print. Only the common fields that are used to define shipping text in the various regulations are available. Define the first print field as a field separator that has a **Sequence** value of *0* (zero), so that it can be used as the separator between other fields. Only one reference to the field separator is required. The following values are available: • **Field separator** – This print field is used as the field separator for the text. Only one field separator is required in the sequence. Usually set the **Sequence** value for this print field to *0* (zero). The system looks for a field separator and uses the first one that it finds in the list. The text value that is used in the string comes from the **Print after** field. • **Identification** – This print field puts the identifying code and/or description in the print text. • **Class** – This print field puts the class code and/or description in the print text. • **Division** – This print field puts the division code and/or description in the print text. • **Packing group** – This print field puts the packing group code and/or description in the print text. • **Tunnel code and/or description** – This print field puts the tunnel code and/or description in the print text. • **Proper shipping name** – This print field puts the proper shipping name in the print text. • **Technical name** – This print field puts the technical name and/or description in the print text. • **Transport category** – This print field puts the transport category code and/or description in the print text. • **Stowage** – This print field puts the stowage code and/or description in the print text. • **Fixed text** – This print field enters the text that is defined in the **Fixed text** field for this row. • **Label code and/or description** – This print field puts the label code and/or description in the print text. • **Aircraft packing** – This print field puts the aircraft packing instructions code and/or description in the print text. • **Limited quantity** – This print field checks whether the item is marked as a limited-quantity item and, if it is, enters the text that is defined in the **Fixed text** field for this row. • **Packing description** – This print field puts the packing description in the print text.
Print before	Enter the text that should be printed before the content that is defined by the **Print field** setting.
Print after	Enter text that should be printed after the content that is defined by the **Print field** setting.
Print with previous	Select this check box to prevent the field separator from being printed between the previous field and this field. Use this check box for print fields that are either optional or included with another print field.
Fixed text	If you set the **Print field** field to **Fixed text** or **Limited quantity**, enter the text that should be

Field	Description
	printed.
Include in print	Select which value or values from the selected print field should be printed for this row. You can print the code, the description, or both the code and the description.

Mode of delivery FastTab

The regulation is a shared table and isn't specific to each legal entity. However, modes of delivery are legal entity–specific. Therefore, when you set up a mode of delivery, select the relationship between the regulation, legal entity, and mode of delivery.

The following table describes the fields that are available on the **Mode of delivery** FastTab.

Expand table

Field	Description
Company	Select a legal entity to associate with this regulation.
Mode of delivery	Based on the legal entity that you selected, select the mode of delivery that should be associated with the regulation.

Country FastTab

For reference purposes, list the countries or regions that the regulation is relevant for. However, when shipment reports are run, only the mode of delivery determines the regulation. When you review a warehouse shipment, there's no direct link to the mode of delivery. The shipment can be associated with a shipping carrier and carrier service. When you define a carrier service, you must associate it with a mode of delivery. This relationship is used on shipment reports such as the bill of lading to find the shipping print text for an item.

The following table describes the field that is available on the **Country** FastTab.

Expand table

Field	Description
Country/region	Select a country/region to associate with the regulation.

Material codes

Material codes establish settings that are related to a specific hazardous component that might be included in a released product. Each material code belongs to a specific hazardous material regulation, and its definition must conform to that regulation. When you apply a material code to a released product by using the **Material code** field, all the material code's hazardous materials settings are automatically applied to that product. Therefore, the process of setting up released products is faster and less prone to error.

To manage your hazardous material definitions, follow these steps:

1. Go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material regulation**.
2. Select the regulation to set up a hazardous material definition for.
3. On the Action Pane, on the **Setup** tab, select **Hazardous materials**.
4. In the **Material code** field, enter a material code for the hazardous material definition. You select this value when you apply the material code to a released product.

   The **Regulation code** field is read-only and shows the regulation that you selected in step 2.
5. Use the remaining fields on this page to create and set up each hazardous material that applies to your selected regulation. The fields that are available are a subset of the hazardous material fields that are available for individual released products. For more information, see Hazardous materials in products, orders, shipments, and loads.

Hazardous material classification groups

Each hazardous material classification group defines a group of field values that establish a template. You can use this template later when you set up hazardous material information for a released item.

When you assign the code for a hazardous material classification group to a released item, the information that is associated with that classification group is copied into the appropriate fields of the item. Therefore, you can simplify the setup processes by establishing a set of related field values that you often use together.

This setup data isn't specific to each legal entity. Therefore, you can have a common set of hazardous material information that is shared among all your legal entities.

To set up hazardous material classification groups, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material classification**

group. On the **Hazardous material classification group** page, you can create any number of groups and configure each by using the fields that are described in the following table.

Expand table

Field	Description
Group code	Enter a code to identify the group.
Description	Enter a description of the group.
Class code	Associate a hazardous material class code with the group.
Division code	Associate a hazardous material division code with the group.
Packing group code	Associate a packing group code with the group.
Transport category code	Associate a transport category code with the group.
Multiplier	Enter the hazardous material multiplier that applies to the selected class and division of hazardous materials, according to the applicable regulation. This multiplier is used as part of the formula that calculates the total *hazardous material points* that are included in a load or shipment. For more information about hazardous material points and this multiplier, see Material management FastTab.

Hazardous material classes

A hazardous material class is typically mapped to the list of classes provided in the regulation that you're conforming to. For example, US regulation CFR 49 lists "class 3" as flammable and combustible liquids. You can set up the classes that are relevant to the materials that you must classify.

Each class is assigned to a material setup in the materials list related to the regulation. Assign a class to each released item as required, to describe the hazardous nature of a product.

This setup data isn't specific to each legal entity. Therefore, you can have a common set of hazardous material information shared among all your legal entities.

Hazardous material classes work together with divisions, groups, and compatibility groups in the following way:

- When you assign a hazardous material class to a released item, you must also assign a hazardous material division.

- Use hazardous material classes together with hazardous material classification groups to establish a template of codes for setting up items.
- Use hazardous material compatibility groups to establish which hazardous material classes and divisions can be shipped together.

To set up hazardous material classes, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material class**. On the **Hazardous material class** page, you can create any number of classes and configure each by using the fields described in the following table.

Expand table

Field	Description
Class code	Enter a code to identify this class. You define this code for the item. It is used in lookup lists when you assign a hazardous material class to a released item.
Description	Enter a description of the class.

Hazardous material divisions

A hazardous material division is a subset of a hazardous material class. You must assign both a division and a class to every product that includes hazardous materials.

For classes that don't have any divisions, create a division where the code is *0*. For example, in the IATA regulation, class-7 radioactive materials have no subdivisions. In this case, you create a *0* division that you can associate with a released product when you assign the class.

This setup data isn't specific to each legal entity. Therefore, you can have a common set of hazardous material information shared among all your legal entities.

Hazardous material divisions work together with classes, groups, and compatibility groups in the following ways:

- When you assign a hazardous material division to a released item, you must also assign a hazardous material class.
- You can use hazardous material divisions together with hazardous material classification groups to establish a template of codes for setting up items.
- Use hazardous material compatibility groups to establish which hazardous material classes and divisions can be shipped together.

To set up hazardous material divisions, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material division**. On the **Hazardous material division** page, you can create any number of divisions and configure each by using the fields that are described in the following table.

Expand table

Field	Description
Division	Enter a code to use as a reference number for the division.
Description	Enter a description of the division.
Class	Look up and assign the class that the division belongs to.

Hazardous material compatibility groups

Hazardous material compatibility groups establish which hazardous material classes and divisions can be shipped together. When operators create warehouse loads or shipments, they can run a compatibility check that issues a warning if the load or shipment includes items that don't all belong to the same compatibility group.

This setup data isn't specific to each legal entity. Therefore, you can have a common set of hazardous material information shared among all your legal entities.

To set up hazardous material compatibility groups, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material compatibility group**. On the **Hazardous material compatibility group** page, you can create any number of compatibility groups and configure each by using the fields that are described in the following subsections.

Hazardous material compatibility group header

Each compatibility group has a code and a description. The following table describes the fields that are available on the header.

Expand table

Field	Description
Compatibility group	Enter a code to identify the compatibility group
Description	Enter a description of the compatibility group.

Compatibility group details

Each compatibility group establishes a list of classes and divisions of hazardous materials that can be shipped together.

Expand table

Field	Description
Class	Select a hazardous material class that's compatible with all other classes in the group.
Division	Select a hazardous material division that belongs to the selected class.

Hazardous material specification values

Microsoft Dynamics 365 Supply Chain Management provides several types of hazardous material specifications. For each type, you must establish a centralized set of values for each relevant field. Users can then select among those values when they configure hazardous material definitions or released products. Supply Chain Management provides a collection of pages where you can establish the values. Each page is dedicated to one type of specification. For a description of each available specification and information about how to establish the values that are available for it, see the following subsections.

One example of a hazardous material specification is the *stowage code*, which specifies how a given material can be stored during transport. By using the information in this section, you establish a central collection of stowage codes. This collection is presented to users in a drop-down list when they set the stowage code for a released product.

These hazardous material specification values aren't specific to each legal entity, Therefore, you can have a set of common values that are shared among all your legal entities.

Use material codes to establish common collections of settings for each specification as it applies to a given regulation. You can then apply the appropriate code to each released product as required. For information about how to apply a hazardous material code to a specific released product, and how to configure individual settings of that product for any specification that is described here, see Hazardous materials in products, orders, shipments, and loads.

Hazardous material emergency response

The *Hazardous material emergency response* specification indicates what to do if something goes wrong while transporting a product that contains a given hazardous material.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material emergency response**. On the **Hazardous material emergency response** page, you can create any number of values and configure each with a classification code and a short description.

Hazardous material identification

The *Hazardous material identification* specification identifies the class or nature of a hazardous material. The value is typically a code that's based on a United Nations (UN) standard. Each class is identified by a code and a description, and it can set limits on transport methods. For example, to identify a flammable item or material, you create a hazardous material class that uses the code *FL* and the description *Flammable*. You also specify that the class must not be transported by air.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material identification**. On the **Hazardous material identification** page, you can create any number of values and configure each by using the fields that are described in the following table.

Expand table

Field	Description
Identification	Enter a code to use as a reference number that identifies this class of hazardous material.
Description	Enter a description of this class.
Restrict from air transport	Select this check box to indicate that this class of hazardous material shouldn't be transported by air.
Restrict from sea transport	Select this check box to indicate that this class of hazardous material shouldn't be transported by sea.

Hazardous material label

The *Hazardous material label* specification identifies the dangerous goods label that must be applied to relevant released products. The labels themselves describe how the product should be handled. For example, you have a product that contains a poisonous gas. In this case, you set up a label code that represents the poisonous gas label. You also build your business process so that it looks up this value when you ship products.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material label**. On the **Hazardous material label** page, you can create any number of labels and configure each with an identifying code and a short description.

Hazardous material packing descriptions

The *Hazardous material packing descriptions* specification indicates how to pack a hazardous item. For example, you might need to pack the item in a specific type of steel drum or use another type of special packaging.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material packing descriptions**. On the **Hazardous material packing descriptions** page, you can create any number of packing descriptions and configure each with an identifying code and a short description.

Hazardous material packing group

The *Hazardous material packing group* specification identifies the packing group for a hazardous item. The packing group lets you define a code and a description to indicate how hazardous material items must be packed during transportation or shipment. The packing group is assigned to the item through the **Item hazardous materials** page.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material packing group**. On the **Hazardous material packing group** page, you can create any number of packing groups and configure each with an identifying code and a short description.

Hazardous material packing instruction

The *Hazardous material packing instruction* specification identifies packing instructions that must be followed when preparing a given hazardous item for transportation by air.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material packing instruction**. On the **Hazardous material packing instruction** page, you can create any number of packing instruction identifiers and configure each with an identifying code and a short description.

Hazardous material stowage

The *Hazardous material stowage* specification indicates how to store a product on a ship when transporting it by sea freight.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material stowage**. On the **Hazardous material stowage** page, you can create any number of stowage identifiers and configure each with an identifying code and a short description.

Hazardous material transport category

The *Hazardous material transport category* specification typically groups similar hazardous products on reports. For example, transport categories are used on the **Shipment summary** report, which you can print from the warehouse shipment record.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material transport category**. On the **Hazardous material transport category** page, you can create any number of transport categories and configure each with a display name and a short description.

Hazardous material technical name

Use the *Hazardous material technical name* specification to provide a commonly used or internal company name that describes each material.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material technical name**. On the **Hazardous material technical name** page, you can create any number of technical names and configure each with a display name and a short description.

Hazardous material tunnel

The *Hazardous material tunnel* specification limits the types of tunnels that a hazardous material can be transported through by identifying the types of tunnels that must be used. Tunnel categories are established by applicable regulations for hazardous material transport. This specification usually applies only to road transport.

To set up values for this specification, go to **Product information management** > **Setup** > **Hazardous material shipping documentation** > **Hazardous material tunnel**. On the **Hazardous material tunnel** page, you can create any number of tunnel identifiers and configure each with an identifying code and a short description.

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Hazardous materials in products, orders, shipments, and loads

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set hazardous material properties for released products, how to put stock limits on hazardous items, and how to include hazardous materials in a sales order, shipment, or load.

Set hazardous material specifications for products

After you define a hazardous material regulation and set up the related reference codes as described in Set up hazardous materials, associate this information with released items. The shipping text for shipping documents is drawn from the hazardous materials information for the released items.

As part of the process of associating a released item with a hazardous material, specify the regulation code and the material. Different regulation codes might be associated with an item, depending on the modes of transport. You can associate multiple regulations and material codes with each item.

To set up a released product as a hazardous material, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select or create a product to open its **Released product details** page.
3. On the **Manage inventory** FastTab, set the **Hazardous material** option to **Yes**. This setting identifies the item as a dangerous good and is used when shipping documentation is printed.
4. On the Action Pane, on the **Manage inventory** tab, in the **Compliance** group, select **Item hazardous material**.

5. Fill in the **Item hazardous materials** page for the selected item by using the fields that are described in the following subsections.

Item hazardous materials header

The following table describes the fields that are available at the top of the **Item hazardous materials** page.

Expand table

Field	Description
Item number	The released product that you're working with.
Regulation code	Select the hazardous material regulation that applies to the product. The regulation defines how the printed shipping text is created for an item and the associated modes of delivery. After it's assigned, the code can't be edited on this page. However, you can assign a new regulation code by selecting **New**.
Material code	(Optional) Select the hazardous material code that applies to the product. The material code provides a template that includes default values for many other fields on this page. When you select a code, all its hazardous materials specifications are copied to the current product. However, the system first prompts you to confirm that item material data should be filled in from the material code.
Group code	(Optional) Select the hazardous material classification group code that applies to the product. As with the **Material code** field, when you select a code, all its hazardous materials specifications are copied to the current product. However, the system first prompts you to confirm that item class group data should be filled in from the group code.

Descriptions FastTab

The following table describes the fields that are available on the **Descriptions** FastTab.

Expand table

Field	Description
Proper shipping name	Enter the standard description for the material, as specified by the applicable regulation. You can provide translations for this value on the **Item ship text translation** FastTab, as described in the next section.
Technical name	Select the common or generic name for the material. This name might be a name that your company uses internally for the material.

Field	Description
N.O.S.	Select this check box to indicate that the **Proper shipping name** value is a not-otherwise-specific (N.O.S.) shipping name for the item. N.O.S. shipping names are used for groups of similar chemicals and materials that have specific end uses, but that might not be listed by name in the hazmat table in a specific regulation.

Item ship text translation FastTab

The **Item ship text translation** FastTab contains a grid that shows translations of the **Proper shipping name** values defined for the primary language on the **Descriptions** FastTab. Use these translations in shipping print text for one or more additional languages.

The following table describes the fields that are available on the **Item ship text translation** FastTab.

Expand table

Field	Description
Language	The language code that the row uses. For example, **pt-br** indicates Brazilian Portuguese.
Shipping print text	The translated **Proper shipping name** value in the language that the row uses.

To add or edit a translation, select **Translations** above the grid to open **Text translations** page. Then follow one of these steps:

- To add a new translation, on the Action Pane, select **Add**. Select the language to add, and then, in the **Translated text** field, enter the translated text.
- To edit an existing translation, select a target language in the **Language** field, and edit the translated text in the **Translated text** field, as required.

Material management FastTab

The following table describes the fields that are available on the **Material management** FastTab.

Expand table

Field	Description
Class	Select the hazardous material class that the product belongs to, as defined by the regulation that you're conforming to. You must assign both a division and a class to

Field	Description
	every product that includes hazardous materials.
Description	The description that is defined for the class that is selected in the **Class** field. This field is read-only.
Division	Select the hazardous material division that the product belongs to, as defined by the regulation that you're conforming to. The division is a subset of the class. You must assign both a division and a class to every product that includes hazardous materials.
Identification	Select the hazardous material identification code. Typically, this code is based on a United Nations (UN) standard.
Packing group	Select the packing group that applies to the current item.
Description	The description that is defined for the group that selected in the **Packing group** field. This field is read-only.
Packing Descriptions	Select the applicable packing description code. This code references a description that indicates how the product must be packed.
Hazardous material labels	Select a code that references the applicable dangerous goods label that should be applied to the product.
Limited quantity	Set this option to **Yes** to report the total product weight that is included in each load and on each load line.
Quantity	Enter the quantity of hazardous material in the product, in the specified unit. This value is used to calculate the total hazardous material score for loads and shipments that include the product.
Multiplier	Enter the multiplier that is applied when the hazardous material score is calculated for each load line that includes the product. This value is specified by the applicable regulation, according to the type of hazardous material that is contained in the product.
Unit	Select the unit of measure that applies to the quantity of hazardous material in the product, as specified in the **Quantity** field. This value is used to calculate the total hazardous material score for loads and shipments that include the product.

How the hazardous material score is calculated

Several of the values specified on the **Material management** FastTab for a product are used to calculate a *hazardous material score* for each load line that includes that product. The score is calculated by using the following formula:

Hazardous material score = <*LineQty*> × <*HazmatQty*> × <*UnitConversion*> × <*Multiplier*>

Here's a key to the formula:

- <*LineQty*> is the quantity of product specified for a load line.
- <*HazmatQty*> is the quantity of hazardous material specified for a product in the **Quantity** field on the **Material management** FastTab.
- <*UnitConversion*> is a conversion factor for converting between the unit used for the load line quantity and the unit specified for a product in the **Unit** field on the **Material management** FastTab.
- <*Multiplier*> is the multiplier specified for a product in the **Multiplier** field on the **Material management** FastTab.

This score is reported for each load line that contains a product where these values are specified. To learn more, see the Shipments that include hazardous materials and Loads that include hazardous materials sections later in this article.

How the hazardous material weight is calculated

Loads and load lines that contain products where the **Limited quantity** option on the **Material management** FastTab is set to **Yes** show the total hazardous material weight, as described in the Shipments that include hazardous materials and Loads that include hazardous materials sections later in this article. The hazardous material weight is calculated by using the following formula:

Hazardous material weight = <*LineQty*> × <*ProductWeight*> × <*UnitConversion*>

Here's a key to the formula:

- <*LineQty*> is the quantity of product specified for a load line.
- <*ProductWeight*> is the net weight specified for the product, in the inventory unit specified for the product.
- <*UnitConversion*> is a conversion factor for converting between the unit used for the load line quantity and the inventory unit that is used for <*ProductWeight*>.

Transport information FastTab

The following table describes the fields that are available on the **Transport information** FastTab.

Expand table

Field	Description
Transport category	Select the related transport category.
Tunnel Code	Select the related tunnel restriction code for the item.

Field	Description
Hazardous material stowage	Select the reference code that is used for sea freight stowage handling when the item is shipped by sea freight.
Aircraft type	Select the aircraft restriction that applies for the material when it's shipped by air freight.
Packing - cargo aircraft only	Based on the value that you selected in the **Aircraft type** field, select the packing instructions code that applies when the product can be shipped only by cargo aircraft.
Packing - passenger and cargo aircraft	Based on the value that you selected in the **Aircraft type** field, select the packing instructions code that applies when the product can be shipped by either cargo aircraft or passenger aircraft.
IATA Star	Set this option to **Yes** to indicate that the air transport specifications that are entered on this FastTab are related to the International Air Transport Association (IATA) dangerous goods standard. This field is for informational purposes only.
Emergence response	Select the code that references instructions for handling the material in an emergency.

Environmental information FastTab

The following table describes the fields that are available on the **Environmental information** FastTab.

Expand table

Field	Description
Environmentally dangerous	Set this option to **Yes** to indicate that the product is environmentally dangerous. Use this field for your own reporting purposes.
Marine pollutant	Set this option to **Yes** to indicate that the product is a marine pollutant. Use this field for your own reporting purposes.

Set stock limits for hazardous products

For safety reasons, you might need to limit the total amount of a given product stocked at a single location. To set stock limits for a released product, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a product to open its **Released product details** page.
3. On the Action Pane, on the **Manage inventory** tab, in the **Compliance** group, select **Reporting details**.

4. In the **Hazardous stock limit** and **Hazardous warning limit** fields, set appropriate values for the selected product.

The **Hazardous material stock limit** report helps you monitor the stock levels of hazardous materials in your warehouse locations, so you can make sure that they stay under the safe limits that are established here. For more information, see Hazardous material stock limit report.

Sales order transactions that include hazardous materials

To include a product that is classified as a hazardous material on a sales order, you must associate the relevant shipping carrier with the sales order. Open the sales order, and then, on the **Delivery** FastTab, set the **Shipping carrier** and **Carrier service** fields as required.

The shipping carrier is also associated with the mode of delivery. Therefore, you must make sure that this information aligns with the hazardous material regulation. In other words, the mode of delivery that is specified in the hazardous material regulation must match the specifications on the sales order header. In this way, the regulation, shipping carrier, and service are connected to the shipment lines that are used on a sales order.

After a sales order is finalized and ready to be shipped, it can be released to the warehouse to indicate the transfer between sales and warehouse operations.

Shipments that include hazardous materials

View hazardous material scores for each shipment line

The **Shipment details** page for a shipment shows the total hazardous material weight and point values that are calculated for each load line that the shipment includes. To view the scores and weights, follow these steps:

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Select a shipment to open its **Shipment details** page.
3. On the **Load lines** FastTab, review the lines. For each line, you see the hazardous material calculations in the following fields:
    - **Hazardous material points** – This field shows the hazardous material score for the load line. The value is calculated according to the rules and values that are established for the applicable regulation and in the released product setup. The calculation takes

the quantity on the load line and references the multiplier in the material management setup for the released product.

- **Limited quantity net weight** – For products marked as limited-quantity products because of their hazardous material content, this field shows the total net weight of hazardous content that the load line includes. The calculation is based on the products marked as hazardous materials in the released product setup. If an item is marked as a limited-quantity item, the calculation takes the quantity on each load line and references the weight in the material management setup for the released product.

Check for compatibility among hazardous materials that a shipment includes

The system can evaluate whether all the hazardous materials that a shipment includes are suitable to be shipped together. To evaluate compatibility, the system checks the compatibility group assigned to each product that the shipment includes. For more information, see Hazardous material compatibility groups.

To run the compatibility check, follow these steps:

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Select a shipment to open its **Shipment details** page.
3. On the Action Pane, on the **Shipments** tab, in the **Actions** group, select **Compatibility check**.

You receive a message that informs you of the results of the check.

Loads that include hazardous materials

View hazardous material scores for each load line

The **Load details** page for a load shows the total hazardous material weight and point values that are calculated for that load and for each load line. To view the scores and weights, follow these steps:

1. Go to **Warehouse management** > **Shipments** > **All loads**.
2. Select a load to open its **Load details** page. (You can also open load details by selecting a link from a related shipment.)

3. On the **Load** FastTab, review the total hazardous material scores and weights for the whole load by inspecting the following fields:

    - **Hazardous material points** – This field shows the hazardous material score for the load. The value is calculated according to the rules and values established for the applicable regulation and in the released product setup. The calculation takes the quantity that is included in the load and references the multiplier in the material management setup for the released product.
    - **Limited quantity net weight** – For products that are marked as limited-quantity products because of their hazardous material content, this field shows the total net weight of hazardous content that the load includes. The calculation is based on the products that are marked as hazardous materials in the released product setup. If an item is marked as a limited-quantity item, the calculation takes the quantity in each load and references the weight in the material management setup for the released product.

4. To review the scores and weights for individual lines, select the **Load lines** FastTab. The values provided for each line resemble the values that it provides for the whole load, as described in the previous step.

Check for compatibility among hazardous materials that a load includes

The system can evaluate whether all the hazardous materials that a load includes are suitable to be shipped together. To evaluate compatibility, the system checks the compatibility group assigned to each product that the load includes. For more information, see Hazardous material compatibility groups.

To run the compatibility check, follow these steps:

1. Go to **Warehouse management** > **Shipments** > **All loads**.
2. Select a shipment to open its **Load details** page. (You can also open load details by selecting a link from a related shipment.)
3. On the Action Pane, on the **Loads** tab, in the **Actions** group, select **Compatibility check**.

You receive a message that informs you of the results of the check.

Last updated on 05/29/2026

Hazardous materials inquiries and reports

Article • 08/23/2022

Microsoft Dynamics 365 Supply Chain Management provides various reports that are related to hazardous materials. Many of these reports are required so that you remain compliant with various hazardous material regulations during shipping and storage.

All these reports, except the **Multimodal dangerous goods** report, use the mode of delivery that is defined for the shipment to find the regulation that should be used to print the shipping text for items. The mode of delivery is associated with the shipping carrier and the carrier service. Therefore, you must set up a shipping carrier and carrier service, and link them to a mode of delivery. The mode of delivery is related to the hazardous materials regulation.

The following illustration shows the sequence of activities that occur when the system generates hazardous materials reports.

Set up hazardous materials reporting

Usually, if you ship items that contain hazardous materials, you must generate specific reports to help preserve safety and comply with hazardous materials regulations. To set up your reports, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Open the **Reports** tab. On the **Hazardous materials report parameter** FastTab, set the following fields.

Expand table

Section	Field	Description
Multimodal Dangerous Goods	Regulation code	Select the regulation that should be used when a **Multimodal dangerous goods** report is generated.
Hazardous Material stock limits	Regulation code	Select the regulation that should be used when stock limits are evaluated.
Carriage of merchandise by road	CMR group product	CMR stands for "carcinogenic, mutagenic, and reprotoxic substances." Set this option to **Yes** to configure the system to print specific warnings and messages that are related to the handling of these substances.
Carriage of merchandise by road	Hazardous material group description	Enter the text of specific warnings that are related to CMR and carriage of merchandise by road. This text will be included on the report.
Shippers declaration	Warning	Enter the text of a warning message that should be printed on the shipper's declaration form (for example, "Warning: Dangerous Goods, Flammable").
Shippers declaration	Footer declaration	Enter the text of a message that should be printed at the bottom of the shipment declaration document.
Hazardous goods report language	Hazardous goods domestic report language	Select the default language for hazardous materials reports that are associated with domestic shipments.
Hazardous goods report language	Hazardous goods export report language	Select the default language for hazardous materials reports that are associated with international shipments.

Hazardous materials report

The **Hazardous materials** report shows a list of all items that have been set up and defined so that they have dangerous goods information. You can use this report to monitor and review the information that you must maintain. The page for the report shows a limited selection of fields from the hazardous material setup. However, you can personalize it to add additional fields as you require.

To view this report, go to **Product information management** > **Inquiries and reports** > **Hazardous material shipping documentation** > **Hazardous materials**.

Hazardous material stock limit report

The **Hazardous material stock limit** report lets you monitor the stock levels of the hazardous materials in your warehouse locations, to make sure that they remain under established, safe limits. These limits come from the limits that are defined for each released product.

To view this report, go to **Product information management** > **Inquiries and reports** > **Hazardous shipping documentation** > **Hazardous material stock limits**.

For more information about how to set stock limits on a released product, see Set stock limits for hazardous products.

The regulation that is used for stock limits is defined on the **Warehouse management parameters** page. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**, and then, on the **Reports** tab, in the **Hazardous materials stock limit**, specify a regulation code. Learn more in the Set up hazardous materials reporting section earlier in this article.

Verified gross mass report

The **Verified gross mass** report lets you print information about the weight of a shipment.

To generate and print this report, go to **Warehouse management** > **Shipments** > **All shipments**, and open the relevant shipment. Then, on the Action Pane, on the **Shipments** tab, in the **Hazardous materials document** group, select **Verified gross mass**.

Multimodal dangerous goods report

The **Multimodal dangerous goods** report is provided for shipments that must be moved by using a combination of transport methods. It's typically used when a shipment is moved first by road and later by sea.

To generate and print this report, go to **Warehouse management** > **Shipments** > **All shipments**, and open the relevant shipment. Then, on the Action Pane, on the **Shipments** tab, in the **Hazardous materials document** group, select **Multi model dangerous goods**.

When you generate this report, the information is saved so that you can edit it and/or reprint the report if you must. To edit a generated report, go to **Warehouse**

management > **Enquiries and reports** > **Hazardous materials shipping documentation** > **Multimodal dangerous goods**, and find the relevant report in the list. After you've finished editing the content as you require, select **Print** on the Action Pane to print the report.

Shippers declaration report

The **Shippers declaration** report lets you print information that is related to a declaration of the materials that are included in the shipment.

To generate and print this report, go to **Warehouse management** > **Shipments** > **All shipments**, and open the relevant shipment. Then, on the Action Pane, on the **Shipments** tab, in the **Hazardous materials document** group, select **Shippers declaration**.

Carriage of merchandise by road report

The **Carriage of merchandise by road** report resembles a bill of lading but is typically used for road transportation in Europe under the Agreement concerning the International Carriage of Dangerous Goods by Road (ADR) regulations. This report uses the shipping print text for an item unless you set the **Hazardous material group description** field on the **Warehouse management parameters** page.

To generate and print this report, go to **Warehouse management** > **Shipments** > **All shipments**, and open the relevant shipment. Then, on the Action Pane, on the **Shipments** tab, in the **Hazardous materials document** group, select **Carriage of merchandise by road**.

When you generate this report, the information is saved so that you can edit it and/or reprint the report if you must. To edit a generated report, go to **Warehouse management** > **Enquiries and reports** > **Hazardous materials shipping documentation** > **Carriage of merchandise by road**, and find the relevant report in the list. After you've finished editing the content as you require, select **Print** on the Action Pane to print the report.

Shipment summary report

The **Shipment summary** report provides information that is summarized by the transport category that is related to the released items.

To generate and print this report, go to **Warehouse management** > **Shipments** > **All shipments**, and open the relevant shipment. Then, on the Action Pane, on the **Shipments** tab, in the **Hazardous materials document** group, select **Shipment summary**.

bill of lading report

When the hazardous materials feature is turned on for your system, the **bill of lading** report includes a **Hazardous materials** column that indicates whether a load includes hazardous materials. This report is available from the **All loads** page, as usual.

Packing list report

When the hazardous materials feature is turned on for your system, packing lists include additional information that is related to the shipping print text for an item. This report is available from the **All loads** page, as usual.

Feedback

Was this page helpful? Yes No

Provide product feedback

Product-related translations FAQ

Article • 06/15/2022

This article describes how to manage translations for products, product dimension values, and product attributes.

What product-related data can be translated?

You can create translations for the following product-related information:

- Names and descriptions of products.
- Descriptions, friendly names, and help text of product attribute values.
- Names and descriptions of product dimension values.

You can translate the product-related information into any language that is available from the **Text translation** page. For more information, see the following section **How do I created translations for product-related information**.

Where can I view the translated information?

You can view translations of product-related information in any external source document, such as an invoice, that uses a language where translations are available.

How do I create translations for product-related information?

To create translations for a product, follow these steps:

1. Click **Product information management** > **Products** > **Released products**.
2. Select a product, and on the Action Pane, in the **Languages** group, click **Translations**.
3. In the **Text translation** page, in the **Language** field, select a language. To add more languages, expand the **Language** field, and then click **OK**.
4. In the **Translated text** group, enter translations in the **Description** and **Product name** fields.

To create translations for product attributes, follow these steps:

1. Click **Product information management** > **Products** > **Released products**.
2. Under **Setup**, click **Attributes**, and then click **Attributes**.

3. In the **Attributes** page, click **Translate**.
4. In the **Text translation** page, in the **Language** field, select a language. To add more languages, expand the **Language** field, and then click **OK**.
5. In the **Translated text** group, enter translations in the **Description**, **Friendly name**, and **Help text** fields.

To create translations for product dimension values, follow these steps:

1. Click **Product information management** > **Products** > **Released products**.
2. Select a product, and then click **Product dimensions**.
3. Select one of the links for the product dimensions: **Configurations**, **Sizes**, **Colors**, or **Style**.
4. Select a dimension value and then click **Translate**.
5. In the **Text translation** page, in the **Language** field, select a language. To add more languages, expand the **Language** field, and then click **OK**.
6. In the **Translated text** group, enter translations in the **Name** and **Description** fields.

Can the names of product variants be translated?

Product variants are based on the dimensions of a released product. Product variant names are based on a combination of dimension values. When the dimension values that are associated with a product variant are translated, the name of the product variant appears in the translated version.

Example

Your product is a T-shirt that comes in different sizes and colors and the variant names are based on the following details:

- Product number: #3
- Dimensions: Size and color
- Size dimension values: Small, Medium, Large
- Color dimension values: Red, Green, Black

The name of a product variant that is based on the dimension values Small and Red is **#3:Small:Red**.

A customer wants to buy some small, red T-shirts and the name of the T-shirt must appear in French on the invoice. You translate the dimension values, Small and Red, into French, and the name of the product variant is **#3:Petit:Rouge**.

Expand table

Tip
To set the preferred language of a customer, follow these steps: 1. Click **Sales and marketing** > **Common** > **Customers** > **All customers**. 2. Double-click a customer to open the **Customers** page. On the **General** tab, in the **Language** field, select the **language**.

What happens if a customer has a preferred language for which no translations are available?

If translations are not available in the customer's preferred language, the names and descriptions are displayed in the global language of your own company.

Can I manage translations for a series of dimension values at the same time?

Dimension values are product specific and you can manage the translations for the dimension values for each product. However, if you create a dimension value group and create translations for the values in the value group, it is easier to manage the translations.

Example

Your company produces T-shirts in different styles, and each style is available in the sizes Small, Medium, and Large. The sizes are collected in one dimension value group. When a new T-shirt style is added, you can associate it with the dimension value group that is used for sizes, so that all the sizes are available for the product. You can also add or change translations for the sizes in the dimension value group at any time.

A dimension value that is associated with a product through a dimension variant group must be maintained from the product variant group.
To create a dimension value group, follow these steps:

1. Click **Product information management** > **Setup** > **Variant groups**.
2. Select **Size groups**, **Color groups**, or **Style groups**.
3. Click **New**, and then enter a name for the group in the **Size group**, **Color group**, or **Style group** field. Click **Sizes**, **Colors**, or **Styles** to create lines for the groups.

4. In the **Size group** lines, **Color group lines**, or **Style group lines** page, click **New**, and then create the sizes, colors, and styles for the groups.

To manage translations for values in a dimension value group, follow these steps:

1. Follow the steps in the previous procedure for creating a dimension value group to open the **Size group lines**, **Color group lines**, or **Style group lines** page.
2. Click **Text translation**. In the **Text translation** page, in the **Translated text** group, enter translations in the **Name** and **Description** fields.

When can translations of product-related information be managed?

Translations of product-related information can be managed at any time. When translations are updated for a dimension value that is associated with a product, the product information is updated, regardless of whether the product has transactions.

Feedback

Was this page helpful? Yes No

Provide product feedback

Search for products and product variants during order entry

Article • 09/29/2021

Use the **Item number** field to search for products and product variants when you manually create a sales order line or a purchase order line. This lets you quickly find product variants when you only have the configuration string or one of the product dimensions available.

Sometimes, having too much of something is not the best situation to be in, and this is especially true if you sell a number of products that are similar, and you are trying to remember item numbers or product search names in order to find the right product to put on a sales order. You can use the **Item number** field on a sales order line or a purchase order line as a search field. You can enter any part of a product name, number, or dimension and get a lookup that displays all the items that match the search word.

How search works

When you search for products or product variants, it is important to understand how the search feature finds the products that match the text that you enter. The key search rules in delivering search results are:

- Search results will return any matching record, disregarding the field that the search text is entered in.
- The search text needs to be present in the matching record in its full length.
- A match will occur even if the search text is found in the middle of a text string in the matching record. It does not have to appear in the beginning of a text string.
- The search text is treated as a single text string even if it contains white space.

Examples

The following examples use products and product variants to illustrate how search is handled in various scenarios. **Prerequisite:** Under **Sales and marketing** > **Setup** > **Search** > **Search parameters** > **Search type**, select the **Full match** option.

Expand table

Product type	Product name	Display product number	Item number	Configuration
Distinct product	SpeakerMidRange	D0001	D0001	NA
Product variant	Active speaker	D0010:::Black:	D0010	000005
Product variant	Active speaker	D0010:::White:	D0010	White

If you type 'speak' in the **Item number** field, you will get all the products above as a result in the lookup. If you type 'black' in the **Item number** field, you will get the second product as a result, because it has the text 'black' in the display product number. These two examples illustrate that the search is not only at the beginning of the field, a match will occur even if the search text is found in the middle of a text string in the matching record.

If you type '05' you will only get the second product variant as a result, because it has '05' in the configuration. This illustrates that the search is across all the enabled fields on the **Search criteria** page.

If you type 'speak 05' you will not get any results. This is because the search looks for the full text that is entered. The search will not try to find 'speak' and then narrow the results to those containing '05'.

You can limit the number of search results by using the **Number of results** field on the **Sales and marketing** > **Setup** > **Search** > **Search parameters** page. If you set this field to 0, all search results will be returned. If you set it to 10, for example, it will return a maximum 10 search results.

Configure the product search

Before you can use the product and product variant search feature, follow these steps to configure the product search.

Step 1: Include all the relevant product and product variant identifiers and dimensions in the search criteria

Examples of product and product variant identifiers and dimensions that you can search by are **Product name, Item number**, **Display product number, Configuration, Color, Size, Style, Search name, etc**.

Go to **Sales and marketing** > **Setup** > **Search** > **Search criteria** page. The **Search criteria** page allows you to define criteria for customer, prospect, and product search. Make sure you filter the page by using product search criteria. You can do this by switching to **Product** in the page's menu.

To add the display product number to the search criteria, click **New** in the page's menu. This will add a new record in the **Search criteria** grid. Open the **Field name** column lookup and chose **DisplayProductNumber**. To add the product's configuration to the search criteria, create a new record in the **Search criteria** grid and chose **configId** in the **Field name** column. In the same manner, create a record with **Field name InventColorId** for the color dimension, **InventSizeId** for the size dimension, and **InventStyleId** for the style dimension.

Step 2: Populate the database table that is used for product search

In the **Search criteria** page, click the **Update search data** button. In the **Update search data** dialog box, make sure that **Source** is set to **Product**, and then click **OK**. The system will aggregate in one table all the selected search criteria specified in step 1. If you have a lot of products and product variants, this operation can be quite lengthy and you may receive a warning. We recommend that you schedule the search table population on the batch server at a time when the server is not too busy.

Until the table is populated, product search will not provide the correct results. If you do not get any search results, make sure that this table is populated.

The table only has to be populated when the search criteria is modified. Newly released products and variants are automatically added to the table. Deleted products and variants are automatically removed from the table.

Step 3: Enable the lookup for product search on sales and purchase order lines

You can enable this functionality by going to **Sales and marketing** > **Setup** > **Search** > **Search parameters** and setting **Enable lookup for search** to **Yes** on the **General** tab.

For sales order line entry, the default behavior is to open the **Product search** page when you start typing in the **Item number** field, and then press the **Tab** key. The **Product**

search page changes the context during order line creation and may be considered unnecessarily intrusive. If you prefer to get the search results in a lookup and not lose context during order line entry, you can use the search lookup instead. If you search for a product or product variant, but you don't select anything in the lookup and press the **Tab** key, the **Product search** page will display.

Feedback

Was this page helpful? Yes No

Provide product feedback

Default order settings for dimensions and product variants

07/10/2025

Default order settings in Dynamics 365 Supply Chain Management define the site and warehouse where items will be sourced from or stored; the minimum, maximum, multiple and standard quantities that will be used for trading or inventory management; the lead times; the stop flag; and the order promising method. Default order settings are used when creating purchase orders, sales orders, transfer orders, inventory journals, and by master planning for generating planned orders. Default order settings can be item specific, site specific, product variant specific, or product dimension specific.

To define the default order settings for a product, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the relevant product in the grid.
3. On the Action Pane, follow one of these steps to open the **Default order settings** page for the selected product:
    - On the **Plan** tab, in the **Order settings** group, select **Default order settings**.
    - On the **Manage inventory** tab, in the **Order settings** group, select **Default order settings**.
4. Configure the settings as described in the rest of this article.

Default order settings

There are three types of default order settings for purchases, sales, and inventory. The default order settings for purchases are used when creating:

- Purchase order lines
- Purchase agreement lines
- Request for quotation lines
- Purchase requisition lines
- Consignment replenishment lines (partially supported, see note)
- Planned purchase orders

> ⓘ **Note**

For consignment replenishment order lines, the only settings from the **Purchase order** FastTab of the **Default order settings** page that apply are the **Default site** field, **Default warehouse** field, and **Stopped** check box.

The default order settings for sales are used when creating:

- Sales order lines
- Sales agreement lines
- Sales quotation lines
- Return order lines and item replacement lines
- Demand forecast lines

The default sales order settings also apply when creating:

- Project item requirements
- Service order item requirements

The default order settings for inventory are used when creating:

- Inventory journals
- Transfer orders
- Planned transfer orders

The default inventory order settings also apply when creating:

- Quarantine orders
- Quality orders
- Production orders
- BOM lines
- Planned production orders

Full definition of a released product

When you create a transaction, you must specify the full definition of a released product on the line, so that Supply Chain Management can try to identify the default order settings. In the full definition of a released product, the item number and all the active product dimensions, such as configuration, size, style, version, and color, are specified on the transaction. For example, if you manually create a purchase order line for a released product variant, you must specify all the required product dimensions before the site, warehouse, quantities, and lead time will appear by default on the order line.

Not all of the default order settings parameters are applied when creating order or journal lines. Quantities and lead times will only display by default when appropriate. For example,

when counting a journal line, only the site and warehouse will display by default when the line is created. For this reason, no default quantity or checks on multiple and minimums are performed when creating the line or posting the journal.

The system always attempts to find a default site and warehouse when an order or journal line is created. The site isn't always displayed by default from the order settings. For example, when creating a sales order or a purchase order, the site from the order header is automatically used on the order lines. When creating a BOM line, the site from the BOM header is used. After the site is determined, it will be used to find any site-specific order settings that can then be used as the default for the warehouse.

The default order type, the purchase, and the inventory lead times can be overridden by the item's coverage rules on the **Item coverage** page. Although the default order settings don't allow for the distinction between the production and transfer lead time, the item coverage rules allow for it. However, the item coverage setup will only be used by master planning when creating planned production and planned transfer orders and won't apply when manually creating production and transfer orders.

Default order settings rules

You can define general default order settings and any number of default order setting rules that apply only in certain conditions, such as site or a specific product dimension or product dimensions combination. You can't define warehouse-specific order settings.

Rank in default order settings

The default order settings rules have ranks. The higher the rank, the more important the rule is, meaning that it will have a higher priority and be used before the rules with lower ranks. The general default order settings have rank zero, which can't be modified. There can only be one rule with rank zero. Rules can have the same rank, provided that the dimensions they apply to differ. This is useful for modeling site-specific order settings. When a new default order settings rule is created, the values for order values, stop flag, etc. are inherited from the rule with rank zero, but can be overridden.

Default order settings for released products

For distinct released products, you can define general order settings or site-specific order settings. The general order settings will always have rank zero. If you set up new sales, purchase, and inventory order settings at the same time, we recommend that you use the **Details view** on the **Default order settings** page. To switch to the details view, go to **Options** > **Page options** > **Change view** > **Details view**.

Site-specific order settings

To create site-specific order settings, select **New**. In **Details view**, enter the site in the **Site** field in the **Settings applicable for** section. In **Grid view**, enter the site in the **Site** column. The new rule is automatically assigned a new rank value that is more than 0 (zero). You can create as many site-specific rules as you require. To indicate that they're equally important, you can assign the same rank value to all the site-specific rules.

If you are in **Details view**, you can't get the overview of the rules created for the item. Use the **Show/Hide list** button to see overview information. When an order line of any type is created and it has no site provided, Supply Chain Management searches for a rule with no site specified. This helps to determine a default site on the order line. This site is then used to search for a site-specific rule, where a default warehouse may have been set. This warehouse is applied to the order line.

Specific order settings for product dimension

You can define order settings rules for any active product dimension or combination of active product dimensions. If a product dimension field is empty, then that the rule applies to all values of the product dimension.

Consider the following example product.

Expand table

Item	Value
Product name	Photoelectric sensor
Item number	XW56
Configuration (used to model the type of light)	C1-Visible red light, C2-Infrared light
Version	V1, V2, V3

For this example, assume that the product is procured and not produced. Also assume that configuration C1 is more commonly used, so it has shorter lead times.

Create the following default order settings to model this scenario.

Expand table

Rank	Site	Configuration	Version	Purchase - Override default settings	Purchase lead time	Purchase - Stopped	Sales - Override default settings	Sales - Stopped
10		C1		Yes	2			
0					5			

When a purchase order line or a planned purchase order is created for item XW56, configuration C1, regardless of the version or site where the line is put, the lead time will be considered 2. Assume that all versions besides V3 are stopped.

You can create the following default order settings rules.

Expand table

Rank	Site	Configuration	Version	Purchase - Override default settings	Purchase lead time	Purchase - Stopped	Sales - Override default settings	Sales - Stopped
20			V2	Yes		Yes	Yes	Yes
20			V1	Yes		Yes	Yes	Yes
10		C1		Yes	2			
0					5			

The two rules for stopping the old versions have the same rank. Therefore, they're equally important. Because both these rules have a higher rank than the rule for configuration C1, they take precedence over the rule for configuration C1.

This example explains the need for the rank. If the rank isn't used, when a purchase order is created for configuration C1 and version V2, the two rules that are defined for V2 and C1 will be ambiguous. To solve the ambiguity, Supply Chain Management will search through the rules in descending order of rank and apply the first applicable rule. In the current example, when a purchase order line is created for configuration C1 and version V2, the user will receive a warning message that states that the item is on hold, and that this hold is caused by the version value. If the rule for the configuration had a higher rank than the rule for the version, a purchase order line would be successfully created for configuration C1 and version V2, and the user would receive no "item on hold" message.

Consider the following default order setting rules.

Expand table

Rank	Site	Configuration	Version	Default site	Default warehouse	Purchase - Override default storage dimensions	Purchase warehouse
20	2					Yes	22
10		C1	V2	2	21		
0				1	11		

The system traverses the set of rules two times to determine the site and warehouse. When a purchase order line is created for configuration C1, version V2, the site is determined based on the rule that has a rank of 10. The system then searches for a rule for site 2 to determine a warehouse. Rule 20 is found, and because it has a higher rank, the warehouse on the purchase order line will be 22, not 21.

As general guidance, specific rules and rules for dimensions that are more important than other dimensions get higher ranks, while more generic rules get lower ranks.

The rule with rank zero serves as a safety net. If no other rules are hit, then the default order settings from rule zero will be used.

Because the rank number is important, on the **Default order settings** Action Pane, there are functions to move a rule up or down and to renumber the rules, so that they're always in increments of 10.

The number of rules created for a released product may be many. In order to get a better understanding of what each rule is overriding and why it's needed, we recommend using the **Grid view** on the **Default order settings** page. You can enable the grid view by going to **Options** > **Page options** > **Change view** > **Grid view**. The number of columns displayed in the grid could be significant, especially for the sales and inventory tabs. To limit the number of columns shown in the grid, groups of columns can be hidden or displayed by using the buttons on the **Default order settings** > **Column display** menu.

Specific order settings for released product variant

If the rule system for default order settings is too cumbersome, then there's the option to define default order settings for each product variant. The following example shows how this will look for the product and the cases described above.

Expand table

Rank	Site	Configuration	Version	Purchase - Override default settings	Purchase lead time	Purchase - Stopped	Sales - Override default settings	Sales - Stopped
10		C2	V3	Yes	5			
10		C2	V2	Yes	5	Yes	Yes	Yes
10		C2	V1	Yes	5	Yes	Yes	Yes
10		C1	V3	Yes	2			
10		C1	V2	Yes	2	Yes	Yes	Yes
10		C1	V1	Yes	2	Yes	Yes	Yes
0					5			

The rank in this case doesn't really matter, so you can choose to hide it. This solution potentially introduces a maintenance issue. However, you may want to consider using this setup if you're considering integrating with Product Lifecycle Management (PLM) systems.

Use strict or standard validation of default order quantities

You can choose how strict the system should be when validating quantities entered in the **Default order settings** for a product. When you use the new strict option, the **Standard order quantity** must always be a multiple of the specified **Multiple** value for purchase orders, inventory, and sales orders. If you're using strict validation, you won't be able to save default order settings that don't meet this requirement (and an error is shown in the message bar).

Strict validation applies to **Standard order quantity** values specified on the **Purchase order**, **Inventory**, and **Sales order** FastTabs of the **Default order settings** page. Each FastTab has its own **Multiple** setting, which is used to validate the **Standard order quantity** value specified for that FastTab.

Turn the strict validation option on or off

To use strict validation, the *Strict validation on default order quantities* feature must be turned on for your system. As of Supply Chain Management version 10.0.21, this feature is turned on by default. As of Supply Chain Management 10.0.25, this feature is mandatory and can't be turned off. If you're running a version older than 10.0.25, then you can turn this functionality on

or off by going to Feature management and searching for the *Strict validation on default order quantities* feature.

Set the validation option

To set the validation option:

1. Go to **Product information management** > **Setup** > **Product information management parameters**.
2. On the **General** tab, set **Validation on default order quantities** to one of the following values:

    - **Strict** - Select this option to ensure that all **Standard order quantity** values will be a multiple of the **Multiple** value for each FastTab (**Purchase order**, **Inventory**, and **Sales order**).
    - **Standard** - Select this option to use standard validation (which works the same as when this feature isn't enabled).

Bills of materials and formulas

07/10/2025

This article provides information about bills of materials (BOMs) and formulas, which are a central part of the definition of products and product variants. BOMs and formulas specify the required materials or ingredients for a specific product. Formulas also specify the co-products and by-products that are received in a specific production context.

Bills of materials

A bill of materials (BOM) defines the components that are required in order to produce a product. The components can be raw materials, semi-finished products, or ingredients. In some cases, services can be referenced in a BOM. However, BOMs typically describe the *material resources* that are required.

When it's combined with a route or production flow that describes the operations and resources that are required in order to build a product, the BOM forms the foundation for calculating the estimated cost of the product.

A BOM is an individual entity that is described by the following information:

- BOM ID
- BOM name
- The BOM lines that describe the components and ingredients
- The BOM versions, which define the product and period that the BOM can be used for

A single BOM describes a single level that is identified by a unique ID. Components might have their own BOMs that are referenced by BOM versions. You can display and edit the complete hierarchy of BOMs for a specific product in the BOM designer.

Formulas, co-products, and by-products

A formula is a subtype of BOM that is typically used for process manufacturing. In addition to components and ingredients, a formula describes co-products and by-products. In the actual version, the definition of co-products and by-products for the formula requires the formula version. A formula is typically defined for one specific finished product (a formula or planning item) that is defined in the formula version.

BOMs in the product lifecycle

In the product lifecycle, many types of BOM might be created for various reasons:

- **Sketching/Draft BOM** – Gives a draft estimation of required materials in an early design phase and helps you do a rough estimate of cost and estimated product attributes. This BOM isn't usually used in enterprise resource planning (ERP).
- **Engineering BOM** – Typically used when you design products that are based on existing product portfolios. Engineering BOMs are structured to simplify the design process and group complex products into engineering modules. For simple products, it might be possible to engineering BOMs for the actual production process. However, for other products, the engineering BOM must be converted to an actual production BOM. Engineering BOMS are typically represented by phantoms in the BOM hierarchy. Although engineering BOMs can be used for the planning and execution of manufacturing operations, this approach can lead to inefficiencies, especially in repetitive operations where many orders are created.
- **Planning BOM** – Used to do planning for material requirements. The demand of components and ingredients is calculated based on the demand of the finished products. Like costing BOMs, planning BOMs might represent a specific mix of material that is used in a period.
- **Production BOM** – The actual BOM that is used for a specific production. A production BOM must take into account the actual resources that are used to produce the product. When a production order, batch order, or kanban is created, the multiple levels of BOMs that are represented by phantoms are collapsed into one level and distributed over the operations for the order.
- **Costing BOM** – Used to calculate the estimated cost of a product. For example, you can use a costing BOM when standard cost is used or the estimated planned cost of a given product is calculated. Costing BOMs can refer to a specific mix of materials and resources that are expected to be used. Therefore, you can use the costing BOM to create a representative estimated cost for a period and help avoid variances over time.

The types of BOM that are actually used in an implementation depend on the implementation, and also on the business scenarios and requirements. In simple implementations, a planning BOM, production BOM, and costing BOM can be modeled as one BOM. In environments that have frequent engineering changes and multiple alternative routes, a larger set of BOM types will probably be required.

Approval of BOMs and formulas

Each BOM and formula can be separately approved or unapproved. Typically, approval of a BOM or formula occurs when the first relevant BOM version is approved. However, in some business scenarios, these approvals might be different steps in the process and might involve different process owners.

If a BOM is unapproved, all related BOM versions are also unapproved.

BOM and formula versions

To relate a specific BOM or formula to a product variant that can be produced, you must create a BOM version or formula version. The validity of BOM versions and formula versions can be constrained by period, quantity, site, specific product dimensions, and other criteria. Formula versions have other important attributes, such as yield, co-product and by-product definitions, and the cost distribution instructions for the formula.

Approval of BOM and formula versions

Before a BOM version can be used in the planning or manufacturing process, it must be approved. When a BOM version is approved, the related BOM can also be approved, depending on the user's selection and authentication rights. A BOM version can be approved only if the related BOM itself is approved.

Activation of the default BOM or formula version

To set a specific BOM or formula as the default BOM version or formula version that will be used by master planning or used to create of production orders, you must activate the version. When a version is activated, the uniqueness of the version for the given constraints (for example, period, site, or quantity) is verified. You receive an error message if the version that you're trying to activate conflicts with a version that is already active. You must then either inactivate the conflicting version or modify the version constraints (usually the period) to prevent an ambiguous activation.

Product change with case management

The product change case for approval and activation of new or changed BOMs and BOM versions provides an easy way to see an overview of the BOM version constraints. You can also approve and activate all BOMs and formulas that are related to a specific change for one activation date.

Alternative BOM versions

Sometimes, the active BOM version or formula version shouldn't be used in forecasts, sales, or a parent product. In this case, you can select a specific approved BOM as part of the requirement (forecast line, sales line, or BOM line) if an approved BOM version or formula version exists for the alternative BOM or formula.

When planned orders, production orders, or kanbans are created, the planner or shop floor supervisor can use any approved BOM version that is valid on the requested planned production date to plan for or produce a specific product. The BOM version that is used doesn't have to be activated as the default BOM version.

BOM and formula lines

A BOM line is created for each material, service, or ingredient. The line defines the planned consumption of the specified product variant and also defines the various attributes that are related to the planned consumption.

BOM lines can have the following line types: **Item**, **Phantom**, **Pegged supply**, **Vendor**.

Item

Select the **Item** line type for materials or services that are directly consumed, and that don't require further explosion or pegged supply.

Phantom

Select the **Phantom** line type when you want to explode any lower-level BOM items that are contained on the BOM line. In Master scheduling, in planned cost calculation, or on estimation of a production order that uses BOM lines of the **Phantom** type, the parent BOM line that refers to a product variant that has a phantom BOM is replaced by the component items that are listed as BOM lines in that BOM, as determined by the applicable active BOM version of that product variant. If the product variant has an applicable active route, the operations of that route are merged into the parent route.

Phantoms are typically used to simplify the engineering process. Extensive use of phantom BOMs in many levels has an effect on performance, especially in highly repetitive manufacturing scenarios. To improve performance, you should avoid deep hierarchies of phantoms. Instead, use pre-exploded production BOMs and routes.

Pegged supply

Select the **Pegged supply** line type when you want to create a subproduction, a BOM line event kanban, or a direct purchase order for any product variant that the BOM line references. The subproduction, event kanban, or purchase order is created when you estimate the production order. The required item quantities are automatically reserved for the consuming production order.

> **Note**
>
> A subproduction order is always created for pegged supply, even if on-hand inventory or other supply is available to cover the demand.

Vendor

Select the **Vendor** line type if the production process uses a subcontractor, and you want a subproduction or purchase order to be created automatically for the subcontractor.

> **Note**
>
> For BOMs that include subcontracted operations, the service or work that is performed by the subcontractor must be created as service item that is tracked in inventory. You must attach the service item to the parent item as a BOM line. The route must contain an operation that is assigned to the subcontractor's operations resource.

BOM designer functionality

Article • 01/31/2025

This article describes how you can use the BOM designer page to design and work with tree structures for bills of materials (BOMs). To choose different configurations and specify what information appears on the lines of the tree, select **Setup**.

When you open the **BOM designer** page from the **Released products** page, it displays the hierarchy of bills of materials (BOMs) that are active and approved for the selected item, the default order site of the item, and the actual date.

To change the initial selection in the view, select **Filter**. By setting the display principle to **Selected/Active or Selected**, you can select individual BOM or route versions to use in the view. You can select non-approved and non-active BOM versions to show or maintain in the BOM designer.

> **Note**
>
> If you open the BOM designer from the **Bills of materials** list page, it doesn't display route information. Currently, the selection of a BOM or route version is a property of the BOM and route version, and applies to all instances of the BOM designer.

The following sections describe the functionality that is available on the various tabs of the BOM designer.

Analyzing a BOM structure by using the BOM designer

The BOM designer has two sections:

- The tree view of the BOM structure.
- The details section, which shows details of the selected data. When you select a node in the tree view, the FastTabs in the details section are updated based on that node:
  - **BOM line details** – Shows the details of the BOM line that is selected in the tree view.
  - **Item data** – Shows the details of the main item or the item that is used in the selected node. To maintain the selected item, select **Edit released product**.
  - **BOM** – Shows the header of the BOM that is related to the selected node.

- **Route** – Shows the header of the route that is related to the selected node.
- **Route operations** – Shows a preview of the operations for the route. When a BOM line that is assigned to a specific operation is selected, the operation is marked as **Component needed at operations**.

Selecting a BOM and route

The filter that is applied for the BOM and route is displayed in the header of the BOM designer. You can change the filter by using the **Filter** dialog box. The following table describes the fields in this dialog box.

Expand table

Field	Description
Product dimensions	If the selected finished product is a product master, you can define the active product dimensions for the main selection. **Note:** If you open the BOM designer for a product that isn't a product master, no product dimensions can be selected in the **Filter** dialog box.
Site	Change the site that the BOM tree is displayed for. The default site is the default inventory site of the finished item.
Display principle	Select the version display principle that applies to the current BOM structure and the current route: • When the principle is set to **Active or Selected/Active**, the valid BOM or route version for this date is found. • When the principle is set to **Selected/Active or Selected**, you can select a BOM version or route version by selecting **BOM** > **BOM versions** or **Route** > **Route versions**.
Version date	Enter the version date for the BOM and route. The version identifies which BOM version is used on a specific date, as determined by the version dates in the BOM version setup.
From quantity	Filter the versions by selecting a specific from quantity. If you set a value, different BOM and route versions might be selected.
Show valid only	When you select the check box, the tree structure shows only BOM lines that have valid dates. Right-click or double-click a BOM line to open the **Edit BOM line** page, where you can see the validity dates for that BOM line.

When you use the BOM designer to review or edit BOMs that consist of one or more levels of phantoms, the route that is associated with the top item typically spans the complete BOM hierarchy. To simplify the overview, you can lock the top-level route in

the display by selecting **View** > **Lock route**. To unlock the route, select **View** > **Unlock route**.

Adding and editing BOMs and BOM lines

Use the **BOM lines** or **BOM** functions to modify the BOM lines or BOM. When you select a node in the tree, the type of the node determines that functions that are available.

Expand table

Function	Description	Node type and conditions
BOM lines > Edit	Open a dialog box where you can edit the BOM line attributes.	This function is available when a BOM line node is selected.
BOM lines > Delete	Delete a BOM line from the selected BOM.	This function is available when a BOM line node is selected, and the BOM isn't locked for editing.
BOM lines > Add before line	Open a dialog box where you can select a product variant to include before the selected BOM line.	This function is available when a BOM line node is selected.
BOM lines > Add to component BOM	Open a dialog box where you can select a product variant to include at the end of the selected BOM.	This function is available when the node that is selected has a selected BOM. If this function isn't available, a BOM version might be missing for the selected item variant. In this case, select **BOM** > **Create version** to create the missing version for the selected node.
BOM lines > Add after line	Open a dialog box where you can select a product variant to include after the selected BOM line.	This function is available when a BOM line node is selected.
BOM > Create version	Create a new BOM version or BOM for the product variant of the selected node.	This function is available when the BOM line node that is selected is linked to an item that has a production type of **BOM** or **Formula**.
BOM > Calculation	Open a dialog box where you can run the cost or sales price calculation for the selected product variant.	This function is available when the node that is selected is related to a BOM version.
BOM > Check	Validate and check the	This function is available when the node that is

Function	Description	Node type and conditions
	selected BOM.	selected is related to a BOM version.

Configuring the tree view

To customize the information that is shown in the tree view of the BOM designer, select **Setup**.

Expand table

Field group	Description
BOM	Use the check boxes to select the criteria that are shown in the tree structure. The BOM designer displays the selected criteria at the bottom of both tabs.
Route	Use the check boxes to select the criteria that are shown for the routes.

Feedback

Was this page helpful? Yes No

Provide product feedback

Formula designer

This article explains how to use the formula designer to analyze and maintain formulas in a tree view.

When you open the **Formula designer** page from the **Released products** page, the tree in the left pane shows the list of co-products and the hierarchy of ingredients for the released product. The structure is derived from the hierarchy of formulas that are active and approved for the selected item and its ingredients, the default order site of the item, and the actual date.

Click **Setup** to select different configurations and specify what information appears on the lines of the tree.

Click **Filter** to change the initial selection in the view. If you set the display principle to **Selected/Active** or **Selected**, you can select individual formula or route versions to use in the view. You can select non-approved and non-active formula versions to show or maintain in the formula designer.

> ⓘ **Note**
>
> If you open the **Formula designer** page from the **Bills of materials** list page, it doesn't show route information. Currently, the selection of a formula or route version applies to all instances of the formula designer.

The following sections describe the functionality that is available in the BOM designer.

Analyze a formula structure by using the formula designer

The formula designer has two sections:

- The tree view of the formula structure.
- The details section, which shows details of the selected data. When you select a node in the tree view, the FastTabs in the details section are updated based on that node:
  - **Formula line details** – View the details of the formula line that is selected in the tree view.

- **Item data** – View the details of the main item or the item that is used in the selected node. You can click **Edit released product** to maintain the selected item.
- **Formula** – View the header of the formula that is related to the selected node.
- **Route** – View the header of the route that is related to the selected node.
- **Route operations** – View a preview of the operations for the route. When a bill of materials (BOM) line that is assigned to a specific operation is selected, the operation is marked as **Component needed at operations**.

Select a formula and route

The filter that is applied for the formula and route is shown in the header of the formula designer. You can change the filter by using the **Filter** dialog box. The following table describes the fields in this dialog box.

Expand table

Field	Description
Product dimensions	If the selected finished product is a product master, you can define the active product dimensions for the main selection. Note that if you open the formula designer for a product that isn't a product master, no product dimensions can be selected in the **Filter** dialog box.
Site	Change the site that the ingredient tree is shown for. The default site is the default inventory site of the finished item.
Display principle	Select the version display principle that applies to the current formula structure and the current route: • When the principle is set to **Active** or **Selected/Active**, the valid formula or route version for this date is found. • When the principle is set to **Selected/Active** or **Selected**, you can select a formula version or route version by clicking **Formula** > **Formula versions** or **Route** > **Route versions**.
Version date	Enter the version date for the formula and route. The version identifies which formula version is used on a specific date, based on the version dates in the formula version setup.
From quantity	Filter the versions by selecting a specific "from" quantity. If you set a value, different formula and route versions might be selected.
Show valid only	When you select the check box, the tree structure shows only formula lines that have valid dates. Right-click or double-click a formula line to open the **Edit formula line** page, where you can see the validity dates for that formula line.

When you use the formula designer to review or edit formulas that consist of one or more levels of phantoms, the route that is associated with the top item typically spans the complete formula hierarchy. To simplify the overview, you can lock the top-level route in the display by clicking **View** > **Lock route**. To unlock the route, click **View** > **Unlock route**.

Add and edit formulas and formula lines

Use the **BOM lines** or **Formula** functions to modify the formula lines or formula. When you select a node in the tree, the type of the node determines the functions that are available.

Expand table

Function	Description	Node type and conditions
BOM lines > Edit	Open a dialog box where you can edit the formula line attributes.	This function is available when a formula line node is selected.
BOM lines > Delete	Delete a formula line from the selected formula.	This function is available when a formula line node is selected, and the formula isn't locked for editing.
BOM lines > Add before line	Open a dialog box where you can select a product variant to include before the selected formula line.	This function is available when a formula line node is selected.
BOM lines > Add to component BOM	Open a dialog box where you can select a product variant to include at the end of the selected formula.	This function is available when the node that is selected has a selected formula. If this function isn't available, a formula version might be missing for the selected item variant. In this case, you can click **Formula** > **Create version** to create the missing version for the selected node.
BOM lines > Add after line	Open a dialog box where you can select a product variant to include after the selected formula line.	This function is available when a formula line node is selected.
Formula > Create version	Create a new formula version or formula for the product variant of the selected node.	This function is available when the formula line node that is selected is linked to an item that has a production type of **BOM** or **Formula**.
Formula > Calculation	Open a dialog box where you can run the cost or sales price calculation for the selected product variant.	This function is available when the node that is selected is related to a formula version.

Function	Description	Node type and conditions
Formula > Check	Validate and check the selected formula.	This function is available when the node that is selected is related to a formula version.

Configuring the tree view

Click **Setup** to customize the information that is shown in the tree view of the formula designer.

Expand table

Field group	Description
BOM	Use the check boxes to select the criteria that are shown in the tree structure. The formula designer shows the selected criteria at the bottom of both tabs.
Route	Use the check boxes to select the criteria that are shown for the routes.

Last updated on 07/01/2026

Override the default reservation principle for materials in production

06/17/2025

The *Override default production reservation* feature lets you set a default reservation principle for each item model group. Therefore, different reservation principles can automatically be applied for each item that is part of a production bill of materials (BOM) or batch order formula. You can select whether each item model group should override the default reservation principle that is set for an order, and what reservation principle should be used instead (*manual*, *estimation*, *scheduling*, *release*, or *start*).

When you create a new production order or batch order, the system prompts you to select the reservation principle for that order and all its BOM or formula lines. If you use the *Override default production reservation* feature, some or all BOM or formula lines can override your selected reservation principle and instead use the reservation principle set for their item model group.

For example, if you have raw materials or ingredients that require pick work, BOM or formula lines that are created for those products require a physical reservation, because physical reservation is a prerequisite for the generation of warehouse work. Typically, if you want the reservation to occur automatically, you select one of the following reservation principles: *estimation*, *scheduling*, *release*, or *start*. On the other hand, if you have materials or ingredients that don't require pick work, because they're consumed directly from a location, you typically select the *manual* reservation principle, which doesn't make any physical reservations or generate any pick work.

Assign a production reservation policy to an item model group

1. Go to **Cost management** > **Inventory accounting policies setup** > **Item model groups**.
2. Create or select an item model group.
3. On the **Inventory policies** FastTab, select the **Override item production reservation** check box.
4. In the **Reservation** field, select the reservation principle for items that belong to the selected model group. (Those items include items that are on a BOM or formula line.)
    - *Manual* – Items in the model group aren't automatically physically reserved for production. However, they can still be manually reserved as required.

- *Estimation* – Items in the model group are physically reserved during estimation of the production order.
- *Scheduling* – Items in the model group are physically reserved during scheduling of the production order.
- *Release* – Items in the model group are physically reserved when the production order is released.
- *Start* – Items in the model group are physically reserved at the start of the production order.

Example: Using reservation principles in a bulk/pack scenario

A bulk lubricant material is produced in a 1,000-liter mixer. After the bulk material is ready, it's pumped out to several filling stations, where bottles of different sizes are filled. After filling is completed, the bottles are packed into boxes. Those boxes are then packed onto pallets.

In this scenario, a batch order to make 1,000 liters of bulk material is created. (This order is the bulk order.) When this batch order is completed, it's reported as finished to the material input location of the filling stations. A batch order to fill and pack each bottle size is then created. (These orders are the pack orders.) The pack orders have a formula that consists of the bulk material, an empty bottle, a label, and other packing materials. Because the bulk material flows directly from the mixer tank to the filling stations, no warehouse work is required to pick this ingredient, and the bulk material is consumed directly from the input location. Therefore, the reservation principle is set to *manual*. The other materials are staged to the filling station by pick work. Therefore, the reservation principle for these lines is set to *release*, for example, so that the reservation automatically occurs when the pack order is released.

Reserve the same batch for a production order (preview)

[This article is prerelease documentation and is subject to change.]

Same-batch selection lets you reserve the consumption of a raw material on a production order or batch order against a single batch of inventory. This capability keeps batch traceability consistent through the manufacturing process. For example, when an ingredient must come from one batch to guarantee a uniform mix, same batch selection reserves the whole bill of materials (BOM) or formula line from one batch instead of spreading it across several batches. It brings the *Same batch selection* capability that's already available for sales orders to production and batch orders.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.
- The feature named *(Preview) Use same batch selection for raw materials in production* must be turned on in feature management.
- The feature supports both warehouse management processes (WMS) and non–WMS scenarios. WMS scenarios are supported only when you use a reservation hierarchy where the batch number dimension is above the location dimension (a *Batch-above [location]* reservation hierarchy).
- This feature applies only to batch-enabled products that aren't tracked by serial numbers. That means that it only supports products assigned a tracking dimension group in which the *Batch number* dimension is active and the *Serial number* dimension is inactive.

Set the Same batch selection field on a BOM or formula line

The **Same batch selection** setting flows through the manufacturing setup, so you can control it at the level that makes sense for your process. The value comes from the **Same batch selection** field on the item model group and flows through the BOM and formula lines to the production order lines, where it becomes read-only.

You can override the default value on individual BOM or formula lines. Follow these steps:

1. Go to **Product information management** > **Bills of materials and formulas** > **Bills of materials** (or **Formulas**).

2. Select or create a BOM or formula version, and open its lines.

3. Select the line for the raw material that you want to configure.

4. On the **Line details** FastTab, select the **Setup** tab.

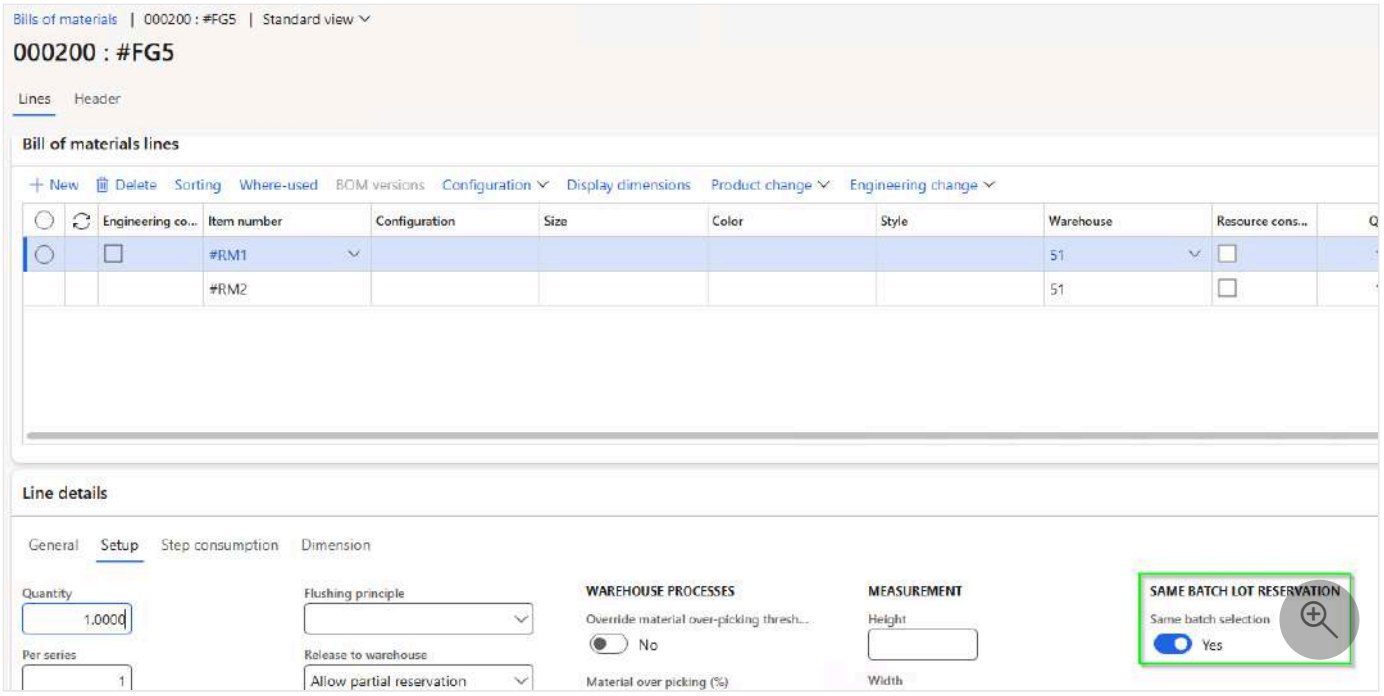
5. In the **Same batch lot reservation** field group, find the **Same batch selection** field. This field has the following values:

   - *Yes* – The raw material must be reserved from a single batch during production. Only inventory from one batch can be used for this line.
   - *No* – No single-batch constraint is applied. The system can reserve inventory from multiple batches.

   The field inherits its default value from the item model group that's assigned to the product. You can override it here for specific BOM or formula lines.

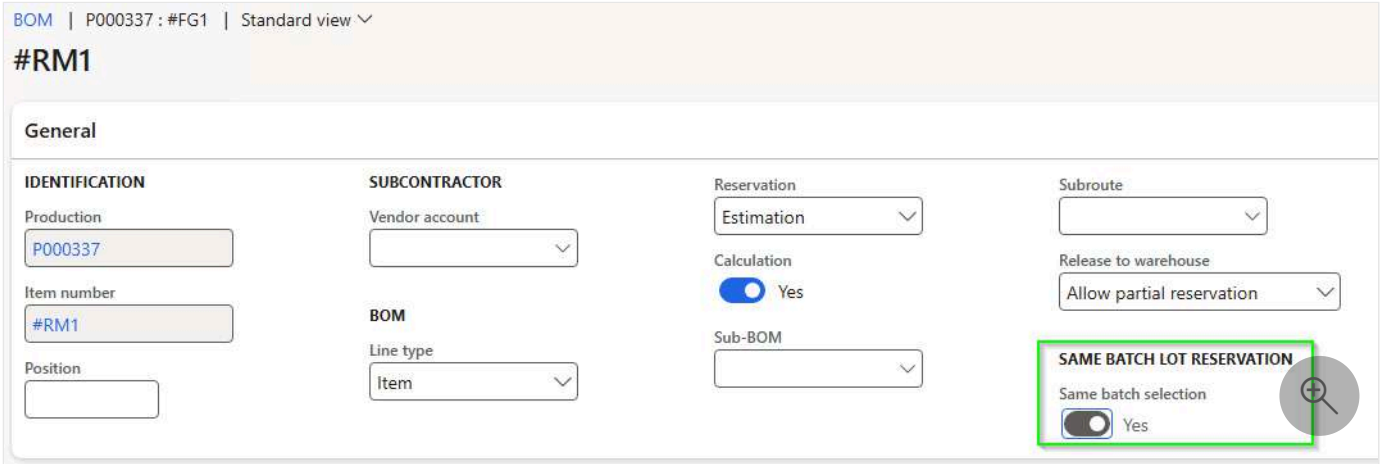
6. Select **Save**.

The following screenshot shows the **Bills of materials** page with the **Same batch selection** field highlighted on the **Setup** tab of the **Line details** FastTab.



When you create a production order or batch order, the BOM or formula line usually copies its **Same batch selection** value to the production order BOM line. On the production order BOM line, the **Same batch selection** field appears in the **Same batch lot reservation** field group on the **General** tab and is read-only.

The following screenshot shows the production order BOM line with the read-only **Same batch selection** field.



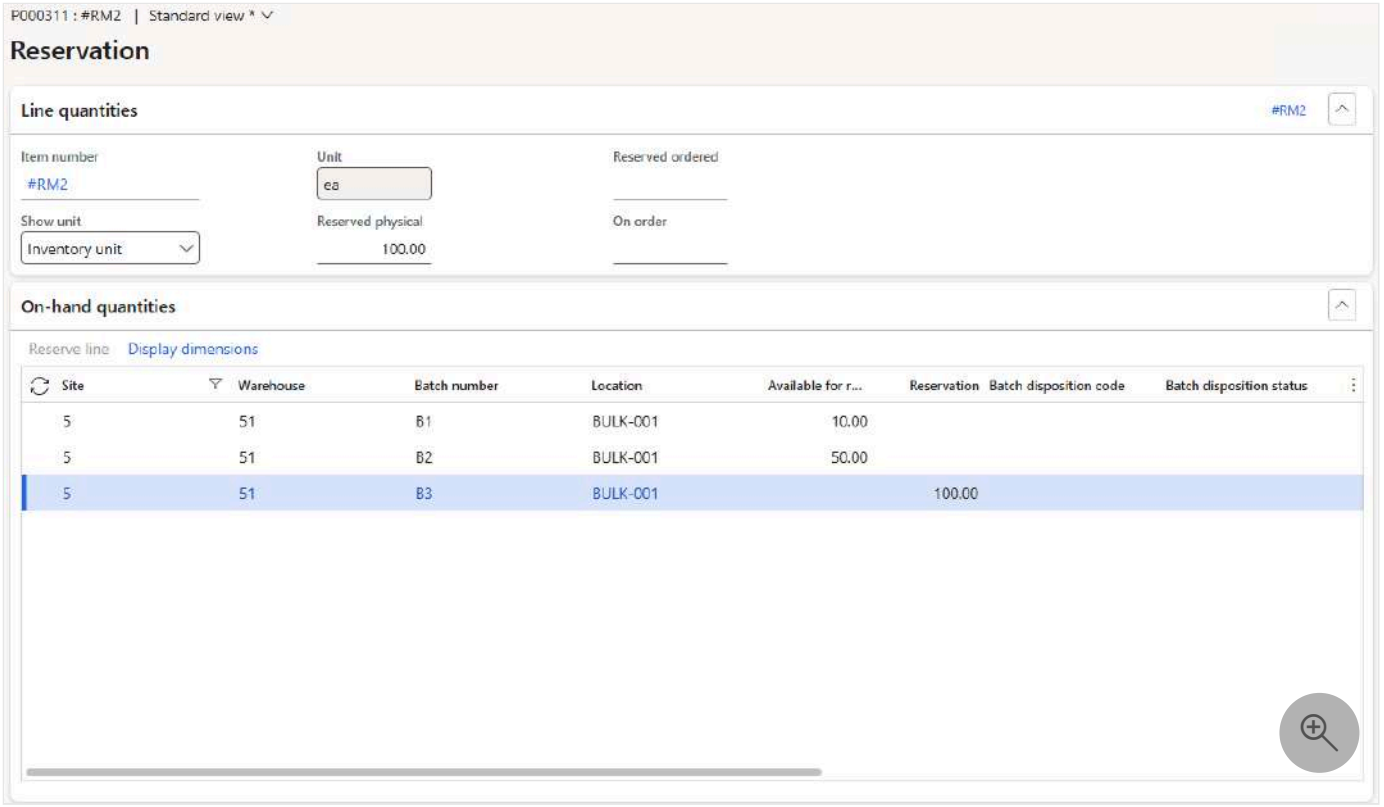
Reservation behavior

When you turn on same-batch selection for a production order or batch order line, the system tries to reserve the required quantity from a single batch during estimation. The outcome depends on available on-hand inventory. Reservation can happen automatically, according to the reservation rules of the BOM or formula line, or manually, when a user reserves against a specific batch directly from the order lines. In either case, same-batch selection constrains the reservation

so that the whole required quantity is drawn from one batch wherever on-hand allows. The outcome depends on available on-hand inventory:

- If a single batch has enough on-hand inventory to fulfill the line, the system reserves that batch. When the item model group enables FEFO, the system selects the batch that has the earliest expiry or best-before date.
- If no single batch can fulfill the line, the system doesn't reserve any batch against the BOM or formula line. The system doesn't show any warning or error message.

The following screenshot shows the **Reservation** page for a production order BOM line where a single batch (B3) was successfully reserved because it had sufficient quantity.



Batch selection sorting priority

When multiple batches can fulfill the requirement, the system selects a batch based on the following sorting priorities:

- **FEFO** – Expiry date or best-before date (ascending), then batch number (ascending).
- **Standard** – Batch number (ascending).

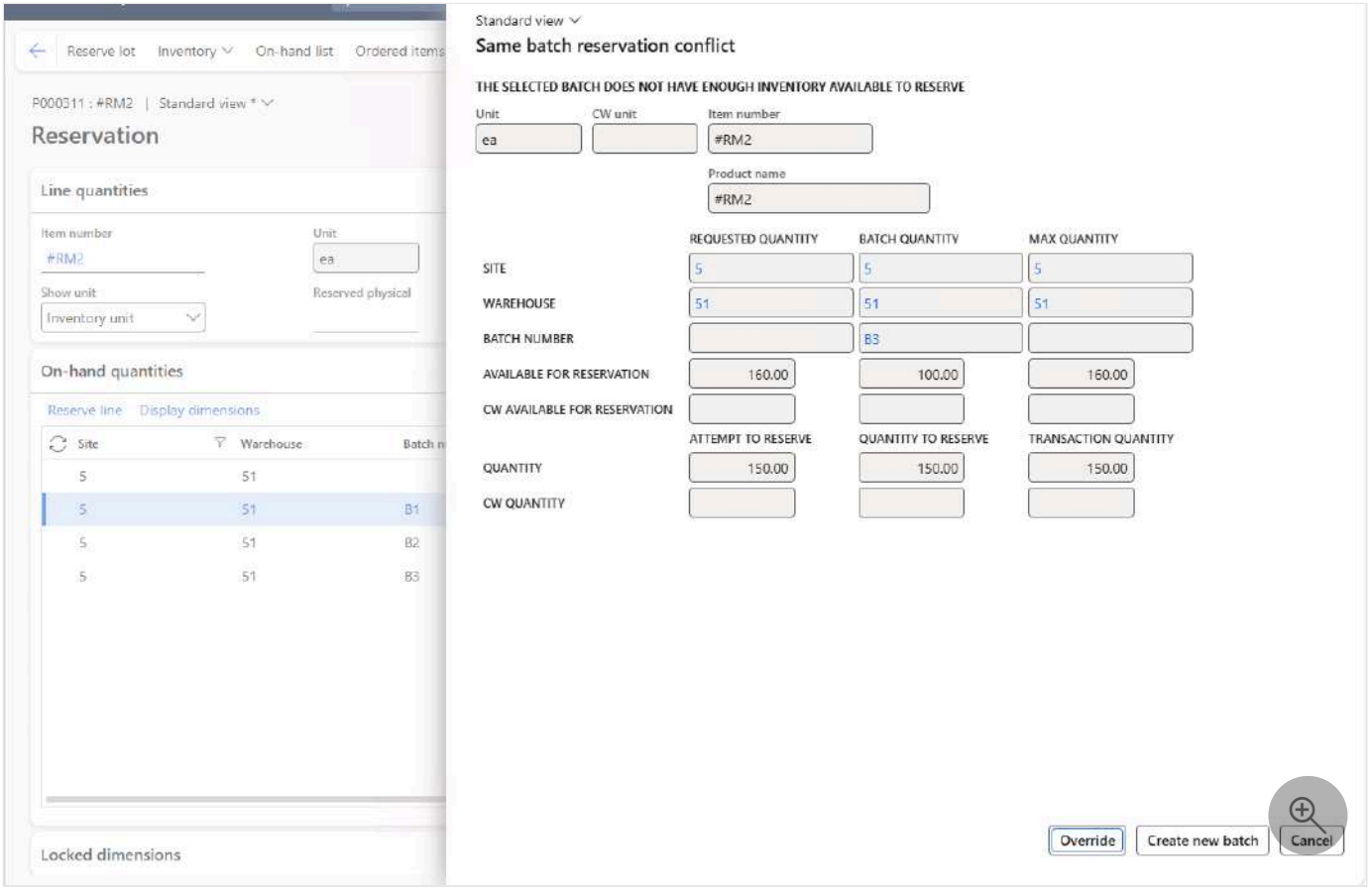
Resolve a same-batch reservation conflict

When you manually reserve a production order BOM line that you can't fulfill from a single batch, the **Same batch reservation conflict** page opens. This page shows the requested quantity, the

available batch quantity, and the maximum quantity available. It offers the following actions:

- **Override** – Turns off single-batch selection for that line and reserves the quantity across multiple batches. The line's **Same batch selection** setting is cleared and marked as overridden.
- **Create new batch** – Generates a new batch number through the standard number sequence and leaves the line on order, so that you can produce or receive inventory into the new batch.
- **Cancel** – Leaves the line on order with no batch reservation.

The following screenshot shows the **Same batch reservation conflict** page with the available actions.



> **Important**
>
> The **Lower quantity** action that's available for sales orders is intentionally turned off for production scenarios, because reducing the consumed quantity would invalidate the production estimation.

Batch reservation policy for non-WMS warehouses

For warehouses that don't use WMS, the **Batch reservation policy for non-advanced warehouses** parameter controls whether the system enforces single-batch reservation when the reservation hierarchy has the batch number dimension below the location dimension (a *Batch-below [location]* reservation hierarchy). To set this option, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Open the **General** tab.
3. On the **Batches** FastTab, set **Batch reservation policy for non-advanced warehouses** to one of the following values:

    - *Simple* – The system doesn't consider same-batch selection constraints when using batch-below reservation hierarchy.
    - *Advanced* – The system applies FEFO rules and batch disposition codes during reservation. Same-batch selection for batch-below items on non-WMS warehouses is blocked, and an error message is shown.

Learn more in Flexible warehouse-level dimension reservation policy.

Related information

- Reserve the same batch for a sales order
- Override the default reservation principle for materials in production

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Last updated on 07/27/2026

Add variant products to purchase orders using variant weights

Article • 09/29/2021

This procedure walks through the steps for using variant weights to auto populate purchase order lines for each variant of a product. When you select the quantity of the product you want to purchase, purchase order lines are created for all the variants of the product with suggested quantities based on the weights configured on the product variants. This procedure doesn't include steps to configure weight values on product dimensions and product variants. This procedure uses the USRT company in demo data.

1. Go to Accounts payable > Purchase orders > All purchase orders.
2. Click New.
3. In the Vendor account field, click the drop-down button to open the lookup.
4. In the list, click the link in the selected row.
5. Toggle the expansion of the General section.
6. In the Site field, click the drop-down button to open the lookup.
7. In the list, click the link in the selected row.
8. In the Warehouse field, click the drop-down button to open the lookup.
9. In the list, find and select the desired record.
10. In the list, click the link in the selected row.
11. Click OK.
12. Toggle the expansion of the Line details section.
13. Click the Variants tab.
14. Click Add line.
15. In the list, mark the selected row.
16. In the Item number field, type '0140'.
17. Set Quantity to '1000'.
18. Click Save.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a sales order for a configurable product

Article • 09/29/2021

This procedure shows how to apply a configuration template to a product on a sales order. This example uses the D0006 speaker model in the USMF demo data company. Typically, a sales order processor uses this procedure.

Create a sales order

1. Go to **Sales and marketing** > **Workspaces** > **Sales order processing and inquiry**.
2. Select **New**.
3. Select **Sales order**.
4. In the **Customer account** field, select *US-001*.
5. Select **OK**.
6. In the **Item number** field, select *D0006*.

    - For this task, you must select a configurable product.

7. Select **Product and supply**.
8. Select **Configure line**.

    - Note that the price has changed, based on the configuration that was selected, and that the **Include cable** field is now set to *True*.
    - Note the default price and the settings that are selected for the cable.

9. Select **Load template**.

    - This example shows how you can apply a template to select a predefined configuration. If you're using this procedure as a task guide and want to see the other attribute values that are available, you must select the **Unlock** button.

10. Select **OK**.
11. Select **OK**.
12. Expand the **Line details** section.
13. Select the **Product** tab.

    - The configuration of the item is now listed under the product dimensions.

14. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create sales price selection criteria

Article • 09/29/2021

This procedure shows how to create a sales price selection criterion for attribute-based sales price models. This procedure requires that at least one sales price model be available. This example uses the price model for the Speaker solution sales price model in the USMF demo data company. Typically, a product manager uses this procedure.

Add a new criterion for an existing sales price model

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, select the row for the Speaker solution product model, but don't select the link for the model name.
3. On the Action Pane, select **Model**.
4. Select **Price model criteria**.
5. Select **New**.
6. In the **Name** field, type 'Customer group 10'.

    - The name of the price model criterion is used to help identify the underlying selection criteria.

7. In the **Price model** field, enter or select a value.
8. In the **Order type** field, select *Sales order*.

    - The order type determines the database fields that will be available for the selection query.

9. In the **Valid from** field, enter a date.
10. In the **Expire by** field, enter a date.
11. Select **Save**.

Create the query for the selection criteria

1. Select **Edit**.
2. In the **Table** field, select *Customers*.
3. In the **Field** field, select *Customer group*.

- In this example, we will use a specific customer group for the selection criteria.

4. In the **Criteria** field, select *Customer group 10*.
5. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up attribute-based pricing for configurable products

Article • 06/03/2022

This article explains how to set up attribute-based pricing. As a prerequisite, you must have a product configuration model that has one or more components and attributes. This example uses the High End Speaker product model in the USMF demo data company. Typically, a product manager uses this procedure.

Create a new price model

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, select the **High End Speaker** line, but don't select the link for the name.
3. On the Action Pane, select **Model**.
4. Select **Price models**.
5. Select **New**.
6. In the **Price model name** field, type a value. Use a name that makes the model easy to identify.
7. In the **Description** field, type a value.
8. Select **Save**.

Add price elements

1. Select **Edit**. Each component in a product model can have a base price element and any number of price expression rules. You can also add prices in different currencies.
2. In the **Base price expression** field, type a value. For example, type 100. A base price expression can be a numerical value, or it can consist of an arithmetic calculation that involves one or more attributes.
3. Select **Add**.
4. In the **Name** field, type `Rosewood`. The price expression name helps identify what the price element represents. In this example, we are creating a price element for the Rosewood speaker cabinet finish option.
5. Select **Edit condition**. A price condition helps guarantee that a price expression element is included in the sales price only if a specific combination of attributes is present.
6. In the **ConstraintBody** field, enter `CabinetFinish=="Rosewood"`.

7. Select **OK**.
8. In the **Expression** field, type a value. For example, type `50`.
9. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

BOM calculations

07/21/2025

The cost roll-up and sales price calculations are known as bill of materials (BOM) calculations, and you initiate them from the Calculations page. This article provides information about BOM calculations.

The cost roll-up and sales price calculations are known as bill of materials (BOM) calculations, and you initiate them from the **Calculations** page. You use the **Calculations** page to perform the following tasks:

- Calculate the cost of a manufactured item, and generate an associated item cost record within a costing version.
- Calculate the sales price of a manufactured item, and generate an associated item sales price record within a costing version.

The way that you use the **Calculations** page varies slightly, depending on how you initiate the BOM calculations. The way that you use the **Calculations** page also depends on whether the BOM calculations involve a costing version for standard costs or planned costs, and on several policies that are defined in the costing version that is used in the BOM calculations. A variation of the **Calculations** page is used in the context of a sales order, sales quotation, or service order line item. These calculations are known as order-specific BOM calculations. An order-specific BOM calculation doesn't generate an item cost record within a costing version. Instead, it generates a calculation record that appears on the **BOM calculation details** page. The calculation record includes a calculated cost and a calculated sales price. The **Calculations** page can be opened for a single manufactured item or for a costing version.

- To calculate costs for a single manufactured item, you initiate BOM calculations from the **Item price** page. The **Calculations** page inherits the item identifier. The costing version, BOM version, route version, calculation quantity, calculation date, and site must be specified.
    - By default, the BOM version and route version are set to the active versions for the item, site, date, and calculation quantity. However, you can override the default values with approved versions.
    - By default, the calculation quantity is set to the item's standard order quantity. However, you can override the default value.
    - The calculation date or site can be mandated by the costing version, or user-specified values can be set when the date or site isn't mandated in the costing version. A future calculation date determines how pending cost records are used. BOM calculations use a pending cost record that has the nearest from-date that is on or before the calculation date.

- To calculate costs for all manufactured items or selected items, or to update items on a where-used basis, you initiate BOM calculations from the **Costing version setup** page or the **Costing version maintenance** page. The **Calculations** page inherits the costing version.
    - For the calculations, it's assumed that the active BOM version and route version are used for a manufactured item (and for the related site, date, and quantity), unless a manufactured component has a specified sub-BOM or subroute.
    - For the calculations, it's assumed that the standard order quantity is used for manufactured items. The standard order quantity provides the basis for calculating component quantities, determining the relevant BOM versions and route versions (when you use quantity-sensitive BOMs and routes), and amortizing constant costs. However, when a manufactured component has a BOM line type of *Production* or *Vendor*, or when you use a make-to-order explosion mode for the BOM calculations, this assumption doesn't apply, because quantities reflect the specified calculation quantity.
    - The calculation date or site can be mandated by the costing version, or user-specified values can be set when the date or site isn't mandated in the costing version.

Other variations in BOM calculations reflect the costing type and restrictions of the costing version:

- BOM calculations that use standard costs must be restricted by costing version policies, because the restrictions help guarantee that standard costing principles are used. Standard costing principles require the enforcement of restrictions about the use of standard costs for purchased items, a single-level explosion mode, and the inclusion of miscellaneous charges in unit costs.
- BOM calculations that use planned costs don't have to follow standard costing principles. These BOM calculations can use different explosion modes, alternative sources of cost data for purchased items, and optional enforcement of restrictions within the costing version.

BOM calculations that use standard costs

Policies within the costing version (for standard costs) can mandate enforcement of five BOM calculation policies. The *Recording restriction* option in the costing version mandates one of these policies, where miscellaneous charges must be included in the unit price. Miscellaneous charges for purchased items can be entered manually, whereas miscellaneous charges for manufactured items reflect the calculated amortization of constant costs. The *Calculation restriction* option in the costing version mandates the other four BOM calculation policies:

- The source of cost contributions for purchased items must be based on standard costs. In other words, BOM calculations must use the item cost records within the specified costing version, or within the fallback that contains standard costs.
- To help guarantee accurate and consistent calculation of standard costs, the explosion mode must be single-level.
- To help guarantee consistent results when the sales price of the items is calculated, the profit setting must be mandated. The profit setting can be used, and the item sales price records can be generated, only if the costing version allows for content of sales prices.
- The fallback principle must be mandated, and can be set to *None*, *Active* (for active cost records), or *Costing version* (for a specified costing version).

BOM calculations that use planned costs

Policies within the costing version (for planned costs) can optionally mandate enforcement of five BOM calculation policies. Alternatively, the policies can just provide default values. The *Recording restriction* option in the costing version determines whether the BOM calculation policy about miscellaneous charges will be mandated or act as a default value. Miscellaneous charges can optionally be included in the unit price. The *Calculation restriction* option in the costing version determines whether the other four BOM calculation policies will be mandated or act as default values:

- The source of cost contributions for a purchased item can be the item cost records within a costing version. Alternatively, the source can be defined by the BOM calculation group that is assigned to the item. For example, the BOM calculation group can define purchase price trade agreements as the source of cost contribution data.
- The explosion mode can be single-level, multilevel, or make-to-order, or it can be based on the BOM line item. The explosion mode for the BOM line type replicates the cost calculation logic for production order estimates.
- The profit setting can be mandated, or it can be a default value. The profit setting can be used, and the item sales price records can be generated, only if the costing version allows for content of sales prices.
- The fallback principle can be mandated, or it can be a default value. The fallback principle can be set to *None*, *Active* (for active cost records), or *Costing version* (for a specified costing version).

BOM calculations generate warning messages and other types of messages. Several BOM calculation policies determine the types of messages. The warning conditions are defined in the BOM calculation group that is assigned to items. However, you can override these warning conditions when you initiate a BOM calculation. When the fallback principle is used, it's often helpful if the fallback is shown as an information message. When you're trying to update

calculated costs for items that have missing cost records, it's also helpful if the information message identifies items that weren't updated.

BOM calculations that use the fallback principle

The following situations illustrate two uses of the fallback principle:

- **Two-version approach to standard cost updates** - A costing version can contain the incremental changes to standard costs, such as pending cost records that represent new items or cost changes. In this situation, the fallback principle can identify the use of the active standard costs that are contained in other costing versions.
- **Simulation of the effect of cost changes by using planned costs** - A costing version for planned costs can contain incremental changes for simulation purposes. This costing version will include pending cost records that represent the simulated cost changes to items, cost categories, and calculation formulas for indirect cost. In this situation, the fallback principle can identify the use of the active standard costs that are contained in other costing versions.

BOM calculation of a suggested sales price

When you use a cost-plus-markup approach, the calculated sales price for an item reflects the set of profit-setting percentages that is specified for the BOM calculation, and the costs that are associated with the item's component items, routing operations, and applicable manufacturing overheads. The markup reflects profit-setting percentages that are assigned to cost groups, and the cost groups that are assigned to items, cost categories for routing operations, and the indirect cost calculation formulas for manufacturing overheads. The sets of profit-setting percentages are labeled *Standard*, *Profit 1*, *Profit 2*, and *Profit 3*. Within the Profit 1 set, for example, a profit-setting percentage of 50 percent can be defined for a cost group that is assigned to purchased material, and a profit-setting percentage of 80 percent can be defined for a cost group that is assigned to cost categories for routing operations. The context of the BOM calculation determines how the results of a calculated sales price are handled:

- **BOM calculation for an item and specified costing version** - The BOM calculation generates a pending sales price record within the costing version. This sales price record provides the starting point for viewing the calculation details (for example, on the **Calculate item cost** page). The sales price record acts mainly as reference information and isn't used as the basis for a sales price on sales orders.
- **Order-specific BOM calculation** - A variation of the **BOM calculation** page is used in the context of a sales order, sales quotation, or service order line item. An order-specific BOM calculation doesn't generate a record within a costing version. Instead, it generates a calculation record that appears on the **BOM calculation results** page. This calculation

record provides the starting point for viewing the calculation details (for example, on the **Calculate item cost** page). Information about a selected calculation record can be transferred to the originating line item. For example, the calculated sales price can be transferred to a sales order line item.

Order-specific BOM calculations

An order-specific BOM calculation represents a variation of a BOM calculation for a manufactured item. The order-specific BOM calculation is performed in the context of a sales order, sales quotation, or service order line item. An order-specific BOM calculation generates a calculation record that appears on the **BOM calculation results** page. The calculation record includes a calculated weight, a calculated cost that is based on active cost records, and a calculated sales price. The calculation record that each order-specific BOM calculation for an item generates on the **BOM calculation results** page is uniquely identified by a calculation number. The results of a calculation record can be optionally transferred to the originating line item. An order-specific BOM calculation differs from a BOM calculation for a manufactured item in two ways:

- An order-specific BOM calculation doesn't generate an item cost record within a costing version. Therefore, the BOM calculation policies aren't applied when an item cost record is created, or when a cost record is overwritten.
- An order-specific BOM calculation always uses the active cost records for components, cost categories, and indirect cost calculation formulas.

Product configuration overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In both business-to-business and business-to-consumer relationships, you often need to configure products to meet special requirements.

A manufacturer that supports configure-to-order scenarios can better address customer needs. By stocking semi-finished goods as generic components instead of finished products, the manufacturer can reduce the capital tied to inventory.

A successful move from a manufacture-to-stock setup to a configure-to-order setup requires careful analysis of the product structures, identification of product families, and componentization. To reduce the number of parts and minimize the number of goods in process, it's important that you understand the product interfaces and design for reusability.

Several product configuration modeling principles exist, such as rule-based, dimension-based, and constraint-based modeling. Studies show that the constraint-based methodology can reduce the number of code lines in models by about 50 percent compared to other modeling principles. Therefore, this methodology can reduce the total cost of ownership (TCO). By moving from a rule-based model that is based on X++ code to a constraint-based model, you no longer need a developer license to maintain product models.

Product configuration

The industrialization period led to great achievements in producing high-quality and feature-rich products at affordable prices. Economies of scale made it possible for most people in the industrialized world to buy cars, TVs, household appliances, and other goods that most of us consider a necessary part of our everyday life.

As many products become commodities, a need arises to differentiate them. The immediate response of manufacturers to this challenge is to create variants of each product, so that

customers have more alternatives. This strategy leads to increased forecast challenges, and also to an increase in inventory cost and unsold products that become obsolete.

By adopting a configure-to-order philosophy, manufacturers can meet customer demand for unique products while reducing or eliminating obsolete inventory items. When shifting from a manufacture-to-stock philosophy to a configure-to-order philosophy, one immediate challenge is to balance the need for short lead times against low inventory levels.

The key to success is to carefully analyze the product portfolio and look for patterns in both product features and processes. The goal is to identify generic components that the same equipment can manufacture and use in all variants.

The product configuration feature set includes a user interface (UI) that provides a visual overview of the product configuration model structure, and also a declarative constraint syntax that doesn't require compilation. Therefore, companies that want to support a configuration practice can get started more easily. As the following sections explain, a product designer no longer requires the support of a developer to build a product configuration model, test it, and release it to the sales organization.

Building a product configuration model

Several approaches exist for building a product configuration model. One option is to follow a sequential flow by first creating all the reference data, such as product masters, distinct products, and operational resources, and then including them as components, bill of materials (BOM) lines, route operations, and other elements of the product configuration model. Alternatively, you can select a more iterative approach by first creating the model and then adding reference data as the need arises.

Components

A product configuration model consists of one or more components that connect through subcomponent relationships. Define components once, and then use them multiple times in one or more product configuration models. Components are the main building blocks of a product configuration model, and nearly all information about the model relates to them.

Attributes

Each component has one or more attributes that identify its properties. Users choose the attributes during the configuration process. Attributes control both inter-component and intra-component relationships through inclusion in constraints or calculations. Through conditions

applied to BOM lines, attributes can determine which physical parts the configured product consists of. Additionally, an attribute can control the property of a BOM line through a mapping mechanism. Similar functionality exists for route operations regarding both inclusion and property settings.

> **Note**
>
> When you create attribute types, avoid creating a high number of values for the attribute type domain. This choice could cause slowdowns in the product configurator.

Expression constraints

Use of a constraint-based product configuration model implies that some limitations exist when the user selects values for the various attributes. Implement such limitations as expression constraints by using the Optimization Modeling Language (OML). Alternatively, implement a constraint in the form of a table constraint.

Table constraints

Table constraints can be user-defined or system-defined.

A user-defined table constraint is built by the user. The user selects a combination of attribute types to represent the columns of the table and then enters values from the domains of the selected attribute types to form the rows in the table constraint.

A system-defined table constraint is defined by selecting which table to use as a reference and then selecting fields from this table to form the columns in the constraint. The rows of the table constraint are the rows of the Supply Chain Management table that are present at configuration time.

Include a table constraint in a product configuration model by referencing the table constraint definition and mapping the relevant attributes in the model to the columns in the table constraint.

Calculations

Calculations represent a mechanism for performing arithmetic operations in a configuration model. For example, a calculation can determine the length of a specific piece of raw material or the processing time for a polishing operation. Calculations are imperative and set the value for a

target attribute after all the attribute values that are included in the calculation expression become available.

Subcomponents

Subcomponents represent the nodes in the product configuration model structure. For each subcomponent relationship, specify a reference to a product master that has the variant configuration technology set to constraint-based configuration.

User requirements

A user requirement has all the constituents of a subcomponent. The only difference is that a user requirement isn't bound to a product master. The practical effect of this difference is that any BOM lines or route operations defined in the context of a user requirement are collapsed into the parent component BOM structure or route. This behavior resembles the behavior of a phantom BOM.

BOM lines

Include BOM lines to identify the manufacturing BOM for each component. A BOM line must reference an item, and all item properties can be set to a fixed value or map them to an attribute.

Route operations

Include route operations to identify the manufacturing route. A route operation must reference a defined operation, and you can set all operation properties to a fixed value. You can map all properties except resource requirements to an attribute instead of a value.

Validating and testing a product configuration model

Validation of a product configuration model can occur on several levels in the model and can therefore cover various scopes. The lowest level is for a single expression constraint. In this case, the product designer typically performs validation to verify that the syntax of an expression is correct.

Similarly, a condition for a BOM line or a route operation can be validated in isolation.

Validation can also be done for a user-defined table constraint definition. In this case, the user verifies that the values entered for each field are inside the domain of the corresponding attribute types.

Finally, validation can be done for a complete product configuration model to verify that the complete syntax is correct, and that all naming and modeling conventions are respected.

Testing

Testing a model is similar to running an actual configuration session. The user can walk through the configuration pages and verify that the model structure supports the configuration process. The user can verify that the attribute values are correct, and that the attribute descriptions guide the user to select the correct values. Finally, after a test session is completed, the system tries to create the BOM and the route that corresponds to the selected attribute values, and presents an error message if anything goes wrong.

The configuration page

To navigate between components, select **Next**, or select a component in the product configuration model tree to set focus on it.

Finalizing a model for configuration

When a product configuration model is ready to use in configure-to-order scenarios, you need to create a version. However, several options can enhance the modeling experience.

User interface

Modify the configuration UI by introducing attribute groups in one or more subcomponents. This grouping highlights the relationships between specific attributes and helps the configuration user identify the area of the product that is currently in focus.

Templates

Create one or more configuration templates to speed up the configuration process. Alternatively, create templates to promote specific attribute combinations, such as when a sales campaign focuses on a specific set of product features.

Translations

If the product will be sold in different countries/regions, translations can be created for all text that appears in the configuration UI. This text includes not only name and description fields, but also attribute text values.

Versions

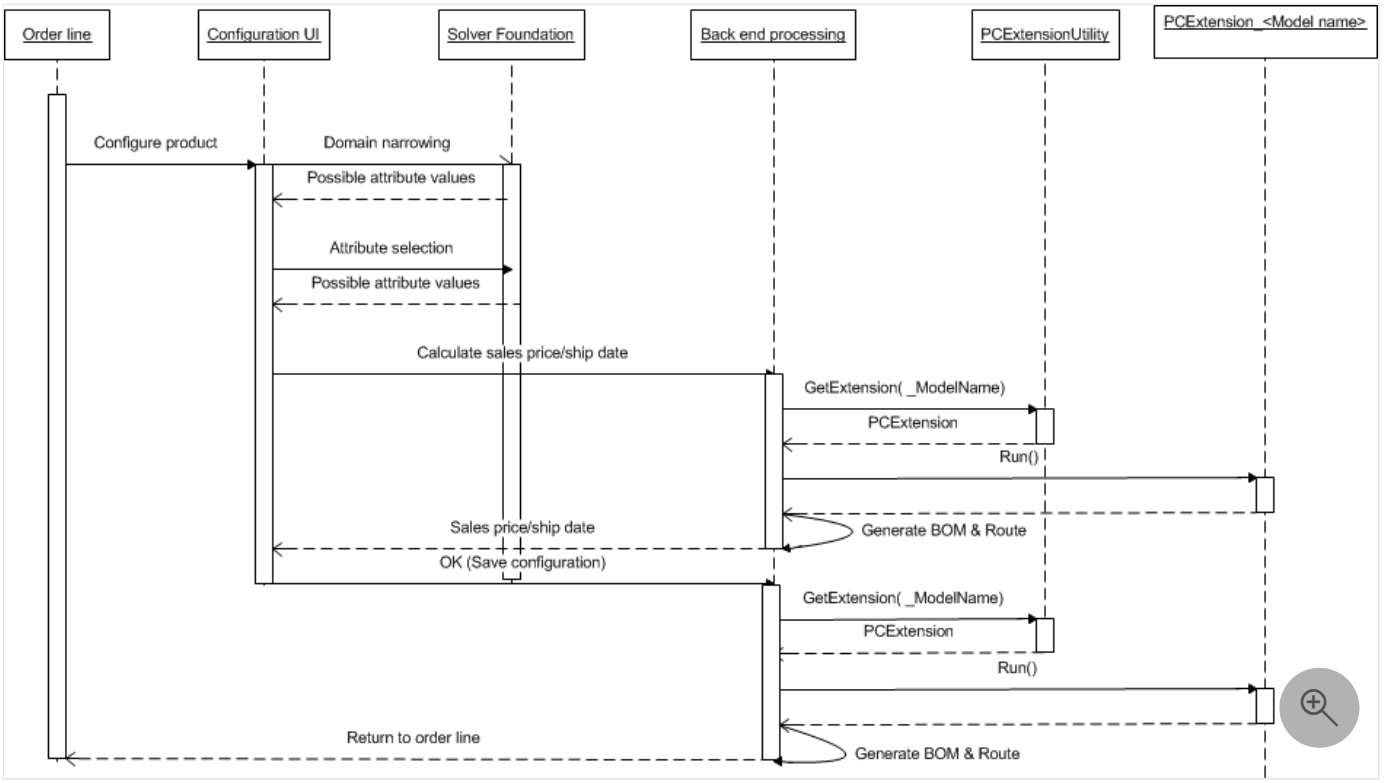
Create a version for the product configuration model. The version represents the relationship between the product master, which can be selected for configuration on an order or quotation line, and the product configuration model. A version must be approved and activated before it can be used in a configuration session.

Extending a product configuration model through the API

Partners and others with a developer license can extend the capabilities of a product configuration model by using a dedicated application programming interface (API). The main goal is to establish a mechanism that lets partners and customers who use the existing Product Builder migrate the code that is embedded in Product Builder models to the API. By using this mechanism, they can migrate their models from Product Builder to a product configuration. However, new partners and customers can also benefit from using the API to extend new product configuration models.

The API is implemented by using a set of `PCAdaptor` classes that expose the data structure of the product configuration models. An instance of the `PCAdaptor` class must be created for each model that can be extended. After a configuration session is completed, the system checks for an instance of this class and runs it if it's found.

The following API flow diagram outlines the process.



Configure products

Configure one or more products

You can configure products from the following places:

- Sales order line
- Sales quotation line
- Purchase order line
- Production order line
- Item requirement line (project)

The purpose of the configuration is to create a distinct variant of the product that meets the customer's requirement. A unique configuration ID is created for each new configuration. This ID enables tracking through inventory.

Multiple sites and intercompany considerations

If the configuration is done at a site, or even a company, that differs from the site or company where production occurs, the BOM and the route are created for and put them at the supplier site in the supplying company. The product variant is released in all companies that participate in the supply chain.

Last updated on 07/01/2026

Dimension-based product configuration overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Dimension-based product configuration represents a simple solution for creating many product variants from a single product master and its bill of materials.

Dimension-based product configuration is one of the three built-in product configuration technologies. The two other technologies are predefined variants and constraint-based configuration. All three technologies use a product master as the starting point and let you create many product variants for one product master.

Key concepts

Dimension-based product configuration is based on the following key concepts:

- Product masters
- Configuration product dimension
- Configuration groups
- Bill of materials (BOM)
- Configuration route
- Configuration rules

Product masters

A product master is the starting point for any product configuration process. For the dimension-based product configuration, you need a product master with this particular configuration technology and a product dimension group that includes the configuration product dimension.

Configuration product dimension

The configuration product dimension identifies the product variants for a product master that uses dimension-based configuration technology. Users enter the configuration dimension value to help identify the individual product variants.

Configuration groups

Configuration groups are defined in a central repository and can be used for all dimension-based product configuration models. Configuration groups are associated with individual BOM lines and hold together a group of lines that are mutually exclusive. This means that only one line in a group can be selected for a single product variant.

Bill of materials (BOM)

The BOM represents the building blocks for a dimension-based product configuration. It must include all the different products that you can use in any product variant. Each line in the BOM can reference a configuration group. If a line doesn't reference a configuration group, it's included in all product variants.

Configuration route

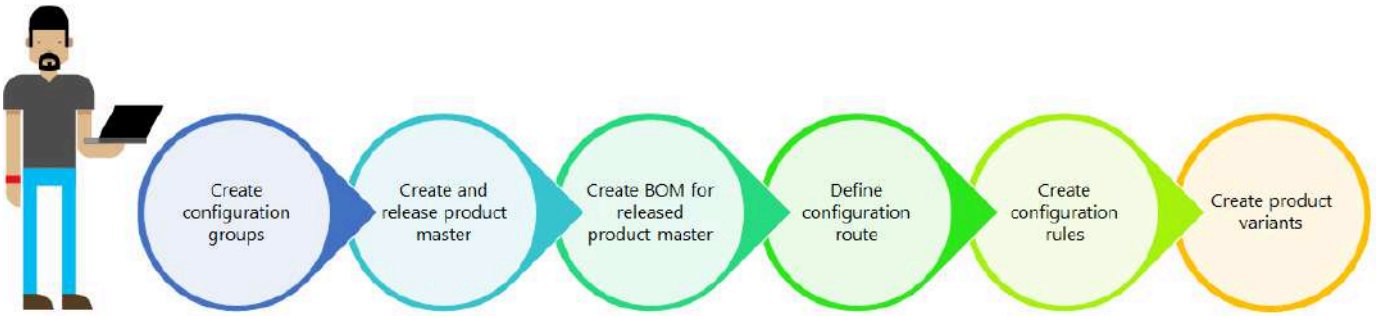
The configuration route determines the sequence of the configuration groups, as they're displayed to the user during the product configuration process.

Configuration rules

The configuration rules represent a mechanism for ensuring that a product included in one configuration group in a BOM enforces either an inclusion or an exclusion of a product in a different configuration group in the same BOM.

Product modeling process

The natural sequence for building a product model for a dimension-based product starts with defining the relevant configuration groups. Ensure that all products used in the BOM have been released to the company that the product model is built for. With these building blocks in place, you can create the BOM and assign configuration groups to all relevant BOM lines. When the BOM is complete, define a configuration route to order the configuration groups in the proper sequence.



If certain products from different configuration groups must or must not be used together, create configuration rules that enforce these product relationships. After the BOM has been tied together with a dimension-based product master through a BOM version and both have been approved and activated, you can create product configurations and enter a name for each configuration. Define the configurations before any transactions are generated, or do it when a specific configuration is needed.

Suggested use

The dimension-based configuration technology works best for products with limited variability when the combination of the standard product dimensions size, color, style, and configuration is unsuitable for identifying a specific product variant. An example is a bicycle with frame height, wheel size, brake types, and different gears.

Next step

The following eight task guides are listed in the order in which you complete them.

1. Create a dimension-based product master
2. Release a dimension-based product master
3. Complete basic setup of a released product master
4. Define configuration groups
5. Create a bill of materials for a dimension-based product master
6. Define configuration routes
7. Create configuration rules
8. Create dimension-based configurations

Last updated on 07/01/2026

Product dimensions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

There are five product dimensions: color, configuration, size, style, and version. You combine product dimensions in dimension groups and assign dimension groups to product masters. The combinations of product dimensions determine how product variants are defined.

Product dimensions are characteristics that serve to identify a product variant. Use combinations of product dimensions to define product variants. A product master requires at least one product dimension to create a product variant.

> **Note**
>
> You can't use physical dimensions to create product variants. Physical dimensions are specified on the product master and are shared by all its variants.

Product variants

Product variants are also referred to as items. An item is a tangible product, which isn't the same as a service. It's also possible to define a product master with the service type. By using the service type, you can specify product variants that include services. For example, you can specify a product master for consultancy work and product variants for work that is performed by senior consultants and junior consultants.

Set up product dimensions

A product variant is generated based on the product dimension values.

Product dimension values for the size, color, and style dimensions are created in the following locations:

- **Size** page (**Product information management** > **Setup** > **Dimension and variant groups** > **Sizes**)
- **Color** page (**Product information management** > **Setup** > **Dimension and variant groups** > **Colors**)
- **Style** page (**Product information management** > **Setup** > **Dimension and variant groups** > **Styles**)

Product dimension values for the configuration dimension are typically created by using either the Product configurator or the Dimension-based configurator.

Product versions are usually created for specific versions as the product evolves during its lifecycle. Product versions are covered in detail later in this article.

You can also create and maintain product dimensions on the **Product dimensions** page from the following locations:

- Go to **Product information management** > **Products** > **Product masters**. On the Action Pane, select **Product dimensions**.
- Go to **Product information management** > **Products** > **All products and product masters**. Select a product master. On the Action Pane, select **Product dimensions**.
- Go to **Product information management** > **Released products**. Select a product master. On the Action Pane, on the **Product** tab, in the **Product master** group, select **Product dimensions**.

The number of variants that you can create for an item is limited by the number of possible product dimension combinations.

> **Tip**
>
> When you use a product on an order line, for example, you select the product dimensions to identify the product variant that you want to work with.

Example

A company sells denim jeans. The item, *Jeans*, uses the color and size product dimensions. The jeans are sold in three different colors and six different sizes. The colors are blue, black, and brown. The sizes are XS, S, M, L, XL, and XXL. Not all sizes are available in all three colors. If all combinations were available, there would be 18 different types of jeans. However, in this example, only the following nine product variant combinations are produced.

Expand table

Color	Size
Blue	XS
Blue	S
Blue	M
Black	M
Black	L
Black	XL
Brown	L
Brown	XL
Brown	XXL

The version product dimension

Version is a product dimension that is intended to help you maintain and track multiple versions of a product throughout the supply chain. Version tracking is essential to the success of manufacturers that operate in a world of constantly shrinking product lifecycles, increased quality and reliability requirements, and increased focus on product safety.

As a standard product dimension, version behaves similarly to the existing product dimensions (size, style, color, and configuration). Therefore, you can use it for other purposes besides tracking product versions.

Turn on the version dimension

As of Supply Chain Management version 10.0.36, the version dimension is turned on by default for new systems, but not for older systems that are upgraded to version 10.0.36 or later. If you need to turn on the version dimension for your system, follow the advice and procedures in this section.

Before you turn on the version dimension

When you turn on the version dimension, some functionality could become broken or stop working as expected if you've installed other solutions that add customizations to the inventory dimensions. For the version dimension to be fully functional, you might have to update those

solutions so that they include the version dimension in their references to the inventory dimensions.

When you're testing your solutions for compatibility with the version dimension, look for the following elements:

1. **Functionality:** Most importantly, any customizations that involve the inventory dimensions must be assessed to ensure that they work with the version dimension.

2. **References to the inventory dimensions:** Look out for references to the inventory dimensions (that is, places where the dimensions are explicitly referenced). References to `InventDimId` should work out of the box, but look out for references to style or color. For example, be sure to check the following elements:

    - API calls in extended classes
    - All references to specific inventory dimensions in extension code (This code must float the version dimension together with the style, color, and size dimensions.)

3. **References to obsolete API calls:** In its introduction of the version dimension, Microsoft has tried to make as few APIs as possible obsolete. The few APIs that are made obsolete issue a warning when you turn on the **Product dimension - version** configuration key. Calls to those APIs must be fixed in your extended solutions before you turn on the version dimension in a production system. Here are the version-specific obsolete APIs:

    - `RetailTransactionServiceInventory::getProductRecordId`
    - `EcoResProductNumberIdentifiedProductVariantEntity::find`
    - `EGAISAlcoholProduction_RU::findByItemDim`
    - `PCVariantConfiguration::findByProductMasterAndDimensions`

4. **Maps:** If any maps use the inventory dimensions, the corresponding relation mapping to these maps must be updated so that they include the version dimension. In the extended model or table extensions, look out for tables where the fields include inventory dimensions.

5. **Microsoft Dynamics 365 Commerce functionality:** After it's turned on, the version dimension appears throughout the Commerce-specific code in Dynamics 365 Supply Chain Management. However, the version dimension isn't yet supported by the Commerce channel database or in the Point of Sale (POS) or e-commerce applications. These Commerce-specific applications don't support users selling/shipping or returning/receiving inventory by version dimension. Inventory availability lookup functions don't discern

inventory by version dimension in Commerce apps. This behavior resembles the current behavior of the config dimension throughout Commerce.

Turn on the version dimension configuration key

Before you can use the version dimension, its configuration key must be turned on for your system. As of Supply Chain Management version 10.0.36, the key is turned on by default for new systems, but not for older systems that are upgraded to version 10.0.36 or later. If you need to turn the version dimension on or off, follow these steps. This task requires admin permissions.

1. Put your system into Maintenance mode.
2. Go to **System administration** > **Setup** > **License configuration**.
3. On the **Configuration keys** tab, expand **Trade**, and select the check box for **Product dimension - Version**.
4. Turn off Maintenance mode.

Areas where the version dimension isn't supported

The following areas don't support the version dimension (you can still use these areas but you won't be able to add versioned products (products where the version dimension is used) to them). For example, you can't add a versioned item to a vendor catalog. This is because adding products with the version dimension to these areas would cause breaking changes.

- Cost object monthly statement
- Cost object statement cache
- MCR sales statistics per item
- Vendor catalogs
- EcoResProductDimensionGroupEntity

In addition, the order creation and order processing features in Commerce (for example, for POS, call center, and e-commerce orders) don't support the version dimension. There's no confirmed timeline for support in Commerce orders.

Functional characteristics of the version dimension

The version dimension works like the other product dimensions. However, because of its specific nature, and because it's intended to maintain and track multiple versions of a product, it behaves slightly differently. Here are some of the differences and similarities:

- **There is no version group.**

Although the concept of groups exists for size, color, and style (color group, size group, and style group), no version group concept exists. Groups let you predefine the applicable values so that when, for example, you assign a color group to a product, the product can use all the colors in that color group. This concept doesn't apply to the version dimension, because the versions that a product takes aren't predefined when the product is created. Instead, versions are created during the lifecycle of the product, as they're required. Typically, if the form, fit, and function of the product remain the same, you create a new version instead of a new product.

- **Product variant suggestions work as they currently do.**

  Product variant suggestions create suggestions for all version dimension values, just as they do for other dimensions. The process of creating and releasing versioned products is the same as it is for products that use other dimensions. When you create a versioned product, the first version (V1) is created as a product dimension, and the variants are released. As the product changes and a new version is needed, the new version value (V2) is added, and the required variants are released. There's no expectation that all the versions (V1, V2, and V3) are created in advance for the product.

> **Important**
>
> If you turn on and use the version dimension, some solutions that reference the inventory dimensions might stop working as expected. To confirm and fix these issues, contact the independent software vendor (ISV) for your affected solutions. Learn more in Enable the version dimension.

Last updated on 07/01/2026

Create a dimension-based product master

Article • 01/06/2023

This procedure shows how to create a new product master with dimension-based configuration technology.

This is the first procedure of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. You should do all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

1. Go to **Product information management** > **Products** > **Product masters**.
2. Select **New**.
3. In the **Product number** field, type a value.

    - Entering a product number is mandatory if no number sequence has been set up for the product number field.

4. In the **Product name** field, type a value.
5. In the **Product dimension group** field, select the drop-down button to open the lookup.
6. In the list, find and select the desired record.

    - Select the configuration dimension for this procedure.

7. In the list, select the link in the selected row.
8. In the **Configuration technology** field, select an option.

    - Select the dimension-based configuration technology.

9. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Release a dimension-based product master

Article • 01/06/2023

This procedure shows how to release a product master, which will be used for the dimension-based configurations. It is a prerequisite that you have created a product master with the dimension-based configuration technology.

This is the second procedure of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. You should do all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

1. Go to **Product information management** > **Products** > **Product masters**.
    - Filter the **Configuration technology** column so that only the dimension-based configuration is displayed. For example, you can filter the column by typing *Dimension*.
2. In the list, mark the selected row.
3. Select **Release products**.
4. Select **Next**.
    - For products that are crated with the dimension-based configuration technology, the product variants must be created in the company where the bill of materials will be created.
5. Select **Next**.
6. In the list, find and select the desired record.
    - Select the company *USMF* for this procedure.
7. Select **Next**.
8. Select **Finish**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Complete basic setup of a released product master

This article shows how to complete the minimum setup that is required before you can use the product master in BOM versions.

This procedure is the third of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. Complete all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified in this article, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, find and select the desired record. Select the product master that you released in the second procedure. This product master is created by using the dimension-based configuration technology.
3. On the Action Pane, select **Product**.
4. Select **Dimension groups** to open the drop dialog.
5. In the **Storage dimension group** field, select the drop-down button to open the lookup.
6. In the list, find and select the desired record. The storage dimension group determines which storage dimensions are used for product transaction. Select **Site** for this procedure.
7. In the **Tracking dimension group** field, select the drop-down button to open the lookup.
8. In the list, find and select the desired record. The tracking dimension group determines which tracking dimensions are used for product transaction. Select **None** for this procedure.
9. Select **OK**.
10. Select **Edit**.
11. In the **Item model group** field, select the drop-down button to open the lookup.
12. In the list, find and select the desired record. Item model groups contain settings that determine how items are controlled and handled on item receipts and issues. They also determine how item consumption is calculated. Select **FIFO** for this procedure.
13. Expand the **Manage costs** section.
14. In the **Item group** field, select the drop-down button to open the lookup.
15. In the list, find and select the desired record. Item groups are used to manage inventory by dividing inventory items into groups. Select **CarAudio** for this procedure.
16. On the Action Pane, select **Plan**.

17. Select **Default order settings**.
18. In the **Default order type field**, select an option. Select **Production** to specify that the default supply option for this product master is to produce it.
19. Select **Save**.
20. Close the page.
21. Close the **Released product details** form.

Last updated on 07/01/2026

Define configuration groups

Article • 01/06/2023

This procedure shows how to define product configuration groups.

This is the fourth procedure of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. You should do all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

1. Go to **Product information management** > **Setup** > **Bills of materials and formulas** > **Configuration groups**.
2. Select **New**.
3. In the **Configuration group** field, type a value.
    - Type a name for the configuration group that will be used for the bill of material lines.
4. In the **Name** field, type a value.
5. Select **New**.
6. In the **Configuration group** field, type a value.
    - Type a name for the configuration group that will be used for the bill of material lines.
7. In the **Name** field, type a value.
8. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a bill of materials for a dimension-based product master

Article • 01/06/2023

This article shows how to create a bill of materials for a dimension-based product master.

This is the fifth procedure of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. You should do all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

This task is typically handled by the product designer.

Select the product

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product you want to work with in the grid.
    - Find the released product master with dimension-based configuration technology that you created in the first procedure in this sequence of eight.
3. On the Action Pane, open the **Engineer** tab and select **BOM versions**.

Create new BOM and BOM version

1. On the Action Pane, select **New** > **BOM and BOM version**.
2. In the **Name** field, type a value.
3. Select **OK**.
4. Select **New** from the **Bill of materials lines** FastTab toolbar.
    - In this procedure we will add four lines to the BOM. Two lines represent cable options and two lines represent cabinet options.
5. Make the following settings for the new line:
    - **Item number** – Select item number A0001, HDMI 6' Cables.

- **Configuration group** – Select the cable configuration group created in the fourth procedure in this sequence.

6. Select **New** from the **Bill of materials lines** FastTab toolbar and then make the following settings for the new line:

   - **Item number** – Select item A0002, HDMI 12' Cables.
   - **Configuration group** – Select the cable configuration group created in the fourth procedure in this sequence.

7. Select **New** from the **Bill of materials lines** FastTab toolbar and then make the following settings for the new line:

   - **Item number** – Select item number D0002, Cabinet.
   - **Configuration group** – Select the cabinet configuration group for this BOM line.

8. Select **New** from the **Bill of materials lines** FastTab toolbar and then make the following settings for the new line:

   - **Item number** – Select item number M0007, StandardCabinet.
   - **Configuration group** – Select the cabinet configuration group for this BOM line.

Approve and activate

1. Close the page.
2. On The Action Pane, open the BOM version tab and select **Approval**.
3. In the **Approved by** field, enter or select a value.
4. Set **Do you also want to approve the bill of materials?** to *Yes*.
5. Select **OK**.
6. Select **Activate**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Define configuration route

Article • 01/06/2023

This procedure focuses on defining a configuration route that determines the sequence in which the configuration groups will be presented.

This is the sixth procedure of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. You should do all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

1. Go to **Product information management** > **Bills of materials and formulas** > **Bills of materials**.
2. In the list, find and select the desired record.
3. On the Action Pane, select **Options**.
4. Select **Change view**.
5. Select **Header view**.
6. Expand or collapse the **Configuration route** section.
7. Select **Add**.
8. In the list, mark the selected row.
9. In the **Configuration group** field, select the drop-down button to open the lookup.
10. In the list, select the link in the selected row.
11. Select **Add**.
12. In the list, mark the selected row.
13. In the **Configuration group** field, select the drop-down button to open the lookup.
14. In the list, find and select the desired record.
15. In the list, select the link in the selected row.
16. Select **Save**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create configuration rules

Article • 01/06/2023

This procedure creates configuration rules that can be used for dimension-based configuration to enforce or prevent certain combinations of BOM lines.

This is the seventh procedure of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. You should do all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

1. Go to **Product information management** > **Bills of materials and formulas** > **Bills of materials**.
2. In the list, find and select the desired record.

    - Find and select the BOM for the dimension-based configuration.

3. On the Action Pane, select **Options**.
4. Select **Change view**.
5. Select **Header view**.

    - Open the header view to access the Configuration route FastTab.

6. Expand or collapse the **Configuration route** section.

    - The **Configuration route** FastTab must be in the expanded mode.

7. Select **Configuration rules**.
8. Select **New**.
9. In the list, mark the selected row.
10. In the **Item number** field, select the drop-down button to open the lookup.

    - The items in the current configuration group are displayed. Select the one that represents the condition in the rule.

11. In the list, select the link in the selected row.
12. In the **Method** field, select an option.

    - It is possible to enforce either a selection or a deselection of an item from another configuration group.

13. In the **Derived group** field, select the drop-down button to open the lookup.

14. In the list, find and select the desired record.
15. In the list, select the link in the selected row.

    - Select the desired configuration group.

16. In the **Derived item number** field, select the drop-down button to open the lookup.
17. In the list, select the link in the selected row.

    - Select the item number that will be either selected or deselected depending on the chosen method.

18. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create dimension-based configurations

Article • 01/06/2023

This procedure shows how to define a configuration for a dimension-based product.

This is the eighth procedure of an eight-procedure sequence that explains how to build combinations for dimension-based configuration. You should do all eight procedures, in order, because each new procedure builds on data created by the previous procedures. To work through this sequence by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

Find the dimension-based product master

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, mark the selected row.

    - Select the dimension-based product master that you created in the first procedure in this sequence of eight.

Create configurations

1. On the **Engineering** Action Pane, select **Maintain configurations**.
2. Select **Configure**.
3. In the list, mark the selected row.
4. In the **Item number** field, enter or select a value.

    - Select any of the items in the first configuration group.

5. In the list, find and select the desired record.
6. In the **Item number** field, enter or select a value.

    - Select any item from the second configuration group.

7. Select **OK**.
8. In the list, mark the selected row.
9. In the **Configuration** field, type a value.

    - Enter a configuration name that will make it easy to identify the configuration.

10. In the **Description** field, type a value.

- Enter a description of the configuration to explain what it contains.

11. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Product configuration models overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article defines terms and concepts that are relevant to product configuration models. Product configuration models let you build a generic product structure that can be used to configure many product variants for a single product.

Product configuration models are created to represent a generic product structure. After you set up a product configuration model, you can configure a distinct product variant that has a unique bill of materials (BOM) and a unique route. Product configuration models use both declarative constraints and imperative calculations to handle the relations and limitations between different product variants. You can configure items on sales orders, sales quotations, purchase orders, and production orders. The following table describes the table constraint–based terms and concepts.

Expand table

Term	Description
Components	Components are the main building blocks of a product configuration model. You see components in a tree structure on the **Constraint-based product configuration model details** page. Components can contain the following elements: • Attributes • Constraints • Calculations • Subcomponents • User requirements • BOM lines • Route operations
Attributes	Attributes describe all the features of the product configuration model. Use attributes to specify the features that can be selected when a distinct product is configured. Attributes are used in constraints and conditions. When attributes are created and added to a product configuration model, the related attribute types are referenced. A default value for an attribute. The default value is used in the configuration user interface (UI) when the

Term	Description
	product configuration model is configured. You can specify that an attribute is mandatory, read-only, or hidden. • **Mandatory** – A value must be set for the attribute when the product is configured. • **Read-only** – The attribute value is displayed during a configuration session, but it can't be changed. • **Hidden** – The attribute value is included in constraints and conditions, but isn't displayed during a configuration session. You can also specify a condition for attributes. If the condition is met, a value must be entered for the mandatory attribute. Conditions are expressions that must be met for attributes, BOM lines, and route operations to be included in a product configuration model. Any attribute that is referenced in a condition becomes mandatory. Select attributes as mandatory on the **Attributes** tab. This selection can make it easier to identify mandatory attributes. Attribute values are an important part of reusing configurations. The system uses attribute values to determine whether a configuration exists that matches the selections that a user made during a configuration session.
Attribute types	Attribute types specify the set of data types for attributes that are used in a product configuration model. The following attribute types are used: • **Integer** with or without a range • **Decimal** • **Text** with or without a fixed list • **Boolean** If the attribute type is **Boolean**, **Integer** with a range, or **Text** with a fixed list, the set of values is available when a product configuration model is set up. **Note:** The Product configuration solver recognizes only the following attribute types: **Boolean**, **Text** with a fixed list, and **Integer** with a range. Therefore, only these attribute types can be used in expression constraints and conditions.
Constraints	Constraints describe the restrictions of the product model configuration. Use constraints to guarantee that only valid values are selected when a product is being configured. Constraints can be either expression constraints or table constraints: • Expression constraints can be used only for the component that they're tied to. The expression constraints for a component can reference attributes of the component's subcomponents. The Product configuration solver is used to solve the constraints, and you must use the solver syntax when you write the constraints. For more information, see the article link about expression constraints and table constraints. • Table constraints must be defined before they can be applied to a component in a product configuration model. Table constraints can be either user-defined or system-defined. A user-defined table constraint is a type of matrix that can be used to describe the set of combinations for the attribute values that are defined by attribute types. For example, if speakers are produced, the matrix for a user-defined table constraint might have columns for the speaker finish and grill. **Example** Speakers are available in four finishes: Black, Oak, Rosewood, and White. The speakers can have one of three front grills: Black, Metal, or White. The Black finish is available for all grills, but the other finishes are limited to specific grills. The following list

Term	Description
	shows an example of the information that is displayed on the **Allowed combinations** tab on the **Edit table constraint** page. • **Cabinet finish**: *Black* | **Front grill**: *Black* • **Cabinet finish**: *Black* | **Front grill**: *Metal* • **Cabinet finish**: *Black* | **Front grill**: *White* • **Cabinet finish**: *Oak* | **Front grill**: *Black* • **Cabinet finish**: *Rosewood* | **Front grill**: *White* • **Cabinet finish**: *White* | **Front grill**: *Black* • **Cabinet finish**: *White* | **Front grill**: *White* A system-defined table constraint represents a mapping between an attribute type and a field in a Supply Chain Management table. A system-defined table constraint dynamically links the attribute type to the field. The link enables the attribute in a product configuration model to reflect the data of the field in the Supply Chain Management table.
Calculations	Calculations represent a supplement to constraints. Use a calculation to perform arithmetic operations on attributes of the **Decimal** and **Integer** types, or logical operations that involve attributes of the **Text** with a fixed list and **Boolean** types. A calculation has a target attribute that holds the result of the calculation expression. The calculation expression is built by using the expression editor.
Subcomponents	Subcomponents reflect the tree structure of the product configuration model. Use subcomponents to build the structure of the product configuration model. Subcomponents reference existing components. Therefore, subcomponents encourage the reuse of components in multiple product configuration models. On the **BOM line details** page for a subcomponent, you can select a distinct value for the subcomponent. Alternatively, you can select an attribute that the value is selected for when the product configuration model is set up. To include a product as a component or subcomponent, you must specify the following information on the **Create product** page when you create the product: • In the **Product type** field, select **Item**. • In the **Product subtype** field, select **Product master**. • In the **Configuration technology** field, select **Constraint-based configuration**. You can view whether a released product can be used as a component or subcomponent on the **General** tab of the **Released product details** page. If **Constraint-based configuration** is selected in the **Configuration technology** field, the product can be used as a component or subcomponent. You can hide subcomponents so that they aren't displayed to the user during a configuration session. Attributes, subcomponents, and user requirements that are related to the subcomponent are also hidden.
User requirements	User requirements represent an abstraction between user requirements and specific components and attributes. You can't map a user requirement to an item. For example, a customer is shopping for a home theater system. The sales representative might ask about the size of the room where the customer plans to install the system, to determine how many watts are required. In this example, the room size can be a user requirement

Term	Description
	that helps determine the appropriate attribute value for a specific component. You can hide user requirements so that they aren't displayed to the user during a configuration session. Attributes, subcomponents, and user requirements that are related to the user requirement are also hidden. You can write a condition to control whether a user requirement can be hidden. You must write the condition by using Optimization Modeling Language (OML) syntax.
BOM lines	BOM lines represent the individual materials of the components in the product configuration model. On the **BOM line details** page, all items are available for selection. A condition can be added to the BOM line, so that the BOM lines that are selected for a distinct product variant can vary, based on the user's selection when the product configuration model is set up. Conditions are expressions that must be met for attributes, BOM lines, and route operations to be included in a product configuration model. On the **BOM line details** page, you can select a distinct value. Alternatively, you can map to an attribute that the value is selected for when the product configuration model is set up.
Route operations	On the **Route operation details** page, you can select a distinct value. Alternatively, you can map to an attribute that the value is selected for when the product configuration model is set up. Conditions are written like expression constraints. Conditions are expressions that must be met for attributes, BOM lines, and route operations to be included in a product configuration model.

Last updated on 05/06/2026

Set up a product configuration model

This article describes the steps for setting up and creating a product configuration model.

Expand table

Task	Description
Create a product master.	Create a product master from the **Product master** list. Release the product master to all relevant companies. For a product master that is used as a version for a product configuration model or as a subcomponent, **Constraint-based configuration** must be selected as the configuration technology, and the configuration dimension must be selected only for the product dimension group.
Create components.	Create components on the **Components** page. Components are the building blocks of a product configuration model and can be reused in multiple product configuration models.
Create attribute types.	Create attribute types on the **Attribute types** page. Attribute types specify the set of data types for all attributes that are used in product configuration models. Attributes of the **Boolean**, **Text** with a fixed list, and **Integer** with a range types list the set of values that are available when you configure a product variant based on a product configuration model.
Create a product configuration model.	Create a product configuration model on the **New product configuration model** page.
Add attributes to a product configuration model.	Create an attribute on the **Attributes** FastTab on the **Constraint-based product configuration model details** page. Attributes describe the features of the product configuration model.
Add constraints to a product configuration model.	Create constraints on the **Constraints** FastTab on the **Constraint-based product configuration model details** page. Constraints are limitations that a product configuration must satisfy. Constraints are either expression constraints or table constraints.
Add subcomponents to a product configuration model.	Create subcomponents on the **Subcomponents** FastTab on the **Constraint-based product configuration model details** page. Subcomponents are related to components and are linked to items that represent the subcomponent.
Add user requirements to a product configuration model.	User requirements are similar to subcomponents, but they don't reference an item. You can think about user requirements as a phantom BOM. Any BOM lines or route operations that are placed directly under the user requirement will be collapsed into the parent component.
Add BOM lines to a product configuration	Create BOM lines on the **BOM lines** FastTab on the **Constraint-based product configuration model details** page. BOM lines represent a part that is required to

Task	Description
model.	build a variant of the product.
Add route operations to a product configuration model.	Create route operations on the **Route operations** FastTab on the **Constraint-based product configuration model details** page. Route operations represent a step in a sequence of steps that are performed to make a variant of the product.
Create an active version for a product configuration model.	Create an active version of the product configuration model on the **Versions** page. A version is the relationship between a product configuration model and a product master. A product configuration model that has an active version can be configured from a sales order, sales quotation, purchase order, or production order.
Test a product configuration model.	Test the product configuration model from either the **Constraint-based product configuration model details** page or the **Product configuration models list** page. Testing the product configuration models simulates the product model configuration process that occurs during order handling.
Create product configuration model template.	Create a product configuration model template on the **Configuration templates** page. A configuration template includes values for attributes in the product configuration model. Select the attribute values on the **Configure line** page. You can select to load a product model configuration template during product model configuration.
Configure an item.	Product configuration models can be configured from a sales order, sales quotation, purchase order, or production order.

Last updated on 07/01/2026

Product configuration model calculations

This article describes how to create calculations for attributes in a product configuration model.

Prerequisites

Calculations are used in a product configuration model to calculate the configuration values for a product. Before you can start to set up calculations, the related product configuration model must exist. For an overview of the setup process for configuration models and the related tasks, see Set up a product configuration model.

Create a calculation

A calculation consists of an expression and a target attribute. Learn more in Calculations for product configuration models FAQ.

To create a calculation for an existing product model, follow these steps:

1. Go to **Product information management** > **Common** > **Product configuration models**.
2. Open a product configuration model, and then select **Edit**.
3. On the **Calculations** FastTab, select **Add** to add a calculation, and then set the following fields:
    - **Name** – Enter a name for the calculation.
    - **Description** – Enter a description of the calculation.
    - **Target attribute** – Select the attribute that you're making the calculation for.
4. Select **Edit expression**.
5. In the **Enter a calculation** dialog box, add the required attributes, operators, and values to the expression. For more information about how to work with these elements, see Expression constraints and table constraints in product configuration models.
6. When your expression is ready, select **OK**.

Calculation examples

This section provides a few examples that show how calculations work.

Example 1

The target attribute is Boolean, and the calculation uses the following conditional expression:

```
If[(decimalAttribute1 / decimalAttribute2) < 1, True, False]
```

This expression returns a value of *True* to the target attribute if `decimalAttribute2` is greater than or equal to `decimalAttribute1`. Otherwise, it returns a value of *False*.

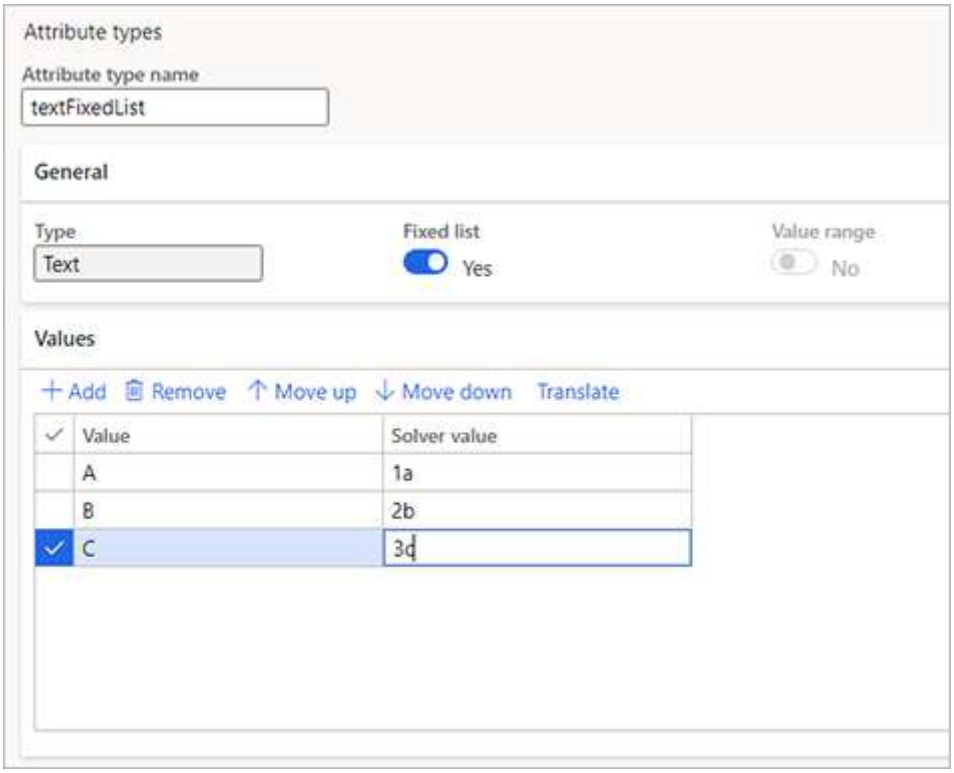
Example 2

This example uses the text attribute `textFixedList` as the target attribute. This attribute contains the following fixed list.

Expand table

Value	Solver value
A	1a
B	2b
C	2c

The following screenshot shows how the settings for this attribute might look in your system.



The attribute is used in the following conditional statement:

```
If[integerAttribute < 150, 0, 2]
```

If `integerAttribute` is less than 150, this statement returns the text value of the first record in the fixed list, *A*. Otherwise, it returns the text value of the third record in the fixed list, *C*.

> **Note**
>
> The fixed list is equivalent to a zero-based enumeration (enum), and its values are accessed by the appropriate integer value. Therefore, the first fixed list value (*A*) is matched to *0*, the second value (*B*) is matched to *1*, and the third value (*C*) is matched to *2*.

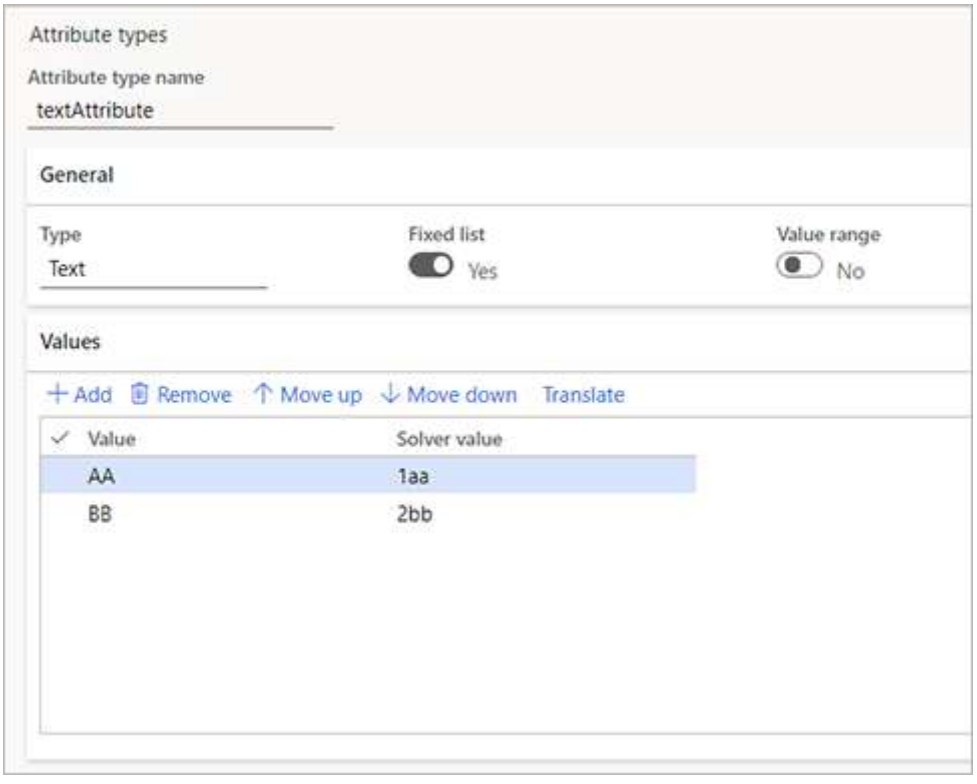
Example 3

This example uses the `textFixedList` target attribute from the previous example. It also uses another text attribute, `textAttribute`, that contains the following fixed list.

Expand table

Value	Solver value
AA	1aa
BB	2bb

The following screenshot shows how the settings for this attribute might look in your system.



The value for the `textFixedList` attribute is calculated by using the following conditional statement:

```
If[textAttribute == "1aa", 0, 2]
```

If the `textAttribute` value has a solver value that equals *1aa*, this expression returns the text value of the first record in the `textFixedList` fixed list, *A*. Otherwise, it returns the text value of the third record in the `textFixedList` fixed list, *C*.

> **Note**
>
> - The conditional statement must use the solver value of the attribute.
> - Only fixed-list text attributes can be used in calculations.

Related information

- Calculations for product configuration models FAQ
- Add an expression constraint to a product configuration model
- Product configuration models overview

Last updated on 12/30/2025

Calculations for product configuration models FAQ

Article • 06/15/2022

This article describes calculations for product configuration models and explains how to use calculations together with constraints.

Calculations can be used for arithmetic or logical operations. They complement expression constraints in product configuration models. You can define calculations on the **Constraint-based product configuration model details** page and then build expressions for the calculations in the expression editor. For more information, see Create calculations.

What is a calculation?

A calculation is an element that you can use in a product configuration model. Calculations complement constraints by letting you use decimal numbers to calculate values when you configure a product. Additionally, calculations have a larger set of available operators than constraints have.

Like a constraint, a calculation is associated with a specific component in a product configuration model, and can't be reused by or shared with another component. One important difference between calculations and constraints is that calculations are imperative (unidirectional), whereas constraints are declarative (bi-directional). For more information about constraints, see Expression constraints and table constraints in product configuration models.

A calculation consists of a target attribute and a calculation expression.

What is a target attribute?

A target attribute is an attribute that receives the result of the calculation expression.

In the following expression, the target attribute is a tablecloth measurement:

Expression: If[decimalAttribute1 <= decimalAttribute2, True, False]

DecimalAttribute1 is the table length, and **decimalAttribute2** is the tablecloth length. The expression returns the value **True** to the target attribute if **decimalAttribute2** is greater than or equal to **decimalAttribute1**. Otherwise, the expression returns **False**.

Therefore, the tablecloth measurement is acceptable if the tablecloth length is the same as or exceeds the length of the table.

What attribute types can be set to target attributes?

All attribute types that the product configurator supports can be set to target attributes, except text without a fixed list.

Can the value of a target attribute restrict the values of the input attributes in a calculation?

No, the value of a target attribute can’t restrict the values of the input attributes, because calculations are unidirectional. Therefore, the value of the target attribute is set based on changes in the value of the input attributes, but a change in the value of the target doesn’t affect the value of the input attributes. This behavior differs from the behavior for constraints. Constraints occur in both directions.

Example

In the following expression, the target for the calculation is the length of a power cord, and the input value is a color:

Expression: [If Color == "Green", 1.5, 1.0]

When you configure the item, the length of the power cord is set to **1.5** if you specify **Green** as the value of color attribute. If you specify any other color, the length is set to **1.0**. However, because calculations are unidirectional, the calculation doesn't set the value of the color attribute to **Green** if you specify a length of **1.5**.

What happens if a calculation has a target attribute of the integer type but a calculation generates a decimal number?

If a target attribute is of the integer type, but a calculation generates a decimal number, only the integer part of the calculated result is returned. The decimal part is removed, and the result isn't rounded. For example, a result of 12.70 is shown as 12.

When do calculations occur?

Calculations occur when a value has been provided for all input attributes.

Can I overwrite the value that is calculated for the target attribute?

You can overwrite the value that is calculated for the target attribute, unless the target attribute is set as hidden or read-only.

How do I set a target attribute as hidden or read-only?

To set an attribute as hidden or read-only, follow these steps.

1. Click **Product information management** > **Products** > **Product configuration models**.
2. Select a product configuration model, and then, on the Action Pane, click **Edit**.
3. On the **Constraint-based product configuration model details** page, select the attribute to use as a target attribute.
4. On the **Attributes** FastTab, select **Hidden** or **Read-only**.

Can a calculation overwrite the values that I set?

No. The values that you set when you configure a product are the values that are used. The calculation that occurs when the input values in a calculation are changed can't overwrite the values that you provide for a specific attribute.

What happens if I remove an input value in a calculation?

If you remove an input value in a calculation, the value of the target attribute is also removed.

Why do I receive an error message that says that my model is in contradiction?

This message is shown when a calculation includes an error, or when a contradiction exists in one or more constraints. For more information about contradictions in constraints, see Expression constraints and table constraints in product configuration models. Here are some situations where errors can occur in calculations:

- A value is divided by 0 (zero).
- A conflict exists between the following two elements:
  - The values that are available for an attribute and are limited by a constraint
  - A value that is generated by a calculation
- The values that are returned by the calculation are outside the domain of the attribute. An example is an integer from [1..10] that is calculated to 0.

Why do I receive an error message even though I successfully validated my product model?

Calculations aren't included in the validation. You must test the product configuration model to find errors in calculations. To test a product configuration model, follow these steps.

1. Click **Product information management** > **Products** > **Product configuration models**.
2. Select a product configuration model, and then, on the Action Pane, in the **Run** group, click **Test**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Configuration rules

This article provides general information about configuration rules. Configuration rules define relationships between items in a bill of materials (BOM) for products that use the dimension-based configuration technology.

Configuration rules are available when you define dimension-based configuration models. Configuration rules are used to either enforce or prohibit specific item combinations in a bill of materials (BOM). After a BOM has been created and the relevant items have been assigned to their respective configuration groups, one or more configuration rules can be defined. If two items belong together, the **Select** operator is used to ensure inclusion. If two items are mutually exclusive, the **Deselect** operator is used to ensure exclusion.

Note: This information applies only to product masters that use the dimension-based configuration technology.

Existing configurations aren't affected by subsequent changes to the configuration rules. However, it's important that you set the rules before you define a new configuration, and that you check the rules if you think they have been changed.

Note: For the **Select** method, the derived configuration group, item number, and configuration are automatically selected. For the **Deselect** method, the derived configuration group, item number, and configuration can't be selected.

Related information

Dimension-based product configuration overview

Last updated on 07/01/2026

Expression constraints and table constraints in product configuration models

Article • 05/30/2023

This article describes the use of expression constraints and table constraints. Constraints control the attribute values that you can select when you configure products for a sales order, sales quotation, purchase order, or production order. You can use expression constraints or table constraints, depending on how you prefer to build the constraints.

What are expression constraints?

Expression constraints are characterized by an expression that uses arithmetic and Boolean operators and functions. An expression constraint is written for a specific component in a product configuration model. It can't be reused by or shared with another component. However, the expression constraints for a component can reference attributes of the component's subcomponents.

What are table constraints?

Table constraints list the combinations of values that are allowed for attributes when you configure a product. Table constraint definitions can be used generically. When you create a table constraint for a component in a product configuration model, you select a table constraint definition. To create the combinations that are allowed, you add attributes of specific types to the components. Each attribute type has a specific value.

Example of a table constraint

This example shows how you can limit the configuration of a speaker to specific cabinet finishes and fronts. The first table shows the cabinet finishes and fronts that are generally available for configuration. The values are defined for the **Cabinet finish** and **Front grill** attribute types.

Expand table

Attribute type	Values
Cabinet finish	Black, Oak, Rosewood, White

Attribute type	Values
Front grill	Black, Metal, White

The next table shows the combinations that are defined by the **Color and finish** table constraint. By using this table constraint, you can configure a speaker that has an oak finish and a black grill, a Rosewood finish and a white grill, and so on.

Expand table

Finish	Grill
Oak	Black
Rosewood	White
White	Black
White	White
Black	Black
Black	Metal

You can create system-defined and user-defined table constraints. Learn more in System-defined and user-defined table constraints.

What syntax should be used to write constraints?

You must use Optimization Modeling Language (OML) syntax when you write constraints. The system uses Microsoft Solver Foundation constraint solver to solve the constraints.

Should I use table constraints or expression constraints?

You can use either expression constraints or table constraints, depending on how you prefer to build the constraints. You build a table constraint as a matrix, whereas an expression constraint is an individual statement. When you configure a product, it doesn't matter what kind of constraint is used. The following example shows how the two methods differ.

When you configure a product by using the following constraint setups, these combinations are allowed:

- A product in the color Black, and in size 30 or 50
- A product in the color Red and in size 20

Table constraint setup

Expand table

Color	Size
Black	30
Black	50
Red	20

Expression constraint setup

(Color == "Black" & (size == "30" | size == "50")) | (color == "Red" & size = "20")

Should I use operators or infix notation when I write expression constraints?

You can write an expression constraint by using either the available prefix operators or infix notation. For the **Min**, **Max**, and **Abs** operators, you can't use infix notation. These operators are included as standard operators in most programming languages.

What operators and infix notation can I use when I write expression constraints?

The following tables list the operators and infix notation that you can use when you write an expression constraint for a component in a product configuration model. The examples in the first table show how to write an expression by using either infix notation or operators.

Expand table

Operator	Description	Syntax	Examples
Implies	If condition a is true, then apply the contraint b.	Implies[a, b], infix: a -: b	• **Operator:** Implies[x != 0, y >= 0] • **Infix notation:** x != 0 -: y >= 0
And	This is true only if all conditions are true. If the number of conditions is 0 (zero), it produces **True**.	And[args], infix: a & b & ... & z	• **Operator:** And[x == 2, y <= 2] • **Infix notation:** x == 2 & y <= 2
Or	This is true if any condition is true. If the number of conditions is 0 (zero), it produces **False**.	Or[args], infix: a | b | ... | z	• **Operator:** Or[x == 2, y <= 2] • **Infix notation:** x == 2 | y <= 2
Plus	This sums its conditions. If the number of conditions is 0 (zero), it produces **0**.	Plus[args], infix: a + b + ... + z	• **Operator:** Plus[x, y, 2] == z • **Infix notation:** x + y + 2 == z
Minus	This negates its argument. It must have exactly one condition.	Minus[expr], infix: -expr	• **Operator:** Minus[x] == y • **Infix notation:** -x == y
Abs	This takes the absolute value of its condition. It must have exactly one condition.	Abs[expr]	**Operator:** Abs[x]
Times	This takes the product of its conditions. If the number of conditions is 0 (zero), it produces **1**.	Times[args], infix: a * b * ... * z	• **Operator:** Times[x, y, 2] == z • **Infix notation:** x * y * 2 == z
Power	This takes an exponential. It applies exponentiation from right to left. (In other words, it's right-	Power[args], infix: a ^ b ^ ... ^ z	• **Operator:** Power[x, 2] == y • **Infix notation:** x ^ 2 == y

Operator	Description	Syntax	Examples
	associative.) Therefore, **Power[a, b, c]** is equivalent to **Power[a, Power[b, c]]**. **Power** can be used only if the exponent is a positive constant.		
Max	This produces the largest condition. If the number of conditions is 0 (zero), it produces **Infinity**.	Max[args]	**Operator:** Max[x, y, 2] == z
Min	This produces the smallest condition. If the number of conditions is 0 (zero), it produces **Infinity**.	Min[args]	**Operator:** Min[x, y, 2] == z
Not	This produces the logical inverse of its condition. It must have exactly one condition.	Not[expr], infix: !expr	• **Operator:** Not[x] & Not[y == 3] • **Infix notation:** !x!(y == 3)

The examples in the next table show how to write infix notation.

Expand table

Infix notation	Description
x + y + z	Addition
x * y * z	Multiplication
x - y	Binary subtraction is translated the same as binary addition where there is a negated second.
x ^ y ^ z	Exponentiation that has right associativity
!x	Boolean not
x -: y	Boolean implication
x	y

Infix notation	Description
x & y & z	Boolean and
x == y == z	Equality
x != y != z	Distinct
x < y < z	Less than
x > y > z	Greater than
x <= y <= z	Less than or equal to
x >= y >= z	Greater than or equal to
(x)	Parentheses override default precedence.

Why aren't my expression constraints validated correctly?

You can't use reserved keywords as solver names for attributes, components, or subcomponents in a product configuration model. Here is a list of the reserved keywords that you can't use:

- Ceiling
- Element
- Equal
- Floor
- If
- Less
- Greater
- Implies
- Log
- Max
- Min
- Minus
- Plus
- Power
- Times
- Slot
- Model
- Decision

- Goal

Related information

Create an expression constraint

Add a calculation to a product configuration model

Feedback

Was this page helpful? Yes No

Provide product feedback

Solver strategy for product configuration

Article • 06/29/2022

This article describes how you can use the solver strategy to improve the performance of product configuration.

The concept of solver strategies was first introduced in Cumulative update 7 (CU7) for Microsoft Dynamics AX 2012 R2. It was extended in Cumulative update 8 (CU8) for Microsoft Dynamics AX 2012 R3 and finance and operations apps, Enterprise edition 7.3.

The solver strategy concept now consists of the following strategies:

- Default
- Minimal domains first
- Top-down
- Z3

Solver strategy

A product configuration model can be formulated as a constraint satisfaction problem (CSP). Microsoft Solver Foundation (MSF) provides two types of solver strategies to solve the CSPs that can be used from product configuration models. These solver strategies rely on heuristics, which are used to determine the order that the variables of the CSPs are considered in when the problem is being solved. Heuristics can significantly affect performance when a problem or class of problems is being solved.

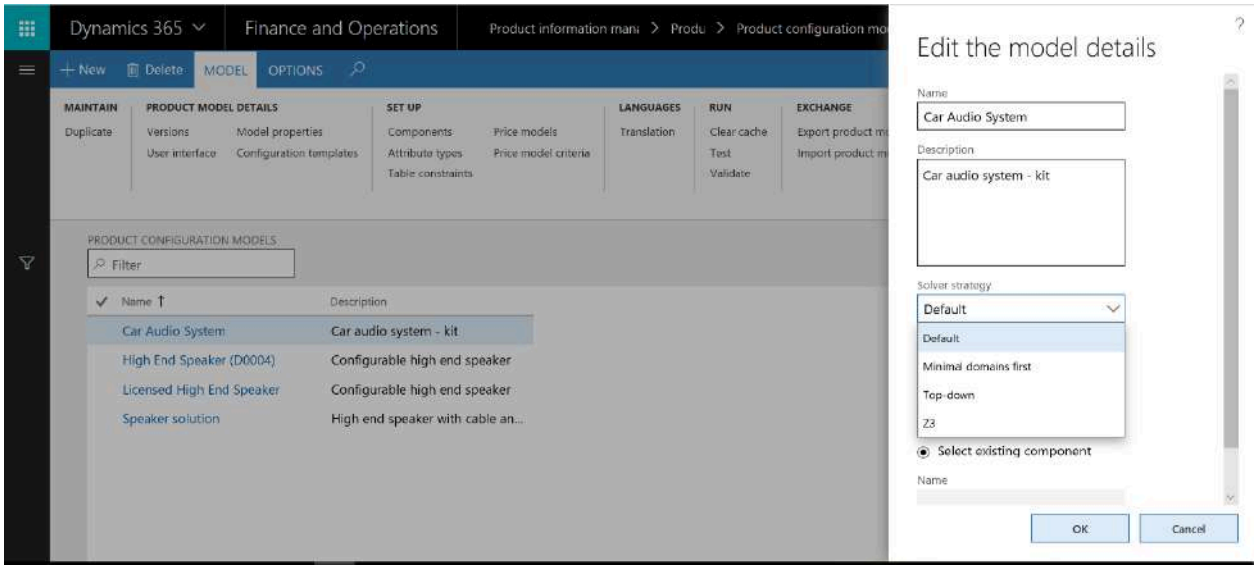
The solver strategy for product configuration models determines which solver is used with heuristics. The **Default**, **Minimal domains first**, and **Top-down** strategies use the two solvers from MSF, whereas the **Z3** strategy uses the Z3 solver.

Real customer implementation studies have shown that a change in the solver strategy for a product configuration model can reduce the response time from minutes to milliseconds. Therefore, it's worth the effort to try different solver strategies to find the most efficient strategy for your product configuration model.

Change the settings for the solver strategy

To change the solver strategy, on the **Product configuration models** page, on the Action Pane, select **Model properties**. Then, in the **Edit the model details** dialog box,

select a solver strategy.



Currently, there is no logic that automatically detects which solver strategy will be the most efficient strategy for constraint-based product configuration. Therefore, you must try the solver strategies one by one.

The following table provides recommendations about the solver strategy to use in various scenarios.

Expand table

Solver strategy	Use the strategy in this scenario
Default	The **Default** strategy has been optimized to solve models that rely on table constraints. Customer implementation studies have shown that this strategy is the most efficient strategy in scenarios where table constraints are used extensively.
Minimal domains first	The **Minimal domains first** and **Top-down** strategies are closely related. Customer implementation studies have shown that the **Top-down** strategy, outperforms the **Minimal domains first** strategy. However, the **Minimal domains first** strategy is kept in the product for backward compatibility. Both these solver strategies have been shown to be more efficient at solving models that contain several arithmetic expressions where no table constraints are used. However, in some cases, the **Default** strategy outperforms these two strategies. Therefore, remember to try each strategy.
Top-down	The **Minimal domains first** and **Top-down** strategies are closely related. Customer implementation studies have shown that the **Top-down** strategy, outperforms the **Minimal domains first** strategy. However, the **Minimal domains first** strategy is kept in the product for backward compatibility. Both these solver strategies have been shown to be more efficient at solving models that contain several arithmetic expressions where no table constraints are used. However, in some cases, the **Default** strategy outperforms these two strategies. Therefore, remember to try each strategy.

Solver strategy	Use the strategy in this scenario
Z3	We recommend that you use the **Z3** strategy as the default solver strategy. If you're concerned about performance and scalability, you can evaluate the other strategies.

Related information

Product configuration overview

Heuristics

Constraint Satisfaction Problem

Feedback

Was this page helpful? Yes No

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Reuse product configurations

You can specify that you want to automatically reuse an existing configuration for a product. Then, when a user has completed a configuration session, the system verifies whether a configuration that matches the user's selections already exists. If a matching configuration is found, the configuration ID, corresponding bill of materials (BOM), and route are reused.

Requirements for reusing configurations

To enable configurations to be reused, you must specify the following information for the components and attributes on the **Product configuration model details** page:

- **Components and subcomponents** – On the **General** FastTab, in the **Reuse configurations** field, select **Yes**.
- **Attributes** – On the **Attributes** FastTab, select the **Include in reuse** option. This option appears only when the related component is enabled for reuse. If you don't select any attributes for reuse, the configuration is always reused, regardless of the user's selections during a configuration session. The attribute values in the existing configuration must match the user's selections. For example, if the user selects **Blue** as the color during a configuration session, the system verifies whether an existing configuration of the component has the color blue.

Resetting configuration reuse

When you reset configuration reuse, previously created configurations are no longer considered. You might want to reset configuration reuse if the BOM or route was changed, but no related attributes were changed. You reset configuration reuse on the **General** FastTab for the component.

Last updated on 07/01/2026

System-defined and user-defined table constraints

This article explains the two types of table constraints for components in a product configuration model - user-defined and system-defined. Table constraints represent matrices of the allowed attribute combinations, where each row defines one set of possible attribute values.

Table constraints represent matrices of the combinations of attributes that are allowed for components in a product configuration model. Each row in the table defines one set of possible attribute values. You can declare two types of constraints in a product configuration model:

- **Expression constraint** – Create an expression that defines relations between attributes to guarantee that only compatible values can be selected during product configuration.
- **Table constraint** – Create a table that defines all the combinations that are allowed for a specified set of attributes. Two types of table constraints are available: user-defined table constraints and system-defined table constraints.

This article describes user-defined and system-defined table constraints for components in a product configuration model.

User-defined table constraints

A user-defined table constraint is a type of matrix that is used to describe the combinations of attribute values that are defined by attribute types. For example, if you produce speakers, you can include columns for the cabinet finish and the front grill in the user-defined table constraint. The attribute type for the cabinet finish has four values, and the attribute type for the front grill has three values. Therefore, if constraints aren't used, there are 4 × 3 = 12 possible combinations. However, in this example, only six combinations are allowed, as shown in the following table. This information is displayed on the **Allowed combinations** tab on the **Edit table constraint** page.

Expand table

Cabinet finish	Front grill
Black	Black
Black	Metal
Oak	Black

Cabinet finish	Front grill
Rosewood	White
White	Black
White	White

User-defined table constraints are defined by static table input that works the same way as an expression constraint. When you use a user-defined table constraint, the advantage is that tables are often easier to create, understand, and maintain than long expression constraints.

System-defined table constraints

A system-defined table constraint creates a dynamic mapping between an attribute type and a field in a table. When a system-defined table constraint is included in a product configuration model, the mapping guarantees that the data in the table is shown instead of the values of the attribute type. The result is a dynamic constraint, because the table contents can be modified (for example, by other modules).

When you create a system-defined table constraint, you select a table, optionally define the query to use, and then associate attribute types to the fields in the selected table. The types of fields must match the types of attribute types.

Before a table constraint can take effect on a product configuration model, the table constraint must be included in a constraint on one of the model's components. The procedure is to create a new constraint, select the table constraint type, and then select the table constraint definition to use. Finally, all fields in the table constraint must be mapped to attributes in the product configuration model.

Related information

Product configuration models overview

Last updated on 07/01/2026

Create product model components

Article • 09/29/2021

Constraint-based product configuration models use components as building blocks to establish the product model structure. The components must be created before a product model can be built. The demo data company used to create this procedure is USMF.

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. On the Action Pane, select **Model**.
3. Select **Components**.
4. Select **New**.
5. In the **Name** field, type a value.
6. In the **Description** field, type a value.
7. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a product configuration model

Article • 09/29/2021

This procedure shows how to create a product configuration model and enter basic information such as attributes and subcomponents. The demo data company used to create this procedure is USMF.

Create a product model

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. Select **New**.
3. In the **Name** field, type a value.
4. In the **Description** field, type a value.
5. In the **Solver strategy** field, select an option.

    - The solver strategy determines how the constraints in a constraint-based product configuration model are processed. This selection can have an impact on the performance of the product configuration model.

6. In the **Name** field, type a value.

    - The root component represents the product configuration model, but it can also be used in other product models.

7. Select **OK**.
8. In the **Reuse configurations** field, select an option.

    - If the reuse configurations parameter is set to Yes, the system will check for identical configurations after each configuration session and reuse if an exact match is found.

Add attributes

1. Expand the **Attributes** section.
2. Select **Add**.
3. In the list, mark the selected row.
4. In the **Name** field, type a value.
5. In the **Solver name** field, type a value.

- The solver name is used by the constraint solver of the product configurator. It must not include spaces or special characters except _ (underscore).

6. In the **Description** field, type a value.

    - The description text is displayed to the configuration user and can therefore serve as help in selecting the right attribute value.

7. In the **Attribute type** field, enter or select a value.

    - The attribute type determines which values are available for the attribute.

8. Select the **Include in reuse** check box.

    - This option is only available when the Reuse configurations option is selected. Including the attribute in the reuse check box means that this attribute will be considered when the system is looking for an exact match.

Add subcomponents

1. Expand the **Subcomponents** section.
2. Select **Add**.
3. In the list, mark the selected row.
4. In the **Name** field, type a value.
5. In the **Solver name** field, type a value.
6. In the **Description** field, type a value.
7. In the **Component** field, enter or select a value.

    - Each subcomponent must reference a component definition. This design supports reusable components and ensures that once a component has been defined it can be used in many product models.

8. Select **Save**.
9. Select **BOM line details**.

    - The BOM line details form enables the user to select the required properties for the subcomponent. Each property can be given a fixed value or mapped to an attribute. Mapping to an attribute will result in the BOM line property getting different values depending on the configuration selection.

10. In the **Item number** field, enter or select a value.

    - Each subcomponent represents a configurable product master with constraint-based configuration technology. The reference is made through the item number.

11. Select the **Set** check box.
12. Select **Yes** in the **Calculation** field.

    - Setting the calculation option ensures that the product will be included when running a cost calculation for the product.

13. Select the **Setup** tab.
14. Select the **Set** check box.
15. In the **Quantity** field, enter a number.

    - The quantity field determines how much of this product will be consumed in the configured product.

16. Select the **Set** check box.
17. In the **Per series** field, enter a number.
18. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add an expression constraint to a product configuration model

Article • 09/29/2021

This procedure shows how you can add a new constraint expression to a product configuration model. It shows how you can mandate that corner protection must be applied to a speaker if the user has selected a front grill in metal. The procedure uses the High end speaker component in the demo company USMF.

Create an expression constraint

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, find and select the desired record.
    - This example uses the high end speaker model.
3. In the list, select the link in the selected row.
4. Expand the **Constraints** section.
5. Select **Add**.
6. Select **Create**.
7. In the **Name** field, type a value.

Enter expression

1. Select **Edit expression**.
    - If you unlock the user interface in the task recording at this stage, you can use IntelliSense and the list of symbols to build the constraint expression .
2. In the **ConstraintBody** field, enter 'Implies[FrontGrill=="Metal", CornerProtection] '.
    - This expression logic states: If the Front grill is metal, then the corner protection option must be selected.
3. Select **Validate**.
    - The validate function runs through the constraint expression and checks for syntax errors.
4. Select **Close**.

5. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add a calculation to a product configuration model

Article • 09/29/2021

This procedure shows how to add a new calculation to a product configuration model. It shows how you can create a logical expression using the "If" operator to set a speaker height to 10 for white speakers and 15 for all other cabinet finishes. The procedure uses the High end speaker component in the demo company USMF.

Add a calculation

Create calculation expression

1. Click Edit expression.
2. In the ConstraintBody field, enter 'If[CabinetFinish=="White", 10, 15]'.
3. Click Validate.
4. Click Close.
5. Click OK.

Feedback

Was this page helpful? Yes No

Provide product feedback

Maintain BOM for a product configuration model

Article • 09/29/2021

Running this procedure requires an existing product configuration model. The High end speaker model in the demo company USMF is used to create this procedure.

Add a BOM line

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, find and select the desired record.
    - Select the High end speaker for this procedure.
3. In the list, select the link in the selected row.
4. Expand the **BOM lines** section.
5. Select **Add**.
6. In the **Name** field, type a value.
7. In the **Description** field, type a value.
8. Select **Save**.

Add BOM line details

1. Select **BOM line details**.
2. In the **Item number** field, enter or select a value.
    - For example, you can select the item M0055.
    - For each BOM line property, you can select if it takes a fixed value or is mapped to an attribute.
3. Select the **Set** check box.
4. Select *Yes* in the **Calculation** field.
    - Setting the **Calculation** property to *Yes* ensures that the BOM line is included in cost calculations.
5. Select the **Setup** tab.
6. Select the **Set** check box.
7. In the **Quantity** field, enter a number.

- The quantity field determines how much of the item that will be included in the BOM. This could be an obvious candidate for an attribute mapping.

8. Select the **Dimension** tab.

    - Verify if any of the product dimensions are active, and therefore must have a value or attribute assigned.

9. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Maintain route for a product model

Article • 09/29/2021

Running this procedure requires that a product configuration model exists. This procedure uses the High end speaker model in the demo company USMF to walk you through the process.

Add a route operation

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, find and select the desired record.

    - Select the High end speaker model for this exercise.

3. In the list, select the link in the selected row.
4. Expand the **Route operations** section.
5. Select **Add**.
6. In the **Name** field, type a value.
7. In the **Description** field, type a value.
8. Select **Save**.

Enter route operation details

1. Select **Route operation details**.
2. In the **Operation** field, enter or select a value.
3. In the **Oper. No.** field, enter a number.

    - Operation numbers determine the route sequence.
    - Each property on a route operation can get a static value or be mapped to an attribute. Mapping to an attribute will result in the value being set as part of the configuration.

4. In the **Route group** field, enter or select a value.

    - The route group determines essential behavior for costing, consumption, and setup.

5. Select the **Setup** tab.
6. Select the **Times** tab.
7. In the **Process qty.** field, enter a number.

- Determine how many will be processed during one operation.

8. In the **Hours/time** field, enter a number.
    - Enter the time ratio.
9. Select the **Set** check box.
10. In the **Run time** field, enter a number.
    - Determine the processing time for the quantity that you have specified.
11. Select the **Resource requirements** tab.
12. Select **Add**.
13. In the list, mark the selected row.
14. In the **Requirement type** field, select an option.
    - Decide if you want to specify specific resources or capabilities that they must possess.
15. In the **Requirement** field, enter or select a value.
16. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Attribute-based sales prices for constraint-based product configuration

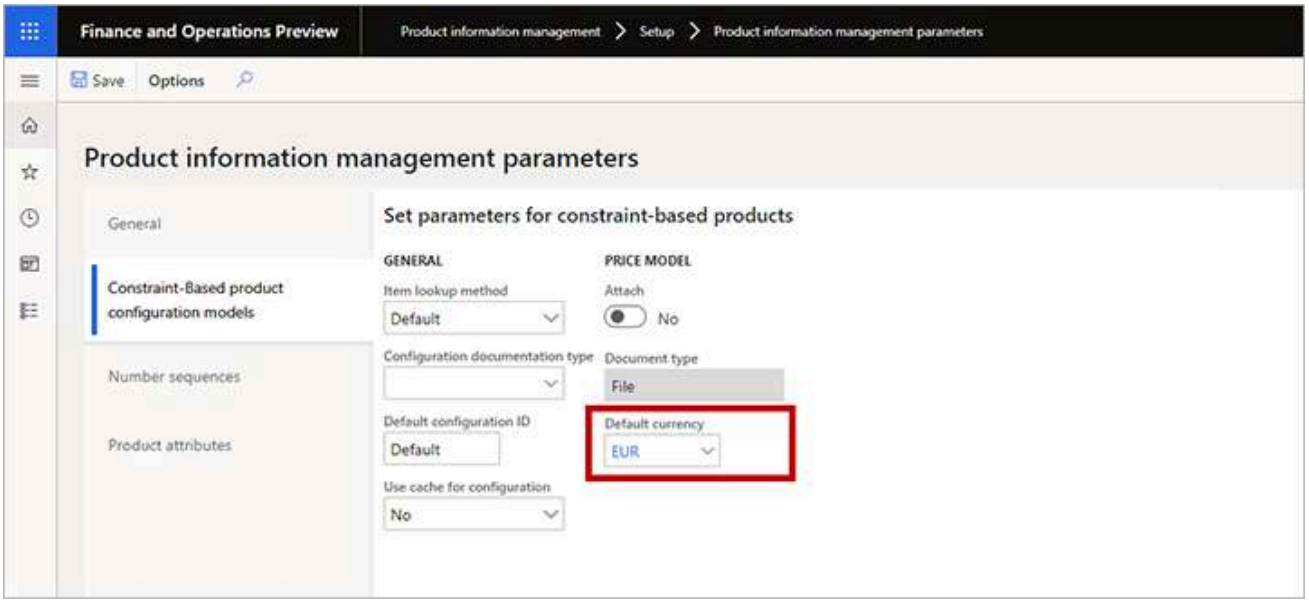
This article describes how to build sales price models with sales prices based on components and attributes rather than on the physical bill of material and the route. You can build several sales price models for each product configuration model.

Set relevant product information management parameters

Before you start building your price models, you must define a default currency, which is used when you build your sales price models. You can also choose whether to attach a Microsoft Excel file with a price breakdown for all order or quotation lines. The price breakdown will enable you to share details with customers about how you arrived at a specific sales price for a configured product.

To set your default currency:

1. Go to **Product information management** > **Setup** > **Product information management parameters**.
2. Open the **Constraint-Based product configuration models** tab.
3. Open the **Default currency** drop-down list and select your currency.



4. If you'd like to attach an Excel file with a price breakdown for all order or quotation lines, then in the **Price model** section, set **Attach** to *Yes*.

Build your sales price models

To build a sales price model:

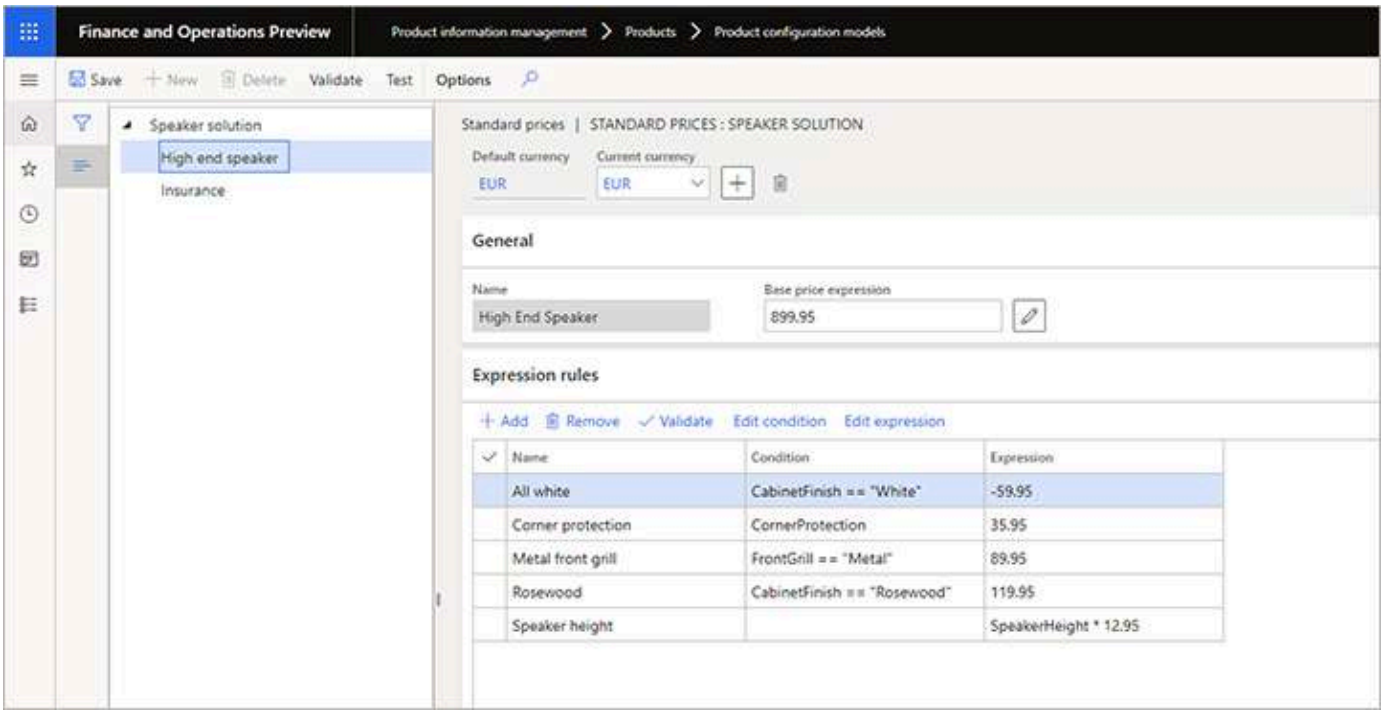
1. Go to **Product information management** > **Products** > **Product configuration models**.
2. Select the target product configuration model.
3. On the Action pane, open the **Model** tab and, from the **Set up** group, select **Price models**.
4. The **Price models** page opens.
5. Select a price model or add a new one to the grid.
6. Select **Edit** to open the edit page for your selected model, which provides the following features:

    - The header of the form shows the default currency and lets you add new currencies for your price setup.
    - The left pane shows all the components and user requirements of the product model. Each node in the product model tree can have one base-price expression and an optional number of expression rules. An expression rule consists of a condition and an expression and each expression rule covers a product option that helps control the price of the product.
    - When you build your conditions and expressions, you have the same operators available as for calculations in a product model. The expression editor also supports both conditions and expressions.

7. Select a node in the left column and then use the features described in the previous step to establish pricing rules for it (see also the example provided after this procedure). Repeat this step for each node as needed.

The following example shows a base price of a static number of 899.95 EUR, which can be modified by one or more of the following five expression rules, depending on the configuration selected by the customer:

- For white cabinet finish, subtract 59.95 EUR.
- For corner protection, add 35.95 EUR.
- For a metal front grill, add 89.95 EUR.
- For rosewood cabinet finish, add 119.95 EUR.
- Add 12.95 EUR for each unit of speaker height.



Add support for multiple currencies

When a configurable product is sold, the system checks whether the prices have been set explicitly in the currency of the customer. If so, the explicit values are used. If not, the system uses the currency exchange rates established for the sales company to convert the default currency value to the customer's currency.

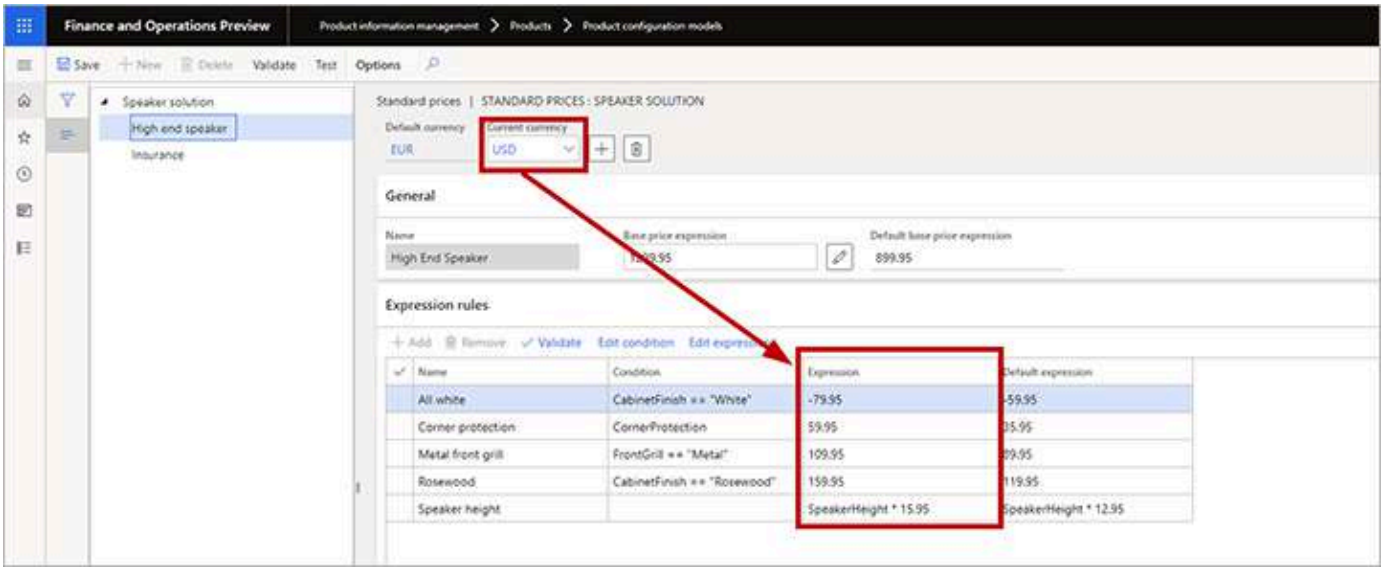
To add explicit prices in an additional currency:

1. Open the edit page for your price model, as described in Build your sales price models.
2. Select the **Add** button in the header of the price model to open the **Currencies** drop-down dialog box, which lists the available currencies.
3. Select the currency you want to add in the **Currencies** drop-down dialog box and then select **OK**.
4. The **Current currency** drop-down list now includes the currency that you just added, plus any other currencies that may have been added previously. Select your new currency and notice that the grid in the **Expression rules** section now includes two expression fields:
    - **Expression** - Shows the expression (or constant value) for finding the price using the currency currently selected for **Current currency**.
    - **Default expressions** - Shows the expression (or constant value) for finding the price using the default currently (shown in the **Default currency** field).

> **Note**
>
> The **Condition** field for the expression rules is "owned" by the default currency, which means that you can't modify the condition for other currencies. You also can't add new expression rules while a currency other than the default currency is selected as the **Current currency**.

5. Edit values in the **Expression** column as needed for the current currency.

In the example below, *EUR* is the default currency, and *USD* has been added as an additional currency.



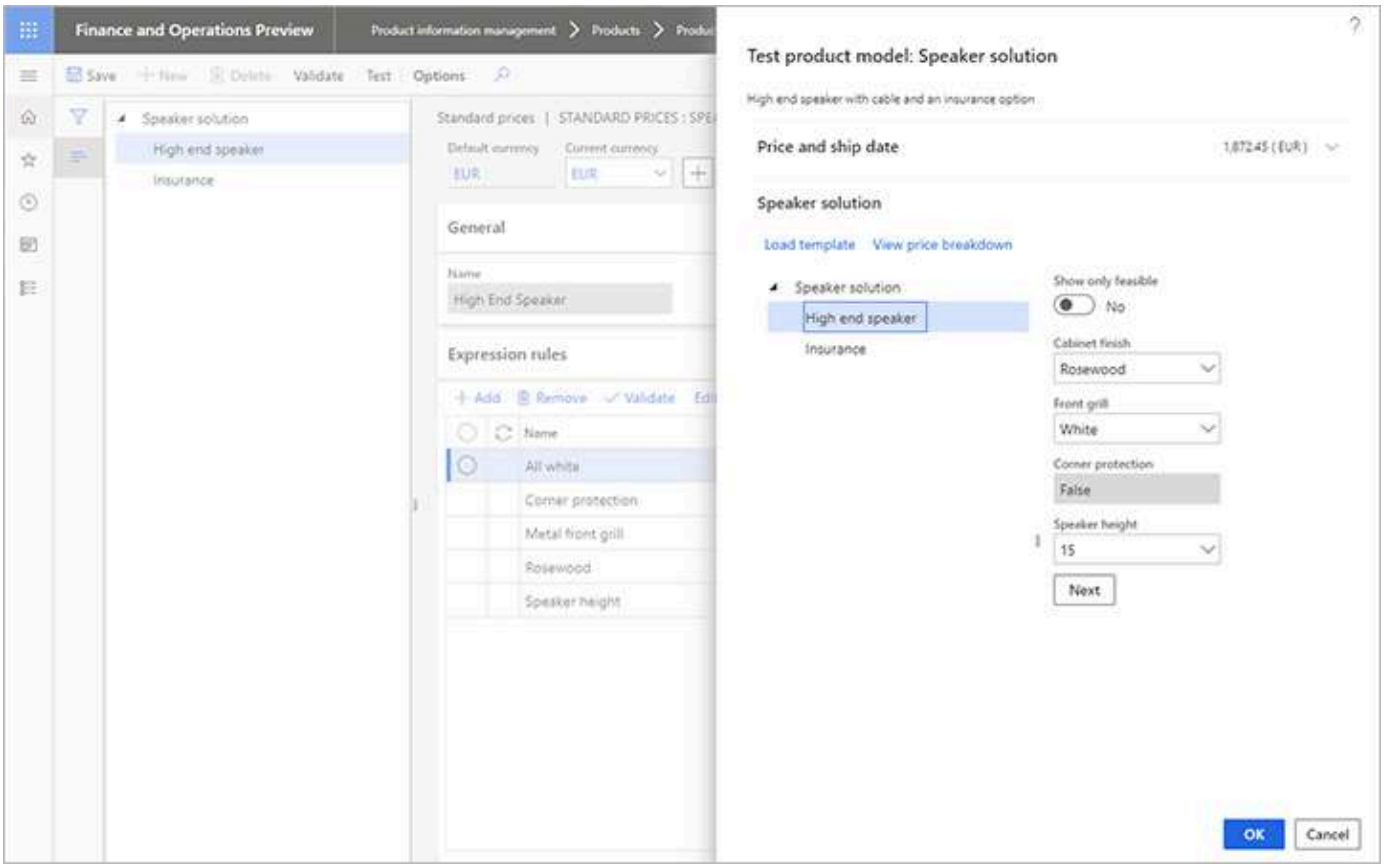
> **Note**
>
> You can't add expression rules that are unique for a non-default currency. To create expression rules that would be relevant only for a currency other than the default currency, set the price expression for the default currency to zero. Then set the appropriate expression for the non-default currency.

Test your price model

To test how the sales prices work in a configuration session, open the edit page for your price model, as described in Build your sales price models and then select **Test** on the Action Pane. The **Test product model** dialog box opens, where you can do the following:

- Use the configuration settings offered here to select product options and then see how they affect the value shown for **Price and ship date**.

- Select **View price breakdown** to download an Excel document that shows full details about how the price was calculated.



The downloaded spreadsheet shows both the absolute value and the contribution as a percentage for each active price element. If you have set the **Attach** price model option on the **Product information management parameters** page, this Excel sheet gets attached to the order or quotation line.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Price breakdown															
2																
3							Test product model: Speaker solution									
4																
5																
6																
7	Time of configuration: 8/17/2020 03:44:19 am										Total sales price		1,872.45	EUR		
8																
9							Price		Price element		By percentage		By percentage			
10	**Test product model: Speaker solution**						1,872.45				100					
11	**Speaker solution**						578.40				30.90					
12		*Base price*							299.95				16.02			
13		*Cable included*							69.95				3.74			
14		*Cable size 30*							208.50				11.14			
15		**High end speaker**					1,214.15				64.85					
16			*Base price*						899.95				48.07			
17			*Rosewood*						119.95				6.41			
18			*Speaker height*						194.25				10.38			
19		**Insurance**					79.90				4.27					
20			*Insurance period*						79.90				4.27			
21																

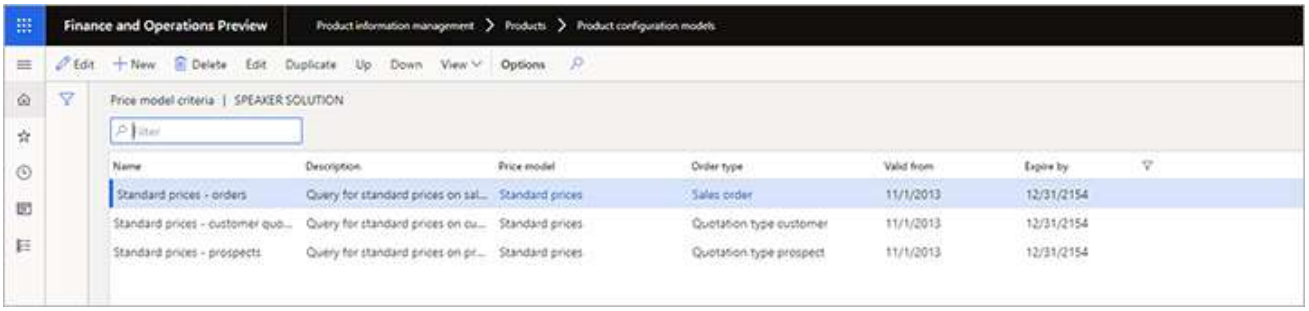
Set up selection criteria for price models

When your price models are in place, you must establish at least one selection criterion to pick up the price model when you configure to quote or to order. You'll do this by setting up one or more queries. In a combination with matching sales price models, the queries provide great flexibility in targeting sales prices for particular customers, regions, periods, and other criteria.

To set up selection criteria for price models:

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. Select the target product configuration model.
3. On the Action pane, open the **Model** tab and, from the **Set up** group, select **Price model criteria**. The **Price model criteria** page opens.
4. If the query row you need doesn't exist yet, select **New** on the Action Pane to add a new row to the grid and make the following settings for it:

    - **Name** - Enter a name for this row.
    - **Description** - Briefly describe the query and what it is for.
    - **Price model** - Select a price model (from the current configuration model) that the query will apply when triggered.
    - **Order type** - Select the type of order where the query will apply.
    - **Valid from** - Specify the first day when the query will apply.
    - **Expire by** - Specify the last date when the query will apply.



5. Select the row for the query you want to define and then select **Edit** on the Action Pane. The query designer dialog box opens. It works like most query designers in Supply Chain Management. Use it to define the conditions under which the price model for the row you selected should be applied.
6. Repeat steps 4-5 for each query you require.

> **Tip**

You can save time by copying an existing row that is already similar to a new one that you need to add. To do this, select a target row and then select **Duplicate** on the Action Pane.

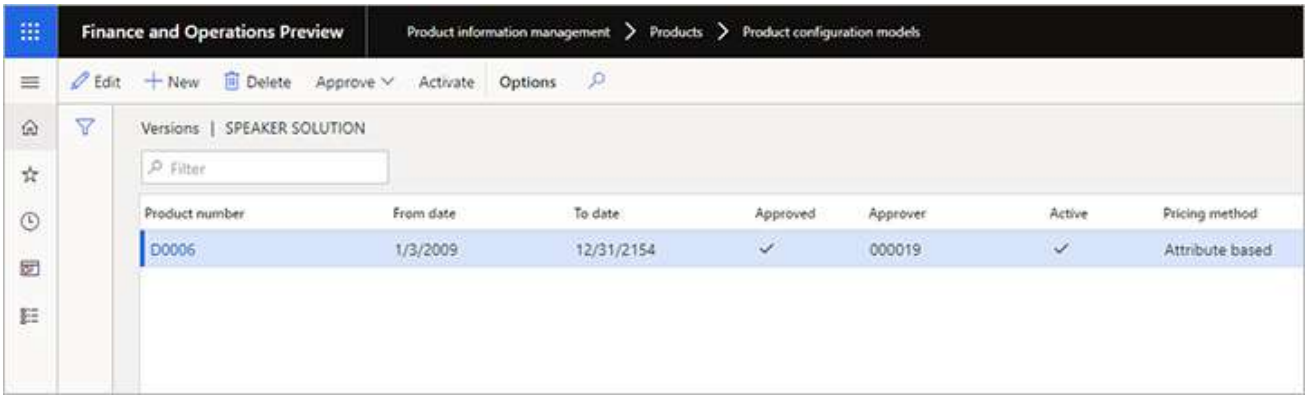
7. When you have finished setting up your criteria, arrange them into the proper order in the **Price model criteria** list. To reposition a row, select the row and then select **Up** or **Down** on the Action Pane.

> **Important**
>
> At configuration time, the system starts searching from the top of the list and uses the first query that matches the data on the quote or the order line. Therefore, you must put your most specific queries on top. If you place a general query at the top of the list, this is the one that will be used even though there might be a query further down the list that targets the exact customer or prospect of the configuration.

Set attribute-based sales prices for the product model version

The final step is to specify attribute-based sales prices for the product model version. To do this:

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. Select the target product configuration model.
3. On the Action Pane, open the **Model** tab and, from the **Product model details** group, select **Versions**.
4. The **Versions** page opens. Make sure the **Pricing method** is set to **Attribute based**.



Last updated on 07/01/2026

Approve a product configuration model

Article • 09/29/2021

Running this procedure requires that at least one product configuration model is available. This procedure uses the High end speaker model in the demo data company USMF. Note that this model has already been approved, but the procedure walks you through the entire process.

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, find and select the desired record.
    - Select the High end speaker model for this procedure.
3. Select **Versions**.
4. Select **New**.
5. In the **Product number** field, enter or select a value.
    - The reference to a product represents a version of a product configuration model. Only product masters which have the constraint-based configuration technology will appear in this list.
6. In the **From date** field, enter a date.
    - Select when the product model version will be available.
7. In the **To date** field, enter a date.
    - Select an end date when this product model version will expire, or select Never.
8. Select **Approve** to open the drop dialog.
9. In the **Approved by** field, enter or select a value.
    - Select the person who is responsible for approving product models for use in operations.
10. Select **OK**.
11. In the **Pricing method** field, select an option.
    - Activate the product model version. It is only possible to have one product active for one product model at a time.
12. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Transfer data for product models

Article • 09/29/2021

Exporting data for a product configuration model requires that you have at least one product model available. This procedure uses the High end speaker model in the demo company USMF to show you how to export data for this model.

1. Go to **Product information management** > **Products** > **Product configuration models**.
2. In the list, find and select the desired record.
    - Select the High end speaker model for this procedure.
3. On the Action Pane, select **Model**.
4. Select **Export product model**.
    - The download behavior varies from browser to browser depending on how the security settings are set up.

Feedback

Was this page helpful? Yes No

Provide product feedback

Engineering change management overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Feature summary

Today's manufacturers need strong product data management, version control, and engineering change management to succeed in a world of constantly shrinking product lifecycles, increased quality and reliability requirements, and an increased focus on product safety.

Engineering change management brings structure and discipline to the product data management process. It enables you to define, release, and revise products in a controlled manner that workflows support. Through product versions and engineering change management, you can document, assess the impact of, and apply engineering changes throughout the whole lifecycle of a product.

Engineering change management helps you plan and manage product versioning, manage product lifecycles, and manage engineering changes. Here's a list of its main features:

- Product versioning
- Enhanced product release functionality that lets you maintain product master data in one legal entity (the engineering company) and publish the fully configured released product to other legal entities
- Rules for validating that required information is entered before a product version is activated (readiness checks)
- Improved product lifecycle management through fine-grained control over when a released product can be used in specific business processes
- Engineering change requests that workflows support
- Engineering change orders that workflows support

https://learn-video.azurefd.net/vod/player?id=5580ff02-3710-47ca-a022-7f9a8ea9c3c3&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-

chain%2Fengineering-change-management%2Fproduct-engineering-overview

The preceding video (Change management capabilities in Dynamics 365 Supply Chain Management) is included in the finance and operations playlist available on YouTube.

Number sequence requirements

To use engineering change management, set both the BOM number sequence and the formula number sequence (if you're using formulas) to *Automatic* on the **Number sequences** page.

Turn the required configuration keys on or off

Starting with Supply Chain Management version 10.0.36, all **Engineering Change Management** keys are turned on by default, but the **Product dimension - Version** key is only turned on by default for new systems (not for older systems that are upgraded to version 10.0.36 or later). If you need to turn any of these keys on or off, follow these steps:

1. Put your system into maintenance mode, as described in Maintenance mode.
2. Go to **System administration** > **Setup** > **License configuration**.
3. Expand the **Trade** node.
4. Enable or disable the configuration key for the main feature by using the **Engineering Change Management** checkbox.
5. Expand the **Engineering Change Management** node, and select or clear the following checkboxes as required (depending on the features that you want to use):
    - **Attribute search** – Select this checkbox to enable the attribute search feature. Enable this feature, but you can clear this checkbox if you won't use it.
    - **Change management for process manufacturing** – Select this checkbox if you want to use Engineering change management features to manage changes in formulas for process manufacturing. If you don't need to manage formulas, clear this checkbox. For more information, see Manage changes in formulas and their ingredients.
6. If you also want to use the version dimension, select the **Product dimension - Version** checkbox. (This checkbox is further down the list, not nested under the **Engineering Change Management** node.) You can clear this check box if you don't need this feature.

    > ⚠ **Warning**

When you turn on the **Product dimension - Version** key for an existing system, some functionality might break or stop working as expected if you installed other solutions that add customizations to the inventory dimensions. For the version dimension to be fully functional, you might have to update those solutions so that they include the version dimension in their references to the inventory dimensions. See **The version product dimension** for details.

7. Turn off maintenance mode, as described in Maintenance mode.

8. The database must be synchronized to ensure that the configuration keys are properly updated to reflect your changes. If you changed any of the key settings, do one of the following steps, depending on which type of environment you're working on:

    - **For Tier 1 (development) environments**: Open your project in Microsoft Visual Studio and then select **Dynamics 365** > **Synchronize database** > **Synchronize**.
    - **For Tier 2 (and higher) environments**: The database syncs automatically after you put the environment in and out of maintenance mode, so you can skip this step.

Last updated on 07/01/2026

Engineering change management feature walkthrough

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an end-to-end walkthrough that shows how to work with engineering change management. It goes through each of the most important scenarios:

- Basic feature configuration
- How an engineering company creates a new engineering product
- How an engineering company releases an engineering product to a local company
- How a local company can review and accept a product that has been released to it by an engineering company
- How a local company can use an engineering product in standard transactions
- How to add an engineering product to a sales order
- How to request changes to an engineering product by creating an engineering change request
- How to schedule and implement requested changes by creating an engineering change order
- How to release a product that has been changed

All the exercises in this article use the standard sample data that's provided for Microsoft Dynamics 365 Supply Chain Management. Additionally, each exercise builds on the previous exercise. Work through the exercises in order, from beginning to end, especially if you've never used the engineering change management feature before. This approach gives you a complete understanding of the feature.

Set up for the sample scenario

To follow the sample scenario in this article, first prepare the feature by making demo data available and adding a few custom records.

Before you try to do any of the exercises in the rest of this article, follow the instructions in all the following subsections. These subsections also introduce several important settings pages that you use when you set up engineering change management for your own organization.

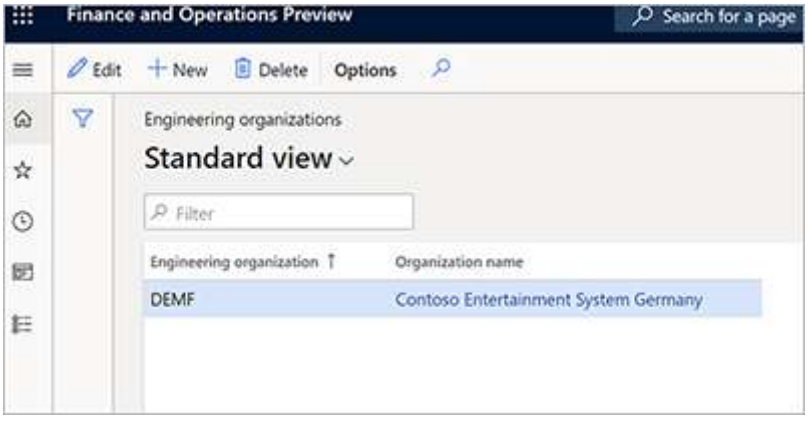
Make standard demo data available

Work on a system where the standard demo data is installed. The standard demo data adds data for several demo legal entities (companies and organizations). As you work through the exercises, use the company picker on the right side of the navigation bar to switch between one company (*DEMF*) that's set up as an *engineering organization* and another company (*USMF*) that's set up as an *operational organization*.

Set up an engineering organization

An engineering organization owns the engineering data, and is responsible for product design and product changes. To set up your engineering organizations, follow these steps:

1. Go to **Engineering change management** > **Setup** > **Engineering organizations**.
2. Select **New** to add a row to the grid, and set the following values for it:
    - **Engineering organization:** *DEMF*
    - **Organization name:** *Contoso Entertainment System Germany*

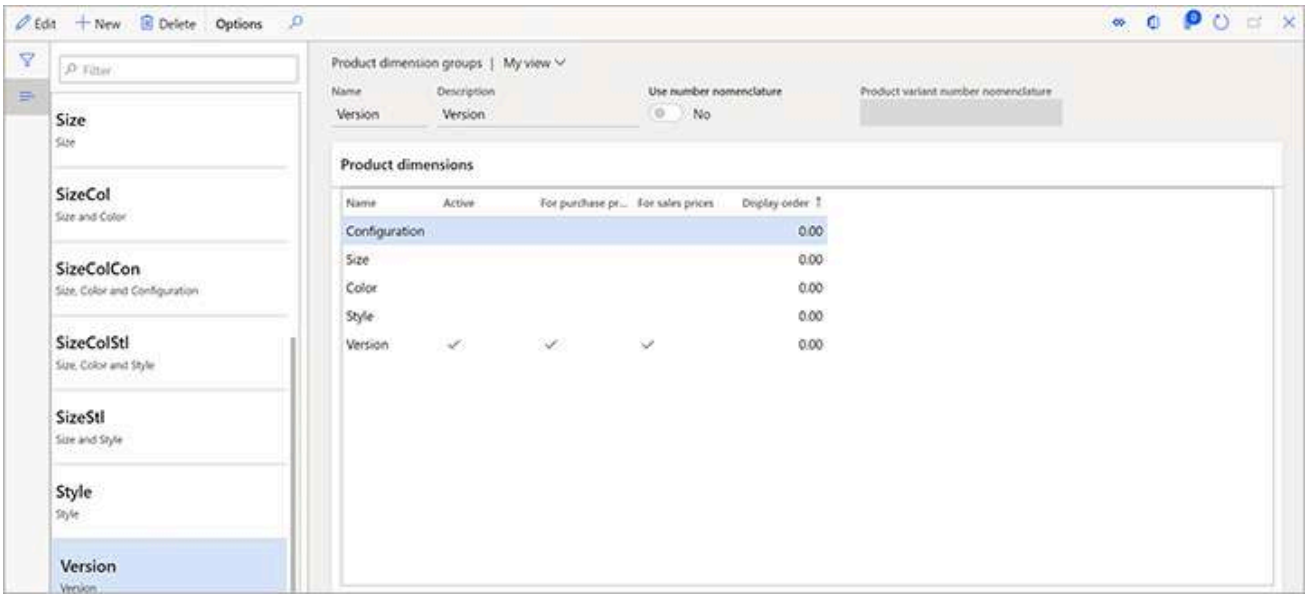


Set up the version product dimension group

1. Go to **Product information management** > **Setup** > **Dimensions and variant groups** > **Product dimension groups**.
2. Select **New** to create a product dimension group.
3. Set the **Name** field to *Version*.

4. Select **Save** to save the new dimension and load values onto the **Product dimensions** FastTab.

5. On the **Product dimensions** FastTab, set **Version** as an active product dimension.



Set up product lifecycle states

As an engineering product goes through its lifecycle, you must be able to control which transactions are allowed for each lifecycle state. To set up the product lifecycle states, follow these steps:

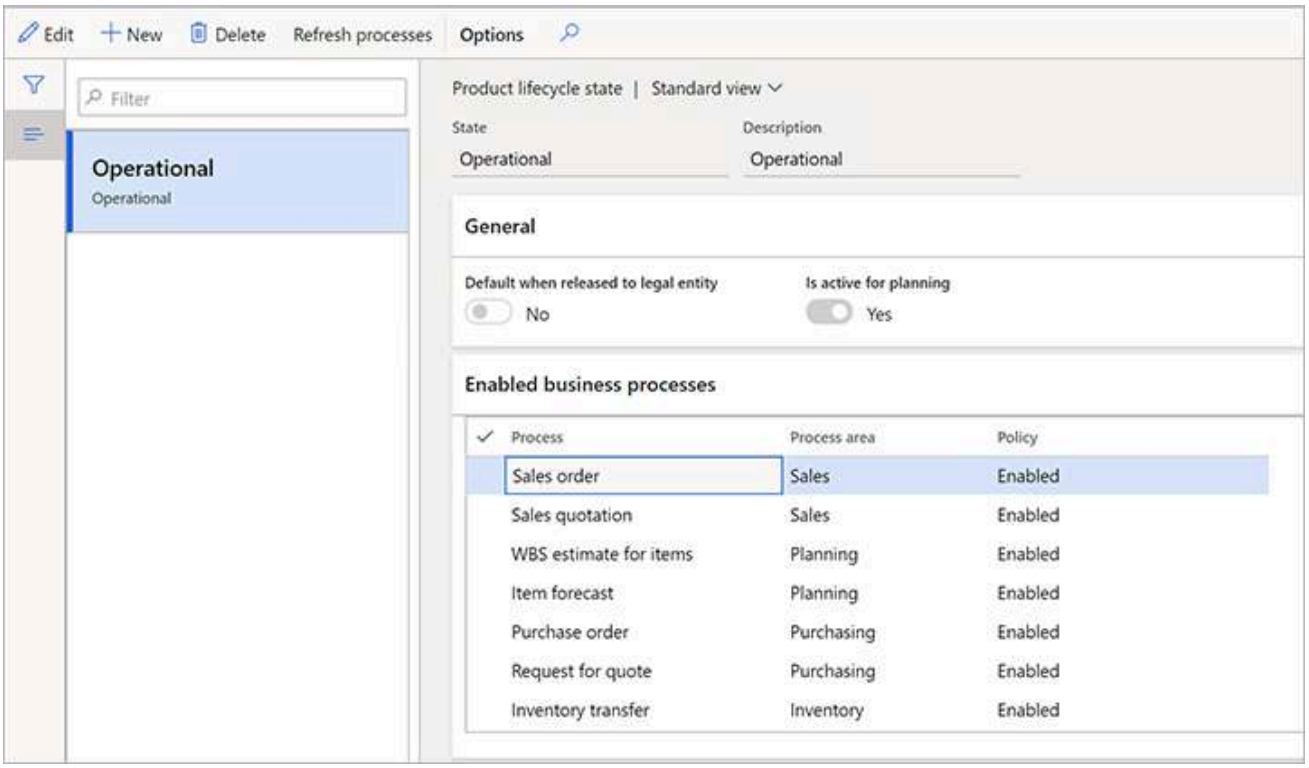
1. Go to **Engineering change management** > **Setup** > **Product lifecycle state**.

2. Select **New** to add a lifecycle state, and set the following values for it:

   - **State:** *Operational*
   - **Description:** *Operational*

3. Select **Save** to save the new lifecycle state and load values onto the **Enabled business processes** FastTab.

4. On the **Enabled business processes** FastTab, select the business processes that should be available. For this example, leave the **Policy** field set to *Enabled* for all business processes.

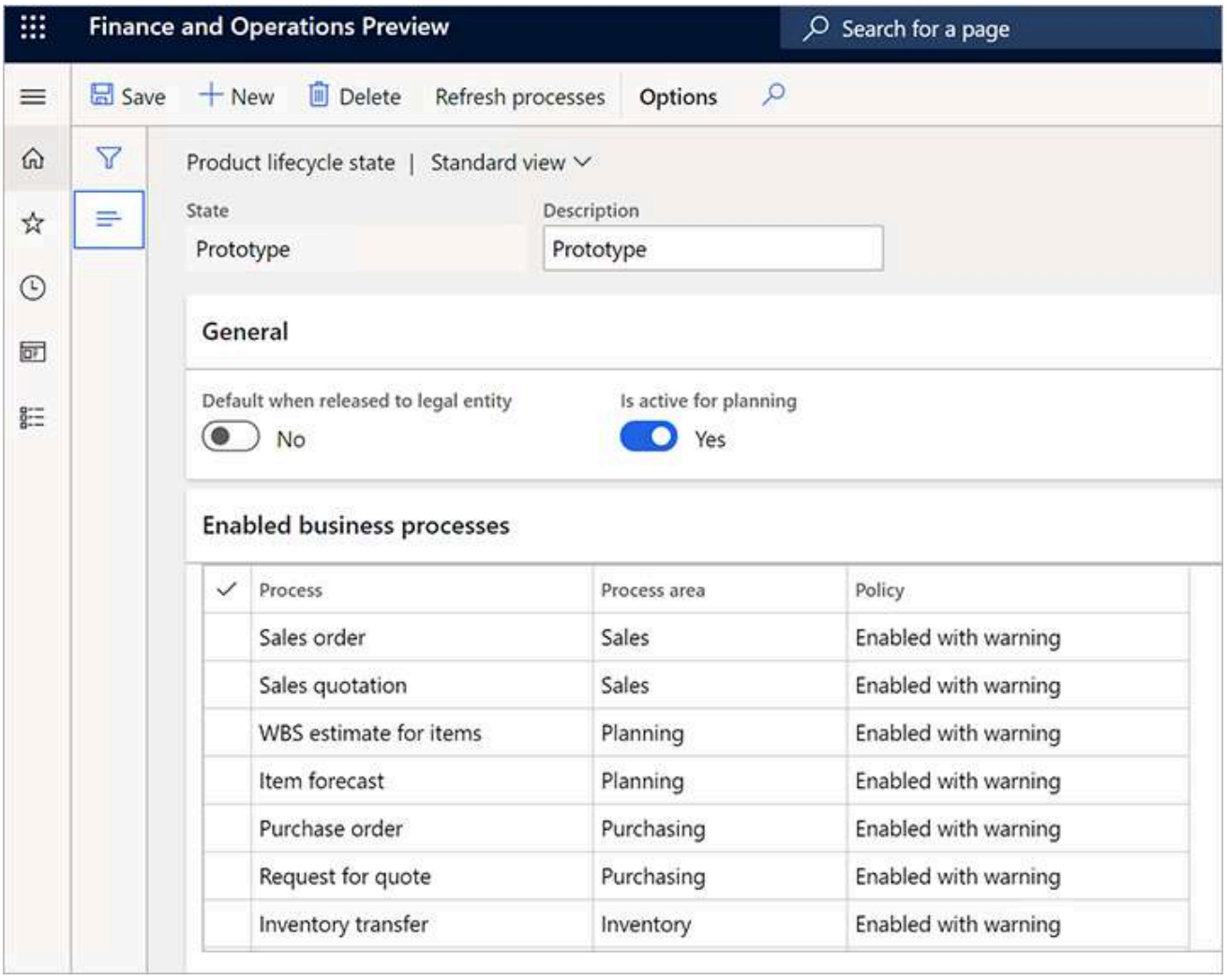


5. Select **New** to add another lifecycle state, and set the following values for it:

    - **State:** *Prototype*
    - **Description:** *Prototype*

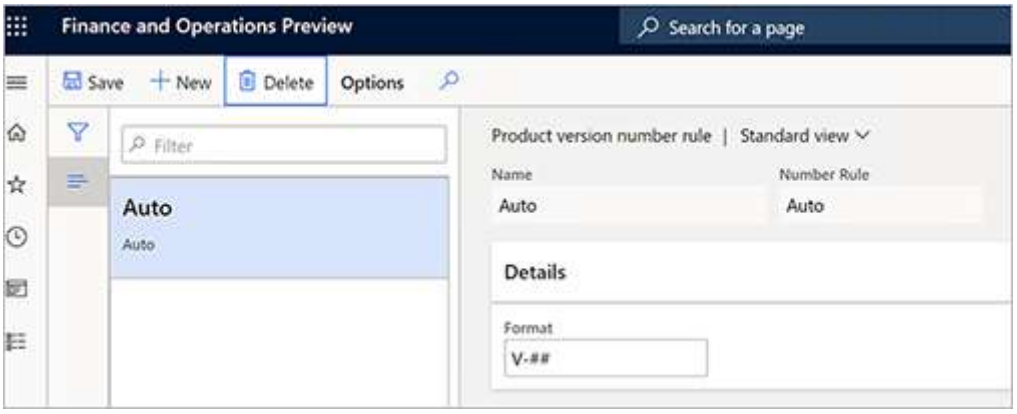
6. Select **Save** to save the new lifecycle state and load values onto the **Enabled business processes** FastTab.

7. On the **Enabled business processes** FastTab, select the business processes that should be available. For this example, set the **Policy** field to *Enabled with warning* for all business processes.



Set up a version number rule

1. Go to **Engineering change management** > **Setup** > **Product version number rule**.
2. Select **New** to add a rule, and set the following values for it:
    - **Name:** *Auto*
    - **Number rule:** *Auto*
    - **Format:** *V-##*



Set up a product release policy

1. Go to **Engineering change management** > **Setup** > **Product release policies**.

2. Select **New** to add a release policy, and set the following values for it:

    - **Name:** *Components*
    - **Description:** *Components*

3. On the **General** FastTab, set the following values:

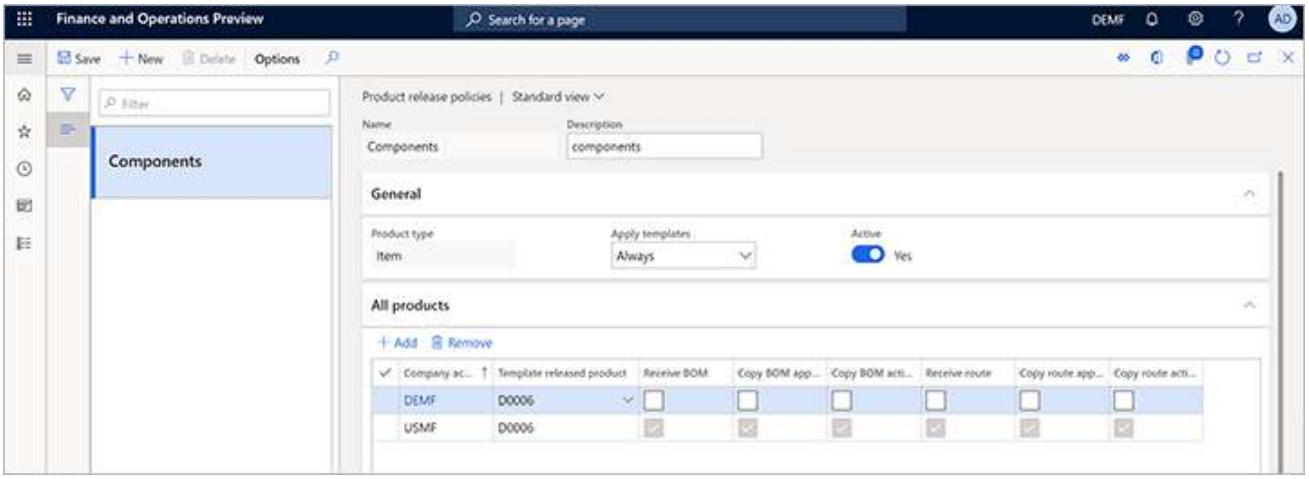
    - **Product type:** *Item*
    - **Apply templates:** *Always*
    - **Active:** *Yes*

4. On the **All products** FastTab, select **Add** to add a line, and set the following values for it:

    - **Company:** *DEMF*
    - **Template released product:** *D0006*

5. Select **Add** to add another line, and set the following values for it:

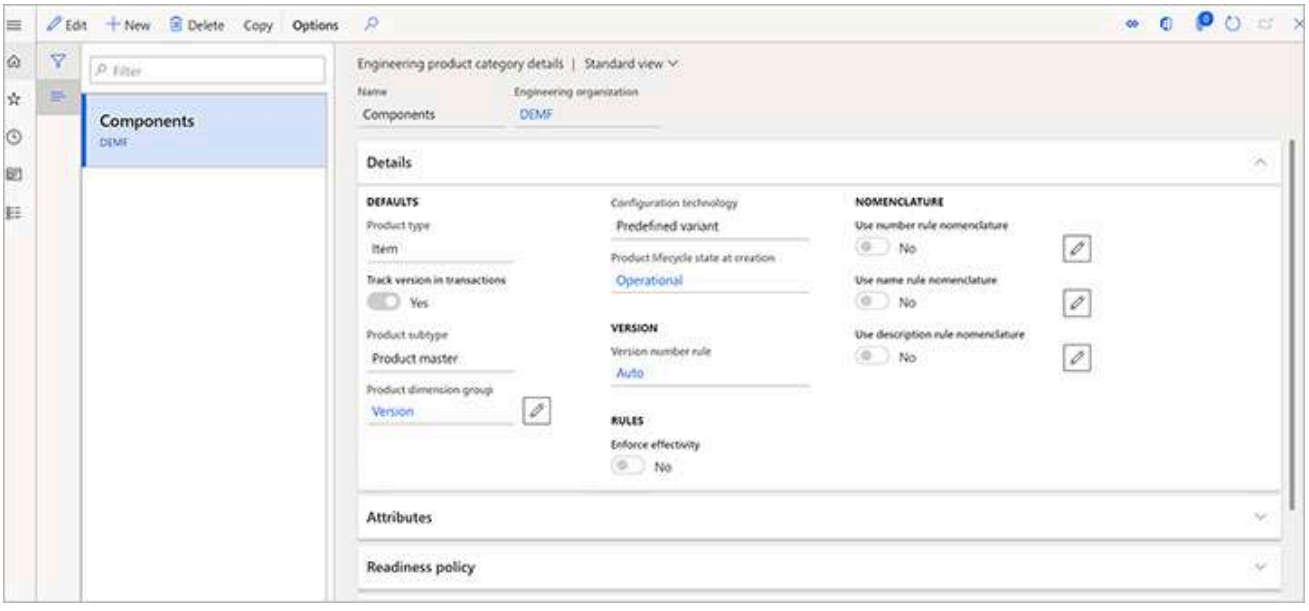
    - **Company accounts ID:** *USMF*
    - **Template released product:** *D0006*
    - **Receive BOM:** Select this check box.
    - **Copy BOM approval:** Select this check box.
    - **Copy BOM activation:** Select this check box.
    - **Receive route:** Select this check box.
    - **Copy route approval:** Select this check box.
    - **Copy route activation:** Select this check box.



Set up an engineering product category

Engineering product categories provide the basis for creating engineering products (that is, products that are versioned and controlled through engineering change management). To set up engineering product categories, follow these steps:

1. Go to **Engineering change management** > **Engineering product category details**.
2. Select **New** to create a category.
3. On the **Details** FastTab, set the following values:
    - **Name:** *Components*
    - **Engineering organization:** *DEMF*
    - **Product type:** *Item*
    - **Track version in transactions:** *Yes*
    - **Product dimension group:** *Version*
    - **Product lifecycle state at creation:** *Operational*
    - **Version number rule:** *Auto*
    - **Enforce effectivity:** *No*
    - **Use number rule nomenclature:** *No*
    - **Use name rule nomenclature:** *No*
    - **Use description rule nomenclature:** *No*
4. On the **Release policy** FastTab, set the **Product release policy** field to *Components*.
5. Select **Save**.

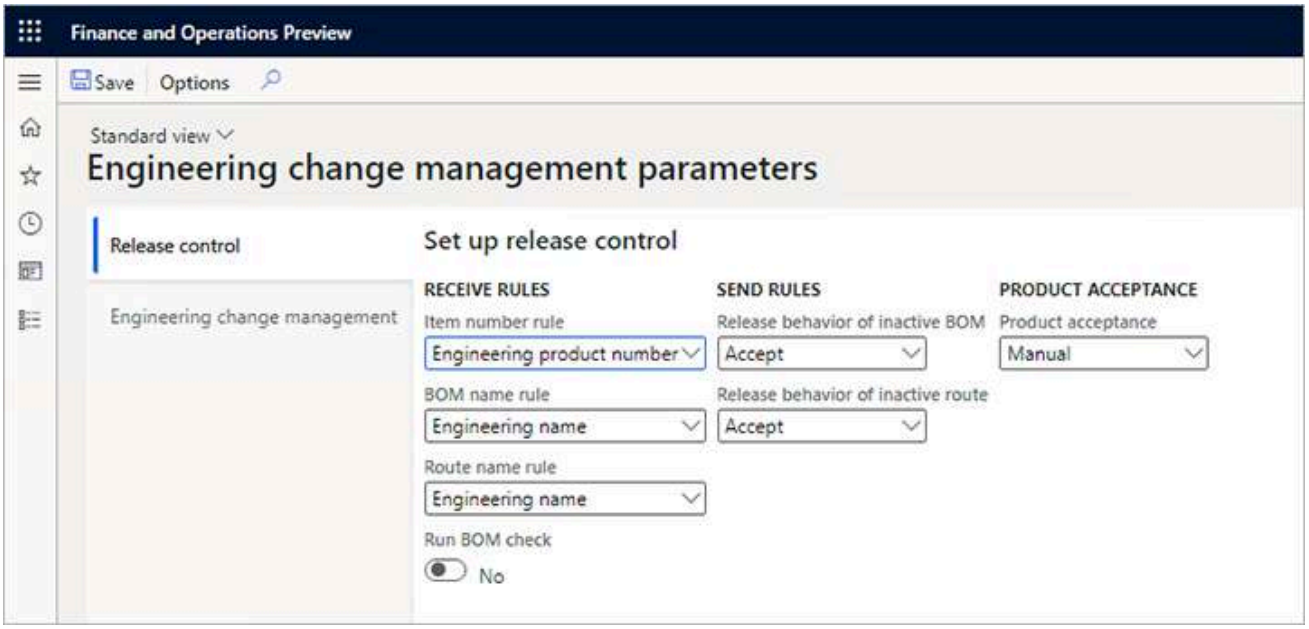


Set up product acceptance conditions

1. Use the company picker on the right side of the navigation bar to switch to the *USMF* legal entity (company).

2. Go to **Engineering change management** > **Setup** > **Engineering change management parameters**.

3. On the **Release control** tab, in the **Product acceptance** section, set the **Product acceptance** field to *Manual*.



Create a new engineering product

An engineering product is a product that is versioned and controlled through engineering change management. In other words, you can control the changes during its life, and the change information is saved by using engineering change orders. To create engineering products, follow these steps:

1. Make sure that you're in the legal entity of your engineering organization (*DEMF* for this example). Use the company picker on the right side of the navigation bar as required.

2. Open the **Released products** page by following one of these steps:

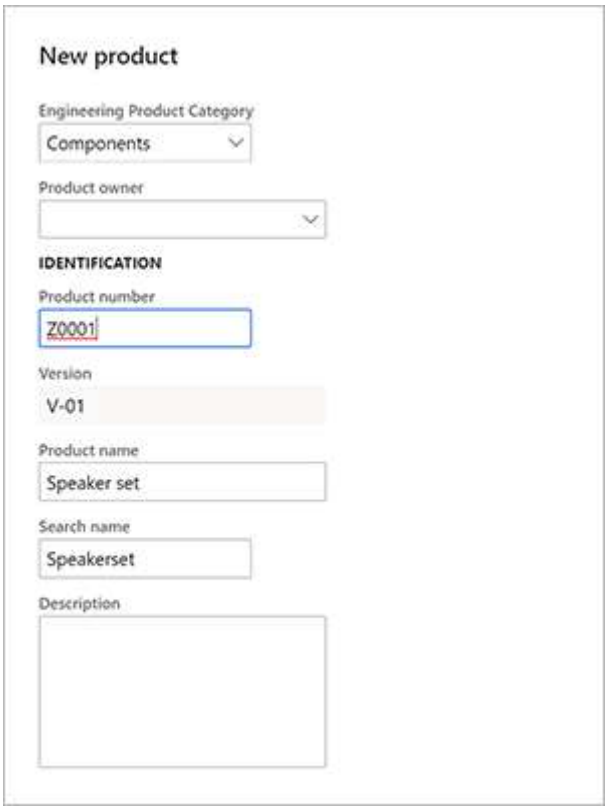
   - Go to **Product information management** > **Products** > **Released products**.
   - Go to **Engineering change management** > **Common** > **Released products**.

3. On the Action Pane, on the **Product** tab, in the **New** group, select **Engineering product**.

4. In the **New product** dialog box, set the following values:

   - **Engineering Product Category:** *Components*

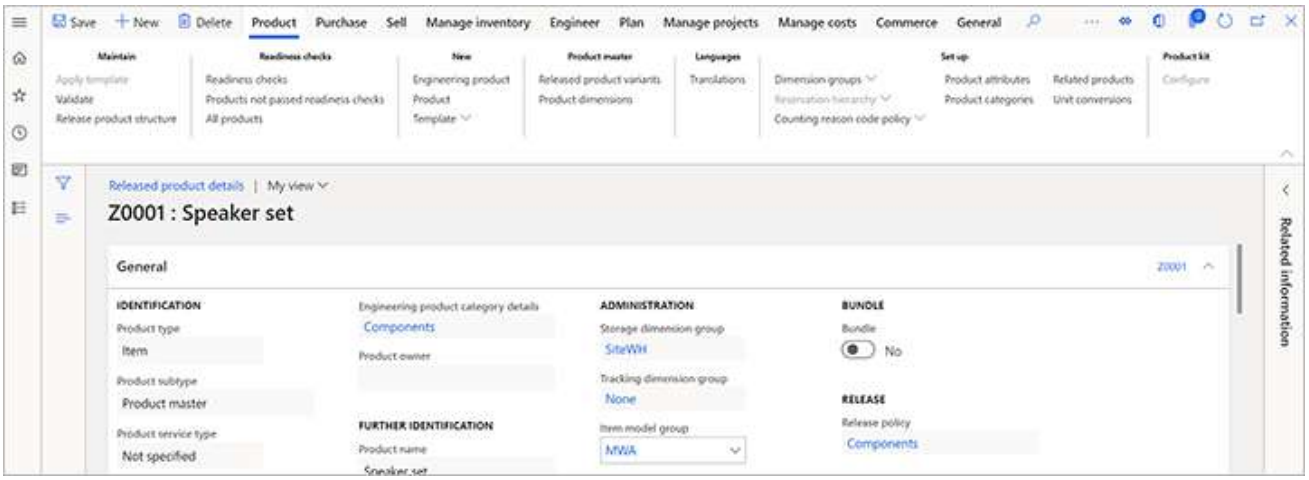
- **Product number:** *Z0001*
- **Product name:** *Speaker set*



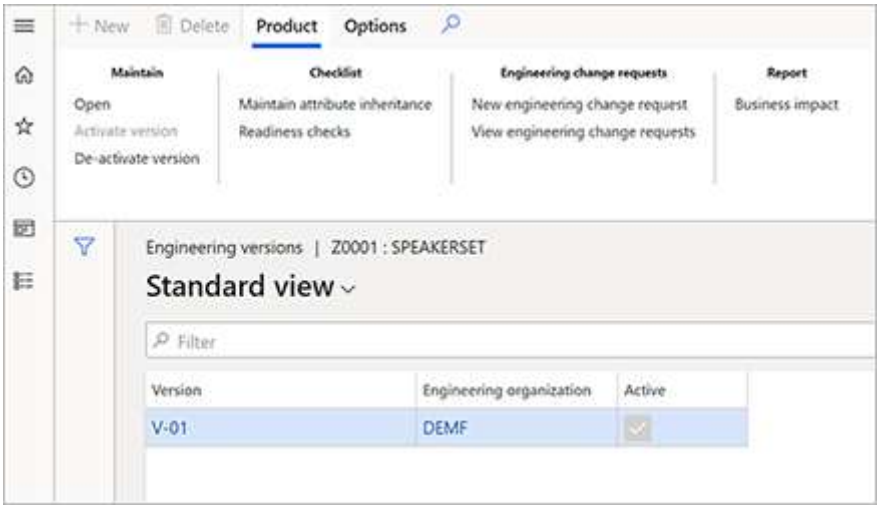
Note that the **Version** field is automatically set by using the product version number rule that you set up earlier.

5. Select **OK** to create the product and close the dialog box.

6. The details page for the new product opens. Notice that values are already filled in for some fields, such as **Storage dimension group**, **Tracking dimension group**, and/or **Item model group**. These fields were automatically set because the product is being released in the *DEMF* legal entity and uses the *Components* product release policy, which is associated with the *Components* engineering product category. Because you previously used item *D0006* as a template to set up a line for the *DEMF* legal entity, the values that were filled in were taken from item *D0006*.

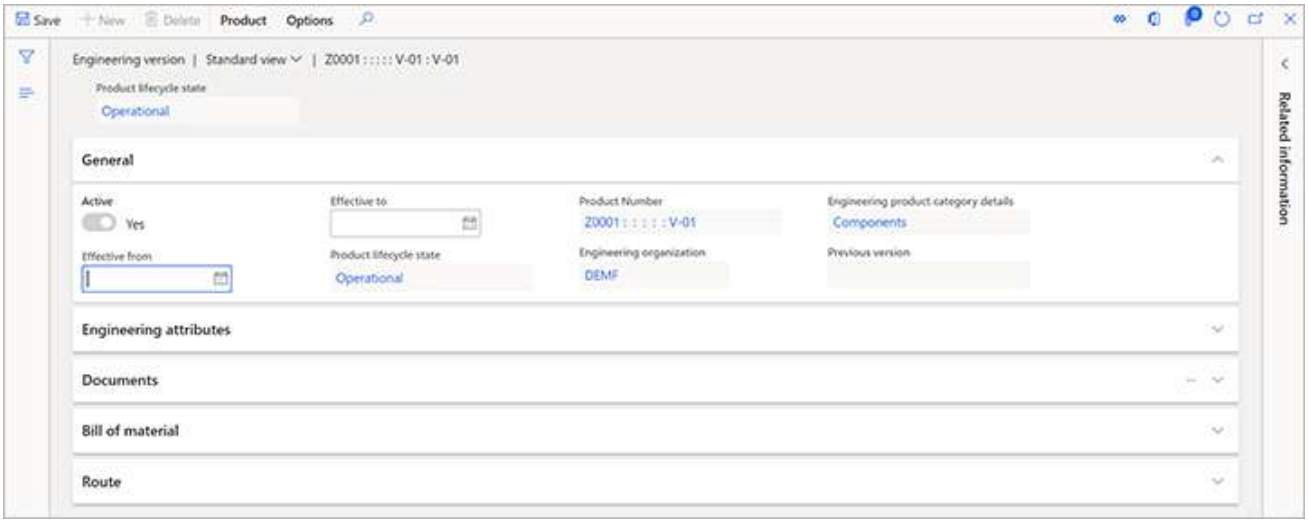


7. On the Action Pane, on the **Engineer** tab, in the **Engineering change management** group, select **Engineering versions** to view the versions of the product.



8. On the **Engineering versions** page, notice that there is only one version for the product, and it's active.

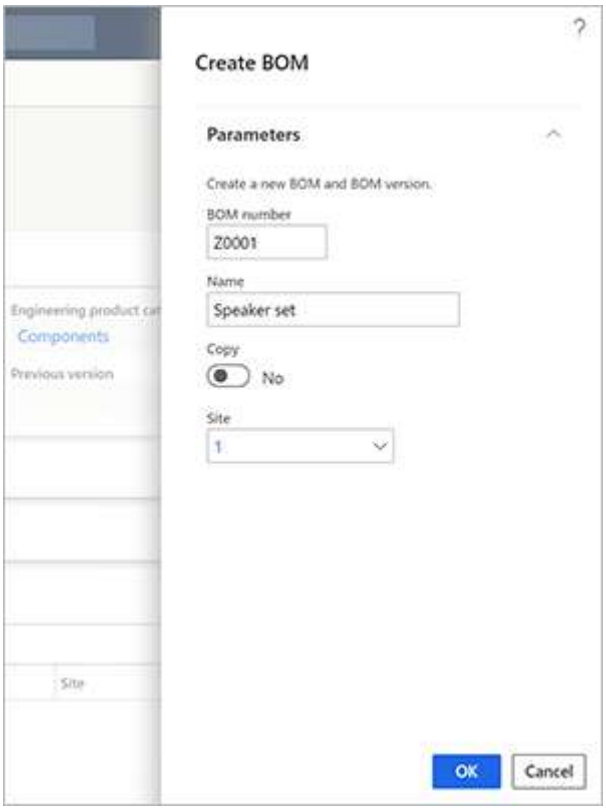
9. Select the version to view its details.



10. On the **Engineering version** page, on the **Bill of material** FastTab, select **Create BOM**.

11. In the **Create BOM** dialog box, set the following values:

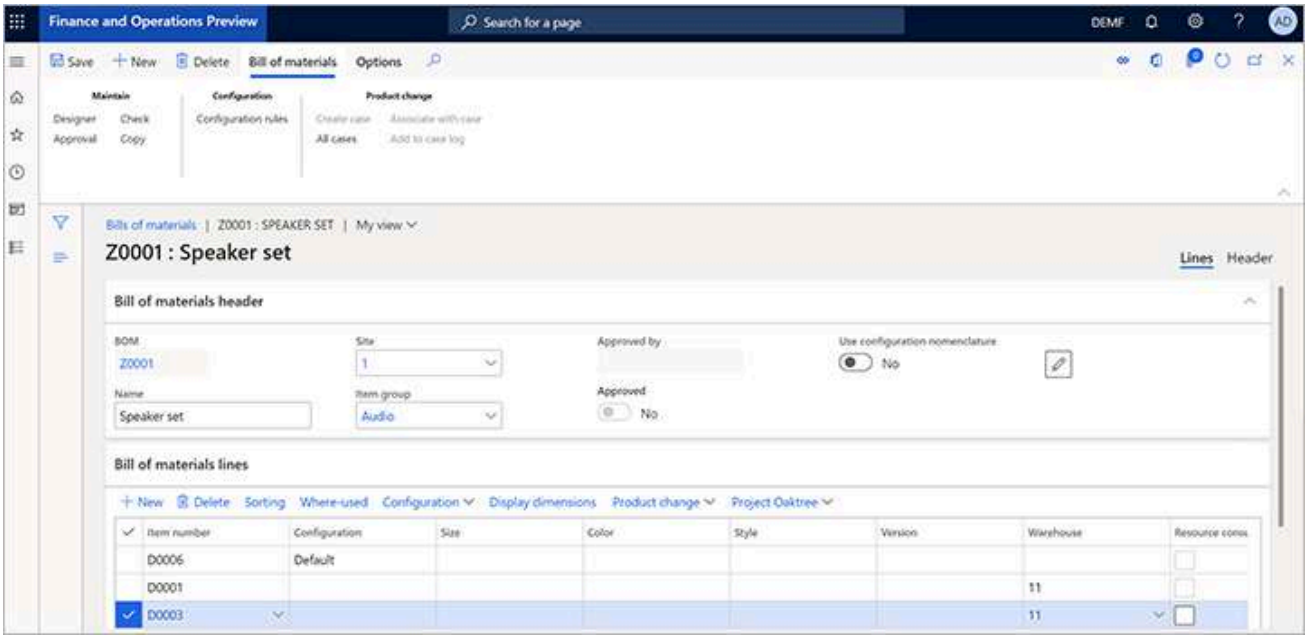
    - **BOM number:** Z0001
    - **Name:** Speaker set
    - **Site:** 1



12. Select **OK** to add the BOM and close the dialog box.

13. On the **Bill of materials** FastTab, select **Bill of material**.

14. On the **Bill of materials** page, on the **Bill of materials lines** FastTab, add three lines, one each for item numbers *D0001*, *D0003*, and*D0006*.

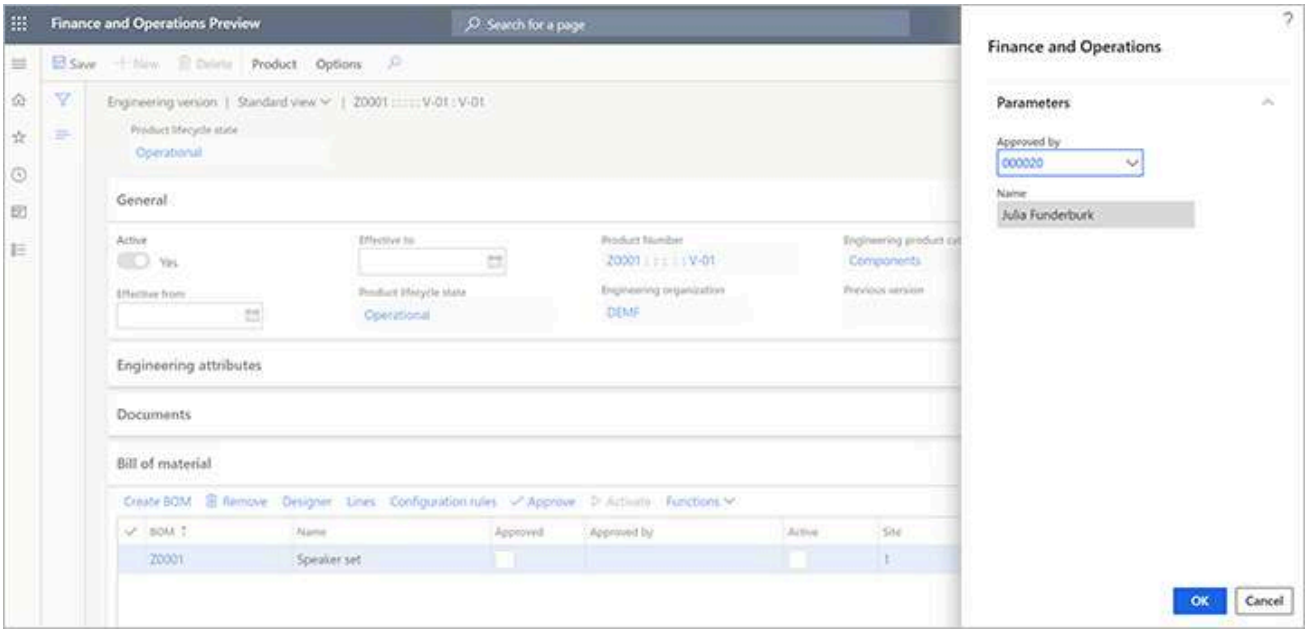


15. Select **Save**.

16. Close the page.

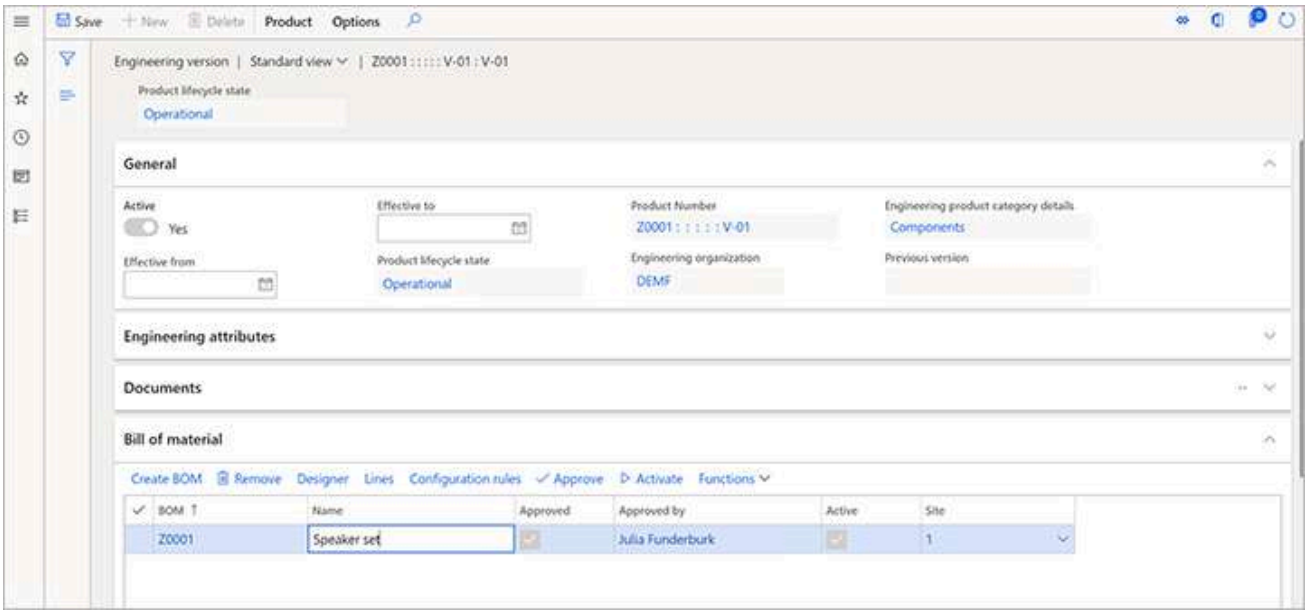
17. On the **Engineering version** page, on the **Bill of material** FastTab, select **Approve**.

18. In the dialog box that appears, select **OK**.



19. On the **Engineering version** page, on the **Bill of material** FastTab, select **Activate**.

20. Notice that the **Active** and **Approved** check boxes are selected for the BOM.



21. Close the page.

Release an engineering product to a local company

The product has now been designed by the Engineering department. For this example, the product is a prototype that engineering has designed for a customer. Because the customer belongs to the *USMF* legal entity, you must release the product to that legal entity.

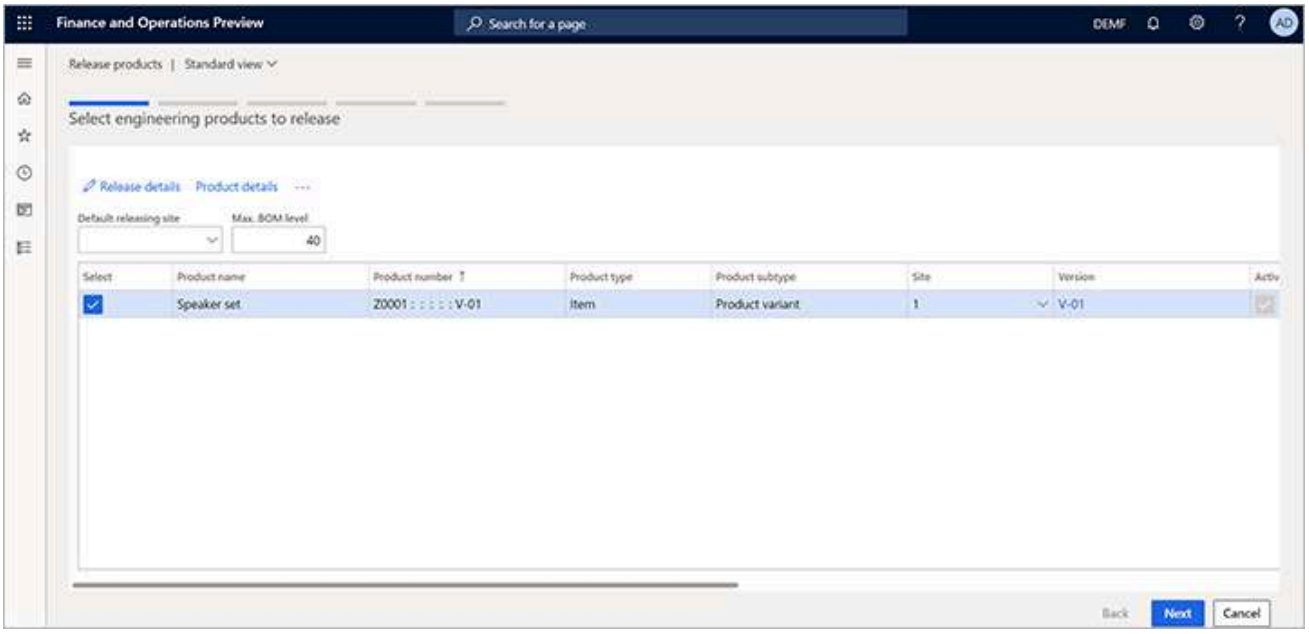
1. Keep the legal entity set to *DEMF*. (Use the company picker on the right side of the navigation bar as required.)

2. Go to **Product information management** > **Products** > **Released products**.

3. Select product *Z0001*.

4. On the Action Pane, on the **Product** tab, in the **Maintain** group, select **Release product structure** to open the **Release products** wizard.

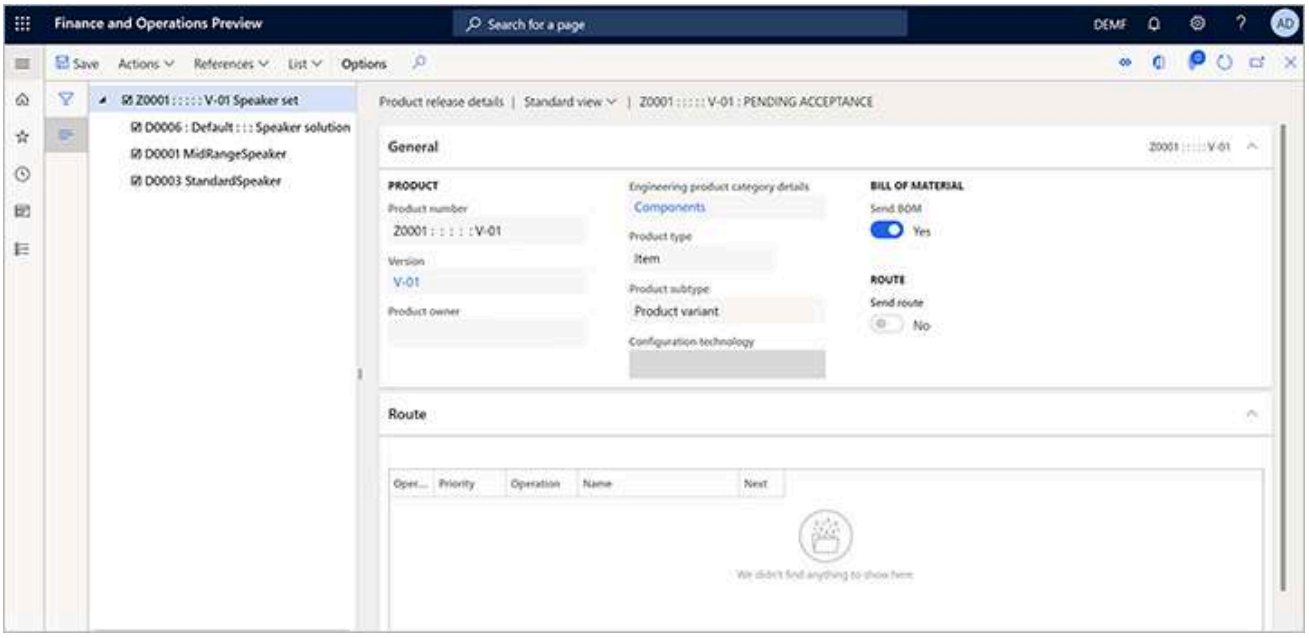
5. On the **Select engineering products to release** page, select the **Select** check box for product *Z0001*.



6. Select **Release details**.

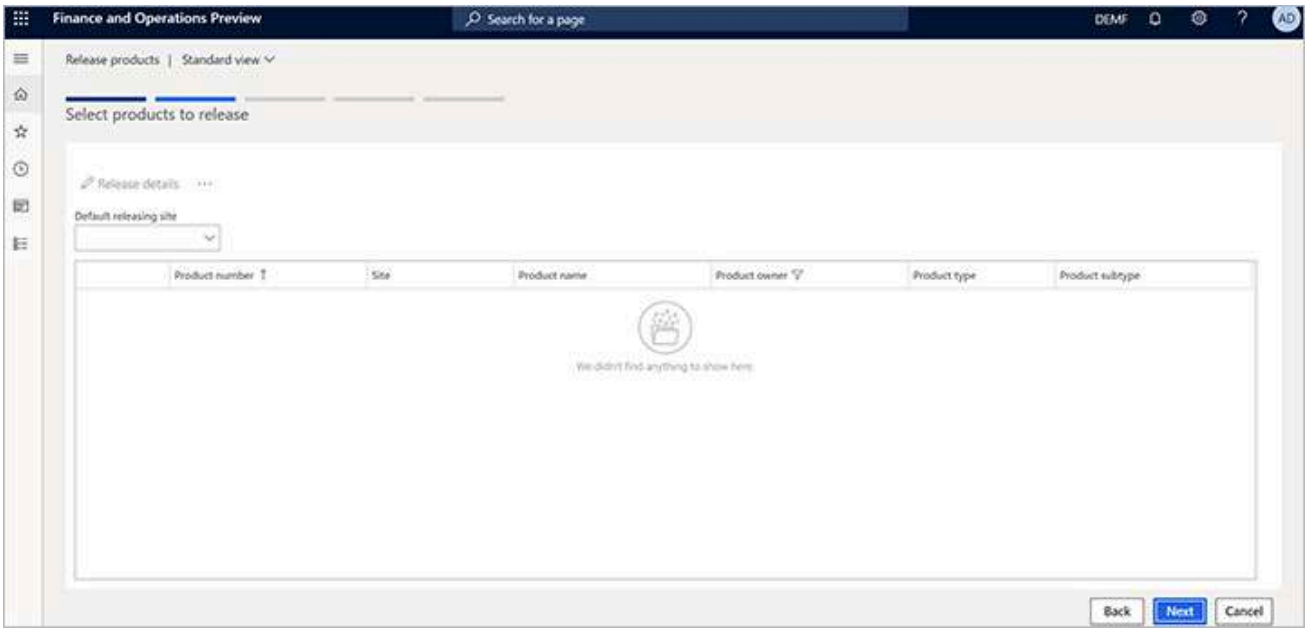
7. The **Product release details** page appears, where you can review the details of the product being released and its product structure. Notice that the **Send BOM** option is set to *Yes*. Therefore, both product *Z0001* and all its child items from the BOM are released.

   You can select any child item in the left pane to review its details. If any child item has a BOM, you can also select to release the BOM of that child item.



8. Close the page to return to the **Release products** wizard.

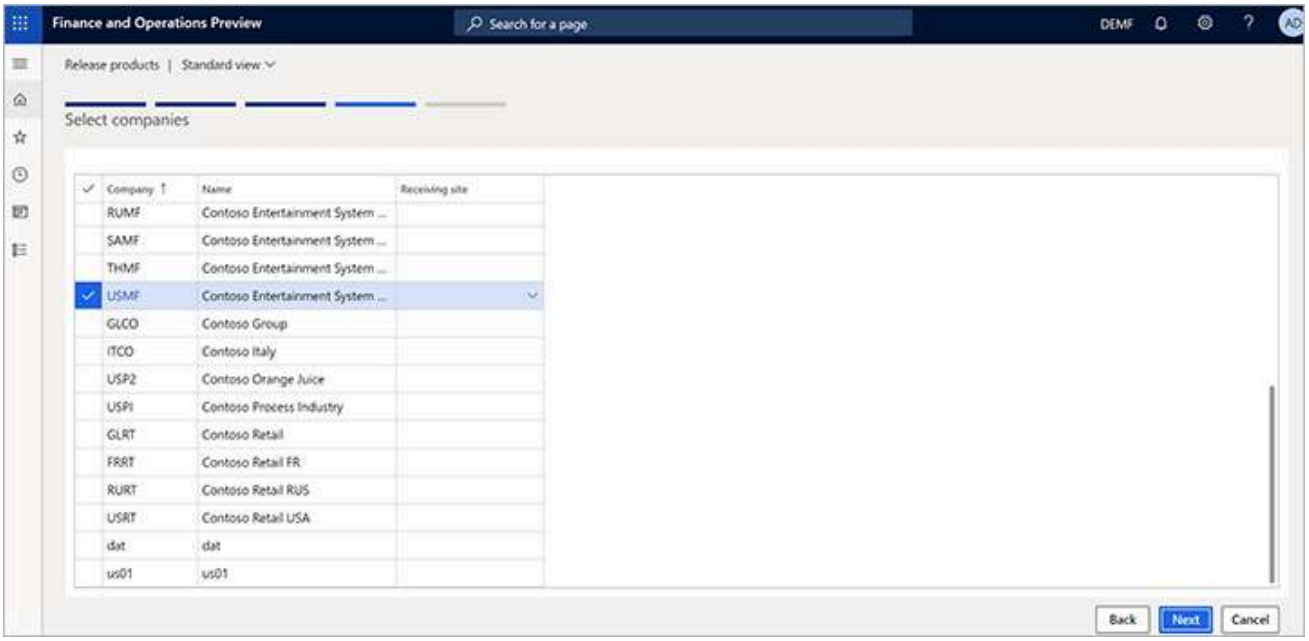
9. Select **Next** to open the **Select products to release** page. If you had selected any standard (non-engineering) products, they would appear on this page. Note that when you release a standard product by selecting **Release product structure**, its BOM and route are also released.



10. Select **Next** to open the **Select product variants to release** page. For this example, there aren't any variants.

11. Select **Next** to open the **Select companies** page.

12. Select the companies that the product should be released to. For this example, select the check box for **USMF**.



13. Select **Next** to open the **Confirm selection** page.

14. Select **Finish**.

Review and accept the product before you release it in the local company

The Engineering department has now released the information to the local companies where the product is used. For this example, the local company is *USMF*.

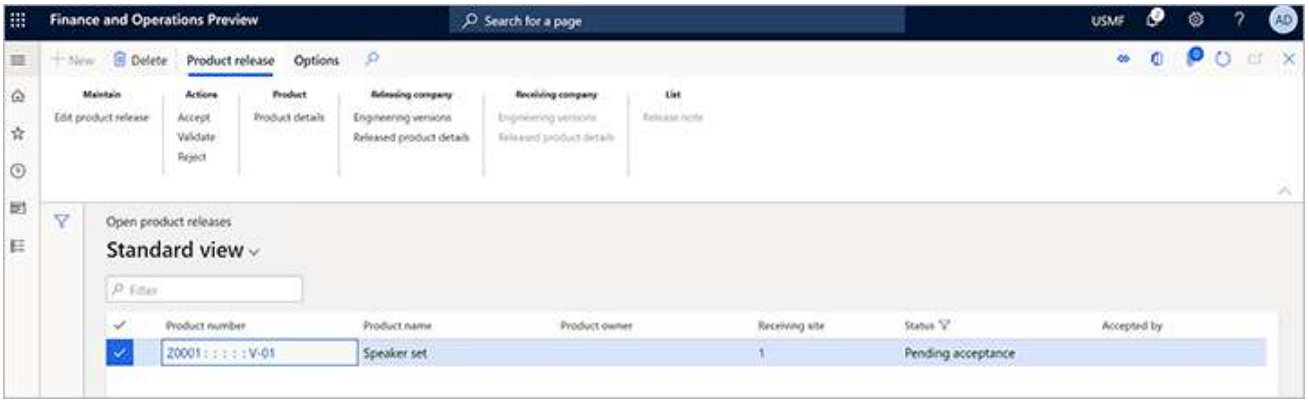
Because you set the **Product acceptance** field to *Manual* on the **Engineering change management parameters** page for the *USMF* company, products must be manually accepted before they are released to that company. In other words, they must be reviewed and accepted before they become released products.

To review the product and release it in the *USMF* company, follow these steps:

1. Set the legal entity to *USMF*. (Use the company picker on the right side of the navigation bar.)

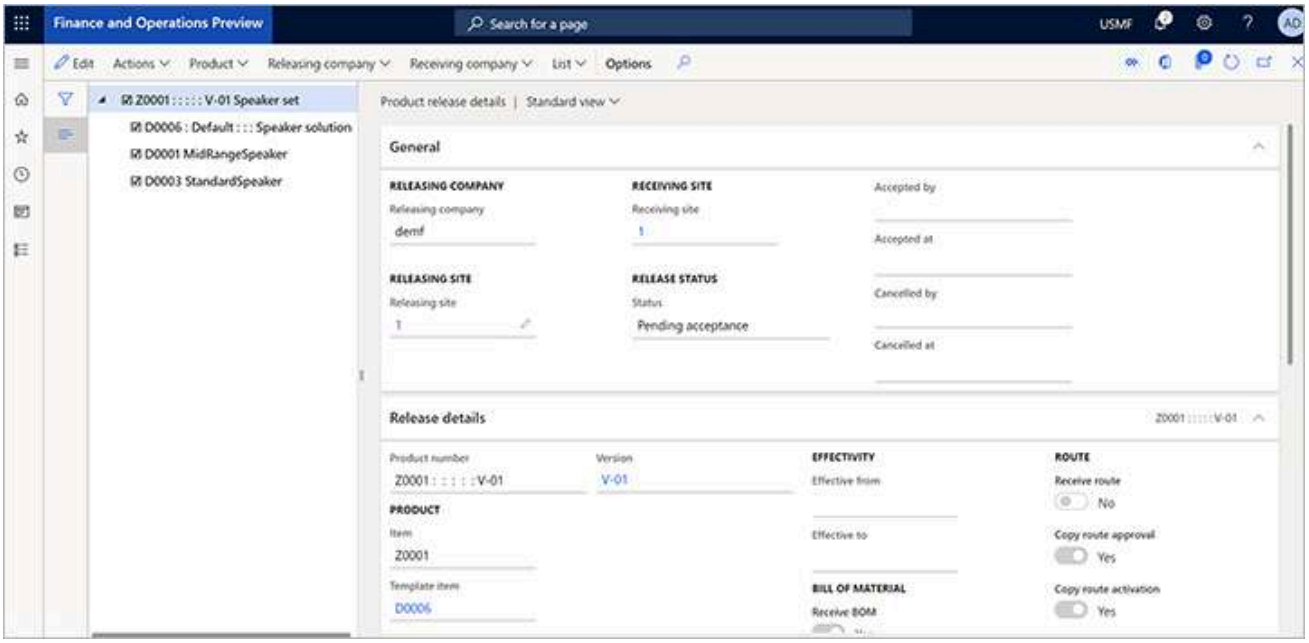
2. Go to **Engineering change management** > **Common** > **Product releases** > **Open product releases**.

   The **Open product releases** page shows product *Z0001*, which has a status of *Pending acceptance*.



3. Select the value in the **Product number** column to open the **Product release details** page. Notice the following details:

   - The **General** FastTab shows information about the product release, such as the releasing company (*DEMF* for this example), the releasing site (*1*), and the receiving site (*1*). Because you didn't specify a receiving site in the **Release products** wizard, the releasing site value is copied to the receiving site.
   - The **Release details** FastTab shows information about the product and the version that was released. Here, you can modify settings such as the effectivity dates.
   - The **Route** FastTab shows the route of the product. However, for this example, you didn't release any routes.



4. When you've finished reviewing the information, you're ready to accept the product and, in this way, release it in the *USMF* company. On the Action Pane, select **Actions** > **Accept**.

5. The product is now released in the *USMF* company. Go to **Product information management** >**Products** > **Released products**. You should see item *Z0001*.

Use the product in transactions in the local company

The master data manager for the *USMF* company wants to make sure that the product is in a *Prototype* state so that users are warned if they accidentally add it to processes that they're working on.

1. Go to **Product information management** > **Products** > **Released products**.

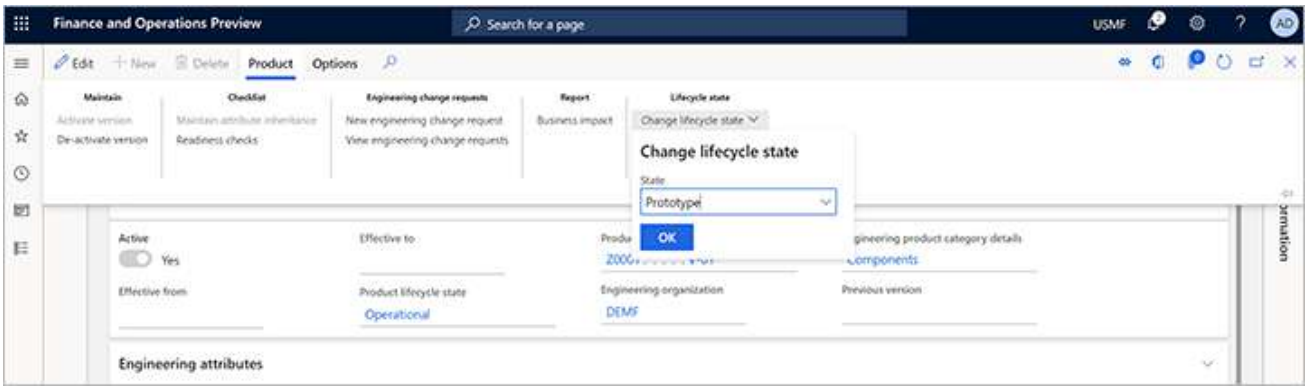
2. Select product *Z0001* to open its details page. (You can use the filter to find the product.)

3. On the Action Pane, on the **Engineer** tab, in the **Engineering change management** group, select **Engineering versions**.

4. On the **Engineering versions** page, select version number *V-01* to open its details page.

5. On the Action Pane, on the **Product** tab, in the **Lifecycle state** group, select **Change lifecycle state**.

6. In the **Change lifecycle state** drop-down dialog box, set the **State** field to *Prototype*, and then select **OK**.

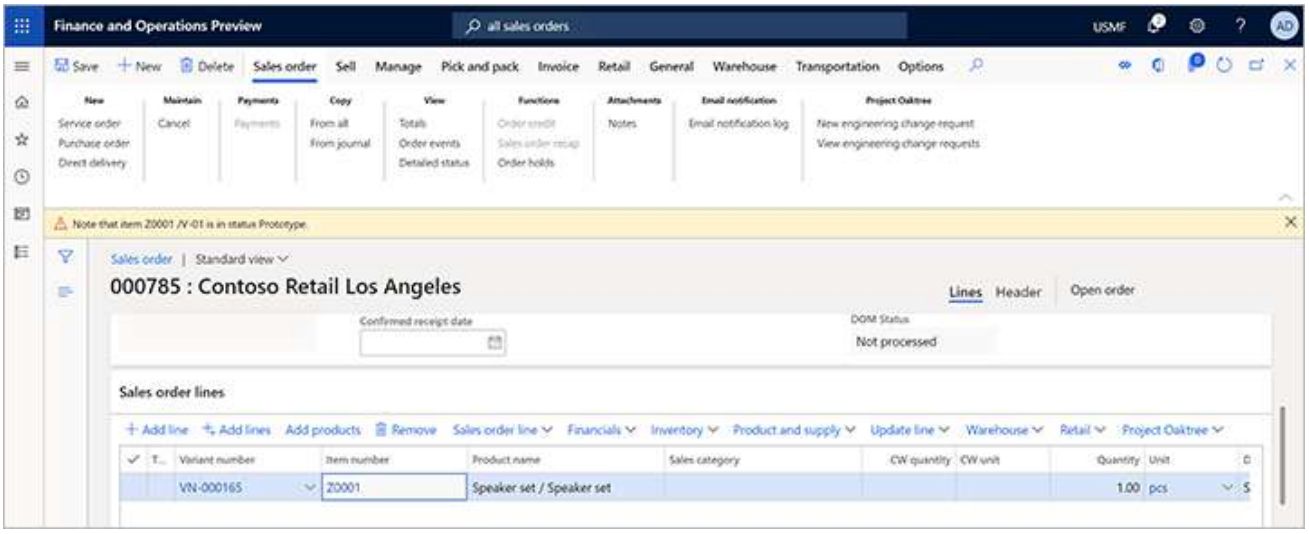


Add the engineering product to a sales order

The product can now be sold to a customer. To add the product to a sales order, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In the **Create sales order** dialog box, set the **Customer account** field to *US-0002*, and then select **OK**.
4. The new sales order is opened. On the **Sales order lines** FastTab, add a row, and set the **Item number** field to *Z000* for it.
5. On the Action Pane, select **Save**.

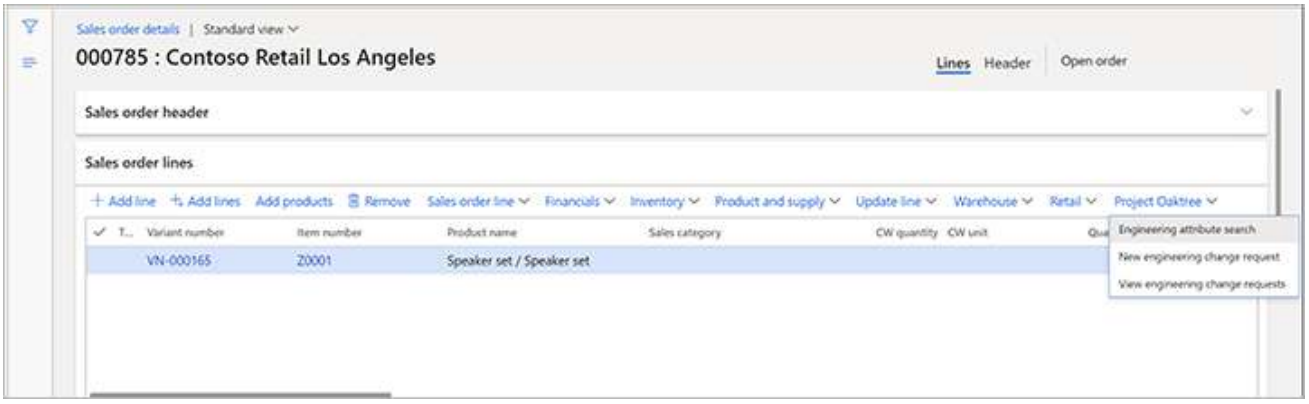
   You receive a warning message that informs you that the item has a status of *Prototype*. However, because the message is just a warning, the sales order was still created.



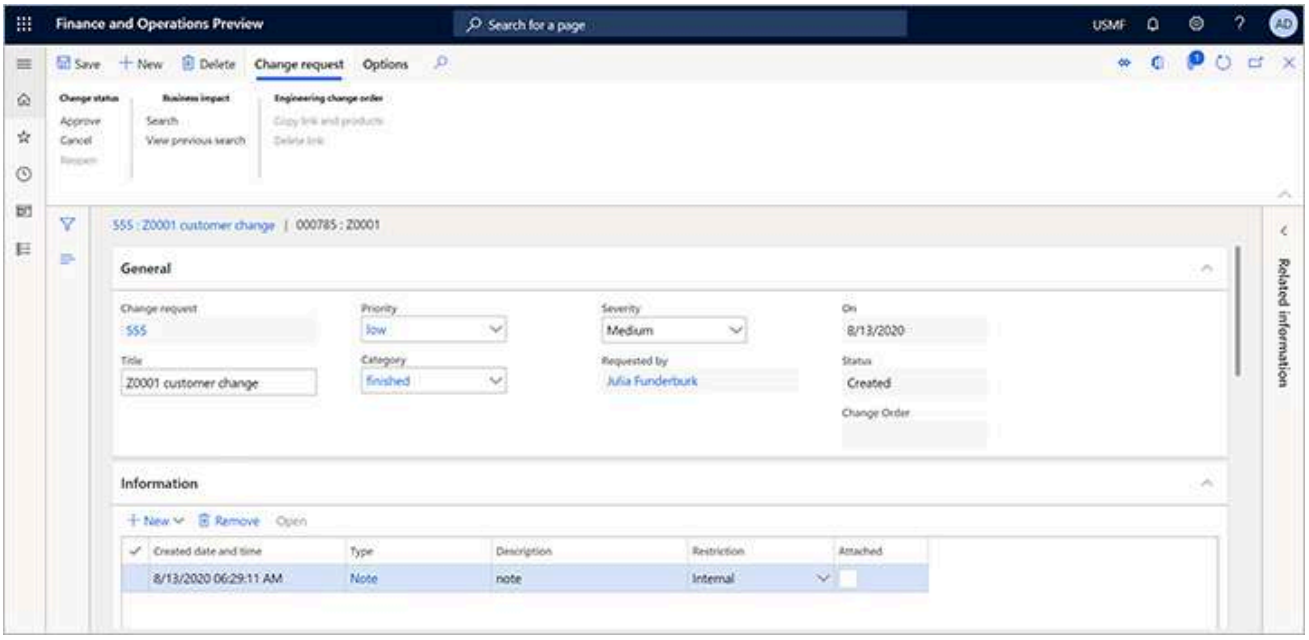
Request changes in the engineering product

The product was sent to a customer, but the customer wasn't completely satisfied and provides feedback that includes suggestions for improvement. While the customer is speaking with a sales clerk on the phone, the sales clerk can request the changes that the customer is describing.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Find and open the sales order that you created in the previous exercise.
3. On the **Sales order lines** FastTab, select **Engineering change management** > **New engineering change request**.



4. Fill in the engineering change request, based on the customer's feedback. For this example, set the following values:
    - **Change request:** *555*
    - **Title:** *Z0001 customer change*
    - **Priority:** *low*
    - **Category:** set change
    - **Severity:** *Medium*
5. On the **Information** FastTab, select **New** > **Note** to add a note to the grid.
6. In the **Description** field for the new note, indicate that item *D0003* should be deleted from the BOM. If you must add more information for the note, you can enter text in the **Notes** field.



7. On the Action Pane, select **Save**.

8. Notice that the item has automatically been added on the **Products** FastTab, and that the source of the engineering change request (the sales order) has been added on the **Source** FastTab.

Make changes to the product by using an engineering change order

The sales clerk knows that the product is important and was designed especially for the customer. Therefore, the sales clerk calls an engineer in the *DEMF* company to notify them about the change request. In this way, the engineer can speed up the process.

The engineer now reviews the request from the customer and creates a change order for the product.

1. Because the engineer works in the *DEMF* company, set the legal entity to *DEMF*. (Use the company picker on the right side of the navigation bar.)

2. Go to **Engineering change management** > **Common** > **Engineering change requests**.

3. Open change request *555*.

4. Review the information, and then approve the change. On the Action Pane, on the **Change request** tab, in the **Change status** group, select **Approve**.

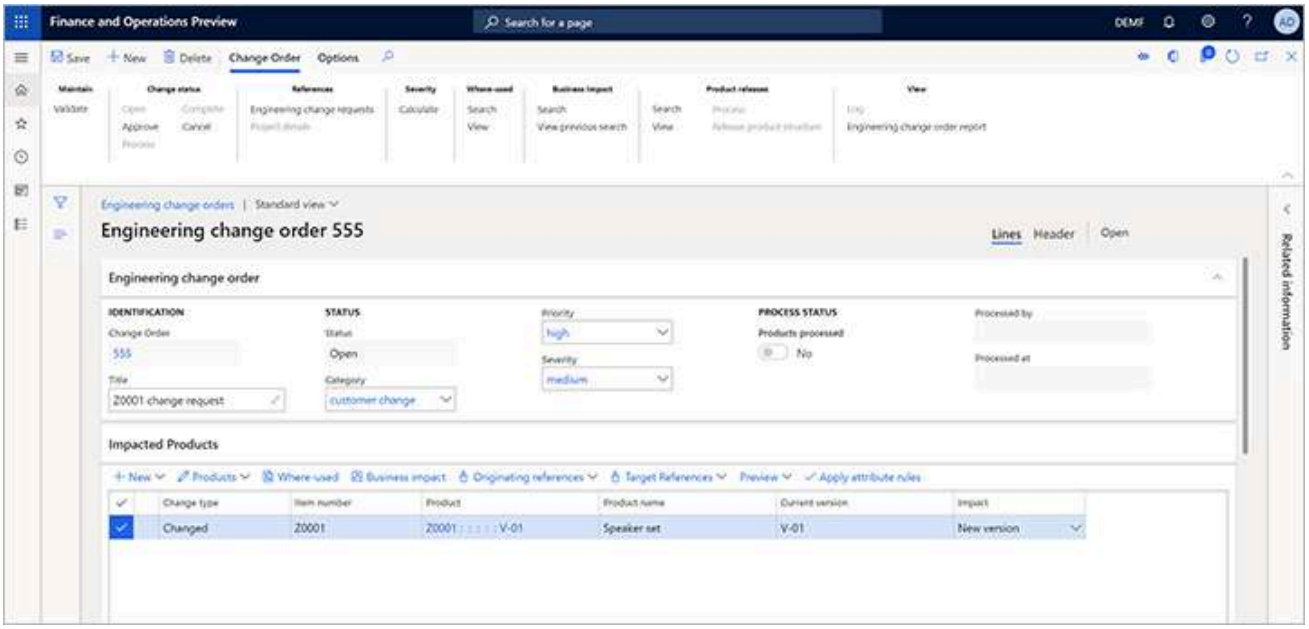
5. Go to **Engineering change management** > **Common** > **Engineering change orders**.

6. On the Action Pane, select **New** to create a change order, and set the following values for it:

    - **Change order:** *555*
    - **Title:** *Z0001 customer change*
    - **Category:** *Customer change*
    - **Priority:** *Low*
    - **Severity:** *Medium*

7. On the **Impacted products** FastTab, select **New** > **Add existing product** to add a row to the grid, and set the following values for it:

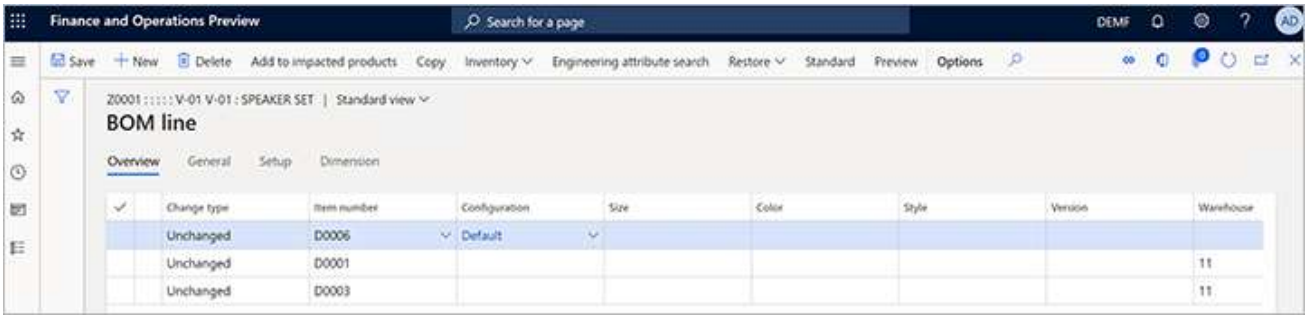
    - **Product:** *Z0001*
    - **Impact:** *New version*



8. Notice that, because you set the **Impact** field to *New version*, the **New version** field on the **Details** tab of the **Product details** FastTab shows the new version number (*V-02* for this example).

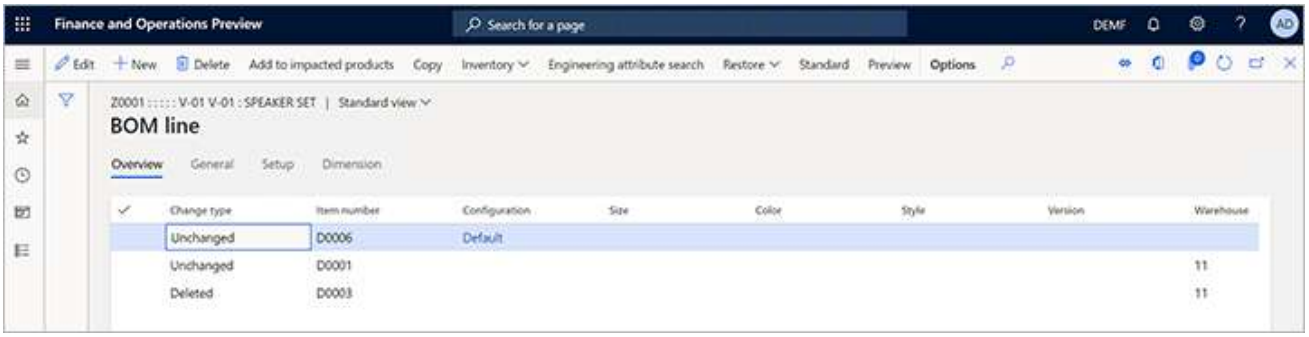
9. On the Action Pane, select **Save**.

10. On the **Product details** FastTab, on the **Bill of material** tab, select **Lines** to open the BOM for version *V-01* of product *Z0001*.



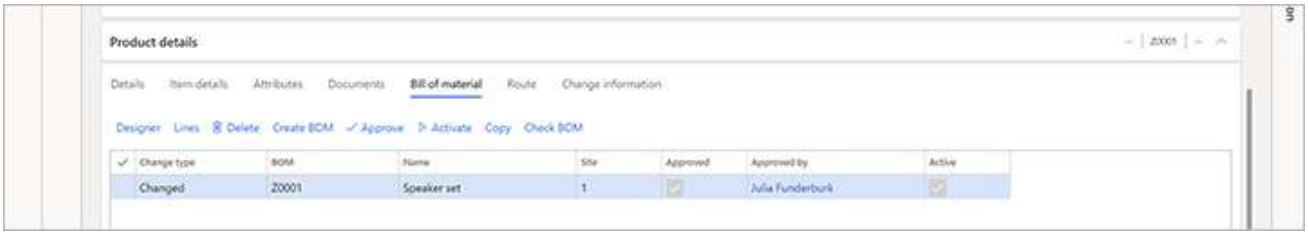
11. Select the line for item number *D0003*, and then, on the Action Pane, select **Delete**. The value of the **Change type** field for this line changes to *Deleted*.

12. On the Action Pane, select **Save**.



13. Close the **BOM line** page to return to the **Engineering change order** page.

14. On the **Product details** FastTab, on the **Bill of material** tab, notice that the value of the **Change type** field for BOM *Z0001* is now *Changed*.



The order must now be approved before the changes can be processed. When the changes are processed, the products are updated with the changes that are included in the engineering change order. For this example, the person who creates the engineering change order has been specified as the approver.

15. On the Action Pane, on the **Change order** tab, in the **Change status** group, select **Approve**.

16. Select **Process** to update the product's information.

Release the changed product

The product can now be released again to the *USMF* company and then sent to the customer. To release the product directly from the engineering change order, follow these steps:

1. Open the engineering change order that you created in the previous exercise, if it isn't already open.
2. On the Action Pane, on the **Change order** tab, in the **Product releases** group, select **Search**.
3. The search results show which companies the affected products have been released to. Close the search results.
4. On the Action Pane, on the **Change order** tab, in the **Product releases** group, select **View** to open the **Releases** dialog box, where you can view the results of the previous search.
5. Select each company that you want to release products to.
6. Select **OK** to close the **Releases** dialog box and return to the change order.
7. On the Action Pane, on the **Change order** tab, in the **Product releases** group, select **Process** to release the affected products to the selected companies. Alternatively, select **Release product structure** to start the release process.

Complete the change order

To mark the change order as completed, which indicates that no further actions remain, select **Complete** on the Action Pane.

Last updated on 06/22/2026

Engineering change management parameters

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Engineering change management parameters** page contains setup parameters that change the default behavior that is related to the release product structure and engineering change management processes.

Open the Engineering change management parameters page

To open the **Engineering change management parameters** page, go to **Engineering change management** > **Setup** > **Engineering change management parameters**. You can then set the fields as described in the remaining sections of this article.

Release control tab

The following table describes the fields that are available on the **Release control** tab of the **Engineering change management parameters** page. These fields affect the release product structure process.

Expand table

Field	Description
Item number rule	Select how the item number should be defined when the product is released to a legal entity. Select *Engineering product number* if the product number in the receiving legal entity should match the product number in the engineering company. Select *Local item number sequence* if the product should take the next number in the number sequence for product numbers in the receiving legal entity.
BOM name rule	Select how the name of the bill of materials (BOM) is defined when the product is received (released) in a legal entity. Select either *Engineering name* or *Receiving item number*.

Field	Description
Route name rule	Select how the route name should be defined when the route of a product is received (released) in a legal entity. Select either *Engineering name* or *Receiving item number*.
Run BOM check	Select whether a BOM check will be run when the product is received (released) in a legal entity.
Release behavior of inactive BOM	Select whether a product can be released if it has an inactive BOM. Select *Accept*, *Warning only*, or *Not allowed*.
Release behavior of inactive route	Select whether a product can be released if it has an inactive route. Select *Accept*, *Warning only*, or *Not allowed*.
Product acceptance	Select whether an additional step for acceptance is required before the product can be released in the legal entity. Select *Manual* to add the acceptance step. In this case, the **Open product releases** page will show the products. Select *Automatic* to show the product directly on the **Released products** page in the target legal entity immediately after the product is released together with its release product structure.

Engineering change management tab

The following table describes the fields that are available on the **Engineering change management** tab of the **Engineering change management parameters** page. These settings affect the engineering change management process.

Expand table

Field	Description
Category	The default category that will be used when an engineering change request is created.
Priority	The default priority that will be used when an engineering change request is created.
Severity rule	Select how the severity of an engineering change order should be established. Select *Manual* if the user is expected to enter a value in the **Severity** field. Select *Calculate* to have the system calculate the value of the **Severity** field when you select **Calculate severity** on the Action Pane of the engineering change order. In this case, the system will use the severity rules that are defined on the **Severity rule set** page. Select *Calculate automatically* to have the value of the **Severity** field automatically calculated and filled in according to the severity rule sets.
Re-release impacted products	This field is applicable when you re-release products via an engineering change order. You can select whether all products or only the affected products should be proposed in the **Releases** dialog box.

Field	Description
BOM levels to release	The depth of the BOM level to release. If the BOM has more levels (that is, if it's deeper) than the value that is specified here, only the levels up through the specified value will be released.

Last updated on 05/07/2025

Establish common values for engineering change management

Article • 06/03/2022

When you set up engineering change management, you must establish several collections of values that will be used to fill in drop-down lists in other parts of the user interface (UI). You should specify these values according to the types of products that you produce and your specific business needs.

Engineering change categories

You use engineering change categories to organize your engineering change orders, so that they are easier to manage and review. For example, you might find it useful to set up a workflow where, depending on the category, a specific department must review the proposed changes. Therefore, the engineering change order includes a **Category** field.

To establish the collection of engineering change categories that is used in your company, go to **Engineering change management** > **Setup** > **Engineering change management** > **Engineering change categories**. You can then use the buttons on the Action Pane to add, remove, and edit values, and to arrange them into the order in which they should appear in the drop-down lists where they are shown.

Engineering change priorities

You use engineering change priorities to indicate the importance or urgency of an engineering change order. They can help you keep track of the importance of an engineering change order, so that you can easily identify which orders should be processed first, and how quickly.

To establish the collection of engineering change priorities that is used in your company, go to **Engineering change management** > **Setup** > **Engineering change management** > **Engineering change priorities**. You can then use the buttons on the Action Pane to add, remove, and edit values, and to arrange them into the order in which they should appear in the drop-down lists where they are shown.

Engineering change reasons

Engineering change reasons indicate the cause or nature of the change in the change order.

To establish the collection of engineering change reasons that is used in your company, go to **Engineering change management** > **Setup** > **Engineering change management** > **Engineering change reasons**. You can then use the buttons on the Action Pane to add, remove, and edit values, and to arrange them into the order in which they should appear in the drop-down lists where they are shown.

Material disposal codes

You use material disposal codes to categorize materials that are used in your finished goods, or components that must be disposed of in a specific way or require some treatment before they can be added to your regular trash. When you add a relevant product to an engineering change order, you can assign a disposal code as part of the change details.

To establish the collection of material disposal codes that is used in your company, go to **Engineering change management** > **Setup** > **Engineering change management** > **Material disposal codes**. You can then use the buttons on the Action Pane to add, remove, and edit values, and to arrange them into the order in which they should appear in the drop-down lists where they are shown.

Received customer approval

When you design products for a specific customer, the design and specifications often must be validated before the product can be set as ready. The **Received customer approval** field lets you indicate how far in the customer approval process the product is and/or whether the approval has been received.

To establish the collection of received customer approval values that is used in your company, go to **Engineering change management** > **Setup** > **Engineering change management** > **Received customer approval**. You can then use the buttons on the Action Pane to add, remove, and edit values, and to arrange them into the order in which they should appear in the drop-down lists where they are shown.

Engineering change – Environmental health and safety codes

If any standard environmental health and safety regulations, or company-specific regulations or procedures, must be considered in the manufacture of a product, you can use the environmental health and safety codes to define them. In the engineering

change order, you can indicate which codes apply to the manufacture of a product while you edit the details of the affected product.

To establish the collection of health and safety values that is used in your company, go to **Engineering change management** > **Setup** > **Engineering change management** > **Engineering change - Environmental health and safety codes**. You can then use the buttons on the Action Pane to add, remove, and edit values, and to arrange them into the order in which they should appear in the drop-down lists where they are shown.

Engineering change severities

You use engineering change severities to indicate the level of impact that applies to the products in an engineering change order.

To establish the collection of engineering change severities that is used in your company, go to **Engineering change management** > **Setup** > **Engineering change management** > **Engineering change severities**. You can then use the buttons on the Action Pane to add, remove, and edit values, and to arrange them into the order in which they should appear in the drop-down lists where they are shown.

You can establish rules that apply to each severity level that you create. For more information about how to assign these rules, see the next section.

Engineering change severity rule sets

You use engineering change severity rule sets to establish a group of rules that you can use to automatically calculate the severity of the change order, based on the type of changes in the affected products. To use the severity rules, open the **Engineering change management parameters** page, and set the **Severity rule** field to *Calculate* or *Calculate automatically*.

When the system evaluates severity, it processes the rules in the order in which they appear on the page, from top to bottom. For a rule to be selected and have its priority established, all the rules in a rule set must be met.

To set up the rules that apply to each change severity level that you've defined, go to **Engineering change management** > **Setup** > **Engineering change management** > **Engineering change severity rule sets**. Then follow one of these steps.

- To create a new rule set, select **New** on the Action Pane, and then set the fields as described later in this section.

- To edit an existing rule set, select it in the list pane, select **Edit** on the Action Pane, and then set the fields as described later in this section.
- To delete an existing rule set, select it in the list pane, and then select **Delete** on the Action Pane.
- To rearrange the list of rule sets, select a rule set in the list pane, and then use the **Up** and **Down** buttons on the Action Pane to reposition it.

For each rule set, set the following field:

- **Severity** – Select the severity level to establish rules for. You use the **Engineering change severities** page to create and name the levels. (For more information, see the previous section.)

Use the buttons on the **Rules** FastTab to add or remove a rule for the current severity setting. Each rule has a **Rule** field and a **Name** field. The rules are established by the system and indicate the types of changes that a product can have. The name indicates the type of change.

Feedback

Was this page helpful? Yes No

Provide product feedback

Engineering companies and data ownership rules

Article • 06/03/2022

Engineering companies and operational companies

To ensure that the master data for products is centrally created and maintained, you can use one or more *engineering companies*. An engineering company owns the engineering products and their engineering-relevant data. It's always connected to (based on) an existing *legal entity*, which is also a company. Through this connection, the system establishes a central entry point for all engineering-relevant data for engineering products in the engineering company. In this central entry point, engineering products are created, and engineering-relevant data is maintained. From it, the engineering products and engineering-relevant data will be released to *operational companies*, which are other legal entities. (For more information about the release management, see Release product structures.) These operational companies will use the engineering data as it has been designed by the engineering company. Any logistical data is locally maintained by each engineering company and operational company.

To create an engineering company, go to **Engineering change management** > **Setup** > **Engineering organizations**. Select **New**, enter a name for the engineering company, and select the existing company (legal entity) that it's based on.

If you're integrating with external product lifecycle management (PLM) systems, you must create a business unit (type of company) that will become an external company.

Engineering product categories and engineering companies

Engineering product categories help ensure that engineering products are created according to your company's business rules and behave as required. For more information about engineering product categories, see Engineering versions and engineering product categories.

Each engineering product category belongs to a specific engineering company and can create only products that belong to that company. Likewise, the right to maintain an

engineering product also belongs to the company that is associated with that product's engineering product category.

Data that is owned by the engineering company

Because the engineering company owns the engineering-relevant data, it controls the following processes:

- **Creation of engineering products:** Each engineering company can only create new engineering products that are based on an engineering product category that it owns. In some cases, operational companies maintain their own local data that is related to those products.
- **Creation of engineering versions:** When a company creates a new engineering product, the system automatically creates an initial engineering version for it. Only the owning engineering company will be able to create new versions of that product.
- **Creation and maintenance of engineering attributes:** When a company creates a new engineering product, the system automatically adds engineering attributes to it. Only the owning engineering company will be able to create and maintain the values for those attributes. For more information about engineering attributes, see Engineering attributes and engineering attribute search.
- **Creation and maintenance of bills of materials (BOMs) that are connected to the engineering versions:** The owning engineering company can directly connect a BOM to an engineering product version. When these BOMs are released to other legal entities, changes to the engineering data on the BOMs are limited in the following ways:
  - The operational company can't remove released BOM lines.
  - The engineering fields on the BOM lines are read-only for the operational company. All other fields are logistical implementation fields and can be edited by the operational company.
  - The operational company can add BOM lines to the same BOM. In this way, it can add local additions, such as packaging materials or lubrication fluids.
  - The operational company can add a completely new, local BOM. This change might be required in cases where, for example, no BOM is provided during the release. The operational company owns and maintains these local BOMs. For more information about release management, see Release product structures.

    - All local BOMs and BOM lines are preserved when the engineering company updates its BOM.

- **Creation and maintenance of routes that are connected to the engineering versions:** The engineering company can directly connect a route to each engineering version. When these routes are released to other legal entities, changes to the data on the routes are limited in the following ways:
    - The other legal entities can't remove the engineering data on the routes.
    - The other legal entities can add operations to the route. In this way, they can add local route steps.
    - Operational companies can add completely new, local routes. This change might be required if, for example, you haven't included routes during the release. The operational companies own and maintain these local routes. For more information about release management, see Release product structures.
    - All changes that are made locally are preserved when updates from the engineering company are released again to the routes.

- **Creation and maintenance of engineering documents:** The engineering company can attach engineering documents to each engineering version.
    - When these documents are released to other legal entities, the documents are protected from being removed by the operational company.
    - Other legal entities can add completely new, local documents. The operational company owns and maintains these local documents.

Feedback

Was this page helpful? Yes No

Provide product feedback

Product owners

The product owner is a group of users who are responsible for specific products. When a product owner group is assigned to a product, only the members of that group can release the product. The product owner can also be used in the approval workflow in engineering change management.

Product owners are global settings. Therefore, they are available to all legal entities.

Create a product owner group

To create a product owner group and add members to it, follow these steps:

1. Go to **Engineering change management** > **Setup** > **Product owners**.
2. On the Action Pane, select **New**.
3. In the **Product owner** field, enter a name for the group.
4. In the **Name** field, enter a description of the group.
5. On the **Members** FastTab, add the workers who should be members of the group.

Assign a product owner to a product

To assign a product owner to a product, follow these steps:

1. Open the **Product details** page for the relevant product or product master.
2. On the **General** FastTab, set the **Product owner** field to the name of the relevant product owner group.

While a product owner is assigned to a product, only the members of the product owner group can edit the **Product owner** setting.

The product owner is also visible on the **Released products** page.

Product owners and product releases

Only users from a product's product owner group can release that product. However, there is an exception when the product is a child item, and its parent is released by the parent's owner. In other words, if the product is part of the BOM of another product, the system doesn't check the product owner of each item in the BOM. It checks only the product owner of the parent item.

For example, product X is assigned to the *Design cabinets* product owner group. Product X is also part of the BOM of product Y, which is assigned to the *Design Speakers* product owner group. If a user from the *Design Speakers* product owner group releases product Y and its BOM, product X will be released together with product Y.

Product owners and approvals

Because product owners know whether specific engineering changes will benefit their products, it often makes sense to include them as part of the approval process in engineering change management. You can implement this approach by setting up the product owners as participant providers in the workflows that are used for engineering change management. The system will then assign approval tasks in the workflows, based on the products that are in engineering change requests and engineering change orders. Learn more in Manage changes to engineering products.

Last updated on 07/01/2026

Engineering attributes and engineering attribute search

Article • 08/15/2024

To ensure that all product master data can be registered in the system, you should use engineering attributes to specify all non-standard characteristics. You can then use engineering attribute search to easily find products, based on those registered characteristics.

Create engineering attributes and attribute types

Typically, engineering products have many characteristics and properties that you must capture. Although you can register some of the properties by using the standard product fields, you can also create new engineering properties as required. You can define your own *engineering attributes* and make them part of the product definition.

Each engineering attribute must belong to an *attribute type*. This requirement exists because each engineering attribute must have a *data type* that defines the types of values that it can hold. An engineering attribute type can be a standard type (such as free text, integer, or decimal) or a custom type (such as text that has a specific set of values to select from). You can reuse each attribute type with any number of engineering attributes.

Set up engineering attribute types

To view, create, or edit an engineering attribute type, follow these steps.

1. Go to **Engineering change management** > **Setup** > **Attributes** > **Attribute types**.
2. Select an existing attribute type in the list pane, or select **New** on the Action Pane to create a new attribute type.
3. Set the following fields:
    - **Attribute type name** – Enter a name for the attribute type.
    - **Type** – Select a standard data type (*Currency*, *DateTime*, *Decimal*, *Integer*, *Text*, *Boolean*, or *Reference*).
    - **Fixed list** – This option is available only if you set the **Type** field to *Text*. Set it to *Yes* to define specific values for attributes of this type. In this case, a drop-

down list will be created. You use the **Value** FastTab to establish the values that are available for this attribute type. Set this option to *No* to allow users to enter any value. In this case, an input field will be created.

- **Value range** – This option is available only if you set the **Type** field to *Integer*, *Decimal*, or *Currency*. Set it to *Yes* to establish minimum and maximum values that will be accepted for attributes of this type. You use the **Range** FastTab to establish the minimum and maximum values, and (for currency) the currency that applies for the limits that you entered. Set this option to *No* to accept any value.
- **Unit of measure** – This field is available only if you set the **Type** field to *Integer* or *Decimal*. Select the unit of measure that applies for this attribute type. If no unit is required, leave this field blank.

Set up engineering attributes

To view, create, or edit an engineering attribute, follow these steps.

1. Go to **Engineering change management** > **Setup** > **Attributes** > **Engineering attributes**.

2. Select an existing attribute in the list pane, or select **New** on the Action Pane to create a new attribute.

3. Set the following fields:

    - **Name** – Enter a unique name for the attribute. Both the **Name** and **Friendly name** are shown throughout the system, though you can decide to hide any of the columns. The **Name** must be unique while **Friendly name** doesn't need to be.
    - **Attribute type** – Select an attribute type that you defined in the previous section.
    - **Friendly name** – Enter a common name for the attribute (except on the **Engineering attributes** page). Unlike the **Name**, this value doesn't need to be unique, which means that there could be two or more different attribute with the same friendly name.
    - **Description** – Enter a description of the attribute.
    - **Help text** – Enter Help text that tells other users what the attribute is for.
    - **Default value** – Enter a default value for the attribute. The options that are presented depend on the attribute type that you selected.
    - **Currency** – If the attribute type that you selected is a currency, select the currency that the attribute will accept and show values in.

4. If the attribute type that you selected is an integer or a decimal, the **Range** FastTab is shown. On this FastTab, set the following fields as required:

    - **Tolerance action** – Select how the system should respond if a user enters a value outside the specified range. If you select *Warning*, a warning is shown, but the user can save the value. If you select *Not allowed*, a warning is shown, and the value can't be saved until the user corrects it.
    - **Minimum** – Enter the minimum recommended or accepted value.
    - **Maximum** – Enter the maximum recommended or accepted value.

Engineering attribute inheritance

For product structures, such as bills of materials (BOMs) or formulas, selected attributes can be passed from the children items up to the parent items. You can think of this process as "reverse inheritance."

Turn engineering attribute inheritance on or off

This feature requires that both the *Engineering Change Management* and *Improved attribute inheritance for Engineering Change Management* features be turned on for your system. For details about how to turn these features on or off, see Engineering change management overview.

Attribute inheritance example

For a food product such as a carrot cake, the system must record each allergen that the product contains. The carrot cake can be modeled in the system as an engineering product that has a formula. This formula contains the carrot cake's ingredients, such as flour, milk, carrots, and nuts. In this example, the company provides two models for carrot cake: one that contains lactose and one that doesn't.

The cake that contains lactose has the following attributes at the ingredient level:

- Ingredient "flour": attribute "gluten" = yes
- Ingredient "milk": attribute " lactose" = yes
- Ingredient "nuts": attribute "nuts" = yes

The cake that doesn't contain lactose uses lactose-free milk and has the following attributes at the ingredient level:

- Ingredient "flour": attribute "gluten" = yes
- Ingredient "milk": attribute "lactose" = no

- Ingredient "nuts": attribute "nuts" = yes

Because these products are mostly similar, it might be convenient to pass these attributes from the children (the two variations) to the parent product (the basic carrot cake). To implement this "reverse inheritance," you can use the *Attribute inheritance* functionality. This functionality is defined for each engineering version.

Connect engineering attributes to an engineering product category

Some engineering attributes apply to all products, whereas others are specific to individual products or product categories. For example, electrical attributes aren't required for mechanical products. Therefore, you can set up *engineering product categories*. An engineering product category establishes the collection of engineering attributes that must be part of the definition for products that belong to that category. You can also specify which engineering attributes are mandatory and whether there is a default value.

For more information about how to work with engineering product categories, including information about how to connect attributes to categories, see Engineering versions and engineering product categories.

Set attribute values for engineering attributes

The engineering attributes that are connected to an engineering product category are presented when you create a new engineering product that is based on that category. At that time, you can set values for the attributes. Later, those values can be changed on the **Engineering version** page or as part of engineering change management in an engineering change order. Learn more in Manage changes to engineering products.

Create an engineering product

To create an engineering product, open the **Released products** page. Then, on the Action Pane, on **Product** tab, in the **New** group, select **Engineering product**.

You must specify the engineering category that the product belongs to. The category will set all the default values and characteristics for the product. It will also set the attributes that are applicable to the product. After the category is selected, values will be set for the attributes. You can then modify those values.

Search for products by using engineering attribute values

You can use engineering attribute search to find products by searching for their engineering attributes values. Therefore, you can easily find engineering products, based on their characteristics. You can search in the products that belong to an engineering product category, or you can search across all engineering products.

The search is available on product master data pages and from transactional items in the system, such as sales orders. For a transactional item, you can use the **Engineering attribute search** page to search for a product. You can then use the **Add as new line** button to add the product to the sales order lines. Products in the search results can also be added directly to the order.

Feedback

Was this page helpful? Yes No

Provide product feedback

Engineering versions and engineering product categories

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Engineering products evolve during their product lifecycle for many reasons. For example, changes might be introduced to improve product serviceability, change a component because the supplier no longer offers it, respond to new insights, or fix mistakes in the initial design. Many reasons exist for storing these changes as part of an ongoing product without overwriting previous data. Here are some of these reasons:

- You want to keep track of the product as it was manufactured and delivered to your customers in previous lifecycle states.
- You need a lead time before you approve and apply the changes.
- You want to have a timestamp on each change, and you want to be able to deliver previously manufactured products separately from each other.

Engineering versions ensure that the various states of a product and its data are current and clear, and that they can be visualized in the system. This concept helps you maintain consistency, lock down the bill of materials (BOM) for production, eliminate variability, and easily identify changes.

Generally, the *form-fit-function* rule is applied to determine whether a change requires a new product, a new version, or an update to an existing version. Each of the three terms in the name of this rule refers to a specific aspect of a part, which helps engineers match parts to needs. The form-fit-function rule increases the flexibility of design changes, because minimal documentation and design cost are required to change a part, provided that the fit, form, and function of the product are maintained.

- **Fit** refers to the ability of the part or feature to connect to, mate with, or join to another feature or part in an assembly. The fit enables the part to meet the required assembly tolerances so that it can be useful.

- **Form** refers to characteristics of a part or assembly, such as the external dimensions, weight, size, and visual appearance. The form is the aspect that's most affected by an engineer's aesthetic choices. It includes the enclosure, chassis, and control panel, which become the outward "face" of the product.
- **Function** is a criterion that's met when the part effectively and reliably performs its stated purpose. For example, in an electronics product, function can depend on the solid-state components that are used, and the software or firmware. Often, it can also depend on the features of the selected enclosure. Two of the most common reasons that an enclosure can fail the function criterion are poorly placed or poorly sized ports, and misleading or missing labeling.

Engineering versions

When you use engineering products, each product has at least one engineering version. The initial engineering version is automatically created when you create an engineering product. Each engineering version stores the engineering-relevant data that is specific to that version. Here are some examples of this data:

- The version number and previous version number (if applicable)
- The effective-from and effective-to dates
- The product version active status, which indicates whether the version can be released and used in transactions (Learn more in Product readiness.)
- The engineering company that created and owns the product (Learn more in Engineering companies and data ownership rules.)
- Related engineering documents, such as an assembly manual, user instructions, pictures, and links
- The engineering attributes (Learn more in Engineering attributes and engineering attribute search.)
- Bill of materials (BOM) for engineering products
- Formulas for process manufacturing products
- The engineering routes

You can update this data on an existing version, or create a new version, by using an *engineering change order*. (Learn more in Manage changes to engineering products.) If you create a new version of a product, the system copies all engineering-relevant data to that new version. You can then modify the data for that new version. In this way, you can track specific data for each consecutive version. To compare the differences between consecutive engineering versions, inspect the engineering change order, which includes change types that indicate all changes.

As stated earlier, the initial engineering version is automatically created when you create an engineering product. The version number for this version follows the version number rule defined in the engineering category for the product. To transition to a subsequent version, add the product to an engineering change order as a line, and set the **Impact** field to *New version*. The engineering change order includes the details of the change from the current version to the next version.

An engineering product can be in only one engineering change order at a time. This restriction ensures data accuracy and helps avoid overlapping or contradictory changes in the product. Also, the **Engineer** field in the **Header** view of the engineering change order shows the engineer who is responsible for the change order. If the engineer belongs to a team that is defined in the system, the **Responsible** field shows the leader of that team.

Track versions in transactions

When you use engineering change management, your product master data always includes one or more engineering versions. In your setup of engineering products, you can choose whether the engineering version is also part of *logistical transactions*. (Learn more in the Set up engineering product categories section later in this article.) If the logistical impact is relevant, it differs per product and per company. Sometimes, only the latest version of a product is used. Therefore, when you introduce a new version, the previous version can no longer be used. In other cases, the previous version is required in logistical transactions to overcome the following challenges:

- The Logistics department must ship two pieces of a product to a customer. In this case, you must decide whether you want or will allow two different versions to be shipped.
- It's later discovered that a problem occurs, and that it's related to a specific change. In this case, it might be beneficial to determine exactly which version was shipped in each order.
- Companies typically want to ship old versions first, to phase them out of inventory. Especially for low-volume products, this approach can often be managed by determining the effectivity dates of the new version in relation to predictions about when stock of the old version will be depleted. However, sometimes you might not be able to make this comparison, or you may consider the uncertainty of stock level predictions to be too high.

The decision about whether to make versions visible in inventory depends on factors such as those that were previously mentioned, plus company practice and other considerations that are specific to each company. You can specify the behavior for the *engineering product category*. It applies to all products that created from that category, for all companies that the product is released to.

For products that are set up so that they have logistical impact, the engineering version must be specified on each transaction. Although the system proposes the *latest active version*, you can select from all the active versions that are available for the company. For products that are set up so that they don't have logistical impact, the engineering version isn't specified on transactions. However, the system uses the latest active version. For example, when you add a product to a production BOM, the latest version is used, and when you run master planning, the latest version is assumed.

Set up engineering product categories

An engineering product category provides a basis for creating a specific engineering product. Each category establishes a set of default values and policies. Therefore, when you create an engineering product, you first select the category to create it from.

A new category hierarchy type (*engineering product hierarchy*) is automatically set up for you. You can manually create the categories by going to **Engineering change management** > **Setup** > **Engineering product category details**.

Each engineering product category establishes the default behavior of the engineering products that you create based on that category. After you create an engineering product, you can't change its engineering product category. However, if you select the incorrect category, you can delete the product and then re-create it.

When an engineering product category is created, you're prevented from changing the following settings:

- Engineering company
- Product type
- Product subtype
- Product dimension group
- Configuration technology
- Version number rule

Other settings might inherit default values set up for the engineering product category. However, according to the system rules, you can change those values.

To work with engineering product categories, go to **Engineering change management** > **Setup** > **Engineering product category details**. Then follow one of these steps:

- To create a new category, select **New** on the Action Pane, and then set the fields as described in the following subsections.
- To edit an existing category, select it in the list pane, select **Edit** on the Action Pane, and then set the fields as described in the following subsections.
- To delete an existing category, select it in the list pane, and then select **Delete** on the Action Pane.

Header

Set the following fields on the header of an engineering product category.

Expand table

Field	Description
Name	Enter a name for the engineering product category.
Engineering company	Select the engineering company where products in this engineering product category can be created and where they're maintained.

Details FastTab

Set the following fields on the **Details** FastTab of an engineering product category.

Expand table

Field	Description
Product type	Select whether the category applies to products or services.
Production type	This field appears only when you enable formula change management in your system. Select the type of production that this engineering product category applies to: • **Planning item** – Use this engineering category to perform formula change management for planning items. Planning items use formulas. They resemble formula items, but they're used to produce only co-products and by-products, not finished products. Formulas are used during process manufacturing. • **BOM** – Use this engineering category to manage engineering products, which don't use formulas and typically (but not necessarily) include BOMs. • **Formula** – Use this engineering category to perform formula change management for finished products. These items have a formula but not a BOM. Formulas are used during process manufacturing.
Catch weight	This option appears only when you enable formula change management in your system. It's available only when the **Production type** field is set to *Planning item* or *Formula*. Set

Field	Description
	this option to *Yes* if you use this engineering category to manage items that require catch-weight support.
Track versions in transactions	Select whether the version of the product should be stamped on all transactions (logistical impact). For example, if you track the version in transactions, each sales order shows which specific version of the product was sold in that sales order. If you don't track the version in transactions, sales orders don't show which specific version was sold. Instead, they always show the latest version. • If this option is set to *Yes*, a product master is created for the product, and each version of the product is a variant that uses the *version* product dimension. The **Product subtype** field is automatically set to *Product master*, and in the **Product dimension group** field, you must select a product dimension group where the *version* dimension is active. Only product dimension groups where *version* is an active dimension are shown. You can create new product dimension groups by selecting the **Edit** button (pencil symbol). • If this option is set to *No*, the *version* product dimension isn't used. You can then select whether to create a product or a product master that uses the other dimensions. This option is often used for products that have a cost difference between versions, or products where different conditions apply in relation to the customer. Therefore, it's important to indicate which version was used in each transaction.
Product subtype	Select whether the category holds products or product masters. For product masters, product dimensions are used.
Product dimension group	The **Track versions in transactions** setting helps you select the product dimension group. If you specify that you want to track the version in transactions, the product dimension groups where the *version* dimension is used are shown. Otherwise, only product dimension groups where the *version* dimension isn't used are shown.
Product lifecycle state at creation	Set up the default product lifecycle state that an engineering product should have when it's first created. Learn more in Product lifecycle states and transactions.
Version number rule	Select the version number rule that applies to the category: • **Manual** – You choose the version number for each new version. • **Automatic** – The system sets the version number, based on a format that you define. When you set up the format, use a number sign (#) to represent a digit and any other character to represent a constant value. For example, if you define the format as *V-##*, the first version is "V-01," the second version is "V-02," and so on. • **List** – The system takes the next number from a predefined list of custom values that you define.
Enforce effectivity	Select whether the effectivity dates of engineering versions must be contiguous, or whether there can be gaps and overlaps. This setting affects the way that you can use the **Effective from** and **Effective to** fields for each engineering version where the category applies.

Field	Description
	• If this option is set to *Yes*, an **Effective from** value must be specified for each version, and neither overlaps nor gaps are allowed between versions. The date range for each engineering version is connected directly to the previous and next engineering versions, if they exist. In this scenario, the newest version is always used, and older versions are no longer used. • If this option is set to **No**, there are no restrictions on the effectivity date fields for engineering versions, and both overlaps and gaps are allowed. In this scenario, multiple versions can be active at the same time, and you can work with any active version. This option also affects BOMs and routes that are connected to a product version. Learn more in the Connect BOMs and routes to engineering versions section later in this article.
Use number rule nomenclature	Set this option to *Yes* to enable rules for defining a product number by using number sequences, engineering attribute names and values, and text constants as segments. To create or modify rules, select the **Edit** button.
Use name rule nomenclature	Set this option to *Yes* to enable rules for defining a name by using the engineering attribute names, engineering attribute values, and text constants as segments. To create or modify rules, select the **Edit** button.
Use description rule nomenclature	Set this option to *Yes* to enable rules for defining the description by using the engineering attribute names, engineering attribute values, and text constants as segments. To create or modify rules, select the **Edit** button.

Attributes FastTab

Use the grid on the **Attributes** FastTab to set up the engineering attributes that apply to products in this category. For information about how to create engineering attributes, see Engineering attributes and engineering attribute search.

Use the buttons on the **Attributes** FastTab to add, remove, and arrange attributes in the grid.

If you change the selection of attributes for an engineering category, and products already exist that are based on that category, you must decide whether to apply your changes to those products. If you want existing products to reflect the changes, select **Update existing products** on the **Attributes** FastTab.

For each row that you add to the grid, set the following fields.

Expand table

Field	Description
Name	Select the attribute to add.

Field	Description
Value	Select the default value for the attribute.
Mandatory	Choose whether the attribute is mandatory, which means users must specify a valid value for the attribute before they can save a product. The effect of this setting varies slightly based on the data type of the selected attribute, as defined in the following list. • **Boolean** – Set this value to *Yes* to require the attribute to have a value of *Yes* (the system refuses to save a product where the attribute is set to *No*). Set this value to *No* to accept a value of *Yes* or *No*. (Attributes of type *Boolean* can't have an empty value.) • **Integer or Decimal** – Set this value to *Yes* to require users to enter a non-zero value for this attribute. Set this value to *No* to allow users to save with a value of zero. (Attributes of these types can't have an empty value.) • **List** – Lists have a data type of *Text*, but also include a predefined list of possible values. Therefore, it's not possible to enter a blank value for attributes of this type, so this setting has no effect and is just informational. • **All other data types** – Set this value to *Yes* to make the attribute mandatory. Set this value to *No* to allow users to save a product without providing a value for this attribute.
Batch attribute	Select whether the attribute should be propagated through the batch functionality.

Readiness policy FastTab

Use the **Product readiness policy** field to select the readiness policy that should be applied to products that are created based on this engineering category. For more information, see Product readiness.

> ⓘ **Note**
>
> The **Product readiness policy** field works slightly differently if you turn on the *Product readiness checks* feature in your system. (That feature lets you apply readiness policies to standard [non-engineering] products). For more information, see Assign readiness policies to standard and engineering products.

Release policy FastTab

Use the **Product release policy** field to select the release policy that applies to products in this category. For more information, see Release product structures.

Connect BOMs and routes to engineering versions

The setting of the **Enforce effectivity** option is important for the connection of BOMs and routes to each engineering version. You can activate multiple BOMs or routes per product only if there's a difference in any of the following settings:

- Product dimension
- Quantity
- Site
- Effectivity dates

Engineering BOMs and routes are created from the engineering version where they apply. They can be recognized by the check mark in the **Engineering controlled** check box. When you work with engineering BOMs and routes, you typically don't design them by using different quantities. You also typically don't design different BOMs per site. Additionally, for engineering BOMs and routes, the effectivity dates always come from the engineering version. Therefore, an engineering version, its BOM, and its route all have the same effectivity dates.

For products where you use the *version* product dimension (together with logistical impact on the transactions), the version is also added to the BOMs and routes. This behavior helps differentiate the BOMs and routes of consecutive versions, regardless of the **Enforce effectivity** setting.

For products where you aren't using the *version* product dimension (without logistical impact on the transactions), the version isn't added to the BOMs or routes. Therefore, there's no difference between the BOMs and routes of consecutive versions. In this case, set the **Enforce effectivity** option to *Yes*. This setting helps prevent engineering versions from overlapping, and you can also activate the BOM and route of a newer version without first deactivating the BOM and route of the previous version. If you set the **Enforce effectivity** option to *Yes* in this case, you must manually deactivate the BOMs and routes of older versions before you can activate the latest version.

Last updated on 05/29/2026

Manage changes in formulas and their ingredients

Article • 08/15/2024

If you're using the process manufacturing capabilities of Microsoft Dynamics 365 Supply Chain Management, you can also use the related formula management capabilities to manage the following changes:

- **Formula and planning items:** Manage changes to ingredients in formulas and their co-products and by-products.
- **Co-products and by-products:** Edit quantities and other information of the co-products and by-products in a formula.
- **Catchweight items:** Manage changes to catch-weight items.

Turn this feature on or off

The functionality described in this article requires that both the *Engineering Change Management* and *Manage changes to formulas and their ingredients* features be turned on for your system. For details about how to turn these features on or off, see Engineering change management overview.

Feature naming conventions

In the Supply Chain Management user interface (UI) and documentation, change management functionality is typically referred to as *engineering change management*, and the manageable products are typically referred to as *engineering products*. Although this terminology isn't usually associated with the management of formulas for process manufacturing, you should think of engineering change management as simply change management, and you should think of engineering products as versioned products.

Work with formula change management features

The following list summarizes how engineering change management features apply to formula management. It also provides links to more information. Because formula change management takes advantage of engineering change management features, you should learn how engineering change management works in general, so that you can use it to manage your formulas and their ingredients.

- **Centralized product data management** – Set up an organization that, through a managed release process, ensures that accurate and relevant product data is available to users across the enterprise. Learn more in Engineering companies and data ownership rules.
- **Product versioning** – Track changes to products through product versions, and control the product throughout all stages of the supply chain. In this way, you can track the changes in your formulations. Learn more in Engineering versions and engineering product categories.
- **Product lifecycle management** – Manage the visibility of product data across the organization, and control the availability of product versions at each stage of the supply chain. You have detailed control over when a product version can be used in specific business processes, and you can create your own lifecycle states to fit your business needs. Learn more in Product lifecycle states and transactions.
- **Formula change management** – Users throughout your organization can request changes to formulas. Use change orders to assess and document the impact of proposed changes. Add workflows to manage the change process and the release of new versions of the product and its formula. Learn more in Manage changes to engineering products.
- **Readiness control** – Use system checks and user guidance (questionnaires and checklists) to ensure that all required product data is fully entered before the product is released. Learn more in Product readiness.
- **Enhanced product release functionality** – Release fully defined versions of a product and its formula from an organization (legal entity) to other legal entities. You can even decide whether the product information must be reviewed or edited before release. Learn more in Release product structures.

Note that most of the articles that are linked to in the previous list provide examples that are based on bills of materials (BOMs). However, formulas work in a similar way. Here are some additional concepts that are useful to know when you use change management (or only formula change management) to manage formulas and BOMs:

- For each product engineering category, you can specify the production type (BOM, formula, or planning item). You can also specify whether catch-weight support is required for products that use that category.
- Co-products and by-products aren't engineering products. Therefore, they aren't versioned. If you must change them, just create a new product. This approach makes maintenance easier.
- You can manage end items that are BOMs and that have child formula items. The functionality will work for any combination of BOMs that contain formulas and/or formulas that contain BOMs.

Feedback

Was this page helpful? Yes No

Provide product feedback

Enable change management on existing products

Article • 08/15/2024

This article explains how you can enable change management for existing products. It also describes cases where your ability to enable change management is limited.

When you enable change management for an existing product, you can create versions of that product and trace changes that are made to it throughout its life. Therefore, you can track those changes by using change orders. To enable change management, you must convert the relevant products to *engineering items* (also referred to as engineering products). Engineering products are products that are versioned and managed through change management. A wizard is provided to guide you through the conversion process.

Turn this feature on or off

The functionality described in this article requires that both the *Engineering Change Management* and *Enable change management on existing products* features be turned on for your system. For details about how to turn these features on or off, see Engineering change management overview.

Restrictions and limitations

Not all product types can be converted to all other types. The following restrictions and limitations apply:

- When you convert a product to an engineering product, it remains a *product*. It doesn't become a *product master*.
- When you convert a product master that has a specific set of dimensions, those dimensions are maintained after the change. For example, if you convert a product master that has the size dimension, it will keep the size dimension.

Therefore, if you have a distinct product, you can change it only to an engineering product that doesn't track the product dimension in transactions (that is, the version dimension isn't used). See the remaining sections of this article for more information about these issues.

Prepare for conversion by creating all required engineering product categories

An *engineering product category* must be assigned to every engineering product. You will do this assignment when you run the **Convert to engineering product** wizard. Engineering product categories must exist for all relevant standard products *before* you can convert those products.

The engineering product category provides a basis for creating an engineering product, and it establishes a set of default values and policies. Engineering attributes and their default values (as defined for the engineering category) are also applied to the resulting engineering product. You can edit the attribute values and/or add more engineering attributes to the resulting product as needed.

The engineering product category must match the product that you assign it to. For example, the product type and dimension group must match both the product and its engineering product category. Learn more in Engineering versions and engineering product categories.

> **ⓘ Important**
>
> The **Convert to engineering product** wizard can convert product only to engineering products where the version isn't tracked in transactions. Therefore, the **Track version in transactions** option must be set to *No* for engineering product categories that you create to convert existing products.

For information about how to create engineering product categories, see Engineering versions and engineering product categories.

Run the Convert to engineering product wizard

The **Convert to engineering product** wizard helps you convert one or more existing products to engineering products. It converts the products into engineering products (versioned products) where the version isn't tracked in transactions (that is, the version dimension isn't used). After the conversion is completed, you can manage the products by using Engineering change management.

The conversion is permanent. Therefore, you won't be able to reverse it later.

Each converted engineering product will continue to be released in the same companies that the original product was released in. However, the engineering bill of materials

(BOM) and routes won't automatically be released to those companies.

Follow these steps to run the **Convert to engineering product** wizard and convert a product to an engineering product.

1. Go to **Product information management** > **Products** > **Released products**.
2. In the grid, select the check box for each product that you want to convert.
3. On the Action Pane, on the **Engineer** tab, in the **Engineering change management** group, select **Convert to engineering product** to open the wizard.
4. The first page of the wizard is the **Welcome** page. If you aren't already familiar with the conversion process, carefully read the information on this page. When you're ready to continue, select **Next**.
5. The **Select details for the products to be converted** page shows each product that you selected before you opened the wizard. Inspect the list. Use the **New** and **Delete** buttons on the toolbar to add or remove products as you require.
6. Use the following fields at the top of the grid to assign default values to every product that is listed. (You will be able to change these values for individual products after the conversion is completed.) Default values won't be assigned to products where a relevant value has already been assigned.

    - **Default engineering category** – Select an initial engineering product category to assign to every listed product. The category that you select will be applied only to products that are compatible with it.
    - **Default version** – Enter the initial product version to assign to every listed product. Every engineering product has at least one engineering version.
    - **Default lifecycle state** – Select the initial product lifecycle state to assign to every listed product.
    - **Current BOM will be part of the engineering product** – Select this check box if the current BOM for each listed product should be used as a BOM for the engineering product.

    For more information about product settings, see the next step.
7. Review each product that is listed in the grid, and evaluate how well the default values that you assigned apply to it. For each row, review the following information and set any relevant fields:

    - **Product number** – The product number.
    - **Product name** – The name of the product.

- **Engineering category** – Select the engineering product category that the product should belong to after it's converted. An appropriate category must already exist for each product, as was explained in the previous section of this article. You must assign a category to every product.
- **Version** – Enter the product version to assign to the product after it's converted. For example, you might select a number that fits in the number sequence that your category already uses. Each engineering version stores the engineering-relevant data that is specific to that version. Learn more in Engineering versions and engineering product categories.
- **Product lifecycle state** – Select the product lifecycle state that the product should be in after it's converted. The product lifecycle state lets you to control which transactions are allowed for a given engineering version. Learn more in Product lifecycle states and transactions.
- **Has BOM** – A selected check box indicates that the product has a BOM. The setting of this check box can help you decide how to set the **Current BOM will be part of the engineering product** check box.
- **Current BOM will be part of the engineering product** – Select this check box if the current BOM of the product should be used as a BOM for the engineering product. That BOM will then be managed by Engineering change management. If the product doesn't have a BOM, or if you prefer to manually create a BOM for the converted product later, clear this check box.
- **Has errors** – A selected check box indicates that the product setup has one or more errors. For example, the product type or the dimension group might not match in the category. Products that have errors will be skipped and won't be converted.

8. When you've finished, select **Validate** on the toolbar to validate the product setup. For each row, the **Has errors** check box will be updated to indicate the product's status. Adjust the values until the setup of every product is free of errors.

9. When all the products are set up correctly, select **Next** to continue.

10. The **Confirm selection** page shows the number of products that have no errors in their setup and are therefore ready for conversion. It also shows the number of products that will be skipped because of errors. To run the conversion as a batch job, set the **Run in batch** option to *Yes*.

11. Select **Finish** to apply your settings and start to convert the products to engineering products.

> ⓘ Note

If your system is set up to manually accept products before they are released, you must accept each converted product by using the **Open product releases** page in the appropriate companies. Learn more in Review and accept the product before you release it in the local company.

Feedback

Was this page helpful? Yes No

Provide product feedback

Product lifecycle states and transactions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

As an engineering product goes through its lifecycle, you need to control which transactions are allowed for each lifecycle state. For example, don't add products that aren't in a mature state to a sales order. Alternatively, if a product is reaching its end-of-life state, you might want to control the inflow of that product.

For an engineering product, changes to the lifecycle state connect to the product's engineering versions. Therefore, the product's lifecycle state can also connect to its engineering versions. When you connect the product lifecycle state to an engineering version, you can use the lifecycle state to control which transactions are allowed for the engineering version.

Create and manage product lifecycle states

To work with product lifecycle states, go to **Engineering change management** > **Setup** > **Product lifecycle state** or **Product information management** > **Setup** > **Product lifecycle state**. Both paths open the same **Product lifecycle state** page. For complete details about how to work with the settings here to create and configure your lifecycle states, go to Product lifecycle states.

Lifecycle states for released products and product variants

For a product that has variants (master and variants), the product (master) has a lifecycle state and each of the variants can have a different lifecycle state.

For specific processes, if either the variant or the product is blocked, the process is blocked. To determine whether a process is blocked, the system makes the following checks:

- For engineering-controlled products:
  - If the current engineering version is blocked, block the process.
  - If the current variant is blocked, block the process.
  - If the released product is blocked, block the process.

- For standard products:
  - If the current variant is blocked, block the process.
  - If the released product is blocked, block the process.

For example, suppose you only want to sell one variant (red) of a given product (t-shirt) and block sales of all other variants for now. You could implement this using the following setup:

- Assign the product a lifecycle state that allows the process. For example, assign the t-shirt product a lifecycle state of *Sellable*, which allows the *Sales order* business process.
- Assign the sellable variant a lifecycle state that allows the process. For example, assign the red variant a lifecycle state of *Sellable*.
- Assign all other variants a lifecycle state that blocks the process. For example, assign the white variant (and all other variants) a lifecycle state of *Not sellable*, which blocks the *Sales order* business process.

Default product lifecycle states

The **Default when released to a legal entity** setting for lifecycle states doesn't apply to engineering products. Instead, the engineering change category specifies the lifecycle state of an engineering product version after it's created in the engineering company. When you release the product to an operational company, the lifecycle state of the product is copied. In other words, when you release an engineering product to an operational company, it has the same lifecycle state that it had in the engineering company. You can overwrite the lifecycle state in the operational company.

The default lifecycle state for an engineering version is specified by its engineering category. The state defaults when you create a new engineering version, including the first version of a new product.

When you create a new product or engineering product, you can also set the default lifecycle state by specifying it on the template released product of the release policy assigned to the product.

In this case, it's possible for the product to have a different lifecycle state than the version when you create a new engineering product.

Last updated on 12/19/2025

Manage changes to engineering products

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Engineering change management provides structured processes for managing changes to engineering products. Use the *engineering change request* process to propose and request changes, and then use the *engineering change order* process to make those changes. Users can create engineering change requests or engineering change orders. A process exists for reviewing and approving them, assessing their impact on existing transactions, and following up on them.

Engineering change requests

The engineering change request process lets you capture requests for changes from all the relevant departments in your company. It doesn't matter whether you work as an engineer, or in the Manufacturing, Sourcing, Warehouse, or Sales department: anybody can use an engineering change request to request a change. This change might be an idea for a new product, an issue that you discovered while working with an existing product, a suggestion for improving an existing product, or something else.

After someone submits a request for change, a workflow manages the *review and approval* process and identifies who must approve the change (for example, the product owner).

To set up a workflow for engineering change orders or engineering change requests, go to **Engineering change management** > **Engineering workflows**. Select **New**, select whether the workflow reviews engineering change orders or engineering change requests, and then configure the workflow.

For more information about workflows, see Workflow system overview. For more information about product owners, see Product owners.

Create a new engineering change request

To create an engineering change request, follow one of these steps.

- Go to **Engineering change management** > **Common** > **Engineering change management** > **Engineering change requests**, and then select **New** on the Action Pane.
- Open the **Product details** page for an existing engineering product. Then, on the Action Pane, on the **Engineer** tab, in the **Engineering change management** group, select **Engineering change request** > **New engineering change request**.

A new change request is created. You can now set the fields on each FastTab, as described in the following subsections.

General FastTab

The **General** FastTab lets you provide a basic description of the change request. The following table describes the fields on this FastTab.

Expand table

Field	Description
Change request	Enter a name for the engineering change request.
Title	Enter text that briefly describes or identifies the changes in the request.
Priority	Select a value to indicate how high the priority of the change is. You can customize the available values for your company, as required. (Learn more in Establish common values for engineering change management.)
Category	Select a value to describe the type of change that you're requesting. You can customize the available values for your company, as required. (Learn more in Establish common values for engineering change management.)
Severity	Select a value to indicate the severity of the issue that should be fixed by implementing the request. You can customize the available values for your company, as required. (Learn more in Establish common values for engineering change management.)
Requested by	The name of the user who created the request.
On	The date when the request was created.
Status	The status of the request. When a request is first created, the status is *Created*. When the request is approved, the status changes to *Approved*. If a related change order has been created for the request, the status changes to *Followed up*.
Change	The change order number, if the change request was followed up on via a change order.

Field	Description
order	

Information FastTab

The **Information** FastTab lets you add more information about the request.

To add a row to the grid, select **New** on the toolbar above the grid, and then select one of the following options:

- **File** – Upload a file.
- **Image** – Upload an image file.
- **Note** – Enter a note directly in the grid.
- **URL** – Enter a URL directly in the grid.

The following table describes the fields on each row.

Expand table

Field	Description
Created date and time	The date and time when the row was created.
Type	The type of information that the row was created for (file, image, note, or URL).
Description	Enter a description for the row.
Restriction	A value that indicates whether the information that you added came from an internal or external source.
Attached	A selected check box indicates that the row includes an attachment (file or image). To download the attachment, select the row, and then select **Open** on the toolbar above the grid.

Products FastTab

The **Products** FastTab lets you list each product that the change request affects. Use the buttons on the toolbar to add products to the grid or to remove products.

This list is for informational purposes only. Therefore, you can add as many related products as you consider relevant. If you create a change request from the **Product details** page for an existing product, that product should be listed on the **Products** FastTab after you save the request record.

Source FastTab

The **Source** FastTab lets you track the starting point of the change request. It's useful if, for example, you want to see whether the change request was created from a sales order, who created it, and which company it was created in.

Evaluate the business impact of a change request and send notifications

When you review a request for change, search for dependencies. This approach helps you assess the impact of the requested change on open transactions, such as sales orders, production orders, and on-hand inventory. As you review change requests, send notifications to the people responsible for fulfilling the various types of related orders.

Review affected transactions, block selected transactions, and send notifications

To review affected transactions, block selected transactions, and send related notifications, follow these steps:

1. Go to **Engineering change management** > **Common** > **Engineering change management** > **Engineering change requests**.

2. Open an existing change request, or select **New** on the Action Pane to create a new change request.

3. On the Action Pane, on the **Change request** tab, in the **Business impact** group, select one of the following buttons:

    - **Search** – Scans all open transactions, and then opens the **Business impact to open transactions** dialog box, which lists all transactions that the change affects.
    - **View previous search** – Opens the **Business impact to open transactions** dialog box, which lists the results of the previous search. (A new search isn't done.)

4. The **Business impact to open transactions** dialog box provides a set of tabs, each of which shows a list of affected transactions of a specific type (**Sales orders**, **Purchase orders**, **Production orders**, **Inventory**, and so on). Each tab also shows a number that indicates the number of affected transactions of that type. Select a tab to view the relevant list.

5. To work with a transaction in the list, select it, and then select one of the following buttons on the toolbar:

- **View transaction** – Opens the selected transaction record.
- **Block order** – This button is available only on the **Sales orders** tab. Select it to block the selected sales order.
- **Block line** – This button is available only on the **Purchase orders** tab. Select it to block the selected purchase order line.
- **Notify responsible** – This button is available only on the **Sales orders** tab. Select it to send a change notification to the user who is set as responsible for the selected sales order. For more information about who can see the notifications and how, see Review and process change notifications for transactions.
- **Notify orderer** – This button is available only on the **Purchase orders** tab. Select it to send a change notification to the user who is set as the orderer for the selected purchase order. For more information about who can see the notifications and how, see Review and process change notifications for transactions.
- **Notify production** – This button is available only on the **Production orders** tab. Unlike sales orders and purchase orders, production orders don't have a single user who is set as responsible for them from end to end. Instead, various supervisors or planners usually take ownership for a specific site or for a specific part of the production (for example, for specific resources or resource groups). Therefore, when you select this button, all users who are responsible for any resource that is related to the selected production order receive a change notification. For more information about who can see the notifications and how, see Review and process change notifications for transactions.
- **Notify preparer** – This button is available only on the **Purchase requisition** tab. Select it to send a change notification to the user who is set as the preparer of the selected purchase requisition. For more information about who can see the notifications and how, see Review and process change notifications for transactions.
- **Notify sales responsible** – This button is available only on the **Quotations** tab. Select it to send a change notification to the user who is set as responsible for the selected quotation. For more information about who can see the notifications and how, see Review and process change notifications for transactions.
- **Scrap** – This button is available only on the **Inventory** tab. Select it to scrap the selected inventory.
- **View history** – Opens a history of actions that are taken on the selected transaction by using the **Business impact to open transactions** dialog box. (For example, the history shows whether notifications are sent or transactions are blocked.)
- **View all transactions** – Opens the full list of all transactions, not just the open transactions.

> **Important**
>
> The **Notify production** button is available only if the *Engineering notifications for production* feature is turned on for your system. For instructions on how to turn this feature and its prerequisites on or off, see Engineering change management overview.

Review and process change notifications for transactions

You can read and process the change notifications that you receive in the following ways:

- Except in the case of production orders, change notifications for the transactions that you're responsible for appear in the Action center. The **Show messages** button (bell symbol) on the right side of the navigation bar indicates when a message is available for you in the Action center. Select the **Show messages** button to open the Action center and review the messages.
- To view all production orders that an engineering notification is sent for, go to **Production orders** > **Production orders** > **All production orders**. Then, on the Action Pane, on the **Production order** tab, in the **Engineering change request** group, select **Engineering notifications** to open the **Engineering notifications** page.
- For production orders, you can choose to review only the change notifications that apply to the production resources that you manage. In the **Production floor management** workspace, on the Action Pane, select **Configure my workspace** to filter the page so it shows only information about the production units, groups, and/or resources that you manage. In the **Summary** section, a tile that is named **Production orders with changed products** shows a count of notifications that match your filter settings. Select this tile to open the **Engineering notifications** page, which shows the full list of transactions that meet the criteria of your filter.

As you're reviewing production order notifications on the **Engineering notifications** page, you can follow links to related change orders or production orders by selecting column values or using the related commands on the Action Pane. After you finish evaluating a change, and after you cancel or modify production orders as required, you can mark a notification as resolved. Select the notification, and then, on the Action Pane, select **Resolve**. The notification is removed from all users' views.

> **Important**

The ability to send notifications for production orders requires that the *Engineering notifications for production* feature be turned on for your system. For instructions on how to turn this feature and its prerequisites on or off, see **Engineering change management overview**.

Create a change order from a change request

An engineer who reviews an engineering change request can create an engineering change order directly from the **Engineering change requests** page. On the Action Pane, on the **Change request** tab, in the **Engineering change order** group, select **Copy link and products**.

Select the correct company for the new engineering change order. If the change order results in the engineering product itself being changed (new version, new product, or new variant), assign the change order to the engineering company. If only a local change is needed (**Impact** is set to *None*), assign the change order to a local company and the changes apply to the current product.

Engineering change orders

Engineering change orders provide a structured process for making changes to engineering products. Propose changes by using a copy of the engineering-relevant data. The real master data isn't affected. For more information about engineering-relevant data, see Engineering versions and engineering product categories.

You can create an engineering change order that is based on an approved engineering change request. Engineers can also create engineering change orders from scratch. You can include multiple products on a single engineering change order by following any of these steps:

- Manually select products.
- Use the bill of materials (BOM) to include products that are lower in the product structure (that is, children).
- Use a where-used search to include products that are higher in the product structure (that is, parents).

After the proposal for changes is completed, a workflow handles the review and approval process. You can set up different workflows, based on priority and severity.

To set up a workflow for engineering change orders or engineering change requests, go to **Engineering change management** > **Engineering workflows**. Select **New**, select whether the

workflow reviews engineering change orders or engineering change requests, and then configure the workflow.

For more information about workflows, see Workflow system overview.

Here are some typical stakeholders who might have to approve an engineering change order:

- **Product owners** – For more information about product owners, see Product owners.
- **Responsible team lead** – The **Engineer** field in the **Header** view of the engineering change order shows the engineer who created the engineering change order. If the engineer belongs to a team that is defined in the system, the **Responsible** field shows the leader of that team.
- **Finance department** – The Finance department might have to review cases where the change involves high costs.

You can choose whether to process the engineering change order directly after it's approved, as part of the workflow, or whether to process it later as a manual step. During processing of an engineering change order, engineering data on the actual product is updated.

While you're reviewing a request for change, on the Action Pane, on the **Change request** tab, in the **Business impact** group, select **Search** to assess the impact of the proposed change on open transactions, such as sales orders, production orders, and inventory on-hand. The results are shown in the **Business impact to open transactions** dialog box, where you can select impacted transactions and then use commands in the toolbar to view more information, notify the responsible user, or block the transaction.

Engineering change orders in engineering or operational companies

As described in Engineering companies and data ownership rules, the product data that you can edit varies depending on the type of legal entity that you're working in (an engineering company versus an operational company). Data ownership rules also apply to engineering change orders. Therefore, depending on the legal entity where you create an engineering change order, you can make different types of changes. Here are some examples:

- For engineering change orders in an *engineering company*, you can make basic changes to the engineering data. For example, you can create new versions of a product, change a product's structure via the BOM, and change engineering attribute values. For each affected product, select one of the following values in the **Impact** field:
  - **None** – Update the existing product version (in-version update).

- **New version** – Create a new version that is based on the selected product version.
- **New product** – Create a completely new product that is based on the selected product version.
- **New variant** – Create a new variant based on the selected product version. Its BOM and route information are copied.

- For engineering change orders in an *operational company*, you can change the logistical data of the product. For example, you can enrich the existing BOM with settings for sourcing, add local routes or local BOMs, and even enrich a BOM by adding new BOM lines for local packaging materials, lubrication fluids, or instructions in the local language. Enrichments that users make in the operational company are preserved when new updates are sent from the engineering company. Learn more in Engineering companies and data ownership rules.

  When engineering change orders are processed in the engineering company, the products are created and/or updated only in the engineering company. Therefore, if you need to update the product master data, you must also release the products to operational companies.

- You can release products directly from engineering change orders. Open the change order, and then, on the Action Pane, on the **Change order** tab, in the **Product releases** group, select **Release product structure**. The process works just as it works when you release products from the **Released products** page. Learn more in Release product structures.

- You can have products automatically released from engineering change orders, based on the following factors:
  - Re-releases to companies where products were previously released. Select **Search** to scan all previous releases, and then select **View** to view the results. The **View** page shows the previous product releases, and you can select which products you want to re-release. Then close the **View** page, and select **Process** to re-release the selected products.
  - Automatic release settings in the release control of the engineering product category. You can do this release as part of the workflow. When the **collect release proposal** block is used, the release proposal is filled with re-releasing proposals (see the previous list item), and products are released to companies if the **Auto release** check box is selected in the release control of the engineering product category. You can select **View** to view the results, as described in the previous list item. The products are also released when the **process release proposal** block is used. If you choose only to collect the release proposal as part of the workflow, you can manually start the release by selecting **Process**, as described in the previous list item.

Follow up on an engineering change request via an engineering change order

As soon as an engineering change request is approved, you can follow up on it via an engineering change order. You can combine multiple engineering change requests into a single engineering change order. A single engineering change order can even include multiple products. (Typically, you use this approach when the same change must be applied to multiple products.) However, you can't create multiple engineering change orders from a single engineering change request.

To follow up on a change request via a change order, open the change request, and then, on the Action Pane, on the **Change order** tab, in the **Engineering change order** group, select **Copy link and products**. You can then select an existing engineering change order to connect the change request to, or you can create a new engineering change order for that specific request.

Engineering change order report

Engineering change order reports describe the changes made in an engineering change order. They're useful both during and after the review and approval process.

To view an engineering change order report, open the relevant change order. Then, on the Action Pane, on the **Change order** tab, in the **View** group, select **Engineering change order report**.

Fields on an engineering change order

Most of the fields on engineering change orders are the same as the fields for released products, engineering versions, documents, BOMs (lines), and routes (operations). However, the fields in the following table are unique to change orders.

Expand table

Field	Description
Engineering change reasons	Select the reason for changing the affected product.
Change description	Enter a description of the change.
Required special tooling	Specify whether special tooling is required to apply the change.
Engineering material disposition	Select a material disposition code for any waste that is produced when the change is applied.

Field	Description
Customer approval required	Specify whether customer approval is required before the change can be applied.
Received customer approval	Specify the status of the customer approval.
Environmental health and safety	Specify whether environmental health and safety rules are applicable to the change. If they are, select the applicable rules.

Use the **Maintain/copy change information** button to copy change information between affected products.

Use electronic signatures to approve and activate BOMs and routes

To use electronic signatures to approve and/or activate bills of material (BOM) and/or route changes, go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature requirements**. Then make sure each of the following items has **Signature required** set to *Yes*:

- Activate engineering change order product bill of materials
- Activate engineering change order product routes
- Approve engineering change order product bill of materials
- Approve engineering change order product routes
- Approve engineering version BOM and BOM versions
- Approve engineering version and route version

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Product readiness

Article • 06/03/2022

You can use readiness checks to help ensure that all the required master data has been specified for a product before it's used in transactions. When readiness checks are used, a user or team is made responsible for validating specific predefined product-related data.

You can mark the **Active** check box for an engineering product, variant, or version after all the required data has been entered and verified, and after all the readiness checks have been processed. If one or more checks haven't been processed for the product, version, or variant, then when you try to mark the **Active** check box, you will receive a prompt warning you that not all checks have been completed.

You can create readiness checks for new engineering products, variants, and versions. You can also apply readiness checks to standard (non-engineering) products. (Learn more in the Readiness checks on standard products section later in this article.)

You can use standard products in transactions even though not all readiness checks have been completed. If you need to block a product from being used in transactions, use its lifecycle state. You can assign a lifecycle state that blocks a product from being used in transactions and then, after all readiness checks have been completed, assign a new lifecycle state that allows the required transactions.

Types of readiness checks

There are three types of readiness checks:

- **System check** – The system verifies whether there is a valid record. For example, the record might be an active bill of materials (BOM).
- **Manual check** – A user verifies whether the record is valid. For example, a readiness check might require validation of the default order settings. In some cases, such as when the product is still being designed and therefore won't be placed in stock, no default order settings are required. However, default order settings might be required for another product of the same type, because the product can be held in stock. The user is responsible for knowing how to correctly decision whether a readiness check is required.
- **Checklist** – The user answers a series of questions from a checklist, and the system determines whether the answers meet expectations. The checklist can have any subject. For example, it can be used to determine whether marketing materials or product documentation is completed.

How readiness checks are created for a new engineering product, variant, or version

Readiness check policies can be applied at the released product level, the released variant level, and the engineering version level.

When you create a new *engineering product*, the system determines whether a readiness check policy applies to it. If a readiness check policy applies, the following events occur:

- Readiness checks are created for the product, according to the applicable policy.
- The engineering version is set to inactive to block the product from being used. All engineering versions for the product are set to inactive.

If a new *variant* is created for a product, the system checks whether a readiness check policy applies to it. (Readiness checks can be applied at the released variant level and the engineering version level.) If a policy applies, the following events occur:

- Readiness checks are created for the product, according to the applicable policy.
- The engineering version and variant are set to inactive to block the product from being used.

If a new engineering *version* is created for a product, the system checks whether a readiness check policy applies to it. (Readiness checks can be applied at the engineering version level.) If a policy applies, the following events occur:

- Readiness checks are created for the product, according to the applicable policy.
- The engineering version is set to inactive to block the product from being used.

> **Note**
>
> You can also set up readiness check policies for standard (non-engineering) products. Learn more in the Readiness checks on standard products section later in this article.

View readiness checks

View open readiness checks for a product or product version

To find the open readiness checks for a product, open the **Released products details** page. Then, on the Action Pane, on the **Product** tab, in the **Readiness checks** group,

select **Readiness checks**.

To find the open readiness checks for a product version, open the **Engineering versions** page for a released product and choose a version. Then, on the Action Pane, on the **Product** tab, in the **Checklist** group, select **Readiness checks**.

View open readiness checks that are assigned to you

To view the open readiness checks that are assigned to you, follow one of these steps.

- Go to **Engineering change management** > **Common** > **Product readiness** > **My open readiness checks**.
- Go to **Product information management** > **Workspaces** > **Product readiness for discrete manufacturing**.

The setup that specifies who a readiness check is assigned to is done for the readiness policy. Readiness checks can be assigned to a person or a team. If a readiness check is assigned to a team, there is one person on the team who must process the readiness check.

Process open readiness checks

Process system and manual readiness checks

After you open the **Readiness checks** page, you can view the subject of system and manual readiness checks by selecting **View related information** on the Action Pane. You can then complete and/or validate the data for the readiness check. Open readiness checks have a **Status** value of *Pending*. This status indicates that the readiness check must still be processed. To process a readiness check, follow one of these steps.

- On the Action Pane, select **Check/complete** to review and complete the readiness check. When you've finished, the **Status** field is updated to *Passed*.
- On the Action Pane, select **Skip** if you want to skip a readiness check that isn't mandatory. For example, you set up a readiness check for price calculations. However, you decide to skip this check while the product is still in the design phase. In this case, the **Status** field is updated to *Skipped*.

Depending on the configuration of the readiness policy, when the **Status** field for a readiness check is updated to *Passed*, an extra step might be required to approve the readiness check. In this case, select **Approval** to complete the readiness check. This approval step is always mandatory when the readiness check is skipped.

When all the open readiness checks for a new product, variant, or version have been processed and approved as required, the item automatically becomes active and therefore ready to use.

Process checklist readiness checks

To open a checklist, open the **Readiness checks** page, and then, on the Action Pane, select **Start checklist**. When you've completed the checklist, the system validates whether the readiness check is passed, based on the settings in the questionnaire. If the check is passed, the **Status** field is updated to *Passed*. If the readiness check isn't mandatory, you can skip it. In this case, the **Status** field is updated to *Skipped*.

Depending on the configuration of the readiness policy, when the **Status** field for a readiness check is updated to *Passed*, an extra step might be required to approve the readiness check. In this case, select **Approval** to complete the readiness check. This approval step is always mandatory when the readiness check is skipped.

When all the open readiness checks for a new product, variant, or version have been processed and approved as needed, the item automatically becomes active and therefore ready to use.

Create and manage product readiness policies

Use product readiness policies to manage the readiness checks that apply to a product. Each readiness policy contains a set of readiness checks. When a readiness policy is assigned to an engineering product category or a shared product, all the products that are related to that category or shared product will have the readiness checks that are included in the readiness policy.

To work with product readiness policies, go to **Engineering change management** > **Setup** > **Product readiness policies**. Then follow one of these steps.

- To create a new policy, select **New** on the Action Pane, and then set the fields as described in the following subsections.
- To edit an existing policy, select it in the list pane, select **Edit** on the Action Pane, and then set the fields as described in the following subsections.
- To delete an existing policy, select it in the list pane, select **Edit** on the Action Pane, and then, on the **General** FastTab, make sure that the **Active** option is set to *No*. Then select **Delete** on the Action Pane.

Header

Set the following fields on the header of a product readiness policy.

Expand table

Field	Description
Name	Enter a name for the policy.
Description	Enter a description of the policy.

General FastTab

Set the following fields on the **General** FastTab of a product readiness policy.

Expand table

Field	Description
Product type	Select whether the policy applies to products of the *Item* or *Service* type. You can't change this setting after you save the record.
Active	Use this option to help maintain your readiness policies. Set it to *Yes* for all readiness policies that you use. Set it to *No* to mark a readiness policy as inactive when it isn't used. Note that you can't inactivate a readiness policy that is assigned to an engineering product category or a shared product, and you can delete only inactive release policies.

Readiness control FastTab

For each type of readiness check that you want to include in the policy, add a row on the **Readiness control** FastTab. Use the following buttons on the FastTab toolbar to add and remove rows as you require:

- **Add check** – Add a standard readiness check to the policy. When you select this button, the **Add check** dialog box appears. There, you can select from a list of available checks.
- **Add existing questionnaire** – Add a blank row to the grid. You can then assign an existing questionnaire by setting the fields in the following table.
- **Copy** – Add a copy of the selected row to the grid.
- **Delete** – Delete the selected row from the grid.

For each row that you add, set the following fields.

Expand table

Field	Description
Process area	Select the area that the check is related to.
Type	Select whether the check is a system check, a manual check, or a checklist (questionnaire).
Name	If the check is a checklist, enter a name. For system and manual checks, this field is automatically set.
Description	If the check is a checklist, enter a description. For system and manual checks, this field is automatically set, and the description explains the focus of the check.
Apply checks on	Select whether the row should generate readiness checks in response to a new released product, released variant, or released version.
Execute in	Select whether readiness checks that the row generates apply to all companies or a single company.
Company	If you set the **Execute in** field to *Single company*, select the company.
Owner type	Select whether readiness checks that the row generates should be assigned to a person or a team.
Owner	Select the person or team that readiness checks that the row generates should be assigned to.
Questionnaire	Select the questionnaire that should be used for the checklist. The checklist is a local checklist in the company where the readiness check is done. The system must be able to evaluate whether the checklist is correctly answered. Therefore, the checklist must be set up so that an evaluation is done based on correct answers. For more information about how to create questionnaires, see Using questionnaires and its related articles.
Automatic approval	Readiness check records include an **Approved** check box that indicates the approval status. Select the **Automatic approval** check box for checks that should be set to approved immediately after the assigned user completes them. Clear this check box to require explicit approval as an extra step.
Mandatory	Select this check box for checks that must be completed by the assigned user. Mandatory checks can't be skipped.

Assign readiness policies to standard and engineering products

When you create a new product based on an engineering category, you create both a *released product* and a related *shared product*. The way that readiness policies are resolved for a released product depends on whether the *Product readiness checks*

feature is turned on for your system (see the Readiness checks on standard products section later in this article for details about this feature and how to turn it on or off).

- When the *Product readiness checks* feature is turned *off* on your system, the readiness policy is set and shown only on engineering category records. To learn which policy applies to a released product, the system checks the **Product readiness policy** field for the related engineering category. You can change the readiness policy for an existing product by editing the related engineering category (not the shared product).
- When the *Product readiness checks* feature is turned *on*, it adds a **Product readiness policy** field to the **Product** page (where shared products are set up) and to the **Released product** page (where the value is read-only and is taken from the related shared product). The system finds the readiness policy for a released product by checking the related shared product. When you use an engineering category to create a new engineering product, the system creates both a shared product and a released product, and copies any **Product readiness policy** setting for the engineering category to the new shared product. You can then change the readiness policy for an existing product by editing the related shared product (not the released engineering category).

To assign a readiness policy to a shared product, follow these steps.

1. Go to **Product information** > **Products** > **Products**.
2. Open or create the product that you want to assign a readiness policy to.
3. On the **General** FastTab, set the **Product readiness policy** field to the name of the policy that should apply to the product.

To assign a readiness policy to an engineering category, follow these steps.

1. Go to **Engineering change management** > **Setup** > **Engineering product category details**.
2. Open or create the engineering category that you want to assign a readiness policy to.
3. On the **Product readiness policy** FastTab, set the **Product readiness policy** field to the name of the policy that should apply to the engineering category.

Readiness checks on standard products

You can enable product readiness checks for standard (non-engineering) products by turning on the *Product readiness checks* feature in Feature management. This feature makes a few small changes to the readiness check system so that it supports standard products.

Enable or disable readiness checks on standard products

This feature requires that both the *Engineering Change Management* and *Product readiness checks* features be turned on for your system. For details about how to turn these features on or off, see Engineering change management overview.

Create readiness policies for standard products

You create readiness policies for standard products just as you do for engineering products. See the information earlier in this article.

Assign readiness policies to standard products

To assign a readiness policy to a standard product, open the related shared product, and set the **Product readiness policy** field to the name of the policy that should apply. Learn more in the Assign readiness policies to standard and engineering products section earlier in this article.

View and process readiness checks on standard products

When this feature is turned on, you view and process readiness checks for standard products just as you do for engineering product. See the information earlier in this article.

Feedback

Was this page helpful? Yes No

Provide product feedback

Release product structures

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To ensure that engineering-relevant product data can be easily reused in different legal entities, release complete product structures in addition to releasing products together with their engineering versions. You can release multilevel bill of materials (BOM) structures together with the parent in a single release action. In this case, the BOM and the lower-level products are also released.

Engineering companies create and maintain engineering products to meet quality requirements as they're designed. Each operational company that manufactures a product needs the same product and underlying BOM. Depending on the production facility, you might create the route locally. In this case, you don't release a route together with the product. For legal entities that sell the products but don't manufacture them, the BOM might not be required.

To make the process more efficient, all engineering-relevant data can be released to other operational companies at the same time. This data includes the product structure. During the release process, you can choose which part of the product data to release.

For more information about engineering companies and operational companies, see Engineering companies and data ownership rules.

You can release both standard products and engineering products together with the release product structure. During this process, the whole product structure is released, even the BOM and route from the company that the products are being released in.

For an example of how to release a product, see Release an engineering product to a local company.

Released data for a product when the release product structure is used

The release of engineering products includes the following data:

- **Product data** – A new released product is created.
- **Engineering version data** – The engineering version and its data are created or updated. If you release the same engineering version again to an operational company, the engineering data is overwritten.
- **Engineering attributes** – The engineering attributes and their values are created or updated.
- **Engineering bill of materials** – The engineering BOM and its lines can be created or updated. For more information about data ownership, see Product owners.
- **Engineering routes** – The engineering routes and their operations can be created or updated. For more information about data ownership, see Product owners.
- **Engineering documents** – The engineering documents that are connected to the engineering version are created or updated.

When you turn on engineering change management on your system, the release product structure is available. In addition, standard products include their BOMs and routes when they're released.

Product acceptance

Product acceptance is a key parameter that influences the release process. Set this parameter for each company by going to **Engineering change management** > **Setup** > **Engineering change management parameters**. For more information, see Engineering change management parameters.

Automatic product acceptance

Each release of engineering products starts when someone from the engineering company selects a product to release. When the **Product acceptance** parameter is set to *Automatic*, the user at the engineering company decides which product data to automatically release to the operational companies. The product is then automatically released to the companies that the user selects in the release wizard.

Manual product acceptance

Each release of engineering products starts when someone from the engineering company selects a product to release. When the **Product acceptance** parameter is set to *Manual*, the user at the engineering company decides which product data to release to the operational companies.

A user from each operational company then reviews the product data and decides whether to accept the release. The user at the operational company can set the following options when the data is received:

- If the products (updates) aren't relevant for the operational company, the user can choose not to accept the release.
- The user can change the item template for new products.
- The user can choose whether the product should be released together with their BOMs and/or routes, and whether they should be released as approved and active.
- The user can change the effective-from dates of the products.

For an example of how to accept a product, see Review and accept the product before you release it in the local company.

> **Note**
>
> For standard products, you can release from any legal entity to any other legal entity. For engineering products, the only legal entity that you can release from is the engineering company.

Release policies

Not all operational companies need the same product data. In general, operational companies that manufacture engineering products require a BOM, whereas operational companies that only sell engineering products don't require a BOM. Use release policies to establish the parameters that are used for the release of products.

For more information about engineering product categories, see Engineering versions and engineering product categories.

During the release process, you can influence the settings.

Create and manage product release policies

To work with product release policies, go to **Engineering change management** > **Setup** > **product release policies**. Then follow one of these steps.

- To create a new policy, select **New** on the Action Pane, and then set the fields as described in the following subsections.

- To edit an existing policy, select it in the list pane, select **Edit** on the Action Pane, and then set the fields as described in the following subsections.
- To delete an existing policy, select it in the list pane, select **Edit** on the Action Pane, and then, on the **General** FastTab, make sure that the **Active** option is set to *No*. Then select **Delete** on the Action Pane.

Header

Set the following fields on the header of a product release policy.

Expand table

Field	Description
Name	Enter a name for the policy.
Description	Enter a description of the policy.

General FastTab

Set the following fields on the **General** FastTab of a product release policy.

Expand table

Field	Description
Product type	Select whether the policy applies to products of the *Item* or *Service* type. You can't change this setting after you save the record.
Production type	This field appears only when you enable formula change management in your system. Select the type of production that this release policy applies to: • **Co-product** – Use this release policy to manage co-products. Co-products are produced during process manufacturing, and aren't versioned or engineering products. Release policies for co-products can help ensure that important settings, such as **Storage dimension group** and **Tracking dimension group**, are set up by using a Released product template before they're released to a company. • **By-product** – Use this release policy to manage by-products. By-products are produced during process manufacturing, and aren't versioned or engineering products. Release policies for by-products can help ensure that important settings, such as **Storage dimension group** and **Tracking dimension group**, are set up by using a Released product template before they're released to a company. • **None** – Use this policy to manage standard products that aren't versioned or engineering products, or co-products or by-products. • **Planning item** – Use this release policy to manage planning items that are produced by using process manufacturing. Planning items use formulas. They resemble formula

Field	Description
	items, but they're used to produce only co-products and by-products, not finished products. • **BOM** – Use this release policy to manage engineering products, which don't use formulas and typically (but not necessarily) include BOMs. • **Formula** – Use this release policy to manage finished items that are produced by using process manufacturing. These items have a formula but not a BOM.
Apply templates	Select one of the following options to specify whether and how product release templates should be applied when the policy is used: • **Always** – Always use a template released product for releases. If you select this option, use the **All products** FastTab to specify the template that is used for each company that you release to. If you don't specify a template for each company that is listed on the **All products** FastTab, you receive an error when you try to save the policy. • **Optional** – If a template released product is specified for a company that is listed on the **All products** FastTab, that template is used when you release to that company. Otherwise, no template is used. If you select this option, you can save the policy without assigning templates to all companies. (No warning is shown.) • **Never** – No template released product is used for any companies that you release to, even if a template is specified for companies that are listed on **All products** FastTab. The template columns are unavailable.
Active	Use this option to help maintain your release policies. Set it to *Yes* for all release policies that you use. Set it to *No* to mark a release policy as inactive when it isn't used. You can't deactivate a release policy that is assigned to an engineering product category, and you can delete only inactive release policies.

All products FastTab

On the **All products** FastTab, add a row for each operational company that you want to use this policy to release to. Use the buttons on the **All products** FastTab to add and remove rows as needed. For each row that you add, set the following fields.

> **Note**
>
> The settings on the **All products** FastTab apply to both engineering products and standard products.

Expand table

Field	Description
Company accounts ID	Select the company that the row applies to. The parameters on the row apply when products are released to this company.
Template released product	Add a template for the product.
Copy BOM approval	Select this check box to copy the BOM approval status to the receiving company.
Copy BOM activation	Select this check box to copy the BOM activation status to the receiving company.
Copy route approval	Select this check box to copy the route approval status to the receiving company.
Copy route activation	Select this check box to copy the route activation status to the receiving company.

Option parameters for engineering products FastTab

Every time you add a row on the **All products** FastTab, a row that has a matching **Company accounts ID** value is also created on the **Option parameters for engineering products** FastTab. If you remove a row from the **All products** FastTab, the row that has a matching **Company accounts ID** value is also removed from the **Option parameters for engineering products** FastTab.

For each row shown on the **Option parameters for engineering products** FastTab, set the following fields.

> **Note**
>
> The settings on the **Option parameters for engineering products** FastTab apply only to engineering products.

Expand table

Field	Description
Template BOM	When a product that has a BOM is released, the lines of the specified template BOM are added. This field is useful for adding local components, such as packaging or instructions in the local language.
Template route	When a product that has a route is released, the lines of the specified template are added.

Field	Description
Copy effectivity	Select whether to copy effectivity dates from the engineering company to the operational company when you release products.
Automatically add to release proposal	Select this check box for products that should automatically be released on the engineering change order. In this way, products that belong to engineering product categories that use this release policy can automatically be released to operational companies where this option is set up. (Learn more in Manage changes to engineering products.)

Review each product when you release it

When engineering products that have BOMs or routes are released, the parameters default to values indicated in the release policy. As a user, you can influence this behavior on the releasing side when you use the release product structure.

To release engineering products, on the **Released products** page, select the products to release, and then select **Release product structure** to open the release wizard. The **Select engineering products to release** page shows the products. Select a single product, and then select **Release details** to review the release details for the product.

On the **Release details** page, you can change the value of the **Receive BOM**, **Copy BOM approval**, **Copy BOM activation**, **Receive BOM**, **Copy route approval**, and **Copy route activation** fields. In the push-pull scenario, you can change the value of the same fields on the receiving side, provided that the BOM and route are released.

Product owners and product releases

Because product owners know which legal entities need their products, only members of a product's product owner group can release that product. Other users can't release products that they don't own.

This behavior applies only when a product is directly selected for release. Non-owner users *can* release products that are part of another product's structure via a BOM when they release the parent product, provided that they own the parent product.

For example, product X is assigned to the *Design cabinets* product owner group. Product X is also part of the BOM of product Y, which is assigned to the *Design speakers* product owner group. If a user from the *Design speakers* product owner group releases product Y and its BOM, product X is released together with product Y.

Learn more in Product owners.

Release multiple BOMs/formulas

In older versions of Supply Chain Management, when you release a product, only the first active BOM or formula (as of the date of release) is released. If there are no active BOMs or formulas, the engineering version isn't released.

Starting in Supply Chain Management version 10.0.34, you can release multiple BOMs or formulas. To enable this new functionality, use the Feature management workspace to enable the *Release multiple BOMs/formulas for Engineering Change Management* feature. As of Supply Chain Management version 10.0.43, this feature is mandatory and can't be turned off.

When the *Release multiple BOMs/formulas for Engineering Change Management* feature is enabled, all BOMs or formulas that are active for a product are released when you release the product. This functionality can be relevant, for example, if you have multiple active formulas that apply for various from quantities. You can view the different BOMs or formulas and their respective routes on the BOM designer when releasing on the **Release product structure** page and when reviewing the release on **Open product releases** page.

A product template is required for release

The release policy for the company where a product is to be released must have a product template assigned. This requirement exists because some fields in the local company (such as item model group) are required to create a BOM, and many of these fields are company specific. These field values are copied to the released product from the release template and can be updated as needed.

Last updated on 05/29/2026

Generate variants for engineering products

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to generate variants for engineering products.

Turn variant generation for engineering products on or off

The functionality described in this article requires that both the *Engineering Change Management* and *Variant generation for engineering products* features be turned on for your system. For details about how to turn these features on or off, see Engineering change management overview.

Generate one or more new variants of an engineering product

You can create one or more new variants of a product without copying information from an existing product. This is useful when you need to create several product variants at the same time.

The following procedure provides an example of how to create several variants that include the color dimension.

1. Open the **Released products** page (for example, go to **Engineering change management** > **Common** > **Released products**).
2. On the Action Pane, open the **Product** tab and, from the **New** group, select **Engineering product**.
3. The **New product** dialog opens. Select the appropriate **Engineering category**.
4. If your selected engineering category includes variant dimensions, then you can set values for them now. For this example, if the category has a **Color** dimension, you can set it to *Blue*.

5. Make other settings as needed. Select **OK** to create the product.
6. The product and the blue V-1 variant are created, and the new product now opens.
7. Add a bill of materials (BOM) and route to the variant as needed.
8. On the Action Pane, open the **Product** tab and, from the **Product master** group, select **Product dimensions**.
9. The **Product dimensions** page opens. This page includes a tab for each available dimension. On each tab, add a row for each value you will support for each relevant dimension. (For this example, you might add rows on the **Color** tab for *White*, *Yellow*, and *Green*).
10. Close the page, and then select **Released product variants**. Notice that the first variant that you created (blue V-1) appears.
11. On the Action Pane, on the **Product variant** tab, select **Variants suggestions**.
12. In the **Variant suggestions** dialog box, follow one of these steps:
    - At the top of the dialog box, there is a section for each available dimension. For each dimension, select the checkbox for each value that you want to generate a variant suggestion for, and then select **Suggest** on the toolbar. Relevant suggestions are added to the **Suggested variants** section.
    - Select **Suggest all** on the toolbar to generate variant suggestions for all available combinations of dimension values. The suggestions are added to the **Suggested variants** section.
13. In the **Suggested variants** section, select the checkbox for each variant that you want to create. Then select **Create** to generate and release the selected variants to the engineering company. The following conditions apply:
    - None of the created variants will have a BOM or route.
    - The attributes for these variants will default from the engineering category and will not be copied from the previous variant.

Generate a variant as a copy of another product variant

You can create a variant of a product based on another product variant. The bill of materials (BOM) and route from the source product variant are copied to the new variant. This may be useful for discrete manufacturing products where it can be helpful to create the BOM based on

a starting BOM and keep track of the changes from the previous variant. To maintain traceability, the system records which variant was copied to create new copy.

The following procedure provides an example of how to create a new variant that includes the color dimension by creating a copy of an existing variant

1. Open the **Released products** page (for example, go to **Engineering change management** > **Common** > **Released products**).
2. On the Action Pane, open the **Product** tab and, from the **New** group, select **Engineering product**.
3. The **New product** dialog opens. Select the appropriate **Engineering category**.
4. If your selected engineering category includes variant dimensions, then you can set values for them now. For this example, if the category has a **Color** dimension, you can set it to *Blue*.)
5. Make other settings as needed. Select **OK** to create the product.
6. The product and the blue V-1 variant are created, and the new product now opens.
7. Add a BOM and route to the variant as needed.
8. Add attributes to the product variant as needed.
9. Add the blue V-1 product variant to an engineering change order.
10. Set **Impact** to *New variant*.
11. Select the new color value (for example, *White*). Note that the following conditions will apply:

    - The new variant has a BOM and route that have been copied from the previous variant.
    - The new variant has the attributes copied from the previous variant.

12. Approve the change order.
13. Process the change order. After the order is processed, the new V-1 variant will be created and released to the engineering company.

Last updated on 05/07/2025

Engineering change management FAQ

Article • 06/03/2022

This article provides answers to frequently asked questions about the engineering change management feature.

Should I track the version in transactions?

When you create an engineering change category, the **Engineering change category details** page provides an option that is named **Track version in transactions**. What is this setting, and what does it do?

- If you set the **Track version in transactions** option to *Yes*, the **Dimension group** field will be filtered so that it shows only dimension groups where version is an active dimension.
- If you set the **Track version in transactions** option to *No*, the **Dimension group** field will be filtered so that it shows only dimension groups where the version dimension is **not** an active dimension.

If you track the version in transactions

For engineering categories where you've selected a dimension group where version is an active dimension, you will track versions in transactions for products in that category. In this case, the engineering product will be a product master, and each version of the product will be a product variant that uses the version dimension. Other dimensions can be used together with the version dimension.

In this case, each engineering version will be treated as a variant in all processes. Therefore, if you have a specific variant in a transaction (so that you can determine which variant was sold or purchased), you must manage stock for each version, master planning will plan supply for each version, and so on. If you retire a version from the market, you must manually adjust its effective-from and effective-to dates so that they reflect the change. You must also manage your inventory to make sure that you don't have unused versions of items in your warehouses.

Although this option requires more management effort, we recommend it if you require high traceability of the specific versions that are used in each transaction.

If you don't track the version in transactions

For engineering categories where you've selected a dimension group where version is **not** an active dimension, you will **not** track versions in transactions for products in that category. In this case, if you don't use any other dimension, the engineering product will be a distinct product. The product will still be versioned, and you can manage information about specific versions (for example, their bill of materials [BOM] and route), but you won't be able to trace which specific version was used in each transaction. The effective-from and effective-to dates indicate the validity of each version.

This option is much easier to manage, because if you want to change from one version to another, you just have to make the required changes in a change order, and then update the effective-from and effective-to dates in the engineering version. The production processes will then pick up the required BOM and route for the product (and its specific version).

Most organizations choose this option because it provides version and change management, but it doesn't add the extra overhead of tracking the version in each transaction, in inventory, and during master planning.

Which fields are copied from the released item template?

When an engineering company creates an engineering product, that product is created as a released product in the engineering company. The released product that is created is based on the selected *released item template*. (The released item template is itself an existing released product.) The released item template is also used when the product is released to an operational company. In each case, the released item template defines most of the field values for the released product, and those values come from the associated **Released product details** page.

The following tables shows the fields that are copied during these processes.

Expand table

FastTab	Fields that are copied on product creation in the engineering company	Fields that are copied on release to an operational company
General	All fields in the **Administration** section	The same fields that are copied for the engineering company
Purchase	All fields	All fields except **Unit**

FastTab	Fields that are copied on product creation in the engineering company	Fields that are copied on release to an operational company
Sell	All fields in the following sections: **Sales order**, **Administration**, **Taxation**, **Price update**, **Base sales price**, **Charges**, **Discounts**, and **Alternative product**	All the same fields that are copied for the engineering company, except **Unit**
Foreign trade	All fields	All fields
Manage inventory	All fields and sections *except* **Physical dimensions** and **Catch weight**	All the same fields that are copied for the engineering company, except **Unit**
Engineer, **Plan**, **Manage costs**, **Manage projects**, **Financial dimensions**, and **Warehouse**	All fields	All fields except **BOM Unit**
Product variants	All fields in the **Default product variant** section	The same fields that are copied for the engineering company

In addition to the fields that are shown in the previous table, all default order settings are copied from the released item template, both when the product is created in the engineering company and when it's released to an operational company. (To view the default order settings for a released item template, open the relevant **Released product details** page, and then, on the Action Pane, on the **Manage inventory** tab, select **Default order settings**.)

> ⓘ Note
>
> - The unit is defaulted from the template.
> - For retailers using Dynamics 365 Commerce functionality, when assigning a retail category to a product, the retail category applies default values for many of the fields for the released product level. These defaults overwrite default values that may have already been set by the template or copied from engineering.

Should I create a separate legal entity for engineering products or use an existing legal

entity?

Your business requirements determine whether you should create a new legal entity for engineering products.

By creating a separate engineering company, you can keep engineering data separate and then add modifications as they are required for your local operational companies. In this way, you can achieve the following goals:

- Keep a separate entity where the engineering products are created and managed.
- Prevent engineering products from appearing together with the rest of your released products until they are ready and released.
- Clearly distinguish data that is dictated by engineering and local data.

On the other hand, you should probably use an existing legal entity as an engineering company if the following conditions apply to you:

- You have only one legal entity in your system, and/or you don't require a clear separation between products that are being engineered.
- You don't want to duplicate some data, such as resource groups, resources, operations, and (perhaps) sites.

What is the nomenclature for engineering versions and variants?

The nomenclature for engineering products and product variants works in the following way:

- For the engineering product (that is, the distinct product if no dimensions are used, or the product master if any dimension is used), the number rule nomenclature is defined in the engineering product category. There, you have the option to use a number sequence, text constants, and attribute names and values.
- For each variant of an engineering product (if your engineering product is a product master, variants are set up in the engineering product category where you specify the dimension group), the number rule nomenclature for each variant is defined in the dimension group. In this case, you can use the product ID of the master, and the dimension values and names.

Feedback

Was this page helpful? Yes No

Provide product feedback

Production process overview

07/10/2025

This article gives an overview of the production processes. It describes the various stages of production orders, batch orders, and kanbans, from order creation to closing of the financial period.

The production of products, a process that is also known as the production life cycle, follows specific steps that are required to complete the manufacture of an item. The life cycle begins with the creation of the production order, batch order, or kanban. It ends with a finished, manufactured item that is ready for either a customer or another phase of production. Each step in the life cycle requires different kinds of information to complete the process. As each step is completed, the production order, batch order, or kanban shows a change in the production status. Different types of products require different manufacturing processes.

The **Production control** module is linked to other modules, such as **Product information management**, **Inventory management**, **General ledger**, **Warehouse management**, **Project accounting**, and **Organization administration**. This integration supports the information flow that is required to complete the manufacturing of a finished item.

The production process is typically influenced by the cost accounting and inventory valuation methods that are chosen for a specific production process. Supply Chain Management supports both actual cost (first in, first out [FIFO]; last in, first out [LIFO]; moving average; and periodic weighted average) and standard cost methods. Lean manufacturing is implemented based on the backflush costing principle.

The choice of the cost measurement methods also defines the requirements for reporting about material and resource consumption during the production process. Typically, actual cost methods require accurate reporting on the job level, whereas periodic costing methods allow for less granular reporting of material and resource consumption.

Mixed mode manufacturing

Different products and production topologies require the application of different order types. Supply Chain Management can apply the various order types in a mixed mode. In other words, all order types can occur during the end-to-end process of producing one finished product.

- **Production order** – This is the classic order type to produce a specific product or product variant in a given quantity on a specific date. Production orders are based on bills of materials (BOMs) and routes.

- **Batch order** – This order type is used for process industries and discrete processes where the manufacturing conversion is based on a formula, or where co-products and by-products can be end products, either in addition to or instead of the main product. Batch orders use **Formula** type BOMs and routes.
- **Kanban** – Kanbans are used to signal repetitive lean manufacturing processes that are based on production flows, kanban rules, and BOMs.
- **Project** – A manufacturing project combines products and services with a given schedule and budget. The manufacturing part of a project can be delivered through any of the other order types.

Manufacturing principles

To select the manufacturing principle that best applies to a particular product and related market, you must consider the requirements of production and logistics, and also customer expectations about delivery lead times.

- **Make to stock** – This is the classic manufacturing principle, where products are produced for stock, based on forecast or minimum stock refill (the latter is typically calculated based on forecast or historic consumption).
- **Make to order** – Standard products are made to order or finished to order. Although pre-production might be done by using the Make to stock principle, expensive steps of the value chain, or steps that create variants, are triggered by a sales order or transfer order.
- **Configure to order** – As for the Make to order principle, the final operations of the value chain are made to order. The actual product variant that is produced isn't predefined but is created at the time of order entry, based on the configuration model of the sales product. The Configure to order principle requires a certain level of process unification for a given product line.
- **Engineer to order** – Engineer to order processes are typically addressed by a project and usually start with the engineering phase. During the engineering phase, the actual products that are required fulfill the order are designed and described. Production orders, batch orders, or kanbans can then be created to produce the products.

Overview of the production life cycle

The following steps in the production life cycle can occur for all order types of mixed mode manufacturing. However, not all of them are represented as an explicit order status.

1. **Created** – You can create a production order, batch order, or kanban manually, or you can configure the system to generate them based on various demand signals. Master planning creates production orders, batch orders, or kanbans by firming planned orders. Other demand signals are sales orders or pegged supply signals from other production

orders or kanbans. For fixed-quantity kanbans, demand signals are generated when kanbans are registered as empty.

2. **Estimated** – You can calculate estimates for material and resource consumption. The estimation generates inventory transactions for raw materials that have a status of **On order**. The receipts for main products, co-products, and by-products are generated when production orders or batch orders are estimated. If the BOM contains lines of the **Pegged supply** type, purchase orders for materials or subcontracted operation services are generated and pegged to the production order or batch order. Items or orders are reserved according to the reservation strategy of the production order, and the price of the finished goods is calculated based on parameter settings.

3. **Scheduled** – You can schedule production based on operations, individual jobs, or both.

    - **Operations scheduling** – This scheduling method provides a rough, long-term plan. By using this method, you can assign start and end dates to production orders. If the production orders are attached to route operations, you can assign them to cost center groups.
    - **Job scheduling** – This scheduling method provides a detailed plan. Each operation is broken down into individual jobs that have specific dates, times, and assigned operations resources. If finite capacity is used, jobs are assigned to operations resources based on availability. You can view and change the schedule in a Gantt chart.
    - **Kanban schedule** – Kanban jobs are scheduled on the kanban schedule board or automatically scheduled based on the automatic planning configuration of the kanban rules.

4. **Released** – You can release the production order or batch order when the schedule is finished and the material is available to be picked or prepared. The material availability check helps the shop floor supervisor assess the availability of material for the production orders or batch orders. You can also print the production order documents, such as the pick lists, job card, route card, and route job. When the production order is released, the status of the order changes to indicate that the production can begin. When warehouse management is used, release of the production order or batch order releases the production BOM lines to warehouse management. Warehouse waves and warehouse work are then generated according to the setup of the warehouse.

5. **Prepared/Picked** – When all materials and resources have been staged at the production location, the production BOM lines or kanban lines are updated to a status of **Picked**. Pegged supply orders and related warehouse work are typically completed at this stage. The kanban cards or job cards that are required to report production progress should be assigned and printed.

6. **Started** – When a production order, batch order, or kanban is started, you can report material and resource consumption against the order. The system can be configured to automatically post the material and resource consumption that are allocated to the order when the order is started. This allocation is known as preflushing, forward flushing, or autoconsumption. You can manually allocate materials to production orders or batch orders by creating additional picking list journals. You can also manually allocate labor and other route costs to the order. If you're using operations scheduling, you can allocate these costs by creating a route card journal. If you're using job scheduling, you can allocate the costs by creating a job card journal. Production orders or batch orders can be started in batches of the requested final quantity. Within a production order, batch order, or kanban, the jobs that are created can be started and reported separately through journals, the manufacturing execution terminal (MES Terminal), or the kanban boards.

7. Report progress/**Complete** jobs – Use the MES Terminal, production journals, kanban boards, or mobile scanning capabilities to report the production progress by job or resource. Material and resource consumption will be posted, and the status of the related kanbans, production orders, and batch orders might be updated to **Received** or **Reported as finished**. Put-away work for the warehouse might be created, depending on the warehouse configuration.

8. **Reported as finished** (the product receipt) – When a production order or batch order is reported as finished, the quantity of the finished goods that were completed is updated in inventory. This quantity includes the quantity of relevant co-products and by-products. If you're using work-in-process (WIP) accounting, a ledger journal is generated to reduce the WIP accounts and increase the inventory of the finished goods. When the cost of a production order is calculated, the actual cost of the production is posted. If the material and labor costs that are associated with a production aren't already allocated in a journal or by preflushing, they can be automatically allocated through back-flushing. Allocation through back-flushing involves the post-deducting of inventory transaction processes. If the production order is completed, select the **End job** check box to change the remaining status to **Ended**. Otherwise, leave the field empty to enable reporting of additional quantities that are produced.

9. **Quality assessment** – A product receipt can trigger the creation of quality orders, depending on the configuration of test processes and the quality rules that are established for specific products. Because a quality order can update the inventory status or the batch attributes of the tested products, quality assessment is a mandatory process in many industries.

10. **Put away** and **Ship to order** – After product receipt and quality assessment, optional put-away work directs the received products to the next point of consumption, to a finished goods warehouse, or to a shipment zone if there are ship-to-order requirements.

11. **Ended** – Before production is ended, actual costs are calculated for the quantity that was produced. All estimated costs of material, labor, and overhead are reversed and replaced with actual costs. If you select the **End job** check box when you run the cost calculation, the production order status changes to **Ended**. This status prevents any additional costs from being posted to a completed production order.

12. **Period closure** – Some cost accounting principles, such as periodic average, back-flush costing, FIFO, or LIFO, require periodic activities to close the inventory or financial period. Typically, the system tries to report all material and resource consumption, and also corrections of inventory and scrap, before the periods are closed. This reporting is typically done by using inventory movement journals or adjustment journals. The goal is to assess the economic performance of operating units per period. In some cases, when long-running production orders are used that span the financial reporting periods, production journals are used to report the production progress and resource consumption by the end of the period.

Next steps

- Production feedback
- Product configuration models overview
- Lean manufacturing overview

Mixed mode planning - Combine discrete, process, and lean sourcing

06/17/2025

This article provides information about mixed mode planning. In mixed mode planning, you can model your supply chain based on the material flow. Dynamics 365 Supply Chain Management makes sure that the material flow follows your models, regardless of the supply policy that is selected (kanbans, production orders, purchase orders, batch orders, or transfer orders).

You can select your overall strategy for supplying a product, regardless of the product structure.

For example, you can have kanban control in the assembly, where materials are sourced for the assembly area by production orders, kanbans, transfers, batch orders, or any combination that is appropriate for the characteristics of your supply chain, but you can still have full visibility across supplies. This capability leads to optimized supply chain processes and enhanced visibility into your supply chain.

The granularity of the supply policies that are used in master scheduling depends on the storage dimensions that are enabled as coverage dimensions. To enable master scheduling to control the replenishment and supply of different types of locations (for example, by separating the production floor for different production units, or by separating different types of material and finished goods warehouses), we recommend that you enable Site and Warehouse as coverage dimensions. Alternatively, Warehouse can be omitted as a coverage dimension. In that case, when you use warehouse management processes (WMS), all movements inside a warehouse are controlled by warehouse work, whereas all movements across warehouses can be controlled by withdrawal kanbans.

Supply policies

Mixed mode planning controls how a product is supplied and, based on the supply, how derived requirements (consumption of items from a bill of materials [BOM]) are issued. Based on the order type, the system automatically sources materials to match the requirements.

Supply policies can be defined at the product level or at any granularity that supports your requirements. You define the granularity of supply policies on the **Default order settings** page.

Supply policies can be controlled by product, item dimensions (configuration, color, and size), site, and warehouse. This setup is done on the **Item coverage** page.

The default order type controls what order master planning generates.

Regardless of how the supply chain is modeled, Supply Chain Management supports your mix of supply policies. You can have production orders that are sourced from kanbans. Alternatively, you can have a batch order that requires a product that is supplied by transfers or by kanbans.

Supply Chain Management makes sure that the material flow follows the model.

The warehouse for picking material is assigned dynamically at run time, after the supply policy has been defined.

Typically, kanbans aren't created for future dates, because a kanban has a short life cycle. To maintain full visibility into the supply chain, we have introduced the new planning concept of a *planned kanban*, which is used to calculate derived requirements and helps guarantee that the requirements are sourced based on the same logic that is used when the actual kanban is created.

The same logic is present for all other supply policy types. Therefore, long-term materials planning is based on the same logic that you expect to run with the actual orders after production and supply are approved.

Materials allocation cross-supply policy – Resource consumption on BOMs

Resource consumption is an important functionality. Resource consumption enables a warehouse for picking materials to be selected dynamically, based on the supply policy (order type), and also makes maintenance of base data easier.

Resource consumption requires that the warehouse that materials are picked from be assigned based on the way that the product is supplied. In other words, at run time, the system finds the resources that should be used for manufacturing. Based on those resources, the system then finds the picking warehouse.

For work that is independent of a supply policy, you don't have to change information on the BOM if the supply is changed. For ad-hoc changes, Supply Chain Management makes sure that materials are sourced from the right warehouse.

Process manufacturing – The production type

For full flexibility in mixed mode, we recommend that you use production type BOMs for all products. You can then use production orders, kanbans, transfer orders, or purchase orders to

supply a product. For process manufacturing, you must use a production type of *Formula*, *Co-product*, *By-product*, or *Planning item*. Kanbans and production orders can't be used for these production types.

Manage subcontracting work in production

08/13/2025

This article explains how subcontracted operations are managed in Dynamics 365 Supply Chain Management. In other words, it explains how production operations that are allocated to a resource are managed by a vendor.

In production processes, work can be done by resources that are owned or administered by vendors. Typically, vendor resources are used to level periodic excess demand that surpasses the available capacity of a company's own resources. The vendor might also be able to offer specific resource capabilities or resources at a lower price.

Depending on the vendor resources that are used in a production process, a route often has additional logistic requirements, because the material and semi-finished products must first be transported to the vendor's site. Then the result of the subcontracted operation must be transported either to the location that is allocated to the next operation or to a finished goods warehouse.

When subcontracting operations or activities are used, they affect all stages of operations, from the definition of the operations that are required in production, costing, forecasting, planning, and scheduling, to the management of logistics for materials, semi-finished products, and finished goods. Finally, these resources require their own processes for accounting and cost control.

For internal resources, a fixed cost rate is typically allocated for a period. By contrast, the cost of subcontracted resources is based on the purchase price of the related service. The service is defined as another product, and is used to drive the procurement and purchase processes for a given subcontracted operation.

Currently, there's no explicit concept of semi-finished products in Supply Chain Management. For a production order that requires more than one operation in order to transform raw materials into a finished good, the finished good is posted back into inventory only in the last operation. The semi-finished products that the earlier operations produce are accounted in work in progress (WIP), but they aren't posted or tracked in inventory. Although you can split the routes and bills of materials (BOMs) into multiple smaller units, this approach increases the number of products, BOMs, and routes that must be managed.

There are two methods for modeling subcontracting work for production operations. These methods differ in the way that the subcontracting process can be modeled, the way that semi-finished products are represented in the process, and the way that cost control is managed.

- Subcontracting of route operations in production orders or batch orders

    - The service product must be a stocked product, and it must be part of the BOM.
    - This method supports first in, first out (FIFO) or standard cost.
    - Semi-finished products are represented by the service product in the process.
    - Cost control allocates the costs that are associated with subcontracted work to the material costs.
- Subcontracting of production flow activities in a lean production flow
    - The service is a non-stocked service product, and it isn't part of the BOM.
    - This method uses purchase agreements as service agreements.
    - This method uses backflush costing.
    - This method allows for aggregated and asynchronous procurement. (Material flow is independent of the procurement process.)
    - Cost control allocates subcontracted work in its own cost breakdown block.

Subcontracting of route operations

To use subcontracting of route operations for production or batch orders, the service product that is used for the procurement of the service must be defined as a product of the *Service* type. Additionally, it must have an item model group that has the **Stocked product** option under **Inventory policy** set to *Yes*. This option defines whether a product is accounted as inventory on product receipt (**Stocked product** = *Yes*), or whether the product is expensed on a profit and loss account (**Stocked product** = *No*). Although this behavior might seem contradictory, it's based on the fact that only products that have this policy will create inventory transactions that can be used in cost control to calculate planned cost and determine the actual cost when a production order is ended.

To be considered in planning and cost calculation, the service must be added to the BOM. The BOM line must be of the *Vendor* type, and it must be allocated to the route operation that the service is allocated to. This route operation must have a costing resource and resource requirement that point to a resource of the *Vendor* type that connects the operation and the related service to the corresponding vendor account.

When this configuration is used, a purchase order is created for the related service product, based on estimation of a production order. The purchase order of the service is used as an anchor for the subcontracted operations. The subcontracted work can be managed through the **Subcontracted work** list page in Production control. The subcontracted work is used to ship raw material and, eventually, a semi-finished product to the vendor in preparation for the operation. It's also used to receive the resulting product of the subcontracted operation in item arrival, where the service product is used to identify the arrival of the semi-finished product. When the purchase order line is received, the production operation is updated as completed.

A production order can have many operations, and each operation can be allocated to a different vendor. Therefore, an end-to-end production order might trigger multiple purchase orders.

Subcontracting of production flow activities

The lean manufacturing solution models the subcontracting work as a service that is related to an activity of a production flow (Task guide article). Therefore, this type of subcontracting is also referred to as activity-based subcontracting. A special cost group type, *Direct outsourcing*, has been introduced, and the subcontracting services aren't part of the BOM of the finished goods. When you use lean manufacturing, all activities are defined by kanbans that can be related to one or multiple production flow activities. So far, that explanation sounds just like an explanation of production orders. However, whereas production orders must always end with a finished product, you can create kanbans to supply a semi-finished product. You don't have to introduce a new product and BOM level.

Because kanban rules can be dynamic, you can model different variants of supply for the same product on a production flow. When you use lean subcontracting, the material flow and the financial flow are strictly separated. All material flow is represented by kanban activities. The purchase orders for the service products and the receipt postings of those services can be automated, based on the status of kanban jobs in the production flow. Kanban jobs can be started and completed even before the purchase orders are created. The subcontracting documents (purchase order and purchase receipt of the service) can be aggregated by period and service. Therefore, the number of purchase documents and lines can be kept small, even in highly repetitive operations where vendors provide subcontracted services in a single-piece flow.

Modeling subcontracting in a production flow

In a lean production flow, a process activity can be defined as subcontracted when it's allocated to a work cell (resource group) that has a single vendor resource. When a work cell is subcontracted, the related process activities must be linked to an active purchase agreement line that contains the service item and the price of the service. The service agreement of the activity also defines the calculation ratio between the product quantity of the kanban job and the resulting service quantity. You can select whether the service quantity is calculated based on the number of jobs, the good product quantity that is reported on the jobs, or the total product quantity (this total quantity includes scrapped products).

Transfer activities can also be defined as subcontracted. This definition occurs implicitly when you select the responsible party for the shipping in the transfer activity. When you select **Shipper** or **Recipient**, if the corresponding source or target warehouse is a vendor-managed

warehouse, the activity is considered subcontracted. When you select **Carrier**, the activity is always subcontracted. Like subcontracted process activities, a subcontracted transfer activity must be connected to a service agreement before the production flow can be activated.

Backflush costing

The cost accounting of subcontracted work is integrated into the costing for the lean manufacturing solution (backflush costing). When the purchase order receipt of the service is posted, or when invoicing occurs, the cost of the service is allocated to the production flow. For backflush costing, the variance of subcontracted services is calculated by offsetting the subcontracting block of the standard cost of the received products against the actual received and invoiced service quantities.

Material supply for subcontracted operations

Semi-finished products and other related materials must be transferred to the location where the work is physically performed. When you use subcontracted operations and activities, this transfer is often related to additional transport to a vendor-operated site. By allocating material in the BOM to the subcontracted operation, you declare that the material must be staged at the input location of the resource group for the allocated resource. Master planning or lean replenishment then provisions the material to that location.

To model the inventory that is located at a vendor site, it's a best practice in the industry to define a vendor-managed warehouse. You can easily define a vendor-managed warehouse by creating a new warehouse and assigning the vendor account. To document that material must be transferred to the vendor before an operation can be performed, you should allocate the vendor-managed warehouse to the input warehouse of the resource group that holds the resource.

To replenish material at this warehouse, you can use multiple strategies:

- Transfer orders
- Transfer journals
- Withdrawal kanbans
- Direct purchase to the vendor location

Semi-finished products are the exception to this rule. To transfer semi-finished products, you're limited to these options:

- For production and batch orders, semi-finished products can only be transferred logically by using the Picking list journal from the **Subcontracted work** list page. This journal will

create a delivery note document that can be used to transfer semi-finished and raw material to the vendor.

- For subcontracted operations in production flows, the transfer of semi-finished products is documented by the receipt of withdrawal or production kanbans at the vendor location. To model an explicit transfer activity, you can end a production kanban with an additional transfer activity.

> **Note**
>
> A production route for a single production order can't cross multiple sites. This rule also applies to the subcontracted work. Therefore, the warehouses that represent the vendor-managed material locations must be defined in the same site as the internal resources that are used in the route. Although production flows can cross sites, they can't transport semi-finished products from one site to another, because that operation implies a change of cost context.

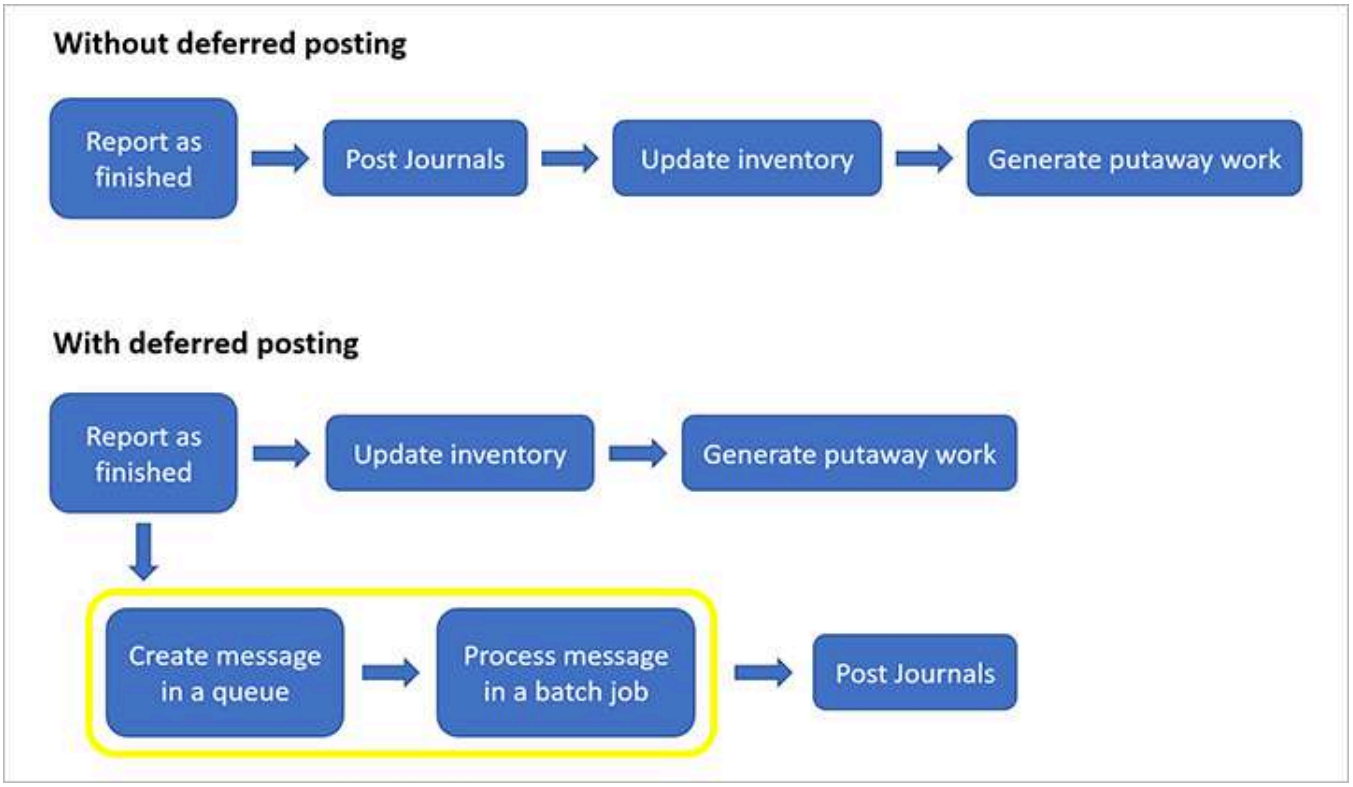
Typically, the output warehouse and location of a subcontracted resource group are directly allocated to the warehouse and location of the next step of the operation in the route or production flow. This setup helps reduce the amount of job reporting that occurs or the number additional transfer operations that must be modeled.

Make finished goods physically available before posting to journals

09/19/2025

When a worker reports a manufactured item as finished, the system registers it as available for further physical processing (such as shipment or putaway). During this process, the system also posts one or more journals (such as the report as the finished journal, picking list journal, and route card journal). If you want to make your items physically available before all postings are processed, you can set up the system to defer the journal postings. A batch job manages deferred postings and processes them as system resources allow.

The following illustration shows how processes for posting journals are invoked both with and without deferred posting.



Set up journal posting options for reporting as finished

Workers can report items as finished by using any of the following clients:

- Web client
- Warehouse Management mobile app
- Production floor execution interface

The web client uses the same posting method configuration as the Warehouse Management mobile app. However, you must configure the posting method that the production floor execution interface uses separately.

Set journal posting options for the web client and the Warehouse Management mobile app

In the mobile app, workers report items as finished by opening a mobile device menu item where the **Work creation process** value is *Report as finished* or *Report as finished and put away*. In the web client, workers report items as finished from the **All production orders** list page or the **Production order (details)** page. You can configure a company-wide default method and set up warehouse-specific overrides as needed.

Use the following procedure to configure the company-wide default posting method for reporting items as finished from the web client or the Warehouse Management mobile app.

1. Go to **Production control** > **Setup** > **Production control parameters**.
2. On the **Journals** tab, in the **Report as finished journal** section, set the **Posting method** field to one of the following values:
    - *Immediate* – When a worker reports an item as finished, the system fully processes all related journal postings before it marks the finished item as physically available.
    - *Deferred* – When a worker reports an item as finished, the system marks the finished item as physically available and adds the journal postings to a message queue. The system doesn't wait until the postings are fully processed before it marks the finished item as physically available.

Use the following procedure to configure warehouse-specific overrides for the default posting method that you set on the **Production control parameters** page.

1. Go to **Warehouse management** > **Setup** > **Inventory breakdown** > **Warehouses**.
2. Select the warehouse that you want to set up.
3. On the Action Pane, select **Edit**.
4. On the **Warehouse** FastTab, in the **Production orders** section, set the **Posting method** field to one of the following values:
    - *Inherit* – The selected warehouse inherits the setting from the **Production control parameters** page.
    - *Immediate* – When a worker reports an item as finished, the system fully processes all related journal postings before it marks the finished item as physically available.

- *Deferred* – When a worker reports an item as finished, the system marks the finished item as physically available and adds the journal postings to a message queue. The system doesn't wait until the postings are fully processed before it marks the finished item as physically available.

Set journal posting options for the production floor execution interface

Use the following procedure to configure the posting method that the system uses when a worker reports items as finished from the production floor execution interface.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Production order defaults**.

2. On the **Report as finished** tab, in the **Report as finished journal** section, set the **Posting method** field to one of the following values:

    - *Immediate* – When a worker reports an item as finished, the system fully processes all related journal postings before it marks the finished item as physically available.
    - *Deferred* – When a worker reports an item as finished, the system marks the finished item as physically available and adds the journal postings to a message queue. The system doesn't wait until the postings are fully processed before it marks the finished item as physically available.

Schedule the message processor batch job to process deferred postings

The message processor batch job processes the journal postings after the system queues them. To ensure that your journal postings are processed, you must configure this job to run at a regular interval. Use the following procedure to set up the required batch job.

1. Go to **System administration** > **Message processor** > **Message processor**.
2. On the **Parameters** FastTab, set the **Message queue** field to *Production*.
3. On the **Run in the background** FastTab, set the **Batch processing** option to *Yes*. Then set up a recurrent schedule, and configure other settings as required. The settings work just as they work for other types of background jobs in Microsoft Dynamics 365 Supply Chain Management.

Track the progress of your deferred postings

Deferred journal postings queue as *message processor messages* that wait until the *message processor* processes them. Set up the message processor to run as a scheduled batch job. To view the deferred posting messages that the message processor has processed or will process, go to **Production control** > **Production orders** > **Deferred production order posting**.

Message grid columns and filters

Use the fields at the top of the **Deferred production order posting** page to find specific messages.

The following filters are available:

- **Message type** – Specify the type of messages that appear in the grid.
- **Message state** – Specify the state of the messages that appear in the grid. The following states exist:
  - *Queued* – The message is ready to be processed by the message processor.
  - *Processed* – The message was successfully processed by the message processor.
  - *Canceled* – The message failed to be processed during the final attempt and was then canceled by the user.
  - *Failed* – The message failed to be processed during the last attempt. The system retries failed messages three times. It then gives up and leaves the message in the *Failed* state. You can't manually cancel or edit a message until after the last of these three attempts.
- **Message content** – This filter does a full-text search of message content. (Message content isn't shown in the grid.) The filter treats most special symbols (such as hyphens) as space characters, and it treats all space characters as Boolean OR operators. For example, if you search for a specific `journalid` value that equals *USMF-123456*, the system finds all messages that contain either "USMF" or "123456." In this case, the list of results is likely to be long. Therefore, it's better to enter just *123456*, because you get more specific results.

You can also sort and filter the list by selecting any of the column headings and entering criteria in the drop-down dialog box.

View the message log, message content, and details

Find detailed information about a message by selecting it in the grid and then selecting the **Log** or **Raw content** tab under the message grid, where each processing event is shown.

The toolbar on the **Log** tab includes the following buttons:

- **Log** – Select this button to show the processing results. This button is especially useful when messages have a **Processing result** value of *Failed*, and you want to learn the reasons for the processing failure.
- **Bundle** – Multiple message processing operations can run as part of the same batch process. Select this button to view this detailed data. For example, you can see whether dependencies exist that require a specific processing order for some messages.

Manually process or cancel a message

You can manually process or cancel a message, depending on its current state. Select the message in the grid, then select **Process** or **Cancel** on the Action Pane.

Set up business events to deliver alerts for failed processing results

You can set up business events that alert you about failed processing results. Go to **System administration** > **Setup** > **Business events** > **Business events catalog**, and activate the business event named *Message processor message processed*.

As part of the activation process, you're prompted to specify whether the event is specific to one legal entity or applies to all legal entities. You're also prompted to provide an **Endpoint name** value. You must define this value first.

> **Note**
>
> If the **When a Business Event occurs** field is set to *Microsoft Power Automate* (instead of *HTTPS*, for example), the **Endpoint name** value is automatically created in Supply Chain Management, based on the *Microsoft Power Automate* setup.

Unified number sequence for job IDs

Article • 06/03/2022

This feature provides a single, unified number sequence that generates job IDs for the Production control, Manufacturing execution, and Time and attendance modules. Previously, you were able to choose a different number sequence for each of these modules, which could result in conflicting job IDs if two or more of these sequences used an identical format. Such a conflict can cause data corruption.

Turn on this feature for your system

If your system doesn't already include the feature described in this article, go to Feature management and turn on the *Unified number sequence for job IDs* feature.

Set up the unified number sequence for job IDs

After enabling this feature, the existing **Job identification** number-sequence settings located on the parameters pages for the Production control, Manufacturing execution, and Time and attendance modules will be deprecated and a new **Unified job ID** field will be added. The **Unified job ID** value is shared by all three modules, thus ensuring that all modules reference the same number sequence. Job IDs will therefore be unique across all three modules and a conflict will never occur.

To set up the unified number sequence for job IDs:

1. Turn on the feature as described in the previous section.
2. Either identify the number sequence you want to use for your unified job IDs, or create a new one. Learn more in Number sequences overview.
3. Go to the **Production control parameters**, **Manufacturing execution parameters**, or **Time and attendance parameters** page. It doesn't matter which one you choose because when you make this setting on any of those pages, all of the other pages will update automatically.
4. Open the **Number sequences** tab on your selected parameters page.
5. Assign the **Number sequence code** that you identified previously to the **Unified job IDs** row of the grid.

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Invoke process automation flows to create quality orders

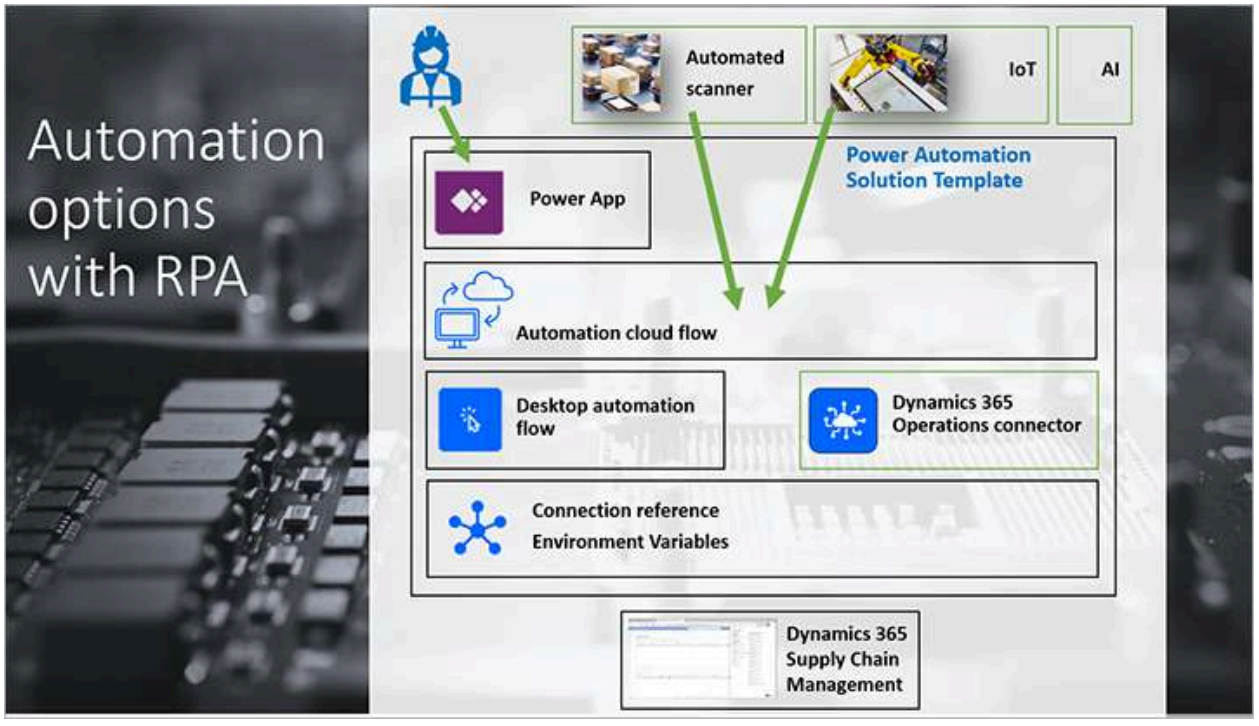
Article • 06/03/2022

Organizations have an increasing demand to automate standard business processes, provide more convenient interactions to the staff, and utilize various data signals and systems to drive business processes automatically. With robotic process automation (RPA) and Microsoft Power Automate, businesses can use a no-code experience to automate repetitive processes, thus gaining efficiency and accuracy.

The downloadable automation solution template for Supply Chain Management showcases how Power Automate can be used to automate business processes using quality orders as an example.

You can download the automation solution template here.

For an overview of this feature and its capabilities, see the following video: Utilize RPA to create quality orders in Dynamics 365 Supply Chain Management

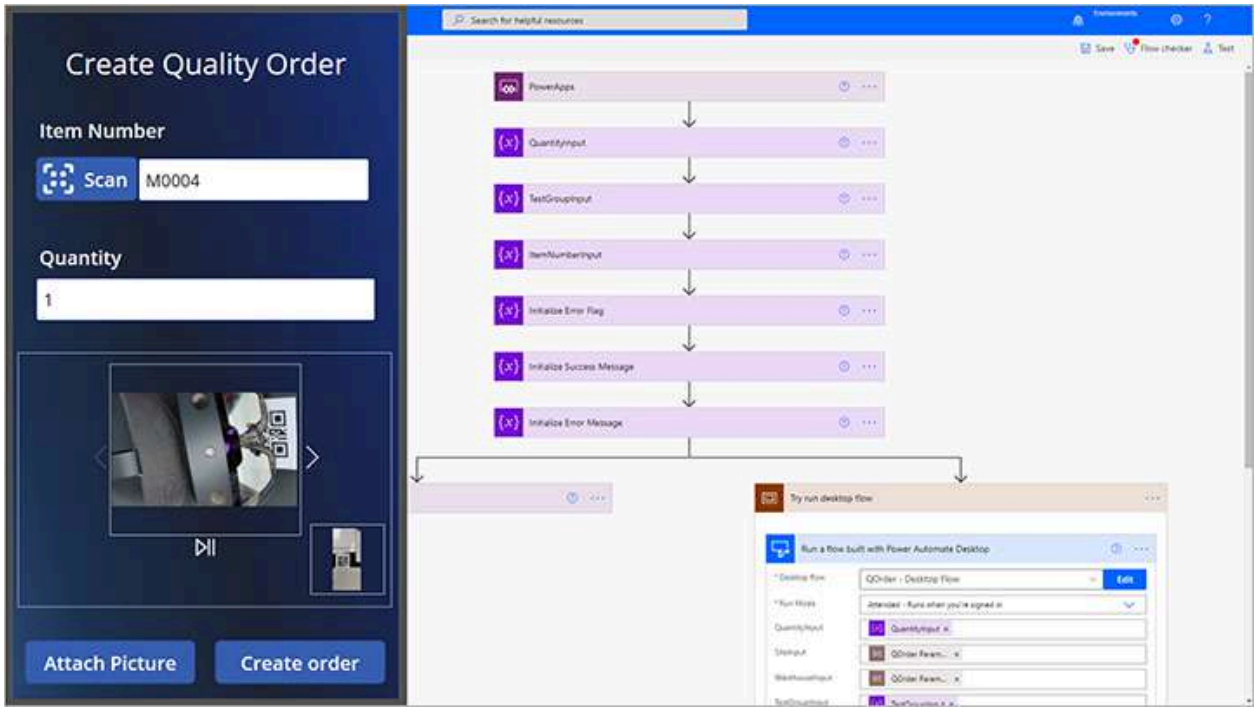


The Power Automate solution template includes a cloud automation flow and a desktop automation flow that automate the creation of quality orders in Supply Chain Management.

The automation can be started in response to many events and signals, including user input or machine signals (including the Internet of Things (IoT)). The *Q-Order* Power

Application example is included in the solution template. It allows the user to scan an item QR code, enter quantity, and attach pictures using a camera.

Solution parameters are included to configure the automation for a specific use case in a production facility.



For a complete step-by-step guide about how to download, install, and use the sample solution for automating quality order creation, see Automate quality order creation on Dynamics 365 Supply Chain Management with Robotic Process Automation using Power Automate Desktop.

Feedback

Was this page helpful? Yes No

Provide product feedback

Integrate with third-party manufacturing execution systems

08/13/2025

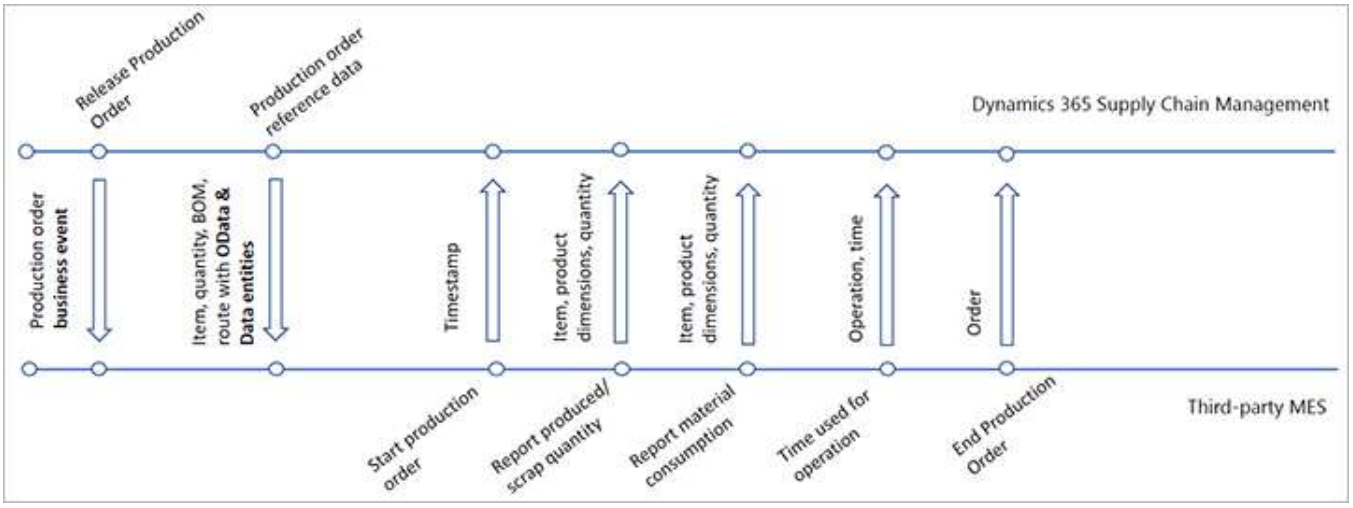
Some manufacturing organizations that use Microsoft Dynamics 365 Supply Chain Management use the native functionality in Dynamics 365 to control their manufacturing activities for machines, equipment, and personnel. However, other manufacturing organizations, especially those that have advanced manufacturing requirements, use a third-party manufacturing execution system (MES) instead. Organizations might choose a third-party MES solution because, for example, it's specifically tailored to their vertical industry.

In the integrated solution, data exchange is fully automated and occurs in near-real time. Therefore, data is kept current in both systems, and no manual data entry is required. For example, when material consumption is registered in the MES, the integration ensures that the same consumption is also registered in Dynamics 365. Therefore, up-to-date inventory records are available to other important processes, such as planning and sales.

The solution makes it faster, easier, and cheaper for Supply Chain Management users to integrate with third-party MESs. It offers the following features:

- Business events and interfaces that support key manufacturing execution processes
- A centralized dashboard where you can track the event processing history, and troubleshoot and fix processes that fail

The following illustration shows a typical collection of business events, processes, and messages that are exchanged in an integrated solution.



Prerequisites

To integrate with a third-party manufacturing execution system, the **Time and attendance** license key must be enabled in Supply Chain Management. Follow these steps to check whether it's enabled and to enable it if necessary:

1. Go to **System administration** > **Setup** > **License configuration**.
2. Make sure that the **Time and attendance** license key is enabled (shows a check mark). This license key is required because it controls the manufacturing execution system's functionality and data. If it isn't enabled, do the following steps:
   a. Put your system into maintenance mode, as described in Maintenance mode.
   b. On the **License configuration** page, select the **Time and attendance** check box.
   c. Turn off maintenance mode, as described in Maintenance mode

Processes available for MES integration

You can enable any or all of the following processes for integration.

Expand table

Process name	Description
Release production orders and production order status change business events	This process provides a business event that the MES can listen to, to get information about the production orders that should be produced. Reference data that is related to the production order is expected to be shared from Supply Chain Management to the MES through Open Data Protocol (OData) or data entities.
Start production order	This process provides Supply Chain Management with information about production orders that are being started by using the MES. It ensures that both systems have an up-to-date view of all manufacturing activities.
Report produced or scrapped quantity	This process provides Supply Chain Management with information about the good and error quantities that are reported on a production job by using the MES. It ensures that shop floor supervisors have an up-to-date view of production plan progress.
Report material consumption	This process provides Supply Chain Management with information from the MES about the quantities of materials that are consumed. It makes up-to-date inventory records available to other important processes, such as planning and sales.
Report time consumed for the operation	This process provides Supply Chain Management with information about the time that is used for a specific operation.
End production order	This process informs Supply Chain Management that the MES has updated a production order to its final status of *Ended*. This status indicates that no more quantities will be produced on the production order.

Monitor incoming messages

To monitor the incoming MES messages to the system, go to **Production control** > **Setup** > **Manufacturing execution** > **Manufacturing execution systems integration**.

All messages for a specific production order are processed in the sequence they're received. However, messages for different production orders might not be processed in the received sequence because batch jobs are processed in parallel. In case of failure, the batch job will attempt to process each message three times before setting it to *Failed* status.

The **Manufacturing execution systems integration** page works in the same way as the **Message processor messages** page and represents very similar functionality. (It even shows MES messages in addition to other types of messages.) For information about how to use either page to review messages, find and fix failed messages, and more, see Monitor and control message processor messages.

Call the API

To call the MES integration API, send a `POST` request to the following endpoint URL:

`/api/services/SysMessageServices/SysMessageService/SendMessage`

The body of the request that you send should resemble the following example. Replace the values for `_companyId`, `_messageType`, and `_messageContent` as required. For information about the various message types that the API supports and how to design their content, see the next section.

JSON

```
{
    "_companyId": "USMF",
    "_messageQueue": "JmgMES3P",
    "_messageType": "ProdProductionOrderReportFinished",
    "_messageContent":
    "{\"ProductionOrderNumber\": \"P000123\", \"ReportFinishedLines\":
[{\"ItemNumber\": \"A0001\", \"ReportedGoodQuantity\": 10,
\"ReportAsFinishedDate\": \"2021-01-01\"}]}"
}
```

API message types and content

This section describes each type of message that can be exchanged through the MES integration API.

Start production order message

For the *start production order* message, the `_messageType` value is `ProdProductionOrderStart`. The following table shows the fields that this message supports.

Expand table

Field name	Status	Type
`ProductionOrderNumber`	Mandatory	String
`StartedQuantity`	Optional	Real
`StartedDate`	Optional	Date
`AutomaticBOMConsumptionRule`	Optional	Enum (FlushingPrincip | Always | Never)
`AutomaticRouteConsumptionRule`	Optional	Enum (RouteDependent | Always | Never)

Report as finished message

For the *report as finished* message, the `_messageType` value is `ProdProductionOrderReportFinished`. The following table shows the fields that this message supports.

Expand table

Field name	Status	Type
`ProductionOrderNumber`	Mandatory	String
`ReportFinishedLines`	Mandatory	A list of lines (at least one), each of which contains the payload that is described in the next table
`PrintLabel`	Optional (Requires Supply Chain Management version 10.0.45 or later)	Enum (Yes | No)

The following table shows the fields that each line in the `ReportFinishedLines` section of the `ProdProductionOrderReportFinished` message supports.

Expand table

Field name	Status	Type
`LineNumber`	Optional	Real
`ItemNumber`	Optional	String
`ProductionType`	Optional	Enum (MainItem | Formula | BOM | Co_Product | By_Product | None), extensible
`ReportedErrorQuantity`	Optional	Real
`ReportedGoodQuantity`	Optional	Real
`ReportedErrorCatchWeightQuantity`	Optional	Real
`ReportedGoodCatchWeightQuantity`	Optional	Real
`AcceptError`	Optional	Enum (Yes | No)
`ErrorCause`	Optional	Enum (None | Material | Machine | OperatingStaff), extensible
`ExecutedDateTime`	Optional	DateTime
`ReportAsFinishedDate`	Optional	Date
`AutomaticBOMConsumptionRule`	Optional	Enum (FlushingPrincip | Always | Never)
`AutomaticRouteConsumptionRule`	Optional	Enum (RouteDependent | Always | Never)
`RespectFlushingPrincipleDuringOverproduction`	Optional	Enum (Yes | No)
`JournalNameId`	Optional	String
`PickingListJournalNameId`	Optional	String
`RouteCardJournalNameId`	Optional	String
`FromOperationNumber`	Optional	Integer
`ToOperationNumber`	Optional	Integer
`InventoryLotId`	Optional	String
`BaseValue`	Optional	String
`EndJob`	Optional	Enum (Yes | No)

Field name	Status	Type
`EndPickingList`	Optional	Enum (Yes | No)
`EndRouteCard`	Optional	Enum (Yes | No)
`PostNow`	Optional	Enum (Yes | No)
`AutoUpdate`	Optional	Enum (Yes | No)
`ProductColorId`	Optional	String
`ProductConfigurationId`	Optional	String
`ProductSizeId`	Optional	String
`ProductStyleId`	Optional	String
`ProductVersionId`	Optional	String
`ItemBatchNumber`	Optional	String
`ProductSerialNumber`	Optional	String
`GenerateLicensePlate`	Optional (Requires Supply Chain Management version 10.0.45 or later)	Enum (Yes | No)
`LicensePlateNumber`	Optional	String
`InventoryStatusId`	Optional	String
`ProductionWarehouseId`	Optional	String
`ProductionSiteId`	Optional	String
`ProductionWarehouseLocationId`	Optional	String
`InventoryDimension1` to `InventoryDimension12`	Optional	String

The 12 extensible dimensions (`InventoryDimension1` through `InventoryDimension12`) require customization and aren't always used. For more information about them, see Add new inventory dimensions through extension.

Material consumption (picking list) message

For the *material consumption (picking list)* message, the `_messageType` value is `ProdProductionOrderPickingList`. The following table shows the fields that this message

supports.

Expand table

Field name	Status	Type
`ProductionOrderNumber`	Mandatory	String
`JournalNameId`	Optional	String
`PickingListLines`	Mandatory	A list of lines (at least one), each of which contains the payload that is described in the next table

The following table shows the fields that each line in the `PickingListLines` section of the `ProdProductionOrderPickingList` message supports.

Expand table

Field name	Status	Type
`ItemNumber`	Mandatory	String
`ConsumptionBOMQuantity`	Optional	Real
`ProposalBOMQuantity`	Optional	Real
`ScrapBOMQuantity`	Optional	Real
`BOMUnitSymbol`	Optional	String
`ConsumptionInventoryQuantity`	Optional	Real
`ProposalInventoryQuantity`	Optional	Real
`ConsumptionCatchWeightQuantity`	Optional	Real
`ProposalCatchWeightQuantity`	Optional	Real
`ConsumptionDate`	Optional	Date
`OperationNumber`	Optional	Integer
`LineNumber`	Optional	Real
`PositionNumber`	Optional	String
`IsConsumptionEnded`	Optional	Enum (Yes | No)
`ErrorCause`	Optional	Enum (None | Material | Machine | OperatingStaff), extensible

Field name	Status	Type
`InventoryLotId`	Optional	String
`ProductColorId`	Optional	String
`ProductConfigurationId`	Optional	String
`ProductSizeId`	Optional	String
`ProductStyleId`	Optional	String
`ProductVersionId`	Optional	String
`ItemBatchNumber`	Optional	String
`ProductSerialNumber`	Optional	String
`LicensePlateNumber`	Optional	String
`InventoryStatusId`	Optional	String
`ProductionWarehouseId`	Optional	String
`ProductionSiteId`	Optional	String
`ProductionWarehouseLocationId`	Optional	String
`InventoryDimension1` to `InventoryDimension12`	Optional	String

Time used for operation (route card) message

For the *time used for operation (route card)* message, the `_messageType` value is `ProdProductionOrderRouteCard`. The following table shows the fields that this message supports.

Expand table

Field name	Status	Type
`ProductionOrderNumber`	Mandatory	String
`JournalNameId`	Optional	String
`RouteCardLines`	Mandatory	A list of lines (at least one), each of which contains the payload that is described in the next table

The following table shows the fields that each line in the `RouteCardLines` section of the `ProdProductionOrderRouteCard` message supports.

Expand table

Field name	Status	Type
`OperationNumber`	Mandatory	Integer
`OperationPriority`	Optional	Enum (Primary | Secondary1 | Secondary2 | ... | Secondary20)
`OperationId`	Optional	String
`OperationsResourceId`	Optional	String
`Worker`	Optional	String
`HoursRouteCostCategoryId`	Optional	String
`QuantityRouteCostCategoryId`	Optional	String
`HourlyRate`	Optional	Real
`Hours`	Optional	Real
`GoodQuantity`	Optional	Real
`ErrorQuantity`	Optional	Real
`CatchWeightGoodQuantity`	Optional	Real
`CatchWeightErrorQuantity`	Optional	Real
`QuantityPrice`	Optional	Real
`ProcessingPercentage`	Optional	Real
`ConsumptionDate`	Optional	Date
`TaskType`	Optional	Enum (QueueBefore | Setup | Process | Overlap | Transport | QueueAfter | Burden)
`ErrorCause`	Optional	Enum (None | Material | Machine | OperatingStaff), extensible
`OperationCompleted`	Optional	Enum (Yes | No)
`BOMConsumption`	Optional	Enum (Yes | No)
`ReportAsFinished`	Optional	Enum (Yes | No)

End production order message

For the *end production order* message, the `_messageType` value is `ProdProductionOrderEnd`. The following table shows the fields that this message supports.

Expand table

Field name	Status	Type
`ProductionOrderNumber`	Mandatory	String
`ExecutedDateTime`	Optional	DateTime
`EndedDate`	Optional	Date
`UseTimeAndAttendanceCost`	Optional	Enum (Yes | No)
`AutoReportAsFinished`	Optional	Enum (Yes | No)
`AutoUpdate`	Optional	Enum (Yes | No)

Other production information

The messages support actions or events that happen on the shop floor. They're processed using the MES integration framework described in this article. The design assumes that other reference information to be shared with the MES (such as product-related information, or the bill of materials or route (with its specific setup and configuration times) used in a specific production order) will be retrieved from the system using data entities via file transfer or OData.

Receive feedback about the state of a message

After the MES has sent a message to Supply Chain Management, it might be relevant for Supply Chain Management to return feedback about the state of the message. Here are some examples of cases where this behavior might be relevant:

- No person is responsible for constantly supervising the MES integration.
- The person who is responsible for supervising the MES integration wants to be notified by email when a message fails, so that they know that they have to take action.
- The MES must show an error message to inform the shop floor operator or somebody from the IT department that they have to take action.
- The MES must recalculate the order schedule after it receives a failure message (for example, because a production order failed to start).

In these cases, you can take advantage of the standard alert feature in Supply Chain Management. For information about how standard alerts work, see the following resources:

- Help article: Alerts overview
- Video: Alert rule options in finance and operations

For example, you might set up the following alerts to provide feedback about the state of a message:

- Create a business event ("Send externally") that is used when a message is *Failed*.
- Send a notification and email to the IT admin or production floor manager.

Clean up processed and canceled message processor messages

Over time, processed and canceled messages related to third-party MES systems can accumulate in the system, potentially affecting performance and data management. Therefore, you should periodically clean up these messages to maintain system efficiency. Learn more in Clean up processed and canceled message processor messages.

React to last-minute changes in production

Components or routes can often change at the last minute of the manufacturing process, just before production orders are released to the shop floor. This article describes a few types of last-minute changes that often occur and explains how to manage them in Microsoft Dynamics 365 Supply Chain Management.

Change BOM items on production orders

Sometimes, a bill of materials (BOM) item must be changed on multiple production orders. These changes are common when a change or revision is applied to a raw item. To allow for such changes, you can change one BOM item to another on estimated or scheduled production orders. Alternatively, you can use all the existing on-hand inventory of an item and then substitute a new item for that item.

Change a BOM item on a production order

To change a BOM item that's used on any production order, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**.

2. Select one or more production orders that you want to change. You can change BOM items only for production orders that have a status of *Estimated* or *Scheduled*. (You can't change them for production orders that have a status of *Created*, for example.)

3. On the Action Pane, on the **Production order** tab, in the **Change** group, select **Change BOM item**.

4. In the **Change production order BOM item** dialog box, on the **Parameters** FastTab, in the **From item** section, specify the item that you want to change from. Include the item's inventory dimensions.

   When you enter the **Item number** value for the "from" item, the system looks up all transactions that have a reference of *Production line*. It shows on-hand inventory from the net requirement for the "from" item. The goal is to easily replace one item with another on the BOM of an item that's being produced.

5. In the **To item** section, specify the item that you want to change to. Include the item's inventory dimensions.

- The "to" item must use the same inventory unit of measure as the "from" item.
- If the new item requires a different quantity, specify the new quantity in the **New quantity** field.
- If the **New quantity** field is set to *0* (zero), the system uses the same quantity as the existing item.

6. The **Production order lines** FastTab shows a list of all the selected production orders that include the specified "from" item on their BOMs. Mark the orders where you want to substitute the specified "to" item for the specified "from" item.

7. A special case occurs when an item must be replaced by a new version or a new product. In this case, you might have to find the time when the on-hand inventory of an item (the "from" item) will be consumed, so that the new item can be swapped in for the production. You'll use the on-hand inventory until it becomes 0 (zero) and then mark all orders so that they use the new item. If this case applies to your current operation, you can select **Proposal to use-up on-hand** on the toolbar of the **Production order lines** FastTab to automatically change the product after its on-hand inventory reaches 0 (in other words, when the accumulated inventory goes negative). In this way, you can see the date when the item will no longer be used, and you can then change the effective dates for the version or item on the BOM.

8. On the **Run in the background** FastTab, either choose to implement the changes right away, or configure settings to control whether and how it runs as a batch job.

9. Select **OK** to apply your settings.

When the change is applied, the system changes the quantity of the BOM line for the "from" item to 0 (zero). It then creates a new BOM line that has the "to" item and the new quantity, and updates the line so that it shows the information for that item. The following fields from the line for the "from" item are copied to the line for the "to" item. Therefore, no manual changes are required.

- Line type
- Vendor
- Site
- Warehouse
- Scrap
- Flushing principle
- Resource consumption
- Per series

Change production order routes

It's often useful to change the setup times for routes when improvements are made or machines are changed for the same process.

To change a production order route, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Select one or more production orders that you want to change.
3. On the Action Pane, on the **Production order** tab, in the **Change** group, select **Change route**.
4. In the **Production order change route item** dialog box, on the **Parameters** FastTab, specify the "from" route and the "to" route.
5. The **Production order lines** FastTab shows a list of the selected production orders that use the "from" route. Mark the orders where you want to substitute the specified "to" route for the specified "from" route.
6. On the **Run in the background** FastTab, either choose to implement the changes right away, or configure settings to control whether and how it runs as a batch job.
7. Select **OK** to apply your settings.

Last updated on 12/30/2025

Production setup requirements

This article provides information about setup requirements before you can work with Production control.

Production control is integrated with features in other modules. This interconnectivity lets you change production orders and make sure that they're automatically updated in all other related processes and calculations in the system. The following setup processes are listed in the order that you should complete them in.

Required baseline setup in other modules

Information in other modules must be set up before you can work with Production control. This setup includes the following tasks:

- Set up general company information.
- Set up the general ledger.
- Define item groups.
- Set up ledger accounts for item groups.
- Set up the inventory item table in Inventory management.
- Create bills of materials (BOMs) and BOM versions in Product information management.

Required calendar and resource setup

Before you use Production control, open Organization administration, and create and define the calendar and operations resources in the following order:

1. **Working time templates** – Set up working time templates to define the times that are available for production scheduling.
2. **Calendars** – Set up working time calendars to define the days of the year that are available for production scheduling.
3. **Resource groups** – Set up resource groups to group the operations resources, so that you can get an overview of any free capacity when you run scheduling. You don't have to set up resource groups before you set up operations resources. However, we recommend that you understand how to group resources when you set up Production control.
4. **Resources** – Set up operations resources to define the resources that are used to complete the production process and plan for capacity.

Required production parameters setup

- **Production control parameters** – Set up basic production parameters to define how the system handles and processes production orders. Define how production orders are created, estimated, scheduled, and consumed. You can also select what kind of feedback you want and how cost accounting is done.

Required journal name identification

- **Production journal names** – Specify the production journal names that are used to record and post transactions.

Setup if you use operations

Operations represent the specific activities that are completed to produce the finished product.

- **Operations** – Set up operations to represent the tasks that must be completed to produce the finished item.
- **Relations** – Set up operation relations to establish detailed properties. To define operation relations, select **Relations** on the **Operations** page.

> ⓘ **Note**
>
> You must know the types of activities that are required in order to produce your item, and the order and priorities of those activities. You must also know which resources are involved, and how many.

Setup if you use routes

If you're working with routes, operations must be defined for every production route that you set up. The route represents the path that the item takes from operation to operation, from the start of the production process to the end.

- **Cost categories** – Set up cost categories to define the cost per hour of specified processes and setup times.
- **Cost groups** – Set up cost groups to create and maintain different types of costing.
- **Route groups** – Set up route groups to define parameters that are related to groups of routes. You must set up route groups before you can create production routes.

- **Routes** – Set up production routes, and define default settings to control scheduling, costing, and pricing of route operations, and to control progress reporting.
- **Route version** – Set up route versions to enable item variations in production.

Optional advanced settings

- **Production groups** – Set up production groups to establish relationships between the production order and ledger accounts. The ledger accounts are used to post or group orders for reporting.
- **Production pools** – Create production pools to group production orders so that you can process urgent production orders, or delete and post groups of orders.
- **Properties** – Define properties to create special attributes that you can assign to your resources to control the order of productions. These attributes are connected to the working time template.

Last updated on 12/19/2025

Bills of materials and formulas

07/10/2025

This article provides information about bills of materials (BOMs) and formulas, which are a central part of the definition of products and product variants. BOMs and formulas specify the required materials or ingredients for a specific product. Formulas also specify the co-products and by-products that are received in a specific production context.

Bills of materials

A bill of materials (BOM) defines the components that are required in order to produce a product. The components can be raw materials, semi-finished products, or ingredients. In some cases, services can be referenced in a BOM. However, BOMs typically describe the *material resources* that are required.

When it's combined with a route or production flow that describes the operations and resources that are required in order to build a product, the BOM forms the foundation for calculating the estimated cost of the product.

A BOM is an individual entity that is described by the following information:

- BOM ID
- BOM name
- The BOM lines that describe the components and ingredients
- The BOM versions, which define the product and period that the BOM can be used for

A single BOM describes a single level that is identified by a unique ID. Components might have their own BOMs that are referenced by BOM versions. You can display and edit the complete hierarchy of BOMs for a specific product in the BOM designer.

Formulas, co-products, and by-products

A formula is a subtype of BOM that is typically used for process manufacturing. In addition to components and ingredients, a formula describes co-products and by-products. In the actual version, the definition of co-products and by-products for the formula requires the formula version. A formula is typically defined for one specific finished product (a formula or planning item) that is defined in the formula version.

BOMs in the product lifecycle

In the product lifecycle, many types of BOM might be created for various reasons:

- **Sketching/Draft BOM** – Gives a draft estimation of required materials in an early design phase and helps you do a rough estimate of cost and estimated product attributes. This BOM isn't usually used in enterprise resource planning (ERP).
- **Engineering BOM** – Typically used when you design products that are based on existing product portfolios. Engineering BOMs are structured to simplify the design process and group complex products into engineering modules. For simple products, it might be possible to engineering BOMs for the actual production process. However, for other products, the engineering BOM must be converted to an actual production BOM. Engineering BOMS are typically represented by phantoms in the BOM hierarchy. Although engineering BOMs can be used for the planning and execution of manufacturing operations, this approach can lead to inefficiencies, especially in repetitive operations where many orders are created.
- **Planning BOM** – Used to do planning for material requirements. The demand of components and ingredients is calculated based on the demand of the finished products. Like costing BOMs, planning BOMs might represent a specific mix of material that is used in a period.
- **Production BOM** – The actual BOM that is used for a specific production. A production BOM must take into account the actual resources that are used to produce the product. When a production order, batch order, or kanban is created, the multiple levels of BOMs that are represented by phantoms are collapsed into one level and distributed over the operations for the order.
- **Costing BOM** – Used to calculate the estimated cost of a product. For example, you can use a costing BOM when standard cost is used or the estimated planned cost of a given product is calculated. Costing BOMs can refer to a specific mix of materials and resources that are expected to be used. Therefore, you can use the costing BOM to create a representative estimated cost for a period and help avoid variances over time.

The types of BOM that are actually used in an implementation depend on the implementation, and also on the business scenarios and requirements. In simple implementations, a planning BOM, production BOM, and costing BOM can be modeled as one BOM. In environments that have frequent engineering changes and multiple alternative routes, a larger set of BOM types will probably be required.

Approval of BOMs and formulas

Each BOM and formula can be separately approved or unapproved. Typically, approval of a BOM or formula occurs when the first relevant BOM version is approved. However, in some business scenarios, these approvals might be different steps in the process and might involve different process owners.

If a BOM is unapproved, all related BOM versions are also unapproved.

BOM and formula versions

To relate a specific BOM or formula to a product variant that can be produced, you must create a BOM version or formula version. The validity of BOM versions and formula versions can be constrained by period, quantity, site, specific product dimensions, and other criteria. Formula versions have other important attributes, such as yield, co-product and by-product definitions, and the cost distribution instructions for the formula.

Approval of BOM and formula versions

Before a BOM version can be used in the planning or manufacturing process, it must be approved. When a BOM version is approved, the related BOM can also be approved, depending on the user's selection and authentication rights. A BOM version can be approved only if the related BOM itself is approved.

Activation of the default BOM or formula version

To set a specific BOM or formula as the default BOM version or formula version that will be used by master planning or used to create of production orders, you must activate the version. When a version is activated, the uniqueness of the version for the given constraints (for example, period, site, or quantity) is verified. You receive an error message if the version that you're trying to activate conflicts with a version that is already active. You must then either inactivate the conflicting version or modify the version constraints (usually the period) to prevent an ambiguous activation.

Product change with case management

The product change case for approval and activation of new or changed BOMs and BOM versions provides an easy way to see an overview of the BOM version constraints. You can also approve and activate all BOMs and formulas that are related to a specific change for one activation date.

Alternative BOM versions

Sometimes, the active BOM version or formula version shouldn't be used in forecasts, sales, or a parent product. In this case, you can select a specific approved BOM as part of the requirement (forecast line, sales line, or BOM line) if an approved BOM version or formula version exists for the alternative BOM or formula.

When planned orders, production orders, or kanbans are created, the planner or shop floor supervisor can use any approved BOM version that is valid on the requested planned production date to plan for or produce a specific product. The BOM version that is used doesn't have to be activated as the default BOM version.

BOM and formula lines

A BOM line is created for each material, service, or ingredient. The line defines the planned consumption of the specified product variant and also defines the various attributes that are related to the planned consumption.

BOM lines can have the following line types: **Item**, **Phantom**, **Pegged supply**, **Vendor**.

Item

Select the **Item** line type for materials or services that are directly consumed, and that don't require further explosion or pegged supply.

Phantom

Select the **Phantom** line type when you want to explode any lower-level BOM items that are contained on the BOM line. In Master scheduling, in planned cost calculation, or on estimation of a production order that uses BOM lines of the **Phantom** type, the parent BOM line that refers to a product variant that has a phantom BOM is replaced by the component items that are listed as BOM lines in that BOM, as determined by the applicable active BOM version of that product variant. If the product variant has an applicable active route, the operations of that route are merged into the parent route.

Phantoms are typically used to simplify the engineering process. Extensive use of phantom BOMs in many levels has an effect on performance, especially in highly repetitive manufacturing scenarios. To improve performance, you should avoid deep hierarchies of phantoms. Instead, use pre-exploded production BOMs and routes.

Pegged supply

Select the **Pegged supply** line type when you want to create a subproduction, a BOM line event kanban, or a direct purchase order for any product variant that the BOM line references. The subproduction, event kanban, or purchase order is created when you estimate the production order. The required item quantities are automatically reserved for the consuming production order.

> **Note**
>
> A subproduction order is always created for pegged supply, even if on-hand inventory or other supply is available to cover the demand.

Vendor

Select the **Vendor** line type if the production process uses a subcontractor, and you want a subproduction or purchase order to be created automatically for the subcontractor.

> **Note**
>
> For BOMs that include subcontracted operations, the service or work that is performed by the subcontractor must be created as service item that is tracked in inventory. You must attach the service item to the parent item as a BOM line. The route must contain an operation that is assigned to the subcontractor's operations resource.

BOM designer functionality

Article • 01/31/2025

This article describes how you can use the BOM designer page to design and work with tree structures for bills of materials (BOMs). To choose different configurations and specify what information appears on the lines of the tree, select **Setup**.

When you open the **BOM designer** page from the **Released products** page, it displays the hierarchy of bills of materials (BOMs) that are active and approved for the selected item, the default order site of the item, and the actual date.

To change the initial selection in the view, select **Filter**. By setting the display principle to **Selected/Active or Selected**, you can select individual BOM or route versions to use in the view. You can select non-approved and non-active BOM versions to show or maintain in the BOM designer.

> **Note**
>
> If you open the BOM designer from the **Bills of materials** list page, it doesn't display route information. Currently, the selection of a BOM or route version is a property of the BOM and route version, and applies to all instances of the BOM designer.

The following sections describe the functionality that is available on the various tabs of the BOM designer.

Analyzing a BOM structure by using the BOM designer

The BOM designer has two sections:

- The tree view of the BOM structure.
- The details section, which shows details of the selected data. When you select a node in the tree view, the FastTabs in the details section are updated based on that node:
  - **BOM line details** – Shows the details of the BOM line that is selected in the tree view.
  - **Item data** – Shows the details of the main item or the item that is used in the selected node. To maintain the selected item, select **Edit released product**.
  - **BOM** – Shows the header of the BOM that is related to the selected node.

- **Route** – Shows the header of the route that is related to the selected node.
- **Route operations** – Shows a preview of the operations for the route. When a BOM line that is assigned to a specific operation is selected, the operation is marked as **Component needed at operations**.

Selecting a BOM and route

The filter that is applied for the BOM and route is displayed in the header of the BOM designer. You can change the filter by using the **Filter** dialog box. The following table describes the fields in this dialog box.

Expand table

Field	Description
Product dimensions	If the selected finished product is a product master, you can define the active product dimensions for the main selection. **Note:** If you open the BOM designer for a product that isn't a product master, no product dimensions can be selected in the **Filter** dialog box.
Site	Change the site that the BOM tree is displayed for. The default site is the default inventory site of the finished item.
Display principle	Select the version display principle that applies to the current BOM structure and the current route: • When the principle is set to **Active or Selected/Active**, the valid BOM or route version for this date is found. • When the principle is set to **Selected/Active or Selected**, you can select a BOM version or route version by selecting **BOM** > **BOM versions** or **Route** > **Route versions**.
Version date	Enter the version date for the BOM and route. The version identifies which BOM version is used on a specific date, as determined by the version dates in the BOM version setup.
From quantity	Filter the versions by selecting a specific from quantity. If you set a value, different BOM and route versions might be selected.
Show valid only	When you select the check box, the tree structure shows only BOM lines that have valid dates. Right-click or double-click a BOM line to open the **Edit BOM line** page, where you can see the validity dates for that BOM line.

When you use the BOM designer to review or edit BOMs that consist of one or more levels of phantoms, the route that is associated with the top item typically spans the complete BOM hierarchy. To simplify the overview, you can lock the top-level route in

the display by selecting **View** > **Lock route**. To unlock the route, select **View** > **Unlock route**.

Adding and editing BOMs and BOM lines

Use the **BOM lines** or **BOM** functions to modify the BOM lines or BOM. When you select a node in the tree, the type of the node determines that functions that are available.

Expand table

Function	Description	Node type and conditions
BOM lines > Edit	Open a dialog box where you can edit the BOM line attributes.	This function is available when a BOM line node is selected.
BOM lines > Delete	Delete a BOM line from the selected BOM.	This function is available when a BOM line node is selected, and the BOM isn't locked for editing.
BOM lines > Add before line	Open a dialog box where you can select a product variant to include before the selected BOM line.	This function is available when a BOM line node is selected.
BOM lines > Add to component BOM	Open a dialog box where you can select a product variant to include at the end of the selected BOM.	This function is available when the node that is selected has a selected BOM. If this function isn't available, a BOM version might be missing for the selected item variant. In this case, select **BOM** > **Create version** to create the missing version for the selected node.
BOM lines > Add after line	Open a dialog box where you can select a product variant to include after the selected BOM line.	This function is available when a BOM line node is selected.
BOM > Create version	Create a new BOM version or BOM for the product variant of the selected node.	This function is available when the BOM line node that is selected is linked to an item that has a production type of **BOM** or **Formula**.
BOM > Calculation	Open a dialog box where you can run the cost or sales price calculation for the selected product variant.	This function is available when the node that is selected is related to a BOM version.
BOM > Check	Validate and check the	This function is available when the node that is

Function	Description	Node type and conditions
	selected BOM.	selected is related to a BOM version.

Configuring the tree view

To customize the information that is shown in the tree view of the BOM designer, select **Setup**.

Expand table

Field group	Description
BOM	Use the check boxes to select the criteria that are shown in the tree structure. The BOM designer displays the selected criteria at the bottom of both tabs.
Route	Use the check boxes to select the criteria that are shown for the routes.

Feedback

Was this page helpful? Yes No

Provide product feedback

Override the default reservation principle for materials in production

06/17/2025

The *Override default production reservation* feature lets you set a default reservation principle for each item model group. Therefore, different reservation principles can automatically be applied for each item that is part of a production bill of materials (BOM) or batch order formula. You can select whether each item model group should override the default reservation principle that is set for an order, and what reservation principle should be used instead (*manual*, *estimation*, *scheduling*, *release*, or *start*).

When you create a new production order or batch order, the system prompts you to select the reservation principle for that order and all its BOM or formula lines. If you use the *Override default production reservation* feature, some or all BOM or formula lines can override your selected reservation principle and instead use the reservation principle set for their item model group.

For example, if you have raw materials or ingredients that require pick work, BOM or formula lines that are created for those products require a physical reservation, because physical reservation is a prerequisite for the generation of warehouse work. Typically, if you want the reservation to occur automatically, you select one of the following reservation principles: *estimation*, *scheduling*, *release*, or *start*. On the other hand, if you have materials or ingredients that don't require pick work, because they're consumed directly from a location, you typically select the *manual* reservation principle, which doesn't make any physical reservations or generate any pick work.

Assign a production reservation policy to an item model group

1. Go to **Cost management** > **Inventory accounting policies setup** > **Item model groups**.
2. Create or select an item model group.
3. On the **Inventory policies** FastTab, select the **Override item production reservation** check box.
4. In the **Reservation** field, select the reservation principle for items that belong to the selected model group. (Those items include items that are on a BOM or formula line.)
    - *Manual* – Items in the model group aren't automatically physically reserved for production. However, they can still be manually reserved as required.

- *Estimation* – Items in the model group are physically reserved during estimation of the production order.
- *Scheduling* – Items in the model group are physically reserved during scheduling of the production order.
- *Release* – Items in the model group are physically reserved when the production order is released.
- *Start* – Items in the model group are physically reserved at the start of the production order.

Example: Using reservation principles in a bulk/pack scenario

A bulk lubricant material is produced in a 1,000-liter mixer. After the bulk material is ready, it's pumped out to several filling stations, where bottles of different sizes are filled. After filling is completed, the bottles are packed into boxes. Those boxes are then packed onto pallets.

In this scenario, a batch order to make 1,000 liters of bulk material is created. (This order is the bulk order.) When this batch order is completed, it's reported as finished to the material input location of the filling stations. A batch order to fill and pack each bottle size is then created. (These orders are the pack orders.) The pack orders have a formula that consists of the bulk material, an empty bottle, a label, and other packing materials. Because the bulk material flows directly from the mixer tank to the filling stations, no warehouse work is required to pick this ingredient, and the bulk material is consumed directly from the input location. Therefore, the reservation principle is set to *manual*. The other materials are staged to the filling station by pick work. Therefore, the reservation principle for these lines is set to *release*, for example, so that the reservation automatically occurs when the pack order is released.

Reserve the same batch for a production order (preview)

[This article is prerelease documentation and is subject to change.]

Same-batch selection lets you reserve the consumption of a raw material on a production order or batch order against a single batch of inventory. This capability keeps batch traceability consistent through the manufacturing process. For example, when an ingredient must come from one batch to guarantee a uniform mix, same batch selection reserves the whole bill of materials (BOM) or formula line from one batch instead of spreading it across several batches. It brings the *Same batch selection* capability that's already available for sales orders to production and batch orders.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.
- The feature named *(Preview) Use same batch selection for raw materials in production* must be turned on in feature management.
- The feature supports both warehouse management processes (WMS) and non–WMS scenarios. WMS scenarios are supported only when you use a reservation hierarchy where the batch number dimension is above the location dimension (a *Batch-above [location]* reservation hierarchy).
- This feature applies only to batch-enabled products that aren't tracked by serial numbers. That means that it only supports products assigned a tracking dimension group in which the *Batch number* dimension is active and the *Serial number* dimension is inactive.

Set the Same batch selection field on a BOM or formula line

The **Same batch selection** setting flows through the manufacturing setup, so you can control it at the level that makes sense for your process. The value comes from the **Same batch selection** field on the item model group and flows through the BOM and formula lines to the production order lines, where it becomes read-only.

You can override the default value on individual BOM or formula lines. Follow these steps:

1. Go to **Product information management** > **Bills of materials and formulas** > **Bills of materials** (or **Formulas**).

2. Select or create a BOM or formula version, and open its lines.

3. Select the line for the raw material that you want to configure.

4. On the **Line details** FastTab, select the **Setup** tab.

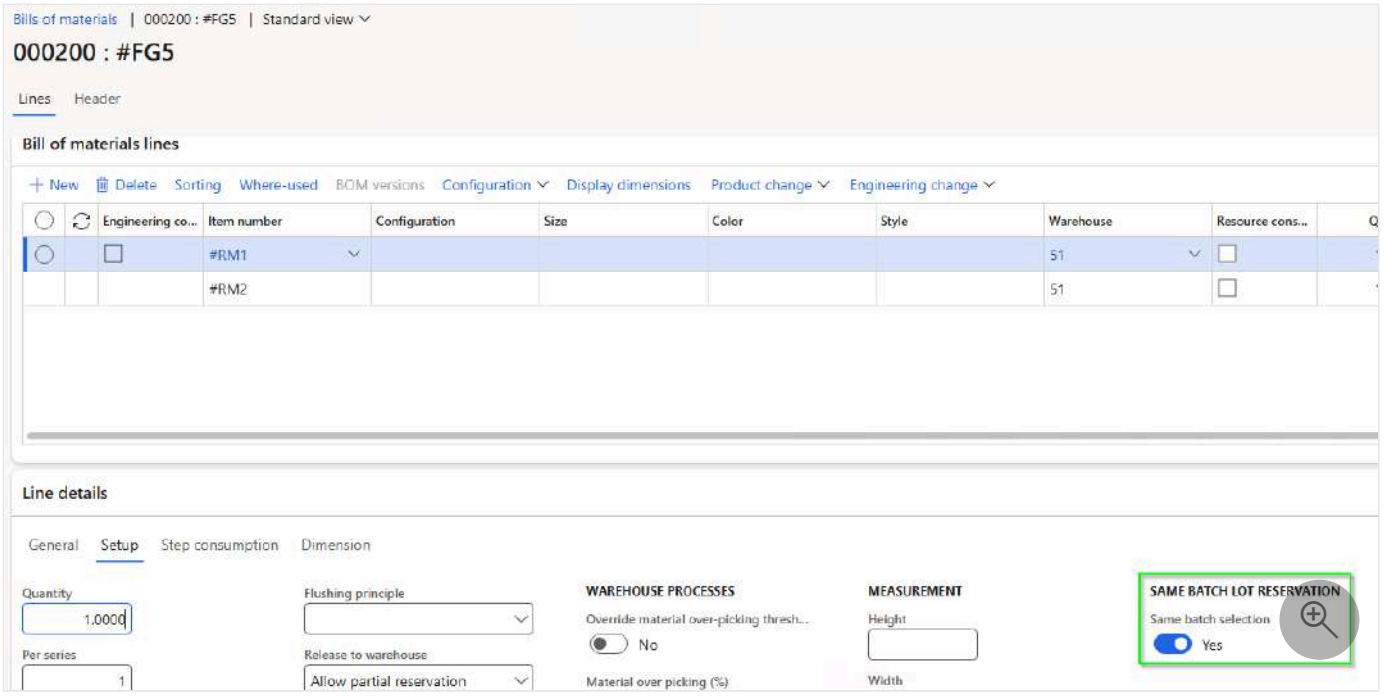
5. In the **Same batch lot reservation** field group, find the **Same batch selection** field. This field has the following values:

   - *Yes* – The raw material must be reserved from a single batch during production. Only inventory from one batch can be used for this line.
   - *No* – No single-batch constraint is applied. The system can reserve inventory from multiple batches.

   The field inherits its default value from the item model group that's assigned to the product. You can override it here for specific BOM or formula lines.

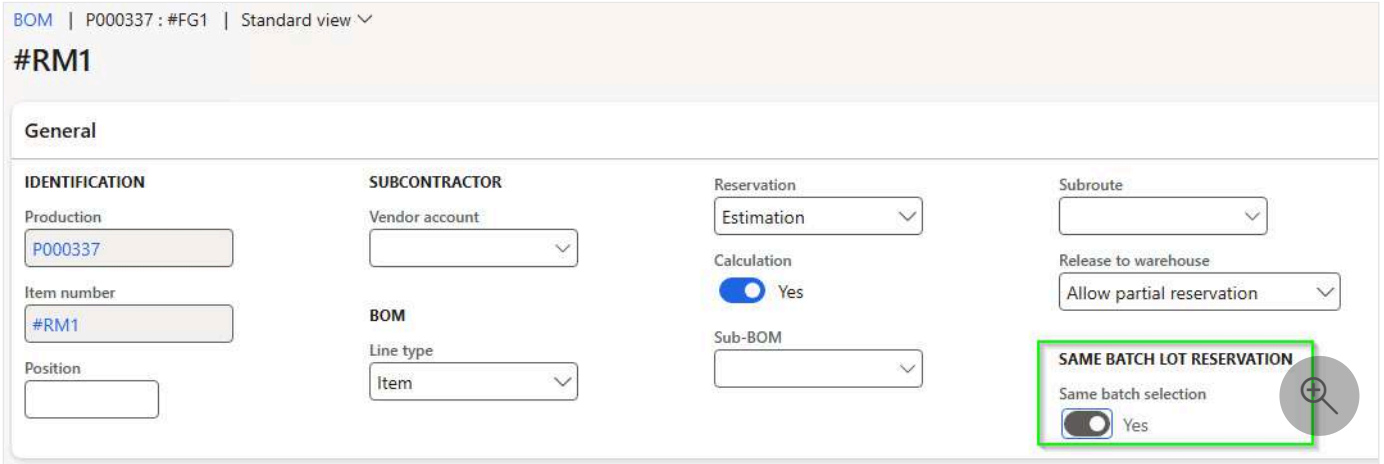
6. Select **Save**.

The following screenshot shows the **Bills of materials** page with the **Same batch selection** field highlighted on the **Setup** tab of the **Line details** FastTab.



When you create a production order or batch order, the BOM or formula line usually copies its **Same batch selection** value to the production order BOM line. On the production order BOM line, the **Same batch selection** field appears in the **Same batch lot reservation** field group on the **General** tab and is read-only.

The following screenshot shows the production order BOM line with the read-only **Same batch selection** field.



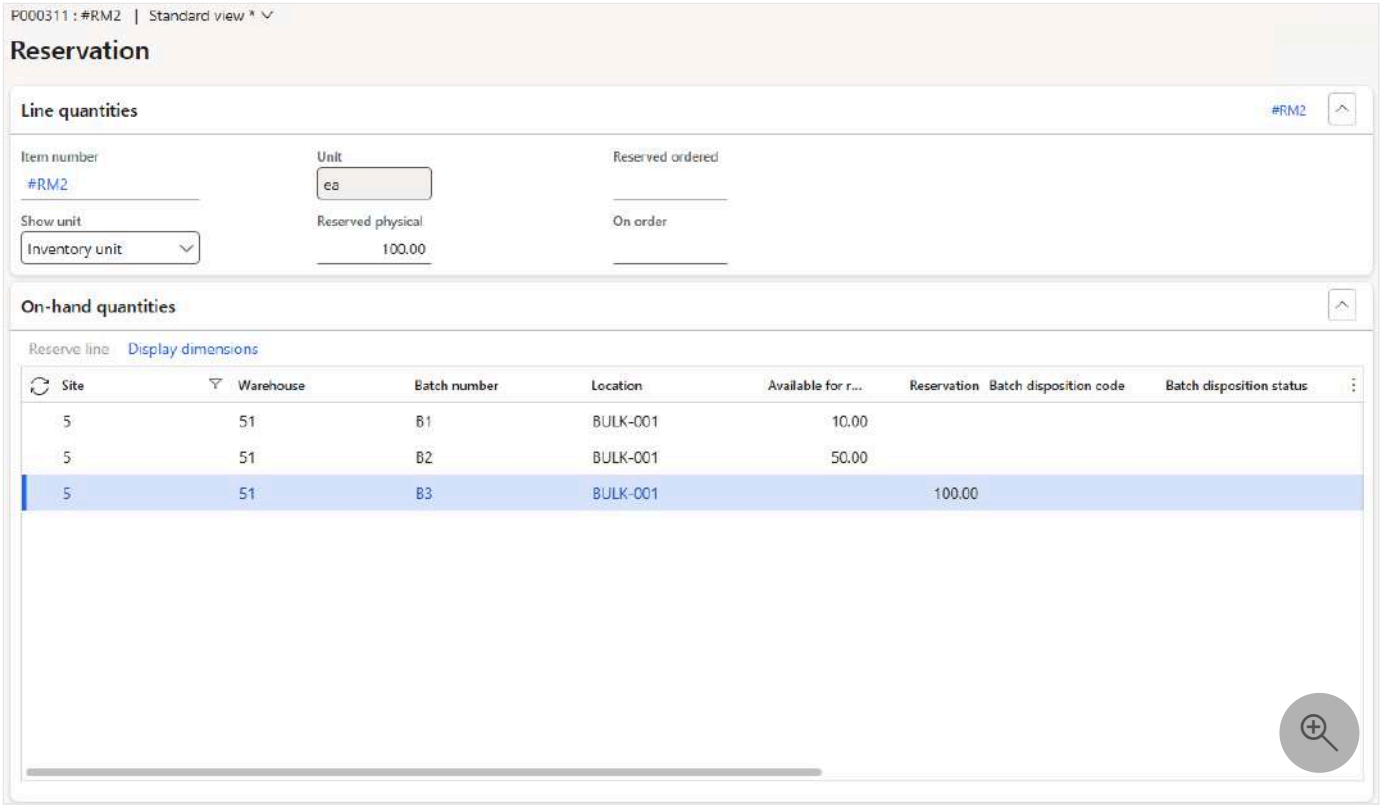
Reservation behavior

When you turn on same-batch selection for a production order or batch order line, the system tries to reserve the required quantity from a single batch during estimation. The outcome depends on available on-hand inventory. Reservation can happen automatically, according to the reservation rules of the BOM or formula line, or manually, when a user reserves against a specific batch directly from the order lines. In either case, same-batch selection constrains the reservation

so that the whole required quantity is drawn from one batch wherever on-hand allows. The outcome depends on available on-hand inventory:

- If a single batch has enough on-hand inventory to fulfill the line, the system reserves that batch. When the item model group enables FEFO, the system selects the batch that has the earliest expiry or best-before date.
- If no single batch can fulfill the line, the system doesn't reserve any batch against the BOM or formula line. The system doesn't show any warning or error message.

The following screenshot shows the **Reservation** page for a production order BOM line where a single batch (B3) was successfully reserved because it had sufficient quantity.



Batch selection sorting priority

When multiple batches can fulfill the requirement, the system selects a batch based on the following sorting priorities:

- **FEFO** – Expiry date or best-before date (ascending), then batch number (ascending).
- **Standard** – Batch number (ascending).

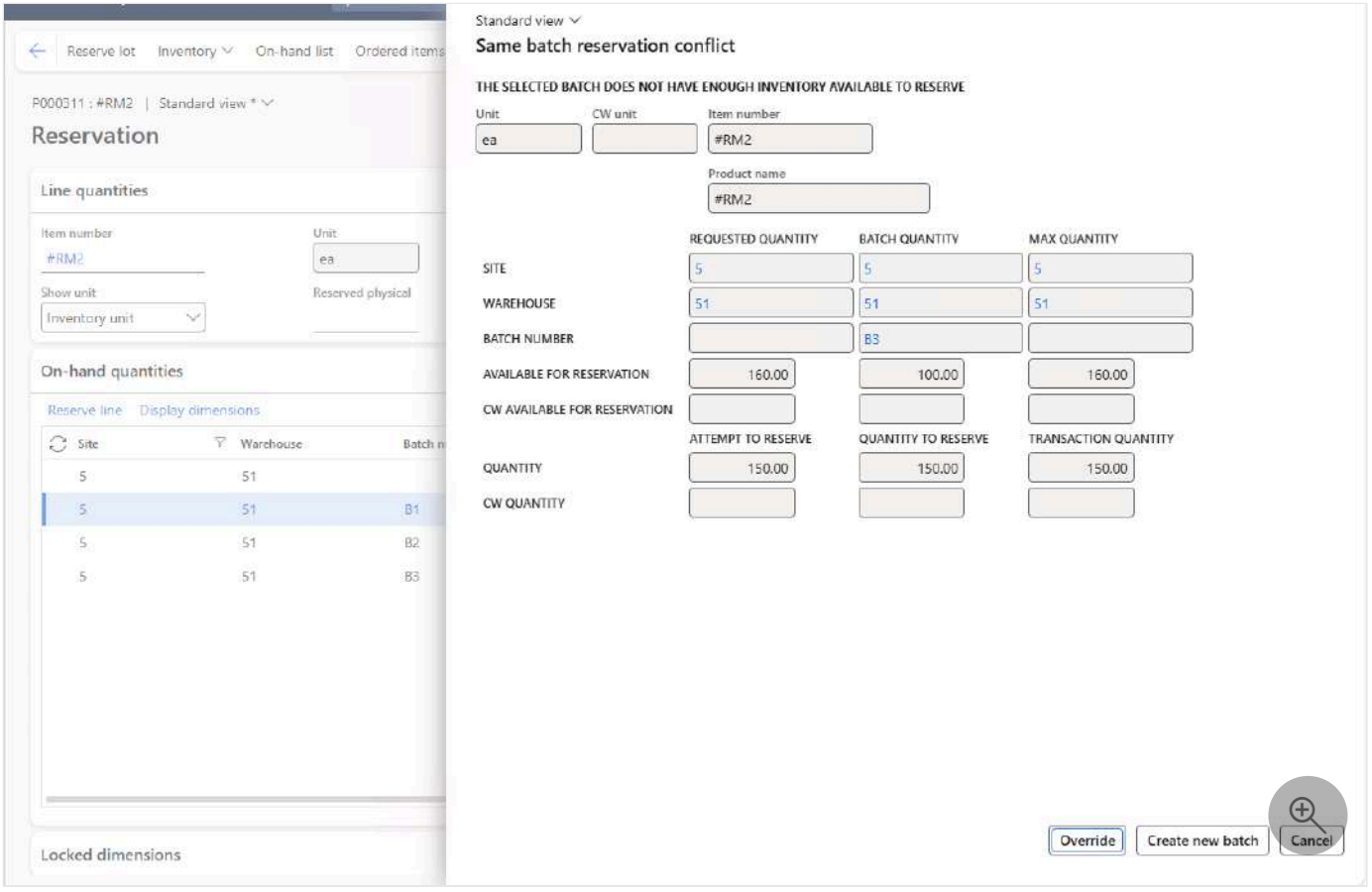
Resolve a same-batch reservation conflict

When you manually reserve a production order BOM line that you can't fulfill from a single batch, the **Same batch reservation conflict** page opens. This page shows the requested quantity, the

available batch quantity, and the maximum quantity available. It offers the following actions:

- **Override** – Turns off single-batch selection for that line and reserves the quantity across multiple batches. The line's **Same batch selection** setting is cleared and marked as overridden.
- **Create new batch** – Generates a new batch number through the standard number sequence and leaves the line on order, so that you can produce or receive inventory into the new batch.
- **Cancel** – Leaves the line on order with no batch reservation.

The following screenshot shows the **Same batch reservation conflict** page with the available actions.



> **Important**
>
> The **Lower quantity** action that's available for sales orders is intentionally turned off for production scenarios, because reducing the consumed quantity would invalidate the production estimation.

Batch reservation policy for non-WMS warehouses

For warehouses that don't use WMS, the **Batch reservation policy for non-advanced warehouses** parameter controls whether the system enforces single-batch reservation when the reservation hierarchy has the batch number dimension below the location dimension (a *Batch-below [location]* reservation hierarchy). To set this option, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Open the **General** tab.
3. On the **Batches** FastTab, set **Batch reservation policy for non-advanced warehouses** to one of the following values:

    - *Simple* – The system doesn't consider same-batch selection constraints when using batch-below reservation hierarchy.
    - *Advanced* – The system applies FEFO rules and batch disposition codes during reservation. Same-batch selection for batch-below items on non-WMS warehouses is blocked, and an error message is shown.

Learn more in Flexible warehouse-level dimension reservation policy.

Related information

- Reserve the same batch for a sales order
- Override the default reservation principle for materials in production

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Last updated on 07/27/2026

Phantom items

08/13/2025

This article describes, in detail, how the *Phantom* line type can be used for the lines of a bill of materials (BOM) and a formula.

In figure 1, (a) is the BOM for product H and parts F and G, and (b) is the route sheet for products H and part F.

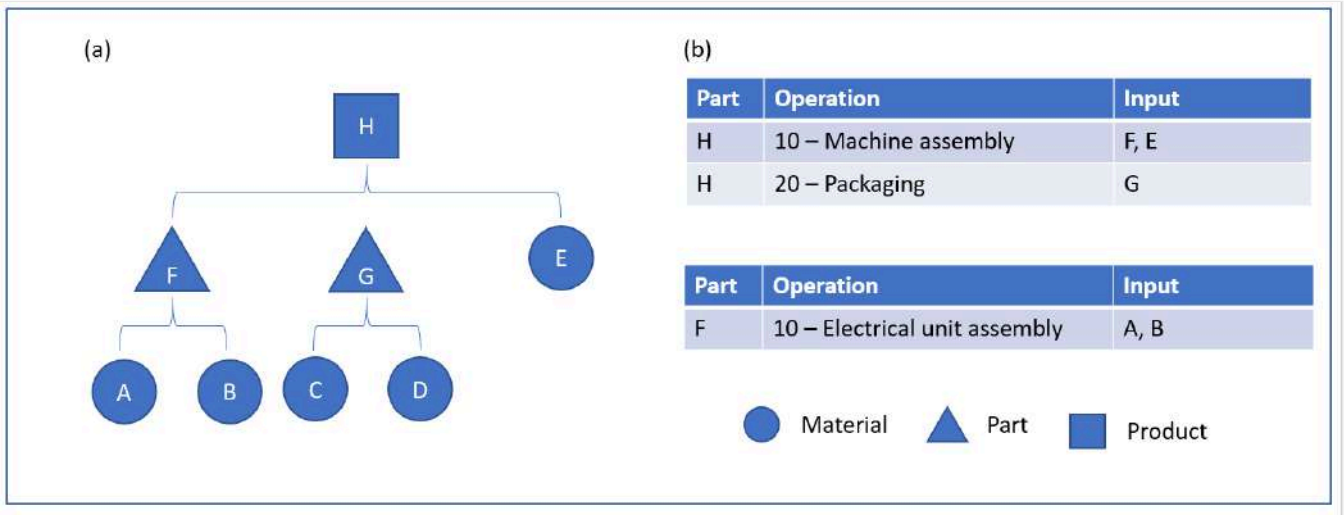


Figure 1: Engineering BOM

Figure 1 shows an example of a BOM structure in two levels. Finished product H represents a product for a machine assembly. The machine assembly consists of two parts, an electrical unit (F) that has two materials (A and B) and a group of packaging materials (G) that also has two materials (C and D). Another material (E) is used during the general assembly of the machine.

Figure 1 represents the Engineering BOM for product H. This structure provides a good overview of the parts and components of the overall machine assembly. However, although product designers might prefer to see the BOM represented in this way, this structure might not correctly represent the way that the machine is built on the shop floor.

For example, the Engineering BOM in the figure 1 indicates that electrical unit F is assembled as a separate part on a separate work order. However, on the shop floor, it might be judged more optimal to assemble the electrical unit as part of the overall machine assembly, not as a separate work order.

This Engineering BOM also indicates that part G is a separate part. However, in this structure, part G doesn't represent a physical part but a collection of packaging materials.

Therefore, although an Engineering BOM provides great value for the design of a product and maintenance of that design, it might not be the most logical way to support the manufacturing

execution process of the product. By contrast, a Manufacturing BOM represents the best way to build a product.

Figure 2 shows how the preceding Engineering BOM is transitioned into a Manufacturing BOM. In figure 2, (a) is the BOM for product H, and b is the route sheet for product H.

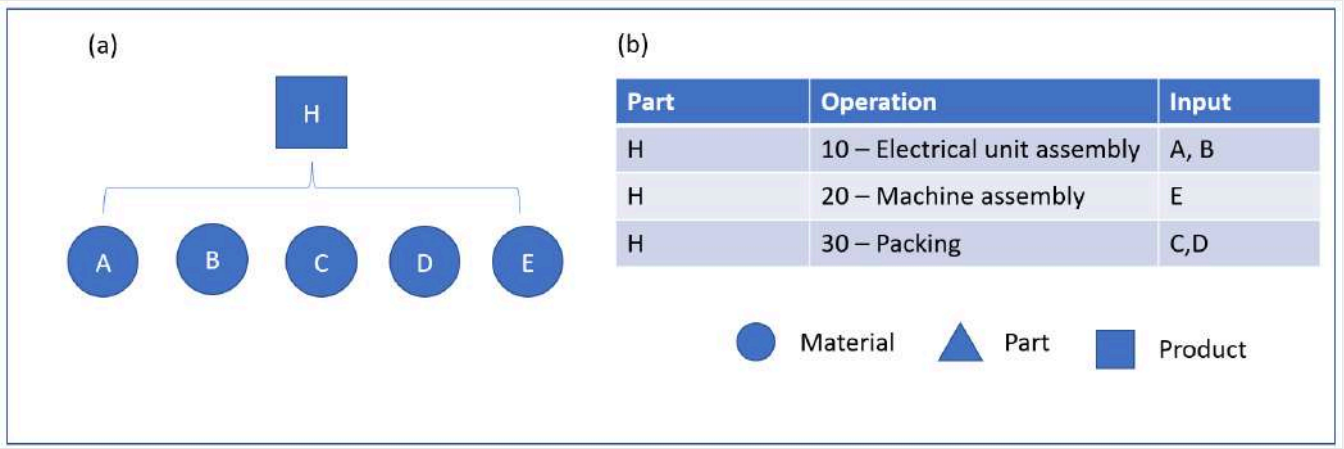


Figure 2: Manufacturing BOM

In this structure, you can see that there's no notion of parts F and G, and the materials that these parts consist of have been elevated to the next BOM level.

Unlike the Engineering BOM, which had two operations sheets, the Manufacturing BOM has only one operations sheet. The packaging operation that was linked to part G has also been elevated and is now part of the operations sheet for product H. The assembly of the electrical unit is the first operation. This order makes good sense, because this unit is used in the next operation, which is the machine assembly. The last operation is the packaging operation, which consumes two packing materials (C and D).

The transition between the Engineering BOM and the Manufacturing BOM is enabled through the *Phantom* BOM line type. As the term *phantom* indicates, parts F and G have disappeared during the transition between the two BOM types. In this example, the *Phantom* line type is applied to the BOM lines for parts F and G in the Engineering BOM. When a production or batch order is created, the Engineering BOM is copied to the production or batch order. Then, when the order is estimated, the transition from the Engineering BOM to the Manufacturing BOM occurs, as shown in figure 2. From the operations sheet in figure 2, packaging materials C and D are input for the operation.

Multilevel phantom BOM structures

The *Phantom* line type can be used in multilevel BOM structures, as shown in figure 3. In figure 3, (a) is the BOM for product G, and (b) is the route sheet for parts E and F and product G.

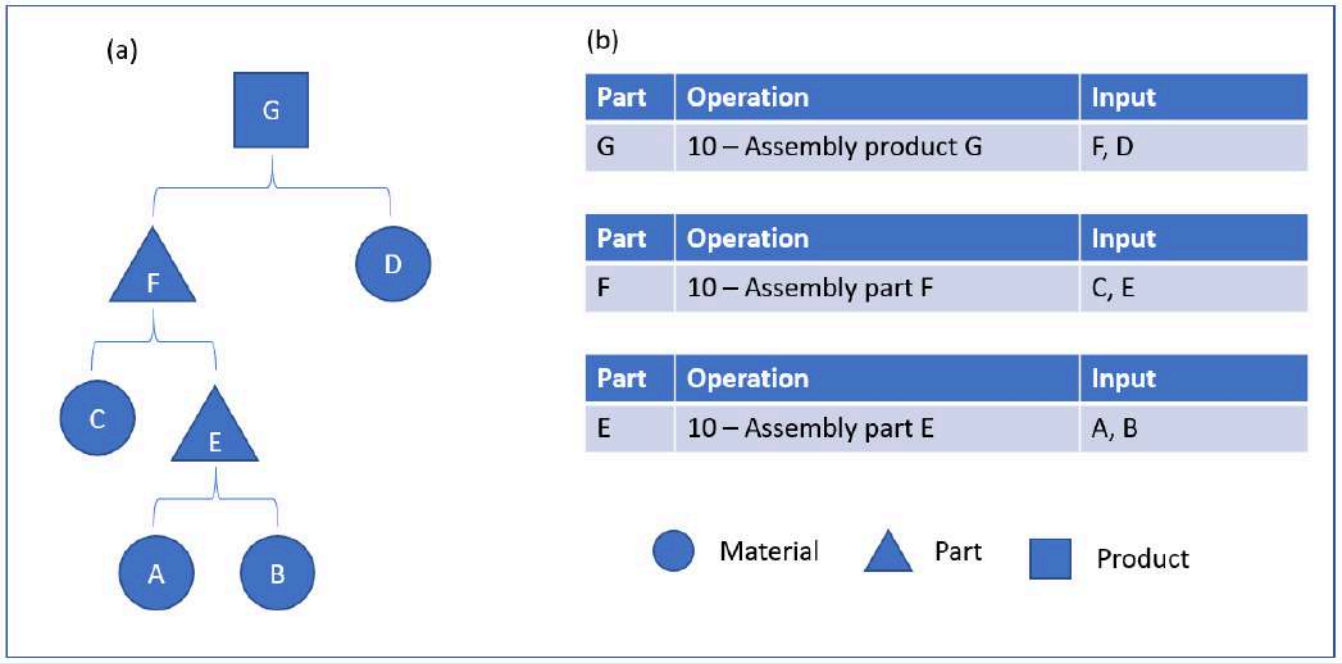


Figure 3: Engineering BOM part G

Figure 4 shows the resulting Manufacturing BOM and route sheet if the BOM lines for parts E and F are configured so that the line type is *Phantom*. In figure 4, (a) is the BOM for product G, and (b) is the route sheet for product G.

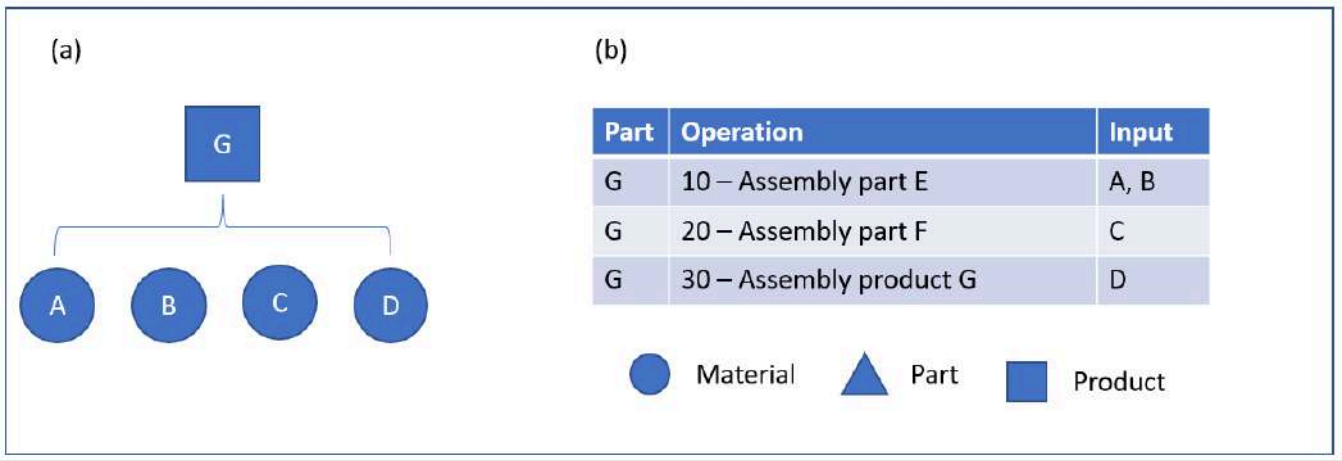


Figure 4: Manufacturing BOM part G

Phantom and route network

Phantom BOMs can also be used for a BOM that has a route network. In a route network, one or more operations run in parallel. Figure 5 shows an example of a route network that is used in a multilevel BOM. In figure 5, (a) is the BOM for product G and part F, and (b) is the route sheet for product G and part F, which has a route network.

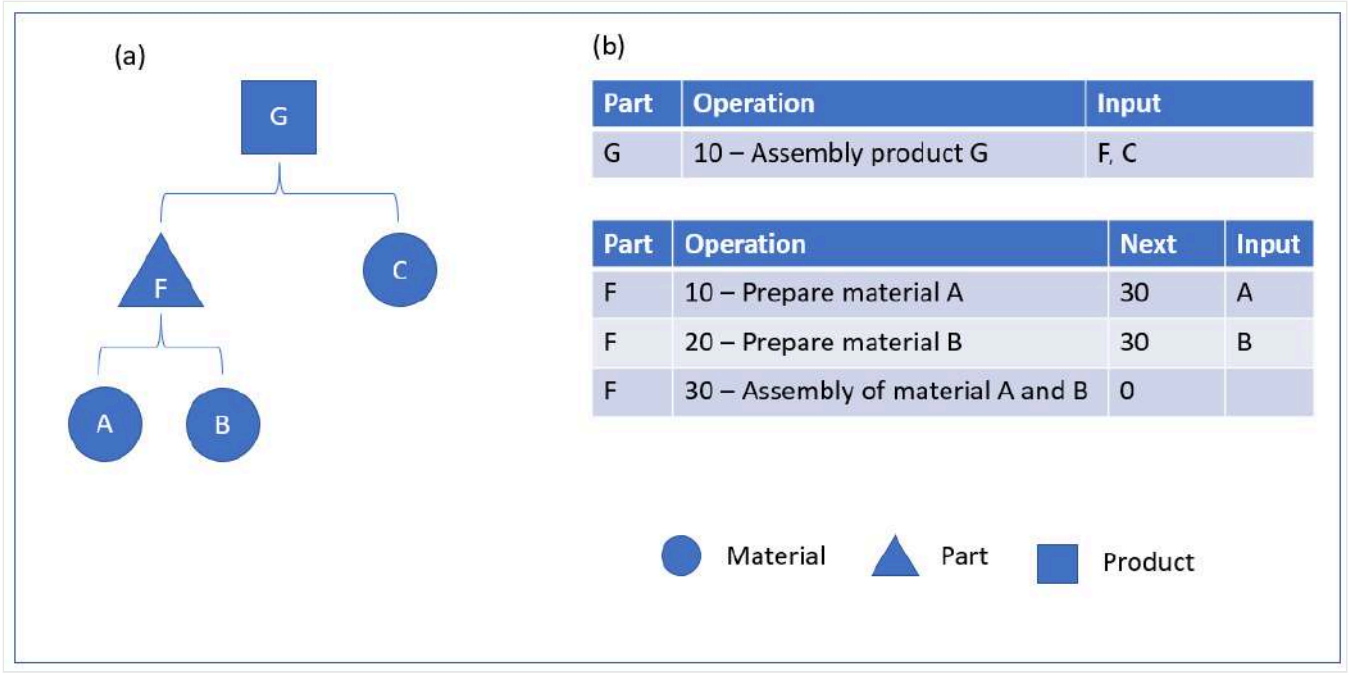


Figure 5: Engineering BOM part G, route network

In figure 6, (a) is the BOM for product G and part F, and (b) is the route sheet for product G and part F.

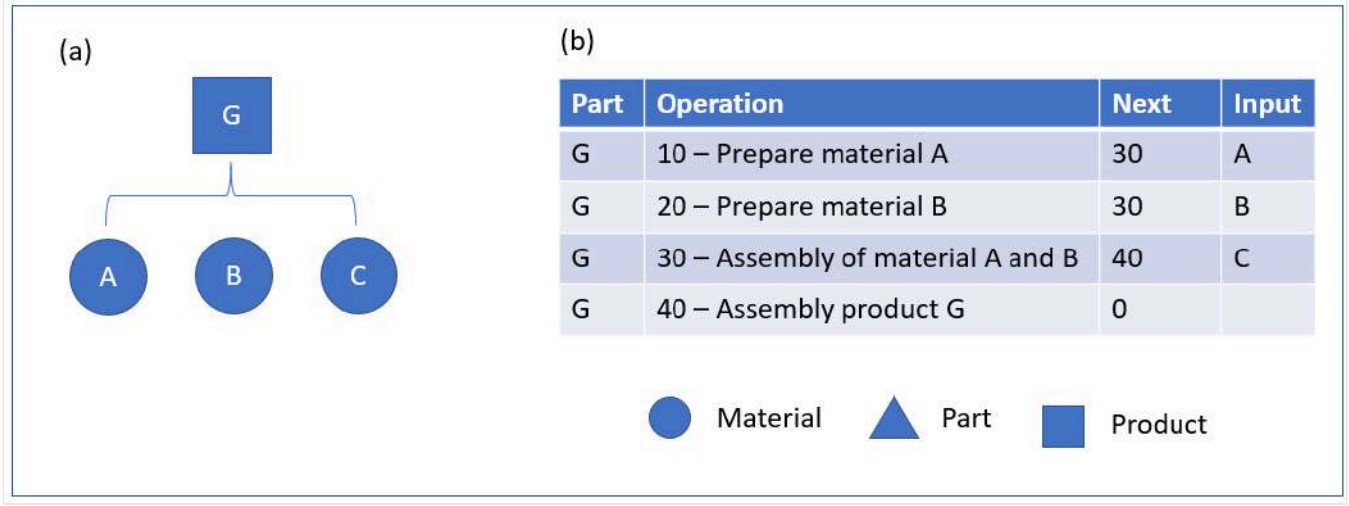


Figure 6: Manufacturing BOM part G, route network

BOM calculations

07/21/2025

The cost roll-up and sales price calculations are known as bill of materials (BOM) calculations, and you initiate them from the Calculations page. This article provides information about BOM calculations.

The cost roll-up and sales price calculations are known as bill of materials (BOM) calculations, and you initiate them from the **Calculations** page. You use the **Calculations** page to perform the following tasks:

- Calculate the cost of a manufactured item, and generate an associated item cost record within a costing version.
- Calculate the sales price of a manufactured item, and generate an associated item sales price record within a costing version.

The way that you use the **Calculations** page varies slightly, depending on how you initiate the BOM calculations. The way that you use the **Calculations** page also depends on whether the BOM calculations involve a costing version for standard costs or planned costs, and on several policies that are defined in the costing version that is used in the BOM calculations. A variation of the **Calculations** page is used in the context of a sales order, sales quotation, or service order line item. These calculations are known as order-specific BOM calculations. An order-specific BOM calculation doesn't generate an item cost record within a costing version. Instead, it generates a calculation record that appears on the **BOM calculation details** page. The calculation record includes a calculated cost and a calculated sales price. The **Calculations** page can be opened for a single manufactured item or for a costing version.

- To calculate costs for a single manufactured item, you initiate BOM calculations from the **Item price** page. The **Calculations** page inherits the item identifier. The costing version, BOM version, route version, calculation quantity, calculation date, and site must be specified.
    - By default, the BOM version and route version are set to the active versions for the item, site, date, and calculation quantity. However, you can override the default values with approved versions.
    - By default, the calculation quantity is set to the item's standard order quantity. However, you can override the default value.
    - The calculation date or site can be mandated by the costing version, or user-specified values can be set when the date or site isn't mandated in the costing version. A future calculation date determines how pending cost records are used. BOM calculations use a pending cost record that has the nearest from-date that is on or before the calculation date.

- To calculate costs for all manufactured items or selected items, or to update items on a where-used basis, you initiate BOM calculations from the **Costing version setup** page or the **Costing version maintenance** page. The **Calculations** page inherits the costing version.
    - For the calculations, it's assumed that the active BOM version and route version are used for a manufactured item (and for the related site, date, and quantity), unless a manufactured component has a specified sub-BOM or subroute.
    - For the calculations, it's assumed that the standard order quantity is used for manufactured items. The standard order quantity provides the basis for calculating component quantities, determining the relevant BOM versions and route versions (when you use quantity-sensitive BOMs and routes), and amortizing constant costs. However, when a manufactured component has a BOM line type of *Production* or *Vendor*, or when you use a make-to-order explosion mode for the BOM calculations, this assumption doesn't apply, because quantities reflect the specified calculation quantity.
    - The calculation date or site can be mandated by the costing version, or user-specified values can be set when the date or site isn't mandated in the costing version.

Other variations in BOM calculations reflect the costing type and restrictions of the costing version:

- BOM calculations that use standard costs must be restricted by costing version policies, because the restrictions help guarantee that standard costing principles are used. Standard costing principles require the enforcement of restrictions about the use of standard costs for purchased items, a single-level explosion mode, and the inclusion of miscellaneous charges in unit costs.
- BOM calculations that use planned costs don't have to follow standard costing principles. These BOM calculations can use different explosion modes, alternative sources of cost data for purchased items, and optional enforcement of restrictions within the costing version.

BOM calculations that use standard costs

Policies within the costing version (for standard costs) can mandate enforcement of five BOM calculation policies. The *Recording restriction* option in the costing version mandates one of these policies, where miscellaneous charges must be included in the unit price. Miscellaneous charges for purchased items can be entered manually, whereas miscellaneous charges for manufactured items reflect the calculated amortization of constant costs. The *Calculation restriction* option in the costing version mandates the other four BOM calculation policies:

- The source of cost contributions for purchased items must be based on standard costs. In other words, BOM calculations must use the item cost records within the specified costing version, or within the fallback that contains standard costs.
- To help guarantee accurate and consistent calculation of standard costs, the explosion mode must be single-level.
- To help guarantee consistent results when the sales price of the items is calculated, the profit setting must be mandated. The profit setting can be used, and the item sales price records can be generated, only if the costing version allows for content of sales prices.
- The fallback principle must be mandated, and can be set to *None*, *Active* (for active cost records), or *Costing version* (for a specified costing version).

BOM calculations that use planned costs

Policies within the costing version (for planned costs) can optionally mandate enforcement of five BOM calculation policies. Alternatively, the policies can just provide default values. The *Recording restriction* option in the costing version determines whether the BOM calculation policy about miscellaneous charges will be mandated or act as a default value. Miscellaneous charges can optionally be included in the unit price. The *Calculation restriction* option in the costing version determines whether the other four BOM calculation policies will be mandated or act as default values:

- The source of cost contributions for a purchased item can be the item cost records within a costing version. Alternatively, the source can be defined by the BOM calculation group that is assigned to the item. For example, the BOM calculation group can define purchase price trade agreements as the source of cost contribution data.
- The explosion mode can be single-level, multilevel, or make-to-order, or it can be based on the BOM line item. The explosion mode for the BOM line type replicates the cost calculation logic for production order estimates.
- The profit setting can be mandated, or it can be a default value. The profit setting can be used, and the item sales price records can be generated, only if the costing version allows for content of sales prices.
- The fallback principle can be mandated, or it can be a default value. The fallback principle can be set to *None*, *Active* (for active cost records), or *Costing version* (for a specified costing version).

BOM calculations generate warning messages and other types of messages. Several BOM calculation policies determine the types of messages. The warning conditions are defined in the BOM calculation group that is assigned to items. However, you can override these warning conditions when you initiate a BOM calculation. When the fallback principle is used, it's often helpful if the fallback is shown as an information message. When you're trying to update

calculated costs for items that have missing cost records, it's also helpful if the information message identifies items that weren't updated.

BOM calculations that use the fallback principle

The following situations illustrate two uses of the fallback principle:

- **Two-version approach to standard cost updates** - A costing version can contain the incremental changes to standard costs, such as pending cost records that represent new items or cost changes. In this situation, the fallback principle can identify the use of the active standard costs that are contained in other costing versions.
- **Simulation of the effect of cost changes by using planned costs** - A costing version for planned costs can contain incremental changes for simulation purposes. This costing version will include pending cost records that represent the simulated cost changes to items, cost categories, and calculation formulas for indirect cost. In this situation, the fallback principle can identify the use of the active standard costs that are contained in other costing versions.

BOM calculation of a suggested sales price

When you use a cost-plus-markup approach, the calculated sales price for an item reflects the set of profit-setting percentages that is specified for the BOM calculation, and the costs that are associated with the item's component items, routing operations, and applicable manufacturing overheads. The markup reflects profit-setting percentages that are assigned to cost groups, and the cost groups that are assigned to items, cost categories for routing operations, and the indirect cost calculation formulas for manufacturing overheads. The sets of profit-setting percentages are labeled *Standard*, *Profit 1*, *Profit 2*, and *Profit 3*. Within the Profit 1 set, for example, a profit-setting percentage of 50 percent can be defined for a cost group that is assigned to purchased material, and a profit-setting percentage of 80 percent can be defined for a cost group that is assigned to cost categories for routing operations. The context of the BOM calculation determines how the results of a calculated sales price are handled:

- **BOM calculation for an item and specified costing version** - The BOM calculation generates a pending sales price record within the costing version. This sales price record provides the starting point for viewing the calculation details (for example, on the **Calculate item cost** page). The sales price record acts mainly as reference information and isn't used as the basis for a sales price on sales orders.
- **Order-specific BOM calculation** - A variation of the **BOM calculation** page is used in the context of a sales order, sales quotation, or service order line item. An order-specific BOM calculation doesn't generate a record within a costing version. Instead, it generates a calculation record that appears on the **BOM calculation results** page. This calculation

record provides the starting point for viewing the calculation details (for example, on the **Calculate item cost** page). Information about a selected calculation record can be transferred to the originating line item. For example, the calculated sales price can be transferred to a sales order line item.

Order-specific BOM calculations

An order-specific BOM calculation represents a variation of a BOM calculation for a manufactured item. The order-specific BOM calculation is performed in the context of a sales order, sales quotation, or service order line item. An order-specific BOM calculation generates a calculation record that appears on the **BOM calculation results** page. The calculation record includes a calculated weight, a calculated cost that is based on active cost records, and a calculated sales price. The calculation record that each order-specific BOM calculation for an item generates on the **BOM calculation results** page is uniquely identified by a calculation number. The results of a calculation record can be optionally transferred to the originating line item. An order-specific BOM calculation differs from a BOM calculation for a manufactured item in two ways:

- An order-specific BOM calculation doesn't generate an item cost record within a costing version. Therefore, the BOM calculation policies aren't applied when an item cost record is created, or when a cost record is overwritten.
- An order-specific BOM calculation always uses the active cost records for components, cost categories, and indirect cost calculation formulas.

Production parameters for manufacturing execution

08/13/2025

This article provides information about the setup of production parameters for manufacturing execution.

Manufacturing execution can be used to register time and item consumption on production jobs or projects. Before you start to use manufacturing execution for job registrations, you must set up various production parameters that define how and when registrations are posted during the production process. The settings of production parameters affect inventory management, production management, and cost calculation.

Before workers start to make registrations on production jobs, you should carefully consider all settings on the **Production order defaults** page. Go to **Production control** > **Setup** > **Manufacturing execution** > **Production order defaults**. If your company uses the multisite functionality, you might want to set up different production parameters for each site. The following tabs are provided on the **Production order defaults** page:

- **General** – General parameter settings for production jobs.
- **Start** – Parameters that are used when production operations are started.
- **Operations** – Parameters that are applied to production operations and feedback about operations during the production process.
- **Report as finished** – Parameters that are used when items are reported as finished on the last operation of a production order.
- **Quantity validation** – Parameters that are used to validate start and feedback quantities on production orders.

Types of production jobs

On the **Operations** tab, you select which types of production jobs require registration.

Typically, workers make registrations on setup jobs and process jobs. However, if job scheduling is applied, you can select other job types that workers must make registrations on when production orders are processed. For example, you can require registrations on transport jobs.

> **Important**

Make sure that you select all relevant job types. Otherwise, jobs might not be available for registration. Your selections should match the selections in the **Job management** column on the **Setup** FastTab of the **Route groups** page (**Production control** > **Setup** > **Routes** > **Route groups**).

If **Job management** is selected on the route group, this job type is reported as finished on the production order when a job is reported as finished. When all job types that **Job management** is selected for have been reported as finished on an operation, the operation is reported as finished.

> ⓘ Note
>
> Some job types can be manually reported through production journals. In this case, select **Job management** for the job type, but don't select the job type for registration on the **Operations** tab on the **Production order defaults** page.

BOM consumption and picking list journals

A consistent setup for bill of materials (BOM) consumption is important because it helps guarantee that inventory management is efficient. For example, if BOM consumption parameters aren't set up correctly, materials might be deducted from inventory two times or not at all.

On the **Production order defaults** page, automatic BOM consumption is set up in three stages:

- At the start of a production. Set up this stage on the **Start** tab.
- During the production process when an operation is completed. Set up this stage on the **Operations** tab.
- When a production order is reported as finished. Set up this stage on the **Report as finished** tab.

For each stage, the **Automatic BOM consumption** field lets you select one of three methods for picking items for a production order:

- *Flushing principle* – This option is used in combination with an option that is defined for the BOM. Select **Production control** > **Production orders** > **All production orders**. On the **All production orders** page, select a production order in the list, and then, on the Action Pane, open the **Production order** tab and select **BOM**. On the **BOM** page, open the BOM line you want to set. Then open the **Setup** FastTab and select one of the following options in **Flushing principle** field:
  - *Start*

- *Finish*
- *Manual*
- *Available at location*
- Blank (No option is selected.)

If *Flushing principle* is selected in the **Automatic BOM consumption** field on the **Start** tab of the **Production order defaults** page, all materials that are set to *Start* in the BOM are deducted from inventory when the operation is started. The *Available at location* option is used for products that are enabled for warehouse management processes (WMS). If you select this flushing principle, material is flushed when warehouse work for raw material picking is completed. Material is also flushed when a BOM line that uses this flushing principle is released to warehouse and the material is available at the production input location.

> ⓘ **Note**
>
> If *Flushing principle* is selected in the **Automatic BOM consumption** field on the **Start** tab of the **Production order defaults** page, you must select the same value on either the **Operations** tab or the **Report as finished** tab. This requirement helps guarantee that materials are deducted from inventory on BOMs that use *Finish* as a flushing principle on the production order. If *Flushing principle* isn't selected on either the **Operations** tab or the **Report as finished** tab, materials might be deducted from inventory two times.

- *Always* – If you select this option for a stage, materials are always deducted from inventory at that stage. For example, materials for the production are deducted when the production order is started. This setting requires that *Never* be selected on the **Operations** and **Report as finished** tabs. This requirement helps prevent items from being deducted from inventory two times.

- *Never* – If you select this option for a stage, no BOM consumption occurs at that stage. For example, if you select *Never* on all three tabs (**Start**, **Operations**, and **Report as finished**), materials must be manually deducted from inventory.

> ⓘ **Important**
>
> Carefully consider your setup of the production order defaults, and make sure that the defaults that are selected on the various tabs of the **Production order defaults** page don't contradict each other.

The following examples illustrate parameter settings that support various BOM consumption principles. The parameters are set up on the **Production order defaults** page.

Example 1: Backflushing on operations

Use the following settings if picking list journals and BOM item consumption should be generated when items are reported as finished on an operation.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status* or *Status + quantity*
Start	Automatic BOM consumption	*Never*
Operations	Automatic BOM consumption	*Always*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status + quantity*

Example 2: Backflushing on production

Use the following settings if picking list journals and BOM item consumption should be generated when items are reported as finished on the production order.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status* or *Status + quantity*
Start	Automatic BOM consumption	*Never*
Operations	Automatic BOM consumption	*Never*
Report as finished	Automatic BOM consumption	*Always*
Report as finished	Update finished report on-line	*Status + quantity*

Example 3: Flushing principle

Use the following settings if picking list journals and BOM item consumption should be generated according to the flushing principle that is set for the BOM items.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status + quantity*
Start	Automatic BOM consumption	*Flushing principle*
Operations	Automatic BOM consumption	*Flushing principle*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status + quantity*

Example 4: Deduction of materials during startup of a production order

Use the following settings if picking list journals and BOM item consumption should be generated when a production is started.

Expand table

Tab	Field	***Setting***
Start	Update start on-line	*Status + quantity*
Start	Automatic BOM consumption	*Always*
Operations	Automatic BOM consumption	*Never*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status* or *Status + quantity*

Based on the selections that are described earlier in this section, picking list journals are posted at various stages of the production order process:

- When an operation is started
- When quantity feedback is reported on an operation
- When items are reported as finished on the production order

Example 5: Manual BOM consumption

You can use the following settings if materials should always be manually deducted from inventory. In this case, picking list journals aren't posted.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status*
Start	Automatic BOM consumption	*Never*
Operations	Automatic BOM consumption	*Never*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status*

Operations resources

07/10/2025

Operations resources perform the activities of a project or a production process. They can be of different types, and can have different capabilities.

Operations resources

Operations resources are the machines, tools, workers, facilities, physical areas, or vendors that perform the activities of a project or a production process. They can be of different types and can have different capabilities.

- **Vendor** – An external resource that performs project activities or production operations. An example is a subcontractor. By linking vendor resources to a vendor account, you can generate purchases for subcontractors, based on the bill of materials (BOM) lines or production lines.
- **Human resources** – A project or production worker that perform an activity, either alone or as an operator of a tool or a machine. If you're using the Human resources functionality, you can link human resources to a worker. The scheduling engine can then allocate the resources, based on the competencies that are defined for the corresponding worker.
- **Machine** – A machine or other production equipment that is required in production.
- **Tool** – An instrument or device that is typically used together with another resource to perform an activity in a project or in production.
- **Location** – A physical location of a specific size that is required in order to perform an activity. An example is an assembly area.
- **Facility** – A building or fixed structure that is required in order to perform an activity.

Calendars

A calendar can be assigned to an operations resource and describes the capacity (in hours) of that resource. The working times of the operations resource are determined by the calendar that is assigned to the resource group that the operations resource is part of. You can set an effective date and an expiration date for the assigned calendar. You can then assign more than one calendar to an operations resource. However, the dates of the assigned calendars can't overlap, and the operations resource can be assigned only one calendar at a time.

> ⓘ **Note**

If there are no effective working calendars for a resource group (for example, if the last assigned calendar or the only assigned calendar has expired), you can no longer access the operations resources that are assigned to the resource group for production planning and operations scheduling. You can also assign a calendar to a resource group to specify the working times for both the resource group and all the operations resources that are assigned to the resource group. The capacity of the resource group is calculated as the sum of the capacities of each operations resource that is assigned to that resource group. The calendar that is assigned to a resource group can change over time.

Resource capabilities

A capability is the ability of an operations resource to perform a particular activity. You can assign one or more capabilities to an operations resource. The scheduling engine will then allocate resources by matching the resource requirements of each activity to the capabilities of the available operations resources. Capabilities can be assigned to all types of operations resources (*Tool*, *Vendor*, *Machine*, *Human resources*, *Location*, or *Facility*). To assign capabilities to operations resources for a limited time, define a start date and an expiration date on the capability assignment. Learn more in Resource capabilities.

Resource groups

A resource group is a set of operations resources that represents the granularity at which you want to schedule resources when you use the operations scheduling method. Therefore, resource groups typically correspond to the physical organization of work cells that is demarcated by yellow lines on the production shop floor. The resource group defines the site, production unit, and warehouse context for operations resources that are assigned to the group. When you assign an operations resource to a resource group, the resource can be scheduled at the site where the resource group is located. You don't have to assign the operations resources that you create to a resource group. However, an operations resource must be assigned to a resource group before it can be scheduled to perform work. An operations resource can be assigned to a resource group for a limited time. You can also assign an operations resource to multiple resource groups, so that you can then share the resource across sites. However, the effective dates and expiration dates can't overlap. In other words, you can't assign an operations resource to two resource groups at the same time. Changes in resource group assignments are effective only for new resource allocations. Capacity reservations for jobs, operations, and project hour forecasts that are already scheduled won't be affected.

Note

When you assign operations resources of the *Vendor* type to a resource group, all operations resources that are assigned to that resource group must be of the *Vendor* type and must be linked to the same vendor account.

Production units

A production unit is an administrative unit that is a collection of resource groups. A production unit can contain multiple resource groups, but a resource group can be assigned to only one production unit. A production unit reflects the physical layout of production resources, and has no effect on transactions or how they're processed. You must associate a production unit with a site. You can also assign a picking warehouse and a storage warehouse to a production unit. You can use a production unit to consolidate and filter production-related data. For example, a shop floor manager can see an overview of the outstanding workload and the available capacity for a particular production unit. You can change the production unit that is assigned to a resource group. You can also delete a production unit. However, these changes to the production unit are effective only for new orders that are created after master scheduling is run. If you want to apply the production unit change to existing orders, you must make this change manually.

Resource scheduling

Operations resources are assigned to activities when a project or a production is scheduled. Two scheduling methods are available: operations scheduling and job scheduling. When you use operations scheduling, each operation or project activity is scheduled on the resource group that contains operations resources that match the resource requirements that are specified for the operation. If a specific operations resource is required for the operation, scheduling reserves capacity only on a specific operations resource in the group. Job scheduling is a more detailed form of scheduling than operations scheduling. It breaks down each operation into its individual tasks or jobs. These jobs are then assigned to the operations resources that will perform them. Scheduling reserves capacity on the resource groups, based on the operation times that are defined on the production route or project activities. If you're working with finite capacity, the schedule depends on the availability of the operations resources that are required in order to complete the activity. For operations scheduling, the capacity of the resource group is the sum of the available capacity of the operations resources that are part of that group. Capacity reservations that already exist for the operations resources are considered unavailable capacity. If there isn't enough available capacity for production, the production orders can be delayed or even stopped. On the **Resources** page, you can define several properties that control how capacity reservations are made:

- **Capacity** – Specify the operations resource's capacity per hour in terms of the capacity unit of measure.
- **Batch capacity** – Specify the maximum number of pieces that the operations resource can process per run.
- **Efficiency percentage** – Specify the efficiency that you expect from the operations resource. The efficiency percentage adjusts the throughput of the operations resource and affects the time that is reserved for the resource. The lead times for the operations that use the operations resource are also adjusted accordingly. Here's the formula that is used for the calculation: Scheduling time = Time × 100 ÷ Efficiency percentage. *Time* includes both the run time and setup time.
- **Operations scheduling percentage** – Specify the maximum percentage of capacity of the operations resource that you want to use in operations scheduling. To allow for flexibility in capacity during job scheduling, you should set this percentage to less than 100 percent.
- **Finite capacity** – Set this option to *Yes* if the operations resource should be scheduled based on the actual capacity that is available, and if existing capacity reservations should be considered. If this option is set to *No*, the operations resource is assumed to have infinite capacity, and the resource might therefore be overbooked.
- **Finite property** – Set this option to *Yes* if you want the operations resource to be scheduled based on the actual capacity that is available with respect to the required working time scheduling properties.
- **Exclusive** – Set this option to *Yes* if you don't want the operations resource to be available for another job or operation until the current production is completed. In this case, the operations resource can't be used even if there are gaps in the resource's run time.
- **Bottleneck resource** – Define the operations resource as a bottleneck resource. A bottleneck resource is scheduled by using finite capacity when the **Finite capacity** and **Bottleneck scheduling** options on the **Master plans** page are selected.

To schedule multiple operations resources at the same time to perform, for example, an operation in production, you must specify the requirements for the various resources by using primary and secondary operations. You can then also reserve multiple operations resources that have different capacity. The operations resource that is scheduled for the primary operation determines the duration of the activity.

Lean work cells

You can specify that a resource group is a lean work cell. The resource group can then be part of a production flow. By specifying a resource group as a lean work cell, you also prevent the resource group and the assigned operations resources from being allocated to the operations of a production order or a project hour forecast. For each resource group that acts as a lean work cell, you must specify the following information:

- **Calendar** – The working calendar of the work cell, which must have the property of a standard workday.
- **Input warehouse and location** – The default location that is used to pick material for an activity.
- **Output warehouse and location** – The default location where all output of the work cell is put.
- **Runtime cost category** – This category must be defined for the work cell if direct labor is tracked in costing.

When a resource group is used as a lean work cell, the capacity of the work cell is specified directly on the resource group. Therefore, you don't have to assign operations resources to the resource group. Only when the work cell is managed by a subcontractor, an operations resource of the *Vendor* type must be assigned to the work cell. If you assign an operations resource to a resource group that is marked as a work cell, the capacity of the work cell isn't affected.

Costing resources

When you define an activity such as a route operation or a project hour forecast, you can specify the requirement for a specific operations resource or resource group. However, you can also specify the requirement for an operations resource of a specific type, or an operations resource that has a specific capability or competency. For this reason, the actual resource assignment isn't made until the activity is scheduled and capacity is reserved. Therefore, on a route operation, you can specify that estimation and BOM calculation must be based on a specific operations resource. This operations resource is referred to as the costing resource. You can also transfer cost categories and operation times from the costing resource to the activity. When the operation is scheduled, estimation and BOM calculation are done by using the operations resource that is actually scheduled.

Create an operations resource

Article • 09/29/2021

An operations resource performs the activities of a project or a production process. This procedures shows you how to define an operations resource. You can walk through this procedure in demo data company USMF, or using your own data.

1. Go to Resources.
2. Click New.
3. In the Resource field, type a value.
4. In the Description field, type a value.

Define capacity and consumption parameters

1. Expand the Operation section.
2. In the Scrap percentage field, enter a number.
3. In the Setup category field, enter or select a value.
    - Specify the cost category that defines how to account for the cost of setup.
4. In the Run time category field, enter or select a value.
    - Specify the cost category that defines how to account for the cost of run time.
5. In the Quantity category field, enter or select a value.
    - Specify the cost category that defines how to account for the resource cost based on the output quantity.
6. In the Capacity unit field, select an option.
    - Specify the unit in which to express the capacity of the operations resource.
7. In the Capacity field, enter a number.
8. In the Efficiency percentage field, enter a number.
    - Specify the efficiency that you expect from the operations resource. The efficiency percentage adjusts the throughput of the operations resource and affects the time that is reserved for the resource.
9. In the Operations scheduling percentage field, enter a number.

- Specify the maximum percentage of capacity of the operations resource that you want to use in operations scheduling.

10. Select Yes in the Finite capacity field.

    - Set this option to Yes if the operations resource should be scheduled based on the actual capacity that is available, and if existing capacity reservations should be considered. If this option is set to No, the operations resource is assumed to have infinite capacity, and the resource might be overbooked.

11. Select Yes in the Bottleneck resource field.

Define working times

1. Expand the Calendars section.
2. Click Add.
3. In the Calendar field, enter or select a value.

    - Specify the working time calendar that defines the capacity (in hours) of the resource.

4. In the list, find and select the desired record.
5. In the list, click the link in the selected row.

Define resource capabilities

1. Expand the Capabilities section.
2. Click Add.

    - A capability is the ability of an operations resource to perform a particular activity. The scheduling engine allocates resources by matching the resource requirements of each activity to the capabilities of the available operations resources.

3. In the Capability field, enter or select a value.
4. In the Level field, enter a number.

    - Specify the level of proficiency by which the resource processes the capability.

5. In the Priority field, enter a number.

    - Specify the priority of the operations resource with respect to the capability. When scheduling by priority, the operations resource with the highest priority

(lowest numeric value) is selected first.

Assign resource to resource group

1. Expand the Resource groups section.
2. Click Add.
    - The resource group defines the site, production unit, and warehouse context for operations resources. The operations resource can only be scheduled when assigned to a resource group, and only on the site where the resource group is located.
3. In the Resource group field, enter or select a value.
4. In the Input location field, enter or select a value.
    - Specify the warehouse location from where the operations resource is consuming materials.

Feedback

Was this page helpful? Yes No

Provide product feedback

Resource capabilities

06/17/2025

This article provides information about resource capabilities. A capability is the ability of an operations resource to perform a particular activity. The article explains how capabilities and related concepts, such as proficiency level and priority, are used to select appropriate resources for an activity.

A capability is the ability of an operations resource to perform a particular activity. An operations resource can have more than one capability assigned to it, and a capability can be assigned to more than one resource. You can temporarily assign capabilities to resources by defining a start date and expiration date on the capability assignment. When the capability for a resource expires, the resource can't be scheduled for a project or a production that requires that capability. Expired capabilities can be renewed. You can delete capabilities provided that they aren't on a route relation or on part of a production route of an active production order. In general, be careful when you delete capabilities. Instead, consider adjusting the expiration date on the resources that have the capability. Capabilities can be assigned to all types of resources: tool, vendor, machine, location, facility, or human resource.

Level

Multiple resources often have the same functional capability but at different levels of proficiency (for example, strength, processing speed, or accuracy). Therefore, when you assign a capability to a resource, you can specify a **Level** value. The level is a simple numeric value. If you specify that a capability is a resource requirement for a production route, you can also specify a **Minimum level needed** value for that capability. The scheduling engine then selects only resources that have the required capability at a level that is equal to or exceeds the minimum level that is specified in the resource requirement.

Priority

Operations resources that have the same capabilities can be assigned a priority. Then, if multiple resources have capabilities that satisfy the scheduling requirements at the required level, and have free capacity, the scheduling engine can select among those resources. If **Priority** is selected in the **Primary resource selection** field on the **Scheduling parameters** page, the resource that has the highest priority (the lowest numeric value in the **Priority** field) is selected first during scheduling.

Resource requirements

When you define resource requirements for a production route, you can specify one or more capabilities as requirements. During production scheduling, the capabilities that are defined in the resource requirements on the route operation are matched with the capabilities that are defined for the resources. Resources that have capabilities that satisfy the requirements are selected. If multiple resources have the same functional capability but different proficiencies (such as strength, processing speed, or accuracy), you can either define multiple capabilities or use the capability level to qualify the capability of the resource. Here's an example:

- On a route, the drilling process time is set to 10 units per hour, but the requirement itself is defined as Drilling.
- You have two drilling machines, M1 and M2.
- The Drilling capability is assigned to both resources, the M1 machine and the M2 machine.
- The M1 machine can drill two units per hour, and the M2 machine can drill 11 units per hour.

In this example, both machines can be selected by the scheduling engine, because both satisfy the basic capability requirement (Drilling). However, the M1 machine can drill only two units per hour. Because this rate is much less than the 10 units per hour that are specified on the route, the M1 machine will cause production problems if it's selected. There are two ways to resolve this issue and make sure that the M2 machine is selected instead:

- Create two separate capabilities: Low-speed drilling and High speed drilling. Define Low-speed drilling as drilling that has a process time of one to nine units per hour. Define High-speed drilling as drilling that has a drilling process time of 10 to 19 units per hour. Then assign the M1 machine the Low-speed drilling capability, and assign the M2 machine the High-speed drilling capability. Finally, change the resource capability requirement on the route to High-speed drilling. The scheduling engine will then select the correct machine (M2).
- Use the capability level to qualify the Drilling capability that is assigned to the drilling machines. Specify that the M1 machine has the Drilling capability at a level of 2.0, and that the M2 machine has the Drilling capability at a level of 11.0. Then specify that 10.0 is the minimum level that is required for the Drilling capability requirement on the route. The scheduling engine will then determine that only the M2 machine satisfies the resource requirements.

Competencies for human resources

When you have operations resources of the **Human resources** type that are linked to workers in Human resources, you can also take advantage of the competencies of workers when you define the resource requirements for a production route. In other words, you can also specify

requirements for specific skills, courses, certificates, or titles. The scheduling engine can then select resources that are linked to workers, and the selection will be based on the competencies of those workers. The competencies are set up in Human resources, not on the **Resource capabilities** page. When you define skills, courses, certificates, or titles as resource requirements, you must use the Human resources functionality and link each resource of the **Human resources** type to a corresponding worker. If you aren't using the Human resources functionality, you can define capabilities on the **Resource capabilities** page that resemble or duplicate the competencies from Human resources. However, the **Resource capabilities** page doesn't contain the functionality that is required in order to maintain skills, courses, certifications, or titles.

Define resource capabilities

Article • 09/29/2021

Resource capabilities describe what operations resources can do. During scheduling, the requirements of each job and operation are matched against the capabilities of the available resources. This task guide will help you create a resource capability and assign it to a resource. The demo data company used to create this task is USMF.

Create a resource capability

1. Go to Resource capabilities.
2. Click New.
3. In the Capability field, type the ID of the resource capability.
    - For a given operation, you use the capability ID to specify that resources must have this capability to perform the operation.
4. In the Description field, enter a description of the capability.

Assign capability to a resource

1. Click Add.
2. In the Resource field, type the ID of the resource.
    - A resource capability can be assigned to one or more resources.
3. In the Expiration field, enter a date.
    - You can use this field to specify that a resource has the capability for only a limited time.
4. In the Priority field, enter a number.
    - When you schedule jobs and operations, you can specify whether to select resources by priority. If you choose to do this, and more than one resource can perform the job or operation by the requested date, the resource that has the lowest priority with respect to the required capability is selected.
5. In the Level field, enter a number.
    - When you specify that a job or operation requires a particular capability, you can also specify the minimum level that is required. Use the capability level to

differentiate resources that can perform the same job, but at different speeds, strengths, sizes, and so on.

Feedback

Was this page helpful? Yes No

Provide product feedback

Routes and operations

07/10/2025

This article provides information about routes and operations. A route defines the process for producing a product or product variant. It describes each step (operation) in the production process and the order that these steps must be performed in. For each step, the route also defines the required operations resources, the required setup time and run time, and how the cost should be calculated.

Overview

A route describes the order of operations that is required in order to produce a product or product variant. For each operation, the route also defines the operations resources that are required, the time that is required in order to set up and perform the operation, and how the cost should be calculated. You can use the same route to produce multiple products, or you can define a unique route for each product or product variant. You can even have multiple routes for the same product. In this case, the route that is used varies, depending on factors such as the quantity that must be produced. The definition of a route in Supply Chain Management consists of four separate elements that, together, describe the production process:

- **Route** – A route defines the structure of the production process. In other words, it defines the order of operations.
- **Operation** – An operation identifies a named step in a route, such as **Assembly**. The same operation can occur in multiple routes and can have different operation numbers.
- **Operation relation** – An operation relation defines the operational properties of an operation, such as the setup time and run time, cost categories, consumption parameters, and resource requirements. The operation relation enables the operational properties of an operation to vary, depending on the route that the operation is used in or the products that are being produced.
- **Route version** – A route version defines the route that is used to produce a product or product variant. Route versions enable routes to be reused across products or changed over time. They also enable different routes to be used to produce the same product. In this case, the route that is used depends on factors such as the location or the quantity that must be produced.

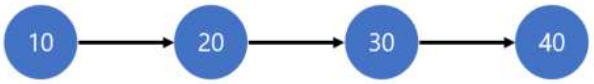
Routes

A route describes the order of operations that is used to produce a product or product variant. Each operation is assigned an operation number and a successor operation. The order of

operations forms a route network that can be represented by a directed chart that has one or more starting points and a single end point. In Supply Chain Management, routes are distinguished based on the type of structure. The two types of routes are simple routes and route networks. In the Production control parameters, you can specify whether only simple routes can be used, or whether the more complex route networks can be used.

Simple routes

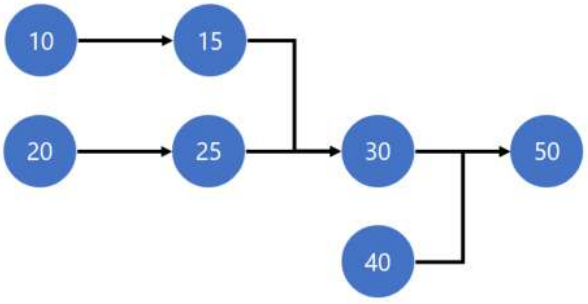
A simple route is sequential, and there's only one starting point for the route.



If you enable only simple routes in the Production control parameters, Supply Chain Management automatically generates the operation numbers (10, 20, 30, and so on) when you define the route.

Route networks

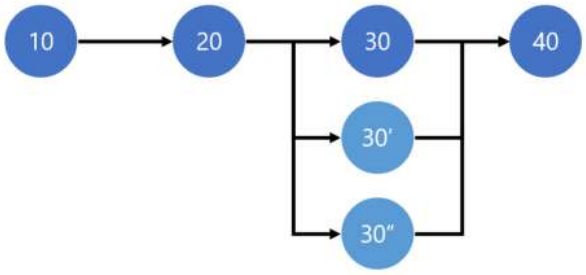
If you enable the more complex route networks in the Production control parameters, you can define routes that have multiple starting points and operations that can be run in parallel.



> **Note**
>
> - Each operation can have only one successor operation, and the entire route must end in a single operation.
> - This does not ensure that multiple operations that have the same successor operation (for example, operations 30 and 40 in the preceding illustration) will actually be run in parallel. The availability and capacity of resources might put constraints on the way that operations are scheduled.
> - You can't use 0 (zero) as the operation number. That number is reserved and is used to specify that the last operation in the route has no successor operation.

Parallel operations

Sometimes, a combination of multiple operations resources that have different characteristics is required in order to perform an operation. For example, an assembly operation might require a machine, a tool, and one worker for every two machines to oversee the operation. This example can be modeled by using parallel operations, where one operation is designated as the primary operation and the others are secondary.



Typically, the primary operation represents the bottleneck resource and dictates the run time for the secondary operations. However, during scheduling that involves finite capacity, the resources that are scheduled for both the primary operation and the secondary operations must be available and have free capacity at the same time.

Both the primary operation and the secondary operations must have the same operation number (30 in the preceding illustration).

In the preceding example, the resource requirement for the primary operation (30) is the machine, whereas the resource requirements for the secondary operations (30' and 30'') are the tool and the worker. A fifty-percent load helps guarantee that the scheduled worker can oversee two machines at the same time.

Approval of routes

A route must be approved before it can be used in the planning or manufacturing process. Approval indicates that the route design has been completed. The same released product or released product variant can have multiple approved routes. Typically, approval of a route occurs when the first relevant route version is approved. However, in some business scenarios, the approval of the route and the route version are separate activities that might involve different process owners.

Each route can be approved or unapproved separately. However, when a route is unapproved, all related route versions are also unapproved. In the Production control parameters, you can specify whether routes can be unapproved, and whether approved routes can be changed.

If you must keep a log of who approves each route, you can require electronic signatures for route approval. Users will then have to confirm their identity by using an electronic signature.

Operations

An operation is a step in the production process. Each operation has an ID and a simple description. The following table shows typical examples of operations from a machine shop.

Expand table

Operation	Description
PipeCut	Pipe cutting
TIGweld	TIG welding
JigAssy	Jig assembly
Inspection	Quality inspection

The operational properties of the operation, such as the setup time and run time, resource requirements, costing information, and consumption calculation, are specified on the operation relation. (For more information about operation relations, see the next section.)

Operation relations

The following operational properties of an operation are maintained on the operation relation:

- Cost categories
- Consumption parameters
- Processing times
- Processing quantities
- Resource requirements
- Notes and instructions

You can define multiple operation relations for the same operation. However, each operation relation is specific to one operation, and stores properties that are specific to a route, a released product, or a set of released products that are related to an item group. Therefore, the same operation can be used in multiple routes that have different operational properties. In addition, you can more easily maintain your master data if you use standard operations that have the same operational properties, regardless of the route that is used and product that is produced. The scope of the operation relation is defined through the **Item code**, **Item relation**, **Route code** and **Route relation** properties, as shown in the following table.

Expand table

Item code	Item relation	Route code	Route relation	Scope of the operation relation
Table	<Item ID>	Route	<Route ID>	The operational properties of an operation when it's used in the route where **Route number**=<*route ID*> to produce the released product where **Item number**=<*Item ID*>.
Table	<Item ID>	All		The default operational properties of an operation when it's used to produce the released product where **Item number**=<*Item ID*>. In other words, these operational properties apply when there's no route-specific operation relation for the released product.
Group	<Item group ID>	Route	<Route ID>	The operational properties of an operation when it's used in the route where **Route number**=<*route ID*> to produce released products that are associated with item group <*Item group ID*>, unless there's a route-specific operation relation for the released product.
Group	<Item group ID>	All		The default operational properties of an operation when it's used to produce released products that are associated with item group <*Item group ID*>, unless a more specific operation relation exists.
All		Route	<route ID>	The default operational properties of the operation when it's used in the route where **Route number**=<*route ID*>. In other words, these operational properties apply when there's no operation relation for this route that is specific to either the released product or its associated item group.
All		All		The default operational properties of an operation. These operational properties apply when a more specific operation relation doesn't exist.

You can also specify that an operation relation is specific to a site. In this way, the operational properties of an operation can vary, depending on the location (that is, the site) where the operation is performed. For configured products, you can also specify different operational properties for each product configuration.

Operation relations give you lots of flexibility when you define your routes. Additionally, the ability to define default properties helps reduce the amount of master data that you must maintain. However, this flexibility also means that you must be aware of the context that you modify an operation relation in.

> **Note**
>
> Because the operational properties are stored in operation relations per operation for each route, all occurrences of the same operation (for example, Assembly) have the same setup time, run time, and resource requirements. Therefore, if two occurrences of an operation must occur in the same route but have different run times, you must create two separate operations, such as Assembly1 and Assembly2.

Modifying product-specific routes

When you open the **Route** page from the **Released product details** page, the route versions that are associated with the selected released product are shown. In this context, for each operation, Supply Chain Management shows the operational properties from the operation relation that best matches the route version. You'll notice that the list of operations includes the **Item code** and **Route code** properties from the operation relation. Therefore, you can determine which operation relation is shown.

On the **Route** page, you can modify the operational properties of the operation, such as the run time or the cost categories. Your changes are stored on the operation relation that is specific to the route and released product that are referenced in the current route version. If the operation relation that is shown isn't specific to the route and the released product, before your changes are stored, the system creates a copy of the operation relation. This copy *is* specific to the route and released product. Therefore, your changes won't affect other routes or released products. To verify which operation relation is being modified on the **Route** page, look at the **Item code** and **Route code** fields.

You can also manually create an operation that is specific to a route and released product by using the **Copy and edit relation** function.

> **Note**
>
> If you add a new operation to a route on the **Route** page, an operation relation is created only for the current released product. Therefore, if the route is also used to produce other released products, no applicable operation relation will exist for those released products, and the route can no longer be used for those released products.

Maintaining operation relations per route

When you open the **Route details** page from the **Routes** list page, a list of all the operation relations that apply to the selected route is shown. Therefore, you can easily verify which

operational properties are used for which products. You can modify both default property values and product-specific property values.

If you add a new operation relation on the **Route details** page, the **Route code** field is automatically set to **Route**, and the **Route relation** field is set to the route number of the current route.

Maintaining operation relations per operation

From the **Operations** page, you can open the **Operation relations** page. On this page, you can modify all operation relations for a specific operation. You can even modify operation relations that contain default values.

If your business uses standard operations, and if the operational parameters are the same across all products and processes, the **Operation relations** page provides a convenient way to maintain the default operational properties of those operations.

Applying operation relations

In some cases, Supply Chain Management must find the operational properties for an operation. For example, when a purchase order is created, the operational properties of each operation must be copied from the operation relations to the production route. In these situations, Supply Chain Management searches the relevant operation relations from the most specific combination to the least specific combination.

When Supply Chain Management searches for the most relevant operation relation for a released product, an operation relation that matches the item ID of the released product is preferred over an operation relation that matches the item group ID. In turn, an operation relation that matches the item group ID is preferred over the default operation relation. The search is done in the following order:

1. **Item code**=*Table* and **Item relation**=<*Item ID*>
2. **Item code**=*Group* and **Item relation**=<*Item group ID*>
3. **Item code**=*All*
4. **Route code**=*Route* and **Route relation**=<*route ID*>
5. **Route code**=*All*
6. **Configuration**=<*configuration ID*>
7. **Configuration**=
8. **Site**=<*site ID*>
9. **Site**=

Therefore, an operation should be used only one time for each route. If the operation occurs multiple times in the same route, all occurrences of that operation will have the same operation relation, and you won't be able to have different properties (for example, run times) for each occurrence.

Route versions

Route versions are used to accommodate variations in the production of products or provide greater control over the production process. They define which route should be used when a specific released product or released product variant is produced. You can use the following constraints to define which route is used for a released product:

- Product dimensions (size, color, style, or configuration)
- Production quantity
- Production site
- Production date

When you're producing the product at a specific site, in a specific quantity, or in a specific period, you can designate a specific route version as the default route version. However, only one active route is allowed for a given released product and a given set of constraints.

On the **Production control parameters** page, you can require that the validity period of a route version must always be specified.

Approval of route versions

Before a route version can be used in the planning or manufacturing process, it must be approved. When you approve a route version, you can also approve the related route. However, a route version can be approved only if the related route is also approved.

Activating the default route version

When you activate a route version, you designate it as the default route version that master planning will use, or that will be used to create production orders. You can have only one active route version for a given set of constraints (for example, period, site, or quantity). If the version that you’re trying to activate conflicts with a version that is already active, you receive an error message. To prevent an ambiguous activation, you must then either inactivate the conflicting version or modify the constraints (usually the period) on the route version.

Electronic signatures

If you must keep a log that records who approves and activates each route version, you can require electronic signatures for these tasks. Users who approve and activate route versions will then have to confirm their identity by using an electronic signature.

Product change that uses case management

The product change case for the approval and activation of new or changed routes and route versions gives you an easy way to see an overview of the route version constraints. You can also approve and activate all routes that are related to a specific change in one operation and document the results in the product change case.

Maintaining routes

Depending on your business requirements, you might be able to reduce the effort that is required in order to maintain your process definitions.

Making routes independent of resources

In many systems, the operations resource or resource group that should perform an operation must be specified in the route. However, in Supply Chain Management, you can define a set of requirements that an operations resource must meet to be applicable for the operation. Therefore, the specific operations resource or resource group that should be used doesn't have to be determined until the operation is actually scheduled. This functionality is especially useful when you have many workers or machines that can perform the same operation.

For example, you specify that an operation requires an operations resource of the **Machine** type that has a **Stamping** capability of 20 tons. The scheduling engine will then resolve these requirements to a specific operations resource or resource group when the operation is scheduled. Because you can just specify these requirements instead of binding the operation to a specific machine, you have much more flexibility. Additionally, maintenance is easier when resources are moved or new resources are added.

For more information about the various types of resource requirements and how to use them, see Operations resource requirements and Resource capabilities.

Sharing routes across sites

If you produce the same product at more than one production site, and if the steps for producing the product are the same at all sites, you can often design a shared route that is used across all production sites. To create a shared route, don't specify a site on the route itself.

However, you must still create a route version that associates the shared route with the product at each site.

You must also make sure that the resource requirements for each operation in the route don't call for specific operations resources or resource groups, but are instead expressed in terms of the characteristics of the required resources. The scheduling engine will then be able to assign the appropriate operations resources from the site that the production is scheduled on. For example, if there are small differences in the run time, or if the setup time for a certain operation is site-specific, you can specify this information by adding an additional operation relation for that site.

To take full advantage of the benefits of shared routes, you should also use resource consumption on the corresponding bill of materials (BOM). When you set the flag for resource consumption on the BOM line, the warehouse and location that raw materials should be consumed from is inferred from the operations resource that the operation is scheduled on. Therefore, the warehouse and location don't have to be determined until the production is actually scheduled. In this way, you can make both the BOM and the route independent of the physical location where the product is produced.

Standard operation relations

If your business uses standardized operations throughout production, and if there's little or no variation in the setup time, run time, consumption calculation, cost calculation, and so on, you might benefit from creating default operation relations for all operations. In this case, avoid creating operation relations that are specific to any route or released product.

If you also express resource requirements in terms of skills and capabilities, and make your routes site-independent, you can help keep the ongoing maintenance of your business processes to a minimum.

When you use this approach, the **Operation relations** page becomes your primary destination for maintaining run times and other properties.

Resource-specific process times

If you don't specify an operations resource or resource group as part of the resource requirements for an operation, the applicable resources might operate at different speeds. Therefore, the time that is required in order to process an operation might vary. To resolve this issue, you can use the **Formula** field on the operation relation to specify how the process time is calculated. The following options are available:

- **Standard** – (Default option) The calculation uses only the fields from the operation relation and multiplies the specified run time by the order quantity.
- **Capacity** – The calculation includes the **Capacity** field from the operations resource. Therefore, the time is resource-dependent. The value that is specified on the operations resource is capacity per hour. The **Process time** is calculated as **Order quantity** divided by **Capacity**. The capacity value isn't specific to a particular unit of measure and therefore isn't converted based on the **Capacity unit** field, which is just a descriptive field that isn't used in calculations.
- **Batch** – A batch capacity is calculated by using information from the operation relation. The number of batches and, therefore, the process time can then be calculated based on the order quantity.
- **Resource batch** – This option is basically the same as the **Batch** option. However, the calculation includes the **Batch capacity** field from the operations resource. Therefore, the time is resource dependent.

Set up route groups

You can define the route groups and the setup for its route or job types under **Production control** > **Setup** > **Routes** > **Route groups**. For each route/job type in the route group, you can select or clear the following options:

- **Activation** – Select this option to enable calculations and scheduling for the selected job type, and to receive job feedback when you run job scheduling. You need to select this option to enable the job type and then, select the rest of the options for that job type. If the activation isn't selected, that job type won't be enabled, regardless of the selection of the other options.
- **Job management** – Select this option to include the job type in job management when you run job scheduling.
- **Working time** – Select this option to schedule the job type according to the working time calendar that is defined for the operations resource, otherwise the Gregorian calendar is used. Working time can be scheduled either according to the Gregorian calendar or the defined working calendar. If you select this option, scheduling is based on the defined working time calendar. Additionally, the job of the job type is scheduled from midnight on the date that is defined as the job's starting date.
- **Capacity** – Select this option to reserve capacity for the job type when you run job scheduling. If you select this option, capacity is reserved when scheduling is run for the selected job type. This gives you an overview of which job types in each route group use the operations resources. For example, in a situation where drying resources are bottleneck resources, these resources must be specified as bottlenecks. Drying operations that are assigned to queue time job types will reserve drying resources.

For each of the job types, you first need to activate or de-activate it. When de-activated, none of the other setup (Job management, working time, and capacity) will be considered, as the job type won't be active.

One of the job types is *Overlap*. Overlap allows different jobs to be performed at the same time. When jobs are overlapping, the resources can be used but can't be reserved for the specific jobs. Therefore, when **Activation** is selected for the row with a **Route/job type** of *Overlap*, the rest of the settings (**Job management**, **Working time**, and **Capacity**) don't have any impact on the route group.

> **Note**
>
> When you upgrade versions, you might encounter the following error: *CRL Error occurred while invoking the scheduling engine*. If you receive this error, go to the **Route groups** page and for all the routes where you have activated the **Route/job type** *Overlap*, clear the **Job management**, **Working time**, and **Capacity** check boxes.

Related information

- Bills of materials and formulas
- Cost categories used in production routing
- Resource capabilities
- Electronic signature overview

Work policies

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up the system and the Warehouse Management mobile app to support work policies. Use this functionality to quickly register inventory without creating putaway work when you receive purchase or transfer orders, or when you complete manufacturing processes. This article provides general information. For detailed information related to license plate receiving, see License plate receiving via the Warehouse Management mobile app.

A work policy controls whether warehouse work is created when a manufactured item is reported as finished, or when goods are received by using the Warehouse Management mobile app. Set up each work policy by defining the conditions where it applies: the work order types and processes, the inventory location, and (optionally) the products. For example, a purchase order for product *A0001* must be received in location *RECV* in warehouse *24*. Later, the product is consumed in another process at location *RECV*. In this case, set up a work policy to prevent putaway work from being created when a worker reports product *A0001* as received in location *RECV*.

> **Note**
>
> - For a work policy to be active, you must define at least one location for it on the **Inventory locations** FastTab of the **Work policies** page.
> - You can't specify the same location for multiple work policies.
> - The **Print label** option for mobile device menu items doesn't print a license plate label unless work was created.

The Work policies page

To set up work policies, go to **Warehouse management** > **Setup** > **Work** > **Work policies**. Then, on each FastTab, set the fields as described in the following subsections.

The Work order types FastTab

On the **Work order types** FastTab, add all the work order types and the related work processes that the work policy applies to. The following work order types and related work processes are supported for work policies.

Expand table

Work order type	Work process
Raw material picking	All related processes
Co-product and by-product put away	All related processes
Finished goods putaway	All related processes
Transfer receipt	License plate receiving (and putaway)
Purchase orders	• License plate receiving (and putaway) • Load item receiving (and putaway) • Purchase order line receiving (and putaway) • Purchase order item receiving (and putaway)
Inbound shipment order	• License plate receiving (and putaway) • Load item receiving (and putaway) • Inbound shipment order line receiving (and putaway) • Inbound shipment order item receiving (and putaway)

To set up a work policy so that it applies to several work processes of the same work order type, add a separate line for each work process to the grid.

For each line in the grid, set the **Work creation method** field to one of the following values:

- **Never** – The work policy prevents warehouse work from being created for the selected work order type and related work process.
- **Cross docking** – The work policy creates cross-docking work by using the policy that you select in the **Cross docking policy name** field.

The Inventory locations FastTab

On the **Inventory locations** FastTab, add all the locations where this work policy should be applied. If you don't associate any location with a work policy, the work policy isn't applied to any process.

You can't specify the same location for multiple work policies.

You can use a warehouse location assigned to a location profile where the **Use license plate tracking** option is turned off. In this case, workers directly register the on-hand inventory.

The Products FastTab

On the **Products** tab, set the **Product selection** field to control which products the policy applies to:

- **All** – The policy applies to all products.
- **Selected** – The policy applies only to products listed in the grid. Use the toolbar on the **Products** FastTab to add products to the grid or remove them from the grid.

Default and custom "to" locations

Mobile device menu items that use the following work creation processes provide the **Use default data** option. This option lets you assign a custom "to" location to one or more menu items. (This option was already available for some other types of menu items.)

- License plate receiving (and putaway)
- Load item receiving (and putaway)
- Purchase order line receiving (and putaway)
- Purchase order item receiving (and putaway)
- Inbound shipment order line receiving (and putaway)
- Inbound shipment order item receiving (and putaway)

The **To location** setting for a menu item overrides the default receiving location for the warehouse, for all orders that are processed by using that menu item.

To set up a mobile device menu item to support receiving at a custom location, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select or create a menu item that uses one of the work creation processes that are listed earlier in this section.

3. On the **General** FastTab, set the **Use default data** option to **Yes**.

4. On the Action Pane, select **Default data**.

5. On the **Default data** page, set the following values:

    - **Default data field:** Set this field to *To location*.
    - **Warehouse:** Select the destination warehouse to use with this menu item.
    - **Location:** This field lists all the location IDs that are available for the selected warehouse. However, the setting of this field doesn't actually have any effect. Therefore, you can leave it blank. Nevertheless, you can use the list to confirm the ID that you must enter in the **Hardcoded value** field.
    - **Hardcoded value:** Enter the location ID for the receiving location that applies to this menu item.

> **Tip**
>
> You can apply a work policy only if the relevant work policy setup lists all the receiving locations. This requirement applies regardless of whether you're using the default warehouse receiving location or a custom "to" location.

Example scenario: Warehouse receiving

All products that are received by the *Purchase order item receiving (and putaway)* process must be registered in location *FL-001*, and they must be available in warehouse *24*. However, work should not be created. Products that are received by any other process (that is, by using other mobile device menu items) should be registered at the default warehouse receiving location (*RECV*), and work should be created as usual. (This scenario doesn't show the default receiving setup.)

This scenario requires the following elements:

- A work policy for the *Purchase order item receiving (and putaway)* process in location *FL-001*, for all products
- A mobile device menu item that has default data, and that sets the **To location** field to *FL-001*

Prerequisites

This scenario uses the standard demo data. Therefore, if you want to work through it by using the values that are provided here, you must work on a system where demo data is installed. Additionally, you must select the **USMF** legal entity.

Set up a work policy

1. Go to **Warehouse management** > **Setup** > **Work** > **Work policies**.
2. Select **New**.
3. In the **Work policy name** field, enter *No purchase item putaway work*.
4. Select **Save**.
5. On the **Work order types** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:
    - **Work order type:** *Purchase orders*
    - **Work process:** *Purchase order item receiving (and putaway)*
    - **Work creation method:** *Never*
    - **Cross docking policy name:** Leave this field blank.
6. On the **Inventory locations** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:
    - **Warehouse:** *24*
    - **Location:** *FL-001*
7. On the **Products** FastTab, set the **Product selection** field to *All*.
8. Select **Save**.

Set up a mobile device menu item to change the receiving location

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. In the left pane, select the existing **Purchase receive** menu item.
3. On the **General** FastTab, set the **Use default data** option to *Yes*.
4. Select **Save**.

5. On the Action Pane, select **Default data**.

6. On the **Default data** page, on the Action Pane, select **New** to add a row to the grid, and then set the following values for the new row:

    - **Default data field:** *To location*
    - **Warehouse:** *24*
    - **Location:** Leave this field blank.
    - **Hardcoded value:** *FL-001*

7. Select **Save**.

Receive a purchase order without creating work

The example in this section shows how to receive a purchase order item, but without creating work, at a location that differs from the default receiving location set up for the warehouse. This example uses the work policy and mobile device item that you created earlier in this scenario.

Create a purchase order

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.

2. Select **New**.

3. In **Create purchase order**, set the following values:

    - **Vendor account:** *US-101*
    - **Site:** *2*
    - **Warehouse:** *24*

4. Select **OK** to close the dialog box and open the new purchase order.

5. On the **Purchase order lines** FastTab, set the following values for the empty row:

    - **Item number:** *A0001*
    - **Quantity:** *1*

6. Select **Save**.

7. Make a note of the purchase order number.

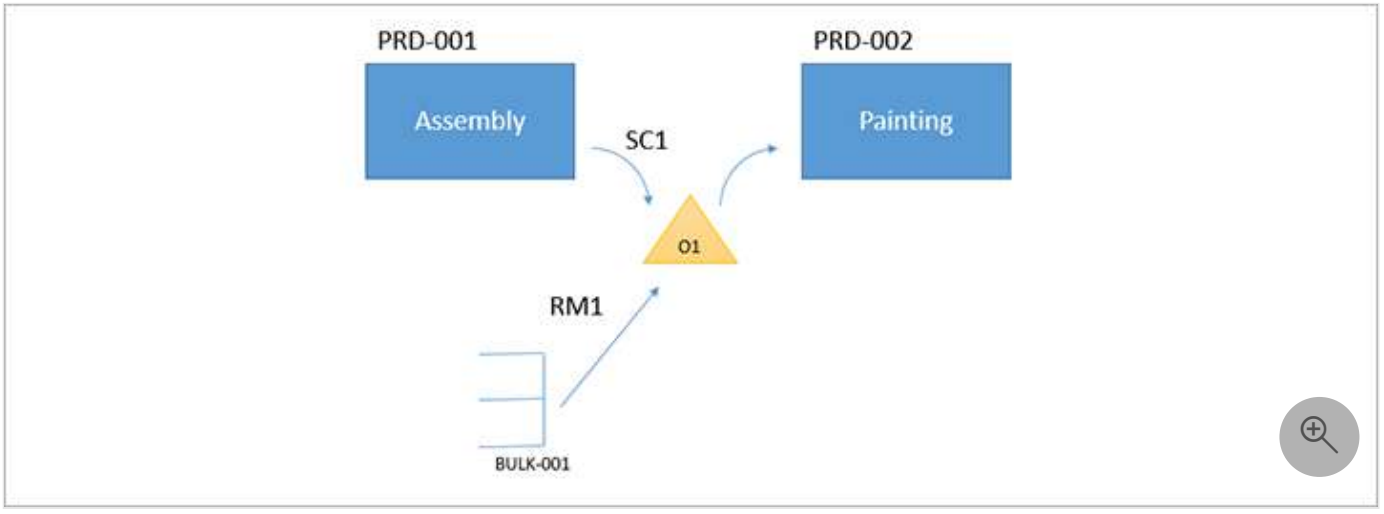
Receive a purchase order

1. On the mobile device, sign in to warehouse *24* by using *24* as the user ID and *1* as the password.
2. Select **Inbound**.
3. Select **Purchase receive**. The **Location** field should be set to *FL-001*.
4. Enter the purchase order number for the purchase order that you created in the previous procedure.
5. In the **Item number** field, enter *A0001*.
6. Select **OK**.
7. In the **Quantity** field, enter *1*.
8. Select **OK**.

The purchase order is now received, but no work is associated with it. The on-hand inventory is updated, and a quantity of *1* of item *A0001* is now available at location *FL-001*.

Example scenario: Manufacturing

In the following example, there are two production orders, *PRD-001* and *PRD-002*. Production order *PRD-001* has an operation named *Assembly*, where product *SC1* is reported as finished to location *001*. Production order *PRD-002* has an operation named *Painting* and consumes product *SC1* from location *001*. Production order *PRD-002* also consumes raw material *RM1* from location *001*. Raw material *RM1* is stored in warehouse location *BULK-001* and is picked to location *001* by warehouse work for raw material picking. The picking work is generated when production *PRD-002* is released.



When you plan to configure a warehouse work policy for this scenario, consider the following points:

- You don't need warehouse work for putaway of finished goods when you report product *SC1* as finished from production order *PRD-001* to location *001*. The reason is that the

Painting operation for production order *PRD-002* consumes product *SC1* at the same location.
- You need warehouse work for raw material picking to move raw material *RM1* from warehouse location *BULK-001* to location *001*.

Here is an example of a work policy that you can set up, based on these considerations:

- **Work policy name:** *No putaway work*
- **Work order types:** *Finished goods put away* and *Co-product and by-product put away*
- **Inventory locations:** Warehouse *51* and location *001*
- **Products:** *SC1*

The following example scenario provides step-by-step instructions for setting up the warehouse work policy for this scenario.

Example scenario: Report as finished to a location that isn't license plate–controlled

This scenario shows an example where a production order is reported as finished to a location that isn't license plate–controlled.

This scenario uses the standard demo data. Therefore, if you want to work through it by using the values that are provided here, you must work on a system where demo data is installed. Additionally, you must select the **USMF** legal entity.

Set up a warehouse work policy

Warehouse processes don't always include warehouse work. By defining a work policy, you can prevent the creation of work for raw material picking and putaway of finished goods for a set of products at specific locations.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work policies**.
2. Select **New**.
3. In the **Work policy name** field, enter *No putaway work*.
4. On the Action Pane, select **Save**.
5. On the **Work order types** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:

- **Work order type:** *Finished goods put away*
- **Work process:** *All related work processes*
- **Work creation method:** *Never*
- **Cross docking policy name:** Leave this field blank.

6. Select **Add** again to add a second row to the grid, and then set the following values for the new row:

   - **Work order type:** *Co-product and by-product put away*
   - **Work process:** *All related work processes*
   - **Work creation method:** *Never*
   - **Cross docking policy name:** Leave this field blank.

7. On the **Inventory locations** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:

   - **Warehouse:** *51*
   - **Location:** *001*

8. On the **Products** FastTab, set the **Product selection** field to *Selected*.

9. On the **Products** FastTab, select **Add** to add a row to the grid.

10. In the new row, set the **Item number** field to *L0101*.

11. On the Action Pane, select **Save**.

Set up an output location

1. Go to **Organization administration** > **Resources** > **Resource groups**.

2. In the left pane, select resource group **5102**.

3. On the **General** FastTab, set the following values:

   - **Output warehouse:** *51*
   - **Output location:** *001*

4. On the Action Pane, select **Save**.

> ⓘ **Note**

Location *001* isn't a license plate–controlled location. You can set up an output location that isn't license plate–controlled only if an applicable work policy exists for the location.

Create a production order and report it as finished

1. Go to **Production control** > **Production orders** > **All production orders**.
2. On the Action Pane, select **New production order**.
3. In the **Create production order** dialog box, set the **Item number** field to *L0101*.
4. Select **Create** to create the order and close the dialog box.

   A new production order is added to the grid on the **All production orders** page.

   Keep the new production order selected.
5. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Estimate**.
6. In the **Estimate** dialog box, read the estimate, and then select **OK** to close the dialog box.
7. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Start**.
8. In the **Start** dialog box, on the **General** tab, set the **Automatic BOM consumption** field to *Never*.
9. Select **OK** to save your setting and close the dialog box.
10. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Report as finished**.
11. In the **Report as finished** dialog box, on the **General** tab, set the **Accept error** option to *Yes*.
12. Select **OK** to save your setting and close the dialog box.
13. On the Action Pane, on the **Warehouse** tab, in the **General** group, select **Work details**.

When the production order is reported as finished, no work is generated for putaway. This behavior occurs because a work policy is defined that prevents work from being generated when product *L0101* is reported as finished to location *001*.

More information

For more information about mobile device menu items, see Set up mobile devices for warehouse work.

For more information about license plate receiving and work policies, see License plate receiving via the Warehouse Management mobile app.

For more information about inbound load management, see Warehouse handling of inbound loads for purchase orders.

Last updated on 05/05/2026

Create working time scheduling properties

Article • 09/29/2021

Working time scheduling properties are used to categorize working time intervals and define the settings that operators must take into account. This procedure shows how to define a working time scheduling property and associated property values. You can walk through this procedure in demo data company USMF, or using your own data.

1. Go to **Workspaces** > **Resource lifecycle management**.
2. Select **Properties**.

Create property

1. Select **New**.
2. In the **Property** field, type a value.
3. In the **Name** field, type a value.

Create property values

1. In the list, mark the selected row.
2. In the **Name** field, type a value.
3. In the **Value** field, type a value.
4. Select **Save**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create working time templates

Article • 09/29/2021

Working time templates define the working hours throughout a week and are used to generate working times for a period of time. This procedure shows you how to define a working time template using working time scheduling properties for categorizing working time intervals. You can walk through this procedure in demo data company USMF, or using your own data.

1. Go to **Workspaces** > **Resource lifecycle management**.
2. Select **Working time templates**.

Create working time template

1. Select **New**.
2. In the **Working time template** field, type a value.
3. In the **Name** field, type a value.
4. Expand the **Monday** section.
5. Select **Add**.
6. In the **From** field, enter a time.

    - Specify the time when work begins in the morning.

7. In the **To** field, enter a time.

    - Specify the time when workers break for lunch.

8. Select **Add**.
9. In the **From** field, enter a time.

    - Specify the time when work resumes after lunch.

10. In the **To** field, enter a time.

    - Specify the end of the work day.

Replicate working times to all week days

1. Select **Copy day**.

    - Copy the working times definitions from Monday to Tuesday.

2. Select **OK**.

3. Select **Copy day**.
    - Copy the working times definitions from Monday to Wednesday.
4. In the **To weekday** field, select an option.
5. Select **OK**.
6. Select **Copy day**.
    - Copy the working times definitions from Monday to Thursday.
7. In the **To weekday** field, select an option.
8. Select **OK**.
9. Select **Copy day**.
    - Copy the working times definitions from Monday to Friday.
10. In the **To weekday** field, select an option.
11. Select **OK**.

Define time slots for special operations

1. Expand the **Friday** section.
2. In the list, find and select the desired record.
3. In the **Property** field, enter or select a value.
4. In the list, find and select the desired record.
5. In the **Property** field, enter or select a value.

Mark weekend days as closed for pickup

1. Expand the **Saturday** section.
2. Select *Yes* in the **Closed for pickup** field.
3. Expand the **Sunday** section.
4. Select *Yes* in the **Closed for pickup** field.

Feedback

Was this page helpful? Yes No

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Define discrete manufacturing resource group

Article • 09/29/2021

A resource group is a set of operations resources that typically correspond to the physical organization of work cells, defined by yellow lines on the production shop floor. This procedure shows you how to define a ressource group for use in discrete production. You can walk through this procedure in demo data company USMF, or use your own data.

1. Go to Resource groups.
2. Click New.
3. In the Resource group field, type a value.
4. In the Description field, type a value.
5. In the Site field, enter or select a value.
6. In the Production unit field, enter or select a value.

Define default operational parameters

1. Expand the Operation section.
2. In the Scrap percentage field, enter a number.
3. In the Setup category field, enter or select a value.
4. In the Run time category field, enter or select a value.
5. In the Operations scheduling percentage field, enter a number.

Define operating hours

1. Expand the Calendars section.
2. Click Add.
3. In the Calendar field, enter or select a value.

Add operations resources

1. Expand the Resources section.
2. Click Add.
3. In the Resource field, enter or select a value.
4. Click Add.
5. In the Resource field, enter or select a value.

6. In the list, find and select the desired record.
7. In the list, click the link in the selected row.

Feedback

Was this page helpful? Yes No

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Add an existing activity to a production flow version

Article • 09/29/2021

When creating new versions of production flows, you can choose to add activities created for the older versions, to the new version. This procedure shows how a new version is created for an existing production flow, without copying the activities. In the next step, an existing activity is added to the new version.

This task requires production flow with version and activities already created.

Create a new production flow version

1. Go to Production control > Setup > Lean production flow > Production flows.
2. In the list, find and select the desired record.
3. In the list, click the link in the selected row.
4. Click Edit.
5. In the list, mark the selected row.
6. In the Expiration date field, enter a date and time.

    - Note that before you create a new production flow version, make sure to check the expiration date and time of the active version. The new version will be created with an effective start date, which connects to the expiry date of the selected version.

7. Click Add.
8. Select No in the Copy from version field.

    - Select No to start with an empty version if most of the activities of the copied version will be replaced by new activities. Add the unchanged activities to the Add existing function in the activity form manually.

9. Select No in the Duplicate kanban rules field.

    - When the activities are not copied to the new version, it is not possible to copy the kanban rules at the time of creation of the new version. Instead you will use the create replacement kanban function later in the kanban rule form, to replace kanban rules of the old production flow version with kanban rules using the activities of the new version.

10. Click OK.

11. In the list, find and select the desired record.

Add an existing activity

1. Click Activities.
2. Click Add existing to open the drop dialog.

    - Find and select an existing activity to be added to the new production flow version. Note that the list shows all activities that have been created for this production flow for all previous versions of the flow.

3. In the Activity field, enter or select a value.
4. Click OK.

Feedback

Was this page helpful? Yes No

Provide product feedback

Deactivate a production flow version

Article • 09/29/2021

When an active production flow version is no longer needed, it can be deactivated. You should only use this option if all kanban rules and activities have ended and will not be activated again. Note that the expiry date of all kanban rules related to this production flow version will be updated with the current date and time.

To modify an active production flow version, consider setting an expiry date for the active version and create a new version. This will allow you to continue your production operations while preparing the new version and related kanban rules.

To expire an active production flow version, you need to set an expiry date. In that sense, deactivation is more like an exception than a rule.

For this procedure you need a production flow with a version that can be deactivated. Do not try this in a production environment unless you are 100% positive that the version is fully obsolete.

Deactivate a production flow version

1. Go to Production control > Setup > Lean production flow > Production flows.
2. In the list, find and select the desired record.
3. In the list, click the link in the selected row.
4. In the list, find and select the desired record.
5. Click Deactivate.
    - Do not proceed if you are not 100% positive that this production flow version is obsolete. Clicking Ok will expire all active kanban rules and put an immediate stop to all production and replenishment activities of this production flow version.
6. Click OK.

Feedback

Was this page helpful? Yes No

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Define an expiry date for a production flow version

Article • 09/29/2021

To end the validity and the processing of a production flow version on a given date, or to plan replacement of an active version with a new version, you have to set an expiry date on the version. It is not necessary to deactivate the version.

Set an expiration date to end a production flow version

1. Go to Production control > Setup > Lean production flow > Production flows.
2. In the list, find and select the desired record.

    - Select any production flow that has a version that is already defined.

3. In the list, click the link in the selected row.
4. Click Edit.
5. In the list, mark the selected row.
6. In the Expiration date field, enter a date and time.

    - For the expiration date, a new version will not start or become activated. It will also no longer be possible to create or start jobs for this production flow. You can still complete started jobs after the expiration date.

Feedback

Was this page helpful? Yes No

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Define production flow models

Article • 09/29/2021

Production flow models describe how the capacity of lean manufacturing work cells is calculated and maintained. Therefore the definition of a production flow model is a prerequisite of the definition of work cells. The demo data company used to create this procedure is USMF.

Define a production flow model.

1. Go to Production control > Setup > Lean production flow > Production flow models.
2. Click New.
3. In the Production flow model field, enter an ID for the production flow model.
4. In the Model type field, select an option.

    - There are two model types: Throughput type and Hours type. For Throughput type, the capacity of work cells that use this production flow model will be expressed and calculated in product quantities. For Hours type, the capacity of work cells that uses this production flow model will be expressed and calculated in hours. Note that this property can't be changed for an existing production flow model. After a work cell has capacity reservations, the production flow model type can't be changed.

5. Enter the number of days in the EPE Cycle.

    - The interval describes the period when every part produced by a production flow or work cell will be produced at least once. The shorter the EPE cycle, the more flexible the production process is. If EPE Cycle = 0, all demand can be produced in the same day. The EPE Cycle is used to schedule lean process jobs. When scheduling a job on a work cell with EPE Cycle = 5, the scheduling process will look for capacity on the work cell at Due date - EPE Cycle (5 days ahead of the due date) to ensure that the product can be produced in that cycle. The inventory lead time of a product is usually equal to or greater than the EPE Cycle of the related production process.

6. In the Planning period type field, select an option.

    - After a work cell has capacity reservations, the planning period type cannot by changed. You can only select production flow models with the same period type for this given work cell.

7. In the Planning time fence field, enter a number.

    - The planning time fence describes the number of days where capacity reservations can be made for the related work cells. In the Planning time fence, enter the number of days. Kanban process jobs that fall outside of this period are not planned with automatic planning. The planning time fence is typically two times the average inventory lead time of the products produced in a production flow or work cell. The EPE Cycle should not be more than half the planning time fence.

8. In the Capacity shortage reaction field, select an option.

    - The options include: Postpone - Postponing the full demand of the scheduling event on the next available production day, with available throughput. Cancel - End the automatic planning for the scheduling event and leave the related jobs unplanned. Add to requested day - Plan the requested jobs for the requested period. This overloads the cell for this day and requires the planner to review and to a manual interaction . Distribute to available periods - Distribute the different jobs of the scheduling event to all available production days, beginning from the first available day. The minimum distribution quantity is the kanban job quantity. The distribution assigns the minimum planning quantity (kanban quantity) to every day with enough available throughput.

Feedback

Was this page helpful? Yes No

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Operations scheduling

07/10/2025

This article provides information about operations scheduling. You can use operations scheduling to provide a general estimate of the production process over time.

You can schedule production at the operation level and the job level. Unlike job scheduling, operations scheduling doesn't explode the operations for the production route into jobs. If you want to include more detail in the scheduling, such as information about current capacity, you can run job scheduling after you run operations scheduling. You can also run job scheduling only. Job scheduling is typically used to schedule individual jobs on the shop floor for an immediate or short-term time frame.

Components of operations scheduling

The main components of operations scheduling are the scheduling direction, the capacity of resources, and materials optimization. By using operations scheduling, you can achieve the following goals:

- Control the planning method by scheduling forward or backward from a given date.
- Optimize the use of resources by scheduling productions based on the capacity of the resources. This approach also helps identify when alternative resources should be used.
- Optimize the use of materials by scheduling productions based on the availability of the required materials.
- Schedule and synchronize reference productions. The dates of the reference productions are adjusted when the production order’s schedule is changed.

You must estimate the cost of a production order before you can run operations scheduling. If you haven't run an estimate, it's automatically run before operations scheduling is started. An operations schedule specifies the following information:

- The product that is planned for production
- The configuration of the product
- The quantities that are involved in the production
- The dates when the production will start and end
- The capacity reservations for the resources that perform the production activities

The setup time, process time, and run time are set for operations in the production. After you run operations scheduling, the status of the production order is *Scheduled*, and all operations are scheduled in the order that is specified by the production route. However, only the duration of the operation is considered. Start times and end times aren't scheduled.

Scheduling direction and date

The scheduling direction is fundamental to the scheduling process. Production can be scheduled forward or backward from any date, depending on timing and scheduling requirements.

- **Forward from the scheduling date** – You can schedule production to start as early as possible. Production can be started today, tomorrow, or on any date in the future. The production is scheduled forward in time to the earliest possible end date.
- **Backward from the scheduling date** – You can schedule production to start as late as possible. Backward scheduling is based on the date when the production must be completed. The schedule counts backward from that date to the latest possible date that the production can be started and still be completed on time.

Resource scheduling

When you run an operations schedule, each operation in the production route is scheduled for the resource that is specified for the operation. Additionally, the duration of each operation is specified on the production route. If a resource group is specified for an operation, the scheduling reserves capacity on the group. However, unlike job scheduling, operations scheduling doesn't select the specific resources in the group. If you're working with finite capacity, the schedule depends on the availability of the resources that are required in order to complete production. Operations scheduling follows the sequence of operations that is specified on the production route. The scheduling reserves capacity on the resource groups, based on the operation times that are defined on the production route. The sum of available capacity on the resources that are involved determines the capacity for the resource group. Capacity reservations that already exist for the resources are considered unavailable capacity. If there isn't enough available capacity for the production, the production orders can be delayed or even stopped. You can also specify the efficiency that you expect from the resources that are involved in the production. You specify the efficiency as a percentage on the resource. The efficiency percentage adjusts the throughput of the resource. This adjustment affects the time that is reserved for the resource. The lead times for the operations that use the resource are also adjusted accordingly.

Operations scheduling and master planning

The operations schedule also drives master planning and determines calculations for material requirements. Operations scheduling considers the following information:

- **Backlogged productions** – Products that are planned, released, or started
- **Material availability** – Inventory, subproductions, suppliers, and vendors

- **Capacity availability** – Resources that are required for production

> **Note**
>
> If you're using multi-threaded master planning and operations scheduling, finite capacity will not be considered.

Cancellations

When you run operations scheduling, you can cancel specific parts of the routing. These parts include the queue time, setup time, process time, overlap time, and transport times.

Finite materials

If you're working with finite materials, scheduling also depends on the availability of the materials that are required for production. If there aren't enough available components for the production, production can be delayed. You can base scheduling on the use of materials by specifying the materials that must be available for production. When you optimize on both resource capacity and the availability of materials, production is calculated according to these restrictions. A production order can't be scheduled to start until capacity and materials are available at the same time and in the required quantities.

Related information

- Operations scheduling options

Job scheduling

09/19/2025

This article provides information about job scheduling, which is a more detailed form of scheduling than operations scheduling. Use job scheduling to schedule individual jobs or shop orders, and to control the manufacturing environment.

You can use job scheduling to schedule individual jobs or shop orders, and to control the manufacturing environment. Job scheduling breaks down each operation into its individual tasks or jobs. Assign these jobs to the operations resources that perform them. Job scheduling also lets you synchronize all jobs that the selected job references. You can specify a start date and time or end date and time for the job, and then run scheduling. The time that you specify can be the start time or the end time, depending on the scheduling direction. This functionality is useful when, for example, a job can run only on one machine at a time, or when you want to optimize the job that runs for each resource.

Tasks in the job scheduling process

The job scheduling process includes the following tasks:

- Split operations into jobs.
- Schedule jobs, based on the dates and times for the resources that are specified for the related operation.
- Calculate start times and end times for each job. Use finite capacity to make sure that there are no overlapping times.
- Determine which resources in the resource group run the job on. This task requires that a resource group be specified for an operation. Job scheduling selects the resources or resource groups based on the shortest lead time, and also considers any previous reservations on the resources.
- Explode operations into jobs when you run job scheduling. The jobs are scheduled by date and time, according to the order that the production route specifies. The setup of the operation determines the jobs that are exploded during the scheduling process. The route group assigned to the operation controls whether jobs are generated. A job is generated only if it has a specific duration. For example, a transport time job is generated if a transport time was specified for the selected operation.

Scheduling direction

You can schedule jobs either forward or backward.

- **Forward** – Use the forward scheduling direction to start the production as early as possible. This method is also known as the push method, because the production is being pushed forward through the production process. The production is scheduled to start and end on the earliest possible dates.
- **Backward** – Use the backward scheduling direction to start the production as late as possible. This method is also known as the pull method, because it's based on the date when the production must be completed. Backward scheduling counts backward to the latest possible date that the production can be started without missing its target deadline.

Finite capacity

You can schedule jobs by using finite capacity. When you use finite capacity, the capacity that is scheduled can't be larger than the capacity that's available for the resource. Available time is defined as the interval when the resource is available and there are no other reservations on capacity. Scheduling that is based on finite capacity makes sure that start times and end times for an operation on a specific date don't overlap. The resource capacity that is already reserved is considered, and overlaps between the start times and end times are also considered. Finite capacity determines the amount of capacity that must be available for a resource in order to achieve optimal use of that resource. This determination is balanced against a calculation of the shortest possible lead time between operations.

Finite materials

Job scheduling that uses finite materials ensures that the required materials are available when the operation starts. The coverage rules for items define these limits. Scheduling uses requirement explosion to determine when the component items are available. If you schedule without setting finite materials, the system assumes that all items are available when they're required.

Finite properties

Job scheduling that uses special properties requires that you specify properties for the operations on the production route. The system must fulfill these properties to reserve capacity.

References

Job scheduling schedules all productions that the current production references. If a production has one or more subproductions, schedule the subproductions at the same time as the main production, because you can't start the main production until the related subproductions are completed.

Schedule resources

The scheduling engine examines combinations of resources to identify those combinations that can satisfy requirements. You can specify selection criteria by selecting one of the following values in the **Primary resource selection** field on the **Scheduling parameters** page:

- *Duration* – The scheduling engine selects the resource that has the shortest lead time.
- *Priority* – The scheduling engine selects the resource that has the highest priority if two or more resources have identical capabilities and levels. The resource that has the lowest numeric value in this field has the highest priority.

> ⓘ **Note**
>
> Scheduling by duration can cause decreased performance when the same resource group contains many resources and secondary operations are used. You can schedule a maximum of 32 resources per operation. If you exceed this quantity, an Action center message is displayed, and job scheduling doesn't find the best alternative resource.

When you run job scheduling, the system plans the resources based on the limitations that you define in the resource parameters. You can control the capacity of the resources by using calendar settings. The system calculates loads for resources during the scheduling process.

> ⓘ **Note**
>
> For productions that use the operations scheduling function, you can run job scheduling after operations scheduling. If you aren't using operations scheduling, you can run job scheduling alone.

Maximum capacities for resources per job order

Job scheduling assigns resources to jobs. You can set maximum capacities for resources per job order. For example, you can set up the system to schedule no more than 50 percent of total capacity for a production order. This setup gives you more control over the scheduling of

resources on the job scheduling level. Therefore, it can help prevent issues if not enough capacity is available to perform simultaneous productions.

Resource efficiency

Job scheduling uses the efficiency percentages that you specify for the resources. Efficiency percentages reduce or increase the time that the system reserves for the resource. Therefore, lead time also increases or decreases. The system uses the following formula for the calculation: `Scheduling time = Time × 100 ÷ Efficiency percentage`. In this formula, *Time* includes both the run time and the setup time.

Gantt chart for job scheduling

07/10/2025

The Gantt chart is designed to empower production planners to control and optimize the production plan. The Gantt chart makes the flow of operations transparent and makes it easy to adjust the production schedule while taking into account material or resource shortages. This helps planners make the best use of available resources, minimize work in progress, and optimize throughput times for production orders.

A Gantt chart is a visual representation of scheduled activities within a defined time interval. The activities are scheduled on resources that have capacity defined by a capacity calendar. The following types of activities can be shown in the Gantt chart.

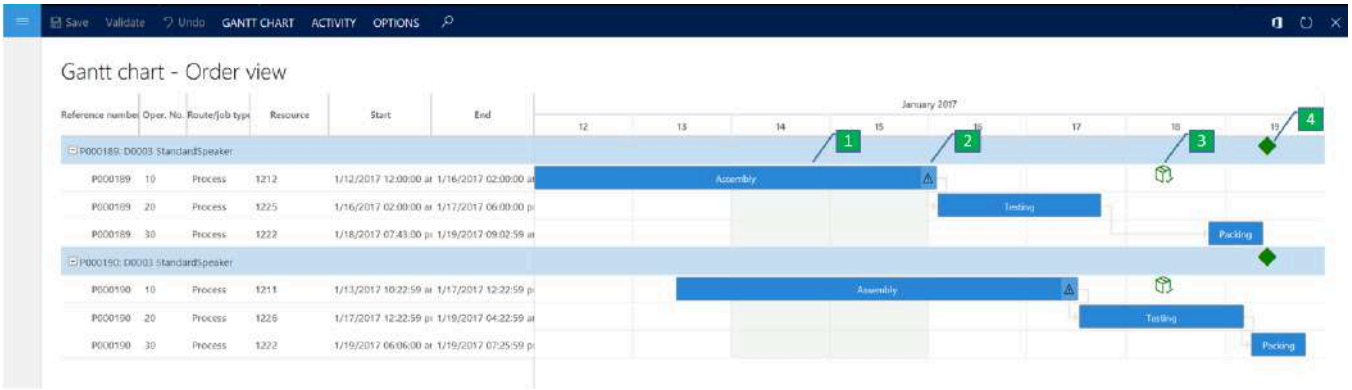
- Jobs from production orders that are job scheduled.
- Jobs from planned production orders.
- Job scheduled project activities of type Hour forecasts.

Gantt chart views

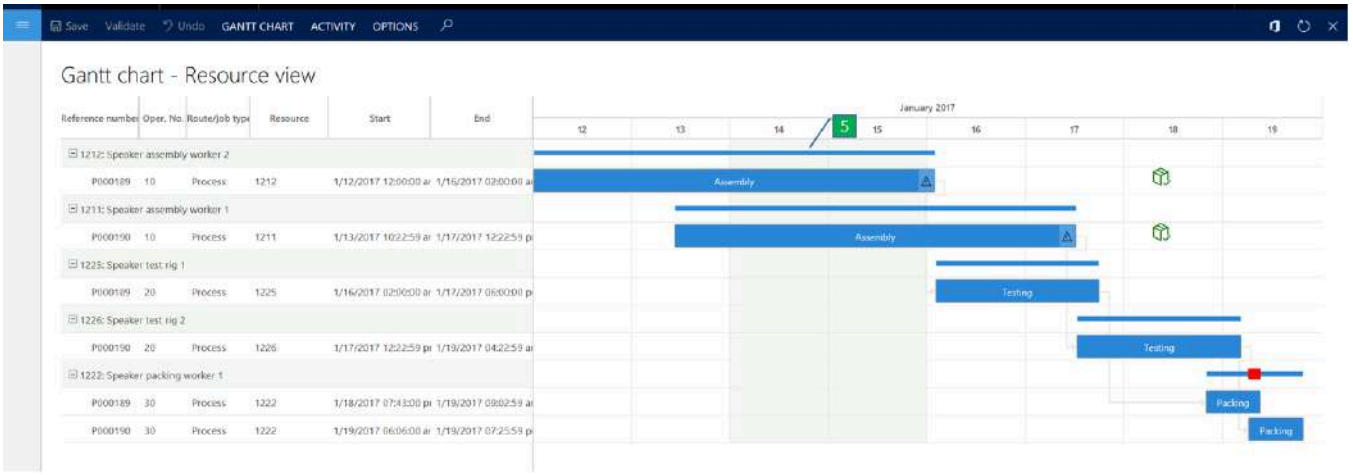
The Gantt chart can be opened in two different views, **Order view** and **Resource view**. In **Order view**, activities are grouped under production orders. This can be useful, for example, if you want to maintain an overview of all the jobs belonging to the same orders. In **Resource view** all jobs are grouped under individual resources. This view can be useful when optimizing the plan at a resource level, for example, a machine or a group of machines. The Gantt charts shown in the illustrations below show **Order view** and **Resource view** with these key elements:

1. Gantt chart activity
2. Material shortage icon
3. Material availability icon
4. Order delivery date icon
5. Capacity bar

The following screenshot shows the order view.



The following screenshot shows the resource view.



Activities

The activities appear as bars and are organized in a time scale grid with a scheduled start and end time, making the length of the bars proportional to the time that is necessary to complete the activity. The activities are shown according to a time scale. You can adjust the time scale on the menu where you select a start and end date and a time unit, for example, hours or days. By adjusting the time scale, you can set focus on a time interval in which you want to manage activities.

To get a better overview, there are different options for controlling the color of the activities. You can configure an individual color for activities, use the theme color that is the general color theme used for the application, or set up the color to be controlled by the color code for production orders.

The time interval for activities has a background shade. Periods with a white shade indicate a time interval with defined capacity on the resource for the activity, whereas periods with a grey shade indicate time intervals with no capacity defined.

On the left side of the chart, there's additional information about the activity, for example, the resource on which the activity is scheduled and production order number. The connection between jobs belonging to the same order is shown with an arrow.

You can get more information about an activity in the activity dialog box. To open the dialog box, double-click the activity or select the **information** menu. In the activity dialog you can see the scheduled start and end date, and time information about which materials the activity is planned to consume.

The activities can be grouped in Grouping levels. The Grouping levels are hierarchical and can be used to make a logical grouping of activities. For example, if you have a layout where manufacturing activities are grouped by site, production units, resource groups, and resources,

you can use the Grouping levels to group the activities according to that layout. The grouping levels can be expanded and collapsed either on the individual grouping level or for all levels in the chart by using the **Expand all** and **Collapse all** buttons on the menu. You can also configure the grouping levels to be expanded or collapsed when the chart is opened.

Material availability

The Gantt chart can be set up to provide the planner with detailed information about material status for the individual activities. For example, this can be helpful if material is delayed and is affecting the production plan. In this case, the material issues will be highlighted in the Gantt chart to help the planner to understand consequences and make necessary adjustments.

A job will appear with a material shortage icon if the schedule start date of the job is later than the material availability date for materials consumed by the job. The material availability date is calculated based on the pegging information in the dynamic master plan. The material shortage icon will, for example, appear on a job that is consuming a material that is pegged against a purchase order that has a receipt that is later than the planned start date of the job.

Indicator of material availability date

When you set up the chart to show jobs with material shortages an icon for showing the material availability date for the job can also be shown. The icon will only be shown if the material availability date is within the defined time interval of the chart. If the material availability date lies outside the defined time interval, then more detailed material availability information can be retrieved from the material list in the job dialog box. In the list, there's also an icon showing late materials for the job. You can reschedule a job using the material availability date as a start date.

Indicator of order delivery date

This icon indicates the delivery date for a production order. The icon is only visible in the order view.

Capacity bar

You can configure the chart to show a resource capacity bar. This bar provides an overview of the resource capacity for an activity in the defined time interval of the chart. The capacity bar isn't shown for periods of the time where the resource isn't booked. In periods where the resource is booked to capacity, the capacity bar is shown as a solid bar. In periods where the resource is overbooked, the bar will appear thicker and in a red color. For example, if two jobs overlap, the capacity bar will indicate an overbooking in the time interval where there's an

overlap. The capacity bar is updated dynamically when you schedule a job. You enable the capacity bar on the **Show capacity bar** menu. It can only be shown in **Resource view**. If you want to get a more detailed view of the capacity load on a resource, use the **Capacity load** chart, which can be opened from the menu or the context menu for a selected activity.

Job scheduling in the Gantt chart

The Gantt chart offers different options for making adjustments to the production plan. In the Gantt chart, you can reschedule activities as a drag-and-drop interaction or from a schedule menu. In the planning process, you can take resource capacity, resource capabilities, and material constraints into account.

Schedule a job as a drag-and-drop interaction

You can reschedule job within the defined time interval as a drag-and-drop interaction. You can only reschedule the job on the same resource, and you can only schedule one job at a time.

Schedule a job from the menu

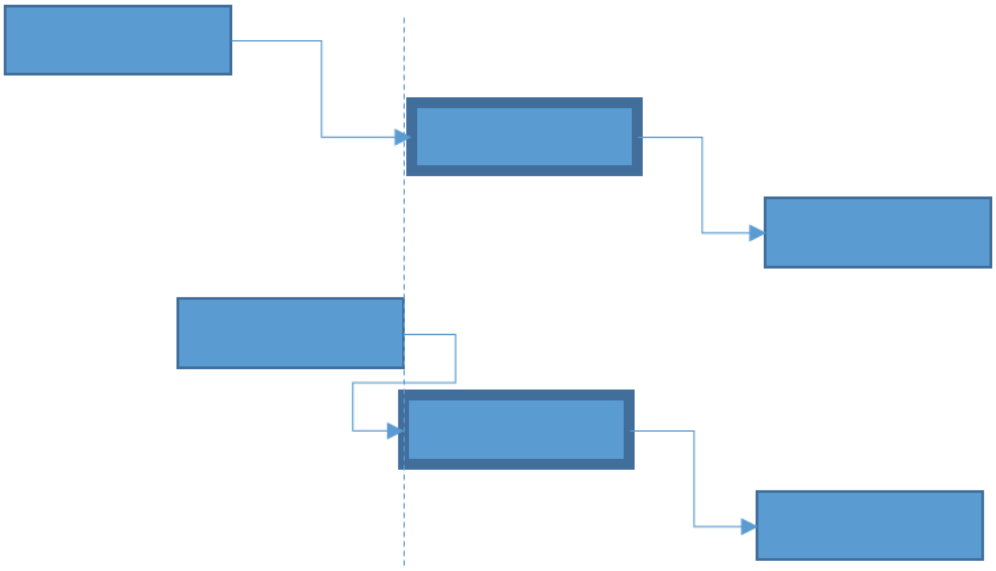
On the **Schedule jobs** menu, you can schedule one or more selected jobs in the chart based on a scheduling direction and a scheduling date time. There are three available schedule directions.

- Forward from scheduling date
- Backward from scheduling date
- Forward from material availability date

It isn't possible to schedule a job outside the defined time interval of the Gantt chart. If you do that, the job will be left unscheduled and you'll receive the error message, "Could not schedule the job within the loaded time period."

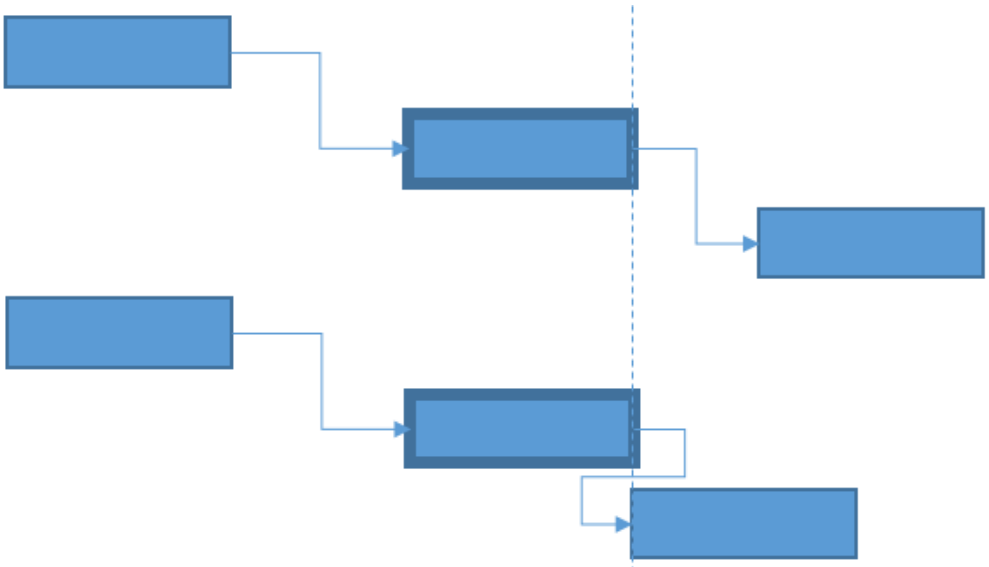
Schedule previous jobs

In a network of activities, such as jobs belonging to the same production order, you can use the **Schedule previous jobs** function to schedule the previous jobs relative to a selected job in the network. In the following example, the highlighted activity is the selected job. The diagram shows before a previous job is scheduled and after the previous job is scheduled.



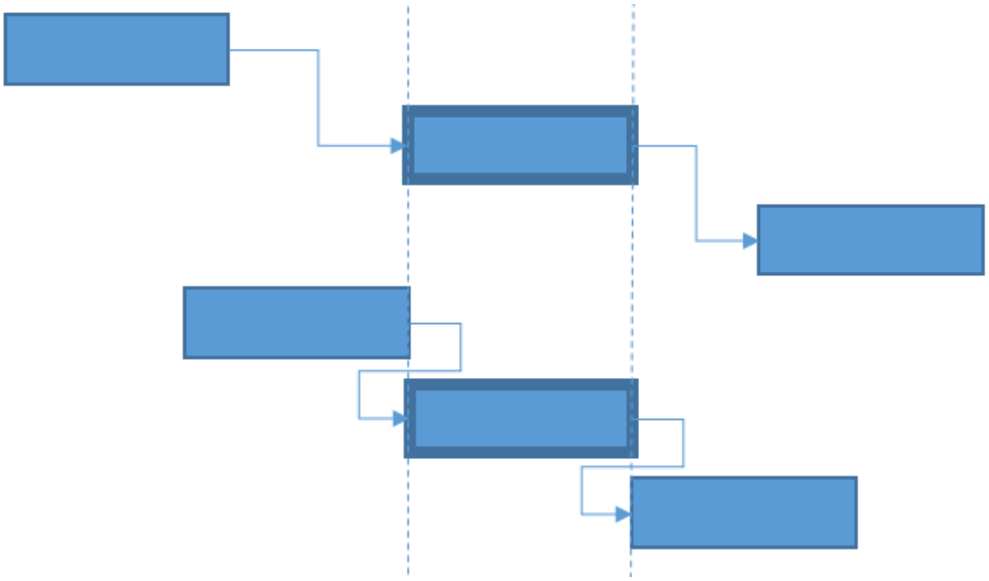
Schedule next jobs

You can use the **Schedule next jobs** function to schedule the next jobs relative to a selected job in a network of activities. In the following example, the highlighted activity is the selected job. The diagram shows before the next job is scheduled and after the next job is scheduled.



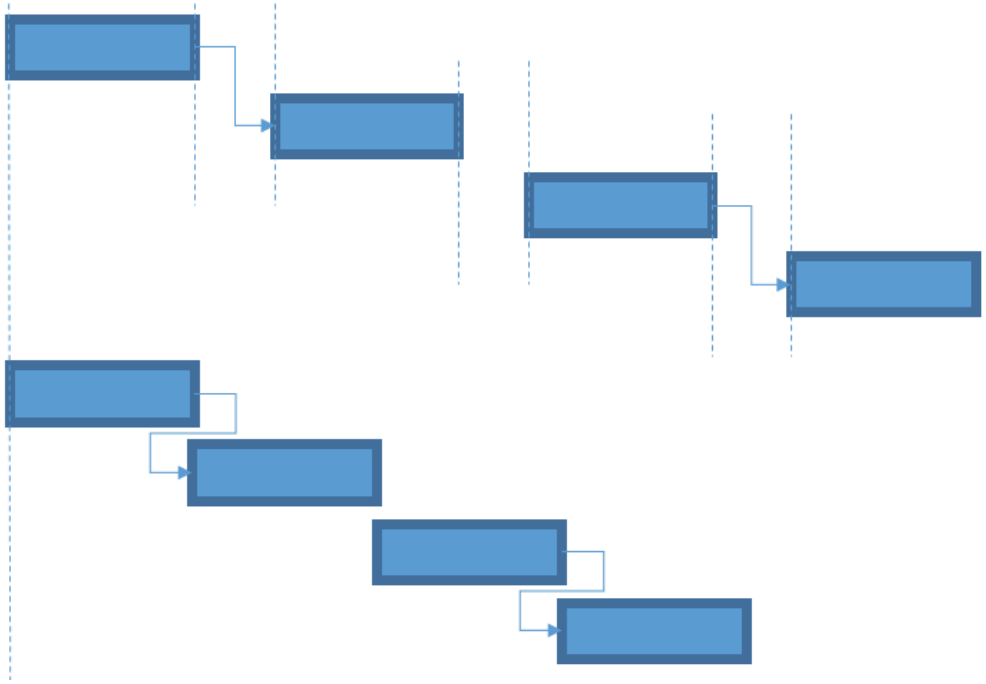
Schedule around job

You can use the **Schedule around job** function to schedule the next job and the previous job relative to a selected job in a network of activities. In the following example, the highlighted activity is the selected job. The diagram shows before a job is scheduled and after the job is scheduled.



Arrange jobs

You can use the **Arrange** function to arrange selected activities on the same resource. These activities can be in the same network of activities, but can also belong to different networks. When you use the arrange function, the time gaps between the selected activities will be eliminated. You can use this function to optimize the capacity utilization of the resources. The diagram shows before a job is scheduled and after the job is scheduled.



Reassign activities from one resource to another

You can reassign a job from one resource to another. This can be useful in situations where a machine is out of order or overbooked, and you need to find another available resource that can do the job.

Reassigning an activity as a drag-and-drop interaction

In the **Resource** view, you can reassign an activity to a different resource in the Gantt chart as a drag-and-drop interaction. You do that by selecting the row in which the activity is scheduled. After the row is selected, you can drag the row to the resource in the chart grouped under a different resource grouping level.

Reassigning an activity from the Schedule jobs menu

You can reassign a job from the **Schedule job** dialog box opened from the **Schedule job** menu. From this menu, you can only reassign a job to a resource that is already loaded to the Gantt chart. If you only select one job, then the drop-down list for the resource will be sorted by applicable resources. If you select more jobs, then there will be no information about applicable resources from the resource list.

Load additional resources to the Gantt chart

In the **Resource view**, you can load additional resources to the Gantt chart. This can be useful if you want to find an alternative resource for a job that is scheduled on a machine that is overbooked or broken down. On the **Load additional resources** page, you'll get a list of resources that are date efficient as of the date the list is opened. Applicable resources, relative to a selected job in the Gantt chart, will be listed first. If you have multiple jobs selected prior to opening the list no indication of applicable resources will be shown. On the **Load additional resources** page, you can select one or more resources that will be loaded to the Gantt chart when you confirm your selection. If there are no jobs scheduled on the selected resource in the time interval of the Gantt chart, then the resource will be placed under a resource grouping level in the bottom of the list of activities in the Gantt chart.

Access the Gantt chart

The Gantt chart can be opened from the following pages.

Expand table

Page	Description
Production order list and detail	On the **Production order list and detail** page, you can open the Gantt chart from one or more selected orders. Opening the chart from the **Gantt chart** menu item will load all jobs related to the selected production orders, but also jobs from other production orders that are scheduled on the same resources. Opening the chart from the **Gantt chart – Fast view** menu item will only load

Page	Description
	jobs related to the selected production orders. In this view, it isn't possible to schedule jobs.
Resource	On the **Resource** page, you can open the Gantt chart from the menu item **Gantt chart**. When selected, all the jobs scheduled on the resource in a selected time interval will be loaded to the chart.
Resource group	On the **Resource group** page, you can open the Gantt chart from the menu item **Gantt** chart. When selected, all the jobs scheduled on the resources in the resource group will be shown in a selected time interval.
Gantt charts	On the **Gantt charts** page, you can configure Gantt charts by resources and resource groups. For example, if you want to control production activities for specific sets of resources or resource groups, then you can make individual configurations of those on the **Gantt charts** page. You can then open the Gantt chart from each configuration.
Hour forecasts (project)	Project activities of type **Hour forecast** can be job scheduled on resources. On the **Hour forecast** page, on the **Scheduling** menu, you can open the Gantt chart on an order to see job scheduled project activities of type hour forecast.
Job to complete (List in **Production floor management** workspace)	The **Jobs to complete list in the Production floor management** workspace shows jobs from production and batch orders that are in progress on the selected resources for the workspace. On the **Gantt chart** menu item you can open the Gantt chart, where all the jobs selected in the list will be loaded to the chart.
Production orders to release (Opened from the **Production floor management** workspace)	The **Production orders to release** page is opened from the **Production floor management** workspace. This page shows scheduled production and batch orders pending release. On this page, you can open the Gantt chart for selected production orders.

Related information

- Visual scheduling with Gantt chart for production and batch orders (Video)
- Visual scheduling for production (demo script)

Operations scheduling options

09/19/2025

This article describes the options for operations scheduling. Use operations scheduling to provide a general estimate of the production process over time.

Operations scheduling calculates the following information for a production order:

- Start and end dates for the production order and each operation.
- Resources assigned to operations.

Several settings determine how production schedules are calculated. You define these settings on the **Operations scheduling** page. The following information describes the scheduling options.

Operations scheduling

Scheduling direction

The scheduling direction is fundamental to the scheduling process. You can schedule a production forward or backward from any date, depending on timing and scheduling requirements.

- **Forward scheduling** – Schedule a production to start as early as possible. The production can start today, tomorrow, or on any specific date in the future. The production is scheduled to start on the earliest possible date and is planned forward in time to the earliest possible end date.
- **Backward scheduling** – Schedule a production to start as late as possible. The schedule is based on the date when the production must be completed and counts backward to the latest possible date that the production can start without missing its target deadline.

The following options are available:

- **Forward from today** – Schedule forward from the current date. (The current date is the system date.)
- **Forward from planned start** – Schedule forward from an earlier start date. If there's no previous scheduling, the scheduling direction is forward from the current date.
- **Forward from scheduling date** – Schedule forward from the date that you specify in the **Scheduling date** field. If you don't specify a scheduling date, the scheduling direction is forward from the current date.

- **Backward from delivery date** – Schedule backward from the delivery date that you specify for the production order. If you select this option, but don't specify a delivery date, the delivery date is the current date.
- **Backward from planned end** – Schedule backward from a previously calculated end date. If there's no previous scheduling, the end date is the current date.
- **Backward from scheduling date** – Schedule backward from the date that you specify in the **Scheduling date** field. If you don't specify a scheduling date, the current date is used. Backward from scheduling date is calculated for the production order the last time that a requirement was calculated. If you don't specify a date for the production order, the current system date is used.
- **Backward from futures date** – Schedule backward from the futures date that was calculated for the production order the last time that a requirement was calculated. If you don't specify a futures date for the production order, the current system date is used.
- **As last scheduling** – For operations scheduling and job scheduling, the selected scheduling direction and date are saved. Therefore, you can select this option for subsequent scheduling. If there's no previous scheduling of the production order, scheduling is backward from the current system date.
- **Forward from tomorrow** – Schedule forward from the current date plus one day.
- **Forward from previous job** – This option is relevant only in job scheduling. If you select this scheduling direction for operations scheduling, the production order is scheduled forward from the current date. In job scheduling, scheduling is established for one job, and all other jobs for the production are scheduled based on that job.
- **Backward from previous job** – This option is relevant only in job scheduling. If you select this scheduling direction for operations scheduling, planned orders are scheduled backward from the current date. In job scheduling, scheduling is established for one job, and all other jobs for the production are scheduled based on that job.

Scheduling date

Use this date for the **Forward from scheduling date** and **Backward from scheduling date** scheduling directions. Scheduling is backward or forward from this date. For more information, see the previous section, "Scheduling direction."

Recalculate BOM levels

When you select **Recalculate BOM levels**, the selected bill of materials (BOM) levels are recalculated to help guarantee the correct scheduling order.

Limitations

Finite capacity

Scheduling depends on the availability of the resources that are required to complete production. If there's not enough capacity, production orders can be delayed or even stopped. If you apply finite capacity to operations scheduling, existing capacity reservations that you made on the resources are considered, and the system sees that capacity as unavailable. The production order is scheduled based on the availability of capacity on the resources that are required. Operations scheduling uses the specified sequence of operations to determine the order of operations in the production route. Operations scheduling reserves capacity on the resource groups, based on the operation times that are defined on the production route. The capacity of the resource groups is the sum of available capacity on all the resources in the resource groups.

Finite material

Scheduling also depends on the availability of the materials that are required for production. Insufficient component availability can also cause production delays. Scheduling can also be based on the use of materials. Just specify the materials that must be available for production instead of the materials that aren't critical. This type of scheduling is known as scheduling with finite material. When you specify finite materials, production is scheduled based on whether materials are available. When you optimize on both capacity and materials, production is calculated to meet both restrictions. The availability of capacity and materials is considered, and the production order’s operations can't be scheduled to start until capacity and materials are available at the same time and in the required quantities. Select this check box if materials should be considered limited during scheduling. If the materials are limited, the material's coverage for that time is considered. In other words, scheduling considers the future dates that are calculated for the items. Scheduling reserves raw materials and explodes the current production. If scheduling occurs several times, only the first scheduling runs an explosion and makes reservations. If you make changes in the production BOM or route, the next scheduling runs an explosion. For the explosion, it's assumed that the materials are required on the same day. Because the production isn't scheduled at the time of the master schedule explosion, the current date is the best estimate of when the items will be available. The explosion then checks whether the items are available. If the items are available, the production requirement can be fulfilled. If the items aren't available by the current date, a planned order is generated, and an offset selection is made for the planned order. If automatic firming is set for the planned order, the planned order is firmed automatically for purchase or production. The production status is automatically changed to the status that is specified in the **Requested production status** field in the **Coverage groups** dialog box. If the check box is cleared, the materials are always considered available. Therefore, scheduling doesn't consider whether the materials are available at the time of requirement.

Finite property

Select this check box if the job scheduling should include property requirements.

Keep production unit

Select whether the scheduling engine should schedule only resources that are already specified on the production unit.

Keep warehouse from resource

Select whether the scheduling engine should schedule only resources that are associated with the input warehouse that is specified on the resource.

References

Schedule references

When references depend on production orders, they're also known as subproductions. You can schedule subproductions as part of the scheduling of a production order. Select this check box if the scheduling of subproductions should be based on the scheduling of the main production. In relation to the main production, overlying productions are scheduled forward, and underlying productions are scheduled backward. You can view production order references in the **Reference level** field on the **Production orders** page.

Synchronize references

You can synchronize references with the production order. If you select this option, the system moves and aligns the dates of the subproductions when you make changes to the production order's schedule. If a production order has one or more subproductions, you might want to schedule the subproductions together with the main production. In this case, you can't start the main production until you complete the related subproductions. Therefore, select this check box if you want the scheduling of subproductions to be based on the start and end times of the selected production. You can select this check box only if you also select the **Schedule references** check box.

Cancellation

Cancel queue time

Select this check box to exclude queue time from the scheduling.

Cancel setup

Select this check box to exclude setup time from the scheduling.

Cancel process

Select this check box to exclude run time from the scheduling.

Cancel overlap

Select this check box to exclude overlap time from the scheduling.

Cancel transport

Select this check box to exclude transit time from the scheduling.

Set default

You can save the current values as default values. Two options are available:

- Set as my default
- Set as default for everyone

Related information

Operations scheduling

Schedule a production order with operations and job scheduling

06/17/2025

This article focuses on scheduling a production order with operations scheduling and job scheduling. No jobs are created with operations scheduling whereas jobs are created with job scheduling. This procedure is intended for the production manager, production planner, or shop floor supervisor working in a discrete manufacturing environment.

Create a production order

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Select **New production order**.
3. In the **Item number** field, enter or select the item number for the production order.
4. Select **Create**.

Schedule operations for the production order

1. Mark the newly created row.
2. On the Action Pane, select **Schedule**.
3. Select **Schedule operations**.
4. In the **Scheduling direction** field, select *Forward from scheduling date*.
5. In the **Scheduling date** field, enter a date. Select a future date, for example, today plus one week. With the selected **Scheduling direction**, the production order will be scheduled forward from this date.
6. Select **OK**.
7. In the list, mark the selected row. The status is changed to *Scheduled*.
8. Select **All jobs**.
9. Close the page.

Schedule jobs for the production order

1. On the Action Pane, select **Schedule**.
2. Select **Schedule jobs**.
3. In the **Scheduling direction** field, select *Forward from scheduling date*.
4. In the **Scheduling date** field, enter a date. Select a date in the future, for example, today plus one week. With the selected **Scheduling direction**, the production order will be scheduled forward from this date.

5. Select **OK**.
6. On the Action Pane, select **Production order**.
7. Select **All jobs**.

Production order lifecycle overview

Article • 01/31/2025

When a production order is created, a request is initiated to start producing an item. The production order contains information about what to produce, the quantity to produce, and the planned finish date. It also contains information about which materials to consume and which process to follow to produce the item.

A production order passes through stages of the production life cycle. When an order is created, it's assigned the status *Created*. When an order is finished, it's assigned the status *Ended*. A parameter setting in each stage allows a user to configure each step. The setting can be set up for a single user or for all users.

The production bill of material and the production route are the main entities of the production order. They're copied to the production order based on the selected item and quantity that are going to be produced. Before the production order is started, the production bill of material and route can be edited.

A production order can be created in the following scenarios:

- Created by master planning execution based on material demand.
- Created directly from a sales order line or when a higher-level production order is created and estimated (pegged supply).
- Created manually.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a production order

07/10/2025

This procedure shows how to create a production order. The demo data company used to create this procedure is USMF. This is the first of seven procedures that together explain the production-order lifecycle.

Create a production order

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Select **New production order**.
3. In the **Item number** field, type *D0001*.
4. In the **Delivery** field, enter a date. The delivery date indicates when the production order should end in order to deliver on time. This date can be used in the scheduling process. For example, you can schedule the order backward from the delivery date.
5. Set **Quantity** to *20*.
6. Select Create.

> **Note**
>
> The **BOM number** field automatically displays the number of any active BOM for the current item, but you can change the BOM for the production order by selecting an active BOM from the list of approved BOM versions. The **Route number** field automatically displays the number of any active route for the current item, but you can change the route for the production order by selecting an active route from the list of approved route versions.

Validate the production order

1. In the list, select the link in the selected row. Select the link for the production order number that you have just created. This opens the details page for the order.
2. Select **Edit**.
3. In the **Delivery** field, enter a date. For example, you can change the delivery date for the production order.
4. Select **Save**.
5. Close the page.

Update the BOM

1. On the Action Pane, select **Production order**.
2. Select **BOM**. Open the BOM page to validate the BOM data that was copied from the default data when the production order was created. In this procedure, you need to update the quantity for a BOM.
3. Select **Edit**.
4. In the **Quantity** field, enter a number. Changing the quantity on the BOM line affects the cost estimate of material consumption for the production order.
5. Select **Save**.
6. Close the page.

Update the production route

1. On the Action Pane, select **Production order**.
2. Select **Route**. Open the **Route** page to validate the data of the production route that was copied from the default data when the order was created. In this procedure, you need to update the quantity for one of the operations in the production route.
3. In the list, find and select the desired record.
4. Select **Edit**.
5. In the **Process qty.** field, enter a number. Changing the process time affects the estimated route consumption and the cost of the production order.
6. Select **Save**.
7. Close the page.

Estimate a production order

Article • 04/01/2021

You can run this procedure by using the USMF demo data company or your own data set. In both cases, you need to have an open production order that has the Created status. This is the second procedure out of seven which explains the production order lifecycle.

Estimate a production order

1. Go to Production control > Production orders > All production orders.
2. Select an order that has the Created status in the grid.
3. On the Action Pane, click Production order.
4. Click Estimate.
    - In this step, the estimated costs of a single production order is calculated.
5. Click OK.

View the calculation details

1. On the Action Pane, click Manage costs.
2. Click View calculation details.
    - This page displays the cost breakdown. For example, you can view the total cost price per unit for the finished product in the first row. The subsequent rows contain costs according to the bill of materials, production route, and indirect costs.

Feedback

Was this page helpful? Yes No

Provide product feedback

Schedule a production order

Article • 09/29/2021

This procedure shows how to schedule a production order. The demo data company used to create this procedure is USMF. This is the third procedure out of seven which explains the production order lifecycle.

Schedule a production order

1. Go to Production control > Production orders > All production orders.
    - Select a production order that has the Estimated status.
2. On the Action Pane, click Schedule.
3. Click Schedule jobs.
    - The parameters for scheduling are set up on this page. You can set up the parameters for specific users or all users.
4. In the Scheduling direction field, select 'Forward from today'.
5. In the Scheduling date field, enter a date.
6. Select or clear the Finite capacity check box.
7. Select or clear the Finite material check box.
8. Click OK.

View the scheduling results

1. On the Action Pane, click Production order.
2. Click All jobs.
    - This page displays the scheduled jobs that you have just generated.
3. Expand or collapse the Scheduling section.
    - On the Scheduling FastTab, you can view the scheduled date and time.
4. Click Inquiries.
5. Click Capacity load.
    - The Capacity load page displays the capacity that is reserved through job scheduling, the total number of hours that are currently reserved on the

resource, and the number of hours that remain available for job scheduling on the resource.

6. Close the page.
7. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Release a production order

Article • 09/29/2021

This procedure shows how to release a production order. The demo data company used to create this procedure is USMF. This is the fourth procedure out of seven which explains the production order lifecycle.

1. Go to Production control > Production orders > All production orders.

    - In the grid, select a production order that has the Scheduled status.

2. On the Action Pane, click Production order.
3. Click Release.

    - On this page, you can confirm the release of the selected range of production orders. Click Select if you want to add orders.
    - The Release step indicates when the production order is released to the production shop floor so that the shop floor operators can report progress on the production jobs. Production papers that contain relevant information about processing the jobs can be issued. The warehouse work for raw material picking is generated for the items that are set up for the warehouse process.

4. Click the General tab.

    - Select which production reports should be printed. The Job card report prints a page for each scheduled job and requires the production order to be scheduled at the job level. The report contains information about the scheduled start and end time, the quantity to produce, and which resource processes the job. The Route job report collects the same information on the same page, but does not print a page for each job. The Route card report only shows the operations but not the jobs. Therefore, this report does not require job scheduling, but can be used when production orders are scheduled at the operation level.

5. Click the Print route card checkbox.
6. Click OK.
7. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Start a production order

06/17/2025

This procedure shows how to start a production order on the shop floor. Time and material consumption are reported in this process. The demo data company used to create this procedure is USMF. This is the fifth procedure out of seven which explains the production order lifecycle.

Start a production order

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Select a production order that has the *Released* status.
3. On the Action Pane, select **Production order**.
4. Select **Start**. On this page, you can confirm the start of the production order.
5. Select the **General** tab.
6. In the **From Oper. No.** field, enter *10*.
7. In the **Automatic route consumption** field, select *Always*.
8. Select the **Post route card now** checkbox.
9. In the **Automatic BOM consumption** field, select *Always*.
10. Select the **Post picking list now** checkbox.
11. Select the **Print picking list** checkbox.
12. Select **OK**. This is the printed picking list that shows the materials used for the production order.
13. Close the page.

Validate the picking list

1. On the Action Pane, select **View**.
2. Select **Picking list**.
3. In the list, find and select the desired record.
4. In the list, select the link in the selected row.
5. Select **Edit**.
6. In the **Consumption** field, enter a number.
7. Select **Post**.
8. Select **OK**. In the picking list journal, the materials consumed by the production order are posted. Before posting the journal, you can make adjustments if there's a difference between the estimated quantity and the actual consumed quantity.
9. Select the **GridPanel** tab.
10. Close the page.

Verify the route card journal

1. On the Action Pane, select **View**.
2. Select **Route card**.
3. In the list, find and select the desired record.
4. In the list, select the link in the selected row.
5. Select **Edit**.
6. In the **Hours** field, enter a number.
7. Select **Post**.
8. Select **OK**.

In the route card journal, the time spent on the individual operations is recorded. Good and error quantity can also be reported.

Report a production order as finished

06/17/2025

This procedure shows how to report a production order as finished. This is the sixth procedure out of seven which explains the production order lifecycle.

Report a production order as finished

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Select a production order that has the *Started* status.
3. On the Action Pane, select **Production order**.
4. Select **Report as finished**. On this page, you can confirm the quantity of the finished product to be reported as finished.
5. Open the **General** tab and make the following settings:
    - **Good quantity** – Enter the quantity of usable items produced.
    - **Error quantity** – Enter the quantity of unusable items produced.
    - **Error cause** – Select the cause of the error for unusable items (for example, *Material*).
    - **End job** – Set as needed.
    - **Accept error** – Set as needed.
6. Select **OK**.

Verify the Report as finished journal

1. On the Action Pane, select **View**.
2. Select **Reported as finished**.
3. In the list, mark the selected row.
4. In the list, select the link in the selected row.
5. The report as finished journal is posted. If you want to make adjustments to the journal, manually create a new journal where you can make changes.

End a production order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to end a production order. It is the final procedure out of seven procedures that explain the production order lifecycle.

End a production order

To end a production order, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Select a production order that has the status *Reported as finished*.
3. On the Action Pane, select **Production order**.
4. Select **End**. On this page, you can confirm that you want to end the production order.
5. Open the **General** tab.
6. In the **Date** field, enter a date.
7. In the **Scrap method** field, select *Allocation*. When you select the allocation method, costs from the scrapped materials are added to the finished goods.
8. Select **OK**.

Validate calculation results

To validate the calculation results after you end the production order, follow these steps:

1. On the Action Pane, select **Manage costs**.
2. Select **View cost comparison**. After you end the production order, you can compare the estimated cost price against the realized cost price to get an overview of the production variances.

Last updated on 05/27/2026

Release production orders

06/17/2025

A released production order is an order that has been authorized for production. The term Released is used to describe a state in the production order life cycle, where the production order is available for execution on the production shop floor and for warehouse processes.

Characteristics of the Released state

The *Released* state is one state in the production order life cycle. Production orders that are in the *Released* state are available for execution on the production shop floor and for warehouse processes. The *Released* state has the following characteristics:

- A production order can be changed to the *Released* state either from the production order or by using a batch process. The production order can also be updated automatically from planned production orders that are firmed by using the **Firming time fence** field on the **Master plan** page.
- The *Released* state is the signal for the shop floor operators (operators) to start executing the production jobs on the shop floor.
- Production papers, such as route cards, route jobs, and jobs cards provide information about production jobs and can be issued.
- For materials that are physically reserved, warehouse work is generated to pick materials for the production order.

Releasing jobs to the shop floor

After a production order is released, production jobs that are related to the order are visible and ready for registration. The operators can make job registrations, such as Start, Stop, and Completion, on the production floor execution interface. The registered time and quantity are automatically transferred from the registration pages to production journals to keep track of the consumed time and quantity.

Route cards

A route card provides an overview of information that comes from route and operation setups, and from operation and job scheduling methods.

Route jobs

A route job lists each job of an operation in detail, and includes setup, process, queue, and transportation times. For example, an operation such as painting might require individual jobs, such as setup time, run time for the painting process, and queue time for drying.

Job cards

A job card lists the individual job numbers of a particular operation. One job appears on each page. The jobs that are included on a job card, and their estimated times, come from the route and operation setup information. From a job card, you can open the **Production journal lines**, **job card** page. People who run operations resources can provide feedback about the production process. There are fields where you can enter consumption statistics and information such as the error quantity.

Warehouse work for raw material picking

Work for raw material picking is generated during release. Work is generated only for the quantity of materials that was physically reserved for the production order before the order was released. The following setup is required to generate warehouse work for raw material picking:

- A location directive for raw materials picking that determines which warehouse location to pick the materials in
- A wave template for raw materials, where policies for the execution of warehouse work are configured
- A production input location that determines where materials are put

When using license plate controlled locations, you can choose whether the ordered quantity or the full quantity available for the item should be picked while processing the raw material picking work. To set this option:

1. Go to **Product information management** > **Products** > **Released products** and open the relevant item.
2. Expand the **Warehouse** FastTab.
3. Select one of the following options for the **Material picking in license plate locations** field:

    - *Order picking* – Only the ordered quantity should be picked.
    - *Staging* – Whenever possible, the full quantity available at the license plate should be picked. For a worker to be able to pick the full license plate quantity, the license plate must not contain mixed items and must not have mixed dimensions. If the

license plate contains mixed dimensions or items, the pick will proceed as if it were set to *Order picking*.

Report production orders as finished

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Reporting as finished is a production stage. At this stage, you report a finished product and move it from the production order to the inventory.

When you report a quantity of finished goods as finished on a production order, you update the inventory as on-hand. You can report partial quantities of the originally planned order quantity as finished. You can also report error quantities with an associated error reason when reporting quantities as finished. When a production order reaches the *Reported as finished* stage, it means that you can't report any more quantity for the production order.

The report as finished process has the following characteristics:

- You can set up consumption of raw material and time that are proportional to the reported quantity (back-flushing).
- You can generate putaway work for items that are enabled for warehouse processes.
- You can set up the planned or standard cost value of the finished goods to be reported to ledger accounts.
- You can create a quality order for the reported quantity based on the setup of a quality association. You report the quantity to the output location. Warehouse work is then generated to move the quantity from the output location to its final destination defined by the location directive for the putaway work.
- You can create a quality order when a production order is reported as finished if a quality association is set up.

Set a production order to Reporting as finished

You can set a production order to report as finished through the standard production order update function, or through the route and job card journals, or through the **Report as finished** journal. Workers can also use the production floor execution interface to update the stage to

Reported as finished when they report on the last job of the production order. Finally, you can add the **Report as finished** option as a process for the Warehouse Management mobile app.

Last updated on 05/27/2026

Register and track batch/serial numbers for finished products and their components

08/13/2025

Workers and managers can register batch/serial numbers for materials and components that are used in manufacturing processes. They can also then associate those numbers with the batch/serial numbers of the products that are produced. In this way, manufacturers can optimize their processes, enhance product quality, and respond quickly to any issues that arise. Managers can use the item tracing report to effectively track batch/serial numbers that are registered through the *Tracked components* feature.

> **Tip**
>
> You can also track batch/serial numbers for materials and components using the Traceability Add-in for Dynamics 365 Supply Chain Management (preview), which offers a graphical browser interface and an API for integration with external systems.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.40 or later.
- The feature that's named *Tracked components* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

Enable registration of tracked component information

To enable registration of tracked component information in production processes, follow these steps.

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. On the **Bills of materials** tab, set the **Enable tracked components** option to *Yes*.

Enable registration of tracked components on products

Before you can use tracked components functionality for a product or its components, the following conditions must be met:

- The product must be defined as serial controlled or batch controlled.
- The product must be associated with a tracked components policy.

To create a tracked components policy, follow these steps.

1. Go to **Production** > **Setup** > **Production** > **Tracked components policy**.

2. Use the buttons on the Action Pane to add a new policy or edit an existing one.

3. For the new or selected policy, set the following fields:

    - **Name** – Enter a name for the policy.
    - **Description** – Enter a description of the policy.
    - **Use tracked components** – Set this option to *Yes* to register tracked information for products that are associated with the policy.
    - **Check pre-allocation for product** – Set this option to *Yes* to validate serial or batch numbers for finished goods when tracked component information is registered.
    - **Check for pre-allocation for components** – Set this option to *Yes* to validate serial or batch numbers for components when tracked component information is registered.
    - **Serial number matching limit** – Set this option to *Yes* to ensure that the number of component serial numbers that are matched to a product serial number doesn't exceed the planned quantity on the production order bill of materials (BOM) line. If the number of matched components exceeds the planned quantity, the system blocks the match to prevent over-allocation.

Register tracked components for finished items

To register tracked components for the finished items for any production order, follow these steps.

1. Start from any of the following places:

    - **All production orders** list or details
    - **Picking list** journal line
    - **Route card** journal line
    - **Job card** journal line

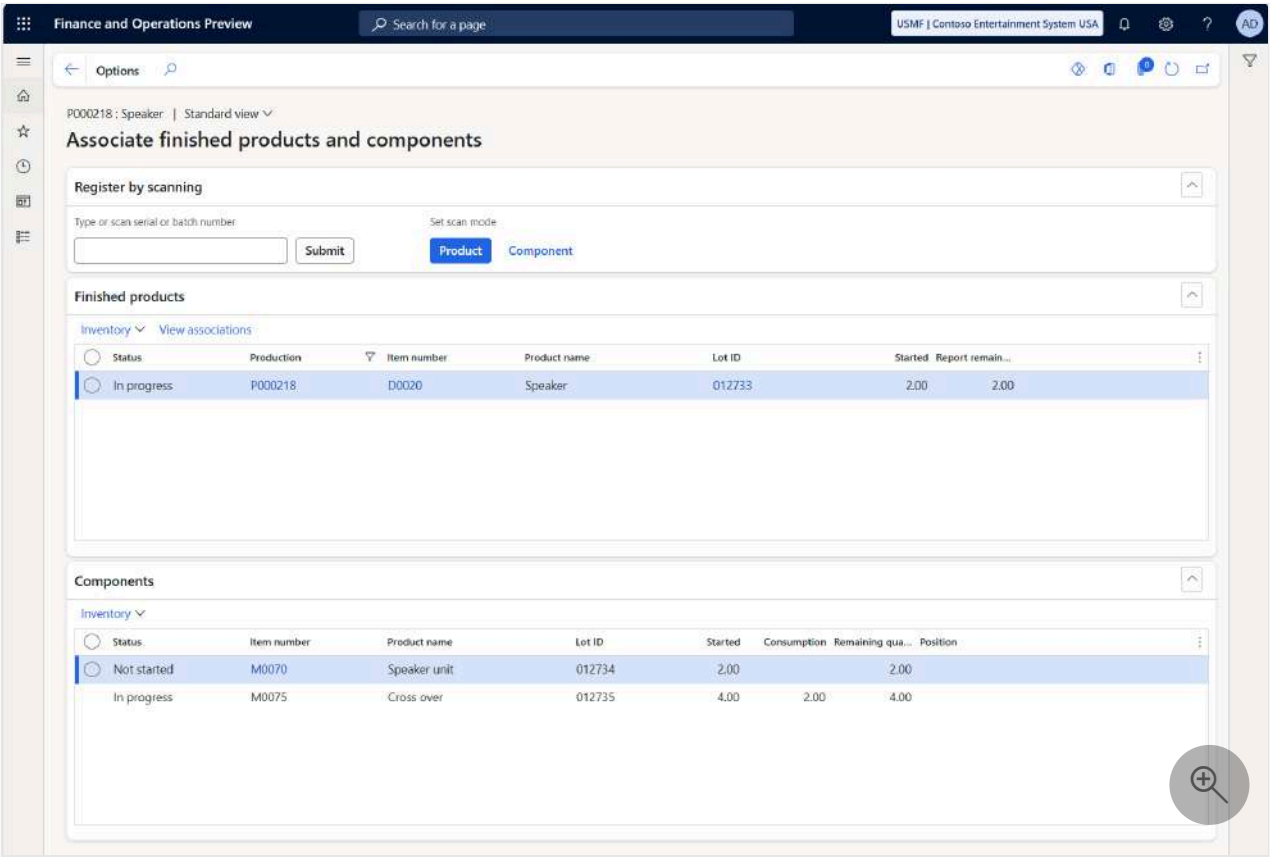
- **Report as finished** journal line
- **Current operations**
- Production order **Start** dialog
- Production order **Report as finished** dialog

2. Select the production order that you want to register batch or serial numbers for.

3. On the Action Pane, find the tab that includes the **Track components** button, and select that button.

4. On the **Tracked components** page, the **Type or scan serial or batch number** field is automatically in focus. In addition, the **Scan mode** field is automatically set to *Product* to indicate that you're about to enter the batch or serial number of a finished product. Follow one of the steps:

   - If you're using a scanner, scan the bar code of the first finished product that you want to register.
   - If you're using a keyboard, enter the batch or serial number of the first finished product that you want to register, and then select **Submit**.



5. The value of the **Scan mode** field is changed to *Component*. You can now register the batch or serial number of a component by following one of these steps:

   - If you're using GS1 bar codes, your system might be able to detect which component you're scanning. In this case, you can just scan the component bar code,

which includes both the component item number and its serial or batch number.
    - If you're using simple bar codes, or if you're using a keyboard, first select a component in the grid on the **Component** FastTab. Then enter or scan the batch or serial number of that component.

6. Repeat the previous step until you finish registering all the components for the current finished product.

7. Change the value of the **Scan mode** field back to *Product*, and repeat this procedure until you finish registering all finished products and their components for the current production order.

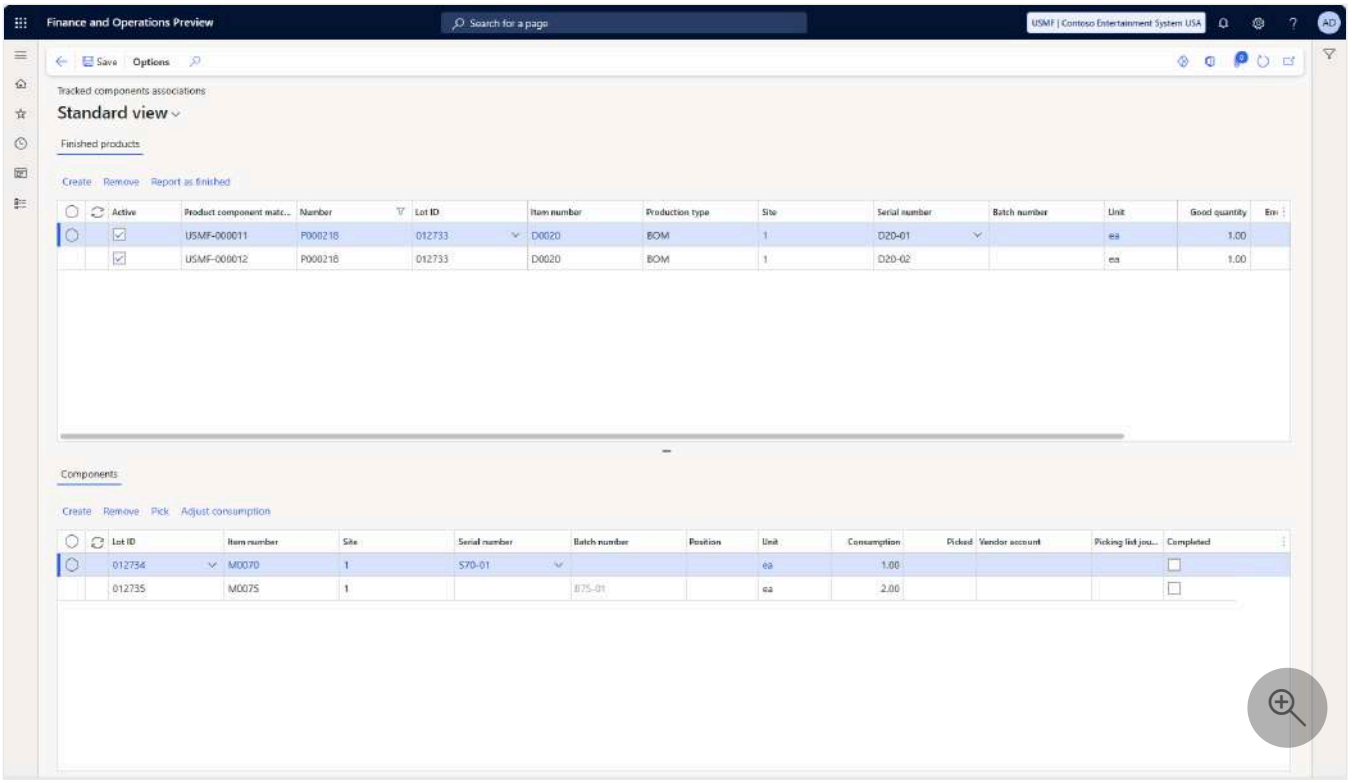
   At any time, you can review and correct the registrations that you've made so far. On the **Finished products** FastTab, on the toolbar, select **View associations** to open the **Tracked components associations** page. For information about how to work with this page, see the next section.

As you work, the **Status** field on the **Finished product** FastTab of the **Tracked components** page indicates the registration status of each finished product. The **Status** field on the **Components** FastTab indicates the registration status of each component in the selected finished product. The **Status** field on each FastTab can have the following values:

- *Not started* – No batch or serial number has been associated with the product or component.
- *In progress* – An association has been established, but only for part of the started quantity for the finished product or the component. On the **Components** FastTab, the **Started**, **Consumption**, and **Remaining quantity** columns let you track the quantity that has been allocated.
- *Completed* – No remaining quantity remains to be allocated for the finished product or the component.

Review and edit batch/serial numbers for finished products and their components

Use the **Tracked components associations** page to review and edit batch/serial numbers for finished products and their components. For information about how to open this page for a selected production order, see the previous section.



Finished products

The **Finished products** section lists batch/serial number registrations for the products that are being produced for the selected production order. The toolbar in this section provides the following buttons:

- **Create** – Add a new row to the grid so that you can register the serial or batch number for a missing finished product.
- **Remove** – Remove one or more selected finished products.
- **Report as finished** – Report a selected row as finished. You can first edit the **Good quantity** and/or **Error quantity** value for batch products as required.

Components

The **Components** section lists the components of the finished product that's selected in the **Finished products** grid. For each component, this section shows the registered batch or serial number, and you can enter or correct these values as required. The toolbar in this section provides the following buttons:

- **Create** – Add a new row to the grid so that you can register the serial or batch number for a missing component.
- **Remove** – Remove one or more selected components.
- **Pick** – Manually pick the quantity that's planned for consumption, as specified in the **Consumption** field.

- **Adjust consumption** – Generate a production picking list journal to manually adjust the consumption. In this case, the quantity that's specified in the **Consumption** field is automatically entered in the picking list journal.

Related information

- How workers use the production floor execution interface
- Traceability overview (preview)

Report as finished from the job card device

07/10/2025

> **Important**
>
> The job card device is no longer supported and has been replaced by the production floor execution interface. We strongly recommend that you use the production floor execution interface instead of the job card device.

Workers use the **Report progress** page on the job card device to report quantities that have been completed for a production job. This article describes how to set up various options that establish how workers can report as finished using this page and what happens next. Options include:

- Control whether and how quantities that are reported as finished are added to inventory.
- Control whether and how batch numbers are generated and applied when reporting as finished.
- Control whether and how serial numbers are generated and applied when reporting as finished.
- Control whether and how to report as finished to a license plate.

Control whether quantities that are reported as finished are added to inventory

To control whether and how the quantities that are reported as finished on the last operation should be added to inventory, follow these steps.

1. Go to **Product control** > **Setup** > **Manufacturing execution** > **Production order defaults**.
2. On the **Report as finished** tab, set the **Update finished report on-line** field to one of the following values:
    - **No** – No quantity will be added to inventory when quantities are reported on the last operation. The status of the production order will never change.
    - **Status + Quantity** – The status of the production order will change to *Reported as finished*, and the quantity will be reported as finished to inventory.
    - **Quantity** – The quantity will be reported as finished to inventory, but the status of the production order will never change.
    - **Status** – Only the status of the production order will change. No quantities will be added to inventory when quantities are reported on the last operation.

> **Note**
>
> Quantities aren't tracked in inventory if the operations that they are reported as finished on aren't defined as the last operation. However, those quantities can be used to view progress. They can also be included in rules that control whether workers can start the next operation before a defined threshold of reported quantities on the previous operation is reached. You can define these rules on the **Quantity validation** tab on the **Production order defaults** page.

For more information about how to work with the **Production order defaults** page, see Production parameters for manufacturing execution.

Report batch-controlled items as finished

The job card device supports three scenarios for reporting on batch items. These scenarios apply both to items that are enabled for warehouse management processes (WMS) and to items that aren't enabled for WMS.

- **Manually assigned batch numbers** - Workers enter a custom batch number. This batch number might come from an external source that isn't known to the system.
- **Predefined batch numbers** - Workers select a batch number in a list of batch numbers that the system automatically generates before the production order is released to the job card device.
- **Fixed batch numbers** - Workers don't enter or select a batch number. Instead, the system automatically assigns a batch number to the production order before it's released.

Configure products that require batch number reporting

To enable a product to support any of the available batch-controlled scenarios, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product to configure.
3. On the **Manage inventory** FastTab, in the **Batch number group** field, select the tracking number group that is set up to support your scenario.

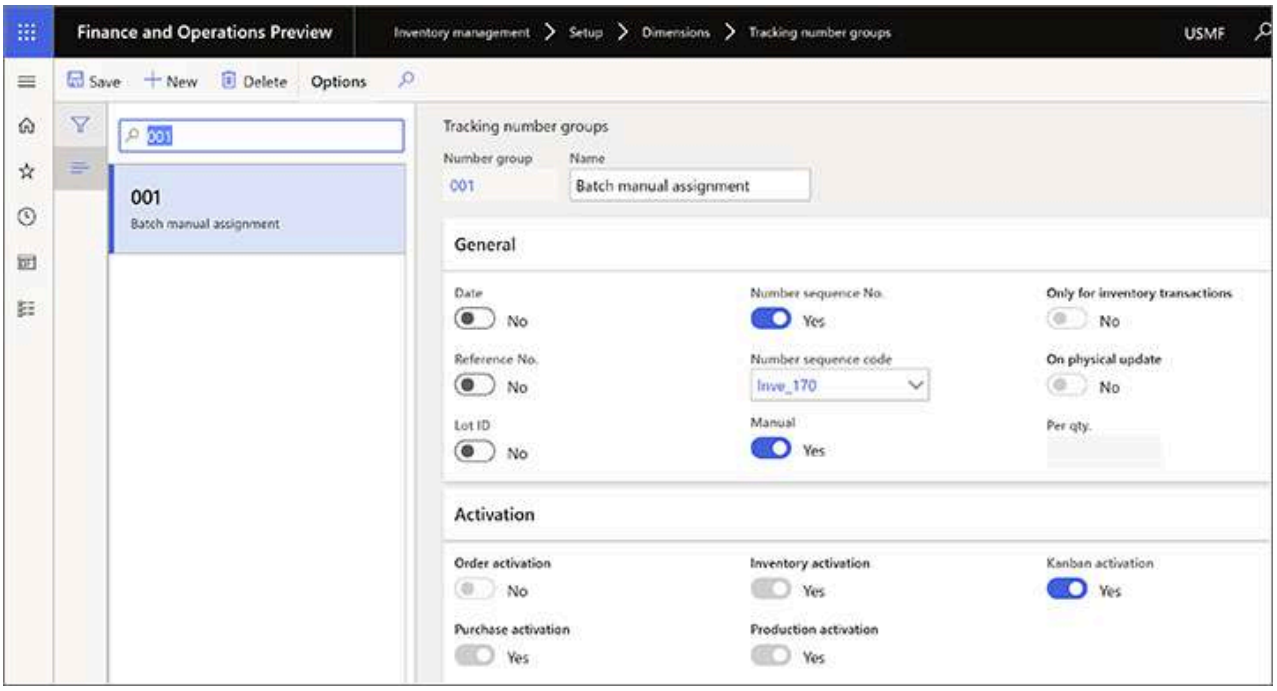
> **Note**
>
> By default, if no batch number group is assigned to a batch-controlled product, the job card device provides manual entry for the batch number during reporting as finished.

The following sections describe how to set up tracking number groups to support each of the three scenarios for reporting on batch items.

Set up a tracking number group that lets workers manually assign a batch number

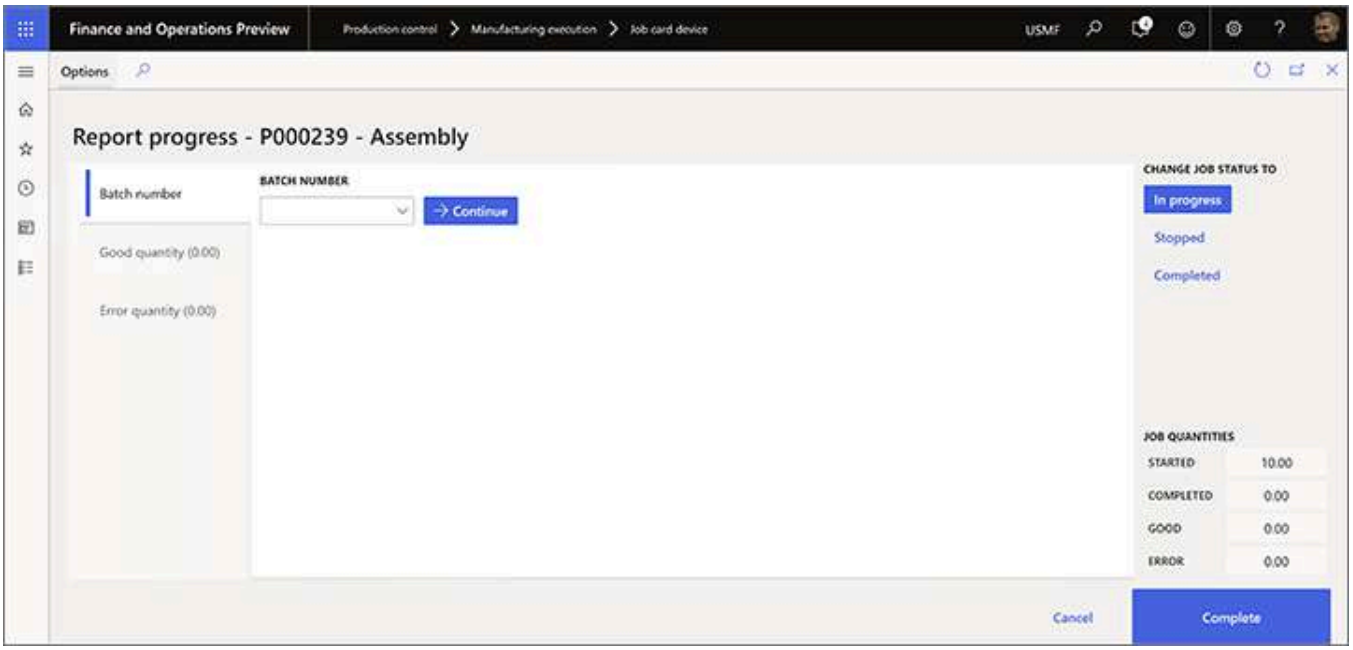
To allow for manually assigned batch numbers, follow these steps to set up a tracking number group.

1. Go to **Inventory management** > **Setup** > **Dimensions** > **Tracking number groups**.
2. Create or select the tracking number group to set up.
3. On the **General** FastTab, set the **Manual** option to **Yes**.



4. Set other values as you require, and then select this tracking number group as the batch number group for released products that you want to use this scenario for.

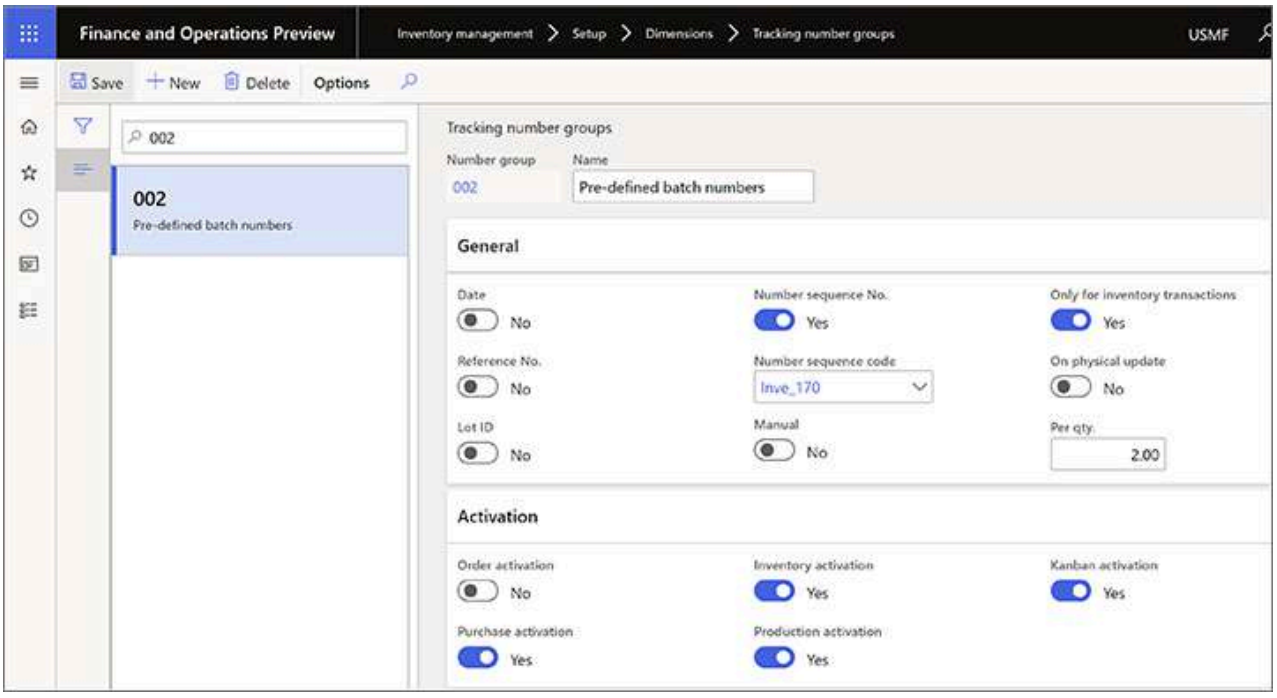
When you use this scenario, the **Batch number** field that the **Report progress** page on the job card device provides is a text box where workers can enter any value.



Set up a tracking number group that provides a list of predefined batch numbers

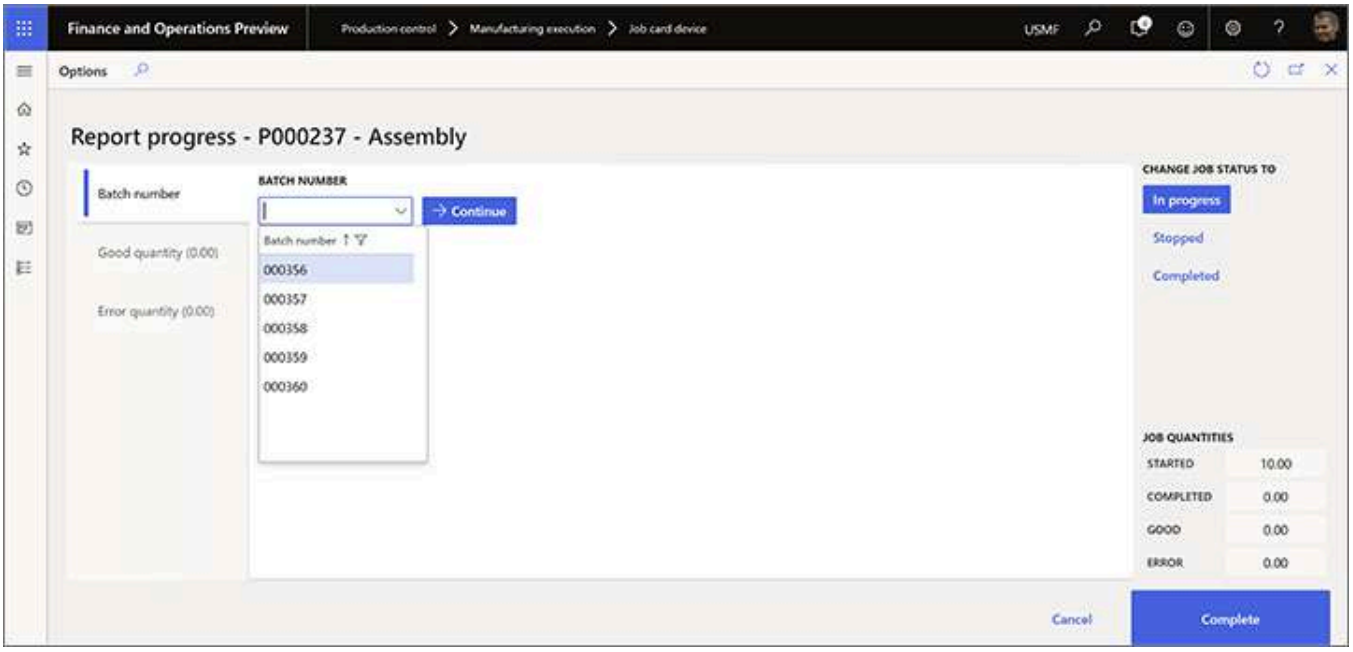
To provide a list of predefined batch numbers, follow these steps to set up a tracking number group.

1. Go to **Inventory management** > **Setup** > **Dimensions** > **Tracking number groups**.
2. Create or select the tracking number group to set up.
3. On the **General** FastTab, set the **Only for inventory transactions** option to **Yes**.
4. Use the **Per qty** field to split batch numbers per quantity, based on the value that you enter. For example, you have a production order for ten pieces, and the **Per qty** field is set to *2*. In this case, five batch numbers will be assigned to the production order when it's created.



5. Set other values as you require, and then select this tracking number group as the batch number group for released products that you want to use this scenario for.

When you use this scenario, the **Batch number** field that the **Report progress** page on the job card device provides is a drop-down list where workers must select a predefined value.



Set up a tracking number group that automatically assigns batch numbers

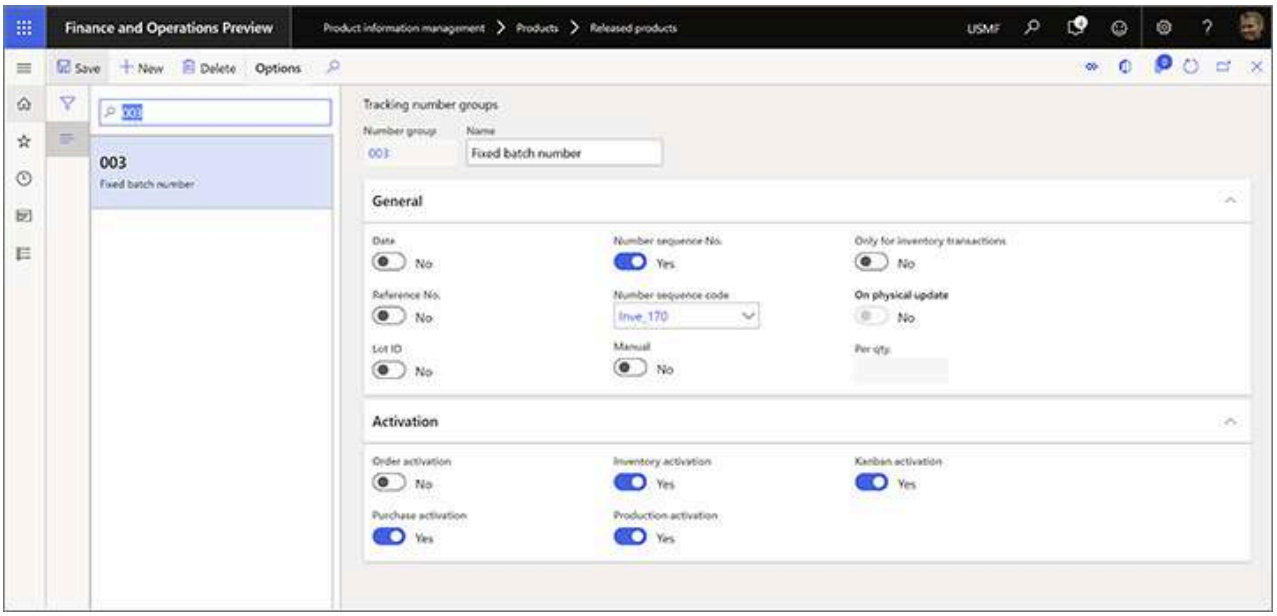
If batch numbers should be assigned automatically, without worker input, follow these steps to set up a tracking number group.

1. Go to **Inventory management** > **Setup** > **Dimensions** > **Tracking number groups**.

2. Create or select the tracking number group to set up.

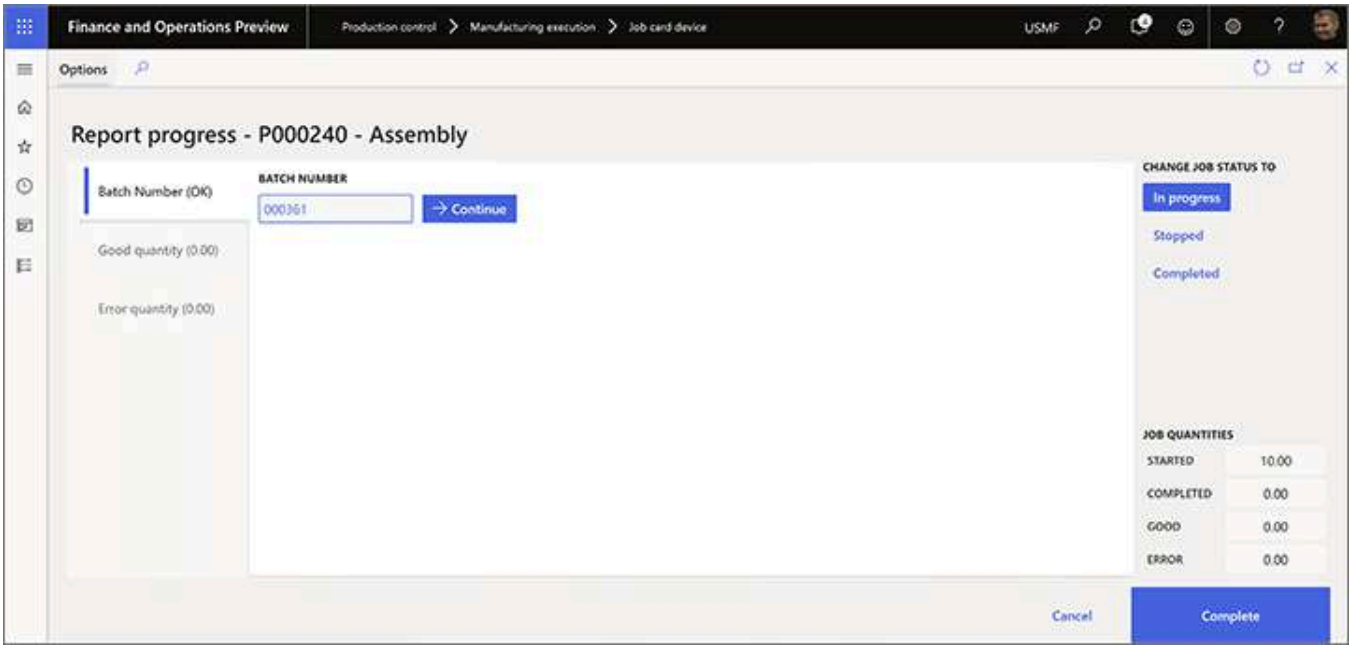
3. On the **General** FastTab, set the **Only for inventory transactions** option to **No**.

4. Set the **Manual** option to **No**.



5. Set other values as you require, and then select this tracking number group as the batch number group for released products that you want to use this scenario for.

When you use this scenario, the **Batch number** field that the **Report progress** page on the job card device provides shows a value, but workers can't edit it.



Report serial-controlled items as finished

The job card device supports three scenarios for reporting on serial-controlled items. These scenarios apply both to items that are enabled for warehouse management processes (WMS)

and to items that aren't enabled for WMS.

- **Manually assigned serial numbers** - Workers enter a custom serial number. This serial number might come from an external source that isn't known to the system.
- **Predefined serial numbers** - Workers select a serial number in a list of serial numbers that the system automatically generates before the production order is released to the job card device.
- **Fixed serial number** - Workers don't enter or select a serial number. Instead, the system automatically assigns a serial number to the production order before it's released.

Configure products that require serial-number reporting

To enable a product to support any of the available serial-controlled scenarios, follow these steps:

To enable each scenario, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product to configure.
3. On the **Manage inventory** FastTab, in the **Serial number group** field, select the tracking number group that is set up to support your scenario.

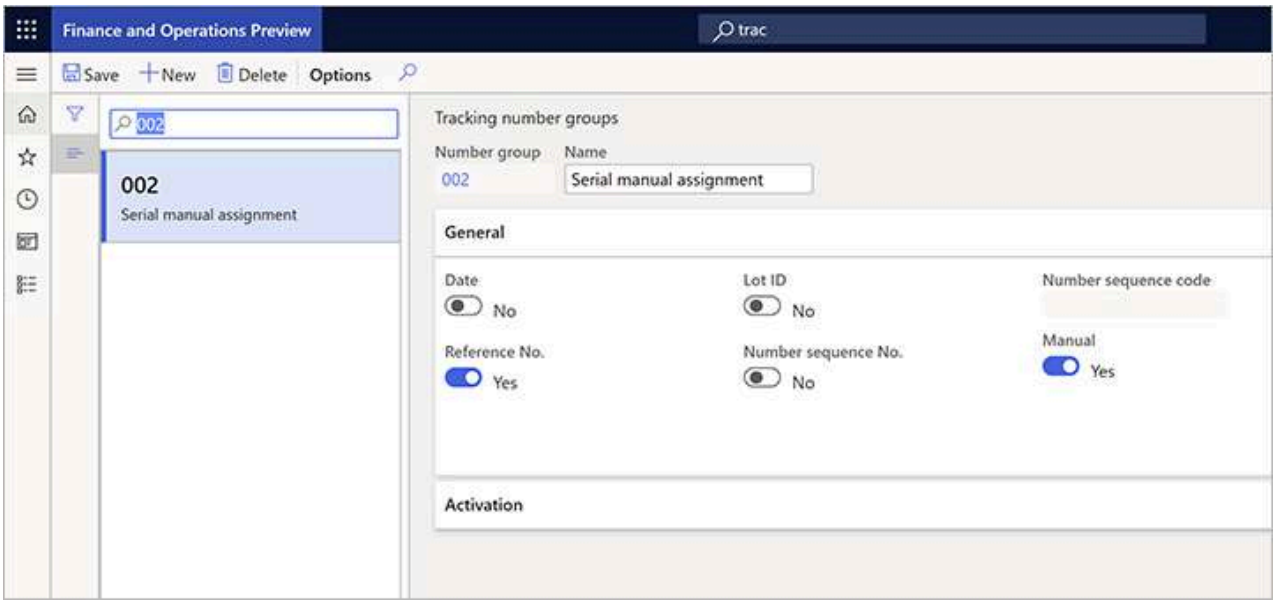
> ⓘ **Note**
>
> By default, if no serial number group is assigned to a serial-controlled product, the job card device provides manual entry for the serial number during reporting as finished.

The following sections describe how to set up tracking number groups to support each of the three scenarios for reporting on serial-controlled items.

Set up a tracking number group that lets workers manually assign a serial number

To allow for manually assigned serial numbers, follow these steps to set up a tracking number group.

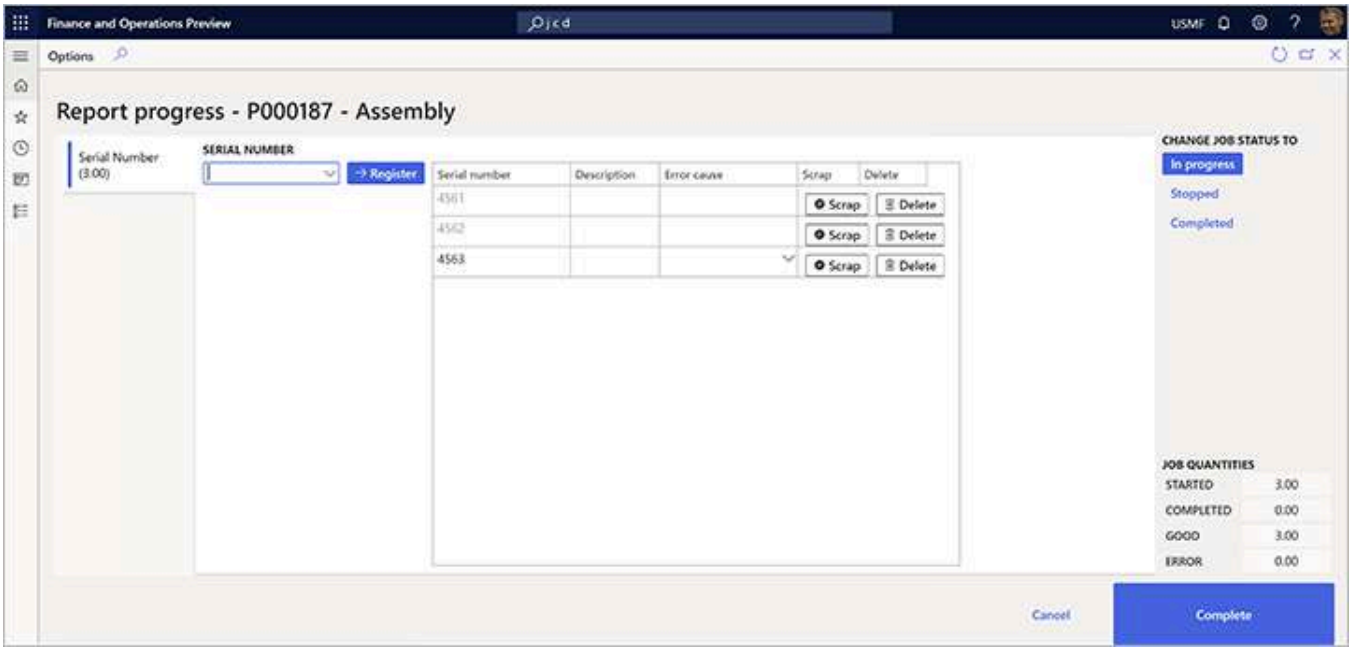
1. Go to **Inventory management** > **Setup** > **Dimensions** > **Tracking number groups**.
2. Create or select the tracking number group to set up.
3. On the **General** FastTab, set the **Manual** option to **Yes**.



4. Set other values as you require, and then select this tracking number group as the serial number group for released products that you want to use this scenario for.

When you use this scenario, the **Serial number** field that the **Report progress** page on the job card device provides is a text box where workers can enter any value for the serial number. On entering a value, it is added to the serial number list. In this list, workers can do the following:

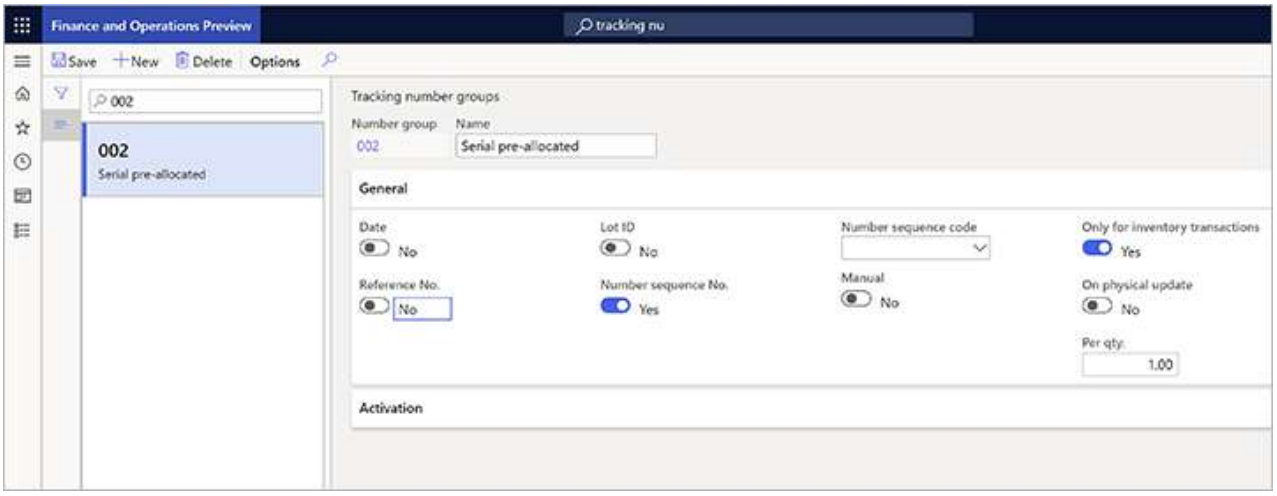
- To mark a serial number as scrapped, select the **Scrap** button for the appropriate row. The worker will be prompted to provide an **Error cause**.
- To delete a serial number, select the **Delete** button for the appropriate row.



Set up a tracking number group that provides a list of predefined serial numbers

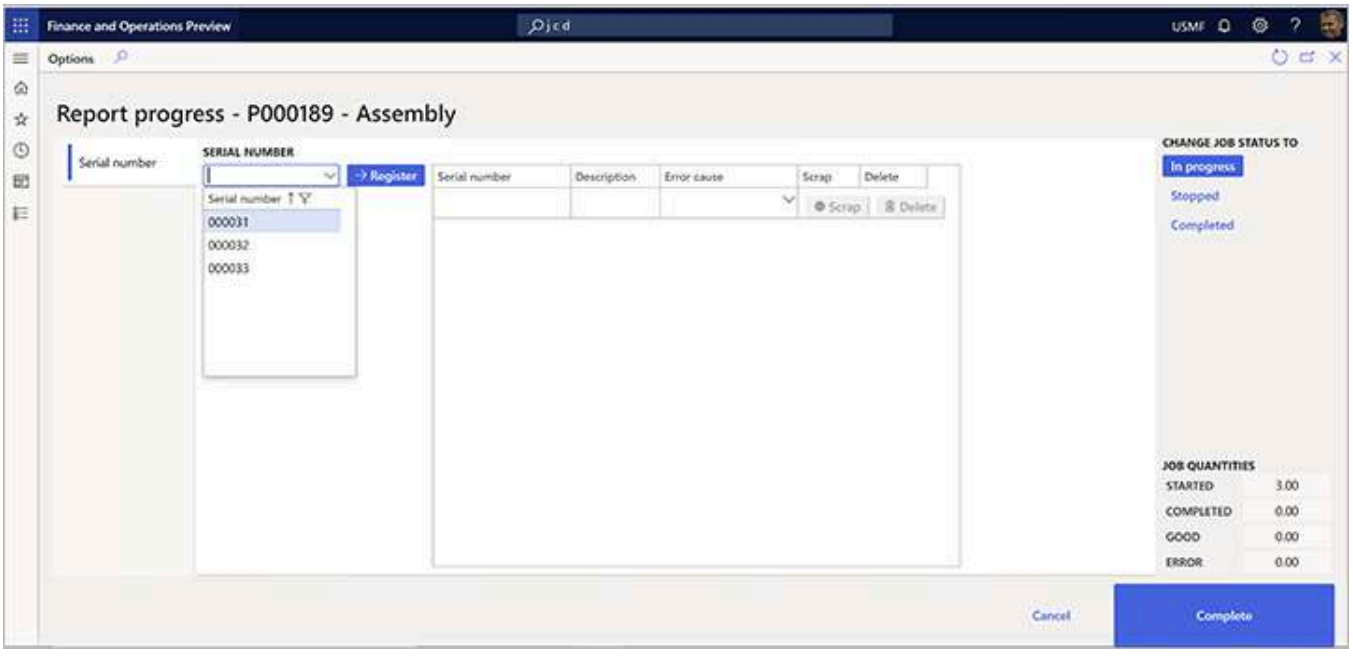
To provide a list of predefined serial numbers, follow these steps to set up a tracking number group.

1. Go to **Inventory management** > **Setup** > **Dimensions** > **Tracking number groups**.
2. Create or select the tracking number group to set up.
3. On the **General** FastTab, set the **Only for inventory transactions** option to **Yes**.
4. Use the **Per qty** field to split serial numbers per quantity of one.



5. Set other values as you require, and then select this tracking number group as the serial number group for released products that you want to use this scenario for.

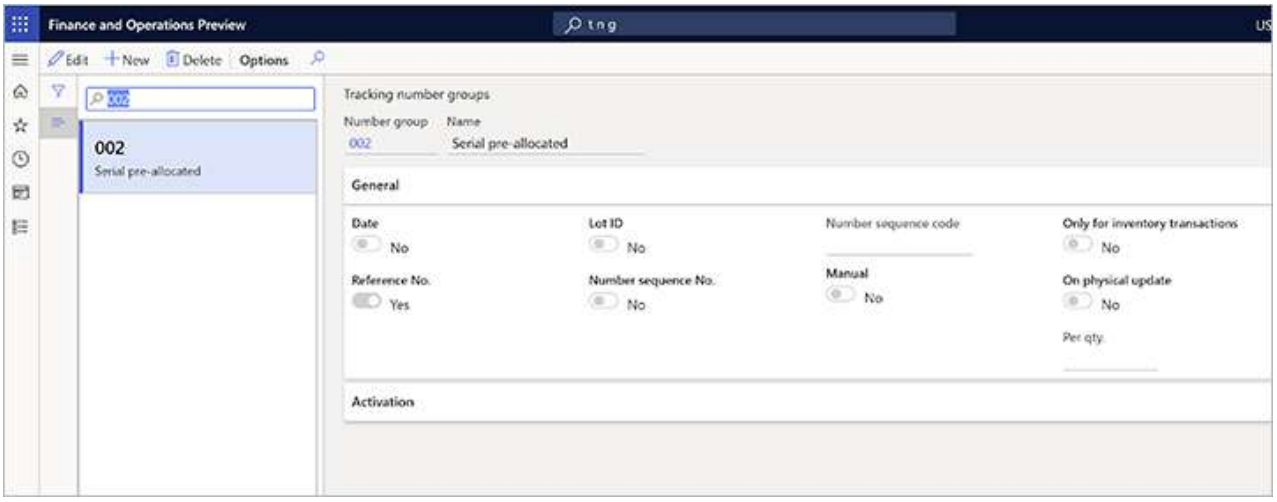
When you use this scenario, the **Serial number** field that the **Report progress** page on the job card device provides is a drop-down list where workers must select a predefined value.



Set up a tracking number group that automatically assigns serial numbers

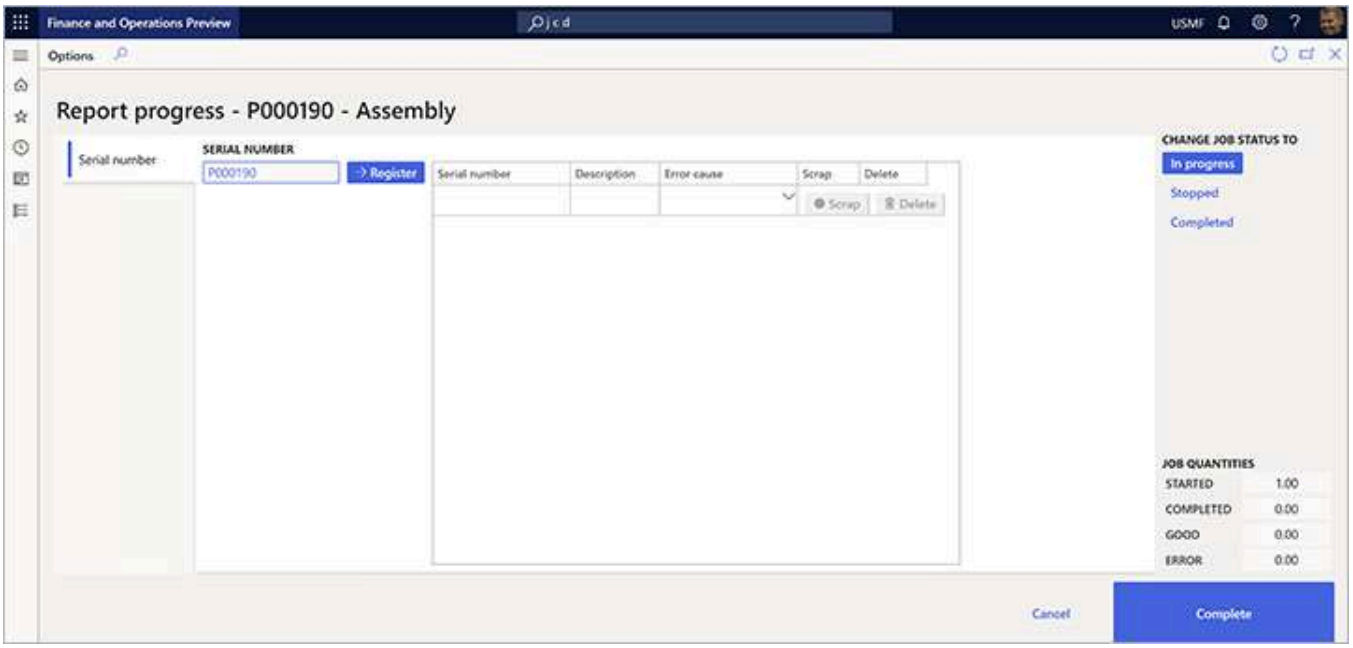
If a serial number should be assigned automatically, without worker input, follow these steps to set up a tracking number group.

1. Go to **Inventory management** > **Setup** > **Dimensions** > **Tracking number groups**.
2. Create or select the tracking number group to set up.
3. On the **General** FastTab, set the **Only for inventory transactions** option to **No**.
4. Set the **Manual** option to **No**.



5. Set other values as you require, and then select this tracking number group as the serial number group for released products that you want to use this scenario for.

When you use this scenario, the **Serial number** field that the **Report progress** page on the job card device provides shows a value, but workers can't edit it. This scenario is only relevant when a production order is created for a quantity of one piece of a serial number-controlled item.



Report as finished to a license plate

Warehouse management processes (WMS) can use the license plate dimension to track inventory on warehouse locations that have been set up for this purpose. In this case, the license plate number is required when a worker reports quantities as finished.

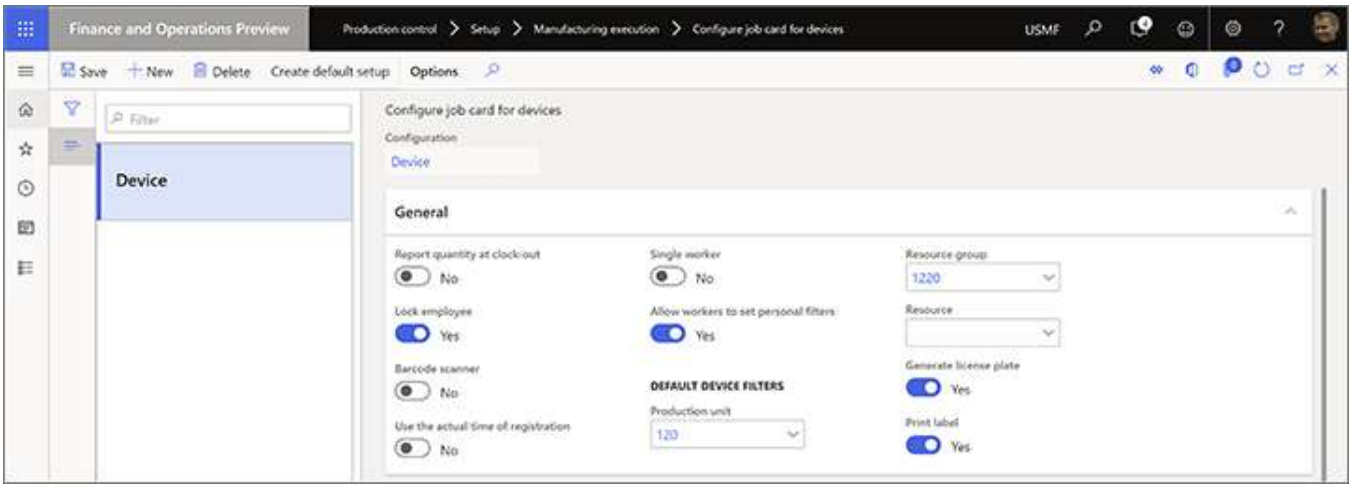
Set up reporting as finished to a license plate

To control whether workers should reuse an existing license plate or generate a new license plate when they report quantities as finished, follow these steps.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Configure job card for devices**.

2. Set the following options for each device:

    - **Generate license plate** – Set this option to **Yes** to generate a new license plate for each report as finished. Set it to **No** if an existing license plate should be used for each report as finished.
    - **Print label** – Set this option to **Yes** if the worker must print a license plate label for each report as finished. Set it to **No** if no label is required.



> **Note**
>
> To configure the label, go to **Warehouse management** > **Setup** > **Document routing** > **Document routing**. Learn more in Document routing label layouts.

Provide mixed-reality Guides for workers in production

Article • 06/03/2022

Workers in production processes will benefit from relevant instructions that are provided at the right time in the context of their work. *Instructions* apply in several domains of work, including: assembly, service, operations, certification, and safety. Across all of these core business functions, ongoing training instructions can help empower workers to achieve more and work better.

Introduction

You can provide instructions in different ways. One efficient system that ships out of the box uses Dynamics 365 Guides.

Dynamics 365 Guides can help empower your employees with hands-on learning. You can define standardized processes with step-by-step instructions that guide your employees to the tools and parts they need and show employees how to use these tools in real work situations.

You can attach guides to various aspects of production control including:

- Resources
- Resource groups
- Released products
- Formulas
- Formula versions
- Bills of material (BOMs)
- BOM versions
- Routes
- Route versions
- Route operation relations

> **Note**
>
> You can also attach Guides with Asset Management. For more information about that option, see Integrate Dynamics 365 Supply Chain Management (Asset Management) with Dynamics 365 Guides.

When a first-line worker chooses a job on the shop floor through Supply Chain Management, the worker can see the relevant guides on the production floor execution interface. When the worker chooses a specific guide, a QR code for that guide is shown on the screen. The worker then uses their HoloLens to scan the QR code, which launches Guides and shows the required instructions.

The following subsections describe a few selected scenarios where companies across industries can see the biggest value when using Guides to present instructions for manufacturing.

Assembly

Instructions in assembly operations show workers the tools and parts they need and how to use them in real work situations.

Production managers can create and assign Guides, for example, for production routes, operation relations, or bills of material. Workers will find the relevant instructions on the respective operation experience on the shop floor.

Service

Equip technicians with guided instructions at the job site, eliminating the need to schedule additional visits.

Service managers can assign Guides, for example, to specific products that walk through routines of quality assessment.

Quality

Rollout new processes and ensure increased consistency by turning employee knowledge into a repeatable tool.

Quality assurance managers can assign guides, for example, to products that walk through routines of quality assessment.

Certifications

Ensure every employee meets high standards by quickly identifying who needs help and where.

Safety

Provide instructions that walk through dangerous procedures virtually before attempting in the physical environment. With a mixed reality approach, workers can experience dangerous procedures virtually.

Production managers can provide dedicated handling instructions for hazardous material handling or delicate handling procedures by assigning instructions to product items, routes, and operations.

Get started with instructions and Guides

To enable instructions in production processes, Supply Chain Management provides an out-of-the-box integration with Dynamics 365 Guides. A licensed and installed application instance of Guides is required to build, maintain, and assign mixed reality instructions to production assets and work.

Prerequisites

To use this feature, your system must include the following:

- Dynamics 365 Supply Chain Management version 10.0.15 or later
- Dual-write for Supply Chain Management apps.
- Dynamics 365 Guides version 400.0.1.48 or later

Turn on the feature

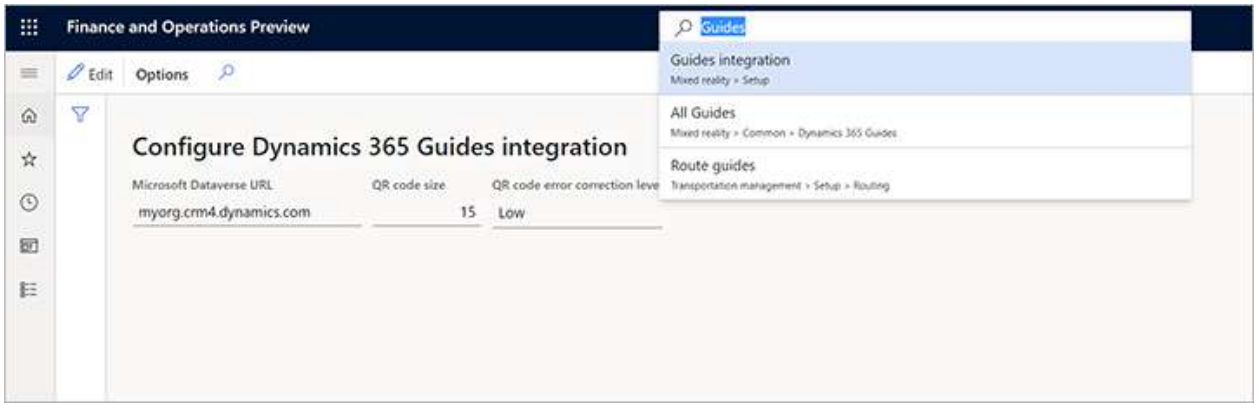
To make the feature available on your system, you must enable its configuration keys. You only need to do this once. To do this, an administrator must do the following:

1. Place your system into maintenance mode as described in Maintenance mode.
2. Go to **System administration** > **Setup** > **License configuration**.
3. Expand the **Mixed reality** section and then select the **Mixed reality guide** check box.

4. Expand the **Production management** section and then select the **Production instructions** check box.
5. Turn off maintenance mode as described in Maintenance mode.

Configure how Guides appear on the shop floor

To configure how Guides appear on the shop floor, go to **Mixed Reality** > **Dynamics 365 Guides** > **Configure Guides integration**.



Set the following fields:

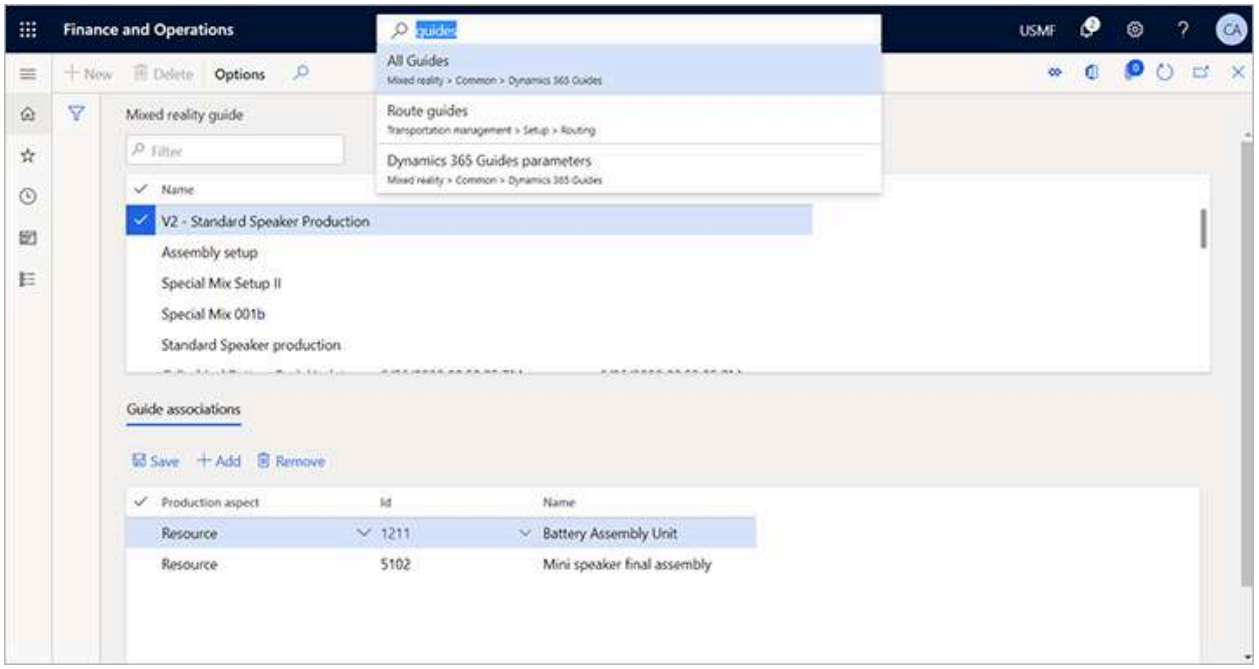
- **Microsoft Dataverse URL** - Specify the URL for the Microsoft Dataverse environment where you create your Guides. The format is "contoso.crm4.dynamics.com", where the first part of the URL is typically named after your organization (such as "contoso."), the second part is specific to the data region of your environment (such as "crm4."), and the last part is the domain (such as "dynamics.com"). One way to find the right URL is to go to the Microsoft 365 apps page and then open your Guides app. When Guides opens, you will see the URL in the address bar of your browser (only take the base URL, which should resemble the previous example). This value is used to compose addresses for your guides and will be encoded into the QR codes."
- **QR code size** - Set the size of the rendered QR code. We recommend choosing a size that will fill most of your display screen, but not more. Typically, *15* is a good value.
- **QR code error correction level** - Set the granularity of the QR code. Higher granularity can help increase the code's reliability, but your **QR code size** must be large enough to support the level of detail required by your selected correction level.

> **Tip**

- QR codes sizes that are too large for your display will take a bit longer to render and then be scaled down to fit your display. These do not provide a benefit.
- QR code sizes that are too small may decrease the ability of HoloLens to read the code properly in some environments.
- We recommend that you test the settings for each device that will display QR codes for HoloLens users. Choose settings that provide sufficient readability in your shop floor environment.

Get an overview of all Guide assignments

Use the **All Guides** page to see the list of all available Guides in your organization and all assignments to your production processes and resources. To open it, go to **Mixed reality** > **Guides** > **All Guides**. The list at the top shows all the available Guides and you can use the field here to filter the list. The list at the bottom shows all Guide assignments and provides a toolbar for managing them.



The following sections describe the types of objects that you can assign Guides to. Each assigned guide provides instructions that are automatically attached to the respective production jobs and will be available on the shop floor.

Associate a Guide to a resource

Add a Guide to a resource to offer the Guide in the context of relevant production jobs.

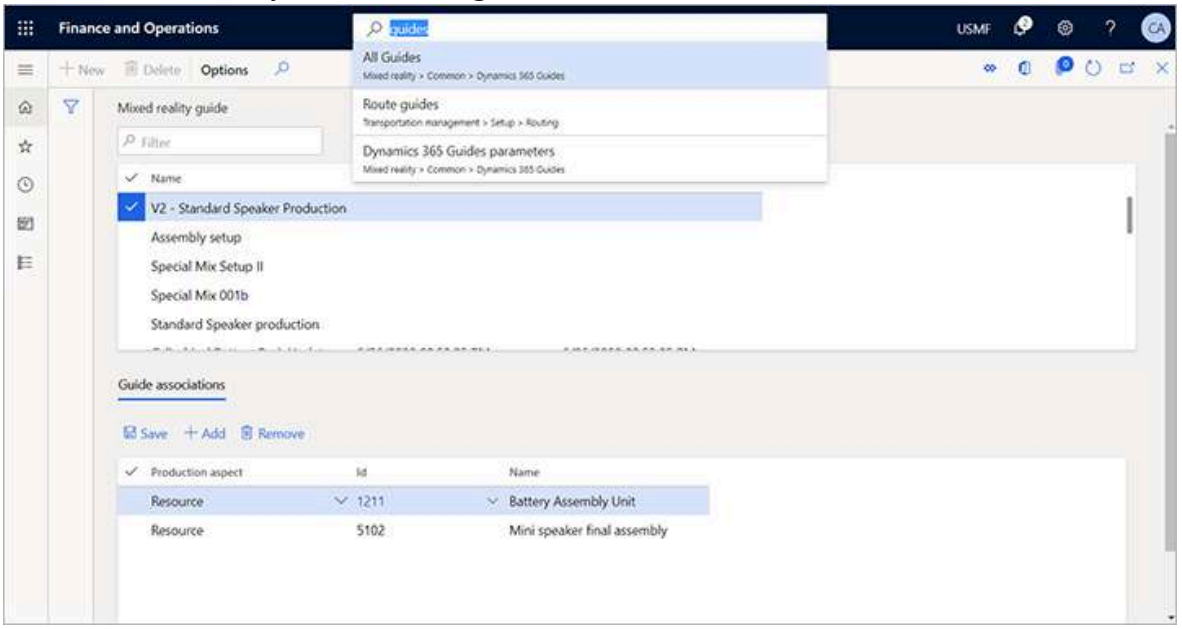
Typical scenario using resources

For example, you could attach a Guide with general machine security or handling instructions to a resource of type machine. Then, the Guide will be available on every job that is performed on the machine.

Add a Guide to a resource

To add a Guide to a resource:

1. Go to **Production control** > **Setup** > **Resources** > **Resources**.
2. From the list pane, select the resource you want to assign a Guide to.
3. Expand the **Associated Guides** FastTab.
4. Select **Add** from the **Associated Guides** toolbar. A new line is added to the grid.
5. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign. If you have a large number of Guides, then you can filter the list to find the one you are looking for.



Associate a Guide to a resource group

You can add a guide to resource groups if you use them to manage groups of machines, production lines, or work cells.

Typical scenarios using resource groups

Example 1: You have defined a resource group for several machines of the same model. Instead of assigning the relevant handling instruction guide for the machine model to

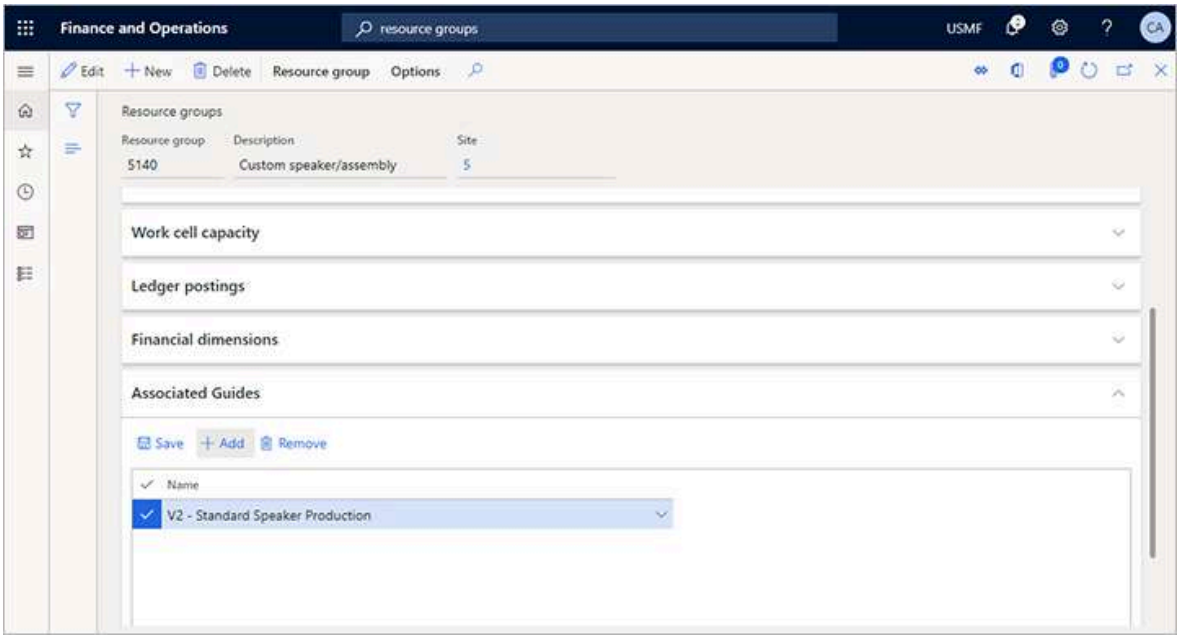
every relevant resource, you could instead assign the Guide to the resource group that reflects that machine model.

Example 2: You have defined a resource group for a work cell that contains different machines and you have a Guide that provides general instructions for how to maintain the work cell. The Guide applies to any production activity in this work cell.

Add a Guide to a resource group

To add a Guide to a resource group:

1. Go to **Production control** > **Setup** > **Resources** > **Resource groups**.
2. From the list pane, select the resource group you want to assign a Guide to.
3. Expand the **Associated Guides** FastTab.
4. Select **Add** from the **Associated Guides** toolbar. A new line is added to the grid.
5. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign. If you have a large number of Guides, then you can filter the list to find the one you are looking for.



Associate a Guide to a released product

You can add a guide to any released product.

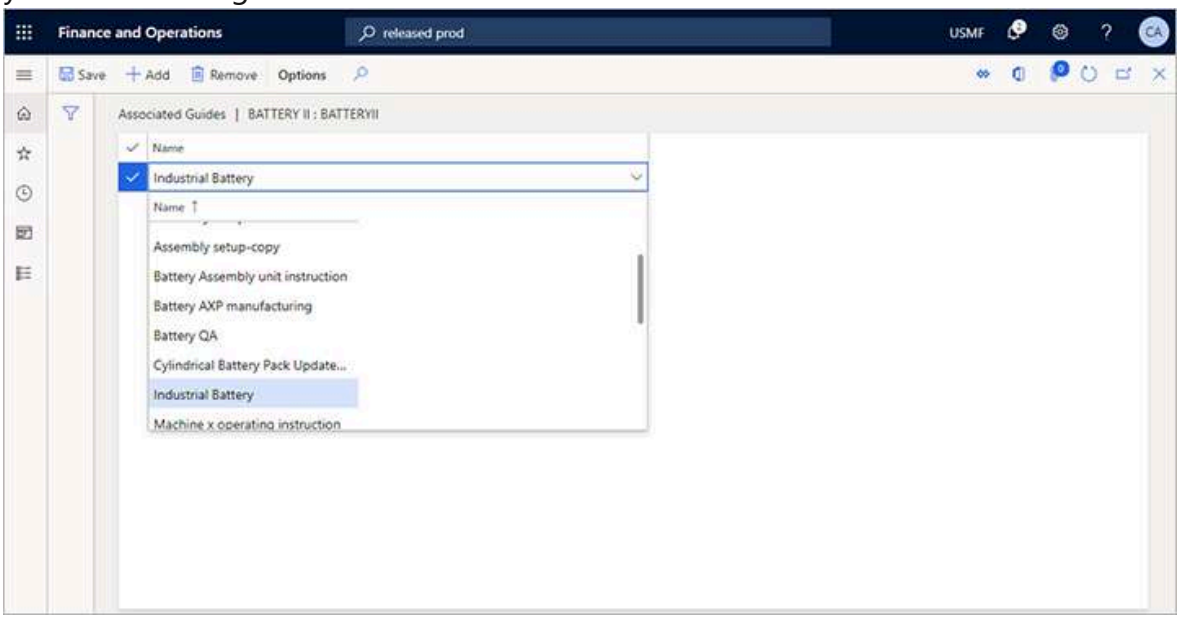
Typical scenario using released products

Product-level Guides help shop floor workers with instructions relevant to operating or handling a specific released product or item.

Add a Guide to a released product

To add a Guide to a released product:

1. Go to **Production information management** > **Products** > **Released products**.
2. Open the product you want to assign a Guide to.
3. On the Action Pane, open the **Engineer** tab and from the **View** group, select **Associated Guides**.
4. The **Associated Guides** page opens for your selected product.
5. Select **Add** on the Action Pane to add a new line to the grid.
6. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign.



Associate a Guide to a formula

You can add a guide to any formula.

Typical scenario using formulas

Formula-level Guides provide shop floor workers with guided handling instructions in the context of a formula or recipe. Guides can also be assigned to versions of a formula.

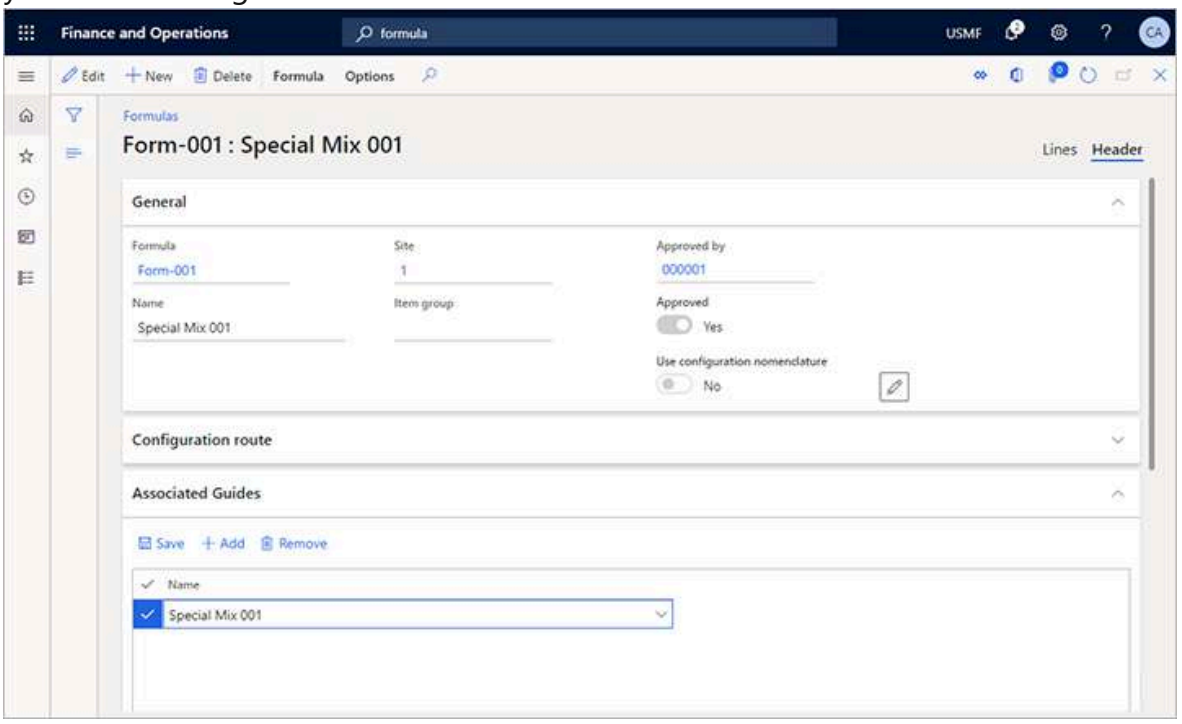
> **Note**
>
> You can assign guidance relevant for production processes based on a formula to a route, route version, or route operation relations.

> Guides can't currently be attached to individual formula lines.

Add a Guide to a formula

To add a Guide to a formula:

1. Go to **Production information management** > **Bills of materials and formulas** > **Formulas**.
2. Open the formula you want to assign a Guide to.
3. Open the **Header** tab above the top FastTab.
4. Expand the **Associated Guides** FastTab.
5. Select **Add** from the **Associated Guides** toolbar. A new line is added to the grid.
6. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign.



Associate a Guide to a formula version

You can add a guide to any formula version.

Typical scenario using formula versions

Guides attached to an individual version of a formula provide shop floor workers with instructions that walk through the production of that version of the formula recipe.

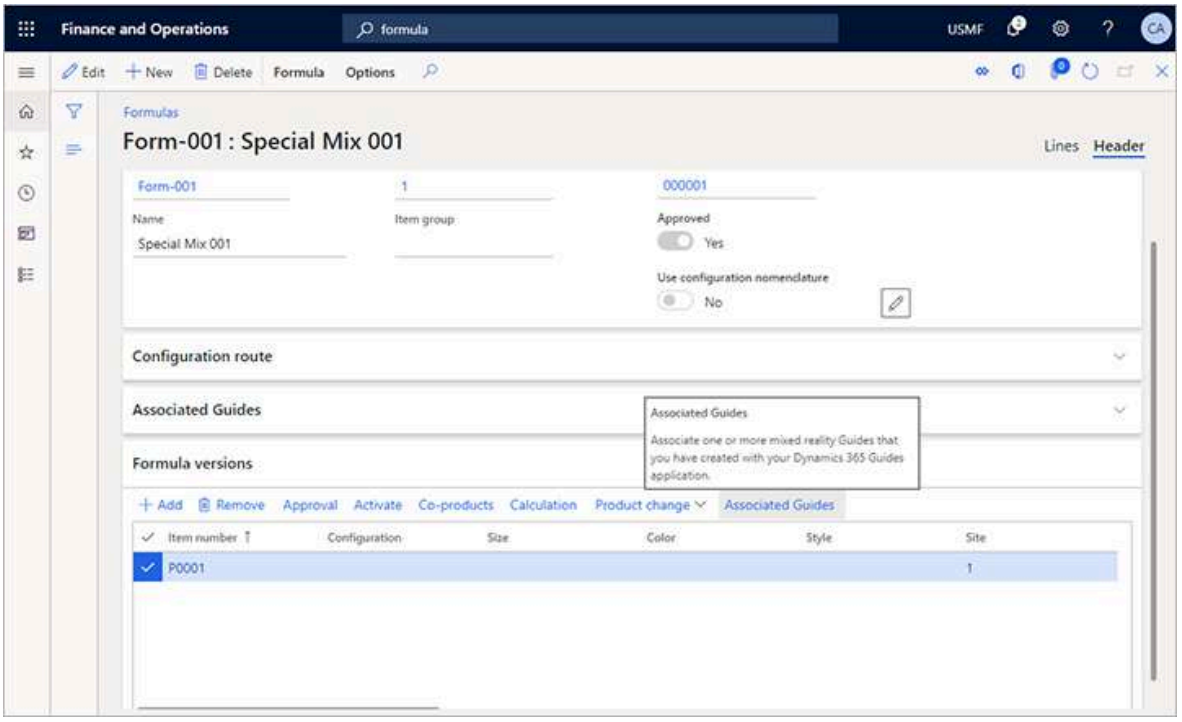
> **Tip**
>
> You can assign guidance relevant for production processes based on this formula version to a route, route version, or route operation relations.

> **Note**
>
> Guides can't currently be attached to individual formula lines.

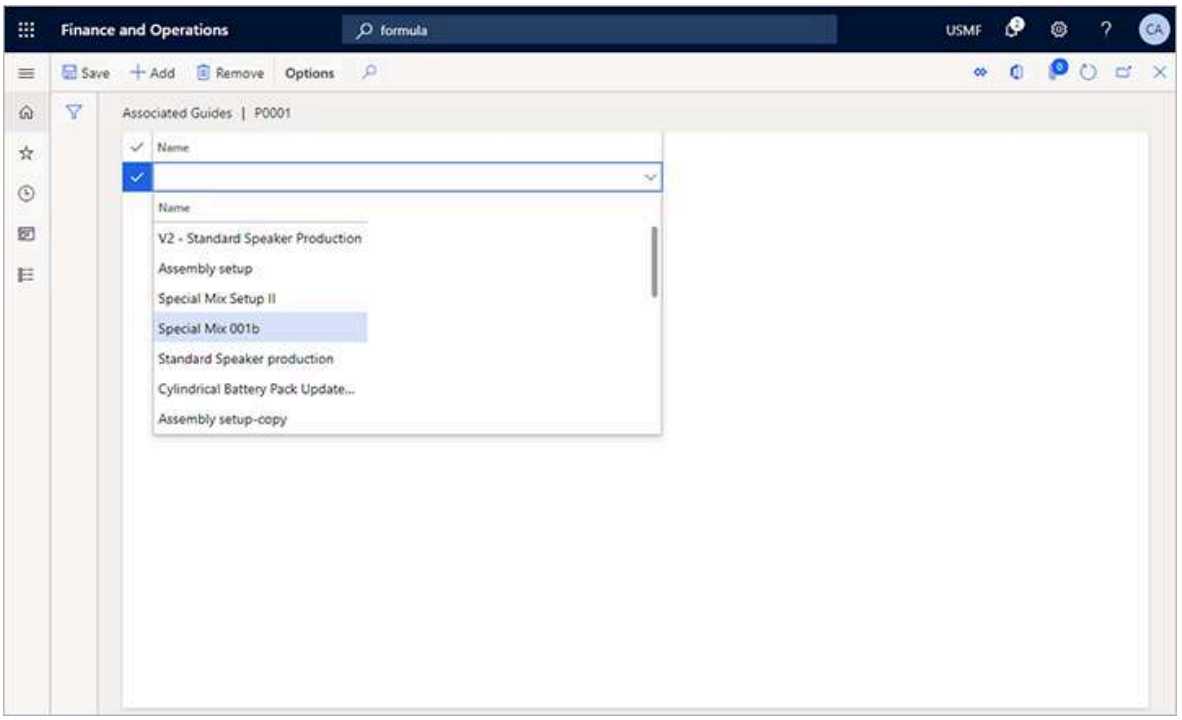
Add a Guide to a formula version

To add a Guide to a formula version:

1. Go to **Production information management** > **Bills of materials and formulas** > **Formulas**.
2. Open the formula that includes a version that you want to assign a Guide to.
3. Open the **Header** tab above the top FastTab.
4. On the **Formula versions** FastTab, select the version you want to assign a Guide to.
5. On the **Formula versions** toolbar, select **Associated Guides**.



6. The **Associated Guides** page opens for your formula version.
7. Select **Add** on the Action Pane to add a new line to the grid.
8. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign.



Associate a Guide to a bill of materials

You can add a guide to any bill of materials (BOM).

Typical scenario using bills of materials

Guides attached to a BOM provide shop floor workers with instructions that explain how to prepare and handle material from a BOM. Guides can also be assigned to versions of a BOM.

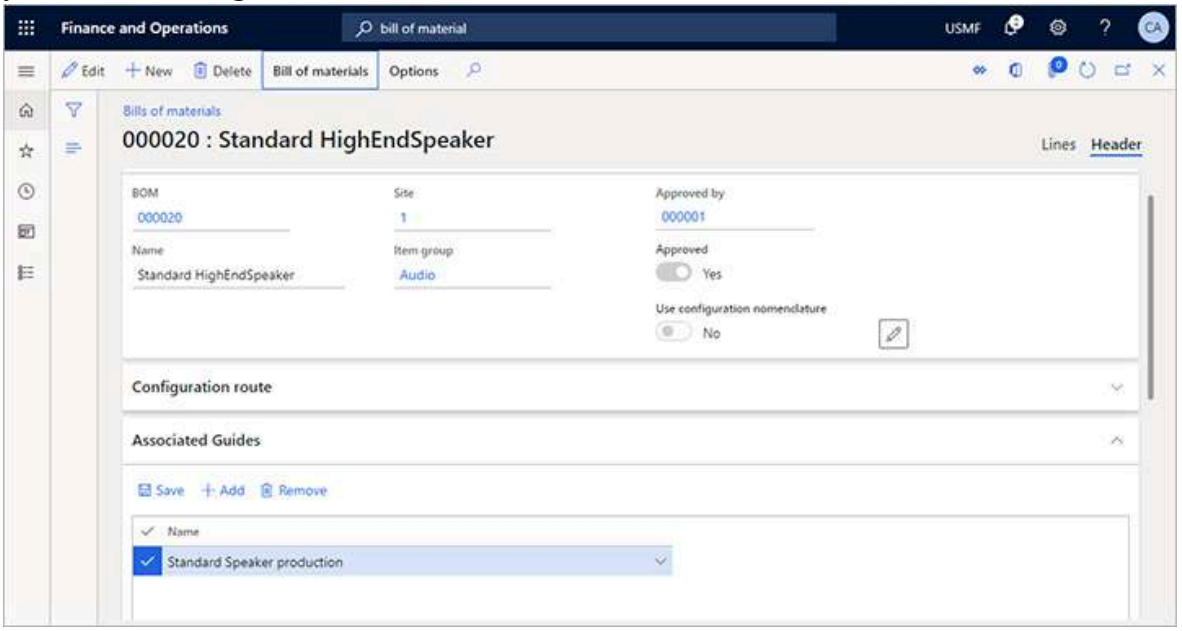
> **Note**
>
> Guides can't currently be attached to individual BOM lines.

Add a Guide to a bill of materials

To add a Guide to a bill of material:

1. Go to **Production information management** > **Bills of materials and formulas** > **Bills of materials**.
2. Open the bill of materials that you want to assign a Guide to.
3. Open the **Header** tab above the top FastTab.
4. Expand the **Associated Guides** FastTab.
5. Select **Add** from the **Associated Guides** toolbar. A new line is added to the grid.

6. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign.



Associate a Guide to a bill of materials version

You can add a guide to any bill of materials version.

Typical scenario using bill of materials versions

Guides attached to an individual BOM version provide shop floor workers with instructions that explain how to prepare and handle material for a version of a BOM that is different from the generic BOM or other versions of it.

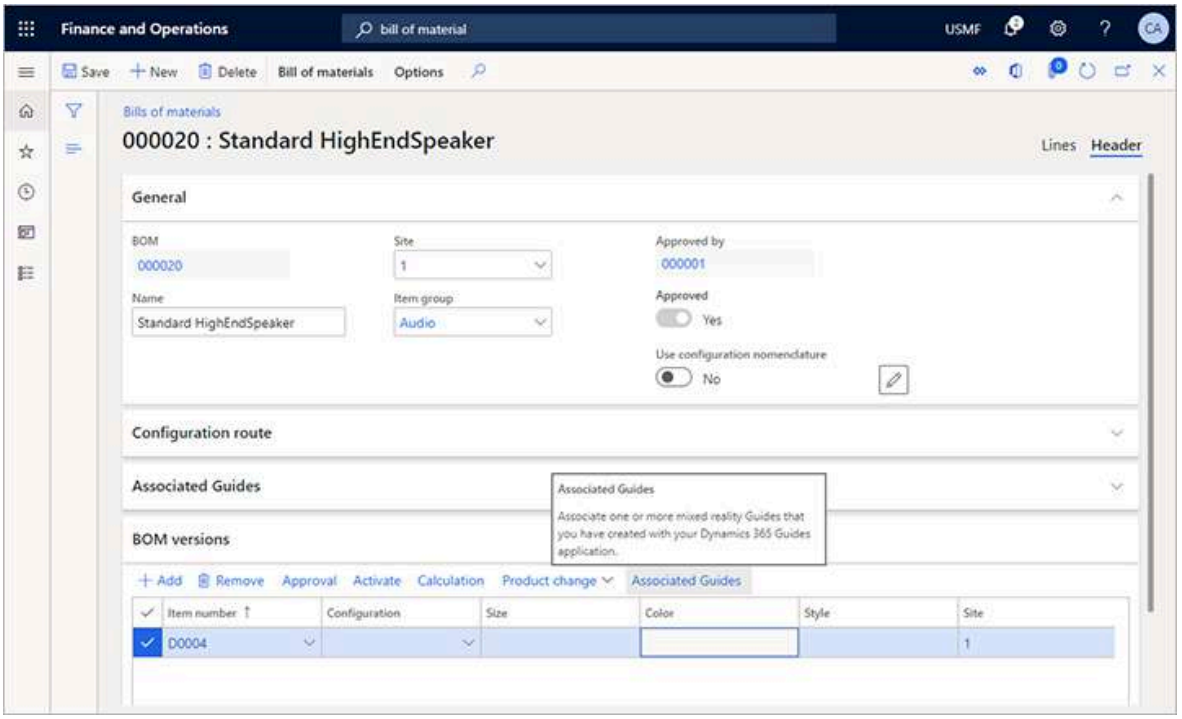
> **Note**
>
> Guides can't currently be attached to individual BOM lines.

Add a Guide to a bill of materials version

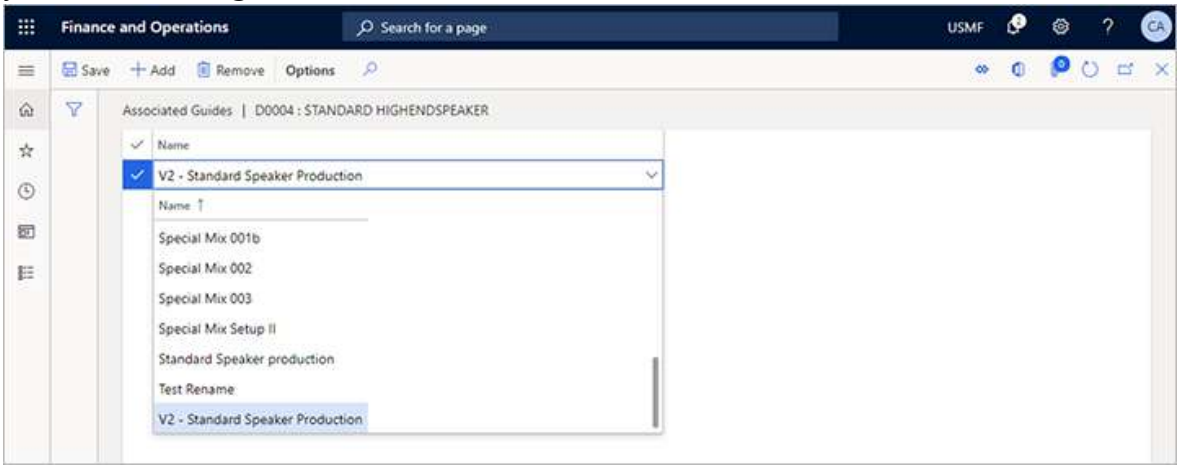
To add a Guide to a bill of materials version:

1. Go to **Production information management** > **Bills of materials and formulas** > **Bills of materials**.
2. Open the BOM that includes a version that you want to assign a Guide to.
3. Open the **Header** tab above the top FastTab.
4. On the **BOM versions** FastTab, select the version you want to assign a Guide to.

5. On the **BOM versions** toolbar, select **Associated Guides**.



6. The **Associated Guides** page opens for your BOM version.
7. Select **Add** on the Action Pane to add a new line to the grid.
8. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign.



Associate a Guide to a route

You can add a guide to any route.

Typical scenario using routes

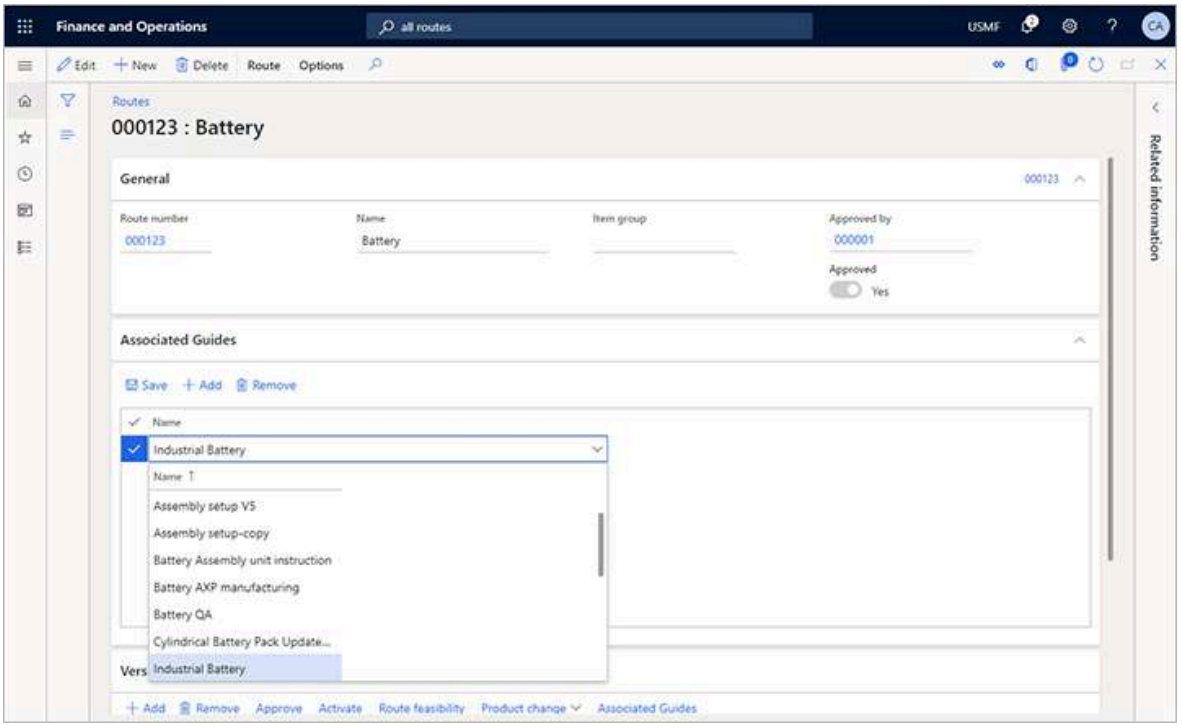
Routes are typically used to specify how a certain released product shall be produced based on a BOM or BOM version and with a set of resources or resource groups.

Assign a Guide to a route to provide step-by-step instructions for the respective production process.

Add a Guide to a route

To add a Guide to a route:

1. Go to **Production control** > **All routes**.
2. Open the route that you want to assign a Guide to.
3. Expand the **Associated Guides** FastTab.
4. Select **Add** from the **Associated Guides** toolbar. A new line is added to the grid.
5. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign.



Associate a Guide to a route version

You can add a guide to any route version.

Typical scenario using route versions

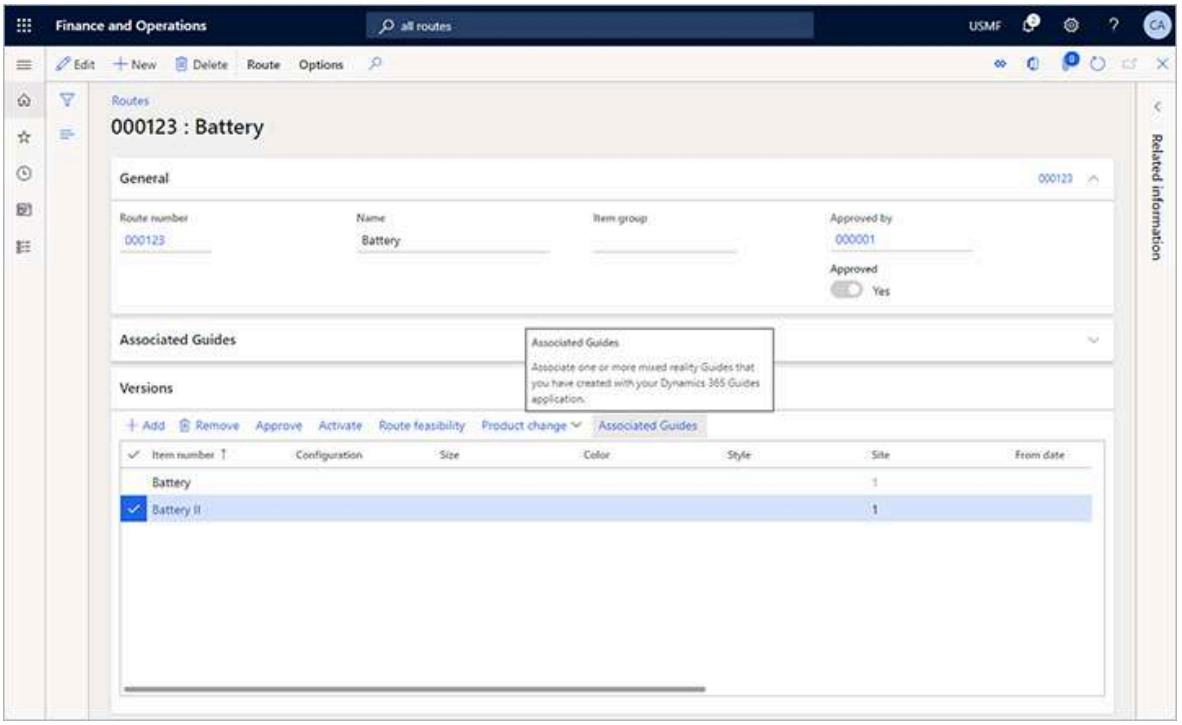
Routes versions are typically used to specify variants of production processes based on an existing route. You can assign different Guides to each route version.

Add a Guide to a route version

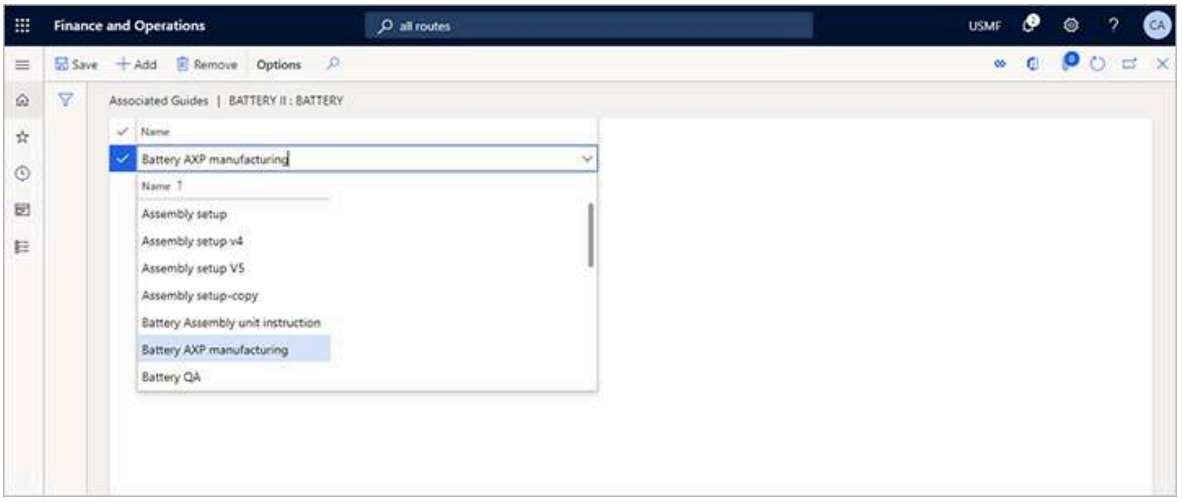
To add a Guide to a route version:

1. Go to **Production control** > **All routes**.
2. Open the route that you want to assign a Guide to.
3. On the **Versions** FastTab, select the version you want to assign a Guide to.

4. On the **Versions** toolbar, select **Associated Guides**.



5. The **Associated Guides** page opens for your BOM version.
6. Select **Add** on the Action Pane to add a new line to the grid.
7. For the new line, use the drop-down list in the **Name** column to choose the Guide you want to assign.



Associate a Guide to a route operation relation

You can add a guide to any route operation relation.

Typical scenario using route operation relations

Operation relations are the most specific way to add guidance to a product process and its related operations. You can specify guidance for each operation in a route and specify different guidance for any type of relation context specified for a route, such as

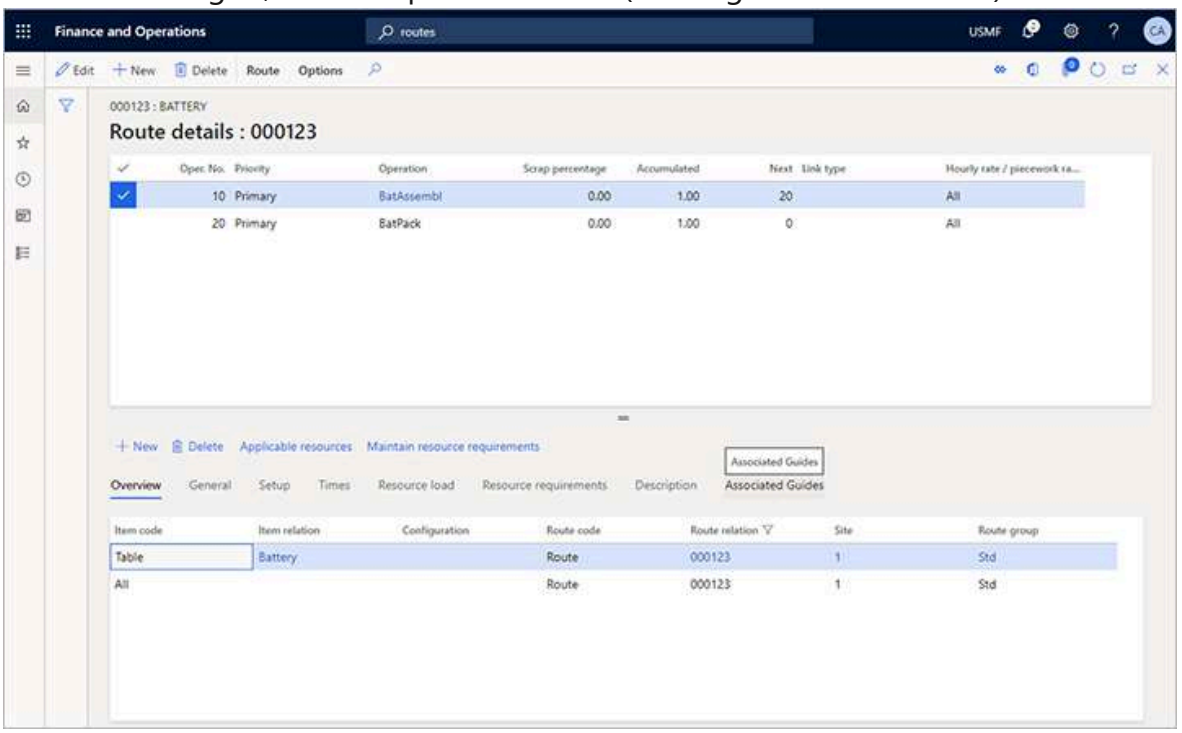
for specific items, configurations, and more. You can also specify to which stages in the operation the guidance applies (such as setup, queueing, process, or transport).

> ⓘ **Note**
>
> If you specify guides for several operation relations of a route, among those guides, only the guide from the most specific relation will be show on the shop floor for the generated job.

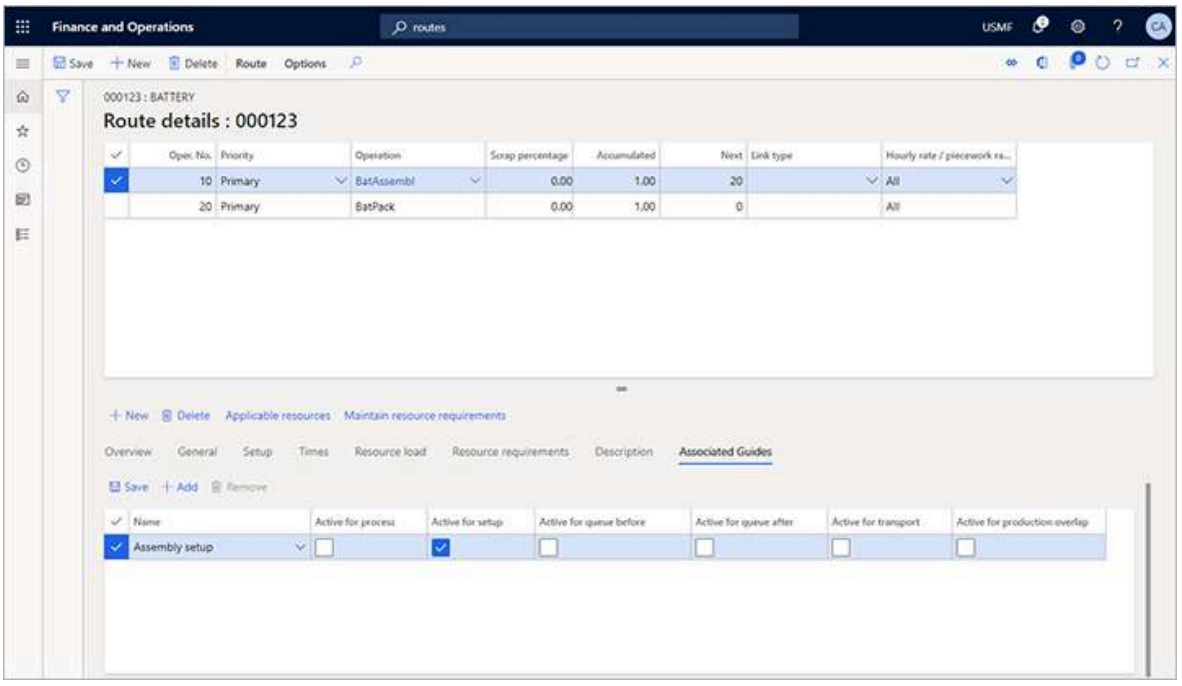
Add a Guide to a route operation relation

To add a Guide to a route operation relation:

1. Go to **Production control** > **All routes**.
2. Open the route that you want to assign a Guide to.
3. On the Action Pane, open the **Route** tab and from the **Maintain** group, select **Route details**.
4. The **Route details** page opens for your selected rout.
5. In the top grid, select the operation you want to provide guidance for.
6. In the bottom grid, select a specific relation (or the generic **All** relation).



7. Above the bottom gird, open the **Associated Guides** tab.

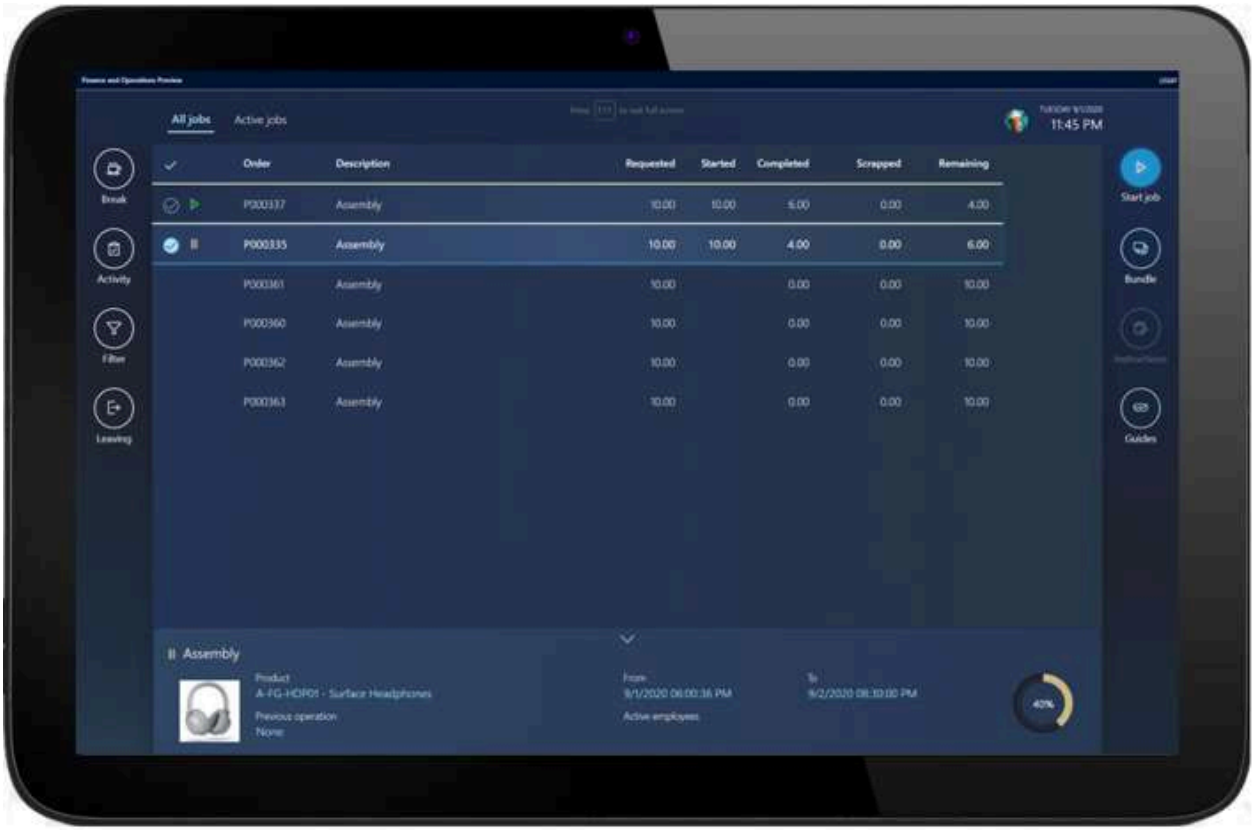


8. Select **Add** from the toolbar at the top of the bottom grid to add a new line to the grid.
9. For the new row, use the drop-down list in the **Name** column to choose the Guide you want to assign. In the rest of the row, select the check box for each context where the selected Guide should be available.

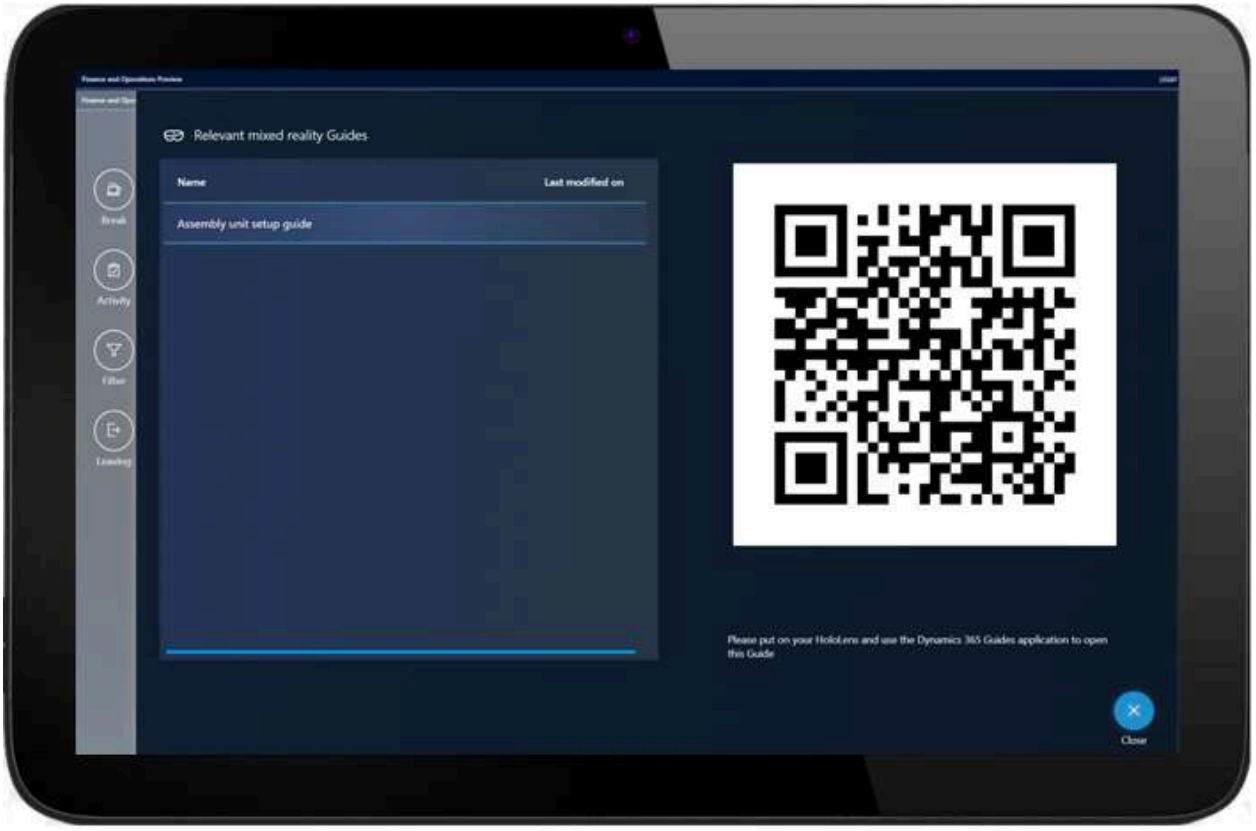
> ⓘ **Note**
>
> You can add one or more guides for each stage of each operation.

Select guides from the shop floor execution interface

When a worker opens a job list on the shop floor execution interface, Supply Chain Management finds the relevant guides for the jobs shown. Use the **Guides** button to view the relevant guides.



Then put on a HoloLens and access the respective guide by glancing at the QR code and activating the respective Guide.



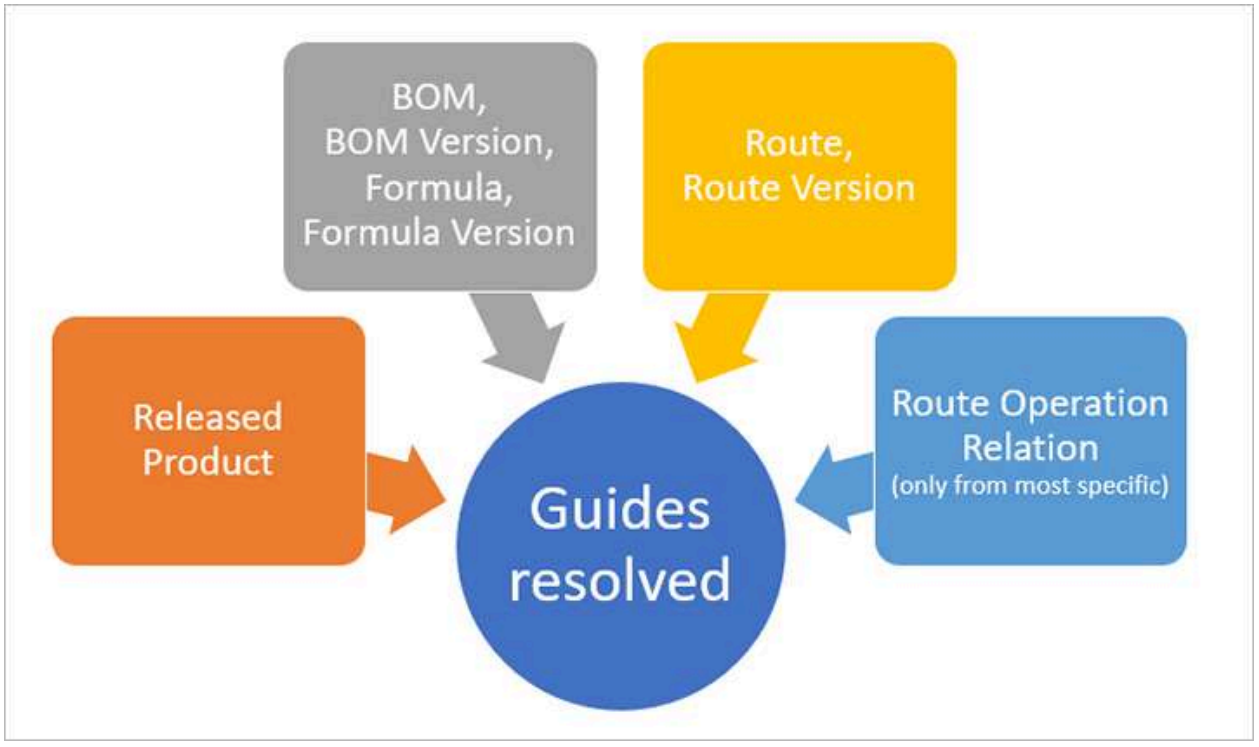
Resolving the logic for selecting Guides

You can add Guides to the following production data:

- Resources
- Resource groups
- Released products
- Formulas
- Formula versions
- Bills of material (BOMs)
- BOM versions
- Routes
- Route versions
- Route operation relations

When Supply Chain Management generates the jobs for the production floor, it will collect the relevant Guides from those sources. Take note of the following important rules.

- If you attach a BOM version or formula version to a route or production order, then any Guides attached to this version, and also the Guides attached to the parent BOM or formula of that version, will be shown on the job.
- If you attach a route version to a production order, then any Guides attached to this version, and also the Guides attached to the parent route of that version, will be shown on the job.
- If you define several route operation relations that include the *All* relation and assign Guides to those, only the Guides from the most specific relation will be shown for the job.



Feedback

Was this page helpful? Yes No

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Reverse the production order status

09/19/2025

This article describes how to reverse production order status.

If you reverse the status of a production order, the order and all operations that are associated with the routes revert to a previous step in the production life cycle. For example, a production order has a status of *Scheduled*, and you change the status back to *Created*. In this case, the system first changes the status to *Estimated*, which is the status that immediately precedes *Scheduled*. It can then change the status to the status that you want, *Created*.

If your order reaches the *Report as finished* status, you can still reverse it to an earlier status. However, you must rerun estimation and operations scheduling, job scheduling, or both types of scheduling, to update the information on the order. This step is required, because any reservations of remaining item consumption and operations resource consumption must also be reset.

This article explains what occurs when you reverse the status of a production order in the following ways:

- From *Estimated* to *Created*
- From *Scheduled* to *Estimated*
- From *Released* to *Scheduled*
- From *Started* to *Released*

From Estimated to Created

When you reverse the status of a production order from *Estimated* to *Created*, the following events occur:

- The system removes the item consumption that it calculated for the items in the bill of materials (BOM).
- The system deletes inventory transactions on the production line, and resets the **Remain status** field on the production order's BOM lines to *Ended*.
- If the **Derived purchases** and **Derived production** options are selected, the system deletes any underlying purchase orders or production orders.
- If you estimated the costs of the production order, or if you manually reserved items so that they could be used in production, the system removes those transactions.

From Scheduled to Estimated

When you reverse the status of a production order from *Scheduled* to *Estimated*, the following events occur:

- The system removes the scheduled start and end dates and times.
- The system removes capacity reservations that were made during scheduling, and deletes jobs that were created during job scheduling.
- The system resets all information that is recorded about operation scheduling and job scheduling on the **Production order details** page.

From Released to Scheduled

When you reverse the status of a production order from *Released* to *Scheduled*, the only change is the value in the status field.

From Started to Released

When you reverse the status of a production order from *Started* to *Released*, the following events occur:

- The system reverts all items that were reported as finished.
- If you picked material, or if you made inbound and outbound deliveries to production, the systems reverses those settings.
- The system changes the **Remain status** field on the production order's BOM lines from *Ended* to *Material consumption*.
- If you registered time, or if you reported quantities as finished for the operations in the production route, the system reverses those settings.
- The system changes the **Remain status** field from *Ended* to *Route consumption* in the production route.
- The system reverses the settings for all items that you posted as in process or work in process.
- On the **Production order details** page, the system resets fields that show a quantity that was started or reported as finished. The system also resets the dates for those transactions.

Visibility into material exceptions

Article • 06/03/2022

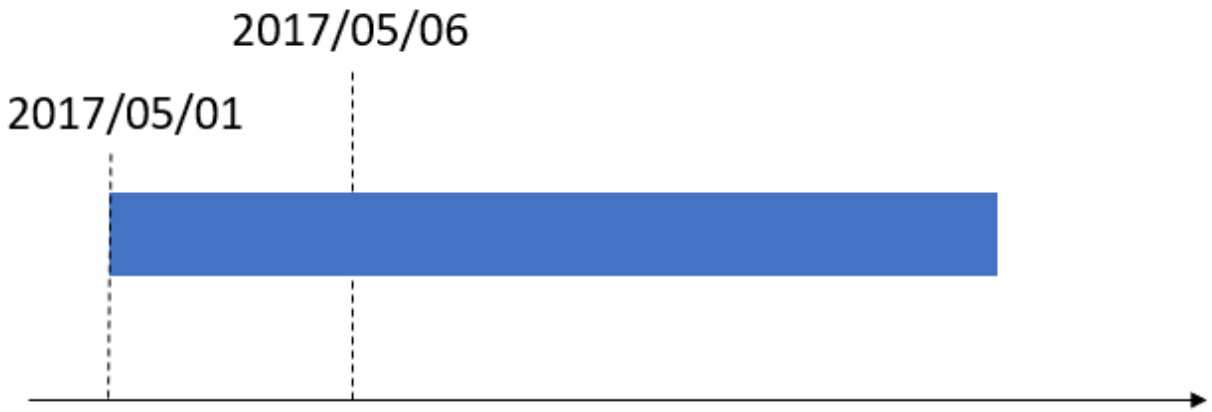
In the **Production floor management** workspace, three tiles give you better visibility into exceptions for raw materials for production orders and batch orders:

- Unreleased material lines needing attention
- Unprocessed waves needing attention
- Open warehouse work needing attention

For all three tiles, the raw material date of the bill of materials (BOM) lines and formula lines is compared against the workspace date, and also against the filters for **Production unit**, **Resource group**, and **Resource** that are set on the **Configure my work space** menu. By default, the workspace date is set to the current date, but you can adjust it.

An unreleased BOM line or formula line requires attention if the raw material date of the line is the same as or earlier than the workspace date, and if it meets the criteria that are defined by the filters in the workspace.

In the following figure, the blue bar represents a production job that is scheduled on a resource. The job is scheduled to start on May 1, 2017 (2017/05/01). This date is the raw material date. In other words, the materials that are assigned to the job on the BOM and formula lines must be ready on this date. The other date in the figure, May 6, 2017 (2017/05/06), represents the workspace date. In this example, the raw material date is earlier than the workspace date. Therefore, the date when consumption of the raw material was supposed to start has passed, and the BOM and formula lines meet the criteria for requiring attention.



Unreleased material lines needing attention

A BOM or formula line can be released to the warehouse in three ways:

- As part of a production order or batch order release
- As a manual release
- Automatically through a batch job

Learn more in Release BOM and formula lines to the warehouse.

If a BOM or formula line hasn't been released or has been only partly released, and if the date and filter criteria of the workspace are met, the line is included in the calculation of the number that appears on the **Unreleased material lines needing attention** tile.

When you select the tile, the **Release to warehouse** page is opened. This page shows the number of unreleased BOM and formula lines that is indicated by the number on the tile. The unreleased lines appear in the upper grid. This grid shows the quantity that was originally estimated for the line, the quantity that has already been released, and the remaining quantity that must still be released. You can add lines from the upper grid to the lower grid. From the lower grid, you can then release the selected lines to the warehouse. From the lower grid, you can also adjust the quantity to release so that only a partial quantity is released.

Unprocessed waves needing attention

When a BOM or formula line is released, it's added to either a new production wave or an existing open wave, depending on the configuration of the production wave template. Through the configuration of the wave template, you can also set up a wave so that it's automatically processed when a BOM or formula line is released. When the wave is processed, warehouse work for raw material picking is generated. If the wave template is configured so that waves aren't processed at the time of release, the wave remains in an unprocessed state. The **Unprocessed waves needing attention** tile shows the number of BOM and formula lines that have been released to the warehouse on unprocessed waves, and that have a raw material date that is earlier than or the same as the workspace date. The lines must also be consumed by an operation resource that applies to the filter of the workspace.

When the tile is selected, the **All production waves** page opens. This page is filtered by the number of open waves that contain wave lines from released BOM and formula lines that meet the criteria for the tile.

Manually maintain production waves

On the **All production waves** page, you can use buttons on **Wave** tab of the Action Pane to manually **Process** and **Release** a wave. You can also use the **Maintain productions**

option to view and maintain **Prod BOM Pool** data, which is used to handle the waving process.

Open warehouse work needing attention

The **Open warehouse work needing attention** tile shows the number of BOM and formula lines that have been released to the warehouse, that have unprocessed work, and that have a raw material date that is earlier than or the same as the workspace date. The lines must also be consumed by an operation resource that applies to the filter of the workspace.

When the tile is selected, the **All work** page is opened. This page is filtered by the number of open work headers that contain work lines from released BOM and formula lines that meet the criteria for the tile. From the **All work** page, you can manually process the work.

Feedback

Was this page helpful? Yes No

Provide product feedback

Release BOM and formula lines to the warehouse

This article describes the process for releasing raw material for bill of materials (BOM) lines and formula lines to the warehouse. When you release a BOM or formula line to the warehouse, the system first determines whether material is already available at the production input location on the shop floor where the material will be consumed for the production process.

- If the material is available at the production input location, it's picked from that location immediately after the signal is given for the release of material to the warehouse.
- If the material isn't available at the production input location, the material release indicates that material must be moved from locations in the warehouse to the production input location. The material is moved via warehouse work for raw material picking. Therefore, warehouse processes for raw material picking must be configured. Learn more in Replenishment overview and Control warehouse work by using work templates and location directives.

Methods for releasing BOM and formula lines

You can configure the release of BOM and formula lines so that it occurs as part of the release of a production order or batch order. Alternatively, the release can be controlled by a batch job or done as a manual interaction.

The method that is used to release BOM and formula lines is controlled by the **Production line release** parameter. You can find this parameter at **Production control** > **Setup** > **Production parameters**.

- **Release BOM and formula lines as part of production or batch order release** – In this method, BOM and formula lines for a production or batch order are released as part of the process of releasing the order. Usually, during the release of a production or batch order, production jobs are released to the shop floor workers, and production papers are printed. During this process, the status of the order is also changed to **Released**.
- **Release BOM and formula lines via a batch job or as a manual interaction** – In this method, BOM and formula lines can be released only through the **Automatic release of BOM and formula lines** batch job or as a manual interaction. To manually release BOM and

formula lines, on the production order list page or the production order details page, on the Action Pane, select **Release to warehouse**.

For a quick demonstration of how to release BOM and formula lines to production by using a batch job, watch this short YouTube video: Release production picking to the warehouse in batch .

Releasing the BOM and formula lines by using a batch job

The **Automatic release of BOM and formula lines** batch job goes through selected BOM and formula lines that have a remaining quantity to release. The job considers only orders that have a status of **Released**, **Started**, or **Reported as finished**. If a BOM or formula line has a remaining quantity to release, the job releases up to the quantity that can be covered by the quantity that has already been physically reserved and the quantity that is physically available.

Example of a batch job release

Expand table

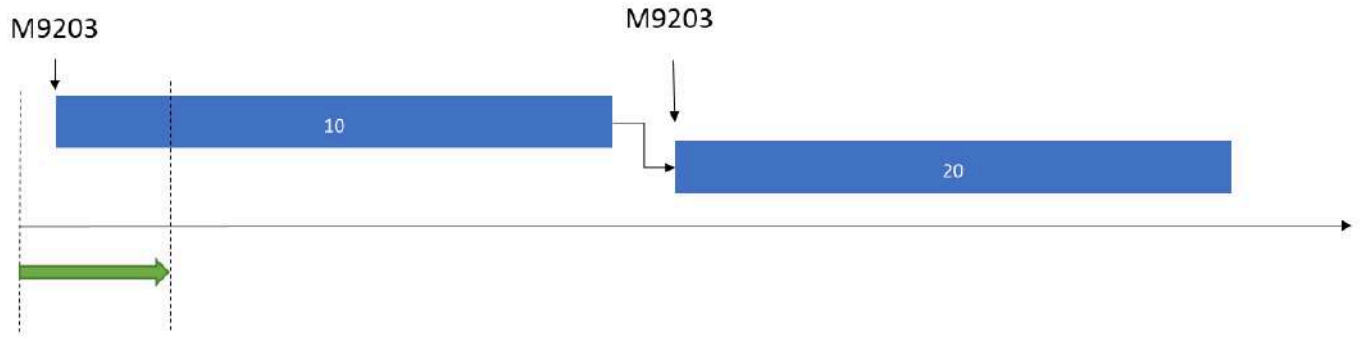
Scenario	Remaining quantity to release	Physically reserved quantity	Physically available quantity	Quantity released by the batch job
1	100	20	90	100
2	100	20	70	90
3	100	0	90	90
4	100	0	110	100
5	100	20	0	20

Batch job setup

In the query for the **Automatic release of BOM and formula lines** batch job, you can set up a filter criterion to specify how many days ahead the job should look for lines that have unreleased quantities. In the query for the job, in the **Raw material date** field, use the **(LessThanDate())** function as a filter criterion.

The following illustration shows a production order that has two jobs, 10 and 20, that cover the assembly and packing for the production order. Each job is set up to consume a quantity of

material. In this illustration, the release time fence that is indicated by the green arrow below the time line equals the number of days that has been specified in the **(LessThanDate())** criterion. For example, **(LessThanDate(2))** indicates that the job should look for unreleased quantities only within a time fence of two days.



Releasing material per operation number or in proportion to the amount of finished goods

If you release materials by using the **On production order release** parameter setting, when you do a manual release, you have two options for controlling the material release:

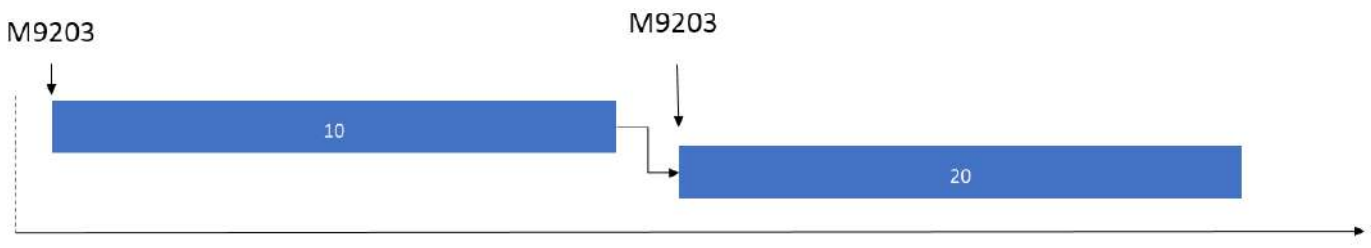
- Release material per operation number.
- Release material in proportion to the amount of finished goods.

Release material per operation number

To control the operations that material should be released to, use the **Release to warehouse** page.

- Select **Production control** > **Production orders** > **All production orders**, select a production order, and then, on the **Warehouse** tab, select **Release to warehouse**. Then use the **From Oper. No.** and **To Oper. No** fields to specify the range of operation numbers.

The following illustration shows a production order that has two operations, 10 and 20. In this example, if you limit the release to operation 10, only material M9203 will be released.



For a quick demonstration of how to release material in proportion to the amount of finished goods, watch this short YouTube video about enhancements to the production order release process.

Release material in proportion to the amount of finished goods

You can release raw material for a partial quantity of finished goods or in a specific unit.

- To release raw material for a partial quantity of finished goods, select **Production control** > **Production orders** > **All production orders**, select a production order, and then, on the **Warehouse** tab, select **Release to warehouse**. Then enter a quantity in the **Quantity** field.

  For example, a production order is created and scheduled for 1,000 pieces (pcs.). The shop floor supervisor is planning the production of 100 pcs. for the next shift and wants to release materials only for that shift. In this case, the supervisor can use the **Quantity** field to release materials for the 100 pcs. that are planned for the next shift.

- To release raw material in a specific unit, select **Production control** > **Production orders** > **All production orders**, select a production order, and then, on the **Warehouse** tab, select **Release to warehouse**. Then use the **Unit** field to select the unit of the finished good to release material in.

  The units that are available are defined in the unit sequence group ID of the finished good.

  For example, a finished good has the following unit conversion between pounds (lbs.) and pallet (PL): 1 PL = 100 lbs. To create a production order for 10,000 lbs. of the finished good, you can release raw materials for the number of pallets that you plan to produce. Select **PL** as the unit, and then select a corresponding number in the **Quantity** field.

Last updated on 07/01/2026

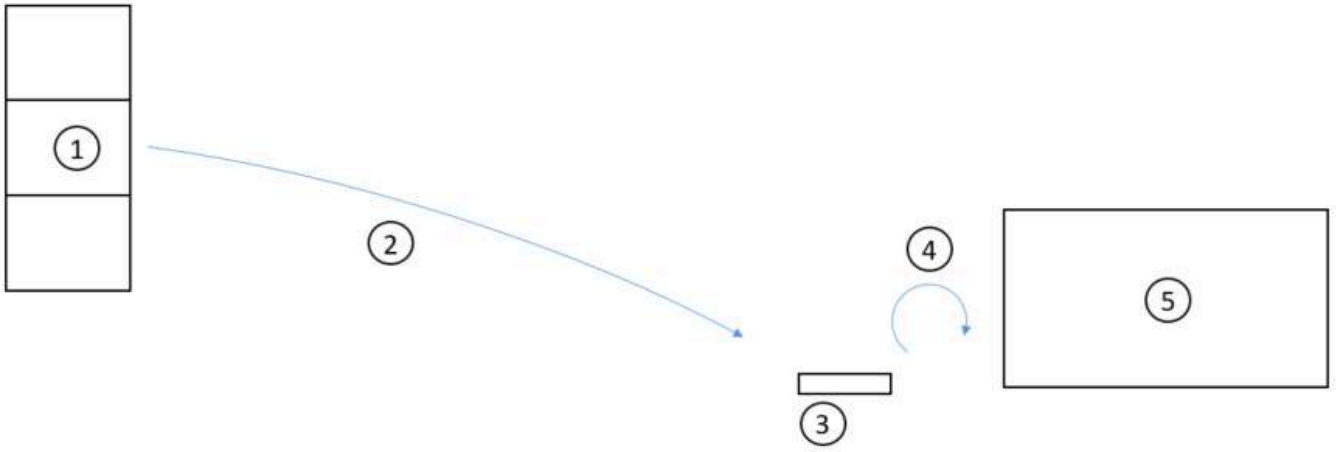
Register material consumption using a mobile device

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes a workflow that enables registration of raw material consumption in production by using a handheld device.

Introduction

This workflow is relevant if there is a strict requirement for material traceability. In those cases, to maintain traceability of the materials, the exact time and quantity must be reported for the consumption. This process can be seen as opposed to pre- or back-flushing operations, where there is an offset between the time of registration and the time when the actual consumption takes place. This explains why a strategy of automatic consumption cannot be used for some materials with traceability requirements. Let's look at a simple scenario that explains how to set up a workflow to enable registration of raw material consumption in production by using a handheld device.



Scenario details

A continuous production process (5) consumes the batch-controlled raw material RM-100. The material is on-hand on location Bulk-001 (1) on license plate PL-1 with two batches, B1 and B2, both with a quantity of 100 lbs. Warehouse work (2) is released and processed for RM-100, and

the material is picked from Bulk-001 to the production input location PIL-01 (3), which is defined as non-license plate controlled. The machine operator weighs out material from the production input location (3) and registers the weight and batch number as consumed (4). From the production input location, a portion of the material is manually added to the production process in defined time intervals. When the machine operator adds material, it is weighed on a scale and the batch number is registered.

Set up the workflow to register consumption using a handheld device

Create a finished-good product, FG-100, with a bill of material that has the batch-controlled raw material RM-100. Add two batches, B1 and B2, of RM-100 in a quantity of 100 to location: Bulk-001 on license plate: PL-1. The flushing principle on the bill of material line for RM-100 is set to **Manual**. Set the production input location to PIL-01. You can do that by selecting this location as the default production input location on warehouse 51.

1. Create a new mobile device menu item:

- **Menu item name** - Register material consumption.
- **Title** - Register material consumption.
- **Mode** - Indirect.
- **Activity code** - Register material consumption.

2. Add the menu item to the **Production Mobile** device menu.
3. Create a production order for the finished product:

- **Item number** - FG-100
- **Site** - 5
- **Warehouse** - 51
- **Quantity** - 150

The production order is **Estimated** and **Released** and warehouse work is created.

4. Complete the work using the workflow for raw material picking for the handheld device.

This will bring the material from the bulk location to the production input location PIL-01. After the work is completed, the material has the status **Picked on the production input location**. The status after work has been processed can be either **Picked** or **Reserved physical**. This is configured with the parameter **Issue status after put on the warehouse form**.

5. Start the production order either from the client or from the handheld device by using the **Production start** menu item.

After the production order has been started, you can register material consumption with the workflow for the handheld device. Let's start by registering consumption of 25 lbs of batch B1.

6. Select the **Register material consumption** menu item in the menu for the hand held device, enter the following details:

- The production order number.
- The location on which the material is going to be consumed, in this case PIL-01.
- Item number RM-100.
- Batch number B1.
- A quantity of 25.

7. Select **OK**.

Note that the message "Journal line is created" appears on the display. On the production order there is an open journal of the type **Production picking list** for item number RM-100 and batch number B1.

You can now choose to continue your registration, for example on batch number B2, and each time you select **OK,** a new journal line is added to the open journal.

After you have finished your registration, select **Done** to post the journal and end the workflow.

Additional comments

- If a user cancels the workflow after a journal line is created, the journal is in an unposted state but if the user at a later point uses the workflow for the same production order, then the lines will be added to the open journal rather than to a new journal.
- The new workflow also supports the registration of serial numbers.
- It is only possible to register an item number that is defined in the bill of material or in the formula for the selected production order or batch order.
- Material can be overconsumed. For example, if the material is estimated to be consumed with the quantity of 100 lbs, then it can be overconsumed with a quantity of, for example, 105 lbs.

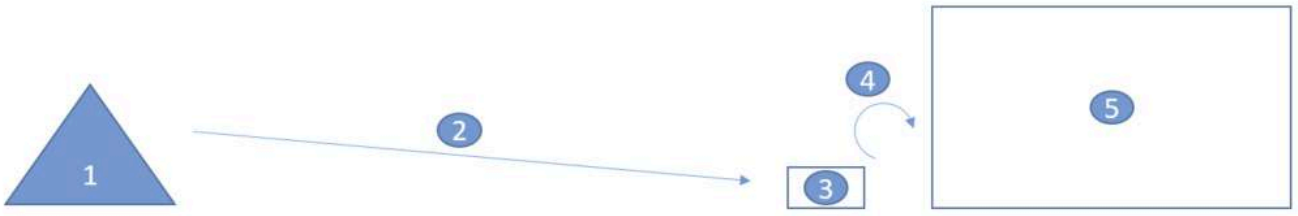
Last updated on 05/07/2025

Control raw material consumption by using flushing principles

07/10/2025

Flushing principles reflect different consumption strategies for raw materials that are used in production processes. Consumption is the process that deducts material from the on-hand inventory and sets the value of the consumed materials to *Work in progress* (WIP) for production orders and batch orders. Raw materials are usually consumed from a location that is configured for the process that consumes the material. This location is known as the production input location.

Before material consumption, the materials are moved to the input location. The following illustration shows the process.



The following elements are shown in the illustration:

1. Material warehouse
2. Raw material picking
3. Production input location
4. Raw material consumption
5. Production process

Material consumption is controlled by the following four flushing principles:

- Manual
- Start
- Finish
- Available at location

The flushing principles are configured in a hierarchy of default values. The hierarchy starts at the released product, where the flushing principle has the value *Start*. On the bill of materials (BOM) or formula line, the flushing principle from the product can be overridden. The default flushing principle on the production BOM lines or batch order formula lines is taken from the product or the overridden value on the BOM or formulas.

Description of the flushing principles

Manual

The *Manual* flushing principle indicates that the registration of material consumption is a manual operation. This principle is relevant if, for example, you want to be able to track time, and the quantity of consumed batch numbers or serial numbers must be accounted for, for tracking purposes. Manual consumption is registered in a production picking list journal. For items that are enabled for warehouse management processes (WMS), a hand-held flow can be applied.

Start

The *Start* flushing principle indicates that material will be automatically consumed when the production order is started. The amount of material that is consumed is proportional to the quantity that is started. When the Start flushing principle is used together with the manufacturing execution system, it can also be used to flush materials when an operation or a process job is started. This principle is relevant if, for example, the variance in the consumption is low, the materials are low-value materials, there are no tracking requirements, or there's a short run time on operations.

Finish

The *Finish* flushing principle indicates that material will be automatically consumed when the production order is reported as finished, or when an operation that is set up to consume the materials is registered as completed. The amount of material that is consumed is proportional to the quantity that is reported as finished. When the *Finish* flushing principle is used together with the manufacturing execution system, it can also be used to flush materials when an operation or a process job is completed. This principle is relevant in the same situations as the Start principle. However, the *Finish* principle is for operations that have a longer run time, where materials shouldn't be set to *Work in progress* before the operation is completed.

> **Note**
>
> You can't use the *Finish* flushing principle together with planning items. Instead, we recommend that you use the *Start* flushing principle. Planning items have a production type of *Planning item*, and only co-products and by-products can be reported as finished on batch orders that are created for planning items.

Available at location

The *Available at location* flushing principle indicates that the material will be automatically consumed when it's registered as picked for production. The material is registered as picked from location when work for the raw material picking is completed, or when material is available on the production input location and the material line is released to the warehouse. The picking list that is generated during the process is posted in a batch job. This principle is relevant if, for example, you have many picking activities against one production order. In this case, you don't have to update the picking list manually, and you can get a current view of the WIP balance.

> **Note**
>
> The *Available at location* flushing principle only applies for products that are enabled for warehouse management processes (WMS).

Production output location

This article describes the hierarchy that is used to identify the production output location.

The production output location is the location where a finished good is first stored after it's produced. Usually, this location is close to the production process that produces the finished good. The production output location is used as intermediate storage for the material before it's moved on to the shipment area, a storage location, a production input location for a downstream production process, and so on.

A default production output location is set when finished goods are reported on a production order or batch order. The following hierarchy is used to identify this output location:

1. Use the output location that is defined on the production order or batch order header.
2. If no location is found there, use the output location that is defined on the resource that is used by the last operation that is defined in the production route.
3. If no location is found there, use the output location that is defined on the resource group that is used by the resource for the last operation that is defined in the production route.
4. If no location is found there, use the output location that is defined on the warehouse that is defined for the production order.

A default production output location is set only for products that are set up by using warehouse management processes (WMS). When this type of item is reported as finished, warehouse work of the **Finished goods put away** or **Co-product and by-product put away** type is created. This type of work uses the production output location as the pick location. The put-away location is determined by the location directives.

Last updated on 07/01/2026

Report BOMs as finished

This article provides information about reporting BOMs as finished.

The **Report as finished** and **Max. report as Finished** pages are used to report bills of materials (BOMs) as finished. Conceptually, the process for reporting a BOM as finished is the same as the process for reporting a production order as finished. This process can be used in, for example, simple assembly and kitting processes, where the more advanced capabilities of production orders aren't required. The **Report as finished** page lets you report multiple BOMs as finished in a batch. The **Max. report as Finished** page lets you report only one BOM as finished at a time. The **Report as finished** page is available from a menu item in Inventory management, and both pages are available as a menu items on the **Released products** page.

Report as finished page

If you open the **Report as finished** page from a released product, the page suggests that you report the standard inventory default quantity as finished. By default, the active BOM version is shown, but you can change the BOM version if there are other approved versions. The page also lets you delete records and create new records for released products that should be reported as finished. To use a query to select products, click the **Select** menu item. You can manually confirm reporting as finished for the selected products by clicking **OK**. Alternatively, you can set up the process to run in a batch. When the report as finished process is confirmed, the system generates a BOM journal where the posting to inventory is processed. This journal consists of one line item for the finished product and a line item for each BOM line. You can control whether the journal is automatically posted or whether it is left open for additional adjustments.

Max. report as finished page

On the **Max. report as finished** page, each BOM line indicates the number of pieces of the product that can be reported as finished. This calculation is based on the physical available on-hand inventory of each material line. In the following example, one piece of item number FG consumes two pieces of raw material RM10 and one piece of raw material RM20. Because there are only 10 pieces of RM10 on hand, the maximum quantity of FG that can be reported as finished is five pieces. This value is shown in the **Max. Report as finished** field.

Expand table

Level	Item number	Quantity	On-hand	Max. Report as finished
0	FG	1	0	5
1	RM10	-2	10	5
1	RM20	-1	8	8

BOMs that have multiple levels

When a BOM has multiple levels, you can control how materials are accounted for at all levels by using the **Explosion** field. This field is available on both the **Report as finished** page and the **Max. report as Finished** page. The following options are available:

- **Never** – Underlying BOMs aren't exploded if there is a material shortage.
- **Always** – All underlying BOMs are fully exploded. Therefore, any free on-hand inventory for semi-finished component items isn't used.
- **Shortage** – Underlying BOMs are exploded only if the full quantity of the material that is wanted isn't available.

Example

In this example, three pieces of the finished product (item number FG) are ready to be reported as finished. There is a two-level BOM, as shown here.

Expand table

Level	Item number	BOM-line quantity	On-hand
0	FG		
1	COMP	1	2
1	RM	1	

The following tables show how the setting of the **Explosion** field affects the way that BOM journal lines are generated. **Explosion: Never**

Expand table

Level	Item number	Quantity
0	FG	3

Level	Item number	Quantity
1	COMP	-3

As the preceding table shows, only item number COMP is considered deducted in the journal. Item number RM, which is part of COMP, isn't exploded to the journal line, and the two on-hand pieces of COMP aren't considered. **Explosion: Always**

Expand table

Level	Item number	Quantity
0	FG	3
1	RM	-3

In this case, item number COMP is exploded into its raw material, item number RM. The two on-hand pieces of COMP aren't considered in the quantity of RM to consume. **Explosion: Shortage**

Expand table

Level	Item number	Quantity
0	FG	3
1	COMP	-2
1	RM	-1

In this case, the two on-hand pieces of item number COMP are considered. However, because three pieces of item number FG are required, one piece of item number RM is also required in order to make the additional one piece of COMP.

Last updated on 07/01/2026

Report as finished to a non-license plate controlled location (Application, May 2016)

Article • 09/29/2021

This task guide shows an example of reporting as finished to a location that isn't license plate–controlled. An applicable work policy is the prerequisite for this task. A previous task guide showed the setup of the work policy. This task guide requires Dynamics AX application 7.0.1 or later.

Set up an output location

1. Go to Organization administration > Resources > Resource groups.
2. In the list, select resource group '5102'.
3. Click Edit.
4. In the Output warehouse field, enter '51'.
5. In the Output location field, enter '001'.

    - Location 001 isn't a license plate–controlled location. You can set up a non–license plate output location only if an applicable work policy exists for the location.

Create a production order and report it as finished

1. Close the page.
2. Go to Production control > Production orders > All production orders.
3. Click New production order.
4. In the Item number field, enter 'L0101'.
5. Click Create.
6. On the Action Pane, click Production order.
7. Click Estimate.
8. Click OK.
9. Click Start.
10. Click the General tab.
11. In the Automatic BOM consumption field, select 'Never'.
12. Click OK.

13. Click Report as finished.
14. Click the General tab.
15. Select Yes in the Accept error field.
16. Click OK.
17. On the Action Pane, click Warehouse.
18. Click Work details.

    - When the production order was reported as finished, no work was generated for put-away. This occurs because a work policy is defined that prevents work from being generated when product L0101 is reported as finished to location 001.

Feedback

Was this page helpful? Yes No

Provide product feedback

Use the safety stock journal to update minimum coverage

Article • 01/06/2025

This procedure shows how to calculate minimum coverage proposals based on historical transactions and then update the item coverage with the proposals. This is done using the safety stock journal. The demo data company used to create this task is USMF. This task is intended for the production planner, to help maintain minimum coverage.

Create a new safety stock journal name

1. Go to **Master planning** > **Setup** > **Safety stock journal names**.
2. Select **New**.
3. In the **Name** field, type *Material*.
4. In the **Description** field, type *Material*.
5. Close the page.

Create a safety stock journal

1. Go to **Master planning** > **Master planning** > **Run** > **Safety stock calculation**.
2. Select **New**.
3. In the **Name** field, enter or select a value. Select the safety stock journal name that you created, for example, Material.
4. Select **Create lines**.
5. In the **From date** field, enter a date.
6. In the **To date** field, enter a date.
7. Select **OK**. This will create lines for the dimensions that have inventory transactions.

Calculate proposal

1. Select **Calculate proposal**.
2. Select the **Use average issue during lead time** option.
3. Set **Multiplication factor** to *10*. The Multiply factor is used to adjust the proposal. Because demo data only has a few transactions, you will need to set the factor to get a realistic proposal.
4. Select **OK**. Scroll down to find M0002 and M0003. View the **Calculated minimum** quantity column.

Update minimum quantity

1. In the **New minimum quantity** field, enter a number. Update the New minimum quantity to match the value in the Calculated minimum quantity. If the Calculated minimum is zero, you can enter the desired future value. For example, you can enter the Calculated minimum quantity in this field for M0002 that has warehouse 12.
2. In the list, find and select the desired record. For example, you can select M0002 that has warehouse 12.
3. In the **New minimum quantity** field, enter a number. Update the New minimum quantity to match the value in the Calculated minimum quantity. If the Calculated minimum is zero you can enter the desired future value.

Post the new minimum quantity and validate the result

1. Select **Post**.
2. Select **OK**.
3. Select to follow the link in the **Item number** field.
4. Select to follow the link in the **Item number** field.
5. On the Action Pane, select **Plan**.
6. Select **Item coverage**. Notice that the **Minimum quantity** has been updated with the new minimum quantity from the safety stock journal.

Feedback

Was this page helpful? Yes No

Provide product feedback

Material substitution in manufacturing

09/19/2025

This article describes how to substitute materials during the production process. There are three methods for doing so:

- By date, when one material is substituted for another after a specific date
- During master planning, when a material in a formula is substituted with a different material because it's out of stock
- During production, when a material is unexpectedly out of stock and is substituted with a different material

Substituting material by date

Consider the following scenario: A machine that a company is manufacturing contains a component that expires from the vendor's catalog in two months. From the expiration date onward, the vendor offers a new component that you can substitute for the old component. Set up **From** and **To** validity dates on bill of materials (BOM) lines. For this example, set up the old component to expire by entering the expiration date in the **To-date** field. Then, on the BOM, set up the new replacement component so that it's valid from the day after the old component expires. To do this, enter the start date in the **From-date** field.

Substituting material by planning

You can substitute materials during planning only when you're using formulas, not when you're using a BOM. Consider the following scenario: A food manufacturing company is making a sauce from a formula that has 20 ingredients. One ingredient in the formula can be substituted by one of two other ingredients. However, because these two ingredients are more expensive than the preferred ingredient, substitution is used only if the preferred ingredient is out of stock. The material that you can substitute is called A, whereas the two materials that can replace it are called B and C. Material substitution by planning is controlled by the **Plan group** and **Priority** fields on the formula lines. For this example, you create formula lines for the three materials, and associate the formula lines with the same plan group. In the setup, the formula line for material A has the highest priority (lowest number), the formula line for material C has the lowest priority, and the formula line for material B has a priority that is between the priority of the other two lines. If you have demand for the finished item, master planning first determines whether the demand for material A can be covered. If the demand can't be covered, master planning looks at materials B and C, in order of priority. If material B is on hand, it will be used after a planned batch order is firmed for the formula. If none of the materials are on hand, master planning creates a planned order for material A.

> **Tip**
>
> When you set up formula lines in a plan group, specify a quantity only on the material that has the highest priority. This quantity calculates the demand of all materials in the plan, even the materials that have lower priority. You can't specify different quantities on lower-priority items in the plan group.

Substituting material during production

Consider the following scenario: A piece of metal plate is required for a welding operation. During the operation, a warehouse worker informs the machine operator that the plate is out of stock. However, it's decided that the plate can be substituted with a plate that is slightly thicker. That way, the operation can be finalized. You can add the material to the production picking list. Then, when the picking list is posted, the system adds the material to the production BOM.

Production feedback

Article • 06/21/2024

This article provides information about production feedback, which lets workers provide feedback about production jobs. The article includes information about the various ways that production feedback can be updated.

Production feedback gives workers the opportunity to provide feedback about production jobs. It lets workers record time and material consumption on production orders, operation quantities and status, and errors that cause a job or operation to fail. Production feedback can be updated in journals that are related to production orders. The **Production job card** and **Production route card** journals are used to report time and quantities per job or operation. For reporting about the last job or operation, quantities on the finished product can be reported as finished.

Workers can also provide production feedback using the production floor execution interface, which is part of the manufacturing execution functionality in the **Production control** module. The production floor execution interface has a configurable user interface that shows a list of the released jobs in a prioritized order for a selected work area. It also offers advanced options such as job bundling and team work. The production floor execution interface has a touch-optimized user interface. Feedback submitted through the production floor execution interface updates the production journals.

Feedback

Was this page helpful? Yes No

Provide product feedback

Allocate time to jobs in a job bundle

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In Manufacturing execution, you can bundle jobs. You can then start multiple jobs at the same time on the Job list page.

If you bundle jobs, you must define how the total registered time for all the jobs should be allocated to each job. You define the allocation by selecting one of the following options in the **Bundle type** field on the **Allocation keys** page:

- **Estimation** – Time is divided among the jobs, based on the estimated time for the jobs.
- **Jobs** – Time is divided according to total jobs that are bundled and how much time was spent finishing all the jobs.
- **Net time** – Time is divided equally among the jobs that are in the bundle at any time.
- **Real time** – Actual job time is allocated. The cost can be calculated based on the actual payroll cost. **Note:** The **Real time** allocation key is available only if your company uses the payroll functionality in Time and attendance.

The following examples show the results of the various allocation keys.

Example scenario

You need to complete three jobs in your job queue. You start the first job, and while that job is in progress, you start the second and third jobs. Therefore, there's a bundle of three jobs. The following table shows the estimated production time for each job.

Expand table

Job	Production time
Job 1	1 hour

Job	Production time
Job 2	3 hours
Job 3	4 hours
Total	8 hours

The following table shows the actual work hours that you spend on each job.

Expand table

Job	Start time	End time	Bundle time
Job 1	09:00	11:00	2 hours
Job 2	10:00	13:00	3 hours
Job 3	10:00	15:00	5 hours
Bundle	09:00	15:00	6 hours

The following sections describe the results of the calculated time for each allocation key.

Estimation allocation key

The following table illustrates the formula for calculating allocated time. Here is the formula: Time per job = Total bundle time × (Estimated job time ÷ Total estimated time)

Expand table

Job	Formula	Allocated time
Job 1	6 × (1 ÷ 8) hours	0.75 hour
Job 2	6 × (3 ÷ 8) hours	2.25 hours
Job 3	6 × (4 ÷ 8) hours	3.00 hours

Jobs allocation key

The following table illustrates the formula for calculating allocated time. Here is the formula: Time per job = Total bundle time ÷ Number of jobs

Expand table

Job	Formula	Allocated time
Job 1	6 ÷ 3	2 hours
Job 2	6 ÷ 3	2 hours
Job 3	6 ÷ 3	2 hours

Net time allocation key

The following table shows the formula for calculating allocated time. The formula is: Calculated time per reporting = Bundle time ÷ Number of jobs

Expand table

Example	09:00–10:00 (1 hour)	10:00–11:00 (1 hour)	11:00–13:00 (2 hours)	13:00–15:00 (2 hours)	Allocated time
Number of jobs in the bundle	1	3	2	1	Not applicable
Job 1	1 ÷ 1 = 1 hour	1 ÷ 3 = 0.33 hour	Not applicable	Not applicable	1.33 hours
Job 2	Not applicable	1 ÷ 3 = 0.33 hour	2 ÷ 2 = 1 hour	Not applicable	1.33 hours
Job 3	Not applicable	1 ÷ 3 = 0.33 hour	2 ÷ 2 = 1 hour	2 ÷ 1 = 2 hours	3.33 hours

Real-time allocation key

To calculate production costs based on real costs, clear the **Cost category** option on the **Production order defaults** page. The following table shows the formula for calculating allocated time. The formula is: Actual time per job = Actual time in bundle

Expand table

Job	Actual time
Job 1	2 hours
Job 2	3 hours
Job 3	5 hours

Consider the three jobs that a worker performs with an hourly wage of USD 12.00. The worker doesn't earn any overtime bonus or premium for the time spent on the jobs. The worker spends six hours working on the three bundled jobs. Therefore, the salary cost is 6 × USD 12.00 = USD 72.00. When you use real-time allocation, the system recalculates the cost per hour by using the factor from the Net time formula. The system then transfers the actual time that was spent on each job together with the corrected cost price per hour. In the example, six hours are spent, although 10 hours were allocated. The following table shows the formula for calculating cost. The formula is: Cost per hour = (Total bundle time per job (Net time) ÷ Actual time per job) × Standard cost price per hour

Expand table

Job	Calculation of corrected cost per hour	Corrected cost per hour	Allocated time	Total cost of job
Job 1	(1.33 ÷ 2) × USD 12.00	USD 8.00	2 hours	USD 16.00
Job 2	(1.33 ÷ 3) × USD 12.00	USD 5.33	3 hours	USD 16.00
Job 3	(3.33 ÷ 5) × USD 12.00	USD 8.00	5 hours	USD 40.00

The corrected cost per hour and the job time are posted in a production journal.

> **Note**
>
> If you select the **Cost category** option on the **General** tab on the **Production order defaults** page, the system transfers the actual time for each job to a production journal, where the cost is applied to the cost category of the specific job.

Last updated on 05/27/2026

Calculate material consumption

08/13/2025

This article provides information about various options that are related to the calculation of material consumption.

The following options that are related to the calculation of material consumption are available on the **Setup** and **Step consumption** tabs on the **Line details** FastTab of the **Bill of materials** page.

Variable and constant consumption

In the **Consumption is** field, you can select whether consumption should be calculated as a constant quantity or a variable quantity. Select *Constant* if a fixed quantity or volume is required for the production, regardless of the quantity that is produced. Select *Variable*, which is the default setting, if the required amount of material in the finished goods is proportional to the number of finished goods that are produced.

Calculating consumption from a formula

In the **Formula** field, you can set up various formulas for calculating material consumption. If you use the default value, *Standard*, the consumption isn't calculated from a formula. The following formulas work together with the **Height**, **Width**, **Depth**, **Density**, and **Constant** fields:

- Height * Constant
- Height * Width * Constant
- Height * Width * Depth * Constant
- (Height * Width * Depth / Density) * Constant

Rounding up and multiples

Together, the **Rounding up** and **Multiples** fields let you round up the material consumption value. For example, you can round up the value according to the handling unit in which the raw material is picked for production. The following options are available in the **Rounding up** field: *Quantity*, *Measurement*, and *Consumption*.

Quantity

If you select *Quantity* as the rounding-up mechanism, the quantity must be a multiple of the specified quantity. For example, if whole numbers are required, select *1* in the **Multiples** field.

Numbers are then rounded up to a quantity that is divisible by 1.

Measurement

Typically, you select *Measurement* as the rounding-up mechanism when the raw material comes in specific dimensions. For example, a piece of 2-meter metal tube is required for a finished good, and the metal tube is stored in 4.5-meter lengths. In this case, the *Measurement* rounding-up mechanism can be used to calculate how many metal tubes are required to produce a specific number of pieces of the finished good. For this example, the **Formula** field is set to *Height * Constant*. The **Height** field is set to *2* to indicate the length of the tube that is required for the finished good. The **Multiple** field is set to *4.5* to indicate that the tube is picked in lengths of 4.5 meters. Here's the calculation:

1. Number of multiples that are required for 10 pieces of the finished good: 10 ÷ 2 = 5 pieces
2. Total consumption: 4.5 × 5 = 22.5 meters of metal tube

It's assumed that 0.5 meters of tube is scrapped for every five pieces of tube that are consumed.

Consumption

Typically, you select *Consumption* as the rounding-up mechanism when raw material must be picked in whole quantities of a specific handling unit of the product. For example, 2 quarts of paint are used to produce one piece of a finished good, and the paint is picked in 25-quart cans. In this case, the *Consumption* rounding-up mechanism can be used to round up consumption to whole numbers of 25-quart cans. Here's the calculation for the amount of paint that is required if 180 pieces of the finished good must be produced:

1. Paint that is required, excluding scrap: 180 × 2 = 360 quarts
2. Number of cans: 360 ÷ 25 = 14.4, which is rounded up to 15
3. Paint that is required, including scrap: 15 × 25 = 375 quarts

Step consumption

Step consumption is used to calculate constant consumption in quantity intervals. If you select *Step consumption* in the **Formula** field on the **Setup** tab, you can add information about the steps on the **Step consumption** tab. The fixed consumed quantity can be set up in intervals of the produced quantity. For example, step consumption is set up as shown in the following table.

Expand table

From series	Quantity
0.00	10.0000
100.00	20.0000
200.00	40.0000

The bill of materials (BOM) quantity is 1, and the production quantity is 110. The formula for the consumption is From series (Quantity) = Consumption. Because the production quantity is 110, it falls into the "From 100 series." Therefore, the quantity is 20.

Production posting

This article provides information about different types of postings in the production process.

Production posting activities follow production processes that are described in the sections below.

Material consumption

Materials are registered as consumed during production when the production picking list journal is posted. This process generates issue transactions that deduct the on-hand inventory. In the production parameters, you can specify whether the value of raw materials that are in progress (work in process [WIP]) should be posted in the ledger. The value of raw materials that are in progress (WIP) is then posted to a dedicated Picking list account and a dedicated Picking list offset account.

Time consumption

The time that workers spend on production jobs is recorded in the Route card journal or the Job card journal. When these journals are posted, ledger posting to a dedicated account for resources that are in progress (WIP) is processed. This posting represents the value of the time that is spent on the production order. After the production order is registered as ended, the WIP accounts are settled.

Reporting finished goods and error quantities

When a production order is reported as finished, the quantity of the finished goods that have been completed is updated in Inventory management through the Report as finished journal. If you're using WIP accounting, which can be set up in the production parameters, a ledger journal is made to reduce the WIP accounts and increase the inventory of the finished goods. The journal uses the standard cost that is defined for the product.

Ending the production order

Before a production order is ended, actual costs are calculated for the quantity that was produced. All estimated costs of material, labor, and overhead are reversed and replaced with actual costs. The overall cost of the finished item is debited from the inventory Receipt account

and credited to the inventory Issues account. If you select the **End job** check box when you run the cost calculation, the status of the production order is changed to *Ended*. This status prevents any additional costs from being unintentionally posted to a completed production order. You can specify that the value of error quantities that are reported during reporting as finished should be allocated to the good quantities that are reported as finished. Alternatively, you can specify that the value of error quantities should be posted to a dedicated scrap account.

Controlling the level of ledger posting

In the **Production control parameters**, you can use the **Ledger posting** field to set the level of ledger posting for production processes. The following values are available:

- *Item and resource* – Use the ledger accounts that are set up on the item groups for raw materials and finished goods. WIP for time consumption is taken from resource or resource group from the route operations.
- *Item and category* – Use the ledger accounts that are set up on the item groups for raw materials and finished goods. WIP for time consumption is taken from the cost categories that are associated with the route operations.
- *Production groups* – Use the ledger accounts that are set up on the production groups for both material and time consumption. The production groups are associated with the released products and copied to the production orders when those orders are created. The posting on the production orders will then follow the production groups that are associated with the production order.

> **Note**
>
> If the standard method for calculating the cost of the finished item was used, the final transactions reflect this fact. If actual costs and the costs that are calculated by using the standard method differ, the difference is posted to the account that shows profit or loss.

Last updated on 11/26/2025

Lean manufacturing overview

Article • 01/31/2025

This article provides an overview and description of the lean manufacturing features in Dynamics 365 Supply Chain Management.

Lean manufacturing offers tools that you can use to model lean operations. These tools support and promote the following concepts and business activities:

- Create a lean manufacturing foundation by modeling manufacturing and logistics processes as production flows.
- Implement a lean pull system by using kanbans to signal demand requirements.
- Monitor and maintain kanban jobs.

The lean manufacturing architecture consists of production flows, activities, and kanban rules. These structures are fully integrated with Supply Chain Management processes. You can use lean manufacturing in a mixed-mode manufacturing environment that combines various supply, production, and sourcing strategies. These strategies include production orders, batch orders for process industries, purchase orders, and transfer orders.

> **ⓘ Important**
>
> You can use Supply Chain Management to support the implementation of lean manufacturing with kanbans. However, a successful implementation of lean principles depends on the internal business processes that you use, and the actual production conditions and environment. |

Modeling manufacturing and logistics processes as production flows

To create a lean manufacturing foundation, model the manufacturing and logistics processes as production flows. This activity consists of the following tasks:

1. Identify the processes for which you want to implement a lean replenishment strategy, and then model these processes as production flows. You can then analyze and streamline the processes. One of the goals of a lean implementation is to reduce waste by improving the flow of material and information.
2. Define a production flow as a sequence of activities that describes the flow of material. A transfer activity defines the movement of a product or products from

one location to another. A process activity defines a value-added operation that is applied to a product.

3. Create a version of the production flow that defines the requirements for takt time. These requirements are used to calculate the cycle times of each activity in the production flow. You can create multiple versions of production flows, and then use these versions to improve processes.

Using kanbans to signal demand requirements

A pull system produces goods only when goods are needed. This practice reduces delivery lead times and excess inventory. You can use kanbans to plan, track, and process requirements that are based on production flows. To create a kanban framework, create kanban rules that define when kanbans are created, and how the requirements are fulfilled. You can create two types of kanban rules. Manufacturing rules create process kanban jobs, and withdrawal kanban rules create transfer kanban jobs. You can set up the following replenishment strategies:

- **Fixed quantity** kanban rules are related to a fixed number of handling units, which means that the numbers of active kanbans are constant. Whenever all the products from a Kanban are consumed and the handling units are manually emptied, a new kanban of the same type is created. When you create fixed quantity kanban rules, you can calculate the optimal kanban quantities and the product quantities that are used. The calculation takes into account forecast, actual demand from open orders, lead time to replenish items, and historical demands.
- **Scheduled** kanban rules replenish requirements that are calculated by master planning. Master planning generates planned kanbans that can be firmed to kanbans.
- **Event** kanban rules replenish requirements that originate from sales order lines, production bill of materials (BOM) lines, kanban lines, or minimum inventory settings. When event kanbans are generated, they're pegged to the source requirements.

When kanbans are created, one or more kanban jobs are generated based on the kanban flow activities that are defined in the kanban rules.

Monitoring and maintaining kanban jobs

Lean manufacturing provides visibility into the current status of manufacturing and logistics activities that are governed by the kanban rules. As a result, you can plan and prioritize the following tasks:

- Gain an overview of the current kanban job schedule.
- Plan and reschedule kanban jobs.
- Track and register the status of kanban jobs.

The following list describes the specialized kanban boards:

- Kanban job scheduling – Provides an overview of the kanban jobs. The board displays kanban jobs and their status for one or multiple work cells. The jobs are listed according to the planning periods (days or weeks) that are defined in the production flow model. The board also displays the capacity consumption for each planning period, so that you can monitor the scheduled load. You can change the status of kanban jobs, reschedule kanban jobs to different planning periods, and perform other tasks.
- Kanban board for transfer jobs – This board provides an overview of the current transfer jobs. You can update and register picking lists, start and complete transfer jobs, and perform other tasks.
- Kanban board for process jobs – This board is designed to support the normal production flow and give an overview of the current situation in one or multiple work cells. From this board Kanbans can be prioritized, picked, or manufactured. The board is also designed to support bar code scanning for the reporting of Kanbans.

Kanban jobs and integration with Supply Chain Management processes

Kanban jobs are fully integrated with current processes for inventory transactions in Supply Chain Management.

- You can perform picking activities to replenish material that is used to fulfill the requirements of kanban jobs.
- You can print kanban cards, circulating kanban cards, and picking lists to support the use of kanbans. These documents are used to represent, track, and register kanban jobs in the warehouse and on the production floor.
- You can register the picking and transfer activities in inventory by scanning bar codes.

In addition, lean manufacturing supports the purchasing and invoicing processes for services that are referenced by subcontracted activities.

- You can assign purchase agreement lines and services to subcontracted activities.
- You can create periodic purchase orders and receipt advices to support the purchase and invoicing of the services.

White papers

The following white papers explore various aspects of lean manufacturing in Dynamics 365 Supply Chain Management:

- Lean manufacturing: Capable to promise and kanban job scheduling
- Lean manufacturing: Production flows and activities

Feedback

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Activity-based subcontracting

Article • 06/03/2022

This article describes, in detail, how to use subcontracted activities in a production flow for lean manufacturing.

In Microsoft Dynamics 365 Supply Chain Management, there are two approaches for subcontracting: production orders and lean manufacturing. In the lean manufacturing approach, the subcontracting work is modeled as a service that is related to an activity of a production flow. A special type of cost group type that is named **Direct outsourcing** has been introduced, and the subcontracting services are no longer part of a bill of materials (BOM). The cost accounting of subcontracted work is fully integrated into the costing solution for lean manufacturing.

Production flows that involve subcontractors

The basic principle of a production flow doesn't change when activities are subcontracted. Material still flows between locations, process activities convert material to products, and transfer activities move material or products from one location to another. You can model locations and work cells as vendor-managed by assigning the vendor account to a warehouse or to a resource of a resource group.

Based on these capabilities, lean manufacturing doesn't require any specific features in order to support the flow of material and products. All possible scenarios that involve vendors as providers of production or transport services can be modeled based on the architecture of production flow and activities.

For example, a subcontractor works out of a supermarket that is located at the subcontractor. When handling units are emptied at the subcontractor, the kanban cards are returned to the assembly cell together with the next shipment. The supermarket at the subcontractor is then replenished. The transfers to and from the subcontractor can be modeled as explicit transfer activities to support a picking and shipment process. If an explicit registration isn't required in order to support the physical transport, the transfer activities can be omitted.

A subcontractor can be used to load balance the overall capacity of the production flow. For example, a production flow is modeled by using scheduled kanban rules. The planner uses the kanban scheduling board to schedule and load level both work cells on demand. The planner also monitors the consolidated supply schedule for the supermarket on the **Supply schedule** page. Multiple subcontractors can be modeled in one or multiple production flows, and there might be multiple kanban rules that can be

used to supply the same product to the same location through different activities. The planner can convert kanbans to an alternative kanban rule to reschedule a kanban that was originally created for internal production to an alternative process. In fact, the subcontracted nature of the work cell has no impact on the production flow. The same working principle applies for two parallel internal work cells or for two subcontracted cells.

Like any other activity in a production flow, subcontracted activities can consume and supply inventoried, non-inventoried (work in process [WIP]), and semi-finished material and products. In all cases, the processes for scheduling and executing subcontracted activities are the same. Additionally, these process the same as the processes for internal work.

Purchase process for subcontracted activities (services)

The purchase process for subcontracted activities is based on the physical material flow that is registered by kanban job progress, for example, Start or Complete. The financial flow, for example, cost of subcontracted work, is a secondary flow that follows the physical flow. At the same time, the purchase process is an independent process that allows for manual adjustment of the purchase documents at every step. Here is the purchase process for subcontracted activities:

1. Create a purchase agreement. The purchase agreement is created for the service and connected to the activity of the production flow.
2. Create a purchase order. A release purchase order can be created for the service, based on the scheduled kanban jobs. Jobs for the same service can be grouped to purchase order lines by day, week, or month. The purchase order lines can be created at any time after the kanban jobs are created. Purchase order lines can even be created after the fact. This option is usually selected if a subcontractor provides services without additional notice, based on the kanbans or kanban cards that the subcontractor receives. In this case, deviations between the purchase order and the invoice can be minimized.
3. Generate kanban cards, material, and a picking list to send to the subcontractor to prepare for processing. Based on the detailed modeling of the production flow, the preparation is done on the kanban board for process activities by using the picking list and the preparation function. Alternatively, the preparation is done on the kanban board for transfer jobs by using the picking list and start or completion. For inventoried material, both processes can be supported by a WMS-Picking and Shipment process. A bill of lading can be created on demand.

4. Generate kanban handling units and kanban cards. After processing, cards are returned from the subcontractor. Usually, the cards include a delivery note that specifies the physical material that has been shipped. A reference to the provided services isn't required. The arrival of the material or product at the customer is registered on the kanban board, depending on the kanban cards. (Either the kanban board for process activities or the kanban board for transfer jobs is used, depending on the modeled activities.).
5. Create a receipt advisory. The receipt advisory can be used to replace a packing slip document for the received services. Receipt advisories can be generated based on the completed kanban jobs for the subcontracting activity for a selected period. For each job receipt, advisories are created for the related purchase order line. The receipt advisory can be printed and sent to the subcontractor as confirmation of receipt.
6. Generate an invoice.

The process ends when the subcontractor is invoiced for a period. The invoice match is done against the receipt advisories that are created. Because the receipt advisories represent the exact physical receipt of material, the three-way matching is simplified.

Configuring activities for subcontracting

The following sections describe how to configure activities for subcontracting.

Subcontracted services

The payment item that is used in activity-based subcontracting must be a product that has the following properties:

- **Product type:** Service
- **Inventory model group:** Non stocked

This requirement enforces the use of the first in, first out (FIFO) inventory model. **Note:** Cost calculation of the products requires that the standard cost of the service be defined. A purchase agreement with the vendor is required. Otherwise, the service can't be used for activity-based subcontracting.

Subcontracted process activities

To configure a process activity as a subcontracted activity, follow these steps.

1. Configure a subcontracted work cell. To configure a work cell as subcontracted, you must create a resource of the **Vendor** type and associate it with the work cell

(resource group). A runtime cost category of the **Direct outsourcing** cost group type should be assigned to the work cell. The cost categories for setup and quantity aren't required.

2. After a process activity is created and related to a subcontracted work cell, you must configure a service for the activity before the production flow version can be activated. You complete this step on the **Activity details** page. For activities that are associated with a subcontracted work cell, the **Service terms** FastTab is shown. On this FastTab, add a default service that is valid for all output items. If specific output items require different services or different service calculation parameters (for example, a different service ratio), you can add other services to the activity.

Subcontracted transfer activities

A transfer activity is configured as a subcontracted activity, depending on the **Freighted by** setting of the transfer activity. The following options are available:

- **Shipper** – The activity is subcontracted if the transfer from the warehouse is managed by a vendor (as defined by a property of the warehouse). All selected purchase agreements for services must have the same vendor ID as the warehouse.
- **Recipient** – The activity is subcontracted if the transfer to the warehouse is managed by a vendor (as defined by a property of the warehouse). All selected purchase agreements for services must have the same vendor ID as the warehouse.
- **Carrier** – The activity is subcontracted to any vendor that provides the service. To be valid, a carrier must be created for warehouse management and must have an assigned vendor account.

As for process activities, you must configure a default service for subcontracted transfer activities on the **Service terms** FastTab of the **Activity details** page.

Service quantity calculation

The whole purchase process is based on an item reference for a service. This item reference is measured in a unit of measure of a service. Services are usually measured either in the number of services (units) or in time. To calculate the service quantity, based on the registered completion of kanban jobs, you can use the following methods:

- **Calculation that is based on the number of jobs** – One kanban job equals *n* units of service, regardless of the product quantity that is supplied. In lean manufacturing, one job corresponds to one handling unit. This calculation method applies to all services that have a fixed price per handling unit. Therefore, this

method usually applies to transfer activities. However, it can also apply to process activities that process whole handling units.

- **Calculation that is based on the product quantity** – The service quantity is relative to the product quantity that is scheduled/supplied. When the supplied product quantity is calculated, error quantities can be either included or excluded. This calculation method applies to all cases where the service price per unit of processed product is agreed upon.
- **Calculation that is based on the activity time** – The theoretical activity times are calculated, based on the processing time of the activity, the total processed quantity, and the throughput ratio of the processed product. This calculation method applies to services that are paid by the hour and have a variance in time per processed product.

Cost accounting of subcontracted services

When the receipt advisory or a vendor packing slip on a purchase order that was created for a production flow (in other words, a purchase order that was generated based on kanban jobs for subcontracted activities) is posted, the value of the receipt is accounted in the WIP accounts of the production flow. Deviations of invoices are also accounted to the production flow. A cost category for subcontracted work has been introduced. This cost category enable transparent tracking of the value of subcontracted work that is allocated to WIP and consumed per period.

The backflush costing for lean manufacturing at the end of a costing period calculates the actual variances of the products that are produced from the production flow during the costing period.

Modeling transfers as subcontracted activities

People often consider transport nonproductive and think that it adds no value. However, when the cost of subcontracting is compared to the cost of internal production, the cost of additional transport activities must be considered. A production flow that spans multiple locations and requires transport services should model the transport cost as part of the cost of supplying the products to the customer.

Activity-based subcontracting in lean manufacturing lets you integrate carriers and transport vendors that move material and products between the locations of a production flow. By modeling a transfer activity, you can assign a carrier or vendor. The transfer activities/job is based on a service and purchase agreement, and you can create purchase orders and receipt advisories, based on the actual transfer jobs. This functionality is the same as the functionality for subcontracted process activities.

Supply Chain Management now supports BOM calculation that includes transport services, the creation of related purchase orders, integrated receipt registration, and the integration of transport service costs into the production flow costing.

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Modeling a lean organization

Article • 09/29/2021

The article provides information about the key concepts in modeling a lean organization.

Typically, a lean manufacturing scenario is more than just a collection of unrelated kanban rules or material supply policies. The flow of material and products throughout work cells and locations for a specific production or supply scenario can be described as a sequence or small network of process or transfer activities. This sequence or network is known as a production flow.

Production flows in lean manufacturing

In production scenarios that are based on production orders, material is issued to a specific production order. During a sequence of operations that is based on a bill of materials (BOM) and routes, products are created and are finally received at the supplied location. The throughput time of production orders varies from minutes to weeks. All related cost, material, and labor are accumulated on the production order.

To reduce the delivery lead times and excess inventory between work centers that batch production causes, lean manufacturing introduces kanban replenishment and supermarkets in manufacturing and warehouse replenishment. Typically, these features disrupt the production of partially independent kanban cycles. The replenishment of a kanban for a semi-finished product is no longer triggered by an order for a finished product.

To re-establish a production and cost context for the various kanban scenarios that are proposed, activity-based production flows have been introduced as the backbone of lean manufacturing. All kanban rules refer to this predefined structure. The activity-based model supports the setup of a wide range of scenarios. However, this model doesn't add complexity for the shop floor workers, because all scenarios use the same activity-based user interface.

Semi-finished products (non-BOM levels)

Lean manufacturing integrates kanbans for inventoried products and semi-finished products in a single framework, and therefore offers a unified user experience for all cases. Because of this architecture, additional BOM levels no longer have to be

introduced to enable kanbans to be used for semi-finished products. This architecture also helps reduce inventory transactions to a minimum.

Products and material in work in progress

The reduction of batch sizes to the ideal state of a single piece flow in lean manufacturing can cause a dramatic increase in inventory transactions if each picking process or kanban registration causes transactions for the consumed items. The production flow architecture allows for the transfer of material to the production flow, together with the withdrawal kanbans in storage or transport handling unit sizes. The value of the issued material is added to the work in progress (WIP) account that is related to the production flow. This behavior resembles the behavior for material that is issued to a production order. The same principle can be applied to products and semi-finished products. Unless these products are created, transferred, or consumed within a production flow, inventory transactions are optional. After the products are posted to inventory, the WIP account for the production flow is reduced by deducting by the related standard cost.

Value streams and value stream mapping

The architecture of Lean manufacturing is inspired by the five Lean principles that were formulated by Womack and Jones: Customer value, Value stream, flow, pull, and perfection. One approved method for implementing lean manufacturing solutions in the physical world of manufacturing is value stream mapping (VSM). This method was introduced by Rother and Shook in their publication “Learning to See” at the Lean Manufacturing Institute.

The future-state value stream can be modeled as a production flow version. All processes of the value stream are modeled as process activities. Movements or transfers can be modeled as transfer activities if the transfer status must be registered, or if integration with inventory picking or consolidated shipments is required.

The value stream itself is modeled as an operating unit. Therefore, the value stream can be used as a financial dimension.

For more information about operating units, see Create an operating unit.

Costing for lean manufacturing based on the production flow

The periodic consolidation of the cost for a production flow corrects the related WIP account and enables variances to be determined for the products that are supplied by the production flow.

Continuous improvement

To better support continuous improvement, the production flows are implemented in time-effective versions. Therefore, an existing production flow version, together with all related kanban rules, can be copied to a future version of the production flow. In addition, the future-state production flow can be modeled before it's validated and activated for production. To help guarantee a seamless material flow on the transition date and beyond, existing kanbans from old production flow versions are automatically related to the new version.

Simplicity

For the implementation of Lean manufacturing, we choose a production flow and activity approach that enables simple and complex production scenarios to be modeled in a single scalable architecture. A closer look at the activity concept reveals a new simplicity for those users who require it: the shop floor and logistics workers. By reporting against activity-based jobs instead of inventory transactions, a unified user interface for all lean manufacturing variants transfers the business complexity from the user interface to where it belongs: the production flow as the backbone of lean manufacturing.

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Operations resources

07/10/2025

Operations resources perform the activities of a project or a production process. They can be of different types, and can have different capabilities.

Operations resources

Operations resources are the machines, tools, workers, facilities, physical areas, or vendors that perform the activities of a project or a production process. They can be of different types and can have different capabilities.

- **Vendor** – An external resource that performs project activities or production operations. An example is a subcontractor. By linking vendor resources to a vendor account, you can generate purchases for subcontractors, based on the bill of materials (BOM) lines or production lines.
- **Human resources** – A project or production worker that perform an activity, either alone or as an operator of a tool or a machine. If you're using the Human resources functionality, you can link human resources to a worker. The scheduling engine can then allocate the resources, based on the competencies that are defined for the corresponding worker.
- **Machine** – A machine or other production equipment that is required in production.
- **Tool** – An instrument or device that is typically used together with another resource to perform an activity in a project or in production.
- **Location** – A physical location of a specific size that is required in order to perform an activity. An example is an assembly area.
- **Facility** – A building or fixed structure that is required in order to perform an activity.

Calendars

A calendar can be assigned to an operations resource and describes the capacity (in hours) of that resource. The working times of the operations resource are determined by the calendar that is assigned to the resource group that the operations resource is part of. You can set an effective date and an expiration date for the assigned calendar. You can then assign more than one calendar to an operations resource. However, the dates of the assigned calendars can't overlap, and the operations resource can be assigned only one calendar at a time.

> ⓘ **Note**

If there are no effective working calendars for a resource group (for example, if the last assigned calendar or the only assigned calendar has expired), you can no longer access the operations resources that are assigned to the resource group for production planning and operations scheduling. You can also assign a calendar to a resource group to specify the working times for both the resource group and all the operations resources that are assigned to the resource group. The capacity of the resource group is calculated as the sum of the capacities of each operations resource that is assigned to that resource group. The calendar that is assigned to a resource group can change over time.

Resource capabilities

A capability is the ability of an operations resource to perform a particular activity. You can assign one or more capabilities to an operations resource. The scheduling engine will then allocate resources by matching the resource requirements of each activity to the capabilities of the available operations resources. Capabilities can be assigned to all types of operations resources (*Tool*, *Vendor*, *Machine*, *Human resources*, *Location*, or *Facility*). To assign capabilities to operations resources for a limited time, define a start date and an expiration date on the capability assignment. Learn more in Resource capabilities.

Resource groups

A resource group is a set of operations resources that represents the granularity at which you want to schedule resources when you use the operations scheduling method. Therefore, resource groups typically correspond to the physical organization of work cells that is demarcated by yellow lines on the production shop floor. The resource group defines the site, production unit, and warehouse context for operations resources that are assigned to the group. When you assign an operations resource to a resource group, the resource can be scheduled at the site where the resource group is located. You don't have to assign the operations resources that you create to a resource group. However, an operations resource must be assigned to a resource group before it can be scheduled to perform work. An operations resource can be assigned to a resource group for a limited time. You can also assign an operations resource to multiple resource groups, so that you can then share the resource across sites. However, the effective dates and expiration dates can't overlap. In other words, you can't assign an operations resource to two resource groups at the same time. Changes in resource group assignments are effective only for new resource allocations. Capacity reservations for jobs, operations, and project hour forecasts that are already scheduled won't be affected.

Note

When you assign operations resources of the *Vendor* type to a resource group, all operations resources that are assigned to that resource group must be of the *Vendor* type and must be linked to the same vendor account.

Production units

A production unit is an administrative unit that is a collection of resource groups. A production unit can contain multiple resource groups, but a resource group can be assigned to only one production unit. A production unit reflects the physical layout of production resources, and has no effect on transactions or how they're processed. You must associate a production unit with a site. You can also assign a picking warehouse and a storage warehouse to a production unit. You can use a production unit to consolidate and filter production-related data. For example, a shop floor manager can see an overview of the outstanding workload and the available capacity for a particular production unit. You can change the production unit that is assigned to a resource group. You can also delete a production unit. However, these changes to the production unit are effective only for new orders that are created after master scheduling is run. If you want to apply the production unit change to existing orders, you must make this change manually.

Resource scheduling

Operations resources are assigned to activities when a project or a production is scheduled. Two scheduling methods are available: operations scheduling and job scheduling. When you use operations scheduling, each operation or project activity is scheduled on the resource group that contains operations resources that match the resource requirements that are specified for the operation. If a specific operations resource is required for the operation, scheduling reserves capacity only on a specific operations resource in the group. Job scheduling is a more detailed form of scheduling than operations scheduling. It breaks down each operation into its individual tasks or jobs. These jobs are then assigned to the operations resources that will perform them. Scheduling reserves capacity on the resource groups, based on the operation times that are defined on the production route or project activities. If you're working with finite capacity, the schedule depends on the availability of the operations resources that are required in order to complete the activity. For operations scheduling, the capacity of the resource group is the sum of the available capacity of the operations resources that are part of that group. Capacity reservations that already exist for the operations resources are considered unavailable capacity. If there isn't enough available capacity for production, the production orders can be delayed or even stopped. On the **Resources** page, you can define several properties that control how capacity reservations are made:

- **Capacity** – Specify the operations resource's capacity per hour in terms of the capacity unit of measure.
- **Batch capacity** – Specify the maximum number of pieces that the operations resource can process per run.
- **Efficiency percentage** – Specify the efficiency that you expect from the operations resource. The efficiency percentage adjusts the throughput of the operations resource and affects the time that is reserved for the resource. The lead times for the operations that use the operations resource are also adjusted accordingly. Here's the formula that is used for the calculation: Scheduling time = Time × 100 ÷ Efficiency percentage. *Time* includes both the run time and setup time.
- **Operations scheduling percentage** – Specify the maximum percentage of capacity of the operations resource that you want to use in operations scheduling. To allow for flexibility in capacity during job scheduling, you should set this percentage to less than 100 percent.
- **Finite capacity** – Set this option to *Yes* if the operations resource should be scheduled based on the actual capacity that is available, and if existing capacity reservations should be considered. If this option is set to *No*, the operations resource is assumed to have infinite capacity, and the resource might therefore be overbooked.
- **Finite property** – Set this option to *Yes* if you want the operations resource to be scheduled based on the actual capacity that is available with respect to the required working time scheduling properties.
- **Exclusive** – Set this option to *Yes* if you don't want the operations resource to be available for another job or operation until the current production is completed. In this case, the operations resource can't be used even if there are gaps in the resource's run time.
- **Bottleneck resource** – Define the operations resource as a bottleneck resource. A bottleneck resource is scheduled by using finite capacity when the **Finite capacity** and **Bottleneck scheduling** options on the **Master plans** page are selected.

To schedule multiple operations resources at the same time to perform, for example, an operation in production, you must specify the requirements for the various resources by using primary and secondary operations. You can then also reserve multiple operations resources that have different capacity. The operations resource that is scheduled for the primary operation determines the duration of the activity.

Lean work cells

You can specify that a resource group is a lean work cell. The resource group can then be part of a production flow. By specifying a resource group as a lean work cell, you also prevent the resource group and the assigned operations resources from being allocated to the operations of a production order or a project hour forecast. For each resource group that acts as a lean work cell, you must specify the following information:

- **Calendar** – The working calendar of the work cell, which must have the property of a standard workday.
- **Input warehouse and location** – The default location that is used to pick material for an activity.
- **Output warehouse and location** – The default location where all output of the work cell is put.
- **Runtime cost category** – This category must be defined for the work cell if direct labor is tracked in costing.

When a resource group is used as a lean work cell, the capacity of the work cell is specified directly on the resource group. Therefore, you don't have to assign operations resources to the resource group. Only when the work cell is managed by a subcontractor, an operations resource of the *Vendor* type must be assigned to the work cell. If you assign an operations resource to a resource group that is marked as a work cell, the capacity of the work cell isn't affected.

Costing resources

When you define an activity such as a route operation or a project hour forecast, you can specify the requirement for a specific operations resource or resource group. However, you can also specify the requirement for an operations resource of a specific type, or an operations resource that has a specific capability or competency. For this reason, the actual resource assignment isn't made until the activity is scheduled and capacity is reserved. Therefore, on a route operation, you can specify that estimation and BOM calculation must be based on a specific operations resource. This operations resource is referred to as the costing resource. You can also transfer cost categories and operation times from the costing resource to the activity. When the operation is scheduled, estimation and BOM calculation are done by using the operations resource that is actually scheduled.

Define lean manufacturing work cells

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

A work cell is a specific form of resource groups that can be used in lean manufacturing process activities. Work cells have input and output locations and a capacity definition based on a production flow model. The demo data company used to create this procedure is USMF

Create a work cell.

1. Go to Organization administration > Resources > Resource groups.
2. Click New.
3. In the Resource group field, type a value.
    - The work cell ID is typically a systematic code and has to be unique for the legal entity.
4. In the Description field, type a value.
    - The description contains the name or title of the work cell.
5. In the Site field, click the drop-down button to open the lookup.
    - A work cell is located at one specific site. Both input and output warehouse and location have to be located on this site.
6. In the list, click the link in the selected row.
7. In the Production unit field, click the drop-down button to open the lookup.
8. In the list, click the link in the selected row.
    - Select a production unit that this work cell belongs to.
9. Select the Work cell check box.
    - To use a resource group as a lean work cell, the Work cell check box has to be selected. Note that this property cannot be changed after resource group is created.
10. In the Input warehouse field, click the drop-down button to open the lookup.

11. In the list, click the link in the selected row.

    - For accounting and material control, the material staged on the shop floor is typically allocated to a specific virtual warehouse. However, if you want to replenish the locations using warehouse work, they must be part of the receptive raw material warehouse.

12. In the Input location field, click the drop-down button to open the lookup.
13. In the list, click the link in the selected row.

    - Note that for a process activity, the input location can by overwritten in general or for a specific product or product variant by defining picking activities that feed to the process activity. The input locations of a work cell cannot be license plate controlled.

14. In the Output warehouse field, click the drop-down button to open the lookup.
15. In the list, find and select the desired record.

    - In multiple activity production flows or production lines, this is often the input warehouse of the next work cell or the sales or transit warehouse where a product is typically transferred to after the production process. Remember when modeling lean manufacturing processes, transport is usually waste, as is reporting transport.

16. In the list, click the link in the selected row.
17. In the Output location field, click the drop-down button to open the lookup.

    - In a production flow with multiple process activities this if often the input location of the next work cell.

18. In the list, find and select the desired record.
19. In the list, click the link in the selected row.
20. Expand or collapse the Operation section.

    - A Run time category must be provided to enable cost calculation and processing of lean kanban jobs.

21. In the Run time category field, click the drop-down button to open the lookup.
22. In the list, find and select the desired record.

    - The run time cost category is used in standard cost calculation and on backflush costing.

23. In the list, click the link in the selected row.
24. Expand or collapse the Calendars section.
25. Click Add.

26. In the Calendar field, click the drop-down button to open the lookup.
27. In the list, find and select the desired record.

    - Typically work cells of a given site use the same working time calendar. If work cells can have individual working times, you might need to create a specific working time calendar for the work cell. Note that the calendar should have a standard working time defined when used for a lean work cell, because the capacity definition is usually related to the standard working time of a work day.

28. In the list, click the link in the selected row.
29. Expand or collapse the Work cell capacity section.
30. Click Add.
31. In the Production flow model field, click the drop-down button to open the lookup.
32. In the list, find and select the desired record.

    - This procedures requires production flow model type Throughput, to show the definition of throughput capacity.

33. In the list, click the link in the selected row.
34. In the Capacity period field, select an option.

    - The options include: Standard workday - The capacity is expressed by the length of the standard workday of the working time calendar for the work cell. For each day, the actual working time is determined from the calendar and the effective available capacity is calculated based on that. Week - Allows a weekly capacity. There is no adjustment done by the actual working time. Month - Allows a monthly capacity. There is no adjustment done by the actual capacity. Typically, the standard workday is used for daily periods and the weekly capacity is used for weekly capacity periods.

35. In the Average throughput quantity field, enter a number.

    - Note that a lean operation is never set up for the maximum possible capacity in an ideal environment. Instead the capacity should always be defined for operations running under typical circumstances.

36. In the Unit field, click the drop-down button to open the lookup.
37. In the list, click the link in the selected row.
38. ResolveChanges the Unit.

Add a financial dimension

1. Expand or collapse the Financial dimensions section.

- Note that financial dimensions defined on the production flow override the financial dimension of a given work cell. The financial dimensions that can be selected depend on the configuration of the financial dimensions of your system. The following steps correspond to the Demo data set in company USMF. When using different data, the steps might not be applicable.

2. In the CostCenter field, click the drop-down button to open the lookup.
3. In the list, find and select the desired record.

    - The dimensions that need to be selected on lean work cells depend on the implementation of financial dimensions in the accounting model for a specific legal entity.

4. In the list, click the link in the selected row.
5. In the ItemGroup field, click the drop-down button to open the lookup.
6. In the list, find and select the desired record.
7. In the list, click the link in the selected row.

Save

1. Click Save.

Last updated on 06/19/2025

Define lean schedule groups

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Lean schedule groups are defined to group and distinguish products in kanban scheduling. The grouping can be done as generic association per company or specific to a work cell. Each group has a color code assigned for visual indication in the kanban scheduling listpage. The demo data company used to create this procedure is USMF.

Define lean scheduling group

1. Go to Product information management > Lean manufacturing > Lean schedule groups.
2. Click New.
3. In the Schedule group field, type a value.
    - A schedule group can be defined as global group or specific to a work cell. In this simple example, we define a global group, and the work cell is kept empty. The settings of this group apply to all work cells that do not have specific schedule groups.
4. Select a color from the color selection.
    - The colors are used to highlight the jobs on the kanban schedule list page or the kanban process board.
5. In the list, mark the selected row.
6. In the list, click the link in the selected row.

Associate product

1. Associate a specific product
    - There are two ways to associate products to lean schedule groups, either as a specific product (Item relation type = Item) or as part of an item allocation key (item relation type = group).
2. In the Item relation type field, select Item

3. In the Item number field, type a value.
4. In the Throughput ratio field, enter a number.

    - The default Throughput ratio is 1, which means that the related products consume exactly the capacity specified in the process activities of the production flows. Throughput ratio > 1 defines a higher resource consumption, Throughput ratio < 1 defines a lower resource consumption. The ratio is used in the cost calculation and in the calculation of the kanban job consumption.

Associate item allocation key

1. Associate an item allocation key

    - Add an association to an item allocation key by using the Item relation type Group. Note that for this process, you need a forecast item alllocation key defined in your data.

2. In the Item relation type field, select Group
3. In the Item allocation key field, click the drop-down button to open the lookup.
4. In the list, click the link in the selected row.

Last updated on 06/19/2025

Visual scheduling for lean manufacturing

Article • 06/03/2022

This article provides information about the Kanban schedule board, which the production planner can use to control and optimize the production plan for kanban jobs.

This article provides information about the Kanban schedule board, which the production planner can use to control and optimize the production plan for kanban jobs.

The Kanban schedule board lets the production planner control and optimize the production plan for kanban jobs. It makes the flow of kanban jobs transparent, and gives the production planner a tool that optimizes and adjusts the production plan for the lean manufacturing work cell.

Visual scheduling of kanban jobs

A kanban job can consist of one or many kanban jobs. There are two types of kanban jobs:

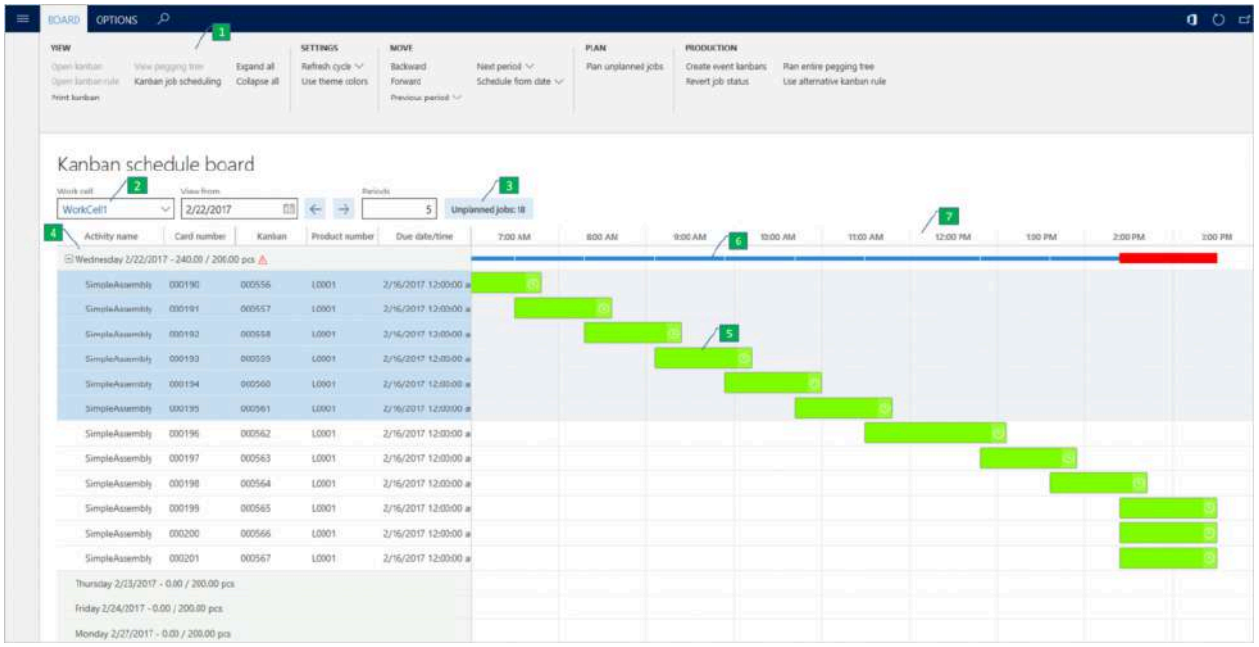
- Process
- Transfer

You can schedule only jobs of the **Process** type. The kanban job and its properties, such as the activity time, are defined in the production kanban flow. In the production kanban flow, the kanban job is also assigned to a work cell. The work cell's daily capacity is calculated based on the work cell capacity that is set on the resource group. It's adjusted by the daily working time in the related calendar. When a kanban job is scheduled, the job loads the capacity of the work cell. The Kanban schedule board provides the following main features:

- A graphical overview of the production plan in a lean work cell. This overview shows the planned kanban process jobs in the defined periods.
- A tool that lets you schedule unplanned kanban jobs and reschedule previously scheduled jobs.

Kanban schedule board

The **Kanban schedule board** page contains seven main elements, as shown in the following illustration.



1. Action Pane
2. Filter fields
3. Button for unplanned jobs
4. Period node
5. Kanban job
6. Capacity bar
7. Time scale

View the time scale

The board is divided into periods, each of which is represented as a node (4). The period nodes are listed on the vertical axis, and the horizontal axis represents a time scale (7) that shows the length of the period. A period has a length of either one day or one week. The period length is determined by the configuration of the work cell that is selected for the Kanban schedule board (2). For each period node, the Kanban schedule board indicates how much the scheduled kanban jobs are loading the period. There is also an indication of the maximum throughput for the period. If the scheduled throughput exceeds the maximum throughput, the period is considered as overloaded, and a red warning symbol appears. A scheduled kanban job appears in a period that has scheduled start and end times (5). The length of the job is equal to the activity time. Kanban jobs appear as overlapping in a period if their activity times exceed the task time of the work cell.

View job status

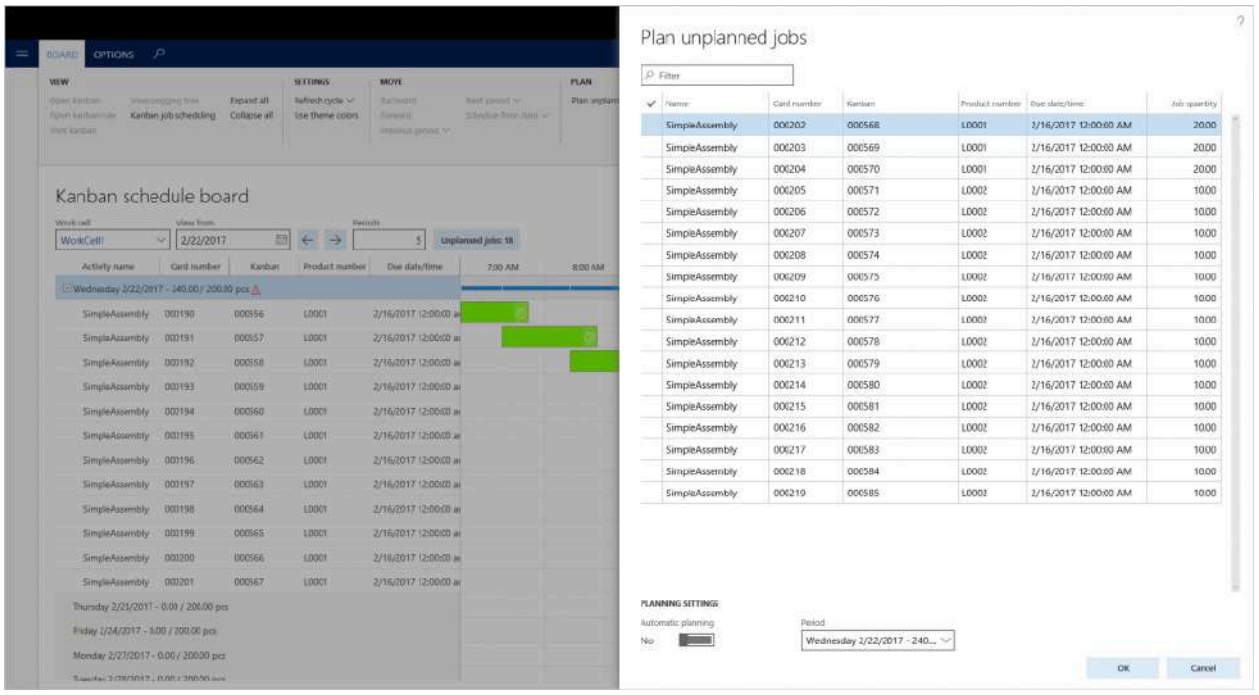
More information about a kanban job is available in the tooltip that appears when you hover the pointer over the job. A symbol provides information about the status of the job. For example, a clock symbol indicates that the kanban job is overdue.

Use colors to view the Kanban schedule board

To enhance the overview that the Kanban schedule board provides, you can use colors to distinguish kanban jobs. The color of a kanban job is configured in the lean schedule group, where you can aggregate the products that should be produced in the same sequence. The **Use theme colors** button on the **Board** tab of the Action Pane lets you switch between the application theme colors and the colors that are configured in the lean schedule group. For each period, a capacity bar (6) indicates how many of the available hours for the period have been loaded with kanban jobs. If the period is overloaded, the capacity bar appears thicker and in red. All these functions are available on the **Board** tab of the Action Pane (1) at the top of the **Kanban schedule board** page.

Plan unplanned jobs

You can schedule unplanned kanban jobs from the **Plan unplanned jobs** dialog box. To open this dialog box, click the **Unplanned jobs** button that shows the current number of unplanned jobs. Alternatively, click **Plan unplanned jobs** on the **Board** tab of the Action Pane. The dialog box shows a list of the unplanned kanban jobs for the work cell. You can use the **Filter** field to filter on all fields in the grid. For example, you can filter on kanban jobs for a specific product. After you have a filtered list of the jobs that you want to schedule, select them in the list, and then click **OK**. To use automatic planning to schedule the jobs, set the **Automatic planning** option to **Yes**. In this case, the jobs are scheduled into a period according to their due date. You can also schedule the jobs by period. Just select a period in the **Period** field. The following illustration shows an example of the **Plan unplanned jobs** dialog box.



Sequence kanban jobs within the same period

You can change the sequence of one or more selected jobs within a period. This capability can be useful if you want to prioritize some jobs within the period. Alternatively, you might want to sequence jobs that have the same product attributes, to optimize job execution. You can change the sequence through a drag-and-drop operation, or by using the **Backward** and **Forward** menu items on the **Board** tab of the Action Pane.

Reassign kanban jobs across periods

Jobs can be reassigned from one period to another. This capability can be useful if a period is overloaded and you want to level the load to periods that have spare capacity. You can reassign jobs through a drag-and-drop operation, or by using the **Next period** and **Previous period** menu items on the **Board** tab of the Action Pane.

Open the Kanban schedule board

You can open the Kanban schedule board by using the menu item on the following pages:

- **Production area** page
- **Kanban jobs scheduling** page
- **Production flow visualization** page

Related information

Kanban job scheduling for lean manufacturing

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Kanban job scheduling for lean manufacturing

Article • 09/29/2021

This article provides information about visual control over kanban job scheduling and various ways to schedule kanban jobs.

The **Kanban job scheduling** page provides visual control over the schedules of lean manufacturing work cells. It gives an overview of all kanban jobs and provides multiple filtering capabilities. From this page, you can move to all other pages that are related to kanban configuration and execution.

Automatic scheduling of kanban jobs

Scheduling can be triggered automatically if you set the **Automatic planning quantity** parameter on the kanban rule. If you set **Automatic planning quantity** to **1**, each kanban job is planned immediately when it's created. The result is a series of first pull, first serve operations. If you set **Automatic planning quantity** to a value that is more than 1, kanban jobs are grouped before they are planned.

This concept enables kanban sizes to be reduced below the actual economic batch sizes. For example, the economic batch size for a specific item (or item family) is 30. Instead of creating kanbans that use the product quantity, 30, you can configure the kanban rule so that it has a product quantity of 10 and an **Automatic planning quantity** value of **3**. Although automatic planning schedules the kanban jobs for the work cell only when three unplanned jobs exist, it's fully transparent to the planner and the shop floor supervisor that two unplanned jobs might be awaiting execution. The planner or shop floor manager can then take those two jobs into production by manually planning them or creating additional kanbans.

Manual scheduling

For manual scheduling, finance and operations introduced the kanban scheduling board. Manual scheduling can be combined with automatic scheduling. The kanban scheduling board lets you plan and unplan jobs, moved them in sequence, or move them from period to period. Jobs that are based on a kanban rule where the **Automatic planning** value is more than **0** can be manually unplanned. However, these jobs will be replanned when the next automatic planning event occurs (that is, when a new kanban is created). The following options are available for manual scheduling:

- **Schedule** schedules the selected jobs according to their due date. (This option resembles automatic planning.)
- **Schedule forward from date** tries to schedule the selected jobs according to their due date but constrains the result by using the specified earliest start date.
- **Backward** moves the selected scheduled jobs back in the sequence within the period.
- **Forward** moves the selected scheduled jobs forward in the sequence with the period.
- **Previous period** moves the selected scheduled jobs to the start or end of the previous period.
- **Next period** moves the selected scheduled jobs to the start or end of the next period.
- **Plan** > **Revert job status** lets you unschedule a scheduled job.

Lean scheduling groups

Each color represents a lean scheduling group. Lean scheduling groups can be freely defined as generic groups or as groups that belong to a single work cell. Items and dimensions can be freely assigned to the scheduling groups. For example, in a Painting cell, a schedule group can represent a color of the product. In work that is driven by specific tooling requirements, items might be grouped by tool requirement, and a packaging work cell might group items by packaging template. The use of colors for lean scheduling groups is optional but recommended. It improves visibility into the status of the plan. For example, it's very easy to see which colors are produced on which day, and you can tell at a glance how this work can be optimized.

Work cell capacity and period capacity

The capacity of a lean work cell is always concurrent capacity. In other words, multiple jobs can be active in a work cell at the same time. The capacity can be tracked in two modes: throughput and hours.

Throughput

The capacity of a work cell is defined by the average throughput quantity of a reference period (standard workday, week, or month). The actual capacity per day or week is then calculated based on the working times of the calendar that is assigned to the work cell. The capacity that is loaded by job is the quantity of the job, adjusted by the throughput ratio of the item in the work cell.

Hours

The available capacity by day or week is defined by the calendar that is assigned to the work cell. The capacity that is loaded by job is the cycle time of the activity, adjusted by the quantity (default activity quantity versus actual job quantity) and the throughput ratio of the item in the work cell.

Period capacity FactBox

The **Kanban job scheduling** list page contains a FactBox that shows the available and booked period capacity for the selected work cell. Depending on the selected schedule periods in the production flow model, the periods show days or weeks.

Related information

Feedback

Was this page helpful? Yes No

Provide product feedback

Job scheduling

09/19/2025

This article provides information about job scheduling, which is a more detailed form of scheduling than operations scheduling. Use job scheduling to schedule individual jobs or shop orders, and to control the manufacturing environment.

You can use job scheduling to schedule individual jobs or shop orders, and to control the manufacturing environment. Job scheduling breaks down each operation into its individual tasks or jobs. Assign these jobs to the operations resources that perform them. Job scheduling also lets you synchronize all jobs that the selected job references. You can specify a start date and time or end date and time for the job, and then run scheduling. The time that you specify can be the start time or the end time, depending on the scheduling direction. This functionality is useful when, for example, a job can run only on one machine at a time, or when you want to optimize the job that runs for each resource.

Tasks in the job scheduling process

The job scheduling process includes the following tasks:

- Split operations into jobs.
- Schedule jobs, based on the dates and times for the resources that are specified for the related operation.
- Calculate start times and end times for each job. Use finite capacity to make sure that there are no overlapping times.
- Determine which resources in the resource group run the job on. This task requires that a resource group be specified for an operation. Job scheduling selects the resources or resource groups based on the shortest lead time, and also considers any previous reservations on the resources.
- Explode operations into jobs when you run job scheduling. The jobs are scheduled by date and time, according to the order that the production route specifies. The setup of the operation determines the jobs that are exploded during the scheduling process. The route group assigned to the operation controls whether jobs are generated. A job is generated only if it has a specific duration. For example, a transport time job is generated if a transport time was specified for the selected operation.

Scheduling direction

You can schedule jobs either forward or backward.

- **Forward** – Use the forward scheduling direction to start the production as early as possible. This method is also known as the push method, because the production is being pushed forward through the production process. The production is scheduled to start and end on the earliest possible dates.
- **Backward** – Use the backward scheduling direction to start the production as late as possible. This method is also known as the pull method, because it's based on the date when the production must be completed. Backward scheduling counts backward to the latest possible date that the production can be started without missing its target deadline.

Finite capacity

You can schedule jobs by using finite capacity. When you use finite capacity, the capacity that is scheduled can't be larger than the capacity that's available for the resource. Available time is defined as the interval when the resource is available and there are no other reservations on capacity. Scheduling that is based on finite capacity makes sure that start times and end times for an operation on a specific date don't overlap. The resource capacity that is already reserved is considered, and overlaps between the start times and end times are also considered. Finite capacity determines the amount of capacity that must be available for a resource in order to achieve optimal use of that resource. This determination is balanced against a calculation of the shortest possible lead time between operations.

Finite materials

Job scheduling that uses finite materials ensures that the required materials are available when the operation starts. The coverage rules for items define these limits. Scheduling uses requirement explosion to determine when the component items are available. If you schedule without setting finite materials, the system assumes that all items are available when they're required.

Finite properties

Job scheduling that uses special properties requires that you specify properties for the operations on the production route. The system must fulfill these properties to reserve capacity.

References

Job scheduling schedules all productions that the current production references. If a production has one or more subproductions, schedule the subproductions at the same time as the main production, because you can't start the main production until the related subproductions are completed.

Schedule resources

The scheduling engine examines combinations of resources to identify those combinations that can satisfy requirements. You can specify selection criteria by selecting one of the following values in the **Primary resource selection** field on the **Scheduling parameters** page:

- *Duration* – The scheduling engine selects the resource that has the shortest lead time.
- *Priority* – The scheduling engine selects the resource that has the highest priority if two or more resources have identical capabilities and levels. The resource that has the lowest numeric value in this field has the highest priority.

> ⓘ **Note**
>
> Scheduling by duration can cause decreased performance when the same resource group contains many resources and secondary operations are used. You can schedule a maximum of 32 resources per operation. If you exceed this quantity, an Action center message is displayed, and job scheduling doesn't find the best alternative resource.

When you run job scheduling, the system plans the resources based on the limitations that you define in the resource parameters. You can control the capacity of the resources by using calendar settings. The system calculates loads for resources during the scheduling process.

> ⓘ **Note**
>
> For productions that use the operations scheduling function, you can run job scheduling after operations scheduling. If you aren't using operations scheduling, you can run job scheduling alone.

Maximum capacities for resources per job order

Job scheduling assigns resources to jobs. You can set maximum capacities for resources per job order. For example, you can set up the system to schedule no more than 50 percent of total capacity for a production order. This setup gives you more control over the scheduling of

resources on the job scheduling level. Therefore, it can help prevent issues if not enough capacity is available to perform simultaneous productions.

Resource efficiency

Job scheduling uses the efficiency percentages that you specify for the resources. Efficiency percentages reduce or increase the time that the system reserves for the resource. Therefore, lead time also increases or decreases. The system uses the following formula for the calculation: `Scheduling time = Time × 100 ÷ Efficiency percentage`. In this formula, *Time* includes both the run time and the setup time.

Kanban transfer board support for bar code scanners

The Kanban transfer board supports scanner input from a widget bar code scanner to Select, Start, Complete, and Empty a kanban job.

Registration modes

On the **Scanner registration** FastTab you can select the registration mode, which controls the action when you scan a kanban card number or manually type the number in the Kanban card number field.

Expand table

Set registration mode	Description
Start	Registers a Kanban transfer job as in progress.
Complete	Registers a Kanban transfer job as completed.
Empty	Registers the material handling unit that is referenced by a Kanban card as empty.
Select	Registers a Kanban card number and automatically selects the referenced job in the Kanban list.

Registration mode Select

When you use a bar code reader to select a job, the display mode of the kanban board changes. In this mode, the following conditions apply:

- Only the scanned kanban job is displayed.
- The details of the selected job are displayed in the **Details** FastTab.
- The **Messages** FastTab displays messages only for the selected job.
- You can change the status of the job by using the functions that are available on the Action Pane. The Kanban transfer board continues to display only a single job during this time.
- You can update the information in the list of jobs manually by clicking **Refresh** (Shift+F5) on the Action Pane. After you refresh the information, the full results for the job filter are

displayed again.

Job status and possible actions

The status of the selected job and the status of any pegged jobs for event kanbans, determine whether you can process the job further. The following table displays information about these statuses and tasks:

- The statuses that are available for jobs, or for the handling units that are referenced by the jobs.
- Each task that you can perform for the job.

Expand table

Job type	Job status or handling unit status	Update picking list	Start	Update registration	Complete	Empty	Create event kanbans
Transfer	• Not planned • No pegged jobs, or pegged jobs are Completed	Yes	Yes	Yes	Yes	No	Yes
Transfer	• Not planned • The pegged job is not Completed	Yes	No	Yes	No	No	No
Transfer	In progress	Yes	No	Yes	Yes	No	No
Transfer	Completed	No	No	No	No	Yes	No
Transfer or process	Empty	No	No	No	No	No	No
Transfer or process	A kanban card is not found	No	No	No	No	No	No

Job type	Job status or handling unit status	Update picking list	Start	Update registration	Complete	Empty	Create event kanbans
Transfer or process	A kanban card is found, but it is not assigned to a kanban	No	No	No	No	No	No
Process	• Not planned • Prepared • In progress	No	No	No	No	No	No
Process	Completed	No	No	No	No	No	No

Last updated on 07/01/2026

Production feedback

Article • 06/21/2024

This article provides information about production feedback, which lets workers provide feedback about production jobs. The article includes information about the various ways that production feedback can be updated.

Production feedback gives workers the opportunity to provide feedback about production jobs. It lets workers record time and material consumption on production orders, operation quantities and status, and errors that cause a job or operation to fail. Production feedback can be updated in journals that are related to production orders. The **Production job card** and **Production route card** journals are used to report time and quantities per job or operation. For reporting about the last job or operation, quantities on the finished product can be reported as finished.

Workers can also provide production feedback using the production floor execution interface, which is part of the manufacturing execution functionality in the **Production control** module. The production floor execution interface has a configurable user interface that shows a list of the released jobs in a prioritized order for a selected work area. It also offers advanced options such as job bundling and team work. The production floor execution interface has a touch-optimized user interface. Feedback submitted through the production floor execution interface updates the production journals.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a production flow version

Article • 04/19/2024

This procedure focuses on creating a new production flow version. For this procedure, the production parameters for lean manufacturing and the units of measurement for class time must be defined. You also need to define a value stream and a production group. The demo data company used to create this procedure is USMF.

Create a production flow

1. Go to Production control > Setup > Lean production flow > Production flows.
2. Click New.
3. In the Name field, type a value.
4. In the Description field, type a value.
5. In the Name field, click the drop-down button to open the lookup.
6. In the list, click the link in the selected row.

    - Select an operating unit of type value stream. A value stream is an operating unit that spans all activities and flows of the value stream. At this stage, operating units are limited to a legal entity and have no further functionality.

7. In the Production group field, click the drop-down button to open the lookup.
8. In the list, click the link in the selected row.

    - Select a production group for the production flow. Production groups allow the definition of material, labor, and WIP accounts for a production process. To associate the accounting context to a production flow, a production group must be selected.

9. Click Save.

Create a production flow version

1. Click Add.
2. In the Expiration date field, enter a date and time.

    - If required, define an Expiration date for the version. You can update it at any given time, even for active versions. You can use it to plan to phase out a version.

3. Click OK.

4. In the list, mark the selected row.
5. In the Unit field, type a value.
6. ResolveChanges the Takt unit.
7. In the Average takt time field, enter a number.

    - Define the Average takt time of the version. For the takt control of the production flow version, define a targeted average takt time. The takt is defined as quantity per time period. In the example, the takt time is 0.2 hours per 10 pieces. For a working time of 8 hours, this corresponds to a daily throughput capacity of 400 pieces.

8. In the Quantity per cycle field, enter a number.

    - Define the quantity per cycle related to the Average takt time.

9. Toggle the expansion of the Version details section.
10. In the Minimum takt time field, enter a number.

    - Define the minimum takt time. The minimum takt time must be less than or equal to the average takt time.

11. In the Maximum takt time field, enter a number.

    - The maximum takt time must be higher than or equal to the Average takt time.

12. In the Period for actual cycle time (days) field, enter a number.

    - Enter the number of days in the Period for actual cycle time. The period for actual cycle time is the number of days that jobs are aggregated from the actual minute backwards to calculate the actual cycle time. The value can be changed at any time and is only used for the calculation of the actual cycle times.

13. Click Save.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create process activities for lean manufacturing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Create a process activity for lean manufacturing.

Prerequisites:

1. A production flow and version that is not active must be created.
2. A work cell must be created.

Find the production flow version

1. Go to Production control > Setup > Lean production flow > Production flows.
2. In the list, find and select the desired record.
3. In the list, click the link in the selected row.

Create a new activity

1. Click Activities.
    - Ensure that the selected production flow has a version in draft and select that version.
2. Click Create new plan activity.
3. Click Next.
4. In the Name field, type a value.
5. In the Name field, type a value.
    - Note that the name must be unique for a given production flow for all versions.
6. In the Process quantity field, enter a number.
    - Note that no matter what job quantity will be processed, this is the minimum quantity per job to calculate the labor cost. If job quantities deviate from this

quantity, labor cost variance will be created.

7. Click Next.

Select the work cell

1. In the Work cell field, click the drop-down button to open the lookup.
2. In the list, click the link in the selected row.

Define the inventory updates

1. Select or clear the Update on hand receipt check box.

    - If Update On hand = No, you can define the activity to receive a semi-finished product (the activity does not reach the next BOM level). You can also select to consume semi-finished products.

Define the picking activities

1. Click Next.
2. In the list, mark the selected row.

    - A default picking activity is created for the input location of the selected work cell.

3. Click Add.

    - You can create additional picking activities for specific products, that are not staged at the work cell input activity.

4. In the list, find and select the desired record.
5. In the Item number field, type a value.
6. In the Warehouse field, click the drop-down button to open the lookup.

    - With this definition, all materials consumed in the activity are picked from the default input warehouse and location except the item that is defined in the second picking activity. This item will be picked from a different warehouse and location.

7. In the list, find and select the desired record.
8. In the list, click the link in the selected row.
9. Select or clear the Register scrap check box.
10. Click Next.

Define the activity times

1. In the list, find and select the desired record.

    - The definition of a Runtime is required. The Runtime is used to calculate costs and the throughput times of the kanban jobs. Runtimes are not used to calculate capacity load and consumption, this is calculated by cycle time, derived from the production flow version task.

2. In the Time field, enter a number.
3. In the Unit field, type a value.
4. Select the Time unit.
5. In the Per quantity field, enter a number.
6. In the list, find and select the desired record.

    - Queue times are optional.

7. In the Time field, enter a number.
8. In the Unit field, type a value.
9. Select the Time unit.
10. In the Per quantity field, enter a number.
11. Click Next.
12. Click Finish.
13. Close the page.

Last updated on 06/19/2025

Create transfer activities for lean manufacturing

Article • 09/29/2021

Create a transfer activity for lean manufacturing.

Prerequisites:

1. A production flow and version that is not active must be created.
2. The from and to warehouse and locations must be created. Optionally, the replenishing or the replenished work cell should be created.

Find the production flow version

1. Go to Production control > Setup > Lean production flow > Production flows.
2. In the list, find and select the desired record.
    - Note that the production flow must have a version in draft status.
3. In the list, click the link in the selected row.

Create a new activity

1. Click Activities.
    - Ensure that the selected production flow has a version in draft and select that version.
2. Click Create new plan activity.
3. Click Next.
4. In the Name field, type a value.
5. In the Activity type field, select 'Transfer'.
6. In the Process quantity field, enter a number.
7. Click Next.

Select the Work cells

1. In the Replenishing field, click the drop-down button to open the lookup.

- To use the work cell output location as the from location in the transfer activity, select a work cell. The same can be done with the replenished work cell, which sets the target location of the transfer activity.

2. In the list, click the link in the selected row.

Define the inventory updates

1. In the Product type field, select an option.

    - Note that a transfer does not change the type of product. You can transfer finished products or semi-finished products (transfer between two activities of a production flow and possibly a kanban flow). When transferring finished products, you can select if picking or receiving results in an inventory transaction.

Define the transfer locations

1. Click Next.
2. In the Warehouse field, click the drop-down button to open the lookup.
3. In the list, find and select the desired record.
4. In the list, click the link in the selected row.
5. In the Location field, click the drop-down button to open the lookup.
6. In the list, click the link in the selected row.
7. In the Freighted by field, select 'Shipper'.

    - Options include: Shipper - the organization operating the shipping warehouse, Recipient - the organization operating the receiving warehouse, Carrier - a third party vendor. If the operating organization is a vendor, the transfer activity requires a subcontracting agreement.

8. Click Next.

Define the activity times

1. In the list, find and select the desired record.

    - The definition of a Runtime is required. The Runtime is used to calculate cost and the throughput times of the kanban jobs. Runtimes are not used to calculate capacity load and consumption, which is calculated by cycle time, derived from the production flow version task.

2. In the Time field, enter a number.
3. In the Unit field, type a value.
4. Select the Time unit.
5. In the Per quantity field, enter a number.
6. In the list, find and select the desired record.

    - Queue times are optional.

7. In the Time field, enter a number.
8. In the Unit field, type a value.
9. Select the Time unit.
10. In the Per quantity field, enter a number.
11. Click Next.
12. Click Finish.
13. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Validate a production flow and version

Article • 09/29/2021

This procedure shows how to create a new production flow and a first version for lean manufacturing. Prerequisites: The production parameters for Lean manufacturing and the units of measure for class time must be defined. You need to define a Value stream and a Production group. This procedure refers to the legal entity USMF in demo data. However, assuming that the legal entity is configured for Lean manufacturing, other legal entities can be used.

Create a production flow

1. Go to Production control > Setup > Lean production flow > Production flows.
2. Click New.
3. In the Name field, type a value.
4. In the Description field, type a value.
5. In the Name field, click the drop-down button to open the lookup.
6. In the list, click the link in the selected row.

    - A value stream is an operating unit that spans over all activities and flows of the value stream. At this stage, operating units are limited to a legal entity and have no further functionality.

7. In the Production group field, click the drop-down button to open the lookup.
8. In the list, click the link in the selected row.

    - Production groups allow the definition of material, labor, and WIP accounts for a production process. To associate the accounting context to a production flow, a production group must be selected.

9. Click Save.

Create a production flow version

1. Click Add.
2. In the Expiration date field, enter a date and time.

    - You can update the Expiration date of the version at any given time, even for active versions. Use the expiration of the version to plan for a phase out of a version.

3. Click OK.
4. In the list, mark the selected row.
5. In the Unit field, type a value.
6. Select the Takt unit.
7. In the Average takt time field, enter a number.

    - For the takt control of the production flow version, define a targeted average takt time. The takt is defined as quantity per time period. In the given example, the takt time is 0.2 hours per 10 pieces. For a working time of 8 hours, this corresponds to a daily throughput capacity of 400 pieces.

8. In the Quantity per cycle field, enter a number.
9. Expand or collapse the Version details section.
10. In the Minimum takt time field, enter a number.

    - The minimum takt time must be less than or equal to the average takt time.

11. In the Maximum takt time field, enter a number.

    - The maximum takt time must be higher than or equal to the Average takt time.

12. Enter a number of days in the Period for actual cycle time

    - The period for actual cycle time is the number of days that jobs are aggregated from the actual minute backwards to calculate the actual cycle time. The value can be changed at any time and is only used for the calculation of the actual cycle times.

13. Click Save.

Feedback

Was this page helpful? Yes No

Provide product feedback

Activate a production flow version

Article • 09/29/2021

Before kanban jobs can be created and activities can be processed for a production flow version, you should activate the version. As a best practice, activate the version before its effective start date and time expire. To complete this task, you must have a production flow version that has the Draft status in order to activate the version.

Find and activate a production flow version

1. Go to Production flows.
2. In the list, find and select the desired record.

    - Select a production flow version that has Draft in the Plan status field.

3. In the list, click the link in the selected row.
4. In the list, find and select the desired record.

    - Select a production flow version that has Draft in the Plan status field.

5. Click Activate.
6. Click OK.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create activity relation - Successor

Article • 09/29/2021

The flow of activities in a lean production flow is documented through activity relations. This recording shows how to create an activity relation.

Prerequisites:

- A production flow and version in draft mode.
- Two activities that follow each other in the production flow are created but not related.

Find the production flow version

1. Go to Production control > Setup > Lean production flow > Production flows.
2. In the list, find and select the desired record.
3. In the list, click the link in the selected row.
4. In the list, mark the selected row.
5. In the list, select a draft version.

    - Activity relations can be added to both draft or active versions of a production flow.

Open the activity overview

1. Click Activities.

    - Note that the form shows all activities of the production flow that are allocated to the Version of the production flows that you are working in.

Add a Successor

1. Click Add successor.
2. In the Activity field, click the drop-down button to open the lookup.
3. In the list, find and select the desired record.
4. In the list, click the link in the selected row.
5. Select the Constraint check box.
6. In the Constraint value field, enter a number.

- The constraint time is the time to be scheduled between the scheduled end of the predecessor (due date and time) and the scheduled start of the successor.

7. In the Units field, type a value.
8. In the Cycle time ratio field, enter a number.

    - If both activities run at the same takt, the cycle time ratio should be 1. If the predecessor runs at the double speed of the successor, the ratio should be 2. The cycle time ratios are used to calculate the individual cycle times of the production flow activities.

9. Click OK.
10. Close the page.
11. Click the GridPanel tab.
12. Close the page.
13. Refresh the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add a predecessor to a production flow activity

Article • 04/19/2024

In a production flow version, all activities must be sequenced. An activity can have one or multiple predecessors or successors.

This procedure shows how to associate a predecessor to an activity.

To perform this task, you need a production flow that has the Draft version with at least two activities that can be connected.

To learn more, read the white paper Lean manufacturing: Production flows and activities.

Find the production flow and version

1. Go to Production control > Setup > Lean production flow > Production flows.
2. In the list, find and select the desired record.
3. In the list, click the link in the selected row.
4. In the list, find and select the desired record.
5. Click Activities.

Select an activity and add a predecessor

1. In the list, find and select the desired record.
2. Click Add predecessor.
3. In the Activity field, enter or select a value.
4. In the Cycle time ratio field, enter a number.

    - The default cycle time ratio of an activity relation is 1. This assumes that both activities run at the same pace or takt time. If the predecessor runs at a higher pace (lower takt time), the ratio should be lower than 1, if the predecessor runs at a slower pace (higher takt time) the cycle time ratio is greater than 1.

5. Click OK.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a value stream

Article • 09/29/2021

Value streams are operating units, that group one or multiple production flows. To define a production flow, you first must define a value stream.

Open the form

1. Go to Production control > Setup > Lean production flow > Value streams.

Create a new value stream

1. Click New to open the drop dialog.
2. In the Name field, type a value.

Select a value stream manager

1. In the Manager field, click the drop-down button to open the lookup.
    - The value stream manager is defined to document the organization.
2. In the list, find and select the desired record.
3. In the list, click the link in the selected row.

Save and close

1. Click Save.
2. Close the page.
3. Refresh the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Lean pegging from sales orders

Article • 09/29/2021

This procedure focuses on validating the pegging tree from a sales line where the item is produced with kanbans. After validating the pegging tree, all the kanban jobs are planned. This is useful for order scenarios where the order taker needs to ensure that production can start right away. The demo data company used to create this procedure is USMF. This procedure is intended for the advanced order taker working in a lean company.

Create a sales order for a kanban controlled item

1. Go to All sales orders.
2. Click New.
3. In the Customer account field, enter or select a value.
    - Use US-001.
4. Click OK.
5. In the Item number field, type 'L0001'.
6. Set Quantity to '30'.
    - It is important that the quantity is higher than 24 in order to trigger the event kanban rule.

Open a pegging tree

1. Click Product and supply.
2. Click View pegging tree.
    - Notice that the pegging tree shows all levels of the pegging needed for the sales order line. In this case, there are two levels of kanbans and all the required components.

Plan the pegging tree

1. In the tree, select 'Sales line 000832\Kanban 000558'.
2. Expand the Kanban jobs section.

- Notice that the job status for the kanban job is Not planned.

3. Click Plan entire pegging tree.

    - This will plan all kanban jobs in the pegging tree, changing the Job status from Not planned to Planned.

4. Refresh the page.

    - Notice that the kanban Job status changed from Not planned to Planned.

5. In the tree, select 'Sales line 000832\Kanban 000558\Issue for L0001\Kanban 000559'.

    - The job for the second kanban is also planned, because the entire pegging tree is planned. Notice that the kanban job status is changed from Not planned to Planned.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a subcontracted work cell for lean manufacturing

Article • 09/29/2021

To model subcontracted work for lean manufacturing, you must create a work cell that is associated with the vendor that provides the work. A subcontracted work cell is linked to the vendor through the association of a resource of the Vendor type. If you play this recording in the USMF demo company, you can select vendor account ID 1002 and site 1.

Create a vendor resource

1. Go to Resources.
2. Click New.
3. In the Resource field, type a value.
4. In the Description field, type a value.
5. In the Type field, select 'Vendor'.
6. In the Vendor field, click the drop-down button to open the lookup.

Create the resource group

1. Go to Resource groups.
2. Click New.
3. In the Resource group field, type a value.
4. In the Description field, type a value.
5. In the Site field, click the drop-down button to open the lookup.

    - Select the site that the work cell should be allocated to. In theory, a site can represent a single site that is operated by a vendor. However, in most cases, subcontracted resources are just allocated to the site that orders the subcontracted work. Note that the input and output warehouses of subcontracted work cells must be on the same site.

6. In the Site field, type a value.
7. @SysTaskRecorder:_RequestClose
8. Select or clear the Work cell check box.
9. In the Input warehouse field, click the drop-down button to open the lookup.

- Select the warehouse and location that are used to stage the material for the vendor-managed work cell. In many cases, the warehouse and location are modeled by using a separate warehouse per vendor and one location per work cell.

10. In the Input location field, click the drop-down button to open the lookup.
11. In the Output warehouse field, click the drop-down button to open the lookup.

    - Define the warehouse and location where the material is posted when the subcontracted activities of the work cell are posted. The warehouse and location can be at the vendor site if the vendor reports the kanban jobs. Alternatively, the warehouse and location can be the receiving location that is associated with the next step of the production flow.

12. In the Output location field, click the drop-down button to open the lookup.
13. Expand or collapse the Calendars section.
14. Click Add.
15. In the Calendar field, click the drop-down button to open the lookup.

    - Associate the working time calendar of the work cell with the resource group. For critical resources, we recommend that you define specific calendars that represent the exact working times and related capacities of the work cell or vendor site.

16. Expand or collapse the Resources section.
17. Click Add.

    - A subcontracted resource group must have an associated resource of the Vendor type that links the resource group to the vendor account.

18. In the Resource field, click the drop-down button to open the lookup.

    - Select or enter the vendor resource that you created in the previous sub-task.

19. Expand or collapse the Work cell capacity section.
20. Click Add.

    - A work cell must have a defined capacity. In this example, we create a throughput capacity of 100 pieces per standard workday.

21. In the Production flow model field, click the drop-down button to open the lookup.
22. In the Capacity period field, select an option.
23. In the Average throughput quantity field, enter a number.
24. In the Unit field, click the drop-down button to open the lookup.

25. ResolveChanges the Unit.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add a kanban quantity calculation policy to a kanban rule

Article • 09/29/2021

This procedure focuses on creating a kanban quantity calculation policy and adding it to a kanban rule to optimize the kanban size and quantities. The demo data company used to create this procedure is USMF. This procedure is intended for the value stream manager. This procedure is a prerequisite for creating the procedure Calculate kanban quantity suggestions.

Create a kanban quantity calculation policy

1. Go to Production control > Periodic tasks > Kanban quantity calculation > Kanban quantity calculation policies.
2. Click New.
3. In the Name field, type a value.
    - For example, type Speaker2016.
4. In the Master plan field, click the drop-down button to open the lookup.
5. In the list, find and select the desired record.
    - Select StaticPlan to calculate demand.
6. In the list, click the link in the selected row.
7. Click Save.
8. In the Minimum kanban quantity field, enter '1'.
    - This is the additional number of kanbans that is included in the kanban quantity calculation.
9. Set Safety factor to '1'.
    - This is the factor that is used to calculate additional quantity of safety stock.
10. In the Days ahead field, enter '30'.
    - This is the number of days prior to the kanban quantity calculation date that is included in the demand calculation.
11. In the Days behind field, enter '30'.

- This is the number of days forward from the kanban quantity calculation date that is included in the demand calculation. The formula used for the calculation is shown with the actual values. For example, Kanban quantity = ((Average daily demand x lead time x 2.00) / Product quantity per handling unit) + 1

12. Close the page.

Add the kanban quantity calculation policy to a kanban rule

1. Go to Product information management > Lean manufacturing > Kanban rules.
2. In the list, find and select the desired record.

    - Select kanban rule 000020 for this procedure.

3. In the list, click the link in the selected row.
4. Toggle the expansion of the Kanban quantity calculation policies section.
5. Click Add.

    - Add the kanban quantity calculation policy that you have just created in the previous sub-task.

6. In the list, mark the selected row.
7. In the Name field, click the drop-down button to open the lookup.
8. In the list, click the link in the selected row.

    - Select the policy Speaker2016 that you have just created in the previous sub-task.

Feedback

Was this page helpful? Yes No

Provide product feedback

Calculate kanban quantity suggestions

Article • 09/29/2021

This procedure focuses on optimizing the kanban size and quantities for a specific kanban rule by using the kanban quantity calculation. The demo data company used to create this procedure is USMF. This procedure is intended for the value stream manager. It is a prerequisite that you have completed the procedure Add a new kanban quantity calculation policy to a kanban rule.

Create a kanban quantity calculation

1. Go to Production control > Periodic tasks > Kanban quantity calculation > Calculate kanban quantity.
2. Click New.
3. In the Name field, type 'Speaker2016'.
4. In the Name field, click the drop-down button to open the lookup.

    - Select the policy that you have created in the procedure Add a new kanban quantity calculation policy to a kanban rule. For example, Speaker2016.

5. In the list, click the link in the selected row.
6. In the Rule active as of date field, set the date and time to '2012-12-17T08:00:00'.

    - This date serves as the basis for determining which fixed kanban rules are included in the kanban quantity calculation.

7. In the Fulfilled demand period start date field, set the date and time to '2012-11-17T09:00:00'.

    - The date from when past demand transactions are included to calculate the kanban quantity.

8. In the Fulfilled demand period end date field, set the date and time to '2012-12-17T07:59:59'.

    - The date until when past demand transactions are included to calculate the kanban quantity.

9. In the Demand period start date field, set the date and time to '2012-12-17T08:00:00'.

    - The date from when current demand transactions are included to calculate the kanban quantity.

10. In the Demand period end date field, set the date and time to '2013-01-16T07:59:59'.

    - The date until when current demand transactions are included to calculate the kanban quantity.

Generate kanban quantity proposal

1. Click Save.
2. Click Generate.

    - This generates a kanban quantity proposal line for the kanban rule 000020.

Run kanban quantity calculation

1. Click Calculate.

    - This calculates the kanban quantity proposal.

2. Click OK.
3. In the list, mark the selected row.

    - Notice the suggested kanban quantity is 2.

Change product quantity and calculate again

1. Set Product quantity to '5'.
2. Click Calculate.
3. Click OK.

    - Notice that with a kanban quantity of 5, the suggestion is changed to a kanban quantity of 4.
    - This is caused by the fact that with a lower product quantity, we need more kanbans to fulfill the demand.

Update kanban rule

1. In the Rule effective date field, enter a date and time.

    - Set the 'Rule active as of date' to a date in the future. For example, today + one year.

2. Click Update.
3. Click OK.
4. Close the page.

Validate change on kanban rule

1. Go to Product information management > Lean manufacturing > Kanban rules.
2. In the list, click the link in the selected row.
    - Select the kanban rule that was created in the previous sub-task. This should be the first kanban rule in the list sorted by number.
3. Toggle the expansion of the Details section.
    - Notice the effective date, which means that this rule is not activated until this date.
4. Toggle the expansion of the Quantities section.
    - Notice this is the default quantity that you entered on the kanban quantity calculation.
    - Notice this is the fixed kanban quantity of 4 from the kanban quantity calculation.
5. Click the ListPanel tab.

Feedback

Was this page helpful? Yes No

Provide product feedback

Change kanban rules for a process job

Article • 09/29/2021

This procedure focuses on changing the used kanban rule for a given kanban. This is useful to level load resources or in case of breakdown. The demo data company used to create this procedure is USMF. This procedure is intended for the planner, working at a lean manufacturing company, responsible for the value stream.

Copy kanban rule

1. Go to Kanban rules.
2. In the list, find and select the desired record.

    - Select Event Kanban rule 000022 for L0001.

3. Click Duplicate kanban rule.
4. Click OK.

Change kanban rule

1. Close the page.
2. Go to Kanban job scheduling.
3. In the list, mark the selected row.

    - Select line with Kanban 000177.

4. Click Use alternative kanban rule.
5. Click Next.
6. In the Kanban rule field, enter or select a value.

    - Select the kanban rule that was created earlier. This is the kanban rule with the highest number.

7. Click Finish.

    - Now the kanban job is using an another kanban rule. This can be useful to level load work cells.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a BOM line event kanban rule

Article • 09/29/2021

This task focuses on the setup needed to create an event kanban rule to ensure supply for production BOM lines in a mixed lean and classic production environment. The demo data company used to create this task is USMF. This task is intended for the process engineer or the value stream manager, as they prepare production of a new or modified product.

Create a new kanban rule

1. Go to Production control > Periodic tasks > Kanban quantity calculation > Kanban rules.
2. Click New.
3. In the Type field, select 'Withdrawal'.
    - The Withdrawal type is used to create transfer kanbans.
4. In the Replenishment strategy field, select 'Event'.
    - The Event strategy is selected to create the transfer of kanbans based on an event. Later in the task guide, we will trigger it by estimating a production order.
5. In the First plan activity field, enter or select a value.
    - Enter or select ReplenishSpeakerComponents. This transfer activity has receipt (output) warehouse and location 12, which means that material will be moved to location 12 in warehouse 12.
6. Expand the Details section.
7. In the Product field, enter or select M0001.
8. Expand the Events section.
9. In the BOM line event field, select 'Automatic'.
    - With the BOM line event field set to Automatic, kanban will be created to fulfill material needs for production order BOM lines.
10. Close the page.

Create and modify a new production order

1. Go to Production control > Production orders > All production orders.
2. Click New production order.
3. In the Item number field, enter or select a value.
    - Enter or select 'L0001'. We use item L0001 because item M0001 is included in the BOM for item L0001.
4. Click Create.
5. In the list, click the link in the row for L0001
6. Click BOM.
7. In the list, click the link in the selected row.
8. Click Edit.
9. In the Line type field, select 'Pegged supply'.
    - Pegged supply is selected to trigger the supply creation of a kanban.
10. Select No in the Resource consumption field.
    - Clearing the check box of Resource consumption lets us change the warehouse.
11. Expand the Inventory dimensions section.
12. In the Warehouse field, type '12'.
    - Warehouse is set to 12 because this is the output warehouse for the withdrawal activity.
13. In the Location field, type '12'.
    - Location is set to 12 because this is the output location of the withdrawal activity.
14. Close the page.
15. Close the page.

Estimate the production order and view the kanban created

1. Click Estimate.
    - Estimating the production order will trigger the creation of the associated kanban to supply item M0001.
2. Click OK.

3. Close the page.
4. Close the page.
5. Go to Product information management > Lean manufacturing > Kanban rules.
6. In the list, click the link in the selected row.

    - Select the event kanban rule created for item M0001.

7. Expand the Kanbans section.
8. In the list, mark the selected row.

    - Notice the kanban created to supply M0001 for the estimated production order.
    - This is the last step!

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a fixed quantity kanban rule for manufacturing

Article • 09/29/2021

This procedure focuses on the setup needed to create a fixed manufacturing kanban rule for triggering transforming activities, at a work cell, in a lean environment. The demo data company used to create this procedure is USMF. This procedure is intended for the Process Engineer or the Value Stream Manager, as they prepare production of a new or modified product.

Create new kanban rule

1. Go to Kanban rules.
2. Click New.
    - Notice that the default Type is Manufacturing and Replenishment strategy is Fixed. You do not have to change these parameters.
3. In the First plan activity field, enter or select a value.
    - This is the activity that will be performed for kanbans created based on this kanban rule. Select 'SpeakerTestAndPackaging'.
4. Expand the Details section.
5. In the Product field, enter or select a value.
    - Select L0050.
6. In the Unit of measure field, enter or select a value.
    - Select minutes.
7. In the Lead time field, enter a number.
    - Enter 5.

Set quantities

1. Expand the Quantities section.
2. Set Default quantity to '10'.
    - This is the quantity that will be transferred for each kanban.

3. Select the Product quantity variance check box.

    - With this, selected kanbans can be completed with a variance from the default quantity. To allow kanbans to be completed with a quantity from 8 to 12, set both variances to 2.

4. Set Variance below to '2'.

    - This will allow completing down to 10 - 2 = 8

5. Set Variance above to '2'.

    - This will allow completing up to 10 + 2 = 12

6. In the Fixed kanban quantity field, enter a number.

    - This is the amount of kanbans that should be active. In this case, 5 kanbans processing 10 each.

7. In the Alert boundary minimum field, enter '2'.

    - Used to keep track of the minimum amount of full kanbans that should be at the destination. For example, this is used on the kanban quantity overview.

8. In the Alert boundary maximum field, enter '4'.

    - Used to keep track of the maximum amount of full kanbans that should be at the destination. For example, this is used on the kanban quantity overview.

9. In the Automatic planning quantity field, enter '1'.

    - Setting this to 1 means that every kanban will be automatically planned. If we entered 3, the kanbans would not be planned until 3 empty kanbans are ready for planning. This is useful for grouping work and avoiding too much setup.

Create Kanbans

1. Expand the Kanbans section.
2. Click Save.

    - The kanban rule needs to be saved before kanbans can be created.

3. Click Add.

- Note that there are no active kanbans, due to this the suggested 'Number of new kanbans' are 5. This is equal to the 'Fixed kanban quantity'.

4. Click Create.

- This will create 5 kanbans.
- Note that 5 kanbans, for 10 each, was created for this manufacturing kanban rule. This is the last step in this procedure.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a kanban rule using a kanban line event

Article • 09/29/2021

This procedure creates a kanban rule by using the kanban line event setting to trigger pull from a process activity. The kanban rule is triggered by a kanban process activity, with a quantity equal to or greater than 25 each. The demo data company used to create this task is USMF. This task is intended for the process engineer or the value stream manager, as they prepare production of a new or modified product in a lean environment.

Create a kanban rule

1. Go to Product information management > Lean manufacturing > Kanban rules.
2. Click New.
3. In the Replenishment strategy field, select 'Event'.

    - This generates kanbans directly from demand. It is used to set up rules that define a make-to-order scenario.

4. In the First plan activity field, enter or select a value.

    - Enter or select SpeakerAssemblyAndPolish. The first activity of a manufacturing kanban rule is a process activity in the production flow. When you select the activity, the validity dates of the activity are copied to the validity dates of the kanban rule.

5. Expand the Details section.
6. In the Product field, type 'L0001'.
7. Expand the Events section.
8. In the Kanban line event field, select 'Automatic'.

    - This generates event kanbans on demand. The field is used to configure kanban rules that replenish material that is required for a downstream process activity. When you select Automatic, the event kanbans are created with the demand. This setting is recommended if you expect to execute production on the same day.

9. Set Minimum event quantity to '25'.

- Event kanbans are generated when the demand quantity is equal to or more than this field. This is useful if you want to produce an order quantity less than this field on one machine and more than this field on another machine.

10. Click Save.

Create sales order and trigger kanban chain

1. Go to Sales and marketing > Sales orders > All sales orders.
2. Click New.
3. In the Customer account field, enter or select a value.

    - Select Customer account US-003, Forest Wholesales.

4. Click OK.
5. In the Item number field, type 'L0001'.

    - L0001 is the item for which you created the kanban rule.

6. Set Quantity to '27'.

    - Because 27 is higher than the minimum quantity of 25 on the kanban rule, this will trigger an event kanban.

7. In the Site field, type '1'.
8. In the Warehouse field, enter or select a value.

    - Select warehouse 13.

9. Click Save.

View the kanban generated by the kanban rule

1. Go to Product information management > Lean manufacturing > Kanban rules.
2. In the list, find and select the desired record.
3. Expand the Kanbans section.

    - Notice that a kanban for 27 was created to process the activity based on the created kanban rule.
    - This is the last step.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a kanban rule using a minimum stock event

Article • 09/29/2021

This procedure focuses on the setup needed to create a kanban rule using a minimum stock event to ensure that a specific product is always available at a specific location. A kanban rule is created to transfer material to the location when the inventory level drops below 200 pieces. By running the Pegging event processing, the needed kanbans are created. The demo data company used to create this task is USMF. This task is intended for the process engineer or the value stream manager, as they prepare production of a new or modified product in a lean environment.

Create a new kanban rule

1. Go to Product information management > Lean manufacturing > Kanban rules.
2. Click New.
3. In the Type field, select 'Withdrawal'.
    - This type is used to create transfer kanbans.
4. In the Replenishment strategy field, select 'Event'.
    - The Event strategy is used to create the transfer kanbans based on an event. Later in the procedure, you will trigger transfer kanbans by using stock replenishment.
5. In the First plan activity field, enter or select a value.
    - Enter or select ReplenishSpeakerComponents. This transfer activity has receipt (output) warehouse and location 12, which means that materials will be moved to location 12 in warehouse 12.
6. Expand the Details section.
7. In the Product field, enter or select a value.
    - Select M0007.
8. Expand the Events section.
9. In the Stock replenishment event field, select 'Batch'.
    - This creates kanbans to fulfill material needs at the related location during Pegging event processing.

Set the minimum quantity for the item

1. Click to follow the link in the Product field.
2. Click to follow the link in the Item number field.
3. Expand the Product image FactBox.
4. On the Action Pane, click Plan.
5. Click Item coverage.
6. Click New.
7. In the list, mark the selected row.
8. In the Warehouse field, enter or select a value.

    - Set Warehouse to 12.

9. Set Minimum to '200'.

Run the batch event creation job

1. Go to Production control > Periodic tasks > Kanban job batch processing > Pegging event processing.
2. Click OK.
3. Go to Product information management > Lean manufacturing > Kanban rules.
4. In the list, click the link in the selected row.

    - Select the kanban rule that you created earlier.

5. Expand the Kanbans section.

    - Notice that a kanban was created to transfer the needed material to warehouse 12.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a kanban rule for multiple activities

Article • 09/29/2021

This procedure shows how to create a kanban rule that includes multiple activities from a production flow. The demo data company used to create this task is USMF. This task is intended for the process engineer or the value stream manager, as they prepare production of a new or modified product in a lean environment.

Create a new kanban rule

1. Go to Product information management > Lean manufacturing > Kanban rules.
2. Click New.
3. In the Replenishment strategy field, select 'Scheduled'.
4. In the First plan activity field, enter or select a value.
    - Select SpeakerAssemblyAndPolish.
5. Select the Multiple activities check box.
    - The purpose is to include more than one activity in the kanban rule. You choose a path in the production flow when you select the last plan activity.
6. In the Last plan activity field, enter or select a value.
    - Select SpeakerTestAndPackaging. After you select the value, a page automatically opens. Select the kanban flow SpeakerAssemblyAndPolish > SpeakerTestAndPackaging. Click OK.
7. Expand the Details section.
8. In the Product field, enter or select a value.
    - Select Item L0006.

Create kanban and view jobs

1. Expand the Kanbans section.
2. Click Add.
3. In the Number of new kanbans field, enter '1'.
    - This will create one kanban.

4. Set Product quantity to '3'.
    - Kanban will process 3 products.
5. In the Due date/time field, enter a date and time.
    - You can enter Today.
6. Click Create.
7. Click Details.
    - Notice that the kanban has two process jobs from the production flow. The first one is SpeakerAssemblyAndPolish, and the second one is SpeakerTestAndPackaging.
    - This is the last step!

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a new kanban rule by duplicating an existing kanban rule

Article • 09/29/2021

This procedure focuses on creating a duplicate of an existing kanban rule. This is useful if you want to create new kanban rules based on existing kanban rules. The demo data company used to create this procedure is USMF. This procedure is intended for the process engineer or the value stream manager as they prepare production for a changed production flow or a new replenishment rule.

Select a kanban rule

1. Go to Kanban rules.
2. In the list, find and select the desired record.

    - Select kanban rule 000017 for Product M0006.

Duplicate a kanban rule

1. Click Duplicate kanban rule.

    - When duplicating a kanban rule, it is possible to change type, dates, activities, and the product selection. Change the product for this procedure in the next step.

2. In the Product field, enter or select a value.

    - Select M0007.

3. Click OK.

    - Note that a duplicate of kanban rule 000017 is created.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a replacement kanban rule

Article • 09/29/2021

This procedure focuses on replacing an existing kanban rule with a new kanban rule on a specific date. This is useful when changes in the production flow or replenishment rules need to be coordinated and scheduled. The demo data company used to create procedure is USMF. This procedure is intended for the process engineer or the value stream manager when they prepare production for a changed production flow or a new replenishment rule. This task replaces kanban rule 000022 with a new rule and increases the maximum quantity from 48 to 100 for the new rule.

Select a kanban rule to replace

1. Go to Kanban rules.
2. In the list, find and select the desired record.

    - Select kanban rule 000022.

Create a replacement kanban rule

1. Click Replace kanban rule.
2. In the Effective date field, enter a date and time.

    - Select a date in the future, such as one week from now. This is the date and time when the new kanban rule becomes active and replaces the old kanban rule.
    - If the kanban rule changes the path in the production flow, another activity can be selected. In this procedure, we will keep the activity untouched.

3. Click OK.

    - Notice that a new kanban rule is created to replace kanban rule 000022.
    - The effective date is set according to the date chosen when you replace the kanban rule.

4. In the list, find and select the desired record.

    - Select the replaced kanban rule 000022.
    - Notice that the replaced kanban rule has the same date as the Expiration date because this is the date when it will expire.

5. In the list, select row 1.

    - Select the new kanban rule on top of the list. This is the kanban rule with the highest kanban rule number.

6. In the list, click the link in the selected row.

    - Click the kanban rule number to open the kanban rule.

Modify maximum quantity for the replacement kanban rule

1. Set Maximum quantity to '100'.

    - Expand the Quantities FastTab to see the Maximum quantity field. Changing the maximum quantity to 100 will allow up to 100 kanbans to be processed. This is the last step in this task.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a sales event kanban rule

Article • 09/29/2021

This procedure focuses on the setup needed to create a kanban rule that is triggered during sales order creation. The event kanban rule replenishes requirements that originate from sales order lines. The demo data company used to create this procedure is USMF. It is intended for the process engineer or the value stream manager as they prepare production of a new or modified product.

Create a new kanban rule

1. Go to Kanban rules.
2. Click New.
3. In the Replenishment strategy field, select 'Event'.

    - Selecting Event means that the kanban rule is triggered by an event, for example, creation of a sales order line. This is applied to areas where each kanban should cover a specific demand.

4. In the First plan activity field, enter or select a value.

    - Select Final assembly.

5. Expand the Details section.
6. In the Product field, enter or select a value.

    - Select L0050.

Define an event

1. Expand the Events section.
2. In the Sales event field, select 'Automatic'.

    - By selecting the sales event Automatic, this kanban rule will be triggered automatically when a sales line matches the product and receipt location. In this procedure, it is product L0050 on warehouse 13.

3. Set Minimum event quantity to '50'.

    - With a minimum event quantity of 50, the kanban rule will only be triggered by events with a quantity of 50 or more.

4. Expand the Production flow section.

    - Notice that the Receipt location is warehouse 13. This means that this kanban rule will be triggered for this location.
    - In this example, a sales line for product L0050, with a quantity of 50 or more, on warehouse 13, will trigger this kanban rule.

Create sales line to trigger event kanban rule

1. Go to All sales orders.

    - A warning is shown when the kanban rule is saved, which means that kanbans will be created in real-time during sales order creation.

2. Click New.
3. In the Customer account field, enter or select a value.

    - For example, select US-003.

4. Click OK.
5. In the Item number field, type 'L0050'.
6. In the Site field, type '1'.

    - Select Site 1 because Warehouse 13 is on Site 1.

7. In the Warehouse field, enter or select a value.

    - Set Warehouse to 13.

8. Set Quantity to '75'.

    - Enter a quantity of 50 or greater, to trigger the created kanban rule.

Verify that kanban is created

1. Click Product and supply.
2. Click View pegging tree.

    - Notice that a kanban with the same quantity as the sales line is created. You can also see the material issues needed to produce L0050. This is the last step in this procedure.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a withdrawal kanban rule

Article • 09/29/2021

This procedure shows the setup that is needed to create a withdrawal kanban rule for transferring material in a lean environment. The demo data company used to create this procedure is USMF. This procedure is intended for the Process Engineer or the Value Stream Manager, as they prepare replenishment of new or modified material.

Create new kanban rule

1. Go to Kanban rules.
2. Click New.
3. In the Type field, select 'Withdrawal'.
    - The Withdrawal type is used for kanban rules to transfer material or goods.
4. In the First plan activity field, enter or select a value.
    - Select ReplenishSpeakerComponents. This activity is set up to move components from warehouse 11, location 11 to warehouse 12, and location 12.
5. In the Product field, enter or select a value.
    - Select M0007.
6. In the Lead time field, enter a number.
    - For example, 60.
7. In the Unit of measure field, enter or select a value.
    - For example, Minutes.

Set quantities for kanban

1. Set Default quantity to '5'.
    - This is the quantity that will be transferred for each kanban.
2. In the Fixed kanban quantity field, enter '2'.

- This is the amount of kanbans that should be active. In this case, 2 kanbans transferring 5 each.

3. In the Alert boundary minimum field, enter '1'.

    - Used to keep track of the minimum amount of full kanbans that should be at the destination. For example, this is used on the kanban quantity overview.

4. In the Alert boundary maximum field, enter '2'.

    - Used to keep track of the maximum amount of full kanbans that should be at the destination. For example, this is used on the kanban quantity overview.

Create kanbans

1. Click Save.

    - The kanban rule needs to be saved before kanbans can be created.

2. Click Add.

    - Note that there are no active kanbans because the suggested 'Number of new kanbans' is 2, which is equal to the 'Fixed kanban quantity'.

3. Click Create.

    - This will create two kanbans.
    - Note that 2 kanbans, for 5 each, was created for this withdrawal kanban rule. This is the last step in this procedure.

Feedback

Was this page helpful? Yes No

Provide product feedback

Execute kanban process jobs

Article • 09/29/2021

This procedure focuses on executing kanban process jobs. The first job is completed with the expected quantity and has no errors. The second job is completed with errors. The demo data company used to create this procedure is USMF. This procedure is intended for the machine operator.

Select a kanban job

1. Go to Production control > Kanban > Kanban board for process jobs.
2. In the Work cell field, click the drop-down button to open the lookup.
3. Click the row with resource group 1250. This filters the Jobs list to display only the jobs for work cell 1250.

    - Mark the row that has the Planned job status.

Complete a job with expected quantity

1. Expand or collapse the Details section.

    - This section displays important information about card number, item number, quantity ordered, and activity name.

2. Expand or collapse the Production instructions section.

    - This section displays production instructions for the activity. The instructions can be text, pictures, drawings, and other documents.

3. Click Start.

    - Select a job that is not completed. Use status icons in the Job status field to view job status.

4. Click Complete.

    - The job is completed with the expected quality.

Complete a job with errors

1. Click Start.

- When a job is completed, the next job on the list is selected automatically. This is why you don't need to select a job before you click Start.

2. On the Action Pane, click Manufacture.
3. Click Complete (details).
4. In the list, mark the selected row.
5. In the Error quantity field, enter a number.
6. In the Good quantity field, enter a number.
7. Click OK.

Feedback

Was this page helpful? Yes No

Provide product feedback

Move scheduled kanban jobs

Article • 09/29/2021

This procedure focuses on moving planned process kanban jobs to a different period. The demo data company used to create this procedure is USMF. This procedure is intended for the shop floor supervisor or production planner working with kanbans.

Select scheduled kanban jobs.

1. Go to **Production control** > **Kanban** > **Kanban job scheduling**.
2. In the **Work cell** field, click the drop-down button to open the lookup.
3. In the list, mark the selected row.
    - Select work cell 1250.
4. Click **Select**.
5. In the **Display job status** field, select **Scheduled**. This filters the job list to display only the scheduled kanban jobs.

Move kanban jobs to a different period.

1. In the list, find and select the desired record. Select a job that has the **Planned job** status, for example, a job scheduled on December 20, 2012 in the **Planned period** field. Then move the job to the previous period.
2. Click **Previous period**.
3. Click **End** to move the job to the end of the job list as the last job in the previous period.
4. In the list, find and select the desired record. Select a job that has the **Planned job** status, for example, a job scheduled on December 18, 2012 in the **Planned period** field. Then move the job to the next period.
5. Click **Next period**.
6. Click **Start** to move the job to the start of the job list as the first job in the previous period.

Move a job within a period.

1. In the list, find and select the desired record. Select a job that has the Planned job status, for example, the second job scheduled on December 19, 2012 in the **Planned period** field. Then move the job within the planned period.
2. Click **Forward**. Notice that the job is moved one line down on the list.
3. Click **Backward**. Notice that the job is moved one line up on the list.

Feedback

Was this page helpful? Yes No

Provide product feedback

Prepare a process kanban job when materials are available for the work cell

Article • 09/29/2021

This task focuses on preparing a process kanban job when all materials are available for the work cell. The demo data company used to create this task is USMF. This task is intended for the machine operator.

1. Go to Kanban board for process jobs.
2. In the Work cell field, click the drop-down button to open the lookup.
3. In the list, click the link in the selected row.

    - Select work cell 1250 and click OK.

4. In the list, select row 4.

    - In the clean demo company, Kanban 000329 in row 4 is the first job that is not completed yet.

5. Toggle the expansion of the Picking list section.

    - Verify that the supply status is available for all items in the picking list.
    - If multiple jobs are selected, the picking list will show the sum of all items needed for the selected jobs.

6. Click Prepare.

    - The preparation process is now completed. The selected check box for all rows in the picking list indicates that the supply status is picked.

Feedback

Was this page helpful? Yes No

Provide product feedback

Prepare a process kanban job when materials are not available for the work cell

Article • 09/29/2021

This procedure focuses on preparing a process kanban job when some materials are not available for the work cell, therefore it's necessary to pick materials from the warehouse. The procedure "Prepare a process kanban job when materials are available" is a prerequisite for creating this procedure. This procedure is intended for the machine operator. The demo data company used to create this procedure is USMF.

1. Go to Production control > Kanban > Kanban board for process jobs.
2. In the Work cell field, click the drop-down button to open the lookup.
3. In the list, click the link in the selected row.

    - Select work cell 1250.

4. In the list, find and select the desired record.

    - Select Kanban 000356.

5. In the list, find and select the desired record.

    - In the list, deselect row 4. or Select row 4 if you haven't completed the task "Prepare a process kanban job when materials are available."

6. Toggle the expansion of the Picking list section.

    - The No entry icon in the supply status indicates that 48 ea of item P0002 are missing for the work cell.

Transfer materials to work cell

1. Toggle the expansion of the Transfer jobs section.
2. Use the Quick Filter to filter on the Item number field with a value of 'P0002'.
3. In the list, find and select the desired record.
4. Click Start.

    - Transfer is in progress.

5. Click Complete.

- Item P0002 is now available in the picking list for the kanban job. This means that we can prepare the kanban with all the needed materials.

6. Click Prepare.

- Notice that an icon in the Job status indicates that the job is now ready.

Feedback

Was this page helpful? Yes No

Provide product feedback

Remove a kanban job from the schedule

Article • 09/29/2021

This procedure focuses on removing a planned process kanban job from the schedule by reverting the job status to Not planned. The demo data company used to create this procedure is USMF. This procedure is intended for the shop floor supervisor or production planner.

Find a planned kanban job

1. Go to Production control > Kanban > Kanban job scheduling.
2. In the Work cell field, click the drop-down button to open the lookup.
3. In the list, click the link in the selected row.

    - Select work cell 1250.

4. Click Select.
5. In the Display job status field, select 'Scheduled'.

Remove the planned kanban job from the schedule

1. In the list, find and select the desired record.

    - Select a job that has the Planned status, for example, a job from December 19, 2012.

2. On the Action Pane, click Plan.
3. Click Revert job status.
4. Click OK.

    - This will revert the current job status from 'Planned' to 'Not planned' and remove it from the process board.

Feedback

Was this page helpful? Yes No

Provide product feedback

Revert kanban job status

Article • 09/29/2021

This procedure focuses on reverting an incorrect kanban job status. This is useful in case the machine operator updates the wrong job, or sets the wrong status by mistake. In this procedure, a kanban job is registered as prepared by mistake, and the status is reverted. The demo data company used to create this procedure is USMF. This procedure is intended for the shop supervisor or machine operator working in a lean manufacturing company.

Open process board for the work cell

1. Go to Kanban board for process jobs.
2. In the Work cell field, enter or select a value.

    - Select work cell 1260.

Prepare kanban job

1. Click Prepare.

    - If you can't click Prepare because it is grayed out, make sure that the selected kanban job has status Planned, which is indicated by the empty kanban icon. If Prepare fails, make sure that all materials in the Picking list are available.

2. In the list, select the prepared job.

    - Select the first job that you have just prepared.
    - Notice that the jobs status is prepared, which is indicated with a triangle inside the kanban icon.

Revert the status of the prepared kanban job

1. In the list, mark the selected row.

    - Select the first job that was prepared.

2. On the Action Pane, click Manufacture.
3. Click Revert status.

- You can use an alternative kanban rule when the following conditions are true: - The replenishment strategy is the same for both rules. - The version of the production flow is the same for both rules. - The product that is supplied is the same for both rules. - Any downstream activities that are configured for the last activity of the kanban rules must be the same for both rules. - The same supplied inventory dimensions must be configured for both rules. - The status of the handling unit must be Not assigned. - The configuration for event kanbans must be the same.
- Ensure that the new status is Planned.

4. Click OK.
5. In the list, unmark the selected row.

- Select the same job.
- Notice that the job status for the kanban job is reverted to Planned, which is indicated by an empty kanban icon.

Feedback

Was this page helpful? Yes No

Provide product feedback

Schedule kanban jobs

Article • 09/29/2021

This procedure focuses on scheduling process kanban jobs for a specific work cell. The procedure "Prepare a process kanban job when materials are not available" is a prerequisite for creating this procedure. The demo data company used to create this procedure is USMF. This task is intended for the shop floor supervisor and production planner working with kanbans.

Select kanban jobs for a work cell

1. Go to Production control > Kanban > Kanban job scheduling.
2. Expand the Period capacity fact box

    - Expand the Kanban FactBox.

3. In the Work cell field, click the drop-down button to open the lookup.
4. In the list, mark the selected row.

    - Select work cell 1250. This will filter the view to display only the jobs for work cell 1250.

5. In the list, click the link in the selected row.
6. Click Select.

Schedule a kanban job in the first available period

1. In the list, mark the selected row.

    - Select the first row in the list that has the Not planned status. The dotted icon in the Job status field indicates not planned.

2. Click Schedule.

    - This will schedule the kanban job in the first available period.
    - Notice that the kanban job is moved to the end of the list because it has been added to the period starting from today.
    - If the period starting from today is fully booked, the job will be moved to the first available period.

Schedule two kanban jobs for a specific day

1. In the list, select row 1.
    - You should see that the first row has the Not planned status in the Job status field.
2. In the list, select row 2.
    - You should see that the second row has the Not planned status in the Job status field. Now you have the first two jobs selected.
3. Click Schedule from date to open the drop dialog.
4. Check or uncheck the Override capacity shortage reaction box.
    - This option allows you to override the default capacity shortage reaction.
5. In the Capacity shortage reaction field, select 'Add to the requested period'.
    - This option will ensure that the job is added to the requested period, regardless if the work cell might be overloaded.
6. Click Schedule.
    - Notice that both jobs are added to the desired period.
    - In the Period capacity section, you can see the load for each period. The Consumption field shows the scheduled consumption in this period. If the scheduled consumption is higher than the available capacity in this period, the overloaded consumption will be selected.

Feedback

Was this page helpful? Yes No

Provide product feedback

Transfer materials with kanban jobs

Article • 09/29/2021

This procedure focuses on executing a withdrawal kanban job to transfer materials. The demo data company used to create this procedure is USMF. This procedure is intended for the warehouse worker.

Display transfer jobs

1. Go to Production control > Kanban > Kanban board for transfer jobs.
2. Expand or collapse the Filters section.

    - In the Filters section, you can specify what jobs you want to see by filtering on Production flow, Activity name, From warehouse and location, and To warehouse and location.

3. In the From warehouse field, type '11'.
4. In the To location field, type '12'.

Start a transfer job

1. In the list, deselect the selected row - if any.
2. In the list, select row 4.

    - Select the first job with status Not planned. Make sure this is the only job selected.

3. Click Start.

    - Notice that an icon indicates that the job is started.

Select a second transfer job and change quantity

1. In the list, find and select the desired record.

    - You can have multiple jobs selected, but for now select row 5.

2. In the list, find and select the desired record.

- Make sure the job in the previous step is the only one selected. Deselect all other jobs.

3. Note the value in the Job quantity field to reference later
4. Set Job quantity to '30'.

    - Notice the warning! You are not allowed to transfer 30. According to the setup of the kanban rule, you can only transfer the original quantity.

5. Use the value noted previously in the Job quantity field

    - Set the Job quantity to the previous value.

Start the second transfer job

1. Click Start.

    - This will start the transfer of the job in row 5.

Complete both transfer jobs

1. In the list, select row 4.

    - Now two transfer jobs are selected on row 4 and row 5.

2. Click Complete.

    - This will complete the transfer of both jobs.

Feedback

Was this page helpful? Yes No

Provide product feedback

Update kanban status

Article • 09/29/2021

When a kanban is emptied by mistake or a received kanban needs to be emptied, you need to update kanban status. The demo data company used to create this procedure is USMF. This procedure is intended for the shop supervisor.

Find the kanban.

1. Go to Production control > Kanban > Kanbans.
2. Open Handling unit status column filter.
3. Click Clear.
    - This resets the filters.
4. Use the Quick Filter to find records. For example, filter on the Card number field with a value of '000149'.

Change emptied status to received status

1. Click Reverse empty handling unit.
2. Click OK.
    - Notice that the Handling unit status is Received.

Change received status to emptied status

1. Click Empty kanban.
2. In the list, mark the selected row.
    - Notice that the Handling unit status is Emptied.

Feedback

Was this page helpful? Yes No

Provide product feedback

Bills of materials and formulas

07/10/2025

This article provides information about bills of materials (BOMs) and formulas, which are a central part of the definition of products and product variants. BOMs and formulas specify the required materials or ingredients for a specific product. Formulas also specify the co-products and by-products that are received in a specific production context.

Bills of materials

A bill of materials (BOM) defines the components that are required in order to produce a product. The components can be raw materials, semi-finished products, or ingredients. In some cases, services can be referenced in a BOM. However, BOMs typically describe the *material resources* that are required.

When it's combined with a route or production flow that describes the operations and resources that are required in order to build a product, the BOM forms the foundation for calculating the estimated cost of the product.

A BOM is an individual entity that is described by the following information:

- BOM ID
- BOM name
- The BOM lines that describe the components and ingredients
- The BOM versions, which define the product and period that the BOM can be used for

A single BOM describes a single level that is identified by a unique ID. Components might have their own BOMs that are referenced by BOM versions. You can display and edit the complete hierarchy of BOMs for a specific product in the BOM designer.

Formulas, co-products, and by-products

A formula is a subtype of BOM that is typically used for process manufacturing. In addition to components and ingredients, a formula describes co-products and by-products. In the actual version, the definition of co-products and by-products for the formula requires the formula version. A formula is typically defined for one specific finished product (a formula or planning item) that is defined in the formula version.

BOMs in the product lifecycle

In the product lifecycle, many types of BOM might be created for various reasons:

- **Sketching/Draft BOM** – Gives a draft estimation of required materials in an early design phase and helps you do a rough estimate of cost and estimated product attributes. This BOM isn't usually used in enterprise resource planning (ERP).
- **Engineering BOM** – Typically used when you design products that are based on existing product portfolios. Engineering BOMs are structured to simplify the design process and group complex products into engineering modules. For simple products, it might be possible to engineering BOMs for the actual production process. However, for other products, the engineering BOM must be converted to an actual production BOM. Engineering BOMS are typically represented by phantoms in the BOM hierarchy. Although engineering BOMs can be used for the planning and execution of manufacturing operations, this approach can lead to inefficiencies, especially in repetitive operations where many orders are created.
- **Planning BOM** – Used to do planning for material requirements. The demand of components and ingredients is calculated based on the demand of the finished products. Like costing BOMs, planning BOMs might represent a specific mix of material that is used in a period.
- **Production BOM** – The actual BOM that is used for a specific production. A production BOM must take into account the actual resources that are used to produce the product. When a production order, batch order, or kanban is created, the multiple levels of BOMs that are represented by phantoms are collapsed into one level and distributed over the operations for the order.
- **Costing BOM** – Used to calculate the estimated cost of a product. For example, you can use a costing BOM when standard cost is used or the estimated planned cost of a given product is calculated. Costing BOMs can refer to a specific mix of materials and resources that are expected to be used. Therefore, you can use the costing BOM to create a representative estimated cost for a period and help avoid variances over time.

The types of BOM that are actually used in an implementation depend on the implementation, and also on the business scenarios and requirements. In simple implementations, a planning BOM, production BOM, and costing BOM can be modeled as one BOM. In environments that have frequent engineering changes and multiple alternative routes, a larger set of BOM types will probably be required.

Approval of BOMs and formulas

Each BOM and formula can be separately approved or unapproved. Typically, approval of a BOM or formula occurs when the first relevant BOM version is approved. However, in some business scenarios, these approvals might be different steps in the process and might involve different process owners.

If a BOM is unapproved, all related BOM versions are also unapproved.

BOM and formula versions

To relate a specific BOM or formula to a product variant that can be produced, you must create a BOM version or formula version. The validity of BOM versions and formula versions can be constrained by period, quantity, site, specific product dimensions, and other criteria. Formula versions have other important attributes, such as yield, co-product and by-product definitions, and the cost distribution instructions for the formula.

Approval of BOM and formula versions

Before a BOM version can be used in the planning or manufacturing process, it must be approved. When a BOM version is approved, the related BOM can also be approved, depending on the user's selection and authentication rights. A BOM version can be approved only if the related BOM itself is approved.

Activation of the default BOM or formula version

To set a specific BOM or formula as the default BOM version or formula version that will be used by master planning or used to create of production orders, you must activate the version. When a version is activated, the uniqueness of the version for the given constraints (for example, period, site, or quantity) is verified. You receive an error message if the version that you're trying to activate conflicts with a version that is already active. You must then either inactivate the conflicting version or modify the version constraints (usually the period) to prevent an ambiguous activation.

Product change with case management

The product change case for approval and activation of new or changed BOMs and BOM versions provides an easy way to see an overview of the BOM version constraints. You can also approve and activate all BOMs and formulas that are related to a specific change for one activation date.

Alternative BOM versions

Sometimes, the active BOM version or formula version shouldn't be used in forecasts, sales, or a parent product. In this case, you can select a specific approved BOM as part of the requirement (forecast line, sales line, or BOM line) if an approved BOM version or formula version exists for the alternative BOM or formula.

When planned orders, production orders, or kanbans are created, the planner or shop floor supervisor can use any approved BOM version that is valid on the requested planned production date to plan for or produce a specific product. The BOM version that is used doesn't have to be activated as the default BOM version.

BOM and formula lines

A BOM line is created for each material, service, or ingredient. The line defines the planned consumption of the specified product variant and also defines the various attributes that are related to the planned consumption.

BOM lines can have the following line types: **Item**, **Phantom**, **Pegged supply**, **Vendor**.

Item

Select the **Item** line type for materials or services that are directly consumed, and that don't require further explosion or pegged supply.

Phantom

Select the **Phantom** line type when you want to explode any lower-level BOM items that are contained on the BOM line. In Master scheduling, in planned cost calculation, or on estimation of a production order that uses BOM lines of the **Phantom** type, the parent BOM line that refers to a product variant that has a phantom BOM is replaced by the component items that are listed as BOM lines in that BOM, as determined by the applicable active BOM version of that product variant. If the product variant has an applicable active route, the operations of that route are merged into the parent route.

Phantoms are typically used to simplify the engineering process. Extensive use of phantom BOMs in many levels has an effect on performance, especially in highly repetitive manufacturing scenarios. To improve performance, you should avoid deep hierarchies of phantoms. Instead, use pre-exploded production BOMs and routes.

Pegged supply

Select the **Pegged supply** line type when you want to create a subproduction, a BOM line event kanban, or a direct purchase order for any product variant that the BOM line references. The subproduction, event kanban, or purchase order is created when you estimate the production order. The required item quantities are automatically reserved for the consuming production order.

> **Note**
>
> A subproduction order is always created for pegged supply, even if on-hand inventory or other supply is available to cover the demand.

Vendor

Select the **Vendor** line type if the production process uses a subcontractor, and you want a subproduction or purchase order to be created automatically for the subcontractor.

> **Note**
>
> For BOMs that include subcontracted operations, the service or work that is performed by the subcontractor must be created as service item that is tracked in inventory. You must attach the service item to the parent item as a BOM line. The route must contain an operation that is assigned to the subcontractor's operations resource.

Formulas and formula versions

07/10/2025

A formula defines the materials, ingredients, and outcomes of a specific process in process manufacturing. Together with the corresponding route, the formula defines the whole process in process manufacturing. Formulas are used to plan and produce products in process manufacturing.

A formula consists of the ingredients and quantities that are required in order to produce a specific quantity of a formula item. Depending on the task that you perform, you can access formula functionality from Inventory and warehouse management or Product information management.

Formulas and formula lines

A formula consists of one or more formula lines that identify the ingredients or items that make up the formula. A formula line can contain Bill of materials (BOM) items, formula items, catch-weight items, purchased items, co-products, or by-products. Because many items are used in multiple products, an item can be used in more than one formula.

An example of a formula is the formula for a chocolate chip cookie. The ingredients for this formula use multiple lines, such as flour, sugar, eggs, butter, and chocolate chips. The formula for the chocolate chip cookie contains ingredients that are likely used in other formulas. While you make the chocolate chip cookies, there might be leftovers, such as crumbs, or some of the cookies might be overbaked or undercooked. These items can be set up as co-products or by-products, depending on the production operations.

When you create a formula line, you use the line type to indicate how the system should handle the line when you run master planning and produce batch orders. Each line type gives a different result. The following table describes the line types that you can select.

Expand table

Line type	Description
Item	Select *Item* when the item is a raw material or a semi-finished item that is picked from inventory or when the item is a service.
Phantom	Select *Phantom* when you want to explode any lower-level formula items that are contained on formula lines. When you estimate the batch order, and the formula items are exploded, the component items are listed as formula lines on the batch order. Additionally, the corresponding routes are added to the production route. Formula items are exploded by

Line type	Description
	using the current configuration. When you use the *Phantom* line type, you can handle production and measurement configurations that occur at different formula levels. If you select *Phantom* for a product on the **Engineer** FastTab of the **Released product details** page and then use this product in a formula, the line type of the formula line is changed to *Phantom*. You can't select *Phantom* for a catch-weight item, or for items where the production type is *Co-product*, *By-product*, or *Planning item*.
Pegged supply	Select *Pegged supply* to create a batch order, production order, kanban, transfer order, or purchase order for the ingredient that is contained on the formula line. The related order is determined based on the default order settings and the production type of the ingredient, and is created when you estimate the batch order. The required ingredient quantities are reserved for the batch order.
Vendor	Select *Vendor* if the production process uses a subcontractor, and you want to create a subproduction or purchase order for the subcontractor. The service or work that the subcontractor performs must be created by using a formula item or service item. You can attach the item to the parent item as a formula line. The route must contain an operation that is assigned to the subcontractor's operations resource. This operation is attached to the formula line by using the **Oper. No.** field.

Formula versions

When you create a new formula, you must first create a formula version before you add the formula line items and their specific characteristics. Every formula must have at least one version. The **Approved** button on a formula version becomes available only after a version record has been successfully saved. Each formula version record is associated with one or many co-products and by-products that can be produced as you produce the finished product. Many products can be made from the same ingredients in different batch sizes, in multiples, or by using different yields. You can create as many versions of a formula as you require.

To manage multiple active formula versions, use effective date ranges or "from" quantity fields. Multiple active formula versions can exist only if the date range and "from" quantity don't overlap.

Unlike BOMs, where one BOM is often associated with many BOM versions, only one formula version typically exists per formula. Remember that only one formula version can be activated for the coverage dimensions and quantity for a given product. However, many formula versions might exist for other reasons, and you can manually select them when you create a batch order.

Approve and activate formulas and formula versions

Formulas and formula versions must be approved before they can be used for planning and production. Formulas are usually activated before they're used. However, during production, you can select a formula version that is approved, but that isn't activated.

To help secure a formula or formula version, you can set the **Block editing** and **Block removal of approval** parameters on the **Production control parameters** page.

If you select **Block editing**, and the formula is approved, no fields on the formula lines can be deleted or edited. However, if you remove the approval of the formula, you can delete and modify the formula lines. You can also create new formulas and new formula versions.

If you select **Block removal of approval**, you can't remove the approval of an approved formula or formula version. However, you can create new formulas and new formula versions, and you can remove the activation of the formula version.

You can add more levels of control by using the electronic signature functionality. When a user is set up so that an electronic signature is required at the time of formula approval, a **Signature** page appears when the formula is activated. The user must be authorized to sign electronically, and the certificate must be successfully validated before the change can be committed. If the signature can't be authenticated, the approval or removal of approval is denied, and the change that initiated the approval or removal of approval is returned to its original state.

Use the Scalable feature

The Scalable feature is available only if all the item components in the formula are set to **Variable consumption**. The feature isn't available if item components are set to **Fixed consumption** or **Step consumption**. When the Scalable feature is used, if you change an ingredient in a formula, the quantity of the other ingredients that you select is adjusted. The size of the formula is also adjusted. Likewise, if you change the formula size, the quantity of all scalable ingredients is changed. This feature is intended specifically for formula creation and maintenance. It doesn't indicate whether the quantity of an ingredient will be scaled up or down on a batch order.

Use Step consumption

Step consumption eliminates the requirement that you must enter a quantity on the **Formula line** tab for an ingredient. Instead, Step consumption is configured so that it has a **From series** value and a **Quantity** value. The information from the Step consumption per series record that

satisfies the quantity on the batch order is selected. Step consumption is useful when the consumption rate isn't linear with respect to the batch order size and only increases the requirement when a specific quantity threshold is met. To enable this feature for a new formula, under the **Consumption calculation** group, change the formula setting for the applicable ingredient from **Standard** to **Step**. You specify this consumption method on the **Setup** tab of the **Formula line** page.

Operations resources

07/10/2025

Operations resources perform the activities of a project or a production process. They can be of different types, and can have different capabilities.

Operations resources

Operations resources are the machines, tools, workers, facilities, physical areas, or vendors that perform the activities of a project or a production process. They can be of different types and can have different capabilities.

- **Vendor** – An external resource that performs project activities or production operations. An example is a subcontractor. By linking vendor resources to a vendor account, you can generate purchases for subcontractors, based on the bill of materials (BOM) lines or production lines.
- **Human resources** – A project or production worker that perform an activity, either alone or as an operator of a tool or a machine. If you're using the Human resources functionality, you can link human resources to a worker. The scheduling engine can then allocate the resources, based on the competencies that are defined for the corresponding worker.
- **Machine** – A machine or other production equipment that is required in production.
- **Tool** – An instrument or device that is typically used together with another resource to perform an activity in a project or in production.
- **Location** – A physical location of a specific size that is required in order to perform an activity. An example is an assembly area.
- **Facility** – A building or fixed structure that is required in order to perform an activity.

Calendars

A calendar can be assigned to an operations resource and describes the capacity (in hours) of that resource. The working times of the operations resource are determined by the calendar that is assigned to the resource group that the operations resource is part of. You can set an effective date and an expiration date for the assigned calendar. You can then assign more than one calendar to an operations resource. However, the dates of the assigned calendars can't overlap, and the operations resource can be assigned only one calendar at a time.

> ⓘ **Note**

If there are no effective working calendars for a resource group (for example, if the last assigned calendar or the only assigned calendar has expired), you can no longer access the operations resources that are assigned to the resource group for production planning and operations scheduling. You can also assign a calendar to a resource group to specify the working times for both the resource group and all the operations resources that are assigned to the resource group. The capacity of the resource group is calculated as the sum of the capacities of each operations resource that is assigned to that resource group. The calendar that is assigned to a resource group can change over time.

Resource capabilities

A capability is the ability of an operations resource to perform a particular activity. You can assign one or more capabilities to an operations resource. The scheduling engine will then allocate resources by matching the resource requirements of each activity to the capabilities of the available operations resources. Capabilities can be assigned to all types of operations resources (*Tool*, *Vendor*, *Machine*, *Human resources*, *Location*, or *Facility*). To assign capabilities to operations resources for a limited time, define a start date and an expiration date on the capability assignment. Learn more in Resource capabilities.

Resource groups

A resource group is a set of operations resources that represents the granularity at which you want to schedule resources when you use the operations scheduling method. Therefore, resource groups typically correspond to the physical organization of work cells that is demarcated by yellow lines on the production shop floor. The resource group defines the site, production unit, and warehouse context for operations resources that are assigned to the group. When you assign an operations resource to a resource group, the resource can be scheduled at the site where the resource group is located. You don't have to assign the operations resources that you create to a resource group. However, an operations resource must be assigned to a resource group before it can be scheduled to perform work. An operations resource can be assigned to a resource group for a limited time. You can also assign an operations resource to multiple resource groups, so that you can then share the resource across sites. However, the effective dates and expiration dates can't overlap. In other words, you can't assign an operations resource to two resource groups at the same time. Changes in resource group assignments are effective only for new resource allocations. Capacity reservations for jobs, operations, and project hour forecasts that are already scheduled won't be affected.

Note

When you assign operations resources of the *Vendor* type to a resource group, all operations resources that are assigned to that resource group must be of the *Vendor* type and must be linked to the same vendor account.

Production units

A production unit is an administrative unit that is a collection of resource groups. A production unit can contain multiple resource groups, but a resource group can be assigned to only one production unit. A production unit reflects the physical layout of production resources, and has no effect on transactions or how they're processed. You must associate a production unit with a site. You can also assign a picking warehouse and a storage warehouse to a production unit. You can use a production unit to consolidate and filter production-related data. For example, a shop floor manager can see an overview of the outstanding workload and the available capacity for a particular production unit. You can change the production unit that is assigned to a resource group. You can also delete a production unit. However, these changes to the production unit are effective only for new orders that are created after master scheduling is run. If you want to apply the production unit change to existing orders, you must make this change manually.

Resource scheduling

Operations resources are assigned to activities when a project or a production is scheduled. Two scheduling methods are available: operations scheduling and job scheduling. When you use operations scheduling, each operation or project activity is scheduled on the resource group that contains operations resources that match the resource requirements that are specified for the operation. If a specific operations resource is required for the operation, scheduling reserves capacity only on a specific operations resource in the group. Job scheduling is a more detailed form of scheduling than operations scheduling. It breaks down each operation into its individual tasks or jobs. These jobs are then assigned to the operations resources that will perform them. Scheduling reserves capacity on the resource groups, based on the operation times that are defined on the production route or project activities. If you're working with finite capacity, the schedule depends on the availability of the operations resources that are required in order to complete the activity. For operations scheduling, the capacity of the resource group is the sum of the available capacity of the operations resources that are part of that group. Capacity reservations that already exist for the operations resources are considered unavailable capacity. If there isn't enough available capacity for production, the production orders can be delayed or even stopped. On the **Resources** page, you can define several properties that control how capacity reservations are made:

- **Capacity** – Specify the operations resource's capacity per hour in terms of the capacity unit of measure.
- **Batch capacity** – Specify the maximum number of pieces that the operations resource can process per run.
- **Efficiency percentage** – Specify the efficiency that you expect from the operations resource. The efficiency percentage adjusts the throughput of the operations resource and affects the time that is reserved for the resource. The lead times for the operations that use the operations resource are also adjusted accordingly. Here's the formula that is used for the calculation: Scheduling time = Time × 100 ÷ Efficiency percentage. *Time* includes both the run time and setup time.
- **Operations scheduling percentage** – Specify the maximum percentage of capacity of the operations resource that you want to use in operations scheduling. To allow for flexibility in capacity during job scheduling, you should set this percentage to less than 100 percent.
- **Finite capacity** – Set this option to *Yes* if the operations resource should be scheduled based on the actual capacity that is available, and if existing capacity reservations should be considered. If this option is set to *No*, the operations resource is assumed to have infinite capacity, and the resource might therefore be overbooked.
- **Finite property** – Set this option to *Yes* if you want the operations resource to be scheduled based on the actual capacity that is available with respect to the required working time scheduling properties.
- **Exclusive** – Set this option to *Yes* if you don't want the operations resource to be available for another job or operation until the current production is completed. In this case, the operations resource can't be used even if there are gaps in the resource's run time.
- **Bottleneck resource** – Define the operations resource as a bottleneck resource. A bottleneck resource is scheduled by using finite capacity when the **Finite capacity** and **Bottleneck scheduling** options on the **Master plans** page are selected.

To schedule multiple operations resources at the same time to perform, for example, an operation in production, you must specify the requirements for the various resources by using primary and secondary operations. You can then also reserve multiple operations resources that have different capacity. The operations resource that is scheduled for the primary operation determines the duration of the activity.

Lean work cells

You can specify that a resource group is a lean work cell. The resource group can then be part of a production flow. By specifying a resource group as a lean work cell, you also prevent the resource group and the assigned operations resources from being allocated to the operations of a production order or a project hour forecast. For each resource group that acts as a lean work cell, you must specify the following information:

- **Calendar** – The working calendar of the work cell, which must have the property of a standard workday.
- **Input warehouse and location** – The default location that is used to pick material for an activity.
- **Output warehouse and location** – The default location where all output of the work cell is put.
- **Runtime cost category** – This category must be defined for the work cell if direct labor is tracked in costing.

When a resource group is used as a lean work cell, the capacity of the work cell is specified directly on the resource group. Therefore, you don't have to assign operations resources to the resource group. Only when the work cell is managed by a subcontractor, an operations resource of the *Vendor* type must be assigned to the work cell. If you assign an operations resource to a resource group that is marked as a work cell, the capacity of the work cell isn't affected.

Costing resources

When you define an activity such as a route operation or a project hour forecast, you can specify the requirement for a specific operations resource or resource group. However, you can also specify the requirement for an operations resource of a specific type, or an operations resource that has a specific capability or competency. For this reason, the actual resource assignment isn't made until the activity is scheduled and capacity is reserved. Therefore, on a route operation, you can specify that estimation and BOM calculation must be based on a specific operations resource. This operations resource is referred to as the costing resource. You can also transfer cost categories and operation times from the costing resource to the activity. When the operation is scheduled, estimation and BOM calculation are done by using the operations resource that is actually scheduled.

Resource capabilities

06/17/2025

This article provides information about resource capabilities. A capability is the ability of an operations resource to perform a particular activity. The article explains how capabilities and related concepts, such as proficiency level and priority, are used to select appropriate resources for an activity.

A capability is the ability of an operations resource to perform a particular activity. An operations resource can have more than one capability assigned to it, and a capability can be assigned to more than one resource. You can temporarily assign capabilities to resources by defining a start date and expiration date on the capability assignment. When the capability for a resource expires, the resource can't be scheduled for a project or a production that requires that capability. Expired capabilities can be renewed. You can delete capabilities provided that they aren't on a route relation or on part of a production route of an active production order. In general, be careful when you delete capabilities. Instead, consider adjusting the expiration date on the resources that have the capability. Capabilities can be assigned to all types of resources: tool, vendor, machine, location, facility, or human resource.

Level

Multiple resources often have the same functional capability but at different levels of proficiency (for example, strength, processing speed, or accuracy). Therefore, when you assign a capability to a resource, you can specify a **Level** value. The level is a simple numeric value. If you specify that a capability is a resource requirement for a production route, you can also specify a **Minimum level needed** value for that capability. The scheduling engine then selects only resources that have the required capability at a level that is equal to or exceeds the minimum level that is specified in the resource requirement.

Priority

Operations resources that have the same capabilities can be assigned a priority. Then, if multiple resources have capabilities that satisfy the scheduling requirements at the required level, and have free capacity, the scheduling engine can select among those resources. If **Priority** is selected in the **Primary resource selection** field on the **Scheduling parameters** page, the resource that has the highest priority (the lowest numeric value in the **Priority** field) is selected first during scheduling.

Resource requirements

When you define resource requirements for a production route, you can specify one or more capabilities as requirements. During production scheduling, the capabilities that are defined in the resource requirements on the route operation are matched with the capabilities that are defined for the resources. Resources that have capabilities that satisfy the requirements are selected. If multiple resources have the same functional capability but different proficiencies (such as strength, processing speed, or accuracy), you can either define multiple capabilities or use the capability level to qualify the capability of the resource. Here's an example:

- On a route, the drilling process time is set to 10 units per hour, but the requirement itself is defined as Drilling.
- You have two drilling machines, M1 and M2.
- The Drilling capability is assigned to both resources, the M1 machine and the M2 machine.
- The M1 machine can drill two units per hour, and the M2 machine can drill 11 units per hour.

In this example, both machines can be selected by the scheduling engine, because both satisfy the basic capability requirement (Drilling). However, the M1 machine can drill only two units per hour. Because this rate is much less than the 10 units per hour that are specified on the route, the M1 machine will cause production problems if it's selected. There are two ways to resolve this issue and make sure that the M2 machine is selected instead:

- Create two separate capabilities: Low-speed drilling and High speed drilling. Define Low-speed drilling as drilling that has a process time of one to nine units per hour. Define High-speed drilling as drilling that has a drilling process time of 10 to 19 units per hour. Then assign the M1 machine the Low-speed drilling capability, and assign the M2 machine the High-speed drilling capability. Finally, change the resource capability requirement on the route to High-speed drilling. The scheduling engine will then select the correct machine (M2).
- Use the capability level to qualify the Drilling capability that is assigned to the drilling machines. Specify that the M1 machine has the Drilling capability at a level of 2.0, and that the M2 machine has the Drilling capability at a level of 11.0. Then specify that 10.0 is the minimum level that is required for the Drilling capability requirement on the route. The scheduling engine will then determine that only the M2 machine satisfies the resource requirements.

Competencies for human resources

When you have operations resources of the **Human resources** type that are linked to workers in Human resources, you can also take advantage of the competencies of workers when you define the resource requirements for a production route. In other words, you can also specify

requirements for specific skills, courses, certificates, or titles. The scheduling engine can then select resources that are linked to workers, and the selection will be based on the competencies of those workers. The competencies are set up in Human resources, not on the **Resource capabilities** page. When you define skills, courses, certificates, or titles as resource requirements, you must use the Human resources functionality and link each resource of the **Human resources** type to a corresponding worker. If you aren't using the Human resources functionality, you can define capabilities on the **Resource capabilities** page that resemble or duplicate the competencies from Human resources. However, the **Resource capabilities** page doesn't contain the functionality that is required in order to maintain skills, courses, certifications, or titles.

Batch attributes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about batch attributes. Batch attributes are characteristics of raw materials and finished products that make up inventory batches. The article also explains how to assign batch attributes, and how you can search on them when you reserve batches.

Batch attributes are characteristics of raw materials and finished products that make up inventory batches. Batch attributes can vary, depending on factors such as environmental conditions, the quality of the raw materials that are used to produce the batch, or the outcome of the finished product. The number and types of batch attributes that are used can vary widely from one industry to another. Here are two examples that show how to use batch attributes:

- In the cheese industry, milk, which is one of the raw materials that are used to produce the cheese, can have attributes such as fat content and percentage weight. The cheese that is produced from the milk can have other attributes, such as moisture and age.
- In the steel industry, the iron that is produced might have attributes such as the percentages of magnesium content, silver content, and zinc content.

To better manage the number and types of attributes, you can use batch attribute groups. In this way, you don't have to add each attribute individually.

Assign batch attributes

You can assign batch attributes to individual products that are held in inventory batches, or you can assign them to products that are associated with specific customers. Before you can assign a batch attribute at the customer level, you must assign it at the product level. The product must have the batch dimension set to *Active* in the tracking dimension group. To assign a batch attribute to an individual product, use the product-specific page. If the attribute is specific to a product for a customer, use the customer-specific page. When you add an attribute to a product, you also define other parameters. Here are some examples:

- The minimum and maximum ranges for an attribute of the *Integer* or *Fraction* type.
- The tolerance actions for an attribute of the *Integer* or *Fraction* type. If the value of the attribute falls outside the minimum and maximum range, the action can be either a

warning message or an error message.
- The target value for the attribute. This value is the optimal value of the attribute, and it applies to all attribute types.

You can access the pages for products that you select on the **Released products** page in Product information management. After you assign batch attributes to a product, you can add specific values to the attributes on the **Inventory batch attributes** page.

Reserve batches

You can search on batch attributes when you do batch reservations for a sales order to fulfill a customer's order, or when you pick and reserve batches for a production order. The search helps locate an inventory batch that contains the product that has the batch attribute that you want. After you locate the batch or batches, you can reserve the product to the originating inventory transaction line.

Last updated on 05/07/2025

Batch balancing

09/19/2025

This article describes how the batch balancing process works.

For more information, watch a video on batch balancing.

In the batch balancing process, you calculate the amount of ingredients to use in a production batch from the concentration of active ingredients in selected product batches.

Products that have an active ingredient

You can define a product by its concentration of an active ingredient. Model the active ingredient of a product by using a product-specific batch attribute that has a minimum value, a maximum value, and a target level.

The target level of a batch attribute represents the estimated percentage of an active ingredient in a product. The minimum and maximum values represent the accepted deviation from the target level. Use these values, for example, as an accepted tolerance for batches at product receipt.

A product can have only one active ingredient. To specify the active ingredient of a product, first define a product-specific batch attribute. Then associate the attribute as a base attribute of the product.

On the product level, specify how the level of the active ingredient for a batch of the product should be recorded: as part of the purchase receipt process or as part of a quality order process.

To associate a base attribute with a product, complete the following setup steps:

1. Make the product batch-controlled by assigning a tracking dimension group to the product that has an active Batch dimension.
2. Define the attribute that indicates the ingredient levels as a product-specific batch attribute for the product.

To look up and edit the actual value of the active ingredient for a batch:

1. Go to **Inventory management** > **Inquiries and reports** > **Tracking dimensions** > **Batches**.
2. Select a batch number from the grid.
3. On the Action Pane, open the **View** tab and then select **Inventory batch attributes**.

Ingredient types and how they interact in the batch balancing process

Formula lines that you create can have one of these ingredient types:

- None
- Active
- Compensating
- Filler

The rest of this section provides examples that show how each ingredient type works. The examples are based on the following formula that has a total batch size of 100 liters.

Expand table

Ingredient type	Item number	Formula line quantity	Unit
None	A	20	Liter
Active	B	30	Liter
Compensating	C	10	Liter
Filler	D	40	Liter

The following table provides an overview of the results of each example.

Expand table

Item number	Ingredient type	Estimated quantity	Balanced quantity	Active quantity	Unit	Base value
A	None	20	20		Liter	
B	Active	30	25.71	9.00	Liter	30.00
C	Compensating	10	14.72		Liter	
D	Filler	40	39.57		Liter	

Active ingredients

When you add a product with a base attribute to a formula line, you add an *active ingredient* to the formula. You can use batch orders with formulas that include active ingredients for the batch balancing process. For each ingredient in the formula, the batch balancing process

estimates the amount needed to produce the product. The estimate is based on the concentration of active ingredients in the selected batches.

Active ingredient example

Ingredient B has base attribute X and a target level of 30. It's included in a formula that requires 30 liters of ingredient B for every 100 liters of the product. You create a batch order with a batch size of 100 liters. You start the batch order, and during the batch balancing process, you select a batch of ingredient B that has a potency level of 35. Because the potency level of 35 is higher than the target level of 30, the batch balancing process reduces the balanced quantity of ingredient B by using the ratio of the potency value to the target level of the batch, which is multiplied by the estimated quantity. The calculation of the balanced quantity looks like this:

(30 ÷ 35) × 30 liters = 25.71 liters

None ingredients

When you apply the batch balancing process and set the **Ingredient type** to *None*, the estimated quantity and the balanced quantity of the formula line in the batch order are the same.

None ingredient example

You assign ingredient A to an ingredient of the *None* type and add it to a formula for a finished product. The formula requires 10 liters of ingredient A for every 100 liters of the finished product. When a batch order requires 200 liters, both the estimated quantity and the balanced quantity of ingredient A are 20 liters.

Compensating ingredients

A compensating ingredient can either offset or complement the effect of the active ingredient in a product. Therefore, the quantity of a compensating ingredient that you consume depends on the potency of the product:

- **Opposing effect** – If the amount of the active ingredient is more than anticipated, you must add less of the compensating ingredient.
- **Complementary effect** – If the amount of the active ingredient is less than anticipated, you must add more of the compensating ingredient.

The relation between an active ingredient and a complementary ingredient is set up on the **Compensating principle** page.

Follow these steps to set up relations between ingredients.

1. Select **Product information management** > **Bills and materials and formulas** > **Formulas**.
2. Open a formula line, and then select **Ingredients** to open the **Compensating principle** page.
3. Select the line that represents a compensating principle, and then select the active ingredient to compensate.

In the compensating principle, enter a positive or negative compensating factor to specify how much to compensate for, and whether the principle should be opposing or complementary. A positive factor indicates a complementary effect, and a negative factor indicates an opposing effect.

Compensating ingredient example

Ingredient B is an active ingredient that has base attribute X and a target level of 30. It's included in a formula that requires 30 liters of ingredient B for every 100 liters of the product. Ingredient C is a compensating ingredient, and a quantity of 10 is included in the same formula. A compensating factor of 1.10 is set up for the compensating principle. Therefore, the balanced quantity of the compensating ingredient is reduced by the difference between the active ingredient's balanced quantity and the estimated required quantity multiplied by 1.10.

In the example for the *Active* ingredient type, the balanced quantity of the required active ingredient is calculated as 25.71, and the estimated required quantity is calculated as 30. In this case, the balanced quantity of the compensating ingredient is calculated like this:

1. Determine the difference between the estimated and the balanced quantity:
   25.71 – 30 = –4.29

2. Multiply the result by the compensating factor:
   4.29 × 1.10 = –4.72

3. Reduce the estimated compensating quantity by –4.72 to determine the balanced compensating quantity:
   10 – (–4.72) = 14.72

Because 1.10 is a positive compensating factor, this compensating principle has a complementary effect. In this case, the active ingredient is more potent than anticipated. Therefore, you need more of the compensating ingredient.

Filler ingredients

A *filler ingredient* is a neutral ingredient that you use to reach the desired output quantity of the finished product. Adjustments to filler quantities are based on variations in the active ingredient and the compensating ingredient compared to the standard quantity.

Filler ingredient example

You formulate a product that includes ingredients A, B, C, and D for a formula size of 100 liters. You calculate the balanced quantity of all the ingredient types except the *Filler* ingredient type that you use on one line. You calculate the balanced quantity of the filler ingredient as the difference between the batch size of 100 liters and the sum of ingredients A, B, and C:

100 – (20 + 25.71 + 14.72) = 39.57

The batch balancing process

You perform the batch balancing process from the **Batch balancing** page. Select **Cost management** > **Batch orders**, then, on the **Process** tab, select **Batch balancing**. Batch balancing is available for batch orders that have a status of *Started*.

In general, you can apply batch balancing to batch orders if the formula has at least one formula line where the **Ingredient type** is *Active*. (For the exception to this rule, see the "Batch orders that aren't applicable for batch balancing" section later in this article.)

You can divide the batch balancing process into two subprocesses:

1. Balance batch ingredients
2. Confirm and release the formula

Balance batch ingredients

In the Balance batch ingredients subprocess, the system calculates the amount of ingredients to use for the production batch based on the selected batches that have active ingredients. As a rule, you can complete the calculation only if there's full coverage of all ingredients. You can't balance only part of the batch that the batch order is set up to produce.

> **Note**
>
> You can't save a calculation and then complete the batch balancing process later. If you close the **Batch balancing** page, you must repeat the calculation to complete the process.

Confirm and release the formula

After you calculate the ingredient quantities, confirm and release the formula. The release process differs, depending on whether the products are enabled for warehouse management processes (WMS):

- For WMS-enabled products, the formula line is released to the warehouse according to the principles for WMS. The formula line is released in quantities that match the balanced quantities, and it's released for the specific batches that are selected for the active ingredients. Formula lines can be released to warehouse only as part of the batch balancing process. Although there are other options for releasing materials for production to warehouse, those options can't be used for formula lines.
- For non-WMS products, a production picking list is created for the product when you confirm and release the formula.

A single formula can include both WMS-enabled products and non-WMS products. In this case, the WMS-enabled products are released to warehouse. For the non-WMS products, a picking list is created when the formula is confirmed and released.

Batch orders that aren't applicable for batch balancing

Batch orders usually apply batch balancing when the formula has at least one formula line where the **Ingredient type** is *Active*. However, the following exceptions apply:

- If a formula contains an active ingredient for a product that is enabled for the WMS, but batch number is below location in the reservation hierarchy, the batch order isn't applicable for batch balancing.
- If the formula unit of measure is different from the inventory unit of measure of the active ingredient, the batch order isn't applicable for batch balancing.

A batch order that isn't applicable for batch balancing goes through the regular process cycle for batch orders.

Catch weight product processing with warehouse management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Prerequisites

To use warehouse management to process catch weight products, turn on the functionality by using a license configuration key. Go to **System administration** > **Setup** > **License configuration**. Then, on the **Configuration keys** tab, expand **Trade** > **Warehouse and Transportation management**, and select the check box for **Catch weight for warehouse**. Turn on both the **Warehouse and Transportation management** license configuration key and the **Process distribution** > **Catch weight** license configuration keys.

After the license configuration key is turned on, when you create a released product, you can select **Catch weight**. You can also associate the released product with a storage dimension group that the **Use warehouse management processes** parameter is selected for.

Before you can use the product in Warehouse management, complete some basic product-specific setup for catch weight:

- Define the nominal weight conversion between the catch weight unit (box) and the inventory unit (kilogram [kg]) as part of a unit conversion definition.
- Define the minimum and maximum weight tolerances as part of the catch weight unit setup.
- Set up a unit sequence group where the catch weight unit is defined as the lowest stock keeping unit (SKU).
- Set up a catch weight item handling policy.

Learn more in Setting up and maintaining catch weight items.

Transaction adjustments

Because the weight of inventory when it comes into a warehouse can differ from the weight when the inventory is issued out of the warehouse, the catch weight product processing must adjust the inventory.

> **Note**
>
> Mobile device activity triggers the transaction adjustments only if the Outbound weight variance method of the item's catch weight item handling policy is **Allow weight variance**.

Example 1

During a **Report as finished** production process, the inbound weight of a license plate that contains eight boxes of a catch weight product is captured as 80.1 kg. The license plate is then stored away in the finished goods area, and during the storage period, some weight is lost into the air.

Later, as part of a sales order picking process, the weight of the same license plate is captured as 79.8 kg. Therefore, in the system, you now have a weight difference as part of the physical dimension set.

In this case, the system automatically adjusts the difference by posting a transaction for the missing 0.3 kg.

Example 2

In its definition, a product is set up to tolerate a minimum weight of 8 kg and a maximum weight of 12 kg for the **Box** catch weight unit.

You have two boxes of the product, and they have a registered weight of 16 kg. If a warehouse worker picks and weighs one of the boxes, and the weight is captured as 9 kg, the remaining box weighs 7 kg. However, because 7 kg is below the minimum weight, the system does an automatic adjustment to increase the weight of the on-hand inventory by 1 kg.

To set up the accounts that these adjustments are posted to, go to **Cost management** > **Ledger integration policies setup** > **Posting**. Then, on the **Inventory** tab, define the following accounts:

- Catch weight loss account
- Catch weight profit account

Catch weight item handling policy

The catch weight item handling policy defines two primary warehouse management flows for the items: when the weight of the items is captured, and how it's captured.

You can define when the weight is captured for sales and transfer order processing. The weight can be captured during either of the following processes:

- **Picking** – The weight is captured during the initial pick work lines of order work.
- **Packing** – The weight is captured during manual packing. (You must send the items to a packing station.)

If the actual weight is captured at the packing station during the container packing processes, warehouse workers aren't prompted to capture the weight during picking work. Instead, the average weight of the physical inventory is used as the weight of the picked inventory that goes to the packing area. This concept also applies to catch weight items that are tracked by tags. For tag-tracked items, these parameters determine when the tag is captured. The tag can be captured either at picking time by using the mobile device or during manual packing.

> ⓘ **Note**
>
> Because the **Packing** option causes inventory to be updated with the average picked weight, this could trigger a discrepancy that could cause a catch weight profit/loss adjustment and/or a difference between on-hand inventory weight and catch weight tag weight.

For internal processes, such as counting and adjustment corrections, you can define whether the weight should be captured. If it isn't captured, the nominal weight is used. Other options let you capture weight per catch weight unit and per counting quantity.

You can also define how the weight is captured. In one of the two main flows, catch weight tags are tracked and used to capture the weight. In the other flow, catch weight tags aren't tracked.

- **Yes** – The item uses catch weight tags. Each tag number represents one catch weight unit (box), and a weight and other information are associated with the tag. For outbound processes, the weight that is associated with the tag is used.
- **No** – The item doesn't use catch weight tags. The inbound and outbound weighing processes are based on the actual weight that is captured during each process.

The process of tracking catch weight tags can be used for items that don't change weight during the storage period. The weight is captured only during the inbound warehouse process. During the outbound process, the catch weight tags are just scanned, and the weights that are associated with the tags are used for the outbound transactional processing.

Another important parameter that is related to the processing of catch weight tags is **Catch weight tag dimension tracking method**. Tags can be either partially tracked or fully tracked. If a tag is partially tracked, it tracks product dimensions, tracking dimensions, and inventory status. If a tag is fully tracked, it tracks product dimensions, tracking dimensions, and **all** storage dimensions.

Additionally, when an item is tag-tracked, there's an **Outbound tag capturing method** parameter. You can set this parameter so that you're always prompted for the tag on outbound transactions from the mobile device. Alternatively, you can set the parameter so that you're prompted for tags only when they're required. For example, there are five catch weight tags in inventory on a given license plate, and you indicate that you want to pick all five tags from the license plate. In this case, if the **Outbound tag capturing method** parameter is set to **Only prompt for tag when needed**, the five tags are automatically picked. You don't have to scan each tag. If the parameter is set to **Always prompt for tag**, you must scan each tag, even if all five tags are being picked.

> ⓘ **Note**
>
> As a rule, tags are captured and updated only from the mobile device menu items. Nevertheless, there are a few scenarios where tags are captured somewhere else (for example, from the manual packing station). However, in general, the mobile device menu items should be used for all warehouse activity if tags are used.

How to capture catch weight

When catch weight tag tracking is used, a tag must always be created for every catch weight unit that is received, and every tag must always be associated with a weight.

For example, **Box** is the catch weight unit, and you receive one pallet of eight boxes. In this case, you must create eight unique catch weight tags, and associate a weight with each tag. Depending on the inbound catch weight tag, you can either capture the weight of all eight boxes and distribute the average weight to each box, or capture a unique weight for each box.

When using the *Use existing catch weight tags when reporting production orders as finished* feature with the process enabled via a mobile device menu item, the inventory is updated based on existing catch weight tag information. As a result, the Warehouse Management mobile app doesn't prompt for capturing the catch weight tag data as part of a production report as a finished operation.

When catch weight tag tracking isn't used, the weight can be captured for each dimension set (for example, for each license plate and tracking dimension). Alternatively, the weight can be captured based on an aggregated level, such as five license plates (pallets).

For the methods for capturing outbound weight, the **Per catch weight unit** option lets you specify that the weighing should be done for each catch weight unit (for example, per box). The **Per picking unit** option lets you specify that the weight should be captured based on the quantity that you pick (for example, three boxes). If you use the **Not captured** option, the average weight is used for the production line picking and internal movement processes.

Define multiple weight capturing methods on the catch weight item handling policy. Various transactions use each weight capturing method parameter. The following table summarizes which parameters are used by which transactions.

Expand table

Method	Transaction
Outbound weight capturing method	Sales picking, Transfer picking
Production picking weight capturing method	Production picking, Production consumption
Movement weight capturing method	Movement
When to capture correction of weight	Adjustments, Counting
Counting weight capturing method	Counting
Warehouse transfer weight capturing method	Warehouse transfer

To prevent the warehouse management picking processes from capturing weights that cause catch weight profit/loss adjustments, you can use the Outbound weight variance method. The Outbound weight variance method applies during the following mobile device processes: sales picking, transfer picking, production picking, movements, counting, and warehouse transfers. You can use the **Restrict weight variance** option if the weight of the catch weight item doesn't fluctuate during warehouse storage, and if catch weight profit/loss adjustments aren't required. You can use the **Allow weight variance** option if the weight can fluctuate, and if catch weight profit/loss adjustments are required when a weight fluctuation is recorded.

Unsupported scenarios

Not all workflows support catch weight product processing with warehouse management. The following restrictions currently apply. They apply to all catch weight items, regardless of whether

they're tagged.

Configuring catch weight products for warehouse management processes

- Only formula processing is supported for catch weight products (not bill of materials).
- Catch weight products can't be associated with a tracking dimension group by using the Owner dimension.
- Catch weight products can't be used as services.
- You can use catch weight products as "stocked products" only as part of the item model group.
- You can't use catch weight products together with the functionality for tracking "Active in sales process."
- Catch weight products can't be used together with the functionality for capturing serial numbers. Therefore, products can't be transferred from a "blank" to a serial number as part of the picking/packing process.
- Catch weight products can't be used together with the functionality for registering serials before consumption.
- You can't use catch weight products that are variant-enabled together with the functionality for converting variant units of measure.
- Catch weight products can't be marked as a commerce "product kit."
- Catch weight products can be used only with a unit sequence group that has catch weight handling units, and that has the catch weight unit as the lowest sequence.
- The setup of bar codes for catch weight products doesn't support a variable weight setup.

Order processing

- The creation of advance ship notice (ASN/packing structures) doesn't support weight information.
- You must maintain the order quantity based on the catch weight unit.

Inbound warehouse processing

- Receiving license plates requires that weights be assigned during registration, because weight information isn't supported as part of the advance ship notice. When catch weight tag processes are used, the tag number must be manually assigned per catch weight unit.
- Inbound quality check work isn't supported for catch weight products. If configured, the quality check work is skipped.

Inventory and warehouse operations

- Manual creation of quarantine orders isn't supported for catch weight products.
- Manual movement of inventory that is related to open work isn't supported for catch weight products.
- License plate loading to initialize warehouse stock isn't supported for catch weight products.
- Batch balancing processes aren't supported for catch weight products.
- Handling of negative physical inventory isn't supported for catch weight products.
- Inventory marking can't be used for catch weight products.

Outbound warehouse processing

- The functionality for cluster picking isn't supported for catch weight products.
- Pick and pack warehouse processing isn't supported for catch weight products.
- For catch weight products, work that is defined in a work template cannot be run automatically.
- For catch weight products, the system doesn't support manual packing station processing where packed container picking work is created after containers are closed.
- The functionality for pcs-by-pcs scanning isn't supported for catch weight products.

Production processing

- For catch weight products, only batch orders for formula products are supported.
- Kanban functionality isn't supported for catch weight products.
- For catch weight products, serial numbers can't be registered before consumption.
- The functionality for reversing license plates from production isn't supported for catch weight products.
- For catch weight products, reporting as finished can't be registered by serial number.

Transportation management processing

- Load building workbench processing isn't supported for catch weight products.
- Transport request lines aren't supported for catch weight products.

Other restrictions and behaviors for catch weight product processing with warehouse management

- During picking processes where the user isn't prompted to identify tracking dimensions, the weight assignment is done based on the average weight. This behavior occurs when, for example, a combination of tracking dimensions is used in the same location and, after a user processes picking, only one tracking dimension value remains in the location.
- When inventory is reserved for a catch weight product that is configured for warehouse management processes (WMS), the reservation is done based on the minimum weight that is defined, even if this quantity is the on-hand last handling quantity. This behavior differs from the behavior for items that aren't configured for WMS. There is one exception to this restriction. For production picking, when the last handling quantity of a catch weight product that is serial number–controlled is picked, the actual weight is used.
- Processes that use the weight as part of capacity calculations (wave thresholds, work maximum breaks, container maximums, location load capacities, and so on) don't use the actual weight of the inventory. Instead, the processes are based on the physical handling weight that is defined for the product.
- In general, Commerce functionality isn't supported for catch weight products.
- For catch weight products, inventory status can't be updated from **Warehouse status change**.

Catch weight tags

A catch weight tag can be created by using a Warehouse Management mobile app process, it can be manually created in the form **Warehouse management** > **Enquiries and reports** > **Catch weight tag** or it can be created by using a data entity process. If a catch weight tag is associated with an inbound source document line, such as purchase order line, the tag will be registered. If the line is used for outbound processing, the tag will be updated as shipped. You can view all the historical catch weight tag registration events via the **Catch weight tag registration** option from the **Catch weight tag** page.

You can use the **Change tag captured weight** option to manually update the weight value for a catch weight tag. Note that the weight for the inventory on-hand won't get adjusted as part of this manual process, but you can easily use the **On-hand discrepancies for catch weight tagged items** page to look up any discrepancies between the currently active catch weight tags and the current inventory.

Other manual options are to **Register tag** to a source document line and **Register work** against an existing warehouse work.

In addition to the restrictions that currently apply for catch weight products, tagged catch weight products have other restrictions that currently apply.

- All manual updates to inventory (that is, updates that aren't done using a mobile device) must include corresponding manual updates to the associated catch weight tags because these updates aren't done automatically. For example, manual adjustment journals will update inventory but not the associated catch weight tags.
- You must manually update catch weight tags to reflect replenishment work movements. This is because the system can't capture weights while processing replenishment work and therefore records the average weight instead.
- Mixed licensing place receiving isn't currently supported for tagged catch weight items.
- The processing of sales return order receiving can record catch weight tags. However, the process doesn't validate that the returned tag is the same tag that was originally shipped for a sales order.
- The mobile device menu item that has the **Register material consumption** activity code doesn't currently support recording catch weight tags.
- Although counting processes are supported for tagged catch weight items, they're limited. For example, you can use the mobile device options for counting tagged catch weight items, and the average weight is used. However, catch weight tags aren't automatically updated by the counting transaction. After the counting transaction is completed, the catch weight tags must be manually updated so that they reflect the inventory. If items that weren't originally in a location are counted into that location, the nominal weight is used.
- License plate consolidation doesn't currently support tagged catch weight items.
- Reverse work functionality isn't supported for catch weight items that are tag number–tracked.

> **Note**
>
> The preceding information about catch weight tags is valid only if the catch weight product has a catch weight tag dimension tracking method that is fully tracked (that is, if the **Catch weight tag dimension tracking method** parameter on the catch weight item handling policy is set to **Product dimensions, tracking dimensions and all storage dimensions**). If the catch weight item is only partially tag-tracked (that is, if the **Catch weight tag dimension tracking method** parameter on the catch weight item handling policy is set to **Product dimensions, tracking dimensions and Inventory Status**), additional restrictions apply. Because visibility is lost between the tag and inventory in this case, some additional scenarios aren't supported.

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Copy a formula

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This procedure focuses on creating a formula that includes the same ingredients as an existing formula, but with minor differences. To create the formula lines, you can use the Copy function to copy an existing formula that has most of the ingredients that you need. You can then make any necessary changes to the individual lines in the new version. By using the Copy function, you do not have to create multiple formulas that are almost identical. The demo data company used to create this task is USP2.

Create a formula

1. Go to Product information management > Bills of materials and formulas > Formulas.
2. Click New.
3. In the Formula field, type a value.
4. In the Name field, type a value.
    - Type a meaningful name for the formula.
5. In the Site field, click the drop-down button to open the lookup.
6. In the list, click the link in the selected row.
7. In the Item group field, click the drop-down button to open the lookup.
8. In the list, find and select the desired record.
9. In the list, click the link in the selected row.
10. Click Save.

Copy formula lines

1. On the Action Pane, click Formula.
2. Click Copy.
3. In the Item number field, click the drop-down button to open the lookup.
4. In the list, click the link in the selected row.
5. In the Formula version field, click the drop-down button to open the lookup.
6. In the list, click the link in the selected row.
7. Click OK.

Adjust copied formula lines

1. In the list, mark the selected row.
2. Click Delete.
3. Click Yes.

Approve formula

1. On the Action Pane, click Formula.
2. Click Approve formula.
3. In the Approved by field, click the drop-down button to open the lookup.
4. In the list, click the link in the selected row.
5. Click Select.
6. Click OK.

Feedback

Was this page helpful? Yes No

Provide product feedback

Copy co-products from an existing formula version

Article • 09/29/2021

This procedure shows how to copy co-products from an existing formula version to a different formula version for a released product. It is a prerequisite that there is at least one formula version associated with co-products. The demo data company USP2 is used to create this procedure.

Find a released product

1. Go to Released products.
2. Click Show filters.
    - You are about to add the field Production type in the filter dialog box.
3. Click Add a filter field to add the field Production type.
    - In the next step, you need to manually enter Formula in the Production type field before you select Apply. This sets the filter on the list of released products.
4. Manually enter Formula in the Production type field.
5. Click Apply.

Select a released product

1. In the list, find and select the desired record.
2. Click Formula versions.
    - On the Engineering Action Pane, click Formula versions.

Copy co-products

1. On the Action Pane, click Formula version.
2. Click Co-products.
3. Click Copy.
4. In the Item number field, enter or select a value.
5. In the Formula version field, enter or select a value.
6. Click OK.

7. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a formula by copying from an existing formula

Article • 09/29/2021

This procedure shows how to quickly create a new formula by copying from an existing formula. You can use the demo data companies USMF, USP2, or USPI to create this procedure.

Create a new formula

1. Go to Formulas.
2. Click New.
3. In the Formula field, type a value.
4. In the Name field, type a value.
5. In the Site field, enter or select a value.
6. In the Item group field, enter or select a value.
7. Click Save.

Copy from an existing formula

1. Click Copy.
2. In the Item number field, enter or select a value.
3. In the Formula version field, enter or select a value.
4. Click OK.

Revise the copied formula

1. In the list, find and select the desired record.
2. Click Delete.
3. Click Yes.

Approve formula

1. Click Approve formula.
2. In the Approved by field, enter or select a value.
3. Click OK.
4. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Production parameters for manufacturing execution

08/13/2025

This article provides information about the setup of production parameters for manufacturing execution.

Manufacturing execution can be used to register time and item consumption on production jobs or projects. Before you start to use manufacturing execution for job registrations, you must set up various production parameters that define how and when registrations are posted during the production process. The settings of production parameters affect inventory management, production management, and cost calculation.

Before workers start to make registrations on production jobs, you should carefully consider all settings on the **Production order defaults** page. Go to **Production control** > **Setup** > **Manufacturing execution** > **Production order defaults**. If your company uses the multisite functionality, you might want to set up different production parameters for each site. The following tabs are provided on the **Production order defaults** page:

- **General** – General parameter settings for production jobs.
- **Start** – Parameters that are used when production operations are started.
- **Operations** – Parameters that are applied to production operations and feedback about operations during the production process.
- **Report as finished** – Parameters that are used when items are reported as finished on the last operation of a production order.
- **Quantity validation** – Parameters that are used to validate start and feedback quantities on production orders.

Types of production jobs

On the **Operations** tab, you select which types of production jobs require registration.

Typically, workers make registrations on setup jobs and process jobs. However, if job scheduling is applied, you can select other job types that workers must make registrations on when production orders are processed. For example, you can require registrations on transport jobs.

> **Important**

Make sure that you select all relevant job types. Otherwise, jobs might not be available for registration. Your selections should match the selections in the **Job management** column on the **Setup** FastTab of the **Route groups** page (**Production control** > **Setup** > **Routes** > **Route groups**).

If **Job management** is selected on the route group, this job type is reported as finished on the production order when a job is reported as finished. When all job types that **Job management** is selected for have been reported as finished on an operation, the operation is reported as finished.

> ⓘ Note
>
> Some job types can be manually reported through production journals. In this case, select **Job management** for the job type, but don't select the job type for registration on the **Operations** tab on the **Production order defaults** page.

BOM consumption and picking list journals

A consistent setup for bill of materials (BOM) consumption is important because it helps guarantee that inventory management is efficient. For example, if BOM consumption parameters aren't set up correctly, materials might be deducted from inventory two times or not at all.

On the **Production order defaults** page, automatic BOM consumption is set up in three stages:

- At the start of a production. Set up this stage on the **Start** tab.
- During the production process when an operation is completed. Set up this stage on the **Operations** tab.
- When a production order is reported as finished. Set up this stage on the **Report as finished** tab.

For each stage, the **Automatic BOM consumption** field lets you select one of three methods for picking items for a production order:

- *Flushing principle* – This option is used in combination with an option that is defined for the BOM. Select **Production control** > **Production orders** > **All production orders**. On the **All production orders** page, select a production order in the list, and then, on the Action Pane, open the **Production order** tab and select **BOM**. On the **BOM** page, open the BOM line you want to set. Then open the **Setup** FastTab and select one of the following options in **Flushing principle** field:
  - *Start*

- *Finish*
- *Manual*
- *Available at location*
- Blank (No option is selected.)

If *Flushing principle* is selected in the **Automatic BOM consumption** field on the **Start** tab of the **Production order defaults** page, all materials that are set to *Start* in the BOM are deducted from inventory when the operation is started. The *Available at location* option is used for products that are enabled for warehouse management processes (WMS). If you select this flushing principle, material is flushed when warehouse work for raw material picking is completed. Material is also flushed when a BOM line that uses this flushing principle is released to warehouse and the material is available at the production input location.

> ⓘ **Note**
>
> If *Flushing principle* is selected in the **Automatic BOM consumption** field on the **Start** tab of the **Production order defaults** page, you must select the same value on either the **Operations** tab or the **Report as finished** tab. This requirement helps guarantee that materials are deducted from inventory on BOMs that use *Finish* as a flushing principle on the production order. If *Flushing principle* isn't selected on either the **Operations** tab or the **Report as finished** tab, materials might be deducted from inventory two times.

- *Always* – If you select this option for a stage, materials are always deducted from inventory at that stage. For example, materials for the production are deducted when the production order is started. This setting requires that *Never* be selected on the **Operations** and **Report as finished** tabs. This requirement helps prevent items from being deducted from inventory two times.

- *Never* – If you select this option for a stage, no BOM consumption occurs at that stage. For example, if you select *Never* on all three tabs (**Start**, **Operations**, and **Report as finished**), materials must be manually deducted from inventory.

> ⓘ **Important**
>
> Carefully consider your setup of the production order defaults, and make sure that the defaults that are selected on the various tabs of the **Production order defaults** page don't contradict each other.

The following examples illustrate parameter settings that support various BOM consumption principles. The parameters are set up on the **Production order defaults** page.

Example 1: Backflushing on operations

Use the following settings if picking list journals and BOM item consumption should be generated when items are reported as finished on an operation.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status* or *Status + quantity*
Start	Automatic BOM consumption	*Never*
Operations	Automatic BOM consumption	*Always*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status + quantity*

Example 2: Backflushing on production

Use the following settings if picking list journals and BOM item consumption should be generated when items are reported as finished on the production order.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status* or *Status + quantity*
Start	Automatic BOM consumption	*Never*
Operations	Automatic BOM consumption	*Never*
Report as finished	Automatic BOM consumption	*Always*
Report as finished	Update finished report on-line	*Status + quantity*

Example 3: Flushing principle

Use the following settings if picking list journals and BOM item consumption should be generated according to the flushing principle that is set for the BOM items.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status + quantity*
Start	Automatic BOM consumption	*Flushing principle*
Operations	Automatic BOM consumption	*Flushing principle*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status + quantity*

Example 4: Deduction of materials during startup of a production order

Use the following settings if picking list journals and BOM item consumption should be generated when a production is started.

Expand table

Tab	Field	***Setting***
Start	Update start on-line	*Status + quantity*
Start	Automatic BOM consumption	*Always*
Operations	Automatic BOM consumption	*Never*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status* or *Status + quantity*

Based on the selections that are described earlier in this section, picking list journals are posted at various stages of the production order process:

- When an operation is started
- When quantity feedback is reported on an operation
- When items are reported as finished on the production order

Example 5: Manual BOM consumption

You can use the following settings if materials should always be manually deducted from inventory. In this case, picking list journals aren't posted.

Expand table

Tab	Field	Setting
Start	Update start on-line	*Status*
Start	Automatic BOM consumption	*Never*
Operations	Automatic BOM consumption	*Never*
Report as finished	Automatic BOM consumption	*Never*
Report as finished	Update finished report on-line	*Status*

Production setup requirements

This article provides information about setup requirements before you can work with Production control.

Production control is integrated with features in other modules. This interconnectivity lets you change production orders and make sure that they're automatically updated in all other related processes and calculations in the system. The following setup processes are listed in the order that you should complete them in.

Required baseline setup in other modules

Information in other modules must be set up before you can work with Production control. This setup includes the following tasks:

- Set up general company information.
- Set up the general ledger.
- Define item groups.
- Set up ledger accounts for item groups.
- Set up the inventory item table in Inventory management.
- Create bills of materials (BOMs) and BOM versions in Product information management.

Required calendar and resource setup

Before you use Production control, open Organization administration, and create and define the calendar and operations resources in the following order:

1. **Working time templates** – Set up working time templates to define the times that are available for production scheduling.
2. **Calendars** – Set up working time calendars to define the days of the year that are available for production scheduling.
3. **Resource groups** – Set up resource groups to group the operations resources, so that you can get an overview of any free capacity when you run scheduling. You don't have to set up resource groups before you set up operations resources. However, we recommend that you understand how to group resources when you set up Production control.
4. **Resources** – Set up operations resources to define the resources that are used to complete the production process and plan for capacity.

Required production parameters setup

- **Production control parameters** – Set up basic production parameters to define how the system handles and processes production orders. Define how production orders are created, estimated, scheduled, and consumed. You can also select what kind of feedback you want and how cost accounting is done.

Required journal name identification

- **Production journal names** – Specify the production journal names that are used to record and post transactions.

Setup if you use operations

Operations represent the specific activities that are completed to produce the finished product.

- **Operations** – Set up operations to represent the tasks that must be completed to produce the finished item.
- **Relations** – Set up operation relations to establish detailed properties. To define operation relations, select **Relations** on the **Operations** page.

> ⓘ **Note**
>
> You must know the types of activities that are required in order to produce your item, and the order and priorities of those activities. You must also know which resources are involved, and how many.

Setup if you use routes

If you're working with routes, operations must be defined for every production route that you set up. The route represents the path that the item takes from operation to operation, from the start of the production process to the end.

- **Cost categories** – Set up cost categories to define the cost per hour of specified processes and setup times.
- **Cost groups** – Set up cost groups to create and maintain different types of costing.
- **Route groups** – Set up route groups to define parameters that are related to groups of routes. You must set up route groups before you can create production routes.

- **Routes** – Set up production routes, and define default settings to control scheduling, costing, and pricing of route operations, and to control progress reporting.
- **Route version** – Set up route versions to enable item variations in production.

Optional advanced settings

- **Production groups** – Set up production groups to establish relationships between the production order and ledger accounts. The ledger accounts are used to post or group orders for reporting.
- **Production pools** – Create production pools to group production orders so that you can process urgent production orders, or delete and post groups of orders.
- **Properties** – Define properties to create special attributes that you can assign to your resources to control the order of productions. These attributes are connected to the working time template.

Last updated on 12/19/2025

Operations scheduling

07/10/2025

This article provides information about operations scheduling. You can use operations scheduling to provide a general estimate of the production process over time.

You can schedule production at the operation level and the job level. Unlike job scheduling, operations scheduling doesn't explode the operations for the production route into jobs. If you want to include more detail in the scheduling, such as information about current capacity, you can run job scheduling after you run operations scheduling. You can also run job scheduling only. Job scheduling is typically used to schedule individual jobs on the shop floor for an immediate or short-term time frame.

Components of operations scheduling

The main components of operations scheduling are the scheduling direction, the capacity of resources, and materials optimization. By using operations scheduling, you can achieve the following goals:

- Control the planning method by scheduling forward or backward from a given date.
- Optimize the use of resources by scheduling productions based on the capacity of the resources. This approach also helps identify when alternative resources should be used.
- Optimize the use of materials by scheduling productions based on the availability of the required materials.
- Schedule and synchronize reference productions. The dates of the reference productions are adjusted when the production order’s schedule is changed.

You must estimate the cost of a production order before you can run operations scheduling. If you haven't run an estimate, it's automatically run before operations scheduling is started. An operations schedule specifies the following information:

- The product that is planned for production
- The configuration of the product
- The quantities that are involved in the production
- The dates when the production will start and end
- The capacity reservations for the resources that perform the production activities

The setup time, process time, and run time are set for operations in the production. After you run operations scheduling, the status of the production order is *Scheduled*, and all operations are scheduled in the order that is specified by the production route. However, only the duration of the operation is considered. Start times and end times aren't scheduled.

Scheduling direction and date

The scheduling direction is fundamental to the scheduling process. Production can be scheduled forward or backward from any date, depending on timing and scheduling requirements.

- **Forward from the scheduling date** – You can schedule production to start as early as possible. Production can be started today, tomorrow, or on any date in the future. The production is scheduled forward in time to the earliest possible end date.
- **Backward from the scheduling date** – You can schedule production to start as late as possible. Backward scheduling is based on the date when the production must be completed. The schedule counts backward from that date to the latest possible date that the production can be started and still be completed on time.

Resource scheduling

When you run an operations schedule, each operation in the production route is scheduled for the resource that is specified for the operation. Additionally, the duration of each operation is specified on the production route. If a resource group is specified for an operation, the scheduling reserves capacity on the group. However, unlike job scheduling, operations scheduling doesn't select the specific resources in the group. If you're working with finite capacity, the schedule depends on the availability of the resources that are required in order to complete production. Operations scheduling follows the sequence of operations that is specified on the production route. The scheduling reserves capacity on the resource groups, based on the operation times that are defined on the production route. The sum of available capacity on the resources that are involved determines the capacity for the resource group. Capacity reservations that already exist for the resources are considered unavailable capacity. If there isn't enough available capacity for the production, the production orders can be delayed or even stopped. You can also specify the efficiency that you expect from the resources that are involved in the production. You specify the efficiency as a percentage on the resource. The efficiency percentage adjusts the throughput of the resource. This adjustment affects the time that is reserved for the resource. The lead times for the operations that use the resource are also adjusted accordingly.

Operations scheduling and master planning

The operations schedule also drives master planning and determines calculations for material requirements. Operations scheduling considers the following information:

- **Backlogged productions** – Products that are planned, released, or started
- **Material availability** – Inventory, subproductions, suppliers, and vendors

- **Capacity availability** – Resources that are required for production

> **Note**
>
> If you're using multi-threaded master planning and operations scheduling, finite capacity will not be considered.

Cancellations

When you run operations scheduling, you can cancel specific parts of the routing. These parts include the queue time, setup time, process time, overlap time, and transport times.

Finite materials

If you're working with finite materials, scheduling also depends on the availability of the materials that are required for production. If there aren't enough available components for the production, production can be delayed. You can base scheduling on the use of materials by specifying the materials that must be available for production. When you optimize on both resource capacity and the availability of materials, production is calculated according to these restrictions. A production order can't be scheduled to start until capacity and materials are available at the same time and in the required quantities.

Related information

- Operations scheduling options

Operations scheduling options

09/19/2025

This article describes the options for operations scheduling. Use operations scheduling to provide a general estimate of the production process over time.

Operations scheduling calculates the following information for a production order:

- Start and end dates for the production order and each operation.
- Resources assigned to operations.

Several settings determine how production schedules are calculated. You define these settings on the **Operations scheduling** page. The following information describes the scheduling options.

Operations scheduling

Scheduling direction

The scheduling direction is fundamental to the scheduling process. You can schedule a production forward or backward from any date, depending on timing and scheduling requirements.

- **Forward scheduling** – Schedule a production to start as early as possible. The production can start today, tomorrow, or on any specific date in the future. The production is scheduled to start on the earliest possible date and is planned forward in time to the earliest possible end date.
- **Backward scheduling** – Schedule a production to start as late as possible. The schedule is based on the date when the production must be completed and counts backward to the latest possible date that the production can start without missing its target deadline.

The following options are available:

- **Forward from today** – Schedule forward from the current date. (The current date is the system date.)
- **Forward from planned start** – Schedule forward from an earlier start date. If there's no previous scheduling, the scheduling direction is forward from the current date.
- **Forward from scheduling date** – Schedule forward from the date that you specify in the **Scheduling date** field. If you don't specify a scheduling date, the scheduling direction is forward from the current date.

- **Backward from delivery date** – Schedule backward from the delivery date that you specify for the production order. If you select this option, but don't specify a delivery date, the delivery date is the current date.
- **Backward from planned end** – Schedule backward from a previously calculated end date. If there's no previous scheduling, the end date is the current date.
- **Backward from scheduling date** – Schedule backward from the date that you specify in the **Scheduling date** field. If you don't specify a scheduling date, the current date is used. Backward from scheduling date is calculated for the production order the last time that a requirement was calculated. If you don't specify a date for the production order, the current system date is used.
- **Backward from futures date** – Schedule backward from the futures date that was calculated for the production order the last time that a requirement was calculated. If you don't specify a futures date for the production order, the current system date is used.
- **As last scheduling** – For operations scheduling and job scheduling, the selected scheduling direction and date are saved. Therefore, you can select this option for subsequent scheduling. If there's no previous scheduling of the production order, scheduling is backward from the current system date.
- **Forward from tomorrow** – Schedule forward from the current date plus one day.
- **Forward from previous job** – This option is relevant only in job scheduling. If you select this scheduling direction for operations scheduling, the production order is scheduled forward from the current date. In job scheduling, scheduling is established for one job, and all other jobs for the production are scheduled based on that job.
- **Backward from previous job** – This option is relevant only in job scheduling. If you select this scheduling direction for operations scheduling, planned orders are scheduled backward from the current date. In job scheduling, scheduling is established for one job, and all other jobs for the production are scheduled based on that job.

Scheduling date

Use this date for the **Forward from scheduling date** and **Backward from scheduling date** scheduling directions. Scheduling is backward or forward from this date. For more information, see the previous section, "Scheduling direction."

Recalculate BOM levels

When you select **Recalculate BOM levels**, the selected bill of materials (BOM) levels are recalculated to help guarantee the correct scheduling order.

Limitations

Finite capacity

Scheduling depends on the availability of the resources that are required to complete production. If there's not enough capacity, production orders can be delayed or even stopped. If you apply finite capacity to operations scheduling, existing capacity reservations that you made on the resources are considered, and the system sees that capacity as unavailable. The production order is scheduled based on the availability of capacity on the resources that are required. Operations scheduling uses the specified sequence of operations to determine the order of operations in the production route. Operations scheduling reserves capacity on the resource groups, based on the operation times that are defined on the production route. The capacity of the resource groups is the sum of available capacity on all the resources in the resource groups.

Finite material

Scheduling also depends on the availability of the materials that are required for production. Insufficient component availability can also cause production delays. Scheduling can also be based on the use of materials. Just specify the materials that must be available for production instead of the materials that aren't critical. This type of scheduling is known as scheduling with finite material. When you specify finite materials, production is scheduled based on whether materials are available. When you optimize on both capacity and materials, production is calculated to meet both restrictions. The availability of capacity and materials is considered, and the production order’s operations can't be scheduled to start until capacity and materials are available at the same time and in the required quantities. Select this check box if materials should be considered limited during scheduling. If the materials are limited, the material's coverage for that time is considered. In other words, scheduling considers the future dates that are calculated for the items. Scheduling reserves raw materials and explodes the current production. If scheduling occurs several times, only the first scheduling runs an explosion and makes reservations. If you make changes in the production BOM or route, the next scheduling runs an explosion. For the explosion, it's assumed that the materials are required on the same day. Because the production isn't scheduled at the time of the master schedule explosion, the current date is the best estimate of when the items will be available. The explosion then checks whether the items are available. If the items are available, the production requirement can be fulfilled. If the items aren't available by the current date, a planned order is generated, and an offset selection is made for the planned order. If automatic firming is set for the planned order, the planned order is firmed automatically for purchase or production. The production status is automatically changed to the status that is specified in the **Requested production status** field in the **Coverage groups** dialog box. If the check box is cleared, the materials are always considered available. Therefore, scheduling doesn't consider whether the materials are available at the time of requirement.

Finite property

Select this check box if the job scheduling should include property requirements.

Keep production unit

Select whether the scheduling engine should schedule only resources that are already specified on the production unit.

Keep warehouse from resource

Select whether the scheduling engine should schedule only resources that are associated with the input warehouse that is specified on the resource.

References

Schedule references

When references depend on production orders, they're also known as subproductions. You can schedule subproductions as part of the scheduling of a production order. Select this check box if the scheduling of subproductions should be based on the scheduling of the main production. In relation to the main production, overlying productions are scheduled forward, and underlying productions are scheduled backward. You can view production order references in the **Reference level** field on the **Production orders** page.

Synchronize references

You can synchronize references with the production order. If you select this option, the system moves and aligns the dates of the subproductions when you make changes to the production order's schedule. If a production order has one or more subproductions, you might want to schedule the subproductions together with the main production. In this case, you can't start the main production until you complete the related subproductions. Therefore, select this check box if you want the scheduling of subproductions to be based on the start and end times of the selected production. You can select this check box only if you also select the **Schedule references** check box.

Cancellation

Cancel queue time

Select this check box to exclude queue time from the scheduling.

Cancel setup

Select this check box to exclude setup time from the scheduling.

Cancel process

Select this check box to exclude run time from the scheduling.

Cancel overlap

Select this check box to exclude overlap time from the scheduling.

Cancel transport

Select this check box to exclude transit time from the scheduling.

Set default

You can save the current values as default values. Two options are available:

- Set as my default
- Set as default for everyone

Related information

Operations scheduling

Job scheduling

09/19/2025

This article provides information about job scheduling, which is a more detailed form of scheduling than operations scheduling. Use job scheduling to schedule individual jobs or shop orders, and to control the manufacturing environment.

You can use job scheduling to schedule individual jobs or shop orders, and to control the manufacturing environment. Job scheduling breaks down each operation into its individual tasks or jobs. Assign these jobs to the operations resources that perform them. Job scheduling also lets you synchronize all jobs that the selected job references. You can specify a start date and time or end date and time for the job, and then run scheduling. The time that you specify can be the start time or the end time, depending on the scheduling direction. This functionality is useful when, for example, a job can run only on one machine at a time, or when you want to optimize the job that runs for each resource.

Tasks in the job scheduling process

The job scheduling process includes the following tasks:

- Split operations into jobs.
- Schedule jobs, based on the dates and times for the resources that are specified for the related operation.
- Calculate start times and end times for each job. Use finite capacity to make sure that there are no overlapping times.
- Determine which resources in the resource group run the job on. This task requires that a resource group be specified for an operation. Job scheduling selects the resources or resource groups based on the shortest lead time, and also considers any previous reservations on the resources.
- Explode operations into jobs when you run job scheduling. The jobs are scheduled by date and time, according to the order that the production route specifies. The setup of the operation determines the jobs that are exploded during the scheduling process. The route group assigned to the operation controls whether jobs are generated. A job is generated only if it has a specific duration. For example, a transport time job is generated if a transport time was specified for the selected operation.

Scheduling direction

You can schedule jobs either forward or backward.

- **Forward** – Use the forward scheduling direction to start the production as early as possible. This method is also known as the push method, because the production is being pushed forward through the production process. The production is scheduled to start and end on the earliest possible dates.
- **Backward** – Use the backward scheduling direction to start the production as late as possible. This method is also known as the pull method, because it's based on the date when the production must be completed. Backward scheduling counts backward to the latest possible date that the production can be started without missing its target deadline.

Finite capacity

You can schedule jobs by using finite capacity. When you use finite capacity, the capacity that is scheduled can't be larger than the capacity that's available for the resource. Available time is defined as the interval when the resource is available and there are no other reservations on capacity. Scheduling that is based on finite capacity makes sure that start times and end times for an operation on a specific date don't overlap. The resource capacity that is already reserved is considered, and overlaps between the start times and end times are also considered. Finite capacity determines the amount of capacity that must be available for a resource in order to achieve optimal use of that resource. This determination is balanced against a calculation of the shortest possible lead time between operations.

Finite materials

Job scheduling that uses finite materials ensures that the required materials are available when the operation starts. The coverage rules for items define these limits. Scheduling uses requirement explosion to determine when the component items are available. If you schedule without setting finite materials, the system assumes that all items are available when they're required.

Finite properties

Job scheduling that uses special properties requires that you specify properties for the operations on the production route. The system must fulfill these properties to reserve capacity.

References

Job scheduling schedules all productions that the current production references. If a production has one or more subproductions, schedule the subproductions at the same time as the main production, because you can't start the main production until the related subproductions are completed.

Schedule resources

The scheduling engine examines combinations of resources to identify those combinations that can satisfy requirements. You can specify selection criteria by selecting one of the following values in the **Primary resource selection** field on the **Scheduling parameters** page:

- *Duration* – The scheduling engine selects the resource that has the shortest lead time.
- *Priority* – The scheduling engine selects the resource that has the highest priority if two or more resources have identical capabilities and levels. The resource that has the lowest numeric value in this field has the highest priority.

> ⓘ **Note**
>
> Scheduling by duration can cause decreased performance when the same resource group contains many resources and secondary operations are used. You can schedule a maximum of 32 resources per operation. If you exceed this quantity, an Action center message is displayed, and job scheduling doesn't find the best alternative resource.

When you run job scheduling, the system plans the resources based on the limitations that you define in the resource parameters. You can control the capacity of the resources by using calendar settings. The system calculates loads for resources during the scheduling process.

> ⓘ **Note**
>
> For productions that use the operations scheduling function, you can run job scheduling after operations scheduling. If you aren't using operations scheduling, you can run job scheduling alone.

Maximum capacities for resources per job order

Job scheduling assigns resources to jobs. You can set maximum capacities for resources per job order. For example, you can set up the system to schedule no more than 50 percent of total capacity for a production order. This setup gives you more control over the scheduling of

resources on the job scheduling level. Therefore, it can help prevent issues if not enough capacity is available to perform simultaneous productions.

Resource efficiency

Job scheduling uses the efficiency percentages that you specify for the resources. Efficiency percentages reduce or increase the time that the system reserves for the resource. Therefore, lead time also increases or decreases. The system uses the following formula for the calculation: `Scheduling time = Time × 100 ÷ Efficiency percentage`. In this formula, *Time* includes both the run time and the setup time.

Consolidated batch orders

This article describes the concept of consolidated batch orders.

A bulk item that is produced is considered a parent item, whereas a packed item is considered a child item. The relation between the bulk item and the packed item is expressed in a bulk item conversion. This bulk item conversion is defined on the bulk item itself.

Packed items can be packaged into containers of either a single size or multiple sizes that are considered one unit. By consolidating the orders for a bulk item, you can see all the related batch orders in a single view that can help you determine any remaining work that must be completed.

A consolidated batch order can contain any combination of the following orders:

- A single bulk order and multiple packed orders
- Multiple bulk orders and multiple packed orders
- Multiple bulk orders and a single packed order
- Only packed orders

Last updated on 07/01/2026

Batch order lifecycle from create to start

Article • 09/29/2021

This procedure takes you through the first part of the life cycle of a batch order.

From creation, cost estimation, and over production job scheduling to the actual start of a batch order.

The demo data company used to create this procedure is USMF.

The prerequisites for running the procedure with another dataset are a released product with an active formula and route version.

Create a batch order

1. Go to All production orders.
2. Click New batch order.
3. In the Item number field, enter or select a value.
4. Click Create.
5. Use the Quick Filter to filter on the Production field with a value of 'b'.

View production formula and expected co-products

1. On the Action Pane, click Production order.
2. Click Formula.
3. Close the page.
4. Click Co-products.
5. Close the page.

Estimate the batch order

1. Click Estimate.
2. Click OK.
3. On the Action Pane, click Manage costs.
4. Click View calculation details.
5. Close the page.

Release the batch order

1. On the Action Pane, click Production order.
2. Click Release.
3. Click OK.

Schedule production jobs

1. On the Action Pane, click Schedule.
2. Click Schedule jobs.
3. Select No in the Finite capacity field.
4. Select No in the Finite material field.
5. Click OK.
6. On the Action Pane, click Production order.
7. Click All jobs.
8. Close the page.

Start the batch order

1. Click Start.
2. Click the General tab.
3. Select No in the Post picking list now field.
4. Click OK.
5. On the Action Pane, click View.
6. Click Picking list.
7. In the list, click the link in the selected row.
8. Close the page.
9. Close the page.
10. Click Route card.
11. In the list, click the link in the selected row.
12. Close the page.
13. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Sequence production jobs for process manufacturing

Article • 09/29/2021

This procedure uses paint products as an example to show how to sequence planned orders according to the priority of color and package size. The demo data company used to create this procedure is USPI. This procedure is intended for the production planner.

Run master planning for USPI

1. Go to Master planning > Master planning > Run > Master planning.
2. In the Master plan field, click the drop-down button to open the lookup.
3. In the list, click the link in the selected row.

    - Select MasterPlan.

4. Click OK.

    - This starts Master Planning, including the sequence process. This process can take a few minutes.

View planned orders for the paint products

1. Go to Master planning > Master planning > Planned orders.
2. Expand the Item details FactBox.
3. Expand the Schedule details FactBox.
4. In the Plan field, click the drop-down button to open the lookup.
5. In the list, find and select the desired record.

    - Select MasterPlan.

6. In the list, click the link in the selected row.
7. Use the Quick Filter to filter on the Item number field with a value of 'P300'.

    - Unlock to scroll to the right to see the order date and delivery date. Notice that these items have an order date of Today and the delivery dates for the planned orders are not sequenced after the priority of color and package size, as shown in the product name. You can hover over an item number to see the product name and priority.

Sequence planned orders for paint

1. Close the page.
2. Go to Master planning > Master planning > Sequenced planned orders.
3. Expand the Item details FactBox.
4. Expand the Schedule details FactBox.

    - Note: Here you see that the Start date/time for the planned orders are sequenced according to color and package size within a bucket period of 28 days. These periods are defined by a number in the field Campaign. The sequenced planned order form acts like the typical planned order form, and the production planner can firm the planned orders here.

5. Mark all rows.
6. Click Accept.

    - This will update the planned orders with the selected sequence action of either Postpone or Advance.

Verify the sequence of the planned orders

1. Close the page.
2. Go to Master planning > Master planning > Planned orders.
3. Expand the Item details FactBox.
4. Expand the Schedule details FactBox.
5. In the Plan field, click the drop-down button to open the lookup.
6. In the list, find and select the desired record.

    - Select MasterPlan.

7. In the list, click the link in the selected row.
8. Use the Quick Filter to filter on the Item number field with a value of 'P300'.

    - Notice that the orders now are sequenced according to the priority of color and size and the planned orders start at the earliest order date and delivery date. Validate the Order date column or the Start date in the Schedule details FactBbox.

Feedback

Was this page helpful? Yes No

Provide product feedback

Material substitution in manufacturing

09/19/2025

This article describes how to substitute materials during the production process. There are three methods for doing so:

- By date, when one material is substituted for another after a specific date
- During master planning, when a material in a formula is substituted with a different material because it's out of stock
- During production, when a material is unexpectedly out of stock and is substituted with a different material

Substituting material by date

Consider the following scenario: A machine that a company is manufacturing contains a component that expires from the vendor's catalog in two months. From the expiration date onward, the vendor offers a new component that you can substitute for the old component. Set up **From** and **To** validity dates on bill of materials (BOM) lines. For this example, set up the old component to expire by entering the expiration date in the **To-date** field. Then, on the BOM, set up the new replacement component so that it's valid from the day after the old component expires. To do this, enter the start date in the **From-date** field.

Substituting material by planning

You can substitute materials during planning only when you're using formulas, not when you're using a BOM. Consider the following scenario: A food manufacturing company is making a sauce from a formula that has 20 ingredients. One ingredient in the formula can be substituted by one of two other ingredients. However, because these two ingredients are more expensive than the preferred ingredient, substitution is used only if the preferred ingredient is out of stock. The material that you can substitute is called A, whereas the two materials that can replace it are called B and C. Material substitution by planning is controlled by the **Plan group** and **Priority** fields on the formula lines. For this example, you create formula lines for the three materials, and associate the formula lines with the same plan group. In the setup, the formula line for material A has the highest priority (lowest number), the formula line for material C has the lowest priority, and the formula line for material B has a priority that is between the priority of the other two lines. If you have demand for the finished item, master planning first determines whether the demand for material A can be covered. If the demand can't be covered, master planning looks at materials B and C, in order of priority. If material B is on hand, it will be used after a planned batch order is firmed for the formula. If none of the materials are on hand, master planning creates a planned order for material A.

> **Tip**
>
> When you set up formula lines in a plan group, specify a quantity only on the material that has the highest priority. This quantity calculates the demand of all materials in the plan, even the materials that have lower priority. You can't specify different quantities on lower-priority items in the plan group.

Substituting material during production

Consider the following scenario: A piece of metal plate is required for a welding operation. During the operation, a warehouse worker informs the machine operator that the plate is out of stock. However, it's decided that the plate can be substituted with a plate that is slightly thicker. That way, the operation can be finalized. You can add the material to the production picking list. Then, when the picking list is posted, the system adds the material to the production BOM.

Register and track batch/serial numbers for finished products and their components

08/13/2025

Workers and managers can register batch/serial numbers for materials and components that are used in manufacturing processes. They can also then associate those numbers with the batch/serial numbers of the products that are produced. In this way, manufacturers can optimize their processes, enhance product quality, and respond quickly to any issues that arise. Managers can use the item tracing report to effectively track batch/serial numbers that are registered through the *Tracked components* feature.

> **Tip**
>
> You can also track batch/serial numbers for materials and components using the Traceability Add-in for Dynamics 365 Supply Chain Management (preview), which offers a graphical browser interface and an API for integration with external systems.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.40 or later.
- The feature that's named *Tracked components* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

Enable registration of tracked component information

To enable registration of tracked component information in production processes, follow these steps.

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. On the **Bills of materials** tab, set the **Enable tracked components** option to *Yes*.

Enable registration of tracked components on products

Before you can use tracked components functionality for a product or its components, the following conditions must be met:

- The product must be defined as serial controlled or batch controlled.
- The product must be associated with a tracked components policy.

To create a tracked components policy, follow these steps.

1. Go to **Production** > **Setup** > **Production** > **Tracked components policy**.

2. Use the buttons on the Action Pane to add a new policy or edit an existing one.

3. For the new or selected policy, set the following fields:

    - **Name** – Enter a name for the policy.
    - **Description** – Enter a description of the policy.
    - **Use tracked components** – Set this option to *Yes* to register tracked information for products that are associated with the policy.
    - **Check pre-allocation for product** – Set this option to *Yes* to validate serial or batch numbers for finished goods when tracked component information is registered.
    - **Check for pre-allocation for components** – Set this option to *Yes* to validate serial or batch numbers for components when tracked component information is registered.
    - **Serial number matching limit** – Set this option to *Yes* to ensure that the number of component serial numbers that are matched to a product serial number doesn't exceed the planned quantity on the production order bill of materials (BOM) line. If the number of matched components exceeds the planned quantity, the system blocks the match to prevent over-allocation.

Register tracked components for finished items

To register tracked components for the finished items for any production order, follow these steps.

1. Start from any of the following places:

    - **All production orders** list or details
    - **Picking list** journal line
    - **Route card** journal line
    - **Job card** journal line

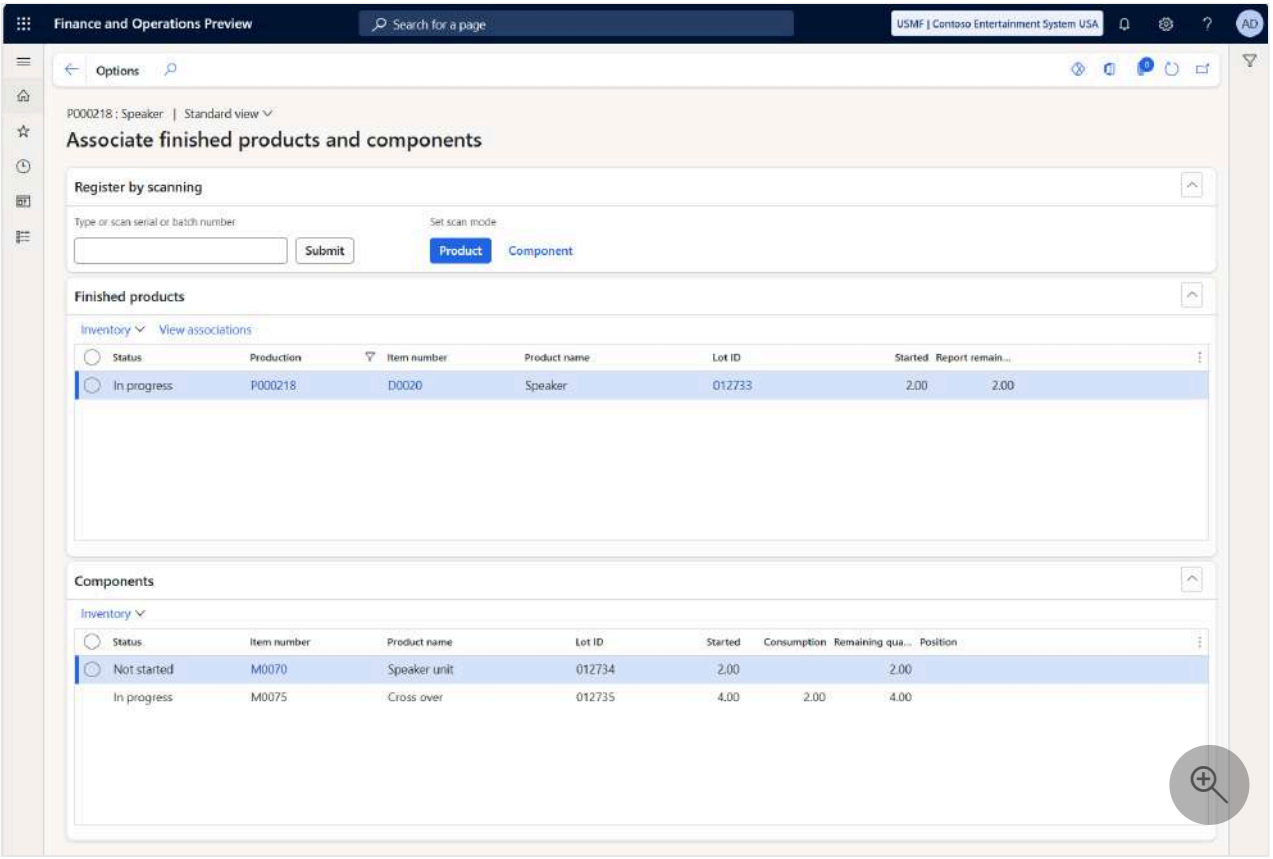
- **Report as finished** journal line
- **Current operations**
- Production order **Start** dialog
- Production order **Report as finished** dialog

2. Select the production order that you want to register batch or serial numbers for.

3. On the Action Pane, find the tab that includes the **Track components** button, and select that button.

4. On the **Tracked components** page, the **Type or scan serial or batch number** field is automatically in focus. In addition, the **Scan mode** field is automatically set to *Product* to indicate that you're about to enter the batch or serial number of a finished product. Follow one of the steps:

   - If you're using a scanner, scan the bar code of the first finished product that you want to register.
   - If you're using a keyboard, enter the batch or serial number of the first finished product that you want to register, and then select **Submit**.



5. The value of the **Scan mode** field is changed to *Component*. You can now register the batch or serial number of a component by following one of these steps:

   - If you're using GS1 bar codes, your system might be able to detect which component you're scanning. In this case, you can just scan the component bar code,

which includes both the component item number and its serial or batch number.
    - If you're using simple bar codes, or if you're using a keyboard, first select a component in the grid on the **Component** FastTab. Then enter or scan the batch or serial number of that component.

6. Repeat the previous step until you finish registering all the components for the current finished product.

7. Change the value of the **Scan mode** field back to *Product*, and repeat this procedure until you finish registering all finished products and their components for the current production order.

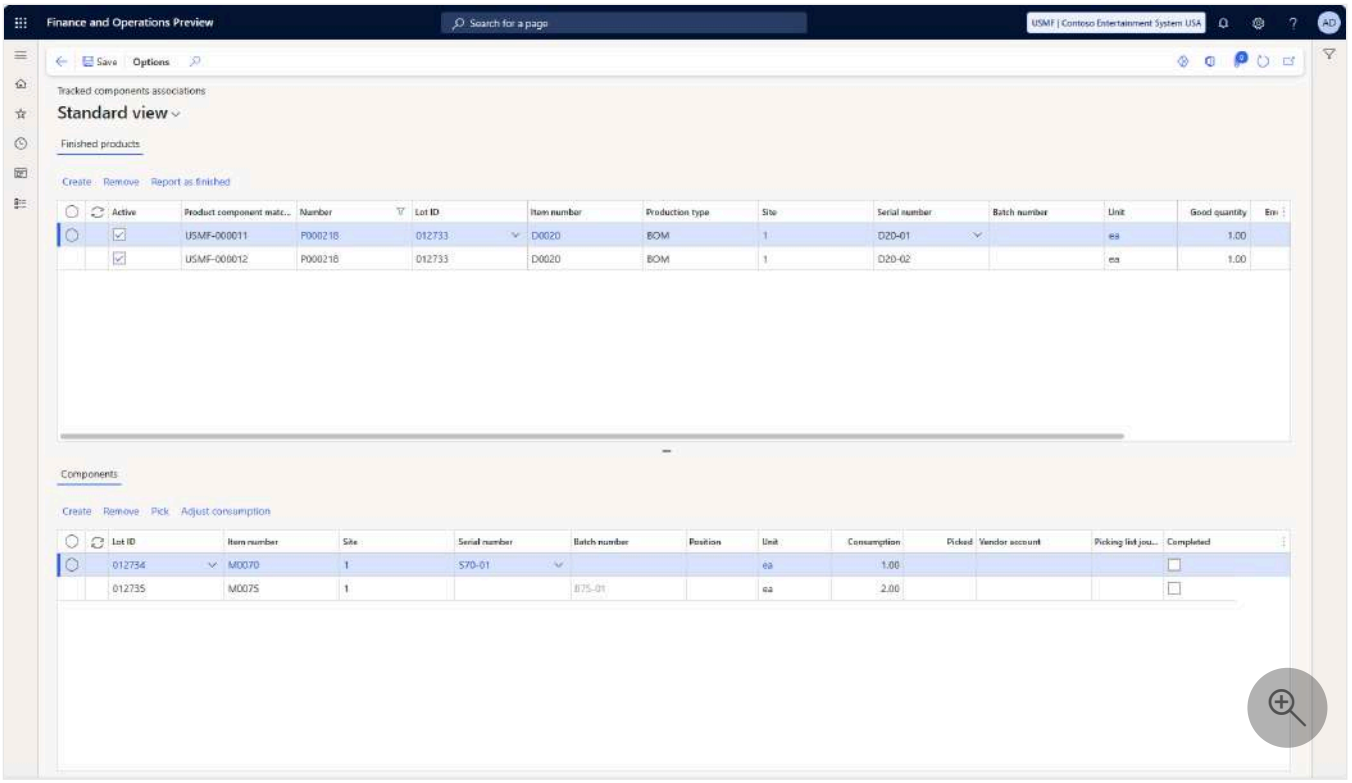
   At any time, you can review and correct the registrations that you've made so far. On the **Finished products** FastTab, on the toolbar, select **View associations** to open the **Tracked components associations** page. For information about how to work with this page, see the next section.

As you work, the **Status** field on the **Finished product** FastTab of the **Tracked components** page indicates the registration status of each finished product. The **Status** field on the **Components** FastTab indicates the registration status of each component in the selected finished product. The **Status** field on each FastTab can have the following values:

- *Not started* – No batch or serial number has been associated with the product or component.
- *In progress* – An association has been established, but only for part of the started quantity for the finished product or the component. On the **Components** FastTab, the **Started**, **Consumption**, and **Remaining quantity** columns let you track the quantity that has been allocated.
- *Completed* – No remaining quantity remains to be allocated for the finished product or the component.

Review and edit batch/serial numbers for finished products and their components

Use the **Tracked components associations** page to review and edit batch/serial numbers for finished products and their components. For information about how to open this page for a selected production order, see the previous section.



Finished products

The **Finished products** section lists batch/serial number registrations for the products that are being produced for the selected production order. The toolbar in this section provides the following buttons:

- **Create** – Add a new row to the grid so that you can register the serial or batch number for a missing finished product.
- **Remove** – Remove one or more selected finished products.
- **Report as finished** – Report a selected row as finished. You can first edit the **Good quantity** and/or **Error quantity** value for batch products as required.

Components

The **Components** section lists the components of the finished product that's selected in the **Finished products** grid. For each component, this section shows the registered batch or serial number, and you can enter or correct these values as required. The toolbar in this section provides the following buttons:

- **Create** – Add a new row to the grid so that you can register the serial or batch number for a missing component.
- **Remove** – Remove one or more selected components.
- **Pick** – Manually pick the quantity that's planned for consumption, as specified in the **Consumption** field.

- **Adjust consumption** – Generate a production picking list journal to manually adjust the consumption. In this case, the quantity that's specified in the **Consumption** field is automatically entered in the picking list journal.

Related information

- How workers use the production floor execution interface
- Traceability overview (preview)

Production feedback

Article • 06/21/2024

This article provides information about production feedback, which lets workers provide feedback about production jobs. The article includes information about the various ways that production feedback can be updated.

Production feedback gives workers the opportunity to provide feedback about production jobs. It lets workers record time and material consumption on production orders, operation quantities and status, and errors that cause a job or operation to fail. Production feedback can be updated in journals that are related to production orders. The **Production job card** and **Production route card** journals are used to report time and quantities per job or operation. For reporting about the last job or operation, quantities on the finished product can be reported as finished.

Workers can also provide production feedback using the production floor execution interface, which is part of the manufacturing execution functionality in the **Production control** module. The production floor execution interface has a configurable user interface that shows a list of the released jobs in a prioritized order for a selected work area. It also offers advanced options such as job bundling and team work. The production floor execution interface has a touch-optimized user interface. Feedback submitted through the production floor execution interface updates the production journals.

Feedback

Was this page helpful? Yes No

Provide product feedback

Time and attendance registration overview

07/10/2025

Time registration workers can enter different types of time registrations, for example, clock in, clock out, register indirect activities, and absence registration. This article describes registrations, their calculation, approval, and the use of workflow to add structure and automated approval to the process of approving timesheets.

Registrations

In companies that use time and attendance, workers must register the time that they spend at work in addition to their attendance. Some companies might only require workers to register clocking-in and clocking-out times. In other companies, workers might also be required to register time consumption on the actual activities they perform and the breaks they take. The intended users of time and attendance are:

- Workers, who are required to register time and attendance at regular intervals, for example daily, weekly or bi-weekly.
- Supervisors, managers, and payroll officers who calculate, approve, and transfer worker registrations for further processing.

> **Note**
>
> If you run Time and attendance together with Manufacturing execution, all registrations on projects, project activities, indirect activities, absence codes, and overtime and flex time will be recorded and are used to calculate payroll in both modules.

Time registrations workers

To be able to register time and absence, workers must be set up as time registration workers in the company they're employed in.

After setup, the workers can enter different types of registrations.

- Clocking in- and out when arriving or leaving work.
- Time and item consumption on production jobs.
- Time used on a machine on the shop floor, if the machine has been defined as a resource.

> **Note**
>
> A worker can be automatically assigned the time registrations that are made on a particular machine on the shop floor, if the worker chooses to work as an assistant to the machine when they start the production job.

- Time registrations on projects and project activities.
- Registering project fees and item consumption via the respective project fee journals and project item journals.
- Planned absence.
- Absence when arriving late to work or leaving earlier than planned.
- Work breaks, either manually registered or automatically calculated by the system.
- Indirect activities, which are nonproductive activities a worker might engage in during a workday. Examples of these activities include meetings or cleaning their workspace.
- Overtime, which can be registered either as extra hours, flextime, or overtime.

Adding clock-out registrations

If a worker forgets to clock-out at the end of their workday, the missing registration can be added by running a batch job. The system will compare the clock-in time and the clock-out time according to the associated profile of the worker, and automatically insert the missing clock-out registration to match the profile's end time. Both the clock-in and clock-out registrations are vital for the subsequent calculation and approval of time registrations before they can be transferred to payroll.

Calculating registrations

When a registration worker is assigned a calculation group that typically relates to a specific team, shift, or work group. The team manager or supervisor typically validates the registrations made by the workers, and is therefore also the responsible person to run the calculation for the respective calculation groups on a daily basis. As part of the calculation process the team manager or supervisor is able to:

- Correct erroneous registrations. For example, change switch codes and adjust feedback on production jobs.

- Add missing registrations. For example, create clock-out registrations and create absence transactions.
- Delete incorrect registrations.

Because the registered time must match the worker's time profile prior to calculating the registrations, you must override the work time profile for any worker who has an exception to their standard work time profile. If the worker profile is day shift, and the worker has agreed to work a night shift with no overtime pay, the team manager or supervisor must override the default worker profile in order to calculate the working time at the standard night rate and not as overtime. The calculation will also display an error if an absence registration is missing. It must be added before the calculation can be completed.

Approving registrations

Just as you assign a calculation group to a time registration worker, you must assign an approval group as well. Typically the group will be specific to a team, shift, or work group. You must approve the time registrations that were calculated correctly – this means doing a calculation without errors – before pay items can be generated that afterward can be transferred to a payroll system. The payroll administrator will typically do the approval of registrations, and prior to the approval they're able to:

- Override pay agreements for individual workers.
- Add manual premiums.
- Enter additional information about absence registrations.

> **Note**
>
> If overtime has been calculated for specific workers, the overtime can be allocated to specific jobs during the day. This is relevant if job cost is calculated based on worker pay.

Approving registrations using workflow

You can set up a workflow approval process that automatically approves registrations that comply with workflow rules, leaving only deviations to be handled manually. If workflow approval is activated, the team manager or supervisor submits the calculated registrations for approval. The workflow process will generate the appropriate approvals and tasks, and then assign them to the right users and roles as identified in the workflow. There are two workflow approvals for time and attendance.

Expand table

Workflow	Purpose	Registration type
Time and attendance days total	The workflow validates registrations against, for example, the expected number of work hours for the day.	
Time and attendance journal registration.	The workflow validates each registration type for the date of the registration.	Time and attendance • Clock-in • Clock-out • Absence • Break • Switch code • Project • Project activity • Indirect activity Production jobs • Queue before • Setup • Process • Overlap • Transport • Queue after • Start assistance • Stop assistance

Transferring approved registrations

After approval of the registrations, you can transfer them to a periodic payroll job. A transferred registration is posted to an activity or job that it relates to, for example, a production order or a project. Payroll transactions are generated for each worker based on the registrations.

Reversing transferred registrations

The task of reversing transactions – rolling them back – can be done until the time when the payroll period's pay transfer is run. This means that payroll data has been transferred to an external file. When reversed, all registrations are withdrawn, and any transactions posted on production orders or projects are offset and neutralized.

> ⓘ **Note**
>
> The external file can be imported into a payroll system.

Registrations in electronic timecards

Workers with job tasks that don't require immediate feedback, as is the case with production jobs, but who work on project activities, can benefit from using the electronic timecard. Electronic timecards offer the flexibility to enter registrations at any time and in the best way for your business schedule (daily, weekly, or when a worker is in the office again after being away). To use electronic timecards or these workers, you must specify, Use timecard, in the worker details. Electronic timecards enable the worker to register:

- Date
- Registration type
- Job reference, such as project, indirect activity, or production order
- Job identification
- Time consumption
- Project fees
- Project items

Electronic batch records (EBRs)

Electronic batch records (EBRs) are required by part 11 of Title 21 of the Code of Federal Regulations (21 CFR Part 11), which applies to life science and pharmaceutical industries. EBRs consist of two major components:

- **Master manufacturing record (MMR)** – This component documents all the methods that might be used to produce an item and the various ways that the item's characteristics might be measured.
- **Batch production record (BPR)** – This component documents the actual method that was used to produce the item and the actual value of the measurement for each of the item's characteristics. It also details the characteristics and quantities of the ingredients that were used to produce the item.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The following features must be turned on in Feature management. As of Supply Chain Management version 10.0.47, both of these features are turned on by default.
  - *Advanced quality management* (As of Supply Chain Management version 10.0.49, this feature is mandatory and can't be turned off.)
  - *Electronic signature improvements*

Turn electronic signature requirements for EBRs on or off

To set up the electronic signature requirements for the types of records that are used for EBRs, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature requirements**.

2. In the left pane, select the row for a feature that you want to turn electronic signature requirements on or off for. For EBRs, the usual expectation is that signatures are required for the following features (as listed in the **Name** column):

    - *Activate BOM version*
    - *Approve bill of materials*
    - *Activate route version*
    - *Approve BOM version*
    - *Approve route*
    - *Approve route version*
    - *Post dispensing ticket production journal*
    - *Post picking list production journal (non-dispensing)*
    - *Post report as finished production journal*
    - *Post route card production journal*
    - *Validate quality order*

3. In the right pane, set the **Signature required** option to *Yes* to require signatures from users who perform actions that are related to the feature. Set the option to *No* to turn off the requirement.

4. Repeat steps 2 and 3 for every other feature that you want to turn electronic signature requirements on or off for.

Set up EBRs

To enable EBRs, follow these steps:

1. Go to **Production control** > **Setup** > **Production control parameters**.

2. On the **Electronic batch record** tab, set the following fields:

    - **Use electronic batch records** – Set this option to *Yes* to enable EBRs in your system. When you change the setting to *Yes*, a message notifies you that you must remove approvals on bills of materials (BOMs), formulas, or routes before you edit those records. In this way, the system can capture electronic signatures of the users who perform the edits. While EBRs are enabled, the *block editing* functions for routes and BOMs/formulas are automatically enabled and can't be disabled. The block editing feature for routes and BOMs/formulas prevents those records from being edited after they are approved. Approval must be removed before the records are edited. Then,

after editing is completed, approval is required before the records can be used in production.

- **Include batch expiration on BPR** – Select whether the batch expiration date should be included on the BPR report.
- **Suppress work instructions on EBR** – Select whether the work instruction section of the BPR report should be omitted if there aren't any work instructions.
- **Allow BPR to print multiple reports at once** – Select whether multiple BPRs can be printed on a single report. If this option is set to *No*, only one BPR can be printed per report.

Set up a number sequence for EBRs

Because EBRs are maintained independently of the batch order or production order, a number sequence must be established.

1. Go to **Production control** > **Setup** > **Production control parameters**.
2. On the **Number sequences** tab, set up a new number sequence for the row where the **Reference** field is set to *Batch production record*.

Set up electronics signature requirements

You can configure electronic signature requirements to support EBRs for the following types of production journals:

- Picking list journal
- Route and job card journals
- Report as finished journal

Each of these journal types provides the following electronic signature options:

- **Post route card production journal** – When this option is set to *Yes*, the system notifies users that route card journals are no longer automatically posted when a batch or production order is started.
- **Post picking list production journal (non-dispensing)** – When this option is set to *Yes*, the system notifies users that picking list journals are no longer automatically posted when a batch or production order is started. In addition, the option for reporting the batch or production order as finished is disabled. Because the system must capture the electronic signature of the users who post those activities, the report as finished production journal must be used to report a batch or production order as finished.

- **Validate quality order** – When this option is set to *Yes*, the system requires a signature when an overall quality order is validated. However, no signature is required when individual tests of a quality order are validated.

Add work instructions to formula items

To add work instructions to a formula line, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Open or select the item that uses the formula that you want to add instructions to.
3. On the Action Pane, on the **Engineer** tab, select **Formula versions**.
4. In the list pane, select the formula version that you want to edit.
5. On the Action Pane, on the **Formulas** tab, select **Formula**.
6. If the **Approved** option for the formula is set to *Yes* on the **Formula header** FastTab, you must follow these steps to remove the approval before you can edit the formula:
   a. On the Action Pane, on the **Formula** tab, select **Approve formula**.
   b. In the dialog, set the **Remove approval** option to *Yes*.
   c. Select **OK**.
7. On the **Formula lines** FastTab, select the line that you want to add instructions to.
8. On the **Line details** FastTab, on the **Work instructions** tab, select **Add** on the toolbar to add work instructions for the selected line.
9. In the **Type** field, select *Note*. Then, in the **Description** field, enter a short description.
10. In the lower text box, enter the required work instructions or other comments.
11. On the Action Pane, select **Save**.

Add work instructions to BOM items

To add work instructions to a BOM line, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Open or select the item that uses the BOM that you want to add instructions to.
3. On the Action Pane, on the **Engineer** tab, select **BOM versions**.

4. In the list pane, select the BOM version that you want to edit.
5. On the Action Pane, on the **Bill of materials** tab, select **Bills of materials**.
6. If the **Approved** option for the BOM is set to *Yes* on the **Bill of materials header** FastTab, you must follow these steps to remove the approval before you can edit the BOM:
   a. On the Action Pane, on the **Bill of materials** tab, select **Approval**.
   b. In the dialog, set the **Remove approval** option to *Yes*.
   c. Select **OK**.
7. On the **Bill of materials lines** FastTab, select the line that you want to add instructions to.
8. On the **Line details** FastTab, on the **Work instructions** tab, select **Add** on the toolbar to add work instructions for the selected line.
9. In the **Type** field, select *Note*. Then, in the **Description** field, enter a short description.
10. In the lower text box, enter the required work instructions or other comments.
11. On the Action Pane, select **Save**.

Add work instructions for routes

To add work instructions for an operation line, you must edit the route. When EBRs are used, formula lines for approved routes can't be edited. To add work instructions to a route, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Open or select the item that uses the route that you want to edit.
3. On the Action Pane, on the **Engineer** tab, in the **View** group, select **Route**.
4. In the top section, select the route version that you want to edit.
5. If a check mark appears in the **Approved** column for the selected route version, you must follow these steps to remove the approval before you can edit the route:
   a. On the Action Pane, on the **Route version** tab, select **Approve**.
   b. In the dialog, set the **Remove approval** option to *Yes*.
   c. Select **OK**.
6. In the bottom section, on the **Overview** tab, select the operation that you want to edit.

7. On the **Work instructions** tab, select **New** on the toolbar to add work instructions for the selected operation.
8. In the **Type** field, select *Note*. Then, in the **Description** field, enter a short description.
9. In the lower text box, enter the required work instructions or other comments.

Configure test plan instructions for test groups

When test plan instructions are associated with a line in a test group, quality orders that reference the test group show the instructions as read-only text. To add test plan instruction to a test group, follow these steps:

1. Go to **Inventory Management** > **Setup** > **Quality control** > **Test groups**.
2. In the top section, on the **Overview** tab, select the test group that you want to add instructions to.
3. In the bottom section, on the **Overview** tab, select the test that you want to add instructions to.
4. On the **Test plan notes** tab, select **Add** on the toolbar to add work instructions for the selected test line.
5. In the **Type** field, select *Note*. Then, in the **Description** field, enter a short description.
6. In the lower text box, enter the required work instructions or other comments.

Work with EBRs

After all the required production data is set up for a BOM or formula item, you can search for that information. This capability is the primary purpose of EBRs.

As was mentioned earlier, EBRs consist of two major components:

- **Master manufacturing record (MMR)** – This component documents all the methods that might be used to produce an item and the various ways that the item's characteristics might be measured. The MMR query contains all the BOM/formula versions that are approved for use in the manufacture of the item. It also includes the various approved route versions. In addition, the information includes quality associations that are referenced by the production process for the item. As a result of these quality associations, quality orders are automatically generated for the item. If the item is an ingredient, quality associations for that ingredient might be referenced and cause quality orders to be generated during an inspection. Other characteristics might be referenced for the ingredients or produced items to indicate the type of characteristic that is being measured. The measured characteristic

might be related to a specific inventory batch dimension or to the vendor that supplied the ingredient.

- **Batch production record (BPR)** – This component documents the actual method that was used to produce the item and the actual value of the measurement for each of the item's characteristics. It also details the characteristics and quantities of the ingredients that were used to produce the item. The BPR query contains the BOM/formula version that was used in the production process, together with the proposed ingredient quantities that were estimated for completion of the production. In addition, the information includes the exact quantities that were used, in comparison to what was estimated or proposed, and any new ingredients that were substituted during the production process.

Whereas the MMR maintains the types of characteristics that might be measured for the item, the BPR maintains the actual values of those measured characteristics. The BPR contains the actual details of the quality orders that were referenced by the quality associations in the MMR.

Run an MMR query

When you run an MMR query, you first select the target item number. You can also select a site to limit the BOM/formula and route versions to that site. In addition, you can select a date and/or quantity.

The query results show all applicable route versions. The BOM/formula versions and the route version number are hyperlinks. You can use them to open the details page for the selected version. The results also show and summarize quality associations that are referenced as production or coproduct production. Select **Details** for a quality association to open the details page for it. Finally, the results show inventory batch attributes that are defined for the produced item, together with the applicable range and target value for each characteristic.

To run an MMR query, follow these steps:

1. Go to **Production control** > **Inquiries** > **Electronic batch records** > **Master manufacturing record**.

2. At the top of the page, set up the query by specifying one or more of the following criteria:

    - **Item number** – Select the item to list the manufacturing information for. The dropdown list includes only items where the **Production type** field is set to *BOM*, *Formula*, or *Planning item*.
    - **Site** – Select a specific site for the selected item.
    - **Date** – Enter a "from" date to filter the formula/BOM version for the selected item.

- **Quantity** – Enter a "from" quantity to filter the formula/BOM version for the selected item.

Results and information are shown on the following FastTabs:

- **General** – This FastTab shows general information about the selected item.
- **BOM/formula versions** – This FastTab lists the formula/BOM versions of the selected item, based on the specified criteria.
- **Route versions** – This FastTab lists the route versions of the selected item, based on the specified criteria.
- **Quality associations** – This FastTab lists all quality association relations for the selected item. Use the **Details** button to open the quality association page for the selected record.
- **Inventory batch attributes** – This FastTab lists all inventory batch attributes that are specified for the selected item.

Run a BPR query

When you run a BPR query, you first select either the target item number or the BPR that is associated with a specific batch or production order.

The query contains the BOM/formula version that was used in the production process, together with the proposed ingredient quantities that were estimated for completion of the production. The information also includes the exact quantities that were used, in comparison to what was estimated or proposed, and any new ingredients that were substituted during the production process. In the BPR, you can view the inventory batch attributes and any quality orders for the produced items. You can also view the inventory batch attributes, vendor batch details, and quality orders for the raw material items that were used in the production process.

To run a BPR query, follow these steps:

1. Go to **Production control** > **Inquiries** > **Electronic batch records** > **Batch production record**.

2. At the top of the page, set up the query by specifying one or more of the following criteria:

    - **Item number** – Select the item to list the manufacturing information for. The dropdown list includes only items where the **Production type** field is set to *BOM*, *Formula*, or *Planning item*.
    - **Batch production record** – Select the BPR ID for a corresponding batch or production order. If you select this criterion before the item number, the **Item number** field is

automatically set to the item number of the batch or production order that is related to the selected BPR ID.

Results and information are shown on the following FastTabs:

- **Production order** – This FastTab shows the batch or production order that is related to the specified BPR ID.
- **Report as finished** – This FastTab shows the items that were produced for the batch or production order that is related to the specified BPR ID. (For formula items, the produced items include coproducts and by-products.) This FastTab is populated when the report as finished production journal is created. Use the **Inventory batch attributes** button on the toolbar to view attribute information for the selected record. (The **Inventory batch attributes** button is available only if there is batch number for the selected record.) Use the **Quality orders** button to view the quality orders for the selected record.
- **BOM/formula** – This FastTab lists the raw material items that were used to produce the selected item.
- **Picking list** – This FastTab shows information from the picking list production journal of the batch or production order. It's populated when the picking list production journal is created. Use the **Inventory batch attributes** button to view attribute information for the selected record. (The **Inventory batch attributes** button is available only if there is a batch number for the selected record.) Use the **Vendor batch details** button to view the vendor batch information for the selected record. (The **Vendor batch details** button is available only if there is a batch number for the selected record.) Use the **Quality orders** button to view the quality orders for the selected record.
- **Route** – This FastTab lists the route version that is used on the batch or production order.
- **Route card** – This FastTab shows the information from the route card production journal of the batch or production order. It's populated when the route card production journal is created.
- **Job card** – This FastTab shows information from the job card production journal of the batch or production order. It's populated when the job card production journal is created.

Generate a BPR report

You can view and/or print a report for a specific BPR ID. A simplified version of the report contains the same information that is reflected in a BPR query.

To generate a BPR report, follow these steps:

1. Run a BPR query to find the relevant BPR, as described in the previous section.

2. On the Action Pane, select **Batch production report (BPR)**.
3. In the dialog, on the **Parameters** FastTab, select the details that you want to include on the report.
4. On the **Destination** FastTab, select whether you want to view the report on screen, print it, save it as a file, or email it.
5. Select **OK** to generate the report.

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Production dispensing

When you deal with hazardous materials or sensitive components in production processes, accurate dispensing of materials is crucial. It helps minimize the risk of contamination, preserve the integrity of the materials, and ensure the safety of both the product and personnel.

The processing of dispensing materials to a production order involves allocating and providing the raw materials or components that are required for the production process. It's essential in some environments, for several reasons:

- In highly regulated industries, such as pharmaceuticals or chemicals, precise dispensing ensures compliance with strict guidelines and standards, and helps maintain product quality and safety.
- When hazardous or sensitive materials are handled, the dispensing process is often performed in a type of controlled environment that is known as a clean room. A sealed box is transferred from the warehouse to the clean room. There, the box is unsealed, and the materials are dispensed. After dispensing is completed, the box is sealed again and returned to the warehouse. This process ensures that the materials remain uncontaminated and safe throughout the handling and dispensing stages.
- For expensive materials, precise dispensing helps prevent waste and fraud. Therefore, it reduces costs and ensures efficient resource use. Additionally, accurate dispensing contributes to consistent product quality. Therefore, it not only helps meet customer expectations but also protects workers and the environment by ensuring that dangerous substances are handled correctly.

To support the regulated dispensing process, you can take these actions:

- Set up thresholds for over-dispensing and under-dispensing. In this way, you help maintain compliance and quality by ensuring that the correct quantities are dispensed.
- Configure the system to automatically return any remaining, undispensed materials to inventory.
- Restrict the dispensing process to workers who are authorized to perform it. In this way, you ensure that only trained and qualified personnel handle the materials.
- Require that workers use an electronic signature to sign off on the dispensed quantities. An electronic signature helps ensure the authenticity and integrity of a process. It provides a secure and verifiable way to confirm that the correct actions were performed by authorized

personnel. Therefore, it helps maintain compliance with regulatory standards. Learn more about electronic signatures in Electronic signatures overview

Workers use the **Dispensing ticket** page to manage the dispensing of materials to the production or batch order. On this page, they register and confirm the amount of material that must be dispensed. After confirmation, the dispensed quantities are registered as consumed in the *dispensing pick journal*. Materials that aren't configured for dispensing on the production or batch order are issued through the production picking list journal.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The following features must be turned on in Feature management. As of Supply Chain Management version 10.0.47, both of these features are turned on by default.
  - *Advanced quality management* (As of Supply Chain Management version 10.0.49, this feature is mandatory and can't be turned off.)
  - *Dispense management* (As of Supply Chain Management version 10.0.49, this feature is mandatory and can't be turned off.)

Set up a dispensing pick journal

To set up a journal for dispensing materials to production and batch orders, follow these steps:

1. Go to **Production control** > **Setup** > **Production journal names**.
2. Select **New** to create a journal name for the dispensing pick journal.
3. Enter a name and a description.
4. Set the **Dispensing tickets** option to *Yes*.
5. Leave the all other fields set at their default value.
6. On the Action Pane, select **Save**.

Grant users access to production dispensing features

All users who should be able to dispense products in the production process must have the *Production dispenser* security role assigned to them. To assign this role to a user, follow these steps:

1. Go to **System administration** > **Users** > **Users**.
2. Open the user that you want to enable for production dispensing.
3. In the **User's roles** section, select **Assign roles**.
4. In the list, select the *Production dispenser* role.
5. Select **OK**.

Require electronic signatures for production dispensing

To require an electronic signature when users post the dispensing pick journal, follow these steps:

1. Go to **Organization administration** > **Setup** > **Electronic signature** > **Electronic signature requirements**.
2. In the list pane, select the row where the **Name** field is set to *Post dispensing ticket production journal*.
3. Set the **Signature required** option to *Yes*.
4. In the list pane, select the row where the **Name** field is set to *Post picking list production journal (non-dispensing)*.
5. Set the **Signature required** option to *Yes*.
6. On the Action Pane, select **Save**.

Set up measuring devices

You can create a catalog of measuring devices that are used in the dispensing process. These devices are typically scales. Every dispensing ticket can have a measuring device assigned to it. In the production control parameters, you can set up a default device. To set up a measuring device, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Measuring devices**.

2. Use the buttons on the Action Pane to add a new measuring device or edit an existing one. (You can also delete an existing device.)

3. For the new or selected measuring device, set the following fields:

    - **Name**
    - **Description**
    - **Test instrument tag** – Specify a unique identifier for the measuring device. You can use this ID to track the device's calibration history. Learn more about instrument calibration in Test instrument calibration.

Set up production control parameters

To enable production dispensing and configure how it works in your system, you must configure several settings on the **Production control parameters** page. To configure the required production control parameters, follow these steps:

1. Go to **Production control** > **Setup** > **Production control parameters**.
2. On the **Standard update** tab, set the following fields:
    - **Enable dispensing for production** – Set this option to *Yes* to enable production dispensing to be used in a production flow.
    - **Allow over-dispensing with reverse pick** – Set this option to *Yes* to enable material to be over-dispensed or over-picked, and any remainder to be returned to inventory through an automatically generated pick list.
3. On the **Journals** tab, in the **Dispensing tickets** field, select the journal name that you created for the dispensing pick journal. This journal is the default journal for dispensed products.
4. On the **General** tab, in the **Default measuring device** field, select the measuring device that should appear by default on the **Dispensing ticket** page.

Set up a product for the dispensing process

To use a dispensing process, you must configure dispensing options for each relevant product.

> ⓘ **Note**
>
> Products that are enabled for warehouse management processes (WMS) can't be enabled for the dispensing process.

To set up a product for the dispensing process, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Create a new product, or open an existing product that you want to set up for dispensing.
3. On the **Manage inventory** FastTab, in the **Dispensing** section, set the following fields:

- **Dispensing control** – Set this option to *Yes* to enable dispensing control for the product.
- **Underdispense** – Define the under-dispending threshold. In other words, specify the percentage by which the dispensed quantity can be less than the proposed quantity. In this way, when workers enter the dispensed quantity in the dispensing ticket on a production order, the system validates that the quantity doesn't fall below the defined threshold.
- **Overdispense** – Define the over-dispensing threshold. In other words, specify the percentage by which the dispensed quantity can exceed the proposed quantity. In this way, when workers enter the dispensed quantity in the dispensing ticket on a production order, the system validates that the quantity doesn't go above the defined threshold.
- **Authorized personnel** – Set this option to *Yes* to allow only workers who have the *Production dispenser* security role to perform dispensing.

> ⓘ **Note**
>
> When you add a bill of materials (BOM) or formula line that includes a product that is enabled for dispensing, that line inherits the item's over-dispensing and under-dispensing thresholds. However, you can override the inherited values on the BOM or formula line. The new values are then used when the dispensed quantity is validated during the dispensing process.

Generate the dispensing ticket and dispensing pick journal

To create the dispensing ticket and dispensing pick journal for a production or batch order where BOM or formula lines are enabled for dispensing, follow these steps:

1. Go to **Production control** > **Production orders** > **All production orders**.
2. Select **New production order** or **New batch order**.
3. In the dialog, in the **Item number** field, select a product that has an approved BOM or formula version where items are enabled for dispensing.
4. Leave all other fields in the dialog set at their default value.
5. Select **OK** to create the order.

6. On the Action Pane, select **Estimate** to move the order to the *Estimated* state.
7. On the Action Pane, select **Release**.
8. In the dialog, on the **General** tab, set the following fields:
    - **Picking list** – Select a name for the picking list.
    - **Dispensing tickets** – Select a name for the dispensing ticket.
9. Select **OK** to close the dialog and move the order to the *Released* state.
10. On the batch order page, on the Action Pane, on the **View** tab, in the **Journals** group, select the following buttons:
    - **Dispensing ticket** – Open the dispensing ticket that was created when the batch order was released.
    - **Dispensing pick journal** – Open the dispensing pick journal that was created when the batch order was released.

Work with the dispensing ticket and dispensing pick journal

After the batch or production order is released as described in previous sections, workers can open the dispensing ticket to register the amount of material that is dispensed for the order.

1. On the production or batch order page, on the Action Pane, on the **View** tab, in the **Journals** group, select **Dispensing ticket**.
2. In the **Journal** column, select the journal that you want to open.
3. In the **Dispensing details** section, select **New** to create a line for registering a dispensed quantity. If you're making multiple measurements of the same portion of the material that you're dispensing, you can create a separate line for each measurement.
4. Select **Post dispensed** to calculate the average value of the registered measurements. In the **Journal lines** section, the **Dispensed** field of the line item for the dispensed material is updated with the calculated average value.
5. On the Action Pane, select **Confirm** to transfer the value for the calculated dispensed quantity to the dispensing pick journal.

6. On the Action Pane, select **Post** to post the dispensing pick journal. The dispensed quantity is registered as consumed in inventory.

> ⓘ Note
>
> Instead of posting the dispensing pick journal from the dispensing ticket, you can post it directly from the dispensing pick journal page.

Automatically create a dispensing pick journal to revert remaining material from the dispensing process

If you set the **Allow over-dispensing with reverse pick** option to *Yes* as described in the Set up production control parameters section, the system can automatically generate a dispensing pick list to reverse material that was over-picked for the dispensing process. This capability can be relevant when you stage materials in full handling units, such as a container or bag, to the location where the dispensing process is performed.

To have any remainder from over-dispensed or over-picked material returned to inventory through an automatically generated pick list, follow these steps:

> ⓘ Note
>
> Before you begin, ensure that the **Allow over-dispensing with reverse pick** option is set to *Yes* on the **Production control parameters** page.

1. Generate the dispensing ticket and dispensing pick journal by releasing the production or batch order as described in previous sections.
2. On the production or batch order page, on the Action Pane, on the **View** tab, in the **Journals** group, select **Dispensing pick journal**.
3. In the **Journal** column, select the journal that you want to open.
4. In the **Journal lines** section, in the **Consumption** field, increase the value so that it matches the quantity of the material that you want to stage to the dispensing area.
5. Close the dispensing pick journal.
6. On the order page, on the **View** tab, select **Dispensing ticket**.
7. In the **Dispensing details** section, select **New** to create lines for the measurements of the dispensed material.

8. Select **Post dispensed**. In the **Journal lines** section, the **Dispensed** field on the material line is updated.
9. On the Action Pane, select **Confirm** to transfer the value for the calculated dispensed quantity to the dispensing pick journal.
10. On the Action Pane, select **Post** to post the dispensing pick journal.
11. Return to the order page.
12. On the Action Pane, on the **View** tab, in the **Journals** group, select **All**.
13. Select the journal where the description starts with "Dispensing adjustment for."
14. Open the journal, and notice that the journal line has a negative quantity. This quantity was calculated as the difference between the quantity that was registered as staged and the quantity that was confirmed and posted as dispensed.
15. Post the journal to update the inventory with the quantity that was over-picked for the dispensing process.

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Allocate time to jobs in a job bundle

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In Manufacturing execution, you can bundle jobs. You can then start multiple jobs at the same time on the Job list page.

If you bundle jobs, you must define how the total registered time for all the jobs should be allocated to each job. You define the allocation by selecting one of the following options in the **Bundle type** field on the **Allocation keys** page:

- **Estimation** – Time is divided among the jobs, based on the estimated time for the jobs.
- **Jobs** – Time is divided according to total jobs that are bundled and how much time was spent finishing all the jobs.
- **Net time** – Time is divided equally among the jobs that are in the bundle at any time.
- **Real time** – Actual job time is allocated. The cost can be calculated based on the actual payroll cost. **Note:** The **Real time** allocation key is available only if your company uses the payroll functionality in Time and attendance.

The following examples show the results of the various allocation keys.

Example scenario

You need to complete three jobs in your job queue. You start the first job, and while that job is in progress, you start the second and third jobs. Therefore, there's a bundle of three jobs. The following table shows the estimated production time for each job.

Expand table

Job	Production time
Job 1	1 hour

Job	Production time
Job 2	3 hours
Job 3	4 hours
Total	8 hours

The following table shows the actual work hours that you spend on each job.

Expand table

Job	Start time	End time	Bundle time
Job 1	09:00	11:00	2 hours
Job 2	10:00	13:00	3 hours
Job 3	10:00	15:00	5 hours
Bundle	09:00	15:00	6 hours

The following sections describe the results of the calculated time for each allocation key.

Estimation allocation key

The following table illustrates the formula for calculating allocated time. Here is the formula: Time per job = Total bundle time × (Estimated job time ÷ Total estimated time)

Expand table

Job	Formula	Allocated time
Job 1	6 × (1 ÷ 8) hours	0.75 hour
Job 2	6 × (3 ÷ 8) hours	2.25 hours
Job 3	6 × (4 ÷ 8) hours	3.00 hours

Jobs allocation key

The following table illustrates the formula for calculating allocated time. Here is the formula: Time per job = Total bundle time ÷ Number of jobs

Expand table

Job	Formula	Allocated time
Job 1	6 ÷ 3	2 hours
Job 2	6 ÷ 3	2 hours
Job 3	6 ÷ 3	2 hours

Net time allocation key

The following table shows the formula for calculating allocated time. The formula is: Calculated time per reporting = Bundle time ÷ Number of jobs

Expand table

Example	09:00–10:00 (1 hour)	10:00–11:00 (1 hour)	11:00–13:00 (2 hours)	13:00–15:00 (2 hours)	Allocated time
Number of jobs in the bundle	1	3	2	1	Not applicable
Job 1	1 ÷ 1 = 1 hour	1 ÷ 3 = 0.33 hour	Not applicable	Not applicable	1.33 hours
Job 2	Not applicable	1 ÷ 3 = 0.33 hour	2 ÷ 2 = 1 hour	Not applicable	1.33 hours
Job 3	Not applicable	1 ÷ 3 = 0.33 hour	2 ÷ 2 = 1 hour	2 ÷ 1 = 2 hours	3.33 hours

Real-time allocation key

To calculate production costs based on real costs, clear the **Cost category** option on the **Production order defaults** page. The following table shows the formula for calculating allocated time. The formula is: Actual time per job = Actual time in bundle

Expand table

Job	Actual time
Job 1	2 hours
Job 2	3 hours
Job 3	5 hours

Consider the three jobs that a worker performs with an hourly wage of USD 12.00. The worker doesn't earn any overtime bonus or premium for the time spent on the jobs. The worker spends six hours working on the three bundled jobs. Therefore, the salary cost is 6 × USD 12.00 = USD 72.00. When you use real-time allocation, the system recalculates the cost per hour by using the factor from the Net time formula. The system then transfers the actual time that was spent on each job together with the corrected cost price per hour. In the example, six hours are spent, although 10 hours were allocated. The following table shows the formula for calculating cost. The formula is: Cost per hour = (Total bundle time per job (Net time) ÷ Actual time per job) × Standard cost price per hour

Expand table

Job	Calculation of corrected cost per hour	Corrected cost per hour	Allocated time	Total cost of job
Job 1	(1.33 ÷ 2) × USD 12.00	USD 8.00	2 hours	USD 16.00
Job 2	(1.33 ÷ 3) × USD 12.00	USD 5.33	3 hours	USD 16.00
Job 3	(3.33 ÷ 5) × USD 12.00	USD 8.00	5 hours	USD 40.00

The corrected cost per hour and the job time are posted in a production journal.

> **Note**
>
> If you select the **Cost category** option on the **General** tab on the **Production order defaults** page, the system transfers the actual time for each job to a production journal, where the cost is applied to the cost category of the specific job.

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Calculate material consumption

08/13/2025

This article provides information about various options that are related to the calculation of material consumption.

The following options that are related to the calculation of material consumption are available on the **Setup** and **Step consumption** tabs on the **Line details** FastTab of the **Bill of materials** page.

Variable and constant consumption

In the **Consumption is** field, you can select whether consumption should be calculated as a constant quantity or a variable quantity. Select *Constant* if a fixed quantity or volume is required for the production, regardless of the quantity that is produced. Select *Variable*, which is the default setting, if the required amount of material in the finished goods is proportional to the number of finished goods that are produced.

Calculating consumption from a formula

In the **Formula** field, you can set up various formulas for calculating material consumption. If you use the default value, *Standard*, the consumption isn't calculated from a formula. The following formulas work together with the **Height**, **Width**, **Depth**, **Density**, and **Constant** fields:

- Height * Constant
- Height * Width * Constant
- Height * Width * Depth * Constant
- (Height * Width * Depth / Density) * Constant

Rounding up and multiples

Together, the **Rounding up** and **Multiples** fields let you round up the material consumption value. For example, you can round up the value according to the handling unit in which the raw material is picked for production. The following options are available in the **Rounding up** field: *Quantity*, *Measurement*, and *Consumption*.

Quantity

If you select *Quantity* as the rounding-up mechanism, the quantity must be a multiple of the specified quantity. For example, if whole numbers are required, select *1* in the **Multiples** field.

Numbers are then rounded up to a quantity that is divisible by 1.

Measurement

Typically, you select *Measurement* as the rounding-up mechanism when the raw material comes in specific dimensions. For example, a piece of 2-meter metal tube is required for a finished good, and the metal tube is stored in 4.5-meter lengths. In this case, the *Measurement* rounding-up mechanism can be used to calculate how many metal tubes are required to produce a specific number of pieces of the finished good. For this example, the **Formula** field is set to *Height * Constant*. The **Height** field is set to *2* to indicate the length of the tube that is required for the finished good. The **Multiple** field is set to *4.5* to indicate that the tube is picked in lengths of 4.5 meters. Here's the calculation:

1. Number of multiples that are required for 10 pieces of the finished good: 10 ÷ 2 = 5 pieces
2. Total consumption: 4.5 × 5 = 22.5 meters of metal tube

It's assumed that 0.5 meters of tube is scrapped for every five pieces of tube that are consumed.

Consumption

Typically, you select *Consumption* as the rounding-up mechanism when raw material must be picked in whole quantities of a specific handling unit of the product. For example, 2 quarts of paint are used to produce one piece of a finished good, and the paint is picked in 25-quart cans. In this case, the *Consumption* rounding-up mechanism can be used to round up consumption to whole numbers of 25-quart cans. Here's the calculation for the amount of paint that is required if 180 pieces of the finished good must be produced:

1. Paint that is required, excluding scrap: 180 × 2 = 360 quarts
2. Number of cans: 360 ÷ 25 = 14.4, which is rounded up to 15
3. Paint that is required, including scrap: 15 × 25 = 375 quarts

Step consumption

Step consumption is used to calculate constant consumption in quantity intervals. If you select *Step consumption* in the **Formula** field on the **Setup** tab, you can add information about the steps on the **Step consumption** tab. The fixed consumed quantity can be set up in intervals of the produced quantity. For example, step consumption is set up as shown in the following table.

Expand table

From series	Quantity
0.00	10.0000
100.00	20.0000
200.00	40.0000

The bill of materials (BOM) quantity is 1, and the production quantity is 110. The formula for the consumption is From series (Quantity) = Consumption. Because the production quantity is 110, it falls into the "From 100 series." Therefore, the quantity is 20.

Production posting

This article provides information about different types of postings in the production process.

Production posting activities follow production processes that are described in the sections below.

Material consumption

Materials are registered as consumed during production when the production picking list journal is posted. This process generates issue transactions that deduct the on-hand inventory. In the production parameters, you can specify whether the value of raw materials that are in progress (work in process [WIP]) should be posted in the ledger. The value of raw materials that are in progress (WIP) is then posted to a dedicated Picking list account and a dedicated Picking list offset account.

Time consumption

The time that workers spend on production jobs is recorded in the Route card journal or the Job card journal. When these journals are posted, ledger posting to a dedicated account for resources that are in progress (WIP) is processed. This posting represents the value of the time that is spent on the production order. After the production order is registered as ended, the WIP accounts are settled.

Reporting finished goods and error quantities

When a production order is reported as finished, the quantity of the finished goods that have been completed is updated in Inventory management through the Report as finished journal. If you're using WIP accounting, which can be set up in the production parameters, a ledger journal is made to reduce the WIP accounts and increase the inventory of the finished goods. The journal uses the standard cost that is defined for the product.

Ending the production order

Before a production order is ended, actual costs are calculated for the quantity that was produced. All estimated costs of material, labor, and overhead are reversed and replaced with actual costs. The overall cost of the finished item is debited from the inventory Receipt account

and credited to the inventory Issues account. If you select the **End job** check box when you run the cost calculation, the status of the production order is changed to *Ended*. This status prevents any additional costs from being unintentionally posted to a completed production order. You can specify that the value of error quantities that are reported during reporting as finished should be allocated to the good quantities that are reported as finished. Alternatively, you can specify that the value of error quantities should be posted to a dedicated scrap account.

Controlling the level of ledger posting

In the **Production control parameters**, you can use the **Ledger posting** field to set the level of ledger posting for production processes. The following values are available:

- *Item and resource* – Use the ledger accounts that are set up on the item groups for raw materials and finished goods. WIP for time consumption is taken from resource or resource group from the route operations.
- *Item and category* – Use the ledger accounts that are set up on the item groups for raw materials and finished goods. WIP for time consumption is taken from the cost categories that are associated with the route operations.
- *Production groups* – Use the ledger accounts that are set up on the production groups for both material and time consumption. The production groups are associated with the released products and copied to the production orders when those orders are created. The posting on the production orders will then follow the production groups that are associated with the production order.

> **Note**
>
> If the standard method for calculating the cost of the finished item was used, the final transactions reflect this fact. If actual costs and the costs that are calculated by using the standard method differ, the difference is posted to the account that shows profit or loss.

Last updated on 11/26/2025

Time and attendance registration overview

07/10/2025

Time registration workers can enter different types of time registrations, for example, clock in, clock out, register indirect activities, and absence registration. This article describes registrations, their calculation, approval, and the use of workflow to add structure and automated approval to the process of approving timesheets.

Registrations

In companies that use time and attendance, workers must register the time that they spend at work in addition to their attendance. Some companies might only require workers to register clocking-in and clocking-out times. In other companies, workers might also be required to register time consumption on the actual activities they perform and the breaks they take. The intended users of time and attendance are:

- Workers, who are required to register time and attendance at regular intervals, for example daily, weekly or bi-weekly.
- Supervisors, managers, and payroll officers who calculate, approve, and transfer worker registrations for further processing.

> **Note**
>
> If you run Time and attendance together with Manufacturing execution, all registrations on projects, project activities, indirect activities, absence codes, and overtime and flex time will be recorded and are used to calculate payroll in both modules.

Time registrations workers

To be able to register time and absence, workers must be set up as time registration workers in the company they're employed in.

After setup, the workers can enter different types of registrations.

- Clocking in- and out when arriving or leaving work.
- Time and item consumption on production jobs.
- Time used on a machine on the shop floor, if the machine has been defined as a resource.

> **Note**
>
> A worker can be automatically assigned the time registrations that are made on a particular machine on the shop floor, if the worker chooses to work as an assistant to the machine when they start the production job.

- Time registrations on projects and project activities.
- Registering project fees and item consumption via the respective project fee journals and project item journals.
- Planned absence.
- Absence when arriving late to work or leaving earlier than planned.
- Work breaks, either manually registered or automatically calculated by the system.
- Indirect activities, which are nonproductive activities a worker might engage in during a workday. Examples of these activities include meetings or cleaning their workspace.
- Overtime, which can be registered either as extra hours, flextime, or overtime.

Adding clock-out registrations

If a worker forgets to clock-out at the end of their workday, the missing registration can be added by running a batch job. The system will compare the clock-in time and the clock-out time according to the associated profile of the worker, and automatically insert the missing clock-out registration to match the profile's end time. Both the clock-in and clock-out registrations are vital for the subsequent calculation and approval of time registrations before they can be transferred to payroll.

Calculating registrations

When a registration worker is assigned a calculation group that typically relates to a specific team, shift, or work group. The team manager or supervisor typically validates the registrations made by the workers, and is therefore also the responsible person to run the calculation for the respective calculation groups on a daily basis. As part of the calculation process the team manager or supervisor is able to:

- Correct erroneous registrations. For example, change switch codes and adjust feedback on production jobs.

- Add missing registrations. For example, create clock-out registrations and create absence transactions.
- Delete incorrect registrations.

Because the registered time must match the worker's time profile prior to calculating the registrations, you must override the work time profile for any worker who has an exception to their standard work time profile. If the worker profile is day shift, and the worker has agreed to work a night shift with no overtime pay, the team manager or supervisor must override the default worker profile in order to calculate the working time at the standard night rate and not as overtime. The calculation will also display an error if an absence registration is missing. It must be added before the calculation can be completed.

Approving registrations

Just as you assign a calculation group to a time registration worker, you must assign an approval group as well. Typically the group will be specific to a team, shift, or work group. You must approve the time registrations that were calculated correctly – this means doing a calculation without errors – before pay items can be generated that afterward can be transferred to a payroll system. The payroll administrator will typically do the approval of registrations, and prior to the approval they're able to:

- Override pay agreements for individual workers.
- Add manual premiums.
- Enter additional information about absence registrations.

> **Note**
>
> If overtime has been calculated for specific workers, the overtime can be allocated to specific jobs during the day. This is relevant if job cost is calculated based on worker pay.

Approving registrations using workflow

You can set up a workflow approval process that automatically approves registrations that comply with workflow rules, leaving only deviations to be handled manually. If workflow approval is activated, the team manager or supervisor submits the calculated registrations for approval. The workflow process will generate the appropriate approvals and tasks, and then assign them to the right users and roles as identified in the workflow. There are two workflow approvals for time and attendance.

Expand table

Workflow	Purpose	Registration type
Time and attendance days total	The workflow validates registrations against, for example, the expected number of work hours for the day.	
Time and attendance journal registration.	The workflow validates each registration type for the date of the registration.	Time and attendance • Clock-in • Clock-out • Absence • Break • Switch code • Project • Project activity • Indirect activity Production jobs • Queue before • Setup • Process • Overlap • Transport • Queue after • Start assistance • Stop assistance

Transferring approved registrations

After approval of the registrations, you can transfer them to a periodic payroll job. A transferred registration is posted to an activity or job that it relates to, for example, a production order or a project. Payroll transactions are generated for each worker based on the registrations.

Reversing transferred registrations

The task of reversing transactions – rolling them back – can be done until the time when the payroll period's pay transfer is run. This means that payroll data has been transferred to an external file. When reversed, all registrations are withdrawn, and any transactions posted on production orders or projects are offset and neutralized.

> ⓘ **Note**
>
> The external file can be imported into a payroll system.

Registrations in electronic timecards

Workers with job tasks that don't require immediate feedback, as is the case with production jobs, but who work on project activities, can benefit from using the electronic timecard. Electronic timecards offer the flexibility to enter registrations at any time and in the best way for your business schedule (daily, weekly, or when a worker is in the office again after being away). To use electronic timecards or these workers, you must specify, Use timecard, in the worker details. Electronic timecards enable the worker to register:

- Date
- Registration type
- Job reference, such as project, indirect activity, or production order
- Job identification
- Time consumption
- Project fees
- Project items

Registration for manufacturing execution

09/19/2025

This article describes key concepts and terms that you need to understand to configure and use manufacturing execution.

Manufacturing execution is primarily for manufacturing companies. Workers register time and item consumption on production jobs by using the production floor execution interface. All registrations go through approval and later transfer to the relevant modules. Continuous approval and transfer of registrations lets managers easily track actual costs on production orders.

Set up operations to use cost for estimated or actual time

On the **Route groups** page, you can configure whether a production job uses estimated or actual time when it's reported in the production floor execution interface.

Follow these steps to configure a route group.

1. Go to **Production control** > **Setup** > **Routes** > **Route groups**.
2. Select or create a route group.
3. On the **General** FastTab, set the **Run time** option to one of the following values:
    - *No* – When a job is stopped or completed, generate a job card journal on the production order. This journal has the actual time that the worker spent on the job.
    - *Yes* – When a job is started, generate a route card journal on the production order. This journal has the estimated time for the operation.

Set up operations to use either estimated or actual time for cost calculations

You can use the **Skip time adjustments** option on the **Production order defaults** page to specify whether calculations of actual cost per production order include time adjustments that are made to actual recorded work times. Adjustments apply when a supervisor uses the **Time and attendance** module to review and edit the recorded work times that each worker registers to record time and attendance. Adjustments can reflect events such as registration errors or

unpaid time off. If you skip time adjustments, you can end production orders without waiting for supervisor approvals.

Prerequisites

Before you can use the **Skip time adjustments** option, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.41 or later.
- You must turn on the feature named *Skip time adjustments when calculating actual cost per production order* in Feature management.

> **Important**
>
> The *Skip time adjustments when calculating actual cost per production order* feature changes the way that automatic route consumption works. When you enable it, route groups that use *estimated production times* to calculate actual production order costs don't include *recorded times* in the calculation.

Set the Skip time adjustments option

Follow these steps to configure whether time adjustments are made to production orders as part of the processing of workers' daily registrations.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Production order defaults**.

2. On the **General** tab, set the **Skip time adjustments** option to one of the following values:

    - *No* – Apply adjustments when a supervisor reviews and edits the recorded work times that each worker registers to record time and attendance.
    - *Yes* – Consider only recorded start and end times. If you skip time adjustments, you can end production orders without waiting for supervisor approvals.

Manufacturing execution and registration terminology

The following table defines terms that pertain to manufacturing execution and related registration tasks.

Expand table

Term	Definition
Manufacturing execution	A function that registers time, material consumption, and costs on production jobs, projects, and indirect activities. Registration happens in a manufacturing execution registration client, such as the production floor execution interface.
Job list	The production floor execution interface shows workers the list of jobs that they must perform on a specific resource, such as a machine. A worker can register time and item consumption on each job or task on the job list.
Job bundling	An approach where a worker starts more than one job at the same time in the production floor execution interface. The time that a worker spends on bundled jobs can be allocated to the individual jobs in various ways by using allocation keys.
Pilot/assistant registrations	A worker can register as an assistant to a resource and create a small team where several workers work on the same production jobs. Resources that workers connect to as assistants are known as *pilots*. Only the pilot resource must make registrations. All assistants then automatically get the same registrations. For example, if a machine acts as the pilot, workers who register as assistants to that machine can make registrations in the production floor execution interface. In this case, both the machine and the workers who are connected as assistants receive the same registrations.
Indirect activity	An activity or task that isn't directly related to a production job or a project. Examples include a department meeting, a cleaning job, and a maintenance job on the shop floor. Workers can make registrations on indirect activities in the same way that they make registrations on production jobs and projects.

Registrations in manufacturing execution

Workers can make various types of registrations in manufacturing execution for work that is performed on production jobs. Depending on the system setup, workers might also be able to make registrations on project activities and nonproductive tasks, such as breaks, absences, and indirect activities. The following list describes the registration types:

- **Clock-in/clock-out** (available with time and attendance) – Workers clock in when they arrive at work, and they clock out when they leave to go home.
- **Register on production jobs** – Workers can make registrations on the production jobs that appear on their job list. Examples of these registrations include starting a job and reporting feedback for a job. Workers can start several jobs at the same time. This approach is known as job bundling.
- **Register on inventory** – Workers can make registrations on materials that they use on the shop floor, but that aren't directly related to production jobs. Examples of these materials include grease, lubricants, and other materials that are used to keep machinery running. Registration is done in an inventory journal.

- **Register on projects** (available with time and attendance) – Workers can make registrations on the projects or project activities that appear on their job list. Examples of these registrations include starting work and finishing work.
- **Register project fees and project items** (available with time and attendance) – Workers can register fees (expenses) that are associated with a project in a project fee journal. Examples of these fees include mileage and a bridge toll. Workers can also register item consumption on projects. This registration is done in a project item journal.
- **Register as assistant to another worker** – If two or more workers work together on a production job or a project, a worker can register as an assistant to a machine or another worker. That machine or other worker acts as the pilot. The pilot can select another worker as the pilot, as required.
- **Register absence** (available with time and attendance) – Workers can register time on various absence codes that are set up. Absence can be indicated if a worker arrives late, requires absence during the workday, or leaves earlier than expected according to the standard work time profile.
- **Register breaks** (available with time and attendance) – During the workday, workers can register that they're leaving their workstation to take a break. Several break types can be set up. When a worker returns and signs in again, the system registers that the worker is back, and the break registration stops.
- **Register indirect activities** (available with time and attendance) – Indirect activities are nonproductive activities that workers might engage in during a workday. Examples include a department meeting, a team meeting, and a maintenance job that is performed on the shop floor. Workers can make registrations on the indirect activities that are set up.
- **Register overtime** (available with time and attendance) – Workers who are asked to work longer hours can select whether the extra hours should be registered as flextime or overtime.

Pay based on registrations

Article • 03/09/2023

This article explains, in detail, how pay is calculated based on worker registrations. It includes examples that show how the various combinations of setup options that are available for the calculation affect the result. Here are some of the areas that will be covered:

- Flex time
- Overtime
- Breaks
- Switch codes
- Pay items
- Pay agreements
- Time and attendance calculation parameters
- Absence

The use of flex time

Periods of flex time are set up in the time profiles that are used in Time and attendance. There are two flex profile types: **Flex+** and **Flex-**. When a worker registers time in a Flex+ period, the worker's flex balance is increased by the hours that were worked. The worker doesn't receive any compensation for the hours that were worked during the Flex+ period. However, the worker can take time off during the Flex- periods and be compensated with the hours from their flex balance. Therefore, time off during the flex periods is considered an absence by the system.

Scenarios based on flex periods

The two scenarios that follow are based on a flex profile that represents a workday. For both scenarios, pay is calculated according to the flex period where the worker clocks in and out.

Flex profile for one workday

Expand table

Profile type	Start	End	Day
Over time	00:00 AM	06:00 AM	Monday
Flex+	06:00 AM	07:00 AM	Monday
Clock in	07:00 AM	07:00 AM	Monday
Standard time	07:00 AM	02:30 PM	Monday
Flex-	02:30 PM	03:30 PM	Monday
Clock out	03:30 PM	03:30 PM	Monday
Over time	03:30 PM	06:00 AM	Tuesday

Scenario 1: A worker registers clock-in during a Flex+ period and clock-out during a Flex- period

The worker's registrations for the day looks like this.

Expand table

Journal registration type	Start	End
Clock in	06:30 AM	06:30 AM
Production job	06:30 AM	02:45 PM
Clock out	02:45 PM	02:45 PM

The worker's registrations for the day are calculated and transferred to pay on the **Approve** page (**Time and attendance** > **Review and approve** > **Approve**). After the registrations have been calculated, the result of the calculation can be viewed on the **Times** tab.

To understand this scenario, see the following fields.

Expand table

Flex +	Flex -	Time	Pay time
0.50	0.75	8.25	8.50

Calculation of Flex+

According to the flex profile, the time between 06:00 AM and 07:00 AM is a Flex+ period. Therefore, if the worker clocks in at 06:30 AM, they earn 0.5 hours. This amount of time is added to the worker's flex account.

Calculation of Flex-

According to the flex profile, the Flex- period starts at 02:30 PM and ends at 03:30 PM. Therefore, if the worker clocks out at 02:45 PM, the 45 minutes (0.75 hours) that remain in the Flex- period are registered as pay time, and the same amount of time is deducted from the worker's flex account. The 45 minutes are included in pay time because the worker will be granted pay for the remaining 45 minutes in the Flex- period. If the worker is absent during the Flex- period, the 45 minutes will be deducted from their flex account.

Calculation of Time

Time is calculated as the time between clock-in and clock-out, that is, 06:30 AM to 14:45 PM with equals 8.25 hours.

Calculation of Pay time

Pay time is the time that a worker is granted pay during. In this scenario, the worker is at work for 8.25 hours (**Time**). However, **Pay time** is calculated as 8.50 hours because the worker is granted pay during the Flex- period after they clocked out. The pay time equals the planned work hours because Flex+ time is added to the worker's flex account, not to the pay time. The absence time during the Flex- period is compensated for by the pay time and deducted from the worker's flex account.

Expand table

Time	Registration type	Pay time (hours)
6:30 AM - 7:00 AM	Flex+	0
7:00 AM – 2:45 PM	Standard time	7.75
2:45 PM – 3:30 PM	Flex-	0.75 (Absence period)
	Total	8.50

Scenario 2: A worker works during the whole Flex- period and also works overtime

The worker's registrations for the day look like this.

Expand table

Journal registration type	Start	End
Clock in	06:30 AM	06:30 AM
Production job	06:30 AM	05:00 PM
Clock out	05:00 PM	05:00 PM

After you've calculated the journal registrations on the **Approve** page, you can view the calculation result on the **Times** tab. To understand this scenario, see the following fields.

Expand table

Flex +	Flex -	Time	Pay time	Pay overtime
0.50	0.00	10.50	10.00	1.50

Calculation of Flex+

According to the flex profile, the time between 06:00 AM and 07:00 AM is a Flex+ period. Therefore, if the worker clocks in at 06:30 AM, they earn 0.5 hours of Flex+ time on their flex balance.

Calculation of Flex-

Because the worker is working during the Flex- period, Flex- isn't calculated. Flex- is calculated only if the worker is absent during the Flex- period. From a payment perspective, if the worker works during the Flex- period, they are granted the pay rate that is defined for standard time. If the worker if is absent during the Flex- period, the 45 minutes are deducted from their flex account.

Calculation of Time

Time is calculated as the time between clock-in at 06:30 AM and clock-out at 05:00 PM, or 10.50 hours.

Calculation of Pay time

In this scenario, the worker works 10.50 hours (**Time**). However, **Pay time** is calculated as 10.00 hours because the worker isn't granted pay during the Flex+ period.

Calculation of Pay overtime

Expand table

Time	Registration type	Pay time (hours)
6:30 AM - 7:00 AM	Flex+	0
7:00 AM – 2:30 PM	Standard time	7.50
2:30 PM – 3:30 PM	Flex-	1.00
3:30 PM – 05:00 PM	Overtime	1.50
	Total	10.00

Generation of pay items

Worker registrations for the day can be transferred from the **Approve** page. During the transfer process, pay items and transferred registrations are generated. Pay items represent a breakdown of pay time into standard time, overtime, paid break time, and so on.

To open the list of pay items, select **Time and attendance** > **Review and approve** > **Approve**. Then select **Inquiry** > **Transferred registrations**.

The pay items are the basis for a worker's pay. You can generate a file that includes the information from the pay items and transfer that file to a payroll system.

As part of the transfer process, time and cost from production and project activities are transferred to production and project journals to account for the time and cost spend. The transferred registrations are the basis for time and cost price per hour for production orders and projects. You can open the transferred registrations by using the **Inquiry** menu on the **Approve** page.

For example, for scenario 2, the following pay items are generated.

Expand table

Wage type	Pay type	Pay units	Rate	Total cost
Standard time	1201	10.0	10	100

Wage type	Pay type	Pay units	Rate	Total cost
Overtime	1301	1.50	5	7.50
			Total	107.50

The pay item for standard time has a pay unit of 10 hours that covers standard time, Flex- time, and overtime. Standard time, Flex- time, and overtime are consolidated into one pay item because all the types are calculated as standard time, based on the default setting of a parameter on the **Calculation parameters** page (**Time and attendance** > **Setup** > **Calculation parameters**). The overtime is calculated on top of the standard time by using an additional rate of 5.

If you want to clearly distinguish standard time and overtime, so that the pay units for the pay types cover only the actual time spend on standard time and overtime, the pay units for standard time must be calculated as 8.50, and the pay units for overtime must be calculated as 1.50.

To configure the system to clearly distinguish standard time and overtime, you must exclude overtime from the standard time. You must also change the setup of the pay type for overtime so that the pay rate for overtime covers all payments for the hours that were spent on overtime.

Exclude overtime from the standard time

On the **Calculation parameters** page, select **Overtime** as the profile specification type, and set the **Pay time** option to **No**, as shown here.

Expand table

Reg. specification	Profile specification type	Calculation	Setting	Paid	Setting
Working time	Overtime	Standard time	Yes	Pay time	No
		Pay time	Yes	Pay overtime	Yes

After you adjust the calculation parameters, the following pay items are generated.

Expand table

Wage type	Pay type	Pay units	Rate	Total cost
Standard time	1201	8.50	10	85.0
Overtime	1301	1.50	15	22.50
			Total	107.50

> **Note**
>
> The calculation parameters have a recommended standard setting. In general, you should be careful when you change these parameters. To restore the recommended standard settings, on the **Calculation parameters** page, select **Restore values**.

Allow a deviation from the standard pay profiles

On the **Profiles** page (**Time and attendance** > **Setup** > **Time profiles** > **Profiles**), you can set up profile types that include switch codes and breaks.

Switch codes

You can use switch codes to allow workers to deviate from their profile type by changing a different profile type. For example, you can allow a worker to change from Flex+ time to overtime. A worker can add a switch code during registration, or you can assign the task of adding a switch code to the approver of the registrations.

Before a switch code can be used, you must define it as a type of indirect activity. You must then add the switch code to the time profile for the period where you want to allow a change of profile type. For example, follow these steps to create a switch code that allows the Flex+ period to be changed from 06:00 AM to 07:00 AM to overtime.

1. Create a switch code that is named **OTBCI (Convert flex to overtime before clock-in)**. Select **Time and attendance** > **Manage indirect activities** > **Indirect activities**.
2. In the **Switch code** column, add OTBCI to the Flex+ line in the time profile.
3. In the **Secondary** column, add the **Overtime** profile type.

Consider the following flex profile that represents a workday.

Expand table

Profile type	Start	End	Day
Over time	00:00 AM	06:00 AM	Monday
Flex+	06:00 AM	07:00 AM	Monday
Clock in	07:00 AM	07:00 AM	Monday
Standard time	07:00 AM	02:30 PM	Monday
Flex-	02:30 PM	03:30 PM	Monday
Clock out	03:30 PM	03:30 PM	Monday
Over time	03:30 PM	06:00 AM	Tuesday

Here are the worker's registrations for the day.

Expand table

Journal registration type	Start	End
Clock in	06:30 AM	06:30 AM
Production job	06:30 AM	02:45 PM
Clock out	05:00 PM	05:00 PM

The following pay items are generated after the transfer.

Expand table

Wage type	Pay type	Pay units	Rate	Total cost
Standard time	1201	8.50	10	85.0
Overtime	1305	1.50	15	22.50
			Total	107.50

On the **Approve** page, undo the transfer, and then use the **Switch code** menu to apply the **OTBCI** switch code. When you transfer the registrations a second time, the following pay items are generated.

Expand table

Wage type	Pay type	Pay units	Rate	Total cost
Standard time	1201	8.50	10	80.0
Overtime	1305	2.00	15	30.0
			Total	107.50

> **Note**
>
> When you apply the switch code, overtime is increased by 0.5 hour, from 1.50 to 2.00. The 0.5 hour is the conversion of the Flex+ time that is registered, from 6:30 AM to 07:00 to overtime.

Breaks

Breaks from work affect the calculation of worker pay. Breaks are defined as a type of indirect activity. They can be defined as either **Paid**, to allow the break to add to the worker's pay, or **Unpaid**, to prevent the break from adding to the worker's pay. A break can also be defined as either **Planned** or **Registered**.

Planned breaks

If a company has a fixed break time, such as a fixed break for lunch, the break can be predefined in the time profile. In this case, the worker doesn't have to register the break on the production floor execution interface. Instead, the break is automatically accounted for when the worker's registrations are calculated on the **Approve** page.

Registered breaks

If a company doesn't use planned breaks, workers can register breaks during the workday. Registered breaks can be used, for example, if a worker is working against a flex time profile that has no defined clock-in and clock-out times. Registered breaks are a type of indirect activity. The worker registers the break on the production floor execution interface. Here, the worker can select the type of break in a list of predefined break activities.

Paid and unpaid breaks

Break activities can be set up as **Paid** or **Unpaid**. A paid break is included in the calculation of pay time, and the system uses the pay type that is defined in the pay

agreement for the **Break** registration type.

Example of a planned break

Consider the following time profile that includes an unpaid break for lunch.

Expand table

Profile type	Start	End	Day
Over time	00:00 AM	06:00 AM	Monday
Flex+	06:00 AM	07:00 AM	Monday
Clock in	07:00 AM	07:00 AM	Monday
Standard time	07:00 AM	12:00 PM	Monday
Break	12:00 PM	12:30 PM	Monday
Standard time	12:30 PM	02:30 PM	Monday
Flex-	02:30 PM	03:30 PM	Monday
Clock out	03:30 PM	03:30 PM	Monday
Over time	03:30 PM	06:00 AM	Tuesday

Here are the worker's registrations for the day.

Expand table

Journal registration type	Start	End
Clock in	06:30 AM	06:30 AM
Production job	06:30 AM	02:45 PM
Clock out	05:00 PM	05:00 PM

After you've calculated the journal registrations on the **Approve** page, you can view the calculation result on the **Times** tab. To understand this scenario, see the following fields.

Expand table

Flex +	Flex -	Time	Pay time	Non-paid break time	Pay overtime
0.50	0.00	10.50	9.50	0.5	1.50

> **Note**
>
> The system calculated 0.5 hours of unpaid break time, and that time isn't part of the pay time.

Example of a registered break

Consider the following time profile that doesn't include planned breaks.

Expand table

Profile type	Start	End	Day
Over time	00:00 AM	06:00 AM	Monday
Flex+	06:00 AM	07:00 AM	Monday
Clock in	07:00 AM	07:00 AM	Monday
Standard time	07:00 AM	02:30 PM	Monday
Flex-	02:30 PM	03:30 PM	Monday
Clock out	03:30 PM	03:30 PM	Monday
Over time	03:30 PM	06:00 AM	Tuesday

Here are the worker's registrations for the day.

Expand table

Journal registration type	Start	End
Clock in	06:30 AM	06:30 AM
Production job	06:30 AM	05:00 PM
Break	12:03 PM	12:32 PM
Clock out	05:00 PM	05:00 PM

When the registrations are calculated, the time for the activities is calculated.

Expand table

Journal registration type	Start	End	Time
Clock in	06:30 AM	06:30 AM	
Production job	06:30 AM	05:00 PM	10.00
Break	12:03 PM	12:32 PM	0.50
Clock out	05:00 PM	05:00 PM	

> **Note**
>
> The time for the break runs in parallel with the time for the activity (a production job, in this example). This behavior is always used for break activities. When the registrations are calculated, the break time is subtracted from the activity time. In this case, the production job has a duration of 10.50 hours, but the time is calculated as 10 because 0.5 hours of break time are subtracted from the activity time.

After you've calculated the journal registrations on the **Approve** page, you can view the calculation result on the **Times** tab. To understand this scenario, see the following fields.

Expand table

Flex +	Flex -	Time	Pay time	Non-paid break time	Pay overtime
0.50	0.00	10.50	9.50	0.5	1.50

If the planned break had been paid instead of unpaid, the calculation result would look like this.

Expand table

Flex +	Flex -	Time	Pay time	Paid break time	Pay overtime
0.50	0.00	10.50	10.00	0.5	1.50

Pay items and paid breaks

When you transfer registrations on the **Approve** page, pay items are generated. A separate pay item is generated for paid breaks.

The pay rate for a paid break is determined by the pay type that is set up in the pay agreements for the break. Instead of using a pay type, you can set up the cost price per

hour on the break for a defined date interval.

Consider the following time profile.

Expand table

Profile type	Start	End	Day
Over time	00:00 AM	06:00 AM	Monday
Flex+	06:00 AM	07:00 AM	Monday
Clock in	07:00 AM	07:00 AM	Monday
Standard time	07:00 AM	02:30 PM	Monday
Flex-	02:30 PM	03:30 PM	Monday
Clock out	03:30 PM	03:30 PM	Monday
Over time	03:30 PM	06:00 AM	Tuesday

Here are the worker's registrations for the day.

Expand table

Journal registration type	Start	End	Time
Clock in	07:00 AM	07:00 AM	
Production job	07:00 AM	03:00 PM	7.5
Break (Paid)	12:00 PM	12:30 PM	0.5
Clock out	03:00 PM	03:00 PM	

For this example, the pay type for standard time is set to **1201**, and a pay rate of **10** is set up in the pay agreement. The paid break has a pay type of **1301** and a pay rate of **8**. When the registrations are transferred, the following pay items are generated.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	7.50	10
Flex-	1201	0.50	10
Break (Paid)	1301	0.50	8

How the cost of paid breaks is allocated to projects and production orders

The hourly cost on project activities and production jobs can be set up so that it's determined either by the pay rates that are calculated in Time and attendance or by the cost categories that are defined for the activities.

To set up the cost category, select **Production control** > **Setup** > **Manufacturing execution** > **Production order defaults**, and set the **Cost category** field to either **Yes** or **No**.

- **No** – Cost is calculated based on pay rates that are defined for Time and attendance registration types.
- **Yes** – Cost is calculated based on cost categories for production and project activities.

Cost calculation based on pay rates that are calculated in Time and attendance

The following example shows how the hourly cost is calculated when the cost is set up so that it's calculated based on pay rates.

The hourly cost rate that is used for production orders and projects is calculated during the transfer process. To view the hourly rate per activity, open the **Approve** page in Time and attendance, and then select **Inquiry** > **Transferred registrations**. You can find the hourly cost rate per registration on the **Cost prices** tab.

Consider the following registrations that use the same time profile as the previous example.

Expand table

Journal registration type	Start	End	Time
Clock in	07:00 AM	07:00 AM	
Process (Order: 4711)	07:00 AM	11:00 AM	4
Process (Order: 4712)	11:00 AM	03:00 PM	3.50
Break (Paid)	12:00 PM	12:30 PM	0.50
Clock out	03:00 PM	03:00 PM	

After the registrations are transferred, the following transferred registrations are generated.

Expand table

Registration type	Time	Cost price per hour
Clock-in	0.00	0.00
Process (Order: 4711)	4.00	10.00
Process (Order: 4712)	3.50	11.14
Break (Paid)	0.50	0.00
Clock out	0.00	0.00

The calculation of the cost price per hour for the paid break depends on a setting for the direct payroll costs. Select **Time and attendance** > **Setup** > **Time and attendance parameters**. On the **Cost price** tab, under **Direct payroll costs**, in the **Standard time** field, you can select **Yes**, **No**, or **Allocation**.

- **Yes** – This value is used for the preceding example. The cost is allocated to the production or project activity that runs in parallel with the paid break activity. In the example, this activity is the production job for order 4712. As you can see, the cost price per hour for the paid break is 0 (zero), and it's allocated to the job that runs in parallel with the break.

  The paid break has a duration of 0.5 hour, and the pay rate is 8. Therefore, the total cost for the paid break is 4. The total cost is then allocated to the 3.5-hour process job. Therefore, the paid break contributes 1.14 per hour to the cost (4 ÷ 3.5 = 1.14).

- **Allocation** – The paid break is equally distributed to the jobs that are registered for the day. If this value is used for the preceding example, the following transferred registrations are generated.

  Expand table

Registration type	Time	Cost price per hour
Clock-in	0.00	0.00
Process (Order: 4711)	4.00	10.53
Process (Order: 4712)	3.50	10.53

Registration type	Time	Cost price per hour
Break (Paid)	0.50	0.00
Clock out	0.00	0.00

The total process time for the two production jobs is 7.5 hours, and the total cost of the paid break is 4. Therefore, the cost of the break is calculated as 0.53 (= 4 ÷ 7.5).

- **No** – The cost of the paid break doesn't increase the hourly cost of the process activities.

Expand table

Registration type	Time	Cost price per hour
Clock-in	0.00	0.00
Process (Order: 4711)	4.00	10.00
Process (Order: 4712)	3.50	10.00
Break (Paid)	0.50	0.00
Clock out	0.00	0.00

Absence

An absence code is used to register the period of a worker's absence. Like breaks and switch codes, an absence code is a type of an indirect activity. Absence time can be either planned or registered, and absence can be either legal or illegal. Examples of legal absence include a doctor's appointment, a seminar, or jury duty. Illegal absence is absence that has no good reason, such as when a worker is late for work. Usually, legal absence doesn't cause a deduction in a worker's pay, whereas illegal absence does cause a deduction.

Planned absence

You can create planned absence for workers on the **Create planned absence** page (**Time and attendance** > **Create planned absence**). There, the planned absence is registered as an absence job for a specified date and time interval.

The job is based on a query. Therefore, you can create planned absence for multiple workers, such as workers who belong to the same calculation group. If the planned absence is for a single worker, the registration can be entered from either the **Attendance** page or the **Time registration workers** page.

- To enter an absence registration from the **Attendance** page, select **Time and attendance** > **Inquiries and reports** > **Attendance** > **Attendance**, and then select **Absence registration**.
- To enter an absence registration from the ***Time registration workers*** page, select **Time and attendance** > **Setup** > **Time registration workers**, and then, on the **Time** tab, under **Time assignment**, select **Absence registrations**.

You can use the **Planned absences** report to see an overview of planned absences for workers. To open this report, select **Time and attendance** > **Inquiries and reports** > **Absence reports** > **Planned absences**.

Registered absence

In general, workers are considered absent if they aren't at work for any period between their planned clock-in time and their planned clock-out time. If workers clock in later than planned, or if they clock out earlier than planned, they are prompted to select an absence code to indicate the reason for the absence. An absence code can be set up so that it's applicable to registration. Only applicable codes will be available for selection in the list.

Scenarios based on various combinations of work hour registrations

The following scenarios show the pay items and entries for approval that are generated for workers based on their registrations. All the scenarios are based on the following time profile.

Expand table

Profile type	Start	End	Day
Over time	00:00 AM	06:00 AM	Monday
Flex+	06:00 AM	07:00 AM	Monday
Clock in	07:00 AM	07:00 AM	Monday

Profile type	Start	End	Day
Standard time	07:00 AM	02:30 PM	Monday
Flex-	02:30 PM	03:30 PM	Monday
Clock out	03:30 PM	03:30 PM	Monday
Over time	03:30 PM	06:00 AM	Tuesday

Scenario 1: The worker clocks in later than planned

The worker clocks in at 08:30 AM. Because their planned clock-in time is 07:00 AM, they are 1.50 hours late for work. Because the 1.50 hour is considered absence time, the worker is prompted to select an absence code. The worker leaves work at 03:30 PM, which is the planned clock-out time. When the worker's registrations are calculated and approved, the absence registration, together with the absence code that the worker selected at clock-in, appears for the time between 07:00 AM and 08:30 AM.

In the time profile, you can configure the **Clock-in** registration type so that there is a tolerance when workers are late for work. For example, if you set up a tolerance of 5, the worker is prompted for an absence code only if they clock in later than 07:05 AM.

In this case, because the worker doesn't have a good reason for being late for work, they select an absence code that is defined for illegal absence. An absence code is considered applicable to illegal absence if the setting for overtime deduction is enabled for the absence group that the absence code belongs to. To set the setting, select **Time and attendance** > **Setup** > **Groups** > **Absence groups**, and then select the **Deduct overtime** check box.

Here is how the worker's registrations for the day appear on the **Approve** page after calculation.

Expand table

Journal registration type	Start	End	Time
Absence (illegal - late for work)	07:00 AM	08:30 AM	1.5
Clock in	08:30 AM	08:30 AM	
Production job	07:30 AM	03:30 PM	7.0
Clock out	03:30 PM	03:30 PM	

Here is the resulting pay item after the registrations are transferred.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	7.00	10

Scenario 2: The worker clocks out before the planned clock-out time during a standard-time period

The worker clocks in at 07:00 AM and clocks out early at 01:00 PM. Because 01:00 PM is before planned clock-out at 03:30 PM, and 01:00 AM is in a standard-time period, the worker is prompted to select an absence code. The worker selects an absence code for a doctor's appointment, which is defined as legal absence. The pay rate for legal absence is defined in the pay agreements for the **Absence** registration type (**Time and attendance** > **Setup** > **Payroll** > **Pay agreements**).

Here is how the worker's registrations for the day appear on the **Approve** page after calculation.

Expand table

Journal registration type	Start	End	Time
Clock in	07:00 AM	07:00 AM	
Production job	07:00 AM	01:00 PM	4.0
Clock out	01:00 PM	01:00 PM	
Absence (legal – doctor's appointment)	01:00 PM	03:30 PM	3.5

Here is the resulting pay item after the registrations are transferred.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	7.50	10

Scenario 3: The worker clocks out before the planned clock-out time during a Flex- period

The worker clocks in at 07:00 AM and clocks out at 02:15 PM, which is in the planned Flex- period. The time between the actual clock-out time and the planned clock-out

time isn't considered absence, and the worker isn't prompted to select an absence code. The amount is deducted from the worker's flex account, and the worker is granted pay during the remaining part of the Flex- period, from 02:15 to 03:30 PM.

Here is how the worker's registrations for the day appear on the **Approve** page after calculation.

Expand table

Journal registration type	Start	End	Time
Clock in	07:00 AM	07:00 AM	
Production job	07:00 AM	02:15 PM	7.25
Clock out	02:15 PM	02:15 PM	

Here is the resulting pay item after the registrations are transferred.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	8.50	10

Scenario 4: The worker clocks in late and clocks out after the planned clock-out time during an overtime period

The worker clocks in late at 09:30 AM and then, to compensate for their late attendance, they work overtime and clocks out at 05:00 PM. Because the worker came in late and compensated by working longer, the company doesn't want to grant the worker overtime pay for the hours that they worked between the planned clock-out at 03:30 PM and their actual clock-out at 05:00 PM, even though this period is defined as overtime in the time profile.

To handle this scenario, the absence code can be set up to reduce overtime hours by any hours of illegal absence that the worker has on the same day. Select **Time and attendance** > **Setup** > **Groups** > **Absence groups**, and select the **Deduct overtime** check box to deduct overtime from hours of illegal absence.

Here is how the worker's registrations for the day appear on the **Approve** page after calculation.

Expand table

Journal registration type	Start	End	Time
Absence (illegal - late for work)	07:00 AM	09:30 AM	1.5
Clock in	09:30 AM	09:30 AM	
Production job	09:30 AM	05:00 PM	7.5
Clock out	05:30 PM	05:30 PM	

If the **Deduct overtime** check box is selected for the selected absence code, the overtime payment is deducted by the hours that the worker was illegally absent. In this case, the following pay items are generated after the registrations are transferred.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	9.00	10
Overtime	1301	0.5	15

Here, the 1.5 hours of illegal absence, from 07:00 AM to 09:30 AM, deduct the 2.0 hours of overtime, from 03:30 PM to 05:30 PM. The result of the registration is 1.5 hours of standard time and 0.5 hours of overtime.

By contrast, if the **Deduct overtime** check box is cleared for the selected absence code, the overtime payment is granted to the worker, even though they were late and had an illegal absence. In this case, the following pay items are generated after the registrations are transferred.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	7.50	10
Overtime	1301	2.0	15

Scenario 5: The worker clocks out before the planned clock-out time and can convert the absence period to a Flex- period

The following example shows how a worker's flex account can be reduced by converting the absence period to a Flex- period.

The worker clocks in at 07:00 AM and clocks out at 01:00 PM. The worker has an agreement that they can go home for the weekend if they deduct these hours from their flex account. When the worker clocks out at 01:00 PM, they are prompted to select an absence code, because the period of absence for the remaining part of the workday that is affected isn't in a planned Flex- period. To convert the remaining part of the workday to a Flex- period, the worker can select an absence code that is set up to reduce their flex account.

To reduce the balance of flexible hours for workers who register absence on a workday, select **Time and attendance** > **Setup** > **Groups** > **Absence groups**, and select the **Reduce flex** check box.

Here is how the worker's registrations for the day appear on the **Approve** page before calculation.

Expand table

Journal registration type	Start	End	Time
Clock in	07:00 AM	07:00 AM	
Production job	07:00 AM	01:00 PM	6.0
Clock out	01:00 PM	01:00 PM	

If the worker selects an absence code for illegal absence, here is how the resulting pay item will look after the registration is transferred.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	6.00	10

If the worker selects an absence code for legal absence, and the absence code is set up to reduce their flex account, here is how the resulting pay items will look after the registrations are transferred.

Expand table

Wage type	Pay type	Pay units	Rate
Standard time	1201	8.50	10

In this case, the worker's flex balance is reduced by the hours between the actual clock-out time and the planned clock-out time (that is, the 2.5 hours from 01:00 PM to 03:30 PM).

> **Note**
>
> We don't recommend that you select both the **Deduct flex** check box and the **Deduct overtime** check box for an absence code, because this setup will deduct the illegal hours from the worker's overtime hours and at the same time reduce the worker's flex account.

Scenario 6: There is no planned absence for the day and no worker attendance for the day

If the worker doesn't show up for work on a workday, and there is no planned absence for the worker on that day, a default absence code is used for the calculation of the worker's registrations. To define default absence codes, select **Time and attendance** > **Time and attendance parameters**. You can then select an absence code in the following fields:

- Auto insert Flex-
- Auto insert absence

When the daily registrations are calculated for a worker who is enabled for flexible hours, the absence code that is specified in the **Auto insert flex-** field is used as a default absence code. If the worker isn't enabled for flexible hours, the absence code that is specified in the **Auto insert absence** field is used. If a company has a combination of workers who are enabled for flexible hours and workers who aren't enabled for flexible hours, both parameters must be set up.

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Absence registration in Time and attendance

This article describes the concepts for absence and explains how to handle absence in Time and attendance.

Absence that is based on regular work hours

Workers are considered absent for any hours that they don't work during their regular work hours. Regular work hours are defined in a worker's standard time profile.

For example, a worker is working on a day profile that has clock-in at 7:00 AM and clock-out at 3:00 PM. If the worker clocks in at 9:00 AM, they are considered absent from 7:00 AM to 9:00 AM on that day.

In this case, workers are prompted to enter a reason for their absence. They can specify a reason by selecting an absence code.

Absence codes

Absence codes define the various types of absence. Absence codes are defined by the company.

Various rules can be applied to absence codes. For example, an absence code can be configured to deduct or grant pay.

For example, a company defines a **Late** absence code that workers use if they come in late and don't have a good reason. The company also defines an **Internal course** absence code that workers use for time that they spend attending internal courses. These absence codes can be set up so that **Late** deducts from a worker's pay but **Internal course** doesn't deduct from a worker's pay.

You can set up automatic absence codes. These absence codes can be used to calculate a worker's time when no absence is registered. The worker's time profile determines whether the absence code for standard time or flex time is used.

Set up standard time and flex time

- Configure the parameters for standard time and flex time by using the **Auto insert absence** and **Auto insert Flex-** options on the **Time and attendance parameters** page.

Absence groups

Absence codes are grouped into absence groups. You use absence groups to group absence codes that have common characteristics. Examples include absence codes for a legal absence, or absence because of a doctor's appointment, jury duty, or a sick child.

Planned absence

If you know that a worker will be absent for a period, such as an upcoming vacation, you can use planned absence. You set up planned absence by configuring the absence code so that it considers the planned absence. When you set up a planned absence, you aren't prompted for an absence code during the absence period when user time registrations are calculated. Planned absence can be defined for a single worker, or you can define a batch job to bulk update the planned absence for workers.

Set up planned absence

1. Select **Human resources** > **Workers** > **Employees**, and select an employee.
2. Select **Time** > **Time assignments** > **Time Absence registration**, and set up the planned absence.

Interrupted planned absence

If you apply the **Interrupt** option when you set up a planned absence, the planned absence will be interrupted if the worker signs in during the planned absence period. The planned absence will be marked as **Interrupted** and won't have any effect on future calculations.

Set up a planned absence for interruption

1. Open the **Time Absence registration** page as described in the procedure for setting up planned absence.
2. Select **Interrupt**.

The **Interrupt** option applies when time is registered through the shop floor terminal or the shop floor device. The option doesn't apply if the registrations are entered on the calculation and approval pages, or on the **Electronic timecard** page.

Examples of the use of absence in a flex profile

The following three examples show how absence is calculated in a profile that has flex periods.

The examples use the following profile.

Expand table

Clock-in	Standard time	Break	Standard time	Flex-	Clock-out	Flex+
8 AM	9 AM to 11:30 AM	11:30 AM to 12 PM	12 PM to 3 PM	3 PM to 4 PM	4 PM	4 PM to 6 PM

Example 1: Signing out during a Flex- period

The worker clocks in at 8:00 AM and clocks out at 3:30 PM. In this case, because the worker clocks out during a Flex- period, no absence is calculated, and half an hour is deducted from the worker's flex balance.

Example 2: Signing out in during Standard time period

The worker clocks in at 8:00 AM and clocks out at 2:30 PM. In this case, because the worker clocks out during the Standard time period, absence is calculated from 2:30 PM to 4 PM, and an absence period of 1.5 hours is registered. An absence code for that period is required.

Example 3: Signing out during a Flex+ period

The worker clocks in at 8:00 AM and clocks out at 4:30 PM. In this case, because the worker clocks out during a Flex+ period, no absence is calculated, and half an hour is added to the worker's flex balance.

Absence in the calculation and approval process

Worker time registrations must be calculated and approved before they can be transferred to a payroll system as pay items.

An approver can change a worker's time registrations. The approver can even change any absence that the worker has registered. If the approver manually enters a time period that has an absence code, the absence code for that period isn't overridden by the default absence code from the Time and attendance parameters.

For example, a worker clocks in at 10 AM and selects an absence code that indicates that they are late. Later, the worker informs their supervisor that they had a doctor's appointment from 8 AM to 10 AM. A doctor's appointment should not cause a deduction in the worker's pay. Therefore, in this case, the supervisor can adjust the two hours of absence from 8 AM to 10 AM by manually entering an absence code that indicates illness for those two hours.

Calculate and approve absence

- Select **Time attendance** > **Review and approve** > **Approve or Calculate**.

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Flex groups

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Flexible working hours let companies minimize payments for overtime by offering workers extra time off during periods when the workload is low. This feature is relevant, for example, in segments that experience seasonal changes in workload.

You can use flex groups to set the following rules and principles for a worker's flexible hours:

- Rules for flex regulations
- Principle for calculating the worker's flex balance

Set up flexible working hours in flex groups

- Select **Time and attendance** > **Setup** > **Groups** > **Flex groups** to set up flex groups for flexible hours.

Associate workers with flex groups

- Select **Time and attendance** > **Setup** > **Time registration workers**.

Rules for flex regulations

You can use rules for flex regulations to define flex limits, or the minimum and maximum number of hours that are allowed in the worker's flex account. The flex limits are set up on the flex group. When the flex limits are exceeded, a worker's flex balance and pay can be adjusted.

If a worker's allowed flex minimum is exceeded (that is, if the number of hours in the flex account is below the specified minimum), you can use these methods to adjust the worker's flex balance by making a flex regulation:

- The worker's flex account can be adjusted to the specified allowed minimum, but without deducting the worker's pay for the number of hours that the flex account is below the allowed minimum.
- The worker's pay can be deducted for the number of hours that the flex account is below the allowed minimum. This deduction is done by generating pay items for a specific pay type that have a negative or positive pay unit.

If the worker's allowed flex maximum is exceeded, you can use these methods to adjust the worker's flex balance by making a flex regulation:

- The worker's flex account can be adjusted back to the specified allowed maximum, but without compensating the worker's pay for the number of hours that the worker worked above the allowed maximum.
- The number of hours that the worker worked above the allowed maximum can be converted to pay. This conversion is done by generating pay items for a specific pay type.

You can adjust a flex balance at the following times:

- When a payment file that is based on payroll data is exported by using the **Transfer pay** job. The payroll data is generated when you transfer the worker's registration from the **Approve** page.
- When the **Adjust flex balance** job is processed.

> ⓘ **Note**
>
> Flex regulations don't occur during the daily approval and transfer of worker registrations on the **Approve** page.

Principle for calculating a worker's flex balance

The principle for calculating the hours that the worker's flex balance are adjusted by is set up on the flex group. Select **Time and attendance** > **Setup** > **Groups** > **Flex groups**.

The following two principles can be used:

- **Time** – The worker's flexible hours are calculated only from the worker's registered time for the day. When the worker's daily registrations are calculated, the number of Flex+ and Flex- hours for the day is calculated from the Flex+ and Flex- zones that are defined in the worker's time profile.
- **Pay types** – The worker's flexible hours are calculated based on earnings of the pay types for Flex+ and Flex- that are defined in the worker's pay agreement. The pay agreement is associated with the time registration worker. You might want to use pay types to manage flex accounts if, for example, you want to increase a worker's flex account by a specific factor in one or more flex zones.

Scenario 1: Adjusting a worker's pay and flex account because the allowed flex minimum is exceeded

A worker who can work flexible hours has a negative flex account.

- **Flex account:** -4

The worker is associated with a flex group that has the following configuration:

- **Flex minimum:** -0.5
- **Minimum pay type:** 1302
- **Pay type factor:** -1.00

As the difference between the worker's flex account and the allowed flex minimum indicates, the worker has exceeded the allowed flex minimum by 3.5 hours.

When the payroll administrator transfers the worker's pay data by running the **Transfer to pay** or **Flex adjustment** job, the following adjustments are made:

- The worker's flex account is adjusted by 3.5 hours. Therefore, the flex balance of -4.0 hours becomes adjusted to the worker's allowed flex minimum of -0.5 hours.
- A pay item for pay type 1302 is created. This pay item has a pay unit of -3.5 hours that will be deducted from the worker's pay. In this case, the pay unit is a negative number, because the positive adjustment of 3.5 hours is multiplied by the negative pay type factor of -1.0 that is defined on the flex group. This pay item will be part of the pay file that is generated by the **Transfer to pay** job.

Scenario 2: Adjusting a worker's pay and flex account because the allowed flex maximum is exceeded

A worker who can work flexible hours has a positive flex account.

- **Flex account:** 6

The worker is associated with a flex group that has the following configuration:

- **Flex maximum:** 2.0
- **Minimum pay type:** 1302
- **Pay type factor:** -1.0

As the difference between the worker's flex account and their allowed flex maximum indicates, the worker has exceeded their allowed flex maximum by 4.0 hours.

When the payroll administrator transfers the worker's pay data by running the **Transfer to pay** or **Flex adjustment** job, the following adjustments are made:

- The worker's flex account is adjusted by -4.0 hours. Therefore, the flex balance of 6.0 hours becomes adjusted to the worker's allowed flex maximum of 2.0 hours.

- A pay item for pay type 1302 is created. This pay item has a pay unit of 4.0 hours that will be added to the worker's pay. In this case, the pay unit is a positive number, because the negative adjustment of 4.0 hours is multiplied by the negative pay type factor of -1.0 that is defined on the flex group. This pay item will be part of the pay file that is generated by the **Transfer to pay** job.

Scenario 3: Managing a worker's flex balance based on pay types

As we explained earlier, flex accounts can be managed based either on the time that is registered in the Flex+ and Flex- zones that are defined the worker's time profile, or on the pay types that are defined in the worker's pay agreements. If pay types are used, a worker's flex account is adjusted by the pay items that are generated when you transfer the worker's registration from the **Approve** page. You might want to use pay types to manage flex accounts if, for example, you want to increase a worker's flex account by a specific factor in one or more flex zones.

This scenario uses the following flex profile that represents a workday.

Expand table

Profile type	Start	End
Flex+	12:00 AM	08:00 AM
Clock in	08:00 AM	08:00 AM
Flex-	08:00 AM	09:00 AM
Standard time	09:00 AM	11:30 AM
Paid break	11:30 AM	12:00 PM
Flex-	12:00 PM	04:00 PM
Clock out	04:00 PM	04:00 PM
Flex+	04:00 PM	12:00 AM

In this case, you want to be able to manage the worker's flex balance based on pay types. Therefore, you must set the **Based on pay types** option to **Yes** on the worker's flex group.

To account for the flexible hours, you must also define a new pay type. For this scenario, the pay type is named **FlexCnt**.

Expand table

Pay type	Description
FlexCnt	Flex counter

Next, follow these steps to set up a pay type and add lines of the new type to a pay profile.

1. Select **Time and attendance** > **Setup** > **Groups** > **Flex groups**, and then select **New**.
2. In both the **Flex+** field and the **Flex-** field, specify the new pay type, **FlexCnt**.
3. Select **Time and attendance** > **Setup** > **Pay agreements**, and then select **Agreement lines**.
4. For **Monday**, for the **Flex+** profile type, add the following three lines.

Expand table

Pay type	Description	From time	To time	Minimum	Maximum	Factor
FlexCnt	Flex counter	12:00 AM	06:00 PM	00.00	00.00	1.00
FlexCnt	Flex counter	06:00 PM	12:00 AM	00.00	02.00	1.50
FlexCnt	Flex counter	06:00 PM	12:00 AM	02.00	06.00	2.00

> **Note**
>
> Each line is used for a different time interval and has a different factor. The flexible hours that the worker works in a time interval are multiplied by the factor for that line. For example, hours that are worked from 06:00 PM to 08:00 PM are multiplied by 1.50. The factor is specified in the **Factor** field on the **General** tab of the **Pay agreement lines** page.

The worker enters the following registrations for the day.

Expand table

Journal registration type	Start	End
Clock in	07:00 AM	07:00 AM

Journal registration type	Start	End
Production job	07:00 AM	09:00 PM
Clock out	09:00 PM	09:00 PM

The amount that must be paid is calculated on the **Approve** page, based on the worker's registration. After the registration is calculated, you can view the result on the **Times** tab. For this scenario, you're interested in the following fields.

Expand table

Flex +	Flex -	Time	Pay time
6.00	0.00	13.50	08.00

The amount of Flex+ time is six hours, and the calculation is based on the flex zones in the time profile. This amount consists of one hour of Flex+ time from 07:00 AM to 08:00 AM and five hours of Flex+ time from 04:00 PM to 09:00 PM.

When you transfer the registrations, you will notice that the amount of Flex+ time is changed from 6.0 hours to 8.0 hours.

Expand table

Flex +	Flex -	Time	Pay time
8.00	0.00	13.50	08.00

This change occurs after the transfer because the flexible hours have been calculated based on pay types instead of time. The following table shows how the eight hours are calculated.

Expand table

From	To	Time	Factor	Flex account
07:00 AM	08:00 AM	1	1	1
04:00 PM	06:00 PM	2	1	2
06:00 PM	08:00 PM	2	1.5	3
08:00 PM	09:00 PM	1	2	2
			Total	**8**

Feedback

Was this page helpful? Yes No

Provide product feedback

Enable the payroll process for time and attendance

Article • 09/29/2021

This procedure shows how to enable the payroll process for time and attendance. The demo data company used to create this procedure is USMF.

Create a pay type with a related pay rate

1. Time and attendance > Setup > Payroll > Pay types
2. Click New.
3. In the Pay type field, type a value.
4. In the Description field, type a value.
5. Click Save.
6. Click Rates.

    - Rates for pay types are set up for specific time intervals, and individual rates can be created for workers. It is not always necessary to create rates for pay types in time and attendance. This information may already exist in the payroll system that is used to generate wages.

7. Click New.
8. In the list, mark the selected row.
9. In the Rate field, enter a number.
10. Click Save.

Create a pay agreement

1. Close the page.
2. Close the page.
3. Go to Pay agreements.

    - Time and attendance > Setup > Pay agreements

4. Click New.
5. In the Pay agreement field, type a value.
6. In the Description field, type a value.
7. Click Save.
8. Click Agreement lines.
9. Click New.

10. In the list, mark the selected row.
11. In the Profile type field, enter or select a value.
12. In the Pay type field, enter or select a value.

Set up pay agreement for time and registration worker

1. Close the page.
2. Close the page.
3. Go to Time registration workers.

    - Time and attendance > Setup > Time registration workers

4. In the list, click the link in the selected row.
5. Click the Employment tab.
6. Expand the Time registration section.
7. Click Edit.
8. In the Pay agreement field, enter or select a value.

Feedback

Was this page helpful? Yes No

Provide product feedback

How workers use the production floor execution interface

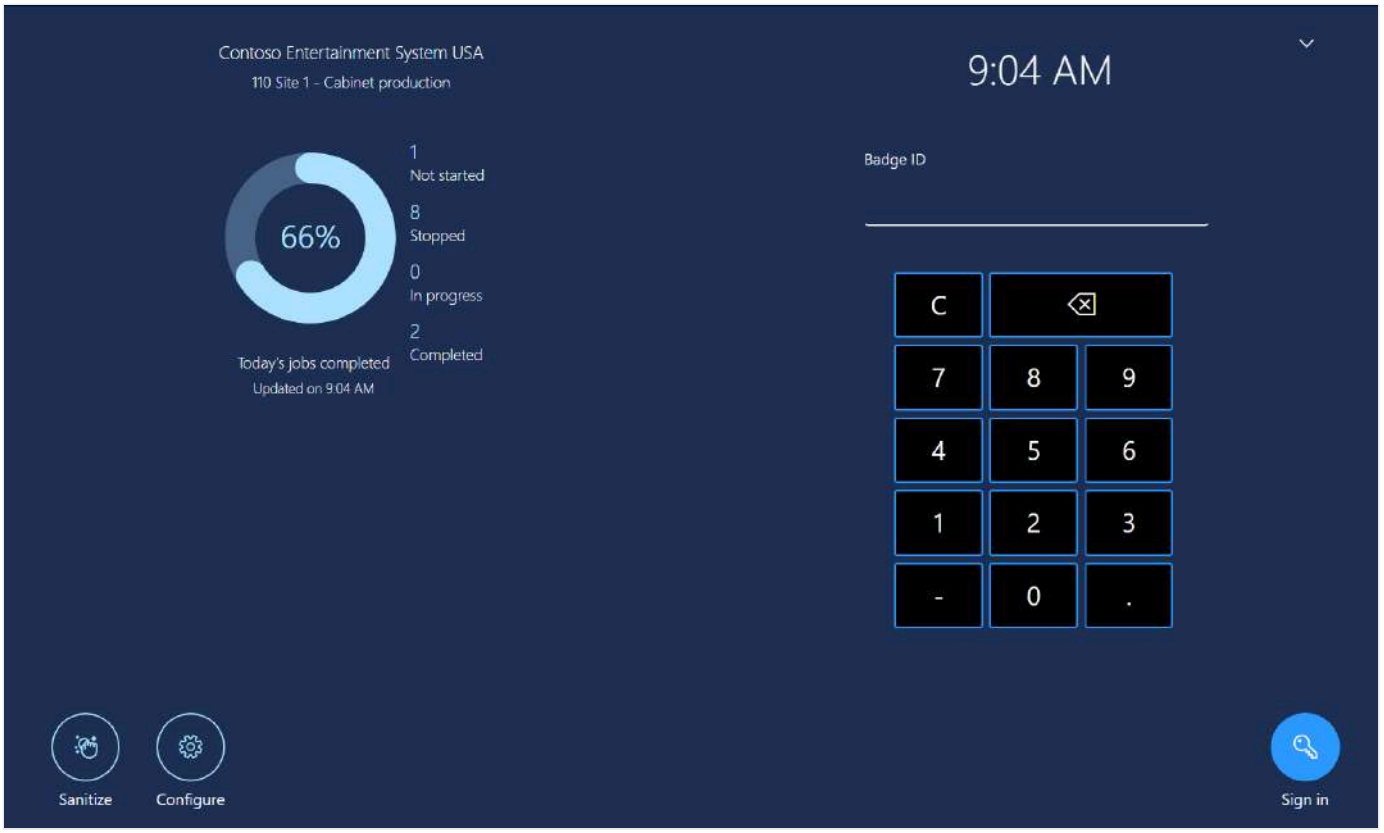
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The production floor execution interface is optimized for touch interaction. Its design provides visual contrast that meets accessibility requirements for shop floor environments. However, it also enables multiple jobs to be started in parallel from a job list. This capability is also known as *job bundling*. Additionally, from a job list, workers can open a guide that was created in Microsoft Dynamics 365 Guide. In this way, they can get visual instructions on a HoloLens.

Sign in to the production floor execution interface as a worker

Before workers can start to use the device, a supervisor or technical staff must prepare it and open the correct page in Dynamics 365 Supply Chain Management. For more information about how to set up the device, see Set up a device to run the production floor execution interface.

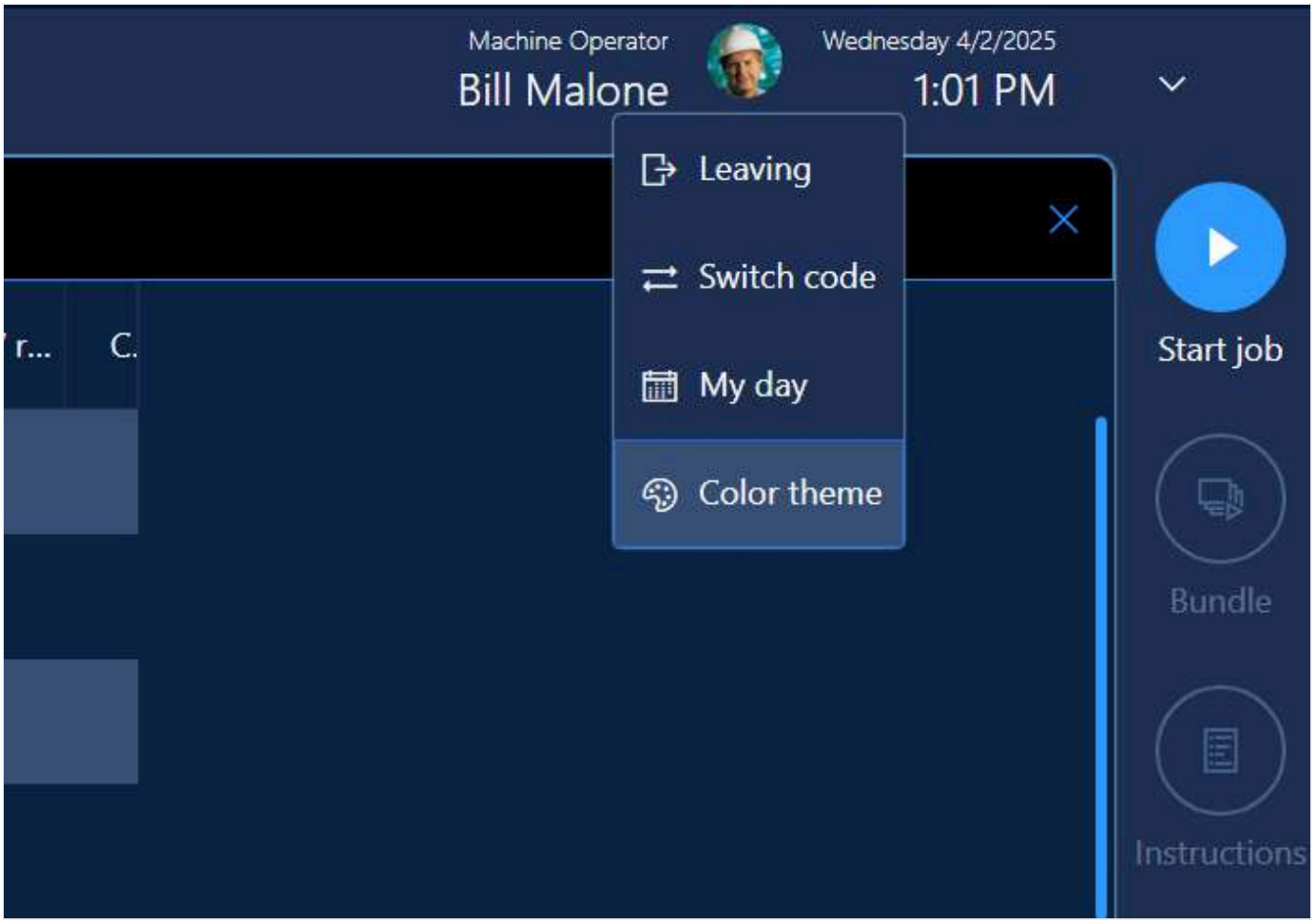
After the device is prepared, the sign-in page appears on it. This page shows information about the status of jobs for the local work cell. This information is updated periodically. On the page, workers use their badge IDs, personnel number, and/or password to sign in. Although workers don't need a user account for Supply Chain Management, they must have a *time registered worker* account that they can use when they sign in. For more information about how to set up sign-in options and worker accounts to use the production floor execution interface, see Set up worker accounts to use the production floor execution interface.



The remaining sections of this article describe how workers interact with the interface.

Choose a display color theme

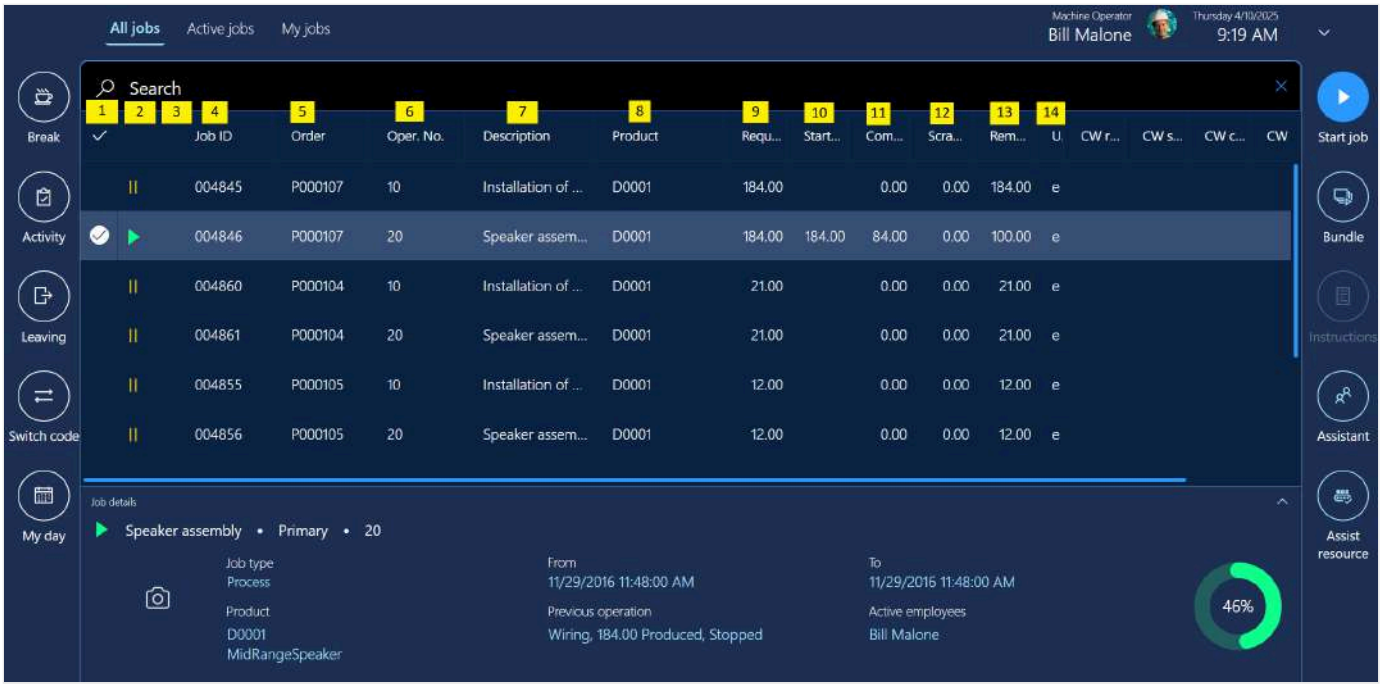
Individual workers can choose their preferred theme by selecting their worker photo (or icon) and then selecting **Color theme** from the dropdown list.



A dialog opens where the worker can select a new theme and preview the colors. The theme is applied immediately. When the worker selects the theme, the system saves it in user preferences. The theme is applied the next time the worker signs in, regardless of the current device's settings.

All jobs tab

The **All jobs** tab provides a job list that shows all the production jobs that have a status of *Not started*, *Stopped*, or *Started*. (You can customize this tab name, so it might be different for your system.)



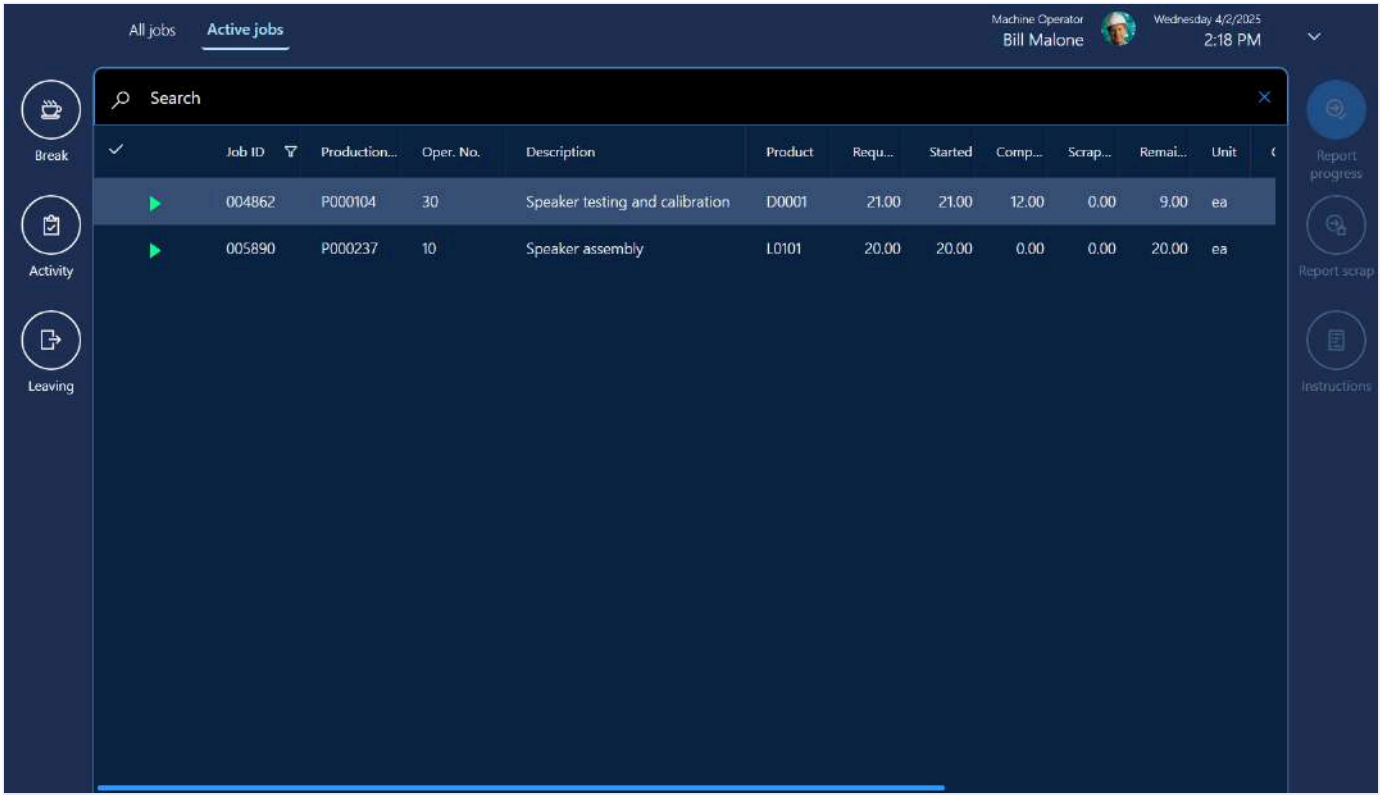
The job list has the following columns. The numbers correspond to the numbers in the previous illustration.

1. **Selection column** – The leftmost column uses check marks to indicate jobs selected by the worker. Workers can select multiple jobs in the list at the same time. To select all the jobs in the list, select the check mark in the column header. When a single job is selected, details about that job appear in the lower part of the page.
2. **Job status column** – This column uses symbols to indicate the status of each job. Jobs that have no symbol in this column have a status of *Not started*. A green triangle indicates jobs that have a status of *Started*. Two yellow vertical lines indicate jobs that have a status of *Stopped*.
3. **High priority column** – This column uses exclamation marks to indicate jobs that have high priority.
4. **Job ID** – This column shows the unique job ID for a job.
5. **Order** – This column shows the production order number for a job.
6. **Oper. No.** – This column shows the operation number of the operation that a job is part of.
7. **Description** – This column shows a description of the operation that a job is part of.
8. **Product** – This column shows the product number of the item that a job produces.
9. **Requested** – This column shows the quantity that a job is planned to produce.
10. **Started** – This column shows the quantity that's already started for a job.
11. **Completed** – This column shows the quantity that's already completed for a job.
12. **Scrapped** – This column shows the quantity that's already scrapped for a job.
13. **Remaining** – This column shows the quantity that remains to be completed for a job.
14. **Unit** – This column shows the unit of measure of quantity for a job.

If the catch weight configuration key is enabled for your system, you might see additional columns to support the catch weight items.

Active jobs tab

The **Active jobs** tab shows a list of all jobs that the signed-in worker already started. (You can customize this tab name, so it might be different for your system.)



The **Active jobs** list shows the same columns as the **All jobs** list.

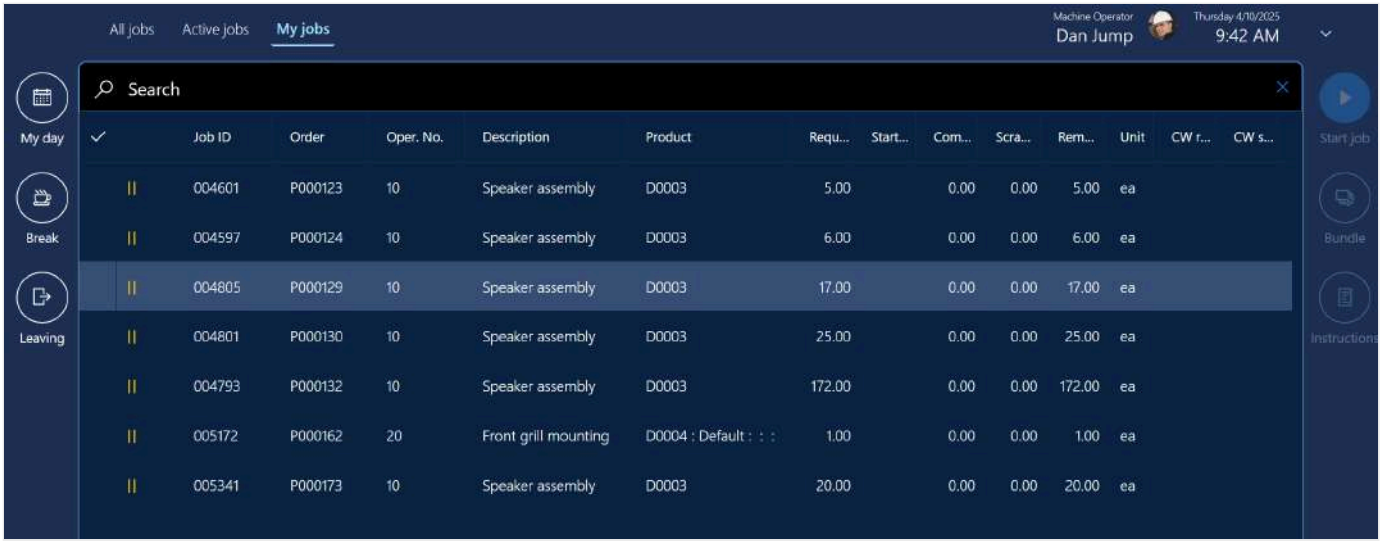
My jobs tab

The **My jobs** tab lets workers easily view all unstarted and unfinished jobs that are assigned specifically to them. It's useful in companies where jobs are sometimes or always assigned to specific workers (human resources) instead of other types of resources (such as machines).

The scheduling system automatically assigns each production job to a specific resource record, and each resource record has a type (such as machine or human). When you set up an employee as a production worker, you can associate the worker account with a unique human resource record.

The **My jobs** tab lists all unstarted and unfinished jobs assigned to signed-in worker's human resource record, if any worker is signed in. It never lists jobs that are assigned to a machine or other type of resource, even if the signed-in worker started to work on those jobs.

To view all jobs that the signed-in worker started, regardless of the type of resource that each job is assigned to, use the **Active jobs** tab. To view all unfinished jobs that match the configuration of the local job filter, regardless of the worker or start status, use the **All jobs** tab.

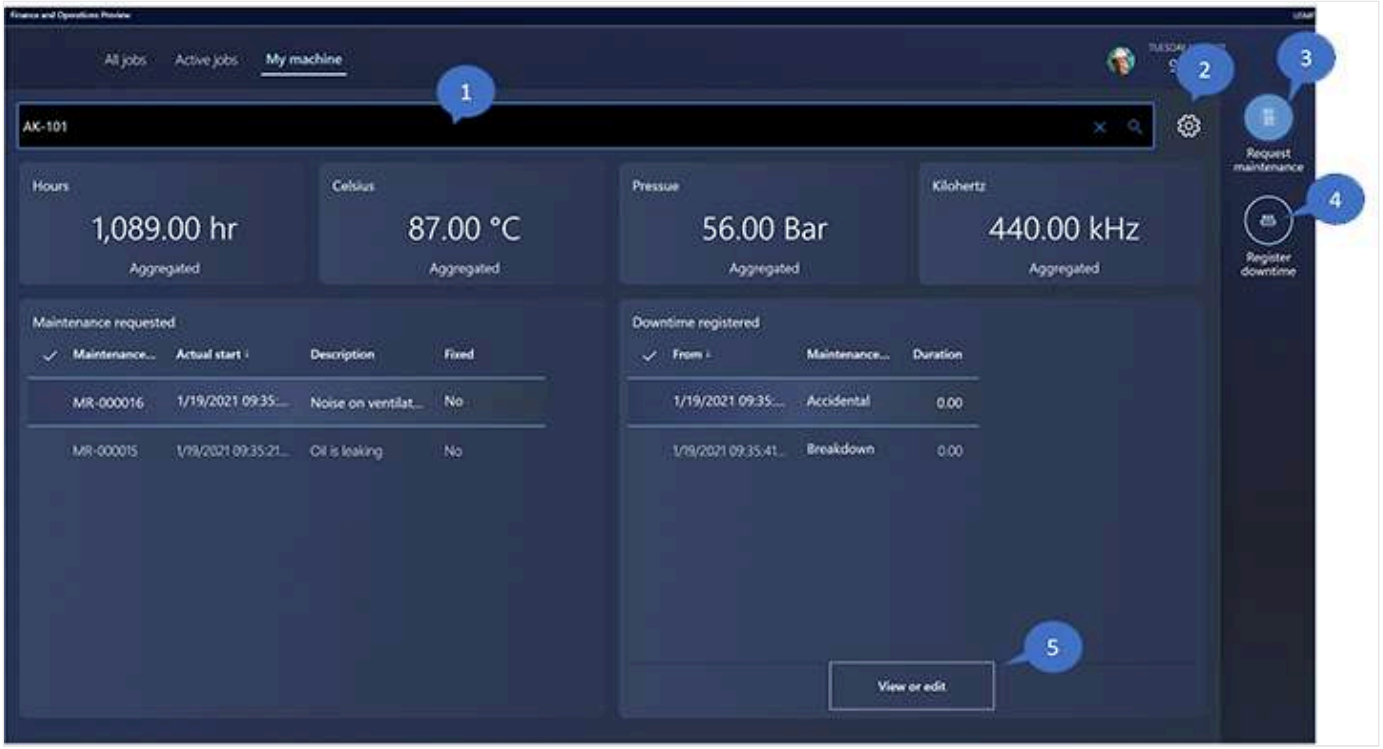


Project tabs

The project tabs **All project jobs**, **Projects**, **Project budget**, and **Project activities** allow workers to register the time spent on production-related activities. Workers can select a project-related activity from the list and log its start. The activity is marked as completed when the worker registers another activity, such as an indirect task or another project activity. The time registered by the worker is recorded under the projects associated with these activities.

My machine tab

The **My machine** tab lets workers select an asset connected to a machine resource within the filter set on the **All jobs** tab. The worker can then view the state and health of the selected asset by reading values for up to four selected counters and lists of recent maintenance requests and registered downtimes. The worker can also request maintenance for the selected asset and register and edit machine downtime. (You can customize this tab name and it might be different for your system.)

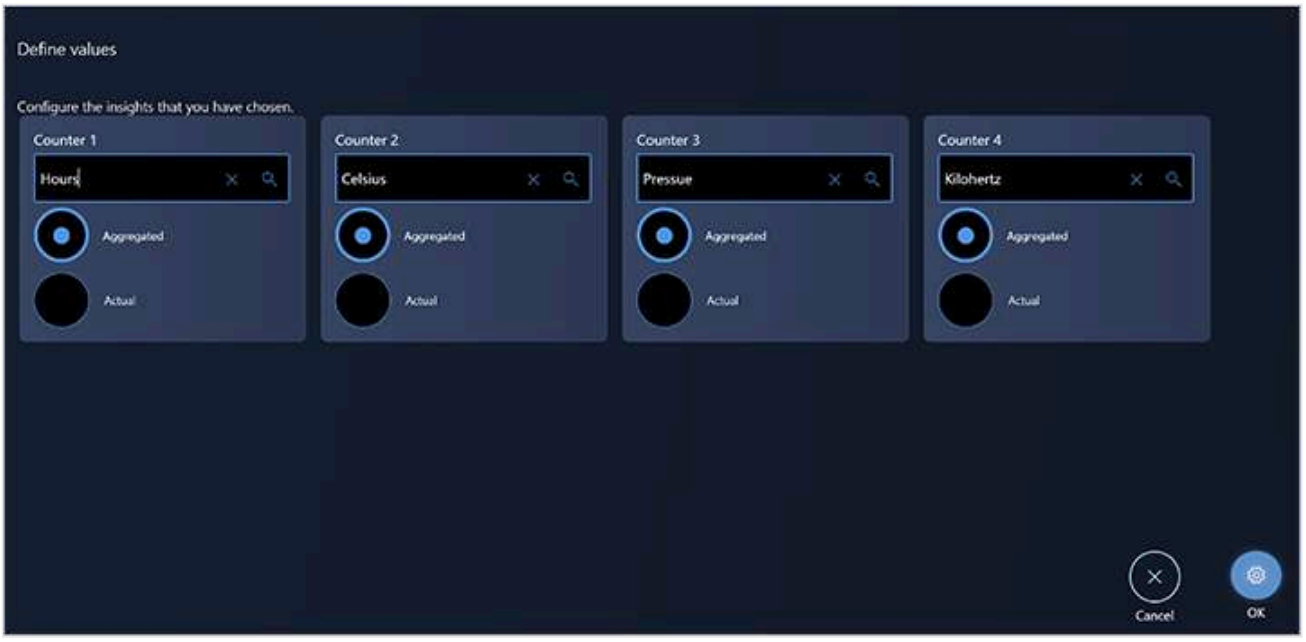


The **My machine** tab has the following columns. The numbers correspond to the numbers in the previous illustration.

1. **Machine asset** – Select the machine asset that you want to track. Start typing a name to select from a list of matching assets, or select the magnifying-glass icon to select from a list of all assets associated with the resources that are within the filter of the job list.

   > ⓘ **Note**
   >
   > Supply Chain Management users can assign a resource to each asset as needed by using the **All assets** page (on the **Fixed asset** tab, by using the **Resource** drop-down list). For more information, see Create an asset.

2. **Settings** – Select the gear icon to open a dialog box where you can choose which counters to view for the selected machine asset. Values for these counters show at the top of the **Asset management** tab. The **Settings** menu (shown in the following screenshot) lets you enable up to four counters. For each counter that you want to enable, use the lookup field at the top of the tile to select a counter. The lookup field lists all of the counters associated to the asset selected at the top of the **Asset management** page. Set each counter to monitor either the **Aggregated** value or the latest **Actual** value for the counter. For example, if you set a counter that tracks how many hours the machine has been running, set it to **Aggregated**. If you set a counter to measure the latest updated temperature or pressure, set it to **Actual**. Select **OK** to save your settings and close the dialog box.



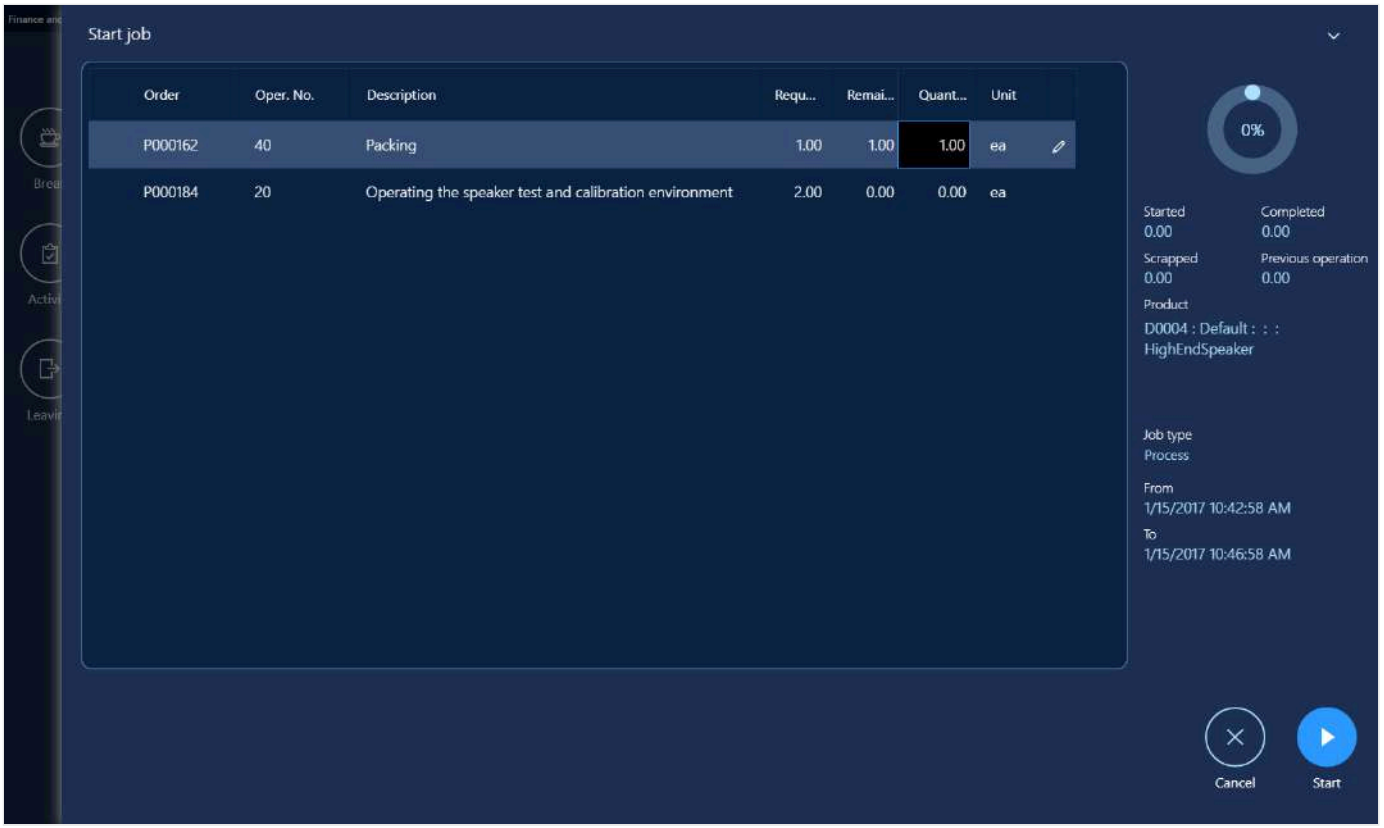
3. **Request maintenance** – Select this button to open a dialog box where you can create a maintenance request. You can provide a description and a note. A Supply Chain Management user sees the request and can convert the maintenance request to a maintenance work order.

4. **Register downtime** – Select this button to open a dialog box where you can register machine downtime. You can select a reason code and enter a date and time span for the downtime. The machine downtime registration is used for calculating the efficiency of the machine asset.

5. **View or edit** – Select this button to open a dialog box where you can edit or view existing downtime records.

Starting and completing production jobs

Workers start a production job by selecting a job on the **All jobs** tab and then selecting **Start job** to open the **Start job** dialog box.

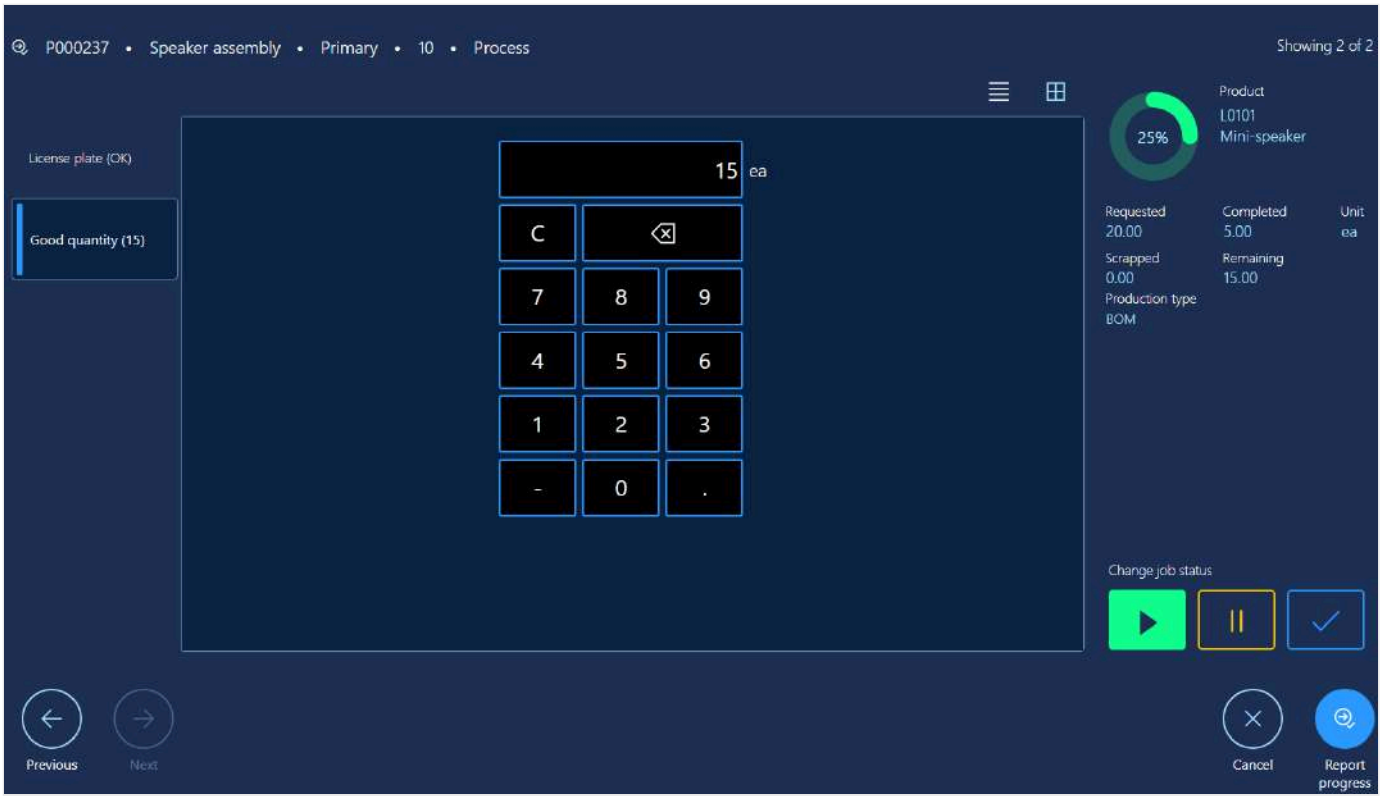


Workers use the **Start job** dialog box to confirm the production quantity and then start the job. Workers can adjust the quantity by selecting the **Quantity** field and then using the numeric keyboard that appears. Workers then select **Start** to start to work on the job. The **Start job** dialog box is closed, and the job is added to the **Active jobs** tab.

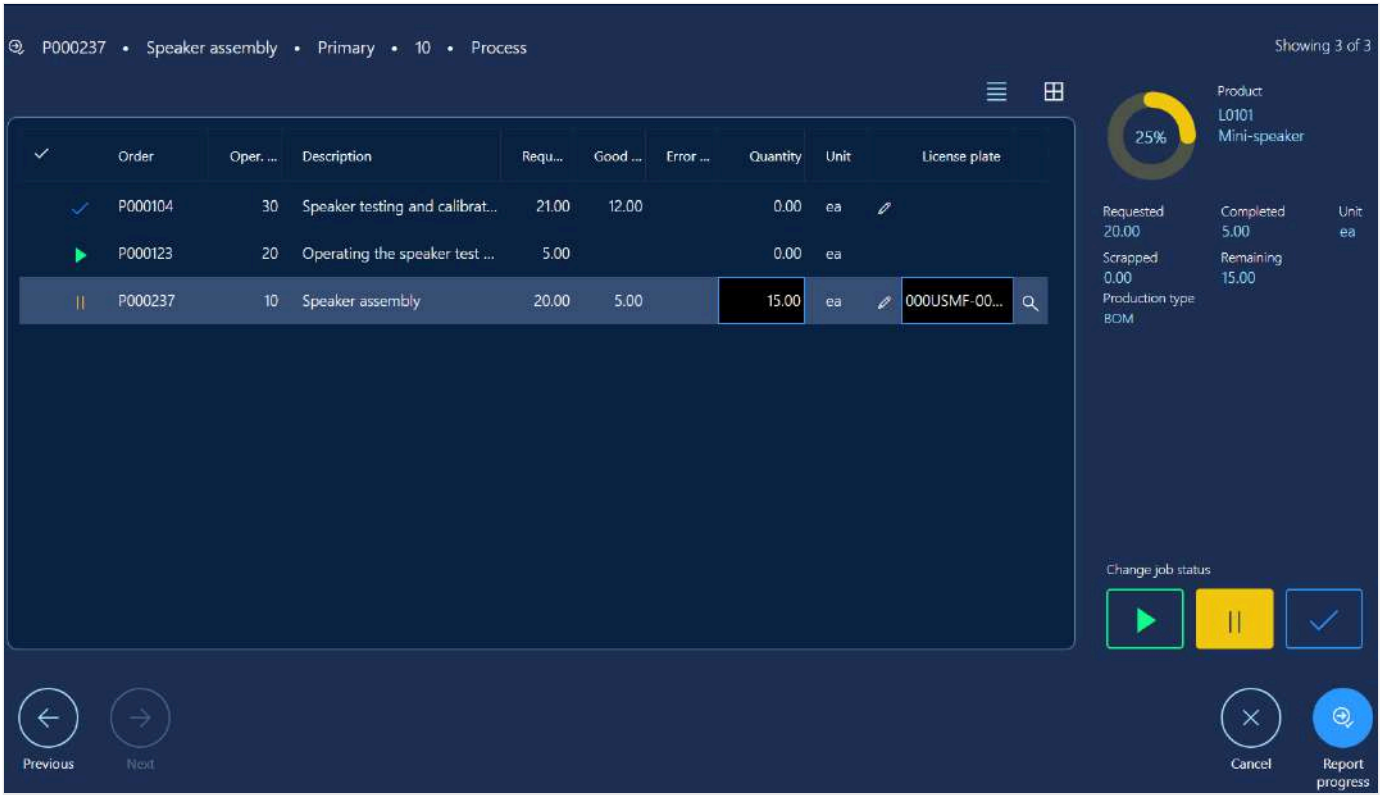
Workers can start a job that is in any status. When a worker starts a job that has a status of *Not started*, the **Quantity** field in the **Start job** dialog box initially shows the full quantity. When a worker starts a job that has a status of *Started* or *Stopped*, the **Quantity** field initially shows the remaining quantity.

Reporting good quantities

When a worker completes or partially completes a job, they can report good quantities that were produced by selecting a job on the **Active jobs** tab and then selecting **Report progress**. Then, in the **Report progress** dialog box, the worker enters the good quantity by using the numeric keyboard. The quantity is blank by default. After a quantity is entered, the worker can update the status of the job to *In progress*, *Stopped*, or *Completed*.



Workers reporting job progress can choose to work from a list view that shows several jobs at once, or from a detail view that shows more information about one job at a time.



Reporting good quantities on batch orders that have co-products and by-products

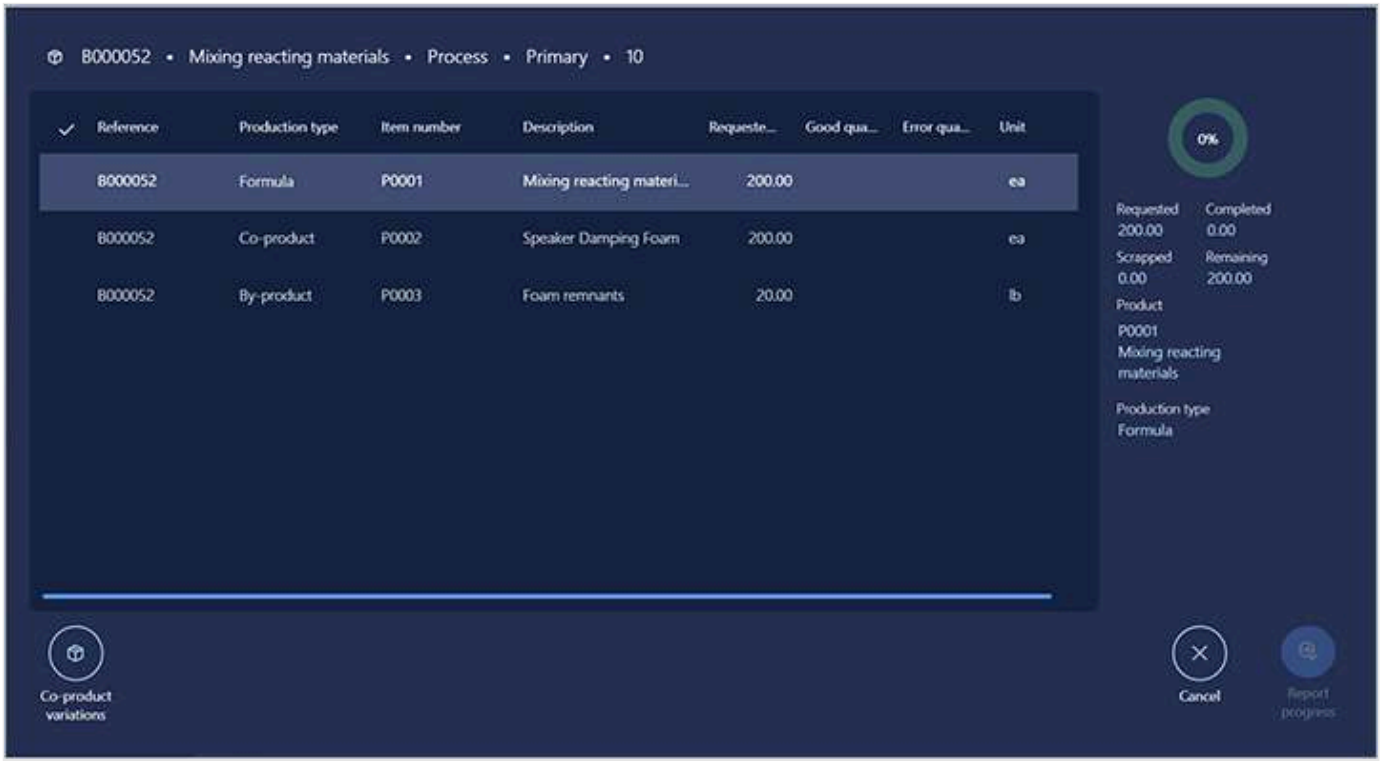
Workers use the production floor execution interface to report progress on batch orders. This reporting includes co-products and by-products.

Some manufacturers, especially in process industries, use batch orders to manage their production processes. Batch orders are created from formulas, and those formulas can be defined so that they have co-products and by-products as output. When feedback about those batch orders is reported, the amount of output must be registered on the formula item, and also on the co-products and by-products.

When a worker completes or partially completes a job on a batch order, they can report good or scrap quantities for each product that is defined as output for the order. Products that are defined as output for a batch order can be of the *Formula*, *Co-product*, or *By-product* type.

To report good quantities on the products, a worker selects a job on the **Active jobs** tab and then selects **Report progress**.

Then, in the **Report progress** dialog box, the worker can select among the products that are defined as output for the batch order to report on. The worker can select one or many products in the list, and then select **Report progress**. For each product, the quantity is blank by default, and the worker can use the numeric keyboard to enter the quantity. The worker can use the **Previous** and **Next** buttons to move between the selected products. After the quantity is entered for each product, the worker can update the status of the job to *In progress*, *Stopped*, or *Completed*.



Reporting on batch orders for planning items

When a worker completes a job on a batch order for a planning item, they report quantities only on co-products and by-products, because planning items don't contain an item of the *Formula* type.

Reporting co-product variation

If a batch order is created from a formula version where the **Co-products variations** option is set to *Yes*, the worker can report on co-products that aren't part of the definition for the batch orders. This functionality is used in scenarios where unexpected product output can occur in the production process.

In this case, the worker can specify the co-product and quantity to report by selecting **Co-products variations** in the report progress dialog box. The worker can then select among all the released products that are defined as co-products.

Reporting catch weight items

Workers can use the production floor execution interface to report progress on batch orders that are created for catch weight items. Batch orders are created from formulas, which can be defined to have catch weight items as formula items, co-products, and by-products. A formula can also be defined to have formula lines for ingredients that are defined for catch weight. Catch weight items use two units of measure to track inventory: catch weight quantity and inventory quantity. For example, in the food industry, boxed meat can be defined as a catch weight item, where the catch weight quantity is used to track the number of boxes and the inventory quantity is used to track the weight of the boxes.

Reporting scrap

When a worker completes or partially completes a job, they can report scrap by selecting a job on the **Active jobs** tab and then selecting **Report scrap**. In the **Report scrap** dialog box, the worker enters the scrap quantity by using the numeric keyboard. The worker also selects a reason (*None*, *Machine*, *Operator*, or *Material*).

Workers reporting job scrap can choose to work from a list view that shows several jobs at once, or from a detail view that shows more information about one job at a time.

Automatically report a job as finished by using sensor signals (preview)

[This section is prerelease documentation and is subject to change.]

You can fit production machines with sensors that report completed operations. By integrating these sensors with Supply Chain Management, you can set up scenarios where the system automatically reports finished quantities when a machine generates enough sensor signals to reach a defined output threshold. The scenario requires fitting the machine with a sensor that emits a signal for each production cycle, and mapping that sensor to the corresponding resource and operation in Supply Chain Management. By defining how many signals correspond to one finished unit, the system can automatically determine when progress should be reported and update the production order and inventory accordingly. If your plant is set up to support this scenario, you can also set up the production floor execution interface to provide real-time visibility into the sensor signals and counters, so workers can monitor the production process more effectively and adjust the counters as needed. For more information about how to set up this scenario and view the counters in the production floor execution interface, see Production auto report scenario (preview).

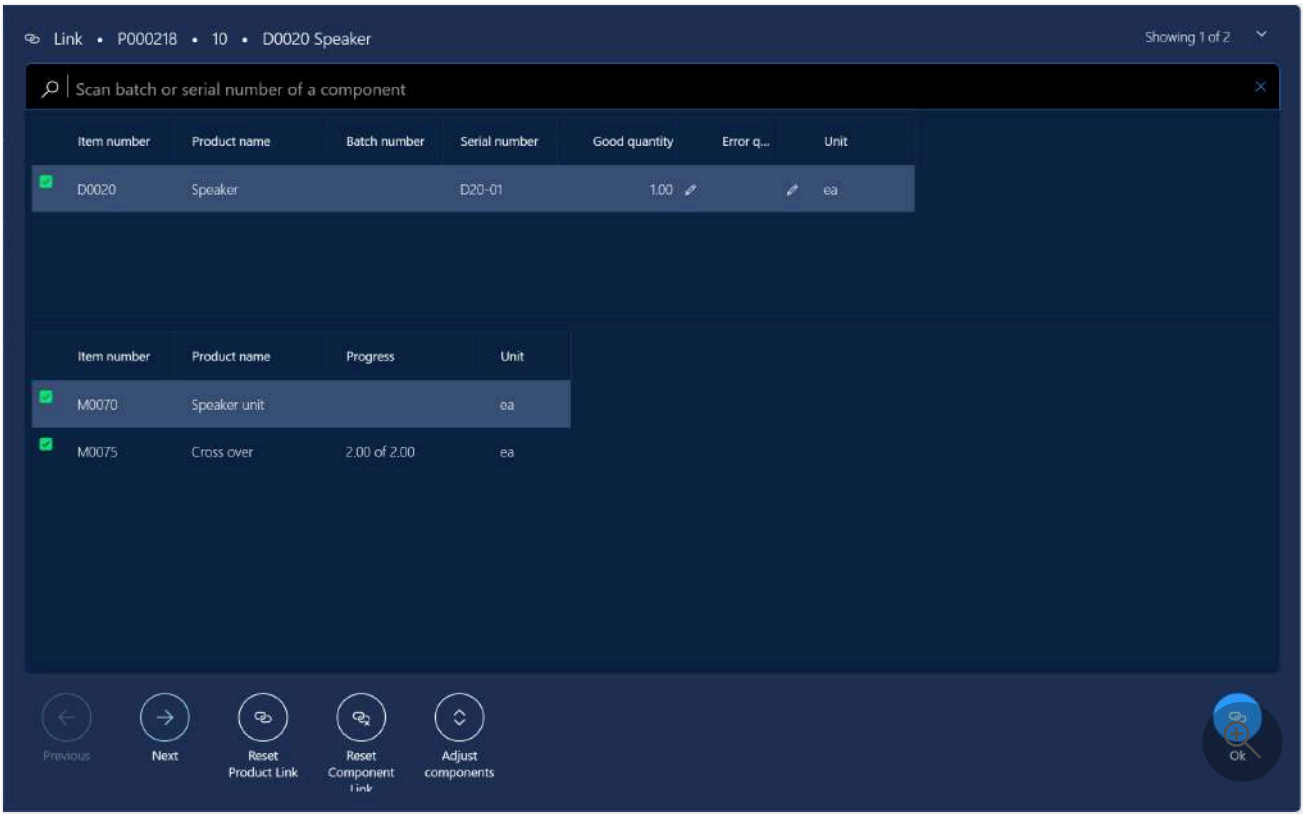
> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Register batch/serial numbers for finished products and their components

Workers can use the production floor execution interface to register batch or serial numbers for tracked components and the finished products that they're producing. For information about how to add this functionality to the interface, see Configure the production floor execution interface.

Workers follow these steps to register tracked components.

1. When you start a production job, select **Tracked component** to start registering batch or serial numbers for the product and its components.



2. When the interface prompts you, scan the batch or serial number of the product that's manufactured.

3. Select a component to add to the product, and then scan the batch/serial number of that component.

4. Continue to add and scan components until all of them are registered.

5. Review the on-screen information, and make any required adjustments to the registrations by using the buttons at the bottom of the page.

6. Select **Next**.

7. Continue to process items on the production order until all products are registered.

Adjust material consumption and make material reservations (preview)

[This section is prerelease documentation and is subject to change.]

> ⓘ **Note**

This feature is generally available (GA) for non-WMS items, but is still in preview for WMS-enabled items.

Workers can adjust material consumption for each production job. Use this functionality when the actual quantity of materials that a production job consumes is different from the planned quantity. Adjust the quantity to keep the inventory levels current.

Workers can also make reservations on the batch or serial numbers of materials. Use this functionality when a worker must manually specify which material batch or serial numbers were consumed, to meet material traceability requirements.

Workers can specify the quantity to adjust by selecting **Adjust material**. This button is available in the following locations:

- In the **Report scrap** dialog box
- In the **Report progress** dialog box
- On the toolbar on the right

For information about how to add this functionality to the interface, see Configure the production floor execution interface.

Important

- This is a preview feature.
- Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Adjust material consumption from the Report scrap and Report progress dialog boxes

After a worker enters the quantity to report in the **Report progress** or **Report scrap** dialog box, the **Adjust material** button becomes available. When the user selects this button, the **Adjust material** dialog box appears. This dialog box lists the items that are planned to be consumed when the good or scrapped quantity is reported for the job.

The list in the dialog box shows the following information:

- **Product number** – The product master and product variant.
- **Product name** – The name of the product.
- **Proposal** – The estimated quantity of material that will be consumed when progress or scrap is reported for the specified quantity for the job.
- **Consumption** – The actual quantity of material that will be consumed when progress or scrap is reported for the specified quantity for the job.
- **Reserved** – The quantity of material that is physically reserved in inventory.
- **Unit** – The bill of materials (BOM) unit.

The right side of the dialog box shows the following information:

- **Product number** – The product master and product variant.
- **Estimated** – The estimated quantity to consume.
- **Started** – The quantity that is started on the production job.
- **Remaining quantity** – Of the estimated quantity, the quantity that remains to be consumed.
- **Released quantity** – The quantity that is consumed.

The following actions can be performed:

- The worker can specify the quantity to adjust for a material by selecting **Adjust consumption**. After the quantity is confirmed, the quantity in the **Consumption** column is updated with the adjusted quantity.
- When the worker selects **Adjust material**, a production picking list journal is created. This journal contains the same items and quantities as the **Adjust material** list.
- When the worker adjusts a quantity in the **Adjust material** dialog box, the **Proposal** field on the corresponding journal line is updated with the same quantity. If the worker selects **Cancel** in the **Adjust material** dialog box, the picking list is deleted.
- If the worker selects **OK**, the picking list isn't deleted. It's posted when the job is reported in the **Report scrap** or **Report progress** dialog box.
- If the worker selects **Cancel** in the **Report progress** or **Report scrap** dialog box, the picking list is deleted.

Adjust material from the primary or secondary toolbar

You can configure the **Adjust material** button to appear on the primary or secondary toolbar. For more information, see Design the production floor execution interface. A worker selects **Adjust material** for a production job that is in progress. The **Adjust material** dialog box appears, where the worker can make the desired adjustments. When the dialog box opens, the system creates a production picking list that contains lines for the adjusted quantities for the production order. If

the worker selects **Post now**, the adjustment is confirmed, and the picking list is posted. If the worker selects **Cancel**, the picking list is deleted, and no adjustment is made.

Adjust material consumption for catch weight items

Workers can adjust material consumption for catch weight items. This functionality is used in scenarios where the actual quantity of a catch weight material consumed by a production job is more or less than the planned quantity. Adjust the quantity to keep the inventory levels current. When a worker adjusts consumption of a catch weight item, they can adjust both the catch weight quantity and the inventory quantity. For example, if a production job is planned to consume five boxes that have an estimated weight of 2 kilograms per box, the worker can adjust both the number of boxes to consume and the weight of the boxes. The system validates that the specified weight of the boxes is within the defined minimum and maximum threshold that is defined on the released product.

Reserve materials

In the **Adjust material** dialog box, a worker can make and adjust material reservations by selecting **Reserve material**. The **Reserve material** dialog box that appears shows the physically available inventory for the item for each storage and tracking dimension.

If the material is enabled for warehouse management processes (WMS), the list shows only the physically available inventory for the production input location for the material. The production input location is defined on the resource where the production job is planned. If the item number is batch or serial number controlled, the full list of physically available batch or serial numbers is shown. To specify a quantity to reserve, the worker selects **Reserve material**. To remove an existing reservation, the worker selects **Remove reservation**.

For more information about how to set up the production input location, see the following blog post: Setting up the production input location.

> **Note**
>
> Reservations that a worker makes in the **Reserve material** dialog box remain when the worker selects **Cancel** in the **Report progress** or **Report scrap** dialog box.
>
> It isn't possible to adjust reservations for catch weight items.

Completing a job and starting a new job

Usually, workers complete a job by selecting one or more current jobs on the **Active jobs** tab and then selecting **Report progress**. They then enter the quantity that was produced (the good quantity) and set the status to *Complete*. If more than one job was selected, a worker then uses the **Previous** and **Next** buttons to move among them. To start a new job, the worker selects it on the **All jobs** tab and then selects **Start job**.

A worker can also start a new job while their previous job is still open. Once again, the worker selects the new job on the **All jobs** tab and then selects **Start job**. However, in this case, the **Start job** dialog box informs the worker that they are currently working on a job, and that they must therefore either stop or complete that job before they start the new job.

Working on multiple jobs in parallel

One worker can work on multiple jobs at the same time (that is, in parallel). In this case, the collection of jobs that the worker is working on is called a *job bundle*. The worker can add new jobs to the bundle, or complete one or more jobs in the bundle. The following two scenarios show how a worker can work on jobs in parallel.

Scenario 1: A worker who has no active jobs wants to start two jobs and work on them in parallel

The worker selects the two jobs on the **All jobs** tab and then selects **Start job**. The **Start job** dialog box shows both selected jobs, and the worker can adjust the quantity to start on each job. The worker then confirms the dialog box and can start both jobs.

Scenario 2: A worker who has two active jobs that are in progress wants to start a third job and work on it in parallel with the other two

The worker selects the third job on the **All jobs** tab and then selects **Bundle**. In the **Bundle** dialog box, the worker can adjust the quantity to start. The worker then confirms the dialog box by selecting **Bundle**.

Working on indirect activities

Indirect activities are activities that aren't directly related to a production order. Indirect activities can be flexibly defined, as described in Set up indirect activities for time and attendance.

For example, Shannon, a floor worker at Contoso, wants to attend a company meeting, and meetings are considered an indirect activity. One of the following two scenarios applies:

- **Shannon is working on one or more active jobs.** Shannon selects **Activity**, identifies the activity (meeting), and confirms the selection. A message that appears informs Shannon about jobs that are in progress. From the message, Shannon can choose to complete or stop the jobs before going to the meeting.
- **Shannon doesn't have any active jobs.** Shannon selects **Activity**, identifies the activity (meeting), and confirms the selection. Shannon is now registered as being at the meeting.

In both scenarios, after confirming the selection, Shannon goes to either the sign-in page or a page that waits for the confirmation that Shannon has returned from the indirect activity. The page that appears depends on the configuration of the production floor execution interface. (Learn more in Configure the production floor execution interface.)

Registering breaks

Workers can register breaks. Breaks can be flexibly defined, as described in Pay based on registrations.

A worker registers a break by selecting **Break** and then selecting the card that represents the break type (such as lunch). After the worker confirms the selection, the device shows either the sign-in page or a page that waits for the worker to confirm that they have returned from the break. The page that appears depends on the configuration of the production floor execution interface. (Learn more in Configure the production floor execution interface.)

View the "My day" dialog

The **My day** dialog provides workers with an overview of their registrations and balances. The dialog is divided into the following three sections:

- The main section lists the registrations that the current worker made on a selected date. It opens showing registrations for the current day, and provides a date picker that lets the worker view other days.
- The **Last calculated daily balance** section shows the worker's current balances for paid time, paid overtime, absence, and paid absence. These values are based on the registrations that are calculated during the approval process.
- The **Balances** section provides an overview of the balances within a defined period for selected categories of registrations, such as vacation, standard time, and overtime. These

balances are based on the way statistical balances are set up in the **Time and attendance** module. For more information about how to set this up, see Show vacation balances in the production floor execution interface.

Admins can add this feature to the interface by placing the **My day** button on a toolbar for each relevant tab as described in Design the production floor execution interface.

Working in teams

When multiple workers are assigned to the same production job, they can form a team. The team can nominate one worker as a pilot. The remaining workers then automatically become assistants of that pilot. For the resulting team, only the pilot registers job status. Time records apply to all team members.

Prerequisites

To use teams, an admin must enable the **Assistant** action for the primary toolbar on the **All jobs** tab of the production floor execution interface. For instructions, see Design the production floor execution interface.

Form a new team that has a pilot and an assistant

A worker registers as an assistant by selecting **Assistant** on the **All jobs** tab. Then, in the **Select an employee to assist** dialog box that appears, the worker selects a pilot in a list of workers who are actively working on a job. After the worker confirms their selection, they become an assistant to the selected worker, who becomes the pilot for the new team.

Assign a new pilot to an existing team

When a team wants to select a new pilot, the current pilot nominates another worker in the team as the new pilot. To nominate a new pilot, the current pilot selects **Assistant** on the **All jobs** tab. Then, in the **Change pilot** dialog box that appears, the pilot selects a new pilot in a list of workers who are already on the team. After the current pilot confirms their selection, they are removed from the team. However, they can rejoin the team as they require.

Assistant clocks out

When a worker who works as an assistant clocks out, they leave the team. If the **Permanent teams** and **Restart at clock in** options are set to *Yes*, a worker who clocks out automatically

rejoins the team the next time that they clock in. You can find these options on the **General** tab of the **Time and attendance parameters** page.

Opening instructions

Workers can open a document attached to a job by selecting **Instructions**. The **Instructions** button appears only if a document is associated with the job in the master data. For example, if a document attached to a product on the **Released products** page in Supply Chain Management is available for workers to open it in the shop floor execution interface.

Opening mixed-reality guides for HoloLens

Dynamics 365 Guides empower workers by providing hands-on learning that uses mixed reality. You can define standardized processes where step-by-step instructions guide workers to the tools and parts that they need and show how to use those tools in real work situations. Here's an overview of the process.

1. Every time a worker opens a job list in the shop floor execution interface, the interface finds all relevant guides for the jobs that are shown.
2. The worker selects **Guides** to view the list of guides.
3. The worker selects a relevant guide in the list.
4. The shop floor execution interface shows a QR code for the selected guide.
5. The worker puts on a HoloLens and glances at the QR code to start the guide.
6. The worker works through the guide to learn the task.

For more information about how to create, assign, and use guides for HoloLens, see Provide mixed-reality Guides for workers in production.

Last updated on 07/01/2026

Configure the production floor execution interface

10/29/2025

Shop floor workers use the production floor execution interface to register their daily work, such as when they start a job, report feedback about jobs, register indirect activities, and report absence. These registrations are the basis for tracking progress and cost on production orders, and for calculating the basis for the workers' pay.

When you open the production floor execution interface, it automatically loads a selected configuration and job filter that are specific to the browser and device. In the configuration, you set the policies that must be applicable for a specific usage. Here are some usage examples:

- On a device in the company hall, employees clock in when they come into the office, and they clock out when they leave for the day.
- On a device on the shop floor, machine operators register when they start and complete jobs. They also register breaks and indirect activities.

This article describes the various options for configuring a production floor execution interface for each device in use at your site.

Turn on the production floor execution interface optional features

The production floor execution interface is always available in Supply Chain Management. But several optional features can be turned on or off for your system. Use the Feature management page to turn on any or all of the features described in the following subsections as required.

Register material consumption

Some manufacturers, especially those in the process industries, must explicitly register the amount of material that is consumed for each batch or production order. For example, workers might use a scale to weigh the amount of material that is consumed as they work. To ensure full material traceability, these organizations must also register the batch numbers that were consumed to produce each product.

There are two versions of this feature. One supports items that *are not* enabled to use warehouse management processes (WMS). The other supports items that *are* enabled to use WMS.

- The *Register material consumption on the production floor execution interface (non-WMS)* feature enables workers to use the production floor execution interface to register material consumption, batch numbers, and serial numbers, but only for non-WMS items.

- The *Register material consumption as complete and edit dimensions on the production floor execution interface* feature enables workers to use the production floor execution interface to adjust material consumption, batch and serial numbers, and license plates for both non-WMS and WMS-enabled items. It also enables workers to mark the consumption of a specific material as completed by marking an **End** flag on the line item for a material.

Register material consumption on the production floor execution interface (non-WMS) functionality is always available as part of Supply Chain Management. To use additional related functionality, turn on one or both of the following features in Feature management:

- *Register material consumption as complete and edit dimensions on the production floor execution interface* (This feature adds support for WMS-enabled items. It requires Supply Chain Management version 10.0.42 or later. As of Supply Chain Management version 10.0.45, this feature is turned on by default.)
- *Add lines to the adjust materials dialog* (Allows workers to add and delete material lines in the **Adjust Materials** dialog on the production floor execution interface. It requires Supply Chain Management version 10.0.44 or later.)

Tracked components

This feature lets workers and managers register batch/serial numbers for materials and components that are used in manufacturing processes. They can then associate those numbers with the batch/serial numbers of the products that are produced. In this way, manufacturers can optimize their processes, enhance product quality, and respond quickly to any issues that arise. Managers can use the item tracing report to effectively track batch/serial numbers that are registered through the *Tracked components* feature.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.40 or later.
- The feature that's named *Tracked components* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default.
- You must make the feature available to workers by adding a **Tracked components** button to the appropriate tab and toolbar, as described in Design the production floor execution interface.

For more information about this feature, see Register batch/serial numbers for finished products and their components (preview).

Select color themes on production floor execution interface

When configuring the production floor execution interface, supervisors can select one of several default color themes, ranging from dark to light. Each worker can also select their own preferred theme, which overrides the default each time that worker signs in.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature that's named *Select color themes on production floor execution interface* must be turned on in Feature management.

List view for reporting job scrap from the production floor execution interface

Workers reporting job scrap from the production floor execution interface can choose to work from a list view that shows several jobs at once, or from a detail view that shows more information about one job at a time. Previously, only the detail view was available.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- The feature that's named *List view for reporting job scrap from the production floor execution interface* must be turned on in Feature management.

Streamline the process for registering indirect activities

This feature streamlines the process for workers to register for indirect activities. Without this feature, when a worker registered for an indirect activity, a dialog box would open, asking them whether they wanted to stop the activity or log it at the end of their shift. This feature removes the dialog. Instead, the system automatically adds the indirect activity to the worker's active job list. The system stops the indirect activity when the worker begins a new task (such as a production job), another indirect activity, or a project job

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.42 or later.
- The feature that's named *Streamlined registration process for indirect activities on the Production floor execution* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

Report progress on multiple jobs in the report-progress list view

This feature lets workers reporting job progress to select the desired status on multiple jobs in the report-progress list view.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.42 or later.
- The feature that's named *Set the desired status on selected jobs in report progress list view in the production floor execution interface* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

Enhanced numpad control

The enhanced numpad input control replaces the previous one. It is now an element of type *Real* instead of type *String*, which makes the decimal separator consistent with the regional settings.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.45 or later.
- The feature that's named *Enhanced numpad control for production floor execution interface* must be turned on in Feature management.

View material list

This feature allows production floor workers to view the list of materials required for a job directly within the production floor execution interface. This feature enhances visibility into material requirements, helping workers ensure that all necessary components are available before starting or continuing a production task.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- The feature that's named *Material list on the production floor execution interface* must be turned on in Feature management.

To show the material list, set **View materials** to *Yes* on the **Report progress** FastTab of the **Configure production floor execution** page for the relevant production floor execution configurations.

Report serial numbers beyond single unit

This feature lets workers use the production floor execution interface to report a quantity greater than one for items that are tracked by serial number but not strictly controlled by it. To enable this functionality, the item must be linked to a tracking dimension group where the **Serial number control setting** is set to *No* for the serial number dimension.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- The feature that's named *Serial number reporting beyond single unit* must be turned on in Feature management.

Record test results from the production floor execution interface (preview)

[This section is prerelease documentation and is subject to change.]

This feature allows workers to record test results on quality orders directly from the production floor execution interface.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- The feature that's named *(Preview) Record test results from the production floor execution interface* must be turned on in Feature management.

Auto-complete secondary operation with primary

This feature completes secondary operations automatically when a worker uses the production floor execution interface to complete their related primary operation. The time recorded for the secondary operation matches the time registered for the primary operation. Supervisors can activate this functionality by enabling the **Complete secondary operation with primary** parameter on the route group associated the secondary operation.

To use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.
- The feature that's named *Auto-complete secondary operation with primary* must be turned on in Feature management.

Work with production floor execution configurations

To create and maintain production floor execution configurations, go to **Production control** > **Setup** > **Manufacturing execution** > **Configure production floor execution**. The **Configure production floor execution** page shows a list of existing configurations. On this page, you can perform the following actions:

- Select any production floor configuration that is listed in the left column to view and edit it.
- On the Action Pane, select **New** to add a new configuration to the list. Then, in the **Configuration** field, enter a name to identify the new configuration. The name that you enter must be unique among all configurations, and you won't be able to edit it later. In the **Description** field, you can optionally enter a description of the configuration.

Next, configure the various settings for the selected configuration, as described in the following subsections.

The General FastTab

The following settings are available on the **General** FastTab:

- **Clock in and out only** – Set this option to *Yes* to create a simplified interface that provides only clock-in and clock-out functionality. This setting disables most of the other options on this page. You must remove all lines from the **Tab selection** FastTab before you can enable this option.
- **Report quantity at clock-out** – Set this option to *Yes* to prompt workers to report feedback about jobs that are in progress when they clock out. When this option is set to *No*, workers won't be prompted.
- **Lock employee** – When this option is set to *No*, workers will be signed out immediately after they make a registration (such as a new job). The interface will then return to the sign-in page. When this option is set to *Yes*, workers will stay signed in to the production floor execution interface. However, a worker can manually sign out so that another worker can sign in while the production floor execution interface continues to run under the same system user account.
- **Use the actual time of registration** – Set this option to *Yes* to set the time for each new registration to the exact time when the worker submitted the registration. When this option is set to *No*, the sign-in time is used instead. You will usually want to set this option to *Yes* if you've set the **Lock employee** and/or **Single worker** option to *Yes* in cases where workers often remain signed in for longer periods.
- **Single worker** – Set this option to *Yes* if only one worker uses each production floor execution interface where this configuration is active. When this option is set to *Yes*, the **Lock employee** option is automatically set to *Yes*. In addition, this setting removes the requirement (and ability) for the worker to sign in by using a badge ID or personnel number. Instead, the worker signs in to Microsoft Dynamics 365 Supply Chain Management by using a system user account that is linked to a *time registered worker*, and gets signed in to the production floor execution interface as that worker at the same time. Learn more in Set up worker accounts to use the production floor execution interface.
- **Suppress numpad keyboard** – Select this option to prevent the numpad keyboard from opening when you need to change or enter a quantity.

The Tab selection FastTab

Use the settings on the **Tab selection** FastTab to select which tabs the production floor execution interface should show when the current configuration is active. You can design as many tabs as you need, and then add and arrange them as you require by using the buttons on the FastTab toolbar. For information about how to design tabs and work with the settings here, see Design the production floor execution interface.

The Login FastTab

The following settings are available on the **Login** FastTab:

- **Enable numpad** – Set this option to *Yes* to add a numpad to the sign-in screen, which allows workers to enter their badge ID or personal number using a touch-screen numpad. Set this option to *No* to hide the numpad.
- **Allow locking the touchscreen** – Set this option to *Yes* to allow workers to lock the touchscreen of the production floor execution interface so that they can sanitize it. When this option is set to *Yes*, a **Lock screen for sanitizing** button is added to the sign-in page. When a worker selects this button, the touchscreen is temporarily locked to prevent unintended input. A countdown timer is also shown. The worker can then safely clean the device and the screen. When the countdown is completed, the touchscreen is automatically unlocked.
- **Screen lock duration** – When the **Allow locking touchscreen** option is set to *Yes*, use this option to specify the number of seconds that the touchscreen should be locked for sanitizing. The duration must be between 5 and 120 seconds.

The Main view FastTab

The following settings are available on the **Main view** FastTab:

- **Enable search** – Set this option to *Yes* to include a search field on the jobs list. Workers can find a specific job by entering the job ID, or they can find all jobs for a specific order by entering the order ID. Workers can enter the ID by using a keypad or scanning a bar code.
- **Enable search by project ID** – Set this option to *Yes* to enable workers to search by project ID (in addition to job ID and order ID) in the search field of the production floor execution interface. You can set this option to *Yes* only when the **Enable search** option is also set to *Yes*.
- **Auto-open start dialog** – When this option is set to *Yes*, the **Start job** dialog box is automatically opened when workers use the search bar to find a job.
- **Auto-open report progress dialog** – When this option is set to *Yes*, the **Report progress** dialog box is automatically opened when workers use the search bar to find a job.

The Report progress FastTab

The following settings are available on the **Report progress** FastTab:

- **Skip product selection** – Set this option to *Yes* to skip the page where workers identify which items they want to report progress on (formula item, co-products, and by-products). If you choose to skip this page, workers will go straight to the page where they

specify the finished quantity and dimensions of each produced formula item, co-product, and by-product.

- **View materials** – Set to *Yes* to allow production floor workers to view the list of materials for a production order from the **Report progress** dialogue. This feature enhances visibility into material requirements, helping workers ensure that all necessary components are available before starting or continuing a production task.
- **Enable adjust material** – Set this option to *Yes* to include the **Adjust material** button in the **Report progress** dialog box. Workers can select this button to adjust material consumption for the job.
- **Default remaining quantity** – Set this option to *Yes* to pre-fill the expected remaining quantity for a production job in the **Report progress** dialog box.
- **Default nominal quantity** – Set to *Yes* to pre-populate the nominal quantity for catch weight items in the report progress dialog. Set to *No* to leave the field blank, which allows workers to enter the quantity manually.
- **Require existing license plate** – Choose whether workers should be required to specify an existing license plate when reporting as finished. Set to *Yes* to require workers to specify a license plate that already exists in the system. Set to *No* to allow the worker to specify any license plate. If the reported license plate doesn't already exist, the system will create it.
- **Generate license plate** – Set this option to *Yes* to generate a new license plate every time that a worker uses the production floor execution interface to report as finished. The license plate number is generated from a number sequence that is set up on the **Warehouse management parameters** page. When this option is set to *No*, workers must specify an existing license plate when they report as finished.
- **Print label** – Set this option to *Yes* to print a license plate label when a worker uses the production floor execution interface to report as finished. The configuration of the label is set up in document routing, as described in Document routing label layouts.
- **Add material line** – Set this option to *Yes* to enable the addition and deletion of material lines within the adjust material dialog.

Clean up job configurations

When the shop floor supervisor sets up the production floor execution interface, they select a configuration and a job filter. These selections are stored in a reference table in Supply Chain Management, and the browser uses an ID that is stored in a local cookie to find the correct row in that table. The table also logs the date and time that a worker last signed in to each device.

A batch job periodically cleans entries in the references table for devices that haven't logged any activity in the last 60 days. You can also manually clean the entries at any time by following these steps.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Configure production floor execution**.
2. On the Action Pane, select **Clean up client configurations**.
3. In the **Clean up client configuration** dialog box, set the **Number of days** field to the number of days of inactivity (before today) to consider. You will remove all configurations and sign-in records for devices that haven't been active during that time.
4. Select **OK** to clean up the relevant configurations, based on the **Number of days** setting.

Show vacation balances in the production floor execution interface

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an example scenario that shows how to set up Microsoft Dynamics 365 Supply Chain Management so that it uses payroll statistics to provide each worker with an overview of their vacation balance for the current year. Workers will be able to see their vacation balance in the **My day** dialog box in the production floor execution interface.

This scenario uses the Danish holiday law, where the vacation year goes from September 1 through August 31. In this scenario, your company has hired a new worker and will grant that worker a balance of 10 vacation days for the rest of the current vacation year.

Make sample data available

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

Create a new pay type

Start by creating a new *pay type* that will track workers' earned vacation days (also referred to as their *vacation balance*). Typically, pay types are used to calculate workers' pay. However, the pay type that you create here will be used to calculate a balance.

1. Go to **Time and attendance** > **Setup** > **Payroll** > **Pay types**.
2. On the Action Pane, select **New** to add a row to the grid.
3. On the new row, set the following values:
    - **Pay type:** *5151-1*
    - **Description:** *Counter for worker vacation days*
    - **Include in export:** *No*

Update the pay agreement

In this section, you will edit an existing *pay agreement* by adding the new pay type and setting the rules that define how each worker's vacation balance is adjusted when vacation days are registered.

1. Go to **Time and attendance** > **Setup** > **Payroll** > **Pay agreements**.
2. Select the pay agreement where you want to set up the vacation policy. (If you're working with standard USMF sample data, there is only one pay agreement, *Man*.)
3. Make sure that the selected pay agreement is valid for the current date. If you're working with standard USMF sample data, the **To date** field of the *Man* pay agreement might be set to a date that is in the past. Change the value so that it's a year or two in the future, as required.
4. On the Action Pane, select **Agreement lines**.
5. On the **Pay agreement lines** page, on the **Overview** FastTab, set the following values on the toolbar:
    - In the first drop-down list, select **Monday**.
    - In the second drop-down list, select **Absence**.
6. On the Action Pane, select **New** to add a row to the grid.
7. On the new row, set the **Pay type** field to the pay type that you created for this scenario (*5151-1*). Leave all other fields set to their default values.
8. While the new row is still selected, on the **General** FastTab, set the following values:
    - **Absence code:** *Vacation*
    - **Use fixed quantity:** *Yes*
    - **Constant:** *-1*
9. On the Action Pane, select **Copy day**.
10. In the **Copy day** dialog box, set the following values:
    - **Tuesday**, **Wednesday**, **Thursday**, and **Friday:** *Yes*
    - **Overwrite:** *Yes*
    - **Absence:** *Yes*
11. Select **OK**.

Create a payroll statistic group

Payroll statistic groups are used to set up statistics for worker registrations during a period. In this scenario, you will use a payroll statistic group to calculate the number of vacation days that workers earn during a vacation period. In Denmark, the vacation period runs from September 1 through August 31.

1. Go to **Time and attendance** > **Setup** > **Payroll** > **Payroll statistics groups**.
2. On the Action Pane, select **New** to add a row to the grid.
3. On the new row, set the following values:
    - **Payroll statistics group:** *VAC*
    - **Description:** *Vacation balance*
    - **Transfer:** *No*
4. While the new row is still selected, select **Setup** on the Action Pane.
5. On the **Setup** page, on the Action Pane, select **New** to add a row to the grid.
6. On the new row, set the **Pay type** field to *5151-1*.
7. Select and hold (or right-click) the **Period code** field, and then select **View details**. You can now set up the period code that you will assign to this field later.
8. On the **Period types** page, on the Action Pane, select **New** to add a row to the grid.
9. On the new row, set the following values:
    - **Period types:** *VacYear*
    - **Description:** *Vacation year*
    - **Frequency:** *Years*
10. While the new row is still selected, select **Generate periods** on the Action Pane.
11. In the **Generate periods** dialog box, set the following values:
    - **Specify start date of the period:** *September 1, 2021*
    - **Length of period:** *5*
12. Select **OK**.
13. Close the **Period types** page to return to the **Payroll statistics groups** page.
14. Set the **Period code** field to the period type that you just created (*VacYear*).

Create a statistical balance setup

In this section, you will create a *statistical balance setup* that is linked to the payroll statistic group that you created in the previous section. Later, you will link this statistical balance setup to the **My day** view in the production floor execution interface. The **My day** view will then be able to show the vacation balances for the pay type that is assigned to the associated payroll statistic group.

1. Go to **Time and attendance** > **Setup** > **Payroll** > **Statistical balance setup**.
2. On the Action Pane, select **New** to add a row to the grid.
3. On the new row, set the following values:
    - **Payroll statistics group:** *VAC*
    - **External name:** *Vacation balance*
    - **Flex:** *No*

Create a manual premium

Manual premiums are typically used to grant workers extra pay for extra work. In this scenario, you will create a manual premium that you can use to set the initial vacation balance for each worker.

1. Go to **Time and attendance** > **Setup** > **Payroll** > **Manual premiums**.
2. On the Action Pane, select **New** to add a record.
3. In the new record, set the following values:
    - **Premiums:** *VAC*
    - **Description:** *Adjustments to workers' vacation balance*
    - **Pay type:** *5151-1*

Set a worker's initial vacation balance and adjust it by one day

Payroll administrators use the **Approve** page to review and approve workers' daily registrations. In this scenario, you will take on the role of an admin so that you can set the initial vacation balance for a worker and register vacation days that the worker takes.

1. Go to **Time and attendance** > **Review and approve** > **Approve**.

2. In the **Approve** dialog box, set the following fields:

    - **Approval group** – Select the approval group that you belong to. (If you're working with standard USMF sample data, there is only one approval group, *AdmMan*.)
    - **View entire group, 1 day** – Select the option, and then set the field to the current date.

3. Select **OK**.

4. The **Approve** page shows a list of records that match your criteria. Select the worker that you want to approve. If you're working with standard USMF sample data, select worker *000496* (*Christina Portra*).

5. Grant the selected worker 10 days of vacation:

    a. While the worker is still selected, select **Premium lines** on the Action Pane.

    b. On the **Premium lines** page, on the Action Pane, select **New** to add a row to the grid.

    c. On the new row, set the following values:

    - **Premiums:** *VAC*
    - **Quantity:** *10*

    d. Close the **Premium lines** page.

6. On the **Approve** page, register the fact that the worker used one of their vacation days on the current date:

    a. While the worker is still selected, in the lower part of the page, on the **Overview** tab, select **New** on the toolbar to add a row to the grid.

    b. On the new row, set the following values:

    - **Journal registration type:** *Absence*
    - **Reference:** *Vacation*

    c. In the upper part of the page, select **Transfer** on the toolbar to apply the vacation balance and calculate the deduction.

Configure the production floor execution interface so that it includes the My day dialog box

In this section, you will add a **My day** button to the production floor execution interface. Workers can then use this button to open the **My day** dialog box.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Configure production floor execution**.
2. Select a configuration, such as *Default*.
3. On the Action Pane, select **Design tabs**.
4. On the **Design tabs** page, in the list pane, select *All jobs* to view the settings for that tab. The **Main view** field should now show a value of *All jobs*.
5. On the Action Pane, select **Edit**.
6. On the **Secondary tool bar** FastTab, in the **Available actions** column, select **My day**. Then select the right arrow button to move it to the **Selected actions** column.

Check your balance in the production floor execution interface

In this section, you will sign in to the production floor execution interface as the worker whose vacation time you set up earlier. You will then open the **My day** dialog box to view your vacation balance.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Production floor execution**.
2. If you're prompted to select a configuration, select the configuration that you set up earlier in this scenario (*Default*).
3. Sign in as the user that you set up earlier in this scenario. If you're working with standard USMF sample data, you should be able to sign in by entering *123* in the **Badge ID** field. This badge ID corresponds to Christina Portra.
4. Select **My day** on the left toolbar.
5. Inspect the **Balances** section of the dialog box. This section should now show that you have a balance of nine vacation days.

Last updated on 10/03/2025

Design the production floor execution interface

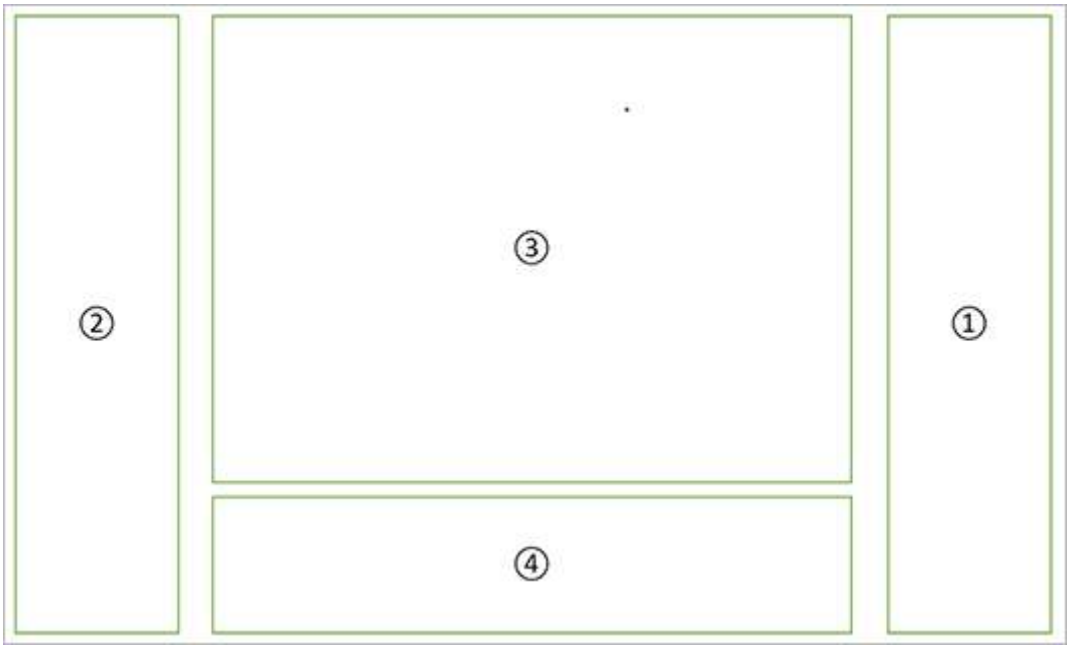
06/17/2025

You can design the content of the user interface for each configuration used by the production floor execution interface. For example, workers in one work cell might need to be able to open job instructions on the production floor, while in another work cell, instructions aren't needed. In that case, two configurations should be created, one with a button for opening document attachments and one without this button.

Design a tab

On the **Configure production floor execution** page, you can create and configure tabs by selecting **Design tabs** on the Action Pane.

Each tab is divided into four sections, as shown in the following illustration.



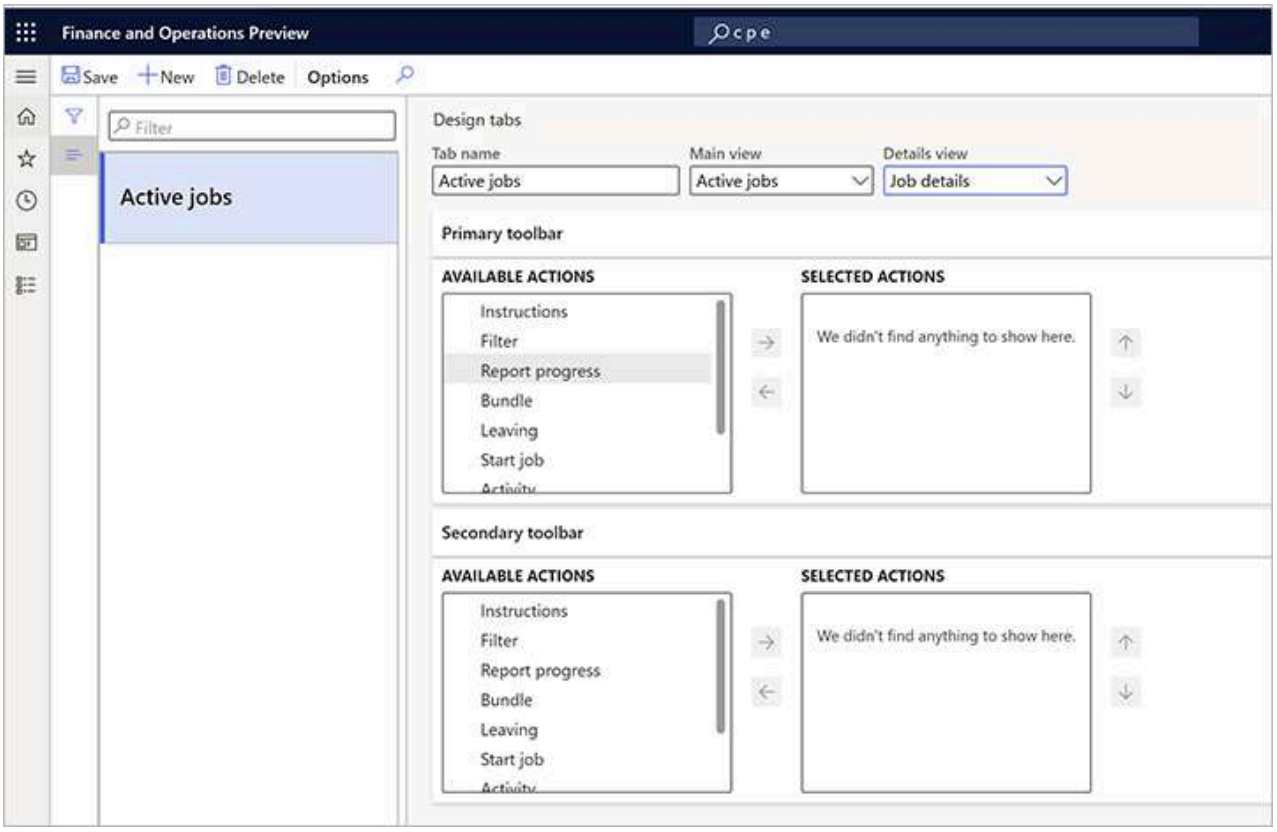
The following elements are shown in the illustration:

1. Primary toolbar
2. Secondary toolbar
3. Main view
4. Detailed view

To create and configure a new tab, follow these steps:

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Configure production floor execution**.

2. Select **Design tabs** on the Action Pane to open the **Design tabs** page.



3. Select **New** on the Action Pane.

4. Make the following settings in the header of the page:

    - **Tab name** – Specify a name for the tab.
    - **Main view** – Select among the predefined job lists (*Active jobs*, *All jobs*, *My jobs*, and *My machine*).
    - **Details view** – Select between a blank value or **Job details**. If you select the blank value, there will be no detailed view in the tab. If you select **Job details**, the detailed view will contain a detailed description of the job selected in the job list in the main view.

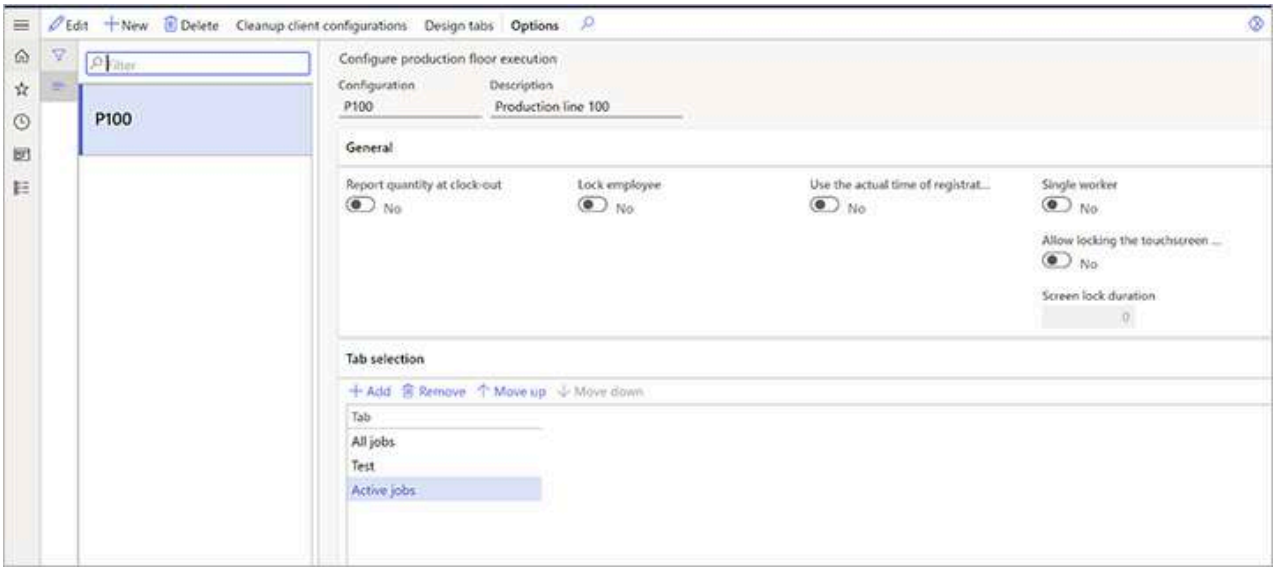
5. In the **Primary toolbar** section, choose which buttons should be available in the primary toolbar. The **Available actions** column shows a list of all the buttons that can be added. The **Selected actions** column shows a list of all the buttons that are included in the current configuration. Use the buttons between the columns to move selected items between the columns as needed. Use the up and down buttons next to the **Selected actions** column to control the order in which the buttons are presented in the user interface.

6. In the **Secondary toolbar** section, choose which buttons should be available in the secondary toolbar. The **Available actions** column shows a list of all the buttons that can be added. The **Selected actions** column shows a list of all the buttons that are included in the current configuration. Use the buttons between the columns to move selected items between the columns as needed. Use the up and down buttons next to the **Selected actions** column to control the order in which the buttons are presented in the user interface.

Associate a tab with a configuration

After you designed all the tabs you need, you can associate them with a configuration.

1. Go to **Production control** > **Setup** > **Manufacturing execution** > **Configure production floor execution**.



2. On the **Tab selection** FastTab, select **Add**.

3. A new row is added to the grid. For this new row, select the name of a tab that you want to add to the configuration.

4. Continue to add additional tabs as needed.

5. Use the **Move up** and **Move down** buttons on the toolbar to arrange the tabs as needed. The tabs will be displayed from left to right in the order shown (the tab at the top is shown on the left).

Set up a device to run the production floor execution interface

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You must set up the production floor execution interface for each device on the production floor. Companies typically set up each device differently, depending on the purpose that the device serves. For example, one device might be in the reception area where workers clock in and clock out, and another device might be on the shop floor where workers manage their jobs.

Set the configuration and filters for a specific device

To set the configuration and job filters for a device, sign in to the **Production floor execution** page by using an account that has a security role that includes the *Maintain time supervision* duty. (Among the out-of-box security roles, only *Shop floor supervisor* has this duty.) Then follow these steps:

1. Go to the device that you want to set up, and sign in to Microsoft Dynamics 365 Supply Chain Management as a shop floor supervisor. (Use an account that includes the *Maintain time supervision* duty.)

2. Make sure that a configuration is available for the device that you're setting up. If no configuration exists, a default configuration is provided. For more information about how to set up a configuration, see Configure the production floor execution interface.

3. Go to **Production control** > **Manufacturing execution** > **Production floor execution**.

   If the production floor execution interface has already been configured at least one time on the current device, a sign-in page appears. Otherwise, a welcome page appears.

4. On either the sign-in page or the welcome page, select **Configure**.

5. Select a configuration in the list.

6. Select **Next**.

7. Select one or more filters to apply to the current device. These filters help ensure that only relevant jobs show on the device. To set a filter, select the filter type to open a list of values, and then select the value to filter on. The following filters are available:

    - **Production unit** – This filter is the highest-level filter. It typically refers to a large work area that has several resource groups and individual resources in it.
    - **Resource group** – This filter is a mid-level filter. It typically refers to a collection of related resources in a limited area of the workspace. If you select a **Production unit** filter first, the list of resource groups shows only groups from that unit. Otherwise, it shows all available resource groups.
    - **Resource** – This filter is the most specific filter. It typically refers to a specific machine or other single resource. If you select a **Resource group** and/or **Production unit** filter first, the list of resources shows only resources from that group and/or unit. Otherwise, it shows all available resources.

8. Select **Next**.

9. Select the display color theme to apply to the current device.

10. Select **OK**.

11. The sign-in page appears, and your device is ready for use.

Allow a worker to override the default filters

You can give specific workers permission to change the filter settings on any device that they use. For workers who have this permission, the production floor execution interface provides a **Filter** button on the **All jobs** and **Active job** pages.

> ⓘ **Note**
>
> If a worker changes a filter, the new filter applies from that point forward, for all users who sign in to the device.

To allow a worker to override the default job filters that you set up for a device, follow these steps:

1. Go to **Time and attendance** > **Setup** > **Time registration workers**.
2. Select a worker in the list to open that worker's **Time registration workers** page.
3. On the **Time registration** tab, set the **Set filters** option to *Yes*.

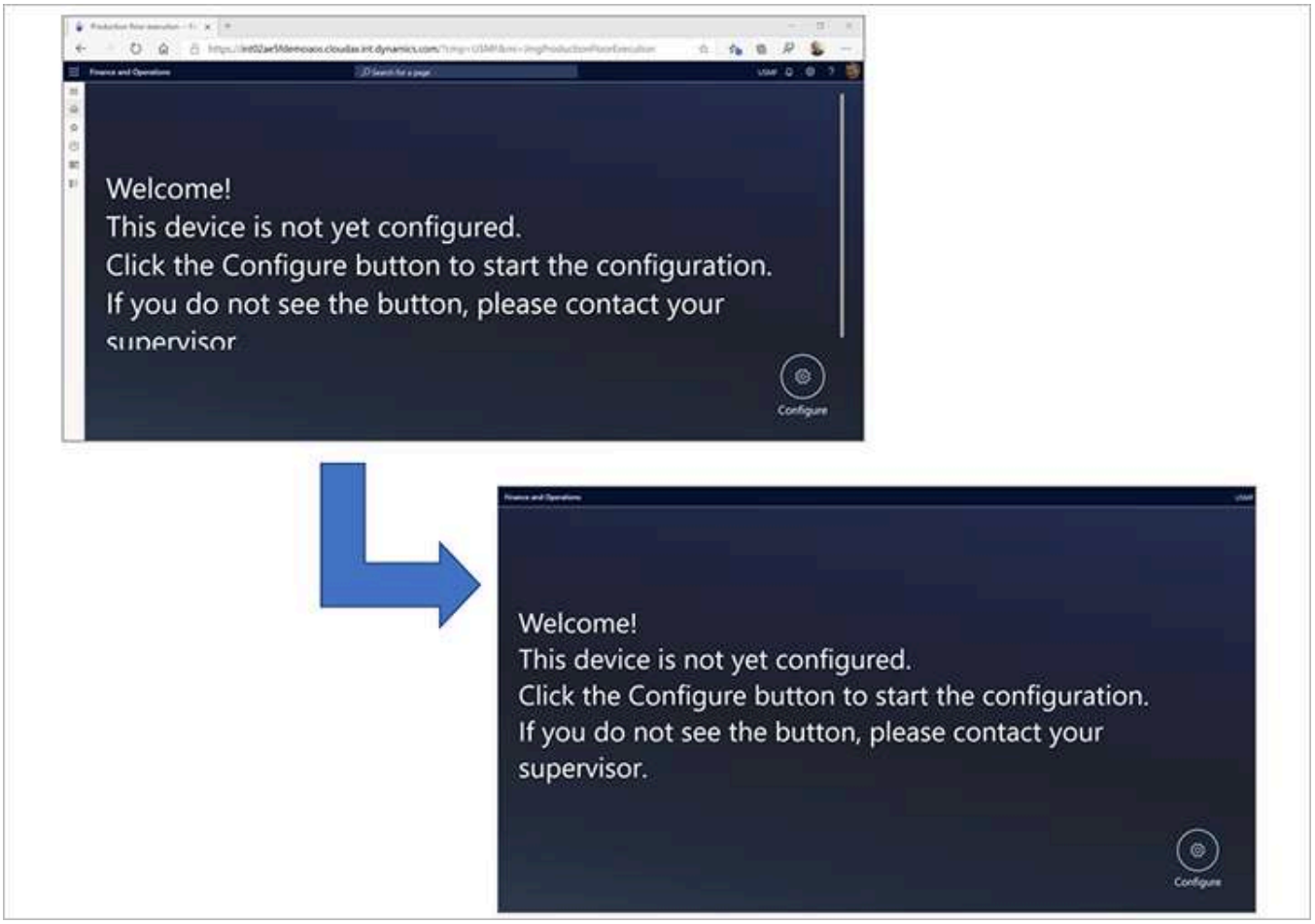
For more information about how to set up worker accounts to access the production floor execution interface, see Set up worker accounts to use the production floor execution interface.

Run the interface in full-screen mode

Often, you'll run the production floor execution interface on a device that is used exclusively for that purpose. Therefore, it might make sense to run the interface in full-screen mode, without showing any navigation or browser chrome.

- To hide the navigation pane that appears in Supply Chain Management, add the following text to the end of the URL in the browser's address bar: `\&limitednav=true`.
- To also hide the browser's address bar, use the browser's native full-screen mode. For instructions, see your browser's documentation.

The upper part of the following illustration shows how the interface looks by default. The lower part shows how it looks in full-screen mode when the navigation pane is hidden.



Extend the session past 12 hours (preview)

[This section is prerelease documentation and is subject to change.]

By default, the production floor execution interface automatically signs out if no one uses it for 12 hours. A Supply Chain Management user must then sign in again. However, you can extend the timeout limit to up to 90 days.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Prerequisites

To allow sessions to extend past 12 hours, the feature named *(Preview) Enables session management for users* must be turned on in feature management.

Set the session timeout for a device

To set the session timeout for a device, follow these steps:

1. Go to **System administration** > **Users** > **User session management**.
2. Do one of the following steps:
    - Find and select the row where **User ID** shows the account that you use to sign in to the device.
    - If the relevant user account isn't listed, select **New** on the Action Pane and then select the relevant **User ID** for the new row.
3. For your new or selected row, set **Maximum session length (hours)** to the desired number of hours, up to 2,160 hours (90 days).
4. On the Action Pane, select **Save**.

Last updated on 12/19/2025

Set up worker accounts to use the production floor execution interface

09/19/2025

This article explains how to set up worker accounts so workers can sign in and use the production floor execution interface.

Choose sign-in requirements

Workers start work by signing in to the production floor execution interface. You can require workers to identify themselves with either their personnel number or badge number, and you can also require a password.

To choose your sign-in requirements, follow these steps:

1. Go to **Time and attendance** > **Setup** > **Time and attendance parameters**.
2. On the **General** tab, make the following settings:

    - **Use password** – If you want to require a password, set this option to *Yes*. Set it to *No* if you want to allow workers to sign in using only their badge ID or personnel number. If you choose to require a password, you must assign one to each worker account as described later in this article.
    - **Use badge ID** – Set this option to *Yes* to require users to sign in with their badge ID. Set it to *No* to require users to sign in with their personnel number. All worker accounts automatically have a unique personnel number. If you choose to require a badge ID, you must assign one to each worker account as described later in this article.

3. On the Action Pane, select **Save**.

Set up a worker account

Workers sign in with credentials linked to their worker account. You can also set each worker account to prevent or allow the worker to change the filter settings on any device they use.

To set up a worker account, follow these steps:

1. Go to **Human resources** > **Workers** > **Workers**.

2. Find and open the account you want to set up or select **New** on the Action Pane to create a new worker account.

3. Open the **Time registration** tab.

4. Find the **Activate on registration terminals** button on the **Time registration** FastTab toolbar.

    - If the button is grayed out, the worker is already set up as a *time registered worker*, so you can skip this step.
    - If the button is available, select it to open the **Create time registration worker** dialog. Make all of the required settings and select **OK**. The worker is now set up as a *time registered worker*.

5. Make the following settings to enable the worker to sign in to the production floor execution interface and choose whether the worker can override default job filters:

    - **Identification** – If you set up your system to require a password when signing in to the production floor execution interface, enter a password for the current worker here.
    - **Badge ID** – If you set up your system to require a badge ID when signing in to the production floor execution interface, enter a badge ID for the current worker here.
    - **Set filters** – To allow a worker to override the default job filters that you set up for a device, set this option to *Yes*. For more information about this option, see Set up a device to run the production floor execution interface.

    > ⓘ **Note**
    >
    > Each worker's personnel number is automatically assigned by the system and can't be changed. You can read each worker's personnel number next to their name on the details page, or in the **Personnel number** column on the list page. This is the ID that workers must use to sign in if you set up your system to require a personnel number instead of a badge ID.

6. Continue to set up the worker account as required. When you're done, select **Save** on the Action Pane.

Single worker configuration

You can configure a production floor execution interface device to be used by a single worker. This option removes the requirement (and ability) for the worker to sign in by using a badge ID or personnel number. Instead, the worker signs in to Microsoft Dynamics 365 Supply Chain Management by using a system user account that's linked to a *time registered worker*, and gets

signed in to the production floor execution interface as that worker at the same time. For details about how to set up this option, see Configure the production floor execution interface.

Style the production floor execution interface

08/13/2025

The article explains how to configure form controls so that the default production floor execution styles are applied to them.

Forms and dialogs

Styles can be applied to a form or dialog only if the following requirements are met:

- If the form should resemble the existing report progress form, the name of your form or dialog must start with `JmgProductionFloorExecutionCustomInputDialog`.
- The form or dialog can contain a detail form part. To apply styles to it, the name of the detail form part must start with `JmgProductionFloorExecutionCustomDetailsDialog`.
- If the form or dialog should have a simple view, then the name of the simple view must start with `JmgProductionFloorExecutionCustomDialog`. Examples of forms that have a simple view include the start form and the indirect activity form.
- All the controls in the dialog must be configured as described in this article.

> **Important**
>
> The features mentioned in the first two bullet points of this list require Supply Chain Management version 10.0.19 or later.

Styles can be applied to the **OK** button in a dialog only if the following requirements are met:

- The button is contained in a form group.
- The group name starts with `OkButtonGroup`.

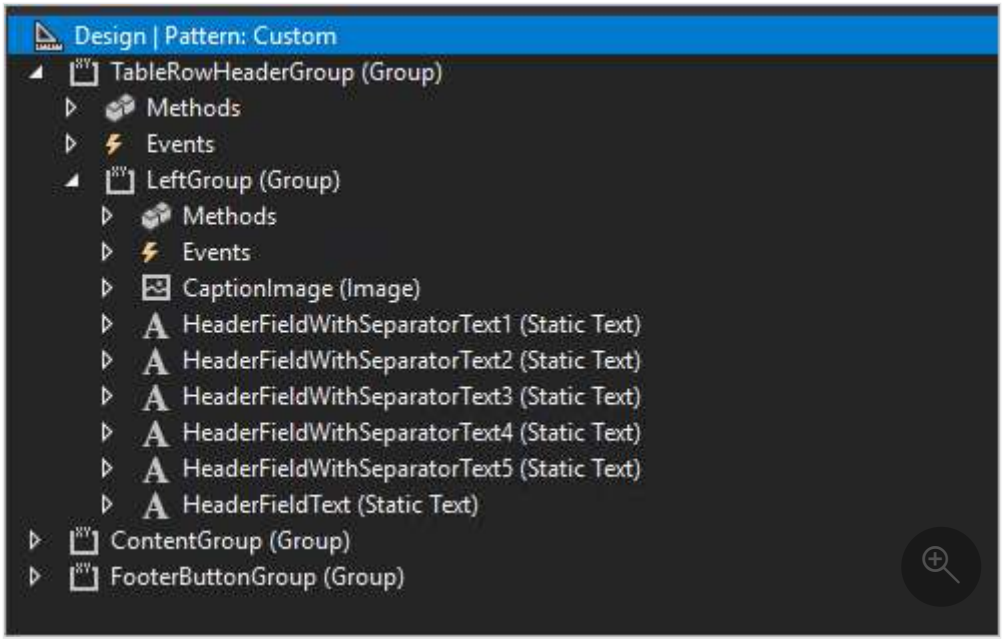
Styles can be applied to the **Cancel** button in a dialog box only if the following requirements are met:

- The button is contained in a form group.
- The group name starts with `CancelButtonGroup`.

Header

The following illustration shows a typical form or dialog header.

In Visual Studio, headers are created by using a structure such as the one that is shown in the following illustration.



To add text to your header, use code such as the following example.

X++

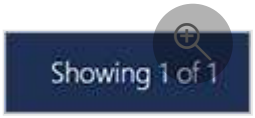
```
private void setCaption()
{
    HeaderFieldWithSeparatorText1.text("Report Progress");
    HeaderFieldWithSeparatorText2.text(ProdId);

    …

    HeaderFieldText.text(OprNum);
}
```

When you write your header code, apply the following rules:

- The name of main group must be `TableRowHeaderGroup`.
- Each block of text (separated by bullets) must start with `HeaderFieldWithSeparatorText`.
- The last text name must start with `HeaderFieldText`.
- `CaptionImage` can be skipped.

Progress indicator

You can include a progress indicator, which is shown to the right of the header. The following illustration shows a progress indicator.



To show the progress indicator, the text field must be named `ShowProgress`.

Grid

Styles are automatically applied. No specific configuration is required.

The grid should have a `TabularView` style, and the `run()` method on the custom form must be overwritten, because a new grid isn't yet supported. Add the following code.

```
X++
```

```
public void run()
{
    super();
    // To opt out a page from the new grid
    this.forceLegacyGrid();
}
```

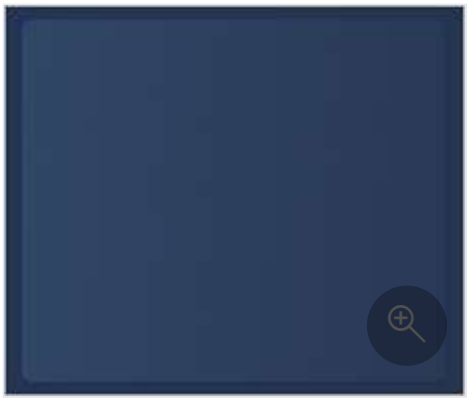
To refresh data in a main view, you might want to use something like `this.parmParentForm().updateLayout();` in a `click` method of your action. (For an example, look at the `JmgProductionFloorExecutionReportFeedbackAction` class.) Just make sure that `parmDataSource` is set in the `init` method of your new form (`formCaller.parmDataSource(this.dataSource(1));`). For an example, look at the `JmgProductionFloorExecutionMainGrid` form.

Card view

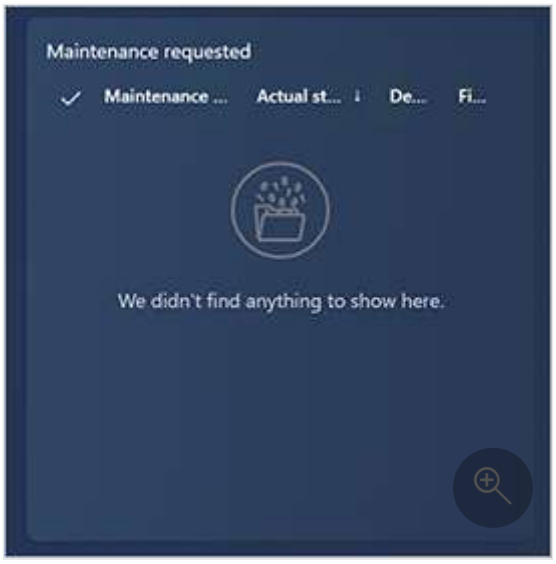
Styles can be applied to card view controls only if the following requirements are met:

- Each card view is contained in a form group.
- The group name starts with `CardGroup` (for example, `CardGroupJobsView`).

The following illustration shows a card view that has no controls inside it.



The following illustrations show card views that have controls inside them.



Business card

Styles can be applied to business card controls only if the following requirements are met:

- Each business card is contained in a form group.
- The group name starts with `BusinessCardGroup` (for example, `BusinessCardGroupJobsList`).

Set the following properties on the business card:

- **Style:** *list*
- **Extended style:** *cardList*
- **Multi Select:** *No*
- **Show Col Labels:** *No*

Radio button

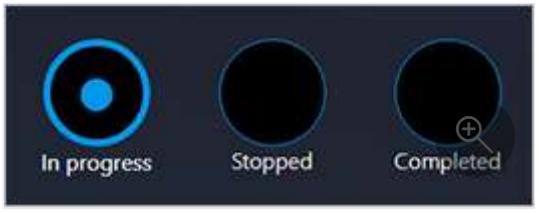
Styles can be applied to radio buttons only if the following requirements are met:

- Each radio button is contained in a form group.
- The group name starts with `RadioTextBelow` or `RadioTextRight`, depending on where you want the text to appear.

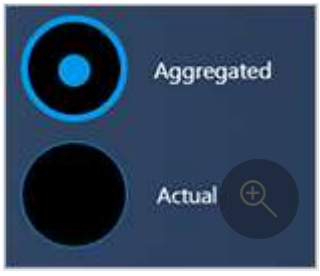
Set the following properties on the radio button:

- **Toggle button:** *Check*
- **Toggle value:** *On* if the radio button should be selected; otherwise, *Off*

The following illustration shows an example where the text appears below the radio buttons.



The following illustration shows an example where the text appears to the right of the radio buttons.



Radio buttons in Internet Explorer

Radio button styles aren't supported in Internet Explorer. The following illustration shows what radio buttons look like in Internet Explorer.

Buttons

Styles can be applied to buttons only if the following requirements are met:

- Each group of buttons is contained in a form group. All the buttons in the group will have the same style.
- There are no requirements about the name of the group.

Set the following properties on the buttons:

- **Button Display:** *TextWithImageLeft*
- **Normal Image:** This property can't be blank. For example, use *CoffeeScript*.
- **Text:** This property can't be blank. For example, use *Start Break*.
- **Width:** *Auto* or *SizeToContent*
- **Height:** *Auto* or *SizeToContent*

Primary button

Styles can be applied to a primary button only if the following requirements are met:

- The button is contained in a form group.
- The group name starts with `DefaultButtonGroup` or `PrimaryButtonGroup` (for example, `DefaultButtonGroup10`).

Secondary button

Styles can be applied to a secondary button only if the following requirements are met:

- The button is contained in a form group.
- The group is named **Right panel**, or the group name starts with `SecondaryButtonGroup`.

Third-group button

Styles can be applied to a third-group button only if the following requirements are met:

- The button is contained in a form group.
- The group is named **Left panel**, or the group name starts with `ThirdButtonGroup`.

Fourth-group button

Styles can be applied to a fourth-group button only if the following requirements are met:

- The button is contained in a form group.
- The group name starts with `FourthButtonGroup`.

Set the following properties on the button:

- **Button Display:** *TextOnly*
- **Normal Image:** This property must be blank.
- **Text:** This property can't be blank. For example, use *View* or *Edit*.
- **Width:** *Auto*
- **Height:** *Auto*

Flat button

Styles can be applied to a flat button only if the following requirements are met:

- The button is contained in a form group.
- The group name starts with `FlatButtonGroup`.

Set the following properties on the button:

- **Button Display:** *ImageOnly*
- **Normal Image:** This property can't be blank. For example, use *CoffeeScript*.
- **Text:** This property must be blank.
- **Width:** *Auto* or *SizeToContent*
- **Height:** *Auto* or *SizeToContent*

Continue button

Styles can be applied to a continue button only if the following requirements are met:

- The button is contained in a form group.
- The group name starts with `ContinueButtonGroup`.

Set the following properties on the button:

- **Button Display:** *ImageOnly*
- **Normal Image:** *Forward*
- **Text:** This property must be blank.
- **Width:** *Auto* or *SizeToContent*
- **Height:** *Auto* or *SizeToContent*

Combo box

A combo box is a combination of three controls: an input control, a button that clears the input control, and a button that runs a search.

Styles can be applied to a combo box only if the following requirements are met:

- The combo box is contained in a form group.
- The group name starts with `Combobox`.
- Inside the group, the first control is an `AxFormStringControl` control. This control shows the current value, and it's where the user enters the required value.
- The second control is a `CommonButton` control, and its name starts with `ClearButton`. This button must contain code that uses the `enable` property to show or hide the button. For example, to show or hide the **Clear** button while the user is typing information in the input control, you can use the following code.

```
X++
```

```
public void textChange()
{
    super();
    ClearButtonSerial.enabled(this.text()? true : false);
}
```

You should have one method where the data is set in the input control. Enable the **Clear** button in that method. Here is an example.

X++

```
public void setSerialId(str _serialId)
{
    JmgTmpJobBundleProdFeedback.InventSerial = _serialId;
    ClearButtonSerial.enabled(_serialId? true : false);

    if (_serialId)
    {
        this.addSerialNumber();
    }
}
```

Use the following code for the `clicked` method of the **Clear** button.

X++

```
public void clicked()
{
    element.setSerialId('');
    InventSerialId.setFocus(); // set focus back to the input box
}
```

Set the value of the input control, `AxFormStringControl`, when the form is initialized by using the `init` method. If the value isn't blank, enable the **Clear** button. If the value is blank, disable the **Clear** button.

- The third control is a `CommonButton` control, and its name starts with `SearchButton`.

The following illustration shows two combo box controls. The combo box on the left has an empty text box, and the **Clear** button is disabled. The combo box on the right has text in the text box, and the **Clear** button is enabled.

Quick filter

The quick filter control adds a search field to the page. You can apply styles to a quick filter provided the following requirements are met:

- The quick filter is contained in a form group.
- The group name starts with `SearchInputGroup`.
- Inside the group, the first control is a `QuickFilter` control. (This control is where the user enters the search string.)
- The second control is a `FormStaticTextControl` that is named `NumberOfResults`. (This control is optional. If it's included, it shows the number of found items.)
- The third control is a `CommonButton` control, and its name starts with `ClearButton`.

The following illustration shows two quick filter controls. The quick filter on the left has an empty quick filter, and the number of results isn't visible. The quick filter on the right contains a search string and shows the number of results.

Center-align elements on a tab

To align elements in the center of a tab, the group name must start with `TabContentGroup`, and the group must have the following properties:

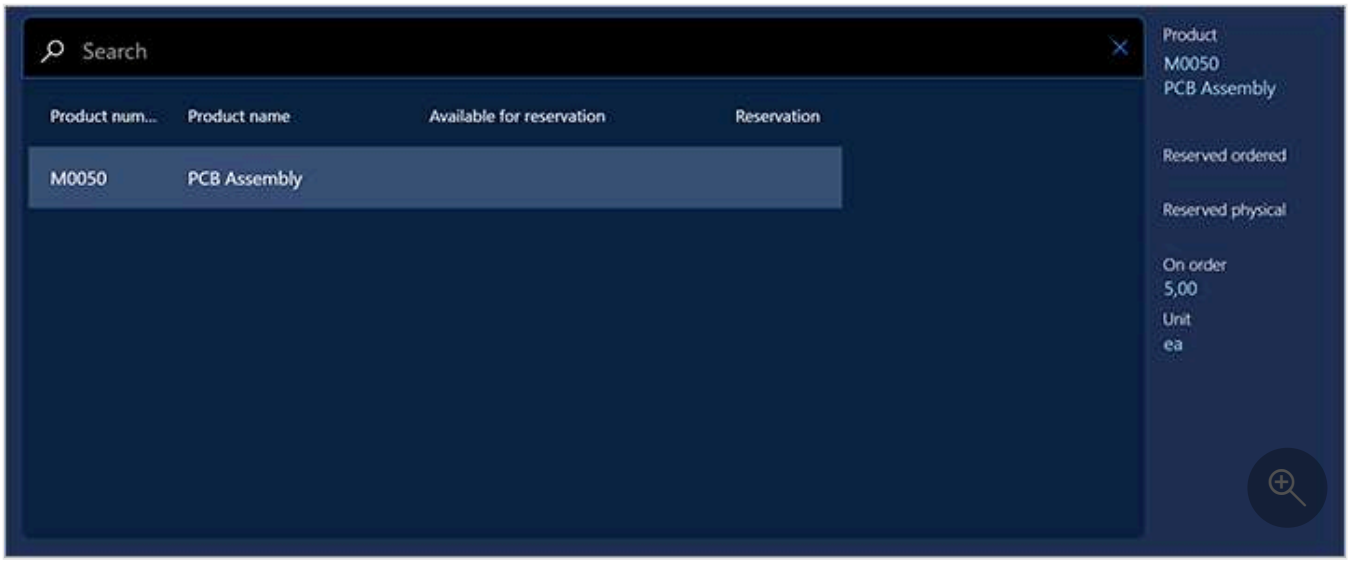
- **Width Mode:** `SizeToAvailable`
- **Height Mode:** `SizeToAvailable`

Align a grid, detail part, and quick filter

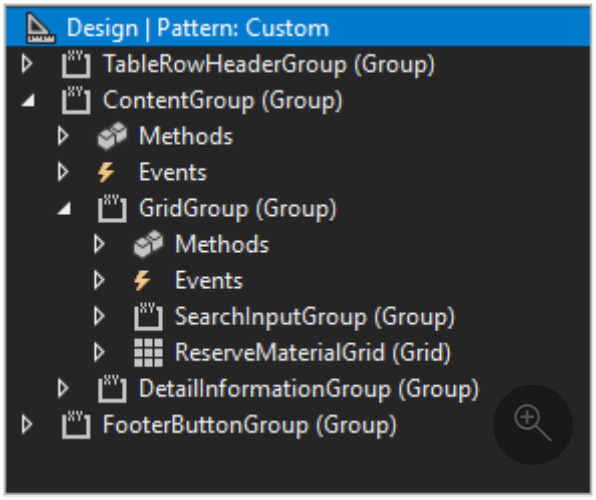
To arrange a customized grid, detail part, and quick filter so that they resemble the standard design, keep the following points in mind when you put them all together:

- If the grid has a quick filter, both the grid and the quick filter should be inside the group that has a name that starts with `GridGroup`.
- To apply styles to a detail part, the group name must start with `DetailInformationGroup`,

The following illustration shows a typical grid that includes a quick filter and a detail part on the right.



In Visual Studio, a grid, detail part, and quick filter can be created by using a structure such as the one that is shown in the following illustration.



Related information

- Customize the production floor execution interface
- Design the production floor execution interface

Customize the production floor execution interface

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Developers can extend current forms or create their own forms and buttons for the production floor execution interface. After you've added the code for these new elements, admins or shop floor managers can easily add them to the interface by using the standard configuration controls.

For example, here are some of the possible solutions if new columns are needed in a main form:

- Extend the `JmgProductionFloorExecutionMainGrid` form, and add the desired fields.
- Create a new form, and add it as a new main view (tab).

Add a new button (action)

To add a new button (action), follow these steps to create a class that implements your custom action.

1. Create a new class that is named `<ExtensionPrefix>_JmgProductionFloorExecution<ActionName>Action`, where:
    - `<ExtensionPrefix>` uniquely identifies your solution, typically by using your company name.
    - `<ActionName>` is a unique name for the class. It typically identifies the kind of action.
2. The new class must extend the `JmgProductionFloorExecutionAction` class.
3. Override all necessary methods.

For examples, look at the code for the following classes:

- `JmgProductionFloorExecutionBreakAction` – A class for a simple action that doesn't need any records.

- `JmgProductionFloorExecutionReportFeedbackAction` – A class that provides more complex functionality.

When you've finished, the new button (action) will automatically be listed on the **Design tabs** page in Microsoft Dynamics 365 Supply Chain Management. There, you (or an admin or floor manager) can easily add it to the primary or secondary toolbar, just as you can add the standard buttons. For instructions, see Design the production floor execution interface.

Add a new main view

1. Create a new form that has the desired elements and functionality. This is a new form, not an extension. Name the form `<ExtensionPrefix>_JmgProductionFloorExecution<FormName>`, where:

    - `<ExtensionPrefix>` uniquely identifies your solution, typically by using your company name.
    - `<FormName>` is a unique name for the form.

2. Create a menu item that is named `<ExtensionPrefix>_JmgProductionFloorExecution<FormName>`.

3. Create an extension that is named `<ExtensionPrefix>_JmgProductionFloorExecution<FormName>_Extension`, where the `getMainMenuItemsList` method is extended by adding the new menu item to the list. The following code shows an example.

    X++

    ```
    [ExtensionOf(classStr(JmgProductionFloorExecutionMenuItemProvider))]
    public final class
    <ExtensionPrefix>_JmgProductionFloorExecutionForm<FormName>_Extension{
        static public List getMainMenuItemsList()
        {
            List menuItemList = next getMainMenuItemsList();

    menuItemList.addEnd(menuItemDisplayStr(<ExtensionPrefix>_JmgProductionFloorExec
    utionForm<FormName>));
            return menuItemList;
        }
    ```

When you've finished, the new main view will automatically be listed in the **Main view** combo box on the **Design tabs** page in Supply Chain Management. There, you (or an admin or floor manager) can easily add it to new or existing tabs, just as you can add the standard main views. For instructions, see Design the production floor execution interface.

Add a details view

1. Create a new form that has the desired elements and functionality. This is a new form, not an extension. Name the form `<ExtensionPrefix>_JmgProductionFloorExecution<FormName>Detail`, where:

    - `<ExtensionPrefix>` uniquely identifies your solution, typically by using your company name.
    - `<FormName>` is a unique name for the form.

2. Create a menu item that is named `<ExtensionPrefix>_JmgProductionFloorExecution<FormName>Detail`.

3. Create an extension that is named `<ExtensionPrefix>_JmgProductionFloorExecution<FormName>_Extension`, where the `getDetailsMenuItemList` method is extended by adding the new menu item to the list.

When you've finished, the new details view will automatically be listed in the **Details view** combo box on the **Design tabs** page in Supply Chain Management. There, you (or an admin or floor manager) can easily add it to new or existing tabs, just as you can add the standard details views. For instructions, see Design the production floor execution interface.

Add a numeric keypad to a form or dialog

The following example shows how to add numeric keypads to a form.

1. The number of numeric keypad controllers that each form contains must equal the number of numeric keypads in that form.

    X++

    ```
    private JmgProductionFloorExecutionNumpadController   numpadController1;
    private JmgProductionFloorExecutionNumpadController   numpadController2;
    private JmgProductionFloorExecutionNumpadController   numpadController3;
    ```

2. Set up the behavior of each numeric keypad controller, and connect each numeric keypad controller to a numeric keypad form part.

    X++

    ```
    /// <summary>
    /// Initializes all numpad controllers, defines their behavior and connects
    them with numpad form parts.
    /// </summary>
    ```

```
public void initializeNumpadController()
{
    numpadController1 = new JmgProductionFloorExecutionNumpadController();
    numpadController1.getValueFromNumpadDelegate +=
eventhandler(this.setQuanityValueFromNumpad_1);
    QuantityNumpad1.getPartFormRun().setNumpadController(numpadController1);

    numpadController2 = new JmgProductionFloorExecutionNumpadController();
    numpadController2.parmNumpadEnabled(false);
    numpadController2.parmNumpadValue(333.56);
    numpadController2.getValueFromNumpadDelegate +=
eventhandler(this.setQuanityValueFromNumpad_2);
    QuantityNumpad2.getPartFormRun().setNumpadController(numpadController2);
}
```

Use a numeric keypad as a pop-up dialog

The following example shows one way to set up a numeric keypad controller for a pop-up dialog.

X++

```
private void setupNumpadController()
{
    numpadController = new JmgProductionFloorExecutionNumpadController();
    numpadController.parmShouldNumpadShowOkCancelButtons(true);
    numpadController.setNumpadValueToParentFormDelegate +=
eventhandler(this.setQtyValueFromNumpad);
}
```

The following example shows one way to call the numeric keypad pop-up dialog.

X++

```
Args args = new Args();
args.name(formstr(JmgProductionFloorExecutionNumpad));
args.menuItemName(menuItemDisplayStr(JmgProductionFloorExecutionNumpad));
args.caller(element);
Object formRun = classfactory.formRunClass(args);
formRun.init();
formRun.setNumpadController(numpadController);
numpadController.setValueToNumpad(333.56);
formRun.run();
```

Add a date and time controls to a form or dialog

This section shows how to add date and time controls to a form or dialog. The touch-friendly date and time controls enable workers to specify dates and times. The following screenshots

show how the controls typically appear on the page. The time control provides both 12-hour and 24-hour versions; the version shown will follow the preference set for the user account under which the interface is running.

The following procedure shows an example of how to add date and time controls to a form.

1. To add date and time controls, add a standard form part to the form. In the initialize method of the form part, set the target name to `JmgProductionFloorExecutionDateControl` for the date control or `JmgProductionFloorExecutionTimeControl` for the time control.

   X++

   ```
   public void initialize()
   {
       super();


   this.targetName(menuItemDisplayStr(JmgProductionFloorExecutionDateControl));
   }
   ```

   X++

   ```
   public void initialize()
   {
       super();


   this.targetName(menuItemDisplayStr(JmgProductionFloorExecutionTimeControl));
   }
   ```

2. Add a controller to the form for each date and time control that the form should contain. (The number of controllers must equal the number of date and time controls in the form.)

```
X++
private JmgProductionFloorExecutionDateTimeController  dateFromController;
private JmgProductionFloorExecutionDateTimeController  dateToController;
private JmgProductionFloorExecutionDateTimeController  timeFromController;
private JmgProductionFloorExecutionDateTimeController  timeToController;
```

3. Declare the required variables (of type `utcdatetime`).

```
X++
private utcdatetime fromDateTime;
private utcdatetime toDateTime;
```

4. Create methods where the datetime will be updated by the datetime controllers. The following example shows one such method.

```
X++
private void setFromDateTime(utcdatetime _value)
    {
        fromDateTime = _value;
    }
```

5. Set up the behavior of each datetime controller and connect each controller to a form part. The following example shows how to set up data for date-from and time-from controls. You could add similar code for date-to and time-to controls (not shown).

```
X++
/// <summary>
/// Initializes all date and time controllers, defines their behavior, and
connects them with the form parts.
/// </summary>
private void initializeDateControlControllers()
{
    dateFromController = new JmgProductionFloorExecutionDateTimeController();
    dateFromController.setDateControlValueToCallerFormDelegate +=
eventhandler(this.setFromDateTime);
    dateFromController.parmDateTimeValue(fromDateTime);

    timeFromController = new JmgProductionFloorExecutionDateTimeController();
    timeFromController.setDateControlValueToCallerFormDelegate +=
eventhandler(this.setFromDateTime);
    timeFromController.parmDateTimeValue(fromDateTime);
```

```
DateFromFormPart.getPartFormRun().setDateControlController(dateFromController,
timeFromController);

TimeFromFormPart.getPartFormRun().setTimeControlController(timeFromController,
dateFromController);

    ...

}
```

If all you need is a date control, you can skip the time control setup and instead just set up the date control as shown in the following example:

X++

```
{
    dateFromController = new JmgProductionFloorExecutionDateTimeController();
    dateFromController.setDateControlValueToCallerFormDelegate +=
eventhandler(this.setFromDateTime);
    dateFromController.parmDateTimeValue(fromDateTime);


DateFromFormPart.getPartFormRun().setDateControlController(dateFromController,
null);
}
```

6. Remember to unsubscribe from events when the form is closed.

X++

```
public void close()
{
   DateFromFormPart.getPartFormRun().close();
   TimeFromFormPart.getPartFormRun().close();

   dateFromController.setDateControlValueToCallerFormDelegate -=
eventhandler(this.setFromDateTime);
   timeFromController.setDateControlValueToCallerFormDelegate -=
eventhandler(this.setFromDateTime);

   super();
}
```

Execute a query on the main form

If you need to run a query to refresh or update the data on the main form after a specific action, use the following approach:

X++

```
parentForm.parmChangeEvent().raiseEvent();
```

Where `parentForm` is `JmgProductionFloorExecution` form.

Related information

- Style the production floor execution interface
- Design the production floor execution interface

Last updated on 10/29/2025

Rebate management module overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can use the **Rebate management** module to create contracts, deals, or agreements between your business and its customers or vendors, so that you can calculate rebates, deductions, and royalties. Rebate management tracks and maintains rebate and deduction transactions in a central location where users can effectively create, review, and process them.

Rebate management supports the creation, maintenance, and processing of *rebates*. A rebate is the return of part of the purchase price by a seller or a buyer. Rebates are typically based on the purchase of a specific quantity or value of goods during a specific period. Unlike discounts, rebates are done after the purchase amount is fully invoiced.

Rebate management also supports *deductions*. A deduction is compensation, a consideration, or a fee that is paid either for a license or the privilege to use intellectual property such as a brand, copyright, or patent, or for the privilege to use a natural resource for a purpose such as for fishing, hunting, or mining. Deductions are usually calculated as a percentage of the revenue or profit from realized use. The more the intellectual property or natural resource is used, the greater the deduction that is realized.

Additionally, Rebate management supports customer *royalties*. Royalties are payments that one party makes to the licensee or franchisee for the right to use an asset.

Rebate management lets you define posting profiles for provisions, rebates, and write-offs. Furthermore, the postings can be defined for a specific customer or vendor. In this way, deals that don't apply to all customer or vendor scenarios can be tracked via postings.

Rebate transactions are automatically generated, based on the calculation method that is selected and scheduled in batch jobs. Additionally, you can set up workflows to manage, review, and approve agreements.

Basis calculation

Customer rebates, customer royalties, and vendor rebates can all use a different basis, depending on your business requirements:

- Customer rebates can be based on sales orders, delivery notes, or invoices.
- Customer royalties can be based on sales orders, delivery notes, or paid invoices.
- Vendor rebates can be based on either purchase orders or sales orders. They can be calculated based on products that are purchased from the rebate vendor and sold to customers via sales invoices.

Flexible calculations

Rebate calculation periods are available for both customer deals and vendor deals. A calculation period defines the length of the deal, the calculation frequency, and the calculation period. The rebates can be accrued based on the sales order quantities or amounts for qualified products during the rebate period.

For each agreement calculation, any of the following periods can be set:

- Invoice
- Day
- Week
- Month
- Quarter
- Year
- Customized period
- Any multiple of any of the previously listed periods

The calculation can be applied to individual customers and products, groups of customers and products, or all customers and products. Rebates that have multiple detail lines can have different qualifying date ranges. The provision and claim periods can differ. For example, provisions can be processed every day, whereas claims are processed once per month.

Rebates can be configured based on many different parameters. For example, they can be configured as a percentage, a rate, or a fixed amount. There are four main calculation methods:

- Stepped
- Cumulative
- Rolling
- Total value

Rebate calculation results can also be reduced by other rebates, depending on whether the rebate is set up to calculate based on the net amount.

On the vendor side, rebates that are based on sales orders can calculate the price based on a first in, first out (FIFO) rule, the latest purchase price, the average purchase price, or the sales price.

Rebate target transactions

The outputs that the rebate deal or agreement generates can be financials or items.

Financial outputs are determined by the payment type that is assigned to the agreement from the posting profile. These outputs can include customer deduction journals, free text invoices, and vendor invoices. For auditing purposes, financial rebate target transactions include a reference to the originating rebate agreement.

Item outputs create a free item sales order for customer rebates and a purchase order for vendor rebates. Item rebate target transactions contain options to determine which rebate reference should be entered on the free item sales order or purchase order.

Accurate rebate calculations

The combination of the associated deals, the frequency of the calculations, the calculation basis, and the calculation method that is selected determines the accuracy and precision of rebate calculations. Rebate provisions can be used to accrue posted and claimed values.

Provisions can be managed daily, weekly, monthly, or according to a custom period. However, the functionality can allocate or pay the rebate, or receive payment of it, at any defined frequency that is the same length as or longer than the provision frequency. Write-off uses the same frequency as the rebate. Users can easily adjust a plan or payment amounts at any time during the payout.

Users no longer have to handle deals or provisions in two steps. Provisions and write-offs are posted directly to the ledger. Additionally, credit notes can be created automatically. Therefore, there's full integration with accounts payable and accounts receivable. During processing, the calculations can consider settlement discounts, paid invoices, trade discounts, and existing credit notes to ensure that amounts and values are accurately calculated.

When rebates are calculated, the process creates transactions that can be reviewed before posting occurs. A separate process posts rebate management transactions. A journal, credit note, or debit transaction can then be created during posting to proposed transactions. Reporting statements and transaction listings can be obtained to ensure compliance, effectiveness, and transparency.

Guaranteed royalty payments

In Rebate management, automatic payment generation enables royalties to be handled quickly and easily, even when guaranteed minimums apply.

Maximizing spend versus rebates

Vendors and products can be grouped by territory, and different offers can be provided, depending on the country/region of the transaction. When users select products, they can define the items that are included and the number of items that will be used in the rebate settlement.

Next steps

- Rebate management deals
- Process, review, and post rebates
- Manage deductions using the deduction workbench

Last updated on 08/05/2025

Rebate management parameters

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Rebate management parameters** page is used to define settings that apply across the **Rebate management** module. These settings affect posting, status updates, number sequences, and other behavior. The setup on this page is shared across legal entities and can be modified by users who have the appropriate security permissions.

To open the **Rebate management parameters** page, go to **Rebate management** > **Setup** > **Rebate management parameters**. Then set the fields as described in the following subsections.

Rebate management tab

The following table describes the fields that are available on the **Rebate management** tab of the **Rebate management parameters** page.

Expand table

Field	Description
Default status	Select the default status for all new deals. To define the set of status values that is available for selection, use the **Rebate statuses** page.
Process by dimension	Select whether provision, rebate, and write-off transactions should be processed by financial dimension. When this option is turned on, the system uses financial dimensions from the source transactions in the target transactions.
Check if previously posted	Select the system behavior if unposted rebate transactions are processed more than once for the same period: • **Warning** – The system allows users to override the original transactions lines, but a warning is shown. • **Error** – The system prevents users from overriding the original transactions lines, and an error message is shown.
Automatically post journals	Select whether the system should automatically post proposed journals. Those journals include daily journals that are used for provisions and customer deductions, and also vendor tax invoice journals.

Field	Description
Automatically post free text invoices	Select whether the system should automatically post free text invoices. This option applies only to free text invoices where the payment type is set to *Tax invoice customer deductions*.
Rebate item order reference	Select the rebate reference to use on sales orders and purchase orders that are generated from the rebate process (*None*, *Rebate management deal*, *Rebate management number*, *Rebate transaction number*, or *Document notes*).
Use claim process	Set this option to *Yes* to use a claims process. In this way, you can mark transactions that Rebate management creates as either claimed or unclaimed, and then post only the claimed transactions. For example, you calculate rebates for a month's worth of transactions, but the customer has left two days unclaimed. In this case, the unclaimed transactions will be re-created the next time that you run the *Process* function for the next period. If you set this option to *No*, all claim transactions are posted.
Include order type journal	For deals or deal lines where the transaction type is set to *Order*, this option controls whether a sales order of the *Journal* type should be included. It provides flexibility if those types of orders are used in scenarios where a rebate shouldn't yet apply.

Number sequences tab

Use the **Number sequences** tab on the **Rebate management parameters** page to assign number sequence codes to the different number sequences that Rebate management uses. The following table describes the purpose of each of those number sequences. For more information about number sequences, see Number sequences overview and its related articles.

Expand table

Reference	Description
Rebate management deal	The number sequence assigns a unique key value to each rebate deal. This key is used when deals are created.
Rebate management number	The number sequence assigns a unique key value to each rebate. This key is used to identify rebate relationships.
Rebate transaction number	The number sequence assigns a unique key value to each rebate transaction. This key is used to identify rebate transactions.
Tax Invoice	The number sequence assigns a unique key value to each rebate invoice. This key is used when rebate journals are automatically posted.

Feature visibility tab

Rebate management adds features (fields and functions) to several frequently used pages in Microsoft Dynamics 365 Supply Chain Management. Those pages include pages that are related to sales orders and sales deals. If you're using Rebate management, you must make those features visible everywhere before you can benefit from them. If you aren't using Rebate management, you can hide the features to keep them out of the way.

On the **Feature visibility** tab of the **Rebate management parameters** page, set the **Activate** option to *Yes* to make Rebate management features visible wherever they're available. Set it to *No* to hide the features on common pages outside the **Rebate management** module. However, even when the option is set to *No*, the module itself, including the **Rebate management parameters** page, will remain available to users who have the appropriate permissions to access it.

Last updated on 08/05/2025

Rebate management deal workflows

To approve rebate deals, Rebate management uses the same workflow platform as other finance and operations apps. Two job processes are associated with every workflow:

- One element of the workflow approves the deal.
- One element of the workflow activates the deal so that the user or workflow process can approve the transactions.

Before you can use a rebate deal, it must be active in the **Rebate management** module. To activate a deal, you must first create and configure a *Rebate management deal workflow*.

Users can't manually approve deals. The workflow must always be used.

Create and manage Rebate management deal workflows

To work with your Rebate management deal workflows, go to **Rebate management** > **Setup** > **Rebate management workflows**. There, you can view, create, and update workflows as required. Only one workflow of this type can be active at a time. For more information about workflows, how to work with the **Rebate management workflows** page, and how to create workflows, see Workflow system overview and its related articles.

Use a workflow to activate a deal

To activate a deal through a workflow, open the deal (for example, on the **All rebate management deals** page). Then, on the Action Pane, select **Workflow** > **Submit**. After the new deal has been processed and approved through the workflow, it will be active and ready to use.

After a deal has been activated, you can't change most of its setup. If you must change an active deal, first deactivate it, and then create a new deal. If the new deal should resemble the old deal, you can create it by copying the old deal.

You can change the following settings for a deal after it has been activated:

- Reconcile by
- Cumulative guarantee
- Posting profile

- Posting profile for guarantee
- Document notes
- Currency
- From date
- To date

In addition, rebate lines can be removed.

Resubmit a workflow

Sometimes, you might have to resubmit a workflow for a deal because of missing or incorrect data, an activated or updated rebate deal, modified rebate rules, or an extension of the original rebate period. To resubmit a workflow, you must first deactivate the active deal, then edit it as required, and finally reactivate it.

Prerequisites

Before you can deactivate and resubmit workflows for vendor rebate deals, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.43 or later.
- The feature named *Enable resubmission of workflows for vendor rebate deals* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

Deactivate and resubmit a workflow

To deactivate and resubmit a workflow for a vendor rebate deal, follow these steps:

1. Go to **Rebate management** > **Rebate management deals** > **All rebate management deals**.
2. Find and select the deal that you want to deactivate, edit, and resubmit. The deal must already be active.
3. On the Action Pane, on the **Rebate management deals** tab, select **Deactivate**.
4. Edit the deal as required. Here are some examples of changes that you might make:
    - Change the due date for one or more deal lines.
    - Add one or more new deal lines.

- Add a new rebate period for an existing deal line.

5. After you finish editing the deal, on the Action Pane, select **Workflow** > **Submit**.

The resubmission works for both customer rebates and vendor rebates.

Last updated on 03/25/2026

Rebate reduction principles

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can group Rebate management deals into *reduction principles* to reduce other provisions that are associated with an item, and/or to reduce the resulting values of rebate calculations. After the rebate reduction principles are set up, you can optionally assign them, as required, to the lines of your Rebate management deals.

The rebate reduction principle rules apply only when rebate deals overlap. Therefore, they might grant multiple rebates in situations where multiple rebates aren't owed.

Manage rebate reduction principles

To work with rebate reduction principles, go to **Rebate management** > **Setup** > **Rebate reduction principles**. Then use the buttons on the Action Pane to add and remove reduction principles as required. For each principle, set the fields as described in the following table.

Expand table

Field	Description
Rebate reduction principle	Enter a unique name for the rebate reduction principle.
Description	Enter a description of the rebate reduction principle.
Apply reduction	Select this check box to apply a reduction basis to rebates that belong to this rebate reduction principle.
Reduction basis	If you selected the **Apply reduction** check box, select the reduction basis (*Provision*, *Rebates*, or *Both*). If you select *Both*, and if both a provision and a rebate have been posted for a previous deal, only the rebate is applied.
Exclude from reduction	If this check box is selected, provisions and rebates that belong to this rebate reduction principle are excluded from subsequent reductions. The setting of the **Reduction basis** field doesn't apply to this setting.

Examples of rebate reduction principle setups

The following table shows some typical examples of rebate reduction principle setups. For each of these rebate reduction principles, the value of the **Description** field describes the purpose of the rebate reduction principle.

Expand table

Rebate reduction principle	Description	Apply reduction	Reduction basis	Exclude from reduction
Deferred	Reduce rebate	Yes	Both	No
Exclreb	Exclude rebate	Yes	Rebate	Yes
Multiple	Multiple rebates	Yes	Both	Yes
None	Prov and Rebate only	No	Both	Yes
Provision	Provision only	Yes	Provision	No
Rebate	Rebate only	Yes	Rebate	Yes

Examples of rebate reduction principle calculations

You have a sales order that has a single sales order line. This line has a value of $1,000. The following deals apply to the order:

- **Deal 1:** 10 percent on $1,000
- **Deal 2:** 15 percent on $1,000
- **Deal 3:** 20 percent on $1,000
- **Deal 4:** 25 percent on $1,000

The order of processing is very important. The processing order is based on the way that you work, as described in Process, review, and post rebates.

For example, the following table summarizes the result if you use the order *1, 2, 3, 4*, and if you process only the provisions.

Expand table

Deal	Description and settings	Provision result
1	The classification doesn't check other deals for reductions. The check is done for both provisions and rebates.	$1,000 × 10% = $100

Deal	Description and settings	Provision result
	• **Apply reduction:** *No* • **Reduction basis:** *Both* • **Exclude from reduction:** *No*	
2	The classification checks other deals for reductions but is flagged so that it itself is ignored. The check is done only for rebates. • **Apply reduction:** *Yes* • **Reduction basis:** *Rebate* • **Exclude from reduction:** *Yes*	$1,000 × 15% = $150
3	The classification checks other deals for reductions. The check is done for both provisions and rebates. • **Apply reduction:** *Yes* • **Reduction basis:** *Both* • **Exclude from reduction:** *No*	($1,000 – Deal 1) × 20% = $180
4	The classification checks other deals for reductions. The check is done for both provisions and rebates. • **Apply reduction:** *Yes* • **Reduction basis:** *Both* • **Exclude from reduction:** *No*	($1,000 – Deal 1 – Deal 3) × 25% = $180

The following table shows some examples where the provision is processed in different orders, and where different totals are therefore achieved. The variance in the provisions depends on the order that the system processes the deals in. It's important that you understand how the order of processing affects the final rebate amount.

Expand table

Sequence	Calculation
1 (1, 2, 3, 4)	• **Deal 1:** $1,000 × 10% = $100 • **Deal 2:** $1,000 × 15% = $150 • **Deal 3:** ($1,000 – Deal 1) × 20% = $180 • **Deal 4:** ($1,000 – Deal 1 – Deal 2) × 25% = $180 • **Total provision:** $610
2 (4, 3, 2, 1)	• **Deal 4:** $1,000 x 25% = $250 • **Deal 3:** ($1,000 – Deal 4) × 20% = $150 • **Deal 2:** $1,000 x 15% = $150

Sequence	Calculation
	• **Deal 1:** $1,000 x 10% = $100 • **Total provision:** $650
3 (3, 2, 1, 4)	• **Deal 3:** $1,000 x 20% = $200 • **Deal 2:** $1,000 x 15% = $150 • **Deal 1:** $1,000 x 10% = $100 • **Deal 4:** ($1,000 – Deal 3 – Deal 1) × 25% = $175 • **Total provision:** $625
4 (2, 4, 1, 3)	• **Deal 2:** $1,000 × 15% = $150 • **Deal 4:** $1,000 × 25% = $250 • **Deal 1:** $1,000 × 10% = $100 • **Deal 3:** ($1,000 – Deal 4 – Deal 1) × 20% = $130 • **Total provision:** $630

Last updated on 08/05/2025

Rebate statuses

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Rebate statuses help you to track the status of each Rebate management deal. You can also use them for reporting and inquiry purposes. You can set up and name as many rebate status values as you require to support your business process.

To set up rebate statuses, go to **Rebate management** > **Setup** > **Status** > **Status**. Then use the buttons on the Action Pane to add and remove status values as required. For each status, set the following fields:

- **Rebate status** – Enter a name for the status (such as *Negotiating*, *Agreed*, or *Completed*).
- **Description** – Enter a description of the status to provide more information about how it should be used.

On the **Rebate management parameters** page, you can select a default status. That status is then assigned to all new Rebate management deals that are created.

Last updated on 08/05/2025

Rebate management groups

07/21/2025

Rebate management calculations can be driven by groups. Rebate management groups can be created for customers, vendors, and items. They can be attached to a master record.

Rebate management customer groups

The calculation for a group of customers is often created for a chain of companies, such as a supermarket chain or a franchise company. This type of calculation is usually related to a rebate.

A deduction is often calculated without considering who the qualifying order was sold to. However, there are exceptions. For example, a licensee might create a deduction scheme where customer group A receives 4 percent, but customer group B receives only 3 percent.

Set up customer groups

To work with Rebate management customer groups, go to **Rebate management** > **Rebate management groups setup** > **Customer groups**. You can use the buttons on the Action Pane to add and remove groups as required. For each group, set the following fields:

- **Rebate management group** – Enter a unique name for the group.
- **Description** – Enter a description of the group to provide more information about how it should be used.

Manage customer group assignments

Each customer can belong to any number of Rebate management groups. To view and assign group members, you can start from either the group list or the customer list.

To view, add, or remove customers for a selected group, follow these steps.

1. Go to **Rebate management** > **Rebate management groups setup** > **Customer groups**.
2. Select the group to manage.
3. On the Action Pane, select **Customers**. The **Rebate management groups** page appears and shows a list of customers that are already members of the selected group.
4. To add a new customer to the group, select **New** on the Action Pane to add a row to the grid. Then set the following field for the new row:

- **Customer account** – Select the customer account ID.

5. To remove a customer from the group, select the customer, and then select **Delete** on the Action Pane.

To view, add, or remove group assignments for a selected customer, follow these steps.

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select the customer to work with.
3. On the Action Pane, on the **Customer** tab, in the **Rebate management** group, select **Rebate management groups**. The **Rebate management groups** page appears and shows a list of groups that the selected customer already belongs to.
4. To add the customer to a new group, select **New** on the Action Pane to add a row to the grid. Then set the following field for the new row:

    - **Rebate management group** – Select the group to add the customer to.

5. To remove a customer from a group, select the group, and then select **Delete** on the Action Pane.

Rebate management vendor groups

The calculation for a group of vendors is often created for a group of delivery points. This type of calculation is usually related to a rebate.

Set up vendor groups

To work with Rebate management vendor groups, go to **Rebate management** > **Rebate management groups setup** > **Vendor groups**. You can use the buttons on the Action Pane to add and remove groups as required. For each group, set the following fields:

- **Rebate management group** – Enter a unique name for the group.
- **Description** – Enter a description of the group to provide more information about how it should be used.

Manage vendor group assignments

Each vendor can belong to any number of Rebate management groups. To view and assign group members, you can start from either the group list or the vendor list.

To view, add, or remove vendors for a selected group, follow these steps.

1. Go to **Rebate management** > **Rebate management groups setup** > **Vendor groups**.
2. Select the group to manage.
3. On the Action Pane, select **Vendors**. The **Rebate management groups** page appears and shows a list of vendors that are already members of the selected group.
4. To add a new vendor to the group, select **New** on the Action Pane to add a row to the grid. Then set the following field for the new row:
    - **Vendor account** – Select the vendor account ID.
5. To remove a vendor from the group, select the vendor, and then select **Delete** on the Action Pane.

To view, add, or remove group assignments for a selected vendor, follow these steps.

1. Go to **Accounts payable** > **Vendors** > **All vendors**.
2. Select the vendor to work with.
3. On the Action Pane, on the **Vendor** tab, in the **Rebate management** group, select **Rebate management groups**. The **Rebate management groups** page appears and shows a list of groups that the selected vendor already belongs to.
4. To add the vendor to a new group, select **New** on the Action Pane to add a row to the grid. Then set the following field for the new row:
    - **Rebate management group** – Select the group to add the vendor to.
5. To remove a vendor from a group, select the group, and then select **Delete** on the Action Pane.

Item rebate groups

By grouping items, you have more flexibility when rebates and deductions are calculated. This approach is often a more efficient way to set up rebates and deductions for customers and vendors.

Set up item groups

To work with Rebate management item groups, go to **Rebate management** > **Rebate management groups setup** > **Item groups**. You can use the buttons on the Action Pane to add and remove groups as required. For each group, set the following fields:

- **Rebate management group** – Enter a unique name for the group.
- **Description** – Enter a description of the group to provide more information about how it should be used.

Manage item group assignments

Each item can belong to any number of Rebate management groups. To view and assign group members, you can start from either the group list or the item list. You can also add items based on your category hierarchy.

To view, add, or remove items for a selected group, follow these steps.

1. Go to **Rebate management** > **Rebate management groups setup** > **Item groups**.
2. Select the group to manage.
3. On the Action Pane, select **Items**. The **Rebate management groups** page appears and shows a list of items that are already members of the selected group.
4. To add a new item to the group, select **New** on the Action Pane to add a row to the grid. Then set the following field for the new row:
    - **Item account** – Select the item account ID.
5. To remove an item from the group, select the item, and then select **Delete** on the Action Pane.

To view, add, or remove group assignments for a selected item, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the item to work with.
3. On the Action Pane, on the **Product** tab, in the **Rebate management** group, select **Rebate management groups**. The **Rebate management groups** page appears and shows a list of groups that the selected item already belongs to.
4. To add the item to a new group, select **New** on the Action Pane to add a row to the grid. Then set the following field for the new row:
    - **Rebate management group** – Select the group to add the item to.
5. To remove an item from a group, select the group, and then select **Delete** on the Action Pane.

To add items to a group based on your category hierarchy, follow these steps.

1. Go to **Rebate management** > **Rebate management groups setup** > **Item groups**.
2. Select the group to manage.
3. On the Action Pane, select **Add items**.
4. In the **Add items** dialog box, in the **Category hierarchy** field, select a top-level category.
5. The item hierarchy is opened in the left pane. Select the hierarchy level that you're looking for.
6. Items from the selected hierarchy and hierarchy level are now listed in the right pane. Follow these steps to work with the pane:
    - Select the check box for each item that you want to add.
    - Select the check box on the grid header to select all the items that are listed.
    - Select the **Show selected** button to filter the grid so that it shows only the items that you've selected so far. Select the button again to return to the full list.
7. Select **OK** to apply your item selection to the selected group.

Rebate management posting setup

Rebate management posting profiles are used to determine posting entries for Rebate management calculation lines. When a posting profile is selected on the deal header, it applies to all deal lines.

This feature works across companies (legal entities). You can specify the company that provisions will be accrued to and that claims will be paid by. You can also set different provision debit accounts and write-off credit accounts per source posting company.

To set up Rebate management posting profiles for customers and vendors, go to **Rebate management** > **Rebate management posting setup** > **Rebate management posting profiles**. The **Rebate management posting profiles** page includes a list pane that shows all your existing profiles. You can use the buttons on the Action Pane to add profiles to the list or remove them.

The remaining sections of this article describe how to use the available fields when you create or edit a profile.

Create a profile that sends rebates to the original transaction account

You can configure rebate management posting profiles to send rebates as output to the original transaction account from the related sales order or purchase order. Before you can use this capability, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.43 or later.
- The feature named *Enable posting of vendor rebate outputs to purchase order vendors* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

To set up this functionality, select the appropriate value in the **Use account source** field of your posting profile, as described later in this article.

Posting profile header

The following table describes the settings that are available in the header section of each Rebate management posting profile.

Expand table

Field	Description
Posting profile	Enter a unique name for the profile.
Description	Enter a description of the profile.
Module	Select the module that the rebates and royalties of the profile are associated with (*Customer* or *Vendor*).
Type	Select the profile type (*Rebate* or *Royalty*).
Payment type	This field determines the format of the posted rebate output. When the **Type** field is set to *Rebate*, the following values are available: • *Pay using accounts payable* – When you post a customer rebate, a vendor invoice for the remittance vendor that is set up on the rebate customer is created. When you post a vendor rebate, a vendor invoice for the rebate vendor account is created. • *Customer deductions* – When you post the rebate, a customer deduction journal for the rebate customer is created. • *Tax invoice customer deductions* – When you post the rebate, a free text invoice for the rebate customer is created. • *Trade spending* – When you post the rebate, a customer deduction journal for the rebate customer is created. • *Reporting* – When you post the rebate, a customer deduction journal for the rebate customer is created. When the **Type** field is set to *Royalty*, the following values are available: • *Pay using accounts payable* – When you post the rebate, a vendor invoice for the rebate vendor account is created. • *Reporting* – When you post the rebate, a vendor invoice for the rebate vendor account is created. Learn more in the Payment types section that follows.
Company	Select the company (legal entity) that provisions will be accrued to and that claims will be paid by.

Payment types

The following table summarizes how the various settings of the **Payment type** field affect where payments are posted. Payments can be posted to a customer account, vendor account, or remittance account, depending on the target transaction and the rebate type.

Expand table

Payment type	Target transaction type	Rebate type	Payment to (invoice account)
Pay using accounts payable	Vendor invoice journal	Customer rebate	Customer's remittance vendor account
Pay using accounts payable	Vendor invoice journal	Customer royalty	Vendor account
Pay using accounts payable	Vendor invoice journal	Vendor rebate	Vendor account
Customer deductions	Daily journal	Customer rebate	Customer account
Tax invoice customer deductions	Free text invoice	Customer rebate	Customer account
Trade spending	Daily journal	Customer rebate	Customer account
Reporting	Daily journal	Customer rebate	Customer account
Reporting	Vendor invoice journal	Customer royalty	Vendor account
Reporting	Vendor invoice journal	Vendor rebate	Vendor account

Note

Consider the following points when you set up Rebate management deals:

- For deals where the **Reconcile by** field is set to *Deal*, you can't use the dynamic deal account during posting. You must use a specified customer or vendor account.
- For deals where the **Reconcile by** field is set to *Line*, you can use a posting profile that offsets to a dynamic deal account on the deal line, because the customer or vendor is set per deal line.

Posting FastTab

The following table describes the fields that are available on the **Posting** FastTab of each Rebate management posting profile.

Expand table

Field	Description
Credit type	Select whether to credit a ledger account or a customer. If the **Payment type** field on the header is set to *Tax invoice customer deductions*, this field is set to *Ledger account*. For vendor rebates, this field is set to *Ledger account*.
Credit account	Select the account that credit amounts are posted to when rebate provisions are made. This account will also be used as an offset account when the rebate is posted to credit the customer or debit the vendor.
Name of journal (In the **Provision** section)	Select the name of the journal to use to record the posted provision.
Type	Select whether to post the rebate to a ledger account, or to a customer or vendor. If the **Payment type** field on the header is set to *Tax invoice customer deductions*, this field is set to *Customer/Vendor*.
Use account source	Select one of the following values: • *Fixed account* – If you select this value, you must specify an account in the **Rebate account** field. • *Deal line account* – Use the customer or vendor account that is specified on the rebate line. You can select this value only for deals where the **Reconcile by** field is set to *Line* and deal lines where the **Account code** field is set to *Table*. It doesn't apply to customer royalty posting profiles or vendor rebates that are based on sales orders. • *Sold-to customer* – Send rebate output to the sold-to customer from the original sales order. This setting is valid only for profiles where the **Module** field is set to *Customer*. • *Purchase order vendor* – Send rebate output to the purchased-from vendor on the original purchase order. This setting is valid only for profiles where the **Module** field is set to *Vendor*.
Rebate account	If you set the **Use account source** field to *Fixed account*, specify the account that actual rebate expenses are posted to.
Name of journal (In the **Rebate management** field group)	Select the name of the journal to use to post a credit note for the rebate amount to the customer or vendor. This field is unavailable when the **Payment type** field on the header is set to *Tax invoice customer deductions*. For customer rebates, journal names of the *Daily* journal type will be available. For customer royalties and vendor rebates, journal names of the *Vendor invoice recording* journal type will be available.
Item sales tax group	Specify whether the rebate is taxable.
Name of journal (In the **Write off** field group)	If the rebate that is posted doesn't equal the provision, the difference can be written off. Select the name of the journal to use to record the posted write-off.

Posting by company FastTab

The **Posting by company** FastTab of each Rebate management posting profile lets you specify the posting account that is used by each company (legal entity) in the grid.

Use the buttons on the toolbar to add companies to the grid and remove them. Each time that you add a row to the grid, use the **Company** field to specify the legal entity for that row. The **Name** field is then set automatically.

Select the row for each company, and then enter the following information by using the fields below the grid:

- **Debit type** – Select whether to debit a ledger account or a vendor. For customer rebates and royalties, this field is set to *Ledger account*.
- **Debit account** – Enter the account that the debit amount is posted to when rebate provisions are made.
- **Main account** – Select the main account for write-offs.

Last updated on 07/27/2026

Rebate management deals

07/21/2025

Rebate management deals are used to control different methods and bases for calculating rebates and royalties. They include rules for inclusions and exclusions. There are three types of Rebate management deals: customer rebates, customer royalties, and vendor rebates. All three types use similar settings. This article points out differences where they exist.

Create a deal

1. Go to **Rebate management** > **Rebate management deals** > **All rebate management deals**. On the **All rebate management deals** list page, you can create and work with deals of all three types.

   Alternatively, follow one of these steps. In each case, the list page that appears provides all the same settings as the **All rebate management deals** list page, but it's filtered to show deals of only one type.

   - Go to **Rebate management** > **Rebate management deals** > **Customer rebate deals** to create only customer rebate deals.
   - Go to **Rebate management** > **Rebate management deals** > **Customer royalty deals** to create only customer royalty deals.
   - Go to **Rebate management** > **Rebate management deals** > **Vendor rebate deals** to create only vendor rebate deals.

2. On the Action Pane, select **New**.

3. In the **Create new deal** dialog box, set the following fields:

   - **Rebate management deal** – If you've set up a number sequence for rebate deals, this field is automatically set to a unique, system-generated ID. Otherwise, enter a unique ID.

   - **Description** – Enter a description of the deal.

   - **Module** – Select the type of partner that the deal applies to (*Customer* or *Vendor*). Depending on the page that you started from, this field might be automatically set based on the selected deal type. In this case, the field is read-only.

   - **Type** – Select the type of deal (*Rebate* or *Royalty*). Depending on the page that you started from, this field might be automatically set based on the selected deal type. In this case, the field is read-only.

- **Reconcile by** – Select how the deal should be calculated and reconciled:
  - *Deal* – A single rebate should be processed for all rebates and deductions in the deal.
  - *Line* – Rebates should be processed for each line in a deal.

- **Posting profile** – Select the posting profile to use for transactions where the deal applies. This field is available only when the **Reconcile by** field is set to *Deal*. If it's set to *Line*, you'll assign the profile later, for each line that you add to the deal.

- **Posting profile for guarantee** – Select the posting profile to use for guarantee transactions. Posting profiles are required for guarantee transactions only if the guarantee applies to a royalty. Otherwise, although a posting profile isn't mandatory, if no posting profile is specified, an error is shown when guarantees are posted. This field is available only when the **Type** field is set to *Royalty*.

- **Rebate output** – Select how the rebate or royalty should be paid:
  - *Financial* – Generate a free text credit note or accounts payable debit note, so that monies can be paid or received later. Provisions *can* be used with rebates where this option is selected. This option is the only available option when the **Type** field is set to *Royalty*.
  - *Item* – Generate a sales order for items according to the deal setup. On this sales order, a price of 0 (zero) is assigned to each item. Provisions *can't* be used with rebates where this option is selected.

- **Rebate currency** – Select the currency to use when the rebate, deduction, or royalty is paid.

- **Document notes** – Enter notes about the deal, as required.

- **Cumulative guarantee** – You can set this option to *Yes* only if the **Type** field is set to *Royalty*. This option affects the calculation of guaranteed deduction payments. Here's an example:
  - The guarantee of the deal is to sell a value that will lead to a payment of $10,000 per quarter.
  - The organization that is selling the goods sells a value that causes a deduction payment of $12,000 in quarter 1. The result is a $12,000 royalty that is paid, minus the $10,000 guarantee.
  - If the **Cumulative guarantee** option is set to *Yes*, the additional $2,000 of royalties that were paid in quarter 1 apply to quarter 2. Therefore, the customer is guaranteed $8,000 (the $10,000 guarantee minus the $2,000 additional payment).
  - In quarter 2, you register sales that trigger a $5,000 royalty.
  - The system identifies a $3,000 payment (the $8,000 guarantee minus the $5,000 of paid royalties).

    - If the **Cumulative guarantee** option is set to *No*, the full $10,000 are guaranteed each quarter. This guarantee corresponds to a suggested payment of $5,000 payment in quarter 2 (the $10,000 guarantee minus the $5,000 of paid royalties).

4. Select **OK** to create the deal and close the dialog box.

This procedure creates the header level of the new deal. Next, you'll add lines to the deal.

Add lines to a deal

After you've created a deal as described in the previous section, you can open it from the list page. You can then add lines to identify the customers or vendors, items, time frames, a basis, and calculation methods for the deal. A deal can have one or multiple lines. If a deal has multiple lines, they're generally of the same type, or the same parameters are associated with them. Until you add lines to a deal, the deal won't actually do anything.

1. Open the appropriate rebate deals list page, as described in the previous section.

2. Find and open the deal that you want to work with.

3. On the **Rebate management** FastTab, select one of the following buttons to add a deal line to the grid:

    - **Add line** – Add a new, blank deal line.
    - **Copy line** – If an existing deal line resembles the deal line that you want to add, you can select this button to copy add a copy of it. The new deal line will include all the settings from the related Rebate management details. You can then edit the settings. For example, you'll typically update the item relation and the rebate percentage.

4. On the new deal line, set the following fields:

    - **Rebate management number** – If you've set up a number sequence for Rebate management numbers, this field is automatically set to a unique, system-generated ID. Otherwise, enter a unique ID.

    - **Type** – The type of deal that the line applies to (*Rebate* or *Royalty*). The value is copied from the header and can't be changed.

    - **Description** – Enter a description of the deal line.

    - **Company** – Select the company (legal entity) that the deal line applies to. During processing, users can process only deal lines that are assigned to the company that they're currently using. The **Customer rebates deals** list page provides a multi-

company view, based on the security access of the user. However, you can edit and process rebates only for the company that you're currently using.

- **Account code** – Select the scope of customers or vendors that the deal line applies to:
  - *Table* – The deal line applies to a specific customer or vendor.
  - *Group* – The deal line applies to a group of customers or vendors. For more information about how to set up groups, see Rebate management groups.
  - *All* – The deal line applies to all customers or vendors.

- **Account relation** – If you selected *Table* in the **Account code** field, select the customer or vendor that the deal applies to. If you selected *Group*, select the customer or vendor group. If you selected *All*, this field is unavailable.

- **Item code** – Select the scope of items that the deal line applies to:
  - *Table* – The deal line applies to a specific item.
  - *Group* – The deal line applies to a group of items. For more information about how to set up groups, see Rebate management groups.
  - *All* – The deal line applies to all items.

- **Item relation** – If you selected *Table* in the **Item code** field, select the item that the deal line applies to. If you selected *Group*, select the item group. If you selected *All*, this field is unavailable.

- **Unit type** – Select the unit type that applies to the deal line (*Inventory unit* or *Catch weight unit*). This field might be blank for older records. In this case, the *Inventory unit* value is assumed.

- **(Inventory management parameters)** – In the remaining fields on the deal line, specify values for the inventory management parameters that will be used to define the items that are included in the deal (such as the item size, color, style, site, and warehouse). To add or remove the dimensions, select **Display dimensions** on the Action Pane.

5. On the Action Pane, select **Save**.

6. If you want to further limit the set of items that a deal line applies to, select the line, and then, on the **Rebate management** FastTab, select **Filter** to open a standard **Inquiry** dialog box.

7. For each deal line that you add, set up the calculation details and other details on the **Rebate management details** FastTab, as described in the next section.

Add Rebate management details to a deal line

For each line in your deal, you must set up details on the **Rebate management details** FastTab. First select the deal line on the **Rebate management** FastTab. Then set values for that deal line by using the various tabs on the **Rebate management details** FastTab. The following subsections describe how to use each tab.

Settings on the General tab

The **General** tab on the **Rebate management details** FastTab lets you set up the calculation method and basis for the selected deal line. This setup controls whether a minimum purchase is required, the posting profiles that are associated with the deal line, and the details of the calculation. The following table describes the fields that are available.

Expand table

Field	Description
Calculation method	Select the method to use when the selected deal line is combined with other deal lines (*Stepped*, *Cumulative*, *Rolling*, or *Total*). The value of this field can dramatically affect the outcome of your rebate calculations. For a full description of each method and examples that show how it affects the rebate calculation, see the Calculation methods for deal lines section later in this article.
Basis	Select whether the rebate is applied based on quantity (that is, the total number of units that are purchased or sold) or value (that is, the total price of goods that are purchased or sold).
Transaction type	Select the point in the process when the calculation should occur: • *Order* – Use the ordered quantity or value as the basis for the calculation. • *Delivered* – Use the delivered quantity or value as the basis for the calculation. • *Invoice* – Use the invoiced quantity or value as the basis for the calculation.
Unit	If you selected *Quantity* in the **Basis** field, select the unit that the quantity must be specified in.
Minimum amount	Enter the minimum amount that must be paid per period. You can enter a negative value to allow for negative rebate amounts if credit notes exceed sales for the period.
Rebate reduction principle	Select the rebate reduction principle that applies to the deal line.
Variant number	If the deal line applies to an item that has variants, select the variant that the deal line applies to.

Field	Description
Vendor rebate basis	This field is shown only for vendor rebates (that is, deals where the **Module** field is set to *Vendor*). Select the basis for the vendor rebate: • *Purchase order* – The rebate that the vendor receives is based on the quantity or value on the purchase order. • *Sales order* – The vendor receives a rebate only after the goods have been sold. The rebate is based on the quantity or value on the sales order.
Price basis	This field is shown only for vendor rebates (deals where the **Module** field is set to *Vendor*). If you selected *Sales order* in the **Vendor rebate basis** field, select the applicable price basis: • *FIFO* – The **Calculate FIFO purchase price** periodic task must be run to calculate the rebate. (To run the task, go to **Rebates management** > **Periodic tasks** > **Calculate FIFO purchase price**.) A first in, first out (FIFO) rule will be used to calculate the value of the stock that is sold. This value will then be used to calculate the vendor rebates. Here's an example: ◦ **Stock sold:** 1,000 units at $3.00 each = $3,000 ◦ **Current purchase price, based on FIFO:** $2.00 ◦ **Working calculation:** *Units sold* × *Current purchase price* = 1,000 × $2.00 = $2,000 • *Latest Purchase Price* – The value from the last purchase transaction will be used to calculate the vendor rebates. Here's an example: ◦ **Stock sold:** 1,000 units at $3.00 each = $3,000 ◦ **Latest purchase price:** $2.00 ◦ **Working calculation:** *Units sold* × *Latest purchase price* = 1,000 × $2.00 = $2,000 • *Average Purchase Price* – The weighted average value for the last *X* months will be used to calculate the vendor rebates. If you select this option, you must enter the number of months in the **Number of months** field (which is available only when the **Price basis** field is set to *Average Purchase Price*). Here's an example: ◦ **Stock sold:** 1,000 units at $3.00 each = $3,000 ◦ **Average purchase price:** $2.00 ◦ **Working calculation:** *Units sold* × *Average purchase price* = 1,000 × $2.00 = $2,000 • *Sales Price* – The sales price will be used to calculate the vendor rebates. Here's an example: ◦ **Stock sold:** 1,000 units at $3.00 each = $3,000 ◦ **Average purchase price:** $2.00 ◦ **Working calculation:** *Units sold* × *Sales price* = 1,000 × $3.00 = $3,000
Number of months	This field is shown only for vendor rebates (deals where the **Module** field is set to *Vendor*). If you selected *Average Purchase Price* in the **Price basis** field, enter the number of months that should be used.
Posting profile	Select the posting profile that applies to the selected deal line. This profile is used only when the **Reconcile by** field on the deal header is set to *Line*. If the **Reconcile by** field is

Field	Description
	set to *Deal*, the posting profile that is set on the deal header is always used. If no posting profile is set on the deal line, the posting profile that is set on the deal header is used.
Posting profile for guarantee	This field works like the **Posting profile** field, but it applies only to royalties.
Payment type	The payment type for the posting profile that is selected for the deal.
Sales tax inclusive	Select whether the transaction is tax-inclusive. Tax inclusiveness is relevant only when the **Basis** field is set to *Value*. The invoice amount will be used as the tax-inclusive amount. If the rebate calculation is based on a percentage, the system will multiply the rebate percentage by the tax-inclusive amount. The calculation uses the sales tax value from the deal line.
Include credit notes	Set this option to *Yes* to include credit notes in the rebate calculation. If you set this option to *Yes*, the effect varies, depending on the value of the **Transaction type** field: • If the **Transaction type** field is set to *Invoice*, the calculation will factor in credit notes. This configuration is the usual configuration. • If the **Transaction type** field is set to *Order*, the calculation will factor in both sales orders that have negative values and open returned orders.
Discounted amount	Set this option to *Yes* to base the calculation on the discounted amount of the item or items in cases where deal line discounts are given.
Paid invoices only	Set this option to *Yes* if the calculation should consider only fully paid invoices. (In other words, rebates won't be calculated for partially paid invoices.) This option is available only when the **Transaction type** field is set to *Invoice*.
Include free text	Set this option to *Yes* to include free text invoices in the calculation. Free text invoices can be included only for deal lines where the **Item code** field is set to *All*.
Include settlement discount	Set this option to *Yes* to reduce the rebate by the first settlement discount. It's assumed that the organization will take the first settlement discount. Set this option to *No* to enable the settlement discount to be used later.
Document notes	Enter notes that should be used as transaction text on rebate transaction journal lines.

Settings on the Dates tab

The settings on the **Dates** tab on the **Rebate management details** FastTab define the frequency and timing of the calculations that are specified on the **General** tab. They determine how calculations occur at processing time.

This tab lets you specify the range of dates that the selected deal line is valid for and the calculation frequency. You can also specify whether the guarantee should be posted, and when.

You can use the buttons on the toolbar to add date lines to the grid or remove them. If multiple date lines exist, the valid date ranges must not overlap. Otherwise, you'll receive an error message when you try to save the deal.

The following table describes the fields that are available for each date line.

Expand table

Field	Description
From date	Enter the first date that the date line applies to. If "from" and "to" dates are specified on the header of the deal, they're used as default values for every new date line.
To date	Enter the last date that the date line applies to. If "from" and "to" dates are specified on the header of the deal, they're used as default values for every new date line.
Per	Specify how often the deal line should be calculated. Enter an integer here, and then select a unit in the **Cumulate by** field. For example, to calculate every other week, set the **Per** field to *2* and the **Cumulate by** field to *Weeks*.
Cumulate by	Select the unit that applies to the **Per** setting. Select *Lifetime* to calculate over the whole lifetime of the deal line. Select *Customized period* to select a period that is defined in the general ledger. In this case, you must also set the **Period type** field.
Period type	This field is available only when the **Cumulate by** field is set to *Customized period*. The values that are available for selection come from the period types that are defined in the general ledger.
First day of the week	Specify how the weeks are identified for the period. This field is available only when the **Cumulate by** field is set to *Weeks*.
Claim per	Specify how often rebate monies can be claimed for the deal line. Enter an integer here, and then select a unit in the **Claim by** field.
Claim by	Select the unit that applies to the **Claim per** setting. Select *Lifetime* to allow claims just once during the whole lifetime of the deal line. Select *Customized period* to select a period that is defined in the general ledger. In this case, you must also set the **Claim period type** field.
Claim period type	This field is available only when the **Claim by** field is set to *Customized period*. The values that are available for selection come from the period types that are defined in the general ledger.
Process at posting	Select this check box if the claim line should be processed at posting time. This option is available only for deal lines where the **Transaction type** field on the **General** tab is set to *Delivered* or *Invoice*. For provisions, the provision will be posted when the delivery note or invoice is generated.

Field	Description
Guarantee per	This field applies only to royalty deals. Specify how often the guarantee of the royalty should be calculated during the period that is defined by the **Cumulate by** setting. Enter an integer here, and then select a payment time in the **Guarantee paid** field.
Guarantee paid	This field applies only to royalty deals. It works with the **Guarantee per** field to define the frequency of the guarantee calculation. Select one of the following values: • *Beginning of period* – The guarantee is paid at the beginning of the agreement period that is defined for the date line. The full guarantee is processed first. Only the value of royalties that exceed the guaranteed amount is posted as a royalty. Here's an example: ◦ **Guarantee minimum:** $10,000 over two months. ◦ **Month 1:** $10,000 were posted as a guarantee, and $0 in royalties were posted. ◦ **Month 2:** $2,000 were posted to royalties, and $0 in guarantees were posted. ◦ **Working calculation:** *Remaining guarantee* – *Royalties posted* = $0 – $2,000 = -$2,000. Because a royalty payment of $2,000 is required, a journal is created for $2,000. **Note:** The guarantee should be calculated and posted together with the deductions provision for the first period. • *End of period* – The guarantee isn't paid until the end of the deductions agreement, as defined on the date line. Here's an example: ◦ **Guarantee minimum:** $10,000 over two months. ◦ **Month 1:** $5,000 were posted as a royalty, and a journal was created. ◦ **Month 2:** $7,000 were posted as a royalty, and a journal was created. ◦ **Working calculation:** Because the royalties exceed the guarantee amount, $0 are posted to the guarantee. **Note:** The guarantee should be calculated and posted only together with the deductions and rebates transaction for the final period.
Guarantee minimum	This field applies only to royalty deals. Enter the minimum amount of the guarantee for a royalty, in the currency that is defined on the deal header.

Settings on the Lines tab

The **Lines** tab of the **Rebate management details** FastTab lets you define calculation lines for the selected deal line. Each calculation line establishes a quantity or value threshold, and a related compensation amount or items. The **Basis** field on the **General** tab specifies whether each calculation line is based on a quantity or a value.

You can use the buttons on the toolbar to add calculation lines to the grid or remove them. You can have any number of calculation lines. However, if multiple calculation lines exist, the "to" and "from" quantity or value ranges must not overlap.

The following table describes the fields that are available for each calculation line.

Expand table

Field	Description
From qty/value	Enter the minimum quantity or value that the calculation line applies to.
To qty/value	Enter the maximum quantity or value that the calculation line applies to.
Rebate management method	This field is available only for deals where the **Rebate output** field on the header is set to *Financial*. Select the method to use to calculate the rebate: • *Fixed amount off* – A fixed price amount is used for the line. • *Percentage off* – A percentage of the sale amount is used. • *Amount off* – A fixed price amount per order is used. **Important:** We strongly recommend that you use the same method for every calculation line on the **Lines** tab. If a new method is required, create a new deal line. Remember that you can use the **Copy line** button on the **Rebate management** FastTab to create a new deal line from an existing deal line. **Note:** If the **Rebate output** field is set to *Item*, this field is always set to *Fixed amount off*. For more information about how to specify the items, see the text that follows this table.
Rebate management amount	This field is available only for deals where the **Rebate output** on the header is set to *Financial*. Enter the amount that should be used to calculate the rebate, based on the method that you selected in the **Rebate management method** field.

As was mentioned in the previous table, for deals where the **Rebate output** field on the header is set to *Item*, you must specify the items that will be provided for every calculation line on the **Lines** tab. To specify the items, follow these steps.

1. On the **Lines** tab, create the required calculation line if it doesn't exist, and select it.
2. If the deal hasn't been saved since you created the calculation line, select **Save** on the Action Pane.
3. On the **Lines** tab, select **Items**.
4. On the **Items** page, use the buttons on the Action Pane to add items to the grid, or remove items, as required. For each row, set the following fields:
    - **Item number** – Select the item that will be provided.
    - **(Other dimensions)** – If you must use more dimensions to define the item (for example, item configuration, size, color, style, site, or warehouse), specify them. To add or remove available dimensions, select **Display dimensions** on the Action Pane.

- **Quantity** – Enter the quantity that will be provided after the threshold for the selected calculation line is met.
- **Unit** – Select the unit that the **Quantity** value is specified in.
- **Multiple** – This field works with the **Quantity** field. For example, on an item line, you set the **Quantity** field to *2* and the **Multiple** field of *100*. If you then have a total sales order quantity of 150, two of the items will be free (two per 100).

Settings on the Inventory dimensions tab

The **Inventory dimensions** tab on the **Rebate management details** FastTab shows all the available dimension values for the product that is specified for the selected deal line. It includes dimensions that aren't shown on the **Rebate management** FastTab. You can edit values only for those dimensions that apply to the selected product.

You can add more dimensions to the grid on the **Rebate management** FastTab by selecting **Display dimensions** on the Action Pane.

Calculation methods for deal lines

Every time that you set up details for one of your deal lines, as described in the previous section, you must select the calculation method for that line. This section explains each calculation method and provides examples that show how each method calculates the total rebate for orders and deal lines. In these examples, the orders and deal lines are identical. Only the calculation methods differ.

Stepped calculation method

The stepped calculation method calculates on a step-by-step basis, where each deal line incrementally contributes to the rebate until the sales quantity or value is reached. The steps can be based on either the quantity or the value of the goods that are sold, depending on the setting of the **Basis** field on the **General** tab of the **Rebate management details** FastTab.

For example, a deal line is set up so that the **Calculation method** field is set to *Stepped* and the **Basis** field is set to *Value*. It includes calculation lines that provide the following rebates:

- **Line A:** 10 percent up to $1,000
- **Line B:** 25 percent up to $2,500

If the value of the goods that were purchased or sold is $2,000, the resulting rebate of $350 is calculated in the following way:

- **Rebate (A):** *Threshold (A)* × *Percentage (A)* = $1,000 × 10 percent = $100

- **Rebate (B):** (*Value sold – Threshold (A)*) × *Percentage (B)* = ($2,000 – $1,000) × 25 percent = $250
- **Total rebate:** *Rebate (A)* + *Rebate (B)* = $100 + $250 = $350

If the **Basis** field is set to *Quantity* instead of *Value*, the stepped calculation method works in a similar way. The first step is used for the identified quantity until the second step is reached. The second step is then used for all quantity above the first step, until the third step is reached. This process then continues through all subsequent steps.

Cumulative calculation method

The cumulative calculation method uses just one calculation line to calculate the rebate. If multiple calculation lines are available for the deal line, the highest-value or highest-quantity calculation line that is reached is used for the whole quantity or value. Like the other calculation methods, the cumulative method uses the setting of the **Basis** field on the **General** tab of the **Rebate management details** FastTab to determine whether the sales quantity or the sales value is used to calculate the rebates.

For example, a deal line is set up so that the **Calculation method** field is set to *Cumulative* and the **Basis** field is set to *Value*. It includes calculation lines that provide the following rebates:

- **Line A:** 10 percent up to $1,000
- **Line B:** 25 percent up to $2,500

If the value of the goods that were purchased or sold is $2,000, the calculation uses only line B. The resulting rebate of $500 is calculated in the following way:

- **Total rebate:** *Value sold* × *Rebate (B)* = $2,000 × 25 percent = $500

Rolling calculation method

The rolling calculation method calculates all possible calculation lines for the deal. If there are multiple calculation lines, and more than one of them is reached, all the calculation lines that are reached will be used, but upper thresholds apply to each percentage. Like the other calculation methods, the rolling method uses the setting of the **Basis** field on the **General** tab of the **Rebate management details** FastTab to determine whether the sales quantity or the sales value is used to calculate the rebates.

For example, a deal line is set up so that the **Calculation method** field is set to *Rolling* and the **Basis** field is set to *Value*. It includes calculation lines that provide the following rebates:

- **Line A:** 10 percent up to $1,000
- **Line B:** 25 percent up to $2,500

If the value of the goods that were purchased or sold is $2,000, the resulting rebate of $600 is calculated in the following way:

- **Rebate (A):** *Threshold (A)* × *Percentage (A)* = $1,000 × 10 percent = $100
- **Rebate (B):** *Value sold* × *Percentage (B)* = $2,000 × 25 percent = $500
- **Total rebate:** *Rebate (A)* + *Rebate (B)* = $100 + $500 = $600

Total calculation method

The total calculation method uses all calculation lines to calculate the rebate. It applies the total quantity to calculate the rebate for each calculation line that is reached, and each calculation line applies its percentage to the full amount. Like the other calculation methods, the total method uses the setting of the **Basis** field on the **General** tab of the **Rebate management details** FastTab to determine whether the sales quantity or the sales value is used to calculate the rebates.

For example, a deal line is set up so that the **Calculation method** field is set to *Total* and the **Basis** field is set to *Value*. It includes calculation lines that provide the following rebates:

- **Line A:** 10 percent up to $1,000
- **Line B:** 25 percent up to $2,500

If the value of the goods that were purchased or sold is $2,000, the resulting rebate of $700 is calculated in the following way:

- **Rebate (A):** *Value sold* × *Percentage (A)* = $2,000 × 10 percent = $200
- **Rebate (B):** *Value sold* × *Percentage (B)* = $2,000 × 25 percent = $500
- **Total rebate:** *Rebate (A)* + *Rebate (B)* = $200 + $500 = $700

Process, review, and post rebates

This article describes how to process your Rebate management deals, calculate their discounts, review the transactions that are generated, post transactions, and review the postings.

Change the status of a deal

You can assign a status to each deal to help track it. The status is for informational purposes only.

1. Select a deal (for example, on the **All rebate management deals** page).
2. On the Action Pane, on the **Rebate management deals** tab, in the **Maintain** group, select **Change status**.
3. In the **Change status** dialog box, set the **Rebate status** field to a new status.
4. Select **OK**.

Calculate FIFO purchase prices

The **Calculate FIFO purchase price** periodic task must be run to calculate the rebates for deals where the **Price basis** field is set to *FIFO*. The system will use a first in, first out (FIFO) rule to calculate the value of the stock that was sold. This value will then be used to calculate the vendor rebates.

Go to **Rebates management** > **Periodic tasks** > **Calculate FIFO purchase price**. In the dialog box that appears, select **OK** to run the calculation.

Create source transactions

You can create the sales orders or purchase orders that have source transactions either before or after you create an applicable Rebate management deal.

You can set up each deal line so that it automatically creates a rebate provision by posting the delivery or invoice for a sales order or purchase order. Set the **Transaction type** field for the deal line to *Delivery* or *Invoice*, and set the **Process at posting** option to *Yes*. If the **Transaction type** field is set to *Order*, processing at posting is disabled. For source transactions that were created after a deal was activated, you can still process the provision as described in the Process Rebate management deals section later in this article.

Enable price details

Before you can create source transactions, you must enable the **Enable price details** option for account receivables.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Prices** tab, on the **Price details** FastTab, set the **Enable price details** option to *Yes*.

Create a source transaction

To create a source transaction, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New**.

   To mimic the way that rebate claims are generated, you must now create a sales order where the product and quantity qualify the customer for a rebate.
3. In the **Customer account** field, enter or select a customer that will qualify for a rebate deal.
4. Select **OK** to create the sales order.
5. On the **Sales order lines** FastTab, add a line, and set the following fields for it:

   - **Item number** – Specify an item that qualifies for a rebate.
   - **Quantity** – Specify a quantity that qualifies for a rebate deal that includes a line where the **Basis** field is set to *Quantity*.
   - **Unit price** – Specify a price that qualifies for a rebate deal that includes a line where the **Basis** field is set to *Value*.
   - **Site** – Select a site where the product is available, and that qualifies for a rebate deal.
   - **Warehouse** – Select a warehouse where the product is available, and that qualifies for a rebate deal.
6. On the **Sales order lines** FastTab, on the toolbar, select **Sales order line** > **Price details**. This command is available only if you enabled price details as described in the previous section.
7. On the **Price details** page, select the **Rebate management** FastTab. This FastTab lists all the rebate management deals that apply to the current order line and shows the estimated rebate amount in the order's currency. The amounts are only estimates of the future rebate claims. The actual rebate amounts might differ. Here are some of the factors that might affect the actual amounts:

   - The total sales volume that the customer achieved under a periodic rebate agreement.

- Whether the customer returned all quantities or partial quantities.
- Whether the applicable sales order achieved the transaction type (*Order, Delivery*, or *Invoice*) that is defined for the Rebate management deal.
- The deal's **Output** value. A blank rebate amount will be shown for deals where the **Output** field is set to *Item*.

8. On the **Detail** FastTab, notice that the **Margin estimation** section includes the following fields. These fields are added by Rebate management. All of them show values per unit (whereas the fields on the **Rebate management** FastTab show total values for the line).

    - **Rebate management rebate amount** (sales orders only)
    - **Rebate management royalty amount** (sales orders only)
    - **Rebate management vendor rebate amount** (sales orders and purchase orders)

9. Close the **Price details** page.

10. If the sales order shouldn't qualify for the rebates that you just viewed, follow these steps to exclude rebates. (However, you usually won't exclude rebates.)
    a. On the **Sales order lines** FastTab, select the relevant line.
    b. On the **Line details** FastTab, on the **Price and discount** tab, set the **Exclude from rebate management** option to *Yes*. This option doesn't apply to purchase orders. Furthermore, only customer rebates are excluded when this option is set to *Yes*. Customer royalty rebates and vendor rebates still apply.

11. On the Action Pane, on the **Pick and pack** tab, in the **Generate** group, select **Post packing slip**.

12. On the **Parameters** FastTab, in the **Quantity** field, select *All*.

13. Select **OK**.

14. If you're prompted to confirm the operation, select **OK**.

15. On the Action Pane, on the **Invoice** tab, in the **Generate** group, select **Invoice**.

16. On the **Parameters** FastTab, in the **Quantity** field, select *All* or *Packing slip*.

17. Select **OK**.

18. If you're prompted to confirm the operation, select **OK**.

Process Rebate management deals

When you process a deal, the system calculates all relevant rebates and royalties that are set up. Only selected deals that have calculation periods that are ready to be calculated and that have a status of *Active* will be processed. After processing is completed, the system generates a set of transactions that you can review and then post.

> ⓘ **Note**
>
> In all cases, rebates are processed at the level that is specified by each deal's **Reconcile by** setting (*Deal* or *Line*). You can do single-line calculations only for deals where the **Reconcile by** field is set to *Line*.

Process all lines for one or more deals

1. Open the appropriate rebate deals list page for the type of deal that you want to work with.
2. Select the row for each deal that you want to process (or open the deal that you want to process).
3. On the Action Pane, on the **Rebate management deals** tab, in the **Generate** group, select one of the following commands:
    - **Process** > **Provision** – Provision a set of accruals for each relevant rebate deal, but don't post them. This menu item is unavailable for deals where the **Rebate output** field is set to *Item*.
    - **Process** > **Rebate management** – Process a series of transactions that provide the value of the rebate for each deal.
    - **Process** > **Write off** – For each source transaction for the rebate deal and specified period, process the variance between the amounts that were posted for a provision and for rebate management. This menu item is unavailable for deals where the **Rebate output** field is set to *Item*.
4. In the dialog box that appears, set the **From date** and **To date** fields to define the date range for the calculation.
5. Select **OK** to run the calculation.

Process one or more specific deal lines for a selected deal

1. Open the appropriate rebate deals list page for the type of deal that you want to work with.

2. Open the deal that you want to process a line from.

3. Select the **Lines** tab in the upper-right corner.

4. On **Rebate management** FastTab, select the row for each deal line that you want to process.

5. On the toolbar of the **Rebate management** FastTab, select one of the following commands. (These commands are available only for deals where the **Reconcile by** field is set to *Line*.)

    - **Process** > **Provision** – Provision a set of accruals for each relevant deal line, but don't post them. This menu item is unavailable for deals where the **Rebate output** field is set to *Item*.
    - **Process** > **Rebate management** – Process a series of transactions that provide the value of the rebate for each deal line.
    - **Process** > **Write off** – For each source transaction for the rebate deal and specified period, process the variance between the amounts that were posted for a provision and for rebate management. This menu item is unavailable for deals where the **Rebate output** field is set to *Item*.

6. In the dialog box that appears, set the **From date** and **To date** fields to define the date range for the calculation.

7. Select **OK** to run the calculation.

Process deals using a batch job

Instead of processing specific deals or deal lines, you can run a batch job to process several deals at the same time. You can optionally apply record filters and/or set up a recurring schedule. To process deals by using a batch job, follow these steps:

1. Follow one of these steps:

    - Go to **Rebate management** > **Periodic tasks** > **Process** > **Provision** to provision a set of accruals for each relevant deal, but without posting them.
    - Go to **Rebate management** > **Periodic tasks** > **Process** > **Rebate management** to process a series of transactions that provide the value of the rebate for each deal.
    - Go to **Rebate management** > **Periodic tasks** > **Process** > **Write off** to reverse previously posted transactions to write them off so that new rebate deal transactions can be calculated.

2. In the dialog box that appears, on the **Parameters** FastTab, in the **Period** section, set the **From date** and **To date** fields to define the date range for transactions for the calculation.

3. In the **Guarantee Period** section, set the **From date** and **To date** fields to define the date range for guarantees for the calculation.

4. On the **Records to include** FastTab, you can set up filters to limit the set of deals that the batch job will process. These settings work in the same way that they work for other types of batch jobs.

5. On the **Run in the background** FastTab, you can set up batch processing and scheduling options as required. These settings work in the same way that they work for other types of batch jobs.

6. Select **OK** to run and/or schedule the calculation.

Process deals by using the rebate workbench

Instead of processing specific deals or deal lines, you can use the *rebate workbench* to process multiple deals at the same time. You can optionally apply record filters and/or set up a recurring schedule. You don't have to select any rows. The system will process all lines that meet the date and filter requirements that you set up.

To process deals by using the rebate workbench, follow these steps:

1. Go to **Rebate management** > **Rebate management deals** > **Rebate workbench**.

2. On the Action Pane, on the **Rebate workbench** tab, in the **Processing** group, select one of the following commands:

    - **Process** > **Provision** – Provision a set of accruals for each relevant deal line, but don't post the accruals.
    - **Process** > **Rebate management** – Process a series of transactions that provide the value of the rebate for each deal line.
    - **Process** > **Write off** – Process the variance between provision and rebate management that is posted for each source transaction per rebate deal and specified period.

3. In the **Rebate management** dialog box, in the **Period** section, set the **From date** and **To date** fields to define the date range for the calculation.

4. In the **Guarantee period** section, set the **From date** and **To date** fields to define the date range for guarantees for the calculation.

5. On the **Records to include** FastTab, you can set up filters to limit the set of deals that the batch job will process. These settings work in the same way that they work for other types of batch jobs.

6. On the **Run in the background** FastTab, you can set up batch processing and scheduling options as required. These settings work in the same way that they work for other types of batch jobs.

7. Select **OK** to run and/or schedule the calculation.

View and edit Rebate management transactions

When you process one or more deals, the system creates transactions that you can view and, perhaps, edit before you post them.

View and edit Rebate management transactions by using the rebate deals list page

To view and edit Rebate management transactions using the rebate deals list page, follow these steps:

1. Follow one of these steps:

    - Select the deal to view transactions for (for example, on the **All rebate management deals** page). On the Action Pane, on the **Rebate management deals** tab, in the **Transactions** group, select **Transactions** or **Guarantee transactions**, depending on the type of transaction that you want to view.
    - Open a deal (for example, on the **All rebate management deals** page) to view its details. On the **Rebate management** FastTab, select the deal line to view transactions for. Then, on the toolbar, select **Transactions** or **Guarantee transactions**, depending on the type of transaction that you want to view. (These buttons are available only for deals where the **Reconcile by** field is set to *Line*.)

2. The **Rebate management date transactions** or **Rebate management guarantee transactions** page appears. It contains a grid, where each line represents a collection of transactions from a single rebate or guarantee period. Select a row in the grid, and then, on the Action Pane, select **Source transactions** to view the transactions that belong to that row.

3. The page that appears shows a list of the transactions of the selected type that belong to the selected row. Each transaction provides relevant details, depending on the transaction

type. For guarantee transactions, you can only view the list. For other types of transactions, you can perform the following actions on this page:

- To verify the total value of all claimed transactions on the page, view the **Claimed amount** field.
- To view more information about any transaction, select it, and then select the **General**, **Financial dimension**, or **Dimension** tab.
- To view any reductions that apply, select **Reduction transactions** on the Action Pane. Learn more in Rebate reduction principles.
- To mark transactions as either claimed or unclaimed if you're using a claims process, select the relevant rows, and then, on the Action Pane, select one of the following commands. (You enable claims processes on the **Rebate management parameters** page.)
  - **Set claimed** > **All** – Mark all transactions as claimed.
  - **Set claimed** > **Selected** – Mark the selected transactions as claimed.
  - **Set unclaimed** > **All** – Mark all transactions as unclaimed.
  - **Set unclaimed** > **Selected** – Mark the selected transactions as unclaimed.
- Select **Post** on the Action Pane to post the claim for all the relevant lines. If you're using a claims process (when the **Use claim process** option is enabled on the **Rebate management parameters** page), only the lines that are marked as **Claimed** are posted. Otherwise, all the source transactions for the selected rebate transaction are posted. The **Post** button is available only for rebate transactions. It's unavailable for provision and write-off transactions. In the **Post** dialog box, the **From date** and **To date** fields are automatically set. Set the **Posting date** field, and then select **OK**.
- To adjust the amount that is shown for any open or unposted transaction, select the transaction, and then follow one of these steps:
  - Edit the value in the **Corrected amount** field.
  - On the Action Pane, select **Set correction**. Then, in the drop-down dialog box that appears, in the **Corrected amount** field, enter a value.

> **Note**

If you're using a claims process, when you process the next period, the transaction list will include any unclaimed transactions from the previous posting, plus any new transactions for the selected period.

View and edit Rebate management transactions by using the rebate workbench

To view and edit Rebate management transactions by using the rebate workbench, follow these steps:

1. Go to **Rebate management** > **Rebate management deals** > **Rebate workbench**.
2. Set the **Show** field to *Not posted*.
3. The **Rebate workbench** page shows a list of the transactions. Each transaction provides relevant details. These details vary, depending on the transaction type. You can perform the following actions on this page:
    - To view more information about any transaction, select it, and then select the **General**, **Financial dimension**, or **Dimension** tab.
    - If you're using a claims process, you can mark transactions as either claimed or unclaimed. Select the relevant rows, and then, on the Action Pane, on the **Rebate workbench** tab, select one of the following commands. (You enable claims processes on the Rebate management parameters page.)
        - **Set claimed** – Mark the selected transactions as claimed.
        - **Set unclaimed** – Mark the selected transactions as unclaimed.
    - To post the claim for one or more lines, select the relevant lines. Then, on the Action Pane, on the **Rebate workbench** tab, select **Post**. The **Post** button is available for provision, rebate, and write-off transactions. In the **Post** dialog box, the **From date** and **To date** fields are automatically set. Set the **Posting date** field, and then select **OK**.
    - To adjust the amount that is shown for any open or unposted transaction, select the transaction, and then follow one of these steps:
        - Edit the value in the **Corrected amount** field.
        - On the Action Pane, on the **Rebate workbench** tab, select **Set correction**. Then, in the drop-down dialog box that appears, in the **Corrected amount** field, enter a value.

> **Note**
>
> If you're using a claims process, when you process the next period, the transaction list will include any unclaimed transactions from the previous posting, plus any new transactions for the selected period.

Consolidate vendor invoices for the same vendor in vendor rebate deals

Sometimes, vendor rebate deals are listed over multiple lines (for example, because the output is split by financial dimension). In these cases, it can be cumbersome to manage multiple entries and vendor accounts for the same vendor. To improve efficiency when payable invoices are generated for vendor accounts, you can consolidate multiple ledger lines for the same vendor into a single vendor line that has several ledger lines.

Turn on vendor invoice consolidation

Before you can set your system to consolidate vendor invoices, it must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.43 or later.
- The feature named *Consolidate vendor invoices for the same vendor in vendor rebate deals* must be turned on in Feature management. As of Supply Chain Management version 10.0.45, this feature is turned on by default. As of version 10.0.49, it's mandatory and can't be turned off.

How vendor invoice consolidation works

You can configure the system to divide rebates by financial dimension for reporting purposes. The system then splits the output into several lines when you post a rebate. The split is based on the financial dimensions of each source transaction (purchase order line or sales order line).

The following procedure explains how to view the effects of vendor invoice consolidation.

1. Go to **Rebate management** > **Setup** > **Rebate management parameters**.
2. On the **Rebate management** tab, set the **Process by dimension** option to *Yes*. Your system is now set to divide rebates by financial dimension for reporting purposes.

3. Go to **Rebate management** > **Rebate management deals** > **All rebate management deals**, and open an active vendor rebate deal.

4. On the Action Pane, on the **Rebate management deals** tab, select **Process** > **Rebate management**.

5. Because of the **Process by dimension** setting, the system divides the output. Each line shows a debit or credit account together with its ledger account code. This situation can cause reconciliation issues, especially when multiple lines point to the same vendor. To view the output, on the Action Pane, on the **Rebate management deals** tab, select **Transactions**.

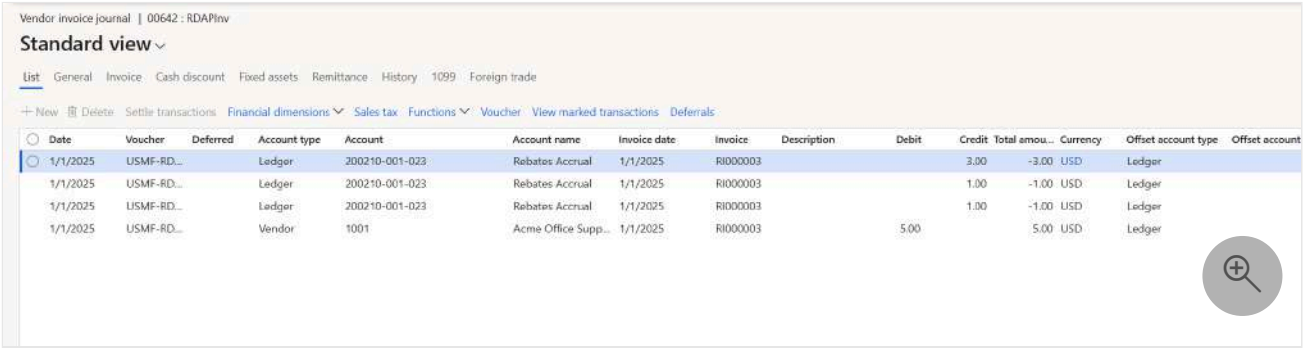
6. At the top of the grid, set the **Show** field to *Posted*.

7. Select a transaction, and then, on the Action Pane, select **Target transactions**.

8. Select the target transaction line, and then select **View details**. The output journal lines are shown.

   Without vendor consolidation, there are multiple vendor invoices for the same vendor in vendor rebate deals, as shown in the following screenshot.

   With vendor consolidation, there's just one vendor line that has multiple ledger lines, as shown in the following screenshot. This consolidation makes it easier for businesses to consolidate and send rebate output to their vendors and to report in the rebate related journals.



Post rebate transactions

To post the value of a processed provision, rebate management amount, and write-off, you must run the posting process. The posting process marks the provision, rebate management, or write-off transactions as posted, and creates the target transaction. If you don't have to review the target transaction, these transactions can be set up so that they're automatically posted.

Set up the system to post all target transactions automatically

To set up your system to post all target transactions as soon as they're generated by a posting provision, rebate management amount, and write-off, turn on the **Automatically post journals** and/or **Automatically post free text invoices** option on the **Rebate management parameters** page. Learn more in Rebate management parameters.

Post transactions for all lines for one or more deals

After you've processed the relevant deals, follow these steps to review and post the generated transactions for all lines for one or more deals.

1. Open the appropriate rebate deals list page for the type of deal that you want to work with.
2. Select the row for each deal that you want to post (or open the deal that you want to post).
3. On the Action Pane, on the **Rebate management deals** tab, in the **Generate** group, select one of the following commands:
    - **Post** > **Provision** – Post available accruals transactions that you've created.
    - **Post** > **Rebate management** – Post available rebate transactions that you've created.
    - **Post** > **Write off** – Post available write-off transactions that you've created.
4. In the dialog box that appears, set the **Posting date** field. Then set the **From date** and **To date** fields to define the date range for the transactions that must be posted.
5. Select **OK** to post the transactions.

Post transactions for one or more specific deal lines for a selected deal

After you've processed the relevant deals, follow these steps to review and post the generated transactions for one or more specific deal lines for a selected deal. This procedure is applicable only for deals where the **Reconcile by** field is set to *Line*.

1. Open the appropriate rebate deals list page for the type of deal that you want to work with.
2. Open the deal that has a line that you want to post transactions for.
3. Select the **Lines** tab in the upper-right corner.
4. On **Rebate management** FastTab, select the row for each deal line that you want to post.
5. On the toolbar of the **Rebate management** FastTab, select one of the following commands. (These commands are available only for deals where **Reconcile by** is set to *Line*.)
    - **Post** > **Provision** – Post available accruals transactions that you've created.
    - **Post** > **Rebate management** – Post available rebate transactions that you've created.
    - **Post** > **Write off** – Post available write-off transactions that you've created.
6. In the dialog box that appears, set the **Posting date** field. Then set the **From date** and **To date** fields to define the date range for the transactions that must be posted.
7. Select **OK** to post the transactions.

Post transactions using a batch job

Instead of posting transactions for specific deals or deal lines, you can run a batch job to post transactions for several deals at the same time. You can optionally apply record filters and/or set up a recurring schedule. To post transactions by using a batch job, follow these steps:

1. Follow one of these steps:
    - Go to **Rebate management** > **Periodic tasks** > **Post** > **Provision** to post available accrual transactions that you've created.
    - Go to **Rebate management** > **Periodic tasks** > **Post** > **Rebate management** to post available rebate transactions that you've created.
    - Go to **Rebate management** > **Periodic tasks** > **Post** > **Write off** to post available write-off transactions that you've created.
2. In the dialog box that appears, on the **Parameters** FastTab, in the **Period** section, set the **Posting date** field. Then set the **From date** and **To date** fields to define the date range for the transactions that must be posted.
3. In the **Guarantee Period** section, set the **From date** and **To date** fields to define the date range for the guarantees that must be posted.

4. On the **Records to include** FastTab, you can set up filters to limit the set of deals that the batch job will process. These settings work in the same way that they work for other types of batch jobs.

5. On the **Run in the background** FastTab, you can set up batch processing and scheduling options as required. These settings work in the same way that they work for other types of batch jobs.

6. Select **OK** to run and/or schedule the calculation.

Post transactions by using the rebate workbench

After you've processed provision, rebate, or write-off transactions, follow these steps to use the rebate workbench to review and post the generated transactions for one or more specific transaction lines for all deals.

1. Go to **Rebate management** > **Rebate management deals** > **Rebate workbench**.

2. In the grid, select the row for each transaction line that you want to post. You can select unposted provision, rebate, and/or write-off transactions. The following rules apply:

    - The system will also post all lines that have the same **Rebate transaction number** value as the lines that you select.
    - The system won't post any lines of the *Rebate* transaction type that aren't marked as claimed.
    - If you select lines that have already been posted, the system will skip the posted lines.
    - The **Post** button is available only if you select at least one unposted line.

3. On the Action Pane, on the **Rebate workbench** tab, in the **Processing** group, select **Post**.

4. In the **Post** dialog box, set the **Posting date** field. The **From date** and **To date** fields are automatically set, based on the earliest **From date** value and the latest **To date** value for the selected rows.

5. Select **OK** to post the transactions.

Review Rebate management journals

After your transactions have been posted, you can review the resulting journals, documents, or items. Target transactions for rebates and royalties are based on the payment type that is set in the posting profile and the rebate's output type. For example, if the rebate output is set to *Item*, a

sales order will be created for a customer rebate, and a purchase order will be created for a vendor rebate. These orders can be viewed via the target transactions. Alternatively, if the payment is set up to use Accounts payable, a vendor invoice for the vendor that is set up on the customer will be created for customer rebates.

Review journals by using the rebate deals list page

To review the journal entries that are associated with a Rebate management deal, follow these steps:

1. Open the appropriate rebate deals list page for the type of deal that you want to work with.
2. Select the deal to inspect journal entries for.
3. On the Action Pane, on the **Rebate management deals** tab, in the **Transactions** group, select either **Transactions** or **Guarantee transactions**, depending on the type of transactions that you want to look at.
4. Set the **Show** field to *All* or *Posted*.
5. Find and select the transaction collection that you want to inspect, and then, on the Action Pane, select one of the following buttons. (These buttons are available only when relevant postings exist for the selected transaction collection.)
    - **Target transactions** – Review relevant journals and other types of documents that were generated by the selected deal.
    - **Items** – Review relevant sales orders or purchase orders that were generated by the selected deal.
6. A list of relevant journals, documents, or items appears. To view more information about any journal, document, or item, select its row, and then, on the Action Pane, select **View details**.

Review journals by using the rebate workbench

To review journals by using the rebate workbench, follow these steps:

1. Go to **Rebate management** > **Rebate management deals** > **Rebate workbench**.
2. Set the **Show** field to *All* or *Posted*.
3. Find and select the line that you want to inspect. Then, on the Action Pane, on the **Rebate workbench** tab, in the **View** group, select **Target transactions**. This button is available only if relevant postings exist for the selected line.

4. A list of relevant journals, documents, or items appears. To view more information about any journal, document, or item, select its row, and then, on the Action Pane, select **View details**.

Rebate management transactions on the deduction workbench

When you post a Rebate management transaction that has one of the following **Payment type** values, the system creates a customer deduction journal or a free text invoice for the relevant customer account:

- Customer deductions
- Tax invoice customer deductions
- Trade spending
- Reporting

After a target transaction is created and posted, it will be available as an open transaction on the **Deduction workbench** page (**Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**). Open transactions have a **Claim type** value of *Rebate management*, and a **Rebate transaction number** value is available to enable traceability. The date is set to the posting date of the Rebate management target transaction. To use the deduction workbench to settle open transactions to existing deductions for the same customer account, select **Maintain** > **Match** on the Action Pane.

Learn more in Manage deductions using the deduction workbench.

Purge unposted transactions

After you've processed provision, rebate, or write-off transactions, follow these steps to purge selected unposted transactions.

1. Go to **Rebate management** > **Rebate management deals** > **Rebate workbench**.
2. Set the **Show** field to *Not posted*.
3. Find and select the transactions to delete. Then, on the Action Pane, on the **Rebate workbench** tab, in the **Processing** group, select **Purge**.
4. Select **OK** to delete the unposted transactions.

Cancel a posted provision

After you've processed and posted a provision, follow these steps to cancel the posted provision transactions.

1. Go to **Rebate management** > **Rebate management deals** > **Rebate workbench**.
2. Set the **Show** field to *Posted*.
3. Find and select the provision transactions to cancel. Then, on the Action Pane, on the **Rebate workbench** tab, in the **Processing** group, select **Cancel provision**.
4. Select **OK** to reverse the transactions.

These provision reversals will also be visible in the relevant Rebate management journals.

Last updated on 07/27/2026

Manage deductions using the deduction workbench

07/21/2025

This article describes how to use the deduction workbench to process customer payments that include deductions.

A customer who is owed a rebate can decide not to wait for a rebate payout. Instead, the customer can send a payment that includes a deduction for the amount of the rebate. To handle this type of transaction, you can use the deduction workbench to match deductions to open credit transactions, split deductions, deny deductions, and write off deductions.

> **Note**
>
> The deduction workbench has been part of the sales and marketing functionality in Microsoft Dynamics 365 Supply Chain Management for a long time. However, it has now been enhanced so that it also works with the newer **Rebate management** module. This article describes how to use both older features and Rebate management features of the deduction workbench.

Prerequisites

Set up the old deduction management system

You can use the deduction workbench together with the old deduction management capabilities of Supply Chain Management, even if you aren't using the **Rebate management** module. However, you must first prepare your system as described in this section.

Before you can use the deduction workbench, you must complete the setup tasks that are described in Set up deduction management. You should also have some type of rebate agreement set up for the customer: either a customer rebate, as described in Setting up and maintaining customer rebates, or a trade allowance rebate.

If you're applying a deduction to a customer rebate, you must complete these tasks:

- Set up your customer rebates.
- Approve and process the rebate, so that you can use the deduction workbench.

If you're applying a deduction to a trade allowance rebate, you must complete these tasks:

- Set up billback rebates.
- Apply billback rebates.

Configure accounts receivable and deductions

The system records all deduction events in a claim journal. Therefore, your system must include a journal that can be used for this purpose. If you don't already have a claim journal, set it up now. This journal is required to create deductions directly on the deduction workbench, customer settlement, or customer page.

To set up a new claim journal for deductions, follow these steps.

1. Go to **General ledger** > **Journal setup** > **Journal names**.

2. Select **New**, and set the following fields for the new journal name:

    - **Name** – Enter a unique name for the claim journal.
    - **Description** – Enter a description of the new journal.
    - **Journal type** – Select *Daily*.
    - **Voucher series** – Assign an existing number sequence. Alternatively, create a new number sequence that has company scope, and assign it to the new journal name.

3. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.

4. On the **Deductions** tab, on the **General** FastTab, set the **Claim journal name** field to the journal name that you just created.

5. On the **Return order** FastTab, set the following fields:

    - **Create return order** – Set this option to specify what the system should do when a quantity-based claim is approved:
        - *Yes* – Create a return order.
        - *No* – Create a negative sales order.

    - **Creation without attached invoice** – Select a value to specify what the system should do when a quantity-based claim is approved but no invoice is attached:
        - *Accept* – Create a return order.
        - *Warning* – Create a return order, but show the following warning message: "Claim is not attached to an invoice."
        - *Error* – Don't create a return order, and show the following error message: "Claim is not attached to an invoice. Update has been canceled."

    - **Create return order prior to deduction approval** – Set this option to *Yes* if return orders can be created before the claim is approved. This setting applies only to

quantity-based claims where the **Create return order** option is set to *Yes*.

Configure General ledger parameters

When the system creates a claim journal for a new deduction, it also creates two new customer transactions: one to offset the amount of the claim against the original invoice and one to register a customer's debt to the amount of the claim (because the claim hasn't yet been approved). Therefore, you must set up your system so that a single voucher can have multiple customer lines.

To enable a single voucher to have multiple customer lines, follow these steps.

1. Go to **General ledger** > **Ledger setup** > **General ledger parameters**.
2. On the **Ledger** tab, on the **General** FastTab, set the **Allow multiple transactions within one voucher** option to *Yes*.
3. If you receive a warning message, select **Close** to accept the change.

Create deduction reasons

The system requires that users specify a reason for each deduction that they enter directly on the deduction workbench, customer settlement, or customer page. The reason determines how the deduction is handled when it's approved.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction reasons**.

2. Select **New** to add a row to the grid, and then set the following fields for it:

    - **Claim reason** – Enter a unique name for the reason.

    - **Description** – Enter a description of the reason to provide more information about how it should be used.

    - **Claim basis** – Select a claim basis for the claim reason:
        - *Price based* – Create a free text credit upon approval.
        - *Quantity based* – Create a negative sales order or return order upon approval.

    - **Return reason code** – Select a return reason code to apply as the **Return reason code** value for the return order.

    - **Type** – Select a deduction type. The **Deduction offset** value for the selected type will be used to set the **Deduction offset** field when a deduction or claim is created. Deduction types are defined on the **Deduction types** page (**Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction types**).

- **Claim posting account** – This field is available only when the **Claim basis** field is set to *Price based*. When a price-based claim is approved, the system assigns the ledger account that you select here as the **Main account** value for the draft free text credit note.

Set up the settle approved deductions periodic task

For deductions that are created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, you can set up the *Settle approved deductions* periodic task to automatically match deductions and credits that have matching **Deduction ID** values and amounts.

To schedule this task, go to **Sales marketing** > **Periodic tasks** > **Settle approved deductions**, and set up options, filters, and a schedule, just as for other types of periodic tasks.

> ⓘ **Note**
>
> If the **Automatic settlement** option is set to *Yes* on the **Settlement** tab of the **Accounts receivable parameters** page (**Account receivable** > **Setup** > **Accounts receivable parameters**), the *Settle approved deductions* periodic task won't do anything, because the credit will automatically be settled.

Create a deduction

Create a deduction journal entry by using the customer payment journal

To create a deduction journal entry, follow these steps.

1. Go to **Accounts receivable** > **Payments** > **Customer payment journal**.
2. Select **New** to add a row to the grid.
3. In the **Name** field for the new row, select the name of the journal.
4. On the Action Pane, select **Lines**.
5. Enter the date, company account, and customer account number.
6. In the **Invoice** field, select the invoice that the deduction is associated with.
7. In the **Credit** field, enter the amount that the customer paid.
8. Select **OK** to confirm that the amount is less than the total amount of the marked transaction.
9. Select the offset account type and the offset account.
10. On the toolbar above the grid, select **Deductions**.

11. On the **Deduction** page, on the Action Pane, select **New** to add a row to the grid. The **Deduction ID** field for the new row is automatically set.
12. In the **Type** field, select the deduction type.
13. In the **Amount** field, enter the amount that is shown in the **Balance** field below the deduction list. This amount represents the amount that the customer deducted from the payment.
14. Close the **Deduction** page. You're returned to the **Customer payments** page, which now shows a new line for the deduction.
15. On the Action Pane, select **Validate** > **Validate**. You should receive the following message: "Journal is OK."
16. On the Action Pane, select **Post**.

Create a deduction by using the deduction workbench

To create a new deduction on the deduction workbench, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.

2. On the Action Pane, select **Maintain** > **New deduction**.

   In the **New deduction** dialog box, the **Deduction ID** field is set based on the **Deduction ID** number sequence that is defined on the **Trade allowance management parameters** page (**Sales and marketing** > **Setup** > **Trade allowances** > **Trade allowance management parameters**).

3. Set the following fields:

   - **Customer account** – Select the customer account that the deduction applies to.
   - **External claim number** – Enter the customer's claim reference.
   - **Claim amount** – Enter the tax-inclusive claim amount. The value must be positive.
   - **Currency** – Select the currency for the deduction. The default value is the currency that is set for the selected customer account.
   - **Claim basis** – Select the claim basis. The claim basis determines the type of credit transaction that is created after the deduction or claim is approved.
     - *Price based* – A draft free text invoice will be created.
     - *Quantity based* – A negative sales order or return order will be created.
   - **Claim date** – Select the date of the claim. The default value is the current date.
   - **Claim reason** – Select the reason code that applies to the current deduction. The claim basis that you selected affects the options that apply. For more information

about how to create and configure the claim reasons that are available for selection here, see the Create deduction reasons section earlier in this article.

- **Notes** – Add any notes that apply. When the claim is approved, the approver will be able to edit or add to the claim's notes.

- **Create claim journal** – Set this option to specify whether the claim journal should be created when the claim or deduction is created:
  - *Yes* – The system will create and post a general journal by using the claim journal that is set up on the **Accounts receivable parameters** page. (Learn more in the Configure accounts receivable and deductions section earlier in this article.) When an invoice is attached to the claim, the claim journal is used to reduce the balance of the applicable invoice. If the claim is later rejected, the claim journal and settlements (if an invoice was attached) will be reversed.
  - *No* – No claim journal is created at this time. It will be created when the claim is approved. An invoice can still be attached to the new claim, even though a claim journal isn't created. However, settlement can't be done without the claim journal.

4. Select **OK**.

   A new deduction is created. If you set the **Create claim journal** option to *Yes*, the following transactions are posted:

   - **Two new customer transactions** – One transaction offsets the amount of the claim against the original invoice. The other transaction registers a customer's debt to the amount of the claim, because the claim hasn't yet been approved. The original invoice transaction and the offsetting claim transaction will automatically be marked for settlement when you attach the invoice by selecting **Maintain** > **Attach invoice** on the Action Pane.
   - **Two offsetting transactions** – These transactions are posted to the **Deduction offset** ledger account.
   - **Claim journal** – To view the claim journal for each deduction that is shown on the deduction workbench, select the **References** tab. The claim journal is shown in the **Journal batch number** field. You can also view the claim journal on the **Deduction events** tab. There, it will have an **Update type** value of *Match*.

Create a deduction from a customer settlement

The process of creating a deduction from a customer settlement resembles the process of creating a deduction via the deduction workbench. However, the customer and invoice currency are automatically set, and the invoice is attached. You don't have to select **Maintain** >

Attach invoice on the Action Pane when you create a claim or deduction via the customer settlement page.

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select the customer to create a deduction for.
3. On the Action Pane, on the **Collect** tab, in the **Settle** group, select **Settle transactions**.
4. In the **Setting transactions** dialog box, on the **Overview** tab, select the invoice to create the deduction against.
5. On the toolbar, select **Deductions** > **New deduction**.

   In the **New deduction** dialog box, the **Deduction ID** field is set based on the **Deduction ID** number sequence that is defined on the **Trade allowance management parameters** page (**Sales and marketing** > **Setup** > **Trade allowances** > **Trade allowance management parameters**). The **Customer account** field is set to the customer account that the deduction applies to.
6. Set the following fields:

   - **External claim number** – Enter the customer's claim reference.
   - **Claim amount** – Enter the tax-inclusive claim amount. The value must be positive.
   - **Currency** – Select the currency for the deduction. The default value is the currency that is set for the selected customer account.
   - **Claim basis** – Select the claim basis. The claim basis determines the type of credit transaction that is created after the deduction or claim is approved.
     - *Price based* – A draft free text invoice will be created.
     - *Quantity based* – A negative sales order or return order will be created.
   - **Claim date** – Select the date of the claim. The default value is the current date.
   - **Claim reason** – Select the reason code that applies to the current deduction. The claim basis that you selected affects the options that apply. For more information about how to create and configure the claim reasons that are available for selection here, see the Create deduction reasons section earlier in this article.
   - **Notes** – Add any notes that apply. When the claim is approved, the approver will be able to edit or add to the claim's notes.
   - **Create claim journal** – Set this option to specify whether the claim journal should be created when the claim or deduction is created:

- *Yes* – The system will create and post a general journal by using the claim journal that is set up on the **Accounts receivable parameters** page. (Learn more in the Configure accounts receivable and deductions section earlier in this article.) When an invoice is attached to the claim, the claim journal is used to reduce the balance of the applicable invoice. If the claim is later rejected, the claim journal and settlements (if an invoice was attached) will be reversed.
- *No* – No claim journal is created at this time. It will be created when the claim is approved. An invoice can still be attached to the new claim, even though a claim journal isn't created. However, settlement can't be done without the claim journal.

7. Select **OK**.

   A new deduction is created. If you set the **Create claim journal** option to *Yes*, the following transactions are posted:

   - **Two new customer transactions** – One transaction offsets the amount of the claim against the original invoice. The other transaction registers a customer's debt to the amount of the claim, because the claim hasn't yet been approved. The original invoice transaction and the offsetting claim transaction will automatically be marked for settlement when you attach the invoice by selecting **Maintain** > **Attach invoice** on the Action Pane.
   - **Two offsetting transactions** – These transactions are posted to the **Deduction offset** ledger account.
   - **Claim journal** – To view the claim journal for each deduction that is shown on the deduction workbench, select the **References** tab. The claim journal is shown in the **Journal batch number** field. You can also view the claim journal on the **Deduction events** tab. There, it will have an **Update type** value of *Match*.

   You're returned to the **Settle transactions** page, which now shows the selected invoice as marked. The **Post** button is available only if you set the **Create claim journal** option to *Yes*. If it's available, select **Post** to reduce the balance on the invoice by the **Claim amount** value.

Create a deduction from a customer page

The process of creating a deduction from a customer page resembles the process of creating a deduction via the deduction workbench. However, the customer is automatically set.

1. Go to **Accounts receivable** > **Customers** > **All customers**.

2. Select the customer to create a deduction for.

3. On the Action Pane, on the **Collect** tab, in the **Deductions** group, select **Create deductions**.

   In the **New deduction** dialog box, the **Deduction ID** field is set based on the **Deduction ID** number sequence that is defined on the **Trade allowance management parameters** page (**Sales and marketing** > **Setup** > **Trade allowances** > **Trade allowance management parameters**). The **Customer account** field is set to the customer account that the deduction applies to.

4. Set the following fields:

   - **External claim number** – Enter the customer's claim reference.
   - **Claim amount** – Enter the tax-inclusive claim amount. The value must be positive.
   - **Currency** – Select the currency for the deduction. The default value is the currency that is set for the selected customer account.
   - **Claim basis** – Select the claim basis. The claim basis determines the type of credit transaction that is created after the deduction or claim is approved.
     - *Price based* – A draft free text invoice will be created.
     - *Quantity based* – A negative sales order or return order will be created.
   - **Claim date** – Select the date of the claim. The default value is the current date.
   - **Claim reason** – Select the reason code that applies to the current deduction. The claim basis that you selected affects the options that apply. For more information about how to create and configure the claim reasons that are available for selection here, see the Create deduction reasons section earlier in this article.
   - **Notes** – Add any notes that apply. When the claim is approved, the approver will be able to edit or add to the claim's notes.
   - **Create claim journal** – Set this option to specify whether the claim journal should be created when the claim or deduction is created:
     - *Yes* – The system will create and post a general journal by using the claim journal that is set up on the **Accounts receivable parameters** page. (Learn more in the Configure accounts receivable and deductions section earlier in this article.) When an invoice is attached to the claim, the claim journal is used to reduce the balance of the applicable invoice. If the claim is later rejected, the claim journal and settlements (if an invoice was attached) will be reversed.
     - *No* – No claim journal is created at this time. It will be created when the claim is approved. An invoice can still be attached to the new claim, even though a claim

journal isn't created. However, settlement can't be done without the claim journal.

5. Select **OK**.

   A new deduction is created. If you set the **Create claim journal** option to *Yes*, the following transactions are posted:

   - **Two new customer transactions** – One transaction offsets the amount of the claim against the original invoice. The other transaction registers a customer's debt to the amount of the claim, because the claim hasn't yet been approved. The original invoice transaction and the offsetting claim transaction will automatically be marked for settlement when you attach the invoice by selecting **Maintain** > **Attach invoice** on the Action Pane.
   - **Two offsetting transactions** – These transactions are posted to the **Deduction offset** ledger account.
   - **Claim journal** – To view the claim journal for each deduction that is shown on the deduction workbench, select the **References** tab. The claim journal is shown in the **Journal batch number** field. You can also view the claim journal on the **Deduction events** tab. There, it will have an **Update type** value of *Match*.

Create a credit note for a customer

After an approved rebate exists for a customer, you can create a credit note on the customer's account to represent the rebate, as required. The credit then appears on the deduction workbench, where it can be matched to a deduction.

To create a credit note, follow these steps.

1. Go to **Sales and marketing** > **Customers** > **All customers**.
2. Select the customer.
3. On the Action Pane, on the **Collect** tab, in the **Settle** group, select **Settle transactions**.
4. In the **Settle transactions** dialog box, select the transaction that the rebate was applied to.
5. On the toolbar, on the **Functions** menu, select the type of rebate program that applies.
6. On the **Rebate** page, on the **Overview** tab, select the **Mark** checkbox next to the relevant rebate ID.
7. On the Action Pane, select **Functions** > **Create credit note**.

Process a deduction on the deduction workbench

On the deduction workbench, you can match deductions to open credit transactions, split deductions, deny deductions, and write off deductions.

Depending on how you want to process a deduction, complete one or more of the procedures in the following subsections. You can combine the procedures as required. For example, you can split a deduction into two parts, and then match one part to a credit but leave the remaining part on the workbench so that it be matched to another credit later. You can also match a deduction to a credit that is smaller than the deduction amount, and then deny or write off the remaining balance of the deduction.

Match a deduction to a credit

To match a deduction to a credit, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the **Mark** checkbox for the deduction to process.
3. In the **Open transactions** section, select the **Mark** checkbox for the credit to match to the deduction. If multiple credits are listed, you can select more than one credit to match to the deduction. If you want the system to automatically select credits that match the amount of the deduction, on the toolbar, select an appropriate option on **Select deduction amount** menu.
4. On the Action Pane, select **Maintain** > **Match**. The system matches the deduction to the credit. If a balance remains in the deduction, it's shown in the **Remaining amount** field on the **Deductions** tab.

   > ⓘ **Note**
   >
   > For deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, the **Maintain** > **Match** command is available only if the **Claim status** field is set to *Accepted*. This command can be used to manually match the price-based or quantity-based transaction to the associated credit in the **Open transactions** section. This credit is created either when the deduction is approved (by using the **Maintain** > **Approve deduction** command), or when it's attached to an existing credit as described in the Credits created outside the approve deduction process section later in this article. The *Settle approved deductions* periodic task (**Sales marketing** > **Periodic tasks** > **Settle approved deductions**) can also be used to automatically match deductions and credits that have matching **Deduction ID** values and amounts.

Split a deduction

To split a deduction, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the **Mark** checkbox for the deduction to process.
3. On the Action Pane, select **Maintain** > **Split**.
4. In the **Split** dialog box, in the **Split amount** field, enter the amount to split from the main deduction. Then select **OK**.
5. On the **Deductions** tab, notice that a new record appears for the split amount. The original deduction record contains the remainder of the deduction balance. You can now manage the two parts of the original rebate separately.
6. Select the original deduction record, and then select the **References** tab. The **Split amount** field shows the amount that was split off from the original amount.

Attach an invoice to a deduction

You can attach an invoice to a deduction if the deduction was created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, and if no invoice is currently attached to it (that is, the **Invoice** column is blank).

To attach an invoice to a deduction, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the **Mark** checkbox for the deduction to process.
3. On the Action Pane, select **Maintain** > **Attach invoice**.
4. In the **Attach invoice** dialog box, select an invoice, and then select **OK**.
5. In the **Settle transactions** dialog box, follow one of these steps, depending on whether a claim journal was posted when the deduction was created:
    - If a claim journal was posted, the invoice that you selected and the claim journal's customer credit transaction show a check mark in the **Mark** column. Select **Post**. The remaining balance on the attached invoice is reduced by the claim amount.
    - If a claim journal wasn't posted, no transaction shows a check mark in the **Mark** column. Because you can't offset against a claim journal until the deduction is approved, select **Cancel**.

Detach an invoice from a deduction

You can detach an invoice from a deduction if the deduction was created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, if an invoice is currently attached to it (that is, the **Invoice** column shows an invoice number), and if the **Claim status** field is set to *Open*. You might complete this task because an incorrect invoice was attached. The invoice is removed from the deduction, and its remaining balance is updated if it was reduced when the invoice was attached.

To detach an invoice, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the **Mark** checkbox for the deduction to process.
3. On the Action Pane, select **Maintain** > **Detach invoice**.

Approve a deduction

You can approve deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page. However, you can approve only deductions where the **Claim status** field is set to *Open*.

To approve a deduction, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the **Mark** checkbox for the deduction to process.
3. On the Action Pane, select **Maintain** > **Approve deduction**.
4. In the **Approve deduction** dialog box, edit or add information to the **Note** value as required.
5. If the deduction is price-based, and an invoice hasn't been attached to it, select an item sales tax group. Typically, the tax item group is set on the invoice. Therefore, the tax item code must be specified when it isn't attached to an invoice.
6. Select **OK**.

   At this point, no further changes can be made to the deduction. If the **Create claim journal** option was set to *No* when the deduction was created, the claim journal is created and posted when the deduction is approved. After the deduction is approved, the credit is automatically created and opened. The type depends on the deduction's **Claim basis** value:

   - *Price based* – If the deduction is price-based, a free text invoice is created for the customer account. You can add a description and post the credit. The following fields are filled in by the deduction when the draft is created:

- **Deduction ID** – This field is added to the header to enable traceability back to the deduction.
- **Main account** – The value is determined by the **Claim posting account** value that is set for the deduction reason that is assigned to the deduction.
- **Item sales tax group** – The value is determined by the attached invoice or by the value that you selected when you approved the deduction.
- **Unit price** – The value is the credit of the deduction's claim amount.
- **Invoice text** – By default, this field is set to the deduction's **Notes** value.

- *Quantity based* – If the deduction is quantity-based, an open sales order or return order is created. The **Create return order** setting on the Accounts receivable parameters page determines whether a sales order or a return order is created when the deduction is approved. The **Copy orders** page appears and is filtered to show rows where the **Invoice account** field is set to the deduction's customer account. Follow these steps:

  a. On the **Invoices** FastTab, the **Headers** section shows sales invoices where the **Invoice account** value matches the deduction's customer account. Select an applicable sales invoice.

  b. The **Lines** section shows lines from the selected sales invoice. Select the **Select** checkbox for each line that you want to copy. Alternatively, in the **Headers** section, select the **Select all** checkbox for the sales order to select all its lines.

  c. Adjust the **Quantity** value for one or more lines as required.

     All the lines that you've selected so far are listed on the **Selected lines or header to be copied** FastTab.

  d. Repeat the previous two steps as required until all the required lines are listed on the **Selected lines or header to be copied** FastTab.

  e. Select **OK**.

     The new return order is opened, and the following fields are automatically set:
     - **Deduction ID** – This field is added to the header to enable traceability back to the deduction.
     - **Return reason code** – By default, this field is set to the **Return reason code** value that is set for the deduction reason that is assigned to the deduction.

After the credit is invoiced, it appears in the **Open transactions** section of the deduction workbench against the applicable **Deduction ID** value, and its **Claim type** field is set to *Other credits*. The credit will be available until it's settled with the deduction in one of the following ways:

- You manually settle it by selecting **Maintain** > **Match** on the Action Pane.
- It's automatically settled by the *Settle approved claims* periodic job (**Sales and marketing** > **Periodic tasks** > **Settle approved claims**).
- It's automatically settled because the **Automatic settlement** option on the **Settlement** tab of the **Accounts receivable parameters** page is set to *Yes*.

To view the credit that is created when the deduction is approved, you can also use the **Open credit** button in the **Open transactions** section of the deduction workbench.

Create a return order

You can create a return order for deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page. However, all the following conditions must be met:

- The **Claim basis** field is set to *Quantity based*.
- The **Claim status** field is set to *Open*.
- The **Create return order** option on the **Deductions** tab of the Accounts receivable parameters page is set to *Yes*.
- The **Create return order prior to deduction approval** option on the **Deductions** tab of the Accounts receivable parameters page is set to *Yes*.

To create a return order, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the **Mark** checkbox for the deduction to process.
3. On the Action Pane, select **Maintain** > **Create return order**.
4. In the **Approve deduction** dialog box, edit or add information to the existing **Notes** value as required, and then select **OK**.
5. In the **Copy orders** dialog box, on the **Invoices** FastTab, the **Headers** section shows sales invoices where the **Invoice account** value matches the deduction's customer account. Select an applicable sales invoice.
6. The **Lines** section shows lines from the selected sales invoice. Select the **Select** checkbox for each line you want to copy. Alternatively, in the **Headers** section, select the **Select all** checkbox for the sales order to select all its lines.
7. Adjust the **Quantity** value for one or more lines as required.

   All the lines that you've selected so far are listed on the **Selected lines or header to be copied** FastTab.

8. Repeat the previous two steps as required until all the required lines are listed on the **Selected lines or header to be copied** FastTab.

9. Select **OK**.

   The new return order is opened, and the following fields are automatically set:

   - **Deduction ID** – This field is added to the header to enable traceability back to the deduction.
   - **Return reason code** – By default, this field is set to the **Return reason code** value that is set for the deduction reason that is assigned to the deduction.

Deny a deduction

To deny a deduction, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.

2. Select the **Mark** checkbox for the deduction to process.

3. On the Action Pane, select **Maintain** > **Deny**.

4. In the **Deny** dialog box, select the reason code for the denial, and then select **OK**.

5. In the **Show** field below the Action Pane, select *Closed*.

   The **Deductions** tab shows the denied deduction, and the **Remaining amount** field for the deduction is set to *0.00*.

   For deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, the following events occur:

   - On the **References** tab, fields in the **Denial** section are updated.
   - If you selected to create the claim journal when the deduction was created, and if an invoice has been attached to the deduction that reduced the invoice's balance, the invoice is detached, and the remaining balance of the previously attached invoice is increased by the rejected claim's amount.
   - The deduction's **Status** field is set to *Closed*.
   - The deduction's **Claim status** field is set to *Rejected*.

To reverse a denial, follow these steps.

1. On the **Deductions** tab, select a denied deduction.

2. On the Action Pane, select **Reverse denial**.

For deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, the following events occur:

- On the **References** tab, fields in the **Denial** section are updated.
- The deduction's **Status** field is set to *Open*.
- The deduction's **Claim status** field is set to *Open*.

Write off a deduction

To write off a deduction, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the **Mark** checkbox for the deduction to process.
3. On the Action Pane, select **Maintain** > **Write-off**.
4. In the **Write-off** dialog box, select the reason code for the write-off, and then select **OK**.
5. In the **Show** field, select *Closed*.

   The **Deductions** tab shows the written-off deduction, and the **Remaining amount** field for the deduction is set to *0.00*.

   For deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, the following events occur:

   - On the **References** tab, fields in the **Write-off** section are updated.
   - If you created the claim journal when the deduction was created, a claim journal is posted to the deduction's write-off reason code. You can view this entry on the **Deduction events** tab, where it has an **Update type** value of *Write-off*.
   - The deduction's **Status** field is set to *Closed*
   - The deduction's **Claim status** field is set to *Write-off*.

To reverse a write-off, follow these steps.

1. On the **Deductions** tab, select a denied deduction.
2. On the Action Pane, select **Reverse write-off**.

   For deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page, the following events occur:

- On the **References** tab, fields in the **Write-off** section are updated.
- If you created the claim journal when the deduction was created, a claim journal is posted to the deduction's write-off reason code. You can view this entry on the **Deduction events** tab, where it has an **Update type** value of *Reverse write-off*.
- The deduction's **Status** field is set to *Open*.
- The deduction's **Claim status** field is set to *Open*.

Credits created outside the Approve deduction process

This section applies only to deductions that were created by using the **New deduction** command on the deduction workbench, customer settlement, or customer page.

Different users might already have created a free text invoice, return order, or negative sales order for a customer's claim outside the **Approve deduction** process. When the existing deduction is approved on the deduction workbench, the system automatically creates another free text invoice, return order, or negative sales order. This section describes how to attach existing credits to a deduction before the deduction is approved, to help prevent duplicate credits.

Attach a credit to deduction

This section describes how you can attach a credit to a deduction from the credit.

After a credit is attached to the deduction, you can view it by using the **Open credit** button on the toolbar in the **Open transactions** section of the deduction workbench.

After a credit is invoiced, and the deduction is approved, the credit appears in the **Open transactions** section of the deduction workbench against the applicable **Deduction ID** value, and its **Claim type** field is set to *Other credits*.

Attach a free text invoice to a deduction

To attach a free text invoice to a deduction, follow these steps.

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. Select the applicable invoice.
3. On the Action Pane, on the **Invoice** tab, select **Attach credit to deduction**. This button is available only if the free text invoice's **Deduction ID** field is blank. A blank field indicates that the free text invoice isn't already attached to a deduction.

4. On the **Attach credit to deduction** page, you can select *one* deduction. Only open *price-based* deductions are available for selection.
5. Select **OK**. The **Deduction ID** field is set on the free text invoice's header.

Attach a return order to a deduction

To attach a return order to a deduction, follow these steps.

1. Go to **Accounts receivable** > **Orders** > **All return orders**.
2. Select the applicable received or open return merchandise authorization (RMA) number.
3. On the Action Pane, on the **Return order** tab, select **Attach credit to deduction**. This button is available only if the return order's **Deduction ID** field is blank. A blank field indicates that the return order isn't already attached to a deduction.
4. On the **Attach credit to deduction** page, you can select *one* deduction. Only open *quantity-based* deductions are available for selection.
5. Select **OK**. The **Deduction ID** field is set on the return order's header.

Attach a sales order to a deduction

To attach a sales order to a deduction, follow these steps.

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Select the applicable open, delivered, or invoiced sales order.
3. On the Action Pane, on the **Invoice** tab, select **Attach credit to deduction**. This button is available only if the sales order's **Deduction ID** field is blank. A blank field indicates that the sales order isn't already attached to a deduction.
4. On the **Attach deduction** page, you can select *one* deduction. Only open *quantity-based* deductions are available for selection.
5. Select **OK**. The **Deduction ID** field is set on the sales order's header.

Detach a deduction from a credit

If the incorrect deduction has been attached, you can detach it from the credit. However, all the following conditions must be met:

- A credit is attached to the deduction.
- The **Claim status** field is set to *Open*.

To detach a deduction from a credit, follow one of these steps, depending on the credit type:

- **Free text invoices:** On the **All free text invoices** page, select an invoice. Then, on the Action Pane, on the **Invoice** tab, select **Detach credit from deduction**.

- **Return orders:** On the **All return orders** page, select an order. Then, on the Action Pane, on the **Return order** tab, select **Detach credit from deduction**.
- **Sales orders:** On the **All sales orders** page, select an order. Then, on the Action Pane, on the **Invoice** tab, select **Detach credit from deduction**.

Attach a deduction to a credit

This section describes how you can attach a deduction to a credit from the deduction.

Attach a deduction to a free text, return order, or sales order credit

To attach a deduction to a free text, return order, or sales order credit, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. Select the applicable open deduction.
3. On the Action Pane, select **Maintain** > **Attach deduction to credit**. This button is available only if the **Claim status** field is set to *Open*.
4. On the **Attach credit** page, you can select *one* credit. The type of credits that is shown depends on the deduction's **Claim basis** value:
    - *Price based* – The page shows free text invoices for the customer account where the **Deduction ID** field is blank. Customer requisitions are also shown because the free text invoice might be unposted. Therefore, it might not have a number that can be referenced.
    - *Quantity based* – The type of credits that is shown depends on the setting of the **Create return order** option on the Accounts receivable parameters page:
        - *Yes* – The page shows return orders for the customer account where the **Deduction ID** field is blank.
        - *No* – The page shows sales orders for the customer account where the **Deduction ID** field is blank.
5. Select **OK**. The **Deduction ID** field is set on the credit's header.

After a credit is attached to the deduction, you can view it by using the **Open credit** button on the toolbar in the **Open transactions** section of the deduction workbench.

After a credit is invoiced, and the deduction is approved, the credit appears in the **Open transactions** section of the deduction workbench against the applicable **Deduction ID** value, and its **Claim type** field is set to *Other credits*.

Detach a credit from the deduction

If the incorrect credit has been attached, you can detach it from the deduction. On the Action Pane, in the **Maintain** group, select **Detach deduction from credit**. The **Deduction ID** value is removed from the credit.

The **Detach deduction from credit** button is available only if the following conditions are met:

- A credit is attached to the deduction.
- The **Claim status** field is set to *Open*.

Create a one-time promotion

Sometimes, you might not have an approved rebate that you can match to a deduction. In this case, you can use the *one-time promotion* feature to match the deduction to a trade allowance that is associated with the customer. The *one-time promotion* feature creates a new trade allowance agreement and a lump sum merchandising event. It then matches the lump sum to the deduction and makes the postings that are required to close the deduction.

This feature is useful if you use trade allowances. For more information about trade allowances, see Trade allowance management.

First, you must set up a template that can be used to create the new trade allowance agreement. To set up a template, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Templates**.
2. On the Action Pane, select **New**.
3. In the fields, enter the information that should appear in the agreements that are created from the template.
4. On the **Customers** FastTab, in the **Hierarchy** field, select a hierarchy level.
5. In the hierarchy list, select the customer that has an unmatched deduction, and then select the right arrow button (>). The customer is added to the **Trade allowance agreement customers** list.
6. Set the remaining fields as you require, and then close the page.
7. Go to **Sales and marketing** > **Setup** > **Trade allowance** > **Trade allowance management parameters**.
8. On the **Overview** tab, in the **One-time promotion template** field, select the name of the template to use to create one-time promotions.

Next, you can create a one-time promotion on the deduction workbench. To create a one-time promotion, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.

2. Select the **Mark** checkbox next to the deduction to process.

3. On the Action Pane, select **Maintain** > **Settle deduction as one-time promotion**.

4. In the **One-time promotion** dialog box, follow these steps to associate the deduction with one or more funds:
   a. Select **New**, and then, in the **Fund ID** field, select a fund ID. Repeat this step to add as many funds as you require.
   b. In the **Percent** field next to each fund ID, enter the percentage of the deduction to allocate to the fund. The amounts that you enter in the **Percent** fields must total 100 percent.

5. Select **OK**. The system creates a new trade allowance agreement that has a lump sum merchandising event and matches the lump sum to the deduction.

Do a mass update of deductions

If you must make the same change to multiple deductions, you can select those deductions and do a mass update of their fields.

To do a mass update, follow these steps.

1. Go to **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**.
2. In the **Show** field below the Action Pane, select the type of deductions to view.
3. Select the checkbox next to each deduction that you want to update. Then, on the Action Pane, select **Maintain** > **Mass update**.
4. The **Mass update** dialog box shows the selected deductions. Update the fields as you require, and then select **OK** to accept the changes.

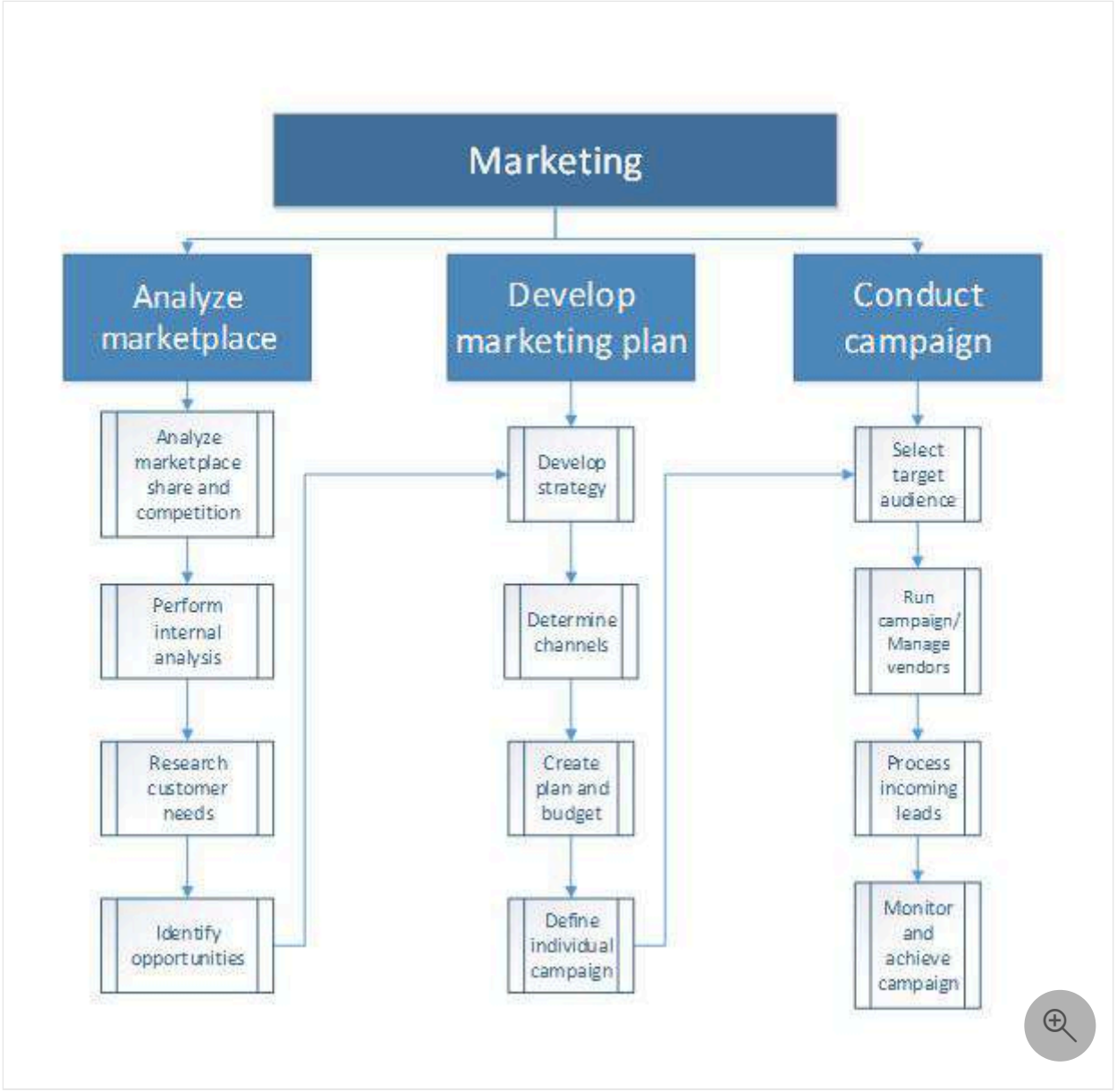
Sales and marketing overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use Sales and marketing to obtain, store, and use various types of data in the sales flow. This data includes the original sales initiative, future follow-up action, and additional sales.

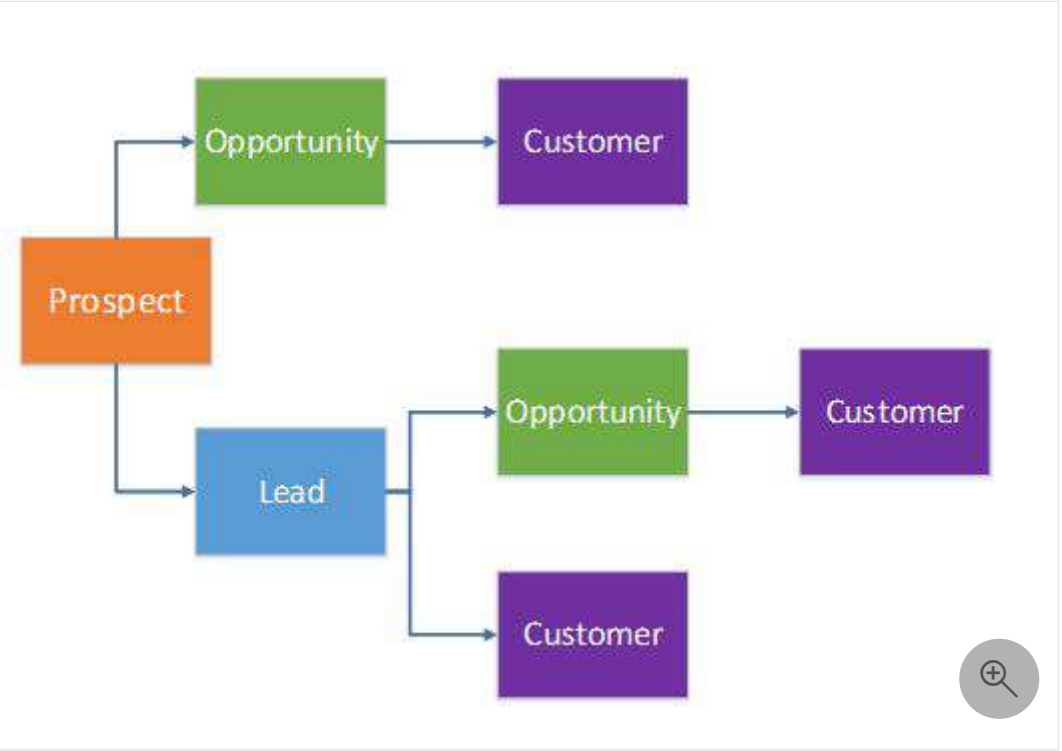
Marketing

Use marketing campaigns and activities to find and build relationships with potential customers, so that initial interactions can develop into sales relationships. The following process flow shows the business process for marketing.



Relationships

In sales and marketing, the initial interactions that you have with potential customers can occur in various situations. For example, you might find a prospective customer while you're attending a trade show, or you might have a possible lead with a customer after your organization runs a mass mailing campaign. It's very important that you understand the flow of a party's entity before that party becomes a customer. The following graphic shows the flow of entity relationships as a potential customer becomes an actual customer.

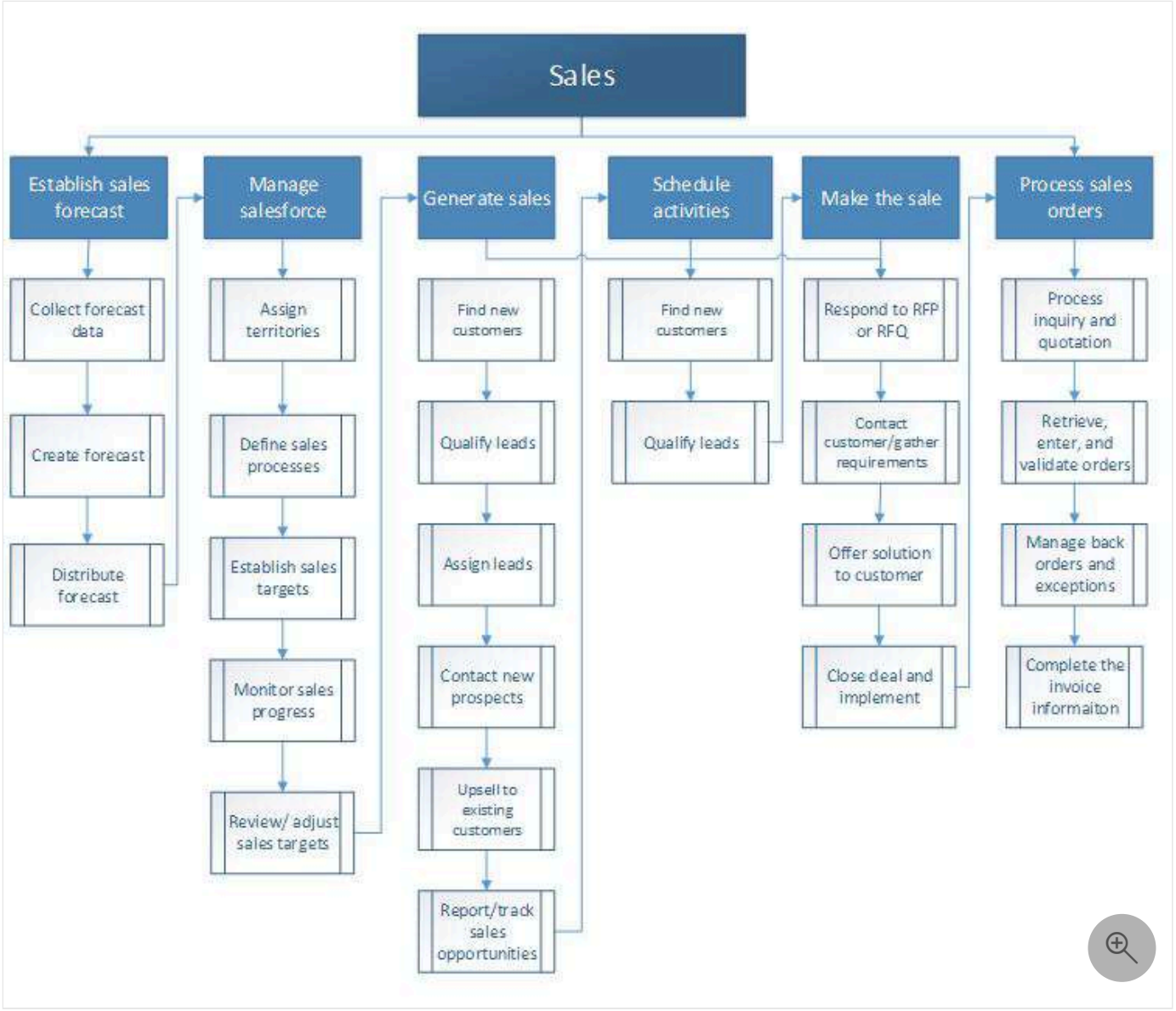


Campaigns

A campaign targets the contacts for prospects, leads, opportunities, and customers that are selected to participate in the campaign. In Supply Chain Management, you can create several types of campaigns, such as telemarketing, mailing, and email campaigns, to maximize your customer potential. As your campaign progresses and you receive positive responses, you can begin the sales process with those recipients who respond positively to the campaign.

Sales

Use the sales functionality to create quotations, up-sell and cross-sell to new and existing customers, create sales orders, and create sales invoices for customers. The following process flow shows the business process for sales.



Sales quotations

Create sales quotations to present customers an offer of the goods or services that you provide. A customer might request a quotation, or you might create a quotation in response to a request from a potential or existing customer. When the customer approves the sales quotation, you can convert it to a sales order.

Up-sell/cross-sell

Up-selling and cross-selling are techniques for selling products when an order is entered for a customer. In up-selling, you suggest another product instead of the current product. In cross-selling, you suggest a product in addition to the current product. When you set up product lists, you can create specific rules to indicate when a product should be suggested as a cross-sell or up-sell product.

Sales orders

When you create a new sales order, select the type of sales order to create. You have five options.
> [!NOTE]

> After you create a sales order, you can change any order type, except the **Items requirements** type if the sales order has a status of **Delivered**.

Expand table

Sales order type	Description
Journal	Use this type as a draft for a sales order. This type has no effect on the stock quantities and doesn't generate item transactions.
Subscription	Use this type for recurring orders. When the order is invoiced, the order status is automatically set to an open order. The delivered quantity that was invoiced and the remaining deliveries are updated. You can't use this sales order type if you're using the Warehouse management functionality.
Sales order	Use this type when a customer places or confirms an order.
Returned order	Use this type when a customer returns an item. A return-item number (RMA number) is assigned automatically.
Item requirements	This type is created automatically when you make an item sale through a project.

Sales agreements

A sales agreement is a contract that commits the customer to buy a product in a specific quantity or for a specific amount over time, in exchange for special prices and discounts. The prices and discounts of the sales agreement overrule any prices and discounts that are stated in any trade agreements that exist. A sales agreement is valid for a defined period. The requested ship date specified for a sale on the **Sales order** page should be in the valid period. By default, a sales agreement is on hold. You can order from a sales agreement only when it's set to **Effective**.

Backorders

When you enter and validate orders, you might need to manage backorders and exceptions before you can complete the sale. Backorders are either purchase orders that a vendor is yet to deliver or sales orders that you're yet to deliver to a customer. It's important that you follow up on backorders. For example, if a vendor delays products, you might need to change the date of

delivery to a customer and then inform the customer of the delay. You can view backorders by item, customer, or vendor.

Viewing backorders by item

When you view backorders by item, you can follow up on the expected future flow of transactions for a specific item. For example, you can check the following information:

- The number of sales orders that are placed for an item
- Whether deliveries of the item from vendors are still missing
- Whether you should order more items so that you can deliver all sales orders in a timely manner

By doing this check, you can respond to requests from customers about the timing of the item delivery. In addition, you can prioritize the sales backorders and split the items that are on hand between the orders.

Viewing backorders by customer

When you view backorders by customer, you can view the status of the customer's remaining orders. This check is useful when you must respond to customers who are waiting for items that are delayed.

Viewing backorders by vendor

When you view backorders by vendor, you can follow up on missing deliveries and expected dates of delivery. This check also helps you prioritize the backorders when products arrive from vendors and the sales orders must be picked for delivery.

Invoices

During the sales process, you can create three types of invoices:

- Customer invoice
- Free text invoice
- Pro forma invoice

Customer invoice

A customer invoice is a bill that an organization gives to a customer in connection with a sale. You create this kind of customer invoice based on a sales order that includes a header and one or

more lines for items or services. The customer invoice completes the sales order, packing slip, and sales invoice cycle.

You can post and print a single customer invoice, based on either a sales order or the packing slip and date. You can also post and print multiple customer invoices together, based on the packing slips and dates. When you post a single customer invoice by using the sales order, the **Invoiced remainder** quantity for each item is updated with the total of the invoiced quantities from the selected sales order.

If both the **Invoiced remainder** and **Deliver remainder** quantities for all items on the sales order are 0 (zero), the status of the sales order changes to **Invoiced**. If the quantity of either field isn't 0, the status of the sales order isn't changed, and you can enter additional invoices. To post and print one or more customer invoices based on the packing slips, you must already have posted at least one packing slip for each sales order. The customer invoice is based on the packing slips and reflects the quantities that are listed.

You can create a customer invoice that is based on the packing slip line items that you shipped to date, even if all the items for a particular sales order aren't yet shipped. You might do this if, for example, your legal entity issues one invoice per customer per month to cover all the deliveries that you ship during that month. Each packing slip represents a partial or complete delivery of the items on the sales order.

When you post the invoice, the **Invoice remainder** quantity for each item is updated with the total of the delivered quantities from the selected packing slips. If the **Invoice remainder** and **Deliver remainder** quantities for all items on the sales order are 0 (zero), the status of the sales order changes to **Invoiced**. If the quantity isn't 0, the status of the sales order isn't changed, and you can enter additional invoices. Inventory transactions are updated with the invoice number, and the status on the sales order line changes to **Invoiced**.

Free text invoice

A free text invoice is an invoice that isn't related to a sales order. It contains order lines that include ledger accounts, free-text descriptions, and a sales amount. You can't enter an item number on this kind of invoice, and you must enter the appropriate sales tax information. A main account for the sale is indicated on each invoice line. The customer balance is posted to the summary account from the posting profile that is used for the free text invoice.

Pro forma invoice

A pro forma invoice is an invoice that you prepare as an estimate of the actual invoice amount before the invoice is posted. You can print a pro forma invoice for either a customer invoice or a free text invoice.

Related information

Blogs

You can find an overview of a sales process in the post How sales work in Dynamics 365 Finance and Operations.

White papers

The following white papers explore various aspects of sales and marketing in Dynamics 365 Supply Chain Management:

- Trade agreements

Last updated on 04/21/2026

Order promising

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about order promising. Order promising helps you reliably promise delivery dates to your customers and gives you flexibility so that you can meet those dates.

Order promising calculates the earliest ship and receipt dates, and is based on the delivery date control method and transport days. You can select among the following delivery date control methods:

- **Sales lead time** – Sales lead time is the time between creation of the sales order and shipment of the items. The delivery date calculation is based on a default number of days, and doesn't consider stock availability, known demand, or planned supply.
- **ATP (available-to-promise)** – ATP is the quantity of an item that is available and can be promised to a customer on a specific date. The ATP calculation includes uncommitted inventory, lead times, planned receipts, and issues.
- **ATP + Issue margin** – The shipping date equals the ATP date plus the issue margin for the item. The issue margin is the time that is required to prepare the items for shipment.
- **CTP (capable-to-promise)** – Availability is calculated through explosion. If you're using Planning Optimization, the *CTP* delivery date control method isn't allowed and, if it's selected, will cause an error when the calculation runs. For more information about how to set up and use CTP, see Calculate sales order delivery dates using CTP.
- **CTP for Planning Optimization** – Use the CTP calculation that is provided by Planning Optimization. This option has no effect if you're using the built-in master planning engine. For more information about how to set up and use CTP, see Calculate sales order delivery dates using CTP.

> **Note**

When a sales order is updated, the order promising information is only updated if the existing order promising date can't be fulfilled, as illustrated in the following examples:

- **Example 1**: The current order promising date is July 20, but due to increased quantity, you can't deliver until July 25. Because the current date can no longer be met, order promising is triggered.
- **Example 2**: The current order promising date is July 20, but due to decreased quantity, it's now possible to deliver on July 15. However, because the current date can still be fulfilled, order promising isn't triggered, and July 20 remains the order promising date.

ATP calculations

The ATP quantity is calculated by using the **cumulative ATP with look-ahead** method. The main advantage of this ATP calculation method is that it can handle cases where the sum of issues among receipts is more than the latest receipt. For example, when a quantity from an earlier receipt must be used to meet a requirement. The **cumulative ATP with look-ahead** calculation method includes all issues until the cumulative quantity to receive exceeds the cumulative quantity to issue. Therefore, this ATP calculation method evaluates whether some of the quantity from an earlier period can be used in a later period.

The ATP quantity is the uncommitted inventory balance in the first period. Typically, it's calculated for each period in which a receipt is scheduled. The program calculates the ATP period in days and calculates the current date as the first date for the ATP quantity. In the first period, ATP includes on-hand inventory minus customer orders that are due and overdue.

ATP is calculated by using the following formula:

ATP = ATP for the previous period + Receipts for the current period – Issues for the current period – Net issue quantity for each future period until the period when the sum of receipts for all future periods, up to and including the future period, exceeds the sum of issues up to and including the future period.

Notice that the ATP calculation doesn't include information around expiry date and beyond the ATP time fence that the system expects when any quantity can be promised.

When there are no more issues or receipts to consider, the ATP quantity for the following dates is the same as the latest calculated ATP quantity.

If all the dimensions used for an item aren't given when the ATP check is completed, you can still specify them on the issue and receipts. In this case, in the ATP calculation, the receipts and issues

must be aggregated to the existing dimensions to reduce the number of receipt and issue lines that are used in the ATP calculation.

The ATP quantity that is shown is always greater than or equal to 0 (zero). If the calculation returns a negative ATP quantity (for example, if the quantity that was previously promised exceeds the available quantity), the quantity is automatically set to 0.

Example

The **ATP backward demand time fence** field controls how far back in time to look for delayed demand orders or inventory issues. The **ATP backward supply time fence** field controls how far back in time to look for delayed supply orders or inventory receipts. For example, if orders that are delayed by only seven days should be considered in the ATP calculation, both fields should be set to **7**.

The **ATP delayed demand offset time** and **ATP delayed supply offset time** fields control when the delayed demand or supply is considered in the ATP calculation. For example, if the delayed supply and demand should be considered in the ATP calculation the day after tomorrow, set both fields to **2**. A value of **2** means that the quantity of an item on a delayed purchase order that should be considered in the ATP calculation is available two days after the current date.

For the following example, **7** is entered in the **ATP backward demand time fence** and **ATP backward supply time fence** fields, and **1** is entered in the **ATP delayed demand offset time** and **ATP delayed supply offset time** fields.

You haven't received a purchase order for 200 pieces of a product that should have been received three days ago. Therefore, a sales order line for 75 pieces of the same product that should have been shipped yesterday isn't shipped.

A customer calls and wants to order 150 pieces of the same product. When you verify the availability of the product, you find that another purchase order for 100 pieces of the product will be delivered 10 days later.

You create a sales order line for the product and enter **150** as the quantity.

Because the delivery date control method is ATP, the ATP data is calculated to find the earliest possible ship date. Based on the settings, the delayed purchase order and sales order are considered, and the resulting ATP quantity for the current date is 0. Tomorrow, when the delayed purchase order is expected to be received, the ATP quantity is calculated as more than 0 (in this

case, it's calculated as 125). However, 10 days from now, when the additional purchase order for 100 pieces is expected to be received, the ATP quantity becomes more than 150.

Therefore, the ship date is set to 10 days from now, based on the ATP calculation. Therefore, you tell the customer that the requested quantity can be delivered 10 days from now.

CTP calculations

CTP functionality lets you give customers realistic dates for when you can promise specific goods. For each sales line, you can provide a date that takes account of existing on-hand inventory, production capacity, and transportation times.

CTP extends ATP functionality by considering capacity information. Whereas ATP considers only material availability and assumes infinite capacity resources, CTP considers availability of both materials and capacity. Therefore, it provides a more realistic picture of whether demand can be satisfied within a given time frame.

CTP works slightly differently, depending on the master planning engine that you're using (Planning Optimization or the built-in engine). CTP for Planning Optimization currently supports only a subset of the CTP scenarios that are supported by the built-in engine.

For detailed information about how to set up and use CTP for each engine, see Calculate sales order delivery dates using CTP.

Next steps

- Calculate sales order delivery dates using CTP
- Delivery alternatives
- Inventory Visibility on-hand change schedules and available to promise

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Last updated on 04/21/2026

Calculate sales order delivery dates using CTP

Capable-to-promise (CTP) functionality lets you give customers realistic dates for when you can promise specific goods. For each sales line, you can provide a date that takes account of existing on-hand inventory, production capacity, and transportation times.

CTP extends available-to-promise (ATP) functionality by considering capacity information. Whereas ATP considers material availability and assumes infinite capacity resources, CTP considers availability of both materials and capacity.

CTP works slightly differently, depending on the master planning engine that you're using (Planning Optimization or the deprecated master planning engine). This article describes how to set it up for each engine.

How CTP compares to ATP

Use ATP when you sell from stock and capacity isn't a constraint. Use CTP when you assemble or make-to-order and capacity constraints must be taken into account. In such cases, CTP can often provide a more accurate result, as the following example shows.

Item A is an item that is composed of items B and C, and the on-hand quantity of item A is 0 (zero). If you do an ATP check, the ATP quantity is also 0 (zero). However, if you do a CTP check, the system checks the availability of items B and C, checks the resources that are required to make them into item A, and calculates how many of item A can be made. In addition, the CTP calculation can check the resources and materials that are required to make more of items B and C, and to use them to make more of item A.

A CTP calculation that considers both materials and resources might show a larger quantity than a calculation that checks only materials, particularly when the item that's being checked is an assemble or make-to-order item. In other words, CTP functionality is based on the explosion function and can be run for a selected sales order line to calculate the expected delivery date.

Near real-time CTP

Near real-time CTP enables the system to calculate CTP confirmed dates in the background, without blocking user interface (UI) interactions and without requiring that you run planning to

update the dates. *Near real-time CTP* also lets you use standard CTP delivery date control with Planning Optimization. This approach removes some of the limitations that apply when you use CTP with either planning engine. Without *Near real-time CTP*, you must use *Batch CTP* instead of *CTP* delivery date control if you're using Planning Optimization.

> **Note**
>
> - Because both Planning Optimization and the deprecated master planning engine can now use standard *CTP* delivery date control, CTP for Planning Optimization delivery date control is renamed in Supply Chain Management version 10.0.41. It's now named *Batch CTP* instead of *CTP for Planning Optimization* to better describe the difference.
> - Because CTP considers material and capacity availability and lead times for the bill of material and route for the item sold, the CTP calculation isn't as fast as the ATP calculation. When deciding which order promising policy to apply for an item, base the policy on whether the item is produced to and sold from stock (ATP, ATP + Issue Margin, or Sales Lead Time) or if it's an assemble or make-to-order item (CTP).

Enable Near real-time CTP

Before you can use *Near real-time CTP*, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.41 or later.
- The following features must be turned on in Feature management (in this order):
    1. *Improve Planning Optimization performance by merging and queueing plan regeneration jobs*
    2. *Near real-time CTP*

Queueing and merging plan regeneration jobs

The prerequisite feature, *Improve Planning Optimization performance by merging and queueing plan regeneration jobs*, enables the system to queue and merge plan regeneration requests. This capability eliminates the "Another job for the same plan is currently in progress" error and minimizes the average time that is required to complete a planning run when multiple users are working at the same time.

Here's how queueing and merging works:

- By default, the system can merge up to 10 planning requests into a single job. When the system reaches that limit, it adds new planning requests to the queue as standalone jobs. The system can also merge those jobs into another new job. If you need to change the maximum number of planning requests, contact Microsoft Support for assistance.

- You can merge planning requests into a single job only if they meet the following requirements:
  - All merged requests must belong to the same organization and master plan.
  - All merged requests must use the same filter attribute (such as `ItemId`).
  - All merged requests must have the same UI language.

- Before the system adds a planning request to the queue as a standalone job, it checks for other existing jobs that aren't yet started and can be merged with it.

- By default, the queue can hold up to 20 jobs. When the queue reaches that limit, all users who submit planning runs receive an error until more space becomes available in the queue. Each job in the queue can include several merged planning jobs. If you need to change the maximum number of jobs that are allowed in the queue, contact Microsoft Support for assistance.

Near real-time CTP functionality

Near real-time CTP provides the following functionality:

- Multiple sellers can create sales lines at the same time and immediately get confirmed delivery dates.
- When you mass-import sales lines, the system automatically calculates confirmed delivery dates.
- You can use the **Calculate confirmed delivery dates** dialog box to make the system recalculate all confirmed dates for a sales order. If you set the **Allow multiple deliveries** option to *No*, the system automatically sets all sales order lines so that they use the same delivery date.

How CTP differs, depending on the master planning engine and delivery date control that you use

The following table summarizes how CTP differs, depending on the master planning engine and delivery date control that you use.

Expand table

Element	Batch CTP	CTP
Planning engine support	Planning Optimization only	Planning Optimization and the deprecated master planning engine
Delivery date control setting for orders, order lines, and products	*Batch CTP*	*CTP*
Calculation time	The calculation is triggered when a dynamic plan is run as a scheduled task.	The calculation is immediately triggered each time that you enter or update a sales order line.
Batch CTP status field value	A value of *Not ready* is shown for orders and order lines where the dynamic plan hasn't been run since the orders and lines were created or last updated. A value of *Ready* is shown for orders and lines where CTP was used to calculate confirmed delivery dates by running the dynamic plan.	A value of *Ready* is always shown.
CTP status field value	The field isn't shown.	A value is shown only if you're using Planning Optimization. The value represents the status of the delivery date calculation.

Set default delivery date control methods

The system can use any of several methods to calculate delivery date estimates for each order and order line. Set the delivery date control method that you want to use most often as the global default method. You can also set individual overrides for each product. Later, you can override the default methods for each order and/or order line as you require.

Set the global default delivery date control

The default delivery date control method applies to all new order lines where an override doesn't apply. To select a method, follow these steps:

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.

2. On the **Shipments** tab, on the **Delivery control** FastTab, in the **Delivery date control** field, select the method that you want to use as the default method for your company:

    - *None* – Don't calculate delivery dates.
    - *Sales lead time* – Sales lead time is the time between creation of the sales order and shipment of the items. The delivery date calculation is based on a default number of days, and doesn't consider stock availability, known demand, or planned supply.
    - *ATP* – ATP is the quantity of an item that's available and can be promised to a customer on a specific date. The ATP calculation includes uncommitted inventory, lead times, planned receipts, and issues.
    - *ATP + Issue margin* – The shipping date equals the ATP date plus the issue margin for the item. The issue margin is the time that is required to prepare the items for shipment.
    - *CTP* – Use the standard CTP calculation, which provides near-real-time calculations. To use this option with Planning Optimization, you must enable *Near real-time CTP* for your system.
    - *Batch CTP* – Use the non-real-time CTP calculation that's available for Planning Optimization. This setting has no effect if you're using the deprecated master planning engine.

Set delivery date control overrides for individual products

Assign overrides for specific products when you want to use a delivery date control method other than the one that you set as your global default method.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product that you want to set up.
3. On the Action Pane, on the **Manage inventory** tab, in the **Order settings** group, select **Default order settings**.
4. On the **Default order settings** page, on the **Sales order** FastTab, set the **Override delivery control** option to *Yes*.
5. In the **Delivery date control** field, select the method to use for the selected product. The same options that are described in the Set the global default delivery date control section are available.

Schedule Batch CTP calculations

When you use Batch CTP, you must run a dynamic plan to trigger the system to do the CTP calculations and then set the confirmed ship and receipt dates for all relevant orders. The plan

must include all items that confirmed ship and receipt dates are required for. (When you use standard CTP delivery date control, the CTP calculations are immediately done locally. Therefore, you don't have to run a dynamic plan to see the CTP results.)

To ensure that the dates are available in good time for all users, set up batch jobs to run the relevant plans on a recurring basis. For example, a batch job that is set up to run a dynamic plan every 30 minutes sets the confirmed ship and receive dates every 30 minutes. Therefore, users who enter and import orders wait a maximum of 30 minutes to get the confirmed ship and receive dates.

To set up a batch job to run a dynamic plan on a regular schedule, follow these steps:

1. Go to **Master planning** > **Master planning** > **Run** > **Master planning**.
2. In the **Master planning** dialog box, on the **Parameters** FastTab, set the **Master plan** field to the dynamic plan that you want to run.
3. On the **Run in the background** FastTab, set the **Batch processing** option to *Yes*.
4. Select **Recurrence**, and set up the schedule as required.
5. Select **OK** to save the schedule.
6. Select **OK** to create the batch job and close the dialog box.

Use standard CTP delivery date control

To use standard CTP delivery date control, you must either use the deprecated master planning engine or use Planning Optimization with *Near real-time CTP* enabled.

Create a new order that uses standard CTP delivery date control

Each time that you add a new sales order or order line, the system assigns a default delivery date control method to it. The order header always starts with the global default method. If you assign an override to an ordered item, the new order line uses that override. Otherwise, the new order line also uses the global default method. Therefore, set your default methods so that they match the delivery date control method that you most often use. After you create an order, you can override the default method at the order header and/or order line level as you require. Learn more in Set default delivery date control methods and Change existing sales orders to use CTP.

View confirmed delivery dates when you use standard CTP delivery date control

CTP calculations apply to orders and order lines where the **Delivery date control** field is set to *CTP*.

For sales lines that use standard CTP delivery date control, the system automatically sets the **Confirmed ship date** and **Confirmed receipt date** fields each time that you save a sales line. If you make a relevant change to a sales line later (for example, by changing its quantity or site), the system immediately recalculates the dates.

- To view the confirmed delivery dates for a sales order line, open the sales order, and select the sales line. Then, on the **Line details** FastTab, on the **Delivery** tab, review the **Confirmed ship date** and **Confirmed receipt date** values.
- To view the confirmed delivery dates for an entire order, open the sales order, and select the **Header** view. Then, on the **Delivery** FastTab, review the **Confirmed ship date** and **Confirmed receipt date** values.

Use Batch CTP

Create a new order that uses Batch CTP

Each time that you add a new sales order or order line, the system assigns a default delivery date control method to it. The order header always starts with the global default method. If you assign an override to an ordered item, the new order line uses that override. Otherwise, the new order line also uses the global default method. Therefore, set your default methods so that they match the delivery date control method that you most often use. After you create an order, you can override the default method at the order header and/or order line level as you require. Learn more in Set default delivery date control methods and Change existing sales orders to use CTP.

View confirmed delivery dates when you use Batch CTP

Batch CTP calculations apply to orders and order lines where the **Delivery date control** field is set to *Batch CTP*.

For sales lines that use Batch CTP, the **Confirmed ship date** and **Confirmed receipt date** fields stay blank until you run the appropriate dynamic plan (typically by using a periodic batch job). The system then sets these fields automatically. Learn more in Schedule Batch CTP calculations.

The **Batch CTP status** field shows whether confirmed dates are calculated for each line that uses Batch CTP. The field is set to *Not ready* for lines and orders where the confirmed delivery dates aren't calculated yet or are no longer valid because of other changes. The field is set to *Ready* for lines and orders that are calculated. You can view the status for each line and for the entire order.

- To view the status for a sales order line, open the sales order, and select the sales line. Then, on the **Line details** FastTab, on the **Delivery** tab, review the **Batch CTP status** value. The **Delivery** FastTab also shows the **Confirmed ship date** and **Confirmed receipt date** values for the line after they're calculated.
- To view the status for an entire order, open the sales order, and select the **Header** view. Then, on the **Delivery** FastTab, review the **Batch CTP status** value. The **Delivery** FastTab also shows the **Confirmed ship date** and **Confirmed receipt date** values for the order after they're calculated.

> ⓘ **Note**
>
> - If you update a sales order line after Batch CTP calculates confirmed dates for it, the system clears those dates until the next time that the appropriate dynamic plan runs.
> - If you edit a related setting that might affect existing confirmed dates (for example, by changing lead times, reservations, or markings), clear the confirmed dates for the relevant existing orders. This action changes the status of each relevant line to *Not ready*. This change causes the system to recalculate the confirmed dates the next time that it runs the dynamic plan.
> - Batch CTP only calculates confirmed dates for sales lines with status *Not Ready*, so subsequent Planning Optimization runs that affect planned supply for sales lines with status *Ready* might cause the confirmed dates to become out of sync with the latest plan result. To prevent this, consider firming planned orders early to make sure that generated supply remains unaffected by future Planning Optimization runs.

Change existing sales orders so that they use CTP

You can change the **Delivery date control** value for any open order at any time. You can change the value at the header level and/or for each line as you require.

Change to CTP at the order header level

To change an order so that it uses CTP at the order header level, follow these steps:

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Open the sales order that you want to set up, or create a new one.
3. Select **Header** to open the header information on the **Sales order details** page.

4. On the **Delivery** FastTab, set the **Delivery date control** field to one of the following values, depending on the planning engine that you're using:

    - *CTP* – Use the standard CTP calculation, which provides near-real-time calculations. To use this option with Planning Optimization, you must enable *Near real-time CTP* for your system.
    - *Batch CTP* – Use the non-real-time CTP calculation that's available for Planning Optimization. This setting has no effect if you're using the deprecated master planning engine.

5. Select **OK** to apply your changes.

Change to CTP at the order line level

If you create an order line by using a different delivery date control method, you can change to CTP at any time. Changes that you make at the line level don't affect any other lines. However, they might cause the overall order delivery dates to move forward or backward, depending on how each updated line calculation changes.

To change an order so that it uses CTP at the line level, follow these steps:

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.

2. Open the sales order that you want to set up, or create a new one.

3. On the **Sales order details** page, on the **Sales order line** FastTab, select the sales order line that you want to set up.

4. On the **Line details** FastTab, on the **Delivery** tab, set the **Delivery date control** field to one of the following values, depending on the planning engine that you're using:

    - *CTP* – Use the standard CTP calculation, which provides near-real-time calculations. To use this option with Planning Optimization, you must enable *Near real-time CTP* for your system.
    - *Batch CTP* – Use the non-real-time CTP calculation that's available for Planning Optimization. This setting has no effect if you're using the deprecated master planning engine.

    The **Available ship and receipt dates** dialog box appears, and shows the available ship and receipt dates. This dialog box works the same way for order lines as it does for the order header, as described in the previous section.

5. On the Action Pane, select **Save**.

Additional resources

- Keep supply for confirmed demand in Planning Optimization
- Optimize confirmed dates for CTP line changes

Last updated on 07/27/2026

Order entry deadlines

This article provides information about order entry deadlines. An order entry deadline is a cut-off time that determines whether a customer order is treated (and fulfilled) as if it was received on the current day or the next day.

In many companies, only sales orders that are received before a certain time of day are treated as if they were received on that day. Any orders that are received after that time are treated as if they are received on the next business day. This cut-off time for orders is known as the order entry deadline.

Order entry deadlines are used as input for order promising. Therefore, they help you manage customer expectations about order deliveries. For example, customers can see that, if they place an order with you before a specific time, you will commit to shipping the goods on the same day. However, if they miss that deadline, they can expect the shipment only on the next business day. You set order entry deadlines based on your warehouse capabilities and shipping carrier schedules.

On the **Order entry deadlines** page, you set up order entry deadline times for all the days of the week. If orders are received after the specified times, they are treated as if they are received on the next day. By default, these times are set to 23:59 (that is, one minute before midnight at the end of the relevant day). You can change the default times so that they coincide with actual ship or receipt deadline times.

You can define order entry deadlines for a specific group of customers. For example, you might want a specific group of customers to have order entry deadlines that are later than those of other customers. In this case, you first define groups for order entry deadlines on the **Order entry deadline groups** page. You then assign the groups to customers on the **Customers** page.

If your company consists of several sites, you can set up order entry deadlines for each site. If the sites are located in different time zones, the order entry deadlines are set up in each site's time zone. However, when you work with sales orders and sales quotations, the order entry deadline is converted to your time zone on the **Available ship and receipt dates** page.

On the **Activate order entry deadline combinations** page, you define the combinations of sites and order entry deadline groups that are allowed.

Example: Order entry deadline

The order entry deadline on Tuesdays has been set to 16:00. On a particular Tuesday, at 17:00, you try to set the current date as the ship date. (Note that there is no lead time for this example.) If the **Delivery date control** check box is selected, you receive a warning that states that the date isn't valid. This warning appears on the **Available ship and receipt dates** page, where you can then select alternative dates.

Example: Different order entry deadlines per site

Your company consists of two sites. The sites are located in different time zones, as shown in the following table.

Expand table

Site A	Site B
California	Florida
PST (Pacific Standard Time)	EST (Eastern Standard Time)

Sites A and B have defined the following order entry deadlines.

Expand table

Day of the week	A: Order entry deadlines (PST)	B: Order entry deadlines (EST)
Monday	13:00	14:00
Tuesday	13:00	14:00
Wednesday	13:00	14:00
Thursday	13:00	14:00
Friday	13:00	14:00

You're an order processor in Utah, where the time zone is MST (Mountain Standard Time). Therefore, provided that you place orders with site A before 14:00 MST and place orders with site B before 12:00 MST, you meet the order entry deadlines for both sites.

The following table shows how the order entry deadlines for sites A and B are converted to MST time.

Expand table

Site A: PST	Site A: MST	Site B: EST	Site B: MST
13:00	14:00	14:00	12:00

Note: If adjustment for daylight saving time is in effect, the order entry deadlines are adjusted accordingly.

Example: Same order entry deadline per site

Your company consists of two sites. The sites are located in different time zones, as shown in the following table.

Expand table

Site A	Site B
California	Florida
PST (Pacific Standard Time)	EST (Eastern Standard Time)

Sites A and B have defined the following order entry deadlines.

Expand table

Day of the week	PST and EST
Monday	13:00
Tuesday	13:00
Wednesday	13:00
Thursday	13:00
Friday	13:00

You're an order processor in Utah, where the time zone is MST. Therefore, provided that you place orders with site A before 14:00 MST and place orders with site B before 11:00 MST, you meet the order entry deadlines for both sites.

The following table shows how the order entry deadlines for sites A and B are converted to MST time.

Expand table

Site A: PST	Site A: MST	Site B: EST	Site B: MST
13:00	14:00	13:00	11:00

Note: If adjustment for daylight saving time is in effect, the order entry deadlines are adjusted accordingly.

Related information

Delivery schedules

Last updated on 07/01/2026

Delivery alternatives

Article • 09/29/2021

Sales order takers can use the **Delivery alternatives** page to discover alternative order fulfillment options.

The **Delivery alternatives** page layout gives an overview of all alternative options. It also lets order takers look beyond the current company for fulfillment opportunities. They can now view both intercompany opportunities and opportunities from external vendors. By sorting the options by delivery date, sales order takers can view an intelligent list of delivery alternatives. In addition, parameters help them better manage the suggested deliveries. Because transport time can affect delivery dates, sales order takers can explore the various transportation choices that carriers offer. Because detailed information is shown for each suggestion, order takers can make informed decisions directly from the **Delivery alternatives** page.

Open the Delivery alternatives page

You can open the **Delivery alternatives** page from the sales order line.

1. Select **Products and supply** > **Delivery alternatives**.
2. Select **Line details** > **Delivery** > **Delivery alternatives.**

You can also open the **Delivery alternatives** page by opening the **Sales order processing and inquiry** workspace, and then selecting **Orders and favorites** > **Delayed order lines** > **Delivery alternatives Note:** You can open the **Delivery alternatives** page only if both the following conditions are met:

- All mandatory sales line information is filled in.
- The **Delivery date control** field is set to a value other than **None**.

Delivery date control methods

The delivery date control method determines how the system establishes delivery dates, how delivery alternatives are calculated, and what information is shown. Note that delivery date control takes calendars into consideration. Therefore, the following calendars can affect the suggested receipt date: Warehouse calendar, Transport calendar, Vendor calendar, and Customer calendar. The following table describes each method for delivery date control.

Expand table

Method	Description
None	• Delivery alternatives for sales lines aren't supported. This option turns off delivery date control.
Sales lead time	• Delivery alternatives are calculated based on the predefined sales lead time. The transport days are calculated based on the mode of delivery. • Delivery alternatives include warehouses that have on-hand inventory, and supply/demand orders.
ATP	• Delivery alternatives are calculated based on available to promise (ATP) logic. The current availability and expected future availability are considered. The transport days are calculated based on the mode of delivery. • Delivery alternatives include warehouses that have on-hand inventory, and supply/demand orders.
ATP + Issue margin	• Delivery alternatives are calculated as for the ATP method, but the issue margin is included in the calculation.

CTP	• Delivery alternatives are calculated based on the capable to promise (CTP) logic. MRP explosion is used to determine availability. Note that, at a minimum, CTP offsets delivery dates to the sales lead time. The transport days are calculated based on the mode of delivery.
	• Delivery alternatives include suggestions for the following warehouses:
	  ◦ Current warehouse
	  ◦ Default warehouse
	  ◦ All warehouses that have available on-hand inventory
	  ◦ All warehouses that have supply orders
	  ◦ All warehouses that have active bill of materials (BOM) versions

View information about delivery alternatives

This section describes the information about delivery alternatives that is available on each FastTab of the **Delivery alternatives** page.

The Product FastTab

This FastTab shows a summary of the product and details of the current sales line.

The Delivery alternatives FastTab

This FastTab shows a list of delivery alternatives that is sorted by receipt date. Above the list, you can select which options to base the suggestions on. You can also select the mode of delivery, which determines the transport days. The following options are available:

- **Include other product variants** - This option is available for products that have product variants. It will include delivery alternatives for other variants of the product. This option isn't available for CTP.
- **Include partial quantity** - By default, only suggestions that fulfill the full quantity of the sales line are included. Select this option to include suggestions that only partially fulfill the order line. This option is useful when the customer requests an earlier delivery date and accepts partial delivery.

- **Include later dates** - By default, only suggestions that are better (earlier) than the current dates on the sales line are shown. Select this option to include later dates. This option can be useful in situations where parameters other than the date have priority. For example, a specific vendor or warehouse might be preferred.
- **Mode of delivery** - Select the preferred mode of delivery to optimize transport time and cost. You will immediately see the effect on the suggested delivery alternatives. Therefore, it's easy to compare the alternatives.
- **Include procurement** - When procurement is selected, the suggested delivery alternatives include options to procure from both external vendors and other companies in the enterprise (intercompany). The **Include procurement** option is supported for ATP and ATP + Issue margin delivery date control. Procurement options from the default purchase vendor for the product and all approved vendors for the product are included.
- For external vendors, the calculation is based on the purchase lead time.
- For intercompany, the calculation considers what is available from the sourcing company, based on delivery date control in the sourcing company.
- **Delivery type** (Relevant for procurement)
  - **Stock** - Products are shipped from the sourcing warehouse to the site/warehouse on the sales line. They are then shipped from that warehouse to the customer.
  - **Direct delivery** - Products are shipped directly from the sourcing warehouse to the customer.

The Availability information FastTab

Information on this FastTab is related to the delivery alternative line that is selected. The following information is shown, depending on the delivery date control for the sales line:

- **Sales lead time**
  - **Available today** - Show the current physical on-hand, physical reserved, and available physical inventory.
  - **Parameters** - Show the inventory unit and sales lead time.

- **ATP and ATP + Issue margin**
  - **Available today** - Show the current physical on-hand, physical reserved, and available physical inventory.
  - **Parameters** - Show the inventory unit and sales lead time.
  - **Future availability** - Show a graphical representation of current and future availability for the selected site and warehouse under **Delivery alternatives**. You can select the chart columns to see more detailed information about the future

availability of the product. The slider shows a list of relevant demand and supply orders within the ATP time fence.

- CTP
  - **Available today** - Show the current physical on-hand, physical reserved, and available physical inventory.
  - **Parameters** - Show the inventory unit and sales lead time.
  - **Explosion** - Show a supply explosion of the selected delivery alternative. You can use **Setup** to change the fields and inventory dimensions that are shown in the explosion.

The Impact of selected alternative FastTab

This FastTab highlights the impact of the selected delivery alternative. If you select **OK**, the sales line is updated with the highlighted values in the SELECTED columns. Note that, if the quantity on the selected delivery alternative is less than quantity on the sales line, a delivery schedule is created, and the order line is split into two lines: one line for the selected quantity and one line for the remaining quantity. You can also update the commercial line so that it matches the schedule lines and affects the pricing.

Related information

Order promising

Feedback

Was this page helpful? Yes No

Provide product feedback

Direct deliveries

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about direct deliveries. Direct deliveries are deliveries that a vendor sends directly to your customer.

Direct deliveries save delivery time and reduce the costs that are associated with carrying inventory, because you don't hold the products in your warehouse before you ship them to the customer.

You can create direct deliveries from the **Sales order** page. First, create a sales order and order lines. Then, on the Action Pane, on the **Sales order** tab, select **Direct delivery**. Finally, specify the lines that must be handled as a direct delivery. A link is now created between the sales order lines for the direct delivery and the corresponding purchase order lines.

> **Note**
>
> If part of the ordered quantity is already delivered, you must split the remaining quantity. Create a new line for the quantity that you must directly deliver, and subtract that quantity from the quantity on the original line. For example, if the original quantity was 15, and five are delivered, you must create a new line for the remaining quantity, 10, and then reduce the original quantity by that amount.

After you create the direct delivery link between the sales order lines and the purchase order lines, the sales order line packing slip is updated automatically through the purchase order line product receipt and can only be updated from the purchase order line product receipt update. Run either a product receipt update or an invoice update from the purchase order. If the invoice update is done on from the purchase order without a previous product receipt, then no automatic packing slip update happens for the sales order. However, the sales order line is still updated to *delivered*. You can then invoice-update the sales order, but don't do a packing slip

update because this action is triggered automatically. You can update the invoice for the sales order from the **Sales order** page. An invoice update can't cause the quantity on the sales order to exceed the quantity that is delivered. For example, a sales order line has 10 pieces, but only five pieces from the sales order line are updated to *delivered*. If you select *All* in the **Quantity** list when you invoice-update the sales order, only the quantities that are delivered and not yet invoiced are invoice-updated.

Delivery date

When you update the **Requested receipt date** field on the sales order line, the **Delivery date** field on the corresponding purchase order line is also updated. Similarly, when you update the **Confirmed** field on the purchase order line, the **Confirmed receipt date** and **Confirmed ship date** fields on the corresponding sales order line are also updated.

Delivery address

Typically, the delivery address for a purchase order is the company's address. However, when you create a direct delivery, you enter the customer's address as the delivery address. If you change the delivery address on a purchase order line that has a delivery type of **Direct delivery**, the delivery address on the corresponding sales order line is also updated. Similarly, if you change the delivery address on the sales order line, the delivery address on the purchase order line is also updated.

Deleting order lines

If you try to delete a sales order line that has a delivery type of **Direct delivery**, a message box states that purchase order lines are attached to the line. If the sales order line is partially delivered, you can't delete the sales order line or the purchase order lines that are attached to it.

Warehouse

When you create a direct delivery, the items that you sell never physically arrive at your warehouse. However, you must still specify a warehouse on the sales order line. Similarly, picking requirements might be specified on the item model group for the item. However, because the items never physically arrive at your warehouse, these requirements are ignored when the sales order is a direct delivery.

Last updated on 05/06/2026

Delivery schedules

Article • 01/27/2025

In Microsoft Dynamics 365 Supply Chain Management, delivery schedules are used to manage and track the delivery of order quantities in multiple shipments. They are particularly useful when a single sales order, sales quotation, or purchase order must be delivered in several parts over time.

Here's a breakdown of how delivery schedules work:

1. **Create delivery schedules.** When you create a delivery schedule, the total quantity of an order line is divided into multiple delivery lines, based on the delivery schedule, quantity, mode of delivery, and storage dimensions (such as site and warehouse). Each delivery line represents a separate shipment.
2. **Manage delivery lines.** Delivery lines can have different delivery dates, quantities, modes of delivery, and storage dimensions. The original order line is converted to a *commercial line*, which acts as a header for the delivery lines. If delivery line quantities are changed, the commercial line is automatically updated so that it reflects the total quantity.
3. **Process orders.** Orders that have a delivery schedule are processed against the delivery lines. This processing includes the posting of packing slips, product receipts, and invoicing. When documents such as confirmations or invoices are printed, only the delivery lines are shown, not the original commercial lines.

When you create a delivery schedule, the type of the original order line is automatically changed to *Order line with multiple deliveries*. A line of this type is referred to as a commercial line and is marked by a symbol. The delivery line is marked by a different symbol. If you change a quantity on a delivery line, the commercial line is updated to the total quantity of the delivery schedule. If a trade agreement defines a total discount for the order, the delivery schedule ensures that your order is eligible for the total order discount, even when the order is split into separate deliveries.

Orders that have a delivery schedule are processed against the delivery lines. This processing includes the posting of packing slips, product receipts, and invoicing.

Document printouts of orders and quotations that have a delivery schedule show only the delivery lines. They don't show the original lines (commercial lines).

> **Note**
>
> Only the delivery lines are shown when you perform the following actions:
>
> - Post
> - Copy pages

- Browse list pages and reports

When you confirm sales quotations, the resulting sales orders show the whole delivery schedule, even the order lines that have multiple deliveries. In addition, the whole delivery schedule is shown on all the major pages, such as sales orders, sales quotations, and purchase orders.

Example of a delivery schedule

In this example, a customer requests delivery of 600 chairs in batches of 100 chairs over a period of six months. To keep track of the delivery requirements, you create a delivery schedule. On the delivery schedule page, you create six separate delivery lines. Each delivery line contains 100 chairs and indicates the delivery date for those 100 chairs. In this case, each line is offset on the first of the month for six consecutive months.

Expand table

Item	Value
Total order (original order line)	600 chairs
Requested delivery schedule	100 chairs per month
Requested time frame for delivery	6 months, on the first day of each month

Create delivery schedule

This procedure demonstrates how to create a delivery schedule for a sales order. A delivery schedule is used when a quantity on an order or a quotation is requested to be delivered in multiple shipments.

To create delivery schedule, follow these steps:

1. Go to **All sales orders**.
2. Select **New**.
3. In the **Customer account** field, enter or select a value.
4. Select **OK**.
5. In the **Item number** field, enter or select a value.
6. In the **Quantity** field, enter a number that is bigger than *1*.
7. Select **Sales order line**.
8. Select **Delivery schedule**.

    - The **Delivery schedule** page is the place where you can specify the number of shipments in which the total quantity of the order line will be delivered to the customer.
    - By default, the system copies the total quantity and other delivery details of the original sales line into the first delivery schedule line. In this example, we'll create a schedule for two shipments, with the second shipment's date offset by a week from the first one.

9. In the **Quantity** field, enter a number that is part of the total quantity.
10. Select **New**.
11. In the **Quantity** field, enter the remaining quantity.
12. In the **Requested ship date** field, enter a date a date that is one week ahead from the date of the first delivery line.

    - The two options on the **Charges conversion** FastTab control how you want the charges to be distributed across the delivery schedule lines, once they've been assigned to the original order line. If you select **Copy gross amounts**, the same charge amount is copied to each line. The **Allocate to delivery lines** option divides the charge equally across the delivery lines.
    - Only fixed charges can be divided, whereas variable charges will still be copied to the lines.

13. Move the cursor away from the second delivery line to update the page. You can keep track of the total quantity that's allocated to the delivery schedule lines by looking at the **Total** and **Remaining** fields. When the remaining quantity is zero, the full quantity from the original line has been allocated to the schedule.
14. Select **OK**.

    - The delivery schedule has now been copied to the order lines.
    - The original order line, referred to as a **Commercial** line, has been converted to an order line with multiple deliveries. It is marked with a distinct icon and acts as a header for the delivery lines.
    - The two new lines, referred to as delivery lines, make up one delivery schedule. The order will be processed against these lines and not the original line. If documents such as confirmation slips, picking lists, packing slips, or invoices are printed, only the delivery lines are shown.
    - The delivery lines can have different delivery dates, quantities, modes of delivery, and storage dimensions, such as site and warehouse. However, the product dimensions must always match the ones on the commercial line and can't be changed.

15. In the **Quantity** field, enter a number that's bigger than the current one.
16. Select the commercial line to see the effect of the quantity recalculation.
17. On the Action Pane, select **Pick and pack**.
18. Select **Post packing slip**.
19. Expand the **Parameters** section.
20. In the **Quantity** field, select *All*. Note that the packing slip will be created for the two delivery schedule lines and not the original order line.
21. Select *Yes* in the **Print packing slip** field.
22. Select **OK**.
23. Select **Yes**.
24. Close the page.

Last updated on 12/30/2025

Working with serialized items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to register serial numbers on packing slips or invoices during the sales process. This functionality is useful if a company wants to capture serial numbers for service and warranty purposes, but doesn't need to maintain serial numbers in inventory from receipt to issue.

Many companies just want to capture serial numbers for service and warranty purposes, and don't need to maintain serial numbers in inventory from receipt to issue. In these scenarios, you can register the serial numbers on the packing slips or invoices when products are sold. If products are later returned, you can trace each product to an invoice to determine whether you sold the product, and whether the service or warranty obligations are valid.

You must enable serial numbers for the sales process by selecting the **Active in sales process** option on the **Tracking dimension groups** page. The following events then occur in Supply Chain Management:

- On the **Serial numbers** FastTab, the **Serial number control** option is set to *Yes*. When you select this option, you must register one serial number for each item on the packing slip or invoice.
- All selections on the tracking dimension group for serial numbers are cleared, except the **Blank issue allowed** option. You can select the **Blank issue allowed** option to override the serial number control, and allow products to be packed and invoiced without registering serial numbers.

> **Important**
>
> When the **Physical inventory** option for the **Serial number** dimension is cleared, including the automatic clearing that occurs when you select **Active in sales process**, on-hand

inventory isn't tracked per serial number. This change can cause discrepancies between received and issued serial numbers for the same item.

When do I register serial numbers during the sales process?

You can register serial numbers either on the packing slip for a sales order or on the invoice. When you prepare an invoice for a serialized item that was shipped together with a packing slip, you can select which serial numbers on the packing slip to invoice. The number of registered serial numbers must not exceed the quantity of items that were shipped. If you're creating a partial invoice, you can select fewer serialized items than were registered on the packing slip. When you print a packing slip or an invoice, the serial numbers that were registered are included.

Can I enter serial numbers by scanning them, or do I have to type them?

You can either scan or type serial numbers. When you use a scanner, the scan mode determines whether the serial numbers are added to or removed from the list of serial numbers on the invoice or packing slip. If you want to scan serial numbers by using, for example, a hand-held bar code scanner, configure the scanner to send an Enter or TAB command after the serial number. This command indicates the end of the data stream. Otherwise, you must press Enter or TAB on the keyboard after you scan each serial number.

If I enable serial numbers for the sales process, do I have to register all serial numbers for all items?

The setup of the tracking dimension group that is assigned to the product determines whether serial numbers must be registered for all items on a packing slip or invoice. When you enable serial numbers for the sales process, the **Serial number control** option is automatically selected. You must then register one serial number, or register a blank registration for an unreadable number, for each item on the packing slip or invoice. If you don't want to require a serial number for each item, select the **Blank issue allowed** option on the tracking dimension group that is assigned to the item. You can then register fewer serial numbers than the quantity of items that are being shipped. If you register more serial numbers than the quantity of items that are being shipped, you won't be able to post the packing slip or invoice.

Can I register serial numbers for partial invoices and partial shipments?

You can create partial invoices and packing slips for sales orders, and register serial numbers only for the items that those invoices and packing slips include. If you want to create a partial invoice and you have more than one packing slip for the sales order, you can include serial numbers from more than one packing slip. However, there can be only one packing slip that doesn't include all serial numbers. For example, if you have three packing slips, and each packing slip includes two serialized items, you can't create a partial invoice for one item from each packing slip.

What do I do when a serial number isn't readable?

If a serial number can't be read or scanned, you can create a blank line for the item by selecting **Not readable** on the **Serial numbers** page. If the serial number becomes available later, you can update the invoice or packing slip. For more information, see the next section, "Can I correct or change the serial numbers that I registered for a sales order?"

Can I correct or change the serial numbers that I registered for a sales order?

Yes, you can correct serial numbers if the following conditions are met:

- **Invoices** – You can change the serial numbers for items that you didn't yet invoice. The packing slip is then also updated. However, if you corrected a sales order line by registering a negative quantity, you can't change serial numbers for the sales order line.
- **Packing slips** – You can't partially correct a packing slip line that contains serialized items. You must reverse the full quantity for the line. If you canceled or corrected a packing slip, you don't have to register the reversed serial numbers again when you create a new packing slip for the same serialized items. The numbers that you registered are used.

Can I view the serial numbers that shipped together with a specific packing slip, or that were included on an invoice?

Yes, you can run an inquiry on the packing slip journal line or invoice journal line to view a list of all serial numbers that the document includes.

Can I view the serialized items that I have on hand?

No, you can't view the serialized items that you have on hand, because serial numbers aren't registered for items until the items are sold.

Can I register serial numbers for catchweight items?

No, you can't register serial numbers for catch-weight items during the sales process. Additionally, if a product is set up as a catch-weight item, you can't assign the product to a tracking dimension group that is set up to use serial numbers only during the sales process.

Can I register serial numbers at the retail POS?

Yes, the retail point of sale (POS) prompts the user to enter a serial number when the user sells an item that is assigned a tracking dimension group that is set up to use serial numbers only during the sales process.

What security roles are required to register serial numbers during the sales process?

All roles that can maintain sales packing slips and sales invoices can use this functionality. The following duties let workers correct serial numbers and register blank entries for serial numbers that can't be read or scanned:

- Maintain serial number corrections
- Maintain registration of non-readable serial numbers

Last updated on 05/27/2026

Reserve the same batch for a sales order

Article • 09/29/2021

This article explains how to set up a product to allow reservation of inventory against a single batch of inventory.

Same batch reservation lets you reserve inventory for a sales order line against a single batch of inventory. For example, a customer who orders wallpaper can request that the whole order be filled from the same batch or lot, to avoid inconsistencies among the rolls. To set up a product to use same batch reservation, the following settings must be active in the item model group, tracking dimension group, and storage dimension group that you assign to the product:

- **Item model groups** – The item model group must have the **Same batch selection** and **Consolidate requirement** fields selected in the **Reservation** field group for inventory policies.
- **Tracking dimensions groups** – The tracking dimension group must have the **Coverage plan by dimension** field selected for the batch number.
- **Storage dimensions groups** – The storage dimension group must have the **Coverage plan by dimension** field selected for **Site** and **Warehouse**.

When you reserve inventory for a product on a sales order line that is set up for same batch selection, the system tries to reserve the ordered quantity from a single inventory batch. Any specific batch attribute requirements are also considered. If the quantity can't be filled from a single batch, the **Same batch reservation conflict** page appears. This page describes the issues and also the actions that you can take to continue with the reservation. The following conditions might prevent the batch from being reserved:

- The batch disposition code has **Block reservation** for sales flagged as **Blocked**.
- The batch has expired, based on the expiration date and any applicable customer sellable days. The item can still be considered for reservation if the item model group for the item is First Expiry First Out (FEFO) date–controlled, and if the best-before date is selected as the pick criterion.
- The batch doesn't have enough shelf-life days remaining, based on the expiration date and best-before date, plus any customer sellable days.

For items associated with a storage dimension group that has **Use warehouse management processes** enabled, you can reserve specific batch numbers by using a reservation hierarchy where the batch number inventory dimension is defined above the location dimension. This type of reservation hierarchy is also known as a *Batch-above [location]* reservation hierarchy. The **Batch reservation** page for sales and transfer order

lines also lets you select and reserve multiple lines based on the available batch numbers. For more information about what to do if you are using a reservation hierarchy that has the batch number dimension below the location (*Batch-below [location]*), see Flexible warehouse-level dimension reservation policy.

Feedback

Was this page helpful? Yes No

Provide product feedback

Ship orders as direct deliveries

07/10/2025

This article demonstrates how to create a direct delivery for a sales order. You use direct delivery when you want to ship goods to the customer directly from your vendor, instead of shipping them to your own warehouse first. To successfully complete the second subtask *Create direct deliveries from the workbench*, make sure that the item that you choose on the sales order has a default vendor specified on the **Purchase** FastTab of the released product master.

Set an individual order for direct delivery

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Select **New**.
3. Enter or select a value in the **Customer account** field, then select **OK**
4. Enter or select values in the **Item number** and **Site** fields, then select **Save**.
5. On the Action Pane, select **Sales order**, then select **Direct delivery**. The Create delivery page lists all the open sales order lines as copied from the sales order. You can review the order details, and if necessary, you can modify details such purchase quantity and pricing terms before you create the direct delivery.
6. Select **Yes** in the **Include all** field.

    - If you want to generate a direct delivery for only a subset of the sales order lines, select these individually.
    - The **Vendor account** field might already be populated with a vendor number. If the default vendor is set up for the product (on the associated Item coverage), then this vendor will be copied to the line. Otherwise, you must enter a vendor manually. In this example, we'll select a new vendor in the next step, even if one is already populated.

7. Enter or select a value in the **Vendor account** field, then select **OK**. The message informs you that the purchase order has now been created.
8. Expand the **Line details** section.
9. Select the **Delivery** tab and verify that the **Direct delivery** field is set to *Yes*.
10. On the Action Pane, select **General**.
11. Select **Related orders**.
12. Select the link in the **Purchase order** field.
13. Expand the **Line details** section and select the **Address** tab.

    - The delivery address for this purchase order line is the customer's delivery address and not your company's address.

- If you change the delivery address on either the purchase order line or the originating sales order line, the address on the corresponding order line will be automatically updated.

14. Select the **Delivery** tab.

    - Like the sales order line, the associated purchase order line type is also set to *Direct delivery*.
    - The purchase order line's **Delivery date** and the **Confirmed delivery date** are set to the **Requested receipt date** and **Confirmed receipt date** of the originating sales order line respectively.
    - If you update any of these dates on either the purchase line or the sales line, the dates on the corresponding order will be automatically updated.
    - The purchase order that is set to deliver goods directly to the customer is linked to the originating sales order using a special association. This association imposes the rule that the packing slip update of the sales order can't be done from the sales order itself and must be done by using the purchase order. However, customer invoicing must be carried out from the sales order.

15. On the Action Pane, select **Purchase**.
16. Select **Confirmation**.
17. Select **OK**.
18. On the Action Pane, select **Receive**.
19. Select **Product receipt**.
20. In the **Product receipt** field, type a value.
21. Select **OK**.
22. On the Action Pane, select **General**.
23. Select **Related orders** and highlight the desired record.

    - After the purchase order has been updated as received, or in other words, after the vendor has shipped the goods to your customer's address, the status of the originating sales order is automatically updated to Delivered.
    - The sales order can now be invoiced.

24. Select **OK**.
25. Close the page.
26. Select **OK**. Close the pages and return to the home page.

Create direct deliveries from the workbench

1. Go to **Navigation** > **Modules** > **Accounts receivable** > **Orders** > **All sales orders**.
2. Select **New**.
3. Enter or select a value in the **Customer account** field, then select **OK**.

4. Enter or select a value in the **Item number** and **Site** fields.
5. Expand the **Line details** section, then select the **Delivery** tab. Instead of creating a direct delivery as part of the sales order processing as in the previous procedure, you can choose to hand over this task to a purchasing professional. To include the sales order line in the direct delivery handling process, you must mark the line for direct delivery.
6. Select *Yes* in the **Direct delivery** field.

    - If the item has already been set up for direct delivery by default, the field will automatically be set to *Yes* at the order line entry. You can set up an item for direct delivery on the Released product's master by setting the Direct delivery option to Yes and selecting a default Direct delivery warehouse.
    - Because the purchase order hasn't yet been created, the **Direct delivery status** is set to *To be direct delivered*.

7. Select **Save**.
8. Close pages until you return to the home page.
9. Enter *Direct delivery* in the search bar.

    - The Direct delivery page acts as a workbench that provides the purchasing agent with an overview of all the sales order lines that are to be direct delivered and it allows them to create the respective purchase orders. In addition, they can view the open direct delivery orders and the confirmed orders on the **Confirmation** and **Delivery** tabs.
    - After you have created a direct delivery order, it automatically moved to the Confirmation tab. You can choose to confirm the order directly from this page. When the purchase is confirmed, it will automatically move to the **Delivery** tab, from which you can register its receipt.

Ship sales orders without warehousing

Article • 01/31/2023

This article explains how to update a sales order when products are shipped to the customer. The guide is applicable to the fulfillment flow that is not set up for warehouse management (neither basic or warehouse management processes (WMS)), and therefore does not require product picking to be registered before shipment. You can run this procedure on your own data or in demo data company USMF. In both cases, before you start this task, create a sales order for an inventoried product with a quantity of greater than 1. To avoid a posting error, you need to check that the product's on-hand quantity in the site and warehouse that you've selected on the order covers the order quantity.

Post packing slip for an order

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. In the list, find and select the order you have created for this task.
3. On the Action Pane, select **Pick and pack**.
4. Select **Post packing slip**.
5. Expand or collapse the **Parameters** section.
6. In the **Quantity** field, select **All**.

    - Other options include **Deliver now** and **Picked**. If the order line is to be shipped partially and the **Deliver now** field on the order line contains a quantity, you would select **Deliver now**. If your organization's fulfillment flow includes picking as a separate process that is managed by and registered with a picking list, you would select **Picked**.
    - Check that the **Posting** option is set to **Yes**.

7. Set the **Print packing slip** option to **Yes**. The **Overview** tab contains a list of packing slips to be generated in this posting. If you are shipping an individual order, there will typically be one packing slip. However, if that order's lines are to be shipped from different sites, posting will automatically be split into the appropriate number of documents. This is a mandatory condition that cannot be changed. Similarly, the posting will also be split into multiple documents if the order's lines are to be shipped to different delivery addresses, and the shipping policy is set up to require a split.
8. On the **Lines** tab, select the row for the order line to be shipped.
9. In the **Update** field, enter a number lower than the original quantity.
10. Select **OK**.

11. Select **Yes**.
12. Close the page.
13. On the Action Pane, select **Options**.
14. Select **Change view**.
15. Select **Header view**.

    - If all of the lines on the order have been fully shipped, the order status changes from Open to Delivered.
    - In this example, the order line has been shipped partially. This is why the order status remains Open.
    - The **Document status** field is set to Packing slip because at least one of the order lines have been shipped.

16. On the Action Pane, select **General**.
17. Select **Line quantity**.
18. Close the page.
19. On the Action Pane, select **Pick and pack**.
20. Select **Packing slip**. The **Packing slip journal** page contains all the packing slip documents that were generated for your order. You can review details of each document and print them, if you wish.

Feedback

Was this page helpful? Yes No

Provide product feedback

Automatic application of charges

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Based on the customer or vendor that you're working with or the item that you're selling, you might want to apply specific additional charges. The *charges* feature in Microsoft Dynamics 365 Supply Chain Management helps you automatically apply charges to purchase orders or sales orders.

Automatic charges (auto charges) are automatically applied when you create a sales order or a purchase order. You can define auto charges for specific vendors, customers, groups of vendors, groups of customers, items, or groups of items. You can also define auto charges that apply to all vendors, customers, or items.

Set up parameters

The **Procurement and sourcing parameters** page has a few settings that are especially relevant when you want to apply charges automatically in the purchase order. To complete this setup, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **Prices** tab, on the **Prices** FastTab, set the following fields:
    - **Find auto charges for header** – Set this option to *Yes* if charges should automatically be applied to purchase order headers.
    - **Find auto charges for line** – Set this option to *Yes* if charges should automatically be applied to purchase order lines.

The **Accounts receivable parameters** page also has a few settings that are especially relevant when you want to apply charges automatically in the sales order. To complete this setup, follow these steps.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Prices** tab, on the **Prices** FastTab, set the following fields:

- **Find auto charges for header** – Set this option to *Yes* if charges should automatically be applied to sales quotation and sales order headers.
- **Find auto charges for line** – Set this option to *Yes* if charges should automatically be applied to sales quotation and sales order lines.

Set up charges codes

To apply charges, you must first define charges codes.

1. Follow one of these steps:

   - For purchase orders: Go to **Procurement and sourcing** > **Setup** > **Charges** > **Charges code**.
   - For sales orders: Go to **Accounts receivable** > **Setup** > **Charges** > **Charges code**.

2. On the Action Pane, select **New** to create a charges code.

3. In the header of the new record, set the following fields:

   - **Charges code** – Enter a code for the charges.
   - **Description** – Enter a description of the charges.
   - **Item sales tax group** – Select an item sales tax group, if applicable.
   - **Prorate** – Set this option to *Yes* if you want to prorate your charges. This option is available only for sales orders.
   - **Maximum amount** – Enter the maximum amount that's allowed for the charges code. This field is used to validate charges for vendor invoices. It's available only for purchase orders.

     > ⓘ **Note**
     >
     > To turn on the functionality for validating charges for purchase orders, go to **Accounts payable** > **Setup** > **Accounts payable parameters**. On the **Invoice validation** FastTab, in the **Invoice validation** section, set the **Enable invoice matching validation** option to *Yes*.

4. The **Posting** FastTab includes **Debit** and **Credit** sections. Set the following fields, depending on the ledger that you want to post the charges to:

   - **Type** – Select the type of account that you're posting to (*Ledger*, *Customer*, or *Item*). If you select *Ledger account* as the debit type or credit type, specify a posting type in

the **Posting** field, and specify the main account in the **Account** field.
- **Posting** – Select the type of postings to create (such as *Broker fee* or *Customer settlement*).
- **Account** – Select the account to post the charge for.

5. On the Action Pane, select **Save**.

Create charge groups

Charge groups automatically apply specific charges to a group of customers or vendors. The following subsections describe how to create and assign these charge groups.

Charge groups for purchase orders

To create charge groups for purchase orders, follow these steps.

1. Go to **Procurement and sourcing** > **Setup** > **Charges** > **Vendor charges group**.
2. On the Action Pane, select **New** to add a row to the grid, and then set the following fields:
    - **Charges group** – Enter the name of the charge group.
    - **Description** – Enter a description of the charge group.
3. On the Action Pane, select **Save**.
4. Go to **Procurement and sourcing** > **Vendors** > **All vendors**, and either open an existing vendor or create a new vendor.
5. On the **Purchase order defaults** FastTab, in the **Purchase order** section, set the **Charges group** field to the charge group that you created.

Charge groups for sales orders

To create charge groups for sales orders, follow these steps.

1. Go to **Accounts receivable** > **Setup** > **Charges** > **Customer charge groups**.
2. On the Action Pane, select **New** to add a row to the grid, and then set the following fields:
    - **Charges group** – Enter the name of the charge group.
    - **Description** – Enter a description of the charge group.
3. On the Action Pane, select **Save**.

4. Go to **Accounts receivable** > **Customers** > **All customers**, and either open an existing customer or create a new customer.

5. On the **Sales order defaults** FastTab, in the **Sales order** section, set the **Charges group** field to the charge group that you created.

Define auto charges

After you set up your charge codes, follow these steps to define the auto charges.

1. Follow one of these steps:

    - For purchase orders: Go to **Procurement and sourcing** > **Setup** > **Charges** > **Automatic charges**.
    - For sales orders: Go to **Accounts receivable** > **Setup** > **Charges setup** > **Auto charges**.

2. In the list pane, in the **Level** field, select the level where your auto charge applies:

    - *Header* – Apply charges to the order header.
    - *Line* – Apply charges to the order lines.

3. Select an existing auto charge to edit it, or select **New** to define a new auto charge.

4. In the **Account code** list, select one of the following values to specify the scope of accounts that the auto charge affects:

    - *Table* – Assign charges to a specific customer or vendor.
    - *Group* – Assign charges to a miscellaneous charges group.
    - *All* – Assign charges to all customers or vendors.

5. In the **Customer relation** or **Vendor relation** field, select a specific customer or vendor if you set the **Account code** field to *Table*. If you set the **Account code** field to *Group*, select a customer or vendor charges group.

6. In the **Item code** field, select one of the following values to specify the scope of items that the auto charge affects. You can select an item code only when you define auto charges at the line level.

    - *Table* – Assign charges to a specific item.
    - *Group* – Assign charges to an item charges group.
    - *All* – Assign charges to all items.

7. In the **Item relation** field, select a specific item if you set the **Item code** field to *Table*. If you set the **Item code** field to *Group*, select an item charges group.

8. *For sales orders only:* In the **Mode of delivery code** field, select one of the following values to specify the scope of delivery modes that the auto charge affects:

    - *Table* – Assign charges to a specific mode of delivery.
    - *Group* – Assign charges to a mode of delivery group.
    - *All* – Assign charges to all modes of delivery.

9. *For sales orders only:* In the **Mode of delivery relation** field, select a specific mode of delivery if you set the **Mode of delivery code** field to *Table*. If you set the **Mode of delivery code** field to *Group*, select a mode of delivery group.

10. On the **Lines** FastTab, define the charges and the charge rates that the current auto charge uses. You can use the toolbar on this FastTab to add as many lines as you require. For each line, set the following fields:

    - **Currency** – Select the currency that you want to use to calculate the charge. This field applies to the *Header* and *Line* levels for sales quotation and sales order charges.

    - **Charges code** – Select the code for the charge.

    - **Category** – Select one of the following values:
        - *Fixed* – Enter the charge as a fixed amount on the line. You can use fixed charges on charges both in the order header and on the order lines.
        - *Pcs* – Base the charge on unit with no unit of measure conversion. *Pcs* in this case represents any unit of measure and not the specific *Pcs* unit of measure. You can use these charges only on order lines. They appear when you calculate the order total.
        - *Percent* – Enter the charge as a percentage on the line. You can use percentage charges on charges both in the order header and on the order lines.
        - *Intercompany percent* – Enter the charge as a percentage on the line for intercompany orders. You can use intercompany percentage charges only on order lines.
        - *External* – A third-party service that's associated with one or more shipping carriers calculates the charge.
        - *Specific unit* – Express the charge value in the unit of measure on the charge line. Unit of measure conversion is used to apply it proportionally to the unit of measure on the sales line. This category applies only to the *Line* level.
        - *Specific unit match* – Express the charge value in the unit of measure on the charge line. The unit of measure on the sales line must match this unit of measure for the charge line to be applied. No unit of measure conversion is applied. This category applies only to the *Line* level.

- **Unit** – Enter the unit that applies when the **Category** field is set to *Specific unit* or *Specific unit match*.
- **Charges value** – Enter the charge value, based on the category that you selected.
- **Charges currency code** – Specify a currency for the charge if you want to use a currency other than the currency that you specified in the **Currency** field. You can use a different currency only if the **Debit type** or **Credit type** field is set to either *Ledger account* or *Item* for the selected charges code.
- **From amount** – Specify a starting amount to apply the auto charge to. In this context, the amount refers to the order total.
- **To amount** – Specify the ending amount to apply the auto charge to. In this context, the amount refers to the order total.
- **Sales tax group** – Specify a sales tax group.
- **Site** and **Warehouse** – Specify a site and warehouse if charges should be applied only for a specific site and warehouse.
- **Keep** – Select this checkbox to keep the charges transactions after invoicing is completed, so that the charge is applied every time that you create a new invoice for the selected customer account.

For sales orders only: If you want to calculate tiered charges, see Tiered charges on sales orders for information.

Supply Chain Management provides the following optional functionality for calculating auto charges. You might need to enable and configure these features before you can use them. For more information, follow the links.

- Auto charge compounding and sequencing
- Units of measure for line-level charges

> **Tip**
>
> When configuring auto charges for accounts payable in Dynamics 365 Finance, you might see duplicate charge calculations if you set up auto charges for both specific vendors (*Table*) and all vendors (*All*). This situation occurs because the system applies all relevant auto charge configurations, regardless of hierarchy, whenever you create a new purchase order. For example, if you set up a charge code with account code *All*, the system applies that charge code to every purchase order, even if you also set up other specific vendor or group configurations.

Allocate charges from the header to a line

The following procedure shows how to allocate header-level charges to a line. Before you start this procedure, you should already have a header-level charge of the *fixed amount* type and an order where you apply that charge. Additionally, the order should already include at least one line item.

1. Open the purchase order or charge order.

2. On the Action Pane, follow one of these steps:

    - For purchase orders: On the **Purchase** tab, in the **Charges** group, select **Allocate charges**.
    - For sales orders: On the **Sell** tab, in the **Charges** group, select **Allocate charges**.

3. In the **Allocate charges to order lines** dialog box, set the following fields:

    - **Charges allocation** – Select one of the following values to specify how the charges should be allocated:
        - *Net amount* – Allocate charges according to each line amount relative to the total net amount.
        - *Quantity* – Allocate charges according to the number of units for each line relative to the total number of units.
        - *Per line* – Allocate charges equally among the total number of lines.

    - **Allocate charges to lines** – Select a value to specify whether charges should be allocated to all lines, to positive lines only, or to negative lines only.

    - **Allocate all** – Select this checkbox to allocate charges to order lines even if the charges code has a debit type other than *Item*.

    - **Received** – Select this checkbox to allocate charges only to received order lines.

    - **Stocked** – Select this checkbox to allocate charges to only inventoried order lines.

    - **Show selections and clear specific lines** – Select this checkbox to exclude specific lines from this allocation. When you select this checkbox, the **Choose lines to exclude from allocation** grid opens. This grid includes only lines that match the criteria that are defined by the **Allocate charges to lines** and **Stocked** settings. For example, if you set the **Allocate charges to lines** field to *Positive lines* and select the **Stocked** checkbox, the grid shows only lines that are both positive and inventoried. In addition, the grid automatically filters out any lines that the full quantity is already received for. While the grid is open, clear the **Include** checkbox for each line that you want to exclude from allocation.

> **Important**
>
> When you work with the **Choose lines to exclude from allocation** grid, be sure to leave the grid open until you select **Allocate**. If you close the grid before you select **Allocate**, your settings in the grid are lost. Therefore, charges are allocated based on the criteria that you previously defined.

4. Select **Allocate** to apply your settings and close the dialog box.

Reapply header auto charges

In some cases, the system reapplies header auto charges by deleting and inserting them after you update a header field.

You can set up auto charges to consider vendor charge groups or modes of delivery. Therefore, when you update the **Charges group** or **Mode of delivery** on a purchase order header, the charge might change based on the auto charge setup. When the charge needs to be recalculated, the system deletes the existing auto charge and inserts a new auto charge. The recalculation doesn't affect charges you add manually to the purchase order.

After the system allocates an auto charge from the header to a line, it doesn't track its origin (there's no reference from the line to the auto charge setup), so the charge becomes a line charge. If you then update the **Charges group** or **Mode of delivery** on a purchase order header, the system reapplies the new, recalculated header charge, but it doesn't affect the line charges. This pattern applies to both purchase order headers and sales order headers.

The following examples show how to update header-level charges and select a new **Mode of delivery** on the purchase order header. Each example assumes that you already have a header-level charge of type *fixed amount* and an order where that charge is applied. The order must already include at least one line item.

Example 1: Header auto charge value resets after changing the mode of delivery

This example shows that if you customize the value of a header auto charge and then update the mode of delivery for the purchase order, the system resets the header auto charge to its original value.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders** and open a purchase order that includes a header auto charge of type *fixed amount*.

2. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.
3. For the existing header auto charge, enter a new value in the **Charges value** field.
4. On the Action Pane, select the **Back** button to return to the purchase order.
5. Open the **Header** tab of the purchase order.
6. Expand the **Delivery** FastTab and select a new value for the **Mode of delivery** field.
7. On the Action Pane, select **Save**.
8. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

   Note that the **Charges value** is now reset to its original auto charge value.

Example 2: Header auto charge is reapplied after allocating it to a line and then changing the mode of delivery

This example shows that if you allocate a header auto charge to a line and then update the mode of delivery for the order, the system reapplies the original header auto charge while leaving the allocated charge applied to the line.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders** and open a purchase order that includes a header auto charge of type *fixed amount*.
2. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

   Note the header auto charges listed here.
3. On the Action Pane, select the **Back** button to return to the purchase order.
4. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Allocate charges**.
5. On the **Allocate charges to order lines** dialog, select the **Allocate all** check box. Then select **Allocate** to allocate the header auto charge to the order lines.
6. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

   Note that the header auto charges are no longer listed because you allocated them to the line level.

7. On the Action Pane, select the **Back** button to return to the purchase order.
8. On the **Purchase order lines** FastTab, select a purchase order line.
9. On the **Purchase order lines** FastTab toolbar, select **Financials** > **Maintain charges**.

   Note that the previous header auto charge is now shown here, at the line level.
10. On the Action Pane, select the **Back** button to return to the purchase order.
11. Open the **Header** tab of the purchase order.
12. Expand the **Delivery** FastTab and select a new value for the **Mode of delivery** field.
13. On the Action Pane, select **Save**.
14. On the Action Pane, open the **Purchase** tab and, in the **Charges** group, select **Maintain charges**.

    Note that the original header auto charge is now reapplied to the order, even though the charge is also listed at the line level.

Last updated on 12/09/2025

Auto charge compounding and sequencing

Article • 12/05/2023

Advanced auto charges let you apply specific charges to order headers and order lines, based on the customer that you're working with and/or the items that you're selling. For sales quotations and sales orders, you can also choose to compound header charges.

Header charge compounding affects the value base that's used to calculate an auto charge. All header charges (where the customer is debit or credit) that have a sequence that's equal to or lower than the sequence for a compound charge are added to the sum of line net amounts. This sum is referred to as the *value base*. The value base can be set either to *Sum of line net amounts only* or *Sum including charge amounts*. Compounding is applied only when the charge category is *Percent*.

Compounded header charges require a *charge sequence*. The charge sequence determines the order that header charges are calculated in. Charge sequences can exist only at the header level. They have an impact only when you have one or more charges that are set to compound.

Depending on how you configure the feature, interim header charge totals can be applied on top of either the sum of line net amounts, including all line charges and sales taxes, or the sum of line net amounts only. As the calculation progresses, interim header charges establish a value base that subsequent compound charges are calculated on.

You can define a default header charge sequence for your system. This charge sequence is then automatically applied to new sales quotations and sales orders.

When several header charges that have the same sequence are applied, the position of the header charges controls the order that they're applied in during compounding. The position has an impact only when one or more charges are set to compound.

Prerequisites

Before you can use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.38 or later.
- The feature that's named *Sequence and compound for customer charges* must be turned on in Feature management.

The *Sequence and compound for customer charges* feature adds new fields to the **Accounts receivable parameters** page to control the compounding. It also adds **Sequence**, **Compound**, and **Position** columns for header charges to sales quotations and sales orders.

Set up auto charge compounding and sequencing

To set up auto charge compounding and sequencing, follow these steps.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Prices** tab, on the **Prices** FastTab, set the **Find auto charges for header** option to *Yes*.
3. On the **Charges** FastTab, set the following fields:

    - **Re-search on posting** – This option controls whether header auto charges on sales orders and sales quotations are re-searched when documents are confirmed or posted. No other manually added charges are affected during the re-search process. Set this option to *Yes* to ensure that the header charges that are set up in auto charges are always applied.
    - **Combine charges on combined invoices** – The option controls how auto charges are calculated when the *Summary update* function is used to combine multiple sales orders into a single invoice. Set this option to *Yes* to first calculate the grand total of all sales orders that are included in the combined invoice and then apply the auto charge to that total. Set it to *No* to calculate a separate auto charge for each sales order that's included in a combined invoice.
    - **Value base for header charges** – Select one of the following values to specify the value base for calculating percentage-based header charges:
        - *Sum of line net amounts only* – Calculate charges based only on the sum of line amounts. This value ensures compatibility with charge calculations that were done before the *Sequence and compound for customer charges* feature was enabled in your system.
        - *Sum including charge amounts* – Calculate charges based on the sum of line amounts, including line charges. If you select this value, you can also include tax amounts in the value base by specifying tax codes that should be used to determine the tax amount. You can define this behavior for an individual auto charge line by selecting the **Include taxes in value base**

button that's added to the toolbar on the **Lines** FastTab of the **Auto charges** page when this value is selected.

The following sections provide example scenarios that show how these settings affect your calculations.

Example scenario 1: Work with auto charge compounding and sequencing

This section provides an example scenario that shows how auto charge compounding and sequencing works. It's based on the standard demo data.

Enable demo data

To work through this scenario and the other scenarios in this article by using the demo records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin.

Create sample auto charges

To apply a charge that's set to compound, you must first set up the auto charge. Use the following procedure to create an example auto charge. (For more information about the settings, see Automatic application of charges.)

1. Go to **Accounts receivable** > **Charges setup** > **Auto charges**.
2. In the list pane, set the **Level** field to *Header*.
3. On the Action Pane, select **New** to create an auto charges record.
4. On the header of the new record, set the following fields:
    - **Account code** – Select *All*.
    - **Item code** – Select *All*.
    - **Mode of delivery code** – Select *All*.
    - **Charge description** – Enter a name for the charge.
5. On the Action Pane, select **Save**.
6. The **Lines** FastTab should now include a blank line. If it doesn't, select **Add** on the toolbar to add one. Set the following values for this line:

- **Sequence** – Enter *1*.
- **Compound** – Clear this option.
- **Currency** – Select *USD*.
- **Charges code** – Select *Freight*.
- **Category** – Select *Fixed*.
- **Charges value** – Enter *100*.
- **From amount** – Leave this field blank.
- **To amount** – Leave this field blank.
- **GST group** – Leave this field blank.
- **Keep** – Clear this option.
- **Site** – Leave this field blank.
- **Warehouse** – Leave this field blank.

7. On the **Lines** FastTab, select **Add** on the toolbar to add a second charge line. Set the following values for it:

    - **Sequence** – Enter *2*.
    - **Compound** – Select this option.
    - **Currency** – Select *USD*.
    - **Charges code** – Select *Handling*.
    - **Category** – Select *Percent*.
    - **Charges value** – Enter *2*.
    - **From amount** – Leave this field blank.
    - **To amount** – Leave this field blank.
    - **GST group** – Leave this field blank.
    - **Keep** – Clear this option.
    - **Site** – Leave this field blank.
    - **Warehouse** – Leave this field blank.

8. On the Action Pane, select **Save**.

Create a sample sales order

The charges that you set up can now be automatically applied to a sales quotation or sales order. To apply the charges to a sales order, follow these steps.

1. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.

2. On the Action Pane, select **New** to create a sales order.

3. In the **Create sales order** dialog box, set the **Customer account** field to *US-004*. Then select **OK** to create the order. The selected customer must use the same currency as the auto charges that you set up (*USD*).

4. The new sales order is opened. On the Action Pane, on the **Sell** tab, in the **Charges** group, select **Maintain charges**.

5. The **Maintain charges** page shows the following two lines, which come from the auto charge setup.

Expand table

Position	Sequence	Compound	Charges code	Category	Charges value
1	1	No	Freight	Fixed	100
2	2	Yes	Handling	Percent	2

Notice that the **Position** values match the **Sequence** values from the auto charge setup. **Position** values differ from **Sequence** values only in cases where there's an overlap in sequence numbering. The position is used to calculate the value base for charges that are set to compound.

In this example, a 2 percent charge is applied on top of the 100 USD fixed charge. Therefore, the total header charges equal 102 USD. The 2 percent charge also applies to each order line.

Users can change the assigned **Position**, **Sequence**, and **Compound** values for a header charge that's applied from auto charges. However, they should be careful, because these types of edits can significantly affect the calculated value base that's used for charges that are set to compound.

For example, a user changes the preceding automatically created lines so that they have the following values.

Expand table

Position	Sequence	Compound	Charges code	Category	Charges value
2	1	No	Freight	Fixed	100
1	2	No	Handling	Percent	2

Because of the updated **Position** values, the 2 percent charge is applied first, while the value base is still zero. However, because 2 percent of zero is zero, the result is no additional charge. The 100 USD fixed charge is then added to give a total header charge of just 100 USD. (The 2 percent charge still applies to each order line.)

The user then changes the lines so that they have the following values.

Expand table

Position	Sequence	Compound	Charges code	Category	Charges value
1	1	No	Freight	Fixed	100
2	2	No	Handling	Percent	2

This time, the **Position** values match the **Sequence** values from the setup. However, the **Compound** option for both charges is set to *No*. In this case, the 100 USD fixed charge is applied first, but the 2 percent handling charge doesn't affect the header charge, because it isn't set to compound, and 2 percent of zero is zero. Therefore, the total sum of header charges is just 100 USD. (The 2 percent charge still applies to each order line.)

6. Close the **Maintain charges** page by selecting the back button on the Action Pane.

7. If necessary, you can revert the edits that you made on the **Maintain charges** page to return to the default setup lines. On the Action Pane, on the **Sell** tab, in the **Calculate** group, select **Header auto charges**. This action restores only those lines that were automatically added from auto charges. It doesn't affect charge lines that were manually added to the sales order header. This action also doesn't restore any order line charges.

Position, Sequence, and Compound fields on header charges

As you saw in example scenario 1, the **Maintain charges** page includes the following fields for charge settings:

- **Position** – Use this field to prioritize the calculation of each charge line. Lines are calculated in sequence, starting with the line that has the lowest **Position** value (because the lowest position has the highest priority). This value is automatically generated for all auto charge lines, based on each line's **Sequence** value on the **Auto charges** page. However, it's still editable and can be manually updated. Each **Position** value must be unique for each order. There can be no overlaps.
- **Sequence** – This field indicates the intended position of the charge as it's set up on the **Auto charges** page. For default charge lines, the value is copied from the auto charge setup. For custom charge lines that are manually added to the **Maintain charges** page, the value is set to *0* (zero). Sequence numbers don't have to be

unique and might therefore overlap. If the auto charge setup includes overlapping sequence numbers, the system assigns unique **Position** values, starting with the most specific (customer) record. Although the **Sequence** field is editable and can be manually updated, we recommend that you keep it aligned with the **Position** value.

- **Compound** – Use this option to specify whether the line should be compounded with other header charges that have been added so far. This option applies only to charges that have a **Category** value of *Percent*. Compounding applies only to charges that are added by the auto charge setup. Although the system allows you to select the **Compound** option for manually added charge lines, compounding isn't actually performed for those lines.

> ⓘ **Note**
>
> Although the system usually prevents you from entering duplicate **Position** values, duplicates can still occur. For example, they might occur if you delete an auto charge line, add new manual charges, and then restore auto charges by selecting the **Header auto charges** action. In these cases, the system uses the record ID of each charge record to resolve the relative positions of the charges when it calculates the value base. We recommend that you avoid this scenario, because you won't be able to see how the value base was calculated.

Example scenario 2: Value base for header charges options

When the system calculates the value of a charge that has a **Category** value of *Percent*, it applies the defined percentage to the value base that it has calculated so far at that position of the calculation. The value base can consist of the net amount across all order lines or the net amount across all order lines that include line changes, both with and without sales taxes.

The following examples show how the **Value base for header charges** field that's set on the **Accounts receivable parameters** page can affect your calculations. This field is also described in the Set up auto charge compounding and sequencing section.

Sum of line amounts only

The following example shows how a value of *Sum of line amounts only* in the **Value base for header charges** field can affect header charge calculations.

1. If you haven't already done so, set up the auto charges that are specified for example scenario 1.

2. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.

3. On the **Prices** tab, on the **Charges** FastTab, set the **Value base for header charges** field to *Sum of line net amounts only*.

4. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.

5. On the Action Pane, select **New** to create a sales order.

6. In the **Create sales order** dialog box, set the **Customer account** field to *US-004*. Then select **OK** to create the order. The selected customer must use the same currency as the auto charges that you set up (*USD*).

7. The new sales order is opened. On the **Sales order lines** FastTab, add an order line for any item, and edit the **Quantity** and **Unit price** fields to ensure that the **Net amount** value for the line is 100 USD.

8. While the new order line is still selected on the **Sales order lines** FastTab, select **Financials** > **Maintain charges** on the toolbar.

9. On the **Maintain charges** page, on the Action Pane, select **New** to add a charge line. Then set the following values for it:

    - **Charge code** – Select *Freight*.
    - **Category** – Select *Fixed*.
    - **Charges value** – Enter *10*.
    - **Currency** – Select *USD*.

10. On the Action Pane, select **Save**.

11. On the Action Pane, select the back button to return to the sales order.

12. On Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.

13. The **Totals** dialog box shows the different calculation results that apply to the current order. Notice the value that's shown for **Total Charges** is 114 USD. This value is calculated in the following way:

    Line charge fixed + *Header charge fixed* + *Header charge percentage and compound* (= 2% of *Line amount* + 2% of *Header charge with lower position*)

    10 USD + 100 USD + (2 USD [= 2% of 100 USD] + 2 USD [= 2% of 100 USD]) = 114 USD

Sum including charge amounts

The following example shows how a value of *Sum including charge amounts* in the **Value base for header charges** field can affect header charge calculations. This setting affects header charges that have a **Category** value of *Percent*, regardless of whether the **Compound** option is enabled.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Prices** tab, on the **Charges** FastTab, change the value of the **Value base for header charges** field to *Sum including charge amounts*.
3. Open the sales order that you created in the previous example.
4. On Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.
5. In the **Totals** dialog box, notice that the value that's shown for **Total Charges** is now 114.20 USD. This value is calculated in the following way:

   Line charge fixed + *Header charge fixed* + *Header charge percentage and compound* (= 2% of *Header charge with lower position* + 2% of *Line amount* + 2% of *Line charge*)

   10 USD + 100 USD + (2 USD [= 2% of 100 USD] + 2 USD [= 2% of 100 USD] + 0.20 USD [= 2% of 10 USD]) = 114.20 USD

When the **Value base for header charges** field is set to *Sum including charge amounts*, you can include selected sales taxes in the value base calculation. To set up this functionality, following these steps.

1. Go to **Accounts receivable** > **Charges setup** > **Auto charges**.
2. In the list pane, set the **Level** field to *Header*.
3. Select the auto charge that you want to set up, or create a new one.
4. On the **Lines** FastTab, select the line that you want to set up, or add a new one.
5. On the toolbar, select **Include taxes in value base**. (This toolbar button is available only when the **Value base for header charges** field is set to *Sum including charge amounts* on the **Accounts receivable parameters** page.)
6. On the **Sales tax codes in value base for auto charge** page, add a line for each sales tax code that you want to include in value base calculations for the selected auto charge line.
7. On the Action Pane, select **Save**.

Example scenario 3: Re-search on posting options

The **Re-search on posting** option ensures that header charges are assessed against the header charge setup when posting occurs. Auto charges that are set up as tiered charges are automatically assessed. The assessment compares automatically applied header charges against the auto charge setup. It ensures that all the header charges that are automatically applied to sales orders and quotations at the time of creation resemble the auto charge setup. All automatically applied header charges that were manually changed or deleted are restored. All manually added header charges are left unchanged. In other words, the option ensures that the auto charge setup is always applied.

> **Note**
>
> Tiered charges are supported only for sales orders. Therefore, they're included in the re-search only for sales orders.

The following example shows the effect of the **Re-search on posting** option.

1. If you haven't already done so, set up the auto charges that are specified for example scenario 1.
2. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
3. On the **Prices** tab, on the **Charges** FastTab, set the following values:
    - **Re-search on posting** – Set this option to *Yes*.
    - **Value base for header charges** – Select *Sum of line net amounts only*.
4. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.
5. On the Action Pane, select **New** to create a sales order.
6. In the **Create sales order** dialog box, set the **Customer account** field to *US-004*. Then select **OK** to create the order. The selected customer must use the same currency as the auto charges that you set up (*USD*).
7. The new sales order is opened. On the Action Pane, on the **Sell** tab, in the **Charges** group, select **Maintain charges**.
8. The **Maintain charges** page shows lines that came from the auto charge setup. Use the **Delete** button on the Action Pane to delete each auto charge line, so that no

charges are listed on the page.

9. On the Action Pane, select **New** to add a charge line. Then set the following values for it:

    - **Position** – Enter *3*.
    - **Sequence** – Enter *3*.
    - **Charges code** – Select *Freight*.
    - **Category** – Select *Fixed*.
    - **Charges value** – Enter *10*.
    - **Currency** – Select *USD*.

10. On the Action Pane, select **Save**.

11. On the Action Pane, select the back button to return to the sales order.

12. On the Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.

13. In the **Totals** dialog box, notice that **Total charges** shows the value of the fixed header charge that you just added (10 USD). Close the dialog box.

14. On the **Sales order lines** FastTab, add an order line for any item, and edit the **Quantity** and **Unit price** fields to ensure that the **Net amount** value for the line is 100 USD.

15. On the Action Pane, select **Save**.

16. On the Action Pane, on the **Sell** tab, in the **Generate** group, select **Confirm sales order**.

17. In the **Confirm sales order** dialog box, select **OK** to confirm the order.

18. You're returned to the sales order. On the Action Pane, on the **Sell** tab, in the **Charges** group, select **Maintain charges**.

19. On the **Maintain charges** page, notice that the two auto charges that you deleted have been reapplied on the sales order header, and that the custom line that you added is still included. This behavior occurs because the **Re-search on posting** option is enabled, and you confirmed (posted) the order. The following three lines should be shown.

Expand table

Position	Sequence	Compound	Charges code	Category	Charges value
1	1	No	Freight	Fixed	100

Position	Sequence	Compound	Charges code	Category	Charges value
2	2	Yes	Handling	Percent	2
3	3	No	Freight	Fixed	10

20. On the Action Pane, select the back button to return to the sales order.

21. On the Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.

22. In the **Totals** dialog box, notice that **Total charges** now shows the calculated value of all three header charges (114 USD). This value is calculated in the following way:

    Line charge fixed + *Header charge fixed* + *Header charge percentage and compound* (= 2% of *Line amount* + 2% of *Header charge with lower position*)

    10 USD + 100 USD + (2 USD [= 2% of 100 USD] + 2 USD [= 2% of 100 USD]) = 114 USD

Example scenario 4: Combine charges on combined invoices options

The setting of the **Combine charges on combined invoices** option on the **Accounts receivable parameters** page controls how header charges are calculated when multiple sales orders are combined into a single invoice by using the *Summary update* function.

- *Yes* – First calculate the grand total of all sales orders that are included in the combined invoice. Then apply the auto charges to that total.
- *No* – Calculate header charges individually for each sales order that's included in the summary invoice. In scenarios where header charges are set up to represent per-invoice charges instead of per-sales order charges, this setting can lead to over-charging.

The following example shows the effects of the **Combine charges on combined invoices** option.

1. If you haven't already done so, set up the auto charges that are specified for example scenario 1.

2. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.

3. On the **Prices** tab, on the **Charges** FastTab, set the following values:

    - **Re-search on posting** – Set this option to *Yes*.

- **Combine charges on combined invoices** – Set this option to *No*.
- **Value base for header charges** – Select *Sum of line net amounts only*.

4. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.

5. Follow these steps to create a sales order that has one sales line:
   a. On the Action Pane, select **New** to create a sales order.
   b. In the **Create sales order** dialog box, set the **Customer account** field to *US-004*. Then select **OK** to create the order. The selected customer must use the same currency as the auto charges that you set up (*USD*).
   c. On the **Sales order lines** FastTab, add an order line for any item, and edit the **Quantity** and **Unit price** fields to ensure that the **Net amount** value for the line is 100 USD.
   d. On the Action Pane, select **Save**.

6. Repeat the previous step to create a second, identical sales order.

   In a real warehouse, warehouse workers would now pick and pack both sales orders and register their work by using the Warehouse Management mobile app. However, these steps aren't required to continue this example scenario.

7. Go back to **Sales and Marketing** > **Sales orders** > **All sales orders**.

8. Find and select the two new sales orders. Then, on the Action Pane, on the **Invoice** tab, in the **Generate** group, select **Invoice**.

9. In the **Posting invoice** dialog box, on the **Parameters** FastTab, set the **Summary update for** field to *Invoice account*.

10. On the toolbar, select **Arrange**.

11. Select **OK**.

12. Select one of the two sales orders. Then, on the Action Pane, on the **Invoice** tab, in the **Journals** group, select **Invoice**.

13. The invoice is opened. On the Action Pane, select **Charges**.

14. Notice that the header charges have been calculated for each sales order. For each sales order, a header charge of 104 USD is applied. This value is calculated as 100 USD + 2% compound (2% of the line amount [100 USD] + 2% of the fixed charge [100 USD]). Therefore, the total charges for the invoice are 2 × 104 USD = 208 USD.

15. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.

16. On the **Prices** tab, on the **Charges** FastTab, change the setting of the **Combine charges on combined invoices** option from *No* to *Yes*.

17. As you did before, go to **Sales and Marketing** > **Sales orders** > **All sales orders**, and create new sales orders for the same account that have identical sales lines. Then pick and pack both sales orders.

18. Go back to **Sales and Marketing** > **Sales orders** > **All sales orders**.

19. Find and select the two new sales orders. Then, on the Action Pane, on the **Invoice** tab, in the **Generate** group, select **Invoice**.

20. In the **Posting invoice** dialog box, set the **Summary update for** field to *Invoice account*.

21. On the toolbar, select **Arrange**.

22. Select **OK**.

23. Select one of the two sales orders. Then, on the Action Pane, on the **Invoice** tab, in the **Journals** group, select **Invoice**.

24. The invoice is opened. On the Action Pane, select **Charges**.

25. Notice that the header auto charges have been re-searched and reapplied per invoice. The customer will be debited just one instance of the fixed auto charge of 100 USD instead of two instances, because the charge is now applied per invoice instead of per sales order. The total calculated charges for the invoice are 106 USD. This value is calculated as one header fixed charge of 100 USD + 2% compound (2% of the total line amounts [100 USD + 100 USD] + 2% of the fixed charge [100 USD]).

    In this setup, the fixed auto charge is applied only once (per invoice), and the percentage-based charge that's set to compound is calculated from a value base that's 100 USD less than the header charges that are calculated per sales order.

26. Go back to **Sales and Marketing** > **Sales orders** > **All sales orders**.

27. Select the first of the two new sales orders. Then, on the Action Pane, on the **Sell** tab, in the **Charges** group, select **Maintain charges**. Make a note of the charges that are shown, and then repeat this step for the second new sales order.

    Notice that, as part of combining charges on combined invoices, the auto charges are now present only on the first sales order. This behavior is by design. It ensures

that the combination of charges on combined invoices provides the correct charge calculation.

> **Note**
>
> Auto charges are applied based on the customer accounts on sales orders, not the invoice accounts on invoices. In cases where a summary update is done for sales orders for different customer accounts that have the same invoice account, and where different header charges are set up for the different customer accounts, the customer account for the *last* sales order for the summary update is the customer account that the header charges are re-searched and re-applied for when invoicing occurs.

Pricing management

The features that are described in this article work together with the Unified pricing management module. Pricing management introduces changes to the setup and search for auto charges. Therefore, it also changes some of the behavior that's described in this article. When the Unified pricing management module is enabled, charges that are set up at **Accounts receivable** > **Charges setup** don't apply. Instead, only charges that are set up on the auto charges page that's specific to the **Pricing management** module apply to sales orders and sales quotations. To set up auto charges in Pricing management, go to **Pricing management** > **During-sales pricing** > **Charges setup** > **Auto charges**.

Limitations

The capabilities for compounding charges and defining a value base that the compounding is applied to are available for sales quotations and sales orders, not purchase orders. This feature also introduces some new capabilities that don't support all scenarios. The following subsections describe the most important limitations.

Intercompany trade

The lack of compounding and sequencing support for purchase orders affects intercompany scenarios, because sales order and purchase order charges would be calculated differently. To prevent intercompany scenarios from being blocked, header charges for any sales order that fulfills a local purchase order are applied and calculated without using the compounding and sequencing features that are described in this

article. Therefore, compounding and sequencing are either applied or not applied, based on context. In intercompany scenarios, compounding and sequencing aren't applied. However, they're applied in non-intercompany scenarios.

Prorate and summary updates

Header charges can be assigned a charges code that's set to *prorate*. The ability to re-search when posting occurs, and to combine charges on combined invoices, prevents the duplication of fixed charges from auto charges. The combination of charges on combined invoices isn't supported for header charges that are set to *prorate*. Therefore, prorated header charges that are applied in a summary update scenario are always re-searched and calculated based on the individual sales orders. They aren't combined.

Tiered charge and summary updates

A header charge can be set to use **From amount** and **To amount** tiers. The tiers are evaluated based on the net amount of a sales order. When you use sequencing and compounding, tiers are always assessed against the net amount of a sales order. If the net amount of a sales order is within a tier, and if a tiered charge is calculated as a percentage, the charge is calculated based on the current value base, even though the tier was assessed based on the net amount. If you make a summary update, such as a summary invoice, when the system is configured to re-search when posting occurs, the tiers are assessed against the net amount only for the *first* sales order that's included in the summary update. The tiers aren't assessed against the net amount across *all* order lines in the summary update.

Feedback

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Units of measure for line-level charges

All line-level auto charges and line charges include a category that controls how and where the charge applies. The available categories include *Specific unit* and *Specific unit match*. These categories let you set up and apply line charges that are specific to the unit of measure for a line item that's either sold or purchased. They work in the following way:

- *Specific unit* – The auto charge is defined by using a specific unit (such as *ea*). The system then converts proportionally from other units if a conversion factor is available. For example, a sales line is for 2 boxes of an item where a box is defined as containing 10 ea. There's a charge of 2 USD per ea. In this case, the sales line produces a charge of 40 USD (= 2 boxes × 10 ea/box × 2 USD/ea). If no conversion factor is available, the charge isn't applied.
- *Specific unit match* – The auto charge is defined by using a specific unit (such as *ea*). However, the system applies that charge only to lines that use the same unit. For example, you set up a charge so that a line charge is added when the item is sold in smaller units (such as *ea*). However, there's no line charge for orders that are specified in larger units (such as cases).

The *Specific unit* and *Specific unit match* categories are supported for sales quotation lines, sales order lines, purchase requisition lines, request for quotation lines, and purchase order lines. Because both categories apply to both sales orders and purchase orders, they're also supported for intercompany trade.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.38 or later.
- The feature named *Unit of measure for line level charges* must be turned on in feature management. As of Supply Chain Management version 10.0.45, it's mandatory and can't be turned off.

Example scenario: Set up line-level auto charges for specific units and unit matching

This example scenario shows how to set up line-level auto charges that apply to a specific unit, or that are calculated based on unit matching.

Enable demo data

To work through this scenario by using the demo records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin.

Set up the auto charges

Follow these steps to set up a line charge in auto charges.

1. Go to **Accounts receivable** > **Setup** > **Charges** > **Auto charges**. (Similar steps can be completed at **Accounts payable** > **Setup** > **Charges** > **Automatic charges** and **Procurement and sourcing** > **Setup** > **Charges** > **Automatic charges**.)

2. In the list pane, set the **Level** field to *Link*.

3. On the Action Pane, select **New** to create an auto charges record.

4. On the header of the new record, set the following values:

    - **Account code** – Select *All*.
    - **Item code** – Select *All*.
    - **Mode of delivery code** – Select *All*.
    - **Charge description** – Enter a short description.

5. On the Action Pane, select **Save**.

6. The **Lines** FastTab should now include a blank line. If it doesn't, select **Add** on the toolbar to add one. Set the following values for this line:

    - **Currency** – Select *USD*.
    - **Charges code** – Select *Freight*.
    - **Category** – Select *Specific unit*.
    - **Unit** - Select *ea*.
    - **Charges value** – Enter *2*.
    - **Sales tax group** – Leave this field blank.
    - **Site** – Leave this field blank.
    - **Warehouse** – Leave this field blank.

- **Keep** – Leave this checkbox cleared.

7. On the **Lines** FastTab, select **Add** on the toolbar to add a second charge line. Set the following values for it:

    - **Currency** – Select *USD*.
    - **Charges code** – Select *Install*.
    - **Category** – Select *Specific unit match*.
    - **Unit** - Select *dz* (dozen).
    - **Charges value** – Enter *5*.
    - **Sales tax group** – Leave this field blank.
    - **Site** – Leave this field blank.
    - **Warehouse** – Leave this field blank.
    - **Keep** – Leave this checkbox cleared.

8. On the Action Pane, select **Save**.

Use the auto charges on a sales order

The charges can now be automatically applied to a sales quotation line or a sales order line. To apply the charges to a sales order line, follow these steps:

1. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.

2. On the Action Pane, select **New** to create a sales order.

3. In the **Create sales order** dialog box, set the **Customer account** field to *US-004*. Then select **OK** to create the order. The selected customer must use the same currency as the auto charges that you set up (*USD*).

4. The new sales order is opened. On the Action Pane, on the **Sell** tab, in the **Charges** group, select **Maintain charges**.

5. If any charges are listed on the **Maintain charges** page, select them, and then select **Delete** on the Action Pane.

6. Close the **Maintain charges** page by selecting the back button on the Action Pane.

7. You're returned to the order. On the **Sales order lines** FastTab, add an order line, and set the following values for it:

    - **Item number** – Select *A0002*.

- **Quantity** – Enter *2*.
- **Unit** – Select *ea*.
- **Site** – Select *6*.
- **Unit price** – Enter *50*.

The **Net amount** field should show a calculated value of *100*.

8. While the new order line is still selected on the **Sales order lines** FastTab, select **Financials** > **Maintain charges** on the toolbar.

9. Confirm that the **Maintain charges** page shows the two auto charges that you set up earlier. Then close the page by selecting the back button on the Action Pane.

10. On Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.

11. The **Totals** dialog box shows the different calculation results that apply to the current order. Notice that the value that's shown for **Total Charges** is 4 USD. This value is calculated in the following way:

    - There's a *Specific unit* charge of 2 USD per ea. Because the line is for the equivalent of 2 ea, this charge is 2 ea × 2 USD/ea = 4 USD.
    - There's also a *Specific unit match* charge of 5 USD per dz. However, because the unit that you're ordering in is ea, not dz, this charge doesn't apply.
    - The total charge for the line combines both charges. Therefore, the total charge is 4 USD.

12. Select **OK** to return to the sales order.

13. On the sales line, change the **Unit** value to *dz*. When you're prompted to overwrite prices and discounts, select *No*, and then select **OK**. (The system is configured to convert 1 dz to 12 ea.)

14. On Action Pane, on the **Sales order** tab, in the **View** group, select **Totals**.

15. In the **Totals** dialog box, notice that the value that's shown for **Total Charges** is now 58 USD. This value is calculated in the following way:

    - The unit is now *dz*, and there's a *Specific unit match* charge of 5 USD per dz. Because you're now ordering two dozens, this charge is 2 dz × 5 USD/dz = 10 USD.
    - There's also a *Specific unit* charge of 2 USD per ea. Because the system is configured to convert 1 dz to 12 ea, and you're now ordering a total of two dozen items, it can

determine that the order line is for 24 ea. Therefore, this charge is 24 ea × 2 USD/ea = 48 USD.

- The total charge for the line combines both charges. Therefore, the total charge is 10 USD + 48 USD = 58 USD.

Pricing management

The features that are described in this article work together with the Unified pricing management module. Pricing management introduces changes to the setup and search of auto charges. Therefore, it also changes some of the behavior that's described in this article. When the Unified pricing management module is enabled, charges that are set up at **Accounts receivable** > **Charges setup** don't apply. Instead, only charges that are set up on the auto charges page that's specific to the **Pricing management** module apply to sales orders and sales quotations. To set up auto charges in Pricing management, go to **Pricing management** > **During-sales pricing** > **Charges setup** > **Auto charges**.

Last updated on 03/25/2026

Period charges

Article • 12/05/2023

Period charges let you charge customers when a collection of orders that were invoiced over a specified period failed to meet specific criteria. The *Period charges* feature supports the setup of period charge rules that identify these invoices and define applicable charges. Charges can be calculated based on a minimum charge amount (such as minimum delivery fee), a minimum order quantity, or a combination of both.

The *Period charges* feature introduces the following changes to your system:

- Fields are added to the **Accounts receivable parameters** page to control the period charge calculations.
- Details are added to the **Free text invoice** details page to accommodate free text invoices that are created for period charges.
- A new page that's named **Period charge rule** is added under **Accounts receivable** > **Charges setup** on the navigation pane.
- A new page that's named **Calculate period charge** is added under **Accounts receivable** > **Periodic tasks** on the navigation pane.

Prerequisites

Before you can use this feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.38 or later.
- The feature that's named *Period charges* must be turned on in Feature management. (As of Supply Chain Management version 10.0.42, it's turned on by default)

Configure period charges

The **Accounts receivable parameters** page provides several fields that control how period charge calculations work in your system. Follow these steps to configure them.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.

2. On the **Prices** tab, on the **Charges** FastTab, set the following fields:

    - **Quantity threshold source** – Select one of the following values to specify which type of lines to consider when quantity-based period charge rules are

calculated:
    - *Invoice lines* – Consider invoice line quantities. We recommend that you select this value unless specific business requirements demand otherwise.
    - *Sales order lines* – Consider sales order line quantities (including canceled sales order lines).

- **Site and warehouse source** – Select one of the following values to specify where to match site and warehouse values that are set as selection criteria in period charge rules:
    - *Invoice line* – Match values that are set on individual invoice lines (in the `CustInvoiceTrans` table). We recommend that you select this value unless specific business requirements demand otherwise.
    - *Sales order* – Match values that are set on the sales order header (in the `SalesTable` table).

- **Required charge code match** – Select whether period charge rule calculations for lines of the *Monetary threshold* or *Quantity threshold minimum amount* type require the creation of at least one invoice line charge that matches the charge code on the rule line for a free text invoice line. We recommend that you set this option to *Yes* unless specific business requirements demand otherwise.

- **Extra batch helpers** – Enter the number of batch helpers (threads) to use for load balancing when period charge calculations are run in a batch.

Manage period charge rules

Period charges are calculated and applied when you run the *Calculate period charge* batch job for a period. You can define period charge rules for a specific customer, site, or warehouse, or for groups of customers, sites, or warehouses. You can also define period charges that apply to all customers, sites, or warehouses. Follow these steps to create, edit, or delete a period charge rule.

1. Go to **Accounts receivable** > **Charges setup** > **Period charge rule**.

2. Follow one of these steps:

    - To add a new rule, select **New** on the Action Pane.
    - To edit an existing rule, select it in the list pane, and then select **Edit** on the Action Pane.
    - To delete an existing rule, select it in the list pane, and then select **Delete** on the Action Pane.

3. On the header of the new or selected rule, set the following fields:

    - **Name** – Enter a name for the rule.
    - **Description** – Enter a short description of the rule.
    - **From invoice date** – Select the first date when the rule is valid.
    - **To invoice date** – Select the last date when the rule is valid.
    - **Account code** – Select one of the following values to define the scope of customers (accounts) that the rule applies to:
        - *Table* – The rule applies only to the customer (account) that's selected in the **Account relation** field.
        - *All* – The rule applies to all customers (accounts).
    - **Account relation** – If you set the **Account code** field to *Table*, select the specific customer (account) that the rule applies to.
    - **Site** – Select the site where the rule applies. If the rule applies to all sites, leave this field blank. If you select a site, the rule applies only to customer invoice lines that have the same site or to invoice lines that are related to a sales order that has the same site on the sales order header.
    - **Warehouse** – Select the warehouse where the rule applies. If the rule applies to all warehouses, leave this field blank. If you select a warehouse, the rule applies only to customer invoice lines that have the same warehouse or to invoice lines that are related to a sales order that has the same warehouse on the sales order header.

4. On the **Lines** FastTab, use the buttons on the toolbar to add and remove rule lines as you require, to define the rule that you want to create. For each line, set the following fields:

    - **Type** – Select one of the following values to specify the type of charge that the rule line creates:
        - *Monetary threshold* – Calculate the monetary value of qualifying charges, establish a minimum monetary threshold for each charge type, and generate a free text invoice line that charges the shortfall to customers who don't meet the thresholds during the period.
        - *Quantity threshold* – Calculate the total purchased quantity of qualifying items, establish a minimum quantity threshold, and generate a free text invoice line that has a fixed charge for customers who don't meet the threshold during the period.

- *Quantity threshold minimum amount* – Calculate a charge based on both quantity and monetary thresholds, and generate free text invoice lines as required.

Thresholds for all three charge types are based on transaction lines. They're based on either line quantities (for sales order or invoice lines) or line monetary values (charge lines for invoice lines).

- **Monetary threshold** – Enter the monetary threshold that the rule line applies at. If the period charge calculation determines that the sum of qualifying line charges that are charged to a customer over the period is below this threshold, the system generates a free text invoice line that has a charge that equals the shortfall.

- **Currency** – Select the currency for the **Monetary threshold** field value. This currency is also used as a selection criterion. Therefore, a period charge rule line is evaluated only against charge lines for invoice lines that use the same currency. The related free text invoice line that's generated for the customer also uses this currency.

- **Charge code** – Select the type of charge that the rule line applies to (such as freight, installation, or handling).

- **Include debit charge amount only** – Control which types of invoice line charges are included in the calculation against the monetary threshold for a rule line. For rule lines where this checkbox is selected, the calculation includes only invoice line charges where the value is a debit (positive). For rule lines where this checkbox is cleared, the calculation includes all invoice line charges, regardless of the value sign (debit/positive or credit/negative).

- **Quantity threshold** – Enter the quantity threshold that the line applies at. If the period charge calculation determines that the sum of qualifying quantities that a customer ordered during the period is below this threshold, the system generates a free text invoice line that has a charge that equals the charge amount that the rule line specifies.

- **Unit** – Select the unit of measure for the **Quantity threshold** field value. This unit of measure is also used as a selection criterion. Therefore, a period charge rule line is evaluated only against order or invoice line quantities that use either the same unit or another unit that can be converted to it by using a conversion factor that's defined in the system. When unit of measure conversion is required, standard unit of measure rounding rules apply.

- **Include debit quantity only** – Control which types of sales order or invoice lines are included in the calculation against the quantity threshold for a rule line. For rule lines where this checkbox is selected, the calculation includes only sales order or invoice lines where the value is a debit (positive). For rule lines where this checkbox is cleared, the calculation includes all sales order or invoice lines, regardless of the value sign (debit/positive or credit/negative).

- **Charge amount** – The amount that a customer will be charged for a period for rule lines of the *Quantity threshold* type when the quantity threshold isn't met for a rule line. This amount appears as a fixed fee on the related free text invoice line for the relevant customer.

- **Currency for charge amount** – The currency of the charge amount for rule lines of the *Quantity threshold* type. This currency applies to the related free text invoice line and isn't used as a selection criterion.

- **Revenue account** – The revenue account that was entered on the free text invoice line that the rule line creates.

5. On the **Line details** FastTab, you can enter or view a description of the rule line that's currently selected on the **Lines** FastTab. This description is added to the related free text invoice lines that the rule line creates.

Run and schedule period charge calculations

To calculate period charges and generate free text invoice lines for customers who don't meet the criteria that your period charge rules define, you must run the *Calculate period charge* batch job. You can manually run this batch job, or you can schedule it to run automatically on a recurring basis. Follow these steps to run or schedule the *Calculate period charge* job.

1. Go to **Accounts receivable** > **Periodic tasks** > **Calculate period charge**.

2. In the **Calculate period charge** dialog box, on the **Parameters** FastTab, set the following fields:

   - **Customer account** – Select the invoice account to calculate period charges for. Only invoices for this invoice account will be considered. To calculate period charges for all accounts, leave this field blank.
   - **Rule name** – To calculate charges only for a specific rule, select the name of that rule. To calculate period charges for all rules, leave this field blank.
   - **From invoice date** – Select the first invoice date to include in the period charge calculation. If you're setting up a recurring job, you should leave this

blank and set the **Max invoice age** field instead.

- **To invoice date** – Select the last invoice date to include in the period charge calculation. If you're setting up a recurring job, you should leave this field blank and set the **Max invoice age** field instead.
- **Max invoice age** – Specify the maximum age (in days) of invoices to include in the period charge calculation. The calculation will consider all invoice lines that are this old or younger on the day when the calculation runs.
- **Days per free text invoice** – Specify the number of days that are covered by each free text invoice that the period charge calculation generates.

3. On the **Run in the background** tab, set up the background options as usual for batch jobs in Supply Chain Management. If you want the job to run periodically, select the **Recurrence** link, and define the run schedule.

4. Select **OK** to run or schedule the job.

Inspect free text invoices created by period charge calculations

The *Calculate period charge* job creates one or more free text invoices when shortfalls are found. Follow these steps to get more information about free text invoices that period charge calculations create.

1. Go to **Accounts receivable** > **Invoices** > **All free text invoices**.
2. Open the free text invoice that you want to inspect.
3. Select the free text invoice line that you want to inspect, and then, on the **Invoice lines** toolbar, select **Line invoice base**. (The **Line invoice base** page is visible only for free text invoices that are created as a result of a period charge calculation.)
4. Select the **Period charge rule line** FastTab to view details about the period charge rule that created the selected free text invoice line.
5. Select the **Lines** FastTab to view a summary of the invoice and sales order transactions that were included in the period charge evaluation for the selected free text invoice line.

For more information about free text invoices, see Create a free text invoice.

Period charge rule example scenarios

This section provides examples that show how you might set up period charge rules.

Example scenario 1

You have a period charge rule that establishes a monetary threshold freight charge of 100 US dollars (USD). It's applicable to all customers, sites, and warehouses, and has no expiration date.

- For customer US-001, multiple sales invoices are posted between January 1 and January 10, 2022. A total overall freight charge amount of 45 USD is posted on multiple sales invoices between January 1 and January 10, 2022.
- For customer US-002, a total overall freight charge amount of 105 USD is posted on multiple sales invoices between January 1 and January 10, 2022.

You now run the *Calculate period charge* batch job for all customers for the period from January 1 through January 10, 2022.

The result is a free text invoice that's generated for customer US-001. It has one invoice line for 55 USD.

Example scenario 2

You have a period charge rule that establishes a quantity threshold of 50 pcs. and a minimum charge of 30 USD. It's applicable to all customers, sites, and warehouses, and has no expiration date.

For customer US-001, multiple sales invoices are posted between January 1 and January 10, 2022. The total overall quantity is 23 pcs.

You now run the *Calculate period charge* batch job for customer US-001 for the period from January 1 through January 10, 2022.

The result is a free text invoice that's generated for customer US-001. It has one invoice line for 30 USD.

Example scenario 3

You have a period charge rule that establishes a quantity threshold of 25 pcs. and a minimum freight charge of 20 USD. It's applicable to all customers, sites, and warehouses, and has no expiration date.

For customer US-001, multiple sales invoices are posted between January 1 and January 5, 2022. The total overall quantity is 15 pcs., and the total overall freight charge is 15 USD.

You now run the *Calculate period charge* batch job for customer US-001 for the period from January 1 through January 10, 2022.

The result is a free text invoice that's generated for customer US-001. It has one invoice line for 5 USD.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create purchase orders from sales orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Set options for how delivery mode and terms of delivery are set for purchase orders created from sales orders

You can configure how the system copies and synchronizes the delivery mode and terms of delivery between purchase orders created from sales orders and the originating sales orders. Set these options separately for sales order lines that are set up for *stocked delivery* and for sales order lines that are set up for *direct delivery*.

Prerequisites

To use the options for delivery mode and terms of delivery described in this section, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.41 or later.
- The feature that is named *Enable purch parameters to control delivery information sync on sales and purchase orders* must be turned on in feature management.

> **Note**
>
> When you turn on the *Enable purch parameters to control delivery information sync on sales and purchase orders* feature, its default settings match the behavior that was in place before the feature was introduced.

Configure options for delivery mode and terms of delivery

To configure how the delivery mode and terms of delivery are set for purchase orders that you create from sales orders in various scenarios, follow these steps:

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.

2. Open the **Delivery** tab.

3. On the **Synchronization of mode and terms of delivery** FastTab, make the following settings in the **For delivery type stock** field group. These settings apply to purchase orders that you create from *stocked delivery* sales order lines.

    - **Synchronize on creation** – Controls how the system sets the delivery mode and terms of delivery on purchase orders that you create from *stocked delivery* sales order lines. Choose one of the following options:
        - *No* – The system sets the delivery mode and terms of delivery for the *purchase order header and lines* to match the *vendor settings*.
        - *Yes* – The system sets the delivery mode and terms of delivery for the *purchase order header* from the *sales order header*. The system sets the delivery mode and terms of delivery for the *purchase order line* from the *sales order line*.
    - **Synchronize on update** – Controls whether the system synchronizes purchase orders that you create from *stocked delivery* sales order lines with the original sales order. Choose one of the following options:
        - *No* – The delivery mode and terms of delivery for the purchase order header and lines *aren't synchronized* with the original sales order. Therefore, changes you make to the purchase order don't affect the original sales order and changes you make to the original sales order don't affect the purchase order.
        - *Yes* – The delivery mode and terms of delivery for the purchase order header and lines *are synchronized* with the original sales order (in both directions). This setting only updates the *header* values when all lines on sales order are *stocked delivery*. At the *line* level, this setting applies to all *stocked delivery* sales lines (only).
    - **Copy from purchase header** – Controls whether changes you make to purchase orders that you create from *stocked delivery* sales order lines also apply to the original sales order. Unlike the **Synchronize on update** setting, this setting only applies changes in one direction—from the purchase order to the sales order. When you set **Synchronize on update** to *No*, the system sets **Copy from purchase header** to *No* and makes it read-only. When you set **Synchronize on update** to *Yes*, the **Copy from purchase header** setting takes precedence when you change the purchase order but has no effect when you change the sales order. Choose one of the following options:
        - *No* – Changes you make to the purchase order don't affect the original sales order. This setting helps to prevent problems that can occur as a result of creating multiple purchase orders from a single sales order. For example, if you update the delivery mode and terms of delivery mode on one such purchase order but not on the others, the original sales order would be out of sync with

some of its related purchase orders. If the system tried to update all related purchase orders after each change, it would cause performance problems.
    - *Yes* – Changes you make to the purchase order are also applied to the original sales order. This setting only updates the *header* values when all lines on sales order are *stocked delivery*. At the *line* level, this setting applies to all *stocked delivery* sales lines (only).

4. On the **Synchronization of mode and terms of delivery** FastTab, the **For delivery direct delivery** field group provides settings that are similar to (and named the same as) the settings provided in the **For delivery type stock** field group, except these settings apply to sales order lines that are set up for *direct delivery* instead of *stocked delivery*. Set these options as needed based on the descriptions provided in the previous step.

Create a purchase order from a sales order

This procedure shows you how to create a purchase order that's based on a sales order. The product quantities on the purchase order fulfill the demand of the originating sales order. Fulfilling sales demand this way is an alternative to a more comprehensive and optimized method of distribution requirements planning. You can run this procedure by using demo data company *USMF* or your own data.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New**.
3. In the **Customer account** field, select the drop-down button to open the lookup.
4. In the list, find and select the desired record.
5. Select **OK**.
6. In the **Item number** field, select the drop-down button to open the lookup.
7. In the list, find and select the desired record. If you're using USMF, select *D0001*.
8. In the **Quantity** field, enter a number.
9. Select **Add line**.
10. In the **Item number** field, select the drop-down button to open the lookup.
11. In the list, find and select the desired record. If you're using USMF, select *T0020*.
12. In the list, select the link in the selected row.
13. In the **Quantity** field, enter a number.
14. Select **Save**.
15. On the Action Pane, select **Sales order**.
16. Select **Purchase order**. The **Create purchase order** page lists all the open sales order lines that the system copied from the sales order. You can review the order details, and if necessary, you can modify selected details such as purchase quantity and pricing terms, before you create the purchases.
17. Select the **Include all option**.

- If you want to generate purchase orders for only a subset of the sales order lines, select these lines individually.
- The **Vendor account** field might or might already be populated with a vendor number. If the default vendor is set up for the product (on the associated Item coverage), the system copies this vendor to the line. Otherwise, you must enter a vendor manually. In this article, regardless of whether the **Vendor account** field already contains a value or not, the following steps instruct you to select a new vendor that's different for each line.

18. In the **Vendor account** field, select the drop-down button to open the lookup.
19. In the list, find and select the desired record.
20. In the list, select the link in the selected row.
21. Select the second order line.
22. In the **Vendor account** field, select the drop-down button to open the lookup.
23. In the list, find and select the desired record.
24. In the list, select the link in the selected row.
25. Select **Validate**.
26. Select **OK**. The message informs you that one or more purchase orders are created. The system generates a separate purchase order for each vendor that you specified for the sales order lines. This means that if several sales order lines are to be supplied by the same vendor, a single purchase order with multiple lines is generated.

Review purchase orders created from sales orders

To review the purchase orders that you created from a sales order, follow these steps:

1. On the Action Pane, select **General**.
2. Select **Related orders**. The **Related orders** page lists all the orders that you created from the sales order. In this example, two purchase orders are generated for two different vendors.
3. Select the link in the **Purchase order** field. Each purchase order line is associated with the sales order line that led to the purchase. The **Product tab** in the **Line details** FastTab shows the relation to the sales order in the **Reference type**, **Reference number**, and **Reference lot** fields.
4. Expand or collapse the **Line details** section.
5. Select the **Product** tab.

    - The **Reference lot** guarantees that the costs from the current purchase are charged on the attached sales order.
    - You can go to the originating sales order by opening the link in the **Reference number** field.

Last updated on 01/14/2026

Create sales orders

10/24/2025

This article describes how to create a sales order. You can use the step-by-step process in demo data company *USMF*. Typically, a sales order processor creates sales orders.

You can create a new sales order either from the **All sales orders** page or directly from the **Sales and marketing** module in the navigator. The second option offers a quicker way to create sales orders without opening the **All sales orders** page. This article describes both methods.

Enter sales order header details

1. Open the **Create sales order** dialog by doing one of the following steps:

    - Go to **Sales and marketing** > **Sales orders** > **Create sales order**.
    - Go to **Sales and marketing** > **Sales orders** > **All sales orders** and, on the Action Pane, select **New**.

2. In the **Customer account** field, select the customer you want to create a sales order for. If you're working with *USMF* demo data, select customer *US-004*.

3. Select **OK**.

Enter sales order line details

The products your organization sells might come in variants differentiated by dimensions, such as configuration, color, size, and style. Also, you might set up products to use storage dimensions, such as site, warehouse, and pallet, and tracking dimensions, such as batch and serial numbers. When you assign these dimensions, you must select the values for those dimensions on the order line. To improve order entry efficiency, consider adding the respective dimension fields to the order grid.

1. Under the **Sales order lines** section, select the **Sales order line**.

2. Select **Dimensions**.

    If you're working with *USMF* demo data, select the *Color*, *Site*, and *Warehouse* dimensions. The dimensions you select here appear in the sales order grid. To save your selections, set the **Save setup** option to *Yes*.

3. Select **OK**.

4. In the **Item number** field, select the drop-down button to open the lookup. If you're working with *USMF* demo data, select item number *T0004*.

    - If the item is part of a sales category, the item name automatically appears in the Sales category field.
    - If product dimension fields already contain a value, the value was copied from the product record where it's defined as a default product dimension. You can change the default value at any time.

5. In the **Color** field, select the drop-down button to open the lookup.

6. In the list, find and select the desired record.

7. In the **Quantity** field, enter a number.

    - If you sell the item in different units than when you purchase, produce, and store it, and you set a sales unit of measure on the product record, this value appears in the **Unit** field. You can change the value at any time.
    - If the **Site** field already contains a value, the value was copied from the order header or from the order settings that are associated with the product. You can change the value at any time. If the field is empty, select a value.
    - If the **Unit price** field already contains a value, the value was copied from a valid trade agreement, or from the product record. (The unit price can also come from a sales agreement, but the process for creating sales orders from sales agreements is different from the one shown here.) If the field is empty, enter a value.
    - The **Discount** field contains a discount amount per product unit. To calculate the total line discount amount, the discount value is multiplied by line quantity. If the **Discount** field already contains a value, the value was copied from a valid trade agreement. If the field is empty, and you want to give the customer a line discount, enter a value.
    - The **Discount percent** field contains a percentage value by which the total line gross amount is reduced. If the **Discount percent** field already contains a value, it was copied from a valid trade agreement. If the field is empty, and you want to give the customer a line discount, enter a value.
    - The **Net** amount field contains a value that's calculated based on the line's quantity and unit price adjusted by discounts. You can override the calculated value to a different one.

Review the order totals

1. On the Action Pane, select **Sales order**.

2. Select **Totals**.

   The **Totals** dialog box displays details about the entire order. This information includes the subtotal amount, which is a sum of all line net amounts adjusted for any line discounts, the total invoice amount, which is a subtotal amount adjusted for any order-level discount, charges, and sales tax, the customer credit limit situation, and more. The invoice amount is the amount that appears on the customer's invoice document.

3. Select **OK**.

> **Tip**
>
> For examples that show how sales totals and discounts are calculated when prices include and exclude sales tax, including information about how related values are shown in the **Totals** dialog box, see Calculate sales totals when prices include sales tax.

Use establishments on sales order lines

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about how to configure, default, and validate establishments on sales order lines to meet invoice compliance requirements.

Overview

The **Establishments** feature adds an establishment field at the sales order line level to support legal and fiscal compliance requirements (notably for France) while remaining globally applicable. An establishment represents an operating unit within a legal entity that buys or sells goods, and is tied to tax registration identifiers, such as SIRET or SIREN.

To enable establishments on sales order lines, enable the parameter highlighted in the following table.

Expand table

Parameter name	Parameter location in Finance	Description
Require establishment on customer invoice checkbox	**Accounts receivable** > **Setup** > **Accounts receivable parameters** > **Updates** tab > **Invoice** FastTab	When enabled, the system enforces establishment requirements on the customer invoice header. • Enables the **Establishment** field on customer invoice documents. • Applies defaulting logic to automatically populate the **Establishment** where possible, based on **Site** setup or **Financial dimensions** on the document. • Validates that an **Establishment** is specified before posting and prevents posting if the field is empty. > **NOTE**: You can't change the **Establishment** value after the invoice is posted.

This parameter controls whether an **Establishment** is required on invoices and determines when establishment-level **Registration IDs** must be enforced during invoice posting. When you enable establishment enforcement, the system validates that an applicable establishment is specified on the invoice and uses the defined applicability rules to resolve the corresponding **Registration IDs**. This validation ensures consistent identification of invoice parties at both legal entity and establishment levels and guarantees that the correct registration information is validated and immutably stored on posted invoices.

Learn more about Establishments and how they're defined in the Establishments article.

Learn more about Registration IDs in the Set up and use registration IDs article.

Why establishments matter

Invoices must include the correct establishment and related tax registration identifiers for audit and compliance. During invoice posting, the system enforces that each invoice contains one and only one establishment unless invoices are split by site. Once posted, the system stores establishment and registration IDs immutably on the invoice for compliance and audit purposes.

Where are establishments stored?

- Sales order lines contain the establishment field, which provides line-level granularity.
- Establishments aren't intended to appear on sales order header list pages because these pages are header-centric views, while establishments are line-driven.

How are establishments determined?

On a sales order line, the establishment can be derived using the following logic:

1. Site-based derivation (primary)
2. Financial dimensions (fallback or complementary)
3. Manual override by the user (allowed if defaulting fails or adjustment is needed)

If the site or relevant dimensions change, the system recalculates the establishment to stay consistent with the new context.

Default behavior

- The system defaults the establishment during sales line insert and update operations.

- For legacy or previously created unposted documents, use the **Backfill Establishments** batch job to populate establishment values instead of relying on form-load logic, to avoid performance problems.

Before an invoice can be posted:

- All sales order lines must have an establishment.
- All lines on a single invoice must share the same establishment.
- If sales order lines span multiple sites (and therefore multiple establishments), invoice posting fails unless you enable the **Split invoice by site** parameter.

Clear validation and error messages guide you toward corrective actions, such as running the backfill job or adjusting parameters.

Error handling and user guidance

You can correct error messages related to establishments by:

- Running the **Backfill Establishment on sales order lines** batch job.
- Enabling **Split invoice by site when multiple establishments are involved**.
- Manually aligning establishments on all relevant sales order lines.

Backfill establishments batch job

Use a dedicated batch job to populate establishment values on open sales order lines within a specified date range. Key characteristics of this backfill batch job include:

- Set-based processing for performance
- Security roles control (for example, Trade Sales Manager)
- Designed to address gaps in historical or legacy data

User interface considerations

- You see establishment at the sales order line level, and you can add it as a column where needed.
- To avoid performance and aggregation complexity, the establishment doesn't appear at the invoice header.
- Line-level visibility ensures clarity while avoiding costly recalculations.

Key takeaways

- Establishments are a line-level compliance construct, not a header attribute.
- Defaulting is automatic but overridable, with strong validations at invoice posting time.
- Batch backfill and actionable error messages ensure operational continuity and compliance.

Last updated on 04/24/2026

Use external item identifiers to add products to orders

When you're adding an item to a sales order, you can search for it using its external item identifier or external description/name. You can define external item identifiers and descriptions for each customer, allowing you to use the customer's terminology when adding products to orders.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.45 or later.
- You must ensure that the feature named *Enable lookup based search for Sales External Item Identifier field* is turned on in feature management. As of Supply Chain Management version 10.0.49, this feature is mandatory and can't be turned off.

Specify external item identifiers for a customer

External item identifiers are set up on the account for the customer that uses them. You can add external item identifiers to a customer account by following these steps:

1. Go to **Account Receivable** > **All customers**.
2. Find and open the customer you want to set up with external item identifiers.
3. On the Action Pane, open the **Sell** tab and then select **Setup** > **External item description**.
4. The **External item description** page opens, showing the existing external item identifiers for the customer. To add a new one, select **New** on the Action Pane and then enter values for the following fields:
    - **ABC code** – If the customer has assigned an ABC inventory classification to the item, you can note that here. ABC analysis is an inventory management technique that determines the value of inventory items based on their importance to the business.
    - **Item number** – Select your standard (internal) item number for the product.

- **External item number** – Enter the external item identifier that this customer uses to identify the item you specified. Users will be able to search for this number when creating a sales order.
- **Description** – Enter the name or other short description that the customer uses to identify the item. Users will be able to search for this description when creating a sales order.
- **External item text** – Enter text that this customer uses to describe the item.
- *Other dimensions* – If the external item number is associated with other specific dimension values (such as configuration, size, color, style, and so on), then enter these dimension values in the remaining columns as needed to finish identifying the item.

5. Continue to add external item identifiers as needed. When you're done, select **Save** on the Action Pane.

Add a product to a sales order using its external item identifier

When a customer has external item identifiers defined, you can use them to quickly find and add products to sales orders. To do this, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Either create a new sales order or open an existing one that you want to add lines to. The sales order must be for a customer that has external item identifiers defined.
3. Make sure that the **External** column is visible in the grid on the **Sales order line** FastTab. If it isn't, you can add it by doing the following steps:
   a. Open the **Grid options** dropdown list, which is the button with three dots on it located at the right side of the heading row of the grid.
   b. Select **Insert columns** to open the **Insert columns** dialog.
   c. Find the row with **Field** name *External* and select the checkbox in that row.
   d. Select **Update** to apply the change and close the dialog.
4. On the **Sales order line** FastTab toolbar, select **Add line**.
5. In the **External** column for the new line, select the item that you want to use based on its external item number or description. You can either type it in the field or select it from the lookup list. The field supports *search as you type*, which filters the list based on the **External item number** and **Description** values from the external item description.
6. On selecting a value, the **Item number**, **Product name**, and other related fields are populated based on the item setup.

Last updated on 07/27/2026

Copy lines between sales orders

Article • 04/03/2025

The *copy from all* feature lets you copy sales order lines between sales orders. This feature is useful when you must create an order that is similar to an existing one.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Either create a new sales order or open an existing sales order that you want to copy lines to.
3. On the Action Pane, on the **Sales order** tab, in the **Copy** group, select **From all**.
4. In the **Copy from all** dialog, on the **Parameters** FastTab, set the following fields:
    - **Quantity factor** – Specify the factor that the quantity for the selected order lines should be increased or decreased by.
    - **Invert sign** – Specify whether inverted (negative) quantities should be created for the copied lines. Set this option to *Yes* to create a return order or a negative-quantity order by reversing the sign of the copied lines.
    - **Copy charges** – Specify whether order line charges should be copied. Set this option to *Yes* to include any additional charges from the original order lines and/or header. If the **Copy order header** option is also set to *Yes*, order header charges are also copied.
    - **Recalculate price** – Specify whether the net amount of the copied lines should be recalculated based on current prices and trade agreements. Set this option to *Yes* to recalculate the prices.
    - **Copy precisely** – Specify whether the copied lines should be exact copies of the selected lines. Exact copies include all order information, such as sales tax, ledger accounts, and dimensions. Set this option to *Yes* to retain all the original details without any modifications.
    - **Delete order lines** – Specify whether previously entered order lines for the current order should be deleted before the new lines are copied. Set this option to *Yes* to replace the existing order lines with the copied lines.
    - **Copy order header** – Specify whether information from the selected order header should be copied to the current order. Set this option to *Yes* to copy order header information together with the selected order. The header information includes customer details, delivery terms, and payment terms.
5. On the **Sales orders** FastTab, specify which order lines (and possibly header information) you want to copy.

- In the **Headers** grid, find and select the sales order that you want to copy lines from. You can copy from only one order at a time.
- In the **Lines** grid, select the checkbox for each line that you want to copy. To select all the lines at once, select the checkbox in **Select all** column for the order that you selected in the **Header** grid.

> **Note**
>
> You can use the other FastTabs on the page to copy other related records and fields in a similar way.

6. Select **OK** to close the dialog.

   The selected sales order lines (and possibly header information) are now copied to the current order.

Feedback

Was this page helpful? Yes No

Provide product feedback

Calculate sales totals when prices include sales tax

This article provides example scenarios that show how sales totals and discounts are calculated when prices include and exclude sales tax.

Prerequisites

Enable sample data

To work through the example scenarios by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

Adjust the tax rate

The example scenarios in this article assume the following setup in Microsoft Dynamics 365 Supply Chain Management:

- A sales tax code that's named *AV_CAST* has a tax rate of 50 percent.
- A sales tax group that's named *CA* includes sales tax code *AV_CAST*.
- An item sales tax group that's named *AU/VI* includes sales tax code *AV_CAST*.

The USMF sample data includes these tax codes and groups, but it uses a different tax rate for the sales tax code. To adjust the tax rate to 50 percent, follow these steps:

1. Go to **Tax** > **Indirect taxes** > **Sales tax** > **Sales tax codes**.
2. In the list pane, select sales tax code *AV_CAST* to open it.
3. On the Action Pane, on the **Sales tax code** tab, select **Values**.
4. For the row where the **Sales tax code** field is set to *AV_CAST*, set the **Value** field to *50*.

Example scenario 1: Sales total calculation when prices exclude sales tax

Follow these steps to see how sales totals are calculated when prices exclude sales tax.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. Create a sales order for customer account *US-001*.

3. Add a sales order line, and set the following values for it:

    - **Item number:** *1000*
    - **Site:** *1*
    - **Warehouse:** *11*
    - **Quantity:** *1*
    - **Unit price:** *1000*

4. While the new sales order line is still selected, on the **Line details** FastTab, on the **Setup** tab, set the following values:

    - **Item sales tax group:** *AU/VI*
    - **Sales tax group:** *CA*

5. On the Action Pane, on the **Sales order** tab, select **Totals**.

6. In the **Totals** dialog box, notice that your tax settings produce the following values:

    - **Invoice amount:** *1500*
    - **Subtotal amount:** *1000*
    - **Sales tax:** *500* (= *Subtotal amount* × 50%)

7. Select **OK** to close the dialog box.

8. Edit the sales line so that the **Discount** field is set to *100*.

9. Open the **Totals** dialog box again, and notice the following new values:

    - **Invoice amount:** *1350* (= *Unit price* − *Discount*)
    - **Subtotal amount:** *900* (= *Unit price* − *Discount*)
    - **Sales tax:** *450* (= *Subtotal amount* × 50%)
    - **Line discount:** *100*

Example scenario 2: Sales total calculation when prices include sales tax

Follow these steps to see how sales totals are calculated when prices include sales tax.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. Create a sales order for customer account *US-001*.

3. On the **Header** tab of the new sales order, on the **Setup** FastTab, set the **Prices include sales tax** option to *Yes*.

4. On the **Lines** tab, add a sales order line, and set the following values for it:

    - **Item number:** *1000*
    - **Site:** *1*
    - **Warehouse:** *11*
    - **Quantity:** *1*
    - **Unit price:** *1000*

5. While the new sales order line is still selected, on the **Line details** FastTab, on the **Setup** tab, set the following values:

    - **Item sales tax group:** *AU/VI*
    - **Sales tax group:** *CA*

6. On the Action Pane, on the **Sales order** tab, select **Totals**.

7. In the **Totals** dialog box, notice that your tax settings produce the following values:

    - **Invoice amount:** *1000*
    - **Subtotal amount:** *666.67* (= *Invoice amount − Sales tax*)
    - **Sales tax:** *333.33* (= *Subtotal amount* × 50%)

8. Select **OK** to close the dialog box.

9. Edit the sales line so that the **Discount** field is set to *100*.

10. Open the **Totals** dialog box again, and notice the following new values:

    - **Invoice amount:** *900* (= *Unit price − Discount*)
    - **Subtotal amount:** *600* (= *Invoice amount − Sales tax*)
    - **Sales tax:** *300* (= *Subtotal amount* × 50%)
    - **Line discount:** *66.67* (= *Original subtotal amount* [666.67] − *New subtotal amount* [600])

    Because tax is included in the price, the line discount doesn't equal 100. Instead, it's calculated as the difference between the gross amount without tax (calculated without the discount) and the line amount without tax (calculated to include the discount).

11. Select **OK** to close the dialog box.

The remaining steps in this example show how to apply a manual adjustment to the sales tax amount.

12. On the Action Pane, on the **Sell** tab, select **Sales tax**.
13. In the **Temporary sales tax transactions** dialog box, select the **Adjustment** tab to view the actual sales tax amount.
14. For the row where the **Sales tax code** field is set to *AV_CAST*, follow these steps:
    a. Set the **Actual sales tax amount** field to *400*.
    b. Make sure that the **Override calculated sales tax** checkbox is selected. (It should be automatically selected when you change the **Actual sales tax amount** value.)
15. Select **OK** to close the dialog box.
16. Open the **Totals** dialog box again, and notice the following new values:

    - **Invoice amount:** *900* (= *Unit price − Discount*)
    - **Subtotal amount:** *500* (= *Invoice amount − Sales tax*)
    - **Sales tax:** *400* (manually adjusted amount)
    - **Line discount:** *166.67* (= *Original subtotal amount* [666.67] − *New subtotal amount* [500])

    The same logic that was used earlier is used to calculate the discount. The discount is the difference between the gross amount without tax (ignoring the manually adjusted tax) and the line amount without tax (calculated to include a discount of 100 and the new manually adjusted tax).

Last updated on 12/30/2025

Manage order holds

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article demonstrates how to place customer sales orders on hold, how to work with order hold checkouts, and how to remove order holds. An order might be placed on hold for various reasons. For example, you might hold an order until a customer address or payment method can be verified or until a manager can review the customer's credit limit. While the order on hold, it can't be processed by the warehouse for shipping.

Set up order hold codes

There could be many reasons why an order should be placed on hold. Use hold codes to set up each of the various types of holds you'd like to be able to apply to sales orders.

1. Go to **Sales and marketing** > **Setup** > **Sales orders** > **Order hold codes**.
2. Do one of the following actions:

    - To create a new hold code, select **New** on the Action Pane.
    - To edit an existing hold code, select **Edit** on the Action Pane, and then select an existing hold code on the list pane.

3. Make the following settings for your new or selected hold code:

    - **Hold code** – Enter a short name for the hold code (for example, *Call back*).
    - **Description** – Enter a short description to provide more information about the hold code (for example, *Order held waiting for customer callback*).
    - **Role** – Select the security role that a user must have to be able to remove this hold code from an order. For example, if you select *Collections agent*, only users with that role can remove this hold code from an order.
    - **Default for sales order** – Select this check box to make this hold code selected by default each time a user adds a new hold to an order.

- **Remove inventory reservations** – Select this check box to remove any physical reservations from an order when this hold code is added to it.

4. Select **Save**.

Place order on hold

To place a sales order on hold, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select the sales order you want to put on hold.
3. On the Action Pane, open the **Sales order** tab and from the **Functions** group, select **Order holds**.
4. On the Action Pane, select **New**.
5. In the **Hold code** field, select the hold code you want to apply.
6. Select **Save**.
7. Go back to the **All sales orders** list page.
8. On the grid, find the order you put on hold and note that it now shows a check mark in the **Do not process** column and a pause symbol in the **Hold** column.
9. On the Action Pane, open the **Pick and pack** tab and note that the buttons for generating picking lists and packing slips are disabled. This restriction is because the order is on hold and can't be processed further until the hold is removed.

Manage orders on hold

To manage orders on hold and remove holds, follow these steps:

1. Go to **Sales and marketing** > **Sales orders** > **Open orders** > **Order holds**. The **Order holds** page functions as a workbench where you can get an overview of all the current and processed holds, and handle them and associated sales orders.
2. Use the **Filter** field and/or the **Show** drop-down list to filter the list by value and/or hold status. Then select the hold you want to change.
3. You can check out order holds. While you have a hold checked out, other users can't clear it. To check an order hold in or out, open the **Hold checkout** tab on the Action Pane and choose one of the following actions:

    - **Check out** – Select this option to check out the selected hold. The **Checked out to** column updates to show your user ID for the selected hold.
    - **Clear checkout** – Select this option to check in a hold that you previously checked out. Your user ID is removed from **Checked out to** column for the selected hold.

- **Override checkout** – Select this option to reassign a checked-out hold to another user (or yourself). The **Checked out to** column updates to show the selected user's ID.

4. When you're ready to remove the hold and allow the order to proceed to the next fulfillment step, clear it. On the Action Pane, open the **Clear hold** tab and select one of the following options:

    - **Clear holds** – If the order requires no changes, select this option to clear the holds on it.
    - **Clear and modify** – If the order requires changes, select this option to clear the holds on it and then open the order so you can modify it.
    - **Clear and submit** – Clear all holds and submit the order for further processing. This action only applies when you're using call center functionality.

Last updated on 05/06/2026

Confirm sales orders

This procedure demonstrates how to confirm sales orders. You see how to confirm a single order, and how to confirm multiple orders at once. A sales order processor typically carries out these tasks.

Confirm a single sales order

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. In the list, find and select the order that you want to confirm.
3. Select the link on the sales order number to open the selected order.
4. On the Action Pane, select *Sell*.
5. Select **Confirm sales order**.
6. Expand the **Parameters** section. Make sure that the **Posting** option is set to *Yes*.
7. Set the **Print confirmation option** to *Yes*. The **Check credit limit** field specifies the method that's used to calculate a customer's remaining credit. By default, it's copied from the **Credit and collections parameters** page. If you want to skip the credit limit check when confirming a specific sales order, set the **Check credit limit** to *None*. However, even when this field is set to *None*, the credit limit check is still performed if the **Mandatory credit limit** option is selected on the customer master data.
8. Select **OK**.
9. Select **Yes**.
10. Close the page.
11. On the Action Pane, select **Options**.
12. Select **Change view**.
13. Select **Header view**. When an order is confirmed, the **Document status** is set to *Confirmation*.
14. On the Action Pane, select **Sell**.
15. Select **Sales order confirmation**.
16. Close the page.

Confirm multiple sales orders at once

1. Go to **Sales and marketing** > **Sales orders** > **Order confirmation** > **Confirm sales order**.
2. Select **Select**.

3. In the list on the **Range** tab, find and select the record that references the **Customer account** field.
4. In the **Criteria** field, find and select the customer account that has multiple orders to confirm.
5. Select **OK**.

    - The **Overview** tab displays a list of the orders that match the query criteria. These orders are included in the confirmation.
    - The **Summary update for** field in the **Parameters** section specifies the parameter by which multiple orders are summarized into one confirmation document. By default, the option is copied from the **Default values for summary update** setting on the **Accounts receivable parameters** page.

6. In the **Summary update for** field, select *Order*. The minimum parameters that are required to create summary updates are **Invoice account** and **Currency**. This requirement means that summary updates that have different invoice accounts and different currencies aren't allowed. You can set up other parameters on the **Summary update parameters** page, which is accessible from the **Accounts receivable parameters** page.
7. In the **Sales order** field, select the order number that you want to be the summary order.
8. Select **Arrange**.
9. Select **OK**.
10. Select **OK**.

Last updated on 07/01/2026

Calculate sales totals

Summarize this article for me

The *Calculate sales totals* batch job recalculates and updates aggregated metrics for sales orders and sales quotations. This critical maintenance task ensures that all summary totals, delivery metrics, and key performance indicators (KPIs) shown on inquiry pages, workspaces, and analytics dashboards remain accurate and up-to-date. The recalculation updates:

- Order-level totals
- Quotation-level totals
- Aggregated values used by workspaces, analytics, and reporting
- KPI caches across sales, retail, and commerce experiences

Use the *Calculate sales totals* job to maintain data integrity and reporting accuracy, especially after bulk data imports, mass updates to sales documents, or as part of regular system maintenance. By running this job on a scheduled basis—such as nightly or as part of month-end closing processes—you ensure that your sales teams, managers, and executives always see current, reliable totals when reviewing orders, quotations, and performance metrics.

This feature provides benefits to many parts of your organization:

- *Sales teams and order managers* get accurate order summaries and delivery information without manual calculation or delays.
- *Finance and management* rely on current KPI data for accurate reporting, forecasting, and business analysis.
- *System administrators* can control the scope and timing of recalculations to balance data accuracy with system performance.
- *E-commerce and retail operations* benefit from fresh aggregated metrics used in commerce workspaces and analytics.

This article walks you through the configurable parameters that let you control which documents to recalculate and how far back to look, explains how to schedule the job to run automatically, and provides performance best practices to ensure the job runs efficiently in your environment.

Learn more about batch jobs in Batch processing overview.

When to recalculate sales totals

Run the *Calculate sales totals* batch job in the following scenarios:

- After *bulk data imports* (data management and integrations).

- After *mass updates* to sales orders or quotations.
- As part of *month-end closing*.
- On a *nightly schedule* to maintain accurate workspace KPI totals.

Performance considerations

To keep your system running smoothly, follow these performance best practices:

- Avoid full recalculation during peak business hours.
- Run targeted recalculations by adjusting the **Ignore documents updated before (days)** parameter.
- If totals appear incorrect, check whether you have any customizations that bypass update triggers.

Set up, run, and schedule sales total calculations

To set up, run, and schedule the *Calculate sales totals* batch job, follow these steps:

1. Go to **Sales and marketing** > **Periodic tasks** > **Calculate sales totals**.

2. In the **Calculate sales totals** dialog, make the following settings on the **Parameters** FastTab:

    - **Ignore documents updated before (days)** – Enter the number of days to look back for recalculation. Only documents that were last updated *after* this date are recalculated. The system uses the following formula to calculate the cutoff date:

      Cutoff date = Today - Entered number of days

      To balance data freshness with performance for routine recalculations, we recommend that you look back 7 to 30 days. Avoid running full historical recalculations in high-volume environments unless necessary.

    - **Calculate totals for sales orders** – Set to *Yes* to recalculate aggregated totals for sales orders, including:
        - Order-level summary totals
        - Line-level aggregated contributions
        - Delivery totals
        - Remaining quantities and values
        - KPI totals used in workspaces and analytics

      Set this option to *No* only in niche scenarios where sales orders aren't part of your workflow.

- **Calculate totals for sales quotations** – Set to *Yes* to recalculate aggregated totals for sales quotations, including:
  - Quotation value summaries
  - Aggregated line-level metrics
  - Totals used in quotation inquiry and reporting

  Set this option to *No* if your organization doesn't use the sales quotation process.

- **Number of threads** – Enter the maximum number of parallel threads to use for the job (or set to zero to let the system decide automatically). If your sales totals calculations take too long, consider increasing this value. To allow more resources for other processes while calculating sales totals, reduce this value. The actual maximum number of used threads never exceeds the maximum batch threads configured for your system. The system distributes orders and quotations selected for the calculation evenly between the threads.

3. On the **Run in the background** FastTab, choose whether to run the job in batch mode and set up a recurrent schedule. Select **Recurrence** to set up the run schedule (or choose to run just once). Tips for when and how often to run this job are provided earlier in this article. Select **Alerts** to configure notifications for job completion or failures. The settings here work just as they do for other types of batch jobs in Supply Chain Management.

Last updated on 02/26/2026

Pay-by-link payment with the Dynamics 365 Payment Connector for Adyen (preview)

[This article is prerelease documentation and is subject to change.]

This article explains how to set up and enable the pay-by-link payment method to capture payments for sales orders by using the Microsoft Dynamics 365 Payment Connector for Adyen. By using pay-by-link functionality, businesses can offer modern payment methods that give their customers the flexibility to choose their preferred way to pay.

By using the pay-by-link payment method, salespeople can generate payment links for their customers, which eliminates the need for customers to share sensitive card details verbally over the phone. This method enhances security and trust. By using these payment links, customers can employ modern payment methods like digital wallets that aren't feasible in traditional phone transactions. Moreover, if customers are undecided about their order, salespeople can create the orders on a payment hold. If customers make the payment within a preconfigured duration, the system releases the order for fulfillment. Otherwise, the order gets canceled.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Prerequisites

Set up and configure the Adyen portal and connector

Before you can set up and use the pay-by-link payment method for Supply Chain Management, the Dynamics 365 Payment Connector for Adyen must already be set up and configured for your system. Learn more in the following articles:

- Dynamics 365 Payment Connector for Adyen overview
- Set up Dynamics 365 Payment Connector for Adyen

Link your Supply Chain Management environment to a Dataverse environment

The payment notification service uses Dataverse. Therefore, to receive payment notifications, you must link your Supply Chain Management environment to a corresponding Dataverse environment. Learn more in Connect finance and operations apps with a new Microsoft Dataverse instance and Connect finance and operations apps with an existing Microsoft Dataverse instance.

Version requirements

Pay-by-link payment requires Supply Chain Management version 10.0.47 or later.

License required to set up the connector

To set up pay-by-link payment for Supply Chain Management, you must have a Dynamics 365 Commerce license.

Security roles required to set up the connector

Some of the steps for setting up pay-by-link payment require you to sign in to Supply Chain Management by using a user account that has either the *Administrator* or *Commerce Payment Administrator* security role in Microsoft Power Platform.

Set up pay-by-link payment

Enable the features required for pay by link

To enable pay-by-link payment, turn on the following features in feature management:

- *Enable unified payments experience in POS* – Must be enabled to use pay-by-link payment, but only applies to POS and doesn't affect sales orders in Supply Chain Management. Learn more in Check out faster with optimized payment flows.
- *Enable Payments Notification feature* – Enables the infrastructure to receive notifications.
- *Enable Pay By Link Payment feature* – Must be enabled to use pay-by-link payment, but only applies to POS and doesn't affect sales orders in Supply Chain Management.
- *Enable asynchronous payments for sales orders* – Enables the pay-by-link capability for sales orders in Supply Chain Management.

Find your payment notification service URL

The payment notification service URL is where Adyen sends the payment notifications. You'll need this value later when you create a webhook to receive payment notifications from Adyen.

To find your payment notification service URL, follow these steps:

1. Go to **Retail and Commerce** > **Headquarters setup** > **Parameters** > **Commerce shared parameters**.
2. Open the **Payment notifications** tab.
3. Copy the value shown in the **Payment notifications end point URL** field.

> **Note**
>
> If the **Payment notifications end point URL** field is blank, then you need to connect your Supply Chain Management environment to a Dataverse environment. Learn more Connect finance and operations apps with a new Microsoft Dataverse instance and Connect finance and operations apps with an existing Microsoft Dataverse instance

Create a Microsoft Entra ID application registration

A Microsoft Entra ID application registration lets you establish OAuth credentials for the payment notification service. Payment notifications from Adyen use these credentials to communicate securely with the payment notification service. You'll need these credentials later when you create a webhook to receive payment notifications from Adyen.

Create a new Microsoft Entra ID application registration in your Azure portal by following these steps:

1. Sign in to the Azure portal by using a work or school account.
2. If your account gives you access to more than one tenant, select your account in the upper-right corner. Then set your portal session to the Microsoft Entra tenant that you want.
3. In the left pane, select the **Microsoft Entra ID** service, and then select **App registrations** > **New registration**.
4. On the **Register an application** page, enter your application's registration information:

- **Name** – Enter a meaningful application name. This name is shown to users of the app.
- **Supported account types** – Any of the available options should work.
- **Redirect URI** – You don't need this field.

5. Select **Register**.

6. After the new application is registered, on the **Overview** page, copy the values of the following properties:

   - **Application (client) ID**
   - **Directory (tenant) ID**

7. Now you must create a client secret for the application. Go to the **Certificates & secrets** section, and then select **New client secret**. Provide a description and an expiration period for the secret. Ensure that you create a business process for rotating the secret, because payment notifications fail after the secret expires. Copy the secret value, and save it for later use.

Create a webhook to receive payment notifications from Adyen

To create a new webhook to receive payment notifications from Adyen, follow these steps.

1. Go to the Adyen portal, and then go to the **Developers** section.

2. Create a standard webhook.

3. In the **Server configuration** section, in the **URL** field, paste the **Payment notifications endpoint URL** value that you copied earlier.

4. Keep the default values for the **Method** field (*JSON*) and the **Encryption protocol** field (*TLS v1.3*).

5. In the **Security** section, select **OAuth**, and enter values for the following properties:

   - **Client ID** – Enter the **Application (client) ID** value that you copied earlier from the Microsoft Entra ID application registration.
   - **Client secret** – Enter the **Secret value** value that you copied earlier from the Microsoft Entra ID application registration.
   - **URL** – Enter `https://login.microsoftonline.com/{tenantid}/oauth2/v2.0/token`. Replace *{tenantid}* with the **Directory (tenant) ID** value that you copied earlier from the Microsoft Entra ID application registration.
   - **Scope** – Enter *a013b12b-2624-40b4-b15b-b7751d733a22/.default*.

6. Generate a new hash-based message authentication code (HMAC) key and copy it. You aren't able to view this key again, so keep it somewhere safe.
7. In the **Events** section, select **Authorization**.
8. Under **Additional settings**, in the **Card** section, select the **Include card bin** and **Include Subvariant** options.
9. Leave the remaining fields set to their default values, and save the configuration.

> **Note**
>
> Don't test the configuration yet because you aren't finished setting up the system. You'll validate the configuration after the next step.

Enter the HMAC and application client ID details

Supply Chain Management uses the HMAC and application client ID details to authenticate the payment notifications sent by Adyen. To enter these details, follow these steps:

> **Important**
>
> To complete this procedure and save the configuration, sign in to Supply Chain Management by using a user account that has either the *Administrator* or *Commerce Payment Administrator* security role in Microsoft Power Platform.

1. Go to **Retail and Commerce** > **Headquarters setup** > **Parameters** > **Commerce shared parameters**.
2. Open the **Payment notifications** tab.
3. Enter the following values:
    - **Data verification HMAC key** – Enter the HMAC key that you copied earlier when setting up the webhook in the Adyen portal.
    - **Payment notifications client ID** – Enter the **Application (client) ID** value that you copied earlier from the Microsoft Entra ID application registration.
4. On the Action Pane, select **Save**.

Test the connection to the payment notification service

To test the connection, follow these steps.

1. Go to the Adyen portal.
2. Edit the webhook that you created earlier, and then select **Test configuration**. If the connection fails, review all the values that you entered when you created the webhook to validate that they're correct. If you make any changes (for example, if you change the keys), wait 20 minutes before you test the connection again. This wait time ensures that caching doesn't affect the test. If the issue persists, contact Microsoft Support.

Enable pay by link functionality for sales orders

In addition to the generic pay-by-link setup described earlier in this article, complete the following steps to enable the pay by link functionality for sales orders.

1. Go to **Accounts receivable** > **Payments setup** > **Methods of payment**.

2. Methods of payment with a **Payment type** of *Credit card* or *Electronic payment* now include a section named **Pay by link** on the **General** FastTab. For each method of payment where you want to enable pay by link, make the following settings:

    - **Allow pay by link** – Set to *Yes* to allow salespeople to create payment links for sales orders that use this method of payment. This setting is required to enable the pay by link functionality for the payment method.
    - **Allow pay later** – Set to *Yes* to allow customers to pay later, enabling scenarios where the customer needs some time to confirm the order but wants to reserve the inventory. Pay-later orders are created with a special hold code and remain on hold for a preconfigured duration (as described in the next step). If the customer makes a payment within this duration, then the order is released for fulfillment; otherwise, the order is systematically canceled. Set to *No* to disable pay-later for a payment method.

3. If you chose to enable the pay-later capability for one or more payment methods, go to **Accounts receivable** > **Setup** > **Account receivable parameters**, open the **General** tab, and make the following settings on the **Sales setup** FastTab:

    - **Hold code for payment confirmation** – Select the hold code you want to apply to pay-later orders that aren't paid yet. This hold code is automatically applied to orders that are placed on hold when the salesperson selects the **Pay later** option while creating the payment link.
    - **Order hold timeout for pending payments (minutes)** – Enter the number of minutes system should wait for payment. If the customer doesn't pay before this limit expires, then the system cancels the order.

4. Go to **Retail and Commerce** > **Retail and Commerce IT** > **Payments** > **Process asynchronous payments for sales orders**. In the dialog, expand the **Run in the background** FastTab and select **Recurrence**. Use the settings here to choose how often the system should check for customer payments (for example, every 15 minutes). Then select **OK** twice to close the dialog and save your setting. These settings schedule the *Process asynchronous payments for sales orders* batch job, which checks for customer payments and updates the status of sales orders accordingly.

Customize the behavior of payment links

This step is optional. Complete it only if you want to change the default behavior of the payment links.

1. Go to **Account receivables** > **Payments setup** > **Payment services**.
2. Open payment service for your Adyen connector.
3. On the **Payment service account** FastTab, in the **Custom settings** property, add key-value pairs for any or all of the following keys (an example is shown later in this step):

   - `PaymentLinkDuration_Orders` – Set the duration for which the payment link is valid. The default value is 24 hours.
   - `Store` – Specify which Adyen store information to include in the payment link. By default, only payment methods that don't depend on the Adyen store are shown.
   - `RequiredShopperFields` – Specify which fields the customers must enter values for before making a payment. By default, the related fields aren't shown to customers before they use the payment link.

   The system uses the default value for any keys that you don't include in the key-value pairs.

   For example, you enter the value `{"PaymentLinkDuration_Orders":"02:00", "Store":"Test_Store", "RequiredShopperFields":"email,name,phone,billingAddress,deliveryAddress"}` for the **Custom settings** property. In this case, the Adyen payment connector sets the payment link expiration to *two hours*, the payment link is associated with the *Test_Store* store, and customers must enter their *name*, *email address*, *phone number*, *delivery address*, and *billing address* before they make the payment.

   > **Tip**

To define the payment link duration to be more than one day, specify the value for the `PaymentLinkDuration` property as described in System.TimeSpan.Parse method.

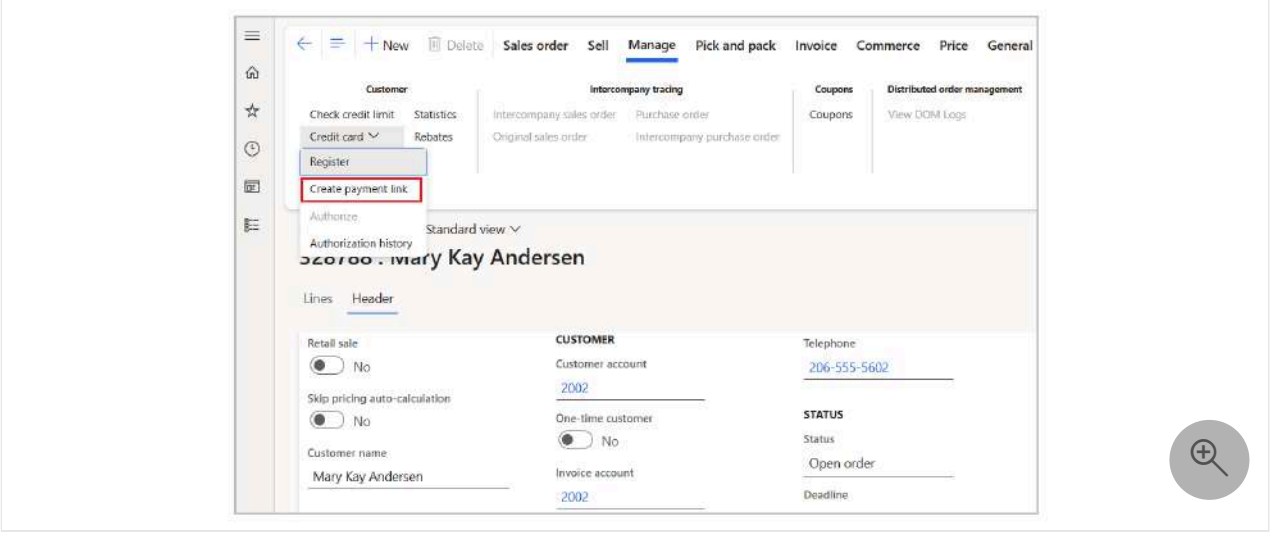
The pay-by-link user experience

The following example describes a typical end-to-end user experience for using pay by link with sales orders in Supply Chain Management.

1. A salesperson takes an order by phone and creates a sales order in Supply Chain Management.

2. When the order is ready for payment, the salesperson goes to the **Header** tab of the sales order and, on the **Price and discount** FastTab, selects a **Method of payment** for which **Allow pay by link** is set to *Yes* (as described earlier in this article). Additionally, as with credit cards, the **Settlement type** field on the **Setup** FastTab might need to be set to *None*.

3. On the Action Pane, the salesperson opens the **Manage** tab, opens the **Credit card** drop-down list, and selects **Create payment link**.



4. A page opens that the salesperson uses to create the payment link. Customer information is prepopulated.

5. After the salesperson updates the customer information as needed, they select **Create link** to create the payment link.

6. The salesperson copies the payment link and can use any available communication channel to send it to the customer.

    - If the customer requests more time to make the payment, the salesperson selects the **Pay later** button on the payment link dialog (if this functionality is enabled).

- If the customer changes their mind about the purchase, the salesperson can cancel the payment link from here.

7. The customer opens the payment link from their phone or computer to complete the payment.

8. The salesperson's screen automatically refreshes every five seconds to check for the payment. As with a credit card authorization, once the payment is received, the system associates the authorization to the order.

Manually check the payment status of a sales order

To manually check the payment status of a sales order, follow these steps.

1. In Supply Chain Management, go to the **All sales orders** page and open the sales order waiting for payment.
2. On Action Pane, open the **Sales order** tab, from the **Payments** group, select **Asynchronous payments**.
3. A page opens showing the payment links associated with the order. If a payment link status is shown as *Active*, then payments against this link are either not yet processed or the customer hasn't yet paid.

    - To refresh the current payment link status display, select **Check status** on the Action Pane.
    - To cancel a selected payment link, select **Cancel** on the Action Pane. You can only cancel a payment link if the customer hasn't yet used the link for payment.

Purge old payment authorization notifications

To save storage and keep the notifications page manageable, delete old notifications manually from the **Payment authorization notifications** page or automatically by running the **Purge payment authorization notifications data** batch job. The batch job allows you to specify the number of days that pass before notifications are deleted.

Enable critical payment notifications for Dynamics 365 Payment Connector for Adyen

You can use the asynchronous payment notification framework to receive crucial and actionable notifications from Adyen, which are then displayed in Commerce headquarters. This functionality provides the operations team with the necessary visibility to take timely actions

based on the notifications received. Learn more in Enable critical payment notifications using the Dynamics 365 Payment Connector for Adyen.

Last updated on 02/11/2026

Sales order FAQs

This article addresses frequently asked questions that come up when working with sales orders in Dynamics 365 Supply Chain Management.

Can I link a purchase order to a sales order to fulfill demand?

You can create a purchase order from a sales order. For more information, see Create a purchase order from a sales order.

Can I cancel or delete a sales order or return order?

You can cancel only sales orders and return orders that are in a *Created* state. For more information, see Cancel a return order.

Can I restore an invoiced sales order that was deleted?

If an invoiced sales order was deleted by mistake, you can restore it or view its details.

Go to **Customer account** > **Transactions** > **Original document** > **View details**. Find the invoice that you're looking for and select it to view its details. These details include the sales order reference. You should also be able to access the sales order details from that page.

How do I delete orphaned sales order records?

To clean up orphaned sales order records, run the *Sales order deletion* periodic job by going to either of the following places:

- Sales and marketing > Periodic tasks > Clean up > Delete sales orders
- Retail and commerce > Retail and commerce IT > Clean up > Delete sales orders

Is there a way to calculate commissions on invoices that are already posted?

Supply Chain Management doesn't currently support the calculation of commissions for posted invoices.

Last updated on 07/01/2026

Sales agreements overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about sales agreements. A sales agreement is a contract that commits the customer to buy products in a specific quantity or for a specific amount over time, in exchange for special prices and discounts.

A sales agreement is a contract that commits the customer to buy products in a specific quantity or for a specific amount over time, in exchange for special prices, special discounts, and other special terms, such as payment and delivery terms. The prices and discounts of the sales agreement override the prices and discounts that are stated in any trade agreements that exist.

The validity period of a sales agreement is defined by the **Effective date** and **Expiration date** fields on the agreement. A customer's sales order qualifies for the agreement terms if the requested ship date of the order is within the validity period. All sales order lines that are linked to a sales agreement contribute to fulfillment of that sales agreement.

You can create a sales order directly from a sales agreement by using the **Release order** action. Alternatively, you can select an effective sales agreement when you're taking orders (see the "Applying sales agreements in the ordering process" section of this article).

> **Note**
>
> In earlier versions, sales agreements were referred to as blanket sales orders.

Commitment types

Each line in a sales agreement expresses a commitment to sell something. In general, two categories of commitment exist:

- **Value commitment** – The customer agrees to buy products for a specific amount.

- **Quantity commitment** – The customer agrees to buy a specific quantity of products.

In addition, a contract can commit the customer to buy a specific product or products in a product category. By combining these two factors (value versus quantity, and specific products versus product categories), you get four types of commitment:

- **Product quantity commitment** – The customer agrees to buy a specific quantity of products. Lines that use this commitment type are defined by an item number, and by the quantity and unit that were agreed on. The **Amount** field isn't available.
- **Product value commitment** – The customer agrees to buy specific products for a specific amount. Lines that use this commitment type are defined by an item number and the amount that was agreed on. The **Quantity** and **Unit** fields aren't available.
- **Product category value commitment** – The customer agrees to buy products from a specific sales category for a specific amount. Lines that use this commitment type are defined by a sales category in the hierarchy of sales categories and an amount. The **Quantity** and **Unit** fields aren't available.
- **Value commitment** – The customer agrees to buy any product or products in any procurement category for a specific amount. The **Quantity** and **Unit** fields aren't available.

Lines in the same sales agreement can have different types of commitments.

Pricing terms for sales agreements

Pricing terms can vary, depending on the type of commitment. On a sales order that is linked to a sales agreement, the pricing terms from that sales agreement override any other pricing terms that apply based on trade agreements. The following table describes the price-related fields that are affected by each commitment type. "Yes" indicates that the field can be updated on an order line.

Expand table

Commitment type	Unit price	Price unit	Discount percent	Cash discount amount
Product quantity commitment	Yes	Yes	Yes	Yes
Product value commitment			Yes	
Product category value commitment			Yes	
Value commitment			Yes	

Policies for sales agreements

The following policies affect how the link between a sales agreement commitment and the corresponding sales order lines works:

- **Max is enforced** – The total quantity or amount for all order lines can't exceed the quantity or amount specified on the related commitment.
- **Price and discount is fixed** – The price on an order line and the price on the related commitment can't differ. If the price is changed on the order line, the link to the commitment is broken. If the link is broken, the order line doesn't contribute to fulfillment of the commitment.
- **Minimum release amount** and **Maximum release amount** – If an amount is specified, a message is displayed if you make any change to an order line that causes the order line to differ from the related commitment.

Fulfillment calculations for sales agreements

The **Fulfillment** tab on the **Line details** FastTab on the **Sales agreements** page shows fulfillment quantities and amounts.

In the **Fulfillment** area, you can view the total quantities and amounts for all order lines that are linked to the specified sales agreement. You can also view the remaining amount or quantity that is required to fulfill the commitment.

In the **Agreement** area, you can view the quantities and amounts from the specified sales agreement. These quantities and amounts are the total quantities and amounts that were committed.

Confirmations and version history for sales agreements

When you confirm a sales agreement, the current version of the sales agreement is stored in a history table. If you change the sales agreement, you can confirm it again to store another version of the sales agreement in the history.

If you don't confirm a sales agreement, you can still use it to create sales orders. However, history information for the sales agreement isn't stored.

You can preview or print all revisions of the confirmations. You can then share the revisions with your customer to obtain approval.

Applying sales agreements during the ordering process

If you don't release sales orders directly for a sales agreement, you can still link a sales agreement to an order during the order entry process. When you're creating a new sales order and select a sales agreement, the terms of that agreement, such as the payment terms, delivery terms, and delivery address, are applied to the order header, and the link between the agreement and the order is created. Then, on the order lines, when you select products and categories that are specified in the sales agreement, the prices and discounts are copied from that agreement. The same sales order can include both lines that aren't related to a sales agreement and lines that have a commitment for a sales agreement.

Modifying sales orders that are linked to sales agreements

If you create (release) a sales order against a sales agreement, you can modify some fields on the sales order lines only if you remove the link to the associated sales agreement lines. The following table lists some of these fields.

Expand table

Field	Description
Requested ship date	If you change the requested ship date to a date that is earlier than the **Effective date** value on the sales agreement line, you must remove the link to the sales agreement line before you can save the changed ship date. If you change the requested ship date to a date that is later than the **Expiration date** value on the sales agreement line, you must remove the link to the sales agreement line before you can save the changed ship date.
CurrencyDiscount, percentDiscountUnit, pricePrice, unitNet amount	If you change the value in any of these fields, and if the **Price and discount is fixed** check box is selected on an associated sales agreement line, a message box prompts you to save the change. Select **Yes** to remove the link to the sales agreement line and recalculate the price. Select **No** to remove the link to the sales agreement line without recalculating the price.
Net amount	If you specify an amount that exceeds the amount that is specified on a sales agreement line where the **Max is enforced** check box is selected, a message box prompts you to save the changed amount. Select **Yes** to remove the link to the sales agreement line and recalculate the price. Select **No** to remove the link to the sales agreement line without recalculating the price.
Quantity	If you specify a quantity that exceeds the quantity that is specified on a sales agreement line where the **Max is enforced** check box is selected, a message box prompts you to save the changed quantity. Select **Yes** to remove the link

Field	Description
	to the sales agreement line and recalculate the price. Select **No** to remove the link to the sales agreement line without recalculating the price.

Returning an item ordered from a sales agreement

When a customer returns a product that they ordered from a sales agreement, Supply Chain Management can find and automatically update the related sales agreement commitment to reflect the change in quantity or amount. By creating a return order that is based on the original sales order that is linked to a sales agreement, you establish a relation between the sales agreement commitment, the sales order line, and the return order invoice.

If you don't want to deduct the returned item quantity from the sales agreement commitment, use the **Remove link** control on the **Return order** page to remove the link between the return order and the sales agreement commitment. If you must reestablish the link later, select **Create link**.

> **Note**
>
> A return order can be linked to only one sales agreement. If a customer returns multiple products that they ordered from multiple sales agreements, you must create a new return order for each product and create a link to the corresponding sales agreement.

Automatic search for sales agreements

In some situations where sales orders are created indirectly, such as when you create a credit note or intercompany sales orders, you can control whether the system automatically searches for applicable sales agreements.

Financial dimensions on sales agreements

You can copy financial dimensions to either document headers or individual lines of a sales agreement. You can change the dimensions on an agreement header or agreement line at any time. In this case, the dimensions are automatically copied to the release header or release line of release orders.

Last updated on 04/27/2026

Enter sales agreements

Article • 01/31/2023

This article explains how to create a sales agreement that commits one of your customers to buy a product for an agreed amount over time in exchange for special discounts. You can run this procedure in demo data company USMF or on your own data.

Set up sales agreement header

1. Go to **Sales and marketing** > **Sales agreements** > **Sales agreements**.
2. Select **New**.
3. In the **Customer account** field, select the desired record from the drop-down menu.
4. In the **Sales agreement classification** field, select the desired record from the drop-down menu.
5. Expand the **General** section.
6. In the **Default commitment** field, select **Product value commitment**. A commitment type is a mandatory criterion that you must assign to the agreement to define how the agreement contract will be fulfilled. The four predefined types let you set up the customer's commitment target, expressed as a quantity or a value. The quantity commitment type can only be applied to a specific product, but the value-based types are applicable to sales of both specific and non-specific products.
7. In the **Expiration date** field, set the date to a future date when you want the agreement to expire.
8. Select **OK**.

Set up product value commitment lines

1. Select **Add line**.
2. In the **Item number** field, select the desired record from the drop-down menu. The type of commitment that you have chosen for the agreement affects the kind of information you can enter for the agreement lines. For example, for a value-based agreement you must specify the total net amount (in the agreed currency) for which the customer commits to buys goods from you. In this example the **Quantity** and **Unit** fields on the line are unavailable because you're creating an agreement for the customer to buy a specific value of a product.

3. In the **Net amount** field, enter the monetary amount that the customer has committed to buying.
4. In the **Discount percent** field, enter a percentage value that will apply to the customer's sales order lines that are linked to this agreement.
5. Expand the **Line details** section.
6. Select **Yes** in the **Max is enforced** field.

    - Selecting **Max is enforced** means that the total amount of all the sales order lines that use the commitment's special prices, discounts and/or payment terms must not exceed the amount specified on the commitment.
    - The minimum and maximum release amounts specify a range of values that must be sold on each sales order that uses the selected agreement.

7. Expand the **Sales agreement header** section. Unless the status of the agreement is set to **Effective**, sales orders cannot be associated with the agreement and can therefore not contribute to the fulfilment of that agreement. You can change the status manually at this stage. However, the status would normally be changed when you confirm the agreement for the customer.
8. On the Action Pane, select **Sales agreement**.
9. Select **Confirmation**. Make sure that the **Mark agreement as effective** option is set to **Yes**.
10. Select **Yes** in the **Print report** field.
11. Select **OK**.
12. Close the page. The agreement is now effective. You can start linking the customer's orders to the agreement to offset against the committed target.

Feedback

Was this page helpful? Yes No

Provide product feedback

Fulfill sales agreements

Article • 09/29/2021

This procedure shows you how to fulfill a sales agreement by associating sales orders with it. You can run this procedure in demo data company USMF or on your own data. Before starting this guide, make sure you have an effective sales agreement of type "Product value commitment". Alternatively, you can run the task guide called "Create sales agreements".

Release a sales order from the agreement

1. Go to Sales and marketing > Sales agreements > Sales agreements.
2. In the list, open the agreement against which you want to release the order.
3. On the Action Pane, click Sales agreement.
4. Click Release order.

    - As the text on top of the Create release order page points out, the details required for the sales order lines will differ depending on whether the agreement is quantity- or value-based.
    - The agreement in this guide is of type "Product value commitment". This is why the Lines section of this page is blank. If the commitment was based on quantity, the line information would be copied from the agreement.

5. Click Create.

    - The message informs you that a sales order has been created. Since the order does not contain any lines, you must add order line details to complete the release process.

6. Close the page.
7. Close the page.
8. Go to Sales and marketing > Sales orders > All sales orders.
9. In the list, find and open the order that was created as the result of the order release in the previous task.
10. Click Add line.
11. In the Item number field, click the drop-down button to open the lookup.
12. In the Item number field, type or select the item that is specified on the associated sales agreement.
13. In the Quantity field, enter a number.

- Make sure that you enter a quantity that brings the Net amount under the value of the associated sales agreement.
- Notice that because the sales order is linked to the agreement, the negotiated discount percent is applied to the order line.

14. Click Update line.
15. Click Attached.

    - The Attached agreement page shows the ID and terms of the agreement from which the line is released.

16. Close the page.
17. On the Action Pane, click General.
18. Click Attached sales agreement.
19. Expand the Line details section.
20. Click the Fulfillment tab.

    - The Fulfillment tab shows a summary of all the sales order lines that are associated with this commitment, and their fulfillment state, as well as the amount or quantity that has not yet been released.

21. Close the page.
22. Close the page.
23. Close the page.

Apply sales agreement in the order process

1. Go to Sales and marketing > Sales orders > All sales orders.
2. Click New.
3. In the Customer account field, click the drop-down button to open the lookup.
4. In the list, find and select the customer specified on the sales agreement.
5. In the list, click the link in the selected row.
6. Expand the General section.
7. In the Sales agreement ID field, click the drop-down button to open the lookup.
8. In the list, click the link in the selected row.
9. Click Yes.
10. Click OK.
11. In the list, mark the selected row.
12. In the Item number field, click the drop-down button to open the lookup.
13. In the Item number field, type or select the item that is specified on the associated sales agreement.
14. In the list, click the link in the selected row.
15. Click Save.

16. On the Action Pane, click Pick and pack.
17. Click Post packing slip.
18. Expand the Parameters section.
19. Select Yes in the Posting field.
20. Click OK.
21. Click OK.
22. On the Action Pane, click General.
23. Click Attached sales agreement.
24. Click the Fulfillment tab.

Feedback

Was this page helpful? Yes No

Provide product feedback

Generate and process customer rebates

This procedure demonstrates how to process customer rebates from claim generation to the point of passing them as accruals to Accounts receivable. It walks you through a specific example to explain how the various conditions on the rebate lines affect the final amounts credited to the customer. Use the USMF demo data company, and complete the following tasks before you start the guide: (1) Go to the **Accounts receivable parameters** page, expand the **Prices** tab, and then the **Price details** tab, and check that the **Enable price details** option is set to **Yes**. (2) Go to the **Rebate agreements** page and select the customer rebate agreement: USMF-000001. If the **Workflow approval status** field isn't set to **Approved**, select **Validation** on the Action Pane to approve it.

Review a customer rebate agreement

1. Go to **Sales and marketing** > **Customer rebates** > **Rebate agreements**.

    - The next few steps look at the conditions of agreement USMF-000001. The conditions makes it easier to understand how the customer credit values are calculated later in the procedure.
    - The agreement is for an individual customer, in this example customer US-009.
    - Rebates are given to the customer when they purchase a specific product. In this case, the product has item number T0020.
    - The customer's sales performance, against which the rebate amounts are estimated, is to be accumulated on a weekly basis.
    - The setting for **Price taken from** is **Gross**, which means that the line's sales amount on which basis the claim is estimated isn't reduced by the line discount.
    - The **Rebate line break type** field shows the method for calculating rebates. In this case, the sales target against which the rebates are to be estimated is set to **Quantity**.
    - The agreement's lines specify the rebate amount type, the actual rebate value, and the thresholds. In this example, the customer qualifies for a rebate of 20 USD per unit sold, if their weekly purchases of the product fall within 1 to 50 units; and a rebate of 40 USD per unit sold, if they purchase above 50 units.

2. Close the page.

Generate rebate claims

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Click **New**. To mimic the way in which rebate claims would be generated, the next task is to create a sales order, where the product and quantity will qualify the customer in question for a rebate.
3. In the **Customer account** field, enter or select a value.
4. Click **OK**.
5. In the **Item number** field, enter or select a value.
6. Set **Quantity** to 40.
7. Under the **Sales order lines** section, select **Sales order line**.
8. Select **Price details**. If you don't see this option, it's because you didn't set the **Enable price details option** to 'Yes' before you started the guide.
9. Expand the **Rebates** section. The **Rebates** tab lists all the rebate agreements that apply to the current order line and shows the estimated rebate amount. The displayed amounts are only indications of what future rebate claims might be. The actual rebate amounts might be different depending on: the total sales volume the customer achieves under a periodic rebate agreement; whether the customer returns all or partial quantities; and whether the applicable sales order is invoiced.
10. Close the page.
11. Select **Add line**.
12. In the **Item number** field, enter or select a value.
13. Set **Quantity** to 60.
14. Click **Save**.
15. On the Action Pane, select **Invoice**.
16. Select **Invoice**.
17. Expand the **Parameters** section.
18. In the **Quantity** field, select **All**.
19. Click **OK**.
20. Click **OK**.

Process rebate claims

1. Go to **Sales and marketing** > **Customer rebates** > **Rebates**.

    - The **Rebates** page acts as a workbench where you can review, approve, and process rebate claims. You can now process the claims that were created as a result of invoicing a sales order for customer US-009, who is the subject of the rebate agreement USMF-000001.

- The first line represents a rebate claim for 800 USD, which is based on the sales of 40 units of product T0020, calculated at 20 USD per unit. This matches the conditions of the first quantity break in the rebate agreement.
- The second claim is for 2,400 USD, which is based on the sales of 60 units of product T0020, calculated at 40 USD per unit, as per the second quantity break in the agreement.
- Both claims are in the **To be calculated** state. This state means that they're associated with an agreement that tracks the customer's sales performance on a periodic basis and that they need to be recalculated to account for the total sales volume within the respective period.

2. Select **Cumulate**.
3. In the **Customer** field, enter or select a value.
4. In the **Start date** field, select today's date.
5. Select **OK**. As a result of running the **Cumulate** function, the estimated claim amount is adjusted to account for the fact that the customer's total sales volume in the relevant period is higher than when the first rebate was generated. More specifically, because the total purchased quantity reaches 100 units, the customer now qualifies for 40 USD per unit (as per the agreement's second quantity break), or 4,000 USD of total rebate amount. The difference is recorded as a new claim "adjustment" for the additional $800 USD. The status of the rebate claims that are included in the **Cumulate** update are now set to **Calculated**.
6. In the list, mark all rows.
7. Select **Approve**.
8. Select **Process**.
9. In the **Customer** field, enter or select a value.
10. Select **OK**. A message shows that the rebate was processed successfully, and the status of the claims changes to **Mark**. This status means that as a result of a Rebate accrual journal being posted:

    - The claims are now transferred to the temporary customer balance as deductions.
    - The Rebate accrual account is credited to represent the future liability towards the customer.
    - The Rebate expense account is debited, in recognition of the cost incurred in connection with the sales.

Last updated on 07/01/2026

Process rebates for payment

Article • 09/29/2021

This procedure demonstrates how to convert approved and processed customer rebates to credit notes. You can use this guide in the USMF demo company. The precondition for this guide is to have one or more rebate claims which have a status of Mark. If you're using USMF you should run the "Generate and process customer rebates" guide before you start this guide.

Convert rebate claims to credit note

1. Go to All customers.
2. In the list, find and select the desired record.
3. In the list, click the link in the selected row.
4. On the Action Pane, click Collect.
5. Click Settle transactions.
6. Click Functions.
7. Click Rebate program.

    - The Rebate page lists the rebate claims that you have processed in the customer rebate workbench and that are in status Mark.

8. Click Edit.

    - Set checkmarks in the Mark field for the claims that you want to include into credit note.

9. Click Functions.
10. Click Create credit note.

    - A message appears to inform you that a journal has been posted (This is the Accounts receivable consumption journal, as specified in the Accounts receivable parameters page). This causes the real liability (credit) amount to be moved to the customer balance. This means that the customer's account has been credited, and the Rebate accrual account has been debited.

11. Close the page.
12. Click Cancel.

    - This refreshes the page so that you can see the updates.

13. On the Action Pane, click Collect.

14. Click Settle transactions.
    - Note that a transaction for negative amount, representing the total rebate amount, without invoice reference has been added to the customer balance.
15. Click Cancel.

Feedback

Was this page helpful? Yes No

Provide product feedback

Look up applicable prices and discounts

Article • 09/29/2021

This procedure shows how to find the price and/or discount for a product which is currently valid for a specific customer, without creating a sales order. The procedure walks through a specific example, and you need follow the example using the USMF demo company in order to select the necessary values.

Find the applicable price

1. Go to Sales and marketing > Prices and discounts > Find prices.
2. In the Customer account field, click the drop-down button to open the lookup.
3. In the list, find and select customer US-001.
4. In the list, click the link in the selected row.
5. In the Item number field, type 'T0004'.

    - By default, the Quantity field is set to 1. However, if you know the size of the order that the customer intends to place for the product in question, then enter this value instead. This information is relevant if the trade agreements with the customer have quantity breaks, that is, the product's price depends on the minimum quantity purchased.

6. In the Date field, enter a date for when the customer expects to place an order.

    - The date can be today's date or any date in the future.
    - The system now returns the price that is valid for the selected product when bought by the selected customer on the selected date with a specified quantity. In this example, if the customer US-001 bought 1 unit of product T0004 today, they would be charged 350 CAD a unit. This price comes from an existing and active trade agreement with the customer. Other fields on the page provide more details about the product price and product cost (if set up on the product master), and calculated profitability.
    - If the Show related product variants option is selected, it means that there are additional trade agreements for product's variants.

7. Click the Show related product variants checkbox.

    - A list of the product variants is shown, with information about their dimensions.

8. In the list, mark the line representing color White.

- Notice, that the product price is now different from the one displayed previously when it was not specified per dimension.

9. Click View sales prices.

    - The Price (sales) page displays all the trade agreements applicable to the product, including its variants.

10. Close the page.

Find the applicable discount

Make sure the Customer account field contains customer number US-001

1. In the Item number field, type 'T0012'.

    - Make sure the Quantity field is set to 1.
    - The following pricing details shown for product T0012 come from one or more trade agreements: The unit price is 1,000 CAD and the discount percentage is 5.

2. Set Quantity to '20'.

    - The increased order quantity causes the line discount that will be offered to the customer to change from 5 to 7 percent.
    - The Net amount is calculated based on the unit price, discount and the total quantity.

3. Click View line discount.

    - There are two line discount agreements for product T0012, specifying a 5 percent discount for an order line quantity from 1 to 10, and 7 percent discount for order quantities above 10. Note that the discounts are applied to a group of products, in this example, Group code 01, of which product T0012 is a member.

4. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Price simulation

Article • 05/03/2022

This article provides information about price simulation for quotations. Price simulation helps you to evaluate the effect of deductions on the future sales price during the quotation process, before you commit to a specific price.

A price simulation for a quotation shows a new total amount, based on a proposed new price. A price simulation can also show a new amount for a specific line that is created in an existing quotation. You can enter a price simulation and apply it later. Alternatively, you can use the original quotation without a price simulation and make more changes as you work through the sales process with your customer.

A price simulation doesn't change the price in the quotation. If the price simulation is applied to a whole quotation, it's treated as a special discount on the quotation header. If the price simulation is applied to specific items, it's treated as a special discount on the quotation lines. The unit sales price on a quotation line that is created doesn't change when a price simulation is applied. Instead, a discount percentage that corresponds to the price reduction of the quotation line is applied. When a price simulation is applied, the unit sales price and the discount percentage are transferred to the quotation line or the quotation header.

> **Note**
>
> When you run a price simulation, only the current sales currency is used to create the simulation. However, when you view the quotation totals, you see a combination of the company currency and the sales currency.

Supplementary items that are added to quotation lines might trigger line discounts or multiline discounts. They might also trigger total discounts that change the contribution margins and contribution ratios of the quotation lines and the whole discount.

You can run multiple price simulations to track the effects of price adjustments on the targets of a quotation. By running these simulations, you can adjust the overall price of the quotation, or the price of one or more specific lines in the quotation, and then apply the price simulation that is most likely to help you close the sale.

When you create a quotation, you can set up an alert. Here are some of the ways that alerts are used:

- They can keep you informed about the status of quotations in the organization.

- They can trigger a review of a specific quotation or inform you when discount limits are exceeded.

Price simulation and discounts

To guarantee that discounts and prices are calculated correctly, be careful when you run price simulations on quotations that have discounts. Because all price simulations are treated as special discounts on the active quotation line or the whole quotation, you must track the differences in the discounts.

Types of discounts in trade agreements

Trade agreements in Supply Chain Management can have four types of price discounts. These discounts can be set up for different items, customers, or price groups, and they can be limited by date. To avoid miscalculations, you must consider trade agreements when you run price simulations. Here are the four types of discounts in trade agreements:

- **Sales price** – Separate sales prices can be specified for items. When quotation lines are created, the program searches for the correct sales price for an item and transfers it to the quotation lines. Therefore, a trade agreement that has this kind of discount doesn't affect the price simulation. The sales price that is used in the quotation line reflects the trade agreement.
- **Line discount** – Special discounts are specified for items, depending on the quantity that is ordered. Line amounts are typically reduced by the line discount before a price simulation is run. Therefore, a trade agreement that has this kind of discount affects the price simulation.
- **Multiline discount** – If the combined quantities exceed the limit that you've defined, predefined combinations of ordered items trigger a discount on the whole order. Line amounts are typically reduced by the line discount before a price simulation is run. Therefore, a trade agreement that has this kind of discount affects the price simulation.
- **Total discount** – If the combined amounts exceed the limit that you've defined, predefined ordered items trigger a discount on the whole order. The total discount is generated by the quotation lines. However, because the total discount is applied to the quotation total as a discount, it reduces the total amount of the quotation. Therefore, a trade agreement that has this kind of discount affects the price simulation.

Quotation lines and trade agreements

When you create or adjust a quotation line, line discounts are automatically calculated. The relevant sales price is found for the item, based on the trade agreement.

Price simulation examples

The following examples use price simulation for quotation headers and single line items.

Price simulation for quotation headers

You create a quotation that has the following lines:

- 10 units of item BR-12 (cost price per unit: USD 9.52, sales price per unit: USD 15.32)
- 12 units of item BR-14 (cost price per unit: USD 7.48, sales price per unit: USD 13.75)

The following table shows the quotation lines.

Expand table

	Calculation	Result
Sales quantity	10 units + 12 units	22 units
Sales value in USD	(10 × 15.32) + (12 × 13.75)	318.20
Cost value in USD	(10 × 9.52) + (12 × 7.48)	184.96
Contribution margin in USD	318.20 – 184.96	133.24
Contribution ratio	([318.20 – 184.96] ÷ 318.20) × 100	41.87%

You run a price simulation and apply a 15-percent total discount for the whole quotation, or the quotation header. The following table shows the new totals of the quotation after the price simulation is run.

Expand table

	Calculation	Result
Sales quantity	10 units + 12 units	22 units
Old sales value in USD	(10 × 15.32) + (12 × 13.75)	318.20
Old cost value in USD	(10 × 9.52) + (12 × 7.48)	184.96

	Calculation	Result
Old contribution margin in USD	318.20 – 184.96	133.24
Old contribution ratio	([318.20 – (10 × 9.52)] ÷ 318.20) × 100	41.87%
Price simulation of 15-percent total discount in USD	(15 × 318.2) ÷ 100	47.73
New sales value in USD	318.20 – 47.73	270.47
New contribution margin in USD	270.47 – 184.96	85.51
New contribution ratio	[(270.47 – 184.96) ÷ 270.47] × 100	31.61%

Price simulation for single line items

You create a quotation that has the following lines:

- 10 units of item BR-12 (cost price per unit: USD 9.52, sales price per unit: USD 15.32)
- 12 units of item BR-14 (cost price per unit: USD 7.48, sales price per unit: USD 13.75)

The following table shows the quotation lines.

Expand table

	Calculation	Result
Sales quantity	10 units + 12 units	22 units
Sales value in USD for BR-12	10 × 15.32	153.20
Sales value in USD for BR-14	12 × 13.75	165.00
Cost value in USD for BR-12	10 × 9.52	95.20
Cost value in USD for BR-14	12 × 7.48	89.76
Contribution margin in USD for BR-12	153.20 – 95.20	58.00
Contribution margin in USD for BR-14	165.00 – 89.76	75.24
Contribution ratio in USD for BR-12	[(153.20 – 95.20) ÷ 153.20] × 100	37.86
Contribution ratio in USD for BR-14	[(165.00 – 89.76) ÷ 165.00] × 100	45.60

	Calculation	Result
Total sales value in USD	(10 × 15.32) + (12 × 13.75)	318.20
Total cost value in USD	(10 × 9.52) + (12 × 7.48)	184.96
Total contribution margin in USD	318.20 – 184.96	133.24
Total contribution ratio	[(318.20 – 184.96) ÷ 318.20] × 100	41.87%

You run a price simulation and apply a 10-percent total discount to the BR-12 units. The following table shows the new totals of the quotation after the price simulation is run for the single line item.

Expand table

	Calculation	Result
Sales quantity	10 units + 12 units	22 units
Old sales value in USD for BR-12	10 × 15.32	153.20
Price simulation of 10-percent discount for BR-12	(10 × 153.2) ÷ 100	15.32
New sales value in USD for BR-12	(10 × 15.32) – 15.32	137.88
Sales value in USD for BR-14	12 × 13.75	165.00
Cost value in USD for BR-12	10 × 9.52	95.20
Cost value in USD for BR-14	12 × 7.48	89.76
New contribution margin in USD for BR-12	137.88 – 95.20	42.68
Contribution margin in USD for BR-14	165.00 – 89.76	75.24
New contribution ratio in USD for BR-12	[(137.88 – 95.20) ÷ 137.88] × 100	30.95
Contribution ratio in USD for BR-14	[(165.00 – 89.76) ÷ 165.00] × 100	45.60
New total sales value in USD	[(10 × 15.32) – 15.32] + (12 × 13.75)	302.88
Total cost value in USD	(10 × 9.52) + (12 × 7.48)	184.96
New total contribution margin in USD	302.88 – 184.96	117.92
New total contribution ratio	[(302.88 – 184.96) ÷ 302.88] × 100	38.93%

The price simulation affects only the line that it's applied to and reduces the total for that line.

Feedback

Was this page helpful? Yes No

Provide product feedback

Mass create sales quotations

Article • 09/29/2021

This procedure demonstrates how to efficiently create quotations offering a set of products or services that are to be sent to multiple customers. This mass quotation creation is based on quotation templates. You can run this procedure on your own data or in demo data company USMF.

Create a quotation template

1. Go to Sales and marketing > Setup > Quotations > Template groups.
2. Click New.
3. In the Group ID field, type an ID of your choice.
4. In the Description field, type a value.
5. Click Save.
6. Close the page.
7. Go to Sales and marketing > Sales quotations > All quotations.
8. Click New.
9. In the Account type field, select 'Customer'.
10. In the Customer account field, enter or select a value.
11. Click OK.

    - For a quotation to become a template you must carry out setup steps on the quotation header. This must be done before you add lines to the quotation.

12. On the Action Pane, click Options.
13. Click Change view.
14. Click Header view.
15. Expand the Setup section.
16. In the Group ID field, enter or select a value.
17. In the Template name field, type a value.
18. Select Yes in the Active field.

    - Only active templates can be used when you apply a template to a new sales quotation.

19. On the Action Pane, click Options.
20. Click Change view.
21. Click Line view.
22. In the Item field, enter or select a value.
23. In the Item field, type a value.

24. Close the page.
25. In the Discount percent field, enter a number.
26. Click Add line.
27. In the Item field, enter or select a value.
28. In the Item field, type a value.
29. Close the page.
30. In the Unit price field, enter a new price or change the current one.
31. Click Add line.
32. In the Item field, enter or select a value.
33. In the Item field, type a value.
34. Close the page.
35. In the Quantity field, enter a number.
36. In the Discount field, enter a number.
37. Click Save.

Apply the template to create a single quotation

1. Go to Sales and marketing > Sales quotations > All quotations.
    - Note that the quotation you have just created is marked as template.
2. Click New.
3. In the Account type field, select 'Customer'.
4. In the Customer account field, enter or select a value.
5. Expand the Template section.
6. In the Group ID field, enter or select a value.
7. In the Template name field, enter or select a value.
8. In the Calculation method field, select 'Based on template values'.
9. Click OK.
    - The new quotation has now been created, based on the data and terms of the template.
10. Close the page.
11. Close the page.

Apply the template to mass create quotations

1. Go to Sales and marketing > Sales quotations > Quotation update > Mass create quotations.
2. In the Account type field, select 'Customer'.

3. In the Group ID field, enter or select a value.
4. In the Template name field, enter or select a value.
5. In the Calculation method field, select 'Based on template values'.
6. Expand the Records to include section.
7. Click Filter.
8. In the Criteria field, set the filter to cover a range of customers you want to include in this mass quotation creation. Use the following format "Customer1..CustomerN.
    - For example, you could set the filter to: US-001..US-004
9. Click OK.
10. Click OK.
11. Go to Sales and marketing > Sales quotations > All quotations.
    - Verify that quotations have been created for all the customers specified in the mass update routine, as based on the selected template.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create and edit sales quotations

This procedure demonstrates how to create and update a sales quotation.

Create a sales quotation

1. Go to **Sales and marketing** > **Sales quotations** > **All quotations**.
2. Select **New**.
3. In the **Account type** field, select 'Prospect'.
4. In the **Prospect** field, enter or select a value.
5. Expand the **General** section. Because you chose to create a quotation from the Sales and Marketing area, the type is automatically set to *Sales quotation*. To create a quotation for a project, you must access it from the **Project management and accounting** module.
6. Select **OK**. The fields and actions on the quotation lines are very similar to the ones on the sales order lines. Like sales orders, quotations can be created for a specific item or, when item number isn't known or doesn't exist at the time of quotation creation, quotations can be created for a sales category.
7. In the **Item** field, enter or select a value.
8. In the **Site** field, type a value.
9. In the **Quantity** field, enter a number. If there are valid trade agreements for the item selected on the line, the applicable price and discounts will be automatically copied to the quotation line. Make sure that the Unit price field contains a value and you can also enter discount values if you want to.
10. Select **Save**.
11. On the Action Pane, select **Sales quotation**.
12. Select **Totals**.
13. Select **OK**.
14. Select the sales quotation line.
15. On the Action Pane, select **Quotation**.
16. Select **Price simulation**.

    - In the **Run price simulation** page, you can experiment with adjusting the expected revenue or profitability of your quotation based on the desired unit price, discount amount, discount percentage, total amount, margin, or contribution ratio. When you're satisfied with the target figures, you apply the suggestion to the quotation line, and its price-related fields will be updated accordingly.

- You can create as many price simulations as you wish. When you select **New**, the price conditions from the current quotation line are copied to the page. You can then modify values in any of the price-related fields to the target ones. A change in one of the fields will trigger recalculation in all the other fields. In order for the system to calculate the sales margin and contribution ratio, the product's unit cost has to be known. Use the **Simulated prices** tab for a detailed view of the original prices, proposed changes, and their effect on the quotation totals. As a general rule, when a simulation that sets a new amount is applied to the quotation line, the system recalculates and enters a new value in the Unit price field. If the simulation is based on a new margin or a new contribution ratio, only the Net amount field is updated, and the Unit price is blank. In both cases, any discounts that were on the quotation line before simulation will be deleted.

17. On the Action Pane, select **Quotation**.
18. Select **Send quotation**.
19. Select 'Yes' in the **Print quotation** field.
20. Select **OK**. The report might take a minute to generate. Don't close the page until it finishes.

Update a sales quotation

1. Go to **Sales and marketing** > **Sales quotations** > **All quotations**.
2. On the Action Pane, select **Follow up**.
3. Select **Convert to customer**.
4. In the **Customer account** field, type a value.
5. Select **Check**. Make sure you see a message that the account number you typed in is free to use.
6. Select **OK**. The system has now created a new customer account for the prospect on the quotation.
7. Close the page.
8. On the Action Pane, select **Follow up**.
9. Select **Confirm**.
10. In the **Reason** field, enter or select a value.
11. Select **OK**.
12. On the Action Pane, select **General**.
13. Select **Sales orders**.
14. Close the page.

Last updated on 07/01/2026

Product bundles overview

Product bundles enable businesses to group multiple items so that they can be priced and sold together. They provide an easy way to ensure that the correct items are always sold and priced together, but that the individual items in the bundle can also be shipped individually and revenue can be recognized against the shipped items.

A product bundle consists of a parent item and multiple component items. Order entry is quick and efficient, because users can add the whole bundle to a sales order just by adding the parent item. Picking lists and packing slips list the component items of a product bundle, but sales confirmations and sales invoices show only the parent item.

Each product bundle is represented by a physical item. However, when the sales order is confirmed, the bundle is exploded into its component items, and the selling price is allocated proportionally from the parent item to the individual component items that make up the bundle. After the bundle is exploded into its component items, the product bundle construct ensures that the sum of the component items equals the bundle net amount, so that the customer is always invoiced the correct amount for the product bundle.

This product bundle feature replaces the previous bundle feature that was part of the revenue recognition module and was deprecated in April 2023.

> **Note**
>
> Product bundles aren't supported in channels that are managed by Microsoft Dynamics 365 Commerce. These channels include e-commerce, point of sale (POS), and call centers. Items that are configured as product bundles shouldn't be added to orders or transactions that were created in Dynamics 365 Commerce channels.

Next steps

- Enable and set up product bundles
- Sell and allocate product bundles

Last updated on 07/01/2026

Enable and set up product bundles

Prerequisites

Before you can use the features described in this article, ensure your system meets the following requirements:

- To use the *Product bundle* feature, you're running Microsoft Dynamics 365 Supply Chain Management version 10.0.36 or later.
- To use the *Product bundles in journals* feature, which extends the *Product bundle* feature, you're running Supply Chain Management version 10.0.40 or later.

Turn on the product bundle features for your system

Turn on the product bundle feature

To make product bundle functionality available, turn it on for your system.

1. Go to **System administration** > **Workspaces** > **Feature management**. (For more information, see Feature management overview.)
2. On the **All** tab, use the **Filter** field to search for the feature named *Revenue recognition*.
3. If the *Revenue recognition* feature is enabled in your system, select it in the list, and then select **Disable** to disable it. You can't use this feature together with the *Product bundle* feature.
4. Use the **Filter** field to search for the feature named *Product bundles*.
5. Select the feature in the list, and then select **Enable now** to enable it.

> **Note**
>
> When you first turn on the *Product bundle* feature, the system checks whether it includes any released products where the **Revenue recognition bundle** option is set to *Yes*. If it does, the system creates a batch job that automatically upgrades the records for those products so that they work with the *Product bundle* feature. Likewise, if the system includes any uncanceled sales orders that have lines that contain revenue recognition bundle items, those sales order lines are also upgraded so that they work with the *Product bundle* feature. This behavior provides a seamless experience as you transition from using revenue recognition

bundles to using product bundles. The upgrade is done as part of a Data maintenance portal action.

Turn on the product bundles in journals feature

The *Product bundles in journals* feature extends the *Product bundle* feature. It enables the system to preserve detailed product bundle information in its database. Therefore, you can reprint original sales order confirmations and invoices even after the related sales order is deleted or archived and purged. This feature also improves the ability to exchange order confirmations and invoices electronically. Because product bundles are now represented in journals, you can electronically exchange order confirmations and invoices that include product bundles.

To make the *Product bundles in journals* feature available, you must turn it on for your system. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

1. Go to **System administration** > **Workspaces** > **Feature management**. (Learn more in Feature management overview.)
2. On the **All** tab, use the **Filter** field to search for the feature that's named *Product bundles in journals*.
3. Select the feature in the list, and then select **Enable now** to enable it.

> **Note**
>
> When you turn on the *Product bundles in journals* feature, the system creates a batch job that automatically upgrades sales confirmation journal lines, sales packing slip journal lines, and sales invoice journal lines that contain product bundles and that sales order lines exist for. As a result of this upgrade, sales confirmations and sales invoices can be reprinted correctly after sales order lines are deleted. The upgrade is done as part of a data maintenance portal action.

Set up product bundles

Bundle items are released products that include components. You set up bundle items by using the bill of materials (BOM) functionality. When a product bundle item is entered on a sales order, revenue is posted for the individual components, not for the product bundle item.

Documents that are printed for the customer (such as sales order confirmations and invoices) show only the product bundle item, not the components. However, documents that are printed

for internal use (such as sales order picking lists and packing slips) show only the components, not the product bundle item. Learn more in Sell and allocate product bundles.

Set up product bundle components

Each component item of a product bundle must be set up as a released product on the **Released products** page (**Product information management** > **Products** > **Released products**). The released products are set up in the same way as products that are included on a BOM. For example, a released product can be an item of either the *Item* type or the *Service* type, but it must be assigned to an item model group where the **Stocked product** option is set to *Yes*. Learn more in Bills of materials and formulas.

In addition, a base sales price must be set for each component. The base sales price is used to calculate the component's revenue contribution percentage. The product bundle's revenue price is then distributed to the components based on their revenue contribution percentages. Learn more in Sell and allocate product bundles.

Set up product bundle items

Set up each product bundle item as a released product on the **Released products** page (**Product information management** > **Products** > **Released products**). To identify a released product as a bundle item, set the following two fields on the released product record:

- On the **General** FastTab, set the **Product Bundle** option to *Yes*.
- On the **Engineer** FastTab, set the **Production type** field to *BOM*.

Then, assign the component items to the product bundle (BOM parent) item on the **BOM versions** page. To open this page, open the product bundle item on the **Released products** page. Then, on the Action Pane, on the **Engineer** tab, select **BOM versions**. For more information, see Bills of materials and formulas.

Related information

- Product bundles overview
- Sell and allocate product bundles

Last updated on 07/01/2026

Sell and allocate product bundles

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to work with product bundles on sales orders and related documents. It also explains how to allocate bundle item prices to each bundle component.

A product bundle consists of a parent item (product bundle item) and multiple component items. The parent item is entered on a sales order line. Therefore, order entry is more efficient. However, the parent item is then exploded into the component items. Internal documents such as picking lists and packing slips list the component items. However, external documents such as sales order confirmations and sales invoices show only the parent item.

> **Note**
>
> Microsoft Dynamics 365 Commerce channels, such as online, point of sale (POS), and call centers, don't support the product bundle feature. The prospect-to-cash solution for Dynamics 365 Supply Chain Management and Dynamics 365 Sales doesn't support it either. Ensure that items configured as product bundles aren't added to orders or transactions that are created in Commerce channels or in the prospect-to-cash solution.

How bundle items are set up and handled

Use the bill of materials (BOM) functionality to set up bundles. For information about how to set up a bundle item, see Enable and set up product bundles.

Parent items that are flagged as product bundles are handled differently than other BOM items. Here's a list of the differences:

- You explode product bundles that are included on a sales order by confirming the order. Open the sales order, and then, on the Action Pane, on the **Sell** tab, select **Confirm sales**

order. Never explode a product bundle item from the **Sales order lines** FastTab (by selecting **Sales order line** > **Explode** > **BOM lines** on the toolbar). Otherwise, the system treats the item as a BOM instead of a product bundle.

- A sales order that contains a product bundle item must be confirmed before the picking list, packing slip, or invoice is created.
- When a product bundle is exploded through sales order confirmation, the parent item is canceled, and its unit price and discounts are allocated to the component items of the bundle.
- The sum of the component items must always equal the price of the parent item. Therefore, there are limitations on the fields that can be updated or changed for component items. For example, the unit price can't be manually changed. It also can't be indirectly changed. To help prevent scenarios such as searching and applying a new price, inventory dimensions can't be changed on the component items.
- When external-facing documents such as sales order confirmations or invoices are printed, only the parent item is shown on the printed document. The component items aren't shown.

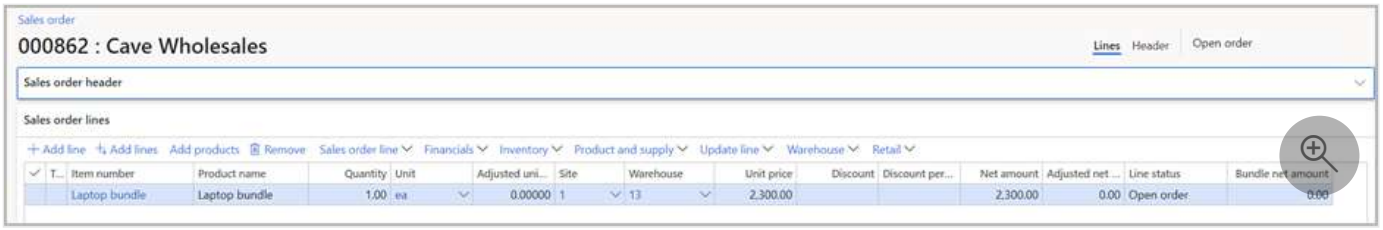
Example of a sales order that includes a product bundle

Set up a product bundle in the following way:

- **Parent item:** *Laptop bundle*
- **Component item:** 1 pcs. of item *1000* (the laptop itself), which has a base sales price of $1,900.00
- **Component item:** 1 pcs. of item *S0021* (insurance), which has a base sales price of $150.00
- **Component item:** 1 pcs. of item *Support* (support), which has a base sales price of $500.00

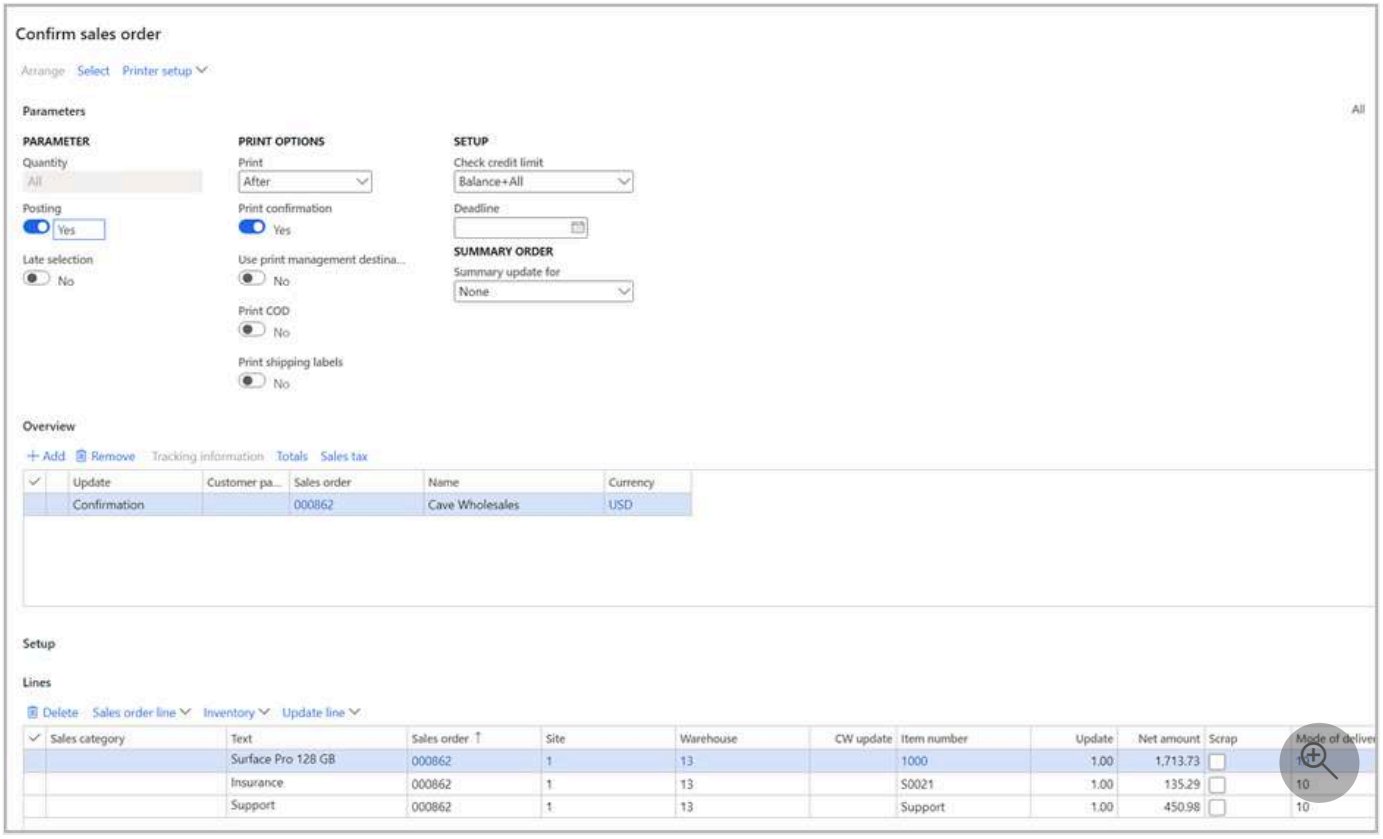
The base sales price of each component item is an essential part of the component setup. It's set on the **Sell** FastTab of the item. The base sales price is used to calculate the allocation factor when the parent item's unit price is allocated to the component items. Trade agreement sale prices are never used for this purpose.

Next, you have a sales order that includes a single sales order line for the *Laptop bundle* item. The default unit price for the parent item can be taken from numerous places, such as the trade agreement or the base sales price. For this example, $2,300 is manually entered as the unit price on the sales order, as shown in the following illustration.

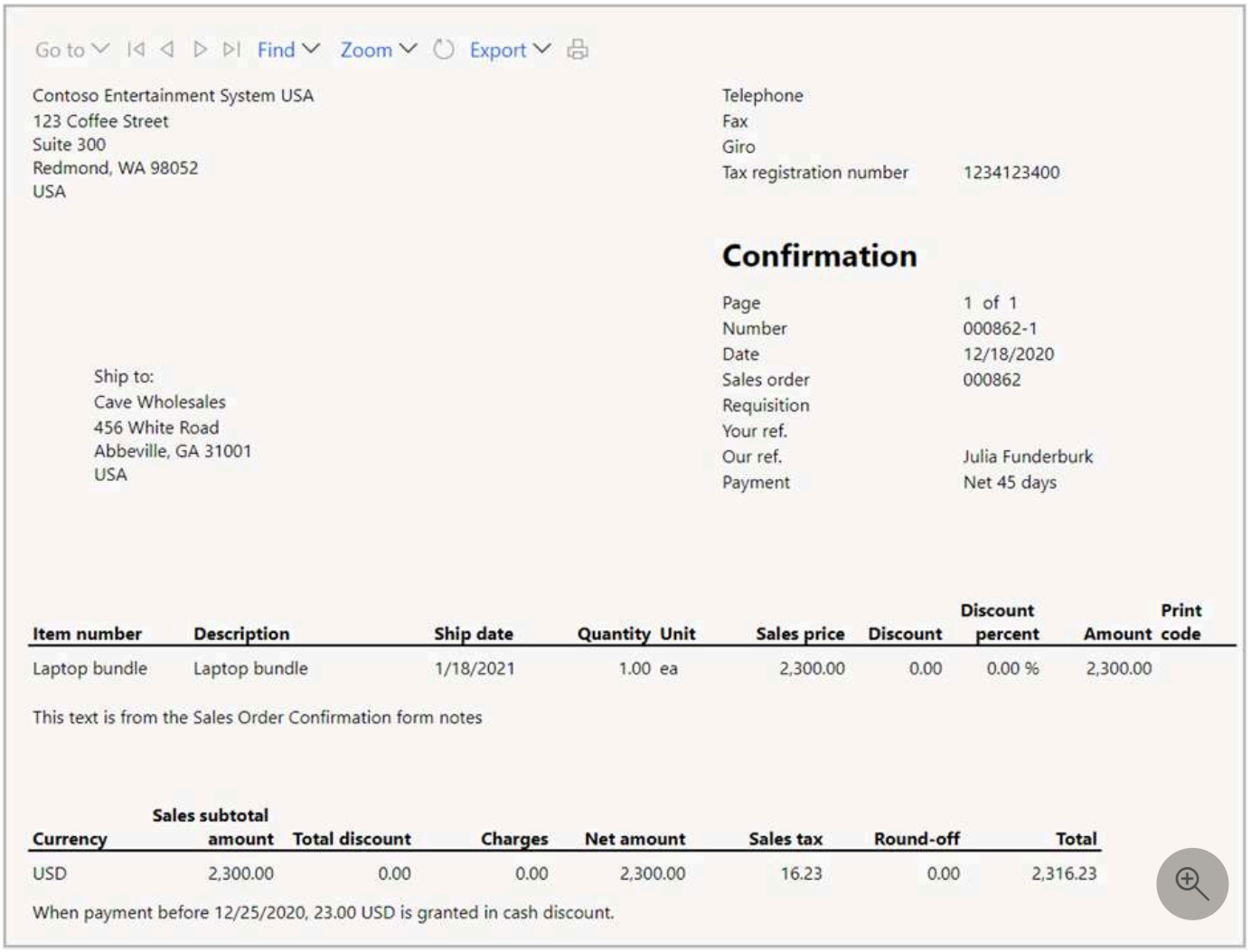


Confirm the order to explode the product bundle

Because the sales order includes a product bundle item, you must confirm it. The **Confirmation sales order** dialog box shows the components of the product bundle, as shown in the following illustration.



However, because the printed confirmation report is the external-facing document for the customer, it shows only the parent item of the product bundle, as shown in the following illustration.



Go to Find Zoom Export

Contoso Entertainment System USA
123 Coffee Street
Suite 300
Redmond, WA 98052
USA

Telephone
Fax
Giro
Tax registration number 1234123400

Confirmation

Page 1 of 1
Number 000862-1
Date 12/18/2020
Sales order 000862
Requisition
Your ref.
Our ref. Julia Funderburk
Payment Net 45 days

Ship to:
Cave Wholesales
456 White Road
Abbeville, GA 31001
USA

Item number	Description	Ship date	Quantity	Unit	Sales price	Discount	Discount percent	Amount	Print code
Laptop bundle	Laptop bundle	1/18/2021	1.00	ea	2,300.00	0.00	0.00 %	2,300.00	

This text is from the Sales Order Confirmation form notes

Currency	Sales subtotal amount	Total discount	Charges	Net amount	Sales tax	Round-off	Total
USD	2,300.00	0.00	0.00	2,300.00	16.23	0.00	2,316.23

When payment before 12/25/2020, 23.00 USD is granted in cash discount.

After you confirm the sales order, the parent item still appears on the sales order, but its status changes to *Canceled*. Additionally, the net amount is tracked in the **Bundle net amount** field. This amount is required to print the invoice, because the invoice shows the parent item, not the component items.

Allocate prices to component items

The sum of the component items must equal the **Bundle net amount** value of the parent item, because that value is the amount that's presented to the customer when the invoice is created. To ensure that the invoice matches the amounts that are posted to the general ledger, edits to the component items are limited. For example, the site and warehouse can't be changed, because those changes might trigger a price change based on a trade agreement.

Because the sum of the components doesn't match the bundle price entered on the sales order in this example, the unit prices are recalculated and allocated to the components in the following way:

- **Product bundle price entered on the sales order:** $2,300
- **Total base sales prices from components:** $1,900 + $500 + $150 = $2,550

- **Allocated price for component 1:** $2,300 × (1,900 ÷ 2,550) = $1,713.73
- **Allocated price for component 2:** $2,300 × (500 ÷ 2,550) = $450.98
- **Allocated price for component 3:** $2,300 × (150 ÷ 2,550) = $135.29

The sum of the components must equal $2,300, and it does ($1,713.73 + $450.98 + $135.29 = $2,300).

If you need to change all component items, remove the parent item. When you remove the parent item, you also remove the component items. Then, add the parent item again, and complete the required edits before you confirm the sales order.

Pick, pack, and ship the order

While the sales order is picked and packed, only the components of the product bundle appear. However, the packing slip and invoice must include a full product bundle. Otherwise, they can't be posted. For example, the dialog box shows three component items. If you try to delete one of them, you receive an error message that states that all products in the product bundle must be shipped before they can be invoiced.

Each product bundle must be shipped and invoiced as a full bundle. For example, if you change the quantity of component item *1000* to *4*, but you leave the quantity of the other component items set to *5*, the packing slip and invoice can't be posted.

You can ship and invoice a partial amount only if the quantity is reduced for all components of the bundle. For example, a quantity of *5* of the *Laptop bundle* item is entered on a sales order. After the sales order is confirmed, the three component items are shown on the sales order, and the quantity of each is *5*. By default, the quantity of each component is set to *5* during shipping and invoicing. However, you can adjust the quantity to *3* for all three component items. In this case, three full product bundles are shipped and invoiced. The remaining two product bundle items (a quantity of *2* of each of the three component items) can be shipped and invoiced later.

Invoice the sales order

The final step is to invoice the sales order. As with the sales order, the invoice dialog box shows the component items, whereas the printed invoice shows only the parent item.

The invoice journal created after posting doesn't include the parent item from the product bundle, because that item has a status of *Canceled*.

The invoice journal must not include the parent item from the product bundle, because any processes that are performed after the invoice is posted are based on that invoice journal. For

example, if you create a credit note from the **Sell** tab of the Action Pane, the credit note that's created includes the component items but not the parent item.

Product bundles in journals

The *Product bundles in journals* feature extends the *Product bundle* feature. (Learn more in Enable and set up product bundles.) It enables the system to preserve detailed product bundle information in its database. Therefore, you can reprint original sales order confirmations and invoices even after the related sales order is deleted or archived and purged. This feature also improves the ability to exchange order confirmations and invoices electronically. Because product bundles are now represented in journals, you can electronically exchange order confirmations and invoices that include product bundles.

This section describes the changes that the *Product bundles in journals* feature introduces. These changes build on the features that were described earlier in this article.

Changes in sales confirmation journals

After a sales order is confirmed, the parent item still appears on the sales order, but its status changes to *Canceled*. An additional status field, **Bundle line status**, represents the overall status across the components that are part of the product bundle. Use it to identify the overall status of the product bundle and its components on the sales order lines. A symbol in the **Type** column helps you identify which order line is a product bundle and which is a component. The net amount is tracked in the **Bundle net amount** field.

Sales confirmation journal lines now contain information that's related to product bundles. This information is presented in the **Bundle parent** column. For each component journal line that's part of a bundle, the **Bundle parent** column shows the lot ID and name of the bundle. Use the **Group by** option on the **Bundle parent** column to see which components belong to which bundle.

Product bundle information on the journal lines makes it possible to reprint sales confirmations correctly even after the sales order is deleted.

Changes in packing slip journals

Sales packing slip journal lines now contain information that's related to product bundles. This information is presented in the **Bundle parent** column. For each component journal line that's part of a bundle, the **Bundle parent** column field shows the lot ID and name of the bundle. Use

the **Group by** option on the **Bundle parent** column to see which components belong to which bundle.

Product bundle information on the journal lines makes it possible to reprint sales packing slips correctly even after the sales order is deleted.

Changes in sales invoice journals

Sales invoice journal lines now contain information related to product bundles. This information appears in the **Bundle parent** column. For each component journal line that's part of a bundle, the **Bundle parent** column field shows the lot ID and name of the bundle. Use the **Group by** option on the **Bundle parent** column to see which components belong to which bundle.

By including product bundle information on the journal lines, you can reprint sales invoices correctly even after the sales order is deleted.

Entity support for exporting sales confirmations and invoiced with bundle information

The following data entities support bundle information on journal lines for sales confirmations and sales invoices. These entities let you send sales order confirmations and sales invoices that include product bundle lines to your customers.

- **Sales order confirmation bundle parent lines** – This entity includes both root and intermediate bundle parent lines. To set criteria that define which lines are included, apply a filter on the *journal line type*. Select *bundle parent* to include only the bundle parent.
- **Sales order confirmation lines including bundle component** – This entity includes standard sales order conformation transactions and bundle components, if they're available. To set criteria that define which lines are included, apply a filter on the *journal line type*. Select *standard* to exclude any components.
- **Sales invoice bundle parent lines** – This entity includes both root and intermediate bundle parent lines. To set criteria that define which lines are included, apply a filter on the *journal line type*. Select *bundle parent* to include only the bundle parent.
- **Sales invoice lines including bundle component** – This entity includes standard invoice transactions and bundle components, if they're available. To set criteria that define which lines are included, apply a filter on the *journal line type*. Select *standard* to exclude any components.

The data that's exported through these entities resembles the data that's printed on sales confirmations and sales invoices.

Important limitations

The following important limitations apply when you use product bundle items:

- The product bundle feature isn't supported in Commerce channels, such as online, point of sale (POS), and call centers. It also isn't supported by the prospect-to-cash solution for Supply Chain Management and Sales. Don't add items that are configured as product bundles to orders or transactions created in Commerce channels or in the prospect-to-cash solution.
- Product bundle items aren't supported in intercompany orders. In a three-legged scenario, product bundle items aren't exploded during sales order confirmation on the original sales order. In a two-legged scenario, the sales order confirmation process is blocked.
- Product bundle items aren't supported in direct deliveries.
- Product bundle items aren't supported in delivery schedules.
- The delivery type of the sales order line for bundle components is always *Stock*, regardless of the direct delivery setting on the released product.
- Unless you're using the *Product bundles in journals* feature, you can reprint an invoice or sales order confirmation only if the related sales order that includes the product bundle order lines still exists. If the sales order lines were deleted, the reprinted invoices and sales order confirmations include the component items but not the parent item.
- Miscellaneous charges added to a parent item line aren't allocated to the component item lines when the parent item is exploded.

Related information

- Product bundles overview
- Enable and set up product bundles

Last updated on 05/06/2026

Set up intercompany trade

To enable Microsoft Dynamics 365 Supply Chain Management to run intercompany trade, set up customers and vendors to run intercompany trade. You must also set up Accounts payable, Accounts receivable, Procurement and sourcing, and Sales and marketing.

Before you set up intercompany trade

Before you set up intercompany trade, decide which customers are intercompany customers, and which vendors are intercompany vendors. For each legal entity in Microsoft Dynamics 365 Supply Chain Management, decide which trading policy to apply to the intercompany trading relation with the specific intercompany customer or vendor.

In particular, we recommend that you and your organization become familiar with the intercompany parameters.

- Discuss the implications of the setup with the managers who are responsible for intercompany trade in each legal entity.
- Set up the appropriate values for each legal entity.

Set up trading relations

To set up intercompany trade for customers and vendors, follow these steps:

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select a customer account.
3. On the Action Pane, on the **General** tab, select **Intercompany**.
4. Specify intercompany setup parameters for the customer account. These parameters include the legal entity that contains the corresponding vendor and the vendor account. The parameters also include the purchase order policies, sales order policies, value mapping, and policies for sales agreements and purchase agreements. You also specify whether to use base data values from the customer account or from the vendor account in the other legal entity.
5. Repeat steps 1 through 4 for the remaining intercompany customers in the legal entity.
6. Go to **Accounts payable** > **Vendors** > **All vendors**.
7. Select a vendor account.
8. On the Action Pane, on the **General** tab, select **Intercompany**.

9. Specify intercompany setup parameters for the vendor account. These parameters include the legal entity that contains the corresponding customer and the customer account. The parameters also include the sales order policies, purchase order policies, value mapping, and policies for purchase agreements and sales agreements. You also specify whether to use base data values from the vendor account or from the customer account in the other legal entity.
10. Repeat steps 6 through 9 for the remaining intercompany vendors in the legal entity.

Set up products

To enable products for intercompany trade, follow these steps:

1. Go to **Product information management** > **Products** > **All products and product masters**.
2. Define the products released to the legal entities where you want to do intercompany trade.

Last updated on 04/23/2026

Intercompany parameters

In an intercompany organization, you can set up parameters that determine how you trade between different legal entities. The fields that you select determine these parameters. You can select different combinations to reflect different trading scenarios.

The following two examples describe scenarios for intercompany organizations, one with two levels and one with three levels.

Example 1: Two-level intercompany chain

The intercompany organization includes the following legal entities:

- Legal entity A – Sells to external customers, but has no stock. This legal entity buys from Legal entity B.
- Legal entity B – Sells only to Legal entity A.

Both legal entities can sell to and buy from each other.

In this example, the pricing on the original sales order, which is directed to the external customer, is always based on the sales price. The pricing on the intercompany sales order and the intercompany purchase order is controlled by the internal sales or transfer pricing on the intercompany sales order in Legal entity B.

The order header information is controlled from the original sales order to the external customer. Any change on the intercompany sales order isn't synchronized to the original sales order.

In Legal entity A, on the **Intercompany** page for vendors, select **Purchase order policies**. Select the following fields in the **Original sales order (direct delivery)** field group:

- **Print packing slip automatically**
- **Post invoice automatically**
- **Print invoice automatically**

In the **Intercompany purchase order (direct delivery)** field group, select the following field:

- **Post invoice automatically**

In the **Original sales order <-> Intercompany purchase order** group, select the following fields:

- **Customer information**
- **RMA number**

In the **Intercompany purchase order <-> Intercompany sales order** field group, select the following fields:

- **Customer information**
- **RMA number**
- **Batch number**
- **Serial number**

In Legal entity B, on the **Intercompany** page for customers, select **Sales order policies**. Select the following fields in the **Intercompany sales order creation** field group:

- **Sales order numbering** : **Company + original number**
- **Allow summary update of documents for original customer**

In the **Intercompany sales order prices** field group, select the following fields:

- **Price and discount search**
- **Allow price edit**
- **Allow discount edit**

In the **Intercompany sales order > Intercompany purchase order** field group, select the following fields:

- **Batch number**
- **Serial number**

Example 2: Three-level intercompany chain

The intercompany organization includes the following legal entities:

- Legal entity A – A sales legal entity that sells to external customers and buys from Legal entity B.
- Legal entity B – A production or distribution legal entity that can't deliver products, and buys from Legal entity C.
- Legal entity C – A production or distribution legal entity that delivers products to Legal entity B.

Internal transfer pricing between Legal entities B and C is at cost from the selling legal entity at the start of the chain. It's also at cost to Legal entity A, which sells to external customers.

However, pricing on the original sales order to the external customer is always based on the sales price.

Pricing on all intercompany sales orders and intercompany purchase orders is controlled on the intercompany sales order. It's controlled at the start of the chain. Therefore, Legal entity C, which sells to Legal entity B, controls the price. Intercompany sales order pricing is based on the internal sales or transfer pricing that is set up in Legal entity C.

The order header information is controlled from the original sales order to the external customer. Any change on the intercompany orders isn't synchronized to the original sales order.

Select the following parameters:

In Legal entity A, on the **Intercompany** page for vendors, select **Purchase order policies**. Select the following fields in the **Original sales order (direct delivery)** field group:

- **Print packing slip automatically**
- **Post invoice automatically**
- **Print invoice automatically**

In the **Intercompany purchase order (direct delivery)** field group, select the following field:

- **Post invoice automatically**

In the **Original sales order** <-> **Intercompany purchase order** field group, select the following fields:

- **Customer information**
- **RMA number**

In the **Intercompany purchase order** <-> **Intercompany sales order** field group, select the following fields:

- **Customer information**
- **RMA number**
- **Batch number**
- **Serial number**

In Legal entity B, on the **Intercompany** page for customers, select **Sales order policies**. Select the following fields in the **Intercompany sales order creation** field group:

- **Sales order numbering** : **Company + original number**
- **Allow summary update of documents for original customer**

In the **Intercompany sales order** > **Intercompany purchase order** field group, select the following fields:

- **Batch number**
- **Serial number**

In the **Intercompany customer invoice posting** field group, select the following fields:

- **Unit price equal to cost price**
- **Initiate original customer invoice posting**

In Legal entity B, on the **Intercompany** page for vendors, select **Purchase order policies**. Select the following fields in the **Original sales order (direct delivery)** field group:

- **Print packing slip automatically**
- **Post invoice automatically**
- **Print invoice automatically**

In the **Intercompany purchase order (direct delivery)** field group, select the following field:

- **Post invoice automatically**

In the **Original sales order<-> Intercompany purchase order** field group, select the following fields:

- **Customer information**
- **RMA number**
- **Price and discount**

In the **Intercompany purchase order <-> Intercompany sales order** field group, select the following fields:

- **Customer information**
- **RMA number**
- **Batch number**
- **Serial number**

In Legal entity C, on the **Intercompany** page for customers, select **Sales order policies**. Select the following fields in the **Intercompany sales order creation** field group:

- **Sales order numbering** : **Number sequence code**
- **Allow summary update of documents for original customer**

In the **Intercompany sales order prices** field group, select the following field:

- **Price and discount search**

In the **Intercompany customer invoice posting** field group, select the following fields:

- **Unit price equal to cost price**
- **Initiate original customer invoice posting**

In the **Intercompany sales order** <-> **Intercompany purchase order** field group, select the following fields:

- **Batch number**
- **Serial number**

Last updated on 06/22/2026

Set up vendors, customers, and items for intercompany trade

Article • 01/07/2025

To prepare your organization for intercompany trade, you must define the vendors and customers with whom you will be trading internally. You must then associate these vendors and customers with the items that you'll be purchasing or selling.

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Select the vendor to define as an intercompany vendor.
3. On the Action Pane, on the **General** tab, select **Intercompany**.
4. Specify intercompany setup parameters for the vendor account. These parameters include the customer legal entity and account, sales order policies, purchase order policies, value mapping, and purchase agreement and sales agreement policies. You also specify whether to use base data values from the vendor account or from the customer account in the other legal entity.
5. Go to **Product information management** > **Products** > **Released products**.
6. On the **Released products** list page, select the items to assign to the vendor, so that the items are available for intercompany trade. For each item, open the **Released product details** page. On the **Purchase** tab, in the **Vendor** field, type the vendor number.
7. Go to **Accounts receivable** > **Customers** > **All customers**.
8. Select the customer to define as an intercompany customer.
9. On the Action Pane, on the **General** tab, select **Intercompany**.
10. Specify intercompany setup parameters for the customer account. These parameters include the vendor legal entity and account, purchase order policies, sales order policies, value mapping, and sales agreement and purchase agreement policies. You also specify whether to use base data values from the customer account or from the vendor account in the other legal entity.
11. When you're done setting up the intercompany parameters, close the **Intercompany** page to return to the selected customer details.

12. Expand the **Miscellaneous details** FastTab and set **Create intercompany orders** to *Yes*. If you also want orders to be delivered directly to customers, set **Direct delivery** to *Yes*.

    > **Note**
    >
    > If there are some items that your organization stocks and delivers to customers, you might not want to create intercompany orders automatically, even when you have the item in stock. To inactivate the automatic generation of orders for items that you might sometimes have in stock, set **Create intercompany orders** to *No*.

13. If you want to allow extra lines to be created indirectly on a sales order, set **Create indirect order lines** to *Yes*. A user can then add lines to the original sales order from the intercompany sales order.

> **Warning**
>
> If you allow order lines to be created indirectly, you're permitting all additions to the original sales order from the intercompany sales order. Each addition is then processed through to the customer, and is added to the order and the invoice. Additionally, every document that is involved in the sale is printed and posted automatically. Users aren't alerted about the addition.

Feedback

Was this page helpful? Yes No

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Set up charges on intercompany orders

Article • 06/03/2022

You can set up charges to be added to intercompany orders. When a charge is added to an intercompany sales order, it is automatically synchronized to the intercompany purchase order. This works both ways—from the intercompany sales order to the purchase order and the other way around.

You can also use charges to add a profit to an intercompany sales order by defining the charge as an intercompany percentage.

When you set up charges to be applied to intercompany orders, you enable the calculation of internal profit for an intercompany sales order from the net amount of the original sales order. You might want to do this if your intercompany vendor sells to you at cost price. The following procedure describes how to do this for intercompany customers. You can use the same procedure for vendors.

1. Go to **Accounts receivable** > **Setup** > **Charges** > **Customer charge groups**.
2. Create a charges group.
3. Go to **Accounts receivable** > **Customers** > **All customers**. Select customer account link. On the Action Pane, select the **Customer** tab, and then select the **Sales order** FastTab.
4. Select **Edit** on the Action Pane to enable edits to the field. In the **Charges group** field, select the intercompany charges group that you set up in step 2.
5. Go to **Accounts receivable** > **Setup** > **Charges** > **Auto charges**.
6. Set up the charges by filling in the relevant fields.
7. In the **Customer relation** field, select the intercompany charges group that you set up in step 2.
8. On the **Lines** FastTab, in the **Category** field, select **Intercompany percent**.

Feedback

Was this page helpful? Yes No

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Set up parameters to post an intercompany order

Article • 06/03/2022

When an intercompany customer invoice is posted, you can set it up to post both the intercompany purchase order and the original customer invoice automatically.

> **Note**
>
> Before you perform this procedure, set up the print management in your organization to point to the correct invoice printer. This will make sure that the invoice for the original sales order is printed on the correct printer.

You must set up the following parameters:

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**. Select sales order that you want to work with.
2. On the sales order, on the Action Pane, select **Header view** and then select the **Intercompany settings** FastTab and open it.
3. Go to the Action Pane, on the **Sales order** tab and then select **Edit**.
4. Go back to the **Intercompany settings** FastTab and select **Direct delivery** to enable direct delivery throughout the intercompany chain (all orders).
5. Select **Save** to save the sales order with the new setting. Then select **Close** to close the sales order.
6. Go to **Procurement and sourcing** > **Vendors** > **All vendors**. On the **General** tab, select **Intercompany**.
7. On the **Intercompany** page, select the **Purchase order policies** link. In the **Intercompany purchase order (direct delivery)** field group, select **Post invoice automatically** and remove the check mark from the **Print invoice automatically** field.
8. In the **Original sales order (direct delivery)** field group, select the **Post invoice automatically** field and the **Print invoice automatically** field.

Feedback

Was this page helpful? Yes No

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Synchronize intercompany prices and discounts

Article • 06/03/2022

Discounts and prices are always synchronized between the intercompany sales order and the intercompany purchase order. You can also synchronize the price and discounts to and from the original sales order, so that all orders have the same prices and discounts. You do this from the **Intercompany** page that is accessed from the **General** tab on the **All customers** list page - either from **Sales and marketing** or from **Procurement and sourcing**.

The **Price and discount** field is synchronized to the intercompany sales order line. This means that by selecting the **Allow price edit** field and the **Allow discount edit** field, you have allowed a change between the intercompany sales order and the intercompany purchase order. A change to the unit price, price unit, or sales charge on the intercompany sales order determines the price on the intercompany purchase order line.

If the **Allow price edit** and **Allow discount edit** fields are enabled on the intercompany sales order or the intercompany purchase order, you can change the price or discount manually on either order. Otherwise, you cannot.

If the **Allow price edit** and **Allow discount edit** fields are not enabled on the intercompany sales order, you cannot make a manual change to the price (or charges) or discount on an intercompany sales order.

If the **Allow price edit** and **Allow discount edit** fields are not enabled on the intercompany purchase order, you cannot make a manual change to the price (or charges) or discount on an intercompany purchase order.

If the **Allow price edit** and **Allow discount edit** fields are enabled on both the intercompany purchase order and the intercompany sales order, you can make manual changes to both orders. However, this can cause a situation in which updates that are made by one person are overruled by updates that are made by another person in another company on the same order.

> ⓘ **Note**
>
> If you have enabled the **Price and discount** field on both the intercompany purchase order and the intercompany sales order, you are not in control of the pricing.

To avoid these types of conflicts, the best practice is to allow prices and discounts to be changed only on the intercompany sales order or the intercompany purchase order. This is accomplished by enabling the **Allow price edit** and **Allow discount edit** fields on either of these orders.

Feedback

Was this page helpful? Yes No

Provide product feedback

Synchronize intercompany charges

Article • 06/03/2022

Charges are synchronized only between the intercompany sales order and the intercompany purchase order, and synchronization always occurs.

If the **Allow price edit** field is selected on the intercompany purchase order or the intercompany sales order, you can add a charge manually to the intercompany sales order or the intercompany purchase order. Otherwise, you cannot.

If the **Allow price edit** field is not selected on the intercompany sales order, you cannot add a charge manually to an intercompany sales order.

If the **Allow price edit** field is not selected on the intercompany purchase order, you cannot add a charge on the intercompany purchase order.

If the **Allow price edit** field is enabled on both the intercompany purchase order and the intercompany sales order, you can add charges manually to both orders. However, this can result in many charges being added incorrectly.

To avoid these types of conflicts, the best practice is to allow charges to be added either to the intercompany sales order or the intercompany purchase order, but not both.

Feedback

Was this page helpful? Yes No

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Synchronize intercompany customer information

Article • 06/03/2022

Customer information is synchronized if the **Customer information** field is enabled when the sales order is created or a change is made to the customer, vendor reference, or customer requisition number.

You can always change the value in these synchronization fields. However, you can only determine whether this value is copied to the other intercompany orders by selecting this parameter. If you have enabled the **Customer information** field on the intercompany sales order, a change to the intercompany sales order is synchronized to the intercompany purchase order. If you have also enabled the **Customer information** field on the intercompany purchase order, it is also synchronized back to the original sales order.

> **Note**
>
> If you have enabled the **Customer information** field on both the intercompany purchase order and the intercompany sales order, updates that are made by one person in one company are overruled by updates that are made by another person in another company on the same order.

The best practice is to enable the **Customer information** field on the intercompany purchase order. This enables synchronization between the original sales order and the intercompany purchase order and from the intercompany purchase order to the intercompany sales order.

Feedback

Was this page helpful? Yes No

Provide product feedback

Synchronize intercompany disposition codes

Article • 06/03/2022

To support intercompany returns, Microsoft Dynamics 365 Supply Chain Management enables you to map externally defined disposition codes to the corresponding internal disposition codes. When an intercompany chain is being set up, the disposition actions in the two companies that are mapped to each other must be the same. If they differ, the synchronization process will be unsuccessful.

About disposition codes for three-legged direct returns

Disposition codes are synchronized from the intercompany sales order line to the original sales order line. However, synchronization has only one immediate effect on the original sales order line. This effect is that miscellaneous line charges are deleted based on the disposition code and miscellaneous charge that are set up in Company A. Company A is the company that is creating the return order.

In a three-legged direct delivery chain, all disposition actions are supported on the intercompany sales order line. However, if a product is being returned to the customer, you should confirm that the delivery address in the return order matches the customer delivery address that is specified in the original order.

> **Note**
>
> Disposition codes do not exist for purchase orders. Therefore, synchronization of disposition codes from the intercompany sales order to the original return order must be performed from sales order to sales order.

Replacing returned items

If a returned item is being replaced, and a replacement order has already been created in Company B, a disposition code cannot be selected, and no additional replacement order is created in Company A.

Rules for intercompany direct deliveries

The following general rules apply to original return orders in scenarios that involve intercompany direct deliveries:

- If the underlying disposition action on the original sales order line is Credit and Scrap, Credit, or Return to Customer, the Credit disposition action is applied. However, the Return to Customer action code sets the net amount to 0 (zero) on the existing line and on the newly synchronized line from the intercompany sales order. In addition, scrap lines are never synchronized. When a line is added to an intercompany sales order, it is never synchronized for a sales order that has a positive quantity and a disposition action of Scrap (for example, Credit and Scrap or Replace and Scrap).
- An underlying disposition action of Credit and Replace or Scrap and Replace is not overruled. It is treated as a disposition action of Credit and Replace or Scrap and Replace. This rule applies even if no scrap line is created in Company A and no replacement order is created in Company B (the company that received the returned item). A replacement order is created in Company A only when a packing slip update is performed.

Feedback

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Creating intercompany purchase and sales orders in several companies

Microsoft Dynamics 365 Supply Chain Management isn't limited to handling only one production company and several sales companies. All companies that are set up for intercompany can be both trading companies and production companies.

If several companies can deliver the same item, you can freely choose whom to buy from, and updates are processed even if one sales order becomes several purchase orders. Usually, when you explode a bill of materials (BOM) version, demand is generated for the BOM lines in the sales order. However, neither product bundle items nor BOM explosions are supported for intercompany sales orders or intercompany purchase orders.

In the same way that you automatically create one intercompany purchase order, you can create an original sales order in your company and then have several intercompany vendor companies fulfill the order by creating more than one intercompany purchase order. Microsoft Dynamics 365 Supply Chain Management automatically creates intercompany sales orders in the vendor companies.

To do this, all the companies must be set up as trading relationships. The vendor companies must be set up in Microsoft Dynamics 365 Supply Chain Management as intercompany vendors, and they must be the primary vendor for the relevant item. On the sales order, in **Header view**, you must select the **Autocreate intercompany orders** field and the **Direct delivery** field on the **Intercompany settings** FastTab. This is the default setting.

Create the sales lines in the usual way. When you leave the record, a message appears informing you that intercompany purchase orders and intercompany sales orders are created. The message contains links to the new orders. To view the intercompany sales orders created in your vendor companies, open the original sales order and on the **General** tab, in the **Related information** group, select **References**.

If you open the intercompany purchase orders for the vendors, you see that Microsoft Dynamics 365 Supply Chain Management automatically fills in the original sales order number and the intercompany sales order number for each vendor.

Similarly, if you open the intercompany sales orders for the vendors, you see that Microsoft Dynamics 365 Supply Chain Management automatically fills in the original sales order number

and the intercompany purchase order number for each vendor.

> **Note**
>
> If the items on order exist in one of your intercompany vendor companies but not in the other, Microsoft Dynamics 365 Supply Chain Management creates the intercompany orders for the vendor company where the items exist but stops the order creation for the other company. A notification message is displayed when this happens.

Last updated on 05/21/2026

Intercompany orders and return orders

This article describes how to create and update intercompany purchase orders, sales orders, return orders, purchase agreements, and sales agreements.

About intercompany orders

When you create an intercompany sales order, Microsoft Dynamics 365 Supply Chain Management automatically creates a corresponding purchase order. Similarly, when you create an intercompany purchase order, a corresponding intercompany sales order is automatically created.

This process applies to two-legged orders (transactions between two related companies) and to most three-legged orders (when an intercompany transaction originates with a sales order for an external customer).

Intercompany order example

You have two related companies. Company A is a sales subsidiary, and Company B is a distribution unit. Intercompany sales orders and purchase orders are created automatically in the following scenarios:

- When Company A creates a purchase order and the vendor is Company B, an intercompany sales order is created automatically in Company B. The two-legged order chain consists of a purchase order in Company A and an intercompany sales order in Company B.
- When Company B creates a sales order and the customer is Company A, an intercompany purchase order is created automatically in Company A. The two-legged order chain consists of a sales order in Company B and an intercompany purchase order in Company A.
- When Company A first creates a sales order for an external customer, a purchase order can be created automatically in Company A. This action prompts the automatic creation of an intercompany sales order in Company B. The three-legged order chain consists of a sales order and a purchase order in Company A and an intercompany sales order in Company B.

About intercompany return orders

You can return intercompany items much like ordinary returns. However, with intercompany items, the return order from the external customer creates an intercompany purchase order. This

purchase order leads to the automatic creation of an intercompany sales return order. You can also return an intercompany item without an external customer return order.

Intercompany return orders, similar to regular return orders, are initiated on the **Create return order** page.

In Microsoft Dynamics 365 Supply Chain Management, intercompany sales and purchase agreements are automatically updated to reflect a returned item. The sales or purchase agreement commitment for that item is automatically adjusted to reflect the change in the quantity or the amount when an item is returned.

Intercompany return order example

Company A creates a purchase order that is linked to a purchase agreement for an intercompany item. An intercompany sales order is automatically created in Company B that is linked to the corresponding sales agreement. The purchase agreement and the sales agreement are related as intercompany agreements.

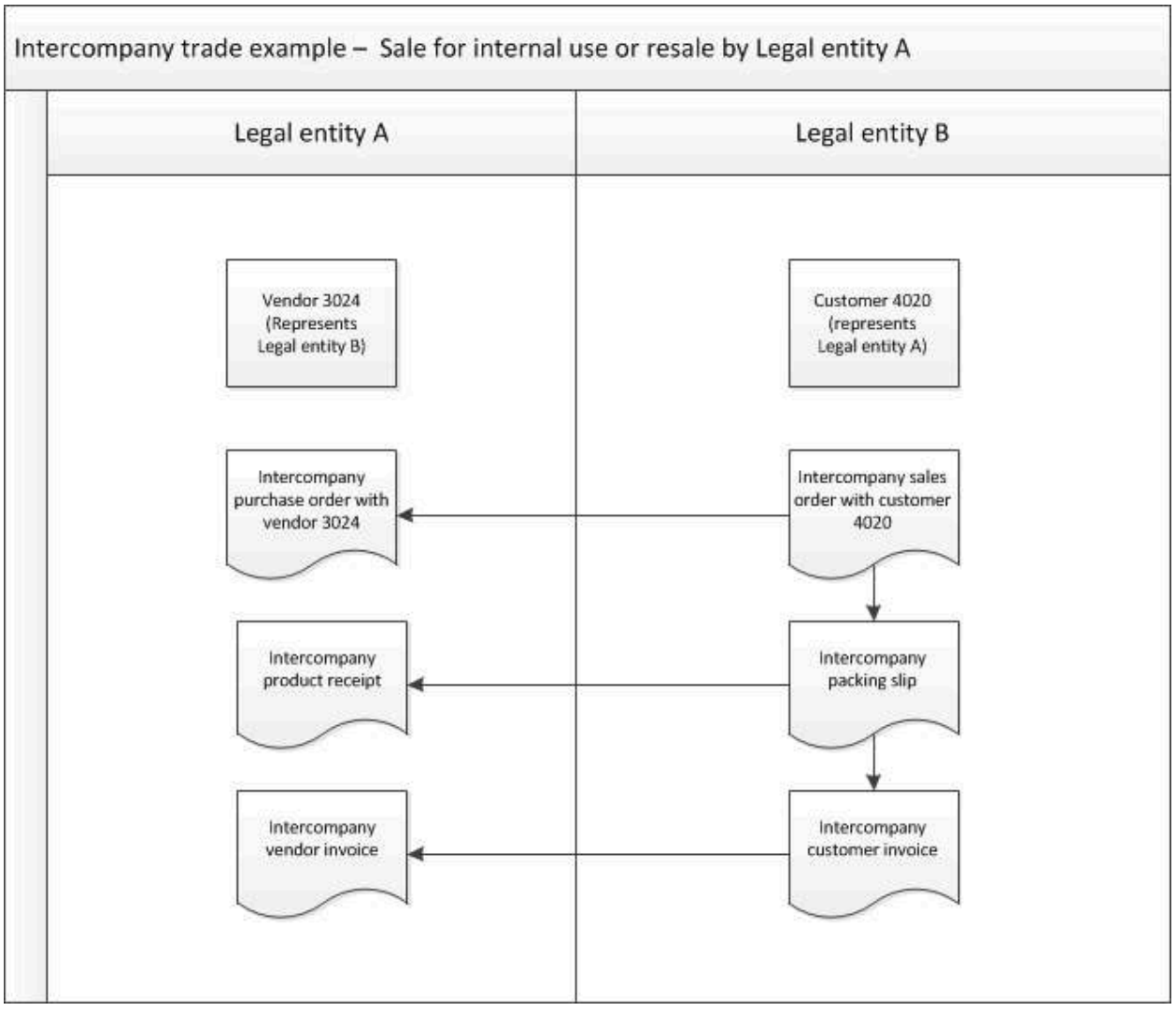
Company A returns the intercompany item to Company B. Company B creates a return order for the item, and a purchase return order is automatically created in Company A. The commitments on both the purchase agreement and the sales agreement that are linked to the original orders are automatically adjusted to reflect the change in quantity or amount.

Last updated on 05/29/2026

Create an intercompany sales order for internal use

Article • 06/03/2022

Typically, an intercompany sales order is created automatically, based on an intercompany purchase order. You can also manually create an intercompany sales order, which then generates an intercompany purchase order in the intercompany customer's legal entity.



Create an intercompany sales order manually

Do these steps in legal entity B, as shown in the illustration.

1. Go to **Accounts receivable** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **Sales order** to create a sales order.

3. On the **Create sales order** page, select a customer account. On the **General** FastTab, make sure that the **Intercompany** check box is selected. This indicates that the selected customer is an intercompany customer.

4. Enter or modify the information, select **OK**, and then complete the order lines as usual.

   The **Delivery address** field value is copied from the intercompany sales order header to the intercompany purchase order header. The **Item number** field value, including product dimensions, and the **Delivery date** and **Currency code** field values are copied from the intercompany sales order lines to the intercompany purchase order lines.

5. In the order header, select **Intercompany** to view the related intercompany purchase order.

6. On the order lines, select **Intercompany** to view information about on-hand inventory for intercompany trade.

> **Tip**
>
> You can view intercompany sales orders on the **Intercompany orders** page.

> **Note**
>
> When you work with intercompany, we recommend that you clear the **Delete order after invoicing** check box on the **Accounts receivable parameters** page. Otherwise, the related purchase order is deleted. We also recommend that you clear the **Delete purchase order after invoicing** check box on the **Accounts payable parameters** page.

Feedback

Was this page helpful? Yes No

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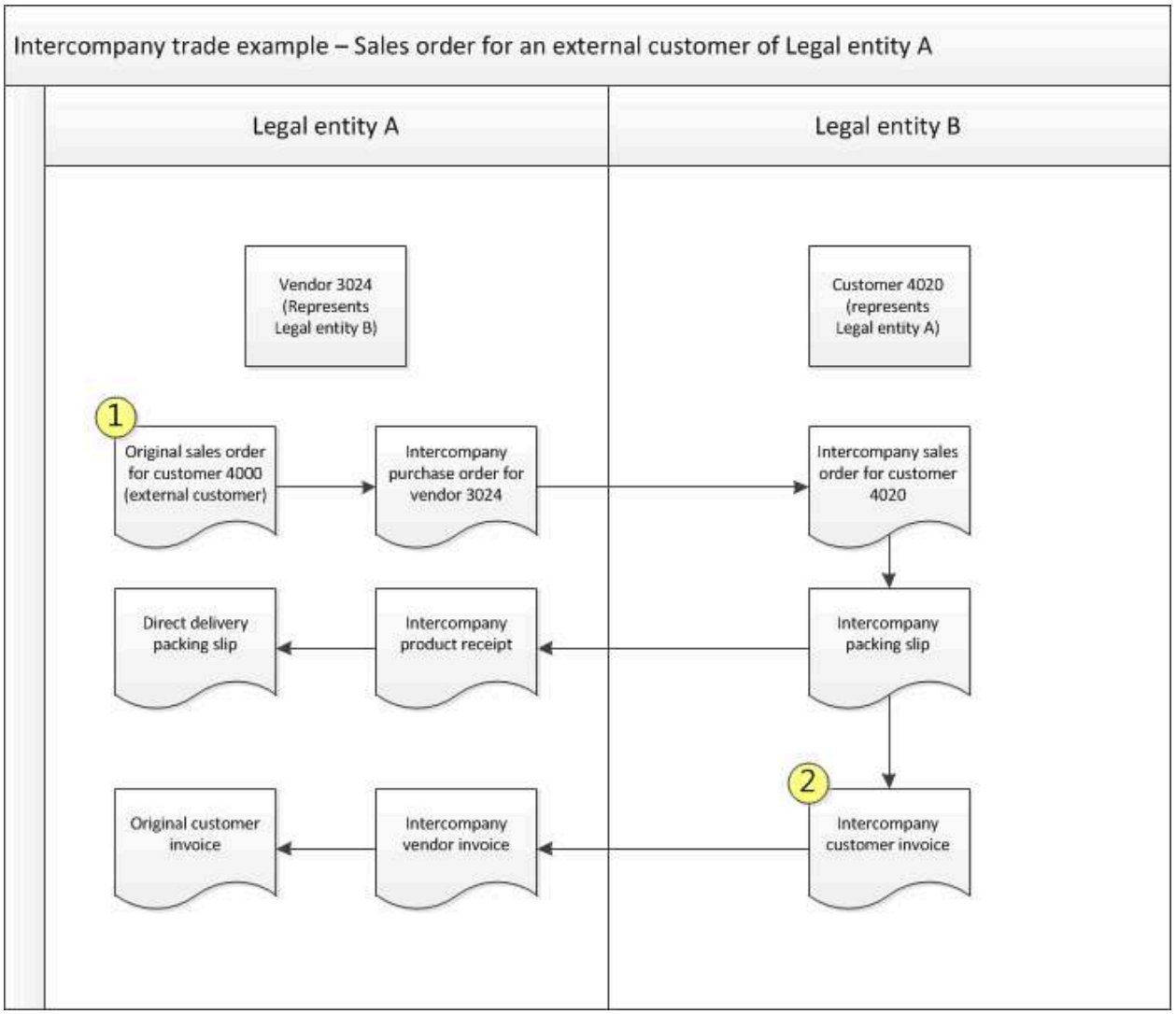
Create and invoice an intercompany sales order for an external customer

Article • 06/03/2022

You can use intercompany trade to record the sale of a product from one legal entity to another legal entity that is in the same organization.

When you create the original sales order, you can automatically create an intercompany purchase order for the intercompany vendor. This automatically creates an intercompany sales order in the intercompany vendor's legal entity.

The following illustration shows the flow of the transactions when you create a sales order for an external customer, but the product must be ordered from a different legal entity in your organization before you can deliver the product to the customer. In this case, an intercompany purchase order is created for the vendor account that represents the other legal entity. In turn, an intercompany sales order is created in the other legal entity for the customer account that represents your legal entity. When an intercompany purchase order is invoiced, the invoice for the original sales order is automatically posted, if it is a direct delivery.



If you are using direct delivery, and you select **Packing slip** in the **Quantity** field on the **Posting invoice** page, the intercompany vendor invoice is based on the intercompany product receipt that was previously posted.

Prerequisites

Before you begin, make sure that the following prerequisites are met to automatically post and print the intercompany invoices.

1. Go to **Sales and marketing** > **Customers** > **All customers**.
2. On the Action Pane, on the **General** tab, select **Intercompany**.
3. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
4. On the Action Pane, on the **General** tab, select **Intercompany**.
5. On the **Intercompany** page for either the vendor or the customer, in the **Purchase order policies** area, in the **Intercompany purchase order (direct delivery)** field group, select the **Post invoice automatically** check box.
6. In the **Purchase order policies** area, in the **Original sales order (direct delivery)** field group, select the **Post invoice automatically** check box. Select this check box

if you want the invoice for the original sales order to be automatically printed when you post the intercompany vendor invoice.

> ⓘ Note
>
> The print settings for the invoice are determined by the setup for the module, document, or transaction on the **Print management setup** page.

Create an original sales order for an external customer

Do these steps in legal entity A. This procedure corresponds to the box labeled 1 in the illustration.

1. Go to **Accounts receivable** > **Sales orders** > **All sales orders**.
2. From the **All sales orders** list page, create the original sales order. The field values are copied from the customer account to the sales order.
3. On the **Sales order** page, select **Header view** on the Action Pane.
4. On the **Intercompany settings** FastTab, select the **Autocreate intercompany orders** check box.
5. If you want the intercompany vendor to deliver the item directly to the external customer, select the **Direct delivery** check box.
6. When you have finished entering the sales order, close the **Sales order** page.

The intercompany purchase order is automatically created for all the items that have an assigned intercompany vendor, and then the intercompany sales order is created. An informational message tells you that an intercompany purchase order and intercompany sales order have been created. The message also contains links to those orders. If an item was not found, a red warning is displayed that means that the order creation process was not completed.

> ⓘ Note
>
> If you are creating several orders in different legal entities, and the items were not found in one of the legal entities, the order creation process stops, even for the legal entities that could fulfill their orders.

Invoice an intercompany sales order

When an intercompany sales order is invoiced, the corresponding intercompany purchase order is automatically invoiced. Then, if the original sales order is a direct delivery order, the original sales order is automatically invoiced.

Do these steps in legal entity B. This procedure corresponds to the box labeled 2 in the illustration.

1. Go to **Accounts receivable** > **Sales orders** > **All sales orders**.
2. On the **All sales orders** list page, select the intercompany sales order.
3. On the Action Pane, select the **Invoice** tab, and then select **Invoice**.
4. Select the **Posting** check box.
5. Select the sales order, and then select **OK**.

The customer invoice for the intercompany sales order is automatically posted in legal entity B. The intercompany vendor invoice then is automatically created and posted in legal entity A. If the original sales order is set up as a direct delivery, the customer invoice is created for the original sales order in legal entity A.

> ⓘ **Note**
>
> Previously, for intercompany sales scenarios, if the vendor invoice workflow was configured in the intercompany purchasing company, the intercompany sales order could not be successfully invoiced. Therefore, the vendor invoice workflow had to be turned off for the intercompany purchasing company.
>
> This limitation has been fixed by a recent feature in release 10.0.25. Intercompany sales orders can now be invoiced when the vendor invoice workflow is configured in the intercompany purchasing company.
>
> To enable this feature, follow these steps.
>
> 1. Select the intercompany sales legal entity.
> 2. Go to **Accounts receivable** > **Customers** > **All customers**.
> 3. Select the customer for the intercompany purchasing company.
> 4. Go to **General** > **Set up** > **Intercompany**.
> 5. On the **Purchase order policies** tab, select the **Bypass vendor invoice workflow for intercompany vendor invoices** parameter.

Feedback

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Intercompany reservations

Article • 06/03/2022

When you create an intercompany purchase order line directly from an original sales order line, the two lines are marked against each other. This marking controls the reservation, cost, and intercompany chain. You can view the marking on the sales order line by selecting **Inventory** > **Marking**, and then closing the dialog box that is displayed.

To remove the marking, select **Inventory** > **Remove link** on the original sales order line.

Feedback

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Change or delete an original intercompany sales order

Article • 06/03/2022

When you change a sales order, your changes are automatically synchronized with both the relevant intercompany purchase order and the intercompany sales order.

> **Note**
>
> Purchase orders for external vendors are not affected by any changes that you make; neither are the original sales lines that are connected to purchase-order lines for external vendors.

The following table summarizes the changes you can make and the expected results.

Expand table

Operation	Result
Delete an original sales order.	The related intercompany purchase order and the intercompany sales order are also deleted. If the status of the order is **Picking list** or later in the process, you cannot delete the sales order.
Delete an original sales-order line.	The related intercompany purchase order line and the intercompany sales order line are deleted. If the status of the order is **Picking list** or later in the process, you cannot delete the sales order line.
Add a new sales line to an original sales order.	The new sales line is created on both the intercompany purchase order and the intercompany sales order.
Change the quantity on an original sales-order line.	The quantity is synchronized with both the intercompany purchase-order line and the intercompany sales-order line. The net amount is recalculated on all the orders.
Disable direct delivery on an original sales order.	You cannot change the **Direct delivery** field on the original sales order if the intercompany sales order line has reached a minimum status of **Picking list**, **Output order**, or **Picking list journal**. The delivery address on the intercompany purchase order is changed to the standard delivery address (standard purchase order delivery address). Also, the intercompany purchase order lines are changed to the standard delivery address for the lines. Both are synchronized with the intercompany sales order and the lines. The delivery type on all lines on the original sales order is changed to **None**, and the

Operation	Result
	field is synchronized with the intercompany purchase-order lines. Purchase orders for external vendors are not affected; neither are the original sales lines that are connected to purchase-order lines for external vendors. The delivery date on the intercompany purchase order and lines – and the Requested receipt/shipping dates on the intercompany sales order and lines – are updated.
Enable direct delivery on an original sales order.	If the intercompany sales order line has reached a minimum status of **Picking list**, **Output order**, or **Picking list journal**, you cannot change the **Direct delivery** field on the original sales order. The delivery address on the intercompany purchase order is changed to the delivery address from the original sales order and synchronized with the intercompany sales order. The delivery type on all lines on the original sales order is changed to **Direct delivery**, and the field is synchronized with the intercompany purchase-order lines. The delivery address on each original sales-order line is synchronized with both the intercompany purchase-order lines and the intercompany sales-order lines. The delivery date on the intercompany purchase order and lines – and the Requested receipt/shipping dates on the intercompany sales order and lines – are updated.
Add a note to both an original sales-order header and the lines.	The note is copied to the intercompany purchase order and the intercompany sales order.
Change or delete a note.	If you change or delete a note, the corresponding note on the intercompany purchase order and the intercompany sales order is changed or deleted accordingly.

Feedback

Was this page helpful? Yes No

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Change intercompany orders

Article • 06/03/2022

If you change an intercompany sales order or purchase order, the corresponding sales order or purchase order in the corresponding company also reflects that change.

Adding new lines

You can add new lines to an intercompany sales order and purchase order if the original sales order is part of the intercompany order chain. You can also add new lines if the original sales order is not a direct delivery. If the original sales order is a direct delivery, you can add new order lines to the intercompany sales order and purchase order only if the **Allow indirect creation** field is selected on the **Intercompany settings** FastTab on the original sales order.

Changing prices and discounts

You can change the prices and discounts if the **Allow price edit** and **Allow discount edit** check boxes are selected on the **Intercompany** page.

If you change the unit price of one of the existing lines on the intercompany sales order line, the unit price on the intercompany purchase order is also changed.

If you change any of the discount fields on the intercompany sales order line, the relevant discount fields on the intercompany purchase order are updated.

Changing and adding new charges

You can change charges if the **Allow price edit** and **Allow discount edit** check boxes are selected on the **Intercompany** page.

If you add a new charge to the intercompany sales order header and a new charge to the intercompany sales line, both charges are copied to the intercompany purchase order. Therefore, the intercompany sales order and the intercompany purchase order have the same total amount. The charges are never copied to the original sales order.

Copying a fee

To copy a fee to the original sales order, create it as a new sales line that has an item of type **Service**.

Changing quantities, and deleting intercompany purchases and sales order lines

You can change the quantity or delete an intercompany purchase order line or sales order line only if the line has been created directly from the **Sales order** or **Purchase order** form. This change makes either the intercompany purchase order or sales order or order lines the source.

Origins of orders and order lines

Intercompany orders and order lines can have one of two origins:

- **Derived** – The order was created automatically from an intercompany order chain.
- **Source** – The order was created manually by a user.

The origin is indicated on the order header of intercompany purchase orders and sales orders in the **Origin** field. This field is located on the **Intercompany settings** FastTab on the sales order and on the **Setup** FastTab on the purchase order.

The **Derived** and **Source** origins apply only to intercompany orders. The **Origin** field is blank for all other sales orders, purchase orders, and order lines. This field is also blank on the original sales order.

Example: Derived origin

You create an original sales order for an external customer. This sales order contains one order line. This original sales order creates an intercompany purchase order that contains one order line, which creates an intercompany sales order that contains one order line. The intercompany purchase order header and the order line are created automatically from the original sales order.

The value of the **Origin** field on the **Setup** FastTab of the intercompany purchase order and the **Intercompany settings** FastTab of the intercompany sales order headers is **Derived**. The value of the **Origin** field on the **Line details** tab of the purchase order and sales order lines is also **Derived**.

If an order line has an origin of **Derived**, you cannot delete the order line directly. You can delete the order line only from the source that created the order line. Also, you can

change the ordered quantity only from the source that created the order line.

If an original sales order and original sales order line is part of the intercompany order chain, you can change ordered quantities from the original sales order, and you can delete order lines from the original sales order.

Example: Source origin

You create a purchase order for an intercompany vendor. The intercompany purchase order and order line both have an origin of **Source**. Therefore, this intercompany purchase order is the controller, and the intercompany sales order that is created automatically in the vendor company has an origin of **Derived**.

Additionally, the ordered quantities of an order line that is created on the intercompany purchase order cannot be changed on the intercompany sales order. The order line can be deleted only from the intercompany purchase order. If a new line is added to the intercompany sales order, the intercompany sales order line then has an origin of **Source**.

When an original sales order is part of the intercompany order chain, you can always control the intercompany orders and intercompany order lines from the original sales order.

Feedback

Was this page helpful? Yes No

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Intercompany batch and serial numbers

Article • 06/03/2022

Companies that use serial numbers or batch numbers to trace their items must also keep track of the serial numbers and batch numbers of picked items. The intercompany functionality automates the push/pull of serial numbers and batch numbers from one company to another. When you register the batch numbers and serial numbers for the items on an intercompany sales order, you can set up the program to push these numbers automatically to the intercompany purchase order and original sales order. You set up the relevant parameters on the **Intercompany** page for the sales order:

- If you select the **Batch number** field on the **Sales order policies** page, the batch number is synchronized to the inventory transactions on the intercompany purchase order lines when you post the packing slip of the intercompany sales order.
- If you select the **Serial number** field, the serial numbers are synchronized to the inventory transactions on the intercompany purchase order lines when you post the packing slip of the intercompany sales order.

Feedback

Was this page helpful? Yes No

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Intercompany packing slips

Article • 06/03/2022

Generate intercompany packing slips

If you work with direct delivery, the packing slip is always generated automatically on both the intercompany purchase order and the original sales order when you generate the packing slip on the intercompany sales order. The intercompany purchase-order invoice is based on the intercompany purchase-order packing slip that was previously generated.

If you make any updates to the packing slip on the intercompany sales order, these updates are reflected on both the intercompany purchase order and the original sales order.

If you do not work with direct delivery, you must manually generate the packing slip on the intercompany sales order, the intercompany purchase order, and the original sales order.

Print intercompany packing slips

If you work with direct delivery, a packing slip can be printed automatically for the intercompany purchase order and the original sales order when you post the packing slip on the intercompany sales order.

To print packing slips automatically, on the **Intercompany** page for purchase orders, select both of the **Print packing slip automatically** fields.

If you update the packing slip on the intercompany sales order, the changes are reflected automatically on the intercompany purchase order and the original sales order.

Feedback

Was this page helpful? Yes No

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Register payments automatically for intercompany customer invoices

Article • 06/03/2022

Microsoft Dynamics 365 Supply Chain Management creates a customer transaction when an intercompany customer invoice is posted. This customer transaction remains open until it is settled, which means that it has been paid. When the corresponding intercompany purchase order is invoice updated, a vendor transaction matching the customer transaction is created. This vendor transaction also remains open until it is settled. To reduce the risk of differences, an accounts receivable payment journal can be automatically created and posted when the accounts payable payment journal is posted.

1. Go to **Sales and marketing** > **Customers** > **All customers**.
2. On the Action Pane, on the **General** tab, select **Intercompany**.
3. To set up the intercompany accounts receivable payments on the **Intercompany** page for sales orders, select the **Sales order policies** link.
4. In the **Payment journal** field, select the accounts receivable payment journal that you want to register intercompany vendor payments to. You set this up on the **Accounts receivable parameters** page.
5. If you want to post to this journal automatically, select the **Post journal automatically** check box.

Feedback

Was this page helpful? Yes No

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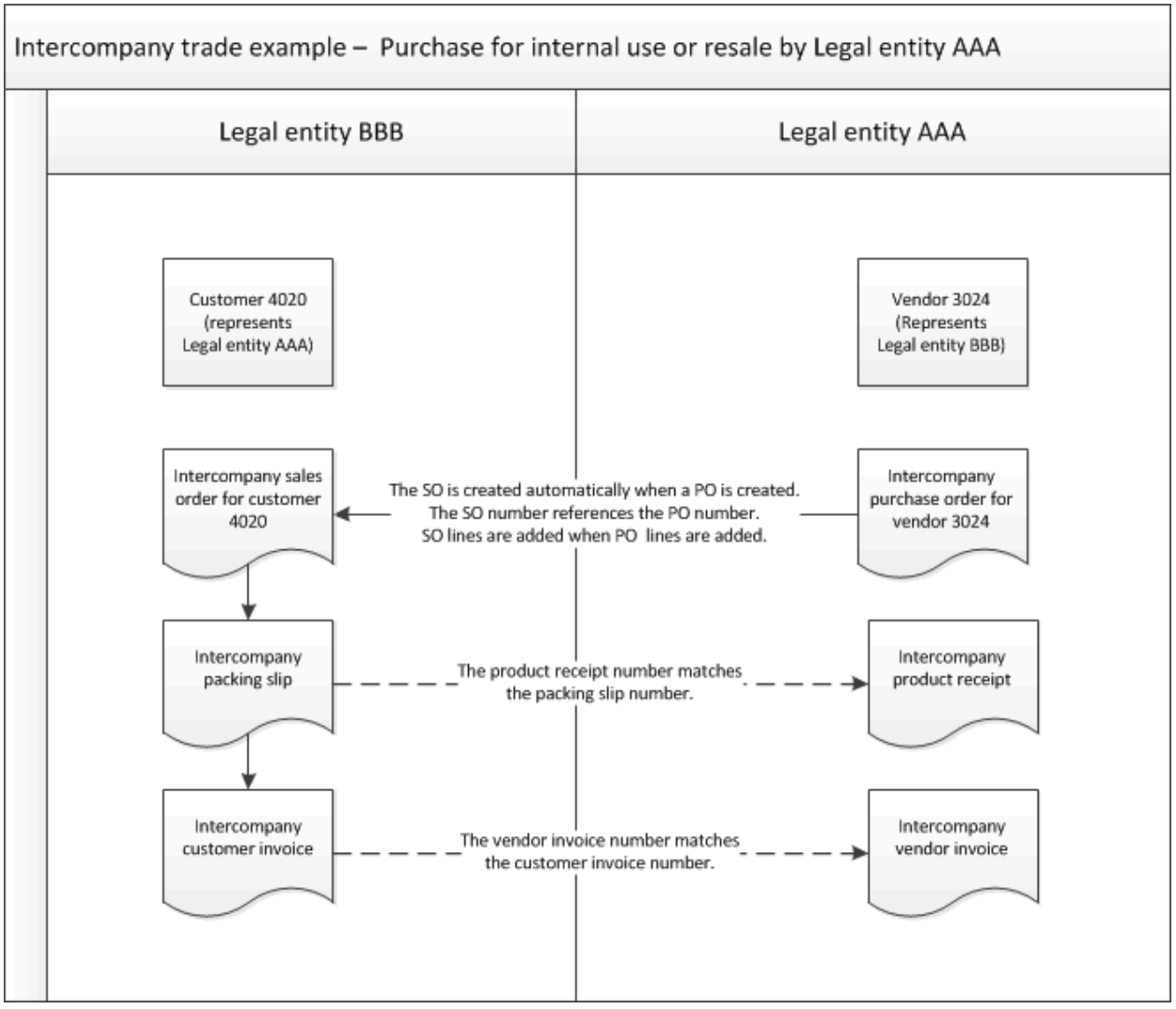
Create and invoice an intercompany purchase order for internal use

Article • 01/07/2025

You can create an intercompany purchase order for an intercompany vendor. This automatically creates an intercompany sales order at the intercompany vendor.



Create an intercompany purchase order and a corresponding intercompany sales order

Do these steps in legal entity AAA, as shown in the illustration.

1. Select **Accounts payable** > **Purchase orders** > **All purchase orders**.
2. On the **All purchase orders** list page, create a purchase order for an intercompany vendor. The field values are copied from the vendor account to the purchase order.

Because you're working with an intercompany vendor, an intercompany sales order is created in the legal entity that corresponds to the vendor. The number of the intercompany sales order can be the same as the number of the intercompany purchase order, and it can include the ID of the legal entity. The number structure that is used depends on the selection in the **Sales order numbering** field in the **Intercompany** page. For example, if you create purchase order 00029_064 in legal entity AAA, the sales order number in legal entity BBB is AAA00029_64.

An informational message tells you that an intercompany purchase order and intercompany sales order have been created. The message includes the intercompany sales order number, for your information.

3. Add line items to the purchase order. The corresponding line items are added automatically to the intercompany sales order. If an item doesn't exist in the other legal entity, a message is displayed and you can't add the item to the purchase order. To fix this problem, switch to the other legal entity and release the product to that legal entity. The item will be available to be added to sales orders in that legal entity. Then, switch back to the legal entity of the purchase order and continue adding line items.

4. When you have finished entering information for the purchase order, confirm it.

Process the intercompany packing slip and customer invoice

Do these steps in legal entity BBB, as shown in the illustration.

1. Go to **Accounts receivable** > **Sales orders** > **All sales orders**.

2. On the **All sales orders** list page, select the intercompany sales order.

3. On the Action Pane, select the **Pick and pack** tab, and then select **Packing slip**.

4. Select the **Posting** check box.

5. Select **OK**. The packing slip is posted in legal entity BBB.

6. On the **All sales orders** list page, select the intercompany sales order.

7. On the Action Pane, select the **Invoice** tab, and then select **Invoice**.

8. Select the **Posting** check box.

9. Select **OK**.

The customer invoice for the intercompany sales order is posted in legal entity BBB.

Process the intercompany product receipt and vendor invoice

Do these steps in legal entity AAA, as shown in the illustration.

1. Go to **Accounts payable** > **Purchase orders** > **All purchase orders**.
2. On the **All purchase orders** list page, select the intercompany purchase order.
3. On the Action Pane, select **Receive**, and then select **Product receipt**. The product receipt is created. The product receipt number is the same as the intercompany packing slip number.
4. Select the **Posting** check box.
5. Select **OK**.
6. On the **All purchase orders** list page, select the intercompany purchase order.
7. On the Action Pane, select **Invoice**, and then select **Invoice**. The vendor invoice is created. The vendor invoice number is the same as the intercompany customer invoice number.
8. Finish entering the vendor invoice, and then post it.

Feedback

Was this page helpful? Yes No

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Check intercompany order price discrepancies

Article • 02/09/2024

You can use this procedure to check for price discrepancies on intercompany orders.

1. Go to **Procurement and sourcing** > **Purchase orders** > **Intercompany** > **Check intercompany order price discrepancies**.

2. Set up a batch job, if you need to, and then select **OK** to check the prices and discounts for intercompany sales orders and purchase orders. Otherwise, select **Cancel** to close the page without performing the check.

   > **Tip**
   >
   > If there are any synchronization issues or issues with the prices, these will be listed in an informational message that will be displayed. You can print the informational message for reference.

3. On the informational message, select the relevant line to open the order in the current legal entity. Open the order in the related legal entity by selecting **Intercompany**, and then **Intercompany purchase order** or **Intercompany sales order**.

   > **Note**
   >
   > To allow an update of the intercompany order:
   >
   > a. Go to **Accounts receivable** > **Customers** > **All customers**.
   > b. On the Action Pane, select the **General** tab, and then select **Intercompany**.
   > c. On the **Intercompany** page, select **Purchase order policies** or **Sales order policies**.
   > d. Select **Allow price edit** and **Allow discount edit** in both areas.

4. Open the relevant intercompany order where you want to keep the price.

5. For each order line, change the **Unit price** field for the line, and then change it back to the original value. Change the **Discount** field for the line back and forth, and change the relevant **Charges on purchases** or **Sales charges** fields back and forth. Changing the values back and forth will trigger a synchronization of the

prices, discounts, and charges from this intercompany order line to the referenced intercompany order line, so they will be automatically aligned.

> **Note**
>
> This synchronization is valid only if the intercompany sales order has not been invoiced. If the intercompany sales order has been invoice-posted and a customer invoice journal has been created, changes must be performed directly on the intercompany purchase order lines by setting the prices, discounts, and charges equal to those on the intercompany customer invoice journal. If this is not done, the intercompany purchase order cannot be invoice-posted, because there will be a mismatch in the order totals.

6. Repeat the procedure until all intercompany purchase and sales orders are synchronized.

Feedback

Was this page helpful? Yes No

Provide product feedback

Intercompany currency conversions

Article • 06/03/2022

When the currency code on the original sales order and the intercompany purchase order differ, the following fields are currency-converted if synchronization is enabled:

- **Unit price**
- **Sales charges** or **Charges on purchases**
- **Discount**
- **Multiline discount**

These fields are then synchronized with the intercompany sales order line.

However, because the currency on the intercompany purchase order and the intercompany sales order is always synchronized, the following fields are synchronized without using currency conversion:

- **Unit price**
- **Sales charges** or **Charges on purchases**
- **Discount**
- **Discount percent**
- **Multiline discount**
- **Multiline discount percentage**

Feedback

Was this page helpful? Yes No

Provide product feedback

Intercompany master scheduling

Article • 06/03/2022

Intercompany master scheduling is the means by which you calculate requirements and generate planned intercompany orders across several internal companies. Intercompany master scheduling is carried out over the number of iterations that you specify. To enable Microsoft Dynamics 365 Supply Chain Management to carry out intercompany master scheduling, you must set up master scheduling in each of your intercompany companies. This entails several iterations in which Microsoft Dynamics 365 Supply Chain Management automatically creates an intercompany purchase order; this, in turn, leads to the automatic creation of an intercompany sales order, which again leads to new demands.

You set up the intercompany master plan and the intercompany scheduling order in the **Master planning** parameters in each of the intercompany companies.

In order to propagate the demand throughout the intercompany chain, you must set parameters to ensure that planned purchase orders are automatically firmed; that is, orders cannot be changed in terms of time or quantity. Set up the **Firming time fence** on the Coverage group, or the **Firming time fence** in the Master plan. If no **Firming time fence** is set up, no intercompany purchase orders are created automatically. Only the first execution of the master scheduling results in planned orders. However, because no intercompany purchase order is created, no intercompany sales order is created, and, therefore, no more intercompany purchase orders are created, and so on.

When you start the intercompany master scheduling, Microsoft Dynamics 365 Supply Chain Management performs one master scheduling in each intercompany company; then, it performs a second master scheduling in each intercompany company, and so on, until the specified number of iterations is reached. A maximum of 30 iterations is possible; however, the more iterations that you choose, the longer the scheduling takes.

Because the master scheduling in the companies is controlled by the intercompany scheduling sequence, that is, the order in which the program prioritizes the master schedules between each intercompany company, you can start the intercompany master scheduling from any of the intercompany companies. Because the intercompany scheduling sequence determines the order in which the master scheduling is performed in the companies, it is not important where the intercompany master scheduling is started. In each iteration, the current company executes last.

> ⓘ **Note**

The best practice is to allow intercompany master scheduling to be initiated from one Microsoft Dynamics 365 Supply Chain Management company.

On the **Intercompany master scheduling** page, you can start a sequence of master scheduling, which executes a master scheduling the first time with the principle for updating the requirement calculation that you have selected for the first iteration for all the intercompany companies. Subsequent iterations use the secondary principle that you select for the next iteration, and this principle is applied until the number of iterations that you have specified is reached.

Feedback

Was this page helpful? Yes No

Provide product feedback

Sales returns

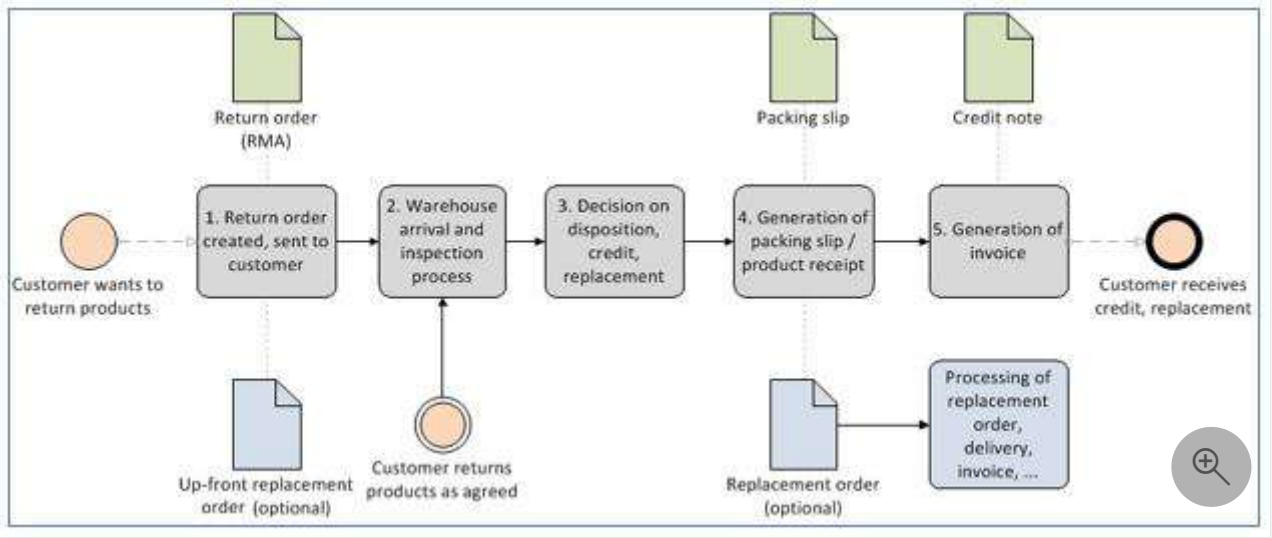
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about the process for return orders. It includes information about customer returns and their effect on costing and on-hand inventory quantities.

Customers can return items for various reasons. For example, an item might be defective, or it might not meet the customer's expectations. The return process starts when a customer issues a request to return an item. After the customer's request is received, a return order is created.

Return order process

The following illustration gives an overview of the return order process.



There are two types of return order process: physical return and credit only.

- **Physical return** – The return order authorizes the physical return of products.
- **Credit only** – The return order authorizes a customer credit but doesn't require that the customer physically return the products.

Physical return order process

1. **Create a return order.** Formally document the authorization for the customer to return any defective or unwanted products. The return order doesn't require that the company accept the returned products or provide a credit for the customer. If the return is accepted, you can authorize a replacement item to be sent before the defective item is returned.
2. **Arrive at warehouse for inspection.** Complete an initial inspection and validation against the return order document. The return order also supports quarantine of the returned items for additional inspection and quality control.
3. **Determine disposition.** Finalize the inspection process, and decide what should be done with the returned products. As part of this step, decide whether you credit the customer, reject the product return, or accept the product return, scrap the product, and then send a replacement product to the customer.
4. **Generate a packing slip.** Generate a packing slip, and commit the disposition decision that you made in step 3. Finalize the logistics processes.
5. **Generate an invoice.** Close the return order.

Credit only process

1. **Create a return order.** Formally document the authorization for the customer to receive a credit without returning the defective or unwanted products. The **Credit only** disposition code authorizes the decision to credit the customer without physical return.
2. **Generate an invoice.** Generate the credit note, and then close the return order.

Return material authorization

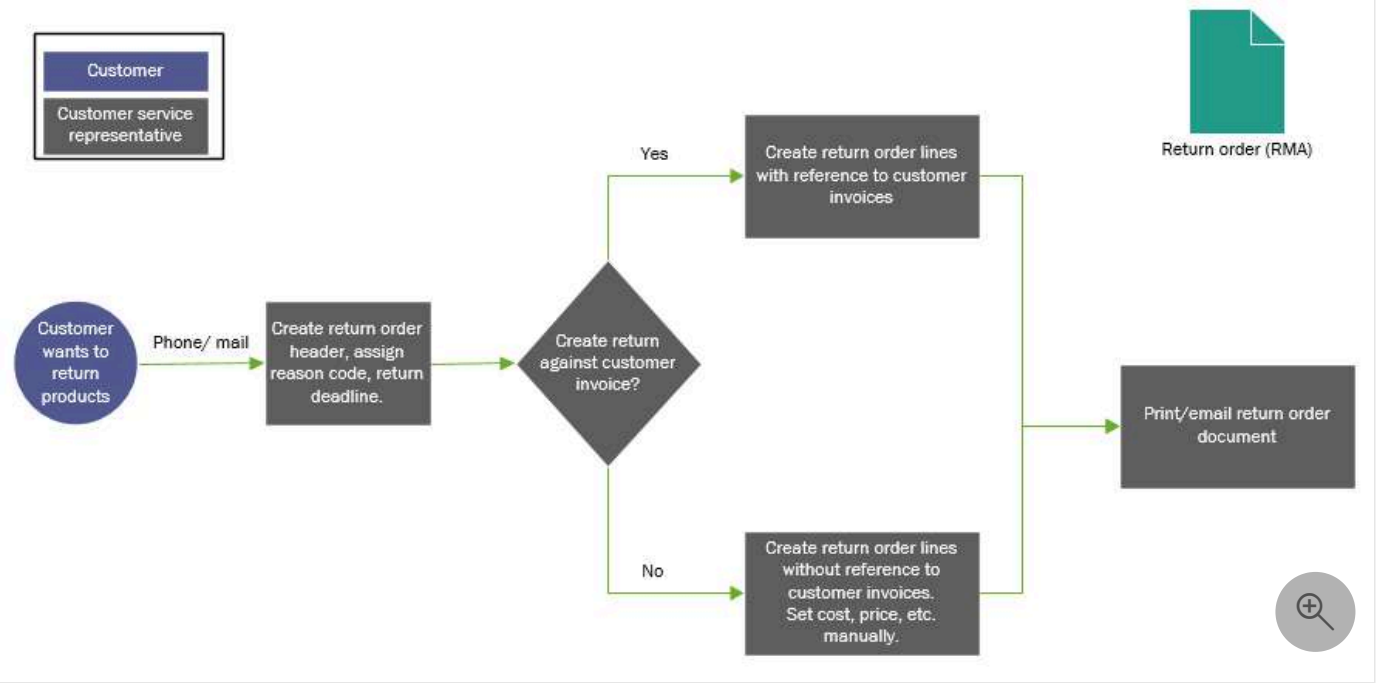
Return Material Authorization (RMA) processing builds on sales order functionality. An RMA is registered as a return order, which is created as a sales order, and may have another sales order associated with it, called a replacement order. Both sales orders link to the originating RMA number.

- **Return order** – To register an RMA, you create a return order, which is a sales order that has the assigned type, **Returned order.** Any changes that you make to the RMA information is automatically updated in the sales order. Until the return order has the status **Open**, it doesn't appear in the list of sales orders. You use RMA to handle the arrival and receipt of the returned items, as well as to authorize a credit only disposition action (see section **Disposition codes and disposition actions**). All other follow-up processes must be handled in the sales order.

- **Replacement order** – When a replacement order must be shipped to the customer, the RMA can include a second associated sales order. You can manually create the replacement order for the RMA to support immediate shipment. Alternatively, the replacement order can be created automatically after the arrival, inspection, and receipt are completed for the RMA line item that has a disposition code that indicates replacement. The replacement order has the same functionality that is associated with a sales order. For example, you can use it to configure a custom product as the replacement item, create a production order to repair a returned item, create a direct delivery purchase order to send the replacement from a vendor, or support other purposes.

Create a return order

The return order process starts when a customer contacts your organization to return a defective or unwanted product and/or to be credited. After your organization accepts the return, the return is documented by a return order. This return order becomes the focal point of the internal processing of the returned product. The following illustration shows the procedure for creating a return order.



Create a return order header

When you create a return order, you need to include the information in the following table.

Expand table

Field	Description	Comments
Customer account	A reference to the Customers table	You must provide an existing customer account.
Delivery address	The address that the item is returned to	By default, the organization's address is used. If you select a specific warehouse on the header, the delivery address changes to the delivery address of the warehouse. You can change this address on the **Return order details** page.
Site/warehouse	The site or warehouse that receives the returned product	The delivery address for the return order is determined based on the delivery address of the site or warehouse.
RMA number	The ID that is assigned to the return order	The RMA number is used as an alternate key throughout the return order process. The assigned RMA number is based on the RMA number sequence that you set up on the **Accounts receivable parameters** page.
Deadline	The last date that an item can be returned	The default value is calculated as the current date plus the period of validity. For example, if a return is valid for only 90 days from date when the return order is created, and the return order was created on May 1, the value in the field is **30-July**. The period of validity is set on the **Accounts receivable parameters** page.
Return reason code	The customer's reason for returning the product	The reason code is selected in the list of user-defined reason codes. You can update this field at any time.

Create return order lines

After you complete the return header, create return lines by using one of the following methods:

- Manually enter the item details, quantity, and other information for every return line.
- Create a return line by using the **Find sales order** function. Use this function when you create a return order. The **Find sales order** function establishes a reference from the return line to the invoiced sales order line, and retrieves line details such as item number, quantity, price, discount, and cost values from the sales line. The reference helps guarantee that, when the product is returned to the company, it's valued at the same unit cost that it was sold at. The reference also validates that return orders aren't created for a quantity that exceeds the quantity that was sold on the invoice.

> ⓘ **Note**

Return lines that have a reference to a sales order are handled as corrections to, or reversals of, the sale. For more information, see the "Post to the ledger" section, later in this article.

Charges

Fees and charges can be added to the return order through one or more of the following methods:

- Manually add charges to the return order header, the return order line, or both.
- Charges can be automatically added to the return order header as a function of the return reason code.
- Charges can be automatically added to the return order line, based on the disposition code of the line.

Charges are automatically added after a return reason code or disposition code is assigned to the line. If the reason code is changed later, the existing charge entry won't be removed, but a new charge entry might be added, based on the new reason code. When you add charges to return order lines, charges that are calculated as a percentage of the line or order value become negative when the line or line order is negative, unless the percentage is also a negative number. A charge that has a negative value represents a credit to the customer.

Return reason codes

By applying reason codes to returns, you can help make return patterns easier to analyze. Reason codes provide information about why a customer wants to return items. Some organizations have many reason codes. These organizations might group the reason codes into reason code groups to gain a better overview and for accumulated reporting.

Disposition codes and disposition actions

An important step in the return order process is assigning a disposition code to the return order line as part of arrival registration. The disposition code determines the following information:

- **The financial implications** – Should the customer be credited for the returned items, and should any charges be added to the return order line?
- **The disposition of the returned item** – Should the item be added back to inventory, scrapped, or returned to the customer?
- **The logistics of the returned item** – Should a replacement item be issued to the customer?

In addition to determining how to dispose of the returned goods, disposition codes can cause charges to be applied to the return line. They can also be used to group returns for statistical analysis. Disposition codes are defined as part of the setup of return orders. However, each disposition code must reference one of the built-in disposition actions. The following table lists the built-in disposition codes and their actions. **Important:** If an item shouldn't be returned, but the customer should still be credited, assign the **Credit only** disposition code to the return line.

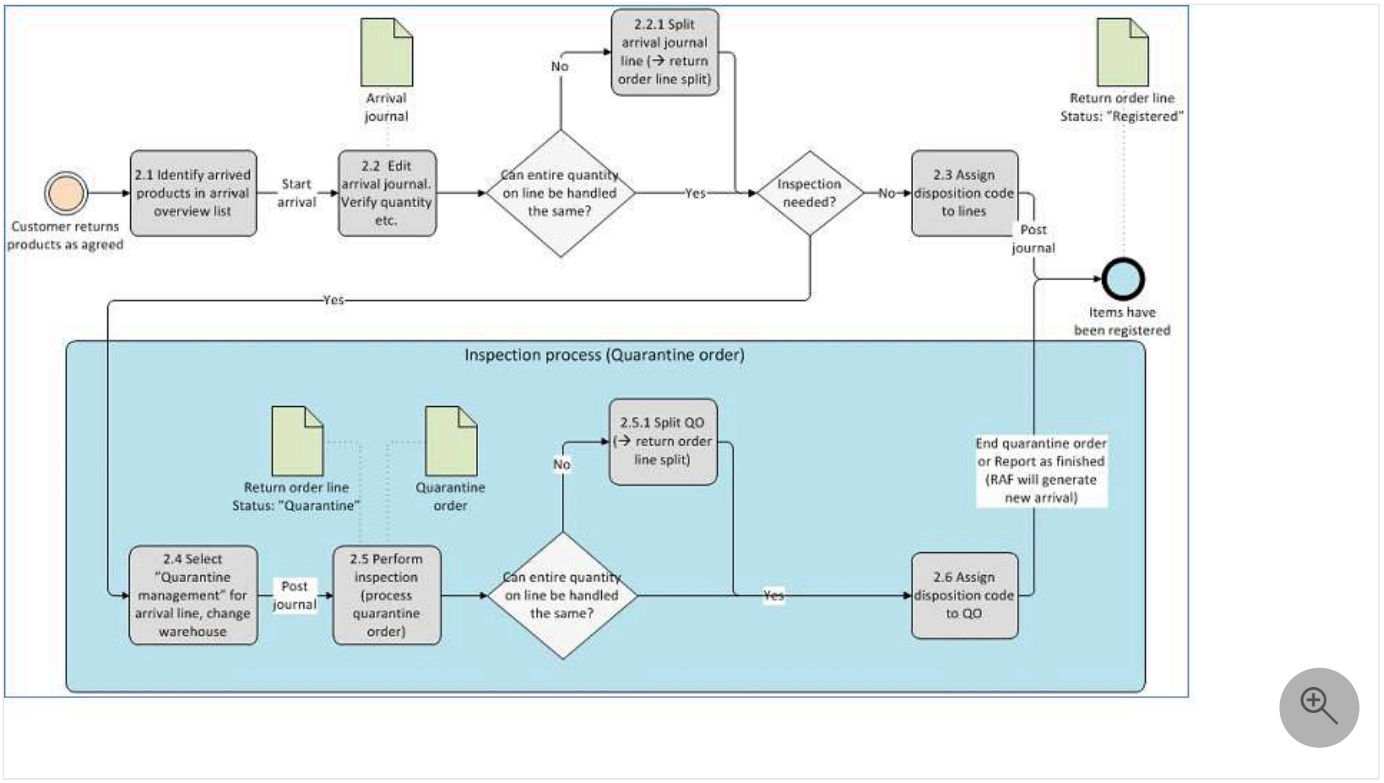
Expand table

Disposition code	Financial implications	Implications for logistics
Credit only	• The customer is credited the sales price minus any fees or charges. • Loss from scrapping the item is posted to the ledger.	The item shouldn't be returned. This disposition action is used for the following cases: • There is sufficient trust among the parties. • The cost of returning the defective item is prohibitive. • The items can't be allowed back into inventory. Because of other conditions, a physical return isn't required.
Credit	• The customer is credited the sales price minus any fees or charges. • The inventory value is increased by the cost of the returned item.	The item is returned and is added back into inventory.
Replace and credit	• The customer is credited the sales price minus any fees or charges. • Inventory value is increased by the cost of the returned item. • A separate sales order for a replacement is created and is handled separately.	The item is returned and is added back into inventory.

Disposition code	Financial implications	Implications for logistics
Replace and scrap	• Customer is credited the sales price, less any fees or charges. • Loss from scrapping the item is posted to the ledger. • A separate sales order for a replacement is created and is handled separately.	The item is returned and scrapped.
Return to customer	None, except any fees or charges.	The item is returned but is sent back to the customer after inspection. This disposition action might be used if the item is deliberately damaged, or if the warranty is voided.
Scrap	• The customer is credited the sales price minus any fees or charges. • Loss from scrapping the item is posted to the ledger.	The item is returned or scrapped.

Arrival at the warehouse for inspection

Before you can physically receive returned items into inventory by posting a packing slip, the items must go through arrival registration and an optional inspection. The following illustration gives an overview of the arrival process. The sections that follow describe each step that is shown in the illustration.



The process has several other variations that aren't covered in this article. Here are some of these variations:

- Don't use the **Arrival overview** list to create an Arrival journal. Instead, manually create the Arrival journal. Return orders have **Sales order** as the reference.
- If you're using Warehouse management, generate pallet transports. The return line has a status of **Arrived** during pallet transport.
- Register the arrival of the returned item directly from the return order line, by using the **Registration** function.

During the arrival process, returns are integrated with the general process for warehouse arrivals. The arrival process also supports the creation of quarantine orders for returned items that must undergo a separate inspection.

Identify products in the Arrival overview list

The **Arrival overview** page lists all the planned incoming arrivals.

> ⓘ **Note**
>
> You must process arrivals from return orders separately from other types of arrival transactions. After you identify an incoming package on the **Arrival overview** page (for

example, by using the accompanying RMA document), on the Action Pane, click **Start arrival** to create and initialize an Arrival journal that matches the arrival.

Edit the Arrival journal

By setting the **Quarantine management** option to **Yes**, you can create a quarantine order for the return line. If a line is sent to quarantine for inspection, you can't specify a disposition code.

If you set the **Quarantine management** option to **Yes** in the item's inventory model group, the **Quarantine management** option on the **Journal lines** page is marked for the Arrival journal line and can't be changed. If the line is sent to quarantine, you must specify the appropriate quarantine warehouse.

If the arrival line isn't sent for inspection, the warehouse arrival clerk must specify the disposition code directly on the Arrival journal line and then post the Arrival journal. If the same disposition code shouldn't be assigned to the whole quantity of the return line, or if the full quantity of the line isn't received, you must split the line. When you split an Arrival journal line, you also split the return line (**SalesLine**) and create a new lot ID. You can split the line by reducing the quantity of the Arrival journal line. When the journal is posted, a new return line is created that has a status of **Expected** for the remaining quantity. You can also split the line by clicking **Functions** > **Split**.

Process the quarantine order

If the returned products are sent for inspection at the quarantine warehouse, any additional processing is completed in a quarantine order. One quarantine order is created for each arrival line that is sent to quarantine. The disposition code indicates the result of the inspection process.

You can split a quarantine order, just as you can split the Arrival journal. If you split the quarantine order, you cause a corresponding split of the return line. After you enter the disposition code, complete the quarantine order by using either the **End** function or the **Report as finished** function. If you select **Report as finished**, a new arrival is created in the designated warehouse. You can then process this arrival by using the **Arrival overview** page.

If the arrival originates from a quarantine order, you can't change the disposition code assigned during inspection. If you complete the quarantine order by using the **End** function, the lot is automatically registered. Sometimes, an item might be sent back from quarantine to the Shipping and receiving department. For example, the quarantine inspector might not know where to store the item in inventory. In this case, you must update the corresponding packing slip to correctly register and act on the disposition code that is specified because of the quarantine.

Acknowledgment of receipt can be sent to the customer when the return line is registered. The **Return acknowledgement** report resembles the return order document. The **Return acknowledgement** report isn't journalized or otherwise registered in the system, and it isn't a required step in the return order process.

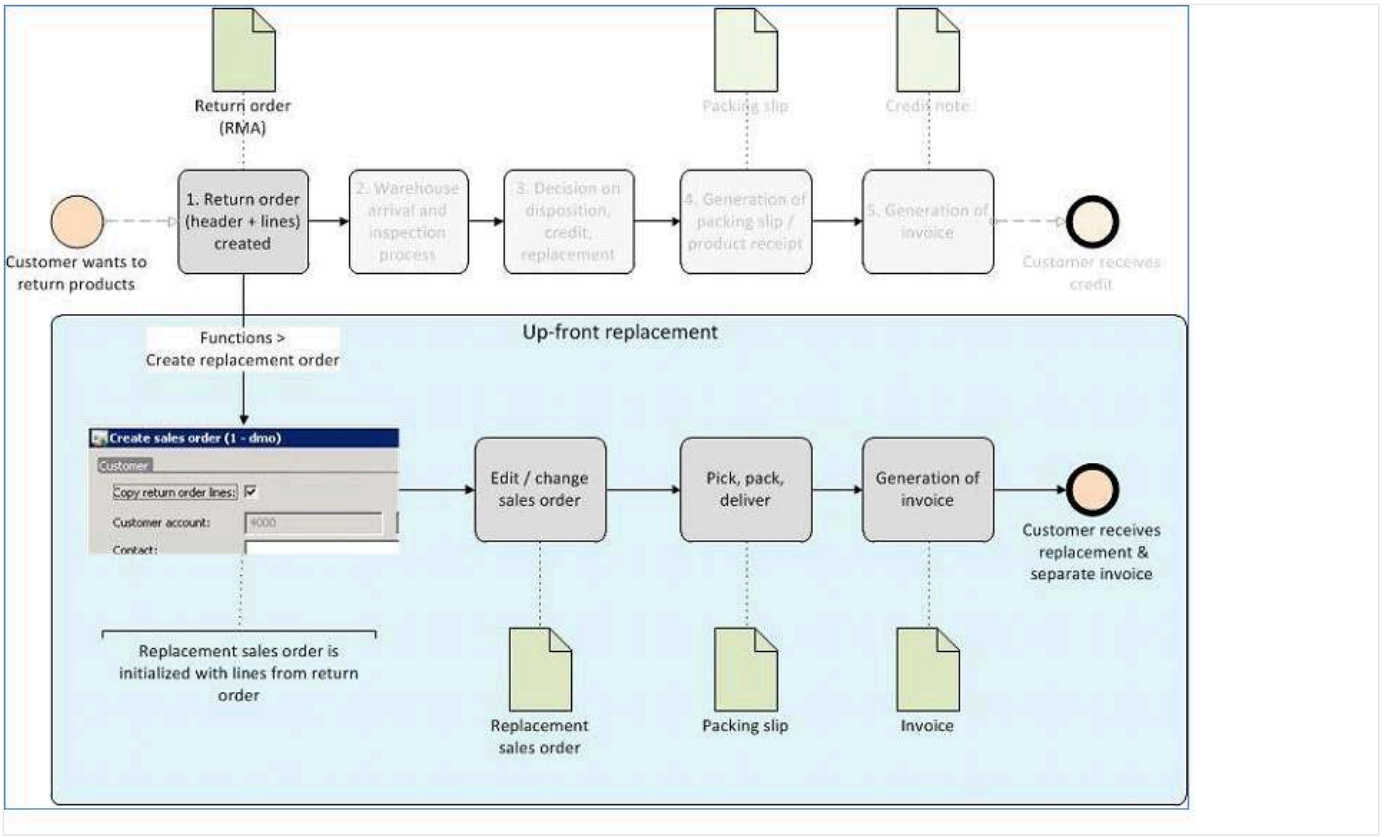
Replace a product

There are two methods for managing product replacement:

- **Up-front replacement** – Replace a product before the returned product is received from the customer.
- **Replacement by disposition code** – Automatically create a new replacement order line.

Up-front replacement

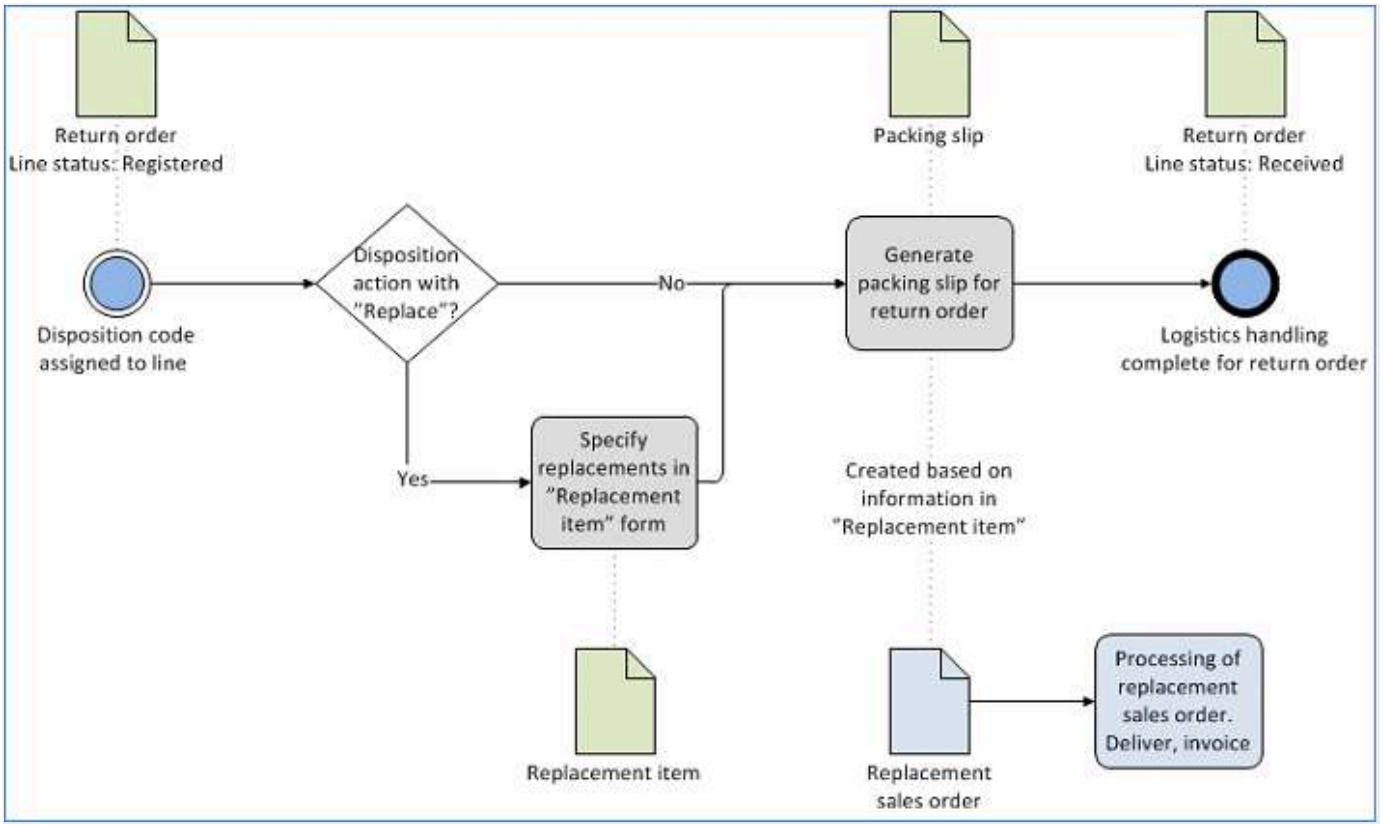
In up-front replacement, the replacement item can be delivered to the customer before the item is returned. This method is useful if, for example, the item is a machine part that can’t be removed unless a spare part is available to take its place, or if you just want your customer to have the replacement product as soon as possible. The up-front replacement order is an independent sales order. The header information is initialized from the customer, and the line information is initialized from the return order. You can edit, process, and delete the replacement order independently of the return order. When you delete a replacement order, you receive a message that the order was created as a replacement order. The following illustration shows the process for up-front replacement.



The return order includes a reference to the replacement order. If you create an up-front replacement order for a return order before the defective item is returned, you can't select disposition codes for replacement after the defective item is returned.

Replacement by disposition code

If you ship a replacement item to the customer and use the **Replace and scrap** or **Replace and credit** disposition action on the return order, use the process shown in the following illustration.



The replacement item is delivered by using an independent sales order, the replacement sales order. This sales order is created when the packing slip for the return order is generated. The order header uses information from the customer that the return order header references. The line information is collected from the data entered on the **Replacement item** page. The **Replacement item** page must be filled in for lines that have disposition actions that start with the word "replace." However, neither the quantity nor the identity of the replacement item is validated or limited. This behavior allows for cases where the customer wants the same item but in a different configuration or size, and also cases where the customers wants a completely different item. By default, you enter an identical item on the **Replacement item** page. However, you can select a different item, provided that the function is set up.

> ⓘ **Note**
>
> You can edit and delete the replacement sales order after it's created.

Generate a packing slip

Before you can receive returned items into inventory, you must update the packing slip for the order that the items belong to. Just as the invoice update process is the update of the financial transaction, the packing slip update process is the physical update of the inventory record. In other words, this process commits the changes to inventory. In the case of returns, the steps

assigned to the disposition action are implemented during the packing slip update. When you generate the packing slip, the following events occur:

- In the warehouse, the standard process performs a physical receipt. Ledger postings are generated if the inventory model group (**Post physical inventory**) and the Accounts receivable parameters (**Post packing slip in ledger**) are set appropriately.
- Items that are marked with a disposition action that contains the word "scrap" are scrapped, and the inventory loss is posted to the ledger.
- Items that are marked with the **Return to customer** disposition action are received and delivered to the customer. These items have no net effect on inventory.
- A replacement sales order is created. This sales order is based on information on the **Replacement item** page.

You can generate the packing slip only for lines that have a return status of **Registered**, and only for the full quantity on the return line. If several lines on the return order have the **Registered** status, you can generate the packing slip for a subset of the lines by deleting the other lines from the **Post packing slip** page.

Partial returns are defined in terms of return order lines, not in terms of return order shipments. Therefore, if you receive the full quantity that is indicated on one return order line, but you receive nothing from the other lines on the return order, the delivery isn't a partial delivery. However, if a return order line requires that 10 units of an item be returned, but you receive only four units, the delivery is a partial delivery. If not all the expected return items arrive, you can set the shipment aside and wait for the rest of the returned quantity to arrive. Alternatively, you can register and post the partial quantity. As part of the process for posting packing slips, you can associate the packing slip reference number from the customer’s shipping documents with the order lines. This association is optional and is for reference only. It doesn't create any transactional updates.

In general, you can skip the packing slip process and go directly to invoicing. In this case, the steps that you would have performed during packing slip generation are completed during invoicing.

Generate an invoice

Although the **Return order** page contains the information and actions that are required to handle the special logistical aspects of the return order, you must use the **Sales order** page to complete the invoicing process. Your organization can then invoice return orders and sales orders at the same time, and the same person can complete the invoicing process, as required. To view the

return order from the **Sales order** page, click the link for the sales order number to open the associated sales order. You can also find the return order on the **All Sales orders** page. Return orders are sales orders that have an order type of **Returned order**.

Credit correction

As part of the invoicing process, verify that any miscellaneous charges are correct. To make the ledger postings corrections (Storno), consider using the **Credit correction** option on the **Other** tab of the **Posting invoice** page when you post the invoice /credit note.

> ⓘ **Note**
>
> By default, the **Credit correction** option is activated if you enable the **Credit note as correction** option on the **Accounts receivable parameters** page. However, we recommend that you don't returns with Storno.

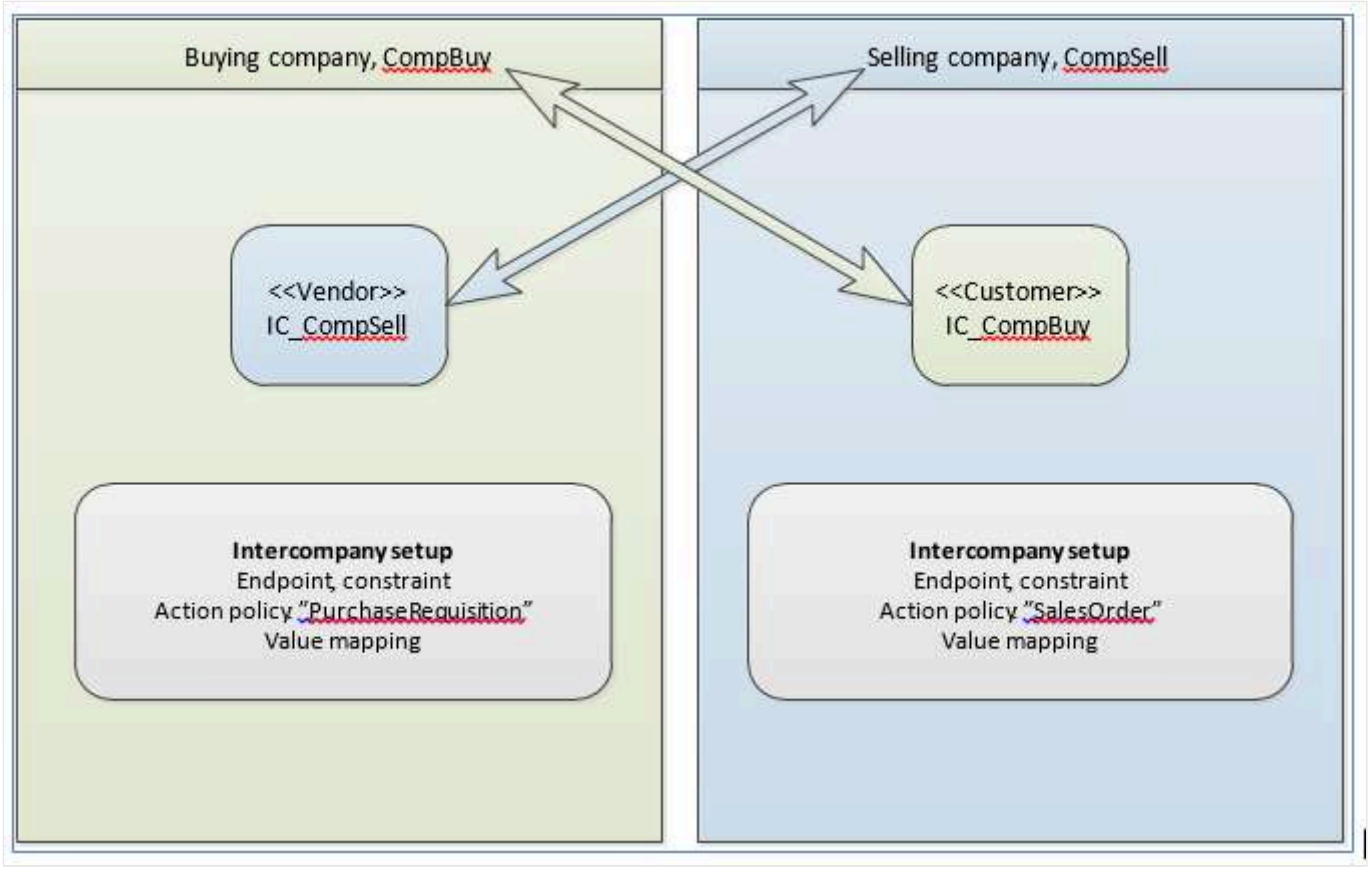
Create intercompany return orders

You can complete return orders between two companies within your organization. The following scenarios are supported:

- Simple intercompany returns between two companies that participate in an intercompany relation
- An intercompany chain that is established when a customer return order is created in the selling company
- An intercompany chain that is established when a vendor return order is created in the buying company
- Direct delivery shipment returns between an external customer and two companies that participate in an intercompany relation

Setup

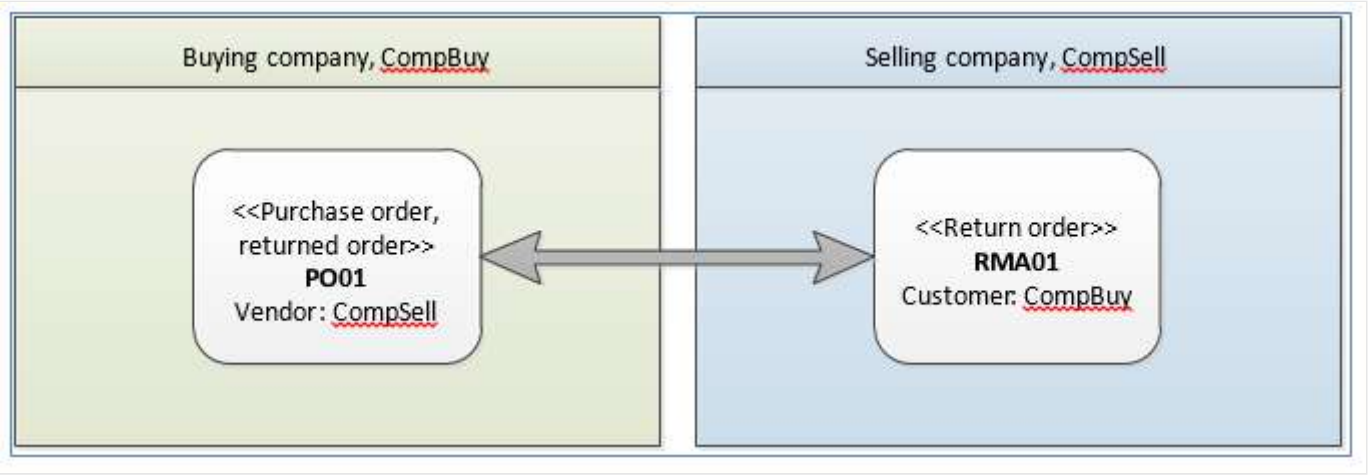
The following illustration shows the minimum setup that is required for two companies to participate in an intercompany relation and take advantage of intercompany trade.



In the following scenario, CompBuy is the buying company, and CompSell is the selling company. Usually, the selling company ships goods either to the buying company or, in direct delivery shipment scenarios, directly to the end customer. In CompBuy, the vendor IC_CompSell is defined as an intercompany endpoint that is associated with the company CompSell. At the same time, in CompSell, the customer IC_CompBuy is defined as an intercompany endpoint that is associated with the company CompBuy. The appropriate action policy details and value mappings must be defined in both companies. In a direct delivery shipment scenario, an intercompany return order, which is also an intercompany sales order, is created in the selling company. The RMA number of the intercompany return order can be picked from the RMA number sequence in CompSell, or it can be copied from the RMA number that is assigned to the original return order in CompBuy. The RMA number settings on the **PurchaseRequisition** action policy in CompBuy determine these actions. If the RMA number is synchronized, you should plan to mitigate the risk of number clashes if the two companies use the same number sequence.

Simple intercompany returns

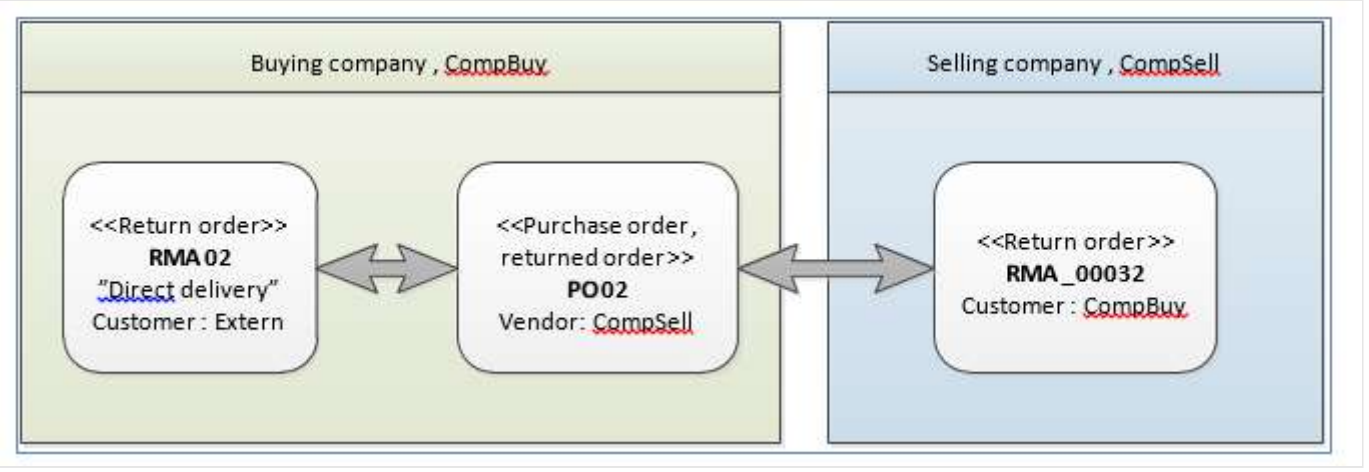
This scenario involves two companies in the same organization, as shown in the following illustration.



The order chain can be established when a vendor return order is created in the buying company or a customer return order is created in the selling company. The corresponding order is created in the other company and makes sure that the header and line information on the vendor return order reflects the settings on the customer return order. The return order that is established can either include or exclude the reference (**Find sales order**) to an existing customer invoice. The packing slips and invoices of the two orders can be processed individually. For example, you don't have to generate a packing slip for the vendor return order before you generate the packing slip for the customer return order.

Direct delivery shipment returns among three parties

This scenario can be established if a previous sale of the **Direct delivery** type is completed, and if an invoice against the customer exists in the company that interacts with the customer. In the following illustration, the company CompBuy previously sold and invoiced products to the customer Extern. The products were shipped directly from the company CompSell to the customer via an intercompany order chain.



If the customer Extern wants to return the products, a return order (RMA02) is created for the customer in the company CompBuy. To establish the intercompany chain, the return order must

be marked for direct delivery. When you use the **Find sales order** function to pick the customer invoice to return, an intercompany order chain that consists of the following documents is established:

- **Original return order:** RMA02 (company CompBuy)
- **Purchase order:** PO02 (company CompBuy)
- **Intercompany return order:** RMA_00032 (company CompSell)

After you create the direct delivery intercompany chain, all physical handling and processing of the returns must occur in context of the intercompany return order, RMA_00032 in the company CompSell. You can't receive the products in the company CompBuy. When you assign a disposition code to the intercompany return order, it synchronizes with the original return order to allow for proper invoicing of the original order.

Post to the ledger

A few important settings and parameters influence the ledger postings that are generated when the return order is invoiced:

- **Return cost price** – For inventory models other than **Standard cost**, the **Return cost price** parameter determines the cost of the item when it's accepted back into inventory or scrap it. To calculate a correct valuation of inventory, set the **Return cost price** parameter correctly. If you use the **Find sales order** function to create a return order line that has a reference to a customer invoice, the **Return cost price** value equals the cost price of the item that is sold. Otherwise, the cost price value comes from the item setup or you can enter it manually.
- **Credit correction/Storno** – The **Credit correction** parameter on the **Posting invoice** page determines whether postings are recorded as positive (DR/CR) entries or as correcting, negative entries.

In the following examples, the return cost price is represented as **Inv. Cost price**.

Example 1: The return order doesn't reference a customer invoice

The return order doesn't reference a customer invoice. The returned item is credited. The **Credit correction** parameter isn't selected when the return order invoice, or credit note, is generated.

Case 1	Qty	Unit price	Inv. Cost price	Line discount	Revenue		Discount		Consumption		Accounts receivable		Inventory issue	
					DR	CR	DR	CR	DR	CR	DR	CR	DR	CR
Sales order invoice	3	10	5	6%		30	1,8		15		28,2			15
Return order invoice	3	10	4	0%	30					12		30	12	

> ⓘ Note
>
> The item master price is used as the default value for the **Return cost price** parameter. The default price differs from the cost price at the time of inventory issue. Therefore, the implication is that a loss of 3 is incurred. Additionally, the return order doesn't include the discount that was given to the customer on the sales order. Therefore, an excessive credit occurs.

Example 2: Credit correction is selected for the return order

Example 2 is the same as example 1, but the **Credit correction** parameter is selected when the return order invoice is generated.

Case 2	Qty	Unit price	Inv. Cost price	Line discount	Revenue		Discount		Consumption		Accounts receivable		Inventory issue	
					DR	CR	DR	CR	DR	CR	DR	CR	DR	CR
Sales order invoice	3	10	5	6%		30	1,8		15		28,2			15
Return order invoice	3	10	4	0%		-30			-12		-30			-12

> ⓘ Note
>
> The ledger postings are entered as negative corrections.

Example 3: The return order line is created by using the Find sales order function

In this example, the return order line is created by using the **Find sales order** function. The **Credit correction** parameter isn't selected when the invoice is created.

Case 3	Qty	Unit price	Inv. Cost price	Line discount	Revenue		Discount		Consumption		Accounts receivable		Inventory issue	
					DR	CR	DR	CR	DR	CR	DR	CR	DR	CR
Sales order invoice	3	10	5	6%		30	1,8		15		28,2			15
Return order invoice	3	10	5	6%	30			1,8		15		28,2	15	

> ⓘ Note
>
> **Discount** and **Return cost price** are set correctly. Therefore, an exact reversal of the customer invoice occurs.

Last updated on 04/20/2026

Receive unannounced sales returns

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In Microsoft Dynamics 365 Supply Chain Management, the sales return process usually starts when you create a return material authorization (RMA) order. The process of creating an RMA order supports a scenario where the reason for the return isn't immediately apparent or disclosed. The RMA order serves as the primary document that guides subsequent steps in the return process, including warehouse arrival and the receiving procedures.

However, in some business scenarios, customers might request to return products without providing advance notification or an associated order. In these scenarios, the standard RMA process is bypassed, and special considerations are made to accommodate the unplanned or unanticipated (unannounced) returns.

This article explains how to set up Supply Chain Management to support unannounced returns, and how to receive unannounced returns in the warehouse.

Prerequisites

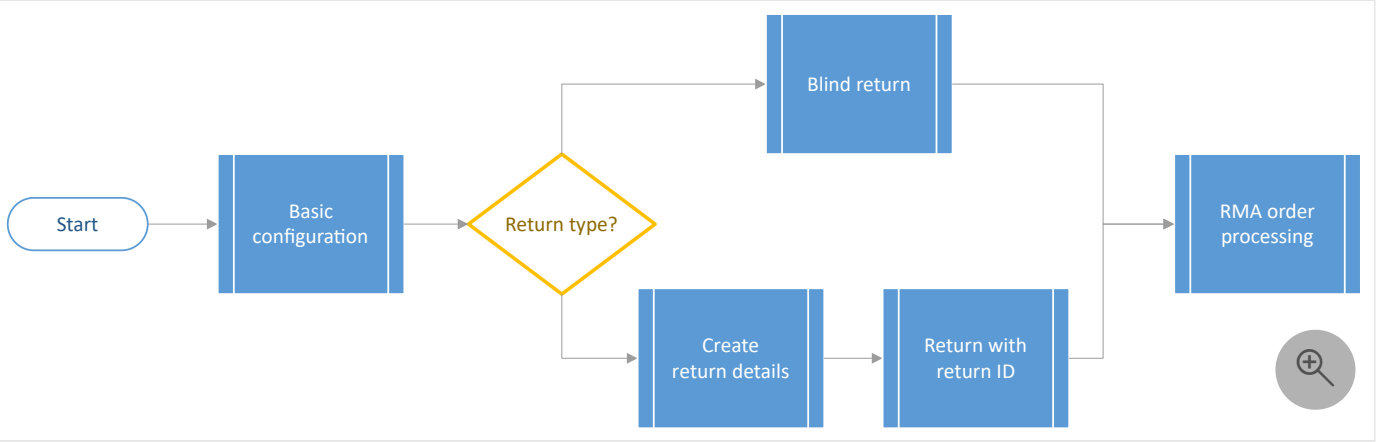
Before you can use the features described in this article, ensure your system meets the following requirements:

- To receive unannounced sales returns, you're running Supply Chain Management version 10.0.39 or later.
- To use the integration with small parcel shipping (SPS), you're running Supply Chain Management version 10.0.39 or later.
- To print license plate labels from a mobile device, you're running Supply Chain Management version 10.0.40 or later.

Receiving process for unannounced returns

The *Return item receiving* flow in the Warehouse Management mobile app supports two processes for receiving unannounced returns: *Blind return* and *Return details*.

The following illustration shows how the flow works for each process.



Blind return process

A blind return is a return where no RMA order or return details record exists in advance, and where no original sales order or shipment must be specified during receiving. During the *Return item receiving* process, workers must use the mobile app to identify and confirm the following information:

- **Customer account or Source system** – The source that returns the item. *Customer account* is used for sales return orders and *Source system* is used when using inbound shipment orders as part of the Warehouse management only mode process.
- **Item number and quantity** – The items that are returned and the quantity that's returned for each item.
- **License plate ID** – The license plate of the location where the incoming items are stored after they're received.
- **Disposition code** – Depending on the setup of the mobile device menu item, workers might be able to select a disposition code to specify what should be done with the returned products. If this option isn't shown in the app during receiving, a Supply Chain Management user must enter a disposition code when they complete the mixed license plate.

> ⓘ **Note**
>
> When running the return process for inbound shipment orders, you must also define source system disposition codes.

Return details process

The *Return details* return process uses a return details record. This record is automatically generated when an order is shipped. It contains all the information required to process a return, such as the return ID, original order ID, shipment ID, order lines, and return-until dates.

Typically, the sending company prints a return label for each outgoing shipment and delivers it together with that shipment. The return label includes a bar code that contains the return ID. Then, if a customer must return an item, they don't have to contact the sending company. Instead, they can just pack the item and attach the return label that they received together with the original shipment.

> ⓘ **Note**
>
> Only the items/variants, tracking dimensions, and quantities that are listed in the specified return details record can be returned through this process.

You can generate a return details record in two ways:

- In a *packing scenario*, the system generates a return details record after each sales order is packed into a container and the container is closed.
- In a *nonpacking scenario*, the system generates a return details record after a worker creates a shipment for a sales order and confirms it.

In both cases, the system fills in the shipment date and return-until date when it confirms the related shipment.

During the *Return item receiving* process, workers must use the mobile app to identify and confirm the following information:

- **Return ID** – The ID number of the return details record.
- **Item number and quantity** – The items that are returned and the quantity that's returned for each item.
- **License plate ID** – The license plate of the location where the incoming items are stored after they're received.
- **Disposition code** – Depending on the setup of the mobile device menu item, workers might be able to select a disposition code to specify what should be done with the returned products. If this option isn't shown in the app during receiving, a Supply Chain Management user must enter a disposition code when they complete the mixed license plate.

Set up unannounced sales return receiving

To prepare your system to receive returns without using an RMA order, you must enable the relevant features and complete several related configurations. The following subsections describe the required settings.

Enable unannounced returns

Before you can use blind returns and/or automatic generation of return details during receiving, you must follow these steps to turn on and set up each feature in your system.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.

2. If you plan to use the *Return details* return process, specify the number sequences that support this process:
   a. On the **Number Sequences** tab, find the row where the **Reference** field is set to *Return ID*. For this row, set the **Number sequence code** field to the number sequence that you want to use.
   b. Find the row where the **Reference** field is set to *Load line inventory pick*. For this row, set the **Number sequence code** field to the number sequence that you want to use.

   For more information about how to set up number sequences for these purposes, see Number sequences overview.

3. If you plan to use the *Return details* process, on the **Loads** tab, set the **Enable sales load line picking route** option to *Yes*. This setting enables load lines to be linked to sales line inventory transactions. The linking is done at the shipment level. This option uses the *Load line inventory pick* number sequence when work is completed and containers are closed.

4. On the **General** tab, on the **Returns** FastTab, set the following options:

   - **Enable return details creation** – Set this option to *Yes* to enable return details to be created when a container is closed and during shipment confirmation. This option is required only for the *Return details* process.
   - **Enable return order creation from mobile device** – Set this option to *Yes* to enable return orders to be created from a mobile device flow. The flow uses mixed license plate receiving planned lines. This option is required for both the *Return details* process and the *Blind return* process.

5. If you're using Warehouse management only mode, set up returns for each source system by following these steps:
    a. Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**.
    b. From the list pane, select the source system you want to set up for returns.
    c. Make the following settings on the **Inbound shipment orders** FastTab:

        - **Enable returns process** – Set to *Yes* to enable the return process for the current source system.
        - **Number sequence code** – Select the number sequence to use to generate order numbers for inbound shipment orders.
        - **Return order type** – Enter text used to stamp the created inbound shipment order. This value is used to distinguish return orders from other inbound shipment orders.

    d. Repeat these steps for each relevant source system.

Set up disposition codes

Disposition codes are collections of rules that you can use when receiving items in Supply Chain Management. They help determine the inventory status, the work template, and the location directive for the items, based on the scanned code.

For example, when a warehouse worker uses a mobile device to receive items that are damaged, the worker scans a disposition code for damaged items.

- If you want to credit the item to your inventory and make it available for repurchase, set up a disposition code that sets the inventory status of the received item to *Available*.
- If you receive items that you can't resell and must therefore dispose of, use a disposition code for RMA scrap. This disposition code then returns a *return disposition code*.

Use disposition codes for different processes, such as *purchase order receiving*, *production order report as finished*, and *sales order return receiving*. For the *sales order return receiving* process, if the items are registered by using a mobile device, the use of disposition codes is mandatory.

Return disposition codes

To view and set up your return disposition codes, go to **Sales and Marketing** > **Setup** > **Returns** > **Disposition codes**. Define each code by specifying an ID number, an action, and a description. The action defines how items that are assigned to the code should be handled.

For more information about return disposition codes and the actions that are available, see Disposition codes and disposition actions.

Mobile device disposition codes

To set up disposition codes that you can use on a mobile device, go to **Warehouse management** > **Setup** > **Mobile device** > **Disposition codes**. Disposition codes that you create here are available to mobile devices and can support several types of operations, including returns, sales orders, and inventory management. Disposition codes for returns must be linked to a return disposition code defined at **Sales and Marketing** > **Setup** > **Returns** > **Disposition codes**, and that assigns a return action for the code.

For more information about mobile disposition codes and how to set them up, see Set up dispositions codes.

Source system disposition codes

If you're using Warehouse management only mode, you must map local dispositions codes with the *source system disposition codes* used in each relevant source systems. These codes are used when processing inbound shipment orders.

To set up source system disposition codes, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source system disposition codes**.
2. On the Action pane, select **New** to add a new code. Then make the following settings for the new code:
    - **Disposition code** – Select the local disposition code that you want to map.
    - **Source system** – Select the source system that you want to map the local disposition code to.
    - **Source system disposition code** – Enter the disposition code that is used in the source system.
3. Repeat the previous step until you set up all of the disposition codes that you need to map.

Set up return item policies

Return item policies let you control the conditions under which the system allows items to be returned. You can set policies for each item, groups of items, or all items, as you require. To set up

your policies, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Return items** > **Return item policies**.
2. On the Action Pane, select **New** to create a policy.
3. In the **Item code** field, select one of the following values to specify the scope that the new policy applies to:
    - *Table* – The return item policy applies only to a specific product.
    - *Group* – The return item policy applies to a specific product group.
    - *All* – The return item policy applies to all products.
4. In the **Item** field, if you set the **Item code** field to *Table* in the previous step, select the item number that the new policy applies to. If you set the **Item code** field to *Group*, select the item group that the new policy applies to. If you set the **Item code** field to *All*, the **Item** field is unavailable.
5. In the **Return acceptance** field, select one of the following values:
    - *Always allowed* – Items that match the selected **Item code** and **Item** values can always be returned.
    - *Never allowed* – Items that match the selected **Item code** and **Item** values can never be returned.
    - *Allowed days after shipment* – Items that match the selected **Item code** and **Item** values can be returned only for a limited time after they were sold, as defined in the **Allowed days for return** field.
6. If you set the **Return acceptance** field to *Allowed days after shipment*, set the **Allow days for return** field to the number of allowed days.
7. Repeat this procedure for each item or item group, as you require. When returns are processed, the most specific return item policy applies to each item in the return.
8. On the Action Pane, select **Save**.

Create return item receiving policies

Before you create the related mobile device menu items, define the required return item receiving policies. These policies enable each menu item to process the return correctly.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Return item receiving policies**.

2. On the Action Pane, select **New** to add a return item receiving policy.

3. Set the following fields for the new policy:

- **Return item receiving policy ID** – Enter a name for the policy, such as *Blind return* or *Return details*.
- **Description** – Enter a short description of the policy.
- **Return process** – Select the type of return process that the policy represents (*Return details* or *Blind return*).
- **Create return order** – Select the type of return order that the system should create to manage returns:
  - *Return order* – The system creates a return order for each return. Select this value if you aren't using Warehouse management only mode.
  - *Inbound shipment order* – The system creates an inbound shipment order for each return. Select this value if you're using Warehouse management only mode.
- **Return order identification** – This field is available only when the **Return process** field is set to *Blind return* and the **Create return order** field is set to *Inbound shipment order*. The way that you use it depends on the needs of your external system. (See the documentation for your external system.) Select the type of extra identifying information that the system should return to the source system when it registers a return order:
  - *None* – The system doesn't return any extra identifying information, and the app doesn't prompt the worker to enter any extra information during receiving.
  - *Account number* – The system returns an account number to help identify the return order. This value is typically the ID of the customer who is returning the items (the customer account number). The mobile app prompts the worker to scan or enter this number during receiving.
- **Return order type** – This field is available only when the **Return order identification** is available and set to *Account number*. Like the account number, the value in this field is returned to the source system when a return order is registered. It's intended to hold the name that identifies the document type for return orders in the source system. The way that you use it depends on the needs of your external system. (See the documentation for your external system.) If you leave this field blank, the value is inherited from the relevant source system configuration.

> **Important**
>
> If your source system is Dynamics 365 Supply Chain Management (as is the case when you run an external shared warehouse scenario), set the following values for your blind return policies:
>
> - **Return order identification** – Select *Account number*, and instruct workers to enter the customer ID (account number) when the mobile app prompts them during receiving.
> - **Return order type** – Enter *Return*, which is the return order type that is used in Supply Chain Management. (Alternatively, if *Return* is already specified as the default return order type for the source system, leave this field blank.)

4. If you support both types of unannounced return processes (*Return details* and *Blind return*), repeat this procedure to add a return item receiving policy for the other process.

Set up mobile device menu items to process unannounced returns

To enable workers to process unannounced returns, create a separate mobile device menu item for each type of unannounced return process (*Return details* or *Blind return*) that you support.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following fields for the new menu item:
    - **Menu item name** – Enter an internal name for the menu item, such as *Blind returns*.
    - **Title** – Enter the name that should appear for the menu item in the mobile app, such as *Blind return item receiving*.
    - **Mode** – Select *Work*.
    - **Use existing work** – Set this option to *No*.
    - **Work creation process** – Select *Return item receiving*.
    - **Barcode data policy** – Select the policy to use if multiple fields are filled in based on a single bar code scan. To learn more, see GS1 bar codes.
    - **Generate license plate** – Set this option to *Yes* to automatically create new license plates as they're needed. Set it to *No* if the worker must always select an existing license plate.

- **Display disposition code** – Select whether workers should be prompted to select a disposition code during the receiving process. The disposition code determines the inventory status, work template, and location directive for the returned items.
- **Print label** – Set this option to *Yes* to always print a license plate label after all steps in the related work template are completed (regardless of whether a print step is included in the work template). If you want to allow the work template to print the license plate label at a different point in the process, set this option to *No*. The license plate label includes a bar code that provides the ID of the license plate where the worker places the returned items. To learn more, see License plate label layouts and printing.
- **Return item receiving policy ID** – Select the item receiving policy that you created for the type of return process (*Return details* and *Blind return*) that this menu item supports.

4. If you support both types of unannounced return processes (*Return details* and *Blind return*), repeat steps 2 and 3 to create a menu item for the other process.

5. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.

6. Add the new menu items to an appropriate place in your mobile device menu structure.

Integrate return label printing with small parcel shipping

Small parcel shipping (SPS) integration lets you retrieve shipping labels from the SPS carriers or carrier hubs that you integrate with your system. However, some customization is required. The integration supports preaddressed and prepaid return labels, which make returns more convenient for your customers. The return labels can include a bar code that identifies the return details record ID. Warehouse workers can then scan this bar code when the package arrives at your warehouse.

For more information about how to set up SPS integration, see Small parcel shipping.

Follow these steps to enable return details with SPS integration.

1. Go to **Transportation management** > **Setup** > **Transportation management parameters**.
2. On the **Shipping carriers** FastTab, set the **Enable shipping label request type** option to *Yes*. This setting adds a new **Print container return shipping label rule** field to the **Container packing policies** page. Use this field to select the conditions under which return labels

should be printed as part of the container packing process. For details about how to use this setting, see the next section.

Print return labels

When you use a *Return details* return process, you must print a return label for each shipment and include that label with the shipment. The *container packing policy* that applies to each shipment establishes the printing options. Depending on the setup of your system, return labels are typically printed as part of the container closing process. However, you can manually print or reprint them for a selected return details record at any time.

The return label includes the shipping address of your returns warehouse and a bar code that contains the return details record ID. Therefore, warehouse workers can scan this bar code when the package arrives at your warehouse.

Follow these steps to configure the label printing options for a container packing policy.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container packing policies**.

2. Select or create the container packing policy that you want to set up.

3. On the **Container manifest** FastTab, set the following fields:

    - **Automatic manifest at container close** – Set this option to *Yes*.
    - **Manifest requirements for container** – Set this field to *Transportation management*.

4. On the **Carrier label printing** FastTab, set the following fields:

    - **Print container shipping label rule** – Select one of the following options to specify the rule for printing outbound shipping labels:
        - *Never* – Users can close containers without printing a shipping label.
        - *Always* – A shipping label is required. Users can't close the container unless the system can retrieve a label.
        - *If setup exists* – For containers where a shipping label setup exists, the system must retrieve the required label before users can close the container. For containers where no shipping label setup exists, users can close the container without printing a label.

    - **Print container return shipping label rule** – Select one of the following options to specify the rule for printing return labels to include in each container:
        - *Never* – Users can close containers without printing a return label.

    - *Always* – A return label is required. Users can't close the container unless the system can retrieve a label.
    - *If setup exists* – For containers where a return label setup exists, the system must retrieve the required label before users can close the container. For containers where no return label setup exists, users can close the container without printing a label.

- **Printer name** – Select the Zebra Programming Language (ZPL) printer where the system should print shipping and/or return labels when this container packing policy is used.

Follow these steps to manually print or reprint a return label for a return details record.

1. Go to **Warehouse management** > **Inquiries and reports** > **Return details**.
2. In the list pane, select the return details record that you want to print a label for.
3. On the Action Pane, select **Print** > **Return shipping label**.

The return label is printed on the printer that's specified in the relevant container packing policy that's associated with the **Container ID** value that's listed for the return details record.

Example scenarios for the sales return order creation process

This section provides example scenarios that show how to set up and process unannounced returns.

Scenario prerequisites

Before you work through the example scenarios, complete all the prerequisites described in the following subsections.

Enable sample data

To work through the example scenarios by using the sample records and values that are specified here, use a system where the standard demo data is installed. Additionally, select the **USMF** legal entity before you begin. The example scenarios use the demo data that's associated with the worker/person *Julia Funderburk*.

You can also use these scenarios as guidance for using the feature on a production system. However, in that case, substitute your own values for each setting that's described here.

Set up number sequences

You must have number sequences that generate unique return IDs and load line inventory pick numbers.

1. Go to **Organization administration** > **Number sequences** > **Number sequences**.
2. On the Action Pane, on the **Number sequence** tab, in the **New** group, select **Number sequence** to create a number sequence that generates return ID numbers.
3. Set the following values for the new number sequence:
    - **Number sequence code:** *ReturnID*
    - **Name:** *ReturnID*
    - **Scope:** *Company*
    - **Company:** *USMF*
4. On the Action Pane, select the **Back** button.
5. On the Action Pane, on the **Number sequence** tab, in the **New** group, select **Number sequence** to create a number sequence that generates load line inventory pick numbers.
6. Set the following values for the new number sequence:
    - **Number sequence code:** *LLIP*
    - **Name:** *LLIP*
    - **Scope:** *Company*
    - **Company:** *USMF*
7. On the Action Pane, select **Save**.

Enable unannounced return features

Follow these steps to enable the features that are required to support unannounced returns.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Number sequences** tab, set the **Number sequence code** field to *ReturnID*.
3. Set the **Number sequence code** field to *LLIP*.
4. On the **General** tab, turn on the **Enable return details creation** option.
5. Turn on the **Enable return order creation from mobile device** option.
6. On the **Loads** tab, turn on the **Enable sales load line picking route** option.

For information about how to configure these settings, see the Enable unannounced returns section.

Create return item policies

Follow these steps to create return item policies for the items that are used in the scenarios.

1. Go to **Warehouse management** > **Setup** > **Return items** > **Return item policies**.
2. Add a return item policy to define the conditions under which the item that's used in the scenarios can be returned.
3. Set the following values for the new policy:

    - **Item code:** *Table*
    - **Item:** *A0001*
    - **Return acceptance:** *Allowed days after shipment*
    - **Allowed days for return:** *28*

Learn more in the Set up return item policies section.

Set up return item receiving policies

Follow these steps to create return item receiving policies for the scenarios.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Return item receiving policies**.
2. Add two return item receiving policies: one for the *Blind return* process and one for the *Return details* process.

Learn more in the Create return item receiving policies section.

Set up mobile device menu items and menus

Follow these steps to create mobile device menu items for the scenarios.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. Create two mobile device menu items for handling unannounced returns: one for the *Blind return* process and one for the *Return details* process.

3. Set up the new menu items in the following way:

    - Give each menu item a name that identifies the type of return that it's intended to process (for example, *Blind return receiving* or *Return details receiving*).
    - For each menu item, set the **Work creation process** field to *Return item receiving*.

- For each menu item, enable both the **Generate license plate** option and the **Display disposition code** option.
- For each menu item, set the **Return item receiving policy ID** field to the policy that matches the type of return that the menu item should handle (*blind returns* or *return details*).

4. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.

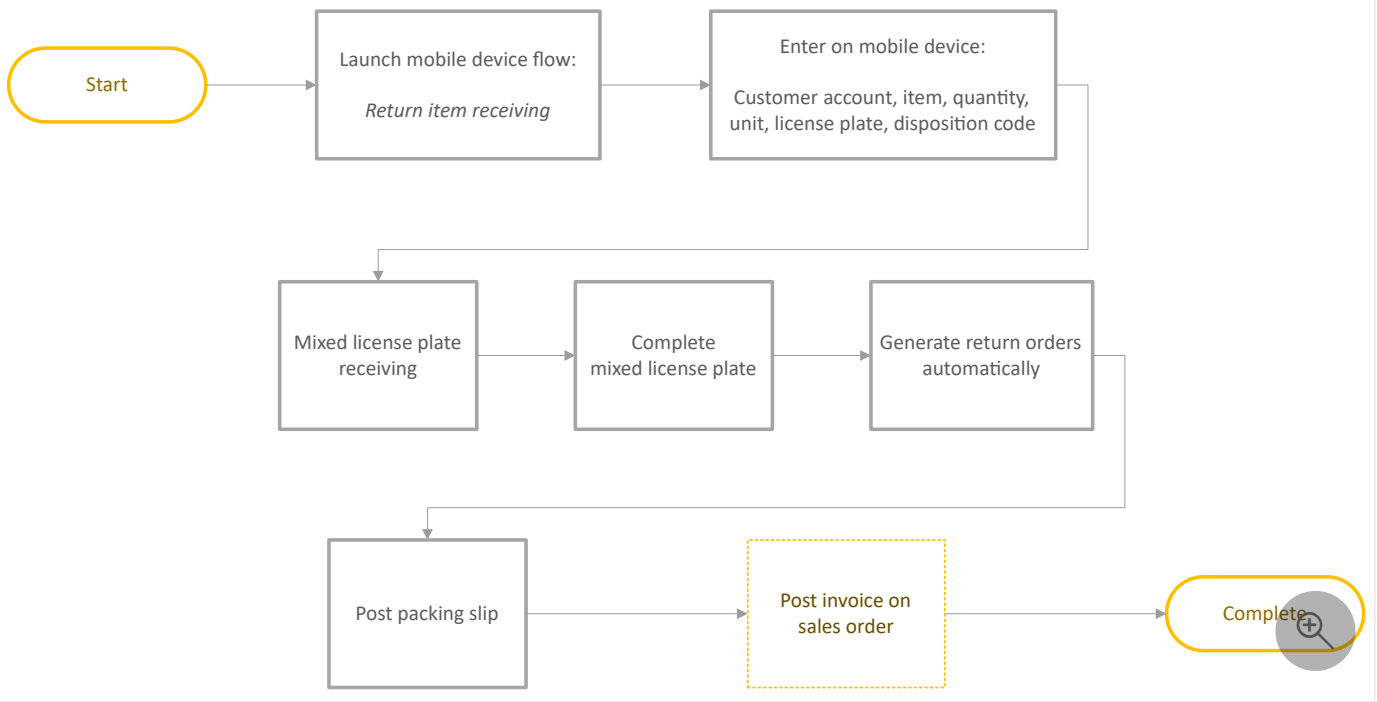
5. Add both menu items to an appropriate place in your mobile device menu structure.

Learn more in the Set up mobile device menu items section.

Example scenario 1: Customer return with a blind return policy

In the blind receiving process, warehouse workers receive return items without matching them to an original sales order or shipment. Warehouse workers just enter the customer account ID, item IDs, and quantities.

The following illustration shows a flowchart for the blind receiving process.



Use the mobile app to receive a blind return to a mixed license plate

Follow these steps to receive products to a mixed license plate in a warehouse when there's a blind return policy.

1. Open the Warehouse Management mobile app, and sign in by using *62* as the user ID and *1* as the password.
2. Open the menu item that you created for receiving blind returns (for example, **Inbound** > **Blind returns receiving**).
3. In the **Customer account** field, enter *US-003*.
4. In the **Item** field, enter or scan item number *A0001*.
5. Accept the license plate number that's suggested.
6. In the **qty** field, enter *1*.
7. In the **Disposition code** field, select *RMA Credit*.
8. Select **OK** to complete the receiving work.
9. Select **Cancel** to exit the flow and close the menu item.

> ⓘ **Note**
>
> You must exit the flow to unlock the license plate. If the license plate is locked, you can't complete mixed license plate receiving in Supply Chain Management, as described in the next section. In some setups, you can finish processing mixed license plates during the *Return item receiving* flow. However, in this scenario, you process the license plate in Supply Chain Management.

Complete mixed license plate receiving

The returned products are now on a mixed license plate for further processing. (Learn more in Mixed license plate receiving.) Follow these steps to finish receiving the returned items.

1. Go to **Warehouse Management** > **Inquiries and reports** > **Mixed license plate receiving**.
2. In the **License plate** grid, find the mixed license plate that you processed by using the mobile app.
3. On the Action Pane, on the **License plate** tab, select **Complete license plate** to complete mixed license plate receiving.

After you complete mixed license plate receiving, Supply Chain Management automatically creates an RMA order that's prepared for further downstream processing, as described in Sales

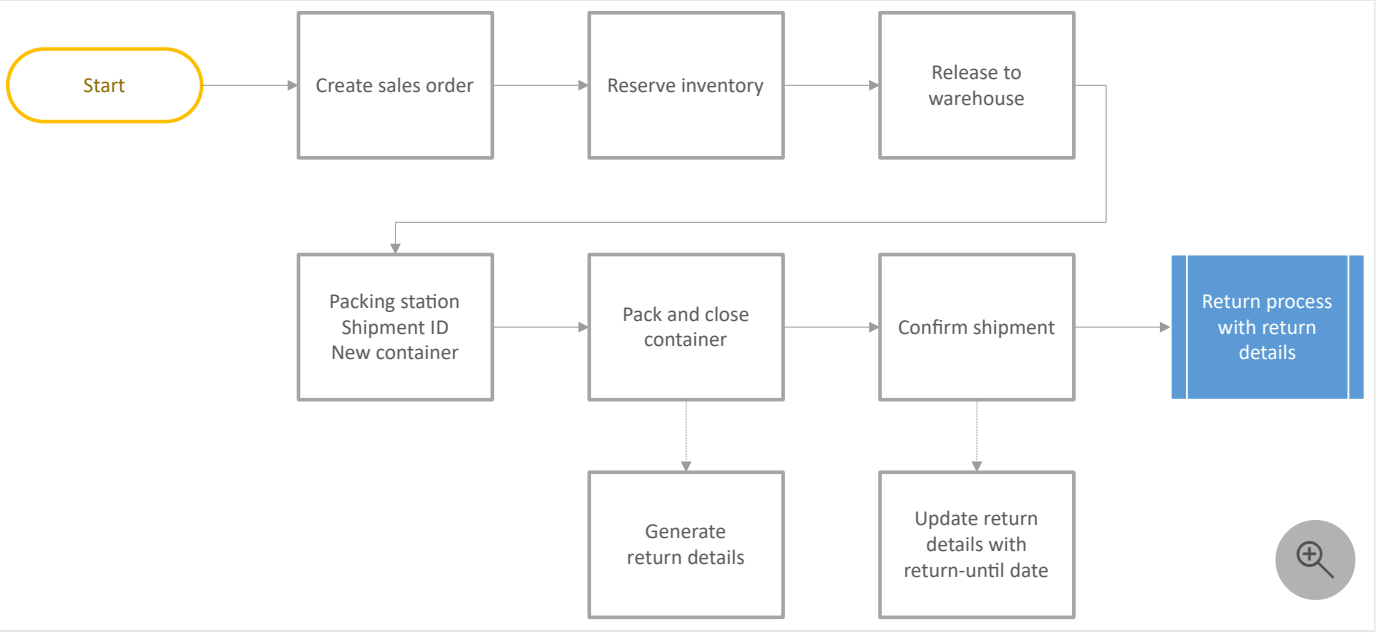
returns.

> **Tip**
>
> You can set up your mobile device menu item for processing returns to include a step that lets workers complete mixed license plate receiving directly from the mobile device while receiving the return. Then you don't need to complete this step later in the Supply Chain Management web client. To do so, create a detour item with the **Activity code** *Complete mixed license plate*. Learn more in Set up a mobile device menu item for completing mixed license plates.

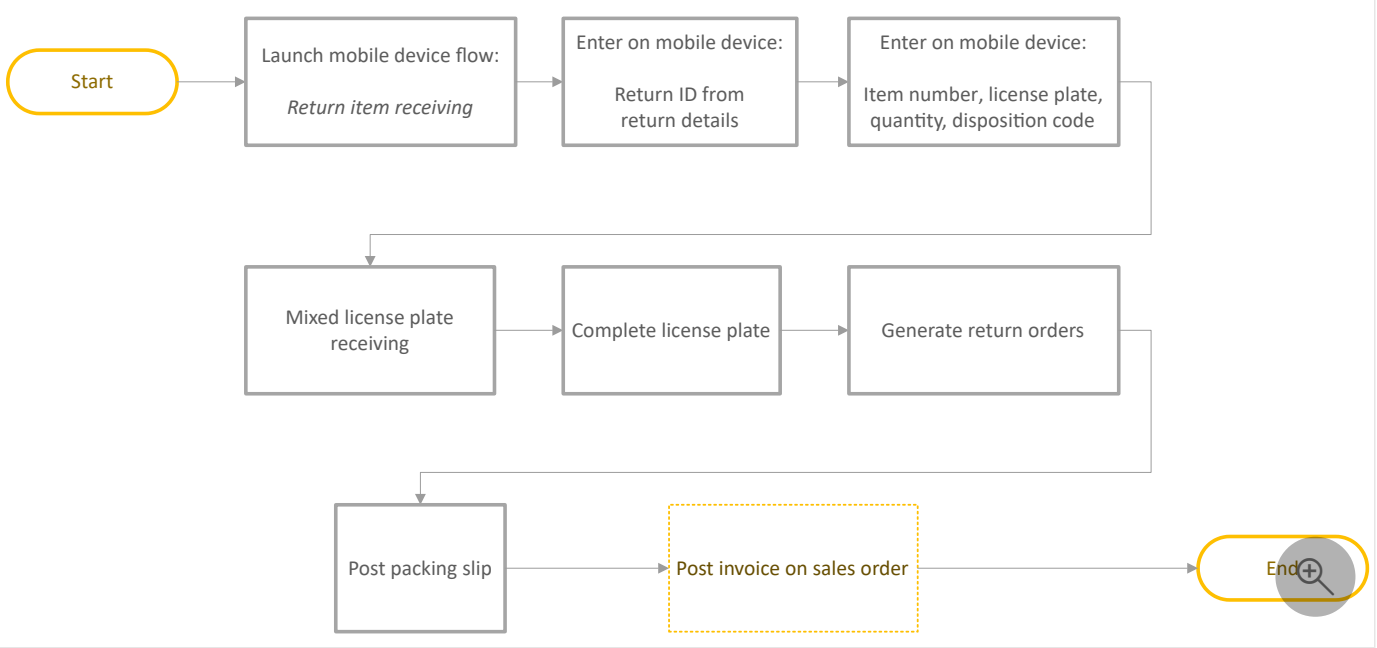
Example scenario 2: Customer return with a return details policy

For this scenario, return details are generated when an order is processed for shipment at the packing station, and matching labels are included with the shipment. If any items must be returned, the customer packs the return items, attaches the return label that they received together with the original shipment, and ships the return items back to the seller. When the returned items arrive at the seller's warehouse, a worker scans the return label and the item to get the return ID. The return ID identifies a return details record that includes the original order details. This process helps prevent fraud, inventory errors, and lost packages.

The following illustration shows the steps that are required to create a sales order and complete the work of moving the ordered items to the packing area. Workers then pack the order, and return details are generated when the container is closed.



The following illustration shows a flowchart of the receiving process where the *Return item receiving* mobile menu item is used to process a return that has return details.



Create a sales order and release it to the warehouse for packing

For this scenario, a related sales order must already exist, and the item must be shipped before the return. Therefore, you must create the required records before you can work through the related return. Follow these steps to create a sales order and complete the work of moving the ordered items to the packing area.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In the **Create sales order** dialog box, set the **Customer account** field to *US-001*.
4. Select **OK** to close the dialog box.
5. The new sales order opens. The **Sales order lines** FastTab includes a single, blank order line. Set the following values for this line:
    - **Item number:** *A0001*
    - **Quantity:** *1*
    - **Site:** *6*
    - **Warehouse:** *62*
6. While the order line is still selected on the **Sales order lines** FastTab, select **Inventory** > **Reservation** on the toolbar.

7. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.

8. On the Action Pane, select the **Back** button to return to the sales order.

9. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**. A message shows the shipment and wave IDs for the order.

10. While the order line is still selected on the **Sales order lines** FastTab, select **Warehouse** > **Work details** on the toolbar. If you use batch processing to run your waves, you might have to wait a short time for the work to be created.

11. On the **Work** page, on the Action Pane, on the **Work** tab, select **Complete work**.

12. On the **Work completion** page, set the **User ID** field to *62*.

13. On the Action Pane, select **Validate work**.

14. When you receive a message that indicates that your work is valid, select **Complete work** on the Action Pane to complete the process of picking the inventory items and putting them in the *Pack* location.

15. Make a note of the **Shipment ID** value that's shown for the work in the upper grid.

Pack the ordered items into a container and generate return details

The inventory items are brought to the packing area and are ready to be packed into containers. Follow these steps to create a new container in the system, and to pack and close the container. After you close the container, the system automatically creates a return details record. You can reference this record later when returned items are received back into the warehouse. At this point, you can print a return label and deliver it together with the container.

1. Go to **Warehouse Management** > **Packing and containerization** > **Pack**.

2. In the **Select packing station** dialog box, set the following values:

    - **Site:** *6*
    - **Warehouse:** *62*
    - **Location:** *Pack*
    - **Packing profile ID:** *WH62*

3. Select **OK** to close the dialog box and return to the **Pack** page.

4. On the **Packing station** FastTab, in the **License plate or shipment** field, enter the shipment ID that you noted at the end of the previous procedure. Then select **Return**.
5. On the Action Pane, select **New container** to start the *container creation* process.
6. In the dialog box, set the **Container type** field to *SmallBox*. Then select **OK** to return to the **Pack** page.
7. On the **Item packing** FastTab, set the **Identifier** field to *A0001*. Then select **Return**. A quantity of *1* of item number *A0001* is moved to the container for the shipment. On the **All lines** FastTab, the line for item number *A0001* now shows a check mark in the **Complete** column.
8. On the Action Pane, select **Close container** to start the *container closing* process.
9. In the dialog box, select the **Get system weight** link to load the weight of the shipment into the **Gross weight** section. Then select **OK** to return to the **Pack** page.
10. Now that the container is closed, the system automatically creates return details. Go to **Warehouse Management** > **Inquiries and reports** > **Return details**.
11. The **Return details** page should list a new return details record that shows the order number and line item that you just packed. Make a note of the **Return ID** value for the new record. You need this value later in this scenario.
12. Go to **Warehouse Management** > **Shipments** > **All shipments**.
13. Select the shipment that has the shipment ID that you noted at the end of the previous procedure.
14. On the Action Pane, on the **Shipments** tab, select **Confirm Shipment**.

    A message shows that the shipment is confirmed.

> **Note**
>
> If you return to the **Return details** page and open the record that has the return ID that you noted earlier, the **Lines** FastTab should now show a **Return until date** value that matches the return item policy that you set for the packed item. The system automatically adds this value after you confirm the shipment.

Receive a product with return details to a mixed license plate

Follow these steps to receive returned items into the warehouse.

1. Open the Warehouse Management mobile app, and sign in by using *62* as the user ID and *1* as the password.
2. Open the menu item that you created for receiving by using the *Return details* process (for example, **Inbound** > **Return details receiving**).
3. In the **Return ID** field, enter the return ID that you noted during the previous procedure. In a real-world scenario, the warehouse worker typically scans this value from the return label that's attached to the package. (This label was sent to the customer together with the original shipment.)
4. In the **Item** field, enter or scan item number *A0001*.
5. Accept the license plate number that's suggested.
6. In the **qty** field, enter *1*.
7. In the **Disposition code** field, select *RMA Credit*.
8. Select **OK** to complete the receiving work.
9. Select **Cancel** to exit the flow and close the menu item.

> **Note**
>
> You must exit the flow to unlock the license plate. If the license plate is locked, you can't complete mixed license plate receiving in Supply Chain Management, as described in the next section. In some setups, you can finish processing mixed license plates during the *Return item receiving* flow. However, in this scenario, you process the license plate in Supply Chain Management.

Complete mixed license plate receiving

The returned products are now on a mixed license plate for further processing. (Learn more in Mixed license plate receiving.) Follow these steps to finish receiving the returned items.

1. Go to **Warehouse Management** > **Inquiries and reports** > **Mixed license plate receiving**.
2. In the **License plate** grid, find the mixed license plate that you processed by using the mobile app.

3. On the Action Pane, on the **License plate** tab, select **Complete license plate** to complete the mixed license plate receiving.

After you complete mixed license plate receiving, Supply Chain Management automatically creates an RMA order that's prepared for further downstream processing, as described in Sales returns.

> **Tip**
>
> You can set up your mobile device menu item for processing returns to include a step that lets workers complete mixed license plate receiving directly from the mobile device while receiving the return. Then you don't need to **complete this step** later in the Supply Chain Management web client. To do so, create a **detour** item with the **Activity code** *Complete mixed license plate*. Learn more in **Set up a mobile device menu item for completing mixed license plates**.

Example scenario 3: Nonpacking scenarios for return details

Use the *Return details* process for scenarios where you don't use a packing station to prepare outgoing shipments. In this case, your outgoing shipments are delivered on a pallet to the bay door, where they're packed and shipped directly. Therefore, the system doesn't generate the return details record after the container is packed and closed (from the **Pack** page in Supply Chain Management). Instead, it generates the return details record when the outbound shipment is confirmed (from the **All loads** page in Supply Chain Management). Otherwise, this scenario works in the same way as the previously described packing scenario.

For this scenario, it's important that your system is configured to create return details, generate load line inventory picking numbers, and use sales load line picking routes, as described in the Scenario prerequisites section.

Last updated on 05/22/2026

Return reason codes and disposition codes

Create and use return reason codes and disposition codes to support the process for returning products.

Use a return reason code to describe the reason that the customer wants to return an item. You can assign a reason code in the **Create return orders** form.

Assign a disposition code an item is received or during the physical inspection of a returned item. Use disposition codes to describe the condition of the item. You can also use disposition codes to indicate whether additional action is required for the transaction. For example, create disposition codes for the following actions:

- Scrap the returned item and provide a replacement item to the customer.
- Return the item to inventory and credit the customer for the cost of the item.
- Repair the item and return it to the customer.

Related information

- Set up return reason codes
- Set up disposition codes

Last updated on 07/01/2026

Set up disposition codes

You can set up disposition codes to specify how to process an item returned by a customer. For example, create a disposition code named *Repair and return* to indicate that the returned item should be repaired and then returned to the customer. For more examples of disposition codes that are typically used for items returned by customers, see Specify how to dispose of returned items.

You can also set up a reason code to help explain why an item was returned. For more information about reason codes, see Set up return reason codes.

To set up a disposition code, follow these steps:

1. Go to **Sales and marketing** > **Setup** > **Sales orders** > **Returns** > **Disposition codes**.
2. Select **New** to create a new disposition code.
3. Enter a unique, descriptive name, select an action, and enter a description for the disposition code.
4. If you want to associate any customer charges with this disposition code, select the **Charges** button to open the **Set up charges** form.
5. If you want to define any external codes to match with the company's own disposition codes, select the **External codes** button to open the **External codes** form.

Related information

- Return reason codes and disposition codes

Last updated on 07/01/2026

Set up return reason codes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Set up return reason code groups

1. Go to **Sales and marketing** > **Setup** > **Sales orders** > **Returns** > **Return reason code groups**.
2. Select **New** to create a line for a new return reason code group.
3. In the **Return reason code group** field, type an identifier for this group.
4. In the **Description** field, type a brief phrase to explain how this group will be used.

Set up return reason codes

This procedure helps you set up return reason codes to indicate why a customer returned a product.

1. Go to **Sales and marketing** > **Setup** > **Sales orders** > **Returns** > **Return reason codes**.
2. Press CTRL + N to create a new return reason code line.
3. In the **Return reason code** field, enter text to identify this code.
4. In the **Description** field, enter text to describe the return reason code.
5. Use the **Return reason code group** drop-down list to add the code to a return reason code group.

Last updated on 07/01/2026

Specify how to dispose of returned items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you handle a return order, you must specify a reason return code to identify why the product is being returned. You must also specify a disposition code and a disposition action to determine what should be done with the returned product itself.

A disposition code can be applied when you create the return order, register item arrival or packing-slip update an item arrival, and end a quarantine order.

You can define any disposition codes that you need in order to support the business processes. The following table provides a set of typically used codes to assign return-item disposition.

Expand table

Disposition type	Common code	Description
Disposal	SC	Scrap/Destroy
Disposal	DC	Donate to Charity
Disposal	TD	Third-Party Disposal
Disposal	SL	Salvage
Disposal	TS	Third-Party Sale (Secondary Markets)
Repair/Modify	RW	Rework
Repair/Modify	RF	Remanufacture/Refurbish
Repair/Modify	MD	Modify
Repair/Modify	RP	Repair
Repair/Modify	RV	Return to Vendor
Other	AI	Use as is
Other	RS	Resale

Disposition type	Common code	Description
Other	EX	Exchange
Other	MS	Miscellaneous

For each disposition code that you define, you must select a disposition action. The disposition action determines the physical and financial implications of the disposition codes. For example, the disposition action determines the physical handling of the returned item, the financial effect of the returned item, and if a replacement item must be sent to the customer. You can define an unlimited number of disposition codes according to your business needs, but there are only six predefined disposition actions that you can select from. The following table provides the disposition actions and their definitions.

Expand table

Disposition action	Description
Credit	Return the item to inventory and credit the customer.
Credit only	Credit the customer without requiring or expecting the item to be returned.
Scrap	Scrap the item and credit the customer.
Replace and credit	Return the item to inventory, create a replacement order, and credit the customer.
Replace and scrap	Scrap the item, create a replacement order, and credit the customer.
Return to customer	Reject the returned item and return it to the customer.

Select a disposition code for a quarantine order

1. Go to **Inventory management** > **Periodic** > **Quality management** > **Quarantine orders**.
2. For an existing quarantine order, select an action from the **Disposition code** field on the **Overview** tab.

Last updated on 05/07/2025

Return cost price and return lot ID

The cost of products returned to inventory is calculated by using the current cost of the products. However, you might want the cost of the returned products to equal the cost of the products at the time when you sold the products to the customer. You can set this cost by using the **Return lot ID** field on the **Line details** FastTab in the **Sales order** form.

For example, consider the following scenario. You invoice a customer. Then, the customer returns the delivered products to you. You return the products to stock. In this case, when you credit the customer for the returned products, the cost of those products is calculated by using the current cost of the products. However, if you use the **Return lot ID** field, the cost of the returned products is calculated by using the cost on the invoice of the original sale to the customer.

To use a cost other than the current cost for returns from a customer, use one of the following methods.

Method 1: Manually enter the return cost price

By default, when you add items to a return order, the items are returned to inventory at the current cost price. To specify a different return cost price, follow these steps:

1. Go to **Sales and marketing** > **Sales returns** > **All return orders**.
2. On the Action Pane, in the **New** group, select **Return order**.
3. In the **Create return order** form, select a customer account, and then select **OK**.
4. In the **Return order - RMA number: %1, %2** form, select an item, and then enter a negative quantity in the **Quantity** field.
5. Select the **Line details** FastTab.
6. On the **General** tab, enter a value in the **Return cost price** field. This value is used when the goods are returned to inventory. If you don't enter a value, the current cost price is used when the goods are returned to inventory.

Method 2: Automatically generate the cost price based on the customer invoice line

Use this method to create return lines. To use the cost of the products at the time when you sold the products to the customer, create a return order and specify a sales line to return.

1. Go to **Sales and marketing** > **Sales returns** > **All return orders**.
2. On the Action Pane, in the **New** group, select **Return order**.
3. In the **Create return order** form, select a customer account, and then select **OK**.
4. In the **Return order - RMA number: %1, %2** form, on the Action Pane, select **Find sales order**.
5. In the **Find sales order** form, select the invoice line to return, and then select **OK**.

   On the **Line details** FastTab, on the **General** tab, the **Return lot ID** field displays the value from the original sales line. Additionally, the **Return cost price** field displays the cost value from the original sales line.

Cost calculation example

When you use the **Return lot ID** field on a return order line to specify the return cost price, the cost on the return order line is used. If you run the inventory close or recalculation functionality, the cost is adjusted on the original sales line. The cost on the return order line is automatically adjusted to reflect the same cost per piece.

1. Create and release a product named Test. In the **Released product details** form, specify the following information:

   a. On the **Manage costs** FastTab, in the **Item group** field, select the **Parts** group.

   b. On the **General** FastTab, in the **Item model group** field, select **DEF**.

   c. On the **Purchase** FastTab, in the **Price** field, type 10.00 as the cost price of the item.

   d. On the Action Pane, select **Dimension groups**. In the **Storage dimension group** field, select **Site and Warehouse only**. In the **Tracking dimension group** field, select **No active tracking dimensions**.

2. Create a purchase order for 10 pieces of the Test item at 6.00 per piece, and then post an invoice for the purchase order.

   Create a second purchase order for 10 pieces of the Test item at 8.00 per piece, and then post an invoice for the purchase order.

3. Post an invoice for a sales order to sell five pieces of the Test item.

   In this case, the sales order line is costed at 35.00 (5 pieces * 7.00 average cost per piece).

4. Create a return order for the customer. In the **Find sales order** form, select the invoice line, and then select **OK**.

5. In the **Return order - RMA number: %1, %2** form, verify that a return order is generated to return the Test item. The quantity of the return order is set to -5.00.

   The **Return lot ID** field displays a lot ID. This lot ID is from the original sales order of the item that sold to the customer. The **Return cost price** field displays the cost price from the original sales line.

6. Register the receipt of the returned items.

7. Post a packing slip for the returned items.

8. Post an invoice for the return order. On the **All sales orders** list page, select a sales order for which **Returned order** is the order type.

9. Open the **Inventory transactions** form. Verify that the return is costed at 7.00 per piece by using the value in the **Return cost price** field, for a total of 35.00 in the **Cost amount** field. You can open the **Inventory transactions** form from the **Return order - RMA number: %1, %2** form. In the **Lines** grid, select **Inventory** > **Transactions**.

10. In Inventory and warehouse management, use the **Closing and adjustment** form to run the **3. Close** procedure.

    This action adjusts the cost on the original sales line that was priced at -35.00 (5 pieces * 7.00) to -30.00 (5 pieces * 6.00). This change happens because the inventory model group uses First in, First out (FIFO), and 6.00 per piece is the FIFO cost from the first purchase order. Additionally, the action adjusts the cost on the return sales line to match the cost per piece on the original sales line. Therefore, the cost of the return line is adjusted from 35.00 to 30.00.

Last updated on 07/01/2026

Cancel a return order

You can cancel a return order (instead of deleting it) when you want to save a record of the planned return.

> **Note**
>
> Only return orders that have a **Return status** of *Created* can be canceled. Return orders with a **Return status** of *Open* or *Closed* can't be canceled.

1. Go to one of the following pages:
    - **Sales and marketing** > **Sales returns** > **All return orders**.
    - **Sales and marketing** > **Sales returns** > **Created return orders**.
2. Select the return order to cancel.
3. On the Action Pane, open the **Return order** tab and, from the **Maintain** group, select **Cancel order**.

Last updated on 07/01/2026

Create an item replacement order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Item replacement orders are usually created after a product is returned and inspected. However, when an item must be replaced before it has been returned, or when the original item won't be returned, you can create an item replacement order immediately after you create a return order.

Create a replacement order after you receive an item that is returned

1. Go to **Sales and marketing** > **Sales returns** > **All return orders**.
2. Create a new return order, or select a returned order from the list to open the **Return order - RMA number: %1, %2** form.
3. Select **Return line**, and then select **Replacement item**.
4. Select the item to replace the returned item with. Enter the item specifications, and then select **Apply**.
5. Select **Packing slip** to generate the packing slip for the return order. A sales order is generated for the replacement item that you selected.

   After the sales order is created for the replacement item, sales agreements are automatically searched and if there's an applicable sales agreement, it's applied to the sales order.

Create a replacement order before you receive an item that will be returned

1. Go to **Sales and marketing** > **Sales returns** > **All return orders**.
2. Create a new return order, or select a return order from the list to open the **Return order - RMA number: %1, %2** form.

3. Select **Find sales order** if you want to identify the sales order for the returned item. Complete the **Find sales order** form and then select **OK** to close the form and return to the **Return order - RMA number: %1, %2** form. The sales order line for the returned item is copied to the return order.

4. Select **Replacement order** to open the **Create sales order** form.

5. Select the **Copy return order lines** check box to transfer details from the return order you selected on the **Return order - RMA number: %1, %2** form to this sales order.

6. Enter or modify details, as required.

   If you identified the sales order in step 3, and if the sales order line for the returned item is linked to a sales agreement, the identifier of the applicable sales agreement for the item replacement order will be automatically displayed in the **Sales agreement ID** field.

7. Select **OK** to close the **Create sales order** form and open the **Sales order** form, where you can continue to enter information for the new sales order. Any applicable return order lines will be copied to the new sales order.

   If the identifier of the sales agreement is automatically displayed in the **Sales agreement ID** field, then the sales agreement has been linked to the sales order header for the item replacement order. If there's an applicable commitment in the sales agreement that hasn't been fulfilled yet, and the sales order is created before the sales agreement expires, a link is established between the sales agreement line and the sales order line. Therefore, information from the sales agreement, such as item price, is copied to the new sales order line.

Last updated on 05/07/2025

Delete a return order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the **Delete return order** form to delete multiple return orders.

1. Go to **Sales and marketing** > **Periodic** > **Clean up** > **Delete return orders**.
2. Select **Select** to open the **Sales update** form.
3. Specify the criteria, and then select **OK** to return to the **Delete return order** form. Return orders that have been created in the legal entity are displayed, based on the specified criteria.
4. Select the return order line to delete, and then select **OK**.
5. Select **Yes** to delete the return orders.

Last updated on 05/07/2025

Pass returned items on to inspection

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When registering a returned item, you may determine that an item should be sent for inspection before it's returned to inventory or disposed of in some other way.

1. Go to **Inventory management** > **Journals** > **Item arrival** > **Item arrival**.

   -or-

   Go to **Inventory management** > **Journals** > **Item arrival** > **Production input**.

2. On the **Location journal** form, register the receipt of an item as usual.

   > **Note**
   >
   > For information about registering the receipt of returned items, see Register the receipt of returned items.

3. On the **Default values** tab, in the **Mode of handling** area, select the **Quarantine management** box.

This prompts the system to create a quarantine order, and the person or department that performs inspections respond to this order using the **Quarantine order** form.

Related information

- Take returned items through inspection
- Specify how to dispose of returned items

Last updated on 05/07/2025

Post arrival journal for returned products

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To process a return, first validate the return quantity, update the quantity field in the item arrival journal. Then select a disposition code or indicate that the returned items have to be inspected. After completing these steps, you can post the item arrival journal for the return order.

1. Go to **Inventory management** > **Periodic** > **Arrival overview**.
2. In the **Setup name** filter, select **Return order**.
3. If the list of receipts is long, use the fields in the **Range** area to narrow the list.
4. Locate the line of the return order that you want to post, select its **Select for arrival** box, and then select **Start arrival**.
5. Go to **Journals** > **Show arrivals from receipts** to open the **Location journal** page.

   > **Tip**
   >
   > To view detailed information, select a journal, and then select **Lines**.

6. Make any necessary updates, and then select **Post**.

After the journal is posted, the returned items are registered in inventory, and the **Return orders** form indicates that the items have arrived at the warehouse.

Last updated on 05/07/2025

Receive partial deliveries of returned items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Partial deliveries are defined in terms of return order lines, not return order shipments.

This means that if you receive the full quantity that is indicated on one return order line, but you receive nothing from the other lines in the return order, the delivery is not a partial delivery. However, if a return order line requires 10 units of a particular item to be returned, but you receive only four, it is a partial delivery.

If a return shipment contains less than the full quantity of a return order line, you can set the shipment aside and wait for the rest of the returned quantity to arrive, or you can register and post the partial quantity.

Register and post a partial quantity

1. After you select a return order for arrival on the **Arrival overview - Warehouse: %1, Dock: %2, Journal name: %3** page, select **Start arrival** to create the arrival journal, and then select **Journals** > **Show arrivals from receipts** to open the **Location journal** page.
2. Select the line of the journal that you want to work with, and then select **Lines** to open the **Journal lines, locations** page.
3. Select the line of the item number for which only a partial quantity has arrived, and then select **Functions** > **Split** to open the **Split** page.
4. In the **Split quantity** field, enter the quantity for the total number of items that have been received, and then select **OK**.
5. On the **Journal lines, locations** page, select the line for the quantity of items that has arrived, and then select **Post**. You can post the line for the additional quantity after the items have arrived.

Related information

- Receive unannounced sales returns

- Receive returned items

Last updated on 05/07/2025

Receive returned items

When a return shipment arrives, you match it to a return order that has already been created.

Depending on company policies, the items in the return shipment can be sent to quarantine for additional inspection or they can be returned to inventory.

Related information

- Register the receipt of returned items
- Post arrival journal for returned products
- Take returned items through inspection
- Receive unannounced sales returns

Last updated on 07/01/2026

Register the receipt of returned items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

There are two methods for registering the receipt of returned items. The first method is a warehouse receiving process that uses the **Arrival overview** page. The second uses the **Registration** page.

Register the receipt of returned items in the Arrival overview page

You can use the **Arrival overview** page to identify a return shipment by its Return Material Authorization (RMA) number. If a journal name is defined on the **Setup** tab, and journal lines that correspond to the lines selected on the **Arrival overview** page exist, a new journal header is created when you select **Start arrival**.

1. Go to **Inventory management** > **Periodic** > **Arrival overview**.
2. In the **Setup name** field, select **Return order**, and then select **Update**.

> **Tip**
>
> You can use the **Range** fields to narrow the search results. You can type or select the RMA number in the **RMA number** field for an exact result. To define and save a set of filters that will restrict which incoming arrivals are displayed, select the **Setup** tab. The available filters include types, warehouses, and docks.

> **Warning**
>
> Return orders cannot be mixed with other arrival types in the **Arrival overview** page. Therefore, the reference will always be sales order, but no sales order ID will be specified on the journal header.

3. In the **Receipts** grid, locate the line that matches the item being returned, and then select the check box in the **Select for arrival** column.
4. To exclude certain lines from the return receipt, such as items from the original order that were not included in the return shipment, clear the associated **Select for arrival** check boxes in the **Lines** table in the lower part of the page.
5. Select the **Start arrival** button to create an arrival journal.

   > ⓘ **Note**
   >
   > You can select multiple return orders and include them all in a single arrival process. Each return order line will be copied into a corresponding item arrival journal line.

   > ⓘ **Note**
   >
   > You can also manually create an arrival journal by using the **Item arrival** page.

6. Go to **Inventory management** > **Journals** > **Item arrival** > **Item arrival**.
7. Select the arrival journal that you just created and then select **Lines** to open the **Journal lines, locations** page.
8. On the **General** tab, adjust the number in the **Quantity** field, if it is required, and then assign a disposition code in the **Disposition code** field.

   Alternatively, you can select the **Quarantine management** check box to have the returned items sent through an inspection process in the context of a quarantine order.

   > ⓘ **Note**
   >
   > If you send the returned items through quarantine, you cannot specify a disposition code.

9. Select the **Validate** button.
10. If there are no errors in the validation process, select **Post**.

Register the receipt of returned items in the Registration page

As an alternative to using the **Arrival overview** page, you can use the **Registration** page to register the arrival of returned items.

1. Go to **Sales and marketing** > **Sales returns** > **All return orders**. Create a new return order or open the return order from the list. On the **Lines** FastTab, select the return order line. Select **Update line**, and then select **Registration**.

2. Assign a disposition code in the **Disposition code** field, and then select **OK**.

   > **Note**
   >
   > It is not possible to send the item for inspection as a quarantine order using the **Registration** page.

3. In the **Transactions** grid, select the transaction to be registered.

4. In the **Register now** grid, select **Add**. Repeat the previous two steps until all of the returned items appear in the **Register now** grid.

5. Select **Post all**.

Last updated on 05/07/2025

Take returned items through inspection

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

1. Go to **Inventory management** > **Periodic** > **Quality management** > **Quarantine orders**.
2. Locate the order line that corresponds to the returned item that you are inspecting.

   > **Note**
   >
   > A quarantine order can be associated with just a single item number. If 10 items that have different item numbers are returned in a single shipment and sent to quarantine, 10 individual quarantine orders are created.

3. After examining the item, make a selection in the **Disposition code** field to indicate what should be done with the item and how to handle the related financial transaction. Examples include returning the item to stock and refunding the customer, scrapping the item and sending a replacement to the customer, or returning the item to the customer without credit.

   > **Note**
   >
   > If multiple returned items in a single item number batch cannot be assigned the same disposition code, you must split the quarantine order (**Functions** > **Split**) to assign a different disposition code to each sub-batch.

4. When you are finished with the inspection, select **Report as finished** to release the returned items and create an item arrival journal entry. The person or department that receives the items then processes the journal for the items to be returned to inventory.

   –or–

   End the quarantine order, and move the items back into inventory directly by using one of the **Inventory** functions.

5. Close the form to save your changes.

Related information

- Specify how to dispose of returned items

Last updated on 05/07/2025

Packing slip updates for returns

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Before returned items can be received into inventory, the packing slip for the order to which they belong must be updated. Just as the invoice update process is the update to the financial transaction, the packing slip update process is the physical update of the inventory record, which means that it commits the changes to inventory. In the case of returns, the steps that are assigned to the disposition action are implemented during the packing slip update.

The packing slip update can be processed in either the item arrival journal or the return order.

As part of the process for posting packing slips, the packing slip reference number from the customer's shipping documents can be associated with the order lines. This association is optional and for reference only. It does not create any transactional updates.

If not all of the expected return items have arrived, you should include only the quantity that has been received in the packing slip update. Leave the remaining items on the order until the rest of the return shipment has arrived.

If an item is sent back from quarantine to the Shipping and Receiving department, such as when the quarantine inspector does not know where to store the item in inventory, the corresponding packing slip must be updated to correctly register and act on the disposition code that is specified as a result of the quarantine.

When you update a packing slip for a returned item that is from a sales agreement, the sales agreement commitment for that item is automatically updated to reflect the change in the quantity or the amount.

Related information

- Perform invoice updates for returns

Last updated on 05/07/2025

Perform invoice updates for returns

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

A return order is a type of sales order that is marked as a returned order. Therefore, the **All sales orders** list page is used to generate invoices for return orders instead of the **Return orders** form. This functionality supports the business processes of organizations that choose to have return orders and sales orders invoiced at the same time and by the same person.

Because the invoice for a returned item is for a negative amount, it is called a credit note.

When you set up the invoice update for batch processing, the sales order of type **Returned order** must have a return line status of **Received**, which indicates that the order's packing slip has been updated.

Post an invoice for a return order

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Select a sales order for which **Returned order** is displayed in the **Order type** field.
3. On the Action Pane, on the **Invoice** tab, in the **Generate** group, select **Invoice**.
4. On the **Parameters** tab, select the **Posting** check box.
5. Review information in the form and make any changes that are needed.
6. Select **OK**. The credit note is posted.

Related information

- Packing slip updates for returns

Last updated on 05/07/2025

Register sales commissions

Article • 01/31/2023

This article explains how sales commissions are calculated and registered. You can run this procedure in demo data company USMF or on your own data. Before starting this guide, run the guide called "Set up sales commission rules" to make sure that you have all the necessary commission calculation setup.

Take note of the customer and item numbers that you have chosen for the commission process and use them when asked to create a sales order in this guide.

Invoice a sales order that qualifies a salesperson for a commission

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New**.
3. In the **Customer account** field, select the desired record from the drop-down menu.
4. Select **OK**.
5. On the Action Pane, select **Options**.
6. Select **Change view**.
7. Select **Header view**.
8. Expand the **Setup** section.

    - The value in the **Sales group** field represents a group with one or more sales representatives assigned to it. The people in the group are the ones who will receive commissions when the order is invoiced, as per predefined rates and distribution.
    - The value is copied from the Customer card, but you can change it if you wish.
    - The Sales group is also copied to the sales order line. You can change it so that it can differ from the one in the header and/or between lines.
    - The value in the **Commission group** field represents a group that you have created for one or more customers with the purpose of tracking commissions.

- The value is copied from the Customer card, but you can change it if you wish.

9. On the Action Pane, select **Options**.
10. Select **Change view**.
11. Select **Line view**.
12. In the drop down menu of the **Item number** field, select the item you have set up for commissions.
13. In the **Quantity** field, enter a number. Take note of the line's Net amount. It represents the sales revenue, which in this example is the basis for commission calculation.
14. Select **Save**.
15. On the Action Pane, select **Invoice**.
16. Select **Invoice**.
17. Expand the **Parameters** section.
18. In the **Quantity** field, select **All**.
19. Select **Yes** in the **Posting** field.
20. Select **OK**, then select **OK** in the next pane. It may take a minute or so to post the transaction. Allow the processing to complete and don't close the page.

Review the registered sales commissions

1. On the Action Pane, select **Invoice**, then select **Invoice** again.
2. On the Action Pane, select **Invoice**, then select **Commission transactions**.

    - The **Overview** tab displays lines representing the commission amounts payable to sales representatives who are associated with the invoiced sales order. Let's review the details.
    - If you used the "Set up sales commission rules" guide to set up the **Commission sales** group, there are two sales people to receive a sales commissions, and the commission is split equally between them.
    - In this example, the total amount of the commission is calculated as a percentage of the sales revenue (the net amount of order line).

3. Close the page.

4. Select **Voucher**. You can review the voucher transactions for the commission amounts that have been posted to the predefined commission expense and commission payable accounts.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up sales commission rules

This procedure shows you how to set up and enable sales commission calculation and tracking. The procedure shows how to create both customer and item commission groups, and then how to link a selected customer and product to the respective groups. Those groups are then used in the commission calculation setup to create a customer, item, and sales representatives combination that must be matched by the sales order to entitle the sales people to a commission. Creating customer and item commission groups are optional, as the calculation of commission can also be done for an individual customer and/or item. You can run this procedure in demo data company USMF or on your own data.

Set up commission groups and commission rates

1. Go to **Sales and marketing** > **Commissions** > **Customer groups for commission**.
2. Select **New**.
3. In the **Group** field, enter a value.
4. In the **Name** field, type a value.
5. Select **Save**.
6. Close the page.
7. Go to **Sales and marketing** > **Commissions** > **Item groups**.
8. Select **New**.
9. In the **Group** field, enter a value.
10. In the **Name** field, type a value.
11. Select **Save**.
12. Close the page.
13. Go to **Sales and marketing** > **Commissions** > **Sales groups**.

    - A Commission sales group specifies the employees in sales representative roles who are eligible to receive a commission when a customer associated with the relevant sales group buys certain items.
    - In the USMF demo data company, there's a sales group called *Sales reps US*.

14. On the Action Pane, select **General**.
15. Select **Sales rep.**. The Sales rep. page displays a list of the company's salespeople who are associated with a specific commission group. You can assign multiple sales representatives to the same group and define their respective share of the total commission fee as a percentage value. The total commission share across all employees must not exceed 100.

16. In the list, mark the selected row.
17. Select **Edit**.
18. Set **Commission share** to *50*.
19. Select **New**.
20. In the **Name** field, select the drop-down button to open the lookup.
21. Use the **Quick Filter** to find records. For example, filter on the Name field with a value of *Susan Burk*.
22. Select **Select**.
23. Set **Commission share** to *50*.
24. Select **Save**.
25. Go to **Sales and marketing** > **Commissions** > **Commission calculation**. In the **Commission calculation** page, you define the commission rate that the employee receives for a sales transaction when it contains the preset combination of customer and product. As part of the commission rate setup, specify the commission calculation basis and whether it should include or exclude discounts. You can also enter a validity period for when the commission rate is active.
26. Select **New**.
27. In the **Item code** field, select *Group*.
28. In the **Item relation** field, select the drop-down button to open the lookup.
29. In the list, find and select the group that you created earlier.
30. In the **Customer code** field, select *Group*.
31. In the **Customer relation** field, select the drop-down button to open the lookup.
32. In the list, select the group that you set up earlier.
33. In the **Sales rep. relation** field, select the drop-down button to open the lookup.
34. In the list, find and select the desired record.

    - Keep the *Before line discount* option.
    - Keep the *Revenue* option as the basis for commission value calculation.

35. In the Commission percentage field, enter a number.
36. Select **Save**.

Setting up commission posting

1. Go to **Sales and marketing** > **Commissions** > **Commission posting**. Employees receive commission fees, so set up this feature to ensure correct financial posting to the appropriate accounts in the **General ledger**. Set up this feature in the **Commission posting** page. Review the setup that's available for the current company. Typically, the commission

amounts are posted to a dedicated expense account and are offset to a dedicated payable account. If you don't have the commission posting rules set up, the system will fail to complete invoicing of a sales order that has eligible commissions.

2. Close the page.

Assign a commission group to a customer and a product

1. Go to **Sales and marketing** > **Customers** > **All customers**.
2. In the list, find and select the desired record.
3. In the list, select the link in the selected row.
4. Select **Edit**.
5. Expand the **Sales order defaults** section.
6. In the **Commission group** field, select the drop-down button to open the lookup.
7. In the list, select the group that you created earlier.
8. In the **Sales group** field, select the drop-down button to open the lookup.
9. In the list, find and select the desired record.
10. Select **Save**.
11. Go to **Product information management** > **Products** > **Released products**.
12. Use the **Filter** to find records. For example, filter on the Item number field with a value of *T0020*.
13. In the list, select the link in the selected row.
14. Select **Edit**.
15. Expand the **Sell** section.
16. In the **Commission group** field, select the drop-down button to open the lookup.
17. In the list, select the commission group that you created earlier.
18. Select **Save**.

Last updated on 07/01/2026

Customer portal for Dynamics 365 Supply Chain Management overview

What is the Customer portal?

Modern supply chain systems rely on integration. They require that inventory, customer demand, and sales departments be integrated instead of residing in separate silos. The Customer portal helps organizations that run Microsoft Dynamics 365 Supply Chain Management enhance this integration and more effectively keep their customers informed.

The Customer portal is a Power Apps portals template that lets companies create an externally facing business-to-business (B2B) website for scenarios that are related to sales order processing. The template uses dual-write, Supply Chain Management, and Power Apps portals to enable external enterprise customers to view and create data from the company's Dynamics 365 environment.

The Customer portal template has all the customization capabilities that the portals feature of Power Apps offers. The template can easily be modified to represent the company's brand, add increased functionality, and change the user experience. All the functionality that the template offers out of the box can be modified as desired.

> **Important**
>
> By itself, the template isn't expected to be completely functional. It just serves as an enabler for customers who want to create an externally facing website so that enterprise customers can engage with data from Supply Chain Management.

> **Note**
>
> The Customer portal documentation is directed at admins, customizers, and system integrators who set up the Customer portal for a Supply Chain Management installation. It uses the terms *customer* and *user* to describe people who are customers of the organization that is running Supply Chain Management, and who use the final portal itself.

Video

https://learn-video.azurefd.net/vod/player?id=b39bd847-38bc-4a21-af49-a86795085fc5&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fsales-marketing%2Fcustomer-portal-overview

The Overview of the Customer portal template in Dynamics 365 Supply Chain Management video is included in the finance and operations playlist available on YouTube.

Who should use it?

The Customer portal is designed for companies that run Supply Chain Management and have these characteristics:

- They want to build an externally facing website that communicates order processing information (such as order status or account information) directly from their Supply Chain Management system to their enterprise customers.
- They're transitioning from Dynamics AX 2012 to Supply Chain Management and previously used the AX 2012 Customer self-service portal.

The following types of organizations aren't good candidates for implementing the Customer portal:

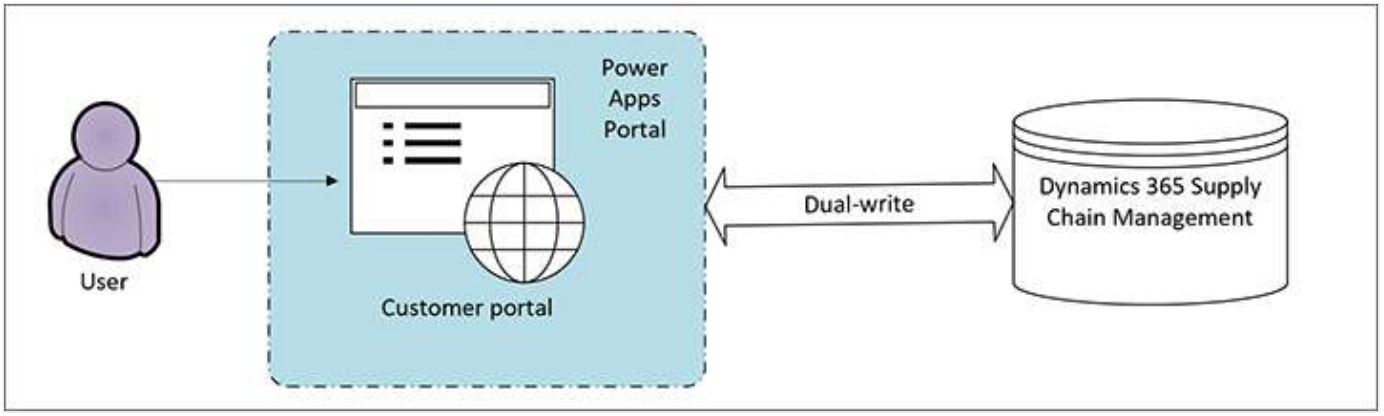
- Companies that want to build a website for non-enterprise customers. These companies should consider creating a Dynamics 365 Commerce e-commerce website.
- Companies that are already using an existing Power Apps portals website for a similar purpose. These companies don't receive any additional benefits from the Customer portal. The Customer portal is delivered as a template that acts as a guide and a starting point for customers who want to "connect the dots" between dual-write, Supply Chain Management, and Power Apps portals. If you already set up a website that serves this purpose, you might not gain much value from using the Customer portal template to reprovision that website.

How does it work?

The Customer portal is provided as a Power Apps portals template. It depends on Power Apps portals and dual-write.

Power Apps portals is a feature that lets users create an externally facing website that people from outside the organization can sign in to. Little to no coding is required to make portals. The Customer portal is one of many Dynamics 365 portal templates that are available from Microsoft.

Dual-write is an out-of-box infrastructure product that provides near-real-time interaction between customer engagements apps and finance and operations apps. Dual-write provides bidirectional integration between finance and operations apps and Microsoft Dataverse. Therefore, it provides an integrated user experience across the apps. The Customer portal depends on tables that are synced with dual-write. Before data from Supply Chain Management can be surfaced in the Customer portal, dual-write must be enabled for all the appropriate tables.



The Customer portal acts as a starting point for organizations that want to use Power Apps portals to build an externally facing website that uses data from their Supply Chain Management installation. It helps organizations connect dual-write, Supply Chain Management, and Power Apps portals.

Last updated on 07/01/2026

Install, set up, and update the Customer portal

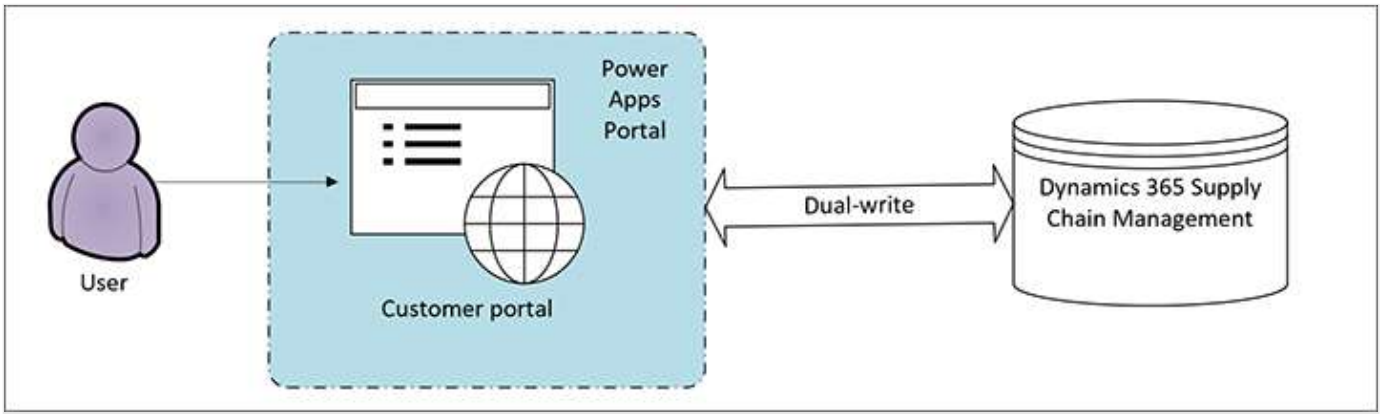
Licensing requirements

To implement the Customer portal, you need the following licenses:

- **Power Apps portals** – This license is required to host the Customer portal. Portals are licensed based on usage. For more information, see the Power Apps portals licensing requirements.
- **Dual-write** – You need the necessary licenses to enable dual-write for Supply Chain Management tables. For more information, see the system requirements for dual-write.

Dependencies on dual-write and Power Apps portals

The Customer portal depends on Power Apps portals and dual-write, as shown in the following illustration.



Unlike other features from Supply Chain Management, the Customer portal template resides in Power Apps portals. Therefore, the Customer portal is limited by the functionality and capabilities that are provided by Power Apps portals and the tables in dual-write.

Required setup to enable the Customer portal

After you ensure that you have the required licenses, set up dual-write as described in the dual-write initial synchronization instructions.

Be sure to enable the following table mappings in dual-write:

- Sales Order Header
- Sales Order Details
- Accounts
- Contacts
- Products

When you complete this setup, you can provision the Customer portal template.

Provision the Customer portal

Before you begin, make sure that you complete the required setup. Then follow these steps to provision the Customer portal.

1. Go to Power Pages.
2. Make sure that you're using the environment where you enabled dual-write.
3. On the **Templates** tab, open the **Dynamics 365** tab, and select the template named **Customer Portal**.
4. Follow the on-screen instructions.

After provisioning is completed, you can access the Customer portal on the **Active Sites** tab of the **Home** page.

> **Note**
>
> If the dual-write solution isn't installed in the environment that you're working in, you receive an error message, and the Customer portal isn't provisioned.

Update the Customer portal

Any changes that Microsoft makes to the underlying solution components automatically appear in your environment. However, the website that is provisioned in your environment doesn't automatically reflect changes made to the configuration data. You must manually apply those changes by getting the code from the new template and merging it with the provisioned website.

Related information

To learn how you can set up and customize the Customer portal, start by reviewing the following documentation for the underlying technologies:

- Power Pages documentation
- Dynamics 365 templates
- Dual-write documentation

To effectively manage your sites, you must understand the Power Pages and Microsoft Dataverse lifecycle. For more information, see the following resources:

- About Power Pages lifecycle
- Upgrade a Power Pages site
- Migrate website configuration
- Solution Lifecycle Management: Dynamics 365 for Customer Engagement apps

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Last updated on 07/01/2026

Create and manage Customer portal users

In the out-of-box implementation, users can't self-register for websites created by using the Customer portal. To sign in and use a website, users must be invited by the admin. Users are intentionally blocked from self-registering.

Before a user can use a website, a contact record must be created for that user. This record indicates which customer account and legal entity the user belongs to. This information is essential for ensuring that the user can create and view sales orders.

When users self-register, contact records are automatically created for them. Therefore, you can't ensure that a user selects the correct customer account and legal entity. On the other hand, the invitation process lets an admin assign the correct customer account and legal entity to the contact record before an invitation is sent. If you're thinking about customizing the solution so that users can self-register, consider the possible consequences.

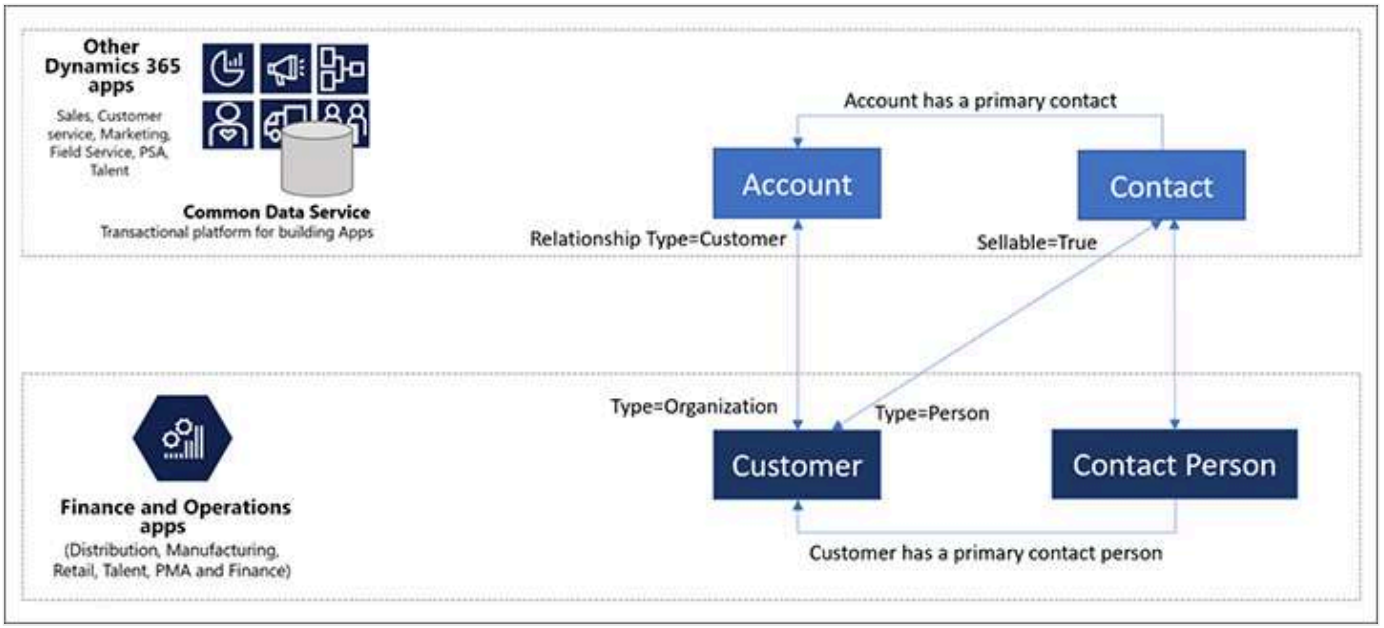
Video

https://learn-video.azurefd.net/vod/player?id=5a9d44db-29b0-429b-ad96-22951664353c&locale=en-us&embedUrl=%2Fdynamics365%2Fsupply-chain%2Fsales-marketing%2Fcustomer-portal-user-admin

The Invite customers to register and use your customer portal video (shown earlier) is included in the finance and operations playlist available on YouTube.

Prerequisite setup

Power Apps portals store contacts as records in the **Contacts** table in Microsoft Dataverse. Dual-write then syncs these records to Microsoft Dynamics 365 Supply Chain Management as required.



Before you start to invite new customers, make sure that you enable the **Contact** table mapping in dual-write.

The invitation process

To invite an existing contact to the Customer portal, follow the steps in Invite contacts to your portals in the Power Apps portals documentation.

Before you invite a customer to join the Customer portal, make sure that the customer's contact record is available and set up in the following way:

1. Set the **Company** field to the legal entity that you want the customer to belong to in Supply Chain Management.
2. Set the **Account Number** field to the customer account number that you want the user to have in Supply Chain Management.

After a contact is created, you should be able to see it in Supply Chain Management.

Learn more in Configure a contact for use on a portal in the Power Apps portals documentation.

Out-of-box web roles and table permissions

User roles in Power Apps portals are defined by web roles and table permissions. A few roles are defined for the Customer portal out of the box. You can create new roles, and you can modify or remove existing roles.

Out-of-box web roles

This section describes the web roles delivered with the Customer portal.

For more information about how to modify the out-of-box user roles, see Create web roles for portals and Add record-based security by using table permissions for portals in the Power Apps portals documentation.

Administrator

The administrator oversees and maintains the website. This person provisions and sets up the Customer portal. The administrator maintains the IT and security aspects of the portal, and makes sure that everything runs smoothly. The administrator might also customize and/or change the portal by adding new functionalities, creating new roles, and more.

Customer representative

A customer representative works for a customer company and is responsible for managing the orders that the company places. The customer representative can see all the orders that the company places and the users who placed them. Additionally, the customer representative can see information about the overall account and which contacts can place orders on behalf of that account.

Authorized users

Authorized users can place orders and view the status of the orders that they place. However, they can't view the status of orders that other users in their company place.

Unauthorized users

Unauthorized users can't view any data. They can see only public information, such as terms and conditions, and details about the company that is running the Customer portal.

Example

The following table shows which sales orders the users in each web role can see in the system.

Expand table

Sales order	Administrator	Customer representative for customer X	Authorized user: Jane	Authorized user: Sam	Unauthorized user: May
Customer X	Yes	Yes	Yes	No	No

Sales order	Administrator	Customer representative for customer X	Authorized user: Jane	Authorized user: Sam	Unauthorized user: May
Orderer: Jane					
Customer X Orderer: Sam	Yes	Yes	No	Yes	No
Customer Y Orderer: May	Yes	No	No	No	No

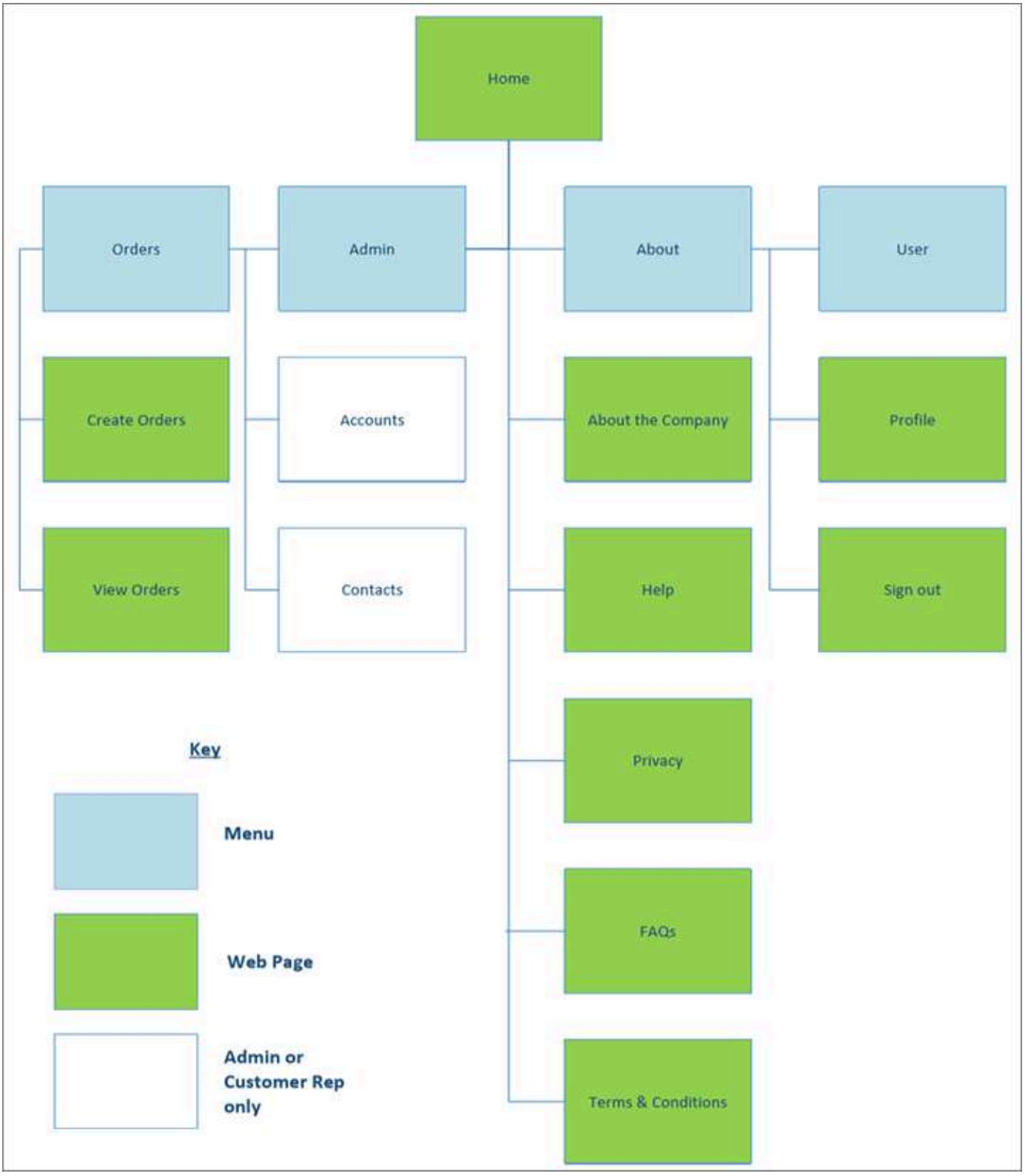
> **Note**
>
> Even though both Sam and Jane are contacts who work for customer X, they can see only the orders that they themselves have placed and nothing else. Although May has an order in the system but can't see that order in the Customer portal, because May is an unauthorized user. (Additionally, May must have placed the order through some channel other than the Customer portal.)

Last updated on 07/01/2026

Customize and use the Customer portal

This article describes the different pages that available in the Customer portal out of the box. It explains what the pages do and how you can customize them.

The Customer portal offers a few webpages and actions out of the box. The following site map provides an overview of those webpages and actions, and the roles that can perform the actions.



Typical customizations

The following articles will help you learn the basics about Power Apps portals and how you can customize portals:

- Work with templates – This article provides a general overview of how Power Apps portals works and how you can do simple customizations of portals.
- Manage portal content – This article explains how you can manage and customize the content that you surface in your portal.
- Edit CSS – This article helps you make more complex customizations to the user interface (UI) of your portal.
- Create a theme for your portal – This article helps you create a UI theme for your portal.
- Create and expose portal content easily – This article helps you manage the underlying data and tables that you use for your portal.
- Configure a contact for use on a portal – This article explains how to create and customize user roles, and how security and authentication work in Power Apps portals.
- Configure notes for table forms and web forms on portals – This article explains how to add documents and additional storage to your portal.
- Error handling for portal website – This article explains how to view portal error logs and store them in your Microsoft Azure Blob storage account.

Customize the order creation process

When a user submits an order by using the Customer portal, the order is automatically synced to the corresponding Dynamics 365 Supply Chain Management environment. Because the user is an external customer, some required information is intentionally hidden from them. This information will automatically be filled in when the form is submitted.

This section shows how you should set up contacts to avoid errors. It explains fields that are automatically set and how you can modify the value of those fields if you must.

The out-of-box order creation process

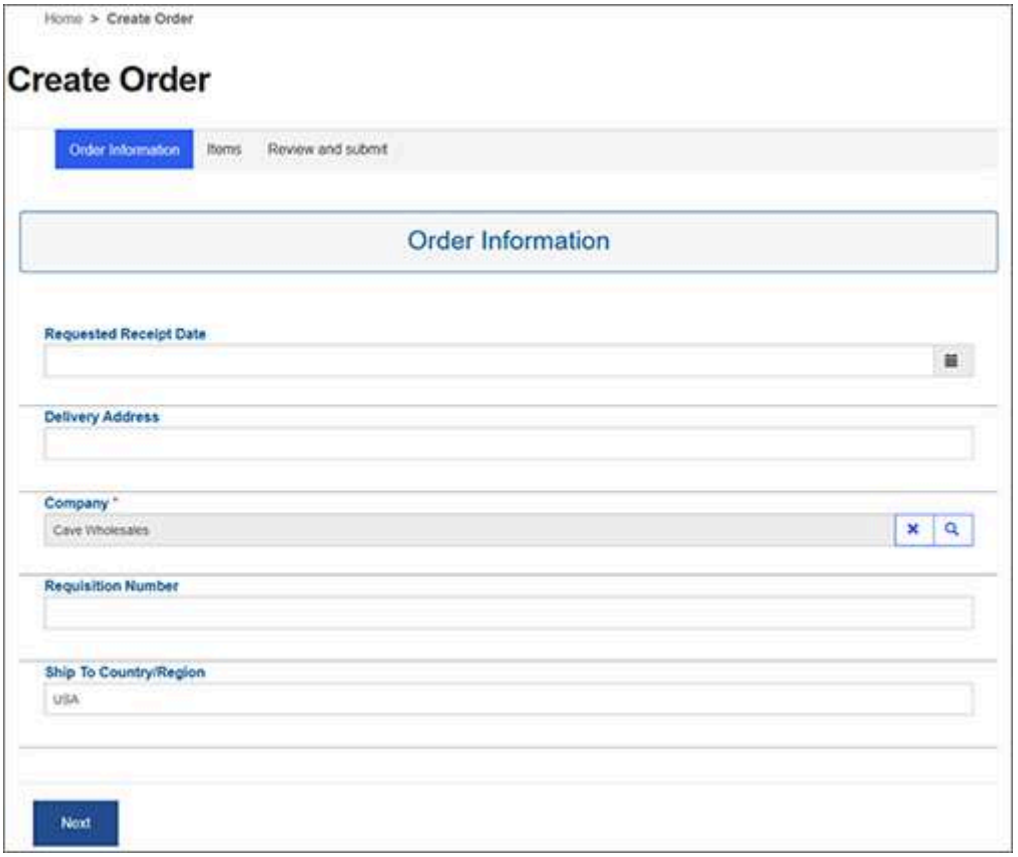
Here are the standard steps for submitting an order from the Customer portal.

1. On the home page, select the **Create order** tile to open the **Create Order** wizard.

2. On the **Order Information** page, set the following fields:

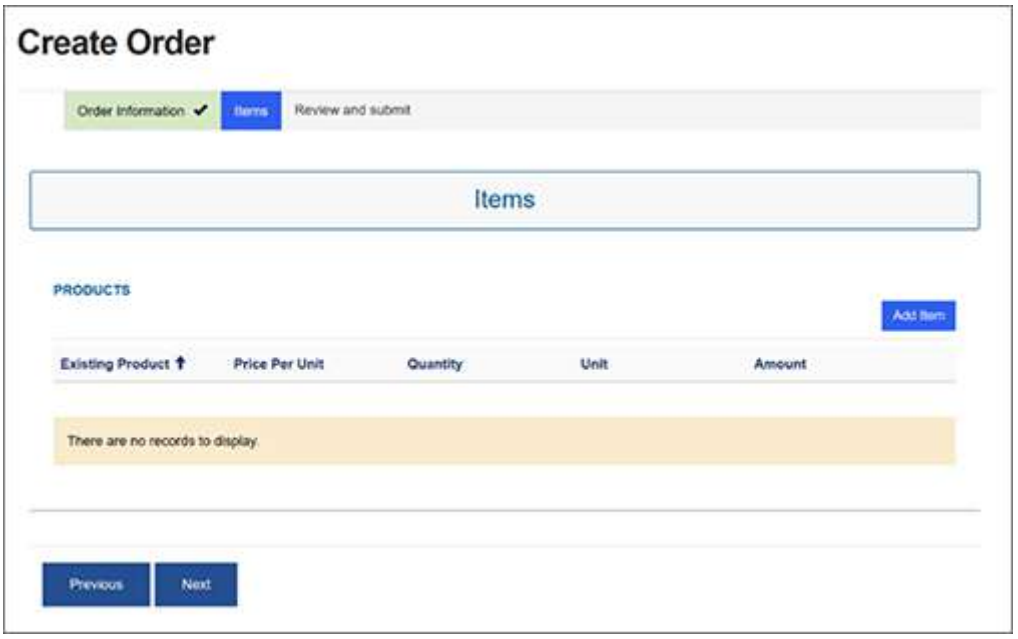
    - **Requested receipt date** – Specify the date of delivery.

- **Delivery address** – Enter the address that the order should be delivered to.
- **Company** – Select the name of the customer company. This field is automatically set for non-admin users.
- **Requisition number** – Enter the requisition number of the order. This field isn't required.
- **Ship to country/region** – Enter the country or region that the items will be delivered to. This field is automatically set for non-admin users.



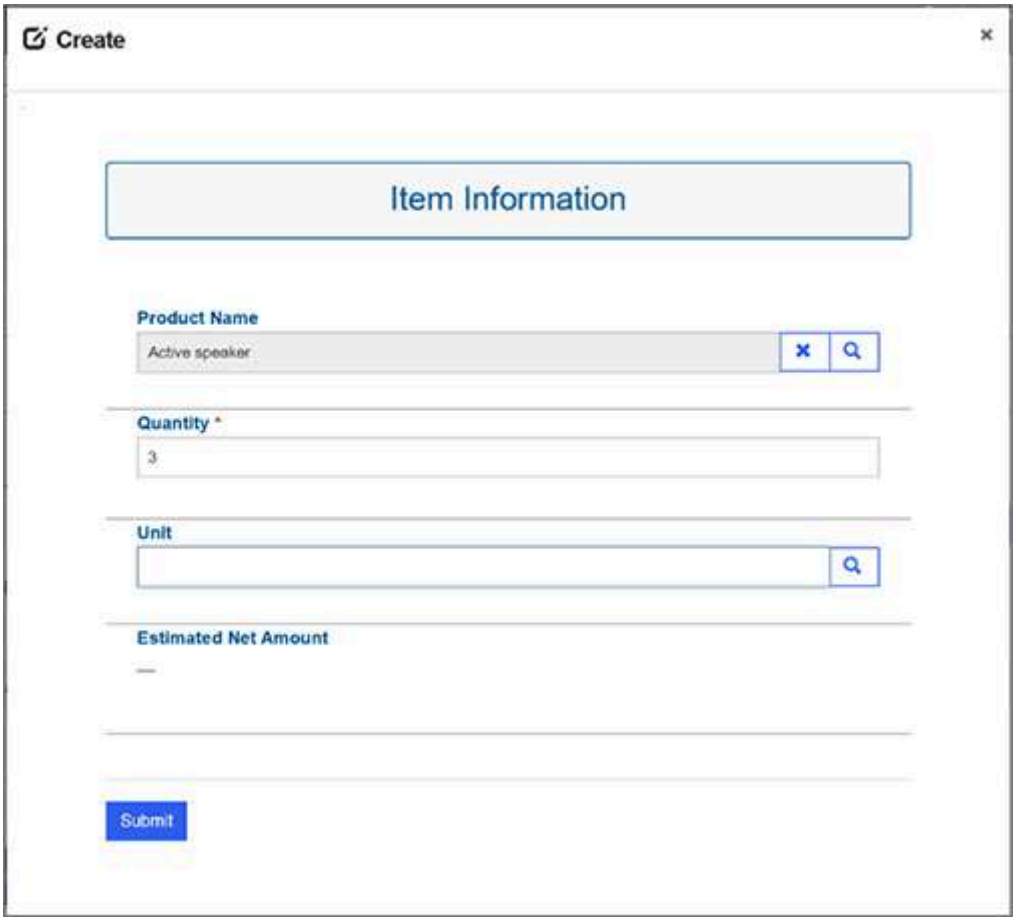
3. Select **Next**.

4. On the **Items** page, select **Add Item**.



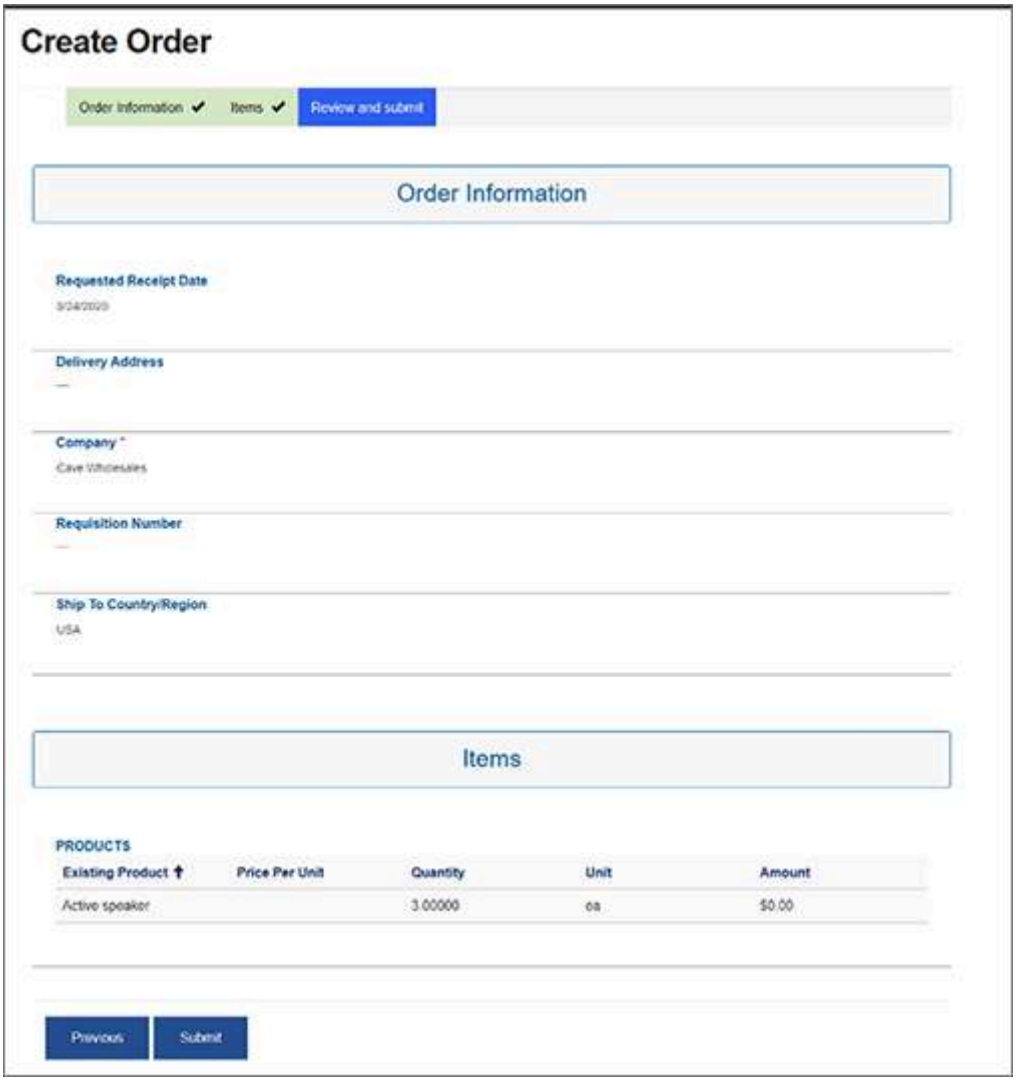
5. In the **Item Information** dialog box, set the following fields:

- **Product Name** – Find and select a product to add to the order.
- **Quantity** – Enter the quantity of the selected product.
- **Unit** – Specify the unit of measure (for example, **ea.**, **kgs**, or **box**).
- **Estimated net amount** – The value is calculated as the estimated price of the item × the quantity for the selected unit.



6. Select **Submit** to add the item to the order.

7. Repeat steps 4 through 6 until you've added all the items that you want to order.
8. When you've finished adding items, select **Next** on the **Items** page.
9. The **Order Information** page provides a summary of the order. Review the order contents and delivery details. If everything looks correct, select **Submit** to submit the order.



Standard data setup

To help ensure a smooth user experience, the Customer portal automatically fills in values for several required fields. These values are based on information in the contact record of the customer who is submitting the order.

For each contact row that belongs to a customer who will use the Customer portal to submit orders, values must be specified for the following required fields. Otherwise, errors will occur.

- **Company** – The legal entity that the order belongs to
- **Potential customer** – The customer account that is associated with the order
- **Price list** – The custom price list for the customer

- **Currency** – The currency of the price
- **Ship to country/region** – The country or region that the items will be delivered to

The following fields are automatically set for the sales order table:

- **Language** – The language of the order (By default, the value is taken from the contact record.)
- **Ship to country/region** – The country or region that the items will be delivered to (By default, the value is taken from the contact record.)
- **Contact person** – The user who can be contacted for information about the order (By default, the value is taken from the contact record.)
- **Company** – The legal entity that the order belongs to (By default, the value is taken from the contact record.)
- **Potential customer** – The customer account that is associated with the order (By default, the value is taken from the contact record.)
- **Invoice customer** – The billing account of the order (The default value is the potential customer from the contact record.)
- **Sales order name** – The name of the sales order (The default value is **sales order**.)
- **Currency** – The currency of the price (By default, the value is taken from the contact record.)
- **Price list** – The custom price list for the customer (By default, the value is taken from the contact record.)
- **Delivery address description** – The delivery address of the sales order (The default value is **delivery address description**.)

Modify the order creation process

You can freely modify the appearance and UI of the Customer portal if you don't change the basic order creation process. If you want to change the order creation process, there are a few points that you must keep in mind.

Don't remove the following columns from the sales order table in Microsoft Dataverse, because they are required to create a sales order in dual-write:

- **Company** – The legal entity that the order belongs to
- **Name** – The name of the sales order
- **Currency** – The currency of the price
- **Price list** – The custom price list for the customer
- **Ship to country/region** – The country or region that the items will be delivered to
- **Potential customer** – The customer account that is associated with the order

- **Language** – The language of the order (Typically, this language is the language of the potential customer.)
- **Delivery address description** – The delivery address of the sales order

For items, the following columns are required:

- **Product** – The product to order
- **Quantity** – The quantity of the selected product
- **Unit** – The unit of measure (for example, **ea.**, **kgs**, or **box**)
- **Ship to country/region** – The country or region of delivery
- **Delivery address description** – The delivery address of the order

You must make sure that your Customer portal somehow submits values for all these columns.

If you want to add columns to the page, or remove columns, see Create or edit quick create forms for a streamlined data entry experience.

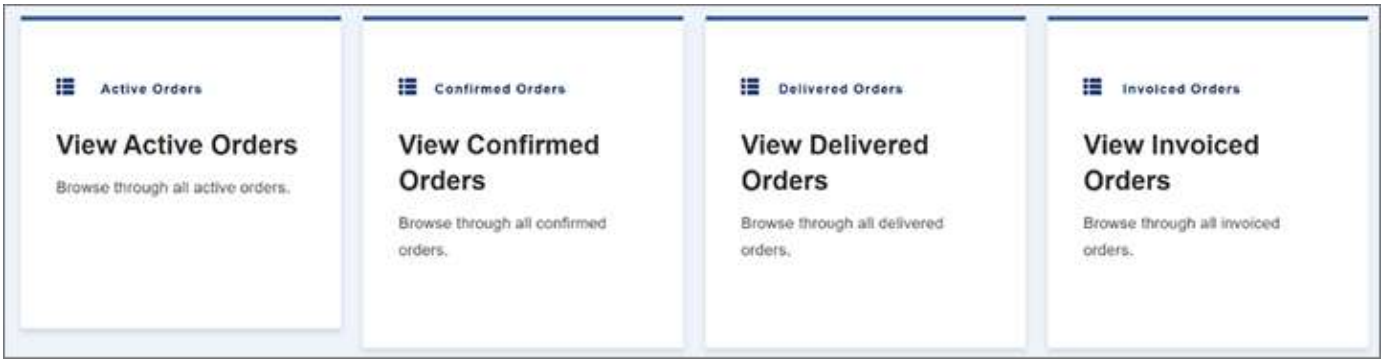
If you want to change how columns are preset and how values are set when the page is saved, see the following information in the Power Apps portals documentation:

- Prepopulate field
- Set Value On Save

Customize the home page

All the controls in the Customer portal are built-in Power Apps portals controls. You can customize them by following the steps in Compose a page in the Power Apps portals documentation.

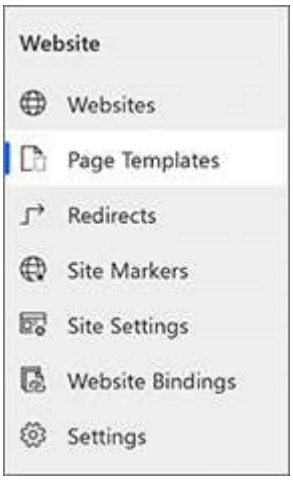
The only custom control that is included in the Customer portal template is used to create the tiles on the home page.



To modify the tiles, follow these steps:

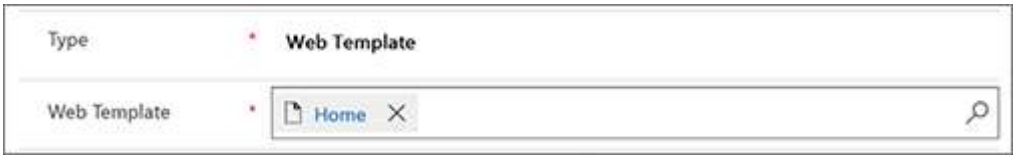
1. Open the Portal Management app.

2. In the navigation pane on the left, select **Page Templates**.



3. Select the page template that is named **Home**.

4. In the **Web Template** field, select the **Home** link to open the source code for that page.



5. You should now see all the source code for the home page and can modify it as you require.

Resources

To learn more about how you can set up and customize the Customer portal, see the following resources:

- Power Apps portals documentation
- Dual-write documentation
- About portal lifecycle
- Upgrade a portal
- Migrate portal configuration
- Solution Lifecycle Management: Dynamics 365 for Customer Engagement apps

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Last updated on 12/30/2025

Approved customer lists

You might sell products and services on a recurring basis to specific customers or groups of customers who are preapproved for these purchases based on their previous and ongoing business relationship. By preapproving customers, you expedite the sales order process and shipment of the product. In addition, you ensure that the product is sold only to customers who are authorized to buy it and prevent it from being sold to other customers.

For example, you're a manufacturer of apple juice. One of your long-time customers, Super Club, can buy only its own privately labeled products. Super Club sells these products at both its Super Club warehouse location and its retail location, Super Mart. Therefore, you sell Super Club's privately labeled apple juice cases only to Super Club Warehouse and Super Mart. This product can't be sold to any other customer.

Prerequisites

To use the features that this article describes, your system must meet the following requirements:

- Run Microsoft Dynamics 365 Supply Chain Management version 10.0.44 or later.
- Turn on the following features in Feature management. As of Supply Chain Management version 10.0.47, both features are turned on by default.
  - *Advanced quality management* (As of Supply Chain Management version 10.0.49, this feature is mandatory and can't be turned off.)
  - *Approved customer list*

Set up approved customers

To authorize an approved customer or customer group for recurring sales, or to limit who you can sell a given product to, use the **Approved customer list setup** page to set up a relationship between the customer or customer group and an item or item group.

As part of the process of setting up a customer/item relationship, you specify the type of relationship. The following types are available:

- A specific customer is authorized to buy a specific item.
- A specific customer is authorized to buy any item in a specific item group.
- A customer group is authorized to buy a specific item.

- A customer group is authorized to buy any item in a specific item group. (A relationship of this type is known as a *group-to-group* relationship.)

In addition, you set up the approved customer list (ACL) check method to define what occurs if a customer tries to buy an item that it isn't authorized to buy.

Set up the ACL check method

When you approve a product for a customer by setting up a customer/item relationship, you also set up the ACL check method at both the customer level and the item level. The ACL check method is used to validate the combination of the customer and the item. In other words, it confirms that the customer is authorized to buy the item.

To set up the ACL check method for products and customers, follow these steps:

- Go to **Product information management** > **Products** > **Released products**. On the **Sell** FastTab, in the **Approved customer list check method** field, select the method.
- Go to **Sales and marketing** > **Customers** > **All customers**. On the **Sales order defaults** FastTab, in the **Approved customer list check method** field, select the method.

The ACL check method defines the action that is taken if you select a customer and/or an item that isn't listed in the item's approved customer list. It also confirms that the transaction date is within the effective approval period that is specified for the customer and the item. You can set up the ACL check method either to issue a warning or to prevent the transaction from being processed.

The following ACI check methods are available:

- *No check* – No validation is done. Therefore, any customer or item that you select is allowed.
- *Warning only* – A warning message is shown, but you can continue with the transaction.
- *Not allowed* – An error message is shown, and you're prevented from continuing with the transaction.

The ACL check method is applicable to the following document types:

- Sales agreements
- Sales quotations
- Sales orders

Based on the ACL check method, only items that are authorized for the selected customer account are permitted for the document.

> **Note**
>
> The system validates sales quotations and sales agreements when they are confirmed. This behavior differs from sales orders, which are validated when they are added.

Set up approved customer lists

To set up and maintain customers who are authorized to buy one or more products, use the **Approved customer list setup** page. To open this page, go to one of the following locations:

- **Product information management** > **Setup** > **Approved customer list** > **Approved customer list setup**
- **Sales and marketing** > **Setup** > **Approved customer list** > **Approved customer list setup**

Then use the buttons and fields described in the following subsections.

Buttons on the approved customer list setup page

- **New** – Add a relationship between an approved customer or customer group and an approved item or item group.
- **Delete** – Remove a relationship between an approved customer or customer group and an approved item or item group.
- **Edit** – Edit the effective and expiration dates for a customer's approval period.
- **View** – Select a view option to find approved items or item groups within the customer's effective date range. The two view options also include approved items that never expire.
  - *Current* – Show records that have an effective date before the current date.
  - *All* – Show all records.

Fields on the approved customer list setup page

- **Item code** – Select or review the item relationship code that the item is approved for. The following values are available:
  - *Table* – A specific item.
  - *Group* – An approved item group.
  - *All* – All items.
- **Item relation** – Select or review the identifier of the approved item or item group.

- **Customer code** – Select or review the customer relationship code that the item is approved for. The following values are available:
  - *Table* – A specific customer.
  - *Group* – An approved customer group.
  - *All* – All customers.
- **Customer relation** – Select or review the identifier of the approved customer or customer group.
- **Effective** – Select or review the effective date when the approved customer list's approval to buy the item or items begins. The default value is the current date.
- **Expiration** – Select or review the date when the approved customer list's approval to buy the item or items expires. The default value is *Never*.

Set up approved customer groups and approved item groups

Set up groups of customers and groups of items to expedite the approval process when the same set of items is approved for a specific customer, a group of customers, or all customers. If the approved item group is associated with an approved customer group, every customer in that customer group is authorized to buy every item in that item group. To set up the groups, go to the following locations:

- **Sales and marketing** > **Setup** > **Approved customer list** > **Approved customer groups**
- **Product information management** > **Setup** > **Approved customer list** > **Approved item groups**

View and maintain approved items from the customer account

From the customer record, you have different options for viewing and maintaining approved items for the customer account.

1. Go to **Sales and marketing** > **Customers** > **All customers**.
2. Select the customer that you want to set up.
3. On the Action Pane, on the **Sell** tab, in the **Approved customer** group, select one of the following buttons:

- **Setup** – Open a view of the **Approved customer list** page that is filtered by the customer account. When you create a line item in this context, you just have to specify approved items or item groups. The customer account is fixed to the account that you opened the list from.
- **Approved items** – Open the **Approved customer list by customer** page, which shows the items that are approved for the customer account. The list on this page resembles the list on the **Approved customer list** page, but it shows the item numbers of all items that are defined in approved item groups. Use the **Effective** and **Expiration** fields to filter the list based on a specific effective date or expiration date.
- **Effective period** – Open the **Approved customer list expiration** page, which shows items that are approved for the customer. By default, the list shows approved items that have an expiration date, and the date in the **As of** field is set to the current date. Therefore, the list shows approved items where the approval is still effective as of the current date. You can change the date to a different date in the future. If you want the list also to show approved items that have no expiration date, select **Include never expired**.

View and maintain approved customers from the released product master

From the product master, you have different options for viewing and maintaining approved customers.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product that you want to set up.
3. On the Action Pane, on the **Sell** tab, in the **Approved customer** group, select one of the following buttons:

    - **Setup** – Open a view of the **Approved customer list** page that is filtered by the customer account. When you create a line item in this context, you only have to specify approved items or item groups. The customer account is fixed to the account that you opened the list from.
    - **Approved customers** – Open the **Approved customer list by item** page, which shows the customers that are approved for the item. The list on this page resembles the list on the **Approved customer list** page, but it shows the customer account numbers of all customers that are defined in approved customer groups. Use the **Effective** and **Expiration** fields to filter the list based on a specific effective date or expiration date.

- **Effective period** – Open the **Approved customer list expiration** page, which shows customers that are approved for the item. By default, the list shows approved customers that have an expiration date, and the date in the **As of** field is set to the current date. Therefore, the list shows approved customers where the approval is still effective as of the current date. You can change the date to a different date in the future. If you want the list also to show approved customers that have no expiration date, select **Include never expired**.

Configure date validation for the approved customer list

When you enter effective and expiration dates for an approved customer list record, validation is done only if the ACL check method is set to *Warning only* or *Not allowed*. (It doesn't validate them if the method is set to *No check*.) The dates are validated for the different documents, based on the setting of the **Date type** field on the **Sales and marketing parameters** page. This field specifies which date on the document is validated against the approved customer list record. The following values are available:

- *Today* – The current date or the system date.
- *Requested ship date* – The requested ship date on the sales order.
- *Requested receipt date* – The requested receipt date on the sales order.
- *Order date* – The date when the order was placed.

Therefore, you can validate the sales order against the approved customer list when the order is expected to be shipped, when it's expected to be received, or when it's placed. The *Today* and *Order date* values usually produce the same result.

Last updated on 07/27/2026

Credit limits for customers

Setting a credit limit lets you specify the maximum amount of credit to extend to your customers. If a credit limit is specified, it's checked automatically when a user attempts to update a document. If the credit limit is exceeded, a message is displayed to the user.

> **Important**
>
> The credit limit functionality described in this article is only available when the *Credit management* feature is *turned off* in Feature management. *Credit management* is on by default and requires blocking rules to be set up to check customer credit limits. For more information about the *Credit management* feature, see Credit management parameters setup.

This article provides an overview of how credit limits work and answers the following questions:

- What documents and processes can I check credit limits for?
- Where do I configure the way that the customer's remaining credit is calculated?
- Where is information about a customer's remaining credit used?
- Where do I specify whether identification is required for credit to be extended to a customer, and also the credit limit amount that requires identification?
- Where do I specify whether to display a warning or error if the credit limit is exceeded?
- How do I specify the credit limit for a specific customer?
- How do I check credit limits manually on sales orders?

What documents and processes can I check credit limits for?

Use the **Accounts receivable parameters** page to specify which documents to check credit limits for. You must be a member of the System administrator (-SYSADMIN-) security role to make changes in this page. You can check credit limits for the following documents and processes:

- Invoices for sales orders, when you post the invoices
- Packing slips for sales orders, when you update packing slips
- Sales orders, when you add lines in the **Sales order** page
- Sales orders, when they're created through a service

- Free text invoices, when you post the invoices

Credit limits are automatically checked if either of the following options is set:

- In the **Accounts receivable parameters** page, the **Credit limit type** field is set to anything other than **None**. Credit limits are checked for all customers.
- In the **Accounts receivable parameters** page, the **Credit limit type** field is set to **None** but **Mandatory credit limit** is selected for a customer in the **Customers** page. Credit limits are checked only for specific customers.

To check credit limits for the following documents, you must specify additional settings.

Expand table

Document	Additional setting
Free text invoice	On the **Accounts receivable parameters** page, in the **Credit rating** area, select **Check credit limit on free text invoice**.
Sales order (manually entered)	On the **Accounts receivable parameters** page, in the **Credit rating** area, select **Check credit limit on sales order**.
Sales order (electronically received)	On the **Accounts receivable parameters** page, in the **AIF** area, select **Check credit limit for sales orders**.

Where do I configure the way that a customer's remaining credit is calculated?

You can configure Supply Chain Management to calculate a customer's remaining credit in any of the following ways:

- Compare the credit limit against the customer balance.
- Compare the credit limit against the customer balance and packing slip amounts.
- Compare the credit limit against the customer balance and all open transaction activity. This includes packing slip amounts and sales order amounts.

Use the **Accounts receivable parameters** page to specify the information to compare to. You must be a member of the System administrator (-SYSADMIN-) security role to make changes in this page. In the **Credit limit type** field, select whether to perform credit limit checks and what transaction information to include when the credit limit is checked. Select from the following options:

- **None** – Don't check credit limits. You can override this option for a specific customer by selecting the **Mandatory credit limit** check box in the **Customers** page. If you do this, the credit limit is checked against the customer balance.
- **Balance** – The credit limit is checked against the customer balance.
- **Balance + packing slip or product receipt** – The credit limit is checked against the customer balance and deliveries.
- **Balance+All** – The credit limit is checked against the customer balance, deliveries, and open orders.

Where is information about a customer's remaining credit used?

Information about a customer's balance and remaining credit amount is calculated and stored when you create an aging snapshot, and is displayed in the **Collections** page. The amounts that are displayed in the **Collections** page might not include all transaction activity until a new aging snapshot is created. Learn more in Collections and credit in Accounts receivable.

Depending on the documents that are selected, information about a customer's balance and remaining credit amount is calculated when sales orders, packing slips, and customer invoices are updated. If the amount of the document that you're working with would cause the credit limit to be exceeded, a message is displayed.

Where do I specify whether identification is required for credit to be extended to a customer, and also the credit limit that requires identification?

Use the **Accounts receivable parameters** page to specify whether to require identification and the credit limit amount limit that requires identification. You must be a member of the System administrator (-SYSADMIN-) security role to make changes in this page.

Expand table

Field	Description
Require identification with credit	Select the type of identification that must be entered for customers to whom your legal entity extends credit. The option that you select in this field determines when and what type of information is required in the **Government identification** fields on the **Customers** page:

Field	Description
	• *No* – No government-issued identification is required, regardless of the customer's credit limit. • *Yes* – A government-issued license number or other government-issued identification is required if the customer's credit limit is higher than or equal to zero. • *Minimum limit* – A government-issued license number or other government-issued identification is required if the customer's credit limit is higher than or equal to the limit that you enter in the **Limit** field in this page.
Limit	Enter the credit limit at which a government-issued license number or other identification is required for customers. For example, type 2000 to require that an identification number, such as a driver's license number, must be entered for customers who have a credit limit of 2,000 or higher. This field is available if you selected Minimum limit in the Require identification with credit field.

Where do I specify whether to display a warning or error if the credit limit is exceeded?

Use the **Accounts receivable parameters** page to specify whether to display a warning or error if the credit limit is exceeded. This warning or error is displayed when a user enters or posts information, or it's included in a log if an electronic service processes the documents. You must be a member of the System administrator (-SYSADMIN-) security role to make changes on this page.

Expand table

Field	Description
Message when exceeding credit limit (in the Credit rating area)	Select how messages about credit limits being exceeded are displayed to users. Select from the following options: • *Error* – An error message is displayed. This usually stops the current operation and the conflict must be resolved before the process can continue. • *Warning* – A warning message is displayed, but the process can continue.
Message when exceeding credit limit (in the AIF area)	Select how messages about credit limits being exceeded are delivered in a log. Select from the following options: • *Error* – An error message is displayed in the **Exceptions** page, and the document isn't processed until the error is resolved. • *Warning* – A warning message is displayed in the **Exceptions** page, but the process can continue.

How do I specify the credit limit amount for a specific customer?

Use the **Customers** page to specify the credit limit amount for a specific customer. You must be a member of a security role that has the Maintain customer master (CustCustomersMaintain) duty assigned to it to make changes in this page.

1. Go to **Accounts receivable** > **Customers** > **All customers** and open a customer account.
2. In the **Customers** page, on the Action Pane, select **Edit**.
3. Enter a currency amount in the **Credit limit** field. This value must be higher than zero (0), and is used as the credit limit amount.
4. If required, enter a license number or other government-issued identification in the **Government identification** field.

> **Note**
>
> In the **Accounts receivable parameters** page, you typically select a credit limit type. However, if the credit limit type is set to **None**, you must also select the **Mandatory credit limit** check box in the **Customers** page to check the customer's credit limit against the customer's balance. For more information about credit limit types, see "What documents and processes can I check credit limits for?" in this article.

How do I manually check credit limits on sales orders?

Sometimes, you need to manually check a customer's credit limit. For example, you might check a customer's credit limit before you start entering a sales order. Use the **Sales order** page to manually check credit limits. You must be a member of a security role that has the Maintain sales order (SalesOrderMaintain) duty assigned to it to make changes in this page.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders** and open a sales order.
2. In the **Sales order** page, on the Action Pane, on the **Manage** tab, select **Check credit limit**.

Last updated on 07/01/2026

Trade allowance management

Article • 01/07/2025

Trade allowance management helps companies manage sales promotion programs that offer "pay-for-performance" monetary rewards to customers that achieve volume and behavioral goals. The feature's capabilities are designed for companies that focus on comprehensive promote-to-profit processes, from promotion fund budgeting and allocation, to allowance contract setup, to claims creation and processing, to payment processing, to promotion effectiveness analysis.

This article will provide a broad overview of the Trade allowance management feature and will familiarize you with the typical set of tasks that are involved in managing a sales promotion program. Several types of users who have operational and decision making responsibilities are expected to use this functionality to achieve their respective goals:

- Allocating discretionary funds to the selected accounts, and setting up trade allowance agreements for promotions, based on bill-backs and one-off lump sum payments (for an agreed service)
- Running the negotiated promotion contracts through ongoing sales and generating bill-back claims
- Calculating, approving, and processing the generated claims, and passing them on to Accounts receivable (A/R) as deductions for payment settlement
- Reconciling the customer's short-pay with the deduction that is due
- Tracking use of the promotional fund, and evaluating program profitability and effectiveness

Audience and purpose

The information in this document is intended for business decision makers in enterprise companies, in positions such as purchase manager, chief financial officer (CFO), and accounting manager, who have the following responsibilities:

- High-level budgets and fund allocation
- Planning and analyzing sales promotions
- Managing staff that processes bill-back claims, runs payments based on Merchandising events, and settles short-pays and deductions

People in these roles are looking for ways to achieve these goals:

- Better use marketing promotional funds.
- Flexibly accommodate different types of promotion programs and allowances.

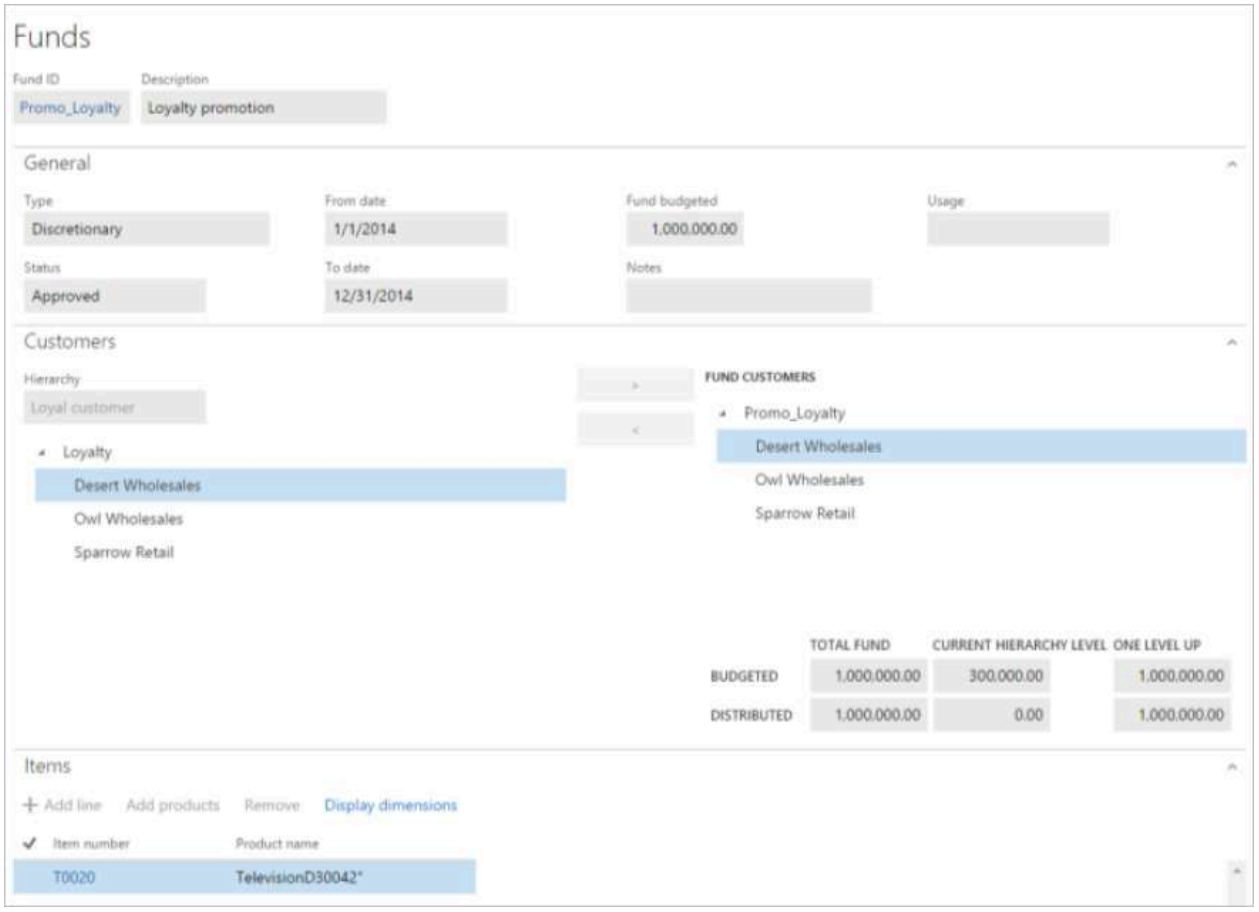
- Reduce the administrative burden and errors that are associated with monitoring promotion performance and processing claims.
- Improve cash flow forecasts by accruing for future liability.
- Have a quantified basis for ongoing and future negotiations with customers about promotions.

Promotional fund and Trade allowance agreement

A trade allowance agreement is an incentive program where pay-for-performance monetary rewards are offered to customers that achieve specific volume targets and/or behavioral goals. Promotional funds are budgeted expenditures. In that way, the promotional campaigns can be captured.

Promotional fund

Funds that are allocated to trade allowance agreements are recorded on the **Funds** page. To open the **Funds** page, select **Sales and marketing** > **Trade allowances** > **Funds** > **Funds**.



On the **Funds** page, you can view the details of promotional funds.

The **General** FastTab shows the period that the fund is valid for and its budgeted amount. In order for the fund to be allocated to the promotion agreement, the **Status** field must have a value of *Approved*.

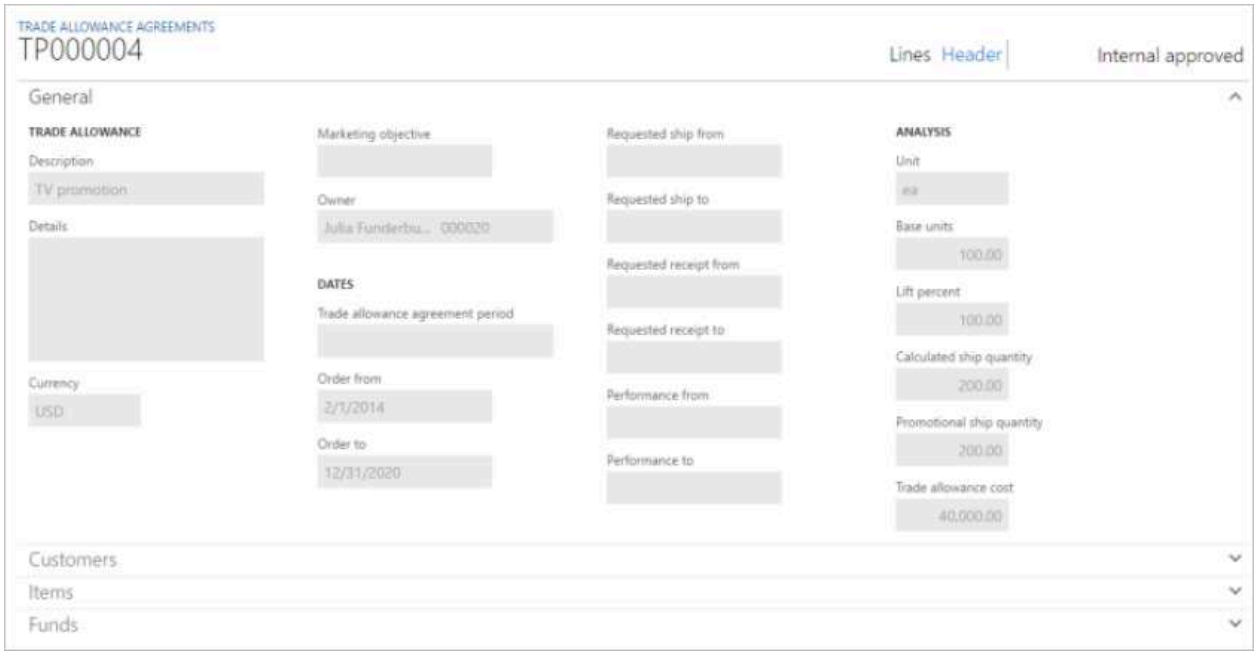
The **Customers** FastTab shows the customer hierarchy. To select the customers that the fund targets, drag them so that they are under **Fund customers**. This FastTab also shows how the total amount of the fund is distributed.

The **Items** FastTab shows the items that are included in the promotion.

Trade allowance agreement

After the fund definition is in place, the next step in promotion planning is to register promotion contracts (which are known as trade allowance agreements), allocate funds, and define performance goals for each Merchandising event.

Trade allowance agreements are recorded on the **Trade allowance agreements** page. To open the **Trade allowance agreements** page, select **Sales and marketing** > **Trade allowances** > **Trade allowance agreements**.



Header

Select **Header** to switch to the Header view.

On the **General** FastTab, the **Order to** and **Order from** fields define the period when the agreement is valid. An approval status of *Internal approved* for the agreement indicates that the agreement isn't yet valid and can't be applied during sales order processing.

The **Analysis** section of the **General** FastTab contains important fields that define the quantities and costs that are used for promotion evaluation:

- The **Base units** field specifies the quantity of products that must be sold to the selected customers before the promotion is applied.
- The **Calculated ship quantity** value is calculated based on the **Lift percent** value, which is a planned target increase for this promotion.
- The **Trade allowance cost** field is calculated based on the quantities of the various events in the trade allowance agreement.

On the **Customers** FastTab, in the list on the left, you can select and view customer groups, which are set up as predefined hierarchies. You can then select the whole hierarchy or specific accounts as targets for the allowance agreement.

On the **Items** FastTab, as on the **Items** FastTab of the **Funds** page, products are added to the agreement to associate its sales with the allowance that was agreed on.

On the **Funds** FastTab, you can view the promotion funds that are associated with this contract. You can also view the contract's event cost allocation. An event cost allocation of 100 percent means that this promotion will be financed exclusively from one fund. Alternatively, a promotion agreement can draw on several funds, and can use equal or differential percentage allocation.

Lines

Next, select **Lines** to switch to the Lines view.

The **Merchandising events** tab shows the types of events covered by an agreement. There are three types: bill back, lump sum, and off-invoice.

When you select the merchandising event and then select the **Amounts** tab, the details for the event are found.

In the **Trade allowance lines** section, you specify the quantity or amount ranges that the customer must achieve for definitions to obtain the rewards.

For merchandising events of the *Bill back* type, the upper section the **Amounts** tab defines the rules that the bill back will be applied, generated, and paid under. For example, the rules might specify the following conditions for the bill back claim:

- It's based on the creation date of the sales order (the **Calculation date type** value is *Created*).
- It's calculated based on the sales order line's amount before discounts, not the net amount, which includes discounts (the **Taken from** value is *Gross*).
- It's based on the volume of the sold products, not the amount (the **Rebate line break type** value is *Quantity*).
- It's calculated per period of a month (the **Cumulate sales by** value is *Month*).
- It's settled as a deduction, not by using A/P (the **Payment type** value is *Customer deductions*).

For merchandising events of the *Lump sum* type, the **Amounts** tab shows the quantity that will be paid to the customer in the form of a deduction when the customer achieves specific performance. An approval status of *Open* indicates that the lump sum hasn't yet been paid.

To apply the agreement to sales orders that meet the agreement's conditions, the agreement's status must be *Confirmed*.

Perform sales under the planned merchandising event and generate bill-back claims

When you create a sales order that has lines that fulfill the requirements of the agreement, you can view the related information on the **Sales order** page by selecting **Sales order line** > **View** > **Price details**.

On the **Price details** page, on the **Rebates** FastTab, the sales clerk can see a bill back from the valid trade allowance agreement (the rebate program ID is shown) and the total amount that is applied to the line. This amount is also shown in the **Rebate amount** field in the **Margin estimation** section of the **Price details** page.

When the sales invoice is posted, a corresponding bill-back claim is generated for each invoice line.

> ⓘ Note
>
> To see the **Price details** page, on the **Accounts receivable parameters** page, on the **Prices** tab, select the **Enable price details** check box. I

Process claims and pass them as deductions to A/R

The next steps in the process for handling bill-backs are to review, calculate, and approve claims, and then convert them into deductions.

The Bill back workbench is where the promotion agreement owner periodically reviews and processes the claims that are generated. It's also where the A/R administrator converts the approved claims into deductions or regular payments, depending on the payment method for the claim.

On the **Bill back workbench** page, you can review the claim lines. If the claims are in *To be recalculated* status, they must be recalculated for any cumulative effect.

Recalculate claims

To recalculate the claims, on the Action Pane, select **Cumulate**. Then, in the **Cumulate rebates** dialog box, select the customer.

As a result of the recalculation, the program generates new claims for the amounts to adjust the original claims to the qualifying amount per unit. One adjustment claim is generated for each unique combination of a customer, an item, a currency, a unit of measure, inventory dimensions, financial dimensions, and a sales tax group. These adjustment claims have the same reference to the sales order and invoice number as the

claims that are being adjusted (that is, the claims that were originally created from the sales document). Unlike the original claim, the adjustment claim doesn't have values in the fields that describe the original sales order line's amounts and quantity.

After the recalculation is completed, the status of the claims is changed to **Calculated**. To approve the claims, on the Action Pane, select **Approve**.

Process claims and pass them to A/R

The claims are now ready for A/R processing. To process them, on the Action Pane, select **Process**.

Upon processing the claims, the status has changed to *Mark* and indicates that a journal posting (the journal that is posted is the Rebate accrual journal, as specified in the A/R parameters) has caused the following events to occur:

- The claims have been transferred to the temporary customer balance as deductions.
- The rebate accrual account has been credited to represent the future liability for the customer.
- The rebate expense account has been debited to recognize the cost that was incurred in connection with the sales.

To complete the process, the A/R clerk must now handle the accrual deductions by transferring them to the customers balance as a credit note (liability).

To start the task, on the Action Pane of the **Customer** page, select **Collect** > **Settle transactions**. Then, on the **Settle transactions** page, select **Functions** > **Bill back program**. This rebate page shows all the bill-back claims that were previously processed.

If you want to create a credit note, select the **Mark** check box for all lines, and then select **Functions** > **Create credit note**.

Upon credit note creation, a journal is posted. (The journal that is posted is the AR consumption journal, as specified in the A/R parameters.) As a result, the real liability (credit) amount has been moved to the customer balance. Financially, this situation implies that the following events have occurred:

- The customer's receivable account has been credited.
- The rebate accrual account has been debited.

To approve a merchandising event of the *Lump sum* type, select the event on the **Trade allowance agreements** page, and then, on the **Amount** tab, select **Approve**.

Settle the deduction that is due and the customer short-pay by using the Deduction workbench

Often, in anticipation of bill-backs, customers choose to short-pay selected invoices. To prevent payment reconciliation issues in the future, the A/R clerk registers those short-pays as deductions when they record the actual customer payments. Then, on the Deduction workbench, those customer deductions can easily be settled against the claim amounts that are due from the company.

To register a customer's short-pay in the Payment journal, select **Accounts receivable** > **Payments** > **Payment journal**, and create a new payment journal. Then, on the Action Pane, select **Deductions**. On the **Deduction** page, you can create and track the amount that was short-paid.

The collection manager is now responsible for settling the open credit note transaction and the short-pay transaction against each other in the Deduction workbench.

To manage deductions, select **Sales and marketing** > **Trade allowances** > **Deductions** > **Deduction workbench**. The upper section of the page contains lines that represent the short-pays from the customer. The lower section of the page contains the customers open credit transactions.

To settle the deduction against the open transaction, mark the deduction line, and then, on the Open transactions tab, mark the line. On the Action Pane, select Maintain > Match.

The status of the originating claims is now set to **Completed**.

Analyze the effectiveness of the promotion and fund consumption

To get an overview of the program's key statistics and fund use, you can use several reports and analytical views that are available in Trade allowance management.

On the **Trade allowance activity** page, the **Overview** tab shows the main details of the trade allowance agreement. The other tabs show more specific details about the associated documents, payments, and other events that are related to the program.

The **Summary** tab shows the total quantity of products that have been sold under the promotion, the sales amount that has been invoiced, and the bill-backs and lump sums

that have been paid.

The **Bill back credits** tab contains the details of individual bill-backs that have been credited to the customer.

To get a more analytical overview of the various performance measures for the promotion, you can use the **Analysis** view. To go to the **Analysis** view, select **Sales and marketing** > **Trade allowances** > **Trade allowance agreements**. On the Action Pane, select **Analysis**.

To get a more analytical overview of the various performance measures for the promotion, you can use the **Analysis** view. To go to the **Analysis** view, select **Sales and marketing** > **Trade allowances** > **Trade allowance agreements**. On the Action Pane, select **Analysis**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a new trade agreement

This procedure shows you how to create a trade agreement where you register a new product sales price that you agreed on with a specific customer. You can run this procedure in demo data company USMF or on your own data. If you're using your own data, before you start this guide, make sure that a trade agreement journal name exists where the **Default relation** is set to *Price (sales)*.

Create and post a new trade agreement journal

1. Go to **Sales and marketing** > **Prices and discounts** > **Trade agreement journals**.
2. Select **New**.
3. In the **Name** field, select the drop-down button to open the lookup.
4. In the list, find and select the desired record.
5. On Action Pane, select **Lines**.
6. In the **Party code type** field, select *Table*. In this example, you're updating the price for a specific customer, which means you must choose *Table*. If you were updating the product's list price, you would select *All*, so that the new price is valid for all customers. If you were differentiating prices among different customer segments, then you would select *Group*. To select *Group*, you must have set up customer price groups.
7. In the **Account selection** field, select the drop-down button to open the lookup.
8. In the list, find and select the desired record.
9. In the **Product code type** field, select *Table*. When you're entering a trade agreement of type *Price (sales)*, you must select *Table* in the **Product code type** field. This requirement exists because a price is an absolute value and can't be the same for all products or a group of products.
10. In the **Item relation** field, select the drop-down button to open the lookup.
11. In the list, select the product you want to include in the agreement. Make a note of which product you selected.
12. In the **From** field, enter a minimum quantity.

    - If the customer has to order a minimum quantity before they can qualify for the new price, specify that quantity here.
    - Enter a value in the **To** field to specify the maximum quantity above which the agreement's price isn't valid. If you offer prices and discounts based on multiple

quantity breaks, specify each quantity bracket as a pair of minimum and maximum quantity in the **From** and **To** fields respectively.

13. In the **Amount in currency field**, enter a price.
14. Under the **Details** section, in the **From date** field, enter a date from which this agreement is valid.
15. Select **Save**.
16. Select **Validate**.
17. Select **Validate selected lines**.
18. Select **OK**.
19. Select **Post**.
20. Select **OK**.

View trade agreements for a product

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, find and select the product whose price you just updated.
3. On the Action Pane, select **Sell**.
4. Select **View trade agreements**. Review the details of the price trade agreement you just created.
5. Close the page.

Related information

White paper

For more information, download the following white paper (written to support AX2012, but still applies for Dynamics 365 Supply Chain Management).

- Trade agreements

Community blogs

- Sales prices in finance and operations

Last updated on 07/01/2026

Export posted trade agreement lines

Article • 05/07/2025

This article provides step-by-step instructions on how to export posted trade agreement lines. Standard data entities don't support the direct export of posted trade agreement lines. This article describes how to convert posted trade agreement lines to unposted lines, which makes them exportable.

Convert posted trade agreement lines

To convert posted trade agreement lines into unposted lines, follow these steps.

1. Open the list of active trade agreement lines for your vendor or customer.

    - For *purchase trade agreements*, go to **Accounts payable** > **Vendors** > **All vendors**. Select the vendor you want to work with. On the Action Pane, open the **Procurement** tab and, from the **Agreements** group, select **Trade agreements**.

    - For *sales trade agreements*, go to **Accounts receivable** > **Customers** > **All customers**. Select the customers you want to work with. On the Action Pane, open the **Sell** tab and, from the **Trade agreements** group, select **Agreements**.

2. Select the lines you want to convert for eventual export.

3. On the Action Pane, select **Edit selected lines**.

4. In the dialog, set the **Name** field to the name of the journal that you want to assign to your selected lines.

5. Select **OK**. This action converts the active price trade agreement lines into open trade agreement lines, which are now exportable.

Export the open trade agreement lines

After you've converted the relevant lines to open trade agreements, you'll be able to export them using the standard data management features of Supply Chain Management. You can export data from this entity to any supported data format (such as comma-separated values (CSV) or Microsoft Excel).

1. Go to **System administration** > **Workspaces** > **Data management**.

2. In the **Import / Export** section, select the **Export** tile.

3. On the **Export** page, enter a **Group name** for the job.

4. On the **Selected entities** FastTab toolbar, select **Add entity**.

5. In the **Add entity** drop-down dialog, set the following fields:

    - **Entity name** – Select *Open purchase price journal lines V2* or *Open sales price journal lines V2*.
    - **Target data format** – Select the format to export data to.

6. Select **Add** to add the new row, and then select **Close** to close the dialog box.

7. On the Action Pane, select **Save**.

8. On the Action Pane, open the **Export options** tab and select **Export now**.

9. The **Execution summary** page opens. Here you can view the status of your export job and a list of the entities that were exported. In the **Entity processing status** section, select the *Open purchase price journal lines V2* or *Open sales price journal lines V2* entity in the list, and then select **Download file** to download the data that was exported from that entity.

For more information about how to use data management to export data, see Data import and export jobs overview.

Clean up resources

After exporting the data, ensure that any temporary journal entries used for the conversion process are deleted or archived to maintain data integrity and avoid clutter in your system.

Recalculate line net amounts when importing sales orders and quotations

Article • 08/23/2023

This article describes whether and how the system recalculates line net amounts when sales orders and quotations are imported. It also explains how you can control the behavior in different versions of Microsoft Dynamics 365 Supply Chain Management.

How updates to net line amounts are calculated on import

Supply Chain Management version 10.0.23 introduced bugfix 604418. This bugfix changed the conditions under which the **Net amount** field on a line can be updated or recalculated when updates to existing sales orders and quotations are imported. In version 10.0.29, you can replace this bugfix by turning on the *Calculate line net amount on import* feature. This feature has a similar effect, but it provides a global setting that lets you return to the old behavior if you must. Although the new behavior makes the system work in a more intuitive manner, it can produce unexpected results in specific scenarios where all the following conditions are met:

- Data that updates existing records is imported through the *Sales order lines V2*, *Sales quotation lines V2*, or *Return order lines* entity by using Open Data Protocol (OData), including situations where you use dual-write, import/export through Excel, and some third-party integrations.
- Trade agreement evaluation policies that are in place establish a change policy that restricts updates to the **Net amount** field on sales order lines, sales quotation lines, and/or return order lines. Note that for return order lines, the **Net amount** field is always calculated and can't be set manually.
- The imported data includes changes to the **Net amount** field on lines, or changes (such as unit price, quantity, or discount) that will cause the value of the **Net amount** field on lines to be recalculated for one or more existing line records.

In these specific scenarios, the effect of the trade agreement evaluation policy is to put a restriction on updates to the **Net amount** field on the line. This restriction is known as a *change policy*. Because of this policy, when you use the user interface to edit or recalculate the field, the system prompts you to confirm whether you want to make the change. However, when you import a record, the system must make the choice for you. Before version 10.0.23, the system always left the line net amount unchanged, unless the incoming line net amount is 0 (zero). However, in newer versions, the system always

updates or recalculates the net amount as needed, unless it's explicitly instructed not to do so. Although the new behavior is more logical, it might cause issues for you if you're already running processes or integrations that assume the older behavior. This article describes how to revert to the old behavior if you must.

Control calculations of line net amounts in versions 10.0.29 and later

Supply Chain Management version 10.0.29 introduced a feature that is named *Calculate line net amount on import*. This feature adds an option that is named **Calculate line net amount** to the **Accounts receivable parameters** page. This option lets you select between the new and legacy behaviors for calculating line net amounts on import.

Turn the Calculate line net amount on import feature on or off

When you update to version 10.0.29, the *Calculate line net amount on import* feature is turned on by default, and the new **Calculate line net amount** option is initially set to *Yes*. The *Yes* setting corresponds to the new standard behavior. It matches the system behavior when the feature is turned off, except in the case of the functionality of the CalculateLineAmount parameter, as described later in this article. The *No* setting matches the system behavior before version 10.0.23 and is provided mainly to support legacy integration scenarios.

As of Supply Chain Management version 10.0.36, the feature is mandatory and can't be turned off. If you're running a version older than 10.0.36, then admins can turn this functionality on or off by searching for the *Calculate line net amount on import* feature in the Feature management workspace.

Set the Calculate line net amount option

When the *Calculate line net amount on import* feature is turned on, you can set the **Calculate line net amount** option by following these steps.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Prices** tab, on the **Line net amount calculation through integration** FastTab, set the **Calculate line net amount** option to one of the following values:
    - *Yes* – The system will always recalculate and update line amounts when needed. (Therefore, it will ignore the trade agreement evaluation policy.)

- *No* – If the existing or incoming net amount for any line is 0 (zero), the value for that line is recalculated based on other values (such as the unit price, quantity, and discount). If the existing or incoming net amount differs from 0 (zero), and a change policy is set on the **Net amount** field on the line, the field isn't recalculated or updated, even when incoming changes to the line price, quantity, and/or discount would imply that the line total should be recalculated. This behavior matches that of version 10.0.22.

How the Calculate line net amount on import feature affects the CalculateLineAmount parameter

When the *Calculate line net amount on import* feature is turned on, the value of the `CalculateLineAmount` parameter for the `SalesLine` and `SalesQuotationLine` tables has no effect. Instead, the behavior is controlled globally by the **Calculate line net amount** option that is described in the previous section. Therefore, when the feature is turned on, you must not take any dependency on the `CalculateLineAmount` value.

When the *Calculate line net amount on import* feature is turned off, the `CalculateLineAmount` parameter for the `SalesLine` and `SalesQuotationLine` tables works as it does in Supply Chain Management versions 10.0.23 through 10.0.28, as described in the next section.

Control line net amount calculations in versions 10.0.28 and earlier

When bugfix 604418 was introduced in version 10.0.23, it became possible to select how each relevant data entity should behave when a line net amount was edited or had to be recalculated because of other changes (such as an updated item price). You could control this behavior by setting the new `CalculateLineAmount` parameter for each line to one of the following values in the imported file:

- **`CalculateLineAmount`** = ***1*** – The **Net amount** field on the line is always recalculated and updated, regardless of whether a change policy is set for the field, and regardless of the value of the incoming or existing line net amount.
- **`CalculateLineAmount`** = ***0*** – If the existing or incoming net amount for any line is 0 (zero), the value for that line is recalculated based on other values (such as the unit price, quantity, and discount). If the existing or incoming net amount differs from 0 (zero), and a change policy is set on the **Net amount** field on the line, the field isn't recalculated or updated.

The system behavior depends on your version of Supply Chain Management:

- In version 10.0.22 and earlier, the system always behaves as though `CalculateLineAmount` is set to *0*, and there is no way to make it behave as though `CalculateLineAmount` is set to *1*.
- In versions 10.0.23 through 10.0.28, the system behaves as though `CalculateLineAmount` is set to *1* for all lines where it isn't explicitly set to *0* in the import file.

Feedback

Was this page helpful? Yes No

Provide product feedback

Schedule sales history data cleanup

As part of its standard operation, Microsoft Dynamics 365 Supply Chain Management generates and stores sales history update data on an ongoing basis. Over time, a large amount of outdated sales history data might accumulate in your system. This accumulated data can cause performance and functional issues when documents that are related to sales orders are posted. (These documents include sales order confirmations, sales packing slips, sales picking lists, and invoices). Therefore, you should set up and schedule the *Sales update history cleanup* periodic task to run at a regular interval. This task removes all sales history update data that is no longer needed.

If you use the *Sales update history cleanup* periodic task, enable the *Sales history cleanup performance improvements* feature, which makes the task run more effectively. (Nevertheless, you can also run the task without enabling this feature.)

Sales history cleanup performance improvements

The **Sales update history cleanup** periodic batch job can take a long time if it runs infrequently on environments with a high volume of sales updates. In these situations, the *Sales history cleanup performance improvements* feature can help reduce the run duration and improve reliability.

The feature improves the existing cleanup job in the following ways:

- It splits the clean-up into batches (you can change the batch size through customizations).
- It runs for a maximum of two hours (you can change the duration through customizations).
- Where possible, it leverages database capabilities to minimize locking contention and avoid joining transactional tables during cleanup.

After enabling the feature, the **Sales update history cleanup** batch job (**Sales and marketing** > **Period tasks** > **Cleanup** > **Sales update history cleanup**) runs as it did before, but with better performance and for a maximum of two hours. This condition means it might need to run several times to clean up all the data for a specific retention time frame.

This feature must be enabled on your system for you to use it. As of version 10.0.41, this feature is turned on by default. As of Supply Chain Management version 10.0.43, it's mandatory and can't be turned off. If you're running a version older than 10.0.43, admins can turn this functionality on

or off by searching for the *Sales history cleanup performance improvements* feature in the Feature management workspace.

Set up and schedule the Sales history cleanup periodic task

To set up and schedule the *Sales history cleanup* periodic task, follow these steps:

1. Analyze your business needs to determine how many days of historic sales order posting data you must keep. Run the cleanup task every three months and a maximum of every six months.

2. Go to **Sales and marketing** > **Period tasks** > **Cleanup** > **Sales update history cleanup**.

3. In the **Clean up sales update history** dialog box, on the **Parameters** FastTab, set the following fields:

    - **Clean up** – Select one of the following values to specify the type of records to clean up:
        - **Executed** – Delete only records that are fully processed. Because you likely have no further use for these records, this option is the safest.
        - **Executed and erroneous** – Delete both fully processed records and records where an error occurred. This option is the most commonly used. You might want to inspect and even fix erroneous records instead of having them automatically cleaned up. However, many companies choose to clean up these records too after about a month, because they're probably no longer relevant by that time.
        - **All** – Delete executed, erroneous, and waiting records. Waiting records are valid but aren't yet fully processed. In most cases, you probably don't want them to be deleted automatically. However, in some situations, you might choose to have them automatically deleted after a specific amount of time has passed.

    - **Retain records based on age** – Specify whether you want to clean up records based on their age on the day when the task runs or delete records that were created before a fixed date. If you're scheduling the cleanup as a recurring task, set this option to *Yes*, because the age is always calculated relative to the date when the task runs.
        - Set this option to *Yes* to enable the **Maximum age** field. Use this field to specify the maximum age of the records to keep each time that the task runs. The **Created until** field is ignored.

    - Set this option to *No* to enable the **Created until** field. Use this field to specify the oldest creation date of the records to keep when the task runs. The **Maximum age** field is ignored.

- **Created until** – This setting applies only when the **Retain records based on age** option is set to *No*. Specify the oldest creation date of the records to keep when the task runs. Records that were created before this date are deleted.

- **Maximum age** – This setting applies only when the **Retain records based on age** option is set to *Yes*. Specify the maximum age (in days) of the records to keep each time that the task runs. Older records are deleted.

4. On the **Run in the background** FastTab, specify how, when, and how often the task runs. Use these settings to implement the schedule that you determined in step 1. The fields work just as they do for other types of batch jobs in Supply Chain Management.

5. Select **OK** to apply your settings and close the dialog box.

Last updated on 07/01/2026

Service management overview

07/10/2025

Use the Service management module to establish service agreements and service subscriptions, handle service orders and customer inquiries, and to manage and analyze the delivery of services to customers. You can use service agreements to define the resources that are used in a typical service visit. You can also use service agreements to view how those resources are invoiced to the customer. A service agreement can also include a service level agreement that specifies standard response times, and offers tools to record the actual time.

You can create service orders to manage information about scheduled and unscheduled visits by a service technician to a customer site. Service orders include information such as:

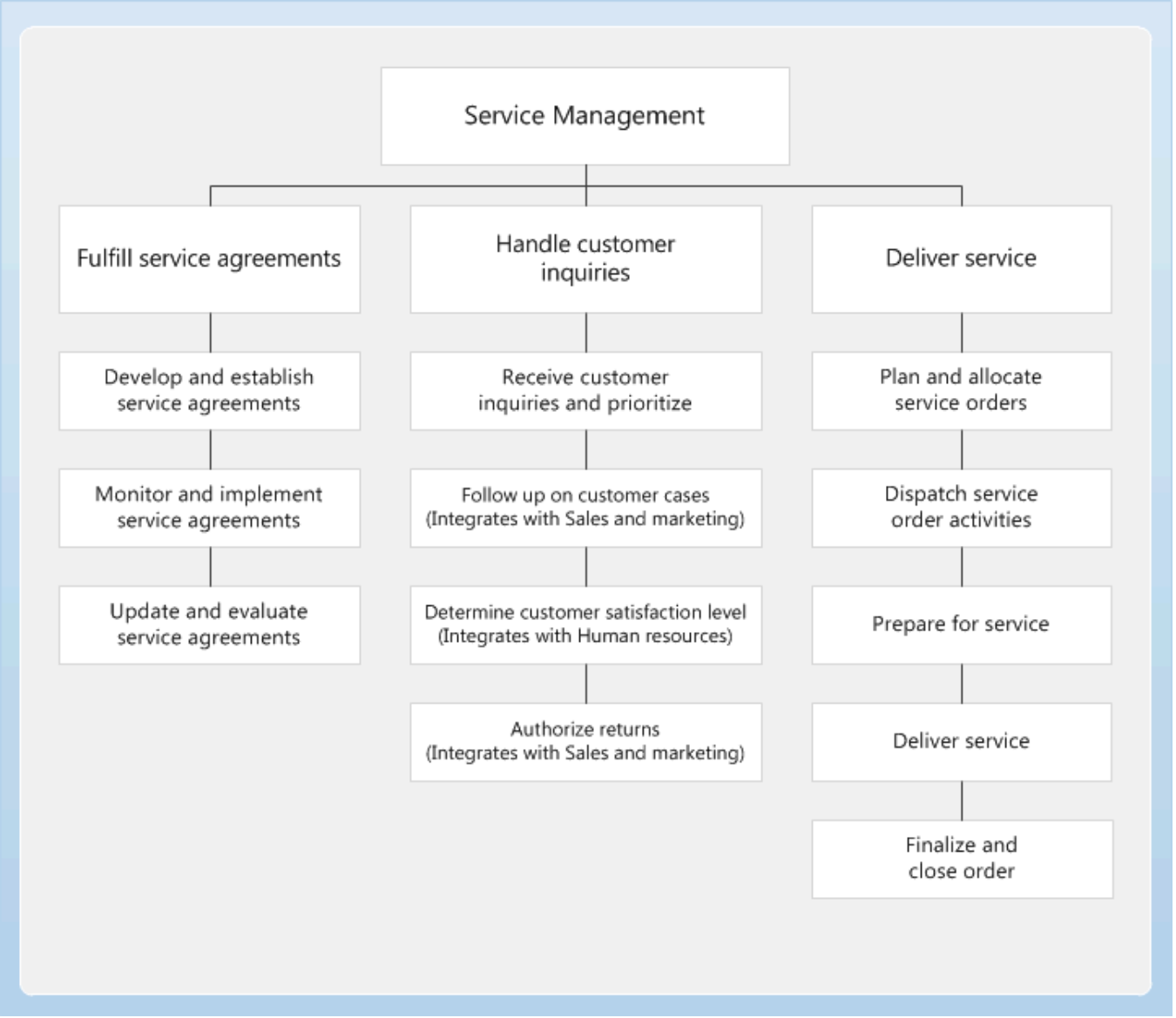
- The number hours required to complete the task
- The type of service or repair
- The item to repair, including details about the symptoms and diagnosis
- Any expenses and fees related to the service or repair

You can receive, process, and dispatch service requests. After you create a service order, you can use service stages to monitor progress and specify rules that control what actions are enabled in each stage. When a service order is complete, you can sign off on the order to confirm that it's complete, and then post the order to start the invoice process.

Use the reporting tools to monitor service order margins and subscription transactions, and print work descriptions and work receipts.

Business processes

The following diagram illustrates the high level business processes for the Service management module, and shows where service processes integrate with other modules.



Service management at a glance

Expand table

Important tasks	Primary pages	Popular reports
Fulfill service agreements	Service agreements	Service order margin
Handle customer inquiries	Service orders	Work description
	Dispatch board	Transaction – subscription
		Subscription fee transactions

Integration of Service management

Service management can be integrated with the following modules:

- Sales and marketing overview
- Human resources

Service agreements overview

07/10/2025

Service agreement groups are useful for sorting and filtering service agreements. Use service agreement groups to group service agreements by, for example, company size, geographical location, or industry type. Alternatively, you can use agreement groups to group service agreements by the level of service offered to a customer.

In the **Service agreement group** field on the **Service management parameters** page, you can define a default service agreement group. This group is suggested when you create a new service agreement.

Example

- You can group your service agreements according to the level of service offered.
- The three agreement groups you have set up in your company are Gold, Silver, and Bronze.
- A new service account manager has been employed in your company. The service manager is responsible for dealing with all clients in the Gold agreement group. You can filter agreement records on the Gold agreement group to find all of the agreements for which you must specify the new service manager as the person in charge.

Next steps

- Create service agreement groups

Create service agreements

07/10/2025

This article describes how to use features in the Service management, Project management, and Accounting modules to create service agreements.

Create a service agreement from Service management

1. Navigate to **Service management**.
2. Select **Service agreements** to create a new service agreement line in the page header.
3. Select **New**. Enter a description, select a reference to a project in the **Project ID** field, and fill in the rest of the fields and lines for the service agreement. Select **Save**.
4. On the **Relations** tab, select **Service objects** or **Service tasks** to create service object relations or service task relations for the service agreement. The service objects and tasks that you have created relations for can be attached on the lines of the service agreement.
5. In the lower half of the page, create service agreement lines by copying lines from a service template, another service agreement, or manually creating the service-agreement lines.

> **Note**
>
> If you copy lines into the service agreement from another service agreement, you can indicate whether you also want to copy service object and service task relations. If you copy these relations, they are added to any existing relations on the service agreement. If you copy service-agreement lines from a service template, the service-object and service-task relations are automatically copied as service-object relations and service-task relations on the new service-agreement lines.

Create service agreement lines manually

1. From the **Service agreements** page, add a service agreement line in the lines grid.
2. Enter the appropriate information for the service agreement line.
3. Select **Save** to save the line, and then close the page.

Create a service agreement from Project

1. Select **Project management and accounting**.

2. Select **All projects**.
3. Select the project from the list.
4. On the Action Pane, select **Manage**. In the **New** Action group, select **Service** and select **Service agreement**.
5. Follow the steps in the section titled **Create a service agreement** as described earlier in this article to enter the project reference.

Related information

- Develop and establish service agreements overview

Create service agreement groups

Article • 01/06/2025

Service agreement groups are useful for sorting and filtering service agreements.

1. Go to **Service management** > **Setup** > **Service agreements** > **Service agreement groups**.
2. On the Action Pane, select **New** to create a new service agreement group.
3. In the **Service agreement group** field, enter an ID for the service agreement group.
4. In the **Description** field, enter a description for the service agreement group.
5. In the **Service level agreement** field, select the agreement that applies to the group.
6. Close the page to save your changes.

> **Note**
>
> If a service agreement group is defined in the **Service agreement group** field of the **Service management parameters** page, the program suggests this group for all new service agreements that you create.

Related information

- Service agreement overview

Feedback

Was this page helpful? Yes No

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Service intervals

Article • 01/06/2025

The service agreement interval indicates the frequency with which service order lines are created for service agreement lines when you create service orders automatically.

When you create service orders automatically, service order lines are created according to the interval that you have specified for the service agreement line from the start date of the agreement line.

If the **Interval** field of a service agreement line on the **Service agreements** page is blank, the line is a one-time event, and it isn't used to create service orders repeatedly.

Example

This example illustrates how a service interval affects service agreement lines and service order lines on a service order.

Create a service agreement

First, you create a service agreement and set the **Combine service orders** option to *By service agreement*.

1. Select **Service agreements**
2. On the Action Pane, on the **Service agreement** tab, in the **New** group, select **Service agreement** to create a new service agreement.
3. Enter a description, select a project in the **Project ID** field, and enter a date in the **Start date** field.
4. In the **Combine service orders** field, select *By service agreement*.

You have now created the following service agreement:

Expand table

Project	Start date
Your project	The date you specified for the project. In this example, the current date is used.

Create a service agreement line

Next, you create a service agreement line that has the transaction type *Hour*.

To complete this part of the example, you must create a service interval of 10 days on the **Service intervals** page.

1. Select the service agreement that you just created.
2. On the **Lines** FastTab, select the **Add** button to create a new line in the lower pane of the **Service agreements** page.
3. In the **Transaction type** field, select *Hour*.
4. In the **Worker** field, select the worker who will deliver the service.
5. In the **Service interval** field, select the 10 days interval.

You have now created a service agreement line with the following information:

Expand table

Transaction type	Start date	Service interval
Hour	The current date.	Every 10 days
Worker	The worker who will perform the service.	

There's no time window specified for the line.

Create planned service orders

You can now create planned service orders and service order lines for the coming month.

1. On the **Service agreements** page, on the Action Pane, on the **Deliver** tab, select **Planned service orders**.
2. On the **Create service orders** page, enter the current date in the **From date** field and a date that is one month from the current date in the **To date** field.
3. Set the **Hour** slider to *Yes*.
4. Select **OK**.

Because there's no grouping on the service order (defined by the **By service agreement** option in the **Combine service orders** field), one service order line is created per service order.

Service orders created

Three service order lines were created within the time frame that you specified in the **Create service orders** dialog box. You can view the service order lines on the **Service agreements** page (Action Pane > **Deliver** tab > **View** button).

Related information

- Set up service intervals

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up service intervals

Article • 01/30/2025

Service interval indicates the frequency with which service order lines are created for service agreement lines when you create service orders.

1. Go to **Service management** > **Setup** > **Service agreements** > **Service intervals**.
2. On the Action Pane, select **New**.
3. Enter the ID and description of the service interval.
4. In the **Range** field, select the range.
5. In the **Frequency** field, type the frequency. The frequency is the factor by which you must multiply the range to obtain the interval for a service agreement.
6. On the Action Pane, select **Save**.

Example

You want to create a service interval of 10 days, so you take the following steps.

1. Go to **Service management** > **Setup** > **Service agreements** > **Service intervals**.
2. On the Action Pane, select **New**.
3. Enter the ID and description of the service interval.
4. In the **Range** field, select *Daily*.
5. In the **Frequency** field, type *10*.
6. On the Action Pane, select **Save**.

Related information

- Service intervals

Feedback

Was this page helpful? Yes No

Provide product feedback

Time windows

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can use time windows to optimize the scheduling of service order lines. You can set up the system so that it automatically creates service orders. Based on the criteria specified by a time window, you can connect as many service order lines as possible to as few service orders as possible.

Time windows specify how far a service order line can move from its calculated date. The calculated date is the date when the service order line was scheduled to occur. The date is based on its interval setting and the service period that you defined on the **Create service orders** page. You define a time window by using the values in the following table.

Expand table

Method	Description
Week	The date that the service order line can be moved to any open day in the same week as the initial calculated date.
Month	The date that the service order line can be moved to any open day in the same month as the initial calculated date. For example, the calculated date for a service order line is February 15, 2017. The service order line can be scheduled for any weekday between February 1 and February 28, 2017.
Manual	You define the maximum number of days before or after the initial calculated date that the service order line can be moved.

If you don't specify a time window for a service agreement line, the service order line that is derived from the service agreement must be on the exact date for which it was originally scheduled.

Related information

- Create time windows

Last updated on 05/07/2025

Create time windows

Article • 01/30/2025

You can use time windows to optimize the scheduling of service order lines.

1. Go to **Service management** > **Setup** > **Service agreements** > **Time window**.
2. On the Action Pane, select **New** to create a new time window.
3. In the **Time window** field, enter a name for the time window.
4. In the **Description** field, enter a description for the time window.
5. In the **Method** field, select the method to be autocreated. If the method is of the **Manual** type, specify the number of days before or after the calculated date that the service order line can be moved.

Related information

- Time windows

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a service template group

07/10/2025

You can use service template groups to sort and filter service templates.

1. Go to **Service management** > **Setup** > **Service agreements** > **Service template groups**.
2. Create a new service template group.
3. Enter the group ID, a description, and a parent service template group, if there is one.
4. Save the service template group.
5. Set the **Tree control** slider to *Yes* to open the tree view and view the service template groups either as a hierarchy in a tree view or as an overview.

Service tasks overview

07/10/2025

Use service tasks to describe the task to be completed during a service order. Both technicians and customers can see this information.

You create service tasks on the **Service tasks** page, and you associate service tasks with a specific service agreement or service order by creating service task relations. Every time that you create a service task relation, you can create the following:

- Internal notes for the task, such as detailed technical requests for the task that are important for the technician to know.
- External notes that the customer can see. These might provide a less technical explanation of the task that is being completed, according to the agreement between the customer and the service company.

When you have set up a service task relation between a service task and a service order or service agreement, you can specify this service task when you create service order lines or service agreement lines for the current service order or service agreement.

When you generate service orders from a service agreement, you can use the service tasks that are assigned to each service agreement line to group service order lines into service orders.

Create a service task

1. Go to **Service management** > **Setup** > **Service tasks**.
2. Create a new line.
3. Enter the service ID and description.

Example

A technician must complete two jobs on a gearbox (service object GB-1234) at a customer site. A service agreement is created for the customer, and several transactions are associated with the jobs. The service agreement and service agreement lines for the two jobs are as follows:

Service agreement

Expand table

Project	Service agreement	Description	Group
9012	000008_001	Inspection and routine replacement – GB-1234	Premium

Service agreement lines

Expand table

Description	Transaction type	Service object	Service task
Inspection and cleaning	Hour	GB-1234	I/C - GB1234
Travel	Expense	GB-1234	I/C - GB1234
Materials	Item	GB-1234	I/C - GB1234
Routine replacement	Hour	GB-1234	RR - GB1234
GR-1	Item	GB-1234	RR - GB1234
GR-5	Item	GB-1234	RR - GB1234

The service agreement lines for the two jobs have two service tasks attached to them. The service tasks determine the transactions that belong to each job. In the first case, service task I/C - GB1234 identifies all the service agreement lines that are involved in the inspection and cleaning of object GB-1234. In the second case, service task RR - GB1234 identifies all the service agreement lines that are involved in completing a routine replacement job.

The service task relations that connect the service tasks to the specific agreement are as follows:

Service tasks

Expand table

Service task	Description	Internal note	External note
I/C - GB1234	Inspection of gearbox GB-1234	Visual and mechanical inspection and cleaning of gearbox GB-1234	Routine inspection of gearbox
RR - GB1234	Routine replacement of parts in GB-1234	Routine service replacement of GR-1 and GR-5 components (for gearboxes manufactured before 2002, also replace GR-2 component)	Routine replacement of parts

Group service orders

When you create service orders automatically, you can use service tasks as grouping criteria. To group service orders by service tasks, define on the service agreement that service orders that are based on the service agreement should be grouped by service task ID as their initial grouping criteria.

Group by service task

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. On the **Setup** tab, select **By service task** in the **Combine service orders** field.

Create service task relations

Article • 01/30/2025

You can associate service tasks with service agreements or service orders in order to describe the service task to be completed for the agreement or order. This information is available to service technicians and customers.

Create a relation with a service agreement

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select an existing service agreement, or create a new service agreement.
3. On the Action Pane, select the **Service tasks** button.
4. On the **Service tasks** page, select **New** to create a new line, and then select a service task from the **Service task** list to attach the service task to the service agreement.
5. On the **Description** tab, enter any internal or external note descriptions in the free text fields.
6. Close the page to save the record.

Repeat this procedure until you have created all the necessary service task relations for the service agreement. You can now specify these service tasks for any attached agreement lines.

A service tasks relation that is created on a service agreement is available from all service orders that are attached to the service agreement.

Create a relation with a service order

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select an existing service order, or create a new service order.
3. On the Action Pane, select the **Service tasks** button.
4. From the **Service tasks** page, select **New** to create a new line, and then select a service task from the **Service task** list to attach the service tasks to the service order.

5. On the **Description** tab, enter any internal or external note descriptions in the free text fields.

6. Close the page to save the record.

Repeat this procedure until you have created all the necessary service task relations for the service order. You can now select the service task for which you have created the relation when you create service order lines.

Service task relations that are created on a service order are available on the specific service order.

Related information

- Service tasks overview

Feedback

Was this page helpful? Yes No

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Service objects overview

07/10/2025

Service objects are a customer's assets and products for which you can perform a service. Depending on the type of service you provide, objects can be tangible or intangible:

- Tangible objects are things, such as a machine or a building, on which you can perform a physical service task.

  A tangible service object can also be an inventory item that you create on the Released product details page. Depending on the inventory dimension group that you attach to the item, you can create a service object to a level of detail that includes the item serial number. This is useful when you must keep track of the exact item that the service object represents.

  A tangible service object can also be an item that isn't directly related to a company's direct production or supply chain. For example, a tool kit that is used for repairs in a service order can be a service object that isn't included in inventory. In this case, you don't register it as an inventory item.

- Intangible objects are nonphysical things, such as a set of accounts or a legal document, on which you can perform a service task.

Example of an intangible service object

Company A maintains the financial records for several small companies. One of Company A's clients is the local football team, for which Company A does the weekly bookkeeping and annual audit of the club's accounts. The club's accounts are set up on the Service objects page and specified as the object in the service agreement. There are two service agreement lines for the object. Line 1 is weekly bookkeeping with a weekly interval assigned to the line, and line 2 is the annual audit with a yearly interval assigned to it.

Related information

- Create service objects

Create service objects

Article • 01/30/2025

Service objects are a customer's assets and products for which you can perform a service.

1. Go to **Service management** > **Setup** > **Service objects** > **Service objects**.
2. On the Action Pane, select **New**.
3. In the **Description** field, enter a description of the object.
4. In the **Service object group** field, select the group that the object belongs to.
5. Optional: Select an item for the service object line, or select a template bill of materials (BOM).

Related information

- Service objects overview

Feedback

Was this page helpful? Yes No

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Attach inventory items to service objects

07/10/2025

This article explains how to attach an inventory item to a service object. When you attach an item to a service object, you can control and report the service activities that are performed for the item.

Before you can attach items to service objects, you must create the items on the **Released product details** page.

Use the following steps to attach an inventory item to a service object:

1. Go to **Service management** > **Setup** > **Service objects** > **Service objects**.
2. In the **Item number** field, select the item to attach to the service object.
3. Save the service object.

The item is now attached to the service object, and any inventory dimensions specified for the item are also copied to the service object.

Related information

- Service objects overview

Service object groups

Article • 01/30/2025

Object groups are useful for sorting and filtering the data about objects for reports and statistics. For example, you can group objects by geographical location or by type.

Group by geographical location

You can use this grouping method to show where the various different objects that your company services are located. Grouping objects by geographical location can also be useful if, for example, you must identify the objects that your company already provides services for in a particular country/region.

Example of grouping by geographical location

A customer from Belgium calls your service center and wants to create a service agreement for an object, ABC. You have attached an object group for geographical location, Belgium, to all objects that are serviced in Belgium. By using this group as a filter, you can quickly search to see whether you already have a record for ABC in the program, or whether you must set up a new object.

Group by type

You can use this grouping method to show the types of objects that your company services. Grouping objects by type can also be useful if, for example, you want to create a new object based on similar objects that already exist in the program.

Example of grouping by type

A customer calls and wants to set up a service agreement for an air conditioning machine, HIJ. You don't already have a record for this machine. However, you have set up an object group titled Air Conditioners, and you have attached this group to all air conditioning objects. Therefore, you can quickly search for and identify all other air conditioning machines and use the template information from these objects to create service agreement lines for HIJ. By using object groups in this manner, you can quickly set up new objects and determine the service tasks that must be performed on them.

Create service object groups

Create groups that you can assign service objects to. Service objects are inventory items and other products for which services are performed. By grouping service objects, you can create reports for similar and related service objects. For example, a service object group might consist of two service objects: One service object is a kit, and the second service object is the service to install the kit.

To create service object groups, follow these steps:

1. Go to **Service management** > **Setup** > **Service objects** > **Service object groups**.
2. Select **New** to create a new service object group.
3. Enter a unique name for the service object group and, optionally, a description.

You can assign service objects to the group by using the **Service objects** page.

Related information

- Create service objects

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up template to represent objects at customer site

07/10/2025

Use this procedure to set up a template that represents objects at a customer site.

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Open a service agreement, or select **Service agreement** to create a new service agreement.
3. Select the **Setup** tab.
4. Select **Service objects** to attach a template bill of materials (BOM) to the service agreement.
5. On the **Service objects** page, select a service object, and then select **Functions** > **Attach template BOM**.
6. On the **Select template BOM** page, select a template BOM in the **Template BOM** field, and then select **OK**.
7. On the **Service objects** page, select **Designer** to modify the template BOM so that the BOM represents the objects that you want.
8. Close the pages to save your changes.

Related information

- Manage template BOMs on object relations

Service object relations

07/10/2025

You can create service object relations between a service object and a service agreement or service order. When you create a relation, you attach the service object to the service agreement or service order.

After the relation is created, you can attach the service object to any lines on the service agreement or service order.

Template BOMs

You can also specify a template BOM for the object relation. The template BOM is a bill of materials for the object on which you perform service. If you replace a component part of the service object during a service visit, you can register this change in the service BOM by using the **Service objects** page.

Example

You create a service agreement for servicing two elevators at a customer site. The service agreement has the identifier ID SAL-001.

The elevators are set up on the Service objects page as objects, EL-S/1000 and EL-L/1000.

You attach the service objects, EL-S/1000 and EL-L/1000, to the service agreement.

You want to register changes in the BOM for the service object as you replace component parts of the object, so you attach a service BOM to the service object relation, as described in the following table.

Expand table

Service object	Relation – Service agreement	Service BOM
EL-S/1000	ID SAL-001	BOM-EL-S/1000
EL-L/1000	ID SAL-001	BOM-EL-L/1000

Because there are service object relations for the agreement, you can now create service agreement lines with these objects, as shown in the following table.

Expand table

Transaction type	Category	Service object
Hour	Inspection	EL-S/1000
Hour	Gear box cleaning	EL-S/1000
Item	Cleaning materials	EL-S/1000
Hour	Inspection	EL-L/1000
Hour	Gearbox cleaning	EL-L/1000
Item	Cleaning materials	EL-L/1000

On a service visit, you have to replace the gearbox for elevator EL-S/1000. To replace a component part of the object, you can access the BOM Designer by using the service object relation that you set up for the current service agreement.

Access the BOM Designer by using a service object relation

1. Go to Service agreements.
2. Open a service agreement, or select **Service agreement** to create a service agreement.
3. Select the **Setup** tab.
4. Select **Service objects** to attach a template BOM to the service agreement.
5. On the **Service objects** page, select **Designer** to open the Designer page to modify the template BOM.

Automatically created service orders

If you automatically create service orders for a service agreement, the service object relations in the agreement are also created in the service orders.

Create service object relations

07/10/2025

This article describes how to create service object relations for a service agreement and a service order. When you a create service object relation, you associate the service object to a service agreement or service order. When a customer requests service for an item that is a service object, you can select the service object from the list of service object relations.

Create a service object relation for a service agreement

Use the following steps to create a service object relation for a service agreement:

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. In the **Service agreements** list, select an existing service agreement, or select **New** to create a new service agreement.
3. On the **Service agreements** page, on the Action Pane, on the **Service agreement** tab, in the **Relations** group, select **Service objects**.
4. On the **Service objects** page, select **New**, and then select a service object for this service agreement.
5. To assign a template bill of materials (BOM) to the service agreement, select **Functions**, and then select **Attach template BOM**. On the **Select template BOM** page, in the **Template BOM** field, select a template.

Create a service object relation for a service order

Use the following steps to create a service object relation for a service order:

1. Go to **Service management** > **Service orders** > **Service orders**.
2. In the **Service orders** list, select an existing service order, or create a new service order.
3. On the **Service orders** page, on the Action Pane, on the **Service order** tab, in the **Relations** group, select **Service objects**.
4. On the **Service objects** page, select **New**, and then select a service object for this service order.

5. To assign a template BOM to the service agreement, select **Functions**, and then select **Attach template BOM**. On the **Select template BOM** page, in the **Template BOM** field, select a template.

Related information

- Service objects overview
- Service object relations
- Template BOMs

Template BOMs

07/10/2025

A template bill of materials (BOM) provides you with a standardized list of components for service objects that are serviced regularly. The components that are listed in the template BOM represent the individual subcomponents of the service object. By applying a template BOM to a service object, you can keep a record of the subcomponents that were replaced on the service object.

To apply a template BOM to a service agreement or a service order, you attach it to a service object relation.

> **Note**
>
> You can apply only one template BOM to a service object.

Create a template BOM

The following table contains information about the various methods that you can use to create a template BOM.

Expand table

Method	Description
Production	The template BOM is based on a production order. This option is applicable only if you operate in a production environment. The benefit is that you have a current, detailed listing of the components that make up an item.
Item BOM	The template BOM is based on an item BOM. The item BOM, unlike the production BOM, is a static list of the components that make up an item.
Existing template	The template is based on an existing template BOM.
Manual	If you know what spare parts are typically replaced on a service object, you can create your template BOM manually. This method helps make sure that the components that are included in the template reflect the actual requirements of your workplace.

Apply the template BOM to a service agreement or service order

You can apply a template BOM to a service agreement, a service order, or both. The service agreement usually covers a long-term relationship with a customer. The history of replacements that is recorded in the service BOM is useful data to have for the service agreement.

You can also apply a template BOM to a service order to record the history of the service that was performed on a service object.

Copy the history of a service BOM

You can copy the history of a service BOM line from one service agreement to another service agreement. By copying the service history between service agreements, you can preserve the record of replacements for an item.

Example

You set up a three-year service agreement for a customer's car. During that period, the customer becomes accustomed to the good service that the company provides. Therefore, after the agreement expires, the customer wants to set up a new one. You're now able to negotiate a more favorable agreement for the company. Because the record of replaced components might be useful in the future, you copy the history of the service BOM to the new agreement.

Modify the template BOM

If a template BOM hasn't been attached to a service object, you can modify or delete lines in it. After the template BOM is attached to a service object, you can modify only the local version of the BOM. If you want to duplicate the setup of a local version of a template BOM, you can create a new template BOM based on the local version. Learn more in Modify a Service BOM.

If you replace an item in the BOM, you can register the replacement on the BOM line in the BOM designer. Optionally, you can create a service order line for the replacement object. If you create a service order line, you can invoice the replacement item. If you don't create a service order line for a replacement, the replacement registration is kept to track the history of the service object.

Change how information on the BOM line is displayed

You can change the way that information on the BOM line is displayed for all template and service BOMs. The changes are applied to all template BOMs and service BOMs.

Set up number sequences for template BOMs

To use template BOMs, you must set up two number sequences. Set up one number sequence for the template BOM and one for the BOM history line number.

> **Note**
>
> Number sequences are used to allocate identifiers to records that require them. Before you can assign a number sequence to a template BOM or a BOM history line number, you must set up number sequences codes.

Set up number sequences

1. On the **Number sequences** list page, create number sequences for template BOMs and the BOM history line number.
2. Go to **Service management** > **Setup** > **Service management parameters**.
3. Select **Number sequences**, and then select a number sequence code for the number sequence references that you created on the **Number sequences** page.
4. Close the page to save your changes.

> **Note**
>
> The BOM history line number is used by the system to associate the transactions in the BOM history with a service agreement or service order. The number is not displayed in the user interface.

Related information

- Create a template BOM
- Manage template BOMs on object relations
- Modify a Service BOM

Create a template BOM

07/10/2025

You can create a template bill of materials (BOM) by using any of the following methods. For all methods, the **From date** and **To date** fields are optional and for information only.

Create a template BOM manually

1. Go to **Service management** > **Setup** > **Service objects** > **Template BOMs**.
2. Select **New** to open the **Create template BOM** page.
3. Under **Copy BOM lines from reference**, select the **Manual** option.
4. In the **Item number** field, enter an item of the type **BOM**.
5. In the **Name** field, enter a name for the template.
6. In the **From date** and **To date** fields, enter a date interval in which the template BOM is active.
7. Select **OK**.

A new, blank template BOM is created.

Create a template BOM based on another template BOM

1. Go to **Service management** > **Setup** > **Service objects** > **Template BOMs**.
2. Select **New** to open the **Create template BOM** page.
3. Under **Copy BOM lines from reference**, select the **Template BOM** option.
4. In the **Reference number** field, select an existing template BOM to copy to your new template BOM.
5. In the **Name** field, enter a name for the template.
6. In the **From date** and **To date** fields, enter a date interval in which the template BOM is active.
7. Select **OK**.

A new template BOM is created by using lines that correspond to the lines in the original template BOM.

Create a template BOM based on an item BOM

1. Go to **Service management** > **Setup** > **Service objects** > **Template BOMs**.
2. Select **New** to open the **Create template BOM** page.

3. Under **Copy BOM lines from reference**, select **BOM**.
4. In the **Reference number** field, select a BOM version. An item of the type BOM is copied to the **Item number**.
5. In the **Name** field, enter a name for the template.
6. In the **From date** and **To date** fields, enter a date interval in which the template BOM is active.
7. Select **OK**.

A new template BOM is created by using lines that correspond to the lines of the BOM listed in **Bills of materials**.

Create a template BOM based on a production BOM

1. Go to **Service management** > **Setup** > **Service objects** > **Template BOMs**.
2. Select **New** to open the **Create template BOM** page.
3. Under **Copy BOM lines from reference**, select **Production**.
4. In the **Reference number** field, select a production BOM.
5. In the **Name** field, enter a name for the template.
6. In the **From date** and **To date** fields, enter a date interval in which the template BOM is active.
7. Select **OK**.

A new template BOM is created by using lines that correspond to the lines of the BOM listed in **BOM**.

Related information

- Template BOMs

Attach templates to service objects

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

A template bill of materials (BOM) can be attached to a service object relation. The service object relation can then be associated with either service orders or service agreements.

1. Go to **Service management** > **Setup** > **Service objects** > **Service objects**.
2. Select a service object.
3. Select **Service agreements** or **Service orders** to display the service agreements or service orders that the service object has been associated with.
4. Select a service agreement or a service order, select **Functions**, and then select **Attach template BOM**.
5. Select the template BOM to attach to the object.

> **Tip**
>
> To move an attached BOM to another service object or to delete an attached BOM, on the **Service objects** page, select **Functions**, and then select **Move service BOM** or **Delete service BOM**.

Related information

- Create service objects

Last updated on 05/07/2025

Reason codes for service orders

07/10/2025

You can use reason codes to help explain the status of a service order when the stage of a service order is updated. For example, if you cancel a service order, you can select a reason code for the cancellation.

To view information about reason codes that are used to track the progress of service orders, run the Service order progress report. This report lists all service orders, regardless of their stage, and the reason codes that are specified when a service order stage is updated.

Turn reason codes on or off

Reason codes are optional. You can decide whether to require a reason code when you update a service order to a specific service stage.

1. Go to **Service management** > **Setup** > **Service orders** > **Service stages**.
2. On the **Service stages** page, select a service stage, and then select the **Reason** check box for the service stage.
3. Close the page to save your changes.

Related information

- Set up service order stages

Service order stages

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can set up stages for a service order to define the tasks that must be completed, the order in which they are completed, and the workers who are responsible for completing them. By defining the stages for a service order and assigning them to workers, you can control the flow of a service order through the tasks that various people perform in the service organization. The sequence of stages must include an initial stage.

You can also define the actions that are permitted at each stage. For example, if you clear the **Post** check box for all stages except the final stage, you prevent any service orders from being posted before the service orders are processed through the complete sequence of stages.

Branching in service order stages

When you set up a service stage, you can create multiple options, or branches, to select from for the next service stage. All the branches that you create are available to select from when the initial stage is completed. For example, you set up *Planning* as an initial stage. You create two stages named *In process* and *Cancel*, and then select *Planning* as the parent for them. You assign the *Planning* stage to sales order. When the planning stage for the sales order is completed, you can select the *In process* stage if the sales order is ready to work on, or you can select the *Cancel* stage if the sales order is canceled.

Related information

- Set up service order stages
- Change the service order stage

Last updated on 05/07/2025

Set up service order stages

07/10/2025

1. Go to **Service management** > **Setup** > **Service orders** > **Service stages**.
2. Select **New** to create a new record.
3. In the **Service stage** and **Description** fields, specify a service stage ID and description.
4. Select the appropriate parameters for the stage.
5. Select the parent stage for the current stage or leave the **Parent** field blank if the stage is the initial stage in the stage setup.

> **Note**
>
> You can't modify the **Parent** field after you save the stage. Instead, you can delete the record and create the record again with a different selection in the **Parent** field.
>
> Also, you can only create one stage with a blank **Parent** field. That is, only one initial stage is permitted.

Create reason codes

07/10/2025

1. Go to **Service management** > **Setup** > **Service orders** > **Stage reason codes**.
2. Select **New** to create a new reason code.
3. Select **Save** to save the reason code.

When you're prompted for a reason code to explain why you're updating a stage of the service order, you can select any of the reason codes that you create here.

Service level agreements overview

07/10/2025

A service level agreement (SLA) is an agreement between a service company and a service customer. In a SLA, the customer agrees to a minimum response time based on when the service company records the issue and when the issue is resolved.

A SLA enforces a standard level of service that is offered to customers, and also makes it transparent to a service company when a service job should be completed.

Any number of SLAs can be created to offer service customers different levels of service.

Create a service level agreement

1. Go to **Service management** > **Setup** > **Service agreements** > **Service level agreements**.
2. Type a name for the service level agreement in the **Service level agreement** field.
3. Type the time that you want to allow for completion of service calls that are attached to the service level agreement. Then select a calendar if you want to base the service level agreement on a specific calendar.

Apply a service level agreement

The SLA is applied directly to a service agreement.

Service orders that you create manually and attach to a service agreement that has an SLA are measured against that SLA.

Service orders that you create automatically aren't attached to an SLA.

Apply the service level agreement to the service agreement

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select the service agreement that you want to apply the SLA to, and then select **Edit** on the Action Pane.
3. In the **Service level agreement** field, select the SLA that you want to assign.

Apply the service level agreement to the service agreement group

1. Go to **Service management** > **Setup** > **Service agreements** > **Service agreement groups**.
2. In the **Service level agreement** field, select the SLA that you want to assign.

Track time on a service order against an SLA

When you create a new service order for a service agreement that an SLA is assigned to, the time interval for the delivery of the service is initiated, and the system starts to track the delivery time. Additionally, you can set the following options:

- You can start and stop time recording on the service order to register the total amount of time that is spent on service orders.
- You can monitor compliance with the time interval that is set in the service level agreement.
- You can define reason codes that must be set if the time interval of the service level agreement is exceeded.

Related information

- View compliance with service level agreements

Subscription sales prices

Article • 01/30/2025

When you create a subscription, the sales price is derived from the sales price setup that was created on the **Sales price (subscription)** page.

On the **Sales price (subscription)** page, you can create sales prices for a specific subscription or you can create sales prices that apply more broadly. For a sales price to be applied to a subscription, the period code and the currency of the subscription must be identical to the period code and the currency of the sales price.

If the period code and currency are identical for the subscription and the sales price, subscription sales prices are selected on the basis of the priorities listed in the following table. A blank cell in the table indicates an empty field and an X indicates a value that is equal to the value in the subscription from which the transaction is generated.

Expand table

Priority	Category	Project ID	Subscription	Sales currency	Period code
1	X	X	X	X	X
2		X	X	X	X
3	X		X	X	X
4			X	X	X
5	X	X		X	X
6		X		X	X
7	X			X	X
8				X	X

When a subscription fee is created, the sales price with the greatest level of detail, as noted in the table above, is selected as the subscription sales price.

Update and index subscription sales prices

You can update the subscription sales price by updating the base price or the index. You can update by a percentage or to a new value.

Subscription fee sales prices

When you create a subscription fee, the sales price is based on the sales price setup of the subscription. You can either use the base price from the subscription price setup or create indexed sales prices.

Example

You want to set up subscription sales prices of EUR 500 for a new project 9030. On the **Sales price (subscription)** page, you create a subscription sales price line as indicated in the following table.

Expand table

Valid from	Category	Project	Subscription	Period code	Sales currency	Sales price
28-08-2006		9030		Month	EUR	500

Note that the **Category** and **Subscription** fields are empty.

You then create the following subscriptions.

Expand table

Service subscription	Project	Subscription group	Category	Currency	Period code
00020_135	9030	Sub1	SubCat1	EUR	Monthly
00021_135	9030	Sub1	SubCat2	EUR	Monthly

Now you create subscription fees for both subscriptions in the subscription group Sub1:

1. Go to **Service management** > **Setup** > **Service subscriptions** > **Subscription groups**.
2. On the **Subscription groups** page, select **Function** > **Create subscription fee**.
3. On the **Create subscription fee** page, enter the appropriate information. Learn more in Create subscription fee transactions.

Subscription fees that have a sales price of EUR 500 are created for both subscriptions, as shown in the following table.

Expand table

Project date	Service subscription	Project	Category	Start date	End date	Sales currency	Sales price
28-08-2006	00020_135	9030	SubCat1	01-01-2007	31-03-2007	EUR	500
28-08-2006	00021_135	9030	SubCat2	01-01-2007	31-03-2007	EUR	500

Later, you decide that you want to specify sales prices for the category SubCat1 for project 9030. Therefore, you create a new sales price line that has a sales price of EUR 550 for the combination of project 9030 and fee category SubCat1. There are now two subscription sales price lines for project 9030, as shown in the following table.

Expand table

Valid from	Category	Project	Subscription	Period code	Currency	Sales price
28-08-2007		9030		Month	EUR	500
28-08-2007	SubCat1	9030		Month	EUR	550

You repeat the procedure described above to create subscription fees for both subscriptions in the subscription group Sub1. The following table shows the transactions that are created for each subscription that is attached to the subscription group.

Expand table

Project date	Subscription	Project	Category	Start date	End date	Sales currency	Sales price
28-07-2007	00020_135	9030	SubCat1	01-01-2008	31-03-2008	EUR	550
28-07-2008	00021_135	9030	SubCat2	01-01-2008	31-03-2008	EUR	500

In the first transaction for subscription 00020_135, the sales price of EUR 550 derives from the subscription sales price that is set up for the combination of the specific project and category. In the second transaction for subscription 00021_135, the sales price of EUR 500 is used as the project subscription sales price because there's no price set up for the combination of project 9030 and category SubCat2.

Related information

- Update and index subscription sales prices

Feedback

Was this page helpful? Yes No

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Subscription groups

Article • 08/26/2024

Subscriptions share the following information, which is defined for the subscription group:

- Number sequence code for the subscription, if any.
- The invoicing period for which subscription transactions are created. For example, if the **Invoicing** period code on the **Subscription groups** page is set to **Quarterly**, transactions for all subscriptions that are attached to the subscription group are created on a quarterly basis.
- How frequently the subscription is to be accrued, if at all.

You can also complete the following tasks for all subscriptions that are attached to the subscription group:

- Create subscription transactions.
- Update the base price or the index for the subscriptions.
- Post accrued revenue for subscription transactions.

Related information

- Set up subscription groups

Feedback

Was this page helpful? Yes No

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Set up subscription groups

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

1. Go to **Service management** > **Setup** > **Service subscriptions** > **Subscription groups**.
2. Select **New** to create a new subscription group.
3. In the **Subscription group** field, enter a name for the new subscription group.
4. In the **Invoicing** field, select an invoicing frequency for the subscription. The invoicing frequency is indicated by a period code that is set up on the **Period types** page.
5. To accrue revenue for subscription transactions, select the **Accrue revenue** check box, and then in the **Accrual** field, select a frequency. The accrual frequency is based on a period code that is set up on the **Periods** page.

Related information

- Subscription groups

Last updated on 05/07/2025

Use stage reason codes

07/10/2025

You use a reason code to indicate why a service level agreement (SLA) has been canceled, or why a service order has exceeded the time limit that is you define in the SLA.

You can also specify that a reason code is required when an SLA is canceled, or when the time limit exceeds the time that is specified in the SLA for the service order.

If you have specified that a reason code is required, you must enter a reason code in the following situations:

- When a service order is moved to a stage that stops time recording against the SLA for the service order
- When the service order is signed off
- When time recording is manually stopped

Set up reason codes

1. Go to **Service management** > **Setup** > **Service orders** > **Stage reason codes**.
2. On the **Stage reason codes** page, select **New** to create a new reason code.
3. In the **Stage reason code** field, enter a unique stage reason code.
4. In the **Description** field, enter a description of the stage reason code.
5. Close the page to save your changes.

Require reason codes when a service level agreement is canceled

1. Go to **Service management** > **Setup** > **Service management parameters**.
2. On the **Service management parameters** page, select the **General** link, and then select the **Reason code on canceling** check box.

Require reason codes when a service order exceeds the time limit that is set by the service level agreement

1. Go to **Service management** > **Setup** > **Service management parameters**.
2. On the **Service management parameters** page, select the **General** link, and then select the **Reason code on exceeding time** check box.

Related information

- Start and stop time recording on a service order

Integration for service agreements and projects

07/10/2025

When you work with service agreements and service agreement lines, you use data that is set up in the following areas in **Project management and accounting**.

Project prices

The cost and the sales price for a service agreement transaction are derived from the cost price setup that applies to the project that is attached to the service agreement. Cost and sales prices can be set up by project, employee, and category.

Project validation

The projects, employees, and categories that are available for selection on a service agreement line can be limited by the validation setup in **Project management and accounting**. By using the validation setup, you can combine employees, projects, and categories for control access.

Project line properties

A line property is entered automatically for a service agreement line.

Line properties are created on the **Line properties** page in **Project management and accounting**. The line property that is entered on a service agreement is attached to the project that is selected for the service agreement and inherited subsequently by the service agreement line.

Default offset accounts

If you enter an expense transaction, a default expense offset account is selected automatically for the transaction. The default expense account is set up for the project that is attached to the current service agreement.

Project categories

The categories that are available for service agreement lines are set up on the **Project categories** page in **Project management and accounting**.

> **Note**
>
> Categories that have the **Active in journals** check box selected on the **Project** tab on the **Project categories** page are available for selection. However, if the **Inactive categories** check box is selected on the **Journals** tab on the **Project management and accounting parameters** page, all categories are available for selection.

Project parameters

If the **Terminated workers** field on the **Journals** tab on the **Project management and accounting parameters** page is selected, you can select inactive employees and active employees on the **Service agreements** and **Service orders** pages.

Also, you can enable the **Start time** and **End time** fields on the **Project** tab on the **Service orders** page to enter starting and ending times on service order lines.

Enable the starting and ending time feature for service orders

1. Select **Project management and accounting** > **Setup** > **Project management and accounting parameters**.
2. Select the **Journals** tab, and then select the **Show start/end times** check box.
3. Select **Project management and accounting** > **Setup** > **Journals** > **Journal names**.
4. Select the journal name that is attached to the service order.
5. Select the **General** tab, and then select the **Show start/end times** check box.

> **Note**
>
> Select the journal name for the service order in the **Hour** field on the **Journals** tab of the **Service management parameters** page.

Repair management

07/10/2025

For repair management you can group problems systematically. This is to help with the suggestion of solutions that have been successful in the past.

You set up symptoms, diagnosis, and resolution settings. All these can later be applied when you receive a similar item for repair. You can also set up repair stages that can follow the progress of a repair issue.

Setting up repair management

Use the following setup pages to enter information that will be used to specify the symptoms, the diagnosis, and the resolution, of the repair.

- **Service management** > **Setup** > **Repair** > **Conditions**.
- **Service management** > **Setup** > **Repair** > **Symptom areas**.
- **Service management** > **Setup** > **Repair** > **Diagnosis areas**.
- **Service management** > **Setup** > **Repair** > **Resolutions**.
- **Service management** > **Setup** > **Repair** > **Repair stages**.

Symptoms and conditions

Symptoms are how the customer characterizes the problem. Symptoms include the customer observations that indicate the need for repair.

You can set up symptom areas, symptom codes, and conditions. The symptom area covers the area of malfunction, and the symptom code covers the malfunction symptom. The condition describes the situation or environment that is present when the problem occurs.

Example

A computer is brought in for repair because the hard drive fails after an extended period of use. The hard drive failure causes a blue screen error. The technician who receives the computer enters the following symptoms and conditions:

- The symptom area is the hard drive.
- The symptom code is the blue screen error.
- The condition is that the hard drive fails after extended use.

Diagnosis and resolutions

The diagnosis and resolution fields are statements from the repair perspective. These are the steps that a technician went through to investigate the problem.

The diagnosis area covers the operation that must occur to solve the issue. The diagnosis code is how the problem was handled, and the resolution could be that the item was repaired, replaced, or the order was canceled by the customer. For example, if the computer is repaired, the diagnosis area could be *defective part*, the diagnosis code could be *new video card installed*, and the resolution could be *replaced*.

Repair stages

Repair stages state the progress of the repair process. The repair stage has a **Finished** sign-off parameter that indicates that the repair line has been completed, and the finishing date and time has been recorded.

Applying repair management

To apply repair management to an item, the item must be set up with a service object relation on a service order. From the service order you can then create a repair line with information about the current issue. On the repair line you define the service object that is in repair and information about symptoms, diagnosis, and execution. You can also create a note for the repair line.

You can create repair lines for each step in the repair process.

Create a repair line on a service order

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select the service order with the service object that needs repair.
3. Select **Repair** > **Repair lines** to open the **Repair lines** page.
4. Select **New** to create a new line.
5. Select a service object. You can select any service object that has been set up with an object relation on the service order.
6. Select any of the preset symptoms, diagnosis, and execution values that are relevant in the repair line and then select the **Note** tab to create a note on the repair line, if needed.
7. Select **Save** to save the new repair line. The **Created date and time** field in the **General** tab of the **Repair lines** page is updated with the time of saving.

Tracking progress and resolving a repair issue

You can set the repair stages of a repair line to track the progress of the repair.

When a repair issue is resolved, you can close the repair line. Set the repair stage to a stage with a **Finished** property enabled. The current date and time is registered as the finish time on the line.

Close a repair line for a resolved issue

1. Open the **Repair lines** page. Follow the procedure earlier in this article to create a repair line.
2. Select the repair line with the repair issue that you want to close.
3. In the **Repair stage** field, select a stage with the **Finished** property enabled.

Work with customer service overview

Article • 01/06/2025

Customer service supports professional, service-based companies and manufacturing service organizations.

Customer service includes the following functionality:

- Key processes for service management, such as service agreements, service orders, and return management
- An advanced setup for repair management that includes tracking of repairs and versioning for bills of materials (BOMs)
- Support for employee roles, such as service manager, dispatcher, and service technician or service consultant
- Service subscriptions, which can generate periodic streams of revenue that are independent of the frequency of service
- Service-level agreements that support a differentiated customer base

The articles in this section provide an overview of the Customer service business process and each business process component. They also list the pages that are associated with each business process component, and they discuss the tasks that are associated with each business process component.

Feedback

Was this page helpful? Yes No

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Fulfill service agreements overview

Article • 01/06/2025

This article provides an overview of the tasks to fulfill service agreements. The article also lists the pages that are associated with each business process component and discusses the tasks that are associated with each business process component.

Pages for the fulfill service contracts business process components

The following table lists the pages that support the fulfill service contracts business process components. The table entries are organized first by task, and then alphabetically by the name of the page.

> **Note**
>
> To open some of the pages in the following table, you must enter information or parameter settings.

Expand table

Business process component task	Form name	Usage
Develop and establish agreement	**Service agreements**	Create and maintain service agreements.
Develop and establish agreement	**Subscription**	Create and maintain service subscriptions.
Monitor and implement agreement	**Change service stage**	Select the next stage in the workflow for a service order.
Monitor and implement agreement	**Service orders**	Create and maintain service orders.
Update and evaluate contract	**Cancel service level agreement**	Cancel the service level agreement on a service order.
Update and evaluate contract	**Cancel service orders**	Cancel service orders, or reverse the Canceled status of service orders.
Update and evaluate	**Index subscription**	Update the base prices and indexes for

Business process component task	Form name	Usage
contract		subscriptions.

Feedback

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Develop and establish service agreements overview

07/10/2025

Service agreements let you define the resources that are used in a typical service visit and how those resources are invoiced to the customer.

Every service agreement is attached to a project through which transactions are posted and invoiced. However, you can also invoice service order transactions directly through the project without first having to connect the service order to a service agreement. You might decide to do this if the service order is for a one-time-only service visit and the need for processing the service transactions quickly outweighs the need for maintaining detailed service-agreement information about the customer over a period of time.

Service agreement

In each service agreement, you must specify a project, a service-agreement ID, and a service-agreement group. The service-agreement group can be used to sort and organize service agreements.

A service agreement header contains settings that apply to all attached agreement lines, including:

- Whether the service agreement is suspended
- The duration of the service agreement
- How service-order lines are to be combined into service orders
- Whether the service agreement is a template

If the service agreement is suspended, you can't generate service orders from the service agreement.

In the service-agreement header, you also set up all the service objects and service tasks that can be used with the service agreement by entering the specific service tasks or service objects that are to be attached to the various lines of the agreement.

From the service-agreement header, you can also copy service-agreement lines or service-template lines into the current service agreement.

You can suspend service agreements and stop individual service agreement lines.

If you select the **Suspended** check box on a service agreement, you can't:

- Create service orders automatically or manually from the service agreement.

If you select the **Stopped** check box on a service agreement line, you can't:

- Create service orders automatically or manually from the service agreement line.
- Copy the service agreement line into another service agreement or service order.

> **Note**
>
> If a service agreement is suspended, all the attached lines are stopped, regardless of their individual status.

Service-agreement lines

Each service-agreement line describes in detail the content of the proposed service work. The lines contain the following settings:

- The transaction type and the description of the transaction type
- The employee who performs the service work
- The objects on which service must be performed for the service agreement
- The frequency with which work is performed and associated item, expense, and fee transactions are registered
- The time window within which service-order lines can be grouped into a service order
- The service tasks that are used to group sets of agreement lines together into work tasks and to summarize for service technicians and customers what service task is to be provided
- Whether a line is stopped (if a line is stopped, you can't create service orders for that individual line)
- Project settings, such as the category and the line property

Related information

- Create service agreements

Create service agreements

07/10/2025

This article describes how to use features in the Service management, Project management, and Accounting modules to create service agreements.

Create a service agreement from Service management

1. Navigate to **Service management**.
2. Select **Service agreements** to create a new service agreement line in the page header.
3. Select **New**. Enter a description, select a reference to a project in the **Project ID** field, and fill in the rest of the fields and lines for the service agreement. Select **Save**.
4. On the **Relations** tab, select **Service objects** or **Service tasks** to create service object relations or service task relations for the service agreement. The service objects and tasks that you have created relations for can be attached on the lines of the service agreement.
5. In the lower half of the page, create service agreement lines by copying lines from a service template, another service agreement, or manually creating the service-agreement lines.

> **Note**
>
> If you copy lines into the service agreement from another service agreement, you can indicate whether you also want to copy service object and service task relations. If you copy these relations, they are added to any existing relations on the service agreement. If you copy service-agreement lines from a service template, the service-object and service-task relations are automatically copied as service-object relations and service-task relations on the new service-agreement lines.

Create service agreement lines manually

1. From the **Service agreements** page, add a service agreement line in the lines grid.
2. Enter the appropriate information for the service agreement line.
3. Select **Save** to save the line, and then close the page.

Create a service agreement from Project

1. Select **Project management and accounting**.

2. Select **All projects**.
3. Select the project from the list.
4. On the Action Pane, select **Manage**. In the **New** Action group, select **Service** and select **Service agreement**.
5. Follow the steps in the section titled **Create a service agreement** as described earlier in this article to enter the project reference.

Related information

- Develop and establish service agreements overview

Service subscriptions

Article • 01/30/2025

Subscriptions are based on a time-and-materials project. You can create subscriptions from a project, from the **Service subscriptions** page, or by using a service subscription group.

For each subscription, you can create an optional number of subscription fees. The subscription fees are the transactions that you invoice to the customer.

A period code indicates the extent of the subscription fee period, or the intervals at which the subscription should be invoiced.

The period code is defined in the service subscription group. It is defined for the subscriptions through the service subscription group. Any new subscription fee that you create has a suggested start date. This is the start date of the period, if it is the first period that is created, or the end of the previous period, if it is the second or later period.

Feedback

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Service templates

Article • 06/03/2022

You can define a service agreement as a template and copy the lines of the template later into another service agreement or into a service order.

Example

A customer for whom you provide service has identical service elevators at five different locations.

You want to set up five service agreements for the customer, one for each site. To limit repetitive setup work, and to make sure that the setup information in the service agreements is identical, you create a service agreement and specify it as a template for the service work on the elevators.

You can now copy the template lines into the five new service agreements, so that each service agreement is populated with the lines from the template.

Create a template

When you create a service template, you create a service agreement, create the required lines on it, and attach the service agreement to a service-template group.

> ⓘ **Note**
>
> A service agreement can be defined as a template only if it has no service orders attached to it. Also, no service orders can be generated from a service agreement that is defined as a template.

Copy template lines

You can copy template lines from a service template into another service agreement or into a service order.

When you copy template lines into your service orders or service agreements, your template groups are displayed in a tree view in which each group can be expanded. This display lets you view each template and template line.

It is easier to determine the service-template lines that you want to copy if you have grouped the templates under names that reflect the use of the templates.

Related information

- Copy service templates lines

Feedback

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Copy service templates lines

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Template information can be copied to either a service agreement or a service order.

Copy service template lines into a service agreement

1. Select **Service agreements**.
2. Select a service agreement to open the **Service agreements** page.
3. Select **Template lines** to open the **Copy service template lines** page.
4. On the **Tree control** tab, expand the tree, find the template that contains the lines that you want to copy to the service agreement, and then select the service template.
5. On the **Lines** tab, select the **Select** check box for the service template lines that you want to copy.
6. Select **OK**.
7. Close the pages to save your changes.

The service template lines are now copied to the service agreement, together with the attached service object and service task relations.

Copy service template lines into a service order

1. Select **Service orders**.
2. Select a service order to open the **Service orders** page.
3. Select Copy > Copy template lines to open the **Copy service template lines** page.
4. On the **Tree control** tab, expand the tree, find the template that contains the lines that you want to copy to the service order, and then select the service template.
5. On the **Lines** tab, select the **Select** check box for the service template lines that you want to copy.
6. Select **OK**.
7. Close the pages to save your changes.

The service template lines are now copied to the new service order, together with the attached service object and service task relations.

Related information

- Service templates

Last updated on 05/07/2025

Change the service order stage

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Follow these steps to move a service order to the next stage in the service workflow or revert it to a previous stage.

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select the service order, and then select the **Dispatch** tab.
3. On the Action Pane, select either **Next stage** or **Previous stage**.
4. On the **Change service stage** page, select the appropriate stage.

Last updated on 05/07/2025

Manage template BOMs on object relations

07/10/2025

Attach a template BOM to a service object

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select the service agreement in which you want to attach a template BOM to an object relation.
3. Select **Setup** > **Service objects**.
4. On the **Service objects** page, select the object to attach a template BOM to, and then select **Functions** > **Attach template BOM**.
5. In the **Select template BOM** dialog box, select a template BOM, and then select **OK**.
6. Close the pages to save your changes.

Delete a service BOM from a service object

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select the service agreement in which you want to delete a service BOM from an object relation.
3. Select **Setup** > **Service objects**.
4. On the **Service objects** page, select the object that has the service BOM to delete, and then select **Functions** > **Delete service BOM**.
5. Close the pages to save your changes.

Move the service BOM history from one service agreement to another

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select the service agreement to move a service BOM from.
3. Select **Setup** > **Service objects**.

4. On the **Service objects** page, select the object that has the service BOM to move, and then select **Functions** > **Move service BOM**.
5. On the **Select service object relation** page, select the service object relation to move the service BOM to, and then select **OK**.
6. Close the pages to save your changes.

Modify the information displayed for a BOM line

1. Go to **Service management** > **Setup** > **Service objects** > **Template BOMs**.
2. Select a template BOM, and then select **Designer**.
3. Select the **Setup** tab. In the **BOM** area, select the check boxes for the information to display on the BOM lines of your service BOMs and template BOMs.
4. Close the pages to save your changes.

Related information

- Delete a service BOM

Delete a service BOM

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

1. Go to **Service management** > **Service agreements** > **Service agreements**. Select the service agreement that you want to delete a service bill of materials (BOM) from.
2. On the Action Pane, select **Service objects**.
3. On the **Service objects** page, select the service object that you want to delete a service BOM from. Select **Functions**, and then select **Delete service BOM**.
4. Select **Yes**.

> **Note**
>
> If you have not applied a template BOM, the service BOM can be deleted from the **Template BOMs** page. If you have applied a template BOM, the service BOM cannot be deleted by using this method.

Related information

- Template BOMs
- Manage template BOMs on object relations

Last updated on 05/07/2025

Indexed sales prices

Article • 01/06/2025

You set up the index for a subscription sales price when you create a subscription fee.

On the **Create subscription fee** page, set the **Get pricing from** field to **Indexed base price**, and then multiply the base price by the percentage in the **Percent price change** field to get the sales price of the subscription transaction.

For example, if the base price is EUR 1,000, and the index is 110, the sales price is EUR 1,100.

Related information

- Create subscription fee transactions

Feedback

Was this page helpful? Yes No

Provide product feedback

Reduce the days on subscription fees

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

To reduce the number of days of an existing subscription fee, you can create a new transaction in which you remove the period of time that should no longer be part of the subscription fee interval.

Reduce the days on a subscription fee

1. Go to **Service management** > **Service subscriptions** > **All service subscriptions**. Select the service subscription, and on the Action Pane, select **Subscription fees**
2. In the **Subscription type** field, select **Reduction days**.
3. Use the **From date** field and the **To date** fields to define the date interval of the subscription fee that you want to remove from the subscription fee period, and then select **OK**.

To view the transaction that was created, on the **Subscription** page, select **Fee transactions**.

Example

If a subscription transaction period runs from January 1 to January 31, and you want to reduce the period by 10 days, create a new transaction in which the reduction period is January 1 to January 10. (The reduction period could also be January 5 to January 15, or any other ten day period).

Also, if the **From date** on the reduction period is January 21 (31 minus 10), you could set the **To date** to any date after January 31, and 10 days will still be removed from the fee transaction period.

Related information

- Reduction days example

Last updated on 05/07/2025

Reduction days example

Article • 08/15/2024

You have created a subscription transaction for a customer's maintenance subscription, as described in the following table.

Expand table

From date	To date	Subscription	Subscription type	Project	Category	Sales currency	Sales price
March 01, 2011	March 31, 2011	NR-2	**Regular**	9013	SubCat2	EUR	200.00

The customer reports that it does not need service coverage for two days (March 10 and March 11). You agree to reduce the subscription by these two days.

You create a new transaction of the **Reduction days** type, as described in the following table.

Expand table

From date	To date	Subscription	Subscription type	Project	Category	Sales currency	Sales price
March 10, 2011	March 11, 2011	NR-2	**Reduction days**	9013	SubCat2	EUR	-12.90

When the transactions for March 2011 are invoiced, the sales price of EUR 200 is reduced by EUR 12.90. The chargeable amount for the subscription transaction is therefore EUR 187.10, and two transactions are invoiced at a total of EUR 187.10.

Related information

- Reduce the days on subscription fees

Feedback

Was this page helpful? Yes No

Provide product feedback

Update and index subscription sales prices

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

1. Go to **Service management** > **Service subscriptions** > **All service subscriptions**.
2. Select the subscription to update the index or base price for.
3. Select **Index**. In the **Change** field, indicate how to update the base price or index, and whether the update must use a percentage or a new value. You can select from the following options:
    - **Index by a percentage**
    - **Index to**
    - **Base price by percentage**
    - **Base price to**
4. Select **OK**.

The base price or index of the subscription is updated with the specified value.

Last updated on 05/07/2025

Fulfill service agreements overview

Article • 01/06/2025

This article provides an overview of the tasks to fulfill service agreements. The article also lists the pages that are associated with each business process component and discusses the tasks that are associated with each business process component.

Pages for the fulfill service contracts business process components

The following table lists the pages that support the fulfill service contracts business process components. The table entries are organized first by task, and then alphabetically by the name of the page.

> **Note**
>
> To open some of the pages in the following table, you must enter information or parameter settings.

Expand table

Business process component task	Form name	Usage
Develop and establish agreement	**Service agreements**	Create and maintain service agreements.
Develop and establish agreement	**Subscription**	Create and maintain service subscriptions.
Monitor and implement agreement	**Change service stage**	Select the next stage in the workflow for a service order.
Monitor and implement agreement	**Service orders**	Create and maintain service orders.
Update and evaluate contract	**Cancel service level agreement**	Cancel the service level agreement on a service order.
Update and evaluate contract	**Cancel service orders**	Cancel service orders, or reverse the Canceled status of service orders.
Update and evaluate	**Index subscription**	Update the base prices and indexes for

Business process component task	Form name	Usage
contract		subscriptions.

Feedback

Was this page helpful? Yes No

Provide product feedback

Start and stop time recording on a service order

07/10/2025

Use this procedure to start and stop time recording for a service order for which a service level agreement is defined.

Start time recording

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select the **Service order** tab. On the Action Pane, in the **Service level agreement** group, select **Start**.
3. Enter the date and time that the time recording should be started.

Stop time recording

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select the **Service order** tab. On the Action Pane, in the **Service level agreement** group, select **Stop**.
3. Enter the date and time that the time recording should be stopped.
4. Select **Add a revocation reason**, and select a reason code in the **Stage reason code** list to provide a reason for stopping the time recording.

> ⓘ **Note**
>
> If **Reason code on exceeding time** is selected on the **Service management parameters** page, you must provide a reason code before you can stop the time recording.

View the start time and duration of a service order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can view when the work on the service order was started and when the service order is going to be completed.

You can also view when the time recording for a service order was started and stopped. When a service order is stopped, the time at which the service order must be completed is postponed.

View the start time for a service order

1. Go to **Service management** > **Service orders** > **Service orders**. Select an order to open the details page.
2. On the **General** tab, view the time that the work was started for a service order in the **Start time** field.

View the time remaining to complete a service order

1. Go to **Service management** > **Service orders** > **Service orders**. Select an order to open the details page.
2. On the **General** tab, view the time remaining to complete a service order in the **Latest completion time** field.

View the start time and stop time recording entries for a service order

1. Go to **Service management** > **Service orders** > **Service orders**. Select an order to open the details page.
2. On the Action Pane, select the **Dispatch** tab > **Time recording** to open the **SLA time recording** page and view the time recording entries for the service order.

Last updated on 05/07/2025

View stage reason codes

View and modify the list of stage reasons for a service order. Use stage reasons to indicate the reason for the status of a service order. For example, you can create a **Canceled** stage that is assigned a reason code of **Canceled by the customer**.

- Go to **Service management** > **Service orders** > **Service orders**. Open a service order.
- On the **Service orders** page, on the Action Pane, on the **Dispatch** tab, select **Stage reasons**.
- On the **Stage reasons** page, select **New** to create a new stage reason code.

Last updated on 07/01/2026

View compliance with service level agreements

07/10/2025

View compliance with a service level agreement

1. Go to **Service management** > **Service orders** > **Service orders**. Open the details page for the order.
2. On the **General** tab, in the **Compliance** field, view the percentage of time that was spent on the service order compared to the time frame specified in the service level agreement.

> **Tip**
>
> If the value in the **Compliance** field is larger than 100, the time limit of the service level agreement was exceeded.

Related information

- Service level agreements overview

Service orders

07/10/2025

A service order represents a visit that a service technician makes to a customer site on a specific date. Each service order consists of one or more service order lines. Service order lines represent the hours of work that must be performed by the service technician, and the related items, expenses, and fees.

You can attach tasks and objects to a service order line. You can then group service order lines by task or by object. You can also attach items that are listed in inventory to service order lines.

Create service orders

You can create service orders based on a service agreement and the lines that are contained in that agreement. However, you can create service orders that are associated with a service agreement only in the period that is specified in the agreement. For example, if a service agreement is valid for the 2011 calendar year, you can create service orders for the agreement from January 1, 2011, and December 31, 2011.

You can also create service orders individually, without associating them with an agreement. These service orders can be used to handle unscheduled or one-time service visits. For example, in the month of March, your customer wants service to be performed on two machines, in addition to the machines that are specified in the service agreement. For this task, you create service orders but do not associate them with an agreement.

> ⓘ **Note**
>
> To create service orders that are not associated with a service agreement, you must select the **Allow without service agreement** check box on the **Service management parameters** page.

Scenario

The following scenario describes another situation where it is useful to create a service order that is not associated with a service agreement.

The company dispatcher receives a call requesting emergency service on an elevator. There is no time to set up a service agreement and a project for the service. Therefore, the dispatcher creates a service order directly on the **Service orders** page, attaches the service order to an existing project, and creates the service order lines. The dispatcher also creates a task or object

relation for an existing service order, to record work that is not related to the service agreement. Learn more in Create service orders manually and Create service task relations.

Monitor the progress of service orders

To monitor the progress of a sales order through the different teams and work processes, you can set up a system of stages and reason codes for service orders. For each stage, you can specify the actions that are allowed. Learn more in Create reason codes.

Example

A service order is approved by the dispatcher. The dispatcher updates the stage of the service order and specifies a reason code that indicates that the service order has been released to the service technician. The technician goes to the customer site and performs the service.

Specify item requirements for service orders

You can specify the inventory items that are required for service orders. However, the service order must be associated with a project. Item requirements for service orders are processed through a project.

Example

The service orders that are created from the service agreement are processed by the dispatcher. For the first service order, the dispatcher realizes that the service technician requires an important spare part that is not in the on-hand inventory. Therefore, the dispatcher creates an item requirement for the spare part directly from the service order.

Move and post lines

A service technician returns from a service visit, and then modifies and updates the service order lines. During the service visit, the technician performed a service job that was scheduled for the next service visit. Therefore, the technician moves the lines from the next service visit to the current service visit. The technician then posts the service order. Learn more in Move service order lines.

Cancel service orders

One of the other service orders that was generated for the month of January becomes obsolete, because the job is canceled. Therefore, the service dispatcher cancels the service order. Learn more in Cancel service orders.

Post from projects

At the end of each week, the dispatcher wants to post all service orders that are attached to a specific project. Therefore, the dispatcher locates the relevant project on the **Projects** page and posts the service orders that have been completed.

Delete service orders

During the second half of the year, your customer decides that the service visits are too infrequent. You must create a new, more frequent series of service visits for the remaining time on the service agreement. Therefore, you must delete the existing service orders and create new service orders. Learn more in Delete service orders.

Automatically create service orders

Article • 03/17/2025

You can generate service orders that are based on a service agreement for the valid period of the service agreement.

The service orders that you generate from a service agreement are all attached to the service agreement.

Service orders are generated automatically from the following settings:

- The service agreement interval that is set up in the service agreement lines. The service agreement interval indicates the frequency that service-order lines are created. Learn more in Service intervals.
- The period that you specify to define the service period in the **From date** and **To date** fields on the **Create service orders** page. Learn more in Create service orders automatically.
- The **Combine service orders** option on the service agreement header. This option defines whether service order lines generated from a service agreement, combines service orders according to employee, service task, service object, or service agreement. Learn more in Combine service orders.
- The **Time window** option on the service agreement line. The time window defines how far a service order line can move relative to its calculated date. Learn more in Time windows.

> **Note**
>
> If the day that is specified for a service order isn't open in the calendar selected on the **Service management parameters** page, a message indicates that there's a calendar conflict. You can safely ignore the message; the service order will still be created even though the day is closed on the calendar.

Example 1

The service agreement lasts from January 1, 2012 until December 31, 2012. If the service period that you specify on the **Create service orders** page is from November 1, 2012 until February 1, 2013, service orders are created from November 1, 2012 until December 31, 2012.

Example 2

The service agreement lasts from January 1, 2012 until December 31, 2012. Two service agreement lines are attached to the service agreement. The first service agreement line has a starting date on January 2, 2012 and an ending date on March 1, 2012. The second service agreement line has a starting date on April 1, 2012 and an ending date on December 31, 2012. You specify a period on the **Create service orders** page that is from October 1, 2012 until December 31, 2012. Therefore, service orders are generated only for the second agreement line, because the starting date and ending date of the first agreement line are before the period that you specified for the service order.

Feedback

Was this page helpful? Yes No

Provide product feedback

Service status and progress field interaction

Article • 05/02/2022

On the **Service orders** page, the **Progress** field on the service order header reflects the status of the whole service order, and the **Status** reports the status of individual service order lines.

Expand table

Progress	Line 1 Status	Line 2 Status	Line 3 Status
In process	Created	Created	Created
In process	Canceled	Created	Created
In process	Created	Canceled	Posted
Canceled	Canceled	Canceled	Canceled
Posted	Posted	Posted	Posted
Posted	Posted	Canceled	Canceled

The progress of a service order is in process if all lines have the status **Created**; it is still in process if some of the lines have a status of **Canceled** or **Posted**.

If all lines in a service order are marked as **Posted**, the progress of the status order is **Posted**. If some lines are **Posted** and some are **Canceled**, the progress is still **Posted**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create service orders automatically

07/10/2025

You can create service orders for one service agreement or for several service agreements. When they're created, you can view your service orders on the **Service orders** page.

Service orders are created only for the valid period of the service agreement. If the interval that you specify on the **Create service orders** page is before the starting date or after the ending date of the service agreement, service orders are created only for the part of the interval that is within the service agreement dates.

When you create service orders manually or automatically from the service agreement line, the service order must be in the time interval that is defined by the starting and ending dates for the line, unless you don't specify an ending date on the line.

Create service orders automatically for a service agreement

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select a service agreement.
3. On the Action Pane, select the **Deliver** tab, and then select **Planned service orders**.
4. Specify dates in the **From date** and **To date** fields to define the service period.
5. Set **Show Infolog** to *Yes* to generate an Action center message that displays a list of the service orders that are created.
6. Select transaction types in the **Include transaction types** field group. The transaction types represent the lines that are created in the service agreement, and each transaction type that you select generates several service orders, depending on the service interval that is specified on the service agreement line.
7. To create any service orders that are missing from continuous series of service orders, set **Continuous** to *Yes*.
8. Select **OK**.

Create service orders automatically for several service agreements

1. Go to **Service management** > **Periodic** > **Service orders** > **Create service orders**.
2. Select **Select** to make selections to add or remove criteria to use to create service orders.
3. Select **OK**.

Related information

- Service orders
- Automatically create service orders

Create item requirements for service orders

Article • 01/06/2025

You can create a service order to track and manage services that you provide to your customers. If you need to reserve specific items for a service order, you can create inventory item requirements for it. An item requirement can be immediately consumed from inventory, or it can initiate a production order for the item.

By using an item requirement instead of an item transaction, you can plan for delivery just before the item is actually used, create a purchase order, include the item in the trade-agreement framework, and include the item requirement in production planning.

Item requirements for service orders are processed through a project. To create an item requirement on a service order, the service order must be assigned to a project. After you create an item requirement for a service order, you can view the item requirement on the **Projects** page for the selected project.

Create an item requirement for a service order

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select the service order that you want to create an item requirement for.
3. On the Action Pane, on the **Dispatch** tab, select **Item requirement**.
4. On the **Item requirements** page, enter information for the required item.

Create an item requirement for a service agreement

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Open the service agreement for which you want to create an item requirement.
3. On the **Lines** FastTab, select **Add** to create a new line.
4. In the **Transaction type** field, select *Item*.
5. In the **Item setup** field, select *Item requirement*.
6. In the **Item number** field, select the item that is required for the service agreement.
7. On the **Line details** FastTab, on the **Product dimensions** tab, in the **Site** field, select the inventory site for the item.
8. To create a service order from the agreement line, on the **Lines** FastTab, select **Create service orders**, and then enter the relevant information on the **Create**

service orders page.

Related information

- Create service orders automatically.

Feedback

Was this page helpful? Yes No

Provide product feedback

Service order item requirements

Article • 01/06/2025

You can create a service order to track and manage services that you provide to your customers. If you need to reserve specific items for a service order, you can create inventory item requirements for it. An item requirement can be immediately consumed from inventory, or it can initiate a production order for the item.

By using an item requirement instead of an item transaction, you can plan for delivery just before the item is actually used, create a purchase order, include the item in the trade-agreement framework, and include the item requirement in production planning.

Item requirements for service orders are processed through a project. To create an item requirement on a service order, the service order must be attached to a project.

As soon as an item requirement is created for a service order, it can be viewed from **Project** in the individual service order or through the **Sales order** page.

View an item requirement from a service order

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select **Dispatch**, and then select **Item requirement** to open the **Item requirements** page.
3. Select the **Project** tab, and check the **Service order** field to see the service orders of the item requirement.

Delete service orders with item requirements

If an item requirement is created on a service order, you can't delete the service order. You must delete the item requirement before you can delete the service order.

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select **Dispatch**, and then select **Item requirement** to open the **Item requirements** page. This page lists the item requirements that are created on the service order.
3. Select the item requirement to delete, and then select **Delete**.

–or–

1. Go to **Project management and accounting** > **Projects** > **All projects**.
2. Open the project that has the service order in which an item requirement is created.

3. On the **Projects** page, in the right pane, select **Item requirements**. The **Item requirements** page lists the item requirements that are associated with the selected project.
4. Select the item requirement to delete, and then select **Delete**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Add an address to a service order

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to add a customer address to a service order. When you create a service order, the address information is transferred from the project that the service order is attached to. However, you can select an alternative location from addresses that are already entered in Microsoft Dynamics AX for customers, vendors, sites, warehouses, service orders, and projects.

You can also create a new address. By default, the new address is transferred to the project. However, you can specify that the new address is only relevant for this instance of the service. If you do, the new address is not transferred to the project.

Create a customer address for a service order

To add an address to a service order, follow these steps:

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Open the service order that you want to create an address for.
3. Open the **Header** tab.
4. Expand the **Address** FastTab, and then select **Add address** from the FastTab toolbar.
5. In the **New address** dialog, enter a unique name for the address and complete the remaining fields.

   > **Warning**
   >
   > If you enter the same name as an existing address, the information that you enter in the remaining fields will overwrite information for the existing address.

6. Select **OK** to copy the new address to your service order.

Specify an alternative address on a service order

To add an alternative address to a service order, follow these steps:

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Open the service order that you want to enter an alternative address for.
3. Open the **Header** tab.
4. Expand the **Address** FastTab, and then select **Other address** from the FastTab toolbar.
5. In the **Address selection** dialog, select **Service orders** from the drop-down list above the grid.
6. Select an address, and then select **OK** to copy it to your service order.

Last updated on 05/07/2025

Subscription workflow overview

Article • 06/03/2022

You are the subscriptions administrator for a light company that offers subscriptions for lighting rig maintenance. A customer contacts your company to purchase a yearly subscription for lighting rig maintenance.

Setting up subscriptions

To set up a subscription, you must first create a subscription group for the new service agreement or verify that a subscription group exists. If a subscription group exists, you must set it up to invoice the customer yearly and to accrue sales revenue every month of the year. For more information about how to set up subscriptions, see Set up subscription groups.

After the subscription group is created, you can create the subscription. For more information about how to create service subscriptions, see Create service subscriptions from a subscription group.

Create and modify subscription transactions

After you set up the subscription, you create the subscription fee transaction for the first invoicing period, which is one year. The transactions are of the **Regular** type. Therefore, you can only create subscription transactions where the from date and the to date correspond to periods that were previously created on the **Period types** page. For more information about fee transactions, see Create subscription fee transactions.

After you set up the subscription for your customer, you remember that they have negotiated a 10 percent discount on all list prices for service offerings. Therefore, you must reduce the price of the transaction fee that you created.

Later in the day, your customer contact calls to say that, although they still want the service agreement for the lighting rig, they plan to introduce a new lighting rig later in the year. Therefore, they only need maintenance coverage until October 2013. To reduce the number of months for the customer's subscription, you create a new subscription fee transaction of the **Reduction days** type. For more information about how to reduce days, see Reduce the days on subscription fees.

Invoice and accrue subscription transactions

You have now finished setting up the subscription, and you are ready to invoice your customer for it. For more information about how to invoice subscriptions, see Invoice subscription transactions.

Two days later, your customer calls to say that the subscription should be invoiced in U.S. dollars, not Euros. Therefore, you create a credit note, and you also create a new subscription transaction in the correct currency. For more information about how to credit subscription transactions, see Credit subscription transactions.

At the end of each month, one month's revenue can be accrued from the customer's subscription to the profit and loss account and the WIP accounts. For more information about how to accrue revenue for subscriptions, see Accrue subscription revenue.

Feedback

Was this page helpful? Yes No

Provide product feedback

Combine service orders

Article • 01/06/2025

When you create service order lines automatically on the **Service agreements** page, you can choose one of the following options to specify how you want to group them:

- **By service agreement**
- **By service task**
- **By employee**
- **By service object**

Example

You create a service agreement that has a start date on 03-31-2007. In the **Combine service orders** field, you specify **By service object**. You then create the following service agreement lines:

Expand table

Agreement line number	Transaction type	Description	Interval	Service object	Start date
1	**Hour**	SAL1	Weekly	X-1	04-01-2007
2	**Hour**	SAL2	Biweekly	X-2	04-01-2007
3	**Hour**	SAL3	Weekly	X-2	04-01-2007

You don't specify time windows for any of the service agreement lines. Therefore, the service order lines won't move from the calculated day on which they fall.

Next, you generate service order lines from the **Create service orders** page from 04-01-2007 until 04-30-2007.

In total, 10 service orders are created. Because the combined setting that you selected was **By service object**, all service orders that are created have only service order lines with one specific service object. Service order lines that are generated from the service agreement and have the same service date and object are combined on the same service order.

> **Note**
>
> In this example, the calendar that is specified on the **Service management parameters** page has no closed days.

Additional grouping of service order lines into service orders occurs according to any time window that you specify on the service agreement lines.

Related information

- Create service orders automatically

Feedback

Was this page helpful? Yes No

Provide product feedback

Create service subscriptions from a subscription group

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

1. Go to **Service management** > **Setup** > **Service subscriptions** > **Subscription groups**.
2. Select a subscription group.
3. Select the **Service subscriptions** button.
4. Select **New** to create a new subscription. The subscription group field displays information from the subscription group on the **Subscription** page.
5. Enter information in the remaining fields.
6. Close the page to save your changes.

Last updated on 05/07/2025

Project attached to service agreement

Article • 05/02/2022

You can attach service agreements to projects of every type except the **Time project** type.

You register and post all service order transactions through the project that you attach to the service agreement. You specify the journals in which these transactions are registered and posted on the **Journals** link on the **Service management parameters** page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create service subscriptions from a project

Article • 01/30/2025

You can create a service subscription from a project in **Project management and accounting**.

1. Go to **Project management and accounting** > **Projects** > **All projects**. Open the project that you want to create a subscription for.
2. On the **Projects** page, on the Action Pane, on the **Manage** tab, select **Service** > **Service subscription**.
3. On the **Subscription** page, in the **Service subscription** field, enter a unique identification number. The project number is entered automatically into the subscription for the selected project.
4. Enter the remaining information on the page.

You can now create subscription transactions for the subscription.

> **Note**
>
> If you have set up a number sequence for the subscription group, a subscription ID is generated automatically for the subscription when you specify the subscription group.

Related information

- Subscription sales prices

Feedback

Was this page helpful? Yes No

Provide product feedback

Service activities

Article • 01/06/2025

Create service activities to define and maintain tasks in a service order. A service order can include one or more service activities to be performed by a worker.

Service activities are defined on the **Activity types** page (go to **Sales and marketing** > **Setup** > **Activities** > **Activity types**). After you set up activities, you can add them to service orders automatically based on the related service agreement, or you can manually add activities when you add lines to a service order.

You can view the status of service activities on the **Dispatch board** page. You can also reassign activities from one worker to another worker in that page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Dispatch board

A dispatch board is the central location from which you can view the status of activities in a service order. You can perform the following tasks in a dispatch board:

- Filter and view service activities in a certain date range.
- Identify the priority of a service activity, and distinguish among service priority levels based on a color scheme that you set up.
- Review the workers that are assigned to a dispatch team.
- Modify the service times and assigned technicians for a service activity.
- Review the list of service activities that have not been dispatched.

To work with the dispatch board, follow these steps:

1. Go to **Service management** > **Perform periodic tasks** > **Dispatch board**.
2. On the Action Pane, select **Adjust view settings** and then use the **Adjust view settings** dialog to select the date range and other options for the service activities you want to see.
3. In the grid, select the service activity you want to work with.
4. Navigate through the tabs in the bottom panel to view and modify the details of the selected service activity.

Related information

- Set up service activity types
- Set up a preferred technician

Last updated on 12/30/2025

Set up service activity types

Article • 01/06/2025

1. Do one of the following steps:

    - Go to **Sales and marketing** > **Setup** > **Activities** > **Activity types**.
    - Go to **Organization administration** > **Setup** > **Activities** > **Activity types**.

2. Add a new activity type line.

3. In the **Type** field, enter a name for the activity type.

4. In the **Description** field, enter text to describe this activity type.

5. Close the page to save your changes.

Related information

- Dispatch board

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up a preferred technician

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can select any worker as a preferred technician for a service agreement or service order. However, it is a good idea to add the worker to the appropriate dispatch team so that the worker is included on the **Dispatch board**.

Assign employee to a dispatch team

1. Go to **Human resources** > **Workers** > **Workers**. Select a worker to open the worker details page. On the Action Pane, select **Setup** >**Dispatch team** to open the **Dispatch workers** page.
2. In the **Dispatch team** field, select the team to assign the worker to.

Assign a preferred technician to a service agreement

1. Go to **Service management** > **Service agreements** > **Service agreements**. Select a service agreement to open the details page.
2. On the **General** tab, select the **Preferred technician** field, and then select a member of the appropriate dispatch team as the preferred technician for the service agreement.

Assign a preferred technician to a service order

1. Go to **Service management** > **Periodic** > **Dispatch board**.

   > **Note**
   >
   > On the **Dispatch board** page, specify a date range for dispatch activities to view. Also, specify whether to display closed activities and whether to limit the dispatch

activity list to teams that you belong to or are authorized to monitor. Select **OK** to open the **Dispatch board**.

2. Select the line of the service activity to modify.
3. On the **Related** tab, use the **Worker** list to assign a member of the appropriate dispatch team as the preferred technician for the service call.

Related information

- Develop and establish service agreements overview
- Create service orders manually

Last updated on 05/07/2025

Set up dispatch teams

07/10/2025

This article describes how to create a dispatch team. After you create a dispatch team, you can assign workers to the team. When you create a service order, you can then relate a member of a dispatch team to an activity in the service order.

1. Go to **Service management** > **Setup** > **Dispatch teams**.
2. On the **Dispatch teams** page, select **New** to create a new team.
3. In the **Dispatch team** field, enter a unique name for the dispatch team.
4. In the **Owner** field, select the employee who is responsible for this team.

   > ⓘ **Note**
   >
   > You can select only employees in this field. You can't select vendors.

5. Optional: In the **Description** field, enter a brief description for this dispatch team.

> ⓘ **Note**
>
> You can assign a worker to this dispatch team on the **Worker** page. Learn more in Set up a preferred technician.

View dispatch teams

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Service orders on the **Dispatch board** page are grouped by dispatch team.

1. Go to **Service management** > **Periodic** > **Dispatch board**.
2. On the Action Pane, select **Adjust view settings**.
3. The **Adjust view settings** dialog opens. Select a date range, and then select **OK**.
4. On the **Dispatch board** page, in the Gantt chart at the top of the page, expand a team list to review the names of the workers on the team. You can also view details about the service activities that are assigned to each worker.

For information about how to assign a worker to a team, see Set up a preferred technician.

Related information

- Prioritize service orders
- View the status of service orders
- Monitor service activities
- Set up dispatch teams

Last updated on 05/07/2025

Prioritize service orders

09/03/2025

1. Go to **Service management** > **Perform periodic tasks** > **Dispatch board**.
2. On the Action Pane, select **Adjust view settings**.
3. Use the **Adjust view settings** dialog to select the date range and other options for the service activities you want to see. Then select **OK**.
4. Expand each dispatch team list.
5. Select the color-coded graphic that represents a service activity.
6. Open the **All**, **Team**, **Undispatched**, or **Related** tab and change the selection in the **Priority** field.
7. Select **Save**.

Related information

- Dispatch board
- View the status of service orders

View the status of service orders

Article • 01/06/2025

You can use the **Dispatch board** page to view the status of service activities in service orders.

More information about service orders

The page also provides tools to track the flow of service orders and includes additional details to help you evaluate the urgency of a service order, such the service agreement and service level agreement that it is associated with.

In order to help you quickly gauge the urgency of a service order, the Gantt chart supports the use of colors. Colors are used to distinguish the priority of a service call. By default, these colors are red for high priority, yellow for typical priority, and green for low priority.

> **Tip**
>
> You can change the colors on the **Dispatching** tab on the **Service management parameters** page.

Related information

- Prioritize service orders
- Monitor service activities

Feedback

Was this page helpful? Yes No

Provide product feedback

Monitor service activities

Article • 01/06/2025

The **Dispatch board** page provides an at-a-glance status update for the various scheduled service activities, and tools for tracking the general flow of service orders in your company.

> **Tip**
>
> To view details about a service activity, select it in the Gantt chart at the top of the **Dispatch board** page, right-select, and then select **Information**.

By using the tools in this page, you can sort service orders by date, priority, or type of service activity, view the priority levels that are assigned to service orders, and review which activities are assigned to individual service orders.

If you must redistribute service activity assignments, you can move the graphical representation of a service activity to assign it to a different employee or a different date and time.

Example

While monitoring the day's service activity, you notice that John, a service technician, is running behind schedule on assigned service calls. You also notice that the last call in John's list must be completed by the end of the day to meet service level agreement requirements. By using the drag-and-drop functionality of the Gantt chart, you can reassign that service call to Meg, who has completed service calls ahead of schedule.

Open the Dispatch board page

Go to **Service management** > **Periodic** > **Dispatch board**.

Related information

- Prioritize service orders
- View the status of service orders

Feedback

Was this page helpful? Yes No

Provide product feedback

Reassign activities

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to reassign service activities from one worker to another worker. You can reassign service activities from one worker to another worker even if those workers are assigned to different dispatch teams.

Use the following steps to reassign an activity to another worker:

1. Go to **Service management** > **Periodic** > **Dispatch board**.
2. In the **From date** and **To date** fields, enter dates to define the time period to view activities for.
3. Select whether to view closed activities and dispatch information for related teams, and then select **OK**.
4. On the **Dispatch board** page, select **Advanced** to display only the Gantt chart at the top of the page. Select **Simple** to display the Gantt chart and the tabs on the page.
5. Expand each dispatch team list.
6. Use either of the following steps to reassign an activity:
    - Select a color-coded graphic that represents the service activity to reassign. Press the SHIFT key, and then move the color-coded graphic to the row for another employee.
    - On the **All** tab, **Team** tab, **Undispatched** tab, or **Related** tab, in the **Worker** field, enter the name of the worker who is replacing the current worker for the activity.

Related information

- Service activities
- Dispatch board

Last updated on 05/07/2025

Reverse changes in a dispatch board

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to reverse unsaved modifications that you make in a dispatch board. For example, you assign a worker to a service activity, save the information, and then later decide to assign a different worker to the service activity. You modify the worker in the dispatch board, and then, before saving the change, learn that the worker just assigned is not available. You can reverse the unsaved modification so that the original worker is reassigned to the service order.

Use the following steps to reverse unsaved changes in a dispatch board:

1. Go to **Service management** > **Periodic** > **Dispatch board**.
2. On the **Dispatch board** page, enter relevant information in the fields, and then select **OK**.
3. To reverse the most recent change that is not saved, select **Undo**.
4. To reverse a series of changes that are not saved, continue selecting **Undo** until each change that you want to discard is reversed.

Related information

- Dispatch board
- Service activities

Last updated on 05/07/2025

View undispatched service orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to view activities in a service order that don't have a worker assigned to them from a dispatch team.

1. Go to **Service management** > **Periodic** > **Dispatch board**.
2. In the **Dispatch board** window, enter the start date and end date for the period to view dispatch activity for.
3. Optional: Select whether to include closed activities and related teams in the results.
4. Select **OK** to view the results.

   > **Note**
   >
   > On the **Dispatch board** page, the **Advanced** view is automatically displayed. In this view, the Gantt chart at the top of the page and the tabs at the bottom of the page are displayed. Select **Simple** to display only the Gantt chart.

5. To view a list of service orders that don't have a dispatch team assigned, select the **Undispatched** tab.

Related information

- Service activities
- Dispatch board
- Reassign activities

Last updated on 05/07/2025

Modify a Service BOM

Article • 01/30/2025

You can record the history of an element in a service bill of materials (BOM). Every time that you update a BOM line, a history line is created in the **History** pane. The history line shows the current state of the BOM line.

Update a service BOM element

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select **Edit** to open the **Service agreements** details page.
3. On the Action Pane, select **Service objects** to open the **Service objects** page.
4. Select the object to update a BOM line for, and then select **Designer**.
5. On the **Designer** page, select the BOM line to update, and then select **Edit BOM line**.

   > ⓘ Note
   >
   > On the **Setup** tab, select the **Edit when adding** check box if you want the **Edit BOM line** page to open when you drag a line into the service BOM.

6. In the **Quantity** field, enter the quantity.
7. If you want to create a service order line for the replacement item, which can then be invoiced, select the **Create service order line** check box.
8. Select **OK** to close the page.

Delete a service BOM line

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select **Edit** to open the **Service agreements** details page.
3. On the Action Pane, select **Service objects** to open the **Service objects** page.
4. Select the object to delete a service BOM line from, and then select **Designer**.

5. On the **Designer** page, select the BOM line to delete, and then select **Delete BOM line**.

Related information

- Template BOMs

Feedback

Was this page helpful? Yes No

Provide product feedback

Credit subscription transactions

Article • 01/06/2025

1. Go to **Service management** > **Service subscriptions** > **All service subscriptions**.
2. Select the subscription attached to the subscription transaction for which you want to create a credit note.
3. Select the **Analyze** tab, and then select the **Fee transactions** button on the Action Pane.
4. From the **Fee transactions** page, select the transaction for which you want to create a credit note.
5. Select **Functions** > **Select for credit note**.
6. From the **Select for credit note** page, select the transaction that you want to credit and then select **OK**.

> **Note**
>
> When you create the credit note, make sure that you select **Credit notes**. This is found in the **Invoicing method** list in the **Create invoice** dialog box.

If the **Reverse accruals on crediting** field on the **Service management parameters** page is set to *Manual*, you have to reverse each accrued revenue transaction individually before you create a credit note proposal for the transaction.

Related information

- Invoice subscription transactions

Feedback

Was this page helpful? Yes No

Provide product feedback

Create subscription fee transactions

07/10/2025

1. Go to **Service management** > **Service subscriptions** > **All service subscriptions**.
2. On the **All service subscriptions** list page, select the subscription for which you want to create a subscription fee transaction.
3. On the Action Pane, on the **Subscription** tab, select **Subscription fees**.
4. On the **Create subscription fee** page, enter the appropriate information in the fields.
5. Select **OK**.

Accruing subscriptions

Article • 03/17/2025

With service subscriptions, you manually accrue revenue in the periods following the date when you invoiced a fee transaction.

Accrual periods are created for the invoice period that you set up for the subscription fee, and the accrual periods are based on the period code of the subscription.

You can accrue and reverse accrued revenue.

Reverse accruals of credit amounts

If you credit invoiced subscription amounts, you can use two methods to reverse the accrual amounts:

- You can reverse each accrued revenue transaction individually before you create the credit note proposal for the transaction. This is the manual method. (manual)
- You can have the accrued amounts reversed on the date where the credit note is posted or on the original posting date of the accrual.

Learn more in Subscription parameters (form).

Setup requirements

To accrue revenue, make sure that the following data requirements are met:

Account setup

The **WIP - subscription** and the **Accrued revenue - subscription** accounts must be set up in the **Project** module.

When you post accrued revenue, the **WIP - subscription** account is debited with the accrual amount, and the **Accrued revenue - subscription** account is credited with the accrual amount.

Set up accounts for accrual of subscription revenue

1. Go to **Project management and accounting** > **Setup** > **Posting** > **Ledger posting setup**.
2. Select the **Revenue accounts** tab, and select **WIP - subscription** or **Accrued revenue - subscription** to set up the accounts.

Subscription group setup

To be able to accrue revenue for subscriptions, the **Accrue revenue** check box must be selected. This is found on the **Subscription groups** page for the group that is attached to the subscription. Go to **Service management** > **Setup** > **Service subscriptions** > **Subscription groups**.

Enable revenue accrual on a subscription group

Go to **Service management** > **Setup** > **Service subscriptions** > **Subscription groups**.

Periods

You must set up an invoicing period code. Unless you want to accrue revenue in the same time intervals as you use for invoicing, you must also set up an accrual period.

The following table provides an overview of which accrual periods can be set up for each invoicing period:

Expand table

Invoicing period	Accrual period
Years	• Years • Quarter • Month • Day
Quarter	• Quarter • Month • Day
Month	• Month • Day

Invoicing period	Accrual period
Week	• Day
Day	• Day

Setting up the invoicing period is a mandatory part of the overall subscription group setup. You can decide whether to also set up an accrual period for the subscription group. If you set up an accrual period for the subscription group, this period is suggested in the **Period code** field. This field is found on the **Accrue subscription revenue** page, when you accrue subscription revenue. However, the accrual period is optional information about the subscription group.

> ⓘ Note
>
> Use the following path to open the **Accrue subscription revenue** page. Go to **Service management** > **Periodic** > **Service subscriptions** > **Accrue subscription revenue**.

Transactions

You can control the number of ledger transactions that are created when you post accrued revenue. On subscriptions, define if the ledger transactions should be created as a total or per line.

Specify the level of posting details to display for accrued transactions

1. Go to **Project management and accounting** > **Setup** > **Project management and accounting parameters**.
2. On the **Financial** tab, in the **Invoice** field, select **Total** or **Line**.

Related information

- Accrue subscription revenue

Feedback

Was this page helpful? Yes No

Provide product feedback

Move service order lines

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Open the service order that contains the lines that you want to move to another service order.
3. On the **Service orders** page, on the Action Pane, click the **Move lines** button.
4. On the **Move service order lines** page, in the **Service order** field, select the service order that you want to move the lines to.
5. Select the **Delete empty service order** check box to delete the service order that you moved the service order lines from.
6. Select **OK**.

> **Note**
>
> You can move service order lines only between service orders that are attached to the same service agreement.

Related information

- Service orders

Last updated on 05/07/2025

Cancel service orders

Article • 01/06/2025

You can cancel a service order or service order line from the service order itself, or you can cancel multiple service orders by running a periodic job.

> **Note**
>
> Service orders cannot be canceled if the stage of the service order does not allow cancellation, if the service order has item requirements, or if the service order has already been posted.

Cancel a service order on the Service orders page

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select the service order.
3. On the Action Pane, open the **Service order** tab and select **Cancel order**.

Cancel a service order line

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Open the service order that contains the line you want to cancel.
3. On the **Lines** FastTab, select the service order line that you want to cancel.
4. On the **Lines** FastTab toolbar, select **Cancel order line** to change the status of the line to *Canceled*.

> **Tip**
>
> To reverse the cancellation of a service order line and change the status back to *Created*, select **Revoke cancel** on the toolbar.

Cancel or uncancel multiple service orders

1. Go to **Service management** > **Perform periodic tasks** > **Service orders** > **Cancel service orders**.
2. Make the following settings on the Parameters FastTab:

- **Show Infolog** – Set to *Yes* to generate an Action center message that lists the canceled or uncanceled service orders.
- **Revoke cancel** – Set to *No* to cancel the selected service orders (already *Canceled* orders aren't affected). Set to *Yes* to uncancel the selected service orders (*In process* orders aren't affected).

3. Select **Filter**.
4. In the **Inquiry** dialog, on the **Range** tab, set the criteria to select the service orders that you want to cancel or uncancel.
5. Select **OK** to close the **Inquiry** dialog.
6. Select **OK**.

The selected service orders are either canceled or uncanceled. Uncanceled orders have their *Canceled* status reversed to *In process*.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create service orders manually

Article • 01/06/2025

You can create service orders manually by using a service agreement or by using the **Service orders** page. You can also create a service order from a project.

> **Tip**
>
> You can use automated processes to create service orders.

Create a service order manually from a service agreement

1. Go to **Service management** > **Service agreements** > **Service agreements**.
2. Select a service agreement or create a new service agreement.
3. Select the **Deliver** tab and in the **Create** group select **Planned service orders** to open the **Create service orders** page.

Create a service order manually on the Service orders page

1. Go to **Service management** > **Service orders** > **Service orders**.
2. Select **New** to create a new service order.
3. Create service order lines for the service order.

> **Note**
>
> If the **Allow without service agreement** check box on the **Service management parameters** page is selected, you can post the transactions from the service order lines without attaching the service order to a service agreement. If the check box is cleared, you must attach the manually created service order to a project before posting the service order lines.

Create a service order from a project

1. Go to **Project management and accounting** > **Projects** > **All projects**.

2. On the **Projects** page, on the Action Pane, select the **Manage** tab > select **Service** > **Service orders**.
3. Follow the previous procedure to create a service order manually on the **Service orders** page. The **Project ID** field displays the project reference.

> ⓘ Note
>
> If the **Allow without service agreement** check box on the **Service management parameters** page is selected, you can post the transactions from the service order lines without attaching the service order to a service agreement. If the check box is cleared, you must attach the manually created service order to a project before posting the service order lines.

Create a service order from the Sales order page

You can create a service order from the **Sales orders** page by using the **Create a new service order based on the sales order** wizard.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Open the relevant sales order.
3. On the **Sales order** tab, select **Service order** to start the **Create a new service order based on the sales order** wizard.
4. Select **Next** >, and then complete the following steps on the **Select agreement for service order** page:
    - Use the **Service agreement** field to select the service agreement with which the new service order should be associated.
    - Optional: Use the **Project ID** field to associate this service order with a particular project.
5. Select **Next** >, and then complete the following steps on the **Create service order** page:
    - Enter a date and time for the service call to begin in the **Preferred service time** field.
    - Optional: Modify the text in the **Description** field. By default, this field contains the description of the service agreement that you selected on the previous page.

- In the **Responsible** field, select the ID of the employee who is responsible for the agreement, and if you know what it is, enter the ID of the customer's preferred technician for the service call.
- In the **Contact ID** field, select the person in the customer's company who should be contacted regarding this service order.

6. Select **Next** >, and then select **Finish**.

Related information

- Service orders
- Create service orders automatically

Feedback

Was this page helpful? Yes No

Provide product feedback

Accrue subscription revenue

Article • 01/06/2025

1. Go to **Service management** > **Periodic** > **Service subscriptions** > **Accrue subscription revenue**.
2. In the **Period code** field, select a period code. Revenue is posted for all subscription fee transactions that have the selected period code.
3. In the **Posting date** field, enter a posting date, and then select **OK**.

> **Note**
>
> You can't post accrued revenue until the fee transactions are invoiced.

Feedback

Was this page helpful? Yes No

Provide product feedback

Delete service orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

1. Go to **Service management** > **Periodic** > **Service orders** > **Delete service orders**.
2. On the **Delete service orders** page, select **Select** to specify the criteria to select the service orders to be deleted, and then select **OK**.
3. Set **Show Infolog** to *Yes* to generate an Action center message that displays the deleted service orders.
4. Select **OK**.

> **Note**
>
> If you do not specify any criteria to select the service orders, all service orders are deleted. However, when you exit the **Delete service orders** page, you will have the option to delete all service orders.
>
> Also, you can only delete service orders with a stage that lets you delete them.

Related information

- Service orders

Last updated on 05/07/2025

Invoice subscription transactions

07/10/2025

1. Go to **Service management** > **Service subscriptions** > **All service subscriptions**.
2. Select a service subscription and then select **Edit** or select **Subscription** to create a service subscription.
3. Select **Analyze** > **Fee transactions**.
4. Select the fee transaction or fee transactions to invoice, and then select **Functions** > **Invoice proposal** to open the **Invoice proposals** page.
5. Select **Create invoice** and enter the appropriate parameters on the **Create invoice** page
6. Select **OK** to create the invoice proposal on the **Invoice proposals** page.
7. Verify the contents of the invoice. Select **Post invoice** to open the **Post invoice** page to post the invoice. You can also print the invoice from the **Post invoice** page.

Traceability overview (preview)

10/28/2025

[This article is prerelease documentation and is subject to change.]

Companies face increasing pressure to comply with legislation, such as the EU and German Supply Chain Act. This pressure is especially high when recalling defective and potentially unsafe products or linking products with certain documents. The Traceability Add-in for Dynamics 365 Supply Chain Management lets you collect business and industry-specific data across the product value chain. You can use this data for regulation audits and compliance processes. Beyond compliance, Traceability also enables companies to meet a wide range of business objectives, including efficiency, resilience, responsiveness, and sustainability.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Features and benefits

Traceability enables you to trace raw materials and subassemblies in a product and track the usage and disposal over the full value chain. As a result, it helps you meet a wider range of business objectives, including efficiency, resilience, responsiveness, and sustainability.

Traceability follows products and goods as they move through the supply chain. It collects information about the provenance of inputs, supplier sourcing practices, and conversion processes. It provides the following features and benefits:

- **Accurate tracking** – Track and document the chain of custody based on individual identifiers, such as serial numbers or batch numbers, across the entire value chain and throughout the full lifecycle of an item.
- **As-built BOM registration** – Document composition genealogy information (as-built bill of materials (BOM)) and activity events, such as procurement, production, and distribution, against each serial and batch number. You can trace all the way back to find the origin of each single part.

- **Forward and backward search** – Use the where-used functionality to track the consumption of materials throughout your entire value chain. You can find finished or semi-finished goods that could be impacted by a defective component.
- **Public API** – Integrate with external systems by using the robust Traceability API. While this integration requires some initial setup and configuration, it offers a high degree of flexibility and customization to cater to specific business needs.
- **Integration with the production floor execution interface** – Production floor workers can register the serial numbers and batch numbers of each component and each finished good.

Join the Traceability group on Microsoft Viva Engage

For more help and support, join the Traceability add-in for Dynamics 365 Supply Chain Management community on Viva Engage. In this group, you can find the following resources:

- On the **Conversations** tab, you can read conversation threads, ask questions, share experiences, and connect with other users. The group includes consultants, Microsoft partners, Microsoft product team members, and other experts and users who can share tips, help answer questions, and provide guidance.
- On the **Files** tab, you can find presentations, videos, tutorials, and hands-on labs that help you learn more about Traceability.

Next steps

- Install, update, or uninstall Traceability (preview)
- Traceability app quick start (preview)

Install, update, or uninstall Traceability (preview)

Summarize this article for me

[This article is prerelease documentation and is subject to change.]

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes how to install, update, or uninstall the various components of the Traceability Add-in for Dynamics 365 Supply Chain Management.

Prerequisites

To use Traceability, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.40 (build 10.0.1935.80) or higher.
- Your Supply Chain Management installation must be running on one of the following types of environments:
  - Tier-2 environment
  - A Unified Developer Experience (UDE) environment copied from a tier-2 environment that didn't have Traceability installed. Copying an environment with Traceability already installed isn't supported.
- The Dynamics 365 app named *Dynamics 365 Finance and Operations Platform Tools* must be installed for your environment in Power Platform admin center.

Turn on features in Supply Chain Management

To use Traceability together with Dynamics 365 Supply Chain Management, you must enable the following features in feature management:

- *Tracked components* – Enables component tracking in Supply Chain Management. As of Supply Chain Management version 10.0.45, this feature is turned on by default.

- *(Preview) Traceability* – Integrates Traceability with the component tracking features in Supply Chain Management.

Microsoft Azure configuration

Register a new Microsoft Entra application

1. Sign in to the Azure portal.
2. Search for or navigate to the **App registrations** page.
3. On the toolbar, select **New registration**.
4. Fill out the **Register and application** page and select **Register**. For more information about these settings, see Quickstart: Register an application with the Microsoft identity platform.
5. Your new app registration opens. On the left navigation pane, select **Overview**. Copy the following values to a temporary text file. You'll need them later.

    - **Application (client) ID**
    - **Directory (tenant) ID**

Create client secrets for the new Microsoft Entra application

1. In the Azure portal, open the application you registered in the previous section.
2. In the left navigation pane, select **Manage** > **Certificates & secrets**.
3. On the **Certificates & secrets** page, select **New client secret**.
4. In the **Add a client secret** dialog, enter a description and choose an expiration date for the secret. Then select **Add**.
5. The **Certificates & secrets** page now includes your new secret. Copy the **Value** and **Secret ID** of the secret and store it in a secure location. You'll need these values later when setting up communication with the API.

    > **Important**
    >
    > The **Value** is only displayed once, so you won't be able to retrieve it again after closing this page.

Install and configure the Traceability app in Power Apps

The Traceability app is part of the Traceability Add-in for Dynamics 365 Supply Chain Management. It's a Power Apps application that provides a user interface for configuring and using the Traceability Add-in. When you install the Traceability app in Power Apps, the system also installs the necessary add-in components in Supply Chain Management.

To install the Traceability app in Power Apps, complete the following steps.

1. Sign in to the Power Platform admin center.
2. On the left navigation pane, select **Resources** > **Dynamics 365 apps**.
3. Find the *Dynamics 365 Supply Chain Traceability* app on the **Dynamics 365 apps** page. Select **More application actions** (the **...** button) for the app and then select **Manage**.
4. The **Install Dynamics 365 Supply Chain Traceability** dialog opens. Make the following settings:
   - **Select an environment** – Select the environment where you want to set up the app.
   - **Enter application ID of service** – Enter the Application (client) ID that you copied after you registered the Microsoft Entra application.
   - **Enter tenant ID of service** – Enter the Directory (tenant) ID that you copied after you registered the Microsoft Entra application.
   - **I agree to the terms of service** – Select this checkbox.

   > **Note**
   >
   > If you don't see the **Enter application ID of service** and/or **Enter tenant ID of service** fields, contact the Traceability product team at d365-sct-team@microsoft.com.

5. Select **Install**.
6. When installation is complete, you should see that an app called *Dynamics 365 Supply Chain Traceability* is shown with a status of *Installed* in the list of **Dynamics 365 apps** for your environment.
7. On the left navigation pane, select **Environments**.
8. Open the environment where you installed the Traceability app.
9. On the **Generative AI features** tile, check to make sure that **Move data across regions** is set to *Allowed*. If it isn't, then follow these steps to allow it:

- Select the **Edit** link on the **Generative AI features** tile.
- In the **Generative AI features** dialog, select the **Move data across regions** checkbox
- Select **Save** to close the dialog.

Update the Traceability app in Power Apps

This section describes how to check for updates to the Traceability app and how to apply them.

1. Sign in to the Power Platform admin center.
2. On the left navigation pane, select **Environments**.
3. Open the environment where you want to update the Traceability app.
4. From the **Resources** tile, select **Dynamics 365 apps**.
5. Find the app called *Dynamics 365 Supply Chain Traceability* in the list and check its **Status**. If the **Status** is *Installed*, then the app is up to date and you can skip the rest of this procedure. If the **Status** is *Update available*, continue with this procedure to apply the update.
6. Select **More application actions** (the **...** button) next to the app name to open a menu, then select **Update**.
7. The **Update Dynamics 365 Supply Chain Traceability** dialog opens. Select **I agree to the terms of service** and then select **Update**.
8. When the update is complete, you should see that the *Dynamics 365 Supply Chain Traceability* app now shows a **Status** of *Installed*.
9. Open the Power Apps maker portal.
10. On the left navigation pane, select **Solutions**.
11. On the **Solutions** page, select **All**.
12. On the toolbar, select **Publish all customizations**.

Uninstall the Traceability Add-in from Supply Chain Management

The Traceability Add-in is automatically installed on your Supply Chain Management environment when you install the Traceability app in Power Apps. However, to uninstall the add-in, you must use Microsoft Dynamics Lifecycle Services by following these steps:

1. Sign in to Lifecycle Services.
2. Open the page for managing your Supply Chain Management environment.
3. In the **Environment add-ins** section, on the **Traceability service** tile, select **Uninstall**.

Uninstall the Traceability app from Power Apps

To remove the Traceability app from your Power Apps environment, complete the following steps:

1. Sign in to Power Platform maker.
2. On the left navigation pane, select **Solutions**.
3. The **Solutions** page opens. Open the **Managed** tab.
4. Find and delete each of the following solutions. To delete a solution, select the **Commands** button next to its name and then select **Delete**.

    - *Dynamics 365 Supply Chain Traceability anchor solution*
    - *Dynamics 365 Supply Chain Traceability portal solution*
    - *Dynamics 365 Supply Chain Traceability controls solution*
    - *Dynamics 365 Supply Chain Traceability Plugins solution*
    - *Supply chain traceability integration - authorization*
    - *Dynamics 365 Supply Chain Traceability common solution*

    > **Important**
    >
    > You must delete *Dynamics 365 Supply Chain Traceability common solution* last.

Last updated on 03/10/2026

Traceability security roles in Power Apps (preview)

Article • 09/13/2024

[This article is prerelease documentation and is subject to change.]

The Traceability Add-in for Dynamics 365 Supply Chain Management uses the *System Administrator* and *Finance and Operations Basic User* Power Platform security roles to control access to Traceability app features.

Use the Power Platform admin center to grant Traceability permissions to the following types of users:

- **Administrator users** – Assign the default *System Administrator* role, as provided, to users who need to configure the Traceability app with full access.
- **Business users** – Assign the *Finance and Operations Basic User* role to business users who only need inquiry access.

For more information about how to complete these steps, see Assign a security role to a user, Save time creating a security role by copying one, and Create or edit a security role to manage access.

Feedback

Was this page helpful? Yes No

Provide product feedback

Configure Traceability (preview)

[This article is prerelease documentation and is subject to change.]

This article describes how to configure the Traceability Add-in.

API endpoint

The Traceability Add-in for Supply Chain Management provides an API endpoint that you can use to exchange messages with external systems. It lets you integrate Traceability with other systems, such as enterprise resource planning (ERP) systems, warehouse management systems, and quality management systems. To use the API, you need to know the URL of the endpoint. To find it, follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **System**.
3. On the **General** tab of the **System** page, you can find the API endpoint URL in the **Endpoint** field. Use the **Copy** button to copy the URL to the clipboard.

Exclude products from Traceability

If you have products that don't need to be traced, you can choose to exclude them from the Traceability Add-in. All products that aren't on the exclude list are traced.

To exclude or reinclude a product, follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **Exclude product**.
3. The **Exclude product** page opens, showing a list of currently excluded products (if any). Use the buttons in the toolbar to add, remove, or edit rows in the list. For each row, make the following settings:

    - **Company** – Specify the legal entity (company) of the product that shouldn't be traced.
    - **Item number** – Specify the item number of the product that shouldn't be traced.

> ⓘ **Note**
>
> Your settings here only affect activity events that occur after you make the change. Activity events that occurred for a product before you added it to the exclude list continue to be included in Traceability. Activity events that occurred for a product while it was on the

exclude list are never included in Traceability, even after you remove the product from the list.

Configure activities that Traceability traces

All activities throughout the logistics process are logged as *activity events*. Activity events include *goods receipt*, *consumption*, *registration*, and so on.

You can make the following configuration settings for each activity:

- Enable and disable activity events traced at the company level.
- Map original activities and activity types to specific activity codes.

> **Note**
>
> Your configuration settings only affect activity events that occur after you make the change. Events that occurred for an activity before you added it to the activities list never appear in Traceability. Events that occurred for an activity while it was on the activities list continue to be included in Traceability even after you remove the activity from the list.

To configure activities that Traceability traces, follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **Activity**.
3. The **Activity** page opens, showing a list of activities currently configured for your environment. Use the buttons in the toolbar to add, remove, or edit rows in the list. For each row, make the following settings:
    - **Activity code** – Enter a name that identifies the activity to users. Traceability displays this name to identify the activity when users make a genealogy trace.
    - **Activity type** – Select the type of activity, as it should be shown to users (for example, *Sales*, *Purchase*, *Production*, *Quality*, *Warehouse*, *Maintenance*, or *Transportation*). Traceability displays this value to identify the activity when users make a genealogy trace.
    - **Source activity type** – Select the activity type reported by the source system (for example, *Sales*, *Purchase*, *Production*, *Quality*, *Warehouse*, *Maintenance*, or *Transportation*). See the table after this procedure for a list of source activity types and codes by Supply Chain Management.
    - **Source activity code** – Enter the activity code reported by the source system. See the table after this procedure for a list of source activity types and codes by Supply

Chain Management.

- **Company** – Specify the legal entity of the activity to be traced. Enter an asterisk (*) to trace the activity for all companies.
- **Trace or Not** – Set to *True* for activities that should be traced. Set to *False*, for activities that shouldn't be traced.

4. On the toolbar, select **Save**.

The following table lists the source activity codes and types supported by the tracked component feature.

Expand table

Source activity code	Source activity type	Integration scenario
PurchaseGoodsReceipt	Purchase	Goods receipt of purchase order.
ProductionReportFinished	Production	Component consumption during manufacturing assembly.
ProductionPickingList	Production	Finished goods produced during production.
ProductionPickingListPartialReverse	Production	Partial reverse picking list of component consumption.
ProductionReportFinishedPartialReverse	Production	Partial reverse report as finished.
ProductionPickingListFullReverse	Production	Full reverse picking list of component consumption.
ProductionReportFinishedFullReverse	Production	Full reverse report as finished.

Configure data collection fields that are traced in Traceability

Data collection captures various types of information at the activity code level, including document reference, order number, site, warehouse, and location. This data enhances track-and-trace capabilities and supports insights in reports like *Find items*. With this functionality, you can do the following actions:

- Create data collection fields.
- Assign data collection fields to specific companies and activity codes.
- Enable or disable data collection tracking.

> **Note**
>
> Certain data collection fields, such as Document reference, Site, and Location, are preassigned to all companies for out-of-the-box integration with Supply Chain Management. To avoid disrupting this integration, don't modify the following predefined fields: Document reference, Journal number, Reference type, Order number, Vendor account, Vendor name, Site, Warehouse, Location, or Operator.

To configure data collection fields that are traced in Traceability, follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **Data collections**.
3. Select the **Add** button on the menu bar to open the initial screen for creating a data collection field.
4. Make the following settings for the new data collection field:

    - **Name** – Enter the name of the data collection field. The integrated systems use this value to identify the specific field.
    - **Description** – Provide a brief description of the data collection field.
    - **Data collection ID** – Enter the internal ID of the data collection configuration record. This value uniquely represents the data collection field in storage. You can specify it manually or allow the system to generate it automatically.
    - **Data type** – Define the type of data the field can store. The following options are available:
        - *Text* – Accepts alphanumeric values.
        - *Number* – Accepts integer and decimal values, both positive and negative.
        - *Boolean* – Maps business values to system values *True* or *False*.
        - *Enum* – Lets you define a list of predefined values.
    - **UoM (unit of measure)** – Specify the unit of measure for the data collection field.
    - **Assignment** – Assign the data collection field to multiple entities. For each assignment, make the following settings:
        - **Activity** – Select a combination of activity type and activity code defined in the activity configuration for the assignment.
        - **Company code** – Specify the company code where the data collection field is assigned.
    - **Status (track or not)** – Set to *On* to track the data collection field. Set to *Off* to disable tracking.

Last updated on 11/12/2025

Open the Traceability app (preview)

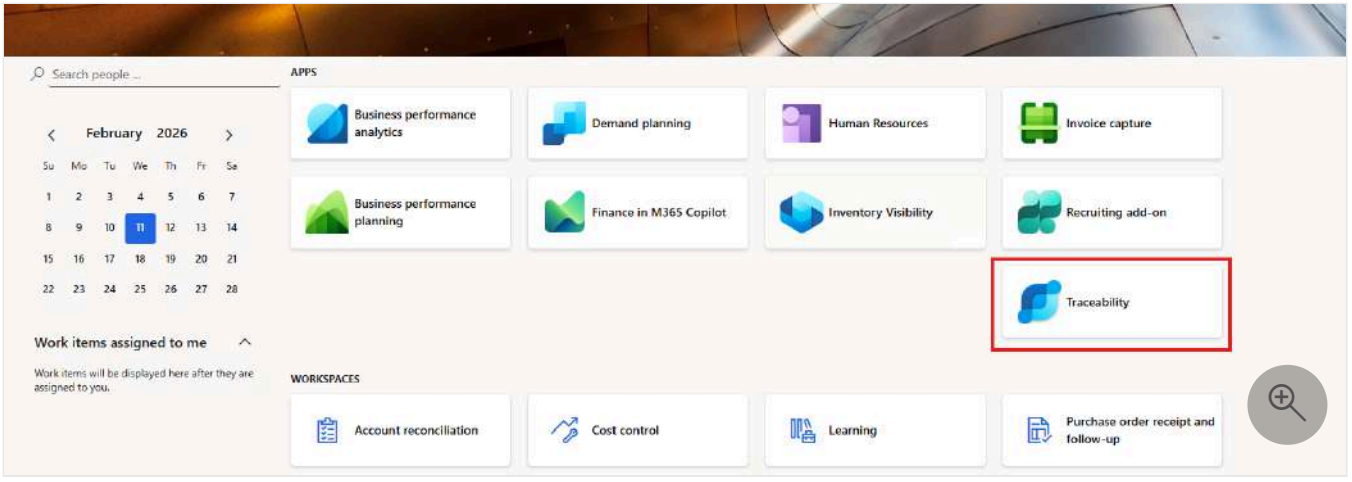
Summarize this article for me

[This article is prerelease documentation and is subject to change.]

You can access the Traceability app in two ways: from Supply Chain Management or from Power Platform. This article describes both methods.

Access Traceability from Supply Chain Management

When you enable the *(Preview) Traceability* feature in feature management, the system adds a **Traceability** tile to the **Apps** section of the **Home** page of Supply Chain Management, as shown in the following image. To open the Traceability app, select the tile.



Access Traceability from Power Platform

Follow these steps to open the Traceability app from Power Platform after you install it in your environment.

1. Sign in to Power Platform maker.
2. On the left navigation pane, select **Apps**. The **Apps** page opens.
3. Open the **My apps** tab. Look for the app with a **Name** of *Traceability*. If you don't see it on the **My apps** tab, switch to the **All** tab and look for it there.
4. Hover your mouse over the *Traceability* app to reveal the **Play** button next to it. Select the **Play** button to run the app.

Last updated on 02/11/2026

Traceability app quick start (preview)

[This article is prerelease documentation and is subject to change.]

This article describes how to verify, import, and remove demo data. It also explains how to use demo data to explore the capabilities of the Traceability app in Power Apps.

Verify demo data

Usually, the installation process automatically installs demo data when you first install the Traceability Add-in for Dynamics 365 Supply Chain Management.

To check whether your system has demo data, follow these steps.

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **Activity**.
3. Verify that the configuration settings shown in the following table appear on the **Activity** page.

Expand table

Company	Activity Code	Activity Type	Source Activity Code	Source Activity Type	Track or not
USMF	Goods Receipt	Production	Goods Receipt	Production	TRUE
USMF	Consumption	Production	Consumption	Production	TRUE

If the configuration settings don't appear, import the demo data as described in the next section.

Import demo data

If you want to explore Traceability by using demo data but don't have the data, follow these steps to import it.

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Home** > **Get started**.
3. In the **Manage environment data** section, on the **Import demo data** tile, select **Import demo data**.
4. Wait until the import process is complete. The process might take several minutes.

Delete all data

When you're done exploring the demo data, you can remove it from your system, along with all other data that's been collected since you started using Traceability.

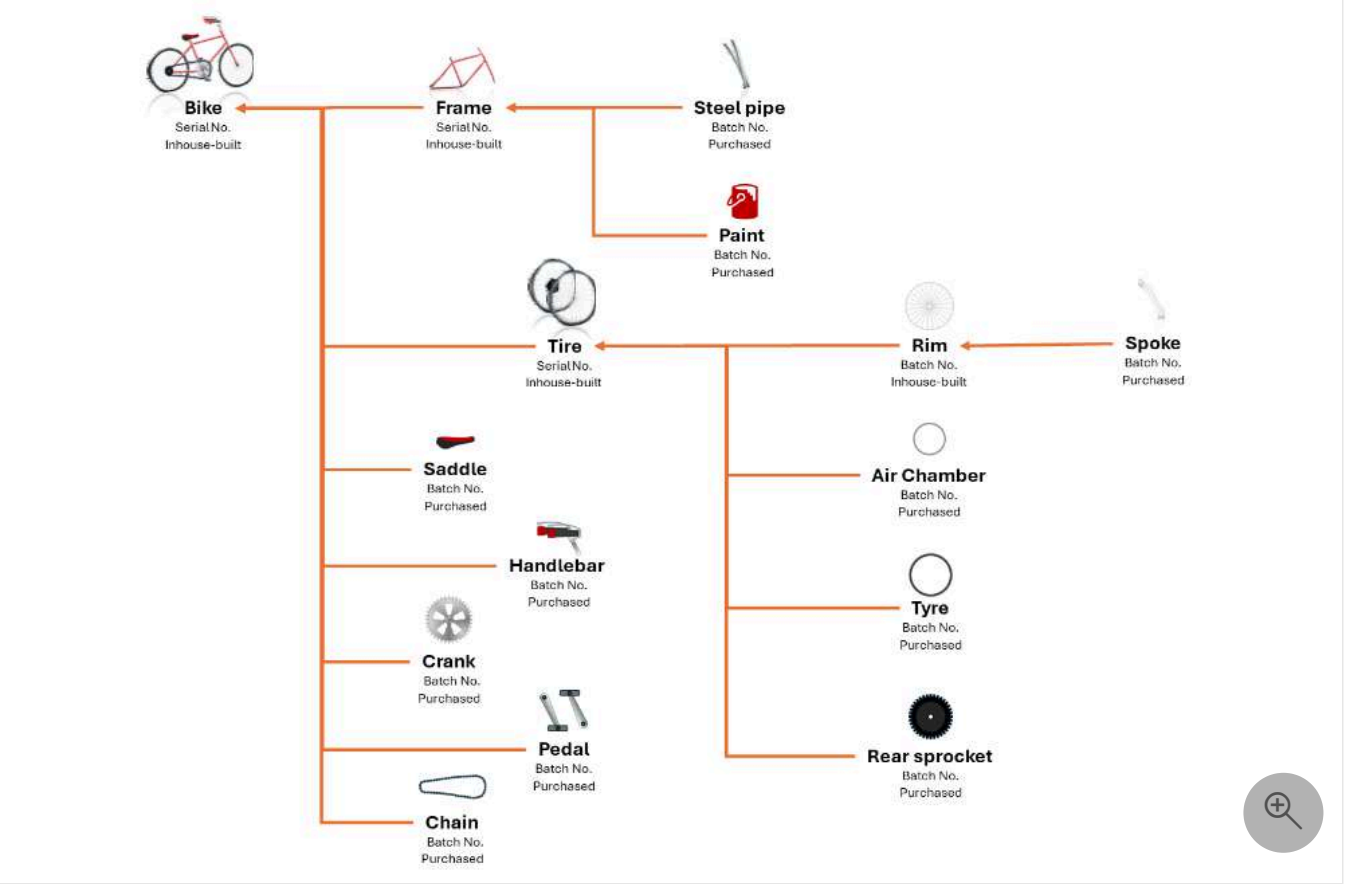
> ⚠ **Warning**
>
> This procedure removes *all* data from the Traceability Add-in for Supply Chain Management, not just the demo data. Be sure you want to delete all data before you proceed.

To delete all data from the Traceability Add-in, follow these steps.

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Home** > **Get started**.
3. In the **Manage environment data** section, on the **Delete all data** tile, select **Delete all data**.
4. Wait until the import process is complete. This process might take several minutes.

Demo data model

The demo data describes the components of a manufactured bicycle. The following illustration shows the bill of materials (BOM) structure of the bicycle.



Explore the demo data

This section shows how to explore the capabilities of the Traceability app by using the demo data. It describes several common scenarios.

1. Open the Traceability app in Power Apps.

2. On the left navigation pane, select **Trace** > **Genealogy trace**.

3. Make a *backwards search*, which lets you enter a unique identifier of a finished good to retrieve its genealogy, activity, and data collection.

   Use the search field to look for a specific serial number of a finished good. For example, use the serial number *bike-s0001* or *bike-s0002* to find the genealogy tree of a bike from the demo data.

4. Traceability shows a genealogy tree for the serial number you entered. Now you can do an *activity verification* by following these steps:
   a. Select the node for the bike in the genealogy tree.
   b. Select the **View detail** button for your selected node to open a dialog.
   c. Open the **Activities** tab in the dialog. Activity events are displayed in reverse chronological order. For example, you might see five activities for *Consumption* and one activity for *Goods receipt*.

5. Do a *forward search* to see the higher-level assembly related to a selected genealogy tree node. Follow these steps:
   a. Select the paint-demo node in the genealogy tree.
   b. Select the **Trace** button for your selected node. The app now shows a high-level genealogy list.

6. Try out the *table views*. Until now, you've been working on the **Graphic view** tab. The table views provide another way to view and explore your data. Try the following options:

   - Select **Table view - composition** to do a backward search.
   - Select **Table view – where used** to do a forward search.

Last updated on 11/20/2025

Work with the Traceability app (preview)

[This article is prerelease documentation and is subject to change.]

The Traceability app lets you search for specific batch and serial numbers and analyze the genealogy of your products. This article describes the basics of how to work with it.

In-app learning

The Traceability app provides in-app learning to help you get started. To access the in-app learning, follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Get started**.
3. A collection of links to videos and tutorials is shown. Select each link that you're interested in.

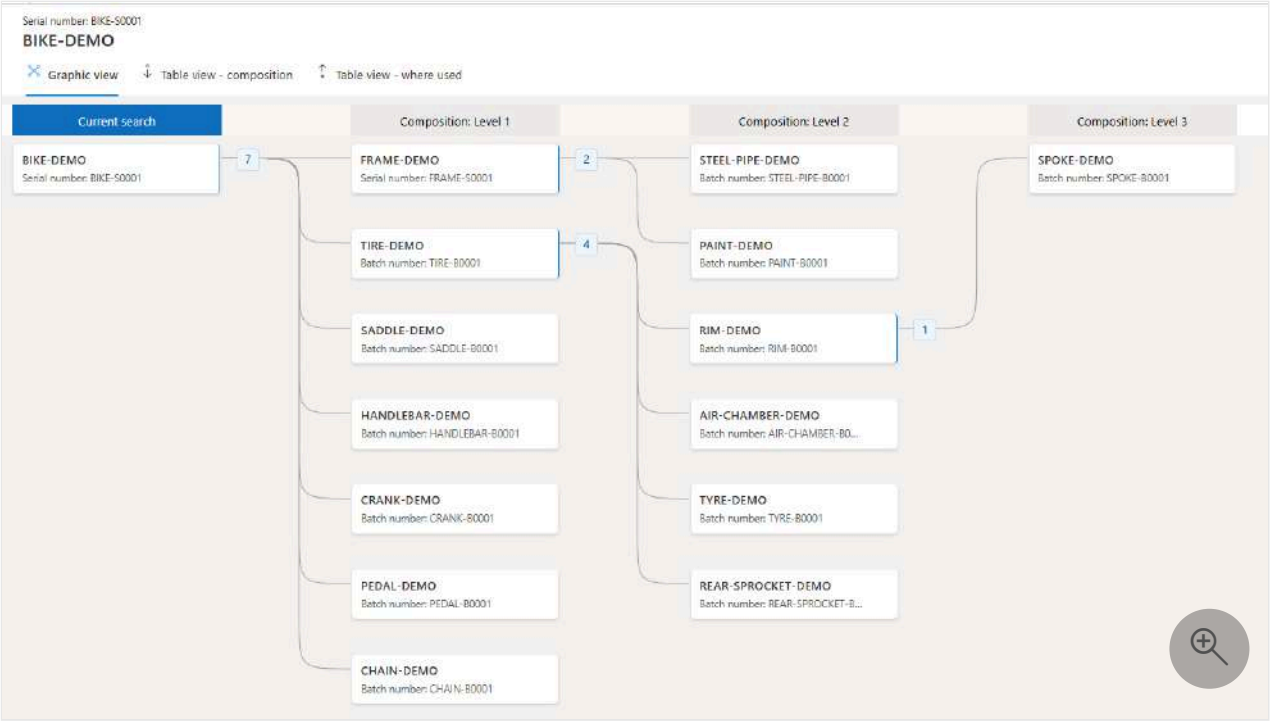
Backward search

Backward searches let you retrieve the full history of a single product.

- In maintenance, repair, and operations (MRO) scenarios, backward search helps maintenance workers retrieve repair logs and component genealogy associated with a specific serial number for root cause analysis.
- Backward search retrieves component genealogy data that can help you decide whether all components included in a specific product were sourced sustainably (environmentally friendly).

To make a backward search, follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Trace** > **Genealogy trace**.
3. In the search form, enter a unique identifier (such as a serial, batch, asset, or lot number) of a finished good to retrieve its genealogy, activity, and data collection. (If you're exploring the app using sample data, search for *bike-s0001*.)
4. Traceability shows a genealogy tree for the serial number you entered.



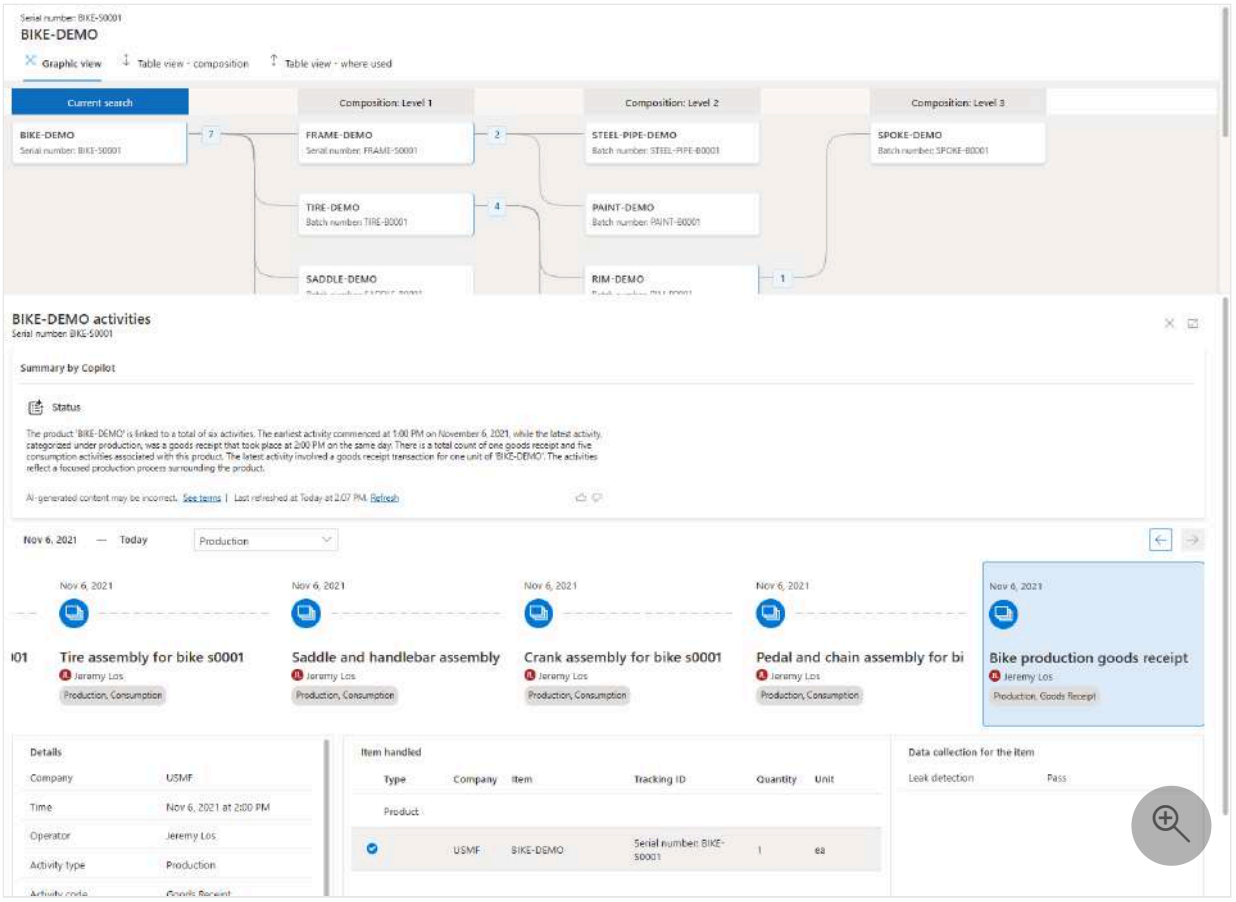
5. Now you can do an *activity verification* by following these steps:

   a. Select a node in the genealogy tree.

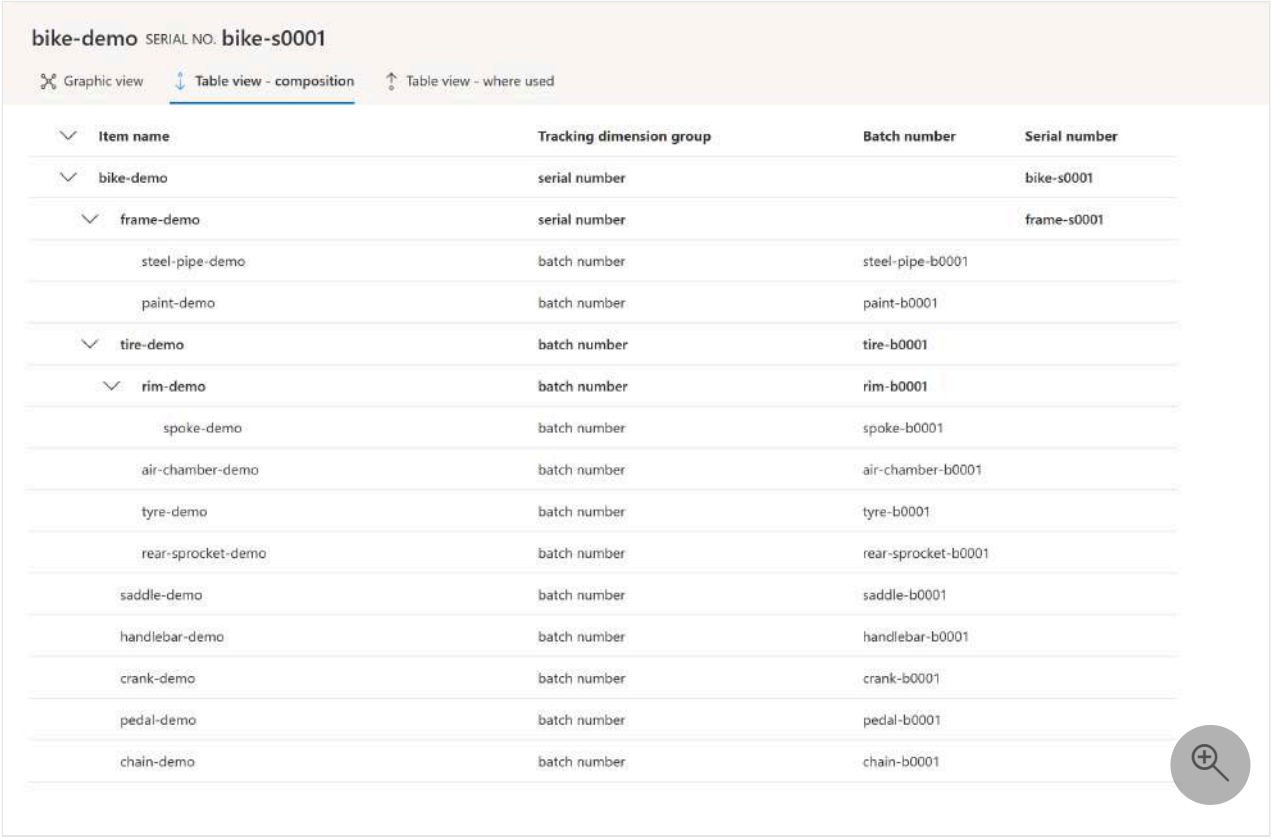
   b. Select the **View activities** button for your selected node to open a dialog.

   c. A Copilot-generated summary gives you a quick overview of the tracked object's history.

   d. Below the summary, all activity events are displayed in reverse chronological order. For example, you might see five activities for *Consumption* and one activity for *Goods receipt*.



6. Until now, you've been working on the **Graphic view** tab. The table views provide another way to view and explore your data. Select **Table view - composition** to see a table view of your backward search.



Forward (where-used) search

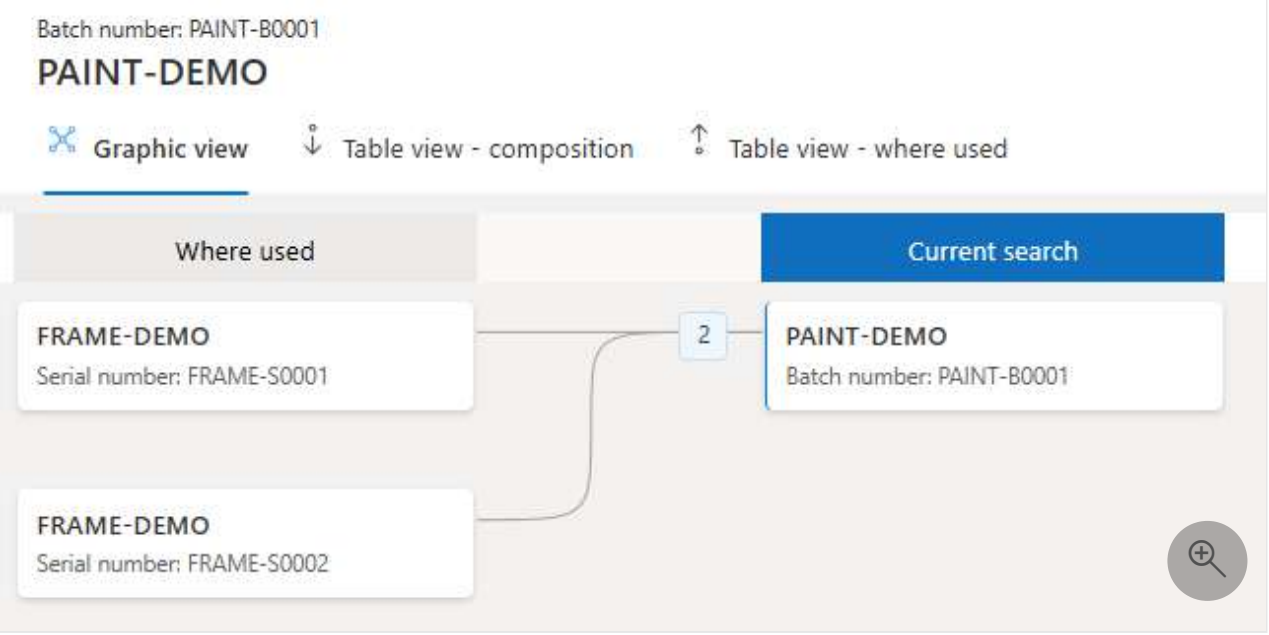
Forward searches help you locate the batch or serial number of finished products that contain semi-products or raw materials with a specific batch or serial number.

In recall scenarios, manufacturers can find the batch or serial number of specific impacted finished goods and then run a specific, accurate recall that minimizes the loss.

For example, in the electric vehicles industry, manufacturers can use the serial number of a battery that's going out of service to locate the cars that use it. They can then offer recycling and warranty services to the affected customers.

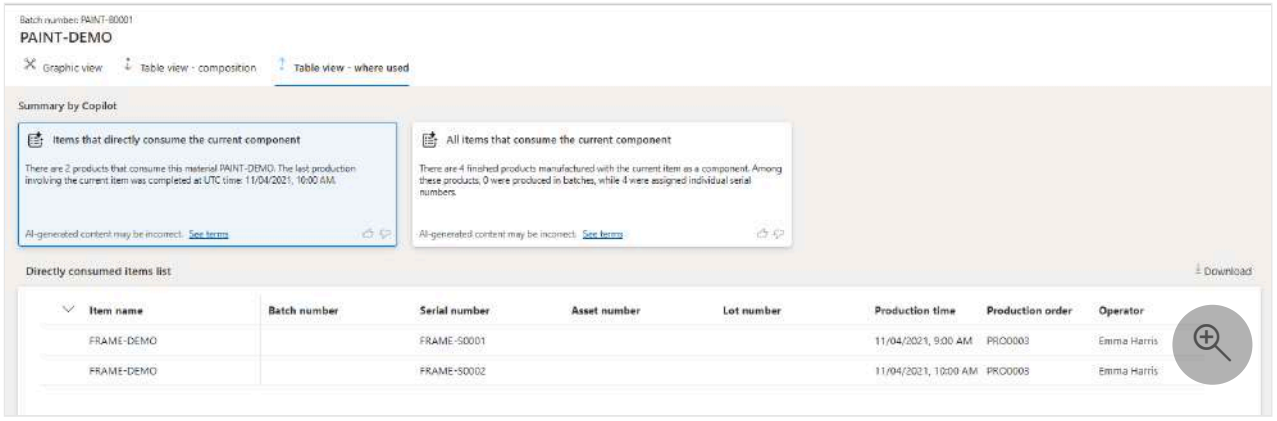
To make a forward search, follow these steps:

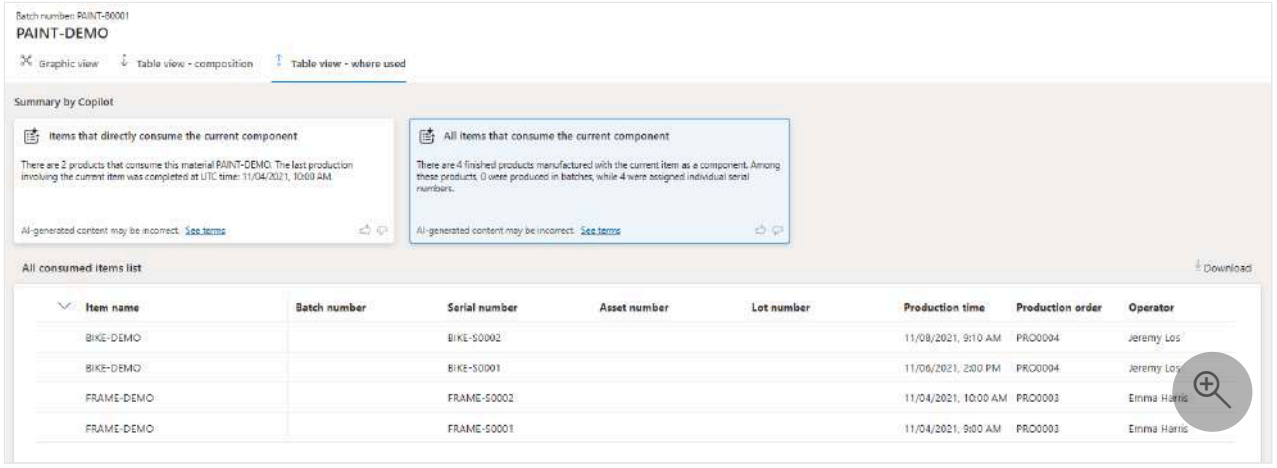
1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Trace** > **Genealogy trace**.
3. Enter a unique identifier of a raw material or semi-product to see the higher-level assembly related to a selected genealogy tree node.



4. Select the relevant node in the genealogy tree.
5. Select the **Trace** button for your selected node. The app now shows a high-level genealogy list.
6. Until now, you've been working on the **Graphic view** tab. The table views provide another way to view and explore your data. Select **Table view – where used** to see a table view of your forward search. Two copilot-generated summaries are provided here: one that summarizes items that directly consume the selected material and another that

summarizes all items that consume the selected material. To see a full list of items associated with either summary, select the appropriate summary.





Find items

The *Find items* report helps you locate serial and batch numbers based on activity codes and data collection values. For example, you can search for finished goods received during a specific period or find all finished goods with a specific data collection value within a defined time range.

To make a find-items search, follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Trace** > **Find items**.
3. Enter values for the following search fields as needed:

    - **Company** – The company associated with the activity you're looking for.
    - **Item** – The item number of the item you're looking for.
    - **Time period** – The time range during which the activity and data collection occurred.
    - **Filters** – Detailed search filters. You can set one or both of the following filters here:

    - **Activity** – The activity code you're looking for.
    - **Data collection** – The data collection field and value you're looking for.

4. Select the **Search** button to search for items that match your search criteria.

5. The system returns the serial and batch numbers along with detailed information.

6. You can verify the results in the **Backward search**.

Monitor and fix failed transactions

Use the *Troubleshooting* report to monitor failed transactions related to the out-of-the-box integration between Traceability and Supply Chain Management.

To monitor failed transactions, follow these steps:

1. Open the Traceability app in Power Apps.

2. On the left navigation pane, select **Trace** > **Failed transaction queue**.

3. A list of failed transactions opens. Each listed transaction shows the following information:

    - **Data** – The content of the failed message.
    - **Error message** – A description of the reason for the failure.
    - **Created on** – The UTC timestamp when the record was created.
    - **Modified on** – The UTC timestamp when the record was last modified.

4. To learn more about a listed transaction, select it and then select **Edit** on the menu bar. The system displays detailed information.

5. Based on the error description, you can either correct the configuration or update the data in the **Data** field.

6. After applying a correction, select the message and then select **Post** on the menu bar to resubmit the failed transaction for processing.

Last updated on 11/12/2025

Integrate Traceability with Supply Chain Management (preview)

Summarize this article for me

[This article is prerelease documentation and is subject to change.]

Traceability can integrate seamlessly with Dynamics 365 Supply Chain Management through configuration only.

Integrate Traceability with the purchase goods receipt

The Traceability Add-in for Dynamics 365 Supply Chain Management provides out-of-the-box integration with the purchase goods receipt in Supply Chain Management.

Prerequisites for integrating with the purchase goods receipt

To integrate Traceability with the purchase goods receipt, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.40 or later.
- The *(Preview) Traceability* feature must be turned on in feature management:

Configure activities in Traceability for integrating with the purchase goods receipt

You must configure Traceability to recognize the activity codes that Supply Chain Management submits. Follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **Activity**.
3. The **Activity** page opens. Make sure the following activity codes are available. Learn more in Configure Traceability.

Expand table

Source activity code	Source activity type	Integration Scenario
PurchaseGoodsReceipt	Purchase	Product receipt of purchase order

4. On the toolbar, select **Save**.

Post product receipts of purchase orders

When you post a purchase order product receipt for batch-managed or serialized products in Supply Chain Management, the system automatically sends the corresponding goods receipt event to Traceability.

Integrate Traceability with tracked components in Supply Chain Management

The Traceability Add-in for Dynamics 365 Supply Chain Management provides out-of-the-box integration with the tracked components feature in Dynamics 365 Supply Chain Management.

Shop floor operators using the production floor execution interface can associate the unique ID (serial or batch number) of finished products with the unique ID (serial or batch number) of raw materials. This association is integrated into Traceability for insight processing. Managers can also use the Supply Chain Management web client to view and register serial and batch number information for components and finished goods.

Prerequisites for integrating with tracked components

To integrate Traceability with the tracked components feature Supply Chain Management, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.40 or later.
- The following features must be turned on in feature management:
  - *(Preview) Traceability*
  - *Tracked components* (As of Supply Chain Management version 10.0.45, this feature is turned on by default.)

Configure activities in Traceability for integrating with tracked components

You must configure Traceability to recognize the activity codes that Supply Chain Management submits. Follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **Activity**.
3. The **Activity** page opens. Make sure the following activity codes are available. For more information about these settings, see Configure Traceability.

Expand table

Source activity code	Source activity type	Integration Scenario
ProductionReportFinished	Production	Components consumed during manufacturing assembly.
ProductionPickingList	Production	Finished goods produced during production.

4. On the toolbar, select **Save**.

Configure Supply Chain Management to record batch and serial numbers

Set up your Supply Chain Management system to record batch and serial numbers for raw materials and finished goods. The following procedure summarizes the configuration settings you should use.

1. Set up a number sequence code for *Product component match ID*.
   a. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters** and open the **Number sequences** tab.
   b. Set up a number sequence code for *Product component match ID*.
2. Enable tracked components for bills of materials.
   a. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters** and open the **Bills of materials** tab.
   b. Set **Enable tracked components** to *Yes*.
3. Create tracked component policy.
   a. Go to **Production control** > **Setup** > **Production** > **Tracked components policy**.
   b. Create a **Tracked components policy**.
   c. Set **Use tracked components** to *Yes*.
4. Associate the tracked component policy to finished products and components.

a. Go to **Product information management** > **Products** > **Released products** .
b. Open the product you want to set up.
c. On the **Manage inventory** FastTab, set **Tracked components policy** to the name of the tracked components policy you created earlier.

5. Add the *Tracked components* action to relevant areas of the production floor execution interface.
   a. Go to **Production control** > **Setup** > **Manufacturing execution** > **Configure production floor execution**.
   b. Select a relevant configuration and, from the Action Pane, select **Design tabs**. Add the *Tracked components* action to each relevant tab and toolbar. Repeat this step for each relevant configuration.

   If you're using *USMF* demo data, open the **Default** configuration, select the *Active jobs* design tab, and add the *Tracked components* action to the **Primary toolbar**.

   For more information about these settings, see Design the production floor execution interface.

Register and view batch and serial numbers for finished products and their components

Workers can register batch and serial numbers for finished products and their components from the production floor execution interface. For details, see Register batch/serial numbers for finished products and their components.

Managers can register and review batch and serial numbers for finished products and their components by using the Supply Chain Management web client. For details, see Register and track batch/serial numbers for finished products and their components.

Integrate Traceability with sales goods issues

The Traceability Add-in for Dynamics 365 Supply Chain Management provides out-of-the-box integration with sales goods issues in Supply Chain Management.

Prerequisites for integrating with sales goods issues

To integrate Traceability with sales goods issues, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.46 or later.

- The following features must be turned on in feature management:
  - *(Preview) Traceability*

Configure activities in Traceability for integrating with sales goods issues

You must configure Traceability to recognize the activity codes that Supply Chain Management submits. Follow these steps:

1. Open the Traceability app in Power Apps.
2. On the left navigation pane, select **Settings** > **Activity**.
3. The **Activity** page opens. Make sure the following activity code is available. Learn more in Configure Traceability.

Expand table

Source activity code	Source activity type	Integration Scenario
SalesGoodsIssue	Sales	Post packing slip of sales order

4. On the toolbar, select **Save**.

Post packing slips of sales orders

When you post a sales order packing slip for a batch-managed or serialized product in Supply Chain Management, the system automatically sends the corresponding goods issue event to Traceability.

Last updated on 02/11/2026

Traceability API (preview)

Summarize this article for me

[This article is prerelease documentation and is subject to change.]

This article describes how to integrate the Traceability Add-in for Dynamics 365 Supply Chain Management with external systems through its API.

Authentication

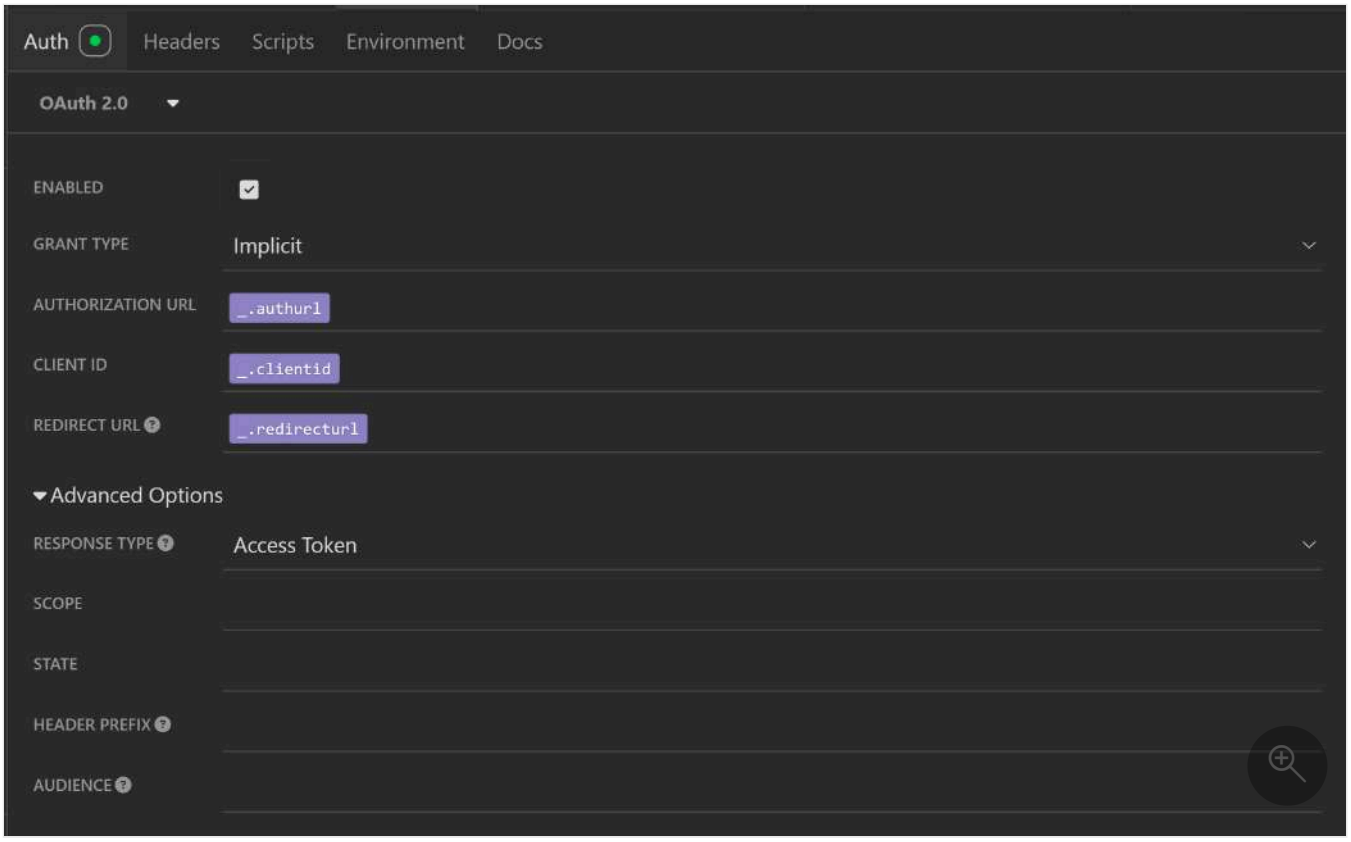
The platform security token is required to call the Traceability public API. The following example uses an Insomnia configuration. Learn more in Use Insomnia with Dataverse Web API.

Use Insomnia to fetch a Microsoft Entra token by submitting an HTTP request that has the following properties:

Expand table

Key	Value
Grant type	Implicit
Authorization URL	`https://login.microsoftonline.com/common/oauth2/authorize?resource={Traceability URL}`. Where *{Traceability URL}* is the URL for your Traceability installation, which resembles `https://operationsxxx.crm.dynamics.com`
Client ID	The Application ID (client ID) registered in Azure portal.
Response type	Access Token

The following image shows an example of an Insomnia configuration. You can configure the keys mentioned in the previous table into the **Auth** tag of a specific HTTP request in Insomnia.



You can create the following JSON file as global environment for an Insomnia project. The `version` and `webapiurl` fields are used to build the path of the specific API.

Available APIs

The following table lists the APIs available for Traceability.

Expand table

Path	Method	Description
/api/data/v9.0/msdyn_sctquerytrace_v1	Post	Query by tracking ID

The remaining sections provide detailed information about each API.

Single query API

This API accepts queries for traceability information and returns genealogy, activity, and data collection information.

- **Path** – Specify the traceability URL. For example: `https://operationsxxx.crm.dynamics.com/api/data/v9.0/msdyn_sctquerytrace_v1`
- **Method** – `POST`

Single query request payload

txt

```
{
    "tracingDirection": "Forward/Backward",
    "eventDetailOption": "EventIdOnly",
    "traceNodeOption": "BuildNodeDictionary",
    "shouldGenerateSummary": true/false,
    "itemNumber": "BIKE-DEMO",
    "serialNumber": "BIKE-S0001",
    "batchNumber": "",
    "company": "USMF",
    "trackingId": "",
    "shouldIncludeEvents": true/false
}
```

Single query request field descriptions

Expand table

Field	Description
`tracingDirection`	Controls the search direction: *Backward* or *Forward*. *Backward* means from top finished good to raw materials; *Forward* means from raw materials to top finished good.
`trackingId`	Key value for the genealogy node. It's a combination of the `itemNumber`, `company`, `batchNumber`, and `serialNumber`.
`company`	The company of the top finished good. For Supply Chain Management, this field maps to the legal entity.
`itemNumber`	The item number of the top finished good.
`serialNumber`	The serial number of the top finished good.
`batchNumber`	The batch number of the top finished good.
`ShouldIncludeEvents`	Controls whether event details should be included. Default is *false*.

Field	Description
`ShouldGenerateSummary`	Set to *True* to include the copilot summary for where-used search.
`ShouldIncludeEvents`	Specify one of the following values: • *EventIdOnly* – Put event ID as a property of the trace node. • *EventInTrace* – Put event details as a property of the trace node. • *EventInDictionary* – Put event IDs as a property of the trace node, return a map whose key is an event ID and the value is the event details.
`TraceNodeOption`	Specify one of the following values: • *BuildNodeGraph* – Render the result as a tree. • *BuildNodeDictionary* – Render the result as a tracking node dictionary, where the key is a tracking ID and the value is the node details.

Single query response header field descriptions

Expand table

Field	Description
`TracingDirection`	Controls the search direction: *Backward* or *Forward*. *Backward* means from top finished good to raw materials; *Forward* means from raw materials to top finished good. Only returns one level above or below the result.

Single query response root element field descriptions

Expand table

Field	Description
`trackingId`	Key value for the genealogy node. It's a combination of the `itemId`, `companyCode`, `batchId`, and `serialId`.

Single query response events element field descriptions

Expand table

Field	Description
`eventId`	Unique identifier for the event (`SerialId`/`BatchId`). Duplicate values aren't allowed. The system generates this value if no value is provided.

Field	Description
`companyCode`	The company code of the top finished good. For Supply Chain Management, this field maps to the legal entity.
`operator`	The operator who executed the activity event. The value can be a user ID, employee ID, or similar.
`description`	Activity event description.
`activityType`	Refers to a predefined activity type (*Purchase*, *Sales*, *Production*, and so on).
`activityCode`	Refers to a configured activity type (*GoodsReceipt*, *Add*, *Remove*, and so on).
`dateTime`	Date and time the activity event occurred.
<data collection name>	These fields are used to collect customization values.

Single query response productTransaction element field descriptions

Expand table

Field	Description
`transactionId`	Unique identifier for the transaction. Duplicate values aren't allowed.
`itemId`	Item number of the top finished good.
`trackingId`	Key value for the genealogy node. It's a combination of `itemId`, `companyCode`, `batchId`, and `serialId`.
`serialId`	Serial number of the top finished good.
`batchId`	Batch number of the top finished good.
`quantity`	Operation quantity of the top finished good.
`unitOfMeasure`	Unit of measure of the received quantity.

Single query response consumptionTransactions element field descriptions

Expand table

Field	Description
`transactionId`	Unique identifier for the transaction. Duplicate values aren't allowed.
`itemId`	Item number of the component.
`trackingId`	Key value for the genealogy node. It's a combination of `itemId`, `companyCode`, `batchId`, and `serialId`.
`serialId`	Serial number of the component.
`batchId`	Batch number of the component.
`quantity`	The consumption quantity of the component.
`unitOfMeasure`	Unit of measure of the consumption quantity.

Single query request example

Query the result of finished goods *BIKE-S001*.

Single query request example request payload

JSON

```
{
    "tracingDirection": "Forward",
    "eventDetailOption": "EventIdOnly",
    "traceNodeOption": "BuildNodeDictionary",
    "shouldGenerateSummary": false,
    "itemNumber": "BIKE-DEMO",
    "serialNumber": "BIKE-S0001",
    "batchNumber": "",
    "company": "USMF",
    "trackingId": "",
    "shouldIncludeEvents": false
}
```

Single query request example API response

JSON

```
{
    "@odata.context":
"https://aurorabapenv2bbcd.crm10.dynamics.com/api/data/v9.0/$metadata#Microsoft.Dyn
amics.CRM.msdyn_sctquerytrace_v1Response",
    "output": "{\"tracingDirection\":\"Forward\",\"root\":{\"trackingId\":\"BIKE-
```

```
DEMO~USMF~~BIKE-S0001~~\",\"next\":[],\"nextIds\":[],\"events\":
[{\"eventId\":\"2026-01-07-09~1~15~3\"},{\"eventId\":\"2026-01-07-09~1~16~3\"},
{\"eventId\":\"2026-01-07-09~1~17~3\"},{\"eventId\":\"2026-01-07-09~1~18~3\"},
{\"eventId\":\"2026-01-07-09~1~19~3\"},{\"eventId\":\"2026-01-07-09~1~20~3\"},
{\"eventId\":\"2026-01-07-09~1~27~4\"},{\"eventId\":\"2026-01-07-
09~1~29~2\"}]},\"traceNodesDictionary\":{\"BIKE-DEMO~USMF~~BIKE-S0001~~\":
{\"trackingId\":\"BIKE-DEMO~USMF~~BIKE-S0001~~\",\"next\":[],\"nextIds\":
[],\"events\":[{\"eventId\":\"2026-01-07-09~1~15~3\"},{\"eventId\":\"2026-01-07-
09~1~16~3\"},{\"eventId\":\"2026-01-07-09~1~17~3\"},{\"eventId\":\"2026-01-07-
09~1~18~3\"},{\"eventId\":\"2026-01-07-09~1~19~3\"},{\"eventId\":\"2026-01-07-
09~1~20~3\"},{\"eventId\":\"2026-01-07-09~1~27~4\"},{\"eventId\":\"2026-01-07-
09~1~29~2\"}]}}}"
}
```

Last updated on 02/11/2026

Transportation management overview

07/31/2025

This article gives an overview of the transportation management functionality in Supply Chain Management.

Transportation management lets you use your company's transportation, and also lets you identify vendor and routing solutions for inbound and outbound orders. For example, you can identify the fastest route or the least expensive rate for a shipment. The following table describes the main scenarios for using Transportation management.

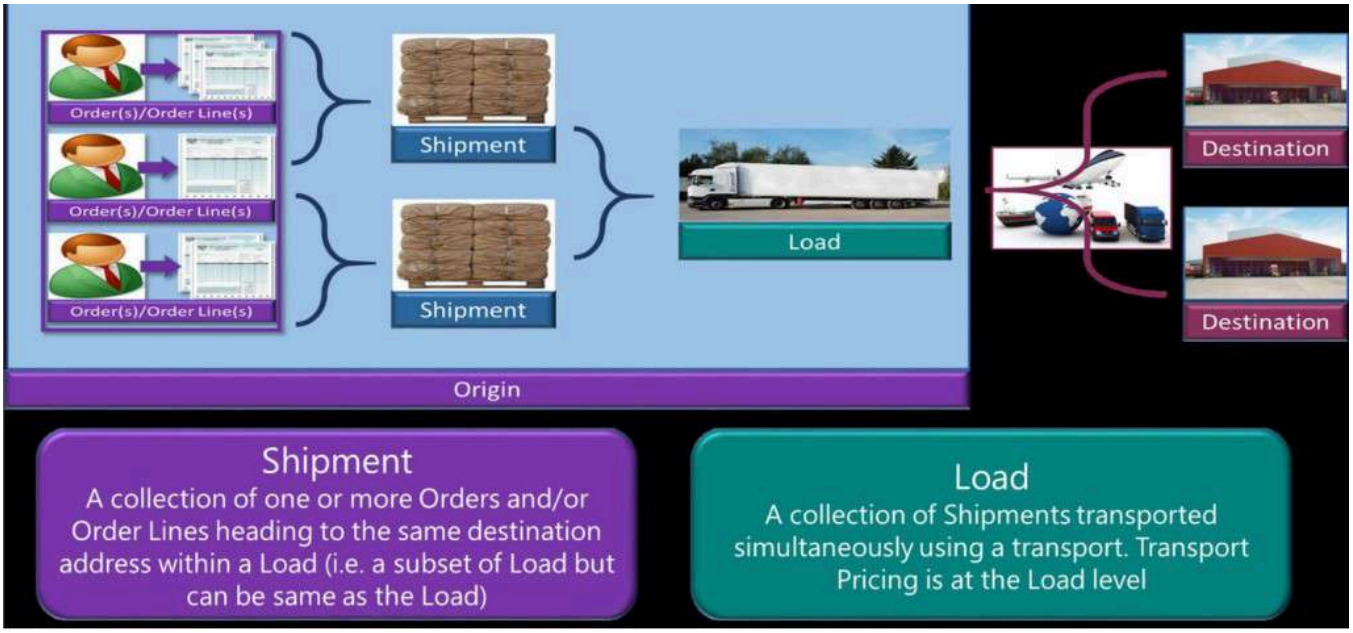
Expand table

Scenario	How Transportation management can help
Use external logistics providers for transportation activities.	Use Transportation management for inbound and/or outbound transportation.
The company's own fleet is available for delivery/pickup, and delivery charges are passed on to customers.	For the outbound processes, you can use Transportation management to determine the transportation charges and pass them on to customers. However, the carrier invoice reconciliation process isn't required.
The company's own fleet is available for delivery/pickup, but delivery charges aren't passed on to customers, because product prices include transportation.	Many of the settings and features of Transportation management aren't required. However, you can use Transportation management to determine the transportation rates and adjust the sales price accordingly.
Logistics service is provided by another legal entity in the same company.	You can use Transportation management by treating the other legal entity like any other shipping carrier. You can't automate the economic transactions between legal entities. Therefore, you must handle these transactions manually (for example, by creating a purchase order). In the legal entity that provides the logistics services, Transportation management can be used to determine transportation rates.

Planning transportation in Supply Chain Management

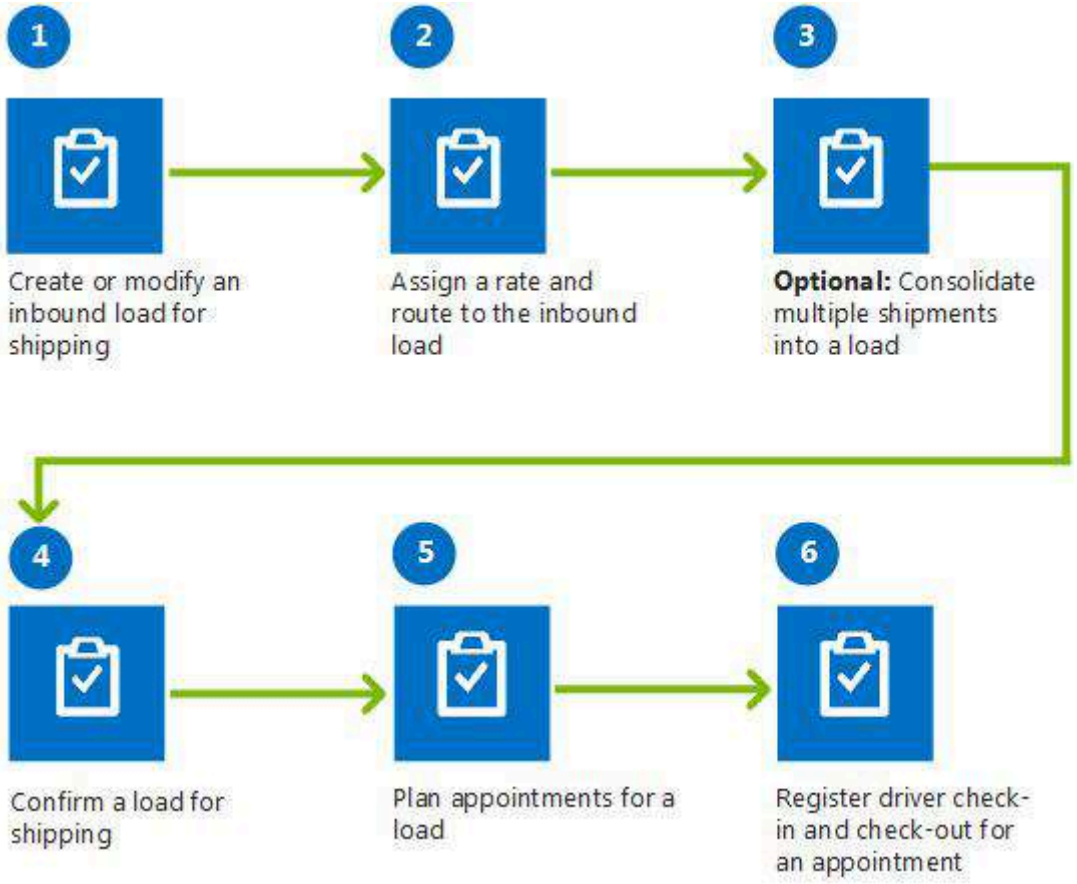
In Transportation management, transportation planning can be based either on orders or on the shipments that are created based on those orders. The shipments always exist at some

point in time but aren't required for transportation planning. Transfer orders are part of the outbound scenario and can be planned together with sales orders.



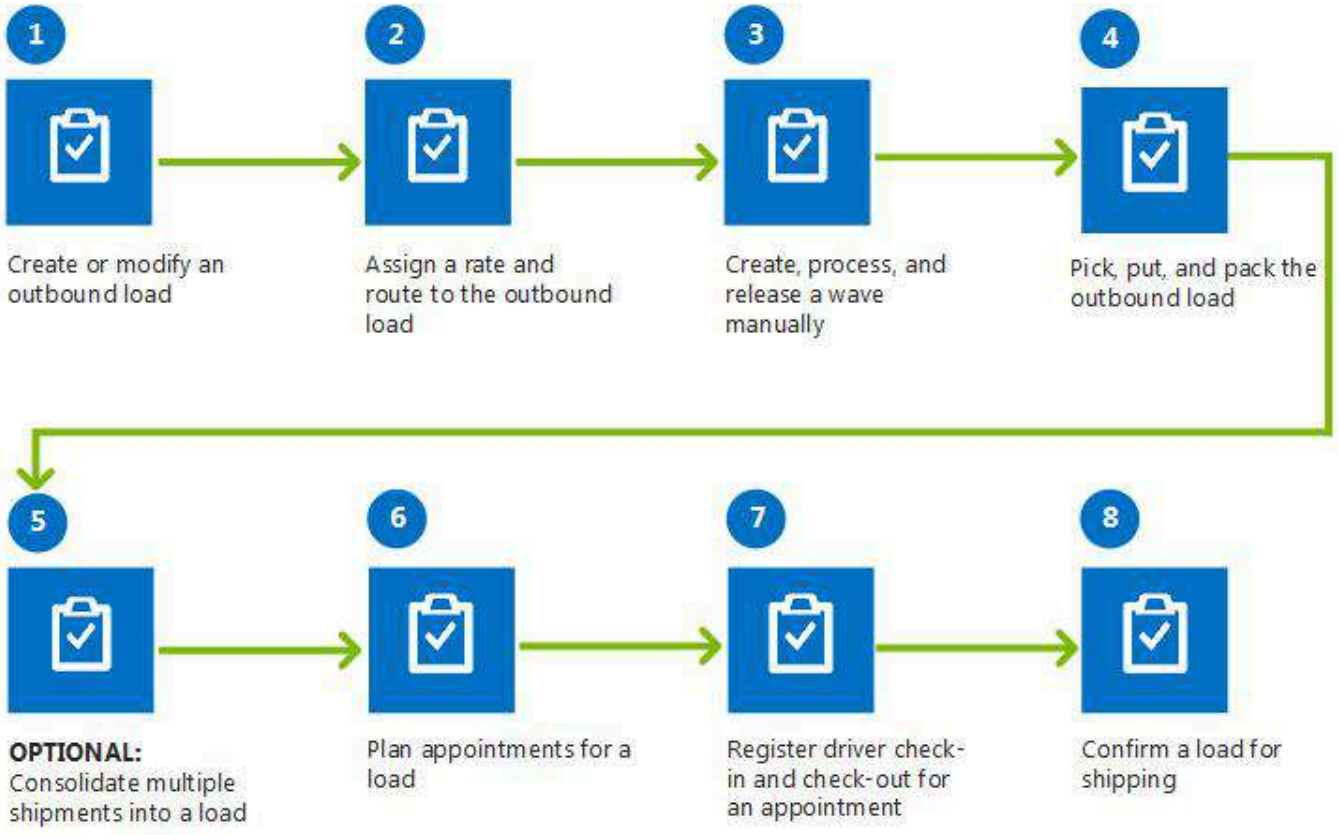
Inbound transportation

When you order items from a vendor, and the items must be delivered to your warehouse, you might want to arrange the transport of the items yourself. You can use Supply Chain Management to plan the transportation and receipt of the inbound load. The following illustration shows the business process flow for planning transportation for an inbound load.



Outbound transportation

You can plan and process an outbound load to ship specific items from a company's warehouse to a customer. You can use Supply Chain Management to plan the transportation and shipping of an outbound load. The following illustration shows the business process flow for planning and processing outbound loads for shipping.



Load building

Supply Chain Management provides a load-building strategy called the *Volume-based load building strategy*. This strategy lets you use the maximum values that are specified for height and weight in the load template, or you can override the settings by entering new values. To use this strategy, select it in the **Load building strategy** field on the **Setup** FastTab on the **Load building workbench** page. In addition, you can add your own load-building strategies by creating a new class in the Application Object Tree (AOT).

White papers

The following white papers explore various aspects of transportation management in Dynamics 365 Supply Chain Management:

- Transportation management engines
- Implementing and deploying Transportation management engines

Transportation management engines

07/31/2025

Transportation management engines define the logic that is used to generate and process transportation rates in Transportation management.

A transportation management engine calculates tasks, such as the carrier's transportation rate. The engine system lets you change calculation strategies at runtime, based on data in Supply Chain Management. A transportation management engine resembles a plug-in that is related to a particular carrier contract.

What engines are available?

The following table shows the transportation management engines that are available.

Expand table

Transportation management engine	Description
Rate engine	Calculates rates.
Generic engine	Simple auxiliary engines that are used by other engines that don't require data from Supply Chain Management, for example, an apportionment engine. Apportionment engines are used to reduce the final costs of transportation to specific orders and lines, based on dimensions, such as volume and weight.
Mileage engine	Calculates the transportation distance.
Transit time engine	Calculates the time that is required to travel from the start to the end destination.
Zone engine	Calculates the zone based on the current address and calculates the number of zones that must be crossed in order to travel from address A to address B.
Freight bill type	Standardizes the freight invoice and the freight bill lines and is used for automatic freight bill matching.

What engines must be configured to rate a shipment?

To rate a shipment using a specific carrier, you must configure multiple transportation management engines. The **Rate engine** is required, but other transportation management

engines might be required to support the **Rate engine**. For example, the **Rate engine** can be used to retrieve data from the **Mileage engine** to calculate the rate based on mileage between the source and the destination.

What's required to initialize a transportation management engine?

A transportation management engine requires that you set up initialization data in order to function in a specific way. The setup can include the following types of data:

- References to other transportation management engines. For details, see the configuration example in this section.
- References to .NET types that are used by the transportation management engine.
- Simple configuration data.

In most cases, you can select the **Parameters** button in the transportation management engine's setup forms to configure the initialization data. **Example of the configuration of a rate engine that references a mileage engine** The following example shows the setup that is required for a rate engine that is based on the .NET engine type `Microsoft.Dynamics.Ax.Tms.Bll.MileageRateEngine` and references a mileage engine.

Expand table

Parameter	Description
`RateBaseAssigner`	The .NET type that interprets the rate base assignment data for a particular schema. The syntax of the parameter value consists of two segments delimited by a vertical bar.
`MileageEngineCode`	Mileage engine code that identifies the mileage engine record in the database.
`ApportionmentEngine`	Generic engine code that identifies the apportionment engine in the database.

How is metadata used in transportation management engines?

Transportation management engines that rely on data that is defined in Supply Chain Management can use different data schemas. The transportation management system enables different transportation management engines to use the same generic physical database tables. To make sure that run-time interpretation of engine data is correct, you can define metadata for the database tables. This reduces the cost of building new transportation

management engines because additional table and form structures aren't required in Operations.

What can be used as search data in rate calculations?

The metadata configuration controls the data you use when calculating rates. For example, if you want to search for rates based on postal codes you must set up metadata based on the lookup type of a postal code.

Do all engine configurations require metadata?

No, transportation management engines that are used to retrieve the data that is required for rate calculation from external systems don't need metadata. The rate data for these engines can be retrieved from external transportation carrier systems, usually through a web service. For example, can use a mileage engine that retrieves data directly from Bing maps so that you don't need a metadata for this engine.

> **Note**
>
> The transportation management engines that are delivered with Supply Chain Management rely on data that is retrieved from the application. Engines that connect to external systems aren't included. However, the engine-based extensibility model lets you build extensions using Visual Studio Tools.

How do I configure metadata for a transportation management engine?

Metadata for transportation management engines is configured differently for the different types of engines.

Expand table

Transportation management engine	Metadata configuration
Rate engine	Requires a **Rate base type**. The rate base type contains metadata for the rate base data and the rate base assignment data. The type of rate engine determines the structure of rate base metadata. The type of rate base assigner associated with rate

Transportation management engine	Metadata configuration
	engine determines the structure of its rate base assignment metadata. You set up the rate base type of a rate engine on the **Rate engine** page and on the **Rate master** page.
Zone engine	Requires metadata to be set up directly on the zone master.
Transit time engine and **Mileage engine**	Retrieves the metadata directly from the mileage engine's configuration setup form.

Example of metadata for a rate engine

The transportation management engine requires identification of the origin address, the destination state and country/region, and the start and end point of the shipment. By using these requirements, the metadata would look like the data in the following table. The table also includes information about what type of input data is required.

Expand table

Sequence	Name	Field type	Data type	Lookup type	Mandatory
1	Origin postal code	Assignment	String	Postal Code	Selected
2	Destination state	Assignment	String	State	
3	Destination start postal code	Assignment	String	Postal Code	Selected
4	Destination end postal code	Assignment	String	Postal Code	Selected
5	Destination country	Assignment	String	Country/region	

> **Note**
>
> The transportation management engines delivered with Supply Chain Management rely on postal codes. They don't support longitude and latitude data. If you need to use longitude and latitude data, then you must develop a custom engine. Learn more in Create a new transportation management engine.

White paper

For more information, download the following white paper (written to support AX2012, but still applies for Dynamics 365 Supply Chain Management)

- Transportation management engines

Create a new transportation management engine

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes how to create a new transportation management engine in Dynamics 365 Supply Chain Management.

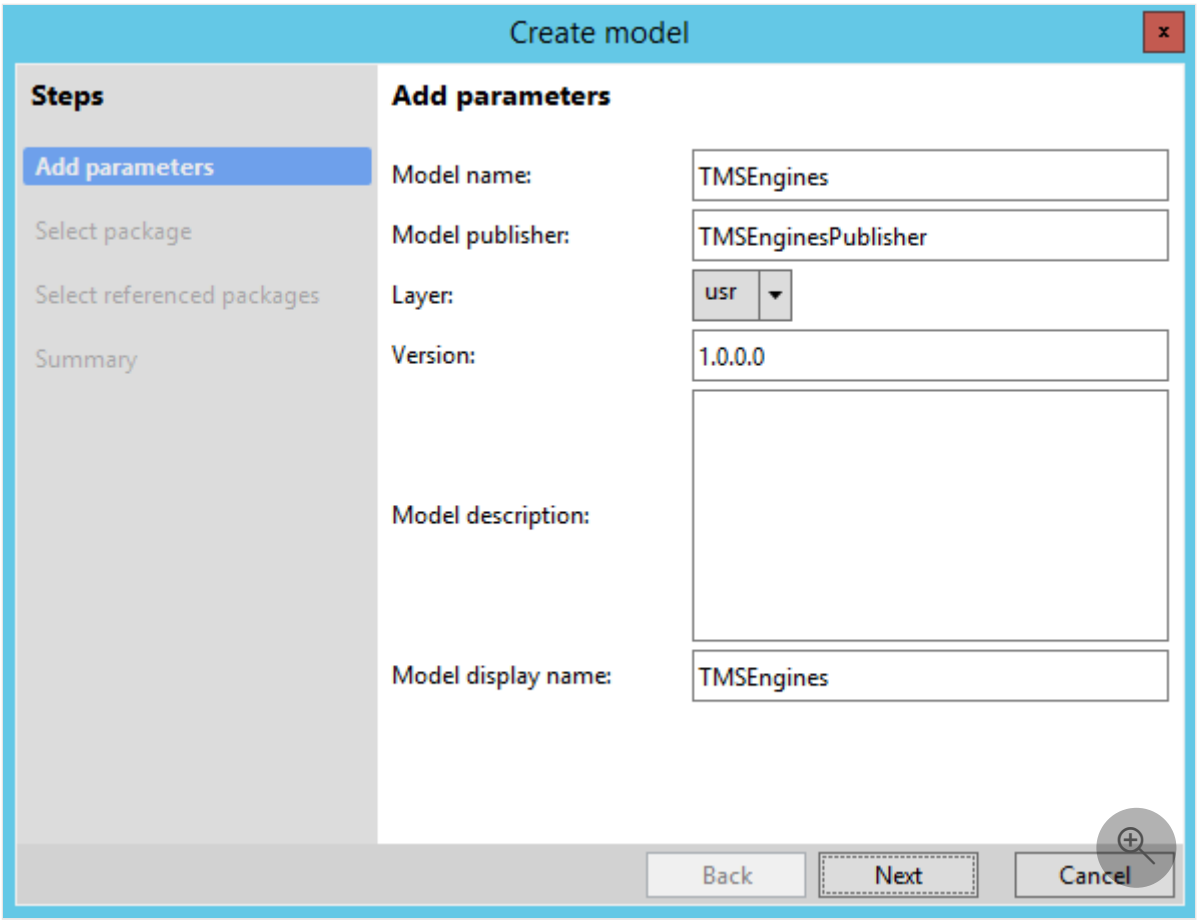
Transportation management (TMS) engines define the logic that generates and processes transportation rates in Transportation management. Supply Chain Management provides several different engine types that calculate different parameters, such as rates, transit times, and the number of zones crossed during transit. This article explains how to use the Microsoft Visual Studio development environment together with Supply Chain Management development tools to create and deploy a new TMS engine, and then how to set up the engine in Operations. For more information about the engines, see Transportation management engines.

Create a new TMS engine

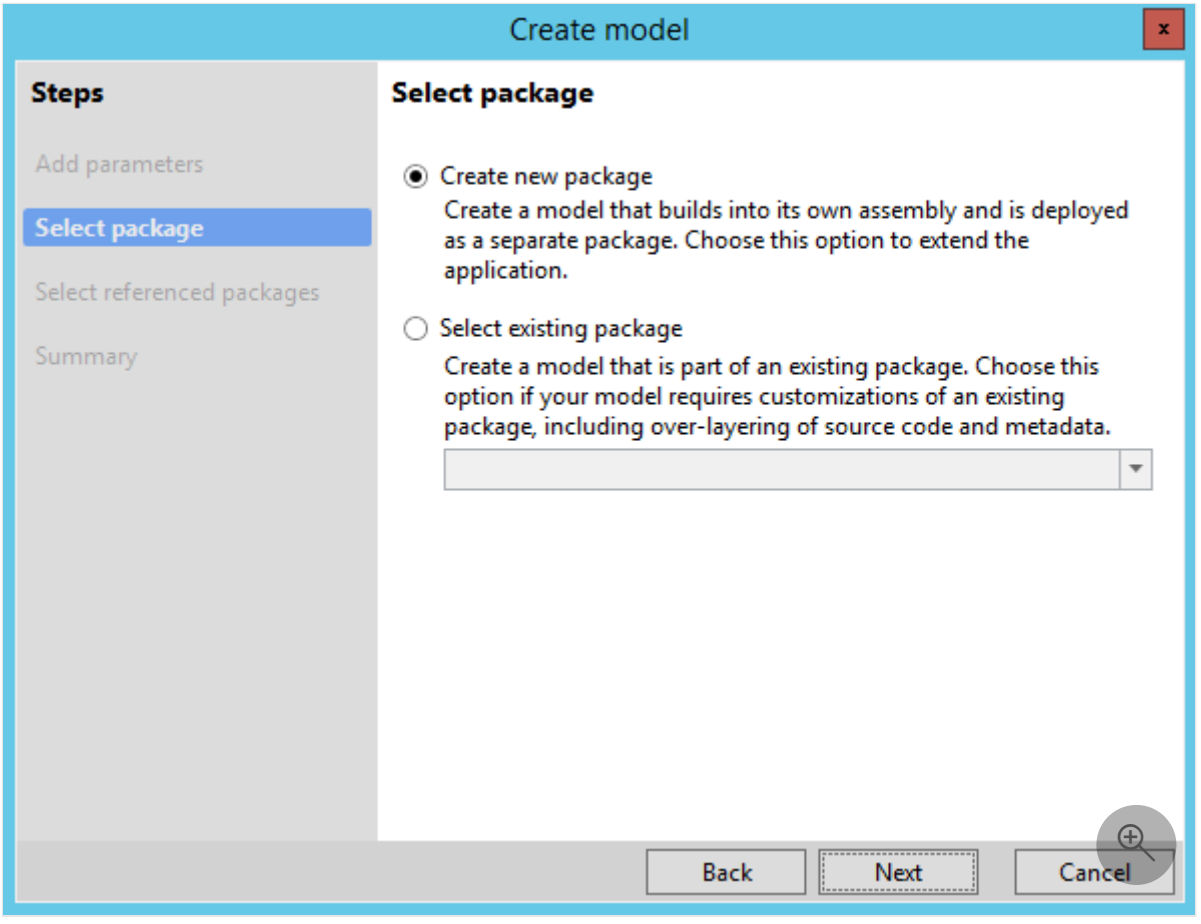
This section explains how to create a class library that has a TMS engine implementation, and how to reference it from a Supply Chain Management model.

1. To deploy your new engines, you must have a model that contains the engines. On the **Dynamics 365** > **Model Management** menu, select **Create model** to create a new model.

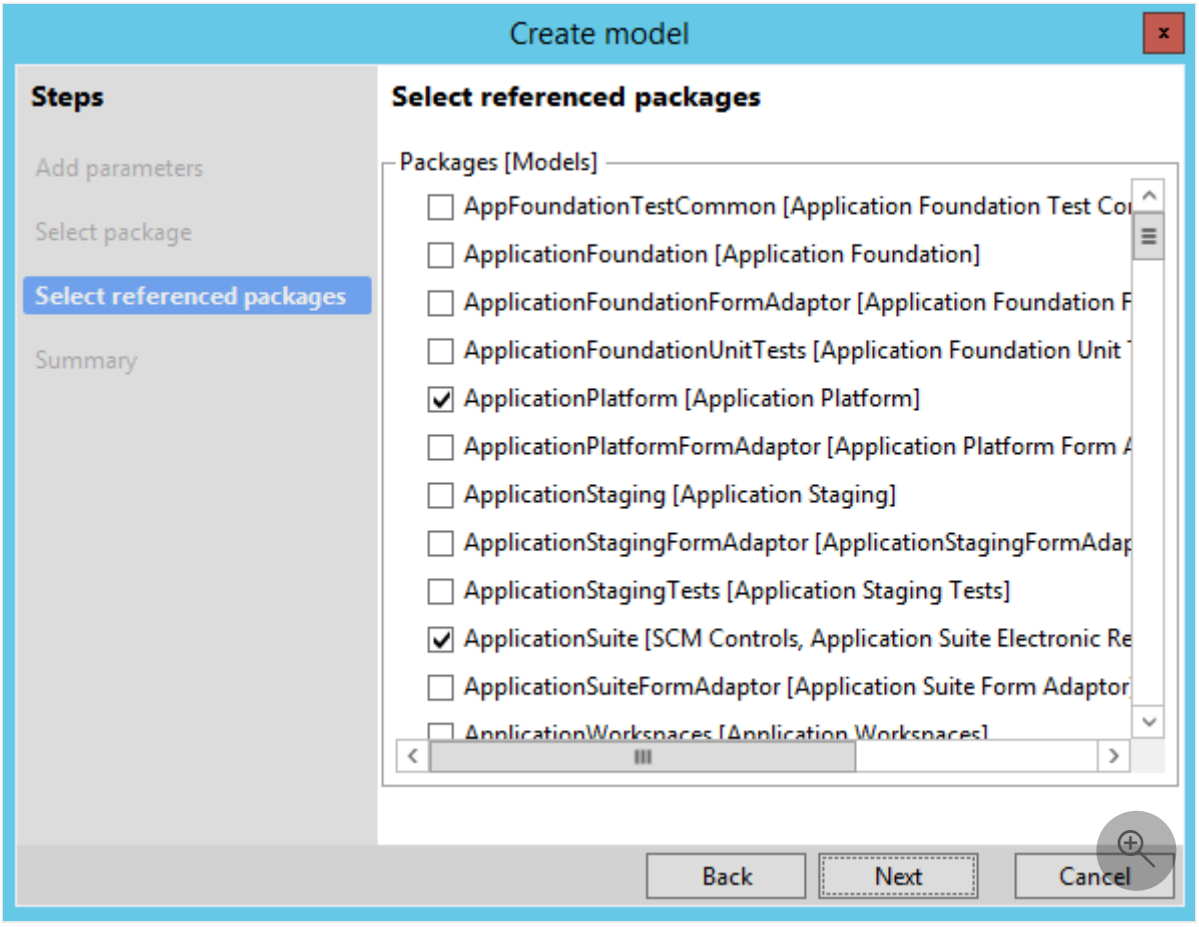
On the first page of the **Create model** wizard, name the model *TMSEngines*.



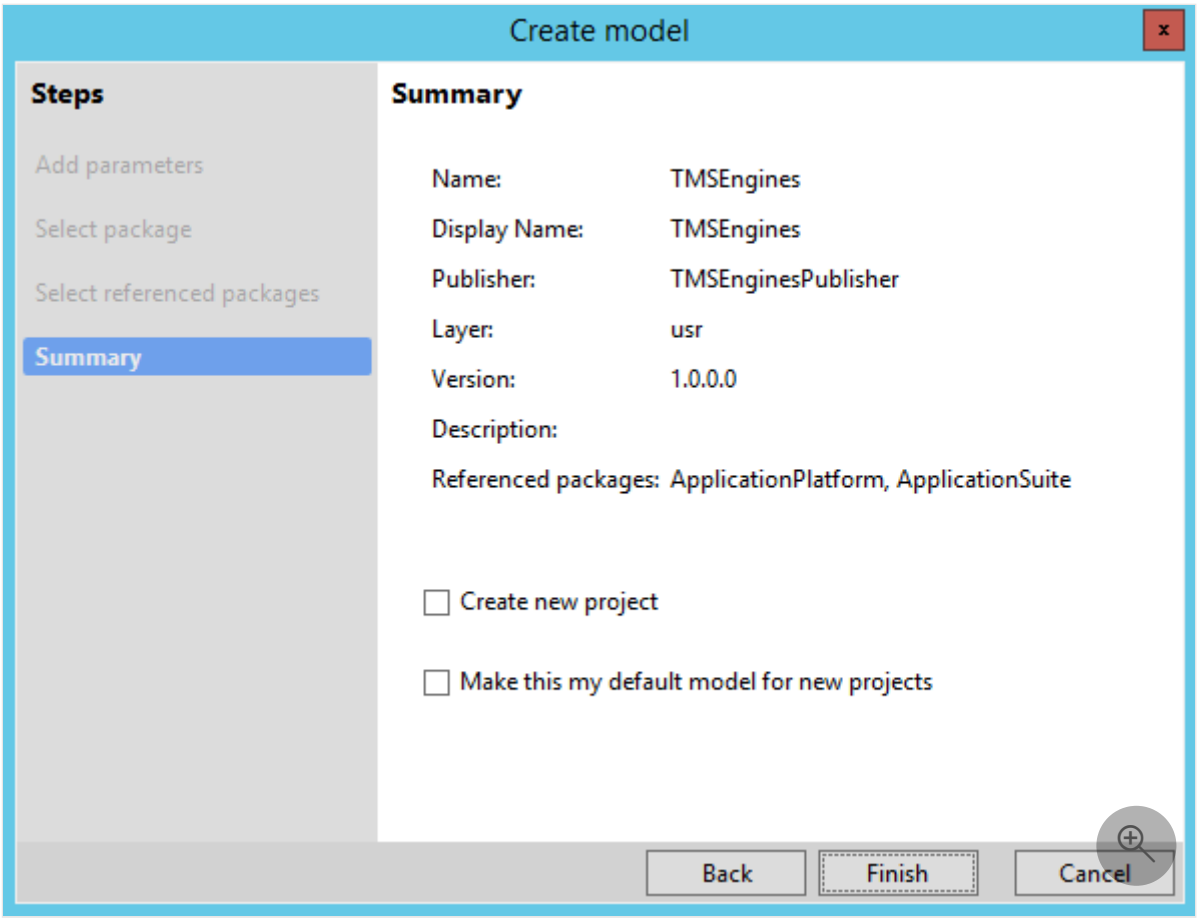
2. On the next page, select **Create new package**.



3. On the next page, select the **ApplicationSuite** model to reference. (The *ApplicationPlatform* model is preselected.)



4. On the next page, select **Finish** to confirm the creation of a new model.



5. In a new solution, create a new Supply Chain Management project, and name it *TMSThirdParty*. The framework must be set to *.NET Framework 4.8* or newer. Double check that the project's file (`TMSThirdParty.rnrproj`) specifies the correct .NET Framework version. If it doesn't, update it directly in the file to `<TargetFrameworkVersion>v4.8</TargetFrameworkVersion>`. In the project properties, set the project's model to *TMSEngines*.

6. Add a new C# *Class Library (.NET Framework)* to your solution, and name it *ThirdPartyTMSEngines*. The framework must be set to *.NET Framework 4.8* or newer and must match the settings on the *TMSThirdParty* project.

7. In the *ThirdPartyTMSEngines* project, add references to Supply Chain Management–specific assemblies specified in the following list.

   > **ⓘ Important**
   >
   > The **Copy Local** property for all the references in this list must be set to *False*. Otherwise, you risk producing runtime issues where the engine class might not be found or the assembly might not load correctly.

   - Application assemblies that enable X++ types to be referenced. These assemblies can be found in the following locations. [Packages root] is the path of the location

where all the deployed assemblies are placed, such as `C:\Packages`.

```text
[Packages
root]\ApplicationPlatform\bin\Dynamics.AX.ApplicationPlatform.dll
[Packages
root]\ApplicationFoundation\bin\Dynamics.AX.ApplicationFoundation.dll
[Packages root]\ApplicationSuite\bin\Dynamics.AX.ApplicationSuite.dll
```

- Framework assemblies that enable access to data, LINQ, and auxiliary functions. All these assemblies can be found in [Packages root]\bin.

```text
Microsoft.Dynamics.ApplicationPlatform.Environment.dll
Microsoft.Dynamics.AX.Data.Core.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.AdoNet.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.Interface.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.Msil.dll
Microsoft.Dynamics.AX.Server.Core.dll
Microsoft.Dynamics.AX.Xpp.AxShared.dll
Microsoft.Dynamics.AX.Xpp.Support.dll
```

- The core TMS assembly (which contains engines) and the TMS base assembly (which contains helpers, constants, data transfer class definitions, and so on). These assemblies can be found in the following locations.

```text
[Packages root]\ApplicationSuite\bin\Microsoft.Dynamics.AX.Tms.dll
[Packages root]\ApplicationSuite\bin\Microsoft.Dynamics.AX.Tms.Base.dll
```

8. Rename the C# class that is automatically generated in the *ThirdPartyTMSEngines* project to `SampleRatingEngine`.

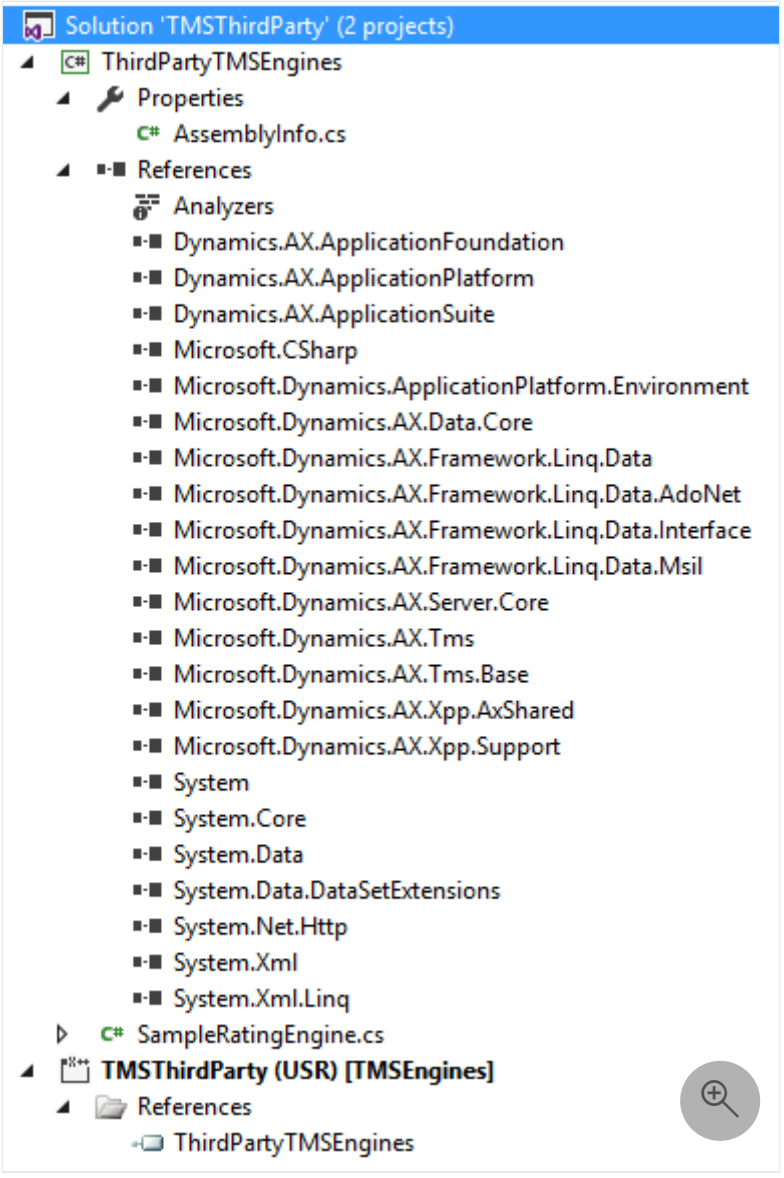
9. Implement the engine. Because you're creating a rate engine in this example, inherit from the base class for rate engines. The base class implements most of the rate engine interface (`TMSFwkIRateEngine`). You just have to implement the rate method. To keep this example simple, make this method register a hard-coded rate of 100. You can create engines that implement any of the engine interfaces, such as `TMSFwkIAccessorialEngine`. All the engine interfaces are defined in X++.

```C#
```

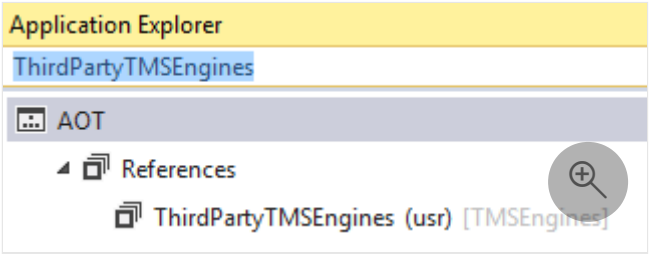
```
namespace ThirdPartyTMSEngines
{
    using Dynamics.AX.Application;
    using Microsoft.Dynamics.Ax.Tms.Base.Data;
    using Microsoft.Dynamics.Ax.Tms.Base.Utility;
    using Microsoft.Dynamics.Ax.Tms.Bll;
    using System.Xml.Linq;
    public class SampleRatingEngine : BaseRateEngine
    {
        public override RatingDto rate(TmsTransactionFacade
_transactionFacade, XElement _shipment, string _rateMasterCode)
        {
           XElement re =
shipment.RetrieveOrCreateRatingEntity(this.RatingDto);
           re.AddRate(TmsRateType.Rate, 100);
           return this.RatingDto;
        }
    }
}
```

10. Build the solution.

11. Add a new reference to the *TMSThirdParty* project. The reference should point to the *ThirdPartyTMSEngines* project. When you finish, your solution should look like this.



12. Build the solution. Verify that the new library appears in the **References** node in Application Explorer.



> **Note**
>
> If you're testing the new engine directly in the development environment, you might also need to build the entire *TMSEngines* model from **Extensions** > **Dynamics 365** > **Build models**. This might resolve the issue if the engine assembly couldn't be found.

Deploy the TMS engine as a package

One way to deploy third-party TMS engines is through a deployment package. To deploy the engine as a package, follow these steps:

1. In Visual Studio, go to **Extensions** > **Dynamics 365** > **Deploy** > **Create Deployment Package** and create a deployment package for the new model you created as described in Create a new TMS engine.

2. In Microsoft Dynamics Lifecycle Services, go to **Asset Library** > **Software deployable package**. Select **Add** to add the deployment package you created to Lifecycle Services.

3. After the validation completes, go to **Environments** > **{Your Environment ID}** > **Maintain** > **Apply updates**, select the deployable package, and apply it.

> **Note**
>
> If the system warns you that you don't have permission to deploy packages on Lifecycle Services, contact your Lifecycle Services administrator for support.

Set up the TMS engine in Supply Chain Management

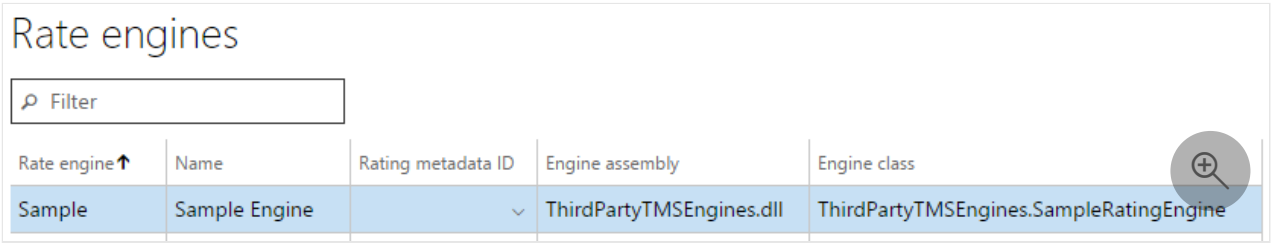
This section explains how to set up Supply Chain Management to use a TMS engine, and shows how the new engine that you created is used in rate shopping. The example in this section uses the USMF demo data company.

1. Create a new engine as described in Create a new TMS engine.

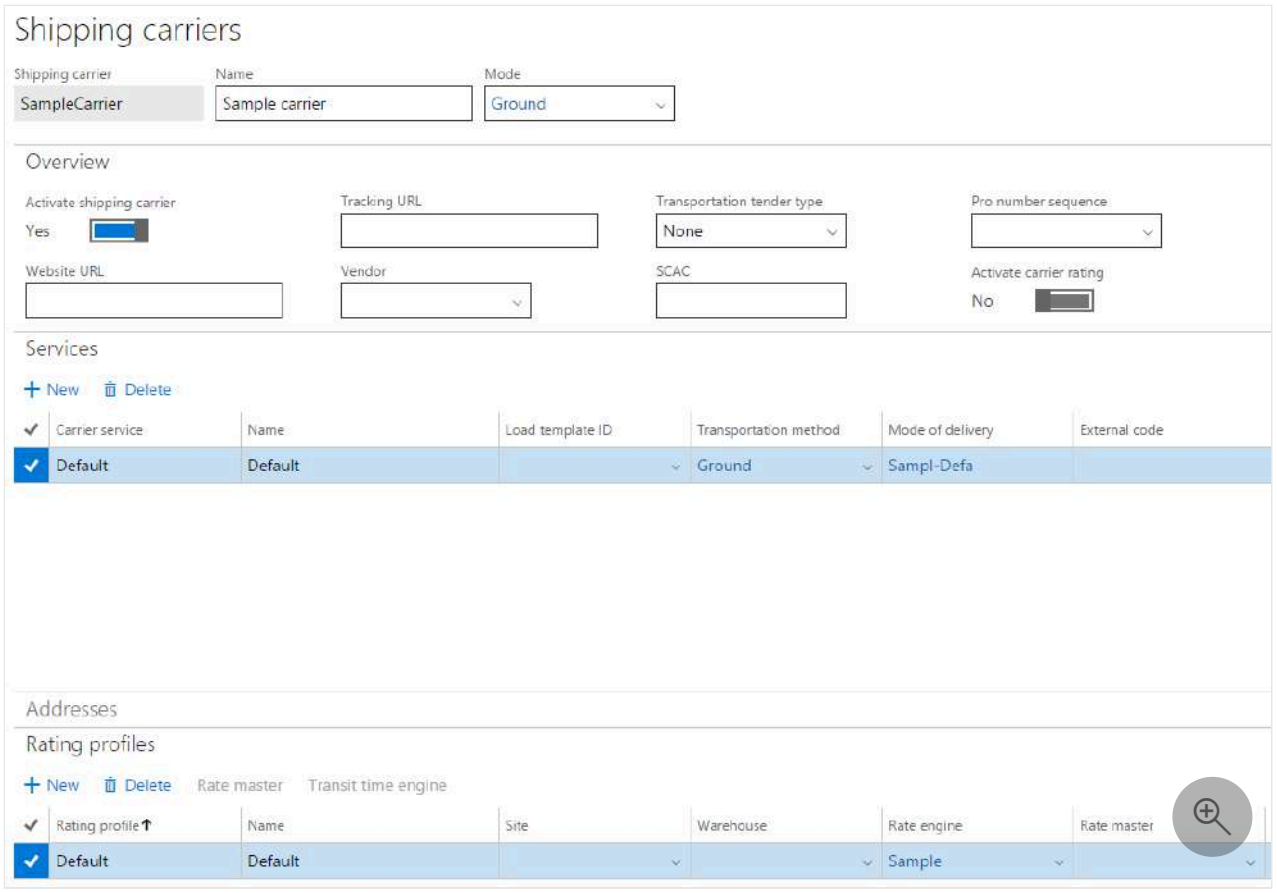
2. Build your solution.

3. Deploy the TMS engine DLL file through a deployment package, as described in Deploy the TMS engine as a package.

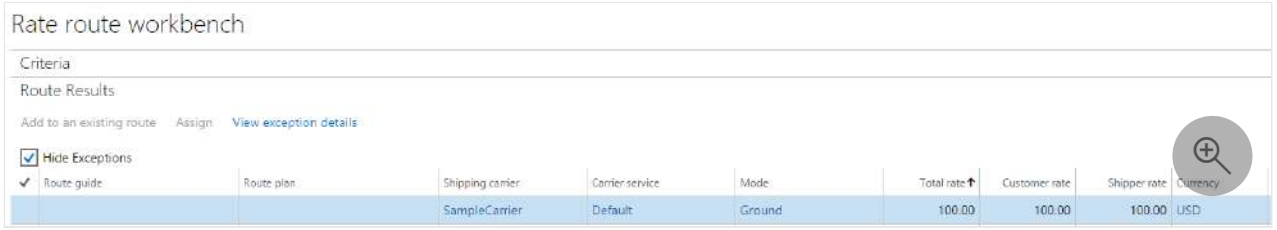
4. In Supply Chain Management, on the **Rate engines** page, create a new rating engine. The engine should point to the engine assembly produced by building the engine class library and the engine class that you implemented.



5. Create a shipping carrier that uses the Sample rate engine. Because our engine doesn't use any data, you don't have to assign a rate master.



6. On the **Rate route workbench** page, select **Rate shop**. You should see a rate of 100.00 from SampleCarrier, as shown in the following screenshot. In this example, you're rate shopping for a route from warehouse 24 to customer US-004. However, but because the rate is hard-coded, you always see a rate of 100.00.



Tips and tricks

- If you're using development tools for Supply Chain Management, it's useful to add a new project to your solution. If you set this project as your startup project and start a debugging session, you can debug both X++ and C# code in the same debugging session.
- Every time that you change and recompile your *ThirdPartyTMSEngines* project, you must deploy through a deployment package. Otherwise, you might run using a stale assembly.

- After you execute TMS-specific operations in Supply Chain Management, the Internet Information Services (IIS) worker process might lock the *ThirdPartyTMSEngines* assembly so that the assembly can't be updated. In this case, restart the w3svc process.

Whitepaper

For more information, download the following white paper (written to support AX2012, but still partially applies for Dynamics 365 Supply Chain Management):

- Implementing and deploying Transportation management engines

---

Last updated on 03/11/2026

Set up shipping carriers

07/31/2025

This article shows how to set up a shipping carrier and define details such as service, shipment mode, transportation tender, transportation constraints, and shipping rate. A transportation coordinator can then assign a shipping carrier to an inbound or outbound load.

Create a new shipping carrier

1. Go to **Transportation management** > **Setup** > **Carriers** > **Shipping carriers**.
2. Select **New** on the Action Pane.
3. In the **Shipping carrier** field, type a value.
4. In the **Name** field, type a value.
5. In the **Mode** field, select an option from the drop-down menu.

Fill in the general information for the shipping carrier

1. Toggle the expansion of the **Overview** section.
2. Check or uncheck the **Activate shipping carrier** checkbox.
3. In the **Vendor account** field, select an option from the drop-down menu. Select the vendor account to assign the shipping carrier to.
4. In the **Transportation tender type** field, select an option. Select **Manual** to use the Transportation Tender page, or select **EDI** to update the tender by using Electronic Data Interchange (EDI).
5. Check or uncheck the **Activate carrier rating** checkbox.

> ⓘ **Note**
>
> It's possible to confirm shipment of a load that doesn't have a rating applied (for example, if no rate is assigned to it).

Create the necessary services for the shipping carrier

1. Toggle the expansion of the **Services** section.
2. Select **New**.

3. In the **Carrier service** field, type a value.
4. In the **Name** field, type a value.
5. In the **Load template ID** field, select a load template to associate with the service. The load template defines the maximum measurements for weight and volume of an entire load. For example, the load template might represent the size of a container or truck. Load template IDs are also specified in load building templates and when using the load building workbench, which helps you apply load building strategies to create loads. As a result, the system is able to match each new load to a suitable shipping carrier service by comparing the specified load template IDs.
6. In the **Transportation method** field, select an option from the drop-down menu.

Set up the address for the carrier (optional)

1. Toggle the expansion of the **Addresses** section.
2. Select **New**.
3. In the **Name or description** field, type a value.
4. In the **Country/region** field, select an option from the drop-down menu.
5. In the **ZIP/postal code** field, select an option from the drop-down menu.
6. In the **Street** field, type a value.
7. Select **OK**.

Set up the rating profile for the shipping carrier

1. Toggle the expansion of the **Rating profiles** section.
2. Select **New**.
3. In the **Rating profile** field, type a value.
4. In the **Name** field, type a value.
5. In the **Site** field, select an option from the drop-down menu.
6. In the **Warehouse** field, select an option from the drop-down menu.
7. In the **Rate engine** field, select an option from the drop-down menu. Select the Rate engine that is in accordance with the contract that you have with the carrier.
8. In the **Rate master** field, select an option from the drop-down menu.
9. In the **Transit time engine** field, select an option from the drop-down menu.
10. Select **Save**.

Associate a fuel index with a carrier as an accessorial charge

06/17/2025

This guide shows how to create an accessorial assignment, carrier accessorial charge, and accessorial master for fuel surcharge. It also shows how to associate a carrier fuel index with a carrier. A carrier fuel index must already exist before you run this guide (learn more in Set up a carrier fuel index). These setup tasks are typically done by a logistics manager.

Create an accessorial master

1. Go to **Transportation management** > **Setup** > **Rating** > **Accessorial masters**.
2. Select **New**.
3. In the **Accessorial master** field, type a value.
4. In the **Name** field, type a value.
5. Select **Save**.

Create a carrier accessorial charge

1. Go to **Transportation management** > **Setup** > **Rating** > **Carrier accessorial charges**.
2. Select **New**.
3. In the **Carrier accessorial ID** field, type a value.
4. In the **Shipping carrier** field, find and select the desired record.
5. In the **Carrier service** field, select the link in the selected row.
6. In the **Accessorial master** field, select the desired record.
7. Select **Save**.

Create an accessorial assignment

1. Select **Accessorial assignments**.
2. Select **New**.
3. In the **Name** field, type a value.
4. Expand the **Criteria** FastTab. Here, you can choose to always apply the fuel surcharge or, for this example, choose that it only applies within a certain region.
5. In the **ZIP/postal code from** field, type a value.
6. In the **ZIP/postal code to** field, type a value.
7. Expand the **Calculation** FastTab. Here, you can specify how to calculate the fuel surcharge. This calculation depends on the accessorial unit that you chose as the base for your

calculation.

8. In the **Accessorial fee type** field, select a fee type (such as *Fuel surcharge*).
9. In the **Accessorial unit** field, select a unit (such as *Mileage*).
10. In the **Region** field, select the link in the selected row.
11. Select **Save**.

Update the carrier rating profile

1. Go to **Transportation management** > **Setup** > **Carriers** > **Shipping carriers**.
2. In the list, find and select the desired record.
3. Expand the **Rating profiles** FastTab.
4. Select **Edit**.
5. In the **Carrier fuel index** field, select the link in the selected row.
6. Select **Save**.

Carrier groups

06/17/2025

A carrier group is a collection of shipping carriers and carrier services. Each carrier group specifies the preferred sequence for the shipping carriers and carrier services that belong to it.

When multiple shipping carriers and carrier services exist for the same route segment, you can specify a carrier group instead of a specific shipping carrier and carrier service in the route plan or route guide.

Create a carrier group

1. Go to **Transportation management** > **Setup** > **Carriers** > **Carrier group**.
2. Select **New**.
3. In the **Carrier group** field, enter a unique identifier (ID) for the group.
4. In the **Name** field, enter a descriptive name for the group.
5. On the **Details** FastTab, add a row, and select a shipping carrier and a carrier service for it. Repeat this step until you've added as many carriers as you require for the group.
6. Close the page.

Transportation management number sequence

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Use the **Number sequences** page in the transportation management module to set up various pro numbers. Carriers use pro numbers to organize and track the progress of each shipment.

Create a number sequence for a pro number

To create a number sequence for a pro number, follow these steps:

1. Go to **Transportation management** > **Setup** > **Carriers** > **Number sequences**.
2. Select **New** to create a new number sequence.
3. Enter a unique ID and descriptive name for the number sequence.
4. In the **Number sequence type** field, select *Pro number*.
5. In the **Check digit** field, select *Check digit*, which is up as a generic engine.
6. On the **Sequence** FastTab, provide information about the sequence.
7. Close the page.

Learn more about how to work with number sequences in Number sequences overview.

Link a number sequence to a shipping carrier

To link a number sequence to a carrier, follow these steps:

1. Go to **Transportation management** > **Setup** > **Carriers** > **Shipping carriers**.
2. Select a shipping carrier.
3. Select **Edit**.
4. On the **Overview** FastTab, select an option in the **Pro number sequence** field.
5. Close the page.

Pro number generation mode

The pro number generation mode controls how the system allocates pro numbers.

> **Tip**
>
> If your high-volume processes (such as release to warehouse) run slowly when you're using pro numbers, use the *Performance optimized* pro number generation mode to help improve throughput.

Prerequisites

You must be running Supply Chain Management version 10.0.47 or higher to use this feature.

Choose the pro number generation mode

To choose the pro number generation mode, follow these steps:

1. Go to **Transportation management** > **Setup** > **Transportation management parameters**.
2. Open the **General** tab.
3. On the **Performance settings** FastTab, set **Pro number generation mode** to one of the following values:

    - *Default* – Generate pro numbers inside the current business transaction (*transaction scoped*). The system keeps the number only if the process finishes successfully. If you cancel, or if an error stops the process, the system puts back the number and reuses it. This option enforces continuous sequencing but can increase locking under high parallel load.
    - *Performance optimized* – Generate the number on a separate connection (*isolated*). This option reduces locking and improves throughput under high load, but numbers aren't recycled if the business process later fails.

Last updated on 01/27/2026

Transportation management statuses

06/17/2025

Set up master codes for transportation statuses to interpret codes that are provided by your shipping carriers. This lets you integrate with shipping carriers to provide a status. The transportation status that you provide for a transportation master status code can help you track the status of a load, shipment, or container. The specific transportation status for a load, shipment, or container can only be updated through data integration and not manually through the user interface.

Create a transportation status

To create a transportation status, follow these steps:

1. Go to **Transportation management** > **Setup** > **Transportation status masters**.
2. Select **New** to create a transportation status master.
3. In the **Transportation status master** field, enter a unique code for the transportation status.
4. In the **Transportation type** field, select either *Shipping carrier* or *Hub* as the type of transportation.
5. Enter a name and transportation status.
6. Close the page.

Map a transportation status to a carrier status

To map a transportation status to a carrier status, follow these steps:

1. Go to **Transportation management** > **Setup** > **Carriers** > **Carrier transportation status**.
2. Select **New** to map a code from a shipping carrier to a transportation status master code.
3. Select the unique ID for the shipping carrier and the carrier service.
4. Select the transportation status code that you want to map to the selected shipping carrier's code.
5. Enter the external code that is used by the shipping carrier.
6. Close the page.

Transportation management zone master

Article • 02/12/2025

Transport management lets you divide geographic locations into zones. Dividing locations into zones can help to:

- **Simplify transportation pricing** – Zone-wise pricing can be simpler than individual location-based pricing, especially when transportation locations are scattered.
- **Optimize load planning** – By consolidating loads by zones.
- **Optimize route planning** – By assigning specific route plans to specific zones.

You define zones based on the metadata field values (such as country/region, zip code range, or carrier service) that qualify each zone. Zone definitions aren't required if your transportation pricing doesn't employ a zone concept.

Feedback

Was this page helpful? Yes No

Provide product feedback

Load templates

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When you create a new load, you can assign a load template. The load template contains information about equipment, and about measures such as the height, width, depth, and volume of the load.

This article describes how to set up load templates, and how to associate a load template with a new load.

Set up a load template

1. Go to **Transportation management** > **Setup** > **Load Building** > **Load template**.
2. On the Action Pane, select **New** to add a new template or **Edit** to edit an existing template.
3. In the row for the new or existing template, set the following fields:
    - **Load template ID** – Enter a unique identifier (ID) for the load template.
    - **Equipment** – Select the equipment that you want to use to ship the load.
    - **Load height**, **Load width**, and **Load depth** – Enter the dimensions of the load.
    - **Max. allowed load volume** and **Max. allowed load weight** – Enter the maximum allowed volume and weight of the load.
    - **Maximum allowed gross weight** – Enter the maximum allowed gross weight of the load. A load's gross weight includes both its tare weight and its loading weight.
    - **Maximum number of freight pieces allowed** – Enter the maximum number of freight pieces that the load can contain.
    - **Stack load on floor** – Select this check box to use floor loading. In a floor loading scenario, boxes are stacked floor to ceiling and wall to wall inside the container, to maximize capacity.
4. On the Action Pane, select **Save**.

Associate a load template with a new load

1. Go to one of the following pages, depending on whether you're setting up an inbound or outbound load:

    - **Transportation management** > **Planning** > **Inbound load planning workbench**.
    - **Transportation management** > **Planning** > **Outbound load planning workbench**.

2. In the upper part of the page, select one of the following tabs, depending on the type of source document that you're creating a load for: **Shipments**, **Sales lines**, **Transfer lines**, or **Purchase order lines**.

3. Select the specific document to plan the load for.

4. On the Action Pane, on the **Supply and demand** tab, in the **Add** group, select **To new load**.

5. In the **Load template** dialog box, in the **Load template ID** field, select the template to apply.

6. Select **OK** to apply the template.

Last updated on 11/19/2025

Set up a carrier fuel index

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This guide shows how to create a fuel index region, a fuel index, and a carrier fuel index. The fuel index region specifies which region the fuel index applies to, and the fuel index specifies a fuel price for a particular period of time. To reflect the change in fuel prices over time, you can associate multiple fuel indexes with a carrier. A transportation coordinator normally performs these tasks. You can use this procedure in demo data company *USMF* or with your own data.

Create a fuel index region

First, you need to create the different regions where you operate and calculate different fuel surcharges by following these steps:

1. Go to **Transportation management** > **Setup** > **Fuel indexes** > **Fuel index regions**.
2. Select **New**.
3. In the **Region** field, type a value.
4. In the **Name** field, type a value.
5. Select **Save**.

Create a fuel index

For the regions you set up, enter the current prices for the fuel by following these steps:

1. Go to **Transportation management** > **Setup** > **Fuel indexes** > **Fuel indexes**.
2. Select **New**.
3. In the **Region** field, select the region.
4. In the **Price per gallon** field, enter a number.
5. In the **Effective start date and time** field, enter a date and time.
6. Select **Save**.

Create a carrier fuel index

1. Go to **Transportation management** > **Setup** > **Fuel indexes** > **Carrier fuel indexes**.

2. Select **New**.
3. In the **Carrier fuel index** field, enter a value.
4. In the **Description** field, enter a value.
5. Select **New**.
6. In the **Effective start date and time** field, enter a date and time.
7. In the **PPG from** field, enter a number. If you're working with *USMF* sample data, enter *1.95*.
8. In the **PPG to** field, enter a number. If you're working with *USMF* sample data, set the PPG To field to *2*.
9. In the **Percentage** field, enter a number. If you're working with *USMF* sample data, set the percentage to *3*.
10. In the **Currency** field, select the currency.
11. Select **Save**.

Last updated on 11/20/2025

Set up a transportation tender

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to set up a transportation tender. This task is typically done by a transportation coordinator.

1. Go to one of the following pages, depending on whether you're setting up a transportation tender for an inbound or outbound load:
    - **Transportation management** > **Planning** > **Inbound load planning workbench**.
    - **Transportation management** > **Planning** > **Outbound load planning workbench**.
2. Clear the **Hide shipped and received** check box.
3. In the **Loads** grid, select a load.
4. On the **Loads** toolbar, select **Rating and routing** > **Routes**.
5. On the Action Pane, select **Transportation tenders**.
6. On the Action Pane, select **New**.
7. Expand the **General** FastTab.
8. In the **Requested rates** field, enter a number.
9. On the Action Pane, select **Save**.
10. On the Action Pane, select **Update status** > **Submit**.
11. Select a route.

Last updated on 11/19/2025

Rating profiles

Article • 05/26/2025

A rating profile resembles a logistics contract (but not a legal contract). It's used to determine transportation tariffs for loads.

Each rating profile is unique to a shipping carrier. In the profile, you associate the shipping carrier with a rate master. The rate master defines the rate base assignment and the rate base. The rate base determines the rate of the carrier.

You can set up a rating profile by using a generic page that shows an overview of all existing rating profiles. Alternatively, you can set up a rating profile directly from the shipping carrier. In both cases, the information that you set up for the rating profile is the same.

Create or edit a rating profile on the Rating profiles page

On the **Rating profiles** page, you can review all available rating profiles. You can also edit existing profiles and create new profiles.

1. Go to **Transportation management** > **Setup** > **Rating** > **Rating profile**.
2. On the Action Pane, select **New** to add a new rating profile to the grid, or select **Edit** to edit an existing profile.
3. In the row for the new or existing rating profile, set the following fields:
    - **Rating profile**: Enter a unique identifier (ID) for the rating profile.
    - **Name**: Enter a descriptive name for the rating profile.
    - **Shipping carrier**: Select a shipping carrier. The rating profile that you're setting up is also shown on the **Shipping carriers** page for the selected carrier.
    - **Site** and **Warehouse**: Select a site and warehouse.
    - **Rate engine**: Select the rate engine for the rating profile.
    - **Rate master**: Select the rate master for the rating profile. You can use the rate master to define a rate base type and a rate base. Learn more in Set up rate masters, rate bases, and break masters.
    - **Transit time engine**: Select the transit time engine for the rating profile.
    - **Carrier fuel index**: Select the carrier fuel index for the rating profile.
    - **Effect start date and time** and **Effective end date and time**: Define the period when the rating profile should be active.
4. On the Action Pane, select **Save**.

Create a rating profile directly on the Shipping carriers page

1. Go to **Transportation management** > **Setup** > **Carriers** > **Shipping carriers**.
2. Select a shipping carrier in the list.
3. On the **Rating profiles** FastTab, select **New** to create a rating profile.
4. Set the fields for the new rating profile. These fields correspond to the fields on the **Rating profiles** page, as described in previous section of this article.

> **Note**
>
> Profiles that are created on the **Shipping carriers** page are also shown on the **Rating profiles** page.

Set up rate masters, rate bases, and break masters

Article • 05/26/2025

This article explains how to set up rates that are returned when using the rate route workbench as part of load planning or building. Rate engines use the configuration of rating profiles, rate masters, rate base assignments, rate bases, and break masters to determine the query structure. The logistics manager usually sets up these options based on the contracts signed with the carriers.

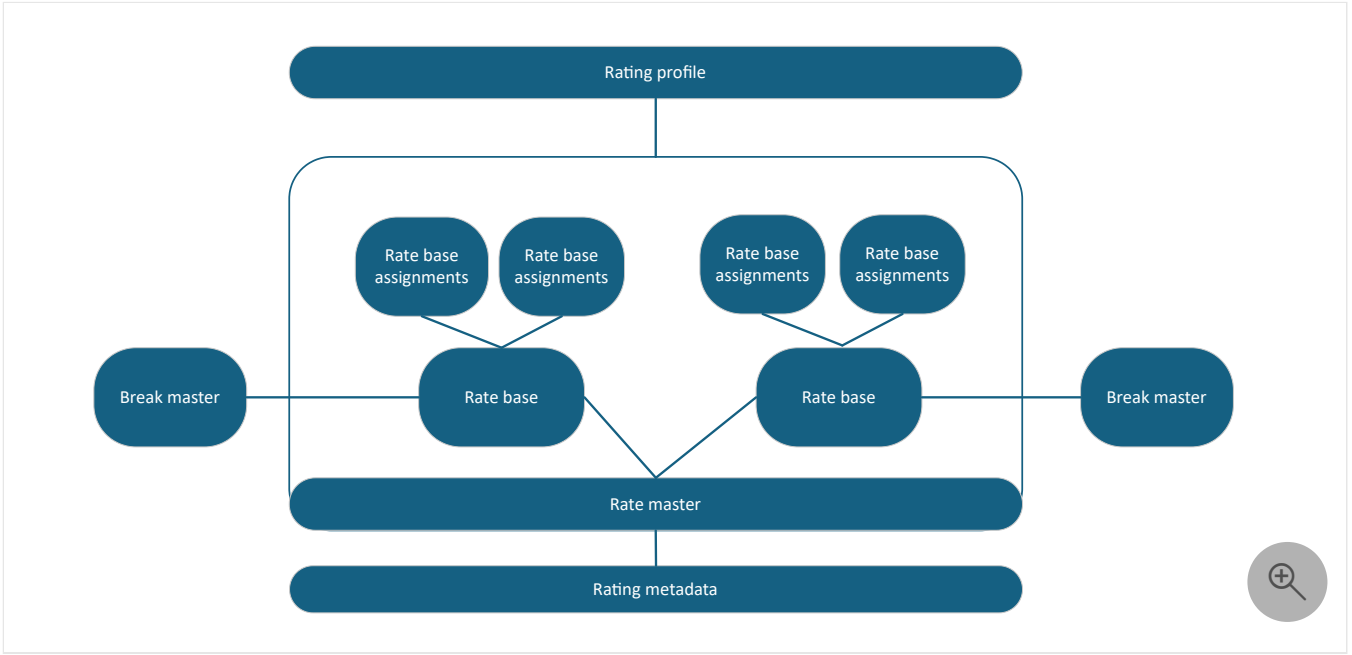
Because of the relationships between each of these rating elements, they must be configured in the following order:

1. Set up rating metadata. Learn more in How is metadata used in transportation management engines?.
2. Set up a break master.
3. Set up a rate master and associate it to one rating metadata ID. For each rate master, you must do the following:

    - Set up one or multiple rate bases, each based on one break master.
    - Set up one or multiple rate base assignments.

After this, you must set up a rating profile, which is unique to a shipping carrier and is associated with one rate master. Learn more in Rating profiles.

The following illustration shows the relationship between and the hierarchy of elements that make up rating in transportation management. You must configure rating from the bottom up.



> ⓘ Note
>
> The descriptions in this article relate to standard engines and might not fully apply to custom engines.

Set up a break master

Break masters define a pricing structure and its breakpoints. The pricing structure uses tiered pricing that is based on the break unit used by the engine. If the shipping carrier charges a certain rate per unit of distance or weight, for example, the break master can reflect this in the rate base.

1. Go to **Transportation management** > **Setup** > **Rating** > **Break master**.
2. On the Action Pane, select **New**.
3. Enter the following values in the header of the new record.

    - **Break master**: Enter a unique identifier for break master.
    - **Name**: Enter a descriptive name for the break master.

4. Enter the following values on the **General** FastTab.

    - **Data type**: Select the data type used to define the breakpoints (*string*, *integer, or *real*).
    - **Comparison**: Enter the operator used when evaluating the breakpoints (typically <, for *less than*, but you can enter any comparison operator in this field).
    - **Break unit**: Enter the unit or measure used to define the breakpoints (for example kgs/lbs or kms/miles). Ideally, the break unit should match the unit of the engine.

5. On the **Details** FastTab, create as many breaks as are needed for your pricing structure. For each row, enter a **Value** using the **Data type** and **Break unit** you specified on the **General** FastTab.
6. Select **Save**.

> ⓘ Note
>
> The **Comparison** and **Break unit** fields don't affect the standard engine behavior. They're only provided for informational purposes.

Set up a rate master

Rate masters represent carrier rates. Each rate master can be associated with only one rating metadata ID but can have multiple rate bases and rate base assignments.

1. Go to **Transportation management** > **Setup** > **Rating** > **Rate master**.
2. On the Action Pane, select **New**.
3. Enter the following values in the header of the new record.

    - **Rate master**: Enter a unique identifier for rate master.
    - **Name**: Enter a descriptive name for the rate master.

4. On the **Overview** FastTab, set **Rating metadata ID** to the rating metadata ID record that applies for this rate master. The rating metadata ID determines the data needed for the rate master. It defines the metadata expected by the transportation management engine using this rate master. Learn more in How do I configure metadata for a transportation management engine?.

Set up a rate base

The rate base defines the prices charged by the carrier at each breakpoint defined in an associated break master. You can set up one or multiple rate bases for each rate master to reflect the carrier pricing structure based on factors such as mileage and/or weight.

1. Go to **Transportation management** > **Setup** > **Rating** > **Rate master**.
2. Select the rate master that you want to add a rate base to.
3. On the Action Pane, select **Rate base**.
4. On the Action Pane, select **New**.
5. Enter the following values in the header of the new record.

    - **Rate base**: Enter a unique identifier for rate base.
    - **Name**: Enter a descriptive name for the rate base.

6. Make the following settings on the **General** FastTab.

    - **Rate master**: Shows the rate master record that you selected previously. This is the master you're creating a rate base for. You can change this value to move a selected rate base to another rate master if needed.
    - **Break master**: Select the break master that defines the pricing structure and breakpoints for this rate base.
    - **Minimum charge**: Specify the lowest amount that can be charged for a shipment, regardless of the calculated rate based on other criteria such as distance, weight, or volume. This ensures that the carrier receives a minimum payment for their services,

even if the calculated rate would otherwise be lower. The minimum charge defined here on the **General** FastTab applies if a minimum charge hasn't been set on the **Details** FastTab for a given line.

- **Maximum charge**: Set an upper limit on the amount that can be charged for a shipment for the rate base. This ensures that the cost for shipping doesn't exceed a specified threshold, regardless of the calculated rate based on other criteria such as distance, weight, or volume. The maximum charge defined here on the **General** FastTab applies if a maximum charge isn't set on the **Details** FastTab for a given line.
- **Currency**: If a currency different from the accounting currency is to be used for the rate base, then select that currency here.

7. On the **Details** FastTab, you can specify a pricing structure for the price break points defined in the break master. The columns available here depend on how the break master and rating metadata for the rate base are configured. For example, if the break master is set to *mileage*, you can set up one rate for less than 100 miles, and another for less than 200, and so on. Select **New** to add a new row to the grid. Then enter values for the following settings for the new row to define the price that applies at each breakpoint. The following are standard columns available in all rate bases, irrespective of the break master and rating metadata configuration:

   - **Minimum charge**: Specify the lowest amount that can be charged for a shipment for the specific rate base line. If a minimum charge is defined here on the **Details** FastTab, it applies even if a different minimum charge is defined on the **General** FastTab.

   - **Maximum charge**: Set an upper limit on the amount that can be charged for a shipment for the rate base. If a maximum charge is defined here on the **Details** FastTab, it applies even if a different maximum charge is defined on the **General** FastTab.

   - **Effective start date and time**: Set the first date and time where the rate applies.

   - **Effective end date and time**: Set the last date and time where the rate applies.

     > **Important**
     >
     > Avoid setting up rate bases with overlapping effective dates and times for matching detail lines.

8. Fill out the remaining fields in the new row to define the rate that applies at each break point.

9. The **Search** FastTab shows information from the **Details** FastTab. Select a line in the **Search** table to view matching values from the **Details** FastTab.

10. On the Action Pane, select **Save**.

Set up rate base assignments

Rate base assignments allow you to create several different price points for each carrier depending on destinations, services, or different rate bases. Rate base assignments define which rate base to apply. Use the **Rate base assignments** FastTab to define the rate base assignment for each rate master.

1. Go to **Transportation management** > **Setup** > **Rating** > **Rate master**.

2. On the **Rate base assignments** FastTab toolbar, select **New** to add a new row to the grid. Then enter values for the following settings for the new row. The columns available here depend on the setup of the rate base assignment metadata, as selected in the **Rating metadata ID** field. The following are standard columns that are always provided.

    - **Name**: Specify a unique identifier for the rate base assignment.
    - **Rate base**: Define the basis on which the rates are calculated.
    - **Service**: Specify the type of service associated with the rate base assignment. It could refer to different levels of service such as standard, expedited, or overnight delivery. The services listed are based on how your shipping carrier services are configured.
    - **Effective start date and time**: Specify the date and time when the rate base assignment becomes active. This setting ensures that the rates are applied correctly from the specified start time.
    - **Effective end date and time**: Specify the date and time when the rate base assignment expires. This setting helps you manage the validity period of the rates and ensures that outdated rates aren't applied.

    > **Important**
    >
    > Avoid setting up rate bases with overlapping effective dates and times for matching detail lines.

3. On the Action Pane, select **Save**.

Transportation management miscellaneous charges

Article • 01/31/2025

As with all miscellaneous charges, transportation-generated charges must be associated with a charge code. Otherwise, they won't be added back to the order as a miscellaneous charge. The **Charges code** determines how the charge is accounted for in relation to the order and order line where it's added.

Go to **Transportation management** > **Setup** > **Rating** > **Miscellaneous charges** to define the qualifying criteria that determine when a specific **Charges code** is applied to a charge.

You should have at least one setup for each relevant **Charges module** setting (*Customer* and *Vendor*) where the **Miscellaneous charge type** is set to *None*. If this setup is missing, the miscellaneous charge won't be added to the order.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up hub accessorial charges and accessorial masters

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to create an accessorial master for a hub and use that master to create a hub accessorial charge. A transportation coordinator typically performs this setup.

Set up a hub master

1. Go to **Transportation management** > **Setup** > **Rating** > **Accessorial masters**.
2. Select **New**.
3. In the **Accessorial master** field, enter a value.
4. In the **Name** field, enter a value.
5. In the **Accessorial type** field, select *Hub*.
6. Select **Save**.

Set up a hub accessorial charge

1. Go to **Transportation management** > **Setup** > **Rating** > **Hub accessorial charges**.
2. Select **New**.
3. In the **Hub accessorial ID field**, enter a value.
4. In the **Hub** field, find and select the desired record.
5. In the **Hub position** field, select an option. You can either create the charge as a pickup or drop-off. Depending on your selection, the charge applies to the corresponding transportation segment on your route.
6. In the **Accessorial master** field, select the master you just created.
7. Select **Save**.

Last updated on 11/20/2025

Set up accessorial assignments

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure shows how to set up an accessorial assignment. A transportation coordinator typically performs this task. Before you follow the procedure, complete the instructions in Set up hub accessorial charges and accessorial masters.

Set up Accessorial assignment

1. Go to **Transportation management** > **Setup** > **Rating** > **Accessorial assignments**.
2. Select **New**.
3. In the **Name** field, type a value.
4. Expand the **Details** section.
5. In the **Hub** field, select the hub that you created an accessorial master for when you completed the procedure in Set up hub accessorial charges and accessorial masters.
6. In the **Hub accessorial ID** field, select the link in the selected row.
7. Expand the **Criteria** section. Here, you can choose the exact criteria for when the charge should apply, based on the different values offered here.
8. Set the **Always apply** option to *Yes*.
9. In the **Accessorial assignment level** field, select an option.
10. Expand the **Calculation** section.
11. In the **Accessorial fee type** field, select *Flat*. The accessorial fee type determines how to calculate the actual charge. In this example, it's a flat charge.
12. In the **Accessorial fee** field, enter a number.
13. Select **Save**.

Last updated on 11/20/2025

Transportation management modes and methods

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The transportation management mode represents the type of transport that the carrier uses for freight deliveries, such as less than load (LTL), truckload (TL), or parcel. The transportation method represents the form of transport that the carrier uses for freight deliveries, such as air, ground, ocean, or rail.

Modes and transportation methods are used in several contexts. Only modes are used in route plans, while both modes and transportation methods are used when setting up shipping carriers and carrier services. No explicit relationship or hierarchy exists between modes and transportation methods.

Create a mode

To create a mode, follow these steps:

1. Go to **Transportation management** > **Setup** > **Carriers** > **Mode**.
2. Select **New** to create a new mode.
3. Enter a unique ID and a descriptive name for the mode.
4. Close the page.

Create a transportation method

To create a transportation method, follow these steps:

1. Go to **Transportation management** > **Setup** > **Carriers** > **Transportation methods**.
2. Select **New** to create a new transportation method.
3. Enter a unique ID and descriptive name for the transportation method.
4. Close the page.

Last updated on 11/19/2025

Assign shipments to related route segments

06/17/2025

This article describes how to configure the system to apportion freight costs for multi-segment scheduled routes that use the inbound and outbound logistic process. It also describes how this feature works.

Enable the feature

To configure the system to apportion freight costs for multi-segment scheduled routes, follow these steps:

1. Go to **Transportation management** > **Setup** > **Transportation management parameters**.
2. Open the **General** tab.
3. On the **Shipment** FastTab, set **Assign shipment to related route segments** to *Yes*.
4. On the Action Pane, select **Save**.

How shipments are assigned to route segments

This feature enables the system to apportion shipment freight costs more accurately, including for loads where multiple shipments are delivered to different segment destinations along a single route. It assigns each shipment to the most suitable route segment, based on the destination addresses of the shipment and segment. The feature then calculates each shipment's freight cost as a proportion of the load's total freight cost, based on the shipment's relative weight, volume, quantity, and distance traveled.

This feature enables correct freight cost apportionment for multi-segment scheduled routes that use the inbound and outbound logistic process. Therefore, you can evaluate the true profitability of sales orders by apportioning freight cost at the item line level. This feature provides a fair and trusted foundation for the billing of freight cost. It also reflects the cost of inbound freight in the material valuation of stock.

This feature applies only to shipments that are managed by using the **Transportation management** (TMS) module.

Set up transportation constraints for an item

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This procedure sets up a transportation constraint to prevent a selected item from being transported through a selected hub. A Transportation coordinator typically carries out this task.

Create an item constraint

1. Go to **Transportation management** > **Setup** > **Routing** > **Constraints**.
2. Open the **Item** tab.
3. On the toolbar, select **New**. Enter values for the following columns of the new row:

    - **Item constraint** – Enter a unique identifier for the constraint.
    - **Name** – Enter a name for the constraint.
    - **Site** – Select the site where the constraint applies.
    - **Warehouse** – Select a warehouse where the constraint applies.
    - **Item number** – Select the item number the constraint applies to.

4. With the new row still selected, make the following settings at the bottom of the page:

    - **Shipping carrier** – Select the shipping carrier the constraint applies to.
    - **Carrier service** – Select the carrier service the constraint applies to.
    - **Mode** – Select the mode of delivery where the constraint applies.
    - **Transportation method** – Select the transportation method where the constraint applies.
    - **Hub** – Select the hub where the constraint applies.
    - **Constraint action** – Select the type of constraint to apply.
    - **Effective start date** – Select the first date where the constraint applies.
    - **Effective end date** – Select the last date where the constraint applies.

5. Select **Save**.
6. Close the page.

Last updated on 11/19/2025

Plan freight transportation routes with multiple stops

Article • 01/31/2025

This article describes the various elements that you use to plan transportation routes in Dynamics 365 Supply Chain Management.

You can use route plans and route guides for complex transportation routes that have multiple stops. If you regularly use the same route, you can set up a scheduled route.

Route plans

A route plan contains route segments that provide information about the stops that are visited on the route and the carriers that are used for each segment. You must define the stops on the route as hubs. A hub can be a vendor, a warehouse, a customer, or even just a reloading place where you change carrier. For each segment, you can define "spot rates" for various charges. Here are some examples:

- Charges for traveling to the given segments
- Charges for picking up the goods
- Charges for dropping off the goods

Each route plan must be associated with a route guide.

Route guides

A route guide defines the criteria for matching a load to a specific route plan. For example, you can specify an origin hub and a destination hub, limits for the container volume or weight, and a shipping carrier, service, or group. Route guides are available on the **Rate route workbench** page, where loads can be matched to routes either manually or automatically. If the route guide is for a scheduled route, it's also available on the **Load building workbench** page.

Scheduled routes

A scheduled route is a predefined route plan that has a schedule for the shipping dates. Scheduled routes and nonscheduled routes differ in the way that loads are assigned to them. If you assign a nonscheduled route by using the **Rate route workbench**, only the load and the route guide are validated. If you assign a scheduled route, the dates and

addresses from the orders and the hubs, and the date on the route plan, are also considered. You don't have to use the **Rate route workbench** page to manually assign loads to a scheduled route. Instead, you can use the Load building workbench to suggest that loads be built based on the customer addresses and delivery dates from sales orders for a given scheduled route. For scheduled routes, the route plan has fixed origin and destination hubs. Typically, the shipping carrier and service are the same for all segments, but they can differ. The destination hubs are created by using the postal codes of the customers that are visited on the route. Several route schedules can be defined for one route plan. The route plan must be associated with a route guide. However, for scheduled routes, the plan can be associated with only one route guide. The route schedule is used only to create the actual routes on the **Route schedule** page. You can use the default load template when you propose loads on the Load building workbench.

Load building workbench

The Load building workbench uses the customer addresses and delivery dates from sales orders, and the scheduled routes that are available, to propose a load. By default, the values from the route are entered on the workbench. However, you can select a "from" date that is earlier than the "from" date on the route. When a load is proposed, the delivery address and delivery date of all open sales orders are checked. A sales order is proposed for a load if the postal code of the delivery address matches the postal code of a hub in the route plan, and if the delivery date is within the range that is selected in the criteria. The capacity of the load template is also considered. Only one load is proposed at a time. If you have a sales order that isn't included, you might have to use a different load template (for example, a load template for a bigger truck or container) or plan an extra delivery.

Sustainable transportation planning

Supply Chain Management offers sustainable transportation planning by integrating with Microsoft Sustainability Manager for the calculation of carbon emissions.

Sustainable transportation planning can help companies reduce their carbon footprint by optimizing routes, reducing fuel consumption, minimizing empty miles, and increasing the use of lower-emission transportation modes. Companies can achieve better visibility, control, and automation for their transportation operations. By reducing fuel consumption and carbon emissions, companies can minimize their environmental impact and play a part in combatting climate change. In addition, by demonstrating a

commitment to sustainability, companies can improve their brand image and attract environmentally conscious customers.

- **System integration** – The integration between Supply Chain Management and Microsoft Sustainability Manager uses REST APIs. Therefore, data can flow seamlessly between the two systems, and organizations can track sustainability metrics alongside their other business processes.
- **Carbon emission calculation** – Microsoft Sustainability Manager uses the route data from Supply Chain Management to offer various data models and services that can help organizations calculate and analyze transportation-related carbon emissions. The route data can be used to calculate emissions from different modes of transportation, including cars, trucks, planes, and ships.
- **Route planning with carbon dioxide equivalent (CO2e) results** – After Microsoft Sustainability Manager returns the calculated emissions for each mode of transportation, you can compare them to determine which mode emits the least carbon dioxide. You can select the most eco-friendly route, based on the emissions calculation, or select the mode of transportation that has the lowest carbon emissions in the transportation management rate and route workbench.

Learn more in Integrate with Microsoft Sustainability Manager.

Feedback

Was this page helpful? Yes No

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Load building workbench

The load building workbench lets you apply load building strategies when you create loads.

Create a load building strategy

Use load building strategies to automatically build loads. This capability is beneficial in the following situations:

- If you regularly ship a specific set of products, load strategies help save time because you don't have to build the same load every time.
- If you want to avoid half-full loads to maximize efficiency, load strategies can help fill each load as much as possible.

A load building strategy class named `TMSLoadBuildingVolumeStrategy` provides a load building strategy named *Volume-based load building strategy*. This strategy lets you use the maximum values that are specified for height and weight in the load template, or you can override the settings by entering new values. This strategy is the only strategy included out of the box. (However, you might have some custom strategies.)

To use the out-of-box *Volume-based load building strategy* strategy, select it in the **Load building strategy** field on the **Load building workbench** page (**Transportation management** > **Planning** > **Load building workbench**).

To create a load building strategy, follow these steps:

1. Go to **Transportation management** > **Setup** > **Load building** > **Load building strategies**.
2. On the Action Pane, select **Generate class list** to make sure that you have the latest versions of all available classes.
3. On the Action Pane, select **New**.
4. Enter a unique name for the strategy, select the load building strategy class for it, and enter a description.
5. On the Action Pane, select **Save**.
6. On the Action Pane, select **Parameters**.
7. On the **Load building strategy parameters** page, select **Volume capacity** in the list, and then, in the **Value** field, enter the percentage of the load's total volume capacity that should be applied for the new load building strategy.

8. Select **Weight capacity** in the list, and then, in the **Value** field, enter the percentage of the load's total weight capacity that should be applied for the new load building strategy.
9. Close the **Load building strategy parameters** page.
10. Close the **Load building strategies** page.

You can now assign the load building strategy to a load building template. Alternatively, you can use it directly in the load building workbench.

Use a load building strategy in the load building workbench

1. Go to **Transportation management** > **Planning** > **Load building workbench**.
2. Follow one of these steps:
    - Select a strategy in the **Load building strategy** field.
    - If you define a load building template and assign the load building strategy to it, on the Action Pane, on the **Manage templates** tab, select **Apply template**. Then, in the **Apply load building template** drop-down dialog box, select a template in the **Load building template name** field.
3. On the **Load templates sequence** FastTab, select one or more load templates. The workbench tries to fit the load into these types of containers, in the sequence that you specify. Typically, put the smallest containers at the top of the list to ensure that the smallest possible container is selected first.
4. On the Action Pane, select **Propose loads**.
5. Review the proposed loads and proposed load lines.
6. On the Action Pane, select **Create loads** to create loads that are based on the source document lines on the **Proposed load lines** FastTab.
7. Close the **Load building workbench** page.

Last updated on 12/30/2025

Integrate with Microsoft Sustainability Manager

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to integrate Microsoft Dynamics 365 Supply Chain Management with Microsoft Sustainability Manager. The integration lets you accurately calculate carbon emissions during rate and route planning in transportation management. Therefore, transportation planners can make informed decisions and strategically assign environmentally friendly transportation service providers to each load.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management version 10.0.38 or later.
- You must have a valid subscription for Microsoft Sustainability Manager.
- You must be running Microsoft Sustainability Manager version 2.14.0.355 or later.
- The feature that's named *Integrate Microsoft Sustainability Manager with transportation management* must be turned on in Feature management. As of Supply Chain Management version 10.0.43, this feature is turned on by default. As of version 10.0.45, it's mandatory and can't be turned off.

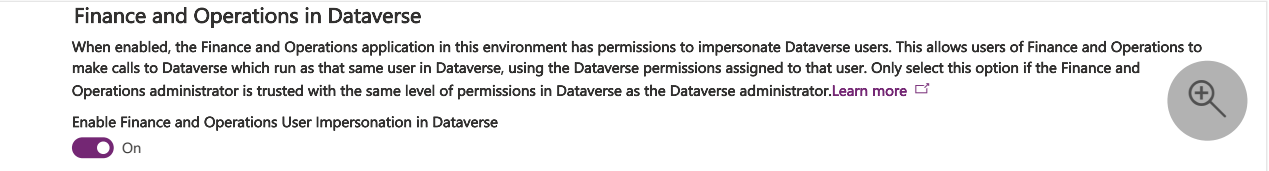
Enable impersonation in Microsoft Power Platform

To enable the two systems to communicate with each other, you must enable impersonation for Microsoft finance and operations apps in Microsoft Power Platform.

1. Sign in to Power Platform admin center.
2. On the left navigation pane, go to **Environment**, and then open the environment where you're running Microsoft Sustainability Manager.
3. On the toolbar, select **Settings**.

4. Expand **Product**, and select **Features**.

5. Under **Finance and Operations in Dataverse**, set the **Enable Finance and Operations User Impersonation in Dataverse** option to *On*.



6. Save your changes.

Map transportation modes in Supply Chain Management

To enable carbon emission calculations, you must map the transportation methods in Supply Chain Management to the transportation modes (vehicle types) in Microsoft Sustainability Manager.

1. Sign in to Supply Chain Management.

2. Go to **Transportation management** > **Setup** > **Carriers** > **Transportation methods**.

3. For each transportation method that's listed on the page, use the **Transportation Mode in Microsoft Sustainability Manager** field to identify the matching transportation mode in Microsoft Sustainability Manager.

   > **Note**
   >
   > The transportation modes that are listed in the **Transportation Mode in Microsoft Sustainability Manager** field are taken from United States Environment Protection Agency Greenhouse Gas Emissions Factors Hub (EPA GHG Emissions Factors Hub). Microsoft Sustainability Manager uses these modes in its transportation and distribution calculation model. Learn more in GHG Emissions Factors Hub.

4. On the Action Pane, select **Save**.

Validate the connection and enable the integration scenario

To complete the setup, you must validate the connection and enable the integration scenario.

1. In Supply Chain Management, go to **Transportation management** > **Setup** > **Transportation management parameters**.
2. On the **Microsoft Cloud for Sustainability integration configuration** tab, select **Validate connection** to check whether your Microsoft Power Platform environment is correctly set up and ready for the integration.
3. When the validation is completed, in the **Model name** field, select the model that best matches the transportation scenario that you want to calculate carbon emissions for:
    - *None* – This value indicates that you haven't yet selected a model. When this value is selected, emission calculations are disabled.
    - *Downstream transportation and distribution* – This model includes emissions that are generated by the transportation and distribution of sold products in vehicles that the reporting organization doesn't own or control.
    - *Upstream transportation and distribution* – This model includes emissions that are generated by third-party transportation and distribution services and emissions that are generated by the transportation and distribution of purchased products.

    > **Note**
    >
    > For information about how Microsoft Sustainability Manager calculates Scope 3 emissions, see Categories 4 and 9: Upstream and downstream transportation and distribution.

4. Set the **Enable McFS integration** option to *Yes*.
5. Select **Save**.

Last updated on 07/29/2025

Plan loads using hub consolidation overview

Article • 01/31/2025

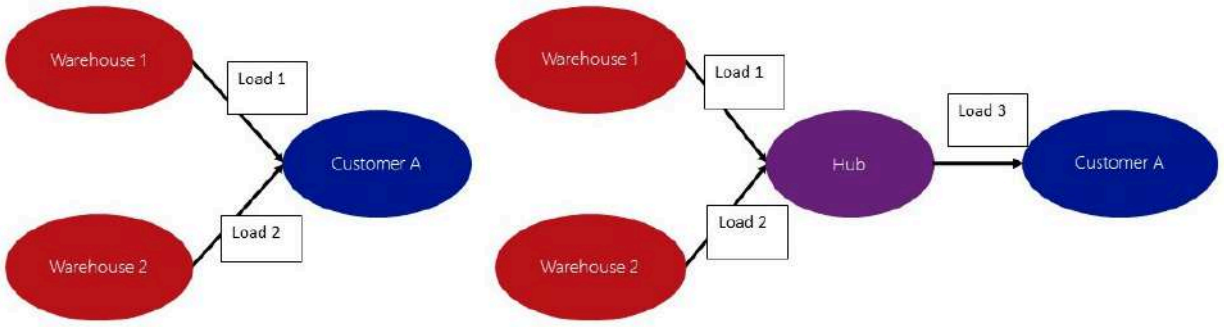
This article describes the feature for consolidating shipments in a hub when you deliver goods from different warehouses to the same customer, or when you receive goods from multiple vendors in the same warehouse.

It can be useful to consolidate shipments in a hub when you deliver goods from different warehouses to the same customer, or when goods are delivered from multiple vendors to the same warehouse.

Building loads

Before you can use hub consolidation, you must enable the **In transit planning** option on the **Transportation management parameters** page. You must also create the hubs where consolidation will occur.

The following diagram shows an example of hub consolidation. In this case, sales orders from different warehouses are going to the same customer. The basic loads are created based on sales orders in the usual way, by using the **Outbound load planning workbench** page.



To consolidate the two loads in a hub before they're delivered to the customer, on the **Outbound load planning workbench** page, in the **Transportation** field, select **Hub consolidation**. When you select the correct hub for each load, the loads will have the hub as the *drop off* destination. You'll also have two *transportation request lines* in the **Supply and Demand** section on the **Outbound load planning workbench** page. You can then add these two lines to a new load. This new load will have both sales order lines, and will also have the hub as the *pick up* address and customer A as the *drop off*

destination. The three loads are then ready to be rated and routed like any other load. You can select whatever shipping carrier the system suggests for each load.

You can use a similar method to consolidate loads for multiple transfer orders. In this case, customer A in the preceding diagram is a warehouse.

Alternatively, you can consolidate loads for multiple purchase orders, where the loads are delivered from different vendors to the same warehouse. In this case, you would work on the **Inbound load planning workbench** page instead of the **Outbound load planning workbench** page.

You can have more than one consolidation hub, and can consolidate in multiple hubs for more loads that come from different warehouses. After you build your basic loads and use the hub consolidation option, you build the new loads by using the consolidated transportation request lines. You then rate and route your loads.

Feedback

Was this page helpful? Yes No

Provide product feedback

Plan loads and shipments by using the outbound load planning workbench

This article shows how to use the outbound load planning workbench to create a load for a sales order. As a prerequisite, create the sales order first. This procedure is part of the daily work for the transportation coordinator. The demo data company used to create this procedure is USMF.

Create a sales order

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Select **New**.
3. In the **Customer account** field, select the drop-down button to open the lookup.
4. Select account *US-004*.
5. Select **OK**.
6. In the **Item number** field, select the drop-down button to open the lookup.
7. Select item *A0001*. *A0001* is enabled for transportation management.
8. In the **Site** field, select the drop-down button to open the lookup, and then select an item.
9. In the **Quantity** field, enter a number.
10. In the **Warehouse** field, type *24* for this example. This warehouse is enabled for transportation management and warehouse management processes (WMS).
11. Select **Save**.
12. Close the page.

Create a new load

1. Go to **Transportation management** > **Planning** > **Outbound load planning workbench**.
2. Select the **Sales lines** tab. Now you build the load for the sales order that you just created. Loads can be built based on supply and demand from purchase orders, transfer orders, and sales orders.
3. On the Action Pane, select **Supply and demand**.
4. Select **To new load**.
5. In the **Load template ID** field, select the drop-down button to open the lookup. The load template defines maximum measurements for weight and volume of the entire load. For example, the load template might represent the size of a container or truck. Select an item.
6. Select **OK**.

Rate and route the load

1. Select **Rating and routing**.
2. Select **Rate route workbench**.
3. Select **Rate shop**.
4. In the list, find and select the desired record.
5. Select **Assign**.
6. Close the page.

Load planning filters

Set up custom filters to control which types of lines show in the **Outbound load planning workbench**. For example, you can choose to only show fully reserved transfer order lines. Follow these steps to set up one or more load planning workbench filters.

1. Go to **Warehouse management** > **Loads** > **Outbound load planning workbench**. (You can also open the workbench from the **Sales orders**, **Transfer orders**, and **Outbound shipment orders** pages.)

2. On the Action Pane, open the **Filters** tab and select **Load planning filters**.

3. On the Action Pane, select **New** to add a new load planning filter to the grid. Then make the following settings for the new line:

    - **Load planning filter code** – Enter a unique name for the filter.
    - **Description** – Enter a short description of the filter.
    - **Load planning filter type** – Choose the type of lines the filter should apply to (*Load*, *Sales order*, *Transfer order*, *Shipment*, or *Outbound shipment orders*). For example, if you want to set up a filter that only shows fully reserved transfer order lines, select *Transfer order*.

4. On the Action Pane, select **Save**.

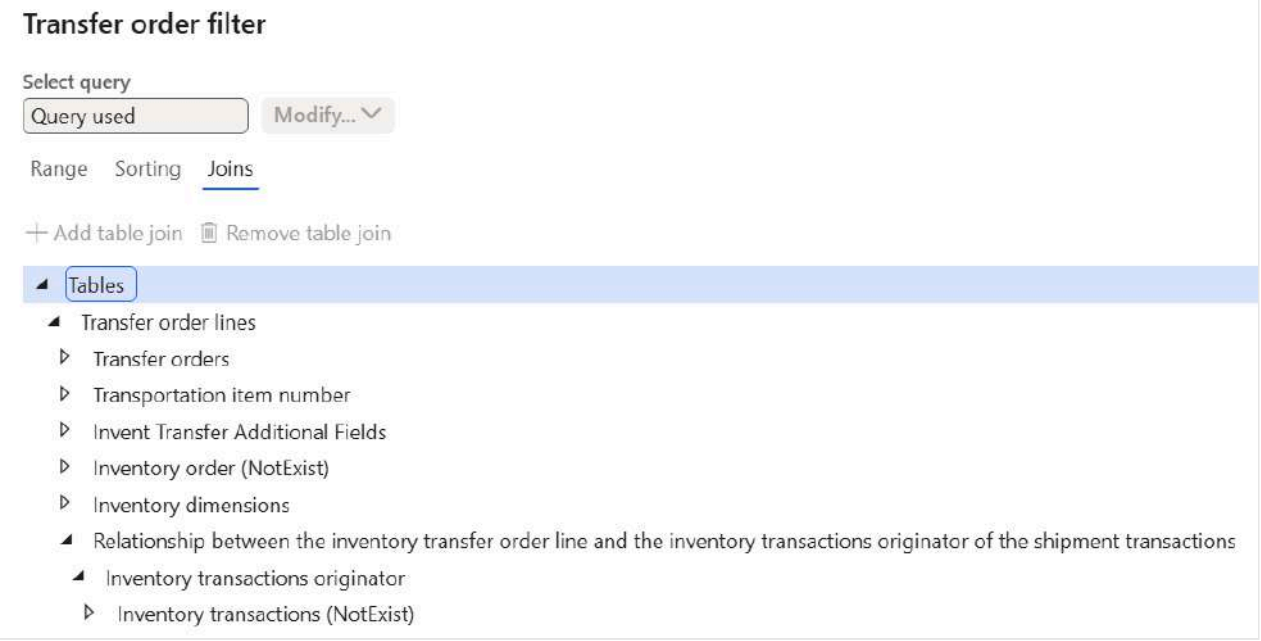
5. With the new filter selected in the grid, select **Edit query** on the Action Pane. A standard query editor opens, where you can define the filter criteria that you want to apply. Here's an example of how to set up a filter to show only fully reserved transfer order lines:

    a. Open the **Joins** tab. Select the *Transfer order lines* node and then select **Add table join** from the toolbar. Find and select the row with a **Relation** of *Relationship between the*

inventory transfer order line and the inventory transactions originator of the shipment transactions (Line number) (with **Join mode** *1:n*). Then select **Select** on the toolbar.

b. Select the new *Relationship between the inventory transfer order line and the inventory transactions originator of the shipment transactions* node and then select **Add table join** from the toolbar. Find and select the row with a **Relation** of *Inventory transactions originator (Inventory transactions originator)*. Then select **Select** on the toolbar.

c. Select the new *Inventory transactions originator* node and then select **Add table join** from the toolbar. Make sure that **Show details** is set to *Yes*. Find and select the row with a **Relation** of *Inventory transactions (NotExist Record-ID)* and a **Relation source** of *InventTrans : InventTransOrigin*. Then select **Select** on the toolbar. Your **Joins** tab should now resemble the following screenshot.



d. Open the **Range** tab. Add a new row to the grid with the following settings:

- **Table** – Select *Inventory transactions (NotExist)*.
- **Field** – Select *Issue status*.
- **Criteria** – Enter *!Reserved physical*, which means "not *Reserved physical*."

These settings result in a filter that only includes transfer order lines with an inventory transactions status of *Reserved Physical*, which are the fully reserved lines. Your **Range** tab should now resemble the following screenshot.

e. Select **OK** to save the query.

6. Select the **Back** button to go back to the **Outbound load planning workbench** page.

7. Open the tab you created your filter for (for example, the **Transfer lines** tab). Then set **Supply and demand filter** to the filter you just created. The page is now filtered according to the criteria you defined for the selected filter.

8. If you want to apply the new filter by default when you open the workbench, then on the Action Pane, open the **Filters** tab and select **Set as default**.

Last updated on 07/01/2026

USMCA certification of origin

07/10/2025

This feature lets you print the certification of origin documents required by the United States-Mexico-Canada Agreement (USMCA).

The *USMCA certification of origin document* contains the minimum data elements required for declaration. Some data elements can be pre-filled before printing while others must be filled in manually after printing. To obtain preferential tariff treatment, the document must be completed and in the possession of the importer at the time the declaration is made. The document may be completed by the importer, exporter, or producer.

You can print the document for a single shipment from the **All shipments** list page or from the **Shipment details** page.

The document is only accessible when the country/region on the primary address for the legal entity is the United States.

Depending on the document print selection, the document can be pre-filled with data from your system. It is possible to change or add data to the printed document by exporting the printed document to an editable format, such as Microsoft Word. After export, you can apply any required changes before a declaration is made.

Turn the USMCA feature on or off

To use this feature, it must be turned on for your system. As of Supply Chain Management version 10.0.29, the feature is turned on by default. Admins can turn this functionality on or off by searching for the *USMCA certification of origin document* feature in the Feature management workspace.

Document content

The USMCA certification of origin document contains the following data elements:

- Address elements
- Role of the certifying party
- Single shipment
- Invoices
- Blanket period
- Item details
- Certifier signature

- Number of pages

For more information about each of these elements and how their values are found, see the remaining sections in this article.

Print a USMCA certification of origin document

To print a USMCA certification of origin document for a shipment, do the following:

1. Do one of the following:

    - Go to **Transportation management** > **Planning** > **Shipments** > **All shipments** and select the shipment you want to print the document for.
    - Open the **Shipment details** page for the shipment you want to print the document for (there are several ways to get here, including from the **All shipments** page).

2. On the Action Pane, open the **Shipments** tab and, from the **Print** group, select **USMCA certificate of origin**.
3. The **Certificate or origin** dialog box opens. Make the settings described in the following subsections and then select **OK** to generate the document.
4. A preview of the document opens. Use the commands provided on the Action Pane to print or export the document as needed.

Certifying party

Use the **Certifying party** drop-down list to identify the type of party that is printing the document. Specify whether the certifying party is the *Exporter*, *Exporter and Producer*, *Producer*, or *Importer*; or leave it blank if the certifying party is none of these. The option you select determines what is printed in the address sections of the document.

The **Certifying party** that you choose will be included in the printed document.

The document can be printed for both inbound and outbound shipments. Select *Importer* as **Certifying party** for inbound shipments only.

The following table describes the types of information that are included in the document based on the **Certifying party** that you choose.

Expand table

Certifying party	Description
[Blank]	Adds the following details to the document:

Certifying party	Description
	• **Certifier details**: Uses the address details for the shipping warehouse, if available; otherwise, it uses the shipping site address, if available; otherwise, it uses the address of the legal entity (company) selected in Supply Chain Management. • **Exporter details**: Blank • **Producer details**: Blank • **Importer details**: Blank
Exporter	Adds the following details to the document: • **Certifier details**: Uses the address details for the shipping warehouse, if available; otherwise, it uses the shipping site address, if available; otherwise, it uses the address of the legal entity (company) selected in Supply Chain Management. • **Exporter details**: Uses the address details for the legal entity. • **Producer details**: Blank • **Importer details**: Uses the invoice account for the related sales order.
Exporter and Producer	Adds the following details to the document: • **Certifier details**: Uses the address details for the shipping warehouse, if available; otherwise, it uses the shipping site address, if available; otherwise, it uses the address of the legal entity (company) selected in Supply Chain Management. • **Exporter details**: Uses the address details for the legal entity. • **Producer details**: Uses the address details for the legal entity. • **Importer details**: Uses the invoice account for the related sales order.
Importer	Adds the following details to the document: • **Certifier details**: Uses the address details for the shipping warehouse, if available; otherwise, it uses the shipping site address, if available; otherwise, it uses the address of the legal entity (company) selected in Supply Chain Management. • **Exporter details**: Blank • **Producer details**: Blank • **Importer details**: Uses the address details for the legal entity.
Producer	Adds the following details to the document: • **Certifier details**: Uses the address details for the shipping warehouse, if available; otherwise, it uses the shipping site address, if available; otherwise, it uses the address of the legal entity selected in Supply Chain Management. • **Exporter details**: Blank • **Producer details**: Uses the address details for the legal entity. • **Importer details**: Blank

Has various producers

The **Certifying party** drop-down list controls the text to be used for the producer details in the document. Choose one of the following:

- *Various producers* - Prints the text "Various" in the producer details.
- *Available upon request* - Prints the text "Available upon request by the importing authorities" in the producer details.

When the **Certifying party** is set to *Exporter and Producer* or *Producer*, then the **Has various producers** setting is overruled, and the producer address details will be the same as the certifier.

Blanket period

Use the **Blanket period from** and **Blanket period to** settings to establish a blanket period, during which the document will cover multiple shipments of identical goods, even though the document is printed for only one shipment. You can set the blanket period dates without any constraints, and it will be added to the document. You can also leave these settings blank or set them in the past.

Is single shipment

In the **Certificate of origin** dialog box, set **Is single shipment** to one of the following:

- *Yes* - Adds "Single Shipment: Yes" next to the invoice number.
- *No* - Adds nothing.

Other information included in the document

In addition to the optional elements that you select using the **Certificate or origin** dialog box, the USMCA certification of origin document will include the information and custom fields summarized in the following subsections. Some of this information must be entered manually after you generate the document.

Invoice number

The IDs of sales invoices related to shipments are printed on the document irrespective of the blanket period. Invoice numbers are printed irrespective of the **Is single shipment** selection.

Item details

The document provides several sections that list specific item details, which are:

- **SKU number**: Prints the item number of the released product.
- **Description**: Prints either the description or name for the released product. If a description in the user's language exists, then this is printed. If no such description exists, then the name in the user's language is printed. If that name doesn't exist, then the item name is printed.
- **Harmonized System (HS) Tariff Classification**: Prints the Harmonized Tariff Schedule associated to the product. You can set up these schedules by going to **Transportation Management** > **Setup** > **Transportation standard** > **Harmonized Tariff Schedules**.
- **Origin criterion**: You must manually enter data in this section the first time you release the document.
- **Country of origin**: Prints the country of origin, which you apply by going to **Product information management** > **Setup** > **Product compliance** > **Country of origin**. (Learn more in Country of origin.) The International Organization for Standardization (ISO) code for the country of origin is printed, based on the country/region of destination in the shipment delivery address and the item. If no country of origin data is set up, the value reverts to the setting that is found at **Released product** > **Foreign trade** > **Origin**. If no country of origin data is found there, you must manually enter the country of origin after you generate the document.

Certifier signature and date

You must enter this manually after generating the document.

Consists of number of pages

The user signing the certification must manually enter the number of pages (for verification) after generating the document.

Page numbers

Current page and number of pages printed at the bottom of the document.

Commercial invoices

This article describes how to create and print a commercial invoice when you use transportation management processes (TMS).

A commercial invoice is required for the export and import clearance process. It's used for customs, the calculation of tariffs, and the declaration of international commercial terms. Commercial invoices aren't usually required for shipments between European Union (EU) countries/regions, but only between EU countries/regions and non-EU countries/regions. A few countries/regions require that the commercial invoice be on a specific form. However, for most countries/regions, either the seller version or the exporter version is acceptable, provided that all the pertinent information is included.

You can print the commercial invoice for a single shipment from the **All shipments** list page or the **Shipment details** page.

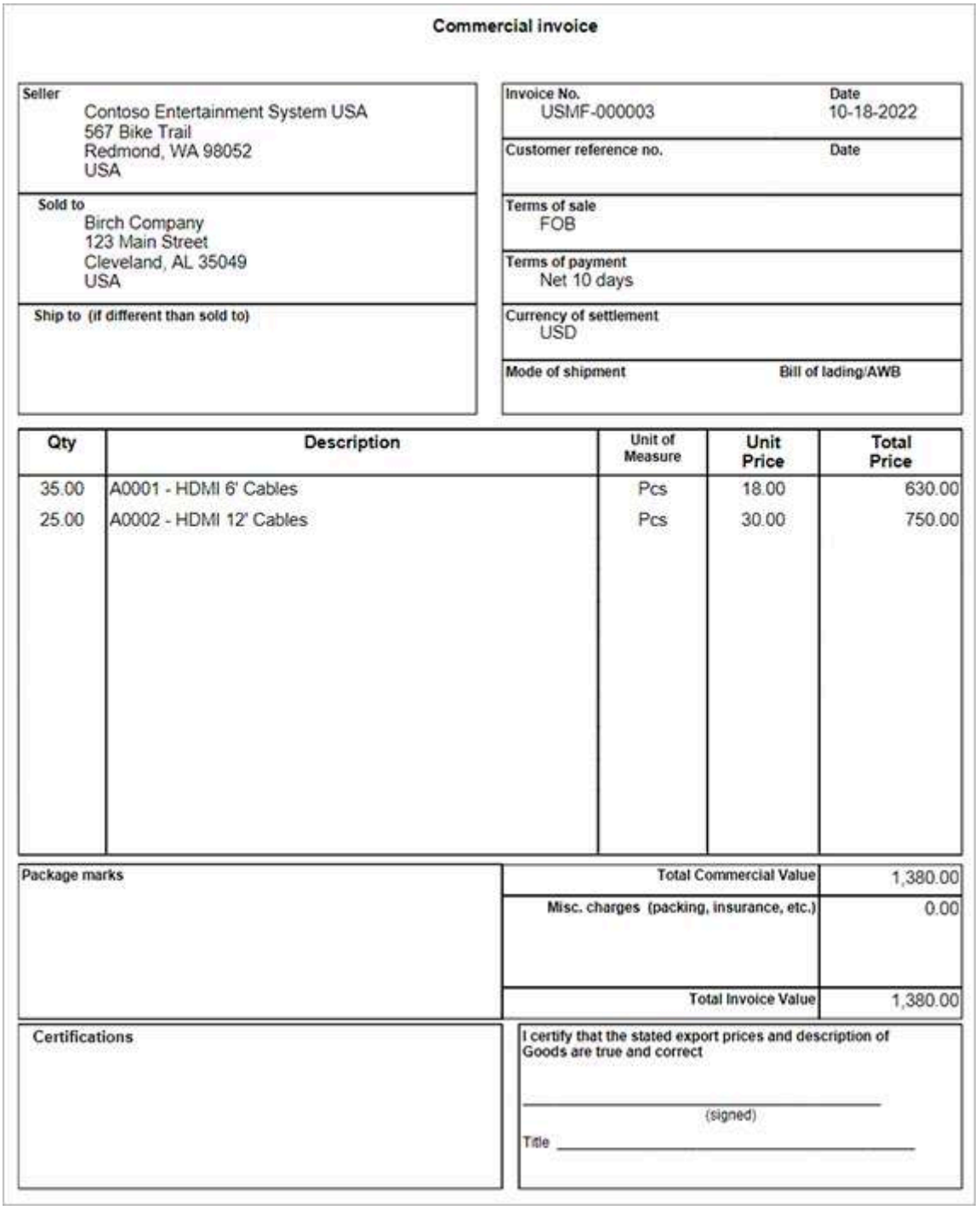
Document content

The commercial invoice includes the following information, which is prefilled from the shipment that the invoice applies to:

- Invoice number and date
- Seller
- Sold-to address
- Ship-to address
- International commercial terms (Incoterms) (listed as **Terms of sale**)
- Payment terms
- Currency of settlement
- Mode of shipment
- Product details (including quantity, description, unit of measure, unit price, and total price)
- Total commercial value
- Miscellaneous charges
- Total invoice value

If you must change or add data to the commercial invoice, export the generated document to an editable format, such as Microsoft Word. You can then apply any required changes before a declaration is made.

The following illustration shows an example of a commercial invoice.



Commercial invoice

Seller Contoso Entertainment System USA 567 Bike Trail Redmond, WA 98052 USA	**Invoice No.** USMF-000003 **Date** 10-18-2022
	Customer reference no. **Date**
Sold to Birch Company 123 Main Street Cleveland, AL 35049 USA	**Terms of sale** FOB
	Terms of payment Net 10 days
Ship to (if different than sold to)	**Currency of settlement** USD
	Mode of shipment **Bill of lading/AWB**

Qty	Description	Unit of Measure	Unit Price	Total Price
35.00	A0001 - HDMI 6' Cables	Pcs	18.00	630.00
25.00	A0002 - HDMI 12' Cables	Pcs	30.00	750.00

Package marks	**Total Commercial Value**	1,380.00
	Misc. charges (packing, insurance, etc.)	0.00
	Total Invoice Value	1,380.00

Certifications	I certify that the stated export prices and description of Goods are true and correct _____ (signed) Title _____

Print a commercial invoice

To print a commercial invoice for a shipment, follow these steps:

1. Follow one of these steps to identify the shipment that you want to print a commercial invoice for:

    - Go to **Transportation management** > **Planning** > **Shipments** > **All shipments**, and select the shipment that you want to print the commercial invoice for.

- Open the **Shipment details** page for the shipment that you want to print the commercial invoice for. There are several ways to open this page (for example, from the **All shipments** page).

2. On the Action Pane, on the **Shipments** tab, in the **Print** group, select **Commercial invoice**.
3. A preview of the document is shown. Use the commands on the Action Pane to print or export the document as you require.

Last updated on 12/30/2025

Create a bill of lading

08/02/2025

This article describes how to create a bill of lading when using warehouse management processes (WMS).

A bill of lading is a legal document between the company that ships the items and the carrier. The document accompanies the shipped items, and it serves as a receipt of shipment when the items are delivered at the destination. If you're using warehouse management, there are two ways to generate a bill of lading:

- Create the report manually, using the **Bill of lading** page.
- Generate the report from the **Outbound load planning workbench** or **Inbound load planning workbench**, depending on whether the load is inbound or outbound.

If you generate the bill of lading from the **Outbound load planning workbench** or **Inbound load planning workbench**, the load status must be *Shipped*. If there's more than one shipment in the load, a bill of lading is created for each shipment. After a bill of lading has been created, you can make changes to it on the **Bill of lading** page.

Master bill of lading

If there's more than one shipment in the load, you can generate a master bill of lading. This has the same layout and information as a bill of lading, but contains the summarized content for all the shipments. If the **Create a master bill of lading when there's more than one shipment on a load** option is set to *Yes* on the **Transportation management parameters** page, a master bill of lading is automatically generated if you create a bill of lading from the **Outbound load planning workbench** or **Inbound load planning workbench**, and there's more than one shipment. You can also get a list of the bills of lading by selecting **Related information** > **Bill of lading**. If you're creating bills of lading manually, you can create a master bill of lading on the **Bill of lading** page.

Include container weight and volume on load

06/17/2025

The functionality for including the container weight and volume on a load provides a clear representation of the total weight and volume of containers and items in a load.

A load contains one or more shipments, and these shipments contain distinct items that belong to one or more sales orders. The items are stored inside a container, and containers are loaded on a load. Items outside a container can also be part of a load. Based on these conditions, the system calculates values for the weight and volume on the load by considering the weight and volume of both containers and items.

Where it's used

The functionality for including the container weight and volume on a load applies in transportation management processes. For example, the values are used on the **Rate route workbench** page, where they help define the rate and route for loads. They're also used for transportation tenders and driver check-in.

When creating a load on the **Rate route workbench** page, you can also see and track capacity utilization against the maximum weight and volume of the chosen load template. On the **Load template assignment** dialog that opens when creating a load, the **Details**, **Weight**, and **Volume** tabs show the total quantity, weight, and volume of the load respectively. The **Weight** and **Volume** tabs also show how close you're to the maximum defined weight and volume for the selected load template, and the system notifies you if you exceed either the maximum allowable weight or volume.

If the calculated values aren't precise, you can adjust them by entering the actual values for the weight and volume on the load.

How it's set up

The number of containers considered for a load is calculated based on the weight, volume, and percentage of the container used.

- To set the weight and volume for a container, go to **Warehouse management** > **Setup** > **Containers** > **Container types**.
- To set the container utilization percentage, go to **Warehouse management** > **Setup** > **Containers** > **Container groups**. Then enter a value in the **Container utilization**

percentage field.

To learn more about how to set up the dimensions of a load template, go to Load templates.

The units used for weight and volume are based on the configured system units for the specific legal entity.

Transportation management discounts

Transportation management lets you set up discounts to be applied on top of the transportation price calculated by the rating engine. You can specify each discount either as a percentage of charges or as a fixed amount.

To set up discounts for transportation management, go to **Transportation management** > **Setup** > **Rating** > **Discounts**.

You can choose whether to pass the discount on to your customer (by setting the discount type to customer) or to retain it in the company (by setting the discount type to shipper). This lets you define a setup in which you earn a margin on the calculated freight charges.

You can set up multiple qualifiers for each discount, such as carrier, carrier service, delivery terms, customer, destination, and more.

Last updated on 07/01/2026

Less than truckload (LTL) classes

A less than truckload (LTL) class is a freight shipping class that classifies items for shipment. Generally, every type of product or commodity has a National Motor Freight Classification (NMFC) code that corresponds to a specific freight class number for LTL shipments. LTL freight classes represent categories of items, whereas NMFC codes relate to specific commodities in each of the 18 freight classes.

This feature lets you use your system to perform the following tasks:

- Define the LTL freight classes that your company uses.
- Assign each LTL class to an NMFC code on the **NMFC Codes** page. Learn more in Work with NMFC codes in Dynamics 365 Supply Chain Management.
- Assign an NMFC code (and therefore its associated LTL code) to each product on the **Products** page.
- Accurately assess the LTL class of each product that you assign an NMFC code to.
- Determine packing requirements for each LTL class by checking the international LTL standards. In this way, you ensure that your products are well protected and safely shipped.
- Get accurate shipping estimates, based on the LTL freight class for each product.

This article describes how to create LTL classes in Microsoft Dynamics 365 Supply Chain Management.

Create an LTL class

To create an LTL class, follow these steps:

1. Follow one of these steps:

    - Go to **Warehouse management** > **Setup** > **Inventory** > **LTL classes**.
    - Go to **Transportation management** > **Setup** > **Transportation standards** > **LTL classes**.

2. On the Action Pane, select **New** to create an LTL class. Then set the following fields:

    - **LTL class** – Enter your company's internal LTL class identifier (ID) for the commodity type. Most companies use the international standard. In that case, the value of this field matches the value of the **Class** field. However, if your company uses its own standard, enter a code that conforms to that standard.

- **Name** – Enter a descriptive name that helps you and other users select the correct LTL class for each NMFC code.
- **Class** – Enter the international standard LTL class ID for the commodity type. The code that you enter here must conform to the official standard.

Example: Set up LTL classes

The following example shows how to set up two different LTL classes that you can use with different types of products.

1. Go to **Warehouse management** > **Setup** > **Inventory** > **LTL classes** or **Transportation management** > **Setup** > **Transportation standards** > **LTL classes**.
2. On the Action Pane, select **New**.
3. On the new line, set the following values:
    - **LTL class:** *92.5*
    - **Name:** *Computers, monitors, refrigerators*
    - **Class:** *92.5*
4. On the Action Pane, select **Save**.
5. On the Action Pane, select **New**.
6. On the new line, set the following values:
    - **LTL class:** *125*
    - **Name:** *Small household appliances*
    - **Class:** *125*
7. On the Action Pane, select **Save**.

Related information

- Work with NMFC codes in Dynamics 365 Supply Chain Management
- Create a bill of lading

Last updated on 12/30/2025

Work with NMFC codes in Dynamics 365 Supply Chain Management

National Motor Freight Classification (NMFC) codes help you classify items that can be shipped. The NMFC code is a designation that is used to group commodities. It enables transport companies to evaluate goods for shipment by classifying items based on considerations such as truck fit, loading issues, handling issues, and perishability. Commodities are grouped into one of 18 freight classes by using a range of numbers from 50 to 500. The class that a commodity is grouped into is based on an evaluation of four transportation characteristics: density, stowability, handling, and liability. Together, these characteristics establish the commodity's transportability.

An NMFC code is associated with every less than truckload (LTL) shipping item. For example, a laptop might be assigned an NMFC that is classed at 2.5, whereas electrical cords might be assigned an NMFC that is classed at 65.

This feature can help workers use NMFC codes to classify LTL shipping items. Here are some examples:

- If your company includes a freight description on the bill of lading (BOL), the carrier will have some idea what the freight is. Freight is an important classification because many transportation companies base their whole business model on the types of freight that they ship.
- This classification might be essential to your company because it's used to determine the cost of a given load.
- Your company can identify the profitability of an LTL logistics and transportation company.

This article describes how to work with NMFC codes in Microsoft Dynamics 365 Supply Chain Management.

Prerequisites

Before you can create NMFC codes, you must set up all the LTL freight classes that must be mapped to them. LTL freight classes represent categories of items, whereas NMFC codes are related to specific commodities in each of the 18 freight classes. For more information about how to work with LTL classes, see Less than truckload (LTL) classes.

Create an NMFC code

To create an NMFC code, follow these steps:

1. Follow one of these steps:

    - Go to **Warehouse management** > **Setup** > **Inventory** > **NMFC codes**.
    - Go to **Transportation management** > **Setup** > **Transportation standards** > **NMFC codes**.

2. Select **New** to create an NMFC code. Then set the following fields:

    - **NMFC code** – Enter the NMFC code for the commodity type.
    - **Name** – Enter a name for the NMFC code.
    - **LTL class** – Select the LTL class that is associated with the NMFC code.
    - **BOL handling unit** – Define the default handling type for the shipment.

Example: Set up NMFC codes

The following example shows how to set up two different NMFC codes that can be used with different products.

1. Follow one of these steps:

    - Go to **Warehouse management** > **Setup** > **Inventory** > **NMFC codes**.
    - Go to **Transportation management** > **Setup** > **Transportation standards** > **NMFC codes**.

2. On the Action Pane, select **New**.

3. On the new line, set the following values:

    - **NMFC code:** *92.5*
    - **Name:** *Computers*
    - **LTL class:** *92.5*
    - **BOL handling unit:** *Unit*

4. On the Action Pane, select **Save**.

5. On the Action Pane, select **New**.

6. On the new line, set the following values:

    - **NMFC code:** *125*
    - **Name:** *Phones*

- **LTL class:** *125*
- **BOL handling unit:** *Unit*

7. On the Action Pane, select **Save**.

Related information

- Less than truckload (LTL) classes
- Create a bill of landing

Last updated on 12/30/2025

Set up an appointment for a load

Article • 02/17/2025

This procedure shows how to set up and plan a dock appointment for an inbound or an outbound load. The appointment reflects a date and time agreed with the carrier for picking up or dropping off a load based on a sales, purchase, or transfer order. Scheduling an appointment can be done after the load has been created and is typically done by a transportation coordinator.

This feature enables transportation and warehouse managers to keep track of appointments and plan ahead to make sure that there's enough capacity to handle upcoming arrival of trucks for loading and unloading.

1. Go to one of the following pages, depending on whether you're setting up an appointment for an inbound or outbound load:

    - **Transportation management** > **Planning** > **Inbound load planning workbench**.
    - **Transportation management** > **Planning** > **Outbound load planning workbench**.

2. Select a load you want to create an appointment for. If the load doesn't appear in the list, clear the **Hide shipped and received** check box. This is especially relevant for creating an appointment for an inbound load that has already been shipped from the vendor's site.
3. From the **Loads** toolbar, select **Transportation** > **Appointment scheduling**.
4. On the Action Pane, select **New**.
5. In the **Appointment rule** field, enter or select a value. This represents the physical loading or unloading dock in your warehouse that you want to schedule the appointment for. Configure your appointment rules by going to **Transport management** > **Setup** > **Appointment scheduling** > **Appointment rules**. Define parameters such as how many docks there are per warehouse and whether they can receive inbound and/or outbound loads.
6. Select a **Planned start date/time at location** value using the local time zone at the facility. This is an appointment that has been agreed upon separately with the carrier. The **Planned end date/time at location** value is automatically populated based on the value defined in the **Duration** field of the selected appointment rule. This is the time frame during which you expect the carrier to arrive at the dock door, the warehouse to complete loading or unloading, the warehouse to print any shipping documentation, and the driver to depart. The **Planned start date/time** and the **Planned end date and time** fields represent the appointment in the

system's time zone and are automatically populated based on the selected **Planned start date/time at location**.

7. Select **Save**.
8. Select **Update status**.
9. Select **Firm**.
10. Select **Save**.
11. Close the page.

After the appointment has been scheduled, the appointment status changes to *Waiting*, which indicates that the driver has yet to check in.

Feedback

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Register driver check-in and check-out for an appointment

This article describes how to register driver check-in and check-out events, which represent drivers arriving to or leaving from a facility. The events can be registered using either the web client or the Warehouse Management mobile app. To learn how to add the required menu items to the Warehouse Management mobile app, go to Set up mobile devices for warehouse work.

This feature lets you keep track of whether loads have been picked up or dropped off, and to confirm whether carriers are arriving and departing within the agreed-upon appointments. It calculates and shows whether carriers are late when checking in and/or out.

Before you start, an appointment must be set up for a load.

Select or open an appointment

To select or open an appointment in the web client, follow these steps:

1. Go to **Transportation management** > **Planning** > **Dock appointment scheduling** > **Driver check-in and check-out**.
2. Set the **Display completed and canceled appointments** check box as needed, depending on whether you'd like to see previous closed appointments in the grid.
3. Select an appointment.

Register driver check-in

You can register driver check-in using either the web client or the Warehouse Management mobile app.

To check in a driver using the web client, follow these steps:

1. Select an appointment, as described previously.
2. On the Action Pane, select **Driver check-in**.
3. Enter the **Actual start date/time at location** to reflect the actual time the driver checked in.
4. Fill out the fields on the **The driver's information** dialog as needed.
5. Select **OK**.

To check in a driver using the Warehouse Management mobile app, follow these steps:

1. Select **Driver check-in**.
2. Scan or enter the load or appointment ID.
3. Select **Enter**.

If the driver check-in was registered within the scheduled appointment window, then the **Appointment status** changes to *Checked in*.

Register driver check-out

After loading or unloading has finished and the driver has left the facility, you can check out the driver using either the web client or the Warehouse Management mobile app.

To check out a driver using the web client, follow these steps:

1. Select an appointment, as described previously.
2. On the Action Pane, select **Driver check-out**.
3. Fill out the fields on the **The driver's information** dialog as needed.
4. Select **OK**.

To check out a driver using the Warehouse Management mobile app, follow these steps:

1. Select **Driver check-out**
2. Scan or enter the load or appointment ID
3. Select **Enter**

If driver check-in and check-out are done within or before the scheduled appointments, the **Appointment status** changes to *Completed*.

Appointment status

For each record listed on the **Driver check-in and check-out** page, the **Appointment status** field indicates how actual driver check-in and check-out times compare to the scheduled times. The feature also calculates and shows if carriers were late checking in and/or out.

> **Tip**
>
> Select the **Display completed and canceled appointments** check box to display closed appointments in the grid.

The following table describes the possible **Appointment status** that an appointment can have.

Expand table

Appointment status	Description
Waiting	The driver hasn't checked in but isn't yet late.
Checked in	The driver checked in but hasn't checked out and isn't yet late checking out.
Dropped trailer	The driver checked in and selected the **Dropped trailer** option.
Completed	The driver checked in and checked out on time or ahead of schedule.
Completed late	The driver checked in and checked out, but after the scheduled deadline.
Check out warning	The driver checked in but hasn't checked out, and the alert interval hasn't yet expired. To specify the alert interval, go to **Transportation Management** > **Setup** > **Transportation Management parameters**, and then, on the **General** tab, on the **Driver check-in and check out** FastTab, set the **Alert interval** field.
Late on check in	The driver hasn't checked in and is now late.
Late on check out	The driver checked in but hasn't checked out, and the alert interval has now expired.

These statuses help warehouse and transportation managers to do the following tasks:

1. Track the status of ongoing shipments, follow up with carriers on loads that deviated from the plan, and mitigate delays.
2. Identify trends in the timeliness of each carrier.
3. Identify potential problems in your own warehouse operations that are resulting in late check-outs caused by delays in loading or unloading.

Last updated on 12/30/2025

Reconcile freight in transportation management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the freight reconciliation process, which is also known as the matching process. During this process, system-generated freight bills that contain estimated freight costs are matched to actual invoices that are received from carriers. Any differences (variances) are reconciled.

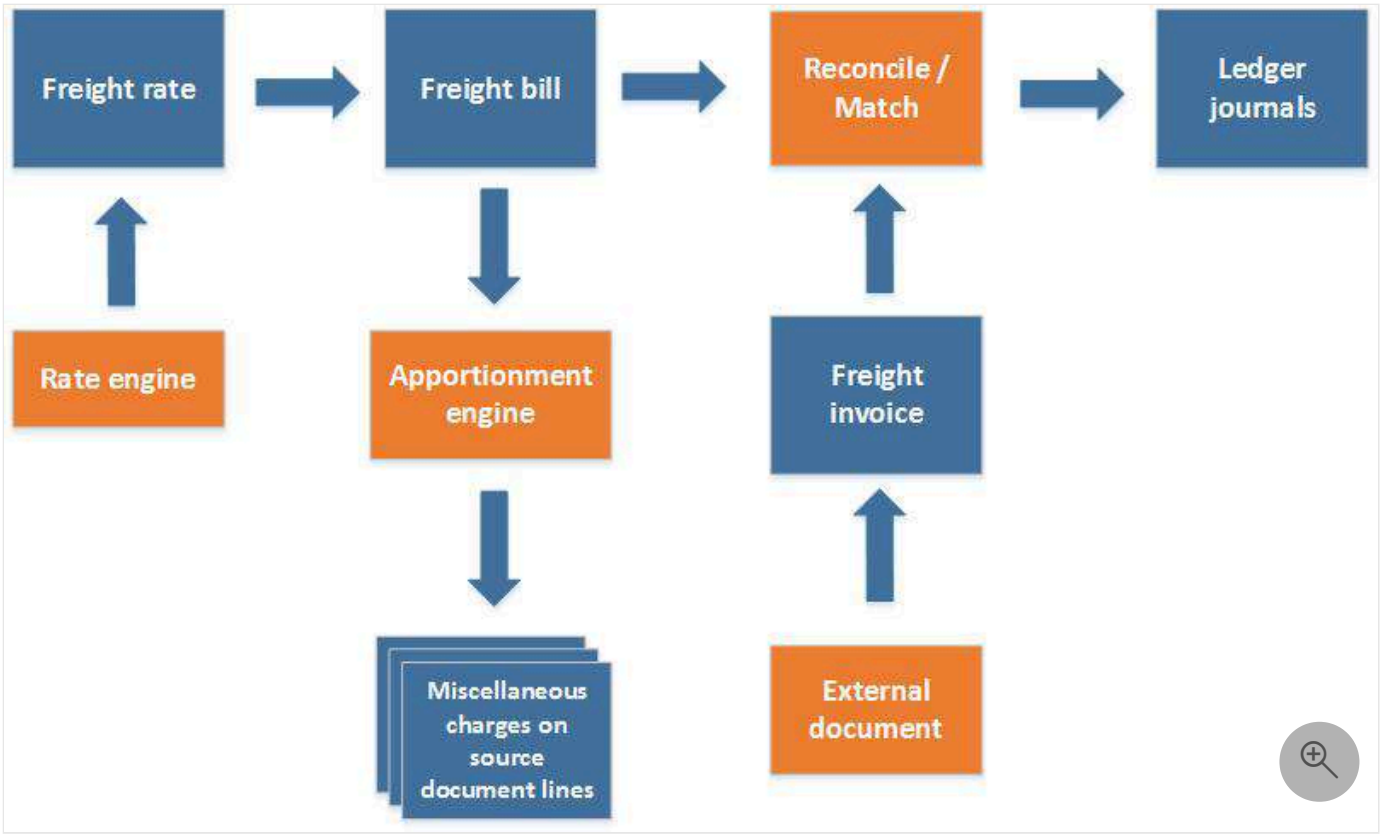
You can perform freight reconciliation manually. Alternatively, to save time and reduce manual errors, you can set it up to occur automatically.

The freight reconciliation process

The relevant shipping carrier's rate engine calculates freight rates. When a load is confirmed, a freight bill is generated, and the freight rates are transferred to it. Depending on the setup used for the regular billing process, the freight rates are apportioned as miscellaneous charges to the relevant source document (purchase order, sales order, or transfer order).

The freight reconciliation process can start as soon as the freight invoice arrives from the shipping carrier. You can receive the invoice electronically or on paper. If it's received on paper, generate an electronic invoice by using the freight bill as a template.

Freight bills and invoices must then be matched. In other words, find the freight bills that correspond to each freight invoice. You can match the invoice lines one by one (manual matching), or you can match all available invoices at once (auto matching). To handle any variances that occur between the freight bill and the invoice, use the reconciliation reason codes that are set up for both manual and automatic reconciliation.



Enable freight reconciliation

To use Transportation management (TMS) to reconcile freight, follow these steps:

1. Go to **Transportation management** > **Setup** > **Transportation management parameters**.
2. On the **General** tab, on the **Freight reconciliation** FastTab, set the **Enable freight reconciliation** option to *Yes*.

Set up reconciliation reasons

Reconciliation reasons (or reason codes) specify accounts that are debited and credited with any variances that occur between the freight bills and invoices as a result of manual or automatic reconciliation. They're set up on the **Reconciliation reasons** page (**Transportation management** > **Setup** > **Reconciliation reasons**).

For some reason codes, you might not want to credit a specific account. Instead, you want any credit to be paid to the vendor. In this case, select the **Pay freight vendor** checkbox. Be sure that you also specify a vendor account for the carrier on the **Shipping carriers** page (**Transportation management** > **Carriers** > **Shipping carriers**).

When the ship confirm action occurs, the freight bill is generated, and freight charges are apportioned back to the original source document as miscellaneous charges. Alternatively, you

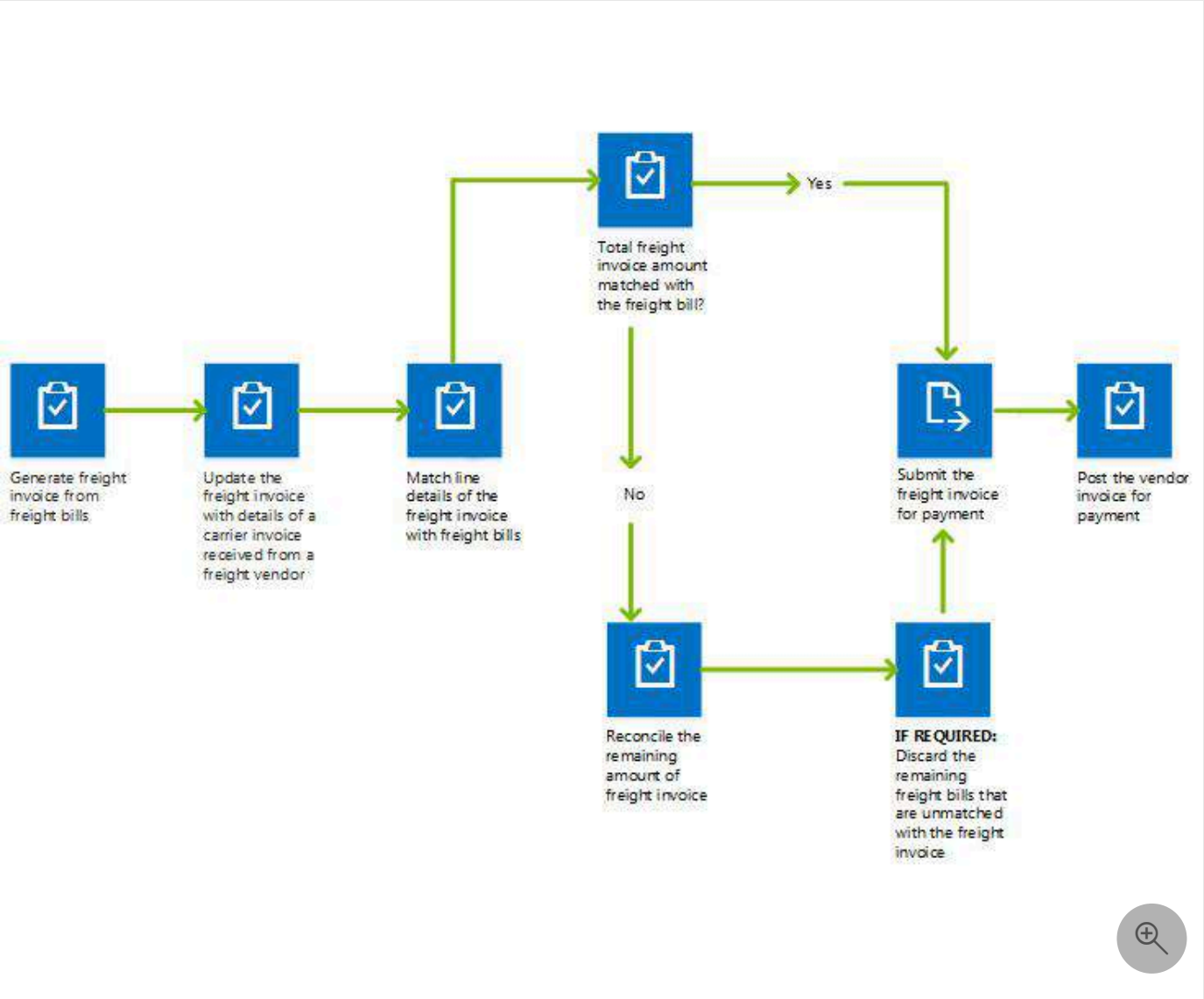
might want to charge the debit and credit accounts that are specified on the **Reconciliation reasons** page. In this case, select the **Override accounts** checkbox.

If you're reconciling freight manually, match each invoice line with the freight bill line or lines for the load that is being invoiced. Do this matching on the **Freight bill and invoice matching** page.

If the amount on the invoice line doesn't match the freight bill amount, select a reconciliation reason for the difference. If multiple reconciliations reasons apply, split the unmatched amount across them. Reconciliation reasons determine how the difference amounts are posted in the general ledger.

After the reconciliation of the whole invoice amount is accounted for, it's submitted for approval. Then the journal is posted.

The following illustration shows how to generate a freight invoice and do freight reconciliation.



Reconcile freight manually

Select a load to reconcile

1. Go to one of the following pages, depending on whether you're reconciling an inbound load or an outbound load:

    - **Transportation management** > **Planning** > **Inbound load planning workbench**.
    - **Transportation management** > **Planning** > **Outbound load planning workbench**.

2. Clear the **Hide shipped and received** checkbox.

3. In the **Load** grid, select the load that you want to reconcile.

Create a carrier invoice

If you reconcile freight manually and don't automatically receive carrier invoices, you can create an invoice based on the freight bill.

1. Select **Related information**.
2. Select **Freight bill details**.
3. Select **Generate freight bill invoice**.
4. In the **Invoice** field, enter a value.
5. Select **OK**.

Reconcile the invoice

When you reconcile a carrier invoice and a freight bill, the reconciliation is done line by line.

1. Select **Match freight bills and invoices**.
2. In the **Invoice details** section, expand the **Unmatched freight bill details** section.
3. In the list, mark the selected row.
4. Select **Match**.
5. Expand the **Matched freight bill details** section.

All freight bills that have positive amounts are available for matching. As in auto matching, you can only match freight invoices that have negative amounts to freight bills that aren't fully matched.

Submit the invoice for approval

1. Select **Submit for approval**.
2. Close the page.

3. Clear the **Hide approved** checkbox.
4. Select **Vendor invoice journals**.
5. In the **Reference journal number** field, select the link.
6. Select **Lines**.

Automatic reconciliation

If you choose to run automatic reconciliation, you must handle any invoices that the system can't match. You must then process those invoices manually before you can post all the invoices for payment.

To use automatic reconciliation, you must specify the schedule for reconciliation. You must also specify the invoices and shipping carriers to use. The automatic matching of the invoice lines and freight bills is done according to the setup of the audit master and freight bill type.

Set up vendor invoice parameters

To enable automatic creation and/or posting of a vendor invoice journal during the automatic reconciliation process, go to **Transportation management** > **Setup** > **Transportation management parameters** > **Vendor invoice**. To enable the system to create an invoice journal to reconcile any differences between the freight bill and freight invoice, turn on the **Write vendor invoice journal** parameter. To enable the system to post the invoice journal after it's approved by the user who is defined in the **Workflow user** field, turn on the **Post journal** parameter.

To run the automated match and pay batch jobs, turn on the **Automatically match and pay the freight invoice** parameter. If the **Automatically match and pay the freight invoice** parameter is turned off, but the **Enable Freight Reconciliation** parameter is turned on, you can run the manual match and pay process, and still submit vendor invoices for approval.

The **Match interval** parameter defines the schedule for automatic reconciliation or matching. If you turn on the **Automatically match and pay the freight invoice** parameter, the minimum standard recommended interval is 120 seconds.

Set up billing groups

Billing groups are user-defined categories. Set them up to reflect the categories of charges that are present on the invoices that you receive from carriers. To set them up, go to **Transportation management** > **Setup** > **Freight reconciliation** > **Billing group**, and specify a billing group ID and name. (Both fields are free-text fields.)

Set up the freight bill type

Freight bill types define how freight bills and carrier invoices should be matched. They define specific criteria, both mandatory and optional. The auto matching process considers a freight bill and an invoice only if defined mandatory criteria are populated and match on both. If optional criteria are blank on the freight bill or the invoice, the process can still consider them.

1. Go to **Transportation management** > **Setup** > **Freight reconciliation** > **Freight bill type**.
2. Select **New**.
3. In the **Freight bill type** field, enter a value.
4. In the **Engine assembly** field, enter *Microsoft.Dynamics.Ax.Tms.dll*. This assembly is the standard TMS matching engine code library.
5. In the **Engine class** field, enter *Microsoft.Dynamics.Ax.Tms.Bll.GenericNormalizer*. This class is the standard TMS matching engine class.
6. Select **New**.
7. In the **Description** field, select the value that should match on the freight bill and the carrier invoice.
8. In the **Match required** field, select one of the following values:
    - *Yes* – The value selected in the **Description** field must match on both the freight bill and the carrier invoice.
    - *No* – The value selected in the **Description** field can be blank on the freight bill or the carrier invoice.
9. Select **Save**.
10. Close the page.

Set up the freight bill type assignment

The freight bill type assignment specifies the freight bill type that is used for a carrier. The system then uses that freight bill type to determine the matching requirements when an invoice is received. If carriers ship from or to multiple sites and warehouses within the same legal entity, leave the **Site** and **Warehouse** fields blank.

1. Go to **Transportation management** > **Setup** > **Freight reconciliation** > **Freight bill type assignments**.
2. Select **New**.
3. In the **Mode** field, enter or select a value.
4. In the **Shipping carrier** field, enter or select a value.
5. In the **Freight bill type** field, select the freight bill type that you created earlier.
6. Close the page.

Set up the audit master

The audit master defines the tolerance limits for automatic freight reconciliation. In other words, it specifies how much the monetary amounts on the freight bill and the carrier invoice can differ. If the difference is outside the defined tolerance limits, reconciliation is prevented from occurring. The audit master also defines how discrepancies are handled.

1. Go to **Transportation management** > **Setup** > **Freight reconciliation** > **Audit master**.
2. Select **New**.
3. In the **Audit master ID** field, enter a value.
4. In the **Shipping carrier** field, select the same shipping carrier that you selected earlier.
5. In the **Freight bill type** field, select the freight bill type that you created earlier.
6. On the **Tolerance** FastTab, in the **Minimum tolerance level** field, enter a number.
7. In the **Maximum tolerance level** field, enter a number.
8. On the **Result** FastTab, in the **Overpayment reason code** field, enter or select a value.
9. In the **Underpayment reason code** field, enter or select a value.

   If the monetary amounts on the freight bill and the carrier invoice differ, and the difference is within the defined tolerance limits, the overpayment and underpayment reason codes specify the accounts that the difference is registered on.
10. Close the page.

You can also define exceptions if you don't want specific types of charges to be evaluated according to the same criteria as the audit master during the automatic reconciliation process.

Auto matching

When multiple freight invoices are matched to the same freight bill, the process for auto matching works in the following way:

1. All unmatched freight invoices are sorted by amount. The invoice that has the largest amount is listed first.
2. The freight invoices are matched one by one until the freight bill has no positive remaining amount.
3. Depending on the setup of the audit master and the remaining amount on the freight invoices, the remaining amount is set.

Example

You have an original freight bill (FB) for the amount of 1,500. You create three freight invoices for FB (Inv1, Inv2, and Inv3). There's one invoice line for each invoice. The following settings are used:

- **Original freight bill FB:** Amount = 1,500
- **Freight invoice Inv1:** Amount = 1,000
- **Freight invoice Inv2:** Amount = 600
- **Freight invoice Inv3:** Amount = −100

Automatic matching result

Auto matching runs in the following order:

1. Sort all freight invoices in descending order by amount: Inv1 → Inv2 → Inv3.
2. Match Inv1 with FB. Inv1 has 1,000 matched, and FB has a remaining amount of 500. Therefore, the status is set to *Partially matched*.
3. Match Inv2 with FB. Inv2 has 500 matched, and FB has a remaining amount of 0 (zero). Therefore, the status is set to *Fully matched*.
4. Because FB is now fully matched, Inv3 isn't processed.

Manual matching result

The results for manual matching vary, depending on the order of matching, as the following example cases show.

Manual matching case 1

One way to do manual matching for this example is to proceed in the following way:

1. Match FB with Inv1. FB has a remaining amount of 500. Therefore, the status is set to *Partially matched*.
2. Match Inv2 with FB. Inv2 has 500 matched, and FB has a remaining amount of 0 (zero). Therefore, the status is set to *Fully matched*.
3. When you try to manually match Inv3, you don't find any unmatched freight bills.

This case is essentially the same as auto matching.

Manual matching case 2

Another way to do manual matching for this example is to proceed in the following way:

1. Match Inv3 with FB. FB has a remaining amount of 1,600, which is the same as subtracting −100 from 1,500. Both FB and Inv3 have a matched quantity of −100.
2. Match Inv1 and Inv2 with FB, one after the other. FB is fully matched.

As this example shows, always match manually if freight invoices have negative amounts. This approach ensures that freight invoices with negative amounts can always be matched to a freight bill that wasn't fully matched, because you can control the matching sequence.

Last updated on 06/22/2026

Vendor invoice journal creation when discarding freight bills

Article • 03/10/2025

This article provides guidance about how to create vendor invoice journals when you discard freight bills that are used in Microsoft Dynamics 365 Supply Chain Management.

As of Supply Chain Management version 10.0.42, you can use the **Freight bill discard vendor invoice journal creation policy** field on the **Transportation management parameters** page to control when vendor invoice journals are created. The default setting, *Only for reconciliation reasons with pay the freight vendor*, creates journals only for reconciliation reasons when functionality for paying the freight vendor is enabled.

Set up the vendor invoice journal creation policy to always create journals

1. Go to **Transportation management** > **Setup** > **Transportation management parameters**.
2. On the **General** tab, on the **Vendor invoice** FastTab, in the **Freight bill discard vendor invoice journal creation policy** field, select *Always*.
3. Save your changes.

Feedback

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Landed cost module overview

The **Landed cost** module helps businesses streamline inbound shipping operations by giving users complete financial and logistical control over imported freight, from the manufacturer to the warehouse. For imported goods, landed costs can account for 40 percent or more of the total cost of each imported item. Therefore, the challenge is to provide accurate estimates for landed costs.

Businesses can use Landed cost to complete the following tasks:

- Estimate landed costs at the time of voyage creation.
- Apportion landed costs to multiple items and purchase orders in a single voyage.
- Support the transfer of goods between physical locations by recognizing landed costs.
- Recognize accruals for goods in transit.

Landed cost provides accurate and timely cost estimates for overhead landed costs. At the same time, it provides increased financial and logistical visibility into the extended supply chain. It also helps reduce the administration of costing and costing errors.

Highlights

Voyages

In Landed cost, a voyage is a distinct movement from an outbound location, through a specific set of destinations over a specified period, to a specified inbound warehouse location. Purchase orders and order lines can be added to either a single container or multiple containers for a voyage, and the costs are correctly allocated back to the item line.

Item ownership

In Microsoft Dynamics 365 Supply Chain Management, you typically receive goods at the warehouse destination and then invoice them. However, under most international trade agreements, a business takes ownership of goods from the time when they leave the original port, before they're physically received. Landed cost enables businesses to invoice goods before they're physically received. This concept is known as a *goods in transit order*. It creates a new type of order to manage the receipt of goods after the original purchase order is invoiced.

Costs

When you create a voyage in Landed cost, costs can automatically be added to it. These costs are set up in Landed cost, and various cost options and cost bases are available for them. Each cost can be set up for different levels of a voyage and apportioned to the item level through apportionment rules that support quantity, volume, weight, amount, and defined volumetric divisors. These estimated costs provide an accurate estimate of all costs for a voyage.

Landed cost then runs a preliminary posting/accrual of the estimated landed costs to ensure that an accurate calculation of estimated costs is available at the time of voyage creation. Estimated landed costs are posted together with the purchase order invoice. Actual costs are posted when the vendor invoice journal is posted.

Actual costs are reverse-estimated costs posted at the time of cost invoicing by using clearing accounts that are set up for each type of cost, such as duty, freight, and insurance. These postings follow the posting behavior that's associated with the specific item. They automatically update inventory posting, regardless of whether the posting type is first in, first out (FIFO), last in, first out (LIFO), moving weighted average, or moving average. All inventory model group posting types are supported. For moving average and standard cost posting, purchase price variance accounts are available to post the differences between estimated costs and actual costs.

Item and order tracking

As a voyage moves from the originating outbound location to the final destination warehouse, users can update each step, or *leg*, of its journey as required. For each leg, a lead time and a shipment status are identified. Confirmed receipt (or delivery) dates for movement to the next leg of the journey are also identified. Together, these delivery dates provide an estimated delivery date of the goods to the inbound warehouse. Users can change these delivery dates. In this case, the estimated delivery date of the goods is then automatically updated, based on the lead times and legs of the journey. Visibility into goods in transit by voyage and vessel is available on a per-container basis before receipt of the goods. Because the dates are updated on each purchase order line, businesses have more control over logistics and warehouse planning.

Landed cost concepts

The following table summarizes some core concepts of landed cost.

Expand table

Concept	Description
Voyage	Typically, a voyage is one vessel. However, depending on your practices and procedures, it can be one vendor or one purchase order. There are no limitations on the use of this concept.
Folio	A folio is often determined by customs regulations. It can consist of one vendor's goods for one entity/company per shipment. The goods in a folio can be in one container or spread among multiple containers.
Shipping container	Shipping containers store purchase order lines. They're a level below the shipment level. They're typically used if costs are apportioned for goods by container, or if receiving is done per container.
Shipping container type	Shipping container types can determine the price for a cost type, such as freight. They also provide useful shipping information.
Cost type	Cost types identify costs that are associated with imports, such as duty, freight, and insurance.
Auto cost	Auto costs work like trade agreements. An auto cost is linked to a voyage level.
Port	Ports are areas where goods are received and transferred from.
Vessel	A vessel is the medium that's used to transport goods, such as a ship or an airplane.
Goods in transit	Depending on settings, goods are put into an in-transit warehouse after an invoice is updated.
Activity	Activities can store each significant step of the journey between two ports. They can update dates.
Journey template	Journey templates are routes that goods take between two ports.

For a comparison of the terminology and features of the **Landed cost** and **Transportation management** (TMS) modules, see Landed cost vs. Transportation management.

Overview of the landed cost process flow

Landed cost can be run so that it generates goods-in-transit orders. Alternatively, it can be run so that it doesn't generate goods-in-transit orders. Both processes start with an inbound demand source document. Next, a new voyage is created, the source document order line is added to a container in the voyage, and costs are added to the voyage. The source document must be invoiced before estimated landed costs can be posted and a goods-in-transit order can be generated, if applicable. A vendor invoice journal must also be posted before the actual costs can be posted. Product receipt isn't always mandatory and can occur at different points in the flow.

You can date-track the delivery of a goods-in-transit order by using the **Tracking control center** page. To learn more, see Track inbound voyages and shipping container journeys. You can use the **Inquiries** feature to look up information across landed cost–enabled orders. For example, you can look up an item or container, and view related information. In addition, you can generate and view various reports by using the **Reports** functionality. To learn more, see Landed cost reports.

Setup and configuration

To use landed cost with goods-in-transit orders enabled, ensure that the following configurations are in place:

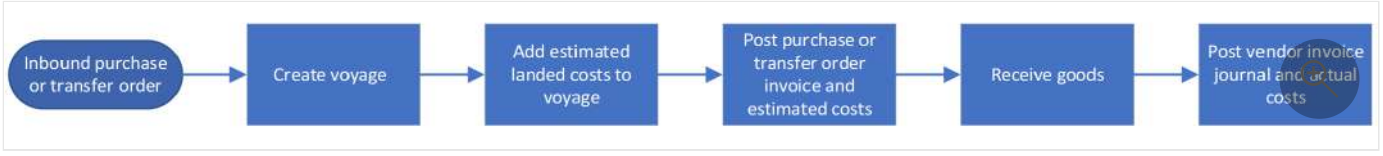
- **Goods-in-transit management** is set to *Yes* on the terms of delivery.
- Terms of delivery are set on the purchase order header.
- Additional goods-in-transit and under-delivery warehouses must be set up.

For more information, see Goods-in-transit processing and receiving.

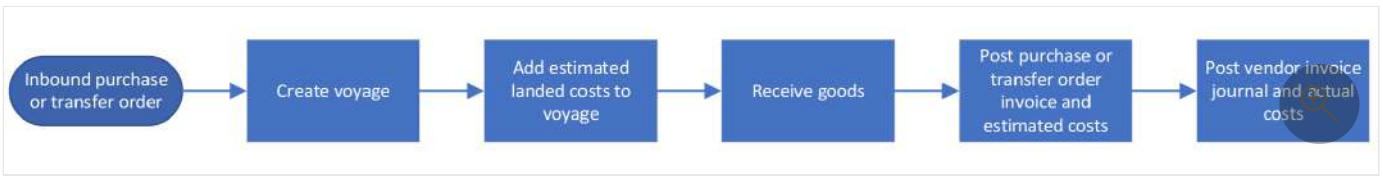
To use landed cost without goods-in-transit orders enabled, set **Goods in transit management** to *No* on the terms of delivery, and don't configure any additional warehouses. Also, you don't need to set terms of delivery on the purchase order.

The costing, financial, and inventory impact of posting estimated and actual costs depends on configurations set in the **Landed cost** module. For more information, see Landed cost parameters setup and Costing parameter values setup.

The following diagram shows the process flow for landed cost when goods-in-transit orders are enabled.



The following diagram shows the process flow for landed cost when goods-in-transit orders aren't enabled.



Create the voyage

After the voyage is created, you can add the purchase order lines to a new or existing shipping container in the voyage.

For more information, see Manage voyages, Manage shipping containers, and Manage folios.

Add the estimated landed cost to the voyage

When you use the **Find auto costs** function, the system automatically finds and adds costs to the voyage, container, item, or order, based on the auto costs setup. You can view the costs that the system adds, and you can manually edit or add any costs.

Learn more in Estimate and manage landed costs and Auto costs setup.

Post estimated costs

Post estimated costs for landed cost with purchase orders

If you use purchase orders as the inbound demand source document, then estimated landed costs are also posted when you post a purchase order invoice. Learn more in Estimate and manage landed costs.

If goods-in-transit is enabled, then the invoicing process generates the required goods-in-transit orders.

Post estimated costs for landed cost with transfer orders

Transfer orders can be used with landed cost only if goods-in-transit orders aren't enabled. In this scenario, instead of posting an invoice, you use the *Ship transfer order* functionality to post estimated costs.

Receive goods

Goods receipt for landed cost when goods-in-transit orders aren't enabled

When goods-in-transit isn't enabled, use the **Post product receipt** functionality on the purchase order. The process resembles the normal process for purchase order receiving.

For more information, see Warehouse handling of inbound loads for purchase and inbound shipment orders.

Goods receipt for landed cost when goods-in-transit orders are enabled

To receive goods when goods-in-transit orders are enabled, use goods-in-transit receiving-specific menu items in the Warehouse Management mobile app, post an arrival journal, or receive the goods via the goods-in-transit order. The receipt method you use depends on whether warehouse management processes (WMS) and serial/batch tracking are enabled.

For more information, see Goods-in-transit processing and receiving.

Processing of over and under transactions

Landed cost processes the over-receipt and under-receipt of goods. Over and under transactions occur if the quantity received from a goods-in-transit-enabled purchase order differs from the ordered quantity, and the difference exceeds configured rules and tolerances.

For more information, see Over/under transactions.

Post the vendor invoice journal

This step reflects an invoice received from the vendor in Supply Chain Management. During this step, actual costs are posted.

Limitations of the Landed cost module

Currently, the **Landed cost** module doesn't support the following items:

- Project purchase orders
- Purchase orders that include catch-weight-enabled items
- Purchase orders that include service/non-stocked items
- Non-deductible tax as part of the voyage costs
- Transfer orders with goods-in-transit orders enabled
- Direct delivery and goods in transit
    - If goods-in-transit isn't enabled for a direct delivery order, direct delivery for landed cost follows the standard behavior.
        - If goods-in-transit is enabled for a direct delivery order, the goods-in-transit processes are skipped. This condition means that the items aren't received in the goods-in-transit warehouse. Posting a purchase order invoice automatically generates the product receipt and also updates the transactions on the sales order. Also, if a purchase order that links to a direct delivery sales order is split across multiple voyages and/or shipping containers, the direct delivery processes don't work.

Last updated on 07/01/2026

Landed cost vs. Transportation management

09/19/2025

Microsoft Dynamics 365 Supply Chain Management provides two different modules for working with transportation: **Transportation management** (TMS) and **Landed cost**. This article summarizes the functionality that the two modules have in common and highlights the differences between them. Use this information to decide which module best fits your business practices. Some business practices work better with TMS, while others work best with Landed cost. Depending on your business requirements, you might choose to use one module exclusively, or you might combine the two modules.

This article isn't a comprehensive review of all the features of either module. Instead, it highlights the available functionality as it pertains to the transportation of goods from a vendor to your business's warehouse, where it can be consumed.

Terminology, reference data, and reporting differences

Terminology comparison

TMS and Landed cost use different terms for similar concepts. The following table summarizes these terms and their usage in the two modules.

Expand table

TMS term	Landed cost term	Notes
Load A *load* is a collection of shipments that are transported simultaneously on the same transportation unit (such as a truck or a ship). Loads can be either inbound or outbound.	**Voyage** Typically, a *voyage* is a single vessel that travels along a single *journey*. A voyage can contain multiple *shipping containers*, and it can also include inbound orders from different legal entities of your organization. Voyages can only be inbound.	TMS also uses the term *voyage number*, which refers to an information field where you can enter an identifier. However, no functionality is associated with the TMS field, and it's unrelated to the *voyage* concept in Landed cost.
Route	**Journey**	

TMS term	Landed cost term	Notes
A *route* is the physical path of a transport from origin to destination.	A *journey* is the physical path of a transport from origin to destination. Each voyage must be assigned to a journey.	
Route segments A route can have multiple *route segments*, each of which represents a step of the journey. A route can include multiple carriers or services, each of which is considered a route segment.	**Legs** Each journey can have multiple *legs*, each of which represents a step of the journey. A leg can be part of transportation, duty handling, or another activity.	
Containers A *container* can be any kind of package or packaging that is used by TMS.	**Shipping containers** A *shipping container* is a literal, physical shipping container, as known from container ships.	
Commodity codes In TMS (and in other places in Supply Chain Management), *commodity codes* identify types of commodities. They can affect tariffs, handling of hazardous materials, and so on.	**Commodity codes** In Landed cost, *commodity codes* are established at the level of the legal entity. They're mostly used for reporting.	Landed cost can also use the commodity codes that are used for TMS and other places in Supply Chain Management. However, the Landed cost commodity codes are used only by Landed cost.

Some reference data isn't shared

TMS and Landed cost don't share reference data for entities such as cost setup, journeys, and legs. If you use both modules, you must recreate the unshared reference data.

Intercompany reports don't work with goods in transit

The following reports don't work with the goods in transit feature that Landed cost provides:

- Intercompany goods in transit totals report
- Intercompany goods in transit transactions report

These reports assume that you put goods in transit as soon as you issue a sales packing slip, and that you take goods into inventory from transit upon receipt. However, goods in transit

aren't processed in this way. Therefore, if you use the goods in transit and intercompany features together, the results for both these reports will be incorrect.

Using the two modules together

You can use TMS for both inbound and outbound operations. Although Landed cost provides most of the same basic functionality as TMS for inbound operations, it also adds some functionality. Therefore, you might want to consider using TMS for outbound operations and Landed cost for inbound operations.

In general, we don't recommend that you use both modules together for inbound operations. Use the module that best meets your requirements. If you use the two modules together, don't share source documents between them. For example, don't use the same purchase order for both a load in TMS and a voyage in Landed cost. In particular, don't set up the system to create inbound loads automatically. If you include items from purchase orders in a voyage, you can't handle them as part of a load.

Goods in transit

One of the primary features of Landed cost is its ability to process goods in transit. Goods in transit processing lets you take financial ownership of shipped items before they're physically received at the destination warehouse. Goods in transit processing is often required for international trade.

TMS goods in transit features

TMS doesn't currently support goods in transit processing.

Landed cost goods in transit features

Goods in transit processing is a primary feature of Landed cost and provides the following functionality:

- **Goods in transit warehouses** – You can associate a goods in transit warehouse with each warehouse in Supply Chain Management. After you invoice the original purchase order, the goods are transferred to goods in transit warehouses. Items that are physically reserved in these warehouses become unavailable for consumption until they're received at their physical warehouse. Therefore, you take financial ownership of goods by invoicing the purchase order.

- **Goods in transit general ledger account** – Landed cost adds a specific general ledger posting account to the purchase order posting profile to handle goods in transit processing. This goods in transit general ledger account acts as an account of the "goods invoiced not received" type. When you invoice the original purchase order and create the goods in transit order, the direct purchase order cost posts to the goods in transit general ledger account until the goods are received at their final physical location.
- **Tracking control updates** – Goods in transit orders support tracking control functionality in Landed cost. Tracking control lets you update the expected date of arrival of goods while they're in transit. Tracking control then updates the voyage and the associated purchase order lines. The expected delivery date is then available for master planning and logistics.
- **Triggering of over/under transactions** – If a variance occurs between the expected goods on a voyage line and the actual goods that are received at the warehouse, Landed cost resolves it by creating over and/or under transactions. You can then manage these transactions, based on that way that you want to process them, through either a purchase order or a movement journal. The over and under transactions provide a link back to the voyage, the original purchase order, and the new movement journal or purchase order for the variance.
- **Goods in transit orders** – Landed cost introduces a new source document that is known as a *goods in transit order*. A goods in transit order lets you invoice the original purchase order to take financial ownership of the items. When you invoice the purchase order, the new source document is created for each purchase order line that has a combination of inventory dimensions. Goods are then physically moved to a goods in transit warehouse, where they're physically reserved against the goods in transit order and can't be consumed unless goods in transit orders are received.

Miscellaneous charges and shipment costs

One of the most important aspects of shipping and shipment management for goods is recognition of the overhead costs that are associated with shipments. These overhead costs aren't the direct inventory costs that are associated with a purchase order and a shipment or voyage. Instead, they're additional costs that the nature of the movement of the goods from the vendor to your organization introduces. These costs can include freight costs that are associated with the shipment of goods from one physical location to another, or duty costs that are associated with the import of goods from a foreign vendor.

Landed cost handles these costs by creating a new type of cost structure that is known as *auto costs*. TMS handles these costs by using miscellaneous charge functionality.

TMS charge and cost features

TMS uses miscellaneous charges to manage extra costs for loads that are allocated to the related source document lines. You can set these miscellaneous charges at the level of either the purchase order line or the purchase order header.

In TMS, you can apportion, or divide, miscellaneous charges by the weight, volume, or quantity of the inventory. Charges can be divided by freight or accessorial charges. Further development will be required to use additional apportionment methods in TMS.

Landed cost charge and cost features

Landed cost provides a set of cost functions that handle extra costs associated with the shipment of goods. These costs are known as auto costs. The system applies these costs at the time of initial shipment invoicing by using the estimated cost amount. Each cost type is managed in its own posting. After you post the actual invoices for overhead costs, the system automatically updates the estimated costs to reflect the actual costs.

You can invoice the overhead costs associated with the shipment of goods during the voyage from the port of origin or anytime after the goods are received. This capability provides greater flexibility for the consumption of goods.

The following list outlines some concepts for the charge and cost features in Landed cost:

- **Cost area** – Assign costs and charges to different levels, or *cost areas*, on a voyage. Apply the cost at the voyage level and break it down to each item in the voyage. You can also apply costs at the level of the container, purchase order, item, or transfer order. Manage each cost charge separately across the various areas and break it down to a per-item cost.
- **Apportionment methods** – Landed cost offers several apportionment methods. Apportion cost charges by quantity, dollar amount, value, weight, volume, measurement, or a user-defined volumetric measure.
- **Clearing/variance control** – Set up cost charges as their own cost code type in Landed cost. Each cost code type lets you define a clearing account for estimated costs charges, and also accrual and purchase price variance accounts for variances in estimated costs versus actual costs. When you initially post the estimated costs, the system credits the clearing account. After you post the actual invoices, the system posts the actual cost charges and debits the estimated costs out of the clearing account.

Matching freight bills and invoices

TMS bill and invoice matching features

TMS can match freight bills that contain estimated costs and invoices with the actual cost of the freight. You can receive invoices either electronically or on paper. Matching variance between the estimated cost and actual cost doesn't affect the inventory valuation.

Learn more in Reconcile freight in transportation management.

Landed cost bill and invoice matching features

Landed cost can match freight bills to estimated cost charges and can invoice the actual cost charges to the estimated costs. At the time of invoicing, the freight charges are in an invoice journal, and the system clears the estimated costs out of the clearing account. Additionally, the system posts the actual cost charges against the cost of goods sold for the item, together with the variances between the estimated charges and actual charges. This behavior allows for flexibility in the invoicing process.

Tracking

An important feature of both TMS and Landed cost is the ability to track goods and identify where they are in the journey from the point of origin to the final destination warehouse. Tracking lets you follow and update where the goods are located and when they're expected to arrive at the warehouse for consumption. In addition to the status of goods during their journey, Landed cost provides expected and actual dates for each step of the journey.

TMS tracking features

TMS provides limited features for tracking incoming loads. It shows dates and enables an integration to be built to find the exact position (for example, on the **Transportation status** page).

For each route segment, you can enter estimated schedules and arrival times.

Landed cost tracking features

Landed cost provides tracking control for each voyage, from the port of origin to the final destination. Tracking control sets the status of the voyage. The status indicates whether the voyage is sailing, at customs for inspection, or in local delivery on its way to the final warehouse. You can set the status at the level of the purchase order line, the container, and the header of the voyage. Therefore, you have more granular control.

Additionally, a confirmed expected date that's based on specified lead times is associated with each step of a journey. The confirmed and expected delivery dates are added to the purchase

order line and goods in transit orders, and you can use them for master planning and logistics. In addition to the expected dates, you can update the actual dates. The subsequent steps in a journey then update accordingly.

Multi-leg journeys

The journey concept identifies where the goods are in the process and controls the status of the goods while they're in transit. Goods can go through several legs of a journey, and you can associate various costs with a specific leg. Examples include a freight cost on the sailing of a vessel and local transport costs on the transport of goods from the port to the destination.

TMS multi-leg journey features

In TMS, you can break down routes into route segments to represent multi-leg journeys. TMS can allocate spot rates and accessorial charges to individual route segments.

Learn more in Plan freight transportation routes with multiple stops.

Landed cost multi-leg journey features

Landed cost provides a framework for moving goods from the point of origin to the destination through a series of legs, which are the stops or steps in a journey. You combine the legs to create journey templates, which define how goods move from the vendor to your organization.

In every leg of a journey, you can further define the status of each purchase order line and the lead times that are associated with it. The combined lead time for all the legs identifies the estimated delivery date. However, goods in transit processing is required for multi-leg journeys to apply. The journey of goods on a voyage is managed in a table that's specific to Landed cost.

Receiving by container

It's important to manage how goods are split and stored on a vessel. On a vessel, you can keep goods in one container or multiple containers. When you receive goods, you receive them in containers. You might receive different containers on a voyage at different times or on different dates.

Both TMS and Landed cost provide functionality for managing the receipt of goods in a container. TMS uses the concept of loads to manage the goods, purchase orders, and transfer orders that are associated with a shipment container. TMS supports receiving based on a

packaging structure received through an advance shipping notice (ASN). Landed cost uses the concept of shipping containers to process purchase orders and control overhead costs that are associated with a container on a vessel.

TMS receiving by container features

TMS supports inbound ASNs, all variants of receiving through the Warehouse Management mobile app, and all methods of receiving through the Supply Chain Management client.

Landed cost receiving by container features

To support receiving by container, Landed cost creates shipping container records and associates purchase orders with a specific shipping container by using its container ID. You can then apply overhead costs to that shipping container and break them down so that they're associated with the relevant purchase orders.

You can receive containers in Landed cost through a new type of receipt known as a *goods in transit receipt*, through arrival journals, or through mobile device receiving. When you use arrival journals, you can initialize the quantities from the goods in transit order or the original purchase order lines in the container. Landed cost provides two work types for receiving through the Warehouse Management mobile app.

Landed cost doesn't provide an ASN for the electronic receipt of goods. Additionally, it doesn't support Warehouse Management mobile app flows that process load receiving, license plate receiving, or mixed license plate receiving.

Rate shopping by vendor

Rate shopping enables a company to select a transportation vendor based on the lowest price, the fastest route, or a combination of both considerations.

TMS rate shopping features

TMS lets you identify vendor and routing solutions for inbound and outbound orders. For example, you can identify the fastest route or the least expensive rate for a shipment.

TMS provides full rate shopping and freight optimization through the rate route workbench, flexible rating options through the rating engine, a rating engine API for integration with external parties, and support for volumetric weight.

Learn more in Transportation management engines.

Landed cost rate shopping features

Landed cost provides only limited support for rate shopping by vendor. Although you can enter freight forwarder values, Landed cost doesn't compare them across multiple vendors.

Driver check-in/check-out with appointment scheduling

Driver check-in/check-out features enable the system to monitor arrivals and prevent congestion at the loading docks.

TMS driver check-in/check-out features

TMS lets you create *appointments*. An appointment represents events that occur at a dock for receiving a purchase order, shipping a sales order, or processing an inbound or outbound load on a specific date and at a specific time. Appointments ensure that docks are available for loading and unloading goods, and they help prevent situations where multiple carriers arrive at a location at the same time.

The system supports allowing drivers to check in at a specific dock door.

Landed cost driver check-in/check-out features

Landed cost can store estimates for the date and time when a container will be delivered.

Transfer orders and additional costs

Often, businesses must be able to transfer goods between warehouses. When this type of transfer occurs, costs are associated with it, and you might want to recognize those costs. In Landed cost, you can add transfer orders to a voyage or vessel to track the movement of goods from one warehouse or site to another, estimate the delivery time and expected delivery date, and add overhead costs to the transfer of goods.

TMS transfer order cost features

TMS supports the creation of freight charges that are connected to transfers. Although you can view these charges from the transfer order, the landed cost isn't added to the item cost. Freight reconciliation is supported through the creation of a freight bill that is based on these charges. This freight bill is then matched against a vendor freight invoice.

Landed cost transfer order cost features

Landed cost lets you add transfer orders to a vessel or voyage. In this way, you can add shipment costs to the voyage as inventory settlements at the time of transfer order receipt. You can invoice the overhead costs for the actual costs and track them against a voyage to update the cost of goods sold. Although transfer orders don't use goods in transit processing in the standard release, you can add them to voyages. The landed cost is added to the total cost of each item.

Turn on the Landed cost module and related features for your system

The **Landed cost** module itself is always turned on and can't be turned off, but several of its add-on features are optional. Admins can use the feature management settings to check the status of each feature and turn it on as needed. In the **Feature management** workspace in Microsoft Dynamics 365 Supply Chain Management, every feature in this table is listed under the **Transportation management** module. Preview features are marked as such; other features are generally available.

Expand table

Feature name and requirements	Feature description
Feature name: *Source document and accounting distribution support for Landed Cost* **Version required:** 10.0.34 or later	This feature for the **Landed cost** module enables landed costs to be included in the accounting distribution of purchased product receipts. Therefore, these costs can easily be identified and tracked. The feature supports the association of freight cost from Landed cost with the corresponding source documents. Therefore, it provides a more accurate and comprehensive view of the total cost of goods that are received. By helping you gain more visibility into the expenses that are associated with your purchases and transportation, this functionality allows for better cost analysis and management. Learn more in Show landed costs in the accounting distribution of product receipts. As of Supply Chain Management version 10.0.41, this feature is turned on by default.
Feature name: *Specify Goods In Transit Order when receiving via mobile device* **Version required:** 10.0.35 or later	This feature for the **Landed cost** module enables workers to select a specific goods-in-transit order for receiving when multiple goods-in-transit orders exist for the same voyage, container, item number, and purchase order number. Learn more in Specify goods-in-transit orders when receiving with a mobile device. As of Supply Chain Management version 10.0.41, this feature is turned on by default. As of version 10.0.43, it's mandatory and can't be turned off.
Feature name: *Split Goods In Transit Order quantity and assign batch/serial number when receiving via mobile device.*	This feature for the **Landed cost** module lets workers who use a mobile device assign multiple batch/serial numbers for different quantities of goods that are received from goods-in-transit orders. During the receiving process, all assigned batch/serial numbers are consolidated into one work record and processed together. Learn more in Assign batch/serial numbers when receiving with a mobile device.

Feature name and requirements	Feature description
Version required: 10.0.35 or later	As of Supply Chain Management version 10.0.41, this feature is turned on by default.
Feature name: *Match vendor invoice journal with voyage cost in different currency.* **Version required:** 10.0.36 or later	This feature for the **Landed cost** module lets users specify a currency code on the vendor invoice journal and match it with voyage cost in another currency. As of Supply Chain Management version 10.0.41, this feature is turned on by default. As of version 10.0.43, it's mandatory and can't be turned off.
Feature name: *Enable Quality Control for Goods In-Transit Order.* **Version required:** 10.0.41 or later	This feature enables organization to initiate and run quality checks against goods-in-transit orders. This proactive approach enables discrepancies or defects that might occur during transportation to be detected early, thereby ensuring that only products that meet predefined quality standards can proceed to their final destination. Learn more in Quality orders.

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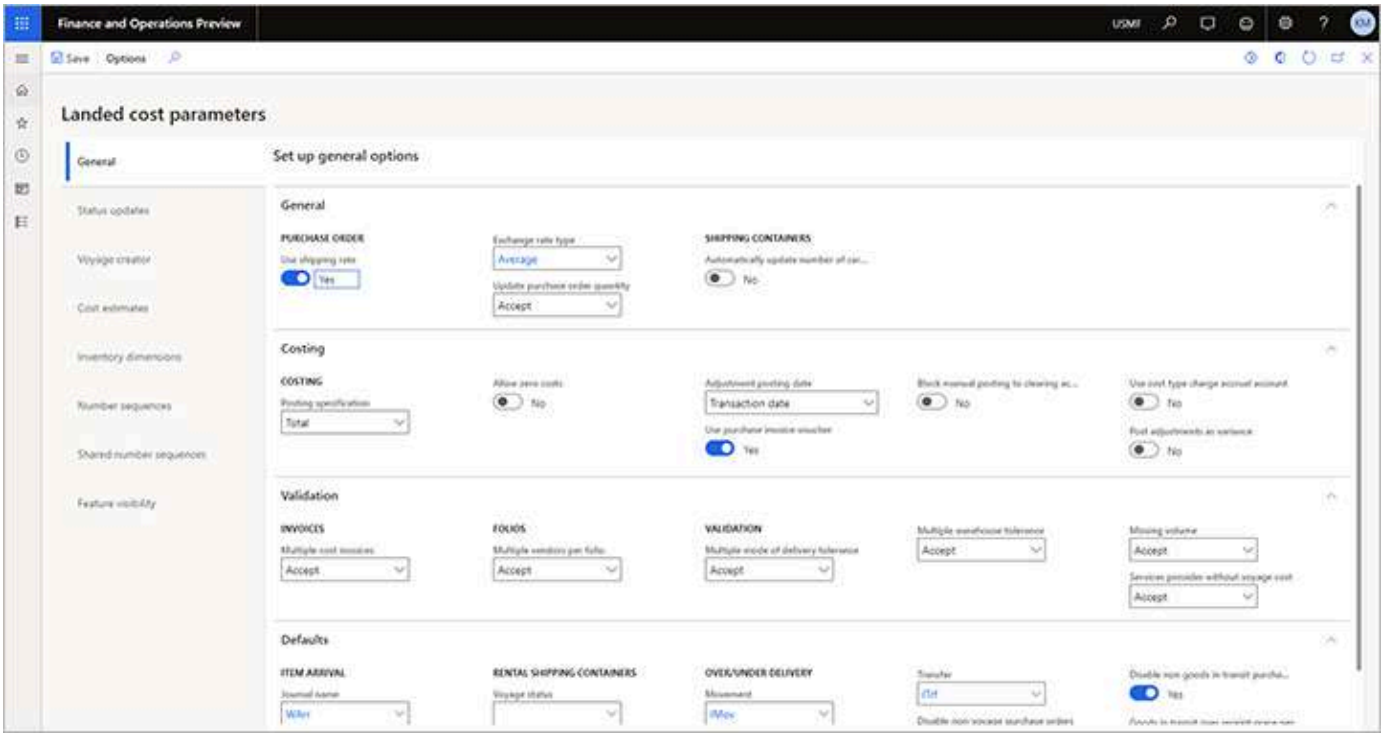
Landed cost parameters setup

You use the **Landed cost parameters** page to set up general information and configuration settings used across the **Landed cost** module for posting, status updates, number sequences, and behavior. The parameter setup is shared across legal entities, and an administrator can modify it.

Open the Landed cost parameters page

To work with the parameters, go to **Landed cost** > **Setup** > **Landed cost parameters**, and then set the fields as described in the following subsections.



General tab

General FastTab

The following table describes the fields that are available on the **General** FastTab of the **General** tab of the **Landed cost parameters** page.

Expand table

Setting	Description
Use shipping rate	A shipping rate is set for a defined period and estimates costs of goods that use multiple currencies. Set this option to *Yes* to use a shipping rate.

Setting	Description
Exchange rate type	The default collection of exchange rates used for multi-currency calculations for a voyage and voyage costs.
Update purchase order quantity	Select what occurs if a user changes the quantity on a purchase order line: • *Accept* – The voyage quantity is automatically adjusted. • *Warning* – If the line is attached to a voyage, a warning is shown, but the voyage quantity is updated. • *Error* – If the line is attached to a voyage, an error message is shown, and the purchase order can't be updated. Therefore, the order line must first be removed from the voyage.
Automatically update number of cartons	If set to *Yes*, units are automatically updated on each voyage line.

Costing FastTab

The following table describes the fields that are available on the **Costing** FastTab of the **General** tab of the **Landed cost parameters** page.

Expand table

Setting	Description
Posting specification	Select the value adjustment in the ledger: • *Total* – A total figure is posted to the ledger. • *Item group* – The adjustment is specified per item group. • *Item number* – The adjustment is specified per item. This value provides the most detail.
Allow zero costs	Set this option to *No* to show an error if a user tries to post a cost estimate for a voyage invoice or purchase order that doesn't include a value for the expected voyage cost. The error message states that a cost of 0 (zero) can't be allocated, and invoice posting will fail. In this case, the user can manually update the estimate (or reconfigure the auto cost), and then either pull in the updated value or delete the cost if it doesn't apply. Set this option to *Yes* to allow the voyage cost to be blank. In this case, a price of 0 (zero) is allocated according to the cost area. A vendor cost invoice can then be processed against the zero-price cost when the voyage is received. Configure the estimate on the auto cost record to prevent a zero-price cost from appearing. Although this estimate isn't completely accurate, it should still be more accurate than an assumed zero cost.

Setting	Description
Adjustment posting date	When you post an Accounts payable voyage cost invoice, the settlements table (inventory adjustments) is also updated. Select the date that is set by default on the **Select voyage costs** page while you're in the invoice journal: • *Transaction date* – Use the date of the journal (posting date). • *Purchase order invoice date* – Use the financial posting date of the stock (purchase order) invoice. • *Selected date* – The user can specify a posting date. Although the date can be left blank, if it's still blank when the cost invoice is posted, the user receives an error.
Use purchase invoice voucher	When this option is set to *Yes*, cost accrual transactions use the same voucher number that is used for the purchase invoice. When it's set to *No*, the system uses the next available number for the **Costs accrual voucher** number sequence that is set up on the **Number sequences** tab of the **Landed cost parameters** page. This option has an effect only when the **Post to charge account in ledger** option is set to *Yes* on the **Invoice** tab of the **Accounts payable parameters** page.
Block manual posting to clearing account	Set this option to *Yes* to prevent posting to clearing accounts where the transaction hasn't been linked to a voyage by selecting **Functions** > **Select voyage costs** on the Action Pane of the vendor invoice journal. Set this option to *Yes* so that the voyage and clearing account are correctly settled.
Use cost type charge accrual account	When this option is set to *Yes*, the charge accrual account configured for the relevant cost type code on the **Cost type codes** page is used to accrue costs as an expense. This option has an effect only when the **Post to charge account in ledger** option is set to *Yes* on the **Invoice** tab of the **Accounts payable parameters** page.
Post adjustments as variance	When this option is set to *Yes*, it overrides the standard functionality and forces the inventory adjustment transactions that are related to variances between estimated costs and actual costs to be posted to a variance account. When this option is set to *No*, the inventory adjustment transactions that are related to variances are handled based on the configuration of the costing method and cost type code. For standard cost, variances are still posted to the variance account. For moving average, variances are posted either to the variance account or to inventory. This option has an effect only when the **Post to charge account in ledger** option is set to *Yes* on the **Invoice** tab of the **Accounts payable parameters** page.
Use cost type credit ledger account for clearing	Use this parameter to control which clearing type and account are used for landed cost posting. If set to *No*, landed cost posting always uses the landed cost clearing type and account. If set to *Yes*, landed cost posting uses the specified posting type and account value on the cost type code when **Credit type** is set to *Ledger Account*.

Validation FastTab

The following table describes the fields that are available on the **Validation** FastTab of the **General** tab of the **Landed cost parameters** page.

Expand table

Setting	Description
Multiple cost invoices	Select what occurs if more than one invoice is processed against a voyage, folio, or container for the same miscellaneous charge. • *Accept* – The system should allow multiple cost invoices. • *Warning* – A warning is shown. • *Error* – An error message is shown.
Multiple vendors per folio	Select what occurs if more than one vendor's purchase orders are added to a folio. • *Accept* – The system should allow the action. • *Warning* – A warning is shown, but the action can still be performed. • *Error* – An error message is shown, and the action is prevented. Your customs broker or local laws might mandate a specific value for this field.
Multiple mode of delivery tolerance	Select what occurs if goods from a purchase order that uses a different mode of delivery than the voyage are added to that voyage. • *Accept* – The system should allow the action. • *Warning* – A warning is shown, but the action can still be performed. • *Error* – An error message is shown, and the action is prevented.
Multiple warehouse tolerance	Select what occurs if a voyage includes several order lines that must be delivered to different warehouses. These order lines might be spread across one or more purchase orders. • *Accept* – The system should allow the action. • *Warning* – A warning is shown. • *Error* – An error message is shown.
Missing volume	Select what occurs if a user adds an item without a volume to a voyage. • *Accept* – The system should accept the item. • *Warning* – A warning is shown. • *Error* – An error message is shown. If volumes are used to calculate and apportion costs, select either *Warning* or *Error*.
Services provider without voyage cost	Select what occurs if a user tries to process an invoice for a service provider that hasn't been linked to a voyage. • *Accept* – The system should allow the action. • *Warning* – A warning is shown. • *Error* – An error message is shown. Select *Warning*.

Defaults FastTab

The following table describes the fields that are available on the **Defaults** FastTab of the **General** tab of the **Landed cost parameters** page.

Expand table

Setting	Description
Journal name	Select the default journal that the *Create arrival journal* function uses.
Voyage status	Select the status that a voyage must have before users can set up a rental shipping container for it in the system. This action typically occurs when the goods are in transit or at the dock.
Journey template	Select the default journey template to use for new rental shipping containers. You'll typically select a journey template that includes rental costs.
Movement	If the over/under quantity for a delivery is within the defined tolerance, a movement journal is automatically processed. Select the default movement journal to use. The **Offset account** field for the selected movement journal name must have a value.
Transfer	When an under-delivery is processed, the short-receipt quantity is transferred to an under-delivery warehouse. Select the default transfer journal to use.
Disable non-voyage purchase orders	Turn off the Landed cost over/under delivery functionality for purchase orders that aren't attached to a voyage.
Disable non goods in transit purchase orders	Turn off the Landed cost over/under delivery functionality for purchase orders that don't use goods in transit functionality.
Goods in transit over receipt grace period	Specify the number of days after the first receipt of a shipping container that additional over-receipts can still be completed for that shipping container.

Status updates tab

The system uses status values to indicate the status of each voyage. Voyage status values can be applied automatically to voyages via voyage tracking or periodic batch jobs. Alternatively, you can apply them manually by opening the voyage and selecting a status in the **Voyage update** group on the **Manage** tab of the Action Pane.

You can create as many voyage status values as you want. However, you must define four of them for a special purpose on the **Status updates** tab of the **Landed cost parameters** page. The following table describes the fields that are available there.

Expand table

Setting	Description
Costed	Select the voyage status that identifies that a voyage has been finalized.
In transit	Select the voyage status that identifies that a voyage is in transit.
Ready for costing	Select the voyage status that identifies that a voyage is ready for costing. This status is used when all stock purchase invoices and voyage cost invoices where the **Credit on the voyage cost** field is set to *Vendor* have been processed for the voyage. Voyages that fail the costed process have a status of *Ready for costing*.
Pre-costed	Select the voyage status that identifies that a voyage is being pre-costed. This status is used when a new cost transaction is added to a voyage after it has already been costed. New cost transactions might be added to a previously costed voyage when a second freight invoice or unexpected demurrage charge is received. This status is automatically applied when a new voyage cost is added to a costed voyage.

Voyage creator tab

The following table describes the sections on the **Voyage creator** tab of the **Landed cost parameters** page.

Expand table

Section	Description
Voyage Editor	The **Voyage Editor** section was added in Supply Chain Management version 10.0.39. It provides the following new settings: • **Generate data manually** – When this option is set to *Yes*, data shown on the **Voyage editor** page isn't updated automatically each time you change a filter value. Instead, you can change several filter values before updating the page manually, which improves page performance and user experience. • **Keep the auto-generated tracking dimension number while splitting source order line** – When this option is set to *Yes*, the system preserves the autogenerated tracking dimension number after splitting a source order line. When this option is set to *No*, the system generates a new tracking dimension number after splitting a source order line. • **Allow creating voyage in batch** – When this option is set to *Yes*, the system can use batch mode to create and update shipping containers. Creating and updating shipping containers in batch mode helps improve system performance when working with shipping containers that have many shipping lines.
Tolerances	The **Outside volume tolerance** and **Outside weight tolerance** fields define thresholds above which goods are considered over-volume and overweight. When a user adds goods on the **Voyage editor** page, if the volume or weight exceeds the value that you set here, the system shows a warning. The value of each field is expressed as a percentage of the maximum volume

Section	Description
	or weight that is set for the relevant shipping container type. Set the value between 5 and 10 percent of the maximum volume or weight. Setting these tolerances to zero means that the total weight and volume of items in lines added to the shipping container must match the values of the container type exactly. To skip the tolerance check, set the maximum volume and weight values to very large values.
Folio creation setup	The system can create multiple folios during the voyage creation process. Use this section to define when a new folio should be created. For each row in this section, the system checks the specified table and field and creates a folio for each unique field value.

Cost estimates tab

The **Cost estimates** tab of the **Landed cost parameters** page provides just one field: **Default costing version**. This field applies only when the costing method is *Standard costing*. Select the default costing version to use for the *Item cost price update* periodic task. You might have to change this setting every time that a new financial year begins.

Inventory dimensions tab

You use the **Inventory dimensions** tab of the **Landed cost parameters** page to control which available inventory dimensions show by default on each Landed cost page where dimensions are used.

Select a dimension, and then set the **Voyage lines**, **Goods in transit**, and/or **Cost estimates** option to *Yes* for each page where that dimension should appear by default. Repeat this step for other dimensions, as required.

The settings on this tab establish the default dimensions for each designated page across legal entities. However, users who are working on one of the designated pages can override the default dimensions by selecting **Inventory** > **Display dimensions** on the Action Pane.

Number sequences tab

The **Number sequences** tab of the **Landed cost parameters** page lists each type of reference number sequence that Landed cost requires, but that isn't shared across legal entities. For each reference in the list, select a number sequence code.

Shared number sequences tab

The **Shared number sequences** tab of the **Landed cost parameters** page lists each type of reference number sequence that is shared across legal entities for Landed cost. Currently, there's just one number sequence in the list. This number sequence is used for the voyage ID.

Feature visibility tab

Landed cost adds features (fields and functions) to several frequently used pages in Microsoft Dynamics 365 Supply Chain Management. These pages include pages that are related to vendor master data, released products, purchase orders, transfer orders, and warehouse setup. If you're using Landed cost, you must make those features visible everywhere before you can benefit from them. If you aren't using Landed cost, you can hide the features to keep them out of the way.

On the **Feature visibility** tab of the **Landed cost parameters** page, set the **Activate** option to *Yes* to make Landed cost features visible wherever they're available. Set it to *No* to hide the features on common pages outside Landed cost. However, even when the option is set to *No*, the module itself, including the **Landed cost parameters** page, will remain available to users who have the correct permissions to access it.

Last updated on 06/22/2026

Voyage status setup

09/19/2025

On the **Voyage statuses** page, you set up the status values that users can assign to voyages. Users can assign voyage status values to all levels of a voyage: voyage, shipping container, folio, purchase order, and item (purchase lines and transfer order lines). You can use these statuses for two purposes:

- Inform the user about the status of the voyage, shipping container, folio, purchase order, or item (purchase lines and transfer order lines).
- Limit the use of the cost area by preventing modification or deletion.

To set up your voyage statuses, go to **Landed cost** > **Setup** > **Voyage statuses**. You can add, remove, and edit statuses by using the buttons on the Action Pane.

Each cost area has its own set and hierarchy of voyage statuses. Therefore, on the **Voyage statuses** page, in the **Cost area** field, first select the cost area that you want to view or create voyage statuses for. Then, for each voyage status, set the fields that are described in the following table, as required. The status of a voyage can also change automatically by system events, such as rules that you establish by using the tracking control center.

Expand table

Field	Description
Voyage status	Enter the name of the voyage status.
Description	Enter a description of the voyage status.
Modify	Select this check box if users are allowed to modify voyages that have this status.
Delete	Select this check box if users are allowed to delete voyages that have this status.
Mandatory	Select this check box to make the voyage status mandatory, so that it can't be skipped.
Parent	Use this field to establish a hierarchy among the status values. You can only change voyage statuses (either manually or automatically) downward in the hierarchy, from a parent status to one of its child statuses.

Note

Set up only the specific voyage statuses that your organization uses. Typical voyage statuses include *Confirmed*, *Goods in transit*, *Received*, *Ready for costing*, and *Costed*. However, other statuses might be present.

Shipping container setup

09/19/2025

This article describes how to set up shipping containers for the **Landed cost** module.

Set up shipping container types

Shipping container types define the types of shipping containers that you can use during shipping and voyages.

To work with the shipping container types, go to **Landed cost** > **Containers setup** > **Shipping container types**. You can view, add, and remove records for your container types. The following table describes the fields that are available for each record.

Expand table

Field	Description
Shipping container type	Enter a unique identification name or number for the shipping container type.
Description	Enter a description of the shipping container type.
Volume	Enter the maximum volume for shipping containers of this type.
Weight	Enter the maximum weight for shipping containers of this type.
Returnable	Specify whether shipping containers of this type can be returned to the vendor after they're used during a voyage. If you set this option to *Yes*, additional costs might apply for the return of shipping containers of this type to the port of origin.

Set up shipping containers

Use shipping container records to identify each container that you use during voyages. When you create a voyage, you can select a specific container for it from the list of all the shipping container records that you define. This feature is especially useful if your company owns its own shipping containers.

You don't need to enter shipping container numbers for shipping containers that you use only one time. Instead, you can add the shipping container number when you create a voyage.

Shipping container records are used only in Landed cost. They aren't available in the standard **Transportation management** module (TMS).

To work with shipping containers, go to **Landed cost** > **Containers setup** > **Shipping containers**. There, you can view, add, and remove records for your shipping containers. The following table describes the fields that are available for each record.

Expand table

Field	Description
Shipping container	Enter a unique identification name or number for the shipping container.
Shipping container type	Select the type of shipping container. Learn more in the Set up shipping container types section earlier in this article.

> **Note**
>
> - The shipping container setup is optional. Typically, you'll use it only if your company owns its own shipping containers or often reuses the same shipping containers.
> - No check digits are calculated for shipping container numbers.

Set up unit types

Unit types provide extra groupings and identification methods for shipping containers. Use the unit type to identify the type of container that goods are packaged in, such as pallets or drums. Select a unit type when you set up a container on the **All shipping containers** page.

Unit types are used only in Landed cost. They aren't available in TMS.

To work with unit types, go to **Landed cost** > **Containers setup** > **Unit types**. You can view, add, and remove records for your unit types. The following table describes the fields that are available for each record.

Expand table

Field	Description
Unit type	Enter a unique identification name or number for the unit type.
Description	Enter a description of the unit type.

Set up refrigeration types

Refrigeration types provide extra groupings and identification methods for shipping containers, usually refrigerated containers. Select a refrigeration type when you set up a container on the **All shipping containers** page.

To work with refrigeration types, go to **Landed cost** > **Containers setup** > **Refrigeration types**. You can view, add, and remove records for your refrigeration types. The following table describes the fields that are available for each record.

Expand table

Field	Description
Refrigeration type	Enter a unique identification name or number for the refrigeration type.
Description	Enter a description of the refrigeration type.

Delivery information setup

09/19/2025

This article describes how to set up delivery information for the **Landed cost** module.

Shipping ports

Shipping ports identify where goods are being shipped from and to. For each voyage, you define a "to" port and a "from" port, based on the journey that you select for the voyage vessel. Shipping ports build the legs of a journey and the voyage activities. Typically, you define them by using the name of the city and the country/region where the physical shipping port is located.

To work with shipping ports, go to **Landed cost** > **Delivery information setup** > **Shipping ports**. There, you can view, add, and remove records for your shipping ports. The following table describes the fields that are available for each record.

Expand table

Field	Description
Shipping port	Enter a unique identification name or number for the shipping port.
Description	Enter a description of the shipping port.

Tracking control center

Use the Tracking control center to set up the lead times, status updates, and flow of information that is associated with a voyage as the shipping containers move from one leg to the next. When you create a tracking control record, you associate it with a specific leg of a journey for a voyage. When a voyage updates a leg, the associated record updates and fills in as defined. You can update tracking information for individual voyages from the **All voyages** page.

To open the Tracking control center, go to **Landed cost** > **Delivery information setup** > **Tracking control center**.

The Tracking control center shows one of three page views, depending on the value that you select in the **Create type** field in the list pane: *Blank*, *Lead time*, or *Status update*. Each create type updates a different set of information that is associated with the progress of a voyage from the point of origin to the final destination.

Common rule settings

The following table shows the fields that are available for all three create types in the Tracking control center.

Expand table

Field	Description
Source table, Source field	These fields identify a source table and field in the database. The rule loads the value in the field and then uses it in the way that other settings of the rule define.
Target table, Target field	These fields identify a destination table and field in the database. The rule loads the value in the field and then uses it (or overwrites it) in the way that other settings of the rule define.
Activity	This field identifies the type of activity that the rule applies to a shipping container that matches the rule.
Matching criteria	This field determines how the system identifies a match for a rule. In each instance, the system reviews the data in the source and destination tables to determine whether and when a field should be updated in the target table. For example, the source table is *Voyages*, and the target table is *Purchase header* or *Purchase lines*. The *Voyages* table has a **From port** value of *Hong Kong Special Administrative Region*, and the *Purchase header* table has a **From port** value of *Shanghai*. A rule is then created that has *Hong Kong Special Administrative Region* as the "from" port. In this case, the values of the **Matching criteria** field work in the following way: • **Both** – The target field isn't updated, because one of the two ports doesn't match. • **Source** – The target field is updated, because the source table's "from" port is *Hong Kong Special Administrative Region*. • **Target** – The target field isn't updated, because the destination table's "from" port is *Shanghai* (not *Hong Kong Special Administrative Region*).
Copy action	The available values are *Copy* and *Default*. Select *Copy* to copy the value in the source field to the destination field. Select *Default* to set a static value for the destination field.
Default	When the **Copy action** field is set to *Default*, the **Default** field defines the default value of the destination field. For example, if the action is related to a port update, and the **Copy action** field is set to *Default*, the **Default** field identifies a port.
Leg	This field identifies the leg of the journey that the specified action occurs for, such as loading or customs.
Service provider	If a specific service provider is used for the current status update/leg of the journey, this field identifies the service provider. Service providers are defined in the vendor account.

Lead time rules

Lead time is the estimated time that a voyage needs to complete a defined leg of a journey. The lead time for each leg of a journey helps calculate the estimated delivery date for the voyage. Enter this date in the **Confirmed delivery date** field on every purchase line in the voyage.

Use lead time rules to manage date updates. When you use a journey template for a voyage, activities are automatically generated.

To add a lead time rule to the Tracking control center, follow these steps.

1. Follow one of these steps:

    - Go to **Landed cost** > **Multi-leg journey setup** > **Legs**. Select the leg that you want to set up lead times for. On the Action Pane, select **Tracking control center**. The Tracking control center is preloaded with information about the selected leg.
    - Go to **Landed cost** > **Delivery information setup** > **Tracking control center**. You must then manually select a leg in step 4 of this procedure.

2. In the list pane, set the **Create type** field to *Lead time*.

3. On the Action Pane, select **New**.

4. If you started from the Tracking control center in step 1, set the **Leg** field to the leg that you want to create a lead time rule for. If you started from the **Legs** page, the **Leg** field is preset to the leg that you selected before you opened the Tracking control center.

    Based on the value of the **Leg** field, several fields are automatically set, as shown here:

    - **Source table:** *Container activities*
    - **Source field:** *Start date*
    - **Target table:** *Container activities*
    - **Target field:** *Estimated end date*

5. In the **Activity** field, select the activity that the current rule should apply to.

6. In the **Lead time** field, enter the lead time (in days) that should be applied when the current rule is triggered.

Status update rules

Records where the **Create type** field is set to *Status update* update the status of purchase order lines, or the status at the voyage, shipping container, or folio level. You can set the statuses

independently. Use them to inform the user or department about the stage of a voyage (such as documents received or goods in transit).

When you set the **Create type** field to *Status update*, the Tracking control center includes a **Lines** FastTab where you can define a cost area and the updated status of the voyage. For example, you have a line where the **Cost area** field is set to *Container*, and the **Voyage status** field is set to *Goods in transit*. In this case, when an order completes the specified leg, the **Voyage status** field of the shipping container updates to *Goods in Transit*.

Status updates provide the status of a voyage throughout the voyage lines and purchase order lines that are associated with that voyage. As a voyage moves through its journey from the port, sailing, and customs, to the destination warehouse, the **Voyage status** field of the voyage record provides a quick view of the stage that the items are at.

To add a status update rule to the Tracking control center, follow these steps.

1. Follow one of these steps:

    - Go to **Landed cost** > **Multi-leg journey setup** > **Legs**. Select the leg that you want to set up a status update for, and then, on the Action Pane, select **Tracking control center**. The Tracking control center is preloaded with information about the selected leg.
    - Go to **Landed cost** > **Delivery information setup** > **Tracking control center**. You must then manually select a leg in step 4 of this procedure.

2. In the list pane, set the **Create type** field to *Status update*.

3. On the Action Pane, select **New**.

4. If you started from the Tracking control center in step 1, set the **Leg** field to the leg that you want to create a status update rule for. If you started from the **Legs** page, the **Leg** field is preset to the leg that you selected before you opened the Tracking control center.

    Based on the value of the **Leg** field, several fields are automatically set, as shown here:

    - **Source table:** *Container activities*
    - **Source field:** *Actual end date*
    - **Target table:** This field is left blank.
    - **Target field:** This field is left blank.

5. In the **Activity** field, select the activity that your rule should apply to.

6. On the **Lines** FastTab, enter the status updates for each cost area.

Blank type rules

You use records that have a **Create type** value of *Blank* to override or enter a field value, based on the data of another field. A field from Landed cost will overwrite data in other areas of Microsoft Dynamics 365 Supply Chain Management, based on rules that are set up in the Tracking control center.

1. Go to **Landed cost** > **Delivery information setup** > **Tracking control center**.
2. In the list pane, set the **Create type** field to *Blank*.
3. On the Action Pane, select **New**.
4. Define the source and destination values, matching criteria, copy action, and other relevant parameters, as required for your rule.

Required tracking control setup

For every journey template, two records must be set up in the Tracking control center. Both these records have a **Create type** value of *Blank* and are used in all Landed cost implementations. They help ensure that the purchase confirmed date and goods in transit expected dates are updated for voyages and on related purchase order lines in the expected way.

The first record is required to update purchase lines. It must have the following settings:

- **Source table:** *Container activities*
- **Source field:** *Estimated end date*
- **Target table:** *Purchase lines*
- **Target field:** *Confirmed or delivery date*

The second record is required to update goods in transit transactions. It must have the following settings:

- **Source table:** *Container activities*
- **Source field:** *Estimated end date*
- **Target table:** *Goods in transit order*
- **Target field:** *Expected date*

Multi-leg journey setup

Article • 07/30/2024

This article describes how to set up multi-leg journeys for the **Landed cost** module.

Legs

Legs are used to identify separate parts of a journey. Each leg is built by selecting the "to" and "from" shipping ports, and the transportation method that is used for that leg. Lead times can be associated with each leg. Lead times can help track a shipment and can also be used to calculate the estimated delivery date of the goods on a voyage. Additionally, when a leg of a journey is completed, the status of the voyage, shipping container, and associated purchase order lines can be updated through the tracking control setup. Legs can be used by a single journey template, or they can be reused by multiple journey templates. Container loading, customs, and local transport are generally set up as legs, and a nonspecific shipping port is used for them.

To work with legs, go to **Landed cost** > **Multi-leg journey setup** > **Legs**. There, you can view, open, create, and delete records for legs. The following table describes the fields that are available for each record.

Expand table

Field	Description
Leg	Enter a unique identification name/number for the journey leg.
Description	Enter a description of the journey leg. Typically, this description identifies the "to" and "from" ports, or the step of the journey.
From port	Enter the point of origin of the goods on the leg.
To port	Enter the point of destination of the goods on the leg.
Delivery method	Enter the method of transport for the leg.

Journey templates

A journey template defines the multi-leg journey between two ports that goods travel during a voyage. The legs of the journey are combined to identify the amount of time that will be required for goods to travel from the vendor's point of origin to the final

warehouse destination. When the legs are put on the journey template in a specific order, the lead times will identify the date of each leg and status of the voyage, container, and purchase lines for the voyage. You use the Tracking control center to set up the lead times that are associated with each leg that makes up the journey template. The journey template is also used when the automatic costs of a voyage are set up. When a journey is defined, the cost that is associated with the transport of the goods can be defined on the auto costs page.

To work with journey templates, go to **Landed cost** > **Multi-leg journey setup** > **Journey templates**. There, you can view, open, create, and delete journey templates.

For each journey template, set the following fields on the header.

Expand table

Field	Description
Journey template	Enter a unique identification name/number for the journey template. The journey template typically describes the point of origin and point of destination for the journey.
Description	Enter a description of the journey template. The description typically indicates the "to" and "from" ports, and the type of travel.

In the **Lines** section, add a line for each leg of the journey, and put the lines in order. Use the **Up** and **Down** arrows on the toolbar to change the order of the lines. The following table describes the fields that are available for each line.

Expand table

Field	Description
Leg	Select a leg to add to the journey.
Description	The description of the leg.
From port	The port that is the point of origin of the goods on the leg. This port is the "to" port that is used to determine the auto costs for a voyage.
To port	The final destination port of the goods on the leg.
Mode of delivery	The mode of delivery that is used for the leg.
Journey from port	If the port that is specified for the leg is used to determine the auto costs, select this check box to identify it as the "journey from" port. This port will be shown on the voyage header.

Field	Description
Journey to port	If the port that is specified for the leg is used to determine the auto costs, select this check box to identify it as the "journey to" port. This port will be shown on the voyage header.
Shipping company	Select the shipping company that is used for the leg.

Activities

The settings on the **Activities** page establish the types of activities that can occur at the destination port of a leg. Users who work on the **All shipping containers** page can select among these values when they estimate the duration of each activity and record the actual duration for comparison purposes.

To set up your activities, go to **Landed cost** > **Multi-leg journeys setup** > **Activities**. There, you can add, remove, and edit activities by using the buttons on the Action Pane.

The following table describes the fields that are available for each activity in the grid.

Expand table

Field	Description
Activity	The name of the activity.
Description	A description of the activity.
Shipping company	The vendor account of the shipping company that is associated with the activity.

Feedback

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Shipping information setup

09/19/2025

This article describes how to set up shipping information for the **Landed cost** module.

Description of goods

Descriptions of goods help identify a voyage, shipping container, or folio of goods, and the goods in it. You can select a description of goods on the shipping container header or the folio header.

To work with descriptions of goods, go to **Landed cost** > **Shipping information setup** > **Description of goods**. There, you can view, open, create, and delete records for descriptions of goods. The following table describes the fields that are available for each record.

Expand table

Field	Description
Description of goods	Enter a unique identification name or number for the type of goods that use this description.
Description	Enter a description of the type of goods in this category.

Vessels

A vessel is the unique name of a ship or vessel that a shipping company or agency uses. When you create a voyage, you must always either select or enter a vessel for it. If you often use the same vessels, you can make it faster and easier to create a new voyage by creating a vessel record for each of them. This approach lets users select the vessel from a list rather than enter the name or number manually each time.

To work with vessels, go to **Landed cost** > **Shipping information setup** > **Vessels**. There, you can view, open, create, and delete records for vessels. The following table describes the fields that are available for each record.

Expand table

Field	Description
Vessel	Enter a unique identification name or number for the ship that transports goods on a voyage.

Field	Description
Description	Enter a description of the vessel. Typically, this description identifies the name of the ship and the shipping company or agent.
Mode of delivery	Select the mode of delivery that the vessel uses, such as *Air*, *Ocean*, or *Train*.

Exporters

Each exporter record identifies a vendor or exporter that you can define as the vendor for a voyage. You can associate the exporter with a voyage and use it for reporting.

To work with exporters, go to **Landed cost** > **Shipping information setup** > **Exporters**. There, you can view, open, create, and delete records for exporters. The following table describes the fields that are available for each record.

Expand table

Field	Description
Exporter	Enter a unique identification name or number for the exporter of goods that are transported on a voyage.
Description	Enter a description of the exporter. Typically, this description identifies the full name of the shipping company or agent.

Commodity codes

Use commodity codes to help with customs identification and the calculation of duty rates for goods. You can select commodity codes on the **Released products** page. Use commodity codes for reporting.

To work with commodity codes, go to **Landed cost** > **Shipping information setup** > **Commodity codes**. There, you can view, open, create, and delete records for commodity codes. The following table describes the fields that are available for each record.

Expand table

Field	Description
Commodity code	Enter a unique code for this type of commodity.
Description	Enter a description of the commodity type.

Customs description

Customs descriptions help identify goods for customs purposes. You can select a customs description on the **Released products** page or on purchase order lines.

To work with customs descriptions, go to **Landed cost** > **Shipping information setup** > **Customs description**. There, you can view, open, create, and delete records for customs descriptions. The following table describes the fields that are available for each record.

Expand table

Field	Description
Customs description	Enter a unique code for this type of customs classification. Often, this code is the official description that a customs authority provides for the description and qualitative value of the goods.
Description	Enter a description of the customs classification.

Auto costs setup

Use the **Auto costs** page to set up cost rules for various cost areas, such as voyages, shipping containers, folios, purchase orders, items, or transfer order lines. Based on these rules and the fields users select when they create records for one of the cost areas, the system calculates the costs and automatically adds them. Therefore, users don't have to manually add the costs.

To work with auto costs, go to **Landed cost** > **Costing setup** > **Auto costs**. Then set up your auto cost rules as described in the rest of this article.

Work with cost areas

Auto costs work like trade agreements or auto miscellaneous charges. Each auto cost record belongs to a specific cost level. Use the **Cost area** field in the list pane of the **Auto costs** page to select the cost area that you want to work with. The list pane shows only auto costs that belong to the currently selected cost area. Any auto cost that you create belongs to the currently selected cost area.

Cost areas enable the system to apportion costs across the purchase order lines in a cost area. For example, a cost for one shipping container apportions the value to all the products in that shipping container. If there are two or more shipping containers, the same cost type can be specified for each shipping container. Therefore, the container type can be used to find the correct auto cost.

Buttons on the Action Pane

The following table describes the buttons that are available on the Action Pane of the **Auto costs** page.

Expand table

Button	Description
Edit	Edit an existing auto cost.
New	Create an auto cost.
Delete	Delete an auto cost.

Button	Description
Copy from	Create a new auto cost record that's based on an existing record. You can then freely edit the new record. This button helps you quickly create new auto costs.
Display dimensions	Open a dialog box where you can select the inventory dimensions that are shown for the cost transactions that you view. If you clear check boxes, fewer inventory dimensions are shown for the cost transactions. Therefore, less detail is shown for the transaction. If an inventory dimension is specified for an auto cost, the auto cost applies only when the specified inventory dimension is used for the item on a voyage.

Settings on the header

The combination of settings in the header section determines when and how a specific auto cost is added to a voyage. An auto cost is applied to a voyage, shipping container, or folio only when the details of the auto cost match the details on the header of that cost area. The sum of all matching auto costs determines the estimated landed cost of a voyage. This cost is apportioned across all the items on the vessel or voyage.

The following table describes all the fields that can appear in the header section. However, the specific fields that are available vary, depending on the currently selected cost area and display dimensions.

Expand table

Field	Description
Account code	Select one of the following values: • **Table** – The cost rule applies only to a specific vendor. • **Group** – The cost rule applies to a specific vendor group. The system looks for the vendor cost type group that is associated with the vendor record. • **All** – The cost rule applies to all vendors.
Shipping company	Select the vendor or vendor group that the cost rule applies to. This field is available only if you select *Table* or *Group* in the **Account code** field.
Customs broker	Select the vendor or vendor group that the cost rule applies to. This field is available only if you select *Table* or *Group* in the **Account code** field.

Field	Description
Item code	Select one of the following values: • **Table** – The cost rule applies only to a specific item. • **Group** – The cost rule applies to a specific item group. The system looks for the item cost type group that is associated with the item record. • **All** – The cost rule applies to all items.
Item relation	Select the item or item group that the cost rule applies to. This field is available only if you select *Table* or *Group* in the **Item code** field.
Mode of delivery	Select the mode of delivery (such as air or sea) that the cost rule applies to.
Container type	The type of shipping container that the goods are shipped in. An auto cost can be applied to a voyage only if the container type in the auto cost setup matches the container type of the voyage.
From port and **To port**	The source ("from") port and destination ("to") port of the voyage. (Some shipping containers might have different "to" ports, because the voyage might stop at multiple ports.) An auto cost can be applied to a voyage only if the "from" and "to" ports in the auto cost setup match the "from" and "to" ports of the voyage.
Auto cost number	The unique identifier of the auto cost rule. The value of this field is automatically generated based on the number sequence set up on the **Landed cost parameters** page.
Inventory dimensions	Some cost areas let you include one or more item dimensions on the header (such as size, color, warehouse, and batch number). Select **Display dimensions** on the Action Pane to select the dimensions that should be included.

Settings on the Lines FastTab

On the **Lines** FastTab, add a row for each currency that can be used when a cost is applied to a purchase order line on a voyage.

For items and purchase orders, the system matches the currency of each purchase order line against the currencies specified on the **Lines** FastTab. It applies only the cost value that is set for the matching line or item. If you don't set up a line for the currency of the line or item, the auto cost isn't applied to that line or item.

For other types of cost areas, all lines on the **Lines** FastTab apply to each voyage, shipping container, folio, or transfer order line that matches the values established on the header.

The following table describes all the fields that can appear on each line. However, the specific fields that are available vary, depending on the currently selected cost area.

Expand table

Field	Description
Currency	Select the currency that the line applies to. This currency is related to the purchase order. When the system tries to find the auto costs that should be applied to a voyage and its voyage lines, it selects the line that has the same currency as the purchase order. This field is available only when the **Cost area** field is set to *Purchase order* or *Item*.
Cost type code	The cost type.
Apportionment method	The method that is used to distribute costs to lines. The following options are available: • **Percent** – The value of the **Cost value** field is a percentage that applies to the total value of goods. For example, if the **Cost value** field is set to *10*, and the total value of goods is $800, the cost value is $80 (= $800 × 10 percent). • **Quantity** – The cost is apportioned based on the quantity of goods. • **Amount** – The cost is apportioned based on value of the goods. • **Volume** – The cost is apportioned based on the volume of goods. Volume is specified on the item master. • **Weight** – The cost is apportioned based on the weight of goods. Weight is specified on the item master. • **Volumetric** – The cost is apportioned based on the volumetric measurement of goods. • **Measurement** – This option enables a measurement to be specified in the **Landed cost** module. It's often used by organizations that don't know the individual volume or weight of the goods, but that require a more accurate apportionment than the **Amount** and **Quantity** options allow for. The freight forwarder or agent provides the organization with the weight or cubic measurement, at either the item level or the purchase order level. For example, sea freight equals $900. Item A has a weight of 800 kilograms (kg) and a volume of 2 cubic meters (m^3). Item B has a weight of 100 kg and a volume of 1 m^3. Here are the results when the costs are apportioned by weight: ◦ Item A = $800 ◦ Item B = $100 Here are the results when the costs are apportioned by volume: ◦ Item A = $600 ◦ Item B = $300 **Note:** When the **Apportionment method** field is set to *Measurement*, the system uses the measurements entered for both the cost area (the shipping container) and the lines. These measurements don't necessarily have to add up to the expected total. However, if you require that they add up to the expected total, you can do a check by using the statistics to review the total

Field	Description
	measurement. The measurement prompt setting and automatic update of the measurement at the shipment or container level are set on the voyage header.
From date	Specify the start of the date range for costing, if there's a range of dates. This date is the first date when the auto cost applies.
To date	Specify the end of the date range for costing, if there's a range of dates. This date is the last date when the auto cost applies.
Category	Select the method that should be used to calculate cost. The following options are available: • **Fixed** – Cost is apportioned based on the apportionment method. • **Pieces** – Cost is allocated per piece. Therefore, the apportionment method isn't used. • **Percent** – A percentage of the goods' value is added. Therefore, the apportionment method isn't used. • **Rate** – Select this option if there are quantity breakdowns. The **Apportionment method** field for the line must be set to *Measurement*.
Broken by quantity	Select this check box to make the **Quantity breaks** button available on the **Lines** FastTab. Quantity breaks enable the cost to be broken down and the different costs to vary, depending on the quantity on the purchase order line of the voyage. This functionality is often used when the mode of delivery is air. The **Category** field for the line must be set to *Rate*. The quantity pertains to the option that is selected in the **Apportionment method** field. The cost value is up to the quantity that is entered in the **Cost value** field. For example, quantities of 45–100 kg produce a charge of $6.00 per measurement (such as kg/m^3). Quantities that exceed 100 kg produce a charge of $5.50 per measurement (such as kg/m^3).
Cost value	Enter the value of the cost.
Cost currency code	Select the currency of the cost.
Item sales tax group	Select the tax code for the cost.
Minimum cost	This field applies only if the **Broken by quantity** check box is selected. If the measurement doesn't reach this value, the minimum value is selected, regardless of the costs that are specified on the **Quantity breaks** page. For example, quantities of 0–45 kg produce a charge of $6 per measurement (kg/$m^3$). Quantities of 46–100 kg produce a charge of $5.50 per measurement (kg/m^3). If 2 kg

Field	Description
	are shipped, the quantity break creates a cost value of $12. However, if the **Minimum cost** field is set to *$100*, the final cost is $100.
Aggregate	Select this check box to allow users to move this cost from the shipping container level to the voyage level. In this case, costs can automatically be calculated at the shipping container level. However, they can also be combined and apportioned across all items in all the containers on a voyage, instead of all items in the individual shipping containers.
Linked cost type	Link the current line to another line in the same auto cost record by specifying the **Cost type code** value of the line that you want to link to. This functionality is available only when **Category** field for the current line is set to *Percent*. When the current line is linked to another line, the cost of the current line is based on the cost of the other line. For example, freight is $1,000, and you want the fuel surcharge to be 10 percent of the freight cost. In this case, the line must have a **Category** value of *Percent*, a **Cost value** value of *10%*, and a **Linked cost type** value of *Freight*.
Value adjustment and **Adjustment amount**	Use these fields to adjust the value and amount for various percentage values. They're available only when the **Cost area** field is set to *Item*. Different costing can be set up for different components of an item. One component of an item might have a different cost percentage than the other components of that item. This approach is sometimes required to value a specific portion of the goods at different rates. For example, duty for a watch is calculated at two rates: one component of the watch has a 10-percent duty rate, and a second component has a 2-percent duty rate. In this case, costing is split between the two components. The cost is calculated for the item, but the cost value is adjusted by the value of the components that have the different cost category. The cost of the remaining components can then be calculated by using the amount that the previous calculation was adjusted by. For example, component A of the watch has a cost of $9.50 and a duty of 10 percent, and component B has a cost of $0.50 and a duty of 2 percent. Therefore, the total cost is $10.00. The first entry is for the 10-percent line. It uses the following values: • **Category:** *Percent* • **Cost value:** *10* • **Value adjusted:** *Adjust by* • **Adjusted value:** *-0.50* This setup specifies that the cost of the item ($10) must be reduced by $0.50 (the price of component B) before a 10-percent duty charge is calculated. Therefore, 10 percent is applied to $9.50.

Field	Description
	The second entry is for the 2-percent line (the $0.50 that the previous entry was adjusted by). It uses the following values: • **Category:** *Percent* • **Cost value:** *2* • **Value adjusted:** *Use* • **Adjustment:** *0.50* This setup calculates the remaining duty that is payable on component B.

Last updated on 05/29/2026

Costing parameter values setup

09/19/2025

When you set up the **Landed cost** module, you can define several sets of common values and related settings for each value. You can then use these values when you select specific types of costing parameter values in other parts of the app. This article explains how to set up these sets of values.

Set up cost type codes

Cost type codes determine the type of cost that you incur when goods arrive at the warehouse, or the landed costs of the voyage. Although they usually increase the value of the goods, you can also use them to accrue amounts into the ledger. Use ledger adjustments when you accrue a cost over time or during a series of voyages and offset it in a single transaction.

> **Note**
>
> If you share the cost type table across legal entities, you must also share the chart of accounts across legal entities. Otherwise, the posting transactions don't work correctly.

To set up your cost type codes, go to **Landed cost** > **Costing setup** > **Cost type codes**. Use the buttons on the Action Pane to create new cost type codes, edit existing codes, or delete a selected cost type.

The following table describes the fields that are available for each cost type code.

Expand table

Field	Description
Cost type code	Enter a name for the cost type code.
Description	Enter a description of the cost type code.
Use shipping rate	Set this option to *Yes* if the voyage exchange rate (sometimes referred to as a management rate) must be used to calculate the value of these costs. In this case, the shipping rate is used instead of the standard default or spot exchange rate to exchange foreign currency invoices.
Reporting category	This field establishes a reporting category for the cost type. You can print reports either by reporting categories or by cost type.

Field	Description
Debit type	Select whether the cost type should debit the item, the ledger account, or the vendor.
Debit posting	If you set the **Debit type** field to *Ledger account*, select the posting description.
Debit account	If you set the **Debit type** field to *Ledger account*, select the ledger account to use.
Credit type	Select whether this cost type should credit the item, the ledger account, or the vendor.
Credit posting	If you set the **Credit type** field to *Ledger account*, select the posting description.
Credit account	If you set the **Credit type** field to *Ledger account*, select the ledger account to use.
Clearing account	Select the clearing account for the cost type. Specify a separate clearing account per cost type to help with reconciliation.
Standard cost posting type	If you're using standard costing, select the posting description.
Standard cost variance account	If you're using standard costing, the account that you specify here is used to post any variance. This account uses the landed cost breakdown on the **Item prices** page. This breakdown is created by using the periodic routine to update prices. For example, the standard cost of an item is $15.00, the FOB is $13.00, and the freight is $2.00. When you receive the invoice for the stock, you receive the item at $15.00, but there's a standard price variance of $2.00 for the item, because the actual FOB is $13.00. This variance is posted to the standard price variance account that is set up in the item posting profile. Because the estimated freight is $2.00, there's no variance when the stock invoice is posted. However, when you receive the invoice for freight, the freight is $2.50 per unit. Therefore, a $0.50 variance is posted to the specified cost.
Moving average variance account	If you're using moving average costing, the account that you specify here is used to post any variance. For example, the estimated freight is $2.00. However, when you receive the invoice for freight, the freight is $2.50 per unit. Therefore, a $0.50 variance must be posted. When you set the **Post adjustments as variance** option to *Yes* on the **Landed cost parameters** page, all variances between estimated and actual shipment costs are posted to the moving average variance account that you specify here. When you set the **Post adjustments as variance** option to *No*, standard functionality is used, and the variance is applied either to inventory or to the moving average variance account that you specify here, depending on stock on hand.
Charge accrual account	Select the account that is used to accrue for cost estimates when the purchase invoice is posted. This field is used only when the **Use cost type charge accrual account** option is set to *Yes* on the **Costing** FastTab on the **General** tab of the **Landed cost parameters** page.

Field	Description
Charge account	Select the account that is used to capture the inbound transportation costs that a supplier invoiced. The amount is posted as a debit. The offset account is the stock variation account. This posting is used only when the **Post to charge account in ledger** option is set to *Yes* on the **Accounts payable parameters** page.
Variance account	The account that offsets the charge accruals when the purchase invoice is posted. This field is used only when the **Use cost type charge accrual account** option is set to *Yes* on the **Costing** FastTab on the **General** tab of the **Landed cost parameters** page.

Vendor cost type groups

Vendor cost type groups help determine how *auto cost* charges are found and applied to a voyage. Vendors that have similar import costs are linked together. For example, all vendors from emerging markets pay the same duty percentage for the same type of product that they purchase from an established market.

You can maintain vendor cost type groups by going to **Landed cost** > **Costing setup** > **Vendor cost type groups**. The **Vendor cost type groups** page provides a grid that lists all existing vendor cost type groups. You can use the buttons on the Action Pane to add, remove, and edit rows in the grid.

The following table describes the fields that are available on each row in the grid.

Expand table

Field	Description
Vendor cost type groups	Enter a unique name for the vendor cost type group, such as *Emerging markets*.
Description	Enter a description of the vendor cost type group. This description can provide details about the level or type of charge that is associated with the vendor group.

Item cost type groups

Item cost type groups help determine how *auto cost* charges are found and applied to a voyage. Similar items are linked together. For example, all items that have a duty rate of 5 percent might belong to a specific cost type group.

You can maintain item cost type groups by going to **Landed cost** > **Costing setup** > **Item cost type groups**. The **Item cost type groups** page provides a grid that lists all existing item cost

type groups. You can use the buttons on the Action Pane to add, remove, and edit rows in the grid.

The following table describes the fields that are available on each row in the grid.

Expand table

Field	Description
Item cost type groups	Enter a unique name for the item cost type group, such as *Duty 5%*.
Description	Enter a description of the item cost type group. This description can provide details about the level or type of charge that is associated with the item group.

> **Note**
>
> The item cost type links to the item through the **Cost type group** field on the **Purchase** FastTab of the item's **Released product** page.

Transfer order cost type groups

Transfer order cost type groups help determine how *auto cost* charges are found. Similar items are linked together. For example, all items that have a duty rate of 7 percent might belong to a specific cost type group.

You can maintain transfer order cost type groups by going to **Landed cost** > **Costing setup** > **Transfer order cost type groups**. The **Transfer order cost type groups** page provides a grid that lists all existing transfer order cost type groups. You can use the buttons on the Action Pane to add, remove, and edit rows in the grid.

The following table describes the settings that are available on each row in the grid.

Expand table

Field	Description
Transfer order cost type groups	Enter a unique name for the transfer order cost type group, such as *Duty 7%*.
Description	Enter a description of the transfer order cost type group. This description can provide details about the level or type of charge that is associated with the transfer order cost type group.

> **Note**
>
> The transfer order cost type links to the item through the **Transfer order cost group** field on the **Purchase** FastTab of the item's **Released product** page.

Cost templates

Use cost templates to set default values for settings that users who get the cost estimate might not know. Cost templates can help reduce complexity in the estimation process by minimizing the selections that users must make to get an accurate estimate.

To work with cost templates, go to **Landed cost** > **Costing setup** > **Cost templates**. On the **Cost templates** page, the list pane on the left shows all current cost templates. Use the buttons on the Action Pane to add, remove, and edit templates.

The following table describes the settings that are available for each template.

Expand table

Field	Description
Cost template	Enter a unique name for the cost template. The name typically describes the factor or cost multiplier for the template.
Description	Enter a description of the cost template.
Shipping company	Select the vendor account of the shipping company to associate with the cost template.
Mode of delivery	Select the mode of delivery that the cost template should use when the estimated cost of an item is calculated. This field helps determine the auto costs that are associated with the goods on the cost estimate.
Shipping container type	Select the type of shipping container type to associate with the cost template. This field helps determine the auto costs that are associated with the goods on the cost estimate.
Customs broker	Select the customs broker (vendor) to associate with the cost template. This field helps determine the auto costs that are associated with the cost estimate.
Factor	Enter a factor to apply to the final cost estimate of goods. For example, to add 10 percent to the calculated cost estimate, enter *1.10*.

Volumetric divisors

Use volumetric divisors to calculate the volumetric weight. Each shipping or freight company sets its own volumetric divisors. In addition, a company's divisors usually vary depending on the mode of delivery. For example, air and sea often have very different divisors. A company can also make its rules more complex depending on where it ships from. The system uses the following formula to find the volumetric weight: VolumetricWeight = Volume ÷ VolumetricDivisor.

For example, a package that you send by air has a volume of 3 cubic meters (m^3). The company charges by volumetric weight and applies a volumetric divisor of 6. This divisor divides the volume to determine the volumetric weight. Therefore, the volumetric weight for this example is 3 ÷ 6 = 0.5 kilograms (kg).

To set up volumetric divisors, go to **Landed cost** > **Costing setup** > **Volumetric divisors**. The **Volumetric divisors** page provides a grid that lists all existing volumetric divisors. You can use the buttons on the Action Pane to add, remove, and edit rows in the grid.

The following table describes the fields that are available on each row in the grid.

Expand table

Field	Description
Shipping company	Select the vendor account of the shipping company that you want to associate with the volumetric divisor.
Cost type code	Select the cost type code that you want to associate with the volumetric divisor. Use this field to put cost types into reporting buckets. You can print reports either by reporting categories or by cost type.
From port	Select the "from" port that you want the volumetric divisor to apply to.
Volumetric divisor	Enter the volumetric divisor value that you want to apply to the row. The volume of each package is divided by this value to determine that package's volumetric weight.

> **Note**
>
> The system uses the maximum value between **actual weight** and **volumetric weight**.

Procurement and sourcing parameters for Landed cost

The **Procurement and sourcing parameters** page has a few settings that are especially relevant when you use the **Landed cost** module. Use the **Update order lines** dialog box that is opened from the **Procurement and sourcing parameters** page to specify whether purchase order lines should automatically be updated when changes are made on the purchase order header.

To complete this setup, follow these steps:

1. Go to **Procurement and sourcing** > **Setup** > **Procurement and sourcing parameters**.
2. On the **General** tab, select the **Update order lines** link.
3. The **Update order lines** dialog box lists the fields on the order header that can also apply automatic updates to related fields on the order lines. Set each field in the list to one of the following values:
    - **Always** – The order lines should automatically be updated when the order header is updated.
    - **Never** – The order lines should never be updated when the order header is updated.
    - **Prompt** – The user will be prompted to select whether the order lines should be updated.

Last updated on 12/30/2025

Vendor settings added for Landed cost

06/17/2025

When you enable the **Landed cost** module, several new fields are added to the existing **Vendors** page. You use these fields to set up the vendors that you'll use together with Landed cost features.

To set the relevant fields, go to **Procurement and sourcing** > **Vendors** > **All vendors**. Open an existing vendor, or create a new vendor, and then select the **Miscellaneous details** FastTab. All the new fields that the **Landed cost** module adds appear under the **Voyages** heading. The following table describes these fields.

Expand table

Field	Description
Shipping type	Select the vendor's role in relation to Landed cost: • **None** – The vendor has no specific role that is related to Landed cost. This value is the default setting, because most vendors will probably have no specific role. • **Shipping company** – The vendor is a shipping company. Vendors that have this shipping type are available for selection in the **Shipping company** field on the **Voyages** page. • **Customs broker** – The vendor is a customs broker. Vendors that have this shipping type are available for selection in the **Customs broker** field on the **Folios** page. • **Agent** – The vendor is an agent. Vendors that have this shipping type are available for selection in the **Agent** field on the **Vendors** and **Purchase orders** pages.
Cost type group	Assign the vendor to a cost type group for the purpose of selecting auto costs.
From port	Select the port of origin for the voyage.
Agent	The default agent when purchases are made from the vendor.
Import costing vendor	Indicate whether the vendor is a Landed cost vendor. **Tip:** You can use this field together with record-level security to limit the purchase orders that are shown when a voyage is created.
Shipping company	Select the default shipping company that is used when purchase orders are created for the vendor.
Services provider	Indicate whether the vendor is services provider.

Field	Description
Over/Under tolerance group	Select the default over/under tolerance group for the vendor.

Default descriptions for the general ledger

Article • 02/12/2025

Default descriptions can be used to update the **Description** field in voucher postings to the general ledger. This functionality has been enhanced to work with Landed cost.

To set up default descriptions for ledger postings, go to **Organizational administration** > **Setup** > **Default descriptions**.

The following table lists the new values that were added for the **Description** field on the **Default descriptions** page to support Landed cost.

Expand table

Description type	Notes
Import costing – Cost accrual	When a purchase order invoice is posted, a cost accrual is processed for estimate voyage costs. Default descriptions can be specified to add the voyage number to the description. Use *%4* for the shipment number.
Import costing – Goods in transit order	If you're using in-transit processing, the purchase order invoice is posted, and the goods are posted to a goods-in-transit account. Default descriptions can be specified to add the in-transit order number to the description. Use *%4* for the in-transit order number.
Inventory – close – adjustment	When the accounts payable (AP) invoice is processed for a voyage cost, an inventory adjustment is processed to estimate voyage costs. Default descriptions can be specified to add the voyage number to the description. Use *%4* for the shipment number.

For more information about how to work with the **Default descriptions** page, see Set up default descriptions for automatic posting.

Feedback

Was this page helpful? Yes No

Provide product feedback

Manage voyages

A voyage typically represents a vessel. However, depending on your practices and procedures, it can represent a vendor, a purchase order, or some other item that makes sense for your organization.

> **Tip**
>
> This article provides reference information that describes all of the settings available on the **Voyages** and **All voyages** pages. For step-by-step guides to creating and managing voyages, see Create a voyage and Manage a voyage, which are training units from the Work with landed cost in Dynamics 365 Supply Chain Management training module.

The All voyages page

The **All voyages** page provides voyage details, delivery and costing information, and information about items, purchase orders, and transfer orders. To open the **All voyages** page, go to **Landed cost** > **Voyages** > **All voyages**. This page shows a list of all current voyages. You can use the buttons on the Action Pane to create, delete, and work with voyages. Select any voyage in the list to view its details.

> **Note**
>
> Shipping containers and folios are linked to a voyage. Purchase lines are linked to a shipping container. In addition, costs that are entered here are apportioned to all attached purchase lines.
>
> Project purchase orders aren't supported in Landed cost.

The Action Pane of the All voyages and voyage details pages

The Action Pane of the **Voyages** page provides buttons that let you work with a selected voyage. Each button performs a single action. The Action Pane also includes tabs, each of which, in turn, provides a set of related buttons. Except where noted, all buttons and tabs are available both in the list view of the **All voyages** page and in the detailed view for a single selected voyage.

Buttons that appear directly on the Action Pane

The following table describes the buttons that are available directly on the Action Pane.

Expand table

Button	Description
New	Create a voyage.
Delete	Delete the current voyage. Only voyages that have a voyage status of *Confirmed* can be deleted. After a voyage leaves port and processes goods in transit, it can no longer be deleted.
Voyage editor	Open the **Voyage editor** page, where you can add or remove purchase lines for the voyage, create new containers, and modify details of the voyage itself. For more information about the **Voyage editor** page, see The Voyage editor page later in this article.
Voyage costs	Open the **Voyage costs** page, where you can view and add voyage costs to all goods in the voyage. When voyage costs are manually added to the voyage, they're automatically added to the **Costs inquiry** page and apportioned to each good according to the method that is specified on the **Voyage costs** page.

Buttons on the Voyage tab

The following table describes the buttons that are available on the **Voyage** tab of the Action Pane. This tab is available only when you're working in the list view of the **All voyages** page.

Expand table

Button	Description
Shipping container	Open a page that shows all the shipping containers associated with the selected voyage. There might be one container or many containers.
Folio	Open a page that shows all the folios associated with the selected voyage. There might be one folio or many folios.

Buttons on the Manage tab

The following table describes the actions that are available on the **Manage** tab of the Action Pane.

Expand table

Button	Description
Documents received	Update the voyage to set the **Documents received** option to *Yes*. You can use this button to lock the item and/or purchase line so that it can't be updated further.
In transit	Update the **Voyage status** field to the in-transit status that is set on the **Landed cost parameters** page. There's no further logic in this process. A voyage can also be updated automatically to the in-transit status, based on settings in the Tracking control center.
Ready for costing	Update the **Voyage status** field to the ready for costing status that is set on the **Landed cost parameters** page. A voyage can be costed when all the invoices have been processed (both stock invoices and voyage cost invoices) and the goods have been received. If the estimated costs associated with a voyage haven't been costed, an error occurs when you try to process the costing of a voyage.
Costed	Clean up any costing irregularities after an invoice exists for all purchase orders and voyage costs. When you select this button, the **Voyage update - Costed** dialog box appears. There, you can select to post on the standard financial date or specify a posting date, and then run the action. You can rerun the action as many times as you want. You can also use the **Voyage update - Costed** dialog box to schedule the action as a periodic task (batch job). Run the action regularly by setting it up as a batch job.
Post receipts list	Post a receipt list for all purchase order lines in the voyage.
Post product receipt	Post a product receipt for all purchase order lines in the voyage. The product receipt process for the purchase order lines associated with a voyage is used only if the goods do **not** go through goods-in-transit processing. If the goods go through goods-in-transit processing, you receive an error when you try to post the product receipt for a purchase order line.
Post invoice	Post an invoice for all purchase order lines in the voyage. If the goods on the voyage go through goods-in-transit processing, the purchase order lines are invoiced before the receiving process is done. When the original purchase order is invoiced, the goods-in-transit orders associated with the original purchase order lines are created. Those orders can then be received by the warehouse.
Ship transfer order	Post a transfer order voyage for all transfer order lines in the voyage. When this button is selected, only transfer orders are available for update.
Receive transfer order	Post a transfer order receipt for all transfer order lines in the voyage.
Receive goods in transit	Receive all order lines that are in transit in the voyage. This button is one of three options available for receiving goods in transit on a voyage. (The other two options are the **Create arrival journal** button described later in this table, and the Warehouse Management mobile app.) This option is the simplest option and processes the goods in transit out of the goods-in-transit warehouse and into the final destination warehouse. If you want more control over the process, use the arrival journal or a mobile device to process the receipt of goods.

Button	Description
Find auto costs	Find any relevant voyage costs. If these costs have already been found or updated, you receive the following message: "Un-invoiced cost lines exist. Do you want to overwrite them?" The system finds any costs that weren't associated with the voyage at the time of creation. Voyage costs that are attached to a voyage and that have been invoiced won't be overwritten.
Create arrival journal	Open the **Create arrival journal** dialog box, where you can create an arrival journal that specifies a location. The dialog box provides the following options: • **Create from goods in transit** or **Create from transfer order** – The label for this option changes depending on whether you're using the goods-in-transit process. Set it to *Yes* to open an arrival journal page that lets you process a standard arrival journal for the goods in transit associated with the voyage. If the item has already been received in the final destination warehouse, it won't be added to the arrival journal lines. • **Initialize quantity** – Set this option to *Yes* to initialize the quantity received, based on the quantity of goods that is specified on the voyage line. If the voyage line has been partially received, this quantity is the remaining quantity. Set this option to *Yes*. • **Create from order lines** – Set this option to *Yes* to take the value from the order lines. This button is one of three options available for receiving goods on a voyage. (The other options are the **Receive goods in transit** button described earlier in this table, and the Warehouse Management mobile app.)
Accrue costs	You can accrue costs where a cost type has a ledger account specified for the debit. This button is typically used when the stock is in transit, or when goods have been received and invoiced.
Aggregate costs	Move costs from the shipping container level to the voyage level. You might use this button in a shared services/shipping scenario, where multiple entities share a shipping container or carton space. For example, the voyage has a 40-foot shipping container and a 20-foot shipping container, and the apportionment is done by volume. In this case, the goods/entities that share or use the space in the 20-foot shipping container might be penalized. To fairly distribute the costs, some organizations might want to transfer the costs to the voyage and distribute them based on the voyage-level apportionment method.
Change journey template	Open a dialog box where you can change the journey template. After you change the template, the voyage costs are deleted. Therefore, you might have to select **Find auto costs** (see the description earlier in this table) or manually add costs again.

Buttons on the General tab

The following table describes the buttons that are available on the **General** tab of the Action Pane.

Expand table

Button	Description
Receipts list	Open a list of product receipts for all purchase order lines in the voyage. If no product receipt lists have been processed, this button is unavailable.
Product receipt	Open the product receipt record for the purchase order lines associated with the voyage, if that record is used. If no product receipts have been posted, this button is unavailable. The product receipt process won't be used if you're using goods-in-transit processing.
Item arrival	Open the item arrival journal, if it's used.
Tracking	Open the **Inbound tracking** page, where you can update the expected arrival date of goods in a shipping container and a voyage, and subsequently update the expected delivery dates of purchase order lines.
Costs inquiry	Open the **Costs inquiry** page, which shows all the costs of a voyage. Select **View** on the Action Pane to adjust the view. You can view any of the levels, plus the item and cost type code. Only cost type codes that are directly related to inventory transactions appear on the **Costs inquiry** page. By removing cost type codes, you can adjust the page by grouping costs together. This capability can be useful if you're using sizes and colors. The **Costs inquiry** page shows only cost type codes where the **Debit** field is set to *Item*.
Goods in transit orders	Open the **Goods in Transit orders** page, which shows the goods-in-transit records for the selected voyage only.

Lines view of the voyage details page

To open the **Lines** view, open the details page for a voyage, and then select the **Lines** tab in the upper left of the voyage heading.

Information on the Voyage header FastTab

The **Voyage header** FastTab in the **Lines** view of a voyage contains basic information that describes the voyage. Many of the fields that appear on this FastTab also appear in the **Header** view, as described later in this article.

Information on the Voyage lines FastTab

The **Voyage lines** FastTab in the **Lines** view of a voyage is related to the voyage details and costing information that apply at the voyage level. Here, you can review the shipping containers, folios, and items attached to the voyage. The price and quantity of the items on the voyage are

also shown. You can view any shipping container or folio that is listed by selecting the relevant link.

Information on the Line details FastTab

The **Line details** FastTab in the **Lines** view of a voyage provides more information about the line currently selected on the **Voyage lines** FastTab. This FastTab includes details about the position that each line occupies in the container, and its declared quantity.

Header view of the voyage details page

To open the **Header** view, open the details page for a voyage, and then select the **Header** tab in the upper left of the voyage heading.

Settings on the General FastTab

The following table describes the fields that are available on the **General** FastTab in the **Header** view of a voyage.

Expand table

Field	Description
Voyage	Enter the voyage or flight number.
Booking reference	If your shipper or shipping center provides a reference number to identify the voyage (that is, the shipper's or shipping center's internal reference), enter it here.
Vessel	Enter or select the vessel. If the vessel isn't listed as a value, you can enter the vessel ID as free text. In that case, the main table isn't updated so that the vessel ID can be selected in this field later.
External voyage ID	Enter the publicly available ID for the voyage (such as the flight number).
Master air waybill/Bill of lading	Enter the master air waybill or bill of lading number. You can specify this value when you ship by air.
House air waybill/Bill of lading	Enter the house air waybill or bill of lading for the shipping container.
Rental	Select this check box to indicate that the container is a rental container that must be returned.

Field	Description
Description	Enter a description of the voyage to make it easier to recognize.
Voyage status	The status of the voyage. Values include *Created*, *In transit*, *Documents received*, and *Costed*. This field isn't calculated based on logic. It's updated only via the posting action. The *Costed* value indicates that an invoice for the stock and all voyage costs has been received. Unless an invoice is received, a voyage can't reach a status of *Costed*.
Purchase order status	The highest status that has been reached among all the purchase orders associated with the voyage.
Documents received	A value that indicates whether your company has received all the documents associated with the voyage. To change the value, select **Documents received** in the **Voyage update** group on the **Manage** tab of the Action Pane.
Shipping company	Select the shipping company for this voyage. This field helps determine auto costs.
Name	The name of the company selected in the **Shipping company** field.
Measurement	This field enables a measurement to be specified in an auto cost. Measurements are often used by organizations that don't know the individual volume or weight of the goods, but that require a more accurate apportionment than the amount or quantity provides. The freight forwarder will provide the weight or cubic measurement, and you can put it at the level of either an item or the purchase order. It can be automatically updated if the parameter is selected, or it can be manually entered.
Measurement unit	The unit of measure that applies to the number in the **Measurement** field.
Number of pallets	Specify the number of pallets on the voyage line. This field is automatically updated if the **Automatically update number of cartons** option is set to *Yes* on the **General** tab of the **Landed cost parameters** page.

Settings on the Delivery FastTab

The following table describes the fields that are available on the **Delivery** FastTab in the **Header** view of a voyage.

Expand table

Field	Description
Created date and time	The date and time when the voyage record was created.
Ex-factory date	This field selects the earliest ex-factory date among the purchase orders that are linked to the voyage. The ex-factory date is available on the purchase order header and is updated

Field	Description
	by using the tracking control feature.
Ship date	The estimated date when the plane or vessel leaves the point of origin.
Departure date	The departure date is usually the date when the plane or vessel actually leaves the overseas port. The ship date and departure date don't have to match. The **Departure date** field is for informational purposes only. It isn't used to estimate the expected delivery date. The tracking control feature is used to estimate the expected delivery date, and this field can be set up to be filled in automatically by the tracking control feature.
In to store date	This field selects the earliest date when the goods from the purchase orders that are linked to the voyage will be available for sale.
Estimated delivery date	The estimated delivery date is usually the date when the goods are due to arrive in the warehouse. This field is for informational purposes only. It isn't used for master planning for goods. The expected delivery date on the purchase order line is used for master planning for goods on a voyage and is updated by using the tracking control feature. This field can be set up so that it's filled in by the tracking control feature.
Actual delivery date	The earliest delivery date, based on the goods that are linked to the voyage.
ETA at shipping port	Enter the date when you expect the voyage to arrive at the shipping port, based on the information provided by your shipping agent.
Original documents received	Enter the date when the original documents were received.
Broker advised	Enter the date when the broker was advised.
Original bill of lading sent	Enter the date when the original bill of lading was sent.
Goods released	Enter the date when the goods were released from customs.
Customer appointment	Enter the customer appointment date, which is the date when you expect the customer to take ownership of the goods.
Delivered at warehouse	Enter the date when the goods were delivered to the warehouse.
Verification date	Enter the verification date.
Delivery instructions	Enter the date when the delivery instructions were received.
Mode of delivery	This field provides information about the method used to deliver the voyage and the cost selection of auto costs associated with that delivery method.

Field	Description
From port	The shipping port that the voyage departs from.
To port	The shipping port where the voyage arrives. The shipping containers can have different ports, because the ship might stop at multiple ports. By verifying the "to" port at the shipping container level, you provide accurate port details for every shipping container on a voyage. The auto cost for the freight is determined based on the combination of the shipping container type, the "to" port, and the "from" port.
Local forwarder	This field is for informational purposes only. The local forwarder should be selectable from the vendor table.
Local transport date	The date that the local transport is booked for. This field can help the warehouse do its planning.
Local transport time	The time slot that the local transport is booked for. This field can help the warehouse do its planning.
Journey template	The journey template specified for the voyage. The journey template is used to enter the "to" port and "from" port of the shipping container and voyage. Those values, in turn, help identify the auto costs for the voyage.

Settings on the Other FastTab

The following table describes the fields that are available on the **Other** FastTab in the **Header** view of a voyage.

Expand table

Field	Description
Remarks	Enter additional information that is related to the voyage.
Valuation date	Select the earliest ex-factory date for the purchase orders that are linked to the voyage.
Pending voyage	Use this to indicate whether your shipment or voyage has left yet. It is for informational purposes only.
Renaming shipping containers	For voyages that have more than one container, the number of containers that haven't yet been received.

The Voyage editor page

Use the **Voyage editor** page to create voyages, create containers for a voyage, and select the purchase order and/or transfer order lines to include in each container. To open the **Voyage**

editor page, follow one of these steps:

- Go to **Landed cost** > **Voyages** > **All voyages**. To create a new voyage, select **New** on the Action Pane, fill in the **Create voyage** dialog box as needed, and then select **OK**. To edit an existing voyage, select it, and then select **Voyage editor** on the Action Pane. In both cases, the **Voyage editor** page opens and shows the new or selected voyage.
- Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**. Select the purchase order to create a voyage and container for, and then, on the Action Pane, on the **Purchase** tab, in the **Landed cost** group, select **Create new voyage**. Fill in the **Create voyage** dialog box as needed, and then select **OK** to open the **Voyage editor** page.

For information about how to use the **Voyage editor** page to create and populate containers for voyages, see Manage shipping containers. The following subsections provide an overview of the settings available on this page.

Action Pane buttons on the Voyage editor page

The following table describes the buttons that are available on the Action Pane of the **Voyage editor** page.

Expand table

Button	Description
Edit voyage header	Open the **Edit voyage** dialog box, where you can view and update header information for the current voyage. These settings are also available on the **General** FastTab in the **Header** view of the voyage details page. For a description of these fields, see Header view of the voyage details page earlier in this article.
Setup	Open a drop-down dialog box where you can select the transaction types and sites to show on the page. These criteria are combined with the criteria on the **Filters** FastTab to control which orders are listed on the **Inbound orders** FastTab.
Generate data	Update the list of orders on the **Inbound orders** FastTab, based on the current **Filters** and **Setup** settings. You must select this button to update the inbound orders list each time that you change these settings.
Display dimensions	Open a dialog box where you can select the inventory dimensions that are shown in the grid on the **Lines to select** FastTab.

The Filters FastTab

Use the **Filters** FastTab to set criteria that control which purchase and transfer orders are shown on the **Inbound orders** FastTab. The criteria that you enter here are combined with the criteria

that are provided in the **Setup** drop-down dialog box on the Action Pane. Each time that you change the **Filters** or **Setup** settings, select **Generate data** on the Action Pane to update the **Inbound orders** FastTab.

The Inbound orders FastTab

The **Inbound orders** FastTab lists the purchase and transfer orders that meet the criteria that you set on the **Filters** FastTab and in the **Setup** drop-down dialog box. Select an order that's listed here to view the lines associated with that order on the **Lines to select** FastTab. Some orders might be listed several times if they include lines that have different delivery dates.

The Lines to select FastTab

The **Lines to select** FastTab lists the lines associated with the order and delivery date that you selected on the **Inbound orders** FastTab. Select the lines that you want to include in a new or existing container for the current voyage, and then select **Add to staging list** on the toolbar to make those lines available for container assignment. Select **View staging list** to open a page where you can actually assign the lines to new or existing containers. For more information about this process, see Manage shipping containers.

The Other dates FastTab

The **Other dates** FastTab lists lines associated with the order that you selected on the **Inbound orders** FastTab, but that have different delivery dates than the selected row. These lines are for information only and can't be added to the staging list from here. To add these lines to the staging list, select the appropriate order and delivery date on the **Inbound orders** FastTab. Those lines then become available on the **Lines to select** FastTab.

Voyage update periodic tasks

The **Landed cost** module includes several voyage-related periodic tasks that can bulk-update several aspects of voyages. To run or schedule these periodic tasks, go to **Landed cost** > **Periodic tasks** > **Voyage updates**, and then select one of the following task types:

- **Documents received** – This task lets you set the **Documents received** option to *Yes* for several voyages at the same time. Use the **Filter** settings to define the set of voyages that you want to update.
- **In transit** – This task lets you set the **Voyage status** field to the in-transit status that's established on the **Landed cost parameters** page for several voyages at the same time. Use

the **Filter** settings to define the set of voyages that you want to update.

- **Ready for costing** – This task lets you set the **Voyage status** field to the ready for costing status that's established on the **Landed cost parameters** page for several voyages at the same time. You'll typically set up this task to run on a regular schedule.
- **Costed** – This task lets you set the **Voyage status** field to the costed status that's established on the **Landed cost parameters** page for several voyages at the same time, provided that they have been costed but haven't yet been updated on the voyage. You'll typically set up this task to run on a regular schedule.

Purchase order clean up periodic task

The *Purchase order clean up* periodic task deletes purchase orders that haven't been added to a voyage. It helps to reduce transaction counts and improve general performance. To run or schedule the task, go to **Landed cost** > **Periodic tasks** > **Clean up** /> **Purchase order clean up**.

Last updated on 06/22/2026

Manage shipping containers

Article • 02/14/2025

Shipping containers are used to group together goods that are physically grouped together. They're also used in cases where costs must be shared only across those goods, usually because they're physically together.

Create shipping containers for voyages

You create a shipping container by opening or creating the voyage where you'll use it, and then assigning the purchase order lines that the shipping container will include. Follow these steps to create a shipping container for a voyage.

1. Follow one of these steps:

    - Go to **Landed cost** > **Voyages** > **All voyages**. To create a new voyage, select **New** on the Action Pane, fill in the **Create voyage** dialog box as needed, and then select **OK**. To edit an existing voyage, select it, and then select **Voyage editor** on the Action Pane. In both cases, the **Voyage editor** page is opened and shows the new or selected voyage.
    - Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**. Select the purchase order to create a voyage and container for, and then, on the Action Pane, on the **Purchase** tab, in the **Landed cost** group, select **Create new voyage**. Fill in the **Create voyage** dialog box as needed, and then select **OK** to open the **Voyage editor** page.

    > **Note**
    >
    > To improve performance of the **Voyage Editor** page, we recommend that you select a narrow date range and filter either by purchase or transfer order ID.

2. Use the filters at the top of the **Voyage editor** page to find the inbound order that you want to create the shipping container for. If you created the voyage from the **All purchase orders** page, the purchase order that you selected there will already be selected in the filter here. However, you can adjust the filters as you want.

3. On the Action Pane, select **Generate data** to update the list of orders on the **Inbound orders** FastTab, based on your current filter settings. (Each time that you change the filter settings, you must select **Generate data** to update the inbound order list.)

4. The **Inbound orders** FastTab shows all the qualifying purchase orders that match your filter settings. Select an inbound order to include in the new shipping container.

5. On the **Lines to select** FastTab, select the checkbox for each purchase order line that you want to include in the new shipping container.

6. Repeat steps 2 through 4 until you've found and selected all the order lines that you want to include in the new shipping container.

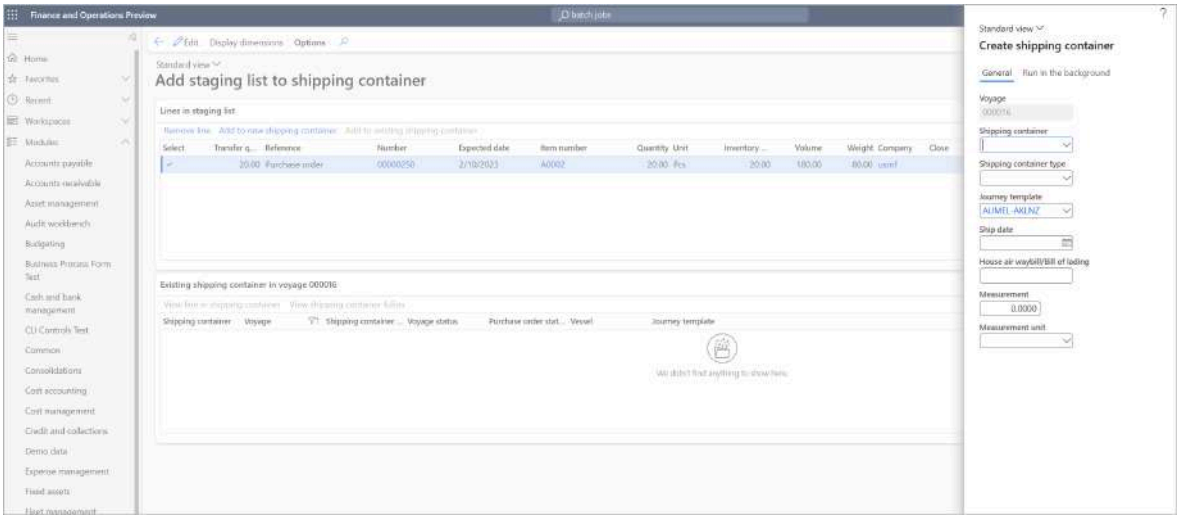
7. On the **Lines to select** FastTab, on the toolbar, select **Add to staging list**.

8. On the toolbar, select **View staging list**.

9. The **Add staging list or shipping container** page is opened. The purchase order lines that you selected are listed on the **Lines in staging list** FastTab. All the purchase order lines in this list will be added to the shipping container that you create. You can remove any line by selecting it and then selecting **Remove line** on the toolbar of the FastTab.

10. On the **Lines in staging list** FastTab, on the toolbar, select **Add to new shipping container**.

11. In the **Create shipping container** dialog box, on the **General** tab, define the new shipping container.



12. If you want to create the shipping container in batch mode, on the **Run in the background** tab, set the **Batch processing** option to *Yes*. If your container includes a large number of lines, batch mode can help improve performance compared to using the **Voyage Editor** page.

> **Note**

The **Run in the background** tab is available only if the *Enable shipping container creation and update in batch mode* feature is turned on for your system. For information about how to turn this feature on or off, see Turn on the Landed cost module and related features for your system.)

Background processing won't be activated if the transfer quantity is less than the total quantity of the related purchase order line.

13. Select **OK** to create the shipping container and add the selected purchase order lines to it.

> ⓘ **Note**
>
> If a container already exists for your selected voyage, you can add new purchase order lines to it by selecting **Add to new existing shipping container** on the **Add staging list or shipping container** page.

View shipping container lists and details

To view and process goods through the shipping container page, go to **Landed cost** > **Shipping containers** > **All shipping containers**. The **All shipping containers** page shows a list of all available shipping containers. You can use the buttons on the Action Pane to delete and manage shipping containers. Select any shipping container in the list to view its details on the **Shipping containers** page.

The upper part of the shipping container details page shows shipping container and costing information. The **Lines** section shows the folios, items, and purchase orders or transfer orders that are attached to the container.

Shipping containers Action Pane

The Action Pane on the **All shipping containers** and **Shipping containers** pages provides buttons that let you work with a selected shipping container. Each button performs a single action. The Action Pane also includes tabs, each of which, in turn, provides a set of related buttons. Except where noted, all buttons and tabs that are described in the following subsections are available both in the list view (that is, on the **All shipping containers** page) and in the detailed view (that is, on the **Shipping containers** page).

Buttons on the Manage tab

The following table describes the buttons that are available on the **Manage** tab of the Action Pane.

Expand table

Button	Descriptions
Post receipts list	Post a receipt list, or view the product receipt lists for all purchase order lines in the shipping container.
Post product receipt	Post a product receipt for all purchase order lines in the shipping container.
Post invoice	Post an invoice for all purchase order lines in the shipping container.
Ship transfer order	Post a transfer order shipment for all transfer order lines in the shipping container. Only those lines in the shipping container that are a type of transfer order appear in the dialog box.
Receive transfer order	Post a transfer order receipt for all transfer order lines in the shipping container. The receive dialog box is the simplest way to receive goods in a shipping container or voyage, and is one of three available options. You can also receive via arrival journals or mobile device processing.
Create arrival journal	You can generate an arrival journal for organizations by using advanced warehouse features. The options are *Initialize quantity* (recommended), and either *Create from goods in transit* or *Create from purchase orders*. The last two options depend on whether goods-in-transit processing is being used.
Rename	Open a dialog box where you can rename a selected shipping container.
Change journey template	Change the journey template. After you change the journey template, you might have to select **Find auto costs** or manually add costs again, because the shipment costs will be deleted.
Convert to rental	Convert a selected shipping container into a rental shipping container.

Buttons on the General tab

The following table describes the buttons that are available on the **General** tab of the Action Pane.

Expand table

Button	Descriptions
Receipts list	Post a receipt list for all purchase order lines in the shipping container.
Product Receipt	View the product receipt record, if it's used. The product receipt process will be used only if the goods don't use goods-in-transit functionality.
Item arrival	View the item arrival journal for the shipping container, if that journal is used.
Legs	Legs are used to identify separate parts of a journey. Lead times can be associated with each leg to help with shipment tracking. Learn more in Multi-leg journey setup.
Tracking	View or update shipment tracking.
Goods in transit orders	You can open the **Goods in transit** page directly from the container. That page shows the goods-in-transit records for the selected shipping container only.

Shipping container Header view

To open the **Header** view, open a shipping container, and then select the **Header** tab in the upper right of the shipping container heading.

Settings on the General FastTab

The following table describes the settings that are available on the **General** FastTab of the **Header** view of a shipping container.

Expand table

Field	Description
Shipping container	The name of the shipping container.
Voyage	The voyage that is associated with the shipping container.
Shipping container type	Enter the shipping container type. This field must be set. You can use it to determine the cost for freight, for example, by selecting the auto cost that is associated with the shipping container type.
Vessel	Enter or select the vessel. If the vessel isn't listed as a value, you can enter the vessel ID as free text. In that case, the main table isn't updated so that the vessel ID can be selected in this field later. Learn more in Vessels.
Unit type	Unit types are used as an additional means of grouping and identifying shipping containers. They're shown and selected on the shipping container

Field	Description
	page. Learn more in Set up unit types.
Refrigeration type	Refrigeration types are used as an additional means of grouping and identifying shipping containers, usually refrigerated containers. They're shown and selected on the shipping container page. Learn more in Set up refrigeration types.
Measurement	This field enables a measurement to be specified in the **Landed cost** module. Measurements are often used by organizations that don't know the individual volume or weight of goods, but that require a more accurate apportionment than the amount or quantity provides. The freight forwarder will provide the weight in kilograms or the cubic measurement, and you can put it at the level of either an item or the purchase order. It can be automatically updated if the parameter is selected, or it can be manually entered.
Measurement unit	The unit of measure that applies to the number in the **Measurement** field.
Actual weight	You can record the actual weight of the carton or container. This value can be used for verification against the maximum weight that is allowed in the setup of a shipping container.
Number of cartons	The number of cartons is automatically updated if the parameter is selected.
Description of goods	A description of goods can be selected on the shipping container or folio header. It's used to help identify a voyage, shipping container, or folio of goods. Learn more in Description of goods.
House air waybill/Bill of lading	You can specify the house air waybill or bill of lading for the shipping container.
Remarks	Additional information that is related to the shipping container.
Returnable	A value that indicates whether the shipping container can be returned after the voyage.
Voyage status	The status of the journey that is associated with the shipping container.
Purchase order status	The status of the purchase order that is associated with the shipping container.

Settings on the Delivery FastTab

The following table describes the settings that are available on the **Delivery** FastTab of the **Header** view of a shipping container.

Expand table

Field	Description
Created date and time	The date and time when the container was created.
Ex-factory date	This date is usually provided to the factory/vendor to indicate when you expect the goods to leave its premises. When you work with a factory in Asia, this date is often required instead of the date that you expect the goods by. (By contrast, for a local delivery, the date that you expect the goods by is required.) This field can be filled in from the purchase order lines in the shipping container list.
Ship date	This date can be printed on the purchase order document. It usually informs the factory/vendor about the date that the goods should be delivered to the port by. This field is for informational purposes only. It isn't used to estimate the expected delivery date of the goods in the shipping container. This field can be set so that it's automatically updated when the tracking control page is updated.
In to store date	The earliest date when the goods from the purchase orders that are linked to the voyage will be available for sale.
Estimated delivery date	Usually, the date when the goods are due to arrive in the warehouse. This field is for informational purposes only. It isn't used to calculate master planning on the purchase order lines in the shipping container. The expected delivery date on the purchase order lines is updated through tracking control. This field can be set up so that it's updated when the tracking control page is updated.
Departure date	Usually, the date when the plane or vessel actually leaves the overseas port.
ETA at shipping port	The estimated arrival date at the destination port ("to" port).
Original documents received	Optional: Update the date when the original documents were received.
Broker advised	Optional: Update the date when the broker was advised.
Original bill of lading sent	Optional: Update the date when the original bill of lading was sent.
Goods released	Optional: Update the date when the goods were released.
Customer appointment	Optional: Update the customer appointment date.
Delivered at warehouse	Optional: Update the date when the goods were delivered to the warehouse.

Field	Description
Verification date	Optional: Update the verification date.
Delivery instructions	Optional: Update the date of the delivery instructions.
From port	The port that the items will be shipped from.
To port	The port that the items will be shipped to.
Local forwarder	This field is for informational purposes only. The local forwarder should be selectable from the vendor table.
Local transport date	Enter the date that the local transport is booked for. This field can help the warehouse do its planning.
Local transport time	Enter the time slot. This field can help the warehouse do its planning.
Journey template	The journey template that is specified for the voyage. The journey template provides the details of the "to" and "from" ports, and the lead times that are associated with the tracking control of the shipping container.

Settings on the Other FastTab

The following table describes the settings that are available on the **Other** FastTab of the **Header** view of a shipping container.

Expand table

Field	Description
Rental	A value that indicates whether the shipping container is a rental shipping container. Rental costs might be associated with rental containers.
Converted to rental	A value that indicates whether the shipping container was converted to a rental shipping container. Rental costs might be associated with rental containers.
Original voyage	If the shipping container was moved to a new voyage, the original voyage.
Used	Use this to record whether a rental shipping container has been used. It is for informational purposes only.
Expected loading date	The date when the shipping container is expected to be loaded with goods.

Field	Description
Our seal number	Enter the seal number that your company uses internally for the shipping container.
Shipping company seal number	Enter the seal number that the shipping company or agent provided for the shipping container.
Examination certificate applied date	The date when an examination was requested for the shipping container.
Examination certificate received date	The date when the examination certificate was received.
Examination certificate expiry date	The date when the examination certificate will expire.
Examination certificate number	The certificate number of the certificate that was issued after an examination was done.

Shipping container Lines view

To open the **Lines** view, open a shipping container, and then select the **Lines** tab in the upper right of the shipping container heading.

Information on the Shipping container FastTab

The **Shipping container** FastTab in the **Lines** view shows information about the folio. Most of this information also appears in the **Header** view, as described earlier in this article.

Information and buttons on the Lines FastTab

The **Lines** FastTab in the **Lines** view shows details about each full or partial purchase order line that is included in the current shipping container.

The following table describes the buttons that are available on the **Lines** FastTab in the **Lines** view.

Expand table

Button	Description
Remove	Remove the selected purchase order line from the voyage.
Inventory > Transactions	View inventory transactions for the selected purchase order line. If you're using goods in transit, the original order and the goods-in-transit orders are also shown.
Inventory > Display dimensions	Open a dialog box where you can select the inventory dimensions that appear for the transactions that you view.
Refresh	Update information that is related to the line amount, weight, or volume of the selected purchase order line.

Information on the Lines details FastTab

The **Lines details** FastTab in the **Lines** view shows details about the purchase order line that is currently selected on the **Lines** FastTab.

Feedback

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Provide product feedback

Manage folios

Article • 02/14/2025

A folio is a subset of a container and is often determined by customs regulations. It can consist of one vendor's goods for one entity or company per shipment. The goods in a folio are managed in one container. One container can have many folios, but a folio can't be spread across several containers.

Use folios to split the goods within a container into different sections. When a container is being loaded, all the goods belonging to a folio are loaded in one go. Separate loads of purchase line items generate different folios. Therefore, a new folio is automatically created when new lines are added to a container.

To open the **All folios** page, go to **Landed cost** > **Folios** > **All folios**. This page shows a list of all current folios. You can use the buttons on the Action Pane to create, delete, and work with folios. Select any folio in the list to view its details on the **Folios** page.

Action Pane

The Action Pane on the **All folios** and **Folios** pages provides buttons that let you work with a selected folio. Each button performs a single action. The Action Pane also includes tabs, each of which, in turn, provides a set of related buttons. Except where noted, all buttons and tabs that are described in the following subsections are available both in the list view (that is, on the **All folios** page) and in the detailed view (that is, on the **Folios** page).

Buttons that appear directly on the Action Pane

The following table describes the buttons that are available directly on the Action Pane.

Expand table

Button	Description
New	Create a folio.
Delete	Delete the open or selected folio.
Voyage costs	Open the **Voyage costs** page, where you can view and add folio-level costs to all the goods in the voyage. When folio costs are manually added to the voyage, they're automatically added to the cost inquiry page and apportioned to every good according to the method that is specified on the **Voyage costs** page.

Buttons on the Manage tab

The following table describes the buttons that are available on the **Manage** tab of the Action Pane.

Expand table

Button	Description
Post receipts list	Post a receipt list for all purchase order lines in the folio.
Post product receipt	Post a product receipt for all purchase order lines in the folio.
Post invoice	Post an invoice for all purchase order lines in the folio.
Ship transfer order	Post a transfer order for all transfer order lines that are related to the current folio in the related shipment.
Receive transfer order	Post a transfer order receipt for all transfer order lines that are related to the current folio in the related shipment.
Receive goods in transit	Receive all order lines that are in transit in the folio.
Documents received	Update the setting of the **Documents received** option to *Yes*. You can use this button to lock the item and/or purchase line so that it can't be updated further.
Find auto costs	Find relevant voyage costs. If these costs have already been found or updated, you receive the following message: "Un-invoiced cost lines exist. Do you want to overwrite them?" Voyage costs that are attached to the folio and that have been invoiced won't be overwritten.
Create arrival journal	Generate an arrival journal for organizations by using advanced warehouse features. You can select **Initialize quantity** (recommended), **Create from goods in transit**, and/or **Create from purchase orders**. The last option depends on whether goods-in-transit processing is being used.
Accrue costs	Accrue costs where a cost type has a ledger account specified for the debit. This button is typically used when the stock is in transit, or when goods have been received and invoiced.

Buttons on the General tab

The following table describes the buttons that are available on the **General** tab of the Action Pane.

Expand table

Button	Description
Receipts list	Post a receipt list for all purchase order lines in the folio.
Product receipt	View the product receipt record, if it's used.
Item arrival	View the item arrival journal, if it's used.
Costs inquiry	Open the cost inquiry page to view all the costs of a voyage, including the shipping container, folio, and purchase order. You can adjust the exact view of the page by using the View action. On the cost inquiry page, you can view any of the areas, plus the item and cost type code. By removing these items, you can adjust the page by grouping together costs. This capability can be useful if you're using sizes and colors. You can change the dimensions that are shown on the page. The **Costs** page shows only cost type codes where the **Dr** entry on the **Posting** tab is set to *Item*.

Header view

To open the **Header** view, open a folio, and then select the **Header** tab in the upper right of the folio heading.

Settings on the General FastTab

The following table describes the fields that are available on the **General** FastTab of the **Header** view of a folio.

Expand table

Field	Description
Folio	The name of the folio. This name is automatically generated when the folio is created.
Voyage	The voyage that is associated with the folio.
Customs broker	Select the customs broker for the folio. Customs brokers are defined on the vendor. They enable created costs to be determined automatically.
House air waybill/Bill of lading	Specify the house air waybill or bill of lading that applies to the folio.

Field	Description
Company	The legal entity (company) that is associated with the folio.
Cargo control number	This field is used by customs departments in some countries/regions.
Measurement	This field enables a measurement to be specified in the **Landed cost** module. Measurements are often used by organizations that don't know the individual volume or weight of goods, but that require a more accurate apportionment than the amount or quantity provides. The freight forwarder will provide the weight or cubic measurement, and you can put it at the level of either an item or the purchase order. It can be automatically updated if the parameter is selected or manually entered.
Measurement unit	The unit that applies to the specified measurement.
Number of cartons	The number of cartons in the folio. This field can be automatically updated, depending on the parameter selection.
Vendor account	Select the vendor that is associated with the folio. This field is for informational purposes only. It doesn't affect any functionality.
Name	The name of the selected vendor account.
Remarks	Enter any additional information that is related to the folio.
Description of goods	Select a goods description to help identify the folio. Learn more in Description of goods.
Valuation date	This field is related to the duty entry page. The **Landed cost** module will use the customs exchange rate for the date that you set here. The default value is the date on the duty entry page.
Customs ID	Enter the customs ID. The customs departments in countries/regions provide this ID.
Tariff code	Enter a tariff code to associate with the folio. This code is typically required (and defined) by the country/region that you're shipping to.

Settings on the Delivery FastTab

The following table describes the settings that are available on the **Delivery** FastTab of the **Header** view of a folio.

Expand table

Field	Description
Folio date	Select a date to associate with the folio. The default value is the creation date of the voyage.
ETA at shipping port	The estimated time of arrival (ETA) date at the destination port ("to" port).
Estimated delivery date	Usually, the date when the goods are due to arrive in the warehouse. This field isn't used when the estimated delivery date is calculated. (The tracking control estimated delivery date is used instead.) To set this field so that the value matches the tracking control estimated delivery date, use the Tracking control center.
Original documents received	The date when the original documents were received.
Broker advised	The date when the broker was advised.
Original bill of landing sent	The date when the original bill of lading was sent.
Goods released	The date when goods were released.
Customer appointment	The customer appointment date.
Delivered at warehouse	The date when goods were delivered to the warehouse.
Verification date	The verification date.
Delivery instructions	The date when the delivery instructions were received.
From port	The port that the voyage departs from.
To port	The port where the voyage arrives. The shipping container might have a different port, because the ship might stop at multiple ports.

Settings on the Export FastTab

The following table describes the settings that are available on the **Export** FastTab of the **Header** view of a folio.

Expand table

Field	Description
Exporter	The exporter can be stored on the folio. In international trade, you might send a purchase order to one company but receive the goods from another company. Tracking and documentation are required by customs. The name and address of the exporter can be stored here.
Name	The name of the selected exporter.

Lines view

To open the **Lines** view, open a folio, and then select the **Lines** tab in the upper right of the folio heading.

Information on the Folio FastTab

The **Folio** FastTab in the **Lines** view shows information about the folio. Most of this information also appears in the **Header** view, as described earlier in this article.

Information and buttons on the Lines FastTab

The **Lines** FastTab in the **Lines** view shows details about each full or partial purchase order line that is included in the current folio.

The following table describes the buttons that are available on the **Lines** FastTab in the **Lines** view.

Expand table

Button	Description
Remove	Remove the selected purchase order line from the voyage.
Inventory > Transactions	View inventory transactions for the selected purchase order line. If you're using goods in transit, the original order and the goods-in-transit orders are also shown.
Inventory > Display dimensions	Open a dialog box where you can select the inventory dimensions that appear for the transactions that you view.
Refresh	Update information that is related to the line amount, weight, or volume of the selected purchase order line.

Information on the Lines details FastTab

The **Lines details** FastTab in the **Lines** view shows details about the purchase order line that is currently selected on the **Lines** FastTab.

Feedback

Was this page helpful? Yes No

Provide product feedback

Track inbound voyages and shipping container journeys

09/19/2025

The **Inbound tracking** page lets you track the progress of your voyages and shipping container journeys. Each voyage and journey is broken down by *activities*, each of which has its own row on the page. You can use the page to view and enter estimated dates and actual dates by activity.

Typically, depending on the setup in the Tracking control center, these activities automatically show the estimated landing date at the final destination. Also depending on the setup, the final date usually updates the delivery date or confirmed date on purchase order lines. For transfer order lines, you can set up the system to update the receipt date.

To open the **Inbound tracking** page, go to **Landed cost** > **Tracking** > **Inbound tracking**.

Filter the activities list

You can use the **Voyage** and **Shipping container** fields at the top of the **Inbound tracking** page to filter the page so that it shows only activities that are associated with the selected voyage and/or shipping container.

Update tracking information

To update the schedule for a voyage or journey, enter the start date for the first activity. The estimated end date of the last activity is then updated. The setup for each leg and journey template in the Tracking control center determines the estimated lead times. The estimated end dates use the lead time from the start date of the activity. Then, when you record the actual end date, the start date of the next activity updates to the same date as the previous activity's actual end date. The actual lead time updates so that comparison and analysis can be done. You can enter additional notes in the **Note** field.

The order of the activities in the grid is determined by the order of the legs in the relevant journey template. If you change the order of the legs in the attached journey, the tracking control also changes.

You can update the dates for all containers by selecting **Update start date** or **Update actual end date** on the Action Pane. Alternatively, you can enter the dates per container on the shipment. This approach allows for greater flexibility, because containers can be split in a multi-journey environment.

> **Note**
>
> The Landed cost tracking control center feature doesn't consider vendor lead times on individual purchase lines when calculating expected delivery dates.

Buttons on the Action Pane

Use the buttons on the Action Pane of the **Inbound tracking** page to work with inbound voyages and journeys. The following table describes the buttons that are available.

Expand table

Button	Description
Edit	Edit a voyage or shipping container journey.
New	Create a new voyage or shipping container journey.
Delete	Delete a selected voyage or shipping container journey.
Update start date	Update the start date of an activity for all containers in a voyage. When you update the start date of a specific activity or leg of a journey, the subsequent estimated start dates are updated based on the specified lead time.
Update actual end date	Update the end date of an activity for all containers in a voyage. When you update the end date of a specific activity, the subsequent activities are updated based on the specified lead time.
Add activities	Manually add an activity to a voyage. For example, you might add a fumigation activity. The addition might cause the confirmed delivery date of the goods in the vessel or voyage to change. When you add an activity to the journey, the estimated days aren't entered until the start date of a journey leg is updated.

Information and settings on the Overview tab

The **Overview** tab shows a list of all the activities that belong to the voyage and/or shipping container selected at the top of the page. The following table describes the fields that are available for each activity.

Expand table

Field	Description
Leg	The leg ID for the activity, as defined by the journey template.

Field	Description
Mode of delivery	The expected delivery method for the activity.
Activity	This field usually identifies the job or task that is associated with the activity. Typical examples include *Sailing* and *Clearance*.
Description	A longer description of the activity.
Start date	This field initially shows the estimated start date of the activity. However, after you enter the previous activity's actual end date, it shows the actual start date. This field is used to determine the estimated end date, based on the lead time.
Estimated end date	The system calculates the value of this field based on the start date and lead time. You usually don't edit the value directly.
Actual end date	Update this field as soon as possible after the activity occurs. The actual lead time is then updated.
Estimated days	The estimated time (in days) that is required to complete the activity.
Actual days	The actual time (in days) that is required to complete the activity.
Note	Use this field to add miscellaneous notes and details about the activity.

Information and settings on the General tab

The **General** tab shows information about the leg selected on the **Overview** tab. Although it repeats some of the information that appears on the **Overview** tab, it also includes additional information about the selected leg.

Estimate and manage landed costs

09/19/2025

The system uses your auto cost setup to estimate your landed cost. You can also define various scenarios to deliver a more accurate estimate. The system stores these scenarios so you can review them later and compare them to actual costs on a report. You can also update the item price.

Set up cost estimates

Set up cost templates

Cost templates establish default settings that users who get the estimate don't necessarily know. Templates can help reduce complexity in the estimating process by minimizing the selections that users must specify to get an accurate estimate. If you're using the standard cost model, you can use cost templates while you set up the cost of goods.

To set up your cost templates, go to **Landed cost** > **Costing setup** > **Cost templates**. On the **Cost templates** page, the list pane on the left shows all current cost templates. You can use the buttons on the Action Pane to create, delete, and edit templates.

The following table describes the fields that are available for each template.

Expand table

Field	Description
Cost template	Enter a unique name for the cost template. The name typically describes the factor or cost multiplier for the template.
Description	Enter a description of the cost template.
Shipping company	Select the shipping company that should be applied when the template is used.
Mode of delivery	Select the mode of delivery, such as sea or air, that should be applied when the template is used. This field helps determine the auto costs that are associated with the goods on a cost estimate.
Shipping container type	Select the type of shipping container that should be applied when the template is used. This field helps determine the auto costs that are associated with the goods on a cost estimate.
Customs broker	The customs broker (vendor) that should be applied when the template is used. This

Field	Description
	field helps determine the auto costs that are associated with the cost estimate.
Factor	Enter the multiplier (percentage) that should automatically be applied to the cost estimate when the template is used. For example, to add 10 percent to the calculated cost estimate, enter *1.10*.

Create a cost estimate

Use the **Cost estimate** dialog box to generate a new cost estimate that is based on a selected cost template, a selected set of items, and other details of a journey. These settings determine the estimated landed costs of goods. Use these cost estimates primarily to work with standard cost items. By adding the estimated landed costs to the standard cost of goods in inventory, you experience smaller variance transactions when you add the goods to a voyage, because the standard cost reflects the estimates of those landed costs.

To open the **Cost estimate** dialog box, go to **Landed cost** > **Periodic tasks** > **Cost estimate**. Then set the fields that are described in the following subsections. When you're finished, select **OK** to create the estimate. The **Cost estimate** page (**Landed cost** > **Inquiries** > **Cost estimates**) then appears and shows your new estimate, as described later in this article.

Settings on the Parameters tab

The following table describes the fields that are available on the **Parameters** tab of the **Cost estimate** dialog box.

Expand table

Field	Description
Cost template	Select a cost template. The settings that are associated with the selected template determine the auto costs that are applied.
Estimate date	By default, this field is set to today's date, but you can change the value. The specified date selects the appropriate sales prices, purchase prices, auto costs, and the exchange rate if a shipping rate is used.
Measurement	Costs might depend on a measurement. For example, when air freight is used, volumetric pricing might apply. If the cost depends on a measurement, enter the value of that measurement. Otherwise, set this field to *1*, unless you're certain that no apportionment occurs by using measurement. Enter a decimal value. The unit is the one that is defined in the applicable auto cost record.
Journey	Select a journey template. This field determines the correct auto costs that should be

Field	Description
template	applied.
From port	The port that the items are shipped from. This field is set based on the selected journey template.
To port	The port that the items are shipped to. This field is set based on the selected journey template.
Currency	Select the currency that the items are purchased in.
Containers	If multiple containers are typically used, specify the number of containers. The system then uses costs for multiple containers when it estimates the costs.
Folios	If multiple folios are typically used, specify the quantity. (The quantity is usually *1*.)
Site	Specify the site where the goods are delivered.

Settings on the Items tab

On the **Items** tab, you can add as many items to the estimate as you need. Use the toolbar to add items to the grid or remove items. Select **Inventory** > **Display dimensions** on the toolbar to open a dialog box where you can add dimension columns to the grid or remove columns.

The following table describes the fields that are available for each item.

Expand table

Field	Description
Item number	Select the item to determine the price for. (If the item doesn't yet exist in the system, create a dummy item, optionally attach it to a voyage item cost group, and then either leave the price blank, or create or change the price.)
Vendor account	Select the vendor to use for the estimate of this item. If a default vendor is assigned to the item, this field is automatically set.
Quantity	Select the quantity that you want to purchase.
Cost price	The system uses its pricing rules to find an initial price, but you can override that price if necessary.
Sales price	Enter the expected sales price of the goods.
Measurement	Enter a decimal value for the measurement of the goods. The unit is the one that is set up for the applicable released product.
Update item	Select this check box to update the weight or volume measurement of the released

Field	Description
weight/volume	product so that it matches the **Measurement** value that you enter for this row.
(Other dimensions)	Depending on the dimensions that you select to show, you might see additional columns of information about each item.

To view or adjust the volume and weight details for an item, select the item in the grid, and then use the fields at the bottom of the page.

Manage estimated costs

To view and edit the cost estimates that you create, go to **Landed cost** > **Inquiries** > **Cost estimates**. On the **Cost estimates** page, the list pane on the left shows all current cost estimates. You can use the buttons on the Action Pane to work with a selected estimate. You can't create a new cost estimate from the **Cost estimates** page. Instead, use the **Cost estimate** dialog box (**Landed cost** > **Periodic tasks** > **Cost estimate**), as described earlier in this article.

The **Cost estimates** page shows how each estimated cost is derived. It also shows the estimated landed cost for each item. You can modify a cost estimate by changing the cost price and currency that is associated with the various goods. You can also modify the associated voyage costs at both the voyage level and the container level. When you use this page to modify the costs, you're prompted to recalculate the estimated costs for the items in the cost estimate. When you're ready, you can use the estimates to update the cost price of the items in the cost template.

Information on the header

The top of the **Cost estimates** page shows the settings that are used to generate the selected cost estimate, as described in the previous section.

Settings and buttons on the Lines FastTab

The **Lines** FastTab lists each item that is included in the current estimate. The following table describes the fields that are available for each row.

Expand table

Field	Description
Vendor account	The vendor account that's associated with the item.
Item number	The item number.

Field	Description
Quantity	The number of items that you use to determine the cost.
Cost price	The cost price according to the trade agreement. This value is shown in the local currency.
Measurement	The individual measurement. The unit is the one that is defined for the applicable released product.
Estimated landed cost	The estimated landed cost for the total quantity.
Per unit	The estimated landed cost divided by the quantity.
(Other dimensions)	Depending on the dimensions that you select to show, you might see additional columns of information about each item.

The following table describes the buttons that are available on the toolbar on the **Lines** FastTab.

Expand table

Button	Description
Add	Add items to the cost estimate. After you add items, you must select **Recalculate** on the Action Pane.
Remove	Remove items from the cost estimate. After you remove items, you must select **Recalculate** on the Action Pane.
Voyage costs	Open a page where you can view and edit various types of voyage costs for the item.
Inventory > Display dimensions	Open a dialog box where you can add dimension columns to the grid or remove columns.

Settings on the General FastTab

The **General** FastTab shows details about the item that you selected on the **Lines** FastTab. Much of this information is repeated from the **Lines** FastTab and you can edit it in either place. You can also see additional details here. The values of the extra fields are based on the setup of the applicable released product.

Settings on the Dimension FastTab

The **Dimension** FastTab shows values for all available inventory dimensions for the item selected on the **Lines** FastTab, regardless of the dimensions that you choose to show there. Any values that you see here come from the applicable cost estimate template. You can choose to include these values in the cost estimate template.

Buttons on the Action Pane

Use the buttons on the Action Pane of the **Cost estimates** page to work with the selected cost estimate. The following table describes the buttons that are available.

Expand table

Action	Description
Cost Inquiry	View all costs for the voyage. You can view costs at the level of the individual item as required.
Update standard cost	Update the standard cost by using the default costing version that is defined on the **Landed cost parameters** page. You can override this version. **Note:** If an item has several item dimensions (for example, various sizes, colors, and configurations), but you don't select these dimensions for the estimate, the system creates a pending price for each combination. **Important:** The price breakdown is created and is used to determine the standard cost variance per landed cost.
Voyage costs	View and edit voyage costs for all goods in the shipment.
Recalculate	Update the estimated landed costs after you update, add, or remove voyage costs.
Lock	Lock the cost estimate record so that no more changes can be made.

Item cost price update

The **Item cost price update** periodic task updates all cost estimates that match the filters that you set when you run the task. The effect is similar to the effect of selecting **Update standard cost** on the Action Pane for a single estimate. However, in this case, the update applies to all matching estimates.

To run the periodic task, follow these steps.

1. Go to **Landed cost** > **Periodic tasks** > **Item cost price update**.
2. In the **Update cost price from estimate** dialog box, set the following fields as required to limit the scope of the update:

- **Version** – Update only estimates that use the specified costing version.
- **Site** – Update only estimates that use the specified site.
- **Cost template** – Update only estimates that use the specified template.
- **To port** – Update only estimates that use the specified "to" port.
- **Estimate date** – Update only estimates that have the specified date.

3. Set the options on the **Records to include** and **Run in the background** FastTab as usual for periodic tasks.

4. Select **OK** to run the task.

> ⓘ **Note**
>
> This periodic task runs successfully only if the following information is available:
>
> - Each relevant product must have a **Gross depth**, **Gross width**, and **Gross height** defined.
> - Each relevant vendor must have a **From port** defined.

Show landed costs in the accounting distribution of product receipts

This feature enables you to include landed costs in the accounting distribution of purchased product receipts. Therefore, you can more easily identify and track those costs. This feature doesn't affect the product receipt accounting logic that's used in other places in Microsoft Dynamics 365 Supply Chain Management.

Prerequisites

To use this feature, you must be running Supply Chain Management 10.0.34 or later, and you must turn on the *Source document and accounting distribution support for Landed Cost* feature in feature management.

View landed costs in the accounting distribution of product receipts

When you post a product receipt against a purchase order, Landed cost creates a source document line for each landed cost amount. The amount from Landed cost appears on the **Account distributions** page, which you can access from the product receipt.

The following illustration shows an example of accounting distributions that include landed costs.

Goods-in-transit processing and receiving

This article describes how to work with goods-in-transit orders. Only the **Landed cost** module uses this type of order. When an order or voyage is set up to use goods-in-transit processing, you don't have to wait until the goods are received in the warehouse before you can invoice them. Instead, the goods are invoiced when they leave the vendor's warehouse or port of origin, and the financial costs are recognized when the voyage begins. This functionality helps you correctly take ownership of inventory, because goods often become the property of your organization when they leave the shipping port.

When you use goods-in-transit orders, you receive the financially updated items in an interim warehouse known as a *goods-in-transit warehouse*. The goods stay in this warehouse until they can be received at the final destination warehouse (that is, the warehouse defined on the purchase order line). You can't manually remove them. When a goods-in-transit order is invoiced, goods are immediately moved to the goods-in-transit warehouse to signify the change in ownership.

As long as the items are in transit, they aren't available in inventory and can't be picked from inventory for a delivery. However, you can view the goods-in-transit inventory. You can also use the goods for master planning. In this case, use the confirmed delivery date on the purchase order line as the expected date when the inventory is available for consumption. The following sections describe the setup that is required to process inventory and voyages by using the goods-in-transit concept and functionality.

Required setup for goods-in-transit orders

Terms of delivery

When you enable the **Landed cost** module, the standard *terms of delivery* entity is enhanced to support the goods-in-transit feature.

When you set the **Goods in transit management** option to *Yes* for the applicable terms of delivery record, the goods are placed in the goods-in-transit warehouse. This action is triggered only if the inventory receipt isn't processed before an invoice is processed. When you set the delivery terms for an order to use goods in transit, users can no longer post a product receipt for the purchase order. If they try, an error occurs. The error message states that they must use the goods-in-transit functionality to proceed.

To work with delivery terms information for goods in transit, go to **Procurement and Sourcing** > **Setup** > **Distribution** > **Terms of delivery**. The following table describes the fields that the **Landed cost** module adds to the **Terms of delivery** page to support the goods-in-transit functionality. Both fields are on the **General** FastTab. For more information about the other fields on this page, see Terms of delivery (form).

Expand table

Field	Description
Shipping port mandatory	Set this option to *Yes* if a shipping port is mandatory when the delivery terms apply.
Goods in transit management	Set this option to *Yes* to use goods-in-transit management when the delivery terms apply.

Warehouses for goods in transit and under-delivery

When you enable the **Landed cost** module, the standard *warehouses* entity is enhanced to enable purchase orders to be invoiced while the goods are in a goods-in-transit warehouse.

The Landed cost module adds two new types of warehouse: *goods in transit* and *under-delivery*. You can select warehouses of both types as default warehouses. To successfully process goods-in-transit orders, you must configure both the goods-in-transit warehouse and the under-delivery warehouse on the **Warehouses** page. Then, for each default warehouse for use with Landed cost and goods in transit, ensure that the goods-in-transit warehouse and under-delivery warehouse are selected for the available warehouses of each type.

The *goods in transit* warehouse type are associated with your goods-in-transit warehouse. That warehouse is used to process the goods on goods-in-transit orders before they're received at the final destination warehouse.

When the purchase order invoice is posted for a voyage that is enabled for goods-in-transit functionality, inventory is purchased in the receiving warehouse. It's then immediately transferred to the configured goods-in-transit warehouse. This transfer signifies the change in ownership of the goods that aren't yet physically arrived at the destination warehouse.

In general, one goods-in-transit warehouse is enough for each site if Site and Warehouse are the only inventory dimensions used for inventory management. If the Location inventory dimension is also used, ensure to set up a goods-in-transit warehouse for each combination of a site and warehouse, so that the default location can also be specified.

A default receipt location must be specified for the under-delivery and goods-in-transit warehouses. Go to **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**, and set the **Default receipt location** field.

The following table describes the fields that the **Landed cost** module adds to the **Warehouses** page to support the goods-in-transit functionality. Both fields appear on the **General** FastTab. For information about the other fields on the page, see Warehouses (form).

Expand table

Field	Description
Goods in transit warehouse	Identify the goods-in-transit warehouse that is related to the main warehouse.
Under delivery warehouse	Identify the under-delivery warehouse that is related to the main warehouse.

> **Note**
>
> Warehouses that have a **Type** value of *Goods in transit* or *Under* can't be enabled for warehouse management processes (WMS). If you try to set the **Use warehouse management processes** option to *Yes* for warehouses of one of these types, you receive the following error message: "Landed cost goods in transit warehouse and under warehouse shouldn't use warehouse management processes. Please replace it with new warehouse if warehouse management processes parameter can't be disabled."

Goods-in-transit orders

You can review and manage goods-in-transit orders directly in the **Landed cost** module. You can process goods-in-transit orders directly from the **Goods in transit orders** page. Alternatively, you can go to the voyage that is associated with the goods-in-transit orders, and process the whole voyage, shipping container, or folio. When you invoice a voyage and create goods-in-transit orders, a new goods-in-transit order is created for each combination of inventory and inventory dimensions that is associated with the purchase order line.

To manage goods in transit, Landed cost uses a two-step procedure:

1. When a purchase order is invoiced, a goods-in-transit order is created. A status of *In transit* is assigned to it.

   > **Note**

After the goods-in-transit order is created, the invoice date can't be changed.

2. The goods-in-transit order is processed on the **Goods in transit orders** page. It's then received in the warehouse that is specified on the purchase order. At that point, the status is changed to *Received*.

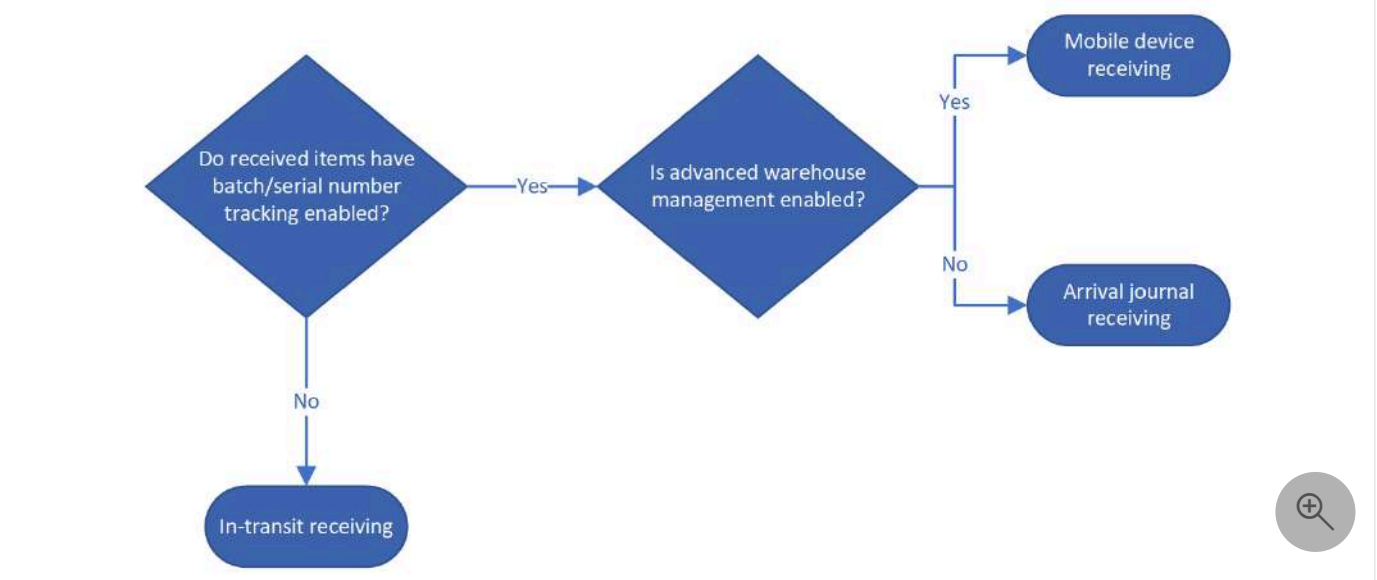
To work with goods-in-transit orders, go to **Landed cost** > **Periodic tasks** > **Goods in transit orders**.

Receiving stock from the goods-in-transit warehouse

You can receive goods from a goods-in-transit order in several ways:

- If warehouse management processes (WMS) isn't enabled for items, you can use in-transit receiving or post an arrival journal.
- If both WMS and batch/serial number tracking are enabled for items, you can use a mobile device to receive goods from a goods-in-transit order.
- If WMS is enabled for items, but batch/serial number tracking isn't, you can use a mobile device or post an arrival journal to register the receipt of goods from a goods-in-transit order.

The following illustration presents the preceding information in the form of a flow diagram to show when you should use each type of receipt method to receive goods-in-transit orders.



In-transit receiving

You can do in-transit receiving from any of the following pages:

- On the **Goods in transit orders** page, select the line, and then, on the Action Pane, select **Receive**.
- On the **All voyages** page, select or open a voyage. Then, on the Action Pane, on the **Manage** tab, in the **Goods in transit** group, select **Receive goods in transit**.
- On the **All shipping containers** page, select or open a shipping container. Then, on the Action Pane, on the **Manage** tab, in the **Goods in transit** group, select **Receive goods in transit**.
- On the **All folios** page, select or open a folio. Then, on the Action Pane, on the **Manage** tab, in the **Goods in transit** group, select **Receive goods in transit**.

> ⓘ Note
>
> In-transit receiving is normally used in situations where locations and batch/serial number tracking aren't used.

Arrival journal

You can also receive goods by creating an arrival journal. You can create an arrival journal directly from the voyage page.

1. Open the voyage, container, or folio.
2. On the Action Pane, on the **Manage** tab, in the **Functions** group, select **Create arrival journal**.
3. In the **Create arrival journal** dialog box, set the following values:
    - **Initialize quantity** – Set this option to *Yes* to set the quantity from the in-transit quantity. If this option is set to *No*, no default quantity is set from the goods-in-transit lines.
    - **Create from goods in transit** – Set this option to *Yes* to take quantities from the selected in-transit lines for the selected voyage, container, or folio.
    - **Create from order lines** – Set this option to *Yes* to set the default quantity in the arrival journal from the purchase order lines. The default quantity in the arrival journal can be set in this way only if the quantity on the purchase order line matches the quantity on the goods-in-transit order.
4. Process the arrival journal as described in Register item receipts with an item arrival journal.

Warehouse mobile device receiving

Goods can be received by using a mobile device, as long as the mobile device menu and the **Warehouse management** module are set up to receive goods on a goods-in-transit order. This section describes the setup that is associated with goods-in-transit receiving.

To set up mobile devices for goods-in-transit processing, go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

Landed cost adds the following work creation processes to the mobile device menu items to support goods-in-transit processing:

- Goods in transit item receiving
- Goods in transit item receiving and putaway

The configuration settings for these processes resemble the settings for the purchase order receive and putaway work creation processes. However, the *Goods in transit item receiving and putaway* process also adds the following field.

- **Enable shipping container complete** – If this option is set to *Yes*, when the putaway work is completed, the Warehouse Management mobile app provides a button named **Shipping container complete**. When that button is selected, the worker is asked to confirm that the container is complete. At that point, all short receipts are processed as an under transaction. The **Shipping container complete** button works just like the **Close** checkbox on the goods-in-transit order receiving page, which is used to depreciate the remaining quantity. The **Shipping container complete** button is available only for under-delivery scenarios and for the **GIT Receive and Putaway** action on the mobile device. When a goods-in-transit order is received in full or as an over-delivery, this button isn't visible.

Specify goods-in-transit orders when receiving with a mobile device

Workers using the Warehouse Management mobile app can register the receipt of goods in transit even when multiple orders are associated with the same voyage, container, item number, and purchase order number. To do so, the worker starts by entering the voyage, container, item, and order numbers, and can then select the relevant goods-in-transit order from a drop-down list.

Differences in mobile device receiving for a goods-in-transit order flow versus a standard purchase order flow

The receipt process for goods-in-transit orders via the **Landed cost** module on a mobile device isn't always consistent with the receipt process for purchase orders when Landed cost and goods-in-transit orders aren't enabled.

Here are some examples of the differences:

- The **Voyage** and **Shipping Container** fields must be set for goods-in-transit receiving. By contrast, these fields aren't needed for standard purchase order receiving.
- *Assign putaway cluster* functionality for the putaway process isn't available for goods-in-transit receiving.
- In the normal receipt process that doesn't involve goods-in-transit orders, the worker can cancel in-process work. The license plate is then automatically deregistered. By contrast, putaway work can't be canceled at mid-stage for goods-in-transit receiving. In this situation, a movement journal must be used to move items back to the receiving location. The registration must then be manually undone.

Receive goods-in-transit orders by using a mobile device when serial/batch numbers are enabled

Workers using the Warehouse Management mobile app can register batch/serial numbers when receiving goods-in-transit orders that include items enabled for batch/serial number tracking. The system consolidates the received quantity for each batch/serial number into one work process and automatically assigns the numbers to the received items.

If batch and serial numbers are known when goods are received, they can be manually specified on the device during the receipt process. When more than one serialized item is received on the Warehouse mobile device, the voyage number, shipping container, item, quantity, and license plate number must be manually entered together with each new serial number.

Alternatively, you can let the system automatically assign batch/serial numbers when you receive items that batch/serial number tracking is enabled for.

Under-receive when batch/serial number control is activated

To under-receive a serial-enabled item, let the system automatically assign batch/serial numbers. If a serial number is manually set for the quantity that is received, the remaining amount on the under-delivery warehouse can't be posted without a serial number.

Location directives

Landed cost adds a new work order type named *Goods in transit* to the **Location directives** page. Configure this work order type in the same manner as the purchase order work order types.

Work templates

This section describes features that the **Landed cost** module adds to work templates.

Goods in transit work order type

Landed cost adds a new work order type named *Goods in transit* to the **Work templates** page. Configure this work order type in the same manner as the purchase order work templates.

Work header breaks

Work templates that have a work order type of *Goods in transit* can be configured to split work headers. On the **Work templates** page, follow one of these steps:

- On the **General** tab for the template, set work header maximums. These maximums work in the same way that they work for purchase order work templates. (Learn more in purchase order work templates.)
- Use the **Work header breaks** button to define when the system should create new work headers, based on fields that are used for sorting. For example, to create a work header for each container ID, select **Edit query** on the Action Pane, and then add the **Container ID** field to the **Sorting** tab of the query editor. Fields added to the **Sorting** tab are available for selection as *grouping fields*. To set up grouping fields, select **Work header breaks** on the Action Pane, and then, for each field that you want to use as a grouping field, select the checkbox in the **Group by this field** column.

Landed cost creates an over transaction if the registered quantity exceeds the original order quantity. When a work header is completed, the system updates the status of the inventory transactions for the principal order quantity. However, it first updates the quantity that is linked to the over transaction after the principal is completely purchased.

If you cancel a work header for an over transaction that is already registered, the over transaction is first reduced by the canceled quantity. After the over transaction is reduced to a quantity of 0 (zero), the record is removed, and any additional quantities are unregistered against the principal order quantity.

Marking for goods-in-transit stock

Goods-in-transit transactions are marked to the purchase order transactions. They're then considered in inventory closing and are reflected in the inventory aging report as zero-on-hand items that have inventory value.

> **Note**
>
> If sales orders are marked to purchase orders to reserve incoming stock, goods-in-transit orders are split if a purchase order line is marked to multiple sales order lines. Any goods-in-transit order that is split must be manually received one by one.

Posting rules for landed cost

Landed cost adds two new posting rules that you can configure. Use these posting rules to financially post the direct purchase order invoice amounts and identify ownership of the goods when they leave the point of origin. This process replaces the concept of *goods received not invoiced*, because the goods are invoiced before they're received.

To work with posting profiles, go to **Inventory management** > **Setup** > **Posting** > **Posting**. On the **Purchase Order** tab, you see the following new posting rules:

- *Landed cost, goods-in-transit* – Specify the posting rules for goods-in-transit management.
- *Landed cost, cost charge accrual* – Specify the posting rules for charge account accrual.

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Process over/under transactions

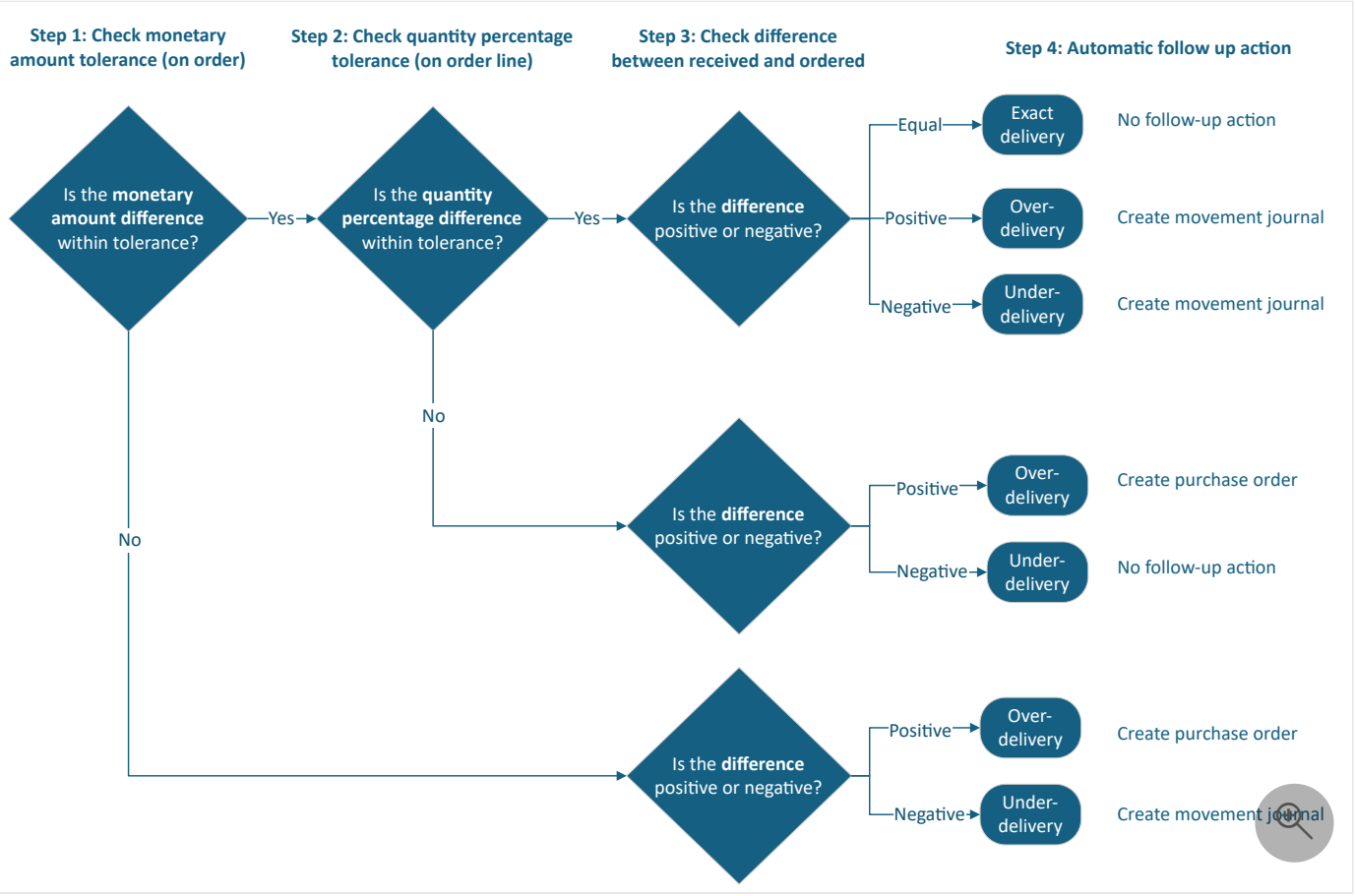
When the orders in a voyage are processed, the system expects the item quantity that is received in the final destination warehouse for consumption to match the quantity specified on the purchase order lines associated with the voyage. However, because the exact quantity on the purchase order lines isn't always received in the warehouse, the **Landed cost** module defines a set of rules that handle over-receiving and under-receiving of goods. These rules are especially important because the original purchase order has been invoiced and can no longer be modified.

By setting up the details of over/under transaction policies, you enable the system to determine how to manage the over-processing and under-processing of goods at the time of receipt. In some cases, an over/under transaction triggers an automatic action. In other cases, resolution requires manual action. You can also manually manage over and under inventory by using the **Over/under transactions** page.

Use a goods-in-transit order to handle over-deliveries and under-deliveries.

Over/under transaction process flow and examples

The following diagram shows a flowchart of the process the system uses to determine whether over/under transactions should be created.



The diagram highlights the following steps of the process:

1. When goods are received, the system compares the total monetary amount of the received purchase with the original purchase order. If there is a difference, the system determines whether it's within or outside the configured amount tolerance.
2. If the difference is within the amount tolerance, the system checks the percentage tolerance against the individual purchase order line.
3. The system determines whether the received amount or quantity is more or less than the ordered amount or quantity.
4. Based on this analysis, the system determines whether an over-delivery or an under-delivery occurred. It then takes the relevant follow-up action as required.

Use the **Over/under transactions** page to track whether the system created an over/under transaction, movement journal, and/or purchase order.

In some cases, a movement journal is automatically created in response to an over-delivery or under-delivery. This movement journal writes off or absorbs the difference between ordered and actual quantities.

In other cases, a purchase order is automatically created. For an over-delivery, the new purchase order records the additional quantity that was received. For an under-delivery, the purchase order

shows a negative quantity, so that the vendor can pay for the missing goods. This negative quantity works as a vendor return or as credit for a future order.

In cases where the system doesn't take any automatic action, you must decide whether to create a movement journal or a purchase order.

Example scenarios

Setup used in the scenarios

All four scenarios use the following setup.

The following over/under tolerances apply. (These tolerances are required so that all items can be used on the purchase order.)

- **Amount tolerance:** $100 (applied to the order)
- **Percentage tolerance:** 10% (applied to the order line)

The following purchase order exists:

- **Line 1:** 10 pcs at $100/ea
- **Line 2:** 20 pcs at $50/ea
- **Total monetary amount of the original purchase order:** $2,000

Because of this setup, any received purchase order where the total value is outside the $1,900–$2,100 threshold is processed as an over/under transaction. Any difference between the quantity of the original and received purchase order lines is considered to be within the percentage tolerance if it's within the following thresholds:

- **Line 1:** 9–11
- **Line 2:** 18–22

Scenario 1

The following items are received and, as a result, the following actions are taken (as shown in the flowchart earlier in this article):

- **Total order:** The total value of the purchase order is $2,050.
  - **Step 1:** The purchase order is *within* the amount tolerance.
- **Line 1:** 10 pcs ($1,000) are received.

    - **Step 2:** Because the exact quantity of the purchase order line was received, the order line is *within* the percentage tolerance threshold.
    - **Step 3:** The received quantity of the purchase order line *equals* the original purchase order line.
    - **Step 4:** The exact quantity of the purchase order line was received. Therefore, the system takes no action.

- **Line 2:** 21 pcs ($1,050) are received.

    - **Step 2:** The difference between the quantity of the original and received purchase order lines is *within* the percentage tolerance threshold.

    - **Step 3:** The received quantity of the purchase order line is *more than* the original purchase order line.

    - **Step 4:** An over-delivery occurred. Therefore, the system creates the following items:
        - An over transaction
        - A movement journal

Scenario 2

The following items are received and, as a result, the following actions are taken (as shown in the flowchart earlier in this article):

- **Total order:** The total value of the purchase order is $2,050.
    - **Step 1:** The purchase order is *within* the amount tolerance.

- **Line 1:** 9 pcs ($900) are received.

    - **Step 2:** The difference between the quantity of the original and received purchase order lines is *within* the percentage threshold.

    - **Step 3:** The received quantity of the purchase order line is *less than* the original purchase order line.

    - **Step 4:** An under-delivery occurred. Therefore, the system creates the following items:
        - An under transaction
        - A movement journal

- **Line 2:** 23 pcs ($1,150) are received.

- **Step 2:** The difference between the quantity of the original and received purchase order lines is *outside* the percentage threshold.
- **Step 3:** The received quantity of the purchase order line is *more than* the original purchase order line.
- **Step 4:** An over-delivery occurred. Therefore, the system creates the following items:
  - An over transaction
  - A purchase order

Scenario 3

The following items are received and, as a result, the following actions are taken (as shown in the flowchart earlier in this article):

- **Total order:** The total value of the purchase order is $1,900.
  - **Step 1:** The purchase order is *within* the amount tolerance.
- **Line 1:** 7 pcs ($700) are received.
  - **Step 2:** The difference between the quantity of the original and received purchase order lines is *outside* the percentage threshold.
  - **Step 3:** The received quantity of the purchase order line is *less than* the original purchase order line.
  - **Step 4:** An under-delivery occurred. Therefore, the system creates an under transaction.
- **Line 2:** 24 pcs ($1,200) are received.
  - **Step 2:** The difference between the quantity of the original and received purchase order lines is *outside* the percentage threshold.
  - **Step 3:** The received quantity of the purchase order line is *more than* the original purchase order line.
  - **Step 4:** An over-delivery occurred. Therefore, the system creates the following items:
    - An over transaction
    - A purchase order

Scenario 4

The following items are received and, as a result, the following actions are taken (as shown in the flowchart earlier in this article):

- **Total order:** The total value of the purchase order is $2,950.
  - **Step 1:** The purchase order is *outside* the amount tolerance. Therefore, step 2 is skipped.
- **Line 1:** 20 pcs ($2,000) are received.
  - **Step 3:** The received quantity of the purchase order line is *more than* the original purchase order line.
  - **Step 4:** An over-delivery occurred. Therefore, the system creates the following items:
    - An over transaction
    - A purchase order
- **Line 2:** 19 pcs ($950) are received.
  - **Step 3:** The received quantity of the purchase order line is *less than* the original purchase order line.
  - **Step 4:** An under-delivery occurred. Therefore, the system creates the following items:
    - An under transaction
    - A movement journal

> **Note**
>
> When an over-delivery automatically triggers the creation of a purchase order, the system also posts a product receipt. However, invoices must always be manually posted for these purchase orders.
>
> Over-delivery and under-delivery quantities don't affect the estimated landed costs. Therefore, automatic costs that are calculated for a voyage or container aren't redistributed across the actual quantities that are received.
>
> Workers who use the Warehouse Management mobile app to receive goods aren't notified when an over-delivery or under-delivery occurs.

Set up over/under transactions

To enable over/under transactions for goods-in-transit orders, you must complete the following setup:

- Set up the good-in-transit feature as described in Goods-in-transit processing and receiving.

- Ensure that a movement journal is set up and specified for the over-delivery or under-delivery in the **Over/Under Delivery** field in the **Defaults** section on the **General** tab of the **Landed cost parameters** page (**Landed cost** > **Setup** > **Landed cost parameters**). As part of this setup, define an offset account.
- Set up over/under tolerances, over/under reasons, and tolerance groups, if applicable, on the **Over/under setup** page (**Landed Cost** > **Setup** > **Over/under setup**).

Over/under tolerances

Over/under tolerances define when a difference between the actual and purchased quantities of goods creates an over/under delivery transaction. You set over-delivery and under-delivery tolerances to specify the over and under quantities or volumes that can be processed on a voyage without causing an error. If the voyage line receipt is outside these tolerances, it must be modified and resolved before the voyage line can be closed for further processing.

By setting tolerance levels, businesses can avoid unexpected expenses that might arise when they receive a smaller quantity of goods than they expected. At the same time, if businesses receive a larger quantity of goods than they expected, they can benefit from the savings and adjust the landed cost calculation accordingly.

To configure the tolerances, go to **Landed cost** > **Over/under setup** > **Over/under tolerances**. There, you can view, edit, add, and remove tolerance records. The following table describes the fields that are available for each record.

Expand table

Field	Description
Account code	Define the scope of vendors that the tolerance applies to by selecting one of the following values: • **Table** – The over/under tolerance applies only to the vendor that is selected for the row. • **Group** – The over/under tolerance applies to all vendors in the vendor tolerance group that is selected for the row. • **All** – The over/under tolerance applies to all vendors.
Account relation	Select the vendor or vendor group that the over/under tolerance applies to, depending on the value that you selected in the **Account code** field.
Item code	Define the scope of items that the tolerance applies to by selecting one of the following values: • **Table** – The over/under tolerance applies only to the item that is selected for the row.

Field	Description
	• **Group** – The over/under tolerance applies to all items in the item tolerance group that is selected for the row. • **All** – The over/under tolerance applies to all items.
Item relation	Select the item or item group that the over/under tolerance applies to, depending on the value that you selected in the **Item code** field.
Amount tolerance	Enter the amount tolerance that should be applied to a whole purchase order. This value is a monetary amount. This field is optional.
Percentage tolerance	Enter the percentage tolerance that should be applied to an individual purchase order line quantity. This field is optional.

> **Note**
>
> The system first checks whether the total amount of the received purchase order falls within the amount tolerance. If it does, the system then checks the percentage tolerance to manage over/under delivery scenarios.

Over/under reasons

When an over or under quantity is associated with a voyage line that is received, you might have to identify the reason for the over or under quantity. In this case, you can select the over-delivery or under-delivery reason on the receiving line when the goods are received.

To set up over-delivery and under-delivery reasons, go to **Landed cost** > **Over/under setup** > **Over/under reasons**. There, you can view, edit, add, and remove the available over-delivery and under-delivery reasons. The following table describes the fields that are available for each reason.

Expand table

Field	Description
Over/under reason	Enter a unique name for the reason for the over-receiving or under-receiving transaction.
Description	Enter a description of the over/under reason.
Movement	Select the movement journal that is associated with the over/under reason. This field lists all available journals that a journal type of *Movement* is associated with on the **Inventory journal names** page (**Inventory management** > **Setup** > **Journal names** > **Inventory**).

Item over/under tolerance groups

Items that have similar tolerances can be grouped together. This setup lets you set the over/under tolerance for all items in that group at the same time.

To set up item over/under tolerance groups, go to **Landed cost** > **Over/under setup** > **Item over/under tolerance groups**. There, you can view, edit, add, and remove over/under tolerance group records. The following table describes the fields that are available for each record.

Expand table

Field	Description
Over/under tolerance group	Enter a unique name for the group. Choose a name that describes the tolerance, such as *1% Var*.
Description	Enter a description of the group.

> **Note**
>
> Configure an item to be part of an over/under tolerance group by using the **Over/under tolerance group** field on the **Purchase** FastTab of the **Released product** page.

Vendor over/under tolerance groups

You can group together vendors that regularly over-deliver or under-deliver. You can then set the over/under tolerance per group.

To set up vendor over/under tolerance groups, go to **Landed cost** > **Over/under setup** > **Vendor over/under tolerance groups**. There, you can view, edit, add, and remove over/under tolerance group records. The following table describes the fields that are available for each record.

Expand table

Field	Description
Over/under groups	Enter a unique name for the group. Choose a name that describes the type of vendors that belong to the group.
Description	Enter a description of the group.

> **Note**

Configure a vendor to be part of an over/under tolerance group by using the **Over/under tolerance group** field on the **Miscellaneous details** FastTab of the **Vendor** page.

View and process over/under transactions on the Over/under transactions page

Over and under transactions are generated when the quantity of goods that is put away doesn't match the initialized quantity. The quantity in the arrival journal should be updated only with the quantity that is put away.

When the receipt of voyage lines is processed, over/under tolerances can be set for a vendor. The items are then reviewed and validated.

If a received item is inside the tolerance, the system generates a movement journal. This journal can be specified on the voyage parameters page. In addition, an over/under transaction is created. For example, the order value is $100, the receipt value is $99, and these values satisfy the tolerance rules. In this case, a negative movement journal for the under-received quantity is created.

If a received item is outside the tolerance, the system generates an over/under transaction for the difference in quantity.

To view and process over/under transactions, go to **Landed cost** > **Inquiries** > **Over/under transactions**. There, an over/under transaction is associated with all the items in a voyage that are over-received or under-received. Use the **Over/under transactions** page to resolve all over/under transactions associated with voyages.

Don't use movement or counting journals to manually resolve under-delivery warehouse transactions. Instead, use the **Over/under transactions** page to manage the under-delivery warehouse quantities.

Upper overview tab

To view your over/under transactions, select the **Overview** tab in the upper part of the **Over/under transactions** page. The following table describes the fields that are available in the grid.

Expand table

Field	Description
Over/under transaction number	The unique name of the over/under transaction record that was automatically created when the arrival journal was processed.
Voyage	The voyage that the purchase line was attached to.
Shipping container	The container that the purchase line was attached to.
Arrival journal	The arrival journal that the over/under line was generated from.
Reference	A reference to the purchase order that is associated with the over/under transaction.
Number	The purchase order that is referenced in the over/under transaction.
Vendor account	The vendor that the over or under is related to.
Goods in transit number	The goods-in-transit number, if applicable.
Item number	The item number that the over or under is related to.
Original quantity	The original quantity of the purchase order line that is attached to the shipment.
Received quantity	The quantity that was actually received.
Difference	The difference between the expected arrival journal amount and the amount that was posted in the arrival journal. A negative value indicates an over-delivery of goods.
Remaining quantity	The quantity that remains on the arrival journal line.
Over/under delivery	A value that indicates whether the quantity that was received is an over or an under.
Status	The status of the over/under transaction. Depending on the settings on the **Landed cost parameters** page, over-delivery can automatically create a movement journal for the over-delivery amount and post the journal. In this case, the over/under transaction will have a status of *Completed*. If action is required to move the goods out of the under-delivery warehouse, the record will have a status of *In process*.
Over/under reason	The reason that is associated with the over/under transaction.
Other inventory dimensions	You can show columns for additional dimensions in the grid as you require. These dimensions include batch number, inventory status, and warehouse. On the Action Pane,

Field	Description
	select **Inventory** > **Display dimensions** to open a dialog where you can select the dimensions to include.

Upper general tab

To view more information about the line that is currently selected on the upper **Overview** tab, select the **General** tab in the upper part of the **Over/under transactions** page. The information on this tab includes the values for all available dimensions. This information is pulled from the originating purchase order.

Lower overview tab

To view the document type for the row that is selected on the upper **Overview** tab, select the **Overview** tab in the lower part of the **Over/under transactions** page. The following table describes the fields that are available.

Expand table

Field	Description
New document type	This field is set to either *Movement journal* or *Purchase order*, depending on the action that is shown on the selected over/under transaction line.
Number	A reference and link to either the purchase order or the movement journal that is associated with the selected over/under transaction line.
Over/under reason	The reason that is associated with the selected over/under transaction line.
Quantity	The quantity that you selected for the purchase order or movement journal that is associated with the selected over/under transaction line.

Lower general tab

To view the over/under transaction number, lot ID, and dimension number that are associated with the selected over/under transaction line, select the **General** tab in the lower part of the **Over/under transactions** page.

Action Pane buttons for processing over/under transactions

The Action Pane on the **Over/under transactions** page provides the following commands for processing the selected transactions. Each command lets you choose how to process each

transaction.

- **Create** > **Movement** – Create and post a movement journal for the selected transaction. For under transactions, a movement journal is automatically created and posted for the difference between the expected and actual received quantity. Select this command for over transactions if you want the vendor to realize the cost difference.
- **Create** > **Purchase order** – Create a purchase order for the difference between the expected and actual received quantity of the selected transaction. Select this command for over transactions if your organization will realize the cost difference.
- **Create** > **Transfer to destination** – This command applies only to transfer orders. Select it if you want to transfer the over or under quantity to the destination warehouse.
- **Create** > **Transfer to origin** – This command applies only to transfer orders. Select it if you want to transfer the over or under quantity to the origin warehouse.

Over-receive or under-receive goods-in-transit orders when quality orders are enabled

When a goods-in-transit order is under-received, if the **Per Updated Quantity** parameter is enabled on the **Item sampling** page (**Inventory management** > **Setup** > **Quality control** > **Item sampling**), a quality order is created for the actual quantity that was received. If the parameter is disabled, a quality order is created based on the order quantity of the goods-in-transit order.

When a goods-in-transit order is over-received, the system automatically creates either a purchase order or a movement journal for the quantity that was over-delivered. In this case, a quality order is created based on the original quantity of the goods-in-transit order.

- If the system creates a purchase order, and you want the over-delivered quantity also to go through the quality check process, you must add a quality association for the new purchase order.
- If the system creates a movement journal, you can manually create a quality order for the over-delivered amount on the goods-in-transit order page.

Learn more about how to set up quality orders for Landed cost in Quality orders.

Last updated on 06/22/2026

Update tracking for put away

The *Update tracking for put away* periodic task is designed to be run as a nightly recurring batch. It identifies which voyages have received all inventory transactions, and which voyages don't have a value for the actual end date. It then sets the actual end date to the current date as required.

Container activities are created for each *leg* of a journey for each *shipping container*. These values are defined by the journey template that is selected when a voyage is created. For each container activities record, values can be set for the **Start date**, **Estimated end date**, and **Actual end date** fields. These fields can be updated when confirmation is received that a leg has begun or has been completed.

Typically, information that is related to confirmed dates is provided by a third party, such as a freight forwarder, because these actions are maintained outside Microsoft Dynamics 365 Supply Chain Management. However, information from a third party isn't required to track the arrival of goods from a journey leg into the warehouse (as marked by the putaway of goods). Therefore, tracking can be determined based on information in Dynamics 365 Supply Chain Management.

To set up and run the *Update tracking for put away* periodic task, follow these steps:

1. Go to **Landed Cost** > **Periodic tasks** > **Update tracking for put away**.

2. In the **Update tracking for put away** dialog box, in the **Parameters** section, set the following fields:

    - **Leg** – Select the unique identification name/number for the part of a journey that you want to update tracking for. The selected value must represent the arrival of the voyage at the warehouse.
    - **Activity** – Select the activity that was undertaken during the selected leg. These values are assigned per journey template line in the tracking control center.

3. To limit the set of records that should be included in the update, on the **Records to include** FastTab, select the **Filter** button. A standard query dialog box appears, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Supply Chain Management. The fields here are read-only and show values that are related to your query.

4. On the **Run in the background** FastTab, set up batch and scheduling options as you require, just as you might do for other types of jobs in Supply Chain Management.

5. Select **OK** to apply your settings, and to run or schedule the update.

Last updated on 11/26/2025

Landed cost inquiries

Article • 02/12/2025

Voyage line inquiries

The **Voyage lines** inquiry shows all voyage lines as they pertain to inventory. This inquiry can be used as a filter to help you find an item, purchase order, or other useful piece of information. It can also be used to identify the expected delivery date of an item that might be on one or more voyages. In this way, it can help you manage expected inventory receiving.

To open this inquiry, go to **Landed cost** > **Inquiries** > **Voyage lines**.

Overview tab

The **Overview** tab of the **Voyage lines** inquiry page includes a grid that shows basic information about each relevant voyage line. The following table describes the columns in the grid.

Expand table

Column	Description
Item number	The item for the voyage line.
Reference	The type of order (purchase order or transfer order).
Number	The purchase order number or transfer order number.
Folio	The folio that is associated with the voyage line.
Shipping container	The shipping container that is associated with the voyage line.
Voyage	The voyage that is associated with the voyage line.
Quantity	The quantity of the item for the voyage line.
(Other dimensions)	You can show columns for additional dimensions as you require. Those dimensions include batch number, inventory status, and warehouse. On the Action Pane, select **Inventory** > **Display dimensions** to open a dialog box where you can select the dimensions to include.

General tab

Select the **General** tab to view more information about the line that is currently selected on the **Overview** tab.

Dimensions tab

Select the **Dimensions** tab to view values for all available dimensions for the line that is currently selected on the **Overview** tab, regardless of the dimensions that you've selected to show on that tab.

Cost estimate inquiries

Every time that you generate a cost estimate, the estimate is added to the **Cost estimates** inquiry page. For complete details about how to generate, view, and work with cost estimates (including estimates on the inquiry page), see Estimate and manage landed costs.

Item tracking

Use the **Item tracking** page to view open purchase orders lines and their current status. Lines don't have to be associated with a voyage to appear here. However, if an item is associated with a voyage, the item tracking record line shows the voyage ID.

To open this page, go to **Landed cost** > **Inquiries** > **Tracking** > **Item tracking**.

Because your system probably contains a very large number of purchase order lines, the **Item tracking** page initially shows no records. Start by using the filter fields at the top of the page to define the set of purchase order lines that you're looking for. Then select **Update** on the Action Pane to generate the list. You can use an asterisk (*) as a wildcard character in any of the filter fields. For example, to find all purchase order lines for items that include the word "DVD" in their name, set the **Type** field to **Product name**, and enter **DVD** in the **Field value** field.

> **Note**
>
> Currently, backorders include only sales orders. Sales quotations aren't shown or considered backorders.

The following table describes the columns in the grid on the **Item tracking** page.

Expand table

Column	Description
To port	The final destination.
Confirmed	The confirmed date of the purchase order line.
Delivery Date	The delivery date of the purchase order line.
Voyage	If the line is associated with a voyage, the voyage ID.
Shipping Container	If the line is attached to a shipping container, the container ID.
Shipping port	The current port, based on the first leg in the tracking form where no actual date is set for the shipping container that is associated with the purchase order line.
Activity	The current activity, based on the first leg in the tracking form where no actual date is set for the shipping container that is associated with the purchase order line.
Estimated end date	The date when the voyage is expected to be received at the destination warehouse.
Name	The name of the vendor.
Voyage Status	The shipment status that is attached to the purchase order line.
Item number	The item number for the purchase order line.
External	The vendor's item name.
Product Name	The name of the item for the purchase order line.
Quantity	The purchase order line quantity for the purchase order line.
Unit	The unit of measure for the purchase order line.
Reference	The type of order (purchase order or transfer order)
Reference number	The purchase order number or transfer order number.
Backorder	A symbol indicates that there are sales backorders for the item.
(Other dimensions)	You can show columns for additional dimensions as you require. Those dimensions include batch number, inventory status, and warehouse. On the Action Pane, select **Display dimensions** to open a dialog box where you can select the dimensions to include.

Related information about backorders

To view more information about backorders, select the **Related information** tab on the right side of the page, and then select **Backorders**. To view even more information about a specific backorder, select the row for it, and then select the **More** link.

Individual shipping container tracking

The **Individual shipping container tracking** inquiry provides a filter that lets you find a shipping container and then identify all the voyage lines in that container.

To open this inquiry, go to **Landed cost** > **Inquiries** > **Tracking** > **Individual shipping container tracking**.

Multiple shipping container tracking

The **Multiple shipping container tracking** inquiry provides a filter that lets you find a collection of shipping containers and then identify all the voyage lines in those containers.

To open this inquiry, go to **Landed cost** > **Inquiries** > **Tracking** > **Multiple shipping container tracking**.

Feedback

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Landed cost reports

Article • 02/12/2025

Outstanding invoices

The outstanding invoices report contains a list of all outstanding invoices of the various cost levels that are associated with every voyage. It's used to reconcile the voyage and voyage costs together with the ledger transactions list on a regular basis. After an overhead cost has been invoiced, it's removed from the report.

To generate an outstanding invoices report, follow these steps.

1. Go to **Landed cost** > **Reports** > **Tracking** > **Outstanding invoices**.
2. In the **Outstanding invoices** dialog box, in the **As at date** field, specify a date. Any invoice that was outstanding on or before that date will be included in the output.
3. Under **Show**, select one of the following options:
    - **Cost not invoiced** – Include costs for invoiced shipments that have an estimated cost value but no actual cost.
    - **Stock not invoiced** – Include costs that have been invoiced, but that the purchase order hasn't yet been received for.
    - **All not invoiced** – Include the results of both the **Cost not invoiced** option and the **Stock not invoiced** option.
4. Set the **Include new costs** option to *Yes* to include costs that don't yet have an actual cost, and that inventory hasn't been received for. If you set it to *No*, the report will include only costs that have been invoiced.
5. In the **View** section, enable each type of detail that you want to include on the report.
6. As you do for other types of reports in Microsoft Dynamics 365 Supply Chain Management, use the **Destination** FastTab to select the output format of the report.
7. As you do for other types of reports in Supply Chain Management, use the **Records to include** FastTab to further limit the records that will be included on the report.
8. Select **OK** to generate the report.

Activity/provider analysis reports

The activity/provider analysis reports let you review the accuracy of your time estimates for each provider.

To generate an activity/provider analysis report, follow these steps.

1. Depending on the type of report that you want to create, follow one of these steps:

    - Go to **Landed cost** > **Reports** > **Analysis of activity/provider by activity**. In this case, the time estimates will be grouped by activity.
    - Go to **Landed cost** > **Reports** > **Analysis of activity/provider by provider**. In this case, the time estimates will be grouped by provider.

    Either the **Analysis of activity/provider by activity** dialog box or the **Analysis of activity/provider by provider** dialog box appears. Both dialog boxes provide the same options.

2. As you do for other types of reports in Supply Chain Management, use the **Destination** FastTab to select the output format of the report.

3. As you do for other types of reports in Supply Chain Management, use the **Records to include** FastTab to further limit the records that will be included on the report.

4. Select **OK** to generate the report.

Voyage costing reports

The voyage costing reports show the cost of the items and the import costs per folio, shipping container, or voyage, depending on the options that you select when you generate the report. Each voyage costing report lets you view the estimated cost of a voyage versus the actual cost, and it can be broken down by the several identifying factors.

To generate a voyage costing report, follow these steps.

1. Depending on the type of report that you want to create, follow one of these steps:

    - Go to **Landed cost** > **Reports** > **Costing** > **Voyage costing by individual cost**. In this case, the individual cost option will show import costs per cost area per cost type code.

- Go to **Landed cost** > **Reports** > **Costing** > **Voyage costing by reporting category**. In this case, the import cost will be broken down into the various reporting categories. Cost types are grouped by reporting categories.

Either the **Voyage costing by individual cost** dialog box or the **Voyage costing by reporting category** dialog box appears. These dialog boxes are similar. Any differences are noted in the rest of this procedure.

2. If you opened the **Voyage costing by reporting category** dialog box, in the **Cost** field, select one of the following values:

   - **Cost value** – The value is either calculated by using auto costs or manually created in the cost area.
   - **Estimated** – After the goods have been received, the estimated cost is set.
   - **Actual** – After the order has been invoiced, the actual cost is set to the cost on the invoice.
   - **Best** – The system will use whichever of the previous three options is most accurate. (We recommend that you select this option.)

3. Set the **Per item** option to *Yes* to show the cost of each item. Set it to *No* to show costs per voyage.

4. In the **View** section, select the categories to break down the cost by.

5. In the **Dimensions** section, select the dimensions to include on the report.

6. As you do for other types of reports in Supply Chain Management, use the **Destination** FastTab to select the output format of the report.

7. As you do for other types of reports in Supply Chain Management, use the **Records to include** FastTab to further limit the records that will be included on the report.

8. Select **OK** to generate the report.

Shipping container receipts list

The shipping container receipts list contains all unreceived quantities that are due on or before an expected date. Warehouse staff can use this report to identify the expected goods on a shipping container. It can also be used to manually validate goods as they are received on a shipping container.

To generate a shipping container receipts list, follow these steps.

1. Go to **Landed cost** > **Reports** > **Tracking** > **Shipping container receipts list**.
2. In the **Shipping container receipts list** dialog box, in the **Expected date** field, specify a date. Any quantities that were unreceived on or before that date will be included in the output.
3. In the **Dimensions** section, select the dimensions to include on the report.
4. As you do for other types of reports in Supply Chain Management, use the **Destination** FastTab to select the output format of the report.
5. As you do for other types of reports in Supply Chain Management, use the **Records to include** FastTab to further limit the records that will be included on the report.
6. Select **OK** to generate the report.

Expected delivery report

The expected delivery report contains basic information about the voyage, shipping container, folio, items, and expected date of delivery.

To generate an expected delivery report, follow these steps.

1. Go to **Landed cost** > **Reports** > **Tracking** > **Expected Delivery**.
2. In the **Expected delivery** dialog box, in the **Expected date** field, select the date when delivery of the goods to the final destination warehouse is expected. Any voyage line that has an expected date on or before that date, and that hasn't yet been received, will be included in the output.
3. Optional: In the **Vendor account** field, select a vendor account to include only deliveries from a specific vendor.
4. In the **Dimensions** section, select the dimensions to include on the report.
5. As you do for other types of reports in Supply Chain Management, use the **Destination** FastTab to select the output format of the report.
6. As you do for other types of reports in Supply Chain Management, use the **Records to include** FastTab to further limit the records that will be included on the report.
7. Select **OK** to generate the report.

Feedback

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Landed cost entities overview

The **Landed cost** module helps businesses streamline inbound shipping operations by providing complete financial and logistical control over imported freight, from the manufacturer to the warehouse. Transaction data entities for Landed cost enable external data sources (such as freight forwarder services) to create voyages and costs, and update container tracking records.

Here are some of the supported operations:

- Create and update voyage, shipping container, and folio records for existing inbound order lines.
- Create and update estimated costs for each cost area.
- Create and update shipping container activity records to capture the start date, estimated end date, and actual end date for each leg of the container's journey.
- Create vendor cost invoice allocations.

Last updated on 11/26/2025

Voyage creation entities

Article • 01/31/2025

The voyage creation data entities group together the data entities that are required to create a working voyage. You or your freight forwarder can use these data entities to create voyage, shipping container, folio, and voyage line records that reference existing purchaser order or transfer order lines.

Voyages (ITMTableEntity)

The voyage represents the journey of inbound goods and serves as the highest-level cost area in Landed cost. It contains header-level information that is related to the vessel, port of origin, and destination port. This functionality is supported by the `ITMTableEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Mode of delivery	ITMTable.DlvModeId	Nvarchar(10)	No	No
Broker advised	ITMTable.ITMBrokerAdvised	Datetime	No	No
Customer appointment	ITMTable.ITMCustomerAppointment	Datetime	No	No
Delivered at warehouse	ITMTable.ITMDelAtWarehouse	Datetime	No	No
Delivery instructions	ITMTable.ITMDeliveryInstructions	Datetime	No	No
Departure date	ITMTable.ITMDepartureDate	Datetime	No	No
Goods released	ITMTable.ITMGoodsReleased	Datetime	No	No
Local transport date	ITMTable.ITMLocalTransportDate	Datetime	No	No
Local transport time	ITMTable.ITMLocalTransportTime	Int	No	No
Original bill of landing sent	ITMTable.ITMOriginalBOLSebt	Datetime	No	No
Original documents received	ITMTable.ITMOriginalDocsReceived	Datetime	No	No

Name	Mapping	Data type	Key	Mandatory
Purchase order status	ITMTable.ITMPurchStatus	Int	No	No
Verification date	ITMTable.ITMVerificationDate	Datetime	No	No
Customs entry identifier	ITMTable.ShipCustomsEntryId	Nvarchar(20)	No	No
Ship date	ITMTable.ShipDate	Datetime	No	No
Description	ITMTable.ShipDescription	Nvarchar(60)	No	No
Documents received	ITMTable.ShipDocReceived	Int	No	No
Estimated delivery date	ITMTable.ShipEstDlvDate	Datetime	No	No
ETA at shipping port	ITMTable.ShipEstPortDate	Datetime	No	No
From port	ITMTable.ShipFromPort	Nvarchar(20)	No	No
House air way/Bill of lading	ITMTable.ShipHAWB	Nvarchar(20)	No	No
Voyage	ITMTable.ShipId	Nvarchar(20)	**Yes**	**Yes**
Booking reference	ITMTable.ShipIdExternal	Nvarchar(20)	No	No
Journey template	ITMTable.ShipJourneyId	Nvarchar(20)	No	No
Local forwarder	ITMTable.ShipLocalForwarder	Nvarchar(20)	No	No
Master air way/Bill of lading	ITMTable.ShipMAWB	Nvarchar(20)	No	No
Measurement	ITMTable.ShipMeasurement	Numeric(32, 6)	No	No
Measurement unit	ITMTable.ShipMeasurementUnit	Int	No	No
Number of pallets	ITMTable.ShipNoOfPallets	Int	No	No
Pending voyage	ITMTable.ShipPending	Int	No	No
Remarks	ITMTable.ShipRemarks	Nvarchar(MAX)	No	No
Rental	ITMTable.ShipRental	Int	No	No
Voyage status	ITMTable.ShipStatusId	Nvarchar(20)	No	No
Tally in number	ITMTable.ShipTallyInNumber	Nvarchar(20)	No	No

Name	Mapping	Data type	Key	Mandatory
To port	ITMTable.ShipToPort	Nvarchar(20)	No	No
Valuation date	ITMTable.ShipValuationDate	Datetime	No	No
Shipping company	ITMTable.ShipVendAccount	Nvarchar(20)	No	No
Vessel	ITMTable.ShipVesselId	Nvarchar(20)	No	**Yes**
External voyage ID	ITMTable.ShipVoyage	Nvarchar(20)	No	No

Number sequences for voyages

The shared number sequence for voyages controls the allocation of voyage IDs.

The value that is passed to the data entity as the **Voyage ID** (`ShipId`) value is used to identify an existing record for update. If that record doesn't exist, the behavior depends on whether the assigned number sequence is configured to allow for manual entry:

- If manual entry is enabled, a new record is created that uses the passed value.
- If manual entry isn't enabled, the next number in the assigned number sequence is used instead of the passed value.

During the import session, the placeholder value that is imported as the voyage ID is retained. This behavior ensures that shipping container and voyage lines that reference the placeholder are included in the voyage, and that they're updated to reflect the value that is assigned from the number sequence.

Automatic cost transactions

Automatic costs can be added to a voyage from the **Auto costs** page in Microsoft Dynamics 365 Supply Chain Management (**Landed cost** > **Costing setup** > **Auto costs**). Records for automatic costs can also be created by using cost transaction data entities for each cost area that Landed cost supports.

Automatic costs that are configured in Supply Chain Management are applied to the voyage when a voyage line is created. No costs will be visible against the record until the header record is associated with line information.

Shipping containers (ITMContainersEntity)

A shipping container represents a physical container that goods are transported in. This physical container might be a freight container for ocean freight or a single pallet for air freight. The shipping container includes header-level information that is related to the shipping container ID, seal number, and shipping container type. (The shipping container type provides dimension information.) This functionality is supported by the `ITMContainersEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Departure date	ITMContainers.ITMDepartureDate	Datetime	No	No
Local transport date	ITMContainers.ITMLocalTransportDate	Datetime	No	No
Local transport time	ITMContainers.ITMLocalTransportTime	Int	No	No
Original voyage	ITMContainers.OrigShipId	Nvarchar(20)	No	No
Actual weight	ITMContainers.ShipActualWeight	Numeric(32, 6)	No	No
Broker advised	ITMContainers.ShipBrokerAdvised	Datetime	No	No
Shipping container	ITMContainers.ShipContainerId	Nvarchar(20)	**Yes**	**Yes**
Shipping container type	ITMContainers.ShipContainerTypeId	Nvarchar(20)	No	No
Unit type	ITMContainers.ShipContainerUnitTypeId	Nvarchar(10)	No	No
Converted to rental	ITMContainers.ShipConvertedToRental	Int	No	No
Customer appointment	ITMContainers.ShipCustomerAppointment	Datetime	No	No
Ship date	ITMContainers.ShipDate	Datetime	No	No
Delivered at warehouse	ITMContainers.ShipDelAtWarehouse	Datetime	No	No
Delivery instructions	ITMContainers.ShipDeliveryInstructions	Datetime	No	No

Name	Mapping	Data type	Key	Mandatory
Examination certificate applied date	ITMContainers.ShipECAppliedDate	Datetime	No	No
Examination certificate expiry date	ITMContainers.ShipECExpiryDate	Datetime	No	No
Examination certificate number	ITMContainers.ShipECNum	Nvarchar(20)	No	No
Examination certificate received date	ITMContainers.ShipECReceivedDate	Datetime	No	No
Estimated delivery date	ITMContainers.ShipEstDlvDate	Datetime	No	No
ETA at shipping port	ITMContainers.ShipEstPortDate	Datetime	No	No
Expected loading date	ITMContainers.ShipExpectedLoadingDate	Datetime	No	No
From port	ITMContainers.ShipFromPort	Nvarchar(20)	No	No
Description of goods	ITMContainers.ShipGoodsDesc	Nvarchar(60)	No	No
Goods released	ITMContainers.ShipGoodsReleased	Datetime	No	No
GPS tracker unit	ITMContainers.ShipGPSUnit	Nvarchar(30)	No	No
House air way/Bill of lading	ITMContainers.ShipHAWB	Nvarchar(20)	No	No
Voyage	ITMContainers.ShipId	Nvarchar(20)	**Yes**	**Yes**
Journey template	ITMContainers.ShipJourneyId	Nvarchar(20)	No	No
Local forwarder	ITMContainers.ShipLocalForwarder	Nvarchar(20)	No	No
Measurement	ITMContainers.ShipMeasurement	Numeric(32, 6)	No	No

Name	Mapping	Data type	Key	Mandatory
Measurement unit	ITMContainers.ShipMeasurementUnit	Int	No	No
Number of cartons	ITMContainers.ShipNoOfCartons	Numeric(32, 6)	No	No
Original bill of landing sent	ITMContainers.ShipOriginalBOLSebt	Datetime	No	No
Original documents received	ITMContainers.ShipOriginalDocsReceived	Datetime	No	No
Our seal number	ITMContainers.ShipOurSealNum	Nvarchar(20)	No	No
Used	ITMContainers.ShipPendingUsed	Int	No	No
Purchase order status	ITMContainers.ShipPurchStatus	Int	No	No
Refrigeration type	ITMContainers.ShipRefrigerationTypeId	Nvarchar(10)	No	No
Remarks	ITMContainers.ShipRemarks	Nvarchar(MAX)	No	No
Rental	ITMContainers.ShipRental	Int	No	No
Returnable	ITMContainers.ShipReturnable	Int	No	No
Voyage status	ITMContainers.ShipStatusId	Nvarchar(20)	No	No
Shipping company seal number	ITMContainers.ShipTheirSealNum	Nvarchar(20)	No	No
To port	ITMContainers.ShipToPort	Nvarchar(20)	No	No
Verification date	ITMContainers.ShipVerificationDate	Datetime	No	No
Vessel	ITMContainers.ShipVesselId	Nvarchar(20)	No	**Yes**

Voyage ID validation

The shipping container ID isn't controlled by a number sequence. It's unique only in the context of the voyage that it's used on.

If the shipping container entity (`ITMContainersEntity`) is used independently of the voyage entity (`ITMTableEntity`), the **Voyage ID** (`ShipId`) value must match an existing record in the voyage table. Otherwise, the import will fail.

If the shipping container entity (`ITMContainersEntity`) and the voyage entity (`ITMTableEntity`) are used as part of the same import session, the voyage entity must precede the creation of a shipping container. Otherwise, the **Voyage ID** (`ShipId`) value can't be successfully validated. If a placeholder value is used for the **Voyage ID** (`ShipId`) value, the association can be created only if the same placeholder is used as the **Voyage ID** (`ShipId`) value in the shipping container entity.

Other field validations

The following table shows the fields in the shipping container table (`ITMContainers`) that are validated against Landed cost setup tables. It also shows the Landed cost data entities that a freight forwarder can use to read the existing values and ensure that valid values are received in your environment.

Expand table

Field in the ITMContainers table	Landed cost data entity
Shipping container type	ITMShippingContainerTypeEntity
Description of goods	ITMGoodsDescriptionEntity
Refrigeration type	ITMShippingContainerRefrigerationTypeEntity

Folios (ITMFolioEntity)

A folio represents a grouping of items in a shipping container for the purposes of customs declarations. The folio includes header-level information that is related to the customs broker and a description of the goods. This functionality is supported by the `ITMFolioEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Broker advised	ITMFolioTable.ShipBrokerAdvised	Datetime	No	No

Name	Mapping	Data type	Key	Mandatory
Cargo control number	ITMFolioTable.ShipCargoControlNumber	Nvarchar(20)	No	No
Customer appointment	ITMFolioTable.ShipCustomerAppointment	Datetime	No	No
Customs broker	ITMFolioTable.ShipCustomsBroker	Nvarchar(20)	No	No
Customs ID	ITMFolioTable.ShipCustomsId	Nvarchar(60)	No	No
Company	ITMFolioTable.ShipDataArea	Nvarchar(4)	No	**Yes**
Delivered at warehouse	ITMFolioTable.ShipDelAtWarehouse	Datetime	No	No
Delivery instructions	ITMFolioTable.ShipDeliveryInstructions	Datetime	No	No
Documents received	ITMFolioTable.ShipDocReceived	Int	No	No
Estimated delivery date	ITMFolioTable.ShipEstDlvDate	Datetime	No	No
ETA at shipping port	ITMFolioTable.ShipEstPortDate	Datetime	No	No
Exporter	ITMFolioTable.ShipExporterId	Nvarchar(20)	No	No
Name	ITMFolioTable.ShipExporterName	Nvarchar(60)	No	No
Folio date	ITMFolioTable.ShipFolioDate	Datetime	No	No
Folio	ITMFolioTable.ShipFolioId	Nvarchar(20)	**Yes**	**Yes**
From port	ITMFolioTable.ShipFromPort	Nvarchar(20)	No	No
Description of goods	ITMFolioTable.ShipGoodsDesc	Nvarchar(60)	No	No
Goods released	ITMFolioTable.ShipGoodsReleased	Datetime	No	No
House air way/Bill of lading	ITMFolioTable.ShipHAWB	Nvarchar(20)	No	No
Voyage	ITMFolioTable.ShipId	Nvarchar(20)	No	**Yes**

Name	Mapping	Data type	Key	Mandatory
Measurement	ITMFolioTable.ShipMeasurement	Numeric(32, 6)	No	No
Measurement unit	ITMFolioTable.ShipMeasurementUnit	Int	No	No
Number of cartons	ITMFolioTable.ShipNoOfCartons	Numeric(32, 6)	No	No
Original bill of landing sent	ITMFolioTable.ShipOriginalBOLSebt	Datetime	No	No
Original documents received	ITMFolioTable.ShipOriginalDocsReceived	Datetime	No	No
Purchase order status	ITMFolioTable.ShipPurchStatus	Int	No	No
Remarks	ITMFolioTable.ShipRemarks	Nvarchar(MAX)	No	No
Voyage status	ITMFolioTable.ShipStatusId	Nvarchar(20)	No	No
Tariff code	ITMFolioTable.ShipTariffCode	Nvarchar(10)	No	No
To port	ITMFolioTable.ShipToPort	Nvarchar(20)	No	No
Valuation date	ITMFolioTable.ShipValuationDate	Datetime	No	No
Verification date	ITMFolioTable.ShipVerificationDate	Datetime	No	No
Vendor account	ITMFolioTable.VendAccount	Nvarchar(20)	No	No

Number sequences for folios

The number sequence for folios controls the allocation of folio IDs.

The value that is passed to the data entity as the **Folio ID** (`FolioId`) value is used to identify an existing record for update. If that record doesn't exist, the behavior depends on whether the assigned number sequence is configured to allow for manual entry:

- If manual entry is enabled, a new record is created that uses the passed value.
- If manual entry isn't enabled, the next number in the assigned number sequence is used instead of the passed value.

During the import session, the placeholder value that is imported as the folio ID is retained. This behavior ensures that voyage lines that reference the placeholder are correctly associated, and that they're updated to reflect the value that is assigned from the number sequence.

Field validations

The **Description of goods** field in the folio table (`ITMFolioTable`) is validated against the Landed cost setup table of the same name. A freight forwarder can use the `ITMGoodsDescriptionEntity` Landed cost data entity to read the existing values and ensure that valid values are received in your environment.

Voyage lines for purchase orders (ITMPurchaseLineEntity)

The voyage line represents a single purchase order line that is included in the voyage. This relationship is established through the **Purchase order number** and **Purchase line number** fields. The voyage line references each previous record that was created for the voyage, shipping container, and folio. This functionality is supported by the `ITMPurchaseLineEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Currency	ITMLine.CurrencyCode	Nvarchar(3)	No	No
Net amount	ITMLine.LineAmountMST	Numeric(32, 6)	No	No
Purchase line number	ITMLine.RefRecId	Numeric(32, 6)	**Yes**	No
Shipping container	ITMLine.ShipContainerId	Int	No	No
Company	ITMLine.ShipDataArea	Nvarchar(20)	**Yes**	No
Declared quantity	ITMLine.ShipDeclaredQty	Nvarchar(4)	No	No
Folio	ITMLine.ShipFolioId	Numeric(32, 6)	No	No
Voyage	ITMLine.ShipId	Nvarchar(20)	**Yes**	No
Item number	ITMLine.ShipItemId	Nvarchar(20)	No	No
Measurement	ITMLine.ShipMeasurement	Nvarchar(20)	No	No

Name	Mapping	Data type	Key	Mandatory
Measurement unit	ITMLine.ShipMeasurementUnit	Numeric(32, 6)	No	No
Number of cartons	ITMLine.ShipNoOfCartons	Int	No	No
Position	ITMLine.ShipPosition	Numeric(32, 6)	No	No
Quantity	ITMLine.ShipQty	Int	No	No
Purchase order number	ITMLine.TransRefId	Numeric(32, 6)	**Yes**	No
Unit	ITMLine.UnitId	Int	No	No
Volume	ITMLine.Volume	Nvarchar(10)	No	No
Weight	ITMLine.Weight	Numeric(32, 6)	No	No

Voyage lines for transfer orders (ITMTransferLineEntity)

The voyage line represents a single transfer order line that is included in the voyage. This relationship is established through the **Transfer order number** and **Transfer line number** fields. The voyage line references each previous record that was created for the voyage, shipping container, and folio. This functionality is supported by the `ITMTransferLineEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Currency	ITMLine.CurrencyCode	Nvarchar(3)	No	No
Net amount	ITMLine.LineAmountMST	Numeric(32, 6)	No	No
Shipping container	ITMLine.ShipContainerId	Int	No	No
Company	ITMLine.ShipDataArea	Nvarchar(20)	**Yes**	No
Declared quantity	ITMLine.ShipDeclaredQty	Nvarchar(4)	No	No
Folio	ITMLine.ShipFolioId	Numeric(32, 6)	No	No
Voyage	ITMLine.ShipId	Nvarchar(20)	**Yes**	No
Item number	ITMLine.ShipItemId	Nvarchar(20)	No	No

Name	Mapping	Data type	Key	Mandatory
Measurement	ITMLine.ShipMeasurement	Nvarchar(20)	No	No
Measurement unit	ITMLine.ShipMeasurementUnit	Numeric(32, 6)	No	No
Number of cartons	ITMLine.ShipNoOfCartons	Int	No	No
Position	ITMLine.ShipPosition	Numeric(32, 6)	No	No
Quantity	ITMLine.ShipQty	Int	No	No
Transfer line number	ITMLine.TransferLineNumber	Numeric(32, 6)	**Yes**	No
Transfer order number	ITMLine.TransRefId	Numeric(32, 6)	**Yes**	No
Unit	ITMLine.UnitId	Int	No	No
Volume	ITMLine.Volume	Nvarchar(10)	No	No
Weight	ITMLine.Weight	Numeric(32, 6)	No	No

Reference table

Voyage lines are created through an association with an open inbound order line. This line can be on a purchase order, or it can be the receive transaction on a transfer order. The `RefTableId` field in the voyage line table (`ITMLine`) specifies which order type the line is related to by referencing the table number. If specific table numbers are referenced in the table where the data is being created, the entities are split based on those values.

The values for the reference table (`RefTableId`) and the transaction type (`TransType`) are implicit in the selection of either the Landed cost purchase line entity or the Landed cost transfer line entity.

Validation

A voyage line directly references a voyage record, a shipping container record, and a folio record. If the purchase line entity (`ITMPurchaseLinesEntity`) or transfer line entity (`ITMPurchaseLinesEntity`) is used independently of the entities that are used to create these reference records, the **Voyage ID** (`ShipId`), **Shipping container** (`ShipContainerId`), and **Folio** (`ShipFolioId`) values must match an existing record in the corresponding tables. Otherwise, the import will fail.

If either line entity is used as part of the same import session, those other entities must precede the creation of a voyage line. Otherwise, the values can't be successfully validated. If a placeholder value is used for the voyage or folio number, the association can be created only if the same placeholder is used for the voyage or folio number in the voyage line entity.

If the purchase order or transfer order line is already assigned to an existing voyage line, the new voyage line won't be created. Before the order line can be assigned to a new voyage, it must be removed from its current voyage.

Order line identification

Voyage lines directly reference the reference **Record ID** (`RefRecId`), **Inventory dimension ID** (`InventDimId`), and **Inventory transaction ID** (`InventTransId`) values. These values no longer have to be included when the data entity is used. Instead, the order line number must now be included. Together, the order line number and the order number enable each of those reference values to be identified.

Voyage line quantity

Because a voyage line is directly associated with an order line, the **Quantity** (`ShipQty`) value that is entered by using the entity requires that the values match. Validation is run against the quantity on either the purchase order line or the transfer line. If the validation fails, the record won't be processed.

Calculated fields

The **Weight** and **Volume** calculated fields accept the values that are received through the data entity. If no values are provided, the system values are used to update these fields, if system values are available. For Landed cost, the values are calculated in the following way:

- *Weight = Quantity × Item gross weight*
- *Volume = Quantity* × (*Item gross depth* × *Item gross height* × *Item gross width*)

Sequencing

Because of dependencies between the tables, the voyage record must be created first. The voyage line can be created only after the voyage, shipping container, and folios have been created.

To create a voyage without validation warnings, the system must process the entities in the following order:

1. Voyages (`ITMTableEntity`)
2. Shipping containers (`ITMContainersEntity`)
3. Folios (`ITMFolioTableEntity`)
4. Voyage lines (`ITMPurchaseLinesEntity` or `ITMTransferLinesEntity`)

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Cost transaction entities

Article • 02/12/2025

Apportionment

Landed cost enables the value of a cost to be split among the contents of a cost area (voyage, shipping container, and so on) through the selection of an apportionment method. The apportionment method is set as part of the configuration of automatic costs that is based on business rules. It's pulled through as a part of the cost when a voyage line is created.

Configure the apportionment mapping for use with data entities

When a cost is created from an external source such as a freight forwarder, that external source can't specify the preferred method for apportioning the cost value. The apportionment mapping defines the default apportionment method for each cost type code. The apportionment mapping table is accessed from the **Apportionment mapping** page in Microsoft Dynamics 365 Supply Chain Management (**Landed Cost** > **Costing setup** > **Apportionment mapping**).

If a mapping combination has been defined, and a cost that matches the cost type code is created by using a cost transaction entity, the mapped apportionment method is used instead of any value was provided to the entity.

If no value is present in the mapping table, but a value has been provided to the entity, the provided value is used.

If no value exists in either the mapping table or the record that was submitted to the entity, creation fails.

Apportionment mapping (ITMApportionmentMapping)

The apportionment mapping entity (`ITMApportionmentMapping`) creates apportionment mapping relationships and enables users to create or update records.

Expand table

Name	Mapping	Data type	Key	Mandatory
Apportionment method	ITMApportionmentMapping.ApportionmentMethod	Int	No	No
Cost type code	ITMApportionmentMapping.ShipCostTypeId	Nvarchar(20)	**Yes**	No

Voyage costs (ITMCostTransVoyageEntity)

The voyage cost entity (`ITMCostTransVoyageEntity`) creates voyage cost transactions that are applied at the voyage level. During the import process, the system uses the **Category** and **Apportionment method** values that are included in the entity to determine how the value of the cost is apportioned across the contents of the voyage.

Expand table

Name	Mapping	Data type	Key	Mandatory
Apportionment method	ITMCostTrans.ApportionmentMethod	Int	No	No
Cost value	ITMCostTrans.CostValue	Numeric(32, 6)	No	No
Currency	ITMCostTrans.CurrencyCode	Nvarchar(3)	No	**Yes**
Line number	ITMCostTrans.LineNum	Numeric(32, 16)	**Yes**	No
Linked cost type	ITMCostTrans.LinkCostTypeId	Nvarchar(20)	No	No
Minimum cost	ITMCostTrans.MinCostAmount	Numeric(32, 6)	No	No
Category	ITMCostTrans.ShipCostCategory	Int	No	No
Cost type code	ITMCostTrans.ShipCostTypeId	Nvarchar(20)	No	No
Company	ITMCostTrans.ShipDataArea	Nvarchar(4)	No	**Yes**
Voyage	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Item sales tax group	ITMCostTrans.TaxItemGroup	Nvarchar(10)	No	No

Shipping container costs (ITMCostTransShippingContainerEntity)

The shipping container cost entity (`ITMCostTransShippingContainerEntity`) creates shipping container costs that are applied at the shipping container level. During the import process, the system uses the **Category** and **Apportionment method** values that are included in the entity to determine how the value of the cost is apportioned across the contents of the shipping container.

The **Aggregate**, **Leg**, and **Linked leg** fields are specific to records where the **Cost area** value is *Shipping container*. Therefore, they aren't present in data entities for other cost areas.

Expand table

Name	Mapping	Data type	Key	Mandatory
Aggregate	ITMCostTrans.AggregatedCost	Int	No	No
Apportionment method	ITMCostTrans.ApportionmentMethod	Int	No	No
Cost value	ITMCostTrans.CostValue	Numeric(32, 6)	No	No
Currency	ITMCostTrans.CurrencyCode	Nvarchar(3)	No	**Yes**
Line number	ITMCostTrans.LineNum	Numeric(32, 16)	**Yes**	No
Linked cost type	ITMCostTrans.LinkCostTypeId	Nvarchar(20)	No	No
Linked leg	ITMCostTrans.LinkLegId	Nvarchar(20)	No	No
Minimum cost	ITMCostTrans.MinCostAmount	Numeric(32, 6)	No	No
Shipping Container	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Category	ITMCostTrans.ShipCostCategory	Int	No	No
Cost type code	ITMCostTrans.ShipCostTypeId	Nvarchar(20)	No	No
Company	ITMCostTrans.ShipDataArea	Nvarchar(4)	No	**Yes**
Voyage	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Leg	ITMCostTrans.ShipLegId	Nvarchar(20)	No	No
Item sales tax group	ITMCostTrans.TaxItemGroup	Nvarchar(10)	No	No

Folio costs (ITMCostTransFolioEntity)

The folio cost entity (`ITMCostTransFolioEntity`) creates cost transaction records that apply to the folio level. During the import process, the system uses the **Category** and **Apportionment method** values that are included in the entity to determine how the value of the cost is apportioned across the contents of each folio.

Expand table

Name	Mapping	Data type	Key	Mandatory
Apportionment method	ITMCostTrans.ApportionmentMethod	Int	No	No
Cost value	ITMCostTrans.CostValue	Numeric(32, 6)	No	No
Currency	ITMCostTrans.CurrencyCode	Nvarchar(3)	No	**Yes**
Line number	ITMCostTrans.LineNum	Numeric(32, 16)	**Yes**	No
Linked cost type	ITMCostTrans.LinkCostTypeId	Nvarchar(20)	No	No
Minimum cost	ITMCostTrans.MinCostAmount	Numeric(32, 6)	No	No
Category	ITMCostTrans.ShipCostCategory	Int	No	No
Cost type code	ITMCostTrans.ShipCostTypeId	Nvarchar(20)	No	No
Company	ITMCostTrans.ShipDataArea	Nvarchar(4)	No	**Yes**
Folio	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Item sales tax group	ITMCostTrans.TaxItemGroup	Nvarchar(10)	No	No

Purchase order costs (ITMCostTransPurchaseEntity)

The purchase order cost entity (`ITMCostTransPurchaseEntity`) creates cost transactions that apply to the purchaser order header. During the import process, the system uses the **Category** and **Apportionment method** values that are included in the entity to determine how the value of the cost is apportioned across the contents of each folio.

Expand table

Name	Mapping	Data type	Key	Mandatory
Apportionment	ITMCostTrans.ApportionmentMethod	Int	No	No

Name	Mapping	Data type	Key	Mandatory
method				
Cost value	ITMCostTrans.CostValue	Numeric(32, 6)	No	No
Currency	ITMCostTrans.CurrencyCode	Nvarchar(3)	No	**Yes**
Line number	ITMCostTrans.LineNum	Numeric(32, 16)	**Yes**	No
Linked cost type	ITMCostTrans.LinkCostTypeId	Nvarchar(20)	No	No
Minimum cost	ITMCostTrans.MinCostAmount	Numeric(32, 6)	No	No
Purchase order	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Category	ITMCostTrans.ShipCostCategory	Int	No	No
Cost type code	ITMCostTrans.ShipCostTypeId	Nvarchar(20)	No	No
Company	ITMCostTrans.ShipDataArea	Nvarchar(4)	No	**Yes**
Item sales tax group	ITMCostTrans.TaxItemGroup	Nvarchar(10)	No	No

Item costs (ITMCostTransItemEntity)

The item cost entity (`ITMCostTransItemEntity`) creates cost transactions that apply to the item level. This entity is specific to items on purchase order lines. The value of the cost is applied in full to the line.

The **Adjustment amount** and **Value adjustment** fields are specific to the line-level cost areas. Therefore, they aren't present in data entities for other cost areas.

Expand table

Name	Mapping	Data type	Key	Mandatory
Adjustment amount	ITMCostTrans.AdjustmentAmount	Numeric(32, 6)	No	No
Cost value	ITMCostTrans.CostValue	Numeric(32, 6)	No	No
Currency	ITMCostTrans.CurrencyCode	Nvarchar(3)	No	**Yes**
Line number	ITMCostTrans.LineNum	Numeric(32, 16)	**Yes**	No

Name	Mapping	Data type	Key	Mandatory
Linked cost type	ITMCostTrans.LinkCostTypeId	Nvarchar(20)	No	No
Minimum cost	ITMCostTrans.MinCostAmount	Numeric(32, 6)	No	No
Purchase order	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Purchase order line number	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Category	ITMCostTrans.ShipCostCategory	Int	No	No
Cost type code	ITMCostTrans.ShipCostTypeId	Nvarchar(20)	No	No
Company	ITMCostTrans.ShipDataArea	Nvarchar(4)	No	**Yes**
Item sales tax group	ITMCostTrans.TaxItemGroup	Nvarchar(10)	No	No
Value adjustment	ITMCostTrans.ValueAdjustmentMethod	Int	No	No

Transfer line costs (ITMCostTransTransferLineEntity)

The transfer line cost entity (`ITMCostTransTransferLineEntity`) creates cost transactions that apply to the transfer order line level. The value of these costs is applied in full to the transfer order line.

The **Adjustment amount** and **Value adjustment** fields are specific to the line-level cost areas. Therefore, they aren't present in data entities for other cost areas.

Expand table

Name	Mapping	Data type	Key	Mandatory
Adjustment amount	ITMCostTrans.AdjustmentAmount	Numeric(32, 6)	No	No
Cost value	ITMCostTrans.CostValue	Numeric(32, 6)	No	No
Currency	ITMCostTrans.CurrencyCode	Nvarchar(3)	No	**Yes**
Line number	ITMCostTrans.LineNum	Numeric(32, 16)	**Yes**	No

Name	Mapping	Data type	Key	Mandatory
Linked cost type	ITMCostTrans.LinkCostTypeId	Nvarchar(20)	No	No
Minimum cost	ITMCostTrans.MinCostAmount	Numeric(32, 6)	No	No
Category	ITMCostTrans.ShipCostCategory	Int	No	No
Cost type code	ITMCostTrans.ShipCostTypeId	Nvarchar(20)	No	No
Company	ITMCostTrans.ShipDataArea	Nvarchar(4)	No	**Yes**
Transfer order	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Transfer order line number	ITMCostTrans.TransRecId	Nvarchar(20)	**Yes**	No
Item sales tax group	ITMCostTrans.TaxItemGroup	Nvarchar(10)	No	No
Value adjustment	ITMCostTrans.ValueAdjustmentMethod	Int	No	No

Transaction table

A cost transaction record is associated with a cost area through the assignment of a transaction table number (`TransTableId`). When specific table identification numbers are required, the entities are split based on these values, so that an entity exists for each cost area.

The value for the transaction table (`TransTableId`) is implicit in the selection of the cost transaction entity.

Calculated fields

The following fields aren't available for insert or update when a cost transaction entity is used, because these fields are set only when specific actions are taken against the voyage record:

- **Estimated cost** – This field is set when an invoice is posted for the voyage (purchase order) or a transfer is received.
- **Estimated cost currency** – This field is set when an invoice is posted for the voyage (purchase order) or a transfer is received.
- **Actual cost** – This field is set when a vendor invoice is posted. It's associated with the cost.

Find auto costs

A **Find auto costs** button that is available for each cost area (voyage, shipping container, and so on) provides a way to update the cost transactions for that cost area from the information in the configuration tables. When you select the button for a cost area, the system clears existing costs for that cost area and creates new records, based on the current configuration of the auto cost setup tables.

Cost transaction records that are created by using a data entity are flagged as imported. These records aren't deleted when you select **Find auto costs**, because they won't be re-created from the auto cost setup tables. However, a cost transaction record that is flagged as imported can still be deleted.

Linked fields

Each cost transaction can be associated with another record in the same cost area. This association is established through the **Linked cost type** field. It enables a cost value to be calculated as a percentage of another cost.

The **Linked cost type** field can be validated only if the record that is being referenced is processed first, or if it already exists in the table. The same requirement applies to the **Linked leg** field that is used for costs in the shipping container cost area only.

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Container activities entity

Container activities are used to track the progress of shipping containers. A record is created for each leg that is assigned to the journey template that is selected at the time of shipping container creation. Records are also created when the shipping container is created via a data entity.

Information about the progress of an in-transit shipping container is often received from an external source. The container activities entity enables an external source such as a freight forwarder to directly update the record. Therefore, no manual update of data is required.

Container activities (ITMContainerActivityEntity)

You can use the container activities entity (`ITMContainerActivityEntity`) to create additional activity records. Alternatively, a freight forwarder can pass updates for a confirmed **Actual end date** value.

Expand table

Name	Mapping	Data type	Key	Mandatory
Actual end date	ITMContainerActivityTable.ActualEndDate	Datetime	No	No
Mode of delivery	ITMContainerActivityTable.DlvModeId	Nvarchar(10)	No	No
Estimated end date	ITMContainerActivityTable.EsimatedDate	Datetime	No	No
Line number	ITMContainerActivityTable.LineNum	Numeric(32, 16)	**Yes**	No
Notes	ITMContainerActivityTable.Notes	nvarchar(MAX)	No	No
Activity	ITMContainerActivityTable.ShipActivityId	Nvarchar(10)	No	**Yes**
Shipping container	ITMContainerActivityTable.ShipContainerId	Nvarchar(20)	**Yes**	**Yes**
Voyage	ITMContainerActivityTable.ShipId	Nvarchar(20)	**Yes**	**Yes**
Leg	ITMContainerActivityTable.ShipLegId	Nvarchar(20)	No	**Yes**
Service provider	ITMContainerActivityTable.ShipServiceProvider	Nvarchar(20)	No	No
Temperature	ITMContainerActivityTable.ShipTemperature	Numeric(32, 6)	No	No
Vessel	ITMContainerActivityTable.ShipVesselId	Nvarchar(20)	No	No

Name	Mapping	Data type	Key	Mandatory
Start date	ITMContainerActivityTable.StartDate	Datetime	No	No

Tracking control

The tracking control center enables an update of a specified source field to be triggered by an update of a specified target field. If both the value of the source field and the value of the target field are updated by using the container activities entity, the target field will show the entity value. This behavior is related to tracking control records that have a **Create type** value of *Lead time*.

For cost areas that have a **Create type** value of *Status update* or *Blank* (which is a user-defined value), the system will update the voyage status or target field according to the tracking control configuration.

Calculated fields

The values of the following fields in a container activity record are calculated based on the values of other fields. Although these calculated fields aren't included in the data entity, they will be set if the fields that the calculations are based on are provided.

- **Estimated days** = **Estimated end date** – **Start date**
- **Actual days** = **Actual end date** – **Start date**

Last updated on 11/26/2025

Vendor invoice entities

06/17/2025

The **Landed cost** module enables cost type codes to be configured for internal costs or externally derived costs. If a cost is external to a business, an invoice is expected from the service provider. This invoice is processed as an invoice journal that can be associated with a voyage, and the value of the invoice can be distributed across one or more costs of the voyage.

The vendor invoice entities enable the value of a journal line to be allocated across one or more costs of a voyage that share the same cost type code.

A cost can be allocated only if the invoice journal header and invoice journal lines exist.

Vendor voyage cost allocations (ITMLedgerJournalCostLinesVoyagesEntity)

The vendor voyage cost allocations data entity enables a vendor invoice line to be allocated across one or more costs that are applied to the voyage cost area. This functionality is supported by the `ITMLedgerJournalCostLinesVoyagesEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Allocated amount	ITMLedgerJournalCostLines.Amount	Numeric(32, 6)	No	No
Cost transaction line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No
Journal line number	ITMLedgerJournalCostLines.RefRecId	Numeric(32, 16)	**Yes**	No
Journal number	ITMLedgerJournalCostLines.RefRecId	Nvarchar(20)	**Yes**	No
Voyage	ITMLedgerJournalCostLines.CostTransRefRecId	Nvarchar(20)	**Yes**	No

Vendor shipping container cost allocations (ITMLedgerJournalCostLinesContainersEntity)

The vendor shipping container cost allocations data entity enables a vendor invoice line to be allocated across one or more costs that are applied to the shipping container cost area. This functionality is supported by the `ITMLedgerJournalCostLinesContainersEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Allocated amount	ITMLedgerJournalCostLines.Amount	Numeric(32, 6)	No	No
Shipping container	ITMLedgerJournalCostLines.CostTransRefRecId	Nvarchar(20)	**Yes**	No
Cost transaction line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No
Journal line number	ITMLedgerJournalCostLines.RefRecId	Numeric(32, 16)	**Yes**	No
Journal number	ITMLedgerJournalCostLines.RefRecId	Nvarchar(20)	**Yes**	No
Voyage	ITMLedgerJournalCostLines.CostTransRefRecId	Nvarchar(20)	**Yes**	No

Vendor folio cost allocations (ITMLedgerJournalCostLinesFoliosEntity)

The vendor folio cost allocations data entity enables a vendor invoice line to be allocated across one or more costs that are applied to the folio cost area. This functionality is supported by the `ITMLedgerJournalCostLinesFoliosEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Allocated amount	ITMLedgerJournalCostLines.Amount	Numeric(32, 6)	No	No
Cost transaction line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No
Folio	ITMLedgerJournalCostLines.CostTransRefRecId	Nvarchar(20)	**Yes**	No
Journal line number	ITMLedgerJournalCostLines.RefRecId	Numeric(32, 16)	**Yes**	No

Name	Mapping	Data type	Key	Mandatory
Journal number	ITMLedgerJournalCostLines.RefRecId	Nvarchar(20)	**Yes**	No

Vendor purchase order cost allocations (ITMLedgerJournalCostLinesPurchTableEntity)

The vendor purchase order cost allocations data entity enables a vendor invoice line to be allocated across one or more costs that are applied to the purchase order cost area. This functionality is supported by the `ITMLedgerJournalCostLinesPurchTableEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Allocated amount	ITMLedgerJournalCostLines.Amount	Numeric(32, 6)	No	No
Cost transaction line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No
Journal line number	ITMLedgerJournalCostLines.RefRecId	Numeric(32, 16)	**Yes**	No
Journal number	ITMLedgerJournalCostLines.RefRecId	Nvarchar(20)	**Yes**	No
Purchase order	ITMLedgerJournalCostLines.CostTransRefRecId	Nvarchar(20)	**Yes**	No

Vendor item cost allocations (ITMLedgerJournalCostLinesPurchLineEntity)

The vendor item cost allocations data entity enables a vendor invoice line to be allocated across one or more costs that are applied to the item cost area. The item cost area covers costs on the purchase order line. This functionality is supported by the `ITMLedgerJournalCostLinesPurchLineEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Allocated amount	ITMLedgerJournalCostLines.Amount	Numeric(32, 6)	No	No
Cost transaction line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No
Journal line number	ITMLedgerJournalCostLines.RefRecId	Numeric(32, 16)	**Yes**	No
Journal number	ITMLedgerJournalCostLines.RefRecId	Nvarchar(20)	**Yes**	No
Purchase order	ITMLedgerJournalCostLines.CostTransRefRecId	Nvarchar(20)	**Yes**	No
Purchase order line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No

Vendor transfer order line cost allocations (ITMLedgerJournalCostLinesTransferLineEntity)

The vendor transfer order line cost allocations data entity enables a vendor invoice line to be allocated across one or more costs that are applied to the transfer line cost area. This functionality is supported by the `ITMLedgerJournalCostLinesTransferLineEntity` entity. The following table lists the fields and mappings that make up this entity.

Expand table

Name	Mapping	Data type	Key	Mandatory
Allocated amount	ITMLedgerJournalCostLines.Amount	Numeric(32, 6)	No	No
Cost transaction line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No
Journal line number	ITMLedgerJournalCostLines.RefRecId	Numeric(32, 16)	**Yes**	No
Journal number	ITMLedgerJournalCostLines.RefRecId	Nvarchar(20)	**Yes**	No
Transfer order	ITMLedgerJournalCostLines.CostTransRefRecId	Nvarchar(20)	**Yes**	No
Transfer order line number	ITMLedgerJournalCostLines.CostTransRefRecId	Numeric(32, 16)	**Yes**	No

Reference table

Voyage cost lines have a direct association with a cost transaction record. This record includes the **Reference table ID** value. This value is unique for each cost area (voyage, shipping container, and so on). When specific table numbers are referenced for data that is created by using data entities, the entities are split based on **Reference table ID** values.

The values for the reference table (`RefTableId`) and the transaction type (`TransType`) are implicit in the selection of either the Landed cost purchase line or the Landed cost transfer line entity.

Vendor invoice journal lines (VendorInvoiceJournalLineEntity)

Before a journal line value can be allocated to one or more costs in the **Landed cost** module, the journal line must be associated with a cost area. To support this functionality, the **Landed cost** module adds some new fields to the journal lines table (`LedgerJournalTrans`).

Fields added to the vendor invoice journal line entity

The following table lists the fields that the **Landed cost** module adds to the vendor invoice journal line entity (`VendorInvoiceJournalLineEntity`). These fields enable the system to create journal lines and allocate costs against them.

Expand table

Name	Mapping	Data type	Key	Mandatory
Cost area	LedgerJournalTrans.ITMCostArea	Int	No	No
Cost type code	LedgerJournalTrans.ITMCostTypeId	Nvarchar(20)	No	No

Main/offset account

The main/offset account for an invoice journal line that is associated with a Landed cost voyage is determined by the cost type code. When the cost type code is included on the invoice journal line, the clearing account for the code will be used as either the main account or the offset account, depending on the scenario:

- **Single-line journal** – If a main account (in other words, the vendor account) is defined, and a cost type code is provided, the clearing account value for that cost type code will

be entered in the offset account.
- **Multiple-line journal** – If a main account isn't defined, but a cost type code is provided, then the clearing account value for that cost type code will be entered in the main account. The journal line that is crediting the vendor won't reference a cost type code.

Sequencing

Because of dependencies between the journal and journal lines tables, the voyage record must be created first. The voyage lines can be created only after the voyage, shipping container, and folios have been created.

To allocate a voyage invoice, the system must process the entities in the following order:

1. Invoice journal header (`VendInvoiceJournalHeaderEntity`)
2. Invoice journal line (`VendInvoiceJournalLineEntity`)
3. Landed cost allocations (`ITMLedgerJournalEntities`)

Warehouse management overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Warehouse management module lets you manage warehouse processes in manufacturing, distribution, and retail companies. This module has a wide range of features to support the warehouse facility at an optimal level, at any time. Warehouse management is fully integrated with other business processes such as transportation, manufacturing, quality control, purchase, transfer, sales, and returns.

Get started

To start working with Warehouse management, you need to complete the setup of the general warehouse parameters to support the business processes of your company.

- Go to the **Warehouse management parameters** page under **Warehouse management** > **Setup** to set up general warehouse parameters.

You must configure components for inbound and outbound warehouse process workflows according to business requirements. The most important components that you must configure are wave templates, work templates, work pools, and location directives.

- Warehouse configuration overview
- Control warehouse work by using work templates and location directives
- Set up mobile devices for warehouse work
- Set up a location directive for purchase order putaway
- Set up a work template for purchase orders

Warehouse management processes (WMS)

- Integrated support for source documents for sales orders, returns, transfer orders, production orders, and kanban
- Flexible, inbound and outbound material workflow support based on queries
- Full integration with the Manufacturing and Transportation offerings
- Full control of location stocking limits and location volumetrics

- Inventory properties controlled by inventory status
- Full batch and serial item support
- Various item receiving capabilities
- Multiple picking strategies
- Out-of-the-box support for the next generation of bar code scanners
- Pallet and container types for warehouse processes
- Advanced counting capabilities
- Label printing and label routing with Zebra ZPL support
- Business intelligence integration into Power BI
- Manual and automatic movement of inventory
- Fully integrated quality control (QMS)
- Full traceability of workers' material handling
- Outbound wave processing
- Manual packing and automatic containerization support
- Cluster picking
- Simple cross docking

Related information

What's new and in development

Go to the Dynamics 365 Release Planner to see what new features are released and what new features are in development.

Blogs

You can find opinions, news, and other information about Warehouse management and other solutions on the Microsoft Dynamics 365 blog.

Articles

The following article explores various aspects of warehouse management in Dynamics 365 Supply Chain Management:

- Reservations in Warehouse management

Last updated on 11/19/2025

Get started with setting up the Warehouse management module

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to work with the built-in wizards and checklists that can help you quickly and efficiently set up and configure the **Warehouse management** module.

Track the Warehouse management configuration processes

Use the **Warehouse implementation tasks** workspace to track your Warehouse management configuration processes. The workspace can be used during a new implementation, after an upgrade, or after a migration.

The workspace lets you create, edit, and import checklists that you can use to identify important implementation tasks and track your progress while you complete them. It lets you perform the following actions:

- Track progress by viewing all current and closed configuration tasks.
- Select hyperlinks to quickly access the configuration page that's required for each step.
- Import a default configuration checklist that includes the tasks that you must complete to get a warehouse up and running.

Open the Warehouse implementation tasks workspace

To open the workspace, go to **Warehouse management** > **Workspaces** > **Warehouse implementation tasks**.

Create an implementation project

You must first create at least one implementation task project. You can add as many projects as you need. All data that's shown in the workspace is filtered by the selected implementation task project. Follow these steps to create a project in the **Warehouse implementation tasks** workspace.

1. In the **Warehouse implementation tasks** workspace, select **New project** at the top of the page.
2. In the drop-down dialog box, set the following fields:
    - **New project** – Enter a name for the project.
    - **Copy existing project** – Select one of the following values to specify whether you want to create a new, blank project or to start from a copy of an existing project:
        - *No* – Create a new, blank project.
        - *Yes* – Copy an existing project.
3. Select **OK** to create the new project and close the drop-down dialog box.

Import the default configuration task list

Microsoft Dynamics 365 Supply Chain Management includes a default task list. This task list is a good place to start when you're planning and implementing your Warehouse management setup. Follow these steps to import it into the **Warehouse implementation tasks** workspace.

1. In the **Warehouse implementation tasks** workspace, in the **Implementation task project** field at the top of the page, select the project that you want to import tasks into.
2. On the **Tasks and status** FastTab, select **Import default tasks** on the toolbar.

    The default tasks should be loaded. However, you might receive the following error instead: "Entity Warehouse implementation tasks not found. Refresh entity list from Data import/export framework parameters form." In that case, follow these steps:

    a. Go to **System administration** > **Workspaces** > **Data management**.

    b. Select the **Framework parameters** tile.

    c. On the **Entity settings** tab, select **Refresh entity list**.

    The system adds a refresh entity job to your batch job queue and runs it as soon as it can. You might have to wait several minutes for the task to complete, depending on the number of other jobs in the queue.

d. When the refresh entity job is completed, return to the **Warehouse implementation tasks** page, and select **Import default tasks** again on the toolbar of the **Tasks and status** FastTab.

Customize the task list

After you import a task list, you can customize it as you require by adding, removing, and rearranging the tasks. You can also create a fully customized task list by starting from scratch. Use the following buttons on the toolbar of the **Tasks and status** FastTab in the **Warehouse implementation tasks** workspace to customize the list:

- **Add task** – Create a new task. This button opens the **Add task** dialog box, where you must set the following fields:
  - **Task category** – Enter the type of task.
  - **Description** – Enter a detailed description of what the user should do to complete the task.
  - **Task link** – Select the page in Supply Chain Management where the user should go to complete the task.
- **Edit task** – Edit a selected task. This button opens the **Edit task** dialog box, where you can edit the **Description** and **Task link** fields for the task. Other settings are read-only for existing tasks.
- **Remove** – Delete all selected tasks. The system prompts you to confirm the operation.
- **Move up** – Move a selected task up in the sequence order.
- **Move down** – Move a selected task down in the sequence order.

Work with the task list

When your task list is ready, follow these steps to work through it.

1. In the **Warehouse implementation tasks** workspace, expand the **Summary** FastTab for an overview of the total number of tasks and the proportion of them that has been marked as completed.
2. Expand the **Tasks and status** FastTab to view the full list and mark tasks as completed while you work.

   For each task, the following information is shown:

- **Sequence number** – The order to complete the task in. Start with the task that has the lowest number, and work upward.
- **Completed** – Select this checkbox for each task as you complete it.
- **Task category** – The type of task. Select the text to open a page where you can configure the settings that are required to complete the task.
- **Description** – Full details about what must be done to complete the task.
- **Status** – A check mark indicates that the task has been completed.
- **Completed by** – If the task has been completed, the name of the user who completed it.
- **Completed date** – If the task has been completed, the date when it was completed.

Use the following fields and buttons on the toolbar of the **Tasks and status** FastTab to find your way around the task list:

- **Filter** – To quickly find a specific task in a long list, enter a value here, and then select the column where that value should be found.
- **Hide completed tasks** or **Show completed tasks** – Select whether to hide or show tasks that have already been completed. The button label changes, depending on whether completed tasks are currently shown or hidden.

To view the full task list (including both completed and uncompleted tasks), on the **Links** FastTab, select the **All tasks** link. The page that appears is the same page that appears when you select the **Warehouse implementation tasks** tile on the **Summary** FastTab. It provides the same information as, and works like, the **Task list** FastTab on the **Warehouse implementation tasks** page.

3. Continue to work until all tasks are completed.

Warehouse management initiation wizard

The *Warehouse management initiation wizard* provides a step-by-step guide that helps you configure the most basic settings that the **Warehouse management** module requires. Use this wizard for new legal entities where you must configure the **Warehouse management** module from scratch. The wizard defines the main settings of Warehouse management parameters for you. The default task list that was described earlier in this article includes steps for setting up the prerequisites for this wizard. It also includes a step that indicates when to run this wizard. Run the wizard when you're instructed to do so in the default task list.

The following table summarizes the configurations that the wizard does.

Expand table

Configuration	Action	Description
Location types	Create	Location types are filtering options that control the different warehouse management processes. The wizard creates several location types and uses the names that you specify. After you complete the wizard, you can add, remove, and/or edit these and other location types by going to **Warehouse management** > **Setup** > **Warehouse** > **Location types**.
Location profiles	Create	Location profiles control the behavior of locations in a warehouse that's enabled for advanced warehouse management. The wizard creates several location profiles and uses the names that you specify. Each profile has a collection of default settings that you might want to customize after you complete the wizard. After you complete the wizard, you can add, remove, and/or edit these and other location profiles by going to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
Location formats	Create	Location formats are a naming system for creating unique and consistent names for the different location bin positions that are used in a warehouse. To make the components of the location, such as the aisle number, easier to identify, you might want to include separators as part of the location format. You can set the wizard to create a separate format for each location profile that it creates. Alternatively, the wizard can create a single format that is shared by all the profiles that it creates. After you complete the wizard, you can add, remove, and/or edit these and other location formats by going to **Warehouse management** > **Setup** > **Warehouse** > **Location formats**.
Inventory status	Create	Inventory statuses categorize and keep track of inventory. The wizard creates an initial inventory status value and uses the name that you specify. After you complete the wizard, you can add, remove, and/or edit this and other inventory statuses by going to **Warehouse management** > **Setup** > **Inventory** > **Inventory statuses**.
Default work user and password	Create	The default work user is used for automated work transactions. The wizard creates an initial work user and uses the user name and password that you specify. After you complete the wizard, you can add, remove, and/or edit this and other work users by going to **Warehouse management** > **Setup** > **Worker**.

Configuration	Action	Description
Mobile device menu	Create	The mobile device menu contains the menu items that the mobile device shows to warehouse workers. The wizard creates an initial mobile device menu and uses the mobile device name and description that you specify. After you complete the wizard, you can add, remove, and/or edit this and other mobile device menus by going to **Warehouse management** > **Setup** >**Mobile device** > **Mobile device menu**.
Load posting methods	Regenerate	Load posting methods are used when a load is released to the warehouse from the outbound load planning workbench. The wizard generates or regenerates the load posting methods. After you complete the wizard, you can manually regenerate and/or edit the method list by going to **Warehouse management** > **Setup** > **Load posting methods**.
Wave processing methods	Regenerate	Wave processing methods perform the actions that a wave template creates. The wizard generates or regenerates the wave processing methods. After you complete the wizard, you can manually regenerate and/or edit the method list by going to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
Warehouse management parameters	Set up	Based on settings that you specify while you run the wizard, the wizard initializes several settings on the **Warehouse management parameters** page (**Warehouse management** > **Setup** > **Warehouse management parameters**). The following values are set: • User location profile • Location types • Default work user ID • Default inventory status

Follow these steps to open and complete the *Warehouse management initiation wizard*.

1. Before you open the wizard, make sure that the following prerequisite is in place. (It should be in place if you run the wizard when you're instructed to do so in the default task list that was described earlier in this article.)

    - The user account that you're using to sign in to Supply Chain Management must be associated with a person record. (Person records are used by the **Human resources** module to manage employees.) To set up this association, go to **System administration** > **Users** > **Users**, open your user account in the list, and use the **Person** field to link the account to the correct person record. The wizard adds this person as a

warehouse worker on the **Worker** page (**Warehouse management** > **Setup** > **Worker**). It also adds the default work user ID that you specify in the wizard to the warehouse worker record. (Learn more in Mobile device user accounts.) The system uses the default work user ID when it runs some automated processes.

2. Go to **Warehouse management** > **Setup** > **Wizards** > **Warehouse management initiation wizard**.

3. The first page of the *Warehouse management initiation wizard* is the **Welcome** page. It summarizes what the wizard does. When you finish reading the summary, select **Next** to continue.

4. On the **Initialize base data** page, use the following fields to define initial settings for the **Warehouse management** module. Default values are suggested, but you can edit them as you require. If you leave any field blank, the wizard doesn't initialize the setting.

    - **Receiving** – Specify the name of the receiving location type and location profile.
    - **Staging** – Specify the name of the staging location type and location profile. The specified staging location type is also identified on the **Warehouse management parameters** page.
    - **Packing** – Specify the name of the packing location type and location profile. The specified packing location type is also identified on the **Warehouse management parameters** page.
    - **Final shipping** – Specify the name of the final shipping location type and location profile. The specified final shipping location type is also identified on the **Warehouse management parameters** page.
    - **Sorting** – Specify the name of the sorting location type and location profile. The specified sorting location type is also identified on the **Warehouse management parameters** page.
    - **User** – Specify the name of the user location type and location profile. The specified user location profile is also identified on the **Warehouse management parameters** page.
    - **Create one generic format instead of each profile** – Set this option to *No* to create a separate location format for each location profile. The name of each format is based on the values that you specify in the **Receiving**, **Staging**, **Packing**, **Final shipping**, **Sorting**, and **User** fields. Set this option to *Yes* to create one location format that applies to all the location profiles that are created. This format uses the name that you specify in the **Location format** field.

- **Location format** – If the **Create one generic format instead of each profile** option is set to *Yes*, specify the name of the shared location format.
- **Default inventory status ID** – Specify the name of the default inventory status. The specified default inventory status ID is also identified on the **Warehouse management parameters** page.
- **Default work user ID** – Specify the name of the default work user. The specified default user ID is also identified on the **Warehouse management parameters** page.
- **Default work user password** – Specify the password of the default work user.
- **Mobile device menu** – Specify the name of the mobile device menu.

5. Select **Next** to continue.

6. The **Base data setup complete** page summarizes the settings that the wizard applies, based on your input. Review the summary, and if everything looks correct, select **Finish** to apply the settings and close the wizard.

Inbound configuration wizard

The *Inbound configuration wizard* provides a step-by-step guide that helps you set up inbound operations for one or more warehouses. After you complete this wizard, the required settings are in place so that you can receive purchase orders and/or inbound shipment orders. Use this wizard for new legal entities or new warehouses where you must configure basic inbound flows. The default task list that was described earlier in this article includes steps for setting up the prerequisites for this wizard. It also includes a step that indicates when to run this wizard. Run the wizard when the default task list instructs you to do so.

The following table summarizes the configurations that the wizard does.

Expand table

Configuration	Action	Description
Locations	Create	Locations identify where items are stored, picked from, and put to in the warehouse. The wizard creates a receiving location for each warehouse that you select at the start of the wizard and uses the name and location profile that you specify in the wizard. After you complete the wizard, you can add, remove, and/or edit these and other locations by going to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.
Work classes	Create	Work classes direct and/or limit the types of work order lines that a warehouse worker can process on a mobile device. The wizard creates a single work class

Configuration	Action	Description
		and uses the name that you specify. After you complete the wizard, you can add, remove, and/or edit this and other work classes by going to **Warehouse management** > **Setup** > **Work** > **Work classes**.
Location directives	Create	Location directives identify pick and put locations for inventory movement. The wizard creates a put location directive for each warehouse that you select at the start of the wizard. It names each location directive by prefixing the name that you specify in the wizard with the name of the warehouse where the location directive applies. Each location directive that's created has a work order type of *Purchase orders* and/or *Inbound shipment order*. After you complete the wizard, you can add, remove, and/or edit these and other location directives by going to **Warehouse management** > **Setup** > **Location directives**. Learn more in Work with location directives.
Work templates	Create	Work templates create warehouse work at different points in the system. The wizard creates a work template and uses the name that you specify in the wizard. The work template has a work order type of *Purchase orders* and/or *Inbound shipment order*. After you complete the wizard, you can add, remove, and/or edit this and other work templates by going to **Warehouse management** > **Setup** > **Work** > **Work templates**.

Follow these steps to open and complete the *Inbound configuration wizard*.

1. Before you open the wizard, make sure that the following prerequisites are in place. (They should already be in place if you run the wizard when you're instructed to do so in the default task list that was described earlier in this article.)

    - Your basic configuration must be set up on the **Warehouse management parameters** page. You can complete this configuration by running the *Warehouse management initiation wizard* as described in the previous section. Alternatively, you can complete it manually by going to **Warehouse management** > **Setup** > **Warehouse management parameters**.
    - Each warehouse that you want to set up by using this wizard must already exist as a record in the system, and the **Use warehouse management processes** option must be set to *Yes* for it. You can create warehouses and set the **Use warehouse management processes** option for each of them by going to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**. Each warehouse must also be assigned to a site that exists

as a record in the system. You can create sites by going to **Warehouse management** > **Setup** > **Warehouse** > **Sites**.

2. Go to **Warehouse management** > **Setup** > **Wizards** > **Inbound configuration wizard**.

3. The first page of the *Inbound configuration wizard* is the **Welcome** page. It summarizes what the wizard does. When you finish reading the summary, select **Next** to continue.

4. The **Warehouse selection** page opens. Use it to specify the warehouses where the wizard applies. Set the **Warehouse selection** field to one of the following values, and then set the associated fields as required:

    - *All* – Configure all warehouses.
    - *Warehouse group* – Configure all the warehouses in a selected warehouse group. After you select this option, use the **Warehouse group** field to select the group that you want to configure. For more information about how to create warehouse groups, see Warehouse groups.
    - *Warehouse* – The wizard configures the warehouses selected in the grid. Only warehouses where the **Use warehouse management processes** option is set to *Yes* are listed. Select the checkbox for each warehouse that you want to set up by using the wizard.

5. Select **Next** to continue.

6. When Warehouse management only mode is enabled, use the **Work order type** section to define *purchase orders* or *inbound shipment orders* for the following setup processes.

7. The **Inbound setup** page opens. Use it to define names for different elements of the purchase or inbound shipment order process. Default values are suggested, but you can edit them as you require. Set the following fields:

    - **Purchase/Inbound work class** – Specify the name of the work class for the receiving process. This value is used in the work template.
    - **Purchase/Inbound work template** – Specify the name of the work template for the receiving process.
    - **Purchase/Inbound put location directive** – Specify the name of the put location directive for the receiving process.
    - **Receiving location** – Specify the name of the receiving location.
    - **Location profile** – Select the location profile that should be assigned to the selected receiving location.

8. Select **Next** to continue.

9. The **Inbound setup complete** page opens. It summarizes the actions that the wizard takes. Review the summary, and if everything looks correct, select **Finish** to apply the settings and close the wizard.

Outbound configuration wizard

The *Outbound configuration wizard* provides a step-by-step guide that helps you set up outbound operations for one or more warehouses. After you complete this wizard, the required settings are in place so that you can ship sales and/or outbound shipment orders. Use this wizard for new legal entities or new warehouses where you must configure basic outbound flows. The default task list that was described earlier in this article includes steps for setting up the prerequisites for this wizard. It also includes a step that indicates when to run this wizard. Run the wizard when the default task list instructs you to do so.

The following table summarizes the configurations that the wizard does.

Expand table

Configuration	Action	Description
Locations	Create	Locations identify where items are stored, picked from, and put to in the warehouse. The wizard creates several locations for each warehouse that you select at the start of the wizard and uses the names that you specify in the wizard. It associates each location with the relevant warehouse. After you complete the wizard, you can add, remove, and/or edit these and other locations by going to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.
Directive codes	Create	Directive codes link the work templates and the location directives. The wizard creates several directive codes and uses the names that you specify in the wizard. After you complete the wizard, you can add, remove, and/or edit these and other directive codes by going to **Warehouse management** > **Setup** > **Directive codes**.
Work classes	Create	Work classes direct and/or limit the types of work order lines that a warehouse worker can process on a mobile device. The wizard creates several work classes and uses the names that you specify in the wizard. After you complete the wizard, you can add, remove, and/or edit these and other work classes by going to **Warehouse management** > **Setup** > **Work** > **Work classes**.

Configuration	Action	Description
Location directives	Create	Location directives identify pick and put locations for inventory movement. The wizard creates pick and put location directives for each warehouse that you select at the start of the wizard. It names each location directive by prefixing the name that you specify in the wizard with the name of the warehouse where the location directive applies. Each location directive that's created has a work order type of *Sales orders* and/or *Outbound shipment order*. After you complete the wizard, you can add, remove, and/or edit these and other location directives by going to **Warehouse management** > **Setup** > **Location directives**. Learn more in Work with location directives.
Wave templates	Create	Wave templates define the wave execution process and set up criteria for when waves are created, run, and released. The wizard creates a wave template for each warehouse that you select at the start of the wizard. It names each wave template by prefixing the name that you specify in the wizard with the name of the warehouse where the wave template applies. Each wave template that's created has a wave template type of *Shipping*. After you complete the wizard, you can add, remove, and/or edit these and other wave templates by going to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
Work templates	Create	Work templates create warehouse work at different points in the system. The wizard creates a work template for each warehouse that you select at the start of the wizard. It names each work template by prefixing the name that you specify in the wizard with the name of the warehouse where the work template applies. Each work template that's created has a work order type of *Sales orders* and/or *Outbound shipment order*. After you complete the wizard, you can add, remove, and/or edit these and other work templates by going to **Warehouse management** > **Setup** > **Work** > **Work templates**.

Follow these steps to open and complete the *Outbound configuration wizard*.

1. Before you open the wizard, make sure that the following prerequisites are in place. (They should already be in place if you run the wizard when you're instructed to do so in the default task list that was described earlier in this article.)

    - Your basic configuration must be set up on the **Warehouse management parameters** page. You can complete this configuration by running the *Warehouse management initiation wizard* as described earlier in this article. Alternatively, you can complete it manually by going to **Warehouse management** > **Setup** > **Warehouse management parameters**.

- Each warehouse that you want to set up by using this wizard must already exist as a record in the system, and the **Use warehouse management processes** option must be set to *Yes* for it. You can create warehouses and set the **Use warehouse management processes** option for each of them by going to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**. Each warehouse must also be assigned to a site that exists as a record in the system. You can create sites by going to **Warehouse management** > **Setup** > **Warehouse** > **Sites**.

2. Go to **Warehouse management** > **Setup** > **Wizards** > **Outbound configuration wizard**.

3. The first page of the **Outbound configuration** wizard is the **Welcome** page. It summarizes what the wizard does. When you finish reading the summary, select **Next** to continue.

4. The **Warehouse selection** page opens. Use it to specify the warehouses where the wizard applies. Set the **Warehouse selection** field to one of the following values, and then set the associated fields as required:

    - *All* – Configure all warehouses.
    - *Warehouse group* – Configure all the warehouses in a selected warehouse group. After you select this option, use the **Warehouse group** field to select the group that you want to configure. For more information about how to create warehouse groups, see Warehouse groups.
    - *Warehouse* – The wizard configures the warehouses selected in the grid. Only warehouses where the **Use warehouse management processes** option is set to *Yes* are listed. Select the checkbox for each warehouse that you want to set up by using the wizard.

5. Select **Next** to continue.

6. When Warehouse management only mode is enabled, use the **Work order type** section to define *sales orders* or *outbound shipment orders* for the following setup processes.

7. On the **General warehouse** page, use the following fields to define names for outbound processes. Default values are suggested, but you can edit them as you require.

    - **Sales/Outbound work template** – Specify the name of the work template for the shipping process.
    - **Shipping wave template** – Specify the name of the wave template for the shipping process.

- **Sales/Outbound pick location directive** – Specify the name of the pick location directive for the shipping process.

8. Select **Next** to continue.

9. On the **Staging area** page, use the following fields to specify whether you use a staging area in your outbound processes and, if you do, to set up the feature. You might have to use the feature if you pick your items and then put them in a staging location where all lines of an order are held. In the staging area, items can be packed, relabeled, or loaded onto a delivery truck, and then sent out the door. Default values are suggested, but you can edit them as you require.

    - **Use staging area** – Select this checkbox if you use a staging area. If you clear this checkbox, a staging area isn't set up, and all the other fields on the page are unavailable.
    - **Location** – Specify the name of the staging location.
    - **Directive code** – Specify the name of the directive code for the shipping process. This value is used in the shipping work template and location directive to link them together.
    - **Sales/Outbound work class** – Specify the name of the work class for the shipping process. This value is used in the work template.
    - **Sales/Outbound put location directive** – Specify the name of the put location directive for the shipping process.

10. Select **Next** to continue.

11. On the **Packing preference** page, use the following fields to specify whether you use packing functionality in your outbound processes and, if you do, to set up the feature. You might have to use the feature if you pack items before you ship them. Default values are suggested, but you can edit them as you require.

    - **Would you like to pack items before shipping** – Select this checkbox if you use packing. If you select this option, you must also select the type of packing functionality that you use (manual packing or automatic packing through wave containerization). If you clear this checkbox, the system doesn't set up packing, and all the other pages that are related to packing are unavailable.

    - **Manually pack items through Pack station** – Select this checkbox if you want to use manual packing. Then set the following fields:
        - **Directive code** – Specify the name of the directive code.

    - **Sales/Outbound work class** – Specify name to use for the work class.
    - **Pack location** – Specify the name of the packing location.
    - **Sales/Outbound put location directive** – Specify the name of the put location directive for the packing process.

- **Pack items through wave containerization** – Select this checkbox if you want to use automatic packing through wave containerization. Then set the following fields:
    - **Container build template** – Specify the name of the container build template.
    - **Wave step code** – Specify the name of the wave step code. This value is used in the container build template and the wave template to link them together.

12. Select **Next** to continue. If you chose not to use packing, you skip ahead to the **Final shipping area** page of the wizard. Therefore, you can skip ahead to step 21 of this procedure. If you chose to use packing, move on to the next step.

13. On the **Container type and group** page, create or select container types for packing.

    - If you don't see the containers that you want to use, select **Add** on the toolbar, and then fill in the columns as required to create and set up the container.
    - If you selected the **Manually pack items through Pack station** option on the **Packing preference** page, you must select exactly one container type that has a maximum net weight that's more than 0 (zero).
    - If you selected the **Pack items through wave containerization** option on the **Packing preference** page, select one or more container types. Then set the **Container group** field to the name of the container group that you want to create to hold the selected container types.

14. Select **Next** to continue. If you selected the **Pack items through wave containerization** option on the **Packing preference** page, you skip ahead to the **Final shipping area** page of the wizard. Therefore, you can skip ahead to step 21 of this procedure. If you selected the **Manually pack items through Pack station** option on the **Packing preference** page, move on to the next step.

15. On the **Release container** page, set the following fields:

    - **Container packing policy** – Specify the name of the container packing policy.
    - **Container weight unit** – Select the container weight unit.
    - **Packing profile** – Specify the name of the packing profile.

16. Select one of the following options to specify what's done with the packed container:

- **Make available at the final shipping location** – Select this checkbox to move the packed container to the final shipping location after it's closed. After you select this option, you must use the **Default location for final shipment** field to define the final shipping location.
- **Create work to move container from packing station to location** – Select this checkbox to create work to move the packed container after it's closed. After you select this option, you must use the **Work template** field to define the name of the work template. Later in this procedure, you can use the **Container staging area** page to set up a staging area for this movement, as required.
- **Make available at sorting location** – Select this checkbox to move the packed container to the sorting area after it's closed. Later in this procedure, use the **Sorting area** page to set up the sorting area.

17. Select **Next** to continue. If you selected the **Make available at sorting location** option on the **Release container** page, the **Sorting area** page opens. In this case, move on to the next step. If you selected one of the other options on the **Release container** page, skip ahead to step 19.

18. On the **Sorting area** page, set the following fields:

    - **Sort location** – Specify the name of the sorting location.
    - **Outbound sorting template** – Specify the name of the outbound sorting template.
    - **Create work on position close** – Select whether you want to create work after the position is closed.
    - **Work template** – This field is shown only if you set the **Create work on position close** option to *Yes*. Use it to specify the name of the work template that's used to create the work.
    - **Sort final location** – This field is shown only if you set the **Create work on position close** option to *No*. Use it to specify the name of the final sort location.

    For more information about sorting, see Outbound sorting.

19. Select **Next** to continue. If you selected the **Create work to move container from packing station to location** option on the **Release container** page, the **Container staging area** page opens. In this case, move on to the next step. If you selected one of the other options on the **Release container** page, skip ahead to step 21.

20. On the **Container staging area** page, use the following fields to set up a staging area for staging packed items before moving them to the final shipping area. Default values are suggested, but you can edit them as you require.

- **Use staging area** – Select this checkbox if you use a staging area. If you clear this checkbox, a staging area isn't set up, and all the other fields on the page are unavailable.
- **Location** – Specify the name of the staging location.
- **Directive code** – Specify the name of the directive code for the shipping process. This value is used in the shipping work template and the location directive to link them together.
- **Packing work class** – Specify the name of the work class for the packing process. This value is used in the work template.
- **Packing put location directive** – Specify the name of the put location directive for the packing process.

21. Select **Next** to continue.

22. The **Final shipping area** page opens, regardless of which options you selected earlier. Use the following fields to define names for the final shipping area in the outbound processes. Default values are suggested, but you can edit them as you require.

    - **Location** – Specify the name of the final shipping location.
    - **Directive code** – Specify the name of the directive code for the shipping process. This value is used in the shipping work template and location directive to link them together.
    - **Packing work class** – Specify the name of the work class for the shipping process. This value is used in the work template.
    - **Packing put location directive** – Specify the name of the put location directive for the shipping process.

23. The **Warehouse setup complete** page summarizes the actions the wizard will take. Review the summary, and if everything looks correct, select **Finish** to apply the settings and close the wizard.

Last updated on 07/01/2026

Reservations in Warehouse management

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the reservation functionality in Warehouse management and provides information about how it's implemented.

The **Warehouse management** module introduces functionality for reserving items and materials. You can use the reservation functionality only for items that are enabled for warehouse management processes (WMS). However, you can use an item that is enabled for WMS in both warehouses that are enabled for WMS and warehouses that aren't enabled for WMS.

The behavior in reservation scenarios varies, depending on the warehouse setup. Later sections of this article provide more details about the differences.

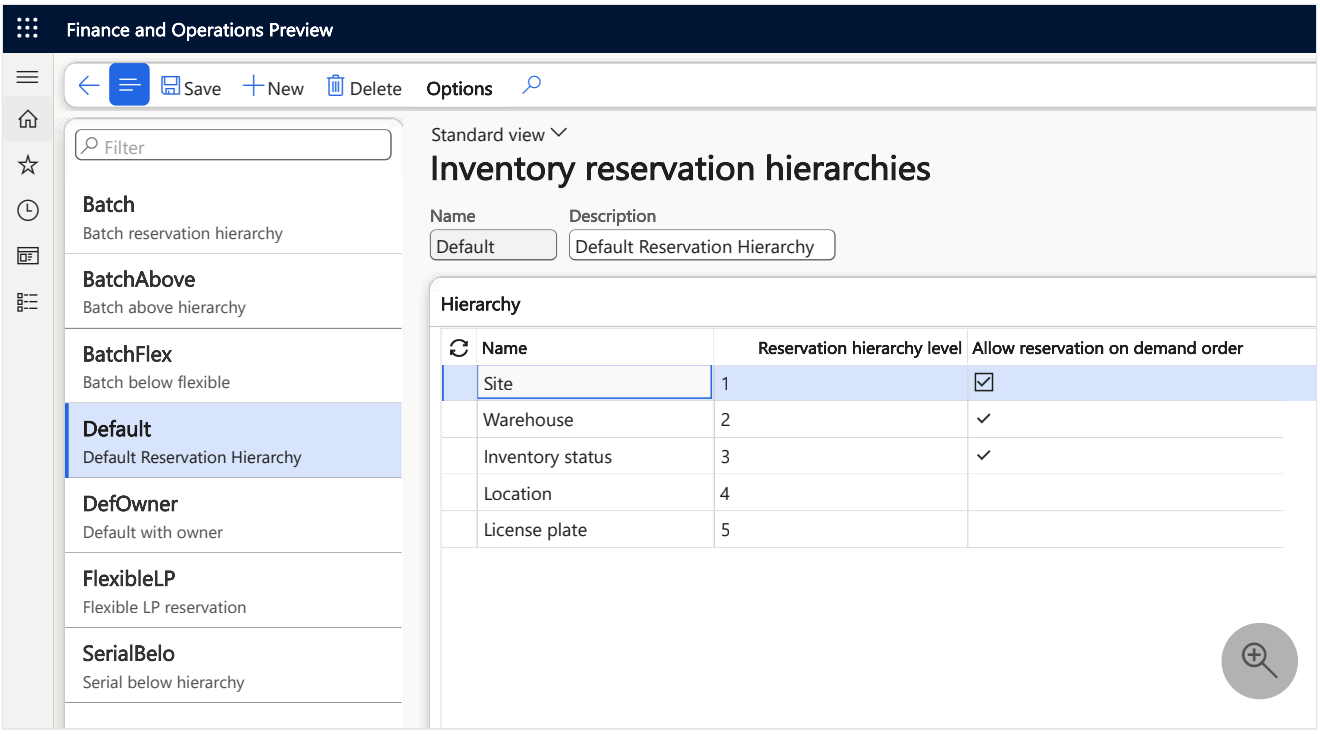
The reservation functionality is built on *reservation hierarchies* and is intended to support the following cases:

- Flexible warehouse operations
- Postponement of reservation details
- Clear separation of which inventory dimensions can be specified, and when they can be specified

Reservation hierarchies

A key component of the reservation system in the **Warehouse management** module is the reservation hierarchy. The reservation hierarchy defines the different levels where reservations can be made. Each level represents a physical inventory dimension.

The following illustration shows an example of a reservation hierarchy on the **Inventory reservation hierarchies** page. *Default* is a typical reservation hierarchy for an item that uses the Site, Warehouse, Inventory status, Location, and License plate dimensions as physical inventory dimensions.

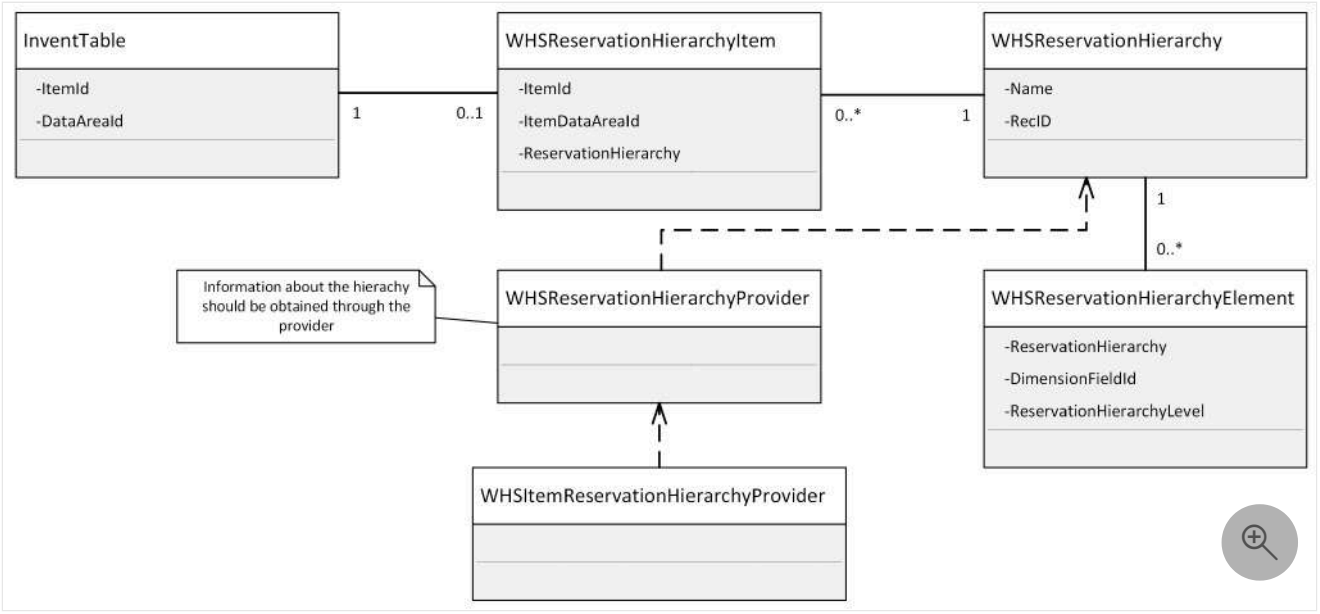


The lower the value in the **Reservation hierarchy level** column of the grid, the less specific the level is. The higher the value, the more details are required to make a reservation on the level. In the previous illustration, the **Site** level has the lowest value (*1*) and is therefore the least specific level. The **License plate** level has the highest value (*5*) and is therefore the most specific level.

The system stores information about on-hand inventory separately for each level in the reservation hierarchy.

Details about the implementation of reservation hierarchies

The following illustration shows the data model for the setup of a reservation hierarchy and its relation to the `inventTable` table.



The `WHSReservationHierarchy` and `WHSReservationHierarchyElement` tables store the definitions of reservation hierarchies. These tables are shared tables. You can associate a reservation hierarchy with one item within a company.

The `WHSReservationHierarchyProvider` class provides many APIs that are useful when you work with and query the reservation hierarchy for an item.

Making reservations on different levels

Based on the reservation hierarchy, you can make reservations on different levels without providing details about where to reserve and what dimensions to reserve on. In this way, you can postpone specific details, such as the location or license plate to make the reservation on. The location to reserve on can be determined later by the location directive.

Here are some examples:

- You make a reservation only on the site level. In this case, you can use inventory blocking to block a quantity of an item on a given site.
- You make reservations for a sales order that you created on site 4 and warehouse 42, and for the *Available* inventory status. Warehouse 42 is enabled for WMS. In this case, the reservation is made only on the inventory status level.

Impact of the reservation hierarchy setup

For WMS-enabled warehouses, the setup of reservation hierarchies greatly affects the processes for outbound orders, such as sales orders and transfer orders.

When you set up a reservation hierarchy, the key is to determine which dimensions should be positioned above the location level in the hierarchy, and which dimensions should be positioned below the location level. The following table describes the effects of arranging dimensions either above or below the location level.

Expand table

Dimension placement	Description
Dimensions above location	You must determine the inventory dimensions above the location level before you can use the Warehouse management functionality. Typically, workers make this determination during order processing, or they let the reservation system make it. If a dimension is above the location level, warehouse workers can't change it, because it's considered a strict picking requirement. For example, if the Batch number dimension is

Dimension placement	Description
	above the location level, a worker can't pick a batch that differs from the one that they were instructed to pick.
Batch above location	Process industry functionality for batches requires that the Batch number dimension is above the Location dimension in the reservation hierarchy. In this case, all functionality for first expiry, first out (FEFO), same batch reservation, batch disposition codes, and batch attributes is supported.
Dimensions below location	Both the system and warehouse workers can determine the Location dimension and the dimensions below it. Don't enter the Location dimension and any dimensions below it on sales and transfer lines if you expect work to be created. For example, if the Batch number dimension is below the Location dimension, don't specify it on the sales line. Otherwise, WMS can't create work to carry out the pick and pack operations.

On-hand representation and calculations

To support reservations on different levels, use a new method to store information about on-hand inventory for WMS-enabled items. Each level in the hierarchy corresponds to a set of physical inventory dimensions. The following information is stored for each level in the reservation hierarchy:

- The quantity that is available physical
- The quantity that is available ordered
- The quantity that is reserved physical
- The quantity that is reserved ordered

The following table shows how the data might look for an item that has 10 pieces available physical and 2 pieces reserved physical on the status level.

Expand table

Level	Site	Warehouse	Inventory status	Location	License plate	Available physical	Available ordered	Reserved physical	Reserve ordered
Item						10	10	2	0
Site	S1					10	10	2	0
Warehouse	S1	WH1				10	10	2	0
Status	S1	WH1	Good			10	10	2	0
Location	S1	WH1	Good	Bulk		12	12	0	0

Level	Site	Warehouse	Inventory status	Location	License plate	Available physical	Available ordered	Reserved physical	Reserve ordered
License plate	S1	WH1	Good	Bulk	LP1	12	12	0	0

As the table shows, reservations are tracked only on the levels that the reservation affects.

Because of the way that the system stores information about availability and reservations, you can't use the corresponding values from the `InventSum` table directly for WMS-enabled items.

On-hand calculation algorithm

The following steps provide a high-level overview of the algorithm that determines the available on-hand quantity for an item and a set of inventory dimensions:

1. Determine the available on-hand quantity based on the exact dimensions. This level is the lowest level in the hierarchy.
2. Determine the smallest available quantity from all levels above the level that was found in step 1.
3. Return the smaller of the quantities that were found in steps 1 and 2. This quantity is the available quantity.

For example, based on the data in the previous table, the available physical quantity is calculated for the following dimensions:

- **Site:** *Site1*
- **Warehouse:** *WH1*
- **Inventory status:** *Good*
- **Location:** *Bulk*

In step 1 of the algorithm, the quantity is 12, because that quantity is available on the location level. However, in step 2, the smallest quantity from the levels above the location level is 10. Therefore, in step 3, the available physical quantity is 10, because that quantity is the smaller of 12 and 10.

Calculations of on-hand quantity for work transactions are done differently. The on-hand quantity is determined only up to the location level, because work affects reservations only on the location level and below. For example, based on the data in the previous table, the available physical quantity is calculated for work on the following dimensions:

- **Site:** *Site1*
- **Warehouse:** *WH1*

- **Inventory status:** *Good*
- **Location:** *Bulk*

In this case, the available physical quantity is 12, because the quantity is determined only up to the location level.

Details about the implementation of on-hand calculations

The on-hand information is stored in a new table named `WHSInventReserve`. The `WHSInventReserveDelta` table tracks changes that the current transaction causes in the on-hand quantity. When the final commit occurs, the system updates the `WHSInventReserve` table based on the changes in the `WHSInventReserveDelta` table.

Because many scenarios involve the calculation of on-hand quantities based on `itemId`, `InventDim`, or `InventSum` records for both WMS-enabled and non-WMS-enabled items, the system provides APIs to support the calculations in a seamless way.

The following example shows how to instantiate the class that you can use to calculate availability for both WMS-enabled and non-WMS-enabled items.

X++

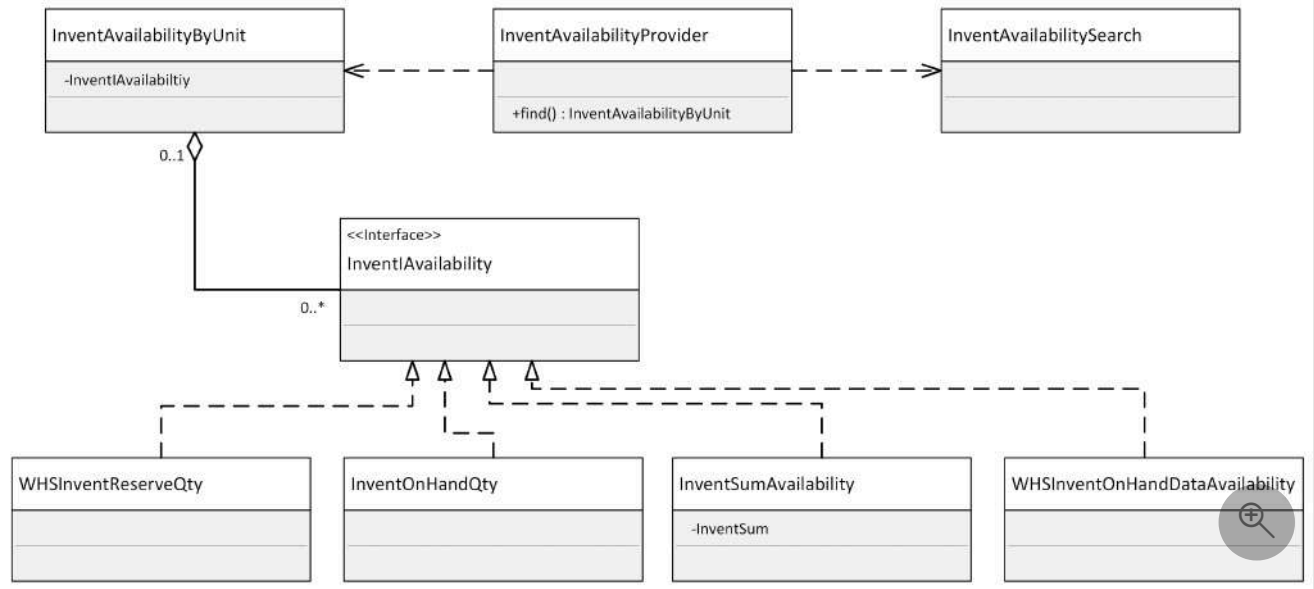
```
InventAvailabilitySearch availabilitySearch;
InventIAvailability availability;

availabilitySearch = InventAvailabilitySearch::construct();
availabilitySearch.setItemId(_itemId);
availabilitySearch.setInventDimCriteria(_inventDimCriteria, _inventDimCriteriaParm);
availabilitySearch.setInventSum(_inventSum);
availability =
InventAvailabilityProvider::construct().find(availabilitySearch).parmInventAvailabilit
y();

this.AvailOrderedCalculated = availability.availTotal();
this.AvailPhysicalCalculated = availability.availPhysical();
this.ReservPhysical = availability.reservPhysical();
this.OrderedSum = availability.orderedSum();
```

The `InventAvailabilityProvider` class provides numerous other APIs that are used to determine availability based on various types of input. The existing `InventOnHandQty` class was updated so that it also provides correct results for WMS-enabled items.

The following illustration shows the various classes and interfaces that are used to provide and encapsulate the calculations. For the sake of simplicity, some details are omitted.



Reservation strategies

To control the levels of the reservation hierarchy, the system uses the concept of *reservation strategies*. A reservation strategy determines the outcome of a reservation and the dimensions used to make the reservation. Reservation strategies are hard coded and aren't user configurable.

Each reservation strategy determines the level for the reservation. If the dimensions that the reservation system receives don't cover all the dimensions to the level that the strategy defines, the reservation system queries the on-hand availability until it reaches the required level. Then, based on the results of this query, it makes the reservation so that it covers all the required dimensions.

The reservation system chooses reservation strategies based on the following factors, among others:

- Warehouse setup
- Order type

The following table describes the reservation strategies that the reservation system uses.

Expand table

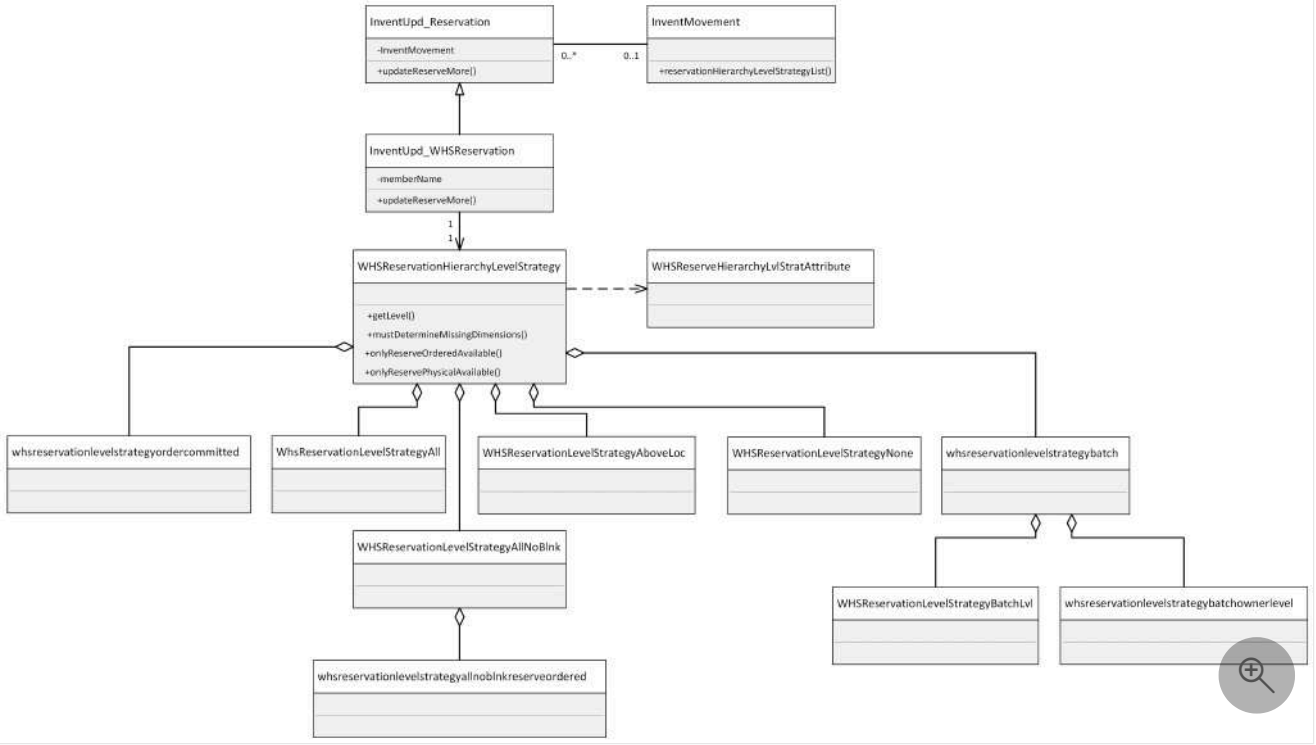
Reservation strategy	Description
None	The reservation system makes the reservation on the dimensions that it receives, if possible. Inventory blocking uses this strategy. It lets you make reservations on, for example, the site or warehouse level.
All	The reservation system makes the reservation on all the dimensions in the reservation hierarchy. This strategy is used, for example, for transfer journals or non-WMS-enabled

Reservation strategy	Description
	warehouses.
Above location	The reservation system makes the reservation only on the dimensions above the location level. This strategy is used, for example, for sales and transfer orders when reservations are made in a WMS-enabled warehouse.
All not allowed blank	The reservation system makes the reservation on the first lowest level that doesn't allow for blank issue for the inventory dimensions. This strategy enables automatic reservations on locations that aren't license plate controlled.
Batch level	This strategy is applied for items that the Batch number dimension is selected for, and that are above the Location dimension in the reservation hierarchy. This strategy is used when only reservations that are reserved ordered can be made. In that case, the reservation system attempts a reservation until the batch level.

The reservation system supports multiple reservation strategies in sequential order when it makes reservations. For example, the system uses the All and All not allowed blank strategies to make reservations for transfer journals.

Details about the implementation of reservation strategies

The following illustration shows the classes that implement reservation strategies and the main consumers of those classes. For simplicity, it omits some details.



You can easily extend the implementation because the instantiation uses the SysExtension framework.

Synchronization of dimensions between receipts and issues

When you mark or reserve order an inventory transaction, the system typically synchronizes the inventory dimensions between receipts and issues. For example, when you modify or receive a purchase order that you created based on a sales order, the system transfers the dimensions to the inventory transactions for the sales line.

For WMS-enabled items, the synchronization differs from the standard behavior. When the system creates work, it can reserve the source line transactions only up to the level above the location level. If all dimensions are synchronized, the system can't create work. Therefore, the system doesn't synchronize dimensions above the location level for all scenarios.

If both the item and the warehouse are enabled for WMS, and if the issue type is one that can generate work, such as a sales line, the system only synchronizes dimensions above the location level. Therefore, if an item uses batch numbers, and the Batch number dimension is below the location level in the reservation hierarchy, the system doesn't synchronize the batch number from receipt to issue transactions.

Details about the implementation of synchronization between receipts and issues

The logic that determines how dimensions are synchronized is implemented in the following methods on the `inventMovement` class. (However, new methods that you add might also require consideration.)

- `getInventDimForReservedTransPhysChange`
- `getInventDimForIssueTransFromReceipt`

> **Note**
>
> When products use an inventory reservation hierarchy of the *Batch-below[location]* type and are batch tracked for WMS, set a *flexible warehouse-level dimension reservation policy*. This policy ensures that you can reserve those products.
>
> Learn more about the flexible warehouse-level dimension reservation policy in **Flexible warehouse-level dimension reservation policy**.

Batch reservation policy for non-WMS warehouses

Items enabled for warehouse management processes (WMS) can be stocked non-WMS-enabled warehouses. However, if you do this, then WMS-enabled items that use a reservation hierarchy where the *Batch number* dimension is below the *Location* dimension prevent the warehouse work engine from enforcing batch-related rules during reservation. By default, this means that the reservation logic doesn't evaluate first expiry, first out (FEFO) rules, or batch disposition codes for those items in those warehouses. As a result, the system might reserve a quantity from a batch that is marked as unavailable through its batch disposition code.

To control this behavior, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Open the **General** tab.
3. Expand the **Batches** FastTab.
4. Set **Batch reservation policy for non advanced warehouses** to one of the following values:

    - *Simple* – The system applies batch disposition code logic only when the batch is explicitly specified on the order line. The reservation engine doesn't apply FEFO rules for non-WMS-enabled warehouses. This behavior matches that of older versions of Supply Chain Management and is the default value for this setting.
    - *Advanced* – The system applies both FEFO rules and batch disposition codes during reservation and during on-hand calculations, even when no batch is specified on the order line. The system excludes batches that are assigned a disposition code with the **Batch disposition status** field set to *Unavailable* (with the *Reserve* block status applied for the order type) from automatic reservation. The user receives the standard "Cannot reserve" feedback when no other on-hand quantity is available.

5. On the Action Pane, select **Save**.

The policy applies in the following situations:

- Sales orders, including automatic reservation, manual reservations from the **Reservation** page, and release to warehouse operations.
- Items that use a reservation hierarchy where the *Batch number* dimension is below the *Location* dimension.
- Lines for warehouses where the **Use warehouse management processes** option is *No*.

The policy does *not* apply in the following situations:

- Retail sales orders.
- Items that use a reservation hierarchy where the *Batch number* dimension is above the *Location* dimension. For those items, standard reservation strategies already apply batch

disposition codes and FEFO, regardless of the **Batch reservation policy for non advanced warehouses** setting.

- WMS-enabled warehouses. For those warehouses, the warehouse work engine enforces batch disposition codes.
- Order types other than sales orders, such as transfer orders or production orders.

There's a small performance overhead when you use the *Advanced* policy because the system performs more validations during reservation and on-hand calculations. There are also scenarios where the advanced criteria can't be applied when on-hand inventory is presented in the user interface.

For information about how to set up batch disposition codes and assign them to batches, see Use batch disposition codes to mark batches as available or unavailable.

Last updated on 06/22/2026

Warehouse handling of inbound loads for purchase and inbound shipment orders

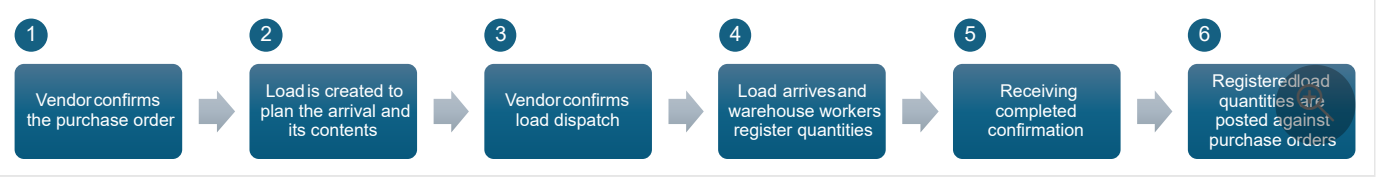
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the warehouse handling process for inbound loads.

Each inbound load can be associated with one or more order line quantities, and your system might also contain transportation plans. For more information about how to create and manage inbound transportation, see Transportation management overview.

Overview: How inbound loads are created, registered, and received

The following high-level illustration shows an example flow for handling inbound loads for a purchase order. Note that if the purchase order originates in an enterprise resource planning (ERP) system other than Microsoft Dynamics 365 Supply Chain Management, it's represented by an *inbound shipment order* in the **Warehouse management** module.



1. **The vendor confirms the purchase order.**

   The process begins when a purchase order is entered into the system and then delivered to a vendor, who confirms the order. The purchase order/inbound shipment order must exist before you can create an inbound load record. However, you can create the inbound load even if the order isn't confirmed. For more information, see Approve and confirm purchase orders.

2. **An inbound load record is created to plan the arrival and its contents.**

   The inbound load record represents a vendor shipment of one or more orders. The load is expected to arrive at the warehouse as one physical transportation unit (such as a truckload). The inbound load record is used for planning purposes and lets the logistics coordinator track the load's progress from the vendor. It's also used to register order line quantities and manage progress through warehouse operations, such as arrival and putaway work. Loads can be created either automatically or manually. Depending on setup, the automatic creation can be done directly based on the order data or advanced shipment notice (ASN) from the vendor. For more information, see Create or modify an inbound load.

3. **The vendor confirms load dispatch.**

   When the vendor dispatches the load, the logistics coordinator at the receiving warehouse confirms the load shipment. If the receiving company is using the **Transportation management** module, inbound shipment confirmation triggers other load management processes that are associated with the inbound loads. For more information, see Confirm a load for shipping.

4. **The load arrives at the warehouse, and workers register quantities.**

   When a truckload arrives at the warehouse receiving dock, warehouse workers register the load quantities. When the **Warehouse management** module is used, workers do the registration by using mobile devices. For more information, see Product receipt against purchase orders - registration and the Register item quantities that arrive on an inbound load section.

5. **The load is updated as receive completed.**

   If the warehouse management parameter setting **Load receiving completed confirmation policy for purchase orders** is enabled, the load is updated to *Receiving completed*.

6. **Registered load quantities are posted against purchase orders.**

   After the load quantities have been registered as arrived, those quantities must be product receipt–posted to the company's inventory ledger to record the physical stock increase. Learn more in Product receipt against purchase orders - product receipt and Post registered product quantities against purchase orders.

Register item quantities that arrive on an inbound load

Dynamics 365 Supply Chain Management supports several operational approaches to recording the arrival of ordered products via loads. You can configure the system to match your specific business requirements. This section describes how to register incoming item quantities by using the Warehouse management mobile app when an inbound load arrives at the warehouse. First, warehouse workers must register the item quantities that are included in the load shipment. This workflow is available only if the following items are present in the system:

- **An inbound load record that describes the item quantities that are expected in the shipment**

Typically, the vendor confirms the inbound load record before the shipment arrives at the warehouse. Therefore, the load has a status of *Shipped*. However, warehouse workers can also register item quantities for loads that have a status of *Open* or *Received*, depending on the setup option.

- **A Warehouse management mobile app that is configured to support load receiving and all related warehouse setup enabled**

The Warehouse Management mobile app for mobile devices supports the following receiving work creation processes:

- License plate receiving (and putaway)
- Load item receiving (and putaway)
- Mixed license plate receiving (and putaway), where the **Source document line identification method** field for the mobile device menu item is set to *Load item receiving*. Learn more in Mixed license plate receiving.

> **Note**
>
> In a typical receiving flow, the system generates inbound warehouse work to take quantities that are registered in the receiving location and put them away in the regular storage locations. When mobile device menu item processes with *put away* are used, the device instructs the worker who registered the load quantity to do the putaway work as part of the receiving task. By contrast, for the other flows that are considered "two-step" processes, another warehouse worker typically processes the movement of the inbound inventory. However, many different configuration options exist to control the flow. In some cases, it might not make sense to create warehouse work. In these cases, you can configure the work policies so that they don't create inbound warehouse work. In other cases, it might make sense to include an inspection/quality process.

Handle discrepancies during registration of inbound load quantities

Warehouse workers can register a partial load quantity receipt. Each partial load quantity receipt creates a separate inventory transaction that has a receipt status of *Registered* for the registered quantity. The lot ID refers to the originating purchase order line or inbound shipment order line.

Load under-receiving

When a load arrives, if the item quantities are less than the quantities that are stated on the load record, warehouse receiving personnel can work directly in the client to acknowledge this discrepancy by reducing the quantity on the load line so that it matches the actual quantity that arrived and was registered.

Load over-receiving

Over-receiving occurs when a load arrives, and the item quantities exceed the expected load line quantity. You can control whether and to what degree over-receiving is allowed during load registration.

Use the **Load over receipt** field for the relevant mobile device menu items to control what occurs when a warehouse worker tries to register an overdelivery. This field is available for mobile device menu items that use the following types of work creation processes:

- Load item receiving (and putaway)
- Mixed license plate receiving (and putaway), where the **Source document line identification method** field is set to *Load item receiving*

The following table explains the options that are available for the **Load over receipt** field.

Expand table

Value	Description
Allow	Workers can register the receipt of quantities that exceed the remaining unregistered quantity for a selected load, but only if the total registered quantity doesn't exceed the quantity of the order line that is associated with the load (after adjustment for the overdelivery percentage).
Block	Workers can't register the receipt of quantities that exceed the remaining unregistered quantity for a selected load (after adjustment for the overdelivery percentage). A worker who tries to register the receipts receives an error and can't continue until they register a quantity that is equal to or less than the remaining unregistered load quantity.

Value	Description
	By default, the value of the overdelivery percentage on a load line is copied from the associated purchase order line. When the **Load over receipt** field is set to *Block*, the system uses the overdelivery percentage value to calculate the total quantity that can be registered for a load line. However, that value can be overwritten for individual loads as required. This behavior becomes relevant during receiving flows where some or all of the excess quantity that represents the order line overdelivery percentage is distributed disproportionately across multiple loads. Here's an example scenario: • There are multiple loads for one purchase order line. • The purchase order line has an overdelivery percentage that is more than 0 (zero). • Quantities are already registered against one or more loads without taking the overdelivery percentage into account. • The overdelivery quantity arrives on the last load. In this scenario, a mobile device can be used to register the excess quantity for the last load only if the warehouse supervisor increases the overdelivery percentage for the relevant load line from the default value to a value that is large enough so that the full overdelivery can be registered with the final load.
Block for closed loads only	Workers can over-receive load line quantities for open loads, but not for loads that have a status of *Received*.

Put away the registered quantities

When you complete the registration on the mobile device, the *Quantity receipt registration* action updates the inventory and warehouse records to show that the quantities are now in the receiving dock location and available for reservation. However, a company's warehouse operations typically require that the quantities be moved from the receiving dock to the regular warehouse storage, so that the subsequent picking processes can occur. Instructions for the putaway are captured in the putaway work that is automatically generated when the inbound load is registered as received.

When the warehouse worker completes the putaway work, the system records and tracks the result by updating several entities, as summarized in the following table.

Expand table

Entity	Updates	Note
Load	The following fields are updated: • The **Load status** value changes to *In process*. • The **Work status** value changes to *100.00% of work completed*.	The **Load status** value changes to *In process* when the worker starts the putaway task for at least one line of putaway work.
Inventory transactions of work that associated quantities have been put away for	The **Receipt** and **Location** fields, and other relevant fields, update to reflect the movement from the receiving location to the storage location.	The **Receipt state** value of the purchase order inventory transaction remains *Registered*.
Warehouse putaway	The **Work status** value changes to *Closed*.	

Mark a load as receive complete

Background processes, workers, and users can run the *Receiving completed* process to indicate that nothing more will be registered against a specific load. For more information, see Mark loads as completely received.

Post registered product quantities against purchase orders

After you register inbound product quantities in the system, you can reserve them for sales and other outbound and internal operations. However, the system doesn't yet update the inventory (interim) accounts. This update can occur only when the operations team posts the registered product receipts.

To open a page where they can post a product receipt, members of the operations team can follow any *one* of these steps:

- Open the relevant load record, and then select the **Product receipt** action.
- Go to **Warehouse management** > **Periodic tasks** > **Update product receipts**, and then, in the **Load ID** field, specify the load to post.
- Open the relevant purchase order, and then select the **Product receipt** action.
- Go to **Procurement and sourcing** > **Purchase orders** > **Receiving products** > **Posting product receipt job**.

The **Product receipt** action that's available on the **Load** page (and on the equivalent page for the update job, the **Update product receipts** page) can update product receipt quantities only on

purchase order quantities that have a status of *Registered*. However, the **Product receipt** action that's available on the **Purchase order** page can include quantities in both processing statuses (*Ordered* and *Registered*). It can also control the scope of product receipt posting through additional parameters, such as *Receive now quantity* and *Registered quantity and services*.

The product receipt posting can occur only for orders that have a status of *Confirmed*. For unconfirmed purchase orders, the **Product receipt** action appears as unavailable.

Post registered quantities from the Load page

For product receipt posting of registered quantities from the **Load** page, the following prerequisites must be in place:

- The load must have at least one quantity unit that has a status of *Registered*.
- The load status must be *Shipped*.
- The purchase order that is associated with the load must have a status of *Confirmed*.

> ⓘ **Note**
>
> If the load status isn't set to *Shipped*, the system automatically confirms the load before it runs the product receipt update. (The load status is set to *Shipped* when a user registers the load's inbound shipment.)

To perform the product receipt posting of the arrival registrations associated with a selected load, the worker selects the **Product receipt** action on the **Load** page. The page that opens has the following key details:

- The **Quantity** field in the **Parameters** section on the **Settings** tab shows the *registered quantity*. No other options are available.
- The grid on the **Overview** FastTab lists all the purchase orders that are included in the selected load.
- The grid on the **Lines** FastTab lists all the order lines that have a registered quantity.

> ⓘ **Note**
>
> Quantities for order lines that appear on the **Line** tab are calculated differently, depending on whether the *Allow multiple product receipt per load* feature is available and turned on in your version of Supply Chain Management.

Expand table

Version	Calculation
Versions before version 10.0.10, and newer versions where the *Allow multiple product receipt per load* feature isn't turned on	The line quantity is the total of all registered quantities *for that purchase order line*, regardless of whether registration was done over multiple loads, independently of the load, from a mobile device, or from the client.
Version 10.0.10 and later, where the *Allow multiple product receipt per load* feature is turned on	The line quantity is the total of all registered quantities *for the load record* that the **Product receipt posting** action was initiated from.

When the user selects **OK** to confirm product receipt posting, the system makes the following key updates on appropriate entities.

Expand table

Entity	Updates
Inventory transaction of the purchase order for which line quantities are included in the posting scope	The following fields are updated (but note that there are also multiple other updates): • The **Receipt** field is set to *Received*. • The **Physical date** field is updated with the date of the posting.
Load that the user posted the product receipt from	Updates to the loads depend on the version that is used and the setting of the **Allow multiple product receipt per load** field. The rules are described in the table that appears later in this section.

The **Allow multiple product receipt per load** field lets companies choose an inbound load receiving policy. Depending on their operational flows, companies might choose to allow or disallow multiple product receipt postings for the same load. Set the **Allow multiple product receipt per load** field to one of the following values:

- **No** – Select this value if warehouse receiving clerks always register all order quantities that arrive with each load within a designated time frame. If any load quantities aren't registered, the system assumes that they didn't arrive or weren't on the load, and therefore shouldn't be considered part of the load. The product receipt posting that runs from a load uses the same assumption. In addition to product receipt–updating all the registered quantities (its main function), it declares the load complete and closed for additional processing. In this case, all load line quantities are automatically updated to the registered quantities, load

lines that have no registered quantities are deleted, and the load status is changed to *Received*.

- **Yes** – Select this value if warehouse receiving clerks require more time to register all the quantities on the load that arrived, but, in the meantime, you must perform a product receipt posting of the quantities that are already registered. In this case, the logistics manager wants to keep a load open even after the product receipt posting job runs, so that remaining load quantities can be registered and product receipt–updated to the ledger on an ongoing basis.
- **Prompt** – Select this value if your load receiving practices are mixed, and a decision is required each time that product receipt posting runs.

The following table summarizes the effects of the **Allow multiple product receipt per load** setting.

Expand table

Allow multiple product receipt per load	Load quantity	Load status	Note
When this field isn't available (versions before 10.0.10)	The load quantity is set so that it equals the registered quantity. If the load quantity is updated to 0 (zero), which means that no registration is done, the load line is deleted. If there are no load lines on the load, the load is deleted.	*Received*	If multiple loads exist for the order line's registered quantity, only the status of the load that the receipt was posted from is updated to *Received*.
No	The load quantity is set so that it equals the registered quantity that is associated with the load ID. If no load ID is recorded for the inventory transaction, the behavior matches the behavior in versions before 10.0.10.	*Received*	
Yes	No updates	*Received*, if the total registered load quantity is equal to or	

Allow multiple product receipt per load	Load quantity	Load status	Note
		more than the load quantity	
Yes	No updates	*Shipped* or *In process*, if the total registered load quantity is less than the load quantity	

After the **Load status** field is set to *Received*, no more product receipt postings can be done for that load. However, the worker can register the remaining order quantity against the received load provided the **Load line quantity over receipt** field on the mobile device menu item for the load item receiving action is set to *Allow*. (Learn more in the Load over-receiving section earlier in this article.)

For product receipt posting of additional registered load quantities against a load that has a status of *Received*, the user must run the posting action from the associated purchase order.

Post registered quantities from the Purchase order page

For product receipt posting of registered quantities from the **Purchase order** page, complete the following tasks before selecting the **Product receipt** action:

- Set the **Quantity** field in the **Parameters** section on the **Settings** tab to *Registered quantity*.
- In the **Product receipt** field, enter the numbers of the purchase orders included in the posting.

> **Note**
>
> The line quantity included in the posting scope is the total of all registered quantities for the order line, regardless of whether quantity registration is done over multiple loads, independently of the load, from a mobile device, or from the client. The same rule applies when product receipt posting runs from a load, if it's done where the **Allow multiple product receipt per load** field either isn't available or isn't enabled.

When the user selects **OK** to confirm product receipt posting, the system makes the following key updates on appropriate entities.

Expand table

Entity	Updates
Inventory transaction of the purchase order for which line quantities are included in the posting scope	The following fields are updated (but note that there are also multiple other updates): • The **Receipt** field is set to *Received*. • The **Physical date** field is updated with the date of the product receipt posting action.
Load	Updates to the loads depend on whether the **Allow multiple product receipt per load** field is available and enabled. The rules are described in the next table.

The following table summarizes the effects of the **Allow multiple product receipt per load** setting.

Expand table

Allow multiple product receipts per load	Load quantity	Load status	Note
When this field is either disabled or unavailable (in versions before 10.0.10)	No updates	No updates are done. (The status remains *Open*, *Shipped*, or *In process*.)	Because the product receipt posting is initiated from a purchase order, the updating logic doesn't have information about the association between the registered quantities within its scope and the loads that the registration was recorded against. Therefore, it can't select the load for the status update.
Enabled	No updates	One of the following actions occurs: • The status is changed to *Received* if the total received and purchased quantities of the purchase order inventory transactions are more than or equal to the quantity of the load that they're associated with. • The status remains *Open*, *Shipped*, or *In process* if the	

Allow multiple product receipts per load	Load quantity	Load status	Note
		previous condition isn't met for all the lines in the load.	

Select the appropriate product receipt posting option for your logistics operations

As described earlier, the system offers two product receipt posting options. You can access these options in the following places:

- On the **Load** page, or from the **Warehouse management** > **Periodic tasks** menu as an update job
- On the **Purchase order** page, or from the **Procurement and sourcing** > **Purchase orders** > **Receiving products** menu as an update job

Use the following functions if your company uses loads to plan and manage transportation and warehouse handling of inbound orders. These functions work with loads:

- **Load item receiving** actions on your warehouse mobile devices, to register the product quantity arrival against the load
- **Product receipt posting** actions initiated from a load, to update the inventory ledger

> **Note**
>
> You can use other quantity registration and product receipt posting functions to support the corresponding inbound operational processes. However, if you use them interchangeably with or instead of the dedicated load-focused functions, you might compromise the data accuracy of the load records and therefore the seamlessness of the load management flows.

Example scenarios

Prepare your system to run the sample scenarios

To work through the sample scenarios described in this section, first make sure that your system has all the required features turned on. The system must also have the required demo data available.

Enable sample data

To work through these scenarios by using the specified sample records and values, use a system where the standard demo data is installed. Also, select the **USMF** legal entity before you begin.

Add a menu item for receiving load items when a mobile device is used

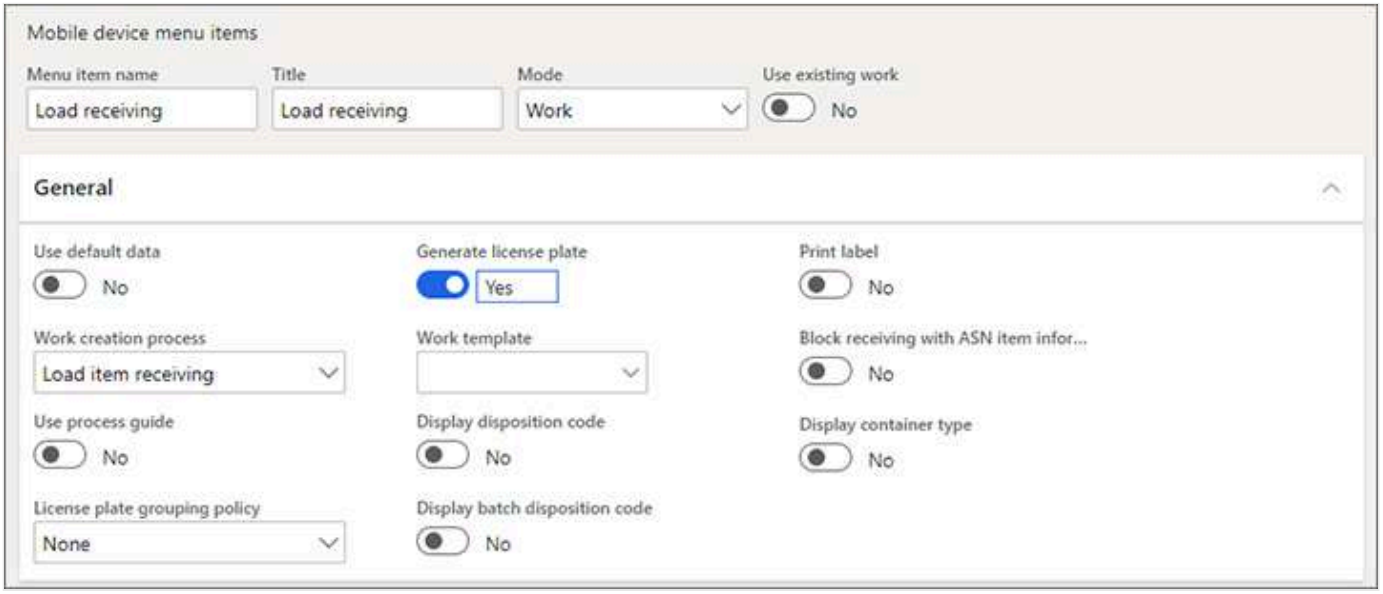
Before warehouse receiving clerks can use a mobile device to register inbound inventory that links to a load, you must create a mobile device menu item for that purpose.

In this section, you create a mobile device menu item and add it to an existing menu. A warehouse worker can then select the menu item in the Warehouse Management mobile app.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**, and make sure that your mobile device menu includes a menu item that has the following settings:

- **Mode:** *Work*
- **Work creation process:** *Load item receiving*
- **Generate license plate:** *Yes*

You can leave all other settings at their default values.



For more information about how to set up mobile device menu items, see Set up mobile devices for warehouse work.

1. After you finish setting up the menu item, go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**, and add the menu item to the menu structure for your mobile devices.

Example scenario 1: Register a load where some items are missing

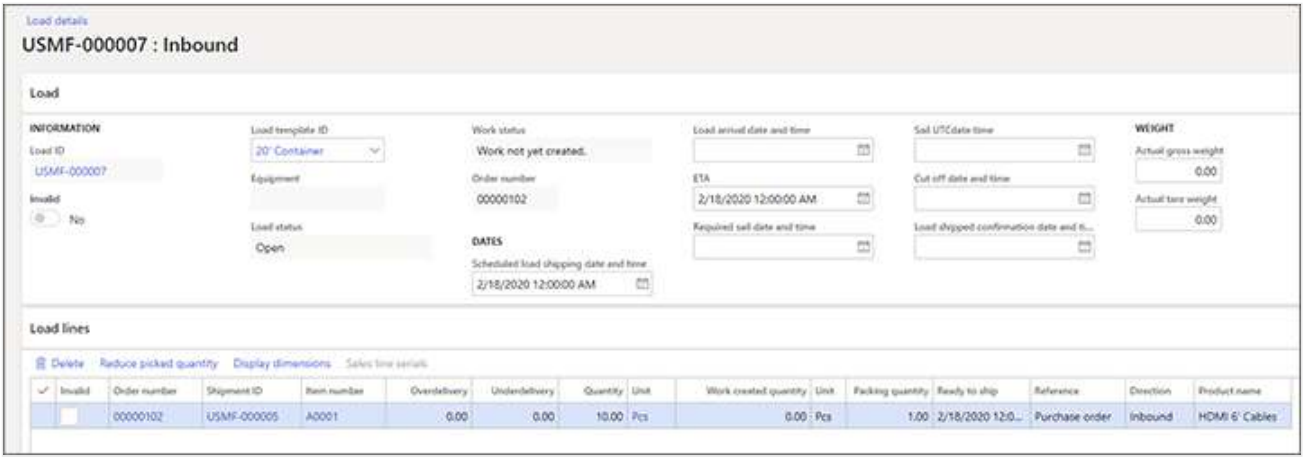
This scenario shows how to register quantities for an inbound load where not all the expected quantities are present. It then shows how to post the product receipt for the purchase order.

Create a load to plan receipt of a purchase order

In this procedure, you manually create a purchase order and an associated load. You then update the load to simulate that it shipped from the vendor (which updates the load status). Warehouse planners can then filter loads by **Load status** to find expected incoming loads.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Select **New**.
3. In **Create purchase order**, set the **Vendor account** field to *1001*.
4. Select **OK** to close the dialog box and create the purchase order.
5. The new purchase order already includes a line under **Purchase order lines**. Set the following values for this line:
    - **Item number:** *A0001*
    - **Warehouse:** *24*
    - **Quantity:** *10*
6. On the Action Pane, on the **Purchase** tab, select **Actions** > **Confirm**. The order status is now *Confirmed*.
7. On the Action Pane, on the **Warehouse** tab, select **Actions** > **Inbound load planning workbench**.
8. On the **Inbound load planning workbench** page, on the Action Pane, on the **Supply and demand** tab, select **Add** > **To new load**.
9. In **Load template assignment**, set the **Load template ID** field to *20' Container*.
10. Select **OK** to close the dialog box and return to the workbench.
11. In the **Loads** section, select **Load ID** to open the newly created load.
12. Review the load header and line details, and notice the following points:

- On the **Load** FastTab, the **Load status** field is set to *Open*.
- In the **Load lines** section, there's a single line where the **Quantity** field is set to *10* and the **Work created quantity** field is set to *0* (zero).



13. On the Action Pane, on the **Ship and receive** tab, select **Confirm** > **Inbound shipment**. Notice that the **Load status** changes to *Shipped*.

14. Make a note of the **Load ID** value, so that you can use it in the next procedure.

Register receipt of the quantities that arrived on the load

When the load arrives at the warehouse receiving dock, a receiving clerk registers the load quantities on a mobile device.

1. Use your mobile device to sign in to warehouse 24. (In the standard demo data, sign in by using *24* as the user ID and *1* as the password.)

2. Select the *Load item receiving* menu item that you created for this scenario.

3. Follow the data entry instructions on the screen to enter the following values. (The order might vary, depending on the mobile device or emulator that you're using.)

    - **Load** – Enter the load ID that you created in the previous procedure.
    - **Item** – Enter *A0001*, which is the item that is expected for this load.
    - **Qty** – Enter *9* as the actual quantity that is present on the load. Note that this quantity is less than the expected quantity.

4. Continue to go through the workflow, leaving all other fields blank or set to their default values, until your device informs you that the work is completed.

The load receiving task is now completed, and the receiving clerk can move on to their next task. However, warehouse receiving personnel eventually review the load record and can see that the

received quantity was less than the expected quantity. They can then complete the following procedure by using the web client.

1. Go to **Warehouse management** > **Loads** > **All loads**.
2. In the list, find the load that you just received. (You might have to select the **Show closed** check box to include the inbound loads that have a load status of *Shipped*.) Then select the link in the **Load ID** column to open the load.
3. In the load record, notice that the **Load status** value remains *Shipped*, but the **Work created quantity** value on the load line changes to *9*.
4. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
5. In the list, find the purchase that you just received, and then select the link in the **Purchase order** column to open the order.
6. On the **Purchase order lines** FastTab, select **Inventory** > **View** > **Transactions**.
7. Review the details of the two purchase order transactions. (You might have to personalize the **Inventory transactions** page to see the **Load ID** field to the grid.) You see two transactions:

- The transaction that has a receipt in *Registered* status represents the registration quantity of *9* that runs against a specific load by using the mobile device. The **Load ID** is associated with the transaction in question.
- The transaction that has a receipt in *Ordered* status represents the remaining unregistered order line quantity of *1*.

Product receipt–post the registered load quantities against purchase orders

In this procedure, you can perform product receipt posting of the inventory that you registered for a load. As a result, the received inventory and its related costs are added to the company's general ledger.

1. Go to **Warehouse management** > **Loads** > **All loads**.

2. In the list, find the load that you received. (You might have to select the **Show closed** check box to include the inbound loads that have a load status of *Shipped*.) Then select the link in the **Load ID** column to open the load.

3. On the Action Pane, on the **Ship and receive** tab, select **Receive** > **Product receipt**. If you're prompted to confirm the action, select **Yes**.

4. In the **Posting product receipt** dialog box, on the **Lines** FastTab, inspect the grid. You should see the purchase order line for which the quantity is registered against the selected load.

5. On the **Overview** FastTab, inspect the **Product receipt** field in the grid. Notice that it's set to a number that includes the ID of the selected load.

6. Select **OK** to post the product receipt and close the **Posting product receipt** dialog box.

7. You're returned to the load details. Notice the following points:

    - The **Load status** field is now set to *Received*.
    - On the load line, the **Quantity** value for the load changes from *10* pcs to *9* pcs to match the registered quantity, but the **Work created quantity** value remains *9*.

If the purchasing team doesn't expect the vendor to deliver the remaining order quantity of 1, it can close the order by updating the line's delivery remainder to *0*. However, if it's soon found that the missing piece arrived on the original load, warehouse personnel can perform one of the following actions:

- Register the quantity against the same load. In this case, the **Load status** field resets to *Shipped*, and the **Work created quantity** value updates to *10*. This choice is only available when the **Load line quantity over receipt** field is set to *Allow*.
- Add the quantity to a new or existing load, and process it in the usual way.
- Register and/or receive the quantity in a way that doesn't involve load handling.

Example scenario 2: Register quantities that arrive on multiple inbound loads where some items are missing

In this scenario, an inbound shipment related to a single purchase order line is split into two loads. For example, a purchase order line might be split into multiple loads because of physical load constraints on weight and/or volume.

This scenario also shows how to process multiple product receipt postings for the same load. You register quantities that arrive on multiple inbound loads, but the quantities don't match the expected quantities. You perform the cost updates that occur through the product receipt posting multiple times for the same load.

Set up load receiving policies

In this procedure, you enable multiple product receipt postings from the same load.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Loads** tab, set the **Allow multiple product receipt per load** field to *Yes*.

Create two loads to plan receipt of a purchase order

In this procedure, you create a purchase order and two loads. You then manually update each load to simulate that the vendor shipped it (which updates the load status). Warehouse planners can then filter loads by **Load status** to find expected incoming loads.

You also learn how to set the purchase order line so that you can receive a quantity that is 20 percent more than the quantity specified for the line.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Select **New**.
3. On the **Vendor** FastTab, set the **Vendor account** field to *1001*, and then select **OK**.
4. Your new purchase order is opened and includes a blank line in the **Purchase order lines** grid. Set the following values for this order line:
    - **Item number:** *A0001*
    - **Warehouse:** *24*
    - **Quantity:** *10*
5. On the **Line details** FastTab, on the **Delivery** tab, set the **Overdelivery** field to *20*.
6. On the Action Pane, on the **Purchase** tab, select **Actions** > **Confirm**. The order status is now *Confirmed*.
7. On the Action Pane, on the **Warehouse** tab, select **Actions** > **Inbound load planning workbench**.
8. On the **Inbound load planning workbench** page, on the Action Pane, on the **Supply and demand** tab, select **Add** > **To new load**.
9. In the **Load template assignment** dialog box, set the **Load template ID** field to *20' Container*. On the **Details** tab, change the **Quantity** value from *10* to *5* to partially add the purchase order line quantity.
10. Select **OK** to apply your settings and close the dialog box.

11. Repeat steps 8 through 10 to create a second load. This time, the **Quantity** field should already be set to *5*.

12. On the **Inbound load planning workbench** page, in the **Loads** grid, select the **Load ID** value for the first load that you created. The **Load details** page appears and shows the selected load. Follow these steps:
    a. On the Action Pane, on the **Ship and receive** tab, select **Confirm** > **Inbound shipment**.
    b. Notice that the **Load status** value changes to *Shipped*.
    c. Select the close button to return to the **Inbound load planning workbench** page.

13. Repeat the previous step for the second load that you created.

14. Make a note of the two **Load ID** values that appear in the **Loads** grid.

Register partial receipt of the quantities that arrived on the first load and post the registered load quantities

When loads arrive at the warehouse receiving dock, a receiving clerk registers the load quantities on a mobile device. The registered inventory that links to a load is then cost-updated in the company's general ledger by posting the purchase order product receipt, based on the load.

This procedure shows how a receiving clerk registers load quantities on a mobile device.

1. Use your mobile device to sign in to warehouse 24. (In the standard demo data, sign in by using *24* as the user ID and *1* as the password.)

2. Select the *Load item receiving* menu item that you created for this scenario.

3. Follow the data entry instructions on the screen to enter the following values. (The order might vary, depending on the mobile device or emulator that you're using.)

    - **Load** – Enter the first load ID that you created in the previous procedure.
    - **Item** – Enter *A0001*, which is the item that is expected for this load.
    - **Qty** – Enter *3*. Note that this quantity is less than the expected quantity. For this scenario, imagine that you, as the receiving clerk, don't have time to register all quantities for this load. Later in this procedure, you register the remaining pieces by repeating this step and setting the **Qty** field to *2*.

4. Continue to go through the workflow, leaving all other fields blank or set to their default values, until your device informs you that the work is completed.

5. In the web client, go to **Warehouse management** > **Loads** > **All loads**.

6. In the list, find the load that you just received, and select the **Load ID** value to open the load. Notice that the **Load status** value remains *Shipped*, but the **Work created quantity** value on the load line changes to *3*.

7. On the Action Pane, on the **Ship and receive** tab, select **Receive** > **Product receipt**. If you're prompted to confirm the action, select **Yes**.

8. In the **Posting product receipt** dialog box, review but don't change the values that are shown, and then select **OK**.

9. You're returned to the **Load details** page for your selected load. Notice the following points:

    - The **Load status** field remains set to *Shipped*.
    - On the load line, the **Quantity** value for the load remains *5* pcs, which is the original load quantity, and the **Work created quantity** value remains *3*.

10. Complete the registration of the remaining quantity on this load by repeating this procedure. However, in step 3, set the **Qty** field to *2*.

The receiving task for the first load is now completed. Two product receipt journals are created, one for each of the two product receipt updates that you ran.

Register receipt of the quantities that arrived on the second load and account for the overdelivered quantity

For this scenario, the receiving clerk inbound-registers a quantity that exceeds the quantity that exists on the load. The system permits over-receiving because it's set up to allow overdelivery.

1. Use your mobile device to sign in to warehouse 24. (In the standard demo data, sign in by using *24* as the user ID and *1* as the password.)

2. Select the *Load item receiving* menu item that you created for this scenario.

3. Follow the data entry instructions on the screen to enter the following values. (The order might vary, depending on the mobile device or emulator that you're using.)

    - **Load** – Enter the second load ID that you created earlier.
    - **Item** – Enter *A0001*, which is the item that is expected for this load.
    - **Qty** – Enter *7*, which is the remaining quantity that the vendor is authorized to deliver as part of total purchase order quantity of 12 (where 10 is the original order quantity, and 2 is the allowed overdelivery quantity of 20 percent). Remember that 5 pcs are already registered against the first load.

The second load is now updated with the quantity of 7 and can be product receipt–updated based on this quantity.

Last updated on 06/15/2026

Mark loads as completely received

Article • 04/23/2025

This article describes the options for marking loads as completely received. It also explains how the *Receiving completed* periodic task works.

Mark a load as completely received

Background processes, workers, and users can run the *Receiving completed* process to indicate that nothing more has to be registered against a specific load. Admins can schedule automatic processing by setting up a batch job as described in the following procedure. Web client users can run the process by using the **Load** or **Inbound load planning workbench** page. Workers who use the Warehouse Management mobile app can run it by using a menu item that is set up with an **Activity code** value of *Receiving completed confirmation*.

During the *Receiving completed* process, workers and users update the **Load status** and **Load receiving completed date and time** fields for the load. In addition, depending on the setting of the **Capture receiving completed packing slip** field on the **Warehouse management parameters** page, they might be able (or required) to enter a packing slip ID and a document date.

Follow these steps to specify how loads that are related to purchase orders are finalized.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Loads** tab, in the **Capture receiving completed packing slip** field, select one of the following options to specify whether users or workers must enter a packing slip ID when they confirm that receiving is completed:
    - *Never* – Workers or users are never required to enter a packing slip ID when they confirm that receiving is completed.
    - *Always* – Worker or users are always required to enter a packing slip ID and a document date for every received shipment.
    - *Optional* – Worker or users can choose whether to enter a packing slip ID when they confirm that receiving is completed. They can also leave the document date blank. (It's automatically set to the current date by default).
3. In the **Load receiving completed confirmation policy for purchase orders** field, select one of the following options:
    - *Disabled* – Loads don't indicate whether inbound receiving is completed. If you select this option, there is a risk that the *Update product receipts* cost update periodic task might run in the middle of an inbound registration process.

- *Enabled* – After the *Receiving completed* process is run, loads are updated with a **Load receiving completed date and time** value, and a **Packing slip ID** value is assigned to related shipments. The *Update product receipts* cost update periodic task checks for these values to ensure that it processes only loads that were received. This check is especially important if you allow over-receiving or under-receiving.
- *Enabled with auto post* – This option works just like the *Enabled* option, but the **Load status** value is also updated to *Received*, and the **Product receipt processing status** field on the load becomes available for use with the subsequent *Product receipt* posting process. You can't select this option if you allow multiple product receipts per load.

4. If you set the **Load receiving completed confirmation policy for purchase orders** field to *Enabled* or *Enabled with auto post*, and you want the *Receiving completed* process to run automatically, set up the *Receiving completed* periodic task as described in the next section.

> **Note**
>
> When you set up the *Update product receipts* cost update periodic task, you can use the **Load status** and **Product receipt processing status** field values as filter criteria. In addition, depending on the setting of the **Capture receiving completed packing slip** option on the **Warehouse management parameters** page, the purchase order product receipt process might be able to use the recorded packing slip ID as part of the *Update product receipts* periodic task.

Automatically run the Receiving completed process

The *Receiving completed* periodic task automatically runs the *Receiving completed* process for loads for purchase orders. To set up the task so that it runs automatically, follow these steps.

1. On the **Warehouse management parameters** page, on the **Loads** tab, set the **Load receiving completed confirmation policy for purchase orders** field to *Enabled* or *Enabled with auto post*. The *Receiving completed* periodic task is available only if the field is set to one of these values. Learn more in the Mark a load as completely received section.
2. Go to **Warehouse management** > **Periodic tasks** > **Receiving completed**.
3. In the **Receiving completed** dialog, on the **Parameters** FastTab, set the **Accept quantity exceptions** option to specify whether differences between the received quantities and the quantities on the load line should be accepted. Set the option to *No* to reject the differences and cancel the confirmation process. Set it to *Yes* to accept the differences and continue the processing.

4. On the **Records to include** FastTab, set up filters and constraints to define which loads to process. Select **Filter** to open a standard query editor dialog, where you can define selection criteria, sorting criteria, and joins. The fields work just as they do for other types of queries in Microsoft Dynamics 365 Supply Chain Management.
5. On the **Run in the background** FastTab, specify how, when, and how often to run the task. The fields work just as they do for other types of background jobs in Supply Chain Management.
6. Select **OK**.

Example: Include only loads with fully completed work

To set up the *Receiving completed* periodic task so that it includes only loads where work was fully completed, follow these steps.

1. Go to **Warehouse management** > **Periodic tasks** > **Receiving completed**.
2. On the **Records to include** FastTab, select **Filter**.
3. In the query editor dialog, on the **Joins** tab, select the **Loads** join in the list.
4. On the toolbar, select **Add table join**.
5. Set the **Show details** option to *Yes*.
6. Select the row where the **Join mode** field is set to *NotExist* and the **Table relation description** field is set to *WHSLoadTable -> WHSWorkTable*.
7. On the toolbar, select **Select**.
8. On the **Range** tab, on the toolbar, select **Add** to add a line.
9. Set the following values for the new line:
    - **Table** – Select *Work (NotExist)*.
    - **Field** – Select *Work status*.
    - **Criteria** – Select all the statuses that denote an open work record. For example, you might select *Open*, *In process*, *Pending review*, and *Combined*.
10. On the toolbar, select **Add** to add another line.
11. Set the following values for the new line:
    - **Table** – Select *Load*.
    - **Field** – Select *Load receiving completed date and time*.

- **Criteria** – Enter two double quotation marks ("") to denote an empty value.

> **Tip**
>
> Because of the **Criteria** setting on this line, the query finds only records that have an empty value in the **Load receiving completed date and time** field. As a result, the task doesn't consider loads that were already received. If you omit this setting, the task becomes slower each time that it runs.

12. Select **OK**.
13. Set other options for the task as described in the previous section.
14. Select **OK** to save and run the task.

Create or modify an inbound load

Article • 01/31/2024

This article explains how to create or modify an inbound load. You can use a purchase order or an inbound shipment order to create an inbound load either automatically or manually. You can also modify an existing inbound load by adding more lines or updating the quantities, for example.

For more information about how to use the inbound load process with transportation planning, see Transportation management overview.

Automatically create inbound loads for new purchase orders

To set up your system so that it automatically creates an inbound load for each new purchase order, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Loads** tab, set the **Automatically create at purchase order entry** option to *Yes*.

An inbound load is now automatically created each time that you create a purchase order.

Automatically create inbound loads when purchase orders are received by using the Warehouse Management mobile app

To set up your system so that it automatically creates an inbound load for purchase order lines that aren't already related to open loads, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Loads** tab, set the **Automatically create at purchase order receiving** option to *Yes*.

This option is valuable when, for example, you use the **Load receiving completed confirmation policy for purchase orders** option, but there's no process to create loads up front. Each of the registered purchase order line transactions is associated with the load ID that's created. In this way, subsequent cost update processes can be correctly

matched with the actual inventory that's registered for the warehouse. For more information about inbound flows, see Warehouse handling of inbound loads for purchase and inbound shipment orders.

Automatically create inbound loads for new inbound shipment orders

To set up your system so that it automatically creates inbound loads for new inbound shipment orders, based on the source system, order type, and/or account, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Source systems**.
2. On the **Source systems** page, you should have one record for each external system that submits inbound shipment orders to Microsoft Dynamics 365 Supply Chain Management. Follow one of these steps:
    - To add a new source system, select **New** on the Action Pane.
    - To edit an existing source system, select it in the list pane, and then select **Edit** on the Action Pane.
3. For the new or selected source system, select the **Inbound shipment order policies** FastTab. The grid might have several rows to define policies for each of several order types and/or accounts. Alternatively, the grid might have just one row that applies to all order types and accounts. Use the toolbar on the FastTab to add or remove rows in the grid as required.
4. For each row where you want to automatically create inbound loads as part of inbound shipment order message processing, set **Load synchronization policy** to *Full synchronization*.

An inbound load is now automatically created each time that you import and process an inbound shipment order for the relevant source system, order type, and account.

If you use the Warehouse management mobile app to run a receiving process against an inbound shipment order line that isn't associated with an open load, the system automatically creates a load as part of the registration process. The load ID is assigned to the related inventory transaction.

Manually create an inbound load from order lines

To manually create an inbound load, you must have at least one order line that isn't already assigned to a load.

1. Go to **Warehouse management** > **Loads** > **Inbound load planning workbench**.
2. Use the filter options at the top of the page to find the order lines that you want to assign to a load.
3. Select the **Purchase order lines** or **Inbound shipment order lines** tab, depending on the type of order lines that you're looking for.
4. Mark each order line that you want to assign to a load.
5. On the Action Pane, on the **Supply and demand** tab, select **To new load**.
6. In the **Load template assignment** dialog box, set the **Load template ID** field to the load template that you want to use.
7. Select **OK**.

Create inbound loads when you import advanced shipping notices

The system automatically creates loads when you import advanced shipping notices (ASNs).

Feedback

Was this page helpful? Yes No

Provide product feedback

Register items enabled for warehouse management processes using an item arrival journal

Article • 08/01/2022

This article presents a scenario that shows how to register items using the item arrival journal when you are using warehouse management processes (WMS). This would usually be done by a receiving clerk.

Enable sample data

To work through this scenario using the sample records and values specified in this article, you must be using a system where the standard demo data is installed, and you must select the *USMF* legal entity before you begin.

You can instead work through this scenario by substituting values from your own data provided that you have the following data available:

- You must have a confirmed purchase order with an open purchase order line.
- The item on the line must be stocked. It must not use product variants, and must not have tracking dimensions.
- The item must be associated with a storage dimension group that has warehouse management process enabled.
- The warehouse that's used must be enabled for WMS and the location that you use for receiving must be license plate controlled.

Create an item arrival journal header that uses warehouse management

The following scenario shows how to create an item arrival journal header that uses warehouse management:

1. Make sure your system contains a confirmed purchase order that fulfils the requirements outlined in the previous section. This scenario uses a purchase order for company *USMF*, vendor account *1001*, warehouse *51*, with an order line for *10 PL* (10 pallets) of item number *M9200*.

2. Make a note of the purchase order number that you will use.

3. Go to **Inventory management** > **Journal entries** > **Item arrival** > **Item arrival**.

4. Select **New** on the Action Pane.

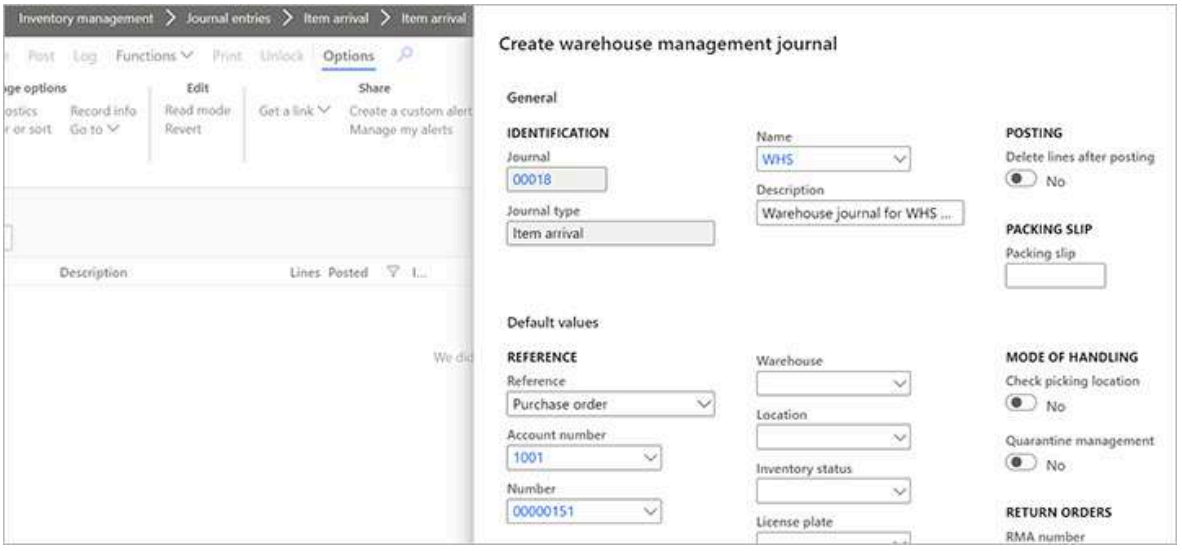
5. The **Create warehouse management journal** dialog box opens. Select a journal name in the **Name** field.

    - If you are using *USMF* sample data, select *WHS*.
    - If you're using your own data, the journal you choose must have **Check picking location** set to *No* and **Quarantine management** set to *No*.

6. Set **Reference** to *Purchase order*.

7. Set **Account number** to *1001*.

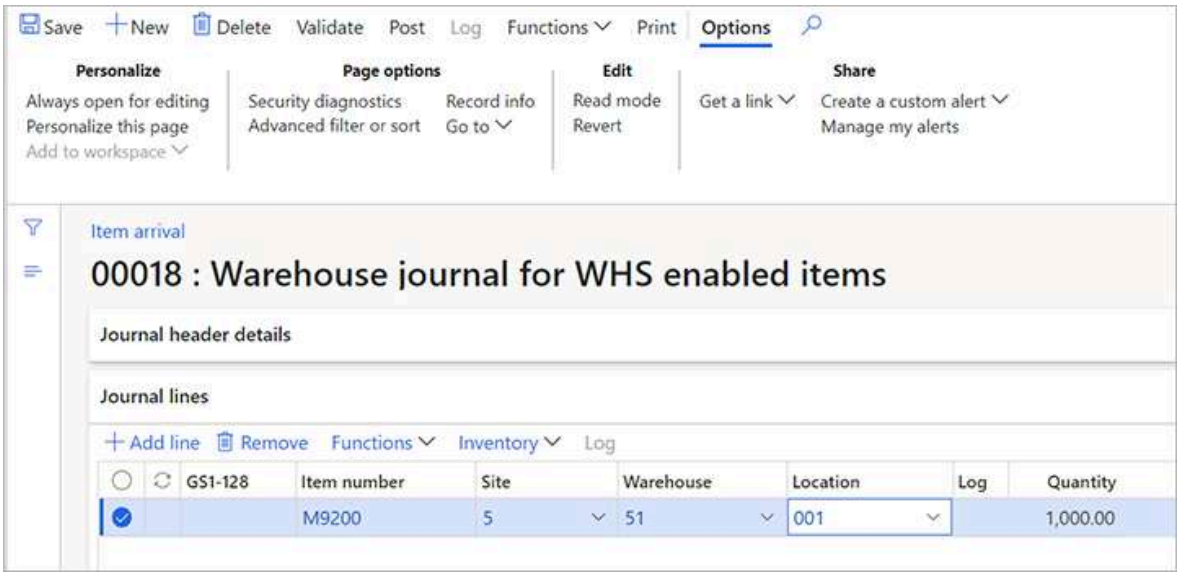
8. Set **Number** to the number of the purchase order that you identified for this exercise.



9. Select the **OK** to create the journal header.

10. In the **Journal lines** section, select **Add line** and enter the following data:

    - **Item number** – Set to *M9200*. The **Site**, **Warehouse**, and **Quantity** will get set based on the inventory transaction data for the 10 pallets (1000 ea.).
    - **Location** – Set to *001*. This specific location does not track license plates.



> **Note**
>
> The **Date** field determines the date on which the on-hand quantity of this item will be registered in the inventory.
>
> The **Lot ID** will be populated by the system if it can be uniquely identified from the information provided. Otherwise you will have to enter this manually. This is a required field, which links this registration to a specific source document line.

11. Select **Validate** on the Action Pane. This checks that the journal is ready to be posted. If the validation fails you will need to fix the errors before you can post the journal.
12. The **Check journal** dialog box opens. Select **OK**.
13. Review the message bar. There should be a message denoting that the operation was completed.
14. Select **Post** on the Action Pane.
15. The **Post journal** dialog box opens. Select **OK**.
16. Review the message bar. There should be messages denoting that the operation completed.
17. Select **Functions** > **Product receipt** on the Action Pane to update the purchase order line and post a product receipt.

Feedback

Was this page helpful? Yes No

Provide product feedback

Register items for a basic warehousing enabled item using an item an item arrival journal

Article • 02/12/2025

This procedure shows you how to register items using the item arrival journal when you are using "basic warehousing" in the Inventory management module. This would usually be done by a receiving clerk. You can run this procedure in demo data company USMF with the example values that are shown. If you are not using USMF, you need to have a confirmed purchase order with an open purchase order line before you start this guide. The item on the line must be stocked. And the item needs to be associated with a storage dimension group, where site and warehouse are active.

Create item arrival journal header

1. Go to **Inventory management** > **Journal entries** > **Item arrival** > **Item arrival**.
2. Select **New**.
3. In the **Name** field, type a value. If you are using USMF, you can type *WHS*. If you're using other data, the journal whose name you choose has to have the following properties: **Cheque picking location** must be set to *No*, and **Quarantine management** must be set to *No*.
4. In the **Packing slip** field, type a value. This is the packing slip ID from the packing slip issued by the vendor. Add a unique number.
5. In the **Number** field, select the purchase order.
6. Select **OK**.

Add lines to item arrival journal

1. Select **Functions**.
2. Select **Create lines**. The lines can be entered manually into this journal or created automatically. This will show you how to create this automatically.
3. Select or clear the **Initialize quantity** checkbox. This initializes the quantity on the journal lines with the quantity not registered from the purchase order line.
4. Select **OK**.

Post the journal

1. Select **Post**.
2. Select **OK**.

Generate the product receipt

1. Select **Functions**.
2. Select **Product receipt**.
3. Select **OK**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Import inbound ASNs through data entities

Advanced shipping notices (ASNs) notify you about vendor deliveries. They help the sender describe the contents of a shipment and provide additional information about it, such as the items and packaging.

ASNs can help warehouse workers learn what is arriving so they can prepare. In addition, warehouse workers can use ASNs to match the details of a shipment to the related purchase order or inbound shipment order that was previously created.

This article presents a collection of scenarios that show, through examples, how to work with ASN files.

> **Important**
>
> *Inbound ASN* import applies only to items that are enabled for warehouse management processes (WMS). Before you can receive an ASN, a purchase or inbound shipment order must be registered in the system.

Inbound ASN

You import inbound ASNs by using the *Inbound ASN V3* and/or *Inbound ASN V5* composite data entities. These entities use the following child entities:

- Inbound load header
- Inbound shipment header
- Inbound load packing structures
- Inbound load packing structure cases
- Inbound load packing structure case lines
- Inbound load packing structure lines

The *Inbound ASN* composite data entities are intended for asynchronous integration scenarios where, for example, you can use XML file–based imports.

> **Note**

Only the *Inbound ASN V5* data entity supports inbound shipment orders. The type of order must be specified as part of the ASN data, which can be either *InboundShipmentOrder* for inbound shipment orders or *Purch* for purchase orders.

XML format for importing V3 ASNs

Supply Chain Management supports the following XML format for importing ASNs. Each node in the XML file represents an attribute from an individual entity.

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSInboundLoadHeaderEntity>
        <WHSInboundShipmentHeaderEntity>
            <WHSInboundLoadPackingStructureEntity>
                <WHSInboundLoadPackingStructureCaseEntity>
                    <WHSInboundPackingStructureCaseLineV3Entity>
                    </WHSInboundPackingStructureCaseLineV3Entity>
                </WHSInboundLoadPackingStructureCaseEntity>
                <WHSInboundLoadPackingStructureLineV3Entity>
                </WHSInboundLoadPackingStructureLineV3Entity>
            </WHSInboundLoadPackingStructureEntity>
        </WHSInboundShipmentHeaderEntity>
    </WHSInboundLoadHeaderEntity>
</Document>
```

Examples of using the Inbound ASN V3 entity (only for purchase orders)

The following subsections provide examples of ASN XML import files for purchase order vendor shipments for *Inbound ASN V3*.

Example 1

The following example shows an XML file for importing vendor shipments for one purchase order when no case details are included.

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSInboundLoadHeaderEntity TRACTORNUMBER="0000101">
        <WHSInboundShipmentHeaderEntity VENDORSHIPMENTID="VendASN_01"
```

```
VENDORADDRESSCOUNTRYREGIONID = "USA" VENDORADDRESSSTREET = "123 Coffee Street"
VENDORADDRESSSTATEID = "WA" VENDORADDRESSCITY = "Redmond" VENDORADDRESSZIPCODE =
"98052">
            <WHSInboundLoadPackingStructureEntity LICENSEPLATENUMBER="LP_ASN_001">
                <WHSInboundLoadPackingStructureLineV3Entity
PURCHASEORDERNUMBER="00000176" ITEMNUMBER="A0001" QUANTITY="1" UNITSYMBOL="pcs" />
            </WHSInboundLoadPackingStructureEntity>
        </WHSInboundShipmentHeaderEntity>
    </WHSInboundLoadHeaderEntity>
</Document>
```

Example 2

The following example shows an XML file for importing vendor shipments for one purchase order when case details are included.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSInboundLoadHeaderEntity ESTIMATEDARRIVALDATETIME="2021-04-25T11:00:00+00:00">
        <WHSInboundShipmentHeaderEntity VENDORSHIPMENTID="MVR_SNN_0004">
            <WHSInboundLoadPackingStructureEntity LICENSEPLATENUMBER="MVR_SNN_0004"
PACKEDTOTALQUANTITY="2.00">
                <WHSInboundLoadPackingStructureCaseEntity
PARENTPACKINGSTRUCTURELICENSEPLATENUMBER="MVR_SNN_0004"
LICENSEPLATENUMBER="MVR_SNN_0004A" PACKEDTOTALQUANTITY="2.00" />
                <WHSInboundLoadPackingStructureLineV3Entity
PURCHASEORDERNUMBER="00000175" ITEMNUMBER="A0001" PURCHASEORDERLINENUMBER="1"
QUANTITY="2.00" UNITSYMBOL="pcs" />
            </WHSInboundLoadPackingStructureEntity>
        </WHSInboundShipmentHeaderEntity>
    </WHSInboundLoadHeaderEntity>
</Document>
```

Example 3

The following example shows an XML file for importing vendor shipments for multiple purchase orders when case details are included.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSInboundLoadHeaderEntity TRACTORNUMBER="0000101">
        <WHSInboundShipmentHeaderEntity VENDORSHIPMENTID="VendASN_01"
VENDORADDRESSCOUNTRYREGIONID = "USA" VendorAddressStreet = "123 Coffee Street"
VENDORADDRESSSTATEID = "WA" VENDORADDRESSCITY = "Redmond" VENDORADDRESSZIPCODE =
"98052">
```

```
            <WHSInboundLoadPackingStructureEntity LICENSEPLATENUMBER="LP_ASN_001">
                <WHSInboundLoadPackingStructureLineV3Entity
PURCHASEORDERNUMBER="00000176" ITEMNUMBER="A0001" QUANTITY="100" UNITSYMBOL="pcs" />
            </WHSInboundLoadPackingStructureEntity>
        </WHSInboundShipmentHeaderEntity>
        <WHSInboundShipmentHeaderEntity VENDORSHIPMENTID="VendASN_02"
VENDORADDRESSCOUNTRYREGIONID = "USA" VendorAddressStreet = "123 Coffee Street"
VENDORADDRESSSTATEID = "WA" VENDORADDRESSCITY = "Redmond" VENDORADDRESSZIPCODE =
"98052">
            <WHSInboundLoadPackingStructureEntity LICENSEPLATENUMBER="LP_ASN_001">
                <WHSInboundLoadPackingStructureLineV3Entity
PURCHASEORDERNUMBER="00000177" ITEMNUMBER="A0001" QUANTITY="200" UNITSYMBOL="pcs" />
                <WHSInboundLoadPackingStructureLineV3Entity
PURCHASEORDERNUMBER="00000177" ITEMNUMBER="P0004" QUANTITY="300" UNITSYMBOL="pcs"
ITEMBATCHNUMBER="BN0001" />
            </WHSInboundLoadPackingStructureEntity>
            <WHSInboundLoadPackingStructureEntity LICENSEPLATENUMBER="LP_ASN_002">
                <WHSInboundLoadPackingStructureCaseEntity
LICENSEPLATENUMBER="LP_ASN_002_C01">
                    <WHSInboundLoadPackingStructureCaseLineV3Entity
PURCHASEORDERNUMBER="00000177" ITEMNUMBER="A0001" QUANTITY="400" UNITSYMBOL="pcs" />
                </WHSInboundLoadPackingStructureCaseEntity>
            </WHSInboundLoadPackingStructureEntity>
        </WHSInboundShipmentHeaderEntity>
    </WHSInboundLoadHeaderEntity>
</Document>
```

Example of using the Inbound ASN V5 entity

This section provides an example of an ASN XML import file for inbound shipment orders that you can use with the *Inbound ASN V5* entity.

> ⓘ **Note**
>
> The *Inbound ASN V5* entity supports the `MODULE` attribute for both the `WHSInboundLoadPackingStructureEntity` element and the `WHSInboundLoadPackingStructureLineV5Entity` element. Use this attribute for both purchase orders (`MODULE="Purch"`) and inbound shipment orders (`MODULE="InboundShipmentOrder"`).

The following example shows an XML file for importing shipments for an inbound shipment order that doesn't include case details.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSInboundLoadHeaderEntity TRACTORNUMBER="0000104">
```

```
        <WHSInboundShipmentHeaderEntity VENDORSHIPMENTID="VendASN_04">
            <WHSInboundLoadPackingStructureEntity MODULE="InboundShipmentOrder"
LICENSEPLATENUMBER="LP_ASN_004">
                <WHSInboundLoadPackingStructureLineV5Entity
MODULE="InboundShipmentOrder" ORDERNUMBER="IO04" ORDERLINENUMBER="1"
ITEMNUMBER="A0001" QUANTITY="2" UNITSYMBOL="pcs"/>
            </WHSInboundLoadPackingStructureEntity>
        </WHSInboundShipmentHeaderEntity>
    </WHSInboundLoadHeaderEntity>
</Document>
```

> **Note**
>
> If you're importing an ASN for an inbound shipment order (`MODULE="InboundShipmentOrder"`), the `ORDERNUMBER` value must correspond to an external (source system) order number.

Inspect the results of importing an ASN file

Follow these steps to inspect the results of importing an ASN file.

1. Go to **Warehouse management** > **Loads** > **All loads**.
2. Find and open a load that you created as part of an ASN import.
3. On the **Load** FastTab, you see values that are based on the XML file.
4. On the **Load lines** FastTab, you see purchase order numbers and item details that are based on the XML file.
5. On the Action Pane, on the **Ship and receive** tab, in the **Receive** group, select **Packing structure** to review the packing structure of the load.
6. On the **Pallets** FastTab, you see license plates that are based on the XML file.
7. On the **Cases** FastTab, you see cases that are based on the XML file.
8. On the **Items** FastTab, you see items and quantities that are based on the XML file.

Last updated on 05/06/2026

Import inbound ASNs as despatch advice messages

Advanced shipping notices (ASNs) notify you about vendor deliveries. They help the sender describe the contents of a shipment—such as the items, quantities, and packaging—so that warehouse workers can prepare to receive it.

A *despatch advice* is an inbound ASN that you import as a *despatch advice message*. Instead of creating warehouse records directly through data entities, the system queues the message and handles it asynchronously by using the message processor. This approach supports integrations where related documents must be processed in sequence, and it provides the following capabilities:

- **Message dependencies** – A despatch advice can depend on a prior message so that a source order is guaranteed to be imported *before* the ASN that references it.
- **Full-document updates** – You can correct a previously imported despatch advice by resubmitting the whole document. The document is replaced in full, and records that are no longer part of the shipment are cleaned up automatically. Learn more in Update or correct a despatch advice.

> **Important**
>
> Despatch advice messages apply only to items that are enabled for warehouse management processes (WMS). Before you can process a despatch advice, the referenced purchase order or inbound shipment order must already be registered in the system.

> **Note**
>
> The *Inbound ASN* data entities remain available for importing ASNs directly, without the message processor. Learn more in Import inbound ASNs through data entities.

Despatch advice message structure

A despatch advice message is a composite document that maps shipment details to warehouse records. It uses three entities that you import together through the *Despatch advice composite*

entity (`WHSDespatchAdviceCompositeEntity`).

Expand table

Document part	Entity	What it represents
Header	`WHSDespatchAdviceMessageEntity`	The shipment notification, including carrier, weights, and shipper address. Becomes an inbound load and shipment.
Logistic unit	`WHSDespatchAdviceLogisticUnitMessageEntity`	A physical container that's identified by a serial shipping container code (SSCC) or license plate—for example, a pallet, case, or carton.
Line item	`WHSDespatchAdviceLineItemMessageEntity`	A product inside a logistic unit, with its quantity and inventory dimensions.

Identify a despatch advice

Each despatch advice is identified by the combination of two fields on the header:

- `ShipperIdentification` – The vendor (shipper) account that the shipment comes from.
- `DespatchAdviceIdentification` – The stable business ID of the despatch advice.

Because the identification is scoped to the shipper, different vendors can independently use the same `DespatchAdviceIdentification` value. The load and shipment IDs are generated automatically during processing—you don't provide them.

Provide header, transport, and address details

Besides the required identification fields, the header (`WHSDespatchAdviceMessageEntity`) carries optional information about the shipment. Use `ReceivingWarehouseId` to specify the warehouse that receives the load, and populate any of the following groups of fields as needed.

Expand table

Group	Fields
Transport and carrier	`ShippingCarrierCode`, `ShippingCarrierGroupId`, `ShippingCarrierServiceId`, `TractorNumber`, `TrailerNumber`, `BrokerId`, `CarNumber`, `TransportationBookingNumber`, `MasterBillOfLadingId`, `ProNumber`, `SealNumber`, `InspectionSealNumber`, `VesselName`, `VoyageNumber`, `EstimatedArrivalDateTime`

Group	Fields
Weights	`ActualGrossWeight`, `ActualNetWeight`, `ActualTareWeight`, `ActualWeightUnitSymbol`
Shipment	`DeliveryTermsCode`, `ShipperReference`, `ShipperDeliveryNoteId`, `ShipperDeliveryNoteDocumentDate`
Shipper address	`ShipperAddressCountryRegionId`, `ShipperAddressStreet`, `ShipperAddressCity`, `ShipperAddressStateId`, `ShipperAddressZipCode`

Nest logistic units by using a flat list

Logistic units are provided as a *flat list* under the header rather than as nested XML elements. You express the container hierarchy (for example, cases on a pallet) by using two fields on each logistic unit:

- `LogisticUnitIdentification` – The SSCC or license plate of this container.
- `ParentLogisticUnitIdentification` – The SSCC of the container that this unit sits inside. Leave it empty for a top-level unit.

This model supports any nesting depth (pallet → case → carton) while keeping the document flat. Line items are attached to the logistic unit that physically holds them.

Provide item details for batches, catch weight, and product variants

Use the following optional fields on each line item (`WHSDespatchAdviceLineItemMessageEntity`) to describe the goods in more detail.

Expand table

Detail	Fields
Batch	`ItemBatchNumber`, `ItemBatchExpirationDate`
Catch weight	`CapturedWeight`, `CapturedWeightUnitSymbol`
Product variant dimensions	`ProductColorId`, `ProductSizeId`, `ProductStyleId`, `ProductConfigurationId`, `ProductVersionId`
Serial number	`ItemSerialNumber`

For example, the following line reports a batch-tracked item that has a batch number, an expiration date, and a color variant.

XML

```
<WHSDespatchAdviceLineItemMessageEntity ITEMNUMBER="A0001" DESPATCHEDQUANTITY="40"
DESPATCHEDQUANTITYUNITSYMBOL="ea" ITEMBATCHNUMBER="BN0001"
ITEMBATCHEXPIRATIONDATE="2027-11-11" PRODUCTCOLORID="Blue" ORDERNUMBER="00000055"
ORDERLINENUMBER="1" />
```

> **Note**
>
> For a catch-weight item, if `CapturedWeightUnitSymbol` differs from the item's catch-weight unit, the weight is converted during processing.

Integrate with a source system

When you integrate an external system with Warehouse management only mode, the despatch advice can reference the source system that sent it. In this case, the external system uses its own item numbers, which differ from the item numbers in the receiving company. The following header field enables the translation:

- `SourceSystemId` – The source system that the despatch advice comes from.

When `SourceSystemId` is specified and the logistic unit belongs to the *Inbound shipment order* module, the `ItemNumber` on each line item is treated as a *source system item number*. During processing, it's translated into the receiving company's item number through the source system item mapping (as seen on the **Source system items** page). The original source system item number is preserved on the message, and the resolved local item number is stored on the ASN item.

> **Note**
>
> If the specified `SourceSystemId` has no item mapping for a line's item number, processing fails with an error. If `SourceSystemId` is empty, the `ItemNumber` value is used as-is, with no translation.

Import a despatch advice message

Import despatch advice messages by using the *Despatch advice composite* entity through any of the following methods:

- Data management import (for example, an XML file–based import), using the *Despatch advice composite* entity
- Open Data Protocol (OData), using the individual message entities (see Import a despatch advice by using OData)
- Dataverse

The following sections show XML import files for the composite entity. Each XML element represents an entity, and each attribute represents a field.

The overall structure nests line items inside logistic units, and logistic units inside the header:

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSDespatchAdviceMessageEntity>
        <WHSDespatchAdviceLogisticUnitMessageEntity>
            <WHSDespatchAdviceLineItemMessageEntity />
        </WHSDespatchAdviceLogisticUnitMessageEntity>
    </WHSDespatchAdviceMessageEntity>
</Document>
```

> **Tip**
>
> For more example XML files, see the despatch advice test resources in the **SCMTestCommon** model.

Example 1: One pallet for a purchase order

This example imports a single license plate that contains one purchase order line. Because no module is specified, the line defaults to the *Purchase order* module, and `ORDERNUMBER` refers to a purchase order.

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSDespatchAdviceMessageEntity MESSAGEID="DA-MSG-0001"
DESPATCHADVICEIDENTIFICATION="DA-1001" SHIPPERIDENTIFICATION="US-101"
RECEIVINGWAREHOUSEID="24" ESTIMATEDARRIVALDATETIME="2026-07-20T10:00:00Z">
        <WHSDespatchAdviceLogisticUnitMessageEntity LOGISTICUNITIDENTIFICATION="LP-
0001">
            <WHSDespatchAdviceLineItemMessageEntity ITEMNUMBER="A0001"
DESPATCHEDQUANTITY="10" DESPATCHEDQUANTITYUNITSYMBOL="pcs" ORDERNUMBER="00000176"
```

```
ORDERLINENUMBER="1" />
        </WHSDespatchAdviceLogisticUnitMessageEntity>
    </WHSDespatchAdviceMessageEntity>
</Document>
```

Example 2: Two cases on a pallet

This example nests two cases inside a pallet. The pallet is a top-level logistic unit, and each case references the pallet through `PARENTLOGISTICUNITIDENTIFICATION`. Items are attached to the cases that hold them, and one line specifies a batch number.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSDespatchAdviceMessageEntity MESSAGEID="DA-MSG-0002"
DESPATCHADVICEIDENTIFICATION="DA-1002" SHIPPERIDENTIFICATION="US-101"
RECEIVINGWAREHOUSEID="24">
        <WHSDespatchAdviceLogisticUnitMessageEntity LOGISTICUNITIDENTIFICATION="PAL-
001" />
        <WHSDespatchAdviceLogisticUnitMessageEntity LOGISTICUNITIDENTIFICATION="CASE-
001" PARENTLOGISTICUNITIDENTIFICATION="PAL-001">
            <WHSDespatchAdviceLineItemMessageEntity ITEMNUMBER="A0001"
DESPATCHEDQUANTITY="12" DESPATCHEDQUANTITYUNITSYMBOL="pcs" ORDERNUMBER="00000177"
ORDERLINENUMBER="1" />
        </WHSDespatchAdviceLogisticUnitMessageEntity>
        <WHSDespatchAdviceLogisticUnitMessageEntity LOGISTICUNITIDENTIFICATION="CASE-
002" PARENTLOGISTICUNITIDENTIFICATION="PAL-001">
            <WHSDespatchAdviceLineItemMessageEntity ITEMNUMBER="P0004"
DESPATCHEDQUANTITY="20" DESPATCHEDQUANTITYUNITSYMBOL="pcs" ITEMBATCHNUMBER="BN0001"
ORDERNUMBER="00000177" ORDERLINENUMBER="2" />
        </WHSDespatchAdviceLogisticUnitMessageEntity>
    </WHSDespatchAdviceMessageEntity>
</Document>
```

Example 3: An inbound shipment order

To import a despatch advice for an inbound shipment order, set the `MODULE` attribute on the logistic unit to `InboundShipmentOrder`. In this case, `ORDERNUMBER` must be the external (source system) order number, which is mapped to the matching inbound shipment order during processing.

This example also specifies `SOURCESYSTEMID` on the header. As a result, the `ITEMNUMBER` value (`ERP-A0001`) is the source system item number and is translated into the receiving company's item number.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSDespatchAdviceMessageEntity MESSAGEID="DA-MSG-0003"
DESPATCHADVICEIDENTIFICATION="DA-1003" SHIPPERIDENTIFICATION="EXT-VENDOR-1"
SOURCESYSTEMID="ERP" RECEIVINGWAREHOUSEID="24">
        <WHSDespatchAdviceLogisticUnitMessageEntity LOGISTICUNITIDENTIFICATION="LP-
9001" MODULE="InboundShipmentOrder">
            <WHSDespatchAdviceLineItemMessageEntity ITEMNUMBER="ERP-A0001"
DESPATCHEDQUANTITY="5" DESPATCHEDQUANTITYUNITSYMBOL="pcs" ORDERNUMBER="IO04"
ORDERLINENUMBER="1" />
        </WHSDespatchAdviceLogisticUnitMessageEntity>
    </WHSDespatchAdviceMessageEntity>
</Document>
```

Import a despatch advice by using OData

Instead of importing the composite XML file, you can post a despatch advice through OData. You post to the individual message entities—first the header, then each logistic unit, and then each line item—and finally post a *Complete* action to queue the message for processing. The message processor then processes the message just as it does for an imported XML file.

Use the following entity collections:

- `DespatchAdviceMessages` – The despatch advice header.
- `DespatchAdviceLogisticUnitMessages` – The logistic units.
- `DespatchAdviceLineItemMessages` – The line items.

Each child message carries the parent keys (`ShipperIdentification`, `DespatchAdviceIdentification`, and `MessageId`, plus `LogisticUnitIdentification` for line items) so that it's linked to the correct header and logistic unit.

The following example posts the header message:

JSON

```
POST {{resource}}/data/DespatchAdviceMessages
{
    "ShipperIdentification": "US-101",
    "DespatchAdviceIdentification": "DA-1001",
    "MessageId": "DA-MSG-0001",
    "dataAreaId": "USMF",
    "ReceivingWarehouseId": "24"
}
```

The following example posts a logistic unit that belongs to the header:

JSON

```
POST {{resource}}/data/DespatchAdviceLogisticUnitMessages
{
    "ShipperIdentification": "US-101",
    "DespatchAdviceIdentification": "DA-1001",
    "MessageId": "DA-MSG-0001",
    "dataAreaId": "USMF",
    "LogisticUnitIdentification": "LP-0001"
}
```

The following example posts a line item that belongs to the logistic unit:

JSON

```
POST {{resource}}/data/DespatchAdviceLineItemMessages
{
    "ShipperIdentification": "US-101",
    "DespatchAdviceIdentification": "DA-1001",
    "MessageId": "DA-MSG-0001",
    "LogisticUnitIdentification": "LP-0001",
    "dataAreaId": "USMF",
    "ItemNumber": "A0001",
    "DespatchedQuantity": 10,
    "DespatchedQuantityUnitSymbol": "pcs",
    "OrderNumber": "00000176",
    "OrderLineNumber": 1
}
```

To queue the message for processing, post the *Complete* action for the header.

JSON

```
POST {{resource}}/data/DespatchAdviceMessages(ShipperIdentification='US-
101',DespatchAdviceIdentification='DA-1001',MessageId='DA-MSG-
0001',dataAreaId='USMF')/Microsoft.Dynamics.DataEntities.Complete?cross-company=true
```

> **Note**
>
> The `dataAreaId` value is part of the key that matches the line messages to the header. Therefore, specify it in each message. Add the `?cross-company=true` suffix only when the company differs from the user's default company that's set up on the **Microsoft Entra ID applications** page.

Update or correct a despatch advice

Despatch advice messages use *replace by new message* semantics. To update a shipment, resubmit the complete, authoritative document with the same `SHIPPERIDENTIFICATION` and `DESPATCHADVICEIDENTIFICATION`, but a new `MESSAGEID`.

When the update is processed, the header and all child records are overwritten from the new message—including empty values—and any logistic units and line items that are no longer part of the shipment are removed. There's no partial or delta update mode, so always send the full document.

Process despatch advice messages

Imported messages are queued and then handled by the message processor. To process them, schedule a batch job by following these steps:

1. Go to **System administration** > **Message processor** > **Message processor**.
2. In the **Message queue** field, select **Despatch advice messages**.
3. Specify a processing interval, and then schedule the batch job.

During processing, each message resolves or creates the inbound load and shipment, rebuilds the logistic unit hierarchy as license plates and packing structures, and creates the ASN item lines. The message status is then set to *Accepted*, or to *Failed* if a validation error occurs. For more information, see Schedule message processing by using the message processor batch job.

Monitor processing results

You can monitor despatch advice processing in the following ways:

- **Message processor messages** – Go to **System administration** > **Message processor** > **Message processor messages** to review message content, find failed messages, and reprocess them. For more information, see Message processor messages for warehouse management processes.
- **Business events** – Set up the *Despatch advice message changed status* business event so that downstream systems are alerted when a message succeeds or fails.

Inspect the results of processing a despatch advice

After a despatch advice message is accepted, follow these steps to inspect the load that it created:

1. Go to **Warehouse management** > **Loads** > **All loads**.
2. Find and open the load that was created for the despatch advice.
3. On the **Load lines** FastTab, review the order numbers, items, and quantities.
4. On the Action Pane, on the **Ship and receive** tab, in the **Receive** group, select **Packing structure** to review the license plates, cases, and items that were built from the logistic units.

Last updated on 07/23/2026

Receive unannounced sales returns

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

In Microsoft Dynamics 365 Supply Chain Management, the sales return process usually starts when you create a return material authorization (RMA) order. The process of creating an RMA order supports a scenario where the reason for the return isn't immediately apparent or disclosed. The RMA order serves as the primary document that guides subsequent steps in the return process, including warehouse arrival and the receiving procedures.

However, in some business scenarios, customers might request to return products without providing advance notification or an associated order. In these scenarios, the standard RMA process is bypassed, and special considerations are made to accommodate the unplanned or unanticipated (unannounced) returns.

This article explains how to set up Supply Chain Management to support unannounced returns, and how to receive unannounced returns in the warehouse.

Prerequisites

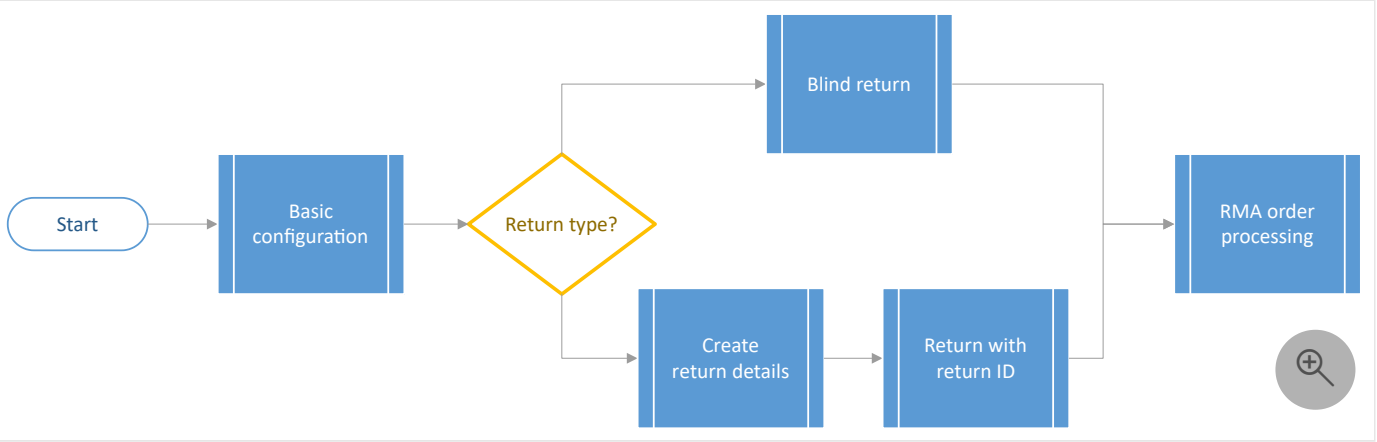
Before you can use the features described in this article, ensure your system meets the following requirements:

- To receive unannounced sales returns, you're running Supply Chain Management version 10.0.39 or later.
- To use the integration with small parcel shipping (SPS), you're running Supply Chain Management version 10.0.39 or later.
- To print license plate labels from a mobile device, you're running Supply Chain Management version 10.0.40 or later.

Receiving process for unannounced returns

The *Return item receiving* flow in the Warehouse Management mobile app supports two processes for receiving unannounced returns: *Blind return* and *Return details*.

The following illustration shows how the flow works for each process.



Blind return process

A blind return is a return where no RMA order or return details record exists in advance, and where no original sales order or shipment must be specified during receiving. During the *Return item receiving* process, workers must use the mobile app to identify and confirm the following information:

- **Customer account or Source system** – The source that returns the item. *Customer account* is used for sales return orders and *Source system* is used when using inbound shipment orders as part of the Warehouse management only mode process.
- **Item number and quantity** – The items that are returned and the quantity that's returned for each item.
- **License plate ID** – The license plate of the location where the incoming items are stored after they're received.
- **Disposition code** – Depending on the setup of the mobile device menu item, workers might be able to select a disposition code to specify what should be done with the returned products. If this option isn't shown in the app during receiving, a Supply Chain Management user must enter a disposition code when they complete the mixed license plate.

> ⓘ **Note**
>
> When running the return process for inbound shipment orders, you must also define source system disposition codes.

Return details process

The *Return details* return process uses a return details record. This record is automatically generated when an order is shipped. It contains all the information required to process a return, such as the return ID, original order ID, shipment ID, order lines, and return-until dates.

Typically, the sending company prints a return label for each outgoing shipment and delivers it together with that shipment. The return label includes a bar code that contains the return ID. Then, if a customer must return an item, they don't have to contact the sending company. Instead, they can just pack the item and attach the return label that they received together with the original shipment.

> ⓘ **Note**
>
> Only the items/variants, tracking dimensions, and quantities that are listed in the specified return details record can be returned through this process.

You can generate a return details record in two ways:

- In a *packing scenario*, the system generates a return details record after each sales order is packed into a container and the container is closed.
- In a *nonpacking scenario*, the system generates a return details record after a worker creates a shipment for a sales order and confirms it.

In both cases, the system fills in the shipment date and return-until date when it confirms the related shipment.

During the *Return item receiving* process, workers must use the mobile app to identify and confirm the following information:

- **Return ID** – The ID number of the return details record.
- **Item number and quantity** – The items that are returned and the quantity that's returned for each item.
- **License plate ID** – The license plate of the location where the incoming items are stored after they're received.
- **Disposition code** – Depending on the setup of the mobile device menu item, workers might be able to select a disposition code to specify what should be done with the returned products. If this option isn't shown in the app during receiving, a Supply Chain Management user must enter a disposition code when they complete the mixed license plate.

Set up unannounced sales return receiving

To prepare your system to receive returns without using an RMA order, you must enable the relevant features and complete several related configurations. The following subsections describe the required settings.

Enable unannounced returns

Before you can use blind returns and/or automatic generation of return details during receiving, you must follow these steps to turn on and set up each feature in your system.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.

2. If you plan to use the *Return details* return process, specify the number sequences that support this process:
   a. On the **Number Sequences** tab, find the row where the **Reference** field is set to *Return ID*. For this row, set the **Number sequence code** field to the number sequence that you want to use.
   b. Find the row where the **Reference** field is set to *Load line inventory pick*. For this row, set the **Number sequence code** field to the number sequence that you want to use.

   For more information about how to set up number sequences for these purposes, see Number sequences overview.

3. If you plan to use the *Return details* process, on the **Loads** tab, set the **Enable sales load line picking route** option to *Yes*. This setting enables load lines to be linked to sales line inventory transactions. The linking is done at the shipment level. This option uses the *Load line inventory pick* number sequence when work is completed and containers are closed.

4. On the **General** tab, on the **Returns** FastTab, set the following options:

   - **Enable return details creation** – Set this option to *Yes* to enable return details to be created when a container is closed and during shipment confirmation. This option is required only for the *Return details* process.
   - **Enable return order creation from mobile device** – Set this option to *Yes* to enable return orders to be created from a mobile device flow. The flow uses mixed license plate receiving planned lines. This option is required for both the *Return details* process and the *Blind return* process.

5. If you're using Warehouse management only mode, set up returns for each source system by following these steps:
    a. Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**.
    b. From the list pane, select the source system you want to set up for returns.
    c. Make the following settings on the **Inbound shipment orders** FastTab:

        - **Enable returns process** – Set to *Yes* to enable the return process for the current source system.
        - **Number sequence code** – Select the number sequence to use to generate order numbers for inbound shipment orders.
        - **Return order type** – Enter text used to stamp the created inbound shipment order. This value is used to distinguish return orders from other inbound shipment orders.

    d. Repeat these steps for each relevant source system.

Set up disposition codes

Disposition codes are collections of rules that you can use when receiving items in Supply Chain Management. They help determine the inventory status, the work template, and the location directive for the items, based on the scanned code.

For example, when a warehouse worker uses a mobile device to receive items that are damaged, the worker scans a disposition code for damaged items.

- If you want to credit the item to your inventory and make it available for repurchase, set up a disposition code that sets the inventory status of the received item to *Available*.
- If you receive items that you can't resell and must therefore dispose of, use a disposition code for RMA scrap. This disposition code then returns a *return disposition code*.

Use disposition codes for different processes, such as *purchase order receiving*, *production order report as finished*, and *sales order return receiving*. For the *sales order return receiving* process, if the items are registered by using a mobile device, the use of disposition codes is mandatory.

Return disposition codes

To view and set up your return disposition codes, go to **Sales and Marketing** > **Setup** > **Returns** > **Disposition codes**. Define each code by specifying an ID number, an action, and a description. The action defines how items that are assigned to the code should be handled.

For more information about return disposition codes and the actions that are available, see Disposition codes and disposition actions.

Mobile device disposition codes

To set up disposition codes that you can use on a mobile device, go to **Warehouse management** > **Setup** > **Mobile device** > **Disposition codes**. Disposition codes that you create here are available to mobile devices and can support several types of operations, including returns, sales orders, and inventory management. Disposition codes for returns must be linked to a return disposition code defined at **Sales and Marketing** > **Setup** > **Returns** > **Disposition codes**, and that assigns a return action for the code.

For more information about mobile disposition codes and how to set them up, see Set up dispositions codes.

Source system disposition codes

If you're using Warehouse management only mode, you must map local dispositions codes with the *source system disposition codes* used in each relevant source systems. These codes are used when processing inbound shipment orders.

To set up source system disposition codes, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source system disposition codes**.
2. On the Action pane, select **New** to add a new code. Then make the following settings for the new code:
    - **Disposition code** – Select the local disposition code that you want to map.
    - **Source system** – Select the source system that you want to map the local disposition code to.
    - **Source system disposition code** – Enter the disposition code that is used in the source system.
3. Repeat the previous step until you set up all of the disposition codes that you need to map.

Set up return item policies

Return item policies let you control the conditions under which the system allows items to be returned. You can set policies for each item, groups of items, or all items, as you require. To set up

your policies, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Return items** > **Return item policies**.
2. On the Action Pane, select **New** to create a policy.
3. In the **Item code** field, select one of the following values to specify the scope that the new policy applies to:
    - *Table* – The return item policy applies only to a specific product.
    - *Group* – The return item policy applies to a specific product group.
    - *All* – The return item policy applies to all products.
4. In the **Item** field, if you set the **Item code** field to *Table* in the previous step, select the item number that the new policy applies to. If you set the **Item code** field to *Group*, select the item group that the new policy applies to. If you set the **Item code** field to *All*, the **Item** field is unavailable.
5. In the **Return acceptance** field, select one of the following values:
    - *Always allowed* – Items that match the selected **Item code** and **Item** values can always be returned.
    - *Never allowed* – Items that match the selected **Item code** and **Item** values can never be returned.
    - *Allowed days after shipment* – Items that match the selected **Item code** and **Item** values can be returned only for a limited time after they were sold, as defined in the **Allowed days for return** field.
6. If you set the **Return acceptance** field to *Allowed days after shipment*, set the **Allow days for return** field to the number of allowed days.
7. Repeat this procedure for each item or item group, as you require. When returns are processed, the most specific return item policy applies to each item in the return.
8. On the Action Pane, select **Save**.

Create return item receiving policies

Before you create the related mobile device menu items, define the required return item receiving policies. These policies enable each menu item to process the return correctly.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Return item receiving policies**.

2. On the Action Pane, select **New** to add a return item receiving policy.

3. Set the following fields for the new policy:

- **Return item receiving policy ID** – Enter a name for the policy, such as *Blind return* or *Return details*.
- **Description** – Enter a short description of the policy.
- **Return process** – Select the type of return process that the policy represents (*Return details* or *Blind return*).
- **Create return order** – Select the type of return order that the system should create to manage returns:
  - *Return order* – The system creates a return order for each return. Select this value if you aren't using Warehouse management only mode.
  - *Inbound shipment order* – The system creates an inbound shipment order for each return. Select this value if you're using Warehouse management only mode.
- **Return order identification** – This field is available only when the **Return process** field is set to *Blind return* and the **Create return order** field is set to *Inbound shipment order*. The way that you use it depends on the needs of your external system. (See the documentation for your external system.) Select the type of extra identifying information that the system should return to the source system when it registers a return order:
  - *None* – The system doesn't return any extra identifying information, and the app doesn't prompt the worker to enter any extra information during receiving.
  - *Account number* – The system returns an account number to help identify the return order. This value is typically the ID of the customer who is returning the items (the customer account number). The mobile app prompts the worker to scan or enter this number during receiving.
- **Return order type** – This field is available only when the **Return order identification** is available and set to *Account number*. Like the account number, the value in this field is returned to the source system when a return order is registered. It's intended to hold the name that identifies the document type for return orders in the source system. The way that you use it depends on the needs of your external system. (See the documentation for your external system.) If you leave this field blank, the value is inherited from the relevant source system configuration.

> **Important**
>
> If your source system is Dynamics 365 Supply Chain Management (as is the case when you run an external shared warehouse scenario), set the following values for your blind return policies:
>
> - **Return order identification** – Select *Account number*, and instruct workers to enter the customer ID (account number) when the mobile app prompts them during receiving.
> - **Return order type** – Enter *Return*, which is the return order type that is used in Supply Chain Management. (Alternatively, if *Return* is already specified as the default return order type for the source system, leave this field blank.)

4. If you support both types of unannounced return processes (*Return details* and *Blind return*), repeat this procedure to add a return item receiving policy for the other process.

Set up mobile device menu items to process unannounced returns

To enable workers to process unannounced returns, create a separate mobile device menu item for each type of unannounced return process (*Return details* or *Blind return*) that you support.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following fields for the new menu item:
    - **Menu item name** – Enter an internal name for the menu item, such as *Blind returns*.
    - **Title** – Enter the name that should appear for the menu item in the mobile app, such as *Blind return item receiving*.
    - **Mode** – Select *Work*.
    - **Use existing work** – Set this option to *No*.
    - **Work creation process** – Select *Return item receiving*.
    - **Barcode data policy** – Select the policy to use if multiple fields are filled in based on a single bar code scan. To learn more, see GS1 bar codes.
    - **Generate license plate** – Set this option to *Yes* to automatically create new license plates as they're needed. Set it to *No* if the worker must always select an existing license plate.

- **Display disposition code** – Select whether workers should be prompted to select a disposition code during the receiving process. The disposition code determines the inventory status, work template, and location directive for the returned items.
- **Print label** – Set this option to *Yes* to always print a license plate label after all steps in the related work template are completed (regardless of whether a print step is included in the work template). If you want to allow the work template to print the license plate label at a different point in the process, set this option to *No*. The license plate label includes a bar code that provides the ID of the license plate where the worker places the returned items. To learn more, see License plate label layouts and printing.
- **Return item receiving policy ID** – Select the item receiving policy that you created for the type of return process (*Return details* and *Blind return*) that this menu item supports.

4. If you support both types of unannounced return processes (*Return details* and *Blind return*), repeat steps 2 and 3 to create a menu item for the other process.

5. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.

6. Add the new menu items to an appropriate place in your mobile device menu structure.

Integrate return label printing with small parcel shipping

Small parcel shipping (SPS) integration lets you retrieve shipping labels from the SPS carriers or carrier hubs that you integrate with your system. However, some customization is required. The integration supports preaddressed and prepaid return labels, which make returns more convenient for your customers. The return labels can include a bar code that identifies the return details record ID. Warehouse workers can then scan this bar code when the package arrives at your warehouse.

For more information about how to set up SPS integration, see Small parcel shipping.

Follow these steps to enable return details with SPS integration.

1. Go to **Transportation management** > **Setup** > **Transportation management parameters**.
2. On the **Shipping carriers** FastTab, set the **Enable shipping label request type** option to *Yes*. This setting adds a new **Print container return shipping label rule** field to the **Container packing policies** page. Use this field to select the conditions under which return labels

should be printed as part of the container packing process. For details about how to use this setting, see the next section.

Print return labels

When you use a *Return details* return process, you must print a return label for each shipment and include that label with the shipment. The *container packing policy* that applies to each shipment establishes the printing options. Depending on the setup of your system, return labels are typically printed as part of the container closing process. However, you can manually print or reprint them for a selected return details record at any time.

The return label includes the shipping address of your returns warehouse and a bar code that contains the return details record ID. Therefore, warehouse workers can scan this bar code when the package arrives at your warehouse.

Follow these steps to configure the label printing options for a container packing policy.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container packing policies**.

2. Select or create the container packing policy that you want to set up.

3. On the **Container manifest** FastTab, set the following fields:

    - **Automatic manifest at container close** – Set this option to *Yes*.
    - **Manifest requirements for container** – Set this field to *Transportation management*.

4. On the **Carrier label printing** FastTab, set the following fields:

    - **Print container shipping label rule** – Select one of the following options to specify the rule for printing outbound shipping labels:
        - *Never* – Users can close containers without printing a shipping label.
        - *Always* – A shipping label is required. Users can't close the container unless the system can retrieve a label.
        - *If setup exists* – For containers where a shipping label setup exists, the system must retrieve the required label before users can close the container. For containers where no shipping label setup exists, users can close the container without printing a label.

    - **Print container return shipping label rule** – Select one of the following options to specify the rule for printing return labels to include in each container:
        - *Never* – Users can close containers without printing a return label.

    - *Always* – A return label is required. Users can't close the container unless the system can retrieve a label.
    - *If setup exists* – For containers where a return label setup exists, the system must retrieve the required label before users can close the container. For containers where no return label setup exists, users can close the container without printing a label.

- **Printer name** – Select the Zebra Programming Language (ZPL) printer where the system should print shipping and/or return labels when this container packing policy is used.

Follow these steps to manually print or reprint a return label for a return details record.

1. Go to **Warehouse management** > **Inquiries and reports** > **Return details**.
2. In the list pane, select the return details record that you want to print a label for.
3. On the Action Pane, select **Print** > **Return shipping label**.

The return label is printed on the printer that's specified in the relevant container packing policy that's associated with the **Container ID** value that's listed for the return details record.

Example scenarios for the sales return order creation process

This section provides example scenarios that show how to set up and process unannounced returns.

Scenario prerequisites

Before you work through the example scenarios, complete all the prerequisites described in the following subsections.

Enable sample data

To work through the example scenarios by using the sample records and values that are specified here, use a system where the standard demo data is installed. Additionally, select the **USMF** legal entity before you begin. The example scenarios use the demo data that's associated with the worker/person *Julia Funderburk*.

You can also use these scenarios as guidance for using the feature on a production system. However, in that case, substitute your own values for each setting that's described here.

Set up number sequences

You must have number sequences that generate unique return IDs and load line inventory pick numbers.

1. Go to **Organization administration** > **Number sequences** > **Number sequences**.
2. On the Action Pane, on the **Number sequence** tab, in the **New** group, select **Number sequence** to create a number sequence that generates return ID numbers.
3. Set the following values for the new number sequence:
    - **Number sequence code:** *ReturnID*
    - **Name:** *ReturnID*
    - **Scope:** *Company*
    - **Company:** *USMF*
4. On the Action Pane, select the **Back** button.
5. On the Action Pane, on the **Number sequence** tab, in the **New** group, select **Number sequence** to create a number sequence that generates load line inventory pick numbers.
6. Set the following values for the new number sequence:
    - **Number sequence code:** *LLIP*
    - **Name:** *LLIP*
    - **Scope:** *Company*
    - **Company:** *USMF*
7. On the Action Pane, select **Save**.

Enable unannounced return features

Follow these steps to enable the features that are required to support unannounced returns.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Number sequences** tab, set the **Number sequence code** field to *ReturnID*.
3. Set the **Number sequence code** field to *LLIP*.
4. On the **General** tab, turn on the **Enable return details creation** option.
5. Turn on the **Enable return order creation from mobile device** option.
6. On the **Loads** tab, turn on the **Enable sales load line picking route** option.

For information about how to configure these settings, see the Enable unannounced returns section.

Create return item policies

Follow these steps to create return item policies for the items that are used in the scenarios.

1. Go to **Warehouse management** > **Setup** > **Return items** > **Return item policies**.
2. Add a return item policy to define the conditions under which the item that's used in the scenarios can be returned.
3. Set the following values for the new policy:

    - **Item code:** *Table*
    - **Item:** *A0001*
    - **Return acceptance:** *Allowed days after shipment*
    - **Allowed days for return:** *28*

Learn more in the Set up return item policies section.

Set up return item receiving policies

Follow these steps to create return item receiving policies for the scenarios.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Return item receiving policies**.
2. Add two return item receiving policies: one for the *Blind return* process and one for the *Return details* process.

Learn more in the Create return item receiving policies section.

Set up mobile device menu items and menus

Follow these steps to create mobile device menu items for the scenarios.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. Create two mobile device menu items for handling unannounced returns: one for the *Blind return* process and one for the *Return details* process.

3. Set up the new menu items in the following way:

    - Give each menu item a name that identifies the type of return that it's intended to process (for example, *Blind return receiving* or *Return details receiving*).
    - For each menu item, set the **Work creation process** field to *Return item receiving*.

- For each menu item, enable both the **Generate license plate** option and the **Display disposition code** option.
- For each menu item, set the **Return item receiving policy ID** field to the policy that matches the type of return that the menu item should handle (*blind returns* or *return details*).

4. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.

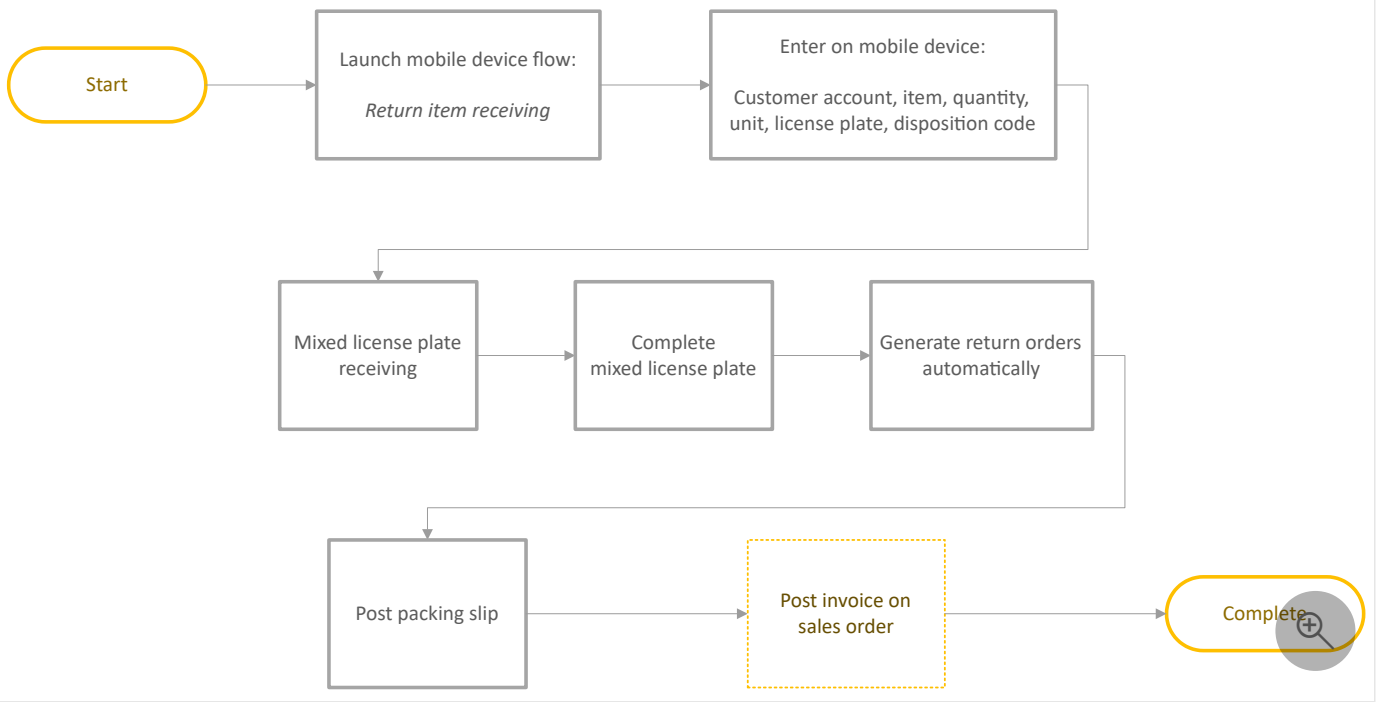
5. Add both menu items to an appropriate place in your mobile device menu structure.

Learn more in the Set up mobile device menu items section.

Example scenario 1: Customer return with a blind return policy

In the blind receiving process, warehouse workers receive return items without matching them to an original sales order or shipment. Warehouse workers just enter the customer account ID, item IDs, and quantities.

The following illustration shows a flowchart for the blind receiving process.



Use the mobile app to receive a blind return to a mixed license plate

Follow these steps to receive products to a mixed license plate in a warehouse when there's a blind return policy.

1. Open the Warehouse Management mobile app, and sign in by using *62* as the user ID and *1* as the password.
2. Open the menu item that you created for receiving blind returns (for example, **Inbound** > **Blind returns receiving**).
3. In the **Customer account** field, enter *US-003*.
4. In the **Item** field, enter or scan item number *A0001*.
5. Accept the license plate number that's suggested.
6. In the **qty** field, enter *1*.
7. In the **Disposition code** field, select *RMA Credit*.
8. Select **OK** to complete the receiving work.
9. Select **Cancel** to exit the flow and close the menu item.

> ⓘ **Note**
>
> You must exit the flow to unlock the license plate. If the license plate is locked, you can't complete mixed license plate receiving in Supply Chain Management, as described in the next section. In some setups, you can finish processing mixed license plates during the *Return item receiving* flow. However, in this scenario, you process the license plate in Supply Chain Management.

Complete mixed license plate receiving

The returned products are now on a mixed license plate for further processing. (Learn more in Mixed license plate receiving.) Follow these steps to finish receiving the returned items.

1. Go to **Warehouse Management** > **Inquiries and reports** > **Mixed license plate receiving**.
2. In the **License plate** grid, find the mixed license plate that you processed by using the mobile app.
3. On the Action Pane, on the **License plate** tab, select **Complete license plate** to complete mixed license plate receiving.

After you complete mixed license plate receiving, Supply Chain Management automatically creates an RMA order that's prepared for further downstream processing, as described in Sales

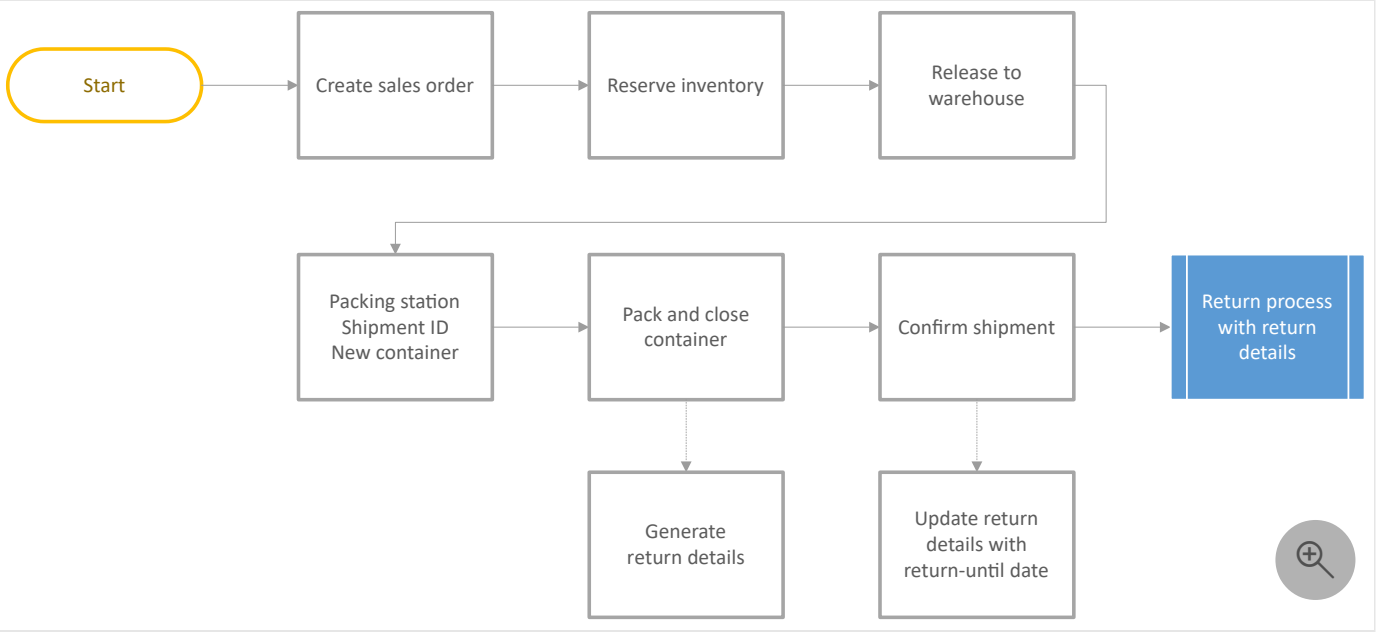
returns.

> **Tip**
>
> You can set up your mobile device menu item for processing returns to include a step that lets workers complete mixed license plate receiving directly from the mobile device while receiving the return. Then you don't need to complete this step later in the Supply Chain Management web client. To do so, create a detour item with the **Activity code** *Complete mixed license plate*. Learn more in Set up a mobile device menu item for completing mixed license plates.

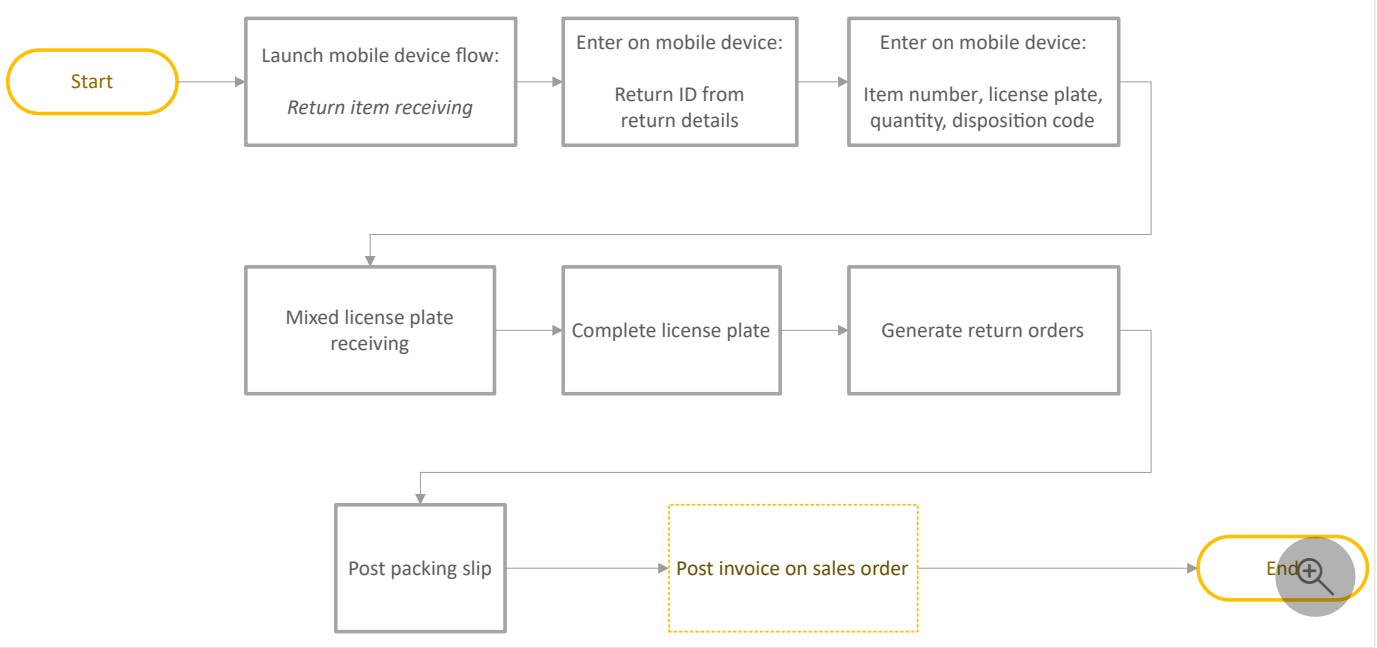
Example scenario 2: Customer return with a return details policy

For this scenario, return details are generated when an order is processed for shipment at the packing station, and matching labels are included with the shipment. If any items must be returned, the customer packs the return items, attaches the return label that they received together with the original shipment, and ships the return items back to the seller. When the returned items arrive at the seller's warehouse, a worker scans the return label and the item to get the return ID. The return ID identifies a return details record that includes the original order details. This process helps prevent fraud, inventory errors, and lost packages.

The following illustration shows the steps that are required to create a sales order and complete the work of moving the ordered items to the packing area. Workers then pack the order, and return details are generated when the container is closed.



The following illustration shows a flowchart of the receiving process where the *Return item receiving* mobile menu item is used to process a return that has return details.



Create a sales order and release it to the warehouse for packing

For this scenario, a related sales order must already exist, and the item must be shipped before the return. Therefore, you must create the required records before you can work through the related return. Follow these steps to create a sales order and complete the work of moving the ordered items to the packing area.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In the **Create sales order** dialog box, set the **Customer account** field to *US-001*.
4. Select **OK** to close the dialog box.
5. The new sales order opens. The **Sales order lines** FastTab includes a single, blank order line. Set the following values for this line:
    - **Item number:** *A0001*
    - **Quantity:** *1*
    - **Site:** *6*
    - **Warehouse:** *62*
6. While the order line is still selected on the **Sales order lines** FastTab, select **Inventory** > **Reservation** on the toolbar.

7. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.

8. On the Action Pane, select the **Back** button to return to the sales order.

9. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**. A message shows the shipment and wave IDs for the order.

10. While the order line is still selected on the **Sales order lines** FastTab, select **Warehouse** > **Work details** on the toolbar. If you use batch processing to run your waves, you might have to wait a short time for the work to be created.

11. On the **Work** page, on the Action Pane, on the **Work** tab, select **Complete work**.

12. On the **Work completion** page, set the **User ID** field to *62*.

13. On the Action Pane, select **Validate work**.

14. When you receive a message that indicates that your work is valid, select **Complete work** on the Action Pane to complete the process of picking the inventory items and putting them in the *Pack* location.

15. Make a note of the **Shipment ID** value that's shown for the work in the upper grid.

Pack the ordered items into a container and generate return details

The inventory items are brought to the packing area and are ready to be packed into containers. Follow these steps to create a new container in the system, and to pack and close the container. After you close the container, the system automatically creates a return details record. You can reference this record later when returned items are received back into the warehouse. At this point, you can print a return label and deliver it together with the container.

1. Go to **Warehouse Management** > **Packing and containerization** > **Pack**.

2. In the **Select packing station** dialog box, set the following values:

    - **Site:** *6*
    - **Warehouse:** *62*
    - **Location:** *Pack*
    - **Packing profile ID:** *WH62*

3. Select **OK** to close the dialog box and return to the **Pack** page.

4. On the **Packing station** FastTab, in the **License plate or shipment** field, enter the shipment ID that you noted at the end of the previous procedure. Then select **Return**.
5. On the Action Pane, select **New container** to start the *container creation* process.
6. In the dialog box, set the **Container type** field to *SmallBox*. Then select **OK** to return to the **Pack** page.
7. On the **Item packing** FastTab, set the **Identifier** field to *A0001*. Then select **Return**. A quantity of *1* of item number *A0001* is moved to the container for the shipment. On the **All lines** FastTab, the line for item number *A0001* now shows a check mark in the **Complete** column.
8. On the Action Pane, select **Close container** to start the *container closing* process.
9. In the dialog box, select the **Get system weight** link to load the weight of the shipment into the **Gross weight** section. Then select **OK** to return to the **Pack** page.
10. Now that the container is closed, the system automatically creates return details. Go to **Warehouse Management** > **Inquiries and reports** > **Return details**.
11. The **Return details** page should list a new return details record that shows the order number and line item that you just packed. Make a note of the **Return ID** value for the new record. You need this value later in this scenario.
12. Go to **Warehouse Management** > **Shipments** > **All shipments**.
13. Select the shipment that has the shipment ID that you noted at the end of the previous procedure.
14. On the Action Pane, on the **Shipments** tab, select **Confirm Shipment**.

    A message shows that the shipment is confirmed.

> **Note**
>
> If you return to the **Return details** page and open the record that has the return ID that you noted earlier, the **Lines** FastTab should now show a **Return until date** value that matches the return item policy that you set for the packed item. The system automatically adds this value after you confirm the shipment.

Receive a product with return details to a mixed license plate

Follow these steps to receive returned items into the warehouse.

1. Open the Warehouse Management mobile app, and sign in by using *62* as the user ID and *1* as the password.
2. Open the menu item that you created for receiving by using the *Return details* process (for example, **Inbound** > **Return details receiving**).
3. In the **Return ID** field, enter the return ID that you noted during the previous procedure. In a real-world scenario, the warehouse worker typically scans this value from the return label that's attached to the package. (This label was sent to the customer together with the original shipment.)
4. In the **Item** field, enter or scan item number *A0001*.
5. Accept the license plate number that's suggested.
6. In the **qty** field, enter *1*.
7. In the **Disposition code** field, select *RMA Credit*.
8. Select **OK** to complete the receiving work.
9. Select **Cancel** to exit the flow and close the menu item.

> **Note**
>
> You must exit the flow to unlock the license plate. If the license plate is locked, you can't complete mixed license plate receiving in Supply Chain Management, as described in the next section. In some setups, you can finish processing mixed license plates during the *Return item receiving* flow. However, in this scenario, you process the license plate in Supply Chain Management.

Complete mixed license plate receiving

The returned products are now on a mixed license plate for further processing. (Learn more in Mixed license plate receiving.) Follow these steps to finish receiving the returned items.

1. Go to **Warehouse Management** > **Inquiries and reports** > **Mixed license plate receiving**.
2. In the **License plate** grid, find the mixed license plate that you processed by using the mobile app.

3. On the Action Pane, on the **License plate** tab, select **Complete license plate** to complete the mixed license plate receiving.

After you complete mixed license plate receiving, Supply Chain Management automatically creates an RMA order that's prepared for further downstream processing, as described in Sales returns.

> **Tip**
>
> You can set up your mobile device menu item for processing returns to include a step that lets workers complete mixed license plate receiving directly from the mobile device while receiving the return. Then you don't need to **complete this step** later in the Supply Chain Management web client. To do so, create a **detour** item with the **Activity code** *Complete mixed license plate*. Learn more in **Set up a mobile device menu item for completing mixed license plates**.

Example scenario 3: Nonpacking scenarios for return details

Use the *Return details* process for scenarios where you don't use a packing station to prepare outgoing shipments. In this case, your outgoing shipments are delivered on a pallet to the bay door, where they're packed and shipped directly. Therefore, the system doesn't generate the return details record after the container is packed and closed (from the **Pack** page in Supply Chain Management). Instead, it generates the return details record when the outbound shipment is confirmed (from the **All loads** page in Supply Chain Management). Otherwise, this scenario works in the same way as the previously described packing scenario.

For this scenario, it's important that your system is configured to create return details, generate load line inventory picking numbers, and use sales load line picking routes, as described in the Scenario prerequisites section.

Last updated on 05/22/2026

Manage inbound putaway based on container types

Article • 01/06/2023

When you run warehouse receiving processes for various types of handling units (such as octabin pallets or Euro-pallets in different sizes), you can use container types to optimize the putaway process. Instead of using the item master's volumetric data to calculate the capacity of the putaway location, you can link each license plate registration to a container type that holds the physical dimension data. That data can then be used to select a valid putaway location.

Set up container types

By associating a container type with a license plate, you can control storage policies based on that container type.

Follow these steps to set up a container type.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container types**.
2. Follow one of these steps:
    - To create a new container type, select **New** on the Action Pane.
    - To edit an existing container type, select it in the list pane, and then select **Edit** on the Action Pane.
3. On the header of the new or selected record, set the following fields:
    - **Container type code** – Enter a unique name or identifier for the container type.
    - **Description** – Enter a short description of the container type.
4. On the **General** FastTab, set the following fields:
    - **Tare weight** – Enter the weight of the container when it's empty.
    - **Maximum net weight** – Enter the maximum weight of the container when it's packed.
    - **Container length**, **Container width**, **Container height**, and **Container volume** – Enter the physical dimensions of the container. These values are required for calculations of volumetric location loads, but they aren't required for calculations of location stocking limits.

- **Flexible volume dimensions** – Set this option to *Yes* if the volume depends on the container volume plus the inventory that's put into the container. Set it to *No* if the volume is considered fixed and doesn't depend on the inventory that's put into the container. This setting doesn't affect existing containerization processes.
- **Unit** – Specify the handling unit for the container, such as *Euro-pallet* or *Octabin*. This unit applies to the container standard itself, not to its dimensions (such as height and length). If you'll configure your system to use this container type to calculate location stocking limits, you must specify a unit here. When the system calculates location stocking limits, it can, as required, convert between the handling units that are used by different container types.
- **Length**, **Width**, **Height**, and **Volume** (in the **Maximums** section) – Enter the maximum physical dimensions that the container can handle.
- **Attributes** – You can define additional attributes for container types that are associated with other operations that use containers. Learn more in Containerization.

5. On the Action Pane, select **Save**.

Define allowed container type groups

Use the **Allowed container type groups** page to prevent locations from holding some container types. To limit locations to specific container types, you must associate the records on this page with a location profile.

Follow these steps to set up an allowed container type group.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Allowed container type groups**.

2. Follow one of these steps:

   - To create a new container type, select **New** on the Action Pane.
   - To edit an existing container type, select it in the list pane, and then select **Edit** on the Action Pane.

3. On the header of the new or selected record, set the following fields:

   - **Allowed container type group name** – Enter a unique name or identifier for the group.
   - **Allow unspecified container types** – Set this option to *Yes* to allow license plates that aren't associated with a container type to be put into locations in

this group. Set it to *No* if all license plates that are stored in locations in this group must be associated with a container type.

4. On the **Details** FastTab, specify the container types that can be put into locations in this group.

    - Use the **New** button to add rows to the grid and the **Delete** button to remove rows.
    - For each row, select a container type.

5. On the Action Pane, select **Save**.

Assign container type groups to location profiles

By assigning a container type group to a location profile, you ensure that only license plates that meet the criteria of that group can be put into locations where the location profile applies. You can assign a container type group only to location profiles where license plate tracking is enabled.

Follow these steps to assign a container type group to a location profile.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. Select the target profile in the list pane, and then select **Edit** on the Action Pane.
3. On the **General** FastTab, set the **Use license plate tracking** option to *Yes*.
4. In the **Allowed container type group** field, select the container type group that applies to the selected location profile.
5. On the Action Pane, select **Save**.

Define location stocking limits based on container types

Location stocking limits let you limit the amount of inventory that the system will try to store at a given location when it generates putaway work. The limits can be based on individual products or product variants, or they can be based on container types. This section described how to set up stocking limits that are based on container types. For more information about other types of location stocking limits, see Location stocking limits.

When you set up location stocking limits that are based on container types, the system checks the number of each type of container that's already stored at a location to

calculate whether there's room to add an incoming container there. During this calculation, the system can convert between different container types based on their handling units, provided that an applicable unit conversion is available (see the note later in this section).

When the system is looking for a location that has room to store an incoming container, it first evaluates the container type limits. If no matching container type constraints are found, it evaluates the settings at the product or variant level. To prevent the product-level settings from being used for a specific combination of a location profile and a container type, set up a container-based limit where the **Allow unlimited quantity** checkbox is selected.

Follow these steps to set up location stocking limits that are based on container types.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location stocking limits**.
2. On the **Container types** tab, each row in the grid defines a stocking limit that's based on a warehouse, a location, and a container type. Use the **New** and **Delete** buttons on the toolbar to add and remove rows as required. For each row, set the following fields:

    - **Warehouse** – Select the warehouse where the stocking limit for the row applies.
    - **Location profile ID** – Select the location profile that's assigned to the locations where the row applies. You can select only location profiles that an allowed container type group is assigned to.
    - **Container type** – Select the container type that the current row applies to. You can select only container types that a unit is assigned to, and that belong to the allowed container type group that's assigned to the selected location profile ID.
    - **Allow unlimited quantity** – Use this checkbox to establish a stocking limit for the warehouse, location profile, and container type that are selected for this row.
        - *Selected* – The location can contain an unlimited number of the specified container type. In this case, the stocking limit settings on the **Products** and **Product variants** tabs aren't evaluated when the current row applies.
        - *Cleared* – The location can hold only a limited number of containers of the selected type. For rows where this checkbox is cleared, you must specify the quantity that can be stocked in the location.

- **Quantity** – If the **Allow unlimited quantity** checkbox is cleared, enter the quantity limit that applies to the current row. The read-only **Unit** field specifies the handling unit that applies to the quantity that you enter. The **Unit** value is based on the unit that's assigned to the selected container type. The system will convert between compatible handling units as required.

3. On the Action Pane, select **Save**.

> **Note**
>
> Even though you can define only one unit in each location stocking limit, locations might be able to store more than one container type, depending on the unit conversion factors that are available. The system applies unit conversion to evaluate the related units.
>
> For example, you have the two handling units, *1-PL* and *1/2-PL*. Each is associated with a container type where the following unit conversion is defined: *1-PL* = 2 × *1/2-PL*. In this case, it doesn't matter whether a row on the **Container types** tab of the **Location stocking limits** page is set up so that **Quantity** = *4* and **Unit** = *1/2-PL*, or **Quantity** = *2* and **Unit** = *1-PL*. In both cases, the row will allow the following combinations of location loads.

Expand table

Number of 1/1-Pallets	Number of 1/2-Pallets	Maximum location load
2	0	Yes
1	0	No
1	1	No
1	2	Yes
0	0	No
0	1	No
0	2	No
0	3	No
0	4	Yes

Define how a default container type is assigned

The logic for assigning a container type to a license plate can be controlled through default values. You can assign a default container type at any of several levels in the system. The system then uses the following hierarchy, from most specific to most general. The most specific level where a default container type is defined will apply.

1. Existing license plate container type
2. Mobile device menu items
3. Unit sequence groups
4. Warehouse
5. Warehouse management parameters
6. (None)

Assign a default container type at the global (Warehouse management parameters) level

The default container type that's assigned at the global level (that is, on the **Warehouse management parameters** page) will apply wherever no other default container type is assigned.

Follow these steps to define the default container type at the Warehouse management parameters level.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **License plates** FastTab, in the **Default container type** field, assign a container type.
3. On the Action Pane, select **Save**.

Assign a default container type to a warehouse

Follow these steps to define the default container type at the warehouse level.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.

2. Follow one of these steps:

    - To create a new warehouse, select **New** on the Action Pane.
    - To edit an existing warehouse, select it in the list pane, and then select **Edit** on the Action Pane.

3. On the **Warehouse** FastTab, in the **Inventory** section, in the **Default container type** field, assign a container type.

4. On the Action Pane, select **Save**.

Assign default container types to units in unit sequence groups

Unit sequence groups establish a collection of units that an inventory item can be counted and stored by. For example, wine might be counted by the bottle, by the case (of 12 bottles), or by the pallet (of 84 cases). Each unit in a sequence is typically used for a different purpose, and a default container type can be assigned to each.

Follow these steps to define the default container type at the unit sequence groups level.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Unit sequence groups**.
2. Follow one of these steps:
    - To create a new group, select **New** on the Action Pane.
    - To edit an existing group, select it in the list pane, and then select **Edit** on the Action Pane.
3. On the **Line details** FastTab, either select the line where you want to assign a container type, or select **New** on the toolbar to create a new line.
4. In the **Default container type** field, assign a container type.
5. On the Action Pane, select **Save**.

Assign default container types to mobile device menu items

When workers receive a shipment, they typically use the Warehouse Management mobile app to register the incoming inventory. To begin, they select a menu item that's configured to receive a specific combination of an item, a warehouse, and/or a location. Administrators can assign a default container type for each menu item, and can control whether warehouse workers can see and edit the container type.

Follow these steps to define the default container type at the mobile device menu item level.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container types**.
2. Make a note of each **Container type code** value that you want to use as a default container type for your menu item. You'll need these codes later, and you'll have to enter them, as free text, exactly as they appear here. You won't be able to select them in a drop-down list.

3. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

4. Follow one of these steps to select or create an inbound menu item. (An inbound menu item is a menu item that's designed to help workers register arriving inventory, as established by its **Work creation process** setting.)

    - To create a new inbound menu item, select **New** on the Action Pane.
    - To edit an existing inbound menu item, select it in the list pane, and then select **Edit** on the Action Pane.

5. On the **General** FastTab, set the following fields:

    - **Use default data** – Set this option to *Yes*. The **Default data** button then becomes available on the Action Pane. You'll use that button to define one or more default container types for the menu item.
    - **Display container type** – Set this option to *Yes* if you want the default container type to be shown (and editable) in the mobile app. Set it to *No* if you want the container type to be hidden (and therefore hardcoded to the applicable default container type).

6. On the Action Pane, select **Default data**.

7. On the **Default data** page, the grid has a row for each default setting that applies to the current menu item. You can set up as many default container types as you want, each for a different combination of a warehouse and a location. Alternatively, you can set up just one default container type that applies to all warehouses and locations. For each row where you want to set up a default container type, set the following fields:

    - **Default data field** – Select *Container type*.
    - **Warehouse** – Select the destination warehouse to use with this row. Leave this field blank to create a row that applies to all warehouses where a specific default container type isn't assigned.
    - **Location** – Select the destination location to use with this row. This field lists all the locations that are available for the selected warehouse. Leave this field blank to create a row that applies to all locations in the selected warehouse where a specific default container type isn't assigned.
    - **Hardcoded value** – Enter the **Container type code** value for the container type that you want to use as the default container type for this row. Use the exact value that you noted in step 1.

8. Continue to work on the **Default data** page until you've set up all the rows that you need for the current menu item.

9. On the Action Pane, select **Save**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Cross-dock products from receiving warehouse to stores

Article • 09/29/2021

This procedure walks through the steps to create and process a Cross-dock to distribute products from the receiving location of a purchase order to one or many stores. The user can define multiple configurations and have the system suggest how to distribute the products, or manually enter where the products are distributed to and how much gets distributed to each store. The procedure doesn't include setup of data that can be used in the Cross-dock, such as replenishment rules, organizational hierarchies, and store weights. The procedure uses the USRT demo company.

1. Go to All purchase orders.
2. Select a purchase order in the list and click the link to open the order.
3. On the Action Pane, click Retail and Commerce.
4. Click Cross docking.
5. Click Edit.
    - The category can be used to filter the items in the Lines section.
6. In the list, find and select the desired record.
7. In the Cross docking quantity field, type a value to specify how much of the quantity being purchased of the selected product should be distributed.
8. In the Additional cross docking quantity field, enter a value to specify the quantities to distribute for the available products being purchased
9. In the Distribution field, enter 'Location weight'.
    - You can select the other types to use different rules for the distribution.
10. In the Replenishment hierarchy field, select a value.
11. Select Yes in the Respect assortments field.
12. Click Calculate quantities.
13. Click Create order.
14. Click Yes.
15. In the list, find and select a warehouse that received products
16. Click Order to view the orders that got created for the selected warehouse

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Feedback

Was this page helpful? Yes No

Provide product feedback

Set up a location directive for purchase order putaway

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up a simple location directive. The example that's shown creates a location directive to be used to determine where to put items that have been received for a purchase order.

Pre-conditions: You must create a disposition code. In this procedure, we use a disposition code called *Relabel*. If you're creating a location directive in your own data, you need to have set up warehouse management processes (WMS) for your warehouse and items.

This procedure is intended for the warehouse manager.

Create a location directive header

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the **Work order type** field, select **Purchase orders**.
3. Select **New**.
4. In the **Sequence number** field, enter a number. This is the sequence in which the location directive is processed for the selected work type. You can also modify the sequence, if needed.
5. In the **Name** field, type a value. This is the unique identifier for this directive.
6. In the **Work type** field, select **Put**. Select the type of work to be performed. For directive with work order type Purchase order, **Put** is the only supported value.
7. In the **Site** field, type a value.
8. In the **Warehouse** field, type a value. Leave the Directive code blank. Directive codes are used to link a work order line of type Put to a specific directive. For purchase orders, the location of the last work order line of type Put is resolved before the work template is determined. Therefore it isn't possible to connect the last line of a work template to a specific directive.
9. In the **Disposition code** field, type a value. The Disposition code limits the use of the location directive, so the location directive is only used if the warehouse worker enters this specific value during registration of the item using a mobile device.

10. Select **Save**.

Edit the query for directive

1. Select **Edit query**. The use of this directive is already limited to items registered in the warehouse that you specified, and with the disposition code that you specified. You can add other constraints using the query.
2. Select **OK**.

Add directive lines

1. Select **New**. This is the sequence in which the location directive lines are processed for the selected work type. You can also modify the sequence, if needed.
2. In the **From quantity** field, enter a number. This is the lowest quantity that this directive line is valid for.
3. In the **To quantity** field, enter a number.
4. In the **Unit** field, type a value. The unit the From quantity and To quantity is expressed in. If you leave this field blank the inventory unit from the item is used.
5. In the **Locate quantity** field, select an option.

    - None, or license plate quantity: The quantity registered on each license plate.
    - Unitized quantity: The entire quantity that's been registered.
    - Remaining quantity: The quantity that is yet to be registered from the purchase order line.
    - Expected quantity: The total quantity that is specified on the purchase order line.

6. Check or uncheck the **Restrict by unit** checkbox. If you select this option, and specify the unit on the **Restrict by unit** page, only items with that unit of measurement can be put into the location. For example, if the unit of measurement is PL (pallets), only items in pallets can be put into the specified location.
7. Check or uncheck the **Allow split** checkbox. This allows the directive to split the quantity across multiple locations.
8. Select **Save**.

Restrict the directive line to a specific unit

1. Select **Restrict by unit**. This button is only available when you press **Save** after you have selected the **Restrict by unit** check box.
2. In the **Unit** field, type a value.
3. Close the page.

Add a location directive action line

1. Select **New**. This is the sequence in which the location directive action lines are processed for the selected work type. You can also modify the sequence, if needed.
2. In the **Name** field, type a value. This is the unique identifier for this directive action.
3. In the **Fixed location usage** field, select an option.

    - *Fixed and non-fixed locations*: All non-fixed locations, including the product's own fixed location, are valid within the range specified in the query.
    - *Only fixed location for the product*: Fixed locations for the product are valid, and all product variants share the same set of fixed locations.
    - *Only fixed location for the product variants*: Only fixed locations specified for each product variant are valid.

4. In the **Strategy** field, select an option. Work orders of type *Purchase order* support the following strategies:

    - *None*: the item is placed at the first location that's found.
    - *Consolidate*: The item is placed in a location where similar items are already available.
    - *Empty location with no incoming work*: the item is placed in the first empty location that's found. A location is considered to be empty if it has no physical inventory and no expected incoming work.

5. Select **Save**.

Edit the query for directive action line

1. Select **Edit query**.
2. Select **Add**.
3. In the **Field** field, type *location profile ID*. In this example, we'll restrict the possible locations using a location profile ID.
4. In the **Criteria** field, type a value.
5. Select **OK**. You can continue to add directive lines and directive actions until you have covered all the possible scenarios in your warehouse.

Last updated on 08/05/2025

Set up a work template for purchase orders

Article • 01/31/2023

This article describes how to set up a simple work template to be used when putting away received items. Work templates determine the set of instructions presented to the warehouse worker on a mobile device when moving items from the receiving area. You can use this procedure with the data mentioned in demo data company USMF. Before you start this guide, create a work pool ID. In this example, a work pool ID called in Inbound is used. This procedure is intended for the warehouse manager.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. In the **Work order type** field, select **Purchase orders**.

Create a work template header

1. Select **New**.
2. In the **Sequence number** field, enter a number. This is the sequence in which the work templates are evaluated. You can modify the sequence, if needed.
3. In the **Work template** field, type a value. This is the unique identifier for this template.
4. In the **Work template description** field, type a value.

    - The **Valid** option will not be checked until you've completed all the information that's needed by the template and have then selected **Save**.
    - A work order of type **Purchase order** cannot be automatically processed, so leave the **Automatically process** option set to **No**.

5. In the **Work pool ID** field, type a value. Work pool IDs allow you to organize work into groups. The value that you set here will be the default value for any work that's created using this template.
6. In the **Work priority** field, enter `1`. This indicates the importance of the work, and the value that you set here will be the default for any work created using this template.
7. Select **Save**. You must save the work template header before the **Edit Query** button becomes available.

Set up the query for the work template

1. Select **Edit query**. We'll set a limitation that the template can only be used within a specific warehouse.
2. Select **Add**.
3. In the **Field** field of the new row, type `warehouse`.
4. In the **Criteria** field, type a value.
5. Select **OK**.
6. Select **Yes**.

Set work template details

1. Select **New**.
2. In the **Work type** field, select **Pick**.
3. In the **Work class ID** field, type `purchase`. The work class that you set here will be the default on all work lines of type Pick that are created using this template. The work class cannot be overridden from the work order lines. Work classes are used to direct and/or limit the type of work order lines a warehouse worker can process on a mobile device.
4. Select **New**.
5. In the **Work type** field, select **Put**.
6. In the **Work class ID** field, type a value. The pick and put instructions are a set. Each pick/put set must have the same work class. Use the same work class that you provided for the pick instruction.
7. Select **Save**. Note that the **Valid** checkbox is now checked.

Feedback

Was this page helpful? Yes No

Provide product feedback

Location product dimension mixing

> **Note**
>
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Location product dimension mixing is location profile functionality that helps improve location management when product variants or products that have dimensions are used, such as in the fashion industry. It lets you decide whether configurations, colors, styles, and sizes can be mixed for a specific location profile, or whether just one of these dimensions or a combination of them can be put to the same location.

Setup

Every location in the warehouse needs to have a location profile associated with it that describes the properties of the location. Therefore, all locations that use the same location profile will be able to allow product dimension mixing after it has been set up.

Configure location profiles to allow product dimension mixing

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. In the list of location profiles, select **BULK**.
3. On the Action Pane, select **Edit**.
4. On the **General** FastTab, set the **Enable location product dimension specific mixing** option to *Yes*.

   > **Note**
   >
   > You can set this option to *Yes* only if the **Allow mixed items** option is set to *No*.

5. On the **Allowed product dimension mixing** FastTab, set the **Size** option to *Yes*. In the scenario that is described in this article, mixing can be done only for products that have different **Size** dimensions. However, other options are also available.
6. Select **Save**.

Create a new product master and product variants

1. Go to **Product information management** > **Products** > **Product masters**.
2. On the Action Pane, select **New** to create a product master.
3. In the **New product** dialog box, set the following values:
    - **Product type:** *Item*
    - **Product subtype:** *Product master*
    - **Product number:** *B0001*
    - **Product name:** *B0001 Size*
    - **Product dimension group:** *Size*
    - **Configuration technology:** *Predefined variant*
4. Select **OK**.
5. On the **Product details** page, on the **General** FastTab, set the following values:
    - **Generate variants automatically:** *Yes*
    - **Size group:** *CASUALDHIR*
6. To view the predefined variants, on the Action Pane, select **Product variants**.

    The **Product variants** page appears and shows a list of variants from the configuration of the size group.

Release products to the USMF company

1. On the Action Pane, select **Release products**.
2. On the **Select products to release** page, confirm that product number *B0001* is in the list, and then select **Next**.
3. Select **Next** to confirm the product variants to release.
4. On the **Select companies to release to** page, select *USMF*, and then select **Next** to confirm the selection.
5. On the **Confirm selection** page, select **Finish** to complete the release.

    You receive an "Operation completed" message.

Update a released product in the USMF company

1. Make sure that you're signed in to the **USMF** company.
2. Go to **Product information management** > **Products** > **Released products** to finish creating the released product.
3. Find and select item number *B0001* to open the **Released product details** page.
4. On the Action Pane, select **Edit**.
5. On the **General** FastTab, make sure that the **Item model group** field is set to *FIFO*.
6. On the Action Pane, on the **Product** tab, in the **Set up** group, select **Dimension groups**.
7. Set the following values:
    - **Storage dimension group:** *Ware*
    - **Tracking dimension group:** *None*
8. Select **OK**.
9. On the Action Pane, on the **Product** tab, in the **Set up** group, select **Reservation hierarchy**.
10. Set the **Reservation hierarchy** field to *Default*, and then select **OK**.
11. On the **General** FastTab, in the **Administration** section, notice that your selections have been updated.
12. On the **Purchase** FastTab, in the **Price** field, enter *10*.
13. On the **Manage costs** FastTab, in the **Item group** field, enter *Audio*.
14. On the **Purchase** FastTab, in the **Price** field, enter *10*.
15. On the **Warehouse** FastTab, in the **Unit sequence group ID** field, enter *ea*.
16. Select **Save**.

Create a location directive

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the left pane, in the **Work order type** field, select *Purchase orders*.
3. In the list, select the location directive that is named *24 PO Direct*.
4. On the **Location Directive Actions** FastTab, select **New** to add a line to the grid.

5. On the new line, set the following values:

    - **Sequence number:** *1*

      The new line should be in front of the previously existing line. To make the change, use the **Move up** and **Move down** buttons on the toolbar, or change the existing line's **Sequence number** value to *2*.

    - **Name:** *Put to BULK Location profile*
    - **Fixed location usage:** *Fixed and non-fixed locations*
    - **Allow negative inventory:** *Clear this check box (=No)*
    - **Batch enabled:** *Clear this check box (=No)*
    - **Strategy:** *None*

6. While the new line is still selected, select **Edit query** above the grid.
7. In the query dialog box, on the **Range** tab, select **Add** to add a line to the grid.
8. On the new line, set the following values:

    - **Table:** *Locations*
    - **Derived table:** *Locations*
    - **Field:** *Location profile ID*
    - **Criteria:** *BULK*

9. Select **OK**.
10. On the **Location directives** page, on the Action Pane, select **Save**.

> ⓘ **Note**
>
> On the **Location Directive Actions** FastTab **Strategy** field, if you use the *Consolidate* location strategy, the setup of the **Allowed product dimension mixing** FastTab on the **Location profiles** will be overridden, and items will be put to the same location even if this behavior isn't allowed by the setup.

Create a mobile device menu item

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to create a menu item to use for sorting.

3. In the header, set the following values:

    - **Menu item name:** *PO line receiving*
    - **Title:** *PO line receiving*
    - **Mode:** *Work*
    - **Use existing work:** *No*

4. On the **General** FastTab, set the following values:

    - **Work creation process:** *Purchase order line receiving and putaway*
    - **Generate license plate:** *Yes*

5. Select **Save**.

Configure the mobile device menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. In the list of menus, select **Inbound**.
3. On the Action Pane, select **Edit**.
4. In the **Available Menus And Menu Items** list, find and select the **PO line receiving** menu item.
5. Select the right arrow button to move the **PO line receiving** menu item to the **Menu Structure** list. In this way, you add your new menu item to the selected menu.
6. Select **Save**.

Scenario

This demo scenario shows how two different product variants can be put to the same location when the location profile doesn't allow for mixed items. In this case, you will use the **Size** variant as the criterion.

Before you begin, make sure that there are empty locations in warehouse *24* that use the *BULK* location profile.

Create a purchase order

You will create a purchase order that has three lines: two lines for the same product number but different **Size** variants, and a third line for a different product that has no variants.

1. Go to **Accounts payable** > **Purchase orders** > **All purchase orders**.

2. On the Action Pane, select **New**.

3. In the **Create purchase order** dialog box, set the following values:

    - **Vendor account:** *1001*
    - **Warehouse:** *24*

4. Select **OK**.

5. The purchase order is created, and a new line is added on the **Purchase order lines** FastTab. Make a note of the purchase order number.

6. On the new line, set the following values:

    - **Item number:** *B0001*
    - **Size** *L*
    - **Quantity:** *2*

7. Select **Add line** above the grid to add a second purchase order line, and set the following values:

    - **Item number:** *B0001*
    - **Size** *XL*
    - **Quantity:** *2*

8. Select **Add line** to add a third purchase order line, and set the following values:

    - **Item number:** *A0001*
    - **Quantity:** *1*

1.Select **Save**.

Receive purchase order lines in the Warehouse Management mobile app

1. Sign in to the Warehouse Management mobile app as a user who is enabled for warehouse *24*.

2. Select the **Inbound** menu.

3. Select **PO Line receiving**.

4. Select the **PONUM** field, and then enter the purchase order number.

5. Confirm your entry by selecting the confirm button (✓) at the bottom of the page.

6. Enter the line number from the purchase order that is being received. Select the **LINENUM** field, and then use the number pad to enter *1*.

7. Confirm your entry.

8. Enter the quantity to receive. Select the plus sign (+) button two times to increase the value in the **Qty** field to *2*.

9. Register your entry by selecting the button (✓) at the bottom of the page, and then confirm your entry by selecting the button (✓) again.

10. View the information on the **Purchase orders: Put** page. This page shows the work that has been created for the put-away (Work 1).

    The location where the received item will be put away, the license plate, the item, the size, and the quantity of the **PO Line receiving** task that you just completed are shown.

11. Confirm the put-away work.

12. Repeat the steps 4 through 11 for the purchase order line 2. However, in step 8, specify a quantity of *1*.

    New put-away work (Work 2) is created for the same location as Work 1.

13. Repeat the steps 4 through 11 again for purchase order line 2. In step 8, specify the remaining quantity of *1*.

    New put-away work (Work 3) is created for the same location as Work 1 and Work 2. This behavior occurs because the *Consolidate* location directive strategy is used, and the **Allowed product dimension mixing** FastTab on the *Bulk* **Location profiles** setup allows the **Size** variant to be mixed on a location.

14. Repeat the steps 4 through 11 for purchase order line 3. In step 8, specify a quantity of *1* for item number *A0001*.

    New put-away work (Work 4) is created for a different location than the location that is used for purchase order lines 1 and 2. This behavior occurs because the location profile doesn't allow for mixed products, but it does allow for mixed dimensions of the same product master.

15. Select the Menu button at the top of the page (sometimes referred to as the hamburger or the hamburger button), and then select **Cancel** to exit **PO Line receiving**.

> **Tip**
>
> You can repeat this scenario, but this time, set **Size** - *No* under the **Allow product dimension mixing** FastTab on the *BULK* **Location profiles**, so that none of the product

dimensions can be mixed. In this case, when you receive the purchase order, each product variant will be put to a new location.

Last updated on 10/03/2025

Putaway clusters

> **Note**
>
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Putaway clusters offer a way to pick multiple license plates at the same time and then take them for putaway in different locations. This process is often referred to as a *milk run*. Putaway clusters can be very useful for retail businesses, where license plates typically aren't full pallets of inventory.

Setup for the example scenario

Cluster profiles

The putaway cluster profile determines where an item goes, based on the location assigned to the item at the time of receipt. If you need different clusters, create different putaway clusters, one for each mobile device menu item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Cluster profiles**.
2. On the Action Pane, select **New**.
3. In the **Header** view, set the following values:
    - **Putaway cluster profile ID:** *Cluster putaway*
    - **Putaway cluster profile ID Name:** *Cluster putaway*
    - **Cluster type:** *Putaway*
    - **Sequence number:** Accept the default value.
4. Select **Save** to make the required fields on the **General** FastTab available.
5. On the **General** FastTab, set the following values:
    - **Cluster assignment timing:** *At receipt*

This field defines whether the putaway cluster is assigned immediately when the inventory is received, or whether it should be sorted later.

- **Cluster assignment rule:** *Manual*

  This field defines whether the system automatically determines the cluster assignment or if the user assigns it manually.

- **Directive code:** Leave this field blank.

- **Putaway cluster locate:** *Receipt*

  The following values are available:
  - **Receipt** – A location is found immediately during receipt.
  - **Cluster close** – A location is found when the cluster is closed.
  - **User directed** – A location is found when the license plate is picked from the cluster for putaway. In this case, no location is specified when the putaway work is created. During the putaway itself, the user must scan the license plate or work ID to initiate the put step. The system then finds the put location again and tells the user where to put the picked quantity.

- **Putaway cluster per user:** *No*

  This field defines whether each cluster is unique per user when the clusters are automatically assigned. It's available only when the **Cluster assignment rule** field is set to *Automatic*.

- **Unit restriction:** Leave this field blank.

  This field defines the unit that must be received for the profile to be valid. If it's left blank, all units are valid.

- **Work unit break:** *Individual*

  This field defines whether all inventory is consolidated (by using the cluster ID and the license plate) onto one license plate when a cluster is closed, and whether it should be put away as a single license plate or separately on the license plates that were received. This field is unavailable when the **Putaway cluster locate** field is set to *Receipt*.

- **Cluster persists as Parent License Plate:** *No*

If this option is set to *Yes*, when the put step is completed, the cluster ID becomes a parent license plate, and all items on the cluster ID link to that parent license plate.

6. On the **Cluster sorting** FastTab, you can define putaway sorting criteria. Select **New** on the toolbar to add a line, and then set the following values:

   - **Sequence number:** Accept the default value.
   - **Field name:** *WMSLocationId*

     This field defines the field that this line uses as a sorting criterion.
   - **Sorting:** *Ascending*

     This field defines whether sorting should be done in ascending or descending order.

7. On the **Cluster work template** FastTab, select **New** on the toolbar to add a line, and then set the following values:

   - **Work order type:** *Purchase orders*
   - **Work template:** *61 PO Direct*

8. On the Action Pane, select **Save**, and then select **Edit query**.

9. In the **Cluster putaway** dialog box, on the **Range** tab, select **Add** to add a second line to the query. Then update the query lines as shown in the following table.

Expand table

Table	Derived table	Field	Criteria
Work	Work	Warehouse	*61*
Work	Work	Work ID	Leave this field blank.

10. Select **OK** to save the query and close the dialog box.

11. On the Action Pane, select **Save**, and close the page.

> **Important**
>
> Fields in the cluster profile that appear dimmed when **Generate cluster ID** is set to *Yes* are unavailable and aren't considered when this feature is used.

Mobile device menu items

Two new mobile device menu items are available for this feature. Use the **Receive and sort cluster** menu item to sort the received inventory into a putaway cluster upon receipt. Use the **Cluster putaway** menu item to put the cluster away after it is assigned.

Receive and sort cluster

Create a new mobile device menu item for receiving inventory and sorting it into a cluster. This menu item creates inbound work after inventory is received, which indicates that the receiving menu item is used for putaway clusters.

> ⓘ **Note**
>
> Use the **Receive and sort cluster** menu item with the following receiving menu items:
>
> - Purchase order line receiving
> - Purchase order item receiving
> - Load item receiving

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New**.
3. In the **Header** view, set the following values:
    - **Menu item name:** *Receive and sort cluster*
    - **Title:** *Receive and sort cluster*
    - **Mode:** *Work*
    - **Use existing work:** *No*
4. On the **General** FastTab, set the following values:
    - **Work creation process:** *Purchase order item receiving*
    - **Generate license plate:** *Yes*
    - **Assign putaway cluster:** *Yes*

        > ⓘ **Note**

> The **Assign putaway cluster** option is available only for the one-step *Work creation process* activity for receiving.

Accept the default values for the remaining fields.

5. On the Action Pane, select **Save**.

Cluster putaway

Create a new mobile device menu item for putting the cluster away after it is assigned.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New**.
3. In the **Header** view, set the following values:
    - **Menu item name:** *Cluster putaway*
    - **Title:** *Cluster putaway*
    - **Mode:** *Work*
    - **Use existing work:** *Yes*
4. On the **General** FastTab, set the **Directed by** field to *Cluster putaway*. Accept the default values for the remaining fields.
5. On the **Work classes** FastTab, set up the valid work class for this mobile device menu item:
    - **Work class ID:** *Purchase*
    - **Work order type:** *Purchase orders*
6. On the Action Pane, select **Save**.

Mobile device menu

Add the menu items that you just created to the inbound menu of the mobile app.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. On the Action Pane, select **Edit**.
3. In the menu list, select **Inbound**.
4. In the **Available menus and menu items** list, find and select **Receive and sort cluster**.

5. Select the right arrow button to move the selected menu item to the **Menu structure** list.
6. Use the up arrow or down arrow button to move the menu item into the desired position in the menu.
7. On the Action Pane, select **Save**.
8. Repeat steps 4 through 7 to add the remaining menu items:
    - Assign cluster
    - Cluster putaway

Example scenario

This scenario simulates putaway cluster processing.

Create a purchase order

1. Go to **Accounts payable** > **Purchase orders** > **All purchase orders**.
2. On the Action Pane, select **New**.
3. In **Create purchase order**, set the following values:
    - **Vendor account:** *1001*
    - **Warehouse:** *61*
4. Select **OK**.

    The **All purchase orders** page appears.
5. On the **All purchase orders** page, on the **Purchase order lines** FastTab, use the **Add line** button to add the following lines:
    - Purchase order line 1:
        - **Item number:** *A0001*
        - **Quantity:** *10*
    - Purchase order line 2:
        - **Item number:** *A0002*
        - **Quantity:** *20*
    - Purchase order line 3:

- **Item number:** *M9215*
- **Quantity:** *30*

6. On the Action Pane, select **Save**.
7. Make a note of the purchase order number.

Receive inventory and put it away from the mobile device

Receive and sort the inventory into a cluster

1. Sign in to the Warehouse Management mobile app as a user who is set up for warehouse *61*.
2. On the main menu, select **Inbound**.
3. On the **Inbound** menu, select **Receive and sort cluster**.
4. In the **Ponum** field, enter the purchase order number.
5. Select **OK** (the check mark button).
6. Select the **Item** field, enter item number *A0001*, and then select **OK**.
7. Select the **Qty** field, enter *10* by using the number pad, and then select the check mark button.
8. On the **Qty** task page, select **OK** (the check mark button) to confirm the quantity that you entered.
9. On the **Item** task page, select **OK** to confirm that item *A0001* was entered.
10. Select the **Cluster ID** field, and enter a value to assign an ID for the cluster that you're creating.

    The ID that you enter here is used when the two remaining items on the purchase order are received.
11. Select **OK**.

    The **Ponum** task page appears and shows a "Work completed" message.

    Item *A0001* is now received into the *RECV* location and assigned to the cluster ID that you entered in step 10.

12. Repeat steps 4 through 11 to receive the remaining two items from the purchase order and assign them to the cluster ID:

    - A quantity of *20* for item *A0002*
    - A quantity of *30* for item *M9215*

Close the cluster

Before you can put away the items in the cluster, ensure that the cluster is closed.

1. In Supply Chain Management, go to **Warehouse management** > **Work** > **Outbound** > **Work clusters**.

2. On the **Work clusters** page, in the **Work cluster** section, search the **Cluster ID** field for the cluster ID that you entered earlier.

3. If the cluster isn't shown, it might already be closed. To determine whether the cluster is closed, select the **Show closed work** check box, and search for the cluster ID that you entered earlier. Then follow one of these steps:

    - If the cluster is closed, skip the remaining steps of this procedure, and move on to the next procedure, Put the cluster away.
    - If the cluster isn't closed, follow the remaining steps of this procedure to manually close the cluster. Then move on to the next procedure.

4. In the **Work cluster** section, select the cluster ID that you entered earlier.

5. On the Action Pane, select **Close cluster**.

    After the cluster is closed, it no longer appears in the **Work cluster** section (unless the **Show closed work** check box is selected).

Put the cluster away

1. Sign in to the Warehouse Management mobile app as a user who is set up for warehouse *61*.

2. On the main menu, select **Inbound**.

3. On the **Inbound** menu, select **Cluster putaway**.

4. Select **Cluster ID**, and enter the cluster ID that you entered earlier for the closed cluster.

5. Select **OK**.

   The **Cluster putaway: Pick** page appears. It shows the cluster ID, the picking location, the items (the value *Multiple* is shown), and the total quantity in the cluster that must be picked.

6. Select **OK**.

   The **Cluster putaway: Put** page appears. The **Put** instructions identify the cluster ID, the put location, the items, the total quantity, and the license plate IDs for the items that are received on the cluster.

   You have the standard options to override or pass this step.

7. Select **OK** to confirm the putaway of the cluster.

   A **Cluster completed** message appears.

Notes and tips

When the cluster ID acts as the parent license plate for a nested pallet, scanning the cluster ID automatically provides the put position. You don't need to scan any other license plate, even if the license plate generation is set to manual.

Last updated on 05/29/2026

Quality management for warehouse processes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The *Quality management for warehouse processes* feature lets you integrate item sampling controls into the warehouse receiving process by using warehouse management processes (WMS). You can automatically generate warehouse work to move inventory to the quality control location, based on a percentage, a fixed quantity, or every *n*th license plate. After you complete a quality order, you can automatically generate work to move inventory to the next location in the process, depending on the quality results.

The *Quality management for warehouse processes* feature extends the capabilities of the basic quality management feature. It provides the option to create quality orders for the inventory that you send to the quality control location, although quality orders aren't always required. Therefore, it allows for a lightweight quality control process that is based on warehouse work.

Key benefits

The *Quality management for warehouse processes* feature automatically generates work as part of the receiving process, to move the inventory quantity that is required for quality control to a quality control location. If the quantity that you receive exceeds the quantity that is required for quality control (according to the item sampling setup), the system moves the excess quantity to an inbound location that is defined in the location directive setup. After the quality order is validated, the system automatically generates work to move the quantity for the quality order to a new inbound or return location, based on the validation result and the location directive setup. The automatic generation of work that has only the quantity that must be moved to and from quality control provides an integrated process experience.

> **Note**

You can still take advantage of the manual process. In the manual process, inventory movement and movement by template are used to have a warehouse worker trigger the creation of warehouse work to move inventory from a quality control location to a new location. You can also still set up an inbound location directive that moves inventory in its entirety from a receiving location to a quality control location without considering the item sampling setup.

Quality management and the Quality management for warehouse processes feature

The following illustration provides an overview of the entities that enable quality orders for warehouse processes.

Enablers: The Quality item sampling and Quality order work order types

The *Quality management for warehouse processes* feature introduces two work order types that enable the work creation process:

- **Quality item sampling** – Use this work order type to create work that moves registered inventory to quality control.
- **Quality order** – Use this work order type to create work that moves inventory from quality control to a new location, based on the location directive setup.

Work classes, location directives, and work templates

The *Quality item sampling* and *Quality order* work order types are consumed by location directives, work classes, and work templates.

Before warehouse work can be automatically generated to move inventory to quality control, you must follow these steps to set up your system.

1. Create separate work classes for the *Quality item sampling* and *Quality order* work order types. In this way, you ensure that appropriate work can be automatically generated based on the two work order types, and that this work can then be run by using the Warehouse Management mobile app.

2. Set up a work template for each work order type:

    - Set up a work template that uses the *Quality item sampling* work order type to automatically move registered inventory to a quality control location.
    - Set up a work template that uses the *Quality order* work order type to move inventory from a quality control location after quality control is completed.

3. For each work order type, set up location directives that apply the correct quality control locations that the inventory should be moved to. After quality control is completed, the location directive for the *Quality order* work order type ensures that a new destination location will be selected so that the inventory can be moved out of the quality control location.

4. Set up the relevant mobile device menu items to support the movement of received inventory to the quality control location, and the movement of inventory that passes or fails quality control from the quality control location to a new location.

For a step-by-step example that shows how to complete this setup, see the example scenario at the end of this article.

Enable a warehouse for quality management

Before the *Quality management for warehouse processes* feature can be applied for a specific warehouse, you must follow these steps to make the feature available for that warehouse.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.
2. Select the warehouse to enable for quality management.
3. On the **Warehouse** FastTab, set the **Enable quality order for warehouse processes** option to *Yes*. (Note that this option can be set to *Yes* only for warehouses that use warehouse management processes (WMS).)

When the **Enable quality order for warehouse processes** option is set to *Yes*, the quality association setup controls whether the *Quality management for warehouse processes* feature is

actually applied for the selected warehouse. You can change the setting of the option to *No* at any time. In that case, the feature will no longer apply for the warehouse, regardless of the quality association setup.

Quality control

The *Quality management for warehouse processes* feature controls several key settings for quality associations and item sampling.

Quality associations

Each quality association record defines the set of tests, the acceptable quality level (AQL), and the sampling plan that apply to the quality orders that are generated. To set up a quality association record, follow these steps:

1. Go to **Inventory management** > **Setup** > **Quality control** > **Quality associations**.
2. Create or select the quality association entry for the item or group that you're working with, or for all items.
3. On the **Conditions** FastTab, set the **Applicable warehouse type** field to one of the following values:
    - **Quality management for warehouse processes only** – Activate the *Quality management for warehouse processes* feature. You can select this value only if the reference type is either *Purchase* or *Production*.
    - **All** – Inactivate the *Quality management for warehouse processes* feature. Select this value for all reference types except *Purchase* and *Production*.

> **Note**
>
> The *Quality management for warehouse processes* feature takes effect only if the item on the source document line uses warehouse management processes (WMS), and if the **Enable quality order for warehouse processes** option is set to *Yes* for the warehouse on the source document line.

As each item is registered (or reported as finished), the system determines which quality associations apply to it.

When the *Quality management for warehouse processes* feature is turned on, the applicable warehouse type is logically inserted into the fourth search group of the quality association search hierarchy. The following table provides a logical representation of the search hierarchy.

Expand table

Search group	Description
Group 1	For each quality association, check the **Reference type**, **Event type**, and **Execution match** values against the item. If there's a match against the source document line, move on to group 2.
Group 2	For each quality association, check the **Item code** value (*Table*, *Group*, or *All*) against the item. *Table* is more specific than *Group*, and *Group* is more specific than *All*. If there's a match for *Table* (a specific item), move on to group 3. If there's no match for *Table*, search for a match for *Group*. If there's no match for *Group*, *All* applies. If there's a match, move on to group 3.
Group 3	For each quality association, check the **Account code** and **Resource code** values against the item. The logic that is applied resembles the logic that is applied for the **Item code** value.
Group 4	For each quality association, check the **Applicable warehouse type** value (*Quality management for warehouse processes only* or *All*) against the item. If the **Enable quality order for warehouse processes** option is set to *Yes* for the warehouse on the source document, and the item on the source document line is set to *Use warehouse management processes*, both associations where there's a match for *Quality management for warehouse processes only* and associations where there's a match for *All* will be applicable in parallel, if both exist. If the **Enable quality order for warehouse processes** option is set to *No* for the warehouse on the source document, and the item on the source document line is set to *Use warehouse management processes*, only quality management will be applicable.

For example, you define a warehouse where the **Enable quality order for warehouse processes** option is set to *Yes*, and you have two quality associations that are defined for the *Purchase* reference type: one for all items and one for the *Registration* event type. The only difference between the two quality associations is the **Applicable warehouse type** value: it's set to *All* for one quality association and *Quality management for warehouse processes only* for the other. In this case, both quality associations are equally specific, and both are applicable.

The value of the **Test group** field for the quality associations is also a factor. This field defines the test procedure that must be applied. If the **Test group** value is the same for both associations, only one quality order is created, for the quality association where the **Applicable warehouse type** value is *Quality management for warehouse processes only*. If the **Test group** value isn't the same for both associations, two quality orders are created. The first quality order is created for the quality association where the **Applicable warehouse type** value is *Quality management for*

warehouse processes only. The second quality order is created for the quality association where the **Applicable warehouse type** value is *All*.

> **Note**
>
> The *Quality management for warehouse processes only* value is considered more specific than *All* when the criteria for the quality associations for groups 1 and 2 are the same, and when the test group is the same. Two quality orders are created only when the test groups differ.

Reference types

When the **Reference type** value is *Purchase*, and the **Applicable warehouse type** value is *Quality management for warehouse processes only*, set the **Event type** field on the **Process** FastTab to *Registration*. *Registration* is the only supported event type for the *Purchase* and *Inbound shipment order* reference types when you're using the *Quality management for warehouse processes* feature.

Quality processing policy

The *Quality management for warehouse processes* feature enables work to be created based only on item sampling. Therefore, it allows for a lightweight process. The inventory that work is created depends on the item sampling that is associated with the quality association. When the lightweight process is used, after a worker puts the quantity in the quality control location, the quality department can manually create a quality order, if a quality order is required.

The **Quality processing policy** field on the **Quality order process** FastTab controls whether a quality order is also created when work is created to move an item to the quality control location. Set this field to *Create quality order* or *Create work only*. The default value is *Create quality order*.

> **Note**
>
> Regardless of whether you create quality orders manually or automatically, the system automatically generates work to move items out of the quality control location when the quality order is marked as validated.

The creation of quality order work is unrelated to the quality association setup. If a work template exists that has a **Work order type** value of *Quality order*, and if the query criteria are met for that

work template, validation of a quality order triggers the creation of quality order work.

Referenced item sampling

Each quality association must reference an item sampling. An item sampling defines the quantity that you send for quality control. You can set it up so that it applies only to quality associations where the **Applicable warehouse type** value is *Quality management for warehouse processes only*. If the **Sampling scope** value for an item sampling is *Load* or *Shipment*, or the **Quantity specification** value is *Full license plate*, the item sampling can be referenced only by quality associations where the **Applicable warehouse type** value is *Quality management for warehouse processes only*.

If you define an item sampling that uses the *Quality management for warehouse processes only* applicable warehouse type, you receive an error if you try to reference it from a quality association that doesn't use the *Quality management for warehouse processes* feature.

> ⓘ **Note**
>
> Item sampling that uses full blocking isn't supported for quality associations where the **Applicable warehouse type** field is set to *Quality management for warehouse processes only*.

Item sampling

Item sampling controls how often you send items for quality control. The *Quality management for warehouse processes* feature introduces the concept of *item sampling scope*. The system uses the item sampling scope when it evaluates whether and how to create quality orders, quality item sampling work, and quality order work.

To set up item sampling, go to **Inventory management** > **Setup** > **Quality control** > **Item sampling**, and set the **Sampling scope** field to one of the following values:

- **Order** – The source document line is the basis for evaluating whether and how to create quality orders, quality item sampling work, and quality order work.
- **Load** – Loads are the basis for evaluating whether and how to create quality orders and work.
- **Shipment** – Shipments are the basis for evaluating whether and how to create quality orders and work.

> ⓘ **Note**

When you set the **Sampling scope** field to *Load* or *Shipment*, the system uses the load entity and shipment entities, if they're available. If they're not available, the system uses the order entity.

The *Quality management for warehouse processes* feature also introduces the *Full license plate* value for the **Quantity specification** field. This value supports the creation of quality order work and quality item sampling work, based on license plates. When you select this value, the following changes occur:

- The **Break count by item** option and the **Per nth license plate** field on the **Process** FastTab become available.
- The **Value** field on the **Sampling quantity** FastTab becomes unavailable.
- The **Per updated quantity**, **Location**, and **License plate** options are all set to *Yes,* and the settings can't be changed.

The **Break count by item** option controls whether the license plate count is evaluated per item or across all items in the sampling scope. Product variants are treated as the same item. This option also controls whether the license plate count is reset for each item.

The value of the **Per nth license plate** field controls how often quality orders are created in relation to the number of items that are registered. For example, a value of *3* sends every third item to quality control, starting with the first item. The value must be more than 0 (zero).

While workers receive items by using the Warehouse Management mobile app, the system validates whether a quality association is set up for each incoming item. If a quality association is set up, the system uses the item sampling record that is configured for that quality association to determine how it creates quality orders, quality item sampling work, and order work.

Note

When you do receipt registration in the web client (by using the small registration page or the item arrival journal), the system doesn't create quality item sampling work or order work, regardless of the setup. Instead, for items that match a quality association, the referenced item sampling controls the creation of quality orders only.

Examples of automatic generation of quality orders

The following examples show how the setup of a quality association and an associated item sampling affects the generation of quality orders when you set the **Applicable warehouse type** field to *Quality management for warehouse processes only*.

When the **Quantity specification** value is *Full license plate*, the **Per nth license plate** field controls which license plates the system creates quality item sampling work for. The first license plate always goes to quality control, and then the value of this field specifies that every *n*th license plate after that license plate should also go.

The **Reference type** value for the following examples is *Purchase*, and the **Event type** value is *Registration*.

Expand table

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
Order	Full license plate	Yes *(locked/not editable)*	Location: Yes License Plate: Yes *(locked/not editable)*	No	3	**Order line quantity: 100 EA** 1. Register receipt in the Warehouse Management mobile app for 20 EA, LP1 Quality item sampling work for 20 EA Quality order 1 for 20 EA 2. Register receipt in the Warehouse Management mobile app for 20 EA, LP2 Purchase order work for 20 EA (putaway) 3. Register receipt in the Warehouse Management mobile app for 20 EA, LP3

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						Purchase order work for 20 EA (putaway) 4. Register receipt in the Warehouse Management mobile app for 20 EA, LP4 Quality item sampling work for 20 EA 5. Register receipt in the Warehouse Management mobile app for 20 EA, LP5 Purchase order work for 20 EA (putaway)
Order	Fixed quantity = 1	Yes	Location: Yes License Plate: Yes	No	Not applicable	**Order line quantity: 100** 1. Register receipt in the Warehouse Management mobile app for 20 EA, LP1 Quality item sampling work for 1 EA Quality order 1 for 1 EA Purchase order work for 19 EA (putaway) 2. Register receipt in the Warehouse Management mobile app for 20 EA, LP2 Quality item sampling work for 1 EA

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						Quality order 1 for 1 EA Purchase order work for 19 (putaway) 3. Register receipt in the Warehouse Management mobile app for 20 EA, LP3 Quality item sampling work for 1 EA Quality order 1 for 1 EA Purchase order work for 19 EA (putaway) 4. Register receipt in the Warehouse Management mobile app for 20 EA, LP4 Quality item sampling work for 1 EA Quality order 1 for 1 EA Purchase order work for 19 EA (putaway) 5. Register receipt in the Warehouse Management mobile app for 20 EA, LP5 Quality item sampling work for 1 EA

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						Quality order 1 for 1 EA Purchase order work for 19 EA (putaway)
Order	Percent = 10	No	Location: No License Plate: No	No	Not applicable	**Order line quantity: 100 EA** 1. Register receipt in the Warehouse Management mobile app for 50 EA, LP1 Quality item sampling work for 10 EA Quality order 1 for 10 EA Purchase order work for 40 EA (putaway) 2. Register receipt in the Warehouse Management mobile app for 50 EA, LP2 Purchase order work for 50 EA (putaway)
Load	Percent = 5	Yes *(locked/not editable)*	Location: No License Plate: No	No	Not applicable	**Order line quantity: 500 EA** **Two loads: first load 200 EA, second load 300 EA** 1. Register receipt in the Warehouse Management mobile app for

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						first load for 100 EA Quality item sampling work for 5 EA Quality order 1 for 5 EA Purchase order work for 95 EA (putaway) 2. Register receipt in the Warehouse Management mobile app for first load for 100 EA Quality item sampling work for 5 EA Quality order 1 for 5 EA Purchase order work for 95 EA (putaway) 3. Register receipt in the Warehouse Management mobile app for second load for 300 EA Quality item sampling work for 15 EA Quality order 1 for 15 EA Purchase order work for 285 EA (putaway)

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
Order	Percent = 10	Yes	Location: Yes License Plate: Yes	No	Not applicable	**Order line quantity: 100** 1. Register receipt in the Warehouse Management mobile app for 50 EA, LP1 Quality item sampling work for 5 EA Quality order 1 for 5 EA Purchase order work for 45 EA (putaway) 2. Register receipt in the Warehouse Management mobile app for 50 EA, LP2 Quality item sampling work for 5 EA Quality order 1 for 5 EA Purchase order work for 45 (putaway)
Load	Full license plate	Yes *(locked/not editable)*	Location: Yes License Plate: Yes *(locked/not editable)*	No	3	**Two items:** • **Order line quantity for item A: 120 EA (4 pallets)** • **Order line quantity for item B: 90 EA (3 pallets)**

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						One load, two load lines with each order line 1. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP1 Quality item sampling work for 30 EA Quality order 1 for 30 EA 2. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP2 Purchase order work for 30 EA (putaway) 3. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP3 Purchase order work for 30 EA (putaway) 4. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP4 Quality item sampling work for 30 EA Quality order 1 for 30 EA 5. Register receipt in the Warehouse Management

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						mobile app for item B, 30 EA, LP5 Purchase order work for 30 EA (putaway) 6. Register receipt in the Warehouse Management mobile app for item B, 30 EA, LP6 Purchase order work for 30 EA (putaway) 7. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP7 Quality item sampling work for 30 EA Quality order 1 for 30 EA
Load	Full license plate	Yes *(locked/not editable)*	Location: Yes License Plate: Yes *(locked/not editable)*	Yes	3	**Two items:** • **Order line quantity for item A: 120 EA (4 pallets)** • **Order line quantity for item B: 90 EA (3 pallets)** **One load, two load lines with each order line** 1. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP1

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						Quality item sampling work for 30 EA Quality order 1 for 30 EA 2. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP2 Purchase order work for 30 EA (putaway) 3. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP3 Purchase order work for 30 EA (putaway) 4. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP4 Quality item sampling work for 30 EA Quality order 1 for 30 EA 5. Register receipt in the Warehouse Management mobile app for item B, 30 EA, LP5 Quality item sampling work for 30 EA Quality order 1 for 30 EA

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						6. Register receipt in the Warehouse Management mobile app for item B, 30 EA, LP6 Purchase order work for 30 EA (putaway) 7. Register receipt in the Warehouse Management mobile app for item A, 30 EA, LP7 Purchase order work for 30 EA (putaway)
Load	Percent = 10	Yes *(locked/not editable)*	Location: No License Plate: No	No	Not applicable	**Order line quantity: 100 EA** **No loads are created. Order scope is applied.** 1. Register receipt in the Warehouse Management mobile app for 50 EA, LP1 Quality item sampling work for 5 EA Quality order 1 for 5 EA Purchase order work for 45 EA (putaway) 2. Register receipt in the Warehouse Management mobile app for 50 EA, LP2 Quality item sampling work for

Sampling scope	Quantity specification	Per updated quantity	Per storage dimension	Break count by item	Per nth license plate	Result
						5 EA Quality order 1 for 5 EA Purchase order work for 45 EA (putaway)

When a worker validates one of the quality orders that are shown in the preceding table, the system automatically generates quality order work to move inventory from the quality control location to the location that is defined in the location directive for the *Quality order* work order type. You can set up any location for this purpose, such as a return or storage location, depending on the test result for the quality order. For an example of this setup, see the example scenario at the end of this article.

You can reopen a quality order that you already validated, as long as the quality order work that relates to moving the inventory from the quality control location doesn't have a **Work status** value of *Closed* or *In progress*.

Process insights when multiple quality associations coexist

You can define and apply more than one quality association to the same source document line. You can set the **Applicable warehouse type** field to *Quality management for warehouse processes only* for some of those quality associations and *All* for others.

In the following example, the **Reference type** value is *Purchase*.

1. The first quality association is set up in the following way:

    - **Applicable warehouse type:** *Quality management for warehouse processes only*
    - **Item code:** *A0001*
    - **Account code:** *All*
    - **Test group:** *Enclosure*
    - **Item sampling:** *5 pcs*

2. The second quality association is set up in the following way:

    - **Applicable warehouse type:** *All*
    - **Item code:** *All*
    - **Account code:** *All*
    - **Test group:** *Enclosure*
    - **Item sampling:** *1 pcs*

3. The third quality association is set up in the following way:

    - **Applicable warehouse type:** *Quality management for warehouse processes only*
    - **Item code:** *All*
    - **Account code:** *104*
    - **Test group:** *Impedance*
    - **Item sampling:** *Every second license plate* (This setting means that the first, third, fifth, and so on, license plates that are received will create a quality order.)

4. The fourth quality association is set up in the following way:

    - **Applicable warehouse type:** *All*
    - **Item code:** *All*
    - **Account code:** *All*
    - **Test group:** *Impedance*
    - **Item sampling:** *5 pcs*

5. The fifth quality association is set up in the following way:

    - **Applicable warehouse type:** *All*
    - **Item code:** *All*
    - **Account code:** *All*
    - **Test group:** *Cone*
    - **Item sampling:** *10%*

You now create a purchase order for a quantity of 10 of item A0001 for vendor 104. Then you register a purchase order line that has a quantity of 10 as received on one license plate by using the Warehouse Management mobile app. Here's the result:

- There's one quality order from the first quality association for the *Enclosure* test group. The quantity is 5. There's no quality order from the second quality association, because the criteria for the first quality association are more specific relative to the *Enclosure* test group.

- There's one quality order for the third quality association for the *Impedance* test group. The quantity is 10. There's no quality order from the fourth quality association, because the criteria for the first quality association are more specific relative to the *Impedance* test group.
- There's one quality order for the fifth quality association for the *Cone* test group. The quantity is 1.

In connection with the creation of one quality order for each of the three quality associations, quality item sampling work is also created. The registered quantity is only 10. However, because of the item sampling setup, the sum of the quality order quantities that are created for the *Quality management for warehouse processes only* applicable warehouse type is 16, which exceeds the physical registered quantity of 10. Therefore, work won't be created for the full quality order quantities (16), because only 10 are physically available for movement to the quality control location. The priority that is used to create quality item sampling work follows the order of quality order creation:

- **First quality order (quantity = 5):** Quality item sampling work is created for 5. A quantity of 5 (10 – 5) now remains for subsequent creation of quality item sampling work.
- **Second quality order (quantity = 10):** Quality item sampling work is created for 5. A quantity of 0 (zero) now remains for subsequent creation of quality item sampling work.
- **Third quality order (quantity = 1):** No quality item sampling work is created.

As part of the process of creating the quality orders, an inventory blocking of a quantity of 10 is created. This inventory blocking is referenced against each of the three quality orders. The sum of the quality order quantities is 16.

When the quality orders are validated, the system tries to create quality order work for each quality order that is validated. Because the sum of the quality order quantities exceeds the quantity that is actually blocked and therefore available for work creation, quality order work can't be created for the full quality order quantities, as shown here. (This example continues the previous example.)

1. **Validate the second quality order that you created (quantity = 10). The system creates quality order work for a quantity of 4.**

   A change in the inventory blocking quantity triggers the creation of quality order work. Because the sum of quality order quantities was 16, validation of a quantity of 10 causes the remaining quality order quantities to be 6. The inventory blocking quantity is reduced from 10 to 6. The reduced quantity of 4 is allotted to quality order work creation.

2. **Validate the first quality order that you created (quantity = 5). The system creates quality order work for a quantity of 5.**

   A change in the inventory blocking quantity triggers the creation of quality order work. Because the sum of quality order quantities was 6, validation of a quantity of 5 causes the remaining quality order quantities to be 1. The inventory blocking quantity is reduced from 6 to 1. The reduced quantity of 5 is allotted to quality order work creation.

3. **Validate the third quality order that you created (quantity = 1). The system creates quality order work for a quantity of 1.**

   A change in the inventory blocking quantity triggers the creation of quality order work. Because the sum of quality order quantities was 1, validation of a quantity of 1 causes the remaining quality order quantities to be 0 (zero). The inventory blocking is removed (that is, the inventory blocking quantity is reduced from 1 to 0). The reduced quantity of 1 is allotted to quality order work creation.

> ⓘ **Note**
>
> The creation of quality order work depends on the inventory blocking quantity that is referenced against one or more quality orders. If the sum of the quality order quantities exceeds the referenced inventory blocking quantity, the order that you validate the quality orders in determines the creation of quality order work.

Canceling quality item sampling work

You can cancel the work that is created for quality item sampling. To control what occurs when you cancel this work, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **Work** FastTab, set the **Unregister receipt when cancelling work** option to one of the following values:

   - **Yes** – When you cancel quality item sampling work, the associated quality order is deleted, and the inventory is unregistered.
   - **No** – When you cancel quality item sampling work, the associated quality order isn't deleted, and the inventory isn't unregistered.

Cross-docking

You can set up a quality association that creates item sampling work. However, when cross-docking exists in parallel with a quality association that creates quality item sampling work, if there's only enough quantity to satisfy cross-docking, the system creates only cross-docking work. In cases where the **Enable quality order for warehouse processes** option is set to *Yes* for the receiving warehouse, and the **Applicable warehouse type** field is set to *Quality management for warehouse processes only* for a quality association, the creation of quality item sampling work takes precedence over the creation of cross-docking work. If the quantity exceeds the requirement for cross-docking, the system still creates only item sampling work.

Destructive testing

You can define a test group that performs destructive testing. In the case of a destructive test, the assumption is that, regardless of the test result, the quantity of the item that is tested is destroyed as part of the test. The way that the *Quality Management for warehouse processes* feature supports destructive testing resembles the way that quality management supports it when the feature isn't turned on. Before the quality order can be validated, the quality controller must specify the pick location for the quantity that is destroyed. You can register picking from the quality order page by selecting **Inventory** > **Pick** on the Action Pane. After the pick for the quality order quantity is registered, validation can be completed.

Example scenario

Prepare the scenario

To work through this scenario, prepare your system as follows:

- Make sure that demo data is installed on the system, and select the *USMF* legal entity.
- Configure warehouse 51 to use the *Quality management for warehouse processes* feature by following these steps:

    1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.
    2. Select warehouse 51.
    3. On the **Warehouse** FastTab, set the **Enable quality order for warehouse processes** option to *Yes*.

Quality-in setup – Move to the quality control location

You must now prepare a basic setup that enables your system to support the *Quality Management for warehouse processes* feature for warehouse 51. (The demo data defines a quality

management location named *QMS*. This location is referenced several times in this scenario.) You'll prepare the following elements, as described in the subsections of this section:

- Work class
- Work template
- Location directive
- Item sampling
- Quality association
- Mobile device menu items

Work class for quality-in

1. Go to **Warehouse management** > **Setup** > **Work** > **Work classes**.
2. Create a work class, and set the following values:
    - **Work class ID:** *QualityIn*
    - **Description:** *Quality item sampling*
    - **Work order type:** *Quality item sampling*

Work template

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. Set the **Work order type** field to *Quality item sampling*.
3. Create a work template, and set the following values:
    - **Work template:** *51 Quality*
    - **Work template description:** *51 Quality*
4. Add a line to the work template, and set the following values:
    - **Work type:** *Pick*
    - **Work class ID:** *QualityIn*
5. Add a second line to the work template, and set the following values:
    - **Work type:** *Put*
    - **Work class ID:** *QualityIn*

Location directive

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. Set the **Work order type** field to *Quality item sampling*.
3. Create a location directive, and set the following values:
    - **Name:** *51 to quality*
    - **Work type:** *Put*
    - **Site:** 5
    - **Warehouse:** *51*
4. Add a line for the location directive, and set the following values:
    - **From quantity:** *1*
    - **To quantity:** *1000000*
5. Create a location directive action, and set the following value:
    - **Name:** *Quality*
6. For the new location directive action, select **Edit query**, and specify a **Range** record that has the following values:
    - **Table:** *Locations*
    - **Field:** *Location profile ID*
    - **Criteria:** *QMS*
7. Select **OK** to save the query, and save the new location directive.

Next, you must change the sequence of the existing purchase order location directives for warehouse 51. The demo data includes two location directives that have a **Work order type** value of *Purchase*: one is named *51 QMS*, and the other is named *51 PO Direct*. To ensure that the *Quality management for warehouse processes* feature is applied for warehouse 51, you must make sure that the *51 QMS* location directive isn't applied. However, instead of deleting that location directive (because you might want to use it in the future), you can just change the sequence.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. Set the **Work order type** field to *Purchase order*.
3. In the sequence list, select sequence number 5, for the *51 PO Direct* location directive.
4. Move the selected sequence up to sequence number 4.
5. Verify that sequence number of the *51 QMS* location directive is now at least 5.

Item sampling

The *Quality management for warehouse processes* feature adds some new item sampling capabilities. The **Sampling scope** value can now be *Order*, *Shipment*, or *Load*, and the **Sampling quantity** value can now be *Full license plate*.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Item sampling**.
2. Create an item sampling record, and set the following values:
    - **Item sampling:** *3rd LP*
    - **Description:** *Every third license plate*
    - **Sampling Scope:** *Order*
3. On the **Sampling quantity** FastTab, set the **Quantity specification** field to *Full license plate*.
4. On the **Process** FastTab, set the **Per nth license plate** field to *3*.
5. In the **Per storage dimension** section, enable both **Warehouse** and **Inventory status**.

Quality associations

Create a quality association that uses the new item sampling.

1. Go to **Inventory management** > **Setup** > **Quality control** > **Quality associations**.
2. Create a quality association record, and set the following values:
    - **Reference type:** *Purchase*
    - **Item code:** *Table*
    - **Item:** *M9201*
    - **Site:** *5*
3. On the **Process** FastTab, set the **Event type** field to *Registration*.
4. On the **Conditions** FastTab, set the **Applicable warehouse type** field to *Quality management for warehouse processes only*.
5. On the **Quality order process** FastTab, set the **Quality processing policy** field to *Create quality order*.
6. On the **Specifications** FastTab, right-click in the **Test Group** field, and then select **View details** to open the **Test groups** page.

7. On the **Test groups** page, on the **Overview** tab of the upper grid, create a test group, and set the following values:

    - **Test Group:** *QMS*
    - **Description:** *QMS test*
    - **Acceptable quantity:** *100*
    - **Item Sampling:** *3rd LP* (Select)

8. On the **Overview** tab of the lower grid, add a record for one test, and set the following values:

    - **Sequence:** *1*
    - **Test:** *Enclosure measuring*

9. On the **Test** tab of the lower grid, set the following values:

    - **Test variables:** *Pass/Fail*
    - **Default outcome:** *Pass*

10. Save the new test group, and close the **Test groups** page.

11. Back on the **Quality associations** page, in the **Test group** field, select **QMS**.

12. Save the record.

Mobile device menu items for quality-in

To complete the setup to move goods to the quality control location, you must make the quality item sampling work available from a mobile device menu item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select the **Purchase Putaway** mobile device menu item.
3. On the **Work classes** FastTab, add the *QualityIn* work class ID.

Summary: Your setup to move goods to quality control

You defined a quality association that uses the *Quality management for warehouse processes* feature to trigger the creation of a quality order. You set up the work and location data for warehouse 51 to ensure that specific work is created when purchase registration is done for item M9201. This setup ensures that every third license plate that you register is moved to a quality location (*QMS*), and that a quality order is created for the license plate quantity. Everything else is moved to putaway instead of the quality control location.

Process quality management work

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Create a purchase order, and set the following values:
    - **Specify Vendor account:** *104*
    - **Warehouse:** *51*
3. Add a purchase order line, and set the following values:
    - **Item:** *M9201*
    - **Quantity:** *20*
    - **UoM:** *ea*
    - **Warehouse:** *51*
4. Write down the purchase order number, so that you can use it later.
5. Go to a mobile device or emulator that is running the Warehouse Management mobile app, and sign in to warehouse 51 by using *51* as the user ID and *1* as the password.
6. Go to **Inbound** > **Purchase Receive**, and enter the following values:
    - **PONum:** The number of the purchase order that you just created
    - **Qty:** *5*
    - **Unit:** *ea*
7. Continue to receive against the line, *5 ea* at a time, until the line is fully received. (A total of four license plates are created.)
8. Sign out of the Warehouse Management mobile app.
9. Back in the web client, go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
10. Find and open your purchase order.
11. In the **Purchase order lines** section, select the line for item number *M9201*, and then select **Purchase order lines** > **Work details**.
12. Notice that the second and third work headers that were created are regular putaway work, whereas the first and the fourth work headers are quality item sampling work. This result is

consistent with the item sampling setup, which is configured to sample every third license plate.

Move to the quality control location

You'll now move the license plates to their designated locations. The first and fourth license plates will go to the quality control location, whereas the second and third license plates will go directly to storage.

1. Go to a mobile device or emulator that is running the Warehouse Management mobile app, and sign in to warehouse 51 by using *51* as the user ID and *1* as the password.
2. Go to **Inbound** > **Purchase put away**, and put away each license plate from the previous procedure until you close all the work.

Summary: Process quality management work

You've now run the quality item sampling work for the first and fourth license plates by moving them to the quality control location. You've also put away the second and third license plates. The next step is to do the quality order testing and control.

Quality-out setup: Move from the quality control location to storage or return

When workers report quality order results, the system automatically generates work.

You'll now continue with the required base setup of the work class, work template, and location directive to enable quality management for warehouse processes. This setup ensures that the required work is created to move the quality order quantity from the quality control location to a designated warehouse location.

Work class for quality-out

1. Go to **Warehouse management** > **Setup** > **Work** > **Work classes**.
2. Create a work class, and set the following values:
    - **Work class ID:** *QualityOut*
    - **Description:** *Quality Out*
    - **Work order type:** *Quality Order*

Work templates

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.

2. Change the **Work order type** value to *Quality order*.

3. Create a work template, and set the following values:

    - **Work template:** *51 quality out*
    - **Work template description:** *51 quality out*

4. Add a line, and set the following values:

    - **Work type:** *Pick*
    - **Work class ID: QualityOut**

5. Add a second line, and set the following values:

    - **Work type:** *Put*
    - **Work class ID:** *QualityOut*

Location directives

1. Go to **Warehouse management** > **Setup** > **Location directives**.

2. Change the **Work order type** value to *Quality order*.

3. Create a location directive, and set the following values:

    - **Name:** *51 Pass*
    - **Work type:** *Put*
    - **Site:** *5*
    - **Warehouse:** *51*

4. On the Action Pane, select **Edit query** to open the query editor dialog box.

5. On the **Range** tab, set the following values:

    - **Table:** *Quality orders*
    - **Field:** *Status*
    - **Criteria:** *Pass*

6. Select **OK** to save the query and close the dialog box.

7. On the **Lines** FastTab, add a line, and set the following values:

- **From quantity:** *1*
- **To quantity:** *1000000*

8. On the **Location directive actions** FastTab, add a row, and set the following value:

    - **Name:** *Pass*

9. On the **Location directive actions** FastTab, select **Edit query** to open the query editor dialog box.

10. On the **Range** tab, set the following values:

    - **Table:** *Locations*
    - **Field:** *Zone ID*
    - **Criteria:** *Bulk*

11. Select **OK** to save the query and close the dialog box.

12. On the Action Pane, select **Save** to save the new location directive.

13. Create a second location directive, and set the following values:

    - **Name:** *51 Fail*
    - **Work type:** *Put*
    - **Site:** *5*
    - **Warehouse:** *51*

14. On the Action Pane, select **Edit query** to open the query editor dialog box.

15. On the **Range** tab, set the following values:

    - **Table:** *Quality orders*
    - **Field:** *Status*
    - **Criteria:** *Fail*

16. Select **OK** to save the query and close the dialog box.

17. On the **Lines** FastTab, add a line, and set the following values:

    - **From quantity:** *1*
    - **To quantity:** *1000000*

18. On the **Location directive actions** FastTab, add a row, and set the following value:

- **Name:** *Fail*

19. On the **Location directive actions** FastTab, select **Edit query** to open the query editor dialog box.

20. On the **Range** tab, set the following values:

    - **Table:** *Locations*
    - **Field:** *Zone ID*
    - **Criteria:** *Return*

21. Select **OK** to save the query and close the dialog box.

22. On the Action Pane, select **Save** to save the new location directive.

Mobile device menu items for quality-out

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select the **QMS putaway** mobile device menu item.
3. On the **Work classes** FastTab, add the *QualityPut* work class ID.

Warehouse workers can now pick quality order work by using the **QMS Putaway** menu item. Goods that fail quality control can be put in a return location, and goods that pass can be put in the bulk-001 location.

Summary: Your setup to move goods from quality control

You've set up the work and location data for warehouse 51, to ensure that work is automatically created when quality orders are completed. This setup ensures that each quality-controlled license plate is moved to either a bulk location or a return location.

Process quality management work

1. Go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.

2. Select the first quality order for the quantities that you registered.

3. Select **Validate**. The status of the test updates to *Fail*.

4. Go to **Warehouse management** > **All work**.

5. Open the work that you just created, and notice that the **Work order type** value is *Quality order*. The work includes a line where the put location is *Return* and the status is *Fail*. (If the

status of the quality order were *Pass*, the put location would be *Bulk* instead.)

6. Go back to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.
7. Select the second quality order for the items that you registered.
8. Select **Results** above the lower grid. Update the **Result quantity** value to *5*, and verify that the **Test result** value changes to a check mark.
9. Select **Validate**, and close the page.
10. Back on the **Quality orders** page, select **Validate**, and do the validation. The status updates to *Pass*.

    > ⓘ **Note**
    >
    > The validation event triggers the creation of the quality order work to move the quantity from the quality control location to a new location.

11. Go to **Warehouse management** > **All work**.
12. Select the work that you just created, and notice that a second quality order work header is created, where the put location is *BULK-001*.
13. Go to a mobile device or emulator that is running the Warehouse Management mobile app, and sign in to warehouse 51 by using *51* as the user ID and *1* as the password.
14. Go to **Quality** > **Put Away from QMS**, and process each of the two license plates that are related to both pieces of work, so that all work is closed.

> ⓘ **Note**
>
> Consider adding the quality-out entry to a mobile device menu item where the activity code is *Display open work list*. For an example, see the mobile device menu item that is named **Work list** in the demo data. First add the *Quality order* work class to a user-directed menu item, because this work class is required for work to be shown in the work list. Then add the *Quality order* work class to the **Work list** menu item. Users who have access to the work list can then pick and process the work that is automatically generated by quality order validation.

Related information

- Quality and nonconformance management overview

Last updated on 05/27/2026

Quality check

Article • 04/08/2025

The *Quality check* feature lets warehouse workers do quick spot checks for quality while they receive items to the inbound dock area. These spot checks are beneficial when workers inspect packaging or other easily recognizable parts of an item. The feature guides workers to take a quick look to see whether anything is obviously faulty or damaged before they stock the inventory in its putaway location.

This feature is an alternative to the standard quality check process. It offers more flexibility and faster processing.

When you use this feature, the arrival and quality check occur in the following way:

1. When a shipment arrives, a warehouse worker registers the arrival. The worker also scans a license plate to register the location.

2. The worker does a quick quality check and records the result (pass or fail) for that license plate.

3. One of the following actions occurs:

    - If the quality check is passed, the license plate is accepted and guided to a receiving location, as usual.
    - If the quality check is failed, the license plate is rejected, and existing putaway work for it is redirected to an alternate location for further inspection. A new quality order is created. To view the quality order that is created from the failed quality check, go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.

This process can also be set up so that all scanned license plates are immediately diverted to the quality check location.

> ⓘ **Note**
>
> For license plate, serial number, or any other dimension to be automatically populated on the **Inventory dimensions** tab of a quality order, they must be turned on for the relevant record on the **Item sampling** page **Process** FastTab (**Inventory management** > **Setup** > **Quality control** > **Item sampling**). Learn more in Quality management item sampling.

Turn the Quality check feature on or off

To use the functionality described in this article, the *Quality check* feature must be turned on for your system. As of Supply Chain Management 10.0.25, this feature is mandatory and can't be turned off. If you're running a version older than 10.0.25, then admins can turn this functionality on or off by searching for the *Quality check* feature in the Feature management workspace.

Set up the feature for the example scenario

This section provides guidelines and an example that shows how to set up the *Quality check* feature and prepare sample data for the example scenario that is provided later in this article.

Make sample data available

To work through the example scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

Quality check template

The quality check template defines the rules for doing quick spot checks for quality at the time of receiving.

1. Go to **Warehouse management** > **Setup** > **Work** > **Quality check template**.
2. Select **New** to add a template to the grid.
3. Set the following values to define the new template:
    - **Quality check template name:** *Dock check*

      Enter a name that identifies the policies applied for this template.
    - **Acceptance policy:** *Prompt user*

      Specify whether users should be prompted to accept or reject the quality of the inventory while they process the work, or whether the quality should automatically be rejected. The available options are *Auto reject* and *Prompt user*.
    - **Quality processing policy:** *Create quality order*

      Select the policy that should be used when the quality of the inventory is rejected. The following options are available:
        - *Create work only* – Just create work to facilitate inventory movement.

    - *Create quality order* – Create a quality order to facilitate quality tests.

- **Test group:** *Enclosure*

  Specify the test group that should be used in the quality order that is created. Test groups are set up in the **Inventory management** module.

  Leave the **Destructive text** option turned off for the test group. This option defines whether the sample will be destroyed as part of the tests in the test group. If a destructive test is used, the system generates an inventory transaction when a quality order is created for an item. The new inventory transaction predicts the inventory reduction for the test quantity. The inventory reduction occurs when the quality order is completed through the validation step. The inventory transaction is identified as a quality order.

Work class – Quality check

Work classes are used to direct and/or limit the type of work order lines that warehouse workers can process on a mobile device.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work classes**.
2. Select **New** to create a work class.
3. In the header, set the following values:

   - **Work class ID:** *QC Check*

     Enter a name that identifies the work class.

   - **Description:** *QC Check*

     Enter a short description that indicates what the work class is used for.

   - **Work order type:** *Quality in quality check*

     Select the type of work order that is created by the work class. When you set up quality control work, always select *Quality in quality check*.

4. On the **Valid put location types** FastTab, leave the **Location type** field blank.

   If you select a location type, you limit the locations where items can be put after they're picked. This field is used when a location directive tries to resolve the location, or when a warehouse worker manually specifies the location for the mobile device menu item.

For more information about work classes and how to create them, see Create a work class.

Work template

Work templates let you define the work operations that must be performed in the warehouse. Typically, warehouse work operations consist of a pair of actions: a warehouse worker picks on-hand inventory up in one location and then puts the picked inventory down in another location. A work template for quality control defines the work operations for doing quality checks.

Purchase orders

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. In the header, set the **Work order type** field to *Purchase orders*.
3. On the Action Pane, select **Edit**.
4. Select a work template that should include a quality check step. In the **Overview** section, in the **Work template** field, select *51 PO Receipt*.
5. In the **Work template details** section, notice that the grid has two existing lines: one for *Pick* and one for *Put*.
6. In the **Work template details** section, select **New** to add a row for quality control to the grid. Notice that the **Line number** field for the new line is set to *3*.
7. On the new line, set the following values. Accept the default values for the remaining fields.

    - **Work type:** *Quality check*
    - **Work class ID:** *Purchase*
    - **Quality check template name:** *Dock check*

      Select the unique ID for the work class. You use this value to configure the menu items on the mobile device and the types of work that those menu items can process.

8. On the Action Pane, select **Save** to save your work so far.

   You receive an informational message that states, "Invalid - Quality check must come directly after a pick." Therefore, you must change the **Line number** value for the line that you just added.

9. Follow these steps to change the **Line number** value for the new line:

a. In the **Work template details** section, select the line where the **Work type** field is set to *Quality check*.
b. Select the **Move up** or **Move down** button to move the *Quality check* line so that it's after the *Pick* line.

10. On the Action Pane, select **Save**.

Quality in quality check

Next, create a work template for the quality check.

1. In the header of the **Work templates** page, change the value of the **Work order type** field to *Quality in quality check*.
2. On the Action Pane, select **New** to add a row to the grid in the **Overview** section.
3. In the new row, set the following values:
    - **Work template:** *51 Quality Check*

      Enter a name for the template.
    - **Work template description:** *51 Quality Check*
4. On the Action Pane, select **Save** to make the **Work template details** section available.
5. While the new template is still selected in the **Overview** section, select **New** in the **Work template details** section to add a row to the grid there.
6. In the new row, set the following values:
    - **Work type:** *Pick*
    - **Work class ID:** *QC Check*

      Select the name of the work class that you created earlier for quality control work.
7. In the **Work template details** section, select **New** again to add another row.
8. In the new row, set the following values:
    - **Work type:** *Put*
    - **Work class ID:** *QC Check*

      Select the name of the work class that you created earlier for quality control work.
9. On the Action Pane, select **Save**.

For more information about work templates, see Control warehouse work by using work templates and location directives

Location directive – Quality failures

Location directives are rules that help identify pick and put locations for inventory movement. For example, in a sales order transaction, a location directive determines where the items will be picked and where the picked items will be put. You must configure a location directive rule to define how failed quality checks will be handled.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the left pane, set the **Work order type** field to *Purchase orders* to work with location directives of that type.
3. On the Action Pane, select **New** to create a location directive for quality checks.
4. In the header, set the following values:
    - **Sequence number:** Accept the default value.
    - **Name:** *51 To Quality*
5. On the **Location directives** FastTab, set the following values. Accept the default values for the remaining fields.
    - **Work type:** *Put*
    - **Site:** *5*
    - **Warehouse:** *51*
6. On the Action Pane, select **Save** to save your directive and make the **Lines** FastTab available.
7. On the **Lines** FastTab, select **New** to add a line to the grid.
8. On the new line, set the following values. Accept the default values for the remaining fields.
    - **From quantity:** *1*
    - **To quantity:** *1000000*
9. On the Action Pane, select **Save** to save the new line and make the **Location directive actions** FastTab available.
10. While the new line is still selected on the **Lines** FastTab, select **New** on the **Location directive actions** FastTab to add a row to the grid there, so that you can set up an action

for the line.

11. In the new row, set the **Name** field to *Quality*. Accept the default values for the remaining fields.

12. On the Action Pane, select **Save** to make the **Edit query** button on the **Location directive actions** FastTab available.

13. While the line that you just added is still selected on the **Location directive actions** FastTab, select **Edit query** to open a dialog box where you can edit the query for the action.

14. On the **Range** tab, select **Add** to add a row to the query.

15. In the new row, set the following values:

    - **Table:** *Locations*
    - **Derived table:** *Locations*
    - **Field:** *Location*
    - **Criteria:** *QMS*

    The *QMS* location is a warehouse location for quality.

16. Select **OK** to close the dialog box.

17. You must now change the sequence of purchase order location directives for warehouse *51*. Save the new *51 To Quality* location directive, refresh the page, and select the location directive in the list. Then use the **Move up** and **Move down** buttons on the Action Pane to put the location directive for warehouse *51* in the following order. (Before you select **Move up** or **Move down**, you must select a location directive in the list.)
    a. 51 To Quality
    b. 51 PO Direct
    c. 51 QMS

Mobile device menu items

Configure a menu item so that mobile devices can perform the **Quality Check** function.

Purchase putaway

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. In the list, select the **Purchase put-away** menu item.

3. On the Action Pane, select **Edit**.

4. In the **Work classes** section, select **New** to add a row to the grid.
5. In the new row, set the following values:
    - **Work class ID:** *QC Check*

      Enter the name of the work class that you created earlier for quality control work.
    - **Work order type:** *Quality in quality check*
6. On the Action Pane, select **Save**.

Purchase order line receiving

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New**.
3. In the header, set the following values:
    - **Menu item name:** *PO line receiving*
    - **Title:** *PO line receiving*
    - **Mode:** *Work*
    - **Use existing work:** *No*
4. On the **General** FastTab, set the following values. Accept the default values for the remaining fields.
    - **Work creation process:** *Purchase order line receiving and put away*
    - **Generate license plate:** *Yes*
    - **Work template:** *51 PO Receipt*
5. On the Action Pane, select **Save**.

Add the menu item to a mobile device menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. In the left pane, select the **Inbound** menu.
3. On the Action Pane, select **Edit**.
4. In the **Available menus and menu items** column, select the new **PO line receiving** menu item.
5. Select the right arrow button to move **PO line receiving** to the **Menu structure** column.
6. In the **Menu structure** column, select **PO line receiving**, and then select the up arrow or down arrow button to move the menu item to the desired position on the mobile device menu.

7. On the Action Pane, select **Save**.

Example scenario

After you've made all the previously described sample data available and set it up, you can work through this scenario to try out the *Quality check* feature. The values that are shown in this scenario assume that you're working with the standard demo data, that you selected the **USMF** legal entity, and that you prepared the sample records that are described earlier in this article. This scenario also serves as an example that shows how the feature can be used in a production setting.

Create a purchase order

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. On the Action Pane, select **New**.
3. In the **Create purchase order** dialog box, set the following values:
    - **Vendor account:** *104*
    - **Warehouse:** *51*
4. Select **OK** to close the dialog box and open the new purchase order.
5. On the **Purchase order lines** FastTab, the grid contains a new, blank line. On this line, set the following values:
    - **Item number:** *M9203*
    - **Quantity:** *3*
    - **Unit:** *PL*
6. On the Action Pane, select **Save**.

Process quality check receiving

After the purchase order has been created, it can be received by using the **PO line receiving** menu item and the functionality of the *Quality check* feature.

Receive pallet 1

1. Sign in to the Warehouse Management mobile app as a user for warehouse *51*. (Enter *51* as the user ID and *1* as the password.)

2. Go to **Inbound** > **PO line receiving**.

3. In the **PONUM** field, enter the purchase order number.

4. Confirm the purchase order number.

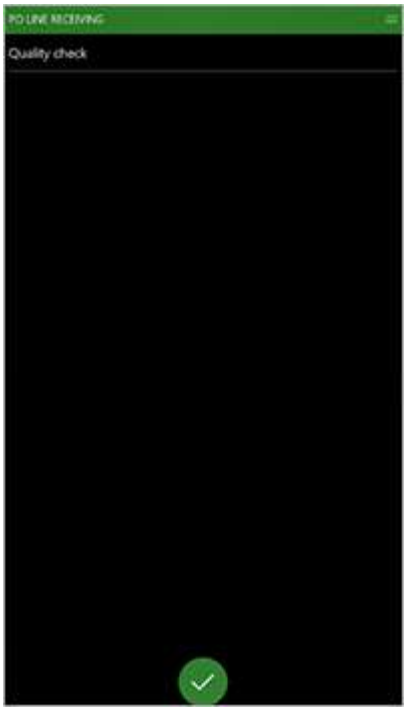
5. In the **LINENUM** field, enter the number of the purchase order line that is being received. Because the order has only one line in this scenario, you'll enter *1* in the **LINENUM** field for each receiving step.

6. Confirm the line number.

7. In the **QTY** field, enter the quantity to receive. Because the purchase order is for three pallets (*PL*) in this scenario, and there are three receiving steps, you'll enter *1* in the **QTY** field for each receiving step.

8. Confirm the quantity.

   The **Quality check** page that appears has no entry fields. It has only the confirmation (check mark) button at the bottom and the Menu button (≡) at the top. (The Menu button is sometimes referred to as the hamburger or the hamburger button.) To expedite the quality check process, when the pallet passes the quality check, the user just confirms the **Quality check** page.



9. Select the confirmation button to pass the quality check for pallet 1 from line 1.

   The **Purchase orders: Put** page that appears shows details of the put work:

   - **LOC:** The system determined location

This location is the designated putaway location for purchase order receiving.

- **LP:** The system-generated License plate ID
- **Item:** *M9203*
- **Qty:** *1 PL: 100 ea*

The item description is also shown.

10. Confirm the putaway work.

    On the **Task** page for purchase order line receiving, you receive a "Work Completed" message. The **LINENUM** field is available so that you can start to receive the next pallet.

Receive pallet 2

For this scenario, pallet 2 will be rejected.

1. In the **LINENUM** field, enter *1*, and confirm the line number.
2. The **QTY** field is now available. Enter *1*, and confirm the quantity.

   The **Quality check** page appears. For this receipt, the pallet will be rejected for quality, and it will be put into the *QMS* quality location.
3. Select the Menu button (≡) at the top of the page, and then, on the menu, select **Reject**.
4. On the **Task** page that appears, enter **QMS** as the *Put* location to send the pallet to for further inspection.

   The **Quality in quality check: Put** page that appears shows details of the put work:

   - **LOC:** *QMS*

     This location is the designated putaway location for rejected quality receiving.
   - **LP:** The system-generated License plate ID
   - **Item:** *M9203*
   - **Qty:** *1 PL: 100 ea*

   The item description is also shown.
5. Confirm the putaway work.

On the **Task** page for purchase order line receiving, you receive a "Work Completed" message. The **LINENUM** field is available so that you can start to receive the next pallet.

You've now completed the quality check and created a quality order for the rejected pallet. To view the order that was created, go to **Inventory management** > **Periodic tasks** > **Quality management** > **Quality orders**.

Quality order testing can now be processed. Quality testing isn't covered in this article.

For more information about quality management, see Quality management overview

Receive pallet 3

For this scenario, pallet 3 will be accepted.

1. In the **LINENUM** field, enter *1*, and confirm the line number.
2. The **QTY** field is now available. Enter *1*, and confirm the quantity.

   The **Quality check** page appears. For this receipt, the pallet will be accepted for quality, and it will be put into a bulk putaway location.
3. Select the confirmation button to pass the quality check.

   The **Purchase orders: Put** page that appears shows details of the put work:

   - **LOC:** The system determined location

     This location is the designated putaway location for purchase order receiving.
   - **LP:** The system-generated License plate ID
   - **Item:** *M9203*
   - **Qty:** *1 PL: 100 ea*

   The item description is also shown.
4. Confirm the putaway work.

   On the **Task** page for purchase order line receiving, you receive a "Work Completed" message. The **LINENUM** field is available so that you can start to receive the next pallet.
5. Select the Menu button (≡) at the top of the page, and then, on the menu, select **Cancel** to return to the menu.

You can now close the mobile app.

Warehouse handling of outbound loads for sales, transfer, and outbound shipment orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the warehouse handling process for outbound loads for sales orders, transfer orders, and outbound shipment orders.

An *outbound load* is a set of shipments from a warehouse that are destined for various locations, such as customer addresses or other warehouses. Usually, outbound loads are linked to a physical delivery vehicle, such as a shipping container or truck.

An outbound load is a component of the warehouse management outbound procedure, which involves organizing, picking, packaging, and sending goods to complete orders. You can form outbound loads manually or automatically. Their creation depends on predefined outbound operations that influence their dependencies and functional effects.

Each outbound load can be associated with one or more order line quantities for sales orders, transfer orders, and outbound shipment orders. Your system might also contain transportation plans. Learn more about how to create and manage outbound transportation in Transportation management overview.

Outbound shipment processing policies

To manage the process of shipping your orders, apply an *outbound shipment processing policy* where you set up the desired flow for your shipments.

The following settings are configured on the **Outbound shipment processing policies** page (**Warehouse management** > **Setup** > **Shipping** > **Outbound shipment processing policies**):

- **Fill entire shipment** – Choose what to do if work creation fails for one or more lines in a shipment (for example, due to location directive failures). This feature only checks for work creation failures and doesn't check whether the full ordered quantity can be fulfilled. Choose one of the following options:
  - *Enabled*: If work creation fails for any line in a shipment, exclude that entire shipment from the wave (create no work for that shipment), regardless of the setting for each customer.
  - *Disabled*: If work creation fails for one or more lines in a shipment, skip those lines but still create work for all of the other lines, regardless of the setting for each customer.
  - *Respect customer settings*: Respect the **Fill entire shipment** setting defined for each customer.

- **Enforce shipment to order matching** – Choose whether the policy should permit just one shipment to be related to each outbound order, or whether multiple shipments per order should be allowed. Select one of the following values:
  - *No* – Allow multiple shipments per outbound order.
  - *Yes* (Supply Chain Management version 10.0.46 and later) – Allow just one shipment per outbound order. In version 10.0.46 and later, when you confirm a shipment, the policy instructs the system to update the delivery remainder on the source order line to reflect the quantity that was actually shipped. If you later reverse the shipment, the delivery remainder is restored to the original ordered quantity.
  - *Yes* (Supply Chain Management version 10.0.45 and earlier) – Allow just one shipment per outbound order. In version 10.0.45 and earlier, if the shipped quantity differs from the ordered quantity, you must manually update the delivery remainder to match the shipped quantity before you confirm the shipment.

  > ⓘ **Note**
  >
  > If you use Warehouse management only mode with external shared warehouses, your source system must be configured to use an outbound shipment processing policy where **Enforce shipment to order matching** is set to *Yes* because that functionality is set up on the source system that is related to the externally managed warehouse.

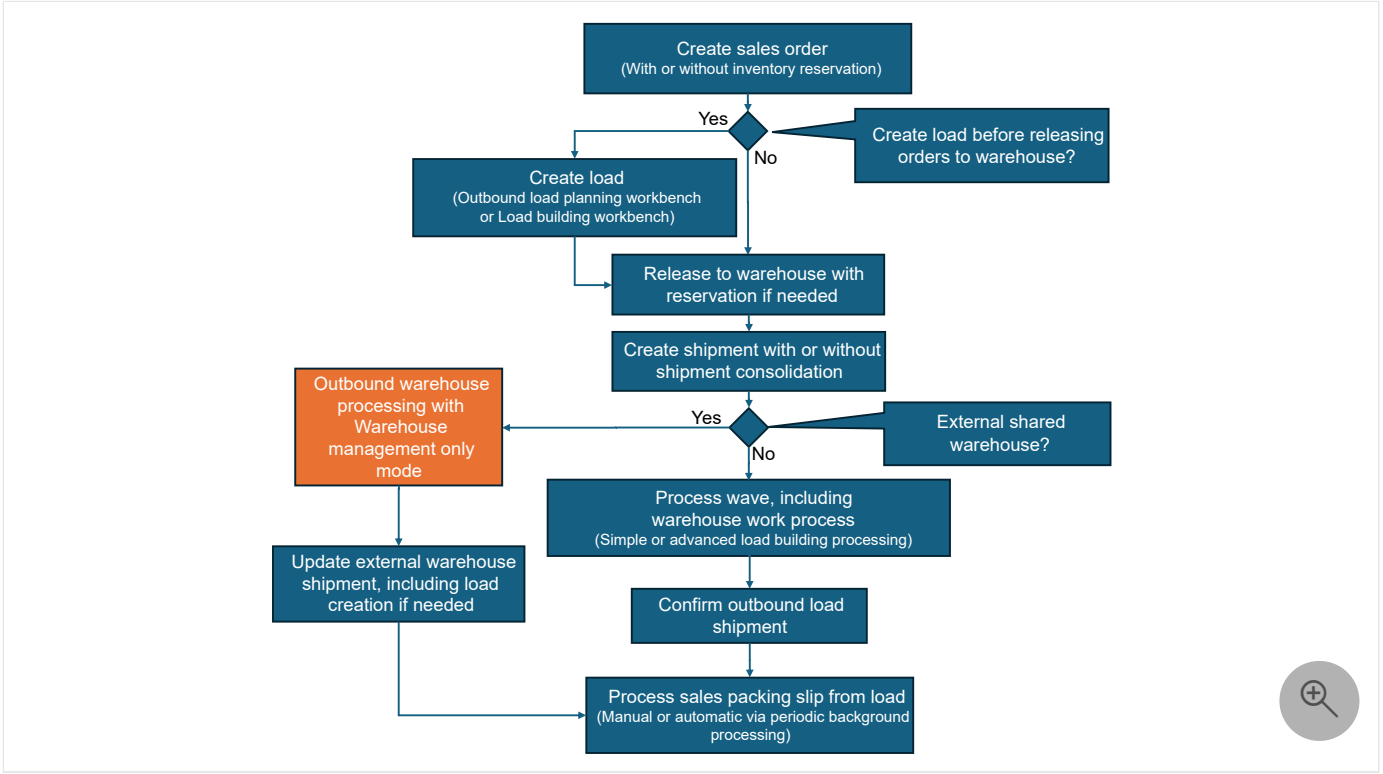
Define default outbound shipment processing policies

Assign an outbound shipment processing policy to each shipment. Assign the default policy in any of the following ways:

- **By order** – Assign policies that you set up on outbound orders to all shipments that you create from those orders.
- **By customer** – Assign a policy to a customer. Outbound orders that you create for that customer inherit the policy. The policy then propagates to shipments that you create from those orders.
- **By source system** – This method is related to Warehouse management only mode. All shipments that come from a source system inherit the policy that you assign to that system.

How outbound loads are created, registered, and shipped

The following illustration provides an overview of the outbound load handling process. It uses outbound loads for sales orders as an example. If the order lines must be processed by an externally managed warehouse, the outbound flow branches off and proceeds exclusively through the Warehouse management only mode process via *outbound shipment orders*. Learn more about this method in Warehouse management only mode with external shared warehouses.



Sales order creation

This process begins when sales orders are created. Reservations for order lines can be set up to occur automatically, they can be performed manually, or they can be delayed according to system setup preferences.

Create a load before the release to a warehouse

A sales order must exist before an outbound load can be generated. Nevertheless, you can define outbound loads before running the release to warehouse procedure.

Use the Outbound load planning workbench and Load building workbench processes to select the order lines and quantities that comprise a load.

Release to warehouse

The release to warehouse process creates *load lines* and groups them into *shipments*. The shipment consolidation process allows for automated shipment consolidation. Shipments that are related to an externally managed warehouse are locked by setting the **Outbound shipment processing ownership** value to *External* until updated order data is returned.

Wave and warehouse work processing

The system generates picking work through *wave processing* and can create loads as required. An available *wave template* must exist to determine whether the advanced load building method is used.

Work templates determine how work is performed for each warehouse process. *Location directives* specify the pick and put locations for inventory movements. Learn more in Control warehouse work by using work templates and location directives.

Warehouse *work* controls any warehouse operation that a warehouse worker or material handling system performs. Typically, warehouse work operations consist of at least two consecutive actions: a *pick* and a *put* of inventory.

Outbound load shipment confirmation

When you complete all the warehouse tasks for a load, run the shipment confirmation procedure. This procedure updates the **Load status** value to *Shipped*. It also changes the **Load packing slip background posting status** value from *None* to *Queued*. Depending on the configuration, you can run this process as part of an automated background process.

> **Note**
>
> When you use the **confirm and transfer** capability, the system can create a new load for any load lines that aren't fully picked.

Generate an outbound ASN

When you ship a load, the system can generate an *outbound advanced shipping notice (ASN)* to notify a customer or downstream warehouse about the shipment. By default, the outbound ASN is generated from the load's *shipments* and *load lines*, which reference the originating sales order lines as they were defined when the order was released. This default generation is true even when the warehouse is enabled for warehouse management processes (WMS). Downstream warehouse execution work, such as picking, sorting, or packing into containers, doesn't change the packing structure of the outbound ASN.

> **Note**
>
> When a transfer order ships from one WMS-enabled warehouse to another, the system uses the outbound shipment data to create a corresponding **inbound ASN** at the destination warehouse. The inbound ASN includes the license plate packing structure to support **license plate receiving** when the goods arrive.

Load packing slip posting

When you process the *packing slip* from a load, the system updates the related sales order line transactions to *Deducted*. At that point, the invoicing process can begin. The *Load packing slip posting* scheduled task (**Warehouse management** > **Periodic** > **Load packing slip posting**) processes *Shipped* outbound loads where the **Load packing slip background posting status** value is set to *Queued*. After successful posting, the system changes the status to *None*. If any errors occur during posting, the status is changed to *Error* instead. For more information about failed postings, you can review the details in the information log that is generated for the batch job.

Although the background posting procedure requires a status of *Queued*, you can manually post a load packing slip even if the status is *Error*. After successful posting, the status is changed to *None*.

To enable the *Load packing slip posting* task to work in the background, you must configure appropriate settings on the **Packing slip posting parameters** page (**Warehouse management** > **Setup** > **Inventory** > **Packing slip posting parameters**). Because this task runs in the background, within a batch job, avoid printing to the screen if printing is included in the process.

> **Tip**
>
> To enable automatic distribution of sales packing slips for each shipment, based on the predetermined values, set the **Packing slip creation policy** value to *Shipment* for the load, and supply **Preallocated packing slip ID** and **Preallocated packing slip document date** values for each associated shipment in the *Packing slip posting parameters* section.
>
> This process automatically applies when the **externally managed warehouse** process is run.

Last updated on 05/13/2026

Flexible warehouse-level dimension reservation policy

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

When an inventory reservation hierarchy of the *Batch-below[location]* type is associated with products, businesses that sell batch-tracked products and run their logistics as operations that are enabled for the warehouse management processes (WMS) can't reserve specific batches of those products for customer sales orders.

In a similar way, specific license plates can't be reserved for products on sales orders when those products are associated with the default reservation hierarchy.

This article describes the inventory reservation policy that lets these businesses reserve specific batches or license plates, even when the products are associated with a *Batch-below[location]* reservation hierarchy.

Inventory reservation hierarchy

This section summarizes the existing inventory reservation hierarchy.

The inventory reservation hierarchy dictates that, as far as storage dimensions are concerned, the demand order carries the mandatory dimensions of site, warehouse, and inventory status. In other words, the mandatory dimensions are all the dimensions above the location dimension in the reservation hierarchy, while the warehouse logic is responsible for assigning a location to the requested quantities and reserving the location. In the interactions between the demand order and the warehouse operations, the demand order is expected to indicate where the order must be shipped from (that is, what site and warehouse). The warehouse then relies on its logic to find the required quantity in the warehouse premises.

However, to reflect the operational model of the business, tracking dimensions (batch and serial numbers) are subject to more flexibility. An inventory reservation hierarchy can accommodate

scenarios where the following conditions apply:

- The business relies on its warehouse operations to manage picking of quantities that have batch or serial numbers *after* the quantities are found in the warehousing storage space. This model is often referred to as *Batch-below[location]* or *Serial-below[location]*. It's typically used when a product's batch or serial number identification isn't important to the customers who place the demand with the selling company.
- The business relies on its warehouse operations to manage picking of quantities that have batch or serial numbers *before* the quantities are found in the warehousing storage space. If batch or serial numbers are necessary as part of a customer's order specification, they are recorded on the demand order, and the warehouse operations that find the quantities in the warehouse aren't allowed to change them. This model is referred to as *Batch-above[location]* or *Serial-above[location]*. Because the dimensions above location are the specific requirements for the demands that must be fulfilled, the warehouse logic doesn't allocate them. These dimensions **must** always be specified on the demand order or in the related reservations.

In these scenarios, the challenge is that only one inventory reservation hierarchy can be assigned to each released product. Therefore, for the WMS to handle tracked items, after the hierarchy assignment determines when the batch or serial number should be reserved (either when the demand order is taken or during the warehouse picking work), this timing can't be changed on an ad-hoc basis.

Flexible reservation for batch-tracked items

Business scenario

In this scenario, a company uses an inventory strategy where finished goods are tracked by batch numbers. This company also uses the WMS workload. Because this workload has well-equipped logic for planning and running warehouse picking and shipping operations for batch-enabled items, most of the finished items are associated with a *Batch-below[location]* inventory reservation hierarchy. The advantage of this type of operational setup is that decisions (which are effectively reservation decisions) about which batches to pick and where to put them in the warehouse are postponed until the warehouse picking operations start. They aren't made when the customer's order is placed.

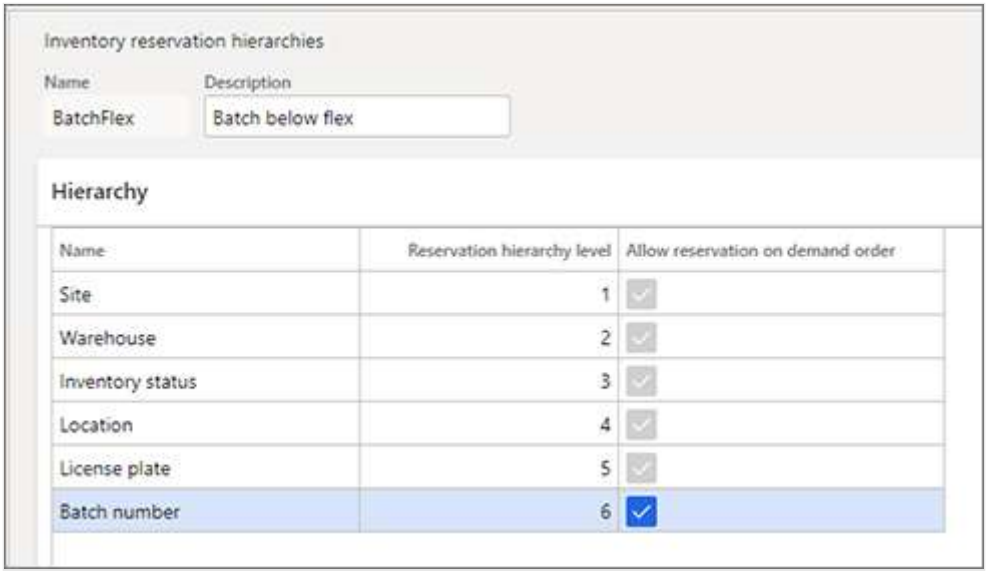
Although the *Batch-below[location]* reservation hierarchy serves the company's business goals well, many of the company's established customers require the same batch that they previously purchased when they reorder products. Therefore, the company is looking for flexibility in the

way that the batch reservation rules are handled, so that, depending on the customers' demand for the same item, the following behaviors occur:

- A batch number can be recorded and reserved when the order is taken by the sales processor, and it can't be changed during warehouse operations and/or taken by other demands. This behavior helps guarantee that the batch number that was ordered is shipped to the customer.
- If the batch number isn't important to the customer, the warehouse operations can determine a batch number during picking work, after sales order registration and reservation have been done.

Allow reservation of a specific batch on the sales order

To accommodate the desired flexibility in the batch reservation behavior for items that are associated with a *Batch-below[location]* inventory reservation hierarchy, inventory managers must select the **Allow reservation on demand order** check box for the **Batch number** level on the **Inventory reservation hierarchies** page.



When the **Batch number** level in the hierarchy is selected, all dimensions above that level and up through the **Location** level are automatically selected. (By default, all dimensions above the **Location** level are preselected.) This behavior reflects the logic where all dimensions in the range between the batch number and location are also automatically reserved after you reserve a specific batch number on the order line.

> ⓘ **Note**
>
> The **Allow reservation on demand order** check box applies only to reservation hierarchy levels that are below the warehouse location dimension.

Batch number and **License plate** are the only levels in the hierarchy that are open for the flexible reservation policy. In other words, you can't select the **Allow reservation on demand order** check box for the **Location** or **Serial number** level.

If your reservation hierarchy includes the serial number dimension (which must always be below the **Batch number** level), and if you turn on batch-specific reservation for the batch number, the system continues to handle serial number reservation and picking operations, based on the rules that apply to the *Serial-below[location]* reservation policy.

At any point, you can allow batch-specific reservation for an existing *Batch-below[location]* reservation hierarchy in your deployment. This change doesn't affect any reservations and open warehouse work that were created before the change occurred. However, you can't clear the **Allow reservation on demand order** check box if inventory transactions of the **Reserved ordered**, **Reserved physical**, or **Ordered** issue type exist for one or more items that are associated with that reservation hierarchy.

> ⓘ **Note**
>
> If an item's existing reservation hierarchy doesn't allow batch specification on the order, you can reassign it to a reservation hierarchy that does allow batch specification, provided that the hierarchy level structure is the same in both hierarchies. Use the **Change reservation hierarchy for items** function to do the reassignment. This change might be relevant when you want to prevent flexible batch reservation for a subset of batch-tracked items but allow it for the rest of the product portfolio.

Regardless of whether you select the **Allow reservation on demand order** check box, if you don't want to reserve a specific batch number for the item on an order line, default warehouse operations logic that is valid for a *Batch-below[location]* reservation hierarchy still applies.

Reserve a specific batch number for a customer order

After a batch-tracked item's *Batch-below[location]* inventory reservation hierarchy is set up to allow reservation of specific batch numbers on sales orders, sales order processors can take customer orders for the same item in one of the following ways, depending on the customer's request:

- **Enter order details without specifying a batch number** – Use this approach when the product's batch specification isn't important to the customer. All existing processes that are

associated with handling an order of this type in the system remain unchanged. No additional considerations are required on the part of users.

- **Enter order details and reserve a specific batch number** – Use this approach when the customer requests a specific batch. Typically, customers request a specific batch when they're reordering a product that they previously purchased. This type of batch-specific reservation is referred to as *order-committed reservation*.

The following set of rules is valid when quantities are processed, and a batch number is committed to a specific order:

- To allow reservation of a specific batch number for an item under the *Batch-below[location]* reservation policy, the system must reserve all dimensions up through location. This range typically includes the license plate dimension.
- Location directives aren't used when picking work is created for a sales line that uses order-committed batch reservation.
- During warehouse processing of work for order-committed batches, neither the user nor the system is allowed to change the batch number. This processing includes exception handling.

The following example shows the end-to-end flow.

Example scenario: Batch number allocation

For this example, install demo data, and use the **USMF** demo data company.

Set up an inventory reservation hierarchy to allow batch-specific reservation

1. Go to **Warehouse management** > **Setup** > **Inventory** > **Reservation hierarchy**.
2. Select **New**.
3. In the **Name** field, enter a name, such as **BatchFlex**.
4. In the **Description** field, enter a description, such as **Batch below flexible**.
5. In the **Selected** field, select **Serial number** and **Owner**, and then select the left arrow button to move them to the **Available** field.
6. Select **OK**.
7. In the row for the **Batch number** dimension level, select the **Allow reservation on demand order** check box. The **License plate** and **Location** levels are automatically selected, and you can't clear the check boxes for them.
8. Select **Save**.

Create a new released product

1. Set the product's three master data parameters by using these values:
    - In the **Storage dimension group** field, select **Ware**.
    - In the **Tracking dimension group** field, select **Batch-Phy**.
    - In the **Reservation hierarchy** field, select **BatchFlex**.
2. Create two batch numbers, such as **B11** and **B22**.
3. Add item quantities to on-hand stock by using the following values.

Expand table

Warehouse	Batch number	Location	License plate	Quantity
24	B11	BULK-001	None	10
24	B11	FL-001	LP11	10
24	B22	FL-002	LP22	10

Enter sales order details

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New**.
3. On the sales order header, in the **Customer account** field, enter **US-003**.
4. Add a line for the new item, and enter **10** as the quantity. Make sure that the **Warehouse** field is set to **24**.
5. On the **Sales order lines** FastTab, select **Inventory**. In the **Maintain** group, select **Batch reservation**. The **Batch reservation** page shows a list of batches that are available for reservation for the order line. For this example, it shows a quantity of **20** for batch number **B11** and a quantity of **10** for batch number **B22**. Note that you can't access the **Batch reservation** page from a line if the item on that line is associated with *Batch-below[location]* reservation hierarchy unless it's set up to allow batch-specific reservation.

    > ⓘ **Note**
    >
    > To reserve a specific batch for a sales order, use the **Batch reservation** page.

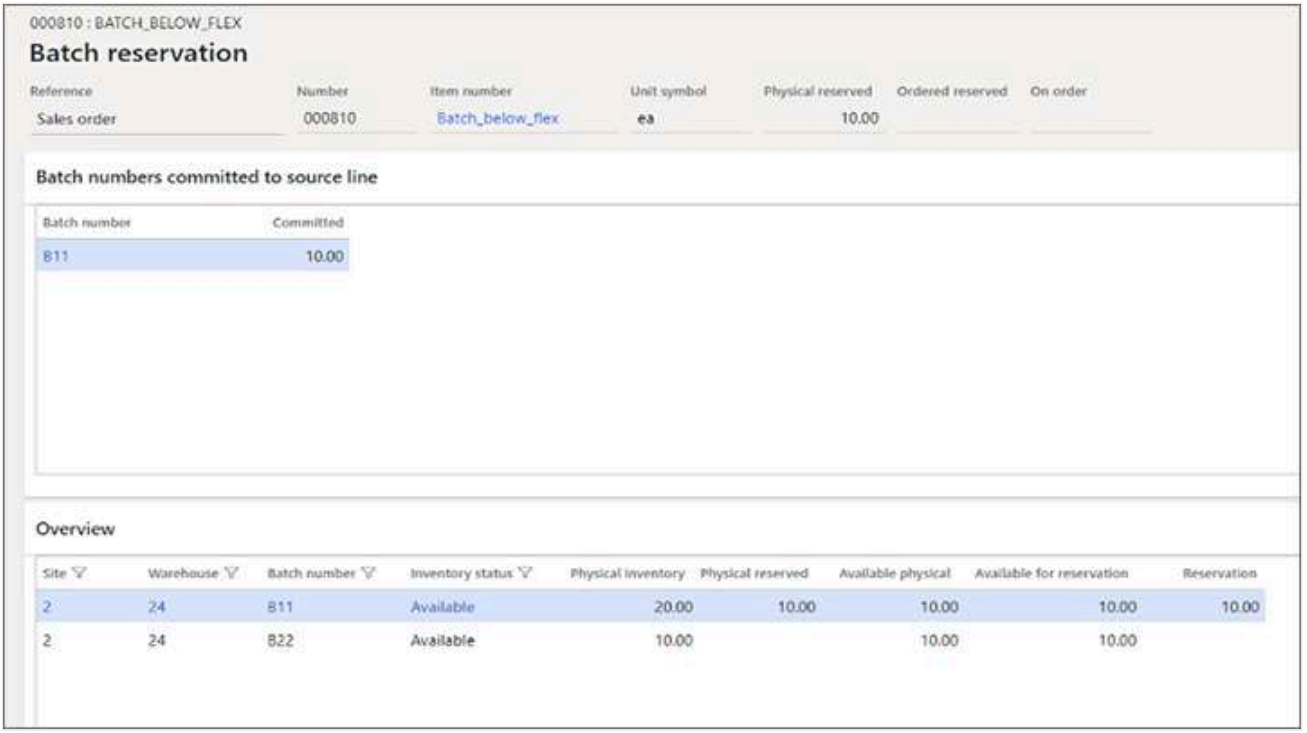
If you enter the batch number directly on the sales order line, the system behaves as though you entered a specific batch value for an item that is subject to the *Batch-below[location]* reservation policy. When you save the line, you receive a warning message. If you confirm that the batch number should be specified directly on the order line, the line isn't handled by the regular warehouse management logic.

If you reserve the quantity from the **Reservation** page, no specific batch is reserved, and the execution of warehouse operations for the line follows the rules that are applicable under the *Batch-below[location]* reservation policy.

In general, you work with this page in the same way for items that have an associated reservation hierarchy of the *Batch-above[location]* type. However, the following exceptions apply:

- The **Batch numbers committed to source line** FastTab shows the batch numbers that are reserved for the order line. The batch values in the grid are shown throughout the fulfillment cycle of the order line, including the warehouse processing stages. By contrast, on the **Overview** FastTab, regular order line reservation (that is, reservation that is done for the dimensions above the **Location** level) is shown in the grid up to the point when warehouse work is created. The work entity then takes over the line reservation, and the line reservation no longer appears on the page. The **Batch numbers committed to source line** FastTab helps guarantee that the sales order processor can view the batch numbers that were committed to the customer's order at any point during its lifecycle, up through invoicing.
- In addition to reserving a specific batch, a user can manually select the batch's specific location and license plate instead of letting the system automatically select them. This capability is related to the design of the order-committed batch reservation mechanism. As mentioned earlier, when a batch number is reserved for an item under the *Batch-below[location]* reservation policy, the system must reserve all dimensions up through location. Therefore, warehouse work carries the same storage dimensions that the users who worked with the orders reserved, and it might not always represent the item storage placement that is convenient, or even possible, for picking operations. If order processors are aware of the warehouse constraints, they might want to manually select the specific locations and license plates when they reserve a batch. In this case, the user must use the **Display dimensions** functionality on the page header, and must add the location and license plate in the grid on the **Overview** FastTab.

6. On the **Batch reservation** page, select the line for batch **B11**, and then select **Reserve line**. There's no designated logic for assigning locations and license plates during automatic reservation. You can manually enter the quantity in the **Reservation** field. Notice that, on the **Batch numbers committed to source line** FastTab, batch **B11** is shown as **Committed**.



> ⓘ **Note**
>
> Reservation of the quantity on a sales order line can be done across multiple batches. Likewise, reservation of the same batch can be done against multiple locations and license plates (if license plates are enabled for the locations).
>
> Reservation of a specific batch for the quantity on a sales order line can also be partial. For example, the total quantity of 100 units can be reserved so that a specific batch is committed to 20 units, whereas 80 units are reserved at the site and warehouse levels for any available batch. In this case, the WMS handles picking operations by using two separate work lines.

7. Go to **Product information management** > **Products** > **Released products**. Select your item, and then select **Manage inventory** > **View** > **Transactions**.

8. Review the item's inventory transactions that are related to the sales order line reservation.

    - A transaction where the **Reference** field is set to **Sales order** and the **Issue** field is set to **Reserved physical** represents the order line reservation for the inventory dimensions above the **Location** level. According to the item's inventory reservation hierarchy, those dimensions are site, warehouse, and inventory status.
    - A transaction where the **Reference** field is set to **Order-committed reservation** and the **Issue** field is set to **Reserved physical** represents the order line reservation for the specific batch and all inventory dimensions above it. According to the item's inventory reservation hierarchy, those dimensions are batch number and location. In this example, the location is **Bulk-001**.

9. On the sales order header, select **Warehouse** > **Actions** > **Release to warehouse**. The order line is now waved, and a load and work are created.

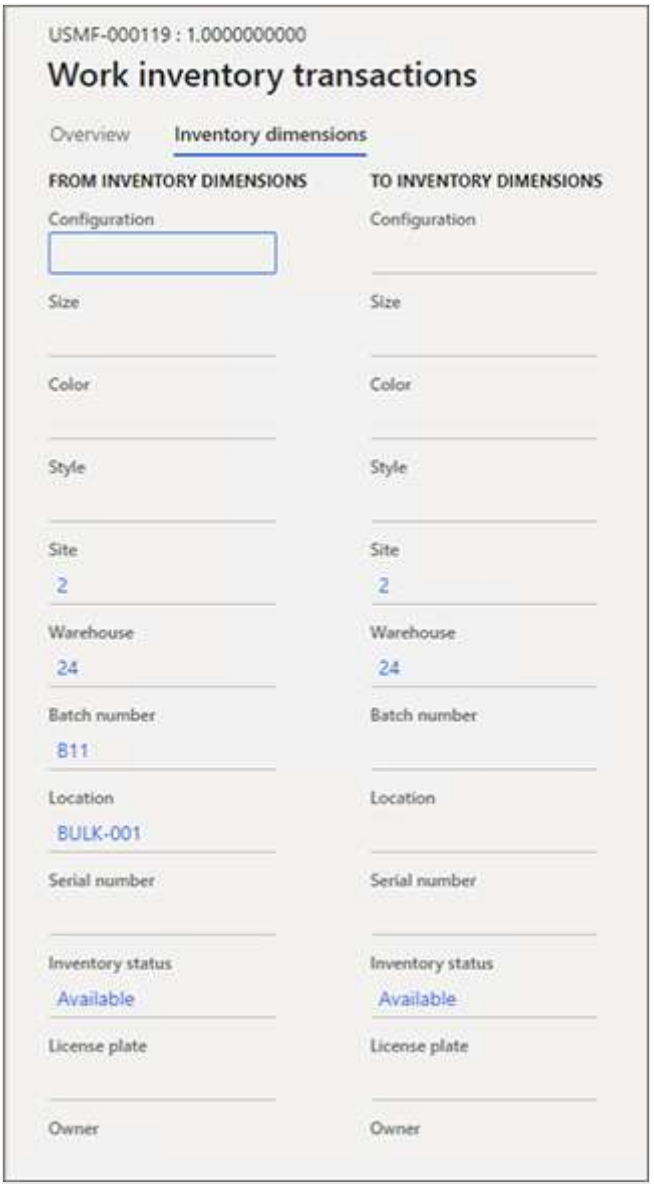
Review and process warehouse work that has order-committed batch numbers

1. On the **Sales order lines** FastTab, select **Warehouse** > **Work details**.

    The work that handles the picking operation for batch quantities that are committed to the sales order line has the following characteristics:

    - To create work, the system uses work templates but not location directives. All the standard settings that are defined for work templates, such as a maximum number of pick lines or a specific unit of measure, are applied to determine when new work should be created. However, the rules that are associated with location directives for identifying pick locations aren't considered, because the order-committed reservation already specifies all the inventory dimensions. Those inventory dimensions include the dimensions at the warehouse storage level. Therefore, the work inherits those dimensions without having to consult location directives.
    - The batch number doesn't appear on the pick line (as is the case for the work line that is created for an item that has an associated *Batch-above[location]* reservation hierarchy). Instead, the **from** batch number and all other storage dimensions appear

on the work line's work inventory transaction that is referenced from the associated inventory transactions.



- After work is created, the item's inventory transaction where the **Reference** field is set to **Order-committed reservation** is removed. The inventory transaction where the **Reference** field is set to **Work** now holds the physical reservation on all the quantity's inventory dimensions.

Warehouse operations can proceed to handle execution of the work in the usual manner. However, the instructions on the mobile device instruct the worker to pick a specific batch number. In warehouse environments where locations are license plate–controlled, after a worker reaches a location that stores the same batch on multiple license plates, they can pick from any license plate that isn't already reserved (for example, by another order-committed reservation or work that originates from a reservation of that type).

If it turns out to be impractical to pick from the location that is specified on the work line, the warehouse operators can use one of the following actions to redirect picking of the specific batch from a more convenient location:

- The standard **Override location** action on a mobile device (provided that the warehouse worker's **Allow pick location override** setting is enabled)
- The **Change location** action on the **Work list details** page.

2. On the mobile device, finish picking and putting the work.

   The quantity of **10** for batch number **B11** is now picked for the sales order line and put in the **Baydoor** location. At this point, it's ready to be loaded onto the truck and dispatched to the customer's address.

Flexible license plate reservation

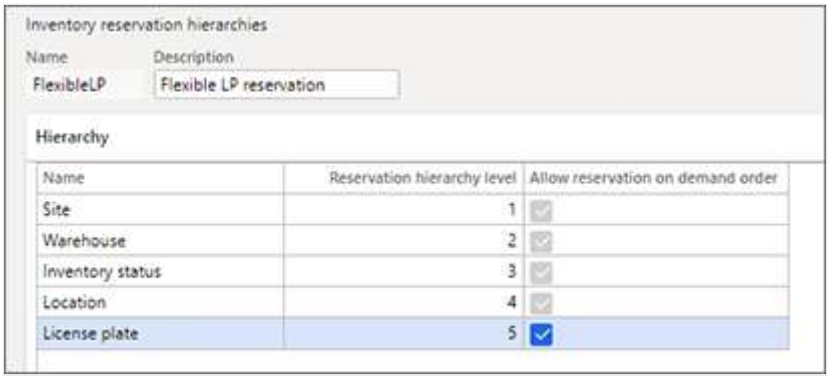
Business scenario

In this scenario, a company uses warehouse management and work processing, and handles load planning at the level of individual pallets/containers outside Supply Chain Management before work is created. These containers are represented by license plates in the inventory dimensions. Therefore, for this approach, specific license plates must be pre-assigned to sales order lines before picking work is done. The company is looking for flexibility in the way that the license plate reservation rules are handled, so that the following behaviors occur:

- A license plate is recorded and reserved when the sales processor takes the order, and other demands can't take it. This behavior helps guarantee that the planned license plate is shipped to the customer.
- If the license plate isn't already assigned to a sales order line, warehouse personnel can select a license plate during picking work, after sales order registration and reservation are completed.

Reserve a specific license plate on the sales order

To enable license plate reservation on an order, select the **Allow reservation on demand order** check box for the **License plate** level on the **Inventory reservation hierarchies** page for the hierarchy that is associated with the relevant item.



You can enable license plate reservation on the order at any point in your deployment. This change doesn't affect any reservations or open warehouse work created before the change. However, you can't clear the **Allow reservation on demand order** check box if open outbound inventory transactions that have an issue status of *On order*, *Reserved ordered*, or *Reserved physical* exist for one or more items that are associated with that reservation hierarchy.

Even if you select the **Allow reservation on demand order** check box for the **License plate** level, it's still possible *not* to reserve a specific license plate on the order. In this case, the default warehouse operations logic that is valid for the reservation hierarchy applies.

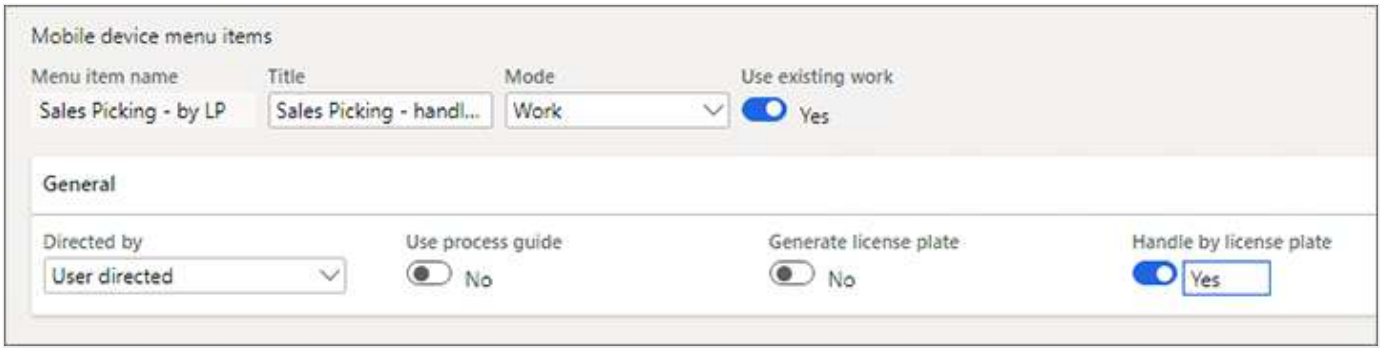
To reserve a specific license plate, use an Open Data Protocol (OData) process. In the application, you can make this reservation directly from a sales order by using the **Order-committed reservations per license plate** option of the **Open in Excel** command. In the entity data that opens in the Excel add-in, enter the following reservation-related data and then select **Publish** to send the data back to Supply Chain Management:

- Reference (Only the *Sales order* value is supported.)
- Order number (The value can be derived from the lot.)
- Lot ID
- License plate
- Quantity

If you must reserve a specific license plate for a batch-tracked item, use the **Batch reservation** page, as described in the Enter sales order details section.

When the sales order line that uses an order-committed license plate reservation is processed by warehouse operations, location directives aren't used.

If a warehouse work item consists of lines that equal a complete pallet and have license plate–committed quantities, you can optimize the picking process by using a mobile device menu item where the **Handle by license plate** option is set to *Yes*. A warehouse worker can then scan a license plate to complete a pick instead of having to scan the items from the work one by one.



Because the **Handle by license plate** functionality doesn't support work that covers multiple pallets, it's better to have a separate work item for different license plates. To use this approach, add the **Order-committed license plate ID** field as a work header break on the **Work template** page.

> **Note**
>
> For the order-committed work creation process, an "order-committed inventory dimension" value is assigned to the picking work lines, and you can't view the license plate value directly. When you set up a mobile device menu item, only the *User directed* process is supported.

Example scenario: Set up and process an order-committed license plate reservation

This scenario shows how to set up and process an order-committed license plate reservation.

Make demo data available

This scenario refers to values and records that are included in the standard demo data that Supply Chain Management provides. To work through the scenario by using the values provided here, use an environment where the demo data is installed. Also, set the legal entity to **USMF** before you begin.

Create an inventory reservation hierarchy that allows for license plate reservation

1. Go to **Warehouse management** > **Setup** > **Inventory** > **Reservation hierarchy**.
2. Select **New**.
3. In the **Name** field, enter a value (for example, *FlexibleLP*).
4. In the **Description** field, enter a value (for example, *Flexible LP reservation*).

5. In the **Selected** list, select **Batch number**, **Serial number**, and **Owner**.
6. Select the **Remove** button to move the selected records to the **Available** list.
7. Select **OK**.
8. In the row for the **License plate** dimension level, select the **Allow reservation on demand order** check box. The **Location** level is automatically selected, and you can't clear the check box for it.
9. Select **Save**.

Create two released products

1. Go to **Product information management** > **Products** > **Released products**.

2. On the Action Pane, select **New**.

3. In the **New released product** dialog box, enter the following values:

   - **Product number:** *Item1*
   - **Item number:** *Item1*
   - **Item model group:** *FIFO*
   - **Item group:** *Audio*
   - **Storage dimension group:** *Ware*
   - **Tracking dimension group:** *None*
   - **Reservation hierarchy:** *FlexibleLP*

4. Select **OK** to create the product and close the dialog box.

5. The new product opens. On the **Warehouse** FastTab, set the **Unit sequence group ID** field to *ea*.

6. Repeat the previous steps to create a second product that has the same settings, but set the **Product number** and **Item number** fields to *Item2*.

7. On the Action Pane, on the **Manage inventory** tab, in the **View** group, select **On-hand inventory**. Then select **Quantity adjustment**.

8. Adjust the on-hand inventory of the new items as specified in the following table.

Expand table

Item	Warehouse	Location	License plate	Quantity
Item1	24	FL-010	LP01	10

Item	Warehouse	Location	License plate	Quantity
Item1	24	FL-011	LP02	10
Item2	24	FL-010	LP01	5
Item2	24	FL-011	LP02	5

> ⓘ **Note**
>
> You must create the two license plates and use locations that allow for mixed items, such as *FL-010* and *FL-011*.

Create a sales order and reserve a specific license plate

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New**.
3. In **Create sales order**, set the following values:
   - **Customer account:** *US-001*
   - **Warehouse:** *24*
4. Select **OK** to close **Create sales order** and open the new sales order.
5. On the **Sales order lines** FastTab, add a line that has the following settings:
   - **Item number:** *Item1*
   - **Quantity:** *10*
6. Add a second sales order line that has the following settings:
   - **Item number:** *Item2*
   - **Quantity:** *5*
7. Select **Save**.
8. On the **Line details** FastTab, on the **Setup** tab, make a note of the **Lot ID** value for each line. You need these values to reserve specific license plates.

> ⓘ **Note**

To reserve a specific license plate, use the **Order-committed reservations per license plate** data entity. To reserve a batch-tracked item on a specific license plate, you can also use the **Batch reservation** page, as described in the Enter sales order details section.

If you enter the license plate directly on the sales order line and confirm it to the system, warehouse management processing isn't used for the line.

9. Select **Open in Microsoft Office**, select **Order-committed reservations per license plate**, and download the file.
10. Open the downloaded file in Excel, and select **Enable editing** to enable the Excel add-in to run.
11. If you're running the Excel add-in for the first time, select **Trust this Add-in**.
12. If you're prompted to sign in, select **Sign in**, and then sign in by using the same credentials that you used to sign in to Supply Chain Management.
13. To reserve an item on a specific license plate, in the Excel add-in, select **New** to add a reservation line, and then set the following values:
    - **Lot ID:** Enter the **Lot ID** value that you found for the sales order line for *Item1*.
    - **License plate:** *LP02*
    - **ReservedInventoryQuantity:** *10*
14. Select **New** to add another reservation line, and set the following values:
    - **Lot ID:** Enter the **Lot ID** value you found for the sales order line for *Item2*.
    - **License plate:** *LP02*
    - **ReservedInventoryQuantity:** *5*
15. In the Excel add-in, select **Publish** to send the data back to Supply Chain Management.

    ⓘ **Note**

    The reservation line appears in the system only if publishing completes without errors.

16. Go back to Supply Chain Management.

17. To review the item's reservation, on the **Sales order lines** FastTab, on the **Inventory** menu, select **Maintain** > **Reservation**. Notice that, for the sales order line for *Item1*, inventory of *10* is reserved, and for the sales order line for *Item2*, inventory of *5* is reserved.

18. To review inventory transactions that are related to the sales order line reservation, on the **Sales order lines** FastTab, on the **Inventory** menu, select **View** > **Transactions**. Notice that there are two transactions that are related to the reservation: one where the **Reference** field is set to *Sales order* and one where the **Reference** field is set to *Order-committed reservation*.

    > ⓘ **Note**
    >
    > A transaction where the **Reference** field is set to *Sales order* represents the order line reservation for inventory dimensions that are above the **Location** level (site, warehouse, and inventory status). A transaction where the **Reference** field is set to *Order-committed reservation* represents the order line reservation for the specific license plate and location.

19. To release the sales order, on the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.

Review and process warehouse work with order-committed license plates assigned

1. On the **Sales order lines** FastTab, on the **Warehouse** menu, select **Work details**.

    When you reserve a specific batch, the system doesn't use location directives when it creates the work for the sales order that uses license plate reservation. Because the order-committed reservation specifies all the inventory dimensions, including the location, you don't need to use location directives, because those inventory dimensions are entered in the work. They appear in the **From inventory dimensions** section on the **Work inventory transactions** page.

    > ⓘ **Note**
    >
    > After the work is created, the item's inventory transaction where the **Reference** field is set to *Order-committed reservation* is removed. The inventory transaction where the **Reference** field is set to *Work* now holds the physical reservation for all the quantity's inventory dimensions.

2. On the mobile device, finish picking and putting the work by using a menu item where the **Handle by license plate** check box is selected.

   > **Note**
   >
   > The **Handle by license plate** functionality helps you process the whole license plate. If you must process part of the license plate, you can't use this functionality.
   >
   > Generate separate work for each license plate. To perform this task, use the **Work header breaks** feature on the **Work template** page.

   License plate *LP02* is now picked for sales order lines and put to the *Baydoor* location. At this point, it's ready to be loaded and dispatched to the customer.

Exception handling of warehouse work that has order-committed batch numbers

Warehouse work for picking order-committed batch numbers is subject to the same standard warehouse exception handling and actions as regular work. In general, the open work or work line can be canceled, interrupted because a user location is full, it can be short-picked, and it can be updated because of a movement. Likewise, the picked quantity of work that is already complete can be reduced, or the work can be reversed.

The following key rule applies to all these exception handling actions: the batch number reserved for the customer can never be replaced with a different batch number, but its storage dimensions (location and license plate) can be changed through either a manual update by the user or an automatic update by the system. The automatic update is based on the same random assignment of storage dimensions that applied when a specific batch was automatically reserved but no storage dimensions were specified.

Example scenario

An example of this scenario is a situation where you unpick previously completed work by using the **Reduce picked quantity** function. This example assumes that you already completed the steps described in Example scenario: Batch number allocation. It continues from that example.

1. Go to **Warehouse management** > **Loads** > **Active loads**.
2. Select the load that was created in connection with the shipment of your sales order.
3. From the **Load order lines** FastTab, select **Reduce picked quantity**.

4. On **Reduce picked quantity**, in the **Move to location** field, select **FL-001**.
5. In the **Move to license plate** field, select **LP33**.
6. In the grid, in the **Inventory quantity to unpick** field, enter **10**.
7. Select **OK**.

Here are the results of the unpicking action:

- The status of the previously closed work is set to **Canceled**.
- New work of the **Inventory movement** type is created for the unpicked quantity of **10** for batch number **B11**. This work represents the movement from the **Baydoor** location to license plate **LP33** in location **FL-001**. The status is set to **Closed**.
- The system re-reserves the batch number that was originally ordered, and assigns the location and license plate IDs. This process is equivalent to running the **Reserve line** function for the order line for a given batch number. As a result, batch **B11** is shown as committed on the **Batch numbers committed to source line** FastTab of the **Batch reservation** page, and the **Reservation** field contains a quantity of **10** for batch number **B11**. Additionally, the **Location** field is set to **FL-001**, and the **License plate** field is set to **LP11**. (You can add these fields to the grid if they aren't visible.)

The following tables provide an overview that shows how the system handles order-committed batch reservation for specific warehouse actions. To interpret the content in the tables, assume that each warehouse action runs in the context of existing warehouse work that originates from an order-committed batch reservation, or that execution of each warehouse action affects work of that type.

> **Note**
>
> In these tables, the **Batch quantity is available** column indicates whether a batch quantity is available in addition to the quantity that is either already reserved for the current order-committed reservations or already reserved by the warehouse work that originates from reservations of that type.

Override the pick location on the open work

Expand table

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
The **Allow pick location override** option is enabled on the worker.	Yes	1. Select the **Override location** menu item on the Warehouse Management mobile app when you start picking work. 2. Select **Suggest**. 3. Confirm the new location that is suggested based on batch quantity availability.	On the current work, the following actions occur: • The location on the pick line is updated to the new location. If the location is license plate–controlled, a random license plate is assigned to the work inventory transaction, and the worker can pick from any license plate that has available quantity. • If the quantity is found on more than one license plate in the new location, the original pick line is split into multiple lines to match each license plate.	Not applicable
	No	1. Select the **Override location** menu item on the Warehouse Management mobile app when you start picking work. 2. Manually enter a location.	The **Override location** action isn't possible. It fails, and an error is thrown.	Not applicable

Full button – Split a work line because of overflow on the user location

Expand table

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
The **Allow splitting of work** option is enabled	Not applicable	1. Select the **Full** menu item on the Warehouse	• On the current work, the quantity is updated to the	Not applicable

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
on the mobile device menu item.		Management mobile app when you process picking work. 2. In the **Pick Qty** field, enter a partial quantity of the required pick to indicate the full capacity.	remaining quantity that must be picked. • New work for the picked quantity is created and closed.	

Reduce the picked quantity of completed work (from a load)

Expand table

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
Not applicable	Yes	1. Open the **Reduce picked quantity** page from the load line. 2. Enter the full quantity to unpick. 3. Select a "move to" location/license plate.	• Work that is associated with the load line is canceled. • New work for the inventory movement is created and closed.	The quantity is re-reserved for the same batch. The system randomly assigns a location and license plate (if the location is license plate–controlled) where the quantity is available.
	No	See the previous row.	See the previous row.	The quantity is re-reserved for the same batch, and for the same location and license plate (if the location is license plate–controlled) that were entered during unpicking.

Move an item within a warehouse

> **Note**

This warehouse action applies only to movement of the **Work creation process** type, not to movement by template.

Expand table

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
The **Allow movement of inventory with associated work** option is enabled on the worker.	Yes	1. Start a movement on the Warehouse Management mobile app. 2. Enter **from** and **to** locations.	• On all existing work that is affected by the move, the pick location is updated to the new "to" location. • New work for the inventory movement is created and closed.	All existing reservations that are affected by the quantity movement from the given location are re-reserved for the same batch. The system randomly assigns a location and license plate (if the location is license plate–controlled) where the quantity is available.
	No	See the previous row.	See the previous row.	All existing reservations that are affected by the quantity movement from the given location are re-reserved for the same batch, and for the new "to" location and license plate (if the location is license plate–controlled).

Reverse the picked quantity of completed work (from a load or a wave)

Expand table

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
Not applicable	Yes	1. Open the **Reverse work** page.	All work that is associated with the load is canceled.	The quantity is re-reserved for the same batch. The system randomly assigns a location and license plate (if

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
		2. On the request page, select the **Leave items at current location** option.		the location is license plate–controlled) where the quantity is available.
	No	See the previous row.	See the previous row.	The quantity is re-reserved for the same batch, and for the location and license plate where the quantity was left upon reversal.
	Yes	1. Open the **Reverse work** page. 2. On the request page, select the **Assign items to this location** option.	• The current work is canceled. • New work for the inventory movement is created and closed.	The quantity is re-reserved for the same batch. The system randomly assigns a location and license plate (if the location is license plate–controlled) where the quantity is available.
	No	See the previous row.	See the previous row.	The quantity is re-reserved for the same batch, and for the location and license plate that the quantity was assigned to upon reversal.
	Yes/No	1. Open the **Reverse work** page. 2. On the request page, select the **Move items to this location** option.	Reversal isn't supported.	Not applicable
	Yes/No	1. Open the **Reverse work** page. 2. On the request page, select the **Move items**	Reversal isn't supported.	Not applicable

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
		based on location directives option.		

Short-pick a quantity – Register the quantity as physically not found at the location/license plate while you perform picking work

Expand table

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
A work exception of the **Short pick** type is set up, where **Item reallocation** = **None**, **Adjust inventory** = **Yes**, and **Remove reservations** = **No**.	Yes	1. Select the **Shortpick** menu item on the Warehouse Management mobile app when you run picking work. 2. In the **Pick Quantity** field, enter **0** (zero). 3. In the **Reason** field, enter **No reallocation**.	• The current work is closed, and the picked quantity is 0 (zero). • An inventory transaction of the **Counting** type and the **Sold** issue type is created to represent the adjustment.	The quantity is re-reserved for the same batch. The system randomly assigns a location and license plate (if the location is license plate–controlled) where the quantity is available.
	No	See the previous row.	• The short-picking action fails, and an error is thrown. • The current work remains open.	Not applicable
A work exception of the **Short pick** type is set up, where **Item reallocation** =	Yes	1. Select the **Shortpick** menu item on the Warehouse Management	• The current work is closed, and the picked quantity is 0 (zero). • An inventory transaction of the	All existing reservations that are affected by the quantity adjustment in the short-picked location are re-

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
None, **Adjust inventory** = **Yes**, and **Remove reservations** = **Yes**.		mobile app when you run picking work. 2. In the **Pick Quantity** field, enter **0** (zero). 3. In the **Reason** field, enter **No reallocation**.	**Counting** type and the **Sold** issue type is created to represent the adjustment.	reserved for the same batch. The system randomly assigns a location and license plate (if the location is license plate–controlled) where the quantity is available.
	No	See the previous row.	See the previous row.	All existing reservations that are affected by the quantity adjustment in the short-picked location are removed.
A work exception of the **Short pick** type is set up, where **Item reallocation** = **Manual**, **Adjust inventory** = **Yes**, and **Remove reservations** = **No/Yes**. Additionally, the **Allow manual item reallocation** option is enabled on the worker.	Yes	1. Select the **Shortpick** menu item on the Warehouse Management mobile app when you run picking work. 2. In the **Shortpick Quantity** field, enter **0** (zero). 3. In the **Reason** field, select **Short Picking with manual reallocation**. 4. Select the location/license plate in the list.	• On the current work, the following actions occur: ◦ The pick line is closed, and the picked quantity is 0 (zero). ◦ The put line is canceled. ◦ A new pick line is created. It uses the location/license plate that the user selected. ◦ A new put line is created. • An inventory transaction of the **Counting** type and the **Sold** issue type is created to represent the adjustment.	Not applicable

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
A work exception of the **Short pick** type is set up, where **Item reallocation** = **Manual**, **Adjust inventory** = **Yes**, and **Remove reservations** = **No**. Additionally, the **Allow manual item reallocation** option is enabled on the worker.	No	1. Select the **Shortpick** menu item on the Warehouse Management mobile app when you run picking work. 2. In the **Shortpick Quantity** field, enter **0** (zero). 3. In the **Reason** field, select **Short Picking with manual reallocation**.	The short-picking action fails, and an error is thrown.	Not applicable

A work exception of the **Short pick** type is set up, where **Item reallocation** = **Manual**, **Adjust inventory** = **Yes**, and **Remove reservations** = **Yes**. Additionally, the **Allow manual item reallocation** option is enabled on the worker.	No	1. Select the **Shortpick** menu item on the Warehouse Management mobile app when you run picking work.
		2. In the **Shortpick Quantity** field, enter **0** (zero).
		3. In the **Reason** field, select **Short Picking with manual reallocation**.
		4. Select the location/license plate in the list.
			• On the current work, the following actions occur:
			  ◦ The pick line is closed, and the picked quantity is 0 (zero).
			  ◦ The put line is canceled.
			• An inventory transaction of the **Counting** type and the **Sold** issue type is created to represent the adjustment.
				All existing reservations that are affected by the quantity adjustment in the short-picked location/license plate are removed.

A work exception of the **Short pick** type is set up, where **Item**	Not applicable	1. Select the **Shortpick** menu item on the Warehouse	Short-picking that involves automatic reallocation isn't supported.	Short-picking that involves automatic reallocation isn't supported.

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
reallocation = **Automatic**, **Adjust inventory** = **Yes/No**, and **Remove reservations** = **Yes/No**.		Management mobile app when you run picking work. 2. In the **Shortpick Quantity** field, enter **0** (zero). 3. In the **Reason** field, select **Short Picking with automatic reallocation**.		

Change the inventory status

> **Note**
>
> You can perform this warehouse action from multiple entry points. The example shown here uses the **Inventory status change** action on the **On-hand by location** page.

Expand table

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
On the **Warehouse** tab, in the **Warehouse** record, the **Remove reservations and markings** field is set to **Reservations** or **Markings and reservations**.	Yes	1. Select a specific location. 2. Select a line that has a specific item, location, and license plate (if the location is license plate–controlled). 3. Select **Inventory status change**. 4. Set the **Inventory status**	Inventory status changes aren't allowed for quantities that are reserved for work.	• The quantity is re-reserved for the same batch. The system randomly assigns a location and license plate (if the location is license plate–controlled) where the quantity is available. • Two inventory transactions of the **Inventory status change** type are created to represent

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
		field to **Blocking**.		the change in the inventory status dimension. • An inventory transaction of the **Inventory blocking** type and the **Reserved physical** issue type is created to represent the reservation of the blocked quantity.
	No	See the previous row.	Inventory status changes aren't allowed for quantities that are reserved for work.	• The reservation is removed. • Two inventory transactions of the **Inventory status change** type are created to represent the change in the inventory status dimension. • An inventory transaction of the **Inventory blocking** type and the **Reserved physical** issue type is created to represent the reservation of the blocked quantity.
On the **Warehouse** tab, in the **Warehouse** record, the **Remove reservations and markings** field is set to **None**.	Yes	1. Select a specific location. 2. Select a line that has a specific item, location, and license plate (if the location is license plate–controlled).	Inventory status changes aren't allowed for quantities that are reserved for work.	The quantity is re-reserved for the same batch. The system randomly assigns a location and license plate (if the location is license plate–controlled) where the quantity is available.

Key setup parameter	Batch quantity is available	Key user steps	Warehouse work	Order-committed batch reservation
		3. Select **Inventory status change**. 4. Set the **Inventory status** field to **Blocking**.		
	No	See the previous row.	Inventory status changes aren't allowed for quantities that are reserved for work.	Inventory status changes aren't allowed.

Limitations

- The flexible warehouse-level dimension reservation functionality doesn't support the following features:
  - Catch weight management
  - Physical negative inventory
  - Reservation against ordered supply
  - Transfer orders and raw material picking
- The container consolidation rule for packing by directive unit has limitations. For order-committed reservations, don't use container build templates where the **Pack by directive unit** field is enabled. In the current design, location directives aren't used when warehouse work is created. Therefore, only the lowest unit in the unit sequence group (the inventory unit) is applied during the containerization wave step.

Next steps

- Batch numbers in Warehouse management
- Reserve the same batch for a sales order
- Pick oldest batch on a mobile device
- Batch, serial, and license plate confirmation

Last updated on 04/28/2026

Release to warehouse

This article provides details about the release to warehouse process. It describes entities that are created when you release an order to the warehouse and options that you can use to initiate the process.

Release to warehouse overview

Release to warehouse is the process of making inventory ready for dispatch processing. When you release an order to the warehouse, the system creates load lines and shipments. If automatic wave processing is set up, loads and required work are also created. The configuration of the entities that are involved depends on the system settings. This section of the article reviews the entities that are created during the release to warehouse process and the system settings that define them.

A *shipment* is a group of order lines for the same customer/receiver or the same delivery address.

A *load* is a group of order lines that are grouped together, and that typically go out on a single truck, rail car, or other mode of transport. A load can have one or many shipments. You can manually create a load by adding order lines to a new load. In this case, order lines are assigned to the load before the release to warehouse process is initiated. Therefore, they can be released only from the **Outbound load planning workbench** page.

Warehouse *work* is any warehouse operation that is performed by a warehouse worker. Typically, warehouse work operations consist of at least two consecutive actions: a warehouse worker picks up on-hand inventory in one location and then puts it away in another location.

When orders are released to the warehouse, the system creates *load lines* and groups them into shipments. The shipment consolidation process allows for automated shipment consolidation during the release to warehouse process. Learn more in Shipment consolidation policies overview.

The system uses *waves* to create picking work and loads for shipment. A *wave template* must be available for the type of wave that you want to create and for the warehouse of the order line. Wave templates of the *Shipping* type are used to ship items for sales orders and transfer orders.

Each wave template contains *wave methods*. Wave methods perform actions such as creating work for the wave or creating loads. For example, a wave template for shipping waves can

contain methods for creating loads, allocating lines to waves, replenishment, and creating picking work for the wave. The wave template establishes settings that define how the wave is generated and processed, which steps users must do manually, and which steps the system does automatically. Learn more in Wave templates. Therefore, depending on the configuration of your wave template, either a wave is automatically created, processed, and released when you release an order to warehouse, or users manually perform some or all of those actions after release to warehouse is complete.

When a wave is processed, the system creates picking work, based on a suitable *work template* and *location directive*, and makes that work available on mobile devices. The work template determines how the work is performed for each warehouse process, and the location directive specifies the pick and put locations for inventory movement. Learn more in Control warehouse work by using work templates and location directives.

> **Note**
>
> By default, waves are processed in batch mode.

During wave processing, the system usually creates a new load for each shipment that no load is assigned to. If you want the system to assign unassigned shipments to existing loads instead, you can use the advanced wave load building functionality. Learn more in Advanced load building during wave.

On the **Sales orders**, **Transfer orders**, and **Outbound shipment orders** pages, you can review the entities created for order lines during the release to warehouse process.

- **All sales orders** page:
  - On the **Sales order lines** FastTab, select **Warehouse**, and then, on the menu, select **Shipment details** to open related shipments, **Load details** to open related loads, or **Work details** to open related work details.
  - On the **Line details** FastTab, select the **Load** tab to review related loads.
- **Transfer orders** page:
  - On the Action Pane, on the **Ship** tab, select **Shipment details** to open related shipments or **Load details** to open related loads.
  - On the **Transfer order lines** FastTab, select **Work details** to open related work details.
- **Outbound shipment orders** page:

- On the **Outbound shipment order lines** FastTab, select **Warehouse**, and then, on the menu, select **Shipment details** to open related shipments, **Load details** to open related loads, or **Work details** to open related work details.

In conclusion, when an order is released to the warehouse, the most automated flow works in the following way:

1. The system creates a shipment for the order and a wave.
2. The wave is processed.
3. The system creates a load and a work ID.

Depending on wave templates, work templates, and location directives settings, some steps in this flow might become manual. However, the overall flow remains the same.

You have several options for how you release an order to the warehouse. You can perform the operation manually, or you can set up a batch job. The remaining sections of this article review, in detail, the various ways that you can perform a release to warehouse operation.

Manual release to the warehouse from the Sales orders, Transfer orders, and Outbound shipment orders pages

You can release an order to the warehouse directly from the **Sales orders**, **Transfer orders**, or **Outbound shipment orders** page, by selecting **Release to warehouse**.

- On the **Sales orders** page, the button is on the **Warehouse** tab of the Action Pane.
- On the **Transfer orders** page, the button is on the **Ship** tab of the Action Pane.
- On the **Outbound shipment orders** page, the button is on the **Warehouse** tab of the Action Pane.

While you're viewing the list pages, you can select multiple records. In this way, you can release several orders simultaneously.

Manual release to the warehouse from the Release to warehouse pages

The system provides pages where you can review lines for multiple orders and release them to the warehouse:

- **Release sales orders to warehouse** (**Warehouse management** > **Release to warehouse** > **Release sales orders to warehouse**) – This page lets you work with sales orders only.

However, it provides better performance than the **Release to warehouse** page.

- **Release transfer orders to warehouse** (**Warehouse management** > **Release to warehouse** > **Release transfer orders to warehouse**) – This page lets you work with transfer orders only. However, it provides better performance than the **Release to warehouse** page.
- **Release outbound shipment orders to warehouse** (**Warehouse management** > **Release outbound shipment orders to warehouse** > **Release to warehouse**) – This page lets you work with outbound shipment orders only.
- **Release to warehouse** (**Warehouse management** > **Release to warehouse** > **Release to warehouse**) – This page lets you work with both sales orders and transfer orders. However, it provides slower performance than the pages that handle specific order types. This page will soon be deprecated.

All the pages provide similar functionality, as described in the rest of this section. All of them let you select order lines or whole orders, and then release them to the warehouse.

Each of the pages consists of an upper section, where you can select order lines, and a lower section, where you can initiate the release to warehouse process. Use filters in the upper section to search for the order lines that you want to release. The **Release to warehouse** page provides a separate tab for sales orders and transfer orders in the upper section, whereas each of the specific pages shows just one type of order.

The toolbar in the upper section includes the following buttons for selecting order lines to release to the warehouse:

- **Add** – Select one or more order lines in the upper section, and then select this button to those lines to the lower section.
- **Add all** – Add all lines from the upper section to the lower section.
- **Add order** – Select an order, and then select this button to add all lines from that order to the lower section.

After you finish adding lines to the lower section, mark each line that you want to release. Then select **Release to warehouse** on the toolbar to release those lines to the warehouse.

Manual release to the warehouse from the Outbound load planning workbench page

You can also manually release orders to the warehouse by using the **Outbound load planning workbench** page. This page lets you organize order lines into loads and then release those loads to the warehouse.

To open the **Outbound load planning workbench** page, go to **Warehouse management** > **Loads** > **Outbound load planning workbench**. You can also open it from the **Sales orders**, **Transfer orders**, and **Outbound shipment orders** pages. In the upper section of the page, you can select to view the following information:

- Shipments
- Sales lines
- Transfer lines
- Outbound shipment order lines
- Transportation request lines (To view this information, you must enable *In transit planning*.)

On each tab, select lines, and then add them to a new or existing load. You can filter the available lines by opening the **Filters** tab in the Action Pane and then selecting **Load planning filters**. Learn more in Load planning filters.

The **Supply and demand** tab on the Action Pane includes the following buttons for adding order lines in the upper section to a load:

- **To new load** – Select order lines in the upper section, and then select this button to create a new load and add those lines to it.
- **To existing load** – Select order lines in the upper section, and then select this button to add those lines to an existing load.
- **Entire order to new load** – Select orders, and then select this button to create a new load and add all lines from those orders to it.
- **Entire order to existing load** – Select an order, and then select this button to add all lines from that order to an existing load.

In the lower section, you can review the loads that are created. To release loads to the warehouse, select them, and then select **Release** > **Release to warehouse** on the toolbar. If you're using automatic wave processing, because loads are already assigned to order lines, the system creates shipments and work IDs when the release to warehouse operation is performed.

Automatic release to the warehouse

To automatically release orders to the warehouse, use the **Automatic release of sales orders**, **Automatic release of transfer orders**, and **Automatic release of outbound shipment orders** jobs.

To set up the batch job that releases sales orders, follow these steps:

1. Go to **Warehouse management** > **Release to warehouse** > **Automatic release of sales orders**.

2. In the **Automatic release of sales orders** dialog box, on the **Parameters** FastTab, set the following fields:

    - **Quantity to release** – Select whether the whole quantity (*All*), only the physically reserved quantity (*Physically reserved*), or only the physically reserved and cross dock quantity (*Reserved physically and cross dock*) should be released to the warehouse.

    - **Allow release of partially released orders** – Specify whether remaining quantities for partially released orders should be released to the warehouse.

    - **Keep reservations on release failure** – Specify whether quantities that were automatically reserved for a sales order should remain reserved if the release to warehouse process fails.

    - **Group releases by customer** – Specify whether the system processes release to warehouse operations separately for each customer or releases all sales orders at the same time. When this option is set to *Yes*, the system collects all the sales order lines for a selected customer, releases those orders to the warehouse, and then processes the next customer. When this option is set to *No*, the system releases all available sales order lines in a single release to warehouse operation. Enabling this option can help improve the performance and resilience of the release to warehouse process. However, be careful when you use this option together with wave templates that are configured to process waves at release to warehouse, because this combination might generate many single-customer waves, each of which has work generated for that customer only. If you want to generate work that combines shipments for multiple customers, either turn off the *Group releases by customer* option or configure your wave templates to use postponed processing.

        > ⓘ **Note**
        >
        > The **Group releases by customer** field appears only in legal entities that have at least one shipment consolidation policy configured. This option affects how the system groups releases when it consolidates shipments, and shipment consolidation policies are company-specific. To make the field appear, configure a shipment consolidation policy by going to **Warehouse management** > **Setup** >

Shipments > **Shipment consolidation policy**. Learn more in Configure shipment consolidation policies.

- **Locked order handling** – Select how the system should handle sales orders that are currently locked because they're being edited by other users or processes:
  - *Wait for orders to unlock* – The system should wait for the orders to become unlocked before it releases them to the warehouse. In this case, the release to warehouse process might take more time.
  - *Skip locked orders* – The system should skip locked orders.
- **Valid order types** – Select the types of sales orders to include in the batch.
- **Credit limit type** – Select whether credit limits checks should be done during the release to warehouse process and, if the checks should be done, the transaction information to include in them.

3. To control which orders should be processed, on the **Records to include** FastTab, select **Filter**. A standard query dialog appears, where you can define selection criteria, sorting criteria, and joins. The fields work just as they work for other types of queries in Microsoft Dynamics 365 Supply Chain Management.
4. On the **Run in the background** FastTab, choose whether the job should run in batch mode, and/or set up a recurrent schedule. The fields work just as they work for other types of background jobs in Supply Chain Management.
5. Select **OK** to apply your settings and initiate the release to warehouse process.

To set up the batch job that releases transfer orders, follow the steps.

1. Go to **Warehouse management** > **Release to warehouse** > **Automatic release of transfer orders**.
2. In the **Automatic release of transfer orders** dialog box, on the **Parameters** FastTab, set the following fields:
   - **Quantity to release** – Select whether the whole quantity (*All*), only the physically reserved quantity (*Physically reserved*), or only the physically reserved and cross dock quantity (*Reserved physically and cross dock*) should be released to the warehouse.
   - **Allow release of partially released orders** – Specify whether remaining quantities for partially released orders should be released to the warehouse.

- **Group releases by destination warehouse** – Specify whether the system should release all transfer orders at the same time, or whether it should group transfer order lines by destination warehouse and then release each group to the warehouse separately.

To set up the batch job that releases outbound shipment orders, follow these steps:

1. Go to **Warehouse management** > **Release to warehouse** > **Automatic release of outbound shipment orders**.

2. In the **Automatic release of outbound shipment orders** dialog box, on the **Parameters** FastTab, set the following fields:

    - **Quantity to release** – elect whether the whole quantity (*All*), only the physically reserved quantity (*Physically reserved*), or only the physically reserved and cross dock quantity (*Reserved physically and cross dock*) should be released to the warehouse.
    - **Allow release of partially released orders** – Specify whether remaining quantities for partially released orders should be released to the warehouse.

3. To control which orders should be processed, on the **Records to include** FastTab, select **Filter**. A standard query dialog appears, where you can define selection criteria, sorting criteria, and joins. The fields work just as they work for other types of queries in Supply Chain Management.

4. On the **Run in the background** FastTab, choose whether the job should run in batch mode, and/or set up a schedule for recurrence. The fields work just as they work for other types of background jobs in Supply Chain Management.

5. Select **OK** to apply your settings and initiate the release to warehouse process.

Last updated on 07/30/2026

Restrict to sales unit

Article • 03/10/2025

This article describes the *Restrict to sales unit* functionality and provides an example scenario.

The *Restrict to sales unit* functionality is designed to ensure that products are sold in a specific, predefined unit. This functionality applies exclusively to sales orders. It specifies the exact unit that a product should be sold in. If the reservation of a product on a sales order leads to incomplete sales units, the release of the sales order to the warehouse is blocked. An error message indicates that the sales order can't be processed because of the incomplete sales units.

To configure the *Restrict to sales unit* functionality for a product, follow these steps.

1. Go to **Product information management** > **Products**.
2. In the list, select a product to view the product details.
3. On the **Warehouse** FastTab, in the **Release to warehouse** section, in the **Restrict to unit** field, select *Sales unit*.
4. Select **Save**.

> ⓘ **Important**
>
> The **Unit** field on the **Sell** FastTab of the product details defines the sales unit that the *Restrict to sales unit* functionality uses.

Example: Releasing a sales order with conversion of units

This example is based on the following scenario:

- Product *item-one* has *Pieces* as the inventory unit.
- The sales unit is *Box*.
- A conversion defines *1 Box* as equal to *5 Pieces*.
- The **Restrict to unit** field is set to *Sales unit*.
- There is a sales order for *10 Pieces* or product *item-one*.

When the sales order is released to the warehouse, the outcome depends on the reserved quantity:

- If the reserved quantity is the full quantity, *10 Pieces*, the sales order is successfully released to the warehouse.
- If the reserved quantity is *7 Pieces*, the result is one full box and almost half of the other box. In this case, the release of the sales order to the warehouse fails, and the customer receives an error message.

> **Note**
>
> There is an exception to the expected behavior of the *Restrict to sales unit* functionality. If the *Automatic release of sales orders* functionality is enabled, and the **Quantity to release** option is set to *All*, the *Restrict to sales unit* functionality is ignored.
>
> Learn more about the *Automatic release of sales orders* functionality in Automatic release to the warehouse.

Feedback

Was this page helpful? Yes No

Provide product feedback

Partial shipment of a transport load

By setting up partial shipment of loads, you can handle loads where the capacity can't be determined until all the sales lines have been added to a load. The process can then be finalized when the exact pallet count is known. Therefore, you don't have to decide which pallets will be assigned to which transport until the moment when a transport is being physically loaded out of the staged inventory.

This feature offers an alternative to the enforcement of a more rigid structure, where you must determine which pallets will be assigned to which transport before picking work or loading work can be created. Instead, users can configure individual loads for a partial shipment confirmation. The transport loading processes for those loads can then occur. Therefore, the transportation planning department can plan loads without having to consider the capacity of individual transports.

At the time of loading, workers can define a new transport load that a pallet can be loaded to. Alternatively, they can identify an existing transport load. This data can be recorded via a mobile device. Therefore, several warehouse workers can load inventory from the same loads or different loads onto the same truck. The loads can then be fully or partially shipped.

> ⓘ **Note**
>
> In order to load inventory from a load to a specific transport load and partially ship the load, work must be generated by using a loading class in a work template. Ordinary picking work of the **Picking** type can't be loaded to a transport load such as a truck.

Set up transport loads for partial shipment

The setup for partial shipment of loads consists of the following two procedures.

Set the loading strategy

You must enable partial loading by setting the loading strategy. You can set the loading strategy after you've created a load.

1. Select **Warehouse management** > **Loads** > **All loads**.
2. Select a load, and then click **Header**.

3. In the **Loading strategy** field, select **Partial load shipping allowed**.

Create a menu item for loading of transport loads

You must create a new menu item that enables transport loads to be loaded. A transport load lets you group work lines from one load or multiple loads. Everything that is added to the transport load can then be shipped by using a mobile scanner.

1. Select **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select **New**, and then, in the **Mode** field, select **Work**.
3. Set the **Use existing work** option to **Yes**.
4. On the **General** tab, in the **Directed by** field, select **Transport loading**.
5. To enable shipment confirmation on a mobile scanner, in the **Allowed ship confirmation type** field, select **Transport load**.

Confirm shipment of a transport load from the client

This setup lets you confirm a transport load that includes a full load or a partially loaded load to be shipped.

1. Select **Warehouse management** > **Loads** > **Transport loads**.
2. On the Action Pane, on the **Ship and receive** tab, in the **Confirm** group, select **Transport**.

Last updated on 07/01/2026

Batch release of partially reserved transfer orders

Article • 07/29/2024

The functionality for batch release of partially reserved transfer orders lets you partially release transfer orders to a warehouse by using a batch job. Because you have the option to release a partial quantity, you don't have to wait for the whole order quantity to be available in the warehouse before you release an order.

The release of orders to a warehouse is a warehouse management processes (WMS) process. This process involves activities, such as picking, packing, and shipping, that a warehouse worker can perform by using a mobile device.

Where it applies

For this functionality, transfer orders are released to a warehouse by using a batch job. This functionality is useful when you don't have enough inventory in the warehouse, but you still want to transfer items from one warehouse to another.

How it is set up

Specify fulfillment criteria for transfer orders and sales orders

Before an order can be partially released to a warehouse in a batch, the fulfillment criteria must be met. These fulfillment criteria are determined by the fulfillment policy.

Fulfillment policies for transfer orders and sales orders are specified at the company level. Depending on the setup of the fulfillment policy, the release of orders in a batch will be accepted or rejected. The orders will then be processed accordingly.

- To create fulfillment policies for transfer orders and sales orders, go to **Warehouse management** > **Setup** > **Release to warehouse** > **Fulfillment policies**, and create a fulfillment policy by entering a name and a description.
- To specify a fulfillment rate, a value type, and the message that is shown if the fulfillment policy is violated, go to **Warehouse management** > **Setup** > **Release to warehouse** > **Fulfillment policies**, and set the **Fulfillment rate**, **Value type**, and **Fulfillment violation message** fields.

Set the fulfillment policies for transfer orders and sales orders

- To set the fulfillment policies for transfer orders, go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**, and then, on the **Transfer orders** tab, in the **Warehouse management** section, select a transfer order fulfillment policy.
- To set the fulfillment policies for sales orders, go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**, and then, on the **Warehouse management** tab, select a sales order fulfillment policy.

Allow release in a batch and specify the quantity that should be released in a batch

A batch job is used to release orders to a warehouse in a batch. The parameters that distinguish the orders that should be run in a batch job are set on the batch job itself.

The **Quantity** parameter specifies whether the whole quantity or the physically reserved quantity should be released in the batch. The **Allow release of partially released orders** parameter determines whether orders in the batch should be accepted or rejected if they were partially released earlier.

- To set the **Quantity** and **Allow release of partially released orders** parameters for transfer orders, go to **Warehouse management** > **Release to warehouse** > **Automatic release of transfer orders**.
- To set the **Quantity** and **Allow release of partially released orders** parameters for sales orders, go to **Warehouse management** > **Release to warehouse** > **Automatic release of sales orders**.
- To set the **Quantity** and **Allow release of partially released orders** parameters for outbound shipment orders, go to **Warehouse management** > **Release to warehouse** > **Automatic release of outbound shipment orders**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up cluster picking

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to enable workers to use their mobile devices to group picking work into clusters, so that they can pick items from a single location for multiple work orders at the same time. This process is called *cluster picking*.

About cluster picking

After you release work orders to the warehouse, the worker can use a mobile device to assign the orders to a cluster. The cluster organizes the picking work for the worker. When you assign a work order to a cluster, the worker must use cluster picking to perform the picking work for the order. The worker can't use other picking methods. If you mistakenly assign a work order to a cluster, the worker must break the cluster and then re-create it.

If needed, a worker can pass a cluster to another worker. This action changes the cluster status to Passed. When the worker uses a mobile device to indicate that the picking and put away work is completed, the shipment or load must be confirmed in the client.

Enable cluster picking

To enable cluster picking, set up the following components:

- **Cluster profiles** – Specify whether to automatically generate cluster IDs, the number of positions to use, when to break clusters, and how to sequence and verify the picking work.
- **Work templates** – Define how to create the picking work for cluster picking.
- **Location directives** – Specify where to pick items from, and where to put them.
- **Mobile device menu items** – Configure a mobile device menu item to use existing work that cluster picking directs. Then add the menu item to a mobile device menu so that it displays on mobile devices.

- **Warehouse management parameters** – Specify the number sequence to use if you want to generate identifiers for clusters.

Set up a cluster profile

To set up a cluster profile, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Cluster profiles**.
2. Select a profile to edit or select **New** to create a new one.
3. In the header of the new or selected record, set a **Cluster profile ID**, **Name**, **Cluster type**, and **Sequence number**. The **Cluster type** specifies for which warehouse operation the cluster profile should be applied. The **Sequence number** determines the order in which the cluster profiles are applied if you have multiple profiles.
4. On the **General** FastTab, set up your profile options. Tooltips explain the purpose of each field. To see a tooltip, hover your mouse pointer over a field. Learn more about the **Cluster picking strategy** setting in the Cluster picking strategy section.
5. On the **Cluster sorting** FastTab, use the toolbar to add and remove sorting criteria for the cluster. The sorting criteria control the order in which the worker performs the work. You can add as many criteria as needed. For each row, make the following settings:

    - **Sequence number** – Enter a number to define the order in which the sorting criteria are processed.
    - **Field name** – Select the field that determines the sorting. For example, if you select the *WMSLocationId* field, the work is sorted by location.
    - **Sorting** – Select one of the following options:
        - *Ascending* – Work is sequenced in ascending order based on the sorting criteria. For example, if you use the *WMSLocationId* field as sorting criteria, and your location IDs are 1, 2, 3, and 4, workers pick from location 4 first.
        - *Descending* – Work is sequenced in descending order based on the sorting criteria. For example, if you use the *WMSLocationId* field as sorting criteria, and your location IDs are 1, 2, 3, and 4, workers pick from location 1 first.

Item confirmation

When you apply cluster picking, item confirmation is crucial to verify the items that you add to clusters. You can verify items in cluster picking during the cluster picking process. The setup is based on the product bar code setup.

Set up item verification with cluster picking

To enable workers to verify each item they pick during the cluster picking process, you must turn on product confirmation for the relevant mobile device menu item. Follow these steps to set it up:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. In the list pane, select the menu item you want to set up.
3. On the Action Pane, select **Work confirmation setup**.
4. Do one of the following actions:

    - If a line exists for the **Work type** you want to set up, select it and then select **Edit** on the Action Pane.
    - If an appropriate line doesn't exist, select **New** on the Action Pane and then set the **Work type** to the appropriate type.

5. Mark the **Product confirm** check box for your new or selected line. This step allows workers to verify each piece of inventory by using the mobile device.

> ⓘ **Note**
>
> If you have multiple work records for an item that has **Material picking in license plate locations** set to *Staging*, the Warehouse Management mobile app might show the error message "Current work is frozen" during the cluster picking process. As a workaround, either make sure that you only pick inventory from locations that aren't license plate tracked for the raw material cluster picking process or set **Material picking in license plate locations** to *Order picking* for these items. Learn more in Release a production order.

Cluster picking strategy

The cluster picking strategy controls whether workers pick inventory for all cluster positions at a location in a single step, or handle each position individually. Set this option as part of the cluster profile.

Prerequisites

To use the cluster picking strategy feature, you must be running Supply Chain Management version 10.0.48 or later.

Choose a cluster picking strategy

To set the cluster picking strategy for a selected cluster profile, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Cluster profiles**.
2. Select the profile you want to set a strategy for. Set up the profile as described in the Set up a cluster profile section.
3. On the **General** FastTab, set the **Cluster picking strategy** to one of the following options:

    - *Process by location* – At each pick location, the worker picks the total quantity across all cluster positions in a single step. The system then presents a sort step where the worker distributes the picked quantity to each position. You can fully customize cluster sort criteria. This strategy is the default.
    - *Process by position (tracked items)* – At each pick location, the worker handles each cluster position one at a time, but only for items tracked by batch or serial number that are below the location level. The *Process by location* strategy still applies to non-tracked items at the same location.
    - *Process by position (all items)* – At each pick location, the worker handles every cluster position one at a time, regardless of whether the item has tracking dimensions.

> ⓘ **Note**
>
> When you select *Process by position (tracked items)*, the system excludes serial-tracked items where the serial number is captured at *packing* from by-position processing. Because the serial number isn't required until packing rather than picking, those items continue to use the *Process by location* flow.

How the strategy affects the picking flow

With the default *Process by location* strategy, the picking sequence at a single location looks like this:

1. The worker arrives at the pick location.
2. The mobile device shows the *total* quantity to pick for all cluster positions at that location.
3. The worker picks the consolidated quantity in a single scan.
4. The device presents the sort step. The worker distributes the picked quantity to each cluster position.
5. The worker moves to the next location.

For items without tracking dimensions, this flow is efficient. For batch- or serial-tracked items, however, the sort step requires the worker to assign specific batch or serial numbers to specific

positions, which can be complex and error-prone.

With either *Process by position* option, the sort step is eliminated:

1. The worker arrives at the pick location.
2. The device shows the pick screen for the *first* cluster position, displaying only the quantity needed for that position.
3. The worker picks the quantity and confirms.
4. The device advances to the next position at the same location, if applicable.
5. Steps 3 and 4 repeat until all positions at the location are complete.
6. The worker moves to the next location.

Because each pick step is already tied to a specific position, batch and serial numbers are captured per position at the moment of picking. No post-pick distribution is needed.

Cluster picking strategy example

Consider a cluster with two positions picking a serial-tracked item:

- **Position 1** — Sales order 1, requires 1 unit
- **Position 2** — Sales order 2, requires 2 units

When you use the *Process by location* strategy, work proceeds as follows:

Expand table

Step	Screen prompt	Worker action
1	Pick 3 units from location A-01-01	Scans item and license plate
2	Sort: assign quantity to Position 1	Enters 1, confirms 1 unit
3	Sort: assign quantity to Position 2	Enters 2, confirms 2 units
4	Put cluster to staging	Confirms location

When you use either *Process by position* strategy (tracked items or all items), work proceeds as follows:

Expand table

Step	Screen prompt	Worker action
1	Position 1 – Pick 1 unit from location A-01-01	Scans item and serial number

Step	Screen prompt	Worker action
2	Confirm Position 1	Confirms position
3	Position 2 – Pick 2 units from location A-01-01	Scans item and serial numbers
4	Confirm Position 2	Confirms position
5	Put cluster to staging	Confirms location

Effect of strategy on cluster sorting criteria

The cluster sort criteria (configured under **Cluster sorting** on the cluster profile) behave differently depending on the selected strategy.

Expand table

Strategy	Sort criteria
Process by location	Fully customizable
Process by position (tracked items)	Auto-set: Location > Item number > Work ID > Line number (all ascending)
Process by position (all items)	Auto-set: Location > Item number > Work ID > Line number (all ascending)

> **Important**
>
> When you save a cluster profile with *Process by position (tracked items)* or *Process by position (all items)* selected, the system automatically deletes any existing sort criteria and creates the following four default sort fields:
>
> 1. Location (ascending)
> 2. Item number (ascending)
> 3. Work ID (ascending)
> 4. Line number (ascending)
>
> You can manually edit these sort criteria after they're created. However, changing the sort order can lead to a suboptimal picking route. For example, if the sort criteria no longer group work by location first, the system might direct the worker to pick one position at a location, then travel to a different location for another position, and then return to the

original location to pick the remaining position there. This change results in unnecessary travel between locations.

If you switch back to *Process by location* after previously using a *Process by position* strategy, the auto-created sort criteria remain in place. You can then edit or remove them as needed.

Last updated on 04/24/2026

Cluster position full

Article • 02/01/2023

The *Cluster position full* feature offers an alternative to more rigid enforcement of work break rules when cluster picking is used, because it enables a larger margin of error in the volumetric constraints of containers or totes. In a common scenario, not all items on a work order fit into a selected container. Warehouse workers who are cluster picking have few options in this scenario: they must either change to a larger container size or work with their supervisor to come up with a different solution.

This feature introduces the ability to run the **Full** button on one of the work units in a cluster. In older versions, this option was available only for regular order picking, not for cluster picking. However, this feature differs from the standard **Full** button in that it cancels the remaining work. It doesn't suggest that the user add another bin to the same cluster, and it doesn't automatically create new work.

Turn the Cluster position full feature on or off

To use the functionality described in this article, the *Cluster position full* feature must be turned on for your system. As of Supply Chain Management 10.0.25, this feature is mandatory and can't be turned off. If you're running a version older than 10.0.25, then admins can turn this functionality on or off by searching for the *Cluster position full* feature in the Feature management workspace.

Setup

This section provides guidelines, and an example that shows how to set up and use the *Cluster position full* feature.

Make sample data available

To work through the example scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

You can also use the example scenario as guidance for working with this feature on a production system. However, in that case, you must substitute your own values for the settings that are described here.

Cluster profiles

You must specify whether cluster IDs are automatically generated, how many positions are used, when clusters are broken, and how the picking work is sequenced and verified.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Cluster profiles**.
2. In the list pane, select the **Create Cluster** record.
3. On the **General** FastTab, verify the following values:
    - **Generate cluster ID:** *Yes*
    - **Activate positions:** *Yes*
    - **Number of positions:** *2*
    - **Position name:** *Numeric*
    - **Break cluster at:** *Put*
    - **Sort verification type:** *Position scan*
4. In the **Cluster sorting** FastTab. The grid should be blank (that is, it should contain no lines).

Work templates

You must define how the picking work for cluster picking is created.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. At the top of the page, set the **Work order type** field to *Sales orders*.
3. Make sure that the following work templates from the demo data are listed. If they aren't available, you won't be able to complete the scenario.
    - 61 SO Stage
    - 61 SO Cluster pick

Location directives

You must specify where items are picked from and where they are put.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the list header, set the **Work order type** field to *Sales orders*.
3. Make sure that the following sales order directives from the demo data are listed. If they aren't available, you won't be able to complete the scenario.

- 61 Cluster pick
- 61 SO Pick order

Mobile device menu items

You must configure a mobile device menu item to use existing work that is directed by cluster picking. In the mobile device menu item for cluster picking, the **Allow splitting of work** parameter must be turned on, and the *SO Pick* work class must be added.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. In the list pane, select the **Cluster Pick Create** record.

3. Select **Edit** in the Action pane.

4. On the **General** FastTab, set the following values:

   - **Directed by:** *Cluster picking*
   - **Generate license plate:** *Yes*
   - **Allow splitting of work:** *Yes*
   - **Cluster profile ID:** *Create Cluster*

   Accept the default values for all other fields.

5. On the **Work classes** FastTab, add the following two lines, as required:

   - Line 1 (usually present in demo data):
     - **Work class ID:** *Sales*
     - **Work order type:** *Sales orders*

   - Line 2 (probably not present in demo data):
     - **Work class ID:** *SO Pick*
     - **Work order type:** *Sales orders*

6. Go to **Work confirmation setup** in the Action pane.

7. Select **Edit**.

8. Enter the following values on the line in grid.

   - **Work type** - *Pick*
   - **Product confirmation** - *Select the check box*

9. Select **Save** in the Action pane and close the page.

Create picking work

Before you can start cluster picking, you must create some outbound work. The cluster profile that you created earlier specifies two cluster positions. Therefore, at least two work IDs must be created for sales order picking. For this scenario, transactions will occur in warehouse *61*, and they will use items *L0101* and *T0100*. The demo data should have enough on-hand inventory of these items. Make sure that you have enough inventory to complete the transactions.

Create sales order 1

1. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create sales order 1.
3. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-010*
    - **Warehouse:** *61*
4. Select **OK**.
5. The new sales order is opened. On the **Sales order lines** FastTab, add a line that has the following settings:
    - **Item number:** *T0100*
    - **Quantity:** *5*
6. On the **Line details** FastTab, on the **Delivery** tab, set the **Confirmed ship date** field to today's date.
7. On the **Sales order lines** FastTab, add a second line that has the following settings:
    - **Item number:** *L0101*
    - **Quantity:** *20*
8. On the **Line details** FastTab, on the **Delivery** tab, set the **Confirmed ship date** field to today's date.
9. For each line that you just added, follow these steps to reserve inventory:
    a. Select the line to reserve.
    b. On the **Sales order lines** FastTab, select **Inventory** > **Reservation**.
    c. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the inventory.

d. Close the **Reservation** page.

10. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

    When the release is completed, you receive informational messages that show the wave and load IDs that were created.

Create sales order 2

1. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create sales order 2.
3. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-011*
    - **Warehouse:** *61*
4. Select **OK**.
5. The new sales order is opened. On the **Sales order lines** FastTab, add a line that has the following settings:
    - **Item number:** *L0101*
    - **Quantity:** *20*
6. On the **Line details** FastTab, on the **Delivery** tab, set the **Confirmed ship date** field to today's date.
7. On the **Sales order lines** FastTab, add a second line that has the following settings:
    - **Item number:** *T0100*
    - **Quantity:** *2*
8. On the **Line details** FastTab, on the **Delivery** tab, set the **Confirmed ship date** field to today's date.
9. For each line that you just added, follow these steps to reserve inventory:
    a. Select the line to reserve.
    b. On the **Sales order lines** FastTab, select **Inventory** > **Reservation**.
    c. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the inventory.
    d. Close the **Reservation** page.
10. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

When the release is completed, you receive informational messages that show the wave and load IDs that were created.

Get work IDs and license plate numbers

Two work IDs should have been created, each of which has two pick lines. Follow these steps to find the work IDs and license plate assignments.

1. Go to **Warehouse management** > **Work** > **Work details**.
2. In the **Overview** grid, search the **Order number** column for the two sales orders that you just created. For each sales order, make a note of the corresponding work ID.
3. Select the row for each sales order to show related information in the **Lines** grid. Make a note of the location that each item will be picked from.
4. Go to **Inventory management** > **Inquiries and reports** > **On-hand list**.
5. On the Action Pane, select **Dimensions** to open the **Dimension display** dialog box.
6. Make sure that the **License plate**, **Warehouse**, and **Item number** check boxes are selected, and then select **OK**.
7. In the **Filter** pane, set the following filters:
    - **Item number** – **is one of** – *L0101* and *T100*
    - **Warehouse** – **begins with** – *61*
8. Make a note of the **License plate** values that are shown.

Example scenario

Mobile device flow execution – Work confirmation setup for the product

1. Sign in to the Warehouse Management mobile app as a user in warehouse *61*.
2. Go to **Outbound** > **Cluster pick create**.

    The **TASK: Assign work to Cluster** page appears.
3. Enter the work ID for sales order 1 to assign it to cluster position 1.

4. Select **OK** (check mark symbol).
5. Enter the work ID for sales order 2 to assign it to cluster position 2.
6. Select **OK** (check mark symbol).

   The **TASK: Cluster Pick Create: Pick** page appears and shows *Item L0101 2 PL*.

Because the cluster profile set the number of positions to 2, the system automatically directs you to the first consolidate pick: two pallets (PL) of item *L0101*.

At any time during the following steps, you can select the **Details** tab to view additional information about the task, such as the picking location.

1. Set the **ITEM** field to *L0101*. This confirms the item number, which is required for this menu item (you configured this earlier by selecting **Work confirmation setup** from the **Mobile device menu item** page when you created this menu item).
2. Enter the license plate number that is associated with the item in the location that is being picked. You will pick two pallets.
3. Set the **LP** field to *LP_PICK_01*.
4. Select **OK** (check mark symbol).

   The **TASK: Sort: Cluster Pick Create** page appears. Here, you will sort the two picked pallets into a pick position. This position might be a tote or container that is used to separate the picked inventory by sales order.

5. View the details that are shown for the item (*L0101*) and quantity (*20* ea) that will be sorted into position 1 (for sales order 1).
6. Set the **POSITION NA** field to *1*.
7. Select **OK** (check mark symbol).
8. View the details that are shown for the item (*L0101*) and quantity (*20* ea) that will be sorted into position 2 (for sales order 2).
9. Set the **POSITION NA** field to *2*.
10. Select **OK** (check mark symbol).

    The **TASK: Cluster Pick Create: Pick** page appears and shows *Item T0100 7 ea*.

In this scenario, position 1 can't accept the full quantity of items that must be picked to fulfill sales order 1. A position must be marked as full. In this scenario, you will do a

partial pick of the second item. The second item will be partially picked for position 1, and new work will be created to pick the remaining quantity to fulfill the order.

1. Select the Menu button (sometimes referred to as the hamburger or the hamburger button), and then select **Position full**.
2. Identify the position that is full, and select *1*.
3. Select **OK** (check mark symbol).
4. Enter the pick quantity that can still be picked into position 1. The system can determine which item number is being picked.
5. Enter *2*.
6. Select **OK** (check mark symbol).
7. Confirm the item number to complete the pick of the remaining item into position 2.
8. Set the **ITEM** field to *T0100*.
9. Select **OK** (check mark symbol).
10. Enter the license plate that the item is being picked from by setting the **LP** field to *LPREPL04*.
11. Select **OK** (check mark symbol).
12. View the details that are shown for the item (*T0100*) and quantity (*2* ea) that will be sorted into position 2 (for sales order 2).
13. Set the **POSITION NA** field to *2*.
14. Select **OK** (check mark symbol).
15. View the details that are shown for the item (*T0100*) and quantity (*2* ea) that will be sorted into position 1 (for sales order 1).
16. Set the **POSITION NA** field to *1*.
17. Select **OK** (check mark symbol).

    The **TASK: Cluster Pick Create: Put** page appears.

In this scenario, the cluster pick has been completed, and the user is directed to put away the picked items from position 1 and position 2 into staging location *STAGE01*.

1. Review the information on the page. It shows that a total quantity of *44* will be put to the staging location.
2. Select **OK** (check mark symbol).

   You receive a "Cluster Completed" message.

You can now use the **Sales Picking** menu item to pick the remaining quantity. You can then use the **Sales loading** menu item to move the items from the staging location to the loading dock.

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System-directed cluster picking

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Cluster picking is a piece picking process that lets you pick items for multiple orders at the same time by clustering them into pick clusters. You then have to visit the pick location only one time. Typically, this functionality is used with small order picking or quantities that are less than case quantities.

When system-directed cluster picking is set up, you can cluster-pick work headers, based on a system-generated cluster. The system cluster-picks orders up to the number of positions that is specified in the cluster profile. Therefore, you can pick multiple orders at the same time without having to manually create a cluster.

System-directed cluster picking offers an alternative to manual cluster building, where a cluster profile is used to create a cluster. Several setup-related lines must be defined in the cluster profile before it's used:

- **Number of positions** corresponds to the number of orders that will be put on a cluster. Therefore, it corresponds to the number of totes.
- **Break cluster** determines when the cluster should be broken.
- **Generate cluster ID** controls whether the cluster ID will be generated by the system or entered by the user.
- **Sort verification type** determines whether position verification is required.

A new mobile device menu item is used for system-directed cluster picking. The **Cluster profile ID** must be specified for the selected **Directed by** option. Additionally, the system-directed work sequence queries can group orders based on company-specific criteria. Therefore, you can further optimize the assignment of work orders by specifying customized sorting criteria using the system-directed work sequence queries.

When system-directed cluster picking is enabled, warehouse workers are presented with a cluster where picking orders have been preassigned to cluster positions. Therefore, workers can start to pick an item for multiple work orders by visiting the pick location only one time. The picking process for system-directed cluster picking is the same as the process for user-directed cluster picking.

Set up system-directed cluster picking

Create cluster profiles

Cluster profiles control how the system creates each cluster. If different clusters are required, a different cluster profile should be created for each mobile device menu item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Cluster profiles**.
2. Select **New**.
3. In the **Cluster profile ID** field, enter **2 Position**.
4. In the **Name** field, enter **2 Position**.
5. On the **General** FastTab, enter the following information:
    - **Generate cluster ID** - Select **Yes**. This option determines whether the cluster ID is automatically created by the system, or whether the user will create it at the start of picking.
    - **Activate positions** - Select **Yes**. This option determines whether the position names are automatically generated based on the position name setup. If this option is set to **No**, the license plate ID for the position is used.
    - **Number of positions** - Select **2**. This field determines the maximum number of positions that the cluster can have (that is, the maximum number of boxes, totes, and so on).
    - **Position name** - Select **Numeric** so that positions are named by using continuous numbers. If you select **Alphabetical**, the positions are named in alphabetical order.
    - **Break cluster at** - Select **Put**. This field determines when the cluster is broken.
    - **Sort verification type** - Select **Position scan**. This field determines whether the put-to-position step is verified.
6. On the **Cluster sorting** FastTab, you define sorting criteria by creating a new line and enter the following information:
    - **Sequence number** - Select **1**. This field determines the sequence that the system sorts by. A value is entered automatically, but you can change it if needed.
    - **Field name** - Enter **WMSLocationId**. This field determines the field that the line uses for sorting criteria.
    - **Sorting** - Select **Ascending**. This field defines whether the sorting is done in ascending or descending order.

Create a mobile device menu item

To create a new mobile device menu item for system-directed cluster picking and link the cluster profile ID to the mobile device menu item, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select **New**.
3. In the header section, enter the following information:
    - **Menu item name** - SD Cluster
    - **Title** - SD Cluster
    - **Mode** - Work
    - **Use existing work** - Yes
4. On the **General** FastTab, enter the following information:
    - **Directed by** - System directed cluster picking
    - **Generate license plate** - Yes
    - **Cluster profile ID** - 2 Position
5. On the **Work classes** FastTab, set up the valid work class for this mobile device menu item by setting the following fields:
    - **Work class ID** - Sales
    - **Work order type** - Sales orders
6. In the **Mobile device menu items** Action Pane, select **System directed work sequence queries** and follow these steps to specify a new system-directed work sequence query:
    - Select **New** in the Action Pane.
    - **Sequence number** - 1
    - **Description** - Work priority – Work ID
7. On the Action Pane select **Edit query**
8. Select the **Sorting** tab
9. Select **Add** to add a new line, then enter the following:
    - **Table** - Work
    - **Derived table** - Work
    - **Field** - Work priority
    - **Search direction** - Ascending
10. Select **Add** to add a second line, then enter the following:
    - **Table** - Work

- **Derived table** - Work
- **Field** - Work ID
- **Search direction** - Ascending

11. The system will now sort work IDs in the cluster, first by work priority and then by work ID.

Set up a mobile device menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. Add the **SD Cluster** menu item that you just created to a Mobile device menu.
3. Select the **Outbound** menu.
4. Select **Edit** from the Action Pane.
5. Scroll until you find **SD Cluster**.
6. Select **SD Cluster**, the arrow pointing to the **Menu Structure** list will be enabled.
7. Select the **arrow** button to move the **SD Cluster** menu item into the **Outbound** menu structure.
8. Select **SD Cluster** from the **Menu Structure** list, then select the **UP** or **DOWN** arrows to move the menu item into the desired position on the mobile device menu.

Scenario

Create picking work

Before you can set up system-directed cluster picking, you must create eligible outbound work. The cluster profile that you created earlier specifies two cluster positions. Therefore, you must create at least two work IDs. In this scenario you will create two sales orders, reserve inventory, release the sales orders to the warehouse, and then manually process the wave.

1. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.
2. Select **New** in the Action Pane to create the first sales order.

    - The **Create sales order** menu opens, enter the following information:
        - On the **Customer** FastTab, enter **Customer account** - **US-004**.
        - On the **General** FastTab, enter **Warehouse** - **62**.
        - Select **OK** to close the menu and create the sales order.
    - On the **Sales order lines** FastTab, select **Add line** if a new line is not automatically added and enter the following:
        - **Item number** - A0001
        - **Quantity** - 1
        - Select **Add line** to add a second line.

    - **Item number** - A0002
    - **Quantity** - 3
  - Reserve inventory for both of the lines that you just created.
    - Select **Line 1**.
    - On the **Sales order lines** Action Pane, select **Inventory**, and then select **Reservation** from the list.
    - On the **Reservation** form, select **Reserve lot** to reserve the inventory.
    - Close the **Reservation** form when the reservation is completed.
    - Repeat these steps to reserve inventory for **Line 2**.
3. Select **New** in the Action Pane to create the second sales order

   - The **Create sales order** menu opens, enter the following information:
     - On the **Customer** FastTab, enter **Customer account** - **US-005**.
     - On the **General** FastTab, enter **Warehouse** - **62**.
     - Select **OK** to close the menu and create the sales order
   - On the **Sales order lines** FastTab, select **Add line** if a new line is not automatically added and enter the following information:
     - **Item number** - A0001
     - **Quantity** - 4
     - Select **Add line** to add a second line.
     - **Item number** - A0002
     - **Quantity** - 2
   - Reserve inventory for both lines that you just created.
     - Select **Line 1**.
     - On the **Sales order lines** Action Pane, select **Inventory**, and then select **Reservation** from the list.
     - On the **Reservation** form, select **Reserve lot** to reserve the inventory.
     - Close the **Reservation** form when the reservation is completed.
     - Repeat these steps to reserve inventory for **Line 2**.
   - Close the sales order and return to the **All sales orders** list page.

4. Find the two sales orders you just created (you may need to refresh the page). In the table, select both sales orders using the section check mark.

   - In the **All sales orders** Action Pane, select the **Warehouse** tab.
   - In the **Actions** group, select **Release to warehouse** to release both sales orders to the warehouse.

5. When the release to warehouse process is completed, an informational message will be displayed.

   - Shipments will be created for each sales order.

- A wave will be created and both shipments will be assigned to the wave. Make note of the **Wave ID**.

6. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.

   - In the **All waves** list find and select the **Wave ID** that you created in the previous step.
   - In the Action Pane, select the **Wave** tab,
   - In the **Wave** group, select **Process** to process the wave and create **Work**.
   - Informational messages will be generated when processing has been completed, indicating that work has been created and the wave has been posted.

7. **Optional**: Go to **Warehouse management** > **Work** > **Work details** to view the work created. Two different work IDs are created. Each work ID has two pick lines.

Run the mobile device flow

1. Sign in to the mobile device for a user in warehouse **62**.

2. On the **Main Menu**, select **Outbound**.

3. On the **Outbound** menu, select **SD Cluster** to initiate the pick.

   - A cluster is created, and the two work IDs that you created earlier are attached. If you created more than two work IDs, only the first two are added to the cluster. Notice that the work IDs are added to the cluster in ascending order, as you specified in the query setup.

   > ⓘ **Note**
   >
   > The new cluster is automatically created only if enough additional work IDs were previously created. Otherwise, the following message is shown: "Not enough work can be found for cluster."

   - After you select the menu, the first pick screen appears. The system aggregates all matching pick lines from the two work IDs and directs you to visit the pick location one time, so that you can satisfy both orders by using one pick. This process is done in the same way as the process for user-directed cluster picking.

4. Confirm the first picking location and item by selecting **OK**.

   - The quantity of the pick will be the total of the item displayed on the sales orders in the cluster.

5. Enter the position name (Numeric or Alphabetical) to confirm that the item quantity picked for the position were put in the correct position.
6. Repeat this process until all item quantities have been picked and put in the correct position.
7. The last step on the mobile device is to **Put** the cluster in the final location. Select **OK**
    - When the put operation is confirmed, the cluster is closed and broken, based on the value that you set for the **Break cluster at** field in the cluster profile. Work IDs are also closed.
8. A "Cluster complete" message is shown on the mobile device.

Last updated on 10/03/2025

Over-picking for sales orders and transfer orders

> **Note**
>
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This article presents a scenario that shows how to enable either a specific worker or all workers to over-pick. The over-picking process allows for controlled over-picking during the picking work.

Warehouse over-picking is a simple concept. The system allows workers to pick more items than are specified for an order. However, it still considers the overdelivery limit that is set at the line level for the transfer order or sales order. If that limit is exceeded, the Warehouse Management app notifies the workers that they are exceeding the overdelivery limit.

The over-pick feature helps minimizes maintenance and also helps your setup remain flexible. You can define one or more mobile device menu items and enable over-picking for only some of them. You can also prevent selected workers from over-picking sales orders and/or transfer orders without having to change the menu items. Instead, you can turn off one or both of those capabilities in the relevant worker setups.

The over-pick feature can help workers save time and effort when they pick and ship items. For example, the feature enables workers to perform these tasks:

- Compensate for shrinkage during picking or shipment.
- Avoid having to unpack some packaging material during the picking process.
- Compensate for item damage during transportation.
- Ship a quantity or unit-of-measure variance.
- Minimize breaking of quantities on license plates.
- Avoid material waste and scarcity of expensive materials.
- Measure the picked quantity after picking (for example, through lorry weighting).

> **Important**
>
> The over-pick feature only applies when picking and processing sales orders, transfer orders, production order raw materials, and batch order raw materials. Replenishment

doesn't support over-picking. When replenishment work is run, the system won't allow workers to over-pick.

This scenario in this article shows how to set up and use the over-pick feature.

Scenario prerequisite: Make demo data available

The scenario in this article references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to *USMF* before you begin.

Scenario setup

Before you work through the example scenario, you must enable over-picking for both the relevant worker and the relevant menu item.

Set up a worker to allow for over-picking

Here is the general procedure for configuring a worker to enable over-picking for sales orders and transfer orders separately. This configuration controls which picking worker can do the over-picking, and whether that worker can do over-picking for both transfer orders and sales orders.

1. Go to **Warehouse management** > **Setup** > **Worker**.
2. In the list pane, select **Julia Funderburk**.
3. On the **Users** FastTab, select the row that has the following values. If no existing row has these values, create it.
    - **User ID:** *24*
    - **User name:** *WH 24*
    - **Default warehouse:** *24*
    - **Menu name:** *Main*
4. On the **Work** FastTab, set the following values for user *24* if they aren't already set:
    - **Allow sales over picking:** *Yes*
    - **Allow transfer order over picking:** *Yes*

Set up a mobile device menu item to allow for over-picking

Here is the general procedure for configuring a mobile device menu item to enable over-picking. Depending on your business requirements for the permission level of the worker who will use the mobile device menu, some parameters might be turned on or off.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. In the list pane, select the record that is named *Sales Picking*. If no existing record has this name, create it. Confirm or set the following values for the record:
    - **Menu item name:** *Sales Picking*
    - **Title:** *Sales Picking*
    - **Mode:** *Work*
    - **Use existing work:** *Yes*
    - **Directed by:** *User directed*
    - **Override target license plate:** *Yes*
    - **Override license plate during put:** *Yes*
    - **Keep work locked by user:** *Yes*
    - **Allow over pick:** *Yes*

> ⓘ **Important**
>
> After the relevant parameters for both the worker and the mobile device menu item are enabled, over-picking can be processed only through the mobile device.

Example scenario

After the prerequisites, worker setup, and menu item setup are in place, you're ready to work with the feature.

Create a sales order that allows for overdelivery

1. Go to **Sales and Marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create a new sales order.
3. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-001*
    - **Warehouse:** *24*

4. Select **OK**.
5. The new sales order is opened. On the **Sales order lines** FastTab, add a line that has the following settings:
    - **Item number:** *A0001*
    - **Quantity:** *10*
6. On the **Line details** FastTab, on the **Delivery** tab, set the **Overdelivery** field to *10*.
7. On the **Sales order lines** FastTab, select **Inventory** > **Reservation**.
8. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the inventory.
9. Close the **Reservation** page.
10. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

When the release is completed, you receive informational messages that show the wave and load IDs that were created.

Get the work ID and license plate number for the new order

When you released the sales order to the warehouse, the system should have created a new work ID that has one pick line. Follow these steps to find the work ID and license plate assignment.

1. Go to **Warehouse management** > **Work** > **Work details**.
2. In the **Overview** grid, search the **Order number** column for the sales order that you just created. For that sales order, make a note of the corresponding work ID.
3. Select the row for the new sales order to show related information in the **Lines** grid. Make a note of the location that the item will be picked from.
4. Go to **Inventory management** > **Inquiries and reports** > **On-hand list**.
5. On the Action Pane, select **Dimensions**.
6. In the **Dimension display** dialog box, make sure that the **License plate**, **Warehouse**, and **Item number** checkboxes are selected, and then select **OK**.
7. In the **Filter** pane, set the following filters:
    - **Item number** – **is one of** – *A0001*
    - **Warehouse** – **begins with** – *24*

8. Select **Apply**.
9. Make a note of the **License plate** values that are shown.

Over-pick the order by using the Warehouse Management mobile app

1. Sign in to the Warehouse Management mobile app as a user in warehouse *24*.
2. Go to **Outbound** > **Sales Picking**.
3. In the **Scan work ID or license plate** field, enter the work ID that was created for the sales order.
4. Select **OK** (check mark symbol).
5. Select **Over pick**.
6. Set the **Pick Qty** field to *14*.
7. Select **OK** (check mark symbol).

   On the **Over pick** page, you receive the following message: "Overdelivery of line is 40.00 percent, but the allowed overdelivery is only 10.00 percent." This message indicates that the pick quantity that you entered exceeds the overdelivery limit. The overdelivery limit for the sales order line is 10 percent. Therefore, for a specified quantity of 10, you can over-pick by only 1, for a total pick quantity of 11.
8. Set the **Pick Qty** field to *11*.
9. Select **OK** (check mark symbol).
10. In the **License plate** field, enter the license plate that you made a note of in the previous section.
11. In the **Target license plate** field, enter a target license plate. Notice that the picking location (*FLOOR-001*), item (*A0001*), and quantity (*11 pcs*) are shown.
12. Select **OK** (check mark symbol).
13. Review the information on the **Sales orders - Put** page. The **Loc** field should indicate that the picked items are going to the *BAYDOOR* location.
14. Select **OK** (check mark symbol).

On the **Scan a work ID / license plate ID** page, you receive a "Work Completed" message. This message indicates that the work ID of the sales order has been completed, and the over-picked quantity doesn't exceed the overdelivery limit of 10 percent.

Last updated on 05/07/2025

Over-pick materials for production and batch orders

09/19/2025

This article explains how warehouse managers can configure Microsoft Dynamics 365 Supply Chain Management to allow and set limits on over-picking of raw materials. Over-picking occurs when a worker picks slightly more material than is specified for a production order. Over-picking can be relevant in situations where it's more efficient for workers to over-pick by rounding up to the nearest packing unit, or where it's more efficient to pick the full content of nearly empty locations to free up space.

This feature lets workers who use the Warehouse Management mobile app over-pick raw materials for production orders when they must. Workers are notified if they exceed the over-picking thresholds that the warehouse manager sets.

> **Important**
>
> - The over-pick feature only applies when picking and processing sales orders, transfer orders, production order raw materials, and batch order raw materials. Replenishment doesn't support over-picking. When replenishment work is run, the system doesn't allow workers to over-pick.
> - All products that use this feature must be enabled for warehouse management processes (WMS).

Set up over-picking for production and batch orders

Use the following procedures to set up over-picking for production and batch orders.

Enable over-picking for a worker

Follow these steps to enable over-picking for a worker.

1. Go to **Warehouse management** > **Setup** > **Worker**.
2. Select the worker that you want to enable the feature for.
3. On the **Users** FastTab, select the user account that should be allowed to over-pick.
4. On the **Work** FastTab, set the **Allow production over picking** option to *Yes*.

Enable over-picking for a mobile device menu item

Follow these steps to enable over-picking for a mobile device menu item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select a mobile device menu item in the list pane, or create a new one. For the new or selected item, the grid on the **Work classes** FastTab must include a row where the **Work order type** field is set to *Raw material picking*. (If you're working with USMF demo data, you can select the mobile device menu item that's named *Production pick*.)
3. On the **General** FastTab, set the **Allow over pick** option to *Yes*.

Set an allowed threshold for over-picking

For each product, you can define an over-picking threshold to control the extent to which a worker can over-pick the product. You can override the threshold for products that are included in a bill of materials (BOM). When you create a production order for a product, the over-picking threshold for each item on that product's BOM comes from either the item itself or the override value set in the product BOM (if available). The threshold is defined as a percentage of the quantity that's designated for the production order.

Set the over-picking threshold for a product

Follow these steps to set the over-picking threshold for a product.

1. Go to **Product Information management** > **Products** > **Released products**.
2. Open the product that you want to set the over-picking threshold for.
3. On the **Warehouse** FastTab, set the **Material over picking (%)** field to the over-picking threshold that you want to allow.

Override the over-picking threshold on a BOM line

When you add a product to a BOM line, the system initially copies that product's over-picking threshold to the BOM line. Follow these steps to override the over-picking threshold for a specific BOM line.

1. Go to **Product Information management** > **Bill of materials and formulas** > **Bill of materials**.
2. Open the BOM where you want to override the over-picking threshold.
3. On the **Bill of material lines** FastTab, select the BOM line that you want to override the over-picking threshold for.

4. On the **Line details** FastTab, on the **Setup** tab, set the following fields:

    - **Override** – Set this option to *Yes*.
    - **Material over-picking (%)** – Enter the new over-picking threshold. When this BOM line is used in a production order, the threshold that you set here overrides the default threshold that's set on the product.

Set policies for staging and over-picking

Warehouse managers can set up policies that define how material should be allocated when it's picked from license-plate-controlled and non-license-plate-controlled locations.

Set the policy for over-picking in license-plate-controlled locations

Follow these steps to set the policy for over-picking in license-plate-controlled locations.

1. Go to **Product Information management** > **Products** > **Released products**.

2. Select the product that you want to set up.

3. On the **Warehouse** FastTab, set the **Material picking in license plate controlled locations** field to one of the following values:

    - *Staging* – The full quantity of the scanned license plate is staged to the production input location. The original estimated quantity is allocated to the production order, and the remaining quantity on the license plate is left unallocated at the production input location. When you use this policy, the over-picking option isn't available in the workflow for raw material picking on the Warehouse Management mobile app.
    - *Order picking* – The full picked quantity is allocated to the production order.

Set the policy for over-picking in non-license-plate-controlled locations

Follow these steps to set the policy for over-picking in non-license-plate-controlled locations.

1. Go to **Product Information management** > **Products** > **Released products**.

2. Select the product that you want to set up.

3. On the **Warehouse** FastTab, set the **Material picking in non-license plate controlled locations** field to one of the following values:

- *Staging* – The original estimated quantity is allocated to the production order, and the remaining quantity is left unallocated at the production input location.
- *Order picking* – The full picked quantity is allocated to the production order.

Set different dimensions for packing and storage

Some items are packed or stored in such a way that you need to track physical dimensions differently for each of several different processes. The *Packaging product dimensions* feature lets you set up one or several types of dimensions for each product. Each dimension type provides a set of physical measurements (weight, width, depth, and height), and establishes the process where those physical measurement values apply. When you enable this feature, your system supports the following types of dimensions:

- *Storage* - Storage dimensions work with location volumetrics to determine how many of each item can be stored in various warehouse locations.
- *Packing* - Packing dimensions work during containerization and the manual packing process to determine how many of each item fits in various container types.
- *Nested packing* - Nested packing dimensions work when the packing process contains multiple levels.

Set up *storage* dimensions by using the **Physical dimension** page in Supply Chain Management. All processes use these dimensions when the packing and nested packing dimensions aren't specified.

Set up *packing* and *nested packing* dimensions by using the **Physical product dimensions** page.

This article provides a scenario that illustrates how to use this feature.

Example scenario

Set up the scenario

Before you run the example scenario, prepare your system as described in this section.

Enable demo data

To work through this scenario using the demo records and values that are specified here, use a system where the standard demo data is installed. Also, select the *USMF* legal entity before you begin.

Add a new physical dimension to a product

Add a new physical dimension for a product by following these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product with **Item number** *A0001*.
3. On the Action Pane, open the **Manage inventory** tab. From the **Warehouse** group, select **Physical product dimensions**.
4. The **Physical product dimensions** page opens. On the Action Pane, select **New** to add a new dimension to the grid with the following settings:

    - **Physical dimension type** - *Packing*
    - **Physical unit** - *pcs*
    - **Weight** - *4*
    - **Weight unit** - *kg*
    - **Depth** - *3*
    - **Height** - *4*
    - **Width** - *3*
    - **Length** - *cm*
    - **Volume unit** - *cm3*

The **Volume** field is automatically calculated based on your **Depth**, **Height**, and **Width** settings.

Create a new container type

Go to **Warehouse management** > **Setup** > **Containers** > **Container types** and create a new record with the following settings:

- **Container type code** - *Short Box*
- **Description** - *Short Box*
- **Maximum net weight** - *50*
- **Volume** - *144*
- **Length** - *6*
- **Width** - *6*
- **Height** - *4*

Create a container group

Go to **Warehouse management** > **Setup** > **Containers** > **Container groups** and create a new record with the following settings:

- **Container group ID** - *Short Box*

- **Description** - *Short Box*

Add a new line to the **Details** section. Set the **Container type** to *Short Box*.

Set up a container build template

Go to **Warehouse management** > **Setup** > **Containers** > **Container build templates** and select **Boxes**. Change the **Container group ID** to *Short Box*.

Run the scenario

After you prepare your system as described in the previous section, you're ready to run the scenario as described in the next section.

Create a sales order and create a shipment

In this process, you create a shipment based on the item *packing* dimensions, for which the height is less than 3.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In **Create sales order**, set the following values:
    - **Customer account:** *US-001*
    - **Warehouse:** *63*
4. Select **OK** to create the sales order and close the dialog box.
5. The new sales order opens. It includes a new, empty line in the grid on the **Sales order lines** FastTab. On this line, set the following values:
    - **Item number:** *A0001*
    - **Quantity:** *5*
6. On the **Sales order lines** FastTab, select **Inventory** > **Reservation**.
7. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the inventory.
8. Close the page.
9. On the Action Pane, open the **Warehouse** tab and select **Release to warehouse** to create work for the warehouse.

10. On the **Sales order lines** FastTab, select **Warehouse** > **Shipment details**.

11. On the Action Pane, open the **Transportation** tab and select **View containers**. Confirm that the item was containerized into the two *Short Box* containers.

Place an item into storage

1. Open the mobile device, sign in to warehouse 63, and go to **Inventory** > **Adjust In**.
2. Enter **Loc** = *SHORT-01*. Make a new license plate with **Item** = *A0001* and **Quantity** = *1 pcs*.
3. Select **OK**. You receive the error "Location SHORT-01 failed because item A0001 doesn't fit in location's specified dimensions." This error occurs because the *Storage* type dimensions of the product are larger than the dimensions specified on the location profile.

Last updated on 05/29/2026

Batch, serial, license plate, and location confirmation

Article • 02/24/2025

Batch confirmation allows workers using a mobile device to confirm that the correct batch is being picked. For *Batch-above[location]* items (where batch-above indicates that batch is placed higher than location in the search hierarchy) the worker must verify that the batch that is picked matches the batch on the work line.

Serial confirmation allows workers using a mobile device to confirm that the correct serial number is being picked. For *Serial-above[location]* items (where the serial-above indicates that serial is placed higher than location in the search hierarchy) the worker must verify that the serial that is picked matches the serial on the work line.

License plate confirmation allows workers using a mobile device to confirm that the correct license plate is being picked. When picking work from a stage location, the worker must verify that the license plate picked matches the license plate that is associated with the work. If the work is started by scanning a license plate, this confirmation step is skipped.

Location confirmation allows workers using a mobile device to confirm that the correct location is being used when performing any of the work types available. During these tasks, the worker must verify that the location matches the one associated with the work. The worker is prompted to enter the check digit for the location, but if no check digit is associated with the location, this step is skipped.

Where it applies

Confirmation applies in the following scenarios:

- Batch confirmation applies to the initial pick work for *Batch-above[location]* items.
- Serial confirmation applies to the initial pick work for *Serial-above[location]* items.
- License plate confirmation applies to picks from stage locations.
- Location confirmation only applies when the location is filled automatically by the system.

> **ⓘ Important**
>
> Replenishment isn't supported for license plate confirmation. When executing replenishment work, no license plate confirmation step is created.

For location confirmation, the location confirmation step is skipped if the location is manually entered on the mobile device. For example, during the manual inventory movement process, the location isn't confirmed even if it's enabled in the menu item.

> **Important**
>
> If the entire quantity from a location or license plate is picked, the worker isn't asked to confirm the batch or serial numbers, regardless of how many batch or serial numbers are available on hand. This is standard tracking behavior that can't be overridden. For more information, see Warehouse mobile device tracking dimensions and license plate picking behavior.

Set up batch, serial, license plate and location confirmation

You can configure batch, serial, license plate, and location confirmation from the **Mobile device menu items** page.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the list pane, select the menu item that you want to set up.
3. On the Action Pane, select **Work confirmation setup**.
4. The **Work confirmation setup** page opens.
5. To add rows for each work type, use the buttons on the Action Pane. Select the check box for each type of confirmation required by the current menu item.
    - For rows with a **Work type** of *Pick*, you can only select the batch, serial, license plate confirmation options, and location confirmation.
    - For rows with any other **Work type**, you can select the location confirmation option, but the batch, serial, and license plate confirmation options aren't available.

> **Important**
>
> If the **Auto submit** option is enabled, the other options can't be activated.

Feedback

Was this page helpful? Yes No

Provide product feedback

Warehouse mobile device tracking dimensions and license plate picking behavior

One of the design goals of the warehouse mobile device picking flows is to minimize the amount scanning and keying required from the worker. To this end, some fields on the mobile device are either prepopulated value or skipped entirely. This article explains some of these cases. Some behaviors differ based on which warehouse mobile device flow you're using and how your on-hand inventory looks in terms of license plates and tracking dimension combinations.

Picking and target license plates

When a worker picks an item, the system moves that inventory to the *user* location. The user location must always be license plate tracked. Several factors influence the system behavior. For example, whether the picking location is license plate tracked, whether the used mobile device menu item has the **Generate license plate** and **Override target license plate** options enabled, and so on. After a target license plate is chosen, it usually can't be changed and it continues to be used for all the subsequent pick work lines from the same work record. The entered target license plate doesn't need to exist in the system before use. If the relevant record didn't exist before, the system creates it. You can use an existing license plate as a target license plate provided it doesn't have any on-hand inventory on it. You can't have the same license plate with on-hand inventory at two different warehouse locations.

If a worker picks from a location that isn't license plate tracked, the worker must enter the target license plate. The system can also automatically generate this target license plate, provided the mobile device menu item being used has the **Generate license plate** option enabled. In that case, the worker is prompted to confirm the generated license plate or to choose a new one. The **Override target license plate** setting isn't used in this case.

If a worker picks from a location that is license plate tracked, the behavior depends on how much of the total license plate quantity is picked.

- If the worker picks the full quantity from the license plate, that license plate is automatically used as the target, and the worker isn't asked to enter or confirm the target license plate. You can change this behavior by using the **Override target license plate** setting on the

mobile device menu item. If that setting is enabled, the worker can change the target license plate. The **Generate license plate** setting isn't used in this case.

- If the worker should only pick part of quantity from the license plate, a different target license plate is needed. This case is basically the same as the one when picking is performed from a location that isn't license plate tracked.

Tracking dimensions (batch number and serial number)

When an item has tracking dimensions enabled (batch number or serial number), the system usually needs to know exactly which batch or serial number is picked, regardless of the reservation hierarchy used. For batch numbers, this requirement is always true, but for serial numbers, it can depend on the **Capture serial number** setting on the item's **Tracking dimension group**. If **Capture serial number** is set to *Picking*, the worker must use the mobile device to capture the serial number during the picking process, even though blank receipt and issue are allowed. If **Capture serial number** is set to *Packing*, the worker captures the serial number later, during the packing process, instead of when picking. If the field is set to *None*, which is the default setting, the system behaves in the standard way, which is described in the next paragraph. A special case occurs when **Active in sales process** is enabled. In this case, serial numbers aren't saved in the inventory transactions, but in a separate table. This behavior is similar to the *Picking* and *Packing* ones described previously.

You can't change the standard tracking behavior described in this paragraph by using configuration settings. It's the same for both batch and serial numbers (provided the serial number doesn't have **Active in sales process** enabled and **Capture serial number** is set to *None*). If the whole quantity from a location or license plate is picked, the worker isn't asked to enter batch or serial numbers, regardless of how many batch or serial numbers are available on-hand. As mentioned previously, this behavior is designed to minimize the number of interactions required from the worker on the mobile device. In case the worker should only pick part of the quantity available at a location or license plate, the worker is prompted to enter each picked batch and/or serial number because the system can't otherwise know which of the available numbers were picked.

Last updated on 05/29/2026

Create transfer orders from the warehouse app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This feature enables warehouse workers to create and process transfer orders directly from the Warehouse Management mobile app. The worker starts by selecting the destination warehouse. Then, they can scan one or more license plates by using the app to add license plates to the transfer order. When the warehouse worker selects **Complete order**, a batch job creates the required transfer order and order lines based on the on-hand inventory registered for those license plates.

Set up a mobile device menu item to create transfer orders

Here are general guidelines for setting up a mobile device menu item for creating a transfer order. Depending on your business requirements for the level of automation to set when users create transfer orders from the floor, different configurations are enabled. This scenario describes one such configuration.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select **New** to add a new menu item. Then make the following settings to get started:
    - **Menu item name** – Assign a name as it should appear in Supply Chain Management.
    - **Title** – Assign a menu name as it should be presented to workers in the Warehouse Management mobile app.
    - **Mode** – Set to *Indirect* (this menu item doesn't create work).
    - **Activity code** – Set to *Create transfer order from license plates* to enable the warehouse workers to create a transfer order based on one or more scanned license plates.

3. Use the **Transfer order line creation policy** setting to control how transfer order lines are created by this menu item. The lines are created or updated based on the on-hand inventory registered for the scanned license plates. Choose one of the following values:

    - *No reservation* – The transfer order lines aren't reserved.
    - *License plate guided with line reservation* – The transfer order lines are reserved and use the License plate guided strategy option, which stores the relevant license plate IDs associated with the order lines. Located license plate ID values can therefore be used as part of the work creation process for the transfer order lines.

4. Use the **Outbound shipment policy** setting to add more automation to the outbound transfer order shipment process, as needed. When a worker selects the **Complete order** button, the app creates the *Complete order* warehouse app event, which saves the value you choose here in the **Outbound shipment policy** field for each line in the current transfer order. Later, when the event queue is processed by a batch job to create the transfer order, the batch job reads the value stored in this field and might control how that job processes each line. Choose one of the following values:

    - *None* – No automated processing is done.
    - *Release to warehouse* – Automates the release to warehouse process.
    - *Ship confirm* – Automates the ship confirmation process.
    - *Release and ship confirm* – Automates both the release to warehouse and ship confirmation processes.

5. Use the **Validate license plate policy** setting to decide whether the system should validate the license plate when creating a transfer order. Choose one of the following values:

    - *None* – Don't validate the license plate.
    - *Error* – Check whether the selected license plate is a valid target for open or in-process work and whether it contains any items with blocking status.

> ⓘ **Note**
>
> The **Validate license plate policy** setting is intended to help speed up the process, which is why it's set to skip license plate validation by default (*None*). The *None* value assumes that the warehouse worker knows whether the license plate can be transferred to another warehouse.

Add the mobile device menu item to a menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. Select **Edit**.
3. Select an existing menu following selection of the new menu item under **Available menus and menu items**. Add the menu item by selecting the right-arrow button.

Create a transfer order based on license plates

The Warehouse Management mobile app has a simple process for creating transfer orders based on license plates. To do this, the worker follows these steps by using the Warehouse Management mobile app:

1. Create the transfer order and identify the destination warehouse.
2. Identify each license plate to ship.
3. Select **Complete order**.

> **Note**
>
> Multiple workers can assign license plates intended for the same transfer order by using the **Select transfer order** button to select an existing, unprocessed, transfer order number from the warehouse app event queue. For information about how to find the transfer order number values, see Inquire into the warehouse app events.

Example scenario

This scenario provides an overview of the process for getting transfer orders created and automatically processed based on the on-hand inventory registered on the selected license plates.

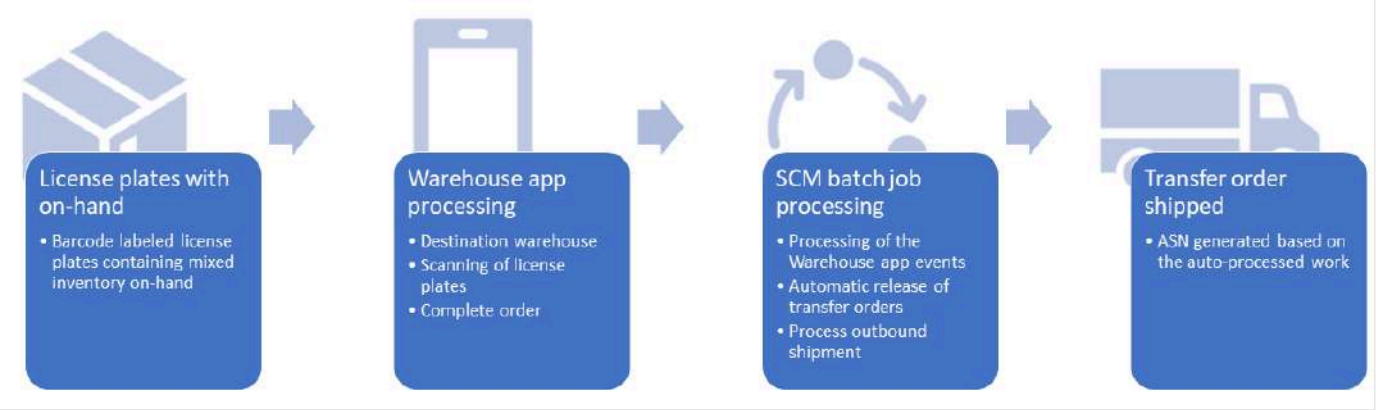
To work through this scenario by using the values suggested, you must work on a system with demo data installed and select the *USMF* legal entity before you begin.

This scenario assumes that you already set up the warehouse app event processing capability.

In addition to setting up the create transfer order in the mobile device menu items, additional templates, location directives, and batch jobs must also be set up and enabled.

Example scenario blueprint

You're a retailer and have multiple license plates, each containing a mix of items placed at a specific location within one of your warehouses (*Warehouse 51*). You want to enable the process that allows workers to create a transfer order to another warehouse (*Warehouse 61*) for a collection of scanned license plates. You automatically ship-update the transfer order as soon as the last license plate for the order is identified.



Create a mobile device menu item for creating transfer orders

This section explains how to create a new mobile device menu item for creating transfer orders. Set the **Mode** to *Indirect* and the **Activity code** to *Create transfer order from license plates*.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select **New**.
3. In the **Menu item name** field, enter the name *Create TO*.
4. In the **Title** field, enter the description *Create TO*.
5. In the **Mode** field, select *Indirect*.
6. In the **Activity code**, select *Create transfer order from license plates*
7. In the **Order line creation policy**, select *License plate guided with line reservation*.
8. In the **Outbound shipment policy**, select *Release and ship confirm*.
9. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
10. Select **Edit**.
11. Select the existing **Inventory** menu. Under **Available menus and menu items**, select the new menu item. Add the menu item into the **Inventory** menu by selecting the right-arrow button.

Set up work templates to auto process and break work by located license plate

This section explains how to enable a work template to automatically process the work created by the template when a wave is released.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. In the **Work order type** field, select *Transfer issue*.
3. Select **New** to create a new work template.
4. In the **Work template** field, enter *51 Auto process LP*.
5. In the **Work template description** field, enter *51 Auto process LP*.
6. Select the **Automatically process** check box. Select this check box to enable automation steps.
7. In the demo data, a work template named *51 Transfer* already exists. Edit the **Sequence number** field so that the new work template has a lower sequence number than the existing work template *51 Transfer*.
8. Select **Save** in the toolbar to enable the **Work Template Details** FastTab.
9. In the **Work Template Details** FastTab, select **New** in the toolbar. You add two lines.
10. In the **Work type** field, select *Pick*.
11. In the **Work class ID** field, select *TransfOut*.
12. Select **New** in the **Work Template Details** toolbar.
13. In the **Work type** field, select *Put*.
14. In the **Work class ID** field, select *TransfOut*.
15. Select **Save** to enable the **Directive code** field.
16. On the **Work type** *Put* line, select **Directive code** *Baydoor*. Make sure this new work template gets the lowest **Sequence number**.
17. In the toolbar, select **Edit query** to open the query editor.
18. In the **Range** tab, select **Add**.
19. On the line added, in **Field** select *Warehouse*.
20. In the **Criteria** field, select *51*.
21. Select the **Sorting** tab.
22. Select **Add** and set **Field** to *Located license plate ID*. Selecting this field enables the toolbar button **Work header breaks**.
23. Select **OK** followed by **Yes** to reset the grouping and return to the **Work templates** page.
24. Select **Work header breaks** and enable the **Group by this field** for the **Located license plate ID** and close.

> ⓘ **Note**
>
> Not all setup can be auto processed. For example, catch weight items and the use of mixed tracking dimensions.

Set up location directives for the license plate guided strategy

This section explains how to set up a location directive pick process to use the **License plate guided** strategy.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. Select **Edit**.
3. In the navigation list header, select the **Work order type** *Transfer issue*.
4. In the navigation list, select the existing location directive **51 TO Pick**.
5. In the **Lines** FastTab, select the **Allow split** checkbox.
6. In the **Location Directive Actions** FastTab select **New** to add a new action line.
7. In the **Name** field, enter *LP Guided*.
8. In the **Strategy** field, select *License plate guided*. This action needs the lowest sequence number.
9. Select **Save** in the toolbar.
10. Select the **Refresh** page icon from the toolbar.
11. In the **Location Directive Actions** FastTab, select the line *TOPick*.
12. In the **Location Directive Actions** toolbar, select **Move down** to change the sequence number to be greater than the sequence number for the *LP Guided* action just created.

> ⓘ **Note**
>
> The License plate guided strategy tries to reserve and create picking work against the locations that hold the requested license plates and are associated with the transfer order lines. If this isn't possible and you still want to create picking work, fall back to another location directive action strategy. You might also need to search for inventory in another area of the warehouse, depending on your business process needs.

Set up a batch job to process warehouse app events

This section explains how to set up a scheduled batch job to process warehouse app events.

1. Go to **Warehouse management** > **Periodic tasks** > **Process warehouse app events**.
2. In the dialog box, enable **Batch processing** under the **Run in background** section.
3. Select **Recurrence** and set up the batch job to process based on the interval needed for your business.
4. Select **OK** to return to the main dialog.
5. Select **OK** in the main dialog to add the job to the batch queue.

Set up a batch job to release transfer orders automatically

This section explains how to set up a scheduled batch job to release the transfer orders that are marked as *Ready to release*.

1. Go to **Warehouse management** > **Release to warehouse** > **Automatic release of transfer orders**.
2. In the dialog box, expand the **Records to include** section.
3. Select **Filter** under the **Records to include** section.
4. In the **WHSTransferAutoRTWQuery** query page, **Range** tab, select **Add** to add a new line to the query.
5. In the new line **Table** field, select the drop-down menu and select the table **Transfer line release to warehouse**.
6. In the **Field** drop-down menu, select **Outbound shipment policy**.
7. In the **Criteria** field, select **Release and ship confirm**.
8. In the line where **Field** is set to *From warehouse*, in the **Criteria** field, select *51*.
9. Select **OK** to return to the main dialog box.
10. Expand the **Run in the background** section to set up batch processing.
11. Enable **Batch processing** under the **Run in background** section.
12. Select **Recurrence** and set up the batch job to process based on the interval needed for your business.
13. Select **OK** to return to the main dialog.
14. Select **OK** in the main dialog to add the batch job to the batch queue.

Set up the "Process outbound shipment" batch job

This section explains how to set up a scheduled batch job to run the outbound shipment confirmation for loads ready to ship related to transfer order lines that are marked as *Ready to ship*.

1. Go to **Warehouse management** > **Periodic tasks** > **Process outbound shipments**.
2. Expand the **Records to include** section.
3. Select **Filter**.
4. In the **WHSLoadShipConfirm** query, select the **Joins** tab.
5. Expand the table hierarchy so that **Loads** and **Load details** are expanded.
6. Select the **Load details** table.
7. Select the **Add table join** button.
8. In the list of table relations, filter or search on the **Relation** column for *Transfer order lines (Reference)*.
9. Focus on the table relation in the list, and then select the **Select** button.

10. Select the **Transfer order lines** table.
11. Select the **Add table join** button.
12. In the list of table relations, filter or search on the **Relation** column for *Invent Transfer Additional Fields (Record-ID)*.
13. Focus on the table relation in the list, and then select the **Select** button.
14. Select the **Range** tab.
15. In the **Range** query tables, you'll set up three query criteria ranges. Select the **Add** button to add a line.
16. Add a range for the table **Loads**. Set **Field** to *Load status* and set **Criteria** to *Loaded*.
17. Add another range for the table **Invent Transfer Additional Fields**. Set **Field** to *Outbound shipment policy* and set **Criteria** to *Release and ship confirm*.
18. Add another range for the table **Load details**. Set **Field** to *Reference* and set **Criteria** to *Transfer order shipment*.
19. Select **OK** to return to the main dialog box.
20. Expand the **Run in the background** section.
21. Enable **Batch processing**.
22. Select **Recurrence** and set up the batch job to process based on the interval needed for your business.
23. Select **OK** to return to the main dialog.
24. Select **OK** in the main dialog to add the batch job to the batch queue.

> **Note**
>
> For more information, see Confirm outbound shipments from batch jobs.

Processing the example for "Create transfer order from the warehouse app"

Add on-hand inventory to a license plate

For this scenario, you need a license plate that contains physical available inventory on hand.

Expand table

Item	Warehouse	Inventory status	Location	License plate	Quantity
A0001	51	Available	LP-010	LP10	1
A0002	51	Available	LP-010	LP10	2

Add physical inventory on-hand quantities by using the following values:

> **Note**
>
> You need to create the license plate and use locations that allow you to carry mixed items, such as LP-010.

Create and process transfer orders from the warehouse app

1. Open the app and sign in as user *51*. The current user warehouse is 51.
2. Select the menu item **Create TO** from the menu location you added it to during setup.
3. Start the creation of a transfer order by entering the destination warehouse (To warehouse) in the **Warehouse** field. Enter *61*. The new transfer order goes from current warehouse 51 (From warehouse) to the destination warehouse *61*.
4. Select **OK**.
5. Scan a license plate ID in the **License plate** field. Enter the license plate of the inventory added in an earlier step, *LP10*.
6. Select **OK**.
7. Select the menu button and then select **Complete order** to finalize the warehouse app transfer order creation.

For the mentioned example, two **Warehouse app events** (*Create transfer order* and *Complete transfer order*) are used.

Inquire into the warehouse app events

You can view the event queue and event messages generated by the Warehouse Management mobile app by going to **Warehouse management** > **Enquiries and reports** > **Mobile device logs** > **Warehouse app events**.

The *Create transfer order* event messages receive the status *Waiting*, which means that the **Process warehouse app events** batch job doesn't pick up and process the event messages. As soon as the event message updates to status *Queued*, the batch job processes the events. This process happens at the same time as the creation of the *Complete transfer order* event (when a worker selects the **Complete order** button on the Warehouse Management mobile app). When the *Create transfer order* event messages are processed, the status updates to *Completed* or *Failed*. When the *Complete transfer order* status updates to *Completed*, all the related events are deleted from the queue.

Because the **Warehouse app events** for the creation of transfer order data aren't processed by the batch job before the messages update to status *Queued*, you need to look up the requested transfer order numbers as part of the **Identifier** field. The **Identifier** field is in the header of the **Warehouse app events** page.

As part of the warehouse event processing, the creation of the transfer order line might fail. In this case, the state of the event message updates to *Failed* and you can use the **Batch log** information to learn why and take action to correct any problems.

Typical issues could be related to missing setup for the process, like a missing transit warehouse for the *Create transfer order* event. In an example like this, you add a transit warehouse to the shipping warehouse and use the **Reset** option to change the status for all the warehouse app event messages from *Failed* to *Queued*, which means that the batch job processes the event messages again after the correction of the setup data.

Within production environments, the exceptions are more process related, such as having a requested license plate, which at the batch job processing time is empty and thereby no transfer order lines are created. This failed event message can either be removed by using the **Delete** option or you can add the needed physical on-hand on the license plate and use the **Reset** option for all the related event messages.

Learn more in Warehouse app event processing.

Follow up on the example scenario processing

During this scenario, the following occurred:

1. Using the Warehouse Management mobile app, you selected a menu item that uses the activity code **Create transfer order from license plates**.
2. The app prompted you to select the destination warehouse for the transfer order. The source warehouse is always the one you currently sign into as a Worker.
3. On the selection of the destination warehouse, the system reserved an ID number for the upcoming transfer order (based on the transfer-order number sequence defined on your system) but didn't create the transfer order yet.
4. When you scanned the license plate *LP10* containing on-hand inventory that should be moved to the new warehouse, a **Warehouse app event** was added to the events queue to be processed later. The warehouse event contained message details about the scan, including the intended transfer-order number.
5. On the Warehouse Management mobile app when the **Complete order** button is selected, a new warehouse app event, **Complete transfer order**, is created and the related existing

event, **Create transfer order**, changes status to **Queued**.

6. On the back end, the **Process warehouse app events batch job** picks up the **Queued** event and collects the on-hand related to the scanned license plate. Based on the on-hand the actual transfer order record and associated lines get created. The job also populates the **Outbound shipment policy** field for the transfer order with the value based on the configured *Release and ship confirm* and linked the license plate against the lines for the **License plate guided** strategy.
7. Based on the transfer order line **Outbound shipment policy** field value the **Automatic release of transfer orders batch job** query now results in releasing the transfer order to the shipping warehouse. And due to the setup for the used **Wave template**, **Work template**, and **Location directives** the work gets auto processes, resulting in the **Load status** being updated to *Loaded*.
8. The **Process outbound shipment batch job** is executed for the load, resulting in the transfer order being shipped and the Advance Shipment Notice (ASN) is generated.
9. The timing of all these events depends on the **Recurrence** settings for the batch jobs created.

Frequently asked questions

Warehouse Management mobile app processes

Why can't I see the menu button "Complete order"?

You must have at least one license plate assigned to the transfer order.

Can several Warehouse Management mobile app users add license plates to the same transfer order at the same time?

Yes, several warehouse workers can scan license plates into the same transfer order.

Can the same license plate be added to different transfer orders?

No, a license plate can only be added to one transfer order at a time.

After I select the "Complete order" button, can I add more license plates for that transfer order?

No, you can't add more license plates to a transfer order that has a **Complete transfer order** warehouse app event.

How can I find existing transfer orders to use through the "Select transfer order" button in the Warehouse Management mobile app, if the order isn't yet created in the backend system?

By using the data inquiry capability, workers can look up transfer order numbers in the Warehouse Management mobile app. For example, you could create a detour mobile device menu item that queries for the data displayed on the web client's **Warehouse app events** page (`WHSMobileDeviceQueueMessageCollection`) as part of the *Select order - MobileDeviceQueueMessageCollectionIdentifierId* step. The transfer order number matches the value shown in the **Identifier** field. To learn more, see Inquire into the warehouse app events.

Can I manually select the transfer order number to use from the Warehouse Management mobile app?

The app supports only autogenerated transfer order numbers through number sequences. See also the answer to the previous question regarding how to set up the **Select transfer order** button. For more information about how to find transfer order numbers, see Inquire into the warehouse app events.

Background processing

How should I clean up records in my warehouse app events queue message tables?

You can view and maintain these records on the **Warehouse app events** page. For more information, see Inquire into the warehouse app events.

Why isn't the transfer order "Receipt date" updated according to my "Delivery date control" setup?

The transfer orders are created without using the **Delivery date control** capabilities.

Can I use a license plate that has physical negative inventory on hand?

This feature supports only positive physical on-hand quantities at the license plate level. However, you can have physical negative on-hand quantities at the higher warehouse and inventory-status levels when you assign license plates to transfer orders.

Last updated on 05/06/2026

Piece picking confirmation

08/18/2025

Piece picking confirmation allows workers to use a mobile device to confirm each piece of inventory registered during picking or counting work. In this context, *confirm* means that the worker verifies the quantity of each item by scanning a bar code or manually entering the information on the device. For picks, workers can confirm the quantity of work to be processed up to the quantity that is specified on the work to be picked. For counting work, workers can scan the inventory they're counting and track the total amount.

When you enable piece picking, product confirmation is also required. For picking work, you can set the maximum number of pieces that workers can be required to confirm while doing picking work. The maximum quantity is based on the current work unit that is being processed. The counting work type doesn't allow a maximum.

The system considers both the quantity and unit of measure (UOM) that is stored in a scanned bar code. This functionality works for receiving inbound flows (including mixed license plate receiving, purchase order item, transfer order item, and load item). It also works for piece picking, where scanning a bar code quantity adds the quantity to the total number of confirmed pieces and converts between the UOM on the bar code and the work unit. When a worker scans a bar code quantity, the system checks the UOM on the bar code, confirms whether the quantity is allowed for counting on the sequence group and, if so, adds the quantity to the total count.

Where piece picking applies

Piece picking works for all counting work and for the initial pick for any type of work. Piece picking doesn't apply for items that are controlled by serial numbers or for production or kanban picks from a license plate location where the item is set to staging.

Ways to confirm a product

There are two ways to confirm a product during piece picking:

- **Scanning a bar code** – Workers can scan a bar code, which can include quantity information depending on the configuration. If the bar code specifies a quantity, the system automatically adds that amount to the total confirmed pieces. If the bar code doesn't include quantity information, the worker is prompted to enter the quantity manually.

- **Manual entry** – Workers can manually enter the item number to confirm the product. In this case, they must also manually enter the quantity being picked or counted. This method is useful when bar codes aren't configured.

Enable bar code scanning for products

To allow workers to scan bar codes to confirm a product, make sure the **Scanning** option is enabled for each relevant product. To enable the **Scanning** option, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. In the list, find and select the desired product.
3. On the Action Pane, open the **Manage inventory** tab and, from the **Warehouse** group, select **Bar codes**.
4. On the Action Pane, select **Edit**.
5. Select the desired bar code.
6. Open the **General** tab.
7. Set **Scanning** to *Yes*.
8. On the Action Pane, select **Save**.

Learn more in Create a bar code for a product.

Set up piece picking

To set up piece picking, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Find and select the mobile device menu item that you want to configure.
3. On the Action Pane, select **Work confirmation setup**.
4. Use the buttons on the Action Pane, to add, remove, or edit confirmation settings for each work type as needed. For rows with a **Work type** of *Pick* or *Counting*, make the following settings:

    - **Piece picking confirmation** – Select this check box to allow workers to confirm each piece of inventory from the mobile device. While this box is selected, the **Product confirmation** checkbox is also selected automatically.
    - **Maximum number of pieces** – Enter the maximum number of pieces that workers can be required to confirm while doing picking work. For example, if the **Maximum number of pieces** is set to *1*, the system will prompt the worker to confirm only one product. After confirming the first piece, the worker can manually enter the remaining quantity without additional confirmation prompts. Set to *0* to remove the maximum limit, which means that workers will be prompted to scan or enter values

as many times as needed until the entire picking quantity has been confirmed. This setting is only available when **Work type** is *Pick* and **Piece picking confirmation** is enabled.

Packing materials and fees

Article • 06/03/2022

Packing material fees are paid to a recycling company at specific intervals. An amount is paid, per unit of weight, for each material that a packing unit consists of. Although packing material fees are calculated and reported, no ledger transactions are posted, because the fees aren't considered taxes that must be paid to an authority.

Packing material weights and fees are calculated for sales order lines and purchase order lines.

You can define one or more packing units for a single item, a group of items (packing group), or all items. A packing unit consists of the packing materials, their weights, and the number of items that are included in the packing unit. A packing material code is assigned to each type of packing material that is defined. Based on the packing material code, you can specify a price for a specific period. The packing material fee is calculated based on this information.

> **Note**
>
> Even if your company doesn't pay packing material fees, you can use the functionality to calculate statistics for the weights of packing materials.

Set up packing material allocation

Before you can calculate packing material weights, packing material fees, or both, you must turn on the calculation and define which materials and fees apply to which items.

1. Go to **Inventory management** > **Setup** > **Inventory and warehouse management parameters**.
2. On the **General** tab, in the **Packing material** section, set the **Calculate packing material fees** option to **Yes**.
3. Go to **Inventory management** > **Setup** > **Packing material** > **Packing groups**, and create all the packing groups that you use. All the items in a packing group will use the same packing material allocation. If you don't use packing groups, you can skip this step.

   > **Tip**

After you've created your packing groups, can assign a group to each product as you require. Go to **Product information management** > **Products** > **Released products**, select a product, and then, on the **Manage inventory** FastTab, in the **Packing group** field, select the appropriate packing group.

4. Go to **Inventory management** > **Setup** > **Packing material** > **Packing material codes**, define each type of packing material that you use, and specify the unit that the packing material is consumed in when you prepare shipments.

5. Go to **Inventory management** > **Setup** > **Packing material** > **Packing material fees**, and set a fee for each type of packing material that you just defined. Fees are calculated based on the price per unit that is consumed.

6. To allocate packing materials to items, go to **Inventory management** > **Setup** > **Packing material** > **Packaging material allocation**, and create allocations. You can create as many allocations as you require. You can allocate packing materials for individual items, groups of items (packing groups), or all items, depending on how detailed your allocations should be.

   For each allocation that you create, follow these steps.

   a. On the **General** FastTab, set the following fields:

   - **Item code** – Select the scope of the allocation:
     - **Table** – Create an allocation for a single specific item.
     - **Group** – Create an allocation for all the items the belong to a packing group that is defined on the **Packing groups** page.
     - **All** – Create an allocation for all items.

     > ⓘ **Note**
     >
     > Usually, you should make all your allocations at the same level (**Table**, **Group**, or **All**). If you use more than one level, the most specific matching allocation will be used for each item. (The **Table** level takes precedence over the **Group** level, and both those levels take precedence over the **All** level.)

   - **Item relation** – If you're allocating for a single item, select the item. If you're allocating for a group of items, select the packing group. If you're allocating for all items, leave this field blank.

- **Configuration**, **Size**, **Color**, and **Style** – Enter values for these dimensions as you require, to further define the item that you're allocating for.
- **Packing unit** – Select the unit that the item is packaged in when the packing material is used. This unit might differ from the unit that the item is purchased and stored in.
- **Packing unit factor** – Enter the conversion factor that is used to convert from the inventory unit to the packing unit. (The conversion uses the formula *Packing units = Item units × Packing unit factor*.)

b. On the **Packing material** FastTab, add a line for each piece of packing material that is required for the current allocation. On each line, specify the type of material (as defined on the **Packing material codes** page) and the amount of it that is consumed for each shipped unit of the current item.

Packing units on sales order lines

After you've turned on the calculation for packing material fees and set up your allocations, the system verifies that packing units are specified for each item that is added to a sales order. It then calculates any fees that are required. When an item is added to a sales order, one of the following steps occurs:

- **If a packing allocation applies to the item:** The system updates the sales order line with the specified packing unit and the packing unit quantity. (The packing unit quantity is calculated by using the formula *Packing unit quantity = Ordered quantity ÷ Number of items in the selected packing unit*.)
- **If no packing allocation applies to the item:** You can manually enter a packing unit and a packing unit quantity on the sales order line.

You can also change the packing unit and the packing unit quantity when you post the sales order line. This capability is relevant if the sales order line is only partly delivered or partly invoiced.

When you invoice the sales order, the system creates packing material transactions. Packing material transactions contain the weights of the packing materials for the sales line. You can modify the transactions after you invoice them. You can then create new transactions.

Packing units on purchase order lines

The system doesn't create packing material transactions for purchase order lines. Instead, you manually create transactions for invoiced purchase order lines on the **Packing material transactions** page.

Set up license numbers for customers that are charged packing material fees

If the sales orders for a specific customer should include charges for the packing materials that are used for each order, follow these steps.

1. Go to **Accounts receivable** > **Customers** > **All customers**.
2. Select the customer that should be charged for packing materials.
3. On the **Invoice and delivery** FastTab, set the following fields:
    - In the **Sales tax** section, in the **Packing duty license number** field, enter the company's license number.
    - In the **Packing material fee** section, in the **License number** field, enter the company's license number.

Now, when you create and invoice a sales order that includes one or more items that use packing materials, the invoice will show the packing material fees.

For customers that should not pay packing material fees, follow these same steps, but clear the license numbers from the **Packing duty license number** and **License number** fields. Invoices for customers that don't pay packing material fees show the weights of the packing materials, but they don't show the fees.

To generate a report that shows all the packing material fees that your company will owe for a specific period, go to **Inventory management** > **Inquiries and reports** > **Packing material reports** > **Packing material fee calculation**, specify a range of dates, and then select **OK**.

Print packing material weights on invoices

You can print packing material weights on an invoice and indicate who pays the related fees. The weights are summarized by packaging code.

1. Go to **Accounts receivable** > **Setup** > **Accounts receivable parameters**.
2. On the **Updates** tab, on the **Invoice** FastTab, set the **Print packing material weight** option to **Yes**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Pack containers for shipment

The container packing process lets you validate inventory items and pack them into containers. During this process, warehouse workers typically pick inventory items from bulk storage areas and move them to a packing area. Multiple warehouse workers check the item quantities and types, and assign them to appropriate container sizes. When a container is fully packed, workers close it and move it to the outbound dock area, where it's made ready for shipment.

Containers represent physical structures that items are packed in, and you can track container information in the system. This information can be useful during transportation planning, especially if you calculate shipping charges based on containers. Before you ship, you can view the number of containers that are used in a shipment, the types of containers that are used, and the physical dimensions of each container, such as the total volume and weight.

You can use several related outbound warehouse capabilities with containers. For more information, see the following articles:

- Wave containerization
- Outbound sorting
- Small parcel shipping
- Confirm and transfer
- Set different dimensions for packing and storage
- Packing work for packing outbound containers and processing shipments
- Packing containers with the Warehouse Management mobile app
- Example scenario – Pack containers with the Warehouse Management mobile app
- Container label layouts and printing

Set up your warehouse to use manual packing operations

Organize your outbound operations in one of two ways:

- **Wave creation and processing** – Workers pick items based on outbound warehouse work. They then put the items directly in an outbound shipping location, where they're ready for immediate shipment. For information about how to set up and use this process, see Wave creation and processing.

- **Manual packing at packing stations** – This process has an additional operation between the picking and shipping operations. Instead of putting the inventory in a *bay door shipping location*, workers put it in a *packing location*. They then manage the packing process at the packing station by using the **Pack** page in the web client. To enable this process, you must configure your system as described in this section.

Set up warehouse locations for packing

You need at least one packing location where workers put inventory items so they can be packed into containers. For more information about how to create warehouse locations, see Configure locations in a WMS-enabled warehouse. The following subsections describe how to create and set up packing locations.

Set up location types for packing

You must define a *location type* for the packing locations. Use location types as filtering options to control the different warehouse management processes. The name of each location type typically describes something about how that location type should be used. Associate the location type that you set up here with each location used for packing operations.

Follow these steps to set up a location type.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location types**.
2. On the Action Pane, select **New** to add a location type to the grid.
3. Set the following fields for the new location type:
    - **Location type** – Enter a name for the location type. Each name must be unique and should describe something about how the location type is intended to be used.
    - **Description** – Enter a short description of the location type.

Inform the system about the packing location type

After you create a location type, set up your system to recognize it as the location type to use for packing operations.

Follow these steps to set the location type that is used for packing operations.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**
2. On the **General** tab, on the **Location types** FastTab, set the **Packing location type** field to the location type that you use to identify packing stations.

> **Note**
>
> If you're upgrading from a version of Microsoft Dynamics AX, the *In packing* status is removed from shipments and loads, because it didn't work consistently and caused redundant data. Therefore, the **In shipments** and **Loads in packing** list pages are deprecated. Track containers that must be packed at the location level.
>
> In previous versions, you defined the packing location by using the **Profile ID for packing location** field on the **Packing** tab of the **Warehouse management parameters** page to specify a *location profile*. In the current version, use the **Packing location type** field on the **General** tab of the **Warehouse management parameters** page to specify a *location type*, as described in this article. The new field is better aligned with the process for identifying staging locations and final shipping locations.
>
> Although you can continue to use the old **Profile ID for packing location** field, start using the new **Packing location type** field instead, because the old field will be eventually deprecated.
>
> If you clear the setting of the old **Profile ID for packing location** field and then save the page, the field is permanently disabled. For installations where the **Profile ID for packing location** field is never used, it's always disabled. After you clear the setting of the **Profile ID for packing location** field, use the settings that are described in this article to set up and identify your packing location.

Set up location profiles for packing locations

Each *location profile* establishes information and rules that apply to the associated locations. Because you need at least one location to use for packing operations, you must create a location profile for it in addition to a location type. Each profile can be used by any number of locations. Locations that are used for packing must be set up to track license plates.

Follow these steps to set up a location profile for a packing location.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. Either select an existing profile from the list pane or select **New** on the Action Pane to create a new one.
3. On the header of the new or selected profile, set the following fields:

- **Location profile ID** – Enter a unique ID for the profile.
- **Name** – Enter a descriptive name for the profile.

4. On the **General** FastTab, set the following fields:

    - **Location format** – Select the location format to use for the current location. Location formats are a naming system that is used to create unique and consistent names for the different location bin positions that are used in a warehouse. If you don't already have the format that you need, you can create it by going to **Warehouse management** > **Setup** > **Warehouse** > **Location formats**. Learn more in Configure locations in a WMS-enabled warehouse.
    - **Location type** – Select the location type that is set up as the packing location type on the **Warehouse management parameters** page, as described earlier in this article.
    - **Use license plate tracking** – For packing locations, set this option to *Yes*. The system must be able to track license plate IDs at packing locations, so that it can control the packing process.
    - **Allow negative inventory** – For packing locations, set this option to *No*.
    - **Allow mixed items** – For packing locations, set this option to *Yes*. This setting is required because many different item numbers are typically brought to the packing locations at the same time.
    - **Allow mixed inventory statuses** – For packing locations, set this option to *Yes*. This setting is required because items that have different inventory statuses are typically brought to the packing locations at the same time.
    - **Allow mixed inventory batches** – For packing locations, set this option to *Yes*. This option is automatically set to *Yes* whenever the **Allow mixed items** option is set to *Yes*.

Set up packing locations

You must have at least one *location* where workers put inventory items that must be packed into containers.

Follow these steps to add a packing location.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.

2. On the Action Pane, select **New** to add a location.

3. Set the following fields for the new location:

    - **Warehouse** – Specify the warehouse where the packing location must exist.
    - **Location** – Enter a name for the new location.

- **Location profile ID** – Specify the ID that you created in the previous section.

Set up the packing process

This section describes how to configure settings that enable the packing process and let workers pack inventory into containers.

Set up container packing policies

Each *container packing policy* defines how to process a container. Each time you create a new container, select a container packing policy to apply to it.

Follow these steps to set up a container packing policy.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container packing policies**.

2. Either select an existing policy from the list pane or select **New** on the Action Pane to create a new one.

3. On the header of the new or selected policy, set the following fields:

    - **Container packing policy** – Enter a unique name for the policy.
    - **Description** – Enter a descriptive name for the policy.

4. On the **Overview** tab, set the following fields. These fields specify what actions to take when the container closes, whether to operate with or without work creation, and when to release the container from the packing station.

    - **Weight unit** – Select the default unit of measure to use when a container is closed and manifested. Typically, this unit of measure is the unit of measure that a scale at the packing station uses. This field applies to policies with or without work creation.

    - **Container closing policy** – Select one of the following options to define what should occur when the container is closed:
        - *Automatic release* – The container is considered released from the packing station, and the action that is specified in the **Container release policy** field is triggered.
        - *Delayed release* – The container isn't immediately released from the packing station. A warehouse worker must manually release it later.
        - *Optional* – While a worker is closing the container, they can choose whether the container should be released.

If the packing station mainly handles single containers that ship directly to customers, the most natural approach is to release the containers immediately. If the packing station handles shipments that consist of multiple containers or even pallets, it's probably best to delay the release until the whole shipment or pallet is packed and ready for pickup.

- **Container release policy** – Select one of the following options to define what occurs when the container is released from the packing location:
  - *Make available at final shipping location* – Update the container to the final shipping location. If you select this option, use the **Default location for final shipment** field to specify a preferred location for the container after it's closed.
  - *Create work to move container from packing station* – Create work to move the container from the packing station to the staging area or directly to the bay door. Use the **Work template** field to specify the work template that should be applied when work is created for the container.
  - *Assign container to outbound sorting position* – This option is used with the outbound sorting capability.

  In most cases, create work to move containers, because this approach better represents the actual manual processes in the warehouse. However, for simple scenarios, or situations where the packing station is located directly in the bay door area, it might be preferable for the container to be immediately available at the final shipping location.

- **Work template** – Select the work template that should be applied when work is created for the container. This field is available only when the **Container release policy** field is set to *Create work to move container from packing station* and related to a work order type that is named *Packed container picking*. Learn more in Work templates and location directives.

5. On the **Warehouse selection** FastTab, set the following fields:

- **Warehouse selection** – Select one of the following values:
  - *All* – Use this policy for all warehouses where a more specific policy isn't assigned.
  - *Warehouse group* – Use this policy for all warehouses in the warehouse group that's selected in the **Warehouse group** field.
  - *Warehouse* – Use this policy only for the specific warehouse that's selected in the **Warehouse** field.

- **Warehouse** – If the **Warehouse selection** field is set to *Warehouse*, select the warehouse where the packing policy applies.
- **Warehouse group** – If the **Warehouse selection** field is set to *Warehouse group*, select the warehouse group where the packing policy applies. For more information about how to set up warehouse groups, see Warehouse groups.
- **Default location for final shipment** – Specify the preferred location for the container after it's closed. This field is available only when the **Container release policy** field is set to *Make available at final shipping location* and the **Warehouse selection** field is set to *Warehouse*.
- **Default location for sorting** – This field is used with the outbound sorting capability. It's available only when the **Warehouse selection** field is set to *Warehouse*.

In the remaining steps, you'll configure settings that are related to *manifesting*. Manifesting is the process of specifying the weight of a container, container group, or shipment, together with the tracking ID that's received from a transportation provider. Dynamics 365 Supply Chain Management doesn't integrate with external transportation provider systems. Instead, warehouse workers must print labels that are received from the transportation provider and then scan a tracking number when they complete the manifest procedure.

Because manifest requirements vary from customer to customer, and even from shipment to shipment, packing policies allow for significant flexibility in the workflow. You can set up manifests for containers, container groups, and shipments in any combination.

If you're using a multilevel manifest procedure, workers must manifest all containers before they manifest the related container group, and they must manifest all container groups before they manifest the related shipment.

6. On the **Container manifest** FastTab, set the following fields:

- **Automatic manifest at container close** – Set this option to *Yes* to apply manifest information as part of the container closing process. If you aren't using transportation integration, you must manually record the information.
- **Manifest requirements for container** – Select one of the following options:
  - *None* – No manifesting process will be used.
  - *Manual* – Manifesting is required by the packing workflow. The system doesn't allow a container to be closed or released until manifesting is completed.

    - *Transportation management* – Manifesting will be done through Transportation management (TMS) rate engines. Because this option requires custom development to implement a specific engine for the transportation provider, it doesn't work out of the box in the current version.

- **Print container contents** – Set this option to *Yes* to automatically print the container contents report when a container is registered as closed. You can also print the report on demand.

7. On the **Container group manifest** FastTab, in the **Manifest requirements for container group** field, select one of the following options:

    - *None* – The container group manifest won't be included as a requirement in the packing workflow.
    - *Manual* – The packing workflow includes the container group manifest as a requirement. All containers that are included in the group must be closed before the group can be manifested. Select this option if you're required to complete a manifest for every container group that is packed at the packing station. Typically, select this option if containers are packed on a pallet and the whole pallet is manifested.

    > ⓘ **Note**
    >
    > The current version doesn't support manifest reports for container groups, and there's no TMS engine support for container groups.

8. On the **Shipment manifest** FastTab, set the following fields:

    - **Manifest requirements for shipment** – Select one of the following options:
        - *None* – The shipment manifest won't be included as a requirement in the packing workflow.
        - *Manual* – The packing workflow includes the shipment manifest as a requirement. No container for a shipment can be released until manifesting is completed.
        - *Transportation management* – Manifesting is done through TMS rate engines. Because this option requires custom development to implement a specific engine for the transportation provider, it won't work out of the box in the current version.

    Shipment manifesting should be enabled if you're required to complete a manifest for the whole shipment that is packed at the packing station. It's typically used when one

consolidated manifest is required even though the shipment consists of multiple containers or container groups.

- **Print packing slip** – Set this option to *Yes* to automatically print the packing slip as part of the shipment manifest. You can also print the packing slip on demand.

- **Print packing slip asynchronously** – This option is available only when the **Print packing slip** option is set to *Yes*. Set this option to *Yes* to process sales packing slips asynchronously by using the message processor. In this case, the system queues sales packing slip postings to the message processors by using messages of the *Run packing slip for container* type to the *Warehouse* queue. This setting requires that you also set the **Packing slip posting parameters ID** field. For more information about how to set up and use the message processor, and how to schedule the batch job that's required to process the messages, see Monitor and control message processor messages.

- **Packing slip posting parameters ID** – This setting applies when the **Print packing slip asynchronously** option is set to *Yes*. You must define at least one set of parameter settings on the **Packing slip posting parameters** page (**Warehouse management** > **Setup** > **Inventory** > **Packing slip posting parameters**) and select the ID for the set of settings that you want to use here. The parameter settings are used when the last container for a shipment is closed. Warehouse workers aren't prompted to confirm the values. If you're using the *Container closing* process for the Warehouse Management mobile app, this process is the only one that's supported for automatic posting of packing slips.

Set up container types

During the manual packing process, containers must be created before items can be packed into them. Each container must be based on a *container type*, which defines a container's maximum physical volume and weight capacity.

Follow these steps to create a container type.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container types**.

2. On the Action Pane, select **New** to add a container type.

3. Set the following fields for the new container type:

    - **Container type code** – Enter a unique ID for the container type.

- **Description** – Enter a description of the new container type.
- **Tare weight** – Enter the actual or estimated weight of the container.
- **Maximum net weight** – Enter the maximum weight that the container can hold.

4. In the **Container dimensions** section, enter the physical dimensions of the container itself in the **Length**, **Width**, **Height**, and **Volume** fields.

5. In the **Maximums** section, enter the maximum physical dimensions that the container can handle in the **Length**, **Width**, **Height**, and **Volume** fields.

6. You can define additional attributes for container types that are associated with other operations that use containers. To learn more, see Containerization.

> **Note**
>
> For the manual packing process, the system uses only the value of the **Maximum net weight** field and the value of the **Volume** field in the **Maximums** section.

Set up packing profiles

Packing profiles are required for the packing process. They can be selected or set by default when you start a packing operation on the **Pack** page.

Follow these steps to set up a packing profile.

1. Go to **Warehouse management** > **Setup** > **Packing** > **Packing profiles**.

2. Either select an existing profile from the list pane or select **New** on the Action Pane to create a new one.

3. Make the following settings for your new or selected profile:

   - **Packing profile ID** – Enter a short ID for the profile.
   - **Description** – Enter a description of the packing profile.
   - **Container packing policy** – Select the packing policy that applies to the profile. For more information, see the Set up container packing policies section of this article.
   - **Container ID mode** – Select whether to automatically generate a container ID on creating a container, or whether to manually create it.
   - **Container type** – Select the container type that's used by default when a new container is created.

- **Autocreate container at container close** – Select this checkbox to automatically create a new container if the previous container is closed, and one or more lines remain in the current shipment.
- **Print container label at container creation** – Select this checkbox to automatically print a container label when a new container is created. For more information about how to set up your container label layouts, see Container label layouts and printing.
- **Prevent editing container ID** – Select this check box to prevent automatically assigned container IDs from being edited.
- **Prevent container creation without items to pack** – Select this checkbox to prevent containers from being created when there aren't any items to pack for a given shipment and packing location.

Set up warehouse workers

Every user or worker who packs containers by using the **Pack** page of the web client or the *Packing* activity code in the Warehouse Management mobile app must have a user account that's associated with a *worker/person* record that has the required security access rights assigned. For more information about how to set up users, see Create new users.

Follow these steps to set up a *worker/person* record for the packing process.

1. Go to **Warehouse management** > **Setup** > **Worker**.
2. On the Action Pane, select **New** to add a work user.
3. Set the **Worker** field to the existing worker record defined in the **Human resources** module for the user who completes the packing process.
4. On the **Profile** FastTab, set the following fields:
    - **Container packing policy** – Select a container packing policy that defines how containers at the packing station should be processed. The container packing policy that you select is preselected for the worker when they open the packing station. For more information, see the following blog post: Improved packing functionality.
    - **Packing profile ID** – Select a packing profile ID that defines the packing policy and container settings that should be used. If the selected packing profile ID is associated with a container packing policy, you can't change the setting of the **Container packing policy** field on this page.

5. On the **Default packing station** FastTab, select the default site, warehouse, and location that apply to the worker.

6. If the worker uses the Warehouse Management mobile app to manage and register their container packing work, on the **Users** FastTab, set up one or more accounts that the worker can use to sign in to the app. Learn more in Mobile device user accounts.

Example scenario

This section provides an example scenario that shows how to create an order and pack the items.

Make sample data available

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

You can also use this scenario as guidance for using the feature on a production system. However, in that case, you must substitute your own values for each setting that is described here.

Sign in as a user that can do packing work

Sign in to Supply Chain Management by using a user account that has the permissions required to pack containers. The user *Julia Funderburk* is included as part of the demo data and has the required permissions. This user has the user ID *Admin*.

Create a sales order and complete the work

Follow these steps to create a sales order and complete the work of moving the ordered items to the packing station.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. On the Action Pane, select **New**.

3. In the **Create sales order** dialog box, in the **Customer account** field, select *US-005*.

4. Select **OK** to close the dialog box.

5. The new sales order is opened and includes a single, empty line on the **Sales order lines** FastTab. Set the following values for the new order line:

- **Item number:** *A0001*
- **Quantity:** *2*
- **Site:** *6*
- **Warehouse:** *62*

6. While the order line is still selected, select **Inventory** > **Reservation** on the toolbar of the **Sales order lines** FastTab.

7. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.

8. Close the **Reservation** page to return to the sales order.

9. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

   A message shows the shipment and wave IDs for the order.

10. While the order line is still selected, select **Warehouse** > **Work details** on the toolbar of the **Sales order lines** FastTab. If you use batch processing to run your waves, you might have to wait a short time for the work to be created.

11. On the **Work** page, on the Action Pane, on the **Work** tab, select **Complete work**.

12. On the **Work completion** page, set the **User ID** field to *62*.

13. On the Action Pane, select **Validate work**.

14. When you receive a message that indicates that your work is valid, select **Complete work** to complete the process of picking the inventory items and putting them in the *Pack* location.

15. Make a note of the **Shipment ID** value that is shown for the work in the upper grid.

Pack the ordered items into a container

The inventory items are now at the packing area and are ready to be packed into containers. Follow these steps to create a new container in the system and pack it.

1. Go to **Warehouse management** > **Packing and containerization** > **Pack**.

2. In the **Select packing station** dialog box, set the following values:

   - **Site:** *6*
   - **Warehouse:** *62*

- **Location:** *Pack*
- **Packing profile ID:** *WH62*

3. Select **OK**.

4. On the **Pack** page, in the **License plate or shipment** field, enter the shipment ID that you noted earlier. You should now see the remaining unpacked items on the **Open lines** FastTab.

5. On the Action Pane, select **New container** to create a container in the system.

6. In the **ID of the new container** dialog box, set the **Container type** field to *SmallBox*.

7. Select **OK** to create the container.

8. Select **OK** to return to the **Pack** page. The **Open containers** FastTab now shows the **Container ID** value of the container that you created.

9. For this scenario, pack just one of the ordered items. On the **Item packing** FastTab, set the following values:

   - **Quantity:** *1.00*
   - **Identifier:** *A0001*

   Immediately after you select the **Identifier** value, the page updates the **Open lines** and **All lines** FastTabs to indicate that one item is packed. You should now have packed one of the two pieces of item *A0001*.

10. On the Action Pane, select **Close container**.

11. In the **Close container** dialog box, select **Get system weight** to fill in the default **Gross weight** value.

12. Select **OK** to close the container.

> **Tip**
>
> You can view containers in various ways based on context. For example, when you pack a shipment, it's often useful to view either containers that are part of the shipment or all containers that are physically at a packing station. The **Packing station** page has buttons that you can use to view all open and closed containers at a packing station. These views aren't restricted to a specific shipment. They can be very helpful in situations where one

worker is packing a container, and another worker is manifesting and releasing the container.

A consolidated view of all containers is also available. This view is useful mostly for users who work outside the context of a single packing station. To see it, go to **Warehouse management** > **Packing and containerization** > **Containers**.

Last updated on 04/22/2026

Packing work for packing outbound containers and processing shipments

This article describes the *Packing* work order type, which manages work for packing containers and supports partial shipments of packed containers that are related to loads where inventory items remain unpacked. Packing work lets you use confirm and transfer functionality to confirm outbound shipments that are associated with containers.

Packing work is automatically created when inventory that is related to source document work is put in locations of the *Packing location* type. The work consists of two lines, one for *Pick* and one for *Put*, and is automatically maintained as part of a container close/reopen process.

> **Note**
>
> In Supply Chain Management version 10.0.45 and later, the Warehouse Management mobile app can record the worker IDs of workers that use it to do packing work. Learn more in Create containers, Close containers, and Pack inventory into containers.

For more information about how to set up and use the container packing process, see Pack containers for shipment.

Set up a location for packing work

Use the following procedure to set up packing locations. For each location, you can select whether the system automatically creates packing work.

1. Go to **Warehouse management** > **Setup** > **Packing** > **Packing station setup**.
2. On the Action Pane, select **New** to add a packing station setup record.
3. In the new record, set the following fields:
    - **Warehouse** – Select or enter the warehouse where the packing location is located.
    - **Location** – Select or enter the packing location. This location must be assigned to a location profile that uses the location type that is configured as the packing location type for your company on the **Warehouse management parameters** page. Learn more in Pack containers for shipment.

- **Create packing work** – Select this checkbox to create packing work each time that items are delivered to the packing location. The work includes links to related load lines, so that partial loads can be packed and shipped.

> ⊗ **Caution**
>
> Before enabling the **Create packing work** option, make sure the packing station is empty (no items are present).

Example scenario

This example scenario shows how to process an outbound sales order flow by packing a container and shipping a partial load.

Make sample data available

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

You can also use this scenario as guidance for using the feature on a production system. However, in that case, you must substitute your own values for each setting that is described here.

Configure packing work for warehouse packing location

To get started, you must configure the *Packing* work process for a specific warehouse and location.

1. Go to **Warehouse management** > **Setup** > **Packing** > **Packing station setup**.
2. On the Action Pane, select **New** to add a setup record.
3. In the new record, set the following values:
    - **Warehouse:** *62*
    - **Location:** *Pack*
    - **Create packing work:** *Yes*
4. Close the **Packing station setup** page.

Create a load template that allows partial shipping

To enable a load to be delivered over multiple shipments, associate it with a load template that allows partial shipping. Follow these steps to create the required template.

1. Go to **Warehouse management** > **Setup** > **Load** > **Load templates**.
2. On the Action Pane, select **New** to add a setup record.
3. In the new record, set the following values:
    - **Load template ID:** *Partial*
    - **Allow load split during ship confirm:** *Yes*
4. Close the **Load templates** page.

Learn more in Confirm and transfer.

Process a sales order

Follow these steps to process a sales order and partially ship it.

1. Complete the example scenario that is provided in Pack containers for shipment. During that scenario, you create a sales order for two pieces of one item. You then pack just one of the pieces into a container and close the container. Make a note of the shipment ID that you create, as instructed in the scenario.
2. Go to **Warehouse management** > **Work** > **All work**.
3. In the filter area, select the **Show closed work** checkbox. Then enter the shipment ID in the **Filter** field, and select to filter by **Shipment ID** value. You should now see three work headers. One is for the sales order picking work and has a status of *Closed*. Two are for the packing process: one is related to the closed container and has a status of *Closed*, and the other is related to the unpacked remaining item and has a status of *Open*.
4. Select the **Load ID** value for any of the work headers to open **Load details** page for the load.
5. Switch to the **Header** view.
6. On the **General** FastTab, select the **Edit** button for the **Load template ID** field. Then select the partial shipping load template that you created for this scenario (*Partial*).
7. On the Action Pane, on the **Ship and receive** tab, in the **Confirm** group, select **Outbound shipment**.
8. In the **Ship confirm** dialog box, select the **Split quantity to new load** option.
9. Select **OK**.

You shipped one container that is related to the original load, and the system creates a new load for the remaining items that must still be packed into containers.

Last updated on 05/21/2026

Packing containers with the Warehouse Management mobile app

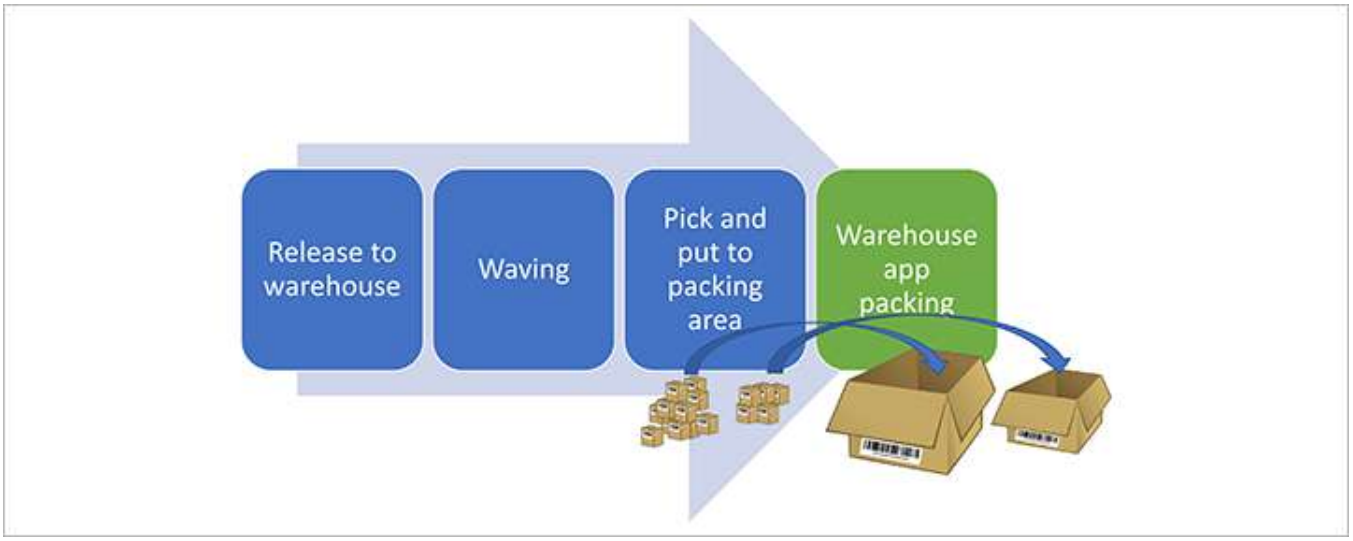
07/29/2025

Traditionally, warehouse workers have performed packing activities at a specific packing station that is configured in Microsoft Dynamics 365 Supply Chain Management, and they've used a process that is optimized for shipments of small to medium-sized parcels. To help improve efficiency in larger packing areas, and to better support the packing and shipment of larger items, the Dynamics 365 Warehouse Management mobile app provides a mobile packing experience that gives workers the freedom to move around while they perform their packing activities.

This article describes how to use the shipment container packing process in the Warehouse Management mobile app. This process can be used in combination with the process on the **Pack** page in the Supply Chain Management web client. For more information about the warehouse management packing process, see Pack containers for shipment.

The Warehouse Management mobile app packing process

As soon as the shipment inventory items have been brought forward to the packing area, workers can start the *Pack inventory into containers* process in the Warehouse Management mobile app.



To take advantage of all the supported packing processes in the Warehouse Management mobile app, you must set up the following mobile device menu items:

- **Pack inventory into containers** – This menu item is used for the main process of packing items into containers.
- **Container creation** – This menu item is used to create containers that shipment items will be packed into.
- **Container closing** – This menu item is used to close the shipment containers.
- **Container deletion** - This menu item is used to delete a container.
- **Print container label** - This menu item is used to manually print a container label.

We recommend that you use the detour functionality. This functionality makes Warehouse Management mobile app operations easier by embedding the *Container creation* and *Container closing* processes into the **Pack inventory into containers** menu item, especially when manual container creation and closing operations are used. We also recommend that you add several lookup options to the Warehouse Management mobile app by using data inquiries in combination with the detour functionality. This approach is especially effective in situations where bar codes are unreadable or missing.

Pack inventory into containers

The *Pack inventory into containers* process is controlled by the mobile device container packing policies that are associated with the relevant mobile device menu items. You can set up the system to require that workers manually specify and confirm the required information during the packing process. Alternatively, you can set up the system so that it automatically finds and applies the required data.

The process requires the following information (at a minimum):

- **Packing location** – This value identifies the location where the container is created. You can assign a default value for each worker by going to **Warehouse management** > **Setup** > **Worker** and/or by setting up a detour to look up the value.
- **Shipment ID** or **License plate ID** – These values are used to validate which inventory items should be packed. The process depends on the mobile device menu item's **Packing policy ID** setting.

Record and view container line packing details (preview)

[This section is prerelease documentation and is subject to change.]

When workers use the Warehouse Management mobile app to pack items into containers, the system can record container line packing details (such as worker IDs and packed quantities). This functionality can be useful, for example, if you have multiple workers processing a container line and you want to keep track of who did what.

> **Note**
>
> The functionality described in this section requires Supply Chain Management version 10.0.45 or later.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

To enable this functionality, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Open the **Packing** tab.
3. Set **Create container line packing details** to *Yes*.

To view the logs created when **Create container line packing details** is set to *Yes*, follow these steps:

1. Go to **Warehouse management** > **Packing and containerization** > **Containers**.
2. On the upper pane, select the container you want to inspect.
3. Do one of the following steps:

    - To view details for all the container lines in the selected container, select **Container lines packing details** on the Action Pane.
    - To view details for a single container line, select the line in the lower pane and then select **Container line packing details** on the lower pane toolbar.

> **Important**
>
> A default container packing policy must be assigned to each relevant worker on the **Worker** page (**Warehouse management** > **Setup** > **Worker**).

Create containers

To create containers by using the Warehouse Management mobile app, workers must have the following information:

- **Packing location** – This value identifies the location where the container is created. You can assign a default value for each worker by going to **Warehouse management** > **Setup** > **Worker** and/or by setting up a detour to look up the value. When the *Create container* function is called from a detour that has a location, you can pass the value.
- **Shipment ID** – This value is used to validate which inventory items should be packed into the container. (You can assign a default value by setting up a detour.)
- **Container type ID** – This value is used to identify the maximum physical volume and maximum weight capacity of the container.
- **Container ID** – This value is a unique number that identifies the shipping container.

In Supply Chain Management version 10.0.45 and later, the system records the ID of the worker who created the container. To view this information, go to **Warehouse management** > **Packing and containerization** > **Containers** and inspect the **Created by** field.

> **Tip**
>
> The **Created by** field might not be shown. If you don't see it on your system, then you can add it by opening the **Grid options** menu (shown as a button with three dots at the right side of the grid heading) and selecting **Insert columns** .

> **Important**
>
> A default container packing policy must be assigned to each relevant worker on the **Worker** page (**Warehouse management** > **Setup** > **Worker**).

Close containers

Workers can trigger the *Container closing* process directly, by using a Warehouse Management mobile app menu. Alternatively, you can set up a detour to embed the process into the **Pack inventory into containers** menu item. During the *Container closing* process, the following values must be specified:

- **Container ID** – The container that is being closed.
- **Weight** – The weight of the container. The system assigns a default value, based on the item master weight definition.

In Supply Chain Management version 10.0.45 and later, the system records the ID of the worker who closed the container. To view this information, go to **Warehouse management** > **Packing and containerization** > **Containers** and inspect the **Closed by** field.

> **Tip**
>
> The **Close by** field might not be shown. If you don't see it on your system, then you can add it by opening the **Grid options** menu (shown as a button with three dots at the right side of the grid heading) and selecting **Insert columns** .

Supported and unsupported processes

The following table shows which processes are and aren't supported when the container packing process in the Warehouse Management mobile app is used instead of the **Pack** page in the Supply Chain management web client. Some rows use parentheses in the *Supported in the mobile app* column. These parentheses indicate either a partial-yes or partial-no status, which is clarified in the *Notes* column.

Expand table

Logic	Supported in the mobile app	Notes
Items enabled for **Catch weight**	No	
Identification of license plate/shipment	Yes	When you use a mobile device menu item without a **Packing policy ID** value (or when you use a packing policy where the **Starting step** field is set to *Scan shipment ID first*), the system will let workers pack physical on-hand inventory from multiple license plates by automatically selecting the "from" license plate. However, if you use a packing policy where the **Starting step** field is set to *Scan license plate ID first*, only physical on-hand inventory that's related to the selected license plate can be packed.
Identification of tracking dimensions	Yes	When you use a mobile device menu item without a **Packing policy ID** value (or when you use a packing policy where the **Capture tracking dimensions** field is set to *Skip capturing*), workers won't be asked to specify the tracking dimensions while they pack. However, if serial numbers are used, and the tracking dimensions group is set to capture serial at packing, workers will always have to specify the serial number. If you use a packing policy where the **Capture tracking dimensions** field is set to *Capture one by one*, the system won't enter default tracking dimensions. Therefore, workers will always have to specify the tracking dimensions.
Identification of product variant	Yes	

Logic	Supported in the mobile app	Notes
dimensions		
Identification via GS1 bar code scanning	Yes	You don't have to provide the quantity to pack (`QtyToPack`) as part of a primary input field. Instead, the system will automatically set the quantity to pack to a default value as part of the packing flow. Therefore, if you're using bar codes that include quantities, the GS1 bar code setup must have the **Allow overwriting** checkbox selected for the *QtyToPack* application identifier in the relevant bar code data policy.
Identification item via bar code setup	(Yes)	However, quantity and unit (piece-by-piece) scanning isn't supported.
New container	Yes	Containers can be created either automatically by using the *Autocreate container at container close* process or manually by using the **Container creation** mobile device menu item. The menu item also captures the container type and enters a default number sequence if the **Container ID mode** field is set to *Auto*.
Print container label	Yes	Labels can be printed either automatically during the *Container creation* process or manually by using either the **Print container label** mobile device menu or the **Containers** page in the web client. For more information about how to set up label printing, see Container label layouts and printing.
Close container	Yes	Containers can be closed by using the **Container closing** mobile device menu item, which also captures the weight.
Release container	No	The release of containers depends on the container closing policy. It isn't available as a mobile device menu option. Workers can't trigger or confirm a release by using the mobile app.
Reopen container	No	
Delete container lines	No	
Delete container	Yes	Containers can be deleted by using the **Container deletion** mobile device menu item. The container must be in the *Open* state, but it can have packed container lines. Those lines will also be deleted.
Change container packing policy	No	The default policy that is assigned to each worker on the **Worker** page (**Warehouse management** > **Setup** > **Worker**) is always used.
Release shipment	No	
Work details	(Yes)	You can make this information visible (read-only) by setting up a data inquiry detour.

Logic	Supported in the mobile app	Notes
Containers for shipment (Open/Closed)	(Yes)	You can make this information visible (read-only) by setting up a data inquiry detour.
Shipment details	(Yes)	You can make this information visible (read-only) by setting up a data inquiry detour.
Display dimensions	(No)	You can make this information visible (read-only) by setting up a data inquiry detour. However, capture of dimensions isn't supported.
Manifest containers	No	Only automatic manifesting setup is supported
Unmanifest containers	No	
Manifest container group	No	No weight capturing process is supported.
Unmanifest container group	No	
Manifest shipment	No	No weight capturing process is supported.
Unmanifest shipment	No	
Transport route rate details	(No)	You can make this information visible (read-only) by setting up a data inquiry detour.
Transport shipment accessorial charges	(No)	You can make this information visible (read-only) by setting up a data inquiry detour.
Transport status	(No)	You can make this information visible (read-only) by setting up a data inquiry detour.
Print packing slip	(Yes)	The setting for automatically processing sales packing slips as part of closing the last container for a shipment is supported by using the *Print packing slip asynchronously* process on the container packing policy.
Print container content	(No)	Only automatic printing (as part of the *Container closing* process) can be set up.
Print shipping label	(No)	Only automatic printing (as part of the *Container closing* process) can be set up.

Logic	Supported in the mobile app	Notes
Pack (all shipment items)	No	Each item number must be identified and packed individually.
Pack items (based on defined quantity)	(Yes)	However, items can be packed only *after* the item number has been identified by updating the quantity as part of the *Pack inventory into containers* process.
Container group license plate ID with delayed work creation	No	No group license plate ID can be specified. It isn't possible to close a container that has a **Container group license plate ID** value assigned.
Container type creation	No	

Next steps

- Learn how to set up mobile device container packing policies, which let you control the packing process that's supported by the Warehouse Management mobile app. See Mobile device container packing policies.
- Work through an example that shows how to set up and use this feature. See Example scenario – Pack containers with the Warehouse Management mobile app.
- Learn how to set up container labels and print them from the mobile app. See Container label layouts and printing.

Mobile device container packing policies

Article • 11/27/2024

Mobile device container packing policies let you control the packing process that's supported by the Warehouse Management mobile app.

For more information about how packing containers work with the Warehouse Management mobile app, see Packing containers with the Warehouse Management mobile app. For an example that shows how to set up and use this functionality, see Example scenario – Pack containers with the Warehouse Management mobile app.

Set up mobile device container packing policies

Follow these steps to set up your mobile device container packing policies.

1. Go to **Warehouse management** > **Setup** > **Packing** > **Mobile device container packing policies**.

2. Follow one of these steps:

    - To create a new policy, select **New** on the Action Pane.
    - To edit an existing policy, select it in the list pane, and then select **Edit** on the Action Pane.
    - To delete an existing policy, select it in the list pane, and then select **Delete** on the Action Pane.

3. If you're creating or editing a policy, on the header of the new or selected policy, set the following fields:

    - **Packing policy ID** – Enter a unique name for the policy.
    - **Description** – Enter a short description of the policy.

4. On the **General** FastTab, set the following fields:

    - **Capture tracking dimensions** – Select one of the following options to indicate whether the worker must specify tracking dimensions in the *Pack inventory into containers* flow on the warehouse mobile device:
        - *Skip capturing* – The worker won't be asked to specify tracking dimensions during packing. However, if serial numbers are used, and the tracking

dimensions group is set to capture serial numbers at packing, the worker will be asked to specify the serial number.
    - *Capture one by one* – Tracking dimensions won't be set by default. Therefore, the worker will always have to specify them.

- **Starting step** – Select the first step that's shown in the *Pack inventory into containers* flow on the warehouse mobile device. (The step will be shown after the worker signs in to the packing station.)
    - *Scan shipment ID first* – The worker will be asked to scan a shipment ID. This option enables workers to pack items from multiple license plates as required.
    - *Scan license plate ID first* – The worker will be asked to scan the license plate ID. When you use this option, all items that will be packed must be found on the specified license plate. If the license plate contains items from multiple shipments, the worker will be prompted to specify which shipment to pack for.

- **Item selection** – Select which items will be packed in the *Pack inventory into containers* flow on the warehouse mobile device:
    - *Select item to pack* – The worker will be asked to scan an item to pack.
    - *Pack all remaining* – The worker will pack items for the specified shipment or license plate into the container. You can't use this option when tracking dimensions are captured.

- **Close container automatically** – This option is available only when the **Item selection** field is set to *Pack all remaining*. Set it to *Yes* to automatically close the container after all the items are packed. Because the worker won't be able to update the weight before they close the container, the weight from the item master data will be used.

- **Create container automatically** – This option is available only when the **Item selection** field is set to *Pack all remaining*. Set it to *Yes* to automatically create a new container at the start of the packing process. For this functionality to work, the following conditions must be met:
    - On the **Worker** page (**Warehouse Management** > **Setup** > **Worker**), in the worker record for each relevant worker, the **Packing profile ID** field must be set.
    - In the selected **Packing profile ID** record for each relevant worker, a **Container type** value must be assigned, and the **Container ID mode** field must be set to *Auto*.

5. On the Action Pane, select **Save**.

Assign a container packing policy to a mobile device menu item

For a mobile device container packing policy to take effect, you must assign it to each mobile device menu item where it applies. Follow these steps to assign a container packing policy a menu item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. Follow one of these steps:

    - To create a new menu item, select **New** on the Action Pane.
    - To edit an existing menu item, select it in the list pane, and then select **Edit** on the Action Pane.

3. On the header of the new or selected menu item, set the following fields:

    - **Menu item name** – Enter a unique internal name for the menu item.
    - **Title** – Enter the display name that will be shown for the menu item in the mobile app.
    - **Mode** – This field must be set to *Indirect*.

4. On the **General** FastTab, set the following fields:

    - **Activity code** – This field must be set to *Pack inventory into containers*.
    - **Packing policy ID** – Select the mobile device container packing policy to use with the menu item.

Related information

- Packing containers with the Warehouse Management mobile app
- Example scenario – Pack containers with the Warehouse Management mobile app

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Feedback

Was this page helpful? Yes No

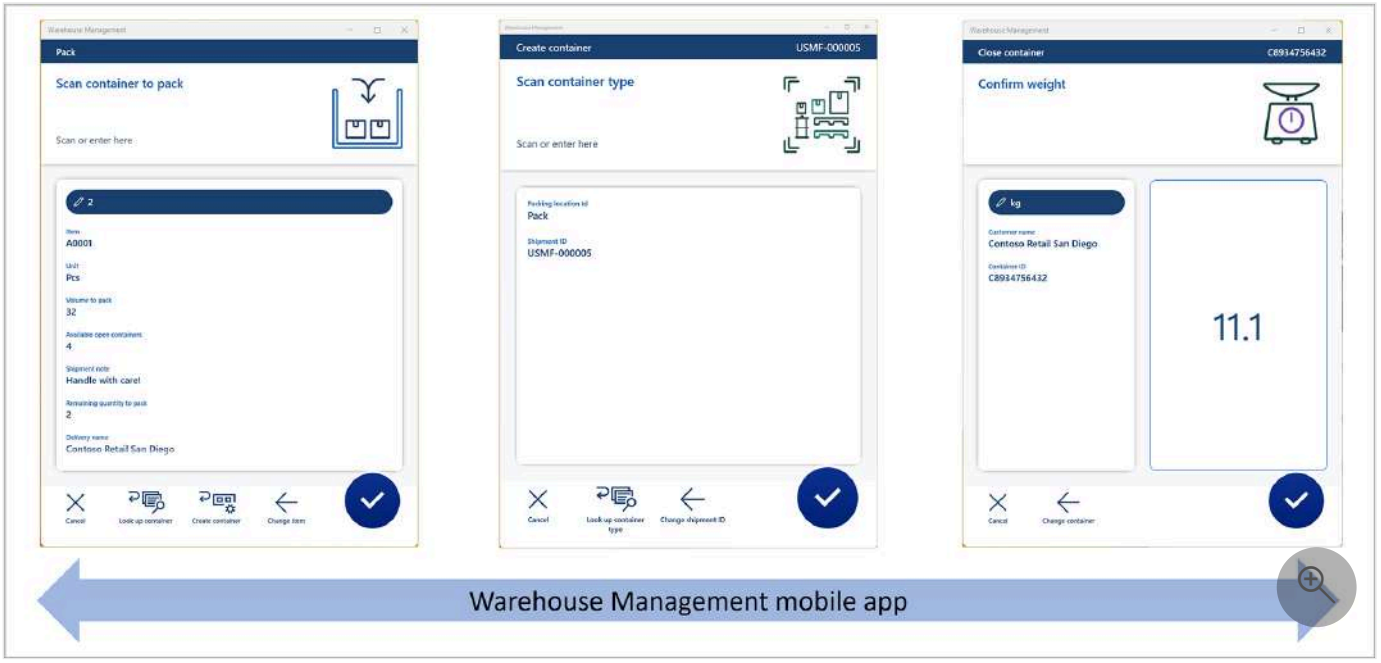
Provide product feedback

Example scenario – Pack containers with the Warehouse Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an example scenario that shows how to set up a Warehouse Management mobile app packing flow, and how to use the flow in the Warehouse Management mobile app to process a simple outbound sales order by packing a container and closing it. The scenario takes advantage of the detour and data inquiry capabilities of the mobile app. For more information about the feature that enables this functionality, see Pack containers using the Warehouse Management mobile app.

The following illustration shows the packing functionality that this scenario adds to the mobile app.



You can establish a packing flow by creating mobile device menu items and detour handling. The example scenario that's provided in this article explains the menu items that aren't enclosed in parentheses in the following table.

Expand table

Pack inventory into containers	Container creation	Container closing	(Print container label)	(Container deletion)
Look up location	Look up container type	Look up container	(Look up container)	(Look up container)
Look up shipment				
Look up item				
Look up container				

Prerequisites

Use the Warehouse Management mobile app version 2.0.37.0 or later

To see the newest icons and user experience (UX) enhancements that are related to the mobile app packing process, you must use Warehouse Management mobile app version 2.0.37.0 or later.

Make sure the required fields are available to the app

Before you start to configure detours for the mobile application, make sure the required mobile application fields are set up on your system. Go to **Warehouse management** > **Setup** > **Mobile device** > **Warehouse app field names** and, if the list is empty, select **Create default setup** on the Action Pane. The default fields are then synchronized and made available for use with detours.

Make sample data available

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin. You can also use this scenario as guidance for using the feature on a production system. However, in that case, you must substitute your own values for each setting that is described here.

The example scenario uses the demo data that is associated with the worker/person *Julia Funderburk*.

Set up a worker to pack containers

For each worker that uses the Warehouse Management mobile app to pack containers, you must set up a container packing policy and a packing profile ID as described in the following procedure.

1. Go to **Warehouse management** > **Setup** > **Worker**.
2. On the Action Pane, select **Edit**.
3. In the list pane, select *Julia Funderburk*.
4. On the **Profile** FastTab, set the following values:
    - **Container packing policy:** Select *WH62Close*. This policy moves containers to the *Baydoor* location when they're closed.
    - **Packing profile ID:** Select *WH62*. This profile doesn't create warehouse work after a container is closed. It also doesn't post a sales packing slip when the last container for a shipment is closed. For more information about how to set up this process, see Set up container packing policies.
5. On the **Default packing station** FastTab, set the following values:
    - **Site:** Select *6*. This site is the site for the warehouse that is used in this scenario.
    - **Warehouse:** Select *62*. This warehouse is already enabled for the packing process in the demo data.
    - **Location:** Select *Pack*. The Warehouse Management mobile app always prompts workers to confirm this value. Workers can edit it as required. If you don't provide a default packing location, workers must always scan or look up the packing location. You can enable that functionality by using the data inquiry capability.
6. On the Action Pane, select **Save**.

> ⓘ **Note**
>
> You can set up the system to automatically print container labels when a new container record is created. Learn more in **Container label layouts and printing**.

Create a mobile device menu item for packing inventory into containers

Follow these steps to create a mobile device menu item that workers can use to pack inventory into containers.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name** – Enter *Packing*. This value is the internal name for the menu item.
    - **Title** – Enter *Packing*. This value is the display name of the menu item in the mobile app.
    - **Mode** – Select *Indirect*.
    - **Activity code** – Select *Pack inventory into containers*.
    - **Packing policy ID** – Leave this field blank. The default process is used.
4. On the Action Pane, select **Save**.

> **Tip**
>
> To control other aspects of the packing process, create a packing policy and use the **Packing policy ID** field to assign it to your menu device menu item. For more information, see Mobile device container packing policies.

Create a mobile device menu item for creating containers

Follow these steps to create a mobile device menu item that workers can use to create containers.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name:** Enter *Create container*.
    - **Title:** Enter *Create container*.
    - **Mode:** Select *Indirect*.
    - **Activity code:** Select *Container creation*.

4. On the Action Pane, select **Save**.

Create a mobile device menu item for closing containers

Follow these steps to create a mobile device menu item that workers can use to close containers.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name:** Enter *Close container*.
    - **Title:** Enter *Close container*.
    - **Mode:** Select *Indirect*.
    - **Activity code:** Select *Container closing*.
4. On the Action Pane, select **Save**.

Add the three new mobile device menu items to the menu

After you create the required mobile device menu items, add them to the mobile device menu, so that workers can access them.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. On the Action Pane, select **Edit**.
3. In the list pane, select *Outbound*.
4. In the **Available menus and menu items** column, select the new *Packing* mobile device menu item, and then select the **Add** button (right arrow) to move it into the **Menu structure** column.
5. Repeat the previous step for the other two new menu items: *Create container* and *Close container*.
6. On the Action Pane, select **Save**.

Add a detour for creating a container

The setup that you completed so far now enables workers to use the mobile app to pack items into containers. However, because workers must be able to access the *Create container* and *Close*

container processes directly from the menu, you can streamline the process further. You add a detour so that workers can create containers from within the packing flow.

Follow these steps to enable new containers to be created on the packing flow page that prompts workers to scan the container that they want to pack.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. In the **Filter** field, enter *ContainerIdToPack* to open a drop-down list that shows possible searches. Select *Step ID:* "*ContainerIdToPack*" in the list.
3. In the grid, select the record. Then, on the Action Pane, select **Add step configuration**.
4. In the drop-down dialog box, set the **Menu item** field to *Packing*, and then select **OK**.
5. The details page for the new step configuration appears. (The name of the page is **Packing : ContainerIdToPack**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
6. In the **Add detour** dialog box, in the **Available detours** list, find and select *Create container* (one of the menu items that you created for this scenario).
7. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.
8. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.
9. In the **Select fields to send** dialog box, in the **Send from packing** section, set the following values for the blank row that's added:

    - **Copy from Packing:** *Location*
    - **Paste in Create container:** *Location*
    - **Auto submit:** *Selected* (Use this setting so that the worker doesn't have to confirm the value.)

    > ⓘ **Note**
    >
    > If the expected fields aren't shown in the dialog box, make sure the default mobile application fields are created in your system, as noted in the **prerequisites**.

10. Select **Add** on the toolbar to add another row. Then set the following values for it:

- **Copy from Packing:** *Shipment*
- **Paste in Create container:** *Shipment*
- **Auto submit:** *Selected*

11. In the **Bring back from create container** section, set the following values for the blank row that's added:

    - **Copy from Create container:** *Container ID*
    - **Paste in Packing:** *Container ID*
    - **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)

12. Select **Add** on the toolbar to add another row. Then set the following values for it:

    - **Copy from Create container:** *Refresh* (This setting automatically updates the page after a return from the detour.)
    - **Paste in Packing:** *Refresh* (This setting triggers the update logic.)
    - **Auto submit:** *Selected* (This setting automatically updates the page so that, for example, the **Available open containers** field is updated.)

13. Select **OK** to close the dialog box.

14. Close the page.

> ⓘ **Note**
>
> Only Warehouse Management mobile app pages that contain the **Refresh** option can be used to trigger an automatic page refresh as part of a detour process. Therefore, you must include a row where both the **Copy from Create container** field and the **Paste in Packing** field are set to *Refresh*, and the **Auto submit** option is selected.

Add a detour for closing a container

Add another detour so that workers can close a container from within the packing flow.

Follow these steps to enable containers to be closed on the packing flow page that prompts workers to scan an item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.

2. In the **Filter** field, enter *ItemId* to open a drop-down list that shows possible searches. Select *Step ID: "ItemId"* in the list.

3. In the grid, select the record. Then, on the Action Pane, select **Add step configuration**.
4. In the drop-down dialog box, set the **Menu item** field to *Packing*, and then select **OK**.
5. The details page for the new step configuration appears. (The name of the page is **Packing : ItemId**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
6. In the **Add detour** dialog box, in the **Available detours** list, find and select *Close container* (one of the menu items that you created for this scenario).
7. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.
8. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.
9. In the **Select fields to send** dialog box, in the **Send from packing** section, set the following values for the blank row that's added:
    - **Copy from Packing:** *Location*
    - **Paste in Close container:** *Location*
    - **Auto submit:** *Cleared* (Use this setting to give workers the opportunity to inspect the location and choose whether to look up a container ID.)
10. Select **Add** on the toolbar to add another row. Then set the following values for it:
    - **Copy from Packing:** *Shipment*
    - **Paste in Close container:** *Shipment*
    - **Auto submit:** *Cleared* (Use this setting to give workers the opportunity to inspect the shipment and choose whether to look up a container ID.)
11. In the **Bring back from create container** section, don't add anything, because you don't want to pass any values back from this detour.
12. Select **OK** to close the dialog box.
13. Close the page.

Create the five lookup menu items

The setup that you completed so far enables workers to use the mobile app to pack inventory into containers. However, you can still add more lookup logic by using the data inquiry capability as part of the process. You add the following lookup mobile device menu items:

- **Look up location** – This menu item inquires about packing locations that inventory should be packed from.
- **Look up shipment** – This menu item inquires about shipments that must be packed.
- **Look up item** – This menu item inquires about item numbers that must be packed.
- **Look up container type** – This menu item inquires about container types for the container creation process.
- **Look up container** – This menu item inquires about containers that must be closed.

The following subsections explain how to set up these menu items.

Create a mobile device menu item to look up a location

Many different lookup capabilities enable lookups for packing locations. This section provides an example that shows how to set up a simple lookup to find locations that are related to a specific location profile. This lookup filters locations that are used for packing operations.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name:** *Look up location*
    - **Title:** *Look up location*
    - **Mode:** *Indirect*
4. On the **General** FastTab, set the following values:
    - **Activity code:** *Data inquiry*
    - **Use process guide:** *Yes* (This value is automatically selected.)
    - **Table name:** *WMSLocation* (You want to look up warehouse locations based on, for example, a location profile ID.)
5. On the Action Pane, select **Edit query** to define a query that's based on the selected base table. In this case, use the locations table. You can join to related tables as required.
6. In the query editor dialog box, on the **Range** tab, add the following lines to the grid.

Expand table

Table	Derived table	Field	Criteria
Locations	Locations	Warehouse	
Locations	Locations	Location profile ID	PACK

In this query, the **Warehouse** field is automatically assigned based on the worker's current warehouse.

7. If you want to specify how the list should be sorted, set up the sorting on the **Sorting** tab.
8. Select **OK**.
9. In addition to defining the query, select which fields to show on the cards on the inquiry list page. Therefore, on the Action Pane, select **Field list**.
10. On the **Field list** page, set the following values:
    - **Display field 1:** *wMSLocationId* (This field value is used as the header for each card.)
    - **Display field 2:** *inventLocationId*
    - **Display field 3:** *LocProfileId*
    - **Display field 4:** *whsDisplayQty()*
    - **Display field 5:** *whsDisplayItemId()*
11. On the Action Pane, select **Save**. Then close the page.

Create a mobile device menu item to look up a shipment

Many different lookup capabilities enable lookups for shipments. This section provides an example that shows how to set up a simple lookup to find active shipments.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name:** *Look up shipment*
    - **Title:** *Look up shipment*
    - **Mode:** *Indirect*
4. On the **General** FastTab, set the following values:
    - **Activity code:** *Data inquiry*

- **Use process guide:** *Yes* (This value is automatically selected.)
- **Table name:** *WHSShipmentTable*

5. On the Action Pane, select **Edit query** to define a query that's based on the selected base table. In this case, use the locations table. You can join to related tables as required.

6. In the query editor, on the **Range** tab, add the following lines to the grid.

Expand table

Table	Derived table	Field	Criteria
Shipments	Shipments	Warehouse	
Shipments	Shipments	Shipment status	Open, Waved, "In process"

In this query, the **Warehouse** field is automatically assigned based on the worker's current warehouse.

7. If you want to specify how the list should be sorted, set up the sorting on the **Sorting** tab.

8. Select **OK**.

9. In addition to defining the query, select which fields to show on the cards on the inquiry list page. Therefore, on the Action Pane, select **Field list**.

10. On the **Field list** page, set the following values:

- **Display field 1:** *ShipmentId* (This field value is used as the header for each card.)
- **Display field 2:** *ShipmentStatus*
- **Display field 3:** *displayDeliveryName()*
- **Display field 4:** *displayNumberOfContainers()*
- **Display field 5:** *displayNumberOfLoadLines()*
- **Display field 6:** *displayTotalVolume()*
- **Display field 7:** *displayTotalWeight()*
- **Display field 8:** *displayShipmentDateTime()*

11. On the Action Pane, select **Save**. Then close the page.

Create a mobile device menu item to look up an item

Many different lookup capabilities enable lookups for item information. This section provides an example that shows how to look up item information based on load lines. However, you can filter

based on a wildcard search instead. You can also use the packing work feature to look up or join to closed put work lines for work that brings inventory to a packing area.

In this example, a simple lookup is filtered based on the shipment ID that was previously captured as part of the packing process.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name:** *Look up item*
    - **Title:** *Look up item*
    - **Mode:** *Indirect*
4. On the **General** FastTab, set the following values:
    - **Activity code:** *Data inquiry*
    - **Use process guide:** *Yes* (This value is automatically selected.)
    - **Table name:** *WHSLoadLine*
5. On the Action Pane, select **Edit query** to define a query that's based on the selected base table. In this case, use the locations table. You can join to related tables as required.
6. In the query editor, on the **Range** tab, make sure that only the following line appears in the grid.

Expand table

Table	Derived table	Field	Criteria
Load details	Load details	Shipment ID	

7. Select **OK**.
8. In addition to defining the query, select which fields to show on the cards on the inquiry list page. Therefore, on the Action Pane, select **Field list**.
9. On the **Field list** page, set the following values:
    - **Display field 1:** *ItemId* (This field value is used as the header for each card.)
    - **Display field 2:** *displayItemName()*
    - **Display field 3:** *displayCustName()*

- **Display field 4:** *displayInventQty()*
- **Display field 5:** *displayInventUOM()*

10. On the Action Pane, select **Save**. Then close the page.

Create a mobile device menu item to look up a container type

Many different lookup capabilities enable lookups for container types. This section provides an example that shows how to set up a lookup to find all container types.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name:** *Look up container type*
    - **Title:** *Look up container type*
    - **Mode:** *Indirect*
4. On the **General** FastTab, set the following values:
    - **Activity code:** *Data inquiry*
    - **Use process guide:** *Yes* (This value is automatically selected.)
    - **Table name:** *WHSContainerType*
5. On the Action Pane, select **Edit query** to define a query that is based on the selected base table. In this case, you use the container type table. You can join to related tables as required.
6. In the query editor, on the **Range** tab, add the following line to the grid.

    Expand table

Table	Derived table	Field	Criteria
Container types	Container types	Container type code	*

    The wildcard character (*) in the **Criteria** field causes all container types to be shown.
7. If you want to specify how the list should be sorted, set up the sorting on the **Sorting** tab. In this example, you might want to sort on the **Maximum volume** field.
8. Select **OK**.

9. In addition to defining the query, select which fields to show on the cards on the inquiry list page. Therefore, on the Action Pane, select **Field list**.

10. On the **Field list** page, set the following values:

    - **Display field 1:** *ContainerTypeCode* (This field value is used as the header for each card.)
    - **Display field 2:** *Description*
    - **Display field 3:** *MaxVolume*
    - **Display field 4:** *MaxWeight*
    - **Display field 5:** *Height*
    - **Display field 6:** *Length*
    - **Display field 7:** *Width*
    - **Display field 8:** *FlexibleVolumeDimensions*

11. On the Action Pane, select **Save**. Then close the page.

Create a mobile device menu item to look up a container

Many different lookup capabilities enable lookups for containers. In this example, a new menu item is configured to find active containers in the current packing location related to the shipment that you're currently packing.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New** to add a mobile device menu item.

3. Set the following values for the new menu item:

    - **Menu item name:** *Look up container*
    - **Title:** *Look up container*
    - **Mode:** *Indirect*

4. On the **General** FastTab, set the following values:

    - **Activity code:** *Data inquiry*
    - **Use process guide:** *Yes* (This value is automatically selected.)
    - **Table name:** *WHSContainerWarehouseLocationView* (By using this view, you enable the lookup to be limited so that it shows only containers for a specific packing location that is related to a specific shipment.)

5. On the Action Pane, select **Edit query** to define a query that's based on the selected base table (in this case, the location table). You can join to related tables as required.

6. In the query editor, on the **Range** tab, add the following lines to the grid.

Expand table

Table	Derived table	Field	Criteria
WHSContainerWarehouseLocationView	WHSContainerWarehouseLocationView	Warehouse	
WHSContainerWarehouseLocationView	WHSContainerWarehouseLocationView	Location	
WHSContainerWarehouseLocationView	WHSContainerWarehouseLocationView	Shipment ID	
WHSContainerWarehouseLocationView	WHSContainerWarehouseLocationView	Container status	!Closed

In this query, the **Warehouse**, **Location**, and **Shipment ID** fields are automatically set based on the detour setup.

7. If you want to specify how the list should be sorted, set up the sorting on the **Sorting** tab.

8. Select **OK**.

9. In addition to defining the query, select which fields to show on the cards on the inquiry list page. Therefore, on the Action Pane, select **Field list**.

10. On the **Field list** page, set the following values:

    - **Display field 1:** *ContainerId* (This field value is used as the header for each card.)
    - **Display field 2:** *ContainerStatus*
    - **Display field 3:** *Weight*
    - **Display field 4:** *WeightUOM*
    - **Display field 5:** *ContainerNum*
    - **Display field 6:** *ContainerReleased*
    - **Display field 7:** *ContainerTypeCode*

11. On the Action Pane, select **Save**. Then close the page.

Add the new mobile device menu items to a menu

The new mobile device menu items can be added to the mobile device menu. You need to complete this task before you can use the menu items as part of a detour process. In this example, you create a new submenu and add the new menu items to it.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. On the Action Pane, select **New**.
3. On the header of the new record, set the following values:
    - **Name:** *Inquire*
    - **Description:** *Inquire*
4. In the **Available menus and menu items** column, select the new *Look up container* mobile device menu item, and then select the **Add** button (right arrow) to move it into the **Menu structure** column.
5. Repeat the previous step for the other four new lookup menu items: *Look up container type*, *Look up item*, *Look up location*, and *Look up shipment*.
6. On the Action Pane, select **Save**.
7. In the list pane on the left, select the *Main* menu.
8. In the **Available menus and menu items** list, in the **Menus** section, select your new *Inquire* menu. Then select the **Add** button (right arrow) to move that item into the **Menu structure** list.

    > **Tip**
    >
    > Workers can only access menu items that are included in the menu assigned to their mobile device user account (or a submenu of that menu). This restriction also applies to menu items that are intended for use as detours, but which you might not want workers to access directly from the menu. In this case, add the detour items to the relevant menus and then hide the items. To hide an item included in a menu, right-click on it in the **Menu structure** column and select **Hide**. To show an item that is currently hidden, right-click on it and select **Show**.

Configure detours in your mobile device steps

To complete the setup, use the detour configuration on the **Mobile device steps** page to add the five new mobile device menu items to existing menu item flows.

In this example, you complete these tasks:

- Add *Look up location* to the *Packing - Scan packing location* step.
- Add *Look up shipment* to the *Packing - Scan shipment* step.
- Add *Look up item* to the *Packing - Scan item* step.
- Add *Look up container* to the *Packing - Scan container to pack* step.
- Add *Look up container type* to the *Create container - Scan container type* step.
- Add *Look up container* to the *Close container - Scan container* step.

Add the Look up location menu item as a detour for packing

Follow these steps to add the **Look up location** menu item as a detour.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. In the **Filter** field, enter *PackingLocationId* to open a drop-down list that shows possible searches. Then select *Step ID: "PackingLocationId"* in the list.
3. In the grid, select the record. Then, on the Action Pane, select **Add step configuration**.
4. In the drop-down dialog box, set the **Menu item** field to *Packing*, and then select **OK**.
5. The details page for the new step configuration appears. (The name of the page is **Packing : PackingLocationId**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
6. In the **Add detour** dialog box, in the **Available detours** list, find and select *Look up location* (one of the lookup menu items that you created for this scenario).
7. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.
8. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.
9. In the **Select fields to send** dialog box, in the **Send from packing** section, don't add anything, because you don't want to pass any values to this detour menu item.
10. In the **Bring back from create container** section, set the following values for the blank row that's added:

- **Copy from Look up location:** *Location*
- **Paste in Packing:** *Location*
- **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)

11. Select **OK** to close the dialog box.

12. Close the page.

Add the Look up shipment menu item as a detour for packing

Follow these steps to add the **Look up shipment** menu item as a detour.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.

2. In the **Filter** field, enter *ShipmentId* to open a drop-down list that shows possible searches. Then select *Step ID: "ShipmentId"* in the list.

3. In the grid, select the record. Then, on the Action Pane, select **Add step configuration**.

4. In the drop-down dialog box, set the **Menu item** field to *Packing*, and then select **OK**.

5. The details page for the new step configuration appears. (The name of the page is **Packing : ShipmentId**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.

6. In the **Add detour** dialog box, in the **Available detours** list, find and select *Look up shipment* (one of the lookup menu items that you created for this scenario).

7. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.

8. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.

9. In the **Select fields to send** dialog box, in the **Send from packing** section, don't add anything, because you don't want to pass any values to this detour menu item.

10. In the **Bring back from create container** section, set the following values for the blank row that's added:

    - **Copy from Look up location:** *Shipment ID*
    - **Paste in Packing:** *Shipment*
    - **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)

11. Select **OK** to close the dialog box.

12. Close the page.

Add the Look up item menu item as a detour for packing

Follow these steps to add the **Look up item** menu item as a detour.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. In the **Filter** field, enter *ItemId* to open a drop-down list that shows possible searches. Then select *Step ID: "ItemId"* in the list.
3. In the grid, find the record that has a value of *Packing* in the **Menu item name** column, and select the link in the **Menu item name** column. This record is present because you already added a detour for this step of the packing process. You add another detour for it.
4. The details page for the new step configuration appears. (The name of the page is **Packing : ItemId**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
5. In the **Add detour** dialog box, in the **Available detours** list, find and select *Look up item* (one of the lookup menu items that you created for this scenario).
6. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.
7. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.
8. In the **Select fields to send** dialog box, in the **Send from packing** section, set the following values for the blank row that's added:
    - **Copy from Packing:** *Shipment*
    - **Paste in Look up item:** *Shipment ID*
    - **Auto submit:** *Selected* (Use this setting so that the worker doesn't have to confirm the value.)
9. In the **Bring back from look up item** section, set the following values for the blank row that's added:
    - **Copy from Look up item:** *Item number*
    - **Paste in Packing:** *Item*
    - **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)
10. Select **OK** to close the dialog box.

11. Close the page.

Add the Look up container menu item as a detour for packing

Follow these steps to add the **Look up container** menu item as a detour.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. In the **Filter** field, enter *ContainerIdToPack* to open a drop-down list that shows possible searches. Then select *Step ID:* "*ContainerIdToPack*" in the list.
3. In the grid, find the record that has a value of *Packing* in the **Menu item name** column, and select the link in the **Menu item name** column. This record is present because you already added a detour for this step of the packing process. You add another detour for it.
4. The details page for the new step configuration appears. (The name of the page is **Packing : ContainerIdToPack**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
5. In the **Add detour** dialog box, in the **Available detours** list, find and select *Look up container* (one of the lookup menu items that you created for this scenario).
6. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.
7. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.
8. In the **Select fields to send** dialog box, in the **Send from packing** section, set the following values for the blank row that's added:
    - **Copy from Packing:** *Location*
    - **Paste in Look up container:** *Location*
    - **Auto submit:** *Selected* (Use this setting so that the worker doesn't have to confirm the value.)
9. Select **Add** on the toolbar to add another row. Then set the following values for it:
    - **Copy from Packing:** *Shipment*
    - **Paste in Look up container:** *Shipment ID*
    - **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)
10. In the **Bring back from look up container** section, set the following values for the blank row that's added:

- **Copy from Look up container:** *Container ID*
- **Paste in Packing:** *Container ID*
- **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)

11. Select **OK** to close the dialog box.

12. Close the page.

Add the Look up container type menu item as a detour for creating containers

Follow these steps to add the *Look up container type* menu item as a detour:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.

2. In the **Filter** field, enter *ContainerTypeToCreateContainer* to open a drop-down list that shows possible searches. Then select *Step ID:* "*ContainerTypeToCreateContainer*" in the list.

3. In the grid, select the record. Then, on the Action Pane, select **Add step configuration**.

4. In the drop-down dialog box that appears, set the **Menu item** field to *Create container*, and then select **OK**.

5. The details page for the new step configuration appears. (The name of the page is **Create container : ContainerTypeToCreateContainer**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.

6. In the **Add detour** dialog box, in the **Available detours** list, find and select *Look up container type* (one of the lookup menu items that you created for this scenario).

7. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.

8. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.

9. In the **Select fields to send** dialog box, in the **Send from packing** section, don't add anything, because you don't want to pass any values to this detour menu item.

10. In the **Bring back from look up container type** section, set the following values for the blank row that's added:

    - **Copy from Look up location:** *Container type code*
    - **Paste in Packing:** *Container type code*

- **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)

11. Select **OK** to close the dialog box.

12. Close the page.

Add the Look up container menu item as a detour for closing containers

Follow these steps to add the *Look up container* menu item as a detour:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.

2. In the **Filter** field, enter *ContainerId* to open a drop-down list that shows possible searches. Then select *Step ID:* "*ContainerId*" in the list.

3. In the grid, select the record that has a value of *ContainerId* in the **Step ID** column. Then, on the Action Pane, select **Add step configuration**.

4. In the drop-down dialog box that appears, set the **Menu item** field to *Close container*, and then select **OK**.

5. The details page for the new step configuration appears. (The name of the page is **Close container : ContainerId**.) On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.

6. In the **Add detour** dialog box, in the **Available detours** list, find and select *Look up container* (one of the lookup menu items that you created for this scenario).

7. Select **OK** to close the dialog box and add the selected menu item to the list of available detours.

8. Select the new detour, and then select **Select fields to send** on the FastTab toolbar.

9. In the **Select fields to send** dialog box, in the **Send from close container** section, set the following values for the blank row that's added:

    - **Copy from Packing:** *Location*
    - **Paste in Look up container:** *Location*
    - **Auto submit:** *Selected* (Use this setting so that the worker doesn't have to confirm the value.)

10. Select **Add** on the toolbar to add another row. Then set the following values for it:

- **Copy from Packing:** *Shipment*
- **Paste in Look up container:** *Shipment ID*
- **Auto submit:** *Selected* (Use this setting so that the worker doesn't have to confirm the value.)

11. In the **Bring back from look up container** section, set the following values for the blank row that's added:

    - **Copy from Look up container:** *Container ID*
    - **Paste in Packing:** *Container ID*
    - **Auto submit:** *Cleared* (Use this setting so that the worker can confirm the value.)

12. Select **OK** to close the dialog box.

13. Close the page.

Create a sales order and complete the work

Follow these steps to create a sales order and complete the work of moving the ordered items to the packing area.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. On the Action Pane, select **New**.

3. In the **Create sales order** dialog box, in the **Customer account** field, select *US-005*.

4. Select **OK** to close the dialog box.

5. The new sales order opens and includes a single, blank line on the **Sales order lines** FastTab. Set the following values for the new order line:

    - **Item number:** *A0001*
    - **Quantity:** *2*
    - **Site:** *6*
    - **Warehouse:** *62*

6. While the order line is still selected on the **Sales order lines** FastTab, select **Inventory** > **Reservation** on the toolbar.

7. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.

8. Close the **Reservation** page to return to the sales order.
9. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**. A message shows the shipment and wave IDs for the order.
10. While the order line is still selected on the **Sales order lines** FastTab, select **Warehouse** > **Work details** on the toolbar. If you use batch processing to run your waves, you might have to wait a short time for the work to be created.
11. On the **Work** page, on the Action Pane, on the **Work** tab, select **Complete work**.
12. On the **Work completion** page, set the **User ID** field to *62*.
13. On the Action Pane, select **Validate work**.
14. When you receive a message that indicates that your work is valid, select **Complete work** on the Action Pane to complete the process of picking the inventory items and putting them in the *Pack* location.
15. Make a note of the **Shipment ID** value that is shown for the work in the upper grid.

Pack the ordered items into a container

The inventory items are brought to the packing area and they're ready to pack into containers. Follow these steps to create a new container in the system, and to pack and close the container.

1. Open the Warehouse Management mobile app, and sign in by using *62* as the user ID.
2. Open the **Packing** menu item.
3. Confirm that the **Packing location** field is set to *Pack*.
4. Enter or scan the **Shipment ID** value that appears when you complete the work.
5. Enter or scan *A0001* as the item.
6. Use the **Create container** detour button to jump to the *Container creation* process.
7. Enter or scan *Box-Large* as the container type.
8. Confirm the number sequence container ID, based on the *Container ID mode* setup. This step creates a new container for the shipment and returns to the main packing flow.
9. Enter or scan the new **Container ID** value to pack the 2 pcs of item *A0001* into the container. If you must update the case quantity, select the **Quantity** field, and edit the value before you confirm the **Container ID** value.
10. Use the **Close container** detour button to jump to the *Container closing* process.
11. Enter or scan the **Container ID** value to select the container to close.

12. Confirm the system weight for the two items that are packed into the box. To update the weight manually, select the weight value, and modify it before you confirm.
13. On the Action Pane, select **Close container**.
14. In the **Close container** dialog box, select **Get system weight** to fill in the default **Gross weight** value.
15. Select **OK** to close the container.

> **Note**
>
> The different field values shown on Warehouse Management mobile app pages can be sorted and highlighted by using the Promoted fields capability.

> **Tip**
>
> As part of the process, consider using packing work to pack outbound containers and process shipments, including adding different types of data inquiry menu items for this data.

Last updated on 05/28/2026

Set up short picking item reallocation

This procedure shows how to enable warehouse workers to quickly find alternative locations isn’t sufficient inventory at the location they’re directed to.

A **Work exception** controls the reallocation process, and the warehouse **worker** uses it.

You can use automatic, manual, or both reallocation processes:

- Automatic reallocation - Location directives determine if the goods are available at another location. If possible, the work is updated and the warehouse user is directed to the alternative location.
- Manual reallocation - The warehouse user can select from one or more locations with unreserved quantities of goods.
- Automatic and manual - If the system can't perform an automatic reallocation, and locations are available with unreserved quantities, the user is prompted to select a location.

Set up work exceptions

Define several work exceptions with different item reallocation policies so the warehouse worker can choose one based on the needs of the shipment that they're processing.

This procedure uses the USMF demo data company.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work exceptions**.
2. Select **New**.
3. In the **Work exception code** field, enter a value. This value is the title of the exception. For example, Short picking manual.
4. In the **Description** field, enter a value. This value is a short description of the usage of this exception. For example, Short picking - item not available.
5. In the **Exception** type field, select **Short pick**.
6. Select the **Adjust inventory** check box. If selected, inventory automatically adjusts to 0 at the short picked location.
7. In the **Default adjustment type code** field, enter or select a value. For example, in USMF you can select **Remove Res Adj Out**. Each Adjustment type code contains four characteristics: name, description, inventory journal name, and **Remove reservations**. If **Remove reservations** is enabled, the short-picked order line's reservations are removed.

8. In the **Item reallocation** field, select a value, such as Manual. If you select Manual, or Automatic and Manual, enable the warehouse worker to use manual reallocation.

Set up a worker to use manual item reallocation

This procedure uses the USMF demo data company.

1. Close the page.
2. Go to **Warehouse management** > **Setup** > **Worker**.
3. Select **Edit**.
4. In the list, select a worker. For example, Julia Funderburk.
5. Expand the **Users** FastTab.
6. In the list, select a **User ID**. For example, 24.
7. Expand the **Work** FastTab.
8. Select **Yes** in the **Allow manual item reallocation** field.

Last updated on 07/01/2026

System-directed work sequencing

10/03/2025

System-directed work sequencing lets you sort and filter the work orders that the system presents to users for execution. It's helpful in scenarios where additional criteria (such as the time of shipping, the picking zone, the location profile, or a combination of various criteria) are required to drive the warehouse picking process.

This functionality extends the current system-directed picking functionality by adding a system-directed query order, where you can set up a sequence and one or more queries that evaluate all work orders that are created. The system captures and presents only work orders that meet the criteria that you specify in the setup of the mobile device menu item.

Therefore, this functionality allows for further optimization of warehouse picking processes as it identifies work orders that match the specified criteria, assigns them to the correct mobile device menu item, and then presents them to a worker, based on a specific skill set, picking equipment, or other requirement.

> **Note**
>
> If you need different criteria, use multiple mobile device menu items.

Setup

Make demo data available

To work through the scenario by using the values that are presented in this article, you must work on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity. The scenario uses warehouse *51* from the demo data.

> **Important**
>
> Before you release the orders to the warehouse, make sure that the pick locations have enough inventory for all the items on the orders.
>
> Default USMF data should support this scenario. If you aren't using demo data, review the **Location directive** setting to learn which picking locations are used for sales order picking. If you must adjust the inventory, you can create manual movements, use replenishment, or use any other flow.

Set up a mobile device menu item

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. In the list of mobile device menu items, select **Sales Picking – System**. The required menu item should already exist.
3. Confirm the following settings:
    - On the **General** FastTab, set the **Directed by** field to *System directed*.
    - On the **Work classes** FastTab, confirm the following settings:

Expand table

Work class ID	Work order type
Sales	Sales orders
SO Pick	Sales orders

4. On the Action Pane, select **System directed work sequence queries**.
5. Select **Edit**.
6. Delete the existing line, and then confirm the action by selecting **Yes**.
7. On the Action Pane, select **New** to create a line.
8. On the new line, set the following values:
    - **Sequence number:** *1*
    - **Description field:** *Work quantity less than 20 and Descending*
9. Select **Save**.
10. On the Action Pane, select **Edit Query**.
11. On the **Joins** tab, expand the join hierarchy to show the **Work lines** table.
12. Select the **Work lines** table join.
13. Select **Add table join**.
14. In the list that appears, find and select the row that has the following settings:
    - **Join mode:** *n:1*
    - **Relation:** *Locations (Location)*

15. Select **Select**.

    The system adds locations to the table join.

16. On the **Sorting** tab, select **Add** to add a line.

17. On the new line, set the following values:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Work quantity* (In the message box that appears, select **Yes** to add sorting to this field.)
    - **Search direction:** *Descending*

18. Select the **Range** tab.

    If you want to include only specific work criteria in the sequencing, specify them on the **Range** tab. In this example, you want to include only work where the quantity is less than 20 ea (the lowest unit of measure).

    Notice that some lines are already included. Don't remove those lines.

19. Select **Add** to add a line.

20. On the new line, set the following values:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Inventory work quantity*
    - **Criteria:** *<20* (less than 20)

21. Select **Add** to add another line.

22. On the new line, set the following values:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Work type*
    - **Criteria:** *Pick*

23. Select **Add** to add another line.

24. On the new line, set the following values:

    - **Table:** *Locations*
    - **Derived table:** *Locations*

- **Field:** *Location profile ID*
- **Criteria:** *!STAGE*

> **Important**
>
> Be sure to include the exclamation point (*!*) in front of *STAGE*.

25. Select **OK** to save and close the query.
26. Select **Save**.
27. Close the page to return to the **Mobile device menu items** page.

> **Note**
>
> This setup defines the criteria that the system uses to feed eligible work to the mobile device menu item. If you add more criteria lines to the query, the system uses the query line with the lowest sequence number first. In other words, the user sees all eligible work for sequence number 1 first, then all work for sequence number 2. Therefore, if a specific range and sorting must be used together, specify them in the same system-directed work sequence query.
>
> This setup captures any work that has at least one line where the quantity is less than 20 ea. Therefore, if the work has a line where the quantity is exactly 20 ea or more than 20 ea, it's valid, provided that it also has at least one line where the quantity is less than 20 ea.

Location directives

If you use default Contoso data, you don't need to change the query for the location directive action. However, to make sure that the location directives capture the items on the sales orders when you apply the feature in a non-Contoso environment, create a new location directive. To verify the settings in the demo environment, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the **Work order type** field, select *Sales orders*.
3. Select the location directive named *51 Pick*.
4. On the **Location Directive Actions** FastTab, select the line for the **Pick** action.
5. Select **Edit query** above the grid.

6. Review the **Range** query.
    a. Find the line where the **Field** field is set to *Location*.
    b. Make sure that the **Criteria** field is blank (that is, there are no restrictions).

Scenario

Create sales order picking work

Before you run system-directed picking, create some outbound work. For this scenario, create four sales orders that are based on the specified system-directed work sequence queries.

You reserve inventory quantities for each sales order. Therefore, you can't withdraw reserved inventory from the warehouse for other orders unless you cancel the inventory reservation, or part of the inventory reservation.

You then release each sales order to the warehouse to create the outbound work.

Sales order 1

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New** to create sales order 1.
3. In the **Create sales order** dialog box, set the following values:
    - In the **Customer** section, set the **Customer account** field to *US-004*.
    - In the **General** section, set the **Warehouse** field to *51*.
4. Select **OK** to close the dialog box. Make a note of the sales order number.
5. Add a line to the new sales order, and set the following values:
    - **Item number:** *M9200*
    - **Quantity:** *20*
6. On the **Inventory** menu above the grid, select **Reservation**.
7. On the **Reservation** page, select **Reserve lot** to reserve the inventory.
8. Close the **Reservation** page.
9. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse** to create work for the warehouse.

You receive informational messages that show the wave ID and shipment IDs that were created for the sales order.

Sales order 2

1. On the Action Pane, select **New** to create sales order 2.
2. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-007*
    - **Warehouse:** *51*
3. Select **OK** to close the dialog box. Make a note of the sales order number.
4. Add a line to the new sales order, and set the following values:
    - **Item number:** *M9200*
    - **Quantity:** *5*
5. Select **Add line** to add a second line, and set the following values:
    - **Item number:** *M9201*
    - **Quantity:** *1*
6. Reserve inventory for both lines.
7. Release the order to the warehouse.

Sales order 3

1. On the Action Pane, select **New** to create sales order 3.
2. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-009*
    - **Warehouse:** *51*
3. Select **OK** to close the dialog box. Make a note of the sales order number.
4. Add a line to the new sales order, and set the following values:
    - **Item number:** *M9200*
    - **Quantity:** *7*
5. Select **Add line** to add a second line, and set the following values:

- **Item number:** *M9202*
- **Quantity:** *8*

6. Reserve inventory for both lines.

7. Release the order to the warehouse.

Sales order 4

1. On the Action Pane, select **New** to create sales order 4.

2. In the **Create sales order** dialog box, set the following values:

   - **Customer account:** *US-010*
   - **Warehouse:** *51*

3. Select **OK** to close the dialog box. Make a note of the sales order number.

4. Add a line to the new sales order, and set the following values:

   - **Item number:** *M9200*
   - **Quantity:** *25*

5. Select **Add line** to add a second line, and set the following values:

   - **Item number:** *M9202*
   - **Quantity:** *10*

6. Reserve inventory for both lines.

7. Release the order to the warehouse.

Get work IDs for the work that you created

1. Go to **Warehouse management** > **Work** > **Work details**.

2. Filter on the **Warehouse** field so that only work for warehouse *51* is shown.

3. Four work IDs are created. Make a note of the work ID for each sales order.

Expand table

Sales order ID	Work ID	Work quantity
Sales Order 1	Work ID 1	20 ea

Sales order ID	Work ID	Work quantity
Sales Order 2	Work ID 2	6 ea (sum of both lines)
Sales Order 3	Work ID 3	15 ea (sum of both lines)
Sales Order 4	Work ID 4	35 ea (sum of both lines)

Before you run the flow on the mobile device, make sure that only the work that you just created is in *Open* status for warehouse *51* and the *Sales order* work order type. Otherwise, the results of the test might vary, because the system-direct picking includes all eligible work.

1. Go to **Warehouse management** > **Work** > **Outbound** > **Open sales work**.
2. In the **Open sales work** grid, filter on the **Warehouse** field so that only work for warehouse *51* is shown.
3. Confirm that only the four work IDs that you created earlier are shown.
4. Close the **Work** page.

Mobile device flow execution

Based on the setup, the system feeds the user work sorted from the highest work line quantity to the lowest. The quantity on every line is less than 20.

Remember that this setup captures any work that has at least one line where the quantity is less than 20. Therefore, if the work has another line where the quantity is exactly 20 or more than 20, it's also valid.

Mobile app

1. Sign in to the warehousing app as a user in warehouse *51*.

2. Go to **Outbound** > **Sales Picking - System**.

   The pick step for work ID *4* is presented. This work ID is presented first because of the setup of the system-directed query order, where you specified that work is sequenced based on descending work line quantity.

3. Complete the required pick and put to close the work ID.

   Next, work ID *3* is presented. One of its work lines is next in the sequence, based on the work line quantity.

4. Complete the pick and put to close the work ID.

   Next, work ID *2* is presented. This work's pick line is next in the sequence.

5. Complete the pick and put to close the work ID.

   No further work is presented to you. Work ID *1* isn't eligible for this mobile device menu item, because the query specifies that work headers are considered only if the quantity on the work lines is less than 20.

Tips

The system-directed work sequence queries are *inclusive*. Remember this fact for some setups. For example, you want a specific menu item to process only work where the work unit is *ea*, and you specify that restriction on the **Range** tab of the query. In this case, the worker receives all work where at least one work line has the work unit set to *ea*. Therefore, this work might also include work where work lines have a work unit other than *ea* (such as *box* or *pallet*). The query excludes only work where no work line has the work unit set to *ea*.

Therefore, in the example from this scenario, the query also captured work ID *4*. When you created it, you added two lines: one for 25 ea and another for 10 ea. The work was still presented to the user, because at least one work line has a quantity of less than 20 ea.

Depending on the scenario, you can prevent this behavior by using work breaks.

Pick line grouping

Pick line grouping enables you to combine multiple work lines that have the same item and location into a single pick. The user sees this pick on the mobile device. Warehouse workers can follow the most efficient instructions, but the system still maintains required work line separation for different containers, orders, and other categories.

Set up pick line grouping

Create a mobile device menu item

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New**.
3. In the **Menu item name** field, enter *Sales group line picking*.
4. In the **Title** field, enter *Sales group line picking*. This title appears on the mobile device menu.
5. In the **Mode** field, select *Work*.
6. Set the **Use existing work** option to *Yes*.
7. On the **General** FastTab, set the following values:
    - In the **Directed by** field, select *User directed*.
    - Set the **Generate license plate** option to *Yes*.
    - Set the **Group pick** option to *Yes*.
    - Accept the default values for the remaining options.
8. Follow these steps to configure the valid work classes for the mobile device menu item:
    a. On the **Work classes** FastTab, select **New**.
    b. In the **Work class ID** field, select either *Sales* or *SO Pick*, depending on the warehouse that you use. For this scenario, select *SO Pick*.

    The **Work order type** field is automatically set to *Sales orders*.

Set up a mobile device menu

Follow these steps to add the menu item that you just created to the **Outbound** menu.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. On the Action Pane, select **Edit**.
3. The list pane shows all existing mobile device menus. Select *Outbound* in the list.
4. In the **Available menus and menu items** list, find and select the *Sales group line picking* menu item that you created.
5. Select the right arrow button to move the *Sales group line picking* menu item to the **Menu structure** list.
6. Use the up arrow and down arrow buttons to move the menu item into the desired position in the menu structure.
7. On the Action Pane, select **Save**.

Set up a work template

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. In the **Work order type** field, select *Sales orders*.
3. In the **Overview** grid, find and select the work template to use with this function. For this scenario, select the *51 Pick to stage* template.
4. On the Action Pane, select **Edit query**.
5. In the query editor dialog box, on the **Sorting** tab, select **Add**, and then set the following values for the new row:
    - In the **Table** column, select *Temporary work transactions*.
    - In the **Derived table** column, select *Temporary work transactions*.
    - In the **Field** column, select *Item number*.
    - In the **Search direction** column, select *Ascending*.
6. Select **OK** to close the dialog box and save your selection.
7. You receive the following message: "Grouping will be reset, continue?" Select **Yes** to continue.

> **ⓘ Important**
>
> For the pick line grouping functionality to work, the work lines must be sorted by item ID. If lines that have the same items aren't sequenced one after another, they aren't grouped.

Example

Create picking work

Before you can set up pick line grouping, you must create some eligible outbound work.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. Select **New** to create a sales order.

3. In the **Customer account** field, select *US-004*.

4. On the **General** FastTab, in the **Warehouse** field, select *51*.

5. Select **OK**.

6. On the **Sales order lines** FastTab, add the following six lines:

   - **Line 1:** In the **Item number** field, select *M9200*. In the **Quantity** field, enter *3*.
   - **Line 2:** In the **Item number** field, select *M9201*. In the **Quantity** field, enter *3*.
   - **Line 3:** In the **Item number** field, select *M9202*. In the **Quantity** field, enter *2*.
   - **Line 4:** In the **Item number** field, select *M9200*. In the **Quantity** field, enter *1*.
   - **Line 5:** In the **Item number** field, select *M9200*. In the **Quantity** field, enter *3*.
   - **Line 6:** In the **Item number** field, select *M9202*. In the **Quantity** field, enter *7*.

   Here is a summary of the total quantities for each item:

   - **Item M9200:** *7* each
   - **Item M9201:** *3* each
   - **Item M9202:** *9* each

7. Before you release the orders to the warehouse, make sure that the pick locations have enough inventory for all the items on all the orders. Review the **Location directive** setting to determine which picking locations are used for sales order picking. If you're using the Contoso demo data environment for warehouse *51*, confirm that there is available inventory.

   You must now reserve the inventory for each line.

8. On the **Sales order lines** FastTab, select one of the lines that must be reserved.

9. On the **Inventory** menu above the grid, select **Reservation**.

10. On the **Reservation** page, on the Action Pane, select **Reserve lot** to apply the reservation. Then close the page.

11. Repeat steps 8 through 10 for the remaining lines that must be reserved.

    You must now release the sales order to the warehouse.

12. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

    You receive a message that states that a shipment and a wave are created, and that the wave is submitted to run in a batch.

    When the wave and all downstream jobs are completed, a work ID is created for work that has six lines. The lines are sorted by item number.

13. Go to **Warehouse management** > **Work** > **All work** to view the work that you created. Make a note of the **Work ID** value, because you need it in the next procedure.

Initiate picking from the mobile device

1. Sign in to the mobile device as a user who is set up for warehouse *51*.

2. On the mobile device, select the menu that includes the new mobile device menu item. For this scenario, select **Outbound**.

3. Select the **Sales group line picking** menu item to initiate the pick.

4. Enter the **Work ID** value that you noted in the previous procedure.

   You see a pick step where all pick lines for item *M9200* are grouped, and you receive an instruction to pick *7* each of item *M9200*.

   > **ⓘ Important**
   >
   > On the mobile device, the pick work for the three pick work lines is aggregated into one picking step for the user.

5. Confirm the pick step.

6. Go to the work page for the work ID, and notice that all three pick lines for item *M9200* are closed simultaneously.

7. Go back to the mobile device, and continue to pick. Work line 4 for item *M9201* is presented. Because there's only one work line on the order, nothing is left to aggregate.
8. Confirm the pick step.
9. The last pick step on the mobile device aggregates the last two pick lines from the work order.
10. Complete the pick step for *9* each of item *M9202*.
11. Confirm the put step and any additional pick/put pairs to complete the work.

> **Important**
>
> - You can group work lines only if they're in sequence.
> - The following functionality isn't supported:
>   - Catch-weight items (If there are any catch-weight items on the work, you receive an error message before you start to pick.)
>   - Piece picking
>   - Work lines that have unfinished replenishment work
>   - Over-picking
>   - Short picking with item reallocation

Last updated on 07/01/2026

Warehouse slotting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Several warehouse slotting features help warehouse managers intelligently plan picking locations before they release orders to the warehouse and create picking work.

The *Warehouse slotting* feature lets you consolidate demand by item and unit of measure from orders that have a status of *Ordered*, *Reserved*, or *Released*. You can apply generated demand to locations used for picking, based on quantity, unit, physical dimensions, fixed locations, and more. After the slotting plan is established, you can create replenishment work to bring the appropriate amount of inventory to each location.

The *Warehouse slotting for transfer orders* feature lets warehouse managers replenish picking locations, based on demand from transfer orders that aren't yet released to the warehouse. It ensures that picking locations include all the items that are required for the transfer orders after they're released to the warehouse.

The *Warehouse slotting allocation enhancements* feature adds an option for the template lines that the *Warehouse slotting* feature uses. The option enables the system to consider existing on-hand inventory at a target location. Therefore, the system might generate fewer replenishments for slotting.

Set up warehouse slotting

To use warehouse slotting, set up the following elements in your system:

- Slotting unit of measure tiers
- Directive codes
- Slotting templates
- Location directives

Create unit-of-measure tiers for slotting

Unit-of-measure tiers enable you to group multiple units of measure together for slotting. For example, if multiple sizes of boxes are all picked from the same box picking area, you can create one tier for all the sizes. **A line must be created for each unit of measure that you want to include in the tier.**

1. Go to **Warehouse management** > **Setup** > **Replenishment** > **Slotting unit of measure tiers**.
2. Select **New**.
3. In the header, set the following values:
    - **Unit of measure tier:** *EaBoxPl*
    - **Description:** *Each box pallet*
4. Select **Save**.
5. On the **Units of measure** FastTab, select **New** to add a line to the grid.
6. On the new line, set the following values:
    - **Unit:** *Box*
    - **Description:** Leave this field blank. It's filled in automatically when you save your changes.
    - **Unit class:** *Quantity*
7. Select **New** to add a second line to the grid.
8. On the new line, set the following values:
    - **Unit:** *ea*
    - **Description:** Leave this field blank. It's filled in automatically when you save your changes.
    - **Unit class:** *Quantity*
9. Select **New** to add a third line to the grid.
10. On the new line, set the following values:
    - **Unit:** *PL*
    - **Description:** Leave this field blank. It's filled in automatically when you save your changes.

- **Unit class:** *Quantity*

11. Select **Save** to save the tier.

Create a directive code for slotting

Select the directive code to associate with a template.

1. Go to **Warehouse management** > **Setup** > **Directive codes**.
2. On the Action Pane, select **New**.
3. In the **Directive code** field, enter *Slotting*.
4. In the **Directive description** field, enter *Slotting*.

Set up slotting templates

Each slotting template controls how inventory is assigned to locations for a specific warehouse. Each template must include a line for each slotting specification. Use the procedures in this section to set up slotting templates.

1. Go to **Warehouse management** > **Setup** > **Replenishment** > **Slotting templates**.
2. Select **New** to create a template.

Next, set up the template header, slotting specifications, and location directives, as explained in the following subsections. The setup for slotting for transfer orders resembles the setup for slotting for sales orders, but the **Demand type** field is set to *Transfer orders* instead of *Sales order*.

Set up the header for a sales order slotting template

1. In the header for the template, set the following values:

    - **Slotting template:** *61*
    - **Description:** *61*
    - **Demand type:** *Sales order*

      > ⓘ **Note**
      >
      > Currently, *Sales orders*, *Transfer orders*, and *Outbound shipment orders* are the only demand types that are supported.

    - **Demand strategy:** *Ordered*

The following values are available in this field:
    - **Ordered** – The full ordered quantity on the sales order is considered demand.
    - **Reserved** – Only the sales order line quantities that are reserved (physical and ordered) are considered demand.
    - **Released** – The released quantity is considered demand.

- **Allow wave demand to use unreserved quantities:** *Yes*

2. Use the **Warehouse selection** fast tab to specify the warehouse where the slotting template applies.

    - **Warehouse selection** – Select one of the following values:
        - *All* – Use the slotting template for all warehouses.
        - *Warehouse group* – Use the slotting template for all warehouses in the warehouse group selected in the **Warehouse group** field.
        - *Warehouse* – Use the slotting template only for the specific warehouse selected in the **Warehouse** field.

    - **Warehouse** – If the **Warehouse selection** field is set to *Warehouse*, select the warehouse where the slotting template applies.

    - **Warehouse group** – If the **Warehouse selection** field is set to *Warehouse group*, select the warehouse group where the slotting template applies. For more information about how to set up warehouse groups, see Warehouse groups.

    For this scenario, set the following values:

    - **Warehouse selection:** *Warehouse*
    - **Warehouse:** *61*

You can also specify a query to narrow the scope of the demand that is evaluated.

Set up slotting specifications for each template

For each sales order template that you create, add a line for each slotting specification.

1. On the **Slotting template details** FastTab, select **New** to create a template line.

2. On the new line, set the following values:

    - **Sequence:** *1*

    - **Description:** *Fixed location*

- **Minimum quantity:** *1*

  This field defines the minimum quantity of demand that is required for the line.

- **Maximum quantity:** *1000000*

  This field defines the maximum quantity of demand that is valid for the line.

- **Unit:** Leave this field blank.

  This field defines the unit of measure that the minimum and maximum quantities refer to.

- **Unit of Measure Tier:** *EaBoxPl*

  This field defines the units of measure of demand that are valid for the line. (Learn more in the Set up unit-of-measure tiers for slotting section earlier in this article.)

- **Assign slot criteria:** *Consider qty*

  The following values are available in this field:
  - **Assume empty** – The system assumes that all locations in the picking area are empty and doesn't check those locations for inventory.
  - **Consider qty** – The system checks the locations in the picking area for inventory and skips any locations that aren't empty.
  - **Consider on-hand** – The system checks whether any target location contains unreserved quantities for the item on the demand line. If the quantity is large enough to satisfy at least one unit of the demand line, the generated slotting plan record is reduced by the available amount. For example, if the demand is 10 cases, and one case is on hand, the located demand is nine cases. If the demand is 10 cases, and one each is on hand, the located demand is 10 cases.

- **Directive code:** *Slotting*

  This field defines the location directive used to find the picking location of the replenishment work.

- **Overflow location:** Leave this field blank.

  This field defines the location that inventory that is put to if a location can't be found for the quantity when the line is processed.

- **Allow let up:** *Yes*

When this option is set to *Yes*, if any demand can't be slotted, movement work is created to take inventory out of locations where there is inventory, but where nothing was slotted. The template is then run again. This time, it ignores the inventory in the locations. This functionality works best when the **Assign slot criteria** field is set to *Consider qty*.

- **Fixed location usage:** *Only fixed locations for the product*

  The following values are available in this field:
  - **Fixed and non-fixed locations** – The system isn't limited to using only fixed locations.
  - **Only fixed locations for the product** – The system slots only to locations that are fixed locations for the product.
  - **Only fixed locations for the product variant** – The system slots only to locations that are fixed locations for the product variant.

> ⓘ **Note**
>
> If the slotting template contains at least one line where the **Assign slot criteria** field is set to *Consider on-hand*, the system no longer allows let-ups for any line in the template.

1. Select **Save**.

2. Select **New** to create a second template line.

3. On the new line, set the following values:

   - **Sequence:** *2*
   - **Description:** *Other*
   - **Minimum Qty:** *1*
   - **Maximum Qty:** *1000000*
   - **Unit:** Leave this field blank.
   - **Unit of measure tier:** *EaBoxPl*
   - **Assign slot criteria:** *Consider qty*
   - **Directive code:** *Slotting*
   - **Overflow location:** Leave this field blank.
   - **Allow let up:** *Yes*
   - **Use fixed locations:** *Fixed and non-fixed locations*

In the query for the second line, specify the criteria that determine what locations the demand for that line can be slotted to.

4. Select the line where the **Sequence** field is set to *2*.
5. Select **Select slot locations** in the FastTab toolbar (this button is labeled **Edit query** in Supply Chain Management version 10.0.43 and earlier).
6. On the **Range** tab, select **Add** to add a line to the grid.
7. On the new line, set the following values:
    - **Table:** *Locations*
    - **Derived table:** *Locations*
    - **Field:** *Location profile ID*
    - **Criteria:** *Pick-06* (Select the double plus sign [++] in the field to expand the list, and then select *Pick-06 - Picking Site 6*.)
8. Select **OK**.

Set up location directives

Set up at least one location directive to support slotting picks. Use the procedures in this section to set up a new *replenishment location directive* for slotting picks.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the left pane, in the **Work order type** field, select *Replenishment*.
3. On the Action Pane, select **New**.
4. In the header for the new location directive, in the **Name** field, enter *61 Slotting pick*.
5. In the **Sequence number** field, accept the default value.

Configure the Location directives FastTab

1. On the **Location directives** FastTab, set the following values. Accept the default values for all other fields.
    - **Work type:** *Pick*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Directive code:** *Slotting*
2. Select **Save** to make the **Lines** FastTab available.

Configure the Lines FastTab

1. On the **Lines** FastTab, select **New** to create a line.
2. On the new line, set the following values.
    - **From quantity:** *0*
    - **To quantity:** *1000000*
3. Accept the default values for the remaining fields.
4. Select **Save** to make the **Location Directive Actions** FastTab available.

Configure the Location Directive Actions FastTab

1. On the **Location Directive Actions** FastTab, select **New** to create a line.
2. On the new line, set the following values. Accept the default values for all other fields.
    - **Sequence number:** Accept the default value.
    - **Name:** *Bulk*
    - **Strategy:** *None*
3. Accept the default values for the remaining fields.
4. Select **Save** to make the **Edit query** button available.

Edit the query

1. On the **Location Directive Actions** FastTab, select **Edit query**.
2. On the **Range** tab, select **Add** to add a line to the grid.
3. On the new line, set the following values:
    - **Table:** *Locations*
    - **Derived table:** *Locations*
    - **Field:** *Zone ID*
    - **Criteria:** *Bulk* (Select the double plus sign [++] in the field to expand the list, and then select *Bulk*.)
4. Select **OK**.

Scenario

Set up the scenario

For this scenario, use the built-in sample data, and create the records described in this section.

Use the USMF sample data

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

Create demand

Follow these steps to create the demand that you apply slotting to.

1. Go to **Sales and marketing** > **Sales orders** > **All sales order**.
2. Select **New** to create a sales order.
3. In the **Create sales order** dialog box, in the **Customer account** field, select *US-007*.
4. In the **Warehouse** field, select *61*.
5. Select **OK**.
6. The new sales order opens. It includes an empty line on the **Sales order lines** FastTab. On this line, set the following values:
    - **Item:** *L0101*
    - **Quantity:** *20*
7. Select **Add line** to add a new line, and set the following values:
    - **Item:** *T0100*
    - **Quantity:** *8*
8. Select **Save**.
9. Select **New** to create a second sales order.
10. In the **Create sales order** dialog box, in the **Customer account** field, select *US-008*.
11. In the **Warehouse** field, select *61*.

12. The new sales order opens. It includes an empty line on the **Sales order lines** FastTab. On this line, set the following values:

    - **Item:** *T0100*
    - **Quantity:** *1*

13. Select **Save**.

Walk through a typical slotting scenario

After all the prerequisite elements are in place, as described in the previous section, you're ready to try out the feature by working through each exercise in this section.

Generate demand

1. Go to **Warehouse management** > **Setup** > **Replenishment** > **Slotting templates**, and select the slotting template that you created earlier.
2. On the Action Pane, select **Generate demand**. This command evaluates all demand in the system that matches the slotting template query. The total demand across all orders is then consolidated onto one line per quantity/unit of measure. An informational message appears when the process is completed.

Slotting demand

The *slotting demand* shows the results of demand generation, based on the setup of the slotting template.

- On the Action Pane, select **Slotting demand** to view the results from the **Generate demand** command. The lines in the slotting demand can be edited. You can delete a line, add a new line, or edit the line details.

> ⓘ **Note**
>
> You can edit demand manually, or you can import it from an external system by using data management. Whatever is in the slotting demand is used in the next step, regardless of where it came from.

Locate demand

After you generate the demand, use the **Locate demand** command to generate the *slotting plan*.

- On the Action Pane, select **Locate demand**. The slotting process runs. An informational message appears when the process is completed.

Slotting plan

The slotting plan shows the location that each item/quantity was assigned to, whether overflow was used, whether let-up work was created, and the template line that was used. *Any demand that couldn't be slotted is highlighted in red.*

- On the Action Pane, select **Slotting plan** to view the results.

> ⓘ **Note**
>
> - The **Generate demand**, **Locate demand**, and **Run replenishment** processes now run in a sandbox. (These processes are available from the Action Pane on the **Slotting templates** page.)
> - The **Generate demand**, **Locate demand**, and **Run replenishment** processes have a lock to ensure that they can't be triggered at the same time. Otherwise, the data use could be deleted.
> - The **Generate demand** and **Locate demand** processes show a warning if the run didn't generate records, or if the records are missing information.
> - When you select **Slotting plan**, the page doesn't have **New**, **Edit**, or **Delete** buttons on the Action Pane, because the data source can't be edited.
> - When you select **Run replenishment**, the system validates the selected slot template and processes.

Create replenishment

After the slotting plan is created, you must create *replenishment work*, based on the plan.

- On the Action Pane, select **Run replenishment**. An informational message appears when the process is completed. This message indicates the number of headers that were created for the work build ID.

Location directives for use are identified based on the directive code specified on each template line.

Tips

Set up automatic slotting

After all the required elements are in place, set up slotting to run automatically by following these steps.

1. Go to **Warehouse management** > **Replenishment** > **Run slotting**.
2. Specify the slotting steps to run. Select one or more of the following slotting steps:
    - Generate demand
    - Locate demand
    - Create replenishment work

    > **Note**
    >
    > The slotting steps are progressive. If you want to select *Locate demand*, you must first select *Generate demand*.

3. Specify the slotting template to use.
4. Set the recurrence to run automatically, if you want.

For the exercises in the scenario, don't set up automatic slotting.

Last updated on 05/29/2026

Location directive inventory picking aging

10/03/2025

This article explains how to use first in, first out (FIFO) and last in, first out (LIFO) location directive strategies during picking. These strategies work in conjunction with the aging dates that are recorded for locations to track when inventory first entered the warehouse. The *Location directive inventory picking aging* feature uses the date on the location to determine aging. The *Warehouse location status* feature updates the date on the location, based on the date from the license plate.

You can use FIFO and LIFO strategies to ship both batch-tracked items and non-batch-tracked items, based on the date when the inventory entered the warehouse. This capability can be especially useful for non-batch-tracked inventory, where an expiration date isn't available to use for sorting.

When inventory is first received or created in the warehouse, the system updates the relevant license plate so that the current date is shown as the aging date. This date is then used by the location directive strategies to identify the oldest or newest inventory in the warehouse. If inventory is moved to a location that isn't tracked by license plate, the location itself is updated with aging information, and this information will then be used by the strategies.

Feature requirements

To use this feature, you must set the **Enable location status** option set to *Yes* for every location profile that is used to store inventory. To set this option for a location profile, go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**, and select the location profile. You can find the option on the **General** FastTab.

Feature scenarios

This section provides examples that show how to set up and use FIFO and LIFO strategies.

> **Tip**
>
> This section provides two scenarios, one for FIFO and one for LIFO. The procedures that are provided assume that you will do both scenarios, in order. We recommend that you do both scenarios, so that you can experience the differences between the two strategies.

Make sample data available

To work through these scenarios by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin.

You can also use these scenarios as guidance for using the feature on a production system. However, in that case, you must substitute your own values for each setting that is described here.

Set up the scenarios

The demo data requires setup and inventory adjustments to support the scenarios. Follow these steps to create the inventory data that is required to work through the FIFO and LIFO scenarios.

1. Sign in to a system where demo data is installed, and select the *USMF* legal entity.
2. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
3. On the Action Pane, select **Edit**.
4. In the list of location profiles, select **FLOOR-05**.
5. On the **General** FastTab, set the **Enable location status** option to *Yes*.
6. Select **Save**.
7. Go to **Warehouse management** > **Setup** > **Location directives**.
8. In the list of location directives, select **63 Pick containerization**.
9. Select **Edit** to put the page into edit mode.
10. On the **Location directive actions** FastTab, find the line where the **Sequence number** field is set to *1*, and follow one of these steps:
    - If you're setting up a FIFO scenario, change the value of the **Strategy** field to *Location aging FIFO*.
    - If you're setting up a LIFO scenario, change the value of the **Strategy** field to *Location aging LIFO*.
11. On the **Location directive actions** FastTab, select **Edit query**.
12. In the query dialog box, on the **Range** tab, select **Add** to add a line, and then set the following values:
    - **Table:** *Locations*

- **Derived table:** *Locations*
- **Field:** *Zone ID*
- **Criteria:** *Floor*

13. Select **OK** to apply your settings and close the query dialog box.

14. Select **Save** to save your changes to the location directive.

15. On a mobile device or in the *Dynamics 365 Supply Chain Management - Warehousing* app on your PC, follow these steps to remove existing inventory from the warehouse location to support the scenarios:
    a. Sign in to warehouse *63* by using the appropriate user ID and password.
    b. On the main menu, select **Quality**.
    c. On the **Quality management** menu, select **Scrap**.
    d. On the **Scrap** page, select the **LOC/LP** field, and then enter *63_1*.
    e. Select **Enter** or **OK**.
    f. Confirm the **Scrap** details for **Adjustment out** by selecting **Enter** or **OK**.

    When the license plate inventory is adjusted out, you receive a "Work Completed" message.

    These steps leave inventory in two locations in the demo data. Each location has a different aging date. Location *FL-001* has an aging date of April 15, 2017, and location *FL-002* has an aging date of January 29, 2017. Both locations contain item *A0001*.

    To view this data, go to **Inventory management** > **Inquiries and reports** > **On-hand list**, and then filter on warehouse *63* and item *A0001*. In the rows where the **Location** field is set to *FL-001* or *FL-002*, select a line that has a positive **Physical inventory** value, and then, on the Action Pane, select **Transactions**. The **Physical date** field will show a date that corresponds to one of the previously mentioned aging dates.

Scenario 1: Set up and use FIFO location aging

The FIFO strategy finds the location that contains the oldest aging date, and it allocates picking based on that aging date.

1. If you haven't already done so, prepare the sample data that is required for this scenario.

2. Go to **Sales and marketing** > **Sales order** > **All sales orders**.

3. Select **New**.

4. In the **Create sales order** dialog box, set the following values:

    - On the **Customer** FastTab, set the **Customer account** field to *US-001*.

- On the **General** FastTab, set the **Warehouse** field to *63*.

5. Select **OK** to create the sales order and close the dialog box.
6. The new sales order is opened. It includes a new, empty row in the grid on the **Sales order lines** FastTab. For this order line, set the **Item number** field to *A0001* and the **Quantity** field to *1*.
7. On the **Inventory** menu above the grid, select **Reservation**.
8. On the **Reservation** page, select **Reserve lot** to reserve the ordered quantity of this item from inventory at the selected warehouse.
9. Close the **Reservation** page.
10. On the **Sales order** page, on the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**. You receive informational messages. The system creates a shipment, adds it to a new load, and creates the required work.
11. On the **Sales order lines** FastTab, on the **Warehouse** menu, select **Work details** to open the work that was created for this sales order. Notice that the line where the **Work type** value is *Pick* shows a **Location** value of *FL-002*. This location contains the license plate that has the oldest aging date (FIFO).
12. Select **Warehouse** > **Shipment details**.
13. On the **General** FastTab, make a note of the wave ID, so that you can use it in scenario 2.

Scenario 2: Set up and use LIFO location aging

The LIFO strategy finds the location that contains the newest aging date, and it allocates picking based on that aging date. In scenario 2, you will edit the setup for scenario 1 (FIFO), and reuse the sales order and wave that were created during that scenario.

1. Before you start this scenario, set up and complete the full FIFO scenario as described in the previous section. In this scenario, you will reuse the wave and most of the setup that were created for that scenario.
2. Edit the **63 Pick containerization** location directive so that it uses the *Location aging LIFO* strategy, as described in the first part of the Set up the scenarios procedure.

   Next, you will modify the wave that was created for the sales order in scenario 1, so that is uses the *Location aging LIFO* strategy.
3. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.

4. Select and open the wave that contains the order that you created for the FIFO scenario.
5. On the Action Pane, on the **Work** tab, select **Cancel** to cancel the work that you created for the FIFO scenario.
6. On the Action Pane, on the **Wave** tab, in the **Wave** group, select **Process**.
7. When the processing is completed, on the Action Pane, on the **Wave** tab, in the **Related information** group, select **Work** to open the work that was created for this wave.
8. On the **Work** page, on the **Overview** tab, there should be two lines. Select the line where the **Work status** field is set to *Open*.
9. Notice that the line where the **Work type** value is *Pick* shows a **Location** value of *FL-001*. This location contains the license plate that has the newest aging date (LIFO).

In these scenarios, you've seen how the location aging strategy directs work to the inventory location that has either the oldest inventory or the newest inventory, depending on the selected strategy.

Anchoring

Article • 06/03/2022

This article provides details about the anchoring process. It describes the configuration that is required and the logic that is run when a warehouse worker changes either the staging location or the loading location.

The anchoring feature lets you override the staging or loading location. All open puts will then be directed to the new staging or loading location that you specify.

This feature can help workers be more efficient while they ship goods. Here are some examples:

- A worker who must put items for order 1 in a staging location by dock 1 can't complete this operation because a previous load hasn't cleared the location. Instead of waiting for the dock 1 staging location to become available, the worker decides to use the staging location for dock 2. Therefore, the worker overrides the suggested staging location. The put location for all remaining items for the work is then updated to the dock 2 staging location.
- A worker who must perform multiple picks for the same delivery can be sure that all put items are assembled in the same place. Therefore, less time is required to load the items into the truck.

You configure anchoring for mobile device menu items by using the **Anchor** option. If you set this option to *Yes*, you can use the **Anchor by** field to specify whether you want to anchor by shipment or load. If you set the **Anchor by** field to *Shipment*, subsequent open puts will be changed to the new location for that shipment. If you set it to *Load*, subsequent open puts will be changed to the new location for that load.

> **ⓘ Important**
>
> The location for subsequent open puts will be changed only on the work lines that are generated from the same work template line. In other words, the system will anchor the put lines that originate from the same work template line.

This article provides a scenario that shows how anchoring works. During the scenario, you will create a set of sales orders and release them to the warehouse. You will then override the suggested staging location and verify that all the remaining put-away work is directed to the new location.

Scenario prerequisite: Make demo data available

The scenario in this article references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to *USMF* before you begin.

Scenario setup

Before you work through the example scenario, you must enable anchoring for the relevant mobile device menu item.

Set up a mobile device menu item to enable anchoring

Follow these steps to enable anchoring for a mobile device menu item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. In the list pane, select the record that is named *Sales Picking*. If no existing record has this name, create it. Confirm or set the following values for the record:
    - **Menu item name:** *Sales Picking*
    - **Title:** *Sales Picking*
    - **Mode:** *Work*
    - **Use existing work:** *Yes*
    - **Directed by:** *User directed*
    - **Anchoring:** *Yes*

      This setting causes the system to anchor multiple work order lines together during sales picking.
    - **Anchor by:** *Load*

      This setting causes the system to anchor by load.
    - **Override target license plate:** *Yes*

- **Override license plate during put:** *Yes*
- **Keep work locked by user:** *Yes*
- **Allow over pick:** *Yes*

Set up a work template to enable staging

Follow these steps to configure a work template to enable staging. This configuration will support the scenario where a worker puts items for an order in a staging location.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. In the **Work order type** field, select *Sales orders*.
3. In the grid, select the **61 SO Stage** work template.
4. In the **Work Template Details** section, make sure that the following lines exist and are configured as shown here:
    - Line 1:
        - **Work type:** *Pick*
        - **Mandatory:** Selected (= *Yes*)
        - **Work class ID:** *SO Pick*
    - Line 2:
        - **Work type:** *Put*
        - **Mandatory:** Selected (= *Yes*)
        - **Directive code:** *Stage*
        - **Work class ID:** *SO Pick*
    - Line 3:
        - **Work type:** *Pick*
        - **Mandatory:** Selected (= *Yes*)
        - **Stop work:** *Yes*
        - **Work class ID:** *SO Load*
    - Line 4:
        - **Work type:** *Put*
        - **Mandatory:** Selected (= *Yes*)
        - **Directive code:** *Baydoor*
        - **Work class ID:** *SO Load*
5. On the Action Pane, select **Edit query** to open the query editor.

6. On the **Range** tab, make sure that the following line is present:

    - **Table:** *Temporary work transactions*
    - **Derived Table:** *Temporary work transactions*
    - **Field:** *Warehouse*
    - **Criteria:** *61*

7. On the **Sorting** tab, make sure that the following line is present:

    - **Table:** *Temporary work transactions*
    - **Derived Table:** *Temporary work transactions*
    - **Field:** *Shipment ID*
    - **Search direction:** *Ascending*

8. Select **OK** to apply your settings and close the query editor. Accept the changes if you're prompted.

9. On the Action Pane, select **Work header breaks**.

10. On the line where the **Field name** field is set to *Shipment ID*, make sure that the **Group by this field** checkbox is selected.

Set up license plates, locations, and inventory

Before you create sales orders and shipments, you must ensure that the required locations, license plates, and inventory exist.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **License plates**, and create two license plates:

    - MyLP1
    - MyLP2

2. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**, and create the following locations if they aren't already present.

Expand table

Warehouse	Location	Location profile ID	Zone ID
61	06A01R2S1B	PICK-06	FLOOR
61	06A01R3S1B	PICK-06	FLOOR
61	STAGE01	STAGE	*(Blank)*

Warehouse	Location	Location profile ID	Zone ID
61	STAGE02	STAGE	*(Blank)*
61	STAGE03	STAGE	*(Blank)*
61	STAGE04	STAGE	*(Blank)*

3. Make sure that the following inventory is available. If you must adjust the inventory, you can create manual movements, use replenishment, or use any other flow.

Expand table

Item number	Quantity	Warehouse	Location	License plate
A0001	100	61	06A01R2S1B	MyLP1
A0002	100	61	06A01R3S1B	MyLP2

Create demand

Before you can try the anchoring functionality, you must create some demand. For this scenario, you will create three sales orders for the same customer.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create the first sales order.
3. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-001*
    - **Warehouse:** *61*
4. Select **OK**.
5. The new sales order is opened. It includes a blank line on the **Sales order lines** FastTab. Set the following values for this line:
    - **Item number:** *A0001*
    - **Quantity:** *1*
6. On the toolbar, select **Add line** to add a second sales order line, and set the following values for it:
    - **Item number:** *A0002*

- **Quantity:** *1*

7. Follow these steps for each sales line on the order to reserve inventory for it:
   a. On the **Sales order lines** FastTab, select a sales order line.
   b. On the toolbar, select **Inventory** > **Reservation**.
   c. On the **Reservation** page, select **Reserve lot**, and then close the page.
   d. On the Action Pane, select **Save**.

8. Repeat steps 2 through 6 to create a second sales order. Set the following values for it:

   - In the **Create sales order** dialog box:
     - **Customer account:** *US-001*
     - **Warehouse:** *61*

   - On sales order line 1:
     - **Item number:** *A0001*
     - **Quantity:** *2*

   - On sales order line 2:
     - **Item number:** *A0002*
     - **Quantity:** *2*

9. Repeat step 7 to reserve each line on sales order 2.

10. Repeat steps 2 through 6 to create a third sales order. Set the following values for it:

    - In the **Create sales order** dialog box:
      - **Customer account:** *US-001*
      - **Warehouse:** *61*

    - On sales order line 1:
      - **Item number:** *A0001*
      - **Quantity:** *3*

    - On sales order line 2:
      - **Item number:** *A0002*
      - **Quantity:** *3*

11. Repeat step 7 to reserve each line on sales order 3.

Use the outbound load planning workbench to create a load and release it to the warehouse

Follow these steps to create a load for the orders that you created for this scenario and then release it to the warehouse.

1. Go to **Warehouse management** > **Loads** > **Outbound load planning workbench**.
2. On the **Sales lines** tab, find and select all the sales order lines for sales order 1 and sales order 2.
3. On the Action Pane, on the **Supply and demand** tab, in the **Add** group, select **To new load**.
4. In the **Load template assignment** dialog box, in the **Load template ID** field, select a load template, such as *Stnd Load Template*.
5. Select **OK** to close the dialog box.
6. In the **Loads** section, find and select the load that you created.
7. On the toolbar, select **Release** > **Release to warehouse**.
8. In the **Release load to warehouse** dialog box, select **OK** to release the selected load to the warehouse.
9. Go to **Warehouse management** > **Work** > **All work** to view the work that was created. You should find two new work IDs, one for each shipment. Each work ID has pick and put lines that bring inventory from the picking locations to the staging location and from the staging location to the baydoor. Make a note of the **Work ID** value for the first shipment, because you will need it in the next procedure.

Sales order picking to the staging location for the first shipment

1. Sign in to the Warehouse Management mobile app as a worker in warehouse *61*. (In the standard demo data, you can sign in by using *61* as the user ID and *1* as the password.)

2. On the main menu, select **Outbound**.

3. On the **Outbound** menu, select **Sales Picking**.

4. Select the **ID** field, and then enter the work ID for the first shipment.

5. Confirm your entry.

6. In the **LP** field, enter a license plate number for the first item (*MyLP1*).

7. Confirm your entry.

8. In the **Target LP** field, enter any number (the targe license plate doesn't have to exist already).

You will pick all lines that were created from the processed wave to the same target license plate.

9. Confirm your entry.
10. In the **LP** field, enter a license plate number for the second item (*MyLP2*).
11. Confirm your entry.

    Imagine that the worker has now collected the order, but when they get to the staging location that specified in the work, they find that the space is already occupied. However, the worker can see that another nearby location (*STAGE03*) is available. Therefore, they will request to change the staging location. Because the anchoring feature is enabled, the system will then automatically update the staging location for this and all related work.

12. Select **Override Loc** to override the suggested staging location.
13. In the **Work exceptions** field, specify *Staging lane changed*.
14. In the **Location** field, enter a new staging location (*STAGE03*).
15. Confirm your entry. You receive a "Work completed" message.
16. Go to **Warehouse management** > **Work** > **All work**.
17. Open the work ID for the first shipment. Verify that the staging location was updated to the new location (*STAGE03*) that was defined by using the mobile device.
18. Open the work ID for the second shipment. Verify that the staging location was updated to new staging location (*STAGE03*) because of the anchoring setup.

> **Note**
>
> The location for all the remaining open put work lines that have the same staging location and that were generated from the same work template line will be updated to the new location.

Use the outbound load planning workbench to add the new sales order to the existing load

Follow these steps to add an order to the load and then release it to the warehouse.

1. Go to **Warehouse management** > **Loads** > **Outbound load planning workbench**.
2. On the **Sales lines** tab, find and select all the sales order lines for sales order 3.
3. On the Action Pane, on the **Supply and demand** tab, in the **Add** group, select **To existing load** to add the selected order lines to an existing load.
4. Select **OK** to close the dialog box.
5. In the **Loads** section, find and select the load from the previous steps.
6. Select **Release** > **Release to warehouse**.
7. In the **Release load to warehouse** dialog box, select **OK** to release the selected load to the warehouse.
8. Go to **Warehouse management** > **Work** > **All work** to view the work that was created. Make a note of the **Work ID** value, because you will need it in the next procedure.

Sales order picking to the staging location for the third shipment

1. Sign in to the mobile app as a worker in warehouse *61*.
2. On the main menu, select **Outbound**.
3. On the **Outbound** menu, select **Sales Picking**.
4. Select the **ID** field, and then enter the work ID for the third shipment.
5. Confirm your entry.
6. In the **LP** field, enter a license plate number for the first item (*MyLP1*).
7. Confirm your entry.
8. In the **Target LP** field, enter any number (the targe license plate doesn't have to exist already).
9. Confirm your entry.
10. In the **LP** field, enter a license plate number for the second item (*MyLP2*).
11. Confirm your entry.

    As for the first load, imagine that the worker finds that the specified staging location isn't available. Therefore, they want to use a different staging location (*STAGE04*).
12. Select **Override Loc** to override the suggested staging location.

13. In the **Work exceptions** field, specify *Staging lane changed*.
14. In the **Location** field, enter a new staging location (*STAGE04*).
15. Confirm your entry. You receive a "Work completed" message.
16. Go to **Warehouse management** > **Work** > **All work**.
17. Open the work ID for the first shipment. Verify that the staging location wasn't updated to the new staging location (*STAGE04*) because the remaining open put line doesn't correspond to the work template line of the completed put line.
18. Open the work ID for the second shipment. Verify that the staging location wasn't updated to the new staging location (*STAGE04*) because the staging location doesn't correspond to the staging location of the completed put line. In other words, the completed put work line and the remaining open work line have different staging locations and, in this case, only lines that have the same staging locations are updated.
19. Open the work ID for the third shipment. Verify that the staging location was updated to the new staging location (*STAGE04*).

Feedback

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Outbound sorting

Article • 03/03/2023

This functionality makes it easier to handle small containers and helps warehouse workers better plan and organize pallet capacity in the truck. When you use outbound sorting, you can sort packed containers to the correct pallet after they have been at a packing station. You can also build a packing hierarchy.

This functionality lets you build pallets from containers that are packed through the packing functionality. The container isn't sent to the final shipping location as it is in the original packing flow. Instead, workers can close the container and move it to a sort type location. They can then sort containers to positions, each of which has a license plate (LP). After the containers have been sorted, work can be created to send the whole LP to the final shipping dock or stage locations, based on location directives or your own requirements. Additionally, the action of closing of the sort position can immediately move the inventory to the final shipping location and pick it to the order.

Turn on the Outbound sorting feature

To use this feature, it must be turned on for your system. As of Supply Chain Management version 10.0.32, this feature is mandatory and can't be turned off. If you are running a version older than 10.0.32, then admins can turn this functionality on or off by searching for the *Outbound sorting* feature in the **Feature management** workspace.

Setup

For this scenario, you must use standard **USMF** demo data and warehouse *62*. You must also complete the setup that is described in the following subsections.

Set up a wave template

This setup automatically processes the wave and creates work when a line is released to the warehouse.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. In the template list, select **Warehouse 62**.
3. On the **General** FastTab, make sure that the **Process wave at release to warehouse** option is set to *Yes*.

Set up a worker

The packing station is considered a location. Warehouse workers who sign in at the packing station see and work on only shipments and containers that are planned at that specific packing location. A user who signs in to Microsoft Dynamics 365 must be set up as a worker in Warehouse management. If the user's name doesn't appear in the list of work users, use the following procedure to add it.

> ⓘ **Note**
>
> These steps assume that the user already exists in the system and has been associated as an employee or worker in the **Human Resources** module.

1. Go to **Warehouse management** > **Setup** > **Worker**.
2. Select **New**.
3. In the **Worker** field, select the target user in the list of employees.
4. Select **Select**.
5. On the Action Pane, select **Save**.
6. On the **Users** FastTab, select **New** to create a mobile device account, and set the following values for it:
    - **User ID:** Enter a unique ID.
    - **User name:** Enter a name for the ID.
    - **Default warehouse:** *62*
    - **Menu name:** *Main*
7. On the Action Pane, select **Save**.
8. The **Set password** dialog box appears, where you can create a simple password that the user can use to sign in to the mobile app. Set the following values:
    - **Password:** Enter a simple password.
    - **Confirm password:** Enter the same password again.
9. Select **Set password**.

    A notification in the Action Center informs you that the password has been set for the user that you created.

Create a location type

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location types**.
2. On the Action Pane, select **New** to create a location type, and set the following values for it:
    - **Location type:** *SORT*
    - **Description:** *Sort*
3. On the Action Pane, select **Save**.

Set up Warehouse management parameters

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **Location types** FastTab, set the **Sorting location type** field to *SORT*.
3. On the Action Pane, select **Save**.

Set up a location profile

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. On the Action Pane, select **New**.
3. In the header, set the following values:
    - **Location profile ID:** *Sorting*
    - **Name:** *Sorting*
4. On the **General** FastTab, set the following values:
    - **Location format:** *ASRB* (Aisle-Rack-Shelf-Bin)
    - **Location type:** *SORT*
    - **Use license plate tracking:** *Yes*
    - **Allow mixed items:** *Yes* (When you set this option to *Yes*, the **Allow mixed inventory batches** option is automatically set to *Yes* and can't be changed independently.)
5. Select **Save**.

Set up a location

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.

2. In the header, clear the **Generate check digits for location** check box.

3. On the Action Pane, select **New** to create a location, and set the following values for it:

    - **Warehouse:** *62*
    - **Location:** *SORT*
    - **Location profile ID:** *SORTING*

4. Select **Save**.

Set up an outbound sorting template

The outbound sorting template determines whether work is created out of the sort location, and whether sorting is done manually or automatically.

For this scenario, you will create an outbound sorting template to build pallets after the packing station.

1. Go to **Warehouse Management** > **Setup** > **Packing** > **Outbound sorting template**.

2. On the Action Pane, select **New**.

3. In the header of the new template, set the following values:

    - **Outbound sorting template ID:** *AutoWork*
    - **Description:** *Auto Work Creation*
    - **Outbound sorting template type:** *Container*

4. Use the **Warehouse selection** FastTab to specify the warehouse and location where the outbound sorting template will apply.

    - **Warehouse selection** – Select one of the following values:
        - *All* – Use the outbound sorting template for all warehouses.
        - *Warehouse group* – Use the outbound sorting template for all warehouses in the warehouse group that's selected in the **Warehouse group** field.
        - *Warehouse* – Use the outbound sorting template only for the specific warehouse that's selected in the **Warehouse** field.

    - **Warehouse** and **Location** – If the **Warehouse selection** field is set to *Warehouse*, select the warehouse and the location where the outbound sorting template applies.

- **Warehouse group** – If the **Warehouse selection** field is set to *Warehouse group*, select the warehouse group where the outbound sorting template applies. For more information about how to set up warehouse groups, see Warehouse groups.

For this scenario, set the following values:

- **Warehouse selection:** *Warehouse*
- **Warehouse:** *62*
- **Location:** *SORT*

5. On the **General** FastTab, set the following values:

   - **Sort verification:** *Position scan*

   - **Create work on position close:** *Yes*

     If this option is set to *Yes*, when the position is closed, work will be created to move inventory to the final shipping location. If it's set to *No*, inventory will immediately be picked to the order when the position is closed.

   - **Position assignment:** *Automatic*

     If this field is set to *Manual*, the user must always indicate which position the inventory should be sorted to. If it's set to *Automatic*, the system will automatically guide the inventory to a position whenever it can, based on the sort template breaks.

6. Select **Save** to make the **Edit query** button on the Action Pane available.

7. On the Action Pane, select **Edit Query**.

8. In the query editor, on the **Sorting** tab, add a line that has the following values:

   - **Table:** *Shipments*

   - **Derived table:** *Shipments*

   - **Field:** *Carrier service*

     When you select this value, you might receive the following message: "Field Carrier service is not an index field. Do you want to add sorting on this?" Select **Yes**.

   - **Search direction:** *Ascending*

9. Select **OK**.

10. You might receive the following message: "Grouping will be reset, continue?" Select **Yes**.

    The **Outbound sorting template breaks** button on the Action Pane becomes available.

11. On the Action Pane, select **Outbound sorting template breaks**.

12. In the **Outbound sorting criteria** dialog box, set the following values:

    - **Reference table name:** *Shipments*
    - **Reference field name:** *Carrier service*
    - **Group by field:** Select this check box to group shipments by carrier service.

13. Select **OK** to save your settings and close the dialog box.

Set up container packing policies

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container packing policies**.

2. On the Action Pane, select **New**.

3. In the header of the new policy, set the following values:

    - **Container packing policy:** *Sort*
    - **Description:** *Sort*

4. On the **Overview** FastTab, set the following values:

    - **Warehouse:** *62*
    - **Default location for sorting:** *SORT*
    - **Weight unit:** *kg*
    - **Container closing policy:** *Automatic release*
    - **Container release policy:** *Assign container to outbound sorting position*

5. Select **Save**.

Set up packing profiles

Create a new packing profile that will be used together with the sorting functionality.

1. Go to **Warehouse management** > **Setup** > **Packing** > **Packing profiles**.

2. On the Action Pane, select **New** to create a line, and set the following values for it:

- **Packing profile ID:** *Sort*
- **Description:** *Sort*
- **Container packing policy:** *Sort*
- **Container ID mode:** *Auto*
- **Container type:** *Box-Large*
- **Auto create container at container close:** Cleared (= *No*)

3. Select **Save**.

Set up work classes

Set up a work class that will be used together with sorting.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work classes**.
2. On the Action Pane, select **New** to create a work class for sorting, and set the following values for it:
    - **Work class ID:** *Sort*
    - **Description:** *Sort*
    - **Work order type:** *Sorted inventory picking*
3. Select **Save**.

Set up mobile device menu items

Set up a new pallet menu item

Create a mobile device menu item to build pallets during sorting.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New**.
3. In the header, set the following values:
    - **Menu item name:** *Pallet build*
    - **Title:** *Pallet build*
    - **Mode:** *Indirect*
    - **Use existing work:** *No*
4. On the **General** FastTab, set the following values:

- **Activity code:** *Outbound sorting*

  When this field is set to *Outbound sorting*, the **Outbound sorting template ID** field is shown.

- **Use process guide:** *Yes*

  When the **Activity code** field is set to *Outbound sorting*, this option is automatically set to *Yes*.

- **Outbound sorting template ID:** *AutoWork*

5. Select **Save**.

Set up a new loading menu item

Next, create a menu item that lets users move the sorted inventory items to the shipping location.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New**.
3. In the header, set the following values:
   - **Menu item name:** *Load from Sorting*
   - **Title:** *Load from Sorting*
   - **Mode:** *Work*
   - **Use existing work:** *Yes*
4. On the **General** FastTab, set the **Directed by** field to *User directed*.
5. On the **Work classes** FastTab, select **New**, and then set the following values:
   - **Work class ID:** *SORT*
   - **Work order type:** *Sorted inventory picking*
6. Select **Save**.

Set up the mobile device menu

You must now add the new menu items to the mobile device menu.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. Select the **Outbound** menu.

3. On the Action Pane, select **Edit**.
4. In the **Available menus and menu items** column, find and select **Pallet build**.
5. Select the right arrow button to move **Pallet build** to the **Menu structure** column.
6. Use the up arrow and down arrow buttons to put the **Pallet build** menu item in the desired position on the mobile device menu.
7. Select **Save**.
8. Repeat this procedure to add the **Load from Sorting** menu item to the **Outbound** menu.

Set up location directives

Location directives are rules that help identify pick and put locations for inventory movement. You must now create a rule to manage the sorting work.

Set up a single-SKU directive

1. Go to **Warehouse management** > **Setup** > **Location directives**.

2. In the left pane, change the value of the **Work order type** field to *Sorted inventory picking*.

3. On the Action Pane, select **New**.

4. In the header, set the following values:

    - **Sequence:** *1*
    - **Name:** *Baydoor*

5. On the **Location directives** FastTab, set the following values:

    - **Work type:** *Put*
    - **Site:** *6*
    - **Warehouse:** *62*
    - **Multiple SKU:** *No*

6. Select **Save** to make the toolbar on the **Lines** FastTab available.

7. On the **Lines** FastTab, select **New**, and then set the following values on the new line. Accept the default values for all the other fields.

    - **Sequence:** *1*
    - **From:** *0*
    - **To:** *1,000,000*

8. Select **Save** to make the toolbar on the **Location Directive Actions** FastTab available.

9. On the **Location Directive Actions** FastTab, select **New**, and then set the following values on the new line. Accept the default values for all the other fields.

    - **Sequence:** *1*
    - **Name:** *Baydoor*

10. Select **Save**.

11. On the **Location Directive Actions** FastTab, select **Edit query**.

12. In the query editor, on the **Range** tab, find the row where the **Field** field is set to *Location*. Set the **Criteria** field for this row to *Baydoor*.

13. Select **OK** to save your settings and close the query editor.

Set up a multiple-SKU directive

1. Go to **Warehouse management** > **Setup** > **Location directives**.

2. In the left pane, change the value of the **Work order type** field to *Sorted inventory picking*.

3. On the Action Pane, select **New**.

4. In the header, set the following values:

    - **Sequence:** *2*
    - **Name:** *Baydoor Multi*

5. On the **Location directives** FastTab, set the following values:

    - **Work type:** *Put*
    - **Site:** *6*
    - **Warehouse:** *62*
    - **Multiple SKU:** *Yes*

6. Select **Save** to make the toolbar on the **Lines** FastTab available.

7. On the **Lines** FastTab, select **New**, and then set the following values on the new line. Accept the default values for all the other fields.

    - **Sequence:** *1*
    - **From:** *0*

- **To:** *1,000,000*

8. Select **Save** to make the toolbar on the **Location Directive Actions** FastTab available.

9. On the **Location Directive Actions** FastTab, select **New**, and then set the following values on the new line. Accept the default values for all the other fields.

    - **Sequence:** *1*
    - **Name:** *Baydoor Multi*

10. Select **Save**.

11. On the **Location Directive Actions** FastTab, select **Edit query**.

12. In the query editor, on the **Range** tab, find the row where the **Field** field is set to *Location*. Set the **Criteria** field for this row to *Baydoor*.

13. Select **OK** to save your settings and close the query editor.

Set up work templates

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.

2. Change the value of the **Work order type** field to *Sorted inventory picking*.

3. On the Action Pane, select **New** to create a work template.

4. On the **Overview** tab, set the following values:

    - **Sequence:** *1*
    - **Work template:** *Sort*
    - **Work template description:** *Sort*

5. Select **Save** to make the **Work Template Details** FastTab available.

6. On the **Work Template Details** FastTab, select **New** to add a line, and then set the following values for it:

    - **Work type:** *Pick*
    - **Work class ID:** *SORT*

7. Select **New** again to add a second line, and then set the following values for it:

    - **Work type:** *Put*
    - **Work class ID:** *SORT*

8. Select **Save**.

Scenario

This scenario simulates a situation where packed containers should automatically be sorted to different positions (pallets) after the packing station, depending on the shipping carrier service. After all items from the load are packed and sorted by address, the pallets will be moved to the bay door.

Create sales orders and picking work

Create sales order 1

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-005*
    - **Warehouse:** *62*
4. Select **OK** to close the dialog box.

    The new sales order is opened.
5. Switch to the **Header** view.
6. On the **Delivery** FastTab, in the **Transportation** section, set the following values:
    - **Shipping carrier:** *Air cargo*
    - **Carrier service:** *Air*
7. Switch to the **Lines** view.
8. If a new, empty line isn't automatically added to the grid on the **Sales order lines** FastTab, select **Add line** to add one.
9. On the new order line, set the following values:
    - **Item number:** *A0001*
    - **Quantity:** *2*
10. While the new order line is still selected on the **Sales order lines** FastTab, on the **Inventory** menu above the grid, select **Reservation**.

11. On the **Reservation** page, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.
12. Close the **Reservation** page to return to the sales order.
13. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.
14. You receive an informational message that shows the shipment and wave for this order. Make a note of the wave ID and shipment ID numbers.

Sales order 2

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-006*
    - **Warehouse:** *62*
4. Select **OK** to close the dialog box.
5. The new sales order is opened. It should include a new, empty line in the grid on the **Sales order lines** FastTab. Set the following values on this order line:
    - **Item:** *A0001*
    - **Quantity:** *1*
6. On the **Line details** FastTab, on the **Delivery** tab, set the **Mode of delivery** field to *Flowe-STD*.
7. On the **Sales order lines** FastTab, select **Add line**, and then set the following values on the second order line:
    - **Item:** *A0002*
    - **Quantity:** *1*
8. On the **Line details** FastTab, on the **Delivery** tab, change the value of the **Mode of delivery** field to *Air C-Air*.
9. On the **Sales order lines** FastTab, select the first order line. Then, on the **Inventory** menu above the grid, select **Reservation**.

10. On the **Reservation** page, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.
11. Close the **Reservation** page to return to the sales order.
12. On the **Sales order lines** FastTab, select the second order line. Then, on the **Inventory** menu above the grid, select **Reservation**.
13. On the **Reservation** page, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.
14. Close the **Reservation** page to return to the sales order.
15. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.
16. You receive an informational message that shows the shipment and wave for this order. Notice that two wave ID numbers and two shipment ID numbers have been created, one for each mode of delivery for the sales order lines.

Get the work IDs from the work details

1. Go to **Warehouse management** > **Work** > **Work details**.
2. The page shows the work IDs that have been created from sales orders. Use the wave IDs and shipment IDs from the sales orders that you created to find the work ID for each wave and shipment. Make a note of those work IDs, because you will need them in the next steps. Notice that two work IDs were created for the second sales order. If different items are picked from different locations, separate word IDs are generated.

Pick items for the sales orders

Complete the created work by using the mobile device to move the items to the pack station.

1. On the mobile device, sign in to warehouse *62* by using the user ID that you created for this scenario (or the user ID of an existing demo data user).
2. On the main menu, select **Outbound**.
3. On the **Outbound** menu, select **Sales Picking**.
4. In the **ID** field, enter the work ID that was created for sales order 1.
5. Select **OK**.

6. On the **Sales orders - Pick** page, enter a target LP that was created for sales order 1. Notice that the picking location (*bulk-001*), item (*A0001*), and quantity (*2 pcs*) are shown.

7. Select **OK**.

8. Review the information on the **Sales orders - Put** page. The **Loc** field should indicate that the picked items are going to the *Pack* location.

9. Select **OK**.

   On the **Scan a work ID / license plate ID** page, you receive a "Work Completed" message, which indicates that the work ID from sales order 1 has been completed.

   You will now pick sales order 2.

10. In the **ID** field, enter the work ID that was created for sales order 2, where line 1 has item *A0001*.

11. Select **OK**.

12. On the **Sales orders - Pick** page, enter a target LP. Notice that the picking location (*bulk-001*), item (*A0001*), and quantity (*1 pcs*) are shown.

13. Select **OK**.

14. Review the information on the **Sales orders - Put** page. The **Loc** field should indicate that the picked items are going to the *Pack* location.

15. Select **OK**.

    On the **Scan a work ID / license plate ID** page, you receive a "Work Completed" message. This message indicates that the work ID from line 1 of sales order 2 has been completed.

16. In the **ID** field, enter the work ID that was created for sales order 2, where line 2 has item *A0002*.

17. Select **OK**.

18. On the **Sales orders - Pick** page, enter a target LP. Notice that the picking location (*bulk-002*), item (*A0001*), and quantity (*1 pcs*) are shown.

19. Select **OK**.

20. Review the information on the **Sales orders - Put** page. The **Loc** field should indicate that the picked items are going to the *Pack* location.

21. Select **OK**.

    On the **Scan a work ID / license plate ID** page, you receive a "Work Completed" message. This message indicates that the work ID from line 2 of sales order 2 has been completed.

Pack sales orders into containers

Pack sales order 1 into containers

1. Go to **Warehouse management** > **Packing and containerization** > **Pack**.

   The **Select packing station** dialog box appears. By default, the **Worker** field should be set to the name of the worker that you set up earlier.

2. Set the following values to view and work on shipments and containers that are planned at the specific packing location:

   - **Site:** *6*
   - **Warehouse:** *62*
   - **Location:** *Pack*
   - **Packing profile ID:** *Sort*

3. Select **OK** to close the dialog box.

4. On the **Pack** page, in the **License plate or shipment** field, enter the target LP for sales order 1. Then select the **Tab** or **Enter** key on your keyboard to move out of the field.

5. On the Action Pane, select **New container**.

6. Accept all the default settings, and select **OK**. Make a note of the container ID.

7. On the **Item packing** FastTab, set the following values:

   - **Quantity:** *1*
   - **Identifier:** Item *A0001*

8. On the Action Pane, select **Close container**.

9. In the **Close container** dialog box, select **Get system weight** to have the system update the **Gross weight** field.

10. Select **OK**. The container is moved to the *SORT* location and is ready for sorting.

11. Create a second container to add the second item from the LP for sales order 1 to a new container.

12. On the Action Pane, select **New container**.

13. Accept all the default settings, and select **OK**. Make a note of the container ID.

14. On the **Item packing** FastTab, set the following values:

    - **Quantity:** *1*
    - **Identifier:** Item *A0001*

15. On the Action Pane, select **Close container**.

16. In the **Close container** dialog box, select **Get system weight** to have the system update the **Gross weight** field.

17. Select **OK**. The container is moved to the *SORT* location and is ready for sorting.

Pack sales order 2 into containers

1. On the **Pack** page, in the **License plate or shipment** field, enter the target LP for line 1 of sales order 2. Then select the **Tab** or **Enter** key on your keyboard to move out of the field.

2. On the Action Pane, select **New container**.

3. Accept all the default settings, and select **OK**. Make a note of the container ID.

4. On the **Item packing** FastTab, set the following values:

    - **Quantity:** *1*
    - **Identifier:** Item *A0001*

5. On the Action Pane, select **Close container**.

6. In the **Close container** dialog box, select **Get system weight** to have the system update the **Gross weight** field.

7. Select **OK**. The container is moved to the *SORT* location and is ready for sorting.

8. In the **License plate or shipment** field, enter the target LP for line 2 of the sales order 2. Then select the **Tab** or **Enter** key on your keyboard to move out of the field.

9. On the Action Pane, select **New container**.

10. Accept all the default settings, and select **OK**. Make a note of the container ID.
11. On the **Item packing** FastTab, set the following values:
    - **Quantity:** *1*
    - **Identifier field:** Item *A0002*
12. On the Action Pane, select **Close container**.
13. In the **Close container** dialog box, select **Get system weight** to have the system update the **Gross weight** field.
14. Select **OK**. The container is moved to the *SORT* location and is ready for sorting.

To view the container details, go to **Warehouse management** > **Packing and containerization** > **Containers**, and search for the container IDs that were created during packing.

Sort the containers

> **Important**
>
> When you access the **Pallet build** menu item on the mobile app to do outbound sorting, you will see a button that is labeled **Full**. *Don't use the **Full** button to sort or close the position.*
>
> The **Full** button is provided by default and can't be disabled or removed from the page. It isn't used for the *Outbound sorting* feature.

Sort the first container

1. On the mobile device, sign in to warehouse *62* by using the user ID that you created for this scenario (or the user ID of an existing demo data user).
2. On the main menu, select **Outbound**.
3. On the **Outbound** menu, select **Pallet build**.
4. In the **LP/Con** field, enter the first container ID that is associated with sales order 1.
5. Select **OK**.
6. Because no sort positions currently exist, you must specify one. In the **Sort position ID** field, enter *SP01*.

7. Because no LP is currently associated with sort position *SP01*, you must specify one. In the **LP** field, enter *PLP01*.
8. Select **OK**.
9. Because sort position validation is turned on, you must enter the sort position ID again. In the **Sort Position ID** field, enter *SP01*.
10. Select **OK**.

    You receive a "Work completed" message.

> **Tip**
>
> To view the sort position and the LP in it, go to **Warehouse management** > **Packing and containerization** > **Outbound sorting position assignments**.
>
> The **Outbound sorting position assignments** page shows all the sort positions that are currently active. The **Sort position transactions** field shows the LP that is associated with each sort position, and the containers that are in the sort position. Notice that one sort position currently exists, and that the **Sort position criteria** FastTab shows a criterion of **Shipment – Carrier service - Air**.

Sort the remaining containers

1. On the mobile device, sign in to warehouse *62* by using the user ID that you created for this scenario (or the user ID of an existing demo data user).
2. On the main menu, select **Outbound**.
3. On the **Outbound** menu, select **Pallet build**.
4. In the **LP/Con** field, enter the second container ID that is associated with sales order 1.
5. Select **OK**. Because the sorting template is set up to sort automatically, and a sort position that has matching criteria already exists, you're automatically directed to the correct sort position.
6. Select **OK**.
7. Confirm the sort position ID to indicate that the inventory is in the correct place. In the **Sort Position ID** field, enter *SP01*.

8. Select **OK**.

   Work is completed on the second container from sales order 1. You will now sort the remaining containers from sales order 2.

9. In the **LP/Con** field, enter the container ID of the container from sales order 2 that holds item *A0001*. Because the carrier service differs, you're prompted to enter a new sort position and assign an LP to that position. Use sort position *SP02* and LP *PLP02*.

10. Select **OK**.

11. Confirm the sort position by entering *SP02* in the **Sort Position ID** field.

12. Select **OK**.

    Work is completed on the container.

13. In the **LP/Con** field, enter the container ID for the remaining container from sales order 2 that holds item *A0002*. Because the carrier service is the same as the carrier service for sales order 1, the system shows the existing sort position that has matching criteria.

14. Select **OK**.

15. Confirm the sort position by entering *SP01* in the **Sort Position ID** field.

16. Select **OK**.

    Work is completed on the container.

Close the outbound sorting positions

When all inventory has been sorted, the position must be closed before work can be created. Sorted inventory picking work will be created to take the inventory to the bay door.

Close a position from the mobile device

1. On the mobile device, sign in to warehouse *62* by using the user ID that you created for this scenario (or the user ID of an existing demo data user).

2. On the main menu, select **Outbound**.

3. On the **Outbound** menu, select **Pallet build**.

4. In the **LP/Con** field, enter a container ID that was sorted to sort position *SP01*.
5. Select **OK**.
6. You receive the following message: "The container is already sorted to position SP01. Close the position?" Select **Close**.

   Work is completed.

Close a position from outbound sorting position assignments

1. Go to **Warehouse management** > **Packing and containerization** > **Outbound sorting position assignments**.
2. In the left column, select **SP02**. This outbound sorting position row is the one that you will close.
3. On the Action Pane, select **Close position**. The sorting position record is closed and is no longer shown.

   > **Tip**
   >
   > To show all closed position records, select the **Show closed** check box.

Sorted inventory picking

You must complete the sorted inventory picking work. When it's completed, the inventory will be picked to the sales order. At that point, all other warehouse processes apply.

1. On the mobile device, sign in to warehouse *62* by using the user ID that you created for this scenario (or the user ID of an existing demo data user).
2. On the main menu, select **Outbound**.
3. On the **Outbound** menu, select **Load from Sorting**.
4. Enter the target LP ID from the first sort position, *SP01*. Set the **ID** field to *PLP01*.
5. Select **OK**.
6. The **Sorted inventory picking: Pick** page shows the pick work that must be done. Pick from the *SORT* location and target LP *PLP01*, which has multiple items and a quantity of *3*.

7. Select **OK**.
8. The **Sorted inventory picking: Put** page shows the put work that must be done. Put to the *Baydoor* location and target LP *PLP01*, which has multiple items and a quantity of *3*.
9. Select **OK**.

   Work is completed.
10. Enter the target license plate ID from the second sort position, *SP02*. Set the **ID** field to *PLP02*.
11. Select **OK**.
12. The **Sorted inventory picking: Pick** page shows the pick work that must be done. Pick from the *SORT* location and target LP *PLP02*, which has multiple items and a quantity of *1*.
13. Select **OK**.
14. The **Sorted inventory picking: Put** page shows the put work that must be done. Put to the *Baydoor* location and target LP *PLP02*, which has multiple items and a quantity of *1*.
15. Select **OK**.

    Work is completed.

From this point forward, all other warehouse processes apply.

Feedback

Was this page helpful? Yes No

Provide product feedback

Put to wall - put to store

Article • 02/01/2023

The *Put to wall - put to store* functionality lets you handle scenarios where you must consolidate a product to a prepack staging area, based on configurable criteria. Because this functionality allows for picking to a single target license plate and can use more put positions than cluster picking, companies that ship to stores or handle small items will benefit from decreased picking time.

The workflow for this functionality directs picked product to a sorting location for distribution into various types of containers. Generally, each sorting location includes multiple sort positions. Each sort position is assigned according to the criteria that are set by the business process. Typical criteria are the final destination, shipment, or load. After a product arrives, it's distributed to each sort position, based on the quantity that is associated with the order, destination, shipment, or load. When a container is full or complete, it's moved to a staging location or a shipping location for further handling, depending on the business process.

This warehousing functionality is also referred to by other names, such as put-to-light and break opens.

Turn on the Outbound sorting feature

To use the *Put to wall - put to store* functionality, two features must be turned on for your system. Admins can use the feature management settings to check the status of these features and turn them on if necessary. Both of the following features are required:

- *Outbound sorting* (As of Supply Chain Management version 10.0.32, this feature is mandatory and can't be turned off.)
- *Organization wide wave step code* (As of Supply Chain Management version 10.0.32, this feature is mandatory and can't be turned off.)

Setup

For this demo, standard Contoso data and warehouse *62* are used. Some additions that are noted later are also used.

Location type

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location types**.
2. On the Action Pane, select **New** to create a location type for sorting.
3. Set the following values:
    - **Location type:** *SORT*
    - **Description:** *Sort*
4. Select **Save**.

Warehouse management parameters

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **Location types** FastTab, in the **Sorting location type** field, enter *SORT*.
3. Select **Save**.

Location profile

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. On the Action Pane, select **New** to create a location profile for the sorting location.
3. In the header, set the following values:
    - **Location profile ID:** *Sort*
    - **Name:** *Sort*
4. On the **General** FastTab, set the following values:
    - **Location format:** *PACK*
    - **Location type:** *SORT*
    - **Use license plate tracking:** *Yes*
    - **Allow mixed items:** *Yes*
    - **Allow mixed inventory statuses:** *Yes*
5. Select **Save**.

Locations

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.
2. Clear the **Generate check digits for location** check box.

3. On the Action Pane, select **New**, and then set the following values:

    - **Warehouse:** *62*
    - **Location:** *Sort*
    - **Location profile ID:** *Sort*

4. Select **Save**.

Packing profiles

1. Go to **Warehouse management** > **Setup** > **Packing** > **Packing profiles**.

2. On the Action Pane, select **New**, and then set the following values:

    - **Packing profile ID:** *Sort*
    - **Description:** *Sort*
    - **Container packing policy:** Leave this field blank.
    - **Container ID mode:** *Auto*
    - **Container type:** *PALLET 48X48*
    - **Autocreate container at container close:** Leave this field blank.

3. Select **Save**.

Wave step codes

If the *Organization wide wave step code* feature is turned on for your system, set up the following code.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave step codes**.

2. On the Action Pane, select **New**, and then set the following values:

    - **Wave step code:** *Sort*
    - **Wave step description:** *Sort*
    - **Wave step type:** *Sort template*

3. Select **Save**.

Outbound sorting template

The sorting template controls whether sort positions are created, what criteria are used, and other attributes of the sorting process.

1. Go to **Warehouse management** > **Setup** > **Packing** > **Outbound sorting template**.

2. On the Action Pane, select **New** to create a sorting template.

3. In the header of the new template, set the following values:

    - **Outbound sorting template ID:** *Wave Sort*
    - **Description:** *Wave sort*
    - **Sort template type:** *Wave demand*

      This field defines the process that the sorting template is used for. The following values are available:
        - **Wave demand** – The sorting template is used for the *Put to wall* process. This template type is used to bypass the pack station and process inventory directly out of the wave. You can use this type only if the **sorting** wave process method is included in the wave template.
        - **Container** – The sorting template is used for the *Pallet building after packing* process. This template type is used to process containers that are closed at the pack station and must be sorted onto pallets.
    - **Warehouse:** *62*
    - **Location:** *Sort*

4. On the **General** FastTab, set the following values:

    - **Sort verification:** *Position scan*

      This field defines the verification that is required during sorting. The following values are available:
        - None
        - License plate scan
        - Position scan
    - **Create work on position close:** *Yes*

      If this option is set to *Yes*, when the position is closed, work will be created to move inventory to the final shipping location. If it's set to *No*, inventory will immediately be picked to the order when the position is closed.
    - **Position assignment:** *Manual*

This field defines the type of position assignment. The following values are available:
  - **Manual** – The user must always indicate which position the inventory should be sorted to.
  - **Automatic** – The system will automatically guide the inventory to a position whenever it can, based on the sort template breaks.

- **Assign sort position criteria:** *Only use empty position*

  This field controls whether inventory that is already present on sort positions is taken into account when a position is assigned for the demand. The following values are available:
  - **Only use empty position** – Positions that already have inventory associated with them will be taken into account
  - **Assume position empty** – Any inventory that is already on the position will be disregarded during assignment. All available positions will be used.

- **Wave step code:** *Sort*

  If the *Organization wide wave step code* feature is turned on, the *Sort* wave step code must also be set up in wave step codes.

- **Auto close sort position:** *Yes*

  If this option is set to *Yes*, the sort position will automatically be closed when all work that comes to the position has been completed.

- **Number of sort positions:** *3*

  This field defines the maximum number of sort positions that the system will create.

- **Sort position prefix:** *SP-*

  This field defines the prefix that the system assigns before the position number.

- **Auto pack sort position:** *Yes*

  If this option is set to *Yes*, the inventory on the sort position will be packed into a container when the position is closed.

- **Packing profile ID:** *Sort*

  This field defines the packing profile that will be used when the sort position is packed into a container.

5. On the Action Pane, select **Edit query** to specify the criteria that are used for this sorting template.
6. In the query dialog box, on the **Sorting** tab, select **New** to add a line, and then set the following values:
    - **Table:** *Load details*
    - **Derived table:** *Load details*
    - **Field:** *Shipment ID*
    - **Search direction:** *Ascending*
7. Select **OK**.
8. You might receive the following message: "Grouping will be reset, continue?" Select **Yes**.

    The **Outbound sorting template breaks** button on the Action Pane becomes available.
9. On the Action Pane, select **Outbound sorting template breaks**.
10. Select the **Group by field** check box to group by shipment ID.

    This setting will create one sort position per shipment that is a container in the wave.
11. Select **OK**.

Wave process methods

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
2. On the Action Pane, select **Regenerate methods**.

    The **sorting** method is added to the list of available methods, and the *Shipping* wave template type is selected for it.

Wave templates

Edit the wave template that is used for wave demand sorting.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. In the **Wave template type** field, select *Shipping*.
3. Select the existing **62 Shipping default** template.

4. On the Action Pane, select **Edit**.

5. On the **General** FastTab, make the following changes:

    - Set the **Process wave at release to warehouse** option to *No*.
    - Set the **Assign to open waves** option to *Yes*.

6. On the **Methods** FastTab, set up the **sorting** method:
    a. In the **Remaining Methods** grid, select **sorting**.
    b. Select the right arrow button to move **sorting** to the **Selected Methods** grid.
    c. In the **Selected Methods** grid, select **sorting**.
    d. Set the **Wave step code** field to *Sort*.

7. Select **Save**.

Mobile device menu items

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New**.

3. In the header, set the following values:

    - **Menu item name:** *Sort*
    - **Title:** *Sort*
    - **Mode:** *Indirect*
    - **Use existing work:** *No*

4. On the **General** FastTab, set the following values:

    - **Activity code:** *Outbound sorting*
    - **Use process guide:** *Yes* (default value)
    - **Outbound sorting template ID:** *Wave Sort*

5. Select **Save**.

Mobile device menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. In the list of menus, select **Outbound**.
3. On the Action Pane, select **Edit**.
4. In the **Available Menus And Menu Items** grid, find and select the **Sort** menu item that you just created.

5. Select the right arrow button to move **Sort** to the **Menu Structure** grid. In this way, you add the menu item to the **Outbound** menu.
6. Select **Save**.

Location directives

You must create location directives to guide the work that is created after the sorting is completed.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the **Work order type** field, select *Sorted inventory picking*.
3. On the Action Pane, select **New**.
4. In the header, set the following values:
    - **Sequence:** *1*
    - **Name:** *Put to Baydoor*
5. On the **Location directives** FastTab, set the following values:
    - **Work type:** *Put*
    - **Site:** *6*
    - **Warehouse:** *62*
    - **Directive code:** Leave this field blank.
    - **Multiple SKU:** *No*
6. Select **Save** to make the **Lines** FastTab available.
7. On the **Lines** FastTab, select **New**, and then set the following values. Accept the default values for all the other fields.
    - **Sequence number:** *1*
    - **From quantity:** *0*
    - **To quantity:** *1000000*
8. Select **Save** to make the **Location Directive Actions** FastTab available.
9. On the **Location Directive Actions** FastTab, select **New**, and then set the following values. Accept the default values for all the other fields.
    - **Sequence number:** *1*
    - **Name:** *Baydoor*

10. Select **Save** to make the **Edit query** button on the **Location Directive Actions** FastTab available.
11. On the **Location Directive Actions** FastTab, select **Edit query**.
12. In the query dialog box, on the **Range** tab, find the row where the **Field** field is set to *Location*. Set the **Criteria** field for this row to *Baydoor*.
13. Select **OK** to confirm the edit.

Work classes

1. Go to **Warehouse management** > **Setup** > **Work** > **Work classes**.
2. On the Action Pane, select **New**.
3. In the header, set the following values:
    - **Work class ID:** *Sorting*
    - **Description:** *Sorting*
    - **Work order type:** *Sorted inventory picking*
4. Select **Save**.

Work templates

1. Go to **Warehouse management** > **Work** > **Work templates**.
2. In the **Work order type** field, select *Sales orders*.
3. In the grid, select the **62 Pick to pack** work template.
4. On the Action Pane, select **Work header breaks**.
5. On the Action Pane, select **Edit**.
6. On the line where the **Field name** field is set to *Shipment ID*, clear the **Group by this field** check box.
7. Select **Save**, and then close the **Work header breaks** dialog box.
8. In the **Work order type** field, select *Sorted inventory picking*.
9. Select **New** to create a work template.

10. In the **Overview** section, set the following values. Accept the default values for all the other fields.

    - **Work template:** *Sorted picking*
    - **Work template description:** *Sorted picking*

11. Select **Save** to make the **Work Template Details** section available.

12. In the **Work Template Details** section, you will create two lines. Select **New**, and then set the following values for line 1:

    - **Work type:** *Pick*
    - **Mandatory:** Selected (= *Yes*)
    - **Work class ID:** *Sorting*

13. Select **New** again, and then set the following values for line 2:

    - **Work type:** *Put*
    - **Mandatory:** Selected (= *Yes*)
    - **Work class ID:** *Sorting*

14. Select **Save**.

Example scenario

This scenario simulates a situation where the warehouse stores small items in locations and must pack them into boxes before they are shipped, and where packing station functionality isn't really suitable. Workers pick orders for multiple customers and different addresses at the same time to increase the picking speed. After picking has been completed, the workers arrive at the sorting location, where the picked items can be sorted to the correct box, based on required criteria. In this example, the shipment ID will be used to determine the correct box, because each shipment has a different address. After all items from the load are packed and sorted by shipment, the boxes will be moved to the staging area and eventually loaded onto a truck.

Before you start the scenario, make sure that all standard warehouse functionality is set up correctly for your warehouse. Standard Contoso warehouse *62* is used for this purpose. Standard configurations haven't been changed, besides what is described in the setup.

Create demand

Before the functionality can be demonstrated, you must create some demand. For this scenario, you will create three sales orders for three different customers to simulate different delivery addresses. In this way, you will create three separate shipments.

Before you create sales orders and shipments, make sure that the pick locations have enough inventory for all the items on the orders. Review the location directive settings to confirm the picking locations that are used for sales order picking. If you must adjust the inventory, you can create manual movements, use replenishment, or use any other flow. Then reserve the inventory.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create a sales order for order 1.
3. In the **Create sales order** dialog box, set the following values:
    - **Customer:** *US-001*
    - **Warehouse:** *62*
4. Select **OK**.
5. A new line is added to the **Sales order lines** FastTab. Set the following values:
    - **Item number:** *A0001*
    - **Quantity:** *5*
6. Select **Add line** to add a second line, and set the following values:
    - **Item number:** *A0002*
    - **Quantity:** *10*
7. Repeat the following steps for each sales line on the order to reserve inventory for it:
    a. On the **Sales order lines** FastTab, on the **Inventory** menu, select **Reservation**.
    b. On the **Reservation** page, select **Reserve lot**, and then close the page.
    c. Select **Save**.
8. Select **New** to create a sales order for order 2.
9. In the **Create sales order** dialog box, set the following values:
    - **Customer:** *US-004*
    - **Warehouse:** *62*
10. Select **OK**.
11. A new line is added to the **Sales order lines** FastTab. Set the following values:

- **Item number:** *A0001*
- **Quantity:** *7*

12. Select **Add line** to add a second line, and set the following values:

    - **Item number:** *A0002*
    - **Quantity:** *3*

13. Repeat the following steps for each sales line on the order to reserve inventory for it:
    a. On the **Sales order lines** FastTab, on the **Inventory** menu, select **Reservation**.
    b. On the **Reservation** page, select **Reserve lot**, and then close the page.
    c. Select **Save**.

14. Select **New** to create a sales order for order 3.

15. In the **Create sales order** dialog box, set the following values:

    - **Customer:** *US-007*
    - **Warehouse:** *62*

16. Select **OK**.

17. A new line is added to the **Sales order lines** FastTab. Set the following values:

    - **Item number:** *A0001*
    - **Quantity:** *8*

18. Follow these steps to reserve inventory for the sales line:
    a. On the **Sales order lines** FastTab, on the **Inventory** menu, select **Reservation**.
    b. On the **Reservation** page, select **Reserve lot**, and then close the page.
    c. Select **Save**.

Complete the following procedure to release each sales order to the warehouse. Three different shipments will be created. You will then add all three shipments to one new wave.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. In the grid, select the first sales order that you created.

3. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

   You receive an informational message that shows the wave ID and shipment ID that were created.

4. Repeat the previous steps to release sales orders 2 and 3 to the warehouse. Notice that the informational message that you receive indicates that a shipment has been added to the wave that was created when you released sales order 1.

5. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.

6. Select the wave ID that was created from the release of the sales orders to open the **Waves** page. This page shows the wave details. The **Wave lines** FastTab shows the shipments that were created.

   You must now create work to bring items from the picking locations to the sorting location.

7. On the Action Pane, select **Process**.

   During wave processing, the sorting method will use the sorting template to assign the inventory to sort positions. When wave processing is completed, you receive an informational message that states that the wave has been posted and work has been created.

8. On the Action Pane, on the **Wave** tab, in the **Related information** group, select **Work** to view the work that was created. Make a note of the work ID.

9. Go to **Warehouse management** > **Packing and containerization** > **Outbound sorting position assignments**.

10. In the left column, you can view the outbound sorting position that was created for each shipment.

11. On the **Sort position criteria** FastTab, you can view the shipment ID for that position.

One work ID has been created to bring inventory from the picking locations to the sorting location. To complete the work, you will need the work ID that was created during wave processing.

Sales order picking to the sorting location

1. Sign in to the mobile app as a worker in warehouse *62*.

2. On the main menu, select **Outbound**.

3. On the **Outbound** menu, select **Sales Picking**.

4. Select the **ID** field, and then enter the work ID from the wave processing.
5. Confirm your entry.

   Next, you're prompted to enter a target license plate (LP). Notice that line 1 from sales order 1 is what must be picked and added to the target license plate. The item number, quantity, item description, and pick location are shown.
6. In the **Target LP** field, enter a license plate number.

   You will pick all lines that were created from the processed wave onto the same target license plate.
7. Confirm your entry.

   The mobile app now presents a series of **Pick** pages that point you to the picking location, and to the item and quantity that must be picked. After the picked item is added to the license plate, you will confirm the pick work. The last page will be the work to put the picked items into the sorting location.
8. Confirm the first pick work.
9. The next pick work is shown. Confirm the pick.
10. Continue to confirm the remaining pick work.
11. The last step is to complete the put work. Confirm the put-away to the sorting location.

    You receive a "Work completed" message.
12. Exit **Sales Picking** on the mobile app.

Sorting/put-to-wall

Now that all inventory has been put to the sorting location, it must be sorted to the correct sort position.

1. Sign in to the mobile app.
2. On the main menu, select **Outbound**.
3. On the **Outbound** menu, select **Sort** to start to sort the items.
4. In the **LP/CON** field, enter the target license plate of the picked sales order work.
5. Confirm your entry.

6. Enter the item number to sort first.

7. The system determines the first sort position that should be shown. Confirm the sort position.

8. You're prompted to assign a license plate to the sort position. Select the **LP** field, enter a license plate number, and then confirm your entry.

   Because the sort position is related to the shipment ID, you will sort the picked items into a license plate that is specific to the outbound shipment and sales order.

   The next page shows the item ID, quantity, sort position ID, and the "from" (picking) and "to" (sorting) license plate IDs. The information on this page instructs you to sort the specified item and quantities from the picking license plate into the sorting license plate.

9. Confirm the sort position.

10. Sort the items in the first sort position. When you've finished, the system directs you to the next sort position.

11. Repeat this process for all picked lines on the work order. You should also use this process when there are multiple pick lines that have the same item number.

    As you repeat this process for all items, the system evaluates the criteria from the next scanned item (work line) and determines which sorting position it should be put to. You're automatically directed to put the item to the correct sort position. Depending on the confirmation setup, you're also directed to confirm this action by entering the position number or license plate ID.

    > ⓘ **Note**
    >
    > If automatic sorting is turned on, manual override isn't available.

12. When you've finished, in Microsoft Dynamics 365 Supply Chain Management, open the **Outbound sorting position assignments** page to review the status of the positions.

    - If positions are closed automatically, select **Show closed** to show the closed positions.
    - Notice that sort position transactions are shown. The item and quantity that were processed through the position are shown.

When you set up the outbound sorting template, you set the **Auto close sort position** option to *Yes*. Therefore, the position is automatically closed after the last expected inventory is put to it. The sort positions are in **Closed** status, and work has been created to move the sorted inventory to *Baydoor* location.

13. Complete the sorted inventory picking work to move the inventory to the shipping location. When the inventory is ready, ship-confirm it.

> ⓘ **Note**
>
> For sorted inventory picking work to be processed correctly, a mobile device menu item that has a work class ID where the **Work order type** field is set to *Sorted inventory picking* should be used for the movement and loading process.

Manually close a position (optional)

If sort positions should be closed manually, the **Auto close sort position** option for the outbound sorting template must be set to *No*, and closing must be done before inventory can be moved to the bay door area. Positions can be closed in various ways:

- Via the Warehouse Management mobile app:
  - The user can scan one of the items that are already on the position and then select **Close** to close the position.
  - If the user scans a container that has already been sorted container, an error message is shown. However, the user can still continue to close the position.
- From the Microsoft Dynamics 365 Supply Chain Management **Outbound sorting position assignments** page:
  - The user can select the outbound sorting position record and then select **Close position** on the Action Pane.

Tips

- Items can't be moved between positions after they have been assigned to one. The system evaluates how many of each item should go to a specific position.
- Sorts template can be configured to automatically create containers when positions are closed. This approach will create standard container ID structure that is based on the specified packing profile.

> ⓘ **Important**

After movement work has been created from the sorting location, you must not cancel the work. Otherwise, the position and the containers in it will be deleted from the system and unavailable for further processing. The inventory will also be removed.

Feedback

Was this page helpful? Yes No

Provide product feedback

Outbound workload visualization

Advanced setup capabilities that are accessible from the **Outbound workload visualization** page let warehouse managers and supervisors create custom workload charts that can be used to monitor the progress of current work and the amount of it that remains. Warehouse managers can create multiple views and set up automatic refresh as they require. Outbound workload visualizations are suitable for display on warehouse performance pages.

This functionality can be used to track the progress of picking work. The feature is integrated with labor management, and if labor management is set up, outbound workload visualizations can show a calculation of the number of hours that remain for the picking work that is shown (filtered).

Set up outbound workload visualizations

To set up your visualizations, you create a collection of filters (views) and design each filter so that it shows a different type of analysis. You use the **Configure filters** page to design the filters.

To set up an outbound workload visualization, follow these steps:

1. Go to **Warehouse management** > **Warehouse monitoring reports** > **Outbound workload visualization**.

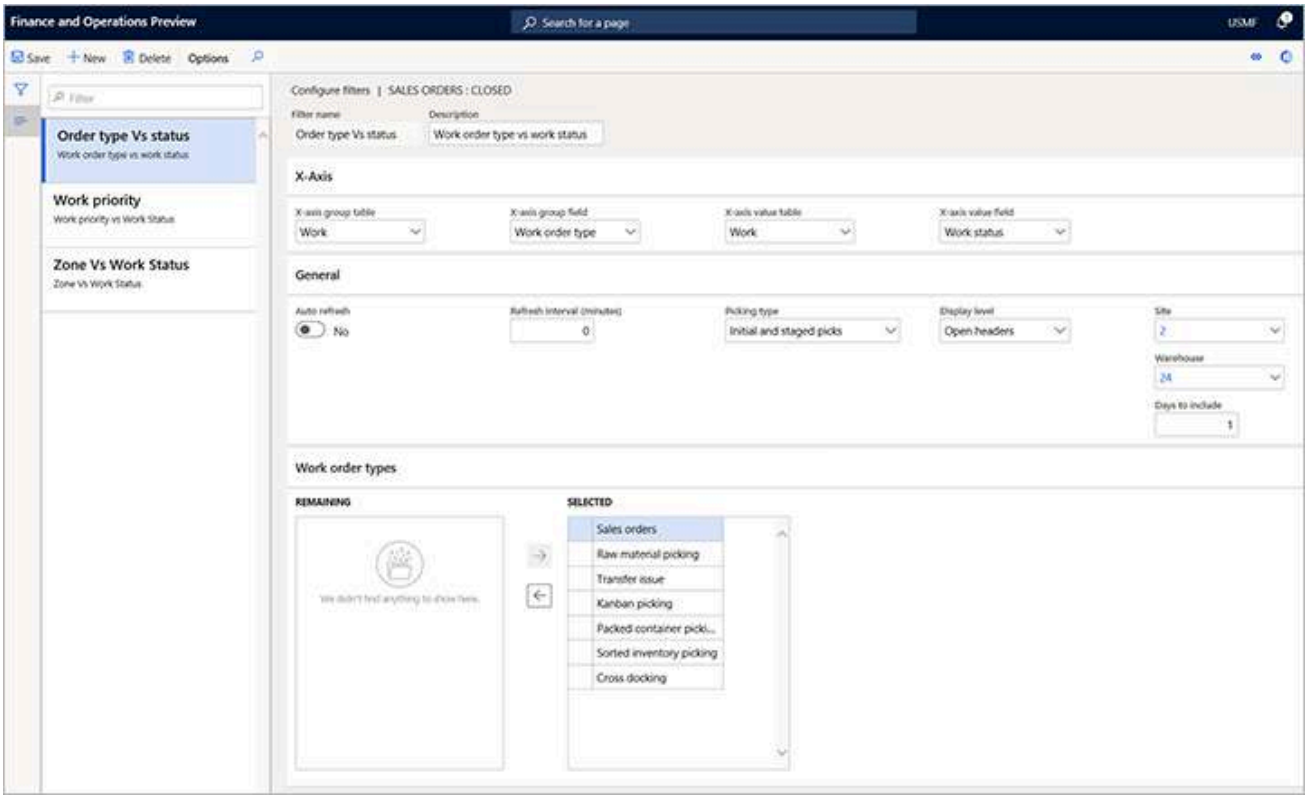
   The **Outbound workload visualization** page appears. After you create some filters, this page will show your visualization. You can create as many filters as you want. All the filters that you create are saved under your user account, so that you can use them later. In other words, each user will have their own set of filters that they created. Those filters won't be shared with other users.

2. On the **Outbound workload visualization** page, on the Action Pane, on the **Filters** tab, select **Configure filters**.

3. On the **Configure filters** page, on the Action Pane, select **New** to add a filter, and then set the following fields for it:

   - **X-axis group table** – Select the table that contains the field that should be used to group the X-axis values.
   - **X-axis group field** – From among the fields of the table that you selected in the **X-axis group table** field, select the field that should be used to group the X-axis values.

- **X-axis value table** – Select the table that contains the field that should be used to further analyze the groups.
- **X-axis value field** – From among the fields of the table that you selected in the **X-axis value table** field, select the field that provides the values that should be analyzed for each group.
- **Auto-refresh** – Select whether the visualization should automatically be refreshed.
- **Refresh interval (minutes)** – Enter the number of minutes between automatic refreshes.
- **Display level** – Select whether the chart should show open lines or open header counts.
- **Picking type** – If you set the **Display level** field to *Open lines*, select whether the count of open work lines in the chart should include initial picks, staged picks, or both initial picks and staged picks.
- **Site** – Select the site to load the chart for.
- **Warehouse** – Select the warehouse to load the chart for.
- **Days to include** – Enter the number of days in the past that the chart should be generated for.
- **Work order type** – Select the outbound work order types to filter on.

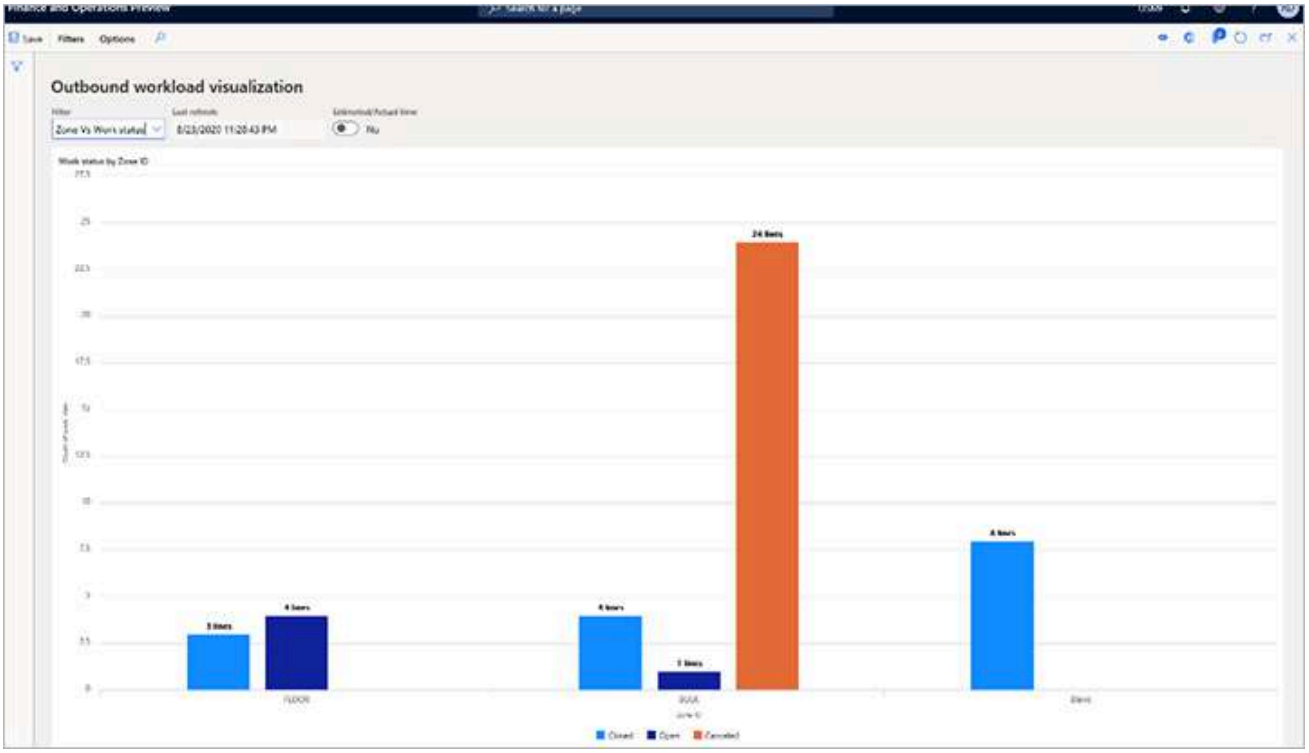


4. Close the **Configure filters** page to return to the **Outbound workload visualizations** page.

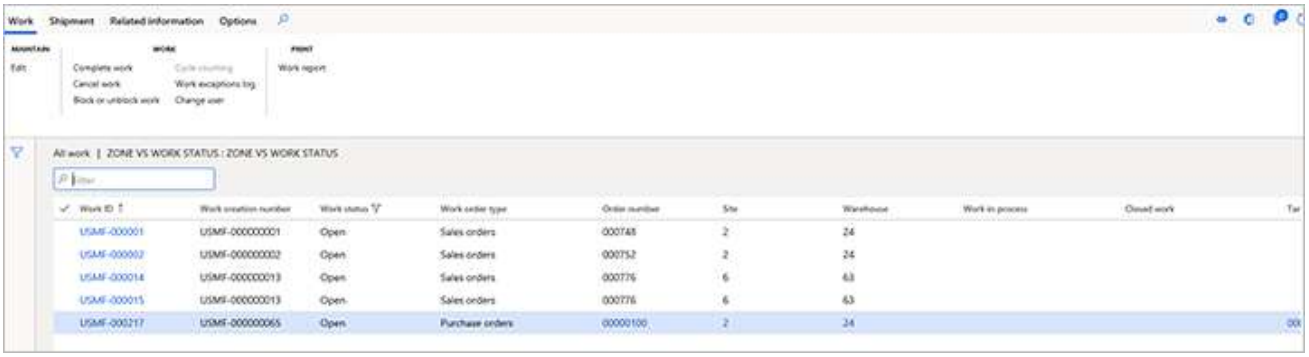
The **Outbound workload visualizations** page now shows data, based on your new filter settings. Your new filter is now available and is selected in the **Filter** field. The following

fields are available at the top of the chart:

- **Filter** – This field includes all the filters that you've created so far. Select a filter to view its data in the chart.
- **Last refreshed** – This field shows the date and time when the information in the chart was last updated.
- **Estimated/actual time** – If labor standards are set up in your system, set this option to *Yes* to show accumulated estimated picking times at the top of each column in the chart. If you aren't using labor standards, this option is unavailable.



5. Select any bar in the chart to view the associated work line details.

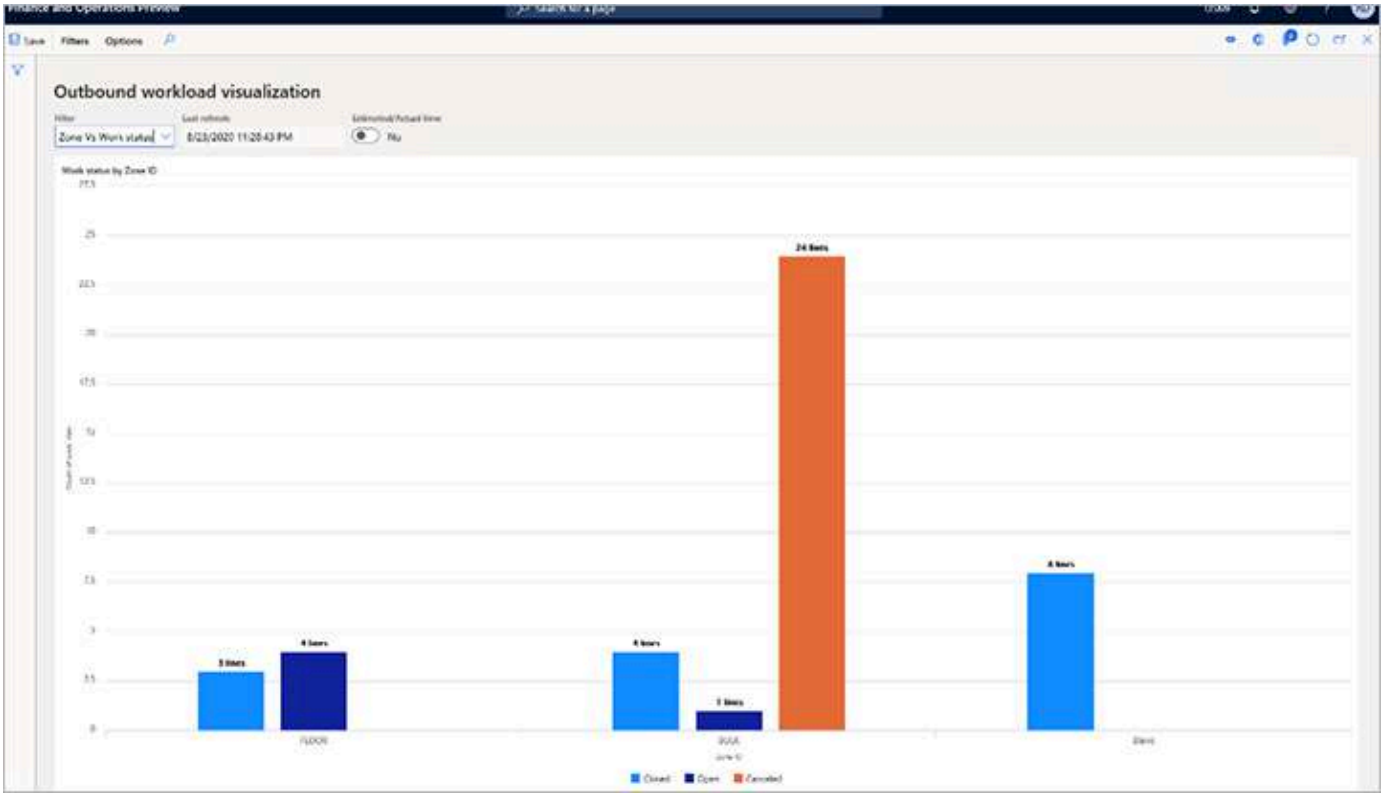


Example: Outbound workload visualization for zones

For this example, you want to set up a visualization that shows work lines for each zone, and the status of each work line (*Open*, *Closed*, or *Canceled*). In this case, you can set up a filter that has the following settings:

- **Filter name** – Enter a name for this filter (such as *Zone vs. work status*).
- **Description** – Enter a short description for this filter (such as *Zone vs. work status*).
- **X-axis group table** – Select *Locations*.
- **X-axis group** – Select *Zone ID*.
- **X-axis value table** – Select *Work*, because you want to view work per zone.
- **X-axis value field** – Select *Work status*, because you want to view work status.
- **Auto refresh** – Select whether the visualization should automatically be refreshed.
- **Picking type** – Select *Initial picks and staged picks*, because you want to include both initial picks and picks from staging locations. In other words, you essentially want to include all the pick work lines that you have.
- **Display level** – Select *Open lines*, because you want to view the information per line, not per work header.

The following illustration shows an example of the resulting chart.



This chart shows two zones that are named **FLOOR** and **BULK**, plus a zone that is named **Blank**. The **Blank** zone represents all work lines that aren't members of any zones. The chart always shows all unrelated filtered data as **Blank**, to provide as much visibility as possible. In the **FLOOR** zone, the chart shows three closed lines and four open lines. In the **BULK** zone, the chart shows four closed lines, one open line, and 24 canceled lines. Finally, the chart shows eight closed lines that aren't part of any zone and are therefore listed as **Blank**.

Last updated on 07/01/2026

Manually handle sales and transfer line picking exceptions

10/03/2025

Manual handling of sales and transfer line picking exceptions enables administrators to fix issues that are caused by corrupted sales and transfer order data. It lets trusted users manually pick (or unpick) inventory transactions that are related to sales and transfer order lines while warehouse processes are already in progress.

The functionality that is described here should be used only in cases where the system can't finish processing the warehouse processes stack in the usual way. By default, only users who have a system admin role are allowed to use it. However, you can grant access to it by assigning the *Pick sales lines manually* privilege to other roles as you require.

Correct transactions related to sales or transfer order lines

Use the following procedure to correct transactions that are related to sales or transfer order lines.

1. Follow one of these steps, depending on the type of order that you must correct:

    - To correct a transaction that is related to a sales order, go to **Warehouse management** > **Periodic tasks** > **Clean up** > **Pick sales line manually**.
    - To correct a transaction that is related to a transfer order, go to **Warehouse management** > **Periodic tasks** > **Clean up** > **Pick transfer lines manually**.

2. On the **Pick sales lines manually** or **Pick transfer lines manually** page, use the filters at the top of the grid to find the line that you're looking for, and then select the target order line in the upper grid.

3. Use the **Inventory transactions**, **Load lines**, **Work lines** and **Container lines** tabs in the lower section to view more information about the selected line. The information on these tabs gives a basic overview of the related warehouse processes and their statuses.

4. On the Action Pane, select **Pick** to open the selected order line on the **Pick** page for inventory transactions. You can complete this step even for order lines that have already been released to the warehouse. (Usually, order lines that have been released to the warehouse can't be opened on the **Pick** page.)

> **Note**
>
> You might receive various warnings when you open the **Pick** page. Take any action that these warnings recommend. For example, you might receive a warning about order lines that are linked to warehouse work. In this case, it makes sense to try to cancel the work by using the standard cancel work functionality before you do manually correction.

5. Select the line in the **Transactions** grid, and then select **Add picking line** on the **Transactions** toolbar.

6. The selected line is moved to the **Picking lines** grid. Set appropriate values for any columns that are missing required information. (For example, specify the location and license plate that the item should be picked/un-picked from.)

7. Select **Confirm pick all** on the **Picking lines** toolbar.

Correct load line quantities

Use the following procedure to correct a load line quantity.

1. Follow one of these steps, depending on the type of order that you must correct:

    - To correct a transaction that is related to a sales order, go to **Warehouse management** > **Periodic tasks** > **Clean up** > **Pick sales line manually**.
    - To correct a transaction that is related to a transfer order, go to **Warehouse management** > **Periodic tasks** > **Clean up** > **Pick transfer lines manually**.

2. On the **Pick sales lines manually** or **Pick transfer lines manually** page, use the filters at the top of the grid to find the line that you're looking for, and then select the target order line in the upper grid.

3. On the **Load lines** tab in the lower section, select **Correct load lines manually** on the toolbar.

4. On the **Correct load lines manually** page, in the **Inventory transactions** grid, review the inventory transactions that are related to the selected load line.

5. In the upper grid, correct the load line quantities in the following columns as required:

    - **Work created quantity**
    - **Picked quantity**
    - **Planned cross docking quantity**

The system will validate any changes that you make to these columns.

Cancel warehouse work for exception handling

The cancel work functionality in Microsoft Dynamics 365 Supply Chain Management lets the admin user cancel specific warehouse work that is currently in progress, but that is blocked by the system or can't be completed because of exceptional circumstances. This functionality is an attractive and secure alternative to SQL corrective scripts that fix inconsistent data. However, whereas these scripts are typically requested from IT professionals, the cancel work functionality can be used by users in the company who have admin rights.

You can access the cancel work functionality at **Warehouse management** > **Periodic tasks** > **Clean up** > **Cancel work**. In the **Cancel work** dialog box, specify the work ID of the work to cancel, and then select **OK**.

Warehouse work that can be canceled

During warehouse picking operations, a worker might encounter situations where they have registered quantities as picked from a storage location to their user location, but then they can't register the put operation. Inconsistent warehouse data is often, but not always, the reason why work is blocked.

Unlike the regular cancel functionality that can be accessed by using the **Cancel** button on the work header, the cancel work functionality doesn't require that the last completed work line be a work line of the **put** type. In other words, for the cancel work functionality, cancellation logic can be run even if the item quantity on a work line is in a user location.

> ⓘ **Note**
>
> For work that must be canceled for operational reasons, warehouse users must continue to use the regular cancel functionality on the work page.

Only work of the **Sales**, **Transfer issue**, **Raw material picking**, or **Replenishment** type can be canceled by using the cancel work functionality. Cancellation logic won't be run for frozen raw material picking work or work that can be canceled by using the regular cancel functionality (see the preceding note).

To unblock the work, the system cancels any remaining work lines and fixes the warehouse data that is associated with the work ID that the user specified. Any regular warehouse-handling operations that involve the affected item quantity can then resume.

To put the affected item in a specific location after the work is canceled, the user must use an inventory movement or quantity adjustment operation on a mobile device.

Last updated on 07/01/2026

Serial number capturing

Article • 03/05/2025

This article explains how to use the **Capture serial number** field on the **Tracking dimension groups** page for items that use the warehouse management processes. Depending on the setting of this field, the actual serial number value on inventory transactions (on-hand inventory) isn't populated until later in the process, even if serial numbers are activated for the items.

The following articles describe a similar, but slightly different, serial number setting for cases where **Active in sales process** is enabled:

- Working with serialized items
- Serial Number Tracking for Service and Warranty in Microsoft Dynamics AX R3 - Microsoft Dynamics 365 Blog
- Enable picking of sales serial number items with R3 warehouse management processes - Microsoft Dynamics 365 Blog
- How to use Serial Sales picking with Warehouse Management processing in Dynamics AX 2012 - Microsoft Dynamics 365 Blog

Capture the serial number during picking

Your business process might require that the exact serial number that is picked for a sales or transfer order is known at the time of picking. To enable this setup, go to the tracking dimension group, select **Serial number**, and set the **Capture serial number** field to *Picking*.

> **Note**
>
> When the field is set to *Picking*, the **Blank receipt allowed** and **Blank issue allowed** fields are automatically selected, and the settings can't be changed.

In this case, when on-hand inventory is added, the user isn't prompted to enter the serial number during warehouse inbound processes such as purchase order receiving on the warehouse mobile device. Instead, the user specifies the serial number during the picking process, such as when sales picking work is completed on the warehouse mobile device. At this point, the serial number is added to the inventory transactions (on-hand inventory).

Capture the serial number during packing

Your business process might require that the exact serial number that is picked for a sales or transfer order is known at the time of packing. To enable this setup, go to the tracking dimension group, select **Serial number**, and set the **Capture serial number** field to *Packing*.

> **Note**
>
> When the field is set to *Packing*, the **Blank receipt allowed** and **Blank issue allowed** fields are automatically selected, and the settings can't be changed.

In this case, when on-hand inventory is added, the user isn't prompted to enter the serial number during warehouse inbound processes such as purchase order receiving on the warehouse mobile device. Instead, the user specifies the serial number during the packing process, such as when the items are packed into containers on the **Pack** page. At this point, the serial number is added to the inventory transactions (on-hand inventory).

Don't capture the serial number

The default setting for the **Capture serial number** field is *None*. In this case, the user isn't prompted to capture the serial number during either the picking process or the packing process. Instead, the serial number is captured during the receiving process, even if **Blank receipt allowed** is enabled. The serial number must be part of the inventory transactions (on-hand inventory) from the very beginning.

> **Note**
>
> Depending on other settings, the user might not be prompted to capture the serial number during the receiving process if it was generated in some other way. For example, the user isn't prompted to capture the serial number if automatic generation of serial numbers is enabled through the tracking number group.

Feedback

Was this page helpful? Yes No

Provide product feedback

Replenishment overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the replenishment strategies that are available for warehouses that use the functionality in Warehouse management. The information in this article doesn't apply to the warehousing solution that's available in Inventory management.

The following replenishment strategies are available:

- **Wave demand replenishment** – This strategy creates replenishment work for outbound orders or loads if inventory isn't available when the wave creates work. For example, replenishment work can be created if the quantity that is required for a sales order isn't available when a wave is processed.
- **Min/Max replenishment** – This strategy uses minimum and maximum stocking limits to determine when locations should be replenished. The item and location criteria define the inventory that is evaluated for replenishment. Min/Max replenishment templates are the primary mechanism for maintaining optimal levels in picking locations. To help guarantee that enough pick face inventory is available to meet wave demand, you can use demand replenishment as a supplement between Min/Max replenishment cycles.
- **Load demand replenishment** – This strategy sums the demand for several loads and creates the replenishment work that is required in order to stock the relevant picking locations. This strategy helps guarantee that the loads that are created can be picked in the warehouse after they're released.
- **Immediate replenishment** – This strategy replenishes inventory before a wave is run if allocation fails for a location directive line that has a replenishment template.

All four strategies create replenishment work, based on a replenishment template.

Wave demand replenishment

Wave demand replenishment creates replenishment work, based on demand, if the quantity that is required for production orders, kanbans, outbound orders, or loads isn't available when a wave creates work. The replenishment template contains information about the item criteria, the unit of measure, the demand increment, and the location.

Location directives are used to determine which location should be replenished. You link these location directives to the replenishment template by using the **Directive code** field. If you don't set the **Directive code** field, queries are used to determine which location directive should be used. If you don't specify a directive code in the replenishment template, and the location directive has a directive code, the location directive is ignored, even if the query on the location directive is correct. Pick location directives are used to determine where to get inventory for the replenishment.

In addition to creating a template, you must specify some replenishment settings in the wave template. The wave template should contain a wave step for replenishment that runs only if an item isn't successfully allocated. This replenishment wave step uses a wave step code to determine which replenishment template should be used. In addition to having a wave step for replenishment, you must make sure that **Replenish** is selected in the **Methods** section of the wave template.

The **Replenishment template** page includes an **Allow wave demand to use unreserved quantities** check box. Select this check box if demand replenishment should be able to deduct unreserved quantities from work that is generated from the selected replenishment template. To enable demand replenishment templates to use this logic, select this check box for every existing replenishment template. When demand replenishment is triggered in the warehouse, it deducts the demand from existing replenishment work that has unreserved quantities, if the work originates from replenishment templates where the **Allow wave demand to use unreserved quantities** check box is selected.

Replenishment unit is the minimum unit to replenish. This must be a whole number that is a multiple of the unit. The system rounds up to the highest unit possible when creating work.

Demand replenishment is supported for sales orders, transfer orders, production orders, and kanbans.

Min/Max replenishment

In Min/Max replenishment, the system replenishes stock so that it's between the minimum and maximum limits that you set. Typically, this process runs once every day to help guarantee that all picking locations are filled to the maximum level before picking starts.

The minimum and maximum amounts are set in a replenishment template. Many of the other settings in the template resemble the settings in templates that are used in Wave demand replenishment. The template should contain one line for each item and location. When you run replenishment by using the batch job, the system evaluates whether replenishment is required by using the sequence that the lines are organized in.

The Min/Max replenishment strategy can't replenish an empty location unless you set the location as the fixed location for the item. If the location that must be replenished isn't a fixed location, the system can't determine which item to replenish. Therefore, at least some on-hand quantity is required before replenishment occurs.

Load demand replenishment

Load demand replenishment sums the demand for several loads and creates the replenishment work that's required to stock the relevant picking locations. Load demand replenishment resembles Wave demand replenishment in many ways. The main difference is how and when Load demand replenishment and Wave demand replenishment run. Like Min/Max replenishment, you run Load demand replenishment by using a batch job. To set up the batch job, on the **Load demand replenishment** page, select the replenishment template to use, and set a filter query to specify which loads are used to determine the demand. The location query defines the locations that the system subtracts any available quantity from to meet the aggregated demand of the loads.

Immediate replenishment

Instead of having to sum demand at the end of an allocation process and do replenishment on the basis of the summed up quantity, you can apply the Immediate replenishment strategy. When you use this strategy, the system can replenish inventory immediately after a location directive line fails. Therefore, you can set up the replenishment so that it's restricted by specific units, and so that it uses quantities that are set for specific locations.

Replenishment prerequisites

Expand table

Prerequisite	Description
Item	You must enable the item for warehouse management processes (WMS).
Warehouse	You must enable the warehouse for warehouse management processes (WMS). To enable a warehouse for WMS, on the **Warehouses** page, select the warehouse, and then select the **Use warehouse management processes** option.
Replenishment templates	You must set up at least one replenishment template for Min/Max replenishment, Wave demand replenishment, or Load demand replenishment.
Locations	You must create locations and connect them to a location profile.

Prerequisite	Description
Location profiles	You need location profiles to create locations.
Location directives	You need location directives to guide work to the locations where replenishment is required and to the locations that inventory is sourced from.
Work templates	You need work templates of the **Replenishment** type to create replenishment work so that inventory can be moved to the desired locations.

Last updated on 11/20/2025

Immediate replenishment

Immediate replenishment lets you replenish inventory immediately after a location directive line fails to allocate inventory. The replenishment is based on a single line in the setup of the location directive. If inventory isn't on hand in the unit of measure that is specified by that line, replenishment of that unit of measure occurs immediately.

Immediate replenishment provides an alternative to the method where the allocation of inventory is based on more lines in the location directive, and where the demand is summed at the end of the allocation and replenished in the unit of measure that is specified by the last line in the location directive.

The benefits of using immediate replenishment are that replenishment can be limited by specific units and the quantity can be directed to specific locations.

Business scenario

For example, you have a warehouse that has separate picking areas for the "box" and "each" units of measure. You want to optimize the picking process by picking as many boxes as possible and then picking any remaining quantity that is less than a box from the "each" area.

In this case, you can use immediate replenishment. In the location directive, you can set up immediate replenishment for boxes so that demand replenishment is used as soon as there is a shortage of boxes that can be picked for the demand quantity. In this way, you optimize the picking process so that picking includes as many boxes as possible. Immediate replenishment will generate replenishment of the boxes, and the demand won't be passed on so that the quantities are picked in the "each" unit of measure. In other words, only the quantities that are supposed to be picked in the "each" unit of measure (that is, quantities that are less than a box) will be picked from the "each" area. If a shortage occurs in the "box" area, you can pick as many boxes as possible out of the total demand. The remaining quantities will then be picked from the "each" area.

Where it applies

Immediate replenishment is used during wave execution if allocation fails for a location directive line that a replenishment template is set for.

Set up immediate replenishment

- Go to **Warehouse management** > **Setup** > **Location directives**, and then, on the **Lines** tab, in the **Immediate replenishment template** list, select a replenishment template for wave demand.

The replenishment template is applied if the location directive line fails to allocate a dedicated unit of measure.

Troubleshooting

If immediate replenishment is selected for a location directive line, but no replenishment work is generated when you use demand replenishment templates for that location directive line, two main causes must be investigated:

- Make sure that the demand replenishment template that is applied is set up to use the correct location templates and work templates of the **Replenishment** type.
- Make sure that there is enough on-hand inventory at the locations where the demand replenishment template searches for on-hand inventory for replenishment.

Last updated on 07/01/2026

Replenishment with withdrawal kanbans

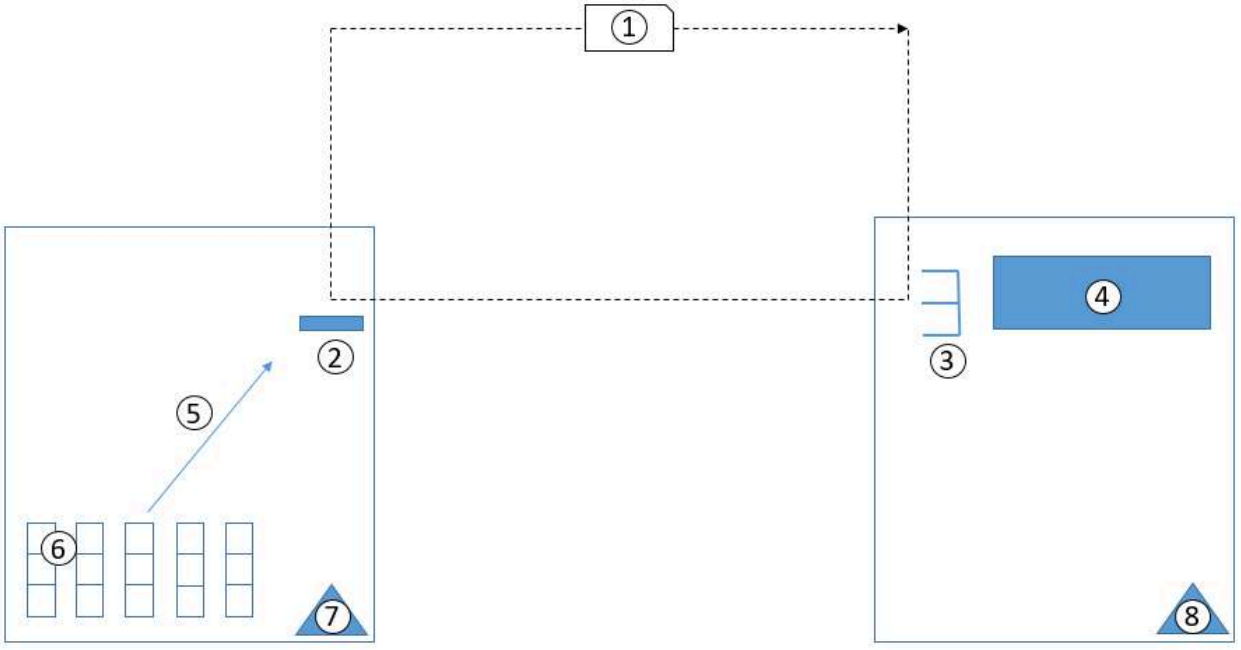
Article • 06/03/2022

This article describes how the withdrawal kanban is used for material replenishment for manufacturing activities.

Workflow for material replenishment that uses the withdrawal kanban

The withdrawal kanban can be used to move a kanban of a single item between warehouses and production locations where the material is consumed. The withdrawal kanban supports a pull-based solution for material replenishment, where a pull signal is required in order to trigger supply for a specific demand.

The following scenario shows a pull-based replenishment system where a pull signal triggers the creation of a kanban to replenish material for a production process.



1. Withdrawal kanban
2. Kanban “from” location and put location for warehouse work
3. Kanban “to” location and production input location
4. Manufacturing process
5. Warehouse work for kanban picking
6. Warehouse locations for raw material
7. Material warehouse
8. Manufacturing warehouse

In this scenario, a manufacturing process (4) consumes material from a production input location (3) in the manufacturing warehouse (8). When a handling unit of the material (kanban) is consumed, it's registered as empty. A replenishment signal is created for the item origin, and a new kanban (1) is created. In this case, the item origin consists of locations in the material warehouse (7). The material for the kanban is picked and put to a location (2) in the same warehouse. When the material is picked, it's ready to be transferred from location 2 to the production input location (3) in the manufacturing warehouse (8).

Configure warehouse work for kanban picking for the withdrawal kanban

To enable raw material picking for the withdrawal kanban, configure wave templates, work templates, and location directives for the **Kanban picking** work order type. This work order type doesn't just support the picking process for the withdrawal kanban. It also supports the picking process for the manufacturing kanban. However, you can configure a separate picking process for each type of kanban by separating the wave templates, work templates, and location directives. To separate the wave templates, work templates, and location directives, set criteria on the activity type (**Process** or **Transfer**) in the queries for those entities.

Configure the withdrawal kanban

The transfer activity that is used for the withdrawal kanban is configured as part of an activated activity plan in a Lean production flow. As part of the configuration of the transfer activity, you specify the "from" and "to" locations for the transfer. After you configure the transfer activity, you can assign it to a kanban rule of the **Withdrawal** type. The kanban rule sets the policies and configurations for the withdrawal kanban. The quantity of the kanban defines how many units of the handling unit the kanban carries during the transfer process. The fixed kanban quantity is used when the Fixed replenishment strategy is selected. This quantity defines how many kanbans that are required in order to prevent stock or build inventory from running out at the source of demand. The fixed quantity can be calculated based on actual demand, historical demand, and service levels. The following two scenarios describe how you can manage material replenishment that uses the withdrawal kanban.

Scenario 1: Replenish a production input location by using a fixed withdrawal kanban

A manufacturing process consumes a purchased raw material from a production input location that is in the production warehouse. When the raw material is received from the vendor, it's stored in locations in the material warehouse. Because the demand for the material is considered stable over a period, it's set up to supply the production in a fixed quantity kanban flow. When a kanban is consumed at the production input location, an empty signal is registered, and a new kanban of the same type is added to the flow.

The empty signal can be configured to occur automatically when a kanban is completed. Alternatively, the empty signal can be set up as a manual interaction that is given either from the Kanban transfer board or from a menu on the hand-held device. The Kanban transfer board is the workspace where all activities in the kanban life cycle are managed.

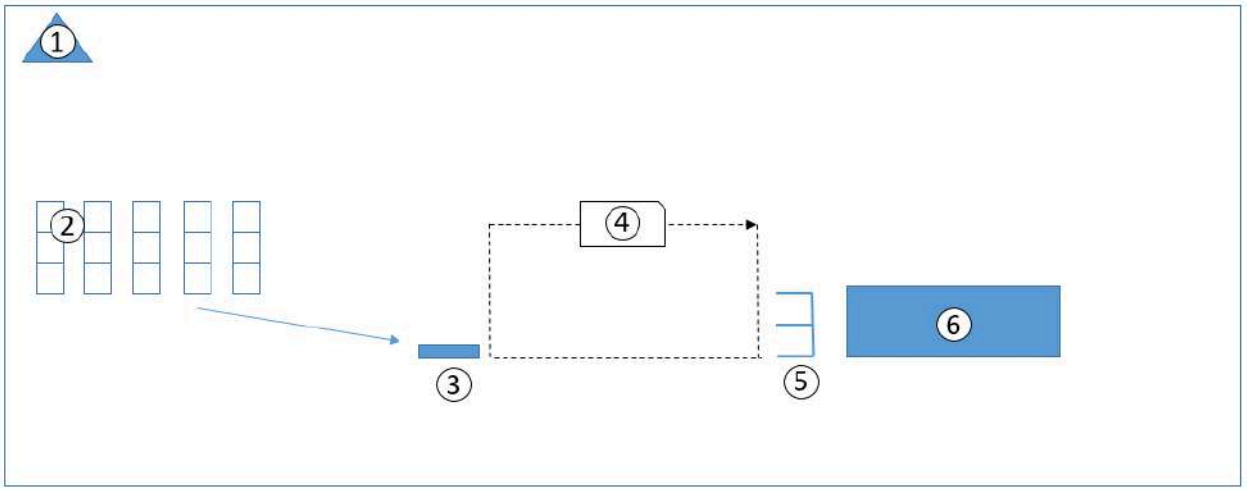
When the kanban is created, a wave line for the raw material is added to a kanban wave for the material warehouse. In the picking list section of the Kanban transfer board, the status of the material and related warehouse processes can be monitored from wave creation to work creation, until the material is on-hand in the "transfer from" location and is ready to be transferred to the production input locations. The kanban can be completed either from the Kanban transfer board or from a menu on the hand-held device.

In this scenario, the picking work in the material warehouse is processed as one activity. The transfer activity between the material warehouse and the production warehouse is processed as a separate activity. This approach can be useful if, for example, there is a large physical distance between the two warehouses. In this case, the kanban transfer activity can represent a truck load.

If the distance between the warehouse locations and the production input location is small, it might be more efficient to include the transfer activity in the picking process. Then, after the material is picked, it can be put directly to the production input location. To support this process, you configure the transfer activity so that it's automatically completed when the pick work of the withdrawal kanban is processed.

Scenario 2: Automatically complete the transfer activity when kanban picking work is processed

In the following scenario, the transfer activity of the withdrawal kanban is configured to transfer between two locations in the same warehouse. The transfer activity of the withdrawal kanban is set up so that it's completed automatically.



1. Shared warehouse for raw materials and production
2. Warehouse locations for raw materials
3. Kanban “from” location and put location for warehouse work
4. Withdrawal kanban
5. Production input location
6. Manufacturing process

After a kanban is consumed at the production input location, the kanban is reported as empty, and a new kanban is added to the flow. When the kanban is created, a wave line is added to a kanban wave. When the kanban wave is processed, warehouse work for kanban picking is created. The warehouse worker processes the work for kanban picking and is directed by the work to pick the material for the kanban in a warehouse location. As this warehouse worker confirms the pick, the kanban is automatically completed, and the warehouse worker is guided to the put the material to the production input location.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up a min-max replenishment process

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article shows how to set up a replenishment process that uses the minimum/maximum replenishment strategy. When inventory falls below the minimum level, the system creates work to replenish the location. This article also shows how to use fixed picking locations to allow restocking even if inventory falls below the minimum level, and how to enable the replenishment process to run regularly using a batch job. A warehouse manager typically carries out these tasks. You can run these procedures by using the USMF demo data company with the example values provided here, or you can run them on your own data. If you're using your own data, make sure that you have a warehouse that's enabled for warehouse management processes (WMS).

> **Caution**
>
> When the following conditions are met, the system might create replenishment work that results in a conflict when processing that work:
>
> - The system is replenishing batch-tracked items.
> - Your reservation hierarchy has the *Batch number* dimension below the *Location* dimension.
> - Mixing of batch numbers isn't allowed at the put location.
>
> The conflict might occur because only *batch above location* items are assigned batch-number information on work lines and the *Replenishment* location directive's *Put* work type isn't used.

Create a fixed picking location

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Fixed locations**. This task is optional for min-max replenishment, but if you use a fixed picking location, it allows the

replenishment of stock even if it falls below the minimum level, because the system can determine which items need replenishment, even when none are left.

2. Select **New**.
3. In the **Item number** field, enter or select a value. If you're using USMF, you can select item *A0001*.
4. In the **Site** field, enter or select a value. If you're using USMF, you can select site *2*.
5. In the **Warehouse** field, enter or select a value. If you're using USMF, you can select warehouse *24*.
6. In the **Location** field, enter or select a value. If you're using USMF, you can select *CP-003*.
7. Close the page.

Create a replenishment location directive

1. Go to **Warehouse management** > **Setup** > **Location directives**. Location directives are used to determine where items should be picked from in the replenishment process.
2. In the **Work order type** field, select *Replenishment*.
3. On the Action Pane, select **New**.
4. In the **Name** field, type a value.
5. In the **Work type** field, select *Pick*.
6. In the **Site** field, enter or select a value. If you're using USMF, you can select site *2*.
7. In the **Warehouse** field, enter or select a value. If you're using USMF, you can select warehouse *24*.
8. Select **Save**.
9. In the **Lines** section, select **New**.
10. In the list, mark the selected row.
11. In the **To quantity** field, enter a number. For example, you can set it to *9999*.
12. Select the **Allow split** checkbox. If you select this option, the work creation process allows work line quantities to be split across multiple locations.
13. Select **Save**.
14. In the **Location directive Actions** section, select **New**.
15. In the list, mark the selected row.
16. In the **Name** field, type a value.
17. Select **Save**.
18. On the Action Pane, select **Edit query**. Edit this query to add restrictions on where inventory can be selected from in the replenishment process. For example, you can use it to limit selection to inventory in the Bulk area of the warehouse.
19. Select **OK**.

20. Close the page.

Create a replenishment work template

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**. The work template guides the system in creating min/max replenishment work. At a minimum, it must include a work template line for a pick and a put. The work template shows as Invalid until all the necessary information has been filled in.
2. In the **Work order type** field, select *Replenishment*.
3. On the Action Pane, select **New**.
4. In the **Work template** field, type a value.
5. Select **Save**.
6. In the **Work template details**, select **New**.
7. In the **Work type** field, select *Pick*.
8. In the **Work class ID** field, enter or select a value. This should be a work class related to replenishment. If you're using USMF, select *Replenish*.
9. In the **Work template details**, select **New**.
10. In the list, mark the selected row.
11. In the **Work type** field, select *Put*.
12. In the **Work class ID** field, enter or select a value.
13. Select **Save**.
14. Close the page.

> ⓘ **Note**
>
> When you use a replenishment template with **Replenishment type** set to *Minimum or maximum*, the location to replenish is defined as part of the replenishment template's setup and not by the location directive's *Put* work type.

Create a new replenishment template

1. Go to **Warehouse management** > **Setup** > **Replenishment** > **Replenishment templates**. The replenishment template is used to define the items and quantities, and the location to replenish.
2. On the Action Pane, select **New**.
3. In the **Replenish template** field, type a value. Give the template a name to indicate that it's for min/max replenishment.

4. In the **Description** field, type a value.
5. Select the **Allow wave demand to use unreserved quantities** checkbox. If you select this option, it enables wave demand replenishment to consume quantities that are related to min/max replenishment. For example, this might be useful if the min/max replenishment work isn't processed immediately, to avoid unnecessary demand replenishment work from being created.
6. In the **Replenishment template details**, select **New**.
7. In the **Sequence number** field, enter a number.
8. In the **Description** field, type a value.
9. In the list, mark the selected row.
10. In the **Replenishment unit** field, enter or select a value. For example, select pcs. This setting is mandatory. It allows you to specify a different unit of measurement for replenishment work compared to the unit specified for the minimum and maximum stock levels in this template.
11. In the **Work template** field, enter or select a value. Choose the work template that you created earlier.
12. In the **Minimum quantity** field, enter a number. Select the minimum quantity that should trigger replenishment. For example, set this to 50. You can leave this set to zero if you're replenishing a fixed location and the **Replenish empty fixed locations** option is set to *Yes*. Also select the **Replenish only fixed locations** option for performance reasons.
13. In the **Maximum quantity** field, enter a number. For example, set this to 100.
14. In the **Unit** field, enter or select a value. Assign the unit for the minimum and maximum quantities. For example, set this to pcs.
15. Select the **Replenish empty fixed locations** checkbox. Select this checkbox to replenish fixed locations when they're empty. Otherwise, the only the locations where there's a quantity on hand are replenished.
16. Select the **Replenish only fixed locations** checkbox.
17. Select **Select products**. Use this to define which products should be replenished. If the Fixed picking locations option is selected, you also need to define the locations in this query. Variant-specific queries are available, as well as product-specific queries.
18. Select the **Items** row.
19. In the **Criteria** field, type a value. Select the items that should be replenished at the fixed locations. For example, type A* to select all item numbers beginning with A.
20. Select **Add**. Add the Location entity (unless it already exists) to be able to restrict the replenishment work to the fixed picking locations within a specific area of the warehouse.
21. In the list, mark the selected row.
22. Set the **Table** field to *Locations*.

23. In the **Field** field, select *Location profile ID*.
24. In the **Criteria** field, enter or select a value.
25. Select **OK**.
26. Close the page.

Set the replenishment process to run as a batch job

1. Go to **Warehouse management** > **Replenishment** > **Replenishments**. The **Replenishments** page allows you to set up replenishment to run as a batch job, or to require that it's started manually.
2. Select **Filter**.
3. In the list, mark the selected row.
4. In the **Criteria** field, enter or select a value.
5. Select **OK**.
6. Expand the **Run in the background** section.
7. Set the **Batch processing** option to *Yes*.
8. Select **Recurrence**.
9. Select the **No end date** option.
10. Set the **Recurrence pattern**. For example, select Days.
11. Select **OK**.
12. Select **OK**.

Last updated on 06/22/2026

Zone threshold replenishment

Article • 08/23/2022

Zone-based replenishment uses a minimum/maximum (min/max) replenishment strategy, but it evaluates whole warehouse zones instead of just individual locations. Therefore, warehouse managers can more quickly learn when additional inventory is required in a picking zone.

The setup for this feature resembles the setup for location-based replenishment. However, when you set up a template for min/max replenishment, you can also specify whether the threshold should be evaluated per location or per zone. If you set up evaluation that is based on zones, you must add specific zones to the zone selection query.

Like location-based min/max replenishment, zone-based min/max replenishment is based the setup of a minimum inventory threshold that triggers the creation of replenishment work for selected items. This replenishment work will increase inventory up to the specified maximum threshold for the zone.

> **Note**
>
> Zone replenishment processes for product variants aren't supported in the current release.

Unlike location-based min/max replenishment, zone-based min/max replenishment doesn't require fixed locations to evaluate whether locations should store a specific item. Therefore, zone-based replenishment lets you use min/max replenishment even if you don't have fixed locations for each item or item variant in the warehouse. When a quantity in the zone falls below the specified minimum threshold, replenishment work is created. Location directives will determine which specific location the inventory should be put into.

Turn on the Zone threshold replenishment feature

Before you can use the *Zone threshold replenishment* feature, it must be turned on for your system. Admins can use the feature management settings to check the status of the feature and turn it on if it's required. In the **Feature management** workspace, the feature is listed in the following way:

- **Module:** *Warehouse management*
- **Feature name:** *Zone threshold replenishment*

Set up zone-based replenishment

To set up zone-based replenishment, you must configure several parts of the system. This section introduces the various settings and provides demo data values that you can enter if you want to work through the scenario at the end of this article.

Set up directive codes

Directive codes let you be more specific when you define the location template that is used in a work template. Each code establishes a common value that you can refer to when you configure each type of template.

View and edit directive codes

To view or edit your directive codes, go to **Warehouse management** > **Setup** > **Directive codes**.

Prepare demo data directive codes

This example shows how to prepare a directive code. If you're planning to work through the scenario at the end of this article, use the demo data values that are provided here. Otherwise, use your own values.

1. Select the **USMF** legal entity to work with the demo data.
2. Go to **Warehouse management** > **Setup** > **Directive codes**.
3. On the Action Pane, select **New** to add a row to the grid.
4. In the new row, set the following values:
    - **Directive code:** *Zone replen*
    - **Directive description:** *Zone replenishment*
5. Select **Save** to save the new code.

Set up replenishment templates

Min/max replenishment templates are the primary mechanism for maintaining optimal levels in picking locations. In these templates, you must set up the rules that will be used to replenish inventory in the warehouse. The replenishment that the templates can be used for includes zone-based replenishment.

View and edit replenishment templates

A replenishment template is a set of rules that control when and how a location is replenished. You select a template to control when and how replenishment is done. To view or edit your replenishment templates, go to **Warehouse management** > **Setup** > **Replenishment** > **Replenishment templates**.

Prepare a demo data replenishment template

This example shows how to prepare a replenishment template. If you're planning to work through the scenario at the end of this article, use the demo data values that are provided here. Otherwise, use your own values.

1. Select the **USMF** legal entity to work with the demo data.
2. Go to **Warehouse management** > **Setup** > **Replenishment** > **Replenishment templates**.
3. Select **Edit** to put the page into edit mode.
4. On the Action Pane, select **New** to add a row to the **Overview** grid.
5. In the new row, set the following values. Accept the default values for all other fields.
    - **Replenish template:** *Zone min/max replen*
    - **Description:** *Zone min/max replenishment*
    - **Replenishment type:** *Minimum or maximum*
6. Select **Save**.
7. While the new row is still selected in the **Overview** grid, select **New** above the **Replenishment Template Details** grid to add a row that is associated with the *Zone Min/Max replen* replenishment template that you just created.
8. In the new row, set the following values:
    - **Sequence number:** Enter *1*.
    - **Description:** Enter *Pick zone replenishment*.

- **Replenishment unit:** Select *ea*.
- **Request type:** Leave this field blank.
- **Directive code:** This field links the replenishment template with a location directive. Select the demo data directive code that you created earlier (*Zone replen*).
- **Work template:** Leave this field blank.
- **Minimum quantity:** This field sets the quantity that replenishment will be triggered at. Enter *50*.
- **Maximum quantity:** This field sets the maximum quantity of an item that can be present in a zone. Generated replenishment work will increase inventory to this quantity. Enter *150*.
- **Unit:** This field sets the unit for the minimum and maximum values. Select *ea*.
- **Demand increment:** Select *Round up*.
- **Replenish empty fixed locations:** Select this check box.
- **Replenish only fixed locations:** Clear this check box.
- **Product query mode:** Select *Product query*.
- **Replenishment threshold scope:** This field defines whether the template should evaluate by zone or by specific location. Select *Zone*.
- **Warehouse:** Select *61*.

9. Select **Select products** above the **Replenishment Template Details** grid.

10. In the **Product query** dialog box, on the **Range** tab, select **Add** to add a row to the grid.

11. In the new row, set the following values:

    - **Table:** *Items*
    - **Derived table:** *Items*
    - **Field:** *Item number*
    - **Criteria:** *A0001*

12. Select **OK** to save your query and close the dialog box.

13. Select **Select zones to replenish** above the **Replenishment Template Details** grid.

14. In the **Zone query** dialog box, on the **Range** tab, add a row to the grid.

15. In the new row, set the following values:

    - **Table:** *Warehouse zone*
    - **Derived table:** *Warehouse zone*
    - **Field:** *Zone ID*
    - **Criteria:** *FLOOR*

16. Select **OK** to save your query and close the dialog box.

Set up location directives

Unlike location-based min/max replenishment, zone-based min/max replenishment requires that you set up both pick location directives and put location directives, because the system evaluates the whole zone instead of just the pick location for outbound work.

View and edit location directives

To view or edit your location directives, go to **Warehouse management** > **Setup** > **Location directives**.

For examples that show how to use the settings to create the required pick location directives and put location directives, see the next section.

Prepare demo data location directives

To prepare demo data so that it can be used in the scenario at the end of this article, you must create two location directives: one for pick and one for put.

Create a replenishment pick directive

1. Select the **USMF** legal entity to work with the demo data.
2. Go to **Warehouse management** > **Setup** > **Location directives**.
3. In the left pane, set the **Work order type** field to *Replenishment*.
4. On the Action Pane, select **New** to create a new directive.
5. Set the following values:
    - **Sequence number:** Accept the default value.
    - **Name:** Enter *Zone pick*.
    - **Work type:** Select *Pick*.
    - **Site:** Select *6*.
    - **Warehouse:** Select *61*.
    - **Directive code:** Leave this field blank.
    - **Multi SKU:** Set this option to *No*.
6. Select **Save** to create a directive that has the settings that you've configured so far.

7. On the **Lines** FastTab, select **New** to add a line to the grid.

8. On the new line, set the following values:

    - **Sequence number:** Enter *1*.
    - **From quantity:** Enter *0*.
    - **To quantity:** Enter *10000000*.
    - **Unit:** Leave this field blank.
    - **Locate quantity:** Select *None*.
    - **Restrict by unit:** Clear this check box.
    - **Round up to unit:** Clear this check box.
    - **Locate packing quantity:** Clear this check box.
    - **Allow split:** Select this check box.

9. Select **Save** to save the new line.

10. While your new line is still selected in the **Lines** grid, select **New** on the **Location Directive Actions** FastTab to add a row to the grid.

11. In the new row, set the following values:

    - **Sequence number:** Enter *1*.
    - **Name:** Enter *Pick from bulk*.
    - **Fixed location usage:** Select *Fixed and non-fixed locations*.
    - **Allow negative inventory:** Clear this check box.
    - **Batch enabled:** Clear this check box.
    - **Strategy:** Select *None*.

12. Select **Save** to save the new action.

13. While your new action still selected, select **Edit query** above the **Location Directive Actions** grid.

14. A query dialog box appears, where you can select the locations to replenish from. On the **Range** tab, select **Add** to add a row to the grid.

15. In the new row, set the following values:

    - **Table:** *Locations*
    - **Derived table:** *Locations*
    - **Field:** *Zone ID*
    - **Criteria:** *BULK*

16. Select **OK** to save your query and close the dialog box.

17. Select **Save** to save your location directive.

Create a replenishment put directive

1. On the **Location directives** page, in the left pane, make sure that the **Work order type** field is still set to *Replenishment*.

2. On the Action Pane, select **New** to create another new directive.

3. Set the following values:

    - **Sequence number:** Accept the default value.
    - **Name:** Enter *Zone put*.
    - **Work order type:** Select *Put*.
    - **Site:** Select *6*.
    - **Warehouse:** Select *61*.
    - **Directive code:** Select *Zone replen* to link this location directive with the replenishment template that you created earlier by using the code that you created earlier.
    - **Multi SKU:** Set this option to *No*.

4. Select **Save** to create a directive that has the settings that you've configured so far.

5. On the **Lines** FastTab, select **New** to add a line to the grid.

6. On the new line, set the following values:

    - **Sequence number:** Enter *1*.
    - **From quantity:** Enter *0*.
    - **To quantity:** Enter *10000000*.
    - **Unit:** Leave this field blank.
    - **Locate quantity:** Select *None*.
    - **Restrict by unit:** Clear this check box.
    - **Round up to unit:** Clear this check box.
    - **Locate packing quantity:** Clear this check box.
    - **Allow split:** Select this check box.

7. Select **Save** to save the new line.

8. While your new line is still selected in the **Lines** grid, select **New** on the **Location Directive Actions** FastTab to add a row to the grid.

9. In the new row, set the following values:

    - **Sequence number:** Enter *1*.

- **Name:** Enter *Zone put*.
- **Fixed location usage:** Select *Fixed and non-fixed locations*.
- **Allow negative inventory:** Clear this check box.
- **Batch enabled:** Clear this check box.
- **Strategy:** Select *Consolidate*.

10. Select **Save** to save the new action.

11. While your new action is still selected, select **Edit query** above the **Location Directive Actions** grid.

12. A query dialog box appears, where you can select the zone to replenish to. This zone should be the same zone that is specified in the replenishment template. On the **Range** tab, select **Add** to add a row to the grid.

13. In the new row, set the following values:

    - **Table:** *Locations*
    - **Derived table:** *Locations*
    - **Field:** *Zone ID*
    - **Criteria:** *FLOOR*

14. Select **OK** to save your query and close the dialog box.

15. Select **Save** to save your location directive.

Scenario

This section provides a sample scenario that shows how to work with the feature.

Prepare the sample data that is required for the sample scenario

Before you start to work through the scenario, you must activate sample data and set up the feature as described in this section and in the previous sections of this article.

Use the USMF legal entity

To work through the scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

Prepare additional sample data

After you've selected the **USMF** legal entity, add the additional sample data that is required, as described in the Set up zone-based replenishment section earlier in this article.

Check your on-hand inventory

Follow these steps to make sure that your system includes enough inventory to support the sample scenario.

1. Make sure that there is on-hand inventory for item *A0001* at two different locations in the pick zone (*FLOOR*) that is specified in the replenishment template. However, the total inventory should be less than the required minimum quantity (*50*) that is specified on the replenishment template. In this way, you can simulate how the calculation occurs for the whole zone instead of just for a single location. **Use any of the warehouse processes to adjust inventory as required.**

2. Make sure that there is enough inventory for item *A0001* at a bulk location that is specified in the zone pick location directive where the replenishment work should pick the items from zone ID *BULK*. The total inventory must be more than the required maximum quantity (*150*) that is specified in the replenishment template.

3. Optional but recommended: Follow these steps to create an inventory adjustment journal:

    a. Go to **Inventory management** > **Journal entries** > **Items** > **Inventory adjustment**.

    b. Select **New**.

    c. In the **Create inventory journal** dialog box, in the **Warehouse** field, select *61*.

    d. Select **OK**.

    e. On the **Journal lines** FastTab, use the **New** button to add three lines to the grid, and set the following values. After you've finished setting up each line, select **Save**.

        - **Line 1:**
            - **Item number:** *A0001*
            - **Site:** *6*
            - **Warehouse:** *61*
            - **Location:** *02A01R1S1B*

    - **License plate:** Select an existing license plate in the list, or create a new license plate.
    - **Quantity:** *1000*

- **Line 2:**
    - **Item number:** *A0001*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Location:** *07A01R2S1B*
    - **License plate:** Select an existing license plate in the list, or create a new license plate.
    - **Quantity:** *15*

- **Line 3:**
    - **Item number:** *A0001*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Location:** *07A01R1S1B*
    - **License plate:** Select an existing license plate in the list, or create a new license plate.
    - **Quantity:** *10*

f. On the Action Pane, select **Validate**. Address any errors that are found before you move on to the next step.

g. On the Action Pane, select **Post** to post the inventory to the warehouse.

Sample scenario: Run zone-based min/max replenishment

After all the prerequisite sample data is in place, you can trigger replenishment by following these steps.

1. Go to **Warehouse management** > **Replenishment** > **Replenishments**.

2. In the **Replenishment** dialog box, on the **Records to include** FastTab, select **Filter**.

3. In the **Inquiry** dialog box, on the **Range** tab, edit the default table row in the following way:

    - **Table:** Select *Replenishment templates*.
    - **Derived table:** Select *Replenishment templates*.
    - **Field:** Select *Replenishment template*.

- **Criteria:** Select *Zone min/max replen*. This replenishment template is the replenishment template that you created while you were preparing the demo data for this scenario.

4. Select **OK** to save the query and go back to the **Replenishment** dialog box.
5. Select **OK** to run the replenishment template.

Replenishment work is now created to pick inventory from the *BULK* zone and replenish it to the *FLOOR* zone.

Notes and tips

Here are a few notes and tips for working with the feature:

- To set up replenishment work that goes to the desired zone, you can link the replenishment template lines and location directives in either of the following ways:
  - Edit the location directive header query, and filter the selected replenishment template lines.
  - Use a directive code on the replenishment template line, and match it to the put location directive.
- If you're using dynamic locations, replenishment work will be created either for the first available location or for a location that already contains inventory, if the location directive action is set up to use the **Consolidate** strategy.
- If you're using fixed locations instead of zones, you should use standard min/max replenishment.

Feedback

Was this page helpful? Yes No

Provide product feedback

Replenishment strategies

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The templates defined on the **Replenishment templates** page include wave demand replenishment template lines that you use to select how replenishment is done. Each line now includes a **Replenishment strategy** field.

The *Wave demand quantity* strategy is the default strategy. It's the replenishment strategy that was used before the introduction of the **Replenishment strategy** field. It uses the replenishment location directives to find locations that can be replenished. It then replenishes those locations until the demand is covered.

The *Maximum location capacity* strategy introduces some new functionality. Like the *Wave demand quantity* strategy, this strategy uses the replenishment location directives to find locations that can be replenished, and then it replenishes those locations until the demand is covered. It differs from the *Wave demand quantity* strategy in that all the replenished locations are replenished to their maximum capacity, as defined by the location stocking limits. The *Maximum location capacity* strategy tries to create work to bring in the requested quantity, plus some extra quantity, to fill the locations that are being replenished. However, in some cases, that attempt might fail. For example, the bulk locations might not have enough inventory to cover the extra quantity. In these cases, the system detects the failure and tries to recover.

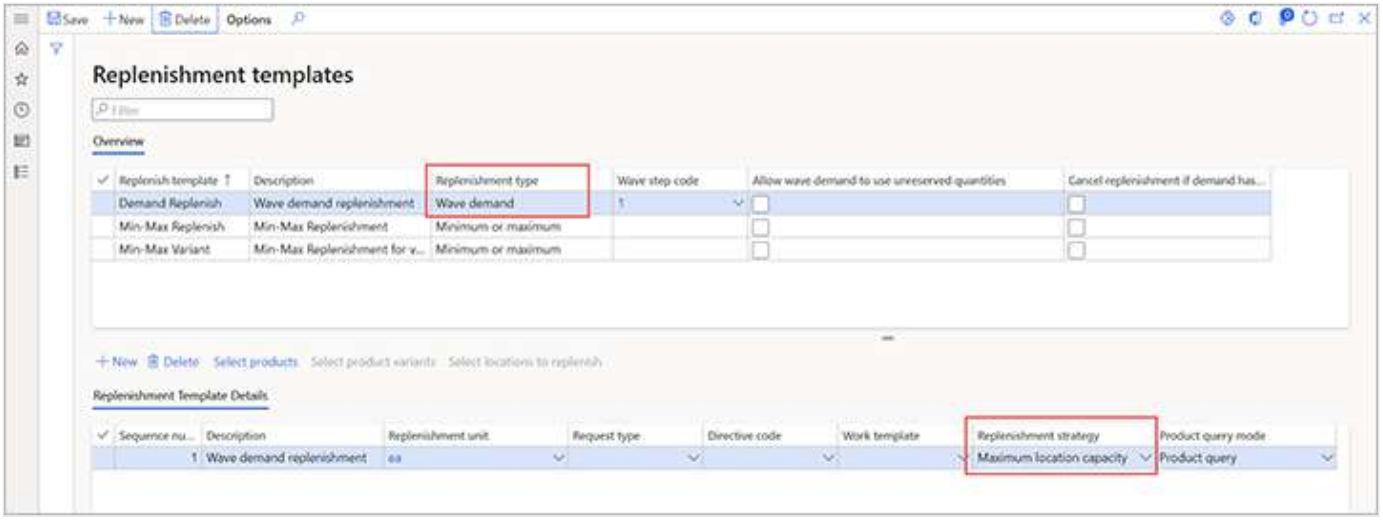
Peak season is one example of a situation where the *Maximum location capacity* strategy is preferable to the *Wave demand quantity* strategy. During peak season, some items might be selling at high volume. Therefore, you might want to proactively replenish the relevant picking locations as much as possible, to reduce the number of work IDs that are created for replenishment.

> **Important**

To take full advantage of the *Maximum location capacity* strategy, you must redefine the stocking limits for the relevant locations. Otherwise, this strategy works just like the *Wave demand quantity* strategy.

Set up replenishment strategies

To access the templates, go to **Warehouse management** > **Setup** > **Replenishment** > **Replenishment templates**. In the **Overview** section, select or create a wave demand replenishment template where the **Replenishment type** field is set to *Wave demand*. Then set up the replenishment template lines in the **Replenishment template details** section. For each line, in the **Replenishment strategy** field, select the replenishment strategy that you want to use.



If the **Replenishment strategy** column doesn't appear in the grid in the **Replenishment template details** section, make sure that the feature is turned on, and that the selected replenishment template has a replenishment type of *Wave demand*.

> **Note**
>
> The *Wave demand quantity* strategy is the default strategy. Therefore, you just need to update the replenishment template lines where you want to use the *Maximum location capacity* strategy instead.

Example scenarios

Example 1

For this example, there's only one replenishment template that has only one replenishment template line.

You create a sales order for 130 pieces (pcs) of item A0001. Before you release the order to the warehouse, ensure to set up the warehouse in the following way:

- There's only one bulk location, and it has 500 pcs of available on-hand inventory.
- There are three pick locations, each of which has a stocking limit of 100 pcs. (Remember that stocking limits are required for the *Maximum location capacity* strategy.)
- The replenishment put locations are the same as the sales pick locations.
- The replenishment unit is a box (1 box = 20 pcs).

At the time of the order, the following inventory is on hand at the sales pick locations:

- **Pick-001:** 20 pcs (1 box)
- **Pick-002:** 0 pcs
- **Pick-003:** 0 pcs

Initially, set the replenishment strategy to *Wave demand quantity*.

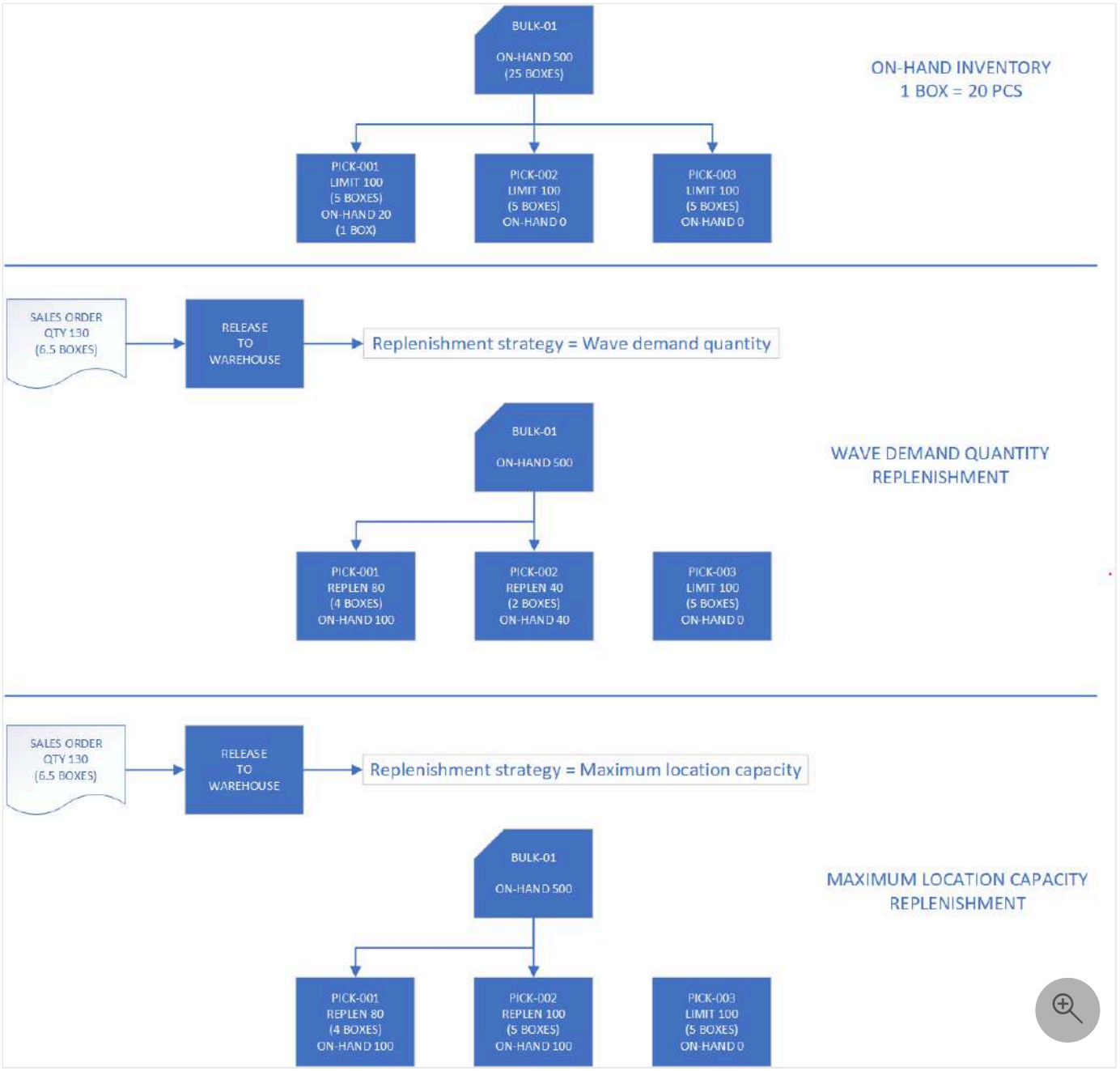
After you release the sales order to the warehouse, and wave processing runs for the wave, you get the following replenishment work:

- **Replenishment work 1:** Pick 4 boxes from the bulk location, and put them in location pick-001.
- **Replenishment work 2:** Pick 2 boxes from the bulk location, and put them in location pick-002.

You get two replenishment work IDs because you must replenish two locations, and multi-puts aren't supported.

If you set the replenishment strategy to *Maximum location capacity* instead, you get the following replenishment work:

- **Replenishment work 1:** Pick 4 boxes from the bulk location, and put them in location pick-001.
- **Replenishment work 2:** Pick 5 boxes from the bulk location, and put them in location pick-002.



Example 2

This example shows what happens when the bulk location doesn't have enough inventory to cover the extra quantity. It uses the same scenario as example 1, but the bulk location has 160 pcs (8 boxes).

The *Wave demand quantity* strategy creates the same work that it did in example 1.

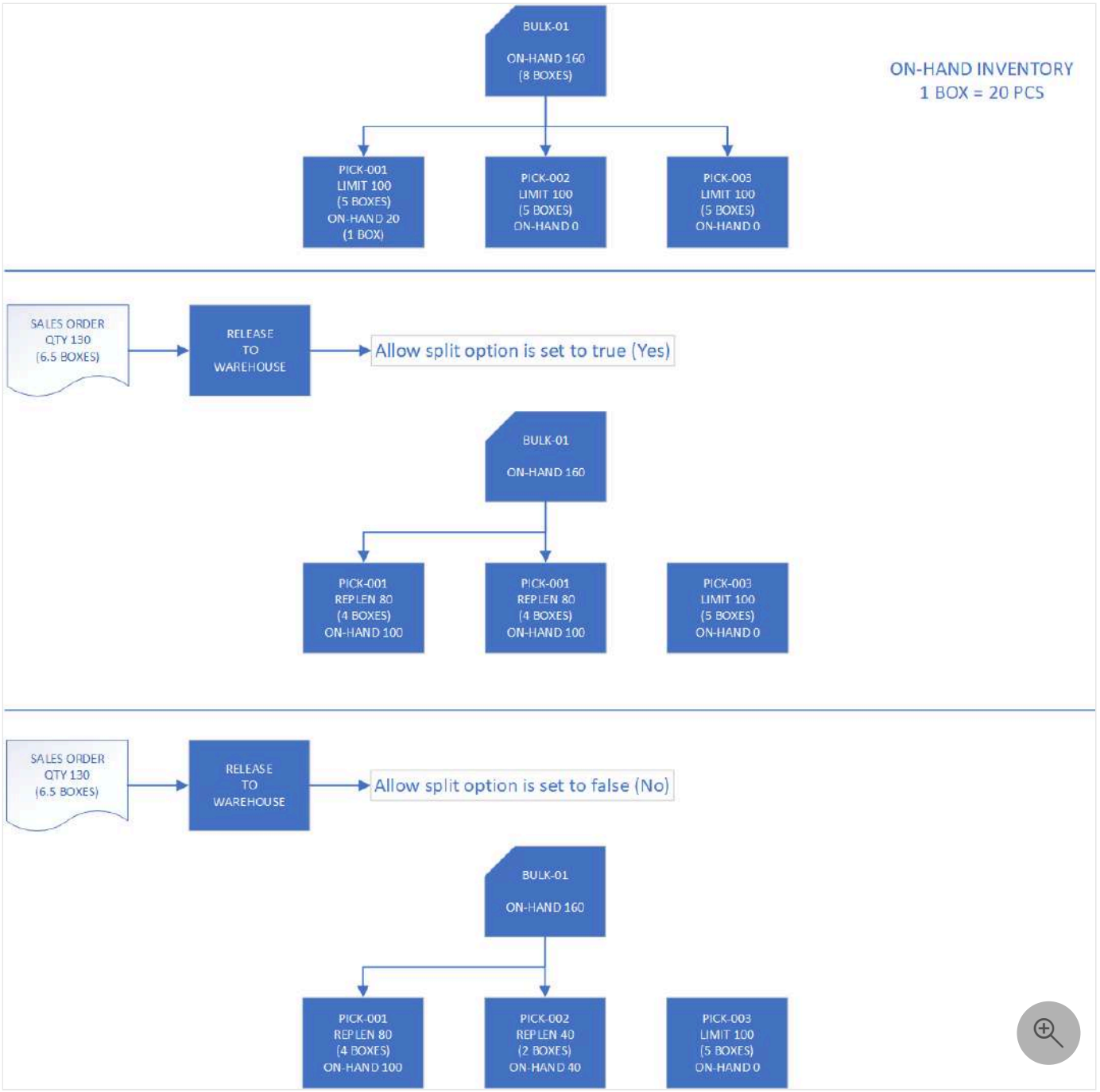
However, because the *Maximum location capacity* strategy tries to create the work IDs as it did in example 1, it might fail. At that point, the system tries to recover.

Depending on the setting of the **Allow split** option on the location directives for replenishment picking, two outcomes are possible:

- If the **Allow split** option is set to *Yes*, the following replenishment work is created:
  - **Replenishment work 1:** Pick 4 boxes from the bulk location, and put them in location pick-001.
  - **Replenishment work 2:** Pick 4 boxes from the bulk location, and put them in location pick-002.

- If the **Allow split** option is set to *No*, the following replenishment work is created:
  - **Replenishment work 1:** Pick 4 boxes from the bulk location, and put them in location pick-001.
  - **Replenishment work 2:** Pick 2 boxes from the bulk location, and put them in location pick-002.

The outcomes differ because of the information that is available when you create the work. When the **Allow split** is set to *Yes* on the location directives for replenishment picking, you know that you managed to find 160 pcs. Therefore, you can create work for that quantity. However, when the **Allow split** option is set to *No*, you don't know about the existence of the 160 pcs. Because the extra quantity that you decided to replenish was 3 boxes, you drop that extra quantity and try the original quantity again.



Therefore, to get the maximum quantity to the replenished locations, set the **Allow split** option to *Yes* on the location directives for replenishment picking.

Last updated on 05/29/2026

Replenishment over location capacity

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Some high-volume or space-constrained warehouses must ship more quantity of an item in a day than can fit in the picking location. The *Replenishment over location capacity* feature enables all replenishment work required for the day to be created and manages the availability of that replenishment work to ensure that the picking location doesn't runs out of inventory or go above capacity.

The feature enables more replenishment work to be created than can fit in a location, and it blocks replenishment work from being completed when the location is full. As inventory in the picking location drops below a configurable threshold, more replenishment work is made available.

Set up the feature for the example scenario

This section provides guidelines and an example that shows how to set up this feature and prepare sample data for the example scenario that is provided later in this article.

Enable sample data

To work through the example scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin.

Location profile

Enable the replenish over capacity functionality on the location profile.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations profiles**.
2. In the left pane, select *PICK-06*.
3. On the Action Pane, select **Edit**.

4. On the **Replenishment** FastTab, set the following values:

- **Exceed Location Capacity:** *Yes*

  When enabled, the maximum capacity of the location is allowed to be exceeded by replenishment work. This also enables other fields on the **Replenishment** FastTab.

  This parameter only enables the feature. For it to have an effect, you must also set the **Work availability threshold type**, **Overflow quantity**, and **Overflow unit**.

- **Work availability threshold type:** *Quantity*

  This field defines the method that is used to determine when more work should be released. You can release by either quantity or a percentage:

  - *Percent* – Select this option to use percentage capacity that is based on stocking limits or volumetrics. Selecting this option enables the **Overflow percentage** field, and disables the two quantity related fields, **Overflow quantity** and **Overflow unit**.

    You can use this option if the picking locations use volumetrics.

    If this option is selected, set the **Overflow percentage** field to the percentage at which more replenishment work will be made available.

  - *Quantity* – Select this option to use a specific quantity value. Selecting this option disables the **Overflow percentage** field and enables **Overflow quantity** and **Overflow unit** fields.

    Use this option when you aren't using volumetrics for the locations that are being replenished, or when you have consistent quantities at which you want more inventory to be brought to the location.

    If this option is selected, set the **Overflow quantity** and **Overflow unit** fields to the quantity and unit at which more replenishment work should be made available.

- **Overflow quantity:** *0.65*

  This field defines the quantity at which the location overflows.

  Work is available whenever the sum of the on-hand quantity in the location and the work quantity is below this value. Any replenishment work above this value is blocked and must be manually unblocked.

  Location stocking limits are considered when the work quantity is calculated.

- **Overflow unit:** *PL*

  This field defines the unit that is associated with the overflow quantity.

  In this case, more replenishment work is made available when the location gets down to 0.65 pallet (PL).

- **Overflow percentage**

  This field defines the percentage at which the location overflows.

  Work is available whenever the sum of the on-hand quantity in the location and the work quantity is below this percentage. Any replenishment work quantity percentage above this value is blocked and must be manually unblocked.

  Location stocking limits are considered when the work quantity percentage is calculated. If no location stocking limits are defined, the system calculates the work quantity percentage by volume if volume constraints are defined for the location profile.

> ⓘ **Important**
>
> If you're using the demo data for the *USMF* legal entity and previously turned on the *Location license plate positioning* feature, you must turn off the **Enable license plate positioning** setting for the *BULK-06* location profile to complete the mobile steps in the example scenario.

> ⓘ **Note**
>
> Location profile replenishment settings only affect replenishment work. Other types of work are still subject to configured stocking limits.

Wave step code

1. Go to **Warehouse Management** > **Setup** > **Waves** > **Wave step codes**.
2. Select **New**, and set the following values:
   - **Wave step code:** *Replen*
   - **Wave step description:** *Replenishment*
   - **Wave step type:** *Replenishment*
3. Select **Save**.

Replenishment template

Replenishment templates are a set of rules that control when and how a location is replenished.

1. Go to **Warehouse management** > **Setup** > **Replenishment** > **Replenishment templates**.
2. On the Action Pane, select **Edit**.
3. In the **Overview** section, select the line where the **Replenish template** field is set to *Demand replenish*.
4. Set the following values:
    - **Wave step code:** *Replen*
    - **Allow wave demand to use unreserved quantities:** *Yes*
5. Select **Save**.

Wave template

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. In the left pane, set the **Wave template type** field to *Shipping*.
3. Select template *61 Shipping* in the list.
4. On the Action Pane, select **Edit**.
5. On the **General** FastTab, set the **Automate replenishment work release** option to *Yes*.

    Set this option to *Yes* to create demand-based replenishment work and release it automatically. You must add the replenishment wave method to the wave template and create a replenishment template of the *Wave demand* type. Set up a replenishment template on the **Replenishment templates** page. To set up a replenishment template, you must add the replenish method to the wave template.
6. On the **Methods** FastTab, in the **Selected methods** column, find the following line:
    - **Method name:** *replenish*
    - **Name:** *Replenishment*
7. Set the **Wave step code** field for this line to *Replen*.
8. Select **Save**.

Example scenario

After you've made all the previously described sample data available and set it up, you can work through this scenario to try out the *Replenishment over location capacity* feature. The values that are shown in this scenario assume that you're working with the standard demo data, that you selected the *USMF* legal entity, and that you prepared the sample records that are described earlier in this article. This scenario also serves as an example that shows how the feature can be used in a production setting.

Create replenishment work

Create sales order 1

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New** to open a dialog box for creating a new sales order.
3. In the dialog box, set the following values:
    - **Customer account:** *US-007*
    - **Warehouse:** *61*
4. Select **OK** to create the sales order and close the dialog box.
5. The new sales order is opened. It includes a new, empty line on the **Sales order lines** FastTab. On this line, set the following values:
    - **Item number:** *T0100*
    - **Quantity:** *40*
6. On the **Sales order lines** FastTab, select **Inventory** > **Reservation**.
7. On the **Reservation** page, select **Reserve lot**.
8. Close the page.
9. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

    You receive informational messages that show the wave and shipment IDs that were created. A replenishment wave is also created.

    You also receive a warning message that states, "Work ID XXXX cannot be unblocked because it has unfinished replenishment work."

Create sales order 2

1. On the **All sales orders** page, on the Action Pane, select **New** to open a dialog box for creating a new sales order.
2. In the dialog box, set the following value:
    - **Customer account:** *US-001*
    - **Warehouse:** *61*
3. Select **OK** to create the sales order and close the dialog box.
4. The new sales order is opened. It includes a new, empty line on the **Sales order lines** FastTab. On this line, set the following values:
    - **Item number:** *T0100*
    - **Quantity:** *60*
5. On the **Sales order lines** FastTab, select **Inventory** > **Reservation**.
6. On the **Reservation** page, select **Reserve lot**.
7. Close the page.
8. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

    You receive informational messages that show the wave and shipment IDs that were created. A replenishment wave is also created.

    You also receive a warning message that states, "Work ID XXXX cannot be unblocked because it has unfinished replenishment work."

Create sales order 3

1. On the **All sales orders** page, on the Action Pane, select **New** to open a dialog box for creating a new sales order.
2. In the dialog box, set the following values:
    - **Customer account:** *US-004*
    - **Warehouse:** *61*
3. Select **OK** to create the sales order and close the dialog box.
4. The new sales order is opened. It includes a new, empty line on the **Sales order lines** FastTab. On this line, set the following values:

- **Item number:** *T0100*
- **Quantity:** *30*

5. On the **Sales order lines** FastTab, select **Inventory** > **Reservation**.

6. On the **Reservation** page, select **Reserve lot**.

7. Close the page.

8. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

   You receive informational messages that show the wave and shipment IDs that were created. A replenishment wave is also created.

   You also receive a warning message that states, "Work ID XXXX cannot be unblocked because it has unfinished replenishment work."

View work details

1. Go to **Warehouse management** > **Work** > **Work details**.

2. In the **Overview** section, filter the **Warehouse** column for warehouse *61*.

3. You should see that seven work IDs were created for the three demand sales orders.

   - Three of the seven work IDs have a **Work order type** value of *Replenishment*, and four have a **Work order type** value of *Sales orders*.

   - All three work IDs that have a **Work order type** value of *Replenishment* have the same *Pick* and *Put* locations in the **Lines** section:
     - **Pick:** *02A01R5S1B*
     - **Put:** *06A01R2S1B*

   - Two work IDs were created for sales order 1.

4. Make a note of the work IDs for the sales orders.

Depending on your on-hand quantities, the work quantities that are created might vary slightly. However, overall, the work headers that are created should match this scenario example.

On-hand inventory license plate ID

Later in this scenario, you'll use the Warehouse Management mobile app (or an emulator), where you must identify the license plate to complete the picking and replenishment scenarios.

To find the license plate IDs that you'll need later, follow these steps.

1. Go to **Inventory management** > **Inquiries and reports** > **On-hand list**.
2. Select the **Show filters** button to open the filter pane.
3. Enter the following filtering criteria to get the license plates for the scenario. Use the *begins with* filter.
    - **Item number:** *T0100*
    - **Warehouse:** *61*
4. Select **Apply**.
5. On the Action Pane, select **Dimensions**.
6. In the **Dimensions display** dialog box, in the **Storage Dimensions** section, select all the values.
7. In the **Transactions** section, select **Item number** and **Quantity <> 0**.
8. When you've finished, select **OK** to close the dialog box.
9. The **On-hand** grid shows the license plate numbers for item *T0100* in each location. Make a note of the license plate that is in each location, because you'll need this information later.
10. Close the page.

Process steps

You'll perform the warehouse location replenishment for the first two work IDs. Work on the third replenishment work is blocked until the inventory level in the picking location falls below the threshold.

Replenishment

1. Sign in to the Warehouse Management mobile app as a user in warehouse *61*. (Enter *61* as the user ID and *1* as the password.)
2. Go to **Inventory** > **Replenishment**.

    You're prompted to complete the first replenishment work. The item number, quantity, and location to pick from are shown.
3. In the **LP** field, enter the license plate number for the item in the location that is shown.

4. Select the **OK** button (check mark symbol).

   The system generates a target license plate number for the new license plate for the picked item.

5. Select **OK** to confirm the value.

   Put work is shown that instructs the user to put the target license plate into the replenishment location. The **Put** location should be *06A01R2S1B*.

6. Confirm the put details, and select **OK**.

   You receive a "Work Completed" message, and the details of the next replenishment pick task are shown: the item number, quantity, and location to pick from. The picking location is the same as the first replenishment location. Therefore, the license plate has the same license plate ID that was used for the first replenishment work task.

7. Repeat the preceding steps to complete the replenishment work for the second work task. The quantity and target license plate differs from the quantity and target license plate for the first work task.

After the second replenishment work is completed, you receive a "Work Completed" message. The mobile device also informs you that there's no work available, even though some replenishment work remains. This behavior occurs because the replenishment work has an availability status of *Held* and is therefore marked as *Blocked*.

The *Held* status was triggered because the location profile for the picking location that the work is being assigned to has an **Overflow quantity** value of *0.65 PL*. The two previous replenishment work tasks filled the location almost to its overflow limit for item *T0100*. (The unit conversion for the item is *1 PL = 100 ea*.) Therefore, the remaining replenishment work would cause the location to exceed its overflow limit.

Until enough inventory is picked from the location to bring it below the work release threshold on the mobile device menu item, this replenishment work will remain blocked.

Sales order pick

Before the remaining replenishment work task can be completed, the picking location must be depleted of inventory to a level where the remaining replenishment work can be unblocked. In other words, the sum of the quantity of on-hand inventory in the location and the replenishment quantity can't exceed the **Overflow quantity** value. When this sum is less than the overflow quantity, the remaining replenishment work is unblocked.

1. Sign in to the Warehouse Management mobile app as a user in warehouse *61*. (Enter *61* as the user ID and *1* as the password.)
2. Go to **Outbound** > **Sales Picking**.
3. Enter the first work ID for sales order 1.

   Refer to the work IDs for sales orders that you made a note of earlier, on the **Work details** page. The work ID that you enter here will generate pick work for a quantity of 10 ea from two separate locations.
4. Select **OK**.

   The **Sales orders: Pick** task page shows the item number, quantity, and location to pick from for the first location.
5. In the **LP** field, enter the license plate number for the item in the location that is shown.
6. Select the **OK** button (check mark symbol).

   The **Sales orders: Pick** task page shows the item number, quantity, and location to pick from for the next location.
7. In the **LP** field, enter the license plate number for the item in the location that is shown.
8. Select the **OK** button (check mark symbol).

   The **Sales orders: Put** page instructs you to put away both the completed picking works to the outbound staging location.
9. Select **OK**.

   You receive a "Work Completed" message.
10. Enter the second work ID for sales order 1.

    There's only one pick task for this work ID.
11. Select **OK**.

    The **Sales orders: Pick** task page shows the item number, quantity, and location to pick from.
12. In the **LP** field, enter the license plate number for the item in the location that is shown.

    The license plate that you specify should be one of the system-generated license plates from the replenishment work tasks. To make sure that you capture the correct license

plate ID, check the inventory on the **On-hand list** page for the item, location, and quantity.

13. Select the **OK** button (check mark symbol).
14. Confirm the instructions for the put task to the outbound staging location.
15. Select **OK**.

    You receive a "Work Completed" message.

Sales order 2 is blocked from picking because the replenishment task that it's linked to isn't completed. Currently, there's still a quantity of 30 ea in the picking location, and the replenishment quantity for sales order 2 is 60 ea. The sum of the on-hand inventory and the replenishment inventory (90 ea) exceeds the overflow quantity of 0.65 PL (or 65 ea). Before the replenishment work can be completed, sales order 3 must be picked.

1. Enter the work ID for sales order 3.

   There's only one pick task for this work ID.

2. Select **OK**.

   The **Sales orders: Pick** task page shows the item number, quantity, and location to pick from.

3. In the **LP** field, enter the license plate number for the item in the location that is shown.

   The license plate that you specify should be one of the system-generated license plates from the replenishment work tasks. To make sure that you capture the correct license plate ID, check the inventory on the **On-hand list** page for the item, location, and quantity.

4. Select the **OK** button (check mark symbol).
5. Confirm the instructions for the put task to the outbound staging location.
6. Select **OK**.

   You receive a "Work Completed" message.

As soon as the sum of the on-hand quantity in the picking location and the replenishment quantity is below the threshold, you'll be able to process the remaining replenishment work.

Return to the **Work details** page, and notice that the replenishment work availability for the final piece of replenishment (for sales order 2) is *Open*, because there's now enough space in the location to accept the replenishment.

You can now process this replenishment work via the mobile device.

1. Go to **Inventory** > **Replenishment**.

   You're prompted to complete the remaining replenishment work. The item number, quantity, and location to pick from are shown.

2. In the **LP** field, enter the license plate number for the item in the location that is shown.

3. Select the **OK** button (check mark symbol).

   The system generates a target license plate number for the new license plate for the picked item.

4. Select **OK** to confirm the value.

   Put work is shown that instructs the user to put the target license plate into the replenishment location. The **Put** location should be *06A01R2S1B*.

5. Confirm the put details, and select **OK**.

   You receive "Work Completed" and "No Work Available" messages.

You can now pick sales order 2. It became unblocked when the replenishment work that is linked to the sales order was completed.

1. Enter the work ID for sales order 2.

   There's only one pick task for this work ID.

2. Select **OK**.

   The **Sales orders: Pick** task page shows the item number, quantity, and location to pick from.

3. In the **LP** field, enter the license plate number for the item in the location that is shown.

   The license plate that you specify should be the system-generated license plate from the replenishment work task. To make sure that you capture the correct license plate ID, check the inventory on the **On-hand list** page for the item, location, and quantity.

4. Select the **OK** button (check mark symbol).

5. Confirm the instructions for the put task to the outbound staging location.

6. Select **OK**.

   You receive a "Work Completed" message.

Notes and tips

- This functionality works with all types of replenishment: wave demand, min/max, load demand, and slotting.
- You can manually override the replenishment work availability for each work header from the **Work details** page if you want.
- When the system sets the replenishment work availability, it considers any inventory that is already in the location before any work is completed.
- Each piece of sales order work is linked to a specific replenishment work. There's no corresponding sales work availability functionality.

Last updated on 10/03/2025

Confirm and transfer

09/24/2025

The *Confirm and transfer* feature lets workers ship loads out of the warehouse before they complete all the work that's associated with those loads. This functionality supports scenarios where the full load doesn't fit onto a single truck, or where some of the load should leave the warehouse before the rest of the load is ready for shipment. In these scenarios, the remaining products can be transferred to a new load and therefore to a new truck. Because you can use this feature with loads that are otherwise intended to require full-load shipment, it provides extra flexibility so that transport managers can solve nonstandard or unforeseen scenarios.

You can choose what the system should do when confirming a load where not all load lines are fully picked.

You can set the system to either ask the worker what to do, or run batch jobs to automatically do the following actions:

- Split the load (add the unfulfilled quantity to a new load)
- Cancel the unfulfilled quantity

Splitting the load

You can select the default split method for loads in the load template details and in the single load details. Systems set up to support load splitting handle scenarios where planned and partially loaded loads need to change because of new or changing circumstances.

The client includes logic that enables a partially loaded load to be closed and confirmed as shipped. If this rollover is required and the setup allows for it, all remaining shipments and load lines (including full or partial line quantities) roll over to a newly created load and shipment. New shipments and loads are automatically created based on the initial criteria for shipment and load creation, and their main attributes remain unchanged. You can also automatically cancel remaining quantities if a work order isn't yet created and you want to cancel these quantities.

When you split a load, the *Confirm and transfer* feature performs the following actions:

- Creates new loads and shipments as required. Each load or shipment has most of the same attributes as the original load or shipment. The exception is the load status, which is recalculated based on the work status.
- Informs you when a new load is created. It also notifies you about the ID of the new load.
- Updates the load lines, work headers, and work lines that were split with the new load and shipment information.

- Maintains the shipment and updates only the load references if a whole shipment is being split. Creates a new shipment if the shipment must be split.

You can split only loads that meet all the following criteria:

- One or more load lines have picked quantities.
- The load status is less than loaded.
- There's no load line data. (This data is created through license plate consolidation on the staging location, and the *Confirm and transfer* feature doesn't support license plate consolidation.)
- No inventory is currently awaiting packing at a packing location. (To be able to split loads with the *Confirm and transfer* feature for inventory that you picked to the pack station but didn't yet pack and move to staging locations, follow the configuration steps detailed in Packing work for packing outbound containers and processing shipments).

Cancelling the load

When you cancel remaining quantities, the process removes any load line quantities that aren't picked and that don't have noncanceled work associated with them from the load. If any work is in process, you can only split the load. You can't cancel remaining quantities.

> **Note**
>
> Use the *Confirm and transfer* feature in situations where you usually plan and create loads ahead of time, but exceptions sometimes occur in which the load doesn't fit the available transport (such as a truck).

Set up confirm and transfer

You must enable *Confirm and transfer* functionality for every load template where you want to use it. Depending on your requirements, you might also want to prepare your work templates to support the feature. This section describes how to set up the feature and also describes the settings you need if you're planning to work through the example scenario that is provided later in this article.

Prepare your load templates

1. Go to **Warehouse management** > **Setup** > **Load** > **Load templates**.
2. Do one of the following steps:

- To edit an existing template, select **Edit** on the Action Pane.
- To create a new template, select **New** on the Action Pane.

3. Set up your new or selected template as usual, and then select the **Allow load split during ship confirm** check box for each template where you want to enable this feature. Every load that you create by using that template inherits this functionality. (If you're planning to work through the scenario provided later in this article, and are working with the *USMF* demo data, select this check box for the *20' Container* load template.)

4. On the Action Pane, select **Save**.

5. For each template that you save with the **Allow load split during ship confirm** checkbox selected, set **Load split shipment confirmation policy** to one of the following values to control what the system should do when confirming a load where not all load lines are fully picked:

   - *Manual selection* – Display a dialog that asks the worker to choose whether to add the remaining quantities to a new load or cancel the incomplete quantities. A batch job can't perform this action.
   - *Split quantity to a new load* – Automatically add the unfulfilled quantity to a new load. A batch job can perform this action.
   - *Cancel unfulfilled quantity* – Automatically cancel the remaining quantity. A batch job can perform this action.

   (If you're planning to work through the scenario provided later in this article, and are working with the *USMF* demo data, then set **Load split shipment confirmation policy** to *Manual selection* for the *20' Container* load template.)

Prepare your work templates

This setup isn't required in all situations. The example that is shown here ensures that work can be broken by shipment to support the example scenario that is provided later in this article. You can also use other methods to achieve this result.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.

2. In the grid in the upper part of the page, select an existing work template where you want to set up the *Confirm and transfer* feature. (If you plan to work through the scenario provided later in this article, and you're working with the *USMF* demo data, select the *51 Pick to stage* work template.) Alternatively, create a new work template.

3. On the Action Pane, select **Edit query** to open the **Sales** dialog box.

4. In the **Sales** dialog box, on the **Sorting** tab, select **Add** to add a row to the grid.

5. In the new row, set the following values:

    - **Table:** *Temporary work transactions*
    - **Derived table:** *Temporary work transactions*
    - **Field:** *Shipment ID*
    - **Search direction:** *Ascending*

6. Select **OK** to save your settings and close the **Sales** dialog box.

7. If you receive a message that states that grouping is reset, select **Yes** to continue.

8. In the grid in the upper part of the **Work templates** page, select the template that you just edited, and then, on the Action Pane, select **Work header breaks**.

9. On the Action Pane, select **Edit** to put the page into edit mode.

10. In the grid, set the following values:

    - **Table name:** *Temporary work transactions*
    - **Field name:** *Shipment ID*
    - **Group by this field:** Select this check box.

11. On the Action Pane, select **Save**.

12. Close the page.

Scenario

This scenario shows an example where the picking process isn't yet completed, but the items that you picked so far must be loaded onto a truck and shipped anyway. Therefore, you can split the unpicked load lines onto a new load. All the data relationships automatically update.

> ⓘ **Note**
>
> The specific values that are described in this scenario are based on the *USMF* demo data. Use this demo data while you're demonstrating or learning how to use the feature. If you don't use the *USMF* demo data, substitute your own values as required.

Step 1: Create a load that has multiple load lines

Before you can use this functionality, you must have a load that contains multiple load lines. You must also make sure that the pick locations have enough inventory for all the items on the sales orders that you create. Review the setup of the location directive (**Warehouse**

management > **Setup** > **Location directives**), and make a note of the picking locations that are used for sales order picking. If you must adjust the inventory, create manual movements, use replenishment, or use any other flow, as required.

To create a qualifying load, first create three sales orders by following these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New** to open the **Create sales order** dialog box.
3. In the **Create sales order** dialog box, set the following values (at a minimum):
    - On the **Customer** FastTab, set the **Customer account** field to *US-004* (*Cave Wholesales*).
    - On the **General** FastTab, set the **Warehouse** field to *51*.
4. Select **OK** to create the sales order and close the **Create sales order** dialog box.
5. Your new sales order opens. In the **Sales order lines** grid, add a line that has the following values:
    - **Item number:** *M9200*
    - **Quantity:** *40*
    - **Unit:** *ea*
6. On the **Inventory** menu above the grid, select **Reservation**.
7. On the Action Pane, select **Reserve lot** to open the **Reservation** page.
8. Reserve the inventory on the sales line, and then close the **Reservation** page.
9. Repeat steps 1 through 4 to add a second sales order for the same customer and warehouse.
10. Add two sales lines that have the following values. After you add each line, reserve inventory for it as described in steps 6 through 8.
    - **Line 1:** Set the **Item number** field to *M9200*, the **Quantity** field to *30*, and the **Unit** field to *ea*.
    - **Line 2:** Set the **Item number** field to *M9201*, the **Quantity** field to *20*, and the **Unit** field to *ea*.
11. Repeat steps 1 through 4 to add a third sales order for the same customer and warehouse.
12. Add two sales lines that have the following values. After you add each line, reserve inventory for it as described in steps 6 through 8.

- **Line 1:** Set the **Item number** field to *M9201*, the **Quantity** field to *20*, and the **Unit** field to *ea*.
- **Line 2:** Set the **Item number** field to *M9202*, the **Quantity** field to *60*, and the **Unit** field to *ea*.

Outbound load planning workbench

The outbound load planning workbench uses the load template ID to build the shipments and release the sales order lines to the warehouse.

1. Go to **Warehouse management** > **Loads** > **Outbound load planning workbench**.
2. In the **Warehouse** field, select *51*.

   You'll now build the load for the sales orders that you created.
3. On the **Sales lines** tab, in the grid, select the sales lines for the new sales orders.
4. On the Action Pane, on the **Supply and demand** tab, in the **Add** group, select **To new load**.
5. In the **Load template assignment** dialog box, in the **Load template ID** field, select *20' Container*.
6. Select **OK**.
7. In the **Loads** section of the **Outbound load planning workbench** page, in the grid, find the newly created load ID for warehouse *51*. Scroll right until you see the **Allow load split during ship confirm** column. The check box should be selected for your load.

   > ⓘ **Note**
   >
   > The default setting for **Allow load split during ship confirm** comes from the load template used to create the load. The **Load split shipment confirmation policy** setting is also inherited from the template but isn't shown on the **Outbound load planning workbench** page. To change the policy for a load, open its **Load details** page (for example, by selecting its **Load ID** in the grid here).
8. Select the load.
9. On the **Release** menu above the grid, select **Release to warehouse** to create work.
10. In the **Release load to warehouse** dialog box, verify that the **Records to include** FastTab shows your load ID.

11. Select **OK**.

    You might receive a "Processing operation" message while the system creates the shipments and work.

12. In the **Loads** section of the **Outbound load planning workbench** page, your load should now have a load status of *Waved*. Select the load, and then, on the **Related information** menu above the grid, select **Wave details** to view the shipment IDs that were created. When you're done, close the **Waves** page.

13. In the **Loads** section of the **Outbound load planning workbench** page, select the load, and then, on the **Related information** menu above the grid, select **Work details** to view the work that was created for the sales orders.

14. Make a note of the work IDs that were created. You might have to scroll right to see the sales order number and shipment ID that are associated with the work ID.

Step 2: Set up the execution flow for mobile devices

Mobile device tasks require worker input, such as the work ID or license plate ID. In the fields, this information typically appears for warehouse workers as bar codes on documentation, packaging, or racking. To complete the mobile device steps for scenarios, make sure that you identify the work IDs for the transactions, and the license plate IDs for the item and location in the transactions.

1. Sign in to the mobile device by using a user ID and password for warehouse *51*.

2. Select **Outbound**.

3. Select **Sales Picking**.

4. Enter the work ID or the license plate ID. Enter the work ID for the first sales order, then select **Enter**.

5. In the **Loc** field, enter the location that is shown to confirm the picking location. Then select **Enter**.

6. In the **LP** field, enter the license plate ID. The license plate ID must match the item, warehouse, and location of the item that you're picking from the selected location. When you're done, select **Enter**.

7. In the **Item** field, enter the item number to confirm the item that you're picking, then select **Enter**.

8. In the **Qty** field, enter the quantity of the item that you're picking, then select **Enter**.

9. In the **Target LP** field, enter a target license plate ID. Target license plates are user-defined. Be sure to enter a license plate ID that you can remember. We recommend that you use the current date plus a two-digit sequence (YYMMDD##) as the format, such as *19112301*. When you're done, select **Enter**.

10. Review the information that is shown. This information is the *Pick* information that now becomes the *Put* data for the put transaction. When you're done, select **Enter**.

    - You receive a **Work Completed** message.

11. Repeat steps 4 through 10 for the work ID of the second sales order.

The next step is to load the two picked license plates to the truck.

1. Sign in to the mobile device by using a user ID and password for warehouse *51*.
2. Select **Outbound**.
3. Select **Sales Loading**.
4. Enter the target license plate ID that you created for the first work ID in the previous procedure. Then select **Enter** to put the target license plate into the *STAGE* location.
5. Enter the target license plate ID again, then select **Enter** to put the license plate into the *BAYDOOR* location.
6. Repeat steps 4 through 5 for the target license plate ID that you created for the second work ID.

The two work IDs are now closed (loaded).

Step 3: Confirm the outbound shipment

In this step, you confirm the two sales orders and the work that you completed for the load to ship the picked items. You also create a new load for the unpicked items. You can confirm the outbound shipment from the **Outbound load planning workbench**, **All loads**, or **Load details** page.

> ⓘ **Note**
>
> If you're running Supply Chain Management version 10.0.41 or older, you can only split loads from the **Load details** page.

1. Go to **Warehouse management** > **Loads** > **Outbound load planning workbench**.

2. In the **Loads** section, in the grid, select the row for the load ID that you created.

3. Select the load ID link to open the **Load details** page.

4. On the Action Pane, on the **Ship and receive** tab, in the **Confirm** group, select **Outbound shipment** to initiate the confirmation.

5. Because the **Load split shipment confirmation policy** for the load template used to create this load is set to *Manual selection*, the **Ship confirm** dialog is shown. In the **Load split method during ship confirm** field, select *Split quantity to new load*.

6. Select **OK**.

   You might receive a *Processing operation* message. You might also receive informational messages that indicate that the shipment for your load is confirmed, and that a new load is created from the split quantity.

7. Refresh the **Outbound load planning workbench** page to see the newly created load.

You can confirm that transaction relations are updated in the following ways:

- The remaining (unprocessed) shipment and shipment lines are updated with the new load ID.
- The sales order is linked to the new load ID.
- The original load doesn't include the remaining load lines.
- The remaining work is updated with the new load ID.
- The wave remains the same.
- The status of the new load is correctly updated. (If the work status is *In process*, the load status is also *In process*.)

Confirm and transfer and Partial shipment of transport load features

The *Confirm and transfer* and *Partial shipment of a transport load* features both focus on flexibility in different aspects of the warehousing process. The *Partial shipment of transport load* feature enables flexibility in planning and loading, while the *Confirm and transfer* feature enables flexibility in ship-confirming a load. You can use these features alone or together.

Both features allow for more adaptable handling of transport loads, accommodating partial shipments and incomplete work scenarios. For example, you can plan loads without considering specific containers or trucks. After releasing these loads to the warehouse, workers can use mobile devices to assign loads in real-time as it makes sense, loading inventory from the same or different loads onto the same truck. This flexibility allows for partial shipments based on real-time loading decisions. Once a truck is full, workers can use the *Confirm and transfer* feature to close and complete an incomplete load and optionally split or cancel remaining quantities. This way, warehouse workers can make real-time decisions on load

assignments and shipment confirmations, optimizing the use of transport capacity, and warehouse operations can proceed without delays.

Notes and tips

- In Supply Chain Management version 10.0.42 and later, you can use a batch job to split loads where the **Allow load split during ship confirm** option is set to *Yes* and **Load split shipment confirmation policy** is set to *Split quantity to a new load* or *Cancel unfulfilled quantity*.
- You can change the load shipment confirmation policy for an individual load after you create it but before the loading process starts. To do so, open its **Load details** page, set **Allow load split during ship confirm** to *Yes* and then select the desired value in the **Load split shipment confirmation policy** field.
- The **Split quantity to new load** option also works when some of the remaining work headers have a status of *In process*. Therefore, you can still use the functionality even if workers are already running the pick orders.
- If you select **Cancel unfulfilled quantity** while there's remaining work that has a status of *Open* or *In progress*, you receive the following error message: "Unable to cancel remaining qty for load. Work exists for load."
- If you select **Cancel unfulfilled quantity** when there's no remaining work but there are unreleased load lines on the load, you receive the following error message: "The shipment for load couldn't be confirmed because the quantity for item exceeds the percentage that is defined for under delivery." To avoid this error, you can set the **Under delivery** percentage on the unreleased load line to *100*. Unreleased lines aren't moved to a new load, but the current load is confirmed with under delivery. In this case, you can't rerelease the original order, so you have to handle it in some other way.

Confirm outbound shipments from batch jobs

10/03/2025

This article describes how to set up a batch job that automatically confirms outbound transfer-order shipments for ready-to-ship loads. The batch job described here only applies to transfer order shipments, not to sales orders.

Process outbound shipments

To set up a scheduled batch job to run the outbound shipment confirmation for loads that are ready to ship:

1. Go to **Warehouse management** > **Periodic tasks** > **Process outbound shipments**.
2. The **Confirm shipment** dialog box opens. On the **Records to include** FastTab, select **Filter**.
3. The query editor dialog box opens. On the **Range** tab, add a row with the following values:

    - **Table** - *Loads*
    - **Derived table** - *Loads*
    - **Field** - *Load status*
    - **Criteria** - *Loaded*

4. Select **OK** to return to the **Confirm shipment** dialog box.
5. On the **Run in the background** FastTab, set **Batch processing** to **Yes**.
6. On the **Run in the background** FastTab, select **Recurrence**.
7. The **Define recurrence** dialog box opens. Set the run schedule as needed for your business.
8. Select **OK** to return to the **Confirm shipment** dialog box.
9. Select **OK** on the **Confirm shipment** dialog box to add the batch job to the batch queue.

Learn more in Batch processing overview.

Advanced load building during wave

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Advanced wave load building automatically assigns shipments to existing waves during wave execution. Therefore, you can create meaningful loads that represent trucks without having to use the outbound load planning workbench. This capability is useful for businesses that want to use the concept of loads to track and plan the shipment of goods in a truck/trailer, but that don't want to manually create those loads every day.

During wave processing, the system usually creates a new load for each shipment that no load is assigned to. However, when advanced wave load building is turned on, the system assigns each unassigned shipment to an existing load (if an appropriate load exists), and new loads are created only when they're required. Advanced wave load building automatically creates the new loads, based on criteria that you define.

To use the feature, you must set up the system in the following way:

- Create *wave templates* that include the *buildLoads* method. This method makes advanced wave load building available for waves that use those templates.
- Set up *load build templates*, each of which is linked to a specific wave template and method. Load build templates control which load (existing or new) the load lines that are being waved are added to. You can combine or separate shipments, based on criteria such as the load template, equipment, and other field values on the load line.
- Define *load mix groups* to control which items should and shouldn't be combined on a single load. You also specify whether the restriction should produce a warning or an error, and whether the volumetric restriction of the load template should be evaluated.

Prepare data to work through the scenario

Make sample data available

To work through this demo by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin.

You can also use this demo as guidance for using this feature when you work on a production system. However, in that case, you must substitute your own values, and you might be missing some types of required records that the standard demo data provides.

Make sure that the scenario setup includes enough available inventory

If you're working with the *USMF* demo data, you must first make sure that your system is set up so that there's enough inventory at each relevant location. For this demo, the expectation is that the following inventory is available in warehouse *62*:

- **Item A0001:** 10 pcs
- **Item A0002:** 10 pcs
- **Item M9200:** 10 ea

Item **M9200** must be added to the warehouse. Complete the procedures in the following subsections to add the item inventory.

Create a new license plate ID

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **License plates**.
2. On the Action Pane, select **New**.
3. In the new row, in the **License plate** field, enter *LP6203*.
4. Select **Save**.

Create a standard cost for item M9200 in site 6

1. Go to **Product information management** > **Products** > **Released products**.
2. Search on **M9200**.
3. Select the row for the item, and then, on the Action Pane, on the **Manage costs** tab, in the **Set up** group, select **Item price**.
4. On the **Item price** page, select the **Pending prices** tab.
5. On the Action Pane, select **New**.
6. On the new line, set the following values:
    - **Costing type:** *Planned cost*
    - **Price type:** *Cost*
    - **Version:** *10*

- **Site:** *6*
- **Price:** *1.60*

7. On the Action Pane, select **Save**.
8. On the Action Pane, select **Activate pending price(s)**.
9. Select the **Active prices** tab to verify that the new cost price for site *6* has been added.

Create inventory in warehouse 62

1. Go to **Inventory management** > **Journal entries** > **Items** > **Inventory adjustment**.
2. On the Action Pane, select **New**.
3. In the **Create inventory journal** dialog box, on the **Overview** FastTab, in the **Warehouse** field, enter *62*. Accept the default values in all the other fields.
4. Select **OK** to close the dialog box.
5. The **Inventory adjustment** page is opened. On the **Journal lines** FastTab, select **New** to add a line.
6. On the new line, set the following values. Accept the default values in all the other fields.

    - **Item number:** *M9200*
    - **Location:** *bulk-003*
    - **Quantity:** Change the value to *10*.

7. On the Action Pane, select **Save**.
8. On the Action Pane, select **Validate** to check for errors.
9. In the **Check journal** dialog box, select **OK** to start the check. You receive a message when the check is completed.
10. On the Action Pane, select **Post** to commit the inventory adjustment.
11. In the **Post journal** dialog box, select **OK** to start the posting. You receive a message when the posting is completed.

Set up advanced wave load building

Regenerate wave process methods

You might have to regenerate your wave process methods to make the load building method (*buildLoads*) available.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
2. Verify that *buildLoads* is present in the list. If it isn't present, select **Regenerate methods** on the Action Pane to add it.

Set up wave templates

To take advantage of advanced wave load building, you must include the *buildLoads* method in each relevant wave template.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Select a wave template.

   If you're working with the *USMF* demo data, select the *62 Shipping Default* template.
3. On the Action Pane, select **Edit** to put the page into edit mode.
4. On the **Methods** FastTab, in the **Remaining methods** grid, select the *buildLoads* method.
5. Select the right arrow button to move the *buildLoads* method to the **Selected methods** grid.
6. To assign a **Wave step code** value for the *buildLoads* method, you must first create a code on the **Wave step codes** page. You can use any value that you want, but be sure to make a note of it, because you'll need it later. Follow these steps to create code *WSC2112*:
   a. In the row for the *buildLoads* method, right-click the down arrow in the **Wave step code** field, and then select **View details**.
   b. On the **Wave step codes** page, on the Action Pane, select **New**.
   c. In the **Wave step code** field, enter *WSC2112*.
   d. In the **Wave step description** field, enter *WSC2112*.
   e. In the **Wave step type** field, select *Load Building*.
7. Select **Save**, and close the page.
8. In the row for the *buildLoads* method, in the **Wave step code** field, select the code that you just created (*WSC2112*).
9. On the Action Pane, select **Save**.

> ⓘ **Note**

Wave step codes are codes that users can set up and use to link specific instances of wave methods to corresponding templates. The templates include templates for replenishment, containerization, label printing, load building, and sorting.

When wave step codes aren't used, users must enter free text to reference a specific template from the wave method instance. Free text is prone to errors, because users must make sure that the wave step text that they add for a specific wave step method in a wave template exactly matches the wave step text in the target template.

Wave step codes for a specific wave step type are set up on a separate page. For every instance of a wave step method in a wave template that requires a wave step code, the wave step code must be selected in a drop-down list. Selection in a drop-down list replaces free text entry and helps reduce the risk and impact of human error. Setup codes are used to link a wave step method in a wave template to a target template for the method.

Set up load mix groups

Load mix groups establish rules for the types of items that can be combined on a single load. You can set up as many load mix groups as you require. However, to use advanced wave load building, you must have at least one. Follow these steps to create a load mix group.

1. Go to **Warehouse management** > **Setup** > **Load** > **Load mix groups**.
2. On the Action Pane, select **New** to create a load group.
3. In the **Load mix group ID** field, enter a name for the new group.

   If you're working with the *USMF* demo data, set the following values:

   - **Load mix group ID:** *TV*
   - **Description:** *TV*
4. On the Action Pane, select **Save** to make the **Load mix group criteria** FastTab available.
5. On the **Load mix group criteria** FastTab, select **New** to add a row to the grid.
6. In the new row, set the desired values in each field. These values determine the item groups that are considered for the load mix.

   If you're working with the *USMF* demo data, select *TV&Video* in the **Item group** field.
7. On the Action Pane, select **Save** to make the **Load mix group constraints** FastTab available.

8. On the **Load mix group constraints** FastTab, select **New** to add a row to the grid.
9. In the new row, set the desired values in each field.

   If you're working with the *USMF* demo data, set the following values:

   - **Item group:** *CarAudio*
   - **Load build action:** *Restrict* (This value prevents items that belong to the *CarAudio* item group from being on the same load as items that belong to the *TV&Video* item group.)
10. Continue to work with the rules until you've added all the criteria and constraints that you require for the load mix group.

If you're working with the *USMF* demo data, you've now finished this setup.

Set up load build templates

You can set up as many load build templates as you require. However, to use advanced wave load building, you must have at least one. Follow these steps to create a load build template.

1. Go to **Warehouse Management** > **Setup** > **Load** > **Wave load building templates**.
2. On the Action Pane, select **New** to add a row to the grid.
3. In the new row, set the following values.

Expand table

Field	Description	Value in the USMF demo data
Sequence number	The order that the template will be evaluated in.	*1*
Load build template name	Enter the unique identifier of the load build template. You should enter the name of the template that you created or updated earlier in this setup.	*62 Shipping Default*
Wave step code	Enter the wave step code to use to link the template to a wave method. You should enter the code that you selected for the *buildLoads* method when you set up the wave template earlier in this setup.	*WSC2112*
Load template ID	Select the load template to use when new loads are created and to match against when assigning to existing loads. The load	*Stnd Load Template*

Field	Description	Value in the USMF demo data
	template defines the maximum weight and volume that are permitted for the whole load.	
Equipment	The equipment to match against when assigning to existing loads and to enter on new loads that are created.	Leave this field blank.
Load mix group ID	Select the load mix group to use if the item is allowed on the load. The mix group establishes rules for the types of items that can be combined on a single load. You should select one of the mix groups that you created earlier in this setup.	*TV*
Use open loads	Select whether existing open loads should be added. The following options are available: • **None** – Don't add open loads to any existing loads. • **Any** – Add open loads to any existing loads that are valid for the line. • **Assigned** – Add open loads to the load that is assigned to the wave.	*Any*
Create loads	Specify whether new loads should be created if no existing loads match the criteria.	Selected (= *Yes*)
Allow load line split	Specify whether a single load line can be split across multiple loads if the full line exceeds the maximum capacity (**Quantity**) of the load template.	Cleared (= *No*)
Validate volumetrics	Specify whether load building should check the weight and volume as each load line is added, to ensure that the volumetric limits of the load template are respected.	Cleared (= *No*)

> **Note**
>
> The **Allow load line split** field only splits based on a quantity limit, not volume or weight limits. It's different from the **Allow load split during ship confirm** field used by load templates. Learn more at Confirm and transfer.

4. On the Action Pane, select **Save** to make the **Edit query** option available.

5. On the Action Pane, select **Edit query** to open a dialog box for editing the query.

6. In the dialog box, on the **Sorting** tab, select **Add** to add a row to the grid.

7. In the new row, define the sorting rules that you want to use. For example, set the following values to sort the search results in ascending order by order number:

    - **Table:** *Load details*
    - **Derived table:** *Load details*
    - **Field:** *Order number*
    - **Search direction:** *Ascending*

8. Select **OK** to save your changes and close the dialog box.

9. On the **Break by** FastTab, set rules to control how your loads are split up. Typically, you might break on custom fields that have been extended onto the load line, such as **Route**, **Tour**, or **Run**. For example, to create one load per order number, select the **Break by** check box for the row that has the following values:

    - **Reference table name:** *Load details*
    - **Reference field name:** *Order number*

Scenario

This scenario shows how the settings that were described earlier in this article affect warehouse operations while a sales order is processed. This scenario uses the *USMF* demo data together with other demo values that are provided in those setup instructions.

Create sales orders

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. On the Action Pane, select **New** to open the **Create sales order** dialog box.

3. In the dialog box, set the following values:

    - On the **Customer** FastTab, set the **Customer account** field to *US-007*.
    - On the **General** FastTab, set the **Warehouse** field to *62*.

4. Select **OK** to create the sales order and close the dialog box.

5. Your new sales order is opened. It should include a new, empty line in the grid on the **Sales order lines** FastTab. On this new line, set the **Item number** field to *A0001* and the **Quantity** field to *1*.

6. On the **Inventory** menu above the grid, select **Reservation**.

7. On the **Reservation** page, on the Action Pane, select **Reserve lot**.

8. Select the **Close** button (**X**) in the upper-right corner of the page to return to the sales order.

9. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**. The system creates a shipment and adds it to a new load, because no existing load contains load lines that have this order number.

   You receive informational messages that indicate the work, wave, and shipment that are created for this order.

10. To confirm the load, shipment, and work details on the sales line, select the line, and then, on the **Warehouse** menu above the grid, select **Load details**, **Shipment details**, or **Work details**.

11. In the sales order that you just created, on the **Sales order lines** FastTab, select **Add line** to add another line.

12. On the new line, set the **Item number** field to *A0002* and the **Quantity** field to *1*.

13. Repeat lines 6 through 9 to reserve the line and release it to the warehouse. The system creates a **new** shipment for the line that you added. However, because you're using advanced wave load building, the system adds that shipment and load line to the existing wave. If you weren't using advanced wave load building, the system would create a new load for the shipment.

14. In the sales order that you just created, on the **Sales order lines** FastTab, select **Add line** to add another line.

15. On the new line, set the **Item number** field to *M9200* and the **Quantity** field to *1*.

16. Repeat lines 6 through 9 to reserve the line and release it to the warehouse. As before, the system creates a **new** shipment for the line that you added. However, because the item is from the **CarAudio** item group, it **fails to pass the constraints that you set up for the load mix group**. Therefore, it's **added to a new load**. If you hadn't specified a load mix group on the load build template, this shipment would have been added to the first load.

Last updated on 10/03/2025

Release to warehouse rule

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The *Release to warehouse rule* feature provides flexibility during release to the warehouse. It adds a configuration option for each warehouse. Use this option to specify whether partially reserved order lines can be released for that warehouse. This feature works together with advanced cross-docking functionality in situations where part of an order line quantity is marked against a supply source, but the remaining part can be processed in the warehouse. Therefore, the line can be released so that warehouse processing of the available inventory quantity can continue.

Set the release to warehouse rule for each warehouse

To set the release to warehouse rule for each warehouse, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.
2. Select the warehouse to configure.
3. Select **Edit** to put the page into edit mode.
4. On the **Warehouse** FastTab, in the **Reservations** section, the **Requirement for inventory reservation** field controls whether partial release of orders is allowed. Select one of the following values:

    - *Not applicable* – When you turn on and initialize the feature, this value is automatically assigned to all warehouses that aren't enabled for warehouse management. You can't change it. This value isn't available for warehouses that are enabled for warehouse management.
    - *Allow partial reservation* – Order lines can be released when any quantity is reserved. The system evaluates whether the unreserved quantity should be marked for advanced cross-docking and marks that quantity as required. If no marking exists, the system creates work only for the reserved quantity. When you turn on and initialize the

feature, the system automatically assigns this value to all warehouses that are enabled for warehouse management. This value isn't available for warehouses that aren't enabled for warehouse management.

- *Require full reservation* – Order lines can be released only if the whole quantity is reserved. They can't be released if the total quantity isn't either physically reserved or planned for cross-docking. This value isn't available for warehouses that aren't enabled for warehouse management.

5. Select **Save**.

Scenarios

This section provides two scenarios that you can work through to learn what the feature does and how to use it.

Make sample data available

To work through these scenarios by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

You can also use these scenarios as guidance for the feature when you work on a production system. However, in that case, you must substitute your own values, and you might be missing some types of required records that the standard demo data provides.

Scenario 1: Require full release (no planned cross-docking)

This scenario shows how the feature works for warehouses that are set to **Require full reservation**.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.
2. For warehouse *62*, set the **Requirement for inventory reservation** field to **Require full reservation**, as described in the Set the release to warehouse rule for each warehouse section earlier in this article.
3. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
4. Select **New** to create a sales order.
5. In **Create sales order**, set the following values:

- On the **Customer** FastTab, set the **Customer account** field to *US-004*.
- On the **General** FastTab, set the **Warehouse** field to *62*.

6. Select **OK** to create the new sales order and close the dialog box.
7. Your new sales order is opened. It includes an empty line in the grid in the **Sales order lines** section. On this line, set the following values:
   - **Item number:** *A0001*
   - **Quantity:** *2*
8. Select **Add line** to add a new line, and set the following values:
   - **Item number:** *A0002*
   - **Quantity:** *2*
9. Select the first order line, and then, on the **Inventory** menu above the grid, select **Reservation**.
10. On the **Reservation** page, select **Reserve lot** to reserve the selected line's full quantity in the warehouse.
11. Close the **Reservation** page to return to the sales order.
12. Don't reserve the second order line.
13. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.
14. Notice that you receive the following error message: "The full quantity must be physically reserved." Therefore, the system doesn't create any work, shipment, or load for the order.

    > ⓘ **Note**
    >
    > You receive the same error message if you partially reserve the second line.

Scenario 2: Allow partial release

This scenario shows how the feature works for warehouses that are set to **Allow partial release**.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.
2. For warehouse *62*, set the **Requirement for inventory reservation** field to **Allow partial reservation**, as described in the Set the release to warehouse rule for each warehouse

section earlier in this article.

3. As you did in the previous scenario, go to **Sales and marketing** > **Sales orders** > **All sales orders**, and create a sales order for customer account *US-004* from warehouse *62*. Add the following two order lines:

    - **Line 1:** Set the **Item number** field to *A0001*, the **Quantity** field to *2*, and the **Unit** field to *Pcs*.
    - **Line 2:** Set the **Item number** field to *A0002*, the **Quantity** field to *2*, and the **Unit** field to *Pcs*.

4. As you did in the previous scenario, reserve the full quantity for order line 1, but don't reserve order line 2.

5. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

6. Notice that you don't receive an error message this time. Instead, the system creates work, shipments, and loads as required, and shows the results.

7. To view the shipment, load, and work information, use the options on the **Warehouse** menu above the grid:

    - **Line 1:** Three options are available: **Shipment details**, **Load details**, and **Work details**. Select each option to view the details of the shipment, load, and work that were created when the order was released to the warehouse.
    - **Line 2:** Only the **Work details** option is available. Select it, and notice that no work has been created, because no inventory was reserved. Therefore, no shipment or load was created either.

> ⓘ **Note**
>
> The same result is expected when the second line is partially reserved. In this case, work is created for the reserved line quantity but not for the unreserved quantity.

Last updated on 05/20/2026

Small parcel shipping

Summarize this article for me

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

The small parcel shipping (SPS) feature enables Microsoft Dynamics 365 Supply Chain Management to interact directly with shipping carriers by providing a framework for communication through carrier APIs. This functionality is useful when you're shipping individual sales orders via commercial shipping carriers instead of using container shipping or less-than-truckload (LTL) shipping.

The SPS feature interacts with your shipping carrier through a dedicated *rate engine*. Your organization must develop this rate engine in collaboration with your carrier or carrier hub service. The rate engine enables Supply Chain Management to submit details about a packed container to your carrier, and then receive a shipping label, shipping rate, and tracking number back from that carrier.

The shipping rate that is returned is added to the associated sales order as a miscellaneous charge. The shipping label that is returned can then automatically be printed by using a Zebra Programming Language (ZPL) printer and applied to the shipment. The carrier will scan this shipping label when it picks up the packages at your warehouse.

Prepare your system to support SPS

Before you can start to use SPS functionality, you must add your rate engine and set up the **Transportation management** and **Warehouse management** modules to support it.

Deploy and set up rate engines

Supply Chain Management doesn't include any rate engines. You must obtain or create any rate engines that you require, and then add them to your system. However, Microsoft provides a demo rate engine that you can use for testing.

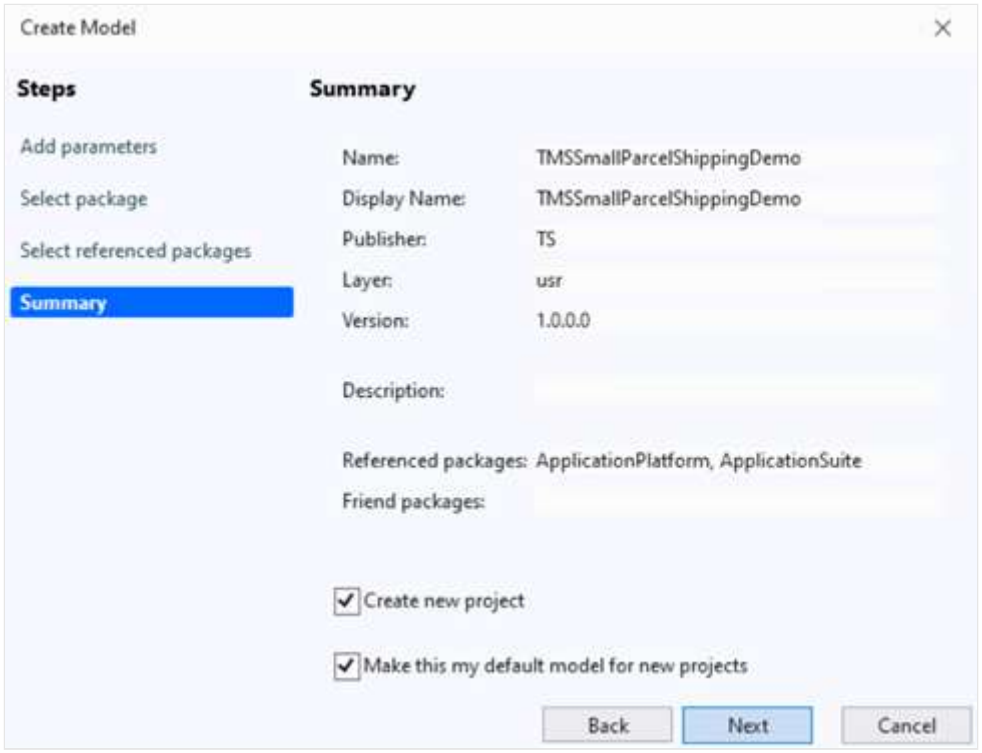
Download and deploy the demo rate engine

Follow these steps to download and deploy the demo rate engine.

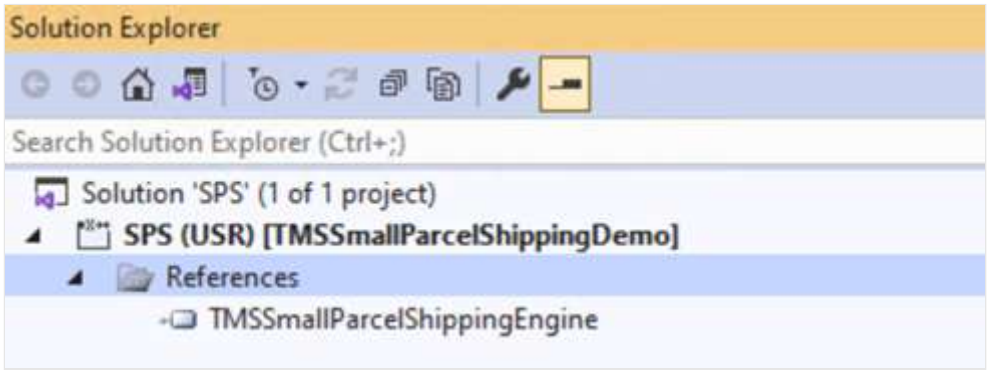
1. On GitHub, download the dynamic-link library (DLL) for the demo rate engine.

2. In Visual Studio, go to **Extensions** > **Dynamics 365** > **Model Management** > **Create Model**.

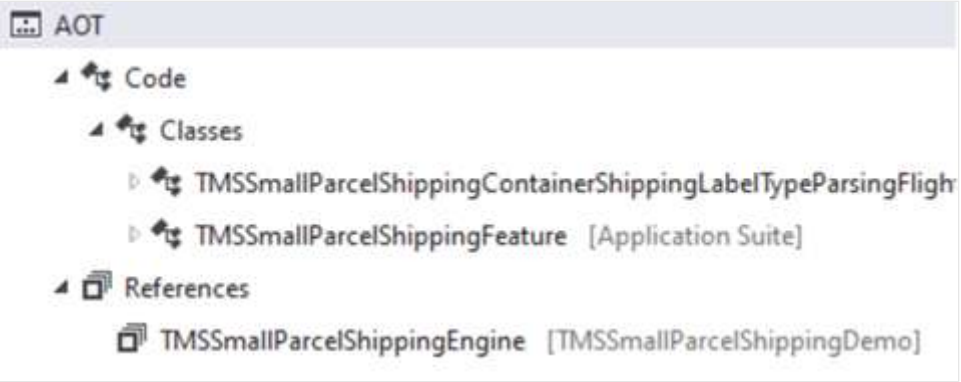
3. Create a new model called *TMSSmallParcelShippingDemo* and make it reference the *ApplicationSuite* and *ApplicationPlatform* models in the select referenced packages step.



4. Create a project in the new model and add the downloaded `TMSSmallParcelShippingEngine.dll` file to the references for the new project.



5. On the **Solution Explorer** pane, right-click and select **Build Solution** to compile the project and make sure the reference is added to the AOT node.



6. Go to **Extensions** > **Dynamics 365** > **Deploy** > **Create Deployment Package** and create a deployment package for the new model.

7. In Microsoft Dynamics Lifecycle Services, go to **Asset Library** > **Software deployable package**. Select **Add** to add the deployment package you created to Lifecycle Services.

8. After the validation completes, go to **Environments** > **{Your Environment ID}** > **Maintain** > **Apply updates**, select the deployable package, and apply.

> **Note**
>
> If the system warns you that you don't have permission to deploy packages on Lifecycle Services, contact your Lifecycle Services administrator for support.

Create and deploy functional rate engines

For information about how to create and deploy functional rate engines so that they can be used in a production or test environment, see the following articles:

- Create a new transportation management engine
- Set up transportation management engines

Set up a rate engine in Supply Chain Management

After you've created and deployed a rate engine for SPS, follow these steps to set it up.

1. Go to **Transportation management** > **Setup** > **Engines** > **Rate engine**.

2. On the Action Pane, select **New** to add a row to the grid.

3. In the new row, set the following fields:

    - **Rating engine** – Enter a unique name for the rate engine. If you're using the demo rate engine, enter *Demo rate engine*.

- **Name** – Enter a short description of the rate engine. If you're using the demo rate engine, enter *Demo rate engine*.
- **Rating metadata ID** – Select the basis that should be used to calculate your rate. For example, your rate might be calculated based on distance. If you're using the demo rate engine, you can leave this field blank.
- **Engine assembly** – Enter the file name of the DLL package that you deployed. If you're using the demo rate engine, enter *TMSSmallParcelShippingEngine.dll*.
- **Engine class** – Enter the class name that was established for your rate engine. If you're using the demo rate engine, enter *TMSSmallParcelShippingEngine.SmallParcelShippingRateEngine*.

Example scenario

This example scenario shows how to set up and use SPS after you've prepared your system as described earlier in this article. This scenario uses the previously mentioned demo rate engine.

Make demo data available

To work through this scenario by using the demo records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

Set up the scenario

For this example scenario, you must have a demo carrier, carrier group, packing policy, and packing profile. The following subsections explain how to prepare the records that are required for the scenario. In a production scenario, the setup process typically resembles the process that is described here. However, you'll set different values.

Set up carriers

Follow these steps to set up a carrier.

1. Go to **Transportation management** > **Setup** > **Carriers** > **Shipping carriers**.
2. On the Action Pane, select **New** to add a carrier.
3. On the header, set the following values:
    - **Shipping carrier:** *Demo Carrier*
    - **Name:** *Demo Carrier*

- **Mode:** *Ground*

4. On the **Overview** FastTab, set the following values:

    - **Activate shipping carrier:** *Yes*
    - **Activate carrier rating:** *Yes*

5. On the **Services** FastTab, select **New** to add a service to the grid.

6. Set the following values for the new service:

    - **Carrier service:** *Demo carrier service*
    - **Name:** *Demo carrier service*
    - **Transportation method:** *Ground*

    Enter arbitrary values for all the other fields, as required. (When you save the new shipping carrier record, a new mode of delivery will be created, and the **Mode of delivery** field will be set automatically.)

7. On the **Rating profiles** FastTab, select **New** to add a rating profile to the grid.

8. Set the following values for the new profile:

    - **Rating profile:** *Demo carrier service*
    - **Name:** *Demo carrier service*
    - **Rate engine:** *Demo rate engine*

    Enter arbitrary values for all the other fields, as required.

9. On the Action Pane, select **Save**.

For more information about how to set up carriers, see Set up shipping carriers.

Set up carrier service accounts

Follow these steps to set up a carrier service account.

1. Go to **Transportation management** > **Setup** > **Rating** > **Carrier service account**.

2. On the Action Pane, select **New** to add a carrier service account.

3. Set the following values for the new account:

    - **Shipping Carrier:** *Demo Carrier*
    - **Carrier service:** *Demo carrier service*

- **Carrier customer account number:** The carrier customer account number that is used to verify and authenticate the connection to the shipping carrier. Your carrier will provide this value. If you're using the demo service, you can enter an arbitrary value.
- **Site:** *6*
- **Warehouse:** *62*

> ⓘ Note
>
> Often, the **Carrier customer account number** value is the only value that is required to authenticate the connection. However, if your carrier requires additional authentication parameters, your organization should customize this page to add extra fields as appropriate.

Set up a container packing policy

Follow these steps to set up a container packing policy.

1. If you haven't already set up a ZPL printer definition, use the Document Routing Agent application to set it up. Learn more in Document printing overview and related articles.
2. Go to **Warehouse Management** > **Setup** > **Containers** > **Container packing policies**.
3. On the Action Pane, select **New** to add a container packing policy.
4. On the header of the new policy, set the following values:
    - **Container packing policy:** *Demo packing policy*
    - **Description:** A description of the policy
5. On the **Overview** FastTab, set the following values:
    - **Warehouse:** *62*
    - **Default location for final shipment:** *Baydoor*
    - **Weight unit:** *KG*
    - **Container closing policy:** *Automatic release*
    - **Container release policy:** *Make available at final shipping location*
6. On the **Container manifest** FastTab, set the following values:
    - **Automatic manifest at container close:** *Yes*
    - **Manifest requirements for container:** *Transportation management*
    - **Print container contents:** *No*

7. On the **Carrier label printing** FastTab, set the following values:

    - **Print container shipping label:** *Always*
    - **Printer name:** The name of the ZPL printer that should print shipping labels

Set up a packing profile

Follow these steps to set up a packing profile.

1. Go to **Warehouse Management** > **Setup** > **Packing** > **Packing profiles**.

2. On the Action Pane, select **New** to add a packing profile to the grid.

3. Set the following values for the new profile:

    - **Packing profile ID:** *Demo packing profile*
    - **Description:** A description of the profile
    - **Container packing policy:** *Demo packing policy*
    - **Container ID mode:** *Auto*
    - **Container type:** *SmallBox*

Set up a customer to use the SPS carrier

Follow these steps to set up a customer so that it can use the carrier that you created.

1. Go to **Accounts receivable** > **customers** > **All customers**.

2. In the grid, find and select customer *US-027*.

3. On the Action Pane, on the **General** tab, in the **Set up** group, select **Carrier customer accounts**.

4. On the **Carrier customer accounts** page, on the Action Pane, select **New** to add an account to the grid.

5. Set the following values for the new account:

    - **Shipping carrier:** *Demo Carrier*
    - **Carrier customer account number:** *12345* (The value isn't important for this scenario, but it will be referred to in the next section.)

Go through the example scenario

After you've set up all the sample data as described in the previous section, you're ready to go through the example scenario.

Create a sales order and process the work

Follow these steps to create a sales order.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create a sales order.
3. In the **Create sales order** dialog box, set the **Customer account** field to *US-027*.
4. Select **OK** to create the sales order and close the dialog box.
5. The new sales order is opened. On the **Sales order header** FastTab, set the **Carrier customer account number** field to the value that you selected for this customer earlier (*12345*).
6. In the **Sales order lines** section, add a sales line, and set the following values for it:
    - **Item number:** *A0002*
    - **Quantity:** *1*
    - **Site:** *6*
    - **Warehouse:** *62*
7. Switch to the **Header** view.
8. On the **Delivery** FastTab, set the following values:
    - **Shipping carrier:** *Demo Carrier*
    - **Carrier service:** *Demo carrier service*
9. Switch back to the **Lines** view. If you're prompted to update the mode of delivery for the sales lines, select **Yes**.
10. In the **Sales order lines** section, select the order line that you set up earlier, and then select **Inventory** > **Reservation**.
11. On the **Reservation** page, select **Reserve lot** to reserve the selected line's full quantity in the warehouse.
12. Close the **Reservation** page to return to the sales order.
13. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

Work is created to move items from the picking location to the packing station.

14. In the **Sales order lines** section, select **Warehouse** > **Shipment details**.
15. On the **Shipment details** page, make a note of the shipment ID. You'll need it when you pack the pack the shipment at the packing station.
16. Close the **Shipment details** page to return to the sales order.
17. Make a note of the sales order number, and then go to **Warehouse management** > **Work** > **All work**.
18. Use the sales order number to find and select the work that was created for the order.
19. On the Action Pane, on the **Work** tab, select **Complete work**.
20. On the **Work completion** page, in the **User ID** field, select a user ID. Then, on the Action Pane, select **Validate work**.
21. If the work passes validation, select **Complete work** on the Action Pane.

    The work is marked as completed to simulate the movement of items to the packing station.

Pack the shipment

Follow these steps to pack the shipment.

1. Go to **Warehouse management** > **Setup** > **Worker**, and make sure that your user account is associated with a worker account for warehouse management.
2. Go to **Warehouse management** > **Picking and containerization** > **Pack**.
3. In the **Select packing station** dialog box, set the following values:

    - **Site:** *6*
    - **Warehouse:** *62*
    - **Location:** *Pack*
    - **Packing profile ID:** *Demo packing profile*

4. Select **OK**.
5. The **Pack** page appears. In a production scenario, a worker will scan a license plate or shipment ID. However, for this scenario, open the **All shipments** page, and look up the shipment number for the shipment that you just created. Then enter this value in the

License plate or shipment field on the **Pack** page. Alternatively, enter the shipment ID that you made a note of earlier.

6. On the Action Pane, select **New container**.
7. The dialog box that appears shows details about the new container. Leave the default values, and then select **OK**.
8. On the **Pack** page, on the **Item packing** FastTab, in the **Identifier** field, select *A0002* to pack that item. The item is added to the container.
9. On the Action Pane, select **Containers for shipment**.

   The **Containers for shipment** page that appears has a row for the container that you just created. However, the **Container manifest ID** field in that row is currently blank, because you haven't yet received the shipping label and tracking number from the carrier.
10. On the Action Pane, select **Close container**.
11. In the **Close container** dialog box, set the **Gross weight** field to *1 kg*, and then select **OK**.

    The shipping label should now be printed on the ZPL printer that you selected earlier. It should resemble the following example.

12. Notice that the **Container manifest ID** and **Total freight** values have been added as received from the carrier.

Last updated on 03/10/2026

Shipment consolidation policies overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Shipment consolidation policies allow for automated shipment consolidation during automated and manual release to the warehouse.

Shipment consolidation policies are used for the following functionality:

- The automated release-to-warehouse batch jobs
- The **Release to warehouse** command on a sales order, transfer order, or outbound shipment order
- The dedicated **Release to warehouse** pages
- The **Release to warehouse** command on the **Outbound load planning workbench** page
- The manual consolidation of shipments on the **Consolidate shipments** and **Shipment consolidation workbench** pages

Before shipment consolidation policies were introduced, the consolidation function existed as a setting at the warehouse level. All orders for all customers from a single warehouse were treated as though they had the same consolidation requirements. Shipment consolidation policies add support for scenarios where different organizations have different requirements for shipment consolidation.

Queries are used to identify the shipment consolidation policy that applies, and then an editable set of fields determines how the load lines are grouped at the shipment level. (This pattern resembles the pattern that wave templates follow.) In addition, a **Consolidate with existing shipments** option has been added to each policy. When this option is turned on, the *Release to warehouse* procedure finds shipments for consolidation by searching among existing shipments that were created based on the same consolidation policy. In this case, the system will select an existing shipment or load instead of creating a new one. However, the system will only consolidate with existing shipments that have a status of *Open*; shipments that belong to a wave release with a status of *Released* or higher won't be considered as targets for consolidation.

You can use the various **Release to warehouse** pages to manually override the applicable consolidation policy in the same way that you can override fulfillment policies.

You can use the **Release** > **Release to warehouse** command on the **Outbound load planning workbench** page to build outbound loads that are based on sales, transfer, and outbound shipment order lines before you do the release to the warehouse. These loads use the same consolidation logic that was introduced together with the consolidation of shipment policies.

You can use the **Shipment consolidation workbench** page to consolidate existing shipments that haven't yet been confirmed but have already been released to the warehouse. This functionality supports scenarios where the automated release process, which has its own shipment consolidation, is run multiple times a day, but potential additional consolidations are manually identified before the shipment to carriers is completed during the confirmation process. This functionality lets you consolidate outbound shipments that are created from sales, transfer, or outbound shipment order lines at any time after the shipments are released to the warehouse but before they are confirmed.

The **Shipment consolidation workbench** page works like the load building workbench, where you can assess multiple shipments at the same time and assign a non-consolidated order to a specific shipment. You can apply shipment consolidation templates to assess proposed consolidations multiple times and confirm them. Some rules are implemented to prevent unauthorized consolidation and to warn you about possible errors.

Shipment consolidation policies functionality

This section describes the pages, commands, and features that are part of the *Shipment consolidation policies* feature.

Shipment consolidation policies page

Policies are differentiated by work order type. The **Sales orders** type represents *Sales order* shipments, the **Transfer orders** type represents *Transfer issue* shipments, and the **Outbound shipment order** type represents *Outbound shipment order* shipments.

Every shipment consolidation policy has a query that defines when it's applied and a sequence number that determines the execution order. Consolidation is applied for each unique combination of the selected fields. An additional parameter, which is provided, is used for consolidation with existing (open) shipments. The policies are evaluated and applied every time that a new shipment is created (before wave processing).

If a policy is missing any mandatory fields, or if it includes any prohibited fields, the policy is marked as not valid in the **Selected** section. The lists of mandatory and prohibited fields are hard-coded and can be extended.

The following list shows the mandatory fields. Because shipments are always split based on these fields, you can't group multiple shipments that have different values for these fields.

- For sales orders:
  - **Account number:** *WHSShipmentTable.AccountNum*
  - **Delivery recipient:** *WHSShipmentTable.DeliveryName*
  - **Postal address (RecId):** *WHSShipmentTable.DeliveryPostalAddress*
  - **Warehouse:** *WHSShipmentTable.InventLocationId*

- For transfer orders:
  - **From warehouse:** *InventTransferTable.InventLocationIdFrom*
  - **To warehouse:** *InventTransferTable.InventLocationIdTo*

- For outbound shipment orders:
  - **Account number:** *WHSShipmentTable.AccountNum*
  - **Delivery recipient:** *WHSShipmentTable.DeliveryName*
  - **Postal address (RecId):** *WHSShipmentTable.DeliveryPostalAddress*
  - **Warehouse:** *WHSShipmentTable.InventLocationId*
  - **Source system:** *WHSShipmentTable.SourceSystem*

The following fields are unavailable for all document types. These fields aren't visible in the user interface (UI), and they can't be used for consolidation.

- **Shipment ID:** *WHSShipmentTable.ShipmentId*
- **Status:** *WHSShipmentTable.ShipmentStatus*
- **Shipment consolidation policy:** *WHSShipmentTable.ShipConsolidationPolicyName*
- **Work transaction type:** *WHSShipmentTable.WorkTransType*
- **Wave ID:** *WHSShipmentTable.WaveId*
- **Load ID:** *WHSShipmentTable.LoadId*
- **Shipment ID:** *WHSLoadLine.ShipmentId*
- **Load ID:** *WHSLoadLine.LoadId*

By default, when you create a policy, the set of mandatory fields is used as the consolidation fields. However, you can modify the list by using left arrow and right arrow buttons. (The process resembles the process for selecting methods in wave templates.)

The values that users select for these fields are used for all newly created shipments, or they're added to existing shipments during consolidation with those shipments. When two shipments have the same value for a field that is selected for consolidation of those shipments, the shipments are consolidated. The same principle applies for all subsequent consolidation fields that are selected. If the values differ, the second shipment is discarded and is selected for a new shipment. The automated consolidation process consists of creating all the unique combinations of values for the shipment consolidation fields and then assigning a shipment to the relevant combination.

Unselected fields are ignored during the consolidation process. If two shipments have different values for an unselected field, the field is cleared (that is, it's set to blank). If both shipments have the same value for an unselected field, the field is filled in.

The list of consolidation fields (that is, fields that are cleared if they have different values) is hard-coded. The list contains all fields that are initialized from a sales order or transfer order line when a new shipment is created. In other words, if a field isn't initialized from a sales order or transfer order line, it's ignored when new data is added to an existing shipment.

Release to warehouse page

- A field in the lower grid shows the applied consolidation policy.
- A button lets you manually select and/or override the consolidation policy.

Release to warehouse command on the Outbound load planning workbench page

- The logic is adjusted so that it applied consolidation policies.
- Shipments are consolidated only within a single load.

Consolidate shipments page

- The search for similar shipments (that is, candidates for consolidation) is changed so that it uses fields selected in the shipment consolidation policy.
- Fields that have different values in different shipments are set to blank.

Shipment consolidation workbench page

- You can open this page from the **Release to warehouse** menu in the **Warehouse management** module.

- The algorithm analyzes existing shipments that aren't yet shipped. It then proposes consolidation, based on fields selected in the consolidation policies.

Comparison of functionality

The following table summarizes how shipment consolidation works when you don't use shipment consolidation policies and when you do use them.

Expand table

Without shipment consolidation policies	**With shipment consolidation policies**
Not applicable	Sales or transfer shipments selected for consolidation must have the same consolidation policy as the shipment being created, or they must be assigned to an open shipment when the **Consolidate with existing shipments** option is turned on.
The *Release to warehouse* procedure doesn't search among existing shipments to find a shipment for consolidation. Only shipments created by a current instance of the *Release to warehouse* procedure creates are used to find a shipment for consolidation.	If the **Consolidate with existing shipments** option is turned on for a consolidation policy that is currently in use, the *Release to warehouse* procedure searches among existing shipments that were created based on the same consolidation policy to find a shipment for consolidation. Therefore, if you have two policies, a shipment being created based on policy 2 is never consolidated with a shipment that was created based on policy 1.
Not applicable	If a list of consolidation policy fields is empty, or if a policy can't be found, a new shipment is created for each sales order or transfer order line.
The following consolidation field defines the unique combination of values that is used to consolidate shipments for a *transfer line*. (All other fields are ignored.) • Order number (OrderNum)	The following consolidation fields define the unique combination of values that is used to consolidate shipments for a *transfer line*. (All other fields are ignored.) • Order number (OrderNum) • Delivery recipient (DeliveryName) • Postal address (DeliveryPostalAddress) • ISO country code (CountryRegionISOCode) • Address (Address) • Site (InventSiteId) • Warehouse (InventLocationId) • Shipping carrier (CarrierCode) • Carrier service (CarrierServiceCode) • Mode of delivery (ModeCode) * • Carrier group (CarrierGroupCode) • Delivery terms (DlvTermId)

Without shipment consolidation policies	With shipment consolidation policies
	These fields are the only fields that are available and initialized when a new shipment is created. * Note: The *ModeCode* is the **Mode** assigned to the **Shipping carrier** selected for a transfer line (not the **Mode of delivery** selected for a transfer line). If you choose to include *Mode of delivery (ModeCode)* among your consolidation criteria, the system only consolidates transfer lines with the same **Mode** value provided **Shipping carrier**, **Carrier service**, and **Mode of delivery** are all defined for the line (regardless of their values). In addition, the system also consolidates all transfer lines where the **Mode** is blank.
The following consolidation fields define the unique combination of values that is used to consolidate shipments for a *sales line*. (All other fields are ignored.) • Order number (OrderNum) • Customer reference (CustomerRef) • Customer requisition (CustomerReq) • Delivery terms (DlvTermId)	The following consolidation fields define the unique combination of values that is used to consolidate shipments for a *sales line*. (All other fields are ignored.) • Order number (OrderNum) • Account number (AccountNum) • Delivery recipient (DeliveryName) • Postal address (DeliveryPostalAddress) • ISO country code (CountryRegionISOCode) • Address (Address) • Site (InventSiteId) • Warehouse (InventLocationId) • Shipping carrier (CarrierCode) • Carrier service (CarrierServiceCode) • Mode of delivery (ModeCode) * • Carrier group (CarrierGroupCode) • Broker ID (BrokerCode) • Direction (LoadDirection) • Delivery terms (DlvTermId) • Customer reference (CustomerRef) • Customer requisition (CustomerReq) These fields are the only fields that are available and initialized when a new shipment is created. * Note: The *ModeCode* is the **Mode** assigned to the **Shipping carrier** selected for a sales line (not the **Mode of delivery** selected for a sales line). If you choose to include *Mode of delivery (ModeCode)* among your consolidation criteria, the system only consolidates sales lines with the same **Mode** value provided **Shipping carrier**, **Carrier service**, and **Mode of delivery** are all defined for the line (regardless of their values). In addition, the system also consolidates all sales lines where the **Mode** is blank.

Without shipment consolidation policies	With shipment consolidation policies
Not applicable	The following consolidation fields are mandatory for a *sales line* and can't be removed: • Account number (AccountNum) • Delivery recipient (DeliveryName) • Postal address (DeliveryPostalAddress) • Warehouse (InventLocationId) By default, these fields are assigned when a new policy is created. They can't be removed.
The *Release of loads to warehouse* procedure on the **Outbound load planning workbench** page uses its own separate code to create shipments and waves.	Shipment consolidation policies are applied to determine which fields should be evaluated for consolidation. Shipments are consolidated only within a single load.
You manually select **Consolidate shipments** on the **All shipments** page and then select a target "base" shipment. The filter suggests any existing shipments that have matching values for several key fields.	You manually select **Consolidate shipments** on the **All shipments** page and then select a target "base" shipment. The system suggests other existing shipments by matching the values of several key fields that are configured for the relevant shipment consolidation policies.
You can use the **Consolidate shipments** command on the **All shipments** page for only a single shipment.	The **Shipment consolidation workbench** page helps you find a set of shipments that aren't yet in a shipped state. These shipments are analyzed based on several key fields that are configured in your shipment consolidation policies. Any shipments where the values of these fields match are suggested for consolidation. You can manually maintain the consolidation by removing shipments from suggested consolidations and/or by adding shipments to them. Several types of errors can occur, but you can override some of them.
Design note: The *Automatic release of sales orders to warehouse* procedure splits sales lines into groups. Each group has its own unique **ReleaseToWarehouseId** value and is processed separately by the *Release to warehouse* procedure. This procedure creates new work regardless of the work break setup.	**Design note:** The *Automatic release of sales orders to warehouse* procedure assigns the same **ReleaseToWarehouseId** value to all sales lines that are being processed. All sales lines are processed by the *Release to warehouse* procedure at the same time. To ensure the earlier behavior, you must configure work break per shipment ID.

Related information

- Configure shipment consolidation policies

Last updated on 05/29/2026

Configure shipment consolidation policies

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The shipment consolidation process that uses shipment consolidation policies automates shipment consolidation during automated and manual release to the warehouse. After you turn on this feature, you must configure your initial policies. If you don't configure any policies, each sales line generates a separate shipment that has a single load line.

The scenarios in this article show how to set up default and custom shipment consolidation policies.

Scenario: Configure custom shipment consolidation policies

This scenario provides an example that shows how to set up custom shipment consolidation policies and then test them by using demo data. Custom policies can support complex business requirements where shipment consolidation depends on several conditions. For each example policy later in this scenario, a short description of the business case is included. Set up these example policies in a sequence that ensures a pyramid-like evaluation of the queries. (In other words, the policies that have the most conditions are evaluated as having the highest priority.)

Make demo data available

This scenario references values and records included in the standard demo data that's provided for Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to *USMF* before you begin.

Prepare master data for this scenario

Before you can go through the exercises in this scenario, you must prepare the master data that is required to do the filtering, as described in the following subsections. These prerequisites also apply to the scenarios that are listed in the Example scenarios of how to use shipment consolidation policies section.

Create two new product filter codes

1. Go to **Warehouse management** > **Setup** > **Product filters** > **Product filters**, and add two product filters:

    - Product filter 1:
        - **Filter code:** *Flammable*
        - **Filter title:** *Code 4*
    - Product filter 2:
        - **Filter code:** *Explosive*
        - **Filter title:** *Code 4*

2. Go to **Product information management** > **Products** > **Released products**.
3. Open the product that has item number *M9200*. The product that you select must be enabled for advanced warehouse (WMS) processes, and this product is pre-enabled for WMS processes in the **USMF** demo data.
4. On the **Warehouse** FastTab, set the **Code 4** field to *Flammable*.
5. Close the page.
6. Open the product that has item number *M9201*. This product is also pre-enabled for WMS processes in the **USMF** demo data.
7. On the **Warehouse** FastTab, set the **Code 4** field to *Explosive*.
8. Close the page.

Create a new transportation mode of delivery

1. Go to **Transportation management** > **Setup** > **Carriers** > **Mode**.
2. Create a transportation mode for use in consolidation queries, and name it *Airways*.
3. Go to **Transportation management** > **Setup** > **Carriers** > **Shipping carriers**.

4. Create a carrier that has the following settings:

    - **Shipping carrier:** *Airways*
    - **Name:** *Airways*
    - **Mode:** *Airways*

5. On the **Services** FastTab for the new carrier, add a row that has the following settings:

    - **Carrier service:** *Air*
    - **Transportation method:** *Air*

6. On the Action Pane, select **Save**.

    > ⓘ **Note**
    >
    > When you save the new carrier, the **Mode of delivery** field for the new row in the **Services** grid is automatically set to *Airwa-Air*. When you use the *Airwa-Air* mode of delivery for a sales order, the *Airways* transportation mode is used for related shipments.

Create an order pool

1. Go to **Sales and marketing** > **Setup** > **Sales orders** > **Order pools**.

2. Create an order pool that you can use for the consolidation query. This order pool should have the following settings:

    - **Pool:** Enter an integer that isn't already used by any other pool.
    - **Name:** *ShipCons*

3. Go to **Sales and marketing** > **Customers** > **All customers**.

4. Open the customer that has account number *US-003*.

5. On the **Sales order defaults** FastTab, set the **Sales order pool** field to the order pool that you just created.

6. Close the page, and then repeat steps 4 and 5 for the customer that has account number *US-004*.

Create example policy 1

In this example, you create a *Customer+Mode* policy for the following business case:

- The policy queries for a specific customer account (*US-001*) and a specific mode of delivery (*Airwa-Air*).
- Consolidation with open shipments is turned off.
- Consolidation is done per order ID. (In other words, there are separate shipments per order, warehouse, and so on.)

Follow these steps to create the shipment consolidation policy for this business case.

1. Go to **Warehouse management** > **Setup** > **Release to warehouse** > **Shipment consolidation policies**.
2. Set the **Policy type** field to *Sales orders*.
3. On the Action Pane, select **New** to create a policy that has the following settings:
    - **Policy name:** *CustomerMode*
    - **Policy description:** *Customer account and mode of delivery*
4. Leave the **Consolidate with open shipments** option set to *No*.
5. On the Action Pane, select **Save**.
6. On the **Consolidation fields** FastTab, in the **Remaining fields** list, select the row where the **Field name** field is set to *Mode of delivery*.
7. Select the **Add** button to move the field to the **Selected fields** list.
8. On the Action Pane, select **Edit query**.
9. In the query editor dialog box, on the **Range** tab, in the grid, find the row where the **Field** field is set to *Customer account*, and set the **Criteria** field for that row to *US-001*.
10. Select **Add** to add a row that has the following settings to the grid:
    - **Table:** *Order lines*
    - **Derived table:** *Order lines*
    - **Field:** *Mode of delivery*
    - **Criteria:** *Airwa-Air*
11. Select **OK** to close the dialog box.

> **Note**
>
> For this business case, order lines for customer *US-001* that use the *Airwa-Air* mode of delivery aren't consolidated across orders. Use this policy first in a sequence, in cases where shipments for all other modes of delivery are consolidated for this customer.

Create example policy 2

In this example, you create a *Hazardous goods* policy for the following business case:

- The policy queries for a specific filter code (*hazardous*) and a specific mode of delivery (*Airwa-Air*).
- Consolidation with open shipments is turned on.
- Consolidation is done across orders. (In other words, there are separate shipments per account, warehouse, and so on, but only within the item group that is specified in the query.)

Follow these steps to create the shipment consolidation policy for this business case.

1. Go to **Warehouse management** > **Setup** > **Release to warehouse** > **Shipment consolidation policies**.
2. Set the **Policy type** field to *Sales orders*.
3. On the Action Pane, select **New** to create a policy that has the following settings:
    - **Policy name:** *Item type*
    - **Policy description:** *Consolidate the same type of item across orders*
4. Set the **Consolidate with open shipments** option to *Yes*.
5. On the Action Pane, select **Save**.
6. On the **Consolidation fields** FastTab, in the **Remaining fields** list, select the row where the **Field name** field is set to *Mode of delivery*.
7. Select the **Add** button to move the field to the **Selected fields** list.
8. On the Action Pane, select **Edit query**.
9. In the query editor dialog box, on the **Joins** tab, expand and select **Tables** > **Load details** in the tree.

10. Select **Add table join**.
11. In the grid of relations that appears, find and select the row where the **Relation** field is set to *Warehouse item number (Item number)*, and then select **Select**.
12. On the **Range** tab, select **Add** to add a row that has the following settings to the grid:
    - **Table:** *Warehouse item number*
    - **Derived table:** *Warehouse item number*
    - **Field:** *Code 4*
    - **Criteria:** *Flammable*
13. Select **OK** to close the dialog box.

> ⓘ **Note**
>
> For this business case, all order lines where items have a specific filter code (that is, the filter code where the **Code 4** field is set to *Flammable*) consolidated with other items of the same type across orders. If there's an open shipment for the same account, warehouse, and group of items, the new lines are attached to it.

Create example policy 3

In this example, you create a *Customers' requirements* policy that you can use for the following business case:

- The policy queries for a specific customer account.
- Consolidation with open shipments is turned on.
- Consolidation is done across orders but is based on customer requisitions. (In other words, multiple orders are grouped into shipments, based on the same customer requisition number and warehouse.)

Follow these steps to create the shipment consolidation policy for this business case.

1. Go to **Warehouse management** > **Setup** > **Release to warehouse** > **Shipment consolidation policies**.
2. Set the **Policy type** field to *Sales orders*.
3. On the Action Pane, select **New** to create a policy that has the following settings:

- **Policy name:** *CustomerOrderNo*
- **Policy description:** *Consolidate lines based on customer PO*

4. Set the **Consolidate with open shipments** option to *Yes*.
5. On the Action Pane, select **Save**.
6. On the **Consolidation fields** FastTab, in the **Remaining fields** list, select the row where the **Field name** field is set to *Customer requisition*.
7. Select the **Add** button to move the field to the **Selected fields** list.
8. In the **Remaining fields** list, select the row where the **Field name** field is set to *Mode of delivery*.
9. Select the **Add** button to move the field to the **Selected fields** list.
10. On the Action Pane, select **Edit query**.
11. In the query editor dialog box, on the **Range** tab, find the row where the **Field** field is set to *Customer account*, and set the **Criteria** field for that row to *US-001*.
12. Select **OK** to close the dialog box.

> ⓘ **Note**
>
> For this business case, all order lines where sales orders have the same customer requisition number are consolidated into one shipment, regardless of the sales order number. (The customer requisition number is used as the customer's purchase order [PO] number.) If there's an open shipment for the same account, warehouse, and customer requisition, the new lines are attached to it. Use this policy if the customer sends additional order lines that have the same PO number several times during a day and wants all the lines to be grouped into one shipment. (In other words, there's one bill of lading and one packing slip.)

Create example policy 4

In this example, you create a *Customers allowing consolidation* policy for the following business case:

- The policy queries a specific order pool to identify customers who accept consolidated shipments.

- Consolidation with open shipments is turned off.
- You can override the rule on a sales order by selecting a different order pool.

Follow these steps to create the shipment consolidation policy for this business case.

1. Go to **Warehouse management** > **Setup** > **Release to warehouse** > **Shipment consolidation policies**.
2. Set the **Policy type** field to *Sales orders*.
3. On the Action Pane, select **New** to create a policy that has the following settings:
    - **Policy name:** *Order pool*
    - **Policy description:** *Consolidate across orders based on order pool*
4. Leave the **Consolidate with open shipments** option set to *No*.
5. On the Action Pane, select **Save**.
6. On the **Consolidation fields** FastTab, in the **Remaining fields** list, select the row where the **Field name** field is set to *Mode of delivery*.
7. Select the **Add** button to move the field to the **Selected fields** list.
8. On the Action Pane, select **Edit query**.
9. In the query editor dialog box, on the **Range** tab, select **Add** to add a row that has the following settings to the grid:
    - **Table:** *Sales orders*
    - **Derived table:** *Sales orders*
    - **Field:** *Pool*
    - **Criteria:** *ShipCons*
10. Select **OK** to close the dialog box.

> ⓘ **Note**
>
> For this business case, all order lines where sales orders belong to the same order pool consolidated into one shipment across sales orders for the same account, warehouse, and transportation mode of delivery. Instead of the order pool, you can use any other field to distinguish a group of customers and use the sales order header by default. Use this

approach if the customer, not the warehouse, drives the need for consolidation. (In the earlier consolidation logic, the warehouse drove the need for consolidation.)

Create example policy 5

In this example, you create a *Warehouses allowing consolidation* policy for the following business case:

- The policy queries a specific order pool to identify warehouses that can consolidate shipments.
- Consolidation with open shipments is turned off.

Typically, you can use default policies to address this business case. However, you can also manually create similar policies by following these steps.

1. Go to **Warehouse management** > **Setup** > **Release to warehouse** > **Shipment consolidation policies**.
2. Set the **Policy type** field to *Sales orders*.
3. On the Action Pane, select **New** to create a policy that has the following settings:
    - **Policy name:** *Cross-order*
    - **Policy description:** *Cross-order consolidation for specific warehouses*
4. Leave the **Consolidate with open shipments** option set to *No*.
5. On the Action Pane, select **Save**.
6. On the **Consolidation fields** FastTab, in the **Remaining fields** field, select the row where the **Field name** field is set to *Mode of delivery*.
7. Select the **Add** button to move the field to the **Selected fields** list.
8. On the Action Pane, select **Edit query**.
9. In the query editor dialog box, on the **Range** tab, find the row where the **Field** field is set to *Warehouse*, and set the **Criteria** field for that row to *61, 63*.
10. Select **OK** to close the dialog box.

Set the order

After you create all your policies, establish the order in which they're applied. To use a pyramid-like approach, where the policies that have the most conditions are evaluated as having the highest priority, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Release to warehouse** > **Shipment consolidation policies**.
2. Set the **Policy type** field to *Sales orders*.
3. Select each policy that's listed in the left column, and then use the **Move up** and **Move down** buttons on the Action Pane to arrange the policies in the following order:
   a. CustomerMode
   b. Item type
   c. CustomerOrderNo
   d. Order pool
   e. Cross-order
   f. Default

Example scenarios of how to use shipment consolidation policies

The following scenarios illustrate how you can use the shipment consolidation policies that you created while reading this article. Each scenario walks you through a shipment consolidation process that uses shipment consolidation policies during automated or manual release to the warehouse:

- Scenario 1: Consolidate shipments when they are released to the warehouse by using Automatic release of sales orders
- Scenario 2: Consolidate shipments when the shipment consolidation policy is overridden from the Release to warehouse page
- Scenario 3: Consolidate shipments by releasing to warehouse from the outbound load planning workbench
- Scenario 4: Consolidate shipments by using the shipment consolidation workbench
- Scenario 5: Consolidate shipments manually by using the Consolidate shipments page

Related information

- Shipment consolidation policies overview

Last updated on 05/29/2026

Consolidate shipments released to the warehouse using automatic release of sales orders

Article • 09/08/2022

This article presents a scenario where multiple orders are released to the warehouse in the same automated release-to-warehouse periodic procedure. The orders will automatically be consolidated into shipments, based on rules that are defined as shipment consolidation policies.

During the scenario, you will create sets of sales orders and release each set to the warehouse. You will then review the shipments that are created or updated during shipment consolidation, based on the configured policies.

Make demo data available

The scenario in this article references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to **USMF** before you begin.

Set up shipment consolidation policies and product filters

The scenario that is described here assumes that you've already turned on the feature, done the exercises in Configure shipment consolidation policies, and created the policies and other records that are described there. Be sure to do those exercises before you continue with this scenario.

Create the sales orders for this scenario

Start by creating a collection of sales orders that you can work with. You must work with a warehouse that is enabled for advanced warehouse (WMS) processes. Unless a different warehouse is explicitly mentioned, that same warehouse must be used for each of the following sets of orders.

Go to **Accounts receivable** > **Orders** > **All sales orders**, and create a collection of sales orders that have the settings that are described in the following subsections.

Create order set 1

Sales order 1-1

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Mode of delivery:** *Airwa-Air*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Sales order 1-2

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Mode of delivery:** *Airwa-Air*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Sales order 1-3

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Mode of delivery:** *10*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Add a second order line that has the following settings:

    - **Item number:** *A0002* (an item that no **Code 4** filter is assigned to)

- **Quantity:** *1.00*
- **Mode of delivery:** *Airwa-Air*

Create order set 2

Sales orders 2-1 and 2-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-002*

2. Add an order line that has the following settings:

    - **Item number:** *M9200* (an item where the **Code 4** filter is set to *Flammable*)
    - **Quantity:** *1.00*

3. Add a second order line that has the following settings:

    - **Item number:** *M9201* (an item where the **Code 4** filter is set to *Explosive*)
    - **Quantity:** *1.00*
    - **Mode of delivery:** *Airwa-Air*

Create order set 3

Sales order 3-1

1. Create a sales order that has the following settings:

    - **Customer account:** *US-002*

2. Add an order line that has the following settings:

    - **Item number:** *M9200* (an item where the **Code 4** filter is set to *Flammable*)
    - **Quantity:** *1.00*

3. Add a second order line that has the following settings:

    - **Item number:** *M9201* (an item where the **Code 4** filter is set to *Explosive*)
    - **Quantity:** *1.00*
    - **Mode of delivery:** *Airwa-Air*

Note

This order is identical to the two orders that you created for order set 2. However, it's listed as its own order set because you will release it separately later in this scenario.

Create order set 4

Sales order 4-1

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Customer requisition:** *1*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Create order set 5

Sales orders 5-1 and 5-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-001*
    - **Customer requisition:** *2*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Sales order 5-3

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Customer requisition:** *1*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)

- **Quantity:** *1.00*

Create order set 6

Sales orders 6-1 and 6-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-003*
    - **Customer requisition:** *2*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Sales orders 6-3 and 6-4

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-004*
    - **Customer requisition:** *1*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Sales orders 6-5 and 6-6

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-007*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Pool:** *ShipCons*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Sales orders 6-7 and 6-8

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-007*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Pool:** Leave this field blank.

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

Automatic release of sales orders to the warehouse

For each set of sales orders that you created earlier, you will complete a procedure for automatic release to the warehouse. In each case, you will work through the basic release-to-warehouse procedure that is provided here.

Basic release-to-warehouse procedure

For each set of sales orders that you created earlier, you will complete the three procedures that are outlined in the following subsections.

Update the wave template that will be used during release

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Set the **Wave template type** field to *Shipping*.
3. Find and select the wave template that is associated with the warehouse that you used in the order sets that you created for this scenario. For example, if you used warehouse *24*, select the **24 Shipping Default** wave template. If you used warehouse *61*, select the **61 Shipping** wave template.
4. On the Action Pane, select **Edit**.
5. Set the **Process wave at release to warehouse** option to *No*.

Release to the warehouse

1. Go to **Warehouse management** > **Release to warehouse** > **Automatic release of sales orders**.

2. Set the **Quantity to release** field to *All*.

3. On the **Records to include** FastTab, select **Filter** to open the query dialog box.
4. On the **Range** tab, select **Add** to add a row that has the following settings to the grid:
    - **Table:** *Sales order*
    - **Derived table:** *Sales order*
    - **Field:** *Sales order*
    - **Criteria:** Enter a comma-separated list of the sales order numbers from the desired order set.
5. Select **OK** to save your query.
6. Select **OK** to start the *Automatic release to warehouse* procedure.

Review the shipment that is created or updated

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Find and select the required shipment.
3. If a consolidation policy was used when the shipment was created or updated, you should see it in the **Shipment consolidation policy** field.

Release sales orders from order set 1

Follow the basic release-to-warehouse procedure to release the sales orders from order set 1.

When you've finished, you should see that two shipments were created:

- The first shipment contains three lines and was created by using the *CustomerMode* shipment consolidation policy.
- The second shipment, which doesn't use the *Airways* transportation mode of delivery, was created by using the *CustomerOrderNo* shipment consolidation policy.

Release sales orders from order set 2

Follow the basic release-to-warehouse procedure to release the sales orders from order set 2.

When you've finished, you should see that three shipments were created:

- The first shipment contains *Flammable* items.

- Each of the other two shipments contains one line that has the *Explosive* item.

Release sales orders from order set 3

Follow the basic release-to-warehouse procedure to release the sales orders from order set 3.

When you've finished, you should see that the following actions occurred:

- One existing shipment (the shipment that was created when order set 2 was released to the warehouse) was updated. A line that has the *Flammable* item was added.
- One new shipment was created that contains the *Explosive* item.

Release sales orders from order set 4

Follow the basic release-to-warehouse procedure to release the sales orders from order set 4.

When you've finished, you should see that one existing shipment (where the **Customer requisition** field is set to *1*) was updated. One new line was added to it.

Release sales orders from order set 5

Follow the basic release-to-warehouse procedure to release the sales orders from order set 5.

When you've finished, you should see that the following actions occurred:

- One existing shipment (where the **Customer requisition** field is set to *1*) was updated. A line from sales order 5-3 (where the **Customer requisition** field is set to *1*) was added to it.
- One new shipment was created, where lines from sales orders 5-1 and 5-2 are grouped into one shipment.

Release sales orders from order set 6

Follow the basic release-to-warehouse procedure to release the sales orders from order set 6.

When you've finished, you should see that four shipments were created:

- Lines from two orders for customer *US-003* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from two orders for customer *US-004* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from sales orders 6-5 and 6-6 for customer *US-007* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from sales orders 6-7 and 6-8 for customer *US-007* were grouped into one shipment by using the *CrossOrder* shipment consolidation policy.

Related information

- Shipment consolidation policies overview
- Configure shipment consolidation policies

Feedback

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Consolidate shipments when the shipment consolidation policy is overridden

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article presents a scenario where one or more sales lines must be manually released to the warehouse from the **Release sales orders to warehouse** page, and the system-defined shipment consolidation policy must be overridden before the release. An override of the shipment consolidation policy might be required if, for example, an order that isn't usually consolidated with open shipments must be consolidated with open shipments.

During the scenario, you will create a set of sales orders and then override the default shipment consolidation policy before you release the orders to the warehouse.

Make demo data available

The scenario in this article references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to **USMF** before you begin.

Set up shipment consolidation policies and product filters

The scenario that is described here assumes that you've already turned on the feature, done the exercises in Configure shipment consolidation policies, and created the policies and other records that are described there. Be sure to do those exercises before you continue with this scenario.

Create the sales orders for this scenario

1. Go to **Accounts receivable** > **Orders** > **All sales orders**, and create three identical sales orders that have the following settings:

    - **Customer account:** *US-002*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Release the sales orders from the Release sales orders to warehouse page

Follow these steps to override the shipment consolidation policy during the release to the warehouse.

1. Go to **Warehouse management** > **Release to warehouse** > **Release sales orders to warehouse**.
2. In the upper pane, select the first sales order that you created for this scenario.
3. Select **Add** to add the line to the release to the warehouse. Notice that the *Default* shipment consolidation policy is applied in the bottom pane.
4. In the bottom pane, select **Select new shipment consolidation policy**.
5. Select a policy that allows for consolidation with other open shipments of the same policy. For example, select the *CustomerOrderNo* policy.
6. Select **Release to warehouse**.
7. Select the second and third sales orders that you created for this scenario.
8. Select **Add** to add the lines to the release to the warehouse. Notice that the *Default* policy is applied in the bottom pane.
9. Select the second line, and then, in the **Select new shipment consolidation policy** field, select the *CustomerOrderNo* policy.
10. Select **Release to warehouse** for both lines.

Verify the shipments

Two shipments should have been created:

- The first shipment contains two lines and was created by using the *CustomerOrderNo* shipment consolidation policy.

- The second shipment contains one line and was created by using the *Default* shipment consolidation policy.

Follow these steps to review the shipments that were created.

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Find and select the required shipment.
3. In the **Shipment consolidation policy** field, review the consolidation policy that was used when the shipment was created.

Related information

- Shipment consolidation policies overview
- Configure shipment consolidation policies

Last updated on 08/14/2025

Consolidate shipments by releasing to warehouse from the outbound load planning workbench

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article presents a scenario where multiple orders are released to the warehouse in the same load and are then automatically consolidated into shipments.

Make demo data available

The scenario in this article references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to **USMF** before you begin.

Set up shipment consolidation policies and product filters

The scenario that is described here assumes that you've already turned on the feature, done the exercises in Configure shipment consolidation policies, and created the policies and other records that are described there. Be sure to do those exercises before you continue with this scenario.

Create the sales orders for this scenario

Start by creating a collection of sales orders that you can work with. You must work with a warehouse that is enabled for advanced warehouse (WMS) processes. Unless a different warehouse is explicitly mentioned, that same warehouse must be used for each of the following sets of orders.

Go to **Accounts receivable** > **Orders** > **All sales orders**, and create a collection of sales orders that have the settings that are described in the following subsections.

Create order set 1

Sales order 1-1

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Mode of delivery:** *Airwa-Air*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales order 1-2

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Mode of delivery:** *Airwa-Air*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales order 1-3

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Mode of delivery:** *10*

2. Add an order line that has the following settings:

- **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
- **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

4. Add a second order line that has the following settings:

    - **Item number:** *A0002* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*
    - **Mode of delivery:** *Airwa-Air*

5. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the second order line.

Create order set 2

Sales orders 2-1 and 2-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-002*

2. Add an order line that has the following settings:

    - **Item number:** *M9200* (an item where the **Code 4** filter is set to *Flammable*)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

4. Add a second order line that has the following settings:

    - **Item number:** *M9201* (an item where the **Code 4** filter is set to *Explosive*)
    - **Quantity:** *1.00*
    - **Mode of delivery:** *Airwa-Air*

5. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the second order line.

Create order set 3

Sales orders 3-1 and 3-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-001*
    - **Customer requisition:** *1*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 3-3 and 3-4

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-001*
    - **Customer requisition:** *2*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Create order set 4

Sales orders 4-1 and 4-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-003*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 4-3 and 4-4

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-004*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 4-5 and 4-6

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-007*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Pool:** *ShipCons*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 4-7 and 4-8

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-007*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Pool:** Leave this field blank.

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Use the Outbound load planning workbench to create loads and release them to the warehouse

Follow these steps to create a load for each order set that you created for this scenario and then release it to the warehouse.

1. Go to **Warehouse management** > **Loads** > **Outbound load planning workbench**.
2. On the **Sales lines** tab, find and select all the sales order lines from one of the order sets that you created for this scenario.
3. On the Action Pane, on the **Supply and demand** tab, select **Add** > **To new load** to add the selected order lines to a new Load.
4. In the **Load template assignment** dialog box, in the **Load template ID** field, select a load template, such as *Stnd Load Template*.
5. Select **OK** to close the dialog box.
6. In the **Loads** section, find and select the load that you just created.
7. Select **Release** > **Release to warehouse** to release the selected load to the warehouse.
8. Repeat this procedure for the other three order sets that you created for this scenario.

Verify the shipments

The following procedure lets you verify the shipments that have been created or updated as a result of shipment consolidation. Use it to review each order set that you created for this scenario, and then review the subsections that follow to make sure that you've obtained the expected results.

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Find and select the required shipment.
3. If a consolidation policy was used when the shipment was created or updated, you should see it in the **Shipment consolidation policy** field.

Release order set 1 in one load

Two shipments should have been created:

- The first shipment contains three lines and was created by using the *CustomerMode* shipment consolidation policy.

- The second shipment, which doesn't use the *Airways* transportation mode of delivery, was created by using the *CustomerOrderNo* shipment consolidation policy.

Release order set 2 in one load

Three shipments should have been created:

- The first shipment contains the *Flammable* items.
- Each of the other two shipments contains one line that has the *Explosive* item.

Release order set 3 in one load

Two shipments should have been created:

- The first shipment contains order lines from the sales order where the **Customer requisition** field is set to *1*.
- The second shipment contains order lines from sales order where the **Customer requisition** field is set to *2*.

Release order set 4 in one load

Four shipments should have been created:

- Lines from two orders for customer account *US-003* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from two orders for customer account *US-004* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from sales orders 4-5 and 4-6 for customer account *US-007* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from sales orders 4-7 and 4-8 for customer account *US-007* were grouped into one shipment by using the *CrossOrder* shipment consolidation policy.

Related information

- Shipment consolidation policies overview
- Configure shipment consolidation policies

Last updated on 05/07/2025

Consolidate shipments by using the shipment consolidation workbench

10/03/2025

This article presents a scenario where multiple orders are released to the warehouse and then consolidated into shipments later by using the shipment consolidation workbench.

Make demo data available

The scenario in this article references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to *USMF* before you begin.

Set up shipment consolidation policies and product filters

The scenario that is described here assumes that you've already turned on the feature, done the exercises in Configure shipment consolidation policies, and created the policies and other records that are described there. Be sure to do those exercises before you continue with this scenario.

Create the sales orders for this scenario

Start by creating a collection of sales orders that you can work with. You must work with a warehouse that is enabled for advanced warehouse (WMS) processes. Unless a different warehouse is explicitly mentioned, that same warehouse must be used for each of the following sets of orders.

Go to **Accounts receivable** > **Orders** > **All sales orders**, and create a collection of sales orders that have the settings that are described in the following subsections.

Create order set 1

Sales orders 1-1 and 1-2

1. Create two identical sales orders that have the following settings:

- **Customer account:** *US-001*
- **Mode of delivery:** *Airwa-Air*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales order 1-3

1. Create a sales order that has the following settings:

    - **Customer account:** *US-001*
    - **Mode of delivery:** *10*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

4. Add a second order line that has the following settings:

    - **Item number:** *A0002* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*
    - **Mode of delivery:** *Airwa-Air*

5. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the second order line.

Create order set 2

Sales orders 2-1 and 2-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-002*
    - **Mode of delivery:** *Airwa-Air*

2. Add an order line that has the following settings:

- **Item number:** *M9200* (an item where the **Code 4** filter is set to *Flammable*)
- **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

4. Add a second order line that has the following settings:

    - **Item number:** *M9201* (an item where the **Code 4** filter is set to *Explosive*)
    - **Quantity:** *1.00*
    - **Mode of delivery:** *Airwa-Air*

5. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the second order line.

Create order set 3

Sales orders 3-1 and 3-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-001*
    - **Customer requisition:** *1*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 3-3 and 3-4

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-001*
    - **Customer requisition:** *2*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Create order set 4

Sales orders 4-1 and 4-2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-003*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 4-3 and 4-4

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-004*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 4-5 and 4-6

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-007*
    - **Site:** *6*
    - **Warehouse:** *61*
    - **Pool:** *ShipCons*

2. Add an order line that has the following settings:

- **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
- **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Sales orders 4-7 and 4-8

1. Create two identical sales orders that have the following settings:

   - **Customer account:** *US-007*
   - **Site:** *6*
   - **Warehouse:** *61*
   - **Pool:** Leave this field blank.

2. Add an order line that has the following settings:

   - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
   - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Release the orders to the warehouse

Follow these steps to release each sales order that you created for this scenario to the warehouse.

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Find and select the sales order to release.
3. On the Action Pane, on the **Warehouse** tab, select **Actions** > **Release to warehouse** to release the selected sales order.
4. Repeat this procedure for every other sales order that you created for this scenario.

Consolidate the shipments by using the shipment consolidation workbench

1. Go to **Warehouse management** > **Release to warehouse** > **Shipment consolidation workbench**.

2. On the Action Pane, select **Edit query**.

3. In the query editor dialog box, on the **Range** tab, select **Add** to add a row that has the following settings to the grid:

    - **Table:** *Sales orders*
    - **Field:** *Sales order*
    - **Criteria:** Enter a comma-separated list of the sales order numbers for each order set that you created for this scenario.

4. Select **OK** to save your query and close the dialog box.

5. On the Action Pane, select **Consolidate shipments**.

6. Select all the shipments, and then, on the Action Pane, select **Consolidate**.

Verify the shipments

The following procedure lets you verify the shipments that have been created or updated as a result of shipment consolidation. Use it to review each order set that you created for this scenario, and then review the subsections that follow to make sure that you've obtained the expected results.

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Find and select the required shipment.
3. If a consolidation policy was used when the shipment was created or updated, you should see it in the **Shipment consolidation policy** field.

Related shipments for order set 1

Two shipments should have been created:

- The first shipment contains three lines and was created by using the *CustomerMode* shipment consolidation policy.
- The second shipment, which doesn't use the *Airways* transportation mode of delivery, was created by using the *CustomerOrderNo* shipment consolidation policy.

Related shipments for order set 2

Three shipments should have been created:

- The first shipment contains *Flammable* items.
- Each of the other two shipments contains one line that has the *Explosive* item.

Related shipments for order set 3

Two shipments should have been created:

- The first shipment contains order lines from the sales order where the **Customer requisition** field is set to *1*.
- The second shipment contains order lines from sales order where the **Customer requisition** field is set to *2*.

Related shipments for order set 4

Four shipments should have been created:

- Lines from two orders for customer *US-003* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from two orders for customer *US-004* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from sales orders 4-5 and 4-6 for customer *US-007* were grouped into one shipment by using the *Order pool* shipment consolidation policy.
- Lines from sales orders 4-7 and 4-8 for customer *US-007* were grouped into one shipment by using the *CrossOrder* shipment consolidation policy.

Related information

- Shipment consolidation policies overview
- Configure shipment consolidation policies

Consolidate shipments manually by using the Consolidate shipments page

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article presents a scenario where multiple orders are released to the warehouse and then consolidated later by using the **Consolidate shipments** page.

Make demo data available

The scenario in this article references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to **USMF** before you begin.

Set up shipment consolidation policies and product filters

The scenario that is described here assumes that you've already turned on the feature, done the exercises in Configure shipment consolidation policies, and created the policies and other records that are described there. Be sure to do those exercises before you continue with this scenario.

Create the sales orders for this scenario

Start by creating a collection of sales orders that you can work with. You must work with a warehouse that is enabled for advanced warehouse (WMS) processes. Unless a different warehouse is explicitly mentioned, that same warehouse must be used for each of the following sets of orders.

Go to **Accounts receivable** > **Orders** > **All sales orders**, and create a collection of sales orders that have the settings that are described in the following subsections.

Create sales orders 1 and 2

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-007*
    - **Pool:** *ShipCons*

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Create sales orders 3 and 4

1. Create two identical sales orders that have the following settings:

    - **Customer account:** *US-007*
    - **Pool:** Leave this field blank.

2. Add an order line that has the following settings:

    - **Item number:** *A0001* (an item that no **Code 4** filter is assigned to)
    - **Quantity:** *1.00*

3. Select **Inventory** > **Reservation**, and then, on the Action Pane, select **Reserve lot** to reserve the order line.

Release the orders to the warehouse

Follow these steps to release each sales order that you created for this scenario to the warehouse.

1. Go to **Accounts receivable** > **Orders** > **All sales orders**.
2. Find and select the sales order to release.
3. On the Action Pane, on the **Warehouse** tab, select **Actions** > **Release to warehouse** to release the selected sales order.
4. Repeat this procedure for every other sales order that you created for this scenario.

Consolidate shipments

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Find and select a shipment that was created when sales order 1 was released to the warehouse. (Its **Shipment consolidation policy** field should be set to *Order pool*.)
3. On the Action Pane, on the **Shipments** tab, select **Shipments** > **Consolidate shipments**.
4. Verify the shipments that are suggested for consolidation. Only one shipment that has the same policy should be suggested for consolidation.
5. Close the **Shipment consolidation** page.
6. Find and select a shipment that was created when sales order 3 was released to the warehouse. (Its **Shipment consolidation policy** field should be set to *Default*.)
7. On the Action Pane, on the **Shipments** tab, select **Shipments** > **Consolidate shipments**.
8. Verify that no shipments are suggested for consolidation.
9. Select **Show filters**.
10. In the **Filters** pane, remove the **Order number** filter, and then select **Apply**.
11. Verify the shipments that are suggested for consolidation. Only one shipment that has the same policy should be suggested for consolidation.

Related information

- Shipment consolidation policies overview
- Configure shipment consolidation policies

Last updated on 05/07/2025

Wave creation and processing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to create, process, and release a wave to create picking work for a load, shipment, production order, or kanban order. You can create waves for the following types of orders:

- **Sales orders**, **Transfer orders**, and **Outbound shipment orders** – Use shipping waves to include lines from sales orders, transfer orders, and outbound shipment orders. When an order is released to the warehouse, you can include its order lines in the wave.
- **Production orders** – Use production waves to include lines from the bill of materials (BOM) for a product.
- **Kanban orders** – Kanban waves include picking list lines from kanban orders.

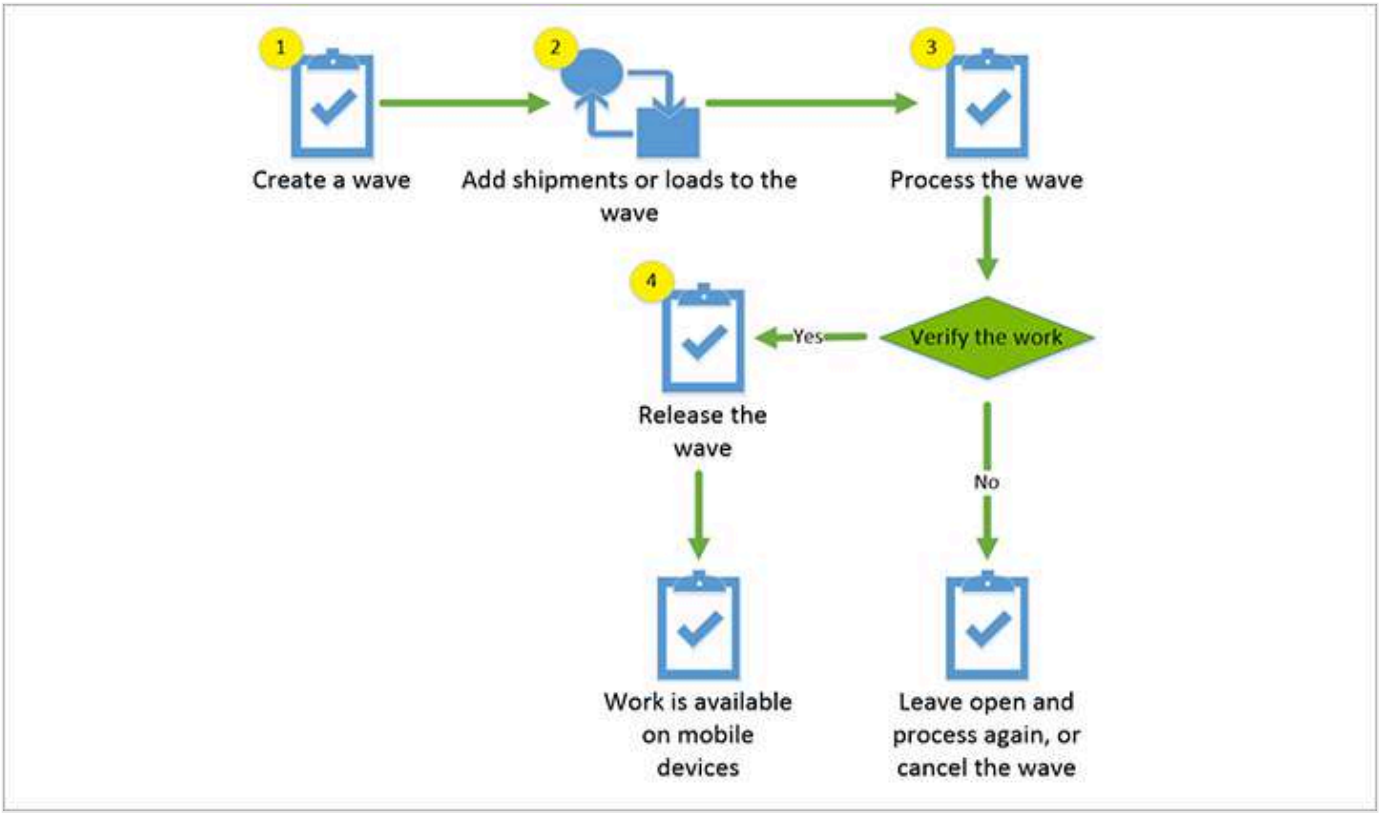
For sales orders and kanban orders, inventory must be reserved before the order is released to the warehouse. Otherwise, the items or allocation lines can't be processed in a wave. However, production orders are slightly more flexible. For production orders, you can choose either of the following options:

- Require that all materials are reserved before an order can be released to the warehouse.
- Allow production orders to be released to the warehouse even though all materials can't be reserved. If you select this option, you must manually repeat the release to warehouse process when the additional materials become available. For example, this option is useful if you have the materials that you need to start a production, and can wait until the additional materials become available.

You can specify which of these production order options to use by default by using the **Requirement for material reservation** field on the **Production control parameters** page. However, you can change the setting for each specific production order as needed. For more information, see Warehouse parameters for wave processing.

Create and process a wave

The following diagram shows the flow for how shipping waves are created, processed, and released. The numbers correspond to the sections later in this article.



Prerequisites

Before you start, a wave template must be available for the type of wave you want to create (shipping, production, or kanban). The wave template establishes many settings for how the wave is generated and processed, including which steps must be done manually and which steps are done automatically. Learn more in Wave templates.

Create a wave

Automatically create waves based on warehouse and order type

To automatically create waves, set up Wave templates that apply to each relevant order type and warehouse. Make sure each template has the **Automate wave creation** option set to *Yes*.

Manually create waves

To manually create a wave, follow these steps:

1. Make sure that the relevant Wave templates aren't set to automatically create a wave for the warehouse and order types where you want to create waves manually.

2. Depending on the type of wave you want to create, go to one of the following locations:

    - Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**. On the Action Pane, select **Wave**.
    - Go to **Warehouse management** > **Outbound waves** > **Production waves** > **All production waves**. On the Action Pane, select **Production wave**.
    - Go to **Warehouse management** > **Outbound waves** > **Kanban waves** > **All kanban waves**. On the Action Pane, select **Create wave**.

3. In the **Description** field, enter a short description of the wave. This description should indicate what you're processing in the wave.

4. In the **Wave template name** field, select the wave template for the type of wave to create. The wave template contains the wave methods that perform actions such as creating work for the wave. For example, the wave template for shipping waves can contain methods for creating loads, allocating lines to waves, replenishment, and creating picking work for the wave.

5. If you want to use wave attributes as additional query criteria for the wave, select the attributes in the **Wave attributes** fields.

Specify what to include in a wave

After you create a wave, add content to it.

> ⓘ **Note**
>
> If needed, you can add lines to a wave even after it’s processed as long as it’s not released.

Automatically specify what to include in a wave

To create waves automatically, set up Wave templates that apply for each relevant order type and warehouse. Make sure each template has the **Automate wave creation** option set to *Yes*. Alternatively, your template could automatically assign lines to any qualifying open wave if the **Assign to open waves** option is set to *Yes*.

Manually specify what to include in a wave

When a wave is created but isn't yet released, you can manually specify what to include in it. To add lines to a wave manually:

1. Depending on the type of wave to add lines to, go to one of the following locations:

    - Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**. On the Action Pane, select **Wave**.
    - Go to **Warehouse management** > **Outbound waves** > **Production waves** > **All production waves**. On the Action Pane, select **Production wave**.
    - Go to **Warehouse management** > **Outbound waves** > **Kanban waves** > **All kanban waves**. On the Action Pane, select **Create wave**.

2. Select the wave. On the Action Pane, select one of the following options:

    - **Maintain shipments**
    - **Maintain productions**
    - **Maintain kanban job picking lists**

3. In the upper part of the window, select the line to add to the wave, and then select **Add to wave**. The line moves to the **Wave lines** FastTab.

    Repeat this step for each line to add. To add all lines, select **Add all**.

    > **Tip**
    >
    > For shipment waves, you can quickly find a specific order by selecting a custom filter in the **Wave filter code** field. Wave filter codes contain query criteria for shipments created in the **Wave filters** form. This field isn't available for production waves or kanban waves. A green check mark in the **On wave** column indicates that the shipment is added to the wave.

Process the wave to create the picking work

After you create a wave and add all required lines, you're ready to process it and create the corresponding picking work.

Automatically process a wave

To automatically process a wave, set up the relevant Wave templates with the automatic processing options you need.

Manually process a wave

You can process a wave only when the **Wave status** is *Created*. After you process a wave, the **Wave status** changes to *Held*.

To manually process a wave that contains all required content, follow these steps:

1. Depending on the type of wave to process, do one of the following steps:

    - Select **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**. On the Action Pane, select **Wave**.
    - Select **Warehouse management** > **Outbound waves** > **Production waves** > **All production waves**. On the Action Pane, select **Production wave**.
    - Select **Warehouse management** > **Outbound waves** > **Kanban waves** > **All kanban waves**. On the Action Pane, select **Create wave**.

2. Select the wave to process. On the Action Pane, select **Process**.

Release the wave to the warehouse to start picking and packing

You must process a wave before you can release it. When you release the wave, the picking work is available in the warehouse. You can cancel a wave after it is released, and add more lines, but you can't change the lines.

Automatically release a wave

To automatically process a wave, set up the relevant Wave templates with the automatic processing options you need.

Manually release a wave

To release a wave manually, follow these steps:

1. Depending on the type of wave to release, do one of the following:

    - Select **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**. On the Action Pane, select **Wave**.
    - Select **Warehouse management** > **Outbound waves** > **Production waves** > **All production waves**. On the Action Pane, select **Production wave**.
    - Select **Warehouse management** > **Outbound waves** > **Kanban waves** > **All kanban waves**. On the Action Pane, select **Create wave**.

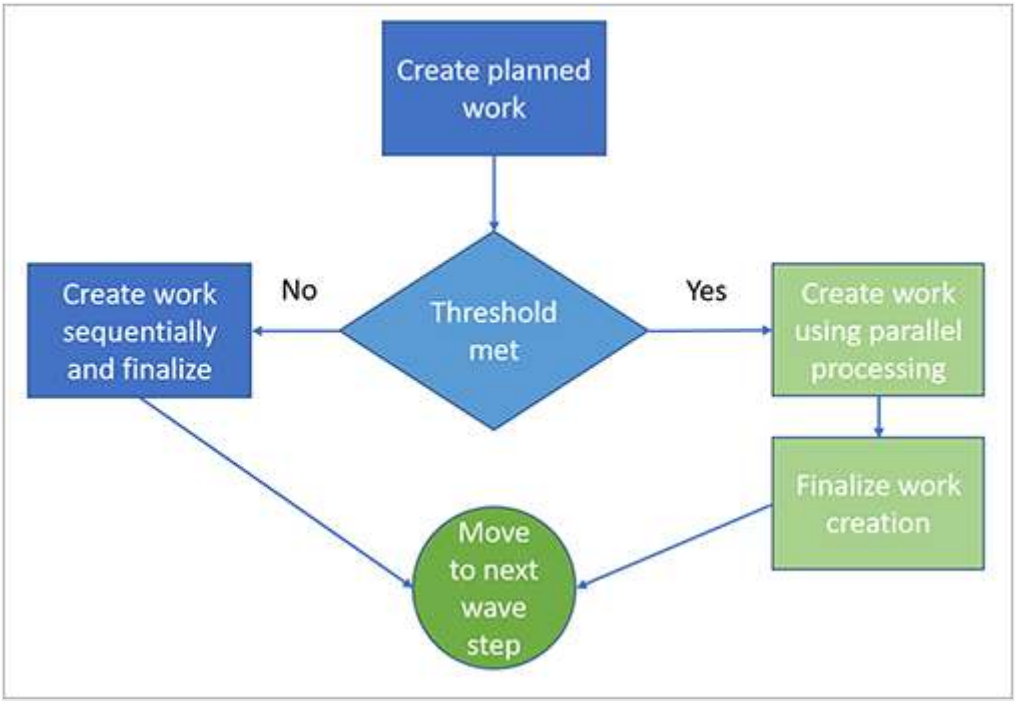
2. Select the wave to release. On the Action Pane, select **Release wave**.

Containerize a wave

Automated containerization creates containers and the picking work for shipments when a wave is processed. For details about how to set it up, see Containerization.

Work with the scheduled work creation

The following flowchart shows how planned work is created during wave processing.



Planned work

The **Planned work details** page (**Warehouse management** > **Work** > **Planned work details**) shows information about the planned work, which is initially created during wave processing. The following **Process status** values are available:

- **Queued** - The planned work is waiting to be used to create work.
- **Completed** - The planned work is used to create work.
- **Failed** – The wave processing failed. The planned work can be in a **Failed** state with or without related actual work. When the actual work creation process fails, the actual work remains in status *Cancelled*.

Batch job for the work creation process

To view the batch jobs for processing waves, select **Batch jobs** on the Action Pane on the **All waves** page.

From here, you can view all the batch task details for each of the batch job IDs.

Cancel a wave

If needed, you can cancel a wave that you processed. To cancel a wave, and the picking work that's created, follow these steps:

1. Depending on the type of wave to cancel, go to one of the following locations:

    - **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
    - **Warehouse management** > **Outbound waves** > **Production waves** > **All production waves**.
    - **Warehouse management** > **Outbound waves** > **Kanban waves** > **All kanban waves**.

2. Select the wave to cancel. On the Action Pane, on the **Work** tab, select **Cancel**.

Review wave batch job details

Use the **Wave batch job details** page to inspect the batch jobs and related tasks associated with any wave. This is especially useful for troubleshooting a wave that has failed. Without this feature, only administrators will typically have access to batch job details. The **Wave batch job details** page can be made available to non-admin users and provides a read-only view of batch jobs and related tasks.

Use the Wave batch job details page

The **Wave batch job details** page combines batch jobs and batch job tasks. You can investigate all the wave steps without needing to navigate back and forth between a single batch job and the batch tasks list. The page also provides access to the batch log and, if you have the required permissions, provides a link to the **Batch jobs** page.

To open this page, select a wave on any of several different wave pages and then select **Wave batch job details** from the Action Pane.

Review load validation and error messages

During wave processing, the system validates and displays the status for each load line in the wave. If no warnings occur, it continues to the next wave step. If warnings do occur, it shows the

following error after it finishes validating the entire wave:

> Found invalid load lines in wave. Please remove the invalid load lines.

You can review the final status of each load line in the wave and correct all warnings before trying again. This process lets you address all the warnings at once before reprocessing the wave. (In previous releases, the system stopped processing the wave after the first warning, so you could only fix warnings one at a time.)

The way the system displays your wave processing status messages depends on how you set the **Create wave processing history log** option on the **Warehouse management parameters** page.

- When **Create wave processing history log** is set to *No*, the load line status messages are shown in the Action center.
- When **Create wave processing history log** is set to *Yes*, the load line status messages are shown on the **Wave processing history log** page. To view the log, go to **Warehouse management** > **Outbound waves** > **Wave processing history log**.

Related information

- Configure wave processing example
- Wave templates
- Containerization

Last updated on 04/27/2026

Reprint and void wave labels

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to manage labels that are generated by wave processing. (For a detailed description and configuration instructions, see Configure wave label printing.)

You can reprint wave labels at any time. For example, you might have to print a single label if an existing label was lost or damaged. Alternatively, a warehouse worker or supervisor might have to reprint a whole roll of labels if the number and/or composition of a whole series of wave labels has changed (for example, because of inventory shortage or other reasons). Often, even if only the number of cartons has changed, the whole roll must be reprinted to keep the total number accurate in the "Carton X of Y" section of each label.

The reprint wave labels feature supports the following functionality:

- Reprint labels from both the warehousing app and the rich client.
- Void labels and simultaneously reprint them. (The ability to void labels is embedded in short pick scenarios, for example.)
- Clean up the wave label history.

This article presents a collection of scenarios that show, through examples, how to work with the reprint wave labels feature.

> **Important**
>
> To work through the scenarios that are presented in this article, you must first turn on and configure the relevant wave printing features, as described in Configure wave label printing. Several of the scenarios in this article also require that you first work through the scenarios in that article to generate prerequisite sample data.

Scenario 1: Reprint labels from the web client

You can view and reprint wave labels from the following pages. On the Action Pane of each page, on the **Shipments** tab, in the **Related information** group, select **Wave labels**.

- All shipments > Shipment details
- All loads > Load details
- All waves
- Wave labels
- Wave label history

To reprint a wave label from the web client, follow these steps.

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave to reprint labels from.
3. On the Action Pane, on the **Wave** tab, in the **Print** group, select **Wave labels**.
4. Follow one or both of the following steps:
    - To reprint the label, select a printer in the **Printer name** field. (Leave this field blank if you just want to update the wave label details, without reprinting the label.)
    - To update the label, select the **Update wave label details** check box. (Leave this check box cleared if you just want to reprint the previous label.)

    > ⓘ **Note**
    >
    > Every time that a wave label is printed or reprinted, its data is converted through the appropriate wave label layout, and all placeholders are replaced with actual values. The resulting string is stored as a record in the wave label history. If the **Update wave label details** check box is cleared, the stored Zebra Programming Language (ZPL) data is used when a label is reprinted. If the **Update wave label details** check box is selected, a new string is generated. The existing wave labels are also recalculated, and any excess labels (for example, if the related work lines have been canceled or modified) are marked as **Voided** and aren't reprinted.

5. Select **OK** to confirm your selection.

Scenario 2: Reprint labels from the warehousing app

This scenario typically applies if a label roll has been lost or damaged. It provides an example that shows how to set up mobile device menu items that will let workers reprint single and multiple labels. It then shows how to use those new menu items while you're working on a mobile device.

Set up the required menu items and menu for the mobile device

Before workers can reprint labels from a mobile device, you must set up menu items to provide this functionality and then add those items to the warehousing app menu.

Create new mobile device menu items

Follow these steps to create a new collection of menu items for reprinting labels from the warehousing app.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. Create a menu item, and set the following values for it:

    - **Menu item name:** *Reprint single wave label*
    - **Title:** *Reprint single wave label*
    - **Mode:** *Indirect*
    - **Activity code:** *Reprint single wave label*

3. Create a second menu item, and set the following values for it:

    - **Menu item name:** *Reprint labels (Item)*
    - **Title:** *Reprint wave labels (Item)*
    - **Mode:** *Indirect*
    - **Activity code:** *Reprint multiple wave labels*
    - **Display work list grouping filter:** *Yes*
    - **System grouping field:** *ShipmentID*
    - **System grouping label:** *Shipment ID*
    - **Print mode:** *Product*

4. On the Action Pane, select **Field list** to open a page where you can select the fields that will be shown to help workers identify the correct label roll.

5. You can show up to seven fields. Use the drop-down lists to select the field that is shown in each available position. Leave any fields that you don't require blank.

    Here is an example:

    - **Display field:** *LabelItemId*
    - **Display field 2:** *LabelItemName*
    - **Display field 3:** *InventQty*
    - **Display field 4:** *LabelUnitId*

6. Close the page.

7. Create a third menu item, and set the following values for it:

    - **Menu item name:** *Reprint labels (Enum)*
    - **Title:** *Reprint wave labels (Enum)*
    - **Mode:** *Indirect**
    - **Activity code:** *Reprint multiple wave labels*
    - **Display work list grouping filter:** *Yes*
    - **System grouping field:** *ShipmentID*
    - **System grouping label:** *Shipment ID*
    - **Print mode:** *Enumeration*

8. On the Action Pane, select **Field list**, and then use the drop-down lists to select the fields that will be shown to help workers identify the correct label roll (for example, *LabelItemId*, *LabelItemName*, *InventQty*, *LabelUnitId*, and *NumberOfLabels*).

9. Close the page.

10. Create a fourth menu item, and set the following values for it:

    - **Menu item name:** *Reprint labels (by last)*
    - **Title:** *Reprint wave labels (by last)*
    - **Mode:** *Indirect*
    - **Activity code:** *Reprint multiple wave labels*
    - **Display work list grouping filter:** *Yes*
    - **System grouping field:** *ShipmentID*
    - **System grouping label:** *Shipment ID*
    - **Print mode:** *Last good wave label ID*

11. On the Action Pane, select **Field list**, and then use the drop-down lists to select the fields that will be shown to help workers identify the correct label roll (for example, *LabelItemId*, *LabelItemName*, *InventQty*, *LabelUnitId*, and *NumberOfLabels*).

12. Close the page.

Set up the mobile device menu

Follow these steps to add your new menu items to the warehousing app menu.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. Select an existing **Outbound** menu.
3. In the list on the left, find the reprint menu items that you just created, and then use the right arrow button to add them to the list on the right.

4. Close the page.

Use cases

The use cases that are provided here give examples that show how to use the various mobile device menu items that you set up to enable workers to reprint wave labels.

Before you work through these use cases, the following prerequisites must be in place:

- Demo data must be installed, and you must select the **USMF** legal entity.
- Wave label printing must be configured, and some labels must be generated, as described in Configure wave label printing.

Use case 2.1: A single wave label is scratched and must be reprinted

1. Sign in to the warehousing app as a user who access to warehouse *62*. (In the standard demo data, you can sign in by using the user ID *62* and the password *1*.)
2. Go to **Outbound** > **Reprint single wave label**.
3. Enter or scan the wave label ID.
4. Select the printer to reprint on.
5. Select **OK** to confirm the action.

Use case 2.2: Several labels for boxes of the same item were damaged and must be reprinted. Each label has a product bar code, but no enumeration or SSCC number

1. Sign in to the warehousing app as a user who has access to warehouse *62*. (In the standard demo data, you can sign in by using the user ID *62* and the password *1*.)
2. Go to **Outbound** > **Reprint labels (Item)**.
3. Enter or scan the shipment ID.
4. Select the tile that has the correct label roll to reprint from.
5. Scan the product bar code from an existing label to confirm that the correct line has been selected.
6. Enter the number of labels to reprint.
7. Select the printer to reprint on.
8. Select **OK** to confirm the action.

Use case 2.3: Several labels for boxes weren't printed because of a printer jam. Because the labels have enumeration, you can define the carton range to reprint

1. Sign in to the warehousing app as a user who has access to warehouse *62*. (In the standard demo data, you can sign in by using the user ID *62* and the password *1*.)
2. Go to **Outbound** > **Reprint labels (Enum)**.
3. Enter or scan the shipment ID.
4. Select the tile that has the correct label roll to reprint from.
5. Enter the first carton to reprint a label for.
6. Enter the last carton to reprint a label for. Alternatively, leave this field blank to reprint labels for all cartons after the specified first carton.
7. Select the printer to reprint on.
8. Select **OK** to confirm the action.

Use case 2.4: Several labels for boxes weren't printed because of a printer jam. The last good label has a wave label ID that is printed as a bar code

1. Sign in to the warehousing app as a user who has access to warehouse *62*. (In the standard demo data, you can sign in by using the user ID *62* and the password *1*.)
2. Go to **Outbound** > **Reprint labels (by last)**.
3. Enter or scan the shipment ID.
4. Select the tile that has the correct label roll to reprint from.
5. Enter or scan the wave label ID of the last good wave label. The app identifies the next label in the sequence as the first carton that a label will be reprinted for.
6. Enter the last carton to reprint a label for. Alternatively, leave this field blank to reprint labels for all cartons after the specified first carton.
7. Select the printer to reprint on.
8. Select **OK** to confirm the action.

Scenario 3: Short pick and reprint

Before you work through this scenario, the following prerequisites must be in place:

- Demo data must be installed, and you must select the **USMF** legal entity.
- Wave label printing must be configured, and some labels must be generated, as described in Configure wave label printing.

Set up work exceptions

Work exceptions control the behavior of short picking. Follow these steps to set up a work exception.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work exceptions**.

2. Create a record that has the following settings:

    - **Work exception code:** *Short pick and print*
    - **Exception type:** *Short pick*
    - **Suggest wave labels reprint:** *Yes*

Void and reprint after a short pick

1. Sign in to the warehousing app as a user who has access to warehouse *62*. (In the standard demo data, you can sign in by using the user ID *62* and the password *1*.)
2. Open a work processing flow for the sales order work that was created when wave labels were originally printed.
3. Select **Short pick**.
4. Select the work exception code that you created for this scenario.
5. If you selected the correct exception, the **Void and reprint** check box should be available. Select this box and confirm. When confirmed, the label roll sequence identified by the **Label build ID** field is recalculated based on the changed work line quantity. It's then reprinted on the specified printer.

Related information

- Wave label printing

Last updated on 08/14/2025

Work policies

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up the system and the Warehouse Management mobile app to support work policies. Use this functionality to quickly register inventory without creating putaway work when you receive purchase or transfer orders, or when you complete manufacturing processes. This article provides general information. For detailed information related to license plate receiving, see License plate receiving via the Warehouse Management mobile app.

A work policy controls whether warehouse work is created when a manufactured item is reported as finished, or when goods are received by using the Warehouse Management mobile app. Set up each work policy by defining the conditions where it applies: the work order types and processes, the inventory location, and (optionally) the products. For example, a purchase order for product *A0001* must be received in location *RECV* in warehouse *24*. Later, the product is consumed in another process at location *RECV*. In this case, set up a work policy to prevent putaway work from being created when a worker reports product *A0001* as received in location *RECV*.

> **Note**
>
> - For a work policy to be active, you must define at least one location for it on the **Inventory locations** FastTab of the **Work policies** page.
> - You can't specify the same location for multiple work policies.
> - The **Print label** option for mobile device menu items doesn't print a license plate label unless work was created.

The Work policies page

To set up work policies, go to **Warehouse management** > **Setup** > **Work** > **Work policies**. Then, on each FastTab, set the fields as described in the following subsections.

The Work order types FastTab

On the **Work order types** FastTab, add all the work order types and the related work processes that the work policy applies to. The following work order types and related work processes are supported for work policies.

Expand table

Work order type	Work process
Raw material picking	All related processes
Co-product and by-product put away	All related processes
Finished goods putaway	All related processes
Transfer receipt	License plate receiving (and putaway)
Purchase orders	• License plate receiving (and putaway) • Load item receiving (and putaway) • Purchase order line receiving (and putaway) • Purchase order item receiving (and putaway)
Inbound shipment order	• License plate receiving (and putaway) • Load item receiving (and putaway) • Inbound shipment order line receiving (and putaway) • Inbound shipment order item receiving (and putaway)

To set up a work policy so that it applies to several work processes of the same work order type, add a separate line for each work process to the grid.

For each line in the grid, set the **Work creation method** field to one of the following values:

- **Never** – The work policy prevents warehouse work from being created for the selected work order type and related work process.
- **Cross docking** – The work policy creates cross-docking work by using the policy that you select in the **Cross docking policy name** field.

The Inventory locations FastTab

On the **Inventory locations** FastTab, add all the locations where this work policy should be applied. If you don't associate any location with a work policy, the work policy isn't applied to any process.

You can't specify the same location for multiple work policies.

You can use a warehouse location assigned to a location profile where the **Use license plate tracking** option is turned off. In this case, workers directly register the on-hand inventory.

The Products FastTab

On the **Products** tab, set the **Product selection** field to control which products the policy applies to:

- **All** – The policy applies to all products.
- **Selected** – The policy applies only to products listed in the grid. Use the toolbar on the **Products** FastTab to add products to the grid or remove them from the grid.

Default and custom "to" locations

Mobile device menu items that use the following work creation processes provide the **Use default data** option. This option lets you assign a custom "to" location to one or more menu items. (This option was already available for some other types of menu items.)

- License plate receiving (and putaway)
- Load item receiving (and putaway)
- Purchase order line receiving (and putaway)
- Purchase order item receiving (and putaway)
- Inbound shipment order line receiving (and putaway)
- Inbound shipment order item receiving (and putaway)

The **To location** setting for a menu item overrides the default receiving location for the warehouse, for all orders that are processed by using that menu item.

To set up a mobile device menu item to support receiving at a custom location, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select or create a menu item that uses one of the work creation processes that are listed earlier in this section.

3. On the **General** FastTab, set the **Use default data** option to **Yes**.

4. On the Action Pane, select **Default data**.

5. On the **Default data** page, set the following values:

    - **Default data field:** Set this field to *To location*.
    - **Warehouse:** Select the destination warehouse to use with this menu item.
    - **Location:** This field lists all the location IDs that are available for the selected warehouse. However, the setting of this field doesn't actually have any effect. Therefore, you can leave it blank. Nevertheless, you can use the list to confirm the ID that you must enter in the **Hardcoded value** field.
    - **Hardcoded value:** Enter the location ID for the receiving location that applies to this menu item.

> **Tip**
>
> You can apply a work policy only if the relevant work policy setup lists all the receiving locations. This requirement applies regardless of whether you're using the default warehouse receiving location or a custom "to" location.

Example scenario: Warehouse receiving

All products that are received by the *Purchase order item receiving (and putaway)* process must be registered in location *FL-001*, and they must be available in warehouse *24*. However, work should not be created. Products that are received by any other process (that is, by using other mobile device menu items) should be registered at the default warehouse receiving location (*RECV*), and work should be created as usual. (This scenario doesn't show the default receiving setup.)

This scenario requires the following elements:

- A work policy for the *Purchase order item receiving (and putaway)* process in location *FL-001*, for all products
- A mobile device menu item that has default data, and that sets the **To location** field to *FL-001*

Prerequisites

This scenario uses the standard demo data. Therefore, if you want to work through it by using the values that are provided here, you must work on a system where demo data is installed. Additionally, you must select the **USMF** legal entity.

Set up a work policy

1. Go to **Warehouse management** > **Setup** > **Work** > **Work policies**.
2. Select **New**.
3. In the **Work policy name** field, enter *No purchase item putaway work*.
4. Select **Save**.
5. On the **Work order types** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:
    - **Work order type:** *Purchase orders*
    - **Work process:** *Purchase order item receiving (and putaway)*
    - **Work creation method:** *Never*
    - **Cross docking policy name:** Leave this field blank.
6. On the **Inventory locations** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:
    - **Warehouse:** *24*
    - **Location:** *FL-001*
7. On the **Products** FastTab, set the **Product selection** field to *All*.
8. Select **Save**.

Set up a mobile device menu item to change the receiving location

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. In the left pane, select the existing **Purchase receive** menu item.
3. On the **General** FastTab, set the **Use default data** option to *Yes*.
4. Select **Save**.

5. On the Action Pane, select **Default data**.

6. On the **Default data** page, on the Action Pane, select **New** to add a row to the grid, and then set the following values for the new row:

    - **Default data field:** *To location*
    - **Warehouse:** *24*
    - **Location:** Leave this field blank.
    - **Hardcoded value:** *FL-001*

7. Select **Save**.

Receive a purchase order without creating work

The example in this section shows how to receive a purchase order item, but without creating work, at a location that differs from the default receiving location set up for the warehouse. This example uses the work policy and mobile device item that you created earlier in this scenario.

Create a purchase order

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.

2. Select **New**.

3. In **Create purchase order**, set the following values:

    - **Vendor account:** *US-101*
    - **Site:** *2*
    - **Warehouse:** *24*

4. Select **OK** to close the dialog box and open the new purchase order.

5. On the **Purchase order lines** FastTab, set the following values for the empty row:

    - **Item number:** *A0001*
    - **Quantity:** *1*

6. Select **Save**.

7. Make a note of the purchase order number.

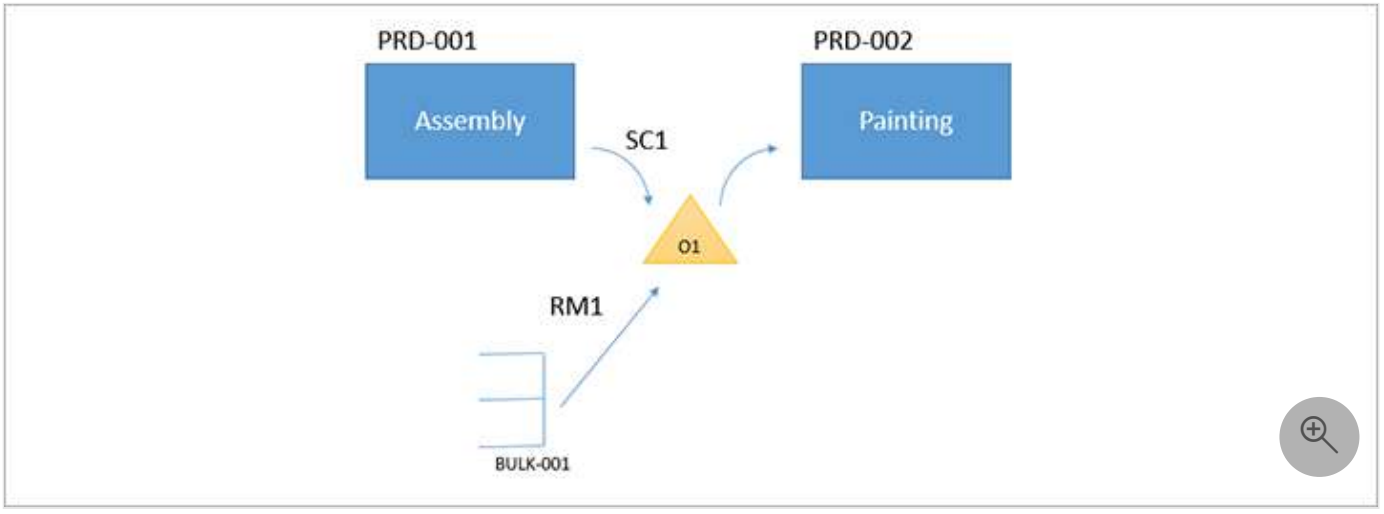
Receive a purchase order

1. On the mobile device, sign in to warehouse *24* by using *24* as the user ID and *1* as the password.
2. Select **Inbound**.
3. Select **Purchase receive**. The **Location** field should be set to *FL-001*.
4. Enter the purchase order number for the purchase order that you created in the previous procedure.
5. In the **Item number** field, enter *A0001*.
6. Select **OK**.
7. In the **Quantity** field, enter *1*.
8. Select **OK**.

The purchase order is now received, but no work is associated with it. The on-hand inventory is updated, and a quantity of *1* of item *A0001* is now available at location *FL-001*.

Example scenario: Manufacturing

In the following example, there are two production orders, *PRD-001* and *PRD-002*. Production order *PRD-001* has an operation named *Assembly*, where product *SC1* is reported as finished to location *001*. Production order *PRD-002* has an operation named *Painting* and consumes product *SC1* from location *001*. Production order *PRD-002* also consumes raw material *RM1* from location *001*. Raw material *RM1* is stored in warehouse location *BULK-001* and is picked to location *001* by warehouse work for raw material picking. The picking work is generated when production *PRD-002* is released.



When you plan to configure a warehouse work policy for this scenario, consider the following points:

- You don't need warehouse work for putaway of finished goods when you report product *SC1* as finished from production order *PRD-001* to location *001*. The reason is that the

Painting operation for production order *PRD-002* consumes product *SC1* at the same location.
- You need warehouse work for raw material picking to move raw material *RM1* from warehouse location *BULK-001* to location *001*.

Here is an example of a work policy that you can set up, based on these considerations:

- **Work policy name:** *No putaway work*
- **Work order types:** *Finished goods put away* and *Co-product and by-product put away*
- **Inventory locations:** Warehouse *51* and location *001*
- **Products:** *SC1*

The following example scenario provides step-by-step instructions for setting up the warehouse work policy for this scenario.

Example scenario: Report as finished to a location that isn't license plate–controlled

This scenario shows an example where a production order is reported as finished to a location that isn't license plate–controlled.

This scenario uses the standard demo data. Therefore, if you want to work through it by using the values that are provided here, you must work on a system where demo data is installed. Additionally, you must select the **USMF** legal entity.

Set up a warehouse work policy

Warehouse processes don't always include warehouse work. By defining a work policy, you can prevent the creation of work for raw material picking and putaway of finished goods for a set of products at specific locations.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work policies**.
2. Select **New**.
3. In the **Work policy name** field, enter *No putaway work*.
4. On the Action Pane, select **Save**.
5. On the **Work order types** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:

- **Work order type:** *Finished goods put away*
- **Work process:** *All related work processes*
- **Work creation method:** *Never*
- **Cross docking policy name:** Leave this field blank.

6. Select **Add** again to add a second row to the grid, and then set the following values for the new row:

   - **Work order type:** *Co-product and by-product put away*
   - **Work process:** *All related work processes*
   - **Work creation method:** *Never*
   - **Cross docking policy name:** Leave this field blank.

7. On the **Inventory locations** FastTab, select **Add** to add a row to the grid, and then set the following values for the new row:

   - **Warehouse:** *51*
   - **Location:** *001*

8. On the **Products** FastTab, set the **Product selection** field to *Selected*.

9. On the **Products** FastTab, select **Add** to add a row to the grid.

10. In the new row, set the **Item number** field to *L0101*.

11. On the Action Pane, select **Save**.

Set up an output location

1. Go to **Organization administration** > **Resources** > **Resource groups**.

2. In the left pane, select resource group **5102**.

3. On the **General** FastTab, set the following values:

   - **Output warehouse:** *51*
   - **Output location:** *001*

4. On the Action Pane, select **Save**.

> ⓘ **Note**

Location *001* isn't a license plate–controlled location. You can set up an output location that isn't license plate–controlled only if an applicable work policy exists for the location.

Create a production order and report it as finished

1. Go to **Production control** > **Production orders** > **All production orders**.
2. On the Action Pane, select **New production order**.
3. In the **Create production order** dialog box, set the **Item number** field to *L0101*.
4. Select **Create** to create the order and close the dialog box.

   A new production order is added to the grid on the **All production orders** page.

   Keep the new production order selected.
5. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Estimate**.
6. In the **Estimate** dialog box, read the estimate, and then select **OK** to close the dialog box.
7. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Start**.
8. In the **Start** dialog box, on the **General** tab, set the **Automatic BOM consumption** field to *Never*.
9. Select **OK** to save your setting and close the dialog box.
10. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Report as finished**.
11. In the **Report as finished** dialog box, on the **General** tab, set the **Accept error** option to *Yes*.
12. Select **OK** to save your setting and close the dialog box.
13. On the Action Pane, on the **Warehouse** tab, in the **General** group, select **Work details**.

When the production order is reported as finished, no work is generated for putaway. This behavior occurs because a work policy is defined that prevents work from being generated when product *L0101* is reported as finished to location *001*.

More information

For more information about mobile device menu items, see Set up mobile devices for warehouse work.

For more information about license plate receiving and work policies, see License plate receiving via the Warehouse Management mobile app.

For more information about inbound load management, see Warehouse handling of inbound loads for purchase orders.

Last updated on 05/05/2026

Movement of inventory with associated work in Warehouse management

Article • 04/01/2021

Using movement of inventory, you can decide which warehouse workers are allowed to move reserved inventory. This provides a flexibility in regulated warehouses where you can decide to not allow a worker to choose a new pick location for pick work that is already created. It also allows a warehouse manager to control which capabilities some less experienced workers should have.

The flexibility to manage the daily operations of warehouse workers can be useful in scenarios such as these:

Scenario 1

A company has a relatively small receiving area, and it's congested with pallets and boxes awaiting put away. A large shipment is expected on the current day, so the receiving clerk decides to clear up the receiving area by moving some of the pallets to a secondary inbound staging area.

Scenario 2

An experienced warehouse worker notices an opportunity in a warehouse to consolidate items in one location instead of having them divided across three nearby locations with little quantity on each. The worker wants to consolidate the quantity by moving items from each of these locations into the same location and onto the same license plate.

Scenario 3

A pallet is awaiting shipment in a staging location, such as STAGE01, which is near BAYDOOR01. However, due to a change of plans the truck is scheduled to arrive at BAYDOOR04. The shipping clerk is aware of this and needs to ensure that the truck does not have to wait to be loaded from STAGE01. The shipping clerk decides to move the items in that shipment from STAGE01 to STAGE04, which is closer to the new destination.

Current limitations

The work reservations that you can move are limited to Sales order, Transfer order issue, Transfer order receipt, Purchase order, and Replenishment work.

Moving items is restricted to prevent splitting of work lines. This means that if you have a work line for 100 pcs of item A from location Loc1, you won't be able to move only 30 pcs of item A from there to another location. This would lead to a split of the existing work line to 30 and 70, because the locations are now different.

For staging scenarios, where the license plate you move the goods from or the license plate you move the goods to, are set as a Target LP for a work order, only movement of the entire LP is allowed, so as not to break up the Target LP.

Only the ad hoc movement is currently supported. That means that you will not be able to move reserved inventory through the movement by template mobile device menu items.

Set up permission to move reserved inventory for individual workers

For the worker who is allowed to move reserved inventory, select the **Allow movement of inventory with work associated** check box under **Warehouse management** > **Setup** > **Worker**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Create a work class

Article • 09/29/2021

This procedure shows you how to set up a work class. Work classes are used to direct and/or limit the type of work order lines that a warehouse worker can process on a mobile device. The lines that a worker can process are determined from the work classes on the mobile device menu items that the warehouse worker has access to and the work class that's specified on the work lines. Work classes can also be used to validate the put location for a work order line. You can run this procedure in demo data company USMF or on your own data. This procedure is intended for the warehouse manager.

1. Go to Warehouse management > Setup > Work > Work classes.
2. Click New.
3. In the Work class ID field, type a value.
4. In the Description field, type a value.
5. In the Work order type field, select an option.
6. Click New.
7. In the Location type field, type a value.

    - If you select a location type, this sets a restriction on where items can be put after they've been picked. This setting is used when a location directive tries to resolve the location, or if a warehouse worker manually provides the location for the mobile device menu item.

8. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Manage warehouse workers

This article describes how to use the Warehouse Management mobile app to help control and monitor the work that employees carry out in your warehouses.

When you use the functionality in Warehouse management, all warehouse worker operations are referred to as *work*. You record work such as picking, moving, and counting on-hand inventory by using mobile devices. Before a warehouse worker can perform work, they must be associated with a worker in Human resources. Each **Worker** account can have multiple warehouse work users associated with it. Those work users can work in different warehouses and can have different levels of access to the various mobile device menus. You can think of the warehouse work users as multiple logons for the selected worker. Each work user has a default warehouse, and specific workflows are exposed by the menu items available to that work user.

To create a new work user, on the **Workers** page, on the **General** tab, in the **Warehouses** section, select **Worker**. You must specify a user ID, a user name, a default warehouse, and a menu name. This menu is loaded when the user signs in to the Warehouse Mobile Device Portal, and it lets you define which menu items the user has access to.

As part of the setup for each work user, you can also define specific process workflows. For example, you can use the **Is a cycle count supervisor** field to specify whether the user can process adjustments to cycle counting discrepancies during a counting operation, or whether these adjustments must first be reviewed by another person.

Defining labor standards

The **Labor standards** page lets you define the calculation methods that the system uses to calculate the estimated time that a particular type of work should require. You can set this definition on a generic level or on a specific level. For example, you can define the time that should be required to process a sales order pick per weight for a specific unit definition when a specific picking process is used. At the same time, you can record the time, based on another calculation method, for the put operation of the on-hand inventory that is picked.

To enable the labor standards that you defined, select the **Allow labor standards** option for each warehouse where labor standards are used.

Monitoring and controlling warehouse work

Use the **All work** page to monitor and maintain all work that is planned, in progress, and completed. From this page, you can update various processes, such as warehouse work user assignments and work priority. You can also drill down into the details that are related to the work header and work lines to gain an understanding of the expected or completed work processes.

If you enable the **Labor standards** option, you can see the calculated estimated time for the work. When the work is processed, the actual time is shown for each work operation. In this way, you can compare the estimated time calculations to the actual time.

You can use the estimated time in the rules for automatically splitting work during work creation. In this way, you can load balance work, based on the expected time to complete the tasks.

Analyzing the time used to process work items can help drive improvements in the warehouse workers' efficiency. The following reports are available to help with this analysis:

- **Labor by user** – This report shows worker productivity, based on actual times against expected times.
- **Labor by work transaction type** – You can use this report to investigate inefficiencies in specific warehouse processes. For example, you notice that picks for transfer orders are taking longer this week than in previous weeks. You can then use this information for further investigation.

Last updated on 07/01/2026

Deferred processing of warehouse work

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes the functionality that makes deferred processing of put operations for warehouse work available in Dynamics 365 Supply Chain Management.

The deferred processing functionality lets warehouse workers continue to do other work while the put operation is processed in the background. Deferred processing is useful when many work lines must be processed and the worker can let that work be processed asynchronously. It's also useful when the server can have ad-hoc or unplanned increases in processing time, and the increased processing time might affect the user's productivity.

Background processing is achieved by using the SysOperation framework. Learn more in SysOperation Framework Overview.

Configuring the work processing policies

To use deferred processing, you must configure and use a work processing policy.

Policies are configured on the **Work processing policies** page. The following table describes the fields on that page.

Expand table

Field	Description
Work processing policy name	The name of the work processing policy.
Work order type	The work order type that the policy is applied to.
Operation	The operation that is processed by using the policy.
Work processing method	The method that is used to process the work line. If the method is set to **Immediate**, the behavior resembles the behavior when no work processing policies are used to process

Field	Description
	the line. If the method is set to **Deferred**, deferred processing that uses the batch framework is used.
Deferred processing threshold	A value of **0** (zero) indicates that there is no threshold. In this case, deferred processing is used if it can be used. If the specific threshold calculation is below the threshold, the Immediate method is used. Otherwise, the Deferred method is used if it can be used. For sales and transfer-related work, the threshold is calculated as the number of associated source load lines that are being processed for the work. For replenishment work, the threshold is calculated as the number of work lines that are being replenished by the work. By setting a threshold of, for example, **5** for sales, smaller works that have fewer than five initial source load lines won't use deferred processing, but larger works will use it. The threshold has an effect only if the work processing method is set to **Deferred**.
Deferred processing batch group	The batch group that is used for processing.

For deferred put-processing, the following work order types are supported: sales order, transfer order issue, replenishment, and packed container picking. (Deferred put-processing for packed container picking work requires Supply Chain Management version 10.0.45 or higher.)

Assigning the work creation policy

The work creation policy can be assigned in the warehouse management parameters. It can also be assigned at the level of individual warehouses. If the policy is assigned to a warehouse, it's applied only to work that is created for that warehouse. If no policy is assigned at the warehouse level, the policy from the warehouse management parameters is applied.

Viewing the deferred put processing tasks

You can view deferred put processing tasks on the **Deferred put processing tasks** page.

By default, the **Completed** tasks are shown.

Expand table

Field	Description
Status	The status of the task.
Batch job status	The status of the related batch job. If the batch job has been processed, the batch history contains the log and information from the batch job.

Here is an explanation of the possible statuses:

- **Awaiting** – The related batch job is awaiting processing on the batch server.
- **Failed** – The processing failed. The task can be reprocessed by using the **Process deferred put** action, or it can be canceled by using the **Cancel deferred put** action.
- **Completed** – The job was completed.

Impact on closed work dates

When deferred put processing is used, the closed work date is set as the created date/time of the deferred put processing tasks. The closed work date is used because it's the best estimate for when the put operation was completed.

Reprocessing a failed task

You can reprocess a failed task by selecting it on the page and then selecting **Process deferred put**. Reprocessing corresponds to a situation where the user completes the put-away from the mobile device as if it was set for immediate processing.

Canceling failed tasks

Only failed tasks can be canceled. When you cancel a task, you can assign it to a new user. Alternatively, the task can remain assigned to the user who processed the work. Cancellation corresponds to a situation where the work is brought back to the mobile device as if the last pick step was just completed and the work must be put away. However, the user will be able to determine that the work is "different," because it will have only a put-away step, and the location will correspond to the location that the first user who processed the work had as a final put location.

When a task is canceled, the work is no longer blocked by the deferred put processing blocking reason. It can be reprocessed by using the mobile device.

The task record is deleted when the task canceled.

Configuring the mobile device menu to skip the deferred processing policy

You can configure the mobile device menu item so that the deferred processing policy isn't used. The work is then processed as it is when no deferred processing policy is used. This functionality lets a supervisor use a specific menu item where deferred put isn't used. To

configure it, set the **Deferred put processing policy** field to **Override settings and process put synchronously**.

Restrictions when the deferred put processing isn't applied

There are several scenarios where deferred put processing isn't applied even though the policy is configured. Here are some examples:

- Manual work completion is used.
- The work is completed by using auto-completion.
- Audit templates are used.

Monitoring the deferred processing tasks from the Outbound work monitoring workspace

The **Outbound work monitoring** workspace has two tiles that help you monitor deferred put processing tasks:

- **Failed deferred put processing tasks** – This tile shows the number of failed tasks. If there are failed tasks, the warehouse manager must either reprocess them or cancel them, because they won't be reprocessed automatically.
- **Awaiting deferred put processing tasks** – This tile shows the number of tasks that have been in the **Awaiting** status for more than 10 minutes. If the tile shows a number, it might indicate that an issue occurred during batch processing. You can manually process the **Awaiting** tasks. If the batch job for a task is processed later, it will just fail, because it has already been processed. There will be no impact.

Deleting completed tasks

You can delete deferred put processing tasks that have been completed by selecting them and deleting them on the page.

Related information

- Deferred processing of manual inventory movement operation

Last updated on 07/16/2025

Deferred processing of manual inventory movement

This article describes how to use deferred processing of manual inventory movement in Microsoft Dynamics 365 Supply Chain Management.

Deferred processing lets warehouse workers continue to do other work while a put operation is processed in the background. Deferred processing is useful when the server can have occasional or unplanned increases in processing time, and the increased processing time might affect worker productivity. The *Inventory movement* work type has now been added to the set of work types that this feature supports.

Background processing is achieved by using the Process warehouse app events feature.

Configure the work processing policies

To use deferred processing, you must set up and use a work processing policy. For deferred put processing, the Deferred processing of warehouse work feature supports the following work types: *Sales order*, *Transfer order issue*, and *Replenishment*. The *Deferred processing of manual inventory movement operation* features adds a new work type: *Inventory movement*.

To set up a work processing policy, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Work** > **Work processing policies**.
2. Either select an existing policy in the list, or create a new policy by selecting **New** on the Action Pane. The header of every policy has the following fields:
    - **Work processing policy name** – The name of the work processing policy.
    - **Description** – A description of the work processing policy.
3. On the **Processing rules** FastTab, set up the collection of rules that the policy will apply. Use the buttons on the toolbar to add or remove rules as you require. For each rule, set the following fields:
    - **Work order type** – Select the work type that the policy applies to.
    - **Operation** – Select the operation that the policy is used to process. If you selected *Inventory movement* in the **Work order type** field, you don't have to set this field, because both pick operations and put operations are processed as a single event.

- **Work processing method** – Select the method that is used to process the work line. If you select *Immediate*, the behavior resembles the behavior when no work processing policies are used to process the line. If you select *Deferred*, the system will apply deferred processing that uses the batch framework.
- **Deferred processing threshold** – If you set this field to *0* (zero), there is no threshold. In this case, the *Deferred* processing method is used if possible. If the specific threshold calculation is below the threshold, the *Immediate* method is used. Otherwise, the *Deferred* method is used if possible. For sales and transfer-related work, the threshold is calculated as the number of associated source load lines that are being processed for the work. For replenishment work, the threshold is calculated as the number of work lines that are being replenished by the work. By setting a threshold of, for example, *5* for sales, you ensure that smaller works that have fewer than five initial source load lines won't use deferred processing, but larger works will use it. The threshold has an effect only if the **Work processing method** field is set to *Deferred*.
- **Deferred processing batch group** – Specify the batch group that is used for processing. If you selected *Inventory movement* in the **Work order type** field, you don't have to set this field, because the batch group is selected in the **Process warehouse app events** dialog box.

Assign the work creation policy

For details about how to assign a work creation policy, see Deferred processing of warehouse work.

Set up a batch job to process warehouse app events

To use the *Deferred processing of manual inventory movement operation* process, set up a scheduled batch job.

1. Go to **Warehouse management** > **Periodic tasks** > **Process warehouse app events**.
2. In the **Process warehouse app events** dialog box, on the **Run in background** FastTab, set the **Batch processing** option to *Yes*.
3. Select **Recurrence**, and set up a run schedule that meets the requirements of your business.
4. Select **OK** in each dialog box.

Inquire about the warehouse app events

You can view the event queue and event messages that the warehouse app generates by going to **Warehouse management** > **Enquiries and reports** > **Mobile device logs** > **Warehouse app events**.

The *Inventory movement* event messages will have a status of *Queued* when they are first created. This status indicates that the **Process warehouse app events** batch job will pick up the event messages and process them. When the status is updated to *Completed*, all the related events are deleted from the queue.

All the *Inventory movement* events are accumulated under one collection per event type and warehouse.

The batch job will process the warehouse app events according to the recurrence that is set up in the **Process warehouse app events** dialog box. Therefore, until a message is processed, you can find the warehouse ID by looking in the **Identifier** field.

The details of the message contain the details of the movement (for example, the "from" and "to" locations).

Learn more in Warehouse app event processing.

Configure the mobile device menu to skip the deferred processing policy

For details about how to configure the mobile device menu to skip the deferred processing policy, see Deferred processing of warehouse work.

Impact on closed work dates

For details about the impact on closed work dates, see Deferred processing of warehouse work.

Related information

- Deferred processing of warehouse work
- Warehouse app event processing
- Set up mobile devices for warehouse work

Last updated on 12/30/2025

Cross-docking from production orders to outbound docks

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to manage the process of cross-docking material that is being reported as finished from a production line to an outbound transportation dock.

Introduction

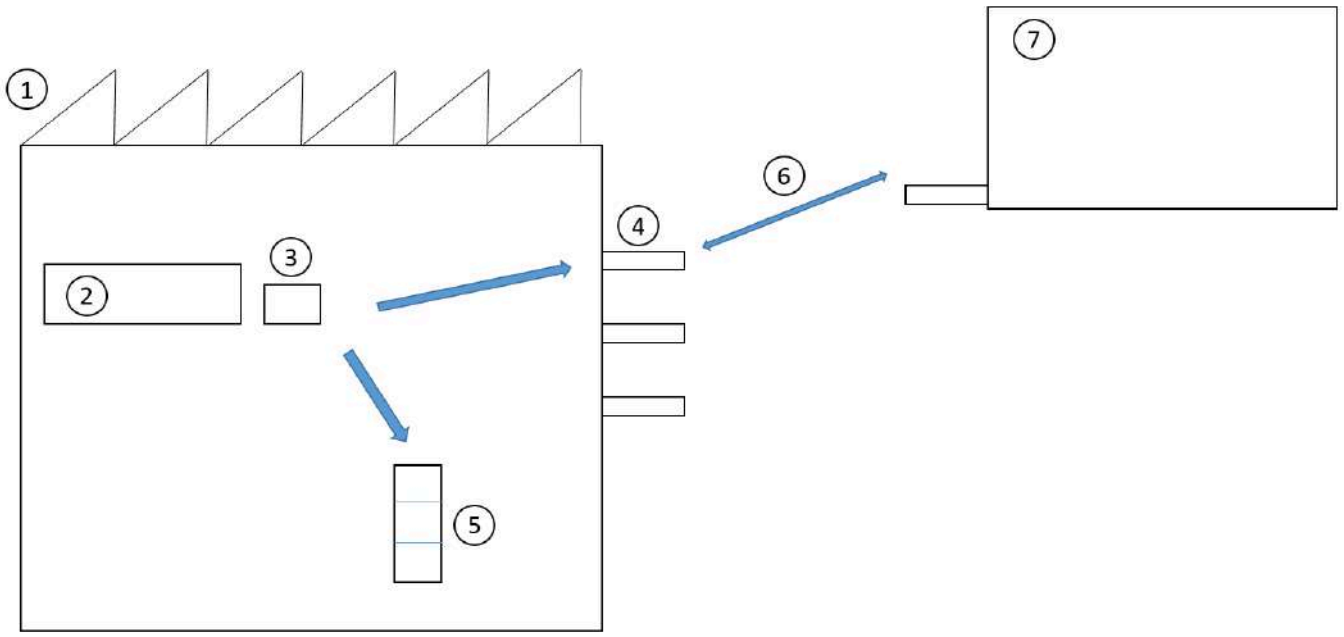
Cross-docking from production to an outbound location is relevant for manufacturers who produce high volume and ideally want to ship the finished products as soon as they are reported as finished from the production lines. The purpose is to ship the products to distribution centers that are physically located close to the customer demand, rather than build up inventory at the manufacturing site.

In case there is no immediate demand for a product, it must be put away to warehouse locations on the manufacturing site. This process is also known as *opportunistic cross-docking*, which indicates that if there is a demand for shipping the product, then this opportunity should be used instead of putting the product away for internal storage.

The following example shows three variations of a flow that starts at the end of the production line (2).

A product is reported as finished to the production output location (3) and a fork lift driver will pick up the pallet at this location (3).

- If there is a planned activity (6) for transferring the product from manufacturing (1) to a distribution center (7), then the truck driver will be directed by the system to put the pallet by a bay-door location (4).
- If a trailer is already assigned to the bay door, the truck driver will be directed to load the product directly to the trailer.
- If there is no planned activity for transferring the product, the fork lift driver will be directed to put the product away to a location in the internal warehouse (5).



Configure cross-docking

You configure the cross-docking process in **work policies**. A work policy includes a work order type, location, and product. In the following example, cross-docking is configured for product X and location Y.

Work order types

- Work order type: Finished goods put way
- Work creation method: Cross docking
- Cross docking policy name: Transfer orders

Inventory locations

- Warehouse: 51
- Location: Y

Products

- Item number: X

Currently, cross-docking can be configured for only two work order types:

- Finished goods put away
- Co-product and by-product put away

In the **cross-docking policy**, you define which document types are applicable for cross-docking. Currently, the only document type that is supported is **Transfer orders**. The following example shows the configuration of a cross-docking policy.

Cross-docking policy name: Transfer order

- Sequence number: 10
  - Work order type: Transfer issue
- Cross docking demand requires location: False
- Cross docking strategy: Date and time

Sequence number

The **sequence number** indicates the priority of the document type. Currently, **Transfer issue** is the only type that is supported. Therefore, the sequence number will become relevant only when more work order types are supported.

Cross-docking policy

The cross-docking policy also sets the policy for the prioritization of transfer order demand. For example, if multiple transfer orders exist for the same product, the scheduled date and time that are set on the load, and associated with the transfer order, determine the prioritization between the orders. The scheduled date and time can be set directly on the load, or they can be set on an **appointment schedule** that is associated with the load. The prioritization is determined by the cross-docking strategy. Currently, there is only one strategy: **Date and time**.

Cross-docking demand requires location

In the cross-docking policy, you can set up a criterion to require that transfer orders have an assigned location in order to be eligible for cross-docking. This criterion is set in the **Cross docking demand requires location** field. The location on the appointment schedule that is associated with the load is used as the final location for the goods that are being cross-docked. The final location for the goods that are being cross-docked is determined by the location directive for **Transfer issue** for the **Put** work order type. You might find it useful to set the **Cross docking demand requires location** field in a scenario where the finished goods should be cross-docked only if a trailer is assigned to a bay door. In this scenario, the goods are moved directly from the production line into the trailer. When a trailer is assigned to the bay door, a user will assign the location to the appointment schedule and will therefore make the location applicable for cross-docking. The following sections walk you through two examples.

Scenario 1 – Cross-docking from production to transfer orders

After a product is reported as finished at the production line it is transferred to a bay-door location where it is loaded to a truck and transferred to a distribution center. Use company USMF.

1. Enable a new number sequence for cross-docking. Go to the **Number sequences** page, and select the **Generate** button. A wizard will guide you through the process.
2. Create a cross-docking policy. Go to the **Cross docking policy** page, and create a new policy that is named **Cross docking to transfer order**. Note that the only work order type that you can select is **Transfer issue**, and the only cross-docking strategy that is available is **Date and time**.
3. Create a work policy. Go to the **Work policies** page, and create a new work policy that is named **Cross Dock L0101**.
4. Set up loads so that they are created automatically for transfer orders. In the warehouse parameters, set up loads so that they are created automatically when transfer orders are created. A load is a prerequisite for making the transfer order eligible for cross-docking.
5. Set up the item load mapping. Go to the **Item load mapping** page, and set up a standard load template for the **CarAudio** item group. This mapping will automatically insert the load template on the load when the transfer order is created.
6. Create a transfer order. Create the transfer order for item number L0101. Quantity = 20.
7. Release the transfer order from the outbound load planning workbench. On the **Ship** tab, select the menu item for the outbound load planning workbench and on the **Release** menu of the load line, select **Release to warehouse**. An open wave line of type **Transfer issue** now exists for the transfer order.
8. Create a production order. Go to the **Production order** list page, and create a production order for product L0101. Quantity = 20. Estimate and start the production order. Note that the **Post picking list now** field remains set to **No**.
9. Report as finished from the mobile device. Go to the mobile device portal and select menu item **Report as finished and put away**. Now report as finished L0101 from the handheld device. Quantity = 10. Note that the put location is **BAYDOOR**. This location is found from the **Transfer issue** location directive for the **Put** work order type. Also notice that work of the type **Transfer issue** has been created and completed. Go to the transfer order work details to verify the work.
10. Now report additional 10 pieces from the mobile device. Note that the put location again is **BAYDOOR**. Also notice that a new work of type **Transfer issue** has been created for the 10 pieces.
11. Now try to start 20 pieces more on the production order and then try to report 20 ea as finished by using the handheld device. This time, location **LP-001** is suggested as the put location. This location is found from the location directive for **Finished goods put away**. This location directive is being used, because no opportunity for cross-docking exists. The

transfer order for LP-001 was completely fulfilled by the two cross-docking activities in step 9 and 10. Notice that work of the type **Finished goods put away** was created and processed.

Scenario 2 - Cross-docking from production to transfer orders with an appointment schedule

After a product is reported as finished at the production line it is transferred to a bay-door location that is identified by an appointment schedule for the bay-door locations. Use company USMF.

1. Change the cross-docking policy. Change the cross-docking policy that you created in scenario 1 by selecting the **Cross docking demand requires location** check box.
2. Create a new transfer order.
3. Open the **Outbound load planning workbench**.
4. From the outbound load planning workbench, go to the **Loads** section, and select **Appointment schedule** on the **Transportation** menu to create a new appointment schedule. Note that the appointment schedule has a reference to the transfer order in the **Order number** field. In the **Planned start date/time at location** field, you can set the date and time for the appointment. This date and time will be used when cross-docking demand is prioritized during the cross-docking process. The date and time that you set in this field will update the **Scheduled load shipping date and time** field on the corresponding load. The location on the **Shipping details** FastTab determines the location that the transfer order is shipped on.
5. On the **Outbound load planning workbench** release to the warehouse.
6. Create a production order for item number **L0101**, and set the status to **Started**, with a quantity of 20.
7. Report as finished from the mobile device.
8. Go to the mobile device portal, and select the **Report as finished and put away** menu item.
9. Report item number **L0101** as finished from the handheld device. Note that the put location is now **BAYDOOR 2**. This location is found from the appointment schedule instead of the **Transfer receipt** location directive.

Related information

- The cross docking scenario is supported for batch and serial controlled items, both with the batch and serial number dimensions defined above and below location in the reservation hierarchy.

Last updated on 05/07/2025

Auto-release shipment for cross-docking

Article • 06/28/2022

This article describes a cross-docking strategy that lets you automatically release a demand order to the warehouse when the production order that supplies the demand quantity is reported as finished. In this way, the quantity that is required for fulfillment of the demand order is moved directly from the production output location to the outbound location.

Cross-docking is a warehouse handling flow where the quantity that is required to fulfill an outbound order is directed to the order's outbound dock or staging area from the location where the inbound order was received. (The inbound order can be a purchase order, a transfer order, or a production order.) Whereas the Advanced cross-docking feature supports all supply and demand orders, and it requires that the outbound demand be released before the cross-dock opportunity is identified, the Auto-release shipment feature has these characteristics:

- It supports only production orders as supply, and only sales orders and transfer orders as demand.
- The cross-docking operation can be started even if the demand order wasn't released to the warehouse before the supply receipt was registered (that is, before the production was reported as finished).

This cross-docking functionality has two advantages:

- The warehouse operations can skip the step of putting away quantities of finished goods to the regular warehouse storage area if those quantities will just be picked up again to fulfill the outbound order. Instead, the quantities can be moved one time, from the output location to a packing/shipping location. In this way, the functionality helps minimize the number of times that stock is handled and, ultimately, helps maximize time and space savings on the warehouse shop floor.
- The warehouse operations can postpone the release of sales orders and transfer orders to the warehouse until the output of finished goods for the associated production order is reported as finished. This advantage might be especially relevant in make-to-order production environments, where manufacturing lead times tend to be longer than the lead times in make-to-stock environments.

Prerequisites

Expand table

Prerequisite	Description
Item	The item must be enabled for warehouse management processes (WMS). **Note:** Catch-weight-enabled items can't be included in the cross-docking processes.
Warehouse	The warehouse must be enabled for warehouse management processes (WMS).
Cross-docking templates	At least one cross-docking template that uses the **At supply receipt** demand release policy must be set up for a given warehouse.
Work class	A cross-docking work class ID must be created for the **Cross docking** work order type.
Work templates	Work templates of the **Cross docking** work order type are required to create cross-docking pick and put work.
Location directives	Location directives of the **Cross docking** work order type are required to guide put work in the locations where sales order quantities will be packed and shipped.
Marking between a demand order and a production order	The warehouse system can trigger automatic release of the outbound order shipment and create cross-docking work from the output location at the report-as-finished action only if sales orders and transfer orders are reserved and marked against a production order.

Example cross-docking flow

A typical cross-docking flow consists of the following main steps.

1. A sales order taker creates a sales order for a make-to-order product. Typically, the requested quantity isn't on hand and must first be produced.

2. The sales order taker creates a production order directly from the sales order line. In this way, the sales order taker reserves and marks the sales order quantity against the production order quantity.

   Alternatively, a production planner specifies that the marking must be updated when planned orders are being firmed.

3. The production order goes through the following steps:
   a. A production planner estimates and releases the production order. (Estimation includes raw material reservation, and the release includes the release to a

warehouse.)
   b. A warehouse worker starts and completes raw material picking from the storage location to the production input location, according to the production pick work.
   c. A shop floor operator starts the production order.
   d. In the last operation, a machine operator uses the mobile device to report the production order as finished.

4. The system uses the setup to identify the cross-docking event for the two linked orders and then completes these tasks:
   a. Automatically release the associated sales order to a warehouse to create a shipment.
   b. Automatically create cross-docking work that has instructions to pick the finished goods from the output location and move them to the outbound location that the cross-docking location directives assigned to the sales order.
   c. Prompt a machine operator to complete the cross-docking work immediately after the production order is reported as finished.

5. After the cross-docking work is completed, and quantities are loaded onto the vehicle, an outbound warehouse planner confirms the sales shipment.

Example scenario

For this scenario, you must have demo data installed, and you must use the **USMF** demo data company.

Set up cross-docking that uses the auto-release shipment feature

Cross-docking template

1. Go to **Warehouse management** > **Setup** > **Work** > **Cross docking templates**.
2. Select **New**.
3. In the **Sequence number** field, enter **1**.
4. In the **Cross docking template ID** field, enter a name, such as **XDock_RAF**.
5. In the **Demand release policy** field, select **At supply receipt**.

6. In the **Warehouse** field, enter the number of the warehouse where you want to set up the cross-docking process. For this example, select **51**.

> ⓘ Note
>
> As soon as you select **At supply receipt** as the demand release policy for the template, all other fields on the page become unavailable. Likewise, you can't define any supply sources. This behavior occurs because cross-docking that uses the auto-release shipment feature supports only production orders as supply sources, and it requires that a marking exist between sales orders and production orders. If you select **Before supply receipt** as the demand release policy, the fields on the **Planning** and **Supply sources** tabs are available and can be edited.

Work classes

1. Go to **Warehouse management** > **Setup** > **Work** > **Work classes**.
2. Select **New**.
3. In the **Work class ID** field, enter a name, such as **CrossDock**.
4. In the **Work order type** field, select **Cross docking**.

To limit the types of locations where cross-docked finished goods can be put, you can specify one or more valid location types.

Work templates

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. In the **Work order type** field, select **Cross docking**.
3. Select **New**.
4. In the **Sequence number** field, enter **1**.
5. In the **Work template** field, enter a name, such as **CrossDock_51**.
6. Select **Save**.
7. In the **Work Template Details** section, select **New**.
8. For the new line, in the **Work type** field, select **Pick**. In the **Work class ID** field, select **CrossDock**.
9. Select **New**.
10. For the new line, in the **Work type** field, select **Put**. In the **Work class ID** field, select **CrossDock**.

Location directives

A standard put-away process for finished goods requires a **Put** location directive to guide the movement of picked production quantities to a regular storage space. Likewise, you must set up a cross-docking **Put** location directive to instruct the work to put the finished quantity in a designated outbound location that services the shipment of the associated sales order.

For cross-docking, as for regular put-away of finished goods, you don't have to create a location directive for the pick work action, because the output location is given. Additionally, this output location is expected to be set up either as the default output location on one of the resource-related records (that is, the resource, resource group relation, or resource group) or as a default production finished goods location for a warehouse.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the **Work order type** field, select **Cross docking**.
3. Select **New**.
4. In the **Sequence number** field, enter **1**.
5. In the **Name** field, enter a name, such as **Baydoor**.
6. In the **Work type** field, select **Put**.
7. In the **Site** field, select **5**.
8. In the **Warehouse** field, select **51**.
9. On the **Lines** FastTab, select **New**.
10. In the **To quantity** field, enter the maximum quantity of the range, such as **1000000**.
11. Select **Save**.
12. On the **Location Directives Actions** FastTab, select **New**.
13. In the **Name** field, enter a name, such as **Baydoor**.
14. Select **Save**.
15. You can use the standard query facility to limit put locations to one or more specific locations. Select **Edit query**, and select **51** as the criterion for the **Warehouse** field in the **Locations** table.

Cross-dock finished goods to the outbound location

To cross-dock the quantity of finished goods to the outbound location of the associated sales order, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. Select **New**.

3. For the sales order header, select customer account **US-001** and a warehouse that is set up for cross-docking that uses the auto-release shipment feature.

4. Add a line for a finished product, and enter **10** as the quantity.

5. In the **Sales order lines** section, select **Product and supply** > **Production order**.

6. In the **Create production order** dialog box, review the default values, and then select **Create**. A new production order is created and linked to the sales order (that is, it's reserved and marked).

7. Optional: Change the value of the **Quantity** field so that it's more than the value that is required to fulfill the sales order. When the production quantity is reported as finished, the system will create cross-docking work for the marked quantity and put-away work for the remaining quantity, according to the regular procedure for handling the put-away of finished goods.

   As was mentioned earlier, a marking must exist between the sales order and the production order. Otherwise, the cross-docking won't work. A marking can be created in the multiple ways:

   - The system can automatically create the marking in the following situations:
     - The production order is manually created directly from the sales order line (see step 6).
     - Planned production orders are firmed, and the **Update marking** field is set to **Standard**.
   - The user can manually create the marking by opening the **Marking** page from the demand order line.

   > ⓘ **Note**
   >
   > A marking can't be manually created for items that are set up to use standard and weighted average as inventory models.

8. On the **Production order** page, on the Action Pane, on the **Production order** tab, in the **Process** group, select **Estimate**, and then select **OK**. The order is estimated, and the raw material quantity is reserved for the production.

9. On the Action Pane, on the **Production order** tab, in the **Process** group, select **Release**, and then select **OK**. Warehouse pick work is created for the raw materials.

10. Open and review the work. On the Action Pane, on the **Warehouse** tab, in the **General** group, select **Work details**. Make a note of the work ID.

11. Sign in to the Warehouse Management mobile app to run work in warehouse 51.
12. Go to **Production** > **Production pick**.
13. Enter the work ID to start and complete the raw material picking.

    After the work is reported as finished, the quantity of raw materials is available in the production input location (**005** in USMF demo data), and execution of the production order can start.
14. On the **Production order** page, on the Action Pane, on the **Production order** tab, in the **Process** group, select **Start**, and then select **OK**.
15. In the app, go to **Production** > **RAF and put away**.
16. In the **Prod ID** field, enter the production order number and other mandatory details, and then select **OK**.

Notice that the following events occur:

- The release to a warehouse is triggered for the linked sales order.
- Based on the release, shipment and cross-docking work is created. This work instructs the warehouse operator to pick the quantities that are required to fulfill the sales order line and put them in the outbound location that specified in the cross-docking location directive.
- If the production order quantity is more than the quantity that is required by the sales order, regular put-away work is created. This work instructs the warehouse operator to pick the quantity of finished goods that remains after cross-docking and move it to regular storage, according to the location directive.

Feedback

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Planned cross-docking

06/17/2025

This article describes advanced planned cross-docking. Cross-docking is a warehouse process where the inventory quantity that is required for an order is directed straight from receipt or creation to the correct outbound dock or staging area. All remaining inventory from the inbound source is directed to the correct storage location through the regular put-away process.

Cross-docking lets workers skip inbound put-away and outbound picking of inventory that is already marked for an outbound order. Therefore, the number of times that inventory is touched is minimized, where possible. Additionally, because there's less interaction with the system, time and space savings on the warehouse shop floor are increased.

Before you can run cross-docking, you must configure a new cross-docking template, where the supply source and other sets of requirements for cross-docking are specified. As the outbound order is created, the line must be marked against an inbound order that contains the same item. You can select the directive code field on the cross-docking template, similar to the way you set up replenishment and purchase orders.

At the time of inbound order receiving, the cross-docking setup automatically identifies the need for cross-docking and creates the movement work for the required quantity, based on the setup of the location directive.

> ⓘ **Note**
>
> Inventory transactions are *not* unregistered when crossing-dock work is canceled, even if the setting for this capability is turned on in Warehouse management parameters.

Setup

Regenerate load posting methods

Planned cross-docking is implemented as a load posting method. After you turn on the feature, you must regenerate the methods.

1. Go to **Warehouse management** > **Setup** > **Load posting methods**.
2. On the Action Pane, select **Regenerate methods**.

When regeneration is completed, you should see a method that has a **Method name** value of *planCrossDocking*.

3. Close the page.

Create a cross-docking template

1. Go to **Warehouse management** > **Setup** > **Work** > **Cross docking templates**.
2. On the Action Pane, select **New** to create a template.
3. In the header, set the following values:

    - **Sequence:** *1*

      This field defines the order that templates are evaluated in.

    - **Cross docking template ID:** *51*
    - **Description:** *Warehouse 51*
    - **Demand release policy:** *Before supply receipt*

4. Use the **Warehouse selection** fast tab to specify the warehouse where the cross docking template will apply.

    - **Warehouse selection** – Select one of the following values:
        - *All* – Use the cross docking template for all warehouses.
        - *Warehouse group* – Use the cross docking template for all warehouses in the warehouse group that's selected in the **Warehouse group** field.
        - *Warehouse* – Use the cross docking template only for the specific warehouse that's selected in the **Warehouse** field.
    - **Warehouse** – If the **Warehouse selection** field is set to *Warehouse*, select the warehouse where the cross docking template applies.
    - **Warehouse group** – If the **Warehouse selection** field is set to *Warehouse group*, select the warehouse group where the cross docking template applies. For more information about how to set up warehouse groups, see Warehouse groups.

    For this scenario, set the following values:

    - **Warehouse selection:** *Warehouse*
    - **Warehouse:** *51*

5. The setup on the **Planning** FastTab controls how the template works. Set the following values:

    - **Demand requirements:** *None*

      This field defines the requirements of the demand inventory. If the demand must be linked to the supply before release, select *Marking*. If the demand must be order-reserved against the supply before release, select *Order reservation*.

    - **Locating type:** *Shipment locations*

      This field defines whether the cross-docking work should use the staging/load locations from the shipment, or whether it should use location directives to find its own staging/load locations.

    - **Work template:** Leave this field blank.

      This field defines the work template that should be used when cross-docking work is created.

    - **Revalidate on supply receipt:** *No*

      This option defines whether the supply should be revalidated during receipt. If this option is set to *Yes*, both the maximum time window and the expiration days range are checked.

    - **Directive code:** Leave this field blank

      This option is enabled by the *Cross docking templates with location directives* feature (as of Supply Chain Management version 10.0.29, the feature is turned on by default). The system uses location directives to help determine the best location to move cross-docking inventory to. You can set it up by assigning a directive code to each relevant cross-docking template. If a directive code is set, the system will search location directives by directive code when work is generated. In this way, you can limit location directives that are used for a particular cross-docking template.

    - **Validate time window:** *Yes*

      This option defines whether the maximum time window should be evaluated when a supply source is selected. If this option is set to *Yes*, the fields that are related to the maximum and minimum time windows become available.

    - **Maximum time window:** *5*

      This field defines the maximum period that is allowed between supply arrival and demand departure.

- **Maximum time window unit:** *Days*
- **Minimum time window:** *0*

  This field defines the minimum period that is allowed between supply arrival and demand departure.

- **Minimum time window unit:** *Days*
- **Expiration days range:** *0*

  First expiry first out (FEFO) criteria: This field defines the maximum number of days between the expiration date of the first-expiring batch that is currently in the warehouse and the batch that is being received.

6. On the **Supply sources** FastTab, you specify the types of supply that are valid for this template. Select **New**, and then set the following values:

   - **Sequence number:** *1*
   - **Supply source:** *Purchase order*

> ⓘ **Note**
>
> You can set up a query to control when a specific cross docking template is used. The query for cross docking templates has only the `InventTable` (standard items) table and the inner-joined `WHSInventTable` (warehouse management processes (WMS) items) table. If you want to add other tables to the query, you can join them by using only *exist joins* or *not exist joins*. When you filter on the joined tables, a record from the main table is retrieved for each matching record in the joined table. If the join type is *exist join*, the search ends after the first match has been found. For example, if you join the sales order line table to the items table, the system validates and returns items for which at least one sales order line has the defined condition. Essentially, the data is fetched from the parent (items) table, not from the child (sales order line) table. Therefore, filtering by source documents such as sales order lines or customers can't be done out of the box.

Create a work class

1. Go to **Warehouse management** > **Setup** > **Work** > **Work classes**.
2. On the Action Pane, select **New** to create a work class.
3. Set the following values:

- **Work class ID:** *CrossDock*
- **Description:** *Cross Dock*
- **Work order type:** *Cross docking*

Create a work template

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. Set the **Work order type** field to *Cross docking*.
3. On the Action Pane, select **New** to add a line to the **Overview** tab.
4. On the new line, set the following values:
    - **Sequence number:** *1*
    - **Work template:** *51 Cross Dock*
    - **Work template description:** *51 Cross Dock*
5. Select **Save** to make the **Work Template Details** FastTab available.
6. On the **Work Template Details** FastTab, select **New** to add a line to the grid.
7. On the new line, set the following values:
    - **Work type:** *Pick*
    - **Work class ID:** *CrossDock*
8. Select **New** to add another line, and set the following values on it:
    - **Work type:** *Put*
    - **Work class ID:** *CrossDock*
9. Select **Save**, and confirm that the **Valid** check box is selected for the *51 Cross Dock* template.
10. Optional: Select **Edit query** if you want to set criteria to control when and where the work template is used.

    You can set up a query to control when a specific work template is used. For example, you can specify that a template can be used for work only at a specific location. If you want the cross docking work template to be applied at a specific location, you must filter on the **Start location** field, not the **Location** field, because work creation for the inbound processes (purchase, cross docking, and replenishment) starts from the put line. When work is created, the location directive sets the **Location** field to the put location. However, the pick location is stored in the **Start location** field.

> ⓘ **Note**
>
> The work class IDs for the *Pick* and *Put* work types must be the same.

Create location directives

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the left pane, set the **Work order type** field to *Cross docking*.
3. On the Action Pane, select **New**, and set the following values:
    - **Sequence number:** *1*
    - **Name:** *51 Cross Dock Put*
    - **Work type:** *Put*
    - **Site:** *5*
    - **Warehouse:** *51*
4. Select **Save** to make the **Lines** FastTab available.
5. On the **Lines** FastTab, select **New** to add a line to the grid.
6. On the new line, set the following values:
    - **From quantity:** *1*
    - **To quantity:** *1,000,000*
7. Select **Save** to make the **Location Directive Actions** FastTab available.
8. On the **Location Directive Actions** FastTab, select **New** to add a line to the grid.
9. On the new line, set the following values:
    - **Name:** *Baydoor*
    - **Fixed location usage:** *Fixed and non-fixed locations*
10. Select **Save** to make the **Edit query** button on the **Location Directive Actions** toolbar available.
11. Select **Edit query** to open the query editor.
12. On the **Range** tab, make sure that the following two lines are configured:
    - Line 1:
        - **Table:** *Locations*

    - **Derived Table:** *Locations*
    - **Field:** *Warehouse*
    - **Criteria:** *51*

- Line 2:
    - **Table:** *Locations*
    - **Derived Table:** *Locations*
    - **Field:** *Location*
    - **Criteria:** *Baydoor*

13. Select **OK** to close the query editor.

Create a mobile device menu item

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. In the list of menu items in the left pane, select **Purchase Put-away**.
3. Select **Edit**.
4. On the **Work classes** FastTab, select **New** to add a line to the grid.
5. On the new line, set the following values:

    - **Work class ID:** *CrossDock*
    - **Work order type:** *Cross docking*

6. Select **Save**.

Scenario

Create a purchase order

Follow these steps to create a purchase order as a source of supply.

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. On the Action Pane, select **New**.
3. In the **Create purchase order** dialog box, set the following values:

    - **Vendor account:** *104*
    - **Warehouse:** *51*

4. Select **OK**, and make a note of the order number.

5. A new line is added to the **Purchase order lines** FastTab. On this line, set the following values:

    - **Item number:** *A0001*
    - **Quantity:** *5*

Create a sales order

Follow these steps to create a sales order as a source of demand.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In the **Create sales order** dialog box, set the following values:

    - **Customer account:** *US-002*
    - **Warehouse:** *51*

4. Select **OK**.
5. A new line is added to the **Sales order lines** FastTab. On this line, set the following values:

    - **Item number:** *A0001*
    - **Quantity:** *3*

Create planned cross-docking

Follow these steps to create the planned cross-docking from the sales order.

1. In the **Sales order details** page for the sales order that you just created, on the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.

    The release to warehouse action creates a shipment and load line for the sales order line, and tries to allocate inventory.

    You receive an informational message. You also receive the following warning message: "No work was created for wave XXXX. See the work creation history log for details." This behavior is expected, because there's no inventory in the warehouse.

2. On the **Sales order lines** FastTab, on the **Warehouse** menu, select **Shipment details**.

    The **Shipment details** page appears and shows the shipment that was created for the sales order.

3. On the **Load lines** FastTab, notice that the **Planned cross docking quantity** field is set to *3*. Because no inventory was available in the warehouse, but a valid supply source will arrive within the time window that is defined in the cross-docking template, the cross-docking quantity was created.

4. On the **Load lines** FastTab, select **Planned cross docking** to view the details of the cross-docking that was created.

Process the cross-docking

Purchase order receiving on the warehousing mobile app

The system will receive the quantity of 5 from the purchase order into the receiving location and create two pieces of work.

The first work ID that is created has a **Work order type** value of *Cross docking* and is linked to the sales order. It has a quantity of 3 and is directed to the final shipping location so that it can be shipped out immediately.

The second work ID that is created has a **Work order type** value of *Purchase orders* and is linked to the purchase order. It has the remaining quantity of 2 that wasn't cross-docked and is directed to put-away to storage.

1. Sign in to the mobile device as a user in warehouse *51*.

2. Go to **Inbound** > **Purchase Receive**.

3. In the **PONum** field, enter your purchase order number.

4. In the **Qty** field, enter *5*.

5. Select **OK**.

6. On the next page, set the **Item** field to *A0001*.

7. Select **OK**.

8. On the next page, confirm the **PONum**, **Item**, and **Qty** values by selecting **OK**.

   You receive a "Work Completed" message.

9. Select **Cancel** to exit.

Put-away to cross-docking and bulk

Currently, both work IDs have the same target license plate. To complete the next steps, you must get the work ID and the target license plate ID. You can get this information from the work details for the purchase order line and the sales order line. Alternately, you can go to **Warehouse management** > **Work** > **Work details** and filter for work where the **Warehouse** value is *51*.

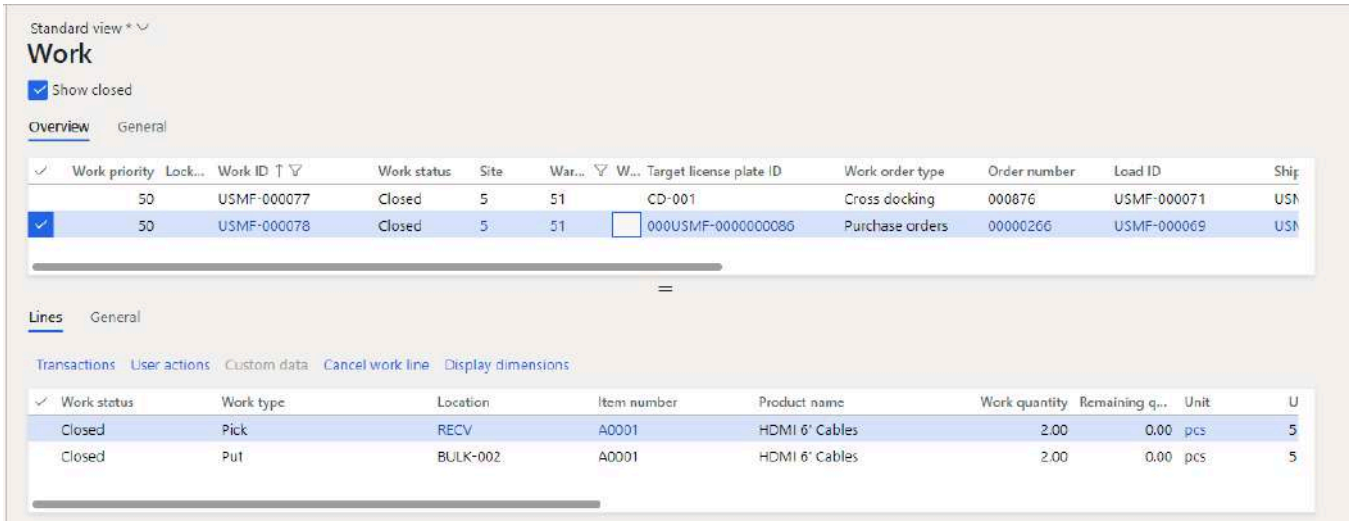
1. On the mobile device, go to **Inbound** > **Purchase put-away**, and enter the target license plate from the work.
2. In the **ID** field, enter the target license plate ID from the work details.

   The cross-docking pick page shows the picking location (*RECV*), target license plate (*license plate*), item (*A0001*), and quantity (*3*).
3. Select **OK**.
4. In the **Target LP** field, enter a target license plate for the license plate ID that should be put (cross-docked) to the shipping location. You can select any license plate ID of your choice.
5. Select **OK**.
6. On the next page, in the **ID** field, enter the target license plate ID.
7. Select **OK**.
8. Confirm the work for picking the remaining quantity of 2, and then select **OK**.
9. On the next page, select **Done** to end the picking process and begin the put-away process.

   The mobile app presents you with the location and license plate to put the item to.
10. Confirm the bulk storage **Put** by selecting **OK**.
11. On the next page, confirm the cross-docking **Put** by selecting **OK**.

    You receive a "Work Completed" message.
12. Select **Cancel** to exit.

The following illustration shows how the completed cross-docking work might appear in Microsoft Dynamics 365 Supply Chain Management.



Note

If you're using warehouse management only mode (which uses inbound and outbound shipment orders), use the source system setup to define whether the release to warehouse process should convert cross docking requests (*Outbound shipment order line cross docking*) to markings before it evaluates planned cross-docking templates where the **Demand requirements** field is set to *Marking*.

To use this logic, specify the requested cross docking quantities as part of the *Outbound shipment order line cross docking* message data (`OutboundShipmentOrderLineCrossDockMessages`) that's linked to the *Inbound shipment order lines* message data (`InboundShipmentOrderLineMessages`) by setting the same value in the `CrossDockSourceID` field. You can monitor this data on the **Cross dock requests** tab on the **Line details** FastTab of the **Outbound shipment orders** page.

Reason codes for inventory counting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Reason codes help you analyze the results of a counting process and any discrepancies that occur during that process. You can specify the reason for doing the count, such as a broken pallet or a stock adjustment that is based on inventory samples. At the same time, use the adjustment functionality to post the value of on-hand inventory adjustments to the appropriate offset account, based on the reason for each inventory adjustment.

Recommendation

Before you set up the system, define a strategy for working with reason codes. For example, try to answer the following questions:

- Should reason codes be mandatory on warehouses?
- Should reason codes be mandatory or optional on some items?
- How many reason codes do you require?
- Do you need to preselect a limited list of reason codes for adjustments?
- How should users of bar code scanners use reason codes? Should the reason codes be preselected, mandatory, or not editable?
- Do warehouse workers require different reason code behavior on mobile scanners? If yes, you can create more menu items and assign them to different people.
- Should the reason codes drive financial offset account posting?

Set up reason codes

Set up reason code policies

Create multiple reason code policies to control when and how counting reason codes are applied. Each reason code policy can have one of two counting reason code types (*Optional* or

Mandatory). Use counting reason code policies at the warehouse level or the item level.

To create a reason code policy, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory** > **Counting reason code policies**.
2. On the Action Pane, select **New** to add a policy to the grid.
3. Set the **Name** field for the new policy.
4. In the **Counting reason code type** field, select either *Mandatory* or *Optional* to specify whether selection of a reason code should be optional or mandatory in one of the following inventory adjustment processes:
    - Cycle Count (mobile device)
    - Spot Count (mobile device)
    - Threshold Count (mobile device)
    - Adjustment In (mobile device)
    - Adjustment Out (mobile device)
    - Counting Journal (rich client)
    - Quantity adjustment/Online counting (rich client)

Set up reason code policies both for individual warehouses and for products. The reason code setup for a product can overrule the setup for the product's warehouse.

> ⓘ **Note**
>
> For warehouses and items where the **Counting reason code policy** field is set to *Mandatory*, the counting journal can't be completed and closed until a reason code is provided. For more information, see the next section.

Assign counting reason code policies to warehouses

To assign a counting reason code policy to a warehouse, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**.
2. Select a warehouse in the list pane.
3. On the Action Pane, on the **Warehouse** tab, in the **Set Up** group, select **Counting reason code policy**. Then, in the **Assign counting reason code policy** drop-down dialog box, follow

one of these steps:

- To use the policy setup for each item to determine whether counting journals are mandatory for it, enter no value (or delete the existing value).
- To require a reason code on counting journals for the warehouse, select a reason policy where the **Counting reason code type** field is set to *Mandatory*.
- If a reason code is optional on counting journals for the warehouse, select a reason policy where the **Counting reason code type** field is set to *Optional*.

Assign counting reason code policies to products

To assign a counting reason code policy to a product, follow these steps:

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a product in the grid.
3. On the Action Pane, on the **Product** tab, in the **Set Up** group, select **Counting reason code policy**. Then, in the **Assign counting reason code policy** drop-down dialog box, follow one of these steps:

    - To use the policy setup for the warehouse to determine whether counting journals are mandatory for the product, enter no value (or delete the existing value).
    - To require a reason code on counting journals for the product, select a reason policy where the **Counting reason code type** field is set to *Mandatory*. This setting overrides any reason code setting at the warehouse level.
    - If a reason code is optional on counting journals for the product, select a reason policy where the **Counting reason code type** field is set to *Optional*. This setting overrides any reason code setting at the warehouse level.

Set up counting reason codes

To set up your counting reason codes, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory** > **Counting reason codes**.
2. On the Action Pane, select **New** to add a row to the grid.
3. Set the **Counting reason code** and **Description** fields for the new row.
4. To assign an offset account, enter or select a value in the **Offset account** field.

> **Note**
>
> If you assign an offset account to a counting reason code, when you post a counting journal that uses the counting reason code, the value is posted against the assigned offset account instead of the default inventory posting profile account.

Set up counting reason code groups

Use *counting reason code groups* as part of the *Adjustment in* and *Adjustment out* menu items in the Warehouse Management mobile app to limit the list of counting reason codes. For more information about counting reason code groups, see the Set up mobile device menu items for adjustment in and adjustment out section later in this article.

1. Go to **Inventory management** > **Setup** > **Inventory** > **Counting reason code groups**.
2. On the Action Pane, select **New** to add a group.
3. Set the **Counting reason group** and **Group description** fields for the new group.
4. On the Action Pane, select **Save**.
5. In the **Details** section, select **New** on the toolbar to add a row to the grid. Then set the **Counting reason code** field for the new row.
6. Repeat the previous step to assign more codes as required. To remove a code from the group, select it, and then select **Delete** on the toolbar.

Set up reason codes for mobile device menu items

You can configure reason codes for the following on-hand adjustment types:

- Cycle Count
- Spot Count
- Threshold Count
- Adjustment In
- Adjustment Out

In most cases, you can define the following information for each relevant mobile device menu item:

- Whether the reason code is shown to the mobile device worker during counting.
- The default reason code that is shown on a mobile device menu item.
- Whether the user can edit the reason code.

Set up mobile device menu items for a counting process

To set up a mobile device menu item for a counting process, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select the relevant menu item in the list pane, or create a new menu item.
3. On the Action Pane, select **Cycle counting**.
4. In the **Default counting reason code** field, set the default reason code that should be recorded when the mobile device menu item is used to do counting.
5. In the **Display counting reason code** field, select one of the following values:
    - *Line* – Show the reason code after each variance is recorded.
    - *Hide* – Don't show the reason code.
6. Set the **Edit counting reason code** to *Yes* to allow the worker to edit the reason code when it's shown on the mobile device during counting. Set it to *No* to prevent the worker from editing the code.

> ⓘ **Note**
>
> You can enable the **Cycle counting** button on any mobile device menu item where counting can be done. Examples include the menu items for spot counts, user-directed work, and system-directed work.

Set up mobile device menu items for adjustment in and adjustment out

To set up a mobile device menu item for adjustment in or adjustment out, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to create a menu item.
3. Set the **Mobile item name** and **Title** fields for the new menu item.
4. Set the **Mode** field to *Work*.
5. Set the **Use existing work** option to *No*.
6. In the **Work creation process** field, select *Adjustment in* or *Adjustment out*.

7. On the **General** FastTab, set the following fields. (All these fields are added when you select *Adjustment in* or *Adjustment out* in the **Work creation process** field.)

    - **Use process guide** – If you're creating an *Adjustment out* process, be sure to set this option to *Yes*. If you're creating an *Adjustment in* process, this option is always set to *Yes*.

    - **Default counting reason code** – Set the default reason code that should be recorded when the mobile device menu item is used to do counting.

    - **Display counting reason code** – Select one of the following values:
        - *Line* – Show the reason code after each variance is recorded.
        - *Hide* – Don't show the reason code.

    - **Edit counting reason code** – Set this option to *Yes* to allow the worker to edit the reason code when it's shown on the mobile device during counting. Set it to *No* to prevent the worker from editing the code.

    - **Counting reason code group** – Select a reason code group if you want to limit the list of options that is presented to workers. For information about how to set up reason code groups, see the Set up counting reason code groups section earlier in this article.

> **Note**
>
> When you assign a counting reason code group to *Adjustment in* and *Adjustment out* menu items where the **Use process guide** option is set to *Yes*, you get a limited list of the counting reason codes as part of the processing in the Warehouse Management mobile app.
>
> The **Use process guide** option can also help prevent large adjustment quantities from occurring by mistake. (For example, a worker might accidentally scan a bar code of an item number instead of a quantity value.) To set up this functionality, set the **Use process guide** option to *Yes* for each relevant menu item. Then go to **Warehouse management** > **Setup** > **Worker**, and set the **Adjustment quantity limit** field for each relevant warehouse worker to specify the maximum adjustment quantity that the worker can register.

Processing that uses counting reason codes

When workers use the Warehouse Management mobile app, the app records the reason codes. Unless a counting approval process is defined, the recorded reason codes are immediately used as part of the counting journal posting that follows.

Cycle count approvals

Before a count is approved, the worker can change the reason code that is associated with the count. When the count is approved, the reason code is entered on the counting journal lines.

Modify reason codes for cycle count approvals

To modify a cycle count approval, follow these steps:

1. Go to **Warehouse management** > **Cycle counting** > **Cycle count work pending review**.
2. Select a cycle count in the grid.
3. On the Action Pane, on the **Work** tab, select **Cycle counting**. Then, in the **Reason code** field, select a new reason code.

Reason codes are added to the journal lines in counting journals of the *Counting journal* type.

1. Go to **Inventory management** > **Journal entries** > **Item counting** > **Counting**.
2. In the line details of the counting journal, in the **Counting reason code** field, select the reason code that matches your current situation.

View the reason codes recorded in the counting history

To view the reason codes that are recorded in the counting history, follow these steps:

1. Go to **Inventory management** > **Inquiries and reports** > **Counting history**.
2. Select an item count record in the list pane.
3. In the **Counting reason code** field, view the counting history that's recorded through a reason code.

Use reason codes for quantity adjustment or online counting

To use a reason code for a quantity adjustment or online counting, follow these steps:

1. Go to **Inventory management** > **Inquiries and reports** > **On-hand list**.
2. On the Action Pane, select **Quantity adjustment**.
3. Select **Quantity adjustment**, and then, in the **Counting reason code** field, select a reason code.

Last updated on 05/21/2026

Inventory blocking

08/14/2025

This article describes how to work with inventory blocking, which is part of the quality inspection process in Supply Chain Management. You can use inventory blocking to prevent items from being processed or consumed.

You can block inventory items in the following ways:

- Manually
- By creating a quality order
- By using a process that generates a quality order
- By using inventory status blocking

Blocking items manually

You can block a quantity of an item by creating a transaction on the **Inventory blocking** page. Only items that are available as on-hand inventory can be blocked manually. For manually blocked quantities, you must decide whether planning activities include expected receipts as an expected on-hand quantity. Expected receipts are blocked items that you expect to be available as on-hand inventory after inspection is completed. You can maintain the expected date. By default, the **Expected receipts** option is selected for items that are blocked through a quality order. You can cancel a manual block on a quantity by deleting the transaction on the **Inventory blocking** page.

Blocking items by creating a quality order

You can specify items that must be inspected by creating a quality order on the **Quality orders** page. When you create a quality order, the quantity that you specify for an item is blocked. The sampling plan that is associated with a quality order controls only the quantity of items that must be inspected, not the quantity that is blocked. The quantity that is entered on the quality order is the quantity that is blocked, regardless of the quantity that the sampling plan specifies should be sent for inspection.

> **Note**
>
> Master planning doesn't support using both the batch expiry date and blocking inventory status features. This could result in double exclusion of on-hand inventory, which can occur during master planning. We recommend that you rely on batch disposition codes, instead of inventory status, for blocking expired batches.

Blocking items by using a process that generates a quality order

If a quality process specifies that an item must be inspected, a quantity of the item is blocked automatically. Therefore, when a quality order is generated automatically, the item sampling plan that is associated with the quality order controls both the quantity of items that is blocked and the quantity that must be inspected. If the **Full blocking** option on the **Item sampling** page is selected, the full quantity of, for example, a purchase order line is blocked during inspection, regardless of the item sampling quantity.

Example

In the following example, a quality order is generated when a purchase order packing slip is posted. On the **Quality associations** page, you specified that posting of a purchase order packing slip is the process that activates a quality order.

Expand table

Setup	User action	Result
A quality association specifies that a quality order must be generated when a purchase order packing slip is posted. The item sampling setup of the quality order specifies that 10 percent of the quantity on the purchase order line must be inspected. Furthermore, because the **Full blocking** option selected in the item sampling setup, the full quantity of the purchase order line must be blocked during inspection, regardless of the quantity that is sent for inspection.	The packing slip is posted.	A quality order is generated. Ten percent of the purchase order quantity for the item is sent to inspection. The full quantity of the purchase order line is blocked.

Blocking items by using inventory status blocking

You can specify which inventory statuses are blocking statuses by using the **Inventory blocking** parameter on the **Inventory statuses** page. You can't use inventory statuses as blocking statuses for production orders, sales orders, transfer orders, outbound transactions, or project integrations. For outbound work, use items that have an available inventory status. If items have a status of **Broken**, and master planning is run on those items, the items are considered missing, and inventory is automatically replenished.

Take care when blocking items that use both inventory status blocking and quality order blocking

You can create a quality order associated with inventory having an inventory status that has its **Inventory blocking** parameter enabled. In this case, the quality order generates an extra inventory blocking record in addition to the one created by the inventory status. Because the quality order inventory blocking will have the parameter **Expected receipts** enabled, this generates an extra *Ordered inventory* transaction that is also blocked by its inventory status. This combination can lead to difficulties in understanding the meaning of generated inventory transactions, because it will appear as though the total blocked quantity is in excess of the total quantity on hand. Let us examine the transactions on an example of a receipt of 10 pieces of item A0001 with a quality order generated to sample 1 piece. The behavior also depends on

whether the option **Reserve ordered items** is enabled on the **Inventory and warehouse management parameters** page.

> **Note**
>
> If you're using inventory status blocking and quality orders together, we strongly recommend having the **Reserve ordered items** option enabled.

Example with "Reserve ordered items" enabled

When **Reserve ordered items** is enabled, all inventory blocking transactions has a status of either *Reserved physical* or *Reserved ordered*.

Expand table

Inventory transaction reference	Receipt	Issue	Quantity	Site	Warehouse	Inventory status	Location	License plate	Comment
Purchase order	Purchased		10 Pcs	2	24	Blocking	RECV	receiptLp1	
Inventory blocking		Reserved physical	-9 Pcs	2	24	Blocking			Transaction generated by the inventory status blocking
Inventory blocking		Reserved physical	-1 Pcs	2	24	Blocking	RECV	receiptLp1	Transaction generated by the quality order sampling blocking
Inventory blocking	Ordered		1 Pcs	2	24	Blocking	RECV	receiptLp1	Expected receipts from the quality order sampling blocking
Inventory blocking		Reserved ordered	1 Pcs	2	24	Blocking			Transaction generated by the inventory status blocking

Example with "Reserve ordered items" disabled

When **Reserve ordered items** is disabled, the expected receipts can't be reserved because they are in status *Ordered*, so some blocking are left in status *On order*.

Expand table

Inventory transaction reference	Receipt	Issue	Quantity	Site	Warehouse	Inventory Status	Location	License Plate	Comment
Purchase order	Purchased		10 Pcs	2	24	Blocking	RECV	receiptLp1	
Inventory blocking		Reserved physical	-9 Pcs	2	24	Blocking			Transaction generated by the inventory status blocking
Inventory blocking		Reserved physical	-1 Pcs	2	24	Blocking	RECV	receiptLp1	Transaction generated by the quality order sampling blocking
Inventory blocking	Ordered		1 Pcs	2	24	Blocking	RECV	receiptLp1	Expected receipts from the quality order sampling blocking
Inventory blocking		On order	1 Pcs	2	24	Blocking	RECV	receiptLp1	Transaction generated by the inventory status blocking

Note the difference in transaction status and dimensions between the two cases. For this reason, we recommend enabling the **Reserve ordered items** option.

Expected receipts from quality orders that sample blocked inventory are disabled

To simplify the inventory transactions when quality orders that sample inventory are blocked as a consequence of inventory status, the system disables expected receipts from such quality orders. Because inventory status blocking immediately blocks the expected receipt, there's no reduction of on-hand inventory because of this.

Related information

- Create and maintain an inventory blocking
- Quality management processes
- Inspect the quality of goods

Cycle counting

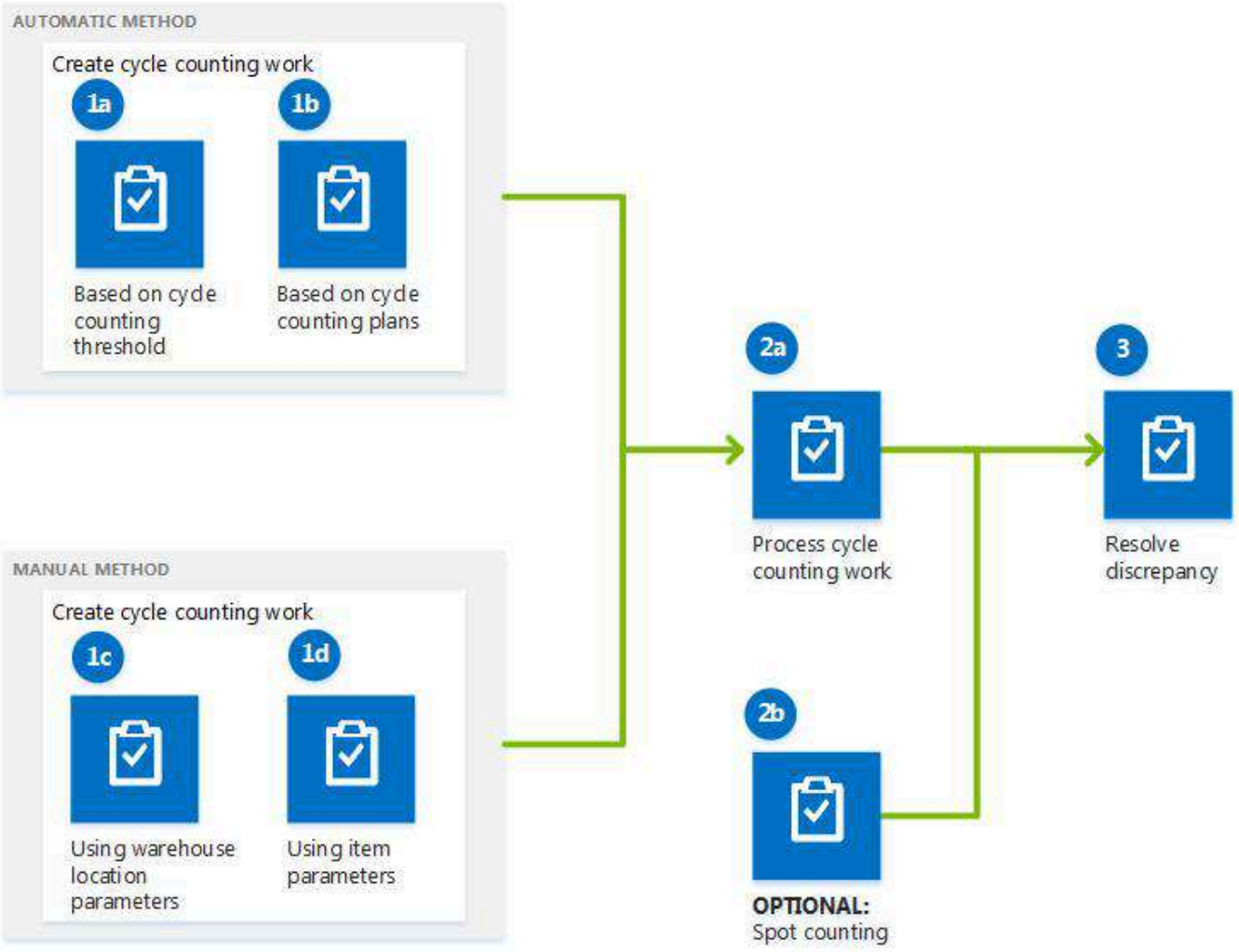
> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how you can use cycle counting with the warehousing solution that's available in Warehouse management. This article doesn't apply to the warehousing solution that's available in Inventory management.

Cycle counting is a warehouse process that you can use to audit on-hand inventory items. The cycle counting process involves three steps:

1. **Create cycle counting work** – You can automatically create cycle counting work based on threshold parameters for items or by using a cycle counting plan. Alternatively, you can manually create cycle counting work by using the item or warehouse parameters on the **Cycle count work by item** page or the **Cycle count work by location** page.
2. **Process the cycle count** – After creating cycle counting work, you count items in a warehouse location and use a mobile device to enter the result in Dynamics 365 Supply Chain Management. Alternatively, you can count items in a warehouse location without creating cycle counting work. This process is referred to as *spot cycle counting*.
3. **Resolve differences in the counted value** – After a cycle count, any items that have differences in the counted value have a work status of *Pending review* on the **All work** page. You can resolve these differences on the **Cycle count work pending review** page.

The following illustration shows the cycle counting process.



Cycle counting prerequisites

The following table shows the prerequisites that must be in place before you can use cycle counting.

Expand table

Prerequisite	Description
Item	The item must be enabled for warehouse management processes (WMS).
Warehouse	The warehouse must be enabled for warehouse management processes (WMS). To enable the warehouse for WMS, on the **Warehouses** page, select the warehouse, and then select the **Use warehouse management processes** option. To enable workers to move pallets during a cycle count, on the **Warehouse management** FastTab, select the **Allow pallet moves during cycle counting** option.
Work pools	Optional: Create a work pool to segregate the warehouse work, based on the type of work (in this case, cycle counting work).

Prerequisite	Description
Locations	Enable cycle counting for locations. To enable cycle counting for a warehouse location, on the **Location profiles** page, select the **Allow cycle counting** option.
Warehouse management parameters	Set up parameters for cycle counting. On the **Warehouse management parameters** page, specify the default adjustment type code, work class ID, and work priority for cycle counting.

Mobile device	• Create a menu item for one of the following methods on the **Mobile device menu items** page:
	  ◦ User directed cycle counting
	  ◦ System directed cycle counting
	  ◦ Cycle count grouping
	  ◦ Spot cycle counting
	• Set up a menu for the mobile device.
	• Create a work user account, and assign a mobile device menu to the work user ID.

Related setup task	Set up a cycle counting plan for a warehouse location.

Automatically create cycle counting work

You can schedule recurring creation of cycle counting work by setting up cycle counting thresholds or cycle counting plans.

- A cycle counting threshold indicates the quantity or percentage limit of inventory items. The system automatically creates cycle counting work when the threshold limit is reached.
- A cycle counting plan creates cycle counting work either immediately or periodically through a batch job. When cycle counting work is created, the counting work line includes information about the location to count. The on-hand inventory that is associated with this location isn't blocked, and is therefore available for reservation and outbound processing, even though open counting work exists.

Create cycle counting work based on threshold parameters for items

You can configure the system to create cycle counting work when the number of items falls below a specific threshold value in a location. For example, there are 60 items in a location that has a cycle counting threshold of 40. During a sales order transaction, a worker picks 25 items from the location and puts them in a staging location. Because the new item count, 35, is less

than the threshold quantity, the system automatically creates cycle counting work for the location.

Schedule cycle counting work

You can schedule cycle counting plans to create cycle counting work immediately or periodically. By setting up cycle counting plans, you can control the work pool that cycle counting work is created for, the maximum number of cycle counts that are created for items in different locations, and the number of days before a warehouse location is counted again. For example, an item is available in three locations in the warehouse, and the maximum number of cycle counts is set to *2*. In this case, when you run the cycle counting plan, two cycle counts are created for the two locations where the item is present. As another example, you set the number of days between cycle counts to *5*. In this case, cycle counting work is created every five days. However, if you process cycle counting work on day 3, the next cycle counting work is created five days after the last cycle counting was processed, on day 8.

Create cycle counting work manually

To create cycle counting work manually, use the **Cycle count work by item** or **Cycle count work by location** page. You can specify the maximum number of cycle counts to create. For example, if the warehouse manager specifies a value of *5*, the system creates cycle counting work for five locations, even if the item is present in 10 locations. You can also select a work pool ID to assign to the cycle counting work IDs that the system creates. When you process a work pool ID for cycle counting, you process the cycle counting work IDs that are assigned to the work pool as a group.

> **Note**
>
> When cycle counting work is created, it's always created for the whole location. This rule applies regardless of which method you use to create cycle counting work (for example, manual or automatic) or which query you use (the query is only used to select which locations to create cycle counting work for). That means that neither the item nor the tracking dimensions are set on the work line. The **Cycle counting transactions** page creates information about the specific item and tracking dimensions when a worker starts the counting process on a mobile device. The only exception is the *Partial location* cycle counting process, which supports counting based on item or product variant (learn more in Define partial location cycle counting process).

Perform a cycle count by using a mobile device

The Warehouse Management mobile app provides the following methods for processing cycle counting work:

- **User directed** – The worker specifies a cycle counting work ID that has a status of *Open*.
- **System directed** – Supply Chain Management assigns a cycle counting work ID to the worker.
- **Cycle count grouping** – The worker groups cycle counting work IDs that are specific to a particular location, zone, or work pool.
- **Spot cycle counting** – The worker counts items in a warehouse location at any time, even if no open cycle counting work exists for that location. To initiate spot cycle counting in a location, the worker enters the location ID. If no open cycle counting work exists for that location, then the system creates a new work record for spot cycle counting. If open cycle counting work does exist for the location, then the existing work record is used for spot cycle counting.

The following example shows how you can perform spot cycle counting by using the Warehouse Management mobile app. The instructions that workers see on the device vary, depending on the setup of the menu item for spot cycle counting.

1. On the mobile device, select the menu item to process spot cycle counting work.
2. Register the location to perform spot cycle counting for.
3. Register and confirm the item number and the counted item quantity. The status of the cycle counting work is updated to either *Pending review* or *Closed* on the **All work** page, depending on the parameters that are set on the **Worker** page.
4. Repeat step 3 as needed to count the remaining items in the location, and confirm that no additional items are available for counting.

> **Note**
>
> - Counting workflows in the mobile app provide an **Add LP or item** button, which lets workers count on-hand inventory that the system doesn't expect, though with a few limitations. For example, in license-plate controlled locations, you can only count each license plate once. As a result, when counting batch-tracked items, workers can only count one batch per license plate. Similarly, if a license plate contains multiple items, only one of those items can be counted.
> - If the unit sequence group for an item has multiple units enabled for cycle counting, workers are asked to provide a quantity for each unit and the system adds these quantities together. For example, if an item has a unit sequence group with *boxes* and *pieces*, then the worker is prompted to enter the number of boxes (for example, 2 boxes) and then the number of pieces (for example, 5 pieces). If the unit

conversion between boxes and pieces is 10, the total counted quantity is 25 pieces. This design enables workers to count full boxes first and then any leftover pieces afterwards. If there are only full boxes, the worker should enter zero when asked for the number of pieces.

- The system never shows the expected quantity to count. This design prevents intentional miscounts. To allow the user to double-check the count, use the **Number of attempts** setting on the mobile device menu item (select **Cycle counting** on the Action Pane). This setting allows a worker to repeat a count when necessary.
- The **Skip** button shown during the *System directed* cycle counting mobile device flow doesn't just skip the item that is currently being counted. Instead, it skips the entire cycle counting work record and goes to the next available record. If no other cycle counting work record is available, the app stays on the current work record and shows an error message stating that no other work is available. This behavior is also true for the *Partial location* cycle counting flow, where the **Skip** button changes to the next available cycle counting work record. Because there can't be two *Open*, *In process*, or *Pending review* cycle counting work records for the same location, selecting the **Skip** button nearly always results in skipping counting at the current location and changing to the next one.
- To prevent accidental selection, the **Cancel** button isn't shown during *cycle counting* mobile workflows.

Resolve cycle counting differences

A cycle counting difference occurs in the following scenarios if the **Is a cycle count supervisor** option is set to *No* for a work user ID:

- The counted value isn't within the deviation limits specified in the **Maximum percentage limit** or **Maximum quantity limit** fields on the **Work users** page. For example, the on-hand inventory quantity in a location is *50*, and the deviation limit for the work user is *10*. If the worker enters a value that isn't between *40* and *60*, a difference occurs.
- The counted value differs from the on-hand inventory quantity, and no deviation limits are set.

You can adjust differences in the counted value and then accept the counted value on the **Cycle count pending review** page. You can verify the modified count of the item quantity on the **On hand by location** page. The counted value is rejected if you don't approve the difference.

Related information

- Set up mobile devices for warehouse work

Last updated on 11/20/2025

Define cycle counting

Cycle counting is a warehouse process that you can use to audit on-hand inventory items. This task recording shows how to set up the default counting work priority, enable a mobile device menu item to process both picking and counting operations, enable a counting threshold trigger when a location becomes empty, and enable a cycle counting plan for a specific item in a specific warehouse. Typically, a warehouse manager performs these tasks. You can go through this procedure in the USMF demo data company or in your own data.

Set the priority of counting work

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Select the **Cycle counting** tab.
3. In the **Default cycle count work priority** field, enter a number. This step changes the priority of cycle counting work compared to other types of work in the warehouse. By entering a number that's lower than the number for other types of work, you raise the priority of the cycle counting work.
4. Select **Save**.
5. Close the page.

Enable the mobile device

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select **New**.
3. In the **Menu item name** field, enter a value.
4. In the **Title** field, enter a value.
5. In the **Mode** field, select **Work**.
6. Set the **Use existing work** option to **Yes**. When you set this option to **Yes**, the system looks for existing work when the mobile device menu item is used.
7. In the **Directed by** field, select **System directed**. When you select **System directed**, the warehouse worker is directed to open work that is in defined work classes. (You create these work classes next.)
8. Expand the **Work classes** FastTab. Next, you create two work classes to use with this mobile device menu item. When the menu item is used, these work classes are queried, and the work with the highest priority is shown to the user.
9. Select **New**.

10. In the **Work class ID** field, select a value.
11. Select **New**.
12. In the **Work class ID** field, select a value.
13. In the Action Pane, select **Save**.
14. Close the page.
15. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
16. In the list, find and select the desired record.
17. In the tree, select **the menu item that you just created**.
18. Select **Edit**.
19. Select the arrow to add the menu item to the menu.
20. Select **Save**.

Create a counting threshold

1. Go to **Warehouse management** > **Setup** > **Cycle counting** > **Cycle count thresholds**.
2. Select **New**.
3. In the **Cycle counting threshold ID** field, enter a value.
4. Set the **Process cycle counting immediately** option to **Yes**.
5. In the **Description** field, enter a value.
6. Select **Save**.
7. Select **Select locations**.
8. In the list, select the row.
9. In the **Criteria** field, select a value.
10. Click **OK**.
11. Close the page.

Create a cycle count plan

1. Go to **Warehouse management** > **Setup** > **Cycle counting** > **Cycle count plans**.
2. Select **New**.
3. In the **Cycle counting plan ID** field, enter a value.
4. In the **Description** field, enter a value.
5. In the **Maximum number of cycle counts** field, enter a number.
6. Select **Save**.
7. Select **Select locations**.
8. In the list, select the row.
9. In the **Criteria** field, select a value.
10. Click **OK**.

11. In the **Days between cycle counting** field, enter a number. For example, if you set the **Days between cycle counting** field to 5, the cycle counting work is created every five days. However, if cycle counting work is processed on day three, the next cycle counting work is created five days after the last cycle counting was processed, on day 8.
12. Select **Save**.
13. Select **New**.
14. In the **Sequence number** field, enter a number. The sort is from the smallest number to the largest number. The value must be more than 0 (zero).
15. In the list, select the row.
16. In the **Description** field, enter a value.
17. Select **Save**.
18. Select **Define product** query.
19. In the list, select the row.
20. In the **Criteria** field, enter or select a value.
21. Click **OK**.
22. Close the page.

Last updated on 07/01/2026

Partial location cycle counting

Cycle count plans guide the actual counting operations. You can request that only specific products and product variants be counted instead of all on-hand inventory in a location.

By using cycle count plans to create counting work, you can guide the actual counting operations. You can request that only specific products and product variants be counted instead of all on-hand inventory in a location. By filtering on specific products, the warehouse manager can reduce review overhead, avoid consolidation mistakes, and save time.

How to configure partial location cycle counting

When you use the warehouse work process for counting operations, a work header is created for each location. When you define the cycle count plan, you can use the **Select locations** query to limit the cycle counting work that is created. When you select products for the cycle count plan, you can select both product and product variant queries to refine what is counted.

You can associate a **work template** with a cycle count plan to define how the cycle count work should be created. The work template for counting operations is directly referenced from the cycle count plan.

When you define the work template details, you can use the **Work line breaks** option to specify whether the counting work lines must be grouped by item number or product variant number. This setup is required if you want to count on-hand inventory only for specific products in a location. The cycle counting work lines that are created have the level of information that you define here, and the guided counting operation is handled based on this level.

If you associate cycle count plans with work templates by using the **Work line breaks** option, the **Partial cycle count** field is selected for the cycle counting work that is created, and multiple cycle counting work lines are created based on the definition of the work template.

Before partial cycle count work can be processed, you must, at a minimum, select **Display item number** for the mobile device menu item as part of the cycle counting setup. The warehouse operator is asked to record only counting information that is related to the counting lines (item numbers and product dimensions). All other on-hand inventory is ignored for this counting process.

> ⓘ **Note**

If your mobile device menu item has **Directed by** set to *System directed*, you must also set up system-directed work sequencing for that menu item. Without at least one system-directed work sequence query, the mobile device returns *No work available* even when open cycle count work exists. The work classes on the menu item only filter the results returned by the sequence queries—they don't retrieve work on their own.

For the partial cycle count process, the **Last cycle count** date and time aren't updated for the location, even though all the items on hand at a given location are counted. The partial cycle count doesn't consider the parameter **Days between cycle counting** on the **Cycle count plans** page. Partial cycle count doesn't support simultaneous counting of multiple items at the same location. Partial cycle count functionality might result in the same location being counted multiple times for an item when **Process cycle counting plan** is run. To avoid that scenario, specify filters in the **Select locations** field.

> **Note**
>
> The Warehouse Management mobile app doesn't provide the **Add LP or item** button when you use the partial cycle count process.

Example

For this example, only item number A0001 is counted in warehouse 61.

1. A new work template for cycle counting is created. Use the **Work line breaks** option to group counting work lines by item number. Therefore, the cycle counting work has lines per item number. You can also group the lines by product variant number.
2. A new cycle counting plan is created that references the newly created work template. The cycle counting plan includes all locations in warehouse 61 (**Select locations** query) that hold inventory for item number A0001. The selection of specific products is defined in the **Cycle count product selections** section.
3. Select products for cycle counting plans by setting the **Empty locations** field to **Exclude empty**. When the cycle counting plan is processed, the partial cycle count work for item number A0001 is created. You can perform the actual counting process by using a mobile device menu item for guided cycle counting.

Related information

- Cycle counting

Last updated on 05/29/2026

Define partial location cycle counting process

Article • 01/22/2025

When you use cycle count plans to create counting work, you can guide the actual counting operations by requesting that only specific products and product variants be counted instead of all on-hand inventory at the location. By filtering on specific products, the warehouse manager can reduce review overhead, help prevent consolidation mistakes, and save time. Typically, a warehouse manager performs the setup tasks.

Create a cycle counting work template

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. In the **Work order type** field, select *Cycle counting*.
3. Select **New**.
4. In the **Sequence number** field, enter a number. The sort order is from the smallest number to the largest number. The value must be more than 0 (zero).
5. In the list, mark the selected row.
6. In the **Work template** field, type a value.
7. In the **Work template** description field, type a value.
8. In the **Work pool ID** field, enter or select a value.
9. In the **Work priority** field, enter a number.
10. Select **Save**.
11. Select **New**.
12. In the list, mark the selected row.
13. In the **Work type** field, select *Counting*.
14. In the **Work class ID** field, enter or select a value.
15. Select **Save**.
16. Select **Work line breaks**.
17. Select **New**.
18. In the **Sequence number** field, enter a number. The sort order is from the smallest number to the largest number. The value must be more than 0 (zero).
19. Select **Save**.
20. Close the page.
21. Close the page.

Create a cycle counting plan

1. Go to **Warehouse management** > **Setup** > **Cycle counting** > **Cycle count plans**.
2. Select **New**.
3. In the **Cycle counting plan ID** field, type a value.
4. In the **Description** field, type a value.
5. In the **Maximum number of cycle counts** field, enter a number.
6. In the **Work template** field, enter or select a value.
7. Select **New**.
8. In the **Sequence number** field, enter a number. The sort order is from the smallest number to the largest number. The value must be more than 0 (zero).
9. In the **Description** field, type a value.
10. Select **Save**.
11. Select **Define product query**.
12. In the list, mark the selected row.
13. In the **Criteria** field, enter or select a value.
14. Select **OK**.
15. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Cycle counting example scenarios

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides a collection of scenarios that explore the cycle counting features of Microsoft Dynamics 365 Supply Chain Management. It first describes the requirements for your existing Supply Chain Management environment. It then explains how to configure cycle counting and describes all the cycle counting stages. When you finish, you should have a good understanding of cycle counting, including guided cycle counting, blind cycle counting, spot cycle counting, cycle count thresholds, and cycle count plans.

Prerequisites

Make demo data available

Each scenario in this article references values and records included in the standard demo data provided for Supply Chain Management. If you want to use the values provided here as you work through the scenarios, be sure to work in an environment where the demo data is installed. Set the legal entity (company) to **USMF** before you begin.

Prepare demo data for the scenarios

Follow these steps to confirm that all the demo data required for the scenarios is available in the USMF company in your system. Create any records or values that are missing.

1. Go to **Warehouse management** > **Setup** > **Worker**.
2. In the list pane, select **Julia Funderburk**.
3. On the **Users** FastTab, select the row that has the following values. If no existing row has these values, create it.
    - **User ID:** *61*

- **User name:** *WH61*
- **Default warehouse:** *61*
- **Menu name:** *Main*

4. On the **Work** FastTab, set the following values for user *61* if they aren't already set:

    - **Is a cycle count supervisor:** *No*
    - **Maximum percentage limit:** *0*
    - **Maximum quantity limit:** *0*
    - **Maximum value limit:** *0*

5. Go to **Warehouse management** > **Setup** > **Work** > **Work pools**.

6. Work pools segregate warehouse work, based on the type of work (in this case, cycle counting work). Make sure that a record exists that has the following settings:

    - **Work pool ID:** *CycleCount*
    - **Description:** *Cycle Count*

7. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

8. In the list pane, select the record that is named *Cycle Count*. If no existing record has this name, create it. Confirm or set the following values for the record:

    - **Menu item name:** *Cycle Count*
    - **Title:** *Cycle Count Guided*
    - **Mode:** *Work*
    - **Use existing work:** *Yes*
    - **Directed by:** *System directed* (This value indicates that Supply Chain Management assigns a cycle counting work ID to the worker.)
    - **Display inventory status:** *Yes*

9. On the Action Pane, select **Cycle counting**.

10. In the **Mobile device cycle counting** dialog box, confirm or set the following values:

    - **Display item number:** *Yes*
    - **Display license plate:** *Yes*
    - **Number of attempts:** *1*

11. Select **OK** to close the dialog box.

12. In the list pane, select the record that is named *Cycle Count Blind*. If no existing record has this name, create it. Confirm or set the following values for the record:

    - **Menu item name:** *Cycle Count Blind*
    - **Title:** *Cycle Count Blind*
    - **Mode:** *Work*
    - **Use existing work:** *Yes*
    - **Directed by:** *Cycle count grouping* (This value indicates that the worker can group cycle counting work IDs that are specific to a location, zone, or work pool.)

13. On the Action Pane, select **Cycle counting**.

14. In the **Mobile device cycle counting** dialog box, confirm or set the following values:

    - **Display item number:** *No*
    - **Display license plate:** *No*
    - **Number of attempts:** *0*

15. Select **OK** to close the dialog box.

16. In the list pane, select the record named *Spot Counting*. If no existing record has this name, create it. Confirm or set the following values for the record:

    - **Menu item name:** *Spot Counting*
    - **Title:** *Spot Counting*
    - **Mode:** *Work*
    - **Use existing work:** *No*
    - **Work creation process:** *Spot cycle counting* (This value indicates that the worker can count items in a warehouse location at any time, even if no open cycle counting work exists for that location. To initiate spot cycle counting in a location, the worker enters the location ID. If no open cycle counting work exists for that location, the system creates a new work record for spot cycle counting. If open cycle counting work exists for the location, the existing work record is used for spot cycle counting.)

17. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.

18. In the list pane, select the record named *Inventory*. If no existing record has this name, create it. Confirm that the following cycle counting menu items appear in the **Menu structure** column:

    - Cycle Count

- Cycle Count Blind
- Spot counting

19. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.

20. On the **Cycle counting** tab, set the following values:

    - **Default cycle counting adjustment type:** *Cycle Count* (This field specifies the journal type that is posted when cycle counting is done.)
    - **Default cycle counting work class ID:** *CCount* (This field specifies the work class that is used for cycle counting.)
    - **Default cycle count work priority:** *50* (This field sets the priority that cycle counting work has relative to other types of work in the warehouse. By entering a number that's lower than the number entered for other types of work, you raise the priority of the cycle counting work.)

21. Go to **Warehouse management** > **Setup** > **Inventory** > **Adjustment types**.

22. The **Adjustment types** page lets you create codes for the different in and out adjustments that might occur. Confirm that a record exists that has the following settings:

    - **Inventory adjustment type:** *Cycle Count*
    - **Description:** *Cycle Count*
    - **Name:** *ICnt*

23. Go to **Warehouse management** > **Setup** > **Warehouse setup** > **Warehouses**.

24. In the list pane, select warehouse *61*. If no existing record has this name, create it.

25. On the **Warehouse** FastTab, set the following values:

    - **Use warehouse management process:** *Yes* (This value enables the warehouse for warehouse management processes (WMS).)
    - **Allow license plate moves during cycle counting:** *Yes* (This value enables workers to move license plates during a cycle count.)

Scenario 1: Guided cycle counting

Before the guided cycle counting can occur, you must create some work. This work guides the assigned person through the warehouse, from location to location, to complete the counts set up in the work.

Create cycle counting work for scenario 1

Follow these steps to create cycle counting work for item location *01A02R2S2B* (BULK-06) in warehouse *61*.

1. Go to **Warehouse management** > **Cycle counting** > **Cycle count work by location**.
2. In the **Create cycle count work by location** dialog box, set the **Work pool ID** field to *CycleCount*.
3. On the **Records to include** FastTab, select **Filter**.
4. In the query editor dialog box, on the **Range** tab, follow these steps:
    - For the row where the **Field** field is set to *Warehouse*, set the **Criteria** field to *61*.
    - For the row where the **Field** field is set to *Location*, set the **Criteria** field to *01A02R2S2B*.
5. Select **OK** to close the query editor dialog box.
6. Select **OK** to close the **Create cycle count work by location** dialog box.

    When the work creation process finishes, a message appears in the Action center.
7. Go to **Warehouse management** > **Work** > **Work details**.
8. Find the newly created work by setting a filter on the **Work pool ID** column to find records that have a value of *CycleCount*.

Does cycle counting work for scenario 1

After you create the cycle counting work, you do the work by counting items in a warehouse location and then using a mobile device to enter the results in Supply Chain Management. Follow these steps to do the cycle counting work in the Warehouse Management mobile app.

1. Sign in to the Warehouse Management mobile app as the work user that you set up in the Prepare demo data for the scenarios section earlier in this article. For the example in this article, the user is named *Julia Funderburk* and is set up for warehouse *61*. (The USMF demo data should let you sign in as this work user by entering *61* as the user ID and *1* as the password.)
2. On the main menu, select **Inventory**.

3. On the **Inventory** menu, select **Cycle Count Guided**.
4. Select the **Qty** field, enter *9* by using the number pad, and then select **OK** (the check mark button).
5. The system expects you to enter a count of 10 pieces for this item, location, and license plate. Therefore, you're asked to recount. Select the **Qty** field, enter *9* again by using the number pad, and then select **OK** (the check mark button). On the second attempt, your count is accepted.

   > ⓘ **Note**
   >
   > When the system detects a difference between the expected on-hand quantity and the quantity that you entered, it asks you to count a second time because the **Number of attempts** field is set to *1* for the **Cycle Count Guided** mobile device menu item. You're guided to a specific item number and license plate number because both the **Display item number** option and the **Display license plate** option are set to *Yes* for the mobile device menu item.

6. Select **OK** (the check mark button).

Review the cycle counting differences for scenario 1

Follow these steps to review the cycle counting differences.

1. Return to Supply Chain Management.
2. Go to **Warehouse management** > **Work** > **Work details**.
3. Find and select the cycle counting work that you looked at earlier. (For example, set a filter on the **Work pool ID** column to find records that have a value of *CycleCount*.) Notice that the **Work status** field for this work is now set to *Pending review*.

   > ⓘ **Note**
   >
   > For the work user account that you used to do the counting work, the **Cycle count supervisor** option is set to *No*, and the **Maximum percentage limit**, **Maximum quantity limit**, and **Maximum value limit** fields are all set to *0* (zero). Therefore, all counting differences that this user reports must be manually approved, and the **Work status** field for the related work is set to *Pending review*. If the counted value were

within the deviation limits (as specified in the **Maximum percentage limit** or **Maximum quantity limit** fields on the **Work users** page), or if the **Cycle count supervisor** option were set to *Yes* for the user, the work would have automatically been closed. When a cycle count supervisor adds a new item or license plate, the count is automatically approved. For other users, the cycle counting work is assigned the *Pending review* work status, regardless of the **Maximum percentage limit**, **Maximum quantity limit**, and **Maximum value limit** settings.

4. On the Action Pane, on the **Work** tab, select **Cycle counting**.
5. On the Action Pane, select **Accept count**.

   The difference is posted by using the standard counting journal, and a new work order is created.
6. On the **Cycle counting transactions** page, on the Action Pane, select **Derived work** to find the work that was created for the approved difference.

Scenario 2: Blind cycle counting

This scenario requires that scenario 1 already be completed in your system.

Create cycle counting work for scenario 2

Before the blind cycle counting can occur, you must create some work. Follow these steps to create cycle counting work for item location *01A02R2S2B* (BULK-06) in warehouse *61*.

1. Go to **Warehouse management** > **Cycle counting** > **Cycle count work by item**.
2. In the **Create cycle count work by item** dialog box, set the **Work pool ID** field to *CycleCount*.
3. On the **Records to include** FastTab, select **Filter**.
4. In the query editor dialog box, on the **Range** tab, add three rows that have the following settings:

   - Row 1:
     - **Table:** *Items*
     - **Field:** *Item number*
     - **Criteria:** *L0101*

- Row 2:
  - **Table:** *Inventory dimensions*
  - **Field:** *Warehouse*
  - **Criteria:** *61*
- Row 3:
  - **Table:** *Inventory dimensions*
  - **Field:** *Location*
  - **Criteria:** *01A02R2S2B*

5. Select **OK** to close the query editor dialog box.
6. Select **OK** to close the **Create cycle count work by item** dialog box.

   When the work creation process finishes, you receive an informational message.

Do cycle counting work for scenario 2

After you create the cycle counting work, follow these steps to do the work in the Warehouse Management mobile app.

1. Sign in to the Warehouse Management mobile app as the work user that you set up in the Prepare demo data for the scenarios section earlier in this article. For the example in this article, the user is named *Julia Funderburk* and is set up for warehouse *61*. (The USMF demo data should let you sign in as this work user by entering *61* as the user ID and *1* as the password.)
2. On the main menu, select **Inventory**.
3. On the **Inventory** menu, select **Cycle Count Blind**.
4. Select the **Zone ID** field, enter *BULK06*, and then select **OK** (the check mark button).
5. Select the **Item** field, enter *L0101*, and then select **OK** (the check mark button).
6. Select the **License plate** field, enter *LP_BULK_06_01*, and then select **OK** (the check mark button).
7. Select the **Qty** field, enter *10*, and then select **OK** (the check mark button).

> ⓘ **Note**

> Even though the system detected a difference between the expected on-hand quantity and the quantity that you scanned, it didn't ask you to count a second time because the **Number of attempts** field is set to *0* (zero) for the **Cycle Count Blind** mobile device menu item. You were asked to scan the item number and license plate because both the **Display item number** option and the **Display license plate** option are set to *No* for the mobile device menu item.

8. Select **OK** (the check mark button).

Review the cycle counting differences for scenario 2

Follow these steps to review the cycle counting differences.

1. Return to Supply Chain Management.
2. Go to **Warehouse management** > **Common** > **Work** > **Cycle count work pending review**.
3. On the Action Pane, on the **Work** tab, select **Cycle counting**.
4. On the Action Pane, select **Reject count**.

   Because the counting difference was rejected, the work is closed.

Scenario 3: Spot cycle counting

The on-hand record states that there's an on-hand quantity of item *L0101* at location *01A02R2S2B*. The warehouse worker is at location *01A02R2S1B*. Although this location should be empty, it's full. Therefore, the warehouse worker immediately does a spot count of this location.

Do cycle counting work for scenario 3

Follow these steps to do the cycle counting work in the Warehouse Management mobile app.

1. Sign in to the Warehouse Management mobile app as the work user that you set up in the Prepare demo data for the scenarios section earlier in this article. For the example in this article, the user is named *Julia Funderburk* and is set up for warehouse *61*. (The USMF demo data should let you sign in as this work user by entering *61* as the user ID and *1* as the password.)
2. On the main menu, select **Inventory**.

3. On the **Inventory** menu, select **Spot counting**.
4. Select the **Location** field, enter *01A02R2S1B*, and then select **OK** (the check mark button).

   The system detects that the location is empty in Supply Chain Management.
5. Select **Add LP or Item**.
6. Select the **Item** field, enter *L0101*, and then select **OK** (the check mark button).
7. Select the **License plate** field, enter *LP_BULK_06_01*, and then select **OK** (the check mark button).
8. Select the **Qty** field, enter *9*, and then select **OK** (the check mark button).

   Because the system detects that the specified license plate is already available at another location in Supply Chain Management, that license plate is moved to the current location. Therefore, the system asks you to confirm the movement.
9. Select **OK** (the check mark button).

Review the cycle counting differences for scenario 3

Follow these steps to review the counting results.

1. Return to Supply Chain Management.
2. Go to **Warehouse management** > **Common** > **Work details**.
3. Select the **Show closed** checkbox at the top of the grid.
4. Set the filter for the **Work order type** column to *Inventory movement*.

   The system automatically detected this count as an inventory movement. This movement is permitted because the **Allow license plate moves during cycle counting** option is set to *Yes* for warehouse *61* on the **Warehouses** page.

Scenario 4: Define cycle count thresholds

One way to create cycle counting work is to use thresholds. A cycle count threshold indicates the quantity or percentage limit of inventory items. The cycle counting work is automatically created when the threshold is reached.

For example, there are 60 items in a location that has a cycle count threshold of 40. During a sales order transaction, 25 items are picked from the location and put in a staging location. Because the new item count, 35, is less than the threshold quantity, the cycle counting work is automatically created for the location.

Follow these steps to set up cycle count thresholds.

1. Go to **Warehouse management** > **Setup** > **Cycle counting** > **Cycle count thresholds**.
2. On the Action Pane, select **New** to create a threshold, and set the following values for it:
    - **Cycle counting threshold ID:** *L0101*
    - **Description:** *Threshold L0101*
    - **Threshold quantity:** *2*
    - **Unit:** *ea*
    - **Capacity threshold based on percentage:** *0.00*
    - **Cycle counting threshold type:** *Quantity*
    - **Process cycle count immediately:** *Yes*
    - **Days between cycle counting:** *1*
    - **Work pool ID:** *CycleCount*
3. On the Action Pane, select **Select items**.
4. In the query editor dialog box, on the **Range** tab, find the row where the **Field** field is set to *Item number*. Set the **Criteria** field for that row to *L0101*.
5. Select **OK** to close the query editor dialog box.

    You defined a cycle count threshold for item *L0101*.

The cycle counting work is created for item *L0101* at any location if the on-hand quantity is less than 2, and if the date of the last cycle count for the location isn't today.

Scenario 5: Define cycle count plans

Cycle count plans let you automate the creation of cycle counting work. Set up each cycle count plan with specific item and location queries. When the batch job runs, it creates cycle counting work for all locations that match the item and location criteria, up to the maximum number of counts specified for the plan. When the cycle counting work is created, the counting work line includes information about the location that needs to count. The on-hand inventory that is

associated with that location isn't blocked. Therefore, it's available for reservation and outbound processing, even though open counting work exists.

Follow these steps to set up a cycle count plan.

1. Go to **Warehouse management** > **Setup** > **Cycle counting** > **Cycle count plans**.
2. On the Action Pane, select **New** to add a row to the grid, and set the following values for it:
    - **Cycle counting plan ID:** *BULK06*
    - **Description:** *Counting of location for BULK06*
    - **Work pool ID:** *CycleCount*
    - **Maximum number of cycle counts:** *10*
    - **Days between cycle counting:** *10*
    - **Empty locations:** *Exclude empty*
    - **Work template:** Leave this field blank.
3. On the Action Pane, select **Select locations**.
4. A standard query editor dialog box appears. On the **Range** tab, add a row, and set the following values for it:
    - **Table:** *Locations*
    - **Field:** *Zone ID*
    - **Criteria:** *BULK06*
5. Select **OK** to close the query editor dialog box.
6. On the Action Pane, select **Process cycle counting plan**.
7. In the **Cycle count plans** dialog box, on the **Run in the background** FastTab, set the **Batch processing** option to *Yes*.
8. Select **Recurrence**.
9. In the **Define recurrence** dialog box, set up the batch job so that it begins immediately and runs once every minute, and so that there's no end date.
10. Select **OK** to close the **Define recurrence** dialog box.
11. Select **OK** to close the **Cycle count plans** dialog box.

    A message informs you that the job was added to the batch queue.

12. Go to **Warehouse management** > **Common** > **Cycle count scheduling**. The plan begins immediately and creates counting work. Because counting work isn't completed, the **Status** field is set to *In process*. After one minute, the value in the **Total cycle counts** column changes to *1*.

    > **Note**
    >
    > The cycle count work isn't created if the number of days since the last cycle counting is less than the value that you set for the **Days between cycle counting** field for the cycle counting plan. For example, if the **Days between cycle counting** field is set to *5*, the cycle counting work is created every five days. However, if the cycle counting work is processed on day 3, the next cycle counting work is created five days after the last cycle counting was processed, on day 8.

13. On Action Pane, select **Work** to view the counting work that was created.

Last updated on 05/18/2026

Warehouse management on-hand entries cleanup job

The performance of queries that calculate on-hand inventory depends on the number of records in the tables. To improve performance, reduce the number of records the database must consider.

This article describes the on-hand entries cleanup job, which deletes unneeded records in the `InventSum` and `WHSInventReserve` tables. These tables store on-hand information for items that are enabled for warehouse management processing. (These items are referred to as WMS items.) Deleting these records can significantly improve the performance of on-hand calculations.

What the cleanup job does

The on-hand entries cleanup job deletes any records in the `WHSInventReserve` and `InventSum` tables where all the field values are *0* (zero). These records can be deleted because they don't contribute to the on-hand information. The job deletes only records that are below the **Location** level.

If you allow negative physical inventory, the cleanup job might not be able to delete all the relevant entries. This limitation exists because the job must account for a special scenario where a license plate has multiple serial numbers, and one of those serial numbers becomes negative. For example, the system shows zero on hand at the license plate level when a license plate has +1 pcs of serial number 1 and –1 pcs of serial number 2. For this special scenario, the job performs a breadth-first deletion, where it tries to delete from lower levels first.

Schedule and configure the cleanup job

You can find the on-hand entries cleanup job at **Inventory management** > **Periodic tasks** > **Clean up** > **Warehouse management on-hand entries cleanup**. Use the standard job settings to control the scope and schedule for running the job. In addition, the following settings are provided:

- **Delete if not updated for this many days** – Enter the minimum number of days that the job should wait before it deletes an on-hand entry that drops to zero quantity. Use this setting to help reduce the risk of deleting on-hand entries that are still in use. If you want cleanup to occur as soon as possible, either enter *0* (zero) or leave the field blank.

- **Maximum execution time (hours)** – Enter the maximum execution time of the cleanup job, in hours. If the job isn't completed before this time passes, it saves the work that it completed so far and then closes itself. This capability is especially relevant for implementations that have high inventory use. In these cases, schedule the job to run at times when the system load is as light as possible. If you want the batch job to continue to run until it's completed, either enter *0* (zero) or leave the field blank.

Although you can run the job during regular business hours, run it outside working hours. In this way, you help prevent conflicts that can occur if a user is working with a record that the job is also cleaning up.

If the job tries to delete a record for an item while another user is using that record, a deadlock error occurs for either the cleanup job or the user.

When the job runs, it has a commit size of 100. In other words, it tries to commit one time for every 100 deletions. However, because some deletions are set-based, there might be scenarios where more than 100 records can be deleted in the same transaction. Therefore, lock escalations can still sometimes occur.

Possible user impact

Users might be affected if the on-hand entries cleanup job deletes all the records for a given level, such as the license plate level. In this case, the functionality for seeing that inventory was previously available on-hand at a license plate might not work as expected because the relevant on-hand entries are no longer available. Users can experience this issue in the following situations:

- On the **On-hand list**, when the user deselects the condition **Quantity <> 0** or selects the condition **Closed transactions** in the **Dimensions display** settings.
- In a **Physical inventory by inventory dimension** report for past periods, when the user sets the **As of date** parameter.

However, the performance improvement that the cleanup job provides should make up for these small losses in functionality.

Last updated on 07/01/2026

Inventory journals

08/29/2025

This article describes how you can use inventory journals to post various types of physical inventory transactions.

In Supply Chain Management, use inventory journals to post physical inventory transactions of various types, such as the posting of issues and receipts, inventory movements, the creation of bills of materials (BOMs), and the reconciliation of physical inventory. All these inventory journals work in a similar way, but they're divided into different types.

Types of inventory journals

The following types of inventory journals are available:

- Movement
- Inventory adjustment
- Transfer
- BOM
- Item arrival
- Production input
- Counting
- Tag counting

Movement

When you use an inventory movement journal, you can add cost to an item when you add inventory, but you must manually allocate the extra cost to a particular general ledger account by specifying a general ledger offset account when you create the journal. This inventory journal type is useful if you want to overwrite the default posting accounts.

Inventory adjustment

When you use an inventory adjustment journal, you can add cost to an item when you add inventory. The extra cost is automatically posted to a specific general ledger account, based on the setup of the item group posting profile. Use this inventory journal type to update gains and losses to inventory quantities when the item should keep its default general ledger offset account. When you post an inventory adjustment journal, you post an inventory receipt or issue, change the inventory values, and create ledger transactions.

Transfer

Use transfer journals to transfer items between stocking locations, batches, or product variants without associating any cost implications. For example, you can transfer items from one warehouse to another warehouse within the same company. When you use a transfer journal, you must specify both the "from" and "to" inventory dimensions (for example, for Site and Warehouse). The on-hand inventory for the defined inventory dimensions changes accordingly. Inventory transfers reflect the immediate movement of material. In-transit inventory isn't tracked. If you need to track in-transit inventory, use a transfer order instead. When you post a transfer journal, two inventory transactions are created for each journal line:

- An inventory issue at the "from" location.
- An inventory receipt at the "to" location.

BOM

When you report a BOM as finished, you can create a BOM journal. By using a BOM journal, you can post the BOM directly. This posting generates an inventory receipt of the product, together with an associated BOM and an inventory issue of the products that are included in the BOM. This inventory journal type is useful in simple or high-volume production scenarios where routes aren't required.

Item arrival

Use the item arrival journal to register the receipt of items (for example, from purchase orders). You can create an item arrival journal as part of arrival management from the **Arrival overview** page, or you can manually create a journal entry from the **Item arrival** page. If you enable the item arrival journal name to check for picking locations, Supply Chain Management looks for a location for received items and, if there's room, generates location destinations for the incoming items.

Production input

Production input journals work like the item arrival journals but are used for production orders.

Counting

Counting journals let you correct the current on-hand inventory that is registered for items or groups of items, and then post the actual physical count, so that you can make the adjustments that are required to reconcile the differences. You can associate counting policies with counting groups to help group items that have various characteristics, so that those items

can be included in a counting journal. For example, you can set up counting groups to count items that have a specific frequency, or to count items when stock falls to a particular level. For information about how to define counting groups, see Define inventory counting processes.

Tag counting

Use tag counting journals to assign a numbered tag to a count lot. The tag contains a tag number, item number, and item quantity. To ensure that a tag is used only one time, and that all tags are used, every item number has a unique set of tags with its own number sequence. You can set three status values for each tag:

- **Used** – The item number is counted for this tag.
- **Voided** – The item number is voided for this tag.
- **Missing** – The item number is missing for this tag.

When you post a tag counting journal, you create a new counting journal based on the tag counting journal lines. For more information about tag counting, see Inventory tag counting.

Working with journals

Only one user at a time can access a journal. If several users need to access journals at the same time to create journal lines, they must select journals that aren't currently being used. This approach prevents information from being overwritten. In situations where multiple departments use the same journal type, it's helpful to create multiple journal names (for example, one per department). It can also be helpful to divide journals so that each posting routine is entered in its own unique inventory journal. For posting routines that are associated with inventory transactions, create one journal for periodic inventory adjustments and another for inventory counting.

Posting journal lines

You can post the journal lines that you create at any time until you lock an item from additional transactions. The data that you enter in a journal remains in that journal, even if you close the journal without posting the lines.

Data entity support for inventory journals

Data entities support the following types of integration scenarios:

- Synchronous service (OData)
- Asynchronous integration

Learn more in Data entities.

> **Note**
>
> Not all inventory journals are OData-enabled, so you can't use the Excel data connector to get data published, updated, and imported back to Supply Chain Management.

Another difference between the journal data entities is the ability to use composite entities that include both the header and line data. Currently, you can use the composite entities for:

- Inventory adjustment journal
- Inventory movement journal

These two inventory journals only support the *Initialize stock* scenario as part of a data management import project:

- When you don't specify a journal header number but specify a number sequence for the journal type, the import job automatically creates journal headers per 1,000 lines. For example, importing 2,020 lines results in the following three journal headers:
  - Header 1: contains 1,000 lines
  - Header 2: contains 1,000 lines
  - Header 3: contains 20 lines
- The import process assumes that unique line information exists per inventory dimension, which can be a product, storage, and tracking dimension. Therefore, you can't import journal lines where only the date field differs on the lines within the same import project.

Related information

- Data entities

Inventory locations

Inventory locations determine where you store and pick items in warehouses that don't use warehouse management processes (WMS).

This article applies to features in the Inventory management module. It doesn't apply to features in the Warehouse management module.

The term location refers to the place that items are stored and drawn from.

For each location, you can also specify the place the item is inserted. By default, they're the same. Items are usually inserted and drawn from the same side of a location, but not always. For example, items that are stored in live storage racks are inserted from one aisle and drawn from another. The main input is given by a location name, which is usually determined by its coordinates: warehouse, aisle, rack, shelf, and bin. You can enter this name or ID manually or generate it from the location coordinates. For example, `01-02-03-4` represents aisle 1, rack 2, shelf 3, and bin 4 in the Inventory locations page.

Location properties

A location has the following characteristics:

- Size (height, width, depth, and thereby volume)
- Warehouse, aisle, rack, shelf, and bin position
- Location type (bulk location, picking location, inbound dock, outbound dock, production input location, inspection location, or kanban supermarket)

Check text can be used in online systems to verify that the operator has selected the correct location for a specific item. This check text can be created manually or by default.

Sort codes

Use sort codes to optimize the handling of picking lines, which describe the information that is required for picking items from inventory, including the picking order. Sort codes can be specified by the aisle and other coordinates, or assigned manually for the location.

Blocked locations

Occasionally, you might want to indicate that a location is blocked for a period of time, for example, to allow for repairs. At other times, you might want to indicate blocking of only the input or only output.

Tree structure

In the Inventory locations page, you can view the warehouse layout in a tree structure based on the coordinates of inventory locations, in a defined display format.

Maintain inventory locations via the warehouse form

You can copy locations from one warehouse to another and create locations through a wizard. Before running the wizard, make sure that you define the default location names on the **Warehouse** page.

Related information

Create a new warehouse layout

Last updated on 07/01/2026

Inventory statuses

This article describes how to use inventory statuses to categorize and keep track of inventory.

Set up and use inventory statuses

You can use inventory statuses to categorize inventory. You can then initiate appropriate actions, such as replenishment or putaway work.

Here are some examples of ways that you can use inventory statuses:

- Create inventory statuses for on-hand inventory, inbound transactions, and outbound transactions.
- Specify a default inventory status for warehouse transactions.
- Change an inventory status for items before arrival, during arrival, or when the items are put away during inventory movement.
- Use an inventory status to price items that are returned and to plan item coverage during master planning.

An inventory status is one of the dimensions in the storage dimension group. Inventory statuses can be categorized as available or unavailable, and you can use the **Inventory blocking** parameter to block items that have an unavailable inventory status. Items that have a blocked status are considered physical inventory, and they can't be used on a production order, sales order, transfer order, or outbound transaction.

You can use warehouse items that have either available or unavailable inventory statuses for inbound work. For example, you create an available status that is named *Ready*, an unavailable status that is named *Damaged*, and a blocked status that is named *Blocked*. When you create a purchase order for received or returned items, if any items are damaged or broken, you can change the inventory status of those items to *Damaged* on the purchase order line. After these items are received, the status is automatically set to *Blocked*. If you scan the damaged items by using a mobile device, Supply Chain Management can use location directives and work templates to show information about an appropriate location or range of locations where you can put away those items. For returned items, an issue type of *Reservation* is created on the **Inventory transactions** page.

You can specify which inventory statuses are blocking statuses by using the **Inventory blocking** check boxes on the **Inventory statuses** page. You can't use inventory statuses as blocking

statuses for sales orders, transfer orders, or project integrations.

For outbound work, you can use different nonblocking inventory statuses to control which inventory to reserve against. If you have items that have a status of *Blocking*, and master planning is run on these items, the items are considered missing, and inventory is automatically replenished. Furthermore, for quality orders associated with outbound work, it isn't possible to update the **Inventory status** as part of the quality order validation.

> **Note**
>
> You can't change the status of inventory at locations where open work exists. For example, if you did a purchase receive for an item, but didn't do the putaway step, then open work would exist for the receiving location and you would get an error if you tried to change the status of inventory at that location. Completing or cancelling the related work would allow you to change the status.
>
> Usually, the status of on-hand inventory related to open warehouse work is only changed by workers using the Warehouse Management mobile app, for example while executing a movement process.

After you set up inventory statuses, you can set the default inventory status for a site, item, and warehouse. You can also set a default status for sales, transfer, and purchase orders. The default status for sales orders and outbound transfer order can't have the **Inventory blocking** option set to *Yes*. The inventory status that is inherited from the default settings on a site, warehouse, item, purchase order, transfer order, or sales order can be changed by using the mobile device, or on the purchase order, sales order, or transfer order line.

To plan coverage for items that have an available inventory status, select the **Coverage plan by dimension** option for a storage dimension on the **Storage dimension groups** page. When you open the **Item Coverage** wizard, items that have an available status appear on the **Status** page. To create coverage settings for these items, select the inventory status ID for the available inventory statuses. Based on the coverage settings, you can calculate the item requirements and forecast the supply and demand of available items during master planning. You can't create an item coverage setup that has a blocked inventory status. Alternatively, use the **Item coverage** page to create or modify the item coverage parameters.

Change inventory statuses

You can change inventory statuses either by using the **On-hand by location** page or by using a batch job.

To change inventory status of individually selected records, follow these steps:

1. Go to **Warehouse management** > **Inquiries and reports** > **On-hand by location**.
2. Use the **Site**, **Warehouse**, and **Location** fields to find the relevant location.
3. On the **Locations** tab, select the location.
4. On the **On-hand** tab, select the relevant records, and then select the **Inventory status change** button.

To set up a batch job task that selects and updates records, follow these steps:

1. Go to **Warehouse management** > **Periodic tasks** > **Warehouse management inventory status change**.
2. On the **Parameters** FastTab, set **To inventory status** to the status value you want to update the selected records to.
3. On the **Records to include** FastTab, select the **Filter** button and set up a query that selects which records to change.
4. On the **Run in background** FastTab, set **Batch processing** to *Yes*. If you want to run the job at a regular interval, select **Recurrence** and assign the interval.

Last updated on 07/01/2026

Inventory tag counting

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about tag counting, which you use to compare the actual contents of a warehouse with the on-hand inventory.

By creating lines on the **Tag counting** page, you place a tag number on each inventory item, such as a number from 1 to 500. During the count, you enter the item number and the quantity on a corresponding tag. This tag can then be used as the basis for input in the tag counting journal.

After you post the tag counting journal, a new counting journal is created on the **Counting** page. The new journal is based on the tag counting journal lines that you created.

To tag-count items by a specific inventory dimension, select the dimension on the **Display dimension** page that is displayed when you create the tag counting journal. For example, to count items in a specific warehouse, select the **Warehouse** check box. If **Lock items during count** on the **General** tab of the **Inventory and warehouse management parameters** page is set to *Yes*, items can't be physically updated during counting. However, items in tag counting journals aren't locked during counting. Inventory transactions aren't created until the tag counting lines are posted and transferred to a counting journal. If tags are entered randomly, and you want to identify missing tags, select the **Tag** column header to sort the lines by tag.

Related information

Cycle counting

Last updated on 04/10/2025

Work line details

10/03/2025

The **Work line details** page shows a comprehensive, sortable, and filterable list of the individual work lines in your system. It provides a complete overview of work that is being planned and executed in the warehouse. You can easily switch between viewing all work lines and viewing only open work lines. Details that are provided for each line include the work status, item number, location, work quantity, load ID, and shipment ID.

Open and use the Work line details page

To view the list of work line details, go to **Warehouse management** > **Work** > **Work line details**. From here, you can perform the following actions:

- Use the **Filter** field to search for lines that have a specific value for any available parameter. (Available parameters include many parameters that aren't shown as columns in the grid.)
- Use the **Show closed** check box to show or hide closed lines.
- Select **Display dimensions** to open the **Dimensions display** dialog box, where you can choose to show or hide various dimension columns in the grid.
- Select any column heading to open a menu where you can choose to sort or filter the list by values in that column.
- Select a work line, and then select **Change location** to open a dialog box where you can change the location for that work line. The location that you specify will override the setup of the location directive.
- Select a work line, and then select **Cancel work line** to open a dialog box where you can partially or fully reduce the quantity of that work line.
- Adjust quantities.
- View the transactions behind any work line.

Try out the feature

This section provides a three-part demo that shows how to work with work line details.

Make sample data available

To work through this demo by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

You can also use this demo as guidance when you work on a production system. However, you must substitute your own values, and you might be missing some types of required records that the standard demo data provides.

Verify that the scenario setup includes enough available inventory

If you're working with the **USMF** demo data, you should first make sure that your system is set up so that enough inventory is available at each relevant pick location. For this demo, the expectation is that you have the following inventory available:

- **Item M9200:** 45 ea. (or more)
- **Item M9202:** 10 ea. (or more)

Follow these steps to verify that enough inventory is available and to make any adjustments that are required.

1. Go to **Warehouse management** > **Setup** > **Location directives**, and determine which picking locations are used for sales order picking at warehouse 51. (Learn more in Control warehouse work by using work templates and location directives.)
2. Check the inventory levels at the relevant locations.
3. Adjust inventory as required. You can create manual movements, use replenishment, or apply any other flow to adjust the inventory.

Part 1: Create picking work

Before you start to create work, make sure that your warehouse is set up to respond to work requests in the expected manner.

Follow these steps to create some picking work.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

2. Select **New** to open the **Create sales order** dialog box.

3. In the **Create sales order** dialog box, set the following values:

    - On the **Customer** FastTab, set the **Customer account** field to *US-001*.
    - On the **General** FastTab, set the **Warehouse** field to *51*.

4. Select **OK** to create the sales order and close the dialog box.

5. Your new sales order is opened. It includes a new, empty row in the **Sales order lines** grid. On this order line, set the following values:

- **Item number:** *M9200*
- **Quantity:** *20*
- **Unit:** *ea*

6. Select the new order line, and then, on the **Inventory** menu, select **Reservation** to open the **Reservation** page.

7. On the **Reservation** page, select **Reserve lot** to reserve the selected line's full quantity in the warehouse.

8. Close the **Reservation** page to return to the sales order.

9. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**. The system creates a shipment, adds it to a new load, and creates the required work.

10. Create a second sales order for the same customer account and warehouse that you used for the first order. Add the following two order lines to this order:

    - **Line 1:** Set the **Item number** field to *M9200*, the **Quantity** field to *25*, and the **Unit** field to *ea*.
    - **Line 2:** Set the **Item number** field to *M9202*, the **Quantity** field to *10*, and the **Unit** field to *ea*.

11. Repeat steps 6 through 8 to reserve the inventory for each order line (one at a time), and then repeat step 9 to release the order to the warehouse.

Part 2: Change the location for a work line

1. Go to **Warehouse management** > **Work** > **Work line details**.
2. Find and select one of the work lines that you created for this demo.
3. Select **Change location** to open the **Select new location** dialog box.
4. In the **Select new location** dialog box, in the **Location** field, select a new location for the work line.
5. Select **OK** to apply your change and close the dialog box.

> ⓘ **Important**
>
> You can submit location changes only if the new location has enough inventory available (for a pick), or if it passes location-type validation (for a put).

Part 3: Change the quantity of a work line or cancel a work line

1. Go to **Warehouse management** > **Work** > **Work line details**.
2. Find and select one of the work lines that you created for this demo. Note that you can cancel or change quantities only for work lines where the work type is *pick*.
3. Select **Cancel work line** to open the **Quantity to cancel** dialog box.
4. In the **Quantity to cancel** dialog box, change the value in the **Quantity** field to specify the quantity that should be *subtracted from* the quantity that is currently specified for the line. By default, the **Quantity** field shows the full quantity.
    - If you cancel the full quantity, the **Work status** value will be changed to *Canceled*, but the **Work quantity** field will still show the original value.
    - If you cancel just part of the quantity, the **Work quantity** field will be updated to show the new value, but the **Work status** value won't change.
5. Select **OK** to apply your change and close the dialog box.

> ⓘ **Important**
>
> If you cancel just part of the quantity for a work line, you must also remove the obsolete quantity from the load line. Otherwise, unless under-delivery is set up correctly, the load line can't be ship-confirmed.

Work blocking and blocking reasons

Work blocking provides a standard way to block or unblock warehouse work. The system prevents blocked work from being processed until it's unblocked. You can block work records automatically when certain conditions are met and you can also block them manually at any time. Multiple reasons can cause work to be blocked, and in some cases, a single work record can be blocked for several reasons at once. The system provides visibility into the reasons why a work record is blocked. You can only block work records with the status *Open*. This article describes the work blocking functionality and reasons.

Supported work blocking reasons

The following work blocking reasons are supported:

- **Unprocessed replenishment work** – The work is blocked because it's linked to an unprocessed replenishment work record.
- **Held wave** – The work is blocked by a wave with status *Held*.
- **Unprocessed production over pick work** – The work is blocked because it's linked to an unprocessed staging over-pick work record.
- **Unprocessed deferred put operation** – The work is blocked by an unprocessed deferred-put processing task.
- **Split work** – The work is blocked because it's being split.
- **Replenishment overflow capacity** – The replenishment work is blocked because it would overflow the capacity of the put location.
- **Processing wave** – The work is blocked by a processing wave.
- **Undefined reason** – The work is blocked for an undefined reason.

No other blocking reasons are supported.

View blocked work and blocking reasons

To find all blocked work and check the reasons why each of them blocked, follow these steps:

1. Go to **Warehouse management** > **Work** > **All work**.
2. Blocked work records show a check mark in the **Blocked wave** column. You can use the column filter to find all blocked work records at once.

3. To see the reasons why a work record is shown as blocked, select the link in the **Work ID** column to open the **Work details** page for that work record. Then open the **Blocking reasons** tab, which shows all blocking reasons for that work record, along with who blocked it and the date and time when each reason was applied.

> **Tip**
>
> If you only want to find work that is blocked by replenishment work, you can go to **Warehouse management** > **Work** > **Outbound** > **Work blocked by replenishment work**.

Manually block or unblock work

You can manually block work at any time. Manually blocked work shows a blocking reason of *Undefined reason*.

You can manually unblock work that is currently blocked for one of the following reasons:

- Held wave
- Undefined reason
- Replenishment overflow capacity

To manually block or unblock work, go to **Warehouse management** > **Work** > **All work** and select one or more work records that you want to block or unblock. Then, on the Action Pane, select **Block work** or **Unblock work**.

The system automatically unblocks work that is blocked for other reasons when the blocking reason is resolved.

Last updated on 12/30/2025

Location license plate positioning

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

License plate location positioning shows where a license plate is located in a multi-pallet location, such as a location that uses double-deep pallet racking.

This feature adds a sequence number to each license plate that is put into a storage location. This sequence number orders the license plates in the storage location. The feature intelligently supports scenarios where customers use a gravity racking system and must know, for picking purposes, which license plate is front-facing.

This article presents a scenario that shows how to set up and use the feature.

Example scenario

Make sample data available

To work through this scenario by using the values that are suggested in this article, you must work on a system where sample data is installed. Additionally, you must select the **USMF** legal entity before you start.

Set up the feature for this scenario

Complete the following procedures to set up the *Location license plate positioning* feature for the scenario presented in this article.

Location profiles

The feature must be turned on in the location profile for every location where it's used.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.

2. In the list of location profiles in the left pane, select **BULK-06**.
3. On the **General** FastTab, the feature adds two new options. Set the following values:
    - **Enable license plate position:** *Yes*

      When this option is set to *Yes*, the license plate position is maintained for license plates in the location.

    - **Display mobile device LP position:** *Yes*

      When this option is set to *Yes*, the license plate position is shown to mobile device users during adjustment and counting. You can change the setting of this option only when the feature is turned on.
4. Select **Save**.

Location directives

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the left pane, make sure that the **Work order type** field is set to *Sales orders*.
3. In the list of location directives, select **61 SO Pick order**.
4. On the Action Pane, select **Edit**.
5. On the **Lines** FastTab, select the line that has a **Sequence number** value of *2*.
6. On the **Location Directive Actions** FastTab, select the line that has a **Name** value of *Pick for less than pallet* (it should be the only line), and change its **Sequence number** value to *2*.
7. Select **New** above the grid to add a line for a new location directive action.
8. On the new line, set the following values:
    - **Sequence number:** *1*
    - **Name:** *Pick position 1*
9. While the new line is still selected, select **Edit query** above the grid.
10. In the query editor, select the **Joins** tab.
11. Expand the **Locations** table join to show the join to the **Inventory dimensions** table.

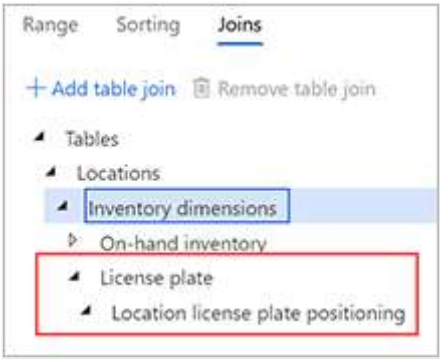
12. Expand the **Inventory dimensions** table join to show the join to the **On-hand inventory** table.

13. Select **Inventory dimensions**, and then select **Add table join**.

14. In the list of tables that appears, in the **Relation** column, select **License plate (License plate)**. Then select **Select** to add **License plate** to the **Inventory dimensions** table join.

15. While **License plate** is still selected, select **Add table join**.

16. In the list of tables that appears, in the **Relation** column, select **Location license plate positioning (License plate)**. Then select **Select** to add **Location license plate positioning** to the **Inventory dimensions** table join.



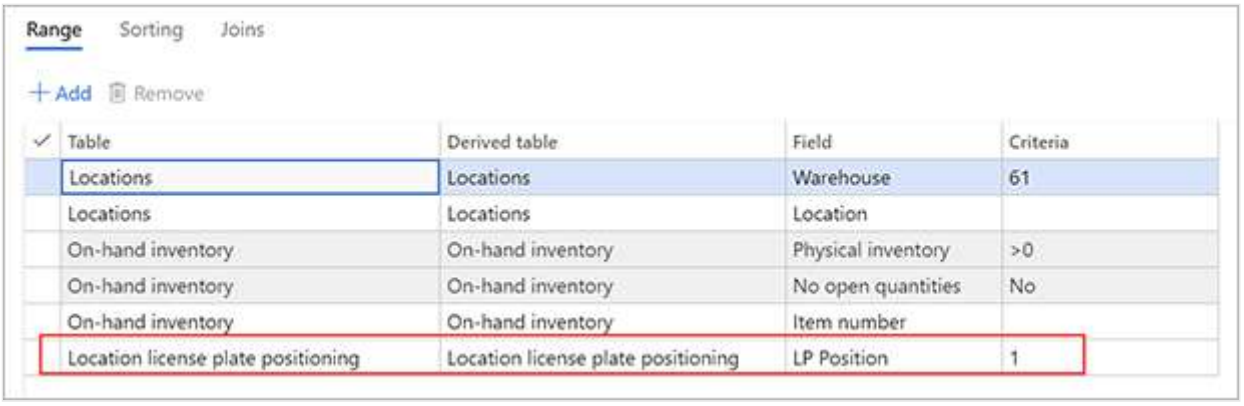
17. Select **OK** to confirm the updated joined tables and close the query editor.

18. On the **Location Directive Actions** FastTab, select **Edit query** again to reopen to the query editor.

19. On the **Range** tab, select **Add** to add a line to the grid.

20. On the new line, set the following values:

    - **Table:** *Location license plate positioning*
    - **Derived table:** *Location license plate positioning*
    - **Field:** *LP Position*
    - **Criteria:** *1*



21. Select **OK** to confirm your changes and close the query editor.

Set up sample data for this scenario

For this scenario, the user must sign in to the warehousing mobile app by using a worker who is set up for warehouse *61* to perform work. The user must also complete transactions.

Because the *Location license plate positioning* feature adds a new identifier for license plate positions in a location, you must first create some data to support the scenario.

Spot-count the first location

1. Open the warehousing mobile app, and sign in to warehouse *61*.
2. Go to **Inventory** > **Spot Counting**.
3. On the **Spot Counting** page, set the **Location** field to *01A01R1S1B*.
4. Select **OK**.

   The page shows the location that you entered. It also shows the following message: "Location complete, add new LP or Item?"
5. Select **Refresh** to add a count in the location.
6. On the **Cycle Count: Add New LP or Item** page, select the **Item** field, and then enter the value *A0001*.
7. Select **OK**.
8. On the **Cycle Count: Add New LP or Item** page, select the **LP** field, and then enter the value *LP1001* (or any other license plate number of your choice).

   The **Cycle Count: Add New LP or Item** page shows **License Plate Position 1**.

9. Select **OK**.

   You must now specify the quantity of the item that is counted on the license plate.

10. Set the **Qty** field to *10*.

11. Select **OK**.

    The page shows the location that you entered. It also shows the following message: "Location complete, add new LP or Item?"

12. Select **Refresh** to add another count in the location.

13. On the **Cycle Count: Add New LP or Item** page, select the **Item** field, and then enter the value *A0002*.

14. Select **OK**.

15. On the **Cycle Count: Add New LP or Item** page, select the **LP** field, and then enter the value *LP1002* (or any other license plate number of your choice, provided that it differs from the license plate number that you specified earlier).

16. Change the license plate position by setting the **LP Position** field to *2*.

17. Select **OK**.

18. Specify the quantity of the item counted on the license plate by setting the **Qty** field to *10*.

19. Select **OK**.

    The page shows the location that you entered. It also shows the following message: "Location complete, add new LP or Item?"

20. Select **OK**.

Work is now completed.

Spot-count the second location

1. On the **Spot Counting** page, set the **Location** field to *01A01R1S2B*.

2. Select **OK**.

   The page shows the location that you entered. It also shows the following message: "Location complete, add new LP or Item?"

3. Select **Refresh** to add a count in the location.
4. On the **Cycle Count: Add New LP or Item** page, select the **Item** field, and then enter the value *A0002*.
5. Select **OK**.
6. On the **Cycle Count: Add New LP or Item** page, select the **LP** field, and then enter the value *LP1003* (or any other license plate number of your choice, provided that it differs from the both the license plate numbers that you specified in the previous procedure).

   The **Cycle Count: Add New LP or Item** page shows **License Plate Position 1**.
7. Select **OK**.
8. Specify the quantity of the item counted on the license plate by setting the **Qty** field to *10*.
9. Select **OK**.

   The page shows the location that you entered. It also shows the following message: "Location complete, add new LP or Item?"
10. Select **OK**.

Work is now completed.

Work details

> **Note**
>
> Spot counts from the mobile app create cycle counting work in Microsoft Dynamics 365. The work requires that the counts be accepted before posting them to inventory.

1. Sign in to Dynamics 365 Supply Chain Management.
2. Go to **Warehouse management** > **Work** > **Work details**.
3. On the **Overview** tab, look for the lines that have the following values:

   - **Work order type:** *Cycle counting*
   - **Warehouse:** *61*
   - **Work status:** *Pending review*

Two work IDs should exist for these lines. The counts for both these work IDs must be accepted.

4. In the grid, select the first work ID for the *Cycle counting* work order type.
5. On the Action Pane, on **Work** tab, in the **Work** group, select **Cycle counting**.

   Two lines are shown, one for each item and license plate. The values in the **Counted quantity**, **Location**, **License plate**, and **Item** fields should match the count entries that you created on the mobile device. If any of these fields aren't visible, select **Display dimensions** on the Action Pane to add them to the grid.
6. Select both lines.
7. On the Action Pane, select **Accept count**.
8. You receive a "Posting - Journal" message. Select **Message details** to view the posted journal number.
9. Close the message details.
10. Refresh the **Work** page.

    The first work ID is closed and no longer shown.

    > **Tip**
    >
    > To view closed work, select the **Show closed** check box above the grid.

    You now accept the work for the license plate in the *01A01R1S2B* location.
11. On the **Overview** tab, select the second work ID for the *Cycle counting* work order type.
12. On the Action Pane, on **Work** tab, in the **Work** group, select **Cycle counting**.

    One line is shown, for the item and license plate. The values in the **Counted quantity**, **Location**, **License plate**, and **Item** fields match the count entries that you created on the mobile device.
13. Select the line.
14. On the Action Pane, select **Accept count**.

15. You receive a "Posting - Journal" message. Select **Message details** to view the posted journal number.
16. Close the message details.
17. Refresh the **Work** page.

    The second work ID is closed and no longer shown.

    > **Tip**
    >
    > To view closed work, select the **Show closed** check box above the grid.

On-hand by location

1. Go to **Warehouse management** > **Enquiries and reports** > **On-hand by location**.
2. Set the following values:

    - **Site:** *6*
    - **Warehouse:** *61*
    - **Refresh across locations:** *Yes*

3. You see that location *01A01R1S1B* has two license plates:

    - **A0001**, where the **LP Position** field is set to *1*
    - **A0002**, where the **LP Position** field is set to *2*

4. You see that location *01A01R1S2B* has one license plate:

    - **A0002**, where the **LP Position** field is set to *1*

Sales order scenario

After the *Location license plate positioning* feature is set up and the inventory is staged, create a sales order to generate picking work. The picking work directs the warehouse worker to pick item *A0002* from the inventory location where the pallet ID is in position *1*.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. In the **Create sales order** dialog box, enter the following values:

- **Customer account:** *US-004*
- **Warehouse:** *61*

4. Select **OK**.

5. A new line is added to the grid on the **Sales order lines** FastTab. On this new line, enter the following values:

   - **Item number:** *A0002*
   - **Quantity:** *1*

6. On the **Inventory** menu above the grid, select **Reservation**.

7. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve inventory for the order line.

8. Close the **Reservation** page.

9. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.

   You receive an informational message that indicates the wave ID and shipment ID that were created for the order.

10. On the **Sales order lines** FastTab, on the **Warehouse** menu above the grid, select **Work details**.

11. The **Work** page appears and shows the work that was created for the sales line. Make a note of the work ID that is shown.

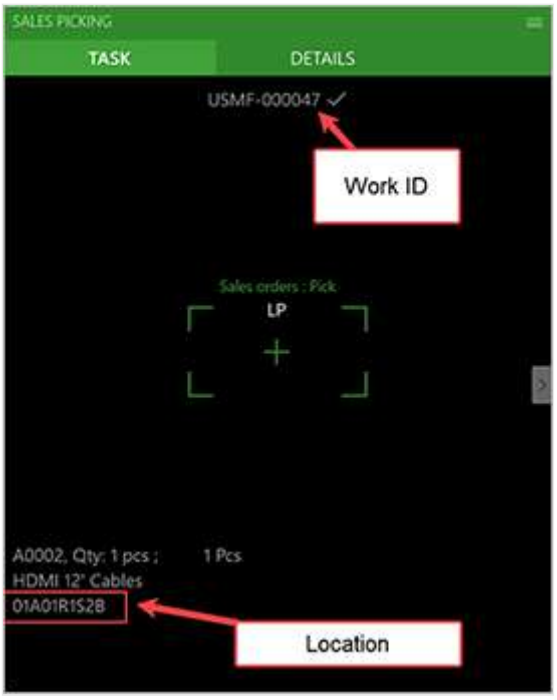
Sales picking scenario

1. Open the mobile app, and sign in to warehouse *61*.

2. Go to **Outbound** > **Sales picking**.

3. On **Scan a work ID / license plate ID**, select the **ID** field, and then enter the work ID from the sales line.

4. The picking work directs you to pick item *A0002* from location *01A01R1S2B*. You receive this instruction because item *A0002* is on a license plate that is in position *1* in that location.



5. Enter the license plate ID that you created for the location, and then follow the prompts to pick the sales order.

Last updated on 05/28/2026

Change work pool on work

10/03/2025

You can use work pools to organize work into groups. For example, you can create a work pool to classify work that occurs in a specific warehouse location.

The *Change work pool on work* feature adds a **Change work pool** button to the Action Pane for work items. Therefore, warehouse managers can easily change the work pool of existing work. This feature lets managers react quickly to changes on the warehouse shop floor, and it helps improve their ability to adapt to changing situations and the need to transfer work to another work pool.

Set up the Change work pool on work feature

To use this feature, you must have some work pools set up. You might also set up your work templates so that they automatically assign a pool. If you want to work through the example scenario that is provided later in this article, set up your system as described in this section.

Set up work pools

Work pools let you organize work items by type. To work with the *Change work pool on work* feature, you must have at least two work pools available. To view and add work pools, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work pools**.
2. If you're working with demo data from the **USMF** company and will work through the example scenario later in this article, add two work pools that have the following settings:
    - Work pool 1:
        - **Work pool ID:** *Webshop*
        - **Description:** *Web Shop*
    - Work pool 2:
        - **Work pool ID:** *CallCenter*
        - **Description:** *Call Center*
3. On the Action Pane, select **Save**.

Set up work templates

For each of your work templates, you can set a default work pool, as you require. For each relevant template, you assign a work pool in the **Work pool ID** column. In this case, all work items that are generated by using a given template automatically inherit the assigned work pool. If you're working with the demo data from the **USMF** company and will work through the example scenario later in this article, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. On the Action Pane, select **Edit** to put the page into editing mode.
3. Edit the template by setting the following values:
    - **Work template:** *62 Pick to pack*
    - **Work pool ID:** *Webshop*
4. Select **Save**.

Example scenario

This scenario shows how to change the stream of processing for an existing work item by changing its work pool. It uses demo data from the **USMF** company and the settings that were suggested earlier in this article.

Create a sales order and release it to the warehouse

1. Confirm that there is enough on-hand inventory for items *A0001* and *A0002* in warehouse *62*. Go to **Inventory management** > **Inquiries and reports** > **On-hand list**, and edit the filters as shown here:
    - The **Warehouse** value begins with *62*.
    - The **Item number** value is either *A001* or *A002*.

    Demo data should have a quantity of 10 each.

    Next, you must create a sales order.
2. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
3. On the Action Pane, select **New**.
4. In the **Create sales order** dialog box, set the following values:
    - **Customer account:** *US-007*
    - **Warehouse:** *62*

5. Select **OK** to create the sales order and close the dialog box.
6. The new sales order is opened. It should include a new, empty line in the grid on the **Sales order lines** FastTab. On this line, set the following values:
    - **Item number:** *A0001*
    - **Quantity:** *2*
7. On the **Inventory** menu above the grid, select **Reservation**.
8. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the inventory.
9. Close the page.
10. On the **Sales order lines** FastTab, select **Add line** to add another line to your sales order. On this line, set the following values:
    - **Item number:** *A0002*
    - **Quantity:** *2*
11. On the **Inventory** menu above the grid, select **Reservation**.
12. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the inventory.
13. Close the page.
14. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.
15. You receive informational messages that show the wave ID and shipment ID that were created from the release. Make a note of the wave ID.

Review the outbound wave

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. In the grid, search for the wave ID that was created from the release of the sales order.
3. Select the wave ID to view the details.
4. On the **Wave lines** FastTab, make sure that a shipment ID is shown for the sales order.

> **Tip**

If the **Process wave at release to warehouse** option is set to *No* in the setup for the shipping wave template, you must manually process the wave by selecting **Process** from the **Wave** tab on the Action Pane to create work.

5. Make sure that the wave has been processed. This processing creates the required work.

View work details and change the work pool

You can use the **Work details** page to view the work that was created and to manage the work pool.

1. Go to **Warehouse management** > **Work** > **Work details**.
2. Select the row for the work that was just created. The **Order number** column will show the sales order number.

   The **Work pool ID** field will be set to the work pool ID that was set up in the work template.

   > **Tip**
   >
   > If you don't see the **Work pool ID** field, add the column to the grid, and then refresh the page.

3. To change the work pool that is associated with the work ID, on the Action Pane, on the **Work** tab, select **Change Work pool ID**.
4. In the **Change work pool** dialog box, on the **Parameters** FastTab, in the **Work pool ID** field, select *CallCenter*.
5. Select **OK** to apply your change.
6. Notice that the value of the **Work pool ID** field has now been changed to *CallCenter*.

> **Important**
>
> When the **Change work pool** dialog box appears, the **Work pool ID** field might be blank by default. If the field is blank when you select **OK** to apply changes, you will remove the work pool completely from the work.

In addition to switching work pools, you can use this procedure to add a work pool to any work item that doesn't have one, or to remove a work pool from any work item that does have one.

Item consolidation - location utilization

10/03/2025

This article provides information about functionality that makes it easy for warehouse managers to view and filter the volumetric utilization of locations across the warehouse. Managers can select locations and create inventory movement work directly from the **Item Consolidation** page to consolidate items and therefore make better use of warehouse space.

Warehouse location status

The following four fields on the **Locations** page track information about the current state of the location:

- **Item number** – The item that is currently in the location. If the location contains multiple items, this field will be blank.
- **Last activity date and time** – The timestamp of the last warehouse transaction that was performed against the location.
- **Aging date** – The date when the inventory in the location was brought into the warehouse. This date is calculated based on the license plate aging date. Although this date is accurate for license plate–tracked locations, it might not be accurate for locations that aren't license plate–tracked.
- **Location status** – The status of the location. Four values are available:
  - *Undetermined* – The location profile doesn't track status. Therefore, the current status is unknown.
  - *Empty* – No inventory is currently in the location.
  - *Picking* – Outbound transactions have been performed against the location after it was last empty.
  - *Storage* – Only inbound transactions have been performed since the location was last empty.

These fields let warehouse managers get a better overview of the status of the locations in the warehouse. They also allow for more advanced reporting and filtering.

Set up item consolidation and location utilization

This section describes how to prepare your system to use item consolidation and location utilization. The procedures use sample values from the standard demo data. If you plan to work through the example scenario that is provided later in this article, select the **USMF** legal entity

(which contains the standard demo data), and create each record that is described in this section. If you don't plan to work through the example scenario, the values that are provided here can be considered examples of the types of setup that you must complete to use the features.

Released product

1. Go to **Product information management** > **Products** > **Released products**.
2. In the **Item number** field, select *M9201*, and open the details page.
3. On the Action Pane, on the **Manage inventory** tab, in the **Warehouse** group, select **Physical dimensions**.
4. On the **Physical dimension** page, on the Action Pane, select **New**.

   A new line is added to the grid. The **Item number** field is preset.
5. In the **Unit** field, select *ea*. The remaining fields on the line are automatically set.
6. Select **Save**, and close the page.

Location profile

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. In the list of location profiles, select **FLOOR-05**.
3. On the Action Pane, select **Edit**.
4. On the **General** FastTab, make sure that both the following options are set to *Yes*:
   - Enable item in location
   - Enable location status
5. Select **Save**.

> **Important**
>
> If the **Enable item in location** and **Enable location status** options were already set to *Yes*, skip ahead to the instructions for setting up the **Dimensions** FastTab in step 10. If the options weren't already set to *Yes*, you must run a consistency check for the **Warehouse management** module after you manually set them. In this case, continue to the next step.

6. To run the consistency check, go to **System administration** > **Periodic tasks** > **Database** > **Consistency check**.

7. In the **Consistency check** dialog box, set the following values:

    - **Module:** *Warehouse management*
    - **Check/Fix:** *Check*
    - **From date:** Leave this field blank.
    - **Warehouse location status consistency check:** Select this check box.

8. Select **OK**.

    > **Tip**
    >
    > When the consistency check is completed, you receive a notification. Open the Action Center to view the message. Select **Message details** to view the details.
    >
    > If the message for the consistency check states, "Incorrect location status information found for location XXXX in warehouse XX," you must run the consistency check again. This time, set the **Check/Fix** field to *Fix error*. View the messages to make sure that no errors were found.

9. You must now finish setting up the location profile. Go back to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**, select location profile **FLOOR-05**, and then, on the Action Pane, select **Edit**.

10. On the **Dimensions** FastTab, set the following values:

    - **Volume utilization percentage:** *100*
    - **Volumetric method used for inventory location:** *Use location volume*
    - **Actual location height:** *10*
    - **Actual location width:** *10*
    - **Actual location depth:** *10*
    - **Maximum weight:** *100*

11. Select **Save**.

Mobile device menu items

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New** to create a menu item for sorting.

3. In the header, set the following values:

- **Menu item name:** *Adjust In*
- **Title:** *Adjust In*
- **Mode:** *Work*
- **Use existing work:** *No*

4. On the **General** FastTab, set the following values:

    - **Work creation process:** *Adjustment in*
    - **Inventory adjustment types:** *Adjust in*

5. Select **Save**.

Mobile device menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. In the list of menus, select **Inventory**.
3. On the Action Pane, select **Edit**.
4. In the **Available Menus And Menu Items** list, find and select the **Adjust In** menu item.
5. Select the right arrow button to move **Adjust In** to the **Menu Structure** list. In this way, you add the new menu item to the selected menu.
6. Select **Save**.

Movement types

1. Go to **Warehouse management** > **Setup** > **Inventory** > **Movement types**.

2. On the Action Pane, select **New**, and then set the following values:

    - **Movement type code:** *CONSOLIDATE*
    - **Description:** *Consolidate locations*
    - **Work class ID:** *InvMov*

3. Select **Save**.

Example scenario

The following scenario uses the Warehouse Management mobile app to make an inventory *adjustment in* to two locations in the warehouse.

Add inventory to locations

1. Sign in to the mobile device as a user who is set up for warehouse *51*.

2. Go to **Inventory** > **Adjust In**.

   You will now enter the first location adjustment.

3. In the **Adjustment in** task, select the location to make the inventory adjustment for. In the **LOC** field, select *LP-001*.

4. Confirm the location.

5. Create a license plate ID for the item that will be added to the location. In the **LP** field, enter *LP00101*.

6. Confirm the license plate.

7. Enter the item that will be added to the license plate. In the **ITEM** field, enter *M9201*.

8. Confirm the item.

9. Enter the quantity of the item that will be added. In the **QTY** field, enter *10*.

10. Confirm the quantity.

    You receive a "Work Completed" message. You will now enter the second location adjustment.

11. In the **Adjustment in** task, select the location to make the inventory adjustment for. In the **LOC** field, select *LP-002*.

12. Confirm the location.

13. Create a license plate ID for the item that will be added to the location. In the **LP** field, enter *LP00201*.

14. Confirm the license plate.

15. Enter the item that will be added to the license plate. In the **ITEM** field, enter *M9201*.

16. Confirm the item.

17. Enter the quantity of the item that will be added. In the **QTY** field, enter *15*.

18. Confirm the quantity.

    You receive a "Work Completed" message.

19. Select the Menu button (sometimes referred to as the hamburger or the hamburger button), and then select **Cancel** to exit the **Adjustment in** task.

Consolidate locations

1. Go to **Warehouse management** > **Periodic tasks** > **Item consolidation**.
2. In the header, select a warehouse to do the consolidation for. In the **Warehouse** field, enter *51*.

   A record is shown for each location where item *M9201* was adjusted. The **Utilization percentage** column shows the volumetric utilization of each location.
3. To consolidate inventory, select all the locations that must be consolidated, and then, on the Action Pane, select **Consolidate Inventory**.
4. In the **Consolidate inventory** dialog box, specify the location and movement type that should be used to create the work for inventory movement. Set the following values:

   - **Location:** *LP-001*
   - **Movement type code:** *CONSOLIDATE*
5. Select **OK**.
6. You receive an informational message that shows the movement work that was created. Make a note of the movement work ID.

View movement work

1. Go to **Warehouse management** > **Work** > **Work details**.
2. View the work that was created. Use the movement work ID from the inventory consolidation to filter or search the work grid.

   In this scenario, an existing location that had inventory was used as the inventory consolidation location. Therefore, only one work ID was created.

   > ⓘ **Note**
   >
   > The system creates one work ID for each move that must be completed. If you specify a location that already contains inventory, only one work ID is created. If you specify a new location, two work IDs are created.

Work split

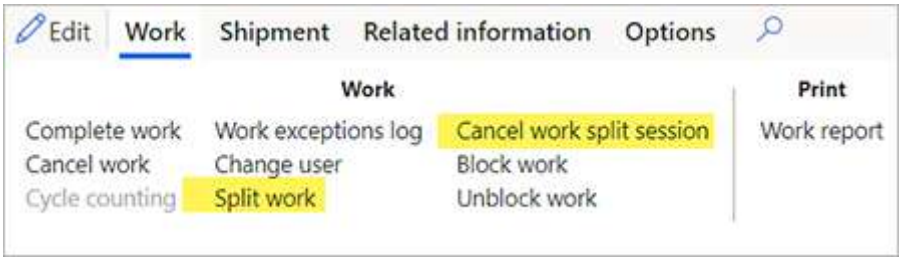
Work split functionality lets you split large work IDs (that is, work orders that have several lines) into several smaller work IDs that you can then assign to multiple warehouse workers. In this way, the same work creation number can be picked simultaneously by several warehouse workers.

> **ⓘ Important**
>
> You can split only work orders that have a status of *Open* or *In-progress*.

Enhancements to the Work details and All work pages

The *Work split* feature adds the following two buttons to the **Work** tab on the Action Pane of the **Work details** and **All work** pages:

- **Split work** – Split the current work ID into multiple smaller work IDs that can be processed by separate workers.
- **Cancel work split session** – Cancel the work split session, and make the work available for processing.



> **ⓘ Important**
>
> The **Split work** button won't be available if any of the following conditions are met:
>
> - The work status is something other than *Open* or *In progress*.
> - A container ID is associated with the work ID. (A container can't be systematically split, because it requires physical actions.)
> - The work is associated with a cluster.
> - The work order type is something other than one of the following types:

    - Sales orders
    - Raw material picking
    - Transfer issue

- The work is currently being split by another user. If you try to open the splitting page for work that is already being split by another user, you receive the following error message: "The work with ID #### is currently being split. Retry in a few minutes. If you continue to receive this message, contact a supervisor."

A new work blocking reason, *Split work*, indicates when the work ID is in the process of being split. It's shown both on the **Split work** page and in the Warehouse Management mobile app if a user tries to run the work. When blocking reasons are used, the name of the **Blocked wave** field from the work ID is changed to **Blocked**.

Initiate a work split

The feature adds a new **Split work** page that lets users split work lines from the work ID. When the page is first opened, it shows lines that have a work status of *Open* and that are available to be split. On the Action Pane, select **Split work** to process the selected work.

To split work, follow these steps:

1. Open one of the following work pages:

    - **Work details** (**Warehouse management** > **Work** > **Work details**)
    - **All work** (**Warehouse management** > **Work** > **All work**)

2. In the grid, select a work ID to split. The **Work order type** field must be set to one of the following values:

    - Sales orders
    - Raw material picking
    - Transfer issue

3. On the Action Pane, on the **Work** tab, in the **Work** group, select **Split work**.

    The **Split work** page appears and shows the work lines that are open and available to be split. By default, only available work lines are shown. To view all lines for the work ID (for example, lines that have a work type of *Put*), select the **Show all lines** check box above the grid.

The following message is shown: "Users can't process lines of the work until you finish splitting and close this page."

The **Work blocking reason** field for the current work will be set to *Split work*, and the work will be blocked.

4. Select the lines to remove from the current work ID and add to a new work ID. The following events occur:

    - When you split the work, the selected line or lines from the original work ID are canceled and then copied to a new work ID.

    - The existing work template structure and the location of the put (and also future pick/put pairs) are preserved. Values for the following work ID fields are copied from the original work to the new work:
        - Load ID
        - Shipment ID
        - Work order type
        - Order number
        - Site
        - Warehouse
        - Work priority
        - Work pool ID
        - Wave ID
        - Work creation number

    - The following fields aren't copied to the new work ID:
        - **Work ID** – A new work ID is generated from the appropriate number sequence.
        - **Work status** – This field is set to *Open*.
        - **Locked by** – This field is initially set to blank.
        - **Target license plate ID** – This field is left blank.
        - **Created date and time** – This field is set to the current date and time.

- **Blocked wave/frozen** – This field is recomputed for the original work ID and the new work ID.

5. On the Action Pane, select **Split work**.

While the work is being split, the following message is shown: "Processing operation - Split work". While this message is visible, you can cancel the operation by selecting **Cancel** in the message box.

If the **Show all lines** check box is cleared, the line that was split off and canceled will no longer appear in the grid. If the check box is selected, you should see that the value of the **Work status** field for that line has changed to *Canceled*.

The following notification is shown to indicate that the new work ID has been created: "Work #### has been created by splitting off from original work ####."

Other work lines from the original work ID (such as *Put* lines) will be adjusted as required to reflect the lines of work that have been canceled. For example, if the original work ID had a *Put* line for a quantity of 15, and *Pick* lines that have a total quantity of 10 were canceled, the new *Put* quantity on the original work ID will now be 5.

The new work won't immediately be assigned to any user. However, you can assign it to a user now, as required, by using the standard functionality of the **Work details** page.

> **Important**
>
> You can split only work IDs that contain two or more available work lines. If you select **Split work** when there is only one work line, you will receive the following error message: "At least one work line must remain on initial work." In this case, no splitting will occur.

Finish a work split

To finish splitting work, the *Split work* blocking reason must be removed. There are two ways to complete this step:

- The user who is splitting the work closes the **Split work** page by selecting the **Close** button (**X**) in the upper-right corner. When the page is closed, the *Split Work* blocking reason will be removed. The *Blocked* state of this work will be recomputed and, if there are no remaining blocking reasons for this work, the work will be unblocked.

- Any user opens the work ID and selects the **Cancel work split session** button on the Action Pane. The *Split work* blocking reason will be removed, and the *Blocked* state of this work will be recomputed, just as when the **Split work** page is closed.

After the *Split work* blocking reason is removed, the work can be run on the mobile device, provided that the **Blocked** state is set to *No* on the work ID.

User blocking on the Warehouse Management mobile app

If you try to use the Warehouse Management mobile app to run picking work against a work ID that is being split, you will receive the following error message: "The work with ID #### is currently being split." If you receive this message, select **Cancel**. You can then continue to process other work.

Other blocked operations

Any operations that modify work lines, work inventory transactions, or replenishment links that are related to work that is being split will fail, and the following error message will be shown: "The work with ID #### is currently being split."

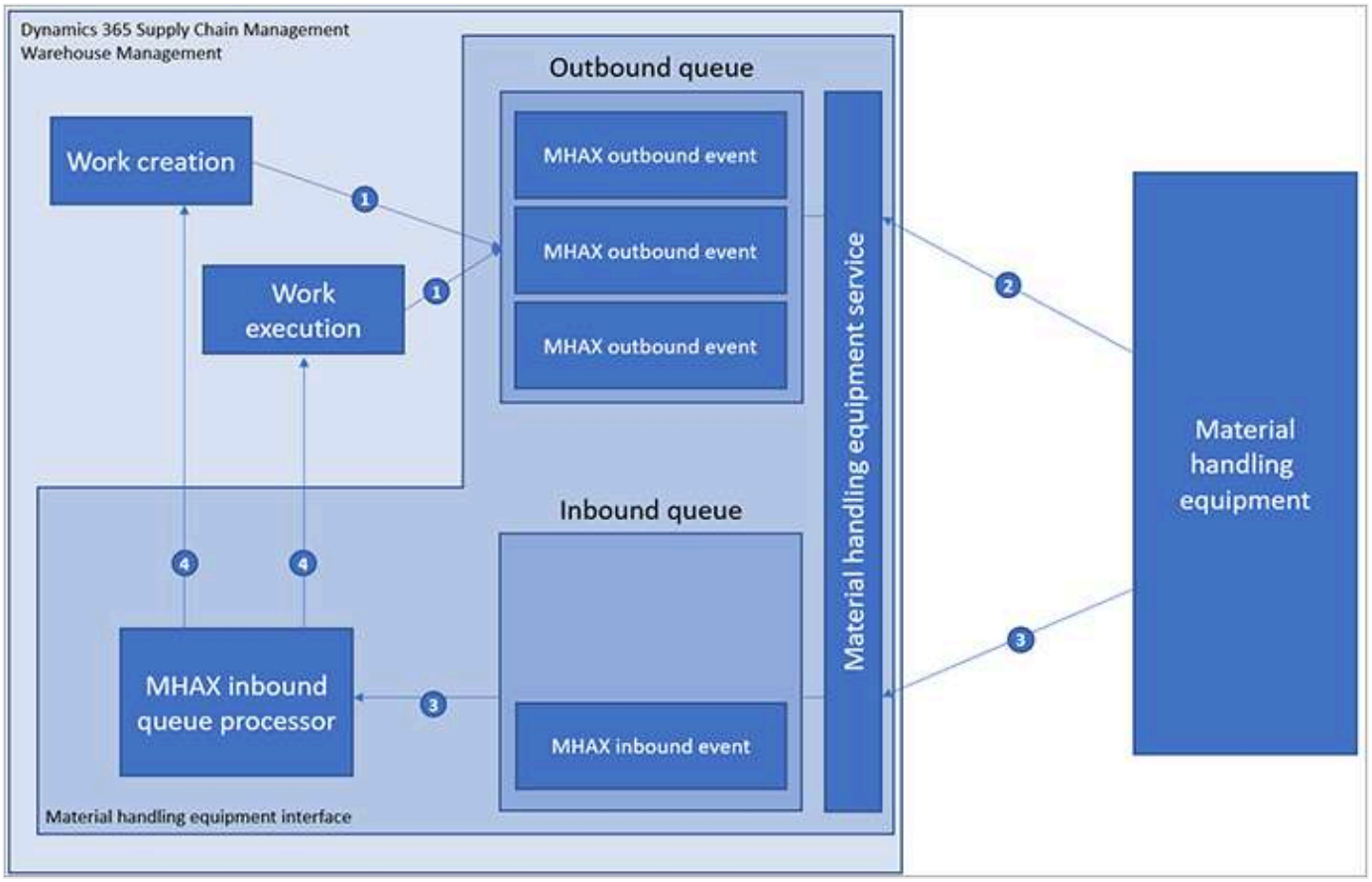
Last updated on 12/30/2025

Material handling equipment interface (MHAX)

You can use the *material handling equipment interface* (MHAX) to connect external physical material handling (MH) systems to a warehouse that is managed by warehouse management processes (WMS) in Microsoft Dynamics 365 Supply Chain Management. The interface between the WMS and MH systems consists of two queues: one for outbound events (WMS to MH) and one for inbound events (MH to WMS). The WMS system generates outbound events based on work lines that are created during various work creation and execution processes. The MH system then regularly polls the WMS system for new events and processes the responses. After the MH system has finished handling the events in accordance with work instructions, it sends inbound events, such as work line completion and short picking.

> **Important**
>
> By enabling this feature, your data might be shared with the third-party services that you select. You control what data is shared with the third-party (if any). Your use of the third-party services is your responsibility and is governed by terms agreed between you and the third party. Your privacy is important to us. To learn more, read our Privacy Statement.

The following illustration shows the various elements and the order that processes occur in when you use MHAX integration.



Here's an explanation of the interactions that are shown in the previous illustration:

1. During work creation or work execution, outbound events are created in the outbound queue.
2. The MH equipment connects to the MH equipment service, polls for any new events that are relevant to it, and processes those events.
3. When the MH equipment is ready to report back, it connects to the service again and submits inbound events. The queue processor immediately processes those events.
4. Based on the inbound event data, the queue processor might run existing work, modify it, or create new work.

Turn on the MHAX feature

Before you can use the MHAX feature, turn on its configuration key.

1. Put your system into maintenance mode, as described in Maintenance mode.
2. Go to **System administration** > **Setup** > **License configuration**.
3. Expand **Trade** > **Warehouse and Transportation management**, and then select the **Material handling equipment interface** check box.
4. Turn off maintenance mode, as described in Maintenance mode.

Set MHAX parameters

Set a few general parameters on the **Material handling equipment interface parameters** page to configure the feature.

1. Go to **Material handling equipment interface > Setup > Material handling equipment interface parameters**.

2. On the **General** tab, set the following fields:

    - **User mappings** – Map system users to their related warehouse workers. The system users call the inbound queue service. Selected workers are assigned the work operations (picks and puts) that are processed through the inbound queue. Work operations are performed in the default warehoues of each worker.
    - **Enable inbound message ID** – When you set this option to *Yes*, if a duplicate inbound message ID is received, the message is rejected and an error message is displayed that states the message already exists. When you set this option to *No*, duplicate inbound message IDs are allowed.
    - **Enable manual inbound message creation** – When you set this option to *Yes*, you can simulate inbound messages by creating a record directly from the **Inbound queue** page.

3. On the **Number sequences** tab, select the system-wide number sequences that generate unique IDs for the inbound queue items, outbound queue items, and work line pairs.

Outbound events

At specific points during work creation or work execution, the system determines whether it must generate outbound events to send to the MH system. If a subscription is configured for a specific point during warehouse processing, the system generates the event according to the setup of the subscription.

Structure of outbound events

Each outbound event is uniquely identified by an outbound queue ID. The outbound transaction type determines the type of the event. The warehouse and the ID of the subscription that generated the event are also recorded on the event.

To carry data to the MH system, the outbound event contains 10 fields for data (**data01** through **data10**). These data fields have a one-to-one (1:1) mapping to existing database fields. Specifically, they are extracted from fields in the work line and work header tables. The fields can be freely selected. You set them up when you create the subscription.

Besides the 10 data fields that have a 1:1 mapping to existing database fields, the event can contain an additional data field that is known as the *payload*. The contents of this field are generated by custom X++ code that is known as a *payload generator*. Any payload generator that should be used is set up in the subscription.

To ensure that the MH system receives each outbound queue ID only one time, a status field is used to specify whether an event is ready to be sent to the external material handling system (*Ready* status) or has already been sent (*Sent* status).

Outbound queue subscriptions

Before any events are generated, a subscription must be set up to tell the MHAX feature whether and how to generate events. Generated events are tagged by the subscription identifier. Therefore, multiple MH systems can connect to the same WMS system but keep their events separate. When the MHAX service is polled for new events, a subscription is one of the available options for retrieving the events.

To create a subscription, go to **Material handling equipment interface** > **Setup** > **Subscriptions**. For each subscription, the following parameters are available:

- **Subscription ID** – A unique name that identifies the subscription.
- **Description** – A free-text description of the subscription.
- **Warehouse** – The specific warehouses that events should be filtered by.
- **Outbound transaction type** – The type of events that the subscription should contain.
- **Payload generator** – An optional code extension that can enter additional information in the **Payload** field of the outbound event.

You can associate a query with each subscription. This query filters work lines and headers to further limit the work that uses the subscription to generate events. To add a query to a subscription, select the **Run query** check box for the relevant subscription on the **Subscriptions** page, and then select **Edit query** on the Action Pane. The standard Supply Chain Management query editor appears.

In addition, the subscription includes a *subscription map* that maps fields from either the work header or the work line to some or all of the 10 free data fields of the outbound event, as required. To return information to the MHAX service, typically include the work line record ID or the *work line pair ID*. (The work line pair ID is a new property that enables the system to use a single return command to process pick and put lines.) The remaining fields depend on the use case. Some examples are provided later in this article.

To set up a subscription map, select the relevant subscription on the **Subscriptions** page, and then select **Subscription map** on the Action Pane. In the **Subscription map** dialog box that appears, you can assign a table and field for each available data field as you require.

Outbound event types

This section describes the various event types that are available. (Event types are also known as *transaction types*.) It also explains when each type of event is created in the WMS system.

Work creation events (WorkCreation)

Work creation events are created after work is generated by the application. This behavior applies to most types of work creation processes, most notably to the creation of picking and replenishment work. In general, if work is created in an *Open* state, which indicates that the work is ready to be run by a worker, a work creation event is generated. In addition, work creation events are generated for basic movement work (not movement by template work), even though that work isn't created as open work.

A notable exception to this behavior is cycle counting work, which isn't currently supported. Stock counts in the MH system are outside the scope of MHAX, and the results of counts must be imported into an inventory counting journal.

After work is created, the MHAX service processes the generated work lines and assigns a work line pair ID to all generated work lines for each work header. The objective is to group all the pick work lines with the successive puts under one work line pair ID. (The groups correspond to pick/put pairs in work templates.) In this way, a single ID can be used to report work completion for all related pick and put lines. The grouping process starts with the first line and then continues with the same ID until it encounters a successive pair of put/pick work lines. The running ID is assigned to the put line of that pair. A new ID is then used for the pick line of the pair onward. This process continues until it processes all lines that belong to the work header.

As a special feature of work creation events, if the **Blocked wave** option is set to *Yes* on the work header, the events that are generated will have a status of *Blocked* instead of the usual status of *Ready* that is used to send them to the MH system. The **Blocked wave** flag on the work header indicates that the work header isn't yet ready for workers to run, perhaps because of unfinished replenishment work. When the **Blocked wave** flag is cleared, events that have already been generated are unblocked and are available for the MH system to retrieve from the queue.

Work initiation events (WorkInitiation)

Work initiation events are triggered when the work status changes from *Open* to *In process* during work update.

Work completion events (WorkCompletion)

Work completion events are triggered when the work status changes from *In process* to *Closed* during work update.

Work cancellation events (WorkCancel)

Work cancellation events are triggered when the work status changes from any status besides *Canceled* to *Canceled* during work update. In addition, all other events that are related to the work header are deleted from the queue for all subscriptions. In this way, external systems are prevented from processing events that aren't required.

Pick/put completion events (PickCompletion/PutCompletion)

Pick/put completion events are triggered when the status of the pick/put line changes from *In process* to *Closed* during work line update.

Monitor the outbound queue

To review your outbound queue, go to **Material handling equipment interface** > **Common** > **Outbound queue**. The **Outbound queue** page lists every outbound queue item and its status. Select a queue item to view its details. These details include the item's transaction type, the subscription that it used, and values for each data field (**data01** through **data10**) and the payload.

Clean up the outbound queue

Eventually, your outbound queue starts to fill with queue items that are already sent. To remove these items, go to **Material handling equipment interface** > **Periodic tasks** > **Cleanup** > **Outbound queue cleanup**.

Inbound events

This section describes the various types of inbound events that the MH system can report back to the WMS system. It also explains data must be supplied by the MH system, and what each inbound event does in the WMS system.

Structure of inbound events

When an inbound event is submitted, the external system must supply the inbound transaction type together with up to 10 parameters (**data01** through **data10**). Optional validation can make sure that the MHAX service doesn't receive the same inbound event more than one time. To enable this validation, each inbound event must have a unique message ID. If a duplicate message ID is received, and if the **Enable inbound message ID** option is set to *Yes* on the **Material handling equipment interface parameters** page, the message is rejected. An error message states that the message already exists.

In addition to the incoming data fields, the system assigns a unique inbound queue ID to the event.

Inbound event types

This section describes the inbound event types (transaction types) that are supported and the data that must be supplied for events to be processed.

Work-confirm events (WorkConfirm)

Work-confirm events require that the incoming data fields include the following information:

- **data01** – The work line pair ID.
- **data02** – The work line record ID (`RecId` value).

  > ⓘ **Note**
  >
  > *Either* the **data01** field *or* the **data02** field must be present.

- **data03** – The ID of the license plate to pick from.
- **data04** – The work header's target license plate ID.

If the work line pair ID is provided, all pick, put, or custom work lines that are marked by the work line pair ID, and that have a status of *Open* or *In process*, are run sequentially. If a work line record ID (`RecId` value) is provided, the work line must be a pick, put, or custom work line with a status of *Open* or *In process*.

Pick lines from license plate–controlled locations require that the **data03** specify the license plate that should be picked from, regardless of whether the lines are marked by the work line record ID or the work line pair ID. The **data04** field must specify the work header's target license plate for the pick.

Put lines don't accept further information. They are run based only on the current work line's location and the work's target license plate. If the put must be done to a different location, change the location of the work line as described in the Override events section later in this article.

Custom work lines don't require or support any additional information in the inbound event.

Short pick events (ShortPick)

Short pick events require that the incoming data fields include the following information:

- **data02** – The work record ID (`RecId` value).
- **data03** – The ID of the license plate to pick from.
- **data04** – The quantity to pick.
- **data05** – The short pick exception code that's linked to a *Work exception* record where the **Exception type** value is *Short pick*.
- **data06** – The work header's target license plate ID. This ID can be left blank when it's set up to be automatically generated.

> ⓘ Note
>
> The **data01** field isn't used for short pick events.

This event resembles the work confirmation event, but it applies only to pick lines.

Override events (Override)

Override events require that the incoming data fields include the following information:

- **data01** – The work record ID (`RecId` value).
- **data02** – The new location ID.

The work line must have a status of either *Open* or *In process*, and the new location must exist.

License plate receipt events (LPReceipt)

License plate receipt events require that the incoming data fields include the following information:

- **data01** – The ID of the inbound license plate to receive.

The system performs a license plate receiving operation, based on the license plate that is passed in as the value of the **data01** field.

Monitor the inbound queue

To review your inbound queue, go to **Material handling equipment interface** > **Common** > **Inbound queue**. The **Inbound queue** page lists every inbound queue item and its status. Select a queue item to view its details. These details include the item's transaction type, the message ID, and values for each data field (**data01** through **data10**).

If an error or another type of log item occurs while processing inbound events, you can inspect the log by selecting **Error log** on the Action Pane.

Inbound event processing

Inbound events are first written to the database and are then immediately (synchronously) run. If an error occurs during processing, the event is still written to the queue, but the status is set to *Errored*. The MHAX service returns an error message to the MH system and stores the error log in the inbound event record for later investigation.

Events that have a status of *Errored* can be reprocessed later if the error condition is fixed. To reprocess them, follow one of these steps:

- Go to **Material handling equipment interface** > **Common** > **Inbound queue**. Select the relevant inbound queue, and then select **Reprocess** on the Action Pane.
- Go to **Material handling equipment interface** > **Common** > **Reprocess errored inbound queue**. A standard batch job dialog box appears. There, you can set up a record filter, and schedule or run a batch job to reprocess the queue.

All work operations (picks and puts) run by using the worker who is selected in the **User ID** field on the **Material handling equipment interface parameters** page.

Clean up the inbound queue

Eventually, your inbound queue fills with queue items that are already processed. To remove these items, go to **Material handling equipment interface** > **Periodic tasks** > **Cleanup** > **Inbound queue cleanup**.

Get a quick overview by using the queue manager

To get a quick overview of all the activity that's related to your inbound and outbound queues, go to **Material handling equipment interface** > **Workspaces** > **Queue manager**. The **Queue manager** page provides a set of tabs and tiles that you can use to monitor and explore your queues. It also provides useful links to most of the other pages that are mentioned in this article.

Connect to the MHAX service

MHAX is implemented as a custom service. Therefore, it's accessible via SOAP and REST calls. Here are the addresses of the SOAP and REST endpoints:

- **SOAP:** `https://base_environment_URL/soap/services/WMHEServices`
- **REST:** `https://base_environment_URL/api/services/WMHEServices/WMHEService`

Retrieve messages from the outbound queue

To retrieve messages from the outbound queue, use one of the following methods:

- Use `readOutboundSubscriptionQueue` to retrieve the events based on the subscription ID.
- Use `readOutboundWarehouseQueue` to retrieve the events based on the event type and warehouse ID across multiple subscriptions.

Last updated on 05/06/2026

What's new or changed in the Warehouse Management mobile app

This article lists new features, fixes, improvements, and known issues for each released version of the Warehouse Management mobile app for Microsoft Dynamics 365 Supply Chain Management. It lists changes for each version released since the general availability (GA) release of version 4. Notes for older versions are available in Warehouse Management mobile app release notes archive.

Starting with version 4.1.1.0, every V4 and later release is tagged with its publication date. Publication dates determine whether a release is within the rolling 12-month support window that applies to V4 and all later releases. Learn more in Support policy for the Warehouse Management mobile app.

> **Note**
>
> The publication date listed for each version marks the day the release becomes available in AppCenter and the day progressive rollout to the app stores (Microsoft Store, Google Play, and Apple App Store) begins. Because rollout is gradual, the update might not appear on every device on that exact date. Availability in each store can take several additional days to reach all users.

> **Tip**
>
> For upcoming release dates, see **Warehouse Management mobile app release schedule**. That article also describes optional ways to validate a release before it reaches your warehouse floor.

Release notes for version 4.1.6.0 (September 2, 2026)

Version 4.1.6.0 is a hotfix release.

Bug fixes in version 4.1.6.0

- **Spinner control** – Fixed an issue where the spinner control reset the entered value to *0* when a worker entered a value that was greater than *20*.

Improvements in version 4.1.6.0

- **External numeric keypads** – You can now use an external numeric keypad (numpad) to enter digits in the app.

Release notes for version 4.1.5.0 (August 25, 2026)

Version 4.1.5.0 is a cumulative update that includes new features, improvements, and a broad set of fixes.

New features in version 4.1.5.0

- **Camera scanning** – The camera scanner now recognizes more barcode formats. No configuration is required because the additional formats are detected automatically. You can also set the color and thickness of the scan frame that outlines the scanning area.
- **Notification sounds** – You can now select a separate sound for success, failure, and warning notifications. A duplicate re-scan sound setting was also removed.
- **ProGlove arm scanners** – Added the first version of the bridge for ProGlove arm scanners. Scanner input, including image capture, is now connected to the form flow. Learn more in Advanced bar code scanner configuration.
- **Client settings** – Added an editable **Client settings** option for connections. Use these settings in enterprise environments that have specific requirements, such as environments that require Transport Layer Security (TLS) 1.2 or that require workers to sign in every time they start the app. New connections now use user name and password authentication by default.

Improvements in version 4.1.5.0

- **App startup** – Reduced the work that the app does while it starts, and reduced the size of the fonts and scripts that it loads. The app now starts faster.
- **Step pages** – Reduced the number of times that a step page redraws itself, so that steps respond faster when values change.
- **Sign-in performance (Android)** – Removed unnecessary authentication token renewals that added delay to routine requests.
- **Network communication** – Improved how the app communicates with the server. Network behavior can now be adjusted centrally, without a new app version.

- **Details cards** – Details cards are now scrollable by default.
- **Android 16** – The app and its native modules now target Android 16 (API level 36), which meets current Google Play requirements.
- **Authentication libraries** – Updated the Microsoft Authentication Library (MSAL) and OneAuth libraries to supported versions.
- **Security** – Applied multiple security fixes and dependency updates.
- **Performance telemetry** – Round-trip telemetry now matches the measurements that Warehouse Management mobile app version 3 reports, so that you can compare performance between the two versions. Telemetry now also measures how long warehouse tasks take, not just how long the server takes to respond.
- **Telemetry quality** – Numeric values are now recorded correctly in flow telemetry, and less irrelevant data is logged. Therefore, meaningful signals are easier to find.
- **Accessibility** – Improved screen reader support and navigation across menus, work cards, the welcome screen, combo boxes, version information, headers, and announcements of selected items.
- **VoiceOver (iOS)** – Improved VoiceOver support, especially in combo box dialog boxes.

Bug fixes in version 4.1.5.0

All platforms

- **Submitted data** – Fixed an issue where values on the current page could be altered when the page was submitted to the server.
- **Duplicate work** – Fixed an issue where the app resent a submitted page after a timeout or a gateway error, without notifying the worker. This behavior could cause warehouse work to be completed twice.
- **Escaped characters** – Fixed an issue where server responses that contained escaped characters were shown incorrectly.
- **Single sign-on (SSO)** – Fixed a *broker no result* error that prevented workers from signing in.
- **App startup** – Fixed an issue where a failure in a single part of the app prevented the whole app from starting.
- **Error messages** – Fixed an issue where the app showed a generic failure message instead of the error message that the server returned.
- **Error page** – Fixed the layout, icon placement, and animation order on the error page. The page now adjusts correctly when the window is resized.
- **Calculator** – Fixed an issue where the calculator dropped digits during fast entry.
- **Spinner control** – Fixed an issue where the spinner didn't respond to tap and drag gestures on desktop devices.

- **Dialog boxes** – Fixed an issue where a dialog box closed when a worker selected the content inside it.
- **Promoted fields** – Fixed the formatting of promoted fields and corrected default values that were outside the allowed range. Both now match version 3 behavior.

Windows

- **Network failures** – Fixed an issue where the app couldn't reach the Supply Chain Management server or the Microsoft sign-in pages.
- **Blank page on startup** – Fixed an issue where the app showed a blank white page when it started, including when several RemoteApp sessions ran at the same time.
- **Camera scanning** – Fixed an issue where the camera scanner incorrectly decoded barcodes that contained group separators.
- **Layout and display** – Fixed the position of the app window, corrected rendering in high-contrast mode, and fixed an issue where the read-only spinner control cut off long numbers.

Android

- **Sign out** – Fixed an issue where signing out didn't sign the worker out completely. The app now clears both the account and the web session of the identity provider. Therefore, the next sign-in correctly prompts for credentials instead of reusing the previous identity.
- **Barcode scanning** – Fixed an issue where some barcodes couldn't be scanned, because several formats were ignored.
- **Diagnostic files** – Fixed an issue where diagnostic files were saved to a location where mobile mass deployment (MDM) solutions couldn't retrieve them from locked-down kiosk devices.

iOS

- **Barcode scanning** – Fixed an issue where some barcodes couldn't be scanned. Barcode detection now uses Apple Vision.
- **Scan frame** – Fixed an issue where the scan frame appeared in the wrong position.

Release notes for version 4.1.4.0 (May 28, 2026)

Version 4.1.4.0 is a minor update that includes the following features and fixes:

New features in version 4.1.4.0

- **Camera improvements** – Added support for Code 39 Mod 43 barcodes.

Bug fixes in version 4.1.4.0

- **Right-to-left (RTL) support** – Corrected layout and formatting issues for right-to-left languages.
- **Submit spinner** – Resolved a UI hang and glitch that occurred when submitting forms.
- **Keyboard focus** – Fixed an issue where the virtual keyboard didn't automatically trigger when typing started.
- **Encoding** – Resolved character rendering and text corruption issues.
- **On-premises** – Fixed issues affecting on-premises deployments on Android and iOS.
- **AppCenter distribution** – Resolved issues experienced by users downloading the `.msixbundle` from AppCenter.

Release notes for version 4.1.3.0 (May 18, 2026)

Version 4.1.3.0 is a minor update that includes the following fixes and improvements:

Bug fixes in version 4.1.3.0

- **Label encoding** – Fixed an encoding issue that caused incorrect signs to display on labels.
- **Quantity selector** – Improved performance of the quantity selector and resolved a concurrency issue.
- **Calculator** – Fixed an issue where typing in the calculator reopened the calculator instead of continuing input.

Improvements in version 4.1.3.0

- **Right-to-left (RTL) support** – Improved support for right-to-left languages.

Release notes for version 4.1.2.0 (May 6, 2026)

Version 4.1.2.0 is a minor update that includes the following fixes and improvements:

Bug fixes in version 4.1.2.0

- **Components fields** – Fixed an issue that caused `data=1` to appear in some edge case scenarios.

Improvements in version 4.1.2.0

- **Diagnostics** – Added the option to clean up authentication data from the **Diagnostics** page.

- **Windows MDM** – Expanded mobile device management (MDM) configuration collection to support a broader range of MDM providers on Windows.

Release notes for version 4.1.1.0 (April 30, 2026)

Version 4.1.1.0 focuses on performance, small-screen layouts, scanning reliability, and a broad set of platform-specific bug fixes.

> **Important**
>
> Starting on **May 1, 2027**, version 4 (V4) and every later release of the Warehouse Management mobile app follow a rolling 12-month support window. Microsoft accepts support cases only for releases that were published within the previous 12 months. This policy applies to every release from V4 onward, regardless of whether it's a major, minor, or patch version. The app continues to run on out-of-window releases, but support cases require an in-window version. This policy is critical to maintain the quality, security, and platform compatibility of the app. Learn more in Support policy for the Warehouse Management mobile app.

Improvements in version 4.1.1.0

- **Performance enhancements** – Faster response times for the calculator, quantity fields, and overall touch interactions.
- **Small-screen optimization** – Improved layout and rendering for displays around 640 px, optimized for truck-mounted devices.
- **Brokered authentication (optional)** – Fixed edge-case issues and updated the technical documentation. This feature remains optional.

Scanning and input improvements

- **New scanning engine** – A dedicated global listener now handles hardware scans, so you get improved reliability. This change also addresses navigation issues on hardware that uses arrow keys.
- **Dialog support** – Scans work correctly while error, confirmation, or option dialogs are displayed.
- **Duplicate scan protection** – Prevents double actions on devices that send the same barcode through multiple channels.

- **GS1 barcode support** – Batch numbers that contain GS1 separators (`0x1D`) are now handled correctly, resolving previous disconnection issues.
- **Navigation fixes** – Restored arrow-key navigation for processes such as Sales picking, and fixed rescanning during quantity confirmation.

Bug fixes in version 4.1.1.0

Layout and display

- **Pallet building** – The **Done** button is now visible on small or nonmaximized windows. You can access the **Correct**, **Cancel**, and **Done** actions without resizing.
- **Calculator** – Fixed layout problems when using landscape mode on small screens.

Windows

- **Spinner control** – Fixed a problem where the spinner could stop between values. It now snaps correctly.
- **Authentication** – Resolved a crash during sign-in caused by an unhandled exception.
- **Clipboard** – Removed the unnecessary *Allow paste* prompt when selecting input fields.
- **MDM** – Implemented mobile device management (MDM) read of configurations for Windows.

iOS

- **Calculator** – Fixed a problem where the calculator opened automatically without user interaction.

Localization

- **Spanish (ES)** – Fixed a problem where trailing or embedded zeros were removed from production and work quantities. Behavior now matches en-US formatting.

Release notes for version 4.0.39.0

This update includes critical fixes for authentication workflows, significant battery life optimizations, and expanded hardware support across Android, iOS, and Windows.

New features in version 4.0.39.0

- **Scrollable details card** – Added the option to turn on scrolling for details cards so workers can view more information on one screen without collapsing sections. The scrollable details card is designed for larger screens; on smaller devices, the feature might not work as expected. On smaller screens, switching to landscape mode might reduce the available space, which could affect scrolling behavior.

Improvements in version 4.0.39.0

- **Battery consumption** – Optimized background processes and resource management, resulting in an approximate 9% reduction in battery usage during standard operation.
- **Authentication and connectivity** – Resolved a caching issue where the application incorrectly reused legacy connection data. This fix prevents missing connections and the need to restart the app manually.
- **MDM detection** – Improved mobile device management (MDM) detection resilience by extending the polling interval for policy checks.
- **Telemetry** – Corrected a reporting error in connection-performance telemetry. Connection speed and latency data are now accurately measured and logged.
- **Device compatibility** – Relaxed hardware and software requirements to enable installation on a broader range of legacy and modern devices.
- **Windows navigation** – Added support for the Esc key on Windows to facilitate quicker navigation and menu exits.
- **Legacy pages** – Implemented targeted performance enhancements for legacy UI pages to reduce load times and input lag.

Fixes in version 4.0.39.0

- **Entra brokered authentication** – Resolved a critical issue on Android that prevented Microsoft Entra brokered authentication from completing successfully.
- **iOS scanning** – Fixed a focus-management glitch where success notifications (toast messages) prevented the scanner from immediately returning to a ready state on iOS devices.
- **iOS UI layout** – Corrected a layout issue on the **Edit connection** page where the system keyboard obscured action buttons at the bottom of the screen on iOS devices.
- **Step banner** – Fixed a bug where the step banner displayed a generic *Scan or enter here* placeholder instead of the context-specific value defined in the step.
- **Stability** – Resolved an issue within the animation library that caused intermittent application crashes, particularly on older hardware.

- **Blind cycle counting** – Fixed a logic error that caused duplicate error messages to be displayed consecutively.

Version 4.0.38.0

Version 4.0.38.0 includes the following fixes and improvements:

- Fixed an issue that caused pages to render incorrectly when users navigated from the **Item inquiry** page to a custom page.
- Fixed an issue where the **Back** button didn't clear previously selected data when navigating between **Item inquiry** pages.
- Fixed an issue where some mobile mass deployment (MDM) solutions couldn't push connection settings, such as those specified in the **ConnectionsJson** field in Microsoft Intune, to the app.
- Fixed an issue where browser shortcuts appeared when users scanned a value.
- Improved the diagnostic file collection flow. Users can now share or save the diagnostic file in a local folder, which is useful when using kiosk mode.
- Fixed an issue where users couldn't scan a license plate on the **Confirm** page when running *Movement* flows. Only the last character of the barcode was shown.
- Fixed an issue in the *Movement* flow where a previous value could be shown in the step banner on the wrong step.
- Fixed a visual issue where **Work list** tile headers that contained special characters weren't displayed correctly.

Version 4.0.37.0

Version 4.0.37.0 includes the following fix:

- Fixed an issue where users had to enter a complete email address when the domain name was present in the connection configuration.

Version 4.0.36.0

Version 4.0.36.0 includes the following fixes and improvements:

- Fixed an issue that caused fields on the **Custom legacy** page to display incorrectly.
- Added support for on-premises environments on iOS.
- Fixed an issue where duplicate names in the XML caused connection failures.
- Fixed an issue where the Copilot summary didn't refresh.

- Fixed an issue where popup list item names were parsed incorrectly, causing an error.
- Fixed an issue where spinner values reset incorrectly after an error occurred.
- Fixed an issue where sending multiple requests caused an application error.
- Improved error messages to provide better clarity for users.
- Improved authentication for on-premises environments on Windows. If you experience problems with a previously working on-premises authentication on Windows, edit the connection and set broker No.
- Fixed an issue where the pull-to-refresh gesture on the work list page didn't refresh correctly.

Version 4.0.35.0

Version 4.0.35.0 includes the following fixes and improvements:

- Fixed an issue in the Sales picking flow where the corresponding quantity value was selected incorrectly.

Version 4.0.34.0

Version 4.0.34.0 includes the following fixes and improvements:

- Fixed an issue that caused installation failure on Windows devices when using a mobile mass deployment (MDM) solution.

Version 4.0.33.0

Version 4.0.33.0 includes the following fixes and improvements:

- Corrected the requirements listed on the Google Play store, which previously stated that near-field communication (NFC) and camera hardware were required. You can install and use the app on devices that don't include these features.
- Resolved a calculator issue that caused a process-guide error when users selected the submit button.

Version 4.0.32.0

Version 4.0.32.0 includes the following fixes and improvements:

- Aligned authentication with Warehouse Management mobile app version 4.0.29.
- Fixed an issue that caused some apps to crash in the previous version.

Version 4.0.31.0

Version 4.0.31.0 includes the following fixes and improvements:

- Fixed an issue that caused some apps to crash in the previous version.
- Added options to check Wi-Fi status, export HAR files, and view logs.
- The app now sends meaningful Wi-Fi telemetry data to Application Insights.
- Fixed several critical crash scenarios, including a gesture-related crash on older devices.
- Fixed an issue where the *Confirm location* placeholder wasn't visible when scanning with the device camera.
- Fixed a visual issue where the focus indicator on work list cards was invisible in light mode.
- Added a **Diagnostics** button to the main menu to improve accessibility.
- Fixed an issue where losing the internet connection redirected users to the sign-in page instead of the page where the connection was lost.
- Resolved an authentication issue that required users to sign in a second time.

Version 4.0.30.0

Version 4.0.30.0 includes the following fixes and improvements:

- Fixed an authentication problem on on-premises Windows installations.
- Improved transitions when navigating between pages.
- Fixed an issue where the work list search field wasn't cleared after refreshing the page.
- Fixed an issue where step instructions didn't remember the *Don't show again* selection when two menu items had the same name.
- Improved Wi-Fi diagnostics through better request handling and clearer error reporting.
- Fixed a concurrency-related problem that could cause sign out failures.
- Fixed an issue on custom legacy pages where fields could appear in the wrong order or show incorrect values after edits.
- Added a loading spinner for long-running sign out operations.

Version 4.0.29.0

Version 4.0.29.0 includes the following fixes and improvements:

- Fixed an authentication failure that occurred when multiple connections used the same client ID. The failure showed an incorrect redirect URL.
- Fixed an issue that affected deployment of the app to iOS devices when using a mobile mass deployment (MDM) solution.

- Improved support for Microsoft Entra Conditional Access.
- Added support for ProGlove devices.
- Added support for haptic feedback through external wearable devices.

Version 4.0.28.0

Version 4.0.28.0 includes the following fixes and improvements:

- Fixed an issue where pulling down to refresh didn't update the work list cards.
- Enabled brokered authentication for Android, iOS, and Windows, which supports features such as Microsoft Entra Conditional Access.
- Added an option on Android to use a new redirect URI, which is required for Microsoft Entra Conditional Access.
- Improved camera-based barcode scanning by adding camera zoom and augmented-reality assisted barcode focusing.
- Fixed problems with hardware keyboard detection.
- Added support for keycode mapping.
- Fixed an issue that prevented non-numerical placeholders from appearing in the step banner.
- Fixed an issue affecting image setup for product and master variants.
- Corrected case-sensitive redirect URI matching on Android.
- Fixed an issue where entering a large digit in the calculator caused the delete button to move offscreen.
- Improved localization across the app.

Version 4.0.27.0

Version 4.0.27.0 includes the following fixes and improvements:

- Fixed an issue where pressing Enter on the scanner hardware keyboard didn't submit the value.
- Fixed a bug that added an extra pipe character when submitting a value.
- Fixed an issue where entering a long digit in the calculator caused the backspace to move unexpectedly.
- Implemented security enhancements.
- Fixed a problem where the device token expired one hour after signing in with device code authentication.
- Improved hardware keyboard functionality.

- Fixed an issue where resizing the screen in Windows changed the order of footer action buttons.
- Added the connection name to the worker sign-in page for better usability.
- Fixed an iOS issue where the calculator appeared unexpectedly during the sales picking flow.
- Fixed a crash that occurred when scrolling through a work list with many cards.
- Fixed an issue where signing out as a default user required reentering the device code for authentication.

Version 4.0.26.0

Version 4.0.26.0 includes the following fixes and improvements:

- Fixed an issue where old values weren't removed after scanning items from a work list.
- Fixed a bug that caused the page to autosubmit when a worker pressed the physical scan button on a device during a summary step.
- Fixed mobile mass deployment (MDM) property retrieval to handle case differences in `connection.json` files.
- Enhanced Windows security.

Version 4.0.25.0

Version 4.0.25.0 includes the following fixes and improvements:

- Resolved a rendering issue on the **Item Inquiry** page.
- Fixed a server request timeout that occurred after the quantity spinner component reset incorrectly when you navigate between pages.
- Stopped the quantity spinner from flickering when it landed between two values.
- Enhanced diagnostic tools for improved troubleshooting.
- Improved translation quality across the app.
- Refined device code error messages for greater clarity.
- Fixed a synchronization issue between the calculator and quantity spinner when using a comma as the decimal separator.

Version 4.0.24.0

Version 4.0.24.0 of the Warehouse Management mobile app is the first general availability (GA) release of version 4 for all supported platforms (Microsoft Windows, Google Android, and Apple iOS) in all supported regions. Version 4 introduces many new features and improvements that

enhance your warehouse management experience. Learn more at Migrate the Warehouse Management mobile app from V3 to V4.

Older versions

Notes for versions older than Version 4 are available in the Warehouse Management mobile app release notes archive.

Last updated on 09/02/2026

Install the Warehouse Management mobile app

This article explains how to download and install the Warehouse Management mobile app on each of your mobile devices, and how to configure the app to connect to your Microsoft Dynamics 365 Supply Chain Management environment. You can configure each device manually, or you can distribute connection settings through your mobile device management (MDM) provider, a file, or a QR code. To compare those options, see Choose how to distribute connection settings.

The Warehouse Management mobile app is only for your internal business use. You may not republish or distribute the Warehouse Management mobile app externally in any app store or similar distribution service.

Operating system requirements

The Warehouse Management mobile app is available for Microsoft Windows, Google Android, and Apple iOS operating systems. To use the app, your mobile devices must have one of the following operating systems installed:

- Windows 10 May 2020 update 1904.1 or later
- Android 7.0 or later
- iOS 13.0 or later

Get the Warehouse Management mobile app

For smaller deployments, you'll typically install the app on each device from the relevant store. Then, manually configure the connection to the environments that you're using.

For larger deployments, automate app deployment and configuration. This approach is more convenient if you manage many devices. For example, use a mobile device management and mobile application management solution such as Microsoft Intune. For information about how to mass deploy installations and updates for the Warehouse Management mobile app, see Mass deploy the mobile app with user-based authentication.

Install the app from an app store

The easiest way to install the app on a single device is to install it from an app store. The store always provides the latest generally available version. Microsoft Intune can also fetch apps from the app stores. Use one of the following links to install the app from an app store:

- **Windows:** Warehouse Management on Microsoft Store
- **Android:** Warehouse Management on Google Play Store
- **iOS:** Warehouse Management on the Apple App Store

Download the app from Microsoft App Center

Instead of installing the app from an app store, you can download it from the Microsoft App Center. The App Center provides installable packages that you can sideload. In addition to the current version, the App Center also lets you download previous versions and might provide preview versions with upcoming features that you can try out. To download current, previous, or preview versions of the Warehouse Management mobile app from Microsoft App Center, use one of the following links:

- **Windows:** Warehouse Management (Windows)

  For instructions about how to install a downloaded package on a Windows device and then set up the required certificates, see Install a Build from App Center.

- **Android:** Warehouse Management (Android)

  A few extra steps might be required to install it. For details, see Testing Android Apps.

- **Android (ARMv7 / armeabi-v7a):** Warehouse Management (Android armeabi-v7a)

  This release targets older Android devices that use 32-bit ARMv7 processors and require a build compiled for the `armeabi-v7a` ABI. Install this package on devices that can't run the standard 64-bit Android release.

- **iOS:** The iOS version of the app is only available through App Store.

For information about how to install a build downloaded from the App Center, see Install a build.

The Warehouse Management mobile app isn't available in app stores in China. However, you can download it from Microsoft App Center and use it along with Dynamics 365 Supply Chain Management operated by 21Vianet in China.

Decide which authentication methods you'll use

Because the Warehouse Management mobile app has read and write access to some of your Supply Chain Management data, each device must authenticate with Supply Chain Management. The app supports several authentication methods. Before you start to deploy the app, learn about the available authentication methods and decide which one you want to use.

After a device authenticates with Supply Chain Management, each worker who uses that device signs in by using their Supply Chain Management worker account. The app loads the worker's personal preferences, such as their default warehouse and app preferences. Therefore, different workers can sign in and out for each shift, while the device itself remains authenticated with Supply Chain Management.

For details about each authentication method and how to set it up, see User-based authentication for the Warehouse Management mobile app.

> **Important**
>
> Use **username/password authentication** for all new and existing deployments. It works without any configuration, and no companion app is required. **Device code flow** remains available for existing deployments, but Microsoft no longer recommends it.

If a device is lost or compromised, see Remove access for a device that uses user-based authentication.

> **Note**
>
> Shared Device Mode authentication isn't currently supported for the Warehouse Management mobile app.

Choose how to distribute connection settings

Every device needs the same connection settings, expressed in JavaScript Object Notation (JSON) format. What differs is how you deliver that JSON to each device. To find your method, ask yourself the following questions:

1. **Do you use a mobile device management (MDM) provider, such as Microsoft Intune?** If you do, use *MDM managed configuration*. It's the only method that requires no setup work on the device itself, and it works on all three platforms.

2. **No MDM provider?** Use a *QR code*. It also works on all three platforms, and it's the practical choice for Android and iOS devices, which restrict file access.
3. **Do you deploy to Windows devices and already push files or run scripts on them?** You can use a *JSON file* instead.
4. **Are you setting up a single device or troubleshooting one?** Enter the settings with *manual input*.

Expand table

Method	When to use it	Platform support	Work required on each device
MDM managed configuration (`ConnectionsJson` key)	You manage devices with an MDM provider. The provider pushes connection settings through app configuration policies, without touching the device file system.	Android, iOS, Windows	None
QR code	You don't use an MDM provider, or you set up devices individually. The app scans a QR code that contains the connection JSON.	Android, iOS, Windows	Scan a code
JSON file (*connections.json*)	You can place files on the device file system.	Windows; Android only through **Add from file** (see the following note)	None, if you use the default file name and location
Manual input	You're setting up a single device or troubleshooting a connection.	Android, iOS, Windows	Type each setting

> **Important**
>
> **Android limitation:** Starting with Android 11, scoped storage prevents external tools (MDM file push, file managers, and USB transfer) from writing to the app's private folder. Therefore, a *connections.json* file can't be delivered to the default path on Android.
>
> On Android, use *MDM managed configuration* or a *QR code* instead. To import a file on a single device, use the app's **Add from file** option, and select a JSON file from an accessible location, such as the downloads folder.

Create a connection settings file or QR code

All the methods except manual input use the same JSON. For the parameters that you can specify, an example file, and the file name and location rules, see Connection settings reference for the Warehouse Management mobile app.

After you create the settings, import them on the device as described in the next section.

Import the connection settings on a device

Follow these steps to import connection settings from a file or a QR code.

1. Start the Warehouse Management mobile app on your mobile device. The first time that you start the app, a welcome message appears. Select **Connect**.
2. If you're importing the connection settings from a file and you used the default name and location when you saved the file, the app might find the file automatically. In this case, skip ahead to step 4. Otherwise, select **Set up connection**, and then continue to step 3.
3. In the **Connection setup** dialog, select **Add from file** or **Add from QR code**, depending on how you want to import the settings:
    - If you're importing the connection settings from a file, select **Add from file**, browse to the file on your local device, and select it. If you select a custom location, the app stores it and automatically uses it the next time.
    - If you're importing the connection settings by scanning a QR code, select **Add from QR code**. The app prompts you for permission to use the device's camera. After you give permission, the camera starts, so that you can use it for scanning. Depending on the quality of the device's camera and the complexity of the QR code, you might find it difficult to get a correct scan. In that case, try to reduce the complexity of the QR code by generating only one connection per QR code. (Currently, you can use only the device's camera to scan the QR code.)
4. After the connection settings load successfully, the selected connection appears.
5. The app connects to your Supply Chain Management server and shows the sign-in page.

Manually configure the application

If you don't have a file or QR code, you can manually configure the app on the device so that it connects to the Supply Chain Management server through the Microsoft Entra ID application.

1. Start the Warehouse Management mobile app on your mobile device.

2. If the app starts in **Demo mode**, select **Connection settings**. If the **Sign-in** page appears when the app starts, select **Change connection**.

3. Select **Set up connection**.

4. Select **Input manually**. The **New Connection** page appears and shows the settings that you need to enter manually.

5. Enter the following information:

   - **Connection name** – Enter a name for the new connection. This name appears in the **Select connection** field the next time you open the connection settings. The name you enter must be unique. (In other words, it must differ from all other connection names that are stored on your device, if any other connection names are stored there.)

   - **Environment URL** – Specify the root URL of Supply Chain Management.

     > **Important**
     > - Don't end this value with a slash (/).
     > - Ensure that the HTTPS (SSL) certificate is valid.

   - **Company** – Enter the legal entity (company) in Supply Chain Management that you want the application to connect to.

   - **Authentication method** – Select one of the following values to specify the method that you use to authenticate with Supply Chain Management. The method that you select here must match the setup of the app in Azure.
     - *Username and password* (recommended) – Ask the worker to enter a user name and password. This option also supports brokered authentication and single sign-on, which are optional.
     - *Device code* (not recommended) – Authenticate by using the device code flow. If a device is still configured this way, reconfigure it to use *Username and password*.

   - **Cloud** – Specify the type of Microsoft Entra ID app registration to authenticate with:
     - *Azure Global* (recommended) – Authenticate by using the global Microsoft Entra ID application that's registered and maintained by Microsoft. This option supports most scenarios, including Microsoft Entra Conditional Access. You don't have to register or maintain your own Microsoft Entra ID app, and you don't have to enter a client ID or tenant.

    - *Manual* – Authenticate through your own custom Microsoft Entra ID app registration. Use this option only when the global application doesn't apply to your deployment because you connect to a Finance + Operations (on-premises) environment, you connect to a cloud other than Azure Global (such as a sovereign cloud), or you have specific requirements that the global application doesn't meet. If you choose this option, you must register and maintain a custom app in Microsoft Entra ID and specify a **Microsoft Entra ID client ID** value for the connection.

- **Microsoft Entra ID client ID** – This field is available only when the **Cloud** field is set to *Manual*. Enter the client ID of your custom app registration. Learn more in User-based authentication.

- **Microsoft Entra ID tenant** – (Optional) This field is available only when the **Cloud** field is set to *Manual*. Enter the Microsoft Entra ID domain name that you're using with the Supply Chain Management server. This value has the form `https://login.windows.net/<your-Microsoft-Entra-ID-domain-name>`. Here's an example: `https://login.windows.net/contosooperations.onmicrosoft.com`. Learn more about how to find your Microsoft Entra ID domain name in Locate important IDs for a user.

  > **Important**
  >
  > Don't end this value with a slash (/).

- **Use Broker** – This option applies only when the **Authentication method** field is set to *Username and password*. It determines whether a broker is used for SSO authentication. Set this option to *Yes* for broker-based authentication and SSO. Set it to *No* to require manual input of a user name and password. Learn more about the broker that each platform requires in Device requirements.

- **Domain name** – This field applies only when the **Authentication method** field is set to *Username and password*. You can use it to make sign-in easier for workers. If you don't set this field, workers must enter their full Microsoft Entra ID user principal name to sign in. A user principal name has the form *<user name>@<domain name>*. If you specify the *<domain name>* part here, workers can sign in by entering only the *<user name>* part. (Nevertheless, workers can still enter their full user principal name.)

6. Select the **Save** button in the upper-right corner of the page.

7. The app connects to your Supply Chain Management server and shows the sign-in page.

Remove access for a lost or compromised device

If a device is lost or compromised, remove its access to Supply Chain Management. The method you use to remove access depends on how the device is configured to authenticate with Supply Chain Management. For instructions, see Remove access for a device that uses user-based authentication.

Related information

- Connection settings reference for the Warehouse Management mobile app
- Warehouse Management mobile app release schedule
- User-based authentication for the Warehouse Management mobile app
- User-based authentication FAQ
- Mobile device user settings
- Assign step icons and titles for the Warehouse Management mobile app

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Last updated on 09/08/2026

Connection settings reference for the Warehouse Management mobile app

Connection settings tell the Warehouse Management mobile app which Microsoft Dynamics 365 Supply Chain Management environment to connect to, which legal entity to use, and how to authenticate. They're expressed in JavaScript Object Notation (JSON) format.

The same JSON is used no matter how you deliver it to your devices. To choose a delivery method, see Choose how to distribute connection settings.

Create a connection settings file or QR code

The JSON must include a connection list that contains the individual connections that you need to add. After you create the JSON, you can save it as a file, deliver it through your mobile device management (MDM) provider, or generate a QR code that has the same content.

The following table summarizes the parameters that you can specify for each connection. Required parameters are listed first, followed by optional parameters.

Expand table

Parameter	Description
`"ConnectionName"`	Specify the name of the connection setting. The maximum length is 20 characters. Because this value is the unique identifier for a connection setting, ensure that it's unique in the list. If a connection that has the same name already exists on the device, the settings from the imported file override it.
`"ActiveDirectoryResource"`	Specify the root URL of Supply Chain Management.
`"Company"`	Specify the legal entity in Supply Chain Management that you want the application to connect to.
`"AuthCloud"`	Specify the type of Microsoft Entra ID app registration to authenticate with: • `"AzureGlobal"` (recommended) – Authenticate by using the global Microsoft Entra ID application that Microsoft registers and maintains. This option supports most scenarios, including Microsoft Entra Conditional Access. You don't have to register or maintain your own Microsoft Entra ID app, and you must not specify an `"ActiveDirectoryClientAppId"` value for the connection.

Parameter	Description
	• `"Manual"` – Authenticate through your own custom Microsoft Entra ID app registration. Use this option only when the global application doesn't apply to your deployment. If you choose this option, you must register and maintain a custom app in Microsoft Entra ID and specify an `"ActiveDirectoryClientAppId"` value for the connection.
`"ActiveDirectoryClientAppId"`	(Optional.) Required only when you set `"AuthCloud": "Manual"`. Specify the client ID of your custom app registration. Learn more in Create a custom application registration.
`"ConnectionType"`	(Optional.) Specify how the connection authenticates with the environment. If you don't specify a value, then `"UsernamePassword"` is assumed. Valid values are: • `"UsernamePassword"` (recommended) – Use username/password authentication. • `"DeviceCode"` (not recommended) – Use device code flow.
`"UseBroker"`	(Optional.) This parameter applies only to the `"UsernamePassword"` connection type. It determines whether a broker is used for single sign-on (SSO) authentication. The default value is `"true"`. Therefore, if you don't set this parameter, the app tries to use a broker. If no broker is available on the device, the app falls back to sign-in through the system browser or a native web view. Set it to `"false"` to always require manual input of a user name and password. Learn more about the broker that each platform requires in Device requirements.
`"DomainName"`	(Optional.) This parameter applies only to the `"UsernamePassword"` connection type. It allows you to implement a simplified sign-in process. If you don't set this field, workers must always enter their full Microsoft Entra ID user principal name (UPN) to sign in. A UPN has the form *<user name>*@*<domain name>*. If you specify the *<domain name>* part here, workers can sign in by entering only the *<user name>* part. (Even if you set the domain name here, workers can still sign in using their full UPN.)
`"ActiveDirectoryTenant"`	(Optional) Applies only when you set `"AuthCloud": "Manual"`. Specify the Microsoft Entra ID domain name that you're using with the Supply Chain Management server. This value has the form `https://login.windows.net/<your-Microsoft-Entra-ID-domain-name>`. Here's an example: `https://login.windows.net/contosooperations.onmicrosoft.com`. Learn more about how to find your Microsoft Entra ID domain name in Locate important IDs for a user.

Example connection settings file

The following example shows a valid connection settings file that contains three connections: *Connection1* uses the global application (no client ID is needed), *Connection2* uses a custom app registration with brokered authentication, and *Connection3* uses a custom app registration where workers enter a user name and password manually.

JSON

```
{
    "ConnectionList": [
        {
            "ConnectionName": "Connection1",
            "ActiveDirectoryResource":
"https://yourenvironment1.cloudax.dynamics.com",
            "Company": "USMF",
            "ConnectionType": "UsernamePassword",
            "UseBroker": true,
            "AuthCloud": "AzureGlobal"
        },
        {
            "ConnectionName": "Connection2",
            "ActiveDirectoryClientAppId": "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
            "ActiveDirectoryResource":
"https://yourenvironment2.cloudax.dynamics.com",
            "Company": "USMF",
            "ConnectionType": "UsernamePassword",
            "UseBroker": true,
            "AuthCloud": "Manual"
        },
        {
            "ConnectionName": "Connection3",
            "ActiveDirectoryClientAppId": "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
            "ActiveDirectoryResource":
"https://yourenvironment3.cloudax.dynamics.com",
            "ActiveDirectoryTenant":
"https://login.windows.net/contosooperations.onmicrosoft.com",
            "Company": "USMF",
            "ConnectionType": "UsernamePassword",
            "UseBroker": false,
            "DomainName": "contosooperations.onmicrosoft.com",
            "AuthCloud": "Manual"
        }
    ]
}
```

File name and location on each device

This section applies when you deliver the settings as a file. If you use MDM managed configuration or a QR code, the app stores the settings itself, and you can skip this section.

If you use the default name and location when you save the connection settings file on each device, the app automatically imports it, even during the first run after the app is installed. If you use a custom name or location for the file, the app user must specify the values during the first run. However, the app continues to use the specified name and location afterward.

The default file name is *connections.json*. The default file location depends on which type of device you're using:

- **Windows:** `C:\Users\<User>\AppData\Local\Packages\Microsoft.WarehouseManagement_8wekyb3d8bbwe\LocalState`
- **Android:** `Android\data\com.Microsoft.WarehouseManagement\files`. Because of Android scoped storage limitations, external tools can't write to this path. Use MDM managed configuration or a QR code instead.
- **iOS:** File sharing isn't supported. Use MDM managed configuration or a QR code instead.

Usually, the paths are automatically created after the first run of the app. However, you can manually create them if you must transfer the connection settings file to the device before installation.

> ⓘ **Note**
>
> If you uninstall the app, the default path and its contents are removed.

How the app applies changes

Every time that the app starts, it reimports the connection settings from their previous location to check for changes. The app updates only connections that have the same names as the connections in the connection settings file. User-created connections that use other names aren't updated.

You can't remove a connection by using the connection settings file.

Related information

- Install the Warehouse Management mobile app
- Choose how to distribute connection settings
- Read connection settings from a QR code
- Mass deploy the mobile app with user-based authentication
- User-based authentication for the Warehouse Management mobile app

- Create a custom application registration

Last updated on 09/08/2026

Migrate the Warehouse Management mobile app from V3 to V4

Version 4 (V4) of the Warehouse Management mobile app brings significant improvements and new features that enhance the user experience and the app's performance.

For V4, Microsoft rewrote the code for the Warehouse Management mobile app to take advantage of the newest software technology. V4 offers the following benefits:

- **Enhanced performance** – Improved application responsiveness and stability.
- **Better customer support capabilities** – Faster issue resolution and customer assistance.
- **Future-ready architecture** – Streamlined development of new features and integrations.

> **Tip**
>
> - The Migration information section provides important advice that can help you avoid unexpected disruptions during the migration process.
> - The V3 support timeline provides the rollout schedule and download details.
> - If you're migrating iOS devices, read the Migrating from V3 to V4 for iOS section before you begin. Connection settings aren't preserved on iOS and Device Code authentication isn't supported.

Rollout

Microsoft has been rolling out V4 globally for several months by progressively increasing coverage of countries and regions. Each expansion wave was carefully tested to ensure stability and readiness. The rollout phase is now nearing completion.

V4 general availability schedule and sources

The general availability release of V4 is available for the following platforms on the following schedule:

- **Google Android** – Available globally starting at the end of January 2026 from Google Play and Microsoft App Center.

- **Microsoft Windows** – Available globally starting in February 2026 from Microsoft App Center.
- **Apple iOS** – The official release of WMA iOS v4 began on 23 February 2026. As of January 2026, it's available through Apple Test Flight.

Migration information

System requirements

The system requirements for V4 are the same as the system requirements for V3, except for Android devices. V3 supports Android 5 and later, but **V4 requires Android 7 or later**. Devices that run older Android versions can continue to use V3 until the May 2026 end-of-support date. However, no new releases or feature updates are available for V3, and all newly reported issues are resolved only in V4.

If you're running a newer version of Android, use V4 because it provides better compatibility than V3 on newer systems.

Compatibility between V3 and V4

V4 supports a smooth transition from V3. The following considerations summarize what stays compatible during the migration and what you should plan for when upgrading.

- **Customizations are preserved** – All customizations and configurations from V3 are fully compatible with V4 and remain functional.
- **On Android and Windows devices, connection settings can be preserved on upgrade** – When you upgrade the Warehouse Management mobile app from version 3.0.8 or higher to V4 on Android or Windows, your existing connection settings are automatically migrated. Connection settings are only preserved during an upgrade, not during a fresh installation. To preserve settings:
  - Don't uninstall V3. Instead, download the V4 installer and select the **Upgrade** option.
  - If you're running V3.0.7 or older, first upgrade to V3.0.8 or V3.0.9, then upgrade to V4.
- **On iOS devices, connection settings aren't preserved on upgrade** – You must reconfigure connections manually after installing V4. To simplify this process, prepare QR codes in advance. Learn more in Read connection settings from a QR code.
- **MDM Deployment** – If you use a mobile device management (MDM) solution to distribute the app, the connection settings are preserved when you migrate from V3.0.9 to V4, or from

V4 to any later version of V4.

- **Concurrent operation** – V3 and V4 can operate simultaneously in the same warehouse environment without conflicts provided they're installed on separate devices. You can use different authentication methods for each version without conflict. This capability allows for a phased rollout of V4 without disrupting ongoing operations. However, you can't run V3 and V4 on the same device at the same time.
- **V3 requests remain active** – Microsoft doesn't block requests coming from V3. You can continue using V3 until you're ready to migrate.

> ⓘ **Important**
>
> Microsoft doesn't force-update any device. When a new version of the Warehouse Management mobile app is released, Microsoft publishes it to the app stores (Microsoft Store, Google Play, and Apple App Store). The update is only downloaded to a device if auto-update is enabled in the device's store settings. If auto-update is disabled, the device continues to run its current version until an administrator or user manually triggers the update.
>
> To ensure a consistent and predictable migration across your device fleet, use a Mobile Device Management (MDM) solution, such as Microsoft Intune. Unlike app stores, an MDM provides a dedicated management channel that gives administrators full control over when and how updates are applied to each device.

Migrating from V3 to V4 for iOS

Microsoft started the official release of WMA iOS V4 on February 23, 2026. The rollout is phased, starting with a limited percentage of users and gradually increasing over time.

> ⓘ **Important**
>
> iOS migration has two key differences from Android and Windows:
>
> - **Connection settings aren't preserved** – When upgrading from V3 to V4 on iOS, you lose existing connection settings. You must manually reconfigure connections after the upgrade. To simplify this process, generate QR codes in advance. Learn more in Read connection settings from a QR code.

- **Device Code authentication isn't supported on iOS V4** – Before upgrading, ensure your environment is configured for username/password authentication.

Before you upgrade iOS devices

1. Verify that username/password authentication is properly configured in your environment.
2. Prepare QR codes or JSON configuration files for all connections that you need to reconfigure.
3. If you want to validate V4 behavior before the full rollout, join Apple TestFlight to test the V4 version.
4. If you want to prevent automatic updates, disable auto-update in your App Store or MDM configurations.

If you need to return to V3

If critical problems arise while you're testing V4, you can return to V3.0.9. The following conditions apply:

- **Regional rollback** – You can't downgrade a region after migrating it.
- **App downgrade** – You can downgrade individual devices from V4 to V3.0.9 without problem to ensure operational continuity.
- **Download links** – Use the following links to download V3.0.9 installers for Android and Windows devices:
  - Downgrade to V3 for Android
  - Downgrade to V3 for Windows
- **To install the Microsoft Certificate from an MSIX bundle file** – Follow these steps:

  1. Go to the folder that contains the MSIX bundle folder.
  2. Right-click the MSIX and select **Properties**.
  3. In the **Properties** window, open the **Digital Signatures** tab.
  4. From the list of signatures, select the Microsoft signature.
  5. Select **Details**, and then select **View Certificate**.
  6. Select **Install Certificate**.
  7. Choose the certificate store location.
  8. Select **Local Machine** and select **Next**.
  9. Select **Trusted Root Certification Authorities**.
  10. Select **Next**, and then **Finish** to complete the certificate installation.

Authentication

Authentication in cloud environments

App users must complete a single authentication process the first time they use the app on each device that is migrated to V4. After a device is successfully migrated, it stays authenticated. You don't need to reauthenticate the device when updating to future versions of V4.

Authentication in on-premises environments

If you use an on-premises environment of Supply Chain Management, you don't need a new Entra ID account or infrastructure configuration to support V4. Continue to use your existing configuration as established for V3. However, to support Windows and Android devices, you must add a new redirect URI to your application registration for the Supply Chain Management tenant in AD FS. For V4 of the Warehouse Management mobile app, the redirect URIs are as follows:

- **Windows** – `ms-appx-web://microsoft.aad.brokerplugin/{clientId}` (where *{clientId}* is your Microsoft Entra client ID).
- **Android** – `msauth://com.Microsoft.WarehouseManagement/hpavxC1xAIAr5u39m1waWrUbsO8=`

For more information and detailed setup instructions, go to User-based authentication for the Warehouse Management mobile app in on-premises deployments.

Supported authentication methods

The following table summarizes the supported authentication methods for each platform and type of deployment environment (cloud or on-premises).

Expand table

Platform	Cloud environment	On-premises environment
Windows	Username/password, Device code[1]	Username/password, Device code[1]
Android	Username/password, Device code[1]	Username/password only
iOS	Username/password only	Username/password only

Brokered authentication isn't a separate authentication method. It's an option of username/password authentication that enables single sign-on (SSO). Learn more in Single sign-on.

[1] For backward compatibility, these platforms still support device code flow, but Microsoft no longer recommends it because it's a frequent target of phishing attacks. It's blocked by default in *new* Microsoft Entra ID tenants and doesn't support single sign-on (SSO) or brokered authentication. Use the migration to V4 as an opportunity to move these devices to username/password authentication. Learn more in Device code flow authentication.

V3 support timeline

Use the following timeline to plan your transition from V3 and ensure devices are migrated to V4 before support ends.

- **End of support** – May 2026 (estimated).
- **Final version** – Version 3.0.9 is the final V3 release. The development team will address any reported issues in V4.
- **Feature development** – The development team won't develop new features for V3.

Last updated on 09/02/2026

User-based authentication for the Warehouse Management mobile app

The Warehouse Management mobile app supports two types of user-based authentication:

- **Username/password authentication** – The standard method for new and existing deployments. It works without any extra configuration and doesn't require a companion app.
- **Device code flow authentication** – A legacy method that remains available for deployments that still use it. Plan a move to username/password authentication, because Microsoft Entra ID security defaults block device code flow in new tenants.

To work out which combination of accounts and sign-in methods fits your warehouse, start with Choose an authentication approach.

> **Note**
>
> Give each Microsoft Entra ID account that signs in to the app only the permissions that it needs for warehousing tasks. Use accounts that are scoped to warehouse mobile device user activities, and don't sign in to devices with admin accounts.

Choose an authentication approach

Two decisions determine how you set up authentication. Answer them in order. The following sections give general guidance about what each combination supports. Your IT department makes the final decision, based on your organization's security policies.

1. **Whose identity does the app sign in with?** Use either a *worker identity*, where every warehouse worker has their own Microsoft Entra ID account, or a *device identity*, where a shared account such as `device1@contoso.com` represents the device itself.
2. **Where is the sign-in completed?** Choose either *locally* (on the warehouse device) or *remotely* (on a separate machine such as an admin's PC). Remote sign-in is typically used when devices are deployed at a site that's far from the people who hold the credentials.

The following table summarizes the four combinations.

Expand table

	Local sign-in (on the device)	Remote sign-in (on another machine)
Worker identity	Recommended for most deployments. Use username/password; QR code and PIN; multifactor authentication (MFA); or any other method that your tenant supports.	Supported. The same methods work. Device code flow is also possible, but it isn't recommended.
Device identity	Supported for shared devices. Workers identify themselves with a mobile device user account after the app is authenticated. Conditional Access signals are less meaningful, because the account represents a device instead of a person.	Use with care. Prefer username/password with MFA, so that you can revoke the session remotely if the device is lost.

In every combination, each warehouse worker still has a *warehouse worker* record with one or more *mobile device user accounts* in the Warehouse management module. The Microsoft Entra ID account authenticates the app with Supply Chain Management. The mobile device user account identifies the person who does the work. Learn more in Mobile device user accounts.

Worker identity, local sign-in

Example: At the Contoso Seattle distribution center, each of the 40 pickers has a Microsoft Entra ID account, such as `maria@contoso.com`. Handhelds are stored in a charging rack. A worker takes any device from the rack and signs in with their own credentials.

It works like this:

1. The worker signs in on the device with their Microsoft Entra ID credentials. Use whichever method suits your security requirements: username/password, QR code and PIN, or MFA.
2. If a default user ID is configured for the worker's warehouse worker record, this single sign-in authenticates the app and signs the worker in as a warehouse worker in one step.
3. Optionally, brokered authentication shares the same Microsoft Entra ID session with other apps on the device, such as Microsoft Teams or Outlook. This step is an advanced option, and it isn't required.

This combination gives you the strongest identity controls, because every action traces back to a named person, and Conditional Access policies apply to a real user.

Worker identity, remote sign-in

Example: Contoso opens a small satellite warehouse with no on-site IT staff. An administrator at headquarters completes the initial sign-in for each device before shipping the device to the site.

The same methods that are available for local sign-in also work here. You can also use device code flow. However, Microsoft no longer recommends it because it's a common target of phishing attacks. Your IT department can decide whether it's acceptable in your environment.

Because the person who enters the credentials often isn't the person who uses the device, plan for the following points:

- Prefer a method that lets the worker complete their own sign-in on the device, even when the device is configured remotely.
- Ensure that you can revoke sessions for a single account without affecting other workers.

Device identity, local sign-in

Example: Contoso mounts terminals on forklifts. Each terminal authenticates with a shared Microsoft Entra ID account, such as `device1@contoso.com`. Drivers then enter their own mobile device user account credentials to start work.

It works like this:

1. The admin configures the app with username/password authentication by using the device's Microsoft Entra ID account.
2. After the app is authenticated, workers sign in with their mobile device user account credentials (user ID and password).
3. When a worker signs out, the app stays authenticated with Supply Chain Management and shows the worker sign-in page for the next person.

This combination works best when many workers share a device, and when you don't want to create a Microsoft Entra ID account for every worker. Be aware of the following limitations:

- Conditional Access policies are harder to apply in a meaningful way. The signed-in identity is a device, not a person, so policies that evaluate user risk or per-user MFA don't produce useful results.
- Supply Chain Management attributes app authentication to the device account. Worker-level traceability comes from the mobile device user account instead.
- Give the device account only the permissions that warehouse mobile device work requires. Learn more in Mobile device user accounts.

Device identity, remote sign-in

Example: Contoso ships preconfigured terminals to a third-party logistics provider. An administrator at headquarters authenticates each device before it leaves.

This combination behaves like worker identity with remote sign-in, but the risk is higher because many people use a single shared account and the device is out of your physical control.

Take the following precautions:

- Use username/password authentication, ideally with MFA. Device code flow isn't recommended for this combination, but whether it's acceptable is a decision for your IT department to make, based on your security policies.
- Use a separate Microsoft Entra ID account for each device, rather than one account for all devices. You can then disable or revoke sessions for a single lost device without disrupting the rest of the fleet.
- Keep the device account's permissions strictly limited to warehouse mobile device work.

Licensing considerations

The Warehouse Management mobile app doesn't introduce its own identity licensing requirements. It requires only that your organization has a Microsoft Entra ID identity provider, and standard Microsoft Entra licensing terms apply.

Features that you layer on top of sign-in have their own licensing requirements. Check the requirements for each feature that you plan to use:

- **Conditional Access** – See Microsoft Entra Conditional Access.
- **QR code and PIN sign-in** – See Prerequisites to enable the QR code authentication method.
- **Microsoft Entra plans in general** – See Microsoft Entra ID licensing.

> **Important**
>
> Using a shared device identity doesn't reduce the number of licenses that you need. A device account such as `device1@contoso.com` authenticates the app, but licensing for Supply Chain Management follows the people who do the work. If you license warehouse workers as frontline workers, you still need a license for each worker who uses the app, even when many workers share one device account.

Confirm your specific entitlements with your licensing contact and the Dynamics 365 Licensing Guide. This article doesn't replace either. Learn more in Microsoft Product Terms.

Username/password authentication

When you use username/password authentication, each human worker must enter the Microsoft Entra ID username and password associated either with the device or with themselves (depending on the approach you chose). They might also need to enter a mobile device user account ID and password, depending on their warehouse worker record setup.

Microsoft recommends username/password authentication for all new and existing deployments. It works without any configuration: the app signs workers in through the system browser or a native web view, and it doesn't need Microsoft Authenticator, Intune Company Portal, or any other companion app. Most frontline deployments need nothing more than this.

If you later want single sign-on or Conditional Access policies that depend on device signals, see Advanced: brokered authentication and single sign-on.

Microsoft Entra ID also offers *QR code and PIN sign-in*, which lets workers sign in quickly on shared devices by scanning a QR code and entering a PIN, instead of typing a full username and password every time. This flow works without Microsoft Authenticator or Intune Company Portal. Licensing and tenant prerequisites are listed in Prerequisites to enable the QR code authentication method.

> **Note**
>
> QR code and PIN sign-in has the following limitations:
>
> - It's available only on Android and iOS/iPadOS devices. It isn't available on Windows devices.
> - In the current version of the Warehouse Management mobile app, it works only when the app authenticates through the browser (browser-based authentication). It isn't supported when a broker handles the sign-in natively on the device. Therefore, you can't combine it with brokered authentication yet. Broader QR code sign-in support is planned for a future version of the app.
> - A worker must complete their first sign-in by using another method before they can use QR code and PIN sign-in.
>
> This QR code is a Microsoft Entra ID sign-in credential for workers. It's not the same as the QR code that you use to distribute connection settings to devices. Learn more in Read connection settings from a QR code.

Use a custom application registration

By default, the app uses a global application that's provided and maintained by Microsoft, and no application registration is required.

You need to register your own application in Microsoft Entra ID only if you connect to an on-premises environment, connect to a cloud other than Azure Global, or have specific requirements that the global application doesn't meet. For the procedure, see Create a custom application registration for the Warehouse Management mobile app.

Set up employee, user, and warehouse worker records in Supply Chain Management

Before workers can sign in by using the mobile app, each Microsoft Entra ID account that you assign to the enterprise app in Azure must have a corresponding employee record, user record, and warehouse worker record in Supply Chain Management. For information about how to set up these records, see Mobile device user accounts.

Advanced: brokered authentication and single sign-on

Brokered authentication is an advanced capability. You don't have to use it. The app signs workers in through the system browser or a native web view, and it doesn't need Microsoft Authenticator, Intune Company Portal, or any other companion app.

Consider it only if you want one of the following things:

- **Single sign-on (SSO)** – Workers sign in to the app without entering a password, because the app reuses credentials from another app on the device.
- **Conditional Access policies that depend on device signals** – For example, a policy that requires a compliant or managed device. A broker supplies those signals.

If you don't need either, you can ignore brokered authentication entirely.

If you already use brokered authentication, keep it. Nothing about it is deprecated, and no change is required. It's described as advanced because it's optional, not because it's going away.

For the concepts, requirements, and setup steps, see Brokered authentication and Conditional Access.

> ⓘ **Note**

SSO requires **username/password** authentication. It doesn't work with **device code flow**.

Two other limitations apply when a broker handles sign-in:

- The Warehouse Management mobile app *doesn't* support **shared device mode**. To let multiple workers share a device, use the **device identity** approach instead.
- QR code and PIN sign-in isn't available. It currently requires browser-based authentication, as described in **Username/password authentication**.

Device registration requirements

The Warehouse Management mobile app works on a device in any Microsoft Entra ID registration state. You can sign in on all of the following types of devices:

- Microsoft Entra joined devices
- Microsoft Entra registered devices
- Devices that aren't joined
- Devices that aren't registered at all, including a device that's new and has never been registered

You don't have to join or register a device before workers can sign in.

Two things can still introduce a registration requirement, and neither comes from the app itself:

- **Your Conditional Access policies.** If a policy requires a specific device state or compliance status, that requirement comes from your policy. Learn more in Brokered authentication and Conditional Access.
- **Brokered authentication.** A broker passes device signals to Microsoft Entra ID by using a device-bound token, so it works with a registered device. The broker can perform the registration itself during the first sign-in, so this requirement usually isn't something that you set up in advance.

Learn more in What is device identity in Microsoft Entra ID?.

Remove access for a device that uses user-based authentication

If a device is lost or compromised, revoke its access to Supply Chain Management immediately. Disabling the associated Microsoft Entra ID user account revokes access for all devices that use that account. This limitation is why you should use a separate account for each device or each

worker, as described in Choose an authentication approach. It lets you isolate and revoke access for a single device without affecting others.

To revoke access, follow these steps:

1. Sign in to the Azure portal.
2. On the left navigation pane, select **Microsoft Entra ID**, and ensure that you're in the correct directory.
3. In the **Manage** list, select **Users**.
4. To open the user's profile, find the user account that's associated with the device or worker, and select the name.
5. On the toolbar, select **Revoke sessions** to revoke the user account's sessions.

> **Note**
>
> Depending on how you set up your authentication system, you might also want to change the user account's password or completely disable the user account.

Device code flow authentication (not recommended)

> **Note**
>
> This section describes a legacy authentication method. It's documented for deployments that already use it. For new deployments, use username/password authentication.

Device code flow is a two-step sign-in method that was originally designed for devices that don't provide a web browser. Microsoft doesn't recommend it for the Warehouse Management mobile app for the following reasons:

- **It's a frequent target of phishing attacks** – A threat actor can ask a victim to sign in with a device code that the attacker generated, and then use the resulting token to access the victim's account from the attacker's own device. Microsoft Entra ID has no reliable way to verify that the person who enters the code is signing in from the device that generated it.
- **It's blocked by default in new tenants** – Starting July 1, 2026, Microsoft Entra ID security default settings block device code flow on new tenants. This behavior commonly affects new tenants that are created for testing purposes. If you create a new tenant to test the Warehouse Management mobile app, expect device code flow to be blocked by default.

Existing tenants aren't automatically affected unless they already have security defaults enabled.

- **It isn't available everywhere** – Device code flow isn't supported on iOS, and it isn't supported for Android devices that connect to on-premises environments.
- **It doesn't support SSO** – It can't be combined with brokered authentication.

If your environment still depends on device code flow (for example, because you use test scripts or workflows that aren't updated yet), an admin can unblock it for the tenant by disabling security defaults. Because security defaults protect the whole tenant, treat this change as a temporary step while you update your setup to use username/password authentication. For more information, see Microsoft Entra security defaults.

How device code flow works

When you use device code authentication, the Warehouse Management mobile app generates and shows a unique device code. The admin who sets up the device enters this device code into an online form, along with the credentials (name and password) for a Microsoft Entra ID user account. This account can represent either the device itself or the human worker who signs in, depending on how the admin implements the system. In some cases, depending on how the Microsoft Entra ID user account is configured, the admin might also have to approve the sign-in. In addition to the unique device code, the mobile app shows the URL where the admin must enter the code and the credentials for the Microsoft Entra ID user account.

Requirements and restrictions

If you must temporarily continue to use device code authentication, be aware of the following extra requirements and restrictions:

- Create a unique Microsoft Entra ID user account for each device or human worker. In addition, *strictly limit these accounts so that they can perform only warehouse mobile device user activities*.
- While a worker is signing in by using the Warehouse Management mobile app, the app shows a generated device code. This code expires after 15 minutes and is then hidden by the app. If the code expires before sign-in is completed, the worker must generate a new code by selecting **Connect** again in the app.
- How long a device stays authenticated is controlled by your Microsoft Entra ID token lifetime and Conditional Access policies, not by the app. When the tokens for a device expire or are revoked, the device must be authenticated again. Learn more in Refresh tokens in the Microsoft identity platform.

- Single sign-on (SSO) isn't supported when you use device code flow authentication together with a mobile mass deployment (MDM) system (such as Intune) to distribute the Warehouse Management mobile app. You can still use an MDM system to deliver the app to each mobile device and deliver a `connections.json` file that sets up connections using device code. The only difference is that workers must manually sign in when they start to use the app. (This step is required only once.)

Move from device code flow to username/password authentication

To move an existing deployment away from device code flow, follow these steps:

1. Confirm that the Microsoft Entra ID accounts that the devices use can sign in with a username and password. If you use the global application that Microsoft provides, nothing else is required. If you use a custom application registration, also confirm that it allows public client flows.
2. Optional: If you want SSO, enable brokered authentication. On Android and iOS, that requires a broker app on the device.
3. Update your connection settings to specify `"ConnectionType": "UsernamePassword"` (and, when you use a broker, `"UseBroker": true`). Then redistribute them through your MDM provider, a `connections.json` file, or a QR code. Learn more in Connection settings reference and Choose how to distribute connection settings.
4. Sign in one time on each device to complete the authentication with the new method.

Related information

- User-based authentication FAQ
- Brokered authentication and Conditional Access
- Create a custom application registration
- Install the Warehouse Management mobile app

Last updated on 09/08/2026

Read connection settings from a QR code

The Warehouse Management mobile app supports QR code configuration to simplify the connection setup process. Instead of manually entering connection details, you can scan a QR code that contains all the required configuration information.

QR codes can contain connection configuration data in JavaScript Object Notation (JSON) format. Therefore, you can quickly deploy and set up the Warehouse Management app across multiple devices. This method is useful for IT administrators who must configure multiple devices. It's also useful when configurations must be shared with warehouse workers.

Other ways to deliver connection settings include your mobile device management (MDM) provider, a connection settings file, and manual entry on the device. To compare them, see Choose how to distribute connection settings.

Supported connection types

The feature for QR code–based connection configuration supports all connection types that the Warehouse Management mobile app supports:

- **UsernamePassword** (recommended) – Username/password authentication. The examples in this article use this connection type.
- **DeviceCode** (not recommended) – Interactive device code flow authentication.

> **Important**
>
> Specify `"UsernamePassword"` in the QR codes that you generate. A QR code can still specify `"ConnectionType": "DeviceCode"`, but that flow is blocked by default in new tenants, so the code might not work. Learn more in Device code flow authentication.

> **Note**
>
> The examples in this article omit the optional `"UseBroker"` parameter, because you don't need to set it. Learn more in Connection settings reference.

Step 1: Prepare your configuration JSON code

Create a JSON configuration that includes your connection details. Follow the instructions in Connection settings reference. The JSON code should have the following structure.

JSON

```
{
    "ConnectionList": [
        {
            "ConnectionName": "Connection1",
            "ActiveDirectoryResource":
"https://yourenvironment1.cloudax.dynamics.com",
            "Company": "USMF",
            "ConnectionType": "UsernamePassword",
            "AuthCloud": "AzureGlobal"
        }
    ]
}
```

Step 2: Generate a QR code

There are several ways to generate a QR code. Use the method that best suits your needs.

Option 1: Use Copilot

Follow these steps to ask Copilot to generate a QR code for your JSON configuration.

1. Open a Copilot chat session. For example, select the **Copilot** button in Microsoft Edge, or open Microsoft Copilot from the Windows taskbar.

2. Enter a prompt that resembles the following example, but that includes your specific JSON configuration.

   text

   ```
   Please generate a QR code for the following JSON configuration:
   {
       "ConnectionList": [
           {
               "ConnectionName": "Production",
               "ActiveDirectoryResource":
   "https://yourenvironment.cloudax.dynamics.com",
               "Company": "USMF",
               "ConnectionType": "UsernamePassword",
               "AuthCloud": "AzureGlobal"
           }
       ]
   }
   ```

3. Copilot generates a QR code image based on the JSON configuration that you provided. Download the image to your device. For example, select and hold (or right-click) the image, and then select to download it. Alternatively, use a drag-and-drop operation.

Option 2: Use an online QR code generator

To generate a QR code by using an online QR code generator, follow these steps:

1. Visit any reputable QR code generator website. Bing can help you find one.
2. Select *Text* or *JSON* format.
3. Paste the complete JSON configuration that you prepared.
4. Generate and download the QR code image.

Option 3: Use PowerShell

You can use PowerShell to generate a QR code from your JSON configuration. The code resembles the following example, but it includes your specific JSON configuration.

PowerShell

```powershell
# Install and import QR code module if not already installed
Install-Module -Name QRCodeGenerator
Import-Module QRCodeGenerator

# Your JSON configuration
$jsonConfig = @"
{
    "ConnectionList": [
        {
            "ConnectionName": "Production",
            "ActiveDirectoryResource": "https://yourenvironment.cloudax.dynamics.com",
            "Company": "USMF",
            "ConnectionType": "UsernamePassword",
            "AuthCloud": "AzureGlobal"
        }
    ]
}
"@

# Generate QR Code
New-QRCodeText $jsonConfig -OutPath "warehouse-config.png"
```

Step 3: Distribute the QR code

After you generate the required QR code, distribute it in any of the following ways:

- Print it for physical distribution.
- Share it digitally via email or collaboration platforms.
- Display it on-screen for easy scanning.
- Include it in documentation or setup guides.

> **Important**
>
> QR codes contain connection configuration data, which might include tenant IDs and environment URLs. Treat QR codes as sensitive information and distribute them only through secure channels. Avoid posting them in publicly accessible locations.

> **Tip**
>
> Copilot can decode a QR code if you enter a prompt such as "Please decode this QR code" and attach the image. This technique can be useful if you want to verify the contents of a QR code before you distribute it.

Step 4: Scan the QR code on each device

To use the generated QR code to configure a device, follow the steps in Import the connection settings on a device.

Last updated on 09/08/2026

Mass deploy the mobile app with user-based authentication

Automated deployment and configuration of Warehouse Management can be more efficient than manual deployment when you manage many devices. One way to achieve this automation is to use a mobile device management (MDM) solution such as Microsoft Intune. For general information about how to use Intune to add apps, see Add apps to Microsoft Intune.

This article explains how to mass deploy the Warehouse Management mobile app with user-based authentication by using Microsoft Intune.

> **Important**
>
> To use MDM, configure the Warehouse Management mobile app to use username/password authentication. You can't distribute authentication tokens to mobile devices by using MDM.
>
> Single sign-on (SSO) isn't required. Without it, MDM still delivers the app and its connection settings to every device, and each worker signs in manually one time on their device. SSO only removes that one-time sign-in.

Prerequisites

To use an MDM solution to deploy the Warehouse Management mobile app and its connection settings, you must have the following resources available:

- Warehouse Management mobile app version 4.0 or later on all mobile platforms. (If you're still running version 3.x, migrate to version 4 before proceeding.)
- A valid store account for each mobile platform that you support (Microsoft account, Google Account, and/or Apple Account)
- Microsoft Entra ID. Mass deployment doesn't itself require a premium Microsoft Entra plan, but features that you choose to enable might. Learn more in Licensing considerations.
- Microsoft Endpoint Manager admin center (the Intune website)

Authentication behavior after mass deployment

After you deploy the app through MDM, the authentication experience depends on whether single sign-on (SSO) is enabled.

- **Without SSO (default)** — Each worker signs in to the app one time on their device. Deployment of the app and its connection settings is still fully automatic; only that first sign-in is manual.
- **With SSO (optional)** — If the worker is already signed in to another app on the device with the same Microsoft Entra ID account, the app reuses that token, so no separate sign-in is needed. SSO requires brokered authentication, which is an advanced option.

Depending on your user account configuration, workers might still sign in with a mobile device user account in either case.

> **Important**
>
> Don't use device code flow for MDM deployments. Configure username/password authentication instead.

Set up the source files for distribution

Each MDM solution offers several methods for sourcing the apps that it delivers to end devices. For example, a solution might use locally stored binary files or fetch binaries from an app store. Use app stores as the preferred method because it's simple and offers the most convenient way to receive updates.

The following subsections provide examples that show how to set up Intune to fetch apps from the different app stores.

Set up Intune to fetch the app from Google Play

Follow these steps to set up Intune to fetch the Warehouse Management mobile app from Google Play.

1. Sign in to the Microsoft Endpoint Manager admin center.
2. Go to **Apps** > **Android**.
3. On the **Android apps** page, on the toolbar, select **Add**.
4. In the **Select app type** dialog box, in the **App type** field, select *Managed Google Play app*. Then select **Select**.

5. On the **Managed Google Play** page, if you're setting up Google Play for the first time, you're prompted to sign in to Google Play. Sign in by using your Google account.
6. In the **Search** field, enter *Warehouse Management*. Then select **Search**.
7. On finding the Warehouse Management app, select **Approve**.
8. In the **Approve settings** dialog box, select an option to specify how updates should be handled when a new version of the app requests more permissions than the current version. We recommend that you select the **Keep approved when app requests new permissions** option. When you've finished, select **Done** to continue.
9. Select **Sync**.
10. You're returned to the **Android apps** page. On the toolbar, select **Refresh** to refresh the list of applications. Then, in the list, select *Warehouse Management*.
11. On the **Warehouse Management** page, on the **Properties** tab, select the **Edit** link next to the **Assignments** heading.
12. On the **Edit application** page, on the **Assignments** tab, add the user groups and devices that the Warehouse Management app should be available and required for. For information about how to use the settings, see Assign apps to groups with Microsoft Intune.
13. Select **Review + save**.
14. On the **Review + save** tab, review your settings. If they look correct, select **Save** to save them.

Set up Intune to fetch the app from Microsoft Store

Follow these steps to set up Intune to fetch the Warehouse Management mobile app from Microsoft Store.

1. Sign in to the Microsoft Endpoint Manager admin center.
2. Go to **Apps** > **Windows**.
3. On the toolbar, select **Add**.
4. In the **Select app type** dialog box, in the **App type** field, select *Microsoft Store app (new)*. Then select **Select**.
5. On the **Add App** page, on the **App information** tab, select the **Search the Microsoft Store app (new)** link.
6. In the **Search the Microsoft Store app (new)** dialog box, in the **Search** field, enter *Warehouse Management*.
7. When you've found the Warehouse Management app, select it, and then select **Select**.
8. The **App information** tab now shows information about the Warehouse Management app. Select **Next** to continue.

9. On the **Assignments** tab, add the user groups and devices that the Warehouse Management app should be available and required for. For information about how to use the settings, see Assign apps to groups with Microsoft Intune.
10. Select **Next** to continue.
11. On the **Review + save** tab, review your settings. If they look correct, select **Create** to save them.

Set up Intune to fetch the app from the Apple App Store

Follow these steps to set up Intune to fetch the Warehouse Management mobile app from the Apple App Store.

1. Sign in to the Microsoft Endpoint Manager admin center.
2. Go to **Devices** > **iOS/iPadOS**.
3. On the **iOS/iPad enrollment** tab, select the **Apple MDM Push certificate** tile.
4. In the **Configure MDM Push Certificate** dialog box, follow the on-screen instructions to create and upload the required Apple MDM push certificate. For more information about this step, see Get an Apple MDM push certificate.
5. Go to **Apps** > **iOS/iPadOS**.
6. On the toolbar, select **Add**.
7. In the **Select app type** dialog box, in the **App type** field, select *iOS store app*. Then select **Select**.
8. On the **Add App** page, on the **App information** tab, select the **Search the App Store** link.
9. In the **Search the App Store** dialog box, in the **Search** field, enter *Warehouse Management*. Then, in the drop-down list next to the **Search** field, select your country or region.
10. When you've found the Warehouse Management app, select it, and then select **Select**.
11. The **App information** tab now shows information about the Warehouse Management app. Select **Next** to continue.
12. On the **Assignments** tab, add the user groups and devices that the Warehouse Management app should be available and required for. For information about how to use the settings, see Assign apps to groups with Microsoft Intune.
13. Select **Next** to continue.
14. On the **Review + save** tab, review your settings. If they look correct, select **Create** to save them.

Manage connection configurations

The Warehouse Management mobile app lets you import connection settings as a managed configuration through an MDM solution. All platforms share the same **ConnectionsJson** configuration key.

> **Tip**
>
> For Android devices, MDM managed configuration is the recommended method for delivering connection settings in enterprise deployments. Due to Android's **scoped storage** restrictions (enforced starting with Android 11), external tools can no longer place a `connections.json` file in the app's private data folder. MDM managed configuration bypasses this limitation by delivering settings through app configuration policies rather than the file system.

The following subsections provide examples that show how to set up Intune to provide managed configuration for each of the supported mobile platforms. Learn more in App configuration policies for Microsoft Intune.

Create a connection JSON file

To set up managed configuration for all mobile platforms, create a connection JSON file as described in Connection settings reference. This file enables the mobile app to connect to and authenticate with your Dynamics 365 Supply Chain Management environment.

Set up Intune to support managed configuration for Android devices

Follow these steps to set up Intune to support managed configuration for Android devices.

1. Sign in to the Microsoft Endpoint Manager admin center.
2. Go to **Apps** > **App configuration policies**.
3. On the **App configuration policies** page, on the toolbar, select **Add** > **Managed devices**.
4. On the **Create app configuration policy** page, on the **Basics** tab, set the following fields:

    - **Name** – Enter a name for the policy.
    - **Platform** – Select *Android Enterprise*.
    - **Profile type** – Select the device profile types that the app configuration profile applies to.
    - **Targeted app** – Select the **Select app** link. In the **Associated app** dialog box, select the Warehouse Management app in the list, and then select **OK** to apply the setting and

close the dialog box.

5. Select **Next** to continue.
6. On the **Settings** tab, in the **Permissions** section, select **Add**.
7. In the **Add permissions** dialog box, select the checkboxes for **Camera**, **External storage (read)**, and **External storage (write)**. Then select **OK** to close the dialog box and add those permissions to the **Settings** tab.
8. In the **Permission state** field for each permission that you just added, select *Auto grant*.
9. In the **Configuration Settings** section, in the **Configuration settings format** field, select *Use configuration designer*.
10. In the **Configuration Settings** section, select **Add**.
11. In the dialog box, select the checkbox for **ConnectionsJson**. Then select **OK** to close the dialog box.
12. A new row is added to the grid in the **Configuration Settings** section of the **Settings** tab. The **Configuration key** field is set to *ConnectionsJson*. In the **Value type** field, select *String*. Then, in the **Configuration value** field, paste the entire contents of the JSON file that you created in the Create a connection JSON file section.
13. Select **Next** to continue.
14. On the **Assignments** tab, add the user groups and devices that the configuration policy should apply to. For information about how to use the settings, see Add app configuration policies for managed Android Enterprise devices.
15. Select **Next** to continue.
16. On the **Review + save** tab, review your settings. If they look correct, select **Create** to save them.

Set up Intune to support managed configuration for Windows devices

Follow these steps to set up Intune to support managed configuration for Windows devices.

1. Sign in to the Microsoft Endpoint Manager admin center.
2. Go to **Devices** > **Windows**.
3. On the **Windows devices** page, on the **Configuration profiles** tab, on the toolbar, select **Create profile**.
4. In the **Create a profile** dialog box, set the following fields:

    - **Platform** – Select *Windows 10 and later*.
    - **Profile type** – Select *Templates*.
    - **Template name** – Select *Custom*.

5. Select **Create** to apply your settings and close the dialog box.
6. On the **Custom** page, on the **Basics** tab, enter a name for the configuration profile, and then select **Next** to continue.
7. On the **Configuration settings** tab, select **Add**.
8. In the **Add Row** dialog box, set the following fields:

    - **Name** – Enter a name for the new row.
    - **Description** – Enter a short description for the new row.
    - **OMA-URI** – Enter the following value:

      ```txt
      ./User/Vendor/MSFT/EnterpriseModernAppManagement/AppManagement/AppStore/Microsoft.WarehouseManagement_8wekyb3d8bbwe/AppSettingPolicy/ConnectionsJson
      ```

    - **Data type** – Select *String*.
    - **Configuration value** – Paste the entire contents of the JSON file that you created in the Create a connection JSON file section.

9. Select **Save** to apply your settings and close the dialog box.
10. Select **Next** to continue.
11. On the **Assignments** tab, add the user groups and devices that the configuration profile should apply to.
12. Select **Next** to continue.
13. On the **Applicability rules** tab, you can limit the set of devices that the configuration profile applies to. To apply the profile to all qualifying Windows devices, leave the fields blank. For more information about how to use the settings, see Create a device profile in Microsoft Intune.
14. Select **Next** to continue.
15. On the **Review + save** tab, review your settings. If they look correct, select **Create** to save them.

Set up Intune to support managed configuration for iOS devices

Follow these steps to set up Intune to support managed configuration for iOS devices.

1. Sign in to the Microsoft Endpoint Manager admin center.

2. Go to **Apps** > **App Configuration policies**.
3. On the **App configuration policies** page, on the toolbar, select **Add** > **Managed devices**.
4. On the **Create app configuration policy** page, on the **Basics** tab, set the following fields:

    - **Name** – Enter a name for the app configuration profile.
    - **Platform** – Select *iOS/iPadOS*.
    - **Profile type** – Select the device profile types that the profile applies to.
    - **Targeted app** – Select the **Select app** link. In the **Associated app** dialog box, select the Warehouse Management app in the list, and then select **OK** to apply the setting and close the dialog box.

5. Select **Next** to continue.
6. On the **Settings** tab, in the **Configuration settings format** field, select *Use configuration designer*.
7. In the grid at the bottom of the page, set the following fields for the first row:

    - **Configuration key** – Enter *ConnectionsJson*.
    - **Value type** – Select *String*.
    - **Configuration value** – Paste the entire contents of the JSON file that you created in the Create a connection JSON file section.

8. Select **Next** to continue.
9. On the **Assignments** tab, add the user groups and devices that the configuration policy should apply to. For information about how to use the settings, see Add app configuration policies for managed iOS/iPadOS devices.
10. Select **Next** to continue.
11. On the **Review + save** tab, review your settings. If they look correct, select **Create** to save them.

Enroll devices with Intune

You must *enroll* each device that you want to manage by using Intune. Enrollment involves registering with Intune and applying organizational policies for security. Depending on the type of device and the platform, you can use the Company Portal app to enroll devices. The enrollment programs provide access to work or school resources.

Android and iOS devices

To enroll an Android or iOS device, install the Intune Company Portal app on the device. The local user must then sign in to the Company Portal app by using their company account.

Windows devices

Several methods are available to enroll a Windows device. For example, you can install the Intune Company Portal app on the device. For information about how to set up the Company Portal app and how to use the other options, see Enroll Windows 10/11 devices in Intune.

Related information

- Install the Warehouse Management mobile app
- User-based authentication for the Warehouse Management mobile app
- User-based authentication FAQ

Last updated on 09/08/2026

Brokered authentication and Conditional Access for the Warehouse Management mobile app

The Warehouse Management mobile app supports *brokered authentication*, a sign-in method where an OS-level identity broker, such as Microsoft Authenticator or Intune Company Portal, handles authentication and token management on behalf of the app. When you use brokered authentication, the broker provides Microsoft Entra ID with device identity, compliance status, and security signals during every authentication request.

> **Important**
>
> Brokered authentication is an advanced capability. It isn't required, and most deployments don't need it. The app signs workers in through the system browser or a native web view, without Microsoft Authenticator, Intune Company Portal, or any other companion app. Read this article only if you want single sign-on (SSO) or Conditional Access policies that depend on device signals. Otherwise, see User-based authentication for the Warehouse Management mobile app.
>
> If you already use brokered authentication, keep it. Nothing about it is deprecated, and no change is required.

Brokered authentication isn't a separate authentication method. It's an option of username/password authentication that does two things: it enables single sign-on (SSO), and it makes the app compatible with Microsoft Entra Conditional Access policies that depend on device signals. This article covers both. Use it as the single reference for brokered authentication, whether or not you use Conditional Access.

Microsoft Entra Conditional Access is an optional policy engine that controls access based on conditions such as user identity, device compliance, location, and risk level. It isn't required to run the Warehouse Management mobile app.

> **Important**

The Warehouse Management mobile app supports *connecting* through brokered authentication. It doesn't define, configure, or enforce Conditional Access policies, and it doesn't change how your policies are evaluated. Responsibilities are divided as follows:

- **The Warehouse Management mobile app (Microsoft)** – Supports brokered authentication so that the broker can pass device identity, compliance status, and security signals to Microsoft Entra ID during sign-in.
- **Your IT department** – Decides whether to use Conditional Access, and designs, configures, tests, and maintains the policies, device registration, compliance rules, and broker deployment in Microsoft Entra ID and your device management solution.

Because policy design depends on your organization's security requirements, the guidance in this article covers only the app-side setup. For policy decisions and troubleshooting of the policies themselves, work with your Microsoft Entra ID and device management administrators.

> **Note**
>
> The `"UseBroker"` connection setting defaults to `true`. Therefore, if you don't set it, the app tries to use a broker. This default doesn't create a requirement—when no broker is available on the device, the app signs workers in through the system browser or a native web view instead. Deploying a broker is a security design decision that your organization owns.

This article explains how to enable brokered authentication on the Warehouse Management mobile app so that your organization can enforce Conditional Access policies.

How brokered authentication works

When you use brokered authentication, the following flow occurs during sign-in:

1. The Warehouse Management mobile app delegates authentication to the broker app (Microsoft Authenticator or Intune Company Portal) installed on the device.
2. The broker validates the app's identity and retrieves the device's primary refresh token (PRT), which is a device-bound token that carries device identity and compliance claims.
3. The broker sends the authentication request to Microsoft Entra ID, including the device signals.
4. Microsoft Entra ID evaluates any Conditional Access policies that apply to the user, device, or app, and either grants or denies access.

5. If access is granted, the broker returns the authentication token to the Warehouse Management mobile app.

This process enables features such as single sign-on (SSO) across apps, MFA enforcement, and device-bound token protection—all without requiring the Warehouse Management mobile app to manage these security concerns directly.

Device requirements

To use brokered authentication, your device must meet the following requirements:

- You must be running Warehouse Management mobile app version 4.0.28 or later.
- Your device must be running a supported version of Windows, Android, or iOS.
- The device must be registered with Microsoft Entra ID. The broker can perform this registration (Workplace Join) as part of the first sign-in, so it isn't necessarily something that you set up in advance. The app itself doesn't require any particular registration state. Learn more in Device registration requirements.
- A broker app must be installed on the device (see the following table). If no broker is installed, the app falls back to browser-based sign-in, and device signals aren't available to Conditional Access.

Expand table

Platform	Broker app required
Windows	No separate app is required. The user must have a work account configured on the device, and the built-in Windows Web Account Manager (WAM) acts as the broker.
Android	Microsoft Authenticator or Intune Company Portal must be installed.
iOS	Microsoft Authenticator must be installed.

Set up Microsoft Authenticator (Android and iOS)

Follow these steps to set up Microsoft Authenticator. The procedure is the same for both Android and iOS.

1. Install **Microsoft Authenticator** from the device's app store (Google Play for Android, App Store for iOS).
2. Open the app and select the **Menu** button.
3. Select **Settings**.
4. Select **Device registration**.

5. Enter your email or organizational account.
6. Select **Register device**.

App registration

Brokered authentication works with the global Microsoft Entra ID application—you don't need a manual app registration. We recommend the global application because it's simpler to set up and maintain.

The global application is provided as a Microsoft first-party application (FPA) and is available on the Azure commercial cloud. If your environment is deployed on a US government cloud—such as US Government Community Cloud (GCC) or GCC High—the global application isn't available, and you must use a manual app registration instead.

If you already use a manual app registration for other reasons (such as on-premises environment requirements or a government cloud deployment), it also works with brokered authentication. Learn more in Create a custom application registration.

Configure devices to use brokered authentication

When you use the global application, the `"UseBroker"` setting defaults to `true` on all platforms, so the app tries to use a broker without extra configuration. A broker still has to be present on the device for brokered authentication to take effect. You can configure devices manually through the app UI or automatically by distributing a JSON file via QR code or MDM.

If you don't want to use brokered authentication, set the **Brokered authentication** option to *No* on the **Edit connection** page (or set `"UseBroker": false` in the JSON configuration). If the device doesn't have Microsoft Authenticator installed, the app falls back to a standard username and password connection.

Configure the connection manually

To manually set up a connection, follow these steps on each device:

1. Open the Warehouse Management mobile app and open the **Edit connection** page by doing one of the following steps:

    - If your device doesn't yet have any defined connections, select **Connect** to create a new one.

- To edit an existing connection, select **Tap to change**, choose the target connection, and then select **Edit connection settings**.
- To add a new connection, select **Set up connection** and then select **Input manually**.

2. Make the following settings on the **Edit connection** page:

   - **Authentication method** – Set to *Username and Password*.
   - **Cloud** – Set to *Azure Global* (recommended). If you use a custom app registration, set this field to *Manual* and provide the **Microsoft Entra ID client ID** value.

   Configure all the other settings as described in Manually configure the application.

3. Select **Save**.

4. Sign in with the worker's Microsoft Entra credentials.

Configure the connection by using a QR code or MDM system

To prepare for automatic connection configurations distributed by using a QR code or MDM system, create a JSON file that contains the connection details. Learn more in Connection settings reference.

For all platforms, the connection must use username/password authentication, which you specify as follows in the JSON file:

- `"ConnectionType": "UsernamePassword"`

When you use the global application (`"AuthCloud": "AzureGlobal"`), brokered authentication is enabled by default, so you don't have to set `"UseBroker"` explicitly.

The following example shows a JSON configuration that uses the global application with brokered authentication enabled:

JSON

```
{
    "ConnectionList": [
        {
            "ConnectionName": "Connection1",
            "ActiveDirectoryResource": "https://yourenvironment.cloudax.dynamics.com",
            "Company": "USMF",
            "ConnectionType": "UsernamePassword",
            "AuthCloud": "AzureGlobal"
        }
```

```
    ]
}
```

> **Note**
>
> If you use a custom app registration instead of the global application, set `"AuthCloud": "Manual"` and include a value for `"ActiveDirectoryClientAppId"`.

Learn more about distributing the JSON file to your devices in Read connection settings from a QR code and Mass deploy the mobile app with user-based authentication.

Conditional Access policy configuration

After you enable brokered authentication on your devices, your organization's IT administrators decide which Conditional Access policies to apply and configure them in the Microsoft Entra admin center. The following examples show the kinds of policies that organizations commonly apply. They're examples only, not recommendations, and the choice depends on your organization's security requirements:

- **Require MFA** – Require multifactor authentication for all users or specific groups.
- **Require compliant device** – Block access from devices that don't meet your organization's compliance requirements.
- **Location-based access** – Restrict access to specific network locations or IP ranges.
- **Risk-based access** – Block or challenge sign-ins that Microsoft Entra ID detects as risky.

The Warehouse Management mobile app doesn't need any extra configuration to work with these policies, and it has no settings that override them. When you use brokered authentication, the broker passes the required device and user signals to Microsoft Entra ID, which evaluates the policies and grants or denies access accordingly. If a sign-in is blocked, the policy evaluation occurred in Microsoft Entra ID, so your IT administrators must review the policy and the sign-in logs there.

For more information, see the Microsoft Entra Conditional Access documentation.

Related information

- Microsoft Entra Conditional Access overview
- Install and configure the Warehouse Management mobile app
- User-based authentication for the Warehouse app

- Mass deploy the mobile app with user-based authentication
- User-based authentication FAQ

Last updated on 09/08/2026

Create a custom application registration for the Warehouse Management mobile app

The Warehouse Management mobile app uses a Microsoft Entra ID application registration to authenticate and connect to your Supply Chain Management environment. By default, it uses a global application that Microsoft provides and maintains, and no extra setup is required.

> **Important**
>
> Most deployments don't need this article. Use the global application if possible. It's easier to set up and maintain, and it supports most scenarios, including Microsoft Entra Conditional Access. When you use the global application, you don't have to provide a client ID or tenant in the app's connection settings.

You need your own app registration (and therefore a client ID) only in the following cases:

- You connect to a Dynamics 365 Finance + Operations (on-premises) environment.
- You connect to a cloud other than Azure Global (for example, a sovereign cloud such as Microsoft Azure operated by 21Vianet in China).
- You have specific requirements that the global application doesn't meet.

If none of these cases apply to you, use the global application instead. For more information, see Install the Warehouse Management mobile app.

When you use a custom app registration, set `"AuthCloud": "Manual"` in your connection settings, and provide your client ID in `"ActiveDirectoryClientAppId"`. Learn more in Connection settings reference.

Register the application in Microsoft Entra ID

The following procedure shows one way to register an application in Microsoft Entra ID. For detailed information and alternatives, use the links at the end of this article.

1. In a web browser, go to https://portal.azure.com.
2. Enter the name and password of the user who has access to the Azure subscription.
3. Use the search field at the top of the page to find and open the **Microsoft Entra ID** service.

4. Make sure that you're working with the instance of Microsoft Entra ID that's used by Supply Chain Management.
5. On the left navigation pane, expand **Manage** and select **App registrations**.
6. On the toolbar, select **New registration** to open the **Register an application** wizard.
7. Enter a name for the application, select the **Accounts in this organizational directory only** option, and then select **Register**.
8. Your new app registration opens. Make a note of the **Application (client) ID** value, because you need it later. This ID is referred to later in this article as the *client ID*.
9. In the **Manage** list, select **Authentication**.
10. On the **Authentication** page for the new app, open the **Settings** tab, set **Allow public client flows** to *Enabled*, and select **Save**.
11. Open the **Redirect URI configuration** tab and select **Add redirect URI**.
12. In the dialog, select **Android**. Then set the following fields:

    - **Package name** – Enter the following value (case sensitive):

      ```text
      com.Microsoft.WarehouseManagement
      ```

    - **Signature hash** – Enter the following value:

      ```text
      hpavxC1xAIAr5u39m1waWrUbsO8=
      ```

    Select **Configure** to save your settings and close the dialog to return to the **Authentication** page, which now shows your new platform configurations.
13. On the **Redirect URI configuration** tab, select **Add redirect URI**.
14. In the dialog, select **iOS / macOS**.
15. Set the **Bundle ID** field to the following value:

    ```text
    ```

```
com.microsoft.WarehouseManagement
```

16. Select **Configure** to save your settings. Close the dialog to return to the **Authentication** page, which now shows your new platform configurations.
17. On the left navigation pane, expand **Manage** and select **API permissions**.
18. Select **Add a permission**.
19. In the **Request API permissions** dialog, on the **Microsoft APIs** tab, select the **Dynamics ERP** tile and then the **Delegated permissions** tile. Under **CustomService**, select the **CustomService.FullAccess** checkbox. Finally, select **Add permissions** to save your changes.
20. Use the search field at the top of the page to find and open the **Microsoft Entra ID** service.
21. On the left navigation pane, expand **Manage** and select **Enterprise applications**. Then, in the new **Manage** list, select **All applications**.
22. In the search form, enter the name that you entered for the app earlier in this procedure. Confirm that the **Application ID** value for the app matches the client ID that you copied earlier. Then select the link in the **Name** column to open the properties for the app.
23. On the left navigation pane, expand **Manage** and select **Properties**.
24. Set the **Assignment required?** option to *Yes* and the **Visible to users?** option to *No*. Then select **Save** on the toolbar.
25. On the left navigation pane, expand **Manage** and select **Users and groups**.
26. On the toolbar, select **Add user/group**.
27. On the **Add Assignment** page, select the link under the **Users** heading.
28. In the **Users** dialog, select each user that you use to authenticate devices with Supply Chain Management.
29. Select **Select** to apply your settings and close the dialog. Then select **Assign** to apply your settings and close the **Add Assignment** page.
30. In the **Security** list, select **Permissions**.
31. Select **Grant admin consent for <*your tenant*>**, and grant admin consent on behalf of your users. If you lack the necessary permissions, return to the **Manage** list, open **Properties**, and

set the **Assignment required?** option to *False*. Each user can then provide consent individually.

More ways to register an application

For more information about how to register an application in Microsoft Entra ID, see the following resources:

- For instructions that show how to use Windows PowerShell to register an application in Microsoft Entra ID, see Use Azure PowerShell to create a service principal with a certificate.
- For complete details about how to manually register an application in Microsoft Entra ID, see the following articles:
  - Register an application in Microsoft Entra ID
  - Register a Microsoft Entra app and create a service principal

Related information

- User-based authentication for the Warehouse Management mobile app
- Install the Warehouse Management mobile app
- Brokered authentication and Conditional Access

Last updated on 09/08/2026

User-based authentication FAQ

This article answers frequently asked questions about user-based authentication for the Warehouse Management mobile app.

Where can I learn more about user-based authentication?

For more information about user-based authentication, read the following articles:

- User-based authentication for the Warehouse Management mobile app
- Mass deploy the mobile app with user-based authentication
- Install the Warehouse Management mobile app
- Removed or deprecated features in Dynamics 365 Supply Chain Management

Why did Microsoft deprecate service-based authentication for the Warehouse Management mobile app?

User-based authentication offers the following advantages over service-based authentication for the Warehouse Management mobile app:

- It's easier to configure and use.
- It's more secure in contexts where the Warehouse Management mobile app is used.
- It provides more robust security and more configuration options in Microsoft Entra ID.
- It enables single sign-on (SSO) authentication across apps on the same device.

Is service-based authentication deprecated everywhere?

No. Service-based authentication is deprecated only for the Warehouse Management mobile app. Other applications continue to support it.

Which authentication method should I use?

Use username/password authentication for all new and existing deployments. It works on all supported platforms, supports mass deployment, and needs no companion app on the device. If you want single sign-on (SSO) or Conditional Access policies that depend on device signals, add brokered authentication, which is an advanced option. Microsoft Entra ID QR code and PIN sign-in currently requires browser-based authentication, so it can't be combined with brokered authentication yet.

Device code flow is still accepted for backward compatibility, but it's no longer recommended. Learn more in What is device code flow, and why is it no longer recommended? later in this article.

How many Microsoft Entra ID apps do I need to register in Azure?

The Warehouse Management mobile app uses a Microsoft Entra ID application to authenticate and connect to your Supply Chain Management environment. You can use a global application that's provided and maintained by Microsoft, or you can register your own application in Microsoft Entra ID.

We recommend that you use the global application, because it's easier to set up, use, and maintain. In this case, you don't need to register any Microsoft Entra ID applications in Azure. All your devices can connect through the global application.

If you have specific requirements that the global application doesn't meet (such as the requirements for some on-premises environments), you can register your own application in Microsoft Entra ID as described in Create a custom application registration. You still need to create just one Microsoft Entra ID app registration. All your devices can then connect through it.

For information about how to configure your devices to connect through either of these methods, see Install the Warehouse Management mobile app.

How many Microsoft Entra ID users do I need?

Choose the approach that fits your warehouse. Your choice depends on how you manage identities and on your risk tolerance. Learn more in Choose an authentication approach.

Expand table

Scenario	Notes
Use the same Microsoft Entra ID account for all devices	*Not recommended.* This scenario, where you use a single Microsoft Entra ID user for all your devices, is less secure than other scenarios. It can lead to confusion in Supply Chain Management. For example, all warehouse processes are associated with a single system user, even when multiple warehouse workers use multiple warehouse devices to process work. You also can't revoke access for one device without affecting every other device.
Use a unique Microsoft Entra ID account for each device	Each worker must manually sign in to the app when they start to use the device. When you set up one Microsoft Entra ID user for each device, it's easy to block the Microsoft Entra ID user for just one device if that device is stolen or damaged. (Learn more in Remove access for a device that uses user-based authentication.)
Use a unique Microsoft Entra ID account for each worker	Single sign-on (SSO) is supported. Therefore, a worker who signs in to a device is automatically signed in to the Warehouse Management mobile app and all other apps that use the same Microsoft Entra ID account. Conditional Access policies are most effective in this scenario, because the signed-in identity represents a person.

> **Note**
>
> The number of Microsoft Entra ID accounts that you create isn't the same as the number of licenses that you need. Sharing one device account among many workers doesn't reduce your license count. Learn more in Licensing considerations.

Do I need to map users on the Microsoft Entra ID applications page in Supply Chain Management when I use user-based authentication?

No, you don't need to map users on the **Microsoft Entra ID applications** page in Supply Chain Management when you use user-based authentication.

How often do workers need to sign in to the Warehouse Management mobile app?

There's no fixed sign-in interval in the app, and the app doesn't apply a session timeout of its own. The amount of time a device or worker stays signed in is controlled entirely by your Microsoft Entra ID configuration, by configurations such as token lifetimes and Conditional

Access policies (for example, sign-in frequency). Workers must sign in again when the tokens on the device expire or are revoked. Learn more in Refresh tokens in the Microsoft identity platform and Configure adaptive session lifetime policies.

Will the current worker sign-in page change or be removed?

Although the worker sign-in page is required in some authentication scenarios, it can be skipped in others (for example, when a default user ID is set for the warehouse worker account). For more information, see Scenarios for managing devices, Microsoft Entra ID users, and mobile device users.

How can I sign out a device?

Follow these steps to sign out the Microsoft Entra ID account that's signed in on a device:

1. Open the Warehouse Management mobile app.
2. Select **Select a connection**.
3. Select **Sign out**.

What's the difference between Log off and Sign out in the Warehouse Management mobile app?

The Warehouse Management mobile app provides two similar-sounding actions that end different sessions. They aren't interchangeable.

- **Log off** is a menu item that you configure in Supply Chain Management on the **Mobile device menu items** page (with **Mode** set to *Activity code* and **Activity code** set to *Log off*). When a worker selects **Log off**, the worker's session ends on the server. The worker's stored state (such as the last screen they were viewing) is cleared, and the next worker who signs in starts from the menu that's configured for their work user.
- **Sign out** is a built-in button in the Warehouse Management mobile app. It signs the Microsoft Entra ID account out of the app on the current device, but it doesn't end the work user session in Supply Chain Management. The worker's server-side state is preserved, so the next time the same worker signs in on any device, the app returns to the screen that they last had open.

> ⓘ **Note**

If a worker uses **Sign out** while a screen other than the main menu is open, that screen reappears on the next sign-in, even if **Menu name** is set to *Main* for the work user on the **Work users** page. To always start at the configured main menu, the worker must use the **Log off** menu item before they sign out of the app. This behavior is by design.

Can I just use the Microsoft Entra ID as a worker and skip the worker sign-in page?

Yes. You can map Microsoft Entra ID users to worker IDs in Supply Chain Management. Workers can then authenticate the app with Supply Chain Management and sign in as a worker at the same time. For more information, see Scenarios for managing devices, Microsoft Entra ID users, and mobile device users.

Can I mass deploy the mobile app with user-based authentication?

Yes. For details and instructions, see Mass deploy the mobile app with user-based authentication.

What about single sign-on? Can I use the mobile app alongside Microsoft Teams to chat with my coworkers?

Yes, single sign-on (SSO) is supported. Among other capabilities, SSO lets you chat with coworkers through Microsoft Teams while you use the same account to sign in to the Warehouse Management mobile app.

If you're using SSO and sign out from one SSO app (such as Microsoft Teams), you also sign out of all other apps that use that same account (including the Warehouse Management mobile app).

Does user-based authentication work with Dynamics 365 Finance + Operations (on-premises)?

Yes. On-premises installations use Active Directory Federation Service (AD FS) instead of Microsoft Entra ID. However, the settings are similar, including the settings for user-based authentication methods such as username/password authentication. Learn more in Configure the Warehousing app for on-premises deployments.

Is it mandatory to have Microsoft Authenticator installed on the same mobile device as the Warehouse Management mobile app?

No, it's not mandatory for normal use. The app authenticates through the system browser or a native web view, without Microsoft Authenticator, Intune Company Portal, or any other companion app. Microsoft Entra ID QR code and PIN sign-in also work without these apps.

You need Microsoft Authenticator on Android and iOS devices if you want either of the following advanced capabilities:

- **Brokered authentication**, which provides single sign-on (SSO). On Android, Intune Company Portal can act as the broker instead.
- **Conditional Access policies that depend on device signals**, such as device compliance. These policies need a broker on the device to supply the signals.

Both options are advanced. Most frontline deployments don't use them. Whether you deploy a broker is a security design decision that your organization owns. Learn more in Brokered authentication and Conditional Access.

What Microsoft Entra ID device registration state does the app require?

The app doesn't require any specific device registration state. The app works on a device in any Microsoft Entra ID registration state: Microsoft Entra joined, Microsoft Entra registered, not joined, and not registered at all. You don't have to join or register a device before workers can sign in, and the app-level authentication flow works even on a device that's new and never registered.

If your organization requires a specific device state, that requirement comes from your own Conditional Access policies, not from the app. Learn more in Device registration requirements.

Can I authenticate by using Microsoft Entra Conditional Access?

Yes. Use Microsoft Entra Conditional Access to enforce policies such as multifactor authentication (MFA), device compliance, and location-based access controls. This feature requires Warehouse Management mobile app version 4.0.28 or later. Policies that depend on device signals, such as

device compliance, also require brokered authentication, because the broker passes those signals to Microsoft Entra ID. Your IT department decides which policies to apply and configures them in Microsoft Entra ID.

Here are some resources to help you work with Conditional Access:

- Learn how to set up Conditional Access in Brokered authentication and Conditional Access for the Warehouse Management mobile app.
- Find general information about Conditional Access in the Microsoft Entra Conditional Access documentation.
- For Intune-specific guidance, see Learn about Conditional Access and Intune and Common ways to use Conditional Access with Intune.

Can I authenticate using Microsoft Entra Guest User Access?

Yes. You can add an external user as a guest in the Microsoft Entra admin center. Learn how in Quickstart: Add a guest user and send an invitation.

After you send the invitation, create a warehouse worker account in Supply Chain Management and link it to the Microsoft Entra ID guest user account. If the guest user uses the Warehouse Management mobile app, you must assign appropriate roles and permissions to the warehouse worker account.

To set up the Warehouse Management mobile app to accept guest users, make the following connection settings:

- **Cloud** – Set to *Manual*.
- **Microsoft Entra ID client ID** – Enter the client ID of the Microsoft Entra ID application that you want to use.
- **Microsoft Entra ID tenant** – Enter the tenant of the host organization, in the form `https://login.windows.net/<host-organization-domain-name>`. Although this setting is optional for most connections, you must specify it for guest user access, so that workers are authenticated against the host organization instead of their home organization.

What is device code flow, and why is it no longer recommended?

Device code flow provides a two-step process that authenticates users on devices or operating systems that don't necessarily provide a web browser. For example, a media streaming app might use device code flow to allow users to sign in on smart TVs, game consoles, and other devices. For devices and operating systems that don't provide a web browser, device code flow lets you sign in by using another device (such as a computer or mobile phone).

Device code flow is a user-based authentication method that lets you enter a Microsoft Entra ID user name and password to sign in from a device. After the app is signed in, individual workers still sign in by entering their Supply Chain Management worker ID.

Microsoft no longer recommends it, mainly because it's a common target of phishing attacks. It's also blocked by default in new tenants, isn't available on iOS, and doesn't support SSO. Use username/password authentication instead.

For the full reasoning, see Device code flow authentication.

Why is device code flow blocked by default in new tenants, and how can I unblock it if I still need it?

Device code flow is vulnerable to phishing, so Microsoft Entra ID security defaults block it in *new* tenants. Existing tenants aren't affected unless they already have security defaults enabled. If you create a new tenant for testing, expect device code flow to fail there.

If a test environment or legacy workflow still depends on it, an admin can unblock it for that tenant by disabling security defaults. Because security defaults protect the whole tenant, treat this change as a temporary step while you update your setup to use username/password authentication. Learn more in Microsoft Entra security defaults.

How do I move devices that still use device code flow to a recommended method?

Follow the steps in Move from device code flow to username/password authentication. In short, verify that the Microsoft Entra ID accounts can sign in with a username and password, optionally install a broker app on Android and iOS devices if you want SSO, update your connection settings to use `"ConnectionType": "UsernamePassword"` (with `"UseBroker": true` when you use a broker), redistribute the settings, and then sign in one time on each device.

Last updated on 09/08/2026

Mobile device user accounts

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Every time that a worker starts to use the warehouse app, they must sign in by using a user name and password. Any number of warehouse app users can be associated with each warehouse worker in the system, and warehouses are typically associated with each of those warehouse app users. You also configure various options for each warehouse worker record to establish default settings and other settings that are relevant to using the warehouse app.

Assign Microsoft Entra ID accounts to the mobile app in Azure

The Warehouse Management mobile app connects to Supply Chain Management through an *enterprise application* that's set up in Azure. Azure provides tools for creating Microsoft Entra ID user accounts and assigning them to the enterprise application.

Depending on the authentication scenario that you're using, you must set up a unique Microsoft Entra ID user account either for each mobile device or for each human worker that authenticates with Supply Chain Management. Make a note of the **User principal name** value of each Microsoft Entra ID account that you use for this purpose. Use these values later when you set up the Supply Chain Management user record for each account.

For more information about how to set up an enterprise application for the Warehouse Management mobile app, see User-based authentication for the Warehouse Management mobile app.

For more information about how to assign Microsoft Entra ID user accounts to enterprise applications in Azure, see Manage users and groups assignment to an application.

Set up employee records for each device or human worker

For each Microsoft Entra ID user account that you set up in the previous section, you must have a matching employee record in Supply Chain Management. Later, link each employee record to the related Supply Chain Management user record and warehouse worker record.

To create, view, and manage employee records, go to **Human resources** > **Workers** > **Employees**.

Set up Supply Chain Management user records for each device or human worker

For each employee record that you set up in the previous section, you must have a matching Supply Chain Management user record that identifies both the Microsoft Entra ID user account and the employee record. Follow these steps to set up a Supply Chain Management user record.

1. Go to **System administration** > **Users** > **Users**.

2. On the Action Pane, select **New**.

3. Set the following fields for the new user record:

    - **User ID** – Enter a unique ID for the user or device.
    - **User name** – Enter a descriptive name for the user or device.
    - **Provider** – Leave this field set to its default value.
    - **Email** – Enter the **User principal name** value of the Microsoft Entra ID account that you set up for the user or device.
    - **Telemetry ID** – Leave this field set to its default value.
    - **Company** – Select the legal entity (company) where the user works or the device is used.
    - **Person** – Select the employee record that corresponds to the user or device.
    - **Enabled** – Set this option to *Yes*.

4. On the **User's roles** FastTab, assign the *Warehouse mobile device user* role to the user account. The user must have this role to sign in to the warehouse app.

Set up mobile device user accounts

For each user record that you created in the previous section, you must create a *warehouse worker record* and assign one or more *mobile device user accounts* to it.

1. Go to **Warehouse management** > **Setup** > **Worker**.

2. To edit an existing worker, select it in the list pane. To add a new record, select **New** on the Action Pane.

3. If you're setting up a new worker, follow these steps:
   a. In the **Worker** field, select the related employee record from the **Human Resources** module. This value must match the **Person** value that's selected for the related Supply Chain Management user record.
   b. Select **Select**.
   c. On the Action Pane, select **Save**.

4. You can use a default profile to guide the warehouse worker at the packing station through the required process. Alternatively, use the default profile to save the worker's preferred profile settings. On the **Profile** FastTab, set the following fields:

   - **Container packing policy** – Select a container packing policy that defines how containers at the packing station should be processed. The container packing policy that you select here is preselected for the worker when they open the packing station. For more information, see the following blog post: Improved packing functionality.
   - **Packing profile ID** – Select a packing profile ID that defines the packing policy and container settings used. If the selected packing profile ID is associated with a container packing policy, you can't change the **Container packing policy** setting on this page.

5. On the **Default packing station** FastTab, set the following fields to define the default packing station that applies when the worker signs in. The worker can still select another packing station as required.

   - **Site** – Select the site where the default packing station is located.
   - **Warehouse** – Select the warehouse where the default packing station is located.
   - **Location** – Select the location of the default packing station.

6. The **Users** FastTab lets you create any number of mobile device user accounts for the selected worker. Each account is associated with a specific warehouse or warehouses. Use the toolbar to add or remove mobile device user accounts, reset the password for a selected account, or assign warehouses to a selected account. For each mobile device user account, set the following fields:

   - **User ID** – Enter a unique ID.
   - **User name** – Enter a name for the ID.

- **Default warehouse** – Set the default warehouse where the worker usually works. You can use the toolbar to assign additional warehouses, and the worker can switch between warehouses by using the **Change warehouse** indirect activity of the mobile device menu item.
- **Menu name** – Select the root menu that is the starting page for the worker. The ability to set up a root menu for each worker is useful because it lets you control the menu structure that each worker can use. For example, the menu for workers that are active only in the outbound area can be tailored for tasks that are related to outbound operations for that area.
- **Inactive** – A selected checkbox indicates that the mobile device user account is inactive. The mobile device user account is automatically deactivated if a worker enters the wrong password five times in a row in the warehouse app. However, you can also manually select this checkbox. Clear the checkbox to make the user active again.
- **Default user** – Select this checkbox for the mobile device user account that should be the default account for the worker, if the worker should have a default account. If you're using a sign-in scenario where you have a unique Microsoft Entra ID user account for each human warehouse worker, the Warehouse Management mobile app automatically signs in by using the default mobile device user account when the human worker signs in to the device by using their Microsoft Entra ID user account. (Learn more in Scenarios for managing devices, Microsoft Entra ID users, and mobile device users.)

7. On the **Work** FastTab, set the following fields:

- **Allow pick location override** – Set this option to *Yes* to allow the worker to override the location for picking steps. This capability can be useful if the physical inventory doesn't match the system-suggested location.
- **Allow put location override** – Set this option to *Yes* to allow the worker to override the location for put steps. This capability can be useful if the suggested put location is currently full or unavailable, or if staging locations changed.
- **Allow sales over picking** – Set this option to *Yes* to allow the worker to over-pick when sales orders are picked. Learn more in Over-picking for sales orders and transfer orders.
- **Allow transfer order over picking** – Set this option to *Yes* to allow the worker to over-pick when transfer orders are picked. Learn more in Over-picking for sales orders and transfer orders.
- **Allow movement of inventory with associated work** – Set this option to *Yes* to allow the worker to move inventory that's already reserved or already associated with other

work. Learn more in Movement of inventory with associated work in Warehouse management.

- **Allow manual item reallocation** – Set this option to *Yes* to enable manual reallocation for the worker during short picking. Item reallocation guides workers to pick inventory from another location. Although automatic reallocation is available for all workers, manual reallocation requires explicit setup for a worker. The ability to control manual reallocation for each worker can be helpful because it lets you control the visibility that each worker has when, for example, item picking from the quarantine or bulk area is limited to trusted workers. For more information, see the following blog post: Automatic and manual item reallocation during short picking.
- **Is a cycle count supervisor** – Set this option to *Yes* to make the worker a cycle count supervisor. In this case, all cycle counts that the worker performs on the warehouse app are immediately approved. The **Maximum percentage limit**, **Maximum quantity limit**, and **Maximum value limit** fields aren't considered for workers where this option is set to *Yes*.
- **Maximum percentage limit** – Enter the highest percentage limit that a cycle count can vary from the expected count without requiring approval by a cycle count supervisor.
- **Maximum quantity limit** – Enter the total quantity that the entered quantity can differ from the expected quantity (in units) without requiring approval by a cycle count supervisor.
- **Maximum value limit** – Enter the maximum amount that the cost of the inventory can differ from the expected cost without requiring approval by a cycle count supervisor.

> **Note**
>
> To skip supervisor approval, all limit checks must succeed. To let all non-supervisor users skip the approval step, set suitably high values for each of the **Maximum percentage limit**, **Maximum quantity limit**, and **Maximum value limit** fields. If all recorded values are below these limits, supervisor approval isn't required.

8. On the Action Pane, select **Save**.

9. If you added a new mobile device user account, the **Set password** dialog box appears, where you can create a simple password that the user can use to sign in to the mobile app. Enter the simple password two times, and then select **Save password** to continue.

Set the language, number formats, and time zone for each warehouse app user

When a worker signs in to the Warehouse Management mobile app, the language, number formats, and time zone change to match that worker's preferences. The account settings for the worker selected in step 3 in the Set up mobile device user accounts section determine the settings that are used. If you require separate settings for each worker, use different worker accounts. The following procedure shows how to change the language, number formats, and time zone for a warehouse app user.

1. Go to **Warehouse management** > **Setup** > **Worker**.
2. Find the worker that you want to set up, and make a note of the value in the **Worker** field.
3. Go to **System administration** > **Users** > **Users**.
4. Open the user record where the **Person** column shows the **Worker** value that you found in step 2.

   > **Important**
   >
   > The **User ID** values that are listed on the **Users** page are *not* related to the values that are shown on the **Users** FastTab of the **Worker** page that you opened in step 1.

5. On the Action Pane, select **User options**.
6. On the **Preferences** tab, set the following fields:

   - **Language** – Select the language that the worker prefers. This field also controls the number format that's shown in the warehouse app.
   - **Date, time, and number format** – Select the date and time format that the worker prefers. The warehouse app uses the number format associated with the language chosen for the **Language** field instead of this setting.
   - **Time zone** – Select the time zone where the worker works. This field affects the time stamp for all registrations that the worker makes by using the app.

> **Note**
>
> In some cases, the warehouse app might not find specific worker settings for language, number formats, and time zone. The following rules apply:

- If the app isn't connected to a Supply Chain Management environment (for example, the first time that the app is started after it's installed), the language of the local device is used. When the device language changes, the app language also changes. For more information about how to configure the language for the local device, see the documentation for your device and/or operating system.
- If the app is connected to a Supply Chain Management environment, but no preferences are set for the signed-in worker, the language, number formats, and time zone are selected based on the account that's associated with the client ID that the device uses to connect to Supply Chain Management. Learn more in **User-based authentication for the Warehouse Management mobile app**.

Tip

If you notice that incorrect time stamps are shown for registrations that are made by using the warehouse app, you might have to adjust the time zone that's set for the mobile device user account that workers use to sign in and/or authenticate with Supply Chain Management. As was previously mentioned, the time zone setting might come from the mobile device user account that's used to sign in to the warehouse app, as set up on the **Worker** page. Alternatively, if the mobile device user account isn't set on the **Worker** page, the time zone comes from the mobile device user account that's associated with the client ID that's used for authentication, as set up on the **Microsoft Entra applications** page.

Last updated on 07/01/2026

Mobile bar code support

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Because Android is an open-source project, any manufacturer of hardware for warehouse bar code scanners can build a device to run the Android operating system. A device is only Android-compatible if it can run apps that are written for the Android execution environment.

However, a hardware vendor can modify and create overlays for the Android version that runs on their hardware. Microsoft can't take any responsibility to ensure that a mobile bar code scanning app for Android is compatible with a manufacturer's bar code scanning hardware and the Android version that runs on it.

The Warehouse Management mobile app has been tested with a selection of Android powered devices for bar code scanning. These tests only cover a sample of the devices that are available on the market.

We recommend that you test the Warehouse mobile scanning app on selected hardware before you decide on the hardware that you want to buy.

For information about how to solve some types of compatibility issues between the Warehouse Management mobile app and your bar code scanner, see Advanced barcode scanner configuration.

Last updated on 05/07/2025

Scan bar codes using a camera in the Warehouse Management mobile app

This article explains how to set up the Warehouse Management mobile app to scan bar codes using a camera on a mobile device.

> **Note**
>
> This article applies to Warehouse Management mobile app version 4.1.5 and later. Earlier versions provide a more limited camera scanning experience, and they support fewer bar code formats.

Setup

In the display settings of the Warehouse Management mobile app, you can choose whether to use the camera for bar code scanning. If you turn on **Use the camera as scanner**, you can use the camera on every input field that has the preferred input mode set to **Scanning**.

To control whether an input field is scannable, on the **Warehouse app field names** page, set **Preferred input mode** to **Scanning**. When you select this option, a camera can be used for scanning in the Warehouse Management mobile app. Learn more in Configure fields for the Warehouse Management mobile app.

Navigation

Each page where the input field has its **Preferred input mode** set to *Scanning* includes a link to the camera page. When you're on the camera page, use the following options to navigate:

- Select the back button to go back to the **Task and details** page.
- Select the pencil on the **Task and details** page to go to a page where you can manually enter input.
- Select the camera on the **Task and details** page to go back to the camera page.

The camera scans continuously while the camera page is open. Every bar code that the camera recognizes is outlined in the preview, and the decoded value is shown under the outline. You don't have to start or restart a scan, and there's no scanning time-out.

The following controls are available on the camera page:

- **Back** – Returns to the **Task and details** page.
- **Flashlight** – Turns the device flashlight on or off. Use it in dark aisles, in freezers, or when a label is in shadow. The button isn't available on devices that don't have a flashlight.
- **Accept** – Submits the bar code. The button shows a green check mark as soon as a bar code is recognized. If more than one bar code is recognized, the app asks you to identify the correct bar code.

Customize the bar code scan marker

When you scan with the device camera, the app draws a highlight around each bar code that it recognizes, labeled with the decoded value. You can choose both the color and the thickness of that highlight.

A thin, pale marker can be hard to pick out against a busy label, a glossy surface, or in bright sunlight. Warehouse lighting and label stock vary from site to site, so a marker that works well in one facility might be hard to see in another.

To change the marker appearance:

1. Go to **Settings** > **Camera**.

2. Select **Scan marker color**, and choose one of the twelve available colors. Available options include: *Peach* (default), *White*, *Black*, *Red*, *Orange*, *Yellow*, *Lime*, *Green*, *Teal*, *Blue*, *Purple*, and *Magenta*.

   Pick a color that contrasts with the labels that you scan most often. Light colors, such as *White*, *Yellow*, and *Peach*, stand out against dark packaging and printed bar codes. *Black* and darker colors work better against pale or reflective label stock. If markers are hard to spot at a glance, increasing the thickness usually helps more than changing the color.

3. Select **Scan marker thickness**, and choose one of the five available widths. Available options include: *Thin*, *Normal* (default), *Medium*, *Thick*, and *Extra thick*.

The app shows a live preview of your settings, so you can see how the combination looks before you commit to it. Your new settings apply the next time you open the camera scanner.

Keep the following points in mind when you set the scan marker color or thickness:

- **The server saves the setting for the specific worker and device combination** – The setting follows the current worker on the current device, but it isn't shared across devices. If the

worker signs in on a different device, that device uses its own setting. Each worker and device combination is configured independently.

- **The setting persists across sessions** – After you set it, your choice is remembered when you close the scanner, sign out, or restart the app.
- **The setting only affects the display** – It doesn't change how bar codes are detected or decoded, and it has no effect on scanning accuracy or on hardware (laser) scanning.

Supported bar code formats

The following table shows the bar code formats that each platform supports.

Expand table

Bar code format	Android	iOS	Windows	Common use cases
Aztec	Yes	Yes	Yes	Transportation tickets, identification documents
Codabar	Yes	Yes	Yes	Libraries, blood banks, shipping labels
Code 39	Yes	Yes	Yes	Automotive, healthcare, government applications
Code 93	Yes	Yes	Yes	Logistics, inventory management
Code 128	Yes	Yes	Yes	Supply chain, shipping, product identification
DataBar (RSS-14)	Yes	Yes	Yes	Fresh foods, healthcare, small items
DataBar Limited	Yes	Yes	Yes	Small item identification, healthcare
DataBar Expanded	Yes	Yes	Yes	Coupons, loyalty cards, variable data
Data Matrix	Yes	Yes	Yes	Electronics, automotive parts, pharmaceuticals
DX Film Edge	No	No	Yes	Photography, film processing
EAN-8	Yes	Yes	Yes	Small retail products, magazines
EAN-13	Yes	Yes	Yes	Retail products, books, international trade
ITF (ITF-14)	Yes	Yes	Yes	Shipping cartons, distribution packaging
MaxiCode	No	No	Yes	UPS shipping, package tracking
MSI Plessey	No	Yes	No	Shelf and inventory labels in retail
PDF417	Yes	Yes	Yes	Driver's licenses, identification cards, forms
Micro PDF417	Yes	Yes	No	Small parts marking, healthcare specimens

Bar code format	Android	iOS	Windows	Common use cases
QR Code	Yes	Yes	Yes	Marketing, inventory, contact information
Micro QR Code	Yes	Yes	Yes	Small components, electronics, jewelry
rMQR Code	No	No	Yes	Rectangular applications, space-constrained items
UPC-A	Yes	Yes	Yes	Retail products, grocery items
UPC-E	Yes	Yes	Yes	Small retail products, compressed UPC

The following platform notes apply:

- On Android, the Google ML Kit bar code scanner decodes bar codes. An extra scanner decodes the DataBar formats, Micro QR Code, and Micro PDF417. The extra scanner only runs when ML Kit doesn't recognize a bar code, so it doesn't slow down everyday scanning. On iOS, the Apple Vision framework decodes bar codes. On Windows, scanning runs in the browser. Because each platform uses a different decoder, the supported formats differ, as shown in the preceding table.
- MSI Plessey is only supported on iOS 17 or later. No decoder is available for it on Android.
- On iOS, Codabar, the DataBar formats, Micro QR Code, and Micro PDF417 require iOS 15 or later.
- On iOS, UPC-A is reported as its EAN-13 equivalent, which is the same 12-digit number with a leading zero. Most hardware scanners behave the same way. If your process compares scanned values with a UPC-A item identifier, allow for the leading zero.
- The app scans all supported formats at the same time. You don't select a format in the app.

Usage tips

Here are a few tips for effective bar code scanning in the Warehouse Management mobile app:

- **Optimal scanning conditions** – For the best scanning results, ensure adequate lighting, and hold the device steady. In poor lighting, turn on the flashlight from the camera page.
- **Distance** – Hold the camera close enough that the bar code fills most of the width of the preview. Dense, one-dimensional formats, such as Code 39 and Code 128, need more detail than QR codes do. If a bar code isn't recognized, the app gradually zooms in to help, but moving closer is faster.
- **Alignment** – Keep one-dimensional bar codes roughly parallel to the focal plane of the camera. Two-dimensional formats, such as QR Code and Data Matrix, can be scanned at any angle.

- **Format selection** – The app automatically detects supported formats. Manual selection isn't required.
- **Compatibility check** – Before deployment, verify that your platform supports your specific bar code format.
- **Performance** – QR codes and standard retail formats (UPC and EAN) typically offer the fastest scanning performance.
- **Damaged or glossy labels** – Change the angle slightly to avoid glare, and clean the camera lens. Torn or wrinkled labels might have to be entered manually. Select the pencil on the **Task and details** page to enter the value.

Last updated on 08/07/2026

Advanced bar code scanner configuration

On most mobile devices, the Warehouse Management mobile app can capture scanned bar code input right out of the box. Therefore, you just have to install the app, set up the connection to Microsoft Dynamics 365 Supply Chain Management, and start scanning. However, some devices require advanced configuration before their bar code scanner hardware can interact correctly with the Warehouse Management app.

One common method for integrating scanning hardware with the Warehouse Management app is to configure your mobile device to provide *intent output* for bar code scanning. The bar code scanner can then submit scanned data directly to the Warehouse Management app. It doesn't have to send the data as simulated keystroke input instead. If you experience compatibility issues with your bar code scanner, you might be able to fix them by setting up your device to provide intent output.

This article explains how to set up a mobile device to provide intent output for bar code scanning. If your bar code scanner already works correctly with the Warehouse Management app, you don't have to read this article.

> **Note**
>
> Intent output often replaces keystroke output. Therefore, you shouldn't configure your device to use both types of output at the same time.

Prerequisites

To use intent output with the Warehouse Management mobile app, you must be running version 2.3.3.0 or later of the app.

Set up your mobile device

The Warehouse Management mobile app automatically supports intent output for bar code scanning. You don't have to configure any settings in the mobile app or Supply Chain Management to use intent output. However, you must adjust your mobile device's native settings to configure the device to use intent output. The names and locations of these settings vary, depending on the device that you're using. Therefore, see your device's documentation for specific details.

To configure your mobile device to use this feature, look for the following settings:

- **Intent output** – Enable the functionality on your device.
- **Intent action** – Set the value to *com.microsoft.warehousemanagement.BARCODE*.
- **Intent type** – Set the value to *Broadcast intent*.
- **Intent category** – Leave the default value.
- **Extra key** – The Warehouse Management mobile app requires that bar code data arrives as a string value for the `com.symbol.datawedge.data_string` key. For Zebra and ProGlove devices, you don't have to specify this value, because the devices use this key name by default. However, you might have to specify the value for other devices.

Verify your configuration

After configuring intent output, test the setup by scanning a known barcode on a field in the Warehouse Management mobile app. If the scanned value appears correctly in the field, the configuration is working. If the field remains empty or shows unexpected characters, review the intent settings on your device and ensure that keystroke output is disabled.

Last updated on 07/01/2026

Customize the company logo shown in the Warehouse Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can display your company logo in the Warehouse Management mobile app by uploading it through the legal entity settings in Supply Chain Management. The app retrieves the logo based on the legal entity the worker signs in to.

How logo priority works

The app checks for a logo in two places on the **Legal entities** page. The setting on the **Dashboard image** FastTab takes priority over the **Report company logo image** FastTab, as summarized in the following table:

Expand table

Source	When the app uses it
Dashboard image FastTab (with **Dashboard company image type** set to *Logo*)	This logo is always used when available, regardless of whether a logo is also defined on the **Report company logo image** FastTab.
Report company logo image FastTab	Used only if no logo image is defined on the **Dashboard image** FastTab. This image is also used for printed reports that include a logo.

> **Note**
>
> The **Dashboard image** FastTab setting also controls the image shown on the Supply Chain Management web client home page. If you want a different image for the web client, upload your logo for the mobile app first. Then change the **Dashboard company image**

type to a different value and upload the web client logo image. The mobile app continues to use the first logo you uploaded.

Upload company logos for the app

To upload a company logo for each legal entity, follow these steps:

1. Go to **Organization Administration** > **Organizations** > **Legal entities**.
2. Select the legal entity you want to upload a logo for.
3. Choose one of the following methods:

    - **Dashboard image method (recommended)** – Expand the **Dashboard image** FastTab, set **Dashboard company image type** to *Logo*, and select **Change** to upload the image.
    - **Report company logo image method** – Expand the **Report company logo image** FastTab and select **Change** to upload the image.

4. On the Action Pane, select **Save**.
5. Repeat from step 2 for each legal entity that needs a logo.

Last updated on 04/08/2026

Warehouse Management mobile app release schedule

Microsoft doesn't force version upgrades for the Warehouse Management mobile app. You can decide when a new version reaches each of your devices, and you can stay on your preferred version for as long as you like. However, Microsoft normally supports each release for one year. Version 4 and later releases follow a rolling 12-month support window, with each release eligible for support cases only while its publication date falls within the previous 12 months. For details, see Support policy for the Warehouse Management mobile app. Therefore, you should plan for at least one update per device per year.

For most deployments, you should install the app from an app store and let it update automatically. This approach is the simplest way to receive security fixes, platform compatibility updates, and bug fixes. It keeps devices inside the support window without extra effort. If your organization wants to validate a release before it reaches production devices, use the release schedule in this article to plan that validation.

Targeted release schedule (dates subject to change)

Each release becomes available in Microsoft App Center first. Progressive rollout to the app stores (Microsoft Store, Google Play, and the Apple App Store) starts later. Because rollout is gradual, a release might not appear on every device on the date that rollout begins. Availability in each store can take several additional days to reach all users. iOS releases can arrive later still because of Apple App Store review times.

The following table lists the current and scheduled releases of the Warehouse Management mobile app. It includes the following columns:

- The **Availability in Microsoft App Center** column indicates when you can download the release for evaluation or for manual distribution. This date is also the publication date that determines support eligibility under the 12-month support window.
- The **App store rollout begins** column indicates when progressive rollout to the app stores starts.

Expand table

Release version	Availability in Microsoft App Center	App store rollout begins	Status
4.1.6.0	September 2, 2026	September 2, 2026	Released
4.1.5.0	August 11, 2026	August 25, 2026	Released
4.1.4.0	May 28, 2026	Complete	Released
4.1.3.0	May 18, 2026	Complete	Released
4.1.2.0	May 6, 2026	Complete	Released
4.1.1.0	April 30, 2026	Complete	Released

For a list of features and fixes included in each release, see What's new or changed in the Warehouse Management mobile app. For releases before version 4.1.1.0, see Warehouse Management mobile app release notes archive.

> **Note**
>
> Microsoft App Center remains available for downloading Warehouse Management mobile app packages. If that changes, Microsoft will communicate the change and the replacement distribution method in advance through What's new or changed in the Warehouse Management mobile app.

Plan your own update process

The schedule gives you a window between App Center availability and app store rollout that you can use for validation. How you use that window is up to you. Every organization has a different device fleet, mobile device management (MDM) solution, and internal change management policy, so develop the process that fits your organization best. The following points are a starting point, not a required process.

- **Track upcoming releases** – Check this schedule and What's new or changed in the Warehouse Management mobile app so that no version reaches your devices unannounced.
- **Review the release notes** – Compare each release against the flows, device models, scanners, and authentication configuration that matter most to your operation.
- **Test on a small set of pilot devices** – Download the release from Microsoft App Center during the window before store rollout begins, and install it on devices that represent your real hardware mix. Preview builds are also available in App Center for earlier evaluation, but they're intended for evaluation only and aren't covered by the support window.

- **Use your MDM solution to control timing** – On Android, Managed Google Play in Microsoft Intune lets you set a different app update mode (*Default*, *High Priority*, or *Postponed*) for each group that the app is assigned to. For details, see Update a Managed Google Play app. On Windows, you can deploy a specific MSIX package from Microsoft App Center as a line-of-business app. For details, see Add a Windows line-of-business app to Microsoft Intune. On iOS, the app is available only through the Apple App Store, so this type of version control isn't available.
- **Report problems that you find** – If validation uncovers a regression, open a support case with a supported version so that the issue can be fixed in a current release.

Things to keep in mind

- **Stay within the support window** – Holding a version indefinitely eventually moves devices outside the rolling 12-month support window for version 4 and later releases. Learn more in Support policy for the Warehouse Management mobile app.
- **Security and platform fixes ship in current releases** – Older releases don't receive backported fixes, and operating system updates can break older builds.
- **Back-end compatibility isn't guaranteed indefinitely** – Supply Chain Management services evolve over time, so a version that you hold for a long time might eventually stop working correctly.
- **Manual distribution adds ownership** – If you take devices off the store update path, your organization becomes responsible for delivering every subsequent update, including security fixes. Make sure that you have a reliable mechanism to get the latest version onto your devices. The Warehouse Management mobile app is for your internal business use only. You may not republish or distribute it externally in any app store or similar distribution service.

Related information

- Install the Warehouse Management mobile app
- Mass deploy the mobile app with user-based authentication
- Support policy for the Warehouse Management mobile app
- What's new or changed in the Warehouse Management mobile app
- Migrate the Warehouse Management mobile app from V3 to V4

Last updated on 09/02/2026

Support policy for the Warehouse Management mobile app

This article describes the support policies for the Warehouse Management mobile app. The policy that applies to you depends on the version you are running:

- **Version 3 (V3)** reaches end of support in May 2026. After that date, Microsoft no longer accepts support cases for V3. Migrate to V4 before May 2026 to keep support coverage.
- **Version 4 (V4) and all later releases** follow a rolling 12-month support window starting May 1, 2027. After that date, a release is eligible for support cases only if its publication date is within the previous 12 months. This rule applies to every V4 release. It also applies to later major, minor, and patch versions.

These policies help keep the app secure, reliable, and compatible with current platforms. By focusing on recent releases, Microsoft can:

- Deliver security fixes faster.
- Keep up with changes in Apple, Google, and Microsoft platforms.
- Provide a consistent baseline for troubleshooting.

Older releases of the app continue to function. The policies define when Microsoft accepts support cases. They don't affect service availability or sign-in.

Version 3 support policy

V3 of the Warehouse Management mobile app reaches end of support in May 2026. For details, see the Migration to V4 Guide. Migrate devices to V4 before that date to keep support coverage.

V3 runs on the *Xamarin* framework. Xamarin is no longer supported, so Microsoft can't ship new features or bug fixes for V3.

Scope of support after May 2026

Back-end APIs stay compatible with V3 during the migration period. This compatibility lets V3 clients keep running while you move to V4. The compatibility duration isn't guaranteed and may end as Supply Chain Management services change.

- **No new releases** – Microsoft can't build, patch, or ship updates for V3. The Xamarin framework is deprecated.
- **Support cases** – Microsoft no longer accepts support cases for the V3 client.

Devices that stay on V3 after May 2026 follow this V3 policy. The 12-month rolling window doesn't apply to V3.

Examples of out-of-scope scenarios

The following table lists examples of issues that Microsoft no longer investigates for V3.

- **Authentication and identity (Microsoft Entra ID)** – V3 uses legacy authentication libraries that Microsoft no longer maintains.
  - **Unsupported** – Failures with modern Microsoft Authentication Library (MSAL) flows, *Device not compliant* errors, and new conditional access policies in Microsoft Entra ID.
  - **Resolution** – Migrate to V4 to use current authentication protocols.

- **Client-side performance and connectivity** – The V3 networking stack and libraries are frozen at their current versions.
  - **Unsupported** – Local latency, connection drops, or timeouts caused by the outdated V3 libraries.
  - **Note** – If a performance issue is verified as a server-side or service-wide problem, Microsoft still investigates it as part of standard cloud service support.

- **Operating system (OS) compatibility** – Apple and Google release new OS versions often. Legacy apps often break on those new OS versions.
  - **Unsupported** – Crashes, UI glitches, or launch failures on mobile OS versions released after the final V3 release.
  - **Resolution** – Use V4 for ongoing OS compatibility. V4 is built on the latest software development kits (SDKs).

Version 4 and later support policy

Starting **May 1, 2027**, V4 and every later release follow a rolling 12-month support window. On or after that date, a release is eligible for support cases only if its publication date is within the previous 12 months. The window is evaluated on the date the support case is opened.

This rule applies to every release from V4 onward. It applies to major, minor, and patch versions equally.

How publication date is defined

The publication date of a release is the date the release first becomes available on Microsoft App Center. Releases on the Microsoft Store, Google Play, and the Apple App Store can appear later. iOS releases often arrive later because of store review times. Even so, the Microsoft App Center date is always the date that determines support eligibility.

Each V4 and later release, starting with version 4.1.1.0, lists its publication date in What's new or changed in the Warehouse Management mobile app.

Examples

- A release published on *April 30, 2026* is eligible for support cases until *April 30, 2027*. Because that date falls before May 1, 2027, the release becomes ineligible on May 1, 2027 (the date the policy takes effect).
- A release published on *June 15, 2026* is eligible for support cases until *June 15, 2027*.
- A release published on *August 1, 2027* is eligible for support cases until *August 1, 2028*.

There's no grace period after the 12-month boundary. Update the device to a supported release before you open a support case.

Why the 12-month window matters

The 12-month window is critical for keeping the app secure and reliable. It lets Microsoft:

- Ship security fixes on a current code base. Backporting across many legacy builds slows fix delivery.
- Keep up with platform changes from Apple, Google, and Microsoft. OS updates, authentication libraries, and store policies often break older releases.
- Diagnose issues against a known, well-tested baseline. This approach makes root-cause analysis faster and more accurate.
- Focus engineering work on changes that help all customers. Issues already fixed in newer releases don't need reinvestigation.

Scope of the 12-month window

- **Support cases** – Microsoft accepts and investigates support cases only when the device runs a V4 or later release with a publication date within the previous 12 months. For older releases, update to a supported version before opening a case.

- **Critical production outages** – Microsoft might still investigate severity-1 incidents that block widespread production operations. The team reviews these cases individually. The standard guidance is still to update to a supported release first. This exception isn't a substitute for staying current.
- **Preview builds** – Preview and beta builds from Microsoft App Center aren't part of the 12-month window. Use them only for evaluation. Report issues through the preview feedback channel.
- **App availability** – The app keeps running on any installed version. Microsoft doesn't block out-of-window clients, and back-end services don't reject their connections. However, Supply Chain Management services change over time. Older releases might eventually stop working with newer back-end behavior. Compatibility for out-of-window releases isn't guaranteed.
- **Security and platform fixes** – Critical fixes ship in current releases. Devices on older releases don't get those fixes.

How to identify the installed version

To check which version is installed, open the app. The version appears on the sign-in screen or in **Settings**. Compare it to the publication dates in What's new or changed in the Warehouse Management mobile app.

How to stay within the support window

- Turn on autoupdate in your app store, or push updates through Microsoft Intune or another mobile device management (MDM) solution. For details, see Mass deploy the mobile app with user-based authentication.
- Track release dates in What's new or changed in the Warehouse Management mobile app. Every V4 and later release, starting with 4.1.1.0, has a publication date.
- Update devices at least once a year. This practice keeps every device inside the 12-month support window.

Related information

- Install the Warehouse Management mobile app
- Migrate the Warehouse Management mobile app from V3 to V4
- What's new or changed in the Warehouse Management mobile app
- Mass deploy the mobile app with user-based authentication

Last updated on 04/30/2026

User settings, color themes, and sound themes

This article explains how to use local user settings to personalize the way your Warehouse Management mobile app works, looks, and sounds.

Workers can set their own preferences locally while working on a device by following the instructions in this article.

Administrators and managers can also set default settings, device-specific settings, and user-specific settings remotely from the web client, as described in Mobile Device User Settings.

Color and sound themes

Warehouse Management mobile app version 4 introduces new color and sound themes that can help make your daily work more enjoyable and visually distinctive.

Color themes

Each worker can select their own color theme, both to match their personal preference and to make it easy to see which user is signed in. This feature is useful when several different workers use the same device throughout the day.

Many color themes are available, including:

- Blue
- Bronze
- Cyan
- Golden
- Green
- High Contrast
- Lavender
- Orange
- Pink
- Platinum
- Red

Each theme also supports several color modes, which are:

- Light – Bright background
- Dark – Dark background
- Auto – Matches your device's default setting

Sound themes

To make your workflow more fun, choose a sound theme that plays a sound after you complete a task or scan an item. For example, you might hear a bird chirp or a robot beep. You can choose from more than 20 sound themes.

Set user preferences on a mobile device

To set your user preferences on a mobile device, including the color and sound theme, follow these steps:

1. Sign in to the Warehouse Management mobile app and go to the **Main menu** page.
2. Select the **Settings** button, which is the gear icon in the top-right corner of the screen.
3. Make the following settings on the **Settings** page as needed:
    - **Product photo placement** – Select how product photos should be shown in the app (learn more about product images in Add an image to a product). Choose one of the following options:
        - *Default (show photo card)* – Display the product image as a large photo card.
        - *Only in step header* – Show the product image within the header section.
        - *Do not show* – Hide the product image in the mobile app.
    - **Color theme** – Select a color theme for the device. Choose from more than 10 color themes.
    - **Color mode** – Select a color mode for the device. The available options are *Light*, *Dark*, and *Auto*. The *Auto* option matches the device's default setting.
    - **Best button position** – Select how buttons should be positioned on the device. Elements in the app move to better fit the preference or handedness of the worker. The available options are *Bottom right (default)*, *Bottom left*, *Top right*, and *Top left*.
    - **Scan with camera** – Set this option to *Yes* to use the device camera to scan input fields where the preferred input mode is set to *Scanning*. If your device has a built-in scanner, set this option to *No* to use the scanner instead.
    - **Field filtering and sorting** – Select which fields are shown on the device and how they're ordered. You can choose to show all fields or just the promoted fields, and

you can choose to sort them or not. Learn more in Configure promoted fields for steps in the Warehouse Management mobile app.

- **Vibration level** – Select the vibration level for the device. Select a value between 0 (zero) and 5. A value of *0* represents no vibration, and a value of *5* represents maximum vibration.
- **Sound level** – Select the sound level for the device. Select a value between 0 (zero) and 10. A value of *0* represents no sound, and a value of *10* represents maximum volume.
- **Sound theme** – Select a sound theme for the device. Choose from more than 20 sound themes.
- **Text scale** – Specify the text size as a percentage of the standard size. Enter a value between 70 and 400. A value of *70* represents the smallest text scale, and a value of *400* represents the largest text scale. *100* is the standard text size.
- **Button scale** – Specify the button size as a percentage of the standard size. Enter a value between 50 and 200. A value of *50* represents the smallest button scale, and a value of *200* represents the largest button scale. *100* is the standard button size.
- **Server request timeout (minutes)** – Set how long the app should wait for a response from the server before timing out. You can select a duration from 1 to 30 minutes.
- **Button style** – Choose how certain buttons should be displayed in the app. The available options are *Buttons* and *Slider*.
- **Quantity spinner mode** – *Spinners* provide a way for workers to enter quantity values by spinning through digits using a sliding gesture, similar to using a combination lock. Use this setting to control how spinners are displayed. Choose one of the following values:
  - *Auto* – Allow the system to choose the appearance based on how much room there is on the screen.
  - *Small* – Always show the small spinner, which presents a compact control.
  - *Don't show* – Don't use spinners. Instead, allow workers to enter numerical values using an on-screen keypad.
- **Show copilot on startup** – Specify whether the Copilot workload page should be shown each time the user signs in to the app. The workload page shows work summaries and AI-generated insights to help warehouse workers better plan their shift. Learn more in Workload insights with Copilot in the Warehouse Management mobile app.
- **Show copilot data as** – Specify whether Copilot summaries should be based on work headers or work lines. The choice depends on how workers prefer to think

about their daily workload. For example, if all your orders typically contain just a few items, workers might prefer to think about their work in terms of whole orders (*Work headers*). However, if quantities vary greatly from order to order, workers might prefer to think about their work in terms of individual items (*Work lines*).

- **Play sound when rescan is required** – Choose whether to play a sound when a rescan is required. A rescan might be required, for example, when a scanned value results in an error, such as an invalid item ID.
- **Configuration mode** – Set this option to *Yes* to explore and understand the relationship between UI controls and their underlying XML. This option can be helpful when you're customizing mobile flow behavior. When this is set to *Yes*, you can press and hold on almost any button or label to drill down into the session XML. This feature isn't available for controls that aren't connected to the session XML.
- **Auto submit** – Controls whether the app or the device handles scanned data submission. Set to *No* (recommended) for new installations. Learn more in Data submission behavior in the Warehouse Management mobile app.

4. Select **Back** when you're done.

Last updated on 04/08/2026

Data submission behavior in the Warehouse Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Warehouse Management mobile app supports two options for how it submits scanned data: *auto submit* and *manual submit*. This article explains the differences between these options and how to configure them.

How the app submits scanned data

The Warehouse Management mobile app supports two approaches for submitting scanned data:

- **Device-controlled (Auto submit = No)** – The app doesn't auto submit. Instead, the device controls submission behavior. This is the *recommended setting* because it provides consistent, predictable behavior across all devices.
- **App-controlled (Auto submit = Yes)** – The app decides how to handle scanned data. Behavior varies by device: some devices (such as Zebra) auto submit, while others (such as Honeywell) require manual confirmation. This option exists for backwards compatibility with versions before 4.0.20.0 and is *not recommended* for new installations.

Configure auto submit in the app

1. Sign in to the Warehouse Management mobile app and go to the **Main menu** page.
2. Select the **Settings** button (gear icon in the top-right corner).
3. Set **Auto submit** to *No* (recommended).

Configure auto submit on the device

When you set **Auto submit** to *No* in the app, you can still achieve auto-submit behavior by configuring your device to simulate an Enter key press after each scan. The exact steps depend on your device model. Refer to your device manufacturer's documentation for details.

> **Tip**
>
> For more granular control, you can use *intents* instead of keystroke simulation. Learn more in **Advanced bar code scanner configuration**.

Last updated on 04/08/2026

Warehouse Management mobile app calculator

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Warehouse Management mobile app includes a built-in calculator that provides essential numerical entry and mathematical computation capabilities. It enables workers to perform calculations without leaving their current workflow.

The calculator supports the following features:

- Standard numerical input and real-time calculations.
- Long press the **9** key to enter negative numbers.
- Long press the decimal point (.) key to open the arithmetical operators menu.

> **Important**
>
> The long-press gestures require a press duration of at least half a second. Brief taps don't trigger the enhanced functionality.

Access the calculator

The workflow automatically shows the calculator when the worker needs it. For example, it appears when workers need to enter values for the following types of fields:

- **Quantity fields**: When entering item quantities during warehouse operations.
- **Number-type inputs**: As defined by the workflow configuration.
- **Measurement values**: When precise numerical input is required.

Basic input and calculator functionality

The calculator functions as a specialized numerical input tool with an enhanced interface. Workers can enter numbers using the numeric keypad and perform mathematical operations in real-time.

Arithmetical operators

The calculator provides access to four basic arithmetical operators. To select an operator, follow these steps:

1. Long press the decimal point (.) key on the keypad.
2. Select your desired operator (addition (+), subtraction (−), multiplication (×), or division (÷)).

Enter a negative number

To enter negative numbers in the calculator, follow these steps:

1. Enter the value using the numeric keypad.
2. Long press the **9** key on the keypad to toggle the entered value from positive to negative.
3. Continue with your calculation as normal.

> ⓘ **Note**
>
> This gesture only works with the **9** key. It provides quick access to negative values without requiring a separate negative sign button.

Usage examples

Enter a quantity value

To enter a numerical value for a quantity field, follow these steps:

1. When the app requires a quantity input, it automatically shows the calculator interface.
2. Enter your desired quantity using the numeric keypad.
3. If you need to convert the value to negative, long press the **9** key to toggle the sign.
4. Confirm the entry to proceed with your workflow.

Perform a calculation when entering a quantity value

To perform a calculation when entering a quantity, follow these steps:

1. Enter the first number by using the numeric keypad.
2. Open the operator menu by long pressing the decimal point (.) key.
3. Select your desired operator (addition (+), subtraction (−), multiplication (×), or division (÷)).
4. Enter the second number.
5. Press the equals key to view the result.
6. Confirm the entry to proceed with your workflow.

Troubleshooting

Expand table

Problem	Solution
Long press gesture doesn't activate	Hold the key for at least half a second. Brief taps enter standard input only.
Arithmetical operators menu doesn't appear	Ensure you're long pressing the decimal point (.) key specifically. Other keys don't open the menu.
Gesture activates unintentionally	Use shorter taps for standard number entry. Reserve longer presses for gesture controls.

Last updated on 04/08/2026

Haptic feedback through external wearable devices

The Warehouse Management mobile can provide haptic feedback through external wearable devices. This capability helps workers stay focused on the task at hand by providing instant, haptic responses without requiring workers to check the mobile device screen. The functionality is hardware-agnostic and supports a growing range of scanning peripherals from multiple manufacturers.

The app can communicate key system responses, such as successful scans or errors, directly through a wearable device using vibration patterns. Workers receive immediate cues while keeping their attention on picking, packing, or movement tasks. Haptic feedback provides the following key benefits:

- **Eyes-on-task productivity** – Workers receive immediate confirmation of a successful action through a tactile pulse, reducing the need to look at the mobile terminal.
- **Immediate error detection** – Distinct vibration patterns notify workers immediately when something is wrong, such as an invalid scan or blocked location.
- **Universal compatibility** – The architecture is hardware-agnostic, ensuring compatibility with a wide range of Bluetooth and tethered scanning hardware.

How it works

The integration uses the standard configuration features provided by the hardware manufacturer. After pairing a compatible wearable device with the mobile terminal (such as a tablet or handheld), the app recognizes the peripheral and routes feedback signals accordingly. You don't need to configure specific settings within the Warehouse Management mobile app or Supply Chain Management to enable feedback.

Haptic feedback signals

The app distinguishes between different operational outcomes by sending unique signals to the connected hardware, as outlined in the following table.

Expand table

Signal type	Signal description	Signal meaning
Success	A short, crisp vibration pattern with a green light.	Confirms a successful scan or task completion.

Signal type	Signal description	Signal meaning
Failure or error	A longer or double-pulse vibration pattern with a red light.	Alerts the user to an error, such as an invalid barcode or blocked location.

Hardware support

The Warehouse Management mobile app supports any human interface device (HID) or scanning peripheral compatible with the host operating system (Google Android or Microsoft Windows).

Wearable devices from ProGlove are the first to be specifically supported with haptic signaling. The Warehouse Management mobile app automatically recognizes ProGlove devices and routes feedback signals to them through their native setup utility, ProGlove Insight.

The app is built to support a growing ecosystem of smart wearables. Future updates will continue to add support for other hardware partners, freeing you to choose the devices that best fit your operational needs.

Scenario: hands-free picking

This scenario walks through a typical hands-free picking flow and shows where haptic feedback is triggered during scanning and validation so that workers can confirm success or spot errors without looking at the screen.

1. **Scan** – A worker picks an item and uses a wearable scanner to scan it.
2. **Validate** – The Warehouse Management mobile app processes the scan.
3. **Tactile response** – The app reacts in one of the following ways:

    - If the scanned item matches the expected item ID in the picking list, the wearable vibrates once, and the worker moves to the next pick immediately.
    - If the scanned item doesn't match the expected item ID in the picking list, the wearable pulses twice, notifying the worker to stop and check the screen for instructions.

Last updated on 02/05/2026

Warehouse Management mobile app accessibility features

This article describes the accessibility features of the Warehouse Management mobile app. Workers can configure these settings locally on the device through its **Settings**. Administrators can also set defaults remotely as described in Mobile device user settings.

Font and button scaling

Workers can adjust text and button sizes to match their visual needs. These settings are independent of the operating system's font and display settings.

Expand table

Feature	Range	Description
Font scaling	50% – 400%	Adjusts the text size throughout the app.
Button scaling	50% – 200%	Adjusts the size of buttons and interactive elements.

Color themes

The app supports five color themes to accommodate different visual needs and environments:

Expand table

Theme	Description
Auto	Matches the operating system's light or dark mode setting.
Light	Light background with dark text. Best for well-lit environments.
Dark	Dark background with light text. Reduces eye strain in low-light environments.
High-contrast white	White background with bold black text and high-contrast interactive elements. Designed for workers who need maximum readability.
High-contrast black	Black background with white text and high-contrast interactive elements. Designed for workers who need maximum readability in dark environments.

In addition to these accessibility-focused themes, the app offers over 10 color themes (such as Blue, Bronze, Cyan, and others) for personalization. Learn more in User settings, color themes, and sound themes.

Button positions

Workers can choose where buttons appear on the screen to match their handedness or preference:

- **Top-left** or **Top-right** — Best for workers who hold the device in one hand and interact with the other.
- **Bottom-left** or **Bottom-right** (default) — Best for thumb-based interaction.

Screen reader support

The app supports platform-specific screen readers on all three supported platforms:

Expand table

Platform	Screen reader
Android	TalkBack (preinstalled)
Windows	Narrator (built-in)
iOS	VoiceOver (built-in)

Next steps

- User settings, color themes, and sound themes
- Mobile device user settings

Last updated on 04/08/2026

Workload insights with Copilot in the Warehouse Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Warehouse Management mobile app provides a *workload* page that shows work summaries and AI-generated insights to help warehouse workers better plan their shift. The information can include the following details:

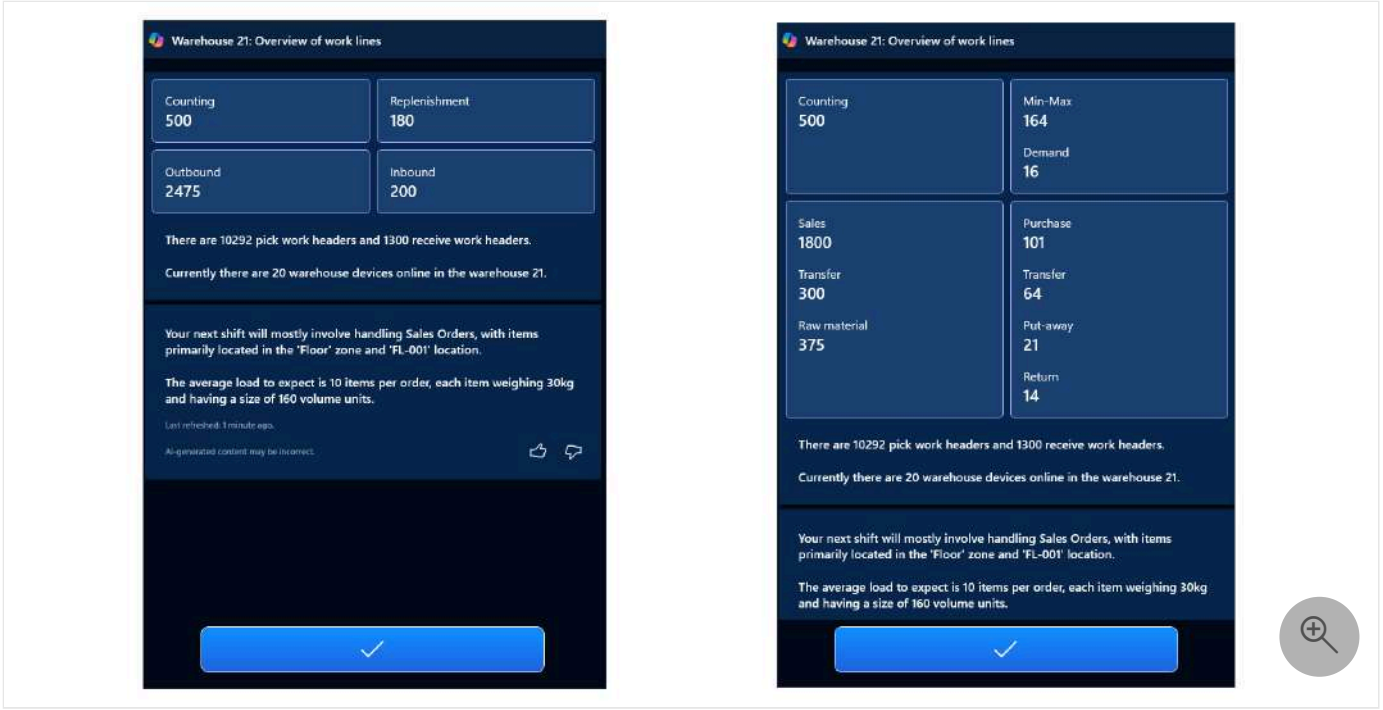
- The number of pick and receive work headers or work lines.
- The number of active warehouse mobile app sessions in the warehouse.
- Insights into the available work types.
- The current workload. This information can be shown either as work headers or work lines.
- Details of the available work per work type.
- More information about the available work. Microsoft Copilot generates this information in natural language.

Workers can use the insights on the workload page to get an overview of their workday and optimize the way that they engage with their daily tasks. Detailed information about available tasks helps enhance operational efficiency, reduce the time that is spent on information retrieval, and optimize resource allocation across the warehouse floor.

The workload page offers insights into relevant warehouse operations, such as the number of active warehouse mobile sessions that are working in the same warehouse. This information fosters a more connected and efficient workplace by promoting awareness and coordination among workers.

The following illustration shows an example of the workload page. When workers first open the page, it shows an overview of unstarted tasks that are scheduled for the current warehouse, together with a natural-language summary that Copilot generates (left). Workers can then tap any tile to view detailed information about open work by type (right). This information helps

them quickly grasp their objectives for the day. To close the workload page and open the **Main menu**, workers select the check mark button.



Prerequisites

Before you can get workload insights with Copilot in the Warehouse Management mobile app, your system must meet the following requirements:

- You must be running Dynamics 365 Supply Chain Management version 10.0.39 or later.
- You must be running Warehouse Management mobile app version 2.3.2.0 or later.
- The feature named *Context-aware worker summary screen in WMA* must be turned on in Feature management. As of Supply Chain Management version 10.0.39, the feature is turned on by default.

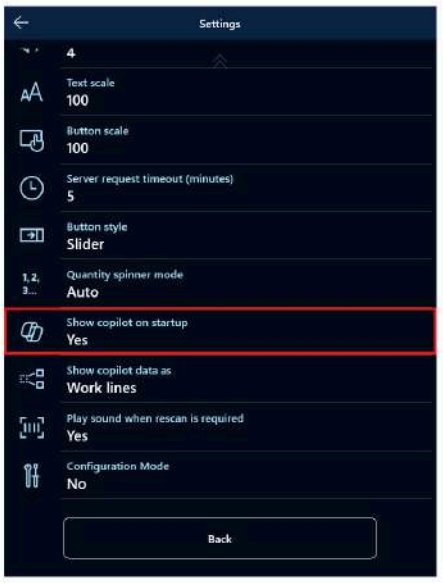
Worker preferences and options

Choose whether the workload page is shown after sign-in

Workers can specify whether they want the workload page to be shown each time they sign in to the app. To set this preference, follow these steps:

1. From the **Main menu**, open the **Settings** menu.

2. Select the **Show copilot on startup** entry.



3. Select one of the following options:

    - *Yes*: Show the workload page each time you sign in (default setting).
    - *No*: Go straight to the **Main menu** each time you sign in, without showing the workload page.

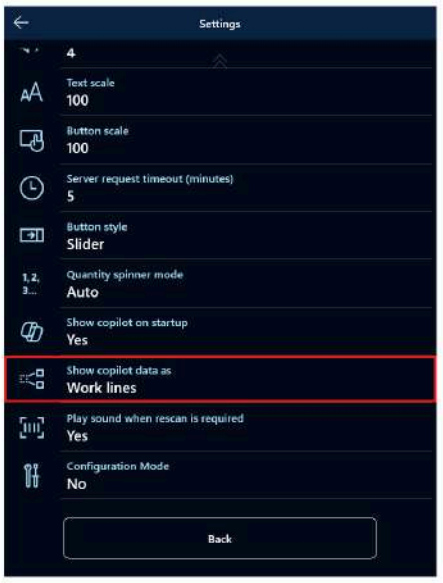
Choose whether workload data is shown as headers or lines

Workers can specify whether they want summaries to be based on work headers or work lines. The choice depends on how the workers prefer to think about their daily workload. For example, if all your orders typically contain just a few items, workers might prefer to think about their work in terms of whole orders (headers). However, if quantities vary greatly from order to order, workers might prefer to think about their work in terms of individual items (lines).

To set this preference, follow these steps:

1. From the **Main menu**, open the **Settings** menu.

2. Select the **Show copilot data as** entry.



3. Select one of the following options:

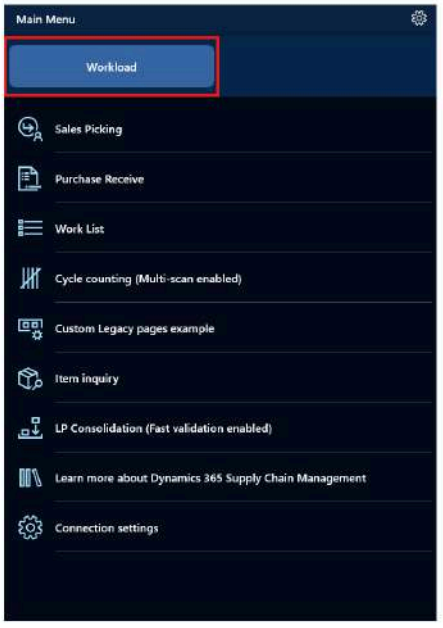
    - *Work lines*: Show the workload data as lines (default setting).
    - *Work headers*: Show the workload data as headers.

Open the workload page

Even if workers specify that the workload page should not be shown after sign-in, they can open the page at any time by following these steps.

1. On the **Main menu**.

2. Select the **Workload** button at the top of the page.



Related information

- Responsible AI FAQ for Workload insights with Copilot

Last updated on 05/08/2026

View the app log on a mobile device

The Warehouse Management mobile app maintains a local log of app operations. Use the **Local log view** page to review activity, identify errors, and export logs for troubleshooting with the Microsoft support team.

What the app log records

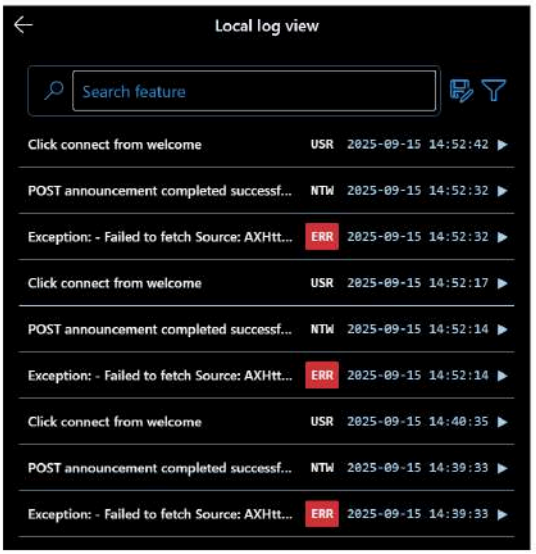
The app log captures events as they happen on the device, including:

- Sign-in attempts and authentication events
- Page navigation and workflow steps
- Connection events and network activity
- Errors and exceptions

The app stores logs locally on the device. You can export the logs as a file to share with support.

Open the Local log view page

1. Open the Warehouse Management mobile app on the affected device.
2. Open the **Diagnostics** page by using one of the following methods:
    - From the **Welcome** page, select **Set up connection** (or select **Connect** and then **Set up connection**), and then select **Diagnostics**.
    - From the **Main menu**, select **Connection settings**, then **Set up connection**, and then **Diagnostics**.
3. Select **Local log view**.



Find and filter log entries

You can find specific log entries and details using any of the following options:

- **Filter by time or type** – Select the filter button next to the **Search** field to narrow the log list by time range or issue type.
- **Search** – Enter text in the **Search** field to find specific log entries, such as an error message or connection event.
- **View details** – Select a log entry to see its full details, including timestamp, type, and message.

Export and share logs

You can export logs from the device in two ways:

- **Save to local file** – Select the **Save** button next to the **Search** field. The app saves all log entries as a text file to the device's local storage.
- **Share** – Use the device's share function to send the log file directly via email or other apps.

When you contact the Microsoft support team, export and attach the log file to your support request. This log file helps the team understand the sequence of events that led to the issue.

Last updated on 04/08/2026

Check Wi-Fi connectivity and performance

The network diagnostics tool in the Warehouse Management mobile app analyzes the most critical endpoints on your network to quickly identify connectivity problems. Use it as a first step when troubleshooting connection issues. It helps detect common problems such as unreachable networks, DNS resolution failures, and slow connections.

What the network diagnostics tool does

The tool runs a series of tests against key network endpoints and displays results that help you identify where problems occur. Based on the results, it shows:

- **Wi-Fi network name** — Confirms which network the device is connected to.
- **Connection status** — Indicates whether the device has an active network connection.
- **Connection speed** — The negotiated link speed with the Wi-Fi access point.
- **Signal strength** — Helps identify if the device is too far from an access point or in an area with poor coverage.
- **IP address** — Confirms the device received a valid IP address from the network.
- **Download speed** — Measures actual data throughput, which might differ from the connection speed due to network congestion or interference.

When to use network diagnostics

Run the diagnostics tool when you encounter any of the following issues:

- The app can't connect to Supply Chain Management.
- Operations are slow or time out.
- Scanning or data submission is unreliable.
- You need to provide network information to the Microsoft support team.

Run a network diagnostic test

1. Open the Warehouse Management mobile app on the affected device.
2. Open the **Diagnostics** page by using one of the following methods:
    - From the **Welcome** page, select **Set up connection** (or select **Connect** and then **Set up connection**), and then select **Diagnostics**.

- From the **Main menu**, select **Connection settings**, then **Set up connection**, and then **Diagnostics**.

3. Select **Analyze Wi-Fi**. The **Wi-Fi diagnostics** page opens and shows details about your current Wi-Fi connection.

4. To run deeper analysis, select **Run network diagnostics**. The tool pings critical endpoints to test DNS resolution, reachability, and response times.

Interpret the results

Use the guidance provided in the following table to identify common issues:

Expand table

Result	What it means	Recommended action
No Wi-Fi network name shown	The device isn't connected to a Wi-Fi network.	Verify Wi-Fi is enabled and the device is in range of an access point.
IP address is missing or invalid	The device didn't receive an IP address (DHCP failure).	Check DHCP server availability and network configuration.
Low signal strength	The device is too far from the access point or there's interference.	Move the device closer to an access point or check for physical obstructions.
Low download speed	Network congestion, interference, or bandwidth limitations.	Check network load and consider dedicated Wi-Fi for warehouse devices.
Endpoint ping failures	DNS resolution or network routing issues.	Verify DNS settings and that required endpoints aren't blocked by a firewall.

Last updated on 04/08/2026

Configure fields for the Warehouse Management mobile app

This article describes how to define and configure the names and priorities of fields shown in the Warehouse Management mobile app.

> **Note**
>
> This article applies to features in Warehouse management. It doesn't apply to features in Inventory management. The Warehouse Management mobile app is an application that you can use to perform warehouse tasks. You can define and configure the input type for the field names that the app uses, as well as configure the priority for the field names. You can't rename mobile app fields. This article explains how to define and configure Warehouse Management mobile app field names and priorities, and how they're used.

Configure warehouse app field names

When you use Warehousing on your mobile device, you can configure how metadata displays on your device on the **Warehouse app field names** page. In a new company, select **Create default setup** to generate all field names that the warehouse mobile device workflows use. Then, assign a preferred input mode and input type to them. After you generate all field names, select the following input options.

Expand table

Option	Description
Preferred input mode	This option defines whether a scanning field or a manual entry input field should show for the selected field name. This option is useful to distinguish fields depending on if bar codes are used for the field. **Note:** For field names with preferred input mode set to **Scanning**, you can enter information manually if the bar code is unreadable or damaged.
Input type	This option defines what input type should be used for the selected field name. Four options are available: **Selection** - Contains a list of options to choose from. Field names with this option aren't editable. **Date** - Field names specified as date show a date format with the label. This format helps warehouse workers see in which format to enter the date. Field names with this option aren't editable. **Alpha** - If selected, the device keyboard is used when entering information manually in the app. The keyboard experience can be changed depending on

Option	Description
	which device is used. **Numeric** - For field names that use numeric input only, select this option to display a custom numeric keypad with the input field instead of the device keyboard.

Configure warehouse app field priority

On the **Warehouse app field priority** page, you can organize field names into different priority groups. This priority system helps you decide what information appears on the main task page when warehouse workers perform tasks using the app. If you select **Create default setup**, a default set of priority groups is generated. You can create as many priority groups as you need, but the task page shows only three priority groups. When the system sends metadata to the app, it assigns each field a relative priority based on its priority group. The app displays the first three priority groups contained in the metadata on the task page. The app displays the rest of the overflowing metadata on a secondary details page. The following table shows an example of five priority groups.

Expand table

Priority group	Assigned fields
Priority 10	Item, Quantity, Unit of measure
Priority 20	Cluster position, Cluster
Priority 30	Item description
Priority 40	Configuration, Color, Size, Style
Priority 50	Location, License plate

For example, when a warehouse worker is performing a task on a mobile device, if the metadata that will be displayed in the app consists of the following fields:

- Item
- Quantity
- Unit of measure
- Item description
- Size and Location

Based on the warehouse app field priority set up in the table above, the following three rows of information will be displayed on the task page:

- Row 1: Item, Quantity, Unit of measure
- Row 2: Item description
- Row 3: Size

The remaining metadata, such as Location, doesn't appear on the task page, but it appears on a details page. To learn more and see examples of the user interface, refer to the blog post Announcing Dynamics 365 Supply Chain Management - Warehousing.

Related information

- Install the Warehouse Management mobile app

Last updated on 07/01/2026

Set up mobile devices for warehouse work

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to configure the menu items that warehouse workers use to perform their work in the Warehouse management mobile app.

For example, you can use the **Mobile device menu items** page to configure menu items for the following tasks:

- Process an inquiry or perform an activity, such as printing a label, generating license plate numbers, starting a production order, or quickly looking up information about items in a location.
- Create work that another process performs. For example, receiving an item for a purchase order can create putaway work for another worker.
- Perform work that another process creates (existing work), such as putaway work that was created when an item was received for a purchase order, or picking work related to a sales order.

Configure menu items for activities and inquiries

To create a menu item for an activity or inquiry, set the **Mode** field to **Indirect**. A list of **Activity code** options then becomes available, so that you can select the type of inquiry or activity that the menu item is for. To create a menu item to generate warehouse work, set the **Mode** field to **Work**. A list of **Work creation process** options then becomes available. To create a menu item to process existing warehouse work, set the **Mode** field to **Work**, and then set the **Use existing work** option to **Yes**.

> **Note**
>
> More fields might be available for menu items, depending on the mode that you select for the menu item, and whether the menu item is used to perform existing work. Learn more about the additional field selections iin the Additional menu item options section later in this article.

If you set the **Mode** field for a menu item to **Indirect**, you can create a menu item to perform a general activity or inquiry that doesn't create work. Examples include reprinting license plate labels and an inquiry about the items in a location. The following table lists the options that are available.

Expand table

Option	Description
None	This default value doesn't enable an activity or inquiry.
About	View information about the system, such as the version number, the warehouse ID, and the worker who is currently logged on.
Change warehouse	Change the warehouse that a worker is logged on to.
Location inquiry	View information about all items and quantities for a location. For license plate controlled locations, data is sorted by license plate number in descending order. The items are usually sorted by physical quantity (for both license plate controlled and non-license plate controlled locations) but the sort order might change depending on what SQL Server finds most optimal.
License plate inquiry	View the quantity of items on a license plate and the location of the license plate.
Start production order	Start a production order.
Production scrap	Enter the quantity of scrap that was created during production for each bill of materials (BOM) line.
Production last pallet	Indicate that the last pallet of items has been produced for a production order, and that the status of the production order must be updated to **Reported as finished**. The status of the raw materials that weren't consumed during production changes back from *Picked* to *On order*, and the items can be returned to inventory.
Item inquiry	Scan an item to determine where it is in the warehouse. The inquiry returns all locations and quantities for the scanned item.
Reprint label	Reprint a license plate label.
License plate build	Create a parent license plate by combining multiple license plates in the same location. This option is useful if you move multiple license plates at the same time. After the parent license plate is moved, you must perform a license plate break before you can pick items from each license plate. **Tip:** To move a parent license plate, you must use a mobile device that is configured to create work for movements.

Option	Description
License plate break	Break up a license plate build so that you can pick items from the license plates that were in the build.
Driver check in	If you're using Transportation management, register the arrival of a driver by scanning the outbound load ID, appointment ID, or shipment ID. For this option, a load must be assigned to the appointment, and the status of the load must be **Loaded**.
Driver check out	Register that a driver has completed their appointment.
Flush number sequence cache	Delete number sequence numbers from the number sequence cache. A system administrator typically performs this activity to resolve caching issues when mobile devices are used.
Change batch disposition	Allow a worker to specify a batch disposition code for an item and batch. This selection updates the disposition code that is specified for the batch.
Display open work list	Show a list of available work to a particular user. The user can then select work to perform and is directed to it. This list is intended to be viewed on tablet devices that have a screen size of 7 inches or more. When you select this option, the **Edit query** and **Field list** menu items become available. The **Edit query** page lets you set up criteria for the work that appears in the list. The **Field list** page lets you select what fields appear in the work list. For example, you can reduce the number of fields that appear, so that the user can more quickly select the most appropriate work item. On the **General** FastTab, in the **Records per page** field, you can also select how many work records are shown per page. If you select the **Allow users to filter work by transaction type** option, the work list includes a **Filter work** control that the user can use to filter by transaction type. In the work list, users see only work that they have permission to access. You must make sure that users have permission for one or more user-directed menu items that support the specific work class types that they should be able to access. Permissions are verified when a user tries to perform work from the list.
Create transfer order from license plates	Allow warehouse workers to create and process transfer orders directly from the Warehouse Management mobile app. A worker starts by selecting the destination warehouse. The worker can then scan one or more license plates by using the app. After the warehouse worker selects **Complete order**, a batch job creates the required transfer order and order lines, based on the on-hand inventory that is registered for those license plates. Learn more in Create transfer orders from the warehouse app.
Data inquiry	Enable the creation of warehouse app menu items that can be used to look up data from the mobile device as an inquiry list. Learn more in Warehouse app data inquiry.
Pack inventory into containers	Enable support for warehouse workers as they pack inventory items into containers. A worker starts by scanning a shipment to identify the inventory items that must be packed. The worker then identifies the destination shipping container by entering its ID or scanning its bar code. Finally, when the container is fully packed, the worker registers it as closed. This step makes the container ready for further processing by Microsoft

Option	Description
	Dynamics 365 Supply Chain Management. Learn more in Packing containers with the Warehouse Management mobile app.
Container creation	Enable the container creation process. This process is typically part of the process of packing containers with the Warehouse Management mobile app.
Container closing	Enable the container closing process. This process is typically part of the process of packing containers by using the Warehouse Management mobile app.
Print container label	Enable container label printing. This process is typically part of the process of packing containers by using the Warehouse Management mobile app. Learn more in Container label layouts and printing.
Receiving completed confirmation	Enable support for receiving clerks to indicate *Receiving completed* for a load.
Complete mixed license plate	Enable support for warehouse workers to complete license plates recorded during mixed license plate receiving.

Configure menu items to create work for another worker or process

You can set up a menu item that creates work for another worker after an initial action is performed on the mobile device. For example, when one worker uses a mobile device to receive an item, the system creates putaway work for another worker. To set up a menu item that creates work, on the **Mobile device menu items** page, in the **Mode** field, select **Work**. In the following table, the options in the **Work creation process** field are arranged by work order type.

Expand table

Work order type	Option	Description
Inbound shipment order	Inbound shipment order line receiving	Register the receipt of a quantity of an item by using the inbound shipment order number and line number, and create putaway work for another worker.
	Inbound shipment order line receiving and put away	Register the receipt of a quantity of an item by using the inbound shipment order number and line number, and put away the items. The same worker performs both actions.

Work order type	Option	Description
	Inbound shipment order item receiving	Register the receipt of a quantity of an item for an inbound shipment order by registering the order number and item number, and create putaway work for another worker.
	Inbound shipment order item receiving and put away	Register the receipt of a quantity of an item for an inbound shipment order by registering the order number, and put away the item. The same worker performs both actions.
	License plate receiving	Receive an inbound advance ship notice (ASN) by using the license plate ID.
	License plate receiving and put away	Receive and put away an inbound ASN by using the license plate ID.
	Load item receiving	Register the receipt of a quantity for a load by using the load ID, and create putaway work for another worker. The item number and product dimensions match the receipt to the order lines.
	Load item receiving and put away	Register the receipt of a load by using the load ID, and put away the items. The item number and product dimensions match the receipt to the order lines. The same worker performs both actions.
	Return item receiving	Record data captured during mixed license plate receiving.
Purchase order	Purchase order line receiving	Register the receipt of a quantity of an item by using the purchase order number and purchase order line number, and create putaway work for another worker.
	Purchase order line receiving and put away	Register the receipt of a quantity of an item by using the purchase order number and purchase order line number, and put away the items. The same worker performs both actions.
	Purchase order item receiving	Register the receipt of a quantity of an item for a purchase order by registering the purchase order number and item number, and create putaway work for another worker.
	Purchase order item receiving and put away	Register the receipt of a quantity of an item for a purchase order by registering the purchase order number, and put away the item. The same worker performs both actions.
	License plate receiving	Receive an inbound ASN by using the license plate ID.
	License plate receiving and put	Receive and put away an inbound ASN by using the license plate ID.

Work order type	Option	Description
	away	
	Load item receiving	Register the receipt of a quantity for a load by using the load ID, and create putaway work for another worker. The item number and product dimensions match the receipt to the purchase order lines.
	Load item receiving and put away	Register the receipt of a load by using the load ID, and put away the items. The item number and product dimensions match the receipt to the purchase order lines. The same worker performs both actions.
Return order	Return order receiving	Register the receipt of a quantity of an item by registering the RMA number, and create putaway work for another worker.
	Return order receiving and put away	Register the receipt of a quantity of an item by registering the RMA number, and put away the items. The same worker performs both actions.
	Return item receiving	Record data captured during mixed license plate receiving.
Transfer order	Transfer order item receiving	Register the receipt of a quantity of an item, and create putaway work for another worker. **Note:** Use this option only if the items were shipped from a warehouse that isn't enabled for warehouse management processes (WMS).
	Transfer order item receiving and put away	Register the receipt of a quantity of an item, and put away the items. The same worker performs both actions. **Note:** Use this option only if the items were shipped from a warehouse that isn't enabled for WMS.
	Transfer order line receiving	Register the receipt of a quantity of an item, and create putaway work for another worker.
	Transfer order line receiving and put away	Register the receipt of a quantity of an item, and put away the items. The same worker performs both actions.
	License plate receiving	Receive an inbound ASN by using the license plate ID.
	License plate receiving and put away	Receive and put away an inbound ASN by using the license plate ID.

Work order type	Option	Description
Production	Report as finished	Register a quantity of a finished item that is finished for a production, and create putaway work for another worker. The quantity can be some or all of the quantity that was planned for production.
	Report as finished and put away	Register a quantity of a finished item that is finished for a production, and put away the items. The quantity can be some or all of the quantity that was planned for production. The same worker performs both actions.
	Kanban	Indicate that a kanban is completed, and create putaway work for another worker.
	Kanban put away	Indicate that a kanban is completed, and put away the items. The same worker performs both actions.
Inventory	Movement	Register that items are moved from one location to another. The worker specifies the location that the items are moved from and where they're moved to.
	Quarantine	Change the status of the on-hand inventory for a license plate or location to make damaged or missing inventory items unavailable.
	Movement by template	Move items from one location to another in a semi-automated manner. The worker selects the location to move items from. The system uses the location directive to determine where to move the items to.
	Warehouse transfer	Register that items are transferred from one warehouse to another. This option requires that the worker be allowed to perform work in both warehouses. **Note:** This menu item requires a default inventory transfer journal where the **Voucher draw** field is set to **Posting**.
	License plate loading	Use this option when you're setting up your warehouse for the first time. Scan all the license plates in all locations in the warehouse. The locations must be license plate–controlled. You can't use this option if **Serial number** or **Batch number** is listed above **Location** in the inventory reservation hierarchy.
Cycle count	Adjustment in	Increase the quantity of items in inventory. Specify the location, license plate, item, quantity, unit of measure, and status.
	Adjustment out	Reduce the quantity of items in inventory. Specify the location, license plate, item, quantity, unit of measure, and status of the inventory.

Work order type	Option	Description
	Spot cycle counting	Start a count for a location. The worker must count all items in the location. When the result of a count is less than the expected quantity, the missing quantity is considered a loss.

> **Note**
>
> - You can set up mobile device menu items, such as ***Mixed license plate receiving (and putaway)***, to embed some of the flows from the previous table in the process.
> - To prevent accidental selection, the **Cancel** button doesn't appear during *cycle counting* mobile workflows.

Configure menu items to process existing work

In addition to setting up menu items to create warehouse work, you can set up menu items to process existing work. Set the **Mode** field to *Work*, and select the **Use existing work** option. Some additional options then become available on the **General** tab. You can control access to the menu item by assigning one or more work classes on the **Work class** FastTab. The work classes define the work that the menu item can process. Use the work class to grant access to specific user roles or to separate processing for different types of operations. The following table describes the available options. Choose the option under the **Directed by** field in the **Mobile device menu items** page.

Expand table

Option	Description
None	This default value doesn't process work.
System directed	Supply Chain Management controls the type of work that it assigns to a worker and the order that the worker performs the work in. When you select this option, you can select **System-directed work** on the Action Pane to open the **System-directed sorting order** page, where you can set up sorting criteria for the work. The sorting criteria control the order that the worker performs the work in. You can add as many criteria as you require.
User directed	The worker selects the work to perform and the order to perform it in.
User grouping	The worker manually groups work. This option is useful when, for example, a worker can pick multiple items at the same time in a location. After the worker finishes picking all the required items, they can put away the items.

Option	Description
System grouping	Supply Chain Management groups work for the worker, based on a specified field. For example, picking work is grouped when a worker scans a shipment ID, load ID, or any value that can link each work unit. If you select this option, the following fields are required: • **System grouping field** – Select the field that the worker scans to group the work. • **System grouping label** – Enter text to instruct the worker what to scan to group the work.
Validated user directed	The worker selects the work to perform when work is associated with a larger entity, such as a load or shipment. The worker determines the order that the items are picked in. If you select this option, the following fields are required: • **Validated user directed field** – Select the field that the worker scans to group the work. • **Validated user directed label** – Enter text that instructs the worker what to scan when the system groups picking work. This option is useful when, for example, multiple pallets are staged for a load. If you select **LoadId** in the **Validated User Directed** field, the worker can pick any pallet that is associated with the load. The worker receives an error message if they scan an item that isn't associated with the load.
Cluster picking	The worker groups work into clusters. Clusters let workers pick items from a single location for multiple work orders at the same time.
Cycle count grouping	The worker selects a zone, work pool, or location, and Supply Chain Management assigns work, based on the selection. If you select this option, you can select **Cycle counting** on the Action Pane to specify additional information to display, and you can also specify the number of times that the worker must repeat the count if a difference is found.
Transport loading	This feature allows several warehouse workers to load inventory from the same or different loads onto the same truck, with loads that are fully or partially shipped.

Additional menu item options

You can find more options for menu items on the **Mobile device menu items** page. The options vary depending on the process that you're configuring the menu item for.

The following table describes these options.

Expand table

Field	Description
Allow splitting of work	Select this option to let workers put items for a work order into more than one target license plate. This option is useful when, for example, a target license plate is full, and the worker must add the remaining items to another license plate. The worker can select **Full** to indicate that the license plate is full and stop receiving picking work for it. The put location for the picked items is then displayed, and the picking work that you already completed is moved to a new work order. The remaining picking work for the target license plate stays on the original work order.
Anchoring	Select this option to let workers specify a location that overrides the suggested staging or loading location. All the remaining putaway work is directed to the new location. This option is useful when, for example, a worker who must put items for order 1 in a staging location by Dock 1 can't, because a previous load hasn't cleared the location. Instead of waiting for the Dock 1 staging location to become available, the worker can decide to use the staging location for Dock 2. In this case, the worker overrides the suggested staging location. The put location for all remaining items for the work order is then updated to the Dock 2 staging location. If you select this option, you must set the **Anchor by** field.
Anchor by	If you're using anchoring, you must specify whether to anchor by shipment or by load.
Audit template ID	Select the work audit template that interrupts the work process for this menu item so that another operation can be performed. For example, if this menu item is for inbound work, the audit template might require that the worker check the temperature in the delivery container. The point at which the process is interrupted is specified on the audit template. This point can be, for example, when work is started or completed, or when its status changes.
Cluster profile ID	Select the cluster profile to use for cluster picking. The cluster profile includes settings such as whether to create clusters automatically, the names of positions and the number of work units that they can be assigned, when to break clusters into individual units, and whether verification is required. This field is available only if **Cluster picking** is selected in the **Directed by** field.
Count total item quantity first	Select this option to require workers to count the total quantity first during a count. If a difference is found, the worker must provide additional information, such as the license plate number, batch number, and serial numbers.
Create movement	Select this option to let a worker create work for a movement, but without requiring the worker to perform the work immediately. This option is useful if, for example, a quality inspection is completed, and the inspector wants the item to be moved from the quality inspection area.
Directive code	To use a specific location directive, select the directive code that is associated with the location directive. This field is available when you create work and the work creation process is **Movement by template**.
Disable cycle count	Select this option to ignore the cycle count thresholds. If you select this option, cycle count work isn't created when threshold values are exceeded.

Field	Description
thresholds	
Display batch disposition code	Select this option to display batch disposition codes. For example, you can display batch disposition codes when you receive a returned batch. Workers can then evaluate the status or quality of a batch, and select the appropriate code. The rules on the batch disposition code determine whether the batch is available to other warehouse processes. If you don't select this option, one of the following batch disposition codes is used: • If you receive a new batch number, the default batch disposition code that is specified on the item model group. • The batch disposition code that is already assigned to the batch.
Display disposition code	Select this option to display disposition codes. For example, you can display disposition codes when you receive return items. Workers can then evaluate the status or quality of the items, and select the appropriate code. The rules on the disposition code determine whether the items are available to other warehouse processes.
Display inventory status	Select this option to display the status of items in inventory. This option is available for all menu items that use existing work, except cycle counting.
Display summary of pick screen	Select this option to display a summary of picking work for the selected work order. The summary is displayed until the first work line is processed for the work order.
Generate license plate	Select this option to generate a unique license plate number, based on the number sequence selection. For example, you can generate a license plate number for items that you receive for purchase orders.
Group put away	Select this option to group the putaway work. This option is available when the work is grouped either by the worker or by Supply Chain Management. When the worker finishes all the picking work in the group, putaway work is created for the same group.
Inventory adjustment types	Select the inventory adjustment type that determines the inventory counting journal that is used to post the adjustment, and whether to remove reservations. This field is available only for the **Adjustment in** or **Adjustment out** work creation process.
Override batch number	Select this option to let workers who report a quantity as finished for a production order enter a batch number that differs from the batch number that is assigned to the production order.
Override target license plate	Select this option to let workers specify a target license plate number that differs from the suggested target license plate. Use this option when the first pick for a work order is for the entire quantity of an item on a license plate. This option is useful when, for example, a pallet is reused.
Pick and pack	Select this option to let workers combine work for a sales order or load into a single work unit. A worker can perform work only for the sales order or load. This option is

Field	Description
	useful when, for example, you must increase a quantity for a sales order after the load, shipment, and work are created for a sales order. This option is available when the menu item uses existing work, and the work is directed by the user or system. Only work headers that contain a single initial pick work line can be combined by using this group picking concept.
Pick oldest batch	Indicate whether the worker must pick the oldest batch in a location first. The following options are available: • **None** – The worker can pick any batch in the location. The worker receives no message. • **Warn** – The worker can pick any batch in the location, but they receive a warning message if a batch isn't the oldest batch. • **Force** – The worker must pick the oldest batch in the location. The worker receives an error message if a batch isn't the oldest batch. **Note:** This option is relevant only if **Batch number** is lower than **Location** in the reservation hierarchy that is assigned to the item.
Print label	Select this option to let workers print license plate labels.
System grouping field	Select the field that determines how Supply Chain Management groups picking work for workers. For example, if you select the **ShipmentId** field, the worker scans the shipment ID to group the picking work. All work for the shipment is then assigned to the worker. This field requires that you create a menu item to use existing work that is grouped by the system. You must also enter text in the **System grouping label** field to instruct the worker what to scan.
System grouping label	Enter the text that instructs the worker what to scan when Supply Chain Management groups picking work. For example, if you're using the **ShipmentId** field to group picking work by shipment, you might enter **Shipment ID** in the field. This field requires that you create a menu item to use existing work that is grouped by the system. You must also select the field to group by in the **System grouping** field.
Use default data	Select this option to enable the **Default data** button on the Action Pane, where you can select fields to display data that a worker typically requires in their daily work. This option is useful if, for example, a worker often picks items from the same location. You can select the **From location** field to display the location by default.
Validated User Directed Field	Select the field that the worker scans to group the work. For example, if you select **LoadId**, a worker can pick any work that's associated with a selected load. You must also enter text in the **Validated User Directed Label** field to instruct the worker what to scan.
Validated User Directed Label	Enter the text that instructs the worker what to scan when picking work is grouped by a validated user-directed field. For example, if you're using the **LoadId** field to group picking work for a load, you might enter **Load ID** in the field.
Work template code	Select the work template that creates the work for a process. For example, if you receive an item for a purchase order, the putaway work is generated based on the work

Field	Description
	template. If you don't select a work template, Supply Chain Management assigns a template, based on query criteria. For more information about work templates, see Controlling warehouse work with work templates and location directives.
Show work line list	Select an option for how workers can view and interact with the lines for the currently selected picking work. For more information about this option, see Set up a mobile device menu item to provide a pick line overview.
Display tracking dimension control for last on-hand	Select one of the following options to specify how the *tracking dimension below location* control, including *batch below number* and *serial below number*, is shown for the last on-hand item. The setting of this field has no impact on the display of *tracking dimension above location*, such as *batch above number* or *serial above number*. • *Default from warehouse management parameters* – The menu item follows the configuration on the **Warehouse management parameters** page. Learn more in Global mobile device parameters. • *Display tracking dimension control* – The current menu item always shows the tracking dimension below location control (batch below number or serial below number), regardless of the settings of warehouse management parameters. • *Hide tracking dimension control* – The current menu item always hides the tracking dimension below location control (batch below number or serial below number), regardless of the setting of warehouse management parameters. **Note:** This setting affects only menu items that process a supported work order type (currently only sales orders). Unsupported work order types (such as transfer orders and purchase orders) aren't affected.

Require workers to confirm the product, location, or quantity when they pick items

You can set up work confirmations that require a worker to use a mobile device to register the location or quantity when they perform work in the warehouse. Work confirmations help ensure that the worker is at the correct location or is handling the correct quantity of items. You can also enable Supply Chain Management to automatically confirm the worker's registration. If you enable automatic confirmation, you can't also require confirmations for location or quantity. Work confirmations also include products and product variants. Additionally, you can register confirmations by scanning a bar code. To confirm products and product variants, you must enter an ID for the product or product variant. This ID can be a product ID, product search ID, external ID, GTIN, or bar code. After you enter the ID or scan the bar code, the dimensions for the product variant are displayed on the mobile device.

The following table describes the various work types that you can use work confirmations with.

Expand table

Option	Description
Pick	Require confirmation when items are picked.
Put	Require confirmation when items are put in a location.
Counting	Require confirmation during cycle counting.
Adjustments	Require confirmation when inventory quantities are adjusted.
Custom	Require confirmation for custom work.
Quarantine	Require confirmation when items are moved to quarantine.
License plate building	Require confirmation when items are consolidated to build a license plate.
Print	Require confirmation when license plate labels are printed.
Status change	Require confirmation when the status of inventory is changed.

> **Note**
>
> You can require product confirmation only for pick and put work types.

Set up mobile device menus

The menu items that you create by using the **Mobile device menu items** page become available to mobile device users after you add them to one or more menus by using the **Mobile device menu** page.

On the **Work users** page (**Warehouse management** > **Setup** > **Worker**), you can select the top-level menu that each worker can access on the mobile app. (Learn more in Mobile device user accounts.) In this way, you can set up a system that supports several different menu structures, where each worker sees only the menu items that are relevant to them.

To set up menus for mobile devices, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. Do one of the following actions:
   - To add a new menu or submenu, on the Action Pane, select **New**.
   - To edit an existing menu or submenu, on the list pane, select the menu you want to edit. Then, on the Action Pane, select **Edit**.

3. In the header of the menu, make the following settings:

    - **Name**: Enter a name for the menu or submenu. This name is internal and must be unique. You can't edit this field after you save the record.
    - **Description**: Enter a short description for the menu or submenu.

4. In the **Mobile device menus** section, find and select a menu item or submenu you'd like to add in the **Available menus and menu items** column. Then select the right arrow button to move selected item to the **Menu structure** column.

5. Use the up arrow and down arrow buttons next to the **Menu structure** column to arrange the items in the menu.

    > **Tip**
    >
    > Workers can only access menu items that are included in the menu that is assigned to their mobile device user account (or a submenu of that menu). This restriction also applies to menu items that are intended for use as detours, but which you might not want workers to access directly from the menu. In this case, add the detour items to the relevant menus and then hide the items. To hide an item included in a menu, right-click on it in the **Menu structure** column and select **Hide**. To show an item that is currently hidden, right-click on it and select **Show**. (Select **Edit** from the Action Pane if you don't see the **Hide** or **Show** options.)

6. On the Action Pane, select **Save**.

Related information

- Set up a mobile device menu item for completing work of type Purchase order
- Set up a mobile device menu item to register received items
- Inventory statuses

Last updated on 11/20/2025

Customize step titles and instructions for the Warehouse Management mobile app

> **Important**
>
> The features described in this article apply only to the new Warehouse Management mobile app. They don't affect the old warehouse app, which is now deprecated.

This article describes how to create and show custom instructions for every step of every task flow that you set up for the Warehouse Management mobile app. When warehouse workers receive well-written instructions, they can immediately start to use new flows without requiring previous training. This feature provides the following benefits:

- **Ramp up workers faster by letting them follow simple instructions for every task step.** Every step of a flow provides instructions that enable front-line workers to understand the task.
- **Provide instructions that match your own processes.** Write your own instructions to match your business and warehouse processes. For example, you can make the terminology fit your physical space and local abbreviations.

Step titles and step instructions in the app

Every step in a task flow on the Warehouse Management mobile app is identified by a step ID. Additionally, every step has a title, icon, and instruction. (Learn more in Assign step icons and titles for the Warehouse Management mobile app.)

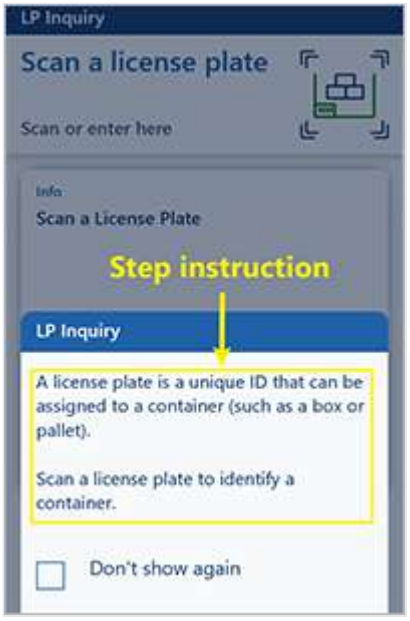
Step titles

A *step title* is a short description of what a worker should do during a step. It appears as large text at the top of the screen, as shown in the following illustration.

> **Tip**
>
> Because of the large text size, keep step titles as short as possible. Otherwise, the text might be cut off. However, for long titles, workers can still press and hold the title to open a dialog box that shows the full text.

Step instructions

A *step instruction* is a longer description that provides more information about what a worker should do during a step. It appears in a pop-up dialog box, as shown in the following illustration.



The step instruction automatically appears when a step is opened. Workers can dismiss it by tapping anywhere outside the pop-up dialog box. The dialog box includes a **Don't show again**

checkbox that workers can select to prevent the instruction from appearing the next time that they perform the same task.

Load the default setup

If your system doesn't already contain default customizable step titles or instructions, you can load a default setup. The default setup provides texts for all the available step IDs in every supported language. To load the default setup, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. On the Action Pane, select **Create default setup**. The page is filled in with the standard steps.

Customize step titles and instructions

To customize the title and/or instruction for a step in any number of languages, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.

   The **Mobile device steps** page lists every step that's available for your system. Each step ID can be shared among any number of mobile device menu items. If a step ID is shared among multiple menu items, the same title and instruction are shown for all those menu items. However, you can create overrides to customize the title and instruction for specific menu items. For more information, see the next section.

   The grid includes the following columns:

   - **Menu item name** – Rows where this column is blank use the default step title and instruction that apply to all mobile device menu items that no override is defined for. Rows where this column is set to the name of a menu item have overrides that apply only to the specified menu item.
   - **Step ID** – The unique ID of the step.
   - **Title for input** – The title that appears when the app requests new information. Typically, the fields on the page are blank (that is, they have no preset values).
   - **Title for confirmation** – The title that appears when the app requests confirmation of a value that's already stored in the system. Typically, the fields on the page have preset values.

2. Find the combination of **Step ID** and **Menu item name** values that you want to edit, and select the value in the **Step ID** column. The page that opens lists all the available translations for the title and instruction of the selected step.

3. Follow one of these steps to customize the text for any language. Both options let you edit texts for existing languages. However, only the first option lets you add texts for new languages, whereas only the second option lets you view and edit texts for all existing menu-specific overrides of the selected language.

    - On the toolbar, select **Add** to open a dialog box where you can add or edit texts for any supported language. Set the **Reference language** field to the language that you want to view values for. The values appear in the left column. Set the **Language of the translations** field to the language that you want to add or customize. In the right column, edit the values of the **Title for input**, **Instruction for input**, **Title for confirmation**, and **Instruction for confirmation** fields as you require. Then select **OK**.
    - In the grid, find and select the row where the **Language** field is set to the language that you want to edit. On the toolbar, select **View <language> translations of this step** to open a dialog box where you can edit texts for all available overrides for the selected language. The dialog box includes a grid that has rows for both the standard texts (where the **Menu item name** field is blank) and for each available override text (where the **Menu item name** field is set to the name of the menu item that the override applies to). Edit the values of the **Title for input**, **Instruction for input**, **Title for confirmation**, and **Instruction for confirmation** fields as you require. Then select **OK**.

4. Continue to work until you define every required title and instruction for every required language.

Add menu-specific overrides

As mentioned in the previous section, you can create any number of menu-specific overrides for each step ID. Use this capability to edit and change the instruction so that it better fits your local business process for a specific menu item. For example, for sales picking, if your company usually provides work IDs to workers on a printed document, you can provide a hint that workers should start by scanning a work ID.

Each override applies to a specific mobile device menu item and can contain any number of translations. If no override exists for a menu item, the menu item uses the standard texts. If no

override translation is defined for a language, the standard texts are used, even for menu items where an override is defined for other languages.

To create and configure an override, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. In the grid, find and select the row to create an override for.
3. On the Action Pane, select **Add step configuration**.
4. In the **Add step configuration** drop-down dialog box, set the **Menu item** field to the name of the mobile device menu item that your override applies to. Then select **OK**.
5. The page that appears shows all the texts that are available for the new override. Initially, just one language is created. All other languages continue to use the standard texts unless you add those languages here. Edit the texts and add new languages as you require, as described in the previous section.

Last updated on 05/29/2026

Configure promoted fields for steps in the Warehouse Management mobile app

06/26/2025

This article describes how to promote and highlight specific information for any step in the task flows for the Warehouse Management mobile app. This capability can help focus workers' attention on the most important fields as they work through a flow. For each step in every process, admins can select which fields to promote and which fields to highlight.

Configure promoted fields from a menu-specific override

Use the following procedure to set up promoted fields.

1. Create a menu-specific override for the relevant menu and step as described in Customize step titles and instructions for the Warehouse Management mobile app.
2. Find the combination of **Step ID** and **Menu item name** values that you want to edit, and then select the value in the **Step ID** column.
3. On the page that appears, on the **Select promoted fields** FastTab, select **Select fields** on the toolbar.
4. In the **Promoted fields** dialog box, select the fields that you want to promote. You can also highlight up to two of the selected fields. Highlighted fields will be shown in bold in the Warehouse Management mobile app. As you select fields, consider the fact that some screens might be large enough to show only the top one or two promoted fields. For an example that shows how to use these settings, see the scenario later in this article.

   > ⓘ **Note**
   >
   > - The **Available fields** list includes all the fields that *could* appear for the menu item. However, other factors (such as item composition) determine which fields *actually do* appear in the Warehouse Management mobile app.
   > - You can only promote the fields that are shown in the **Available fields** list. You can't add new fields from here.
   > - For a given step, you should only promote fields that are displayed by default in the mobile app for that step. If you promote a field that isn't shown in the app, that setting has no effect.

- If you've promoted one or more fields, then only those fields will be shown on the main page of the app. However, workers can still view the remaining fields by opening the details page.
- If the promoted field has a null value, it won't be displayed.
- Non-promoted fields aren't removed from the mobile app page, just from the main card. Workers can still access them from the main card by selecting the card. This mechanism provides a way to hide less relevant fields. Non-promoted fields that are required for a workflow are always shown when relevant.

5. Select **OK** to apply your settings. Your selected fields are now listed on the **Select promoted fields** FastTab.

Example scenario

Enable sample data

To use the specified sample records and values to work through this scenario, you must be using a system where the standard demo data is installed. You must also select the **USMF** legal entity before you begin.

Configure sales picking with promoted steps on the license plate step

In this procedure, you'll set up promoted and highlighted fields for the **Sales picking** menu item in the license plate step.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. Find the step ID that is named *LicensePlateId*, and select it.
3. On the Action Pane, select **Add step configuration**.
4. In the drop-down dialog box, use the **Menu item** field to find and select *Sales picking*.
5. Select **OK**.
6. The details page for the override that you just created appears. On the **Select promoted fields** FastTab, select **Select fields** on the toolbar.

7. In the **Promoted fields** dialog box, you can select the fields to promote and highlight. In the **Available fields** list, find and select the following fields, and use the buttons to move them to the **Selected fields** list:

    - Location
    - Item
    - Product name
    - Inventory status

8. Use the up arrow and down arrow buttons to customize the order of the fields in the **Selected fields** list. In the Warehouse Management mobile app, the fields will appear in the order that you set here.

9. Select the **Location** field, and then select **Highlight**.

10. Select **OK** to save the configuration.

View the promoted fields in the Warehouse Management mobile app

In this procedure, you'll open the Warehouse Management mobile app and go through the steps to view the fields that you promoted and highlighted in the previous procedure.

1. In Microsoft Dynamics 365 Supply Chain Management, create a sales order that will require a pick step to pick from a location that is license plate tracked. Then release the sales order to the warehouse. Make a note of the work ID that is generated.
2. Open the Warehouse Management mobile app, and sign in to warehouse 24. (In the standard demo data, sign in by using *24* as the user ID and *1* as the password.)
3. Select the **Outbound** menu, and then select the **Sales picking** menu item.
4. Follow the data entry instructions on the screen to enter the work ID that was generated in step 1.
5. On the step that contains the text **Scan a license plate**, you should see the changes that you made on the details page. The fields appear in the order that you set for the promoted fields, and the **Location** field is shown in bold.

Configure detours for steps in mobile device menu items

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to configure detours for menu items so that workers can "park" the current task, perform another task, and then return to the original task without losing any information.

A detour is a separate menu item that workers can open from a step in a main task. At the end of the detour, the worker returns to the place where they left the main task. During the configuration, you specify the menu item that should act as a detour. You also select which field values from the main task should automatically be forwarded (copied) to the detour and entered there. Therefore, you must understand where in the task flow you want the detour to be available to workers. You must also ensure that the information you need to copy to the detour is available for that step of the task flow.

Configure a detour from a menu-specific override

Use the following procedure to set up a detour from a menu-specific override.

1. Create a menu-specific override for the relevant menu and step as described in Customize step titles and instructions for the Warehouse Management mobile app.
2. Find the combination of **Step ID** and **Menu item name** values that you want to edit, and then select the value in the **Step ID** column.
3. On the page that appears, on the **Available detours (menu items)** FastTab, specify the menu item that should act as a detour. You can also select which field values from the main task should automatically be copied to and from the detour. For examples that show how to use these settings, see the scenarios later in this article.

> **Important**
>
> Workers can only access menu items that are included in the menu that is assigned to their mobile device user account (or a submenu of that menu). This restriction also applies to menu items that are intended for use as detours, but which you might not want workers to access directly from the menu. In this case, add the detour items to the relevant menus and then hide the items. Learn more about how to hide menu items in Mobile device menu.

Sample scenario 1: Sales picking where a location inquiry acts as a detour

This scenario shows how to configure a location inquiry as a detour in a worker-directed sales picking task flow. This detour enables workers to look up all the license plates in the location that they're picking from and pick the license plate that they want to use to complete the pick. This type of detour might be useful if the bar code is damaged and therefore unreadable by the scanner device. Alternatively, it might be useful if a worker must learn what is actually on hand in the system. This scenario works only if you're picking from license plate–controlled locations.

Enable sample data

To use the specified sample records and values to work through this scenario, you must use a system where the standard demo data is installed. You must also select the **USMF** legal entity before you begin.

Create a menu-specific override and configure the detour for scenario 1

In this procedure, you configure a detour for the **Sales picking** menu item in the license plate step.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. Find the step ID named *LicensePlateId*, and select it.
3. On the Action Pane, select **Add step configuration**.
4. In the drop-down dialog, use the **Menu item** field to find and select *Sales picking*.

5. Select **OK**.

6. The details page for the override that you just created appears. On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.

7. In the **Add detour** dialog, select **Location inquiry** as the detour that the Warehouse Management mobile app makes available.

8. Select **OK**.

9. On the **Available detours (menu items)** FastTab, select the detour that you just added, and then select **Select fields to send** on the toolbar.

10. In the **Select fields to send** dialog, specify the information that should be sent to and from the detour. In this scenario, you enable workers to use the location that they're supposed to pick from as input for the location inquiry detour. Therefore, in the **Send from sales picking** section, select **Add** on the toolbar to add a row to the grid. Then set the following values for the new row:

    - **Copy from Sales Picking:** *Location*
    - **Paste in Location Inquiry:** *Location*
    - **Auto submit:** *Selected* (the page refreshes with the pasted *Location* value)

11. Because the detour in this scenario is configured on the license plate step, it helps if workers can bring the license plate from the inquiry back to the main flow. Therefore, in the **Bring back from location inquiry** section, select **Add** on the toolbar to add a row to the grid. Then set the following values for the new row:

    - **Copy from Location Inquiry:** *License plate*
    - **Paste in Sales Picking:** *License plate*
    - **Auto submit:** *Cleared* (no auto update occurs when returning from the detour with a *License plate* value)

12. Select **OK**.

The detour is now fully configured. A button to start the **Location inquiry** detour appears on the license plate step for the **Sales picking** menu item.

Complete a sales pick on a mobile device and use the detour

In this procedure, you complete a sales pick by using the Warehouse Management mobile app. You use the detour that you just configured to find the license plate that you use to complete

the pick step.

1. In Microsoft Dynamics 365 Supply Chain Management, create a sales order that requires a pick step to pick from a location that is license plate tracked. Then release the sales order to the warehouse. Make a note of the work ID that is generated.
2. Open the Warehouse Management mobile app, and sign in to warehouse 24. (In the standard demo data, sign in by using *24* as the user ID and *1* as the password.)
3. Select the **Outbound** menu, and then select the **Sales picking** menu item.
4. Follow the data entry instructions on the screen to enter the work ID that you generated in step 1.
5. On the step that contains the text **Scan a license plate**, open the actions menu. You see a new button that is labeled **Location inquiry**. An arrow in the upper-left corner indicates that the button starts a detour. Select **Location inquiry**.
6. The detour starts. On the next page, confirm the location that was copied from the main flow.
7. In the list of available license plates in the location, select the license plate that you want to copy back to the main flow. Then select the **Back** button.
8. You're returned to the sales picking main flow, and the license plate is copied back to it from the detour. Confirm the value to continue to the next step.
9. You can now complete the rest of the standard flow by entering the requested data entry instructions.

The warehouse work is now completed. The worker successfully opens a detour to perform a secondary task (location inquiry) without losing their place in the main task, and brings back the information that is required to complete the main task.

Sample scenario 2: Item inquiry with movement and adjustment detours

This scenario shows how to configure movement and adjustments as multiple detours in the location inquiry task flow. This capability can be useful in situations where a worker arrives at a location, finds that the inventory in the location differs from what is registered in the system, and must therefore adjust or move goods.

You can replace the location inquiry with a license plate inquiry or an item inquiry as you require.

Enable sample data

To use the specified sample records and values to work through this scenario, you must use a system where the standard demo data is installed. You must also select the **USMF** legal entity before you begin.

Create a menu-specific override and configure the detour for scenario 2

In this procedure, you configure a detour for the **Sales picking** menu item in the license plate step.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. Find and select the step ID named *LocationInquiryList*.

   > **Note**
   >
   > To use an item inquiry instead of a location inquiry, select a row where the step ID is named *ItemInquiryList*. To use a license plate inquiry, select a row where the step ID is named *LPInquiryList*.

3. On the Action Pane, select **Add step configuration**.
4. In the drop-down dialog, use the **Menu item** field to find and select *Location inquiry*.
5. Select **OK**.
6. The details page for the override that you just created appears. On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
7. In the **Add detour** dialog, select **Movement** as the detour that the Warehouse Management mobile app makes available.
8. Select **OK**.
9. On the **Available detours (menu items)** FastTab, select the detour that you just added, and then select **Select fields to send** on the toolbar.
10. In the **Select fields to send** dialog, specify the information that you send to and from the detour. In this scenario, workers look for license plate–controlled locations. Therefore, it helps to copy the license plate from the inquiry to the **Movement** detour. In the **Send**

from sales picking section, select **Add** on the toolbar to add a row to the grid. Then set the following values for the new row:

- **Copy from Location Inquiry:** *Location*
- **Paste in Movement:** *Loc / LP*
- **Auto submit:** *Cleared* (no auto update occurs)

In this detour, you don't expect any information to be copied back, because the main flow was an inquiry where no additional steps are required.

11. Select **OK**.

The detour is now fully configured. A button to start the **Movement** detour appears on the license plate step for the **Location inquiry** menu item.

Do a location inquiry on a mobile device and use the detour

In this procedure, you use the Warehouse Management mobile app to do a location inquiry. You then use the detour to complete a movement of goods.

1. Open the Warehouse Management mobile app, and sign in to warehouse 24. (In the standard demo data, sign in by using *24* as the user ID and *1* as the password.)
2. Select the **Inventory** menu, and then select the **Location inquiry** menu item.
3. Enter a location to inquire about, such as *FL-001*.
4. When the list appears, select and hold one of the cards in it to show the available detours.
5. Select **Movement**.
6. Notice that the license plate is copied from the card that you selected. Confirm the value.
7. Follow the standard task flow to complete the movement. After the work is completed, open the actions menu, and select **Cancel**.
8. You're returned to the **Location inquiry** page. The values aren't automatically updated. Therefore, you must manually refresh the page to see the changes from the movement detour.

Sample scenario 3: Movement by template with a data inquiry detour

This scenario shows how to configure movement by template with a data inquiry detour.

Enable sample data

To use the specified sample records and values to work through this scenario, you must use a system where the standard demo data is installed. You must also select the **USMF** legal entity before you begin.

In this procedure, you configure a data inquiry detour for the **Movement by template** menu item in the **Scan location or license plate** step.

Configure an Item data inquire mobile device menu item for scenario 3

To create the **Item data inquire** mobile device menu item, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - In the **Menu item name** field, enter *Item data inquire*.
    - In the **Title** field, look up the inventory dimensions by items.
    - In the **Mode** field, select *Indirect*.
4. On the **General** FastTab, set the following values:
    - In the **Activity code** field, select *Data inquiry*.
    - In the **Use process guide** field, ensure that *Yes* is selected. (*Yes* is the default value.)
    - In the **Table name** field, select *InventSum*, because you want to look up inventory dimensions from this table.
5. Select the fields that appear on the cards on the **Inquiry list** page. On the Action Pane, select **Field list**.
6. On the **Field list** page, set the following values:
    - **Display field 1:** *ItemId* (This field appears as the header for each card.)
    - **Display field 2:** *InventColorId*
    - **Display field 3:** *InventSizeId*
    - **Display field 4:** *WMSLocationId*
7. On the Action Pane, select **Save**.
8. Close the page.

> ⓘ **Note**
>
> You must add the **Item data inquire** mobile device menu item to the mobile device menu before you can use it as part of a detour process.

Create a menu-specific override and configure the detour for scenario 3

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. Find and select the step ID named *LocOrLP*.
3. On the Action Pane, select **Add step configuration**.
4. In the dropdown dialog, in the **Menu item** field, select *Movement by template*.
5. Select **OK**.
6. The details page for the override that you just created appears. On the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
7. In the **Add detour** dialog, select **Item data inquiry** as the detour that the Warehouse Management mobile app will provide.
8. Select **OK**.
9. On the **Available detours (menu items)** FastTab, select the detour that you just added.
10. In the **Bring back from Items data inquiry** section, select **Add** on the toolbar to add a row to the grid. Then set the following values for the new row:
    - **Copy from Items Inquiry:** *Item number*
    - **Paste in Movement by template:** *Item*
    - **Auto submit:** *Yes*
11. Select **Add** on the toolbar to add another row to the grid. Then set the following values for the new row:
    - **Copy from Items Inquiry:** *Size*
    - **Paste in Movement by template:** *Size*
    - **Auto submit:** *Yes*

12. Select **Add** on the toolbar to add another row to the grid. Then set the following values for the new row:

    - **Copy from Items Inquiry:** *Color*
    - **Paste in Movement by template:** *Color*
    - **Auto submit:** *Yes*

13. Select **Add** on the toolbar to add another row to the grid. Then set the following values for the new row:

    - **Copy from Items Inquiry:** *Location*
    - **Paste in Movement by template:** *Loc / LP*
    - **Auto submit:** *No*

14. Select **OK**.

The detour is now fully configured. A button to start the **Item data inquiry** detour now appears in the **Scan location or license plate** step for the **Movement by template** menu item.

Create a new product master item T-Shirt and add on-hand for scenario 3

Create a new *T-shirt* product master item that has size and color variants. Add one on-hand entry for one specific size and color of this item for warehouse 24.

Do a movement by template on a mobile device and use the detour

In this procedure, use the Warehouse Management mobile app to do a movement by template. Then, complete the movement by using the detour to find the item, color, size, and location.

1. Open the Warehouse Management mobile app, and sign in to warehouse 24. (In the standard demo data, sign in by using *24* as the user ID and *1* as the password.)
2. Select the **Movement by template** menu item.
3. In the first step that contains the text **Scan location or license plate**, select **Item data inquiry**.
4. The detour starts, and the first detour step is **Item number**. Specify *T-shirt*, and then confirm the entry.

5. Select the card that contains information about the location, color, and size of the existing on-hand entry. You're returned to the movement by template flow and receive a "Work completion" message.

The detour is configured to auto-submit only the item, size, and color. The system also auto-populates the location and quantity. This behavior occurs because the system can determine the size, location, and quantity from the specified color, based on the on-hand data distribution. The detour contains three auto-submit controls that correspond to three **OK** buttons. Because a default location and a default quantity are entered after the return from the detour, they're automatically confirmed.

> **Note**
>
> Multi-level detours are detours within detours, which allow workers to jump from an existing detour to a second one and then back again. The feature supports two levels of detours out of the box and, if necessary, you can customize your system to support three or more levels of detours by creating code extensions on the `WHSWorkUserSessionState` table.
>
> Auto-submit detour steps can make it faster and easier for workers to complete detour flows in the Warehouse Management mobile app. It enables some flow steps to be skipped by letting the app populate detour data on the back end and then move automatically to the next step by auto submitting the page, as shown in Sample scenario 1: Sales picking where a location inquiry acts as a detour earlier in the article.
>
> Essentially, the number of auto-submits corresponds to the number of times that the **OK** button is selected. If a mobile app flow has defaulting mechanics, auto-submit for detours can cause variations in behavior, based on the on-hand data composition. An example is shown in Sample scenario 3: Movement by template with data inquiry detour.

Last updated on 04/08/2026

Query data by using Warehouse Management mobile app detours

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Feature introduction

By providing bar code scanning capability, the Warehouse Management mobile app gives you an easy and accurate way to capture data as part of your warehouse processes. However, bar codes can sometimes get damaged and become unreadable, or the required data might not exist as a bar code in your business process flows. In these cases, manual entry of the data can take a long time and can even cause incorrect data to be captured. The results can be reduced effectiveness and a lower service level.

When you use a flexible data inquiry process, workers can easily look up the required information as part of the embedded Warehouse Management mobile app flows and apply filtering options so that only the relevant data is shown. Therefore, manual selection is faster and more accurate.

For example, in the purchase order receiving flow, a purchase order number is required to match arriving inventory. As part of this process, you can easily configure menu items to provide a card list view of the relevant purchase order numbers. In this way, you can continue the receiving flow by using a quick point-to-select approach. This article provides example scenarios, but the functionality can also be used within any or all your Warehouse Management mobile app flows.

The work user session query range utility tool

Supply Chain Management version 10.0.37 (and later) includes a *work user session query range utility tool*, which adds the following functionality for setting up queries to help workers find specific records by using the Warehouse Management mobile app:

- **Filter records by warehouse** – Set up queries that use the worker's current warehouse as a value. Use this value as a criterion for range filters (on the **Range** tab of the standard query designer). This filter is useful, for example, if you want to display a list of purchase orders expected to arrive at the warehouse where the worker is currently working. To set up a mobile device menu item to use this feature, open the query designer and, on the **Range** tab, set up a row that uses a **Criteria** of

`(WhsWorkUserSessionSysQueryRangeUtil::whsWorkUserSessionCurrentWarehouse())`. An example of how to use this functionality is provided later in this article.

- **Filter records by app user** – Set up queries that use the current worker's worker ID as a value. Use this value as a criterion for range filters (on the **Range** tab of the standard query designer). To set up a mobile device menu item to use this feature, open the query designer and, on the **Range** tab, set up a row that uses a **Criteria** of `(WhsWorkUserSessionSysQueryRangeUtil::whsWorkUserSessionCurrentWorkUserId())`.
- **Show or hide filters in the mobile app** – Choose whether to show a page with filter values during a data inquiry flow for mobile device menu items. When you're using the **Mobile device menu items** page to set up a mobile device menu item where this feature is relevant, you find a **Show filters page** setting, which lets you choose whether to show the filters *Always* or *Only when filters can be changed*.

Example scenarios

This article uses example scenarios to show how to query data by using detours to improve the purchase receipt flow. The scenarios use the standard sample data, which includes a flow named *Purchase receive*.

This flow starts by prompting workers to identify a purchase order number that they receive against. To help workers more easily identify the purchase order, enhance the first page of the flow by adding the following new query options as detours:

- **Look up POs by vendor** – Open a page that prompts workers to enter a vendor name or part of a vendor name. Use wildcard characters. For example, if a worker expects an inbound delivery today from a vendor that includes *Tailspin* in its name, they can enter **Tail*** to view a set of cards for open purchase orders that include this text. Every card has several fields that provide information about each purchase order. In addition to the vendor's name, design the cards so that they show the vendor account number, delivery date, and document status.
- **Look up POs for today** – Open a page that doesn't prompt workers to enter data, but shows a set of cards that match the "hardcoded" filter. Workers proceed by selecting a card for the purchase order that they want to register inventory items against. This process is enabled with the option **Show filter page** set to *Only when filters can be changed*. This setting means the app shows results immediately (without showing the filters page).
- **Look up POs by item** – Open a page that prompts workers to scan the bar code of any item in the arrived inventory. The flow then lists all open purchase orders that contain lines for the scanned item number. To cover situations where a bar code can't be read, add another detour lookup to this page that lets workers search for item numbers within a specific purchase order.

In each case, the worker identifies a purchase order by selecting a card and is then returned to the first page, which shows the selected purchase order number. The worker can then continue the inbound inventory item registration flow.

Enable sample data

To work through the example scenarios described in this article, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity (company) before you begin.

Configure the mobile device menu items

To create each of the new query options that you add to the first page of the flow, set up each option as a mobile device menu item. Later, make the query options available as detours to the *Purchase receive* flow.

Create the "Look up POs by vendor" menu item

Create the **Look up POs by vendor** menu item by following these steps.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name:** *Look up POs by vendor*
    - **Title:** *Look up POs by vendor*
    - **Mode:** *Indirect*
4. On the **General** FastTab, set the following values:
    - **Activity code:** *Data inquiry*
    - **Use process guide:** *Yes* (This value is automatically selected.)
    - **Table name:** *PurchTable* (You want to look up purchase order numbers from this table.)
    - **Show filter page:** *Only when filters can be changed* (Filters are shown due to the *Vendor name*.)
5. On the Action Pane, select **Edit query** to define a query that's based on the selected base table (in this case, the purchase orders table).
6. In the query editor, on the **Range** tab, add the following lines to the grid.

Expand table

Table	Derived table	Field	Criteria
Purchase orders	Purchase orders	Purchase order status	Open order
Purchase orders	Purchase orders	Delivery date	(dayRange(-10,10))
Purchase orders	Purchase orders	Vendor name	

> **Tip**
>
> In this example, the filter doesn't include any warehouse criteria and therefore returns open purchase orders for all warehouses within the current legal entity. To limit the search results to only find purchase orders for the current worker's warehouse, add a query line with **Field** set to *Warehouse* and **Criteria** set to `(WhsWorkUserSessionSysQueryRangeUtil::whsWorkUserSessionCurrentWarehouse())`. You can also use this value for fields from joined tables (such as the *Inventory dimensions* table). For more information, see **The work user session query range utility tool**.

7. Select **OK**.

   In this example, the new menu item is configured to find open purchase orders that are expected to arrive anytime between 10 days in the past and 10 days in the future.

   In the query, the **Criteria** column for *Vendor name* is left blank. Therefore, workers can specify this value while they use the Warehouse Management mobile app.

   If you want to specify how the list is sorted, set up the sorting on the **Sorting** tab.

8. In addition to defining the query, select which fields show on the cards on the inquiry list page. On the Action Pane, select **Field list**.

9. On the **Field list** page, set the following values:

   - **Display field 1:** *PurchId* (This field shows as the header for each card.)
   - **Display field 2:** *PurchStatus*
   - **Display field 3:** *PurchName*
   - **Display field 4:** *OrderAccount*
   - **Display field 5:** *DeliveryDate*
   - **Display field 6:** *displayDocumentStatus()* (This value is a display method, as the "()" at the end indicates.)

10. On the Action Pane, select **Save**. Then close the page.

Create the "Look up POs for today" menu item

Create the **Look up POs for today** menu item by following these steps.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New** to add a mobile device menu item.

3. Set the following values for the new item:

   - **Menu item name:** *Look up POs for today*
   - **Title:** *Look up POs for today*

- **Mode:** *Indirect*

4. On the **General** FastTab, set the following values:

   - **Activity code:** *Data inquiry*
   - **Use process guide:** *Yes* (This value is automatically selected.)
   - **Table name:** *PurchTable* (You want to look up purchase order numbers from this table.)
   - **Show filter page:** *Only when filters can be changed* (Show results immediately.)

5. On the Action Pane, select **Edit query** to define a query that's based on the selected base table (in this case, the purchase orders table).

6. In the query editor, on the **Range** tab, add the following lines to the grid.

Expand table

Table	Derived table	Field	Criteria
Purchase order	Purchase order	Purchase order status	Open order
Purchase order	Purchase order	Confirmed receipt date	(Day(0))
Purchase order	Purchase order	Warehouse	(WhsWorkUserSessionSysQueryRangeUtil::whsWorkUserSessionCurrentWareh

> **Note**
>
> In this example, the query filters search results by warehouse because it includes a line with the *Warehouse* field. If you set this line with a blank value for **Criteria**, the system shows a filter page that automatically displays the worker's current warehouse and allows the worker to update the value as needed. If you set this line to have a **Criteria** value of `(WhsWorkUserSessionSysQueryRangeUtil::whsWorkUserSessionCurrentWarehouse())`, the application automatically uses this value without showing it to the worker for confirmation. Learn more in The work user session query range utility tool.

7. Select **OK**.

   In this example, the new menu item is configured to find open purchase orders that are expected to arrive today.

   If you want to specify how the list is sorted, set up the sorting on the **Sorting** tab.

8. In addition to defining the query, select which fields show on the cards on the inquiry list page. On the Action Pane, select **Field list**.

9. On the **Field list** page, set the following values:

    - **Display field 1:** *PurchName* (This field shows as the header for each card.)
    - **Display field 2:** *PurchId*
    - **Display field 3:** *PurchStatus*
    - **Display field 4:** *DlvMode*
    - **Display field 5:** *DlvTerm*
    - **Display field 6:** *OrderAccount*
    - **Display field 7:** *VendorName()* (This value is a display method, as the "()" at the end indicates.)

10. On the Action Pane, select **Save**. Then close the page.

Create the "Look up POs by item" menu item

Create the **Look up POs by item** menu item by following these steps.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New** to add a mobile device menu item.

3. Set the following values for the new item:

    - **Menu item name:** *Look up POs by item*
    - **Title:** *Look up POs by item*
    - **Mode:** *Indirect*

4. On the **General** FastTab, set the following values:

    - **Activity code:** *Data inquiry*
    - **Use process guide:** *Yes* (This value is automatically selected.)
    - **Table name:** *PurchLine* (You want to look up purchase order numbers based on item number via the line data.)
    - **Show filter page:** *Only when filters can be changed* (Filters will be shown due to the *Item number*.)

5. On the Action Pane, select **Edit query** to define a query that is based on the selected base table (in this case, the purchase order lines table, but you can use any of the values that are related to the header by joining to the *PurchTable*).

6. In the query editor, on the **Range** tab, add the following lines to the grid.

Expand table

Table	Derived table	Field	Criteria
Purchase order lines	Purchase order lines	Line status	Open order
Purchase order lines	Purchase order lines	Delivery date	(dayRange(-10,10))
Purchase order lines	Purchase order lines	Item number	

> **Tip**
>
> In this example, the filter doesn't include any warehouse criteria and returns open purchase orders for all warehouses within the current legal entity. To limit the search results to the current worker's warehouse, you can join to the *Inventory dimensions* table, then include the *Warehouse* field as part of the range query, and enter `(WhsWorkUserSessionSysQueryRangeUtil::whsWorkUserSessionCurrentWarehouse())` as the **Criteria** for the *Warehouse* field. For more information, see **The work user session query range utility tool**.

7. Select **OK**.

   In this example, the new menu item is configured to find open purchase order lines that are expected to arrive anytime between 10 days in the past and 10 days in the future.

   In the query, the **Criteria** column for *Item number* is left blank. Therefore, workers can specify this value while they use the Warehouse Management mobile app.

   If you want to specify how the list is sorted, set up the sorting on the **Sorting** tab.

8. In addition to defining the query, select which fields show on the cards on the inquiry list page. On the Action Pane, select **Field list**.

9. On the **Field list** page, set the following values:

   - **Display field 1:** *PurchId* (This field value will be used as the header for each card.)
   - **Display field 2:** *VendAccount*
   - **Display field 3:** *PurchQty*
   - **Display field 4:** *PurchUnit*
   - **Display field 5:** *PurchStatus*

10. On the Action Pane, select **Save**. Then close the page.

Add the new mobile device menu items to a menu

Add your three new mobile device menu items to the mobile device menu. You must complete this task before the menu items can be used as part of a detour process. In this example, you create a new submenu and add the new menu items to it.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. On the Action Pane, select **New**.
3. Set the following values on the header of the new record:
    - **Name:** *Inquire*
    - **Description:** *Inquire*
4. In the **Available menus and menu items** list, select the first of the mobile device menu items that you just created. Then select the right arrow button to move that item into the **Menu structure** list.
5. Repeat the previous step for the other two new menu items.
6. In the list pane on the left, select the **Main** menu.
7. In the **Available menus and menu items** list, scroll down to the **Menus** section, and select your new **Inquire** menu. Then select the right arrow button to move that item into the **Menu structure** list.

Configure detours in your mobile device steps

To complete the setup, use the detour configuration on the **Mobile device steps** page to add the three new mobile device menu items to the existing purchase order identification step in the *Purchase receive* flow.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device steps**.
2. In the **Filter** field, enter *PONum*. Then select *Step ID: "PONum"* in the drop-down list.
3. While the record that is found is selected in the grid, select **Add step configuration** on the Action Pane. In the drop-down dialog box that appears, set the **Menu item** field to *Purchase Receive*. Then select **OK** to close the dialog box.
4. On the details page for the new step configuration (**Purchase Receive : PONum**), on the **Available detours (menu items)** FastTab, select **Add** on the toolbar.
5. In the **Add detour** dialog box, find and select the **Look up POs by vendor** menu item that you previously created.
6. Select **OK** to close the dialog box and add the selected menu item to the detours list.
7. Select the new detour, and then select **Select fields to send** on the toolbar.
8. In the **Select fields to send** dialog box, don't add anything to the **Send from purchase receive** section, because you don't want to pass any values to the detour menu item. However, in the **Bring back from look up POs by vendor** section, set the following value for the empty row that is already added:

- **Copy from Look up POs by vendor:** *Purchase order*
- **Paste in Purchase Receive:** *Purchase order*

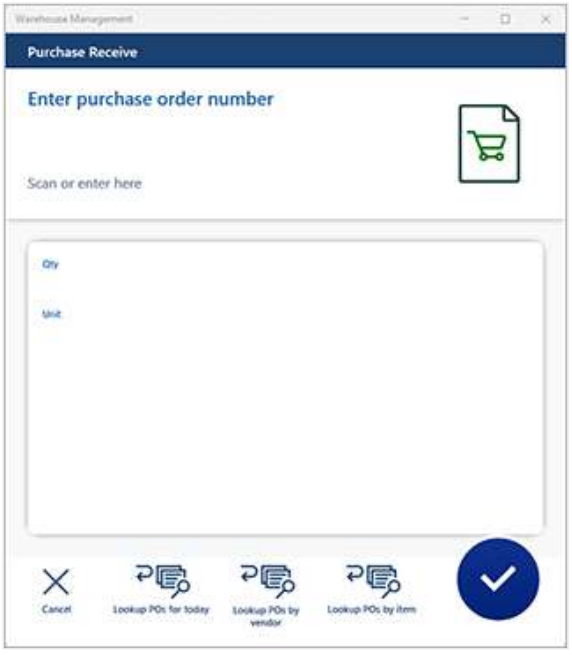
9. Select **OK** to close the dialog box.
10. Repeat steps 4 through 9 for the other two new menu items (**Look up POs for today** and **Look up POs by item**). As for the **Look up POs by vendor** menu item, you don't want to send any data to these detours, but you do want to return a purchase order number.
11. Close the page.

Try a purchase receiving flow that has a data inquiry as part of a detour

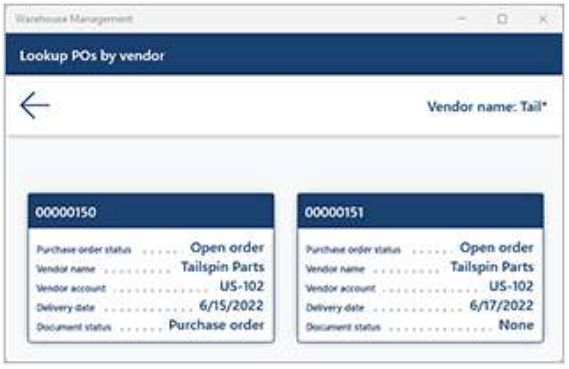
Follow these steps to test your new mobile app setup.

1. Create several purchase orders that have lines for warehouse 51.
2. Go to a mobile device or emulator that is running the Warehouse Management mobile app, and sign in to warehouse 51 by using *51* as the user ID and *1* as the password.
3. On the mobile app menu, select **Inbound** and then **Purchase receive**.

   You should see the following page, which includes the three new menu items.



4. Try out the different capabilities, and notice that you can send back a purchase order number by selecting one of the cards in the list.



> **Tip**
>
> Instead of running the receiving flow by doing a lookup from the **Purchase receive** menu item, you can start from an inquiry flow (**Main** > **Inquire** > **Look up POs by vendor**) and invoke a detour to run the desired flow by selecting one of the cards in the list. To use this approach, you can define a detour on the **Mobile device steps** page for the step that has a **Step ID** value of *GenericDataInquiryList*. Provided the ***Multi-level detours for the Warehouse Management mobile app*** feature is turned on for your system, you can also add an additional detour if needed (this feature adds support for up to two levels of detours and can be customized to support additional levels).

Last updated on 05/29/2026

License plate receiving via the Warehouse Management mobile app

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up the Warehouse Management mobile app to support a license plate receiving process for physical inventory.

This functionality lets you quickly record the receipt of inbound inventory related to an advance ship notice (ASN). The system automatically creates an ASN when warehouse management processes (WMS) are used to ship a transfer order. For the purchase order and inbound shipment order processes, an ASN can be recorded manually or imported automatically by using an inbound ASN data entity process.

The ASN data is linked to loads and shipments via the *packing structures*, where pallets (parent license plates) can contain cases (nested license plates).

> **Note**
>
> To reduce the number of inventory transactions when packing structures that have nested license plates are used, the system records the physical on-hand inventory on the parent license plate. To trigger the movement of the physical on-hand inventory from the parent license plate to the nested license plates, based on the packing structure data, the mobile device must provide a menu item that is based on the *Pack to nested license plates* work creation process.

Warehousing mobile device app processing

When a worker scans an incoming license plate ID, the system initializes a license plate receiving process. Based on this information, the content of the license plate (data coming from the ASN) is

physically registered at the inbound dock location. The flows that follow depend on your business process needs.

Work policies

As with the *Report as finished* mobile device menu item process, the license plate receiving process supports several workflows based on the setup.

Work policies with work creation

When you register incoming items using a work policy that creates work, the system generates and saves put-away work records for each registration. If you use the *License plate receiving and put away* work process, then registration and put away are handled as a single operation using a single mobile device menu item. If you use the *License plate receiving* process, then the receiving and put-away processes are handled as two different warehouse operations, each with their own mobile device menu item.

Work policies without work creation

You can use the license plate receiving process without creating work. If you define work policies that have a work order type of *Transfer receipt*, *Purchase orders*, and/or *Inbound shipment order*, and you use the process for *License plate receiving (and put away)*, the following two Warehousing mobile app processes don't create work. Instead, they just register the inbound physical inventory on the license plate at the inbound receiving dock.

- *License plate receiving*
- *License plate receiving and put away*

> **Note**
>
> - You must define at least one location for a work policy in the **Inventory locations** section. You can't specify the same location for multiple work policies.
> - The **Print label** option for mobile device menu items doesn't print a license plate label without work creation. However, you can print a license plate label as part of the deferred receiving process.

Receive inventory on a location that doesn't track license plates

You can use a warehouse location assigned to a location profile even when **Use license plate tracking** isn't turned on. Therefore, when you receive inventory, you can directly register the on-hand inventory on a location without work creation.

Add mobile device menu items for each receiving location in a warehouse

The *License plate receiving enhancements* feature lets you receive at any location in a warehouse by adding location-specific license plate receiving (and put away) menu items to the Warehousing mobile app. Previously, the system supported receiving only at the default location defined for each warehouse. However, this feature lets mobile device menu items for license plate receiving (and put away) provide the **Use default data** option, which lets you select a custom "to" location for each menu item.

Show or skip the receiving summary page

You can use the *Control whether to display a receiving summary page on mobile devices* feature to take advantage of an additional detailed Warehouse Management mobile app flow as part of the license plate receiving process. With this feature, mobile device menu items for license plate receiving or license plate receiving and put-away provide a **Display receiving summary page** setting. This setting has the following options:

- **Display a detailed summary** – During license plate receiving, workers see an extra page that shows the full ASN information.
- **Skip the summary** – Workers don't see the full ASN information. Warehouse workers also can't set a disposition code or add exceptions during the receiving process.

Prevent transfer order–shipped license plates from being used at warehouses other than the destination warehouse

A license plate receiving process can't be used if an ASN contains a license plate ID that already exists and has physical on-hand data at a warehouse location other than the warehouse location where the license plate registration occurs. For transfer order scenarios where the transit warehouse doesn't track license plates (and therefore also doesn't track physical on-hand inventory per license plate), you can use the *Prevent transfer order shipped license plates from being used on other warehouses than the destination warehouse* feature to prevent physical on-

hand updates of license plates that are in transit. To manage the functionality provided by this feature, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **License plates** FastTab, set the **Transit warehouse license plate policy** field to one of the following values:
    - **Allow reuse of non-tracked license plate** – Non-tracked license plates can be reused.
    - **Prevent reuse of non-tracked license plate** – Only on-hand updates related to a shipped license plate are allowed at the destination warehouse until the transfer order has been received.

Deferred receiving processing

For businesses that use the *License plate receiving* process to inbound receive thousands of lines related to the same license plate, the Warehouse Management mobile app receiving process must wait until all lines and related work are created as part of the registration process. The exception is when the deferred receiving option is used. This option postpones registration as a background process. Therefore, warehouse workers can immediately continue other work.

More information

- For more information about mobile device menu items, see Set up mobile devices for warehouse work.
- For more information about work policies, see work policies.
- For more information about deferred receiving, see mixed license plate receiving.
- For more information about the *Report as finished* production scenario, see Warehouse work policies overview.
- For more information about inbound load management, see Warehouse handling of inbound loads for purchase orders.

Last updated on 06/22/2026

License plate build on the warehouse app

This article explains how to set up the Warehouse Management mobile app to build license plates. A license plate is a unique identifier that is assigned to a group of items. License plates help streamline the process of managing items, because items don't have to be handled individually. Instead, all the items that share a license plate can be tracked as a single unit.

This functionality lets you assign multiple child license plates to a single parent license plate. Therefore, in effect, the license plates become nested. License plate nesting is beneficial because a warehouse worker can move a larger group of inventory in one action. Therefore, the worker's processes are streamlined.

The **License plate build** menu item in the Warehouse Management mobile app helps with this process by letting users select a parent license plate and assign child license plates to it. However, license plate nesting works only if the following conditions are met:

- No on-hand inventory is assigned to the parent license plate. If the license plate that you try to use as a parent has on-hand inventory, you receive an error message.
- Any license plates that have on-hand inventory are the final members of the sequence. If you think of the whole sequence of license plates as a tree, these license plates are the leaves.

Configure the License plate build menu item

To configure the **License plate build** menu item, follow these steps:

1. In Supply Chain Management, go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New**.

3. On the header of the new record, set the following fields:

    - **Menu item name** – Enter a unique name to identify the menu item. For example, enter *License plate building*.
    - **Title** – Enter the title that should be shown on the mobile device menu. The title doesn't have to be unique.
    - **Mode** – Select *Indirect*, because this flow isn't directly related to work.

4. On the **General** FastTab, set the following fields:

    - **Use process guide** – Set this option to *No*, because process guide isn't supported for this flow.
    - **Barcode data policy** – Leave this field blank.

5. On the Action Pane, select **Save**.

To create a standalone menu item that can be used to assign license plates to a parent, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. Add the new menu item to an appropriate place in your mobile device menu structure.

Example

This example shows how a parent license plate can be used to move on-hand inventory.

For this example, the following setup is in place:

- A parent license plate, *LP-PARENT*, has two nested license plates, *LP-CHILD1* and *LP-CHILD2*.
- There are two items, *item-1* and *item-2*.
- Both items have on-hand inventory in the same location, *location-1*.
- In location *location-1*, item *item-1* has on-hand inventory in license plate *LP-CHILD1*. Item *item-2* has on-hand inventory in license plate *LP-CHILD2*.

You want to move the on-hand inventory of items *item-1* and *item-2* to a new location, *new-location*. Therefore, you create an inventory movement on the parent license plate, *LP-PARENT*. By moving the parent license plate, you transfer all the nested on-hand inventory in the sequence of license plates to the new location in a single action.

To verify the on-hand movement, follow these steps:

1. In Supply Chain Management, go to **Inventory and management** > **Inquiries and reports** > **On-hand list**.
2. Filter the results by the two items, *item-1* and *item-2*.
3. For each item, select **Quantity adjustment** on the Action Pane. The page shows the on-hand inventory quantity that was put in the new location, *new-location*.

> **Important**

Parent license plates are intended to serve as links to groups of nested license plates. Therefore, for optimal performance, we recommend that you never assign on-hand inventory to parent license plates. Although some workflows might continue to work if a parent license plate has on-hand inventory, this scenario isn't fully supported.

Learn about parent license plates when advance ship notices (ASNs) are used in License plate receiving via the Warehouse Management mobile app.

Last updated on 12/30/2025

Set up a mobile device menu item for moving items in the warehouse

Article • 12/20/2023

This article explains how to set up a mobile device menu item that lets workers register manual movements of items in the warehouse. The worker who uses this mobile device menu item decides what to move and where to move it.

Follow these steps to create and configure a mobile device menu item for registering movements in the warehouse.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New**.

3. On the header of the new record, set the following fields:

    - **Menu item name** – Enter a unique name to identify the menu item.
    - **Title** – The title that you enter will be shown in the user interface (UI). It doesn't have to be unique.
    - **Mode** – Select *Work*. This value is required because this mobile flow involves work.
    - **Use existing work** – Set this option to *No*, because the work is created as the movement is done.

4. On the **General** FastTab, set the following fields:

    - **Work creation process** – Select *Movement*, because this type of work is what the menu item supports. The page is updated to provide settings that are relevant to this type of work.

    - **Display inventory status** – Set this option to *Yes* to show the inventory status on the device. Set it to *No* to use the default inventory status. If the items that are moved have only one inventory status, that status is the default status. If you use only one inventory status, you typically won't have to show this information.

    - **Use default data** – Set this option to *Yes* if you want specific data fields to be shown by default to workers who use the mobile app. Default data field values can provide information that workers typically need in their daily work. After you set this option to *Yes*, select **Default data** on the Action Pane to

open a page where you can select the fields to show. For example, the *From location* field is often useful. Set this option to *No* if you don't want to select any default data for the menu item.

- **Use process guide** – Set this option to *No*, because process guide isn't currently supported for movements of items.

- **Barcode data policy** – Select the policy to use when multiple fields are filled in based on a single bar code scan. Learn more in GS1 bar codes.

- **Generate license plate** – Set this option to *Yes* to automatically create new license plates as they're needed. Set it to *No* if the worker must always select an existing license plate. A license plate is needed if part of the quantity is moved from a license plate. A license plate is also needed if the "from" location isn't license plate controlled, but the "to" location is.

- **Confirm cancel** – Select whether workers should be asked for confirmation when they cancel a movement operation that's in progress. Confirmation adds an extra step but can help prevent loss of information if a worker unintentionally cancels the operation.

- **Display container type** – Select whether to show the container type in the app. The container type must typically be shown if workers require information about the type of container that the license plate is associated with.

- **Deferred operation processing policy** – Select one of the following values to specify whether deferred processing should be used for movements that are registered by using this menu item:
  - *Respect settings* – Use deferred operation processing as defined on the work processing policy.
  - *Override settings and process operation synchronously* – Override the setting, and process the operation synchronously. This value allows for configurations where the mobile device processes the movement synchronously, regardless of the configuration for the type.

- **Full LP movement policy** – This field can help make workers more efficient when they move full license plates. The best value depends on how you manage your warehouse movements. Select one of the following options:
  - *Use specification for single item movement* – The worker is prompted to specify all relevant details (such as quantities) before they move a selected license plate.

- *Automatically move full LP for single item LP* – If a license plate contains a single item, the worker isn't prompted to scan that item before they move the license plate. However, if a license plate contains multiple items, the worker is prompted to scan each item after they identify the license plate.
- *Automatically move full LP* – The system assumes that all warehouse movements are for entire license plates, regardless of the number items that a license plate contains. The worker is never prompted to scan the items. This option works well when, for example, all warehouse movements are for entire pallets.

5. On the Action Pane, select **Save**.

6. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**, and add the new mobile device menu item to the appropriate new or existing menu.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up a mobile device menu item for moving items by template

07/16/2025

This article explains how to set up a mobile device menu item that lets workers register movements of items in the warehouse. The worker doesn't decide where to move the items, but instead relies on the system to determine the target location based on the work template and/or location directive configured for the menu item.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New**.

3. On the header of the new record, set the following fields:

    - **Menu item name** – Enter a unique name to identify the menu item.
    - **Title** – The title that you enter is shown in the user interface (UI). It doesn't have to be unique.
    - **Mode** – Select *Work*. This value is required because this mobile flow involves work.
    - **Use existing work** – Set this option to *No*, because the work is created as the movement is done.

4. On the **General** FastTab, set the following fields:

    - **Work creation process** – Select *Movement by template*, because this type of work is what the menu item supports. The page is updated to provide settings that are relevant to this type of work.

    - **Display inventory status** – Select one of the following options:
        - *Yes* – Show inventory status on the device. In this case, if default values are set for the **From inventory status** and/or **To inventory status** fields, then those values are used as defaults. To set these default data field values, select **Default data** on the Action Pane.
        - *No* – Use the default inventory status specified on **Warehouse management parameters** page or at the warehouse level (available at **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**). The **Default data** settings for inventory status aren't used in this case.

        > ⓘ **Note**
        >
        > If the moved items have only one inventory status, that status is the default status. If you only use one inventory status, you typically won't need to show

this information on the device.

- **Use default data** – Set this option to *Yes* if you want specific data fields to be shown by default to workers who use the mobile app. Default data field values can provide information that workers typically need in their daily work. After you set this option to *Yes*, select **Default data** on the Action Pane to open a page where you can select the fields to show. For example, the *From location* field is often useful. Set this option to *No* if you don't want to select any default data for the menu item.
- **Use process guide** – Set this option to *No*, because process guide isn't currently supported for movements of items.
- **Barcode data policy** – Select the policy to use when multiple fields are filled in based on a single bar code scan. Learn more in GS1 bar codes.
- **Generate license plate** – Set this option to *Yes* to automatically create new license plates as they're needed. Set it to *No* if the worker must always select an existing license plate. A license plate is needed if part of the quantity is moved from a license plate. A license plate is also needed if the "from" location isn't license plate controlled, but the "to" location is.
- **Create movement** – Set to *Yes* to all workers to create work for a movement without requiring that the worker do the work immediately. This option is useful, for example, after a quality inspection has been completed and the inspector wants the item to be moved from the quality inspection area.
- **Work template** – Select a work template to use when creating the movement work.
- **Directive code** – To use a specific location directive, select the directive code that is associated the location directive.
- **Override license plate during put** – Set to *Yes* to allow workers to consolidate the quantity on the license plate already at the location.
- **Group put away** – Set to *Yes* to group the putaway work. This option is available when work is grouped either by the worker or by the system. When the worker has finished all the picking work in the group, putaway work is created for the same group.
- **Audit template ID** – Select the work audit template that will interrupt the work process for this menu item so that another operation can be performed. For example, if this menu item is for inbound work, the audit template might require that the worker checks the temperature in the delivery container. The point at which

the process is interrupted is specified on the audit template and can be, for example, when work is started or completed, or when its status changes.

- **Full LP movement policy** – This field can help make workers more efficient when they move full license plates. The best value depends on how you manage your warehouse movements. Select one of the following options:
  - *Use specification for single item movement* – The worker is prompted to specify all relevant details (such as quantities) before they move a selected license plate.
  - *Automatically move full LP for single item LP* – If a license plate contains a single item, the worker isn't prompted to scan that item before they move the license plate. However, if a license plate contains multiple items, the worker is prompted to scan each item after they identify the license plate.
  - *Automatically move full LP* – The system assumes that all warehouse movements are for entire license plates, regardless of the number items that a license plate contains. The worker is never prompted to scan the items. This option works well when, for example, all warehouse movements are for entire pallets.

5. On the Action Pane, select **Save**.

6. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**, and add the new mobile device menu item to the appropriate new or existing menu.

Set up a mobile device menu item for completing work of type Purchase order

Article • 06/03/2022

This article shows how to set up a Mobile device menu item. In this example, the menu item is used for performing work of type Purchase order. The work class that's associated with the menu item determines which work is valid. You can use this guide in demo data company USMF. This procedure is typically carried out by a warehouse manager.

Create a mobile device menu item

1. Go to **Mobile device menu items** by entering it in the search bar.
2. Select **New**.
3. In the **Menu item name** field, type a value. Enter a unique value. For example, you could type `POMove`. Remember the value, you'll need it later.
4. In the **Title** field, type a value. This is the title which will be displayed on the mobile device. For example, you could type `PO Move`.
5. In the **Mode** field, select **Work**.
6. Select **Yes** in the **Use existing work** field.

    - This mobile device menu item is used to perform existing work. Therefore you must set this value to **Yes**.
    - The **Display inventory status** field determines whether the inventory status of the on-hand inventory will be displayed to the warehouse worker on the mobile device.

7. In the **Directed by** field, select **System grouping**. When you select something in the **Directed by** field, additional fields appear in the **General** section on this page. The fields that appear depend on what you selected. When you select **System grouping**, two new fields are added. These are explained below.
8. In the **System grouping** field, select **WorkPoolId**. When warehouse workers open this menu item, they'll be asked to scan a Work pool ID. All work orders with this Work pool ID and open work order lines with one of the work classes added to this menu item will be pushed to the user.
9. In the **System grouping label** field, type a value. This is the text displayed to the user on the mobile device. For example, you could type **Work pool**.
10. Select **Yes** in the **Override license plate during put** field. This option allows warehouse workers to override the target license plate when items are put down

on a license plate controlled location.

11. Select **Yes** in the **Group put away** field.

    - If all the Put lines on the work order share the same location, the user will receive one combined Put instruction for all lines.
    - Audit template ID: A work audit template allows you to specify that the work process for a menu item should be interrupted so that another operation can be performed. For example, if this menu item is for inbound work, the audit template might require that the worker checks the temperature. The point at which the process is interrupted is specified on the audit template and can be, for example, when work is started or completed, or when its status changes.

12. Expand the **Work classes** section.
13. Select **New**.
14. In the **Work class ID** field, type `Purchase`. The work pool restricts the work that the menu item can be used for. In this case it will be used for open work order lines that have the Purchase work class ID.
15. Select **Save**.

Set up work confirmation

1. Select **Work confirmation setup**.
2. In the **Work type** field, select **Pick**.
3. Select the **Auto confirm** check box. The work instruction with work type Pick will be auto-confirmed. This instruction will not be presented to the user.
4. Select **New**.
5. In the **Work type** field, select 'Put'.
6. Select the **Location confirmation** check box. The warehouse worker will be asked to perform a confirmation scan of the location, when the item is put down.
7. Select **Save**.

Add the menu item to a mobile device menu

1. Go to **Mobile device** menu by entering it in the search bar.
2. Select **Edit**.
3. Use the Quick Filter to find records. For example, filter on the **Name** field with a value of **inbound**. You want to find the menu you use for inbound menu items. In USMF this is called **Inbound**.
4. In the tree, select **a value**.
5. Select the arrow that points to the right.

6. Select **Save**.
7. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Set up a mobile device menu item to register received items

Article • 01/31/2023

This article focuses on the setup of a mobile device menu item. This menu item is used for registration of the receipt of items ordered via purchase orders.

You can use this guide in demo data company USMF. This procedure is intended for the warehouse manager.

Create a mobile device menu item

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select **New**.
3. In the **Menu item name** field, type a value. This is the unique identifier for this mobile device menu item. For example, you could type `My PO registration`.
4. In the **Title** field, type a value. This is the title, which will be displayed to the user on the mobile device. For example, you could type `PO registration`.
5. In the **Mode** field, select **Work**. Registration of on-hand quantities received for a purchase order line will create work to move the items from the receiving area into the inventory. Work isn't created until the items are registered. Therefore, leave the **Use existing work** option set to **No**.
6. In the **Work creation process** field of the **General** section, select **Purchase order item receiving**.

    - A purchase order line must be uniquely identified before on-hand can be registered in the warehouse. In this scenario, the mobile device will register the purchase order number and item number, and this will allow the system to identify the PO line. Put away work will be created and can be picked up by another worker. The work creation method that you select determines which fields become available on the **General** FastTab.
    - If you select the **Use default data** option, the **Default data** button is enabled. Here you can select fields to display data that a worker typically needs in their daily work, so that these values are shown on the mobile device.
    - The **License plate grouping** parameter works in combination with the unit sequence group that's assigned to the item that's being received. You can specify whether receipts of less than or more than one pallet should be

grouped into one license plate, or divided into a separate license plate for each unit.

- If you select the **Generate license plate** option, this generates a unique license plate number based on the number sequence selection.
- You can select the template that will be used when work is created. For example, if you register an item for a purchase order, the put away work will be generated based on the work template. If you don't select a work template here, the system will assign a template based on the query criteria that are associated with the templates.
- If disposition codes are displayed on the mobile device, workers can evaluate the status or quality of the items, and select the appropriate code. The rules for the disposition code determine whether the items will be available to other warehouse processes. The rules also determine which location directive is used for the work that's created.
- If you select the **Batch disposition codes** option, workers can evaluate the status or quality of a batch, and select the appropriate disposition code. The rules that are set on the batch disposition code determine whether the batch will be available to other warehouse processes.
- If you select the **Print labels** option, a license plate label will be printed automatically when items are received.

7. Select **Save**.
8. Close the page.

Add the menu item to a mobile device menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. Use the **Quick Filter** to filter on the **Name** field with a value of `inbound`.
3. Select **Edit**.
4. In the Available menus and items tree, select the menu item that you created before.
5. Select the arrow that points to the right.
6. Select **Save**.
7. Close the page.

Feedback

Was this page helpful? Yes No

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Set up a mobile device menu item to provide a pick line overview

This article explains how to configure options that are related to the pick line overview for mobile device menu items that are used to process picking work. The pick line overview lets warehouse workers view and select from a list of all the work lines that are related to their current task. This capability can help workers optimize their picking sequence. The feature provides options that replace the standard **Skip** button that lets workers cycle through the lines one at a time, in a fixed order. (However, the option to use that button is still available.)

Admins can configure each menu item individually to control how, when, and where the Warehouse Management mobile app presents the pick line overview.

Configure menu items to show a list of all work lines

To set up a mobile device menu item to provide a pick line overview, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. Select or create a menu item that is related to picking work, and set the following values:

   - **Mode:** *Work*
   - **Use existing work:** *Yes*
   - **Directed by:** *User directed* or *System directed*

   For more information about how to create menu items and use the various settings that are available on the **Mobile device menu items** page, see Set up mobile devices for warehouse work.

3. On the **General** FastTab, configure the feature by setting the **Show work line list** field to one of the following values:

   - **Show only upon request** – Workers can choose to view the pick line list by selecting the **Skip to** button in the Warehouse Management mobile app.
   - **Show at the start of every pick** – Workers see the list every time that they start or finish a pick line. They can also view the list again by selecting the **Skip to** button in the Warehouse Management mobile app.

- **Show at the start of the first pick only** – Workers see the list every time that they start new picking work, but not after each line. They can also view the list again by selecting the **Skip to** button in the Warehouse Management mobile app.
- **Never show** – The standard **Skip** button appears in the Warehouse Management mobile app, and display of the work line list is turned off. The **Skip** button lets workers cycle through the lines one at a time, in a fixed order. They can also cycle through the list as many times as they require, until all lines have been processed.

4. On the Action Pane, select **Save**.

   If you set the **Show work line list** field to any value except *Never show*, the **Field list** button on the Action Pane becomes available.

5. On the Action Pane, select **Field list**.

6. On the **Field list** page, configure the information that the Warehouse Management mobile app shows for each line in the list.

   - The **Primary control** field is always set to *LineNum*. Therefore, each row in the list begins with a line number.
   - Use the remaining **Display field** fields to add up to seven additional display fields, as you require. In each **Display field** field, select the name of a work line field. Each line will then show a value for that field. The values will be shown in the order that you select here. You can leave some of the **Display field** fields blank if you don't require all seven values.

7. On the Action Pane, select **Save**, and then close the **Field list** page.

Last updated on 12/30/2025

Create a mobile device menu item for license plate consolidation

Article • 09/29/2021

This procedure shows you how to create a mobile device menu item for license plate consolidation work. This enables warehouse workers to consolidate items on one license plate with items on another license place within the same location. For example, they might use this if subsequent staging steps were the same on both work orders, so that the work only needs to be performed once for the merged items. You can use this procedure in demo data company USMF. The task would typically be carried out by a warehouse manager. This procedure is for a feature that was added in Dynamics 365 for Operations, version 1611.

1. Go to Warehouse management > Setup > Mobile device > Mobile device menu items.
2. Click New.
3. In the Menu item name field, type a value.
4. In the Title field, type a value.
5. In the Mode field, select 'Indirect'.
6. In the Activity code field, select 'Consolidate license plates'.

Feedback

Was this page helpful? Yes No

Provide product feedback

Mixed license plate receiving

Mixed license plate receiving enables you to build a license plate consisting of multiple items before you register it and create putaway work.

A license plate that consists of multiple items doesn't have to be split at the receiving dock for you to register each item.

When using an item-related flow to identify the source document lines, you can scan bar codes on the item control. If the bar code has a quantity and a unit of measure (UOM) configured on it, the item and quantity are automatically added to the mixed license plate, and you're returned to the screen to scan another item. This process lets you quickly scan all the items without having to make a confirmation at each step.

In the flow for mixed license plate receiving, you can display the list of items that are already scanned to the license plate. From this list, you can modify or correct the quantity of an item.

Where it applies

Mixed license plate receiving is a mobile device receiving flow to register and create work for multiple lines or items at the same time. This process is useful if you receive inbound license plates with multiple items.

The *deferred receiving* process uses both the specific *Mixed license plate receiving* mobile device flow and the data entities that are part of the **Mixed license plate receiving** page.

How to set up mixed license plate receiving

Mixed license plate receiving is set up as a mobile device menu item.

Create a new menu item with mode **work** that doesn't use existing work. Use one of the following work creation processes:

- Mixed license plate receiving
- Mixed license plate receiving and put away

Use the following options for identifying the source document lines:

- Purchase order item receiving

- Purchase order line receiving
- Transfer order item receiving
- Transfer order line receiving
- Inbound shipment order item receiving
- Inbound shipment order line receiving
- Return order receiving
- Load item receiving

> **Note**
>
> You can add multiple items to a license plate for all the source document line identification methods. However, although you can change the receiving order on a single license plate during the receiving process for all the other methods, you can't change the selected load for the *Load item receiving* method.

Deferred receiving processing

The deferred receiving process lets you, for example, optimize the process of receiving shipments that include order lines for many items and/or serial numbers. To use this process, assign a **Deferred receiving policy ID** value to your mobile device menu item. In a scenario that involves several thousand serial numbers, the system quickly records the serial numbers and stores the information in the *mixed license plate receiving* entities. It then processes on-hand inventory updates and creates work in the background, so that warehouse workers can continue to do other work. Deferred receiving processing is useful for warehouse processes where different workers are responsible for inbound receiving and putaway.

To run a deferred receiving process that quickly records incoming inventory and group-registers on-hand inventory against license plates, configure mobile device menu items that use one of the following **Work creation process** settings:

- *Purchase order item receiving*
- *Purchase order line receiving*
- *Transfer order item receiving*
- *Transfer order line receiving*
- *Inbound shipment order item receiving*
- *Inbound shipment order line receiving*
- *Load item receiving*
- *License plate receiving*

Prerequisites

Deferred receiving processing requires Supply Chain Management 10.0.38 or higher.

Create and assign deferred receiving policies

To use deferred receiving processing, first create a deferred receiving policy and assign it to the relevant mobile device menu items.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Deferred receiving policies**.
2. Identify the policy that you want to use, or create a new one. Each policy defines when a receiving flow uses deferred receiving processing. It also sets a few other options that control how the process works. Use the tooltips that are provided on the page to learn how to use each setting.
3. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
4. In the list pane, select the menu item that you want to set up. The selected menu item must use one of the previously listed work creation policies.
5. Set the **Deferred receiving policy ID** field to the policy that you identified in step 2.

Print license plate labels as part of the receiving process

For other mobile menu item receiving processes, the system can generate and print license plate labels when you run a deferred receiving process. You can enable this capability by using the settings established by the deferred receiving policy that's selected for each menu item.

Set the system to generate up to two license plate labels by using the following options:

- **Label printing on confirming receiving** – Generate and print labels immediately (before registration and work creation).
- **Label printing on deferred receiving async processing** – Generate and print labels based on the work data.

If you set the **Label printing on confirming receiving** option so that labels are printed before work is created, the labels don't include details such as the "to" location for the license plate. This information is omitted because it isn't available until after the system finishes running the deferred receiving asynchronous process.

> ⓘ **Note**

You can use the **Reprint label** mobile device menu item as a detour process to print the label that's generated by the **Label printing on deferred receiving async processing** option as part of the putaway process.

Deferred receiving asynchronous processing

Deferred asynchronous receiving processing automatically runs immediately after a mobile device menu item completes a *Receiving completed* process, which indicates that all the required *Mixed license plate receiving* data is available. You can monitor the status of the process on the **Mixed license plate receiving** page, where the **Mixed license plate receiving status** field shows one of the following values:

- *Building license plate*
- *Processing*
- *Error*
- *Received*

If an *Execute deferred receiving* batch job fails, the related **Mixed license plate receiving status** field is set to *Error*. To view the reason for the failure, select **Mixed license plate receiving batch jobs**, and then select **Batch job log**. Take the appropriate action to address the error (for example, by updating a line quantity or deleting a line). Then manually rerun the process by selecting **Complete license plate**.

Tip

If you want the page to show the **Receiving processing ID** value that's used for each background processing task, use the personalization tools to add the column to the **Mixed license plate receiving** grid.

Last updated on 07/01/2026

Set up a mobile device menu item for completing mixed license plates

Article • 05/20/2024

This article explains how to set up the *Complete mixed license plate* mobile device flow. This flow helps streamline warehouse operations by letting workers continue to process an incoming shipment immediately after they register its arrival.

Mixed license plate receiving helps warehouse managers achieve these goals:

- **Increase efficiency and accuracy.** Workers can receive various items onto a single license plate before they register and create putaway work.
- **Handle inbound license plates.** The inbound process is simplified because multiple lines/items can be handled simultaneously.
- **Optimize workflows.** The system can be configured to create work records for each registration. Alternatively, the license plate receiving and putaway process can be used as a single operation.

You can add this capability directly to the mobile device menu, so that workers can run the flow by itself. Alternatively, you can set up a *detour* to make the flow available immediately after a related flow (such as the flow for receiving a purchase order or sales return). A detour is a menu item that can be opened from a step in a main task. Detours let workers "park" the current task, perform another task, and then return to the original task without losing any information.

Prerequisites

To use the features describe in this article, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.40 or later.

Create a mobile device menu item for completing mixed license plates

Follow these steps to create a mobile device menu item that runs the *Complete mixed license plate* mobile device flow.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. On the Action Pane, select **New**.

3. On the header of the new record, set the following fields:

    - **Menu item name** – Enter a unique name to identify the menu item (for example, *Complete mixed license plate*).
    - **Title** – Enter the title that should be shown on the mobile device menu. The title doesn't have to be unique.
    - **Mode** – Select *Indirect*. This value is required because this flow isn't directly related to work.
    - **Use existing work** – Set this option to *No* because this flow creates new work when the movement is done.

4. On the **General** FastTab, set the following fields:

    - **Activity Code** – Select *Complete mixed license plate*.
    - **Barcode data policy** – Leave this field blank.

5. On the Action Pane, select **Save**.

6. Follow one of these steps:

    - To create a standalone menu item that workers can use to complete any existing license plate: Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**, and add the new menu item to an appropriate place in your mobile device menu structure.
    - To add the capability to complete a license plate as part of a larger, existing flow (such as the flow for sales returns): Add the new menu item as a detour of the existing flow. For instructions, see Configure detours for steps in mobile device menu items.

Related information

- For information about how to receive shipments by using mixed license plates, see Mixed license plate receiving.
- For information about to manage return orders, see Receive unannounced sales returns.

Feedback

Was this page helpful? Yes No

Provide product feedback

Configure Display older batches within warehouse on a mobile device

The **Display older batches within warehouse** configuration lets you display a list of locations with batches older than the current location of the work line. The list of locations that are displayed includes information about the older batches in the location with the expiration date and the physical inventory of each batch. You can choose to pick from a new location or to continue picking from the current location.

- Pick from a new location - If you select a new location to pick from, the current work line will be updated to use the new location and work will continue as usual with the new location. For the new location to be valid, it must have enough available quantity for the whole work line. If the required quantity is not available, the work line will not be updated, and the list will display.
- Continue picking from the current location - If you continue with the current work line location, the quantities for the work line will continue to be picked from the original location.

Where it applies

Display older batches within warehouse is configured on mobile device menu items and affects the pick for batch below items.

Set up Display older batches within warehouse

The **Display older batches within warehouse** configuration is available on mobile device menu items when the **Pick oldest batch** option is set to **Warn**.

- Under **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**, set **Use existing work** to **Yes** for the menu item, and select **Warn** in the **Pick oldest batch** field.

Last updated on 07/01/2026

Pick oldest batch on a mobile device

Article • 09/24/2024

You can configure mobile device menu items to allow workers to pick any batch, or to force or warn them to pick the oldest batch in their current location. The option effects pick work for batch below items.

To set this option for a mobile device menu item, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select or create a menu item that has **Use existing work** set to *Yes*.
3. On the **General** FastTab, set **Pick oldest batch** to one of the following values:

    - *None* – Workers aren't shown any messages and can pick any batch in their location.
    - *Warn* – When picking batch-controlled items, the app shows a list of the batch(es) with the oldest expiration date. If the location is license plate controlled, the app shows a list of license plates that have the oldest batch. If a worker chooses an unlisted license plate or batch, the app shows a warning that tells the worker that an older batch is available. The worker can choose to continue with the selected batch or license plate, even though it isn't the oldest, by confirming the action.
    - *Force* – Works the same way as a *Warn*, except the worker is only allowed to continue after selecting the oldest batch or license plate.

Feedback

Was this page helpful? Yes No

Provide product feedback

Query item data on a mobile device

Item inquiry mobile device menu items enable warehouse workers to view key inventory details for specific items, including location, available quantity, and reserved quantity. This feature improves visibility into warehouse stock and supports efficient decision-making during daily operations.

> **Note**
>
> This functionality doesn't display batch numbers or serial IDs. If an item is a product variant, the app shows information for the captured product dimensions, such as color or size.

Key capabilities of the item inquiry feature

Item inquiry mobile device menu items provide the following key capabilities:

- **Fast access to inventory details** – Workers can view where an item is stored and see how much stock is available or reserved.
- **Support for product variants** – If an item uses product dimensions such as color or size, the app shows data for the relevant variant based on the scanned or entered information.

Example scenario

A warehouse worker wants to verify stock levels of a particular item before picking. The worker selects the **Item Inquiry** menu item on the mobile app and then scans the item barcode. In response, the app displays the following information:

- Warehouse locations where the item is stored.
- Available and reserved quantities.
- Details based on the product variant (such as color or size values).

This information allows the worker to confirm inventory without interrupting active picking or receiving workflows.

Set up the mobile device menu item

To enable this functionality, you must configure the mobile device menu item in the system. Create the **Item inquiry** menu item by following these steps.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. Set the following values for the new menu item:
    - **Menu item name** – Enter an internal name for the menu item, such as *Item inquiry*.
    - **Title** – Enter the title for the menu item as workers should see it, such as *Item inquiry*.
    - **Mode** – Select *Indirect*.
4. On the **General** FastTab, set the following values:
    - **Activity code** – Select *Item inquiry*.
    - **Use process guide** – This setting is automatically set to *Yes* and can't be changed.
5. Go to **Warehouse management** > **Setup** > **Mobile device menu** and add the new menu item to each menu where workers should access it. Learn more in Set up mobile devices for warehouse work.

Last updated on 04/08/2026

Global mobile device parameters

06/04/2025

This article explains how to set up global warehouse management parameters that affect how the system interacts with mobile devices.

For more information about how to set up warehouse workers, see Manage warehouse worker. For more information about license plate handling on mobile devices, see License plate receiving via the Warehouse Management mobile app. For more information about the cycle counting processes, see Cycle counting and Cycle counting example scenarios.

Open the Warehouse management parameters page

To open the **Warehouse management parameters** page, go to **Warehouse management** > **Setup** > **Warehouse management parameters**. You can then set the fields that are related to mobile device work.

Global mobile device settings

The global mobile device settings are found on the **Mobile device** FastTab on the **General** tab of the **Warehouse management parameters** page. The following fields are available.

Expand table

Field	Description
Mobile device note type	Select the type of information that is shown to workers during sales order picking.
Scanned quantity limit	Enter the maximum item quantity that a worker can scan during each session by using a mobile device menu item that has a **Work creation process** value of *Adjustment in*. This field doesn't affect scans that are done by using any other type of menu item.
Use mobile device session logging	Set this option to *Yes* to keep a log of worker sign-in history. To view the log, go to **Work users sessions** > **Inquiries and reports** > **Mobile device logs** > **Work user sessions**.
Number of stored errors	Enter the maximum number of error records that the system should store. To view the error log, go to **Work users sessions** > **Inquiries and reports** > **Mobile device logs** > **Work user sessions**.

Field	Description
Default warehouse transfer journal	Select the journal that is used when workers use a mobile device to move products from one warehouse to another.
Allow purchase order line registration when in external review	Set this option to *Yes* if workers should be able to use a mobile device to register order lines for purchase orders that have an **Approval status** value of *In external review*. Set this option to *No* to prevent workers from registering lines for purchase orders that are in external review.
Enable RSAT support	The field enables the Warehouse Management mobile app task validator, which logs and validates each step that the app performs. Because this process can significantly slow down system performance, you should enable the validator only during testing. You should never enable it in a production environment.

Inventory dimension display options

On the **Mobile device menu item display inventory dimensions** FastTab, you can use the **Display tracking dimension control for last on-hand** field to specify how the mobile app shows the *tracking dimension below location* control, including *batch below number* and *serial below number*. The setting has no impact on the display of *tracking dimension above location*, such as *batch above number* or *serial above number*.

This feature requires Supply Chain Management version 10.0.44 or later.

The setting of the **Display tracking dimension control for last on-hand** field applies to the entire work order type. Therefore, it affects all mobile device menu items that are used to process the work order type. Currently, this feature is supported only for sales orders. No other work order types (such as transfer orders and purchase orders) are supported.

Two options are available for this field:

- *Display tracking dimension control* – The mobile app always shows the tracking dimension below location control (batch below number or serial below number) when the work order type is processed.
- *Hide tracking dimension control* – The mobile app always hides the tracking dimension below location control (batch below number or serial below number) when the work order type is processed.

Consider the following scenario:

- You have a batch below item that is named *whsbb*.
- A sales order is placed for a quantity of two pieces of item *whsbb*.
- The available on-hand inventory for item *whsbb* is two pieces (the last batch in stock).

If the **Display tracking dimension control for last on-hand** field is set to *Hide tracking dimension control* (the default value), the following behavior occurs:

- The system automatically assigns the batch number behind the scenes and then proceeds.
- The mobile app doesn't show batch number control.
- The worker can't confirm or edit the batch number in the mobile app, even if batch number confirmation is enabled.

If the field is set to *Display tracking dimension control*, the following behavior occurs:

- The mobile app shows an *empty* batch number control.
- The user must manually enter the batch number.

You can override the default setting for any individual mobile device menu item that supports it by going to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**. Learn more in Additional menu item options.

> **Note**
>
> This setting affects only menu items that process a supported work order type (currently only sales orders). Unsupported work order types (such as transfer orders and purchase orders) aren't affected.

Warehouse mobile devices workspace

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The **Warehouse mobile devices** workspace lets you monitor the health and license status of all handheld devices that are used in your warehouses.

The first time that a user signs in to the Warehouse Management mobile app on a new device, the system collects information about that device and adds it to the **Warehouse mobile devices** workspace. The system then continuously monitors the status of each device to make sure that it's running a supported operation system and the latest version of the mobile app. Devices remain listed in the workspace until they're manually deleted.

> **Note**
>
> If the Warehouse Management mobile app is reinstalled on the same device, you might have multiple device IDs for the same physical device. In this case, you must delete the inactive device IDs.

The **Warehouse mobile devices** workspace also helps you confirm that you're in compliance with licensing requirements by showing how many devices you're running and letting you assign a device license status to each device. However, the system doesn't enforce licensing requirements. Your organization is responsible for ensuring that you've acquired the correct number of licenses for your devices.

Prerequisites

This feature requires Microsoft Dynamics 365 Supply Chain Management version 10.0.37 or later.

Open the Warehouse mobile device workspace

To open the **Warehouse mobile devices** workspace, go to **Warehouse management** > **Workspaces** > **Warehouse mobile devices**.

Summary information

The **Summary** FastTab of the **Warehouse mobile devices** workspace provides tiles that show and link to the following information:

- **Device licenses used across all companies** – This tile shows the number of device licenses that are required to cover all the devices that are currently used across all companies (legal entities). If a single device is used across multiple companies, it's counted only once here. This total includes previously used devices that are now blocked. You can delete blocked devices to stop them from being counted here. Use this information to confirm how many licenses you need. This tile doesn't provide a drill-down link.
- **Device licenses** – This tile shows the number of device licenses that are required to cover all the devices that are currently used in the current company (legal entity). The devices that are counted here have a licensing state of *License required* or *Licensed*. Select this tile to view a list that provides details about each device in this category.
- **Devices supported** – This tile shows the number of devices that are running current versions of the Warehouse Mobile app on a fully supported operating system. Select this tile to view a list that provides details about each supported device.
- **Soon out of support devices** – This tile shows the number of devices that are still healthy and compliant, but will soon become unsupported (typically because of the mobile app or operating system version that they're running). To help keep your warehouses running smoothly, be sure to upgrade or replace these devices before it's too late. Select this tile to view a list that provides details about each device that will soon become unsupported.
- **Unsupported devices** – This tile shows the number of devices that are no longer supported and might not work correctly. These devices might soon become blocked without further warning from Microsoft. To help keep your warehouses running smoothly, be sure to upgrade or replace these devices as soon as possible. Select this tile to view a list that provides details about each device that's no longer unsupported.
- **Blocked devices** – This tile shows the number of devices that are now blocked by Microsoft. Any attempt to sign in to one of these devices will be rejected by the system. Microsoft controls which devices are blocked and will use this mechanism as a last resort to help protect system integrity and compliance.

Devices needing attention

The **Devices needing attention** FastTab of the **Warehouse mobile devices** workspace provides a quick overview of devices that require attention. It provides the following two tabs:

- **Devices to update** – Select this tab to view a list of devices that are running an old version of the Warehouse Management application. You should update each of these devices to the latest version to ensure optimal compliance, performance, and user

experience. For details about the latest version of the app and a change history, see What's new or changed in the Warehouse Management mobile app. For information about how to mass deploy installations and updates for the Warehouse Management mobile app, see Mass deploy the mobile app with user-based authentication.

- **Devices to replace** – Select this tab to view a list of devices that are running an operation system that's no longer supported. These devices might pose a risk to system integrity and compliance. Therefore, you should replace them as soon as possible. If the list includes devices that you're no longer using, you can delete them.

Manage device licenses

How devices are licensed

The following concepts apply to the way that the Warehouse Management mobile app is licensed:

- **Supply Chain Management user** – This type of user has a full Supply Chain Management user license. The user can access the web client and is also automatically licensed to use the Warehouse Management mobile app. No additional user license or device license is required. This type of user is usually an administrator, office worker, or warehouse manager instead of a warehouse worker.
- **Warehouse worker (shared work user)** – This type of user can sign in only by using the Warehouse Management mobile app (not the web client). Therefore, the user doesn't require a Supply Chain Management user license. This type of user is usually a warehouse worker. In Supply Chain Management, you typically set up users of this type by using a single work user record and then assigning multiple *user* accounts to it (one for each worker). This type of user is also referred to as a *shared work user*.
- **Warehouse Management mobile app user license** – A user license allows a specific warehouse worker to use a specific device. You can't view or manage user licenses in the **Warehouse mobile devices** workspace.
- **Warehouse Management mobile app device license** – A device license allows any number of shared work users (warehouse workers) to use a specific device. You can view and manage device licenses in the **Warehouse mobile devices** workspace.

For more information about how to set up mobile device user accounts, including both Supply Chain Management users and shared work users, see Mobile device user accounts.

For complete licensing details, see the Dynamics 365 Licensing Guide.

Update licensing information for a device

The **Warehouse mobile devices** workspace lets you view information about each device that's connected to the system. It also lets you view and manage the device license status for each device. To update licensing information for a device, follow these steps:

1. Open the **Warehouse mobile devices** workspace.
2. On the **Summary** FastTab, select a tile to open the relevant device list (such as **Device licenses** or **Devices supported**).
3. Find the target device in the list, and select it.
4. On the Action Pane, select **Change device licensing**, and then select a new license state to assign to the selected device. The following license states are used:
    - *Auto* – This value is assigned to new devices and persists until the system has enough information to determine the correct license state.
    - *License required* – The device needs a license that hasn't yet been acquired. Usually, this value is automatically assigned the first time that a shared work user signs into the device.
    - *Licensed* – The device is licensed. Select this value to indicate that you've acquired the required licenses for the device.
    - *Block share usage* – The device doesn't require a device license. Devices in this license state can be used only by Supply Chain Management users (who don't require a user or device license to use the mobile app). This value prevents shared work users from signing in to the device.

Rename a device

Devices are primarily identified using a unique device ID, which is, due to privacy reasons, unrelated to any other device identifier, such as a device serial number or device name. To make it easier to identify which device ID correlates to which physical device, you can give each device a friendly name in addition to the device ID. To rename a device, follow these steps:

1. Open the **Warehouse mobile devices** workspace.
2. On the **Summary** FastTab, select a tile to open the relevant device list (such as **Devices supported**). Alternatively, go to **Warehouse management** > **Inquiries and reports** > **Mobile device logs** > **Mobile devices**.
3. Find the target device in the list, and select it.
4. On the Action Pane, select **Edit**.
5. Change the **Friendly name** field.

> ⓘ **Note**

The Warehouse Management mobile app can also provide a hardware identification of the device in some cases. If so, this information is shown in the **Hardware identification** field and a friendly name is generated using the device brand, model, and hardware identification.

Remove a device

After devices are discovered, they remain listed in the system until they're manually deleted. When you delete a device, its license is released and can be used with other devices. Follow these steps to remove a device from the system.

1. Open the **Warehouse mobile devices** workspace.
2. On the **Summary** FastTab, select a tile to open the relevant device list (such as **Blocked devices** or **Unsupported devices**).
3. Find the target device in the list, and select it.
4. On the Action Pane, select **Delete**.

Last updated on 01/09/2026

Mobile device user settings

09/17/2025

The new Warehouse Management mobile app has a set of app-specific settings that help tailor the user experience. Because the app can be used on devices of different screen sizes and configurations (such as tablet, phone, or arm-held), it can be useful to centrally manage these settings from Microsoft Dynamics 365 Supply Chain Management.

The *mobile device user settings* feature lets you define global user settings that will be used on all devices. You can also define more granular user settings for individual device brands, device models, and/or workers. When a worker signs in to the Warehouse Management mobile app, the app fetches and applies the most specific settings profile that is stored in Supply Chain Management for the matching brand, device, and/or user ID.

This feature can help workers get started more quickly whenever they begin to use a new device. Here are some examples:

- Admins can establish a standard set of preferences that work best for devices from a specific manufacturer and/or for specific device models. Therefore, workers can get started with a new device without necessarily having to change the settings.
- If your company has a pool of identical devices that are shared among workers, workers will see their preferred setup every time that they sign in, even if they have never used the specific physical device that they selected on a given day.
- In Supply Chain Management, admins can view and edit all stored settings, even for individual workers. This capability can help them troubleshoot or fine-tune new features.

Create and manage user settings

Use the **Mobile device user settings** page to create, view, and manage settings profiles at all levels of granularity. The first time that a worker edits their user settings on a new device, a new profile is automatically added on this page, if it doesn't already exist. That profile is then updated every time that the worker makes a change.

You can also define a settings profile that applies to all device brands, device models, and/or workers. You can then increase the granularity by applying more specific settings for individual brands, models, and/or workers.

Follow these steps to create and manage user settings for your mobile devices.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device user settings**.

2. Select an existing user settings profile in the list pane to open its record. Alternatively, select **New** on the Action Pane to create a new profile.

   Each profile in the list pane is labeled to indicate the brand, model, and/or user ID that the profile applies to. More general profiles have a value of *All* for some or all of these characteristics.

3. In the header section of the record for the new or selected settings profile, set the following fields:

   - **Device brand name** – Select the device brand name that the profile should apply to. If the profile should apply to all brands, leave this field blank. The list of values includes all the brands that have been defined in your system. For information about how to define the list of brands, see the next section.
   - **Device model ID** – Select the device model that the profile should apply to. If the profile should apply to all models of the selected brand, leave this field blank. The list of values includes all the models that have been defined for the brand that is selected in the **Device brand name** field. For information about how to define the list of models for each brand, see the next section.
   - **User ID** – Select the user ID of the warehouse worker that the setting profile should apply to. If the profile should apply to all workers, leave this field blank.

4. On the **General** FastTab, set the user settings values that you want to apply. These settings are the same as those offered in the app itself, but the app also provides some extra settings that aren't available here in the web client (including **Server request timeout** and **Configuration mode**). Learn more about the available settings in User settings, color themes, and sound themes.

Set local user settings in the Warehouse Management mobile app

Workers can set their own local user settings in the Warehouse Management mobile app. These settings override the default settings set by the admin. For example, a worker might prefer to use a different theme, sound level, or text scale. When a worker signs in to the Warehouse Management mobile app, the app fetches and applies the most specific settings profile that is stored in Supply Chain Management for the matching brand, device, and user ID. When a worker changes their local settings, the app updates the settings profile that is stored in Supply Chain Management for the matching brand, device, and user ID. If no matching profile exists, a new one is created.

To set local user settings in the Warehouse Management mobile app, sign in to the app and then select the **Settings** button, which is the gear icon in the top-right corner of the screen.

Learn more in User settings, color themes, and sound themes.

Create and manage mobile device brands

Use the **Mobile device brands** page to view, create, and manage the device brands and models that can be used with your settings profiles. The mobile app automatically fetches and submits the local brand name and model ID the first time that a worker edits their settings on a given device. Therefore, most of these records will usually be automatically generated. However, you can also manage all the records on this page. For example, you can give more useful descriptions to each brand and model to help distinguish similar or cryptic model IDs.

Follow these steps to create and manage your mobile device brands and models.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device brands**.
2. In the list pane, select a mobile device brand to open its record. Alternatively, select **New** on the Action Pane to create a new brand.
3. In the header section of the record for the new or selected device brand, set the following fields:
    - **Device brand name** – The brand name of the device, such as *Microsoft Corporation*.
    - **Description** – You can enter a description to help distinguish brand names.
4. The **Mobile device models** FastTab shows all the models for the selected device brand. Use the buttons on the toolbar to add rows to the grid or remove rows. For each row, you can set the following fields:
    - **Device model ID** – The device model ID, such as *Surface Book 2*.
    - **Description** – You can enter a description to help distinguish model IDs.

Related information

- Install the Warehouse Management mobile app
- Assign step icons and titles for the Warehouse Management mobile app

Warehouse app event processing

10/03/2025

Batch jobs running in Supply Chain Management can use data from a queue for processing events issued by the Warehouse Management mobile app to react as needed to the signaled events. This feature adds relevant events to the queue in response to certain types of actions taken by workers using the app. An example is when using the *Create and process transfer orders from the warehouse app* feature, the transfer order header and lines get created and updated in the back end when the system is running the **Process warehouse app events** batch job.

Set up a batch job to process warehouse app events

Process warehouse app events

Set up a scheduled batch job to process the warehouse app events for the transfer order creation and line updates.

1. Go to **Warehouse management** > **Periodic tasks** > **Process warehouse app events**.
2. The Process warehouse app events dialog box opens. Expand the **Run in background** FastTab and set **Batch processing** to **Yes**.
3. On the **Run in the background** FastTab, select **Recurrence**.
4. The **Define recurrence** dialog box opens. Set the run schedule as needed for your business.
5. Select **OK** to return to the **Process warehouse app events** dialog box.
6. Select **OK** in the **Process warehouse app events** dialog box to add the batch job to the batch queue.

Query warehouse app events

You can view the event queue and events messages generated by the Warehouse Management mobile app by going to **Warehouse management** > **Enquiries and reports** > **Mobile device logs** > **Warehouse app events**.

The standard event queue process

The warehouse app events queue will typically be used with the following described flow:

1. An event gets added to the queue with an event message. The new message initially has an Event state of **Waiting**, which means that the **Process warehouse app events** batch job will not pick up and process this message.
2. As soon as the message state is updated to **Queued**, the **Process warehouse app** events batch job will pick up and process the event.
3. The batch job updates the event states to either **Completed** or **Failed**, depending on whether the requested process was possible.
4. When all the related event messages are **Completed**, the event is deleted from the queue.

For a detailed example, see Create transfer order from warehouse app process.

Handle event errors

As part of the warehouse event processing, the requested message activity may not receive processes from the batch job. In this case, the event message will change to **Failed**. You can use the **Batch log** information to learn why and take needed action to correct any problems.

For a detailed example, see Create transfer order from warehouse app.

To reset a failed warehouse app event message:

1. Go to **Warehouse management** > **Enquiries and reports** > **Mobile device logs** > **Warehouse app events**.
2. On the **Warehouse app event messages** FastTab, find and select a relevant event that is showing **Failed** in the **Event state** column.
3. Select **Reset** from the **Warehouse app event messages** toolbar.
4. Continue working until all relevant messages are reset.

You can also remove a **Failed** event message by using the **Delete** option on the **Warehouse app event messages** toolbar.

Inspect details of active Warehouse Management mobile app sessions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The whole time that a warehouse worker user is using the Warehouse Management mobile app, the system maintains a *work user session* for that user. You can view the details of each active work user session by going to **Warehouse management** > **Inquiries and reports** > **Mobile device logs** > **Work user sessions**. For each session, the **Work user sessions** page shows associated **User name**, **User ID**, **Login date and time**, and **Last user action date and time** values, together with a detailed history of XML communication messages and logged errors.

> **Note**
>
> If no historical data appears on the **Work user sessions** page, a cleanup job, such as Work user session log cleanup, might be running.

The XML messages can be useful if, for example, you must look up mobile device step IDs when you're setting up a detour process. They can also be useful if you must look up field names to use in your barcode data policy definitions. Those fields name must exactly match the field names that are used in the relevant XML messages.

> **Important**
>
> To view the correct XML so that you can find information such as field names and step IDs, you must perform work by using the actual Warehouse Management mobile app, not an alternative tool such as the browser-based emulator.

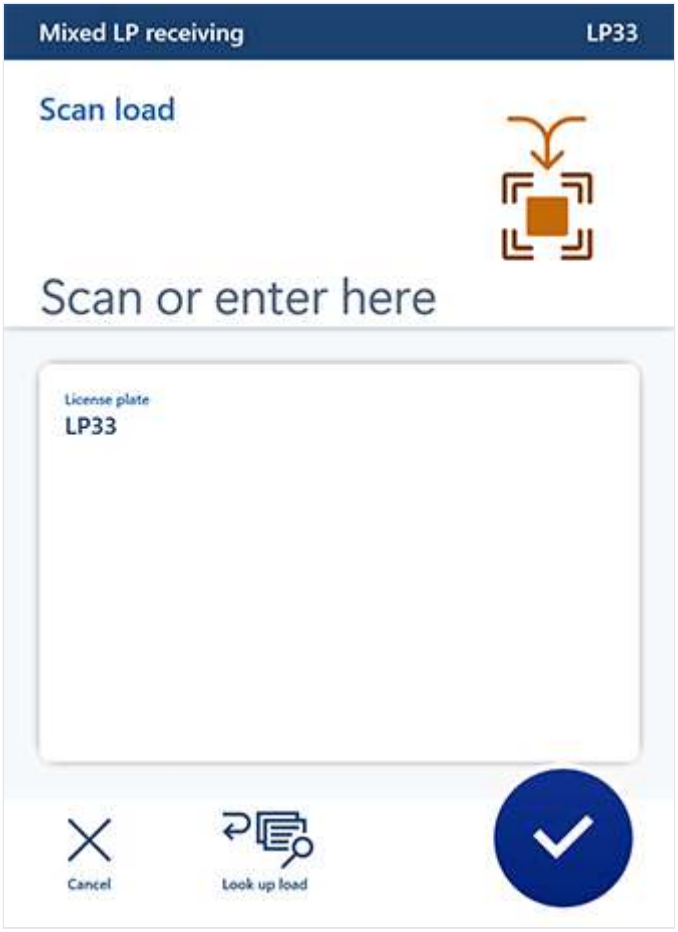
The following example shows XML content that was copied from a *Mixed license plate receiving* process. Here, the Warehouse Management mobile app is currently processing a step where `Step Id="LoadId"`. This step includes a control where `DisplayArea="PrimaryInputArea"` (the display area where controls that are awaiting input or confirmation are typically found) and `name="LoadId"` (the field name that's associated with the control).

XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<ParentNode>
    <Controls PagePattern="Default" PageTitle="Mixed LP receiving"
MenuItemName="Mixed LP receiving">
        <Control controlType="label" name="MixedLPReceiving" label="Mixed LP
receiving" newLine="1" data="" type="Undefined" length="-1" error="0"
defaultButton="0" enabled="1" selected="" color="#000000" Status="1"
NumDecimals="-1" DisplayArea="SubHeaderArea" PreferredInputMode=""
PreferredInputType="" DisplayPriority="0" DisplaySubPriority="0" DataSequence="3"
AttachedTo="" InstructionControl="" Footer1="" Footer2="" InputType="16806" />
        <Control controlType="text" name="LicensePlateId" label="License plate"
newLine="1" data="LP33" type="String" length="25" error="0" defaultButton="0"
enabled="0" selected="" color="#000000" Status="1" NumDecimals="-1"
DisplayArea="InfoAndSecondaryInputArea" PreferredInputMode="Scanning"
PreferredInputType="Alpha" DisplayPriority="50" DisplaySubPriority="22"
DataSequence="4" AttachedTo="" InstructionControl="" Footer1="" Footer2=""
InputType="2694" />
        <Control controlType="text" name="LoadId" label="Load" newLine="1" data=""
type="String" length="20" error="0" defaultButton="0" enabled="1" selected=""
color="#000000" Status="1" NumDecimals="-1" DisplayArea="PrimaryInputArea"
PreferredInputMode="Scanning" PreferredInputType="Alpha" DisplayPriority="70"
DisplaySubPriority="11" DataSequence="5" AttachedTo="" InstructionControl=""
Footer1="" Footer2="" InputType="14265" />
        <Control controlType="button" name="OK" label="OK" newLine="1" data=""
Icon="USMF|ActionIcon|OK" type="Undefined" length="-1" error="0" defaultButton="1"
enabled="1" selected="" color="#000000" Status="1" NumDecimals="-1"
DisplayArea="PrimaryActionArea" PreferredInputMode="" PreferredInputType=""
DisplayPriority="0" DisplaySubPriority="0" DataSequence="6" AttachedTo=""
InstructionControl="" Footer1="" Footer2="" InputType="16806" />
        <Control controlType="button" name="Cancel" label="Cancel" newLine="1"
data="" Icon="USMF|ActionIcon|Cancel" type="Undefined" length="-1" error="0"
defaultButton="0" enabled="1" selected="" color="#000000" Status="1"
NumDecimals="-1" DisplayArea="" PreferredInputMode="" PreferredInputType=""
DisplayPriority="0" DisplaySubPriority="0" DataSequence="7" AttachedTo=""
InstructionControl="" Footer1="" Footer2="" InputType="16806" />
        <Control controlType="detourButton" name="Look up load" label="Look up load"
newLine="1" data="" Icon="USMF|MenuIcon|GenericDataInquiry" type="16806" length="0"
error="0" defaultButton="0" enabled="1" selected="" color="0" Status="0"
NumDecimals="-1" DisplayArea="" PreferredInputMode="" PreferredInputType=""
DisplayPriority="0" DisplaySubPriority="0" DataSequence="8" AttachedTo=""
InstructionControl="" Footer1="" Footer2="" InputType="0" />
    </Controls>
    <Step Id="LoadId" Icon="USMF|StepIcon|LoadID" Title="Scan load" />
    <Auth userId="51" userGUID="{00aa00aa-bb11-cc22-dd33-44ee44ee44ee}" sessionId="
{XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX}" />
    <UserCulture>en-us</UserCulture>
    <UserDateCulture>en-us</UserDateCulture>
    <OperationalInsightsInstrumentationKey />
    <ServerAadTenantId>ffffaaaa-5555-bbbb-6666-cccc7777dddd</ServerAadTenantId>
    <ServerEnvironmentId /><ServerAzureRegion /><ServerVersion />
<BatchFlightsEnabled />
    <Device DeviceId="{C1DEE34C-69FA-44DD-BE17-2655931016CC}" /><ServerActivity
```

```
ServerActivityId="{B2AAD7A2-7674-0006-E259-ABB27476D901}" />
</ParentNode>
```

The following illustration shows what this step looks like in the Warehouse Management mobile app.



Last updated on 05/07/2025

Process guide framework

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides information about the process guide framework for developers who are extending the warehouse mobile processes in X++. The warehouse mobile processes are extensible because the processes are broken into small steps. The business logic and user interface building of each step is extracted into individual classes, which allows for extensibility.

Overview of the existing design

The warehouse mobile execution flows are exposed through a single custom service endpoint. The request arrives from the mobile app in the form of an XML string, which contains the metadata of the user interface presented in the mobile app, and the values that the user enters.

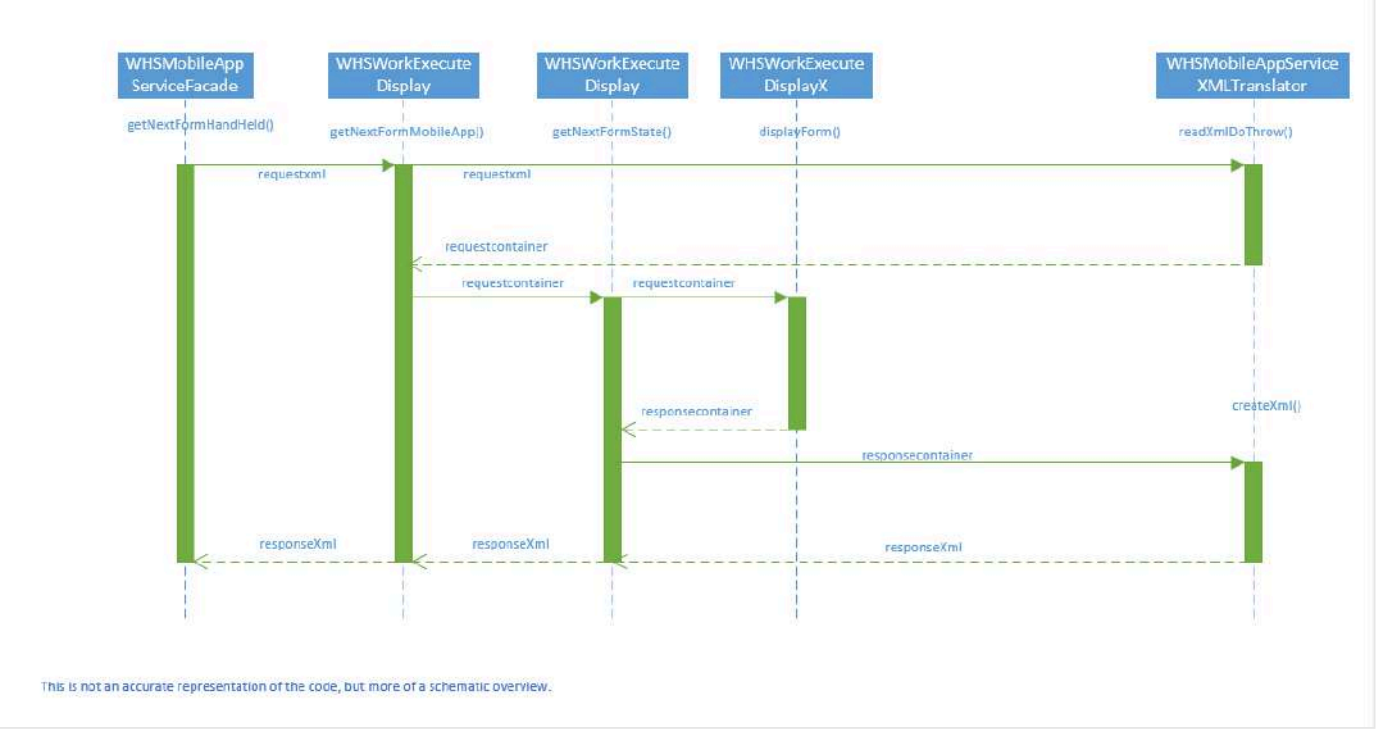
When the request is received, the first step is to deserialize the XML. The `WHSMobileAppServiceXMLTranslator` class converts the XML into a container, which holds both the control information and session information.

Next, information in the container is used to determine which warehouse process the user is working on or about to start. This process is represented by the `WHSWorkExecuteMode` enumeration. Based on this information, a derived class of `WHSWorkExecuteDisplay` is instantiated. The `displayform()` method is invoked, which performs the following actions:

- Processes the data from the user (delegated to the `WHSRFControlData` class, but some processes implement specific logic by overriding the `processControl()` method).
- Executes business logic.
- Increments the step.
- Builds the container representing the new user interface (typically in a `build…()` method).

The container is then returned to the translator, which then serializes the XML and sends it back as a response to the mobile device.

The following sequence diagram shows an overview of the execution flow. The diagram is more of a schematic overview and isn't a 1:1 representation of the actual code.



Reason for the redesign

The design offers a simple framework for building processes used in mobile flows. However, the `displayform()` methods take on multiple responsibilities. They delegate these responsibilities to other methods and classes, but without concrete class responsibilities, this delegation is inconsistent across classes. As the number of supported scenarios grows organically, some of these classes become complex. In addition, some of those classes/methods are overridden and reused in multiple modes. The result is long methods with high cyclomatic complexity. These methods have posed maintenance issues in the past. Fixing bugs in these methods is risky and regression prone. For example, the `processWorkLine()` method in the `WhsWorkExecuteDisplay` class is referred to from multiple processes (basically, anywhere where work execution is performed).

To make these methods extensible, you could split the `displayForm` methods into smaller methods and introduce extensibility points. However, because of the scenario matrix, it would be challenging for partners to write extensions and validate against regressions. Not only that, but because of the lack of structured responsibility distribution noted previously, the code would keep growing in unpredictable ways over time, posing challenges in building quality extensions.

As a result, the redesign is the sustainable option, with a goal to have clearly defined classes with independent responsibilities. A class should have one responsibility, one reason to change, and

one reason to be extended.

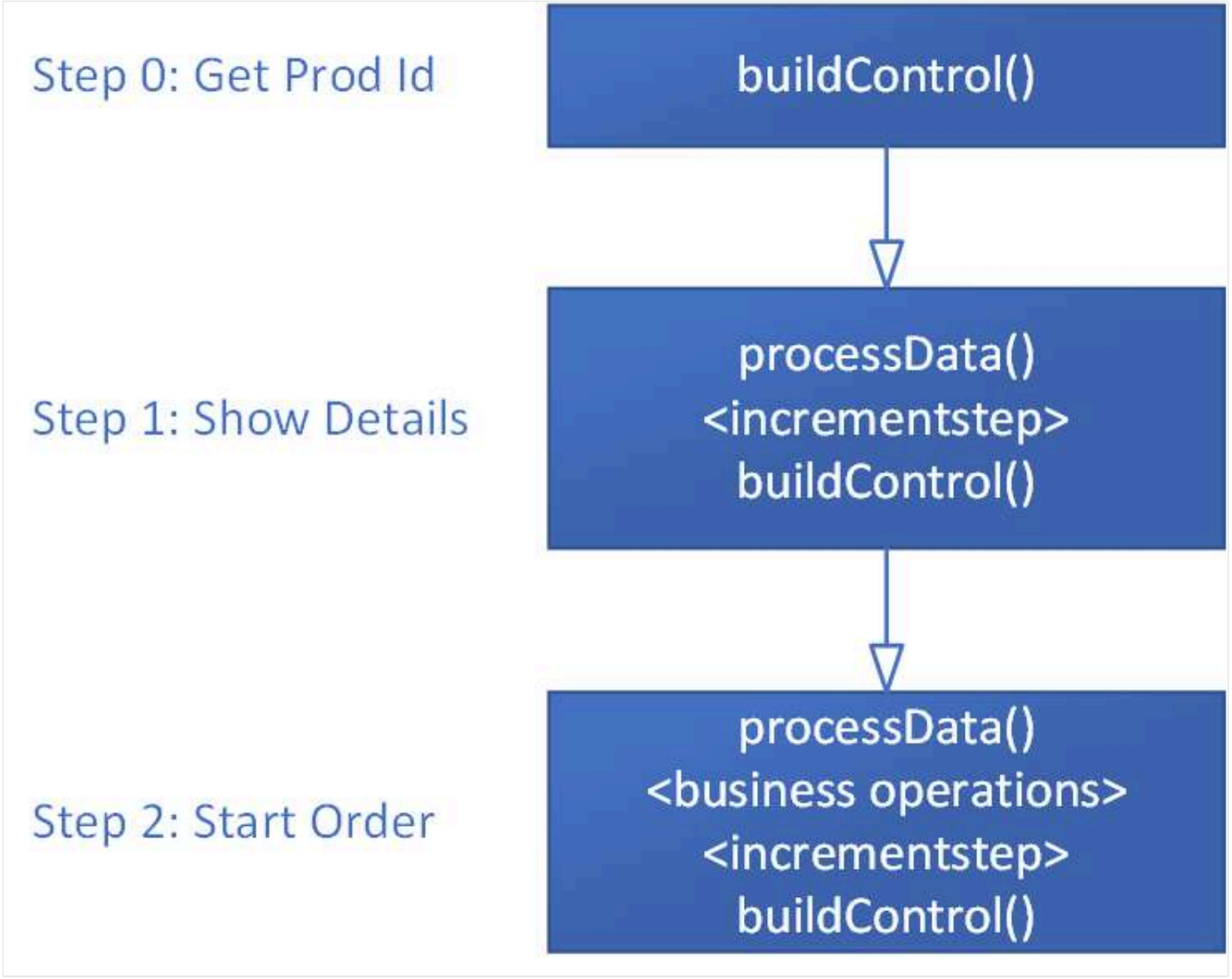
Design overview

The redesigned framework uses two core principles: divide the execution flow into individual components with well-defined responsibilities, and include well-defined extension points in each component.

The new framework is named `ProcessGuide`. The aim of these classes is to guide a user through a business process, as opposed to the rich client, which is a form-based experience where the user has more flexibility in how they interact with the data or in which order they perform tasks.

> **Note**
>
> One notable detail is the deliberate omission of the `WHS` prefix. While the mobile processes were initially introduced for warehousing, they now support various production and inventory management processes. As a result, the warehouse reference was excluded in the name of the framework.

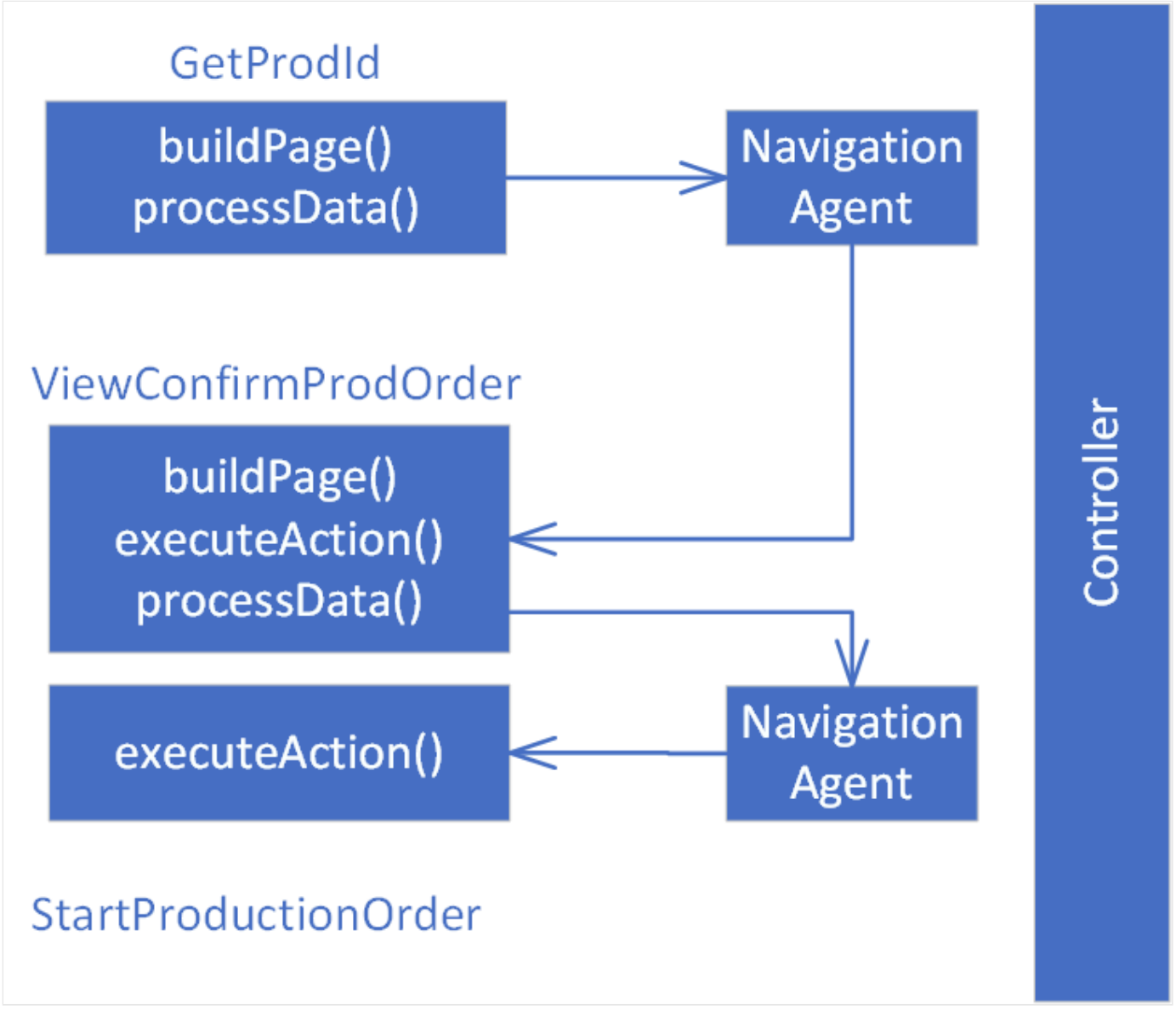
To identify the components, look at the Production Start process (`WhsWorkExecuteDisplayProdStart` class). Here's a schematic of the process.



Looking at the control flow, you need the following components:

- A controller to manage the entire business process.
- A step responsible for execution of a step in the process.
- A data processor for processing the data in a step.
- A page builder responsible for building the user interface for a step.
- A navigation agent responsible for step transition.
- A class responsible for executing the business process.

In the process flow diagram, if you begin at step 1 and start processing the data from the previous step, and then end with building a UI, data continues to be processed in the next step. This approach introduces a tight coupling between consecutive steps. As a result, the new high-level schematic looks like the following:



The key components in the redesigned process are as follows:

- `ProcessGuideController` - This class orchestrates the overall execution of the business process. It defines the factories that instantiate the step and the navigation agent, which later constitutes the process execution, and the clean-up logic for cancellation or exiting the process.
- `ProcessGuideStep` - This class represents one single step in the business process. It contains a definition of the factories that instantiate a page builder, actions, and data processors. It's responsible for invoking them in the correct sequence.
- `ProcessGuideNavigationAgent` - This class is responsible for navigation between the steps. When a step is completed, the navigation agent defines the next step and passes any parameters that the previous step might need to communicate to the next one.
- `ProcessGuidePageBuilder` - This class is responsible for instantiating the user interface.
- `ProcessGuideAction` - This class represents an action, shown as a button to the user.

- `ProcessGuideDataProcessor` - This class is responsible for processing the user entered data in a field.

Execution flow

The starting point of the execution flow remains unchanged. So, the request still arrives through the same endpoints, followed by deserializing the XML into the container. This container is then passed to `getNextFormState()`.

There are three important classes to note:

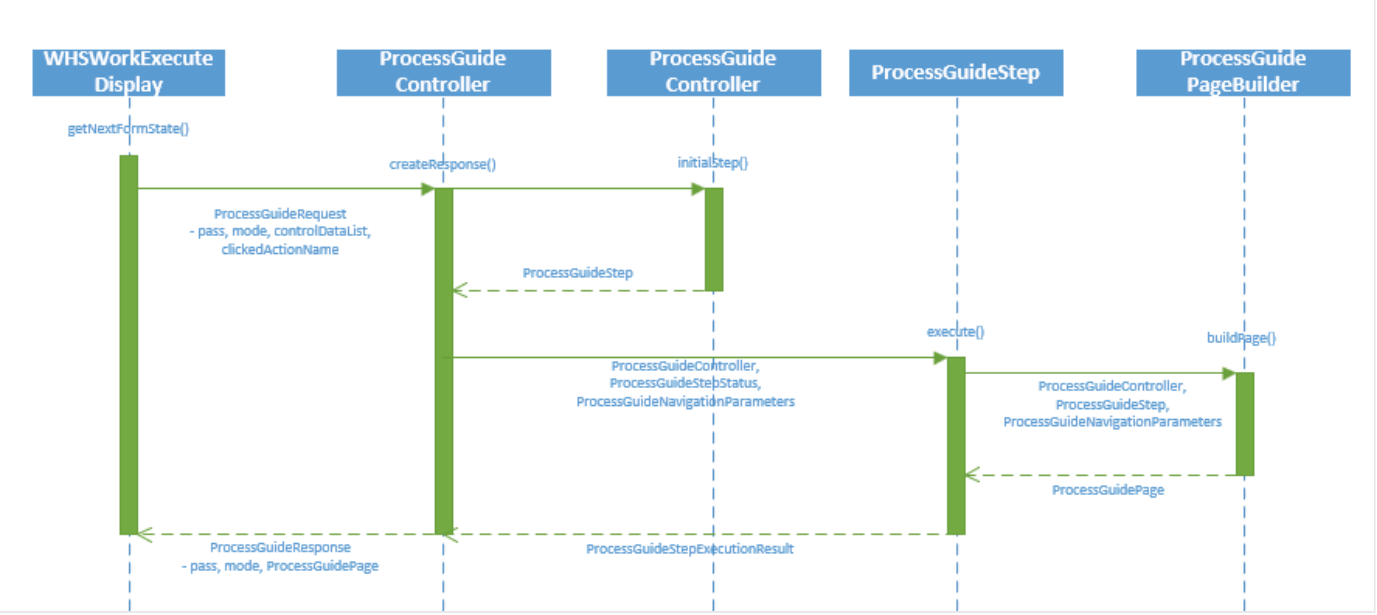
- `ProcessGuideSessionState` – This contains the session state information (such as the mode, pass, controller, and the step being executed).
- `ProcessGuidePage` – This contains a strongly typed representation of the user interface metadata.
- `ProcessGuideRequest` – This contains the above two as members and is a strongly typed representation of the request received from the mobile device.

These classes are created using the container information (both state and user entered control data). This provides a type-safe way to access and manipulate the values. Compared to repeated access of the container during the process, this provides benefit both in terms of readability and performance.

The session state information is used to instantiate the correct `ProcessGuideController` class. Once instantiated, the `createResponse()` method in the `ProcessGuideController` class is invoked. This method is the entry point to the process guide logic, and after execution, comes back with the response (represented in the `ProcessGuideResponse` class). The response is then converted back to the container and handed back to the legacy logic, which then serializes it to the XML and sends the response back to the mobile device.

Next, the controller needs to find the next step to execute. If this is the start of a new process, the controller will call `initialStep()` to get the first step in the process. After that, it would call `execute()` method in the `ProcessGuideStep`. This method would then instantiate a `ProcessGuidePageBuilder` class and call `buildPage()`, which would return with a `ProcessGuidePage` object, which is a virtual representation of the user interface to be presented to the user. The step would then send the result back to the controller, which would then save the current session state and then return the result back to `getNextFormState()` in the form of the `ProcessGuideResponse` class. Thereafter, the response is converted back to the container, and later serializes to XML and sends back the response to the mobile device.

The following sequence diagram explains this control flow. This is the most common control flow, simplified for explaining the design.



When the user takes an action on the mobile device by selecting a button (or scanning a value – which typically triggers the default action) – the request arrives at the `createResponse()` method in the `ProcessGuideController` class through the same route. This time, however, the controller knows from the session state information which step the user is in. Accordingly, it instantiates the appropriate `ProcessGuideStep` class and invokes the execute method. The `ProcessGuideStep`, in turn, reads the action name invoked by the user and then instantiates the appropriate `ProcessGuideAction` class and calls `execute()`.

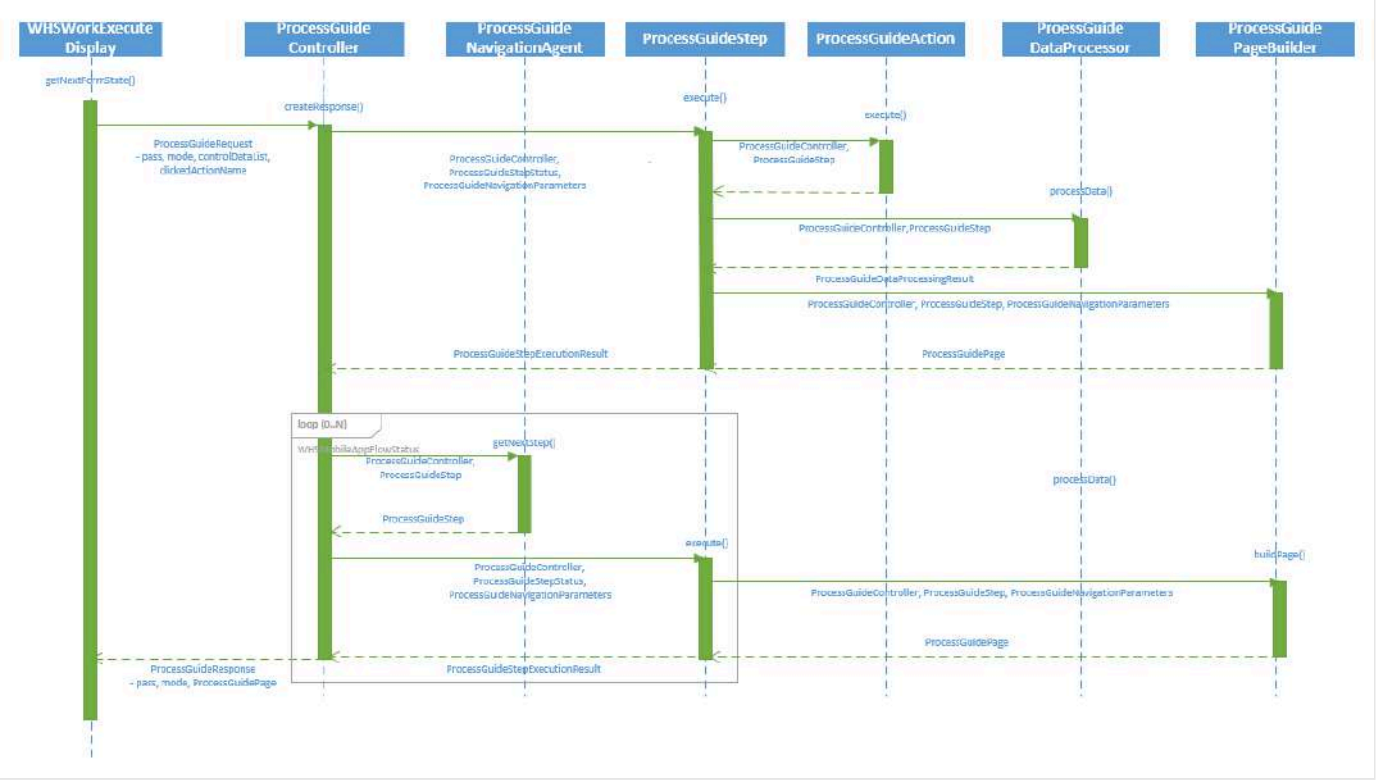
The `ProcessGuideAction` class is responsible for executing the specific action, however there are two notable exceptions.

The first one is the `ProcessGuideOKAction` class. This action implies that the user wants to confirm and move forward in the process. In accordance to that – this method actually does a callback to the ProcessGuideStep class, which means that the step invokes `processData()` in `ProcessGuideDataProcessor`. This processes the data that the user has entered, and then updates the state of the step and sends the result back to the controller. Depending on the outcome of the processor, the step invokes the page builder to build the appropriate user interface or sets the status of the step as completed. This is reflected in the top half of the following sequence diagram.

The other exception is the cancellation action, implemented in the `ProcessGuideCancelResetProcessAction` and `ProcessGuideCancelExitProcessAction` classes. These actions represent an intent to cancel the process and go back to either the start of the process or exit the process altogether. Similar to the **OK** action, these actions also perform a callback to the

step, which signals the intent to the `ProcessGuideController`. The controller then performs the necessary cleanup of state variables and either moves control to the initial step in the process or terminates the process altogether.

After the step is completed, if the status of the step is set to *Completed*, then the controller instantiates the `ProcessGuideNavigationAgent`, which returns the name of the next step. The controller then instantiates this step and invokes the `execute()` method – and the cycle continues. Most commonly, the new step invokes the corresponding `ProcessGuidePageBuilder` to build the user interface for the next screen to be presented to the user, which is then sent back. This flow is depicted in the lower half of the following sequence diagram.

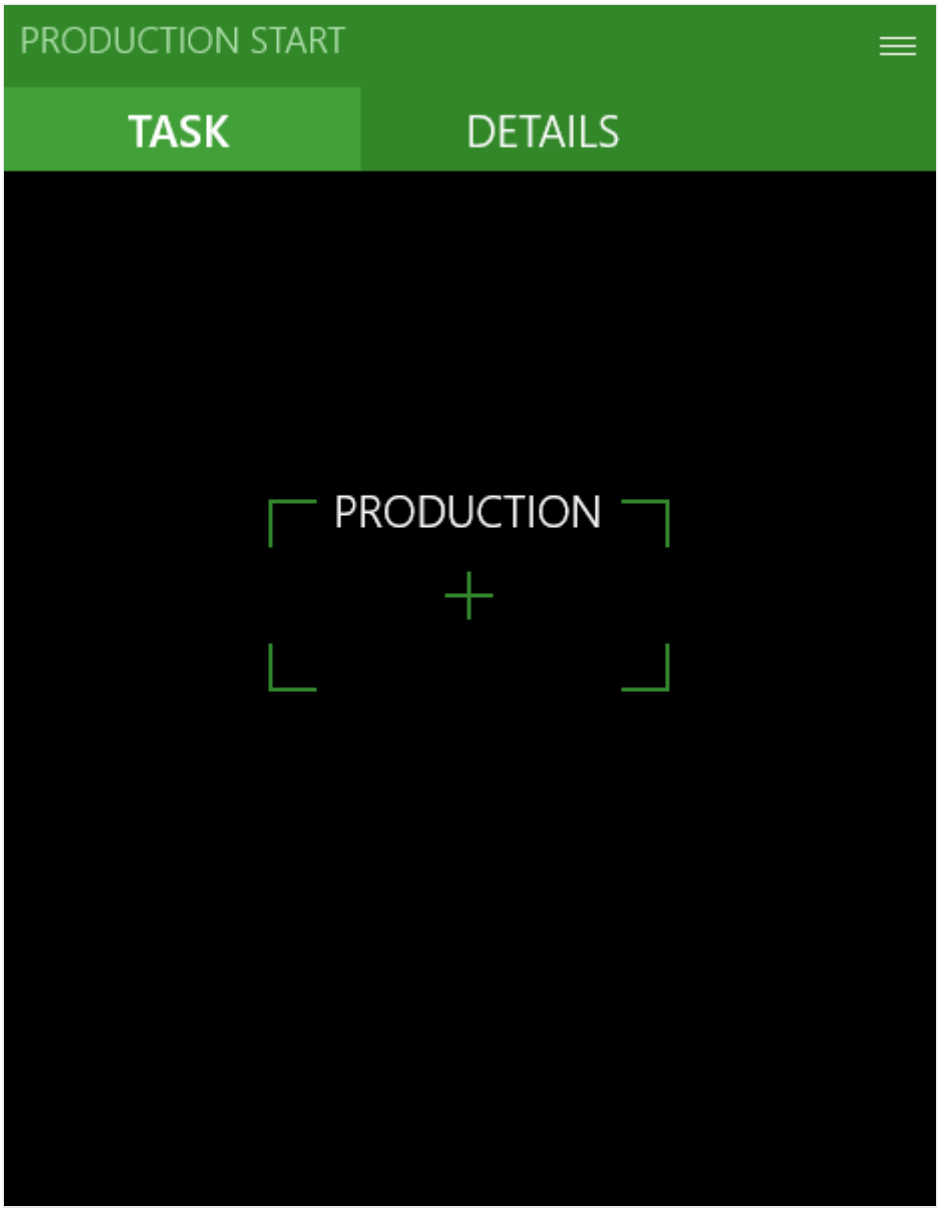


Building a new process using the ProcessGuide framework

The best way to explain the control flow is by using an example that exists in the application – the Production Start process.

Overview of the production start process

Let's start by understanding the process flow. In the first step, the user is prompted for production order ID.



When the user enters the production order ID, the order number is validated. Some of the validations that run are based on whether the order is in the same warehouse as the user is signed in to, and the status of the order. If the validation fails, the user sees an error message. If the validation succeeds, the user sees details of the production order and item.

The user can either cancel to go back to the start of the process or select **OK** to confirm. In the latter case, the production order is set to *Started* status, the corresponding journals are posted, the control moves back to the first step, and the *Work Completed* message is shown to the user.

Creating the controller

The first step in building the business process is creating the controller class, extending from the `ProcessGuideController` abstract class which implements the default behaviors of a controller. The new class name is `ProdProcessGuideProductionStartController` and decorated with the `WHSWorkExecuteMode` value of `StartProdOrder`. The same `SysExtension` based instantiation already

used in the `WHSWorkExecuteDisplay` classes is used here, which helps instantiate the controller when the user executes a menu item for this mode.

X++

```
[WHSWorkExecuteMode(WHSWorkExecuteMode::StartProdOrder)]
public class ProdProcessGuideProductionStartController extends ProcessGuideController
```

> ⓘ Note
>
> The naming pattern of the class is `\<FunctionalArea\>ProcessGuide\<Businessprocessname\>Controller`. This is the pattern used for the controller classes and to extend to other classes

Building the first step

Next, define the first step by creating the `ProdProcessGuidePromptProductionIdStep` class, which extends `ProcessGuideStep`.

A step factory instantiates the class. The `ProcessGuideController` base class invokes the factory. The default implementation of the factory instantiates the step based on name. To instantiate `ProdProcessGuidePromptProductionIdStep` as the first step in the controller, complete the following two tasks:

- Decorate the `ProdProcessGuidePromptProductionIdStep` class with a `ProcessGuideStepName` attribute.

  X++

  ```
  [ProcessGuideStepName(classStr(ProdProcessGuidePromptProductionIdStep))] public
  class ProdProcessGuidePromptProductionIdStep extends ProcessGuideStep
  ```

- In the controller class, implement the abstract method `initialStepName()` to return the step name.

  X++

  ```
  protected final ProcessGuideStepName initialStepName()
  {
      return classStr(ProdProcessGuidePromptProductionIdStep);
  }
  ```

> **Note**
>
> The value in the `ProcessGuideStepName` attribute doesn't need to exactly match the class name as shown in the preceding code. However, implementing this attribute allows for uniformity and type-safety around cross-references when using the class. Use this naming convention.
>
> The `ProcessGuideStepName` based instantiation of the step is implemented in the `ProcessGuideStepDefaultFactory` class. In the rare case that you want a different strategy for instantiating the step, complete the following tasks:
>
> - Create a new factory class that inherits from `ProcessGuidStepAbstractFactory`.
> - Optionally, create a new parameter class that implements the `ProcessGuideIStepCreationParameters` interface, containing the parameters the factory needs.
> - In your controller class, override the `stepFactory()` and `stepCreationParameters()` methods to return the preceding factory and parameters.

Next, implement the functionality of the `ProdProcessGuidePromptProductionIdStep` class. Implement the logic for building the user interface, processing the user-entered data, and determining when the step is complete.

Building the user interface for the first step

Build the user interface by creating a class that inherits from the `ProcessGuidePageBuilder` abstract class. For this step, name the class to represent what it does – `ProdProcessGuidePromptProductionIdPageBuilder`.

The instantiation mechanism of the class is similar to how the step is instantiated from the controller.

- Decorate the `ProdProcessGuidePromptProductionIdPageBuilder` class with a `ProcessGuidePageBuilderName` attribute.

```
X++
```

```
[ProcessGuidePageBuilderName(classStr(ProdProcessGuidePromptProductionIdPageBuilder))] public class ProdProcessGuidePromptProductionIdPageBuilder extends ProcessGuidePageBuilder
```

- In the `ProdProcessGuidePromptProductionIdStep` class, implement the abstract method `pageBuilderName()` to return this name.

```
X++
protected final ProcessGuidePageBuilderName pageBuilderName()
{
    return classStr(ProdProcessGuidePromptProductionIdPageBuilder);
}
```

> **Tip**
>
> Similar to the step factory, there's also an abstract factory pattern implemented for the page builder factory. So, in the rare case that you want a different strategy for instantiating the page builder, you can do the following:
>
> - Create a new factory class that inherits from `ProcessGuidePageBuilderAbstractFactory`.
> - Optionally, create a new parameter class that implements the `ProcessGuideIPageBuilderCreationParameters` interface, containing the parameters the factory needs.
> - In your step class, override the `pageBuilderFactory()` and `pageBuilderCreationParameters()` methods to return the factory and parameters.

To implement the user interface, you need a page with one text box to enter the production order ID, plus an **OK** button and a **Cancel** button. The **Cancel** button should exit the process.

To implement this UI, override two methods in the `ProdProcessGuidePromptProductionIdPageBuilder` class:

- Use the `addDataControls()` method to add the text box.

```
X++
protected final void addDataControls(ProcessGuidePage _page)
{
    _page.addTextBox(ProcessGuideDataTypeNames::ProdId, "@SYS4398",
extendedTypeNum(ProdId));
}
```

- Use the `addActionControls()` method to add the **OK** and **Cancel** buttons.

```
X++
```

```
protected final void addActionControls(ProcessGuidePage _page)
{
    #ProcessGuideActionNames
    _page.addButton(step.createAction(#ActionOK), true);
    _page.addButton(step.createAction(#ActionCancelExitProcess));
}
```

This approach adds the data controls, followed by the buttons. However, if you want to build a screen with interspersed data controls and buttons, you can override the `addControls()` method instead for flexibility.

Another scenario to consider is how to rebuild the page if validation failures occur, such as when the user enters an incorrect production order ID. The `ProcessGuidePageBuilder` base class implements the default behavior, which rebuilds the user interface, clears out the scanned value, and adds the error control with the error message. Because this is the default behavior that you want to use, you don't need to add any code for handling errors.

> **Tip**
>
> If you want to implement custom UI behavior for error situations, you can override one or more of the methods `rebuildFromRequestPage()`, `isErrorState()`, and `reuseRequestPageOnError()`.

Processing the user-entered data in the first step

The processing of the data is done in the `ProcessGuideDataProcessorDefault` class, which in turn invokes the legacy `WhsRfControlData` class. No changes are needed to this default behavior, and `WhsRfControlData` already has logic for validating the `ProdId` field, so you don't need to write any logic for handling this. If extensions are required for the validation logic, consider using the `WhsControl` based extension mechanism.

Determine if the first step is complete

When validation is successful, mark the step as completed. The base class handles this action, but you need to implement the condition that determines step completion. The following overridden method implements this condition.

X++

```
protected final boolean isComplete()
{
```

```
    WhsrfPassthrough pass = controller.parmSessionState().parmPass();
    ProdId prodId = pass.lookup(ProcessGuideDataTypeNames::ProdId);
        return (prodId != '');
}
```

View order details and confirm

In the second step in the process, the user sees a screen with details about the order. The user can either select the **OK** button to confirm the start of the production order, or select **Cancel** to go back to the start of the process. For this example, name the step `ProdProcessGuideConfirmProductionOrderStep` and the page builder class `ProdProcessGuideConfirmProductionOrderPageBuilder`.

The `ProdProcessGuideConfirmProductionOrderStep` class looks like the following.

X++

```
[ProcessGuideStepName(classStr(ProdProcessGuideConfirmProductionOrderStep))]
public class ProdProcessGuideConfirmProductionOrderStep extends ProcessGuideStep
{
    protected final ProcessGuidePageBuilderName pageBuilderName()
    {
        return classStr(ProdProcessGuideConfirmProductionOrderPageBuilder);
     }
}
```

Because the user doesn't enter any values here, you don't need to override the `isComplete()` method. The step is complete when the user selects **OK**.

The page builder class overrides the `addDataControls()` method to add three labels. The first label shows the production order ID. The second label contains item information, such as item ID, dimensions, or description. The third label contains the quantity and unit of measure.

The `addActionControls()` method is overridden to add two buttons – the **OK** button, and the **Cancel** button to cancel the process and go back to the start of the process.

X++

```
/// <summary>
/// The <c>ProdProcessGuideConfirmProductionOrderPageBuilder</c> builds a page that
allows the user to see details of a production order
/// and then confirm.
/// </summary>
[ProcessGuidePageBuilderName(classStr(ProdProcessGuideConfirmProductionOrderPageBuilde
r))]
public class ProdProcessGuideConfirmProductionOrderPageBuilder extends
```

```
ProcessGuidePageBuilder
{
    protected void addDataControls(ProcessGuidePage _page)
    {
        WhsrfPassthrough pass = controller.parmSessionState().parmPass();
        ProdTable prodTable =
ProdTable::find(pass.lookup(ProcessGuideDataTypeNames::ProdId));
        UnitOfMeasureSymbol inventUOM = InventTableModule::find(prodTable.ItemId,
ModuleInventPurchSales::Invent).UnitId;

        _page.addLabel(ProcessGuideDataTypeNames::ProdIdLabelName, strFmt("@WAX1684",
prodTable.ProdId), extendedTypeNum(ProdId));
        _page.addLabel(ProcessGuideDataTypeNames::ItemInfo,
this.generateItemInfoForProdId(pass.lookup(ProcessGuideDataTypeNames::ProdId)),
extendedTypeNum(WHSRFUndefinedDataType));
        _page.addLabel(ProcessGuideDataTypeNames::QtyLabelName, strFmt("@WAX1685",
WHSWorkExecuteDisplay::num2StrDisplay(ProdUpdStartUp::proposalStartUpQty(prodTable.Pro
dId)), inventUOM), extendedTypeNum(WHSRFQuantityAndUOM));

        if (PdsGlobal::pdsIsCWItem(prodTable.ItemId))
        {
            _page.addLabel(ProcessGuideDataTypeNames::InventQtyLabelName,
strFmt("@WAX1685",
WHSWorkExecuteDisplay::num2StrDisplay(ProdUpdStartUp::pdsCWProposalStartupQty(prodTabl
e.ProdId)), PdsCatchWeightItem::pdsCWUnitId(prodTable.ItemId)),
extendedTypeNum(WHSRFQuantityAndUOM));
        }
    }

    protected void addActionControls(ProcessGuidePage _page)
    {
        #ProcessGuideActionNames
        _page.addButton(step.createAction(#ActionOK), true);
        _page.addButton(step.createAction(#ActionCancelResetProcess));
    }
```

> **Note**
>
> You can find the same source code for the X++ methods in this article by using the Application Explorer. Filter on the class name, and then right-click the class name and select **View code**.

Start the production order

The third step executes the business logic for starting the production order. This step is different from the previous steps because it doesn't have a user interface. This step runs silently when the user selects **OK** in the previous step.

The `ProcessGuideStepWithoutPrompt` abstract class implements the default behavior for such steps. Therefore, the current step should extend the `ProcessGuideStepWithoutPrompt` class and override the `doExecute()` method.

The following code example shows the class and the `doExecute()` method implementation. The method retrieves the order ID and user ID from the session state and invokes the method to start this production order.

X++

```
/// <summary>
/// The <c>ProdProcessGuideStartProductionOrderStep</c> represents a step that starts
a production order.
/// </summary>
[ProcessGuideStepName(classStr(ProdProcessGuideStartProductionOrderStep))]
public class ProdProcessGuideStartProductionOrderStep extends
ProcessGuideStepWithoutPrompt
{
    protected final void doExecute()
    {
        WhsrfPassthrough pass = controller.parmSessionState().parmPass();
        WHSUserId userId = pass.lookup(ProcessGuideDataTypeNames::UserId);
        ProdTable prodTable =
ProdTable::find(pass.lookup(ProcessGuideDataTypeNames::ProdId));
        WhsWorkExecute workExecute = WhsWorkExecute::construct();
        workExecute.prodStartUp(prodTable.ProdId,
ProdUpdStartUp::proposalStartUpQty(prodTable.ProdId), userId);

        this.addProcessCompletionMessage();

        super();
    }

}
```

If an exception occurs, the framework exception handling logic rolls back the process to the previous step.

> **Note**
>
> The invocation of `addProcessCompletionMessage()` adds the *Work completed* message to the navigation parameters. The next step (assuming it has a user interface) displays this message. The base classes handle this logic, and no specific code needs to be added to the process classes to achieve this behavior.

Building the navigation through the steps

The `ProcessGuideController` base class instantiates the `ProcessGuideNavigationAgentDefault` class, which relies on a predefined navigation route. This route is a simple map of source and destination steps. For the production start scenario, because there's no conditional branching, this implementation works. Therefore, you only need to override the `initializeNavigationRoute()` method to define the navigation route.

X++

```
    protected ProcessGuideNavigationRoute initializeNavigationRoute()
    {
        ProcessGuideNavigationRoute navigationRoute = new
ProcessGuideNavigationRoute();

navigationRoute.addFollowingStep(classStr(ProdProcessGuidePromptProductionIdStep),
classStr(ProdProcessGuideConfirmProductionOrderStep));

navigationRoute.addFollowingStep(classStr(ProdProcessGuideConfirmProductionOrderStep),
classStr(ProdProcessGuideStartProductionOrderStep));

navigationRoute.addFollowingStep(classStr(ProdProcessGuideStartProductionOrderStep),
classStr(ProdProcessGuidePromptProductionIdStep));

        return navigationRoute;
    }
```

Some processes require conditional branching (based on user actions, or any other conditions). Such processes need to do the following tasks:

- Implement specific navigation agents inherited from the `ProcessGuideNavigationAgent` class.
- Implement the specific navigation agent factory inherited from the `ProcessGuideNavigationAgentAbstractFactory` class, containing logic to instantiate the correct navigation agent based on current step, session state, user action, or other logic.
- Optionally, override `navigationAgentCreationParameters()` in the controller class to pass suitable parameters.
- Override `navigationAgentFactory()` in the controller to instantiate the navigation agent factory created previously.

Action classes

Action classes represent user actions, so this example uses the **OK** action to show how the actions are created.

X++

```
[ProcessGuideActionName(#ActionOK)]
public class ProcessGuideOKAction extends ProcessGuideAction
{
    public final str label()
    {
        return "@SYS5473";
     }
     protected final void doExecute()
     {
        step.executeOKAction();
      }
}
```

The class must implement two abstract methods:

- `label()`, which returns the label to be displayed in a button control tied to this action.
- `doExecute()`, which performs the action. As mentioned earlier, the **OK** button simply performs a callback to the step. However, other actions might have more complex logic.

The actions are instantiated using `SysExtension` framework based on the `ProcessGuideActionName` attribute. Similar to the instantiation of page builders, the step class implements the default action factory, and it's possible to override that. The page builder adds a button control like this.

X++

```
_page.addButton(step.createAction(#ActionOK), true);
```

When you add the button, it asks the step to create an action class for the passed name and ties that action to the button.

Summary

To summarize everything explained in this article, here's a comprehensive summary of the code needed for the process:

1. `ProdProcessGuideProductionStartController`

    a. Override `initialStepName()` to provide the name of the first step.

    b. Override `initializeNavigationRoute()` to construct the navigation map.

    X++

    ```
    /// <summary>
    /// The <c>ProdProcessGuideProductionStartController</c> class is the
    ```

```
controller class for the production order start process guide.
/// </summary>
[WHSWorkExecuteMode(WHSWorkExecuteMode::StartProdOrder)]
public class ProdProcessGuideProductionStartController extends
ProcessGuideController
{
    protected ProcessGuideStepName initialStepName()
    {
        return classStr(ProdProcessGuidePromptProductionIdStep);
    }

    protected ProcessGuideNavigationRoute initializeNavigationRoute()
    {
        ProcessGuideNavigationRoute navigationRoute = new
ProcessGuideNavigationRoute();

navigationRoute.addFollowingStep(classStr(ProdProcessGuidePromptProductionIdSt
ep), classStr(ProdProcessGuideConfirmProductionOrderStep));

navigationRoute.addFollowingStep(classStr(ProdProcessGuideConfirmProductionOrd
erStep), classStr(ProdProcessGuideStartProductionOrderStep));

navigationRoute.addFollowingStep(classStr(ProdProcessGuideStartProductionOrder
Step), classStr(ProdProcessGuidePromptProductionIdStep));

        return navigationRoute;
    }

}
```

2. `ProdProcessGuidePromptProductionIdStep`

   a. Override `isComplete()` to specify when the step is considered complete.

   b. Override `pageBuilderName()` to specify the page builder to use.

X++

```
/// <summary>
/// The <c>ProdProcessGuidePromptProductionIdStep</c> represents a step that
/// that prompts the user for a production order id.
/// </summary>
[ProcessGuideStepName(classStr(ProdProcessGuidePromptProductionIdStep))]
public class ProdProcessGuidePromptProductionIdStep extends ProcessGuideStep
{
    protected boolean isComplete()
    {
        WhsrfPassthrough pass = controller.parmSessionState().parmPass();
        ProdId prodId = pass.lookup(ProcessGuideDataTypeNames::ProdId);

        return (prodId != '');
    }
```

```
    protected ProcessGuidePageBuilderName pageBuilderName()
    {
        return classStr(ProdProcessGuidePromptProductionIdPageBuilder);
    }

}
```

3. `ProdProcessGuidePromptProductionIdPageBuilder`

    a. Override `addDataControls()` to add the `Prod ID` textbox.

    b. Override `addActionControls()` to add the **OK** and **Cancel** buttons.

X++

```
/// <summary>
/// The <c>ProdProcessGuidePromptProductionIdPageBuilder</c> class builds a
page
/// that prompts the user for a production order id.
/// </summary>
[ProcessGuidePageBuilderName(classStr(ProdProcessGuidePromptProductionIdPageBu
ilder))]
public class ProdProcessGuidePromptProductionIdPageBuilder extends
ProcessGuidePageBuilder
{
    protected void addDataControls(ProcessGuidePage _page)
    {
        _page.addTextBox(ProcessGuideDataTypeNames::ProdId, "@SYS4398",
extendedTypeNum(ProdId));
    }

    protected void addActionControls(ProcessGuidePage _page)
    {
        #ProcessGuideActionNames
        _page.addButton(step.createAction(#ActionOK), true);
        _page.addButton(step.createAction(#ActionCancelExitProcess));
    }

}
```

4. `ProdProcessGuideConfirmProductionOrderStep`

    a. Override `pageBuilderName()` to specify the page builder to use.

X++

```
/// <summary>
/// The <c>ProdProcessGuideConfirmProductionOrderStep</c> class represents the
step for viewing production order
/// details and confirming the same.
/// </summary>
```

```
[ProcessGuideStepName(classStr(ProdProcessGuideConfirmProductionOrderStep))]
public class ProdProcessGuideConfirmProductionOrderStep extends
ProcessGuideStep
{
    protected ProcessGuidePageBuilderName pageBuilderName()
    {
        return classStr(ProdProcessGuideConfirmProductionOrderPageBuilder);
    }

}
```

5. `ProdProcessGuideConfirmProductionOrderPageBuilder`

   a. Override `addDataControls()` to add the order, item, and quantity information labels.

   b. Override `addActionControls()` to add the **OK** and **Cancel** buttons.

X++

```
/// <summary>
/// The <c>ProdProcessGuideConfirmProductionOrderPageBuilder</c> builds a page
that allows the user to see details of a production order
/// and then confirm.
/// </summary>
[ProcessGuidePageBuilderName(classStr(ProdProcessGuideConfirmProductionOrderPa
geBuilder))]
public class ProdProcessGuideConfirmProductionOrderPageBuilder extends
ProcessGuidePageBuilder
{
    protected void addDataControls(ProcessGuidePage _page)
    {
        WhsrfPassthrough pass = controller.parmSessionState().parmPass();
        ProdTable prodTable =
ProdTable::find(pass.lookup(ProcessGuideDataTypeNames::ProdId));
        UnitOfMeasureSymbol inventUOM =
InventTableModule::find(prodTable.ItemId,
ModuleInventPurchSales::Invent).UnitId;

        _page.addLabel(ProcessGuideDataTypeNames::ProdIdLabelName,
strFmt("@WAX1684", prodTable.ProdId), extendedTypeNum(ProdId));
        _page.addLabel(ProcessGuideDataTypeNames::ItemInfo,
this.generateItemInfoForProdId(pass.lookup(ProcessGuideDataTypeNames::ProdId))
, extendedTypeNum(WHSRFUndefinedDataType));
        _page.addLabel(ProcessGuideDataTypeNames::QtyLabelName,
strFmt("@WAX1685",
WHSWorkExecuteDisplay::num2StrDisplay(ProdUpdStartUp::proposalStartUpQty(prodT
able.ProdId)), inventUOM), extendedTypeNum(WHSRFQuantityAndUOM));

        if (PdsGlobal::pdsIsCWItem(prodTable.ItemId))
        {
            _page.addLabel(ProcessGuideDataTypeNames::InventQtyLabelName,
strFmt("@WAX1685",
WHSWorkExecuteDisplay::num2StrDisplay(ProdUpdStartUp::pdsCWProposalStartupQty(
```

```
prodTable.ProdId)), PdsCatchWeightItem::pdsCWUnitId(prodTable.ItemId)),
extendedTypeNum(WHSRFQuantityAndUOM));
        }
    }

    protected void addActionControls(ProcessGuidePage _page)
    {
        #ProcessGuideActionNames
        _page.addButton(step.createAction(#ActionOK), true);
        _page.addButton(step.createAction(#ActionCancelResetProcess));
    }
```

> ⓘ **Note**
>
> The `generateItemInfoForProdId()` method, which is used for generating the item information labels, is excluded from this article. This method queries a few tables to get item ID, description, and dimensions. If you want a better understanding of `generateItemInfoForProdId()`, look at the source code.

6. `ProdProcessGuideStartProductionOrderStep`

   a. Override `doExecute()` to perform the production start process and add the process completion message.

X++

```
/// <summary>
/// The <c>ProdProcessGuideStartProductionOrderStep</c> represents a step that
starts a production order.
/// </summary>
[ProcessGuideStepName(classStr(ProdProcessGuideStartProductionOrderStep))]
public class ProdProcessGuideStartProductionOrderStep extends
ProcessGuideStepWithoutPrompt
{
    protected final void doExecute()
    {
        WhsrfPassthrough pass = controller.parmSessionState().parmPass();
        WHSUserId userId = pass.lookup(ProcessGuideDataTypeNames::UserId);
        ProdTable prodTable =
ProdTable::find(pass.lookup(ProcessGuideDataTypeNames::ProdId));
        WhsWorkExecute workExecute = WhsWorkExecute::construct();
        workExecute.prodStartUp(prodTable.ProdId,
ProdUpdStartUp::proposalStartUpQty(prodTable.ProdId), userId);

        this.addProcessCompletionMessage();

        super();
    }
```

```
}
```

> ⓘ **Note**
>
> Many of the common patterns (regeneration of UI on error, setting process completion message, **OK** and **Cancel** behavior) are now part of the framework. This change saves the application developer from writing boilerplate code that is both error prone and has a risk of inconsistent behavior across processes. If the scenario needs to deviate from the common path, the application developer can override suitable methods. This deviation is intentional, explicit, and trackable.

Extending a business process

So far, this article highlights how to build a new process by using the `ProcessGuide` framework. In this final section, you find some examples of how you can extend this business process.

Add a step in a flow (using ProcessGuideNavigationAgentDefault)

Where to extend:

- Child of `ProcessGuideController` class for the process.

How to extend:

- Extend the `initializeNavigationRoute()` method in the controller class, and invoke `addFollowingStep()` in the `ProcessGuideNavigationRoute` class.

Add a step in a flow (using custom navigation agent)

Where to extend:

- Child of `ProdProcessGuideNavigationAgentFactory/ProdProcessGuideNavigationAgent`.

How to extend:

- Create a new child class of `ProcessGuideNavigationAgent` that returns the desired step name.
- Create a new class deriving from `ProcessGuideNavigationAgentFactory` that conditionally returns the navigation agent created previously.

- Extend the `navigationAgentFactory()` method in the controller class to return the factory created previously.

Add a new control in the UI of an existing step

Where to extend:

- Child of `ProdProcessGuidePageBuilder` for the step.

How to extend:

- Extend the `addDataControls()` method and add the extra control.

Complete overhaul of the user interface in an existing step

Where to extend:

- Child of `ProdProcessGuideStep`.

How to extend:

- Create a new child class of `ProdProcessGuidePageBuilder` class, and implement the desired user interface.
- Extend the `pageBuilderName()` method in the step class to return the `ProcessGuidePageBuilderNameAttribute` for the class created previously.

Alter logic when a step is considered complete

Where to extend:

- Child of `ProdProcessGuideStep`.

How to extend:

- Extend the `isComplete()` method to build the additional logic.

Last updated on 06/17/2026

Assign step icons and titles for the Warehouse Management mobile app

10/03/2025

This article describes how to assign step icons and step titles for new or customized task flows for the Warehouse Management mobile app.

The following illustrations shows how step icons and titles appear in the Warehouse Management mobile app.

Standard step IDs, classes, and icons

Each step in a task flow is identified by a step ID, and each step ID has a corresponding step class. The step icon and title are specified in each step class.

Step IDs and step classes

The following table lists every step ID that is currently available, and its corresponding step class. The control name of the primary input field is used as the step ID.

For an example that shows how these step IDs and classes are used, see the implementation of the `WHSMobileAppStepInfoBuilder.stepId()` method in the Example: Assign step icons and titles for a custom flow section later in this article.

Expand table

Step ID	Step Class
BatchDisposition	WHSMobileAppStepBatchDisposition
Carrier	WHSMobileAppStepCarrier
CatchWeight	WHSMobileAppStepCatchWeight
CatchWeightQtyOutboundWeight	WHSMobileAppStepCatchWeight
CatchWeightTag	WHSMobileAppStepCatchWeightTag
CatchWeightTagWeight	WHSMobileAppStepCatchWeightTagWeight
ChangeWarehouseSuccess	WHSMobileAppStepChangeWarehouseSuccess
CheckDigit	WHSMobileAppStepCheckDigit
ClusterId	WHSMobileAppStepClusterId
ClusterPickQtyVerification	WHSMobileAppStepQtyVerification
ClusterPosition	WHSMobileAppStepClusterPosition
ConfigId	WHSMobileAppStepConfigId
Confirmation	WHSMobileAppStepConfirmation
ConsolidateFromLicensePlateId	WHSMobileAppStepConsolidateFromLicensePlateId
ConsolidateLPConfirmation	WHSMobileAppStepConsolidateLPConfirmation
ConsolidateToLicensePlateId	WHSMobileAppStepConsolidateToLicensePlateId
ContainerType	WHSMobileAppStepContainerType
CountingReasonCode	WHSMobileAppStepCountingReasonCode
CycleCountingAddLPOrFinish	WHSMobileAppStepCycleCountingAddLPOrFinish
CycleCountQty1	WHSMobileAppStepCycleCountQty
CycleCountQty2	WHSMobileAppStepCycleCountQty
CycleCountQty3	WHSMobileAppStepCycleCountQty
CycleCountQty4	WHSMobileAppStepCycleCountQty
Disposition	WHSMobileAppStepDisposition
DriverCheckInConfirmation	WHSMobileAppStepDriverCheckInConfirmation
DriverCheckInId	WHSMobileAppStepDriverCheckInId
DriverCheckOutConfirmation	WHSMobileAppStepDriverCheckOutConfirmation
DriverCheckOutId	WHSMobileAppStepDriverCheckOutId

Step ID	Step Class
ExpDate	WHSMobileAppStepExpDate
FromBatchDisposition	WHSMobileAppStepFromBatchDisposition
FromInventoryStatus	WHSMobileAppStepInventoryStatusFrom
FullQty	WHSMobileAppStepFullQty
InboundPut	WHSMobileAppStepInboundPut
InventBatchId	WHSMobileAppStepBatch
InventColorId	WHSMobileAppStepInventColorId
InventLocation	WHSMobileAppStepInventLocation
InventLocationId	WHSMobileAppStepWarehouse
InventSerialId	WHSMobileAppStepInventSerialId
InventSizeId	WHSMobileAppStepInventSizeId
InventStatusId	WHSMobileAppStepInventStatus
InventStyleId	WHSMobileAppStepInventStyleId
InventVersionId	WHSMobileAppStepInventVersionId
ItemId	WHSMobileAppStepItem
ITMContainerID	ITMMobileAppStepContainerId
ITMShipmentID	ITMMobileAppStepShipmentId
KanbanCardId	WHSMobileAppStepKanbanCard
KanbanCardToEmpty	WHSMobileAppStepKanbanCardToEmpty
KanbanOrCardId	WHSMobileAppStepKanbanCard
LicensePlateId	WHSMobileAppStepLicensePlate
LoadId	WHSMobileAppStepLoadId
LocationLicensePlatePosition	WHSMobileAppStepLocationLicensePlatePosition
LocOrLP	WHSMobileAppStepLocOrLP
LocOrLP_From	WHSMobileAppStepLocOrLPFrom
LocOrLP_To	WHSMobileAppStepLocOrLPTo
LocOrLPCheck	WHSMobileAppStepLocOrLPCheck
LocVerification	WHSMobileAppStepLocVerification

Step ID	Step Class
LPAdjustIn	WHSMobileAppStepLPAdjustIn
LPBreakChildLP	WHSMobileAppStepLPBreakChildLP
LPBreakParentLP	WHSMobileAppStepLPBreakParentLP
LPBuildChildLP	WHSMobileAppStepLPBuildChildLP
LPBuildParentLP	WHSMobileAppStepLPBuildParentLP
LPVerification	WHSMobileAppStepLPVerification
MergeContainerId	WHSMobileAppStepMergeContainerId
MixedLPLineNum	WHSMobileAppStepMixedLPLineNum
MobileDeviceQueueMessageCollectionIdentifierId	WHSMobileAppStepSelectOrder
MovementConfirmCancel	WHSMobileAppStepMovementConfirmCancel
NewCaptureWeight	WHSMobileAppStepCatchWeight
NewQty	WHSMobileAppStepNewQty
OutboundCatchWeightTag	WHSMobileAppStepCatchWeightTag
OutboundPut	WHSMobileAppStepOutboundPut
OutboundWeight	WHSMobileAppStepCatchWeight
OverridePutNewLocation	WHSMobileAppStepOverridePutNewLocation
PieceByPieceConfirmation	WHSMobileAppStepQtyVerification
POLineNum	WHSMobileAppStepPOLineNum
PONum	WHSMobileAppStepPONum
PositionFull	WHSMobileAppStepPositionFull
PositionFullQty	WHSMobileAppStepPositionFullQty
Potency	WHSMobileAppStepPotency
PrinterName	WHSMobileAppStepPrinterName
ProdId	WHSMobileAppStepProdId
ProdLastPalletConfirmation	WHSMobileAppStepProdLastPalletConfirmation
ProductConfirmation	WHSMobileAppStepProductConfirmation
ProductionScrapConfirmation	WHSMobileAppStepProductionScrapConfirmation
Put	WHSMobileAppStepPut

Step ID	Step Class
PutawayClusterId	WHSMobileAppStepPutawayClusterId
Qty	WHSMobileAppStepQty
QtyAdjust	WHSMobileAppStepQtyAdjust
QtyShort	WHSMobileAppStepQtyShort
QtyToConsume	WHSMobileAppStepQtyToConsume
QtyToPick	WHSMobileAppStepQtyToPick
QtyToPut	WHSMobileAppStepQtyToPut
QtyToScrap	WHSMobileAppStepQtyToScrap
QtyVerification	WHSMobileAppStepQtyVerification
QtyWithScanningLimit	WHSMobileAppStepQtyAdjust
ReasonString	WHSMobileAppStepReasonString
RecvLocationId	WHSMobileAppStepRecvLocationId
RemoveContainerId	WHSMobileAppStepRemoveContainerId
ReprintLabelConfirmation	WHSMobileAppStepReprintLabelConfirmation
RMANum	WHSMobileAppStepRMANum
ShortPickReason	WHSMobileAppStepShortPickReason
SortConOrLP	WHSMobileAppStepSortConOrLP
SortLicensePlateId	WHSMobileAppStepSortLicensePlateId
SortPositionId	WHSMobileAppStepSortPositionId
SortVerification	WHSMobileAppStepSortVerification
StartLocationId	WHSMobileAppStepStartLocationId
StartProdOrderConfirmation	WHSMobileAppStepStartProdOrderConfirmation
TargetLicensePlateId	WHSMobileAppStepTargetLicensePlateId
TOLineNum	WHSMobileAppStepTOLineNum
ToLocation	WHSMobileAppStepToLocation
TONum	WHSMobileAppStepTONum
ToWarehouse	WHSMobileAppStepWarehouseTo
TransportLoadId	WHSMobileAppStepTransportLoadId

Step ID	Step Class
WaveLabelId	WHSMobileAppStepWaveLabelId
WaveLblQty	WHSMobileAppStepWaveLblQty
Weight	WHSMobileAppStepWeight
WeightToConsume	WHSMobileAppStepWeightToConsume
WHSAdjustmentType	WHSMobileAppStepWHSAdjustmentType
WHSReceivingException	WHSMobileAppStepWHSReceivingException
WHSWorkException	WHSMobileAppStepWHSWorkException
WHSWorkLicensePlateId	WHSMobileAppStepWorkLicensePlateId
WMSLocationId	WHSMobileAppStepLocation
WorkId	WHSMobileAppStepWorkId
WorkIdToCancel	WHSMobileAppStepWorkIdToCancel
WorkLPIdPutawayCluster	WHSMobileAppStepWorkLPIdPutawayCluster
WorkPoolId	WHSMobileAppStepWorkPoolId
ZoneId	WHSMobileAppStepZoneId

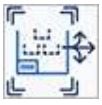
Available step icons

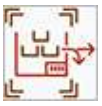
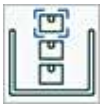
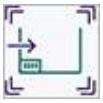
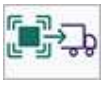
The system includes a collection of standard step icons that you can also use for your custom steps. You can't currently upload custom step icons. Therefore, you must always select one of the standard step icons.

The following table shows every standard step icon that is currently available, and its name.

Expand table

About	AddLpOrItem	BatchDisposition	Carrier
CatchWeightTag	CatchWeightTagWeight	CheckDigit	CheckInOutId
ChildLP	ClusterId	ClusterPosition	ConfigId

ConfiguredField	ConOrLP	ConsolidateFromLicensePlateID	ConsolidateToLicensePlateID
ContainerType	Counting	CountingReasonCode	CountryOfOrigin
Disposition	Done	DriverCheckInConfirmation	DriverCheckInId
DriverCheckOutId	ExpDate	Field	FromBatchDisposition
FromInventoryStatus	IdAttribute	InventBatchID	InventColorID
InventLocation	InventSerialID	InventSizeID	InventStatusID
InventStyleID	InventVersionID	ItemID	ITMContainerID
ITMShipmentID	KanbanCardID	KanbanOrCardID	LicensePlateID
LoadId	LocationLicensePlatePosition	LocOrLP	LocOrLPCheck
LocOrLPFrom	LocOrLPTo	LongProcessCompleted	 LPBreakParentLP
MergeContainerId	MixedLPLineNum	OutboundWeight	Owner
	PleaseConfirm	POLineNum	PONum

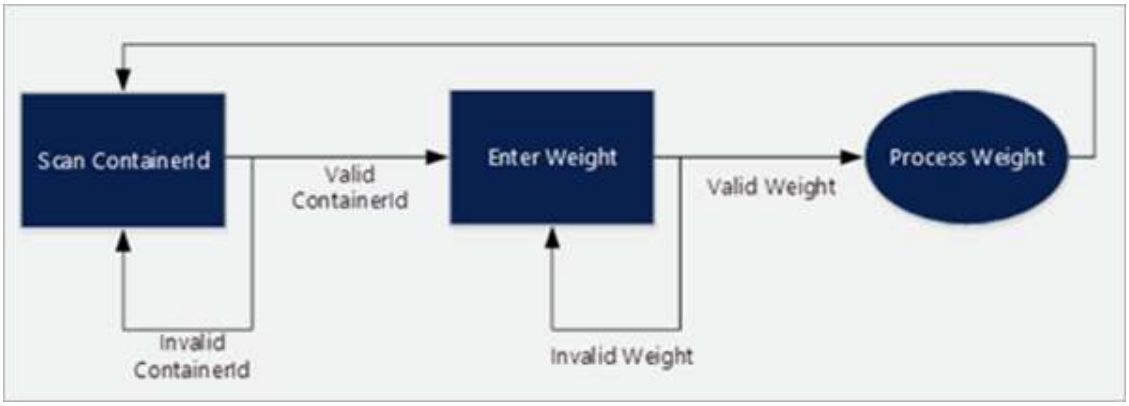
ParentLP			
PositionFull	Potency	PrinterName	ProdId
ProductConfirmation	Put	 PutawayClusterId	Qty
QtyAdjustIn	QtyShort	QtyToConsume	QtyToPut
QtyToScrap	QuantityConfirmation	RAFEndJob	RecvLocationID
 RemoveContainerID	RMANum	SelectOrder	ShortPickReason
SortPositionId	 TargetLicensePlateId	ToLineNum	ToLocation
ToNum	ToWarehouse	 TransportLoadId	VendBatchId
WaveLabelId	WaveLblQty	Weight	WeightToConsume
WHSAdjustmentType	WHSReceivingException	WMSLocationID	WorkId
WorkIdToCancel	WorkLicensePlateId	WorkLPIDPutawayCluster	WorkPoolID
ZoneID			

Example: Assign step icons and titles for a custom flow

This example explains how to set up step icons and titles for a custom task flow. The scenario is built on an example of a custom task flow that is presented and explored in more detail in the following blog post: Customizing the Warehousing Mobile App. The task flow works in the following way:

1. The app shows a page that prompts the worker to provide a container ID (for example, by scanning a bar code).
2. If the container ID is valid, the app opens a new page that prompts the worker for the weight. (If the container ID isn't valid, the worker is returned to the first page.)
3. When the worker enters a valid weight, the system stores the weight and returns the worker to the first page.

The following illustration shows this task flow.



Create a step class for the container input page

The container input page lets the worker scan or enter a container ID.

On the container input page, the control name of the input field is `ContainerId`. Because this control name isn't in the list of step IDs, you won't find an existing step that is based on it. Therefore, you must create a step class that represents the step. Here is an example.

X++

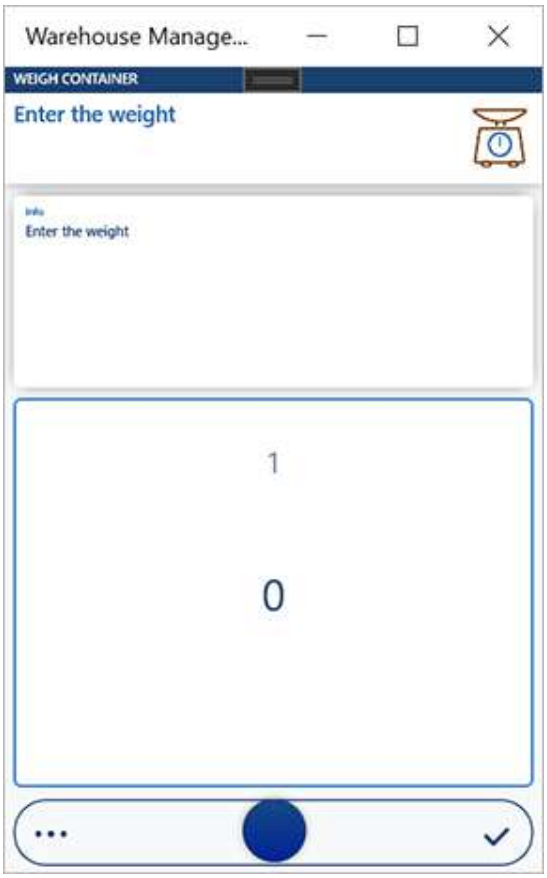
```
[WHSMobileAppStepId('ContainerId')]
final internal class WHSMobileAppStepContainerId extends WHSMobileAppStep
{
    private const WHSMobileAppStepIcon PopulationIcon = 'InventBatchID';
    private const WHSMobileAppStepTitle InputNotFilledTitle =
"@WAX:WHSMobileAppStepContainerID_InputNotFilled"; //Scan a container
    protected void initValues()
    {
        defaultStepIcon = PopulationIcon;
        defaultStepTitle = InputNotFilledTitle;
    }
}
```

The identifier of the step icon is stored in the `defaultStepIcon` class member, and the step title is stored in the `defaultStepTitle` class member.

To assign a step icon, set `defaultStepIcon` to one of the icon IDs that are listed in the Available step icons section earlier in this article.

Use a standard or custom step icon and title for the weight input

The weight input page lets the worker enter a weight.



On the weight input page, the control name of the input field is `Weight`, which is in the list of step IDs. Therefore, if the step icon and title that are defined in the `WHSMobileAppStepWeight` class are acceptable to you, you don't have to change anything for this step.

However, if you prefer to use a different icon or title for this step, you can override either the `stepId()` method or the `stepInfo()` method in the builder class. Each task flow has its own step info builder.

Override the stepId() method

The following example shows one way that you can modify a builder class by overriding the `stepId()` method.

X++

```
[WHSWorkExecuteMode(WHSWorkExecuteMode:: WeighContainer)]
public class WHSMobileAppStepInfoBuilderWeighContainer extends
WHSMobileAppStepInfoBuilder
{
    protected WHSMobileAppStepId stepId()
    {
        WHSMobileAppStepId stepIdLocal = super();
        if (stepIdLocal == 'Weight')
        {
            return 'NewWeight';
        }
        return stepIdLocal;
```

```
    }
}
```

You then create a step class for the `NewWeight` step. The code should resemble the code for the `ContainerId` example that was shown earlier in this article.

Override the stepInfo() method

The following example shows one way that you can modify a builder class by overriding the `stepInfo()` method.

```
X++

[WHSWorkExecuteMode(WHSWorkExecuteMode:: WeighContainer)]
public class WHSMobileAppStepInfoBuilderWeighContainer extends
WHSMobileAppStepInfoBuilder
{
    protected WHSMobileAppStepInfo stepInfo()
    {
        if (stepId != 'Weight')
        {
            return super();
        }
        WHSMobileAppStepInfo stepInfo = WHSMobileAppStepInfo::construct();
        stepInfo.parmStepIcon('NewIcon');
        stepInfo.parmStepTitle('NewTitle');
        return stepInfo;
    }
}
```

You then construct a `WHSMobileAppStepInfo` object, and set the icon and/or title directly.

Related information

- Install the Warehouse Management mobile app
- Mobile device user settings

Warehouse configuration overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to configure a warehouse. It includes information about how to enable a warehouse layout and warehouse processes.

> **Note**
>
> This article applies to features in the Warehouse management module. It doesn't apply to warehouse features in the Inventory management module.

Warehouse layout

Warehouse management processes (WMS) in Supply Chain Management give you flexible ways to define your warehouse layout to meet changing needs, so that you can achieve optimal warehouse efficiency.

- You can establish high-priority and low-priority storage areas for optimum placement of goods.
- You can divide your warehouse into zones to accommodate various storage needs, such as temperature requirements, or various turnover rates for items.
- You can specify warehouse locations on any level (for example, site, warehouse, aisle, rack, shelf, and bin position).
- You can group locations by using physical capacity constraint settings.
- You can control how items are stored and picked, based on query-defined rules.

To use WMS in Supply Chain Management, you must create a warehouse and enable it for WMS. On the **Warehouses** page, select the **Use warehouse management processes** option.

Zone groups, zones, location types, and locations

As part of the process for enabling a warehouse layout, you must define warehouse zone groups, and zones, location profiles, location types, and locations.

- **Zone groups** – A logical or physical grouping of zones within a warehouse.
- **Zones** – A logical or physical grouping of locations within a warehouse.
- **Location profiles** – A logical or physical grouping of locations that have the same warehouse location process policies (for example, a mix of different item numbers can be stored there, and the same physical capacity constraints apply).
- **Locations types** – A logical or physical grouping of the warehouse locations. For example, you can create a location type for all staging locations. Mandatory settings on the **Warehouse management parameters** page drive the process of defining staging location types and the final shipping location type.
- **Locations** – The lowest level of location information. Use locations to track where the on-hand inventory is stored and picked in a warehouse.

The entities that you create to define your warehouse layout are used in the queries that you set up in work templates to drive work orders in the warehouse. Therefore, when you define the zones, location types, and so on, consider how different areas in the warehouse are used for different processes. Additionally, consider factors such as the physical characteristics of a particular area. For example, there might be areas where you can use only a certain type of forklift truck. Or, if your company has both production and finished goods within the same facility, you might want to create a single warehouse in Supply Chain Management but then separate the two operations by creating two zone groups. Give your entities descriptive names, so that it's easy to identify them when you use them in template queries.

Location stocking limits, location profiles, and fixed picking locations

You must consider the physical layout of the warehouse, both to determine storage capacities (location stocking limits and location profiles) and as part of your attempts to achieve optimal warehouse processes.

Location stocking limits help guarantee that work isn't created to request inventory to be put in a location that doesn't have the physical capacity to carry the inventory. For example, if some locations within a warehouse can hold only one pallet per location, enable location stocking limits. Set the **Quantity** value to *1*, and set the **Unit** value to *PL* within a specific location profile grouping.

If you need more advanced calculations to control the location capacity constraints, use location profiles. In this case, the weight and volume are considered when capacity calculations are done.

To achieve optimal outbound processes, evaluate whether to use fixed picking locations and fixed packing locations. Often, minimum/maximum replenishment is used for replenishment processes from a bulk area to the fixed picking locations, and multiple fixed picking locations

can be enabled within the same warehouse and for product variants. Consider the flexibility that you can achieve by enabling dedicated demand replenishment overflow locations that are used only for wave/load replenishment processing.

Location setup wizard

To quickly create the locations within a warehouse, use the **Location setup** wizard. As part of this process, you can easily maintain the format of the location names.

Warehouse processes

As part of the configuration of a warehouse, enable warehouse processes according to your business requirements. The most important components to configure are wave templates, work templates, work pools, and location directives.

Wave templates

Wave templates help enable the outbound "Release to warehouse" process. Use the wave template functionality as soon as you release order lines (either directly from source documents, via batch job processes, or via loads that you already created).

You can create three types of wave templates:

- **Shipping**
- **Production order**
- **Kanban**

Use parameters to define how far the system should automatically go in the outbound work processing. The system selects a wave template based on the wave template sequence and criteria that you specify in the template. The system checks the criteria in the template listed at the top of the sequence first. If the criteria are met, the system processes the wave template. Otherwise, the system checks the criteria in the next template, and so on. Therefore, put the template that has the most specific criteria at the top of the wave template sequence list, so that it's processed first. For example, you want to process all the work for a specific carrier today and temporarily delay processing of the work for other carriers. In this case, list the wave template that selects work for that carrier higher in the sequence than other templates. Otherwise, the work for other carriers might be processed before the work for that carrier is completed.

You must specify the wave process methods in each wave template. The available methods vary, depending on the wave template type.

Work templates

Work template definitions play an important role in the definition of warehouse management work processes. They define what work is performed, and how the work is done. Templates can also contain a directive code that links to a location directive to determine where work is performed. Work templates include a query that specifies the criteria for the work. Each template must include at least one Pick operation and one Put operation to drive the basic work operation of transferring on-hand inventory from one location to another.

If multiple workers must be able to process work for some of your warehouse operations, consider using the concept of *staging* for the inventory and separating the work execution into different work classes.

General warehouse work policies

General warehouse work policies let you influence how warehouse work is created and handled. Set up policies to control options such as default work priority values, deferred work processing, work transaction dates, and more. To view and set these policies, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Open the **General** tab.
3. Expand the **Work** FastTab.
4. Select policies and set other work options available on this FastTab. Each field has tooltip help. To see the tooltip for a field, hover your mouse pointer over the relevant field label.

Work pools

Use work pools to organize work into groups. For example, create a work pool to classify work that occurs in a particular warehouse location. For all work types except counting, assign a work pool to a work template. For cycle counting, assign a work pool on the following pages:

- Cycle count plans
- Cycle count thresholds
- Cycle count work by location
- Cycle count work by item

When you use work templates to create work, the work pool is automatically assigned to the work.

You can also use work pool IDs to limit the type of work that is directed to a particular warehouse worker, provided that you configure this functionality on the relevant mobile device

menu item.

Location directives

As the name suggests, use location directives to direct the work transactions to the appropriate locations in the warehouse. In other words, they define where to pick and put.

To make it easier and quicker to define the actions that are associated with each location directive line, use one of the predefined strategies. For example, you can use the *Empty location with no incoming work* strategy to search for free locations in a warehouse, or you can use *FEFO batch reservation* strategy for outbound sales picking.

Next steps

- Configure locations in a WMS-enabled warehouse
- Get started with setting up the Warehouse management module

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Last updated on 11/20/2025

Set up a warehouse by using a warehouse configuration template

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up a warehouse by using a warehouse configuration template. There are several predefined configuration templates that you can use. For information about how to use these templates, see Configuration data templates.

Scenarios where configuration templates can be helpful

Configuration templates can be helpful in many scenarios. Here are some examples:

- You've completed and tested a configuration setup in a test environment, and you now want to copy the setup to a production environment.
- You want to roll the warehouse setup out to several legal entities or create a new warehouse in the same legal entity.
- You want to quickly prepare for a demo of the warehouse functionality.
- You want existing items and warehouses to use the functionality in Warehouse management instead of the functionality in Inventory management.

This article focuses on the first of these scenarios. It shows how you can use a configuration template to copy a configuration setup from a test environment to a production environment.

Copy a configuration setup from a test environment to a production environment

For this scenario, the configuration setup for a warehouse and some transaction processes already exist in a test environment. You now want to copy the configuration setup for the warehouse from the test environment to a production environment.

> **Note**

It's important that you include other related setup data when you copy a configuration setup. For example, you want to set up products by copying the setup from a test environment. However, you can't set up a fixed picking location for a product before that product is created. Although individual configuration templates don't support this type of dependency, there are default data templates that support it. You can easily include these default data templates in a configuration process.

Export a default warehouse template

1. Open the **Data management** workspace.

   > ⓘ **Note**
   >
   > If you're using this workspace for the first time, you must load all the data entities before you continue.

2. Select the **Templates** tile, and then select **Load default templates** to load the default templates.

   > ⓘ **Note**
   >
   > **Load default templates** is available only in the **Enhanced** view. Select **Framework parameters**, and then, in the **View defaults** field, select **Enhanced view**.

3. After the default templates are loaded, you can change them so that they meet your business requirements.

   > ⓘ **Note**
   >
   > To load default templates and import templates, you must have system administrator access. This requirement helps guarantee that all entities are correctly loaded into the template.

4. Make sure that you're in the legal entity that you want to export the company-specific data from.

5. In the workspace, select **Export**.

6. Create a new export project.

7. Select + **Add template**, and find the **400 - WMS** default warehouse template. This template adds data entities for warehouse configuration.

   > ⓘ **Note**
   >
   > If the data that you're exporting must be filtered (for example, you want to export only the data that is related to a specific warehouse), you must evaluate each data entity and add filtering via a query. Alternatively, you can export all data and then delete the records that aren't required in the destination files.

8. Select **Export**. Data that is related to all the data entities in the project is exported.

You can download a zip file for the data package. This file contains all the data in the selected format (for example, Excel format). As has been mentioned, some records in the data package files might have to be updated before you can import them into the production environment. If you update a record, make sure that you don't change the file name.

Import a warehouse configuration setup

1. In the destination environment, make sure that you're in the company that you want to import the warehouse data into.

   > ⓘ **Note**
   >
   > Before you do the import, you should identify any data dependencies. For example, the **Warehouse management** template includes a data entity that is named **Warehouse disposition codes**. This entity contains data that is related to the **Disposition codes** setup page (**Warehouse management** > **Setup** > **Mobile device** > **Disposition codes**). If an existing setup already handles the return process for sales orders, the **Return disposition code** field contains a value. The disposition code in this field is related to the **Disposition code** data entity, which is part of the **Sales and marketing** template. If the data from the **Disposition code** data entity isn't imported before the data from the **Warehouse disposition codes** field, the import will fail.

2. In the **Data management** workspace, select **Import**.

3. Create a new import project.

4. Select + **Add file**, and upload the zip file for the data package.

5. Select **Import**. In the **Enhanced** view, you can use the **Filter** option to quickly get an overview of issues that might occur during the import.

The **View execution** log provides detailed information about each data entity that is imported. You can use the staging data view to quickly get to the target data. In this way, you can see what the imported data looks like on the related pages in the application. When you use the default data templates, the import sequence for each data entity works in the predefined manner, to help guarantee that all dependent data is imported first. If custom data entities are part of the project, you must make sure that the correct sequence is defined. Learn more in Configuration data templates.

To learn more about how to use warehouse template to copy the configuration of a warehouse from one company to a new company within the same instance, see this 3-minute video on YouTube about how to use warehouse template to copy the configuration for finance and operations.

Related article

Configuration data templates

Last updated on 04/10/2025

Warehouse groups

10/03/2025

Warehouse groups let you establish groups of warehouses that you can associate with various other records in Microsoft Dynamics 365 Supply Chain Management.

> ⓘ **Note**
>
> - Each warehouse can be assigned to any number of warehouse groups.
> - Only warehouses that are enabled for advanced warehouse processes (WMS) can be assigned to warehouse groups.

Create and manage warehouse groups

To create and manage your warehouse groups, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Warehouse groups**.
2. Follow one of these steps:
    - To create a new group, select **New** on the Action Pane.
    - To edit an existing group, select it in the list pane, and then select **Edit** on the Action Pane.
    - To delete an existing group, select it in the list pane, and then select **Delete** on the Action Pane.
3. If you're creating or editing a warehouse group, set the following fields on the header of the new or selected group:
    - **Warehouse group** – Enter a unique name for the group.
    - **Name** – Enter a short description of the group.
4. On the **Warehouses** FastTab, define the list of warehouses that should be members of the current group.
    - To add a warehouse to the group, select **Add** on the FastTab toolbar to add a new row to the grid. Then, in the **Warehouse** column for the new row, select a warehouse. (Only WMS-enabled warehouses are available for selection.) The other column values will be updated automatically, based on the warehouse that you selected.

- To remove a warehouse from the group, select it in the grid, and then select **Remove** on the FastTab toolbar.
- To replace one warehouse in the group with another warehouse, find the warehouse that you want to replace in the grid. Then, in the **Warehouse** column, select the warehouse that you want to use instead.

Use warehouse groups

When you're setting up some types of warehousing features (such as location directives, wave templates, outbound sorting templates, cross docking templates, slotting templates or container packing policies), you can specify whether the feature applies to a single warehouse, a group of warehouses, or all warehouses. Usually, you'll see two fields: one where you select the warehouse scope (*Warehouse*, *Group*, or *All*) and one where you make a selection, based on that scope (a single warehouse ID, a warehouse group ID, or all warehouses).

Warehouse spatial location overview (preview)

[This article is prerelease documentation and is subject to change.]

The *warehouse spatial location* feature lets you assign X, Y, and Z coordinates to warehouse locations. The system uses these coordinates to calculate optimized picking routes that minimize the travel distance required for warehouse workers to move between locations during pick work. When you use this feature, the *Sort picking work lines* wave processing step reorders the pick work lines within a work record so that workers follow an efficient route through the warehouse. The step uses the assigned coordinates, your selected distance calculation strategy, and your selected sorting algorithm to determine the optimal order.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Key concepts

The *warehouse spatial location* feature employs the following key concepts:

- **Spatial coordinates** – The X, Y, and Z values that represent a location's physical position in the warehouse. The X and Y axes represent the horizontal position, and the Z axis represents the vertical position (for example, a rack level).
- **Pick work line** – A line in a work that instructs a worker to pick items from a specific warehouse location. A work record typically contains multiple pick work lines followed by a put work line.
- **Put work line** – A line in a work record that instructs a worker to place (put) the picked items at a destination location, such as a bay door or staging area.
- **Location sorting algorithm** – The algorithm that determines the order in which pick work lines are sorted. You can choose between a fast calculation or an optimized route. Both

algorithms produce a route that starts at the farthest pick location and ends at the pick location nearest to the put location.

- **Distance calculation strategy** – The method that the system uses to calculate the distance between two locations. You can choose between a straight-line calculation or a city-block calculation. The choice affects the relative distances between locations, which in turn affects the sort order.
- **Coordinate retrieval strategy** – The method that the system uses to retrieve coordinate values for a location. Currently, the only available method retrieves the X, Y, and Z values directly from the location record.
- **Sort picking work lines** – A wave processing step that reorders pick work lines based on the spatial coordinates of the locations involved. This step only sorts pick lines that appear before the first put line in a work.

How sorting works

When the *Sort picking work lines* step runs during wave processing, the system performs the following steps for each work record:

1. *Identify the first put line in the work.* This line is the destination where the worker delivers the picked items.
2. *Collect all pick lines before the first put line.* The system sorts only these lines. Any lines after the first put line (such as additional pick and put cycles in the same work) remain in their original positions.
3. *Validate the work.* The system checks that there are at least two pick lines to sort, that the put location has valid coordinates, and that at least one pick location has valid coordinates. If validation fails, the system skips the work and keeps the pick lines in their original order.
4. *Calculate distances between locations* by using the warehouse's configured distance calculation strategy.
5. *Sort the pick lines* by using the warehouse's configured sorting algorithm so that the farthest pick location is first and the nearest pick location (closest to the put location) is last.
6. *Renumber the sorted pick lines* in the database so that the new order is preserved when the work is done.

Sorting algorithms

The *location sorting algorithm* controls how the system determines the order of pick work lines. Both algorithms produce a route where the worker starts at the farthest pick location from the

put location and finishes at the nearest pick location, so the last pick is close to the put destination.

Fast calculation algorithm

The *fast calculation algorithm* works best for warehouses that need quick route calculation with good (but not always optimal) results. It works well when pick locations are spread evenly or arranged in lines. It uses a nearest-neighbor approach:

1. Starting at the put location, the system finds the closest unvisited pick location.
2. From that location, it finds the next closest unvisited pick location.
3. This process continues until all pick locations are visited.
4. The resulting route visits locations from nearest-to-farthest (relative to the put). The system then reverses the route so that the worker starts at the farthest location and ends at the nearest.

This algorithm is fast because it makes a single pass through the pick locations. However, because it always chooses the next closest location, it can produce routes with crossing paths when locations are spread across the warehouse in complex patterns.

Optimized route algorithm

The *optimized route algorithm* works best in warehouses where minimizing total travel distance is critical. This algorithm is especially useful when pick locations are clustered in groups or scattered in patterns where the nearest-neighbor approach produces crossing paths. It starts with the same nearest-neighbor route as fast calculation, and then improves it:

1. The system builds the same initial route as fast calculation (as described in the previous section).
2. It repeatedly examines pairs of segments in the route and checks whether reversing the section between them produces a shorter total distance.
3. When it finds a shorter route, the system applies the change and continues checking.
4. The process repeats until no further improvements are found or the maximum number of improvement passes is reached.

The additional optimization step can eliminate crossing paths and produce noticeably shorter routes, especially for complex pick patterns. The trade-off is slightly longer calculation time.

When to choose each algorithm

Use the following guidelines to choose the best algorithm for your warehouse:

- *Fast calculation* is best for most scenarios. It produces near-optimal routes with minimal processing time and is suitable for high-volume wave processing.
- *Optimized route* is best when pick locations are scattered across the warehouse in patterns where the nearest-neighbor approach might create inefficient crossing paths. The optimization adds processing time but can reduce travel distance for complex pick patterns.

Example: Sorting algorithm comparison

Consider a warehouse with a put location (bay door) at coordinates (1, 1, 1) and five pick locations arranged in a pattern that challenges the nearest-neighbor approach:

Expand table

Location	X	Y	Z
PICK-A	1	50	1
PICK-B	50	1	1
PICK-C	50	50	1
PICK-D	25	25	1
PICK-E	25	50	1

- **Fast calculation result** – The nearest-neighbor approach starts at the put (1, 1, 1) and repeatedly selects the closest unvisited location. It then reverses the result. This approach can produce a route where the worker's path crosses itself, such as: *PICK-C* > *PICK-A* > *PICK-E* > *PICK-B* > *PICK-D* > *Put*.
- **Optimized route result** – The algorithm starts with the same route, then detects that reversing certain segments eliminates crossing paths. The result is a loop-free route with shorter total distance, such as: *PICK-C* > *PICK-B* > *PICK-D* > *PICK-E* > *PICK-A* > *Put*.

> **Tip**
>
> For warehouses with aisle-based layouts where pick locations fall along predictable lines (such as locations within the same aisle), *fast calculation* typically produces routes that are close to optimal. The *optimized route* algorithm provides the greatest benefit in open or irregularly laid-out warehouses where locations are scattered across multiple aisles and levels.

Distance calculation strategies

The *distance calculation strategy* controls how the system measures the distance between two locations. The sorting algorithm uses these distances to determine which pick locations are closer or farther from the put location and from each other. Choose a strategy that matches your warehouse layout so that the calculated distances reflect the actual paths that workers follow.

Straight line calculation strategy

The *straight-line calculation strategy* works best for open warehouse layouts where workers can move diagonally, such as open floor areas or wide-aisle warehouses. It calculates the direct (Euclidean) distance between two points. It assumes that workers can move freely in any direction between locations. It's calculated using the formula:

$$distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (|z_2| + |z_1|)^2}$$

The horizontal component uses the difference in X and Y coordinates. The vertical component uses the sum of the absolute heights of both locations (as explained in the Vertical distance calculation section).

City block calculation strategy

The *city-block calculation strategy* works best for warehouses with aisle-based layouts where workers walk along aisles and turn corners rather than cutting across open space. It calculates the distance along a grid-like path (also known as Manhattan distance). It assumes that workers must follow aisles and can't cut diagonally. It's calculated using the formula:

$$distance = |x_2 - x_1| + |y_2 - y_1| + |z_2| + |z_1|$$

The horizontal component sums the absolute differences in X and Y separately, which means the distance reflects the path a worker takes along aisles (first walking along one axis, then the other). The vertical component uses the sum of the absolute heights of both locations (as explained in the Vertical distance calculation section).

Vertical distance calculation

Both calculation strategies handle vertical movement (the Z axis) the same way: the system adds the absolute height of the starting location to the absolute height of the destination location.

For example, to move from shelf level 1 to shelf level 4, the vertical distance is *|1| + |4| = 5*, not *|4 - 1| = 3*.

This approach models the real cost of vertical movement in a warehouse. A worker can't move directly from one shelf level to another. Instead, the worker must climb down from the current level to the ground (costing the full height of that level) and then climb up to the target level (costing the full height of that level). For example, using a ladder or lift to descend from level 1 and then ascend to level 4 requires more effort than a simple height difference of 3 suggests.

> **Important**
>
> Two locations at the same shelf level (for example, both at Z = 2) still have a vertical distance of |2| + |2| = 4, not 0. This result reflects the reality that even on the same level, a worker must reach up to that shelf height at both the source and destination. If your warehouse has all locations on the ground level, set Z = 0 for all locations to eliminate the vertical component from distance calculations.

Distance calculation examples

The following examples show how the two strategies calculate distance differently for the same pair of locations.

Example 1: Two locations on the same level

You have two locations on the same shelf level (Z = 1) but different X and Y coordinates:

- Location A: (2, 3, 1)
- Location B: (8, 7, 1)

The following table shows the distance calculations for both strategies:

Expand table

Strategy	Calculation	Result
Straight line	$\sqrt{(8-2)^2+(7-3)^2+(1+1)^2} = \sqrt{36+16+4}$	**7.48**
City block	$\|8-2\|+\|7-3\|+\|1\|+\|1\| = 6+4+1+1$	**12**

The straight-line distance (7.48) is shorter because it assumes the worker walks diagonally. The city-block distance (12) is larger because the worker must follow the aisle grid: 6 units along one axis, then 4 units along the other, plus the vertical component at both locations.

Example 2: Two locations on different levels

You have two locations with the same X and Y coordinates but different shelf levels:

- Location A: (2, 3, 1)
- Location B: (2, 3, 4)

The following table shows the distance calculations for both strategies:

Expand table

Strategy	Calculation	Result
Straight line	$\sqrt{(2-2)^2 + (3-3)^2 + (4+1)^2} = \sqrt{0+0+25}$	5
City block	$\|2-2\| + \|3-3\| + \|4\| + \|1\| = 0+0+4+1$	5

When the locations differ only in shelf level, both strategies produce the same result (5 units) because vertical distance is calculated the same way in both: the worker climbs down from level 1 (1 unit) and climbs up to level 4 (4 units).

Example 3: Diagonal movement across a warehouse

You have two locations that differ significantly in X, Y, and Z coordinates:

- Location A: (1, 1, 1)
- Location B: (10, 10, 3)

The following table shows the distance calculations for both strategies:

Expand table

Strategy	Calculation	Result
Straight line	$\sqrt{(10-1)^2 + (10-1)^2 + (3+1)^2} = \sqrt{81+81+16}$	13.34
City block	$\|10-1\| + \|10-1\| + \|3\| + \|1\| = 9+9+3+1$	22

The difference between the strategies is most pronounced when the worker must cover large horizontal distances. The straight-line distance (13.34) assumes a direct diagonal path, while the city-block distance (22) reflects walking 9 units in one direction, 9 units in another direction, and then handling the vertical change at both levels.

Related information

- Set up warehouse spatial locations

- Wave processing

Last updated on 04/24/2026

Set up warehouse spatial locations (preview)

[This article is prerelease documentation and is subject to change.]

Warehouse spatial location sorting optimizes picking routes by assigning X, Y, and Z coordinates to warehouse locations. You can configure the sorting algorithms, add the sorting step to wave processing, and set up batch processing for large-volume warehouses.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Prerequisites

To use warehouse spatial locations, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.48 or later.
- The feature named *Warehouse spatial location* must be turned on in feature management.
- Warehouse locations must already be defined for your warehouse.

> **Note**
>
> After you turn on the *Warehouse spatial location* feature, you can turn it off if needed. However, all coordinate values and configurations that you entered remain in the database.

Configure warehouse-level sorting settings

Before you assign coordinates to individual locations, configure the sorting algorithms and strategies at the warehouse level.

To configure warehouse-level spatial location settings, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.

2. Select the warehouse that you want to configure.

3. On the **Warehouse** FastTab, in the **Warehouse coordinates configuration** section, make the following settings:

    - **Distance calculation strategy** – Select the method that the system uses to calculate the distance between two locations. The choice affects how distances are measured, which in turn affects the sort order of pick work lines. You can learn more about how each strategy calculates distance in Distance calculation strategies. Choose one of the following options:
        - *Straight Line* – Calculates the direct distance between two points, assuming the worker can move freely in any direction. Use this strategy for open warehouse layouts where workers can walk diagonally.
        - *City Block* – Calculates the distance along a grid-like path (walking along one axis and then another), assuming the worker must follow aisles. Use this strategy for warehouses with aisle-based layouts where workers can't cut across aisles.
    - **Coordinate retrieval strategy** – Select the method that the system uses to retrieve a location's coordinates. Currently, only one setting is available and is selected by default:
        - *Location coordinates* – Retrieves the X, Y, and Z coordinate values directly from the fields on the location record.
    - **Location sorting algorithm** – Select the algorithm that the system uses to sort pick work lines. Both algorithms sort picks so that the worker starts at the farthest location from the put (destination) location and finishes at the nearest. You can learn more about how each algorithm works in Sorting algorithms. Choose one of the following options:
        - *Fast calculation* – Builds a route by starting at the put location and repeatedly selecting the next nearest unvisited pick location, then reverses the result. Generates a route quickly and produces good results in most scenarios. Recommended for high-volume wave processing where calculation speed is the priority.
        - *Optimized route* – Starts with the same route as the fast calculation, then improves it by checking whether rearranging segments produces a shorter path. Recommended when minimizing travel distance is the priority. This option might take slightly longer to calculate but can produce noticeably shorter routes for complex pick patterns.

4. Repeat these steps for each warehouse where you want to use spatial location sorting. You can have different settings for different warehouses based on their layouts and operational priorities.

Assign coordinates to warehouse locations

After you configure the warehouse-level settings, assign X, Y, and Z coordinates to each warehouse location. The system uses these coordinates to calculate distances between locations and determine the optimal pick route.

To assign coordinates to warehouse locations, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.
2. Select a location record.
3. Make the following settings:
    - **X coordinate** – Enter the position of the location along the X axis of the warehouse. This value typically represents the aisle or horizontal position.
    - **Y coordinate** – Enter the position of the location along the Y axis of the warehouse. This value typically represents the position within an aisle (depth).
    - **Z coordinate** – Enter the position of the location along the Z axis of the warehouse. This value typically represents the vertical level (for example, the shelf or rack level). Set this value to 0 for ground-level locations to exclude vertical distance from calculations.
4. Repeat these steps for each location in each warehouse where you want to use spatial location sorting.

> **Tip**
>
> Use a consistent coordinate system across all locations in a warehouse. For example, you might use the aisle number for X, the bay number for Y, and the shelf level for Z.

Add the *Sort picking work lines* step to your wave templates

To use spatial location sorting during wave processing, add the *Sort picking work lines* step to the relevant wave templates. Learn more about wave templates in Wave templates.

To add the wave step, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Select the wave template that you want to modify.
3. On the **Methods** FastTab, in the **Remaining methods** list, find and select the method with **Name** *Sort picking work lines*.
4. Select the **Add** button (right arrow) to move it to the **Selected methods** list.
5. Use the **Move up** and **Move down** arrows to position the step after the work creation step (such as *Create work*) and before the release step (if there is one). The sort must run after work is created so that there are pick lines to sort.

> **Note**
>
> The *Sort picking work lines* method only appears in the lists on the **Methods** FastTab after the *Warehouse spatial location* feature is turned on in **Feature management**. If the method still doesn't appear, go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**, and select **Regenerate methods** from the Action Pane.

Configure batch processing for *Sort picking work lines*

By default, the *Sort picking work lines* step runs synchronously during wave processing. For high-volume warehouses, you can improve performance by configuring the step to run as parallel batch tasks. When you enable batch processing, the system splits work across multiple parallel batch tasks, which can significantly reduce sorting time during wave processing.

To configure batch processing, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
2. Select the *sortPickingWorkLines* method in the list.
3. On the Action Pane, select **Task configuration**. The **Wave post method task configuration** page opens.
4. On the Action Pane, select **New** to add a configuration row.
5. In the **Warehouse** field, select the warehouse that you want to set up to use batch processing.
6. Make the following settings:

- **Maximum number of batch tasks** – Enter the maximum number of parallel batch tasks that the system creates to process the sort step. The actual number of tasks is the lesser of this value and the number of work records to sort. A higher value allows more parallel processing but uses more batch server capacity.
- **Wave processing batch group** – Enter the batch group to use for the sorting batch tasks. If you leave this field blank, the system uses the default wave processing batch group that you configure on the **Warehouse management parameters** page.

7. Repeat these steps for each warehouse where you want to use batch processing for the sort step.

> **Tip**
>
> Start with a small number of batch tasks (for example, 2-4) and increase the value based on your wave volume and batch server capacity. Each batch task processes one work record at a time from a shared queue, so the work is automatically distributed across available tasks.

> **Important**
>
> The system uses batch processing only when a task configuration record exists for the warehouse. If you don't set up a configuration, the sort step runs synchronously during wave execution.

Example scenario

This scenario shows how spatial location sorting reorders pick work lines to reduce travel distance, and how the distance calculation strategy affects the results.

Set up the scenario

This scenario uses a warehouse with four pick locations and one bay door (put location). The pick locations are arranged so that without sorting, the worker would zigzag across the warehouse. After sorting, the pick lines are reordered so the worker starts at the farthest location and works back toward the bay door in a more efficient path.

To set up the scenario so you can run it, follow these steps to set up the warehouse and wave template:

1. Configure a warehouse with the following settings:

    - **Distance calculation strategy** – *Straight Line*
    - **Coordinate retrieval strategy** – *Location coordinates*
    - **Location sorting algorithm** – *Fast calculation*

2. Create four pick locations with the following coordinates:

Expand table

Location	X	Y	Z	Description
PICK-001	3	2	1	Near the bay door, ground level
PICK-002	10	8	1	Far side of warehouse, ground level
PICK-003	6	5	1	Middle of warehouse, ground level
PICK-004	10	8	3	Far side of warehouse, shelf level 3

3. Create a bay door (put) location at coordinates (1, 1, 1). This location is where the worker delivers the picked items.

4. Add the *Sort picking work lines* step to the wave template, positioned after the work creation step.

Run the scenario

To run the scenario, follow these steps:

1. Create a sales order that requires items from all four pick locations.
2. Process the wave. The system creates a work record with four pick lines and one put line.
3. The *Sort picking work lines* step runs and reorders the pick lines.

Result

Before sorting, the pick work lines are in the order the system created them, which doesn't consider location proximity, as shown in the following table:

Expand table

Line	Type	Location	Coordinates
1	Pick	PICK-002	(10, 8, 1)

Line	Type	Location	Coordinates
2	Pick	PICK-001	(3, 2, 1)
3	Pick	PICK-004	(10, 8, 3)
4	Pick	PICK-003	(6, 5, 1)
5	Put	BAY-DOOR	(1, 1, 1)

With this order, the worker would zigzag across the warehouse: *far side* > *near the door* > *far side again* > *middle* > *bay door*.

During sorting, the system calculates the distance from the put location to each pick location and reorders the pick lines so the worker starts at the farthest location and works back toward the bay door. The following table shows how the distances from the put location (1, 1, 1) are calculated using the *straight-line strategy*:

Expand table

Location	Coordinates	Calculation	Distance from put
PICK-001	(3, 2, 1)	$\sqrt{(3-1)^2+(2-1)^2+(1+1)^2} = \sqrt{4+1+4}$	3.00
PICK-003	(6, 5, 1)	$\sqrt{(6-1)^2+(5-1)^2+(1+1)^2} = \sqrt{25+16+4}$	6.71
PICK-002	(10, 8, 1)	$\sqrt{(10-1)^2+(8-1)^2+(1+1)^2} = \sqrt{81+49+4}$	11.58
PICK-004	(10, 8, 3)	$\sqrt{(10-1)^2+(8-1)^2+(3+1)^2} = \sqrt{81+49+16}$	12.08

The sorted work now looks like this:

Expand table

Line	Type	Location	Distance from put
1	Pick	PICK-004	12.08 (farthest)
2	Pick	PICK-002	11.58
3	Pick	PICK-003	6.71
4	Pick	PICK-001	3.00 (nearest)
5	Put	BAY-DOOR	--

The worker now follows an efficient path: *far shelf* > *far ground* > *middle* > *near the door* > *bay door*. Each pick is closer to the destination than the previous one, eliminating the back-and-forth travel.

> **Note**
>
> The sorting step doesn't move the put line (line 5). It only reorders the pick lines before the first put. If the work contained additional pick and put cycles after the first put, those lines would also stay in their original positions.

How the distance strategy changes the result

If you change the **Distance calculation strategy** to *City Block*, the system calculates distances differently. The city-block strategy assumes workers follow aisles instead of walking diagonally:

The following table shows how the city-block strategy calculates distances from the put location (1, 1, 1):

Expand table

Location	Coordinates	Calculation	Distance from put
PICK-001	(3, 2, 1)	$\|3-1\| + \|2-1\| + \|1\| + \|1\| = 2+1+1+1$	5
PICK-003	(6, 5, 1)	$\|6-1\| + \|5-1\| + \|1\| + \|1\| = 5+4+1+1$	11
PICK-002	(10, 8, 1)	$\|10-1\| + \|8-1\| + \|1\| + \|1\| = 9+7+1+1$	18
PICK-004	(10, 8, 3)	$\|10-1\| + \|8-1\| + \|3\| + \|1\| = 9+7+3+1$	20

The sorted pick order (farthest first) is now:

Expand table

Line	Type	Location	Distance from put
1	Pick	PICK-004	20 (farthest)
2	Pick	PICK-002	18
3	Pick	PICK-003	11
4	Pick	PICK-001	5 (nearest)

Line	Type	Location	Distance from put
5	Put	BAY-DOOR	--

In this example, both strategies produce the same sorted order because the relative distances between locations are similar. However, the city-block distances are larger because they account for aisle-following movement. Notice that the gap between PICK-004 (distance 20) and PICK-002 (distance 18) is larger with city block than with straight line (12.08 vs 11.58). This difference exists because the city-block strategy adds the full shelf height at both levels (level 3 + level 1 = 4) as a separate cost, whereas the straight-line strategy combines vertical and horizontal distance into a single direct measurement.

> **Tip**
>
> The distance strategy doesn't always change the sorted order, but it can. When pick locations have similar horizontal positions but different shelf levels, the city-block strategy gives more weight to vertical distance, which can cause the sort order to differ from the straight-line strategy. Choose the strategy that best reflects how workers actually move through your warehouse.

Related information

- Warehouse spatial location overview
- Warehouse configuration overview
- Wave processing
- Wave templates

Last updated on 04/24/2026

Control warehouse work by using work templates and location directives

08/06/2025

This article describes how to use work templates and location directives to determine how and where work is carried out in the warehouse.

The instructions that warehouse workers receive on a mobile device are determined by the Dynamics 365 Supply Chain Management work templates that you set up to define the various warehouse processes and tasks. Work templates determine how the work is performed for each warehouse process. By linking a location directive to work templates, you can help guarantee that work occurs in specific physical areas of the warehouses.

Work templates

The **Work templates** page lets you define the work operations that must be performed in the warehouse. Typically, warehouse work operations consist of a pair of actions: a warehouse worker picks up on-hand inventory in one location and then puts the picked inventory down in another location.

Work templates consist of a header and associated lines. Each work template is for a specific *work order type*. Many work order types are associated with source documents, such as purchase or sales orders. However, other work order types represent separate warehouse processes, such as cycle counting. The *work pool ID* lets you organize work into groups.

Use the settings in the work header definition to determine when a new work record should be created. For example, you can set a maximum number of pick lines and a maximum expected pick time. Then, if the work for a sales order picking process exceeds either of those values, that work is split into two pieces of work. Use the **General** tab of the work template to set limits associated with each work record and set up work-split criteria such as **Estimated pick time**, **Volume**, **Weight**, **Quantity**, and **Unit**.

> **Note**
>
> Items associated with multiple units of measure (such as pieces, boxes, and pallets) should be set up with a unit sequence group and relevant conversion ratios. These settings allow you, for example, to create an order using one unit (such as boxes) and then use a different unit (such as pieces) to split the work associated with the order. Learn more in Unit of measure and stocking policies.

Use the **Work header breaks** button to define when the system should create new work headers. For example, to create a work header for each *order number*, select **Edit query** on the Action Pane, and then add the **Order number** field to the **Sorting** tab of the query editor. Fields that are added to the **Sorting** tab are available for selection as *grouping fields*. To set your grouping fields, select **Work header breaks** on the Action Pane, and then, for each field that you want to use as a grouping field, select the check box in the **Group by this field** column.

The work lines represent the physical tasks that are required to process the work. For example, for an outbound warehouse process, there might be one line for picking up the items in the warehouse and another line for putting those items into a staging area. There can then be an additional line for picking the items from staging and another line for putting the items into a truck as part of the loading process. You can set a *directive code* on work template lines. A directive code is linked to a location directive, and therefore helps ensure that the warehouse work is processed in the correct location in the warehouse.

You can set up a query to control when a particular work template is used. For example, you can set a limitation so that a particular template can be used for work only in a specific warehouse. Alternatively, you might have several templates that create work for outbound sales order processing, depending on the sales origin. The system uses the **Sequence number** field to determine the order that the available work templates are assessed in. Therefore, if you have a very specific query for a particular work template, you should give it a low sequence number. That query will then be evaluated before the other, more general queries.

> ⓘ **Note**
>
> To prevent the system from automatically overwriting work template *sequence numbers* after a template has been deleted, turn on the *Preserve work template sequence numbers on delete* feature in **Feature management**.

To stop or pause a work process, you can use the **Stop work** setting on the work line. In that case, the worker who is performing the work won't be asked to perform the next work line step. To move on to the next step, that worker or another worker must select the work again. You can also separate the tasks within a piece of work by using a different *work class ID* on the work template lines.

Location directives

Location directives are rules that help identify pick and put locations for inventory movement. For example, in a sales order transaction, a location directive determines where the items will

be picked, and where the picked items will be put. Location directives consist of a header and associated lines, and you create them on the **Location directives** page.

On the header, each location directive must be associated with a *work order type* that specifies the type of inventory transaction that the directive will be used for, such as sales orders, replenishment, or raw material picking. The *work type* specifies whether the location directive will be used for picking or putting work, or for some other warehouse process, such as counting or inventory status changes. You must also specify a *site* and a *warehouse*. A *directive code* that you specify on the header can be used to link the location directive to one or more work templates.

As for work templates, you can set up a query to determine when a particular location directive is used. For example, you can specify that when e-commerce is the origin of a sales order, the inventory must be picked up from a dedicated area in the warehouse. The system uses the **Sequence number** field to determine the order that the available location directives are assessed in.

The location directive lines set additional restrictions on the application of the location finding rules. You can specify a minimum quantity and a maximum quantity that the directive should apply to, and you can specify that the directive should be for a specific inventory unit. For example, if the unit of measure is pallets, the items in pallets can be put in a specific location. You can also specify whether the quantity can be split across multiple locations. Like the location directive header, each location directive line has a sequence number that determines the order that the lines are assessed in.

Location directives have one additional level of detail: *location directive actions*. You can define multiple location directive actions for each line. Once again, a sequence number is used to determine the order that the actions are assessed in. On this level, you can set up a query to define how to find the best location in the warehouse. You can also use predefined **Strategy** settings to find an optimal location.

For more information about how to create and configure location directives, see Create a location directive.

How location directives work

Location directives determine *where* items should be picked and *where* they should be put. The system evaluates a location directive against each work line and then selects a location, based on the work line details. The system first finds all location directives that match a particular work line (for example, they are for the correct warehouse and match the query). It then sequentially evaluates the directives that it has found.

> **Note**
>
> There are special cases where the pick location or put location is pre-selected. For example, during *purchase registration*, the first pick is always from the location where the registration occurs. Another example is *inventory movement by template*, where the warehouse worker selects the pick location, and only the put locations are found through location directives.

Related information

- Video: Warehouse management configuration deep dive
- Help article: Work with location directives

Dynamic work classification

Warehouse managers often need different work pools, priorities, or bay door assignments depending on the carrier, shipping deadline, or type of items being picked. Without dynamic classification, each combination requires its own work template, which can lead to a large and complex configuration.

Dynamic work classification lets you use Power Fx formulas to determine these work parameters at runtime instead of configuring them statically on each work template. A single work template combined with a classification rule can replace many static templates, making your configuration simpler to maintain and more adaptable to changing business requirements.

Dynamic work classification provides two main capabilities:

- **Classification at work creation** – When warehouse work is created, the system evaluates a Power Fx formula to override the work pool, work priority, location directive codes, and work classes on the generated work. The formula can look up values from the work header and associated records, including the load, shipment, wave, and transportation appointment.
- **Reclassification after load changes** – If a load changes after work has been created (for example, the carrier is reassigned), the system can reevaluate the formula and update the associated work. Work can also be reevaluated on a schedule if the formula depends on the current date or time.

To start using dynamic work classification, you must configure how loads trigger reclassification, create one or more classification rules that contain your Power Fx formulas, and associate each rule with the work templates that should use it.

Prerequisites

To use the feature described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.48 or later.
- The feature named *(Production Ready Preview) Dynamic work classification* must be turned on in feature management. As of Supply Chain Management version 10.0.49, this feature is turned on by default.

Configure work reclassification on load update

To control whether dynamic work classification rules can reclassify work when loads change, configure the **Work classification on load update** option. Work is only reclassified by rules that are explicitly set to **Reclassify work on load update**, so you can choose which rules should apply when a load changes. The system always classifies work at creation if a rule is associated with the work template, regardless of this setting.

To configure this setting, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, open the **Work** FastTab.
3. Set the **Work classification on load update** field to one of the following values:

    - *Disabled* – No action is performed when a load is updated.
    - *Synchronous* – All work associated with an updated load is reclassified immediately after the load is updated.

> ⓘ **Note**
>
> When using synchronous reclassification, the user might experience a brief wait during load updates while the system processes all associated work headers.

Create dynamic work classification rules

Classification rules define the Power Fx formulas that determine how work should be classified. Each rule specifies which fields to override and the logic for calculating the new values.

To create a rule, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Work** > **Dynamic work classification rules**.

2. Select **New** to create a rule.

3. Make the following settings in the header of the new rule:

    - **Dynamic work classification rule** – Enter a name for the rule.
    - **Description** – Enter a short description for the rule.
    - **Rule scope** – Specify when the rule's formulas run during work creation and which data they can reference. The scope determines whether classification is applied only at

the work header level or also at the initial work line level. Choose one of the following values:
    - *Work* – The formula runs once, after the work header is created. Use this scope when classification depends only on header-level properties such as the carrier or shipping time.
    - *Work and initial work lines* – The formula runs twice. First, as the initial pick work line is selected, the system evaluates a separate *initial work line formula* and applies overrides to that line. Then, after all initial work lines are collected, the system runs the main formula against the work header. Use this scope when classification depends on item-level properties such as the pick zone.

4. On the **Recalculation** FastTab, make the following settings to control how and when the rule is reevaluated if a load changes after the initial work creation:

    - **Reclassification schedule ID** – This setting is reserved for future use and has no effect in the current release.
    - **Reclassify work on load update** – Set this option to *Yes* to run the classification again each time a load associated with the work header changes. This setting ensures that any updates to the load are reflected in the work classification. This setting only applies if the **Work classification on load update** parameter is set to allow reclassification (see the previous section).

5. On the **Power Fx formula** FastTab, enter a formula that returns a record with the fields you want to override. See the next sections for details on writing formulas.

6. If the **Rule scope** is set to *Work and initial work lines*, the page includes a **Initial work line formula** FastTab. This formula runs when the first work line is assigned to a new work record. See the next sections for details on writing formulas.

> **Tip**
>
> - Use the **Copy** button on the Action Pane to duplicate an existing rule as a starting point for a new one.
> - Use the **Preview** button to test a rule against an existing work header and see the results without making changes.
> - The **Work reclassification schedules** button is reserved for future use and has no effect in the current release.

Write work classification formulas

On the **Power Fx formula** FastTab of each classification rule, enter a formula that returns a record with one or more of the following fields.

The following table describes the fields that a work classification formula can return.

Expand table

Field	Type	Description
`WorkPoolId`	String	Overrides the work pool on the created work.
`DefaultWorkPriority`	Number	Overrides the priority on the created work.
`DirectiveCodeOverrides`	Record	A record where each field name is a location directive code to replace and the value is the new directive code. Applies to all pick/put pairs.
`WorkClassOverrides`	Record	A record where each field name is a work class to replace and the value is the new work class. Applies only to the second and subsequent pick/put pairs. To change the work class of the first pick/put pair, use the initial work line formula instead.

The following example shows a formula that returns all four fields.

powerfx

```
{
    WorkPoolId: "PoolA",
    DefaultWorkPriority: 100,
    DirectiveCodeOverrides: {
        BAYDOOR: "BAYDOOR-C1",
        STAGING: "STAGING-C1"
    },
    WorkClassOverrides: {
        'Standard Class': "Express Class"
    }
}
```

> **Important**
>
> Returning `Blank()` for a field (or omitting it) preserves the original value from the work template. Returning an empty string (`""`) actively clears the value. For example, returning `Blank()` for `WorkPoolId` keeps whatever work pool was set on the work template, while returning `""` removes the work pool entirely.

Available data objects for work formulas

The following table lists the objects that you can reference in a work classification formula.

Expand table

Object	Description
`workTable`	The work header that was generated.
`workTable.Load`	The load record associated with the work header.
`workTable.Shipment`	The shipment record associated with the work header.
`workTable.Wave`	The wave record associated with the work header.
`workTable.Appointment`	The first transportation management (TMS) appointment associated with the load.

Write initial work line formulas

If you set the **Rule scope** to *Work and initial work lines*, enter an additional formula on the **Initial work line formula** FastTab. This formula runs as the first pick work line is assigned to a new work record and can override the following fields.

Expand table

Field	Type	Description
`WorkPoolId`	String	Overrides the work pool on the created work header.
`WorkClassId`	String	Overrides the work class of the initial pick/put work lines.

Available data objects for initial work line formulas

The following table lists the objects that you can reference in an initial work line formula.

Expand table

Object	Description
`workLine`	The initial pick work line that was generated.
`workLine.InventTable`	The item master record for the item being picked.
`workLine.WHSInventTable`	The warehouse item settings for the item being picked.

Object	Description
`workLine.Load`	The load record associated with the work line.
`workLine.LoadLine`	The load line associated with the initial pick line.
`workLine.Shipment`	The shipment record associated with the work line.

Associate rules with work templates

After creating a classification rule, associate it with the work template that should use it. During work creation, the system evaluates the rule's formula and applies the returned overrides to the generated work.

Go to **Warehouse management** > **Setup** > **Work** > **Work templates**, select a template, and set the **Dynamic work classification rule** field to the rule you want to use for that template.

> **Note**
>
> When you use dynamic work classification with a work template, make sure the template's grouping settings align with your formula logic. For example, if your formula classifies work based on zone, the work template should group work creation by zone so that each work header contains lines from only one zone.

Examples

The following examples illustrate common scenarios where dynamic work classification can simplify your configuration.

Example 1: Set work pool and priority by carrier

A warehouse uses three carriers that pick up shipments at different times during the day. The warehouse manager wants to route each carrier's work to a separate work pool and assign higher priority to the earliest pickup.

Without dynamic classification, this setup requires three work templates. Each template filters on a different carrier code, each with its own work pool and priority settings.

With dynamic classification, a single work template and one rule handle all three carriers.

powerfx

```
Switch(workTable.Load.CarrierCode,
    "Carrier1", {WorkPoolId: "Carrier1", DefaultWorkPriority: 100},
    "Carrier2", {WorkPoolId: "Carrier2", DefaultWorkPriority: 50},
    "Carrier3", {WorkPoolId: "Carrier3", DefaultWorkPriority: 10}
)
```

If you later reassign the load to a different carrier, the system can automatically reclassify the work when reclassification on load update is turned on.

Example 2: Direct put-away to carrier-specific bay doors

Building on example 1, the warehouse manager also wants outbound put lines directed to different bay doors depending on the carrier. With static configuration, each work template needs a different location directive code.

With dynamic classification, you can add directive code overrides to the same formula.

powerfx

```
Switch(workTable.Load.CarrierCode,
    "Carrier1", {
        WorkPoolId: "Carrier1",
        DefaultWorkPriority: 100,
        DirectiveCodeOverrides: {BAYDOOR: "BAYDOOR-C1"}
    },
    "Carrier2", {
        WorkPoolId: "Carrier2",
        DefaultWorkPriority: 50,
        DirectiveCodeOverrides: {BAYDOOR: "BAYDOOR-C2"}
    },
    "Carrier3", {
        WorkPoolId: "Carrier3",
        DefaultWorkPriority: 10,
        DirectiveCodeOverrides: {BAYDOOR: "BAYDOOR-C3"}
    }
)
```

Example 3: Prioritize work by shipping deadline

Instead of classifying by carrier, this example prioritizes work based on how close the load is to its scheduled shipping time. Loads with earlier deadlines receive higher priority, ensuring workers pick the most time-sensitive orders first.

The following formula converts the scheduled shipping time into a numeric priority. A shipment scheduled for 9:00 AM becomes priority 900, while a 3:00 PM shipment becomes priority 1500.

powerfx

```
{
    DefaultWorkPriority:
        If(Not(IsBlank(workTable.Load.LoadSchedShipUTCDateTime)),
            If(IsToday(workTable.Load.LoadSchedShipUTCDateTime),
                Hour(workTable.Load.LoadSchedShipUTCDateTime) * 100
                    + Minute(workTable.Load.LoadSchedShipUTCDateTime),
                3000),
            Blank())
}
```

The formula assigns a priority of 3000 to loads scheduled for a future day (beyond the maximum time-based priority of 2359) and returns `Blank()` for loads with no scheduled date, which preserves the work template's default priority.

An alternative approach calculates priority based on the number of minutes remaining until the shipping deadline.

powerfx

```
{
    DefaultWorkPriority:
        If(Not(IsBlank(workTable.Load.LoadSchedShipUTCDateTime)),
            If(IsToday(workTable.Load.LoadSchedShipUTCDateTime),
                Max(0, DateDiff(Now(), workTable.Load.LoadSchedShipUTCDateTime,
                    TimeUnit.Minutes)),
                24 * 60),
            Blank())
}
```

> **Important**
>
> The time-remaining approach requires the reclassification batch job to run on a recurring schedule (for example, every hour) to keep priorities current. Depending on the number of open work headers, this requirement can add load to the system. The time-of-day approach (first formula) is more efficient because work only needs reclassification when the load's scheduled shipping time changes.

Example 4: Classify work by pick zone

This example assigns a different work class to the initial pick/put pair based on the zone where items are picked. For example, picks from a freezer zone can be restricted to workers with appropriate cold-weather gear.

To use this approach, set the **Rule scope** to *Work and initial work lines* and configure the work template to group work by zone. This configuration ensures each work header contains lines from only one zone.

The initial work line formula concatenates a prefix with the zone ID to create a zone-specific work class.

powerfx

```
{
    WorkClassId: Concatenate("S-", workLine.ZoneId)
}
```

Alternatively, you can set the work pool instead of the work class.

powerfx

```
{
    WorkPoolId: Concatenate("S-", workLine.ZoneId)
}
```

Last updated on 07/27/2026

Work with location directives

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Location directives are rules that help identify pick and put locations for inventory movement. For example, in a sales order transaction, a location directive determines where the items are picked, and where the picked items are put. Location directives consist of a header and associated lines. You create them for specific *work order types*.

> **Note**
>
> This article applies to features in the **Warehouse management** module. It doesn't apply to features in the Inventory management module.

Use location directives to perform the following tasks:

- Put away incoming items.
- Pick and stage items for outbound transactions.
- Pick and put raw materials for production.
- Replenish locations.

Prerequisites

Before you create a location directive, complete the following steps to make sure that the prerequisites are in place.

1. Turn on the required license key. Go to **System administration** > **Setup** > **License configuration**, expand the **Trade** license key, and then select the **Warehouse and Transportation management** configuration key. You need admin access for this step.
2. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.
3. Create a warehouse.
4. On the **Warehouse** FastTab, set the **Use warehouse management processes** option to *Yes*.
5. Create locations, location types, location profiles, and location formats. Learn more in Configure locations in a WMS-enabled warehouse.

6. Create sites, zones, and zone groups. Learn more in Warehouse set up and Configure locations in a WMS-enabled warehouse.

Work order types for location directives

Many of the fields that you can set for location directives are common to all work order types. However, other fields are specific to particular work order types.

> **Note**
>
> Two work order types, *Canceled work* and *Cycle counting*, are used only by the system. You can't create location directives for these work order types.

The tables in the following subsections list the common and work order type–specific fields that are available when you set up a location directive.

Fields common to all work order types

The following table lists the fields common to all work order types.

Expand table

FastTab	Field
Location directives	Work type
Location directives	Site
Location directives	Warehouse
Location directives	Directive code
Location directives	Scope
Lines	Sequence number
Lines	From quantity
Lines	To quantity
Lines	Unit
Lines	Locate quantity
Lines	Restrict by unit

FastTab	Field
Lines	Round up to unit
Lines	Locate packing quantity
Lines	Allow split
Location Directive Actions	Sequence number
Location Directive Actions	Name
Location Directive Actions	Fixed location usage
Location Directive Actions	Allow negative inventory
Location Directive Actions	Batch enabled
Location Directive Actions	Strategy

Fields that are specific to work order types

The following table lists the fields that are specific to particular work order types.

Expand table

FastTab	Field	Work order type
Location directives	Locate by	Purchase orders
Location directives	Locate by	Transfer receipt
Location directives	Applicable disposition code	Purchase orders
Location directives	Disposition code	Purchase orders
Location directives	Applicable disposition code	Finished goods put away
Location directives	Disposition code	Finished goods put away
Location directives	Applicable disposition code	Return orders
Location directives	Disposition code	Return orders
Location directives	Applicable disposition code	Kanban put away
Location directives	Applicable disposition code	Kanban picking
Lines	Immediate replenishment template	Sales orders
Lines	Immediate replenishment template	Raw material picking

FastTab	Field	Work order type
Lines	Immediate replenishment template	Transfer issue
Lines	Immediate replenishment template	Kanban picking

Open the Location directives page

To open the **Location directives** page, go to **Warehouse management** > **Setup** > **Location directives**.

From there, you can view, create, and edit your location directives by using the commands on the Action Pane. For information about how to use all the fields that are available on the page, see the remaining sections of this article.

Action Pane

The Action Pane on the **Location directives** page contains buttons that you can use to create, edit, and delete directives (**Edit**, **New**, and **Delete**). It also contains the following buttons that let you adjust the sequence that the location directive is processed in and configure a query that defines the criteria for applying the location directive:

- **Move up** – Move up the selected location directive in the sequence. For example, you can move it from sequence number 4 to sequence number 3.
- **Move down** – Move the selected location directive down in the sequence. For example, you can move it from sequence number 4 to sequence number 5.
- **Copy** – Open a dialog box where you can create an exact copy of the current location directive.
- **Edit query** – Open a dialog box where you can define the conditions that the selected location directive should be processed under. For example, you might want it to apply only to a specific warehouse.
- **Acceptance tests** – Open a page where you can set up automated tests to determine how your location directives behave under different starting conditions. In this way, you can quickly validate your directives as you create and maintain them. Learn more in Test location directives with acceptance tests.

Location directives header

The location directive header includes the following fields for the sequence number and descriptive name of the location directive:

- **Sequence number** – This field indicates the sequence that the system tries to apply each location directive in for the selected work order type. Low numbers are applied first. You can change the sequence by using the **Move Up** and **Move Down** buttons on the Action Pane.
- **Name** – Enter a descriptive name for the location directive. This name should help identify the general purpose of the directive. For example, enter *Sales order picking in warehouse 24*.

Location directives FastTab

The fields on the **Location directives** FastTab are specific to the work order type that you select in the **Work order type** field in the list pane.

- **Work type** – Select the type of work that must be performed. The available values depend on the type of inventory transaction that you selected in the **Work order type** field. Select one of the following values:
  - *Put* – The location directive tries to find the most ideal location to put or locate inventory that comes into the system from receiving, production, or inventory adjustments. It can also be used to define the put to the stage location or the final bay door shipping location.
  - *Pick* – The location directive tries to find the most ideal locations to physically reserve inventory from (that is, create work). The pick can be completed (that is, the pick work line can be closed) even if the work isn't completed. The user can complete physical picking. In the system, that action is a pick step. The user can then cancel from the mobile device and complete the work later. However, the work header is first closed when the final put is completed.

  > **Important**
  >
  > The other values in the **Work type** field aren't relevant for location directives. They appear only because the field isn't filtered to match the selected work order type.

- **Directive code** – Select the directive code to associate with a work template or replenishment template. On the **Directive code** page, you can create new codes for connecting work templates or replenishment templates to location directives. You can also use directive codes to establish a link between any work template line and a location directive (such as the bay door or stage location).

  > **Tip**

If you set a directive code, the system doesn't search location directives by sequence number when it needs to generate work. Instead, it searches by directive code. In this way, you can be more specific about the location directive that is used for a particular step in a work template, such as the step for staging the materials.

- **Scope** – Use this option to specify the scenarios that the location directive applies to.

Expand table

Scope setting	Single order with one item	Multiple orders with the same item	Single order with multiple items	Multiple orders with multiple items
Single item	Yes	Yes	No	No
Multiple items	No	No	Yes	Yes
Single item or order	Yes	Yes	Yes	No
All	Yes	Yes	Yes	Yes

The following table describes when the scopes are available, and whether they allow for the **Edit query** function.

Expand table

Scope	Supported work type	Supported work order types	Allow for Edit query
Single item	All	All	Yes
Multiple items	All	All	No
Single item or order	Puts	Co-product and by-product put away, finished goods put away, kanban put away, purchase orders, quality orders, replenishment, return orders, sales orders, transfer issue, and transfer receipt	Yes
All	Puts	All	No

Note

- To do puts for both multiple items and single items, you must ensure that location directives exist that cover both scenarios. For example, you might set up one or more *Single item or order* location directives to cover scenarios that require fine-tuning (such as through edits to the query), and then one or more *All* location directives to cover the remaining scenarios.
- Although *Single item* and *Multiple items* scopes can be used for puts, this approach typically leads to redundant configurations. Consider using *Single item or order* and *All* scopes instead, because this approach produces a cleaner setup.
- When you use the *Single item or order* scope, receiving flows (like purchase orders) add the order number to the location directive query during work creation. To prevent unexpected behavior, don't add query ranges for the lines in such cases.

- **Applicable disposition code** – Specify whether the disposition code of the location directive must match the disposition code that is applied when the item is received, or whether the location directive can be selected based on any disposition code. If you select *Exact match*, and the **Disposition code** field is blank, only blank disposition codes are considered for this location directive.

  > ⓘ **Note**
  >
  > This field is available only for selected work order types. For a complete list, see the Fields that are specific to work order types section earlier in this article.

- **Locate by** – Specify whether the putaway quantity should be the whole quantity on the license plate, or whether it should be item by item. Use this field to help ensure that all the contents on a license plate are put into one location, and that the system doesn't suggest that you split the contents into several locations for *ASN* (license plate receiving), *Mixed license plate* receiving, and *Cluster* receiving processes. The behavior of the location directive query, the lines, and the location directive actions vary, depending on the value that you select. The **Lines** FastTab is only used when the **Locate by** is set to *Item*.

  > ⓘ **Note**
  >
  > Depending on the order type, this field has different options and is only accessible for some work order types. You can use the *ASN* option for both transfer receipts and purchase orders.

- **Disposition code** – Use this field for location directives that have a work order type of *Purchase orders*, *Finished goods putaway*, or *Return orders*, and a work type of *Put*. Use it to guide the flow to use a specific location directive, depending on the disposition code that a worker selects in the Warehouse Management mobile app. For example, you can direct returned goods to an inspection location before they're returned to stock. A disposition code can be linked to an inventory status. In this way, it can be used to change the inventory status as part of a receiving process. For example, you have a disposition code, *QA*, that sets the inventory status to *QA*. You can then have a separate location directive to move that inventory to a quarantine location.

  > **Note**
  >
  > This field is available only for selected work order types. For a complete list, see the Fields that are specific to work order types section.

Warehouse selection FastTab

Use the **Warehouse selection** FastTab to specify the warehouse and site where the location directive applies.

- **Warehouse selection** – Select one of the following values:
  - *All* – Use the location directive for all warehouses where a more specific location directive isn't assigned.
    - *Warehouse group* – Use the location directive for all warehouses in the warehouse group selected in the **Warehouse group** field.
    - *Warehouse* – Use the location directive only for the specific warehouse selected in the **Warehouse** field.

- **Site** and **Warehouse** – If you set the **Warehouse selection** field to *Warehouse*, select the site and warehouse where the location directive applies. If you select the warehouse first, the site is filled in automatically. If you select the site first, the warehouse list is filtered so that it shows only warehouses at that site.

- **Warehouse group** – If you set the **Warehouse selection** field to *Warehouse group*, select the warehouse group where the location directive applies. For more information about how to set up warehouse groups, see Warehouse groups.

Lines FastTab

Use the **Lines** FastTab to set conditions for applying the related actions that you specify on the **Location directive actions** FastTab. For each line, you can specify the minimum quantity and maximum quantity that the actions should apply to. You can also specify that the actions should apply to a specific inventory unit.

- **Sequence number** – Enter the sequence that each location directive line should be processed in for the selected work type. You can change the sequence as you require by using the **Move up** and **Move down** buttons on the toolbar.
- **From quantity** – Specify the start of the range of quantities that the line applies to. Specify the quantity in the unit of measure selected in the **Unit** field.
- **To quantity** – Specify the end of the range of quantities that the line applies to. Specify the quantity in the unit of measure selected in the **Unit** field.
- **Unit** – Select the unit of measure for the items. You can specify a minimum quantity and a maximum quantity that the directive should apply to, and you can specify that the directive should be for a specific inventory unit. The **Unit** field is used *only* for quantity evaluation. To determine whether the location directive line is applicable at all, the system uses the quantity in the unit that is specified on that line. Every time that it reaches a location directive line, the system tries to convert the demand unit to the unit that is specified on the line. If the unit of measure conversion doesn't exist, the system moves on to the next line.
- **Locate quantity** – This field is used only during attempts to put or locate items in the warehouse. (Therefore, it applies only when the **Work type** field is set to *Put*). Select one of the following values to specify the quantity that is used to evaluate whether a quantity is within the **From quantity** to **To quantity** range:
  - *License plate quantity* – Use the quantity on the license plate that's being received.
  - *Unitized quantity* – Use the quantity that is used during the transaction. For example, you receive a quantity of 1,000 in a warehouse and break it into 10 license plates, each of which has a quantity of 100. In this case, you can use a quantity of 1,000 items instead of the license plate quantity of 100.
  - *Remaining quantity* – Use the quantity that must still be received on the purchase order line that's being processed.
  - *Expected quantity* – Use the total quantity of the purchase order line, regardless of what was already received.
- **Restrict by unit** – Select this check box to make the location directive line specific to a unit of measure or multiple units of measure. Select it to restrict the units of measure that are considered valid selection criteria for the location directive lines. This functionality works only for location directives where the **Work type** field is set to *Pick*.

For example, when you reserve quantities, you want to reserve pallets only from a specific set of locations. In this case, the lines restrict that sequence to pallets in such a way that any quantity that's less than a pallet isn't selected for the location directive.

The **Restrict by unit** check box doesn't control the unit or units that are applied on work lines. The unit restriction applies only to the units that are made available via the unit sequence group. For example, an item is associated with a unit sequence group that includes both the *pallets* unit and the *pcs* units. A unit of measure is defined, where 1 pallet = 100 pcs, and the location directive uses the **Restrict by unit** functionality only for pallets. Furthermore, a work template is defined that limits the work header creation to 50 pcs. In this case, work lines of 50 pcs are created. To specify the unit of measure for restriction, follow these steps:

1. On the **Lines** FastTab, select **Restrict by unit** on the toolbar. (This button becomes available only after you select the **Restrict by unit** check box on the line and then select **Save**.)
2. On the **Restrict by units** page, in the **Unit** field, select the unit of measure that you want to restrict by for the pick and put processes.

- **Round up to unit** – This field works together with the **Restrict by unit** check box. For example, if **Restrict by unit** and **Round up to unit** are selected on the location directive line, the work that is generated from the directive for raw material picking is rounded up to a multiple of one of the handling units specified on the **Restrict by unit** page.

  > ⓘ **Note**
  >
  > This **Round up to unit** setup works only for the *Raw material picking* work order type, and only for location directives where the **Work type** field is set to *Pick*.

- **Locate Packing Qty** – If you specify a packing quantity on a sales order, transfer order, or production order, this check box lets you restrict the system so that it can select only locations that have at least that packing quantity.

  > ⓘ **Note**
  >
  > This functionality works only with location directives of the *Pick* type.

- **Allow Split** – Specify whether the location directive can split the quantity that's being received or reserved across multiple warehouse locations, or whether the whole quantity must be located in a single location or reserved from a single location to create work.

- **Immediate replenishment template** – Use this field to create a connection to a replenishment template, so that replenishment starts immediately if items aren't allocated. If you leave this field blank, item replenishment doesn't start until all lines of the location directive are processed.

  > ⓘ **Note**
  >
  > This field is available only for selected work order types where replenishment is permitted. For a complete list, see the Fields that are specific to work order types section.

Location directive actions FastTab

You can define multiple location directive actions for each line. Use a sequence number to determine the order that the actions are assessed in. At this level, you can set up a query to define how the best location in the warehouse is found. You can also use predefined **Strategy** values to find an optimal location.

- **Sequence number** – This field shows the sequence that the actions are processed in for the selected work type. You can change the sequence by using the **Move Up** and **Move Down** buttons on the toolbar.

- **Name** – Enter the name of the location directive action. Be specific, so that the action that is performed is clear from the name.

- **Fixed location usage** – Specify which locations the location directive should consider. Select one of the following values:
  - *Fixed and non-fixed locations* – The location directive considers all locations.
  - *Only fixed locations for the product* – The location directive considers only fixed locations for products.
  - *Only fixed locations for the product variant* – The location directive considers only fixed locations for product variants.

- **Allow negative inventory** – Select this check box to allow negative inventory at the specified warehouse location in location directives.

- **Batch Enabled** – Select this check box to use batch strategies for items that are batch-enabled. Select this check box for processes that use location directives to find locations to pick batch number–tracked items from. In this way, the search for locations that hold batch number–tracked items is included. If you select this check box, and set the **Strategy** field to *None*, the system moves on to the next action line.

- **Strategy** – To more easily and quickly define the actions that are associated with each location directive line, select one of the following predefined strategies:
  - *None* – No strategy is used.
  - *Match packing quantity* – This strategy verifies whether a pick location has the specified packing quantity. This strategy is valid only when the **Work type** field is set to *Pick*.
  - *Consolidate* – This strategy consolidates items in a specific location where similar items are already available. This strategy is valid only when the **Work type** field is set to *Put*. A typical setup for put tries to consolidate on the first action line and then, on the second line, tries to put without consolidation. Consolidation of goods makes later picking more efficient.
  - *Consolidate including incoming work* – This strategy works like the **Consolidate** strategy, but it also considers locations where items on incoming put-work lines aren't already available. This consideration is beneficial when receiving multiple license plates with the same item and the items should be moved to the same location. A typical example could occur when receiving based on advance shipping notices (ASNs). This strategy also helps the system find locations when it's necessary to split work line quantities due to location stocking limits or other constraints.
  - *FEFO batch reservation* – This strategy uses first expiry, first out (FEFO) batch reservations. Use it when inventory is located by using a batch expiration date and allocated for batch reservation. You can use this strategy only for batch-enabled items. It's valid only when the **Work type** field is set to *Pick*.
  - *Round up to the full LP and FEFO batch* – This strategy combines the elements of the *FEFO batch reservation* and *Round up to a full LP* strategies. It's valid only for batch-enabled items and location directives that have a work type of *Pick*. The line must be batch-enabled to use the *FEFO batch reservation* strategy, and the *Round up to a full LP* strategy can be used only for replenishment. If you configure this strategy together with a location stocking limit, it can cause the selected put work location to be overloaded and stocking limits to be ignored.
  - *Round up to a full LP* – This strategy rounds up the inventory quantity so that it matches the license plate quantity that is assigned to the items that must be picked. You can use this strategy only for replenishment location directives of the *Pick* type. If you configure this strategy together with a location stocking limit, it can cause the selected put work location to be overloaded and stocking limits to be ignored.
  - *License plate guided* – Use this strategy when you release the order to the warehouse to create the pick-and-put work. You can use this approach for multiple license plates. This strategy tries to reserve and create picking work against the locations that hold the requested license plates that are associated with the transfer order lines. However, if these actions can't be completed, but you still want to create picking work, you should fall back to another strategy for location directive actions. Depending on your business process requirements, you might also want to search for inventory in another area of the warehouse.

- *Empty location with no incoming work* – Use this strategy to locate empty locations. A location is considered empty if it has no physical inventory and no expected incoming work. You can use this strategy only for location directives that have a work type of *Put*.
- *Location aging FIFO* – Use the first in, first out (FIFO) strategy to ship both batch-tracked items and non-batch-tracked items, based on the date when the inventory entered the warehouse. This capability can be especially useful for non-batch-tracked inventory, where no expiration date is available to use for sorting. The FIFO strategy finds the location that contains the oldest aging date, and then allocates picking based on that aging date.
- *Location aging LIFO* – Use the last in, last out (LIFO) strategy to ship both batch-tracked items and non-batch-tracked items, based on the date when the inventory entered the warehouse. This capability can be especially useful for non-batch-tracked inventory, where no expiration date is available to use for sorting. The LIFO strategy finds the location that contains the newest aging date, and then allocates picking based on that aging date.

Example: Using location directives

For this example, consider a purchase order process where the location directive must find free capacity within a warehouse for inventory items that you just registered at the receiving dock. First, you need to find free capacity within the warehouse by consolidating with existing on-hand inventory. If consolidation isn't possible, then you need to find an empty location.

For this scenario, you must define two location directive actions. The first action in the sequence must use the *Consolidate* strategy, and the second should use the *Empty location with no incoming work* strategy. Unless you define a third action to handle an overflow scenario, two outcomes are possible when there's no more capacity in the warehouse: work can be created even though no locations are defined, or the work creation process can fail. The setup on the **Location directive failures** page determines the outcome. You can choose whether to select the **Stop work on location directive failure** option for each work order type.

Next step

After you create location directives, you can associate each directive code with a work template code for work creation. Learn more in Control warehouse work by using work templates and location directives.

Related information

- Video: Warehouse management configuration deep dive

- Help article: Control warehouse work by using work templates and location directives

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Last updated on 11/19/2025

Test location directives with acceptance tests

Acceptance tests let you define a set of *given-when-then* tests to verify that location directives behave as expected. In this type of test, *given* a set of conditions, *when* something happens, *then* some specified result should occur. This terminology is known from quality assurance in engineering and can be traced back to the scientific method.

Acceptance tests provide two main benefits:

- **Faster initial setup:** You can verify the outcomes of your location directives without having to go through the regular warehouse processes and inspect the work creation history log.
- **Lower maintenance:** You'll be able to have more confidence when you modify your location directives later, because you can have the impact of each change automatically validated by running your defined acceptance tests.

Acceptance tests for location directives have no operational impact on the warehouse.

> **Note**
>
> You can't create acceptance tests for location directives where the **Scope** option is set to *Multiple items* (or the **Multiple items** option is set to *Yes*). We recommend that, whenever possible, you use location directive scopes that can be tested with single items, such as *Single item or order* and *All*.

For more information about how to set up location directives, including how to use location directive scopes, see Work with location directives.

Set up an acceptance test

Follow these steps to set up an acceptance test.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. On the Action Pane, select **Acceptance tests**.
3. On the **Location directive acceptance tests** page, follow one of these steps:
    - To create a new test, select **New** on the Action Pane.

- To edit an existing test, select it in the list pane, and then select **Edit** on the Action Pane.
- To copy an existing test, select the source test in the list pane, and then select **Copy** on the Action Pane. This step can be useful when you must create a new acceptance test that is a variation of an existing test.

4. On the header of the new or selected test, set the following fields:

- **Name** – Enter a name for the test.
- **Description** – Enter a short description of the test.
- **Inactive** – Set this option to *Yes* to make the test inactive. Set it to *No* to make the test active. Inactive tests can't be run and will be skipped if you choose to run all tests.

5. On the **Given** FastTab, specify the starting conditions for the test. Set the following fields:

- **Inventory levels** – Specify whether you want to run the test based on your actual inventory or simulate empty inventory. Select one of the following values:
  - *Current on-hand inventory* – Run the test by using whatever inventory is available when the test is run. This approach can make the test result less predictable and subject to arbitrary failures (for example, if the item that is used by the test isn't available).
  - *No inventory* – Simulate an empty warehouse. The simulation will clear inventory of the item that is used in the test, and for all locations that are specified under **Additional inventory**.

- **Additional inventory** – Simulate on-hand quantities for one or more items. The test will add these quantities to the inventory that is specified by the **Inventory levels** field.
  - Select **New** on the toolbar to add a row to the grid.
  - Select **Delete** on the toolbar to remove a row from the grid.
  - Select **Display dimensions** to open a dialog box where you can add dimension columns to the grid, or remove them, as you require.
  - For each row, enter values to specify an item, dimension values, location, and quantity.

> **ⓘ Important**
>
> The settings on the **Given** FastTab don't influence the actual inventory levels in the warehouse. Instead, they simulate conditions that are in effect only temporarily, during test execution.

6. On the **When** FastTab, specify what you want to test. The values that you enter are the inputs to the location directive engine. This approach is simpler than building manual tests by creating orders. Set the following fields:

    - **Work order type** – Specify the type of order to simulate (for example, *Sales orders* or *Purchase orders*).
    - **Work type** – Specify the work type to simulate. Typically, you'll select *Pick* or *Put*.
    - **Order number** – Specify the order number to use during the test. This information can be useful if the location directive query has ranges related to the order table.
    - **Disposition code** – Specify the disposition code that is used to handle return orders.
    - **Directive code** – Specify the directive code that drives the location directives.
    - **Item number** – Specify the item to locate.
    - **Quantity** – Specify the quantity to locate.
    - **Unit** – Specify the unit of measure for the **Quantity** field.
    - **Dimensions** – Specify the storage, product, and tracking dimensions for the item to locate.

7. On the **Then** FastTab, specify the expected outcome of the acceptance test. You **must** set one (and only one) of the following fields:

    - **Exact location** – Select a precise location. The test will be marked as passed if this location is the result of the location directive.
    - **Location matching regular expression** – Enter a regular expression that will be validated against the resulting location, even if the resulting location is blank (no result). The test will be marked as passed if the regular expression matches the name of the resulting location. For more information about regular expressions, see .NET regular expressions.
    - **Location with profile** – Select a location profile. The test will be marked as passed if the resulting location has this profile.
    - **Location in zone** – Select a location zone. The test will be marked as passed if the resulting location has this zone.

8. Select **Save** to save your test. The **Results** FastTab stores a record of any test results for each test. For information about how to run tests and interpret the results, see the next section.

Run acceptance tests

After you set up your tests, you can run them, either one by one or all at once. To run one or more tests, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Location directives**.

2. On the Action Pane, select **Acceptance tests**.

3. On the **Location directive acceptance tests** page, follow one of these steps:

    - To run a single, specific test, select it in the list pane, and then select **Run** on the Action Pane.
    - To run all active tests, select **Run all** on the Action Pane.

4. After the tests have been run, the list pane is updated to indicate the most recent result of each test. To inspect the results of a test, select the test in the list pane, and then select the **Results** FastTab. The grid shows the result of each run of the test. For each result, the following information is provided:

    - **Result** – The result of the test: *Passed*, *Failed* or *Skipped*.
    - **Resulting location** – The location that was found by the test. If no location was found, this field is blank.
    - **Duration (ms)** – The duration of the test in milliseconds (ms). This field also indicates how quickly the system will be able to process your directives during daily operation. For efficient warehouse operations, you should design your location directives so that they can be processed as quickly as possible. One typical cause of a slow response is the use of location directive queries where the defined ranges or sort orders don't match an index on the table. (The system will warn you if you try to save a query that is configured in this way.)
    - **Locations evaluated** – The number of locations that were evaluated during the test. For efficient warehouse operations, you should try to have as few locations as possible evaluated. One way to minimize the number of evaluated locations is to have many location directives, the first of which are the most specific and the last of which are the most general. You can also segment your warehouse by keeping certain types of items in dedicated zones. This approach can help you avoid scanning the whole warehouse every time.
    - **Created date and time** – The date and time when the test was run.
    - **Created by** – The name of the person who ran the test.

5. To view more information about any test run, select it in the grid, and then review the **Log** field. The log includes a work creation history that will help you understand the outcome.

    > **Important**

To determine the result of the test, the system compares the **Resulting location** value with the **Then** condition. The log for a failing test might indicate that a location directive did find a location, but that location didn't match the expected location that is defined in the **Then** condition.

Troubleshoot location directives and acceptance tests

Definition of acceptance tests is typically an iterative process. If a test fails, you should find out why it failed. Perhaps the test isn't set up correctly, or perhaps a location directive must be adjusted so that it better meets your requirements.

To troubleshoot your location directives and acceptance tests, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the list pane, select a location directive that you've designed an acceptance test for.
3. Expand the FactBox pane. This pane is on the right side of the page and is labeled **Related information**.
4. In the FactBox pane, expand the **Acceptance tests** FactBox. The grid in this FactBox lists the acceptance tests that are relevant to the selected location directive. It also shows the most recent result for each of those tests. From this FactBox, you can perform the following actions:

    - Select **Run all** to run all the listed tests.
    - Select **Coverage** to switch to and from the coverage view. The coverage view uses colored highlighting to indicate which location directives, lines, and actions were used to determine the result of a selected test. Records that are marked yellow were evaluated but didn't find a location. Records that are marked green did find a location. To change the test that coverage is shown for, select the symbol in the **Result** column for the relevant test.
    - Hover over a test to view more information about it, including the log.
    - In the **Name** column, select the name of a test to open that test on the **Location directive acceptance tests** page. There, you can inspect and adjust the test as you require.
    - Select **Run tests after change** to turn on and off the setting that will automatically run all tests each time that you change a location directive. Use this functionality to provide immediate feedback about the impact of changes as you make them.

5. Based on the results indicated by the coverage view and test log, adjust your tests and/or location directives until they produce the expected results for each test.

Last updated on 12/30/2025

Configure product filters for warehouse transactions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to configure product filters and filter codes to categorize inventory items in a warehouse. You can also use filters to specify which customers can order a particular item and which items can be purchased from a particular vendor.

By setting up and using product filters, you can automatically organize inventory items in a warehouse and combine filtered items into filter groups. Use filters to put items into categories for handling, purchasing, and selling processes. You might want to group items together or separate them from each other when the way that you handle them is based on weight or handling restrictions. You can also specify which customers or vendors an item can be purchased from or sold to.

Prerequisites

The following table shows the prerequisites that must be in place before you start.

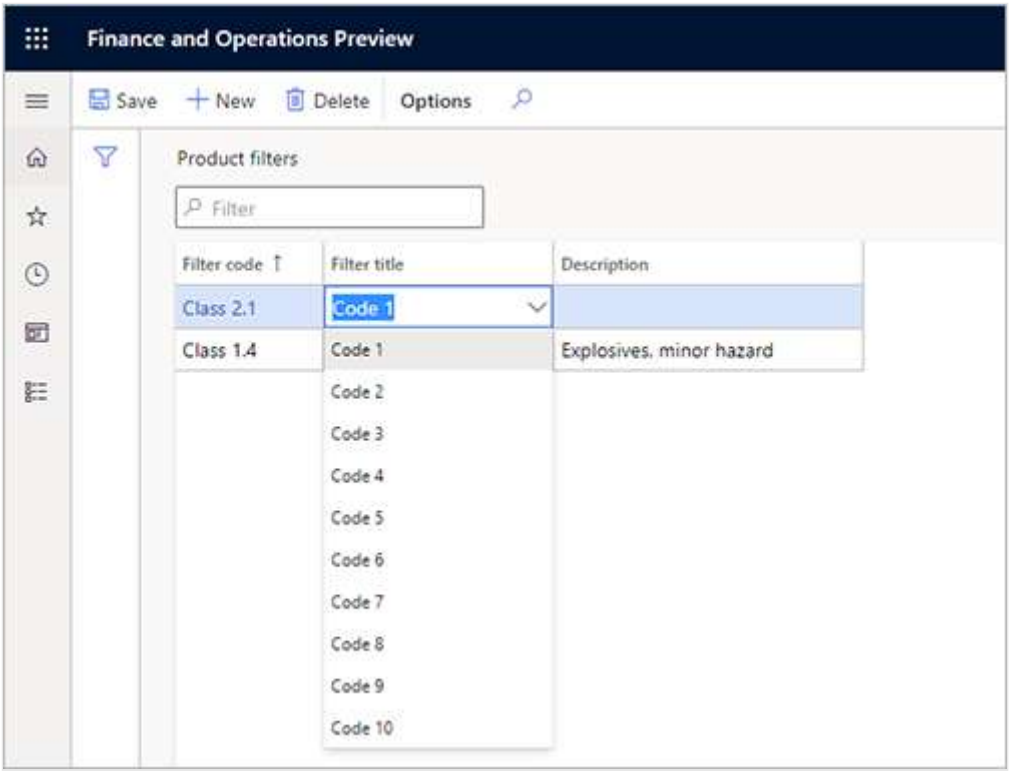
Expand table

Prerequisite	Instructions
Before you start to configure products on the **Released product details** page, turn on warehouse processing for the product's storage dimension group.	Go to **Product information management** > **Setup** > **Dimension and variant groups** > **Storage dimension groups**, and select or create a storage dimension group where the **Use warehouse management processes** option is set to *Yes*.
If you use customer filters and/or vendor filters, enable their use in Warehouse management parameters.	Go to **Warehouse management** > **Setup** > **Warehouse management parameters**. On the **Product filters** tab, set the **Enable customer filters** and/or **Enable vendor filters** options to *Yes*.

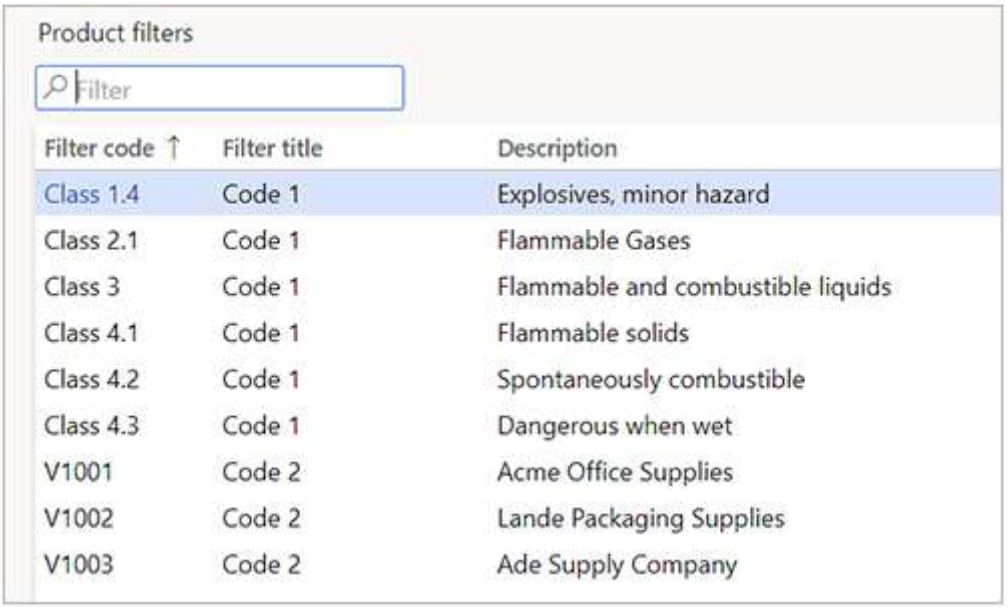
Set up product filters

Product filters provide up to 10 **Filter title** characteristics, which are enumeration (enum) values. You can select these enum values when you create a product filter. The enum values *Code 1* through *Code 10* are system-defined and represent a specific characteristic or attribute of an item. For example, *Code 1* might represent items that have a hazardous material classification. *Code 2* might represent items that only vendors can purchase. Product filters then define the specific **Filter code** value that is associated with a **Filter title** value.

1. Go to **Warehouse management** > **Setup** > **Product filters** > **Product filters**.
2. On the Action Pane, select **New** to add a product filter to the grid.
3. In the **Filter title** field, select a value.
4. In the **Filter code** field, enter a value.



5. In the **Description** field, enter a name for the code. For example, *Code 2* might represent vendors. You can then create a product filter for a specific vendor or group of vendors. To learn more, see the Setup vendor filter codes section later in this article.



Set up product filter groups

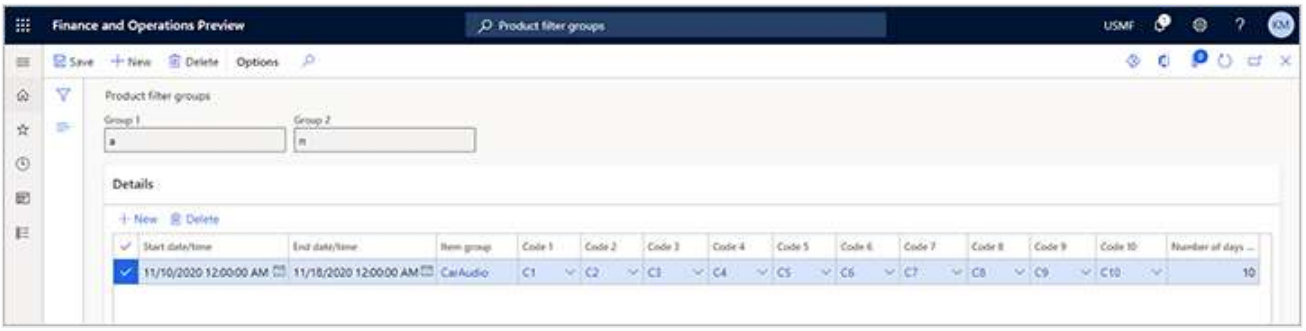
Use filter groups to organize filter codes. This capability is helpful when a group is used in a query in a location directive and you want to search for the group instead of a series of codes. Each filter group is associated with an item group.

> **Important**
>
> Not all product filter groups are available for filter codes that are higher than *Code 4* (that is, *Code 5* through *Code 10*). For example, for released products, although all the product filter codes are added, the grouping of filter codes isn't updated. This behavior might be updated in later releases.

To set up filter groups, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Product filters** > **Product filter groups**.
2. On the Action Pane, select **New**.
3. In the **Group 1** and **Group 2** fields, enter the names that you can use to categorize items.
4. On the **Details** FastTab, select **New** to add a line.
5. In the **Start date/time** and **End date/time** fields, enter start and end dates for the filter group.
6. In the **Item group** field, select the item group that the product filter should apply to.

7. In the **Code 1** through **Code 10** fields, select the filter codes to include in the group, as required.



> **Note**
>
> If you receive an error message when you close the page, a code setup might be missing. On the **Item groups** page, you can make the codes mandatory for an item group by selecting the **Assign filter code 1 for item group**, **Assign filter code 2 for item group**, and so on, check boxes.

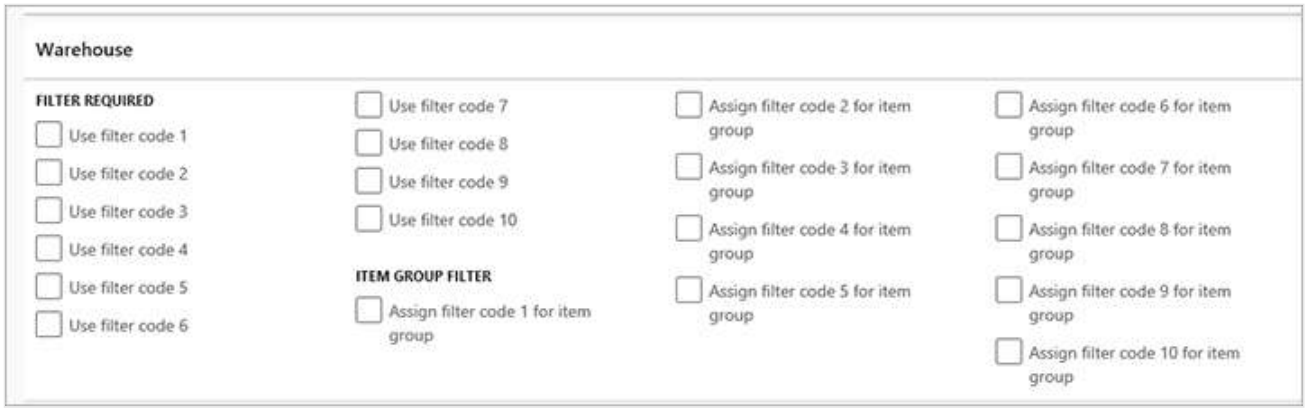
Set up filter codes on item groups

When you set up filter codes on an item group, you specify the codes required for products attached to that item group.

To set up filter codes on item groups, follow these steps:

1. Go to **Inventory management** > **Setup** > **Inventory** > **Item groups**.
2. On the Action Pane, select **New** to create an item group.
3. In the **Item group** field, enter a name.
4. In the **Name** field, enter a description.
5. On the **Warehouse** FastTab, in the **Filter required** section, select the check boxes for the filter codes that you must specify for products associated with the item group.

   To update a released product, open its **Released product details** page, and then, on the Action Pane, select **Edit**. The filters that are associated with codes then become available on the **Warehouse** FastTab.



6. In the **Item group filter** section, select the check boxes for the filters that must match for the filter group to be the default filter group for an item.

   For example, if you select the **Use filter code 1** and **Use filter code 2** check boxes, both filter code 1 and filter code 2 of the item must match the setup of the filter group for the item group before you can select the filter group. When you create a new item, the selected filter group is the default filter group in the **Group 1** and **Group 2** fields on the **Warehouse** FastTab of the **Released product details** page.

> ⓘ **Important**
>
> You can enable product filter codes only for items that use warehouse management processes (WMS).

Specify filter codes for released products

Follow these steps to specify filter codes for released products. For example, use filter codes to group hazardous products that specific vendors purchase.

1. Go to **Product information management** > **Products** > **Released products**.
2. On the Action Pane, select **New** to create a product.
3. In the **New released product** dialog box, enter the data required to create the base of a new product, and then select **OK**.

   Product filter codes aren't enabled unless the item group attached to the product is configured for filter codes.

4. On the **Warehouse** FastTab, in the **Product filter codes** section, select filter codes for the **Code 1** through **Code 10** fields, as required, to specify filter codes for the product.

Set up generally available items

You can make specific inventory items available only for customers, only for vendors, or for both customers and vendors.

> **Note**
>
> Customer and vendor filters don't apply to any item set up as generally available.

To set up generally available items, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Product filters** > **Generally available products**.
2. On the Action Pane, select **New** to create a record.
3. In the **Customer or vendor** field, select *Customer*, *Vendor*, or *All* to make the items available for customers, vendors, or both.
4. In the **Start date/time** field, enter the date and time when the item becomes available.
5. In the **Item group** field, select an item group.
6. In the **Code 1** through **Code 10** fields, select the filter codes to limit the items that are generally available.

   When you select an item group, you set that group of items as generally available. By selecting filter codes in these fields, you limit the items that are available.

Set up customer product filters

Use this optional procedure to specify items that customers can access, in addition to the items available through the filter setup on the **Generally available items** page. You can set up multiple filters for a single customer.

To set up customer filter codes, follow these steps:

1. Go to **Sales and marketing** > **Customers** > **All customers**.
2. Select a customer.
3. On the Action Pane, on the **Customer** tab, in the **Set up** group, select **Product filters**.
4. On the **Product filter codes** page, on the Action Pane, select **New**.

5. In the **Start date/time** and **End date/time** fields, enter start and end dates for the selected item group.
6. In the **Item group** field, select an item group.
7. In the **Code 1** through **Code 10** fields, select the filter codes to use as criteria to limit the items that are available for customers in the selected item group. You must make a selection for every filter code set up for the item group.

Set up vendor product filters

Use this optional procedure to specify items that a vendor can access, in addition to the items made available through the filter setup on the **Generally available items** page. You can set up multiple filters for a single vendor.

To set up vendor filter codes, follow these steps:

1. Go to **Procurement and sourcing** > **Vendors** > **All vendors**.
2. Select a vendor.
3. On the Action Pane, on the **Vendor** tab, in the **Set up** group, select **Product filters**.
4. On the **Filter codes** page, on the Action Pane, select **New**.
5. In the **Start date/time** and **End date/time** fields, enter start and end dates for the selected item group.
6. In the **Item group** field, select an item group.
7. In the **Code 1** through **Code 10** fields, select the filter codes to use as criteria to limit the items that are available for vendors in the selected item group. You must make a selection for every filter code set up for the item group.

> ⓘ **Note**
>
> The setup of vendor product filters applies to released products where warehouse management processes (WMS) are enabled for the associated storage dimension group. The filter codes determine whether the system allows users to purchase a given item from a given vendor when they create purchase order lines. Microsoft Dynamics 365 Supply Chain Management has two methods for handling vendor approval. If one or more released products exist where the **Approved vendor check method** field is set to *Warning only* or *Not allowed*, both vendor approval methods can be enabled for those items. This situation might cause issues when users create purchase order lines.

Related information

- WMS Filter codes (blog post)

Last updated on 05/29/2026

Configure the transfer order receiving process

This article explains how to configure the receiving process for a warehouse. It includes information about the parameters that are available for warehouse receiving and how to configure them.

Prerequisites

Two fields are used to configure the receiving process. The **Transfer order receiving process** field is a standard part of all current versions of Microsoft Dynamics 365 Supply Chain Management. However, the **Transfer order receiving ASN cleanup** field requires Supply Chain Management version 10.0.44 or later.

Configure the receiving process for a warehouse

To configure the receiving process for a warehouse, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **Receiving** FastTab, set the following fields:
    - **Transfer order receiving process** – Specify whether the registration and cost-update processes should be split for transfer orders.

      When the processes are split, receiving clerks can use the Warehouse Management mobile app to update a transfer order receipt without waiting for any financial background processing, regardless of the receiving flow. After the registration is completed, transfer order receiving can be done either manually or as part of a scheduled batch job.

      Select one of the following options:
        - *Combine the registration and receiving* – Run registration and receiving together.
        - *Split the registration and receiving for license plate receiving* – Run registration and receiving separately for license plate receiving.
        - *Split the registration and receiving for all flows* – Run registration and receiving separately for all flows.

> **Note**
>
> We recommend that you select the *Split the registration and receiving for all flows* option, because it covers all flows. By contrast, the *Split the registration and receiving for license plate receiving* option covers only license plate receiving.

- **Transfer order receiving ASN cleanup** – Specify when associated advance shipping notice (ASN) data should be deleted during transfer order receiving.

  This field covers scenarios where ASN data is required beyond the default settings of the receiving process. For example, it's applicable when you use both the Warehouse Management mobile app and the web client to partially receive a transfer order that has multiple license plates.

  Select one of the following options:

  - *Default* – ASN data is deleted when the following events occur:
    - Putaway work creation or cancellation
    - Mobile application receiving, if the work policy is set not to create work
    - Transfer order line registration from the web client
    - Transfer order receipt posting

  - *On transfer order status received* – ASN data is deleted when the following events occur:
    - Putaway work creation or cancellation
    - Mobile application receiving, if the work policy is set not to create work
    - Update of the transfer order status to *Received*

Related information

- Learn more about how to use a mobile device for license plate receiving in License plate receiving via the Warehouse Management mobile app.

Last updated on 12/30/2025

Location stocking limits

You can use the **Location stocking limits** page (**Warehouse management** > **Setup** > **Warehouse** > **Location stocking limits**) to control the load capacity at warehouse locations without having to use the more advanced processes for volumetric calculations of physical products.

The purpose of location stocking limits is to evaluate the maximum quantity that a location can contain. You can set up the feature on any of three levels, each of which has its own tab on the **Location stocking limits** page:

- Products
- Product variants
- Container types

For each level, you can define different field values. The system evaluates the capacity of a specific location, based on the **Quantity** and **Unit** values for each row. It selects the most detailed matching record first. For example, a row that has a value in the **Location profile ID** field is evaluated before a row that has a value only in the **Warehouse** field. The remaining capacity is also evaluated based on the **Quantity** value for the location stocking limit record that is used.

In the **Products** section of the page, you can define the following field values for the search for location stocking limits:

- Warehouse
- Location profile ID
- Location
- Pack size category ID
- Item number
- Unit

> ⓘ **Note**
>
> You don't have to define a **Unit** value for each location stocking limit record. The location capacity calculations are based on the inventory unit quantities. If no unit conversion is defined for a value that is used, the location stocking limit record is skipped as though another **Item number** value were defined for it.

For more information about how to set up stocking limits based on container types, see Manage inbound putaway based on container types.

> **Note**
>
> It's possible to allow stocking limits for specific locations to be exceeded for replenishment work. Learn more in Replenishment over location capacity.

Example – Purchase order receiving

This example is based on a clean *USMF* demo data set, where the following values are set on the **Product variants** tab of the **Location stocking limits** page.

Expand table

Warehouse	Location profile ID	Item number	Size	Quantity	Unit
24	FLOOR	D0013	M	300	Ea
24	FLOOR	D0013	L	240	Ea
24	FLOOR	D0013	S	360	Ea

Different unit of measure product variants are set up for the products. These variants are aligned with the location stocking limits for three pallets (PL):

- **Size M:** 1 PL = 100 each (Ea)
- **Size L:** 1 PL = 80 Ea
- **Size S:** 1 PL = 120 Ea

Therefore, each location where the **Location profile ID** value is set to *FLOOR* can carry *3 PL* of item number *D0013*.

Prepare for the example

In this example, you'll run a purchase order receiving flow for two lines. However, you must first update the demo data in the following way to ensure that the locations allow mixed items to be carried, only the empty locations *FL-002* through *FL-004* are used, and there's no open inbound work.

1. For location *FL-001*, change the value of the **Location profile** field from *FLOOR* to *FLOOR-05*.
2. For the *FLOOR* location profile, set the **Allow mixed items** option to *Yes*.
3. Create a purchase order that has the following two lines.

Expand table

Warehouse	Item number	Size	Quantity	Unit
24	D0013	S	4	PL
24	D0013	L	4	PL

Process the example

You'll first receive a quantity of *4* of unit *PL* in size *S*, and review the put line locations for the work that is created. You'll then receive a quantity of *4* of unit *PL* in size *L*, and review the put line locations for the work that is created.

1. In the Warehouse Management mobile app, sign in by using *24* as the user ID and *1* as the password.
2. Select **Inbound** > **Purchase Receive**.
3. Receive *4 PL* of item number *D0013* in size *S*.
4. Review the putaway work that was created. You should see the following result:
    - 3 PL -> FL-002
    - 1 PL -> FL-003
5. Receive *4 PL* of item number *D0013* in size *L*.
6. Review the putaway work that was created. You should see the following result:
    - 1 PL -> FL-003
    - 3 PL -> FL-004

Based on the results, you might conclude that the system failed to allocate the correct putaway locations. Otherwise, why did the system add only *1 PL* of size *L* to location *FL-003* in the last step, not *2 PL*? That is, why isn't there a total of *3 PL* for putaway to that location?

To explain this apparent failure, you must understand the selection criteria for the location stocking limits. The system selects the most detailed matching record. In this example, the most detailed matching record is the line where the **Size** value is *L*, the **Quantity** value is *240*, and the **Unit** value is *Ea*. Because you already have open work from the previous receipt of a quantity of *120 Ea* of size *S*, the remaining capacity is calculated as *240 – 120 = 120 Ea*. Therefore, the remaining capacity can carry only *1 PL* of *80 Ea*.

> **Note**
>
> You can't use location stocking limits to control, for example, the replenishment of items that have different quantities in the same location. In this case, use a *replenishment template*.

Last updated on 07/01/2026

Prevent multiple owners in warehouse locations

Overview

The *prevent multiple owners* feature prevents inventory owned by different legal entities or consigned from different parties from being stored in the same warehouse location. It helps you enforce inventory owner segregation in your warehouse by making sure that all the items in a given location have the same owner. This feature is essential for warehouses that manage inventory on behalf of multiple owners, such as third-party logistics (3PL) operations, consignment warehouses, warehouses running in warehouse management only mode, or warehouses supporting multiple legal entities with distinct ownership tracking requirements.

What is inventory owner tracking?

In Dynamics 365 Supply Chain Management, the **Owner** field in the inventory dimensions tracks inventory ownership of each item. This field identifies which legal entity or external party owns a specific inventory lot, enabling proper tracking and settlement in multi-owner scenarios. Location profiles that use the **Prevent multiple owners** constraint enforce rules that make sure that all inventory items in a location share the same **Owner** value.

When to use this feature

Consider using the *prevent multiple owners* feature in any of the following scenarios:

- **Warehouse management only (WMS-only) mode** – You manage inventory for multiple legal entities or external parties and need to enforce segregation of ownership at the location level.
- **Multi-owner warehouses** – You manage physical inventory owned by multiple legal entities and need to segregate their stock.
- **Third-party logistics (3PL)** – You manage consigned inventory from multiple suppliers or customers and need to prevent cross-mingling.
- **Consignment operations** – You track and manage consigned inventory separately from owned inventory and need location-level guarantees.
- **Regulatory compliance** – Your industry or operational model requires documented segregation of ownership.

Benefits

The *prevent multiple owners* feature provides the following benefits:

- **Prevents ownership disputes** – Eliminates ambiguity about which owner inventory belongs to when stock is physically co-located.
- **Operational clarity** – Warehouse operators receive clear, immediate feedback when they attempt to violate the constraint.
- **Compliance support** – Simplifies audit trails and regulatory compliance by enforcing segregation rules consistently.
- **Automated constraints** – Reduces the need for manual workarounds or bypass procedures.

How the feature works

Core behavior

When you turn on the **Prevent multiple owners** constraint for a location profile, the system enforces the following rules for locations that use that profile:

- Each location can only store inventory for a single owner.
- An empty (undefined) owner value is treated as a distinct owner, so inventory with an empty owner value can only be mixed with other inventory that also has an empty owner value.

When the system evaluates ownership

The system evaluates the **Prevent multiple owners** constraint at three key stages:

1. **Putaway operations** – When the system selects a location for received or internal-order inventory, it confirms that the location's current inventory (if any) belongs to the same owner.
2. **Manual location changes** – When a warehouse operator manually reassigns inventory to a new location using a mobile device, the system verifies the owner compatibility before it allows the move.
3. **Replenishment operations** – When the system triggers replenishment to refill a picking location, it respects the owner constraint by selecting inventory of the same owner.

Error handling and operator feedback

If a manual location change violates the constraint, the system rejects the change with a label-driven error before any posting occurs.

If automatic location selection fails due to owner mismatches, the system triggers immediate replenishment to build new capacity for the incoming owner's inventory.

When an operator attempts to putaway inventory to a location that violates the **Prevent multiple owners** constraint, the mobile device displays a dedicated error message. The message informs the operator that the location already contains inventory owned by a different party, and suggests alternative locations.

Relationship to other mixing constraints

The **Prevent multiple owners** constraint works alongside the other mixing constraints available for location profiles:

- **Allow mixed items** – Controls whether different products can coexist in a location.
- **Allow mixed batches** – Controls whether different batch lots can coexist in a location.
- **Allow mixed status** – Controls whether inventory with different statuses (such as *Available*, *On-order*, or *Damaged*) can coexist in a location.

The system evaluates all constraints together. If any constraint is violated, the location isn't considered as a putaway candidate.

Prerequisites

To use the *prevent multiple owners* feature, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.49 or later.
- Your warehouse must be enabled for warehouse management processes (WMS) and use location-directive-driven operations.
- The tracking dimension groups that apply to the relevant items must include the **Owner** dimension.

Configure a location profile to prevent multiple owners

Enable the owner-segregation constraint as needed for each relevant location profile. After you complete this procedure, any location that uses the updated profile is subject to the owner-mixing checks described in the How the feature works section, both during putaway and during manual location changes. Repeat this procedure for each location profile that should enforce single-owner storage.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. In the list of location profiles, select the profile you want to configure. If it doesn't exist, select **New** to create one.
3. On the **Action Pane**, select **Edit**.
4. On the **General** FastTab, set **Prevent multiple owners** to *Yes* to enforce single-owner segregation for locations that use this profile.

   > ⓘ **Note**
   >
   > The **Prevent multiple owners** option is independent of the other mixing constraints (**Allow mixed items**, **Allow mixed batches**, **Allow mixed status**). You can freely combine these constraints as needed.

   > ⓘ **Important**
   >
   > When you change **Prevent multiple owners** from *No* to *Yes* for a location profile that already contains inventory, the system checks to make sure that no existing inventory violates the new constraint. If it detects violations, you see an error dialog that lists locations and their current inventory owners. Perform one or more of the following actions based on the content of the error message. Then try to enable the option again.
   >
   > - **Relocate inventory** – Move inventory with different owners to separate locations manually or according to a transfer order.
   > - **Reserve capacity** – Create new locations dedicated to specific owners.
   > - **Schedule the change** – Enable the constraint during a maintenance window when relevant locations are empty.

5. Select **Save**.

Set up the Inventory owner dimension on a tracking dimension group

Inventory ownership is tracked through the **Owner** dimension on the tracking dimension group that is assigned to your items. If this dimension isn't active, the **Owner** field stays blank on

inventory transactions, and the *prevent multiple owners* checks described in the How the feature works section have nothing to compare against. Use this procedure to enable the **Owner** dimension on each tracking dimension group that applies to items that you want to segregate by owner.

> **Important**
>
> Activating the **Owner** dimension changes the inventory tracking model for any item that uses the group. Plan the change carefully for items that already have on-hand inventory or open transactions, and follow your organization's standard process for modifying tracking dimension groups.

1. Go to **Product information management** > **Setup** > **Dimension and variant groups** > **Tracking dimension groups**.
2. Follow one of these steps:
   - On the Action Pane, select **New** to create a tracking dimension group. Enter a name and description, and then select **Save** on the Action Pane.
   - In the list pane, select the existing tracking dimension group that you want to update.
3. On the **Tracking dimension** FastTab, in the **Owner** row, select the checkboxes in the **Active** and **Physical inventory** columns.
4. Select **Save**.
5. Assign the tracking dimension group to each released product that you want to track by owner. To do this, go to **Product information management** > **Products** > **Released products**, where you can assign a **Tracking dimension group** to each new or existing released products.
   - For new products, assign a **Tracking dimension group** in the dialog when you create the released product.
   - For existing products, go to the Action Pane, open the **Product** tab and, from the **Set up** group, select **Dimension groups**. In the drop-down dialog, set the **Tracking dimension group** field.

Assign an inventory owner to inventory

After you activate the **Owner** dimension on a tracking dimension group, you must also assign an actual owner value to the inventory that flows through your warehouse. The *prevent multiple owners* feature compares these owner values when it evaluates location candidates, so any inventory that doesn't have an owner assigned is treated as belonging to a separate, *empty owner* group, as described in the How the feature works section.

Assigning an owner to inventory has two parts: registering each party that can own inventory, and then making sure that owner is set on inbound and internal transactions.

Register inventory owners

Register inventory owners before referencing them on transactions. Inventory owners are typically either vendors that consign stock to your warehouse or other legal entities for which you manage inventory.

- If you're using warehouse management only (WMS-only) mode, each source system setup must have a *Warehouse inventory owner* configuration that contains a mapping for every **Owner** value that the source system might send in an order line. The mapping ensures that the system can recognize the owner and assign it to inventory transactions. To make the configuration, open the relevant source system record and select **Warehouse inventory owner** on the Action Pane. Learn more in Configure your source systems.

- If you're not using WMS-only mode, register inventory owners on the **Inventory owners** page. Go to **Inventory management** > **Setup** > **Dimensions** > **Inventory owners** and, for each owner, make the following settings:
  - **Vendor account** – Select the vendor that owns the inventory. The **Name** and **Owner** fields are populated automatically.
  - **Owner** – Optionally, enter an owner value that's easier for external parties (such as the vendor) to recognize. You can edit this field only until you save the record.

Set the owner on inventory transactions

After you register owners, the **Owner** value flows onto inventory through the transaction that brings the stock on hand or changes who owns it. Use the appropriate method for your scenario:

- **Consignment replenishment orders** – When you create a consignment replenishment order for a vendor that's registered as an inventory owner, the vendor is set as the owner on each order line automatically. The product receipt update then registers physical on-hand inventory under that owner. For details, see Set up consignment.

- **Inventory ownership change journal** – Use a journal of type *Ownership change* to transfer existing on-hand inventory from one owner to another (for example, from a vendor to your own legal entity when you consume consignment stock). Create the journal from **Inventory management** > **Journal entries** > **Items** > **Inventory ownership change**.
- **Other inbound transactions** – For purchase orders, transfer orders, and inbound shipment orders, open the order, select a line, and on the **Line details** FastTab, open the **Product** tab and assign the **Owner**.

> ⓘ Note
>
> If you post an inventory transaction without an owner value, the resulting on-hand inventory is tracked under an empty owner. Locations that use a profile where **Prevent multiple owners** is set to *Yes* then treat that inventory as incompatible with any inventory that has a non-empty owner, and the reverse.

Example configuration

The following end-to-end example shows how the *prevent multiple owners* feature works. It illustrates one of the scenarios where the feature is most useful—a third-party logistics (3PL) warehouse—and demonstrates how location profiles, zones, and location directives work in combination with the **Prevent multiple owners** constraint to keep each owner's inventory segregated.

In this example, your 3PL warehouse stores consigned inventory from two customer partners (Partner A and Partner B). You want to prevent their stock from commingling. Therefore, you might configure your warehouse as follows:

1. Create two location profiles:
    - Name the first profile *3PL-PARTNER-A* and set **Prevent multiple owners** to *Yes*.
    - Name the second profile *3PL-PARTNER-B* and set **Prevent multiple owners** to *Yes*.
2. Assign locations in zone A to use the *3PL-PARTNER-A* profile and locations in zone B to use the *3PL-PARTNER-B* profile.
3. Configure location directives and putaway templates to route Partner A inventory to zone A and Partner B inventory to zone B.
4. When receiving Partner A inventory:

- The system automatically assigns it to a zone A location using the *3PL-PARTNER-A* profile.
- If a location is full, the system finds alternate locations within zone A with the same owner already present.
- If no available capacity exists for Partner A's owner, the system either suggests a new empty location or initiates replenishment to free capacity.

Day-to-day warehouse operations

After you configure location profiles to prevent multiple owners, day-to-day warehouse operations must align with the new constraint so that putaway, replenishment, and manual moves continue to flow smoothly. The following sections describe what warehouse managers should plan for, how operators should respond on the mobile device, and how to recover when the system can't find a compatible location.

For warehouse managers

As a warehouse manager, your role is to make sure the location-profile configuration translates into a workable physical layout and that operators have enough capacity per owner to keep putaway flowing. The following planning and monitoring activities help you avoid *owner mismatch* errors and capacity issues:

- **Monitor owner segregation** – Use the warehouse location master data to confirm that locations assigned to constrained profiles contain expected inventory owners.
- **Plan location allocation** – Allocate sufficient location capacity to each owner based on forecasted volumes.
- **Coordinate with 3PL partners** – If you manage multiple partners' inventory, agree on owner-assignment conventions to avoid empty-owner ambiguity.

For warehouse operators

Warehouse operators typically use a mobile device or terminal and barcode scanner to manage putaway, picking, and manual relocation tasks. The system handles owner segregation automatically in most cases, but operators need to know how to respond when a location is rejected so they don't bypass the **Prevent multiple owners** constraint or accidentally commingle stock.

During putaway:

1. Scan the item and owner code as directed by your receiving process.

2. The system automatically selects a location respecting the owner constraint.
3. If you see an error message that the location already has inventory from a different owner, *do not manually override it*. The constraint exists to prevent commingling. Instead, take these steps:

    - Confirm with your supervisor that the inventory owner is correct.
    - Wait for the system to suggest an alternate location (usually automatic).
    - If prompted, manually select an alternate location suggested by the system.

When manually relocating inventory:

1. Always verify the inventory owner before reassigning to a new location.
2. If the system rejects your location change due to owner mismatch, it means that the location is incompatible. Choose a location that already contains inventory of the same owner, or an empty location if available.

When putaway fails

If the system can't find an available location due to owner constraints, it performs one of the following actions:

- **Recover automatically** – The system might trigger immediate replenishment to consolidate and free capacity for the new owner.
- **Provide operator guidance** – The mobile device or work queue displays a message that explains the constraint and suggests next steps.
- **Wait for manual intervention** – In rare cases, a supervisor might need to manually relocate existing low-priority inventory to free capacity.

Interaction with location directives

Location directives control where inventory is placed during putaway and picking. When the **Prevent multiple owners** constraint is active, the system implements the following behaviors to ensure that directives respect owner segregation:

- **Strategy refinement** – The system augments location-directive logic to include owner as a grouping dimension alongside item, batch, and status.
- **Multi-step filtering** – Location selection first applies traditional directive criteria (zone, aisle, and so on), then filters by compatible owners.
- **Replenishment triggers** – If the directive is set to enable immediate replenishment, a failed putaway due to owner mismatch triggers replenishment, which is similar to item or batch

mismatches.

For example, suppose you have a directive that routes all purchase order receipts to a bulk location. The system evaluates the directive as follows depending on whether the **Prevent multiple owners** constraint is enabled:

- **Without owner constraint** – Each item is placed in any bulk location that has available capacity.
- **With owner constraint** – Items are only placed in bulk locations that exclusively contain items for the same owner. If no suitable bulk location exists, the directive might select a different location class or trigger replenishment.

Troubleshooting

Inventory owner information isn't available

Symptom

The system doesn't show inventory owner information on location or work displays.

Resolution

Confirm that the tracking dimension group for your items includes the **Owner** field. If it's missing, add it to the tracking dimension group for the items, and then perform a new receipt to generate inventory with owner tracking.

Changing Prevent multiple owners from *No* to *Yes* fails

Symptom

When you try to enable **Prevent multiple owners** on a location profile, the system displays an error dialog that lists conflicting inventory.

Resolution

To resolve the conflict, do one of the following actions:

- Relocate the conflicting inventory to other locations that don't use this profile.
- Create separate location profiles for each owner and adjust your location assignments.

When the location is empty or contains only single-owner inventory, try again to enable the constraint.

Operators frequently receive "owner mismatch" errors

Symptom

Warehouse operators report frequent constraint-violation errors during putaway.

Possible causes

This problem can occur for any of the following reasons:

- **Insufficient segregated capacity** – You allocated too few locations to each owner. Expand the location pool or reduce expected volumes.
- **Incorrect owner assignment** – Incoming inventory is being tagged with the wrong owner code. Review the receiving process and incoming document data.
- **Directive misconfiguration** – Location directives might not be routing inventory to the correct owner-dedicated locations. Review directive sequencing and query criteria.

Resolution

To resolve the issue, consider these steps:

- Work with your warehouse planner to rebalance capacity allocation per owner.
- Audit incoming data quality and owner-assignment rules.
- Review location directives to ensure they route inventory by owner correctly.

Frequently asked questions

Does the "Prevent multiple owners" constraint apply during cycle counting?

No. Cycle counting is a physical inventory verification process and doesn't enforce putaway constraints. However, if cycle counting reveals that a location contains inventory with multiple owners (from inventory that was placed before the constraint was enabled), you should relocate one owner's stock to another location to comply with the constraint going forward.

What happens if an owner code is empty or null?

The system treats empty owner values as a distinct owner and only mixes them with other inventory that also has an empty owner. This approach prevents accidentally commingling unowned inventory with owner-specific inventory.

Can I enable both "Prevent multiple owners" and "Allow mixed items" on the same location profile?

Yes. Both constraints work independently. The location allows multiple items but only from the same owner.

Does this feature support multi-owner inventory transfers?

The constraint prevents storing multi-owner inventory in the same location. To transfer inventory between owners, you must use a separate transfer order and ensure inventory passes through a quarantine or staging location configured to allow multiple owners (that is, with the constraint disabled).

If I disable "Prevent multiple owners", what happens to existing inventory?

Disabling the constraint doesn't automatically move inventory. Existing multi-owner inventory (if any) remains in place. Disabling simply stops enforcing the constraint for future putaway operations.

Related information

- **Location profiles** – For general information on setting up location profiles and constraints, see Create location profiles.
- **Location product dimension mixing** – To learn about mixing constraints for product dimensions, see Location product dimension mixing.
- **Location directives** – For guidance on setting up location directives, see Create location directives and Control work by using work templates and location directives.
- **Warehouse management overview** – For a broader introduction to warehouse management concepts, see Warehouse management overview.

Last updated on 07/27/2026

Configure wave processing example

This article provides an example of how to set up the criteria that determine what work is generated for a warehouse when a wave is processed, and whether waves are processed manually or automatically. You specify the criteria by setting up wave templates and queries that match a wave with released lines in sales orders, production orders, or kanban orders.

Enable sample data

To work through this scenario using the sample records and values specified here, use a system where the standard demo data is installed. You must also select the **USMF** legal entity before you begin.

Example scenario: configure wave processing

This example scenario walks through many of the diverse settings that affect how waves are created, populated, processed, and released.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Select **New**.
3. In the **Wave template name** field, type a value. When you set up a wave template, specify the sequence in which the templates are matched to released lines on sales orders, production orders, or kanbans. When a line is released to the warehouse or to production, it uses the first wave template that it meets the criteria for. Put templates with the most specific criteria at the top of the list. The broader the criteria, the more likely it is for a line to meet the criteria, and this condition could lead to lines being assigned to the wrong wave.
4. In the **Wave template description** field, type a value.
5. In the **Site** field, enter or select a value. If you're using USMF, you can select site 2.
6. In the **Warehouse** field, enter or select a value. If you're using USMF, you can select warehouse 24.
7. Set the **Automate wave creation** field to **Yes**. Select this option to automatically create a wave when a sales order, production order, or kanban is released to the warehouse.

8. Set the **Process wave at release to warehouse** option to **Yes**. Select this option to automatically process the wave and create work when a line is released to the warehouse.

9. Set the **Automate wave release option** to **Yes**. Select this option to automatically release the wave. The picking work is created and made available on mobile devices.

10. Set the **Assign to open waves option** to **Yes**. Lines are assigned to waves based on the query filter for the wave template.

11. Set the **Process wave automatically at threshold option** to **Yes**. Select this option to automatically process the wave when its values reach the thresholds for weight, shipment, and lines specified in the **Wave thresholds** field group. This option is available only if **Shipping** is selected in the **Wave template type** field.

12. Set the **Automate replenishment work release option** to **Yes**. Select this option to create demand-based replenishment work and release it automatically. You must add the replenishment wave method to the wave template, and create a replenishment template using the **Wave demand** type.

13. Use settings in the **Default values** field group to assign wave attributes. Wave attributes act as filters, to restrict the kind of items that can use the wave. For example, you could specify an item group.

14. Expand the **Methods** section and set the actions taken by the wave template. Wave template methods allow you to control the sequence of activities that each wave goes through when it's processed. For example, you might have a method for wave replenishment. When you add a method, it's automatically listed in the appropriate location in the sequence of steps. If you set the **Automate replenishment work release** option to **Yes**, you need to add the replenish method here.

15. Select **Save**.

16. Close the page.

17. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.

18. Expand the **Wave processing** section.

19. In the **Wave processing batch group** field, enter or select a value.

20. Set the **Process waves in batch option** to **Yes**.

21. In the **Wait for lock (ms)** field, enter a number. Enter the time, in milliseconds, that an allocation step waits for a system resource that is locked by another allocation step. When this time is exceeded, the wave isn't processed and an error message is displayed.
22. Select **Save**.
23. Close the page.
24. Go to **Production control** > **Setup** > **Production control parameters**.
25. In the **Release to warehouse** field, select an option.

    For sales orders and kanban orders, inventory must be reserved before the order is released to the warehouse. Otherwise, items or allocation lines can't be processed. For production orders, you can also select **Allow partial reservation**. For example, this option is useful if you have the materials that you need to start production, and can wait until the additional materials become available to finish the process. If you select this option, you must manually repeat the release to warehouse process when the additional materials become available.
26. Close the page.

Related information

- Wave creation and processing

Last updated on 07/01/2026

Warehouse parameters for wave processing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to set up warehouse parameters for wave processing. You can use wave processing to group picking work for multiple work orders into a single wave.

To use wave processing, specify the following on the **Warehouse management parameters** page:

- If you can process waves by using a batch job.
- If you collect information in a log file when waves are processed.

Set up warehouse management parameters for wave processing

To set up warehouse parameters for wave processing, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.

2. On the **Wave processing** FastTab, make the following settings:

    - **Wave processing batch group** - Select the batch group to use when you process waves by using batch jobs. The batch group specifies the server that batch jobs will run on.
    - **Process waves in batch** - Choose whether to enable waves to be automatically processed by a batch job. You must set this to *Yes* to use parallel processing. You set up the batch job on the **Process waves** page. (See also the note at the end of this list.)
    - **Create wave progress log** - Choose whether the system should generate a log record every time allocation for an item and its dimensions begins and ends. You should only enable this log when you need it, for example, during initial testing or for troubleshooting. Learn more in Wave allocation.
    - **Create wave processing history log** - Choose whether to automatically save information about a wave in a log file after the wave is processed, including during the parallel processing of pending allocations. Typically you should only enable this

during troubleshooting because it adds extra overhead. To view the log, go to **Warehouse management** > **Outbound waves** > **Wave processing history log**. Learn more in Wave creation and processing.

- **Create containerization history log** - Choose whether to automatically save information about containerization for a wave in a log file after the wave is processed. To view the log, go to **Warehouse management** > **Packing and containerization** > **Containerization history**.
- **Wait for lock (ms)** - Enter the time, in milliseconds, that an allocation step will wait for a system resource that is locked by another allocation step. When this time is exceeded, the wave is not processed and an error message is displayed.

3. On the **Release to warehouse** FastTab, make the following setting:

    - **Error on batch failure** - Choose whether to generate an error when a release to warehouse batch job fails. If this is enabled, failed jobs will end with a status of *Error*. If this is disabled, failed jobs will end with a status of *Ended*.

> ⓘ **Note**
>
> On the wave template that is used to process the wave, you can specify the settings that automate wave processing. If you set up a schedule for the batch job, you should coordinate the timing with the settings for automation in the wave template. Learn more in Create a wave template.
>
> If you are using *Transportation management* and *warehouse management processes (WMS)*, you can specify whether to consolidate loads when you process a wave. For example, this is useful when several small loads can be shipped at the same time. To consolidate loads when you process a wave, on the **Loads** tab, select the **Consolidate loads during wave processing** check box.

Set up full or partial reservation for production waves

For sales orders and kanban orders, inventory must be reserved before the order is released to the warehouse. Otherwise, the items or allocation lines cannot be processed in a wave. However, production orders are slightly more flexible. For production orders, you can specify the following:

- Allow production orders to be released to the warehouse although all materials cannot be reserved. If you select this option, you must manually repeat the release to warehouse

process when the additional materials become available. For example, this is useful if you have the materials that you need to start a production, and can wait until the additional materials become available.
- Require that all materials are reserved before an order can be released to the warehouse.

To require full reservation or allow partial reservation, follow these steps:

1. Go to **Production control** > **Setup** > **Production control parameters**.
2. On the **General** tab, in the **Release to warehouse** field, select the default setting.

Last updated on 05/07/2025

Wave templates

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to set up the criteria that determine whether waves are processed manually or automatically, and the work that is generated for a warehouse when a wave is processed. You specify the criteria by setting up wave templates and queries that match a wave with released lines in sales orders, production orders, and kanbans.

Settings for wave templates

When you set up a wave template, you specify the following:

- The **Site** and **Warehouse** that the template creates work for.
- The order in which the system evaluates the templates. This order determines how the templates are matched to released lines on sales orders, production orders, and kanbans. When a line is released, the system applies the first wave template whose criteria the line meets. The broader the criteria, the more likely a line is to meet them, so put the templates with the most specific criteria at the top of the list. Use the **Move up** and **Move down** buttons on the Action Pane to arrange templates in the list.
- The actions taken by each template. The wave **Methods** perform the actions that the template creates, such as creating or distributing work for each type of wave template.
- The wave attributes (filters) that must apply to use the wave template.

> **Note**
>
> If needed, a developer can create additional methods. You can view the full list of methods on the **Wave process methods** page.

Some settings represent strategic decisions for wave processing, as follows:

- If the template is used to ship items for sales, transfer, and outbound shipment orders, or to move items to production for production orders or kanbans.

- If a wave is processed manually or automatically, as follows:
  - **Manual processing** – The line is added to a wave, and the inventory is reserved. However, you must select **Process** on the **All waves** list page to create the picking work for the order.
  - **Automatic processing** – You can fully or partially automate wave processing. If you fully automate wave processing, a wave is created that includes the line from the sales order, production order, or kanban when a sales order, production order, or kanban is created. The items are deducted from on-hand inventory, and the picking work is created. If you partially automate wave processing, you can specify values that will trigger wave processing. For example, you can specify a maximum weight for shipments. Wave processing is then triggered when the total weight of the lines in the wave reaches that value.

- If you assign shipments to an open wave. For example, if you promise customers that an order placed by 12:00 PM will ship within 24 hours, you can set up the wave template to automatically assign order lines to an open wave until 12:00 PM. At that time, the wave is automatically processed.

When a wave is processed, the system creates picking work based on the work template and location directive specified for the warehouse. The work template specifies how picking work is created, and the location directive specifies the pick and put locations.

Create a wave template

To set up a wave template, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.

2. Select **New** to create a new wave template.

3. In the **Wave template type** field, select one of the following options:

   - *Shipping* - Use the wave template for shipping items for sales orders, transfer orders, and outbound shipment orders.
   - *Production orders* - Use the wave template to move items for production orders.
   - *Kanban* - Use the wave template to move items for kanban orders.

4. In the **Wave template name** and **Wave template description** fields, enter a name and a description for the wave template.

   > **Note**
   >
   > If more than one template exists for a warehouse, the number in the **Wave template sequence** field indicates the template's position in the sequence that is applied when the template's criteria are met. You can select **Move up** or **Move down** to rearrange the sequence of templates.

5. On the **Warehouse selection** FastTab, set the following fields to specify the warehouse and site where the wave template applies:

   - **Warehouse selection** – Select one of the following values:
     - *All* – Use the wave template for all warehouses where a more specific wave template hasn't been assigned.
     - *Warehouse group* – Use the wave template for all warehouses in the warehouse group that's selected in the **Warehouse group** field.
     - *Warehouse* – Use the wave template only for the specific warehouse that's selected in the **Warehouse** field.

   - **Site** and **Warehouse** – If the **Warehouse selection** field is set to *Warehouse*, select the site and warehouse where the wave template applies. If you select the warehouse first, the site is filled in automatically. If you select the site first, the warehouse list is filtered to show only warehouses at that site.

   - **Warehouse group** – If the **Warehouse selection** field is set to *Warehouse group*, select the warehouse group where the wave template applies. For more information about how to set up warehouse groups, see Warehouse groups.

6. If you want to automate wave processing, make the following settings as needed:

   - **Automate wave creation** - Set this to *Yes* to automatically create a wave when an order or kanban is released to the warehouse.
   - **Assign to open waves** - Set this to *Yes* to automatically assign lines to an open wave when the lines are released. Lines are assigned to waves based on the query filter for the wave template.
   - **Process wave at release to warehouse** - Set this to *Yes* to automatically process the wave and create work when a line is released to the warehouse.

- **Process wave automatically at threshold** - Set this to *Yes* to automatically process the wave when its values reach the thresholds for weight, shipment, and lines specified in the **Wave thresholds** field group. This setting is only active if *Shipping* is selected in the **Wave template type** field.
- **Automate wave release** - Set this to *Yes* to automatically release the wave. The picking work is created and made available on mobile devices.
- **Automate replenishment work release** - Set this to *Yes* to create demand-based replenishment work and release it automatically. You must add the replenishment wave method to the wave template, and create a replenishment template using the *Wave demand* type. Set up a replenishment template on the **Replenishment templates** page. This requires that you add the replenish method to the wave template.
- **Continue wave processing when work creation fails** - When set to *Yes*, the system uses a blank location if it can't reserve inventory at the location proposed by the location directive (for example, because the inventory is no longer available). When set to *No*, the wave fails if the system can't reserve the inventory.

7. Set the **Wave thresholds** field group as needed:

   - **Wave weight threshold** - Enter the maximum weight a wave can contain.
   - **Shipment threshold** - Enter the maximum number of shipments that can be included in a wave.
   - **Line threshold** - Enter the maximum number of lines that can be included in a wave.

8. In the **Default values** field group, select the wave attributes to use as additional criteria for the wave template. Wave attributes are useful for assigning additional criteria, such as a specific customer name, to a wave template. You create these attributes on the **Wave attributes** page.

9. Set **Wave notification policy** to the policy you want to use for generating notifications related to waves that use this template. For an example of a wave notification policy, see Wave execution notifications.

10. On the **Methods** FastTab, the **Selected methods** pane lists the methods for the selected wave template. The wave methods perform the actions that the template creates, such as creating or distributing work. These actions are also referred to as wave steps. Wave methods are predefined for each type of wave template. You can't remove the predefined wave methods. However, you can rearrange the order of the methods and add additional methods. For example, if you're creating a wave template for shipping, you can add methods for replenishment and containerization. Wave containerization can be added to a

sequence of wave methods to define the containerization of the lines processed in a wave template. To add an additional method, do the following:

- Select a method on the **Remaining methods** pane, and then select the **Add** button (right arrow) to add it to the **Selected methods** pane.
- To change the sequence, select a method, and then select **Move up** or **Move down** arrows.

> **Note**
>
> When you add a method, it's automatically listed in the appropriate location in the sequence of steps.

11. To set up the query that will match released lines to an appropriate wave, select **Edit query** on the Action Pane.
12. To verify that the wave template settings are valid, select **Validate template**.

Last updated on 07/01/2026

Wave step codes

10/03/2025

Wave step codes are codes that users can set up and use to link specific instances of wave methods to a corresponding template. The templates include templates for replenishment, containerization, label printing, load building, and sorting.

When wave step codes aren't used, users must enter free text to reference a specific template from the wave method instance. Free text is prone to errors because users must make sure that the wave step text that they add for a specific wave step method in a wave template exactly matches the wave step text in the target template.

Wave step codes for a specific wave step type are set up on a separate page. For every wave step method instance in a wave template that requires a wave step code, the wave step code must be selected in a drop-down list. Selection in a drop-down list replaces free text entry and helps reduce the risk and impact of human error. Setup codes are used to link a wave step method in a wave template to a target template for the method.

> **Note**
>
> Use of the wave step codes feature is optional. It is enabled organization-wide for all legal entities.

Setup demo

For this demo, demo data must be installed, and you must use the *USMF* demo data company.

Create new wave step codes

You can use the **Wave step codes** page to create and delete wave step codes.

Every new wave step code and its associated ID must be unique across all wave step types, and they must be defined for a specific wave step type.

Apply wave step codes

After you've defined the appropriate wave step codes, they can be applied to the wave process methods.

To apply wave step codes, go to the appropriate target template. Here are the target templates that the wave step codes are designated to point to:

- **Replenishment templates:** Warehouse management > Setup > Replenishment > Replenishment templates
- **Load build templates:** Warehouse management > Setup > Load > Load build templates
- **Sort templates:** Warehouse management > Setup > Packing > Outbound sorting templates
- **Containerization templates:** Warehouse management > Setup > Containers > Container build templates
- **Label printing templates:** Warehouse management > Setup > Document routing > Wave label templates

The templates in this list are applied when they are referenced from a wave process method that is selected in a wave template. When the wave step code on a wave process method in a wave template matches the wave step code in one of the templates types, the template is applied.

Sample scenario

The following procedure helps guarantee that the replenishment template that you created will be applied for the wave template.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave step codes**, and create a wave step code for the **Replenishment** type.
2. Go to **Warehouse management** > **Setup** > **Replenishment** > **Replenishment templates**, and create a replenishment template.
3. In the replenishment template, select the wave step code that you created for the **Replenishment** type.
4. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**, and select the wave template that you intend to use.
5. In the template, on the **Methods** FastTab, select the **Replenishment** method.
6. In the **Wave step code** field, select the wave step code that you selected in the replenishment template.

You perform these steps for each legal entity.

Wave template grouping

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Wave template grouping enables the system to use wave template setups to determine, based on criteria that you define, how it should split released lines and assign them to new or existing waves. This feature can be useful in warehouses where waves are created based on specific criteria, but where managers prefer to create waves automatically instead of manually. It enables the system to add each newly released shipment to the first wave that it finds that has matching grouping field values. If no match is found, the system creates a new wave for the new shipment.

> **Important**
>
> Wave template grouping isn't supported for the work types *production raw material picking* or *Kanban picking*. This is because wave grouping is based on shipments and these work types don't use shipments.

Set a wave template to use wave template grouping

To make wave template grouping available, follow these steps to set up your wave template.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. In the left pane, select the wave template to set up. If you're preparing to work through the scenario later in this article by using demo data, select the **62 Shipping default** template.
3. Select **Edit** to put the page into edit mode.
4. On the **General** FastTab, set the following values:
    - **Automate wave creation:** *Yes*
    - **Assign to open waves:** *Yes*

- **Process wave at release to warehouse:** *No*

5. On the Action Pane, select **Edit query** to open the query dialog box.
6. In the query dialog box, on the **Sorting** tab, review the sorting criteria, and make sure that there is a rule that includes the field that you want to use to group your waves.

   If you're preparing to work through the scenario by using demo data, add a row that has the following values:

   - **Table:** *Shipments*
   - **Derived table:** *Shipments*
   - **Field:** *Carrier service*

     > ⓘ **Note**
     >
     > After you select this value, you might receive the following message: "Field Carrier service is not an index field. Do you want to add sorting on this?" Select **Yes** to add sorting.

   - **Search direction:** *Ascending*
7. Select **OK** to save your changes and close the query dialog box.
8. On the Action Pane, select **Wave template grouping**.
9. On the **Wave template grouping** page, select the **Group by** check box for each row that you want to use to group your order lines into waves, as required. If you're preparing to work through the scenario by using demo data, select the **Group by** check box for the *Carrier service* row.
10. Select **Save**.
11. Close the **Wave template grouping** page.
12. Select **Save** to save your template.

Scenario

This section provides a script that you can work through to learn what this feature does and how to work with it.

Make sample data available

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the **USMF** legal entity before you begin.

You can also use this scenario as guidance for using the feature when you work on a production system. However, in that case, you must substitute your own values, and you might be missing some types of required records that the standard demo data provides.

Scenario: Wave template grouping

This scenario shows how to use wave template grouping to automatically create multiple waves, based on grouping criteria that are defined in a wave template. In this scenario, the wave template is set up in the system to create one wave per carrier service.

Before you begin, prepare your wave template as described in the Set a wave template to use wave template grouping section earlier in this article. If you will be working with demo data for this scenario, be sure to use the demo data values that are suggested in that procedure. This setup will group your waves according to the carrier service that is set for each sales order.

Create sales order 1

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create a sales order.
3. In the **Create sales order** dialog box, set the following values:
    - On the **Customer** FastTab, set the **Customer account** field to *US-004*.
    - On the **General** FastTab, set the **Warehouse** field to *62*.
4. Select **OK** to create the sales order and close the **Create sales order** dialog box.
5. The new sales order is opened in the **Lines** view. Make a note of the sales order number.
6. Switch to the **Header** view.
7. On the **Delivery** FastTab, in the **Transportation** section, set the following values:
    - **Shipping carrier:** *Air cargo*
    - **Carrier service:** *Air*
8. Switch back to the **Lines** view.

9. In the **Sales order lines** section, select **Add line** to add a line to the grid.
10. On the new line, set the following values:
    - **Item number:** *A0002*
    - **Quantity:** *2*
11. Select the new order line, and then, on the **Inventory** menu above the grid, select **Reservation**.
12. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.
13. Close the **Reservation** page to return to the sales order.
14. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.
15. You receive an informational message that shows the shipment and wave for this order. Make a note of the wave ID number and the shipment ID numbers.

View the wave that was created from sales order 1

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave ID that was created from the sales order.
3. Select the wave ID link to open the wave details page.
4. Notice that the shipment has been added to the **Wave lines** FastTab.

Create sales order 2

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create a sales order.
3. In the **Create sales order** dialog box, set the following values:
    - On the **Customer** FastTab, set the **Customer account** field to *US-005*.
    - On the **General** FastTab, set the **Warehouse** field to *62*.
4. Select **OK** to create the sales order and close the **Create sales order** dialog box.
5. The new sales order is opened in the **Lines** view. Make a note of the sales order number.
6. Switch to the **Header** view.
7. On the **Delivery** FastTab, in the **Transportation** section, set the following values:

- **Shipping carrier:** *Flower moving*
- **Carrier service:** *Std*

8. Switch back to the **Lines** view.
9. In the **Sales order lines** section, select **Add line** to add a line to the grid.
10. On the new line, set the following values:

    - **Item number:** *A0001*
    - **Quantity:** *1*

11. Select the new order line, and then, on the **Inventory** menu above the grid, select **Reservation**.
12. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.
13. Close the **Reservation** page to return to the sales order.
14. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.
15. You receive an informational message that shows the shipment and wave for this order. Make a note of the wave ID number and the shipment ID numbers. Notice that the wave ID differs from the wave ID of the first sales order.

View the wave that was created from sales order 2

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave ID that was created from the second sales order.
3. Select the wave ID link to open the wave details page.
4. Notice that the shipment has been added to the **Wave lines** FastTab.

A new wave was created for this shipment, because it uses a different carrier service than the first sales order.

Create sales order 3

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select **New** to create a sales order.
3. In the **Create sales order** dialog box, set the following values:

- On the **Customer** FastTab, set the **Customer account** field to *US-006*.
- On the **General** FastTab, set the **Warehouse** field to *62*.

4. Select **OK** to create the sales order and close the **Create sales order** dialog box.
5. The new sales order is opened in the **Lines** view. Make a note of the sales order number.
6. Switch to the **Header** view.
7. On the **Delivery** FastTab, in the **Transportation** section, set the following values:

    - **Shipping carrier:** *Air Cargo*
    - **Carrier service:** *Air*

8. Switch back to the **Lines** view.
9. In the **Sales order lines** section, select **Add line** to add a line to the grid.
10. On the new line, set the following values:

    - **Item number:** *A0001*
    - **Quantity:** *1*

11. Select the new order line, and then, on the **Inventory** menu above the grid, select **Reservation**.
12. On the **Reservation** page, on the Action Pane, select **Reserve lot** to reserve the full quantity of the selected line in the warehouse.
13. Close the **Reservation** page to return to the sales order.
14. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.
15. You receive an informational message that shows the shipment and wave for this order. Make a note of the wave ID number and the shipment ID numbers. The shipment has been assigned to the existing wave from the first sales order.

View the wave for sales orders 1 and 3

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave ID that was created from the third sales order.
3. Select the wave ID link to open the wave details page.
4. Notice that the shipment has been added to the **Wave lines** FastTab, together with the shipment for the first sales order.

Last updated on 10/03/2025

Wave allocation

Article • 01/23/2023

Wave processing can be time consuming, and most of the processing time is spent in the allocation step and in the work creation step.

It is now possible to run each of these steps in parallel, which can improve the performance of the wave processing, and allow for a larger throughput of waves in the same warehouse. This article explains how to set up the wave allocation method to run in parallel. For more information about how to set up work creation to run in parallel, see Schedule work creation during wave.

Previously it was only possible to allocate one wave at a warehouse at a time. This constraint has been removed and replaced by a new constraint that only locks the item and dimensions that are above location in the reservation hierarchy. Dimensions above the location always include product dimensions. For example, if an item is configured using *Color*, then variants for *Red*, *Blue*, and *Yellow* could each be processed in parallel.

This means that if the same item with the same dimensions above the location is being allocated by one wave, other waves will have to wait to acquire a lock on the same item and dimensions. If the lock can't be acquired in a timely manner, an error will occur and the wave processing will fail.

In order to utilize parallel processing, the wave must run in batch.

Performance improvements

Performance benefits of parallel processing fall in two categories:

- **Improved throughput** - The throughput of waves will typically improve even if parallel processing isn't configured, especially for scenarios where there is no overlap of items within the waves.
- **Improvement of the allocation for a single wave** - Testing on customer data has shown a near 50% performance improvement after switching to parallel allocation. The parallel processing is done per items and dimensions above the location, so the improvements depend on how many different items a wave contains, the infrastructure available, and the duration of the allocation versus the duration of the work creation.

Configure parallel allocation

Warehouse management parameters

To use parallel allocation processing, go to **Warehouse management** > **Setup** > **Warehouse management parameters**, open the **Wave processing** tab, and make the following settings:

- **Wave processing batch group** - Select the batch group that the initial processing of the waves should use. The subsequent processing of allocation can be done using different batch groups.
- **Process waves in batch** - Set this to *Yes* to use parallel processing.
- **Wait for lock (ms)** - Enter the time, in milliseconds, that an allocation step will wait for a system resource that is locked by another allocation step. When this time is exceeded, the wave is not processed and an error message is displayed. We recommend that you allow at least a few seconds to allow for allocation of one logical unit to finish.

For information about these and other wave processing options on the **Warehouse management parameters** page, see Warehouse parameters for wave processing.

Wave process methods

To set up parallel processing:

1. Go to **Warehouse Management** > **Setup** > **Waves** > **Wave process methods**.
2. Select the `allocateWave` method in the grid.
3. On the Action Pane, select **Task configuration**.
4. The **Wave post method task configuration** page opens. This grid lists each warehouse where you have configured the `allocateWave` method. Parallel processing will only be used for warehouses that are listed. Use the Action Pane buttons to add or remove warehouses from the grid as needed.
5. For each warehouse, make the following settings:

    - **Maximum number of batch tasks** - Specify the number of batch tasks that should be used for the allocation for the selected warehouse. The optimal number of batch tasks depends on the infrastructure available and what other batch jobs are being processed on the server. Tests done on a four core environment that was dedicated to wave processing showed that using eight tasks produced good results.
    - **Wave processing batch group** - Specific batch groups can be used for different warehouses to allow the allocation processing to scale out per warehouse.

Enable or disable parallelization across all legal entities

We recommend that you set the `allocateWave` method to run in parallel across all legal entities because this helps to improve the performance of the wave processing. Starting in Supply Chain Management version 10.0.17, the *Wave parallelization for Allocate Wave method* feature is turned on by default for all new and updated installations, and can't be turned off again. After enabling this feature, the following occurs:

- The `allocateWave` method is updated to include a task configuration setting that lets you use the **Wave process methods** page to define the number of tasks that will run simultaneously, equivalent to the number of parallel processes. As a result, the time used on the allocate-wave step (which is typically 30% to 60% of the total processing time) is reduced by a factor roughly equivalent to the number of tasks. It's also possible to select which batch will be assigned to process these tasks. It's important to note that all of your legal entities will be configured to process waves in batch. For the warehouses that are already configured to process waves in batch, and for the warehouses that are already configured to use the `allocateWave` method in parallel, the existing configuration will be kept.
- By default, all the new legal entities are configured to process waves in batch. All new warehouses with the **Warehouse management processes** option enabled will have the `allocateWave` method configured to run in parallel by default.
- On the **Warehouse management parameters** page, **Process saves in batch** is set to *Yes* and **Wait for lock (ms)** set to a default 15 seconds. This means that all waves will be executed in batch. When a wave is running, it acquires a lock on the item and dimensions above location during the allocation step. When another wave processing task tries to acquire the same lock for the identical record, it is blocked until the current process is finished. The **Wait for lock (ms)** settings establishes the maximum time the system will wait before the lock is released.

Parallel allocation processing requires wave processing to run in batch. Therefore, you will reduce your wave processing performance if you turn off the **Process saves in batch** setting, especially if wave processing is using a parallel process as defined by the task configuration for the relevant wave methods.

If necessary, you can undo each of the settings made by default when the *Wave parallelization for Allocate Wave method* feature is automatically enabled for your instance. To do this:

- Go to **Warehouse management** > **Setup** > **Warehouse management parameters**. On the **Wave processing** tab, apply your preferred values for **Process waves in**

batch and **Wait for lock (ms)**.

- Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**. Select the `allocateWave` method. On the Action Pane, select **Task configuration** to open a page that lists each warehouse where the method is set to run in parallel. Modify or delete the number of batch tasks and the assigned wave group for each listed warehouse as needed.

Troubleshooting

Troubleshoot using the Action center

Because the batch framework is used, errors that occur during wave processing will be captured in Action center messages generated by each batch job. To read the batch jobs related to a wave:

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave you want to inspect.
3. On the Action Pane, open the **Wave** tab and, from the **Wave** group, select **Batch jobs**.

Wave processing is self-correcting, so any error detected during processing should be reported using the Action center.

A typical error related to parallel processing could be that two waves try to allocate the same item at the same time and one does not complete so that the other wave is unable to acquire a lock within the specified time. If this situation occurs, the batch jobs log will contain information stating that the lock for the item could not be acquired, in which case the wave that failed must be processed again.

Because the processing is happening in parallel, data must be maintained in different tables to track the state of the processing. This means that the logs for the batch jobs might contain errors such as duplicate key errors.

The errors from the batch tasks are also part of the batch jobs log. The most important information is typically at the bottom.

In rare cases, for example if the SQL connection ended, it is possible for the wave processing to end in an inconsistent state where the batch job appears to be running but the processing is stopped. The wave can't handle errors like this, so an attempt to clean up failed waves is done when the next wave runs. Alternatively, if the current wave is in an inconsistent state, perform the following steps:

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave you need to clean up.
3. On the Action Pane, open the **Wave** tab and, in the **Wave** group, select **Cleanup wave data**.

Troubleshoot using the wave progress log

If the **Create wave progress log** option is enabled on the **Warehouse management parameters** page, then a log record is created every time allocation for an item and its dimensions begins and ends. You should only enable this log when you need it, for example, during initial testing or for troubleshooting. When this option is enabled, you can view the log by taking the following steps:

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave you want to inspect.
3. On the Action Pane, open the **Wave** tab and, in the **Wave** group, select **Progress**.

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Automatic rewaving of nonallocated shipment lines

Article • 04/26/2024

Automatic rewaving of nonallocated shipment lines automatically creates work orders for failed shipment lines. Therefore, those work orders don't have to be manually monitored and created.

Use of this feature requires only that warehouse administrators schedule the *Auto add shipments to wave* job to run as often as it's required. Each time that the job runs, it checks for uncompleted work that can be automatically rewaved. This feature boosts operational efficiency by reducing downtime and initiating work that couldn't previously be completed.

This feature is especially beneficial for high-volume businesses, where it can significantly enhance operational speed and efficiency. Here are some examples:

- A retailer can support continuous packing and shipping operations, even during inventory shortages.
- In high-velocity sales and production environments, goods can be sold before they reach the storage facility from the production line.

The rewaving feature collects shipments lines for inventory that couldn't be allocated during initial wave processing. It adds a **Failed shipment lines** page, where you can review the lines that are ready to be rewaved. A scheduled batch job regularly processes these lines, tries again to allocate inventory for them, and adds the lines to a new wave if inventory is available.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- To run the *Auto add shipments to wave* batch job, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.37 or later.
- To monitor the status of failed shipment lines by using the **Failed shipment lines** page, you must be running Supply Chain Management version 10.0.40 or later.

Wave and rewaving workflows

The standard wave workflow

The wave management process involves three key steps:

1. **Create a wave** – Group work orders, based on criteria such as the shipment date, priority, or item type, and add the group to a wave.
2. **Process the wave** – Pick items from inventory, pack them, and prepare them for shipment.
3. **Release the wave** – Ship the packed items to customers.

The rewaving workflow

The rewaving workflow ensures continuous, uninterrupted warehouse operations, even when a shipment line fails. It ends the need for manual monitoring of failed shipment lines and enables work to be automatically created and added to each wave. It proceeds in the following way:

1. **A shipment line fails while a wave is being processed** – The system removes the failed shipment line from the wave and continues to process the rest of the wave. A failed shipment line can occur if, for example, you don't have enough inventory at a particular picking location.
2. **Create a work placeholder for the failed shipment line** – The system creates a placeholder for the work that's required to process the failed shipment. It stores this placeholder in the *Failed shipment lines* table. (If you aren't using this feature on your system, a user must manually identify failed shipments and then create new work for them.)
3. **Replenish inventory** – Inventory at the picking location is replenished according to standard warehouse operations. That inventory then becomes available for shipment in the next wave.
4. **Automatically rewave** – At the next scheduled run of the *Auto add shipments to wave* batch job, the system checks the *Failed shipment lines* table for previously failed shipments and creates work for them.
5. **Add work to the next wave** – The *Auto add shipments to wave* batch job adds the newly created work to the next wave for processing.
6. **Control and remove placeholders for completed shipment lines** – The *Auto add shipments to wave* batch job ends by checking the *Failed shipment lines* table for records that were successfully processed (that is, their status is no longer *Open*) and that the *Shipment* table includes a shipment for. These records indicate shipment lines that were successfully rewaved. The system cleans up these placeholders by removing them from the table. The system tries to rewave a

shipment line up to five times. If a line fails on the fifth attempt, the system stops trying and marks that line as *Failed*.

7. **Cleanup shipment lines that failed rewaving** - The *Wave processing removed shipments cleanup* job runs and removes *Failed* shipment lines from the *Failed shipment lines* table. These shipment lines are removed from the *Failed shipment lines* table according to the configuration of the *Wave processing removed shipments cleanup* job.

To restart the rewaving process from the beginning, you must allow the cleanup job to remove the failed shipment lines that couldn't be rewaved and then manually add them to a new wave. Then, if a shipment line fails again, the rewave process starts over with the retry count reset to zero.

Enable rewaving for your system

To enable rewaving for your system, you must set up your system to process waves in batches. You must also schedule the *Auto add shipments to wave* batch job.

Enable wave processing in batches to allow for rewaving

Before you can use rewaving, your system must be set up to process waves in batch.

1. In the system settings, go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **Wave Processing** FastTab, set the **Process waves in batch** option to *Yes*.
3. On the Action Pane, select **Save** to apply the new configuration.

Schedule the Auto add shipments to wave batch job

The rewaving process runs as a batch job that you must schedule to run as often as is required for your system. Follow these steps to set up the batch job.

1. Go to **Warehouse Management** > **Outbound Waves** > **Auto add shipments to wave**.

2. On the **Run in the background** FastTab, set the following fields:

    - Set the **Batch processing** option to *Yes*.
    - Make a note of the **Task description** value. The default value is *Auto add shipments to wave*, but you can change it. You can use this value to monitor and edit the job on the **Batch jobs** page.

3. Select the **Recurrence** link.

4. Use the fields that are provided to set up a schedule that defines how often the rewaving job runs, and how long it runs for.

5. Select **OK** to save your schedule for the job.

6. Select **OK** to create the job.

7. On the **Records to include** FastTab, you can define selection criteria to limit the set of shipment lines that's processed.

    - Select the **Filter** link to open a standard query editor dialog box, where you can add or remove filter criteria.
    - The **Records to include** FastTab lists each field name and value that you added by using the query editor.
    - The **Number of retries** filter is always included, and its value (*5*) is read-only. This setting indicates that the system will try to rewave each failed shipment line up to five times. If a line fails on the fifth attempt, the system stops trying and marks that line as *failed*.

Schedule the cleanup job

The cleanup job removes failed and old shipment lines from the *Failed shipment lines* table. You must schedule it to run as often as is required for your system. To set it up, follow these steps.

1. Go to **Warehouse Management** > **Outbound Waves** > **Wave processing removed shipment cleanup**.

2. On the **Parameters** FastTab, set the following fields:

    - **Cleanup rewave threshold** – Enter the maximum number of retries that the cleanup job allows for a shipping line before it removes that line from the *Failed shipment lines* table. The *Auto add shipments to wave* batch job allows for maximum of five retries, but you can remove shipping lines sooner by entering a lower value here. Values that are more than *5* have no effect.
    - **Last update older than given number** – Enter the maximum number of days that the cleanup job should allow a shipping line to remain in the *Failed shipment lines* table. It removes all shipping lines that are older than this value.

3. On the **Run in the background** FastTab, set the following fields:

- **Batch processing** – Set this option to *Yes*.
- **Task description** – The default value is *Wave Processing Removed Shipment*, but you can change it. Make a note of the value. You can use it to monitor and edit the job on the **Batch jobs** page.

4. Select the **Recurrence** link, and use the fields that are provided to set up a schedule that defines how often the cleanup job runs, and how long it runs for.
5. Select **OK** to save your schedule for the job.
6. Select **OK** to create the job.

Monitor the rewave process

Monitor failed shipment lines

To monitor shipment lines that must be rewaved, go to **Warehouse Management** > **Shipments** > **Failed shipment lines**. For each failed shipment line, the page shows the line ID, the line's current status, the reason for the failure, and the number of times that the system tried to rewave the line.

Monitor batch job status

To monitor the status of all batch jobs, adjust their schedules, and fix any issues that arise, follow these steps.

1. Go to **System administration** > **Inquiries** > **Batch jobs**.
2. Use the **Filter** field to find the relevant jobs, based on the value that you entered in the **Task description** field when you set up the job. (On this page, the field is named **Job description**.) The default description for this type of job is *Auto add shipments to wave*.
3. View basic status information about the job in the grid.
4. Select the **Job ID** value to view more details about the job. From the details, you can edit the job's status, change its recurrence schedule, set alerts, and more.

Unsupported processes

The rewaving feature doesn't support the following processes:

- Automatic work creation for inventory that's stored in locations that aren't configured for wave processing

- Continuous checks on inventory levels

Feedback

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Wave execution notifications

10/03/2025

The *Wave execution notifications* feature uses business events and the Action center to deliver notifications that are related to wave execution. It lets you specify the types of events that generate notifications, the warehouses that generate them, and the users who receive them.

The **Show messages** button (bell symbol) on the right side of the navigation bar indicates when an Action center message is available for the current user. The user can select the **Show messages** button to open the Action center and review the messages.

Business events occur when business processes are run. Business processes are made up of tasks. During a business process, the users who participate in it perform business actions to complete those tasks. Business events provide a mechanism that lets external systems receive notifications from finance and operations applications. In this way, the systems can perform business actions in response to the business events. Learn more in Business events overview.

Scenario: Send wave batch execution notifications to the Action center

This scenario shows the end-to-end flow for sending notifications about wave batch execution errors to a specific role via the Action center.

Make demo data available

To follow this scenario, you must have demo data installed, and you must select the **USMF** legal entity.

Make sure that waves are run in batch mode

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Wave processing** FastTab, set the **Process waves in batch** option to *Yes*.

> **Note**
>
> If you want to disable wave batch execution notifications, set the **Disable wave processing batch notifications** option to *Yes*.

Configure a wave notification policy

Wave notification policies define the types of notifications that are sent and the users who are notified.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave notification policies**.
2. Create a record that has the following settings:
    - **Wave notification policy:** *24BatchError*
    - **Description:** *Warehouse 24 wave batch error*
    - **Send notification on:** *Error only*
    - **To role:** *System administrator*

    > **Note**
    >
    > Because this scenario uses demo data, the *System administrator* role is selected for the sake of simplicity. Therefore, because you're signed in as a system administrator user, you will receive the notifications. However, in practice, you should usually select a more specific role to notify about wave batch execution errors, such as *Warehouse manager*.

3. On the Action Pane, select **Save**.

Configure a wave template

Wave templates let you link specific instances of wave methods to corresponding wave label templates.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. In the list pane, set the **Wave template type** field to *Shipping*, and then select the *24 Shipping Default* wave template for warehouse 24.
3. On the **General** FastTab, set the **Wave notification policy** field to *24BatchError*.

Configure a work template

Work templates are used during wave execution to generate work. For this scenario, wave execution should trigger an error. By setting the work template query to use a nonexistent warehouse, you will ensure that wave execution fails and therefore sends a notification.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.

2. In the list pane, set the **Work template type** field to *Sales orders* and then select the *24 SO Stage* work template for warehouse 24.
3. On the Action Pane, select **Edit query**.
4. In the query editor dialog box, on the **Range** tab, edit the following row (or add it if it doesn't exist):
    - **Table:** *Temporary work transactions*
    - **Derived table:** *Temporary work transactions*
    - **Field:** *Warehouse*
    - **Criteria:** Change the value from *24* to *Error*.
5. Select **OK**, and confirm the change.

Create a sales order and release it to the warehouse

1. Go to **Sales and marketing** > **Sales order** > **All sales orders**.
2. Create a sales order that has the following settings:
    - **Customer account:** *US-001*
    - **Warehouse:** *24*
3. On the **Sales order lines** FastTab, add a sales order line that has the following settings:
    - **Item number:** *A0001*
    - **Quantity:** *10*

    > **Note**
    >
    > These items and quantities are only examples. The specified warehouse must contain enough stock.

4. While the new line is still selected on the **Sales order lines** FastTab, select **Inventory** > **Reservation** on the toolbar.
5. On the **Reservation** page, on the Action Pane, select **Reserve lot**. Then close the page.
6. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

Notifications from wave batch job execution

Depending on the setup of your business events, you will eventually receive a notification about wave execution failure. The Action center message will resemble the following example and will include a link to the wave that failed.

> **Error during wave execution**
> An error occurred while executing wave USMF-000000001.
> Last messages: No Work was created for Wave USMF-000000001.
>
> **STATUS**
> Active
>
> Open wave details

Wave label printing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Wave label printing offers an alternative approach to label printing by introducing a new wave step method that lets you create and print labels directly from the wave template during wave execution. Therefore, the labels are available before workers run the work order on a mobile device. Workers can then attach the required labels during picking instead of after picking.

Wave label printing uses Zebra Programming Language (ZPL) to create label layouts. A label layout is divided into three sections (header, body, and footer) to allow for labels that have repeating structure. Wave label templates tell the system which label layout to use. You can specify which printer to use. You can also print labels on several printers at the same time, as you require. The **Wave label history** page shows a record of all labels that are created by using this setup.

You can print and collate labels based on work headers, you can print break labels per work header, and you can print container content labels, case labels, and other similar labels.

> **Note**
>
> This functionality doesn't replace existing label printing functionality that is based on document routing.

Wave label printing offers the following enhancements:

- Print labels according to the number of cartons on a single work line, without using containerization. (A "carton" is a unit designated on unit sequence group lines.)
- Print several different label sequences (for example, carton and pallet labels).
- Include label enumeration (for example, 1/124, 2/124, ... 124/124), and define the range of enumeration (for example, work line, load line, or shipment).

- Create and print a bill of lading ID on labels before the bill of lading is generated.
- Create a unique serial shipping container code (SSCC) for each carton, and include it on each label.
- Create GS1-compliant number sequences for bill of lading IDs and SSCCs.
- Reprint labels from both mobile devices and the rich client.
- Void labels (for example, in short pick scenarios), and reprint them.
- Clean up the wave label history.
- Improvements to document routing layouts are shared between document routing layouts and wave label layouts. (Learn more in Document routing label layouts.)

These enhancements make it more efficient to label cartons before palletization. They especially benefit companies that ship to large retailers that automatically confirm order receipts by scanning each carton separately.

> **Note**
>
> You can implement the configuration scenarios that are described in this article either separately or in combination, depending on your business requirements. You can design several wave label templates that work in sequence (as illustrated in scenario 3). For example, you can use scenario 1 to print carton labels and scenario 2 to print pallet labels (if pallets in stock vary in size and composition).

Scenario 1: Wave label printing where a single wave label is generated

This scenario shows how a company can print shipping labels for a large retailer that automatically confirms order receipts by scanning each carton separately.

This scenario shows the end-to-end flow.

Make demo data available for scenario 1

To follow this scenario, ensure that you have the demo data installed, and select the **USMF** legal entity.

Make sure that the wave label method is available for scenario 1

You might need to regenerate the wave process methods to make the wave label printing method available.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
2. Confirm that **waveLabelPrinting** is in the list. If it isn't, select **Regenerate methods** on the Action Pane to add it.

Set up a wave template for scenario 1

Wave templates let you link specific instances of wave methods to a corresponding wave label template.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Select a template, such as **62 Shipping Default**.
3. On the **Methods** FastTab, move the **Wave label printing** method to the **Selected methods** column.
4. In the **Selected methods** column, select the **Wave label printing** method, and set its **Wave step code** field to *PrintLabel*. For more information about wave step codes, see Wave step codes.

Create a wave label layout for scenario 1

The label layout controls what information is printed on the label and how it's laid out. Here, you enter the ZPL code that is sent to the printer.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label layouts**.
2. Create a record with the following settings:
    - **Label layout ID:** *Carton*
    - **Description:** *Carton (SSCC)*
3. On the Action Pane, select **Save**.
4. On the Action Pane, select **Wave label row settings**.

    The **Wave label row settings** page appears. Here, you can configure the dynamic part of the label.
5. Add a row with the following settings:
    - **Row Id:** *WaveLabel*

- **Row table name:** *WHSWaveLabel*
- **Row start position:** *0*

  This field defines the vertical position where the row begins on the label.

- **Row height:** *0*

  This field defines the height of each row (in points), according to the ZPL standard. The row height is positive for horizontal labels and negative for vertical labels. Because there's just one row in this example, set the value to *0* (zero).

- **Rows per page:** *1*

  This field defines the number of rows that can print on each label.

  > ⓘ **Note**
  >
  > This setup prints a separate ZPL label for each record in the wave labels table.

6. Close the page.
7. On the Action Pane, select **Edit query**.
8. In the query editor dialog box, on the **Range** tab, add a row with the following settings:

   - **Table:** *Work lines*
   - **Derived table:** *Work lines*
   - **Field:** *Work type*
   - **Criteria:** *Pick*

   This query ensures that only pick-type work lines print on the label, not put-type work lines.

9. If you want to print the bill of lading ID, on the **Joins** tab, select the **Work lines** table, and join the **Shipments** table to it.
10. Close the query editor dialog box.
11. The **Printer text Layout** FastTab has three sections where you can write printer code: **Header section**, **Body section**, and **Footer section**. In the **Header section**, in the **Label header** field, enter code for the required header. For example, if you're using Zebra printers, you can use the following code.

plaintext

```
CT~~CD,~CC^~CT~
^XA~TA000~JSN^LT0^MNW^MTT^PON^PMN^LH0,0^JMA^PR8,8~SD15^JUS^LRN^CI0^XZ
^XA
^MMT
^PW812
^LL1218
^LS0
^FT85,505^A0N,28,28^FH\^FD$WHSShipmentTable.CustomerReq$^FS
^FO1,173^GB809,0,1^FS
^FO0,391^GB809,0,1^FS
^FO3,599^GB809,0,2^FS
^FO420,176^GB0,216,1^FS
^FO313,3^GB0,169,1^FS
^FO0,807^GB809,0,1^FS
^FT529,370^A0N,28,26^FH\^FD$WHSShipmentTable.BillOfLadingId$^FS
^BY2,3,132^FT25,344^BCN,,N,N
^FD>:(420)>38102>63^FS
^FT526,315^A0N,28,28^FH\^FD ^FS
^FT437,248^A0N,28,28^FH\^FDCARR: $WHSShipmentTable.SCAC$^FS
^FT425,201^A0N,23,24^FH\^FDCARRIER:^FS
^FT17,68^A0N,20,19^FH\^FDIntershipping, Inc.^FS
^FT15,99^A0N,20,19^FH\^FD1000 Shipping Lane^FS
^FT16,158^A0N,20,19^FH\^FD ^FS
^FT438,368^A0N,28,28^FH\^FDB/L#^FS
^FT15,128^A0N,20,19^FH\^FDShelbyville TN 38102^FS
^FT19,203^A0N,23,24^FH\^FD(420) SHIP TO POSTAL CODE^FS
^FT331,39^A0N,28,28^FH\^FDShip To:^FS
^FT14,39^A0N,28,28^FH\^FDShip From:^FS
^FT331,67^A0N,23,24^FH\^FDWAL-MART DC 1111A-ABC DIS^FS
^FT330,98^A0N,23,24^FH\^FDDEPT 10^FS
^FT329,166^A0N,23,24^FH\^FDSpringfield TN 39021^FS
^FT330,134^A0N,23,24^FH\^FD100 Main Street^FS
^FT19,504^A0N,28,28^FH\^FDPO#:^FS
^FT437,316^A0N,28,28^FH\^FDPRO#^FS
^FT105,371^A0N,28,28^FB130,1,0,C^FH\^FD(420)39021^FS
```

12. In **Body section**, in the **Label body** field, enter ZPL code for the required body. Here's an example.

plaintext

```
<Row name="WaveLabel">
^FT127,439^A0N,28,28^FH\^FD$WHSWaveLabel.SeqNum$^FS
^FT256,439^A0N,28,28^FH\^FD$WHSWaveLabel.NumberOfLabels$^FS
^FT17,439^A0N,28,28^FH\^FDCARTON^FS
^FT522,422^A0N,23,24^FH\^FDVPN:^FS
^FT74,1156^A0N,28,28^FH\^FDSSCC-18^FS
^FT21,579^A0N,28,28^FH\^FDItem name:^FS
^FT107,580^A0N,28,28^FH\^FD$WHSWaveLabel.LabelItemName$^FS
^FT576,423^A0N,23,21^FH\^FD$WHSWaveLabel.LabelItemId$^FS
^FT252,1155^A0N,32,31^FH\^FD(00)$WHSWaveLabel.WaveLabelId$^FS
```

```
^BY4,3,283^FT66,1115^BCN,,N,N
^FD>;>800$WHSWaveLabel.WaveLabelId$^FS
^FT194,439^A0N,28,28^FH\^FDof^FS
</Row>
```

13. In **Body section**, in the **Label footer** field, enter ZPL code for the required footer. Here's an example.

```
plaintext
^PQ1^XZ
```

> ⓘ Note
>
> This setup prints one copy of each label. If you require more copies (for example, one copy for each side of the pallet), set the **n** value for the **^PQn** section in the footer to the required number of copies. For example, to print four copies of each label, specify **^PQ4**.

Your label is now ready to use.

Create a wave label type

Use wave label types to link wave label templates to a unit on unit sequence group lines.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label types**.
2. Add a wave label type that has the following settings:
   - **Label type:** *Carton*
   - **Description:** *Carton*

Set up unit sequence groups for scenario 1

Next, set up the unit sequence group for the wave label type.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Unit sequence groups**.
2. Select the **Ea Box PL** group.
3. For the **Box** line, set the **Wave level type** field to *Carton*.

> ⓘ Note

If a **Wave level type** is set on a unit of measure and a wave label template references the same type, wave label records are created for the total quantity on the work line, one for each unit. For example, setting the type on unit *Box* and given a work line for five boxes, then five wave label records are created. Another example is if the work line is for a single box, and one box equals ten Ea, and the wave label type is set on the unit *Ea*, then ten wave label records are created.

Create a wave label template for scenario 1

Next, create the wave label template for the wave label type.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label templates**.
2. Add a wave level template, and set the following values in the header:
    - **Label template name:** *Carton labels*
    - **Description:** *Carton labels*
    - **Wave step code:** *PrintLabel*
    - **Warehouse:** *62*
3. On the **General** FastTab, set the **Wave label type** field to *Carton*.
4. On the **Wave label template details** FastTab, add a new row that has the following settings:
    - **Label layout ID:** *Carton*
    - **Printer name:** Select an appropriate ZPL printer.
    - **Run query:** *Yes* (This setting is optional, but it's recommended for optimal performance.)
5. On the Action Pane, select **Save**.
6. Optional: If you're setting up a customer-specific label design, create a query to find the customer's account. On the **Wave label template details** FastTab, select **Edit query**. Then, in the query editor dialog box, on the **Range** tab, add a row with the following settings:
    - **Table:** *Shipments*
    - **Derived table:** *Shipments*
    - **Field:** *Account number*
    - **Criteria:** Enter the relevant customer account number.

    When you finish, select **OK** to close the query editor dialog box.

7. On the Action Pane, select **Edit query** to open the query editor dialog box for the whole label template.
8. In the query editor dialog box, on the **Sorting** tab, add a row with the following settings:
    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Reference load line id (Record-ID)*
    - **Search direction:** *Ascending*
9. Select **OK** to close the query editor dialog box.
10. A message box prompts you to confirm the grouping reset operation. Select **Yes** to continue.
11. On the Action Pane, select **Wave label template group**.
12. In the **Wave label template group** dialog box, select the row where the **Reference field name** field is set to *Reference load line id*, and then select the **Label build ID** check box for this row.

    > ⓘ **Note**
    >
    > This setup creates one label sequence ("Carton 1 of X") per load line throughout the wave, regardless of the work grouping setup. You can print this label sequence on the label layout.

Configure number sequence extensions for scenario 1

Number sequence extensions control the GS1 compliance of specific number sequences. This configuration is optional for the current scenario. For more information and configuration instructions, see Configure number sequence extensions.

Create a sales order and release it to the warehouse for scenario 1

1. Go to **Sales and marketing** > **Sales order** > **All sales orders**.
2. Create a sales order with the following settings:
    - **Customer account:** *US-001*

- **Warehouse:** *62*

3. Add two sales order lines with the following settings:

    - Sales order line 1:
        - **Item number:** *A0001*
        - **Quantity:** *9024*
        - **Unit:** *ea* (9024 ea = 376 Box = 47 PL)

    - Sales order line 2:
        - **Item number:** *A0002*
        - **Quantity:** *9016*
        - **Unit:** *ea* (9016 ea = 322 Box = 46 PL)

    > ⓘ **Note**
    >
    > The items and quantities that are provided here are only examples. You must use the unit sequence group that you defined earlier, define appropriate unit conversions from *ea* to *Box* to *PL* for them, and ensure they have stock in warehouse *62*. For more information, see Unit of measure and stocking policies.

4. Select sales order line 1. Then, in the **Sales order line** section, on the **Inventory** menu, select **Reservations**.

5. On the **Reservation** page, on the Action Pane, select **Reserve lot**, and then close the page.

6. Repeat steps 4 and 5 for sales order line 2.

7. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

    The following events occur:

    - The system processes the created shipment by using the template that includes the label printing step. The label layout defines the format of the label, and the result is a label that is printed on the printer, which is selected in the label template.
    - Wave labels are generated and printed. The number of labels equals the number of cartons (in this example, 376 Box labels for line 1 and 322 Box labels for line 2).
    - A new bill of lading ID is generated for the shipments. If you configured the number sequence extensions, the wave label IDs follow the **SSCC-18** number format.

You can view and reprint wave labels from the following pages. On the Action Pane of each page, on the **Shipments** tab, in the **Related information** group, select **Wave labels**.

- All shipments > Shipment details
- All loads > Load details
- All waves
- Wave labels
- Wave label history

Scenario 2: Wave label printing for containerization (without wave label records)

This scenario shows how to print wave labels when you use containerization to automatically split items into cartons and therefore don't require a wave label record. In this case, the container ID acts as a placeholder for the SSCC.

Here are the main differences between this scenario and scenario 1:

- **Wave label templates:** You don't select a wave label type on the wave label template, and you don't require a label build grouping. Otherwise, you configure the wave label template and link to the wave template in the same way that is described in scenario 1. You must leave the wave label type blank to prevent wave labels from being generated.
- **Wave label layouts:** You configure the wave label layout row settings for work lines instead of wave label records. You must configure the row setting for the label layout by using the **WHSWorkLine** table instead of the **WHSWaveLabel** table. The **Rows per page** setting controls the number of rows that the body section has.

This configuration is also suitable for business scenarios where multiple different items are packed into one labeled box or into a pallet, and this packing process can be defined by work creation (for example, work that is grouped by shipment).

This scenario shows the end-to-end flow.

Make demo data available for scenario 2

To follow this scenario, ensure that you have the demo data installed, and select the **USMF** legal entity.

Make sure that the wave label method is available for scenario 2

You might need to regenerate the wave process methods to make the wave label printing method available.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
2. Confirm that **waveLabelPrinting** is in the list. If it isn't, select **Regenerate methods** on the Action Pane to add it.

Set up a wave template for scenario 2

Wave templates let you link specific instances of wave methods to a corresponding wave label template.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Select a template, such as **63 Containerization**.
3. On the **Methods** FastTab, move the **Wave label printing** method to the **Selected methods** column.
4. In the **Selected methods** column, select the **Wave label printing** method, and set its **Wave step code** field to *PrintLabel*. For more information about wave step codes, see Wave step codes.

Create a wave label layout for scenario 2

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label layouts**.
2. Create a record with the following settings:
    - **Label layout ID:** *Carton*
    - **Description:** *Carton (SSCC)*
3. On the Action Pane, select **Save**.
4. On the Action Pane, select **Wave label row settings**.

    The **Wave label row settings** page appears. Here, you can configure the dynamic part of the label.
5. Add a row with the following settings:
    - **Row Id:** *WorkLine*
    - **Row table name:** *WHSWorkLine*
    - **Row start position:** *500*

This field defines the vertical position where the row begins on the label.

- **Row height:** *-50*

  This field defines the height of each row. The row height is positive for horizontal labels and negative for vertical labels.

- **Rows per page:** *5*

  This field defines the number of rows that can print on each label.

  > ⓘ **Note**
  >
  > This setup prints several ZPL labels per work, where each page can hold up to five work lines. For example, if a label is printed for a container that has 12 lines, three labels are printed. If you want to print a separate label for each pick line, set this value to *1*.

6. Close the page.
7. On the Action Pane, select **Edit query**.
8. In the query editor dialog box, on the **Range** tab, add a row with the following settings:
   - **Table:** *Work lines*
   - **Derived table:** *Work lines*
   - **Field:** *Work type*
   - **Criteria:** *Pick*
9. If you want to print the bill of lading ID, on the **Joins** tab, select the **Work lines** table, and join the **Shipments** table to it.
10. Close the query editor dialog box.
11. The **Printer text Layout** FastTab has three sections where you can write printer code: **Header section**, **Body section**, and **Footer section**. In the **Header section**, in the **Label header** field, enter code for the required header. For example, if you're using Zebra printers, you can use the following code.

```plaintext
CT~~CD,~CC^~CT~
^XA
```

```
^LH10,10
^FO0,0 ^AT ^FD$WHSWorkTable.ContainerId$ ^FS
^FO0,75 ^AT ^FD$WHSWorkLine.ShipmentId$ ^FS
^FO0,150 ^AT ^FD$WHSShipmentTable.BillOfLadingId$ ^FS
```

12. In **Body section**, in the **Label body** field, enter ZPL code for the required body. Here's an example.

    plaintext

    ```
    <Row name="WorkLine">
    <Heading>
    //Optional heading for section of rows
    </Heading>
    ^FO0,450 ^AT ^FDItem ^FS
    ^FO200,450 ^AT ^FDQuantity ^FS
    ^FO0,[[YPos]] ^AT ^FD$WHSWorkLine.ItemId$ ^FS
    ^FO200,[[YPos]] ^AT ^FD$WHSWorkLine.QtyWork$ ^FS
    </Row>
    ```

13. In **Body section**, in the **Label footer** field, enter ZPL code for the required footer. Here's an example.

    plaintext

    ```
    ^PQ1^XZ
    ```

    > ⓘ **Note**
    >
    > This setup prints one copy of each label. If you require more copies (for example, one copy for each side of the pallet), set the **n** value for the **^PQn** section in the footer to the required number of copies. For example, to print four copies of each label, specify **^PQ4**.

Your label is now ready to use.

Create a wave label template for scenario 2

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label templates**.
2. Add a wave level template, and set the following values in the header:
    - **Label template name:** *Container labels*
    - **Description:** *Container labels*

- **Wave step code:** *PrintLabel*
- **Warehouse:** *63*

3. On the **Wave label template details** FastTab, add a row with the following settings:

    - **Label layout ID:** *Container*
    - **Printer name:** Select an appropriate ZPL printer.
    - **Run query:** *Yes* (This setting is optional, but it's recommended for optimal performance.)

4. On the Action Pane, select **Save**.

5. Optional: If you're setting up a customer-specific label design, create a query to find the customer's account. On the **Wave label template details** FastTab, select **Edit query**. Then, in the query editor dialog box, on the **Range** tab, add a row with the following settings:

    - **Table:** *Shipments*
    - **Derived table:** *Shipments*
    - **Field:** *Account number*
    - **Criteria:** Enter the relevant customer account number.

    When you finish, select **OK** to close the query editor dialog box.

Configure number sequence extensions for scenario 2

Number sequence extensions control the GS1 compliance of specific number sequences. This configuration is optional for the current scenario. For more information and configuration instructions, see Configure number sequence extensions.

Create a sales order and release it to the warehouse for scenario 2

1. Go to **Sales and marketing** > **Sales order** > **All sales orders**.

2. Create a sales order with the following settings:

    - **Customer account:** *US-001*
    - **Warehouse:** *63*

3. Add five sales order lines:

    - Sales order line 1:

    - **Item number:** *A0001*
    - **Quantity:** *10*

- Sales order line 2:
    - **Item number:** *A0002*
    - **Quantity:** *20*

- Sales order line 3:
    - **Item number:** *L0006*
    - **Quantity:** *20*

- Sales order line 4:
    - **Item number:** *L0100*
    - **Quantity:** *40*

- Sales order line 5:
    - **Item number:** *L0101*
    - **Quantity:** *50*

> ⓘ **Note**
>
> The items and quantities that are provided here are only examples. They must have stock in the specified warehouse.

4. Select sales order line 1. Then, in the **Sales order line** section, on the **Inventory** menu, select **Reservations**.

5. On the **Reservation** page, on the Action Pane, select **Reserve lot**, and then close the page.

6. Repeat steps 4 and 5 for each additional sales order line.

7. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

   The following events occur:

   - The system processes the created shipment by using the template that includes the label printing step. The label layout defines the format of the label, and the end result is a label that has five lines and that is printed on the printer, which is selected in the label template.
   - A new bill of lading ID is generated for the shipments. If you configured the number sequence extensions, the wave label IDs follow the **SSCC-18** number format.

You can reprint these wave labels by going to **Warehouse management** > **Enquiries and reports** > **Wave label history**.

Scenario 3: Wave label printing for multilevel labels

This scenario shows how to use the wave label printing functionality when the warehousing processes require several tiers of shipping labels. For example, you might need to print separate labels for cartons and pallets, and you might need to print a break label for a whole shipment. Break labels are a separate type of label that can be used as a divider between rolls and containers, such as labels for the shipment ID and a bar code, so that the labels can easily be sorted after they're printed.

The main difference between the configuration of this scenario and the configuration of scenario 1, besides the fact that break labels are enabled, is that multiple wave label types must be associated with wave label templates and unit sequence group lines. To accomplish this configuration, you set up the following elements for this scenario:

- **Wave processing methods:** Mark the wave label method as repeatable, add it two or more times to the wave template, and set different wave step codes.
- **Wave label templates:** Configure the wave label templates and link them to the wave template. Each wave label template has its own wave label type.
- **Wave label layouts:** Create multiple wave label layouts. There is a separate label layout for each tier of labels, and there is also a break label layout.

This scenario shows the end-to-end flow.

Make demo data available for scenario 3

To follow this scenario, ensure that you have the demo data installed, and select the **USMF** legal entity.

Set up a wave process method

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
2. Confirm that **waveLabelPrinting** is in the list. If it isn't, select **Regenerate methods** on the Action Pane to add it.
3. For the **waveLabelPrinting** method, select the **Make method repeatable** check box.

Set up a wave template for scenario 3

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Select a template, such as **62 Shipping Default**.
3. On the **Methods** FastTab, move the **Wave label printing** method to the **Selected methods** column.
4. In the **Selected methods** column, assign a **Wave step code** value, such as *Carton*, to the **Wave label printing** method. For more information about wave step codes, see Wave step codes.
5. Move the **Wave label printing** method to the **Selected methods** column a second time.
6. In the **Selected methods** column, assign a different **Wave step code** value, such as *Pallet*, to the second **Wave label printing** method. For more information about wave step codes, see Wave step codes.

Create three wave label layouts for scenario 3

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label layouts**.

2. Create a record with the following settings:

   - **Label layout ID:** *Carton*
   - **Description:** *Carton (SSCC)*

3. On the Action Pane, select **Save**.

4. On the Action Pane, select **Wave label row settings**.

   The **Wave label row settings** page appears. Here, you can configure the dynamic part of the label.

5. Add a row with the following settings:

   - **Row Id:** *WaveLabel*
   - **Row table name:** *WHSWaveLabel*
   - **Row start position:** *0*

     This field defines the vertical position where the row begins on the label.

   - **Row height:** *0*

     This field defines the height of each row (in points), according to the ZPL standard. The row height is positive for horizontal labels and negative for vertical labels. Because there's just one row in this example, set the value to *0* (zero).

- **Rows per page:** *1*

  This field defines the number of rows that can print on each label.

  > ⓘ **Note**
  >
  > This setup prints a separate ZPL label for each record in the wave labels table.

6. Close the page.
7. On the Action Pane, select **Edit query**.
8. In the query editor dialog box, on the **Range** tab, add a row with the following settings:

   - **Table:** *Work lines*
   - **Derived table:** *Work lines*
   - **Field:** *Work type*
   - **Criteria:** *Pick*

   This query ensures that only pick-type work lines print on the label, not put-type work lines.
9. If you want to print the bill of lading ID, on the **Joins** tab, select the **Work lines** table, and join the **Shipments** table to it.
10. Close the query editor dialog box.
11. The **Printer text Layout** FastTab has three sections where you can write printer code: **Header section**, **Body section**, and **Footer section**. In the **Header section**, in the **Label header** field, enter code for the required header. For example, if you're using Zebra printers, you can use the following code.

```plaintext
CT~~CD,~CC^~CT~
^XA~TA000~JSN^LT0^MNW^MTT^PON^PMN^LH0,0^JMA^PR8,8~SD15^JUS^LRN^CI0^XZ
^XA
^MMT
^PW812
^LL1218
^LS0
^FT85,505^A0N,28,28^FH\^FD$WHSShipmentTable.CustomerReq$^FS
^FO1,173^GB809,0,1^FS
^FO0,391^GB809,0,1^FS
^FO3,599^GB809,0,2^FS
^FO420,176^GB0,216,1^FS
```

```
^FO313,3^GB0,169,1^FS
^FO0,807^GB809,0,1^FS
^FT529,370^A0N,28,26^FH\^FD$WHSShipmentTable.BillOfLadingId$^FS
^BY2,3,132^FT25,344^BCN,,N,N
^FD>:(420)>38102>63^FS
^FT526,315^A0N,28,28^FH\^FD ^FS
^FT437,248^A0N,28,28^FH\^FDCARR: $WHSShipmentTable.SCAC$^FS
^FT425,201^A0N,23,24^FH\^FDCARRIER:^FS
^FT17,68^A0N,20,19^FH\^FDIntershipping, Inc.^FS
^FT15,99^A0N,20,19^FH\^FD1000 Shipping Lane^FS
^FT16,158^A0N,20,19^FH\^FD ^FS
^FT438,368^A0N,28,28^FH\^FDB/L#^FS
^FT15,128^A0N,20,19^FH\^FDShelbyville TN 38102^FS
^FT19,203^A0N,23,24^FH\^FD(420) SHIP TO POSTAL CODE^FS
^FT331,39^A0N,28,28^FH\^FDShip To:^FS
^FT14,39^A0N,28,28^FH\^FDShip From:^FS
^FT331,67^A0N,23,24^FH\^FDWAL-MART DC 1111A-ABC DIS^FS
^FT330,98^A0N,23,24^FH\^FDDEPT 10^FS
^FT329,166^A0N,23,24^FH\^FDSpringfield TN 39021^FS
^FT330,134^A0N,23,24^FH\^FD100 Main Street^FS
^FT19,504^A0N,28,28^FH\^FDPO#:^FS
^FT437,316^A0N,28,28^FH\^FDPRO#^FS
^FT105,371^A0N,28,28^FB130,1,0,C^FH\^FD(420)39021^FS
```

12. In **Body section**, in the **Label body** field, enter ZPL code for the required body. Here's an example.

plaintext

```
<Row name="WaveLabel">
^FT127,439^A0N,28,28^FH\^FD$WHSWaveLabel.SeqNum$^FS
^FT256,439^A0N,28,28^FH\^FD$WHSWaveLabel.NumberOfLabels$^FS
^FT17,439^A0N,28,28^FH\^FDCARTON^FS
^FT522,422^A0N,23,24^FH\^FDVPN:^FS
^FT74,1156^A0N,28,28^FH\^FDSSCC-18^FS
^FT21,579^A0N,28,28^FH\^FDItem name:^FS
^FT107,580^A0N,28,28^FH\^FD$WHSWaveLabel.LabelItemName$^FS
^FT576,423^A0N,23,21^FH\^FD$WHSWaveLabel.LabelItemId$^FS
^FT252,1155^A0N,32,31^FH\^FD(00)$WHSWaveLabel.WaveLabelId$^FS
^BY4,3,283^FT66,1115^BCN,,N,N
^FD>;>800$WHSWaveLabel.WaveLabelId$^FS
^FT194,439^A0N,28,28^FH\^FDof^FS
</Row>
```

13. In **Body section**, in the **Label footer** field, enter ZPL code for the required footer. Here's an example.

plaintext

```
^PQ1^XZ
```

> **Note**
>
> This setup prints one copy of each label. If you require more copies (for example, one copy for each side of the pallet), set the **n** value for the **^PQn** section in the footer to the required number of copies. For example, to print four copies of each label, specify **^PQ4**.

14. The first label is now ready to use.

15. Create a second layout record with the following settings:

    - **Label layout ID:** *Pallet*
    - **Description:** *Pallet*

16. On the Action Pane, select **Save**.

17. On the Action Pane, select **Wave label row settings**.

    The **Wave label row settings** page appears. Here, you can configure the dynamic part of the label.

18. Add a row with the following settings:

    - **Row Id:** *WaveLabel*
    - **Row table name:** *WHSWaveLabel*
    - **Row start position:** *0*

      This field defines the vertical position where the row begins on the label.

    - **Row height:** *0*

      This field defines the height of each row (in points), according to the ZPL standard. The row height is positive for horizontal labels and negative for vertical labels. Because there's just one row in this example, set the value to *0* (zero).

    - **Rows per page:** *1*

      This field defines the number of rows that can print on each label.

      > **Note**

This setup causes a separate ZPL label to be printed for each record in the wave labels table.

19. Close the page.

20. On the Action Pane, select **Edit query**.

21. In the query editor dialog box, on the **Range** tab, add a row with the following settings:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Work type*
    - **Criteria:** *Pick*

    This query ensures that only pick-type work lines print on the label, not put-type work lines.

22. If you want to print the bill of lading ID, on the **Joins** tab, select the **Work lines** table, and join the **Shipments** table to it.

23. Close the query editor dialog box.

24. The **Printer text Layout** FastTab has three sections where you can write printer code: **Header section**, **Body section**, and **Footer section**. In the **Header section**, in the **Label header** field, enter code for the required header. For example, if you're using Zebra printers, you can use the following code.

    plaintext

    ```
    CT~~CD,~CC^~CT~
    ^XA
    ^LH10,10
    ^FO0,0 ^AT ^FD$WHSWaveLabel.WaveLabelId$ ^FS
    ^FO0,75 ^AT ^FD$WHSWorkLine.ShipmentId$ ^FS
    ^FO0,150 ^AT ^FD$WHSShipmentTable.BillOfLadingId$ ^FS
    ```

25. In **Body section**, in the **Label body** field, enter ZPL code for the required body. Here's an example.

    plaintext

    ```
    <Row name="WaveLabel">
    ^FO0,450 ^AT ^FDItem ^FS
    ^FO200,450 ^AT ^FDQuantity ^FS
    ^FO0,[[YPos]] ^AT ^FD$WHSWaveLabel.LabelItemId$ ^FS
    ```

```
^FO200,[[YPos]] ^AT ^FD$WHSWaveLabel.QtyWork$ ^FS
</Row>
```

26. In **Body section**, in the **Label footer** field, enter ZPL code for the required footer. Here's an example.

    plaintext

    ```
    ^PQ1^XZ
    ```

    > ⓘ **Note**
    >
    > This setup prints one copy of each label. If you require more copies (for example, one copy for each side of the pallet), set the **n** value for the **^PQn** section in the footer to the required number of copies. For example, to print four copies of each label, specify **^PQ4**.

27. The second label is now ready to use.

28. Create a third layout record that has the following settings:

    - **Label layout ID:** *Break*
    - **Description:** *Break label*

29. On the Action Pane, select **Save**.

30. The **Printer text Layout** FastTab has three sections where you can write printer code: **Header section**, **Body section**, and **Footer section**. In the **Header section**, in the **Label header** field, enter ZPL code for the required header. Here's an example.

    plaintext

    ```
    CT~~CD,~CC^~CT~
    ^XA
    ^LH10,10
    ^FO0,0 ^AT ^FD$WHSWorkLine.ShipmentId$ ^FS
    ```

31. This time, no body is required. Therefore, just enter the required text in the **Footer section**. Here's an example.

    plaintext

```
^XZ
```

The third label is now ready to use.

> **Note**
>
> This third label is a break label used as a divider between label rolls.

Create two wave label types

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label types**.
2. Create a record with the following settings:
    - **Label type:** *Carton*
    - **Description:** *Carton*
3. Create a second record that has the following settings:
    - **Label type:** *Pallet*
    - **Description:** *Pallet*

Set up unit sequence groups for scenario 3

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Unit sequence groups**.
2. Select or create an **Ea Box PL** group.
3. For the **Box** line, set the **Wave level type** field to *Carton*.
4. For the **PL** line, set the **Wave level type** field to *Pallet*.

Create wave label templates

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Wave label templates**.
2. Create a label template with the following settings:
    - **Label template name:** *Carton labels*
    - **Description:** *Carton labels*
    - **Wave step code:** *Carton*
    - **Warehouse:** *62*
3. On the **General** FastTab, in the **Wave label type** field, select a value, such as *Carton*.

4. On the **Wave label template details** FastTab, add a row with the following settings:

    - **Label layout ID:** *Carton*
    - **Printer name:** Select an appropriate ZPL printer.
    - **Run query:** *Yes* (This setting is optional, but it's recommended for optimal performance.)

5. On the Action Pane, select **Save**.

6. Optional: If you're setting up a customer-specific label design, create a query to find the customer's account. On the **Wave label template details** FastTab, select **Edit query**. Then, in the query editor dialog box, on the **Range** tab, add a row with the following settings:

    - **Table:** *Shipments*
    - **Derived table:** *Shipments*
    - **Field:** *Account number*
    - **Criteria:** Enter the relevant customer account number.

    When you finish, select **OK** to close the query editor dialog box.

7. On the Action Pane, select **Edit query** to open the query editor dialog box for the whole label template.

8. In the query editor dialog box, on the **Sorting** tab, add a row with the following settings:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Reference load line id (Record-ID)*
    - **Search direction:** *Ascending*

9. Add a second row with the following settings:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Shipment ID*
    - **Search direction:** *Ascending*

10. Select **OK** to close the query editor dialog box.

11. A message box prompts you to confirm the grouping reset operation. Select **Yes** to continue.

12. On the Action Pane, select **Wave label template group**.

13. In the **Wave label template group** dialog box, for the row where the **Reference field name** field is set to *Shipment ID*, set the following values:

    - **Print break label:** Select this check box.
    - **Label layout ID:** Select a break label. (For example, select the *Break* label layout that you created earlier in this scenario.)
    - **Printer name:** Select the printer for the break label. (Typically, when splitting label rolls, you select the same printer that is selected on the **Wave label template details** FastTab. However, other scenarios are possible.)

14. For the row where the **Reference field name** field is set to *Reference load line id*, select the **Label build ID** check box.

    > **Note**
    >
    > This setup creates one label sequence ("Carton 1 of X") per load line throughout the wave, regardless of the work grouping setup. You can print this label sequence on a label layout. Additionally, labels for different shipments are separated by the selected break label.

15. Select **OK** to close the **Wave label template group** dialog box.

16. Create a second label template with the following settings:

    - **Label template name:** *Pallet labels*
    - **Description:** *Pallet labels*
    - **Wave step code:** *Pallet*
    - **Warehouse:** *62*

17. On the **General** FastTab, in the **Wave label type** field, select a value, such as *Pallet*.

18. On the **Wave label template details** FastTab, add a row with the following settings:

    - **Label layout ID:** *Pallet*
    - **Printer name:** Select an appropriate ZPL printer.
    - **Run query:** *Yes* (This setting is optional, but it's recommended for optimal performance.)

19. On the Action Pane, select **Save**.

20. Optional: If you're setting up a customer-specific label design, create a query to find the customer's account. On the **Wave label template details** FastTab, select **Edit query**. Then, in the query editor dialog box, on the **Range** tab, add a row with the following settings:

    - **Table:** *Shipments*
    - **Derived table:** *Shipments*
    - **Field:** *Account number*
    - **Criteria:** Enter the relevant customer account number.

    When you finish, select **OK** to close the query editor dialog box.

21. On the Action Pane, select **Edit query** to open the query editor dialog box for the whole label template.

22. In the query editor dialog box, on the **Sorting** tab, add a row with the following settings:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Reference load line id (Record-ID)*
    - **Search direction:** *Ascending*

23. Add a second row with the following settings:

    - **Table:** *Work lines*
    - **Derived table:** *Work lines*
    - **Field:** *Shipment ID*
    - **Search direction:** *Ascending*

24. Select **OK** to close the query editor dialog box.

25. A message box prompts you to confirm the grouping reset operation. Select **Yes** to continue.

26. On the Action Pane, select **Wave label template group**.

27. In the **Wave label template group** dialog box, for the row where the **Reference field name** field is set to *Shipment ID*, set the following values:

    - **Print break label:** Select this check box.
    - **Label layout ID:** Select a break label. (For example, select the *Break* label layout that you created earlier in this scenario.)

- **Printer name:** Select the printer for the break label. (Typically, when splitting label rolls, you select the same printer that is selected on the **Wave label template details** FastTab. However, other scenarios are possible.)

28. For the row where the **Reference field name** field is set to *Reference load line id*, select the **Label build ID** check box.

> ⓘ **Note**
>
> This setup creates one label sequence ("Carton 1 of X") per load line throughout the wave, regardless of the work grouping setup. You can print this label sequence on a label layout. Additionally, labels for different shipments are separated by the selected break label.

Configure number sequence extensions for scenario 3

Number sequence extensions control the GS1 compliance of specific number sequences. This configuration is optional for the current scenario. For more information and configuration instructions, see Configure number sequence extensions.

Create a sales order and release it to the warehouse for scenario 3

1. Go to **Sales and marketing** > **Sales order** > **All sales orders**.

2. Create a sales order with the following settings:

   - **Customer account:** *US-001*
   - **Warehouse:** *62*

3. Add two sales order lines:

   - Sales order line 1:
     - **Item number:** *A0001*
     - **Quantity:** *9024*
     - **Unit:** *ea* (9024 ea = 376 Box = 47 PL)

   - Sales order line 2:
     - **Item number:** *A0002*
     - **Quantity:** *9016*

- **Unit:** *ea* (9016 ea = 322 Box = 46 PL)

> ⓘ **Note**
>
> The items and quantities that are provided here are only examples. You must use the unit sequence group that you defined earlier, define appropriate unit conversions from *ea* to *Box* to *PL* for them, and ensure they have stock in warehouse *62*. For more information, see Unit of measure and stocking policies.

4. Select sales order line 1. Then, in the **Sales order line** section, on the **Inventory** menu, select **Reservations**.

5. On the **Reservation** page, on the Action Pane, select **Reserve lot**, and then close the page.

6. Repeat steps 4 and 5 for sales order line 2.

7. On the Action Pane, on the **Warehouse** tab, select **Release to warehouse**.

   The following events occur:

   - The system processes the created shipment by using the template that includes the label printing step. The label layout defines the format of the label, and the result is a label that is printed on the printer, which is selected in the label template.
   - Wave labels are generated and printed. The number of labels equals the number of cartons (in this example, 376 Box labels for line 1, 322 Box labels for line 2, 47 PL labels for line 1, 47 PL labels for line 2, and two break labels that have the shipment ID).
   - A new bill of lading ID is generated for the shipments. If you configured the number sequence extensions, the wave label IDs follow the **SSCC-18** number format.

You can view and reprint wave labels from the following pages:

- All shipments > Shipment details
- All loads > Load details
- All waves
- Wave labels
- Wave label history

For most of these pages, you can find the relevant function by selecting **Wave labels** in the **Related information** group on the **Shipments** tab of the Action Pane.

Related information

- Reprint and void wave labels
- Schedule wave label printing during wave

Last updated on 05/06/2026

Schedule wave label printing during wave

10/03/2025

Use the *Task based wave label printing* feature as part of your waving process to help improve efficiency, and to have the system create wave labels and work in separate tasks.

The process of configuring wave label printing is complex and relies on accurate configuration and master data. It isn't uncommon for the generation of wave label records to fail, and when it does, the whole wave processing is rolled back. The *Task based wave label printing* feature helps you avoid having to re-create work and work lines every time that a wave label is incorrectly printed.

When you use the *Task based wave label printing* feature, the system first creates work and work lines. It then creates and prints wave labels. Finally, if the wave labels are correctly created, it releases the work and wave for picking.

Manually enable the new wave step method

You must first create the new wave step method and enable it for parallel, asynchronous task processing.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave process methods**.
2. On the Action Pane, select **Regenerate method**. Notice that *waveLabelPrinting* is added to the list of wave process methods that you can use in your shipping wave templates.
3. Select the record where the **Method name** field is set to *waveLabelPrinting*, and then, on the Action Pane, select **Task configuration**.
4. On the Action Pane, select **New** to add a row to the grid. Then set the following fields for the new row:

    - **Warehouse** – Select the warehouse that you'll use to schedule work creation processing. (If you're using demo data for testing purposes, you can select warehouse *24*.)
    - **Maximum number of batch tasks** – Specify a maximum number of batch tasks. In most cases, the value should be from *8* through *16*. However, we recommend that you experiment to find the optimal setting for your scenarios.
    - **Wave processing batch group** – Select a dedicated wave processing batch group to optimize batch queue processing.

You can now update an existing wave template so that it uses the *Wave label printing* wave processing method. Alternatively, you can create a new wave template that uses it.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. On the Action Pane, select **Edit**.
3. In the list pane, select the wave template to update. (If you're using demo data for testing purposes, you can select *24 Shipping default*.)
4. On the **Methods** FastTab, in the **Remaining methods** column, select the row where the **Name** field is set to *waveLabelPrinting*.
5. Select **add** (right arrow button) to move the selected row to the **Selected methods** column.
6. In the **Wave step code** field, enter the wave step code that connects the wave template with a wave label template.

Set wave task processing threshold data

The first time that a wave process runs by using any task-based processing, the system creates default wave task processing threshold data. This data is used to control whether wave processing should run asynchronously and be task-based, so that it can process and create wave labels in parallel.

The default data initially uses a threshold value of *1* for the minimum number of work IDs (`MinimumWorkThresholdForLabelPrinting`). Therefore, when the system processes a wave that has more than one work ID, it uses task-based processing of wave labels in a separate transaction. You can manually insert or update this data in the `WHSWaveTaskProcessingThresholdParameters` table in your test environments. To update the table, create and run a custom X++ script.

Changes to the wave processing logic when task-based wave label printing is used

If the wave label processing exceeds the wave task processing threshold, task-based processing is initiated. In the next wave processing that fits the wave template, wave label printing will run in a standalone *ttsbegin/ttscommit* transaction immediately after work creation. If work release (unblocking) is configured on the wave template to run automatically, it occurs only after the wave label printing process is successfully completed.

If wave label generation fails (for example, if conversion of the work quantity to the wave label quantity fails and throws an error), only the appropriate transaction fails. The work that was previously created remains frozen. To correct errors and print the wave labels, follow these steps.

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.

2. Select the relevant wave in the grid.
3. On the Action Pane, on the **Wave** tab, in the **Print** group, select **Wave labels**.
4. Follow the on-screen instructions to send the labels for printing.
5. On the Action Pane, on the **Wave** tab, in the **Wave** group, select **Release** to manually release the work for the selected wave.

Related information

- Wave label printing

Containerization

This article describes how to automate the containerization of loads. Automated containerization creates containers and the picking work for shipments when a wave is processed.

To set up containerization, you must create the following components:

- **Container type** — Define the physical characteristics of the containers. Use container types to pack inventory items into a specific type and size of packaging, such as bins or pallets.
- **Container groups** — Create groups of container types that are similar. For example, a container group can include container types that have similar size dimensions. A container group specifies the sequence in which containers are packed, and the fill percentage of each container.
- **Container build template** — Create templates that define rules for containerization. For example, rules for mixing inventory and other packing strategies.
- **Wave templates** – Set up one or more wave templates to create the picking work for containerization.

Create wave templates for containerization

Set up one or more shipping wave templates for containerization. Wave templates include criteria that determine the following:

- How to process waves. You can choose either manual or automatic processing.
- How to create picking work. This is determined by the wave methods. The wave template must include the **containerization** method.
- How to match items or allocation lines with a wave.

For more information, see Wave templates.

Create container types

Use container types to create descriptions of containers, including maximum values for physical size dimensions and weight capacity. When a containerized wave is processed, the containers are created based on the container type information.

To set up a container type, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container type**.

2. Select **New** to create a new container type.
3. Enter a unique identifier (ID) and description for the container type.
4. In the **Tare weight** field, enter the actual or estimated weight of the container.
5. In the **Maximum weight** field, enter the maximum weight that the container can hold.
6. In the **Containerization maximum volume**, **Containerization maximum length**, **Containerization maximum width**, and **Containerization maximum height** fields, specify the physical dimensions of the container.

   > ⓘ **Note**
   >
   > The values for length and width are interchangeable. This interchangeability means that you can rotate items laterally, or on the x-axis, if needed. For example, if the length is 2 feet, and the width is 1 foot, you can change the length to 1 foot and the width to 2 feet. This interchangeability doesn't apply to the height dimension. You can't change the height value to rotate an item vertically.

7. Select custom attribute values for the container in the fields for attributes. Attributes are custom values that help to filter or sort items based on a value that isn't otherwise available. For example, if you want to pack items for a particular customer, you can create an attribute for the customer name. You create attributes on the **Container attributes** page.

Create container groups

Set up logical groups of container types. For each group, specify the sequence in which to pack the containers and the percentage of the containers to fill. The size dimensions of the item determine whether it fits in a container. The container that is closest to the size dimensions of the item is used. If you have multiple container types in a group, arrange the sequence by size, so that the largest container is first, number 1 in the sequence, and the smallest container is last.

To set up a container group, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container groups**.
2. Select **New** to create a container group.
3. Enter a unique ID and description for the container group.
4. On the **Details** FastTab, select **New** to add a container type to the group.
5. In the **Container type** field, select a container type.

6. Select **Move up** or **Move down** to specify the sequence in which the container types are packed.

Create container build templates

Set up rules for the containerization process, such as inventory mixing rules and other packing strategies. For example, set up a rule so that workers can't pack allocation lines that represent sales orders from different customers in the same container.

To set up a container build template, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container build template**.
2. Create a new container build template.
3. In the **Containerization name** and **Sequence number** fields, enter the name of the containerization template and the order that matches allocation lines.

   > ⓘ **Note**
   >
   > The system applies the first template that the allocation line meets the query criteria for. Therefore, put the template with the most specific criteria at the top of the list. The broader the criteria, the more likely that an allocation line meets the criteria. This condition could lead to lines being assigned to the wrong container. If needed, select **Move up** or **Move down** to change the sequence of templates.

4. In the **Container group ID** field, select the container group from which to create containers.
5. In the **Base query types** field, select the query type that determines what to pack and what to base the filter query on. The following options are available:

   - **Sales allocation line** — Pack allocation lines that are created for sales orders.
   - **Transfer allocation line** — Pack allocation lines that are created for transfer orders.
   - **Container** — Pack a container that the containerization process already created. For example, this option is used for nesting containers.
   - **Outbound order allocation line** — Pack allocation lines that are created for outbound shipment orders.

> **Note**
>
> To use nesting containers, you must make the containerization method repeatable. Learn more in Wave templates.

6. On the **General** FastTab, in the **Wave step code** field, enter the unique identifier of the wave process method that links the container build template to steps in a wave template.

7. Select the **Allow split picks** check box to allow workers to pack items from a work order in separate containers. This condition requires that the entire quantity fits in the container. The largest unit of measure in the allocation line is always used.

8. In the **Pack by unit** field, select the unit of measure that represents the container. For example, you can indicate that a case is a container. If you pack by the unit of measure, you can't also specify a container group because the unit of measure is the container.

9. In the **Container packing strategy** field, select the packing strategy to use. The following options are available:

   > **Note**
   >
   > The strategy applies only to sales allocation lines and transfer allocation lines.

   - **Pack into all open containers** – The system evaluates whether the allocation line fits in any container that's created during the containerization cycle.
   - **Pack into current container only** – The system only evaluates whether the allocation line fits in the most recently created container.

   For more information and examples that show how to work with container packing strategies, see Container packing strategies.

10. To set up rules for packing allocation lines in containers, select **Mixing Logic Breaks**. For example, you can create a rule that allows workers to pack allocation lines for two different items in the same container. To define a mixing rule, follow these steps:
    a. In the **Mixing Logic Breaks** page, select **New**.
    b. In the **Table** field, select the table that contains the field to use as a criterion for the mixing logic break.
    c. In the **Field Select** field, select the field to use as a criterion for the mixing logic break.
    d. Repeat these steps until you add all of the criteria for the mixing logic break.

> **Note**
>
> If you use mixing logic, sort by the same fields in the filter criteria query. This action reduces the number of containers that are created.

Last updated on 07/01/2026

Container packing strategies

10/03/2025

A *container packing strategy* is a strategy that you can use to define item allocations across containers. This article explains the differences between the *Pack into all open containers* and *Pack into current container only* strategies.

- **Pack into all open containers** – During packing, the system checks each item to determine whether it fits into any of the previously created containers. If an item doesn't fit into an existing container, the system creates a new container and continues until it finishes packing the whole order.

  For example, *n* ordered items require containerization. In the worst case, every time that the system processes an item that doesn't fit into any existing container, it does a total of ([(*n* – 1) × (*n* + 1)] ÷ 2) checks to evaluate whether the item fits into an existing container.

- **Pack into current container only** – During packing, the system checks each item to determine whether it fits into the most recently created container. If the item doesn't fit into that container, the system creates a new container and continues until it finishes packing the whole order.

  For example, *n* ordered items require containerization. In the worst case, the system does a total of (*n* – 1) checks to evaluate whether the item fits into an existing container.

Example of the flow for container packing strategies

You set up the following items for containerization.

Expand table

Item	Physical dimensions (width × depth × height)	Weight
HDMI Cable 6'	1 × 1 × 1	1
HDMI Cable 12'	2 × 1 × 1	1
HDMI Cable 18'	3 × 1 × 1	2

You also set up the following box to be used for packaging.

Expand table

Container	Physical dimensions (length × width × height)	Weight	Volume
Medium Box	6 × 3 × 2	10	100

Finally, you set up an order that has the following products and quantities.

Expand table

Sales order line	Quantity
HDMI Cable 12'	9
HDMI Cable 18'	8
HDMI Cable 6'	13

The following table summarizes how containerization works when you use the *Pack into all open containers* strategy and when you use the *Pack into current container only* strategy.

Expand table

Pack into all open containers	Pack into current container
HDMI Cable 12': 1. Create a new container, CONT0001. 2. Put 9 ea into container CONT0001.	HDMI Cable 12': 1. Create a new container, CONT0001. 2. Put 9 ea into container CONT0001.
HDMI Cable 18': 1. Check whether the item can fit into container CONT0001. 2. Create a new container, CONT0002. 3. Put 5 ea into container CONT0002. 4. Create a new container, CONT0003. 5. Put 3 ea into container CONT0003.	HDMI Cable 18': 1. Check whether the item can fit into container CONT0001. 2. Create a new container, CONT0002. 3. Put 5 ea into container CONT0002. 4. Create a new container, CONT0003. 5. Put 3 ea into container CONT0003.
HDMI Cable 6': 1. Check whether the item can fit into container CONT0001. 2. Put 1 ea into container CONT0001. 3. Check whether the item can fit into container CONT0002. 4. Check whether the item can fit into container CONT0003. 5. Put 4 ea into container CONT0003. 6. Create a new container, CONT0004.	HDMI Cable 6': 1. Check whether the item can fit into container CONT0003. 2. Put 4 ea into container CONT0003. 3. Create a new container, CONT0004. 4. Put 9 ea into container CONT0004.

Pack into all open containers	Pack into current container
7. Put 8 ea into container CONT0004.	

Example scenario: Pack single orders per container

This section presents a scenario where the system consolidates multiple orders into one shipment. Therefore, the system performs containerization from the sales order to ensure that every order that contains multiple products is packed into its own container.

This functionality lets you handle scenarios where you must pack only one sales order into each container, so that the distribution center can cross-dock full containers between retail stores. In addition to the retail scenarios (order per retail store and shipping to a distribution center for cross-docking) this technique is also commonly used in lean supply chains (sales order per just-in-time production line).

This scenario shows how you can decrease the number of containers that the system evaluates during packing by using the *Pack into current container only* strategy for containerization.

Prerequisites

This scenario references values and records that are included in the standard demo data that is provided for Microsoft Dynamics 365 Supply Chain Management. If you want to use the values that are provided here as you do the exercises, be sure to work in an environment where the demo data is installed, and set the legal entity to **USMF** before you begin.

Inspect or create container types

To inspect your container types, or to create new container types if they're required, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container types**.
2. Make sure that each of the following container types is available in your demo data. Edit or create container types as required.

    - Container type 1:
        - **Container type code:** *Box-Large*
        - **Description:** *Large Box*
        - **Maximum net weight:** *100*
        - **Volume:** *400*
        - **Length:** *4*

    - **Width:** *10*
    - **Height:** *10*

- Container type 2:
    - **Container type code:** *Box-Medium*
    - **Description:** *Medium Box*
    - **Maximum net weight:** *50*
    - **Volume:** *200*
    - **Length:** *2*
    - **Width:** *10*
    - **Height:** *10*

- Container type 3:
    - **Container type code:** *Box-Small*
    - **Description:** *Small Box*
    - **Maximum net weight:** *20*
    - **Volume:** *100*
    - **Length:** *1*
    - **Width:** *10*
    - **Height:** *10*

Inspect or create container groups

To inspect your container groups, or to create new container groups if they're required, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container groups**.

2. Make sure that the following container group is available in your demo data. If it isn't available, select **New** to create it.

    - **Container group ID:** *Boxes*
    - **Description:** *Box sizes*

3. On the **Details** FastTab for the *Boxes* container group, make sure that the following lines exist. If they don't, select **New** to add them.

    - Line 1:
        - **Sequence number:** *1*
        - **Container type:** *Box-Large*
        - **Container utilization percentage:** *100*

    - Line 2:

    - **Sequence number:** *2*
    - **Container type:** *Box-Medium*
    - **Container utilization percentage:** *100*

- Line 3:
    - **Sequence number:** *3*
    - **Container type:** *Box-Small*
    - **Container utilization percentage:** *100*

Create a new container build template

To create a new container build template, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container build template**.
2. Select **New** to create a container build template that has the following settings:

    - **Sequence number:** *1*
    - **Container template ID:** *Box*
    - **Container group ID:** *Boxes*
    - **Base query types:** *Sales allocation line*
    - **Wave step code:** *234*
    - **Allow split picks:** *Yes*
    - **Container packing strategy:** *Pack into current container only*
    - **Pack by directive unit:** *No*

3. While the row for the new template is still selected, select **Edit query** on the Action Pane.
4. A standard query editor dialog box appears. On the **Sorting** tab, select **Add** to add a row that has the following settings:

    - **Table:** *Temporary work transactions*
    - **Derived table:** *Temporary work transactions*
    - **Field:** *Order number*
    - **Search direction:** *Ascending*

    > **ⓘ Important**
    >
    > To avoid cycling over all the other open containers, and to speed up the process by checking one container at a time, use the *Pack into current container only* strategy in addition to sorting by order number. This combination works like a work break on a work template.

5. Select **OK** to close the query editor dialog box.

6. While the row for the new template is still selected, select **Container mixing constraints** on the Action Pane.

   You now add a constraint that puts items from a single order into a single container. Items from any other order go into a separate container.

7. Select **New** to create a mixing constraint that has the following settings:

   - **Table:** *Sales orders*
   - **Field select:** *SalesId* (The field appears as *Sales order* in the grid.)

8. Select **OK** to add the constraint.

9. Close the page.

Set up a wave template for containerization

To set up a wave template, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.

2. In the list pane, set the **Wave template type** field to *Shipping*.

3. Select the **63 Containerization** template in the list.

4. On the Action Pane, select **Edit**.

5. On the **Methods** FastTab, in the **Selected methods** column, find the following line:

   - **Method name:** *containerization*
   - **Name:** *Containerization*

6. Set the **Wave step code** field for the line to *234*.

Set up a work template

To set up a work template, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.

2. Set the **Work order type** field to *Sales orders*.

3. In the **Overview** grid, find and select the work template that you want to use for packing single orders per container. For this scenario, select the **63 Pick to container** template.

4. On the Action Pane, select **Edit query**.

5. A standard query editor dialog box appears. On the **Sorting** tab, add the following lines:

    - Line 1:
        - **Table:** *Temporary work transactions*
        - **Derived table:** *Temporary work transactions*
        - **Field:** *Shipment ID*
        - **Search direction:** *Ascending*

    - Line 2:
        - **Table:** *Temporary work transactions*
        - **Derived table:** *Temporary work transactions*
        - **Field:** *Order number*
        - **Search direction:** *Ascending*

    - Line 3:
        - **Table:** *Temporary work transactions*
        - **Derived table:** *Temporary work transactions*
        - **Field:** *Container ID*
        - **Search direction:** *Ascending*

6. Select **OK** to close the query editor dialog box.

7. You receive the following message: "Grouping will be reset, continue?" Select **Yes** to continue.

8. While the **63 Pick to container** template is still selected, select **Work header breaks** on the Action Pane.

    You now apply settings to break the work so that each container in the order is linked to one work order.

9. Select the **Group by this field** checkbox for each row on the **Work header breaks** page (**Shipment ID**, **Order number**, and **Container ID**).

10. Close the page.

Set up shipment consolidation policies

To set up a shipment consolidation policy, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Release to warehouse** > **Shipment consolidation policies**.
2. In the list pane, set the **Policy type** field to *Sales orders*.

3. Select the **Default** policy in the list.
4. On the Action Pane, select **Edit**.
5. On the **Consolidation fields** FastTab, in the **Selected fields** list, select the row where the **Field name** field is set to *Order number*.
6. Select the **Remove** button to move the field to the **Remaining fields** list.
7. On the Action Pane, select **Save**.

Set up physical dimensions for the product

To set up physical dimensions for the products in this scenario, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select the product where the **Item number** field is set to *A0001*.
3. On the Action Pane, on the **Manage inventory** tab, in the **Warehouse** group, select **Physical dimensions**.
4. On the **Physical dimensions** page, you should see the following line in the grid:
    - **Unit:** *pcs*
    - **Gross weight:** *3.00*
    - **Width:** *2.00*
    - **Depth:** *2.00*
    - **Height:** *4.00*
    - **Volume:** *16.00*
5. Close the page.
6. Select the product where the **Item number** field is set to *A0002*.
7. On the Action Pane, on the **Manage inventory** tab, in the **Warehouse** group, select **Physical dimensions**.
8. On the **Physical dimensions** page, you should see the following line in the grid:
    - **Unit:** *pcs*
    - **Gross weight:** *4.00*
    - **Width:** *3.00*
    - **Depth:** *1.00*
    - **Height:** *3.00*
    - **Volume:** *9.00*

Create sales order 1

To create a sales order, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. A dialog box for creating a new sales order appears. Set the following values:
    - **Customer account:** *US-001*
    - **Warehouse:** *63*
4. Select **OK** to create the sales order and close the dialog box.
5. The new sales order opens. On the **Sales order lines** FastTab, add the following sales lines:
    - Line 1:
        - **Item number:** *A0001*
        - **Quantity:** *2*
    - Line 2:
        - **Item number:** *A0002*
        - **Quantity:** *2*
6. Select the first line, then select **Inventory** > **Reservation**.
7. On the **Reservation** page, select **Reserve lot**. Then close the page.
8. Repeat the previous two steps for the second line.
9. Close the page.

Create sales order 2

To create a second sales order, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. On the Action Pane, select **New**.
3. A dialog box for creating a new sales order appears. Set the following values:
    - **Customer account:** *US-001*
    - **Warehouse:** *63*
4. Select **OK** to create the sales order and close the dialog box.
5. The new sales order opens. On the **Sales order lines** FastTab, add the following sales lines:

- Line 1:
  - **Item number:** *A0001*
  - **Quantity:** *4*
- Line 2:
  - **Item number:** *A0002*
  - **Quantity:** *4*

6. Select the first line, then select **Inventory** > **Reservation**.
7. On the **Reservation** page, select **Reserve lot**. Then close the page.
8. Repeat the previous two steps for the second line.
9. Close the page.

Create the load

To create a load for each order that you created for this scenario and then release it to the warehouse, follow these steps.

1. Go to **Warehouse management** > **Loads** > **Outbound load planning workbench**.
2. On the **Sales lines** tab, find and select all the sales order lines from the sales orders that you created for this scenario.
3. On the Action Pane, on the **Supply and demand** tab, in the **Add** group, select **To new load**. The selected order lines are added to a new load.
4. In the **Load template assignment** dialog box, in the **Load template ID** field, select a load template, such as *40' Container*.
5. Select **OK** to close the dialog box.
6. In the **Loads** section, find and select the load that you just created.
7. Select **Release** > **Release to warehouse**.
8. In the **Release to warehouse** dialog box, select **OK** to release the selected load to the warehouse.

Verify the shipments and containers

The following procedure lets you verify the shipments that you created. Use it to review the order that you created for this scenario, to make sure that you got the expected results.

1. Go to **Warehouse management** > **Shipments** > **All shipments**.
2. Find and select the shipment that was created for the load that you just released.
3. On the Action Pane, on the **Transportation** tab, select **View containers**.

4. Confirm that the items from the sales orders were containerized into two different containers.

Related information

- Containerization

Container label layouts and printing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Container labels provide information about a container and the related shipment data. A typical scenario that might involve this type of label is one where a worker is creating and packing containers by using the Warehouse Management mobile app. The worker can print a container label that includes a bar code of the container ID and apply it to the physical container.

As with license plate labels, use the Zebra Programming Language (ZPL) to create label layouts for container labels.

Learn more in Packing containers with the Warehouse Management mobile app.

Example scenario: Print container labels when creating containers by using the Warehouse Management mobile app

This example scenario shows how you can set up your system to print container labels when a worker creates a container by using the web client and/or the Warehouse Management mobile app. The Warehouse Management mobile app scenario builds on the information that is provided in Packing containers with the Warehouse Management mobile app. That article provides more details about the full process of packing containers by using the Warehouse Management mobile app.

Make sample data available

To work through this scenario by using the sample records and values that are specified here, you must be on a system where the standard demo data is installed. Additionally, you must select the *USMF* legal entity before you begin.

You can also use this scenario as guidance for using the feature on a production system. However, in that case, you must substitute your own values for each setting that is described here.

Create a container label layout

The label layout controls what information is printed on the label and how it's laid out. Enter the ZPL code that is sent to the printer. Typically, you copy this code from a label designer program.

As the system generates a label, it replaces field and method names that you use in the label layout with actual values. You can easily find text that the system replaces by looking for dollar signs (`) in the code.

Create a basic label layout

Follow these steps to create a container label layout.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.
2. At the top of the list pane, set the **Label layout type** field to *Container label*.
3. On the Action Pane, select **New** to create a label.
4. Set the following values for the new label:
    - **Label layout ID** – Enter *Container*.
    - **Description** – Enter *Container ID barcode*.
    - **Definition type** – Select the method that's used to define the label layout:
        - *ZPL* – Define the label layout by using ZPL.
        - *Variables* – Define a label layout that can be used with an external service. (Learn more in Print labels using an external service.) If you select this option, the **Printer text layout** FastTab provides grids where you define system variables and data variables. Under **System variables**, set the **LabelFile** field to the path of the label design in the external system, and set the **Quantity** field to the number of labels to print. Under **Data variables**, define the values that Supply Chain Management sends by mapping them to the corresponding placeholders in the label design from the external system.

- *Variables (script)* – Use the script format to define a label layout that can be used with an external service. (Learn more in Print labels using an external service.) On the **Printer text layout** FastTab, define the label file, the quantity, and the values to send. Here's an example.

  plaintext

  ```
  "filePath": "/Instant Print/GS1-128.nlbl",
  "quantity": "1",
  "dataSources": [
  {
      "GTIN of Contained Trade Items": "06183928726611",
      "Product_name": "D365FO: $ItemName$"
  }
  ]
  ```

- **Label layout data source ID** – Leave this field blank if you use only container data. If you must include data from other tables, select a label layout data source that has the required joins. For more information about how to set up and use a label layout data source, see the next section in this article.

- **Enable label template support** – Leave this option set to *No* for now. (When it's set to *Yes*, you can add header, row, and footer elements to your layout, as described later in this article.)

- **Date, time, and number format** – Select the language to use when date, time, and number values that are shown in a label layout are formatted.

- **Printer stock type** – Select a *printer stock type*. A printer stock type typically describes the type of paper that a specific printer uses. It's also used to specify the type of paper that a specific label layout should be printed to. For information about how to set up printer stock types, see Set up printer stock types.

5. On the **Printer text Layout** FastTab, enter label code in a way that's appropriate for the selected definition type. The following example shows code that you can copy and paste for testing if the **Definition type** field is set to *ZPL*.

   ZPL

   ```
   CT~~CD,~CC^~CT~
   ^XA~TA000~JSN^LT0^MNW^MTT^PON^PMN^LH0,0^JMA^PR8,8~SD15^JUS^LRN^CI0^XZ
   ^XA
   ^MMT
   ^PW812
   ```

```
^LL0609
^LS0
^BY3,3,262^FT658,186^BAI,,Y,N
^FD$WHSContainerTable.ContainerId$^FS
^FT660,457^A0I,39,38^FH\^FDContainer ID^FS
^FT660,515^A0I,39,38^FH\^FDShipment: $WHSContainerTable.ShipmentId$^FS
^PQ1,0,1,Y^XZ
```

> ⓘ **Note**
>
> While you're customizing the label code on the **Printer text layout** FastTab, you can add valid field and method names by following these steps:
>
> a. In the **Tables** list, select the table.
> b. Depending on the type of item that you want to add, select either the **Fields** tab or the **Methods** tab, and then select the name of the field or method to add.
> c. If the **Definition type** field is set to *ZPL* or *Variables (script)*, select **Insert at end of text** to add the field or method to the end of the code. As you require, move the new field or method to the place in the code where you want to use it.
> d. If the **Definition type** field is set to *Variables*, select a row in the **Data variables** table, and then select **Insert field reference** to add the field or method as a field value.

6. On the Action Pane, select **Save**.

Set up and use a label layout data source

In the label layout in the preceding example, only the container ID (`$WHSContainerTable.ContainerId$`) is used, and this value is available directly in the container table. If you want to include related information (such as the delivery name that's related to a shipment), and the required layout label data source doesn't already exist, follow these steps to create it and then select it in your label layout.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout data source**.

2. On the Action Pane, select **New**.

3. Set the following values for the new label layout data source:

   - **Label layout data source ID** – Enter a name for the data source.
   - **Description** – Enter a short description of the data source.
   - **Label layout type** – Select *Container label*.

4. On the Action Pane, select **Save**.
5. On the Action Pane, select **Edit query**.
6. A standard query editor dialog box appears. On the **Joins** tab, add joins to the required tables. (For example, you might make a join to the shipment table if you want your label to show the delivery name that's related to a shipment.)
7. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.
8. Create or select a label layout, and then, in the **Label layout data source ID** field, select the record that you just created.
9. Add the new field values to the print layout code. Reference the correct *table.field-names* values in the ZPL code. The additional tables include a number as a suffix (_#).

> ⊗ **Caution**
>
> On the **Label layout data source** page, be careful about removing a table from the query for an existing record. You might remove field and/or method names that are already used in existing label layouts.

Enable label template support

Label templates let you design labels that have more advanced layouts, which can include header, row, and footer elements. Follow these steps to format a label that includes label template elements.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.
2. At the top of the list pane, set the **Label layout type** field to *Container Label*.
3. Follow one of these steps:
    - To create a new layout, select **New** on the Action Pane.
    - To edit an existing layout, select the layout in the list pane, and then select **Edit** on the Action Pane.
4. In the **Label layout data source ID** field, select a data source. You need a data source to enable label template support. However, if you need only container table data, you can select a very simple data source where no joins are defined.

5. Set the **Enable label template support** option to *Yes*.

6. Use the `{{Header ... }}`, `{{Row ... }}`, and `{{Footer ... }}` elements in your code. The following example shows a label that includes all these elements. Because it prints data about items that are packed in a container, you need to use a **Label layout data source ID** value that uses a query that joins to the container lines (container details). The data can be split among multiple pages to ensure that all data is printed if you have many container lines. In this example, a container ID bar code and six container lines are printed on the first page. Ten lines are printed per page. Each line contains information about the item, quantity, and unit. This setup is controlled by the `RowsPerLabelFirst=6` and `RowsPerLabel=10` attributes.

ZPL

```
{{LabelStart
^FX ... ZPL commands to start the label ...

^XA
~TA000
~JSN
^LT0
^MNW
^MTT
^PON
^PMN
^LH0,0
^JMA
^PR6,6
~SD15
^JUS
^LRN
^CI27
^PA0,1,1,0
^XZ
^XA
^MMT
^PW800
^LL900
^LS0
}}

{{HeaderFirst
^FX ... Header on the first label only ...

^BY3,3,220
^FO150,120^BC
^FD$WHSContainerTable.ContainerId$^FS
^FT80,420^A0N,33,33^FH\^CI28^FDItem^FS^CI27
^FT579,420^A0N,33,33^FH\^CI28^FDQuantity^FS^CI27
^FT720,420^A0N,33,33^FH\^CI28^FDUnit^FS^CI27
```

```
^FT80,100^A0N,58,58^FH\^CI28
^FDShipment: $WHSContainerTable.ShipmentId$^FS^CI27
}}

{{Header
^FX ... Header on every label after the first ...

^FT80,100^A0N,58,58^FH\^CI28
^FDShipment: $WHSContainerTable.ShipmentId$^FS^CI27
^FT80,150^A0N,40,40^FH\^CI28
^FDContainer: $WHSContainerTable.ContainerId$^FS^CI27
^FT80,220^A0N,33,33^FH\^CI28^FDItem^FS^CI27
^FT579,220^A0N,33,33^FH\^CI28^FDQuantity^FS^CI27
^FT720,220^A0N,33,33^FH\^CI28^FDUnit^FS^CI27
}}

{{Row Table=WHSContainerLine_1 RowsPerLabelFirst=6 RowsPerLabel=10
StartYFirst=500 StartY=300 IncY=50
^FX... ZPL commands to format the row using *$position.YPos$* to position the
location of the text fields ...

^FT80,$position.YPos$^A0N,30,30^TBN,480,30^FH\^CI28^FD$WHSContainerLine_1.ItemId$
^FS^CI27
^FT579,$position.YPos$^A0N,30,30^TBN,120,30^FH\^CI28^FD$WHSContainerLine_1.Qty$^F
S^CI27
^FT720,$position.YPos$^A0N,30,30^TBN,100,30^FH\^CI28^FD$WHSContainerLine_1.UnitId
$^FS^CI27
}}

{{FooterFirst
^FX ... Footer on the first label only ...

^FT550,800^A0N,58,58^FH\^CI28^FDLabel:
$position.labelNumber$/$position.labelCount$^FS^CI27
^PQ1,0,1,Y
}}

{{Footer
^FX ... Footer on every label after the first...

^FT550,800^A0N,58,58^FH\^CI28^FDLabel:
$position.labelNumber$/$position.labelCount$^FS^CI27
^PQ1,0,1,Y
}}

{{LabelEnd
^FX ... ZPL commands to end the label ...

^XZ
}}
```

Note

Because of the `RowsPerLabel=10` attribute, this setup loops over container lines and splits out a label for each set of 10 container lines. If you change the attribute to `RowsPerLabel=1`, a label is generated for each line.

This setup prints one copy of each label. If you require more copies (for example, one copy for each side of the container), set the `n` value for the `\^PQn` section in the footer to the required number of copies. For example, to print two copies of each label, specify `\^PQ2`.

Set up container label routing

To specify the container label layouts that are used and where they're printed, define a **Container label routing** record by following these steps:

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Container label routing**.
2. On the Action Pane, select **New** to create a routing record.
3. On the header of the new routing record, set the following fields:
    - **Sequence number** – Enter an integer to define the order in which to evalute the routing record. Each routing must have a unique sequence number. The system evaluates routings in order of ascending sequence numbers and uses the first routing for which the the criteria are met. If you're using demo data, enter *1*.
    - **Name** – Enter a name for the routing record. For example, enter *Container packing*.
4. On the **Overview** FastTab, use the following fields to define the criteria that select the label routing:
    - **Warehouse** – Specify the warehouse where the routing should be used. If you're using demo data, enter *62*.
    - **Location** – Specify the location where the routing should be used. If you're using demo data, select *Pack*, based on the assumption that the target printer is physically placed at the packing location.
    - **Worker** – Specify the worker that the routing should be used for. To use the routing for any worker, leave this field blank.
    - **Mobile device user ID** – Specify the user ID that the routing should be used for. To use the routing for any worker, leave this field blank.
    - **Container type** – Specify the container type that the routing should be used for.

- **Container status** - Specify the container status (*Open* or *Closed*) that the routing should be used for.
- **Account number** – Specify the customer account that the routing should be used for. To use the routing for any customer, leave this field blank.
- **Carrier** – Specify the shipping carrier that the routing should be used for. To use the routing for any carrier, leave this field blank.
- **Run query** – To add custom selection criteria to a routing record, set this option to *Yes*, and then select **Edit query** on the Action Pane. A standard query editor dialog box appears, where you can add more selection criteria.

> ⓘ Note
>
> - When you print a container label from the Warehouse Management mobile app, the current user's warehouse, location, worker ID, and user ID are passed as possible filter values for selecting the printer and layout. Other values are found based on the selected shipment.
>
> - When a worker prints a container label from the **Close container** dialog, their worker ID, packing station warehouse, and location are passed as possible filter values for selecting the printer and layout. Other values are found based on the selected shipment.

5. On the **Container label routing printer** FastTab, assign the printer and label layout to use when the criteria for the routing record are met. Select **New** on the toolbar to add a line to the grid. Then set the following fields for the new line:

   - **Name** – Select an appropriate ZPL printer. To learn more, see Install the Document Routing Agent to enable network printing. Leave this field blank if you want to use dynamic printer selection, or select *As fallback printer* in the **Use specified printer** field.
   - **Use specified printer** – If you want to use dynamic printer selection and use the configured printer as a fallback when dynamic printer selection doesn't find a more specific printer, set this field to *As fallback printer*. If this field is set to *Always* and a printer is configured, then dynamic printer selection is disabled.
   - **Label layout ID** – Select the label layout to use. The example label layout ID value that was suggested earlier in this scenario was *Container*.

   > ⓘ Note

To reduce the number of label routings, consider using dynamic printer selection with printers associated with packing station locations, mobile device users, or mobile devices. In those cases, label routings only specify the label layouts to be printed based on the properties of the container and you don't need to duplicate the setup when new packing stations and printers are added.

Set container labels to be printed automatically when new containers are created

If you want a container label to be printed automatically each time that a new container is created, configure each packing profile as described in the following procedure.

1. Go to **Warehouse management** > **Setup** > **Packing** > **Packing profiles**.
2. On the Action Pane, select **Edit**.
3. Select the profile that container labels should automatically be printed for. If you're working with sample data, select the row where the **Packing profile ID** field is set to *WHS62*.
4. Select the **Print container label at container creation** checkbox for the selected row.
5. Close the page.

> **Note**
>
> The **Container ID mode** field for packing profile *WH62* is set to *Auto*. Therefore, the number sequence defined for the *Container ID* reference is used as part of the container creation process.

Set container labels to be printed automatically when containers are closed

If you want a container label to be printed automatically each time a container is closed, configure the container packing policy as described in the following procedure.

1. Go to **Warehouse management** > **Setup** > **Containers** > **Container packing policies**.
2. On the Action Pane, select **Edit**.
3. Select the policy that container labels should automatically be printed for. If you're working with sample data, select the row where the **Packing profile ID** field is set to *WHS62Close*.
4. Set **Print container labels on container close** to *Always*.
5. Close the page.

Create a new mobile device menu item for printing container labels

To enable workers to print container labels manually, create a new mobile device menu item for the Warehouse Management mobile app.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. On the Action Pane, select **New** to add a mobile device menu item.
3. For the new menu item, set the following fields:
    - **Menu item name** – Enter an internal name for the new menu item. For example, enter *Print container label*.
    - **Title** – Enter the item name as it should appear in the Warehouse Management mobile app. For example, enter *Print container label*.
    - **Mode** – Select *Indirect*.
    - **Activity code** – Select *Print container label*.
4. Close the page.

Add the new mobile device menu item to the menu

After you create the mobile device menu item, add it to the mobile device menu. In this example, add it to the existing **Outbound** mobile device menu.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. On the Action Pane, select **Edit**.
3. In the list pane, select the **Outbound** menu.
4. In the **Available menus and menu items** column, select the mobile device menu item that you created (for example, *Print container label*).
5. Select the **Add** button (right arrow) to move the menu item into the **Menu structure** column.
6. Close the page.

Run a scenario to print container labels

For an example that shows how to print bar codes automatically as part of a container creation process, see Packing containers with the Warehouse Management mobile app. Follow the instructions in that article, and confirm that the scenario described in this article is also supported

when you use a packing profile where the **Print container label at container creation** checkbox is selected.

To manually print a container label, follow one of these steps:

- In the web client, go to **Warehouse management** > **Packing and containerization** > **Containers**, and select **Print** > **Container label** on the Action Pane.
- In the Warehouse Management mobile app, use the **Print container label** mobile device menu item.

To help reduce the number of steps that workers must perform when they print container labels, consider these suggestions for customizing and fine-tuning this scenario:

- Set up the mobile device menu item to query data by using Warehouse Management mobile app detours. By using this method, the menu item can look up a container ID instead of prompting the worker to enter it manually.
- When a worker selects the **Print container label** mobile device menu item on the **Outbound** menu, the app automatically submits the current **User ID** and **Warehouse** values. If workers want to specify a **Location** value, they can do so in the app.
- If you want the **Location** value to be assigned automatically when a worker selects **Print container label** from the **Pack inventory into containers** menu item, set up a detour.

Related information

- Pack containers for shipment
- Document routing label layouts

Last updated on 05/29/2026

License plate label layouts and printing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Label layouts control what information prints on a label and how it's arranged. You can define a license plate label layout in two ways:

- **Document routing label layouts** – These layouts provide basic layout capabilities.
- **Label layouts** – These layouts let you build more advanced layouts. Your layouts can have repeating structures and include header, body, and footer elements. You can print information from the related tables and define custom date, time, and number formats.

This article describes how to create and use *label layouts* for license plate labels. For more information about *document routing label layouts*, see Document routing label layouts.

Enable license plate label layouts

To enable license plate label layouts, set up the following elements (as described later in this article):

- **Warehouse management parameters** – Define whether to use document routing label layouts or label layouts.
- **License plate label layout** – Define the label layout to use for the license plate labels.
- **License plate label routing** – Define which Zebra Programming Language (ZPL) layouts should be printed to which network printer and under which conditions.

Set up warehouse management parameters

Follow these steps to set up warehouse parameters for license plate label printing.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **License plates** FastTab, set the **Use label layouts for license plate labels** option to *Yes* to use label layouts for your license plates (as described in this article). Set it to *No* to use document routing label layouts instead.

Create a license plate label layout

The label layout controls what information is printed on the label and how it's laid out. Enter the ZPL code that's sent to the printer. Typically, you copy this code from a label designer program.

As the system generates a label, it can replace field and method names that are used in the label layout with actual values. You can easily find text that the system replaces by looking for dollar signs (`) in the code.

Create a basic label layout

Follow these steps to create a license plate label layout.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.
2. At the top of the list pane, set the **Label layout type** field to *License Plate Label*.
3. On the Action Pane, select **New** to create a label.
4. Set the following fields for the new label:
    - **Label layout ID** – Enter a name for the layout (for example, *License plate*).
    - **Description** – Enter a short description of the layout (for example, *License plate*).
        - **Definition type** – Select the method used to define the label layout:
        - *ZPL* – Define the label layout by using ZPL.
        - *Variables* – Define a label layout that you can use with an external service. (Learn more in Print labels using an external service.) If you select this option, the **Printer text layout** FastTab provides grids where you define system variables and data variables. Under **System variables**, set the **LabelFile** field to the path of the label design in the external system, and set the **Quantity** field to the number of labels to print. Under **Data variables**, define the values that Microsoft Dynamics 365 Supply Chain Management sends by mapping them to the corresponding placeholders in the label design from the external system.
        - *Variables (script)* – Use the script format to define a label layout that you can use with an external service. (Learn more in Print labels using an external service.) On the **Printer text layout** FastTab, define the label file, the quantity, and the values to send. Here's an example.

plaintext

```
"filePath": "/Instant Print/GS1-128.nlbl",
"quantity": "1",
"dataSources": [
{
    "GTIN of Contained Trade Items": "06183928726611",
    "Product_name": "D365FO: $ItemName$"
}
]
```

- **Label layout data source ID** – Leave this field blank if you use only license plate data. If you must include data from other tables, select a label layout data source that has the required joins. For more information about how to set up and use a label layout data source, see the next section in this article.
- **Enable label template support** – Leave this option set to *No* for now. (When it's set to *Yes*, you can add header, row, and footer elements to your layout, as described later in this article.)
- **Date, time, and number format** – Select the language to use when date, time, and number values that appear in the label layout are formatted.
- **Printer stock type** – Select a *printer stock type*. A printer stock type typically describes the type of paper that a specific printer uses. It's also used to specify the type of paper that a specific label layout should be printed to. For information about how to set up printer stock types, see Set up printer stock types.

5. On the **Printer text Layout** FastTab, enter label code in a way that's appropriate for the selected definition type. The following example shows code that you can copy and paste for testing if the **Definition type** field is set to *ZPL*.

ZPL

```
CT~~CD,~CC^~CT~
^XA~TA000~JSN^LT0^MNM,0^MTT^PON^PMN^LH0,0^JMA^PR8,8~SD15^JUS^LRN^CI27^PA0,1,1,0
^XZ
^XA
^MMT
^PW831
^LL609
^LS0
^FT19,59^A0N,28,28^FH\^CI28^FDLicense plate label^FS^CI27
^FT19,148^A0N,42,43^FH\^CI28^FDItem: ^FS^CI27
^FT128,148^A0N,42,43^FH\^CI28^FD$ItemId$ ^FS^CI27
^BY3,3,180^FT116,525^BCN,,Y,N
^FH\^FD$LicensePlateId$^FS
^FT19,206^A0N,42,43^FH\^CI28^FDQty:^FS^CI27
```

```
^FT128,206^A0N,42,43^FH\^CI28^FD$Qty$^FS^CI27
^PQ1,0,1,Y^XZ
```

> **Note**
>
> While you're customizing the label code on the **Printer text layout** FastTab, you can add valid field and method names by following these steps:
>
> a. In the **Tables** list, select the table.
> b. Depending on the type of item that you want to add, select either the **Fields** tab or the **Methods** tab, then select the name of the field or method to add.
> c. If the **Definition type** field is set to *ZPL* or *Variables (script)*, select **Insert at end of text** to add the field or method to the end of the code. As you require, move the new field or method to the place in the code where you want to use it.
> d. If the **Definition type** field is set to *Variables*, select a row in the **Data variables** table, then select **Insert field reference** to add the field or method as a field value.

6. On the Action Pane, select **Save**.

Set up and use a label layout data source

In the label layout in the preceding example, only the license plate ID (`$LicensePlateId) is used, and this value is available directly in the license plate table. If you want to include related information (such as the order number that's related to a license plate), and the required layout label data source doesn't already exist, follow these steps to create it and then select it in your label layout.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout data source**.

2. On the Action Pane, select **New**.

3. Set the following fields for the new label layout data source:

   - **Label layout data source ID** – Enter a name for the data source (for example, *LPPlusPurchOrder*).
   - **Description** – Enter a short description of the data source (for example, *License plate + Purchase order*).
   - **Label layout type** – Select *License plate label*.
   - **Join type** – Select the type of joins to use in the data source. Choose one of the following options:

- *Inner* – Use inner joins, which only return rows where there's a match in both tables. This join type creates more specialized joined tables. This option is the default behavior.
- *Outer* – Use outer joins, which return all rows from the left table and the matched rows from the right table. This join type creates larger joined tables that contain more information.

Using outer joins can sometimes help reduce the number of label layout data sources that you need to set up. For example, a license plate label with an inner join to batch requires two label layout data sources: one for items that are batch enabled and one for items that aren't. However, an outer join would let you use just one label layout data source because it returns all items, regardless of whether they're batch enabled. If you instead only want to print labels for non-batch items, then an inner join might work better.

4. On the Action Pane, select **Save**.
5. On the Action Pane, select **Edit query**.
6. A standard query editor dialog box appears. On the **Joins** tab, add joins to the required tables. (For example, if you want your label to show the order number, you might make a join to the purchase order table.)
7. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.
8. Create or select a label layout, and then, in the **Label layout data source ID** field, select the record that you just created.
9. You can now add the new field values to the print layout code. Be sure to reference the correct *table.field-names* values in the ZPL code. The additional tables include a number as a suffix (_#).

> ⊗ **Caution**
>
> On the **Label layout data source** page, be careful about removing a table from the query for an existing record. You might remove field and method names that are already used in existing label layouts.

Enable label template support

Label templates let you design labels that have more advanced layouts, which can include header, row, and footer elements. Follow these steps to format a label that includes label

template elements.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.
2. At the top of the list pane, set the **Label layout type** field to *License Plate Label*.
3. Follow one of these steps:
    - To create a new layout, select **New** on the Action Pane.
    - To edit an existing layout, select the layout on the list pane, then select **Edit** on the Action Pane.
4. In the **Label layout data source ID** field, select a data source. (You need a data source to enable label template support. However, if you need only license plate data, you can select a simple data source where no joins are defined.)
5. Set the **Enable label template support** option to *Yes*.
6. Use the `{{Header ... }}`, `{{Row ... }}`, and `{{Footer ... }}` elements in your code. The following example shows a label that includes all these elements. It prints data about items that are put in a license plate.

    ZPL

    ```
    {{LabelStart
    ^FX ... ZPL commands that will be printed on every label ...
    CT~~CD,~CC^~CT~
    ^XA
    ~TA000
    ~JSN
    ^LT0
    ^MNM,0
    ^MTT
    ^PON
    ^PMN
    ^LH0,0
    ^JMA
    ^PR8,8
    ~SD15
    ^JUS
    ^LRN
    ^CI27
    ^PA0,1,1,0
    ^XZ
    ^XA
    ^MMT
    ^PW831
    ^LL609
    ^LS0
    }}
    ```

```
{{Header
^FT31,59^A0N,28,28^FH\^CI28^FDLicense plate label with work lines^FS^CI27
^BY3,3,180^FT116,300^BCN,,Y,N
^FH\^FD$WHSLicensePlateLabel.LicensePlateId$^FS
}}
{{Row Table=WHsWorkLine_1 StartY=400 IncY=68 RowsPerLabel=5
^FT19,$position.YPos$^A0N,42,43^FH\^CI28^FDLine:^FS^CI27
^FT128,$position.YPos$^A0N,42,43^FH\^CI28^FD$WHSWorkLine_1.ItemID$^FS^CI27
^FT250,$position.YPos$^A0N,42,43^FH\^CI28^FD$WHSWorkLine_1.QtyWork$^FS^CI27
}}
{{LabelEnd
^PQ1,0,1,Y
^XZ
}}
```

> ⓘ **Note**
>
> Because of the `RowsPerLabel=5` attribute, this example loops over license plate lines and splits out a label for each set of five license plate lines. If you change the attribute to `RowsPerLabel=1`, a label is generated for each line.
>
> This example prints one copy of each label. If you need more copies (for example, one copy for each side of the license plate), set the `n` value for the `\^PQn` section in the footer to the required number of copies. For example, to print two copies of each label, specify `\^PQ2`.

Set up license plate label routing

To specify the license plate label layouts and where they're printed, you must define a document routing record, as described in the following procedure.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Document routing**.
2. At the top of the list pane, set the **Work order type** field to *Purchase orders*.
3. On the Action Pane, select **New** to create a routing record.
4. On the header of the new routing record, set the following fields:
    - **Sequence number** – Enter an integer to define the order that the routing record is evaluated in. Each routing must have a unique sequence number. The system evaluates routings in order of ascending sequence numbers and uses the first routing that meets the criteria.
    - **Name** – Enter a name for the routing record. For example, enter *License plate*.

5. On the **Overview** FastTab, use the following fields to define the criteria that are used to select the label routing:

    - **Warehouse** – Specify the warehouse where the routing is used.
    - **Mobile device user ID** – Specify the user ID that the routing is used for. To use the routing for any worker, leave this field blank.
    - **Account number** – Specify the vendor account that the routing is used for. To use the routing for any vendor, leave this field blank.
    - **Carrier** – Specify the carrier that the routing is used for. To use the routing for any carrier, leave this field blank.
    - **Work template** – Specify the work template that the routing is used for. To use the routing for any work template, leave this field blank.
    - **From zone ID** and **To zone ID** - Specify the range of zones that the routing is used for. To use the routing for any zone, leave these fields blank.
    - **Run query** – To add custom selection criteria to a routing record, set this option to *Yes*, then select **Edit query** on the Action Pane. A standard query editor dialog box appears, where you can add more selection criteria.

6. On the **Document routing printers** FastTab, assign the printer and label layout that should be used when the criteria for the routing record are met. Select **New** on the toolbar to add a line to the grid. Then set the following fields for the new line:

    - **Name** – Select an appropriate ZPL printer. Learn more in Install the Document Routing Agent to enable network printing. Leave this field blank if you want to use dynamic printer selection.
    - **Use specified printer** – If you want to use dynamic printer selection and use the configured printer as a fallback when dynamic printer selection doesn't find a more specific printer, set this field to *As fallback printer*. If this field is set to *Always* and a printer is configured, then dynamic printer selection is disabled.
    - **Label layout ID** – Select the label layout to use. The example label layout ID value that was suggested earlier in this article was *License plate*.

Automatically print labels when purchase orders are received by using the mobile app

If you want a license plate label to be printed automatically each time that a new purchase order is received, configure mobile device menu items as described in the following procedure.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.

2. Follow one of these steps:

- To create a new menu item, select **New** on the Action Pane.
- To edit an existing menu item, select it in the list pane, then select **Edit** on the Action Pane.

3. Set the following fields for the new or selected menu item:

    - **Menu item name** – Enter an internal name for the new menu item. For example, enter *Mixed*.
    - **Title** – Enter the item name as it should appear in the Warehouse Management mobile app. For example, enter *Mixed LP Receiving*.
    - **Mode** – Select *Work*.
    - **Work creation process** – Select *Mixed license plate receiving*.
    - **Print label** – Set this option to *Yes*.

4. Close the page.

If you create a new mobile device menu item, you must add it to the mobile device menu, as shown in the following example. In this example, you add it to the existing **Inbound** mobile device menu.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
2. On the Action Pane, select **Edit**.
3. In the list pane, select the **Inbound** menu.
4. In the **Available menus and menu items** column, select the mobile device menu item that you created (for example, *Mixed*).
5. Select the **Add** button (right arrow) to move the menu item into the **Menu structure** column.
6. Close the page.

Run a scenario to print license plate labels

To experiment with printing license plate labels, set up a scenario for mixed license plate receiving through the Warehouse Management mobile app. Learn more in Mixed license plate receiving. Follow the instructions in that article, and confirm that the scenario described in this article is supported.

Related information

- Document routing label layouts

Last updated on 11/20/2025

Enable license plate label printing

Article • 01/31/2023

This article shows how to enable the automatic printing of a Serial shipping container code (SSCC) label after the last item is picked from inventory in a sales picking work process. You can run this procedure in demo data company USMF. If you're run it using your own data, you need to have a number sequence set up for license plates. You need to set up a label printer before you begin this task. Go to Organization administration > Setup > Network printers. On the Action Pane, click Options, and then click the Download document routing agent installer button. Run the installer and make sure that you have a working network printer set to Active before you continue with the procedure.

Set up the GS1 company prefix

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. In the **GS1 company prefix** field, enter the 7 numbers for your GS1 company number.
3. Select **Save**.
4. Close the page.

Setup the SSCC license plate number sequence

1. Go to **Organization administration** > **Number sequences** > **Number sequences**.
2. In the **Area** field, select an option.
3. In the **Reference** field, select an option.
4. In the **Company** field, type a value.
5. Expand the **Segments** section.
6. Select **Edit**.
7. In the **Segments** table, select the first row
8. Select **Remove**.
9. Select **Remove**.
10. Select **Save**.
11. Close the page.

Create the document route layout

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Document routing layouts**. Enable the SSCC layout.

2. Select **New**.
3. In the **Layout ID** field, type a value.
4. In the **Description** field, type a value.
5. Select **Insert at end of text**.
6. Select **Save**.
7. Close the page.

Set up the document routing

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Document routing**.
2. In the **Work order type** field, select an option.
3. Select **New**.
4. In the **Warehouse** field, type a value.
5. In the **Name** field, type a value.
6. Select **New**.
7. In the **Layout ID** field, enter or select a value.
8. In the **Name** field, enter the printer name that you want to use.
9. Select **Save**.
10. Close the page.

Create mobile device menu

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Select **New**.
3. In the **Menu item name** field, type a value.
4. In the **Title** field, type a value.
5. In the **Mode** field, select an option.
6. Select **Yes** in the **Use existing work** field.
7. Select **Yes** in the **Generate license plate** field.
8. Expand the **Work classes** section.
9. Select **New**.
10. In the **Work class ID** field, type a value.
11. Select **Save**.
12. Close the page.
13. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu**.
14. In the tree, select the menu item that you created before.
15. Select **Edit**.
16. Select the arrow to add the menu item to the menu.

17. Select **Save**.
18. Close the page.

Update a work template

1. Go to **Warehouse management** > **Setup** > **Work** > **Work templates**.
2. Select **Edit**.
3. Select **New**.
4. In the **Work type** field, select **Print**.
5. In the **Work class ID** field, enter or select a value.
6. Select **Move up**.
7. Select **Save**.
8. Close the page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Custom label layouts and printing

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to create and use *custom labels*. With custom labels, you can print labels for any type of data that you set up. For example, you can print product labels, location labels, customer labels, and more. When you define one or more custom label layouts, the system automatically shows a **Print** button on the relevant pages.

Set up your system to show a Print button for custom labels

Follow these steps to set up your system to automatically show a **Print** button on pages that handle data for which a custom label is available.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **Reports** tab, on the **Custom labels** FastTab, set the **Display custom label print buttons** option to *On all forms*.

Set up the data source for a custom label

Follow these steps to define the data source for a custom label and then select it in your label layout.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout data source**.

2. Follow one of these steps:

   - To create a new record, select **New** on the Action Pane to add a row to the grid.
   - To edit an existing record, select **Edit** on the Action Pane, then select the row for the label that you want to edit.
   - To remove a record, select it in the grid, then select **Delete** on the Action Pane.

3. If you're creating or editing a record, set the following fields for the new or selected label:

- **Label layout data source ID** – Enter a name for the data source (for example, *Locations*).
- **Description** – Enter a short description of the data source (for example, *Locations*).
- **Label layout type** – Select *Custom label*.
- **Custom label root table** – Select the table that should be used as the main data source for the label (for example, *WMSLocation* to print labels for locations, *InventTable* to print labels for products, or *CustTable* to print labels for customers).
- **Join type** – Select the type of joins used when adding related tables. Selecting *Inner join* retrieves data only if data exists in both joined tables, while selecting *Outer join* retrieves data from the root table even if there's no data in the related table.

> ⓘ **Note**
>
> Forms that don't have a data source or that use a temporary table as a data source can't use custom labels (for example, the **Registration** page for purchase orders uses `TmpInventTransWMS` as a data source, so custom labels can't be printed for that table). Instead, you must use a standard table.

4. On the Action Pane, select **Save**.

5. If you want the custom label to include information from related tables, follow these steps to set up the required joins:
   a. On the Action Pane, select **Edit query**.
   b. A standard query editor dialog box appears. On the **Joins** tab, add the required joins to the required tables.
   c. Select **OK** to apply your settings and close the dialog box.

6. If you want to use parameters specified at printing time, follow these steps to set up the parameters:

   a. On the Action Pane, select **Parameters**.

   b. The **Label layout data source parameters** page opens. Use the Action Pane buttons to add or delete rows as needed. For each parameter, specify the following fields:

      - **Name** – Enter the name of the parameter. This name is used in the label layout to refer to the parameter (for example, *label_quantity*). It can only include letters A-Z (upper and lower case), numbers, and the underscore (_) character.
      - **Data types** – Specify the type of data the parameter can hold. Currently, only *String* is supported, which means that no special number or date formatting is available in label layouts when replacing the placeholder with the value. Using

the *String* value shouldn't pose any issues with whole numbers, such as quantity of labels printed.
- **Display name** – Enter the name that will be shown to the user on the custom label printing dialog (for example, *Quantity of labels*).
- **Mandatory** – Specify whether a value must be given for the parameter on the custom label printing dialog.
- **Default value** – Enter the value that will be initially presented to the user on the custom label printing dialog (for example, *1*).

c. Close the page.

Create and manage custom label layouts

Follow these steps to create a custom label layout that uses data from a label layout data source that you set up.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.

2. At the top of the list pane, set the **Label layout type** field to *Custom Label*.

3. Follow one of these steps:

   - To create a new record, select **New** on the Action Pane.
   - To edit an existing record, select **Edit** on the Action Pane, then select the record that you want to edit in the list pane.
   - To remove a record, select it in the list pane, then select **Delete** on the Action Pane.

4. If you're creating or editing a record, set the following fields on the header:

   - **Label layout ID** – Enter a name for the layout (for example, *Locations*).
   - **Description** – Enter a short description of the layout (for example, *Locations*).
   - **Label layout data source ID** – Select the data source for the custom label (for example, *Locations*).
   - **Enable label template support** – Set this option to *No* if you're creating a simple label layout. Set it to *Yes* to create more advanced layouts that include `{{Header ... }}`, `{{Row ... }}`, and `{{Footer ... }}` elements. For more information about how to use these elements, see Enable label template support.
   - **Date, time, and number format** – Select the language to use when date, time, and number values that are shown in the label layout are formatted.

5. On the **Printer text Layout** FastTab, enter your label content code. Here's an example of code for printing a location label. You can copy and paste it for testing.

ZPL

```
CT~~CD,~CC^~CT~
^XA~TA000~JSN^LT0^MNM,0^MTT^PON^PMN^LH0,0^JMA^PR8,8~SD15^JUS^LRN^CI27^PA0,1,1,0
^XZ
^XA
^MMT
^PW831
^LL609
^LS0
^FT40,59^A0N,28,28^FH\^CI28^FDLocation^FS^CI27
^FT19,148^A0N,42,43^FH\^CI28^FDLocationID: ^FS^CI27
^FT230,148^A0N,42,43^FH\^CI28^FD$WMSLocation_1.wMSLocationId$ ^FS^CI27
^BY3,3,180^FT116,525^BCN,,Y,N
^FH\^FD^FS
^PQ$dsparams.label_quantity$^XZ
```

> ⓘ **Note**
>
> Use `dsparams.parameter-name` to refer to label layout data source parameters.
>
> In this example, the print dialog provides a **Quantity of labels** field, which lets the user specify the number of labels to print.

6. On the Action Pane, select **Save**.

Print a custom label

Follow these steps to print a custom label.

1. In Supply Chain Management, open the page that's associated with the data source that's used for your custom label. Here are some examples:

   - To print a location label (which uses the *WMSLocation* table), go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.
   - To print a product label (which uses the *InventTable* table), go to **Product information management** > **Products** > **Released products**.
   - To print a customer label (which uses the *CustTable* table), go to **Sales and marketing** > **Customers** > **All customers**.

2. Select one or more records that you want to print a label for.

3. On the Action pane, on the **Options** tab, in the **Custom labels** group, select the print option for the type of label that you're printing (for example, **Print locations**).

4. In the **Print custom labels** dialog box, set the following fields:

- **Label layout ID** – Select the layout that you want to use (for example, *Locations*).
- **Printer name** – Select the printer that you want to print from.
- **Label layout data source parameters** - Specify any parameters defined on the label layout data source (for example, *Quantity of labels*).

5. Select **OK**.

Print Arabic language labels

To print Arabic language descriptions on a Zebra Programming Language (ZPL) printer, set up the electronic reporting (ER) framework to generate ZPL labels. Follow these steps:

1. Set up the electronic reporting (ER) Framework.

   Configure the ER parameters in Supply Chain Management to enable label printing. Learn more in Design a new ER solution to print ZPL labels.

2. Design a data model.

   Create an ER configuration that includes a warehouse management domain for label printing.

3. Use Unicode-compatible fonts.

   Make sure that your ZPL printer supports Arabic fonts such as *TT0003M_.TTF* or *ZEBRA Swiss Unicode*.

4. Generate ZPL code.

   Use ER to format the label in ZPL II and include Arabic text by using the `^CI28` command for character encoding.

5. Print the labels.

Related information

- Document routing label layouts

Last updated on 11/20/2025

Document routing label layouts

This article describes how to create layouts for license plate, container, custom, and wave labels. It also provides guidelines for using the Zebra Programming Language (ZPL) used to create the layouts.

Document routing label layouts define the way that labels are laid out and the data that is printed on them. You configure the printing trigger points when you set up mobile device menu items and work templates.

The information in this article applies to all document routing label layouts, including the layouts for license plate labels, container labels, custom labels, and wave labels.

You can print highly complex labels, provided that the printing device can interpret the text that is sent to it. For example, a ZPL layout that includes a bar code might resemble the following example.

ZPL

```
^XA~TA000~JSN^LT0^MNW^MTD^PON^PMN^LH0,0^JMA^PR2,2~SD15^JUS^LRN^CI0^XZ
^XA
^MMT
^PW320
^LL0160
^LS0
^FT20,58^A0N,28,28^FH\^FDLabel:^FS
^FT20,81^AAN,18,10^FH\^FD$LicensePlateId$^FS
^BY1,3,17^FT20,106^BCN,,Y,N,N,A
^FD$LicensePlateId$^FS
^PQ1,,,Y^XZ
```

As part of the label printing process, the text `$LicensePlateId$` in this example is replaced with a data value. Several widely available label generation tools can help you format the text for the label layout. Many of these tools support the `$FieldName$` format. In addition, Microsoft Dynamics 365 Supply Chain Management uses special formatting logic as part of the field mapping for the document routing layout.

To see the values that are printed, go to **Warehouse management** > **Enquiries and reports** > **License plate labels**.

Custom number formats

You can customize the formatting of numerical field values that are printed by using codes that have the following format.

ZPL

```
$FieldName:FormatString$
```

Here is an explanation of this format:

- `FieldName` is the name of the data field (such as **Qty**).
- `FormatString` defines how the data must be printed.

The following examples show how you can customize the work quantity (**Qty**) field:

- To always show four digits (by using zeros as placeholders), use `$Qty:0000$`. For example, if the quantity is 10, the label will show "0010."
- To always show two decimal places, use `$Qty:0.00$`. For example, if the quantity is 10, the label will show "10.00."

For a complete list of the available number format strings, see Custom numeric format strings.

Custom string formats

You can remove the first characters of a string by using the following field and format code.

ZPL

```
$FieldName:#..$
```

Here, `#` specifies the number of characters to skip. For example, to print a Serial Shipping Container Code (SSCC) license plate number that doesn't include the first two characters, use `$LicensePlateId:2..$`. In this case, the license plate number 00111111111111222221 will be printed as "11111111111222221."

Custom date/time formats

The following example shows how you can control the format used to print dates.

ZPL

```
$PrintedDate:dd-MM-yyyy$
```

In this example, the date April 30, 2020, is printed as "30-04-2020."

For a complete list of the available date/time formats, see Custom date and time format strings.

Print individual lines from multiline data

If a data field contains multiple lines (that is, lines that are separated by line breaks), you can print an individual line by using the following format.

ZPL

```
$FieldName[#]$
```

Here, `#` is the line number that you want to print. Use 1 for the first line.

For example, your system has an `AdditionalAddress` field that stores the following multiline address:

Contoso Inc.
123 Street Name
Some City, Some State

You can print this address, one line at a time, by using the following codes.

Expand table

Code	Text that is printed
`$AdditionalAddress[1]$`	Contoso Inc.
`$AdditionalAddress[2]$`	123 Street Name
`$AdditionalAddress[3]$`	Some City, Some State

Print and format from a display method

You can print from a display method by using the following format.

ZPL

```
$DisplayMethod()$
```

You can combine this format with other types that were described earlier in this article. For example, you have a display method that is named `DisplayListOfItemsNumbers()`, and you want to print the first item number of this method. In this case, you can use the following code.

ZPL

```
$DisplayListOfItemsNumbers()[1]$
```

Related information

For more information about how to set up and print labels, see the following articles:

- License plate label layouts and printing
- Container label layouts and printing
- Custom label layouts and printing
- Wave label printing

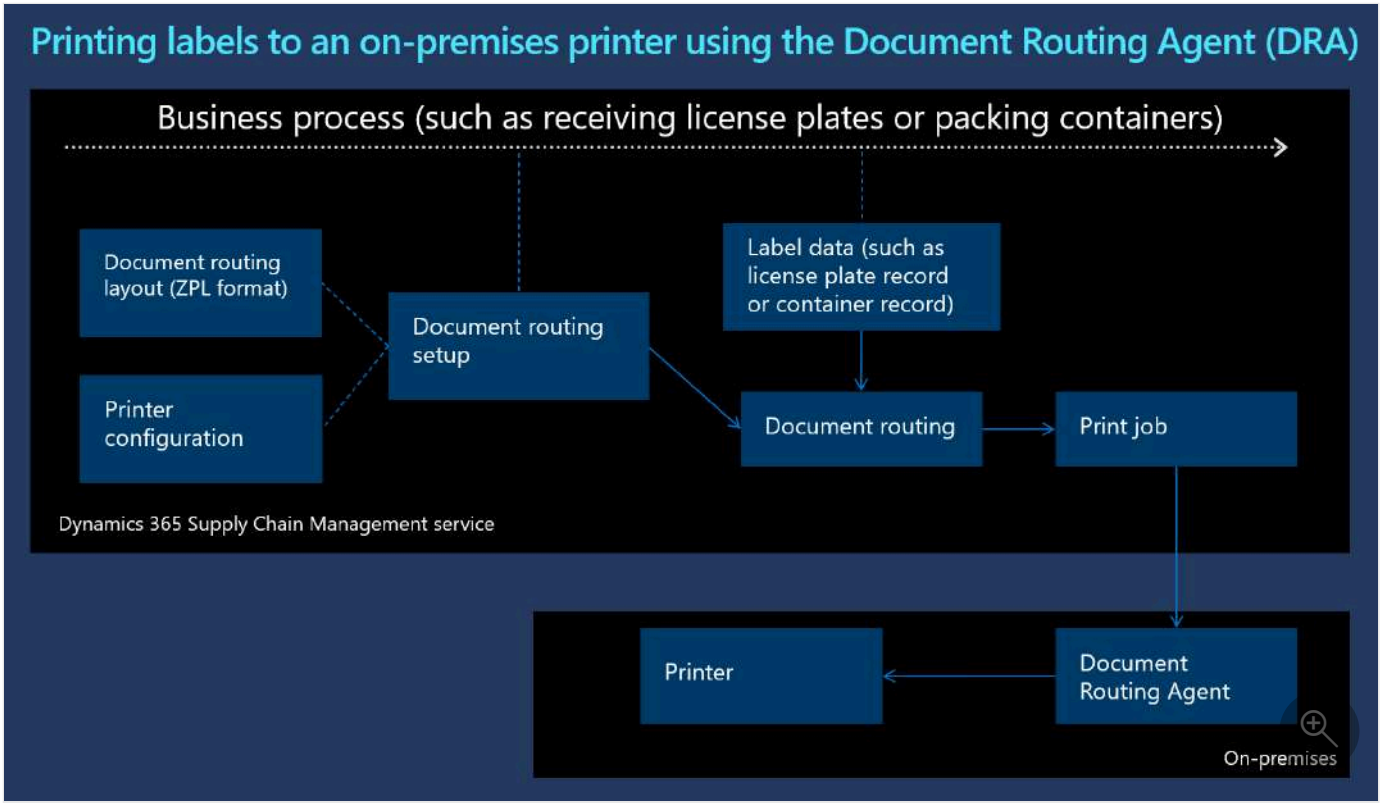
Last updated on 07/01/2026

Print labels by using an external service

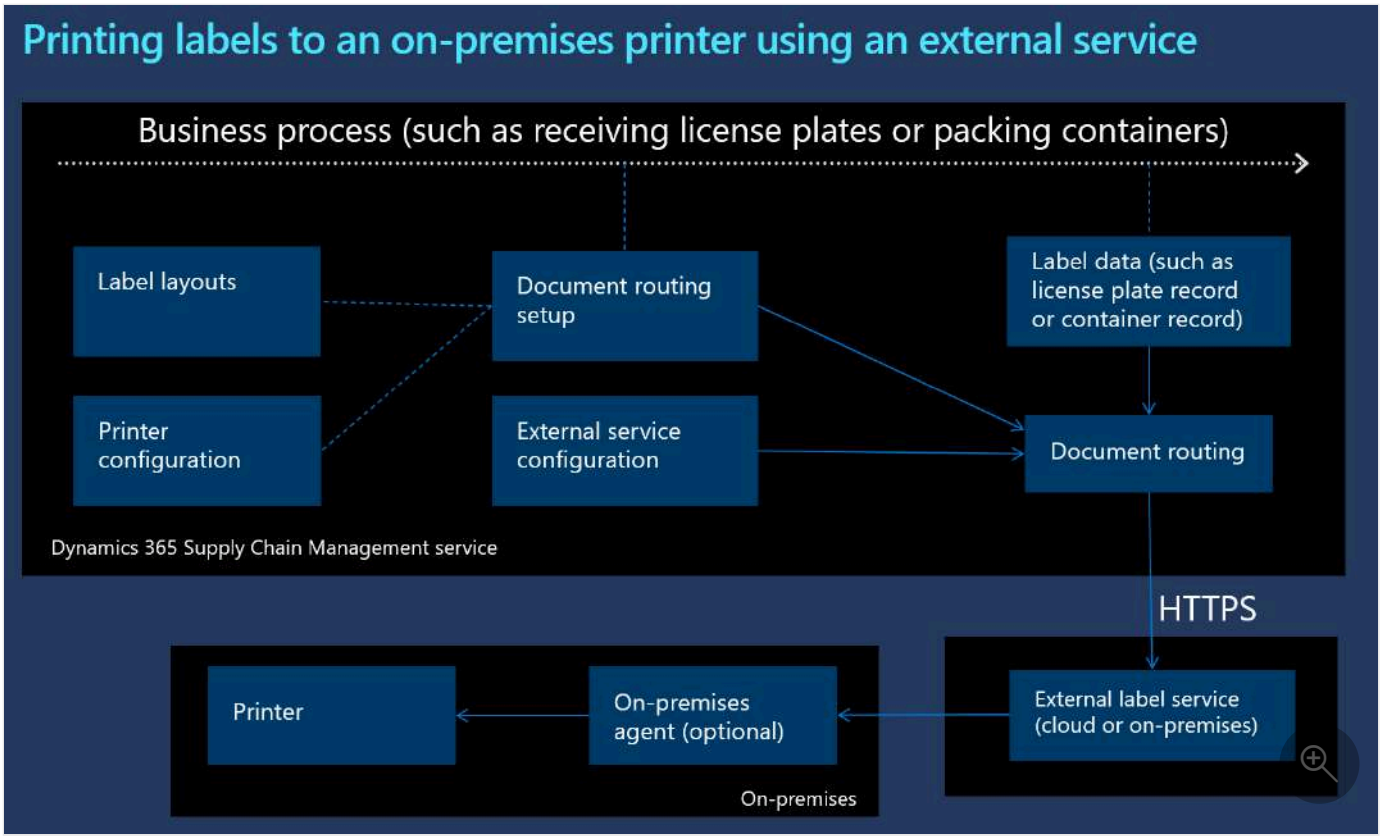
This article describes how to set up and print labels by using an external service. This feature enables direct interaction between Microsoft Dynamics 365 Supply Chain Management and third-party solutions by providing a framework for communicating via HTTP APIs, without requiring the Document Routing Agent (DRA). You can design and print different types of labels by using third-party labeling products, such as Seagull Scientific BarTender® and the Loftware label service solution.

The following illustrations show how printing through the DRA compares to printing through an external service. They highlight the benefits and drawbacks of each method, and can help you decide which printing solution best fits your business needs.

Printing labels by using the DRA



Printing labels by using an external service



For more information about the document routing framework, see Install the Document Routing Agent to enable network printing.

For an example that shows how to set up this feature for the Loftware third-party label printing service, see Print labels using the Loftware label service solution.

For an example that shows how to set up this feature for the Seagull Scientific BarTender third-party label printing service, see Print labels using the Seagull Scientific BarTender label service solution.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- You must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.34 or later.
- You must design your labels using the newer *label layouts* feature instead of the older *document routing label layouts* feature. See the next section for guidelines.

Use label layouts instead of document routing label layouts

Classic *document routing label layouts* treat the layout body as plain text and apply string escaping, which can produce malformed JSON or XML payloads. Therefore, you must use the newer *label layouts* when you print labels through an external service.

- To view and create layouts that use the new *label layouts* feature, go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**. Use these layouts when you print labels through an external service.
- To see the older *document routing label layouts*, go to **Warehouse management** > **Setup** > **Document routing** > **Document routing layout**. Don't use these layouts to print labels through an external service.

Learn more about how to work with label layouts in Container label layouts and printing, License plate label layouts and printing, and Custom label layouts and printing.

To enable label layouts for license plate labels, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. Open the **General** tab.
3. On the **License plates** FastTab, set **Use label layouts for license plate labels** to *Yes*.

If you need to migrate existing document routing label layouts, use the Data entities framework to export and then import them by using the following entities:

- **Document routing printer layouts** — to export your existing document routing layout definitions, including the template body content.
- **Label layouts with active versions** — to import and export label layout definitions.
- **Label layout data sources** — to import and export custom data source configurations used by label layouts.

Overview of required configurations

To enable label printing through an external service, set up the following elements:

- **External service definition** – Defines how to call the external API. The definition consists of one or more operations.
- **External service instance** – Represents a specific deployment of the service described by the external service definition. Specifies the URL of the service and the authentication secret to use. For each external service definition, you can use one or more instances. For example, you can set up multiple warehouses, each of which has its own on-premises installation of labeling software.

- **Label printers** – Defines the label printers that the labeling service manages.
- **Document routing setup** – Defines which printer is used for which layout in a specific business process.
- **Label layout** – Defines a label layout.

The remaining sections of this article describe how to configure each of these elements.

Set up an external service definition

An external service definition specifies how to call a specific service. This configuration depends on the API definitions that the external service supports. Typically, the external service provider supplies these API definitions.

Follow these steps to set up an external service definition.

1. Go to **Warehouse management** > **Setup** > **External services** > **External service definitions**.
2. On the Action Pane, select **New** to create an external service definition.
3. Set the following fields for the new service definition:
    - **External service definition** – Enter a name for the definition, such as *External*.
    - **Description** – Enter a short description of the definition, such as *External*.
4. On the Action Pane, select **Save**.
5. On the **External service operations** FastTab, select **Edit operations** on the toolbar.
6. The **External service operations** page appears. External service operations define how the system formats its HTTP requests to the external service. You must add at least one operation, which the system uses to print labels. On the Action Pane, select **New** to add a new operation, and then follow these steps:
    - On the header of the new record, set the following fields:
        - **External service definition** – This read-only field shows a value of *External*.
        - **External service operation** – Enter a name for the operation.
        - **Description** – Enter a short description of the operation.
    - On the **General** FastTab, set the following fields:
        - **HTTP method** – Select the HTTP method to use (*GET*, *POST*, or *PUT*).

    - **Operation timeout** – Select the timeout period for the operation, in milliseconds (for example, *500*).
    - **Request body type** – Select the type of body that the service expects:
        - *Raw* – The service expects an XML or JSON message.
        - *form-data* – The service expects data that's formatted as an HTTP form.
    - **Relative URL** – Enter the relative part of the URL for calling the service. Your external service provider must provide this value.
- If you want to express a request entirely in a URL, use the **Request query string** FastTab to define query parameters for authenticating requests.
- Use the **HTTP request headers** FastTab to define the HTTP authorization header, which is the most common method for providing authentication information.
- The **Request body (form data)** FastTab appears when the **Request body type** field is set to *Form-data*. Use it to provide the names of the form data fields and their values. Use the toolbar to add or remove rows in the grid as required. For each row, set the following fields:
    - **Key** – Enter the name of the form data field.
    - **Value** – Enter the value of the form data field.
    - **File name** – This field is required only in special cases where the service expects that the value resembles an uploaded file. For example, it's required by Zebra's `SendFileToPrinter` service.
- The **Request body (raw)** FastTab appears when the **Request body type** field is set to *Raw*. Use it to specify the body content and the content type. Set the following fields:
    - **Content type** – Specify the MIME content type identification (for example *application/xml*, *application/json*, or *text/plain*).
    - **Request body** – Enter the body content.

You can use the following substitution placeholders in the header values, the form data values, or the request body:

- `$auth.secret` – This placeholder is replaced by the authentication secret configured on the external service instance. Use it to store a shared key, token, or password that's required to authenticate with the external service instance.
- `$label.printer` – This placeholder is replaced by the name of the printer that you configure later. Use it to signal to the external service where the label should be

printed.
- `$label.body` or `$label.body:base64` – This placeholder is replaced by the label generated from a label layout or document routing layout. Use the base64 formatting version (`$label.body:base64`) if the externals service expects a base64-encoded Zebra Programming Language (ZPL) script.

7. If you require additional service operations (for example, one for ZPL-based layouts and one for variable-based layouts), repeat the previous step to add them.

8. On the Action Pane, select **Save**. Then select the **Close** button to return to the **External service definitions** page.

9. On the **Label print service** tab, set the following fields:

    - **Print operation** – If you created an operation for printing ZPL-based layouts, select the name of that operation. The operation must already be defined for the current service definition, as described earlier in this procedure.
    - **Variables print operation** – If you created an operation for printing variable-based layouts, select the name of that operation. This operation must already be defined for the current service definition, as described earlier in this procedure.
    - **Variable label layout template** – If you're using a variable-based layout, enter the content for the variable-based layout template.

10. On the Action Pane, select **Save**.

Set up an external service instance

Each external service instance defines a specific instance of an external service.

1. Go to **Warehouse Management** > **Setup** > **External services** > **External service instances**.

2. On the Action Pane, select **New** to create an external service instance.

3. Set the following fields for the new service instance:

    - **External service instance** – Enter a name for the instance, such as *External*.
    - **Description** – Enter a short description of the definition, such as *External*.
    - **External service definition** – Select the service definition to use with the instance. The example service definition value that was suggested earlier in this article was *External*.

4. On the **General** tab, set the following fields:

- **Base URL** – Enter the host name of the external service.
- **Authentication secret** – Enter the authentication secret (password or shared key) used to authenticate with the service.
- **Logging level** – Select the level of log entries to generate:
  - *Errors only* – Log only errors.
  - *Successes and errors* – Log both successes and errors (recommended).
  - *None* – Don't generate any log entries.
- **Log request bodies** – Select how much extra detail to include in log entries:
  - *Errors only* – Include details about errors.
  - *Successes and errors* – Include details about both successes and errors (recommended).
  - *None* – Don't include any details. Show only log headers.
- **Service offline** – If you know that the service will be offline for some time, set this option to *Yes* to prevent the system from trying to connect to it. This setting can help save system resources, because it prevents the system from waiting for the service to respond to requests. Set this option back to *No* when the service is back online.

5. On the Action Pane, select **Save**.

For information about how to read the log entries, see the Review the external service request log section later in this article.

Set up a printer

Use the following procedure to link a printer with the external print service.

1. Go to **Warehouse Management** > **Setup** > **Document routing** > **Label printers**.
2. On the Action Pane, select **New** to add a printer to the grid.
3. Set the following fields for the new printer:
   - **Connection type** – Select the type of connection that's used to communicate with the printer:
     - *External label service* – Connect to a cloud printer through an external service. Cloud printers include printers that are accessible only through the external label service.
     - *Document routing agent/Hybrid* – Connect to a DRA printer, or use the DRA as a fallback for the external service if the service can't print the label.

- **Printer name** – Enter or select the printer name. This value represents an internal Supply Chain Management printer name. If the printer is already configured as a DRA printer, you can select its name in the lookup dialog box. To set up an external service, you must specify the printer name. That name must not already be used for a DRA printer. When you enter a name for an external service printer, use a prefix or another method to keep the names of external service printers separate from printers that you might register through the DRA in the future.

- **Printer stock type** – Select a *printer stock type*. A printer stock type typically describes the type of paper that a specific printer uses. It's also used to specify the type of paper that a specific label layout should be printed to. For information about how to set up printer stock types, see Set up printer stock types.

- **Batch print** – Select this checkbox to enable the system to send multiple labels to the selected printer as a single batch job. Clear this checkbox to send one label at a time. Batch jobs print labels all at once, without being interrupted by other print jobs.

- **Maximum file size** – If you enabled batch printing for the printer, enter the maximum size, in megabytes, for batch files that are sent to it. The value must be more than 0 (zero). Batches that exceed the maximum size are split into smaller batches before they're sent. A large file size lets you include more labels in each batch. However, large batches might take a long time to print and therefore delay other jobs. Alternatively, they might fail if they exceed the memory capacity of your printer. To find the maximum supported value, see your printer's specifications.

- **Label print service instance** – Select the service instance to use. The example service instance value that was suggested earlier in this article was *External*.

- **Label print service printer name** – Enter the printer name as it's defined in the external service. This value represents an external printer name that's used in the external service.

- **Label print service execution policy** – If the **Connection type** field is set to *Document routing agent/Hybrid*, select one of the following values to specify what the system should do if a request to the external service fails:
  - *Only use the label print service* – The system won't try to print the label by using the DRA, even for printers where using the DRA would otherwise be an option.
  - *Fallback to the Document Routing Agent* – The system tries to print the label by using the DRA.

Configure a label layout

You can configure a label layout in either of the following ways:

- Use ZPL or another printer command language directly.
- Specify XML, JSON, or a similar data format that the external service uses to generate and print label layouts.

For more information about how to set up label layouts, see License plate label layouts and printing.

The rest of this section provides an example that shows how to use a JSON request body for an external label service. For this example, the external service has an operation that requires the following JSON body (including several variables) to print a label to a specific printer.

JSON

```
{
    "printer": "LABELPRINTER",
    "labelfile": "/Labels/SimpleLicensePlateLabel1",
    "variables": {
        "Item": "ITEM1",
        "Container": "CONTAINER1"
    }
}
```

Part of this JSON request body is generated by the label layout and includes data for a specific label. It's embedded in the external service operation's request body definition, which specifies a printer and other data that isn't specific to the label. Because the printer is provided by the `$label.printer` substitution, this part of the request goes into the external service operation. Because the label file and the variables are specific to each label, this part of the request goes into the label layout.

The following example shows what the external service operation's request body might look like.

JSON

```
{
    "printer": "$label.printer$",
    $label.body$
}
```

The following example shows what the label layout (in this case, the layout for a license plate label) might look like.

JSON

```
"labelfile": "/Labels/SimpleLicensePlateLabel1",
"variables": {
    "Item": "$ItemId$",
    "Container": "$LicensePlateId$"
}
```

Set up document routing

For information about how to set up document routing, see License plate label layouts and printing.

Run a scenario to print custom labels

If you want to experiment with printing labels, you can set up a scenario for printing location labels. For more information, see Custom label layouts printing. Follow the instructions in this article, and confirm that the scenario is supported.

Review the external service request log

The external service request log contains information about each request that the system sends to an external service. It can help you troubleshoot problems.

The amount of detail that the log records depends on the **Log level** and **Log request bodies** settings that you specify for each external service instance. Learn more in the External service instance configuration section earlier in this article.

For each request, the system checks if the external service returns the HTTP success code (200). If the external service returns a different HTTP code, such as 4xx or 5xx, the system reports an error, provided that you enable the relevant logging options.

To review the request log that's generated while labels are printed to the external service, follow these steps:

1. Go to **Warehouse management** > **Enquiries and reports** > **External service request log**. Alternatively, to review the log for a specific instance, go to **Warehouse Management** > **Setup** > **External service instances** > **Request log**.
2. In the grid, select the request log to review.
3. On the Action Pane, select **Request details**.
4. Review the log.

Related information

- Label Layouts
- Document routing label layouts
- Print labels using the Loftware label service solution
- Print labels using the Seagull Scientific BarTender label service solution

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Last updated on 07/14/2026

Print labels by using the Loftware label service solution

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to set up and print labels in Microsoft Dynamics 365 Supply Chain Management by using the Loftware solution. It's one example that shows how to use the Supply Chain Management external service label printing feature. For general information about how this feature works, see Print labels using an external service.

Loftware lets you create, manage, and print standardized, compliant bar code labels. It also lets you print labels to supported cloud-connected printers or a wide variety of classic on-premises printers. Loftware has several products that can be used together with the external label printing service, including a cloud solution (Loftware Cloud) and several versions of their on-premises products. This article focuses on how to integrate Supply Chain Management with the Loftware Cloud solution.

For more information about Loftware Cloud, see Loftware Cloud & Label Management System.

> **Important**
>
> By enabling the external service integration, you affirm that you understand that the data handling, privacy, and compliance standards of the external service might not be the same as the standards that Dynamics 365 Supply Chain Management provides. To learn whether the external service meets your organization's security and privacy requirements, including requirements for the handling of personal data and geo-residency, consult the external service's documentation and terms. Your privacy is important to us. To learn more, read the Microsoft Privacy Statement.

Loftware Cloud supports two integration approaches for other cloud applications:

- **Cloud Print API** – Loftware Cloud generates the labels and sends print jobs to the printer. This printer can be either a so-called *cloud printer* (a printer that's connected directly to the Loftware Cloud instance, without using any locally installed software or printer driver) or any printer that's connected to a workstation where a print gateway component is installed. For the current list of supported cloud printers, see the Loftware website.
- **Cloud Trigger API** – Loftware Cloud sends messages to an on-premises Loftware Automation Service instance, which generates the labels and sends them to the printer. You must install a Windows printer driver for each of your printers.

If you don't want any local software footprint, the preferred option is to use the Cloud Print API together with supported printers. Otherwise, use the Cloud Trigger API. Nevertheless, both options are available.

Here are some of the pros and cons of each integration option:

- The Cloud Trigger API lets you use an out-of-box cloud integration pack or completely customize the processing of print requests.
- The Cloud Print API is synchronous. In other words, the server waits for the print engine to print the label. Although the Cloud Print API reports the outcome of the print job, Supply Chain Management doesn't use that result. Therefore, you must wait a little longer when you print labels, or you must set a time-out on the external service operation. If a time-out occurs, the Cloud Print API continues to print, but the request is logged as a time-out.
- On-premises printer naming is simpler in the Cloud Trigger API, because you can use the names of the printers as they're configured on the servers that run the Loftware Automation Service. To use the Cloud Print API, you must specify a workstation ID to indicate which workstation you want to print to.
- For the Cloud Trigger API, you can install multiple instances of the Loftware Automation Service on the local network to achieve load balancing and high availability. As was noted in the previous point, you must specify which workstation the printing request should be routed to. If that workstation isn't available, the label isn't printed. Conversely, multiple Loftware Automation Service instances can handle requests to the same cloud trigger.
- The Cloud Print API has a smaller installation footprint.

> **Important**
>
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product at any time without notice. If you experience any issues or have any additional questions about the Loftware product, contact Loftware directly.

Prerequisites

To use the features that this article describes, you must be running Supply Chain Management version 10.0.34 or later.

Prepare for Loftware Cloud integration

Before you can access Loftware Cloud by using either the Cloud Print API or the Cloud Trigger API, you must register in the Loftware Developer Portal and then link your Developer Portal subscription with your Loftware Cloud instance. For instructions, see the Loftware Help Center.

> **Important**
>
> When you register in the Loftware Developer Portal, be sure to make a copy of your primary or secondary subscription key. (Use the primary key by default.) You need this key to configure the external service definition later.

If you want to use printers that are enabled for the Internet of Things (IoT), you must connect them to your Loftware Cloud instance. For instructions, see the Loftware Help Center.

Configure Supply Chain Management to print to cloud printers by using the Cloud Print API

By using the Cloud Print API, you can print labels that Loftware Cloud generates based on variables that are sent from Supply Chain Management. You can also send a Zebra Programming Language (ZPL) label, such as an existing label layout or a shipping label that's provided by a shipping carrier for small parcel shipping (SPS) integration.

This section describes how to set up an external service definition that has two operations: one for printing variable-based layouts and one for printing ZPL-based label layouts. If you don't need to use both types of layout, you can set up just one of the two operations that are described here.

Set up an external service definition for printing through the Cloud Print API

Follow these steps to set up an external service definition.

1. Go to **Warehouse management** > **Setup** > **External services** > **External service definitions**.
2. On the Action Pane, select **New** to create an external service definition.
3. Set the following fields for the new service definition:
    - **External service definition** – Enter a name for the service definition, such as *NLPrint*.
    - **Description** – Enter a short description of the service definition, such as *Loftware Cloud Print API*.
4. On the Action Pane, select **Save**.
5. On the **External service operations** FastTab, select **Edit operations** on the toolbar.
6. On the **External service operations** page, on the Action Pane, select **New** to add an operation for printing variable-based layouts. Then follow these steps for the new record. If you don't use variable-based layouts, you can skip ahead to step 8.
    - On the header, set the following fields:
        - **External service operation** – Enter a name for the operation, such as *Print*.
        - **Description** – Enter a short description of the operation, such as *Print to cloud printer*.
    - On the **General** FastTab, set the following fields:
        - **HTTP method** – Select *POST*.
        - **Operation timeout** – Enter a timeout period for the operation, in milliseconds, such as *500*.
        - **Request body type** – Select *Raw*.
        - **Relative URL** – Enter */Print/v2/Print*.

        > ⓘ **Note**
        >
        > Cloud Print is a synchronous API. Therefore, it waits until the label is printed and then provides immediate feedback about whether printing was successful. However, because this process can take several seconds, the system might provide a bad user experience for mobile device users. Additionally, the result isn't currently used. Therefore, set a timeout period of 500 milliseconds. Even though Supply Chain Management stops waiting after this timeout expires, printing continues in the background.

- On the **Request HTTP headers** FastTab, add a row, and set the following fields for it:
  - **Key** – Enter *Ocp-Apim-Subscription-Key*.
  - **Value** – Enter *$auth.secret$*. Be sure to enter the value exactly as it appears here.

- On the **Request body** FastTab, set the following fields:

  - **Content type** – Enter *application/json*.

  - **Body** – Enter the content of the request body. Here's an example.

    JSON

    ```
    {
        "deviceType": "CloudPrinter",
        "deviceId": {
            "printerName": "$label.printer$",
            "workstation": null
        },
        $label.body$
    }
    ```

7. On the Action Pane, select **Save**.

8. On the Action Pane, select **New** to add an operation for printing ZPL-based layouts. Then follow these steps for the new record. If you don't use ZPL-based layouts, you can skip ahead to step 10.

   - On the header, set the following fields:
     - **External service operation** – Enter a name for the operation (for example, *SendData*).
     - **Description** – Enter a short description of the operation (for example, *Send ZPL to cloud printer*).

   - On the **General** FastTab, set the following fields:
     - **HTTP method** – Select *POST*.
     - **Operation timeout** – Enter a timeout period for the operation, in milliseconds, such as *500*.
     - **Request body type** – Select *Raw*.
     - **Relative URL** – Enter */Print/v2/SendData*.

   - On the **HTTP request headers** FastTab, add a row, and set the following fields for it:
     - **Key** – Enter *Ocp-Apim-Subscription-Key*.
     - **Value** – Enter *$auth.secret$*. Be sure to enter the value exactly as it appears here.

- On the **Request body** FastTab, set the following fields:
  - **Content type** – Enter *application/json*.
  - **Body** – Enter the content of the request body. Here's an example.

```
JSON
{
    "deviceType": "CloudPrinter",
    "deviceId": {
        "printerName": "$label.printer$",
        "workstation": null
    },
    "data": "$label.body:base64$"
}
```

9. On the Action Pane, select **Save**.
10. Select the **Close** button to return to the **External service definition** page.
11. In the list pane, select the record that you were working on at the beginning of this procedure.
12. On the **Label print service** FastTab, set the following fields:
    - **Print operation** – If you created an operation for printing ZPL-based layouts, select the name of that operation (for example, *SendData*).
    - **Variables print operation** – If you created an operation for printing variable-based layouts, select the name of that operation (for example, *Print*).
    - **Variable label layout template** – If you're using a variable-based layout, enter the content for your variable-based layout template. Here's an example.

```
JSON
"filePath": "$SystemVariables.LabelFile$",
"quantity": "$SystemVariables.Quantity$",
"dataSources": [
    {
        {{Row Table=LabelLayoutVariable
            "$LabelLayoutVariable.Variable$": "$LabelLayoutVariable.Value$",
        }}
    }
]
```

Set up an external service instance for printing through the Cloud Print API

Follow these steps to set up an external service instance for printing through the Cloud Print API.

1. Go to **Warehouse Management** > **Setup** > **External services** > **External service instances**.
2. On the Action Pane, select **New** to create an external service instance.
3. Set the following fields for the new service instance:
    - **External service instance** – Enter a name for the instance, such as *NLPrintProd*.
    - **Description** – Enter a short description of the instance, such as *Loftware Cloud Print API Production*.
    - **External service definition** – Select the service definition to use with the instance. The example service definition value suggested earlier in this article is *NLPrint*.
4. On the **General** tab, set the following fields:
    - **Base URL** – Enter `https://labelcloudapi.onnicelabel.com`.
    - **Authentication secret** – Enter your primary or secondary subscription key from the Loftware Developer Portal. You should have made a note of this key earlier, when you were preparing for Loftware integration.
    - **Logging level** – Select *Successes and errors*.
    - **Log request bodies** – Select *Successes and errors*.

You can now create label printers and label layouts by using either variables or ZPL.

> ⓘ **Note**
>
> - When you create label printers on the **Label printers** page in Supply Chain Management, in the **Label print service printer name** field, enter the name of the printer from the **Cloud printers** tab in Loftware Cloud Control Center.
> - When you create a variable-based label layout on the **Label layout** page of Supply Chain Management, in the **System Variables** grid, include a row where the **Variable name** field is set to *LabelFile* and the **Value** field is set to the full path and file name of the label. To find the full path and file name, sign in to Loftware Control Center, and then, on the **Documents** tab, select the label file, and copy the full value of the **Path** field (under **File properties**).

Configure Supply Chain Management to print to local network printers by using the Cloud Trigger API

By using the Cloud Trigger API, you can call an on-premises installation of the Loftware Automation Service, which then processes the request and performs preprogrammed actions. Before you move on to the next steps, install Loftware on the computers that run the Loftware Automation Service, and set up the required printers for those computers. For guidance that helps you install and set up printers, see the Loftware documentation.

To simplify the configuration of the Loftware Automation Service, use the Loftware Cloud Data Integration pack, and follow the instructions provided in the Loftware Help Center. The following procedure summarizes the required steps. More details are provided in the Loftware Help Center.

1. Download the integration pack by using the link that's provided.
2. Sign in to Loftware Control Center.
3. On the **Documents** tab, create the following folder: `/Demo/LabelCloudDataIntegration`.
4. Open the zip file that you downloaded, and extract the files.
5. Upload the contents of your document storage folder into the `LabelCloudDataIntegration` folder in Loftware Control Center.
6. On the computer where the Loftware Automation Service is installed, open Loftware Automation Manager.
7. If you didn't already, connect the Loftware Automation Service to your Loftware Cloud instance.
8. In Loftware Automation Manager, select **Add**, and browse to the `LabelCloudDataIntegration` folder in Loftware Cloud.
9. Open the `CloudIntegration-CloudTrigger.misx` file.
10. Verify that the print trigger uses the unique identifier *Api-CloudIntegrationDemo-Print*. Then start only this trigger. Supply Chain Management doesn't currently support other triggers.

After you install and start the print trigger, you're ready to configure the external service definition and external service instance that are required to access the Loftware Automation Service by using the Cloud Trigger API. Unless you customize the demo automation configuration, you can print only variable-based layouts, not ZPL-based layouts. In this case, you configure only one operation on the external service definition.

Set up an external service definition for printing through the Cloud Trigger API

Follow these steps to set up an external service definition for printing through the Cloud Trigger API.

1. Go to **Warehouse management** > **Setup** > **External services** > **External service definitions**.
2. On the Action Pane, select **New** to create an external service definition.
3. Set the following fields for the new service definition:
    - **External service definition** – Enter *NLTrigger*.
    - **Description** – Enter *Loftware Label Cloud Trigger API*.
4. On the Action Pane, select **Save**.
5. On the **External service operations** FastTab, select **Edit operations** on the toolbar.
6. On the **External service operations** page, on the Action Pane, select **New** to add an operation for the *Print* service operation. Then follow these steps for the new record:
    - On the header, set the following fields:
        - **External service operation** – Enter a name for the operation, such as *Print*.
        - **Description** – Enter a short description of the operation (for example, *Print to local printer*).
    - On the **General** FastTab, set the following fields:
        - **HTTP method** – Select *POST*.
        - **Operation timeout** – Enter a timeout period for the operation, in milliseconds, such as *500*.
        - **Request body type** – Select *Raw*.
        - **Relative URL** – Enter */Trigger/v1/CloudTrigger/Api-CloudIntegrationDemo-Print*.
    - On the **HTTP request headers** FastTab, add a row, and set the following fields for it:
        - **Key** – Enter *Ocp-Apim-Subscription-Key*.
        - **Value** – Enter *$auth.secret$*. Be sure to enter the value exactly as it appears here.
    - On the **Request body** FastTab, set the following fields:
        - **Content type** – Enter *application/json*.
        - **Body** – Enter the content of the request body. Here's an example.

            JSON

```
{
    "Printer": "$label.printer$",
    $label.body$
}
```

7. On the Action Pane, select **Save**.

8. Select the **Close** button to return to the **External service definition** page.

9. On the **Label print service** FastTab, set the following fields:

    - **Print operation** – Leave this field blank.

    - **Variables print operation** – Select the print operation that you created earlier in this procedure (for example, *Print*).

    - **Variable label layout template** – Enter the content for your variable-based layout template. Here's an example.

      JSON

      ```
      "FilePath": "$SystemVariables.LabelFile$",
      "Quantity": "$SystemVariables.Quantity$",
      "Variables": [
          {{Row Table=LabelLayoutVariable
              {
                  "Name": "$LabelLayoutVariable.Variable$",
                  "Value": "$LabelLayoutVariable.Value$"
              },
          }}
      ]
      ```

Set up an external service instance for printing through the Cloud Trigger API

Follow these steps to set up an external service instance for printing through the Cloud Trigger API.

1. Go to **Warehouse Management** > **Setup** > **External services** > **External service instances**.

2. On the Action Pane, select **New** to create an external service instance.

3. Set the following fields for the new service instance:

    - **External service instance** – Enter a name for the instance, such as *NLTriggerProd*.

- **Description** – Enter a short description of the instance, such as *Loftware Cloud Trigger API Production*.
- **External service definition** – Select the service definition to use. The example service definition value suggested earlier in this article is *NLTrigger*.

4. On the **General** tab, set the following fields:

   - **Base URL** – Enter `https://labelcloudapi.onnicelabel.com`.
   - **Authentication secret** – Enter your primary or secondary subscription key from the Loftware Developer Portal. You should have made a note of this key earlier, when you were preparing for Loftware integration.
   - **Logging level** – Select *Successes and errors*.
   - **Log request bodies** – Select *Successes and errors*.

You can now create label printers and label layouts by using variables.

> **Note**
>
> - When you create label printers on the **Label printers** page in Supply Chain Management, in the **Label print service printer name** field, enter the name of the printer from the **Cloud printers** tab in Loftware Cloud Control Center.
> - When you create a variable-based label layout on the **Label layout** page of Supply Chain Management, in the **System Variables** grid, include a row where the **Variable name** field is set to *LabelFile* and the **Value** field is set to the full path and file name of the label. To find the full path and file name, sign in to Loftware Control Center, and then, on the **Documents** tab, select the label file, and copy the full value of the **Path** field (under **File properties**).
> - When you create a variable-based label layout, **Variable name** in the **Data variables** grid can't be the same as **Variable name** in the **System variables** grid.

Related information

- Print labels using an external service
- Label Layouts
- Document routing label layouts

Last updated on 05/06/2026

Print labels by using the Seagull Scientific BarTender® label service solution

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to set up and print labels in Microsoft Dynamics 365 Supply Chain Management by using the BarTender® labeling solution by Seagull Scientific. It's one example that shows how to use the Supply Chain Management external service label printing feature. For general information about how this feature works, see Print labels using an external service.

The BarTender labeling solution is available both as on-premises software and as a cloud-based software-as-a-service (SaaS) offering that's named BarTender Cloud™. Currently, only BarTender Cloud can be used together with the external label printing service for Supply Chain Management.

This article describes how to integrate Supply Chain Management with BarTender Cloud. It includes information about how to communicate with BarTender Cloud by exchanging documents in YAML format through the BarTender Cloud REST API. For more information about BarTender Cloud, see BarTender Cloud.

> **Important**
>
> By enabling the external service integration, you affirm that you understand that the data handling, privacy, and compliance standards of the external service might not be the same as the standards that Dynamics 365 Supply Chain Management provides. To learn whether the external service meets your organization's security and privacy requirements, including requirements for the handling of personal data and geo-residency, consult the external service's documentation and terms. Your privacy is important to us. To learn more, read the Microsoft Privacy Statement.

The information in this article is for general informational purposes only. Although Microsoft tries to keep the information up to date and correct, Microsoft makes no representation or warranties of any kind, express or implied, about accuracy, reliability, or completeness with respect to the BarTender product. Seagull Scientific might change the functionality of the BarTender product at any time without notice. If you have any additional questions about the BarTender product, contact Seagull Scientific.

Prerequisites

To use the features that this article describes, you must be running Supply Chain Management version 10.0.34 or later.

Get a copy of your personal access token from BarTender Cloud

Before you can access BarTender Cloud through its REST API, you must get a copy of your personal access token from BarTender Cloud. Supply Chain Management uses this token to authenticate with BarTender Cloud. For more information about how to get this token, see BarTender Cloud REST API in the BarTender help center. Be sure to make a note of the token, because you need to enter it later.

> **Note**
>
> Personal access tokens expire after a fixed number of days. You need to enter a new token before the old one expires. You can manually refresh the token to obtain a new one.

Configure Supply Chain Management to print to printers managed by BarTender Cloud

For the scripts that the BarTender Cloud service must run, the BarTender Cloud Actions REST API accepts several formats, including YAML-based and JSON-based action scripts and scripts that use the older BarTender XML (BTXML) format. In this article, the label that's printed is based on a design that's saved in your BarTender Cloud account. The article describes how to use the `PrintBTWAction` action to print the label to a printer that's accessed through a BarTender Cloud Print Gateway that's installed on your local network.

Set up an external service definition for printing through the Actions API

Follow these steps to set up an external service definition.

1. Go to **Warehouse management** > **Setup** > **External services** > **External service definitions**.
2. On the Action Pane, select **New** to create an external service definition.
3. Set the following fields for the new service definition:
    - **External service definition** – Enter a name for the service definition, such as *BTCloud*.
    - **Description** – Enter a short description of the service definition, such as *BarTender Cloud*.
4. On the Action Pane, select **Save**.
5. On the **External service operations** FastTab, select **Edit operations** on the toolbar.
6. On the **External service operations** page, on the Action Pane, select **New** to add an operation for printing variable-based layouts. Then follow these steps for the new record:
    - On the header, set the following fields:
        - **External service operation** – Enter a name for the operation, such as *Actions*.
        - **Description** – Enter a short description of the operation, such as *Actions API*.
    - On the **General** FastTab, set the following fields:
        - **HTTP method** – Select *POST*.
        - **Operation timeout** – Enter a timeout period for the operation, in milliseconds, such as *500*.
        - **Request body type** – Select *Raw*.
        - **Relative URL** – Enter */api/actions*.

        > ⓘ **Note**
        >
        > Cloud Print is a synchronous API. Therefore, it waits until the label is printed and then provides immediate feedback about whether printing was successful. However, because this process can take several seconds, the system might provide a bad user experience for mobile device users. Additionally, the result isn't currently used. Therefore, set a timeout period of 500 milliseconds. Even

though Supply Chain Management stops waiting after this timeout expires, printing continues in the background.

- On the **Request HTTP headers** FastTab, add a row, and set the following fields for it:
  - **Key** – Enter *Authorization*.
  - **Value** – Enter *Bearer $auth.secret$*. Be sure to enter the value exactly as it appears here.
- On the **Request body** FastTab, set the following fields:
  - **Content type** – Enter *text/yaml*.
  - **Body** – Enter the content of the request body. Here's an example.

```
YAML
- SetVariableAction:
    VariableName: D365PrinterName
    VariableValue: $label.printer$
$label.body$
```

> **ⓘ Important**
>
> Paste the body text exactly as it appears here. Indentation is important. Be sure to use four spaces to indent. Don't use tabs.

7. On the Action Pane, select **Save**.
8. Select the **Close** button to return to the **External service definition** page.
9. In the list pane, select the record that you were working on at the beginning of this procedure.
10. On the **Label print service** FastTab, set the following fields:
    - **Print operation** – Leave this field blank.
    - **Variables print operation** – Select the name of the operation that you created for printing variable-based layouts, such as *Actions*.
    - **Variable label layout template** – Enter the content for your variable-based layout template. Here's an example.

YAML

```
- PrintBTWAction:
    DocumentFile: librarian://Main/$SystemVariables.LabelFile$
    Printer: "%D365PrinterName%"
    Copies: $SystemVariables.Quantity$
    NamedDataSources:
{{Row Table=LabelLayoutVariable
      $LabelLayoutVariable.Variable$: $LabelLayoutVariable.Value$
}}
```

> **Important**
>
> Paste the body text exactly as it appears here. Indentation is important. Be sure to use four spaces to indent. For the `NamedDataSources` variables, be sure to use six spaces to indent. Don't use tabs.

Set up an external service instance for printing through the Actions API

For this procedure, you need the personal access token that you previously obtained from BarTender Cloud.

Follow these steps to set up an external service instance for printing through the Actions API.

1. Go to **Warehouse Management** > **Setup** > **External services** > **External service instances**.
2. On the Action Pane, select **New** to create an external service instance.
3. Set the following fields for the new service instance:
   - **External service instance** – Enter a name for the instance, such as *BTCloud*.
   - **Description** – Enter a short description of the instance, such as *BTCloud*.
   - **External service definition** – Select the service definition to use with the instance. The example service definition value that was suggested earlier in this article is *BTCloud*.
4. On the **General** tab, set the following fields:
   - **Base URL** – Enter one of the following instance URLs, depending on the region:
     - **For the AMER region:** `https://am1.bartendercloud.com`
     - **For the EMEA region:** `https://eu1.bartendercloud.com`
     - **For the APAC region:** `https://ap1.bartendercloud.com`

- **Authentication secret** – Enter the personal access token that you obtained from BarTender Cloud. You should have made a note of this token earlier, when you were preparing for BarTender integration.
- **Logging level** – Select *Successes and errors*.
- **Log request bodies** – Select *Successes and errors*.

You can now create label printers and label layouts.

> **Note**
>
> - When you create label printers on the **Label printers** page in Supply Chain Management, in the **Label print service printer name** field, enter the name of the printer as it appears in the **Print** dialog box in BarTender Cloud, or as it's retrieved from the Printer API. Use the following format: *printer:workstation/printer*.
> - When you create a variable-based label layout on the **Label layout** page of Supply Chain Management, in the **System Variables** grid, include a row where the **Variable name** field is set to *LabelFile* and the **Value** field is set to the full path and file name of the label (including any folders, if the label file is stored in a folder). The BarTender Cloud REST API uses a Librarian path, but the beginning of the full Librarian name is already coded in the variable label template.
> - When you create a variable-based label layout, **Variable name** in the **Data variables** grid can't be the same as **Variable name** in the **System variables** grid.

Related information

- Print labels using an external service
- Label Layouts
- Document routing label layouts

Last updated on 05/06/2026

Dynamic printer selection

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up your system to dynamically select a printer when printing license plate labels and container labels.

When dynamic printer selection is used, and a worker requests printing, the system selects a printer based on the current work user, warehouse, location, and/or zone. The warehouse manager can set up default printers for each work user, warehouse, location, and/or zone.

> **Important**
>
> The location is only considered when printing container labels.

Prerequisites

To use the features described in this article, your system must meet the following requirements:

- Dynamic printer selection requires Microsoft Dynamics 365 Supply Chain Management version 10.0.36 or later.
- The added ability to set up default printers for specific mobile devices (on the **Devices** tab of the **Default label printers** page) requires Supply Chain Management version 10.0.41.

Basic printing and label layout setup

Before you can set up dynamic printer selection, you must configure your printers and labels in the system, as summarized in the following subsections.

Set up printer stock types

A *printer stock type* typically describes the type of paper, such as the size, that a specific printer uses. Use it to specify the type of paper that a specific label layout should print to.

> **Important**
>
> To use dynamic printer selection, you *must* set up a printer stock type for each relevant printer. You must also set up a printer stock type for each label layout that you want to use for dynamic printer selection.

Follow these steps to set up printer stock types.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Printer stock types**.
2. Create or select a stock type.
3. For the new or selected stock type, set the following fields:
    - **Printer stock type** – Enter a name for the printer stock type, such as *A1*.
    - **Description** – Enter a short description of the printer stock type.
4. Repeat steps 2 and 3 until you set up all the stock types that you need.
5. On the Action Pane, select **Save**.

Set the printer stock type for a label printer

Follow these steps to set the printer stock type for a label printer.

1. Go to **Warehouse management*** > **Setup** > **Document routing** > **Label printers**.
2. Create or select a printer.
3. For the new or selected printer, set the **Printer stock type** field to the relevant printer stock type. For example, select one of the stock types that you set up in the previous section. For information about the other settings that are available for a label printer, see Set up a printer.
4. Repeat steps 2 and 3 until you set up all the stock types that you need.
5. On the Action Pane, select **Save**.

Set up a label layout, including the printer stock type

Follow these steps to set the printer stock type for a label layout.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Label layout**.
2. At the top of the list pane, set the **Label layout type** field to *License Plate Label*.
3. Create or select a label layout.
4. For the new or selected label layout, set the **Printer stock type** field to a printer stock type that you set up earlier in this article. For information about the other settings that are available for a label layout, see License plate label layouts and printing.
5. On the Action Pane, select **Save**.

Configure the license plate label routing

Follow these steps to configure the license plate label routing.

1. Go to **Warehouse management** > **Setup** > **Document routing** > **Document routing**.

2. Set up license plate label routing as described in Set up license plate label routing. However, on the **Document routing printers** FastTab, include only one row, and configure it in the following way:

   - **Name** – Leave this field blank.
   - **Label layout ID** – Select the label layout that you set up in the previous section.

   > ⓘ Note
   >
   > If the **Document routing printers** FastTab includes a row where the printer name is specified (that is, the **Name** field isn't blank), dynamic printer selection isn't used. Instead, the specified printer is used.

3. On the Action Pane, select **Save**.

Set up default label printers for specific workers and locations

After your printers and label layouts are set up, you can set up default label printers for specific workers and locations. The settings define the printer that the system selects for the current worker and/or location when you use dynamic printer selection. Learn more in the How the system selects a printer section later in this article.

Follow these steps to set up default label printers for each worker and/or location.

1. Go to **Warehouse management** > **Document routing** > **Default label printers**.

2. On the **Locations** tab, the grid shows the default label printers assigned to each location. For flows where the system can determine the location of a worker who requests printing, the printers that you set in this grid have priority (provided that the worker didn't manually override the default printer selection). Use the toolbar buttons to add or remove rows as required. For each row, set the following fields:

    - **Printer stock type** – Select the printer stock type that the row applies to.
    - **Warehouse** – Select the warehouse that includes the location.
    - **Location scope type** – Select *Location* if the row applies to one specific location. Select *Zone* if the row applies to a zone that includes multiple locations.
    - **Location scope** – Depending on the **Location scope type** value, select either the location or the zone that the row applies to.
    - **Printer name** – Select the default printer to use for the location or zone that the row applies to.

3. On the **Users** tab, the grid shows the default label printers that assigned to each worker in each warehouse. Use the toolbar buttons to add or remove rows as required. For each row, set the following fields:

    - **Printer stock type** – Select the printer stock type that the row applies to.
    - **User ID** – Select the user ID of the worker that the row applies to.
    - **Warehouse** – Select the warehouse that the row applies to. You can set up multiple rows for workers who work in more than one warehouse. Leave this field blank to set the default printer that's used for a worker if no other, more specific row applies. When the worker requests printing, the system selects the printer that's most specific to the warehouse where the worker is signed in.
    - **Printer name** – Select the default printer to use for the combination of a worker, warehouse, and stock type on the row.

4. On the **Devices** tab, the grid shows the default label printers assigned to mobile devices. Use the toolbar buttons to add or remove rows as required. For each row, set the following fields:

    - **Printer stock type** – Select the printer stock type that the row applies to.
    - **Mobile device** – Select the ID or friendly name of the device that the row applies to.
    - **Printer name** – Select the default printer to use for the device that the row applies to.

5. On the Action Pane, select **Save**.

Create a menu item that lets workers override their default printer

Set up the Warehouse Management mobile app to let workers override the default printer that's assigned to them. Workers can use their mobile device to scan a new printer name.

Create a new mobile device menu item, set the **Mode** field to *Indirect*, and set the **Activity code** field to *Override label printer*.

How the system selects a printer

When a print job is run, the system uses the following sequence to identify which printer to use:

1. If the current worker manually overrides the printer, and the selected printer uses the printer stock type of the layout that's currently being printed, the system uses that printer (and then skips the remaining steps in this sequence).

2. If no printer override exists, the system checks the default printer setup.
   a. If a default label printer that uses the required stock type is set up for the location or zone that's specified in the current flow (for example, when a container label is printed from the packing station), the system uses that printer.
   b. If a default label printer that uses the required stock type is set up for the mobile device the current flow is executing on (for example, forklift mounted mobile device and printer), the system uses that printer.
   c. Otherwise, the system uses the default label printer that uses the required stock type and that's set up for the current user and warehouse.

3. If no printer is found, a fallback printer, if provided, is used. Fallback printers are configured for specific label printing scenarios, such as license plate label printing or container label printing. For more information, see Related information.

4. If no override or default printer is found, no label is printed.

Advanced scenarios

You can also use the functionality described in this article for more advanced scenarios, such as reprinting a license plate label created in one warehouse on a printer located in another warehouse.

Reprint a label in a different warehouse

One way to reprint a label in a different warehouse is to use the **Default label printers** page to set up a default label printer for a user, location, or device in another warehouse. For example, for a given **Printer stock type**, **User ID** *25* has **Printer name** set to *Printer25*. This means that when user *25* is reprinting labels that were originally created in warehouse *24*, they will now be sent to printer *Printer25* instead of the printer defined on the **Document routing** page for warehouse *24*.

> **Important**
>
> A valid document routing setup must exist based on the field values on the license plate label (for example, *Warehouse*), even when when a different printer is used later. For example, if the label was originally created in warehouse *24*, then there must be a document routing record for warehouse *24* with the same **Layout ID** and **Printer stock type** as the default printer for the other user, location, or device.

Related information

- License plate label layouts and printing
- Container label layouts and printing
- Document routing label layouts

Last updated on 05/04/2026

Shipment auto-updates

Article • 06/28/2022

The auto-update shipment functionality automatically updates quantities (both increases and decreases) on a load line that is associated with a shipment, after the load has been released to a warehouse. This functionality remains turned on until the load line on the shipment or load is processed on a wave. When it's used, order updates can automatically flow through to the warehouse, without requiring manual intervention, until warehouse work is created.

When the auto-update shipment functionality isn't used, only quantity decreases automatically flow until warehouse work is created. Users must manually update or delete lines, and they must then re-release lines if order quantities are increased or new order lines are added. By using the auto-update shipment functionality, businesses can seamlessly provide updates to the warehouse without having to worry that related shipments and loads won't reflect order line updates.

The auto-update shipment functionality applies to both sales order lines and transfer order lines, and it's turned on for a specific warehouse. Therefore, companies can apply different auto-update shipment policies across warehouses, as they require. By default, the auto-update shipment policy for quantity decreases is applied for all warehouses that use warehouse management processes (WMS). When this default policy setting is used, only quantity decreases automatically flow through to a shipment and load until warehouse work is created. This behavior resembles the behavior that was used before the auto-update shipment functionality was introduced.

Main elements of the functionality

The auto-update shipment functionality relies primarily on the shipment status to determine whether the quantity on a load line should be changed when a change is made on a sales order line or transfer order line. It also relies primarily on the shipment status to determine when a new load line should automatically be added to an existing load. When the shipment status is **Waved** or higher, no automatic update occurs.

Wave status is also considered for automatic updates. When the wave that is related to the load line has a status of **Held**, **Executing**, **Released**, **Picked**, or **Shipped**, if a user tries to reduce the quantity on a load line (via a quantity decrease on the sales order line or transfer order line), the following error message is shown: "Reservations cannot be removed because there is work created which relies on the reservations." Additionally, when the wave has one of the previously mentioned wave statuses, if a user tries to

indirectly increase the load line quantity by increasing the quantity on the sales order line or transfer order line, the quantity on the load line isn't automatically increased. In this case, the load line must be manually updated.

Scenarios

The auto-update shipment functionality supports four scenarios: adding a new order line, increasing the quantity on an order line, decreasing the quantity on an order line, and removing an order line.

- **Add a new order line** – When the **Auto update shipment** field on the **Warehouse** FastTab of the **Warehouses** page (**Warehouse management** > **Setup** > **Warehouse** > **Warehouses**) is set to **Always**, if a shipment exists for the order, and a new order line is added to a sales order or transfer order after a load has already been created for the sales order, the existing load isn't updated. A new load line that has no reference to the existing load is created and associated with the existing shipment. The new line is added to the load and released.
- **Increase the quantity on an order line** – When the **Auto update shipment** field is set to **Always**, if a shipment exists for the order, and the quantity on an existing sales order line or transfer order line is increased after a load has already been created for the sales order, the load line is increased by the same quantity as the order line. If the load was released, but no work was created, the load line is increased by the same quantity as the order line.
- **Decrease the quantity on an order line** – When the **Auto update shipment** field is set to **Always** or **On quantity decrease**, if a shipment exists for the order, and the quantity on an existing sales order line or transfer order line is decreased after a load has already been created for the sales order, the quantity on the associated load line is updated to match, unless the quantity on that load line already equals or is less than the new quantity on the order line. In that case, the load line isn't affected. If the load was released, but no work was created, the quantity on the associated load line is updated to match, unless the quantity on the load line already equals or is less than the new quantity on the order line. In that case, the load line is affected.
- **Remove an order line** – When the **Auto update shipment** field is set to **Always** or **On quantity decrease**, if the user tries to remove an order line that a load line exists for, an error message is shown.

Example scenario

For this scenario, you must have demo data installed, and you must use the **USMF** demo data company.

Turn on the auto-update shipment functionality

To turn on the auto-update shipment functionality, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses**.
2. Select warehouse **24**.
3. On the **Warehouse** FastTab, in the **Auto update shipment** field, change the value from **On quantity decrease** to **Always**.

After you change the value to **Always**, any increases or decreases in the quantities on sales order lines and transfer order lines, and any additions of new lines, are reflected on shipments and loads for the selected warehouse, given the previously mentioned update constraints.

Change the wave template so that load lines aren't automatically processed

To configure the wave template so that it doesn't automatically process load lines, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Waves** > **Wave templates**.
2. Select wave template **24 Shipping default**.
3. Select **Edit**.
4. On the **General** FastTab, set the **Automate wave creation** option to **Yes**, and make sure that all other options are set to **No**.

It's important that no work be automatically created and released as part of the wave creation process. After work is created that is related to the load line that was created for the sales order line, the load line is no longer automatically updated if the quantity on the sales order line is changed.

Create a sales order

To create a sales order, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select customer **US-003**.
3. Create a line for item number **A0001**.
4. Enter a quantity of **10**. (Make sure that you're using warehouse **24**.)

5. Select **Save**.
6. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**. A shipment and a wave are created.

Because you changed the wave template in the previous procedure, no load or work is created. The shipment status is **Open**, and the wave status is **Created**.

Decrease the quantity on a sales order line

To decrease the quantity on a sales order line, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select the sales order that you just released to the warehouse.
3. Select the sales order line. In the **Quantity** field, change the value from **10** to **8**.
4. From the sales order line, select **Warehouse** > **Shipment details**. On the **Shipment details** page, on the **Load lines** FastTab, the quantity reflects the change on the sales order line.

Increase the quantity on a sales order line

To increase the quantity on a sales order line, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select the sales order that you previously released to the warehouse.
3. Change the line quantity from **8** to **12**.
4. Select **Save**.
5. Go back to the **All sales orders** page, and select the sales order again.
6. On the Action Pane, on the **Warehouse** tab, in the **Related information** group, select **Shipment details**. On the **Shipment details** page, on the **Load lines** FastTab, the quantity reflects the change on the sales order line.

Although the quantity on the load line was increased from 8 to 12, only eight items remain reserved, unless automatic reservation is turned on. Because the quantity that was added to the existing shipment hasn't been reserved, if the wave is processed at this point, without reservation, work is created only for the quantity that has already been reserved.

> **Note**
>
> When the quantity on an order line is decreased, the quantity on the load line isn't affected if it already equals or is less than the new quantity on the order line. When the quantity on an order line is increased, the load line is increased by the same

quantity as the order line. If the quantity on the order line differs from the quantity on the load line, the difference remains.

Add a sales order line

To add a sales order line, follow these steps.

1. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.
2. Select the sales order that you previously released to the warehouse.
3. Create a line for item number **A0002**.
4. In the **Quantity** field, enter **10**. (Make sure that you're using warehouse **24**.) The new line is automatically added to the existing shipment.
5. Select **Save**.
6. Go back to the **All sales orders** page, and select the sales order again.
7. On the Action Pane, on the **Warehouse** tab, in the **Related information** group, select **Shipment details**. On the **Shipment details** page, on the **Load lines** FastTab, notice the second load line.

Because the sales order line that you just added to the existing shipment hasn't been reserved, if the wave is processed at this point, work is created only for the quantity on the first sales order line and the first load line.

Process a wave

To process the wave, follow these steps.

1. Go to **Warehouse management** > **Outbound waves** > **Shipment waves** > **All waves**.
2. Select the wave that you previously created.
3. On the Action Pane, on the **Wave** tab, in the **Wave** group, select **Process**.

The wave is processed and creates work for the reserved quantities on the load lines. The shipment status is updated from **Open** to **Waved**. As the shipment status is updated to **Waved**, any changes that occur, such as decreases or increases in line quantities, or the addition of new lines to the sales order, don't affect the existing load lines that are associated with the waved shipment.

If a shipment has a status of **Waved** or higher, updates to the quantity on a sales order line aren't reflected on or validated against a load line that is associated with the shipment. Changes to the quantity on a load line must be made directly on the load line.

Validation is done after work has been created for the load line and a reservation has been made. A decrease in the quantity on the sales order line is then validated against the work line reservation.

Feedback

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Create a location profile

Article • 01/31/2023

This article explains how to create a location profile in Dynamics 365 Supply Chain Management. Every location in the warehouse needs to have a location profile associated with it that describes the properties of the location, for example, whether the location allows mixed items. In this procedure we'll create a profile for a location that doesn't require license plate control. We'll enable mixed items, and mixed inventory statuses, and allow cycle counting. You can use this procedure in the USMF demo data company.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. Select **New**.
3. In the **Location profile ID** field, type a value.
4. In the **Name** field, type a value.
5. In the **Location format** field, enter or select a value.
6. In the **Location type** field, enter or select a value.
7. In the **Dock management profile ID** field, enter or select a value.
8. Select **Yes** in the **Allow mixed items** field.
9. Select **Yes** in the **Allow mixed inventory statuses** field.
10. Select **Yes** in the **Allow cycle counting** field.
11. Select **Save**.

Feedback

Was this page helpful? Yes No

Provide product feedback

Configure locations in a WMS-enabled warehouse

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This guide shows you how to configure the location setup for a new WMS-enabled warehouse (a warehouse that uses warehouse management processes (WMS)). A warehouse manager typically performs this process. You can run this guide in demo data company USMF or on your own data. Ensure that you have at least one configured site.

> **Note**
>
> For a full guide on how to set up the Warehouse management module, see Get started with setting up the Warehouse management module.

Create a new warehouse

1. Go to **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**.
2. Select **New**.
3. In the **Warehouse** field, enter a value.
4. In the **Name** field, enter a value.
5. In the **Site** field, select or enter an existing site value.
6. Expand the **Warehouse** section.
7. Set the **Use warehouse management processes** option to *Yes*. This setting enables you to run warehouse management processes (WMS) by using warehouse work and mobile devices.
8. Close the page.

Define a location format

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location formats**. Location formats are a naming system that you use to create unique and consistent names for the different location bin positions within a warehouse. Use separators as part of the location format to make it easier to identify components of the location, such as the aisle number. In this example, you create a name with four components. For example, these components could be *aisle*, *rack*, *shelf*, and *bin*.
2. Select **New**.
3. In the **Location format** field, enter a value.
4. In the **Name** field, enter a value.
5. In the **Segment description** field, enter a value. This field describes what the first component of the location name represents. For example, it could be *Aisle*.
6. In the **Length** field, enter a number. This field determines how many characters this part of the location name must have. The total of all components in the name, including the separators, can't exceed 10 characters.
7. In the **Separator** field, enter a value. This field determines which character or symbol is used between the first and second component of the name.
8. In the **Details** section, select **New**.
9. In the **Segment description** field, enter a value.
10. In the **Length** field, enter a number.
11. In the **Separator** field, enter a value.
12. In the **Details** section, select **New**.
13. In the **Segment description** field, enter a value.
14. In the **Length** field, enter a number.
15. In the **Separator** field, enter a value.
16. In the **Details** section, select **New**.
17. In the **Segment description** field, enter a value.
18. In the **Length** field, enter a number.
19. Select **Save**.
20. Close the page.

Define location types

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location types**. Use location types as filtering options to control different warehouse management processes. At a minimum, create staging and final shipping location types to define the outbound warehouse management process.
2. Select **New**.

3. In the **Location** type field, enter a value.
4. In the **Description** field, enter a value.
5. Close the page.

Define location profile

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**. The definition of location profiles is very important. You can control grouped locations capacity and the policies related to what inventory gets stored and how it's stored. Use location profiles as filtering options to control different warehouse management processes. At a minimum, create a user location profile to enable WMS.
2. Select **New**.
3. In the **Location profile ID** field, enter a value.
4. In the **Name** field, enter a value.
5. In the **Location format** field, select the dropdown to open the lookup.
6. In the list, find and select the desired record.
7. In the list, select the link in the selected row.
8. In the **Location type** field, select the dropdown to open the lookup.
9. In the list, find and select the desired record.
10. In the list, select the link in the selected row.
11. Select or clear the **Allow mixed inventory statuses** check box. Enable this option if you want to allow mixed inventory status values in the locations grouped by this location profile.
12. Select or clear the **Override rules for batch days** check box. Enable this option to override the rule for how many days the inventory batch expiration dates can differ and to allow mixing of inventory batches that don't obey this rule.
13. Select or clear the **Allow cycle counting** check box. Enable this option to allow cycle counting processing in all the locations grouped by this location profile.
14. Expand or collapse the **Dimensions** section. The **Dimensions** tab allows you to define parameters and methods to enable precise calculations of the load capacity within each of the locations.
15. Close the page.

Enable warehouse management parameters

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**. To process warehouse work, set parameters for the user location profile, the staging location type, and the final shipping location type. When the outbound process ends at the final

shipping location type that you define, the related outbound transactions update to "Picked".

2. Expand the **Location profiles** section.
3. In the **User location** field, select the dropdown to open the lookup.
4. In the list, select the link in the selected row.
5. Expand the **Location types** section.
6. In the **Staging location type** field, select the dropdown to open the lookup.
7. In the list, select the link in the selected row.
8. In the **Final shipping location type** field, select the dropdown to open the lookup.
9. In the list, select the link in the selected row.
10. Close the page.

Define warehouse zone groups

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouse zone groups**. Warehouse zones can be filters for options to control the different warehouse management processes. You need to create a zone group before you can define a zone.
2. Select **New**.
3. In the **Zone group ID** field, type a value.
4. In the **Zone group name** field, type a value.
5. Close the page.

Define warehouse zones

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Zones**.
2. Select **New**.
3. In the **Zone ID** field, enter a value.
4. In the **Zone name** field, enter a value.
5. In the **Zone group ID** field, select the dropdown to open the lookup.
6. In the list, find and select the desired record.
7. In the list, select the link in the selected row.
8. Close the page.

Create locations using the Location setup wizard

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location setup wizard**.
2. In the **Warehouse** field, select the dropdown to open the lookup.
3. In the list, find and select the desired record.

4. In the list, select the link in the selected row.
5. In the **Zone ID** field, select the dropdown to open the lookup.
6. In the list, find and select the desired record.
7. In the list, select the link in the selected row.
8. In the **Location profile ID** field, select the dropdown to open the lookup.
9. In the list, find and select the desired record.
10. In the list, select the link in the selected row.
11. In the list, mark the selected row.
12. In the **From number** field, enter a number. The From number and To number fields define how many locations will be created. For example, if you set From number to 1 and To number to 3 for all four lines in the location format, 81 locations will be created (3x3x3x3).
13. In the **To number** field, enter a number.
14. In the list, find and select the desired record.
15. In the **From number** field, enter a number.
16. In the **To number** field, enter a number.
17. In the list, find and select the desired record.
18. In the **From number** field, enter a number.
19. In the **To number** field, enter a number.
20. In the list, find and select the desired record.
21. In the **From number** field, enter a number.
22. In the **To number** field, enter a number.
23. Select Create.

Create locations manually

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**. You can easily create locations within a warehouse. You must enter the location name and the location profile ID.
2. Select **New**.
3. In the **Warehouse** field, enter a value.
4. In the **Location** field, enter a value. You're creating a new location, so enter a new unique name instead of selecting an existing one.
5. In the **Location profile ID** field, enter a value.
6. Close the page.

Define pack size categories

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Pack size categories**. Use pack size categories to group items that have similar physical packing sizes. In this example, the

pack size category controls the capacity at the picking locations within a specific zone of the warehouse. Assign the pack size category ID to the released product entity to use it as part of the stocking limits processing.

2. Select **New**.
3. In the **Pack size category ID** field, enter a value.
4. In the **Pack size category name** field, enter a value.
5. Close the page.

Define location stocking limits

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location stocking limits**. Location stocking limits help ensure that work isn't created that requests inventory to be put in a location that doesn't have the physical capacity to carry the inventory.
2. Select **New**.
3. In the **Warehouse** field, enter a value.
4. In the **Location profile ID** field, enter a value.
5. In the **Pack size category ID** field, enter a value.
6. In the **Quantity** field, enter a number.
7. Select **Save**.
8. Close the page.

> **Note**
>
> You can allow stocking limits to be exceeded for specific locations for replenishment work. For more information, see Location stocking limits and Replenishment over location capacity.

Define fixed picking locations

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Fixed locations**. You can define the locations to use for each product or each product variant. You can create multiple fixed locations for the same product within the same warehouse.
2. Select **New**.
3. In the **Item number** field, enter a value.
4. In the **Warehouse** field, enter a value.
5. In the **Location** field, select the dropdown to open the lookup.
6. In the list, select the link in the selected row.

7. Close the page.

Define location blocking causes

Assign each location **Input blocked** and **Output blocked** causes. These causes affect warehouse management business processes and determine whether you can use that location for input and/or output. One reason to block a location is that it requires maintenance. To set up the blocking causes that you can use for locations, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Inventory** > **Blocking causes**.
2. Select **New**.
3. In the **Cause of blocking** field, enter a short name that identifies the cause. This value appears when setting up the location.
4. In the **Description** field, enter a short description.
5. In the **Policy** field, select one of the following values:

    - *Do not use for warehouse work* – Locations blocked for this cause aren't blocked when processing warehouse work.
    - *Also use for warehouse work* – Locations blocked for this cause are also blocked when processing warehouse work.

6. Close the page.

Last updated on 05/06/2026

Warehouse location status

Microsoft Dynamics 365 Supply Chain Management includes several location fields that give you flexibility when you work with and maintain locations. You can include location statuses in the location directive query to provide better control over warehouse flow.

The following four fields on the **Locations** page track information about the current status of a location. These fields let warehouse managers get an overview of the status of the warehouse locations. They also allow for advanced reporting and filtering.

- **Item number** – The item that is currently in the location. If the location contains multiple items, this field is blank.
- **Last activity date and time** – The timestamp of the last warehouse transaction that was performed against the location.
- **Aging date** – The date when the inventory in the location was brought into the warehouse. This value is calculated based on the aging date of the license plate. It's accurate for locations that are license plate–tracked, but it might not be accurate for locations that aren't license plate–tracked.
- **Location status** – The status of the location. There are four possible values:
  - **Undetermined** – The location profile can't track status. Therefore, the current status is unknown.
  - **Empty** – There's currently no inventory in the location.
  - **Picking** – Outbound transactions have been performed against the location since it was last empty.
  - **Storage** – Only inbound transactions have been performed against the location since the location was last empty.

Set up warehouse location status

Prepare the sample data required for the example scenario

Before you start working through the scenario, activate sample data and set up the feature as described in this section. To complete the example scenario, use either the Warehouse Management mobile app or the browser-based emulator. The steps that are provided here use the Warehouse Management mobile app. The steps for the browser-based emulator are similar.

Use the USMF legal entity

To work through the example scenario by using the sample records and values that are specified here, use a system where the standard demo data is installed. Additionally, select the *USMF* legal entity before you begin.

Set up location profiles

The example scenario requires that you prepare two location profiles.

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location profiles**.
2. Select **Edit** to put the page into edit mode.
3. Select the *BULK-06* profile.
4. On the **General** FastTab, set the following values:
    - **Enable item in location:** Set this option to *Yes*.
    - **Enable location activity date and time:** Set this option to *Yes*.
    - **Enable location status:** Set this option to *Yes*.

    These options control whether the reference fields on the location are active.
5. Repeat steps 3 through 4 for the *PICK-06* profile.

> **Note**
>
> When the parameters on the location profile (**Enable item in location**, **Enable location activity**, **Enable location status**) are set to *Yes*, the system immediately updates the relevant locations by executing the *warehouse location status consistency check* job.

Scenario

1. Go to **Procurement and sourcing** > **Purchase orders** > **All purchase orders**.
2. Select **New**.
3. In **Create purchase order**, on the **Vendor** FastTab, in the **Vendor account** field, select *104*.
4. On the **General** FastTab, in the **Warehouse** field, select *61*.
5. Select **OK**.

6. Your new purchase order (PO) opens. It includes an empty line in the **Purchase order lines** grid. On this line, set the following values:

    - **Item number:** *A0002*
    - **Quantity:** *5*

7. On the Action Pane, on the **Purchase** tab, in the **Actions** group, select **Confirm** to confirm the purchase order.

8. On the mobile device, go to **Inbound** > **Purchase Receive**.

9. Select the **PONUM** field, enter the PO number, and confirm.

10. Select the **ITEM** field, enter *A0002* as the item number, and confirm.

11. On the **QTY** page, enter *5* as the quantity, and confirm.

    You can enter the quantity in either of the following ways:

    - Select the plus sign (+) or minus sign (–) button to add or subtract a numerical value.
    - Select the blank field between the plus sign (+) and minus sign (–) buttons to open the number pad.

12. Confirm your selection of item number *A0002* and a quantity of *5*. A "Work Completed" message appears at the bottom of the page.

13. Select the Menu button (sometimes referred to as the hamburger or the hamburger button) in the upper-right corner, and then select **Cancel** to exit **Purchase Receive** and return to the **Inbound** menu.

14. On the purchase order page, select **Work details** above the **Purchase order lines** grid.

15. On the **General** tab, notice the **Work ID** and **Target license plate ID** values that were created.

16. In the **Lines** section, notice the **Location** values for the *Pick* and *Put* work types.

17. On the mobile device, go to **Inbound** > **Purchase Put-away**.

18. Select the **ID** field, enter the work ID, and confirm.

19. Confirm once more to complete the *Pick* entry.

20. Select the Menu button in the upper-right corner, and then select **Done** to complete the *Pick* work.

21. Make a note of the Putaway location, and confirm. A "Work Completed" message appears at the bottom of the page.

22. Select the Menu button in the upper-right corner, and then select **Cancel** to exit **Purchase Put-away** and return to the **Inbound** menu.

23. Select **Back** to return to the main menu.

24. In Dynamics 365 Supply Chain Management, go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.

25. Filter on **Location**, and enter the putaway location from the purchase order work. You see the following results:

    - The **Location status** column shows a value of *Storage*, because the last transaction against this location was a put.
    - The **Item number** column shows a value of *A0002*, because that item was received and put to the location.
    - The **Last activity date and time** column shows the timestamp for the date and time when the work was completed at the location.

26. On the mobile device, go to **Quality** > **Movement**.

27. Select the **LOC/LP** field, and enter the location you made note of in the previous steps.

28. Confirm the information that is shown. Make a note of the license plate number that is generated.

29. On the **To Information** screen, select the **LOC/LP** field, and enter *06A07R2S1B* as the location to move the item to.

30. On the **To Information** screen, confirm the **LP** value (the target license plate ID), which is automatically generated. A "Work Completed" message appears at the bottom of the page.

31. Select the Menu button in the upper-right corner, and then select **Cancel** to exit **Movement** and return to the **Quality Management** menu.

32. Select **Back** to return to the main menu.

33. In Dynamics 365 Supply Chain Management, go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.

34. Refresh the **Locations** page, and view the original putaway location again. Notice that the **Location status** field is now set to *Empty*, and the **Item number** column is blank.

35. View the record for location *06A07R2S1B*, and notice that the **Status** value has changed to *Storage*, and the **Item number** and **Last activity date and time** fields are updated.

36. Go to **Sales and marketing** > **Sales orders** > **All sales orders**.

37. Select **New**.

38. In **Create sales order**, in the **Customer account** field, select *US-002*.

39. In the **Warehouse** field, select *61*.

40. Select **OK**.

41. Your new sales order opens. It includes an empty line in the **Sales order lines** grid. On this line, set the following values:

    - **Item number:** *A0002*
    - **Quantity:** *1*

42. On the **Sales order lines** FastTab, on the **Inventory** menu, select **Reservation**.

43. On the **Reservation** page, select **Reserve lot** to reserve the order line. Then select the **Close** button (**X**) in the upper-right corner to close the page.

44. On the Action Pane, on the **Warehouse** tab, in the **Actions** group, select **Release to warehouse**.

45. In the **Sales order lines** section, on the **Warehouse** menu, select **Work details**.

46. Copy the **Work ID** value that was created.

47. On the mobile device, go to **Outbound** > **Sales picking**.

48. Select the **ID** field, enter the work ID that you copied earlier, and confirm.

49. On the **Sales orders: Pick** page, the **LOC** field suggests the picking location as the putaway location that was created earlier. Make a note of the location.

50. Select the **LOC** field, enter the location, and confirm.

51. Select the **LP** field, enter the license plate number that you made a note of during the Movement activity, and confirm.

52. Select the **Item** field, enter the *A0002* as the item number, and confirm.

53. On the **QTY** page, enter *1* as the quantity, and confirm.

    You can enter the quantity in either of the following ways:

    - Select the plus sign (+) or minus sign (–) button to add or subtract a numerical value.
    - Select the blank field between the plus sign (+) and minus sign (–) buttons to open the number pad.

54. Select the **TARGET LP** field, enter a user-defined target license plate ID, and confirm.

55. Confirm once more to complete the picking work. A "Work Completed" message appears at the bottom of the page.

56. Select the Menu button in the upper-right corner, and then select **Cancel** to complete the picking activity and return to the **Outbound** menu.

57. In Dynamics 365 Supply Chain Management, go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**.

58. Filter on **Location**, and enter the pick location from the sales order work.

59. Notice that the **Location status** field for the location that the sales order work picked from is now set to *Picking*, and the **Last activity date and time** field is updated.

> ⓘ **Note**
>
> You can update the location fields only through warehouse transactions. If you move inventory by using a journal or other non-WMS processes, the fields aren't updated.

Last updated on 07/01/2026

Additional location zones

Three new zone fields are available in Microsoft Dynamics 365 Supply Chain Management. Warehouse managers can use them to define additional warehouse organizations or layouts. The new zone fields can be set either manually or by using the **Location setup** wizard. They can be used in any query or filtering that uses the Locations table.

No additional setup is required to use the zone fields.

Use location zones

1. Go to **Warehouse management** > **Setup** > **Warehouse** > **Location setup wizard**.
2. Set the following values:
    - In the **Warehouse** field, select *62*.
    - In the **Zone ID** field, select *FLOOR*.
    - In the **Additional Zone 1** field, select *PICKZONE1*.
    - In the **Additional Zone 2** field, select *WEBSHOP1*.
    - In the **Location profile ID** field, select *FLOOR*.
3. Select the **Floor** line.
4. In the **From number** field, enter *1*. In the **To number** field, enter *3*.
5. Select the **Aisle** line.
6. In the **From number** field, enter *1*. In the **To number** field, enter *5*.
7. Select **Create**.
8. You receive messages that state that new locations have been added. Select the **Show messages** button to view the messages.
9. Go to **Warehouse management** > **Setup** > **Warehouse** > **Locations**. The new locations appear in the list, and all zone fields are available (that is, the existing zone field and the new additional zone fields).

Last updated on 07/01/2026

Schedule load utilization

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can schedule load utilization for selected location types, and you can also project the current and future load utilization. You can project the load for one or more sites, for the load units (zone or warehouse), or for a combination of a zone and a warehouse.

Schedule and view the load for a warehouse or site

To schedule the load for sites, warehouses, or zones, you create a space utilization setup and associate it with a master plan.

In the space utilization setup, you use location types, such as **Bulk location** and **Picking location**, to specify how space utilization should be projected. You also specify a storage load mode, such as **Zone**.

The projection of future space utilization is based on information that is calculated on the associated master plan. Master plans forecast the resource planning for incoming and outgoing orders for production and operations. The projection of available space is based on the relation between the space utilization setup and the selected master plan.

By using the storage load mode that you selected in the space utilization setup, you can specify whether the load should be projected for each warehouse or zone, or whether the projections should include information about both warehouses and zones. You also specify whether blocked locations should be excluded from the calculation of load utilization.

You can project the space utilization by generating the **Warehouse load utilization** report. When you generate this report, you can specify whether the load utilization should be projected for each site, across sites, or for the selected load unit, such as zone or warehouse.

Create a space utilization setup for a warehouse

1. Select **Inventory management** > **Setup** > **Warehouse monitoring** > **Space utilization**.
2. Select **New** to create a space utilization setup. Specify an ID and a name for the new setup.

3. In the **Storage load mode** field, select whether the overview of space utilization should show information by warehouse, zone, or warehouse and zone.
4. Set the **Exclude blocked locations** option to **Yes** to exclude blocked inventory locations from the calculation of available space. You can block an inventory location for input and output by specifying a blocking cause for the location in the **Input blocked** or **Output blocked** field on the **Other** FastTab on the **Inventory locations** page.
5. On the **Location type** FastTab, select the location types to include in the space utilization calculation. You must select at least one location type for the projection.

Associate a space utilization setup with a master plan

1. Select **Inventory management** > **Periodic tasks** > **Schedule load utilization**.
2. In the **Master plan** field, select a master plan.
3. In the **Number of days** field, specify the number of days to include in the projection of current and future workloads.
4. In the **Space utilization** field, select the space utilization setup to use for the projection of current and future workloads.

Specify the load utilization projection and view information

1. Select **Inventory management** > **Inquiries and reports** > **Physical inventory reports** > **Warehouse load utilization**.

2. In the **Show by** field, select the level of the space utilization projection:

    - **Site** – Project the space utilization for each site. This projection is useful if, for example, you want to view all the warehouses for a site so that you can balance the space utilization between the warehouses.
    - **Load unit** – Project the space utilization for zones or warehouses. The available space is then projected according to the value that you selected in the **Storage load mode** field on the **Space utilization** page when you created the space utilization setup.

3. Follow one of these steps, depending on the value that you selected in the previous step:

    - If you selected **Site** in the **Show by** field, the **Site** field is available. Select one or more sites to include in the projection.
    - If you selected **Load unit** in the **Show by** field, the **Load unit** field is available. Select the load unit.

4. In the **Load type** field, select **Volume** or **Weight** to specify the warehouse operating unit to project space for.

5. In the **Space utilization setup** field, select the space utilization setup that the projection should be based on.

Last updated on 05/07/2025

Schedule workload capacity

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

You can schedule workload capacity for warehouses, and you can also project the current and future workloads for the workers in individual warehouses. You can project the workload for the whole warehouse, or you can project the workload separately for incoming and outgoing workloads.

To project workload output for selected warehouses, master scheduling data must be available for those warehouses. Learn more in Master plans overview.

Schedule and view workloads for a warehouse

To schedule workload capacity for a warehouse, you create a workload setup for one or more warehouses, and then associate the workload setup with a master plan. In the workload capacity setup, you can define limits for the weight or volume for incoming and outgoing transactions. You can also create more than one setup for each warehouse and then associate the setup with individual master plans. For example, you might use this approach to accommodate seasonal changes in the workforce.

If the workers for a warehouse work with transactions for both incoming and outgoing workloads, you can configure the warehouse capacity setup so that the workload is projected in a combined view.

To schedule and view workloads for warehouses, you must complete the following tasks:

1. Create a workload capacity setup and define workload capacity limits for one or more warehouses.
2. Associate the workload capacity setup with a master plan to create workload projections and specify how long those projections will apply.

Create a workload capacity setup for a warehouse

1. Select **Inventory management** > **Setup** > **Warehouse monitoring** > **Workload capacity**.

2. On the Action Pane, select **New** to create a workload capacity setup.
3. On the **Warehouses** FastTab, select **New**, and then enter values on the line to associate a warehouse with the workload capacity setup.
4. Select the **Combined inbound and outbound workload** check box if the **Workload capacity** report should show the total workload for incoming and outgoing transactions in one view.
5. On the **Transaction types** FastTab, select the types of incoming and outgoing transactions that the workload limits will apply to.

> ⓘ **Note**
>
> You must select at least one transaction type for both the incoming workload and the outgoing workload.

Define limits for volume or weight

You can set up limits for volume or weight, depending on the limitation that is relevant for the warehouse workforce. The limits that you specify are included in the workload capacity projection that you can view on the **Workload capacity** report.

To project information about volume and weight for items, the standard volume of one inventory item and the weight of one inventory item must be specified for all products. The fields that are required are available in the following field groups on the **Manage inventory** FastTab of the **Released product details** page:

- Handling
- Physical dimensions
- Weight measurements

If this information isn't specified correctly, you receive a message when you generate the **Workload capacity** report. From the report, you can then drill down to identify the missing information that is required in order to project the future workload.

Associate a workload capacity setup with a master plan

1. Select **Inventory management** > **Periodic** > **Schedule workload**.
2. In the **Master plan** field, select the master plan to use for workload projections.
3. In the **Number of days** field, specify the number of days that the workload projection covers.

4. In the **Workload** field, select the workload setup to associate with the master plan.

View workload capacity

1. Select **Inventory management** > **Inquiries and reports** > **Physical inventory reports** > **Workload capacity**.
2. In the **Number of columns** field, specify the number of columns to show on the report.
3. In the **Order type** field, select **Planned and confirmed**, **Planned**, or **Confirmed** to indicate the type of orders to project on the report.
4. In the **Load type** field, select a load type to specify whether the workload capacity should be projected for volume or weight.
5. In the **Workload capacity** field, select a workload capacity setup.

Last updated on 05/07/2025

Set up dispositions codes

Article • 09/29/2021

This procedure focuses on the setup of a disposition code that can be used on a mobile device for the return order receiving process. Disposition codes are a collection of rules that can be used when items are received. For example, when a work user uses a mobile device to receive items that were damaged, the user must scan a disposition code for damaged items. The inventory status of the goods received, the work template, and the location directive can be determined from the scanned disposition code. For the purchase order receiving process and the production order report as finished process, the use of a disposition code is optional. For the sales order return receiving process, if the items are registered using a mobile device, the use of disposition code is mandatory. This guide was created using the demo data company USMF. This procedure is intended for the warehouse manager.

1. Go to Warehouse management > Setup > Mobile device > Disposition codes.
2. Click New.
    - Create a new disposition code to use for customer returns.
3. In the Disposition code field, type a value.
4. In the Inventory status field, select an inventory status where there is inventory blocking.
    - If you're using USMF, select 'Blocking'. You can add an inventory status to the disposition code to override the default status that's on the order lines.
5. In the Work template field, type a value.
    - Optional: Select a work template code that is associated with a return order. If no value is provided, the work template will be resolved using the standard rules configured in your system. Selecting a work template will limit the processes this disposition code can be used with. For example, if a disposition code has a work template with a work order of type purchase order, it can't be used to register items that are returned by customers.
6. In the Return disposition code field, type a value.
    - The return disposition code determines the remainder of the return order process for the items registered. In this example, the customer should receive a credit note. Add a returns disposition code that contains an action Credit.

Feedback

Was this page helpful? Yes No

Provide product feedback

Unit of measure and stocking policies

This article describes how default units, unit sequences, and unit conversions are used in warehouse processes.

Unit sequence groups define the sequence of units that can be used in warehouse operations. You create them on the **Unit sequence groups** page. The sequence shows the relationship of the various units. For example, you store pallets that contain boxes that contain individual pieces of items. In this case, you must provide the three different units and the logical order of the layers. Unit sequence groups let you define the policies for the grouping of license plates, and the default units that should be used for various warehouse processes. This article applies to both the warehouse management processes (WMS) available in the Warehouse management module and the more basic warehousing solution that is available in the Inventory management module.

Unit sequence groups for released products

To use released products in warehouse work processes, assign a unit sequence group to them. If you validate a product that is associated with a Storage dimension group, and the **Use warehouse management processes** option for the Storage dimension group is set to **Yes**, you receive an error message if a unit sequence group ID isn't defined for the product. If the unit sequence group that you use contains multiple lines (and therefore multiple units), you must set up a unit conversion between the units. You complete this setup on the **Unit conversions** page. The smallest unit in a sequence group that you associate with a released product must match the inventory unit that is defined for the corresponding product. The inventory unit is the unit that is used for base calculations of the on-hand inventory. You can also set up unit of measure conversions for product variants of product masters by using the **Enable unit of measure conversions** option.

License plate grouping

You can specify whether receipts of less than or more than a specific unit should be grouped into one license plate or divided into a license plate for each unit. To set up this behavior, use the **License plate grouping** option on the **Line details** tab of the **Unit sequence groups** page. If you want to use the license plate grouping when you process work by using a mobile device, select the **License plate grouping** option on the **Mobile device** menu item. For example, you're using a mobile device to register an item that is associated with a unit sequence group that has two lines.

The first line is for Pieces, and the **License plate grouping** option is set to **Yes**. The second line is for the Pallet unit, and the **License plate grouping** option is set to **No**. In this case, the system can automatically guide the split and creation of license plates per 100 pieces.

Units for cycle counting

To define which units should be used as part of the cycle counting processes, select the **Use unit for cycle counting** option on the **Unit sequence groups** page. You can select up to four units in the sequence group. If you select more than four units, the additional units don't appear on the mobile device.

Default units for mobile device receiving processes

To set the default units that should be used for receiving processes on mobile devices, enable the **Default unit for purchase and transfer** and **Default unit for production** options on the **Unit sequence groups** page. For example, you can specify that, by default, the system should use Pallet quantities when purchase order on-hand inventory is received by using a mobile device, but that the stock-keeping unit should be Pieces. To get the conversion for the number of pieces that each pallet contains, you must define a unit conversion, such as 100 Pcs = 1 PL.

Default order settings

As part of the creation of released products, select default units for purchases, sales, and inventory to process the various orders. Set the default units and quantities for the various source documents by using the **Default order settings** and **Site specific order settings** pages. You can access these pages from the **Released products** page.

Last updated on 07/01/2026

System grouping on an open work list

By using a system grouping field, you can filter an open work list without having to edit the mobile device menu item. Where it applies, system grouping works for filtering a work list on a single work header field. You cannot use system grouping to filter on line level fields.

Set up system grouping on an open work list

Use these steps to set up system grouping on an open work list.

- From a mobile device menu item, select **Mode: Indirect** and **Activity Code: Display open work list**. The following options become available. These options are required for system grouping on an open work list.

Expand table

Option	Description
Allow system grouping	Enables system groping for a selected work list menu item.
System grouping field	Available only if **Allow system work** is set to **Yes**. Select the field that determines how picking work will be grouped for workers. For example, if you select the **ShipmentId** field, the worker will scan the shipment ID to group the picking work. All work for the shipment is then assigned to the worker. This field requires that you create a menu item to use existing work that is grouped by the system. Use the **System grouping label** field to inform the worker what to scan.
System grouping label	Available only if **Allow system work** is set to **Yes**. Enter information for the worker about what to scan when picking work is grouped. For example, if you use the **ShipmentId** field to group picking work by shipment, you might enter Shipment ID in the field. This field requires that you create a menu item to use existing work that is grouped by the system. You must also select the field to group by in the **System grouping** field.

Last updated on 07/01/2026

GS1 bar codes and QR codes

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the **Request access to Finance and Operations Viva Engage Community** form and choose the community you want to join.

Warehouse workers often complete several tasks when they use a mobile device scanner to register movements of an item, pallet, or container. These tasks can include both scanning bar codes and manually entering information on the mobile device. The bar codes use a company-specific format that are defined and managed by using Microsoft Dynamics 365 Supply Chain Management.

GS1 bar codes for shipping labels provide a global standard for the exchange of data between companies. They're available in both linear (1D) symbologies (bar code formats), such as GS1-128, and 2D symbologies, such as GS1 DataMatrix and GS1 QR codes. GS1 bar codes not only encode data but also let you use a predefined list of *application identifiers* to define the meaning of that data. The GS1 standard defines the data format and the various kinds of data that it can encode. Unlike older bar code standards, GS1 bar codes can have multiple data elements. Therefore, a single bar code scan can capture several types of product information, such as the batch and the expiration date.

GS1 support in Supply Chain Management dramatically simplifies the scanning process in warehouses where pallets and containers are labeled by using bar codes in GS1 format. Warehouse workers can extract all the required information through a single scan of a GS1 bar code. By eliminating the need to do multiple scans and/or manually enter information, GS1 bar codes help reduce the time that is associated with tasks. At the same time, they also help improve accuracy.

Logistics managers must set up the required list of application identifiers and associate each of them with the appropriate mobile device menu items. The application identifiers can then be used across warehouses as a global setting for moving and packing purposes. Therefore, all shipping labels take a unified form.

Unless otherwise stated, this article uses the term *bar code* to refer to both linear (1D) bar codes and 2D bar codes.

The GS1 bar code format

The GS1 General Specifications specify which symbologies can be used for GS1 bar codes and how to encode the data in the bar code. This section provides a short introduction to the article. For full details, see the GS1 General Specifications published by GS1. The GS1 specifications document is regularly updated, and the information that it provides is up to date with GS1 General Specifications release 22.0.

GS1 bar codes use the following symbologies:

- **Linear or 1D bar codes** – GS1-128 and GS1 DataBar
- **2D bar codes** – GS1 DataMatrix, GS1 QR Code, and GS1 Dotcode

GS1 is mentioned in GS1-128, which is a special case of the ordinary Code-128 linear bar code, GS1 DataMatrix, and GS1 QR Code. The difference between the GS1 version and the non-GS1 version is the presence of a special character (FNC1) as the first character in the bar code data. The presence of an FNC1 character indicates that the data in the bar code should be interpreted according to the GS1 specification.

The data in the bar code itself consists of multiple data elements, each of which is identified by an application identifier at the start of the field. Usually, the data is also presented under the bar code in a human-readable format, where the application identifier is shown in parenthesis. Here's an example: `(01) 09521101530001 (17) 210119 (10) AB-123`. This bar code contains three elements:

- **Application identifier 01** – The GS1 global trade item number (GTIN) of the item.
- **Application identifier 17** – The expiration date.
- **Application identifier 10** – The batch number.

For each element, the data might have either a predefined length or a variable length. There's a fixed list of application identifiers that have predefined lengths. All other application identifiers have variable length, and the GS1 application identifier list specifies the maximum length and format of data. For example, application identifier 01 has a predefined length of 16 characters (two for the application identifier itself and then 14 for the GTIN), and application identifier 17 has a predefined length of eight characters (two for the application identifier itself and then six for the date). However, application identifier 10 has two numbers for the application identifier itself and then up to 20 alphanumeric characters.

When elements are put together, if an element follows a variable length element, a separator character must be used. This separator can be either the special FNC1 character or the group separator character (a nonprintable character that has ASCII code 29 and hexadecimal code 1D). The separator shouldn't be used after the last element. Although the separator also shouldn't be used after elements that have predefined length, its presence isn't a critical error.

In the bar code data from the previous example of a bar code that contains application identifiers 01, 17, and 10, the data in a Code-128, QR Code, or DataMatrix symbol is encoded as `<FNC1>` **`01`** `09521101530001` **`17`** `210119` **`10`** `AB-123` (application identifiers are shown in bold). As a best practice, put any element that has variable length at the end, to eliminate the need for an extra group separator character. However, the bar code can also have a different order of elements, where the separator is mandatory. Here's an example: `<FNC1>` **`01`** `09521101530001` **`10`** `AB-123<GS>` **`17`** `210119`.

Dates and decimal numbers

You must always represent dates in *YYMMDD* format, where GS1 specifications determine the century of the year. You can represent only dates from 49 years in the past to 50 years in the future (relative to the current year).

Some data elements contain decimal numbers. For example, application identifiers 3100, 3101, ... 3105 represent a net weight in kilograms. Because these application identifiers have a predefined length of 10, six numbers are available for the quantity. The position of the decimal point is specified by the last number of the application identifier. Therefore, this family of application identifiers can also be represented as *310n*. Because the GS1 standard specifies that there must always be at least one number to the left of the decimal point, a maximum of five decimal places is allowed.

The following examples show how different application identifiers (shown in bold) interpret the number *123456*:

- **`3100`** `123456` → 123456 (integer)
- **`3101`** `123456` → 12345.6 (one decimal place)
- **`3102`** `123456` → 1234.56 (two decimal places)
- **`3103`** `123456` → 123.456 (three decimal places)
- **`3104`** `123456` → 12.3456 (four decimal places)
- **`3105`** `123456` → 1.23456 (five decimal places)

Scanning GS1 bar codes in Supply Chain Management

To scan GS1 bar codes, warehouse workers use a scanner that's built into or connected to a mobile device. The scanner then transmits the scanned bar code to the Warehouse Management mobile app as a series of keyboard events. This mode of operation is also known as a *keyboard wedge* or *wedge*. The mobile app then sends the received text as-is to Supply Chain Management. When the system receives input data, it first determines whether the data begins with one of the configured prefixes that indicate that the data is actually a GS1 bar code (see the Set up global GS1 options section). If the scanned data does begin with one of those prefixes, the system uses a GS1 parser to parse the data and extract individual data elements according to their application identifiers. After the data is parsed, either the current input field or multiple fields are filled in with the scanned data.

Configuration of bar code scanner hardware and software

For Supply Chain Management to correctly recognize and decode GS1 bar codes, the hardware scanner or supporting software must be configured to perform the following actions:

- Add a prefix to the scanned bar codes, so that the system can recognize a GS1 bar code.
- Convert the nonprintable ASCII group separator character (ASCII code 29 or hexadecimal code 1D) to a printable character, such as a tilde (~).

Although you can add any prefix to the scanned bar code, one option is to add an ISO/IEC 15424 symbology identifier, also known as an *AIM identifier*. This three-character identifier starts with `]`, then has one character that identifies the symbology that is used, and then has a number that is used as a further modifier. For example, the AIM identifier `]C1` specifies a Code 128 bar code (because of the character `C`), and the modifier `1` specifies that there's an FNC1 character in the first position of the data. On the other hand, `]C0` is a Code 128 bar code that has any other character as the first character of the data.

The following five symbology identifiers correspond to GS1 bar codes that have application identifier elements:

- `]C1` – Code 128 (`C`) with FNC1 character in first position (`1`), also known as GS1-128.
- `]e0` – GS1 DataBar.
- `]d2` – DataMatrix (`d`) with ECC 200 and FNC1 in first position (`2`), also known as GS1 DataMatrix.
- `]Q3` – QR Code (`Q`) Model 2 symbol with FNC1 in first position (`3`), also known as GS1 QR Code.
- `]J1` – GS1 DotCode.

If you use these identifiers, compatibility with non-GS1 bar codes requires that the scanners or scanning software b econfigured to remove any identifiers that don't correspond to the GS1 identifiers. For example, if you scan a "normal" Code 39 bar code, the prefix `]A0` is added. Because the system doesn't understand this prefix as one of the GS1 prefixes, it interprets it as data and produces unexpected results.

> ⓘ **Note**
>
> For convenience, version 2.0.17.0 and later of the Warehouse Management mobile app removes any AIM prefixes that aren't included in the previous list. This behavior supports cases where you can configure the scanner to add the AIM prefix but not to remove the unwanted prefixes.

Single and multiple field scanning

After the data is parsed from the bar code, it's fed into the mobile device flow controls. Two methods are processed:

- **Single field scanning** – This method fills in only the field where you scanned the bar code. For example, if you scan the bar code `<FNC1>` **`01`** `09521101530001` **`17`** `210119` **`10`** `AB-123` while the cursor is in the **Item** field, the GTIN `09521101530001` from the bar code is entered in that field. If you scan the same bar code while the cursor is in the **Batch ID** field, the batch number `AB-123` from the bar code is entered. This mode works for all fields in all flows and requires only that the generic bar code data setup be configured. If a bar code contains multiple elements, you must still scan it multiple times, because only one piece of the bar code at a time is entered into the mobile device flow. The generic bar code data setup controls this behavior, as described in the Establish the generic bar code data setup section.
- **Multiple field scanning** – This method fills in multiple fields when you scan one bar code, by pushing extra data into the mobile device flow state. For example, the policy is configured to push application identifier 01 into the `ItemId` control and application identifier 10 into the `InventBatchId` field. In this case, if you scan the bar code `<FNC1>` **`01`** `09521101530001` **`17`** `210119` **`10`** `AB-123`, data for both variables is pushed at the same time. Therefore, the system doesn't prompt you for the item and batch number in the flow. Bar code data policies that link to menu items control this behavior, as described in the Set up bar code data policies for mobile device menu items section.

> ⚠ **Warning**

The default bar code data policies work without unexpected behavior. However, customization of bar code data policies linked to menu items can cause unexpected behavior, because the flow might not expect some data to be available at a particular time.

Set up global GS1 options

The **Warehouse management parameters** page provides a few settings that establish global GS1 options.

To set up global GS1 options, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **Bar codes** FastTab, set the following fields:
    - **FNC1 Character**, **Datamatrix character**, and **QR code character** – Specify characters that should be interpreted as a prefix for each type of GS1 bar code.
    - **Group separator** – Specify the character that replaces the ASCII group separator character.
    - **Maximum length of identifier** – Specify the maximum number of characters that is permitted for the application identifier. This field isn't required if the *Enhanced GS1 Parser* feature is turned on for your system.
    - **Unknown application identifier policy** – Select one of the following values to specify what the system should do if it encounters an unknown application identifier when it parses a GS1 bar code:
        - *Error* – The system reports an error and doesn't scan any part of the bar code. To scan this type of bar code, you must add a matching entry in the GS1 application identifier table.
        - *Skip the data element* – The system continues to parse the bar code without reporting an error or a warning. The data from the unknown application identifier isn't available to the application for further processing.

Note

Prefixes tell the system that a bar code is encoded according to the GS1 standard. Up to three prefixes (**FNC1 Character**, **Datamatrix character**, and **QR code character**) can be used simultaneously and for various purposes.

Bar code data application identifiers

GS1 formats not only encode the data but also let you use a predefined list of application identifiers to define the meaning of the data. Logistics managers must set up the required list of application identifiers and associate each of them with the appropriate mobile device menu items. Use the identifiers across warehouses as a global setting for moving and packing purposes. Therefore, all shipping labels take a unified form.

The system uses the data, especially the predefined application identifiers, to establish the rules that apply to the relevant piece of scanned information.

Each application identifier signals to the system that subsequent characters in the scanned bar code should be considered a block of encrypted information. The setup of the application identifiers defines how the system interprets a bar code and saves it as a value in the system.

Logistics managers can use standard international application identifiers and/or create their own.

Load the standard application identifiers

To get started quickly, load a list of common international application identifiers. You can then extend or edit the list later, as needed.

To load the standard application identifiers, follow these steps:

1. Go to **Warehouse management** > **Setup** > **GS1** > **Barcode data application identifiers**.
2. On the Action Pane, select **Create default setup**.

> ⚠ **Warning**
>
> The **Create default setup** command deletes all currently defined application identifiers and replaces them with the standard list. However, after you load the default setup, you can customize the list as needed.

Set up custom application identifiers

If some or all application identifiers that your company uses differ from the standard set, you can create your own codes from scratch or customize the standard set as you require.

To set up and customize your GS1 own application identifiers, follow these steps:

1. Go to **Warehouse management** > **Setup** > **GS1** > **Barcode data application identifiers**.

2. Follow one of these steps:

    - To create a new identifier: On the Action Pane, select **New**.
    - To edit an existing identifier: Select the identifier, and then, on the Action Pane, select **Edit**.

3. Set the following fields for the new or selected identifier:

    - **Application identifier** – Enter the identification code for the application identifier. Typically, this code is a two-digit integer, but it can be longer. For decimal values, the last digit indicates the number of decimal places. For more information, see the description of the **Decimal** checkbox later in this list. You can create a single application identifier for all decimal place variants by using the letter *n* as the last character in the application identifier. For example, you can configure just one application identifier (*310n*) instead of a separate application identifier for each number of decimal places (*3101*, *3102*, *3103*, and so on).
    - **Description** – Enter a short description of the identifier.
    - **Fixed length** – Select this checkbox if values that are scanned by using this application identifier have a fixed number of characters. Clear this checkbox if the length of values is variable. In this case, you must indicate the end of the value by using the group separator character that you specified on the **Warehouse management parameters** page.
    - **Length** – Enter the maximum number of characters that can appear in the values that are scanned by using this application identifier. If the **Fixed length** checkbox is selected, exactly this number of characters is expected.
    - **Type** – Select the type of value that is scanned by using this application identifier (*Numeric*, *Alphanumeric*, or *Date*). For more information about how dates and numbers are represented in bar code data, see the Dates and decimal numbers section.
    - **Decimal** – Select this checkbox if the value includes an implied decimal point. If this box is selected, the system uses the last digit of the application identifier to determine the number of decimal places. For more information about how dates and numbers are represented in bar code data, see the Dates and decimal numbers section.

> ⚠ **Warning**
>
> Although the system allows you to set the **Fixed length** checkbox for any application identifier, use it only for the subset of application identifiers that have a predefined length per the GS1 General Specifications. The enhanced GS1 parser already contains the list of all application identifiers that have predefined lengths.

> ⓘ **Note**
>
> The **Group separator** value specified on the **Warehouse management parameters** page is optional if a value that is followed by an application identifier has a fixed length.

Establish the generic bar code data setup

The generic bar code data setup establishes a collection of common mappings. These mappings match each relevant input field in the mobile app to the application identifier that controls how values from scanned bar codes are interpreted and stored in that field. By default, these settings apply to all scans for all mobile device menu items. However, a bar code data policy assigned to a specific menu item can overwrite these settings for one or more specific fields.

The generic bar code data setup supports scanning only one value at a time. Therefore, if you want to load multiple field values from a single scan, you must set up a bar code data policy for the relevant menu items.

For more information about bar code data policies, see the next section.

Load the standard generic bar code data setup

The **Barcode data generic setup** page lets you load a standard set of mappings between mobile device fields and standard application identifiers that the default setup creates.

To establish the generic bar code data setup, go to **Warehouse management** > **Setup** > **GS1** > **Barcode data generic setup**. Then select **Create default setup** to automatically assign a suitable application identifier to each field that mobile device menu items typically use.

> ⚠ **Warning**

If a generic bar code data setup already exists, the **Create default setup** command deletes it and loads the standard setup.

Customize the standard generic bar code data setup

To customize the generic bar code data setup, follow these steps:

1. Go to **Warehouse management** > **Setup** > **GS1** > **Barcode data generic setup**.

2. Follow one of these steps:

    - To create a new mapping, on the Action Pane, select **New**.
    - To edit an existing mapping, select the mapping, and then, on the Action Pane, select **Edit**.

3. Set the following fields for the new or selected mapping:

    - **Field** – Select or enter the mobile app input field that the incoming value should be assigned to. The value isn't the display name that workers see. Instead, it's the key name that is assigned to the field in the underlying code. The default setup provides a collection of fields that are likely to be useful, and includes intuitive key names for each field and matching programmed functionality. However, you might need to talk to your development partners to find the correct selections for your implementation.
    - **Application identifier** – Select the applicable application identifier, as defined on the **Barcode data application identifiers** page. The identifier establishes how the bar code is interpreted and stored as a value for the named field. After you select an application identifier, the **Description** field shows the description of it.

Set up bar code data policies for mobile device menu items

The purpose of the GS1 standard is to enable workers to load several values when they scan a single bar code one time. To achieve this purpose, logistics managers must set up bar code data policies that tell the system how to interpret multivalue bar codes. Later, you can assign policies to mobile device menu items to control how a bar code is interpreted when workers scan it while they're using a specific menu item.

If no bar code data policy is assigned to a menu item, the system can capture only a single value. This value is applied to the mobile app input that is selected when the worker takes the scan, as

specified by the generic bar code data setup. If a bar code data policy is assigned to the menu item, the system still uses the generic bar code data setup to map the first bar code value to the selected field. However, it can then capture additional field values, as specified by the applicable policy.

Load the standard specific bar code data policies

To get started quickly, load a set of standard bar code data policies. You can later extend or edit the policies as needed.

To load the standard application identifiers, follow these steps:

1. Go to **Warehouse management** > **Setup** > **GS1** > **Barcode data policy**.
2. On the Action Pane, select **Create default setup**.

> ⚠ **Warning**
>
> The **Create default setup** command deletes all currently defined policies and replaces them with the standard set of policies. However, after the default setup is loaded, you can customize the policies as needed.

Set up custom specific bar code data policies

> ⚠ **Warning**
>
> Some bar code data policies might not work with every mobile flow that you use. When you configure custom bar code data policies, test the mobile device flow by using different pieces of information that are scanned at different points in the flow. In this way, you can determine whether the flow behaves as you expect.

To set up and customize your bar code data policies, follow these steps:

1. Go to **Warehouse management** > **Setup** > **GS1** > **Barcode data policy**.
2. Follow one of these steps:
   - To create a new policy: On the Action Pane, select **New**.
   - To edit an existing policy: Select the policy in the list pane.
3. On the header of the new or selected policy, set the following fields:

- **Policy name** – Enter a name for the policy.
- **Description** – Enter a short description of the policy.
- **Field value capturing method** – Select one of the following values to specify how the individual bar code values should be handled as part of the mobile device flow:
  - *Process immediately* – The application identifier values are immediately passed to the next mobile device step, even though the recorded values aren't required for capture as part of a later mobile device step. Depending on the recorded values, this value can cause unexpected mobile device flow processing.
  - *Save as default* – The application identifier values are stored throughout the whole mobile device menu item flow and are used only in mobile device steps that require the defined fields. Therefore, the exact mobile device step field control names must be specified as part of the bar code data policy setup, to link the application identifier values from the bar codes to the Warehouse Management mobile app fields. For more information about how to find the correct field names, see Inspect details of active Warehouse Management mobile app sessions.
- **Auto submit** – This option is available only when the **Field value capturing method** field is set to *Save as default*. Set it to *Yes* to automatically submit the mobile device step if all the fields are set.

4. On the **Policy lines** FastTab, map field names to application identifiers as required for the current policy. Use the buttons on the toolbar to add or remove rows as you require. For each row, set the following fields:
   - **Field** – Select or enter the mobile app input field that the incoming value should be assigned to. The value isn't the display name that workers see. Instead, it's the key name that is assigned to the field in the underlying code. The default setup provides a collection of fields that are likely to be useful, and includes intuitive key names for each field and matching programmed functionality. However, you might have to talk to your development partners to find the correct selections for your implementation. For more information about how to look up field names, see Inspect details of active Warehouse Management mobile app sessions.
   - **Application identifier** – Select the applicable application identifier, as defined on the **Barcode data application identifiers** page. The identifier establishes how the bar code is interpreted and stored as a value for the named field. After you select an application identifier, the **Description** field shows the description of it.

- **Sorting** – This column is available only when the **Field value capturing method** field is set to *Process immediately*. Every multivalue bar code includes a series of application identifiers, each of which is followed by a value. The applicable bar code data policy identifies which application identifier is mapped to each database field. However, if a bar code uses the same application identifier more than once, the system uses the order in which application identifiers appear in the code to map them to fields. For rows that share an application identifier with one or more other rows, use this field to establish the order that the matching rows should be processed in. The row that has the lowest sorting value is processed first.
- **Allow overwriting** – This column is available only on the **Policy lines** grid when **Field value capturing method** in the policy details header is set to *Save as default*. Select the checkbox in the **Policy lines** grid for rows where scanned data should replace the default data related to the specified **Application identifier** in the system. Only field values that are enabled for editing in the mobile application step are overwritten when this option is selected.

> ⓘ **Note**
>
> - For bar codes that include more than one identical application identifier, you *must* use the **Sorting** field to establish the order of the fields.
> - When **Auto Submit** is set to *Yes*, the mobile device immediately sends scanned information to the system. After the information is received, the mobile device displays a new screen, ready for the next input. This feature allows for a quick and efficient scanning process. The device only displays data after scanning when user interaction is required.
> - Select the **Allow overwriting** checkbox for rows where scanned information should replace the default data in the system. For example, if there's a purchase order for 10 pieces, but you only need to scan 5 pieces, this option allows the scanned data to override the default quantity.

Assign bar code data policies to mobile device menu items

By default, all mobile device menu items provide input fields where workers can scan a single value, according to the generic bar code data setup. If you want workers to be able to scan more

than one field value in a single scan for any mobile device menu item, assign a bar code data policy by following these steps.

1. Go to **Warehouse management** > **Setup** > **Mobile device** > **Mobile device menu items**.
2. Create or open a menu item.
3. On the **General** FastTab, set the **Barcode data policy** field to the policy that applies to the menu item.

GS1 setup example

This example applies to a system where the GS1 options are set up in the following way:

- On the **Warehouse management parameters** page, the following global settings are available:
  - **FNC1 character:** *]C1*
  - **Group separator:** ~
- On the **Barcode data application identifiers** page, the following application identifiers are relevant to this example.

Expand table

Application identifier	Description	Fixed length	Length	Type	Decimal
01	GTIN	Selected	14	Numeric	Cleared
10	Batch number	Cleared	20	Alphanumeric	Cleared
17	Expiry date	Selected	6	Date	Cleared
30	Receiving quantity	Cleared	8	Numeric	Cleared

- On the **Barcode data generic setup** page, the following settings for the generic bar code data policy are relevant to this example.

Expand table

Field	Application identifier	Description
ItemId	01	GTIN

- On the **Barcode data policy** page, there's a policy where the **Policy name** field is set to *Purchase receiving*. This policy includes the following lines.

Expand table

Field	Application identifier	Description	Sorting
ExpDate	17	Expiry date	0
InventBatchId	10	Batch number	0
Qty	30	Receiving quantity	0

- On the **Mobile device menu items** page, there's a menu item that is named *Purchase receiving*. Its **Barcode data policy** field is set to *Purchase receiving*.

After goods for a purchase order arrive at the warehouse, the worker follows these steps.

1. On the mobile device, select the **Purchase receiving** menu item.
2. Enter the purchase order number.
3. Select the **Item** field, and scan the following bar code: `]C10100000012345678~3030~10b1~17220215`.

Because of the settings that are established for this example, the system parses the scanned bar code in the following way.

Expand table

Field key	Application ID	Value	Note
ItemId	01	00000012345678	Because the worker scanned to the **Item** field, the first value in the bar code is mapped to that field. The mapping is taken from the generic bar code data setup.
Qty	30	30	Because several field values are being captured in a single scan, this mapping and all remaining mappings are taken from the bar code data policy that is assigned to the **Purchase receiving** menu item. This value is the quantity that was received.
InventBatchId	10	b1	This value is the batch ID.
ExpDate	17	220215	The date format is YYMMDD. Therefore, the expiry date is February 15, 2022.

The receipt is then registered, and the relevant database values are entered after the single scan.

> **Tip**

If GS1 bar codes that contain multiple values don't work as part of a Warehouse management mobile app flow, check the following settings:

- On the **Barcode data generic setup** and **Barcode data application identifiers** pages, make sure that the settings are aligned with the bar codes that you're using and with the bar code data policy that's assigned on the **Mobile device menu item** page.
- On the **Barcode data policy** page, select the policy that you're having trouble with, and check the following settings:
  - Set the **Field value capturing method** field to *Save as default*.
  - In the grid, make sure that all the field names are aligned with the step ID input control names for the actual mobile device. For more information about how to look up field names, see **Inspect details of active Warehouse Management mobile app sessions**. For some typical examples, see the table after this list.
- On **Warehouse management parameters**, on the **General** tab, on the **Bar codes** FastTab, set the **Unknown application identifier policy** field to *Skip the data element*.

The following table shows some typical examples where the step ID field mapping fails, so that a scanned GS1 bar code can't be processed.

Expand table

Used field name	Correct field name	Process
ItemId	ProductConfirmation	Item confirmation step
CatchWeight	OutboundWeight	Catch weight picking step

Last updated on 06/12/2026

Set up warehouses for transfer orders

Use warehouse levels to create a hierarchy that supports transfer orders between warehouses. With this setup, master scheduling calculates item requirements at the individual warehouse level and generates planned transfer orders from an assigned source warehouse to fulfill them.

1. Select **Inventory management** > **Setup** > **Inventory breakdown** > **Warehouses**.
2. Select the warehouse that you want to refill.
3. On the **Master planning** FastTab, select the **Refilling** check box.
4. In the **Main warehouse** field, select the warehouse that you want to assign as the refilling warehouse. Master scheduling calculates a transfer requirement for the selected warehouse and generates a planned transfer order from the assigned **Main warehouse**.

   > **Note**
   >
   > If you clear the **Refilling** check box, the selected warehouse is assigned a warehouse level in regard to the **Main warehouse**, but the **Main warehouse** isn't set up as a refilling warehouse.

5. Close the page to apply the new setup.

> **Tip**
>
> To assign a warehouse for refilling, first set up the warehouse as a storage dimension on the **Storage dimension groups** page. On this page, select the **Active** field and the **Coverage plan by dimension** field for the warehouse.

Set up transport lead time

Set up the transport lead time between the warehouses on the **Transport days** page.

1. Go to **Inventory management** > **Setup** > **Distribution** > **Transport days**.
2. In the **Receiving point** field, select **Warehouse**.
3. Select the **Shipping warehouse**, **Receiving warehouse**, and **Transport days**.

4. (Optional) Under the **Transport days per mode of delivery** tab, set the transport time depending on the mode of delivery.

Last updated on 07/01/2026

View and manage the work exceptions log

Work exceptions are work-related errors that can occur during warehouse operations (such as discrepancies in inventory or missing goods at a given location). The system registers these exceptions in the work exceptions log, which lets managers track and diagnose issues related to warehouse workflows such as pick or pack procedures.

Example of how a worker can create a work exception

The Warehouse Management mobile app automatically creates exceptions during some types some workflows in response to certain events. For example, one possible way that a worker can trigger a work exception is by completing the following procedure to perform a short pick (assuming the required work record already exists):

1. Open the Warehouse Management mobile app.
2. Go to **Outbound** > **Sales Picking**.
3. Enter the **Work ID** and select **OK**.
4. Enter the **Location** and select **OK**.
5. Select **Short Pick**.
6. Specify the following values:

    - The target license plate.
    - The quantity picked.
    - The reason for the short pick (for example, the item wasn't available at the expected location).

7. Select **OK** and complete the work.

The short pick event generates a work exception for the specified **Location**.

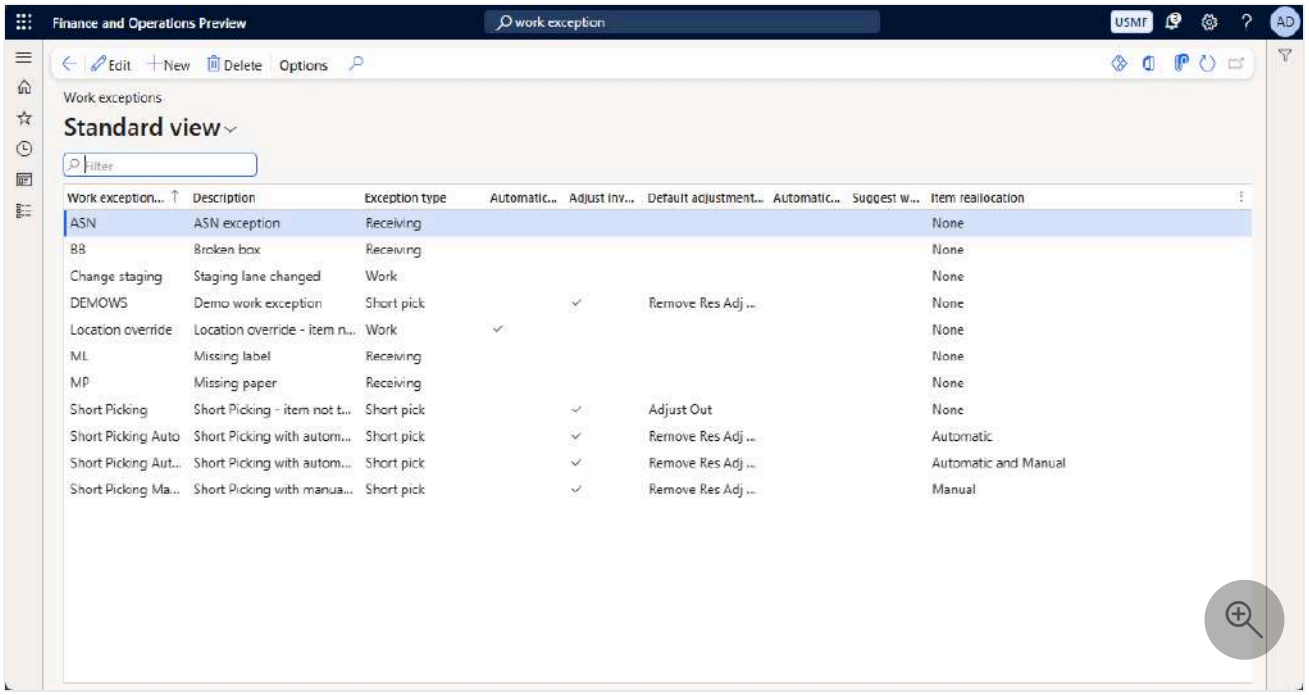
Configure behaviors for handling each type of work exception

You can configure the way the system should handle each relevant type of work exception. For example, you could set the system to adjust inventory at an affected license plate or allow automatic item reallocation.

To configure system behaviors, follow these steps:

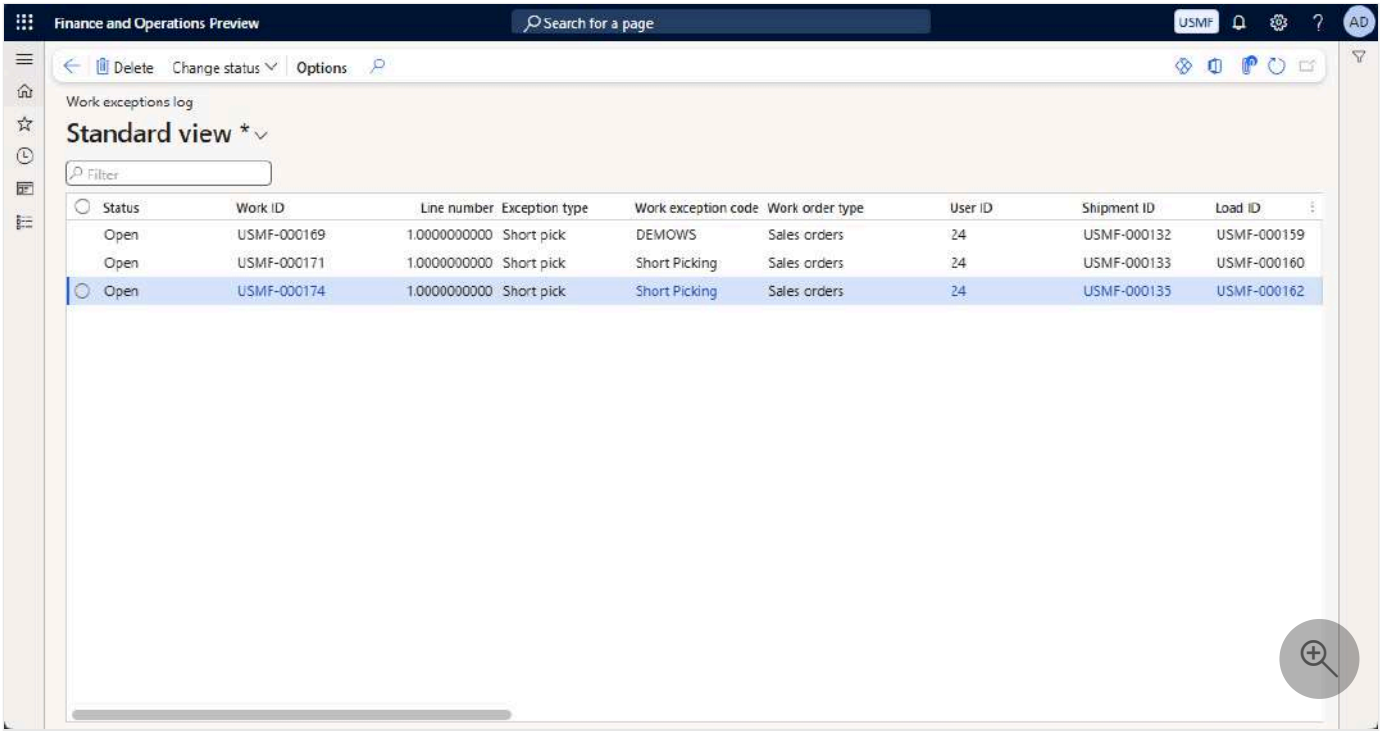
1. Go to **Warehouse management** > **Setup** > **Work** > **Work exceptions**.

2. Use the buttons on the Action Pane to add, edit, and delete rows as needed. To learn how to use the settings in each column, hover your mouse pointer over the column header to see a tooltip.

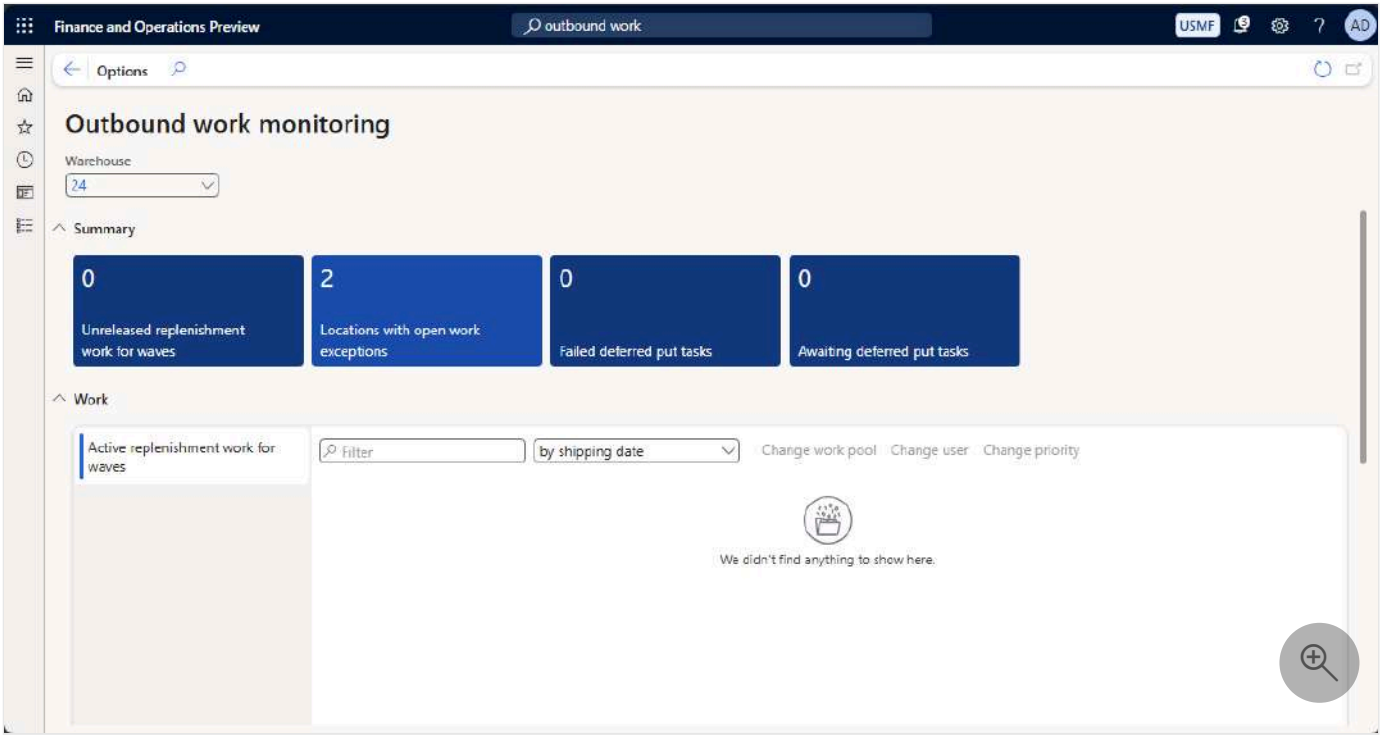


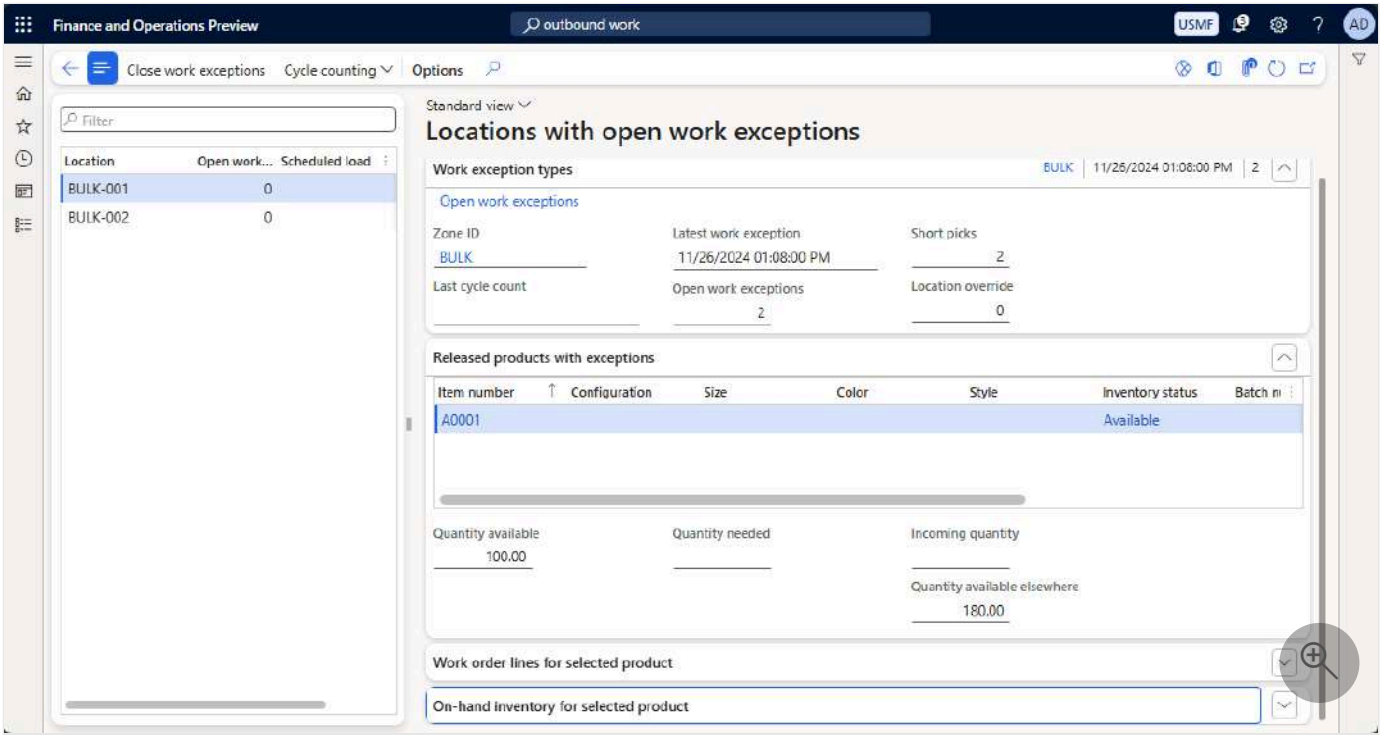
View the work exceptions log

To view the work exceptions log, go to **Warehouse management** > **Work** > **Work exceptions log**. You can sort and filter the list using the column headers or by entering a value in the **Filter** field and selecting where to search for it (for example by **Status**).



Work exceptions can also be shown on other pages, such as the **Outbound work monitoring** workspace (available at **Warehouse management** > **Workspaces** > **Outbound work monitoring**), which provides a tile that shows the number of **Locations with open work exceptions**. Select that tile to view and explore full details about the exceptions at each location.





Clean up the work exceptions log

Even after they're resolved, the system keeps all log entries until they're explicitly removed. Cleaning out old work exceptions helps make it easier for users to search for locations with open

work exceptions and improves the performance of pages that show work exceptions.

The system provides a clean-up batch job to help you delete multiple entries in the work exceptions log once they're resolved or no longer necessary. When you set up the job, you set the criteria for selecting which entries should be deleted (for example, according to the status and/or age of each entry). When the job runs, it removes all work exceptions that match the criteria.

This feature requires Supply Chain Management version 10.0.43 or later.

To clean up the work exceptions log, follow these steps:

1. Go to **Warehouse management** > **Periodic tasks** > **Clean up** > **Clean up work exceptions logs**.
2. In the dialog, expand the **Parameters** FastTab and make the following settings:

    - **The number of days to keep** – Specify the age (in days) of the oldest entries to keep. Entries older than this are removed.
    - **Status** – Select the status of the exception logs to delete (*Open* or *Closed*).
    - **Maximum cleanup records count** – Specify the maximum number of records to delete. Setting a limit here can improve system performance by preventing too many records from being deleted in a single operation. (Default is 100,000.)

3. On the **Run in the background** FastTab, set up batch, batch group, scheduling, and alert options as you require, just as you might do for other batch jobs in Supply Chain Management.

When the job finishes execution, the system shows a notification of how many records were removed.

Last updated on 07/01/2026

User-configurable queries in warehouse management

Article • 03/10/2025

The purpose of user-configurable queries is to limit the data that is processed. You might want to limit this data either because of business requirements or for performance reasons. The exact behavior of a query and the way that the results are used depend on the functional areas where that query is used. However, most user-configurable queries have the same basic structure and behavior. The logic resembles the logic of regular SQL queries.

Many places in the **Warehouse management** module must be set up for user-configurable queries. Here are some typical examples:

- The **Edit query** button on the **Work templates** page
- The **Location directives** page
- The **Select products** button on the **Replenishment templates** page

Here are some more examples of pages:

- SysQueryForm
- Wave template
- Work templates
- Location directives
- Labor standards
- Document routing
- Wave label templates
- Wave label layouts
- Label layout data source
- Container label routing
- Wave filters
- Mobile device menu items
- Cluster profiles
- Wave load building templates
- Cross docking templates
- Container build templates
- Replenishment templates
- Slotting templates
- Outbound sorting template
- Cycle count plans

- Cycle counting threshold
- Shipment consolidation policies
- Shipment consolidation templates

Here are some more examples of buttons:

- Edit query
- Select items
- Select locations
- Select products
- Select product variants
- Select locations to replenish
- Select zones to replenish
- Define product query
- Define product variant query

> **ⓘ Important**
>
> The most common mistake that users make in queries is to add table joins and expect the functionality to apply only to the added records. For example, the user adds a join to a sales table in the *Location directive action* query. The *Location directive action* query applies only when sales orders that meet the user-defined criteria are processed.
>
> Filters are added only to the tables that are in the default query. If the user adds a join to sales orders, the record doesn't add a filter to the sales order that is being released.

Basic query structure

Ranges

Ranges are filters that are used to limit which records are returned from the query. They are explicitly applied to individual table fields. In addition, they are implicitly applied when join relations are used.

Joins

Joins are applied between two tables to define how the data between them is filtered and potentially combined. There can be multiple joins to the same table.

The queries in X++ support several types of joins: *InnerJoin*, *OuterJoin*, *ExistsJoin*, and *NoExistsJoin* (*NotExist*). However, only *ExistsJoin* and *NoExistsJoin* are available in user-configurable queries. *ExistsJoin* is the default join type, regardless of the join mode (for example, *1:n* or *n:1*).

> **Note**
>
> In some special cases (for example, on the **Label layout data source** page), dedicated functionality was developed to support inner or outer joins in user-configurable queries, but those joins aren't available. If the default query has those types of joins, the user can add ranges to them. New table joins that are added through the user interface (UI) can be joins of the *ExistsJoin* or *NoExistsJoin* type only. Users can't view which type of join is used in the default query.

- *InnerJoin* returns the combined records from both joined tables. Therefore, it's a subset of the two table records. The join is made based on the specified relations between the two tables.
- *OuterJoin* returns the records from the first table, but any records that have a match in the other table will have those fields populated. The join is made based on the specified relations between the two tables.
- *ExistsJoin* resembles *InnerJoin*, but no record information from the second table is returned in the result.
- *NoExistsJoin* resembles *ExistsJoin* in that only record information from the first table is returned. However, the filtering is the opposite. Therefore, the record is returned if the match does **not** exist.

Sorting

Sorting is applied after the results are returned from the ranges and joins. It's applicable only to the returned records and fields. Sorting can't be applied to the second table in a join of the *ExistsJoin* type.

> **Important**
>
> The addition of further ranges, sorting, and especially table joins negatively affects performance. In some cases, ranges and joins must be added to meet business requirements. However, users should not create queries that are too complex.

Examples

Work templates page

On the **Work templates** page, each work order type has a slightly different default query. The purpose of the query is to select the work template that is used to create warehouse work and remaining work lines from the initial temporary work transactions during wave processing. The temporary work transactions are initialized from the order lines and headers.

In the case of sales orders, for example, the top table is `Temporary work transactions`. Two tables are joined to it, `Order lines` and `Locations`. You can't view the join type from the UI. You can view it only from the development environment. In this case, both joins are inner joins. The `Locations` table has no further joins. However, the `Order lines` table has one inner join to the `Sales orders` table and another inner join to the `Items` table. The `Sales orders` table has no further joins. However, the `Items` table has an inner join to the `Warehouse item number` table. The `Warehouse item number` table has no further joins. No default ranges are added to the query.

Users can add ranges to any of the previously mentioned tables. Only records that meet all the conditions can use this work template during wave processing.

For example, the user selects the `Warehouse item number` table and sets the **Field** field to *Code 1* and the **Criteria** field to *A*. In this case, only items where product filter code *Code 1* is set to *A* can use this work template.

Next, the user adds a relation to the `Carrier (Shipping carrier)` table to the `Temporary work transactions` table. The join is an *ExistsJoin* join. The user can now add a range on that table. For example, the user adds a range on the `Carrier` table and the **Shipping carrier** field, and specifies the name of some carrier in the **Criteria** field. In this case, the work template applies only to temporary work transactions where the specified carrier name is set.

> **Note**
>
> Although many tables are listed as available for joins, the related fields aren't populated in all cases. This behavior might lead to unexpected results.

Location directives page

On the **Location directives** page, there are two different user-configurable queries. One is related to the location directive (header), and the other is related to the location directive actions. The action-related query has different default values, depending on

the work type or the setting of the **Batch enabled** checkbox. The purpose of this query is to find a suitable warehouse location for the work line to pick from or put to.

If you use a sales order pick action where the **Batch enabled** checkbox is cleared, the top table is `Locations`. One table is joined to it, `On-hand inventory`. The join is an inner join. Ranges are predefined. The **Warehouse**, **Physical inventory**, and **No open quantities** fields have values set in the **Criteria** field and aren't editable. The **Location** and **Item number** fields are present, but they have no value in the **Criteria** field and are editable.

> **Note**
>
> In older versions, there might be a third table (`Inventory dimensions`) between `Locations` and `On-hand inventory`.

The user can specify the desired **Location profile ID** ranges on the `Locations` table. In this case, only locations that match the criteria are evaluated when the query tries to find a valid location to pick the item from.

The user might also add sorting, such a sort on the **Location** field of the `Locations` table in the *Ascending* search direction. In this case, the locations are evaluated based on the order of their location IDs. For example, the location that has the ID *Location123* is evaluated before the location that has the ID *Location321*.

> **Note**
>
> In some cases, additional ranges or tables are applied during processing but aren't visible to the user. In the preceding example, the item is always applied for the pick location directive action, because on-hand inventory is picked for a specific item.

Next, the user adds a relation to the `Order lines (Item number)` table to the `On-hand inventory` table. The join is an *ExistsJoin* join. The user can now add a range on that table. For example, the user adds a range on the `Order lines` table and the **Customer** field, and specifies some customer account in the **Criteria** field.

> **Important**
>
> You might think that the preceding query applies only when sales orders are processed for the specified customer account. However, the query actually has the following meaning: *find a location that has an item for which there is a sales line for*

the specified customer account. If the item was ever added to a sales line for that customer account, the additional table join and range have no functional effect. However, they add performance overhead.

Feedback

Was this page helpful? Yes No

Provide product feedback

Data entity support for user-configurable queries in warehouse management

Article • 03/07/2025

One challenge of user-configurable queries is the interaction with data entities. Because the queries are packed in a way that isn't readable by users, it's difficult to make changes to them through the data entities. To address this issue, special JavaScript Object Notation (JSON) fields are added to the warehouse data entities that have user-configurable queries. All relevant warehouse entities are updated with support for these JSON fields.

Example

This article uses the example of a newly created *Location directive actions* query.

Set up a location directive action

To set up a location directive action, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Location directives**.
2. In the **Work order type** field, select *Sales orders*.
3. On the Action Pane, select **New** to create a location directive.
4. In the **Name** field, enter *JSON test*.
5. In the **Work type** field, select *Pick*.
6. Select **Save**.
7. On the **Lines** FastTab, select **New**.
8. In the **To quantity** field, enter *999*.
9. Select **Save**.
10. On the **Location Directive Actions** FastTab, select **New**.
11. In the **Name** field, enter *JSON test action*.
12. Select **Save**.
13. Set up the data entities. Learn more in Data entities overview.
14. Export the new location directive action with the *Warehouse location directive line actions V3* data entity, for example, in the *XML-Element* format.

Review the exported file

If you open the exported XML file, its content should resemble the following example.

XML

```
<?xml version="1.0" encoding="utf-8"?>
<Document>
    <WHSWAREHOUSELOCATIONDIRECTIVELINEACTIONV3ENTITY>
        <ACTIONNAME>JSON test action</ACTIONNAME>
        <WAREHOUSELOCATIONDIRECTIVEID>JSON
test</WAREHOUSELOCATIONDIRECTIVEID>

<WAREHOUSELOCATIONDIRECTIVEWORKTYPE>Pick</WAREHOUSELOCATIONDIRECTIVEWORKTYPE
>

<WAREHOUSELOCATIONDIRECTIVEWORKORDERTYPE>Sales</WAREHOUSELOCATIONDIRECTIVEWO
RKORDERTYPE>
        <WAREHOUSELOCATIONDIRECTIVEINVENTORYSITEID/>
        <WAREHOUSELOCATIONDIRECTIVEWAREHOUSEID/>

<WAREHOUSELOCATIONDIRECTIVELINESEQUENCENUMBER>1</WAREHOUSELOCATIONDIRECTIVEL
INESEQUENCENUMBER>
        <ACTIONQUERYJSON><!
[CDATA[{"table":"WMSLocation","name":"WMSLocation","firstOnly":false,"fetchM
ode":1,"type":"InnerJoin","fields":
["inventLocationId","LocProfileId","wMSLocationId"],"links":[],"ranges":
[{"field":"wMSLocationId","rangeStatus":"Open"},
{"field":"inventLocationId","rangeStatus":"Locked"}],"joins":
[{"table":"InventSum","name":"InventSum","firstOnly":false,"fetchMode":0,"ty
pe":"InnerJoin","fields":
["configId","InventBatchId","InventColorId","InventDimension1","InventDimens
ion10","InventDimension11","InventDimension12","InventDimension2","InventDim
ension3","InventDimension4","InventDimension5","InventDimension6","InventDim
ension7","InventDimension8","InventDimension9","InventDimId","InventGtdId_RU
","InventLocationId","InventOwnerId_RU","InventProfileId_RU","InventSerialId
","InventSiteId","InventSizeId","InventStatusId","InventStyleId","InventVers
ionId","ItemId","LicensePlateId","PhysicalInvent","wMSLocationId","wMSPallet
Id"],"links":
[{"field":"inventLocationId","relatedField":"InventLocationId","table":"WMSL
ocation","relatedTable":"InventSum","joinRelation":""},
{"field":"wMSLocationId","relatedField":"wMSLocationId","table":"WMSLocation
","relatedTable":"InventSum","joinRelation":""}],"ranges":
[{"field":"PhysicalInvent","value":">0","rangeStatus":"Locked"},
{"field":"ItemId","rangeStatus":"Open"},
{"field":"ClosedQty","value":"No","rangeStatus":"Locked"}]}]}]]>
</ACTIONQUERYJSON>
        <ACTIONSEQUENCENUMBER>1</ACTIONSEQUENCENUMBER>
        <ACTIONSTRATEGY>None</ACTIONSTRATEGY>
        <FIXEDLOCATIONUSAGEMETHOD>Any</FIXEDLOCATIONUSAGEMETHOD>
        <ISITEMBATCHENABLED>No</ISITEMBATCHENABLED>
        <ISNEGATIVEINVENTORYALLOWED>No</ISNEGATIVEINVENTORYALLOWED>
    </WHSWAREHOUSELOCATIONDIRECTIVELINEACTIONV3ENTITY>
</Document>
```

Notice the `ACTIONQUERYJSON` field. This new JSON field contains all the tables, relations, sortings, and ranges in a user-readable format. Learn more in User-configurable queries in warehouse management.

Feedback

Was this page helpful? Yes No

Provide product feedback

Message processor messages for warehouse management processes

07/29/2025

Message processor messages are used for processes that run asynchronously. For example, you can use a message processor message to post a sales packing slip when the last shipment container is closed as part of a manual packing process or by Warehouse management only mode to create *inbound shipment orders* and *outbound shipment orders*.

Example message type: Run packing slip for container

For example, the *Run packing slip for container* message type is used to create and post a sales packing slip when a warehouse worker closes the last container in a shipment. The message that's inserted will be in a *Queued* message state. This state indicates that the message is waiting to be picked up by the **Message processor** batch job.

View the message log, message content, and details

To view the messages that are processed by the message processor, go to **System administration** > **Message processor** > **Message processor messages**. For information about how to use the **Message processor messages** page to review messages, find and fix failed messages, and more, see Message processor messages page.

Message processor batch job

You should schedule a periodic batch job to process the messages, by going to **System administration** > **Message processor** > **Message processor**. In the dialog box that appears, you must select a value in the **Message queue** field (for example, *Warehouse*), and you must specify a processing interval. Learn more in Schedule message processing by using the message processor batch job.

Set up business events to deliver alerts for failed processing results

On the **Message processor messages** page, you can filter on the *Failed* value of the **Message state** field to inquire about failed messages. In addition, you can set up business events to alert you about failed processing results. Learn more in Set up business events to deliver alerts for failed processing results.

Clean up processed and canceled message processor messages

Over time, processed and canceled messages, which are no longer needed, can accumulate in the system, potentially affecting performance and data management. Therefore, you should periodically clean up these messages to maintain system efficiency. Learn more in Clean up processed and canceled message processor messages.

Warehouse-specific inventory transactions

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Warehouse-specific inventory transactions (referred to as *warehouse transactions* in this article) are database records that store information about how inventory is stored and moved around in the warehouse. Warehouse transaction records are highly optimized for warehouse operations.

In versions of Microsoft Dynamics 365 Supply Chain Management before 10.0.32, the system used a single type of standard inventory transactions (referred to as *inventory transactions* in this article) for both warehouse operations and other types of inventory transactions. However, in newer systems, use warehouse transactions where they apply (though inventory transactions still support other functionality). For warehouse operations, warehouse transactions provide many benefits in comparison to inventory transactions. Here are some examples:

- Significantly improved warehouse operation performance, especially for scenarios that include large numbers of serial or batch numbers (tracking dimensions).
- Improved scalability that's made possible by a reduction in the number of inventory transaction records that are required to support warehouse operations. Standard inventory transactions generate many more records and can therefore cause a heavy load on the database.

How warehouse transactions compare to inventory transactions

Warehouse transactions differ from inventory transactions in several ways. Here are the key differences:

- Warehouse transactions can operate on items in bulk. In other words, a single warehouse transaction can reserve, issue, or receive a set of items that have different serial or batch numbers. This capability enables the system to avoid repeated validations and use efficient set-based operations. It also reduces the amount of data that the system generates. Because the item sets can be reused, the same information doesn't have to be recorded multiple times.

- Warehouse transactions don't hold any financial or cost information. They're used only in warehouse operations, which don't have any financial impact. This change significantly reduces the number of fields, indexes, and validations that are required in the database. For warehouse operations that have a financial impact (such as counting, adjust-in, or adjust-out), the system still relies on inventory transactions.
- Warehouse transactions are immutable. Therefore, except in a few system processes, they're never updated or deleted. This property allows for more predictable system performance, because it avoids splitting the transactions. It also allows for a more detailed audit log of what occurred in the system. The extra details can be useful during investigations.
- Warehouse transactions are frequently archived. This feature helps keep the size of the warehouse transactions table small. It also enables the system to implement different data models for active and completed transactions. The data model for active transaction is optimized for inserts. (This optimization is critical for fast mobile device operations.) By contrast, the data model for completed transactions is optimized for queries.

Prerequisites

Before you can use this feature, the following prerequisites must be in place:

- You must update any customizations that rely on inventory transactions that will be replaced by warehouse transactions. Learn more in the Audit and update your customizations to support warehouse transactions section of this article.
- You must be running Dynamics 365 Supply Chain Management version 10.0.32 or later.
- If you're running Supply Chain Management version 10.0.35 or earlier, the feature that's named *Warehouse-specific inventory transactions* must be turned on in Feature management. In newer versions, this feature is fully integrated into the product, so it's either mandatory or no longer listed in Feature management.

Audit and update your customizations to support warehouse transactions

When you enable the *Warehouse-specific inventory transactions* feature, your system changes as follows:

- The system stops generating inventory transactions for warehouse operations. Instead, it uses warehouse transactions. Therefore, you must update any customizations and extensions that rely on the relevant inventory transactions for warehouse operations. This change affects only those warehouse scenarios that you configure to use warehouse

transactions. Learn more in the Choose which scenarios should use warehouse transactions section of this article.
- The system stops calling several existing methods when it processes warehouse operations.

When you try to enable the feature, the system checks for extensions that might break because of these changes. If it finds any issues, it sends you an error message that identifies the extension and explains how you must update it. If you receive such a message when you try to enable the feature, follow these steps.

1. Contact your developer or Microsoft partner, and ask them to review the relevant extensions and make the required fixes.
2. Declare the module that contains the problematic extension as *validated*. You can make this change by extending the `buildValidatedModuleSet()` method in the `WHSWarehouseInventoryTransactionFeatureExtensionValidator` class and adding the name of the module that contains a flagged extension into the set.
3. Deploy the updated module. The feature validator should now allow you to activate the feature.

> **⊗ Caution**
>
> The feature validator can't detect all possible issues. For example, it can't detect whether an extension is reading inventory transactions that are related to warehouse work. (Those inventory transactions will no longer exist after you enable the feature.)

Choose which scenarios should use warehouse transactions

The system lets you choose which scenarios use warehouse transactions. For example, you can use the new warehouse transactions for warehouse movements, but continue to use inventory transactions for all other operations. Therefore, organizations that use a system that was customized before enabling the *Warehouse-specific inventory transactions* feature can gradually move toward using warehouse transactions. They can start with the least customized scenarios or the scenarios where performance improvements are most required.

To choose which scenarios use warehouse transactions, follow these steps.

1. Go to **Warehouse management** > **Setup** > **Warehouse management parameters**.
2. On the **General** tab, on the **Warehouse inventory transaction mechanism** FastTab, select the **Use warehouse inventory transactions** checkbox for each scenario where you want to

use warehouse transactions. Scenarios that you leave this checkbox cleared for will continue to use inventory transactions.
3. On the Action Pane, select **Save**.

Some warehouse work types are just wrappers around other system processes (such as adjust-in/adjust-out and counting operations) and continue to use inventory transactions. Those operations change the on-hand inventory on all the inventory dimensions and therefore have a financial impact.

Source document transactions still use inventory transactions (records in the `InventTrans` table) and continue to use the inventory stack. For example, in the sales picking scenario, the reservation that's related to the sales line is still represented by an inventory transaction. However, when you release a sales order to the warehouse, the work that's created uses warehouse transactions. When you complete the work, the sales line's inventory transaction is updated to *Picked* status, and all specific dimension values are assigned, just as if the work had used inventory transactions.

Review warehouse transactions

The **Warehouse transactions** page shows both warehouse transactions and inventory transactions. By contrast, the **Inventory transactions** page shows only inventory transactions. Therefore, only the **Warehouse transactions** page shows all the transactions that are related to your warehouse operations. For example, in a sales picking scenario, if you want to view all transactions that are related to the sales order, you must open the **Warehouse transactions** page. It shows both the sales order inventory transactions and work-related warehouse transactions. The **Inventory transactions** page shows only the sales order transactions.

The purpose of the **Warehouse transactions** page is to show all the transactions that are relevant to managing the warehouse. It doesn't show inventory transactions that are in *Ordered* or *On order* status. You can open it from several places in the system, including the **On-hand** pages and the **Warehouse inventory transactions** page.

For example, to open the **Warehouse transactions** page from the **On-hand** page for a selected product, follow these steps.

1. Go to **Product information management** > **Products** > **Released products**.
2. Select a released product.
3. On the Action Pane, on the **Manage inventory** tab, in the **View** group, select **On-hand inventory**.

> **Note**
>
> To view *all* transactions for the selected product, you can select **Warehouse transactions** instead.

4. On the **On-hand** page, select the row for the on-hand inventory that you want to inspect.
5. On the Action Pane, select **Warehouse transactions**.

The **Warehouse transactions** and **Inventory transactions** pages provide similar functionality. However, the **Warehouse transactions** page is limited to functionality that applies to warehouse operations. It lets you perform the following actions:

- To choose which inventory dimensions to show, select **Display dimensions** on the Action Pane.
- To view all transactions that have the same inventory dimensions as a selected row, select **Warehouse transactions for dimensions** on the **Inventory** tab of the Action Pane.
- To trace inventory dimensions for a selected row, select **Trace** on the **Inventory** tab of the Action Pane to open the **Trace inventory dimensions** page.
- To view batch attribute details for a selected row, select **Inventory batch attributes** on the **Inventory** tab of the Action Pane.

Archive warehouse transactions

Because warehouse transactions and inventory transactions differ slightly in nature, the system can archive warehouse transactions more often than inventory transactions. Whenever warehouse work is completed (closed or canceled), the system archives all the related warehouse transactions. The archived warehouse transactions are still available through the user interface (UI), just as nonarchived transactions are. However, they're moved from the storage that's optimized for fast inserts (active transactions) to the storage that's optimized for queries (archived transactions). The system can detect this distinction, and it takes the archived warehouse transactions into account when, for example, on-hand recalculation consistency checks are done or the **Trace inventory dimensions** page is used.

The archival procedure is implemented by the *Archive warehouse inventory transactions* process automation background process. This process is automatically registered when you enable the *Warehouse-specific inventory transactions* feature. By default, it runs every 10 minutes. However, system administrators can change the recurrence properties, based on the actual system use.

To change the recurrence properties of the *Archive warehouse inventory transactions* process, follow these steps.

1. Go to **System Administration** > **Setup** > **Process Automations**.
2. On the **Background Processes** tab, in the grid, select the row where the **Name** field is set to *Archive warehouse inventory transactions*.
3. Select **Edit** on the toolbar.
4. The **Edit background process** dialog opens. Review and edit the configuration as needed and select **OK**.

To review the execution history of the *Archive warehouse inventory transactions* process, follow these steps.

1. Go to **System Administration** > **Setup** > **Process Automations**.
2. On the **Background Processes** tab, in the grid, select the row where the **Name** field is set to *Archive warehouse inventory transactions*.
3. Select **View most recent results** on the toolbar.
4. The **Execution results** dialog box that appears shows a list of each process execution. Here, you can see whether each process succeeded and view its execution log.

Frequently asked questions

This section provides answers to a few frequently asked questions about warehouse-specific inventory transactions.

Is Microsoft planning to decouple on-hand inventory from the InventTrans and WHSInventReserve tables?

There are no current plans to decouple on-hand inventory from these tables.

What do the new transaction types represent and how are they used in the WHSInventoryTransactionTable table?

The `WHSInventoryTransactionTable` table includes the following four types of transactions (as defined by `WHSInventoryTransactionTypeEnum`).

- **Registered issue** – Represents an item being physically issued.
- **Registered receipt** – Represents an item being physically received.
- **Physical reservation** – Represents item reservations.
- **Removed physical reservation** – Represents an explicit, separate transaction type for removing a reservation. All reservation-related transactions can be dropped during the

archival process to save space, depending on your settings.

How does ItemSetId differ from a license plate, and is there any guidance on how to proceed with both going forward?

`InventTrans`-based inventory wasn't designed to represent inventory operations (such as *issue*, *receipt*, and *reservation*) over a set of items. License plates (such as target license plates in warehouse work) are one example of such a *set of items*. From the perspective of the `InventTrans`-based inventory stack, a license plate is just a dimension similar to other storage dimensions in `InventDim`. The concept of item sets in the new inventory stack aims to address this gap. During initial picks, an item set is constructed, and subsequent operations (such as *put to stage*, *pick from stage*, and *put-away to bay door*) reuse that item set. This approach avoids recording redundant information about items being moved. However, license plates are still used in the new system.

Related information

- For more information about this feature, consider joining the Warehouse Inventory Transactions community on Microsoft Viva Engage, where you can find useful guides, get our latest updates, and post any questions you have about using warehouse transactions.

Last updated on 11/19/2025

Optimize location directive queries

Article • 05/26/2023

Because of recent improvements to the on-hand inventory data model, location directive queries can now run much faster than before. However, if you've been running Microsoft Dynamics 365 Supply Chain Management for a while, you might have many older location directives that still use queries that were designed for the old data model. The *Optimize location directive queries* tool identifies location directive queries that use a suboptimal query pattern and transforms them so that they use the optimal querying approach that the new data model enables.

The optimization advisor now includes a rule that periodically checks whether your queries can be optimized. If it identifies the potential for improvement, it generates an optimization advisor opportunity that highlights the changes and suggests that you to run the tool as described in this article.

> **Important**
>
> - We recommend that you first run the tool on a user acceptance testing (UAT) environment. You should run it on your production environment only after you're sure that all location directives still behave as expected in the UAT environment, and that all **location directive acceptance tests** still pass.
> - Avoid running the tool during peak hours. It will lock your location directives and flush them from all application object servers. Therefore, waving and work creation will run slower for the duration of the job.

Prerequisites

To use the *Optimize location directive queries* tool, you must be running Supply Chain Management 10.0.35 or later.

Run the tool

Follow these steps to run the *Optimize location directive queries* tool.

1. Go to **Warehouse management** > **Periodic tasks** > **Clean up** > **Optimize location directive queries**.

2. The **Optimize location directive queries** dialog box appears. If you want to limit the scope of the optimization, on the **Records to include** FastTab, select **Filter** to open a standard query editor dialog box, where you can define selection criteria. The fields work just as they do for other types of queries in Supply Chain Management.

3. On the **Run in the background** FastTab, specify how, when, and how often the tool should run. The fields work just as they do for other types of background jobs in Supply Chain Management.

    - We recommend that you run the tool only as it's needed instead of setting up a recurring schedule.
    - As was mentioned, we also recommend that you run the tool only during off-peak hours. Therefore, instead of running it immediately, consider setting it up as a batch job that's scheduled to run just once at an appropriate time (such as Sunday at 8 PM).

4. Select **OK** to run the tool by using the options that you selected.

How it works

Location directives for picks typically rely on on-hand inventory information that's available in the `InventSum` table to determine where items should be picked from. Because location directive queries usually include inventory dimensions to ensure that product variants are considered when this determination is made, these queries previously required a join between the `InventSum` and `InventDim` tables. However, the data model has now been denormalized to optimize this common query pattern. All the required information is now available directly in the `InventSum` table. The elimination of a join operation typically boosts the performance of location directive queries by 30 to 50 percent.

The *Optimize location directive queries* tool transfers all ranges from `InventDim` to `InventSum` and removes the join to `InventDim`.

Feedback

Was this page helpful? Yes No

Provide product feedback

Analyze warehouse material movement through process mining

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The Microsoft Power Automate Process Mining feature offers *task mining* and *process mining* capabilities to help you better understand your business processes, so that you can optimize them. Dynamics 365 Supply Chain Management offers a *Warehouse material movement analysis* template for the Power Automate Process Mining. Use this template to create processes that help warehouse and operations managers gain insights into the material flow in the warehouse. It helps find inefficiencies that you can remove to improve the performance of the warehouse.

The *Warehouse material movement analysis* template uses warehousing, product, and transactional data that's stored in your system. It lets you visualize and analyze material movements on the warehouse floor, based on closed warehouse work records. In process mining terminology, each warehouse work record corresponds to a *case*, and each pick or put work line corresponds to an *event*. Depending on your configuration, an *activity* can be any one of the following items:

- Warehouse location
- Warehouse location profile
- Warehouse zone

For more information about Power Automate Process Mining, see Overview of Power Automate Process Mining.

This article describes what's included in the *Warehouse material movement analysis* template, and explains how to deploy and access Power Automate Process Mining processes.

Prerequisites

The *Warehouse material movement analysis* template for Power Automate Process Mining has the following prerequisites:

- You're running Supply Chain Management version 10.0.35 or later.
- Your Supply Chain Management environment must be linked to a Dataverse environment.
- The user who deploys the process in Power Automate Process Mining must have the *Warehouse manager*, *Warehouse planner*, and/or *System administrator* role.
- The Supply Chain Manager user who deploys the process in Power Automate Process Mining must also be configured as a user in the linked Dataverse environment. To deploy the process, a user must have the *Environment maker* and/or *System administrator* role. After the process is deployed, it can be shared with other users. Those users automatically get the permissions required to use the process.

Deploy a new Power Automate Process Mining process based on the template

Set up as many processes as you need, each with a different name and filters. Follow these steps to deploy a *Warehouse material movement analysis* process.

1. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**.

2. On the Action Pane, select **Deploy process** to open the deployment wizard.

3. Follow the on-screen instructions. Provide the following information:

    - **Process name** – Enter a name for the process. Include the legal entity (company) and other filter information in the name, so you can easily identify the process later.
    - **Company** – Select the legal entity that you want to use this process to analyze. Only warehouse work records in the selected legal entity are included in Power Automate Process Mining process.
    - **Number of months to load** – Specify the number of months to load at a time. This filter helps prevent excessive or irrelevant data from being loaded into Power Automate Process Mining. Only warehouse work records that were closed after the specified time threshold are included in Power Automate Process Mining process.
    - **Activity** – Specify which column Power Automate Process Mining should treat as an *activity*. To learn more about process mining terminology, see Glossary in Power Automate Process Mining.

> **Note**
>
> Try to limit the data by optimizing the **Number of months to load** filter value. Power Query Online includes several threshold checks. Therefore, if you try to load too much data, time-out or quota violation exceptions might occur, and the system might fail. Load less than 25 months of data into the *Warehouse material movement analysis* process and ensure that the total number of events in the process is less than 4 million.

When you complete the deployment wizard, the following actions occur:

- **Initial data is loaded.** The system loads 200 *cases* into the staging table. In the context of the *Warehouse material movement analysis* process template, each case corresponds to a single closed warehouse work record. Most likely, many more than 200 records meet the filtering criteria set in the deployment wizard. By loading only 200 cases, the system enhances performance and the user experience, so that you can access the visualization more quickly.
- **You're prompted to complete the deployment process.** Power Automate Process Mining user interface (UI) opens on a new tab in your browser. You might be asked to sign in. Power Automate Process Mining uses your credentials to read data from Supply Chain Management. Don't close the page right away. Otherwise, you interrupt the deployment. You don't have to do anything else on the page. Power Automate Process Mining lets you know when deployment is completed.

Refresh data in the staging table and Power Automate Process Mining

As mentioned earlier, after you deploy a new process, the system loads an initial set of 200 cases into the staging table and loads that data into Power Automate Process Mining. This behavior lets you get started quickly. However, to work with your full data set, you must refresh the staging table and Power Automate Process Mining. This section describes how to manually trigger this process.

> **Tip**
>
> Although it can be useful to refresh the data on demand from time to time, you probably want to set up each process so that it automatically refreshes its data on a regular basis. For

instructions, see the Configure the warehouse material movement process section of this article.

Follow these steps to manually refresh data in the staging table and Power Automate Process Mining.

1. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**.
2. In the list pane, select the process to refresh the data for.
3. On the Action Pane, select **Refresh staging table**. The batch job that's responsible for refreshing the staging table runs every five minutes. The next time that it runs, it picks up the request to refresh the table. It then loads all the data that matches the filters specified during deployment.
4. On the **General** FastTab, the **Last staging table refresh** field is updated to indicate that the table is refreshed. However, the new data isn't available in Power Automate Process Mining until you refresh it there. Select **View process details** on the Action Pane, and then select **Refresh data** in Power Automate Process Mining.

A staging table cleanup batch job runs daily and removes any records from the staging table that no longer satisfy the filter criteria. For example, it removes any records that are now too old.

Configure warehouse material movement processes

After you deploy a process, you can configure it at any time by following these steps.

1. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**.

2. In the list pane, select the process to refresh the data for.

3. On the **General** FastTab, set the following fields:

    - **Automatic refresh recurrence** – Select how often the system should update the data table (*None*, *Once per day*, or *Once per week*). Although this update refreshes data in the staging table, it doesn't automatically load that data into Power Automate Process Mining. To set Power Automate Process Mining to periodically load this data from the staging table, select **View process details** on the Action Pane. Then, in Power Automate Process Mining, select **Schedule refresh**, and select a refresh schedule. You can also manually refresh the data in Power Automate Process Mining by selecting **Refresh data** in Power Automate Process Mining.

- **Last staging table refresh** – This read-only field shows the date and time when the staging table was last refreshed.
- **Enforce offline hours** – Set this option to *Yes* to prevent the system from refreshing the staging table during the specified hours. This option can be useful if you want to avoid refreshing the table during peak hours.

4. On the **Process parameters** FastTab, review the following fields:

    - **Legal entity** – This read-only field shows the legal entity specified during deployment.
    - **Number of months to load** – This read-only field shows the number of months to load, as specified during deployment.
    - **Activity** – This read-only field shows the activity specified during deployment.

    The **Processing log** FastTab shows a history of data refresh and cleanup operations. This information can be useful during troubleshooting.

Access the warehouse material movement analysis process

You can access the warehouse material movement analysis process from both Supply Chain Management and Power Automate Process Mining UI.

To access the process from Supply Chain Management, follow these steps:

1. Go to **Warehouse management** > **Enquiries and reports** > **Warehouse performance analysis** > **Warehouse material movement analysis**.

2. In the **Process name** field, select the name of the process to view. The list includes only processes that meet both the following conditions:

    - The process is ready for analysis. (In other words, it has been deployed and analyzed in Power Automate Process Mining.)
    - The current user can access the process. (In other words, the current user created or shared the process.)

3. The Power BI report for your selected process opens. You can interact with the report to view the process and analyze it in various ways.

4. To view the selected process in Power Automate Process Mining, select **View process report** on the Action Pane. Power Automate Process Mining UI opens on a new tab in your browser.

5. To perform advanced process analysis, assess process behavior with business rules, detect rework, and build your own business metrics, select **Open Power Automate Process Mining** to open process in Power Automate Process Mining desktop app. For more information about its capabilities, see Power Automate Process Mining overview.

FAQ

Why don't I see all the events (work lines) and cases (work headers) that I expect in Power Automate Process Mining process?

Several reasons might prevent you from seeing all the events and cases that you expect. Check the following items:

- Confirm that the staging table is recently refreshed. Go to **Warehouse management** > **Setup** > **Process mining** > **Warehouse material movement process configuration**, and review the **Last staging table refresh** field on the **General** tab and the information on the **Processing log** FastTab.
- Confirm that the warehouse work that you expect to see in the process is closed. The warehouse work closed date and legal entity (company) should meet the process filtering criteria that are specified during process deployment. You can check these details on the **Warehouse material movement process configuration** page.
- Confirm that Power Automate Process Mining process is recently refreshed. For information about how to view refresh information, and how to set up schedules to automatically refresh the staging table and Power Automate Process Mining process, see the Configure the warehouse material movement process section of this article.

Why don't I see the process that I'm looking for on the Warehouse material movement analysis page?

Several reasons might prevent you from seeing a process on the **Warehouse material movement analysis** page. Check the following items:

- Confirm that you have access to the process. Usually, to view a process, you must have created it yourself, or the creator must share it with you. To confirm that you have access, sign in to Power Automate Process Mining in Power Automate by using the same user account, and determine whether you can access the process.
- Confirm that the process is deployed and ready for analysis. The **Warehouse material movement analysis** page shows only processes that are ready for analysis. If the process is

still being deployed, it might not yet appear on the page. To confirm that the process is ready for analysis, try to open the Power BI report for the process in Power Automate Process Mining UI.

Why do I receive an error during process deployment?

The system performs several checks while it's provisioning a process. Most failures occur either because the Supply Chain Management user account that's used isn't registered in Dataverse or because the account doesn't have the required privileges. If deployment takes a long time (about four hours), you're probably exceeding the threshold limits for Power Automate Process Mining integration. Consider changing the filters to reduce the number of events that are loaded into Power Automate Process Mining. You can typically lower the **Number of months to load** filter value.

Last updated on 05/29/2026

Enable warehousing telemetry with Application Insights

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes how to set up Microsoft Dynamics 365 Supply Chain Management to send warehousing telemetry data to Application Insights.

Turn the telemetry feature on or off for your system

To use this feature, it must be turned on for your system. As of Supply Chain Management version 10.0.43, it's turned on by default. Admins can turn this functionality on or off by searching for the *Monitoring and telemetry* feature in the **Feature management** workspace.

Set up Application Insights

Start by setting up Application Insights on your Azure subscription.

1. If you don't already have a subscription to Azure, get one.

   > **Note**

You can set up the Application Insights resource on any Azure tenant that can be accessed by your system. For example, you might set up Application Insights on a tenant that belongs to your Microsoft partner. In this case, your Microsoft partner will be able to analyze your telemetry data even if they don't have access to your Azure tenant.

2. Sign in to the Azure portal for the account where you want to install Application Insights.
3. Create an Application Insights resource by following the instructions in Create an Application Insights resource.
4. Keep a copy of the **Instrumentation key** or **Connection string** value for your Application Insights resource. (For more information, see Create an Application Insights resource.) You'll need this value later, when you set up Supply Chain Management to submit the data.

Set up your apps to send telemetry data to Application Insights

After you've set up Application Insights and have a copy of its instrumentation key, you're ready to enable Supply Chain Management to send telemetry data.

1. Sign in to Supply Chain Management as a user who has system admin privileges.
2. Go to **System administration** > **Setup** > **Monitoring and telemetry parameters**.
3. On the Configure tab, choose which types of telemetry you'd like to capture. Only the selected types are sent to Application Insights. Of the options listed, only the following are relevant for warehousing telemetry:

    - **User sessions (custom events)** – This must be enabled to capture warehouse events. You should set this option to *Yes*.
    - **Warehouse events** – Specify whether you want to send warehousing telemetry data to Application Insights. You should set this option to *Yes*.

4. On the **Environments** tab, identify the environment mode (*Development*, *Test*, and *Production*) of each environment that you want to send telemetry from. Use the **New** and **Delete** buttons to add and remove rows as needed. You can create as many rows as you want, but we recommend that you create no more than one for each **Environment mode** value (if more than one environment ID is mapped to the same mode, the system selects an ID at random). Make the following settings for each row:

- **LCS Environment ID** – Enter the ID of the Microsoft Dynamics Lifecycle Services environment that you want to send telemetry data from. To find this ID, sign in to Lifecycle Services, and open the environment details page for your environment. In the **Environment Details** section, look for the **Environment ID** field.
- **Environment mode** – Select the mode of your selected environment.

5. On the **Application Insights registry** tab, map each environment mode that you use to a target Application Insights connection string or instrumentation key. If a database is copied from one environment to another, the mode will be auto-detected and fail over to the new target endpoint. Environments that aren't mapped default to the *Development* mode. Use the **New** and **Delete** buttons to add and remove rows as needed. You can create up to three rows here (one for each **Environment mode** value). Make the following settings for each row:

   - **Environment mode** – Specify the mode you want to map.
   - **Connection string** or **Instrumentation key** – Enter the value that you copied when you set up Application Insights in Azure.

   > **Note**
   >
   > You can use the same instrumentation key for different environment modes.

6. On the Action Pane, select **Save** to save your settings.

Pricing

Application Insights is billed based on the volume of telemetry data that your application sends (data ingestion) and the length of time that you want data to be available (data retention). For up-to-date information about pricing, see Azure Monitor pricing.

Next steps

- Monitor Warehouse Management usage and performance

Last updated on 03/10/2026

Monitor Warehouse Management usage and performance

Microsoft Dynamics 365 Supply Chain Management emits telemetry data for various Warehouse Management activities and operations, including both the Supply Chain Management tenant and the Warehouse Management mobile app. Telemetry data helps provide insight into the activities and general health of your tenants and devices, so that you can diagnose problems and analyze operations that affect performance. Use Application Insights, an Azure service that's optimized for this purpose, to collect and process telemetry data.

Prerequisites

Before you can collect and analyze Warehouse Management telemetry data, ensure that your system meets the following prerequisites:

- **Supply Chain Management version** – Telemetry with Application Insights requires Supply Chain Management version 10.0.29 or later. Additional telemetry features require later versions. See the tables later in this article for details about which features require which versions of Supply Chain Management.
- **Warehouse Management mobile app version** – Telemetry with Application Insights requires Warehouse Management mobile app version 2.0.28 or later. Additional telemetry features require later versions of the app. See the tables later in this article for details about which features require which versions of Supply Chain Management and the Warehouse Management mobile app.
- **Application Insights** – You must have an Application Insights resource in Azure, and you must configure your Supply Chain Management environment to send telemetry data to it. For instructions, see Enable warehousing telemetry with Application Insights.

View telemetry data in Power BI

The fastest and easiest way to get started viewing your warehousing telemetry is to download and set up the Power BI reports provided by Microsoft. The downloadable reports and instructions for how to set them up are available in the Supply Chain Management telemetry repository on GitHub.

Technical details of all available telemetry data

In Application Insights, telemetry data from Supply Chain Management tenants and the Warehouse Management mobile app are logged as custom events. Technical details of all custom events and all data that could be logged for each event are available in the Supply Chain Management telemetry repository on GitHub. This information can help you design your own custom Power BI reports and explore the data directly in Application Insights. The following sample query files list all of the available event IDs, field names, and version requirements:

- For Supply Chain Management tenant telemetry details, see the WarehouseManagement.kql example query on GitHub.
- For Warehouse Management mobile app telemetry details, see the WarehouseMobileApp.kql example query on GitHub.

Analyze how long work takes in the mobile app (preview)

[This section is prerelease documentation and is subject to change.]

Warehouse Management mobile app version 4.1.5.0 and later emits telemetry that describes how long work takes in the app, rather than only how long the server takes to answer. Use this telemetry to find the steps where workers lose the most time, and to tell a slow server apart from a page that's simply hard to complete.

> **Important**
>
> - This is a preview feature.
> - Preview features aren't meant for production use and might have restricted functionality. These features are subject to supplemental terms of use, and are available before an official release so that customers can get early access and provide feedback.

Time spent on each page

Every request that the app sends to the server reports how long the worker spent on the page that the request replaces. The app splits this time into two separate measurements:

- **Active time** – Time when the app is in the foreground and the worker can act on the page.
- **Idle time** – Time when the app is in the background, or the device is locked or asleep.

The app reports these measurements separately because a page that's left open on a locked device during a break otherwise looks the same as a page that a worker struggles to complete. Only the active time indicates that a page is hard to fill in.

A page is new when its structure changes. The structure refers to the page layout and the set of fields and buttons on it. Retyping a value or correcting an entry doesn't start a new measurement because the worker is still completing the same step.

Completed operations

The app reports a summary for each unit of work that a worker completes, such as a full putaway or a picking run, instead of a single page or request. Each summary covers the whole flow, from the moment the worker starts the operation until the server confirms it. The summary describes the following aspects of the flow:

- How many pages the worker moved through and how many requests the app sent.
- How many buttons the worker pressed and how often the worker had to open a dialog box to enter or edit a value.
- How many error messages the server returned during the flow.
- How the worker entered data, including how much of the flow the worker spent scanning with the camera compared to a dedicated scanner such as a ring scanner.
- How the elapsed time divided between waiting for the server, filling in pages, and idle time.

The app also reports operations that a worker abandons. These operations indicate how the flow ended, such as whether the worker canceled it or signed out. The app includes abandoned operations so that the data doesn't reflect only the flows that succeeded, which are the flows that are least likely to need attention.

Because each summary covers a complete flow, you can compare the same operation across sites, shifts, and devices. You can see the effect of a configuration change on the time that the operation takes end to end.

Correlate app telemetry with server telemetry

Mobile app events include the identifier of the server session that the worker signed in to. Because Supply Chain Management also records the identifier, you can join the events that the app emits to the corresponding server-side activity. You can follow a single worker's session across both.

Use this identifier when the app reports a slow step, so you can determine whether the worker spent the time on the device or in Supply Chain Management. Keep the detailed diagnostic information in the server-side telemetry, and use the session identifier to find it, instead of increasing the volume of data that each device sends.

For the field names that carry this data, and the app and Supply Chain Management versions that are required for each field, see the WarehouseMobileApp.kql example query on GitHub.

View telemetry data in Application Insights

As an alternative to Power BI, you can view your telemetry data directly in Application Insights. Telemetry is stored in Azure Monitor Logs in the `customEvents` table. You can view the collected data by writing log queries in the Kusto Query Language (KQL). For more information, see Azure Monitor Logs overview and Log queries in Azure Monitor.

As a simple example, follow these steps:

1. In the Azure portal, open your Application Insights resource.
2. On the **Monitoring** menu, select **Logs**.
3. On the **New Query** tab, enter the following code to view the last 100 custom events.

```plaintext
kql
customEvents
| take 100
| sort by timestamp desc
```

Review sample code, FAQs, and more in the Supply Chain Management telemetry GitHub repository

For more examples of how to work with KQL, plus answers to frequently asked questions, and tips about using Supply Chain Management telemetry with Excel, Power Automate, Power BI, PowerShell, and more, see the Supply Chain Management telemetry repository on GitHub.

Set up alerts on telemetry events

You can configure the system to send you an alert message if something occurs in your environment or app that requires immediate action. Application Insights makes it easy to define

these alerts. For more information and instructions, see What are Azure Monitor Alerts.

Pricing

Application Insights is billed based on the volume of telemetry data that your application sends (data ingestion) and the length of time that you want data to be available (data retention). For up-to-date information about pricing, see Azure Monitor pricing.

Last updated on 08/10/2026

Upgrade warehouse management from Microsoft Dynamics AX 2012 to Supply Chain Management

Article • 02/07/2024

This article provides an overview of the process of upgrading from Microsoft Dynamics AX 2012 R3, running the *WMSII* module, to Dynamics 365 Supply Chain Management.

Supply Chain Management no longer supports the legacy *WMSII* module from Dynamics AX 2012. Instead, you can use the *Warehouse management* module. In the *WMSII* module, the Location and Pallet ID inventory dimensions can be selected for financial inventory. However, the Pallet ID inventory dimension can't be used for financial inventory in Supply Chain Management.

During an upgrade, all products that are associated with a storage dimension group that uses the Pallet ID inventory dimension are identified, marked as blocked, and not processed for upgrade.

Upgrading to Supply Chain Management when AX 2012 R3 WMSII is used

After the upgrade, you can use a set of options on the **Change storage dimension group for items** page to unblock products that were blocked during upgrade, and then process transactions for those products.

Enabling items in Supply Chain Management

This change is required because in Supply Chain Management, item tracking is part of the warehouse management processes (WMS). For these processes, all warehouses and their locations must be associated with a location profile. If you want to use WMS, the following configuration must be done:

- Existing warehouses must be enabled to use WMS.
- Existing released products must be associated with a storage dimension group that uses WMS.

If the source storage dimension groups use the Pallet ID inventory dimension, the locations of existing on-hand inventory that used the Pallet ID inventory dimension must be associated with a location profile where the **Use license plate tracking** parameter is

selected. If the existing warehouses should not be enabled to use WMS, you can change the storage dimension groups of the existing on-hand inventory to groups that handle only the Site, Warehouse, and Location inventory dimensions.

> ⓘ **Note**
>
> You can change the storage dimension group for items even if open inventory transactions exist.

Find products that were blocked because of the pallet ID

To view the list of released products that were blocked during upgrade and can't be processed, go to **Inventory management** > **Setup** > **Inventory** > **Items blocked for inventory updates**.

Change the storage dimension group for blocked products

To be used as part of a warehouse management process, an item must be associated with a storage dimension group in which the Location inventory dimension is active, and the **Use warehouse management processes** parameter is selected. When this setting is selected, the Site, Warehouse, Inventory status, Location, and License plate inventory dimensions become active.

To unblock products that were blocked during upgrade, you must select a new storage dimension group for the products. Note that you can change the storage dimension group even if open inventory transactions exist. To use items that were blocked during upgrade, you have two options:

- Change the storage dimension group for the item to a storage dimension group that uses only the Site, Warehouse, and Location inventory dimensions. As a result of this change, the Pallet ID inventory dimension is no longer used.
- Change the storage dimension group for the item to a storage dimension group that uses WMS. As a result of this change, the License plate inventory dimension is used.

Enable warehouses to use WMS

Follow these steps to enable a warehouse to use WMS.

1. Create at least one new location profile.
2. Go to **Warehouse management** > **Setup** > **Enable warehouse management processes** > **Enable warehouse setup**.
3. On the **Enable warehouse setup** page, add the warehouses that should be enabled to use WMS. You can complete this step either directly on the page or by using the Microsoft Office integration.
4. Assign a location profile to all the locations. You can easily complete this step directly from the page by using the Microsoft Office integration. You can either export and import the data, or use the data entity processing in Data management.
5. Validate the changes. As part of the validation process, various validations of data integrity occur. As part of a larger upgrade process, issues that occur might have to be adjusted on the source implementation. In this case, an additional data upgrade will be required.
6. Process the changes.

Change the storage dimension group for items to use WMS

This section describes how to change product storage dimension groups for items as part of the process of upgrading warehouse management from AX 2012 to Supply Chain Management.

Purpose of changing storage dimension groups for items

By changing storage dimension groups for items, you can migrate existing items that have open inventory transactions so that they use the new storage dimension that's required to use WMS in Supply Chain Management.

In a typical migration scenario, you have existing items where the location dimension is enabled. The migration process enables those items to fulfill the WMS item requirements and ensures that all the related data matches the new settings. In this way, business processes can proceed after the migration.

To use WMS, your items must meet the following requirements:

- The storage dimension group for each item must have the **Use warehouse management processes** parameter set to *Yes*. The storage dimension represents how you store or retrieve items in your inventory.
- An inventory reservation hierarchy must be assigned.

- A unit sequence group must be assigned. The unit sequence group defines the sequence of units that can be used in warehouse operations.

Prerequisite settings

You must configure the following settings before you start to migrate items:

1. Create a new **Inventory status** value, and assign it as the **Default inventory status ID** value in the **Warehouse management parameters** settings.
2. Create a new storage dimension group where the **Use warehouse management processes** parameter is set to *Yes*.
3. On the **Reservation hierarchy** page, define a new reservation hierarchy to accommodate each item's storage and tracking dimension groups.
4. Create one or more unit sequence groups that include the units that are used for each item's inventory units.

Change storage dimension groups for items

Supply Chain Management provides a migration tool to help you change the storage dimension group for items. Follow these steps to migrate an item.

1. Go to one of the following locations:

    - **Inventory Management** > **Setup** > **Inventory** > **Change storage dimension group for items**
    - **Warehouse management** > **Setup** > **Enable warehouse management processes** > **Change storage dimension group for items**

2. On the Action Pane, select **New** to add a row to the grid. Set the following fields for the new row:

    - **Item number** – Select the item number to migrate.
    - **Storage dimension group** – Select the storage dimension group that you plan to migrate the item to.
    - **Reservation hierarchy** – Select the reservation hierarchy that applies to the migrated item.
    - **Unit sequence groups** – Select the unit sequence group that applies to the migrated item.

3. Repeat step 2 until you've added all the rows that you need.

    Tip

Instead of adding one row one at a time, as described here, you can import many rows at the same time by using either Microsoft Office integration or the data entity process that's described in **Data management**.

4. On the Action Pane, select **Validate changes**. The system validates all the rows on the page to ensure that data integrity will be maintained. As part of a large migration process, you might have to adjust data on the source implementation to fix issues. Therefore, several additional rounds of data migration might be required. The validation checks for the following conditions, among others:

    - A default inventory status must be defined.
    - On-hand inventory with pallets must exist only on license-plate-tracked locations.
    - On-hand inventory without pallets must not exist on license-plate-tracked locations.
    - All combinations of storage dimension groups and reservation hierarchies must be valid.
    - Items that are migrated to use WMS must not be enabled to use catch weight.

5. On the Action Pane, select **Process changes**. The update of all the inventory dimensions can take a while. You can monitor the progress by checking the batch job tasks. The process uses both parallel processing and the batch framework, and the difference is handled by different batch tasks.

    > ⓘ **Note**
    >
    > If you upgrade from a version that supports pallets, the upgrade process identifies any items where the pallet dimension is active. Because these items use unsupported settings, the upgrade process creates a record for each of them on the **Items blocked for inventory updates** page. In this way, the items are blocked from all inventory processes. Blocked items must be migrated.

Unsupported scenarios

The following scenarios aren't supported:

- **Catch-weight-enabled items:** Because the pallet dimension is active for catch-weight–enabled items, those items must be downgraded to a dimensions group where only the location dimension is active.

- **Reserve ordered transactions:** Because of the way that reservations work for WMS-enabled items, there are specific constraints on the state of the inventory transactions. If any reserved ordered transactions have a "hole" in the dimensions, the item can't be converted. A hole can occur for items where the batch dimension is active and is below the location dimension in the reservation hierarchy. If a reservation exists on the site, warehouse, or batch, the location is missing. We refer to this situation as a hole. Holes aren't supported. To fix the issue, remove the hole by either assigning the missing dimensions or clearing the dimensions.

Feedback

Was this page helpful? Yes No

Provide product feedback

Warehouse management only mode overview

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Warehouse management only mode lets you set up a legal entity in Microsoft Dynamics 365 Supply Chain Management that's dedicated to warehouse management processes. This legal entity can then provide warehousing services to other legal entities in Supply Chain Management. Alternatively, it can provide warehousing services to external enterprise resource planning (ERP) systems or order management systems. There are two basic scenarios:

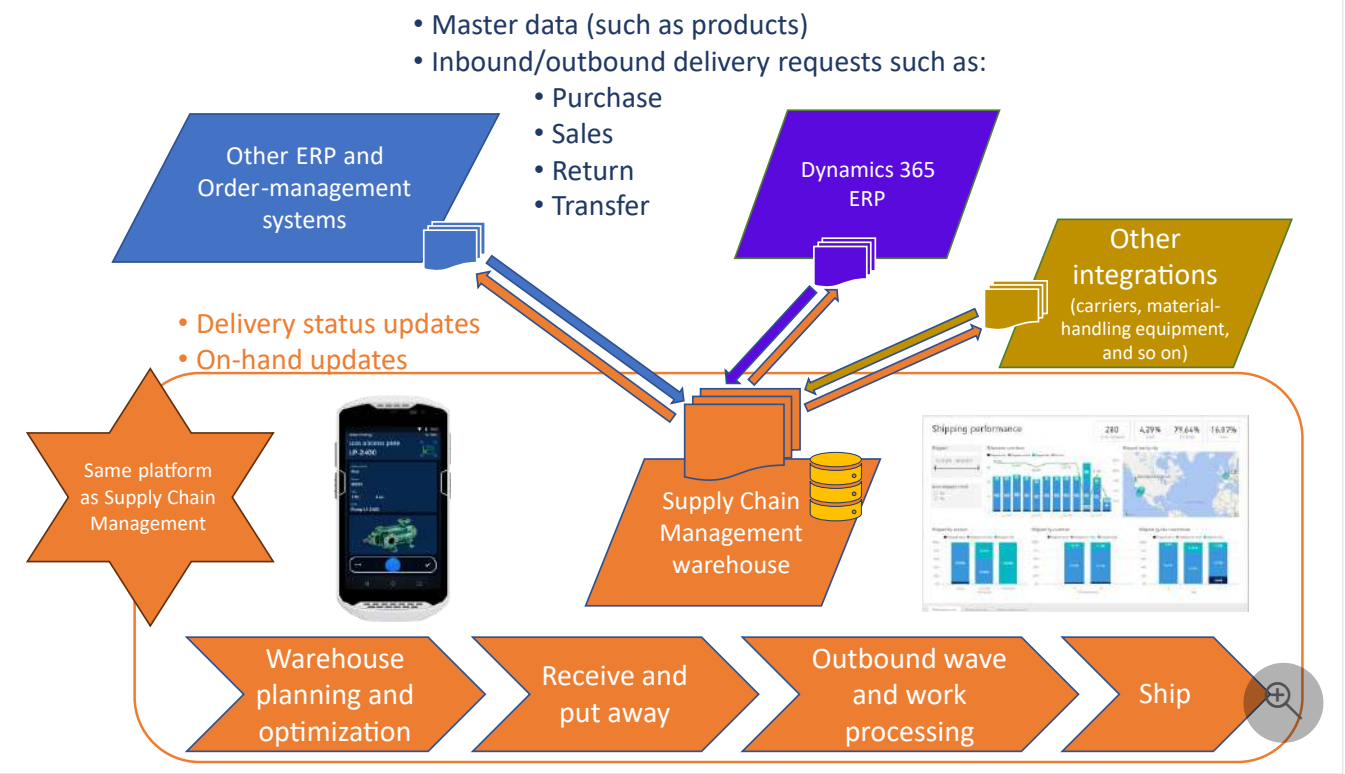
- **Integrate Warehouse management only mode with an external ERP system** – Take advantage of the core warehouse management (WMS) functionality that Supply Chain Management offers while you also continue to take advantage of your existing investments in third-party ERP and order management systems. Regardless of the ERP or ordering systems that you have in place, you can now quickly deploy our advanced WMS functionality without having to set up or maintain areas of Supply Chain Management that you don't need. You're then ready to benefit from advanced WMS features such as automation integration, carrier integration, and the Warehouse Management mobile app.
- **Set up an external shared warehouse for other legal entities in Supply Chain Management** – If you use an external shared warehouse, logistic operations run in a separate legal entity. This legal entity shares warehousing services with other legal entities that manage all the order and financial processing.

Warehouse management only mode uses lightweight source documents that are dedicated to inbound and outbound shipment orders. Because these documents focus exclusively on warehouse management, they can replace multiple types of more general-purpose documents (such as sales orders, purchase orders, and transfer orders) from a pure warehouse management perspective.

Deployment options

Warehouse management only mode provides several deployment options to support the business needs of running your warehouse management processes.

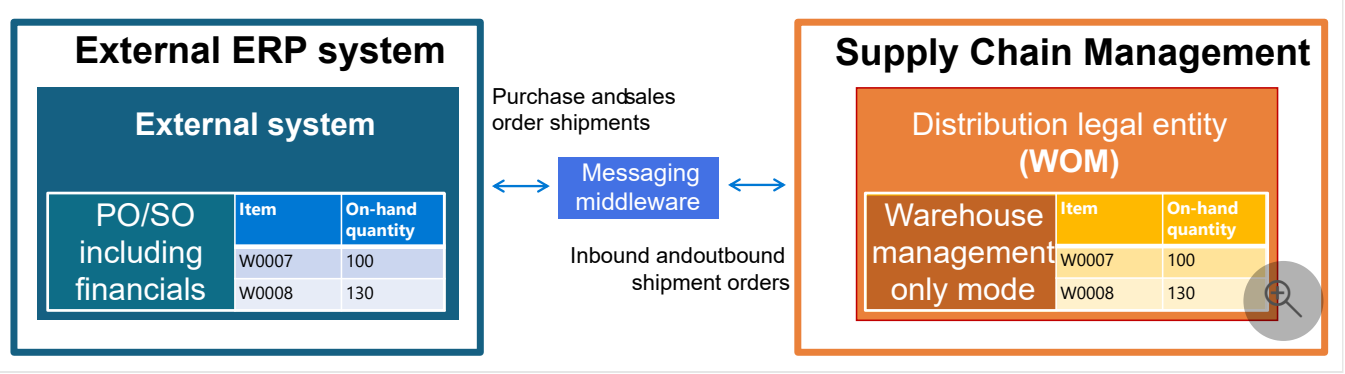
You can start a free trial of Dynamics 365 Supply Chain Management via the unified admin experience for finance and operations apps. You can then try out an implementation as shown in the following high-level diagram of the elements and processes of an integrated system. Learn more in this example that shows how to use inbound and outbound shipment orders.



The solution is very flexible. Therefore, you can choose the configuration of features and systems that best fits your business needs. The following subsections provide a few examples that show how you can integrate Supply Chain Management warehouse management mode.

Handle all orders and financial processing in an external system

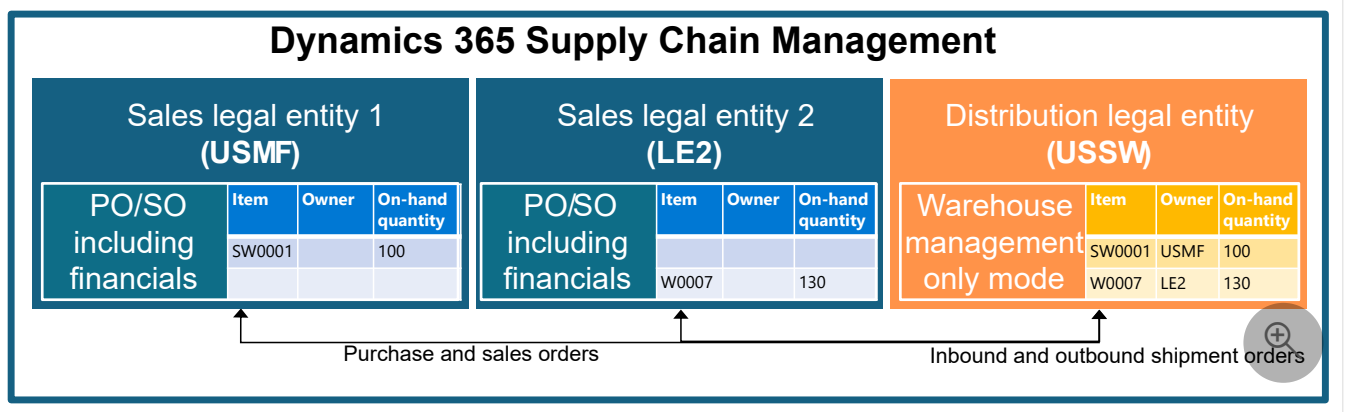
The following high-level diagram shows an example where Supply Chain Management is used to handle warehouse operations only. An external system is used to handle all orders and financial processing.



For this type of implementation, you must configure the warehouse management processes as part of Supply Chain Management. For a more detailed description of this process and the related processes, see Warehouse management only mode with external ERP system.

Use a separate legal entity to manage warehouse management processes for other legal entities in Supply Chain Management

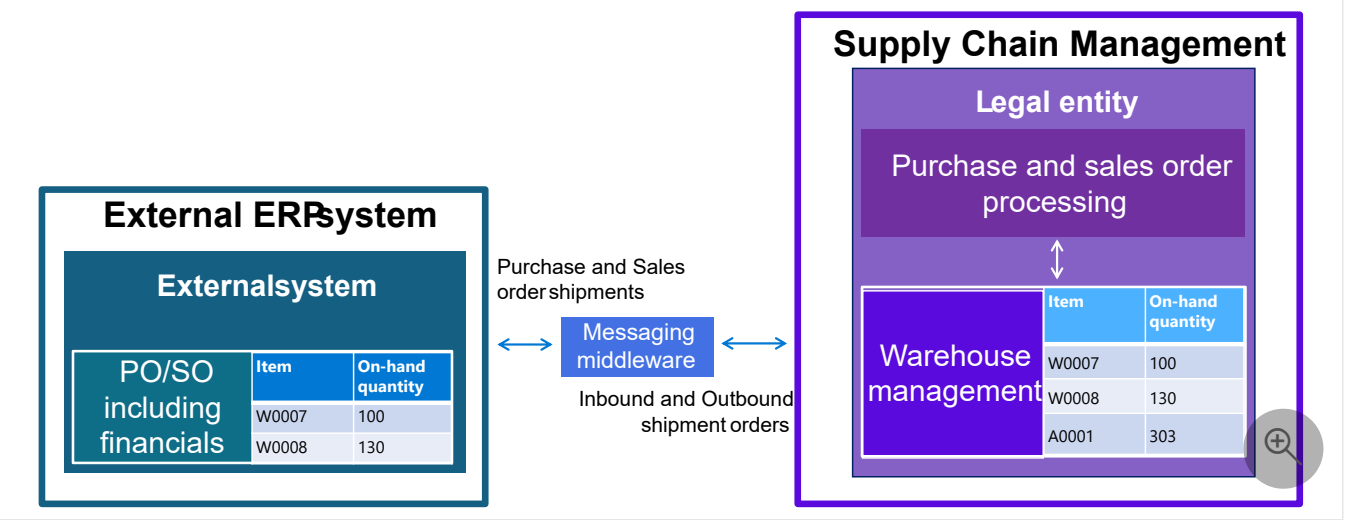
The following high-level diagram shows an example where a system uses a separate Supply Chain Management legal entity to manage warehouse management processes for other legal entities. As part of this management, it tracks the ownership of shared items by using an owner inventory dimension.



For a more detailed description of this process and the related processes, see Warehouse management only mode with external shared warehouse.

Handle logistic warehouse processes for both internal and external integrations

The following high-level diagram shows an example where a system uses Supply Chain Management to handle warehousing, plus a wider range of processes (such as sales, purchase, and production orders). At the same time, it also uses Supply Chain Management to handle warehouse operations for other ERP and order processing systems.



For this type of implementation, the same warehouse instance can handle all the logistic warehouse processes for both internal and external integrations.

Unsupported processes

The following high-level processes aren't supported out of the box when Warehouse management only mode is processed. The list is most relevant to existing customers who already run warehouse management processes and are considering adopting the *Warehouse management only mode* functionality.

Expand table

Process	Description
Inbound Warehouse Management mobile app flows	The Warehouse Management mobile app flows for inbound shipment orders don't support: • Goods in transit where the receiving process is handled against a container. • Sales return orders that support return reason codes and disposition codes as part of the flow for the **Return order receiving (and put away)** mobile device menu item. Instead, you must use the blind return process.
Production flows	Inbound and outbound shipment orders don't support production order, batch order, or kanban processing, including material consumption and reporting as finished via the Warehouse Management mobile app. In addition, you can't use cross-docking from production orders to outbound docks in combination with inbound and outbound shipment orders.
Transportation management processes	The transportation management engines that are currently supported for purchase order loads aren't supported for the inbound shipment order processes. Note that charges can't be assigned, and direct invoicing can't be processed, for either inbound shipment orders or outbound shipment orders. Therefore, the apportionment weight engine can't be used to generate freight bills.

Process	Description
Creation of orders from the warehouse app	The process of creating outbound shipment orders from the Warehouse Management mobile app isn't supported. (That process resembles the *Create transfer order from license plates* process for mobile devices.)
Internal order processing information that's provided to external systems	When you're using the supported orders in Supply Chain Management (such as transfer, sales, purchase, and production orders), all the related business process data is automatically maintained in Supply Chain Management. However, no business events or related inbound and outbound on-hand information is provided to external systems for these types of processes. For example, if you create a transfer order, ship inventory out of one warehouse, and receive it in another warehouse in Supply Chain Management, you can't use the method that's described for inbound and outbound shipment orders to inform the external systems about the operations. You must instead use a different method, such as Warehouse inventory update logs.
Order-committed reservations as part of the *allow reservation on demand order* capability	• In Supply Chain Management version 10.0.46 and later, *order-committed reservations* are fully supported as part of the *allow reservation on demand order* capability. • In Supply Chain Management version 10.0.45 and earlier, *outbound shipment order line* transaction reservations don't support reservations on inventory dimensions below the location in the reservation hierarchy. However, these reservations are supported for *sales order line* transactions.
Items enabled for catch weight processing	Items that are enabled for catch weight processing aren't supported for inbound or outbound shipping orders.
Policies and processes around vendor or customer accounts	Representations of vendors and customers aren't used for inbound or outbound shipping orders. Therefore, you can't use related order processing policies with this type of setup. For example, you can't use customer-specific or vendor-specific product filters or nonconformance management.
Order line registration and pick update processing	Inbound and outbound shipment order lines don't support the manual registration and un-registration processes that are supported by other types of order lines (such as purchase, sales, and transfer order lines).

Last updated on 11/20/2025

Warehouse management only mode with external ERP systems

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

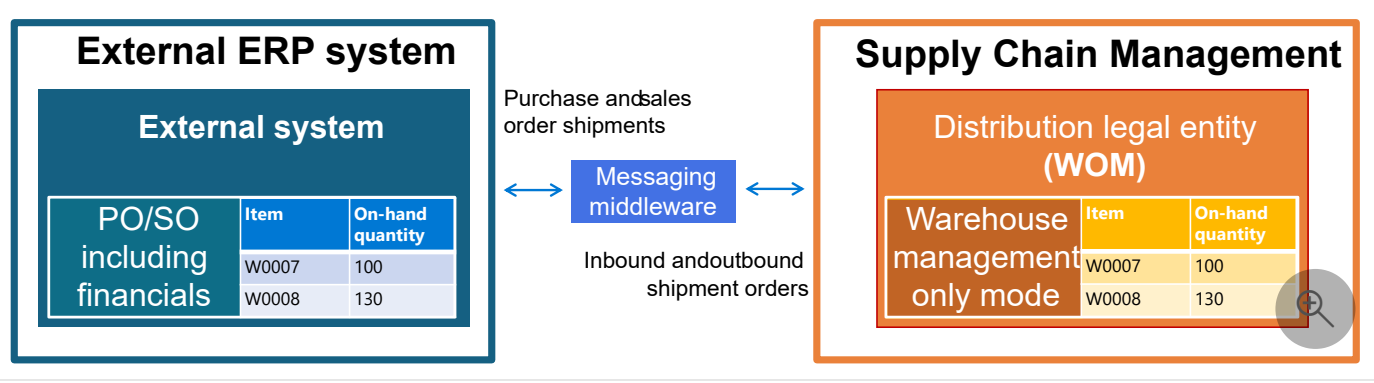
This article explains how to perform day-to-day warehousing tasks when you use Warehouse only mode to integrate with an external enterprise resource planning (ERP) system.

There are numerous ways to use Warehouse management only mode. For example, you can enable Microsoft Dynamics 365 Supply Chain Management to handle logistics operations, and then connect warehouses to external ERP systems that do all the order and financial processing.

In addition, the warehouse management processes can use an owner inventory dimension to track the ownership of inventory for items that are shared across source systems.

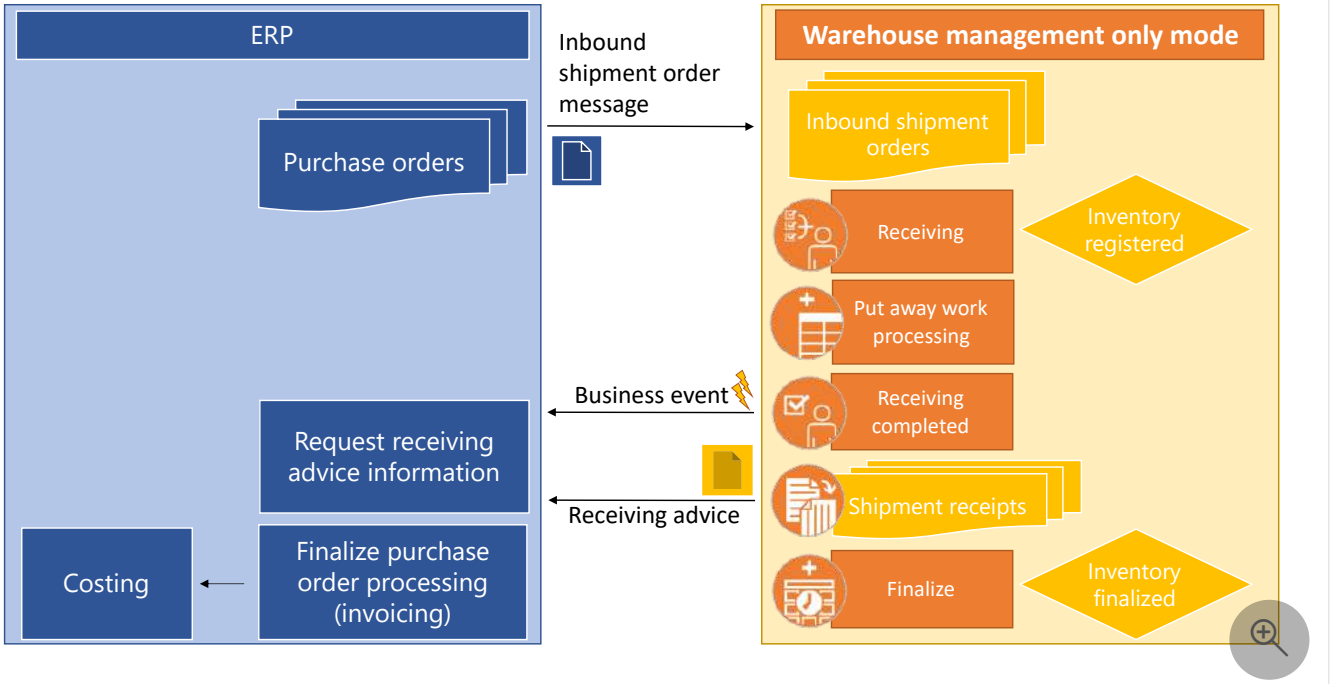
High-level implementation example

The following illustration shows an example where Warehouse management only mode (WOM) is running in the *WOM* Supply Chain Management legal entity. This legal entity handles logistics warehouse operations for an external ERP system that manages order and financial processing.



Inbound process example (external ERP system integration)

The following illustration highlights the elements of the inbound process.



Here's a high-level description of the inbound process. Steps that start with *ERP* are done by the ERP system. Steps that start with *WOM* are done by Supply Chain Management in Warehouse management only mode.

1. *ERP:* An external system submits an inbound shipment order message to Supply Chain Management.

2. *WOM:* Supply Chain Management processes the message in Warehouse management only mode and creates orders.

3. *WOM:* Inbound loads are created in one of four ways, as established by the Source systems settings in Supply Chain Management:

    - Manually, by using the Inbound load planning workbench
    - By importing advanced shipping notices (ASNs)
    - Automatically during message processing
    - Automatically during the Warehouse Management mobile app receiving process

4. *WOM:* Warehouse workers use the Warehouse Management mobile app to *register* the inbound shipment transactions.

5. *WOM:* Supply Chain Management runs receiving completed processes that are related to each relevant load. These processes update the load status to *Received*, generate shipment

receipts, and trigger *business events* for the external systems.

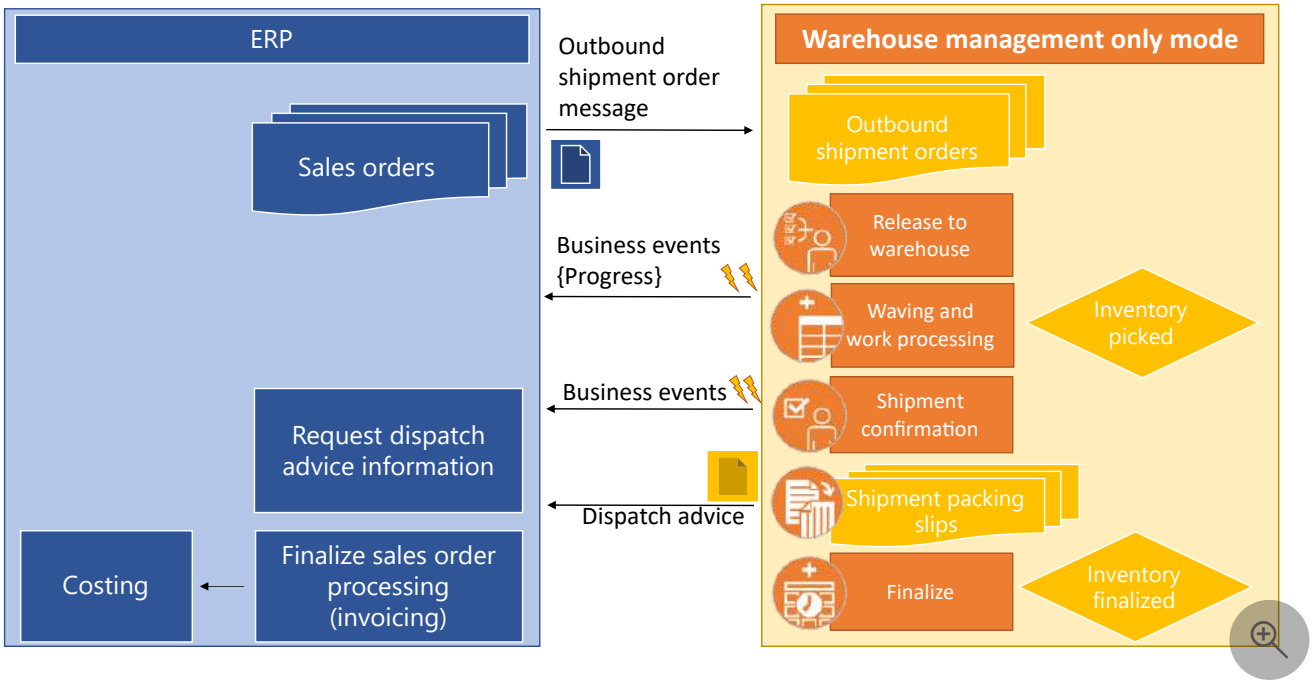
6. *ERP:* The external system reads and uses the shipment receipt data for further processing. For example, if purchase orders are associated with the inbound shipment orders in the external system, this processing involves purchase order invoicing.

7. *WOM:* Supply Chain Management finalizes the inbound shipment orders by running the *Post shipment receipts* batch job.

For a more detailed description of this process and the related processes, see Work with warehouse management only mode in Supply Chain Management.

Outbound process example (external ERP system integration)

The following illustration highlights the elements of the outbound process.



Here's a high-level description of the outbound process. Steps that start with *ERP* are done by the ERP system. Steps that start with *WOM* are done by Supply Chain Management in Warehouse management only mode.

1. *ERP:* An external system submits an outbound shipment order message.

2. *WOM:* Supply Chain Management processes the message in Warehouse management only mode and creates orders.

3. *WOM:* Inventory reservations are created in one of two ways, as established by the Source systems settings in Supply Chain Management:

    - Automatically by the message processor
    - Manually, as part of the release process

4. *WOM:* The orders are released for further warehouse processing, either manually or automatically (via the *Automatic release of outbound shipment orders* batch job).

5. *WOM:* Depending on the setup of your wave template definitions, warehouse work might be created and released immediately.

6. *WOM:* The outbound warehouse work is processed, and the status of the related outbound shipment order line transactions is updated to *Picked*.

7. *WOM:* The loads are outbound ship confirmed. As a result, *business events* and a *shipment packing slip* are created for the external system.

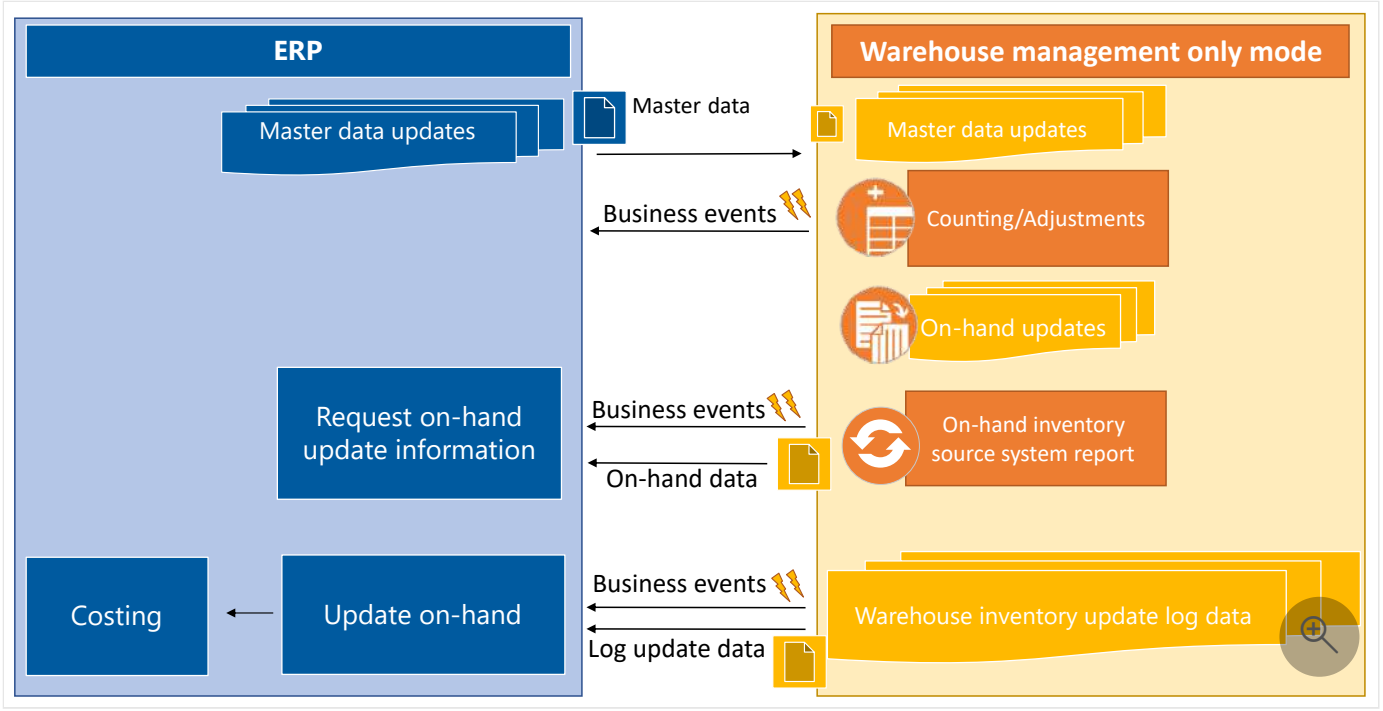
8. *ERP:* The external system reads the shipment packing slip and uses its data for further processing (such as sales order invoicing for sales orders that are associated with outbound shipment orders).

9. *WOM:* Supply Chain Management finalizes the outbound shipment order by running the *Post shipment packing slips* batch job.

For a more detailed description of this process and the related processes, see Work with warehouse management only mode in Supply Chain Management.

On-hand inventory updates between systems

The following illustration shows the internal processes that are used for Warehouse management only mode.



The Warehouse management module uses the counting journal to support multiple on-hand inventory update processes. For more information about the counting process, see Cycle counting.

As part of the journal posting process, Supply Chain Management triggers a business event. External systems can read about the updates through the counting journal entities. It's important to act only on the updated quantities. Otherwise, the systems can go out of sync because of the updates. The following scenario provides an example.

Example scenario: Updating on-hand inventory between systems

At the start of this scenario, on-hand information about item number A0001 is in sync between the external ERP system (ERP) and the Supply Chain Management warehouse management system (WMS), as the following table shows.

Expand table

Item number	ERP on-hand	WMS on-hand
A0001	0 pcs	0 pcs

The following subsections show how different events cause these values to change.

On-hand update 1

Supply Chain Management receives 10 pcs of item number A0001 against an inbound shipment order without running *receiving completed* processing. Therefore, the external system isn't yet informed about this update. As a result, the external system and Supply Chain Management are now out of sync, as the following table shows.

Expand table

Item number	ERP on-hand	WMS on-hand
A0001	0 pcs	10 pcs

On-hand update 2

In Supply Chain Management, an on-hand adjustment (counting journal) that's posted for item A0001 adds 1 pcs of on-hand inventory. The following table shows the result.

Expand table

Item number	ERP on-hand	WMS on-hand
A0001	0 pcs	11 pcs

A business event informs the external system about the on-hand adjustment. As part of this process, the journal posting changes from 10 pcs to 11 pcs in Supply Chain Management. The external system considers only the updated quantity of 1 pcs. The following table shows the result.

Expand table

Item number	ERP on-hand	WMS on-hand
A0001	1 pcs	11 pcs

On-hand update 3

Supply Chain Management runs a *receiving completed* process that's related to the received 10 pcs of item number A0001. Therefore, the external system is informed via a business event. It then reads the shipment receipt information and updates its on-hand quantity with an additional 10 pcs. The following table shows the result.

Expand table

Item number	ERP on-hand	WMS on-hand
A0001	11 pcs	11 pcs

> **Note**
>
> Make sure that each of your items is assigned to an item model group that's configured as described in Master and reference data. In this way, you don't have to configure inventory postings and fiscal calendars when you make adjustments by using the counting journal.

On-hand inventory reconciliation

Warehouse management only mode generates data for an on-hand inventory reconciliation process when you generate a **Create source system on-hand inventory** report (**Warehouse management** > **Inquiries and reports** > **Physical inventory reconciliation** > **Create source system on-hand inventory report**).

To create the header and line data, specify **Source system** and **As of date** values. You must also select the level of inventory dimensions that the report should be generated for.

When inventory that's related to inbound shipment orders is received, on-hand inventory is physically updated based on the status of the registered inventory transactions. When inventory is shipped via outbound shipment orders, the physical on-hand inventory is reduced based on the *Picked* inventory transactions. This physical inventory on-hand representation of the registered and picked items remains until the related *Shipment receipt* and *Shipment packing slip* journals are posted as part of the background finalization processes. To include this part of the physical on-hand inventory in the export, be sure to enable the **Include Registered and Picked inventory quantities** parameter.

The external system is informed about the available data via the `WHSSourceSystemInventoryOnhandReportBusinessEvent` business event. It can read the data via the `WarehouseInventoryOnhandReports` and `WarehouseInventoryOnhandReportLines` data entities.

> **Note**
>
> If you run the **Create source system on-hand inventory** report as a recurring batch job, the **As of date** value is ignored, and data is generated based on the current processing date. For example, you set up the recurrence so that it has a **Start date** value of yesterday, and you set

the job to run once per day. In this case, every day, the batch job automatically generates on-hand inventory data for the previous day.

Warehouse inventory update logs

For integrations that require very quick on-hand inventory synchronization processes, use the Warehouse inventory update log (**Warehouse management** > **Inquiries and reports** > **Physical inventory reconciliation** > **Warehouse inventory update log**). This log can collect all the inventory transaction updates that lead to on-hand updates that are of interest for the external systems. For example, you might have an external system that handles information about inventory status changes.

To keep the external systems updated about inventory transaction updates that are related to inbound and outbound shipment orders, set the **Enable warehouse inventory update logs** option to *Yes* for the relevant source systems, for both inbound and outbound shipment orders.

To view the update log, go to **Warehouse management** > **Inquiries and reports** > **Physical inventory reconciliation** > **Warehouse inventory update log**.

> **Important**
>
> When the **Enable warehouse inventory update logs** option is enabled, be sure to uptake the updates in the external systems in such a way that they don't cause double updates in combination with the data that's used as part of the ***Shipment receipts*** and ***Shipment packing slips*** messages.

By default, the *Publish warehouse inventory update log updates* background process runs every 10 minutes. It creates data that external systems can consume by using the `WarehouseInventoryUpdateLogs` entity. The `WHSInventoryUpdateLogBusinessEvent` business event can be used as part of this process.

Warehouse inventory owner

Warehouse management processes can use an owner inventory dimension to track the ownership of inventory for items shared across multiple source systems.

To use the owner inventory dimension, products must have a tracking dimension group where the *Owner* dimension is enabled. You must also create a record on the **Warehouse inventory**

owner page (**Warehouse management** > **Setup** > **Warehouse management integration** > **Warehouse inventory owner**).

> **Important**
>
> The *Warehouse inventory owner* configuration must contain a mapping for every *Owner* value that a source system sends in the order line. If a mapping is missing, the order message fails to process. This requirement also applies when the source system sends an empty owner value. In such cases, the configuration must map the empty value from the source system to the appropriate value in the *WOM* Supply Chain Management legal entity system.

Last updated on 05/22/2026

Warehouse management only mode with external shared warehouses

09/17/2025

This article explains how to use Warehouse management only mode to support external shared warehouse processing. This approach allows warehouse management functionality to be consolidated into a separate legal entity that handles requests from multiple sales subsidiaries (legal entities) in Microsoft Dynamics 365 Supply Chain Management.

You can use Warehouse management only mode to handle logistics operations in a legal entity solely dedicated to warehousing operations. You can then connect warehouses between that legal entity and other legal entities that do all the order and financial processing. In addition, warehouse management processes can use an *owner* inventory dimension to track which legal entity owns the inventory for items that are shared across legal entities.

This feature works without modifications in a single-instance deployment model. If you want to link between multiple instances, one of which handles logistics and warehouse processes, you must set up an integration process like the one described in Warehouse management only mode with external ERP system.

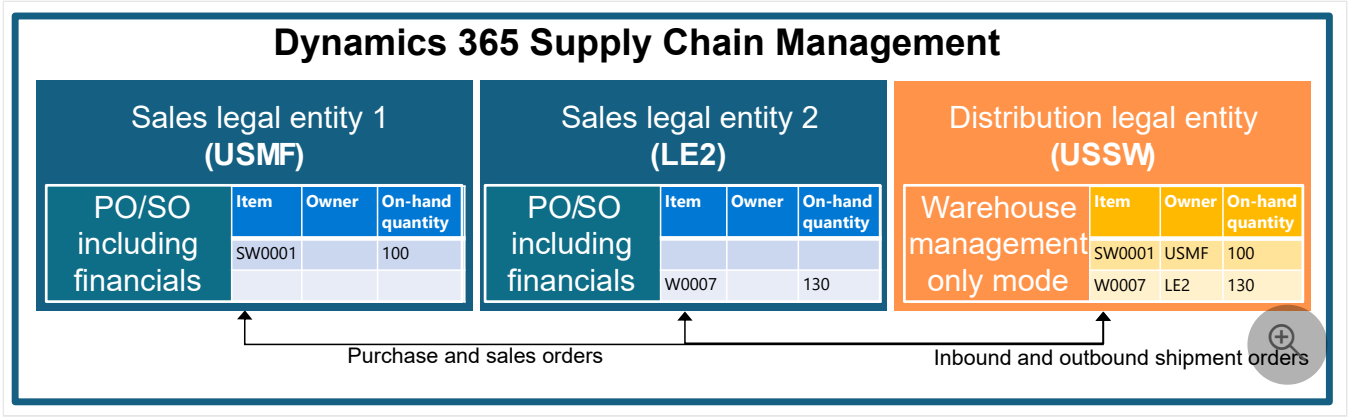
To use *Warehouse management only mode* in a multi-instance deployment, you must use the standard integration process.

The following articles provide detailed information about how *Warehouse management only mode* works:

- Warehouse management only mode with external ERP system
- Exchange data between systems
- Monitor and control message processor messages

High-level implementation example

The following illustration shows an example system where Warehouse management only mode is active with an external shared warehouse in the *USSW* legal entity. The *USSW* legal entity handles logistics warehouse operations for two sales subsidiaries that manage order and financial processing in the *USMF* and *LE2* legal entities.



The *USMF* and *USSW* legal entities are included in the standard demo data. The *LE2* legal entity shown in the illustration is just an example.

The scenarios described in this article use sample records and values from the demo data. To follow along, ensure your system has the standard demo data installed. Otherwise, substitute the sample values with your own configuration settings.

Warehouse management only mode demo data

The standard demo data includes the following data to support scenarios involving *Warehouse management only mode with external shared warehouses*:

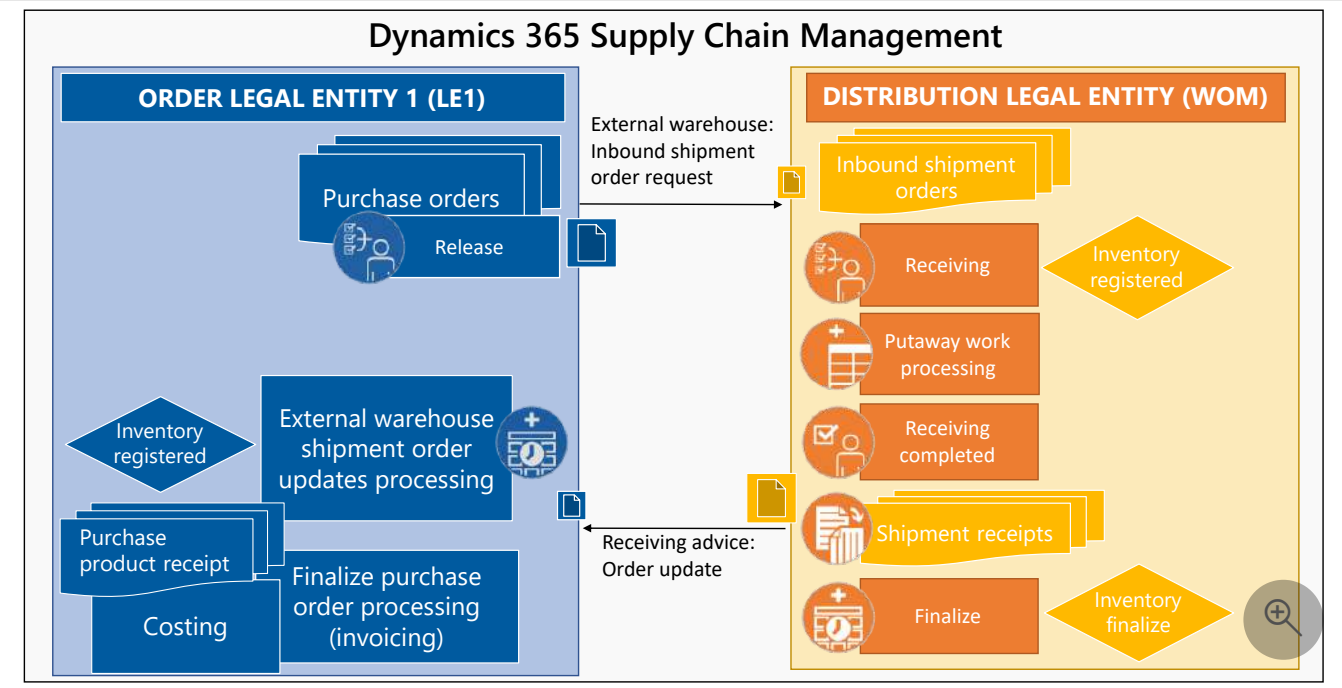
- The legal entity *USSW*, which is set up with a site *1* and warehouse *SW-11*.
- Work templates and location directives for inbound and outbound shipment orders.
- The source system *Product Information Management (PIM)*
- The source system *Contoso Entertainment System USA (USMF)*
- The item *SW0001*, which is released to both the *USMF* and the *USSW* legal entities.
  - In the *USSW* legal entity, the *SW0001* item uses the *Non-valuated* item model group and has the *Owner* tracking dimension enabled.
- Warehouse *2-EXT* in the *USMF* legal entity is marked as an external warehouse that is mapped to warehouse *SW-11* in the *USSW* legal entity.

Other demo data is also included to support the scenarios. The previous list just highlights the most important elements.

> **Important**
>
> The demo data for these scenarios is only available in Supply Chain Management version 10.0.45 and later.

Inbound process example (shared warehouse)

The following illustration highlights the elements of the inbound process when you use a shared warehouse.



Here's a high-level description of the inbound process. Steps that start with *USMF* are done by a sales subsidiary. Steps that start with *USSW* are done by the warehousing entity.

1. *USMF:* Purchase orders are created and released to the warehouse. The system then creates *external warehouse inbound shipment order requests*. As a result, inbound shipment order messages are delivered to the *USSW* legal entity. Purchase orders that are released to the externally managed warehouse can be processed via a periodic background process. (This background process is set up at **Warehouse management** > **Periodic tasks** > **Create external warehouse inbound shipment order requests**.) Alternatively, they can be manually processed by using the **Release to warehouse** option on the purchase orders page.
2. *USSW:* The warehousing entity processes the inbound shipment order messages. As a result of this processing, *inbound shipment orders* are created.
3. *USSW:* Inbound loads are created manually, automatically, or through import (depending on your configuration).
4. *USSW:* Warehouse workers use the Warehouse Management mobile app to register the inbound shipment order transactions.
5. *USSW:* The warehousing entity processes receiving completed operations for the related loads. These operations update the load status to *Received* and generate *external warehouse inbound shipment order updates* for *USMF*. You can turn on automatic message processing for the *External warehouse shipment order updates* message queue.
6. *USMF:* During the processing of the external warehouse inbound shipment order updates, inbound loads and shipments are created, and the related purchase order line

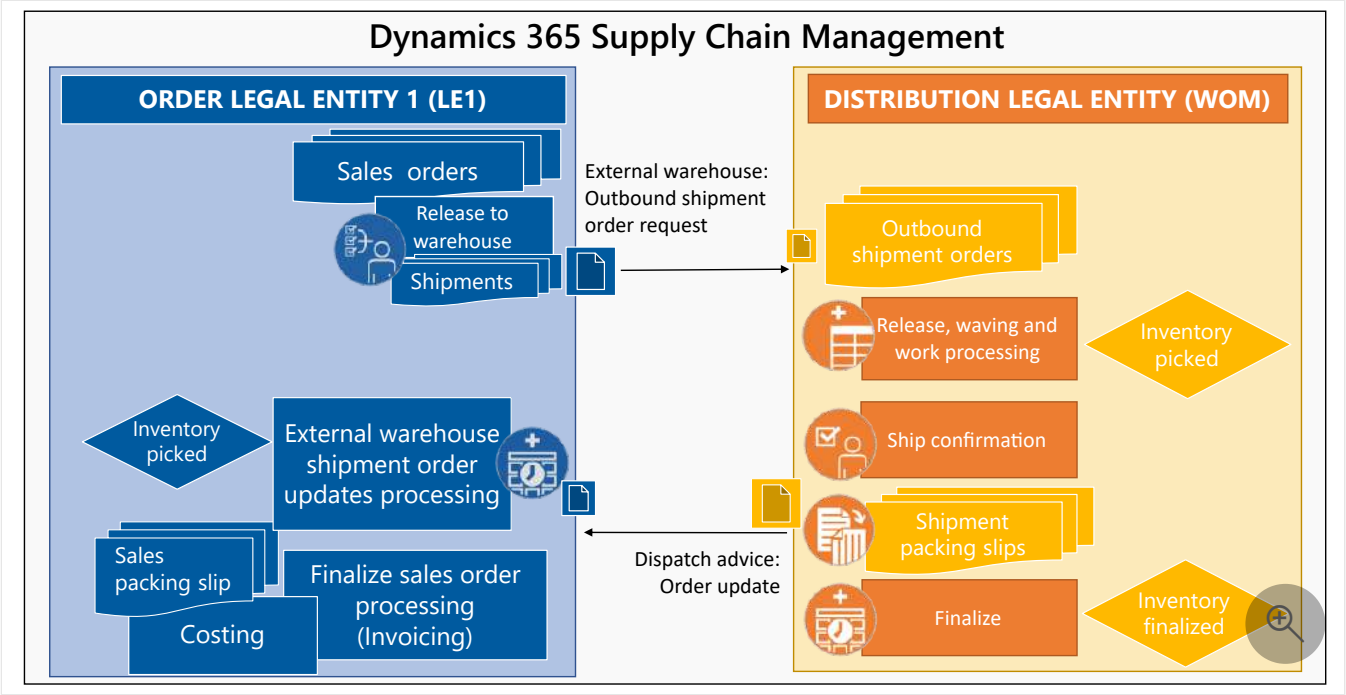
transactions are updated to *Registered*. This status update allows further processing of product receipts and invoices.

7. *USSW:* The warehousing entity finalizes the inbound shipment order line transactions by running the *Post shipment receipts* batch job.

For a more detailed description of this process and the related processes, go to Work with warehouse management only mode in Supply Chain Management.

Outbound example process (shared warehouse)

The following illustration highlights the elements of the outbound process when you use a shared warehouse.



Here's a high-level description of the outbound process. Steps that start with *USMF* are done by a sales subsidiary. Steps that start with *USSW* are done by the warehousing entity.

1. *USMF:* Sales orders are created and released to the warehouse. The system then creates *external warehouse outbound shipment order requests*. As a result, outbound shipment order messages are delivered to the *USSW* legal entity.
2. *USSW:* The *Outbound shipment order messages* are processed. As a result of this processing, *outbound shipment orders* are created.
3. *USSW:* Inventory reservations are created either manually or automatically (depending on your configuration).
4. *USSW:* The orders are released for further warehouse processing, either manually or automatically. If you use outbound load planning processes, you can create loads by using the Outbound load planning workbench before you release the orders.

5. *USSW:* Depending on the setup of your wave template definitions, warehouse work might be created and released immediately.
6. *USSW:* Outbound warehouse work is processed, and the status of the related outbound shipment order line transactions is updated to *Picked*.
7. *USSW:* The loads are outbound ship confirmed. As a result, external warehouse outbound shipment order updates are generated for *USMF*.
8. *USMF:* During the processing of the external warehouse outbound shipment order updates, outbound shipment data and the related sales order line transactions are updated. The transactions become *Picked*. This update allows further processing of the packing slip and invoice. You can turn on automatic message processing for the *External warehouse shipment order updates* message queue.
9. *USSW:* The warehousing entity finalizes the outbound shipment order line transactions by running the *Post shipment packing slips* batch job.

For a more detailed description of this process and the related processes, go to Work with warehouse management only mode in Supply Chain Management.

> **Important**
>
> For the preceding flow, the **Allow load split during ship confirm** option for outbound loads isn't supported in the *USSW* legal entity. Likewise, planned cross-docking information isn't automatically transferred from the sales subsidiary legal entities to the *Warehouse management only mode* process.

> **Note**
>
> When you release orders or loads to the warehouse in the *USMF* legal entity, the shipments that are created are locked with **Outbound shipment processing ownership** set to *External* until order data from the *USSW* legal entity is returned. If order data isn't returned, or if other issues occur, you can get update rights for the shipments by selecting the **Claim processing ownership** action on the shipments. Only admin roles should be granted permission to use this action.

Transfer orders (shared warehouse)

Transfer orders with external shared warehousing mean that either the source, destination, or both warehouses in a transfer order are external. Most aspects of the transfer order processing work in the same way, regardless of whether you use external shared warehouses.

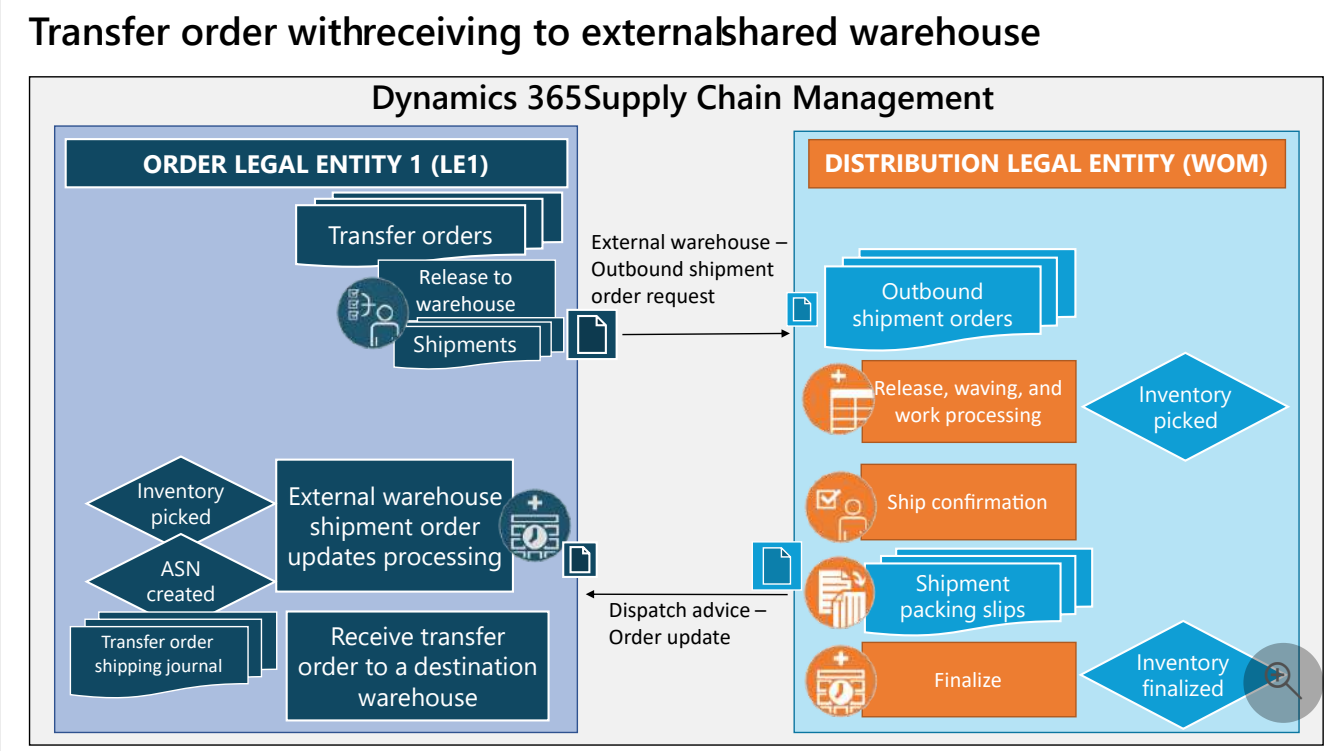
Prerequisites

Before you can use this feature, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.44 or later.
- In the warehouse configuration of the externally managed warehouse, you must set both a *default location* and a *default license plate location*. This location is used as a temporary location when processing a transfer order. When shipping from an external warehouse, the system sends an update to communicate what was shipped and what license plate was used. Inventory is then moved from a *default location* to a *default license plate location* and the transfer order is processed against that location.

Transfer order shipped from an external warehouse

The following illustration highlights the elements of the transfer order process when shipping from an external warehouse.



Here's a high-level description of the process. Steps that start with *USMF* are done by a sales subsidiary. Steps that start with *USSW* are done by the warehousing entity.

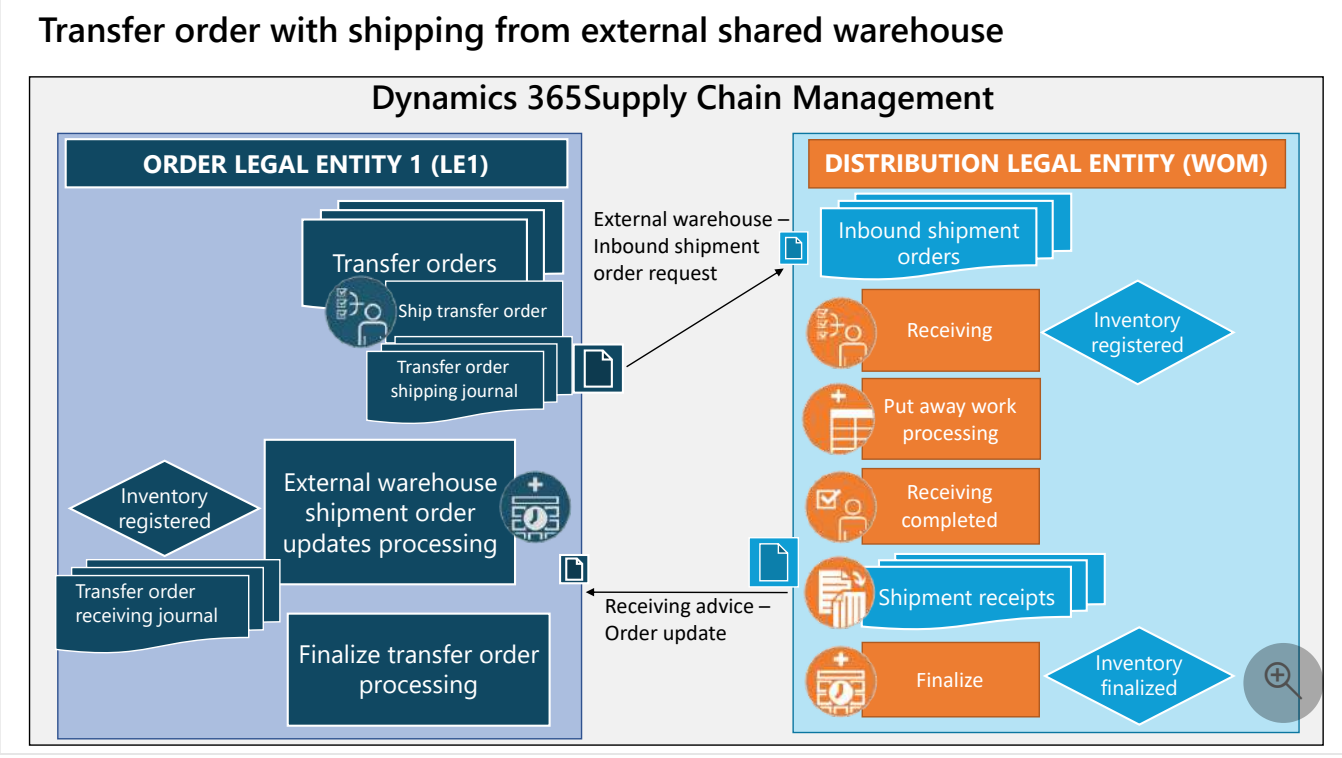
1. *USMF:* The transfer order is created and released to the warehouse. The system then creates *external warehouse outbound shipment order requests*. As a result, outbound shipment order messages are delivered to the *USSW* legal entity.
2. *USSW:* The *Outbound shipment order messages* are processed. As a result of this processing, *outbound shipment orders* are created.

3. *USSW:* The orders are released for further warehouse processing, either manually or automatically. The rest of the process proceeds as described in Outbound example process (shared warehouse).
4. *USSW:* The loads are outbound ship confirmed. As a result, external warehouse outbound shipment order updates are generated for *USMF*.
5. *USMF:* As the external warehouse outbound shipment order updates are processed, the system updates outbound shipment data and the related transfer order transactions. The transfer order becomes *Shipped* if everything from the order is shipped. This update allows further processing of the transfer order when received at the destination warehouse.

> **Note**
>
> When the outbound shipment order update is processed in *USMF*, an inbound advance shipment notice (ASN) is created for every license plate that was shipped from the external warehouse. This allows for license plate receiving in the destination warehouse.

Transfer order received at an external warehouse

The following illustration highlights the elements of the transfer order process when receiving at an external warehouse.



Here's a high-level description of the process. Steps that start with *USMF* are done by a sales subsidiary. Steps that start with *USSW* are done by the warehousing entity.

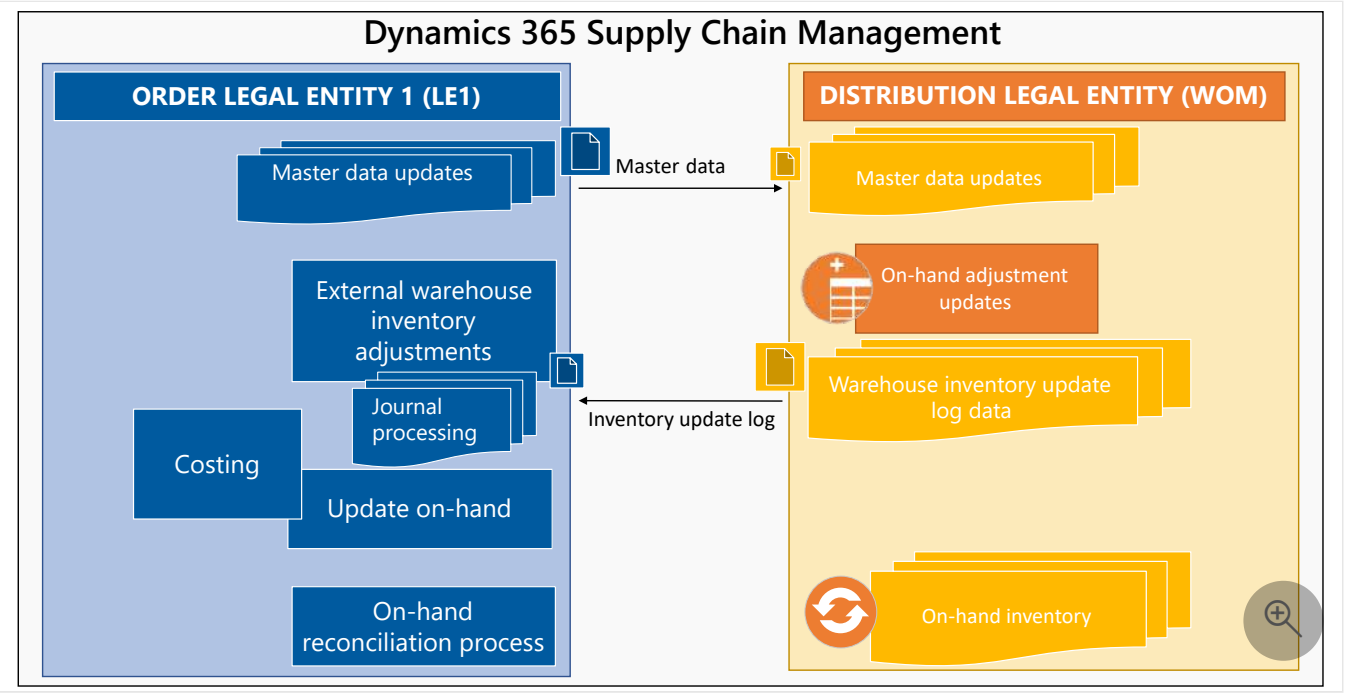
1. *USMF:* The transfer order is created and released to the warehouse. The transfer order is shipped and the system creates a *transfer order shipping journal*. The system then creates *external warehouse inbound shipment order requests* from the shipping journal. As a result, inbound shipment order messages are delivered to the *USSW* legal entity.
2. *USSW:* The warehousing entity processes the inbound shipment order messages. As a result of this processing, *inbound shipment orders* are created.
3. *USSW:* Inbound loads are created manually, automatically, or through import (depending on your configuration). The rest of the processing in the *USSW* proceeds as described in Inbound example process (shared warehouse).
4. *USSW:* The warehousing entity processes receiving completed operations for the related loads. These operations update the load status to *Received* and generate *external warehouse inbound shipment order updates* for *USMF*.
5. *USMF:* As the external warehouse inbound shipment order updates are processed, the system creates inbound loads and shipments, and updates the related transfer order line transactions. If everything from the transfer order is received, the transfer order moves to the *Received* status.

Transfer order with shipping from and receiving to an external warehouse

It's possible to have both source and destination warehouses set as external. This scenario works the same as the previous two scenarios combined.

On-hand adjustments

The following illustration shows how on-hand inventory adjustments are handled when Warehouse management only mode is run.



Changes in on-hand inventory that are the result of inbound and outbound shipment orders are handled when *Receive external warehouse inbound shipment order update* and *Receive external warehouse outbound shipment order update* operations are processed through the message processor in the *USMF* legal entity. However, other warehouse movements, such as warehouse counting operations, must also ensure that on-hand inventory is the same between the *USSW* legal entity and any related order processing legal entities, such as *USMF*. Therefore, Warehouse management only mode can record the changes in warehouse inventory in the warehouse inventory update log (**Warehouse management** > **Inquiries and reports** > **Physical inventory reconciliation** > **Warehouse inventory update log**). This data is used to automatically create external inventory adjustments for the relevant legal entities via the *Publish warehouse inventory update log updates* process automation background process. (By default, this background process runs every 10 minutes.)

You can use the *Create external inventory adjustment journals* process (**Warehouse management** > **Periodic tasks** > **Create external inventory adjustment journals**) to generate the inventory adjustment journals that are used to update the on-hand inventory. In this way, you can keep inventory information synced between the two legal entities.

To automatically post the inventory adjustment journals, go to **Warehouse management** > **Periodic tasks** > **Post external inventory adjustment journals**. This process posts the journals that the *Create external inventory adjustment journals* process creates.

Warehouse inventory update logs

The Warehouse inventory update log (**Warehouse management** > **Inquiries and reports** > **Physical inventory reconciliation** > **Warehouse inventory update log**) collects all the inventory

transaction updates that lead to on-hand updates that are of interest for the related legal entities. For example, you might want to handle information about inventory status changes.

> **Tip**
>
> For example, if you want to adjust the inventory in the warehouse only mode legal entity without affecting the order legal entity or causing duplicate adjustments through the warehouse inventory log update process, you can use an Inventory journal name with the *Exclude from warehouse inventory update logs* setting turned on. This prevents the journal posting from being logged.

> **Important**
>
> For source systems that are related to external warehouse management systems, keep the **Enable warehouse inventory update logs** option turned off for inbound and outbound shipment orders. In this way, you prevent inventory adjustment journal processing from updating your on-hand inventory in the sales subsidiary legal entities.

Example setup that uses external shared warehouse processing in Supply Chain Management

This section provides an example that shows how to set up and use Warehouse management only mode with external shared warehouse processing in Supply Chain Management.

Legal entities and source systems

To use Warehouse management only mode as described in this article, you must have at least two legal entities. For this example, the two legal entities are named as *USMF* and *USSW*.

In the *USSW* legal entity, you must create a source system to handle shipment orders and the inventory on-hand update processes. For this example, this source system is named *SS-USMF*.

> **Important**
>
> When you use Warehouse management only mode with external shared warehouses, your source system must be set up to use an outbound shipment processing policy that has

its **Enforce shipment to order matching** option set to *Yes*. The behavior of this setting varies slightly based on the version of Supply Chain Management you're using.

- Supply Chain Management version 10.0.46 and later – When you confirm a shipment, the policy instructs the system to update the delivery remainder on the source order line to reflect the quantity that was actually shipped. If you later reverse the shipment the shipment, the delivery remainder is restored to the original ordered quantity.
- Supply Chain Management version 10.0.45 and earlier – If the shipped quantity differs from the ordered quantity, you must manually update the delivery remainder to match the shipped quantity before you confirm the shipment. Otherwise, the update fails on the *USMF* legal entity.

In the *USMF* legal entity, you must set up an external warehouse management system of the *Legal entity* type and link it to the *SS-USMF* source system in the *USSW* legal entity. You can complete this setup by going to **Warehouse management** > **Setup** > **Warehouse management integration** > **External warehouse management systems**.

In the *USMF* legal entity, you can now go to **Warehouse management** > **Setup** > **Warehouse** > **Warehouses** and select the warehouses that you want to handle externally. Specify the external warehouse management system and the external warehouse name from the *USSW* legal entity. You should select a default location that doesn't use license plates. This location is used for all inventory changes from the external warehouses in the *USMF* legal entity.

> **Note**
>
> The **Use warehouse management processes** option must be turned on for the warehouses in both legal entities.

Product master and reference data

Warehouse management must be fully set up in all legal entities that run warehouse management processes (such as *USSW*). This setup includes product master data. The same products and product variants can be used for the released products in all legal entities. In this case, you must use a separate source system as the source for managing product master data. If you share products among multiple sales subsidiaries or legal entities, you must create the necessary source system item data. For example, you might create a source system named *PIM-D365* and assign it as the *product master source system* for the *SS-USMF* source system.

You must also ensure that the products have a tracking dimension group where the *Owner* dimension is enabled. In this way, the ownership of the inventory can be tracked for each of the related legal entities. For this example setup, you don't have to enable the *Owner* tracking dimension for the released product in the *USMF* legal entity. However, if you enable it, the legal entity owner dimension value is automatically assigned to the purchase and sales order lines, and you can't change it.

You can find more details about the product master data in Master and reference data. However, be sure to use the *Non-valuated* inventory model for the products in the *USSW* legal entity. In addition, be sure to enable the *Owner* inventory tracking dimension.

> **Note**
>
> The products must be linked to a storage dimension group where the **Use warehouse management processes** option is turned on.

To use the *Owner* dimension, you must create a record on the **Warehouse inventory owner** page (**Warehouse management** > **Setup** > **Warehouse management integration** > **Warehouse inventory owner**). Learn more in Warehouse inventory owner.

If you use this example setup in an environment where you already have some released products, ensure that you create the data for the released products and variants on the **Source system items** page.

> **Tip**
>
> Set the **Product maintenance rule** value to *Maintain source system items* on the *PIM-D365* source system in the *USSW* legal entity. In this way, the source system item data is automatically created for you when you release products to the *USSW* legal entity.

Exchange data between systems

Warehouse management only mode requires that you set up integration between external systems and the Microsoft Dynamics 365 Supply Chain Management system. The following categories of interactions are required:

- Master data (such as product information)
- Document data (such as purchase orders and sales orders)
- Progress data (such as receiving, dispatch, and on-hand inventory information)

You can use many different integration methodologies for these three categories. This article describes the recommended integration process.

Master and reference data

To keep communication consistent, both systems need to sync and have access to several types of master and reference data. One example is the product master data. You can import this type of data into Supply Chain Management by using the following messages that relate to product master data:

- `WHSSourceSystemProductMessageEntity` – Create products and released products, including product masters for variants.
- `WHSSourceSystemProductVariantMessageEntity` – Create variants for product masters where **ProductSubtype** = *ProductMaster*.
- `WHSSourceSystemProductSpecificUnitOfMeasureConversionMessageEntity` – Create product-specific unit-of-measure conversions.
- `WHSSourceSystemProductBarcodeMessageEntity` – Create the product bar code setup.
- `WHSSourceSystemProductGlobalTradeItemNumberMessageEntity` – Create the Global Trade Item Number (GTIN) for the products.
- `WHSSourceSystemProductDocumentAttachmentMessageEntity` – Attach product documents, product images, and so on.

> **Tip**
>
> When you import products, use **record templates** to include the **TemplateName** value in your messages. You can also make sure that the required reference fields for the released products are assigned.

Like shipment orders, the system validates these messages during message processing. The process automatically links the product information to a source system record through the *Source system items* entity. The external system can use business events to monitor how the status of messages changes during message processing.

You can mark only one source system record as the external system that maintains the product master data related to the unique reference for a released product or item number. Use the **Source system items** page to view and maintain this data.

> **Note**
>
> The message processor handles each message related to creating product master data separately, based on its `MessageId`. Some messages have dependencies, such as the requirement to create the released product before giving it a barcode. If you use number sequences instead of external item numbers for products, then `SourceSystemProductMessages` processes create new products when no released product or source system item data exists.

> **Tip**
>
> Use the **Source system item number** field during communication between the systems. It's useful when, for example, an external system uses a European Article Number (EAN) bar code as the unique identification number that's linked to an **Item/variant number** field that has a different value. The *Source system item number* data is automatically created when you use the previously listed messages.
>
> When you use the Warehouse Management mobile app, you can use the **Source system item number** value to look up the **Item/variant number** value that's used internally.

You can import the required master data into Supply Chain Management by using data entities. The following types of master and reference data are required to create the **Release product/Item number** value that's used in warehouse management processes:

- **Item model groups** – Each released product must be assigned to an item model group in Supply Chain Management. Therefore, at least one group must be available. The group can control business processes for batch-tracked items. The following settings are recommended for every item model group that's used with Warehouse management only mode. These settings eliminate the need to set up any costing data for the products.
  - **Inventory model** – Set this field to *Non-valuated*.
  - **Post physical inventory** – Turn off this option. You can select this option only if you've already set up at least one source system record.
  - **Post financial inventory** – Turn off this option. You can select this option only if you've already set up at least one source system record.
- **Item groups** – Use these groups to organize business processes, especially when you use product filter codes. No account setup is required when you use *Non-valuated* inventory model groups.
- **Storage dimension groups** – Enable the use of storage inventory dimensions such as sites, warehouses, locations, and license plates. Be sure to enable the **Use warehouse management processes** parameter.
- **Tracking dimension groups** – Enable the use of tracking inventory dimensions such as owner, batch, and serial numbers. Note that the *Owner* dimension value must equal the company that a warehouse is associated with. Learn more in Unsupported processes.
- **Reservation hierarchy** – Defines which dimensions are reserved during the outbound shipment order reservation process. Dimensions that are put below the *Location* dimension are controlled by the warehouse management processes.

- **Units** – Each quantity that a warehouse process handles must be associated with a unit. When multiple units (such as each, box, and pallet) are used for an item, define the *inventory unit* as the smallest unit for the item.
- **Unit sequence groups** – Define the sequence of units that can be used in warehouse operations. For more information about the required setup, see Unit of measure and stocking policies.

> **Note**
>
> The messages that create product master data use the ***product data entities***. You can use these entities by themselves, or you can use them to maintain product master data.

View and maintain source system product messages

In Warehouse management only mode, you can view, update, and create product messages. This capability lets you quickly test integrations during the implementation process. When an externally created message is in a *Failed* message state, you can update field values and assign the updated message back into the message queue. The original message is versioned and non-editable. To view and maintain the messages, go to one of the following pages:

- **Warehouse management** > **Source system products** > **Source system product messages**
- **Warehouse management** > **Source system products** > **Source system product variant messages**
- **Warehouse management** > **Source system products** > **Source system product barcode messages**
- **Warehouse management** > **Source system products** > **Source system product document attachment messages**
- **Warehouse management** > **Source system products** > **Source system product global trade item number messages**
- **Warehouse management** > **Source system products** > **Source system product specific unit of measure conversion messages**

The **Warehouse integration monitoring** workspace lets you track the number of source system product messages that are *Queued* and *Failed*.

> **Note**
>
> You can set each source system to either allow or prevent users from manually creating messages on the listed pages. Open the relevant source system record, and set the **Enable manual source system product message creation** option to *Yes* to allow manual messages or *No* to prevent them. Unlike messages that are imported via integration, manually created messages aren't versioned.

Source system inventory status mapping

When you use Warehouse management only mode with external shared warehouses, inventory statuses in your external source system might have different identifiers than the inventory

statuses that you set up in Supply Chain Management. The *source system inventory status mapping* feature lets you create a mapping between your internal inventory status values and the values that your source systems use. The system then automatically translates inventory status values during inbound and outbound shipment order processing, including when inventory status is applied during order registration.

This mapping works in a similar way to source system items. While source system items map item numbers between systems, inventory status mapping maps inventory status identifiers between systems.

Prerequisites

To use source system inventory status mapping, you must be running Supply Chain Management version 10.0.49 or later.

Set up inventory status mapping

To set up the mapping between source system inventory statuses and Supply Chain Management inventory statuses, follow these steps:

1. Do one of the following:
    - Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source system inventory statuses**.
    - Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**. Select the source system that you want to set up inventory status mapping for, and then select **Source system inventory statuses** on the Action Pane.
2. On the Action Pane, select **New** to create a new mapping record.
3. Set the following fields:
    - **Source system** – Select the source system that the mapping applies to.
    - **Inventory status** – Select the Supply Chain Management inventory status that this record maps to.
    - **Source system inventory status** – Enter the inventory status identifier that the external source system uses.
4. Repeat the previous steps for each inventory status that you need to map for each source system.

> **Note**
>
> Each combination of source system and inventory status must be unique. Likewise, each combination of source system and source system inventory status must be unique. This uniqueness ensures that the mapping can reliably convert values in both directions.

Import inventory status mapping by using data entities

You can import the inventory status mapping data by using the `WHSSourceSystemInventoryStatusEntity` data entity. Use the Data management framework to

import records in bulk. The entity exposes the following fields:

Expand table

Field	Description
`SourceSystemId`	The source system identifier.
`InventStatusId`	The Supply Chain Management inventory status ID.
`SourceSystemInventoryStatusId`	The inventory status identifier used by the external source system.

How the mapping is applied

The system applies inventory status mapping in the following scenarios:

- **Inbound shipment order updates** – When an external warehouse sends an update that includes an inventory status (for example, as part of a receiving completed operation), the system uses the mapping to convert the source system inventory status to the corresponding Supply Chain Management inventory status. The mapped value is then used to register the purchase order or transfer order line transaction. Learn more in Inventory status in inbound shipment order update registration.
- **Outbound shipment order updates** – When an external warehouse ships inventory and generates a packing slip, the transaction data that's sent to the source system includes inventory status information. The system converts the Supply Chain Management inventory status to the corresponding source system value in the update message.
- **Inbound and outbound shipment order requests** – When shipment order request data includes inventory status dimensions, the system converts between source system and Supply Chain Management values, depending on the direction of the message.

If the system doesn't find a mapping for a given inventory status value, it processes the message without applying an inventory status conversion.

Inventory status in inbound shipment order update registration

In Supply Chain Management version 10.0.49 and later, when an external warehouse sends an inbound shipment order update that includes inventory status information, the system applies that status during registration processing in the source system legal entity. This process means that the registered purchase order or transfer order line transaction uses the inventory status from the external warehouse update, rather than the inventory status that was originally specified on the order line.

This behavior is important in scenarios where the external warehouse assigns a different inventory status during receiving. For example, if a purchase order line has an inventory status of *Available*, but the external warehouse determines during inspection that the goods should be marked as *QualityInspection*, the inbound shipment order update from the external warehouse includes the *QualityInspection* status. When the update is processed, the system registers the purchase order line with the *QualityInspection* status (after applying source system inventory status mapping, if configured).

> **Important**

When using this feature with source system inventory status mapping, ensure that you configure all inventory statuses that the external warehouse might send back in updates in the mapping table. If the external warehouse sends an inventory status that isn't mapped, the system uses the value as-is without conversion.

The following example shows the end-to-end flow:

1. A purchase order line in the *USMF* legal entity has inventory status *Available*.
2. The order is released to the external warehouse (*USSW*) via an inbound shipment order request.
3. In *USSW*, warehouse workers receive the goods and assign inventory status *QualityInspection* during the receiving process.
4. Receiving is completed and an inbound shipment order update is sent back to *USMF*. The update includes inventory status *QualityInspection* (or its source system equivalent, if mapping is configured).
5. *USMF* processes the update. If source system inventory status mapping is configured, the system maps the source system status to the Supply Chain Management status. The purchase order line transaction is registered with the mapped inventory status.

> **Note**
>
> The inventory status field is visible on the **External warehouse inbound shipment order request** and **External warehouse outbound shipment order request** pages when the source system inventory status mapping feature is enabled. The `WHSShipmentReceiptTransactionInventoryDimensionV2Entity` and `WHSShipmentPackingSlipTransactionInventoryDimensionV2Entity` entities also expose the **SourceSystemInventoryStatus** field, which shows the mapped source system inventory status value.

Consigner and consignee information

To make it easier to set up your warehouse operation, create and use data for *consigners* and *consignees* and their related group definitions. For example, use this approach for a process that's related to setting up a quality order creation process for a specific consigner or consigner group.

Neither the *Inbound shipment order policies* (which are part of the **Source systems** setup) nor inbound shipment order message processing requires that the fields for the **Consigner's account number** value exist in the entity for the **Consigners** page (**Warehouse management** > **Setup** > **Warehouse management integration** > **Consigners**). The same "free text" concept exists for the outbound shipment order process that's related to the **Consigner's account number** value.

Country/region

To create a new legal entity for your warehouses and import outbound shipment orders, you must define **country/region** values in Supply Chain Management. Use these records in outbound shipment orders to create addresses. Depending on your address setup and the way that you use address fields in order messages, you might have to create additional data before you can import order messages (for example, to support state/province and county combinations).

Inbound and outbound shipment order messages

Use inbound and outbound shipment order messages to inform Supply Chain Management about which physical inventory to receive and ship. These messages include both header data and lines data.

- `WHSInboundShipmentOrderMessageEntity` – Create inbound shipment order header.
- `WHSInboundShipmentOrderLineMessageEntity` – Create inbound shipment order lines.
- `WHSOutboundShipmentOrderMessageEntity` – Create outbound shipment order header.
- `WHSOutboundShipmentOrderLineMessageEntity` – Create outbound shipment order lines.

> **Note**
>
> - You can update an inbound or outbound shipment order by submitting a message that includes a matching order number and source system in the header data.
> - You can delete an inbound or outbound shipment order by submitting a message that includes a matching order number and source system in the header data and omits all line data.

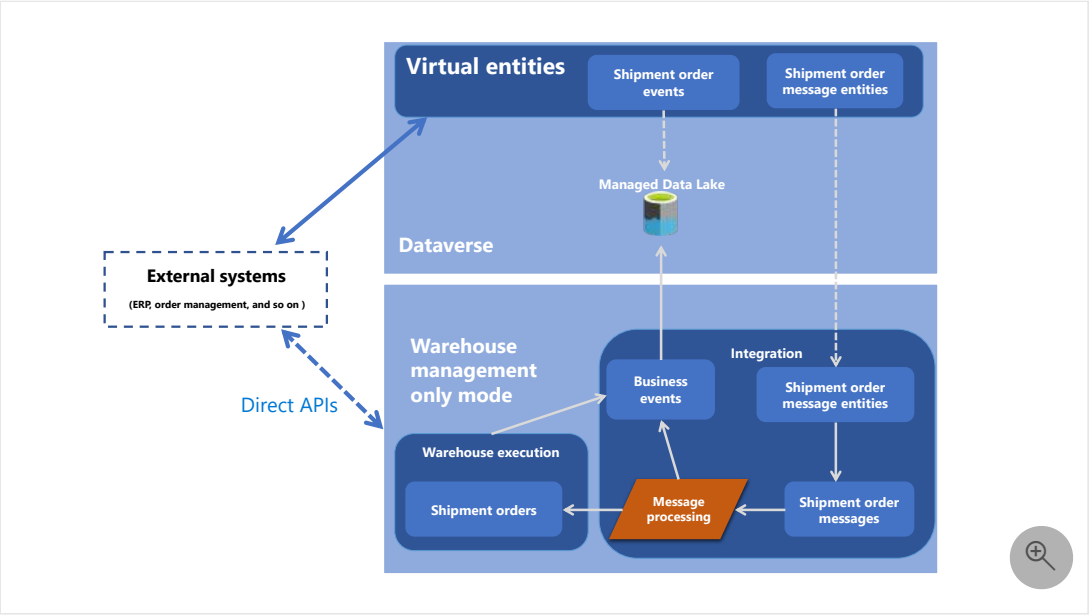
Messages between systems are exchanged by using lightweight *inbound shipment order* and *outbound shipment order* documents. These documents eliminate the need to use several other types of documents that Supply Chain Management typically uses (such as sales orders, purchase orders, and transfer orders). Therefore, they have several benefits. For example, they simplify integration with enterprise resource planning (ERP) and order management systems. They also make Supply Chain Management warehouse management functionality available to a wide range of external ERP and order management systems.

You can exchange inbound and outbound shipment order messages by using Dataverse. Alternatively, you can exchange these messages through Open Data Protocol (OData) by using shipment order message entities. You can also use the Data management import process. For example, you can use the *Inbound shipment order messages composite entity* (`WHSInboundShipmentOrderMessageCompositeEntity`) and *Outbound shipment order messages composite entity* (`WHSOutboundShipmentOrderMessageCompositeEntity`).

You can attach documents to inbound and outbound shipment order messages by using the *Inbound shipment order message document attachment entity* (`WHSInboundShipmentOrderMessageDocumentAttachmentEntity`) and *Outbound shipment order message document attachment entity* (`WHSOutboundShipmentOrderMessageDocumentAttachmentEntity`).

Supply Chain Management queues the incoming documents and then processes them by using the message processor. This approach ensures consistent data between the systems, both for master data (such as products) and order progress status. Therefore, Supply Chain Management inbound and outbound shipment orders are prevented from creating or updating nonvalid or unsupported order data. Process the messages as part of a periodic batch job that the message processor triggers by using the *Shipment orders* message queue.

The following illustration shows how the message processor fits into an integrated system.



Progress data and business events

External systems can send many different business process requests to the warehouse management system. For example, each external system can continuously check for the progress of a sales order. To support this process, you can set up Supply Chain Management to deliver business events as required. Business events keep external systems informed about the progress and actions that are occurring in Supply Chain Management. When you set up this process, the external systems don't have to continue to check for information that might not have changed since the last request. Instead, they can react only when they're informed.

Warehouse integrations support several out-of-the-box business events. The following table lists some of them.

Some business events don't include all journal details directly. In these cases, the event contains only the journal ID. You can use this ID to retrieve the full information through related data entities listed in the following table.

Expand table

Business event ID	Description	Data entities
`WHSSourceSystemProductMessageChangedStatusBusinessEvent`	Source system product message changed status	
`InventCountingJournalPostedBusinessEvent`	Counting journal posted	`InventInventoryCountingJournalHeaderEntity` `InventInventoryCountingJournalHeaderLineEnti` `InventInventoryCountingJournalNameEntity`
`WHSSourceSystemInventoryOnhandReportBusinessEvent`	Source system on-hand inventory report created	`WHSInventoryOnhandReportEntity` `WHSSourceSystemInventoryOnhandReportLineEnti`

Business event ID	Description	Data entities
`WHSInventoryUpdateLogBusinessEvent`	Warehouse inventory update log updated	`WHSSourceSystemItemInventoryUpdateLogEntity`
`WHSOutboundNotificationCreatedBusinessEvent`	Outbound warehouse notification created	`WHSOutboundNotificationEntity` `WHSOutboundShipmentNotificationEntity` `WHSOutboundShipmentLineNotificationEntity`
`WHSShipmentOrderMessageChangedStatusBusinessEvent`	Shipment order message status updated	
`WHSShipmentPackingSlipJournalModifiedBusinessEvent`	Shipment packing slip updated	`WHSShipmentPackingSlipJournalHeaderEntity` `WHSShipmentPackingSlipJournalLineEntity` `WHSShipmentPackingSlipTransactionInventoryDi`
`WHSShipmentPackingSlipJournalFailedBusinessEvent`	Shipment packing slips update failed	
`WHSShipmentReceivingJournalModifiedBusinessEvent`	Shipment receipts updated	`WHSShipmentReceiptJournalHeaderEntity` `WHSShipmentReceiptJournalLineEntity` `WHSShipmentReceiptTransactionInventoryDimens` (Supply Chain Management version 10.0.45 and `WHSShipmentReceiptTransactionInventoryDimens` (Supply Chain Management version 10.0.46 and
`WHSShipmentReceivingJournalFailedBusinessEvent`	Shipment receipts update failed	
`SysMessageProcessorMessageProcessedBusinessEvent`	Message processor message failed	
`WhsWaveExecutedBusinessEvent`	Wave executed	
`WHSQualityOrderValidatedBusinessEvent`	Quality order validated	
`WHSEWInboundShipmentOrderRequestCreatedBusinessEvent`	Inbound shipment order request created (can be used to integrate Supply Chain Management with another warehouse management system)	`WHSEWInboundShipmentOrderRequestEntity`, `WHSEWInboundShipmentOrderLineRequestEntity` `WHSEWInboundShipmentOrderRequestStatusEntity`
`WHSEWOutboundShipmentOrderRequestCreatedBusinessEvent`	Outbound shipment	`WHSEWOutboundShipmentOrderRequestEntity` `WHSEWOutboundShipmentOrderLineRequestEntity`

Business event ID	Description	Data entities
	order request created (can be used to integrate Supply Chain Management with another warehouse management system)	`WHSEWOutboundShipmentOrderRequestStatusEntit`
`WHSEWInboundShipmentOrderUpdateChangedStatusBusinessEvent`	Inbound shipment order update is being processed and has therefore changed status (can be used to integrate Supply Chain Management with another warehouse management system)	
`WHSEWOutboundShipmentOrderUpdateChangedStatusBusinessEvent`	Outbound shipment order update is being processed and has therefore changed status (can be used to integrate Supply Chain Management with another warehouse management system)	

At a minimum, use the following business events for integration with an external ERP system:

- `InventCountingJournalPostedBusinessEvent` – Announces that an on-hand inventory adjustment occurred and indicates where detailed information about the update can be found.
- `WHSSourceSystemInventoryOnhandReportBusinessEvent` – Announces that an on-hand inventory report was generated and indicates where detailed information about the update can be found.
- `WHSShipmentPackingSlipJournalModifiedBusinessEvent` – Announces that an outbound shipment confirmation process occurred and indicates where the detailed dispatch advice

data can be found. This data can be used for a sales invoicing process, for example.

- `WHSShipmentReceivingJournalModifiedBusinessEvent` – Announces that an inbound receiving completion process occurred and indicates where the detailed receiving advice data can be found. This data can be used for a purchase order invoicing process, for example.

On-hand adjustments

When you integrate an ERP system and a warehouse management system, keep on-hand inventory data aligned. Several processes can help maintain this alignment as part of the Warehouse management only mode implementation approach. For more information about how the inventory on-hand update process works, see On-hand inventory updates between systems.

Warehouse management only mode with external shared warehouses

You can set up an integration between Supply Chain Management and an external warehouse management system. To enable this integration, use external warehouse shipment order requests and updates for communication between the systems. Learn more in Warehouse management only mode with external shared warehouses.

Inbound and outbound shipment order requests

Use external warehouse inbound and outbound shipment order requests to import orders from Supply Chain Management into your warehouse management system. When you release an order to the warehouse, the system creates a business event with a request ID. Use this ID to retrieve full details through related data entities.

Expand table

Business event ID	Description	Data entities

`WHSEWInboundShipmentOrderRequestCreatedBusinessEvent`	Inbound shipment order request created	`WHSEWInboundShipmentOrderRequestEntity`
		`WHSEWInboundShipmentOrderLineRequestEntity`
		`WHSEWInboundShipmentOrderRequestStatusEntity`

`WHSEWOutboundShipmentOrderRequestCreatedBusinessEvent`	Outbound shipment order request created	`WHSEWOutboundShipmentOrderRequestEntity`
		`WHSEWOutboundShipmentOrderLineRequestEntity`
		`WHSEWOutboundShipmentOrderRequestStatusEntity`

Inbound and outbound shipment order updates

Use external warehouse inbound and outbound shipment order updates to send shipment updates from your warehouse management system to Supply Chain Management. The message processor handles these updates in the same way as inbound and outbound shipment order messages.

- `WHSEWInboundShipmentOrderUpdateEntity`
- `WHSEWInboundShipmentOrderLineUpdateEntity`

- `WHSEWInboundShipmentOrderLineInventDimUpdateEntity`
- `WHSEWOutboundShipmentOrderUpdateEntity`
- `WHSEWOutboundShipmentOrderLineUpdateEntity`
- `WHSEWOutboundShipmentOrderLineInventDimUpdateEntity`

Unannounced sales returns from another warehouse management system

If your warehouse management system receives an unannounced sales return, it must update the Supply Chain Management system accordingly. To achieve this goal, send an external warehouse inbound shipment order update message to the Supply Chain Management system. Ensure that you include all required details and entities so the return update can be processed correctly.

- `WHSEWInboundShipmentOrderUpdateEntity`
  - Set the value of the `InboundShipmentOrderOriginLinkType` field to *1*, which represents the `ExternalOrigin` enum value of the `WHSInboundShipmentOrderOriginLinkType` enum.
- `WHSEWInboundShipmentOrderLineUpdateEntity`
- `WHSEWInboundShipmentOrderLineInventDimUpdateEntity`
- `WHSEWInboundShipmentOrderExternalOriginEntity`
  - This entity provides additional required information.
  - The value of the `InboundShipmentOrderExternalOriginId` field must match the `InboundShipmentOrderOriginId` field on the `WHSEWInboundShipmentOrderUpdateEntity` entity. You can choose and maintain this ID.
  - Set the `ExternalWarehouseManagementSystemId` field to the external warehouse management system ID configured in your Supply Chain Management system.

Last updated on 07/23/2026

Enable and configure Warehouse management only mode

This article explains how to configure Warehouse management only mode by setting up source systems, master data, and business events.

Turn on Warehouse management only mode

To use the Warehouse management only mode capability, your system must meet the following requirements:

- You're running Microsoft Dynamics 365 Supply Chain Management version 10.0.36 or later.
- The feature named *Warehouse management only mode* is turned on in feature management. As of Supply Chain Management version 10.0.47, this feature is mandatory and can't be turned off.

Configure your source systems

Use the **Source systems** page to set up each external system that you want to use Warehouse management only mode to integrate with. To use Warehouse management only mode, you must have at least one record on this page. To set up a source system, follow these steps:

1. Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**.

2. Follow one of these steps:

    - To create a new record, select **New** on the Action Pane.
    - To edit an existing record, select it in the list pane, and then select **Edit** on the Action Pane.
    - To delete an existing record, select it in the list pane, and then select **Delete** on the Action Pane.

3. On the header of the new or selected record, enter a name for the source system. The source system must include this name in each message that it sends to Supply Chain Management, and the name in the message must exactly match the value that you specify here.

4. Configure other settings as required to control the shipment order import processes. Tooltip help is provided for each field. Hover your mouse pointer over a field's label to view the tooltip.

5. If you use Warehouse management only mode with external shared warehouses, then on the **Outbound shipment orders** FastTab, you must select an **Outbound shipment processing policy** that has its **Enforce shipment to order matching** option set to *Yes*. Learn more in Outbound shipment processing policies.

6. (Optional) If your external source system uses different inventory statuses than Supply Chain Management, select **Source system inventory statuses** on the Action Pane to set up the mapping. Learn more in Source system inventory status mapping.

7. (Optional) Warehouse management processes (WMS) can use an *Owner* inventory dimension to track the ownership of inventory for items shared across multiple source systems. To use the owner inventory dimension, products must have a tracking dimension group where the *Owner* dimension is enabled, and you must also create a *Warehouse inventory owner* configuration that contains a mapping for every *Owner* value that a source system sends in an order line. If a mapping is missing, the order message fails to process. This requirement also applies when the source system sends an empty owner value. In such cases, the configuration must map the empty value from the source system to the appropriate value in the *WOM* Supply Chain Management legal entity system. To create a warehouse inventory owner configuration, select **Warehouse inventory owners** on the Action Pane. Learn more in Product master and reference data.

8. If another source system maintains the product master data, specify that source system in the **Product master source system** field. To view and manually maintain the product master data mapping integration, select **Source system items** on the Action Pane.

9. Use the **Source system items** FastTab to define how the product creation message processing must create the internal item numbers.

10. On the Action Pane, select **Save**.

> **Note**
>
> You can't find or search for the menu items that are related to Warehouse management only mode until you create at least one record on the **Source systems** page. In addition,

searching for pages requires that the current user's default company has a **Source system** record.

Background processes

As part of the Source system creation process, the system automatically creates the following automated background processes:

- **Process source system product messages** – Processes product master data messages. By default, the repeat interval is one minute.
- **Process shipment order messages** – Processes inbound and outbound shipment order messages. By default, the repeat interval is one minute.
- **Publish warehouse inventory update log updates** – Makes inventory update log data available to external systems via the `WarehouseInventoryUpdateLogs` entity. By default, the repeat interval is 10 minutes.
- **Post shipment packing slips** – Finalizes outbound shipment orders. By default, the repeat interval is 10 minutes if the **Shipment packing slips posting delay** value is set to one day.
- **Post shipment receipts** – Finalizes inbound shipment orders. By default, the repeat interval is 10 minutes if the **Shipment receipts posting delay** value is set to one day.

Prevent multiple owners in warehouse locations

If you're running warehouses that contain inventory from multiple owners, use the *prevent multiple owners* feature to ensure that all the items stored in a given warehouse location have the same owner. The feature prevents cross-mingling of inventory and allows each owner to have its own segregated storage space. Learn more in Prevent multiple owners in warehouse locations.

Set up automatic release of outbound shipment orders

Release to warehouse is the process of making inventory ready for dispatch processing. When you release an order to the warehouse, the system creates load lines and shipments. If automatic wave processing is set up, loads and required work are also created.

To automatically release outbound shipment orders to the warehouse, set up a batch job to process the orders on a regular basis. Go to **Warehouse management** > **Release to warehouse** > **Automatic release of outbound shipment orders** to set up and schedule the job.

For more information about the *release to warehouse* process, see Release to warehouse.

Set up number sequences

If the order numbers that the external system provides don't match the order numbers used in Supply Chain Management, set up number sequences for outbound shipment orders and inbound shipment orders. Define all the number sequences that are associated with the **Number sequences** page (**Warehouse management** > **Setup** > **Warehouse management parameters** > **Number sequences**). These number sequences include the following extra number sequences that are added for Warehouse Management only mode:

- *Outbound shipment order*
- *Inbound shipment order*
- *Shipment packing slip*
- *Shipment receipt*
- *Warehouse outbound notification ID*
- *Load line inventory pick*
- *Internal consigner's account number*
- *Internal consignee's account number*
- *External warehouse outbound shipment order origin ID*
- *External warehouse inbound shipment order origin*
- *External warehouse outbound shipment order update ID*
- *External warehouse inbound shipment order update*

> **Tip**
>
> You can quickly enable all number sequences by using the **Generate** option on the **Number sequences** page.

Set up mobile device menu items

To enable warehouse workers to use the Warehouse Management mobile app to register inbound shipment transactions, set up one or more mobile device menu items that use one of the following processes. Each of these processes requires an inbound load.

- *License plate receiving (and put away)*
- *Load item receiving (and put away)*
- *Mixed license plate receiving (and put away)*
- *Inbound shipment order line receiving*
- *Inbound shipment item receiving*

- *Inbound shipment order line receiving (and put away)*
- *Inbound shipment order item receiving (and put away)*

For more information, see Set up mobile devices for warehouse work.

Set up master data and business events

To enable Supply Chain Management to exchange meaningful information with an external system, sync some master and reference data between the two systems. This data includes released products, item model groups, and countries/regions. You must also set up business events to manage the exchange of information between the two systems. For more information, see Exchange data between systems.

Last updated on 07/23/2026

Enable and configure unannounced returns in Warehouse management only mode

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article explains how to set up Microsoft Dynamics 365 Supply Chain Management to handle unannounced returns when you use Warehouse management only mode. Most aspects of the unannounced returns process work in the same way, regardless of whether you use Warehouse management only mode. This article highlights the differences. Learn more about the unannounced returns process and how to set it up in Receive unannounced sales returns.

Prerequisites

Before you can use the features that are described in this article, your system must meet the following requirements:

- You must be running Supply Chain Management version 10.0.41 or later.

Set up source systems

To use Warehouse management only mode, you must have at least one *source system*. Each source system record configures the way that a specific external system integrates with Supply Chain Management in Warehouse management only mode.

In Configure your source systems, you can find general information about how to set up source systems so that they can be used with Warehouse management only mode. The following procedure highlights the settings that are important for unannounced returns.

1. Go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**.
2. In the list pane, select the source system that you want to set up. Alternatively, create a new source system

3. On the **Inbound shipment orders** FastTab, set the following fields to configure the options for unannounced returns:

    - **Enable returns process** – Set this option to *Yes*.
    - **Number sequence code** – Select the number sequence that should be used to generate order numbers for inbound shipment orders that are created during the returns process.
    - **Return order type** – Enter the name that identifies the document type for return orders in the source system. To find the value, consult your external system's documentation.

    > **Important**
    >
    > If your source system is Dynamics 365 Supply Chain Management (as is the case when you run an external shared warehouse scenario), you must set the **Return order type** field to *Return*, which is the return order type that is used in Supply Chain Management.

4. On the **Outbound shipment orders** FastTab, set the following fields to configure the options for unannounced returns:

    - **Outbound shipment processing policy** – Select a policy where the **Enforce shipment to order matching** option is set to *Yes*. Learn more in Outbound shipment processing policies.
    - **Enable return details creation** – If you use return details receiving, set this option to *Yes*. If you use only blind returns, you can save system resources by setting this option to *No*.

Set up return item receiving policies and mobile device menu items

To enable workers to process unannounced returns in the warehouse, you must set up mobile device menu items and return item receiving policies. The required settings are almost the same as the settings for receiving unannounced returns without Warehouse management only mode. This section points out the differences.

Return item receiving policies

Return item receiving policies enable each menu item to process the return correctly. You set them up by going to **Warehouse management** > **Setup** > **Mobile device** > **Return item**

receiving policies. Policies that are used with Warehouse management only mode must have the following settings:

- **Return process** – Select either *Blind return* or *Return details*, depending on the type of return that your menu item should process.
- **Create return order** – Select *Inbound shipment order*.
- **Return order identification** – This field is available only when the **Return process** field is set to *Blind return*. Select the type of extra identifying information that the system should return when it registers a return order with your external system:
  - *None* – The system doesn't return any extra identifying information, and the app doesn't prompt the worker to enter any additional information during receiving.
  - *Account number* – When Supply Chain Management registers a return with your external system, it returns an account number to help identify the return order. The way that you use this value depends on the needs of your external system. (See the documentation for your external system.) It's typically the ID of the customer who is returning the items (their customer account number). The mobile app prompts the worker to scan or enter this number during receiving.
- **Return order type** – This field is available only when the **Return order identification** field is set to *Account number*. Like the account number, the value in this field is returned to the source system when a return is registered with your external system. It's intended to hold the name that identifies the document type for return orders in the source system. The way that you use it depends on the needs of your external system. (See the documentation for your external system.) If you leave this field blank, the value is inherited from the relevant source system configuration.

> **ⓘ Important**
>
> If your source system is Dynamics 365 Supply Chain Management (as is the case when you run an external shared warehouse scenario), you must set the following values for your blind return policies:
>
> - **Return order identification** – Select *Account number*, and instruct workers to enter the customer ID (account number) when the mobile app prompts them during receiving.
> - **Return order type** – Enter *Return*, which is the return order type that is used in Supply Chain Management. (Alternatively, if *Return* is already specified as the default return order type for the source system, leave this field blank.)

Learn more in Create return item receiving policies.

Mobile device menu items

To create the mobile device menu items that you need, follow the instructions in Set up mobile device menu items to process unannounced returns. Be sure to configure each menu item so that it uses a return item receiving policy that was set up as described in the previous section.

Last updated on 09/10/2025

Work with Warehouse management only mode in Supply Chain Management

Article • 04/30/2024

This article explains how to use Warehouse only mode to perform day-to-day warehousing tasks.

Monitor the integration

To monitor the integration between the external systems or legal entities with Microsoft Dynamics 365 Supply Chain Management, go to **Warehouse management** > **Workspaces** > **Warehouse integration monitoring**. From the **Warehouse integration monitoring** page, you can perform the following tasks:

- Get an overview of integration messages.
- Go to pages that have related information and functionality, such as the **Message processor messages** page.

Maintain inbound shipment orders

To review and maintain your inbound shipment orders, go to **Warehouse management** > **Inbound shipment orders** > **Inbound shipment orders**. The documents that are listed on the **Inbound shipment orders** page provide information about the expected product receipts. The page provides a **Header** view and a **Lines** view of the available shipment orders as seen from Supply Chain Management. For each line, you can view detailed information about the receiving status, inventory transactions, and associated warehouse work. You can also view the receiving status on the order header and lines, and use it to follow the inbound progress of the orders.

The following **Receiving status** values are available:

- *Open* – No quantities have been fully received.
- *Partially received* – Some, but not all, quantities have been fully received.
- *Received* – All quantities have been fully received.

If you're already familiar with Supply Chain Management, you might recognize that this document resembles a simplified purchase order. However, inbound shipment orders don't make any financial postings for the general ledger.

> ⚠ **Warning**
>
> For inbound shipment order lines, you can select **Update line** > **Delivery remainder** to update expected order line transaction quantities. Make sure that you have the correct user role security privilege for this process, because this update (like message editing) allows for potential inconsistencies between the external systems and Supply Chain Management.

> ⓘ **Note**
>
> The internal inbound shipment order number must be unique. You can configure the system to use the external order numbers as internal numbers. In this way, you don't have to use a number sequence for the order. To ensure unique numbers across external systems, consider using the **Order number prefix/suffix** options.

Enable manual inbound shipment order creation

In addition to using entity messages to create inbound shipment orders, you can enable manual inbound shipment order creation for one or more source systems. Users can then create inbound shipment orders directly on the **Inbound shipment orders** page. To enable manual inbound shipment order creation, go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**, and set the **Enable manual inbound shipment order creation** option to *Yes* for the relevant source system. This setting activates the **New** button on the **Inbound shipment orders** page. For shipment orders that are created without message processing, the **Order type** field has the value *Manual order* instead of *Inbound shipment order*.

Maintain outbound shipment orders

To review and maintain documents that contain information about products that have been requested for dispatch, go to **Warehouse management** > **Outbound shipment orders** > **Outbound shipment orders**. The **Outbound shipment orders** page provides a **Header** view and a **Lines** view of the available shipment orders as seen from Supply Chain Management. For each line, you can view detailed information about the release status, shipment status, inventory transactions, and associated warehouse work. You can also view and edit the release status and shipment status on the order header, and use them to follow the outbound progress of the orders.

The following **Release status** values are available:

- *Open* – No quantities have been released to the warehouse.
- *Partially released* – Some, but not all, quantities have been released to the warehouse.
- *Released* – All quantities have been released to the warehouse.

The following **Shipment status** values are available:

- *Open* – No quantities have been shipped confirmed.
- *Partially shipped* – Some, but not all, quantities have been shipped confirmed.
- *Shipped* – All quantities have been shipped confirmed.

If you're already familiar with Supply Chain Management, you might recognize that this document resembles a simplified sales order that uses some of the same reservation and release to warehouse processes. Use the Source systems settings to control whether you want to automatically trigger reservations when documents are imported, and/or automatically reject shipment orders that can't be partly or fully reserved.

> ⚠ **Warning**
>
> For outbound shipment order lines, you can select **Update line** > **Delivery remainder** to update expected order line transaction quantities. Make sure that you have the correct user role security privilege for this process, because this update (like message editing) allows for potential inconsistencies between the external systems and Supply Chain Management.

> ⓘ **Note**
>
> The internal outbound shipment order number must be unique. You can configure the system to use the external order numbers as internal numbers. In this way, you don't have to use a **number sequence** for the order. To ensure unique numbers across external systems, consider using the **Order number prefix/suffix** options.

Enable manual outbound shipment order creation

In addition to using entity messages to create outbound shipment orders, you can enable manual outbound shipment order creation for one or more source systems. Users can then create inbound shipment orders directly on the **Outbound shipment orders** page. To enable manual outbound shipment order creation, go to **Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**, and

set the **Enable manual outbound shipment order creation** option to *Yes* for the relevant source system. This setting activates the **New** button on the **Inbound shipment orders** page. For shipment orders that are created without message processing, the **External order type** field has the value *Manual order* instead of *Outbound shipment order*.

Receiving completed process

The *receiving completed* process updates the load status to *Received* and generates shipment receipts. The shipment receipts then trigger a business event for the external systems.

Workers can manually trigger the *receiving completed* process from the load by using either the web client or the Warehouse Management mobile app. To enable workers to use the Warehouse Management mobile app, set up a **Receiving completed confirmation** mobile device menu item. You can add this menu item as part of a detour in the inbound receiving flows.

You can specify whether workers should capture a packing slip ID and date for each shipment that's associated with an inbound load. On the **Warehouse management parameters** page, on the **Loads** tab, in the **Capture receiving completed packing slip** field, select one of the following values to specify the system behavior:

- *Never* – Don't prompt for the packing slip ID and date.
- *Always* – Prompt for the packing slip ID and date, and allow the worker to proceed only after values they're specified.
- *Optional* – Prompt for the packing slip ID and date, but allow the worker to proceed without specifying them.

> **Note**
>
> Depending on your setup on the **Inbound shipment order policies** FastTab of the **Source systems** page (**Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**), inbound loads might be created in any of the following ways:
>
> - Automatically, when an inbound shipment order is imported
> - Automatically, when an ASN is imported
> - Via a manual process
> - As part of the Warehouse Management mobile app receiving process
> - As part of an *Order receiving completed* process, which creates loads for registered inventory transactions that aren't associated with a load. This

process is useful when the item arrival journal process is used. It can be triggered directly from an inbound shipment order or via the **Inbound shipment order receiving completed** background process.

When the system creates load data as a result of processing an inbound shipment order message, the delivery policy controls whether load quantities are adjusted to the received quantities, or whether the received quantities must match the load line quantities, as part of the *Receiving completed* process.

Shipment receipts

Go to **Warehouse management** > **Inbound shipment orders** > **Shipment receipts** to view the detailed line transactions that are related to received inventory. The data is version controlled, and you can follow the posting status on the header.

The header status is changed from *Ready for posting* to *Posted* as part of the *Post shipment receipts* process automation job. This job is automatically initialized as part of the Source systems setup. The Source systems setup also lets you schedule the batch job to ensure that the related inbound shipment order line transactions are moved to a finalized transaction state for the **Warehouse management** module.

To view all the background processes that you have running, go to **System administration** > **Setup** > **Process automations**.

> ⓘ **Note**
>
> You can reverse receiving confirmations only if the related inbound shipment order line transactions haven't yet been finalized.

> ⚠ **Warning**
>
> If you enable Warehouse management only mode and are already running a periodic *Update product receipts* batch job for loads that are associated with purchase orders, you probably have to update the query for the batch job to exclude inventory transaction updates for inbound shipment orders. To update the query, add the *Load details* entity, and specify a *NotExist* join to the *Loads* entity. Then add a range definition for the **Reference** field, where **Criteria** = *Inbound shipment order*.

External systems can be informed via business events when new data is available. They can read the data via the following data entities:

- `ShipmentReceiptJournalHeaders` – The shipment receipt header data.
- `ShipmentReceiptJournalLines` – The shipment receipt line data.
- `ShipmentReceiptTransactionDimensions` – The detailed shipment receipt line data.

Shipment packing slips

To view detailed line transactions that are related to shipped inventory, and to print a report, go to **Warehouse management** > **Outbound shipment orders** > **Shipment packing slips**, and select **Preview/Print**. To control the printed inventory dimension values, go to **Warehouse management** > **Setup** > **Warehouse management parameters**, and then, on the **Reports** tab, select an option for **Shipment packing slip**.

Shipment packing slip data is version controlled, and you can follow the posting status on the header. The **Posting status** value is changed from *Ready for posting* to *Posted* as part of the *Post shipment packing slips* process automation job. This job is automatically initialized as part of the Source systems setup. The Source systems setup also lets you schedule the batch job to ensure that related outbound shipment order line transactions are moved to a finalized transaction state for the **Warehouse management** module.

To view all the background processes that you have running, go to **System administration** > **Setup** > **Process automations**.

> **Note**
>
> If no language is defined for an order, the report uses the company-specific language settings.

External systems can be informed via business events when new data is available. They can read the data via the following data entities:

- `ShipmentPackingSlipJournalHeaders` – The shipment packing slip header data.
- `ShipmentPackingSlipJournalLines` – The shipment packing slip line data.
- `ShipmentPackingSlipTransactionDimensions` – The detailed shipment packing slip line data.

> **Note**

You can reverse the shipment confirmation only if the related outbound shipment order line transactions haven't yet been finalized.

View and maintain shipment order messages

In Warehouse management only mode, you can view, update, and create shipment order messages. Therefore, you can quickly test integrations during the implementation process. When an externally created message is in a *Failed* message state, you can update all field values except the order header and line numbers. Go to one of the following pages to view and maintain the messages:

- **Warehouse management** > **Inbound shipment orders** > **Inbound shipment order messages**
- **Warehouse management** > **Outbound shipment orders** > **Outbound shipment order messages**

To try out the process of creating inbound and outbound shipment orders via messages, set the **Enable manual outbound shipment order message creation** and **Enable manual inbound shipment order message creation** options to *Yes* for the relevant **Source system** record. You can then create shipment order messages directly on the **Outbound shipment order messages** and **Inbound shipment order messages** pages. In addition, you can follow the **Message origin** value, which can be *Manual entry* or *Integration*.

Use the **Processing status** field to monitor the progress of each shipment order message. The following **Processing status** values are available:

- *Receiving* – The message is in the process of being imported.
- *Received* – The message has been received and is in a *Queued* state in the message processor. It's now ready to be picked up for processing.
- *Accepted* – The message processor state is *Processed*. Therefore, a shipment order has been created.
- *Failed* – The message processor processed the message, but one or more errors occurred. You can create a copy of the message when you save it after you edit it.
- *Draft* – The message is a manually copied or created message that can be updated. To reprocess the message, move it into the *Queued* message state by selecting **Queue** on the Action Pane.
- *Canceled* – The message has been manually canceled.

Tip

Select **Show old versions** to follow manual message updates based on the value of the **Replaced by message** field.

> ⚠ **Warning**
>
> In a production environment, manually entered messages and field updates can cause data inconsistencies between the external source system and Supply Chain Management. For example, you might change an item number to a value that isn't known to the external system. This type of update will probably cause issues in progress information flows. The issues might even be impossible to correct in the external system. We recommend that you use user roles and security privileges to restrict access to this functionality to as few people as possible.

Feedback

Was this page helpful? Yes No

Provide product feedback

Example of using inbound and outbound shipment orders

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article provides an example scenario that shows how to create inbound and outbound shipment orders via message processing. It uses the standard sample data that's associated with the *USMF* example legal entity (company).

How to test the creation process

To try out the creation process for inbound and outbound shipment orders via messages, set the **Enable manual outbound shipment order message creation** and **Enable manual inbound shipment order message creation** options to *Yes* for a **Source system** record. You can then create shipment order messages directly on the **Outbound shipment order messages** and **Inbound shipment order messages** pages.

Another quick way to post example messages is to use Open Data Protocol (OData) requests.

In both example cases, the message processor in Microsoft Dynamics 365 Supply Chain Management processes the messages and creates the orders in the warehouse system.

> **Tip**
>
> To completely skip the shipment order creation processes via messages, you can create the inbound shipment orders and outbound shipment orders directly on the order pages by allowing the *Enable manual inbound shipment order creation* and *Enable manual outbound shipment order creation* settings for a source system.

The same message structure logic applies to both the inbound and outbound shipment order messages:

- Order header
    - Order line 1

    - Order line 2

      ...

    - Order line *n*

- Complete

> **Tip**
>
> For more examples of HTTP requests for creating messages and integrating inventory and product master data, go to the **dynamics365scm-warehouse repository on GitHub**.

Prerequisites

Before you can work through this example by using a Supply Chain Management environment, you must prepare your system in the following way:

- Check the version requirements, and enable the feature as described in Enable and configure warehouse management only mode.
- Work in a system where the standard demo data is installed, and select the *USMF* legal entity.
- Set up at least one record on the **Source systems** page. This example scenario assumes that you've set up a source system where the **Source system** field is set to *ERP*. Learn more in Configure your source systems.
- Set up the required number sequences as described in Set up number sequences.

Set up authentication for the example

On the **Microsoft Entra ID applications** page, assign the *Admin* user to the client that's used for authentication during interaction with the Supply Chain Management environment from an external source. Alternatively, assign another user who has authentication access to the integration messages, such as the default *Warehouse System Integration Operator* role. If you use the same user as part of the product master data import, more privileges that are related to product master data entities must be added to the *Warehouse System Integration Operator* role.

When you post entities via OData, you must ensure either that the user's default company matches the company that the entity will be posted to, or that the company (`dataAreaId` value) is specified in the request payload messages. Either way, shipment order messages can be completed only if the company (`dataAreaId` value) is specified.

Create shipment order messages

Example of a simple inbound shipment order message

For the `InboundShipmentOrderMessages` inbound shipment order header message, you must provide the following data at a minimum:

- `MessageId`: *M1*
- `dataAreaId`: *USMF* (Optional, depending on the default authorization user company)
- `SourceSystemId`: *ERP*
- `OrderNumber`: *IO1*
- `ReceivingWarehouseId`: *51*

When you use variables, the `InboundShipmentOrderMessages` message looks like the following example.

JSON

```
POST {{resource}}/data/InboundShipmentOrderMessages
{
"MessageId": "{{MessageId}}",
"dataAreaId": "{{dataAreaId}}",
"SourceSystemId": "{{SourceSystem}}",
"OrderNumber": "{{OrderNumber}}",
"ReceivingWarehouseId": "{{Warehouse}}"
}
```

The `InboundShipmentOrderLineMessages` message looks like the following example.

JSON

```
POST {{resource}}/data/InboundShipmentOrderLineMessages
{
"MessageId": "{{MessageId}}",
"dataAreaId": "{{dataAreaId}}",
"SourceSystemId": "{{SourceSystem}}",
"OrderNumber": "{{OrderNumber}}",
"OrderLineNumber": 1,
"ItemNumber": "A0001",
"ExpectedQuantity": 10,
"ExpectedUnitSymbol": "Pcs"
}
```

To commit the messages, post a *complete* message for the header and lines. The complete message looks something like the following example.

JSON

```
POST {{resource}}/data/InboundShipmentOrderMessages(MessageId='{{MessageId}}',
dataAreaId='{{dataAreaId}}',SourceSystemId='{{SourceSystem}}',
OrderNumber='{{OrderNumber}}')/Microsoft.Dynamics.DataEntities.Complete?cross-
company=true
```

> **Note**
>
> The `dataAreaId` value is used as part of the key to match against the released header and line messages. Therefore, the `dataAreaId` value must be specified. The suffix `?cross-company=true` is required only for messages where the company differs from the user's default company that's set up on the **Microsoft Entra ID applications** page.

Example of a simple outbound shipment order message

For the `OutboundShipmentOrderMessages` outbound shipment order header message, you must provide the following data at a minimum:

- `MessageId`: *M2*
- `dataAreaId`: *USMF* (Optional, depending on the default authorization user company)
- `SourceSystemId`: *ERP*
- `OrderNumber`: *OO1*
- `ShipFromWarehouseId`: *51*
- `ConsigneeName` or `ReceiverName`: *Microsoft*
- `ConsigneeCountryRegionId` or `ReceiverCountryRegionId`: *USA*

When you use variables, the `OutboundShipmentOrderMessages` message looks like the following example.

JSON

```
POST {{resource}}/data/OutboundShipmentOrderMessages
{
"MessageId": "{{MessageId}}",
"dataAreaId": "{{dataAreaId}}",
"SourceSystemId": "{{SourceSystem}}",
"OrderNumber": "{{OrderNumber}}",
"ShipFromWarehouseId": "{{Warehouse}}",
"ConsigneeName": "{{ConsigneeName}}",
"ConsigneeCountryRegionId": "{{ConsigneeCountryRegionId}}"
}
```

The `OutboundShipmentOrderLineMessages` message looks like the following example.

JSON

```
POST {{resource}}/data/OutboundShipmentOrderLineMessages
{
"MessageId": "{{MessageId}}",
"dataAreaId": "{{dataAreaId}}",
"SourceSystemId": "{{SourceSystem}}",
"OrderNumber": "{{OrderNumber}}",
"OrderLineNumber": 1,
"ItemNumber": "A0001",
"OrderedQuantity": 10,
"OrderedUnitSymbol": "Pcs"
}
```

To commit the messages, post a *complete* message for the header. The complete message looks something like the following example.

JSON

```
POST {{resource}}/data/OutboundShipmentOrderMessages(MessageId='{{MessageId}}',
dataAreaId='{{dataAreaId}}',SourceSystemId='{{SourceSystem}}',
OrderNumber='{{OrderNumber}}')/Microsoft.Dynamics.DataEntities.Complete?cross-
company=true
```

> **Note**
>
> The `dataAreaId` value is used as part of the key to match against the released header and line messages. Therefore, the `dataAreaId` value must be specified. The suffix `?cross-company=true` is required only for messages where the company differs from the user's default company that's set up on the **Microsoft Entra ID applications** page.

Message processor messages for shipment orders

After the two documents are imported into the message queue, you must use the message processor to process them and create the actual inbound and outbound shipment orders.

Last updated on 09/26/2025

Warehouse management only mode FAQ

Article • 04/30/2024

This article provides answers to frequently asked questions about Warehouse management only mode.

What should I do when a message is shown as Failed in the message processor?

The message processor retries three times before it fails. You can use business events to be notified about this failure. Follow the view log information for the **Message processor messages** page, and use that information to take the next appropriate action: queue the message (that is, move it back into the processing queue), cancel it, or manually update it.

> **Note**
>
> Usually, you must correct errors in the data before it makes sense to try to reprocess a failed message.

Why does my message stay in the Queued state instead of being processed?

If multiple messages that have dependencies are in the *Failed* state (for example, for the same order number), you must either correct the data and move the message back to the *Queued* state, or cancel the dependent messages.

Why do I have to provide a Country/Region field value for my outbound shipment order?

Outbound shipment orders must be able to create a shipping address. Therefore, you must specify at least two field values: **Country/Region** and **Name**. The **Country/Region** value must match existing data in the address setup. You can use the following message fields to provide the data:

- `ConsigneeName`
- `ConsigneeCountryRegionId`
- `ReceiverName`
- `ReceiverCountryRegionId`

Learn more in Master and reference data.

Why does Supply Chain Management show all the modules instead of just those that are related to warehouse management?

Warehouse management only mode is part of a larger deployment of Microsoft Dynamics 365 Supply Chain Management. Therefore, it lets you use all the integration points. To limit a user's access to other modules, assign only those security roles that are related to warehouse management processes:

- *Warehouse manager* – Enable and review processes, maintain master data, and respond to inquiries.
- *Warehouse mobile device user* – Access the *Warehouse Mobile Devices Portal* service.
- *Warehouse planner* – Work with setup and planning processes.
- *Warehouse system integration operator* – Set up integration processes that are related to inbound/outbound shipment orders and on-hand inventory synchronization.
- *Receiving clerk* – Work with inbound processes.
- *Shipping clerk* – Work with outbound processes.
- *Warehouse worker* – Work with daily warehouse processes.

Why do I receive the following error when I post a counting journal: "No fiscal calendar has been defined for the ledger. In general ledger setup, select a fiscal calendar for the ledger"?

Usually, Supply Chain Management requires that you configure several costing and general ledger settings, including fiscal calendars. However, you can remove this requirement by setting up item model groups that don't use these features and associating those item model groups with the relevant released products. Learn more in Master and reference data.

Why do I receive the following error when I process warehouse operations: "The accounting currency hasn't been defined for the ledger. You must define the currency in the Ledger form"?

Usually, Supply Chain Management requires that you configure several costing and general ledger settings, including a ledger currency. However, you can remove this requirement by setting up item model groups that don't use these features and associating those item model groups with the relevant released products. Learn more in Master and reference data.

Why do I receive the following error when I process different order types: "The accounting currency hasn't been defined for the ledger. You must define the currency in the Ledger form"?

Warehouse management only mode uses just two types of documents: *inbound shipment orders* and *outbound shipment orders*. All other types of documents (such as *sales orders* or *purchase orders*) can be processed only if you've configured the required costing and general ledger options, regardless of your item model group setup.

Why can't I just Receive complete what I have partly inbound registered?

Depending on your setup on the **Inbound shipment order policies** FastTab of the **Source systems** page (**Warehouse management** > **Setup** > **Warehouse management integration** > **Source systems**), inbound loads might be created in any of the following ways:

- Automatically, when an inbound shipment order is imported
- Automatically, when an advanced shipping notice (ASN) is imported
- Via a manual process
- As part of the Warehouse Management mobile app receiving process
- As part of an *Order receiving completed* process

When the system creates load data as a result of processing an inbound shipment order message, the delivery policy controls whether load quantities are adjusted to the received quantities, or whether the received quantities must match the load line quantities, as part of the *Receiving completed* process.

Why do I receive the following error for my existing Update product receipts process: "Unable to update product receipt for a load with inbound shipment order lines"?

If you enable Warehouse management only mode and are already running a periodic *Update product receipts* batch job for loads that are associated with purchase orders, you must update the query for the batch job to exclude inventory transaction updates for inbound shipment orders. To update the query, add the *Load details* entity, and specify a *NotExist* join to the *Loads* entity. Then add a range definition for the **Reference** field, where **Criteria** = *Inbound shipment order*.

Why do I receive the following error when I process a transfer order: "The accounting currency has not been defined for the ledger. You must define the currency in the Ledger form"?

In addition to setting up item model groups that don't use costing and general ledger settings (see Master and reference data), you must enable Warehouse-specific inventory transactions for the *Transfer issue* and *Transfer receipt* warehouse scenarios.

Can I attach documents and notes to inbound and outbound shipment orders?

Yes. Both inbound and outbound shipment order messages support document attachments. For more information, see the descriptions of *Inbound shipment order messages composite entity* and *Outbound shipment order messages composite entity* in Data management. These entities contain the child entities *Inbound shipment order document attachment messages* and *Outbound shipment order document attachment messages*.

Why can't I see and search for page names that are related to Warehouse management only mode?

Even if the *Warehouse management only mode* feature is enabled, you must create a record on the **Source systems** page before the system can look up the related pages. In addition, the current user's default company must have a source system set up to search for the page names. Learn more in Enable and configure Warehouse management only mode.

Why doesn't my registered inventory transaction have a load ID?

If you use the item arrival journal process to receive updated inbound shipment order line quantities, inventory transactions won't be associated with a load. To achieve this result, you must use the *Order receiving completed* process, which creates loads for registered inventory transactions that aren't associated with a load. This process can be triggered directly from an inbound shipment order or via the **Inbound shipment order receiving completed** background process (available at **Warehouse management** > **Periodic tasks** > **Inbound shipment order receiving completed**).

Why do I receive the following error when processing shipment orders: "Not able to resolve released item/variant for external item ID [ItemId]"?

When processing shipment orders, you may receive the following error, even though a related item or variant exists:

> Not able to resolve released item/variant for external item ID [ItemId].

This error occurs because released products and variants must be linked to a source system through the Source system items data entity. This is crucial for recording how item and variant numbers must be handled for each external source system.

Why aren't any external inventory updates recorded after I run external shared warehouse processing?

In one typical case, you create a counting journal for an item, which adjusts the on-hand inventory for the selected warehouse. Before you proceed, make sure that the **Warehouse inventory update log** page shows the information that you need for this adjustment. If it doesn't, you must verify that all the required **Source system** and **Source system item** settings are in place.

Even if the warehouse inventory update log has the correct information, it sends the data to the **External inventory updates** page only for the warehouse inventory dimensions that are connected through the external warehouse management system and the relevant warehouses. For more information about how to configure this process, see Warehouse management only mode with external shared warehouses.

Feedback

Was this page helpful? Yes No

Provide product feedback

Monitor and control message processor messages

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

The message processor is a framework for processing messages that represent events. It has the following properties:

- It processes messages in the correct order. (Dependent messages are processed in sequence.)
- It's scalable. (Independent messages can be processed in parallel.)
- It uses the required system resources.
- It avoids exhaustion of system resources if a spike in messages occurs.
- It's reliable.
- It's traceable.

You might use this framework, for example, to develop and manage custom integration with external systems, and to process other custom functionality. Microsoft Dynamics 365 Supply Chain Management includes, for example, several out-of-box features that use predefined message types and message queues. These features include third-party manufacturing execution system (MES) integration, deferred posting, and packing slip posting during container close. Warehouse management only mode uses the message processor framework to manage *inbound and outbound shipment orders*.

This article describes how to monitor and control the processing of all message types by using the **Message processor messages** page.

Message processor messages page

Use the **Message processor messages** page to view the list of incoming messages, view the message log, manually process messages, and troubleshoot issues.

Open the Message processor messages page

To view the list of messages processed by the message processor, go to **System administration** > **Message processor** > **Message processor messages**.

Grid columns and filters on the Message processor messages page

Use the fields at the top of the **Message processor messages** page to find specific messages. Most of these filters match the column headings in the message grid. The following filters and column headings are available:

- **Message type** – The type of message.
- **Message queue** – The name of the queue that the message is processed in. The following queues are available:
  - *Manufacturing Execution 3rd Party* – This queue holds messages that are created as part of the *Manufacturing execution system integration* feature. These messages also appear on the **Manufacturing execution systems integration** page, which is like the **Message processor messages** page but is focused exclusively on that feature. For more information, see Integrate with third-party manufacturing execution systems.
  - *Production* – This queue holds messages that are created as part of the *Make finished goods physically available before posting to journals* feature. These messages also appear on the **Deferred production order posting** page, which is like the **Message processor messages** page but is focused exclusively on that feature. For more information, see Make finished goods physically available before posting to journals.
  - *Warehouse* – This queue holds messages that are created for warehouse management, such as to post a sales packing slip when the last shipment container is closed as part of a manual packing process. (This message has a message type of *Run packing slip for container*.)
  - *Shipment Orders* – This queue holds messages that support Warehouse management only mode.
  - *Source System Products* – This queue holds messages that support source product master data.
  - *External warehouse shipment order updates* – This queue holds messages that support external shared warehouse processing.
  - *Dynamics 365 Sales Integration* – This queue holds messages that integrate with Dynamics 365 Sales. For more information about this feature and the messages that it

might add to this queue, see Work with added efficiency in quote-to-cash with Dynamics 365 Sales.
    - *<Custom queues>* – If your system is customzied to support additional types of queues, they appear here. For more information about how to add custom queues, see Implement a new queue.

- **Message state** – The state of the message. The following states exist:
    - *Queued* – The message is ready to be processed by the message processor.
    - *Processed* – The message was successfully processed by the message processor.
    - *Canceled* – The message is canceled by a user.
    - *Failed* – The message failed to be processed.

- **Message content** – This filter does a full-text search of message content. (The grid doesn't show message content.) The filter treats most special symbols, such as hyphens, as spaces, and it treats all space characters as Boolean OR operators. For example, if you search for a specific `journalid` value that equals *USMF-123456*, the system finds all messages that contain either "USMF" or "123456," and the list is likely to be long. Therefore, it's better to enter just *123456* in this case, because more specific results are returned.

View the message log, message content, and details

To view detailed information about a message, select it in the grid. Then, select the **Log** or **Message content** tab under the message grid, where each processing event is shown.

The text on the **Message content** tab depends on the **Message type** value. Therefore, the text length varies. A typical message content text starts with an opening brace ({) and ends with a closing brace (}). In between are field names (for example, `journalId`), each followed by a colon and a value (for example, *USMF-123456*).

The toolbar on the **Log** tab includes the following buttons:

- **Log** – Select this button to show the processing results. This function is especially helpful when messages have a **Processing result** value of *Failed*, and you want to understand the reasons for the processing failure.
- **Bundle** – Multiple message processing operations can run as part of the same batch process. Select this button to view the detailed data. For example, you can see whether dependencies exist that require the system to process some messages in a specific sequence.

Manually process, cancel, or requeue a message

Depending on the current state of a message, you can manually process or cancel it. Select the message in the grid, and then select **Process** or **Cancel** on the Action Pane.

If you want to requeue a previously canceled message, select it in the grid. Then, select **queue** on the Action Pane. The system processes the message as usual.

Schedule message processing by using the message processor batch job

To process a message queue, set up a batch job to run it. Usually, you set up a fixed, regular schedule for processing each queue. However, you can also run any queue on demand. To create and schedule the required batch jobs, follow these steps:

1. Go to **System administration** > **Message processor** > **Message processor**.
2. In the **Message processor** dialog box, in the **Message queue** field, select the message queue that's associated with the messages that you want to process. The queue that you select depends on the feature or system that generated the messages.
3. On the **Run in the background** FastTab, set up batch and scheduling options as you require, just as you would do for other types of jobs in Supply Chain Management.
4. Select **OK** to run or schedule the job based on your settings.

Message processor queue setup

You can configure the number of processor tasks dedicated to each message processor queue and set rules for how often the queue cleans processed and canceled messages. Unconfigured queues use a default value that you can override as needed. Follow these steps to customize one or more queues.

1. Go to **System administration** > **Message processor** > **Message queue setup**.

2. Follow one of these steps:

    - To edit an existing queue, select **Edit** on the Action Pane, and then select the target queue in the grid.
    - To add a new configuration, select **Add** on the Action Pane to add a new row to the grid. Then, in the **Message queue** field for the new row, select the name of the queue that you want to configure.

3. For the new or selected row, make the following settings:

- **Number of processor tasks** – Specify the number of processor tasks dedicated to the specified queue. The maximum value is *8*. The minimum value depends on the minimum number of batch threads that are configured for your system (typically *2*).
- **Days before processed message deletion** – Specify the number of days before processed messages are cleaned up (deleted). Set this field to zero (0) to turn off processed message cleanup. For more information, see Clean up processed and canceled message processor messages.
- **Days before canceled message deletion** – Specify the number of days before canceled messages are cleaned up (deleted). Set this field to zero (0) to turn off canceled message cleanup. For more information, see Clean up processed and canceled message processor messages.

4. On the Action Pane, select **Save**.

Related information

- Clean up processed and canceled message processor messages
- Business events, custom message queues, and custom message types
- Message processor messages for warehouse management processes
- Exchange data between systems
- Integrate with third-party manufacturing execution systems
- Enable and configure extra efficiency in quote-to-cash with Dynamics 365 Sales
- Make finished goods physically available before posting to journals

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Last updated on 05/05/2026

Clean up processed and canceled message processor messages

The message processor is a framework that processes messages in the system, such as those related to inventory transactions, sales orders, and more. Over time, processed and canceled messages, that are no longer needed, can accumulate in the system, potentially affecting performance and data management. Therefore, you should periodically clean up these messages to maintain system efficiency.

Prerequisites

To use the cleanup functionality described in this article, you must be running Microsoft Dynamics 365 Supply Chain Management version 10.0.45 or later. No feature management settings are required.

Set up message cleanup for each message processor queue

You set up message cleanup using the **Message queue setup** page, which allows you to specify how long processed and canceled messages should be retained for each queue before they're deleted.

To configure message cleanup, follow these steps:

1. Go to **System administration** > **Message processor** > **Message queue setup**.

2. Follow one of these steps:

    - To edit an existing queue, select **Edit** on the Action Pane, and then select the target queue in the grid.
    - To add a new configuration, select **Add** on the Action Pane to add a new row to the grid. Then, in the **Message queue** field for the new row, select the name of the queue that you want to configure.

3. For the new or selected row, make the following settings:

    - **Number of processor tasks** – Specify the number of processor tasks that should be dedicated to the specified queue. The maximum value is *8*. The minimum value

depends on the minimum number of batch threads that are configured for your system (typically *2*). This setting isn't related to the cleanup job, but ensures that the queue can process messages efficiently.

- **Days before processed message deletion** – Specify the number of days to retain processed messages before deletion. To turn off processed message cleanup, set this field to 0.
- **Days before canceled message deletion** – Specify the number of days to retain cancelled messages before deletion. To turn off cleanup, set this field to 0.

4. On the Action Pane, select **Save**. The system automatically creates a background process to run the cleanup job. By default, the process runs daily, but you can modify this schedule in the Process automation workspace, as described in the next section.

Review and manage cleanup jobs

Whenever you have at least one message queue set to be cleaned up, the system automatically creates a background process to run the cleanup operation regularly. Follow these steps to review, schedule, and manage this process in the Process automation workspace:

1. Go to **System administration** > **Setup** > **Process automation**.
2. Open the **Background processes** tab to view the list of background processes.
3. Use the **Filter** field to search for the process that has a **Name** of *Cleanup job for the message processor*.
4. Select the process in the grid and then select **Edit** from the toolbar.
5. The **Edit background process** dialog opens. Use the settings here to view and modify the schedule and other settings for the cleanup job.
6. Select **OK** when you're done.

Optimization advisor

The Optimization advisor can remind you to manage outdated messages when they exist in the message processor. It includes a diagnostic rule called *Check for aged processed or canceled messages*, which identifies messages older than 30 days and creates an optimisation opportunity for them. You can take action directly from the **Optimization advisor** workspace. By default, this rule runs monthly.

To turn the rule on or off, or to change the frequency of the rule, follow these steps:

1. Go to **System administration** > **Periodic tasks** > **Maintain diagnostic validation rules**.

2. Find the row where **Rule name** is *Check for aged processed or canceled messages*.
3. Make the following settings for the rule as needed:

    - **Status** – Set to *Active* or *Inactive*.
    - **Run frequency** – Choose how often to run the check (*Daily*, *Weekly*, *Monthly*, or *Unscheduled*).

To view and act on your optimization opportunities, including those related to cleaning up aged messages, follow these steps:

1. Go to **System administration** > **Workspaces** > **Optimization advisor**. Here, you can see a list of optimization opportunities that the system has identified, including those related to message processor cleanup.
2. Select the row where **Area** is *SCM* and **Optimization opportunity** is *Cleanup job for message processor*.
3. From the toolbar, select **More information** to learn more about the opportunity and recommended actions.
4. If you want to act on the opportunity, select **Take action** from the toolbar to open the **Message queue setup** page, and configure the cleanup settings as described previously in this article.

Related information

- Monitor and control message processor messages
- Business events, custom message queues, and custom message types
- Message processor messages for warehouse management processes
- Exchange data between systems
- Integrate with third-party manufacturing execution systems
- Enable and configure extra efficiency in quote-to-cash with Dynamics 365 Sales
- Make finished goods physically available before posting to journals

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Last updated on 12/30/2025

Business events, custom message queues, and custom message types

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

This article describes how to design your own custom message queues and message types for the message processor by using Visual Studio. It also describes how to set up business events to deliver alerts for failed processing results.

Set up business events to deliver alerts for failed processing results

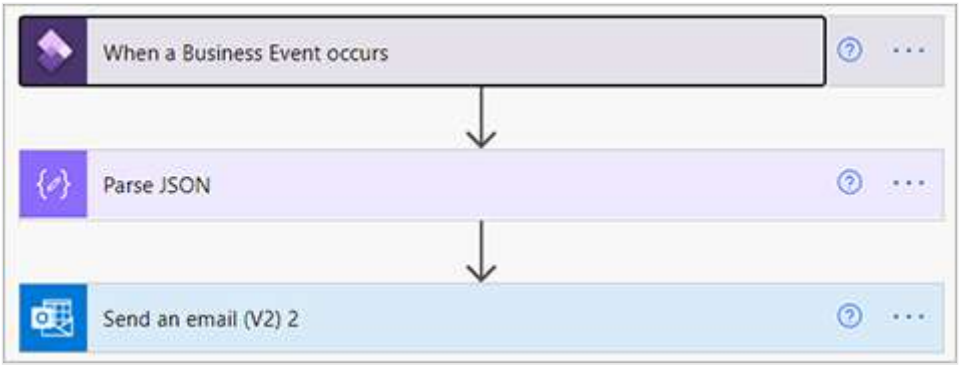
You can set up business events to alert you about failed processing results. To complete this setup, activate the *Message processor message processed* business event on the **Business events catalog** page (**System administration** > **Setup** > **Business events** > **Business events catalog**). As part of the activation process, you'll be directed to specify whether the event is specific to one legal entity or all legal entities. You'll also be directed to provide an endpoint name, which must be defined in advance.

> **Note**
>
> If **When a Business Event occurs** is set to *Microsoft Power Automate* (instead of *HTTPS*, for example), the endpoint name will automatically be created in Supply Chain Management, based on the *Microsoft Power Automate* setup.

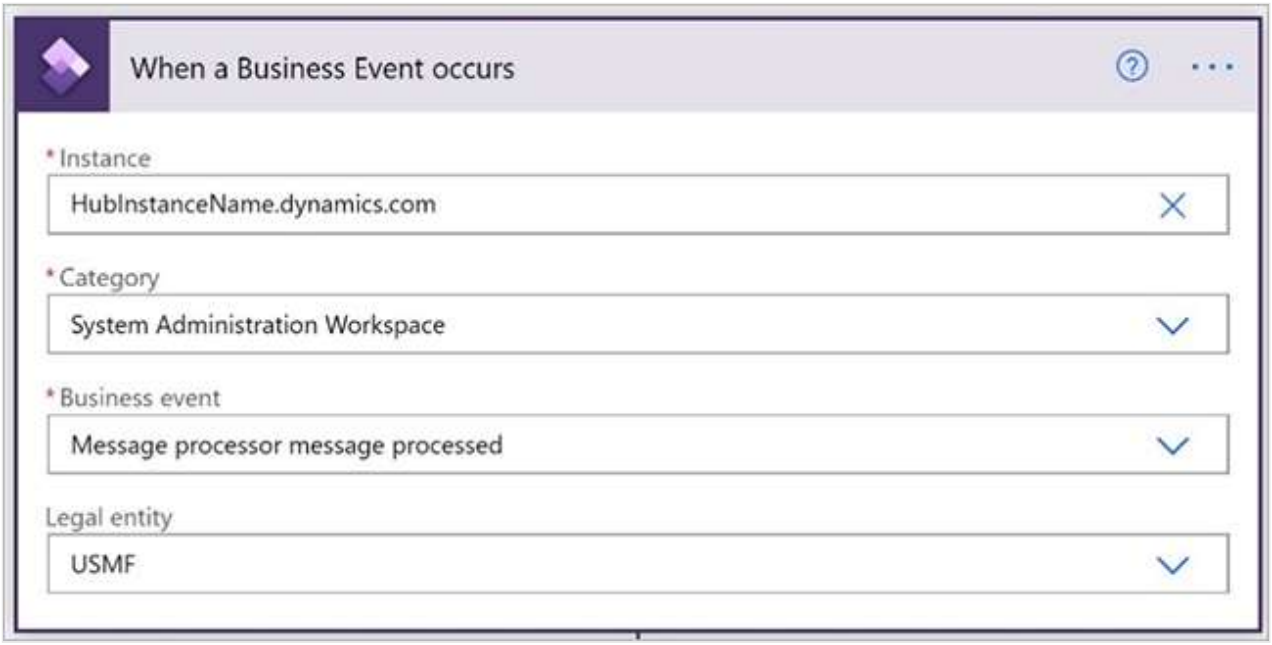
Business events Power Automate example

In this example, **When a Business Event occurs** is set to *Microsoft Power Automate* to send email notifications that containing Action center (formerly Infolog) messages and hyperlinks that open the **Message processor messages** page for a specific failed message. As you require, you can add extra logic to send the notifications in parallel via different channels and control the recipients based on the event data.

1. In Power Automate, create an automated cloud flow for the **When a Business Event occurs - Fin & Ops App (Dynamics 365)** flow trigger, followed by the **Parse JSON** and **Send an email** steps.



2. In the **When a Business Event occurs** step, you can look up or enter values in the **Instance** and **Category** fields, and then set the **Business event** field to *Message processor message processed*.



3. In the **Parse JSON** step, in the **Schema** field, enter a schema that defines the extended fields. You can use the **Download schema** option on the **Business events catalog** page in Supply Chain Management, or you can start by pasting in the following example schema text.

JSON

```
{
    "properties": {
        "BusinessEventId": {
            "type": "string"
        },
        "ControlNumber": {
            "type": "integer"
```

```
        },
        "EventId": {
            "type": "string"
        },
        "EventTime": {
            "type": "string"
        },
        "MajorVersion": {
            "type": "integer"
        },
        "MessageContent": {
            "type": "string"
        },
        "MessageDestinationCompanyId": {
            "type": "string"
        },
        "MessageDestinationOperationalSiteId": {
            "type": "string"
        },
        "MessageDestinationWarehouseId": {
            "type": "string"
        },
        "MessageDestinationWorkloadName": {
            "type": "string"
        },
        "MessageInfolog": {
            "type": "string"
        },
        "MessageProcessingResult": {
            "type": "string"
        },
        "MessageProcessingResultLabel": {
            "type": "string"
        },
        "MessageProcessorMessagePageUrl": {
            "type": "string"
        },
        "MessageQueue": {
            "type": "string"
        },
        "MessageQueueLabel": {
            "type": "string"
        },
        "MessageSourceCompanyId": {
            "type": "string"
        },
        "MessageSourceOperationalSiteId": {
            "type": "string"
        },
        "MessageSourceWarehouseId": {
            "type": "string"
        },
        "MessageSourceWorkloadName": {
            "type": "string"
        },
```

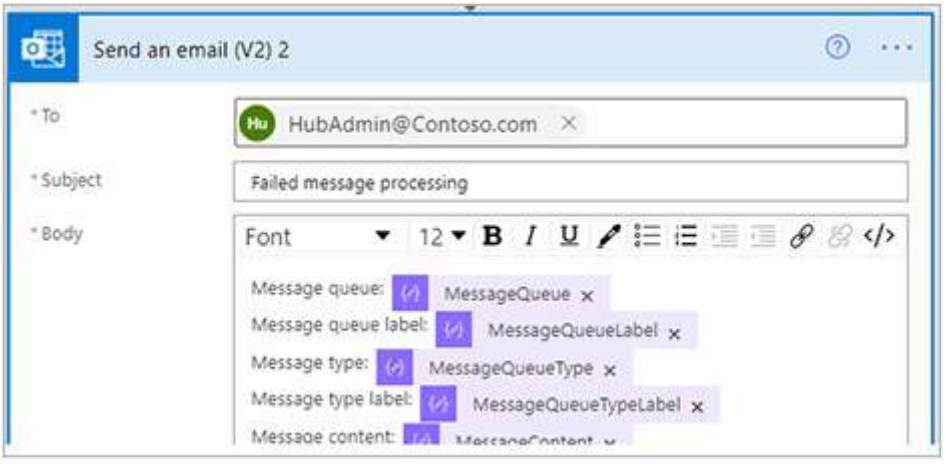
```
        "MessageState": {
            "type": "string"
        },
        "MessageStateLabel": {
            "type": "string"
        },
        "MessageType": {
            "type": "string"
        },
        "MessageTypeLabel": {
            "type": "string"
        },
        "MinorVersion": {
            "type": "integer"
        }
    },
    "type": "object"
}
```

4. In the **Send an email** step, you can select the individual fields, or you can start by pasting the following example email body into the **Body** field.

plaintext

```
Message queue: @{body('Parse_JSON')?['MessageQueue']}
Message queue label: @{body('Parse_JSON')?['MessageQueueLabel']}
Message type: @{body('Parse_JSON')?['MessageQueueType']}
Message type label: @{body('Parse_JSON')?['MessageQueueTypeLabel']}
Message content: @{body('Parse_JSON')?['MessageContent']}
Message state: @{body('Parse_JSON')?['MessageState']}
Message state label: @{body('Parse_JSON')?['MessageStateLabel']}
Message processing result: @{body('Parse_JSON')?['MessageProcessingResult']}
Message processing result label: @{body('Parse_JSON')?
['MessageProcessingResultLabel']}
Message infolog: @{body('Parse_JSON')?['MessageInfolog']}
Message source company ID: @{body('Parse_JSON')?['MessageSourceCompanyId']}
Message source workload name: @{body('Parse_JSON')?
```

```
['MessageSourceWorkloadName']}
Message source site ID: @{body('Parse_JSON')?
['MessageSourceOperationalSiteId']}
Message source warehouse ID: @{body('Parse_JSON')?['MessageSourceWarehouseId']}
Message destination company ID: @{body('Parse_JSON')?
['MessageDestinationCompanyId']}
Message destination workload name: @{body('Parse_JSON')?
['MessageDestinationWorkloadName']}
Message destination site ID: @{body('Parse_JSON')?
['MessageDestinationOperationalSiteId']}
Message destination warehouse ID: @{body('Parse_JSON')?
['MessageDestinationWarehouseId']}
Message processor message page URL: @{body('Parse_JSON')?
['MessageProcessorMessagePageUrl']}
```



When you save the business event, it's automatically activated and is ready to be used as part of Supply Chain Management.

Message processor scheduler

The message processor has one scheduler. The `SysMessageKeyDateTimeSequenceProcessorScheduler` class schedules messages that have dependencies that are based on a key, date, and time, so that the messages are processed in the correct order. Messages that must be processed are stored in the `SysMessageProcessorTaskBundle` and `SysMessageProcessorTaskBundleMessage` tables. Dependent messages must be in the same bundle.

The `SysMessageKeyDateTimeSequenceProcessorScheduler` class is used by the third-party MES integration feature to secure messages that are related to a production order, and that are processed in the order that they're received or created in. The dependencies are defined by a key (production order number) and the time.

For example, the system receives or creates the following messages:

- Production order 1 – Start message

- Production order 2 – Start message
- Production order 2 – Report as finished message
- Production order 1 – Report as finished message

In this case, the `SysMessageKeyDateTimeSequenceProcessorScheduler` class will schedule the processing of messages in the following order:

1. Production order 1 – Start message
2. Production order 1 – Report as finished message
3. Production order 2 – Start message
4. Production order 2 – Report as finished message

Each report-as-finished message depends on the successful processing of the related start message.

Because a bottleneck can occur when records are picked up for processing for multiple tasks, and because there might be dependencies between the messages, the system bundles the messages. For example, if the bundle size is 2, the system will create the following bundles:

- Bundle 1 - Production order 1 – Start message
- Bundle 1 - Production order 1 – Report as finished message
- Bundle 2 - Production order 2 – Start message
- Bundle 2 - Production order 2 – Report as finished message

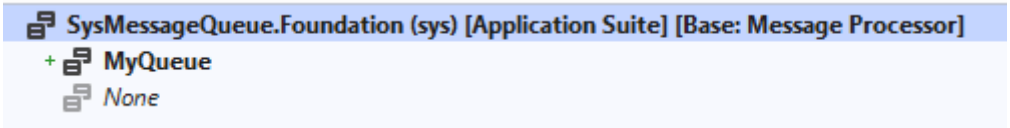
For scalability, when you develop a new queue, you can configure the number of tasks that should process the bundles. (Learn more in Implement a new queue.) If the configuration uses two message processor tasks, the two bundles can be processed in parallel.

Development example: Implement a custom message queue

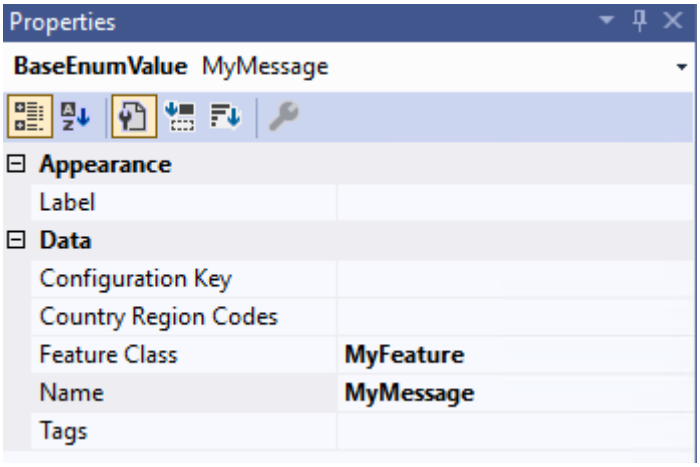
This example shows how to add a new queue. After the queue is created, you'll be able to submit messages to it. Those messages will then be processed by the message processor according to the settings that were described earlier.

1. In Visual Studio, create an extension for the `SysMessageQueue` enum.

2. Add a new enum value for the new queue.



3. Set a feature class on the enum value.



4. Create a class that extends `SysMessageQueueMetadata`, as shown in the following example.

X++

```
[SysMessageQueueFactoryAttribute(SysMessageQueue::MyQueue)]
public final class MyQueue extends SysMessageQueueMetadata
{

    /// <summary>
    /// Gets the default number of message processor tasks.
    /// </summary>
    /// <returns>The default number of message processor tasks.</returns>
    public SysMessageProcessorsTasks getDefaultMessageProcessorsTasks()
    {
        return 1;
    }

    /// <summary>
    /// Get the maximum number of messages in a bundle.
    /// </summary>
    /// <returns>The maximum number of messages in a bundle.</returns>
    public SysMessageProcessorMaxBundleSize getMaxBundleSize()
    {
        return 10;
    }

    /// <summary>
    /// Determines if messages should be queued when sending or processing
messages.
    /// </summary>
    /// <returns>true if messages should be queued when sending a message;
otherwise false.</returns>
    public boolean queueMessageOnSend()
    {
        return true;
    }

    /// <summary>
```

```
    /// Determines if dependent messages should be committed per created
transaction ID.
    /// </summary>
    /// <returns>true if dependent messages should be committed per created
transaction ID.; otherwise false.</returns>
    public boolean commitPerCreatedTransactionID()
    {
        return false;
    }

}
```

5. Create an extension for the `SysMessageProcessorToggle` class to enable the message processor in the user interface (UI).

X++

```
[ExtensionOf(classStr(SysMessageProcessorToggle))]
final class MySysMessageProcessorToggle_Extension
{
    /// <summary>
    /// Determines if the toggle is enabled.
    /// </summary>
    /// <returns>
    /// true if the toggle is enabled; otherwise, false.
    /// </returns>
    internal boolean isEnabled()
    {
        return next isEnabled() || MyFeature::instance().isEnabled();
    }

}
```

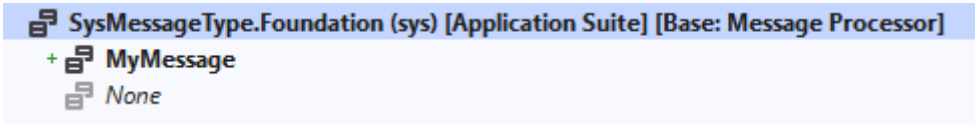
Development example: Implement a custom message type

This example shows how to create a message type for each type of process that you want to make available to the message processor. The message type establishes which processes can be run by messages of that type. The name of the message type will be shown on the **Message processor messages** page to identify the purpose of each message in the queue.

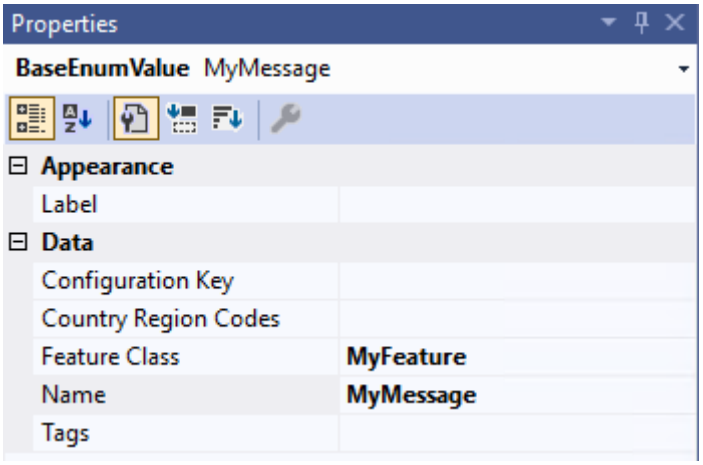
Follow these steps to create a new message type.

1. In Visual Studio, create an extension for the `SysMessageType` enum.

2. Add a new enum value for the new message type.



3. Set a feature class on the enum value.



4. Create a class that extends `SysMessageTypeMetadata`, as shown in the following example.

X++

```
[SysMessageTypeFactoryAttribute(SysMessageType::MyMessage)]
public final class MyMessage extends SysMessageTypeMetadata
{
    /// <summary>
    /// Validates if it is allowed to send a message on a given queue.
    /// </summary>
    /// <param name = "_messageQueue">A queue to send message to.</param>
    /// <param name = "_messageTarget">A message target.</param>
    /// <param name = "_sourceMessageTarget">A message source.</param>
    /// <returns>true if it is allowed to send a message; false, otherwise.
</returns>
    public boolean canSendMessage(SysMessageQueue _messageQueue,
SysMessageTargetRecId _messageTarget, SysMessageTargetRecId
_sourceMessageTarget)
    {
        if (_messageQueue == SysMessageQueue::MyQueue)
        {
            return true;
        }

        return false;
    }

    /// <summary>
    /// Returns a list of allowed message state transitions.
    /// </summary>
    /// <returns>A list of allowed message state transitions.</returns>
    public Enumerator getAllowedMessageStateTransitions()
    {
        // Allow the following state transitions:
```

```
        // SysMessageState::None -> SysMessageState::Queued
        // SysMessageState::None -> SysMessageState::Processed

        // SysMessageState::Queued -> SysMessageState::Processed

        // SysMessageState::Failed -> SysMessageState::Queued
        // SysMessageState::Failed -> SysMessageState::Cancelled
        // SysMessageState::Failed -> SysMessageState::Processed

        return
SysMessageStateTransition::getProcessedCancelledStateTransitions();
    }

    /// <summary>
    /// Returns dependencies related to a message.
    /// </summary>
    /// <param name = "_message">The message to add dependencies to.</param>
    /// <returns>A list of dependencies related to a message.</returns>
    public List getDependencies(SysMessage _message)
    {
        List dependencies = new List(Types::Record);

        // Add a dependency to the message if you need to process messages in
order.
        // For example, process a start message before an end message.
        // The message will not be processed before the dependent message is
processed or cancelled.

        // The default scheduler,
SysMessageKeyDateTimeSequenceProcessorScheduler only supports one dependency
per message.
        // Messages with different keys may be processed in parallel.

        SysMessageDependencyKey dependencyKey = strFmt('Mykey:%1',
this.getContract(_message).parmMyKey());

dependencies.addEnd(SysMessageKeyDateTimeSequenceDependency::create(dependencyK
ey));

        return dependencies;
    }

    /// <summary>
    /// Execute the logic associated with the given message state transition.
    /// </summary>
    /// <param name = "_message">The message.</param>
    /// <param name = "_messageStateTransition">The state transition to
execute.</param>
    public void processMessageStateTransition(SysMessage _message,
SysMessageStateTransition _messageStateTransition)
    {
        if
(_messageStateTransition.isEqual(SysMessageStateTransition::newFromStates(SysMe
ssageState::Queued, SysMessageState::Processed)))
```

```
        {
            this.processMessage(_message);
        }
        else if
(_messageStateTransition.isEqual(SysMessageStateTransition::newFromStates(SysMe
ssageState::Failed, SysMessageState::Processed)))
        {
            this.cancelMessage(_message);
        }
    }

    private void processMessage(SysMessage _message)
    {
        // Add logic to process the message.
    }

    private void cancelMessage(SysMessage _message)
    {
        // Add logic to cancel the message if needed.
    }

    private ClassId contractClass()
    {
        return classNum(MyContract);
    }

    protected MyContract getContract(SysMessage _message, ClassId _objectTypeId
= this.contractClass())
    {
        var contract = FormJsonSerializer::deserializeObject(_objectTypeId,
_message.Content) as MyContract;

        return contract;
    }

}
```

Additional resources

- Monitor and control message processor messages
- Clean up processed and canceled message processor messages

Last updated on 09/25/2025

Gantt control development guide

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

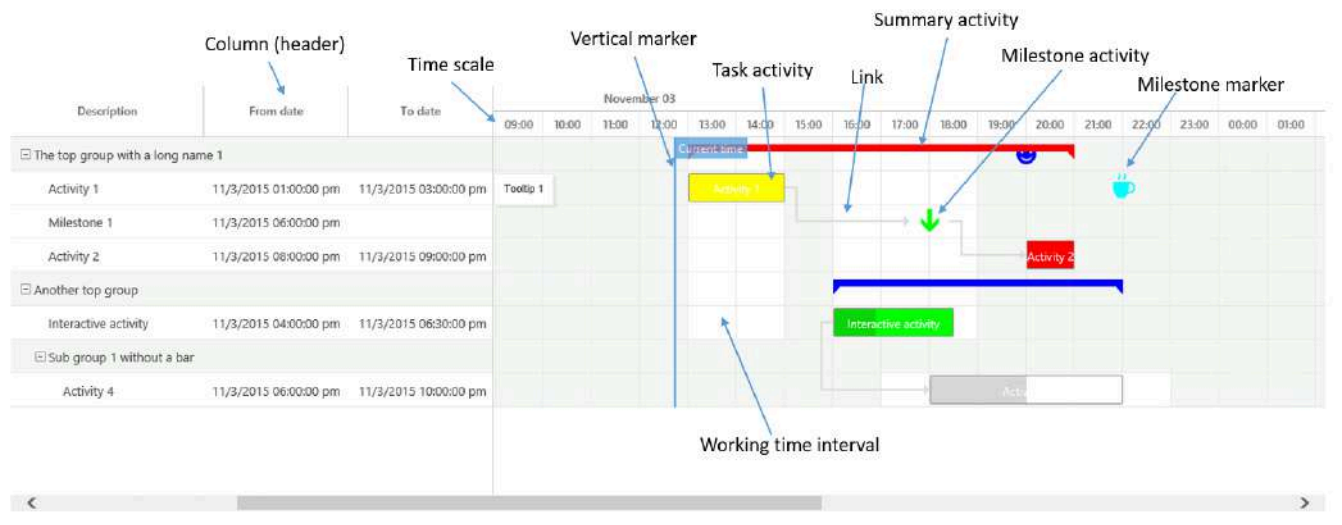
This article describes how to create new forms by using the Gantt control. Review the code in the Tutorial_Gantt form. This code demonstrates the full capabilities of the Gantt control, and shows how to load data and work with the application programming interface (API).

What's new for Gantt

In Microsoft Dynamics AX 2012, the client was a Win32 application, and extensions used Microsoft ActiveX, WinForm, or Microsoft Windows Presentation Foundation (WPF) controls. ActiveX and ManagedHost controls are incompatible with the HTML-based platform, so you can't use them to add custom controls. Instead, use the new extensible control framework to add controls by using HTML and JavaScript. This framework implements the new Gantt control. Unlike earlier versions, you don't have to pay an extra license fee to use the control in your own forms or to extend the control.

High-level overview of the control

The following illustration shows the visual elements of the Gantt control.



To add the control to a new form, right-click the top node in the form design, and then select **Add new** > **Gantt**. After you set the **Height** and **Width** properties to **SizeToAvailable**, you don't need to set any other properties in the designer. Unlike many other controls, Gantt controls can't be bound to data sources. Instead, you must add all data to the control from code.

Adding activities, links, and milestone markers

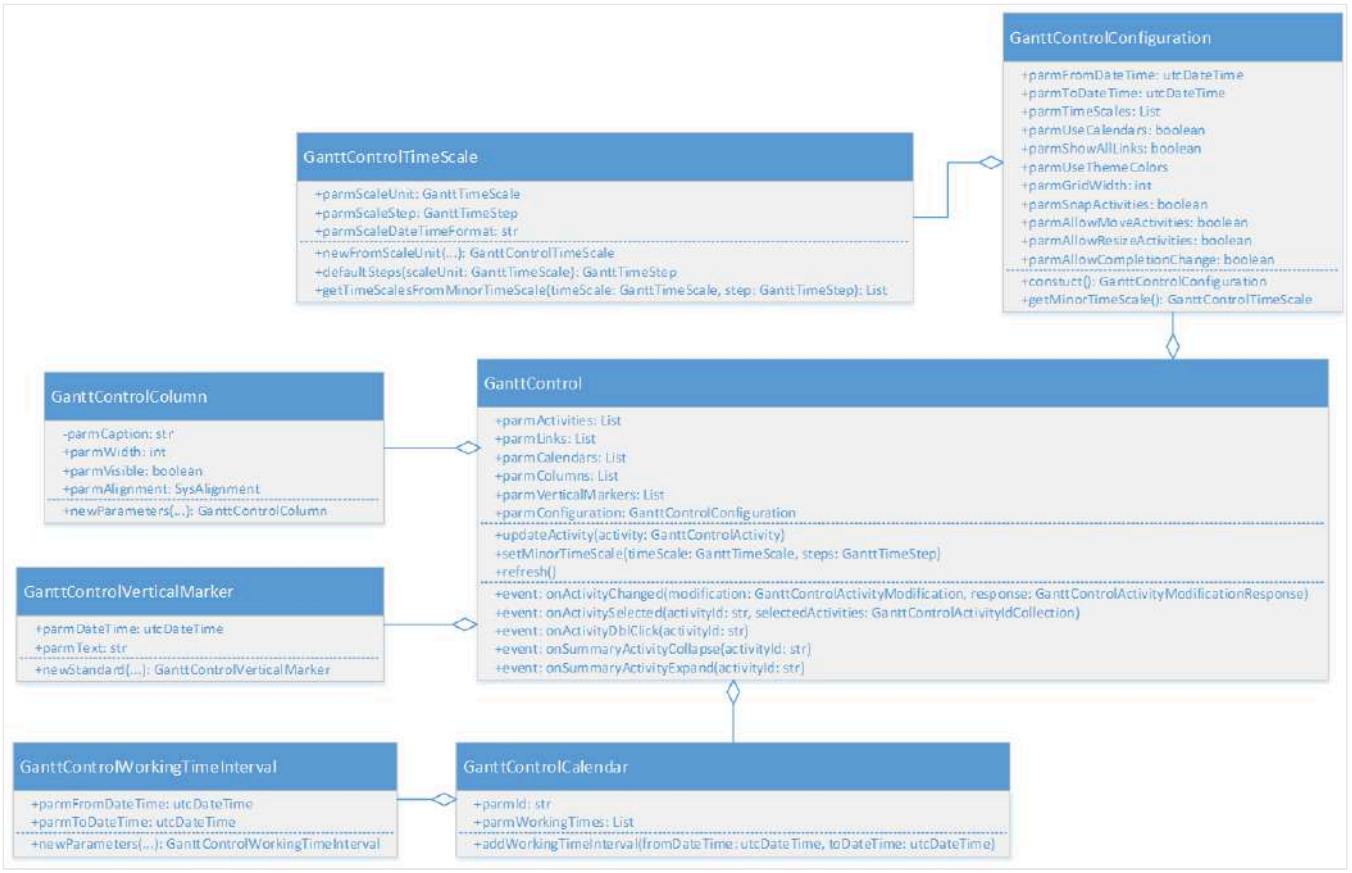
The most basic element of a Gantt chart is the task activity. Each activity gets its own row in the chart. Activities can form a hierarchy, like a tree, by referencing a parent activity. Additionally, you can connect any two activities to each other through links (across hierarchies). To add data to the control, build up a list that contains the instances of the element that you want to add. Then assign the list to the control by using a **parm** method. When you modify data, don't just modify the content of the list. To notify the client that data changed and to trigger a refresh, you must reassign the list to the control by using the **parm** method.

Each activity must have a unique ID across all activity types. This ID is required to build the hierarchy and the links. The ID also identifies the activity when the server is notified after a user interaction. There are two types of milestones: The milestone activity is a stand-alone activity that gets its own row in the chart and can be referenced from the links. The milestone markers appear on the same row as the related activity. Both types of markers use symbols from the Dynamics Symbol font.

Column headers and content

In the grid on the left side of the control, you can arrange the data into columns. Provide the column definitions through the **GanttControl.parmColumns** property, which is a list of **GanttControlColumn** objects. Give the content of the columns as a simple **List** object that

contains the text strings for an activity through the **GanttControlActivity.parmColumnTexts** property. The values in the column list must be strings. Therefore, format any numbers, enumerations, dates, and so on on the server. When formatting **utcdatetime**, take into account the user's preferred time zone because the Gantt chart uses that time zone when rendering.



Working times

An activity can reference a calendar. A calendar is basically a list of working time intervals. Use a calendar to visually separate working time from non-working time by giving the cells (intersections of a row and a time scale interval) a different background color. These working time intervals don't have to come from the standard calendar tables. The intervals are just data that you can load from anywhere. Although calendars are a nice feature that can give you a good overview, you must consider some performance issues. When you turn calendars on (by setting **GanttControlConfiguration.parmUseCalendars** to **true**), the control renders each cell in HTML by using a DIV element. If a calendar shows many activities for a long period at a very granular minor timescale, many DIV elements exist. Depending on the browser and hardware, the large number of DIV elements can cause performance problems for the user. In this case, you might need to turn calendars off. By using this approach, you significantly reduce the number of DIV elements that the control must render. The control can't turn off calendars by itself to improve performance. Based on your specific scenario, determine what works best for your users.

Color use

Elements such as the activities and milestone markers have a color property. Typically, you calculate the int value for a specific color by using the **WinAPI::RGB2int** method, or the user sets the value by using the color picker (which you open by calling **ColorSelection::selectColor**). Instead of controlling the colors yourself, you can specify that the colors adhere to the current theme. When you set the **GanttControlConfiguration.parmUseThemeColors** option to **true**, the control overrides any color values you set that aren't part of the current theme. You don't need to specify colors manually. When you use theme colors, the **parmIsActive** flag on an activity determines whether the activity appears in a dark color or a light color.

Interactions

Whenever the user interacts with the Gantt, the control raises an event to notify the server-side code. The following events are currently available:

- onActivitySelected(str _activityId, GanttControlActivityIdCollection _allSelectedActivityIds)
- onActivityDblClick(str _activityId)
- onSummaryActivityExpand(str _activityId)
- onSummaryActivityCollapse(str _activityId)
- onActivityChanged(GanttControlActivityModification _modification, GanttControlActivityModificationResponse _response)

All these events are one-way notifications from the client. However, the **onActivityChanged** event is somewhat special, because you can set a response. Typically, when a user makes a modification (for example, dragging an activity to a new time), you must either update other activities or make some adjustments to the activity itself (for example, change the column texts). In the **GanttControlActivityModificationResponse** response, you can provide a list of activities that must be updated. If the change that the user makes is just accepted as is, don't set a response. However, at the very least, you must update the column texts of the current activity to guarantee that the new from and to dates appear. On an activity, the **parmAllowMove** flag determines whether the activity can be moved. However, higher-level flags on **GanttControlConfiguration** determine whether you can move any activity (**parmAllowMoveActivities**), resize any activity (**parmAllowResizeActivities**), or change the completion percentage for the activity (**parmAllowCompletionChange**).

Last updated on 03/14/2026

Create a new transportation management engine

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article describes how to create a new transportation management engine in Dynamics 365 Supply Chain Management.

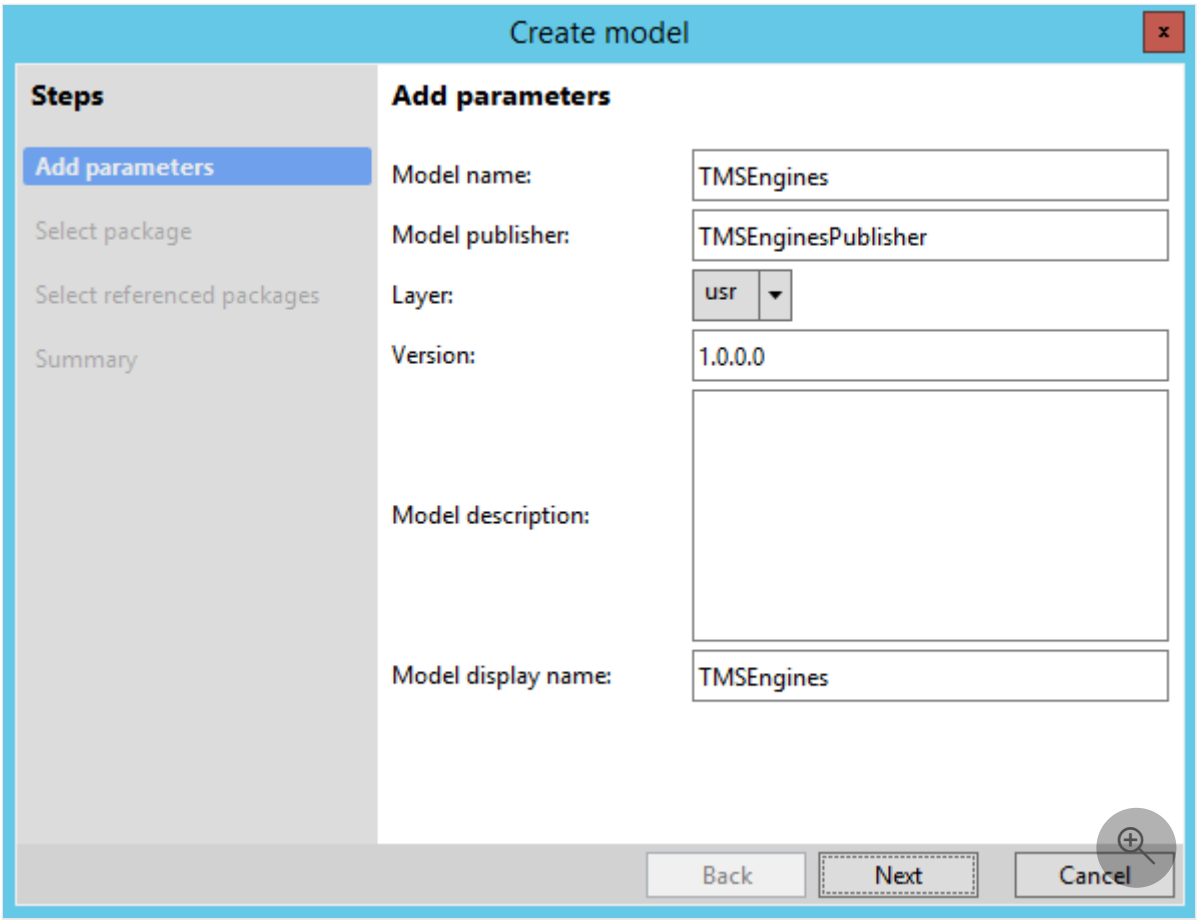
Transportation management (TMS) engines define the logic that generates and processes transportation rates in Transportation management. Supply Chain Management provides several different engine types that calculate different parameters, such as rates, transit times, and the number of zones crossed during transit. This article explains how to use the Microsoft Visual Studio development environment together with Supply Chain Management development tools to create and deploy a new TMS engine, and then how to set up the engine in Operations. For more information about the engines, see Transportation management engines.

Create a new TMS engine

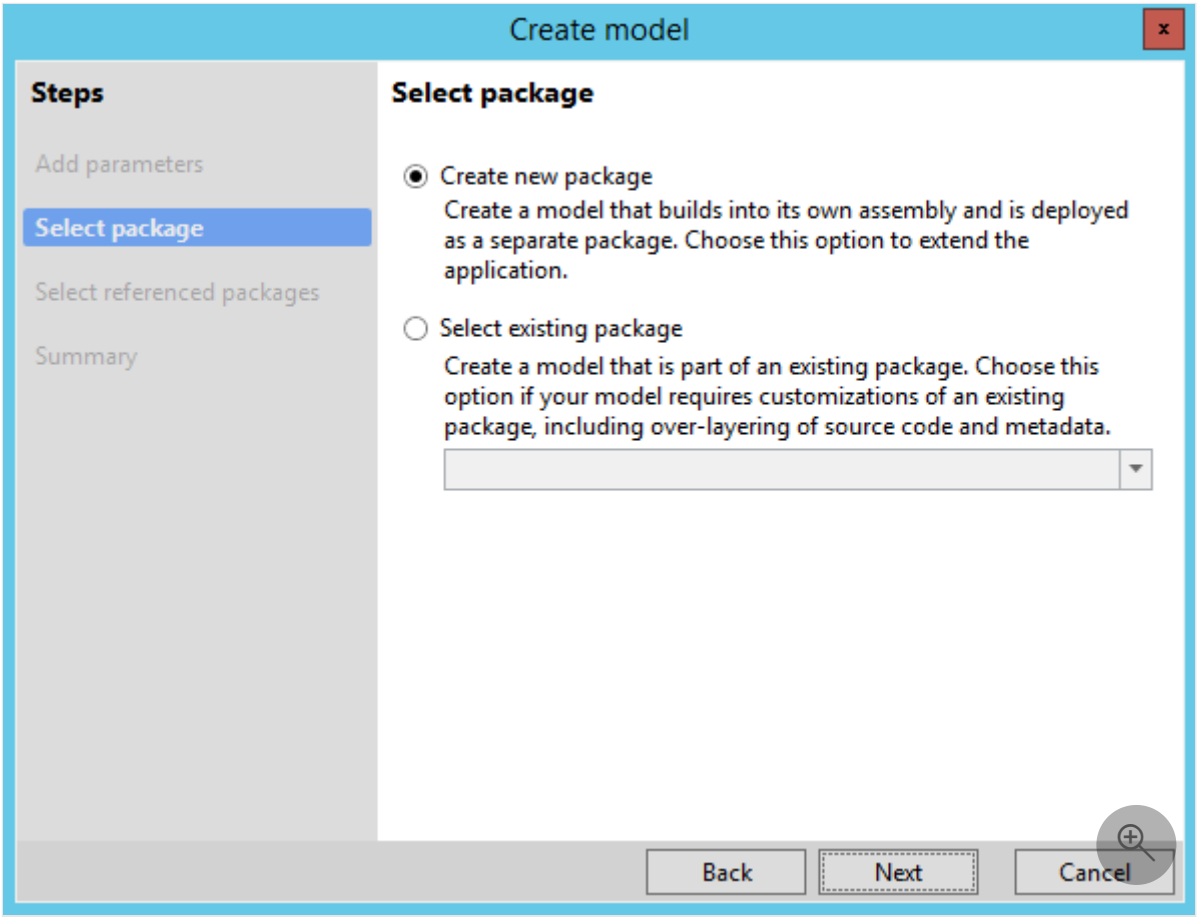
This section explains how to create a class library that has a TMS engine implementation, and how to reference it from a Supply Chain Management model.

1. To deploy your new engines, you must have a model that contains the engines. On the **Dynamics 365** > **Model Management** menu, select **Create model** to create a new model.

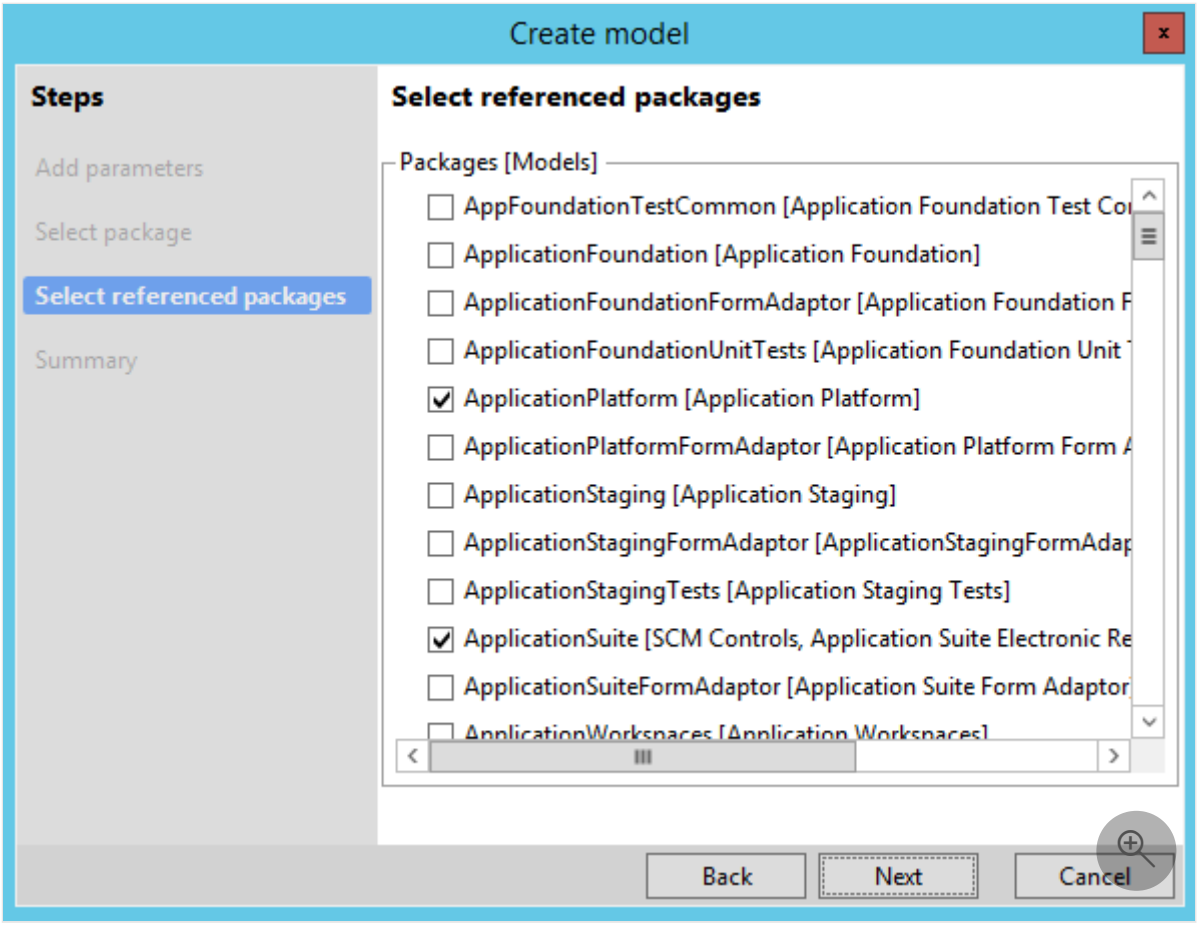
On the first page of the **Create model** wizard, name the model *TMSEngines*.



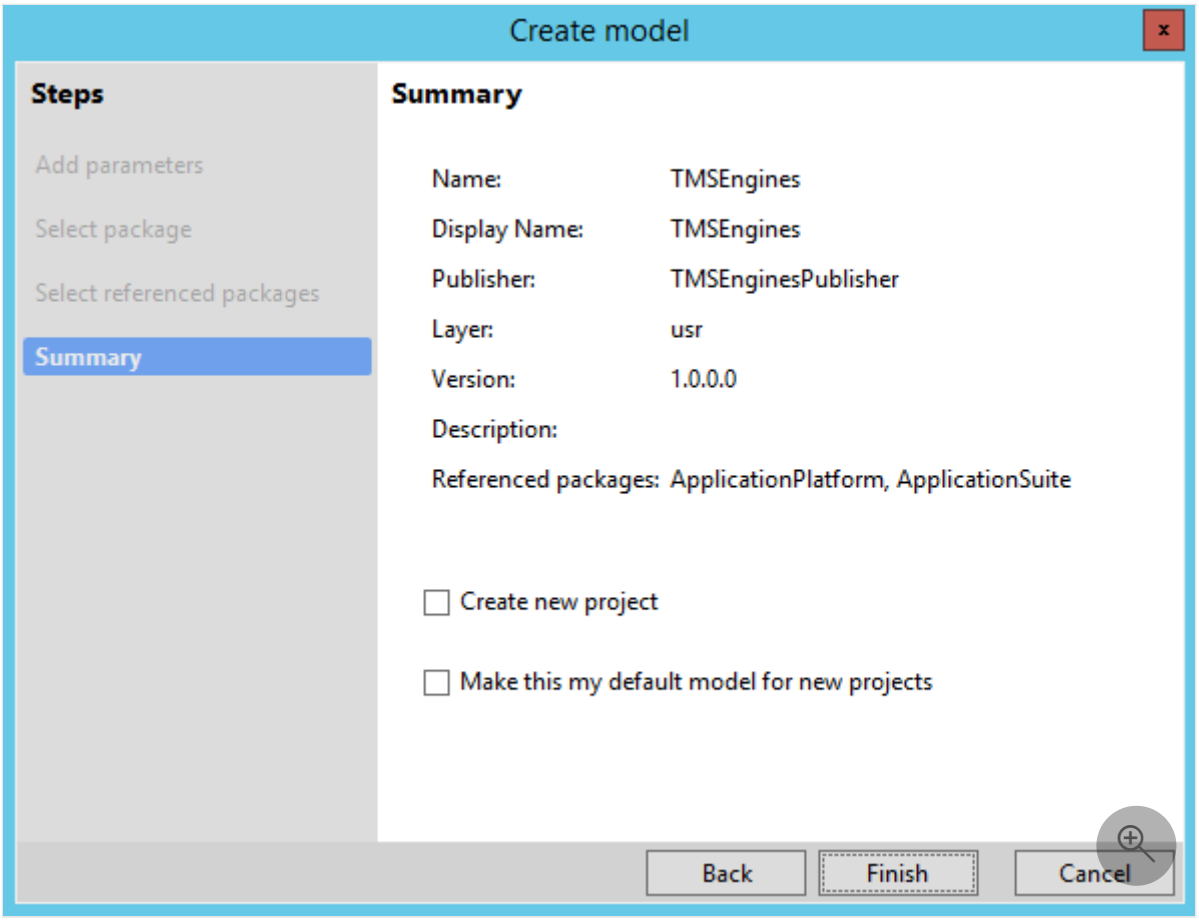
2. On the next page, select **Create new package**.



3. On the next page, select the **ApplicationSuite** model to reference. (The *ApplicationPlatform* model is preselected.)



4. On the next page, select **Finish** to confirm the creation of a new model.



5. In a new solution, create a new Supply Chain Management project, and name it *TMSThirdParty*. The framework must be set to *.NET Framework 4.8* or newer. Double check that the project's file (`TMSThirdParty.rnrproj`) specifies the correct .NET Framework version. If it doesn't, update it directly in the file to `<TargetFrameworkVersion>v4.8</TargetFrameworkVersion>`. In the project properties, set the project's model to *TMSEngines*.

6. Add a new C# *Class Library (.NET Framework)* to your solution, and name it *ThirdPartyTMSEngines*. The framework must be set to *.NET Framework 4.8* or newer and must match the settings on the *TMSThirdParty* project.

7. In the *ThirdPartyTMSEngines* project, add references to Supply Chain Management–specific assemblies specified in the following list.

   > **ⓘ Important**
   >
   > The **Copy Local** property for all the references in this list must be set to *False*. Otherwise, you risk producing runtime issues where the engine class might not be found or the assembly might not load correctly.

   - Application assemblies that enable X++ types to be referenced. These assemblies can be found in the following locations. [Packages root] is the path of the location

where all the deployed assemblies are placed, such as `C:\Packages`.

```text
[Packages
root]\ApplicationPlatform\bin\Dynamics.AX.ApplicationPlatform.dll
[Packages
root]\ApplicationFoundation\bin\Dynamics.AX.ApplicationFoundation.dll
[Packages root]\ApplicationSuite\bin\Dynamics.AX.ApplicationSuite.dll
```

- Framework assemblies that enable access to data, LINQ, and auxiliary functions. All these assemblies can be found in [Packages root]\bin.

```text
Microsoft.Dynamics.ApplicationPlatform.Environment.dll
Microsoft.Dynamics.AX.Data.Core.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.AdoNet.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.Interface.dll
Microsoft.Dynamics.AX.Framework.Linq.Data.Msil.dll
Microsoft.Dynamics.AX.Server.Core.dll
Microsoft.Dynamics.AX.Xpp.AxShared.dll
Microsoft.Dynamics.AX.Xpp.Support.dll
```

- The core TMS assembly (which contains engines) and the TMS base assembly (which contains helpers, constants, data transfer class definitions, and so on). These assemblies can be found in the following locations.

```text
[Packages root]\ApplicationSuite\bin\Microsoft.Dynamics.AX.Tms.dll
[Packages root]\ApplicationSuite\bin\Microsoft.Dynamics.AX.Tms.Base.dll
```

8. Rename the C# class that is automatically generated in the *ThirdPartyTMSEngines* project to `SampleRatingEngine`.

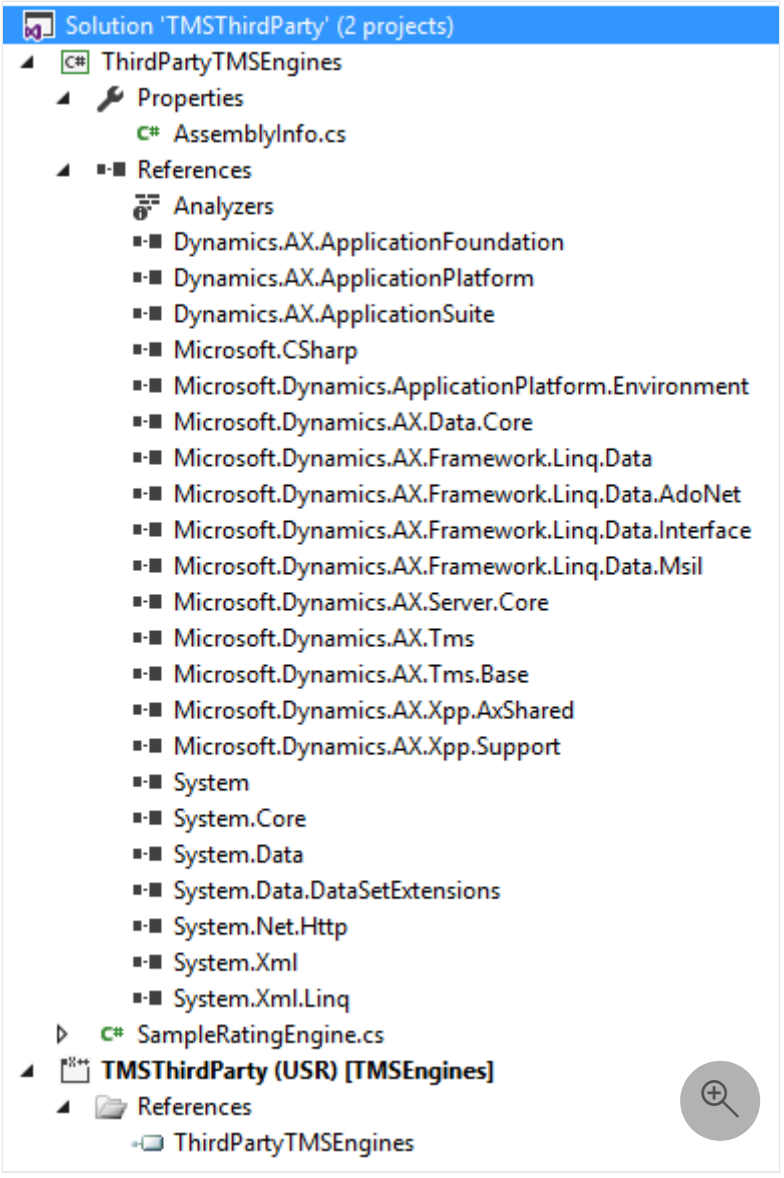
9. Implement the engine. Because you're creating a rate engine in this example, inherit from the base class for rate engines. The base class implements most of the rate engine interface (`TMSFwkIRateEngine`). You just have to implement the rate method. To keep this example simple, make this method register a hard-coded rate of 100. You can create engines that implement any of the engine interfaces, such as `TMSFwkIAccessorialEngine`. All the engine interfaces are defined in X++.

```C#
```

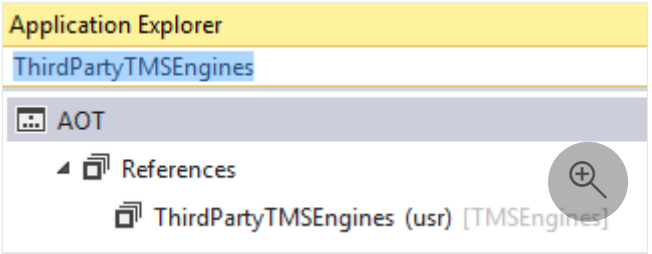
```
namespace ThirdPartyTMSEngines
{
    using Dynamics.AX.Application;
    using Microsoft.Dynamics.Ax.Tms.Base.Data;
    using Microsoft.Dynamics.Ax.Tms.Base.Utility;
    using Microsoft.Dynamics.Ax.Tms.Bll;
    using System.Xml.Linq;
    public class SampleRatingEngine : BaseRateEngine
    {
        public override RatingDto rate(TmsTransactionFacade
_transactionFacade, XElement _shipment, string _rateMasterCode)
        {
           XElement re =
shipment.RetrieveOrCreateRatingEntity(this.RatingDto);
           re.AddRate(TmsRateType.Rate, 100);
           return this.RatingDto;
        }
    }
}
```

10. Build the solution.

11. Add a new reference to the *TMSThirdParty* project. The reference should point to the *ThirdPartyTMSEngines* project. When you finish, your solution should look like this.



12. Build the solution. Verify that the new library appears in the **References** node in Application Explorer.



> **Note**
>
> If you're testing the new engine directly in the development environment, you might also need to build the entire *TMSEngines* model from **Extensions** > **Dynamics 365** > **Build models**. This might resolve the issue if the engine assembly couldn't be found.

Deploy the TMS engine as a package

One way to deploy third-party TMS engines is through a deployment package. To deploy the engine as a package, follow these steps:

1. In Visual Studio, go to **Extensions** > **Dynamics 365** > **Deploy** > **Create Deployment Package** and create a deployment package for the new model you created as described in Create a new TMS engine.

2. In Microsoft Dynamics Lifecycle Services, go to **Asset Library** > **Software deployable package**. Select **Add** to add the deployment package you created to Lifecycle Services.

3. After the validation completes, go to **Environments** > **{Your Environment ID}** > **Maintain** > **Apply updates**, select the deployable package, and apply it.

> **Note**
>
> If the system warns you that you don't have permission to deploy packages on Lifecycle Services, contact your Lifecycle Services administrator for support.

Set up the TMS engine in Supply Chain Management

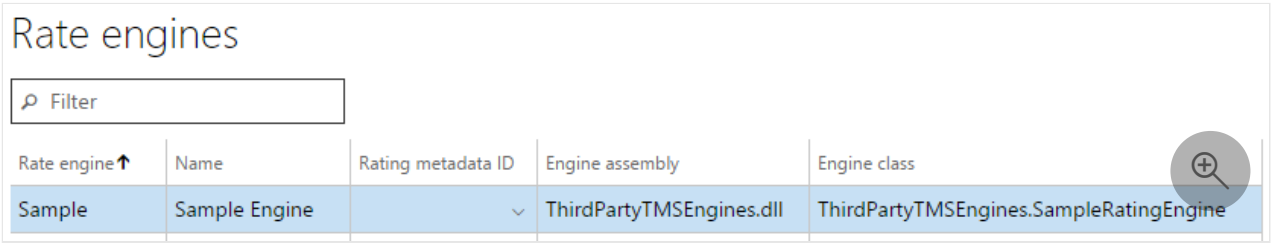
This section explains how to set up Supply Chain Management to use a TMS engine, and shows how the new engine that you created is used in rate shopping. The example in this section uses the USMF demo data company.

1. Create a new engine as described in Create a new TMS engine.

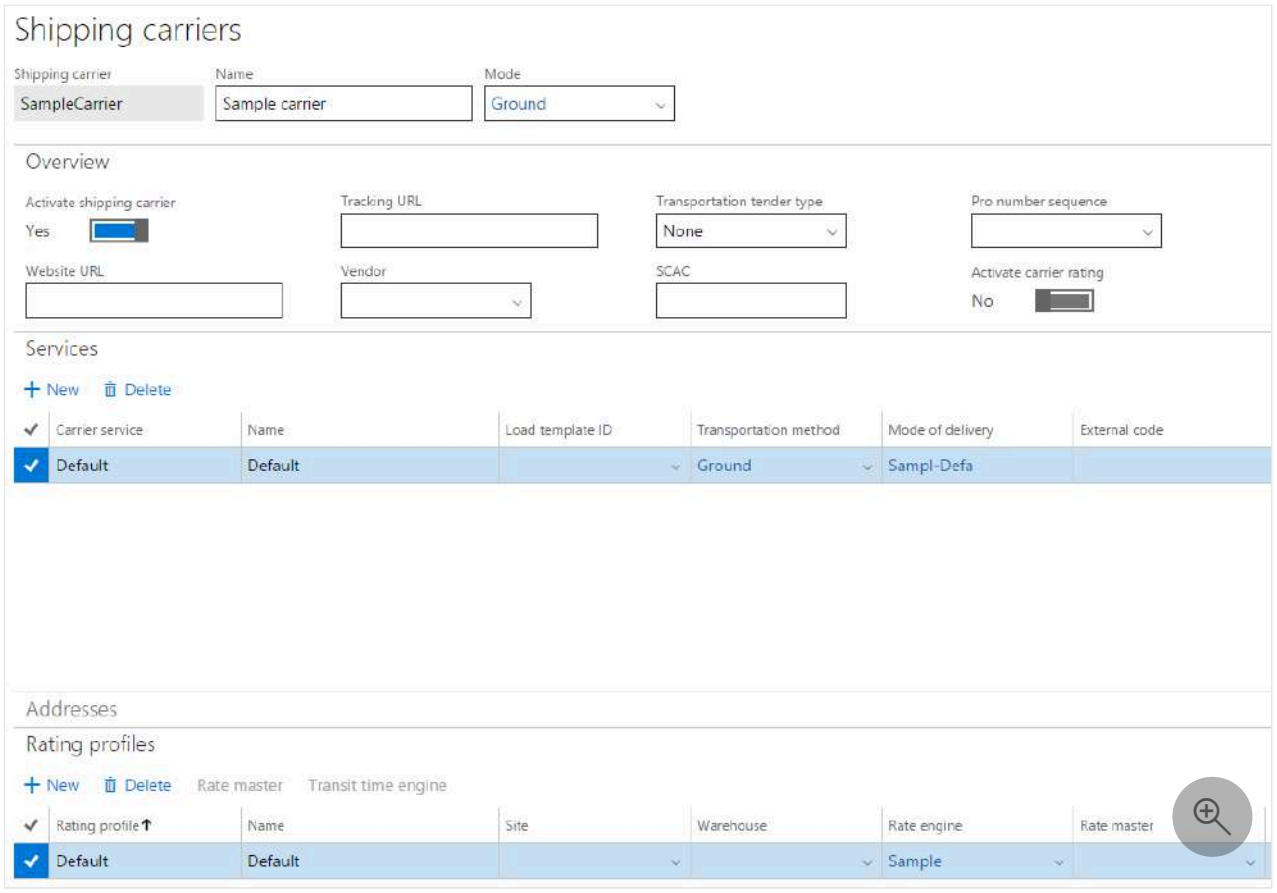
2. Build your solution.

3. Deploy the TMS engine DLL file through a deployment package, as described in Deploy the TMS engine as a package.

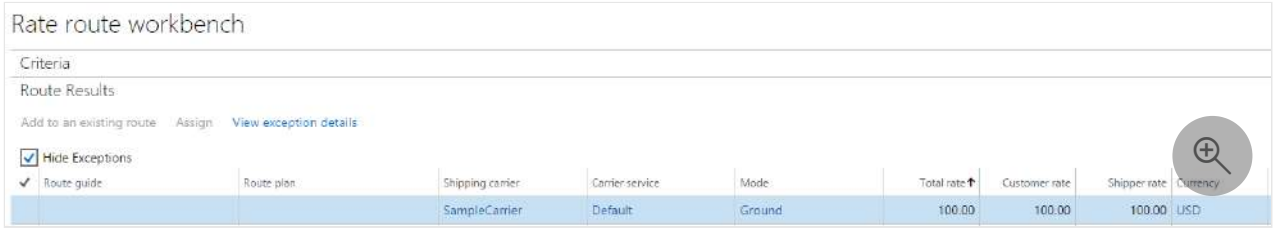
4. In Supply Chain Management, on the **Rate engines** page, create a new rating engine. The engine should point to the engine assembly produced by building the engine class library and the engine class that you implemented.



5. Create a shipping carrier that uses the Sample rate engine. Because our engine doesn't use any data, you don't have to assign a rate master.



6. On the **Rate route workbench** page, select **Rate shop**. You should see a rate of 100.00 from SampleCarrier, as shown in the following screenshot. In this example, you're rate shopping for a route from warehouse 24 to customer US-004. However, but because the rate is hard-coded, you always see a rate of 100.00.



Tips and tricks

- If you're using development tools for Supply Chain Management, it's useful to add a new project to your solution. If you set this project as your startup project and start a debugging session, you can debug both X++ and C# code in the same debugging session.
- Every time that you change and recompile your *ThirdPartyTMSEngines* project, you must deploy through a deployment package. Otherwise, you might run using a stale assembly.

- After you execute TMS-specific operations in Supply Chain Management, the Internet Information Services (IIS) worker process might lock the *ThirdPartyTMSEngines* assembly so that the assembly can't be updated. In this case, restart the w3svc process.

Whitepaper

For more information, download the following white paper (written to support AX2012, but still partially applies for Dynamics 365 Supply Chain Management):

- Implementing and deploying Transportation management engines

---

Last updated on 03/11/2026

Extend inventory on-hand data entities

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Microsoft Dynamics 365 Supply Chain Management provides extensibility features that let you add fields to tables through extension. This article provides an example that shows how to add extended fields to the `INVENTORSITEONHANDENTITY` and `INVENTWAREHOUSEONHANDENTITY` views, so that the capabilities of the inventory on-hand data entities can work with the extensions. For more information about data entities, see Data management overview.

> **Note**
>
> Here's a list of some of the inventory on-hand data entities:
>
> - Inventory on-hand by site
> - Inventory on-hand by site V2
> - Inventory on-hand by warehouse
> - Inventory on-hand by warehouse v2

After you add fields to tables that are used by the `inventSiteOnHandView` view, you must sync the engine so that the extensions are correctly recognized.

1. Extend the `InventSiteOnHandView` view by adding the extension field.
2. Extend the `InventSiteOnHandAggregatedView` view with the extension fields.
3. Extend the `InventSiteOnHandAggregatedViewBuilder` viewBuilder class by modifying the `getExtensionFields()` method. In this way, you map old view fields to new view fields when viewBuilder synchronization is run.

For example, you added the following four fields to the `InventTable` table through extension:

- Custom field 1
- Custom field 2
- Custom field 3
- Custom field 4

In this case, you must modify the `getExtensionFields()` method in the following way.

X++

```
[ExtensionOf(classStr(InventSiteOnHandAggregatedViewBuilder))]
public final class InventOnHandAggregatedViewBuilder\_Extension
{
    protected Map getExtensionFields()
    {
        next getExtensionFields();
        Map extensionFields = new Map(Types::Int64, Types::Int64);
        extensionFields.insert(fieldNum(InventSiteOnHandView, Custom field 1),
fieldNum(InventSiteOnHandAggregatedView, Custom field 1));
        extensionFields.insert(fieldNum(InventSiteOnHandView, Custom field 2),
fieldNum(InventSiteOnHandAggregatedView, Custom field 2));
        extensionFields.insert(fieldNum(InventSiteOnHandView, Custom field 3),
fieldNum(InventSiteOnHandAggregatedView, Custom field 3));
        extensionFields.insert(fieldNum(InventSiteOnHandView, Custom field 4),
fieldNum(InventSiteOnHandAggregatedView, Custom field 4));
        return extensionFields;
    }
}
```

After you complete these steps, you can extend the inventory on-hand by site and inventory on-hand by warehouse data entities by adding the new fields. In this way, you ensure that the extended fields are recognized and included during data migration that uses those data entities.

Last updated on 04/10/2025

Global Inventory Accounting home page

> **Important**
>
> Global Inventory Accounting for Dynamics 365 Supply Chain Management is currently deprecated and will reach end of support on July 1, 2027. Learn more in Removed or deprecated features in Dynamics 365 Supply Chain Management.

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>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Human Resources, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

International organizations are under increasing pressure from authorities to comply with local and global accounting standards. The valuation of inventory plays a significant role in ensuring compliance. The Global Inventory Accounting Add-in for Microsoft Dynamics 365 Supply Chain Management provides a comprehensive solution that enables organizations (especially international organizations) to use multiple costing ledgers to do inventory accounting. Therefore, those organizations can comply with multiple accounting standards and internal management accounting at the same time.

Global Inventory Accounting lets you account inventory in multiple representations by applying the appropriate valuation method (standard cost, average, or specific identification) and the selected accounting currency per instance. Global Inventory Accounting enables organizations to report inventory statements and subledger accounting values (also known as the inventory

balance and the cost of goods sold) in what is often referred to as dual valuation and/or dual currency.

Inventory accounting is done in individual ledgers. Several Global Inventory Accounting ledgers can be created for each legal entity in an organization, as required. Therefore, multiple inventory representations can be obtained. All documents that are posted in a legal entity (such as purchase orders, sales orders, and transfer orders) will be accounted in all the costing ledgers that are associated with that legal entity.

A Global Inventory Accounting ledger is defined by a combination of the following elements:

- Calendar
- Currency
- Exchange rate table
- Convention

A convention is a collection of inventory accounting policies that can be associated with one or more ledgers. Conventions let you share a common set of policies within your organization.

Availability

Global Inventory Accounting is currently available in the following Azure geographic regions:

- Australia
- Canada
- China
- Europe
- France
- Norway
- South Africa
- Switzerland
- UAE
- United Kingdom
- United States

If you try to install the add-in from another geographic region, Microsoft Dynamics Lifecycle Services will show a message that your geographic region isn't supported. Global Inventory Accounting doesn't support on-premises deployments of Supply Chain Management.

If you have any issues enabling Global Inventory Accounting in one of the supported geographic regions listed here, log the issuee through Microsoft Customer Service and Support channels including your environment ID, tenant ID, and error details in the request.

Licensing

Global Inventory Accounting is licensed together with the standard cost management features available for Supply Chain Management. You don't have to purchase an additional license to use Global Inventory Accounting.

Last updated on 06/24/2026

Get started with Global Inventory Accounting

> **Important**
>
> Global Inventory Accounting for Dynamics 365 Supply Chain Management is currently deprecated and will reach end of support on July 1, 2027. Learn more in Removed or deprecated features in Dynamics 365 Supply Chain Management.

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

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Global Inventory Accounting lets you do multiple inventory accountings in the Global Inventory Accounting ledgers that you've set up. You must associate each Global Inventory Accounting ledger with a *convention*. A convention is a collection of the following types of accounting policies:

- Cost object
- Input measurement basis
- Cost flow assumption
- Cost element

> **Note**
>
> Even after you turn on Global Inventory Accounting, you can still do inventory accounting in Microsoft Dynamics 365 Supply Chain Management in the usual way.

Global Inventory Accounting is an add-in. To make its features available, you must install it from Microsoft Dynamics Lifecycle Services. You can choose to evaluate it in a test environment before you turn it on for production environments.

Global Inventory Accounting doesn't currently support all the cost management features that are built into Supply Chain Management. Therefore, it's important that you evaluate whether the set of features that is currently available will meet your requirements.

How to get the Global Inventory Accounting add-in

> **Important**
>
> To use Global Inventory Accounting, you must have a Lifecycle Services–enabled high-availability environment (not a OneBox environment). Additionally, you must be running Supply Chain Management version 10.0.19 or later.

To install Global Inventory Accounting, install the add-in. The Global Inventory Accounting service endpoints are automatically set up, so you don't need to find them manually. If you have any issues enabling Global Inventory Accounting, log the issue through Microsoft customer service and support channels including your environment ID, tenant ID, and error details in the request.

Licensing

Global Inventory Accounting is licensed together with the standard features of inventory accounting that are available for Supply Chain Management. You don't have to purchase an additional license to use Global Inventory Accounting.

Prerequisites

Set up Microsoft Power Platform integration

Before you can enable add-in functionality, you must integrate with Microsoft Power Platform by following these steps.

1. Open the Lifecycle Services environment where you want to add the service.
2. Go to **Full details**.
3. In the **Power Platform Integration** section, select **Setup**.
4. In the **Power platform environment setup** dialog box, select the checkbox, and then select **Setup**. Typically, setup takes between 60 and 90 minutes.
5. After setup of the Microsoft Power Platform environment is completed, the page shows the name of your environment. Additionally, the **Power Platform Integration** section shows the statement, "Power Platform environment setup is complete." Global Inventory Accounting doesn't require a dual-write application.

Learn more in Enable Power Platform Integration, and follow the steps to connect either to a new Power Platform environment or to an existing Dataverse instance that you already manage.

Install or update the add-in and solution

Use the following procedure to install or update the Global Inventory Accounting add-in and solution. The part of the procedure you should follow depends on whether you're installing for the first time or just need to update the solution for an existing installation.

- If you have never installed the add-in before, follow the full procedure to install both the add-in and the solution.
- If you're already using Global Inventory Accounting but need to update the solution in the Power Platform admin center, then do step 6 only and skip all of the other steps.

To install or update the add-in and solution:

1. Sign in to Lifecycle Services.
2. Open the Lifecycle Services environment where you want to add the service.
3. Go to **Full details**.
4. Go to **Power Platform Integration** and select **Setup**.
5. In the **Power platform environment setup** dialog box, select the checkbox, and then select **Setup**. Typically, setup takes between 60 and 90 minutes.
6. After the Microsoft Power Platform environment setup is complete, sign in to the Power Platform admin center and then install or update the Global Inventory Accounting solution by doing the following steps:
   a. Select the environment where you want to install or update the solution.
   b. Select **Dynamics 365 apps**.
   c. Select **Install App**.
   d. Select **Dynamics 365 Global Inventory Accounting**.

e. Select **Next** to install.

7. After the solution is completely installed, go back to the Lifecycle Services environment. On the **Environment add-ins** FastTab, you should see that Global Inventory Accounting is being installed. After a few minutes, the status should change from *Installing* to *Installed*. (You might have to refresh the page to see this change.) At that point, Global Inventory Accounting is ready to use.

If the default language of your Dataverse installation isn't English, follow these steps:

1. Go to **Advanced Setting** > **Administration** > **Languages**.
2. Select *English* (*LanguageCode=1033*), and then select **Apply**.

Set up the integration

Follow these steps to set up the integration between Global Inventory Accounting and Supply Chain Management.

1. Sign in to Supply Chain Management.
2. Go to **System administration** > **Feature Management**.
3. Select **Check for updates**.
4. On the **All** tab, search for the feature named *Global inventory accounting*.
5. Select **Enable now**.
6. Go to **Global inventory accounting** > **Setup** > **Global inventory accounting parameters** > **Integrations parameters**.
7. Depending on which version of Supply Chain Management you're running, follow one of these steps:

    - **Supply Chain Management version 10.0.19 to 10.0.26**: In the **Data service endpoint** and **Global inventory accounting endpoint** fields, enter the URLs that the Global Inventory Accounting team sent to you by email. (Learn more in the How to get the Global Inventory Accounting add-in section.)
    - **Supply Chain Management version 10.0.27 and newer**: You don't need to enter the endpoints, so you can skip this step.

Global Inventory Accounting is now ready to use.

Last updated on 06/24/2026

Business documents supported by Global Inventory Accounting

> **Important**
>
> Global Inventory Accounting for Dynamics 365 Supply Chain Management is currently deprecated and will reach end of support on July 1, 2027. Learn more in Removed or deprecated features in Dynamics 365 Supply Chain Management.

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

After the Global Inventory Accounting Add-in is fully set up, it's ready to process inventory-related documents that are entered in Microsoft Dynamics 365 Supply Chain Management.

Supported business documents

There are two types of business documents:

- **Documents that have a journal** – These documents include product receipt, purchase invoice, packing slip, and sales invoice documents.
- **Documents that don't have a journal** – These documents include counting, movement, and inventory adjustment documents.

Later in this article, purchase orders are used as an example to illustrate the process.

The following table lists the documents that the current release supports.

Expand table

Document type	Document
Purchase Order	Product Receipt

Document type	Document
Purchase Order	Invoice
Sales Order	Packing slip
Sales Order	Invoice
Inventory Journals	Movement
Inventory Journals	Adjustment
Inventory Journals	Counting
Inventory Journals	Transfer
Transfer Order	Shipment
Transfer Order	Receive

> ⓘ **Important**
>
> Global Inventory Accounting asynchronously processes the documents that are entered in Supply Chain Management. No error messages are shown for problematic documents.

Example: Purchase order

Product receipt

Post product receipts in the usual way. On the **All purchase orders** page, select a purchase order, and then, on the Action Pane, on the **Receive** tab, select **Product receipt** to open the **Product receipt journal** page. An operation event and a Global Inventory Accounting event are generated for each line. Therefore, select the **Lines** tab, and then select **Inventory** > **Events and measurements** to open the **Events and measurements** page.

Global Inventory Accounting is an accounting system that is based on events and measurements. The measurement line grid on the **Events and measurements** page shows a list of measurements. Each measurement has a list of dimensions.

For each operation event, there are two types of measurement:

- **Operations measurements** – These measurements describe business documents in generic terms. One measurement is the product quantity using the unit of measure that is used in

the document. There are also measurements that affect the inventory value, such as extended price, discount, discount percent, and line charge.
- **Resource accounting measurements** – These measurements are from the inventory register perspective. They describe the document's impact on the inventory in the inventory unit of measure. If there are multiple inventory registrations, there are multiple lines of resource accounting measurements.

The dimensions are organized in the following way:

- **Duality** – Duality describes the agents who participate in the economic events. For external business documents, the primal dimensions describe the legal entity where the document is entered, whereas the dual dimensions describe the vendor, customer, and so on. For internal business documents (for example, transfer orders), the dual dimensions describe the counterpart warehouse.
- **Dimension type** – Dimension types categorize the dimensions.
- **Dimension** – The name of the dimension.
- **Dimension value** – The value of the dimension.

Global Inventory Accounting event

Global Inventory Accounting takes the operational events and measurements as input. It then does the appropriate accounting for each related ledger, based on the attached currency and convention. You can select **Global inventory accounting events and measurements** to view the Global Inventory Accounting event. The Global Inventory Accounting event follows the same methodology as operation events, but it uses different measurements. There are three major measurement types: product cost quantity, product cost, and variance.

Subledger accounts are used to further classify the measurements. There might be multiple ledgers. These ledgers are linked to the legal entity where the document is entered. You can view the events and measurements for each ledger by changing the value of the **Ledger** field.

Cost element

Based on the cost element policy that is attached to the ledger, the system assigns a cost element to each accounted operation event measurement that affects the inventory cost. These measurements include extended price, discount, discount percent, and line charge.

Measurement line details

The **Overview** tab shows the details of the attached policies. The cost object dimensions show the cost object, based on the cost object policy. Specific identification dimensions show the batch number if the cost flow assumption is *Specific – Batch*.

Purchase invoice

Post the invoice in the usual way. Then, on the Action Pane, on the **Invoice** tab, in the **Journals** group, select **Invoice** to open the invoice journal. An operation event and a Global Inventory Accounting event are generated for each line. Therefore, select the **Lines** tab, and then select **Inventory** > **Events and measurements** to open the **Events and measurements** page. The **Events and measurements** page shows the same measure type. However, because the page shows a different measure role and measure modifier, the impact on the agent (legal entity) differs.

Select **Global inventory accounting events and measurements** to view the Global Inventory Accounting event. The inventory quantity and cost now flow out from the *Goods received not invoiced inventory* subledger account and into the *Cost of goods purchased* subledger account.

Last updated on 06/24/2026

Enable Power BI for Global Inventory Accounting

> **Important**
>
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> **Note**
>
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>
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You can pin tiles, dashboards, and reports from your PowerBI.com account to your Microsoft Dynamics 365 application workspace.

Prerequisites

The following prerequisites must be in place before you can enable Power BI reporting:

- You must be a system administrator in the Dynamics 365 application.
- You must have a PowerBI.com account.
- You must have at least one dashboard and one report in your Power BI account.
- You must be an administrator for your Microsoft Entra account.

- The Microsoft Entra domain must be the same domain that you used for your PowerBI.com account.

Setup

To set up the Power BI integration, follow these steps:

1. Go to the *Global Inventory Accounting Add-in Templates* repo on GitHub and download the latest *Global inventory accounting- production costing* file (for example, `Global.inventory.accounting-.production.costing.Preview.-.v2.1.pbix`).
2. Sign in to PowerBI.com, upload and configure the **Global Inventory Accounting** Power BI report file by following these steps:
   a. Upload **Global Inventory Accounting** Power BI file by following these steps:
      i. Select **New**.
      ii. Select **Upload a file**.
      iii. Select the **Global Inventory Accounting** Power BI report file.
   b. Configure the **Global Inventory Accounting** Power BI report by following these steps:
      i. Go to **My workspace**, find the data set for Global Inventory Accounting, and then, from the **Options** menu, select **Settings**.
      ii. In **Settings for Global Inventory Accounting**, expand **Parameters**, and update all parameters as required. These settings let Power BI report use the data in your Dataverse. In particular, be sure to check the following settings:

          - Overwrite the default **Dataverse URL** values using the values found under **Power platform environment information** in Microsoft Dynamics Lifecycle Services (in the **Power platform integration** section).
          - Overwrite the default **Environment ID** values using the values found under **Environment details** in Lifecycle Services (in the **Manage environment** section).
          - Select the **Edit credentials** link next to the **CDS** label in the **Data source credentials** section. Then sign in to your Dataverse account using the **OAuth2** authentication method.

      iii. Verify that the Power BI reports found at **My workspace** > **Reports** > **Global Inventory Accounting** are now working correctly and display content from your system. If after completing this procedure, you're still having trouble enabling Power BI for Global Inventory Accounting, please submit a support request to Microsoft Customer Service and Support. Include your environment ID, tenant ID, and error details in the request.
3. Register the application as described in Configure PowerBI.com integration. By completing this step, you'll get the **application ID** and **application key** that is required for the next step.

4. Integrate the **Global Inventory Accounting** Power BI report file into Dynamics 365 Supply Chain Management by following these steps:
    a. Go to **System administration** > **Setup** > **PowerBI.com configuration**.
    b. Select **Edit**.
    c. Select **Enable PowerBI.Com integration**.
    d. In the **Application ID** field, enter the application ID that you found in step 4.
    e. In the **Application key** field, enter the application key that you found in step 4.
    f. Select **Save**.
    g. Refresh the page so that the Power BI sign-in dialog box appears.
    h. On the **Options** tab, select **Open report catalog**, and then select the **Global Inventory Accounting** Power BI report file that you uploaded earlier.
    i. Refresh the page. You should see that your report has been added.
    j. Under **Power BI reports**, select the **Global Inventory Accounting** link.

Learn more in Configure PowerBI.com integration.

Last updated on 06/24/2026

Conventions

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A convention is a container for a set of policies that affect system behavior. Based on your business requirements, you must define conventions by using a combination of the various policies that establish how costs should be accounted in Global Inventory Accounting. You can associate each convention with one or more ledgers to ensure consistency in accounting policies that are applied across ledgers.

To set up your conventions, go to **Global inventory accounting** > **Setup** > **Conventions**. For each convention, set the following fields:

- **Name** – Enter the name of the convention.
- **Description** – Enter a description of the convention.
- **Cost Object policy** – Select a cost object policy. These policies determine the level of granularity that the system applies to calculate and maintain the inventory value. The following predefined options are available:
  - Product – Site
  - Product – Site – Warehouse

  For example, if you select *Product – Site*, for every inventory movement (inflow or outflow), the system calculates and maintains the inventory cost of each product at the site level. Therefore, inventory movements at lower levels, such as the warehouse level, don't affect

the inventory value. (An example of a warehouse-level transfer is an item transfer between two warehouses in one site.) Likewise, you can't view the inventory cost at a lower level, such as the warehouse level.

- **Input measurement basis policy** – Select an input measurement basis policy. These policies determine the costs that should flow into the inventory account and the costs that should be charged. The following options are relevant for trading companies:
  - **Normal historical** – All the cost components flow into the inventory account.
  - **Standard** – Standard cost flows into the inventory accounts, and the difference between the applied cost and the actual costs is charged to variance accounts. If you want to create a *Standard* input measurement basis policy, you must first create a price list where the policy can look up the item's standard cost.
  - **Price lists** – Global Inventory Accounting supports fetching item prices from multiple legal entities. You can define a price list that the input measurement basis policy uses. In this way, the system knows where to look up the item price. Follow these steps to set up price lists:
    1. In the **Name** field, enter a name.
    2. In the **Description** field, enter a description.
    3. In the **Costing type** field, select a costing type (*Standard cost* or *Planned cost*).
    4. In the **Price type** field, select a price type (*Cost*, *Purchase*, or *Sales price*).
    5. Add a costing version.
    6. On the Action Pane, select **Price** to validate the item prices on the price list.
- **Cost flow assumption policy** – Select a cost flow assumption policy. These policies determine how costs are removed from inventory and reported as the cost of goods sold. The following predefined options are available:
  - Average
  - Specific – Batch

  > **Note**
  >
  > Global Inventory Accounting is a perpetual inventory system. Therefore, the system tracks the inventory value on a transaction-by-transaction basis.

- **Cost element policy** – You can define cost element policies and link them to this field. A *cost element* is the cost of a resource consumed by an event. You can use cost elements to track

and categorize costs. To create cost element policies, enter information in the following places:

- **Name** field
- **Description** field
- **Cost element** list
- **Rules** grid

If you don't want to break down the inventory value further, you must still create a cost element list that has a single cost element. You must then create a cost element policy to map all the relevant measurement types (cost components) to that cost element.

Last updated on 06/24/2026

Global Inventory Accounting ledger

> **Important**
>
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Global Inventory Accounting has its own set of ledgers. Each time an inventory-related transaction is processed for a relevant legal entity, the system can account for that transaction in any number of Global Inventory Accounting ledgers, as required.

A ledger is a register of debit and credit measures. These measures are classified by using cost elements and subledger accounts. A Global Inventory Accounting ledger is defined by its combination of a currency, a calendar, a convention, and an association with a legal entity.

To set up your Global Inventory Accounting ledgers, go to **Global inventory accounting** > **Setup** > **Global inventory accounting ledgers**. For each ledger, set the following fields:

- **Name** – Enter the name of the ledger.
- **Description** – Enter a description of the ledger.
- **Fiscal calendar** – Specify the fiscal calendar for the ledger.
- **Currency and exchange rate type** – Use the fields on this FastTab to select the accounting currency that the ledger uses, and the exchange rate type. The currency that you select can differ from the currency that the legal entity uses. In Global Inventory Accounting, users can define as many costing ledgers as they require. Global Inventory Accounting supports

inventory accounting in multiple currencies and in multiple valuations. Set the following fields:

- **Currency** – Select the costing currency that inventory-related values are maintained in. These values include the inventory value, cost of goods sold, inventory in transit, and all kinds of variance.
- **Exchange rate type** – Select the exchange rate that the system should use to convert the transaction amount in the transaction currency and the standard cost of the items into the costing currency.

- **Convention name** – Specify a convention. A convention is a collection of policies that establish how costs are accounted in this ledger.

- **Legal entity** – The ledger accounts the documents that are posted to the selected legal entity.

- **Priming** – Select whether inventory transactions that were created before the ledger was created should be processed according to the currency and convention in the ledger. Select one of the following values:
  - **Activated** – The ledger should process transactions that were created before the ledger was created.
  - **Deactivated** – The ledger should process only transactions that are created after the ledger was created.

  > ⓘ **Important**
  >
  > You will **not** be able to come back and set this field after you enter new documents.

- **Status** – The system automatically sets the status of newly created ledgers to *Prepare*.

Last updated on 06/24/2026

Scale unit capability for Supply Chain Management is deprecated and no longer available to new customers

Article • 05/20/2024

Important

The scale unit capability for Microsoft Dynamics 365 Supply Chain Management has been paused for new customers since July 2022 and is now deprecated. Warehouse management only mode replaces some of the functionality planned for scale units and adds many new architectural and integration possibilities. For more information about this and other feature deprecations, see Removed or deprecated features in Dynamics 365 Supply Chain Management.

The Commerce Scale Unit (CSU) Core platform for Microsoft Dynamics 365 Commerce isn't related to the scale unit capability for Dynamics 365 Supply Chain Management in any way. CSU Core is not affected by this announcement and is still fully supported for the Dynamics 365 Commerce application. For more information about that functionality, see Introduction to Commerce Scale Unit (CSU) Core.

Feedback

Was this page helpful? Yes No

Provide product feedback

The Pricing management module is deprecated and no longer available to new customers

Important

The Pricing management module (previously in preview) is deprecated and will be removed in a future release. It's replaced by the Unified pricing management module, which offers nearly all of the same functionality as the deprecated Pricing management module, plus several important enhancements. Both the deprecated pricing management module and the Unified pricing management module use similar or identical navigation paths in the Supply Chain Management user interface, so only one of these modules can be active at a time. As of Supply Chain Management version 10.0.41, if you never enabled the deprecated pricing management module in your environment, you can only use the Unified pricing management module. If you enabled the deprecated pricing management module, you can continue to use it until its removal, but transition to the Unified pricing management module as soon as possible to take advantage of the new features and enhancements that it offers. For help with transitioning from the deprecated pricing management module to the Unified pricing management module, contact Microsoft Support.

Last updated on 07/01/2026

Mobile app home page

> **Important**
>
> The finance and operations (Dynamics 365) mobile app and platform are no longer supported. The platform components supporting the mobile app will be removed in a future update. The mobile app has also been removed from app stores. Previously installed instances of the app will continue to work. For more information, see Removed or deprecated platform features.

This article describes the **Finance and operations (Dynamics 365)** mobile app and provides links to resources that can help you implement it in your organization.

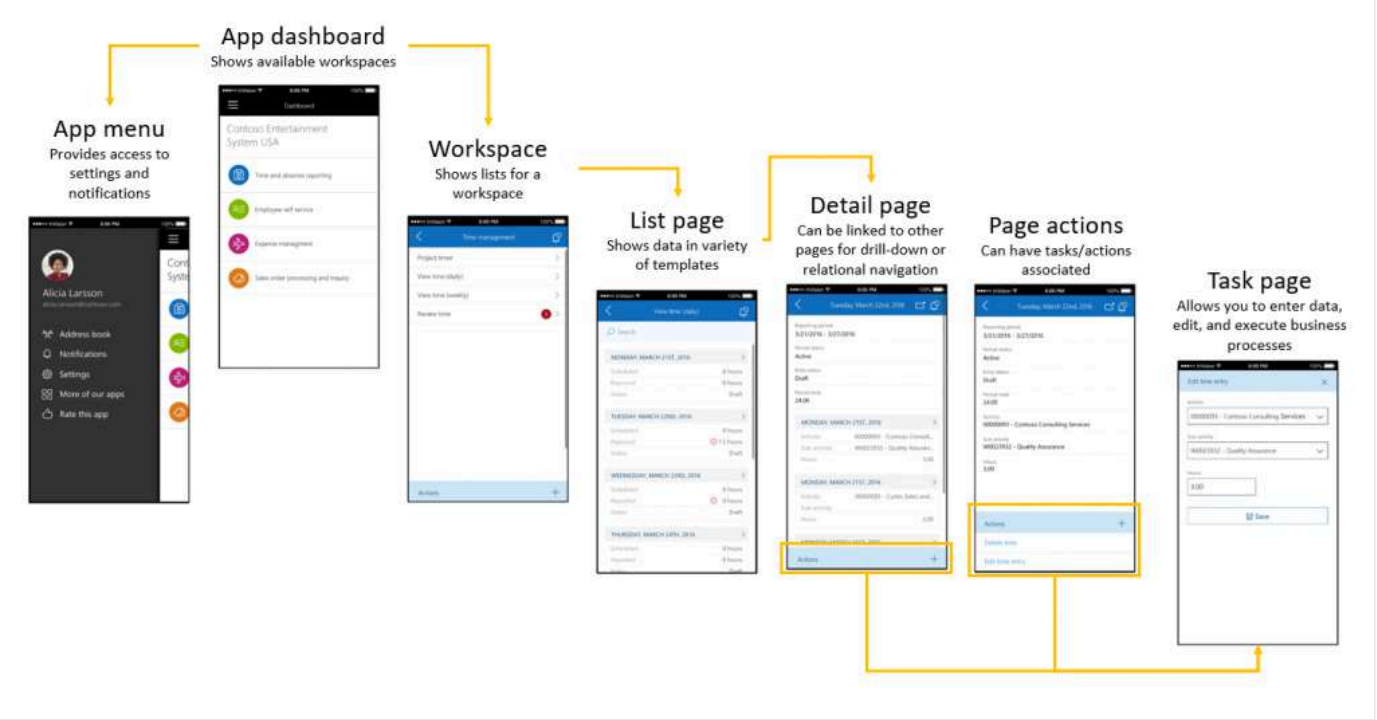
Overview

The mobile app enables your organization to make its business processes available on mobile devices. After your IT admin enables the mobile workspaces for your organization, users can sign in to the app and immediately begin to run business processes from their mobile devices. The mobile app includes the following features that can help increase productivity:

- Users can view, edit, and act on business data, even if they have intermittent network connectivity or their mobile devices are offline. When a device reestablishes a network connection, offline data operations are automatically synchronized.
- IT admins or developers can build and publish mobile workspaces that are tailored to their organization. The app uses your existing code assets. Therefore, you don't have to reimplement your validation procedures, business logic, or security configuration.
- IT admins or developers can easily design mobile workspaces by using the point-and-click workspace designer that's included with the web client.
- IT admins or developers can optionally optimize the offline capabilities of workspaces by using the Business logic extensibility framework. Because data continues to be processed while a device is offline, your mobile scenarios remain rich and fluid, even if devices don't have constant network connectivity.

Elements of the mobile app

Navigation in the mobile app consists of four basic concepts: the dashboard, workspaces, pages, and actions.



1. When you start the app, you go to the **dashboard**.

2. On the dashboard, you can see a list of **workspaces** that are published.

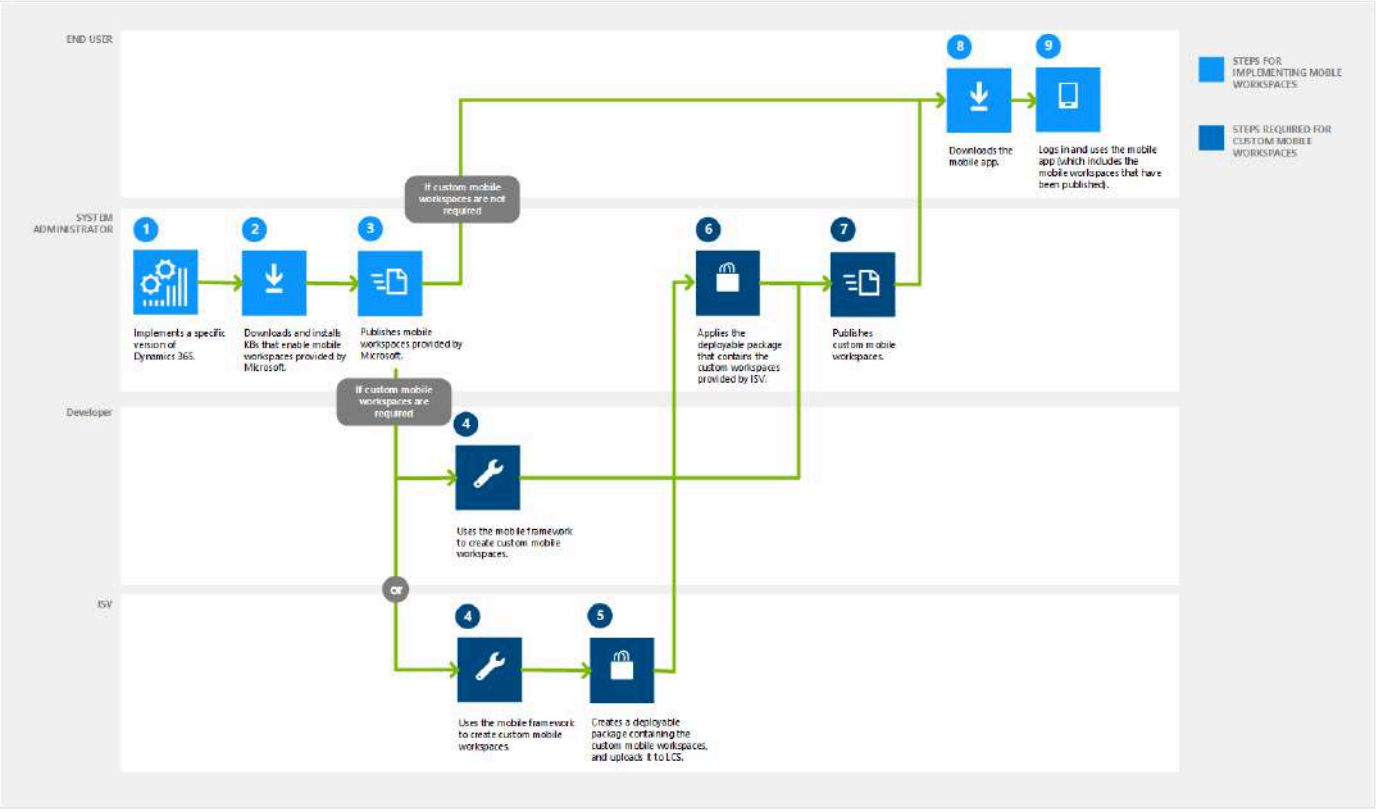
3. In each workspace, you can see a list of **pages** that are available for that workspace.

4. After you're on a page, you can perform several actions. Here are some examples:

    - View detailed data.
    - Navigate to other pages for related data, such as entity details or lines.
    - See a list of **actions** that are available for that page. Actions let you create or edit existing data.

Implementation process

The following illustration shows the process for implementing both mobile workspaces that are provided by Microsoft and custom mobile workspaces.



The following table includes links to resources that can help you implement both mobile workspaces that are provided by Microsoft and custom mobile workspaces. The numbers in the first column correspond to the numbered steps in the previous illustration.

Expand table

Step	Role	Action	Resources to help you complete the action
1	System administrator	Implement the finance and operations app in your organization.	• If you haven't yet deployed a version of Microsoft Dynamics 365, see Deploy a demo environment. • To see a list of mobile workspaces that can be used, see Mobile workspaces recently released.
2	System administrator	**If you're using Microsoft Dynamics 365 for Operations version 1611:** Download and install KBs that enable the mobile workspaces that are provided by Microsoft.	See the following articles for more information: • Cost controlling mobile workspaces • Inventory on-hand mobile workspace • Sales orders mobile workspaces

Step	Role	Action	Resources to help you complete the action
			• Vendor collaboration mobile workspace • Project time entry mobile workspace • Expense management mobile workspace
3	System administrator	Publish the mobile workspaces that are provided by Microsoft.	Publish a mobile workspace
4	Developer or independent software vendor (ISV)	Use the mobile platform to create custom mobile workspaces.	Mobile platform
5	ISV	Create a deployable package that contains custom mobile workspaces, and upload the package to Microsoft Dynamics Lifecycle Services.	Create a deployable package
6	System administrator	Apply the deployable package that contains the custom workspaces that are provided by the independent software vendor (ISV).	Apply a deployable package
7	System administrator	Publish the custom mobile workspaces that are provided by the ISV.	Publish a mobile workspace
9	User	Sign in, and use the mobile app. The app includes the mobile workspaces that have been published by the system administrator.	To see a list of mobile workspaces that are provided by Microsoft, see Mobile workspaces recently released.

Troubleshooting

Mobile platform resources

Last updated on 06/26/2026

Available mobile workspaces

Summarize this article for me

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>
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> **Important**
>
> The finance and operations (Dynamics 365) mobile app and platform are no longer supported. The platform components supporting the mobile app will be removed in a future update. The mobile app has also been removed from app stores. Previously installed instances of the app will continue to work. For more information, see Removed or deprecated platform features.

This article lists the mobile workspaces that you can use with the finance and operations (Dynamics 365) mobile app. The following table shows the deprecation plan for each of these mobile workspaces and whether a replacement experience is planned.

Expand table

Mobile workspace	Description	Deprecation plan
Company directory	View and contact other employees in your organization.	Retire
My team	View your direct reports and extended staff, and send praise for individuals in your reporting chain.	Retire
Invoice approval	Provides a listing of invoices that are assigned to you through the vendor invoice header workflow process.	Retire
Expense management	Capture and upload a receipt, so that you can attach it to an expense report later. The mobile workspace also lets you quickly create an expense line by using an attached receipt.	Replace
Purchase order approval	View and respond to purchase orders with actions such as approve or reject.	Replace

Mobile workspace	Description	Deprecation plan
Project time entry	Enter and save time against a project by using your mobile device.	Replacement available Microsoft Dynamics 365 Project Timesheet
Cost controlling	Cost center managers can see the performance of the cost center.	Retire
Inventory on-hand	Gain insights into reserved and available inventory.	Replace
Sales orders	Stay up to date on your sales orders.	Retire
Vendor collaboration	Vendors can stay up to date on the purchase orders that are sent to them for approval. They can also view information about new and updated purchase orders and contacts.	Retire
Asset management	This workspace lets users view and create maintenance requests and work orders. Users can also view the assigned work order jobs in a calendar or list view. Users can also view and search for assets and functional locations.	Replace

Last updated on 03/10/2026

Set up the Asset management mobile workspace

Article • 08/06/2024

> **Important**
>
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This article describes how to set up Microsoft Dynamics 365 Supply Chain Management and the finance and operations (Dynamics 365) mobile app to run an **Asset management** mobile workspace that workers can use to perform asset management tasks.

Set up maintenance worker users in Supply Chain Management

For each user that requires access to the **Asset management** mobile workspace, follow these steps.

1. In Supply Chain Management, go to **Human resources** > **Workers**, and make sure that a worker record exists for the user that you want to set up. Create a new worker record as required.
2. Go to **Asset management** > **Setup** > **Workers** > **Workers**, and make sure that the worker record that you identified (or created) in the previous step is mapped to a maintenance worker record. Create a new maintenance worker record as required, and set the **Worker** field to the worker record from the previous step.
3. Go to **Asset management** > **Setup** > **Workers** > **Maintenance worker groups**, and make sure that the maintenance worker record that you identified (or created) in the previous step belongs to a maintenance worker group.
4. Go to **System administration** > **Users**.
5. Select the relevant user in the grid.
6. On the **User Details** FastTab, set the **Person** field to the worker account that you want to associate with the current user account. This worker account should be the

worker record that you identified (or created) in step 1 and mapped to a maintenance worker record in step 2.

> ⓘ **Note**
>
> User permissions and security roles apply to the features of the **Asset management** mobile workspace just as they apply to the features of the Supply Chain Management user interface. Therefore, every user that you set up to access the **Asset management** mobile workspace must have the security roles that are required to perform similar operations directly in Supply Chain Management.

Publish the Asset management mobile workspace

To make asset management features available in the finance and operations (Dynamics 365) mobile app, you must publish the **Asset management** mobile workspace.

1. In Supply Chain Management, select the **Settings** button (the gear symbol in upper-right corner), and then select **Mobile app** on the menu.
2. In the **Manage mobile app** dialog box, find the **Asset Management** tile. If it contains the text "In metadata - not published," the workspace hasn't yet been published. If it contains the text "In metadata - published," the workspace has already been published, and you can skip the rest of this procedure.

3. Select the **Asset Management** tile, and then select **Publish** on the toolbar. After a few seconds, you should receive a notification that states that the workspace has been successfully published. Additionally, the text on the tile should change to "In metadata - published."

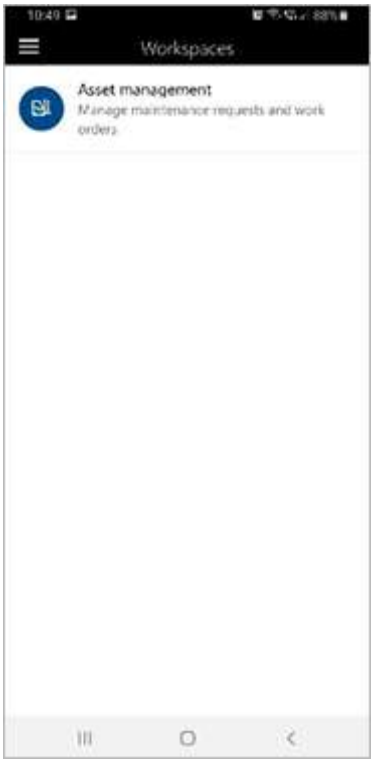
Set up the finance and operations (Dynamics 365) mobile app

1. Open the finance and operations (Dynamics 365) app. The sign-in page should appear. In the **Sign in** field, enter your Supply Chain Management URL, or select a recent URL in the **Recent environments** list, and then tap **Connect**.

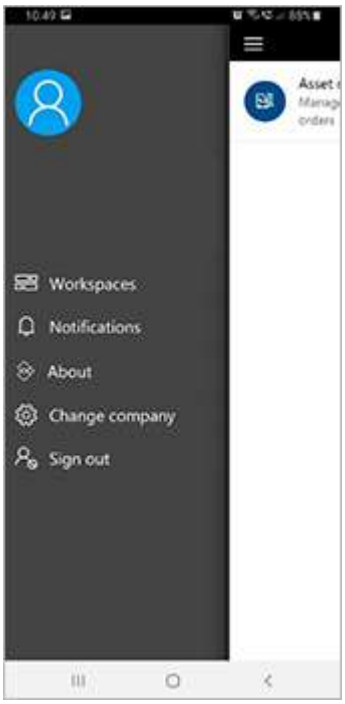
2. If you're prompted to confirm the connection, select the **I understand** check box, and then tap **Connect**.

3. On the **Pick an account** page, use your Microsoft account to sign in to the mobile application.

   The **Workspaces** page appears. It lists every mobile workspace that has been published by your Supply Chain Management instance.



4. If you must change the legal entity (company), tap the Menu button (sometimes referred to as the hamburger or the hamburger button) in the upper-left corner, and then tap **Change company**.



5. On the **Workspaces** page, select the workspace that you want to work with to open it.

Work with the Asset management mobile workspace

For more information about how to work with the **Asset management** mobile workspace, see Use the Asset management mobile workspace.

For more information about the finance and operations (Dynamics 365) mobile app, see the Mobile app home page.

Feedback

Was this page helpful? Yes No

Provide product feedback

Use the Asset management mobile workspace

Article • 06/29/2022

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This article provides information about the **Asset management** mobile workspace. This workspace lets users view and create maintenance requests and work orders. Users can also view the assigned work order jobs in a calendar or list view. Assets and functional locations can also be viewed and searched for.

Overview

Asset Management is an advanced module for managing assets and work order jobs in Dynamics 365 Supply Chain Management. The **Asset management** mobile workspace lets users quickly view assigned work order jobs on the mobile device of their choice. Users can also create and manage maintenance requests, update lifecycle state, and view asset and functional location details by using their mobile device.

Specifically, the **Asset management** mobile workspace lets users perform these tasks:

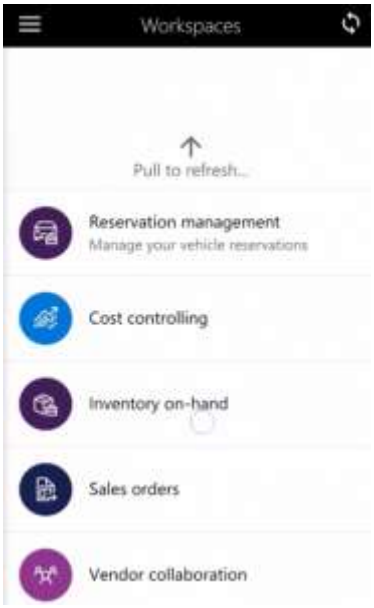
- Create, view, and edit maintenance requests, take a photo or attach an existing image to the maintenance request, change the maintenance request lifecycle state.
- Create, view, and edit work orders, take a photo or attach an existing image to the work order, change the work order lifecycle state, view work order jobs.
- View assigned work order jobs in a calendar view.
- Create, view, and edit work order job, update asset counters, view maintenance checklist, view and edit work order job notes, view the tools required for the work order job.
- View or search for a specific asset or functional location.

Prerequisites

Before you can use the **Asset management** mobile workspace, your admin must set up the required user and worker accounts, and publish the workspace. Learn more in Set up the Asset management mobile workspace.

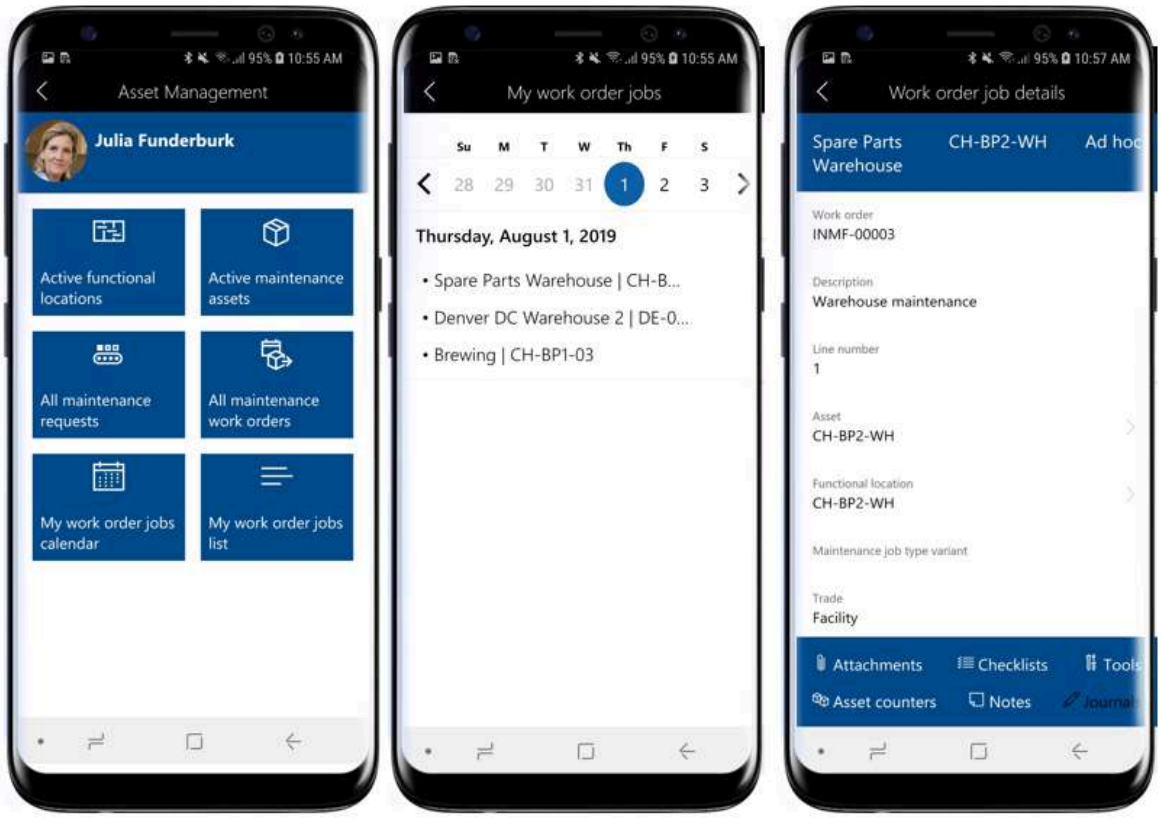
Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the available workspaces for your company are shown. Note that if your system administrator publishes a new workspace later, you'll have to refresh the list of mobile workspaces.



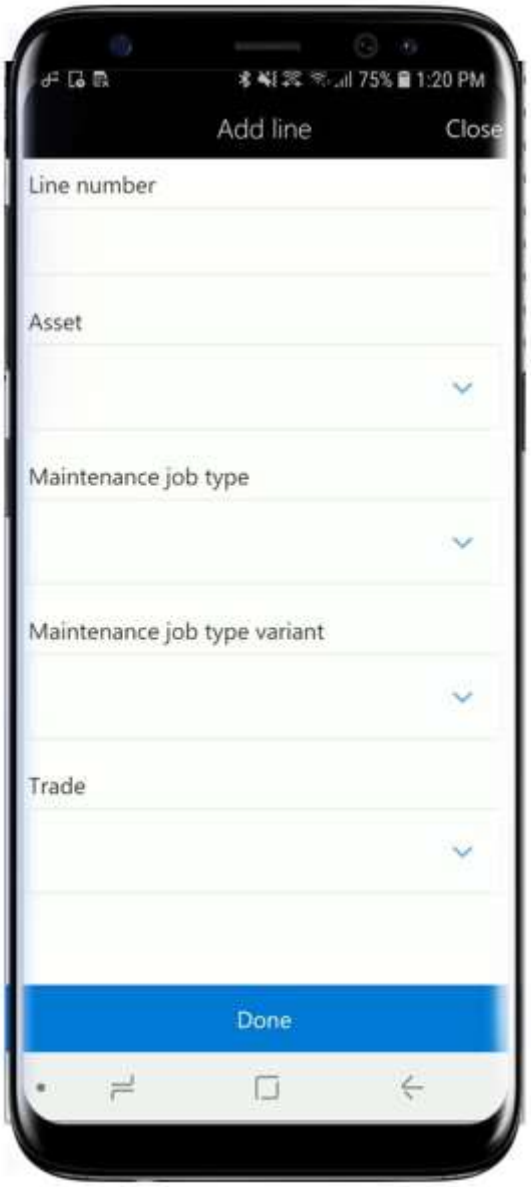
View assigned work order jobs in calendar view

1. On your mobile device, open the **Asset management** workspace.
2. Select **My work order jobs calendar**.
3. Select the date you want to view work order jobs for. In the list, you'll see the asset ID and functional location ID for each work order job.
4. Select a work order job in the list to see job details: Asset and functional location details as well as other navigation links to view **Attachments**, **Checklists**, **Tools**, **Asset counters**, **Notes**, **Journals**.



Create a work order job

1. On your mobile device, open the **Asset management** workspace.
2. Select **All maintenance work orders**.
3. Select the work order you want to create a new work order job for.
4. Select the **Add line** button.
5. Select the **Asset** you want to create a work order job for.
6. Select **Maintenance job type**, **Maintenance job type variant** and **Trade**.
7. Select **Done**.

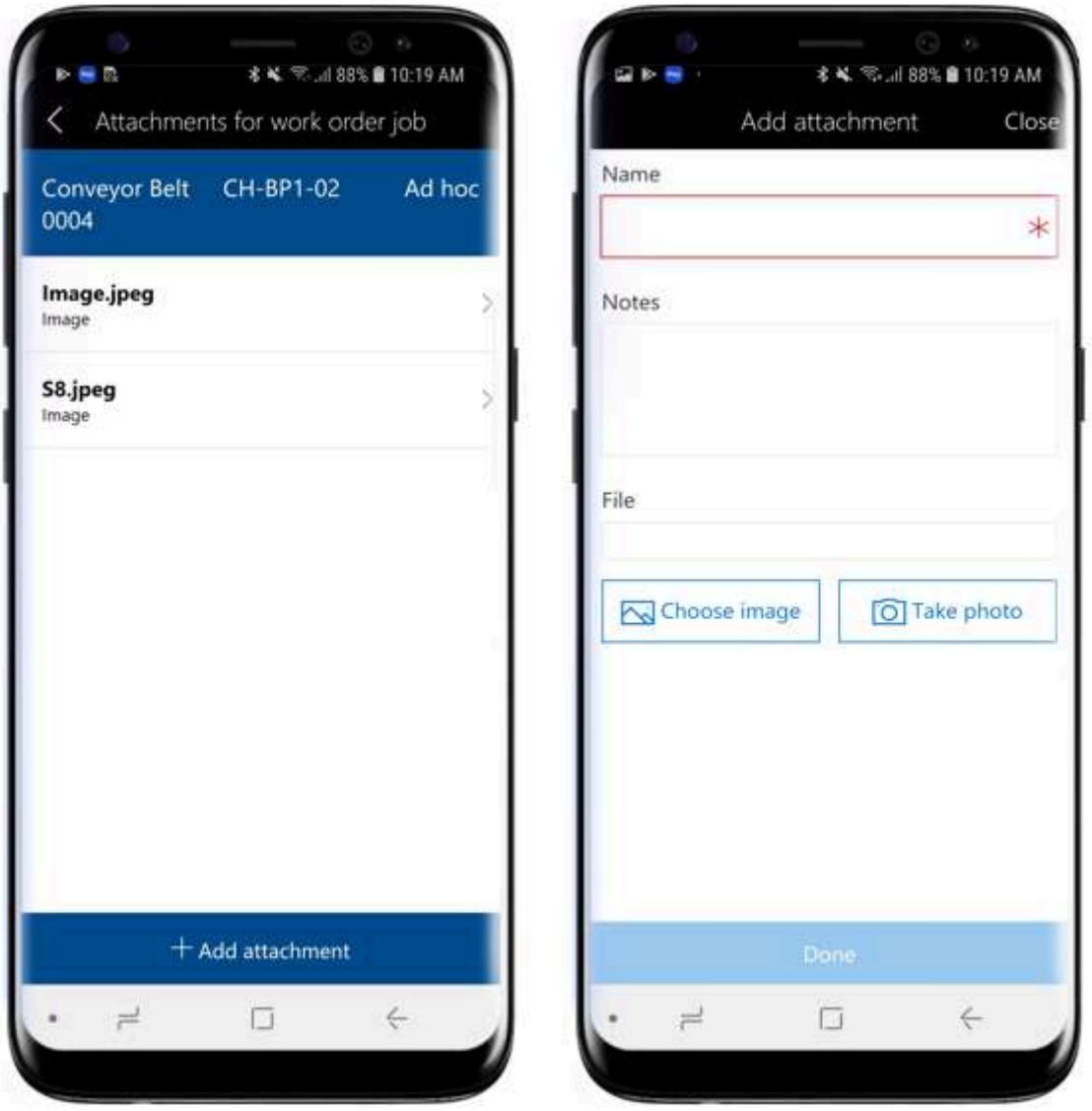


Add attachment to a work order job

1. On your mobile device, open the **Asset management** workspace.
2. Select **All maintenance work orders**.
3. Select the work order > work order job you want to add an attachment to.
    - Alternatively, you can also select **My work order jobs calendar** or **My work order jobs list** on the home page to navigate to the **Work order job details** page.
4. Select **Attachments** on the **Work order job details** page.
5. You'll see existing attachments on the work order job. Select **Add attachment**.
6. Enter **Name** and **Notes** for the attachment.

7. Select **Choose image** to select a photo from the mobile gallery, or **Take photo** to take a photo.

8. Select **Done**.



View maintenance checklist on a work order job

1. On your mobile device, open the **Asset management** workspace.

2. Select **All maintenance work orders**.

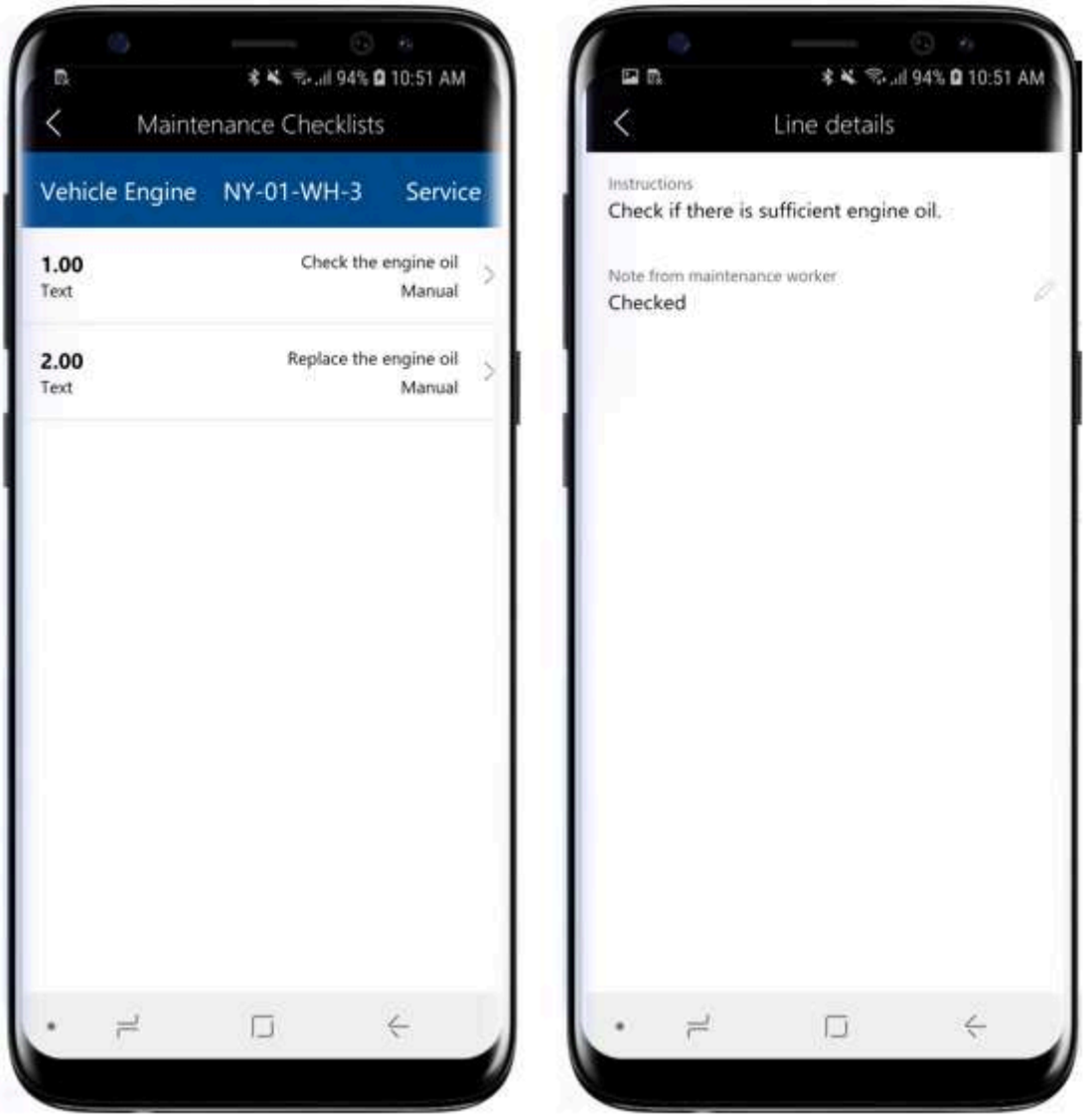
3. Select the work order > work order job you want to view checklists for.

    - Alternatively, you can also select **My work order jobs calendar** or **My work order jobs list** on the home page to navigate to the **work order job details** page.

4. Select **Checklists** on the **Work order job details** page.

5. You'll see a list of checklist lines related to the work order job. Select a checklist line to view **Instructions** and add **Notes**.

6. Select the back button (<) to return to the previous page.



View and update asset counters on a work order job

1. On your mobile device, open the **Asset management** workspace.

2. Select **All maintenance work orders**.

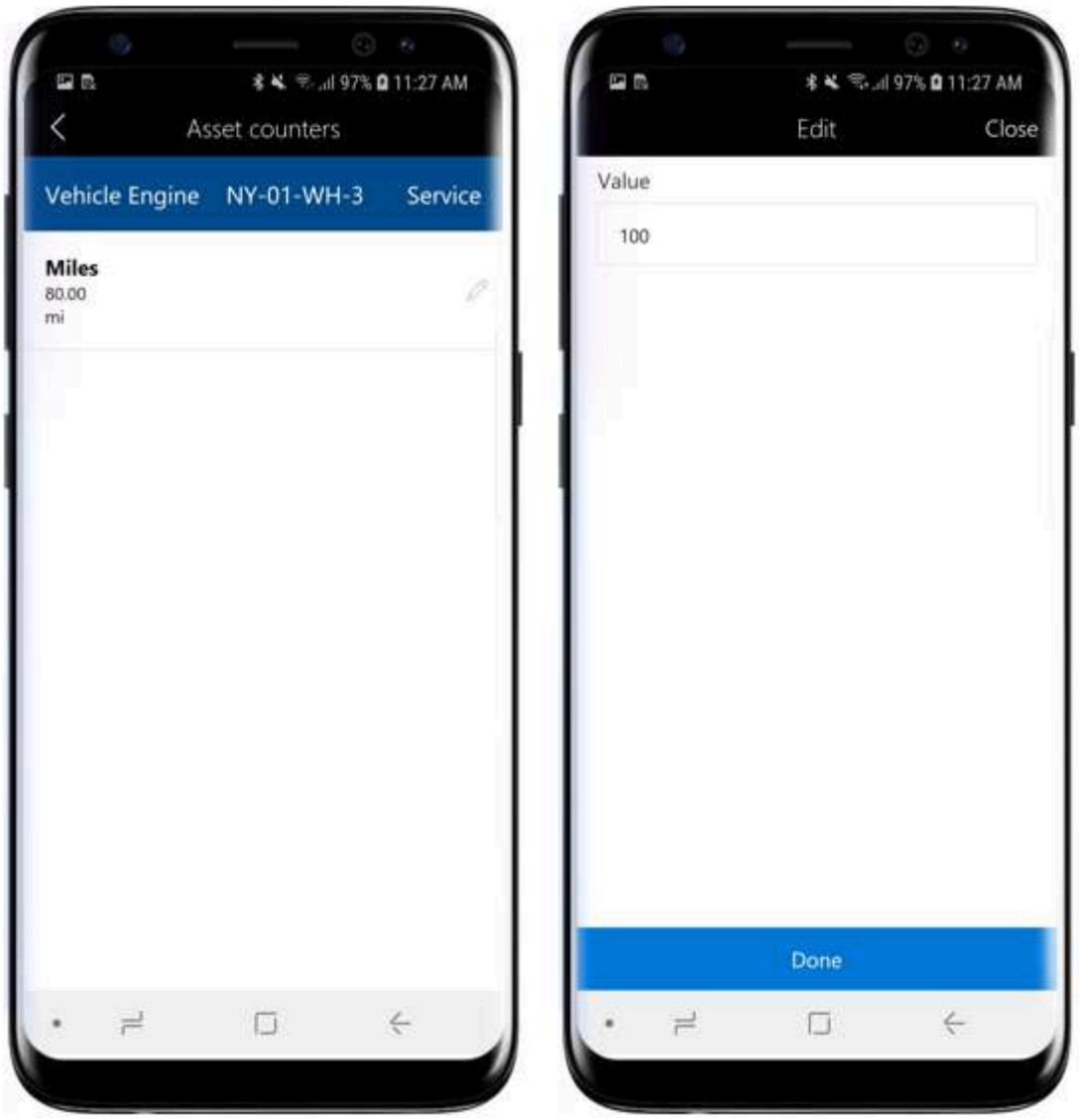
3. Select the work order > work order job you want to view asset counters for.

    - Alternatively, you can also select **My work order jobs calendar** or **My work order jobs list** on the home page to navigate to the **work order job details** page.

4. Select **Asset counters** on the **Work order job details** page.

5. You see a list of asset counters related to the work order job. Select the pencil icon on an asset counter line to update the counter value.

6. Enter a new counter value, and select **Done**.



Register consumption on a work order job

1. On your mobile device, open the **Asset management** workspace.

2. Select **All maintenance work orders**.

3. Select the work order > work order job you want to add consumption registrations for.

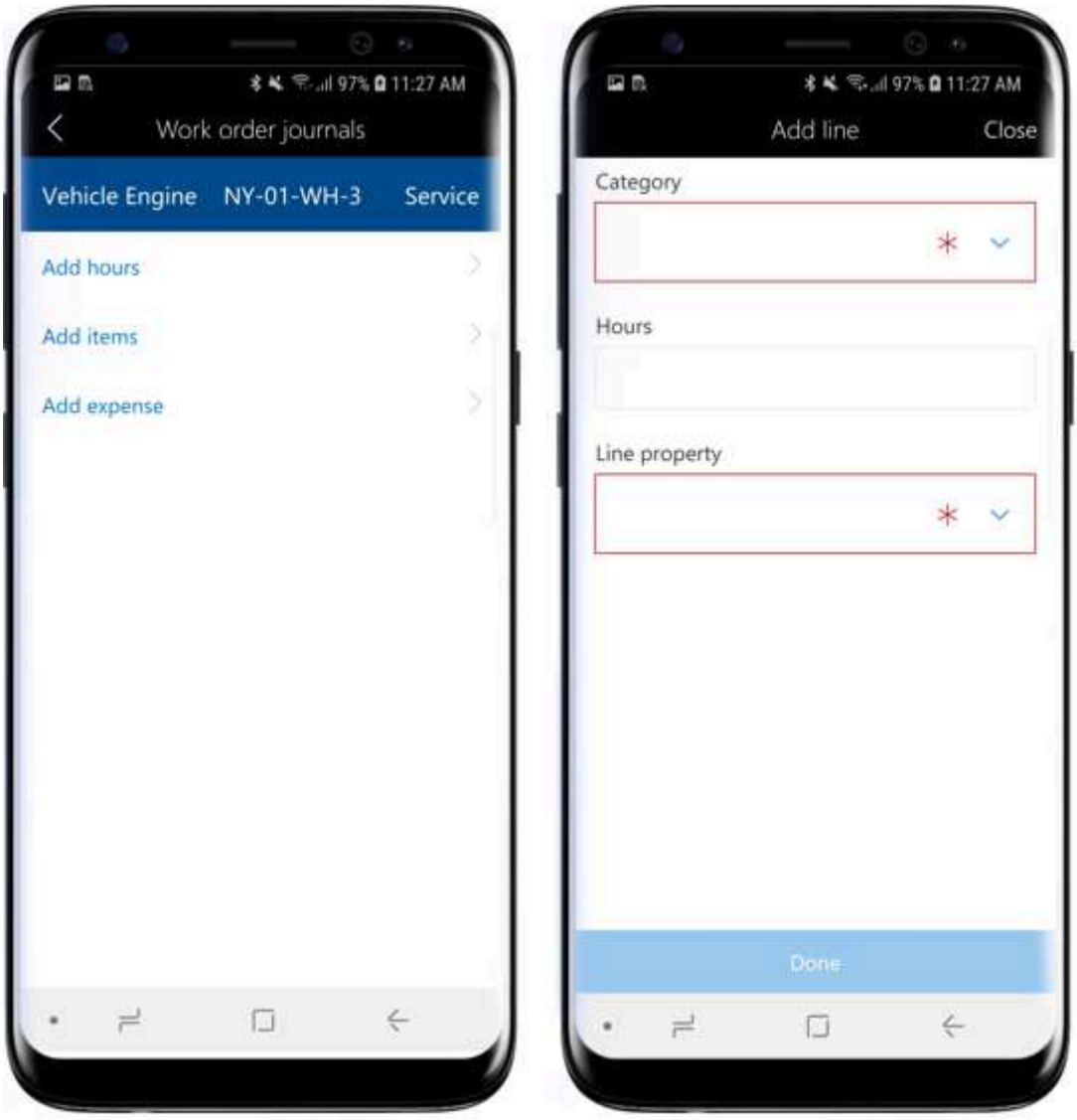
    - Alternatively, you can also select **My work order jobs calendar** or **My work order jobs list** on the home page to navigate to the **work order job details** page.

4. Select **Journals** on the **Work order job details** page.

5. Select **Add hours** to create work hour registrations.
    a. Select the **Category** from the lookup.
    b. In the **Hours** field, enter the number of work hours spent on the work order job.
    c. Select the appropriate **Line property**.
    d. Select **Done**.

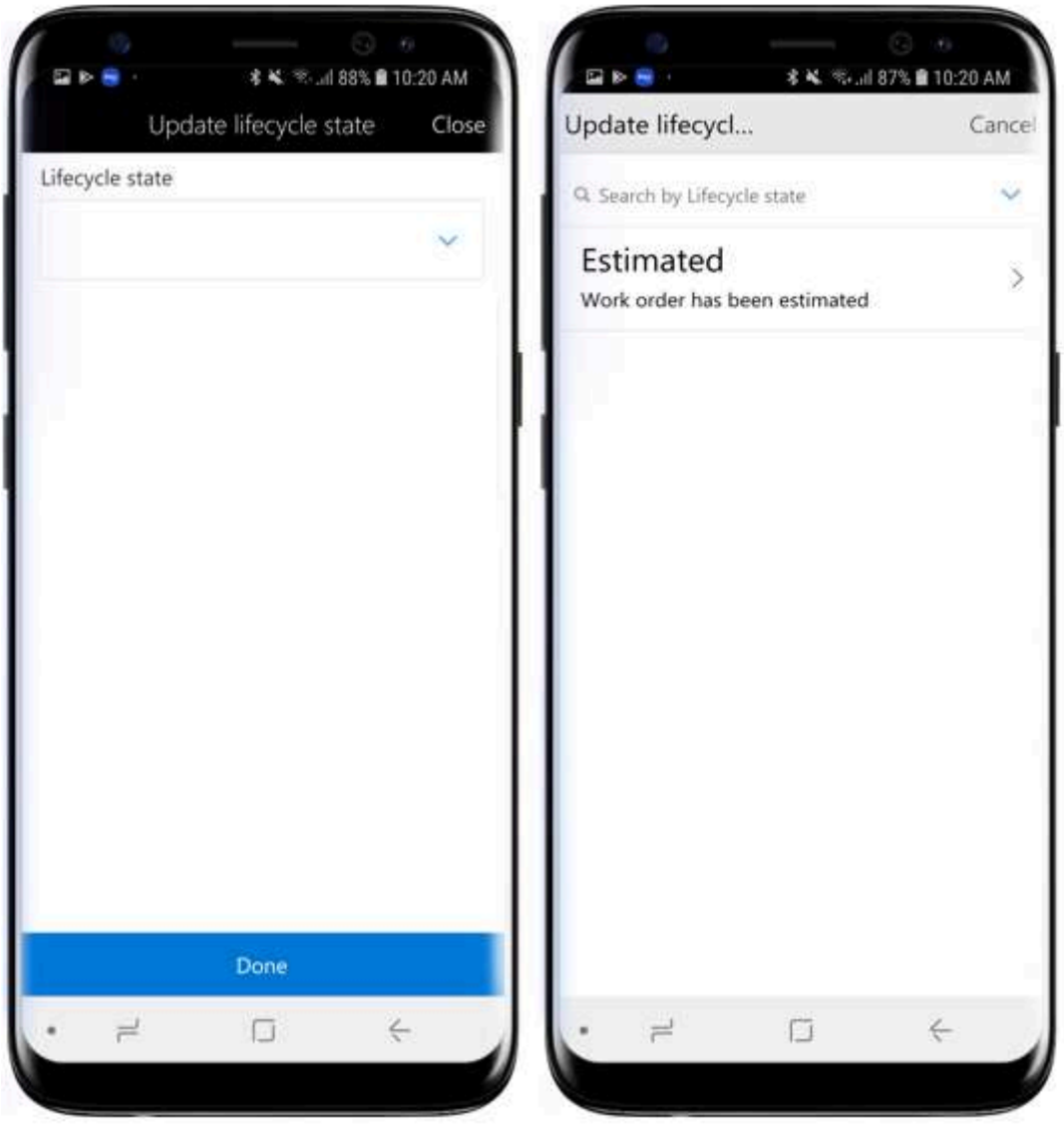
6. Select **Add items** to create item registrations.
    a. Select the **Item number** from the lookup.
    b. Select the **Site** from the lookup.
    c. Enter the **Quantity** of items consumed.
    d. Select **Done**.

7. Select **Add expense** to create expense registrations.
    a. Select the **Category** from the lookup.
    b. Enter the quantity for the expense registration.
    c. Select the **Sales currency** from the lookup.
    d. Enter the **Cost price** for the expense registration.
    e. Select **Done**.



Update lifecycle state on a work order

1. On your mobile device, open the **Asset management** workspace.
2. Select **All maintenance work orders**.
3. Select the work order you want to update lifecycle state for.
4. Select the **Update state** button at the bottom of the screen.
5. Select a new lifecycle state from the list.
6. Select **Done**.

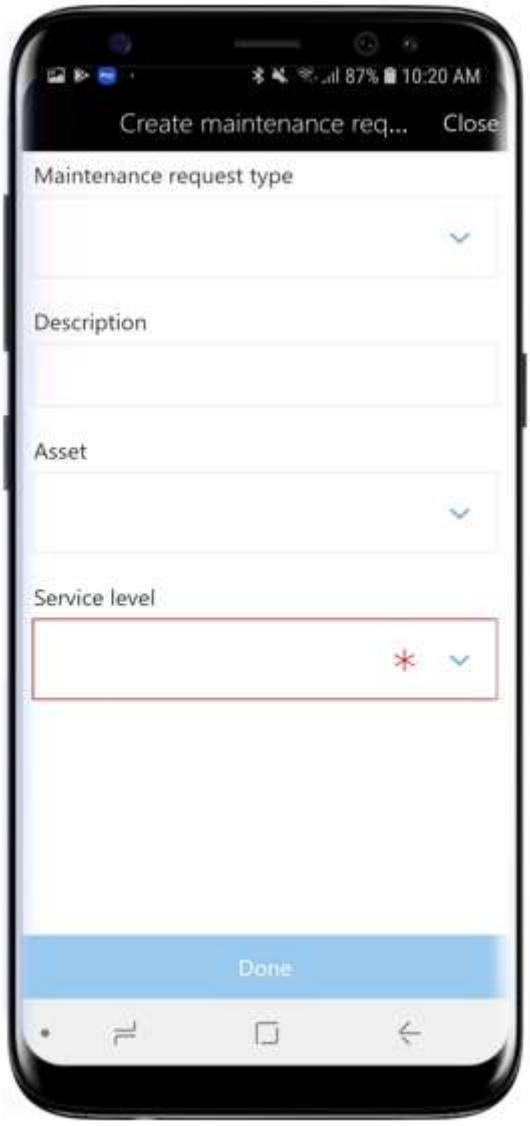


Create a maintenance request

1. On your mobile device, open the **Asset management** workspace.
2. Select **All maintenance requests**.
3. Select **Actions** at the bottom of the screen, and select **Create maintenance request**.
4. If number sequence is enabled for maintenance requests in **Asset management**, the **Maintenance request** field is hidden because it is automatically filled out. If the **Maintenance request** field is visible, enter a maintenance request ID.
5. Select a **Maintenance request type**.
6. Enter a **Description** for the maintenance request.
7. Select the **Asset** you want to create the request for.

8. Select the **Service level** for the maintenance request.

9. Select **Done**.



Add attachment to a maintenance request

1. On your mobile device, open the **Asset management** workspace.

2. Select **All maintenance requests**.

3. Select the maintenance request you want to add an attachment to.

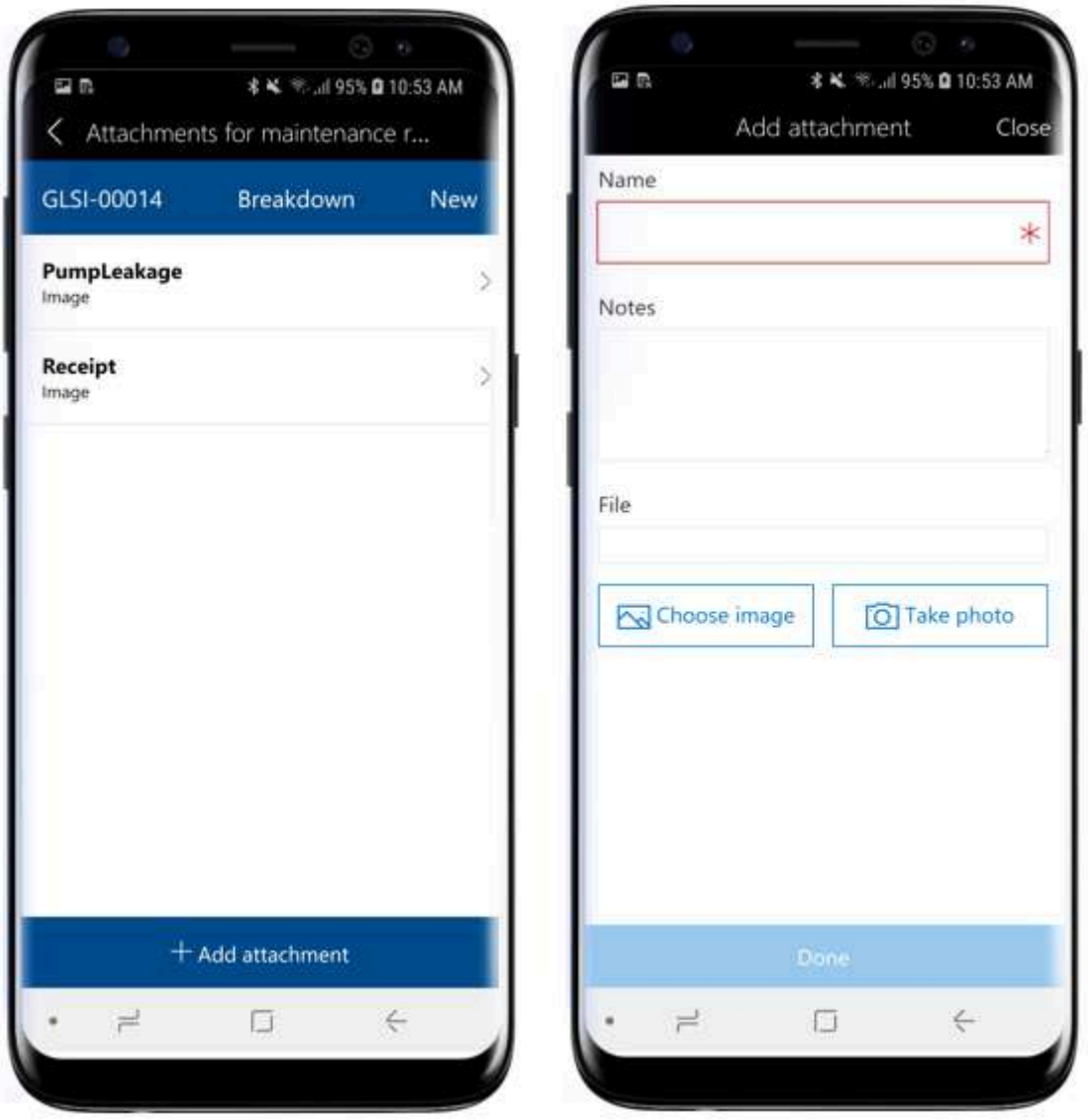
4. Select **Attachments** at the bottom of the screen.

5. Select **Add attachments**.

6. Enter **Name** and **Notes** for the attachment.

7. Select **Choose image** to select a photo from the mobile gallery or **Take photo** to take a photo.

8. Select **Done**.



Feedback

Was this page helpful? Yes No

Provide product feedback

Company directory mobile workspace

> **Note**
>
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This article provides information about the **Company directory** mobile workspace. This workspace lets users view and contact other employees in their organization.

This mobile workspace can be used with the finance and operations mobile app.

Overview

The **Company directory** mobile workspace lets users perform these tasks:

- View a list of employees in the organization.
- Search for employees in the organization.
- View contact information for employees.
- Contact employees from the profile information.

Prerequisites

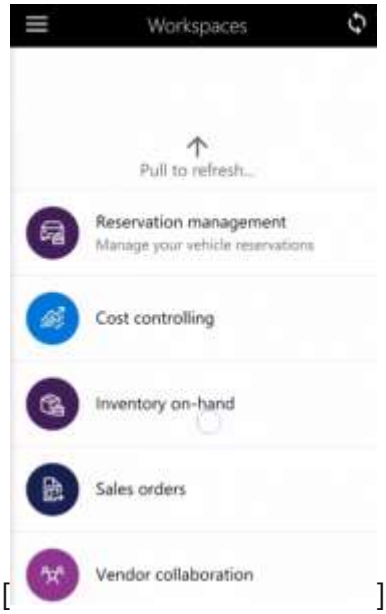
Before you can use this mobile workspace, the following prerequisites must be met.

Expand table

Prerequisite	Role	Description
One of the following products must be deployed in your organization: • A finance and operations app • Microsoft Dynamics 365 Human Resources	System administrator	If you don't already have a finance and operations app deployed in your organization, see Deploy a demo environment. If you don't already have Human Resources deployed in your organization, the system administrator can access a trial version from the Human Resources webpage.
The **Company directory** mobile workspace must be published.	System administrator	See Publish a mobile workspace.

Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Microsoft Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the available workspaces for your company are shown. Note that if your system administrator publishes a new workspace later, you will have to refresh the list of mobile workspaces.



View the company directory by using the mobile workspace

1. In the mobile app, select the **Company directory** workspace. A list of employees is shown.
2. Select an employee. The **Employee profile** page appears. The information on this page includes the employee's first name, last name, title, and department.

Search the company directory by using the mobile workspace

1. In the mobile app, select the **Company directory** workspace.
2. In the **Search** field, enter an employee's first name, last name, title, or department to start the search.
3. Select an employee. The **Employee profile** page appears. The information on this page includes the employee's first name, last name, title, and department.

Last updated on 09/25/2025

Cost controlling mobile workspace

> **Note**
>
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This article provides information about the **Cost controlling** mobile workspace. Cost center managers use this workspace to view information about cost center performance anytime and anywhere.

Use this mobile workspace with the finance and operations (Dynamics 365) mobile app.

Overview

The **Cost controlling** mobile workspace provides an instant view of the current performance of cost centers by comparing actual costs against the budgeted costs. You can drill down to the

status of individual cost elements.

For example, an employee receives an invitation to an international conference, but the organization must cover all the travel expenses. The employee asks their manager whether they can attend the conference. The manager opens the **Cost controlling** mobile workspace on their mobile device to see whether there's budget for the employee to attend the conference.

Data security

The data in the **Cost controlling** mobile workspace is secured through user credentials. Cost center managers are allowed to see data only for their own cost center. The access-level security is managed in the **Cost accounting** module.

Cost accountants define the configuration of the **Cost controlling** mobile workspace in the **Cost accounting** module. After the workspace is published to the mobile app, it's available in the app. Therefore, all cost center managers in the organization can view data in the same format.

Actions, views, and links

The **Cost controlling** mobile workspace provides the following actions, views, and links:

- **Actions:**
  - Use **Select configuration** to select a layout.
  - Use **Select cost object** to select the cost centers to filter data on.

    ```
    > [!NOTE]
    > The cost centers that appear in the list depend on the access that you grant in the **Cost accounting** module.
    ```

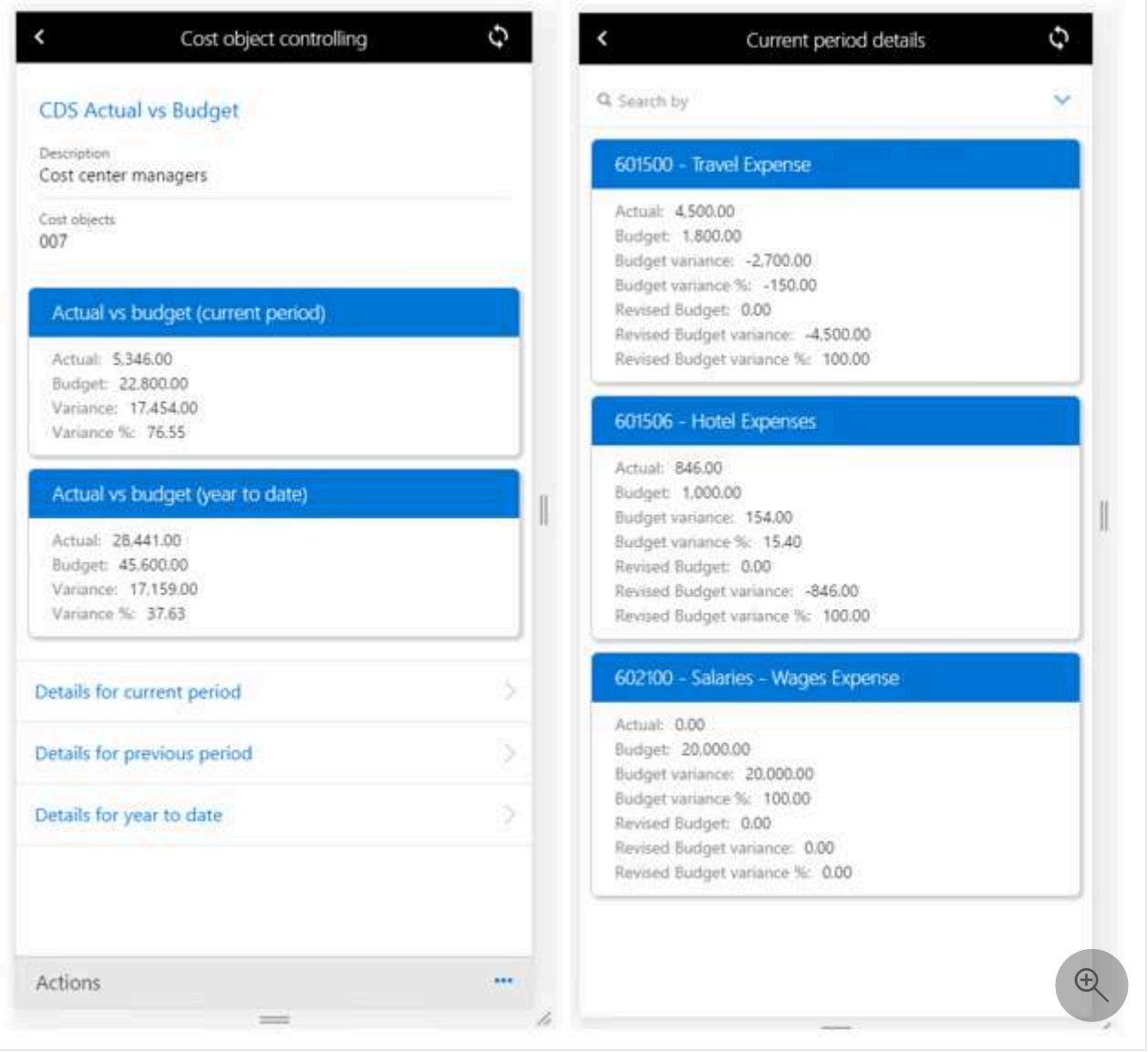
- **Views:** Based on the actions that you select and the configuration in the **Cost accounting** module, you can view the following information on the cards:
  - Actual vs budget (current period)
  - Actual vs revised budget (current period)
  - Actual vs budget (previous period)
  - Actual vs revised budget (previous period)

- Actual vs budget (year to date)
- Actual vs revised budget (year to date)

  The following amounts show on every card: Actual, Budget, Variance, and Variance %.

- **Links:**
  - Details for current period
  - Details for previous period
  - Details for year to date

  When you select a link, a card shows for each cost element. The following amounts show on every card: Actual, Budget, Budget variance, Budget variance %, Revised budget, Revised budget variance, and Revised budget variance %.



Prerequisites

The prerequisites differ, based on the version of Microsoft Dynamics 365 that you deploy for your organization.

Prerequisites if you use Microsoft Dynamics 365 Finance

If you deploy Finance for your organization, the system administrator must publish the **Cost controlling** mobile workspace. For instructions, see Publish a mobile workspace.

Prerequisites if you use version 1611 with Platform update 3 or later

If your organization uses version 1611 with Platform update 3 or later, the system administrator must complete the following prerequisites.

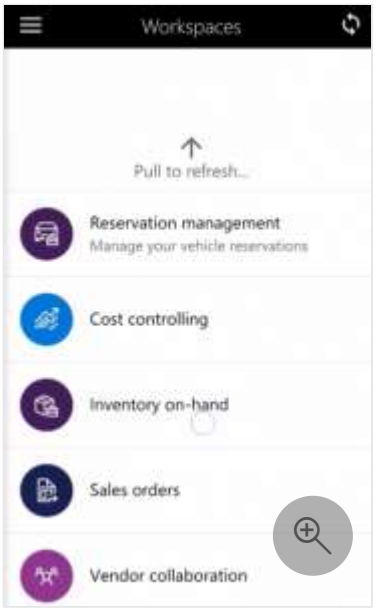
Expand table

Prerequisite	Role	Description
Implement KB 4013633.	System administrator	KB 4013633 is an X++ update or metadata hotfix that contains the **Cost controlling** mobile workspace. To implement KB 4013633, your system administrator must follow these steps: 1. Download the metadata hotfix from Microsoft Dynamics Lifecycle Services (LCS). 2. Install the metadata hotfix. 3. Create a deployable package that contains the **SCMMobile** model, and then upload the deployable package to LCS. 4. Apply the deployable package.
Publish the **Cost controlling** mobile workspace.	System administrator	See Publish a mobile workspace.

Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the available workspaces for your company are shown. If your system administrator publishes a new workspace later, you need to refresh the list of mobile

workspaces.



View the performance of your cost center by using the Cost controlling mobile workspace

1. On your mobile device, select the **Cost controlling** workspace.
2. Select **Cost object controlling**.
3. Select **Actions**.
4. Select **Select configuration** to select a cost controlling layout.
5. Select **Done**.
6. Select **Actions**.
7. Select **Select cost object** to select the cost centers that you have access to.
8. Select **Done**.
9. View the overall performance of your cost center.
10. Select the **Details for current period** link.
11. View the performance of individual cost elements.
12. You can also search for specific cost elements.

Last updated on 06/23/2026

Inventory on-hand mobile workspace

Summarize this article for me

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>
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This article provides information about the **Inventory on-hand** mobile workspace. This workspace helps you gain insights into reserved and available inventory anytime and anywhere.

Use this mobile workspace with the finance and operations (Dynamics 365) mobile app.

Overview

Typically, companies ship and receive inventory multiple times each day. These movements constantly change the on-hand inventory status. The **Inventory on-hand** mobile workspace lets you see the cross-company on-hand inventory status, so that you can gain the latest insights into inventory data on the mobile device of your choice. Regardless of whether you work in the

warehouse, purchasing, sales, manufacturing, or management, or have other roles, you can access on-hand inventory data anytime and anywhere.

The mobile workspace provides an instant view of the on-hand status across facilities. It lets you view on-hand inventory across facilities, current material reservations, and unreserved on-hand inventory. You can also enter item numbers to query on-hand inventory, and can do a filtered search for on-hand products or variants.

Specifically, the mobile workspace provides these features:

- You can search by product number or product name to find products to view the on-hand inventory status for.
- For the selected products, you can view the following information:
  - On-hand inventory per site
  - On-hand inventory per warehouse
  - On-hand inventory per location
  - On-hand inventory per batch (for batch-controlled products)
  - On-hand inventory per inventory status
- Product on-hand inventory is shown in the following ways:
  - By physical inventory (This view represents the total amount.)
  - By physical reserved (This view represents the reserved amount.)
  - By available physical (This view represents available amount that has no reservations.)

Prerequisites

The prerequisites differ, based on the version of Supply Chain Management that you deployed for your organization.

Prerequisites if you use Supply Chain Management

If Supply Chain Management is deployed for your organization, the system administrator must publish the **Inventory on-hand** mobile workspace. Learn more in Publish a mobile workspace.

Prerequisites if you use Platform update 3 or later

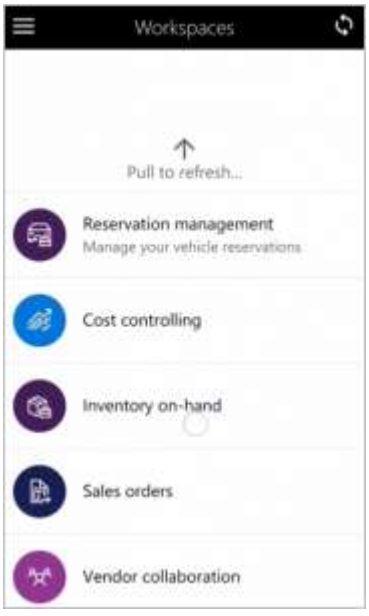
If you deploy platform update 3 or later for your organization, the system administrator must complete the following prerequisites.

Expand table

Prerequisite	Role	Description
Implement KB 4013633.	System administrator	KB 4013633 is an X++ update or metadata hotfix that contains the **Inventory on-hand** mobile workspace. To implement KB 4013633, your system administrator must follow these steps: 1. Download the metadata hotfix from Microsoft Dynamics Lifecycle Services. 2. Install the metadata hotfix. 3. Create a deployable package that contains the **SCMMobile** model, and then upload the deployable package to Lifecycle Services. 4. Apply the deployable package.
Publish the **Inventory on-hand** mobile workspace.	System administrator	See Publish a mobile workspace.

Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Supply Chain Management URL.
3. When you sign in for the first time, enter your user name and password.
4. After you sign in, the app shows the available workspaces for your company. If your system administrator publishes a new workspace later, you need to refresh the list of mobile workspaces.



View the on-hand inventory for a product by using the Inventory on-hand mobile workspace

1. On your mobile device, select the **Inventory on-hand** workspace.
2. Select **Check on-hand for an item**. You see a list of the products that are loaded into your app for offline use. By default, 50 items are loaded, but a developer can change this number. For more information, developers should see Mobile platform.
3. If your item isn't in the list, select **Search more**. Search by product number, or switch to a search by product name.
4. Select a product. If the item has an image, the image is shown.
5. Select one of the following options to view the status of on-hand inventory:

   - View on-hand per site
   - View on-hand per warehouse
   - View on-hand per location
   - View on-hand per batch (for batch-controlled products)
   - View on-hand per inventory status

   Product on-hand inventory is shown in the following ways:

   - By physical inventory (This view represents the total amount.)
   - By physical reserved (This view represents the reserved amount.)
   - By available physical (This view represents the available amount that has no reservations.)

Last updated on 03/10/2026

Invoice approvals mobile workspace

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> The finance and operations (Dynamics 365) mobile app and platform are no longer supported. The platform components supporting the mobile app will be removed in a future update. The mobile app has also been removed from app stores. Previously installed instances of the app will continue to work. For more information, see Removed or deprecated platform features.

This article provides information about the **Invoice approvals** mobile workspace. This workspace provides a list of invoices that you receive through the vendor invoice header workflow process.

Use this mobile workspace with the finance and operations mobile app.

Overview

The **Invoice approvals** mobile workspace lets Accounts payable clerks and managers view invoices that the vendor invoice header workflow process assigns to them. You can view the invoice information, and even the line and distribution details, to help you make informed approval decisions. From the workspace, you can take action to move the invoice through the workflow process.

Prerequisites

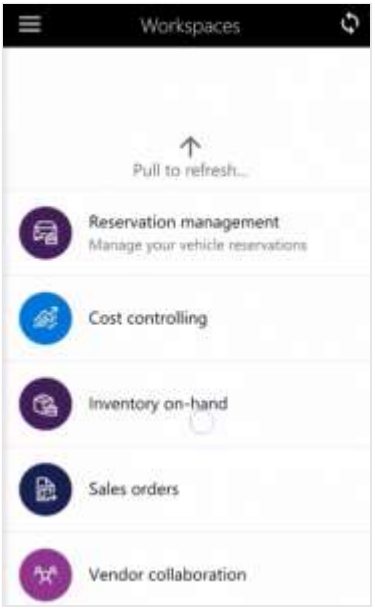
Before you can use this mobile workspace, the following prerequisites must be met.

Expand table

Prerequisite	Role	Description
Microsoft Dynamics 365 Finance must be deployed in your organization.	System administrator	See Deploy a demo environment.
The **Invoice approvals** mobile workspace must be published.	System administrator	See Publish a mobile workspace.

Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Microsoft Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the app shows the available workspaces for your company. If your system administrator publishes a new workspace later, you need to refresh the list of mobile workspaces.



Approve invoices by using the Invoice approvals mobile workspace

1. On your mobile device, select the **Invoice approvals** workspace.
2. Select the invoice that the vendor invoice header workflow process assigned to you.
3. On the **Invoice details** page, review the invoice header information, such as the vendor and date information.

4. Select a line on the invoice to view more detailed information about it in the **Invoice line details** view.
5. In the **Invoice line details** view, select **Distributions** to show the line distributions. Here, you can view the accounting for the invoice line. The information that is shown includes the financial dimensions and the main account.
6. On the **Invoice details** page, select **Distributions** to show all distributions. Here, you can view the accounting for the whole invoice. The information that is shown includes the financial dimensions and the main accounts.
7. Select **Attachments** to view any notes or files that are attached to the invoice.
8. On the **Invoice details** page, select the appropriate workflow action to complete your review process.
9. Select **Done**.

Last updated on 11/26/2025

My team mobile workspace

> **Important**
>
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This article provides information about the **My team** mobile workspace. This workspace lets managers view their direct reports and extended staff. They can also send praise to individuals in their reporting chain.

Use this mobile workspace with the finance and operations mobile app.

Overview

The **My team** mobile workspace lets managers perform these tasks:

- View a list of the manager's direct reports.
- View a list of the manager's extended reporting team.
- View detailed information for each team member, such as birth date, seniority date, years of service, and compensation and performance information.
- Send praise to any individual in the manager's extended reporting team.

Prerequisites

Before you can use this mobile workspace, the following prerequisites must be met.

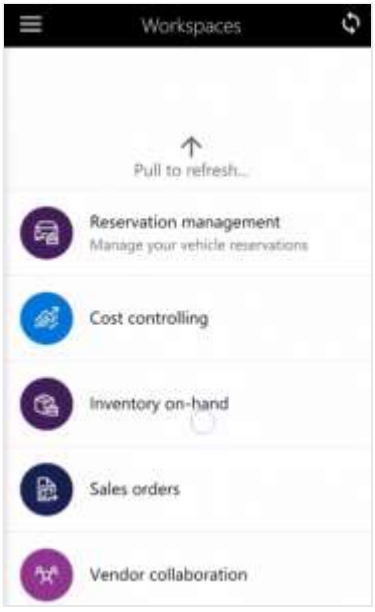
Expand table

Prerequisite	Role	Description
One of the following products must be deployed in your organization:	System administrator	If you don't already have a finance and operations app deployed in your organization, see Deploy a demo environment. If you don't already have Human Resources deployed in your organization, the system administrator can access a trial version from the Human Resources webpage.

Prerequisite	Role	Description
• A finance and operations app • Microsoft Dynamics 365 Human Resources		
The **My team** mobile workspace must be published.	System administrator	See Publish a mobile workspace.

Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Microsoft Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the available workspaces for your company are shown. Note that if your system administrator publishes a new workspace later, you will have to refresh the list of mobile workspaces.



View team members by using the My team mobile workspace

1. In the mobile app, select the **My team** workspace. A list of team members is shown. The list also shows each team member's title, and any direct reports that the member has.

2. Select a team member. The **Team member summary** page appears. The information on this page includes the team member's birth date, seniority date, years of service, years in the current position, and compensation information.

View extended team members by using the My team mobile workspace

1. In the mobile app, select the **My team** workspace. A list of team members is shown. The list also shows each team member's title, and any direct reports that the member has.
2. Select the **Direct reports** link. A list of your extended team is shown.
3. Select a team member. The **Team member summary** page appears. The information on this page includes the team member's birth date, seniority date, years of service, years in the current position, and compensation information.

Send praise about team members by using the My team mobile workspace

1. In the mobile app, select the **My team** workspace. A list of team members is shown. The list also shows each team member's title, and any direct reports that the member has.
2. Select a team member. The **Team member summary** page appears.
3. Select **Send praise**.
4. Enter the text of the praise that you want to send.
5. Select **Done**.

Last updated on 06/26/2026

Project mobile applications overview

Three mobile applications related to project time are available for Microsoft Dynamics 365: Project Timesheet, Project Time Entry, and My timesheets/Timesheets for my review (Optimized for mobile).

Project Timesheet mobile app

The Dynamics 365 Project Timesheet mobile app lets users submit and approve timesheets for projects on their mobile device. This mobile app surfaces the timesheet functionality in the Project management and accounting area of Dynamics 365 Finance. It helps improve user productivity and efficiency, and also helps with the timely entry and approval of project timesheets.

Project Time Entry workspace

The Project Time Entry workspace lets users enter and save time against a project. However, it doesn't let users submit or approve project timesheets. It can be used only for time entry. This mobile workspace is intended to be used together with the finance and operations mobile application.

My timesheets/Timesheets for my review

My timesheets (Optimized for mobile) and Timesheets for my review (Optimized for mobile) make up a solution that lets employees create, submit, and approve project timesheets on a mobile device. Workers can access the Finance functionality through the URL for the application. This solution doesn't require the finance and operations mobile application.

For more information

- Project timesheet mobile application
- Project time entry
- Project timesheets on a mobile device
- Implement custom fields for the Microsoft Dynamics 365 Project Timesheet mobile app on iOS and Android

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Last updated on 01/31/2026

Project timesheet mobile application

Article • 03/19/2025

Overview

The Microsoft Dynamics 365 Project Timesheet mobile app lets users submit and approve timesheets for projects on their mobile device (iPhone or Android). This mobile app surfaces the timesheet functionality that resides in the Project management and accounting area of Dynamics 365 Finance. It helps improve user productivity and efficiency, and also enables timely entry and approval of project timesheets.

Deprecation of mobile app (Dec 2025)

Microsoft ended support for Xamarin on **May 1, 2024**. As a result, the Dynamics 365 Project Timesheet mobile app (which is based on Xamarin) will be **deprecated by Dec 31, 2025**.

You can use Timesheets optimized for mobile as an alternative mobile experience that supports creation, submission, and reviewing of timesheets.

> **Important**
>
> Learn about the end of Microsoft Xamarin support in the Xamarin Support Policy. Microsoft recommends that customers switch to Timesheets optimized for mobile to avoid potential disruptions caused by security issues that result from ending Xamarin support.

Download and install the mobile app

Download and install the Microsoft Dynamics 365 Project Timesheet mobile app for Android or iPhone from the mobile store for your device.

Enable the app

In Finance, the Project Timesheet mobile app must be enabled. To enable the functionality, go to **Project management and accounting parameters** > **Timesheet** and select the **Enable Microsoft Dynamics 365 Project Timesheet** parameter.

Resolve sign-in issues

Issue: During sign-in to the Project Timesheet Mobile app, users receive an error message that states that they "can't access the application '2bc50526-cdc3-4e36-a970-c284c34cbd6e' in that tenant."

Issue: During sign-in to the Project Timesheet Mobile app, users receive an error that resembles one of the following examples:

- "AADSTS50020: User account '[user name]' from identity provider 'https://sts.windows.net/[app ID]' doesn't exist in tenant '[tenant ID]' and can't access the application '[app ID]' in that tenant."
- "Selected user account doesn't exist in tenant '[tenant ID]' and can't access the application '[app ID]' in that tenant."

Explanation: These issues are caused by a change that was made to Microsoft Entra ID in May 2022 and that is related to external users. Because this change wasn't made to finance and operations apps, it can affect customers on any version of the platform or application.

Fix: All external users must be invited to the tenant through Microsoft Entra ID. For more information, see Invite users with Microsoft Entra B2B collaboration.

Sign in to the app

1. Start the app on your mobile device.
2. Enter your Finance URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. You'll be signed into your default company.

Submit a project timesheet

You can create and submit a project timesheet in the app. You can base a new timesheet on information from a previous timesheet, saved lines, or project assignments. If you're designated as a delegate, you can also enter a timesheet for another worker. To create a timesheet as a delegate, select the **Menu** button and then select a resource name.

The timesheet page creates a new timesheet for the timesheet period, based on the current date. The work week is displayed. If the timesheet period covers multiple weeks,

you can select another work week from the work week tabs. If a timesheet exists for the current date, it's displayed. If you need to create a new timesheet in a different timesheet period, select the **Menu** button and then select **New timesheet**.

You can enter project information by clicking the **Add time** action or the **Copy time from** action. The **Copy time from** action copies project line information, but not the hours. The **Copy time from** menu includes the following options:

- **Copy from existing timesheet** - Copy timesheet lines from an existing timesheet.
- **Copy from favorite** - Create new timesheet lines by using the timesheet settings that you selected as favorites.
- **Copy from assignment** - Create new timesheet lines from assigned projects.

The project information that is displayed is dependent on the mobile parameters that you defined on the **Project management and accounting parameters** page.

In the **Legal entity** field, select the legal entity for which you performed project work. The **Legal entity** field is available only if intercompany timesheet support has been enabled for your legal entity.

Select the customer who is associated with the project for the timesheet. For the initial release on the Android, entry by customer isn't supported, as you must select the project first. If you selected the project first, the **Customer** field is filled in automatically.

In the **Project** field, select the project that you're entering time for. The **Customer** field is filled in automatically.

The customer and project lookups enable searching across both customers and projects.

Select information in the **Category**, **Activity**, **Line property**, **Sales tax group**, and **Item sales tax group** fields as required. These fields can be overridden.

The **Line property** field is set to a default value, based on project management and accounting parameters. When the project/category and category/resource parameters are enabled, the **Line property** value is set to the default value you have defined for this validation. When the project/category and category/resource parameters aren't enabled, the **Line property** value defaults according to the **Enable default line property** field on the **Project management and accounting parameters** page. The **Line property** value can be overridden.

Select a day to add time. Enter the number of hours that you worked each day.

To add comments about the hours you're entering, click **Add comments**, and then enter comments for an internal audience, customer audience, or both. Internal comments can be viewed by project managers. Customer comments are included on invoices.

To save the line as a favorite, select the check box, and then click **Save as favorite**.

Financial dimension and attachment support aren't provided in the mobile application.

Continue adding project lines as needed to complete your timesheet.

Click **Submit** to send the timesheet to the approval workflow.

Review timesheets

A list of the timesheets that need to be reviewed is available on the menu. This option is only available if you've been designated as a workflow approver. Both header and line approval are supported. Line level approval offers the ability to mark one or more lines for approval. After reviewing the timesheet information, click **Approve**, **Delegate**, or **Return** to continue the workflow.

Feedback

Was this page helpful? Yes No

Provide product feedback

Purchase order approval mobile workspace

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

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>
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> **Important**
>
> Starting February 2026, new customers can't create projects in Microsoft Dynamics Lifecycle Services for Microsoft Dynamics 365 Finance, Microsoft Dynamics 365 Supply Chain Management, and Microsoft Dynamics 365 Project Operations. New customers should use the Power Platform admin center instead. For more information, see Lifecycle Services project creation freeze.

This article provides information about the **Purchase order approval** mobile workspace. This workspace lets you view purchase orders and respond to them through actions. For example, you can approve or reject a purchase order.

Overview

Purchase orders that require approval go through an approval workflow. The workflow can include various steps that require that one or more people take action. For example, a person might have to complete a task or approve the purchase order.

The **Purchase order approval** mobile workspace lets you easily view and respond to purchase orders from your mobile device. This workspace also lets you take the same workflow actions

that you can take from the web client.

Prerequisites

The prerequisites vary, depending on the version of Supply Chain Management that has been deployed for your organization.

Prerequisites if you use Supply Chain Management

If Supply Chain Management has been deployed for your organization, the system administrator must publish the **Purchase order approval** mobile workspace. For instructions, see Publish a mobile workspace.

Prerequisites if you use Microsoft Dynamics 365 for Operations version 1611 with Platform update 3 or later

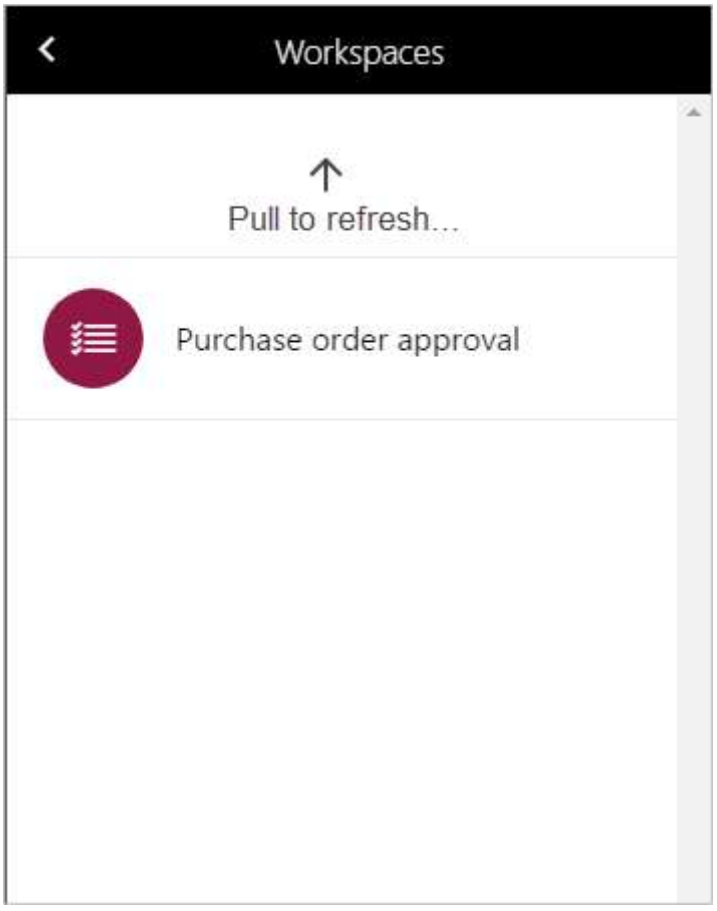
If Microsoft Dynamics 365 for Operations version 1611 with Platform update 3 or later has been deployed for your organization, the system administrator must complete the following prerequisites.

Expand table

Prerequisite	Role	Description
Implement KB 4017918.	System administrator	KB 4017918 is an X++ update or metadata hotfix that contains the **Purchase order approval** mobile workspace. To implement KB 4017918, your system administrator must follow these steps: 1. Download the metadata hotfix from Microsoft Dynamics Lifecycle Services. 2. Install the metadata hotfix. 3. Create a deployable package that contains the **SCMMobile** model, and then upload the deployable package to Lifecycle Services. 4. Apply the deployable package.
Publish the **Purchase order approval** mobile workspace.	System administrator	See Publish a mobile workspace.

Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Microsoft Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the available workspaces for your company are shown. If your system administrator publishes a new workspace later, you'll have to refresh the list of mobile workspaces.



View orders that are assigned to you

1. On your mobile device, select the **Purchase order approval** workspace.
2. Select **Orders assigned to me** to view all the purchase orders for which you've been asked to take action in the purchase order approval workflow.
3. Select an order. On the **Order details** page, you'll see the order header information and lines. You can also find guidelines from the workflow task.
4. Select **Accounting distributions** to open the **Header accounting distributions** page.
5. Return to the **Order details** page, and select a line. From the order line details, you can also explore the line-specific accounting distributions.

Complete an action on the purchase order

After you've viewed the purchase order that is assigned to you and read the workflow instructions, you should be ready to take action.

1. On your mobile device, select the **Purchase order approval** workspace.
2. Select **Orders assigned to me** to view all the purchase orders for which you've been asked to take action in the purchase order approval workflow.
3. Select an order, and view the details page.
4. Select **Actions** to show the available actions. The actions that are available depend on the task that has been assigned to you.

Expand table

Task action	Approval action
Complete	Approve
Return	Reject
Request change	Request change
Delegate	Delegate

5. Select the appropriate action.
6. On the **Complete task** page, enter a comment. If you select the **Delegate** action, you must select a user to delegate the task to.
7. Select **Done**. After you refresh your workspace, the purchase order will no longer be in your list.

Last updated on 03/10/2026

Sales orders mobile workspace

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

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> **Important**
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This article provides information about the **Sales orders** mobile workspace. This workspace helps you stay up to date about your sales orders anywhere and anytime.

This mobile workspace is intended to be used with the finance and operations (Dynamics 365) mobile app.

Overview

The **Sales orders** mobile workspace lets you view detailed information about each sales order. This information includes the status of the order, contact information for the customer, and contact information for the order taker. The **Sales orders** mobile workspace provides an instant

view of sales orders. You can view all sales orders, view sales orders by customer, or view information about a specific sales order.

The mobile workspace provides two views to help you analyze sale orders in depth.

View all sales orders

This view lists all sales orders.

- Use one of the following filters to select the sales orders to view:
  - Search by sales order
  - Search by customer account
  - Search by customer name
  - Search by status
  - Search by release status
  - Search by created date and time

- After you select sales orders, you can view the details of specific orders. Specifically, you can view the following information:
  - Customer name and address information
  - Various dates for the sales order, such as the requested ship date and the confirmed ship date
  - Contact information for the order taker
  - Customer contact information
  - Order lines
  - Shipments that show how and when a sales order was shipped

View orders for a customer

This view lists sales orders by customer.

- Use one of the following filters to view orders for a customer:
  - Search by name
  - Search by account

- After you select a customer, you can view the following information:

  - Customer name and group

  - Customer contact information

  - Customer sales orders and details about those sales orders:
    - Customer name and address information

- Various sales order dates
- Contact information for the order taker
- Customer contact information
- Order lines
- Shipments that show how and when a sales order was shipped

Prerequisites

The prerequisites differ, based on the version of Microsoft Dynamics 365 that has been deployed for your organization.

Prerequisites if you use Supply Chain Management

If Supply Chain Management has been deployed for your organization, the system administrator must publish the **Sales orders** mobile workspace. For instructions, see Publish a mobile workspace.

Prerequisites if you use Dynamics 365 for Operations version 1611 with platform update 3 or later

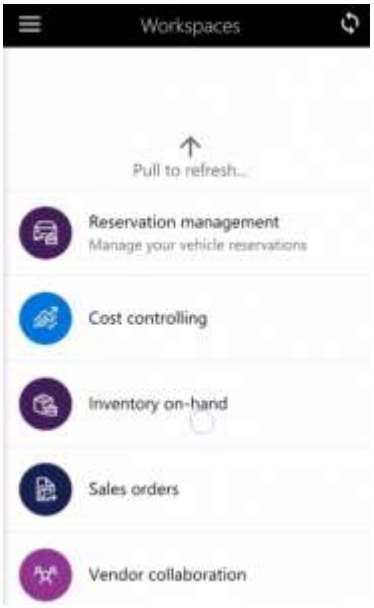
If Dynamics 365 for Operations version 1611 with platform update 3 or later has been deployed for your organization, the system administrator must complete the following prerequisites.

Expand table

Prerequisite	Role	Description
Implement KB 4013633.	System administrator	KB 4013633 is an X++ update or metadata hotfix that contains the **Sales orders** mobile workspace. To implement KB 4013633, your system administrator must follow these steps: 1. Download the metadata hotfix from Microsoft Dynamics Lifecycle Services. 2. Install the metadata hotfix. 3. Create a deployable package that contains the **SCMMobile** model, and then upload the deployable package to Lifecycle Services. 4. Apply the deployable package.
Publish the **Sales orders** mobile workspace.	System administrator	See Publish a mobile workspace.

Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the available workspaces for your company is shown. Note that if your system administrator publishes a new workspace later, you will have to refresh the list of mobile workspaces.



View information about sales orders for a customer by using the Sales order mobile workspace

1. On your mobile device, select the **Sales orders** workspace.
2. Select **View orders for a customer**.
3. Use account or customer name information to find the customer.
4. Select the customer.
5. Select **Contact information** or **Sales orders**. If you select **Sales orders**, a list of sales orders for the customer is shown.
6. Select **Sales order**. You can now view information about sales order lines, information about shipments, customer contact information, and contact information for the order taker.

Last updated on 03/10/2026

Vendor collaboration mobile workspace

Summarize this article for me

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

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This article provides information about the **Vendor collaboration** mobile workspace. This workspace helps your vendors stay up to date about the purchase orders that have been sent to them for approval. They can also view information about new and updated purchase orders and contacts.

This mobile workspace is intended to be used with the finance and operations (Dynamics 365) mobile app.

Overview

The **Vendor collaboration** mobile workspace keeps vendors informed about new purchase orders, so that they can view purchase orders and then respond to them in the web client.

> ⓘ **Note**
>
> The mobile workspace should be used as a supplement to the vendor collaboration web interface, not a replacement for it.

Your vendors can use the **Vendor collaboration** mobile workspace to view new purchase orders that are sent to them for approval. It shows purchase order information, such as products, quantities, and requested receipt dates. Price information is also available, depending on the configuration of each vendor.

A user who signs in as a vendor will see which purchase orders have been responded to, and which purchase orders are still awaiting customer action. For example, a purchase order might be awaiting customer action because the vendor suggested another receipt date, but the customer hasn't yet agreed to that date. The vendor will also see a list of purchase orders that have been confirmed but haven't yet been delivered.

To respond to a purchase order, the vendor must use the vendor collaboration web interface that is available in the web client. There, the vendor can also get more information about the order, such as document attachments, the delivery address per line, and charges that are associated with the vendor.

Vendors that have a special security role can see which contact persons are registered for a vendor account. The same security role lets a vendor view the status of any user request that has been submitted.

The vendor collaboration web interface in the web client must be used to create new contacts and submit new user requests.

The **Vendor collaboration** mobile workspace lets a vendor perform these tasks:

- View new purchase orders that are sent to the vendor.
- View purchase orders that the vendor has responded to, and that are awaiting customer action.
- View purchase orders that have been confirmed but haven't yet been fully received.
- View contact person information that is registered for the vendor account. (This task requires an additional security role.)
- View information about a user request that was submitted by the vendor, and follow the status of the request. (This task requires an additional security role.)

Prerequisites

The prerequisites vary, depending on the version of Microsoft Dynamics 365 that has been deployed for your organization.

Prerequisites if you use Supply Chain Management

If Supply Chain Management has been deployed for your organization, the system administrator must publish the **Vendor collaboration** mobile workspace. For instructions, see Publish a mobile workspace.

Prerequisites if you use Microsoft Dynamics 365 for Operations version 1611 with Platform update 3 or later

If Microsoft Dynamics 365 for Operations version 1611 with Platform update 3 or later has been deployed for your organization, the system administrator must complete the following prerequisites.

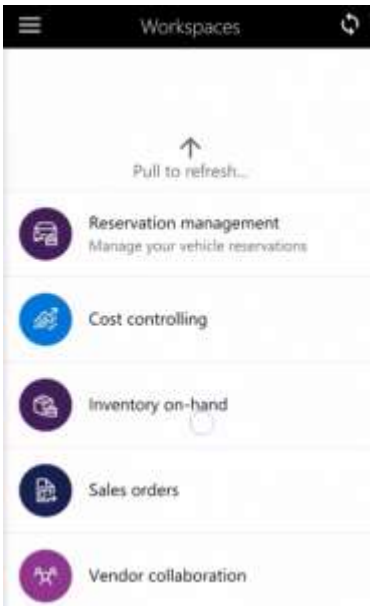
Expand table

Prerequisite	Role	Description
KB 3216943 must be implemented if you're using Platform update 3.	System administrator	KB 3216943 is a binary update that is required if you're using Platform update 3. To implement this KB, the system administrator must follow these steps: 1. Download KB 3216943 from Microsoft Dynamics Lifecycle Services. 2. Install the binary update, which is delivered as a deployable package. For information about how to apply a deployable package, see Apply a deployable package.
KB 4013633 must be implemented.	System administrator	KB 4013633 is an X++ update or metadata hotfix that contains the **Inventory on-hand** mobile workspace. To implement KB 4013633, your system administrator must follow these steps: 1. Download the metadata hotfix from Lifecycle Services. 2. Install the metadata hotfix. 3. Create a deployable package that contains the **SCMMobile** model, and

Prerequisite	Role	Description
		then upload the deployable package to Lifecycle Services. 4. Apply the deployable package.
The **Vendor collaboration** mobile workspace must be published.	System administrator	See Publish a mobile workspace.
The vendor user must have access to the vendor collaboration web interface in the web client and must set up a vendor collaboration user.	Purchasing professionals and the system administrator	Follow the steps in the following articles to set up and work with the vendor collaboration web interface. • Use vendor collaboration to work with external vendors • Manage vendor collaboration users • Set up and maintain vendor collaboration • Use vendor collaboration to work with customers in Supply Chain Managements

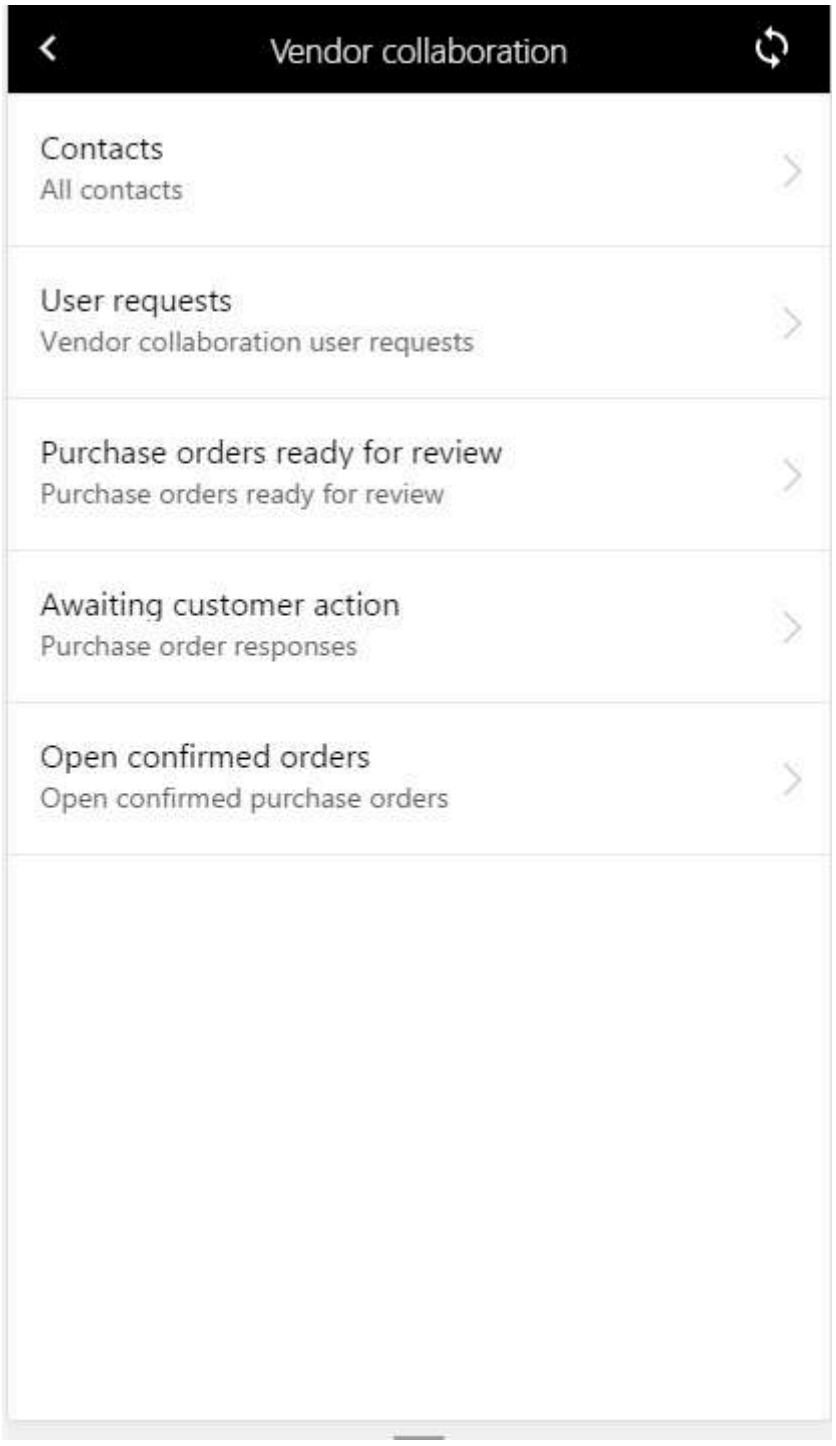
Sign in to the mobile app

1. Start the app on your mobile device.
2. Enter your Microsoft Dynamics 365 URL.
3. The first time that you sign in, you're prompted for your user name and password. Enter your credentials.
4. After you sign in, the available workspaces for your company are shown. Note that if your system administrator publishes a new workspace later, you will have to refresh the list of mobile workspaces.



Use the Vendor collaboration mobile workspace

When you select the **Vendor collaboration** workspace, you'll see the following options.



The **Vendor collaboration** workspace includes the following pages.

Contacts

The **Contacts** page lets you see all the contacts that have been set up for the vendor account. It shows the contact person's name, primary email address, and user alias, if the contact person has an alias. This page also shows whether the contact person's user account is active. When you select a contact, you see contact details, such as the legal entities that the person is a contact for. You also see contact information, such as a telephone number or an alternative email address.

User requests

The **User requests** page lets you see all the user requests that you've submitted via the vendor collaboration web interface. You can also follow the status of those requests. When you select a user request, you can see what was requested, add or deactivate a user, change security, and see which security roles were requested for the user.

Purchase orders ready for review

The **Purchase orders ready for review** page lets you see all the purchase orders that the customer has sent, but that haven't yet been responded to. You can view selected information about the order, such as which products were requested and when those products should be delivered. Price information is also available, depending on the configuration of the vendor.

You can also see whether the purchase order has notes or attachments. However, to open notes and attachments, you must use vendor collaboration web interface in the web client. Select **Purchase order line** to see all the lines together with their details. For each line, an indicator will show whether there are notes or attachments, or whether the delivery address differs from the delivery address that is shown on the header.

To respond to the purchase order, you must use the vendor collaboration web interface in the web client.

Awaiting customer action

The **Awaiting customer action** page lets you find purchase orders that you or another person in your company who has access to vendor collaboration has responded to. The purchase orders are visible in this list only if the customer must take one of the following actions on the purchase order:

- If the purchase order was rejected, the customer must either update or cancel the original order, and then send it again. When the purchase order is sent again, it no longer appears on the **Awaiting customer action** page.
- If the purchase order was accepted with changes, the customer must either update the original order and then send it again for review, or update the order per the requested changes and then confirm it immediately. In both cases, the purchase order no longer appears on the **Awaiting customer action** page.
- If the purchase order was accepted but still appears on the **Awaiting customer action** page, the purchase order wasn't automatically confirmed when it was accepted. It's now waiting for a purchasing agent to change the order status to **Confirmed**. Typically, a purchase order is considered an agreement between the customer and the vendor as

soon as the vendor accepts the order. Therefore, the update to **Confirmed** status is usually just a formality.

When you select a purchase order, additional details appear about the response. You can see the line details and response for every line. The line status shows which of the following responses has been given:

- Accepted
- Rejected
- Accepted with changes
- Substituted/Substitute
- Split into schedule/Schedule line

Note that the **Delivering** field is set to either **Yes** or **No** to indicate whether the lines will be delivered. A line might not be delivered because for the following reasons:

- The line was rejected.
- A substitution was made, and the original line isn't expected to be delivered as requested in the received order.
- The line was split into multiple schedule lines, and the original line isn't expected to be delivered as requested in the received order.

Any changes that have been made to the order line response are shown. However, uploaded notes and attachments aren't shown. To view notes and attachments, you must use the vendor collaboration web interface in the web client.

Open confirmed orders

When the purchase order is confirmed by the customer (that is, the status of the purchase order is changed to **Confirmed**), it appears in the open confirmed order. It will remain in the list until it's registered as received by the customer.

Last updated on 03/10/2026

Publish mobile workspaces

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

> **Important**
>
> The finance and operations (Dynamics 365) mobile app and platform are no longer supported. The platform components supporting the mobile app will be removed in a future update. The mobile app has also been removed from app stores. Previously installed instances of the app will continue to work. For more information, see Removed or deprecated platform features.

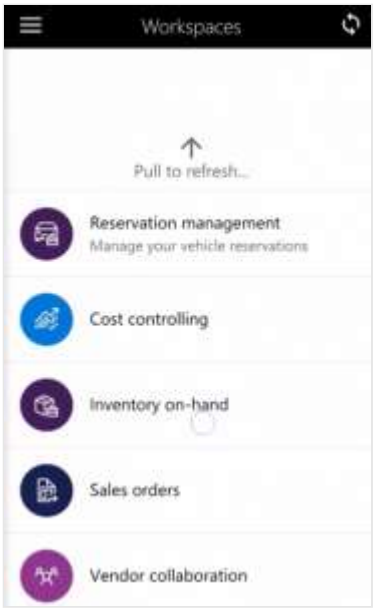
This article describes the steps that system administrators must follow to publish a mobile workspace. You must publish a mobile workspace so that users can access it in the mobile app.

Publish a mobile workspace

To publish a mobile workspace, follow these steps:

1. In your browser, start your web client.
2. Select **Settings** > **Mobile app**.
3. Select the mobile workspace to publish.
4. Select **Publish**.

After you publish a new workspace, users must pull to refresh the list of mobile workspaces.



Last updated on 11/26/2025

Responsible AI FAQs for Dynamics 365 Supply Chain Management

An AI system includes not only the technology, but also the people who use it, the people affected by it, and the environment in which it's deployed. Microsoft's Responsible AI FAQs are intended to help you understand how AI technology works, the choices system owners and users can make that influence system performance and behavior, and the importance of thinking about the whole system, including the technology, the people, and the environment. You can use Responsible AI FAQs to better understand specific AI systems and features that Microsoft develops.

Responsible AI FAQs are part of a broader effort to put Microsoft's AI principles into practice. To find out more, see Microsoft AI principles.

AI-driven features in this app

This app contains a growing list of AI-driven features. To learn about the capabilities and impact of specific features, select a feature name from the list.

- Responsible AI FAQ for AI summaries with Copilot
- Responsible AI FAQ for Analyze demand plans with Copilot
- Responsible AI FAQ for the Confirmed purchase orders with changes workspace
- Responsible AI FAQ for Workload insights with Copilot
- Responsible AI FAQ for the Procurement Agent (production-ready preview)
- Responsible AI FAQ for Generative insights for Demand planning (production-ready preview)

Related information

- Overview of Copilot capabilities in finance and operations apps
- Enable Copilot capabilities in finance and operations apps
- Responsible AI FAQs for the Microsoft Dynamics 365 finance and operations platform
- Responsible AI FAQs for Microsoft Dynamics 365 Commerce
- Responsible AI FAQs for Dynamics 365 Finance
- Responsible AI FAQs for Dynamics 365 Human Resources
- Responsible AI FAQs for Microsoft Dynamics 365 Project Operations

Last updated on 05/13/2026

Responsible AI FAQ for AI summaries with Copilot

This FAQ provides answers to frequently asked questions about the AI technology that's used in AI summaries with Microsoft Copilot in Dynamics 365 Supply Chain Management. It includes key considerations and details about how the AI is used, how it was tested and evaluated, and any specific limitations.

What are AI summaries with Copilot?

AI summaries with Copilot are available on many of the most-used pages in Supply Chain Management. These summaries provide a quick overview of the most important information that's related to the page, personalized for the current user. Summaries can include information such as the number of lines on a purchase order, the number of items in a warehouse, or the number of overdue invoices for a vendor.

The information that Copilot provides depends on the current page and user context. For example, the information can vary based on the other pages that the user works with the most, and it's limited based on the user's security roles and permissions.

What are the capabilities of AI summaries with Copilot?

AI summaries with Copilot use Copilot to generate natural-language summaries of information that's available on the pages where they appear and other related pages. Copilot uses the *gpt-3.5-turbo* generative AI model to generate the natural-language content. Summaries and content are generated from data records that are stored in the system, including products, purchase orders, and vendors.

You can share your feedback directly with Microsoft if an administrator enables user feedback for Copilot and related experiences. Learn more about how Microsoft's in-product feedback system works in Learn about Microsoft feedback for your organization. Learn more about how to enable user feedback for Copilot and related experiences in finance and operations apps in Enable enhanced user feedback for Copilot and related experiences.

What is the intended use of AI summaries with Copilot?

This section describes the intended use for each of the available summaries.

Product hover and released product summaries

Many different people need product information, including sellers, planners, and purchasers. Unlike product information managers, people who work in these roles often lack deep knowledge of each product and where all its information can be found in the system. Product summaries provide high-level information about what each product is, its availability, and its purchase and sale information.

Purchase order summaries

Each purchase order summary shows an overview of a selected purchase order's status. For example, it can show the number of partially or fully confirmed, received, and invoiced lines. It can also show other selected insights, such as a summary of lines that are overdue or nearly overdue. Each summary helps focus users' attention on the most important or unusual things about an order.

Sales order summaries

Sales order summaries are similar to purchase order summaries, but for sales orders.

Vendor summaries

Intelligent vendor summaries include information about vendor on-hold status, active contracts, rebates, open purchase orders, outliers, and risks. Each summary indicates the vendor's account number, vendor currency, and accounting currency. Purchase orders, posted vendor invoices, and payment status information are provided. Overdue vendor invoices, purchasing agreements, and active rebates are also shown. Risks arise from overdue purchase lines, delivery trends, vendor history, and potential foreign exchange losses from outstanding invoices.

Traceability activity summaries

The Traceability Add-in for Dynamics 365 Supply Chain Management collects activity events across the supply chain and product lifecycle, but it isn't always easy for users to get a quick picture of a tracked object's history. Copilot analyzes the many activity events associated with a selected tracked object and then provides a natural-language summary that helps you quickly understand the object's history. This summary is shown at the top of the full list of activity events.

Traceability where-used summaries

Traceability provides a comprehensive where-used insight report, which lets you track the usage of items across various processes. Copilot offers detailed where-used summaries at both the item-number and quantity levels, allowing for precise analysis.

How were AI summaries with Copilot evaluated? What metrics are used to measure performance?

AI summaries with Copilot underwent substantial testing before they were released. They rely on user feedback to report inappropriate content.

If you encounter inappropriate generated content, report it to Microsoft by using this feedback form: Report abuse. Your feedback helps improve the functionality.

Microsoft might disable Copilot-driven features for selected customers if abuse of the functionality is detected.

What are the limitations of AI summaries with Copilot? How can users minimize the impact of these limitations?

AI summaries with Copilot use Copilot to generate summaries of information on the page where they appear and other related pages. Not all pages are used.

The generated content should never be used without manual review or supervision.

What operational factors and settings allow for effective and responsible use of AI summaries with Copilot?

When you use the feature, follow these recommendations:

- Make sure that your company has enough control over data permissions to ensure that business data can't be manipulated to influence AI data processing in an undesirable way.
- Always review the generated summaries before you make decisions based on the information that's provided.

Related information

- AI summaries with Copilot

- Transparency note for Copilot data security and privacy in Microsoft Power Platform

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Last updated on 05/07/2026

Responsible AI FAQ for Analyze demand plans with Copilot

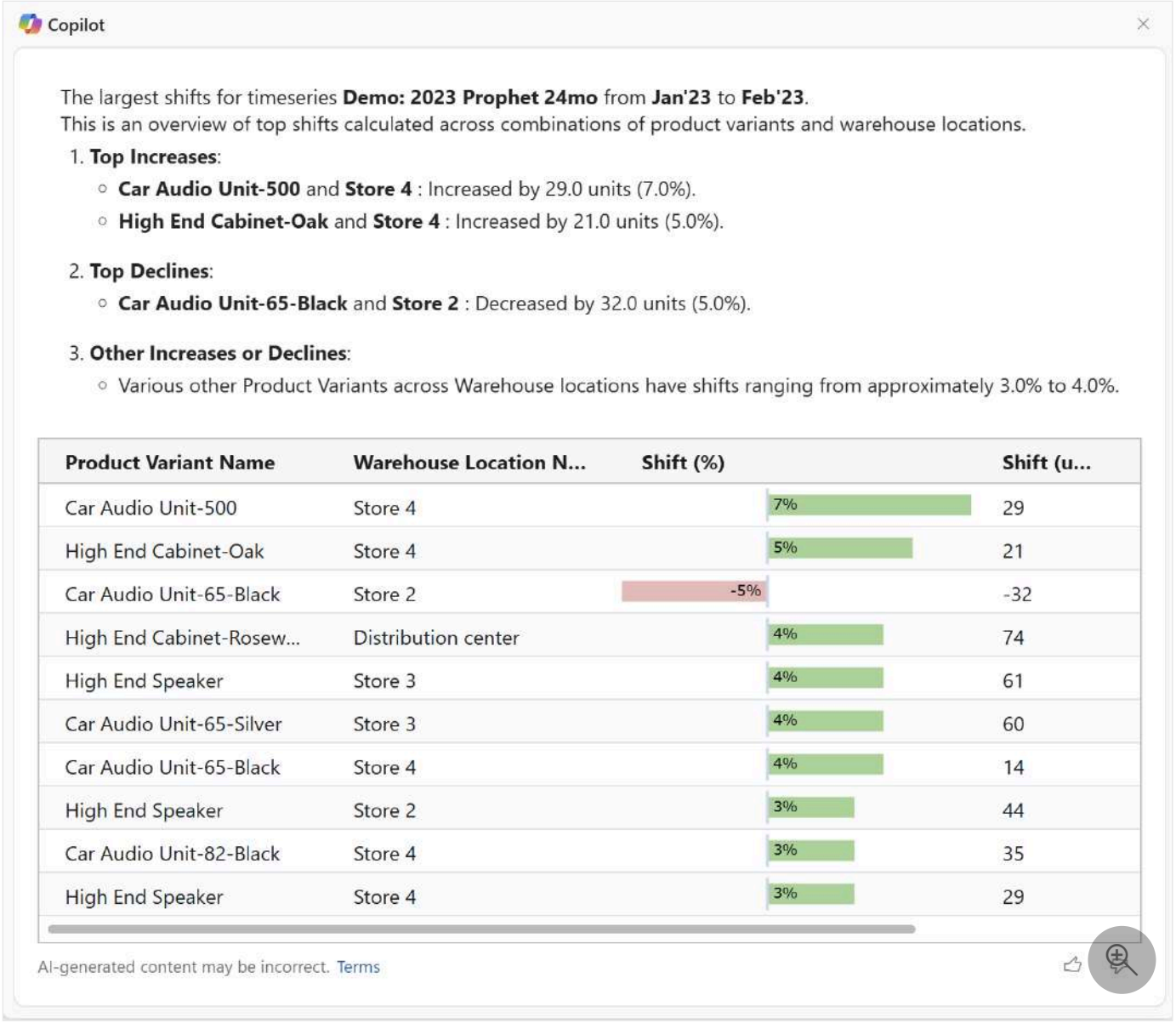
This FAQ provides answers to frequently asked questions about the AI technology that's used in the *Analyze demand plans with Copilot* feature for Demand planning in Microsoft Dynamics 365 Supply Chain Management. It includes key considerations and details about how the AI is used, how it was tested and evaluated, and any specific limitations.

What is Analyze demand plans with Copilot?

Demand planning in Supply Chain Management lets you use Copilot to analyze your demand plans. You can select from a set of predefined questions to analyze data for shifts, trends, anomalies, and accuracy. The results are delivered as a summary using natural language and visualizations.

By using Copilot, you can quickly gain valuable insights across multiple dimensions. You can use these insights to improve operational efficiency, increase profitability, and improve customer satisfaction.

For example, a demand planner is interested in learning the most significant factors that contributed to a notable shift between two consecutive periods. The planner opens the relevant worksheet, selects a point of interest on the chart, and then selects a predefined question to ask Copilot. Copilot's response might resemble the following example.



What are the capabilities of Analyze demand plans with Copilot?

This feature lets users select from a set of predefined questions to analyze their multidimensional demand plans. The system queries the data in each dimension and for all dimensions. The results are delivered as a summary using natural language and visualizations.

Copilot respects all of the filters applied to the data set, so only the filtered data is analyzed. Increasing and decreasing results are highlighted when found.

Copilot allows you to analyze data for the following metrics:

- Shifts *(Compares two consecutive periods of days, weeks, or months)*
- Trends *(Compares six consecutive periods of days, weeks, or months)*
- Deviations *(Compares two time series)*
- Anomalies

What is the intended use of Analyze demand plans with Copilot?

Demand planners spend considerable time analyzing demand plans. This feature helps these users gain valuable insights into demand plans. It helps them make informed decisions by providing a single result that summarizes the most significant shifts, trends, anomalies, or accuracies that occurred across several dimensions. Planners can then adjust their plans based on the detected results.

How was Analyze demand plans with Copilot evaluated? What metrics are used to measure performance?

The system underwent substantial testing before it was released. It relies on user feedback to report inappropriate content.

If you encounter inappropriate generated content, report it to Microsoft by using this feedback form: Report abuse. Your feedback helps improve the functionality.

Microsoft might disable Copilot-driven features for selected customers if abuse of the functionality is detected.

What are the limitations of Analyze demand plans with Copilot? How can users minimize the impact of these limitations when they use the system?

Copilot shows a maximum of ten results. These results are ordered by significance. Results of equal significance are ordered arbitrarily. Although there might be additional results that have the same significance as the tenth result, Copilot doesn't indicate that those results exist.

As of January 2025, AI-generated results are available in all languages supported by Demand planning.

AI-generated content should not be used without manual review or supervision.

Summaries and terminology provided by Copilot in languages other than English may be less accurate than their English language versions.

What operational factors and settings allow for effective and responsible use of Analyze demand plans with Copilot?

When you use this feature, follow these recommendations:

- Make sure that your company has enough control over data permissions to ensure that business data can't be manipulated to influence AI data processing in an undesirable way.
- Before you make decisions about modifications to the demand plan, always review and study the generated response, and compare it with actual demand plan data.

Related information

- Analyze demand plans with Copilot
- Transparency note for Copilot data security and privacy in Microsoft Power Platform

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Last updated on 05/07/2026

Responsible AI FAQ for the Confirmed purchase orders with changes workspace

09/11/2025

This Responsible AI FAQ provides answers to frequently asked questions about the AI technology that's used in the **Confirmed purchase orders with changes** workspace in Microsoft Dynamics 365 Supply Chain Management. It includes key considerations and details about how the AI is used, how it was tested and evaluated, and any specific limitations.

What is the Confirmed purchase orders with changes workspace?

The **Confirmed purchase orders with changes** workspace helps procurement professional quickly and easily assess, manage, and follow up on changes that are made to purchase orders after they're confirmed. It uses Copilot to generate natural-language change summaries and draft communications for vendors.

What are capabilities of the Confirmed purchase orders with changes workspace?

The feature takes changes to confirmed purchase orders as input and then identifies the downstream orders that are related to those purchase orders. The feature uses Copilot to generate natural-language summaries and draft communications for vendors. Copilot uses the *gpt-3.5-turbo* generative AI model to generate the natural-language content. Summaries and content are generated from data records stored in the system, including products, purchase orders, and vendors.

What is the intended use of the Confirmed purchase orders with changes workspace?

The **Confirmed purchase orders with changes** workspace is intended to help procurement professional review changes to confirmed purchase orders and mitigate the downstream impacts. It supports a workflow where the purchaser first reviews and reconfirms changed purchase orders that have little or no downstream impact. It then provides tools to help the purchaser assess and mitigate the impact of changes that have significant downstream impact.

Messaging text that Copilot generates isn't intended to be used without manual review or supervision.

How was the Confirmed purchase orders with changes workspace evaluated? What metrics are used to measure performance?

The **Confirmed purchase orders with changes** workspace underwent substantial testing before it was released. It relies on user feedback to report inappropriate content.

If you encounter inappropriate generated content, report it to Microsoft by using this feedback form: Report abuse. Your feedback helps improve the functionality.

Microsoft might disable the Copilot-driven features for selected customers if abuse of the functionality is detected.

What are the limitations of the Confirmed purchase orders with changes workspace? How can users minimize the impact of its limitations when they use it?

The **Confirmed purchase orders with changes** workspace uses Copilot to generate a summary of impact and proposed communication content.

The generated content should never be used without manual review or supervision.

What operational factors and settings allow for effective and responsible use of the Confirmed purchase orders with changes workspace?

When you use the feature, follow these recommendations:

- Make sure that your company has sufficient control over data permissions to ensure that business data can't be manipulated to influence AI data processing in an undesirable way.
- Always review generated summaries, and study the detailed lists of changes and impacts, before you make decisions about reconfirming purchase orders.
- Always review generated communication drafts before you send them by email or post them to a Teams message.

Related information

- Review and accept changes to confirmed purchase orders
- Transparency note for Copilot data security and privacy in Microsoft Power Platform

Responsible AI FAQ for Workload insights with Copilot

This FAQ provides answers to frequently asked questions about the AI technology that is used in the *Workload insights with Copilot in the Warehouse Management mobile app* feature in Microsoft Dynamics 365 Supply Chain Management. It includes key considerations and details about how the AI is used, how it was tested and evaluated, and any specific limitations.

What is the Workload insights with Copilot feature?

The *Workload insights with Copilot* feature provides visibility into the warehouse workload when workers start their shifts. It uses Copilot to generate natural-language work insights.

What are the capabilities of Workload insights with Copilot?

The *Workload insights with Copilot* feature provides insights into available work headers, pick lines, and receiving lines. It also provides information about available workforce and Copilot-generated insights into pending warehouse work.

What is the intended use of Workload insights with Copilot?

The *Workload insights with Copilot* feature is intended to help workers understand the current warehouse workload through Copilot-generated analyses and summaries of warehouse trends. It helps workers plan their day, so that they can be as effective as possible during their shift.

How was the Workload insights with Copilot feature evaluated? What metrics are used to measure performance?

The *Workload insights with Copilot* feature underwent substantial testing before it was released. It relies on user feedback to report inappropriate content.

If you encounter inappropriate generated content, report it to Microsoft by using this feedback form: Report abuse. Your feedback helps improve the functionality.

Microsoft might disable the Copilot-driven features for selected customers if abuse of the functionality is detected.

What are the limitations of Workload insights with Copilot? How can users minimize the impact of those limitations when they use the feature?

The *Workload insights with Copilot* feature uses Copilot to generate a summary of warehouse insights into the workload.

The generated content should never be used without manual review or supervision.

What operational factors and settings allow for effective and responsible use of Workload insights with Copilot?

When you use the *Workload insights with Copilot* feature, follow these recommendations:

- Make sure that your company has sufficient control over data permissions to ensure that business data can't be manipulated to influence AI data processing in an undesirable way.
- Always review generated summaries, and study the detailed information before you make decisions about workload actions.

Related information

- Workload insights with Copilot in the Warehouse Management mobile app
- Transparency note for Copilot data security and privacy in Microsoft Power Platform

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Last updated on 05/08/2026

Responsible AI FAQ for the Procurement Agent (production-ready preview)

[This article is prerelease documentation and is subject to change.]

This FAQ provides answers to frequently asked questions about the AI technology that is used in the Procurement Agent that is available with Copilot in Microsoft Dynamics 365 Supply Chain Management. It includes key considerations and details about how the AI is used, how it was tested and evaluated, and any specific limitations.

> **Important**
>
> - This is a production-ready preview feature.
> - Production-ready previews are subject to supplemental terms of use.

What is the Procurement Agent?

The Procurement Agent provides the following capabilities that help purchasers communicate with vendors and handle change requests as part of their daily work:

- *Send follow-up emails to vendors on purchase orders*
- *Speed up updates in purchase orders based on incoming vendor emails*
- *Review the impact of purchase order changes from vendors* - automatically triggering the impact analysis for incoming change requests
- *Simulate if purchase order changes have a downstream impact* - manually triggering the impact analysis

Send follow-up emails to vendors on purchase orders overview

It can be tedious to follow up on purchase orders that are sent to vendors if the vendors haven't yet confirmed or delivery is late. It can take significant time to find the correct orders in the system and then write a specific email for each order.

To streamline this process, the system helps you perform the following actions:

- Create a query to find the purchase orders to follow up on. Configuration is user specific.

- Configure the email generation process for each group of purchase orders that the query finds. The system provides two default queries. One query is used to follow up on unconfirmed purchase orders, and the other query is used to follow up on late deliveries. Queries are unique for each user. Therefore, you can create and adjust them to meet your specific business needs.
- Review and modify drafted emails, copy them to your email client, and/or send them. The data that is used to generate the emails comes from the purchase order lines and/or related tables.

Speed up updates in purchase orders based on incoming vendor emails overview

This capability helps speed up communications with vendors about purchase orders. It enables Copilot to read emails from all or specific vendors.

The Procurement Agent can determine what each email is about (for example, whether it's a purchase order confirmation or a purchase order change request) and which purchase order numbers it applies to. Copilot then matches the information that it extracts from the email to fields in the system and indicates whether there are any changes. Therefore, the purchaser can review just the information that Copilot provides and update the fields in the system as required. They don't have to manually open the purchase order and do all the work themselves. They can then spend the time that they save on other tasks that add more value.

Review the impact of purchase order changes from vendors overview

Impact analysis automatically analyzes the downstream impact of change requests that arrive through e-mails (Procurement Agent - Supplier communications) and through the vendor collaboration interface if configured and enabled.

Purchasing teams regularly receive change requests from suppliers, including delivery delays, quantity reductions, and purchase order cancellations. Each change request can potentially affect safety stock, production schedules, internal transfers, and customer commitments.

In most cases, purchasers don't want to spend time on changes to purchase orders with no downstream impact. Impact analysis gives purchasers the confidence, visibility, and ability to quickly identify these changes that can be safely accepted, apply them right away, and move on.

For changes with downstream impact on orders, or ones that cause inventory levels to go below certain thresholds, purchasers need to quickly understand the details, consequences, and decide how to respond. Purchasers often follow up with the supplier. If purchasers must accept a change with impact, they now know straight away which internal stakeholders to be informed.

Simulate whether purchase order changes have impact overview

The impact simulation capability allows purchasers and planners to test the impact of purchase orders changes that weren't directly communicated through supported channels. This capability could be used to proactively test changes on high-priority or at-risk purchase orders, or to analyze the impact of changes communicated through non-supported channels such as telephone.

What are the capabilities of the Procurement Agent?

Send follow-up emails to vendors on purchase orders

The *Send follow-up emails to vendors on purchase orders* capability uses Copilot to generate natural language email texts and subjects, based on information from purchase orders that are found through user-specified queries. User-defined settings such as language tone, length, and urgency are used to adjust the resulting text. Copilot uses the gpt-4o generative AI model to generate the natural language content.

Speed up updates in purchase orders based on incoming vendor emails

The *Speed up updates in purchase orders based on incoming vendor emails* capability must be set up to access the relevant users' email and analyze emails from either all or specifically selected vendor email addresses. Each relevant vendor record must include an email address to enable Copilot to recognize it.

You must also set the frequency of the email analysis.

Copilot classifies each vendor email that it reads into one of the following categories, based on the intent that it detects:

- *Purchase order confirmation* – Copilot detected that the purpose of the email is to confirm one or more purchase orders (all the purchase orders that are mentioned in the email).

- *Requested changes* – Copilot detected that the email contains at least one requested change to a purchase order. For example, the vendor can't deliver the full quantity that was requested, or they can't meet the requested delivery date.
- *Rejection* – Copilot detected that the vendor can't deliver the purchase order as a whole.
- *Other* – Copilot couldn't identify any of the previous intents.

For the first two categories, the system identifies which purchase orders the email is related to and matches the information in the email to information in the system.

For messages where the vendor requested a change, the system indicates the current information plus the new information in the purchase order. Therefore, the changes are easier to understand.

A review page lists the emails that were identified and classified, and summarizes the information in each message. From this page, you can apply the changes directly to the purchase order that is stored in the system.

Review the impact of purchase order changes from vendors

The *Review the impact of purchase order changes from vendors* capability only runs automatically on change requests, and only on sources selected during setup.

If you choose to enable impact analysis based on changes received in email, the analysis only runs on emails classified as *Requested changes*, and during the specific conditions set up for the *Speed up updates in purchase orders based on incoming vendor emails* capability. In other words, the analysis only runs for specified mailboxes and vendors.

If you enable the vendor collaboration interface, this capability runs on *In external review requires action* messages only.

Simulate whether purchase order changes have impact

To use the *Simulate whether purchase order changes have impact* capability, enable the *Procurement agent - Impact analysis* feature in feature management. This capability runs only when you change a date or quantity of a purchase order and select *Simulate impact*.

What is the intended use of the Procurement Agent?

The Procurement Agent has the following intended uses:

- The *Send follow-up emails to vendors on purchase orders* capability streamlines the process of following up on purchase orders. By default, it identifies orders that are delayed or

unconfirmed. It presents a draft message to purchasers, who can review and modify it, copy it to their email client, and send it.

- The *Speed up updates in purchase orders based on incoming vendor emails* capability saves purchasers time by automatically reading emails, extracting information from them, and mapping that information to information in the system.
- The *Review the impact of purchase order changes from vendors* capability saves purchasers time in manually analyzing planning data to understand if a change request has downstream impact.
- The *Simulate whether purchase order changes have impact* capability enables purchasers and planners to stay on top of potential purchase order changes and to test different scenarios.

How was the Procurement Agent evaluated? What metrics are used to measure performance?

The Procurement Agent underwent substantial testing with demo data examples before it was released.

Microsoft relies on users to report inappropriate generated content. If you encounter inappropriate content, report it to Microsoft by using the following feedback form: Report abuse. Your feedback helps improve the functionality.

Microsoft might disable Copilot-driven features for selected customers if it detects abuse of the functionality.

What are the limitations of the Procurement Agent? How can users minimize the impact of these limitations?

The *Send follow-up emails to vendors on purchase orders* capability uses Copilot to generate an email based on information that it retrieves through user-specified queries. You can select which fields to include in the email text. The information in the email is limited to data that comes from purchase orders, purchase order lines, and related information (such as vendor records). Primary contacts that are linked to emails and vendors are used to identify the recipient's name and company. User data is used to identify the sender's name and company.

The generated content should never be used without manual review or supervision.

The system doesn't send emails automatically, unless you explicitly configure it to do so. You're responsible for reviewing and sending each email.

To configure the system to send emails automatically, an admin must turn on the *(Production ready preview) Procurement Agent – Supplier communications - automatically sending follow-up emails* feature in Feature management. Use this capability only after you test and review drafted emails over a long period.

The *Speed up updates in purchase orders based on incoming vendor emails* capability is intended only for vendor collaboration (classification of emails that are related to purchase orders). Therefore, during setup, you must specify which vendors' email addresses (either all or only specific ones) the system can read emails from.

What operational factors and settings allow for effective and responsible use of the Procurement Agent?

When you use the feature, follow these recommendations:

- Make sure that your company has enough control over data permissions to ensure that business data can't be manipulated to influence AI data processing in an undesirable way.
- When you use the *Send follow-up emails to vendors on purchase orders* capability, always review each email and subject before you use the information that is provided or make decisions based on it.
- When you use the *Speed up updates in purchase orders based on incoming vendor emails* capability, always review each email, intent extraction, and data before you use the information that is provided or make decisions based on it.
- When you use the *Review the impact of purchase order changes from vendors* capability, review the detailed impact before determining if a change is safe to accept and apply to the purchase order.

Related information

- Supplier communications features of the Procurement Agent

Last updated on 05/13/2026

Responsible AI FAQ for Generative insights for Demand planning (production ready preview)

08/18/2025

[This article is prerelease documentation and is subject to change.]

This FAQ provides answers to frequently asked questions about the AI technology that's used in Generative insights for Demand planning in Microsoft Dynamics 365 Supply Chain Management. It includes key considerations and details about how the AI is used, how it was tested and evaluated, and any specific limitations.

> **Important**
>
> - This is a production-ready preview feature.
> - Production-ready previews are subject to supplemental terms of use.

What are Generative insights for Demand planning?

Generative insights for Demand planning analyzes your demand plans to detect patterns (such as seasonality and signal correlations) across products and locations. The feature provides answers using natural language and visualizations.

By using Generative insights, you can quickly gain valuable insights across multiple dimensions. You can use these insights to improve operational efficiency, increase profitability, and improve customer satisfaction.

Learn more about this feature in Generative insights for Demand planning (preview).

What are the capabilities of Generative insights?

This feature lets you analyze your multidimensional demand plans for patterns such as seasonality and signal correlations. The system queries the data for all dimensions, and returns natural-language summaries, graphs, and other visualizations.

What is the intended use of Generative insights?

Demand planners spend considerable time analyzing demand plans. This feature helps them gain valuable insights into the patterns that can be found in their data. These insights help planners make informed decisions about when to initiate promotions and also help them to predict the potential impact of those promotions.

How was Generative insights evaluated? What metrics are used to measure performance?

The system underwent substantial testing before it was released. It relies on user feedback to report inappropriate content.

If you encounter inappropriate generated content, report it to Microsoft by using this feedback form: <Report abuse. Your feedback helps improve the functionality.

Microsoft might disable Copilot-driven features for selected customers if the abuse of the functionality is detected.

What are the limitations of Generative insights? How can users minimize the impact of these limitations when they use the system?

A maximum of fifteen seasonal patterns can be detected. Generative insights calculates a confidence score for each seasonal pattern and sorts them according to that score. Results of equal confidence are ordered arbitrarily.

Products with insufficient data are grouped into a system-generated seasonal pattern named *Insufficient data*. This approach ensures that all data is analyzed and assigned a classification.

The AI-generated results are always presented in US English.

AI-generated content should not be used without manual review or supervision.

What operational factors and settings allow for effective and responsible use of Generative insights?

When you use this feature, follow these recommendations:

- Make sure that your company has enough control over data permissions to ensure that business data can't be manipulated to influence AI data processing in an undesirable way.
- Before you make decisions about modifying a demand plan, always review and study the generated response, and compare it with actual demand plan data.

Related information

- Generative insights for Demand planning (preview)
- Transparency note for Copilot data security and privacy in Microsoft Power Platform

Development and administration for finance and operations apps

> **Note**
>
> Community interest groups have now moved from Yammer to Microsoft Viva Engage. To join a Viva Engage community and take part in the latest discussions, fill out the Request access to Finance and Operations Viva Engage Community form and choose the community you want to join.

Development and administration for finance and operations apps includes:

- Administrator experience and Lifecycle Services
- Developer experience
- Intelligence
- Mobile apps
- Data management and data entities
- Office integration

Developer experience

The developer experience is based on modern tooling that uses Visual Studio and .NET components.

- The development tools are decoupled from any running environment. You develop against local, XML-based files, not the online database.
- Microsoft Visual Studio is the development environment. Finance and operations customizes the Visual Studio environment to provide you with a smooth and familiar experience.
- The X++ compiler generates Common Intermediate Language (CIL) for all features. CIL is the same intermediate language used by other .NET-based (managed) languages, such as the C# programming language.
- You can leverage the browser-based client and design patterns for forms to provide an improved end-user experience.
- The Application Lifecycle Management (ALM) system supports build automation, test automation, and deployment of models to the cloud.

For more information, see Develop and customize home page.

Administrator experience and Lifecycle Services

Finance, Supply Chain Management, and Commerce are cloud-hosted. As an IT professional, use Microsoft Dynamics 365 Lifecycle Services to monitor and tune your environments, deploy features, and stay up to date with recent hotfixes. Within your deployment, you can configure security, and manage when processes run. You can also use Lifecycle Services when you're called on to support business intelligence and reporting, mobile apps, Office, and other integrations.

BI and reporting

Finance and operations provides five distinct reporting experiences. It offers specialized tools to meet the complex and diverse reporting needs of various functions throughout the organization.

- Operational views – Designed to address the specific needs of a given business persona.
- Business documents – Static documents used to capture and exchange processed business data.
- Analytical tools and visualizations – Personalized presentations of logical calculations that allow the user to explore their data.
- Electronic reporting – Tool used to configure formats for electronic documents.
- Financial reporting – Designed to provide in-depth accounting management tools based on standard views of financial activities across legal entities.

Mobile apps

The finance and operations mobile app empowers your organization to mobilize its business processes. After your IT admin enables the mobile workspaces feature for your organization, users can sign in to the app and immediately begin to run business processes from their mobile devices. The Dynamics 365 for finance and operations mobile app includes the following features that can help increase productivity:

- Users can view, edit, and act on business data, even if they have intermittent network connectivity or their mobile devices are offline. When a device reestablishes a network connection, offline data operations automatically synchronize with finance and operations.

- IT admins or developers can build and publish mobile workspaces that are tailored to their organization. The app uses your existing code assets, so you don't have to re-implement your validation procedures, business logic, or security configuration.
- IT admins or developers can easily design mobile workspaces by using the point-and-click workspace designer that's included with the finance and operations web client.
- IT admins or developers can optionally optimize the offline capabilities of workspaces by using the Business logic extensibility framework. Because data continues to be processed while a device is offline, your mobile scenarios remain rich and fluid, even if devices don't have constant network connectivity.

Data management and data entities

You can easily integrate data from finance and operations with Microsoft and non-Microsoft data sources by using Dataverse, Power Apps, and Power BI. For more information, see Data entities overview.

Office integration

The Microsoft Office integration capabilities provide users with a productive environment that helps them get the job done by using Office products. For more information, see Office integration overview.

eLearning courses

For online courses and training, see Dynamics 365 finance and operations.

Last updated on 04/13/2026

Welcome to Dynamics 365 Finance

This article provides a list of the help topics and other resources for the financial management features in Microsoft Dynamics 365 Finance.

Select a feature area to learn more about it.

- Accounts payable
- Accounts receivable
- Asset leasing
- Business performance analytics
- Business performance planning
- Budgeting
- Cash and bank management
- Cost accounting
- Financial reporting
- Fixed assets
- Finance insights
- Public sector

Additional resources

Blogs

- Microsoft Dynamics 365 blog
- Financials blog
- Microsoft Dynamics Operations Partner Community Blog

Videos

Check out the how-to videos that are now available on the Microsoft Dynamics 365 YouTube Channel.

Country/region functionality

Country/region regulations affect tax setup and other areas of financial management. To learn about country/region-specific functionality, see Localization and regulatory features.

Additional content

Supply chain management functionality covers parts of the procure-to-pay process that include:

- Requisitioning
- Ordering
- Receiving
- Invoicing
- Paying for the goods and services your organization purchases

For information about the capabilities for managing purchases, inventory, and manufacturing, see Welcome to Dynamics 365 Supply Chain Management.

eLearning courses

For online courses and training, check out Dynamics 365 Finance learning paths.

Last updated on 03/11/2026

Welcome to Dynamics 365 Commerce

This article provides an overview of Microsoft Dynamics 365 Commerce.

Dynamics 365 Commerce, built on the proven Dynamics 365 Retail capabilities, delivers a comprehensive omnichannel solution that unifies back office, in-store, call center, and digital experiences. Dynamics 365 Commerce enables you to build brand loyalty through personalized customer engagements, increase revenue with improved employee productivity, optimize operations to reduce costs and drive supply chain efficiencies, and ultimately deliver better business outcomes.

This release enables the creation of digital experiences by using built-in web authoring and development tools to produce engaging and intelligent digital storefronts. A connected marketing and headless commerce platform further enable the seamless management of content, assets, promotions, inventory, and pricing across all channels.

- **Everything to build and run digital commerce:** Streamline your business with an end-to-end commerce solution that scales to your needs across traditional and emerging channels. Built-in web authoring and development tools enable you to create engaging intelligent digital storefronts, while a connected marketing and headless commerce platform enables seamless management of content, assets, promotions, inventory, and pricing across channels.

- **Build loyalty and exceed customer expectations:** Use clienteling tools to gain a comprehensive view of your customer and respond to their needs at every level of engagement, based on customer profile, history, and preferences that flow across physical and digital channels. Empower your employees to foster lasting relationships through AI-driven recommendations, customer insights, and loyalty programs that elevate brand appeal.

- **Flexible and intelligent omnichannel experience:** Unify physical and digital commerce by providing consistent experiences to customers across cloud search and discovery, product reviews, wish lists, inventory, gift cards, and loyalty. Allow customers to purchase when, how, and where they want, on any device, while providing choice around modern payment methods and product collection or delivery.

- **Streamline operations using AI in the cloud:** Drive omnichannel commerce experiences and integrated, optimized back-office operations through ingrained, pervasive, and context-

aware cloud intelligence. Use advanced merchandising, inventory management, distributed order management, and pricing and promotion to innovate and stay ahead of competition. Derive insights by visualizing and analyzing comprehensive and consistent data across all aspects of your business. Use AI-driven technologies to provide accessible websites, protect your business against payment fraud, and efficiently moderate user-generated content like ratings and reviews.

Core concepts and tasks

Select a feature area to learn more about it.

- Configure a Commerce preview environment
- Commerce architecture
- Set up your channels
- Merchandizing your products and services
- Manage your orders
- Manage your customers
- Manage your financials
- Manage your e-Commerce site
- Commerce development and extensibility

eLearning courses

For online courses and training, check out Dynamics 365 Commerce.

Last updated on 03/13/2026

Human Resources home page

This article provides a list of the help topics and other resources in Microsoft Dynamics 365 Human resources.

Select a feature area to learn more about it.

- Employee self service
- Personnel management
- Task management
- Leave and absence
- Recruiting
- Benefits
- Benefits management
- Compensation management
- Employee development
- Learning
- Workflow system

Human Resources connects people and operations data to help you optimize workforce costs and take care of your employees.

Dynamics 365 Finance

For information about Dynamics 365 Finance, see Finance home page.

Additional resources

eLearning courses

For online courses and training, check out Dynamics 365 Human Resources.

Last updated on 07/01/2026